BROADCOM SOFTWARE

CA Service Management - 17.4

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Release Information

Explore CA Service Management's latest enhancements, features, and cumulative patches.

- CA Service Management 17.4 Release Notes
- CA Service Management 17.4.5.0 Release Notes
- CA Service Management 17.4.4.0 Release Notes
- CA Service Management 17.4.3.0 Release Notes
- CA Service Management 17.4.2.0 Release Notes
- CA Service Management 17.4.1.0 Release Notes

CA Service Management 17.4 Release Notes

Learn about the new features, enhancements, and deprecated items with CA Service Management 17.4.

What's New

CA Service Management 17.4 includes the following new features:

- CA SDM and Jira Agile Management Integration
- CA Virtual Analyst Integration with MS Teams
- **Catalog Cache versioning for Service Point**: Service Point can now identify the availability of a new version of a file by using the unique file version identifier, saving business users' efforts to delete the browser cache whenever new roll-ups or T-fixes are applied.
- MS Graph API Support for OAuth 2.0: SDM Maileater supports IMAP/POP3 as one of the protocols for the inbound emails. This feature is to introduce MS Graph API as a protocol for the inbound emails processed by SDM Maileater.
- Lift and Shift cloud migration of CA Service Management. For more information, look into Lift and Shift Cloud Migration.

What's Deprecated?

The following items are deprecated and not supported on CA Service Management 17.4:

- Support Automation
- Service Catalog integration with CA Business Service Insight
- We recommend customers add CXF Web Services. Support for Axis-based web services would be dropped from June 30, 2025. For more information on CXF Usage, see the following topics:
  - Web Services Management
  - Tips for SOAP Web Services Clients
  - Contents of the Samples Directory

Quick CheckPoints

To install, upgrade, and configure Service Management, check out the following helpful URL links:

- Review the supported platform requirements, before you proceed with installing or upgrading CA Service Management 17.4. For more information, see Supportability Matrix.
- Installation
- Upgrade
- Administration
- Configure
  - Service Point
  - xFlow Interface
- Using
  - Service Point
  - xFlow Interface
- Integration

Fixed Issues - 17.4

Learn about the Fixed Issues in CA Service Management 17.4.

The following issues were fixed and are included in CA Service Management 17.4:

xFlow Interface and Service Point

S. No	Fixed Issue ID	Description
1	Case #33359741	Service Point does not display KD images login when SP is installed on secondary server
2	Case #33381955	CV-GEN-INQUIRY_CATALOG_17.3_Xflow-When clicked 'Cancel Request' of a Request in S

CA Service Desk Manager

S. No.	Fixed Issue ID	Description
1	Case #33083350	Visualizer: Save map
2	Case #33319405	Export functionality stops working frequently
3	Case #33310522	SDM 17.3 RU18 - Failed to Copy SDM Ticket
4	Case #33358665	GRLoader removes line breaks
5	DE68660	Spell Check Not Working Correctly
6	Case #33374738	Assignee field empty after save Knowledge Document
7	DE68967	17.3 RU21 breaks SDM IIS Web Interface (exception in w3wp.exe)
8	Case #33433506	[SDM-Attachments] unable to upload attachments from employee interface
9	Case #33396607	Visualizer Tomcat fails to start
10	Case #33442176	SDM_17.3.22-33442176-A ""Knowledge Manager"" cannot create new ""Top"" Knowledge Categorie

CA Service Catalog

S.No	Fixed Issue ID	Description
1	DE68779	Service Catalog - news message - announcements
2	Case #33382286	Service Catalog - Form - Table Column widths

3	Case #33440375	CV-GEN-INQUIRY_CATALOG_17.3-The label text overlaps in Service Point

Jasper Reports Server 9.0.0

For the vulnerability of ***Log4j1 Removal*** in Datamart installed location, apply the patch with APAR No : 99112251

Known Issues - 17.4

This page talks about the Known Issues for the CA Service Management 17.4 release.

The CA Service Management 17.4 release includes the following known issues:

CA Service Management

Login Issue after installing CA EEM on Linux

Problem: Unable to log in into CA EEM application server after installing CA EEM on Linux (RH Linux or Suse) using the CA Service Management Common Installer 17.4.

Solution: Perform the following steps:

1. Uninstall existing CA EEM. Run the following command from where you have installed CA EEM:
   ```
   command eiamuninstall -DFORCE_UNINSTALL=true
   ```
2. Install CA EEM server (RH Linux or Suse) manually from CA Service Management DVD.
   For more information, see Install CA EEM.

CA Service Desk Manager

Post Installation task fails with pdm_webcache -H option

Problem: Post-Installation task fails with pdm_webcache -H option on SDM Application Server on a Slow network connection.

Solution: Workaround:

1. Extract the CA Service Desk Manager Installer DVD, remove pdm_webcache -H from SDM_patch.xml file from the installation folder location.
2. Retry the installation. SDM Application server is installed successfully.

Unable to establish connection with Microsoft Exchange SMTP server

Problem: Recent changes in JavaMail API has changed certain authentication defaults and the following authentication error message is displayed:

```
No authentication mechanisms supported by both server and client.
```

Workaround: This issue occurs when username and password are provided for the SMTP server which is configured with no authentication mechanism. To work around the issue, follow these steps:

1. Remove the existing mail server configuration.
2. Create a new one leaving **username** and **password** empty.
3. Restart the server.

Post Installation task fails with pdm_webcache -H option

Problem: Post-Installation task fails with pdm_webcache -H option on SDM Application Server on a Slow network connection.

Solution: Workaround:

1. Extract the CA Service Desk Manager Installer DVD, remove **pdm_webcache -H** from **SDM_patch.xml** file from the installation folder location.
2. Retry the installation. SDM Application server is installed successfully.

Unable to edit and save an email Address in EditList

Problem : The edit list functionality fails when attempting to edit a field using the **Edit in List Option.** The following error message is displayed: `error AHD04606: 0 records updated; n attempted update(s) failed - see message(s) below. AHD04605:Can't update - someone else has changed email_address`

Workaround: If you are applying the "**fmtFunc**" attribute to any field in the list htmpl files and have enabled the edit in list functionality for that list, ensure to add a name attribute to the anchor tag. The **fmtFunc** attribute converts the field into a hyperlink (wrapped inside HTML anchor tags): <a name='dis_lnk_in_listedit_<any_value>'

Here **<any_value>** must be replaced with a suitable value, such as field name.

Unable to establish connection with Microsoft Exchange SMTP server

Problem: Recent changes in JavaMail API has changed certain authentication defaults and the following authentication error message is displayed:

```
No authentication mechanisms supported by both server and client.
```

Workaround: This issue occurs when username and password are provided for the SMTP server which is configured with no authentication mechanism. To work around the issue, follow these steps:

1. Remove the existing mail server configuration.
2. Create a new one leaving **username** and **password** empty.
3. Restart the server.

After applying 17.4 GA domsrvr memory growth is observed

Problem: After applying 17.4 GA domsrvr memory growth is observed

Solution: Apply the required hotfix:

Windows APAR no : 99111862

Linux APAR no : 99111863

JasperReports® Server 8.1.1

Derived Tables are incompatible for Fields from Different Objects

Problem: When creating an ad-hoc view, you ideally select fields that belong to an object and derived tables that are linked to this same object. However, derived tables show incompatibility when fields from different objects that are not directly joined to the derived table are pulled in. For example, you can pull in fields belonging to "*chg*" object and derived table fields that are linked to "*chg*". However, you cannot pull "*chg*" fields with derived tables that are linked to "*nr*" object.

Workaround: Close and reopen the ad hoc view. Avoid any incompatible field.

Error message in the Insights log file

Problem: The following error message is displayed in the InsightMSs.log file: [*c.t.c.FreshnessCalculator] - calculateFreshnessFromExpires: HTTP date parsing failed Error.*

Solution: This does not affect the Insights functionality and hence no action is required.

CA Service Catalog

Any new or update to existing SRFs are not loading after upgrading the CA Service Catalog to 17.4.RU2

Problem: Any new or update to existing SRFs are not loading after upgrading the CA Service Catalog to 17.4.RU2

Steps to reproduce the problem:

1. Go to Administration > Events-Rules-Actions > Request/Subscription Item Create
2. Select an existing Rule or create a new Rule.
3. Click the Add button and this will bring you to the Add Action screen
4. In the Action Information Section select:
   a. Type: Process Automation
   b. Click the Magnifying glass next to the Start Request Form field
   c. Click an SRF (or the location of the offering you imported into ITPAM earlier.)
5. Start Request Form Field is populated but the popup window does not disappear.
6. Cursor is stuck as a circle.

Solution:

Download the fix for version 17.4 RU2 from here.

Cannot display charts in Data Views

Problem: Graph type Data Views cannot be rendered in the dashboards due to removed support for Applets.

Solution: As of now there is no workaround.

Case sensitivity for user name in Oracle database

Problem: The user name for CA Service Catalog is case-sensitive in Oracle database.

Solution: Enter the correct user name.

Form Designer style attribute is not supported for Dual List

Problem: Form Designer style attribute is not supported for Dual list.

Solution: There is no workaround available for this issue.

While configuring CA Service Catalog, if the database level collation is changed, the MDB task fails during installation

Problem: MDB database installation fails.

Workaround: Follow these steps:

1. After Configuration of CA Service Catalog - MDB task failure in the installer, navigate to USM_HOME\temp\mdb (for example: C:\Program Files\CA\Service Catalog\temp\mdb1.5\windows\Unicenter_Service_Management-1.0_mssql/DBTable) and modify the xml files for MDB tables as following:
   a. Search for the following: @COLLATE_SENSITIVE@
   b. Replace it with: @COLLATE_INSENSITIVE@
2. Retry upgrading or installing CA Service Management 17.4

Unable to execute CA EEM safex commands and ant commands like 'backup-eem-app' and 'restore-eem-app' in CA Service Catalog machine installed on Windows 2k22 operating system.

Problem: Unable to execute CA EEM safex commands and ant commands like 'backup-eem-app' and 'restore-eem-app' in CA Service Catalog machine that is installed on Windows 2k22 operating system.

Solution: Install Microsoft Visual C++ 2010 x64 Redistributable Package in CA Service Catalog installed machine which can be downloaded from here https://www.microsoft.com/en-us/download/details.aspx?id=26999

Submenu is missing under Catalog Tab with Administrator Privileged Role.

Problem: Submenu is missing under Catalog Tab with Administrator Privileged Role.

Solution: Follow these steps to resolve the issue:

1. Login to EEM using Eiamadmin super user account and select Service Catalog as an Application from the dropdown.
2. Click Manage Access Policies Tab - under Access Policies.
3. Click **Explicit Denies** Tab and click **Dashboard.**
   a. Edit and update filters value for “ACL_restrict_unwanted_Dashboards” access policy by adding “,17.4” to the list.
4. In **Explicit Denies** tab, click **Guinode**.
   a. Edit and update filters value for “ACL_restrict_unwanted_Guinodes” access policy by adding “,17.4” to the list.
   b. Edit and update filters value for “ACL_restrict_r11_Guinodes” access policy by adding “,17.4” to the list.
5. In Explicit Denies tab and click Launchpad.
   a. Edit and update filters value for “ACL_restrict_r11_Launchpads” access policy by adding “,17.4” to the list.
   b. Edit and update filters value for “ACL_restrict_unwanted_Launchpads” access policy by adding “,17.4” to the list.

      NOTE
      Ensure that there are no spaces between the values.

After Configuration of CA Service Catalog-MDB task failure

Problem: After Configuration of CA Service Catalog - MDB task failure in the installer, navigate.

- The following compilation errors occur after deploying the CustomJavaScript customization:
  a. Custom scripts of the customer can fail after jQuery upgrade.
  b. Administrators may have customizations in xsl files such as catalogbrowse.xsl, catalogbrowse_layout8.xsl, cataloggetchildren.xsl, and writing their scripts in custom_form_lib.js or by adding new files under filestore.
  c. Users write form scripts in the form designer page.

     NOTE

     - We recommend that administrators use OOTB Form APIs and JavaScript code.
     - jQuery has been upgraded to the latest version (from 1.6.4 to 3.6.2) as part of 17.4 release. If jQuery APIs that are being used are deprecated or deleted from the latest versions of jQuery, this may break their custom functionality.

Resolution: The customers must validate their custom code and ensure deprecated jQuery or removed APIs are not being used; if they are in use, resolve them by following the jQuery upgrade guidelines at https://jquery.com/upgrade-guide.

Mobile App

Ticket Details Not Displayed

Problem: Analysts cannot view the details of tickets in the mobile app.

Solution: For attributes to appear in the mobile app, administrators must configure and enable them.**Perform the following steps**

1. Configure mobile attributes. Follow Configure the Mobile Attributes for Tickets.
2. To enable the mobile attribute in the mobile app, navigate to **Administration** >**xFlow Interface** > **Object Attributes**.
   a. Select the mobile attribute.
   b. Click Edit.
   c. Select the **Show in Mobile Attributes** checkbox.
   d. Select Save.
3. Restart the CA xFlow Interface service.

CA Asset Portfolio Management

Unable to add cost to an asset in a localized environment

Problem: Unable to add cost to an asset with Data Importer in any localized environment.

Solution: There is no workaround available for this issue.

CA Asset Portfolio Management (APM) Installation Fails.

Problem: CA APM installation fails during the 'Preparing installation of CA APM' step, and the following error message is displayed: "**Exit code is non 0**".

Solution: This issue occurs when any other software is installed on the same server, that requires a system restart. Restart the server and install CA APM again.

Tenancy Management Page Does not Display on Google Chrome and Microsoft Edge

Problem: This is valid on all supported operating environments with Google Chrome and Microsoft Edge browsers.

Solution: Common Administration for Service Management (CASM) component provides multi-tenancy administration for CA APM. If you plan to perform Tenancy Management in CA APM, use a browser other than Google Chrome and Microsoft Edge.

CA Asset Portfolio Management shows incorrect version information

Problem: After CA Asset Portfolio Management is installed, the version information is displayed as 17.0.0.1034, instead of 17.4.

Solution: There is no workaround available for this issue.

Incorrect Microsoft SQL Server/ Oracle Database Version is Displayed

Problem: Login to CA APM. Navigate to **Administration**, **system configuration**, **Database**, MS SQL Server Microsoft SQL Server version is always displayed as '**MS SQL SERVER 2008'** instead of the correct installed Microsoft SQL Server versions (for example, 'MS SQL SERVER 2016/'MS SQL SERVER 2018)

Solution:

Perform the following steps for Microsoft SQL Server:

1. Launch the Microsoft SQL Developer tool to connect to MDB database,
2. Open a new SQL Worksheet and execute the following query:
   ```
   select * from AL_CDB_CONFIGURATIONPARAMETERS  where  CONFIGVALUE='MS SQL SERVER 2008' and
    componentkey='Database' and configkey='SqlServerVersionddl';
   ```
3. If the search result shows a record finding, run the following query.
   ```
   update AL_CDB_CONFIGURATIONPARAMETERS set  CONFIGVALUE='MS SQL SERVER 2016' where CONFIGVALUE='MS SQL
    SERVER 2008' and componentkey='Database' and configkey='SqlServerVersionddl';
   ```
4. In above query, locate **set CONFIGVALUE='MS SQL SERVER 2016'** replace it with **CONFIGVALUE='<MS SQL SERVER VERSION>'** as per the SQL Server version installed in your environment.
5. Perform an IISReset.

Perform the following steps for Oracle Server:

1. Launch the SQL Developer tool to connect to MDB database.
2. Open a new SQL Worksheet and execute the following query:
   ```
   select * from AL_CDB_CONFIGURATIONPARAMETERS  where  componentkey='Database' and
   configkey='SqlServerVersionddl';
   ```
3. If the search result shows a record finding, run the following query as per the Oracle version installed in your environment, for Oracle19c:-
   ```
   update AL_CDB_CONFIGURATIONPARAMETERS set  CONFIGVALUE='Oracle 19c' where componentkey='Database' and
    configkey='SqlServerVersionddl';
   commit;
   ```
   For 21c:

```
update AL_CDB_CONFIGURATIONPARAMETERS set  CONFIGVALUE='Oracle 21c' where componentkey='Database' and
 configkey='SqlServerVersionddl';
commit;
```

4. Perform an IISReset

Add attachments by cut/copy and paste in a ticket in xFlow and ServicePoint.

Problem: The copy/cut and paste of files does not work as expected on Firefox and Microsoft Edge browsers.

Solution: No solution

Supportability Matrix

The Supportability Matrix contains information related to the current release.

The CA Service Management 17.4 Supportability Matrix contains information that is related to the following:

NOTE

CA Service Management supports service packs and point releases not necessarily noted on this matrix as long as the problem reported is reproducible with versions that are listed on the support matrix. CA Technologies reserves the right to refuse support of new point releases should the reported problem require a major redesign to function properly. If the resolution to a problem is determined to be outside the realm of CA Support responsibilities, they may ask that you escalate your request for certification to your local account team.

NOTE
From 17.4.2.0 and later, the following versions are supported:

- JasperReports Server 9.0.0: Refer JasperReports Server
- JasperSoft Studio 9.0.0: Refer JasperSoft Studio

NOTE

From 17.4.2.0 to 17.4.4.0:

- EEM 12.6.7.0: Installers are available in 17.4 RU2 DVD ex: CASM_17.4.2.0\filestore\EEM, for more information refer EEM 12.6.7.
- ITPAM 04.4 CP03: Refer PAM 4.4 CP03.

From 17.4.5.0:

- If EEM is integrated with CASM, It is mandatory to upgrade to EEM 12.7.2.0 before installing CA Service Management 17.4 RU5. For more information refer EEM 12.7.2.0. Installers are available in CASM_17.4.5.0\filestore\EEM.
- From CA Service Management 17.4 RU5 onward, only CA PAM CP04 HF01 is supported. Refer CP04 - HF01. Installers are available in CASM_17.4.5.0\filestore\PAM_4.4_CP04_HF01.

Operating System

- The certified edition of the Microsoft Windows Server platform is Standard, Enterprise, and Data left.
- For configuration on a Linux or Unix server platform, we recommend that you install the database of the product on Linux or Unix platform and access the web client interface on the Windows platform.

Operating System - CA Service Management Solution

Operating System	Version	CA Service Desk Manager	CA Service Catalog	CA Asset Portfolio Management	xFlow / Service Point
Microsoft Windows Server	2022, 2019, 2016	Yes	Yes	Yes	Yes
RHEL 7.2, 8.6 (x86; 64-bit only)		Yes	No	No	Yes
RHEL 9.4 (64-bit only)		Yes (17.4 GA, and 17.4.5.0)	No	No	Yes (17.4 GA, and 17.4.5.0)
SUSE Linux Enterprise Server 12 SP2 (64-bit only)		Yes	No	No	Yes
SUSE Linux Enterprise Server 15.6 (64-bit only)		Yes (17.4 GA, and 17.4.3.0 and above.)	No	No	Yes (17.4 GA, and 17.4.3.0 and above.)

Operating System - Other Components

Operating System	CA EEM (12.6.6, 12.6.7, 12.7.2.0*)	CA Process Auto CP04, 4.4 CP04 H
Microsoft Windows Server 2022	No	Yes
Microsoft Windows Server 2019	Yes	Yes
Microsoft Windows Server 2016	Yes (12.6)	Yes
RHEL 8.1 (x86; 64-bit only)	Yes	Yes
RHEL 9.4	Yes (from 17.4 RU5)	Yes (from 17.4 RU
SUSE Linux Enterprise Server 12 SP2 (64-bit only)	Yes	No
SUSE Linux Enterprise Server 15.6 (64-bit only)	Yes	No

NOTE

- *EEM12.7.2.0 is supported from CASM17.4.5.0 and above.
- *CA PAM 4.4 CP 04 HF01 is supported from CASM17.4.5.0 and above.

Database

The following databases are certified on 64-bit:

IMPORTANT

To enable **SQL Always ON** with listener for Microsoft SQL Server, perform the below post-installation steps for CA Service Catalog ONLY:

1. After configuring the availability group listener, ensure to update the connection string to connect to the **Always On** availability group listener.
2. Open*USM_HOME\config.properties* and update the connection string of DBSource.url key with:

   ```
   ;MultiSubnetFailover=True
   ```

   Use semicolon as a separator in step 2.

   DB Source URL:

```
jdbc:sqlserver://
XX.XX.XXX.XXX:XXXX;instanceName=MSSQLSERVER;databaseName=mdb;SelectMethod=cursor;MultiSubnetFailover=True
```

3. Restart the Service Catalog services.

Database - CA Service Management Solution

Database	CA SDM	CA Service Catalog	CA APM	xFlow int
Microsoft SQL Server 2022	Yes	Yes	Yes	Yes
Microsoft SQL Server 2019	Yes	Yes	Yes	Yes
Microsoft SQL Server 2017	Yes	Yes	Yes	Yes
Microsoft SQL Server 2016	Yes	Yes	Yes	Yes
Oracle 21c	Yes	Yes	Yes*	Yes
Oracle 19c	Yes	Yes	Yes**	Yes
Oracle 19c RAC	Yes	Yes	Yes**	Yes

NOTE
Prerequisite for MS SQL Server 2022: Microsoft has removed the SQL Native Client 11 for out of the box SQL 2022. CA Service Management uses SQL Native Client 11, you must manually install it from the Microsoft website. Download the 64-bit SQL Native Client 11", named as Microsoft® SQL Server® 2012 Native Client from: https://www.microsoft.com/en-us/download/confirmation.aspx?id=50402

NOTE

- * CA APM - For installing CA APM on Oracle 21c database with Oracle 21c 32-bit client, add a new system environment variable "TNS_ADMIN" on the CA APM Web/Application servers. This variable value must point to the Oracle 21c client installed location. Set the variable value as <oracle_client_installed_location>\product \21.0.0\client_1\network\admin
- **CA APM uses Oracle 19c 32-bit client.

Database - Other Components

Database	CA Process Automation (4.3.5 CP08, 4.4, 4.4 CP02, 4.4 CP03, 4.4 CP04, 4.4 CP04 HF01)	JasperReports Server 9.0.0, 8.1.1/ Studio 9.0.0, 8.1.0
Microsoft SQL Server 2022	Yes	No
Microsoft SQL Server 2019	Yes	Yes
Microsoft SQL Server 2017	Yes	Yes
Microsoft SQL Server 2016	Yes	Yes
Oracle 21c	Yes	No
Oracle 19c, 19c RAC	Yes	Yes

Web Browser

Web Browser - CA Service Management Solution

Web Browser	CA SDM	CA Service Catalog	CA APM	xFlow/ Service Point
Microsoft Edge 114.0.1823.67 (Official build) (64-bit) on Windows 10 and above	Yes	Yes	Yes	Yes
Mozilla Firefox ESR 102.11.0(64-bit)	Yes	Yes	Yes	Yes
Google Chrome Version 114.0.5735.199 (Official Build) (64-bit)	Yes	Yes	Yes	Yes
Apple Safari 16.5.1 (18615.2.9.11.7)	Yes	Yes	Yes	Yes

NOTE
Apple Safari is not supported on Windows operating systems.

Web Browser - Other Components

Web Browser	CA EEM 12.6.6, 12.6.7, 12.7.2.0.0	CA Process Automation 4.3.5 CP08, 4.4, 4.4 CP02, 4.4 CP03, 4.4 CP04, 4.4 CP04 HF01	JasperReports Server 9.0.0, 8.1.1	Xtraction (2023.1, 2021.3, 2019.1)
Microsoft Edge 114.0.1823.67 (Official build) (x86_64) on Windows 10 and above	Yes	Yes	Yes	No
Mozilla Firefox ESR 102.11.0(64-bit)	Yes	Yes	Yes	Yes
Google Chrome Version 114.0.5735.199 (Official Build) (64-bit)	Yes	Yes	Yes	Yes
Apple Safari 16.5.1 (18615.2.9.11.7)	Yes	No	Yes	No

CA Common Components

The CA Service Management solution includes a limited entitlement to CA Process Automation and CABI. The entitlement is restricted to use within the context of workflows that are associated with the products.

CA Common Components - CA Service Management Solution

CA Common Components - CA Service Management Solution

CA Common Components	CA SDM	CA Service Catalog	CA APM
CA EEM (12.6.6, 12.6.7, 12.7.2.0*)	Yes	Yes	Yes
CA Process Automation (4.3.5 CP08, 4.4, 4.4 CP02, 4.4 CP03, 4.4 CP04, 4.4 CP04 HF01*)	Yes	Yes	Yes
JasperReports Server 9.0.0, 8.1.1, JasperStudio ,9.0.0, 8.1.0	Yes	No	No

NOTE

- *EEM12.7.2.0 is supported from CASM17.4.5.0 and above.
- *CA PAM 4.4 CP 04 HF01 is supported from CASM17.4.5.0 and above.
- For information on JRS 8.1.1, refer CA Business Intelligence JasperReports Server 8.1.1.
- For information on JRS 9.0.0, refer CA Business Intelligence JasperReports Server 9.0.0.

Third-Party Common Components

Third-Party Components - CASM Solution

Third-Party Common Components	CA SDM	CA Service Catalog	CA AI
.NET Framework 3.5, 4.5, 4.6, 4.8 and .NET Extensibility 4.7	Yes	NA	Yes
Internet Information Server 7.0, 8.0, 10.0	Yes	NA	Yes
Apache Tomcat (9.0.71)	Yes	Yes	Yes
Apache Tomcat (9.0.108)	Yes (from 17.4.5.0)	Yes (from 17.4.5.0)	Yes (f
AdoptOpenJDK 11.0.18	Yes	Yes	Yes
AdoptOpenJDK 11.0.27	Yes (from 17.4.5.0)	Yes (from 17.4.5.0)	Yes (f
Search Server 7.17.6	Yes	Yes	No
RASA 3.0.6	No	No	No
RASA 3.6.19	No	No	No

SMTP Mail Server

SMTP Mail Server	CA SDM	CA
Microsoft Exchange Server 2019	Yes	Ye
Microsoft Exchange Server 2016	Yes	Ye
Microsoft Windows SMTP Service	Yes	No
Microsoft Exchange Online (Office 365)	Yes	No

Integration

Integration - CA Service Management Solution

Product	CA SDM	CA Service Catalog	CA APM
Automic 21.0.6	No	Yes	No
ITCM (CA IT Client Automation) 14.5.CU5	Yes	No	No
CA SiteMinder (SSO) 12.8.7	Yes	Yes	Yes
CA SiteMinder (SSO) 12.9	Yes (From 17.4 RU5)	Yes (From 17.4 RU5)	Yes (From 17.4 RU5)
CA Spectrum Automation 23.3.13	Yes	No	No
CA Spectrum Automation 25.04.1	Yes (From 17.4 RU5)	No	No
CA Service Operations Insight (SOI) 4.0 SP2	Yes	No	No
CA Configuration Automation (CCA) 12.9.0.54	Yes	No	No
CA Identity Manager 12.6.8	Yes	No	No
Microsoft SharePoint 2016	Yes	No	No

Interoperability

CA recommends upgrading all the Service Management capabilities in a customer environment using the common installer and common administration capabilities that are available with CA Service Management 17.4 solution. If you plan to upgrade product by product in a phased manner, the information in below tables can be used as a guideline to see the interoperability among the Service Management capabilities.

NOTE
Upgrade to EEM-12.6.6.0 and ITPAM-4.4 CP2 before upgrading to CASM 17.4 GA.

NOTE
If you have xFlow and CA Service Point installed in your environment, upgrade the 17.4 CA Service Catalog. Then upgrade CA SDM and its sub-components (i.e., xFlow/ CA Service Point and Search Server) concurrently.

Interoperability for CA Service Management

CA SDM	XFlow/ Service Point and Search Server	CA Catalog	ITAM
17.3 GA to 17.3 RU11	17.3 GA to 17.3 RU11	17.3 GA to 17.3 RU11	17.4 G
17.3 RU12 to RU23	17.3 RU12 to RU23	17.3 RU12 to RU23	17.4 G
17.3 GA to 17.3 RU11	17.3 GA to 17.3 RU11	17.4 GA	17.4 G
17.3 RU12 to RU23	17.3 RU12 to RU23	17.4 GA	17.4 G
17.4 GA	17.4 GA	17.4 GA	17.3 G
17.4 GA	17.4 GA	17.4 GA	17.3 R

17.4 RU5	17.4 RU5	17.4 RU5	17.4 R

Compliance

Compliance - CA Service Management Solution

Accessibility	CA SDM	CA Service Catalog	CA APM	xFlow/ Service Point
Section 508 Compliance	Yes	Yes *	Yes *	Yes *
Web Content Accessibility Guidelines (WCAG) 2.0 and 2.1	Yes	Yes *	Yes *	Yes *
Internet Protocol version 6 (IPv6)	Yes	No	No	No
Federal Information Processing Standards (FIPS)	Yes	No	No	No
Secure Sockets Layer (SSL)	Yes	Yes	Yes	Yes
Transport Layer Security (TLS) 1.2.	Yes	Yes	Yes	Yes

NOTE

***** indicates compliance with a few exceptions. For more information, see Section 508 Compliance.

Load Balancing

Load Balancing - CA Service Management Solution

Load Balancer	CA SDM	CA Service Catalog	CA APM
Apache Web Server	Yes	Yes	Yes
F5 Networks	Yes	Yes	Yes
Microsoft NLB	No	No	Yes

Product Documentation

The CA Service Management solution documentation is available in the following languages. Some of the products in the solution support other languages such as Canadian French, Traditional Chinese, Danish, and so on, which are not supported by documentation. Refer to the English documentation for such languages.

Documentation - CASM Solution

- English
- French
- Spanish
- Brazilian Portuguese
- German

Internationalization and Localization

An Internationalized product (also referred to as a language-certified product) is an English-language product that runs on local language versions of the supported operating system and required third-party products.

Internationalized products support local language data for input and output, and also support the ability to specify local language conventions for date, time, currency, and number formats.

> **NOTE**
> CA Service Management 17.4 will support partial translations for German and Italian languages. New Features strings will not be translated in these languages and will appear in English.

The following must be considered:

- CA Business Intelligence JasperReports Server is supported only in English.
- The reports are localized in the following languages:
  - CA Service Desk Manager 17.4: French, German, Italian, Japanese, Spanish, and Brazilian Portuguese

Localized Versions of The Product Supported Based on Operating System

Language	Operating System
English	Windows and Linux
French	Windows and Linux
German	Windows and Linux
Spanish	Windows
Brazilian Portuguese	Windows

Internalization - CASM Solution

Internationalization	CA SDM	CA Service Catalog	CA APM	xFlow/ Service Point	JasperReports Server 8.1.1, 9.0.0	
English	Yes	Yes	Yes	Yes	Yes	
French	Yes	Yes	Yes	Yes	No	
German	Yes	Yes	Yes	Yes	No	
Spanish	Yes	Yes	Yes	Yes	No	
Brazilian Portuguese	Yes	Yes	Yes	Yes	No	

Localization - CASM Solution

Localization	CA SDM	CA Service Catalog	CA APM	xFlow/ Service Point	JasperReports Se 9.0.0
English	Yes	Yes	Yes	Yes	Yes
French	Yes	Yes	Yes	Yes	No
German	Yes	Yes	Yes	Yes	No
Spanish	Yes	Yes	Yes	Yes	No
Brazilian Portuguese	Yes	Yes	Yes	Yes	No

Check out where we're going; attend a CA Product Roadmap Session

Latest updates about CA Service Management.

We are always listening to our customers, and one of the most common requests we receive is for insight into our Product roadmap. You can now join your industry peers to hear directly from our product management team about the future direction of our products. By attending the product roadmap and feedback sessions, you will gain firsthand knowledge about upcoming product functionality and features along with architectural enhancements. At Broadcom, we strive to eliminate the barriers between your ideas and the business outcomes you seek.

Product management regularly hosts Office Hours, bimonthly on Thursdays . You can see the upcoming event on our community page.

Defect Resolution Process

A customer reported issue is characterized as a defect. This article outlines the process for defect triage, defect resolution, and the delivery cadence.

In CA Service Management, customer reported issue is characterized as a defect. This article outlines the process for defect triage, defect resolution, and the delivery cadence.

How to Report a Defect

CA Support is a one-point destination for reporting a product defect.

For other queries, ideas, tips, and feature enhancements, that do not fit the definition of a defect, can be submitted in the CA Service Management community. The community provides a unique opportunity to network and help you maximize your software investment by tapping into a community of expertise, open 24/7.

Defect Triage

Each defect is triaged and subsequently a technical severity is assigned to the defect, by the Customer Voice team (CV Team).

The technical severity mentioned above is different from the CA Support case assignment severity. The technical severity helps prioritize workload within the CV Team for the purpose of delivering solutions to reported defects.

NOTE

All defects that are triaged and determined to be an enhancement request should be submitted as idea for consideration in future releases. The idea must be submitted by the customer in the CA Service Management community.

Severity	Description	Criteria
S1	System-down or a product-inoperative condition impacting production environment, and an immediate workaround is not available.	• Causes downtime • Irrecoverable data corruption • Daily application crash • Critical security exploits • Blocks a critical business functionality.

S2	A high-impact business condition possibly endangering a production environment.	• Blocks a product functionality with no available workaround • Impacts but does not block a critical business functionality • Causes recoverable data corruption or loss • Pertains to significant performance degradation • Causes localization defects that make the functionality unusable • Causes application crash weekly or less frequently
S3	A low-impact business condition with most software functions still usable; however, some circumvention might be required to provide service.	• Impairs users ability to use full functionality set of the product • Might have a reasonable work-around • Insignificant performance degradation • Incomplete/Incorrect un-installation of the application • Incorrect error or log messages • Accessibility or usability defects causing minor inconvenience
S4	A minor problem or question that does not affect the software function.	• Causes confusion or inconvenience but does not result in loss of functionality • Low business impact • Low visibility or impacts small set of users

Defect Communication

All customer reported defects can be tracked through Broadcom Support Online. An update is provided when the triage is complete. The frequency of updates for an S1 is high. For lower severity defect, the updates are available based on the progress milestones.

Cadence for Delivery Method

Defect fixes are delivered quarterly through either a cumulative patch, a roll-up patch, or GA release.

TIP

- `We recommend that you review and meet the prerequisites before you install either a cumulative patch, a roll-up patch, or GA release.`
- Before applying either a cumulative patch, roll-up patch, or a release, we recommend that you verify them in a test environment that is a representative of your production environment in terms of the configuration and data volume.
- A cumulative patch, and roll-up patch is available for download from the Solutions and Patches page. The GA release is available through the CA Service Management Product Download page.
- Mobile App updates are made available through Google Play or Apple App Store.

A cumulative patch or the next release will contain fixes from the previous cumulative patches, roll-up patches, and test fixes (released, and unreleased). A cumulative patch ensures optimal performance and seamless system maintenance. Also, it reduces the overall patch level complexity and ensures that the collection of fixes are interoperable as they undergo a complete QA lifecycle.

A roll-up patch contains the fixes for S1, S2 defects since the last cumulative patch or roll-up patch. The roll-up patch undergoes QA testing that is limited to the following:

- Verification of the fix that is resolved in the patch.
- Regression and integration test coverage that is limited to the impacted product area.

Severity	Delivery Method	Additional Information
S1	Test fix*	A work-around or test fix is provided to the affected customers to support business continuity based on the latest cumulative patch or roll-up patch.
S2	Cumulative patch	Resolution for a S2 defect will be provided through the next available cumulative patch, roll-up patch, or product release. In critical cases, a test fix may be provided to the customer based on the latest cumulative/roll-up patch level.
S3	Cumulative patch or future release	Resolution of a S3 defect will be provided on a case-by-case basis either through a cumulative patch or a future release based on the complexity and risk.
S4	Selected defects in a cumulative patch or a future release	Low severity defects that affect several customers would be considered for resolution in a cumulative patch or a future release. Other S4 defects are closed.

* indicates Test fixes are not provided for xFlow Interface, CA ITAM, and Mobile App.

Product Accessibility Features

CA Technologies is committed to helping users with a wide range of abilities to use our products and supporting documentation to accomplish vital business tasks. This section outlines the accessibility features that are part of the CA Service Management product solution.

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This section contains the following topics:

Product Enhancements

CA Service Management offers accessibility enhancements in the following areas:

- Display
- Sound
- Keyboard
- Mouse

NOTE

The following information applies to Windows-based and Macintosh-based applications. Java applications processed on many host operating systems, some of which already have assistive technologies available to

them. For these existing assistive technologies to provide access to programs written in JPL, they need a bridge between themselves in their native environments and the Java Accessibility support that is available from within the Java virtual machine (or Java VM). This bridge has one end in the Java VM and the other on the native operating environment, so it will be slightly different for each operating environment it bridges to. Sun is currently developing both the JPL and the Win32 sides of this bridge.

Display

To increase visibility on your computer display, you can adjust the following options:

Font style, color, and size of items

Lets you select font color, size, and other visual combinations.

Screen resolution

Lets you change the pixel count to enlarge objects on the screen.

Cursor width and blink rate

Lets you make the cursor easier to find or minimize blinking.

Icon size

Lets you make icons larger for visibility or smaller for increased screen space.

High contrast schemes

Lets you select color combinations that are easier to see.

Sound

Use sound as a visual alternative or to make computer sounds easier to hear or distinguish by adjusting the following options:

Volume

Lets you turn the computer sound up or down.

Text-to-Speech

Lets you hear command options and text read aloud.

Warnings

Lets you display visual warnings.

Notices

Gives you aural or visual cues when accessibility features are turned on or off.

Schemes

Lets you associate computer sounds with specific system events.

Captions

Lets you display captions for speech and sounds.

Keyboard

You can make the following keyboard adjustments:

Repeat Rate

Lets you set how quickly a character repeats when a key is struck.

Tones

Lets you hear tones when pressing certain keys.

Sticky Keys

Lets those who type with one hand or finger select alternative keyboard layouts.

Mouse

You can use the following options to make your mouse faster and easier to use:

Click Speed

Lets you select how fast to click the mouse button to make a selection.

Click Lock

Lets you highlight or drag without holding down the mouse button.

Reverse Action

Lets you reverse the functions controlled by the left and right mouse keys.

Blink Rate

Lets you select how fast the cursor blinks or if it blinks at all.

Pointer Options

Lets you complete the following tasks:

- Hide the pointer while typing
- Show the location of the pointer
- Set the speed that the pointer moves on the screen
- Select the pointer's size and color for increased visibility
- Move the pointer to a default location in a dialog box

Keyboard Shortcuts

The following table lists the supported keyboard shortcuts. These keyboard shortcuts are applicable to all CA Service Management products - CA Service Desk Manager, CA Service Catalog, and CA Asset Portfolio Management.

Keyboard	Description
Ctrl+X	Cut
Ctrl+C	Copy
Ctrl+K	Find Next
Ctrl+F	Find and Replace
Ctrl+V	Paste
Ctrl+S	Save
Ctrl+Shift+S	Save All
Ctrl+D	Delete Line
Ctrl+Right	Next Word
Ctrl+Down	Scroll Line Down
End	Line End

xFlow Interface Shortcuts	
Alt+m	Notification icon. Displays the Notification overlay with the focus on *Mark All As Read* option.
Alt+t	Brings the focus to Tickets in Streams.
Alt+s	Save stream. In the stream builder, when you create or edit a stream, click Alt+s to save the stream.
Alt+c	Cancel stream. In the stream builder, when you create or edit a stream, click Alt+c to cancel creating or editing the stream.
Alt+p	New Message button in Team Messages. When a new message is added while the Team Messages page is open, a pop-up appears displaying the New Messages button. Click on the button to display the latest messages.
xFlow Interface Calendar Shortcuts	
Left arrow	Focus on the previous date.
Right arrow	Focus on the next date.
Up arrow	Focus on the same day in the previous week.
Down arrow	Focus on the same day in the next week.
Page Up	Focus on the same date in the previous month.
Page Down	Focus on the same date in the next month.
Home	First date of the view
End	Last date of the view
Enter	Select the date
Space	Select the date
Esc	Close the calendar and focus on the input
CA Service Desk Manager Keyboard Shortcuts	
List Page	
Alt+C	Activates the List Edit Cancel button to cancel edit mode and discard all unsaved updates. Does nothing if not in List Edit mode.
Alt+E	Activates the Edit in List button to switch to List Edit mode.
Alt+G	Switches focus to the object type dropdown select in the Go button frame.
Alt+M	Switches focus to the first menu in the main menu bar (the File menu).
Alt+P	Switches focus to the first tab in the product toolbar (the Service Desk tab).
Alt+S	Activates the Search button to request a new search.

Alt+X	Switches focus to the top of the tree control in the left frame of the current page. There are left-frame tree controls on all tabs of the main page, the Knowledge Categories form, and the Knowledge Tree Designer. Alt+X has no effect if the current page does not include a left frame containing a tree control.
Ctrl+W	Closes the browser.
Detail Pages	
Alt+C	In edit mode, activates the Cancel button to return to read-only mode and discard all unsaved changes. Does nothing in read-only mode.
Alt+E	In read-only mode activates the Edit button to switch to edit mode.
Alt+G	Switches focus to the object type dropdown select in the Go button frame.
Alt+M	Switches focus to the first menu in the main menu bar (the File menu).
Alt+N	Switches focus to the first link on the index of the page's notebook (if present).
Alt+P	Switches focus to the first tab in the product toolbar (the Service Desk tab).
Alt+S	In edit mode, submits the form by activating the Save, Continue, OK, or Finished button. In read-only mode, activates the Search button on the first deferred tab containing a list page or does nothing if no exposed deferred tabs contain a list page.
Alt+number	Switches focus to the notebook index link for one of the ten tabs in the notebook. Alt+1 switches focus to the index link of the first tab; Alt+2 to that of second tab; and so on to Alt+0, which switches focus to the index link of the tenth tab. In Internet Explorer, you must activate the link by pressing Enter after the pressing the access key. In Mozilla, the link is automatically activated when you press the access key.
Alt+X	Switches focus to the top of the tree control in the left frame of the current page. There are left-frame tree controls on all tabs of the main page, the Knowledge Categories form, and the Knowledge Tree Designer. Alt+X has no effect if the current page does not include a left frame containing a tree control.

CA Service Catalog Accessibility

Note: Accessible Shortcut keys provided with the CA Service Catalog application may not work if your system or toolbar plugins are using a similar set of shortcut keys. Ensure to close such plugins before using the access keys provided with CA Service Catalog.

Note: The JAWS screen reader by design reads outs the first data column associated with a CA Service Catalog Form Designer table. When you focus on a checkbox associated with a form designer table, JAWS reads outs the data column associated with the checkbox. If you have customized the data columns and have added images, icons, or HTML values, JAWS will read these. Hence, it is recommended that you enter only plain text or numerical values in the table's first data column so that meaningful context is read out by JAWS.

Note: The JAWS Screen Reader while navigating through the CA Service Catalog Service Offering tree view may read out either '**Blank**' at the beginning of each service designer folder or may read out '**List End**' as you navigate to the next service level folder.

CA Service Catalog Keyboard Shortcuts	Description

Ctrl+Shift+Z	Reach the first service offering from the category tree in the browse widget.
v (small letter)	Open the popup window when you click the links in the list widget.
Ctrl+Shift+O	Reach the first button in button area when a request form is loaded and the focus is at any place in between the form.
Ctrl+Alt+A	Adds a form, folder, or service offering.
Ctrl+Shift+X	Cut a folder or a service offering
Ctrl+Shift+C	Copy a folder or a service offering
Ctrl+Shift+V	Paste a folder or a service offering
Ctrl + Enter	Opens the Attribute value editor popup window
Alt+S	To focus on the forms and folders search text field
Ctrl+Alt+V	Save
Ctrl+Alt+D	Delete
Ctrl+Alt+E	Rename
Ctrl+Alt+Z	Customize
Ctrl+Alt+H	Paste As Inherited
Ctrl+Alt+L	Opens the Localization Dialog.
Ctrl+Alt+J	Opens the Script dialog if a form is selected.
Ctrl+Alt+R	Restores the unsaved changes with the saved changes.
Ctrl+Alt+A	To add a Folder or add a service offering
CA Service Catalog Date (Calendar) Keyboard Shortcuts	Description
Left arrow	Move focus to the previous day.
Right arrow	Move focus to the next day.
Up arrow	Move focus to the same day of the previous week.
Down arrow	Move focus to the same day of the following week.
Page Up	Move focus to the same date of the previous month.
Page Down	Move focus to the same date of the following month.
Shift+Page Up	Move focus to the same date of the previous year.
Shift+Page Down	Move focus to the same date of the following year.
Home	Move to the first day of the month.
End	Move to the last day of the month
Tab	Navigate between calendar grid and other elements in the calendar.
Enter/Space	Select date if there is no time component in a calendar or else use the OK button to select a date.
Esc	Close the calendar and focus on the input
Note: The A, C, D, E, F, H, T, V, X, Z access keys are browser specific (IE and other browsers). Numeric keys 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9 can also be used as Access Keys.	
Access or Hot Keys	**Button Name**

Alt + B	Back BI Launch Pad Change the Business Unit Browse Offerings
Alt + G	GO and Next Configure Define SOG
Alt + I	Edit Edit Tasks Edit account Edit-group Finish Define Dependencies Definition Search
Alt + J	Test Create Data Object Save & Test
Alt + K	• OK • Checkout • Tracking
Alt + L	• Delete • Delete Items • Delete Tasks
Alt + M	• Add • Add User • Add News Messages • Add Cart • Add Sub Unit • Email • Column Rules • Add Option • Add Sub Types • Resume link • CA CMDB Association
Alt + N	• Cancel • Close • Close Window
Alt + O	• Done • Set Folders • Organize options • Options • Hold link • Define offering Selection
Alt + P	• Set Permissions • Publish To Dashboard Delivery • Push Through Request
Alt + Q	• Request

Alt + R	• Refresh • Reset • Remove • Create • Erase All Rules • Search
Alt + S	• Save • Send • Share • Search • Create variable • Save Variable • Save Dependencies
Alt + U	• Submit • Save and Submit Cart • Launch • Feature • Update Configuration • Unshared • Upload • Update • Audit Trail
Alt + W	• Create Data view • View Scheduler system alerts • Preview
Alt + Y	• Copy • Modify • Modify Settings • Modify the account list • Save as Copy • CA EEM

Numeric Access Keys to Switch Tabs	**Tab Name**
Alt+1	Dashboard
Alt+2	Messages
Alt+3	Requests
Alt+4	Subscription
Alt+5	Invoice
Alt+6	Home
Alt+7	Catalog
Alt+8	Accounting
Alt+9	Administration
CA Asset Portfolio Management Keyboard Shortcuts	
Alt+Shift+Ctrl+D	Resize a Dialog Box in the Downward direction
Alt+Shift+Ctrl+U	Resize a Dialog Box in the Upward direction
Alt+Shift+Ctrl+R	Resize a Dialog Box to the Right

Alt+Shift+Ctrl+L	Resize a Dialog Box to the Left
ESC	Close a Dialog Box
Ctrl+Shift+Down	View the Contents of a Drop-Down List

"Skip to Main Content" Navigation

The user interface has a **Skip to Main Content** link that allows users to quickly navigate to the first data field in the user interface without having to tab multiple times to get to the main content on the page.

In CA SDM, this link exists as **Skip Navigation**. This link allows the Screen Reader users to skip the in-between links and navigate to the main content page. This link is available by default for the Customer and Employee roles. For the Analyst role, the link is available only when the Screen Reader mode is enabled.

Product Names and Abbreviations

This page lists all the product names and abbreviations in CA Service Management.

This documentation references the following products and abbreviations:

- CA Service Desk Manager (CA SDM)
- CA Asset Portfolio Management (CA APM)
- CA Service Catalog (CA SC)
- CA Embedded Entitlements Manager (CA EEM)
- CA Process Automation (ITPAM)
- CA Business Intelligence (CABI JasperSoft Reports)
- CA Service Desk Manager Advanced Availability (AA)

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S.No	Software Component Name	License Acknowledgement Legal Document Details
1.	Abdera 1.1.	Legal_Doc_00002556_8.pdf
2.	ActiveMQ 5.13.0	Legal_Doc_00001418_13.pdf
3.	Adaptive Communication Environment (ACE) 5.5.10	None

4.	Adaptive Communication Environment (ACE) 5.7.8	ACE 5.7.8 - This product includes Adaptive Communication Environment (ACE)5.7.8, which is distributed in accordance with the following terms: Copyright and Licensing Information for ACE(TM), TAO(TM), CIAO(TM), and CoSMIC(TM) [1]ACE(TM), [2]TAO(TM), [3]CIAO(TM), and [4]CoSMIC(TM) (henceforth referred to as "DOC software") are copyrighted by [5]Douglas C. Schmidt and his [6]research group at [7]Washington University, [8]University of California, Irvine, and [9]Vanderbilt University, Copyright (c) 1993-2010, all rights reserved. Since DOC software is open-source, freely available software, you are free to use, modify, copy, and distribute--perpetually and irrevocably--the DOC software source code and object code produced from the source, as well as copy and distribute modified versions of this software. You must, however, include this copyright statement along with any code built using DOC software that you release. No copyright statement needs to be provided if you just ship binary executables of your software products. You can use DOC software in commercial and/or binary software releases and are under no obligation to redistribute any of your source code that is built using DOC software. Note, however, that you may not do anything to the DOC software code, such as copyrighting it yourself or claiming authorship of the DOC software code, that will prevent DOC software from being distributed freely using an open-source development model. You needn't inform anyone that you're using DOC software in your software, though we encourage you to let [10]us know so we can promote your project in the [11]DOC software success stories. The [12]ACE, [13]TAO, [14]CIAO, and [15]CoSMIC web sites are maintained by the [16]DOC Group at the [17]Institute for Software Integrated Systems (ISIS) and the [18]Center for Distributed Object Computing of Washington University, St. Louis for the development of open-source software as part of the open-source software community. Submissions are provided by the submitter ``as is'' with no warranties whatsoever, including any warranty of merchantability, noninfringement of third party intellectual property, or fitness for any particular purpose. In no event shall the submitter be liable for any direct, indirect, special, exemplary, punitive, or consequential damages, including without limitation, lost profits, even if advised of the possibility of such damages. Likewise, DOC software is provided as is with no warranties of any kind, including the warranties of design, merchantability, and fitness for a particular purpose, noninfringement, or arising from a course of dealing, usage or trade practice. Washington University, UC Irvine, Vanderbilt University, their employees, and students shall have no liability with respect to the infringement of copyrights, trade secrets or any patents by DOC software or any part thereof. Moreover, in no event will Washington University, UC Irvine, or Vanderbilt University, their employees, or students be liable for any lost revenue or profits or other special, indirect and consequential damages. DOC software is provided with no support and without any obligation on the part of Washington University, UC Irvine, Vanderbilt University, their employees, or students to assist in its use, correction, modification, or enhancement. A [19]number of companies around the world provide commercial support for DOC software, however. DOC software is Y2K-compliant, as long as the underlying OS platform is Y2K-compliant. Likewise, DOC software is compliant with the new US daylight savings rule passed by Congress as "The Energy Policy Act of 2005," which established new daylight savings times (DST) rules for the United States that expand DST as of March 2007. Since DOC software obtains time/date and calendaring information from operating systems users will not be affected by the new DST rules as long as they upgrade their operating systems accordingly. The names ACE(TM), TAO(TM), CIAO(TM), CoSMIC(TM), Washington University, UC Irvine, and Vanderbilt University, may not be used to endorse or promote products or services derived from this source without express written permission from Washington University, UC Irvine, or Vanderbilt University. This license grants no permission to call products or services derived from this source ACE(TM), TAO(TM), CIAO(TM), or CoSMIC(TM), nor does it grant permission for the name Washington University, UC Irvine, or Vanderbilt University to appear in their names. If you have any suggestions, additions, comments, or questions, please let [20]me know. [21]Douglas C. Schmidt License URL: http://www.cs.wustl.edu/~schmidt/ACE-copying.html

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6	AdoptOpenJDK 1.8.0_212	AdoptOpenJDK_v8.pdf
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12	alpine 3.10	Legal_Doc_00002754_63.pdf
13	analytics 2.1.1	00002528_71_doc_requirements.pdf
14	analytics-core 2.1.1	00001658_4_doc_requirements.pdf
15	angular.io 6.0.6	Legal_Doc_00002528_17.pdf
16	angular2-draggable 2.3.2	Legal_Doc_00002754_76.pdf
17	angular2-moment 1.8.0	Legal_Doc_00002528_1.pdf

18	angular-animate 1.4.9	Legal_Doc_00002754_42.pdf
19	angular-aria 1.4.9	Legal_Doc_00001457_26.pdf
20	angular-bootstrap 2.1.4	Legal_Doc_00000545_12.pdf
21	angular-busy 4.1.3	:Legal_Doc_00001457_2.pdf
22	angular-debounce 1.0.3	Legal_Doc_00001457_22.pdf
`23	angular-drag-and-drop-lists 2.1.0	Legal_Doc_00001738_4.pdf
24	angular-elastic 2.5.1	Legal_Doc_00001438_29.pdf
25	AngularJS 1.4.9	Legal_Doc_00001531_11.pdf
26	angular-messages 1.4.9	Legal_Doc_00001457_25.pdf
27	angular-mocks 1.4.9	Legal_Doc_00001527_3.pdf
28	angular-once 0.1.9	Legal_Doc_00002528_21.pdf
29	angular-resource 1.4.9	Legal_Doc_00001531_9.pdf
30	angular-sanitize 1.4.9	Legal_Doc_00001527_5.pdf
31	angular-toastr 2.1.1	Legal_Doc_00002353_5.pdf
32	angular-translate 2.15.1	Legal_Doc_00002053_4.pdf
33	angular-translate-loader-static-files 2.15.1	00001531_48_doc_requirements.pdf
34	angular-tree-component 8.2.0	Legal_Doc_00002754_75.pdf
35	angular-tree-component 8.4.1	Legal_Doc_00002754_75.pdf
34	angular-ui-router 0.2.17	Legal_Doc_00001457_3.pdf
35	angular-websocket 1.0.14	Legal_Doc_00002754_75.pdf
36	angular-xeditable 0.1.10	Legal_Doc_00001457_17.pdf

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268	react-native-vector-icons 5.0.0	00002528_54_doc_requirements.pdf
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CA Service Management 17.4.5.0 Release Notes

CA Service Management 17.4.5.0 is a cumulative patch for CA Service Management 17.4. For more information about the contents of previous cumulative patches, see Release Information.

This roll-up patch contains quality and security improvements, for more information review the Fixed Issues section.

Enhancements

- **Axis removal**:

  As announced in CA Service Management 17.4, the CASM suite of products now fully supports **CXF-based web services**, and **AXIS-based web services have been deprecated and are no longer supported**. For more information refer Axis to CXF Migration.
- **Improved Knowledge suggestions in xFlow**: Analysts can now view real-time knowledge and solution suggestions while typing the problem description on the xFlow ticket creation page. For more information refer, View Knowledge Base Documents from Search Results.
- **Direct Link/ Access to Catalog Services or Folders**: You can now access Catalog Services or Folders directly through bookmarked links, enabling faster navigation to frequently used requests or offerings. For more information refer, Bookmark Service Folders/ SDM Service Categories.
- **xFlow Rate Limiting Configuration**: You can now control the number of API requests allowed within a specific time frame using the Rate Limiter Filter. For more information, refer Enable Rate Limiting Configuration.
- **Announcements in xFlow**: This enhancement enables the rendering of announcements within xFlow, extending a feature previously available only in Classic Service Desk and Service Point.
- **Manual Notifications for xFlow**: The Manual Notify feature has been enhanced to provide a more intuitive and flexible notification experience. For more information, refer Manual Notifications for xFlow.
- **POST to GET conversion in CA SDM web interface:** Conversion or simulation of a POST operation using a GET operation is prohibited or strongly discouraged due to the critical security and data integrity risks. For more information, see Options Manager.
- A new FormDesigner JS API (ca_fdClearFieldValue) is now available in CA Service Catalog. This API clears any previously saved values in the DualList component and reloads the component.

  NOTE
  Key Considerations:

  - For CASM 17.4.5.0, you must install EEM 12.7.2.0 first, for installation you must follow the steps mentioned here. Installers are available in CASM_17.4.5.0\filestore\EEM. For more information, refer EEM 12.7.2.0.
  - **ITPAM 4.4 CP04 HF01** must be installed after EEM12.7.2.0 installers are available in ***CASM_17.4.5.0\filestore\PAM_4.4_CP04_HF01***. For installation, refer ITPAM CP04-HF01.
  - **Component Support:** From **CASM 17.4.5.0**, only **EEM 12.7.2.0** and **ITPAM CP04 HF01** are supported. These versions are **not compatible** with CASM 17.4 RU4 or earlier releases.
  - **Rollback Option:** Once upgraded to CASM 17.4.5.0, rollback to earlier versions is not supported. Ensure a full backup before upgrading.
  - **Interoperability**: Cross-version interoperability within CASM 17.4 is not supported. Upgrade all CA Service Management components to 17.4.5.0 to ensure compatibility.

NOTE

- To install CA Service Management 17.4.5.0, see Installing CA Service Management 17.4.5.0
- Uninstallation is not supported for CASM17.4.5.0.
- For troubleshooting of CASM 17.4.5.0, refer Troubleshooting for CASM 17.4.5.0

Certification

CA Service Management is now certified with:

- EEM 12.7.2.0
- ITPAM 4.4 CP04 HF01
- RHEL 9.4
- Siteminder 12.9
- CA Spectrum Automation 25.04.1

Fixed Issues

Fixed Issue ID	Description
Case #36119033	"POST" to "GET" Conversion -A5 - Security Misconfiguration
Case #36555258	Sanitization Tool gives out of memory issues, java.exe using 4.5 GB memory and then throw ""outofmemory"" exception. For more information refer, Admin insights for RTE.
Case #36456991	SDM 17.4.4_OpenJDK Multiple Vulnerabilities
Case #36448333	CASM 17.4.4 - SLCM - Searchserver synch
Case #36386604	In CA SDM, in the "Edit in List" view for workflows, the 'Assignee' and 'Group' fields are now linked. Selecting a group will now correctly filter the assignee list to only show members of that group and vice-versa.

For the complete list, go to Fixed Issues for 17.4.5.0

Fixed Issues for 17.4.5.0

Fixed Issues

Fixed Issue ID	Product	Description
Case #33679939	CA Service Desk Manager	CA SDM 17.3/17.4: Edit-in-List for Workflow Tasks
Case #36140819	CA Service Desk Manager	SDM - getListValues is returning more rows than stated in the documentation
Case #36172525	CA Service Desk Manager	SDM_17.4.03-36172525-incorrect parent child ticket relationship under edit-in-list
Case #36194886	CA Service Desk Manager	SDM - CACF Incident Template
Case #36198988	CA Service Desk Manager	Changing Incident ticket Category through WebServices generates ""Error setting attr 'category' on object '<TICKETPERSID>' to value '<PCAT ID>'
Case #36246397	xFlow/ ServicePoint	Issues in xFlow /ServicePoint functionalities

Fixed Issue ID	Product	Description
Case #36247637	CA Service Desk Manager	CV-GEN-INQUIRY_SDM_17.4 RU3-Ticket Web form is broken when a ticket is created by maileater
Case #36318942	CA Service Point	Certain service offerings not loading in Service Point
Case #36353822	CA Service Point	ServicePoint_17.4.4_Add to Cart and Check out buttons are missing for certain offerings option groups
Case #36374376	CA Service Desk Manager	Maileater does not import html-body to html_description
Case #36386604	xFlow	CASM 17.4.2 xFlow - Activity Feed - system-description missing. For more information, see Troubleshooting xFlow Interface.
Case #36448333	CA Service Desk Manager	CASM 17.4.4 - SLCM - Searchserver synch
Case #36456991	CA Service Desk Manager	SDM 17.4.4_OpenJDK Multiple Vulnerabilities
Case #36477139	CA Service Desk Manager	SDM_17.4.04-36477139-Cannot lock html description field
Case #36517730	CA Service Desk Manager	SDM_17.4.04-36517730-xFlow is not filtering for Affected Service CI's correctly
Case #36519372	xFlow	Lack of Rate Limiting on Sensitive Endpoints – CA Service Point. For more information refer Enable Rate Limiting Configuration.
Case #36555258	CA Service Desk Manager	Sanitization Tool gives out of memory issues, java.exe using 4.5 GB memory and then throw ""outofmemory"" exception. For more information refer, Admin insights for RTE.
Case #36400012	CA Service Desk Manager	The command pdm_replace -f input-file takes the input file from the NX_ROOT folder ONLY. The input file in any other location will lead a input file not found message.
Case #36119033	CA Service Desk Manager	"POST" to "GET" Conversion -A5 - Security Misconfiguration
Case #36422907	CA Service Desk Manager	Either in Incidents, Changes, and Problems, the words / Texts in pointed brackets <...> are deleted
Case #36601047	CA Service Desk Manager	Issue with Knowledge Attachments via WebService.
Case #36451946	ServicePoint	switchScope error in browser console (Service Point)
Case #36443867	ServicePoint	A date type field in Service Point is clearing the value after Chekout
Case #36443739	ServicePoint	Service Point shows This widget does not support CA Workflow based forms.
Case #36385532	ServicePoint	After creating a service request in Service Point the Offering tab disappears.
Case #36400880	ServicePoint	After clicking “Checkout” the request forms are not displayed.
Case #36285144	ServicePoint	setTimeout causing issues in dual list behavior in Service Point Catalog offerings
Case #36415118	ServicePoint	Expand the Comments field width under "Request Details" in Service Point when checking out offering
Case #36353824	ServicePoint	ServicePoint_17.4.4_Requests can not be checked out for certain offering options/forms
Case #36613252	ServicePoint	RU4_Virtual Analyst loses the chat icon reference

Troubleshooting for CASM 17.4.5.0

This page contains troubleshooting issues for CASM 17.4.5.0.

Troubleshooting

1. **Problem**:
   While upgrading to 17.4 RU5, the migration task to CXF endpoint fails in an SSL environment.
   Resolution:
   a. Login to ITPAM environment and navigate to SDM>> Configuration >> SDM Dataset
   b. Update the value of port from non- SSL to SSL port. (for example: from 8080 to 8443)
   c. Click **Retry Install** in the Common Installer.
2. **Problem**:
   The CA Asset Portfolio Management post-installation task (17.4.5.0) encounters a failure, with the following error: ***type=itam_sdm_cxf_integ.***
   Resolution:
   a. Start the CA SDM Services.
   b. Click **Retry Install** in the Common Installer.
3. **Problem**:
   When clicking the **Gold brick** icon on the CA Service Catalog request page, the ITAM URL opens using a non-SSL protocol, with the port number and the short hostname of the ITAM server.
   Resolution:
   Run the following query to update the ITAM URL to use the SSL protocol, FQDN, and SSL port:

   ```
   UPDATE ca_application_registration
   SET product_specific_data = '<SSL Protocol>://<ITAM FQDN>:<ITAM SSL Port>/ITAM'
   WHERE product_name = 'CA Asset Portfolio Management';
   ```

   For example:

   ```
   UPDATE ca_application_registration
   SET product_specific_data = 'https://test.itam.com:199/ITAM'
   WHERE product_name = 'CA Asset Portfolio Management';
   ```

4. **Problem**:
   CA Service Catalog Featured offerings are not accessible from **More** button/option in the ServicePoint application.
   Resolution:
   1. Access the USM_HOME folder of the CA Service Catalog computer and verify that the following certificate files exist:
      a. USMcertfile.key
      b. USMcertfile.pem
   2. Copy both the files and paste to this location **xFlow_Home\APPS\Services** location
   3. Restart xFlow Analyst Interface services.
5. **Problem**:
   An error occurs in the Viewlog file: "**Unable to find valid certification path to requested target**".
   Resolution:
   Open the command prompt, and enter the path:
   ```
   cd %USM_HOME%\embedded\jdk\bin
   ```

Now run the following command:

```
keytool -import -alias <alias_name> -file <certfile> -keystore %USM_HOME%\embedded\jdk\lib\security
\cacerts
```

For example:

```
keytool -import -alias server -file casmapp.cer -keystore "C:\Program Files\CA\Service Catalog\embedded
\jdk\lib\security\cacerts"
```

6. **Problem**:
   CA Service Desk Manager failed to connect to **Maileater**, **PAM**, or **Catalog** due to SSL configuration issues. The logs displayed the error *"Unable to find valid certification path to requested target."*
   Resolution:
   Follow this KB article.

CA Service Management 17.4.4.0 Release Notes

CA Service Management 17.4.4.0 is a cumulative patch for CA Service Management 17.4. For more information about the contents of previous cumulative patches, see Release Information.

This roll-up patch contains quality and security improvements, for more information review the Fixed Issues section.

Enhancements

- **Reporting API for Service Desk Manager**:
  Reporting is a crucial feature of CA SDM, enabling administrators, service owners, and service delivery managers to analyze service desk performance, ticket management, and workload trends. These reports are key for assessing performance and identifying areas for improvement. Previously, we integrated JasperSoft Reporting to provide visualizations, drill-down capabilities, and custom dashboards. However, as JasperSoft will no longer be supported, we are transitioning to a more flexible and extensible reporting solution based on APIs.
- Enable HTTP Strict Transport Security (HSTS) support across the solution.
- The SSL Configurator Utility Wizard for CA Service Management has been improved to enable secure configurations across the solution. Upgrading to 17.4RU4 will also maintain existing SSL settings across CA SDM, xFlow/ ServicePoint, and CA Service Catalog.
- When a Knowledge Document is accepted as a solution for an Request/Incident/Problem, a short resolution is displayed. For more information refer Post Installation page.
- xFlow Usability Enhancement: Add attachments by cut/copy and paste in a ticket, and attach a file using the drag and drop functionality or by clicking attach button and selecting the file.

NOTE
More Information

- To install CA Service Management 17.4.4.0, see Installing CA Service Management 17.4.4.0
- For updated information on uninstallation, see Uninstall CA Service Management 17.4.4.0

Certification

CA SDM 17.4.4.0 is now certified with CA ITPAM 04.4 CP 04.

Fixed Issues

CA Service Management 17.4.4.0 includes the following defect fixes.

Fixed Issue ID	Product	Description
Case #35797333	CA Service Desk Manager	Autosuggest doesn't work when searching for '%23' string in a lookup field.
Case #35938955	CA Service Desk Manager	The description field contains additional (unwanted) Linebreaks.
Case #36155888	CA Service Desk Manager	SOAP with CXF - When invoking the getGroupMemberListValues method, an exception is being generated, if the query does not return a result.
Case #33598936	CA Service Desk Manager	The PDM_D_MGR process is crashing on a single CA SDM APP server, about two or three times in a week.
Case #35791986	CA Service Catalog	Javascript functions are not populating the Category as expected for specific Service options.
Case #35994948	CA Service Desk Manager	CASM 17.4.2 – xFlow SAVE-Button does not come up.
Case #35974874	CA Service Desk Manager	Incident Impact should not be changed by Blackout Window.
Case #35994962	CA Service Desk Manager	In CASM 17.4.2 – xFlow Button solve (Lösung) does not work as expected.
Case #35924208	CA Service Desk Manager	In SDM 17.4 RU2 and RU3, the Spell-Checker and Knowledge-Search buttons are missing in the HTML Description.
Case #35973579	CA Service Desk Manager	When editing a knowledge document, links to other documents are created with an OpenDocumentViewer function, but saving the document removes this function, making the link non-functional.
Case #35957268	CA Service Desk Manager	When an Incident or Request is created in the Service Desk interface and the summary field is autofilled from the description field, it may contain blank spaces at the beginning and end.
Case #35845254	CA Service Desk Manager	REST API is not respecting category validation: REST generated requests are allowing non-request categories, and allowing the application to create tickets.
Case #35958350	CA Service Desk Manager	Cannot Create Incidents via WebServices: Axis Web Services method createRequest does not work when using type="crt:182".
Case #35842358	CA Service Desk Manager	No publish action in history when publishing knowledge document: When publishing a document, changing the Owner during the process doesn't create an entry in the document's history, whereas not changing the Owner does trigger an event entry in the history.
Case #35744820	CA Service Desk Manager	Issues with the new (CXF) Webservice logs.
Case #35721052	CA Service Desk Manager	Knowledge Is not showing Comments, Doc versions: Only one event is recorded in the history of a knowledge document in Service Desk Manager, whereas multiple events (like creation, publishing, and updates) are expected to be logged.

Fixed Issue ID	Product	Description
Case #35919217	CA Service Desk Manager	CASM 17.4 RU2 - maileater images in htm_description shown as empty frame: When maileater is importing an email containing the image in the body this image is shown as an empty frame in html_description.
Case #36046173	CA Service Desk Manager	GRLoader Model lookup doesn't work: If the model is specified as a lookup field in the input file for GRloader, it fails to update or create the CI with the corresponding model information.
Case #35892013	CA Service Desk Manager	HTML formatting issues in the Maileater email replies.
Case #35603597	CA Service Desk Manager	List-type property fields do not appear as expected in xFlow.
Case #35904750	CA Service Desk Manager	Issues with font colour change (font color and background font) in RTE.
Case #36122937	CA Service Desk Manager	Undesired Data partition messages.
Case #35911348	CA Service Desk Manager	Issue in CA SDM: Inactive Contacts with Email Addresses in CA Contacts - GRloader Assigns Configuration Items to Inactive Accounts.
Case #36149070	CA Service Desk Manager	Errors in the stdlog when using the REST APIs, from 17.4 RU3.
Case #35995195	CA Service Desk Manager	Temp files are not deleted from CATALINA_BASE_REST temp folder.
Case #35995597	CA Service Desk Manager	Login to Visualizer fails when accessed from the CI List by a domain user.
Case #36166062	CA Service Desk Manager	An error occurs when attempting to change the Virtual Analyst icon.
Case #35794772	CA Service Desk Manager	Vulnerabilities due to insufficient session ID entropy and cookies not being transmitted over SSL.
Case #35695195	CA Service Desk Manager	Vulnerability in attachment uploading with .exe in employee interface.
Case #35792488	CA Service Desk Manager	Issue in Appsec Point on servicedesk Uat for 2024-25
Case #36040543	CA Service Desk Manager	Path traversal vulnerability in file upload functionality: During the attachment upload process, the customer is modifying certain HTTP request parameters in the POST request related to the database update for attachment information.
Case #36049575	CA Service Desk Manager	Unrestricted file upload in CA SDM.
Case #36092728	CA Service Desk Manager	Issue with the size of the Change Order effort attribute.
Case #36012949	CA Service Desk Manager	In 17.4.2.0, xFlow - Description Field is no longer resizable.
Case #36075295	CA Service Desk Manager	Unable to log in to NTLM Xflow.
Case #36062739/ 36246286/ 36272268	CA Service Desk Manager	Search of areas and offerings does not work after upgrade to 17.4 GA. ServicePoint not searching keywords.
Case #36096506	CA Service Desk Manager	PAM 4.4 - Error - Policy Limit Exceeded
Case #35549332	CA Service Desk Manager	Email notification issues in CA SDM: Post upgrade from 4.3 SP5 Process Automaton Manager (PAM) to 4.4 CP2 Process Automation Manager, our notifications changed.

Fixed Issue ID	Product	Description
Case #35835937	CA Service Catalog	Cannot read properties of undefined (reading '$select') - ca_fdFetchSelectData
Case #35959770	CA Service Catalog	In CASM 17.4 RU3, report id does not display the values as expected: In CA Service Catalog, the select type fields that contain a report id do not display the values from the list.
Case #35972087	CA Service Catalog	Approval emails to Groups are not delivered to members for certain/most groups in in EEM/Catalog.
Case #35943312	CA Service Catalog	Grant all access roles permissions does not apply to all option groups items.
Case #36123023	CA Service Catalog	In 17.4 RU3, the table label appears in front of the date picker, causing a visibility issue.
Case #36117563	CA ServicePoint	No data partition message is displayed for employee workflow task.
Case #35728701	CA ServicePoint	Insights - repository loading
Case #36117553	CA xFlow/ ServicePoint	While switching between xFlow and ServicePoint, user is unable to access the reports and other objects while using insights.
Case #35898007	CA APM	Import SAM data into ITAM.

Hot Fix

1. For 17.4 RU2 and later, integrated with the JasperReports 9.0.0 environment, apply Hotfix T5UG596.caz with APAR No: 99112519 to resolve a performance issue in ad-hoc views when adding additional fields to crosstab rows.
2. JasperReports Server Directory/Path Traversal Vulnerability: A directory traversal vulnerability has been identified that potentially allows unauthenticated users to access sensitive information on the host system. An attacker could send a specially-crafted URL request to view files on the server.
   This fix should be applied on the 17.4 RU4 and above, integrated with JasperReports 9.0.0 environment.
   To eliminate the CVE-2024-3325, apply the hotfix T5UG584.caz with APAR No : 99112426.

Known Issues

1. **Issue**: With some mail clients, Maileater fails to generate the description using HTML content when the email body contains only HTML and does not include a Maileater tag.
   Solution: To resolve the issue, apply patch T5UG591 (for Windows with APAR No: 99112469) or T5UG592 (for Linux with APAR No: 99112470) on top of RU4.

For a list of existing issues, see Known Issues.

CA Service Management 17.4.3.0 Release Notes

CA Service Management 17.4.3.0 is a cumulative patch for CA Service Management 17.4. For more information about the contents of previous cumulative patches, see Release Information.

This roll-up patch contains bug fixes, for more information review the Fixed Issues mentioned below.

Enhancements

CA SDM

- **Enhancements to Change Calendar and Change Order Details**

- **Non-Global Windows Added**: The Change Calendar now displays both global and non-global change windows for a complete view.
- **Description Field Included**: The description field from the Change Window detail page is now visible in the Change Calendar.
- **Detailed Blackout Alerts**: The Change Order detail page now shows which blackout windows apply, including their names and total count.

These updates improve visibility and clarity for users managing change schedules. For more information, refer to Change Window Management.

CA ITAM

- ITAM now supports both SDM AXIS and CXF web services, for more information please refer CA Service Catalog CXF webservices for CA APM

Certification

CA SDM 17.4.3.0 is now certified with SUSE Enterprise Linux 15.6.

Fixed Issues

CA Service Management 17.4.3.0 includes the following defect fixes. For more information, see Steps to reproduce document.

Fixed Issue ID	Product	Description
Case #35618464	CA Service Desk Manager	The Description field is not coming through on the Change Orders created by Employees.
Case #35587746	CA Service Desk Manager	Unable to change the template of Manual Notify after the CA Service Desk 17.4.2.0 Rich Text-/HTML implementation.
Case #35581473	CA Service Desk Manager	Issues arise while creating a new CMDB model.
Case #35546587	CA Service Desk Manager	Personalized reply does not work in the “Manual Notify” activity.
Case #35540546	CA Service Desk Manager	Scratch Pad information does not transfer to the Incident Detail - Description.
Case #35532769	CA Service Desk Manager	Service Management v17.4.2.0 Upgrade installer is overwriting an existing option record.
Case #35490287	CA Service Desk Manager	If a CI's Status is set to inactive, the IP-address (alarm_id) should be released.
Case #35489449	CA Service Desk Manager	After applying RU2 in the body of maileater, new lines of text are not copied to description.
Case #35470183	CA Service Desk Manager	After installing 17.4 RU2, in the Incident Detail report the Manual Notify act log displays html code instead of plain text.
Case #35375329	CA Service Desk Manager	After applying RU2 in the mailbox Email Type GRAPH is displaying as DIAGRAMM in German install.
Case #35448547	CA Service Desk Manager	After applying RU2 the maileater is processing plain text emails incorrectly.
Case #35348496 and 35332768	CA Service Desk Manager	Issues in maileater and email templates in 17.4.2.0.
Case #35333039	CA Service Desk Manager	Manual Notify not showing in HTMPL format in Activates Log.

Fixed Issue ID	Product	Description
Case #35326026	CA Service Desk Manager	In the Employee Interface, an incident ticket is not create after searching for knowledge.
Case #35318053	CA Service Desk Manager	When you run the pdm_pki command utility, it does not log error/info messages anywhere.
Case #35230272	CA Service Desk Manager	Line break issues in Description text.
Case #35170567	CA Service Desk Manager	SDM Sanitation does not replace \n by
Case #35047759	CA Service Desk Manager	Configuration in CA SDM Advanced Availability fails.
Case #35044812	CA Service Desk Manager	When the Type field is empty, the ticket is created as a Request, even when the Category does not allow the creation of Request ticket type.
Case #35020216	CA Service Desk Manager	Cannot read properties of undefined (reading 'value') at copy_field_val.
Case #33723379	CA Service Desk Manager	In REST Tomcat Appsec Vulnerability - HTTP “OPTIONS” Method Enabled.
Case #33723368	CA Service Desk Manager	Stored cross-site scripting in CA SDM.
Case #33723330	CA Service Desk Manager	After upgrading to 17.4 RU2 from RU1, SDM IIS SSL URL does not load.
Case #33722876	CA Service Desk Manager	Vulnerability: Java script library.
Case #33701408	CA Service Desk Manager	In SDM_17.3RU22_Error message Alias for session was not indexed.
Case #33684723	CA Service Desk Manager	SDM_17.2RU1_Editing OOTB and running reports leave suspended querys.
Case #33643149	CA Service Desk Manager	Attached events continue to fire even after they are set to be inactive.
Case #33598973	CA Service Desk Manager	Service Catalog integration with SDM fields goes empty.
Case #35332786	CA Service Desk Manager	Maileater is updating tickets with wrong data.
Case #35485998	CA Service Desk Manager	Jasper 7.9 error in CA SDM.
Case #35499329	CA Service Desk Manager	When hovering over the Manual Notification in the Activity Log, it displays HTMPL code.
Case #35461689	CA Service Desk Manager	After applying RU2 newlines in the HTML-Part of the notification-template are no longer in the received email.
Case #33665082	CA Service Desk Manager	Vulnerabilities in SDM.
Case #33643260 and #35083411	CA Service Desk Manager	Vulnerabilities in datamart log4j.
Case #33593356	CA Service Desk Manager	CA SDM attachment issue.
Case #35124582	CA Service Desk Manager	RASA and Python vulnerabilities.
Case #35157933	CA XFlow	After upgrading CASM 17.4.1 to 17.4.2 xFlow no longer allows to work with Tickets.
Case #33638949	CA Service Catalog	One Service Offering is not displayed in the Service Catalog.
Case #33729964	CA Service Catalog	USS integration error occurs while upgrading Swingbox from 17.1 to 17.4.

Fixed Issue ID	Product	Description
Case #33712946	CA Service Catalog	Table header disappears in request form table component.
Case #33708646	CA Service Catalog	Changes in table behavior after upgrading to 17.4.1.0
Case #33705519	CA Service Catalog	CV-CONFIRM-DEFECT_Catalog_17.4RU1-Datasource ODBC connection does not work.
Case #33689588	CA Service Catalog	Issue with listing subscriptions on certain BUs.
Case #33622749	CA Service Catalog	End users are facing problem to see their pending request, assigned to their respective groups.
Case #35614393	CA ServicePoint	The lookup for Requester is not working in ServicePoint for the Catalog Offering.
Case #35627952	CA ServicePoint	In the ServicePoint, the Catalog forms are taking longer to refresh.
Case #35617982	CA ServicePoint	Service Point mouse over/hover and complete ticket activity frame does not line break after 17.4.2.
Case #35406022	CA ServicePoint	Default value for the custom object attribute not working.
Case #35535085	CA ServicePoint	Issues in Service Point "My Approvals" functionality.
Case #35358525	CA ServicePoint	You can create tickets in Service Portal with inactive categories if the link is bookmarked.
Case #33630542	CA ServicePoint	Large logo size for Catalog service offerings in ServicePoint.
Case #33710557	CABI	On SDM and Jasper integrated environment, unable to run the Adhoc views when locale is set to en_ZA.
Case #33701429	CA APM	Field Label Changes on Location Configuration won't Save (Legal Document -->Cost -->Payment).
Case #33602808	CA APM	Asset family and Class field are not always populated.
Case #33586589	CA APM	The Stored Procedure blocks DB tables for several seconds.

Known Issues

For a list of existing issues, see Known Issues.

NOTE
More Information

- To install CA Service Management 17.4.1.0, see Installing CA Service Management 17.4.3.0.
- For updated information on uninstallation, see Uninstall CA Service Management 17.4.3.0

CA Service Management 17.4.2.0 Release Notes

Release Notes for CA Service Management 17.4.2.0

CA Service Management 17.4.2.0 is a cumulative patch for CA Service Management 17.4. For more information about the contents of previous cumulative patches, see Release Information.

This roll-up patch contains bug fixes, for more information review the Fixed Issues mentioned below.

What's new?

CA Service Desk Manager

- We have made capturing details of a ticket (incident/request/problem/change order/issue) easier by including support for rich text in the description for ticket types. For further information, refer to Rich Text Editor Capability in CA SDM.

CA ServicePoint

- Supports classic workflow approval process. For further information, refer to ServicePoint Workflow Approval Process for CA SDM.
- Now supports browsing through the list of Service Desk Manager Categories. For more information, refer to Ability to Browse SDM Categories.
- Virtual Analyst - The RASA installation process has been automated for Windows. For more information, refer to Automated Installation of Virtual Analyst for Windows.

CA Service Catalog

From 17.4 RU2, CA Service Catalog supports CXF based webservices, which helps the product to move away from AXIS based web services. AXIS based webservices are deprecated and they will eventually get removed in the future versions of CA Service Catalog. For more information, have a look at the Axis to CXF Migration guide.

CA APM

ITAM now supports both SDM AXIS and CXF web services, for more information please refer SDM CXF support for CA APM.

Certifications

- CA Service Desk Manager is now certified with TIBCO Jaspersoft 9.0.0 (Jaspersoft® Studio 9.0.0 and JasperReports® Server 9.0.0). For more information, refer to For more information refer Jaspersoft Studio and JasperReports Server.
- Embedded Entitlements Manager (EEM) 12.6.7.0. For more information, refer to CA EEM 12.6.7.0.
- IT Process Automation Manager 4.4 CP03. For further information, refer to CA IT PAM 4.4 CP03.

Fixed Issues

CA Service Management 17.4.2.0 includes the following defect fixes. For more information, see Steps to reproduce document.

Fixed Issue ID	Product	Description
Case #33416126	CA Service Desk Manager	"Unknown value for status" error is received in Spectrum while trying to create a ticket in SDM.
Case #33544230	CA Service Desk Manager	Knowledge Management Custom Approval Process does not populate Assignee.
Case #33546477	CA Service Desk Manager	The specified error message in the Pre-Update DP does not show in Service Point.
Case #33557830	CA Service Catalog	Offering is stuck in Approved state.
Case #33562159	CA Service Desk Manager	17.3.23-SOAP Call provides ambiguous fault message without additional tracing
Case #33586849	CA Service Desk Manager	When multiple files are attached to the same ticket, the attachments are not being written to correct locations.
Case #33590312	CA Service Desk Manager	Few HSTS vulnerabilities were reported.
Case #33592625	CA Service Desk Manager	German localization
Case #33593356	CA Service Desk Manager	Intermittently the upload and download of attachments are not working.

Fixed Issue ID	Product	Description
Case #33595872	CA Service Desk Manager	In a multi-tenanted environment, Activity Notifications are sent indiscriminately to all members of a named Contact Type in the Activity Notification configuration.
Case #33596447	CA Service Desk Manager	If a user accesses the URL with Category, the displayed ticket creation screen does not have the specified Category. However, the created ticket had the expected Category.
Case #33606408	CA Service Desk Manager	pdm_cgireport.exe on IIS exposes the hidden arguments, which present vulnerability that unexpected parties can use to access SDM data.
Case #33609483	CA Service Desk Manager	Reported by vulnerability tool.
Case #33627232	CA Service Catalog	End users are facing problem to see their pending requests assigned to their respective groups.
Case #33639815	CA Service Desk Manager	ACAS Nessus scan found a few vulnerabilities on the SDM 17.4.1 server.
Case #33654916	CA Service Desk Manager	Creating an Incident with a Request category is not enforced in SOAP webservices.
Case #33660564	CA Service Desk Manager	Issues in the Announcement Page HTML on Employee Interface.
Case #33666776	CA Service Catalog	After upgrading to 17.4 RU1 from 17.3 RU1, there were discrepancies in fetching the report data.
Case #33681660	CA Service Catalog	In CMDB CI Association window, the remove button next to each CI does not match the CI UUID in Service Catalog CI association.
Case #33684464	CA Service Desk Manager	In USRD table in CA SDM the tenant is not auto-populated when it is a lookup field.
Case #33701831	CA Service Desk Manager	The SDM upgrade script always resets the repository paths back to defaults.
Case #33668407	CA Service Desk Manager	Page loads without the ServicePoint log.

Mandatory Cumulative Test fixes after applying 17.4 RU2

NOTE
Install all the patches mentioned below. To install the patches, follow the steps mentioned in the respective ReadMe files.

Support Site Links

Windows

Fix	File type	Solution Description
99112110	CAZ	WIN-xFlow\BUI Activity log entries not showing in HTML format.
99112111	CAZ	WIN-Activity log entries not showing in HTML format.
99112220	CAZ	WIN-Personalized Responses in Manual Notify

Linux

Fix	File type	Solution Description
99112110	tar.Z	LNX-xFlow\BUI Activity log entries not showing in HTML format.
99112112	tar.Z	LNX-Activity log entries not showing in HTML format.
99112221	tar.Z	LNX-Personalized Responses in Manual Notify

Fixed Issues

After installing the CA Servicement Management 17.4 RU2, a few issues were present. The following are a list of test fixes. For more information see Steps to reproduce document.

Fixed Issue ID	Product	Description
Case #35170567	CA Service Desk Manager	CA SDM Sanitation does not replace \n by
Case #35230272	CA Service Desk Manager	The description text no longer breaks the lines properly. It looks like the text is on one line. The issue only occurs with old tickets from before 17.4 RU2.
Case #35348496	CA Service Desk Manager	Issues with Maileater and email templates.
Case #35448547	CA Service Desk Manager	Maileater does not read text-emails correctly.
Case #35470183	CA Service Desk Manager	The incident detail report's Manual Notify act log has html code instead of the plain text.
Case #35461689	CA Service Desk Manager	Empty lines in an email notification are missing, and everything is displayed as a single line.
Case #35187945	CA Service Desk Manager	Manual notification displays html tags instead of html content in Activity log.
Case #35532769	CA Service Desk Manager	Service Management v17.4.2.0 Upgrade installer is overwriting an existing option record.
Case #35546587	CA Service Desk Manager	Personalized reply does not work in the “Manual Notify” activity.

JasperReports Server 9.0.0 TFix

In JasperReports Server 9.0.0 DVD integrated environment, the ***Allow.jar*** file is visible in the UI. To eliminate the CVE-2024-3325, apply the hotfix T5UG571.caz with APAR No : 99112250.

Known Issues

1. **Issue:**
   Start Request Form is not loading in Service Catalog-17.4 RU2
   Solution:
   Refer the KB Article and follow the steps.

For a list of existing issues, see Known Issues.

NOTE
More Information

- To install CA Service Management 17.4.2.0, see Installing CA Service Management 17.4.2.0.
- For updated information on uninstallation, see Uninstall CA Service Management 17.4.2.0

CA Service Management 17.4.1.0 Release Notes

Release Notes of 17.4.1.0

CA Service Management 17.4.1.0 is a cumulative patch for CA Service Management 17.4. For more information about the contents of previous cumulative patches, see Release Information.

This roll-up patch contains bug fixes, for more information review Fixed Issues mentioned below.

Security Vulnerability

Addressed the security vulnerability Apache ActiveMQ CVE-2023-46604. Upgraded Apache ActiveMQ version from 5.15.15 to 5.15.16

Fixed Issues

CA Service Management 17.4.1.0 includes the following defect fixes. For more information, see Steps to reproduce document.

Fixed Issue ID	Product	Description
DE574567	CA Service Desk Manager	The -m option for the maileater does not work as expected.
DE574845	CA Service Desk Manager	Same search query for knowledge documents search is giving different search results in SDM/xFlow/ServicePoint.
DE574831	CA Service Desk Manager	When we transfer a ticket to a new group and contact who is not in the new group, a log is generated saying the transfer is made.
DE574918	CA Service Desk Manager	Unable to sync LDAP status of a user with the SDM user.
DE576471	CA Service Desk Manager	GRLoader removes line breaks, and the description fields display text in a single line.
DE575776	CA Service Desk Manager	SAML login fails if username has special characters.
DE575200	CA Service Desk Manager	The html injection given in the scratch pad is running the command.
DE575768	CA Service Desk Manager	Cross site scripting fails.
DE575770	CA Service Desk Manager	DB Connections remain in open state and sleeping, until DB Server runs on errors and CA Application Server Environment (Advanced Availability) crashes.
DE576682	CA Service Desk Manager	WebServices and GRLoader do not work after upgrading to 17.3 RU 22.
DE577420	CA Service Desk Manager	A few related tables are not getting archived along with the SDM tickets, and when they are manually added to itil_arcpur_cfg.xml, the other tables are archived twice.
DE580333	CA Service Desk Manager	Maileater is stuck trying to process the given mail and it remains in the Inbox.
DE575766	CA Service Desk Manager	High memory utilization of Domsrvr.

Fixed Issue ID	Product	Description
DE579182	CA Service Desk Manager	When creating a CI via a script or the GUI, a usp_owned_resource record is created but not a ca_owned_resource record.
DE579622	CA Service Desk Manager	When emails are sent with the new pdm_mail_assembly.jar file provided, the original Unicode encoding is lost (Serbian Latin encoding).
DE576743	CA Service Desk Manager	Performance issues related to usp_lock_release_multiple SP
DE580441	CA Service Desk Manager	Unable to export incident list after the upgrade in 17.4.
DE576481	CA Service Point	Search on ticket confirmation screen does not work.
DE576467	CA Service Point	Cross-Site Scripting (XSS) vulnerabilities occur in CA Service Point when using CA Catalog offerings.
DE576461	CA Service Point	The label text overlaps in Service Point, if a Box Label of a checkbox has long text in a form.
DE576456	CA Service Point	In 17.3 RU22, when the subject of the ticket is more than 90 characters, the label overlaps the link for reopening the ticket. If the status of the ticket has more than 28 characters, the label covers the last modifed date. Both occur on the Service Point home page.
DE576455	CA Service Point	The "Close Ticket Status" message in Aria Chatbot is incorrectly displayed as NONE instead of “Closed”.
DE575198	CA Service Catalog	For a notification to be sent from Catalog, a new field "Related Ticket" details is added.
DE576465	CA Service Catalog	Procedures and scripts to archive or delete data do not work and throws errors.
DE576720	CABI	Ad Hoc views using fields like "Duration Resolve to Close in Days" error out with "already closed" error.
DE575194	CABI	Reports which query a Double type field from CA SDM's object schema displays incorrect values.
DE576459	CA APM	In ITAM_17.3_the admin_password field from al_process_account table is created with a space at the end.

Mobile App Fixed Issues

CA ServicePoint Mobile app 1.7.2 includes the following defect fixes.

Fixed Issue ID	Product	Description
DE581035	Service Point Mobile App	Service Point Mobile Attribute is not showing custom fields in the drop-down.

Resolved Known Issues

Fixed Issue ID	Product	Description
DE583305	CA Service Management	CA service Management now supports SiteMinder 12.8.7 integration.

Known Issues

For a list of existing issues, see Known Issues.

NOTE
The issue- "Siteminder integration with XFlow/Servicepoint is not working if we configure SSOArtifactType as "header"" is now fixed in CA Service Management 17.4.1.0

NOTE
More Information

- To install CA Service Management 17.4.1.0, see Installing CA Service Management 17.4.1.0.
- For updated information on uninstallation, see Uninstall CA Service Management 17.4.1.0

Getting Started

An overview of features and basic use of CA Service Management.

This section includes information about CA Service Management architecture, capabilities, and benefits. Review the information in this section before you implement the solution.

Service Desk Manager

CA Service Desk Manager offers a comprehensive set of customizable services to suit the needs of the organization. CA SDM integrates all stages and elements of service desk operations into a single platform, including service requests, incident, change, knowledge and configuration management.

CA SDM is designed to improve service levels provided to end users and efficiency of the overall department, and also includes reporting and advanced dashboard features, that facilitate better decisions and lower operational costs.

Following are the benefits of Service Desk Manager:

- Multi-tenant architecture that supports service desk consolidation for global enterprises.
- Empowers service desk analysts to be more efficient and effective.
- Minimize service disruptions and risks resulting from unforeseen changes thus, improving the day-to-day operations of the organization.

xFlow Interface

In today's world, IT organizations must adapt and adopt changes to meet the growing business demands, unfortunately, the tools available to service desk analyst typically fall short and mostly use ticketing based solutions developed decades ago. Thus, business users are disconnected and a service desk analyst lacks the knowledge and context to make the right decision.

xFlow Interface provides comprehensive and intelligence-based views that analysts, teams, and the service desk to provide first-class customer service.

For a service desk analyst, the xFlow Interface empowers to stay connected, provides a clear-and-contextual view of the work.

Following are the benefits of xFlow Interface:

- Prioritize workload.
- Understand the state of the IT environment.
- Get the context of issues.
- Quickly triage and resolve issues.

Service Point

Service Point for CA Service Management provides a self-service experience with embedded assistive intelligence for business-users. Service Point uses Natural Language Processing (NLP) that provides business users with context-based solutions and suggestions.

Business users can perform the following actions using Service Point:

- Access knowledge
- Create, view, and modify tickets
- Contextually collaborate on tickets using live chat
- View allocated resources
- Manage ticket notifications

With a business-centric focus, Service Point helps an organization achieve the following:

- Broad adoption of services
- Context-based solutions
- Improved productivity
- Reduced costs
- Improved collaboration and knowledge sharing
- Elevated perception of IT organization

CA Service Management Virtual Analyst

Few quick checkpoints are listed below for CA Service Management Administrators to get started with the implementation and deployment of Virtual Analyst in their organization.

1. Install Virtual Analyst and RASA Server. RASA is the NLU engine used in CA Service Management Virtual Analyst. For more information, see Install Virtual Analyst. In case, if you do not have internet connectivity, you do have the option to install the virtual analyst in an offline mode. For more information, see Offline Installation - CA Service Management Virtual Analyst.
2. After successful installation, training the NLU server is mandatory for the Virtual Analyst to be up and running. Train the NLU server to train the virtual analyst with out-of-the-box (OOTB) data. For more information, see Train the NLU Server
3. RASA server configuration details must be updated in CA Service Desk Manager (CA SDM) to enable the Virtual Analyst services. Perform this step in CA Service Desk Manager Interface. For more information, see Configure Virtual Analyst in Service Desk Manager
4. Enable the Virtual Analyst capabilities via the CA Service Desk Manager Interface. Login to CA SDM, navigate to **Administration**, **xFlow Interface**, **Service Point** configurations. For more information, see Manage Features
5. Administrators can configure and create tickets via the Virtual Analyst through template-based ticket creation or through auto-categorization. For more information, see Manage Features.
6. If you have custom attributes in your tickets, and want these to be a part of Virtual Analyst ticket creation process, see Custom Attributes for Ticket Creation.
7. Administrators can customize the Virtual Analyst responses to suit implementation requirements. For more information, see Virtual Analyst Intents.
8. Customize the look and feel of the Virtual Analyst through Manage Features. CA SDM Administrators can update the Virtual Analyst Name, Logo, Launch Text via the Manage Features capability in Service Point.

Insights

Insights (CABI JasperReports 6.4.3) is built on TIBCO Jaspersoft Business Intelligence (BI) Technology and is a light-weight embeddable BI tech stack. It supports Out-of-the-Box Reports, creation of complex reports and dashboards for Aggregate and Transactional Reports. Supports Multi-tenant architecture with easy to use self-service BI. CA SDM along with xFlow Interface/Service Point provides access to your reports via the Insights App and CA SDM Interface.

Insights enable decision-makers with self-service analysis capability to understand and analyze the key metrics and historical trends from CA SDM operational data. The interactive reports and dashboards allow to view all the key metrics of the IT environment in one place and perform adhoc analysis of relevant metrics.

Following are the benefits of Insights:

- Self-service analysis of KPIs for decision makers.
- Easy monitoring and access.
- Data driven decisions.
- Measure the effectiveness of your IT process.
- View all the key metrics in one place.

To help you with compatibility related information, see Supportability Matrix.

To help you with pre-installation steps, see CABI JasperReports Server (Insights) Considerations.

For step-by-step instructions to Install and configure CABI JasperReports for CA Service Management, see JasperReports Server.

To administer, see Administer Insights.

For using, see Using Insights.

For troubleshooting issues, see Troubleshooting Insights (CABI JasperReports)

Service Catalog

CA Service Catalog helps create a one-stop storefront experience to manage and automate the entire request lifecycle, including the initial request, approval, and fulfillment of services. Behind the scenes, workflow automates and manages fulfillment and related processes, simplifying power user tasks and increasing the productivity of all the involved users.

Following are the benefits of Service Catalog:

- Define service offerings with complete cost transparency and associated SLAs.
- Reduce the cost and effort of delivering services.
- Automate time-consuming, manual, and scripted tasks.

Asset Portfolio Management

CA Asset Portfolio Management enables you to holistically manage your company's IT assets and determine what assets are available and in use, improve the organizational distribution of those assets, and increase productivity around license reporting and audit responses.

Following are the benefits of Asset Portfolio Management:

- Deliver business value.
- Reduce risks and costs.
- Maximize the value of IT assets.
- Increase software compliance.

To help you with install planning of CA Service Management, see CA Service Management 17.4 Release Notes.

For step-by-step install instructions, see Installing.

To help you with upgrade planning of CA Service Management, see CA Service Management 17.4 Release Notes.

For step-by-step upgrade on upgrading CA Service Management, see Upgrading.

Lift and Shift Cloud Migration

Lift and Shift for CA Service Management on AWS, Azure, GCP.

The lift and shift migration approach involves migrating your application and associated data to the cloud with minimal or no changes, effectively "lifting" applications from their existing environments and "shifting" them as-is to a new hosting premises, typically in the cloud.

Typically no significant changes are made to the application architecture, data flow, or authentication mechanisms.

Key considerations in a lift and shift migration include the application's compute, storage, and network requirements, which should be mapped from the current infrastructure to equivalent resources in the cloud. This strategy takes the path of least resistance, allowing you to "re-host" your existing workload without modifying the application.

> **NOTE**
> CA Service Management is certified for Oracle database on Google cloud platform.

For more information refer Moving the CA MDB Data from the Source to the Target Systems.

For steps on cloud migration, refer Moving the CA MDB Database on Microsoft SQL Server.

CMDB Visualizer

CA CMDB Visualizer updates.

Check out the CMDB Visualizer updates in this article.

- CMDB Visualizer Administration Interface
- CMDB Visualizer Interface
- Configuring CMDB Visualizer

CMDB Visualizer Administration Interface

This article describes the CA CMDB Visualizer Administration Interface optimizations from an end-user perspective.

Use the CMDB Visualizer administration interface to perform the following functions:

> **NOTE**
> For Oracle 21c, set the **isClusterEnv** setting to "true" in the application.conf file of CMDB viz and restart the CA SDM Services.:
>
> - For Windows: <SDM installed location>\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\WEB-INF\classes\application.conf)
> - For LINUX: <SDM installed location>/bopcfg/www/CATALINA_BASE_VIZ/webapps/CMDBVisualizer/WEB-INF/classes/application.conf).

> **NOTE**
> The functions described in this article are only available to roles with administrative privileges.

> **IMPORTANT**
> To access CA CMDB Visualizer, ensure that the URL name is consistent for all CA Service Management products (including FQDN, if any) as '**vizserver**' value in **casm.conf.js** (NX_ROOT\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\conf\casm.conf.js).

1. Install and configure the Visualizer server.
2. Modify the display of CI status and relationships.
3. Create and manage administrative filters.
4. Create and manage Visualizer CI graphics.

> **NOTE**
> In case of CA SDM secondary servers, the Database (SQL Server or Oracle) Client software must be installed. A local domsrvr must be used for the **sdm.domsrvr.viz** setting in **application.conf** and ensure that you have opened the database port for CMDB Visualizer. Visualizer uses the database connection for initial cache loading and performance improvement. If you do not want to expose the database connection to an external network, it is recommended that you plan to have the CA SDM secondary server in a DMZ network.

The Visualizer Web App comes with a configuration file called **application.conf**. The following settings can be configured by an Administrator:

- **sessionTimeout - User session timeout**, in minutes. **Default value = 60**
- **expirationCheckInterval - Interval time**, in minutes, to check for expired sessions. Expired sessions are not removed at the exact time but instead when the expiration check finds it. **Default value = 5**

NOTE
The color is not changed on the hue or color sliding bar in graph, you must select the color in the large palette above the hue slider bar. This is an expected behavior with the slider when the initial color is #000000 or #ffffff (grey scale colors). This is a limitation with the angular component.

CMDB Visualizer Administration Interface provides the following functions:

Find CIs by Assets and Attributes

NOTE
To disable CIs and Asset search, disable the **Enable Asset Search** toggle option on the Visualizer administrative interface.

You can search for CI (and asset) attributes to locate and manage CIs in your database. It mainly consists of the following: Perform the following steps:

1. Type the search criteria in "**Find CIs**" Search pane on the top right pane of Visualizer graph.
   The Search Results tab displays all CIs/Assets that match your search criteria.

- **Primary**:
  Primary fields focus your search that is based on the following basic CI attributes.
- **Network**:
  Network fields search on the following common CI attributes for network addressing.
- **Assets**:
  Asset fields search on the following basic CI/asset attributes.
- **Maintenance**:
  Maintenance fields provide searching on CI attributes for the following maintenance tasks.

For more information, see the following topics.

- Configuration
- Filters

Configuration

This article talks about the configuration of CA CMDB Visualizer.

The CMDB Visualizer Configuration tab allows for modifying server settings and displaying Configuration Items (CIs). Login to CA CMDB. Select the **Configuration** option from the Left Menu tree to configure the following options in a graph:

1. **Display Settings**: Provides the following options to set CI labels in a graph:
   a. **Number of Lines to Display for CI Labels**:
      Based on the specified number, the CI Label displays in the graph. Example: If 3 is specified, the label displays in three lines in a graph.
   b. **Characters per Line for CI Labels**: Specify the Characters for CI labels in a graph. For example: If 10 is specified, after 10 characters, the label will display in a new line.
   c. **Display Relationship Label**: Enable or disable this option as per your requirement. If you disable this option, CI relationships are not displayed in a graph.
   d. **Relationship Label Length**: Specify the CI relationship label length - maximum number of characters in which a graph relationship label will be displayed.
   e. **CI Relationships to (Default value)**: Used to specify the default number of levels with which the graph will be displayed.
   f. **Enable Asset Search**: If enabled, you can search for assets or CIs.

Relationship Style
Use this option to define relationship characteristics in a graph.
Select the Relationship from the list. Select the option to differentiate a relationship in a graph via **line color** and **line thickness**. Use **Preview** to visualize how the graph will be displayed in real time with the specified options.

CI Status

Use this option to define CI graphical characteristics.

The user can select CI configuration Status Image along with Image color. User can **Preview** CI configuration status with selected color and category.

Filters

This article describes how built-in filters work in CA CMDB Visualizer.

CA CMDB Visualizer provides built-in analysis filters, and an ability to create filters.
Filters support CI families, relationship types, and other attributes. Admin and Non-Admin Filters are supported. However, in administrator mode, only admin filters are shown, while all other filters are shown in non-admin mode.

> **NOTE**
> While rendering a filter with "**populate from graph**" peer attributes is shown only when there is a peer relationship among nodes. To apply peer relationship on a selected graph, uncheck "**populate from graph**" and add the peer attributes and apply it on a graph. The graph is displayed with peer relations.
>
> **IMPORTANT**
> CMDB Visualizer supports both **JSON** and **FormData** for creating and editing filters. By default, **JSON** is used. If administrators have a requirement to switch to **FormData**, set the **isFilterObjJsonTyp** value to **false**. By default, it is set to **true** in the **casm.conf** file in <NX_ROOT>*bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\conf\casm.conf.js* .

Admin and Non-Admin Filters

Admin and Non-Admin filters support the following:

- Create Filters
- View Filters
- Edit Filters
- Delete Filters

Non-Admin Supported Filters
Non-Admin Filters are of the following types:

- Load Filter Options from Graph
- Apply Saved Filter on Graph

Filters support the following attributes:

- Family
- Classes
- Status
- Location
- Provider Relationships
- Dependent Relationships
- Peer Relationships
- Other Relationship
- Model
- Manufacturer
- Maintenance Vendor

Create Filters

CA CMDB Visualizer provides built-in analysis filters, and you can create filters for your business needs.

NOTE
Users can create a maximum of 200 filters in the filter pane. Even if you create more than 200 filters, the filter popup shows only 200 filters. The additional filters cannot be accessed.

NOTE
The color is not changed on the hue or color sliding bar, you must select the color in the large palette above the hue slider bar. This is an expected behavior with the slider when the initial color is #000000 or #fffff (gray scale colors). This is a limitation with the angular component.

Follow these steps:

1. Click the New Filter button.
   The Filter Wizard launches.
2. Name the filter and select one of the following filter types:
   a. **Browse**
      Displays the unfiltered display of all related CIs.
   b. **Impact Analysis**
      With the focal CI as provider, shows the CIs that are affected when any change occurs to the focal CI.
   c. **Root Cause**
      With the focal CI as dependent, shows the CIs that could be the root cause for the focal CI to go down or can affect the focal CI's behavior.
   d. **Cause and Effect CIs**
      Combination of Root Cause and Impact Analysis. From the focal CI, shows all the CIs that could be the root cause. Or shows the CIs that could be affected when a change occurs to the focal CI.
   e. **Trace Relation**
      Displays all possible relationships among focal CIs, based on the Levels setting. If the graph has only one focal CI, applying this filter displays only the focal CI.
   f. **Shortest Path**
      Displays the shortest chain of relationships among focal CIs, based on the Levels setting. If the graph has only one focal CI, applying this filter displays only the focal CI.

      NOTE
      In addition to the predefined filters, any available administrator-defined and user-defined filters display.
3. Click Save & Finish.
   The filter is created.

CMDB Visualizer Interface

This article describes the CA CMDB Visualizer Interface optimizations from an end-user perspective.

CA CMDB Visualizer provides a graphical interface with toolbars and a graphical canvas. You can click the button on the left side of the canvas to display the retractable Search pane.

> **NOTE**
> Visualizer uses the space bar to select or confirm an operation. For example, if the focus is on a button, then instead of pressing Enter, press the space bar.

Graphical Interface

CDMB Visualizer Graphical interface provides the following capabilities:

1. A way to create, manage and apply analysis filters.
2. Change the graphical layout and number of CI levels to be displayed.
3. Select CIs for graphical viewing and right-click to provide various functions, including the ability to view and modify CI properties.
4. Use the Toolbar icons to provide various functions.
5. Zoom in to view areas of interest, or zoom out to see the larger context.
6. Use the Full Screen Mode to display a larger work area.

   > **NOTE**
   > When you enable Full-Screen mode, you cannot use the keyboard functionality. Search for CIs by Attributes.

For more information, see the following topics:

- Visualizer Graphs
- Change Icons of CIs on Visualizer

Visualizer Graphs

This article gives an overview on how to use Graphs in CA CMDB Visualizer.

This article contains the following topics:

Open Graph

1. Click Open Graph on the toolbar.
   The Open Graph list displays graphs with saved metadata.
2. From the list, complete one of the following actions:
   - Double-click a graph item.
   - Select a graph and click Open.

   The graph is rebuilt based on its saved metadata and its current CMDB data.

Follow these steps:

1. Click Open Graph on the toolbar.
   The Open Graph list displays graphs with saved metadata.
2. Select the graph from the Open Graph list.
3. Click Delete.
   The graph is deleted.

Invert Graph

By default, Visualizer displays Provider CIs at the top and Dependent CIs at the bottom, relative to the Focal CI.

Follow these steps:

1. Click the Invert Graph button to reverse this arrangement to a vertical axis.
   You can view the data from a different perspective.

Find CIs on the Graph

You can find specific CIs to display as nodes on the graphical display. Use the Find CIs button to open a new window over the graph to prompt for what string to match. The Find string is matched using graphical node attributes such as the CI name and family. Find shows the count of CIs which matched the users input criteria. The count is displayed in the bottom left corner.

Follow these steps:

1. Select the nodes to use from the Find CI results.
2. Click Select On Graph.
   The selected CIs display.

For more information about finding CIs by Assets and Attributes, see CMDB Visualizer Interface.

Show Graph for a Single CI

Follow these steps:

1. On the Visualizer graph, select a CI to serve as the focus of the analysis.
2. Right-click the CIs menu and select Make Focal CI.
   The selected CI becomes the *Focal CI* in the Visualizer. Its associated relations display, including the configured number of levels.

Add Multiple CIs to Graph

Follow these steps:

Make Focal CI, Add to Graph, or drag-and-drop CIs from the following sources:

- Search Results
- Scratchpad
- Saved graphs
- Hidden CI list

> **NOTE**
> Users can add maximum of 10 focal CIs or 10 graphs.

Create Relationships

CMDB Visualizer now provides the ability to define Relationship between existing CIs via the CMDB application without the need to be taken away from the context to define relationships from the CA SDM Interface. The established/defined relationship is reflected immediately in the graphical interface without reloading the graph. For more information, see Working with CIs in Visualizer
Perform the following steps:

1. Select one or multiple (only two) CIs that are displayed on the graph, right click the context menu and select Create Relationships to define relationships between CIs.
2. From the New Configuration Item Relationship dialog box, fill in the following: For example: Relationship Type: deploys between the selected two CIs.
3. Click Save.
   Relationship is created between two CIs.

Filters - Adhoc Analysis
CMDB visualizer now provides the ability to preview and perform Adhoc analysis of graphs. Users can now filter the graph on multiple attributes of a CI and view a preview of the graph with the applied filter. This capability enables users to filter large graphs and perform Adhoc analysis without being forced to save the filter.

- Open a CI in Visualizer or search for CI and drag it on to the graph.
- Click **Filter** to select the attributes that are needed for your analysis. By default, the attributes are populated from the underlying graph. You can unselect the option to view all the attribute values.
- Click **Preview** to view a graph based on the selected attribute values.
- To modify your selection, click **Filter** again and preview the changes.
- To reset the selections to the default state, click **Reset**.
- To save a filter, click filter and provide the name of the filter and click **Save and Apply**.

> **NOTE**
> Actions like addition of CIs to a graph, display levels change, switching of the view can be done while the filter is still not saved. Visualizer provides alerts for unsaved filters.

Layouts

This article talks about

Layout

The Layout drop-down list lets you switch your CI display between two different layout options, depending on your analysis needs. The following types of layouts are available:

- Hierarchical
- Circular

Hierarchical Layout

Use a hierarchical layout when you want to view related CIs that can be logically distant from the focal CI.

Circular Layout

You use a circular layout when you want to emphasize all relationships closest to a focal CI.

Change Levels

You can change the number of levels of CIs that are displayed relative to the focal CI. The maximum number of levels that are allowed is nine. The default value is two levels, which you can modify by using the Administration interface.

Follow these steps:

1. Click the Levels drop-down list.
   The numbered levels appear.
2. Select the number of levels that you want.
   The graph adjusts to reflect the new setting.

Change the Number of Levels of CIs
You can change the number of levels of CIs that are displayed relative to the focal CI. The maximum number of levels that are allowed is nine. The default value is two levels, which you can modify by using the Administration interface.
For more information about how to create Filers, see Filters.

Using the Visualizer Toolbar

This article contains the following topics:

The topic describes the Visualizer toolbar features that are available when the CA SDM server with Visualizer is running.

> **NOTE**
> Select **Overview** to zoom in on a certain area of the graph. Overview section helps users to focus and highlight particular CI/CIs from rest of the CIs on a graph when graphs have multiple CIs (ranging from 2 to 2000 or more).

Launch Visualizer in CI Context

Follow these steps:

1. In the CA SDM web interface, navigate to a CI List, or Search for the CIs to analyze.
   The CI List displays.
2. Click a CI of interest.
   The CI Detail page is displayed and the Visualizer button is available. If CA CMDB Visualizer was not installed, the Visualizer button is inactive.
3. Click Visualizer.
   Visualizer starts and the relationships for the focal CI display.
   Default: Cause and Effect CIs

Launch Visualizer with Applied Filter

Follow these steps:

1. Navigate to the CI List, or search for the CIs you want to analyze.
   The CI List displays.
2. Click a CI of interest.
   The Configuration Item Detail page displays.
3. On the Filters list, select a filter.
4. Click Visualizer.
   Visualizer starts and the graphical relationships for the focal CI display according to the selected filter. If CA CMDB Visualizer was not installed, the Visualizer button is inactive.

Trace Relation with Focal CIs

You can trace all relationships between the selected CI and any focal CIs on the graph.

Follow these steps:

1. Select a CI on the graph.
2. Right-click and select Trace Relations with Focal CIs.
   Any relationships are displayed.

> **NOTE**
> If the selected CI is also the only focal CI, only that CI is displayed.

Launch MDRs

Follow these steps:

1. Right-click a CI on the Visualizer graphical display.
   A pop-up menu displays.
2. Select the Launch MDR option on the pop-up menu.

All external data providers that are associated with the CI appear.

3. Select an application name and click it to start the associated application.
   Visualizer launches the external MDR application as it is defined in the CMDB.

   NOTE
   Visualizer does not validate the external MDR applications credentials and URLs.

Hide/Display CIs

To promote clearer viewing, you can hide and then redisplay graphed CIs. Hidden CIs are available only for the current graph. When a new graph is generated, or when a Focal CI is set, the Hidden CIs List is cleared.

Follow these steps:

1. To select a CI, click it.
2. Right-click the CI and select Hide from the drop-down list.
   The graph updates so that the CI is hidden.

Follow these steps:

1. Click Hidden CIs on the toolbar or status bar.
   The Hidden CIs list displays.
2. Select one or more hidden CIs from the list and perform one of the following actions:
   - Click Unhide.
   - Drag to the graph canvas.
   - Right-click and select Unhide.

   The graph updates so that the CIs display.

View Change History for a CI

Follow these steps:

1. Select the CI you want to view.
2. Click Properties.
   CA CMDB is launched in context, displaying the CI Detail page.
3. Select the Versioning tab and click Show Log.
   The tab displays the details of the changes that were made to the CI.

Apply a Filter

Follow these steps:

1. Build the desired graphical display.
2. Click the Filter drop-down list.
   The Filter list is displayed.
3. Select a filter from the list.
   The filter is applied to the graph, which reorients to display the new perspective.

Use the Scratchpad

To view the Scratchpad, click the Scratchpad button on the Visualizer toolbar.

The Scratchpad displays.

Follow these steps:

1. Select one or more CIs on the Scratchpad.

2. Complete one of the following actions:
   – Click the Make Focal CI button on the Scratchpad toolbar.
   – Drag and drop one or more CIs from the Scratchpad to the graph.
   – Right-click on a CI and select Make Focal CI.

Follow these steps:

1. Select one or more CIs on the Scratchpad.
2. Complete one of the following actions:
   – Click the Add to Graph button on the Scratchpad toolbar.
   – Right-click on the CI and select Add To Graph.

Follow these steps:

1. Select one or more CIs to delete.
2. Click the Remove button on the Scratchpad toolbar.
   The CI is deleted from Scratchpad.

To synchronize the Scratchpad to the Database, click Refresh. The Scratchpad removes all the CIs that were deleted from the database.

To save a CI on the Scratchpad, right-click the CI and select Add to Scratchpad.

The CI is added to the Scratchpad.

Change Icons of CIs on Visualizer

You can change the icon of a CI on the Visualizer. Complete the following:

Follow these steps:

1. Navigate to $NX_ROOT \bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\assets
2. Open the serverjpegfamilyimagemap_OOTB.properties file and comment out the line (with a #) that corresponds to the family you would like to change the icon.
3. Navigate to $NX_ROOT \bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\assets
4. Open the serverjpegfamilyimagemap.properties file and add the family class and image name (300002=new_hardware_server.jpg).
5. Add new Image under $NX_ROOT \bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\assets \images\viz_icons.
6. Restart the CA Service Desk Manager service.
   You have successfully changed the icon of a CI on the Visualizer.

Configuring CMDB Visualizer

Configure CMDB Visualizer by performing the following:

- CMDB Visualizer Resource Management and Performance Tuning
- Security Configurations
- Tomcat Logging
- Troubleshooting CMDB Visualizer

CMDB Visualizer Resource Management and Performance Tuning

This article describes how CA SDM Administrators can manage CMDB resources for optimizing the performance of CMDB.

This article describes how CA SDM Administrators can manage CMDB resources (like database connection, user-sessions, nodes, data retrieval, configurations and so on) and thereby optimizing the performance of CMDB. To change the default configurations, open the application.conf file from where you have installed CMDB: Web-INF/classes/ application.conf .

Few of these attributes that Administrators can change in the **application.conf** file are explained below:

1. **app**:
   a. **name** = "CMDB Visualizer"
   Specifies the name of the application in log file. for example, CMDB Visualizer.

   NOTE
   This string only shows up in the log files and is not intended for Customization.

   b. **sessionTimeout = 60 (in minutes)**:
   Specifies the user idle time for logged-in user in the visualizer app before the session times out. Session timeout is specified in minutes.
   c. **expirationCheckInterval** = 5
   Interval time (in minutes) to check for expired sessions. Expired sessions will not be logged out at the exact time but instead when the expiration check interval is reached.

   IMPORTANT
   The recommended value is **5** minutes. Anything less than this value impact system performance, so this value must be changed judiciously. This value will add up with the timeout value (max time lag).
2. **Graph Node Configurations**: This is a Configuration Item (CI). Specifies the following values to be applied to a CMDB Graph Node.
   a. **filter = "inactive=0"**
   Specifies the filter to be applied while fetching the nodes and is applied only during initial loading when index-based loading is disabled.
   When system starts up we preload some CMDB graph nodes data into cache and this is a filter condition to filter out what nodes will be part of pre-load data. If '**indexBasedLoading**' is '**on**', this value is ignored.
   b. **maxFetchSize = 1000**
   Specifies maximum number of records in a single query during initial load. During system startup, the database query is split for node initial load based on this value.
   c. **indexBasedLoading = off**
   To persist the most used Nodes data, so that system restart can pre-load these nodes.

   NOTE
   This requires the **indexFilePath** value to be configured by Administrators. Ensure that the file exists.
   d. **Index File Path**: Complete path must be provided, relative path does not work. For example: indexFilePath = "C:\ \Program Files (x86)\\CA\\Service Desk Manager\\site\\nodes.index". To be configured by administrators and they must make sure that the file exists.
   e. **maxIndexEntries = 100000**
   Maximum number of nodes in the index file '. The maximum number of nodes that will be persisted to thendexFilePath' file
   f. **indexFileWriteInterval = 30**
   Specifies the interval to write the cached ID values to index file (in Minutes). It is frequency of writing the nodes data to the 'indexFilePath' file.
   g. **maxNodeLoadOnStartUp = 100000**
   Specifies the maximum number of nodes to load into cache on system startup. if 'indexBasedLoading' is 'on' then this value will be ignored.
3. **sdm.domsrvr.viz = domsrvr**
   Specifies the domsrvr list to connect to Visualizer. If the number of active visualizer users are more than the specified supported user count per domsrvr, then multiple domsrvrs must be used.

For example: If a single domsrvr supports 75 Visualizer Analysts, you get a better response when the number of users are under 75. If you plan to use more than 150 Visualizer Analysts, then you need two domsrvrs (specified with comma separated values).

NOTE
Domsrvr configured in this variable must be local to that machine. For all SDM servers "domsrvr" is local except in case of secondary servers.

4. **sdm.log4jRefreshInterval = 300000 (in milliseconds)**
   Specifies the time interval value to check for log level changes for Slump logging. Value is specified in seconds.
5. **db**: Contains database details.
   a. **defaultDB = "default"**

CAUTION
Do not change this value, else system will fail to start.

   b. **maxFetchSize = 100000**
      Maximum number of records that can be fetched from the database in a single query. It can be fine tuned as appropriate. **However, It is not recommended to change this value**.
   c. **inClausePlaceHonderCount = 1000**
      Maximum number of items to be part of **IN CLAUSE** when fetching data from a database.
   d. **initialPoolSize = 8**
      Number of active database connections to be used by Visualizer during system startup.
   e. **maxPoolSize = 12**
      Maximum number of active database connections to be used by Visualizer (can be increased or decreased based on data load requirements).
   f. **cleanupSchedulerInterval = 15 (minutes)**
      Frequency to check for idle database connections for cleanup.Value is specified in minutes.

Security Configurations

This article describes configuring and mitigating the cross-site request forgery for CMDB visualiser.

This article describes how to configure "X-XSS-Protection" header,"X-Content-Type-Options" header, "Content-Security-Policy" header and to mitigate Cross-Site Request Forgery for CMDB Visualizer.
Perform the following steps:

1. Navigate to <NX_ROOT>\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\WEB-INF.
2. Open the **web.xml** file in a text editor.
3. Search for the filter name **httpHeaderSecurity**, uncomment the below lines:
   Example:

```
<filter>
         <filter-name>httpHeaderSecurity</filter-name>
         <filter-class>org.apache.catalina.filters.HttpHeaderSecurityFilter</filter-class>
         <async-supported>true</async-supported>
    </filter>
```

4. Search for filter-mapping of **httpHeaderSecurity**,uncomment the below lines:
   Example:

```
<filter-mapping>
        <filter-name>httpHeaderSecurity</filter-name>
        <url-pattern>/*</url-pattern>
        <dispatcher>REQUEST</dispatcher>
    </filter-mapping>
```

5. Search for the filter name, **CrossSiteSecurityFilter** and uncomment the below section:
   Example:

```
<filter>
   <filter-name>CrossSiteSecurityFilter</filter-name>
  <filter-class>com.broadcom.visualizer.rest.security.filters.CrossSiteSecurityFilter</filter-class>
  <async-supported>true</async-supported>
<init-param>
<param-name>contentSecurityPolicy</param-name>
<param-value>default-src 'self' 'unsafe-inline' 'unsafe-eval'; font-src 'self'; style-src 'self' 'unsafe-
inline'; img-src 'self' data:</param-value>
          </init-param>
<init-param>
  <param-name>refererValidation</param-name>
  <param-value>true</param-value>
          </init-param>
<init-param>
 <param-name>isCSPEnable</param-name>
  <param-value>true</param-value>
   </init-param>
   </filter>
```

6. Search for filter-mapping of **CrossSiteSecurityFilter**, uncomment the below lines:
   Example:

```
  <filter-mapping>
        <filter-name>CrossSiteSecurityFilter</filter-name>
        <url-pattern>/*</url-pattern>
    </filter-mapping>
```

7. Restart the CA Service Desk Manager server.

Tomcat Logging

Tomcat Logging for CA Service Desk Manager.

CA SDM uses a separate **log4j.properties** file for its web components such as Servlets and PDM_RPC. Support Automation, CMDB Visualizer, and REST also have a separate l**og4j.properties** file. Starting or stopping the Tomcat logging does not require you to recycle the Tomcat daemons.

The following list describes how the **log4j.properties** and **log4j2.properties** file for CA SDM components differ:

- CA SDM monitors the **log4j2.properties** in the NX_ROOT\site\cfg and in the NX_ROOT\bopcfg\www \CATALINA_BASE\webapps\CAisd\WEB-INF for changes.
- REST monitors the rest.log4j2.properties in the NX_ROOT\site\cfg.

Use the pdm_log4j_config utility for changing variables in the log4j2.properties, logback.properties, and rest.log42j.properties files. You cannot use the pdm_log4j_config utility to modify the polling interval. CA SDM checks the log4j2.properties file for any changes periodically using the monitorInterval attribute.

Servlet Defaults

The following servlets log Info level messages to the **jsrvr.log** file by default:

- **PDMweb**: Log entry from operations on the user interface.
- **UploadServlet**: Log entry generated when you attach files to a ticket.
- **PDMExport**: Log entry generated when you click Export on list forms.
- **PDMReport**: Log entry generated when you click the Report menu on list forms.
- **Page_cache**: Log entry that is generated from operations on the user interface.

Enable Additional Logging for Visualizer

To enable additional logging for Visualizer, follow these steps:

1. Increase the log level for visualizer_slump.log and visualizer_bop.log under the NX_ROOT\bopcfg\www \CATALINA_BASE_VIZ\logs directory. directory.
   a. Edit the NX_ROOT\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\WEB-INF\classes \log4j2.properties file.
   b. For rootLogger.level, change from INFO to DEBUG and for logger.bop_logging.level, change from INFO to DEBUG
      NOTE
      Example: rootLogger.level=DEBUG logger.bop_logging.level=DEBUG
   c. You can also increase the log level for the slump package and bop package by changing the level to debug for logger.slump.name and logger.bop.name packages.
      NOTE
      Example: logger.slump.level=DEBUG logger.bop.level = INFO
2. Increase the log level for VisualizerMS_debug.log under NX_ROOT\log directory
   a. Edit the `NX_ROOT\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\WEB-INF\classes \logback.properties` file.
   b. For **rootLogger.level**, change from INFO to DEBUG. For example:

      ```
      rootLogger.level=DEBUG
      ```

   c. Save the changes and restart the Visualizer tomcat using following commands:

      ```
      pdm_tomcat_nxd -c stop -t VIZ
      pdm_tomcat_nxd -c start -t V
      ```

Troubleshooting CMDB Visualizer

This article describes troubleshooting steps for CMDB Visualizer issues.

This page contains:

- Unable to login to CMDB Visualizer
- CMDB Visualizer Interface may fail to launch with HTTP 404 Error

Unable to login to CMDB Visualizer

Problem: Unable login into CMDB Visualizer Interface URL with valid credentials when value of “vizserver” is not in the same format in the casm.conf file: <NX_ROOT>\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux \conf\casm.conf.js.

NOTE
Before accessing the CMDB URL, ensure to cross verify the CMDB entries specified in **casm.conf** and **nx.env**.

Perform the following steps:

1. Login to CA Service Desk Manager (SDM) browser Interface and access the URL: https://<hostname>:<portnumber> CA SDM is accessed successfully with a valid login role/user (Example: ServiceDesk).
2. Now, try to access the CMDB URL: http://<hostname>:<portnumber>/CMDBVisualizer

3. CMDB Visualizer Interface is launched but login access is denied even to valid users/role like ServiceDesk.

Solution: To resolve this issue, verify the value of "**vizserver**" and access the URL interface with the same format as specified in **casm.conf** file.

CMDB Visualizer Interface may fail to launch with HTTP 404 Error

Problem: Visualizer does not launch from the CA Service Desk Manager Web Interface and shows HTTP 404 error.

Solution: Update the **context.xml** file in <NX_ROOT>\bopcfg\www\CATALINA_BASE_VIZ\conf\context.XML. Update the following in the beginning or topmost portion of this file:

```
<Valve className="org.apache.catalina.valves.rewrite.RewriteValve" />
```

For example, after adding the above lines, the **context.xml** file is updated as shown below:

```
<Valve className="org.apache.catalina.valves.rewrite.RewriteValve" />
    <WatchedResource>WEB-INF/web.xml/<WatchedResource>
    <WatchedResource>${catalina.base}/conf/web.xml/<WatchedResource>
```

Installing

Learn how to plan, and install the product.

CA Service Management products have different prerequisites, pre-installation considerations, installation procedures and post installation configurations.

> **NOTE**
> Always create a new directory to place the CA Service Management 17.4 installers at the directory location that does not contain any spaces. For more information, refer to the KB Article.
> **Example:** E:\ProgramFiles(x86)\CA\ServiceDeskManager

To successfully install the CA Service Management solution review the following.

- CA Service Management 17.4 Solution Architecture
- CA Asset Portfolio Management Product Architecture
- CA Service Catalog Product Architecture
- CA Service Desk Manager Product Architecture
- Pre-Installation Considerations
- Installation Prerequisites
- Installing CA Service Management 17.4
- Post Installation Tasks
- CA Service Management Solution Integration

Patches

The following cumulative patches are available for CA Service Management 17.4.

- Installing CA Service Management 17.4.5.0
- Installing CA Service Management 17.4.4.0
- Installing CA Service Management 17.4.3.0
- Installing CA Service Management 17.4.2.0
- Installing CA Service Management 17.4.1.0

Installing CA Service Management 17.4.5.0

Based on your CA Service Management implementation, review the following topics for a successful installation.

- Install Prerequisites for CA Service Management 17.4.5.0
- Install CA Service Management 17.4.5.0 (Windows)
- Install CA Service Management 17.4.5.0 (Linux)
- Validate CA Service Management 17.4.5.0 Installation
- Post-Installation Tasks for CA Service Management 17.4.5.0
- Post-Installation Tasks - CA SDM Advance Availability 17.4.5.0
- Uninstallation is no longer supported from CASM 17.4.5.0

Install Prerequisites for CA Service Management 17.4.5.0

Installation prerequisites for CASM 17.4.5.0

Verify and complete the following prerequisites before installation of CA Service Desk Manager, CA Service Catalog, and CA Asset Portfolio Management.

NOTE

CA Service Management (CASM) solution is a distributed environment, follow the order mentioned for a seamless installation:

1. CA Embedded Entitlements Manager 12.7.2.0
2. CA Process Automation 4.4CP4+Hotfix
3. 17.4.5.0 CA Service Catalog
4. 17.4.5.0 CA SDM and its sub-components.
5. 17.4.5.0 CA Asset Portfolio Management.

NOTE
Pre-Upgrade Recommendation:
Take a snapshot or image of all CASM components (CA SDM, CA Service Catalog, ITAM, CA PAM, and EEM) and back up the mdb before installing CASM 17.4.5.0 , as rollback to earlier versions is not supported.

NOTE

- If EEM is integrated with CASM, It is mandatory to upgrade to EEM 12.7.2.0, before installing CA Service Management 17.4.5.0.
  Installers are available in CASM_17.4.5.0\filestore\EEM.
  Before upgrading EEM to 12.7.2.0, ensure you follow the **mandatory** steps mentioned here.
- From CA Service Management 17.4.5.0 onward, only ITPAM CP04 HF01 is supported. Refer ITPAM CP04 - HF01.
  Installers are available in CASM_17.4.5.0\filestore\PAM_4.4_CP04_HF01.

Installation Prerequisites for CA Service Desk Manager

Review the following CA SDM installation prerequisites before you start the installation process:

NOTE
We recommend that you place the installer in a shorter directory path, containing no spaces, to avoid errors when running the setup file. For example, unzip the file at: C:\WIN_17_4_5_0.

- Ensure that CA SDM and xFlow Interface 17.4 is installed in your environment.
- Back up the system database (MDB). The system database (MDB) backup is mandatory and ensures safe data recovery if any problems are encountered during the installation process.
- Do not invoke ApplyPTF manually in your system while running the CA Service Management CA Service Management 17.4.5.0 installer.
- In the case of, CA Service Desk Manager - Conventional configuration, if the Primary and Secondary servers are on separate nodes, and the xFlow Interface is deployed on the Secondary Server, then the Database Client (Microsoft SQL Native Client or Oracle Database Client) is required on the Secondary Server for a successful installation.
- CA SDM must be installed for deploying the xFlow Interface. In the conventional configuration, it1s recommended to install the xFlow Interface server on a secondary server. Advanced Availability users can install the xFlow Interface only on an application server.

IMPORTANT
After upgrading to CA Service Management 17.4.5.0, Maileater does not work with POP4 on a non-TLS port with Clear Text as a security level. This is a limitation of the upgraded Javamail (Third-party jar) used in connection to the mail server. Users are required to use **LOGIN** as a security level.

(Optional) Backing up the CR_Stored_Queries Table

CA Service Management provides a fix to address a vulnerability related to SQL injection. As part of this fix, **Cr_Stored_Queries** table is updated. If you have made customizations or changes to the Out-of-the-box (OOTB) stored queries, perform the below steps to take a backup before applying the CA Service Management patch:

1. Open a command prompt window and navigate to *$NX_ROOT\patches.* and create a folder **Backup_Data**. Navigate to this folder and take the Backup of the Cr_Stored_Queries table by running the following command:
   ```
   pdm_extract Cr_Stored_Queries> Cr_Stored_Queries_17.4_backup.dat
   ```
2. Verify that the **Cr_Stored_Queries_17.4_backup.dat** file is successfully created.
3. Now, proceed with CA Service Management 17.4.5.0 installation.
4. After applying the patch, if required based on your customizations, you can update the database files created in **Step 1**. Navigate to $NX_ROOT\patches\Backup_Data location and run the following command:

   ```
   pdm_load -f Cr_Stored_Queries_17.4_backup.dat
   ```

Installation Prerequisites for CA Service Catalog

Verify and complete the following prerequisites before proceeding with CA Service Catalog 17.4.5.0 installation:

- Ensure that CA Service Catalog 17.4 is installed in your environment.
- Back up the system database (MDB). The system database (MDB) backup is mandatory and ensures safe data recovery if any problems are encountered during the installation process.

Installation Prerequisites for CA Asset Portfolio Management

Consider the following prerequisites:

> **WARNING**
> CA Asset Portfolio Management (CA APM) architecture is highly database-driven and has no rollback mechanism for database-related changes. Hence, the uninstallation of the CA APM patch is not recommended and is not supported. The only way to revert the database changes, if required, is by restoring the database backup which can lead to data loss.

Verify and complete the following prerequisites before proceeding with the CA Asset Portfolio Management installation:

- Ensure that CA Asset Portfolio Management 17.4 is installed in your environment.
- Back up the system database (MDB). The system database (MDB) backup is mandatory and ensures safe data recovery if any problems are encountered during the installation process.

Install CA Service Management 17.4.5.0 (Windows)

> **NOTE**
> **Pre-Upgrade Recommendation:**
>
> - Take a snapshot or image of all CASM components (CA SDM, CA Service Catalog, ITAM, CA PAM, and EEM) and back up the mdb before installing CASM 17.4 RU5, as rollback to earlier versions is not supported.
> - If the EEM certificate key length was not updated to 2048-bit key length before upgrading to EEM 12.7.2.0, update it now before installing CASM 17.4 RU5. For related commands refer here.

CA Service Management 17.4.5.0 installer performs prerequisite checks and mandatory validations before installing the patch.

NOTE

- We recommend that you place the installer in a **shorter directory path**, containing no spaces, to avoid errors when running the setup file. For example, unzip the file at: C:\WIN_17_4_5_0
- In CA Service Desk Manager (SDM) Availability mode, the xFlow Interface can only be installed on an Application Server. If you have converted CA SDM conventional configuration to advanced availability configuration, and you have a previously installed version of xFlow Interface, ensure that you uninstall xFlow Interface and then, reinstall xFlow Interface on an Application server.

Perform the following steps to install 17.4.5.0 on Windows:

1. From the Broadcom Support portal, download the **WIN_17_4_5_0.zip** file.
   NOTE
   CA Service Management 17.4.5.0 (Windows) installer is provided as a ZIP file and not as a CAZ file. Extract/unzip the content of the ZIP file to your local folder.
2. Copy **WIN_17_4_5_0.zip** on all servers in your environment.
3. Extract the contents of the ZIP file to the folder named WIN_17_4_5_0.
   NOTE
   In the extracted folder, CA EEM 12.7.2.0.0.0 installer is available in *CASM_17.4.5.0\filestore\EEM\Windows\12.7.2.0.0.0*
4. In CA SDM conventional configuration, stop CA SDM services in your environment. In the case of Advance Availability configuration, stop CA SDM services only on the server on which CA SDM will be installed.
5. Log in to the server as a local admin to install the patch.
6. Navigate to **WIN_17_4_5_0** folder. Right-click **Setup.exe** and select **run as administrator**.
7. Review and Accept the License Agreement Information.
8. Select **Microsoft SQL Server** or **Oracle** as the database type in the Database Configuration screen.
   For Windows Microsoft SQL Server database, *mdbadmin* is the default user.
9. Provide the database details:
   a. If you have selected **Microsoft SQL Server**, provide the following field information:
      - **Database Server**: Enter the hostname of the database server.
        If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
      - **Database Name**: Enter the Database name (MDB), the name of the target DBMS. The default value is mdb.
      - **Database Port**: Provide the port identifier for the target DBMS.
      - **Database Server Instance**: The database instance name. Leave this field blank if you are using the default instance.
        If you are using the SQL named instance and a non-default port number, the **Database Server Instance** field should look like this: ***<database instance name>,<non-default port number>***
        For example:
        ```
        sdmcasmqa,10555
        ```
      - **Database Admin User**: For Microsoft SQL Server, **sa** is the default value.
        The **sa** user is the admin user with permission to create a user and schema.
      - **Database Admin Password**: Enter the database password of the user specified by a database admin user. No default value is assigned to this parameter.
      - **mdbadmin Password**: Enter the password for the user specified for the MDB admin user.
      - **Confirm mdbadmin Password**: Confirm the password.
   b. If you have selected **Oracle** database, provide the following field information:
      - **Database Server**: Enter the hostname of the Oracle database server.
        If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
      - **Oracle Service Name**: Enter the Oracle Service name.
      - **Listener Port**: Enter the listener port for the database.
      - **Net Service Name**: Enter the Net Service Name of the Oracle database where the database resides.

If the database is remote, use the Net Service Name that is defined in the Oracle client on the local computer. CA SDM has access to the database through a local installation of the Oracle client.

- **DBA User Name**: Provide the name of an Oracle user with DBA access. The default value for DBA user is **SYS**.
- **DBA User Password**: Enter the password for the DBA user.
- **mdbadmin Password**: Enter the password for MDB admin.
- **Confirm mdbadmin Password**: Confirm the password.
- **Tablespace Path (on DB Server)**: Provide the tablespace path on the database server.
- **Oracle Home Path**: Provide the Oracle Client 64-bit path. Provide Oracle client 42-bit path. For CA Asset Portfolio Management, provide the Oracle 42-bit client.
- **Data Tablespace Name**: Provide the name of the data tablespace. The default value is MDB_DATA.
- **Index Tablespace Name**: Provide the index Tablespace Name: The default value is MDB_INDEX.

Click **Next**.

10. On **Select the Products and Integrations** page, patch list is displayed. Select the required patch to install.

    WARNING

    CA Service Management does not support the uninstallation of the CA Asset Portfolio Management patch. Uninstalling the CA APM patch does not uninstall the MDB patch, as there is no rollback mechanism for the same. For more information, see Installation Prerequisites.

    a. **CA Service Desk Manager**
       - CA Service Desk Manager Patch
       - xFlow Interface Patch
       - Search Server Patch: Search server patch must be installed on a Server where an earlier version from previous releases is already installed.

    b. **CA Service Catalog**
       - CA Service Catalog Patch

    c. **CA Asset Portfolio Management**
       - CA Asset Portfolio Management Patch

    Click **Next**.

11. Review the Pre-Installation Configuration page. Click **Next**.

    NOTE

    Select the option to perform the automated post-install steps for CA SDM. If this option is not selected, you must manually perform the post-install steps for CA SDM. Note that for CA Service Catalog and CA Asset Portfolio Management, you need to perform the manual post-install steps.

    NOTE
    From 17.4 RU5, when ITPAM is integrated with any CASM product, a new alert (*Out of the box CA Process Automation Sample data willbe automatically re-imported*") pops-up in the common installer.

12. Click **Install** to proceed with the installation. Review the Installation progress and when the installation is complete, click **Next.**

    NOTE
    From 17.4 RU5, a new task, **Migrating Solution Products to use CXF endpoints** is created, when any of the CASM products are integrated.

    NOTE
    If Migration Task to CXF endpoints ***fails*** in an SSL environment during RU5 upgrade, follow these steps:

    1. Login to ITPAM environment and open SDM Dataset located at /SDM/Configuration/
    2. Update the value of port from <non ssl port of SDM tomcate> to <ssl port of SDM tomcate> example:- 8080 to 8443.
    3. Click Retry Install in the Common Installer.

NOTE
If "Performing CA Asset Portfolio Management Post install steps: 17.4.5.0" task fails with "type=itam_sdm_cxf_integ", follow these steps:

1. Start the CA SDM Services.
2. Click Retry Install in the Common Installer.

13. To review the installation logs, navigate to the path shown in the installation summary screen.
14. Click **Finish** to exit the installer
15. After installing the Key definition for "xflow" not found in the DITA map., start the **CA Key definition for "xflow" not found in the DITA map.** service from Windows Services (**Control Panel, Administrative Tools, Services**) as this service is not started by default.
16. After installing the search server, if the **Search Server** service is not started by default or if it stops, start the services from **Control Panel, Administrative Tools, Services.**
17. Validate the Installation.
18. Complete the Post-Installation Tasks.

After installing the Key definition for "xflow" not found in the DITA map. and Search Server, you can configure the cache size for Key definition for "xflow" not found in the DITA map.. For more information about how to configure cache size, see Perform initial load and refresh index on the search server.

Install CA Service Management 17.4.5.0 (Linux)

Installing CASM 17.4.5.0 on Linux.

NOTE
Pre-Upgrade Recommendation:

- Take a snapshot or image of all CASM components (CA SDM, CA Service Catalog, ITAM, CA PAM, and EEM) and back up the mdb before installing CASM 17.4 RU5, as rollback to earlier versions is not supported.
- If the EEM certificate key length was not updated to key length 2048-bit key length before upgrading to EEM 12.7.2.0, update it now before installing CASM 17.4 RU5. For related commands refer here.

NOTE
In CA SDM Advanced Availability mode, the xFlow Interface can only be installed on an Application Server. If you have converted CA SDM conventional configuration to advanced availability configuration, and you have a previously installed version of xFlow Interface, ensure that you uninstall xFlow Interface and then reinstall xFlow Interface on an Application server.

CA Service Management 17.4.5.0 installer performs prerequisite checks and mandatory validations before installing the patch.

IMPORTANT
We recommend that you place the installer in a **shorter directory path**, containing no spaces, to avoid errors when running the setup file. For example, unzip the file at: *<root>\casm_dvd.x86Linux\<installer_file>*

Perform the following steps to install 17.4.5.0 on Linux:

1. From the Broadcom Support portal, download the **LNX_17_4_5_0.tar.gz** file.
2. To extract the LNX_17_4_5_0.tar.gz, run the following command:

```
tar -xvzf LNX_17_4_5_0.tar.gz
```

NOTE
In the extracted folder, CA EEM 12.7.2.0.0.0 installer is available in CASM_17.4.5.0\filestore\EEM\Linux \12.7.2.0.0.0

3. Set CA SDM bin and library environment variables.
   For example:

```
export PATH=/opt/CA/ServiceDeskManager/bin:$PATH
export LD_LIBRARY_PATH=/opt/CA/ServiceDeskManager/lib:$LD_LIBRARY_PATH
```

4. In SDM conventional configuration, stop CA SDM services in your environment. In the case of Advance Availability, stop CA SDM services only on the server on which CA SDM will be installed.
5. Run the setup file from the casm_dvd.x86Linux folder to install the application(s) using the following command:
   ```
   ./setup.sh
   ```
6. Review and accept the license agreement terms.
7. In Database Configuration, select **Oracle** as the database type. Ensure that you have installed the Oracle database client in your system.
   Provide the following Oracle database configuration details:
   - **Database Server**: Enter the hostname of the Oracle database server.
     If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
   - **Oracle Service Name**: Enter the Oracle Service name.
   - **Listener Port**: Enter the listener port for the database.
   - **Net Service Name**: Enter the Net Service Name of the Oracle database where the database resides.
     If the database is remote, use the Net Service Name that is defined in the Oracle client on the local computer. CA SDM has access to the database through a local installation of the Oracle client.
   - **DBA User Name**: Authenticates the name of the user with DBA access. The default value for DBA user is **SYS.**
   - **DBA User Password**: Authenticates the password for DBA user.
   - **mdbadmin Password**: Specify the password for the mdbadmin user.
   - **Confirm mdbadmin Password**: Confirm the mdbadmin user password.
   - **Tablespace Path (on DB Server)**: Enter the tablespace directory on the database server.
   - **Oracle Home Path**: Provide the Oracle Client 64-bit path. Provide Oracle client 42-bit path for fresh installation
   - **Data Tablespace Name**: Enter the tablespace name. The default value is **MDB_DATA**.
   - **Index Tablespace Name**: Enter the index tablespace name: The default value is **MDB_INDEX**.
8. Click **Next.**
9. On **Select the Products and Integrations** page, patch list is displayed. Select the required patch to install.
   a. **CA Service Desk Manager**
      - CA Service Desk Manager Patch
      - xFlow Interface Patch
      - Search Server Patch: A search server patch must be installed on a server where an earlier version from previous releases is already installed.
10. Click **Next**.
11. Review the Pre-Installation Summary page. Click **Next.**

    NOTE
    From 17.4 RU5, when ITPAM is integrated with any CASM product, a new alert (*Out of the box CA Process Automation Sample data willbe automatically re-imported*") pops-up in the common installer.
12. Click **Install** to proceed with the installation. Review the Installation progress and when the installation is complete, click **Next.**

    NOTE
    From 17.4 RU5, a new task, **Migrating Solution Products to use CXF endpoints** is created, when any of the CASM products are integrated.

    NOTE
    If Migration Task to CXF endpoints ***fails*** in an SSL environment during RU5 upgrade, follow these steps:

    1. Login to ITPAM environment and open SDM Dataset located at /SDM/Configuration/
    2. Update the value of port from <non ssl port of SDM tomcate> to <ssl port of SDM tomcate> example:- 8080 to 8443.

3. Click Retry Install in the Common Installer.

13. To review the installation logs, navigate to the path shown in the installation summary screen.
14. Click **Finish** to exit the installer.
    If patch installation fails for a product, retry the installation by using the **Retry** option in the wizard where it fails. If retry does not work, exit the installer and relaunch it. If the problem persists, contact Broadcom Support.
15. After installing XFlow Interface, start the **CA XFlow Interface** service. Navigate to /opt/CA/xFlow/Apps and run the following command from a command prompt window:
    ```
    ./xFlowServer start
    ```
    To stop the server, run the following command:
    ```
    ./xFlowServer Stop.sh
    ```
16. Navigate to /opt/CA/SC/SearchServer/elasticsearch/bin and run the following command from the command prompt window:
    ```
    ./ServicesStart.sh
    ```
    NOTE
    If you get the below message while restarting the search server service, ignore it and continue:
    ```
    groupadd: group 1elasticsearch1 already exists
    useradd: user 1elasticuser1 already exists
    runuser: warning: cannot change directory to /home/elasticuser: No such file or directory
    OpenJDK 64-Bit Server VM warning: Option UseConcMarkSweepGC was deprecated in version 9.0 and
     will likely be removed in a future release.
    ```
    To stop the server, run the following command:
    ```
    /ServicesStop.sh
    ```
17. Validate the Installation.
18. Complete the Post-Installation Tasks.

After installing the XFlow Interface and Search Server, you can configure the cache size for XFlow Interface. For more information about how to configure cache size, see Perform initial load and refresh index on the search server.

Validate CA Service Management 17.4.5.0 Installation

Validate the CASM 17.4.5.0 installation

To verify the successful installation of CA Service Management 17.4.5.0, validate the installation for the corresponding applications in the following location. You can also navigate to the root folder where you have locally installed the application and select the "env.info" file for validating your installation:

- **CA ITAM:**
  NOTE
  CORA version: 17.4.0.57
  - C:\Program Files (x86)\CA\ITAM\env.info
  - product=ITAM
  - major=17
  - minor=4
  - cum=5
  - roll-up=0

  NOTE
  CA CASM uses Apache Tomcat version 9.0.108.
- **CA Service Catalog:**

  - %USM_HOME%*CA_SLCM_<hostname>.his* file
  - %USM_HOME%\view\webapps\usm\locale\about_info.xml
  - %USM_HOME%\build.xml
  - %USM_HOME%\env.info
- **xFlow Interface:**
  - *C:\Program Files\CA\xFlow\<hostname>.his* file
  - NX_ROOT\env.info
  - NX_ROOT\version.xml
- **CA SDM:**
  - *C:\Program Files (x86)\CA\Service Desk Manager\<hostname>.his* file
  - NX_ROOT\pdmconf\version
  - NX_ROOT\env.info
  - HELP, ABOUT

NOTE
For CA SDM, the Binary and Language Combo patches are installed irrespective of the environment. The language patch for the corresponding CA SDM environment is installed.

To validate the successful installation of the patches, follow these steps:

1. Open the <sdm_installation>\<hostname>.his file in a text editor.
2. Search for the instances of "gol-351". In a successful install, you should come across these three entries (Binary, Language Combo, and corresponding language patch).

Application	Windows	Linux
CA Asset Portfolio Management	T6EE1745.CAZ	NA
CA Service Catalog	T6D91745.CAZ	NA
CA Service Desk Manager - Binary	gol-351_cum_C	gol-351_x86_64Linux_cum_C
CA Service Desk Manager - Languag504e Comb	gol-351_langcombo	gol-351_langcombo
CA Service Desk Manager - English	gol-351_en_US	gol-351_x86_64Linux_en_US
CA Service Desk Manager - German	gol-351_de_DE	gol-351_x86_64Linux_en_DE
CA Service Desk Manager - Spanish	gol-351_es_ES	NA
CA Service Desk Manager - French	gol-351_fr_FR	gol-351_x86_64Linux_fr_FR
CA Service Desk Manager - Italian	gol-351_it_IT	NA
CA Service Desk Manager - Japanese	gol-351_ja_JP	gol-351_x86_64Linux_ja_JP
CA Service Desk Manager - Brazilian Portuguese	gol-351_pt_BR	NA

Post-Installation Tasks for CA Service Management 17.4.5.0

Post-installation steps and checklist for CASM 17.4.5.0

After successful installation of CA Service Management 17.4.5.0, perform the following post-installation steps.

CA Service Desk Manager

Manual Steps required for Automated Post-Installation

CA Service Management provides an option in the installer to automate the post-install tasks.

If you have selected **Yes** to automate the post-installation steps while running the CA Service Management Installer Wizard, you are not required to perform any manual post-install steps for CA Service Desk Manager (SDM).

If you have selected **No** for automating the post-install steps while running the CA Service Management Installer wizard, perform the below manual post-install tasks as per your deployment requirements:

Manual Post-Install Tasks After Installing CA Service Desk Manager (SDM) 17.4.5.0 on CA SDM 17.4.4.0

Perform the following steps:

1. Stop CA SDM services.
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0.
3. Open a command prompt window and run the following command:
   ```
   pdm_configure
   ```
4. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_05_insert.dat
   pdm_load -u -f r17_4_05_update.dat
   ```

Manual Post-Install Tasks After Installing CA Service Desk Manager (SDM) 17.4.5.0 on CA SDM 17.4.3.0
Perform the following steps:

1. Stop CA SDM services.
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0.
3. Open a command prompt window and run the following command:
   ```
   pdm_configure
   ```
4. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_04_insert.dat
   pdm_load -u -f r17_4_04_update.dat
   pdm_load -f r17_4_05_insert.dat
     pdm_load -u -f r17_4_05_update.dat
   ```

Manual Post-Install Tasks After Installing CA Service Desk Manager (SDM) 17.4.5.0 on CA SDM 17.4.2.0
Perform the following steps:

1. Stop CA SDM services.
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0.
3. Open a command prompt window and run the following command:
   ```
   pdm_configure
   ```
4. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_03_insert.dat

   pdm_load -u -f r17_4_03_update.dat
   pdm_load -f r17_4_04_insert.dat
   pdm_load -u -f r17_4_04_update.dat
    pdm_load -f r17_4_05_insert.dat
    pdm_load -u -f r17_4_05_update.dat
   ```
5. Notification message template is modified to update the existing email templates with the html_description field. **Tool location**: NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate This has to be executed one time when the SDM service is up and running. **For e.g:**
   Windows
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.bat
   ```
   Linux

```
NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.sh
```

Manual Post-Install Tasks After Installing CA Service Desk Manager (SDM) 17.4.5.0 on CA SDM 17.4 or CA SDM 17.4.1.0

Perform the following steps:

1. Stop CA SDM services.
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0.
3. Open a command prompt window and run the following command:
   ```
   pdm_configure
   ```
4. Change the directory to NX_ROOT/data, and execute the following command:

   ```
   pdm_load -f r17_4_02_insert.dat
   pdm_load -u -f r17_4_02_update.dat
   pdm_load -f r17_4_03_insert.dat
   pdm_load -u -f r17_4_03_update.dat
   pdm_load -f r17_4_04_insert.dat
   pdm_load -u -f r17_4_04_update.dat
   pdm_load -f r17_4_05_insert.dat
   pdm_load -u -f r17_4_05_update.dat
   ```

5. Notification message template is modified to update the existing email templates with the html_description field.
   Tool location: NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate This has to be executed one time when the SDM service is up and running. **For e.g:**
   Windows
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.bat
   ```
   Linux
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.sh
   ```

Manual Post-Installation Task for CMDB

> **NOTE**
> For Oracle 21c, set the **isClusterEnv** setting to "true" in the application.conf file of CMDB viz and restart the CA SDM Services:
>
> - For Windows: <SDM installed location>\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\WEB-INF\classes\application.conf)
> - For LINUX: <SDM installed location>/bopcfg/www/CATALINA_BASE_VIZ/webapps/CMDBVisualizer/WEB-INF/classes/application.conf).

If you are applying CA Service Management 17.4.4.0 in an SSL environment, CA SDM Administrators must perform the following post-installation steps for CMDB Visualizer Interface to open using the **HTTPS** protocol when launching from CA SDM UI:

1. In the NX.env file, change the **NX_CMDB_VISUALIZER** entry to provide **HTTPS protocol** and **HTTPS port**.
   For example: @NX_CMDB_VISUALIZER=https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer
2. In **casm.conf.js** file from NX_root\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\conf, provide the HTTPS as protocol and HTTP port.
   For example:

   ```
   sdm: {

   vizserver: 'https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer/viz',
   ```

```
server: 'https:// sdmurl:<SDM_SSL_PORT>/CAisd/pdmweb.exe'
}
```

Manual Post-Installation Task for OAuth 2.0 (Maileater/ Mailer)

After generating an access token for OAuth 2.0 to connect to a mail server, perform the mandatory post-install tasks on all servers in your environment:

1. Create a backup of web.xml from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF in your local directory to perform backout if required.
2. Open the web.xml file from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF.
3. Search for the following statement:
   ```
   <!--USD servlet mappings start-->
   ```
4. Copy and paste the following configuration under the statement:

   ```
   <servlet>
   <servlet-name>OAuthProcessor</servlet-name>
   <servlet-class>com.broadcom.oAuth.OAuthProcessor</servlet-class>
   </servlet>
   ```
5. Copy and paste the following configuration after any <servlet-mapping>:

   ```
   <servlet-mapping>
   <servlet-name>OAuthProcessor</servlet-name>
   <url-pattern>/OAuthProcessor</url-pattern>
   </servlet-mapping>
   ```
6. Save and close the web.xml file.
7. For OAuth servlet preservation in a primary and secondary environment, the secondary server must also have the OAuth processor entries in the following file: *<NX_ROOT>\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF\web.xml.last*.
8. Repeat Steps 1-8 to update **web.xml.tpl** file in the NX_ROOT\samples\pdmconf\web.xml.tpl folder to avoid losing these changes when pdm_configure is run at a later point of time.
9. Run the following command:
   ```
   pdm_tomcat_nxd -c STOP command
   ```
10. Wait for some time (a minute or so) and run the following command:
    ```
    pdm_tomcat_nxd -c START command
    ```

For more information, see Define a mailbox.

Manual Post-Installation Task for Excel file (.xlsx) export*

1. Execute the following command:
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst
   ```
2. Execute the following command to add the new NX variable to NX.env.tpl file.
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst -t
   ```
3. Restart CA SDM services.

HTML Tags Sanitization

NOTE
This step is mandatory when you are updating from 17.GA or 17.4.1.0 to 17.4.5.0. For information, refer Admin Insights for RTE.

The sanitization tool can be run any number of times. Administrators can choose if they wish to run the tool for each ticket type separately or for all ticket types at one go.

Before running the Sanitization tool ensure you take a backup of all the ticket objects using the following commands or by taking a backup of the sdm databse :

```
"Call_Req": pdm_extract Call_Req > cr_backup.txt
"Issue": pdm_extract Issue > Issue_backup.txt
"Change Orders": pdm_extract Change_Request >change_orders_backup.txt.
```

Both active and archived tickets can be sanitized.
For Windows:

```
NX_ROOT\samples\sdk\SanitizationTool\sanitizeTool.bat
```

For Linux:

```
NX_ROOT\samples\sdk\SanitizationTool\sanitizeTool.sh
```

NOTE
We recommend executing the sanitization during slack time on any Service Desk Manager server.

After executing the sanitization tool, execute the following command to update domsrvr memory cache:

```
pdm_cache_refresh -t Call_Req
pdm_cache_refresh -t Issue
pdm_cache_refresh -t Change_Request
```

(Optional) Enable HTML content for Ticket Description in the Activity log

NOTE

- This step is mandatory when you are updating from 17.GA or 17.4.1.0 to 17.4.5.0.
- A new Field Update activity association (**HTML Description** has been introduced)
- Either **HTML Description** or **Description** activity association has to be enbaled. Do not enable both the options.
- Out of the box Description Field Update activity association is enabled in the activity log for all ticket types (Incident,Request,Problem,ChangeOrder and Issue).

Enabling HTML content for Description in the Activity Log:

1. Login into CA SDM with Administrator access type.
2. Administration --> Notifications --> Activity Associations.
3. Click **Show Filter**.
4. Enter "HTML Description" in the Symbol text box in the Activity Association List page.
5. Click **Search**
6. For every **HTML Description Field Update association**, click **Edit** and check the **Log Me** checkbox, then save the changes made.

Disabling plain text for Description in the Activity log:

1. Click **Show Filter** in the Activity Association List page.
2. Enter "Description" in the Symbol text box and Select Object Type as "Request/Incident/Problem" and click **Search**.
3. Now see the one row and click the Description link in the Activity Association List page.
4. Unselect the **Log Me** check box and save the page.
5. For Change Order and Issue repeat the steps from 1 to 4.

(Optional) Manual Post-Installation Task for Knowledge Document export with German fields

If you are exporting Knowledge Document with German Fields (included in the export field list), then perform the below post-install steps for successful export of knowledge documents:

1. Take the backup of NX_ROOT\NX.env file (NX_ROOT is the installation directory for CA Service Desk Manager).
2. Open NX_ROOT\pdmconf\NX.env_nt.tpl file using any text editor and take the value of NX variable @NX_KEIT_AVAILABLE_FIELDS which is COPIED_DATA and update Nx.env in NX_ROOT by executing below command:
   ```
   pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA>  -a  pdm_option.inst
   ```
   Where COPIED_DATA is the value copied from NX.env_nt.tpl (step 2).
   For example, " pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <provide the exact string here> -a pdm_option.inst "
   Note to avoid losing the updated changes when you run pdm_configure, execute the above command with the '-t' flag as follows:
   ```
   pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA> -a pdm_option.inst -t
   ```

(Optional) Handle HTML Mails that contains Tables in MailEater

Java based Maileater already has the ability to process the emails that were sent in plain format or HTML format, to support the processing of emails that contain Tables in the HTML body. But the format in which the email was sent is lost when received by Maileater.

1. Run the following commands from NX_ROOT/bin from the command prompt/ terminal:
   ```
   pdm_options_mgr -c -s NX_ENFORCE_HTML_EMAIL_PARSING -v "Yes" -a pdm_option.inst
   pdm_options_mgr -c -s NX_ENFORCE_HTML_EMAIL_PARSING -v "Yes" -a pdm_option.inst -t
   ```
2. Restart CA SDM and XFlow applications.

(Optional) Comments are not allowed when a Data Partition (O_COMMENTS) is created with Customer, Employee and ServiceDesk Analyst

To disable comments:

1. Login to SDM and navigate to Administration -> Security and Role Management-> Data partition-> Data Partition List.
2. In Data Partition List, create a constraint (O_COMMENTS with the condition id=0) and link it to Customer, Employee and ServiceDesk Analyst data partitions.
3. Run the following commands from NX_ROOT/bin from the command prompt/ terminal:
   ```
   pdm_options_mgr -c -s KT_COMMENT_USE_SUPERUSER -v 0 -a pdm_option.inst
   pdm_options_mgr -c -s KT_COMMENT_USE_SUPERUSER -v 0 -a pdm_option.inst -t
   ```
4. Restart CA SDM and XFlow applications.

(Optional) POST to GET conversion in CA SDM web interface

Conversion or simulation of a POST operation using a GET operation is prohibited or strongly discouraged due to the critical security and data integrity risks.

To enable the OP codes, follow these steps:

1. Login to CA SDM with the Administrator user type.
2. Navigate to Administration >> Options Manager >> Web >> block_opcodes_for_get.
3. Click *block_opcodes_for_get* link.
4. In *block_opcodes_for_get Options Details* window, click **Edit >> Install.**

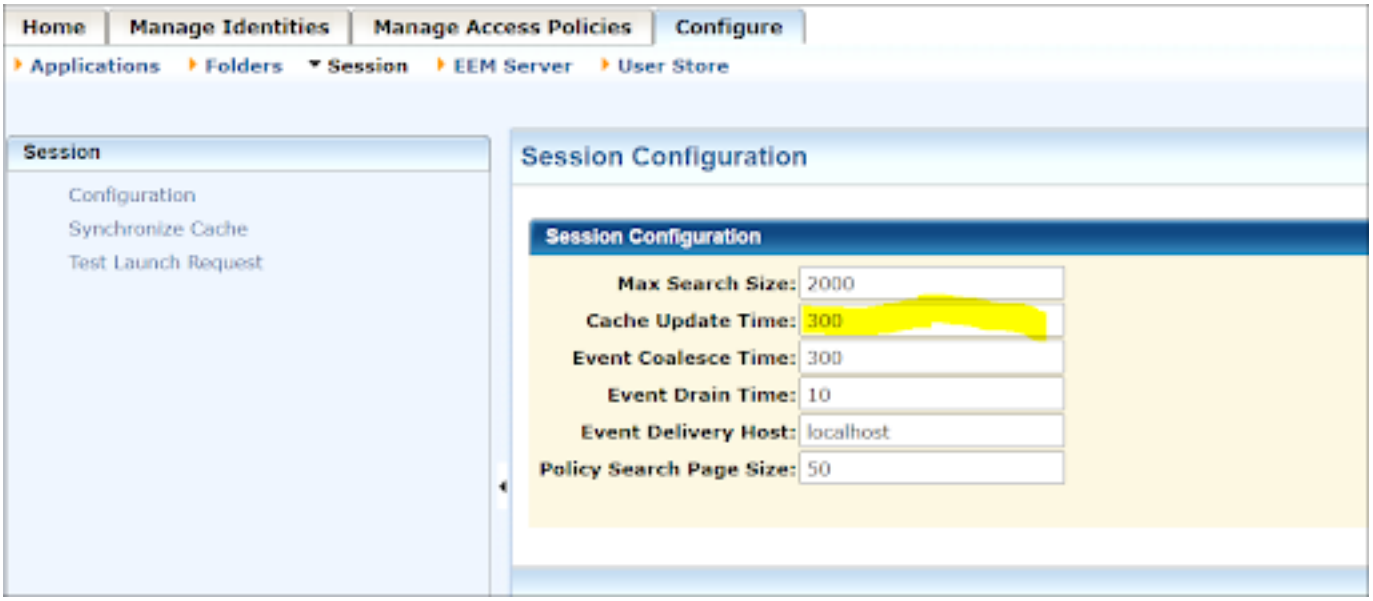
   NOTE
   The default values are:
   ```
   ACCUMULATE,ACCUMULATE_ALL,ATTACH_STYPE_EVENT,CLOSE_ALL_CHILD,KT_ES_SAVE_TREE,KT_SAVE_REPORT_CARD,LOGIN,LogA
   ```
5. Now restart CA SDM service.

(Optional) Manual Post-Installation Task for EEM Cache Timeout Configuration Update

You may face issues while logging in to EEM or SDM using the EEM user accounts due to a session cache timeout mismatch between EEM and SDM.

You can update the EEM cache timeout configuration by adding a new NX variable, "NX_EEM_CACHE_UPDATE_TIME" in SDM.

Keep the value the same as the EEM cache timeout configuration value mentioned in the following screenshot.



To set the EEM cache update time to 300 seconds, perform the following steps:

To add the variable in the **nx.env** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst
```

To add the variable in the **nxenv_nt.tpl** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst -t
```

(Optional)casc_impersnate_user

Once both CA SDM and CA Service Catalog are integrated, do the following:

1. Login to CA SDM with the Administrator user type.
2. Navigate to Administration >> Options Manager >> CA Service Catalog >> in the Options List, install casc_impersnate_user
3. Restart the SDM service.

(Optional)Incident/Request/Problem Solution should display Short Resolution

To display a short solution for tickets, do the following:

1. Login to CA SDM with the Administrator user type.
2. Navigate to Administration >> Options Manager >> Knowledge >> use_resolution_short_for_soln.
3. In the use_resolution_short_for_soln window displayed, click **Edit,** and click **Install**.
4. Restart CA SDM.

xFlow

(Optional)Enable Rate Limiting Configuration

Each incoming request to the backend passes through multiple filters before being fulfilled. To protect against potential misuse or malicious activity, a Rate Limiter Filter has been introduced. The Rate Limiter Filter is a comprehensive API protection mechanism that safeguards against API abuse by enforcing configurable request limits. It supports both token-based and IP-based identification strategies to ensure secure and controlled access.

To enable the rate- limiting functionality:

- Locate the file ***application.conf*** at the default location. (Default location: ***C:\Program Files\CA\xFlow\APPS\Services\incidentmicroservice-17.0.479\conf***).
- After updating the below configurations, restart xFlow service.

Configuration	Description
ratelimit.enabled = true	Enables the rate limiting for API endpoints.
ratelimit.defaultPerUserLimit = 50	Number of requests allowed per API endpoint per X-Access Token within the defined time window. Applies individually to each API Example: 50 for /api/tickets, 50 for /api/emailtemplate.
ratelimit.timeWindowMinutes = 1	Duration of the rate-limiting time window (in minutes). Users can make the configured number of requests within this window. The cleanup interval is automatically calculated as max(2 * timeWindow, 5 minutes)
ratelimit.customIpHeader = ""	Specifies a custom header for extracting the client IP address for rate limiting. If configured, this header is checked before standard headers (X-Forwarded-For, X-Real-IP, Remote-Addr). Example: ratelimit.customIpHeader = "X-Custom-Client-IP"

Service Point

After successful installation of Service Point, refresh index and data on the Search Server.

NOTE
In case of Linux Search Server environment, ensure to run the following commands on the terminal:

```
sysctl -w vm.max_map_count=262144
ulimit -Hn 65535
```

(Optional) Manual Post-Installation Task for Rebuild Index

NOTE
This step is mandatory when you are updating from 17.GA or 17.4.1.0 to 17.4.5.0.

If you have upgraded or are planning to install CA EEM 12.6 with Service Point and CA Service Catalog, ensure that you rebuild the CA Service Catalog index after completing the CA EEM upgrade/installation process. For more information, see Perform Initial Load and Refresh Index Data on Search Server.
Run the following command to rebuild the index:

```
$ES_HOME/search/bin/pdm_es_rebuild_index.bat index=catalog_index
```

CA Service Catalog

Perform the following CA Service Catalog post-installation steps as per your environment requirements:

- Post-Installation Tasks after installing CA Service Catalog 17.4.3.0 on CA Service Catalog 17.4 or later
- gMSA Authentication

Post-Install Tasks After Installing CA Service Catalog 17.4.3.0 on CA Service Catalog 17.4 or later

Review and perform the post-installation steps based on your deployment requirements.

NOTE
If any SSL certificate(s) chain was imported to the Oracle Java's JAVA_HOME\lib\security\cacerts Keystore, those certificate(s) chain again need to be exported explicitly and imported properly into the new AdoptOpenJDK's %USM_HOME%\embedded\jdk\lib\security\cacerts Keystore.

Follow these steps

- (Optional) If Service Catalog is accessed from applications like Service Point, SharePoint and so on, then you must perform this step. Open the "**web.xml**" file located in *USM_HOME%\view\webapps\usm\WEB-INF* and update the respective parameter values for the parameter name "cors.allowed.origins" and "cors.support.credentials". For example, if you are using Catalog Widgets in Service Point
- (Optional) Unable to approve any workflow requests through email. The email sent for approvals is corrupted with special characters. Follow the KB article steps.

(Optional) Approve workflow requests through email

The workflow email sent for approvals is corrupted with special characters. To approve any workflow requests through email, follow the KB article steps.

gMSA Authentication Post-Install Steps for CA Service Catalog (Automated steps)

CA Service Catalog provides automated post-installation steps for gMSA Authentication. Review the following before you proceed:

- Prerequisites and Considerations
- How to Automate gMSA Post-Installation Steps
- Unconfigure gMSA

Prerequisites and Considerations:

Ensure that you have taken care of the below considerations on all nodes of CA Service Catalog where you want to automate the gMSA post-installation procedure.

1. If the accounting service is available, perform the automated post-install steps by updating serviceName to ServiceAccounting and providing the appropriate gMSA account name in the **gmsa.properties** file.
2. If the filestore path is external to the server from where the **GMSAPostInstall.bat** is running, ensure to provide appropriate access to the server for setting the gMSA account permissions.
3. While preparing the properties file, check for any copy/paste issues, special characters, spaces, and so on.

How to Automate gMSA Post-Installation Steps

Perform the following steps to enable the batch file **GMSAPostInstall.bat** and to automate the gMSA post-installation steps:

1. Create a gMSA account for CA Service Catalog.
2. Ensure that appropriate DB (database) permission is given to the newly created account(s).
3. Enter the below values in the **gmsa.properties** file and save it.

   NOTE
   A template file (**gmsa.properties**) is added in the common installer in *"\filestore\SLCM\gMSAProperties"* . Optionally. use this file as a template file to fill gmsa properties. This can be copied to "USM_HOME/scripts/GMSA" where Catalog is installed.

   NOTE
   All values are mandatory. Make sure that there are no leading or trialing spaces while providing these values.

   a. **serviceName**: Application service name to enable GMSA. In the case of Catalog, it will be ServiceCatalog and for accounting, it will be ServiceAccounting.

b. **folderName**: Absolute path of the application installation directory.
c. **gMSAName**: Name of the GMSA service account (domain\accountName$).
d. **filestorePath**: Shared network path (\\Server_name\filestore) or local filestore path

4. Navigate to %USM_HOME% and launch **usm.cmd**
5. Navigate to %USM_HOME%/scripts/GMSA/ folder and run the "GMSAPostInstall.bat" in command prompt.

NOTE
If the command executes without any error, "Post-installation steps for GMSA executed successfully" message is displayed with exit code 0.
If there are errors, the error message is shown with a non-zero exit code. Correct the error to continue and perform step **5** again.

Unconfigure gMSA for CA Service Catalog

To disable or unconfigure gMSA for CA Service Catalog, follow the steps below:

1. Replace **config.properties** file from*%USM_HOME%*\ folder.
2. Revert back all the folder permissions which were assigned to domain users.
3. Start Catalog and Accounting Services with the local system account.

NOTE
In the case of load balancers, repeat steps 1-3 to unconfigure gMSA on all Catalog servers.

CA Asset Portfolio Management

Manual Post-Installation Task for Removing jQuery, log4net, and log4j Vulnerabilities

After installing 17.4.5.0, delete jquery-1.3.2.min.js, jquery-ui-1.7.2.custom.min.js , log4net.dll, and log4j from their respective installed folders.

NOTE
Perform the below steps if you have not done this earlier.

Perform the following steps:

1. Log on to CA APM Web Server, navigate to the <ITAM_install_dir>\ITAM\Web Server\castylesr5\javascript.
   Example: C:\Program Files (x86)\CA\ITAM\Web Server\castylesr5\javascript
   Locate and delete the jquery-1.3.2.min.js and jquery-ui-1.7.2.custom.min.js files from the directory.
2. Navigate to the ITAM_install_dir\CA\ITAM\Data Importer.
   Locate and delete the log4net.dll file from the directory.
3. Navigate to the C:\Program Files (x86)\CA\SC\CORA.
   Locate and delete the log4j-1.2.15.jar, log4j.properties, and ojdbc6.jar files from the directory.
4. Navigate to the ITAM_install_dir\CA\ITAM.
   Backup the following folders and manually delete them:
   - CDB
   - MDB
   - jre8

   NOTE
   jre8 folder has to be deleted manually for every RU installation.
5. Perform an IISRESET and delete the browser's cache to load the upgraded files.

For ITAM installation on French Operating System

Following the successful installation of version 17.4.3.0 on a French Windows operating system with the browser set to French, we are unable to log in to ITAM.

An error message is displayed: ***Une valeur Request.Form potentiellement dangereuse a été détectée à partir du client (_UPLOAD_NON_TXT_CSV_DATA_FILE_="...ectionné n'est pas de type...").***

To overcome this, follow these steps:

1. Navigate to *C:\Program Files (x86)\CA\ITAM\Web Server, edit web.config*, and search for *<pages enableEventValidation="false" viewStateEncryptionMode="Never" controlRenderingCompatibilityVersion="4.0" clientIDMode="AutoID">* , then add *validateRequest="false"* to it.
2. After adding, it looks like this:
   <pages enableEventValidation="false" viewStateEncryptionMode="Never" controlRenderingCompatibilityVersion="4.0" clientIDMode="AutoID" validateRequest="false">
3. Do an IISRESET.
4. You can now successfully login to ITAM.

Post-Installation Tasks - CA SDM Advance Availability 17.4.5.0

The installer wizard provides an option in the installer to automate the post-install tasks. Perform the following post-installation tasks for CA SDM Advanced Availability Configuration. For the Advanced Availability configuration, the following host names are considered as an example:

- The background server hostname is sdmcahost1.
- The standby server hostname is sdmcahost2.
- The application server hostname is sdmcahost3.

Manual Steps required for Automated Post-Installation

If you have selected **Yes** for automating the post-install steps while running the installer wizard, then perform manually the following post-install tasks:

Manual Post-Install Tasks - After Installing CA Service Desk Manager (SDM) 17.4.5.0 on CA SDM 17.4.4.0

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Start the CA SDM service on the standby server (sdmcahost2). Open the command prompt and run the following command:
   ```
   pdm_server_control -b
   ```
3. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server.
4. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```
5. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_05_insert.dat
   pdm_load -u -f r17_4_05_update.dat
   ```
6. Clear the browser cache by the following command:
   ```
   pdm_webcache -H
   ```
7. Ensure that the patch is applied on all the application servers and the remaining standby servers if any. If you selected YES to automate the post-install steps, then there are no more manual post-install steps to be performed.

Manual Post-Install Tasks - After Installing CA Service Desk Manager (SDM) 17.4.5.0 on CA SDM 17.4.3.0

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Start the CA SDM service on the standby server (sdmcahost2). Open the command prompt and run the following command:
   ```
   pdm_server_control -b
   ```
3. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server.

4. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```
5. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_04_insert.dat
   pdm_load -u -f r17_4_04_update.dat
   pdm_load -f r17_4_05_insert.dat
   pdm_load -u -f r17_4_05_update.dat
   ```
6. Clear the browser cache by the following command:
   ```
   pdm_webcache -H
   ```
7. Ensure that the patch is applied on all the application servers and the remaining standby servers if any. If you selected YES to automate the post-install steps, then there are no more manual post-install steps to be performed.

Manual Post-Install Tasks - After Installing CA Service Desk Manager (SDM) 17.4.5.0 on CA SDM 17.4.2.0

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Start the CA SDM service on the standby server (sdmcahost2). Open the command prompt and run the following command:
   ```
   pdm_server_control -b
   ```
3. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server.
4. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```
5. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_03_insert.dat
   pdm_load -u -f r17_4_03_update.dat
              pdm_load -f r17_4_04_insert.dat
   pdm_load -u -f r17_4_04_update.dat
   pdm_load -f r17_4_05_insert.dat
   pdm_load -u -f r17_4_05_update.dat
   ```
6. Clear the browser cache by the following command:
   ```
   pdm_webcache -H
   ```
7. Ensure that the patch is applied on all the application servers and the remaining standby servers if any. If you selected YES to automate the post-install steps, then there are no more manual post-install steps to be performed.

Manual Post-Install Tasks - After Installing CA Service Desk Manager (SDM) 17.4.5.0 on CA SDM 17.4 GA or CA SDM 17.4.1.0

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Start the CA SDM service on the standby server (sdmcahost2). Open the command prompt and run the following command:
   ```
   pdm_server_control -b
   ```
3. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server.
4. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```
5. Change the directory to NX_ROOT/data, and execute the following command:

   ```
   pdm_load -f r17_4_02_insert.dat
       pdm_load -u -f r17_4_02_update.dat
       pdm_load -f r17_4_03_insert.dat
       pdm_load -u -f r17_4_03_update.dat
     pdm_load -f r17_4_04_insert.dat
   ```

```
  pdm_load -u -f r17_4_04_update.dat
pdm_load -f r17_4_05_insert.dat
pdm_load -u -f r17_4_05_update.dat
```

6. Clear the browser cache by the following command:
   ```
   pdm_webcache -H
   ```
7. Ensure that the patch is applied on all the application servers and the remaining standby servers if any. If you selected YES to automate the post-install steps, then there are no more manual post-install steps to be performed.

Manual Steps required for Non-Automated Post-Installation

If you have selected **No** for automating the post-install steps while running the CA Service Management installer wizard, perform manually the appropriate post-install after this task as per your deployment requirements.

Post-Install Tasks After Installing CA SDM 17.4.5.0 on CA SDM 17.4.4.0 for Advanced Availability Configuration

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0. Open a command prompt and run the following command:
   ```
   pdm_configure
   ```
   On the installer configuration page, before clicking Finish, uncheck the Start service when completed checkbox. Failure to uncheck the box may result in the loss of custom schema changes and configurations that were changed on the server.
3. Promote the standby server as the new background server by performing the following steps. To suppress version control on the standby server and later on the background server, run the following command on each server:
   ```
   pdm_server_control -v
   ```
4. Start SDM service on the standby server. Run the following command:
   ```
   pdm_server_control -b
   ```
5. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server, and perform the following steps. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
   ```
   pdm_configure
   ```
6. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```
7. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_05_insert.dat
   pdm_load -u -f r17_4_05_update.dat
   ```
8. Ensure that the patch is applied on all application servers and the remaining standby server if any. Perform the following steps on all application servers and standby servers.
9. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
   ```
   pdm_configure
   ```
10. Clear the browser cache by running the following command:
    ```
    pdm_webcache -H
    ```

Post-Install Tasks After Installing CA SDM 17.4.5.0 on CA SDM 17.4.3.0 for Advanced Availability Configuration

1. Ensure that the patch is applied on the standby server (sdmcahost2).

2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0. Open a command prompt and run the following command:
   ```
   pdm_configure
   ```
   On the installer configuration page, before clicking Finish, uncheck the Start service when completed checkbox. Failure to uncheck the box may result in the loss of custom schema changes and configurations that were changed on the server.
3. Promote the standby server as the new background server by performing the following steps. To suppress version control on the standby server and later on the background server, run the following command on each server:
   ```
   pdm_server_control -v
   ```
4. Start SDM service on the standby server. Run the following command:
   ```
   pdm_server_control -b
   ```
5. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server, and perform the following steps. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
   ```
   pdm_configure
   ```
6. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```
7. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_04_insert.dat
   pdm_load -u -f r17_4_04_update.dat
   pdm_load -f r17_4_05_insert.dat
   pdm_load -u -f r17_4_05_update.dat
   ```
8. Ensure that the patch is applied on all application servers and the remaining standby server if any. Perform the following steps on all application servers and standby servers.
9. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
   ```
   pdm_configure
   ```
10. Clear the browser cache by running the following command:
    ```
    pdm_webcache -H
    ```

Post-Install Tasks After Installing CA SDM 17.4.5.0 on CA SDM 17.4.2.0 for Advanced Availability Configuration

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0. Open a command prompt and run the following command:
   ```
   pdm_configure
   ```
   On the installer configuration page, before clicking Finish, uncheck the Start service when completed checkbox. Failure to uncheck the box may result in the loss of custom schema changes and configurations that were changed on the server.
3. Promote the standby server as the new background server by performing the following steps. To suppress version control on the standby server and later on the background server, run the following command on each server:
   ```
   pdm_server_control -v
   ```
4. Start SDM service on the standby server. Run the following command:
   ```
   pdm_server_control -b
   ```
5. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server, and perform the following steps. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
   ```
   pdm_configure
   ```
6. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```

7. Change the directory to NX_ROOT/data, and execute the following command:

```
pdm_load -f r17_4_03_insert.dat
pdm_load -u -f r17_4_03_update.dat
                    pdm_load -f r17_4_04_insert.dat
pdm_load -u -f r17_4_04_update.dat
             pdm_load -f r17_4_05_insert.dat
pdm_load -u -f r17_4_05_update.dat
```

8. Ensure that the patch is applied on all application servers and the remaining standby server if any. Perform the following steps on all application servers and standby servers.
9. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:

```
pdm_configure
```

10. Clear the browser cache by running the following command:

```
pdm_webcache -H
```

Post-Install Tasks After Installing CA SDM 17.4.5.0 on CA SDM 17.4 GA or CA SDM 17.4.1.0 for Advanced Availability Configuration

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0. Open a command prompt and run the following command:

```
pdm_configure
```

   On the installer configuration page, before clicking Finish, uncheck the Start service when completed checkbox. Failure to uncheck the box may result in the loss of custom schema changes and configurations that were changed on the server.
3. Promote the standby server as the new background server by performing the following steps. To suppress version control on the standby server and later on the background server, run the following command on each server:

```
pdm_server_control -v
```

4. Start SDM service on the standby server. Run the following command:

```
pdm_server_control -b
```

5. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server, and perform the following steps. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:

```
pdm_configure
```

6. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:

```
pdm_server_control -b
```

7. Change the directory to NX_ROOT/data, and execute the following command:

```
pdm_load -f r17_4_02_insert.dat
pdm_load -u -f r17_4_02_update.dat
pdm_load -f r17_4_03_insert.dat
pdm_load -u -f r17_4_03_update.dat
                   pdm_load -f r17_4_04_insert.dat
pdm_load -u -f r17_4_04_update.dat
                 pdm_load -f r17_4_05_insert.dat
pdm_load -u -f r17_4_05_update.dat
```

8. Notification message template is modified to update the existing email templates with the html_description field. **Tool location**: NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate This has to be executed one time when the SDM service is up and running. **For e.g:**
   Windows

```
NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.bat
```

   Linux

```
NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.sh
```

9. Ensure that the patch is applied on all application servers and the remaining standby server if any. Perform the following steps on all application servers and standby servers.
10. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
    ```
    pdm_configure
    ```
11. Clear the browser cache by running the following command:
    ```
    pdm_webcache -H
    ```
12. Execute the following command.
    ```
    pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst
    ```
13. Execute the following command to add the new NX variable to NX.env.tpl file.
    ```
    pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst -t
    ```
14. Restart CA SDM Services.

Installing CA Service Management 17.4.4.0

Based on your CA Service Management implementation, review the following topics for a successful installation.

- Install Prerequisites for CA Service Management 17.4.4.0
- Install CA Service Management 17.4.4.0 (Windows)
- Install CA Service Management 17.4.4.0 (Linux)
- Validate CA Service Management 17.4.4.0 Installation
- Post-Installation Tasks for CA Service Management 17.4.4.0
- Post-Installation Tasks - CA SDM Advance Availability 17.4.4.0
- Uninstall CA Service Management 17.4.4.0

Install Prerequisites for CA Service Management 17.4.4.0

Installation prerequisites for patch 17.4.4.0

Verify and complete the following prerequisites before installation of CA Service Desk Manager, CA Service Catalog, and CA Asset Portfolio Management.

Installation Prerequisites for CA Service Desk Manager

Review the following CA SDM installation prerequisites before you start the installation process:

> **NOTE**
> We recommend that you place the installer in a shorter directory path, containing no spaces, to avoid errors when running the setup file. For example, unzip the file at: C:\WIN_17_4_4_0.

- Ensure that CA SDM and xFlow Interface 17.4 is installed in your environment.
- Back up the system database (MDB). The system database (MDB) backup is mandatory and ensures safe data recovery if any problems are encountered during the installation process.
- Do not invoke ApplyPTF manually in your system while running the CA Service Management CA Service Management 17.4.4.0 installer.
- In the case of, CA Service Desk Manager - Conventional configuration, if the Primary and Secondary servers are on separate nodes, and the xFlow Interface is deployed on the Secondary Server, then the Database Client (Microsoft SQL Native Client or Oracle Database Client) is required on the Secondary Server for a successful installation.
- CA SDM must be installed for deploying the xFlow Interface. In the conventional configuration, it1s recommended to install the xFlow Interface server on a secondary server. Advanced Availability users can install the xFlow Interface only on an application server.

IMPORTANT
After upgrading to CA Service Management 17.4.4.0, Maileater does not work with POP4 on a non-TLS port with Clear Text as a security level. This is a limitation of the upgraded Javamail (Third-party jar) used in connection to the mail server. Users are required to use **LOGIN** as a security level.

(Optional) Backing up the CR_Stored_Queries Table

CA Service Management provides a fix to address a vulnerability related to SQL injection. As part of this fix, **Cr_Stored_Queries** table is updated. If you have made customizations or changes to the Out-of-the-box (OOTB) stored queries, perform the below steps to take a backup before applying the CA Service Management patch:

1. Open a command prompt window and navigate to *$NX_ROOT\patches.* and create a folder **Backup_Data**. Navigate to this folder and take the Backup of the Cr_Stored_Queries table by running the following command:
   ```
   pdm_extract Cr_Stored_Queries> Cr_Stored_Queries_17.4_backup.dat
   ```
2. Verify that the **Cr_Stored_Queries_17.4_backup.dat** file is successfully created.
3. Now, proceed with CA Service Management 17.4.4.0 installation.
4. After applying the patch, if required based on your customizations, you can update the database files created in **Step 1**. Navigate to $NX_ROOT\patches\Backup_Data location and run the following command:

   ```
   pdm_load -f Cr_Stored_Queries_17.4_backup.dat
   ```

Installation Prerequisites for CA Service Catalog
Verify and complete the following prerequisites before proceeding with CA Service Catalog 17.4.4.0 installation:

- Ensure that CA Service Catalog 17.4 is installed in your environment.
- Back up the system database (MDB). The system database (MDB) backup is mandatory and ensures safe data recovery if any problems are encountered during the installation process.

Installation Prerequisites for CA Asset Portfolio Management
Consider the following prerequisites:

WARNING
CA Asset Portfolio Management (CA APM) architecture is highly database-driven and has no rollback mechanism for database-related changes. Hence, the uninstallation of the CA APM patch is not recommended and is not supported. The only way to revert the database changes, if required, is by restoring the database backup which can lead to data loss.

Verify and complete the following prerequisites before proceeding with the CA Asset Portfolio Management installation:

- Ensure that CA Asset Portfolio Management 17.4 is installed in your environment.
- Back up the system database (MDB). The system database (MDB) backup is mandatory and ensures safe data recovery if any problems are encountered during the installation process.

Install CA Service Management 17.4.4.0 (Windows)

CA Service Management 17.4.4.0 installer performs prerequisite checks and mandatory validations before installing the patch.

NOTE

- We recommend that you place the installer in a **shorter directory path**, containing no spaces, to avoid errors when running the setup file. For example, unzip the file at: C:\WIN_17_4_4_0
- In CA Service Desk Manager (SDM) Availability mode, the xFlow Interface can only be installed on an Application Server. If you have converted CA SDM conventional configuration to advanced availability

configuration, and you have a previously installed version of xFlow Interface, ensure that you uninstall xFlow Interface and then, reinstall xFlow Interface on an Application server.

Perform the following steps to install 17.4.4.0 on Windows:

1. From the Broadcom Support portal, download the **WIN_17_4_4_0.zip** file.
   NOTE
   CA Service Management 17.4.4.0 (Windows) installer is provided as a ZIP file and not as a CAZ file. Extract/unzip the content of the ZIP file to your local folder.
2. Copy **WIN_17_4_4_0.zip** on all servers in your environment.
3. Extract the contents of the ZIP file to the folder named WIN_17_4_4_0.
   NOTE
   In the extracted folder, CA EEM 12.6.7.0 installer is available in *CASM_17.4.4.0\filestore\EEM\Windows\12.6.7.0*
4. In CA SDM conventional configuration, stop CA SDM services in your environment. In the case of Advance Availability configuration, stop CA SDM services only on the server on which CA SDM will be installed.
5. Log in to the server as a local admin to install the patch.
6. Navigate to **WIN_17_4_4_0** folder. Right-click **Setup.exe** and select **run as administrator**.
7. Review and Accept the License Agreement Information.
8. Select **Microsoft SQL Server** or **Oracle** as the database type in the Database Configuration screen.
   For Windows Microsoft SQL Server database, *mdbadmin* is the default user.
9. Provide the database details:
   a. If you have selected **Microsoft SQL Server**, provide the following field information:
      - **Database Server**: Enter the hostname of the database server.
        If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
      - **Database Name**: Enter the Database name (MDB), the name of the target DBMS. The default value is mdb.
      - **Database Port**: Provide the port identifier for the target DBMS.
      - **Database Server Instance**: The database instance name. Leave this field blank if you are using the default instance.
        If you are using the SQL named instance and a non-default port number, the **Database Server Instance** field should look like this: ***<database instance name>,<non-default port number>***
        For example:
        ```
        sdmcasmqa,10555
        ```
      - **Database Admin User**: For Microsoft SQL Server, **sa** is the default value.
        The **sa** user is the admin user with permission to create a user and schema.
      - **Database Admin Password**: Enter the database password of the user specified by a database admin user. No default value is assigned to this parameter.
      - **mdbadmin Password**: Enter the password for the user specified for the MDB admin user.
      - **Confirm mdbadmin Password**: Confirm the password.
   b. If you have selected **Oracle** database, provide the following field information:
      - **Database Server**: Enter the hostname of the Oracle database server.
        If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
      - **Oracle Service Name**: Enter the Oracle Service name.
      - **Listener Port**: Enter the listener port for the database.
      - **Net Service Name**: Enter the Net Service Name of the Oracle database where the database resides.

If the database is remote, use the Net Service Name that is defined in the Oracle client on the local computer. CA SDM has access to the database through a local installation of the Oracle client.

- **DBA User Name**: Provide the name of an Oracle user with DBA access. The default value for DBA user is **SYS**.
- **DBA User Password**: Enter the password for the DBA user.
- **mdbadmin Password**: Enter the password for MDB admin.
- **Confirm mdbadmin Password**: Confirm the password.
- **Tablespace Path (on DB Server)**: Provide the tablespace path on the database server.
- **Oracle Home Path**: Provide the Oracle Client 64-bit path. Provide Oracle client 42-bit path. For CA Asset Portfolio Management, provide the Oracle 42-bit client.
- **Data Tablespace Name**: Provide the name of the data tablespace. The default value is MDB_DATA.
- **Index Tablespace Name**: Provide the index Tablespace Name: The default value is MDB_INDEX.

Click **Next**.

10. On **Select the Products and Integrations** page, patch list is displayed. Select the required patch to install.

    WARNING

    CA Service Management does not support the uninstallation of the CA Asset Portfolio Management patch. Uninstalling the CA APM patch does not uninstall the MDB patch, as there is no rollback mechanism for the same. For more information, see Installation Prerequisites.

    a. **CA Service Desk Manager**
       - CA Service Desk Manager Patch
       - xFlow Interface Patch
       - Search Server Patch: Search server patch must be installed on a Server where an earlier version from previous releases is already installed.

    b. **CA Service Catalog**
       - CA Service Catalog Patch

    c. **CA Asset Portfolio Management**
       - CA Asset Portfolio Management Patch

    Click **Next**.

11. Review the Pre-Installation Configuration page. Click **Next**.

    NOTE

    Select the option to perform the automated post-install steps for CA SDM. If this option is not selected, you must manually perform the post-install steps for CA SDM. Note that for CA Service Catalog and CA Asset Portfolio Management, you need to perform the manual post-install steps.

12. Click **Install** to proceed with the installation. Review the Installation progress and when the installation is complete, click **Next.**
13. To review the installation logs, navigate to the path shown in the installation summary screen.
14. Click **Finish** to exit the installer
15. After installing the Key definition for "xflow" not found in the DITA map., start the **CA Key definition for "xflow" not found in the DITA map.** service from Windows Services (**Control Panel, Administrative Tools, Services**) as this service is not started by default.
16. After installing the search server, if the **Search Server** service is not started by default or if it stops, start the services from **Control Panel, Administrative Tools, Services.**
17. Validate the Installation.
18. Complete the Post-Installation Tasks.

After installing the Key definition for "xflow" not found in the DITA map. and Search Server, you can configure the cache size for Key definition for "xflow" not found in the DITA map.. For more information about how to configure cache size, see Perform initial load and refresh index on the search server.

Install CA Service Management 17.4.4.0 (Linux)

Installing CASM 17.4.4.0 on Linux.

> **NOTE**
> In CA SDM Advanced Availability mode, the xFlow Interface can only be installed on an Application Server. If you have converted CA SDM conventional configuration to advanced availability configuration, and you have a previously installed version of xFlow Interface, ensure that you uninstall xFlow Interface and then reinstall xFlow Interface on an Application server.

CA Service Management 17.4.4.0 installer performs prerequisite checks and mandatory validations before installing the patch.

> **IMPORTANT**
> We recommend that you place the installer in a **shorter directory path**, containing no spaces, to avoid errors when running the setup file. For example, unzip the file at: *<root>\casm_dvd.x86Linux\<installer_file>*

Perform the following steps to install 17.4.4.0 on Linux:

1. From the Broadcom Support portal, download the **LNX_17_4_4_0.tar.gz** file.
2. To extract the LNX_17_4_4_0.tar.gz, run the following command:
   ```
   tar -xvzf LNX_17_4_4_0.tar.gz
   ```
   > **NOTE**
   > In the extracted folder, CA EEM 12.6.7.0 installer is available in CASM_17.4.4.0\filestore\EEM\Linux\12.6.7.0
3. Set CA SDM bin and library environment variables.
   For example:
   ```
   export PATH=/opt/CA/ServiceDeskManager/bin:$PATH
   export LD_LIBRARY_PATH=/opt/CA/ServiceDeskManager/lib:$LD_LIBRARY_PATH
   ```
4. In SDM conventional configuration, stop CA SDM services in your environment. In the case of Advance Availability, stop CA SDM services only on the server on which CA SDM will be installed.
5. Run the setup file from the casm_dvd.x86Linux folder to install the application(s) using the following command:
   ```
   ./setup.sh
   ```
6. Review and accept the license agreement terms.
7. In Database Configuration, select **Oracle** as the database type. Ensure that you have installed the Oracle database client in your system.
   Provide the following Oracle database configuration details:
   - **Database Server**: Enter the hostname of the Oracle database server.
     If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
   - **Oracle Service Name**: Enter the Oracle Service name.
   - **Listener Port**: Enter the listener port for the database.
   - **Net Service Name**: Enter the Net Service Name of the Oracle database where the database resides.
     If the database is remote, use the Net Service Name that is defined in the Oracle client on the local computer. CA SDM has access to the database through a local installation of the Oracle client.
   - **DBA User Name**: Authenticates the name of the user with DBA access. The default value for DBA user is **SYS.**
   - **DBA User Password**: Authenticates the password for DBA user.
   - **mdbadmin Password**: Specify the password for the mdbadmin user.
   - **Confirm mdbadmin Password**: Confirm the mdbadmin user password.
   - **Tablespace Path (on DB Server)**: Enter the tablespace directory on the database server.
   - **Oracle Home Path**: Provide the Oracle Client 64-bit path. Provide Oracle client 42-bit path for fresh installation
   - **Data Tablespace Name**: Enter the tablespace name. The default value is **MDB_DATA**.
   - **Index Tablespace Name**: Enter the index tablespace name: The default value is **MDB_INDEX**.
8. Click **Next.**

9. On **Select the Products and Integrations** page, patch list is displayed. Select the required patch to install.
   a. **CA Service Desk Manager**
      - CA Service Desk Manager Patch
      - xFlow Interface Patch
      - Search Server Patch: A search server patch must be installed on a server where an earlier version from previous releases is already installed.
10. Click **Next**.
11. Review the Pre-Installation Summary page. Click **Next.**
12. Click **Install** to proceed with the installation. Review the Installation progress and when the installation is complete, click **Next.**
13. To review the installation logs, navigate to the path shown in the installation summary screen.
14. Click **Finish** to exit the installer.
    If patch installation fails for a product, retry the installation by using the **Retry** option in the wizard where it fails. If retry does not work, exit the installer and relaunch it. If the problem persists, contact Broadcom Support.
15. After installing XFlow Interface, start the**CA XFlow Interface**service. Navigate to /opt/CA/xFlow/Apps and run the following command from a command prompt window:
    ```
    ./xFlowServer start
    ```
    To stop the server, run the following command:
    ```
    ./xFlowServer Stop.sh
    ```
16. Navigate to /opt/CA/SC/SearchServer/elasticsearch/bin and run the following command from the command prompt window:
    ```
    ./ServicesStart.sh
    ```
    NOTE
    If you get the below message while restarting the search server service, ignore it and continue:
    ```
    groupadd: group 1elasticsearch1 already exists
    useradd: user 1elasticuser1 already exists
    runuser: warning: cannot change directory to /home/elasticuser: No such file or directory
    OpenJDK 64-Bit Server VM warning: Option UseConcMarkSweepGC was deprecated in version 9.0 and
     will likely be removed in a future release.
    ```
    To stop the server, run the following command:
    ```
    /ServicesStop.sh
    ```
17. Validate the Installation.
18. Complete the Post-Installation Tasks.

After installing the XFlow Interface and Search Server, you can configure the cache size for XFlow Interface. For more information about how to configure cache size, see Perform initial load and refresh index on the search server.

Validate CA Service Management 17.4.4.0 Installation

Validate CASM 17.4.4.0 installation

To verify the successful installation of CA Service Management 17.4.4.0, validate the installation for the corresponding applications in the following location. You can also navigate to the root folder where you have locally installed the application and select the "env.info" file for validating your installation:

- **CA ITAM:**
  NOTE
  CORA version: 17.4.0.57

  - C:\Program Files (x86)\CA\ITAM\env.info
  - product=ITAM
  - major=17
  - minor=4
  - cum=4
  - roll-up=0

    NOTE
    CA CASM uses Apache Tomcat version 9.0.102.
- **CA Service Catalog:**
  - %USM_HOME%*CA_SLCM_<hostname>.his* file
  - %USM_HOME%\view\webapps\usm\locale\about_info.xml
  - %USM_HOME%\build.xml
  - %USM_HOME%\env.info
- **xFlow Interface:**
  - *C:\Program Files\CA\xFlow\<hostname>.his* file
  - NX_ROOT\env.info
  - NX_ROOT\version.xml
- **CA SDM:**
  - *C:\Program Files (x86)\CA\Service Desk Manager\<hostname>.his* file
  - NX_ROOT\pdmconf\version
  - NX_ROOT\env.info
  - HELP, ABOUT

    NOTE
    For CA SDM, the Binary and Language Combo patches are installed irrespective of the environment. The language patch for the corresponding CA SDM environment is installed.

    To validate the successful installation of the patches, follow these steps:

    1. Open the <sdm_installation>\<hostname>.his file in a text editor.
    2. Search for the instances of "gol-310". In a successful install, you should come across these three entries (Binary, Language Combo, and corresponding language patch).

Application	Windows	Linux
CA Asset Portfolio Management	T6EE1744	NA
CA Service Catalog	T6D91744	NA
CA Service Desk Manager - Binary	gol-310_cum_C	gol-310_x86_64Linux_cum_C
CA Service Desk Manager - Languag504e Comb	gol-310_langcombo	gol-310_langcombo
CA Service Desk Manager - English	gol-310_en_US	gol-310_x86_64Linux_en_US
CA Service Desk Manager - German	gol-310_de_DE	gol-310_x86_64Linux_en_DE
CA Service Desk Manager - Spanish	gol-310_es_ES	NA
CA Service Desk Manager - French	gol-310_fr_FR	gol-310_x86_64Linux_fr_FR
CA Service Desk Manager - Italian	gol-310_it_IT	NA
CA Service Desk Manager - Japanese	gol-310_ja_JP	gol-310_x86_64Linux_ja_JP
CA Service Desk Manager - Brazilian Portuguese	gol-310_pt_BR	NA

Post-Installation Tasks for CA Service Management 17.4.4.0

After successful installation of CA Service Management 17.4.4.0, perform the following post-installation steps.

CA Service Desk Manager

Manual Steps required for Automated Post-Installation

CA Service Management provides an option in the installer to automate the post-install tasks.
If you have selected **Yes** to automate the post-installation steps while running the CA Service Management Installer Wizard, you are not required to perform any manual post-install steps for CA Service Desk Manager (SDM).

If you have selected **No** for automating the post-install steps while running the CA Service Management Installer wizard, perform the below manual post-install tasks as per your deployment requirements:

Manual Post-Install Tasks After Installing CA Service Desk Manager (SDM) 17.4.4.0 on CA SDM 17.4.3.0
Perform the following steps:

1. Stop CA SDM services.
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0.
3. Open a command prompt window and run the following command:
   ```
   pdm_configure
   ```
4. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_04_insert.dat
   pdm_load -u -f r17_4_04_update.dat
   ```

Manual Post-Install Tasks After Installing CA Service Desk Manager (SDM) 17.4.4.0 on CA SDM 17.4.2.0
Perform the following steps:

1. Stop CA SDM services.
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0.
3. Open a command prompt window and run the following command:
   ```
   pdm_configure
   ```
4. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_03_insert.dat
   pdm_load -u -f r17_4_03_update.dat                                                    pdm_load -f
   r17_4_04_insert.dat
   pdm_load -u -f r17_4_04_update.dat
   ```
5. Notification message template is modified to update the existing email templates with the html_description field.
   Tool location: NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate This has to be executed one time when the SDM service is up and running. **For e.g:**
   Windows
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.bat
   ```
   Linux
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.sh
   ```

Manual Post-Install Tasks After Installing CA Service Desk Manager (SDM) 17.4.4.0 on CA SDM 17.4 or CA SDM 17.4.1.0

Perform the following steps:

1. Stop CA SDM services.
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0.

3. Open a command prompt window and run the following command:
   ```
   pdm_configure
   ```
4. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_02_insert.dat
   pdm_load -u -f r17_4_02_update.dat
   pdm_load -f r17_4_03_insert.dat
   pdm_load -u -f r17_4_03_update.dat                                                    pdm_load -f
   r17_4_04_insert.dat
   pdm_load -u -f r17_4_04_update.dat
   ```
5. Notification message template is modified to update the existing email templates with the html_description field.
   Tool location: NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate This has to be executed one time when the SDM service is up and running. **For e.g:**
   Windows
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.bat
   ```
   Linux
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.sh
   ```

Manual Post-Installation Task for CMDB

> **NOTE**
> For Oracle 21c, set the **isClusterEnv** setting to "true" in the application.conf file of CMDB viz and restart the CA SDM Services.:
>
> - For Windows: <SDM installed location>\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\WEB-INF\classes\application.conf)
> - For LINUX: <SDM installed location>/bopcfg/www/CATALINA_BASE_VIZ/webapps/CMDBVisualizer/WEB-INF/classes/application.conf).

If you are applying CA Service Management 17.4.4.0 in an SSL environment, CA SDM Administrators must perform the following post-installation steps for CMDB Visualizer Interface to open using the **HTTPS** protocol when launching from CA SDM UI:

1. In the NX.env file, change the **NX_CMDB_VISUALIZER** entry to provide **HTTPS protocol** and **HTTPS port**.
   For example: @NX_CMDB_VISUALIZER=https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer
2. In **casm.conf.js** file from NX_root\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\conf, provide the HTTPS as protocol and HTTP port.
   For example:

```
sdm: {

     vizserver: 'https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer/viz',
     server: 'https:// sdmurl:<SDM_SSL_PORT>/CAisd/pdmweb.exe'
   }
```

Manual Post-Installation Task for OAuth 2.0 (Maileater/ Mailer)

After generating an access token for OAuth 2.0 to connect to a mail server, perform the mandatory post-install tasks on all servers in your environment:

1. Create a backup of web.xml from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF in your local directory to perform backout if required.
2. Open the web.xml file from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF.
3. Search for the following statement:

```
<!--USD servlet mappings start-->
```

4. Copy and paste the following configuration under the statement:

```
<servlet>
  <servlet-name>OAuthProcessor</servlet-name>
  <servlet-class>com.broadcom.oAuth.OAuthProcessor</servlet-class>
 </servlet>
```

5. Copy and paste the following configuration after any <servlet-mapping>:

```
<servlet-mapping>
  <servlet-name>OAuthProcessor</servlet-name>
  <url-pattern>/OAuthProcessor</url-pattern>
</servlet-mapping>
```

6. Save and close the web.xml file.
7. For OAuth servlet preservation in a primary and secondary environment, the secondary server must also have the OAuth processor entries in the following file: *<NX_ROOT>\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF\web.xml.last*.
8. Repeat Steps 1-8 to update **web.xml.tpl** file in the NX_ROOT\samples\pdmconf\web.xml.tpl folder to avoid losing these changes when pdm_configure is run at a later point of time.
9. Run the following command:
   ```
   pdm_tomcat_nxd -c STOP command
   ```
10. Wait for some time (a minute or so) and run the following command:
    ```
    pdm_tomcat_nxd -c START command
    ```

For more information, see Define a mailbox.

Manual Post-Installation Task for Excel file (.xlsx) export*

1. Execute the following command:
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst
   ```
2. Execute the following command to add the new NX variable to NX.env.tpl file.
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst -t
   ```
3. Restart CA SDM services.

HTML Tags Sanitization

The sanitization tool can be run any number of times. Administrators can choose if they wish to run the tool for each ticket type separately or for all ticket types at one go.

Before running the Sanitization tool ensure you take a backup of all the ticket objects using the following commands or by taking a backup of the sdm databse :

```
"Call_Req": pdm_extract Call_Req > cr_backup.txt
"Issue": pdm_extract Issue > Issue_backup.txt
"Change Orders": pdm_extract Change_Request >change_orders_backup.txt.
```

Both active and archived tickets can be sanitized.
For Windows:

```
NX_ROOT\samples\sdk\SanitizationTool\sanitizeTool.bat
```

For Linux:

```
NX_ROOT\samples\sdk\SanitizationTool\sanitizeTool.sh
```

NOTE
We recommend executing the sanitization during slack time on any Service Desk Manager server.

After executing the sanitization tool, execute the following command to update domsrvr memory cache:

```
pdm_cache_refresh -t Call_Req
pdm_cache_refresh -t Issue
pdm_cache_refresh -t Change_Request
```

(Optional) Enable HTML content for Ticket Description in the Activity log

NOTE

- A new Field Update activity association (**HTML Description** has been introduced)
- Either **HTML Description** or **Description** activity association has to be enbaled. Do not enable both the options.
- Out of the box Description Field Update activity association is enabled in the activity log for all ticket types (Incident,Request,Problem,ChangeOrder and Issue).

Enabling HTML content for Description in the Activity Log:

1. Login into CA SDM with Administrator access type.
2. Administration --> Notifications --> Activity Associations.
3. Click **Show Filter**.
4. Enter "HTML Description" in the Symbol text box in the Activity Association List page.
5. Click **Search**
6. For every **HTML Description Field Update association**, click **Edit** and check the **Log Me** checkbox, then save the changes made.

Disabling plain text for Description in the Activity log:

1. Click **Show Filter** in the Activity Association List page.
2. Enter "Description" in the Symbol text box and Select Object Type as "Request/Incident/Problem" and click **Search**.
3. Now see the one row and click the Description link in the Activity Association List page.
4. Unselect the **Log Me** check box and save the page.
5. For Change Order and Issue repeat the steps from 1 to 4.

(Optional) Manual Post-Installation Task for Knowledge Document export with German fields

If you are exporting Knowledge Document with German Fields (included in the export field list), then perform the below post-install steps for successful export of knowledge documents:

1. Take the backup of NX_ROOT\NX.env file (NX_ROOT is the installation directory for CA Service Desk Manager).
2. Open NX_ROOT\pdmconf\NX.env_nt.tpl file using any text editor and take the value of NX variable @NX_KEIT_AVAILABLE_FIELDS which is COPIED_DATA and update Nx.env in NX_ROOT by executing below command:

   ```
   pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA>  -a  pdm_option.inst
   ```

   Where COPIED_DATA is the value copied from NX.env_nt.tpl (step 2).
   For example, " pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <provide the exact string here> -a pdm_option.inst "
   Note to avoid losing the updated changes when you run pdm_configure, execute the above command with the '-t' flag as follows:

   ```
   pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA> -a pdm_option.inst -t
   ```

(Optional) Handle HTML Mails that contains Tables in MailEater

Java based Maileater already has the ability to process the emails that were sent in plain format or HTML format, to support the processing of emails that contain Tables in the HTML body. But the format in which the email was sent is lost when received by Maileater.

1. Run the following commands from NX_ROOT/bin from the command prompt/ terminal:
   ```
   pdm_options_mgr -c -s NX_ENFORCE_HTML_EMAIL_PARSING -v "Yes" -a pdm_option.inst
   pdm_options_mgr -c -s NX_ENFORCE_HTML_EMAIL_PARSING -v "Yes" -a pdm_option.inst -t
   ```
2. Restart CA SDM and XFlow applications.

(Optional) Comments are not allowed when a Data Partition (O_COMMENTS) is created with Customer, Employee and ServiceDesk Analyst

To disable comments:

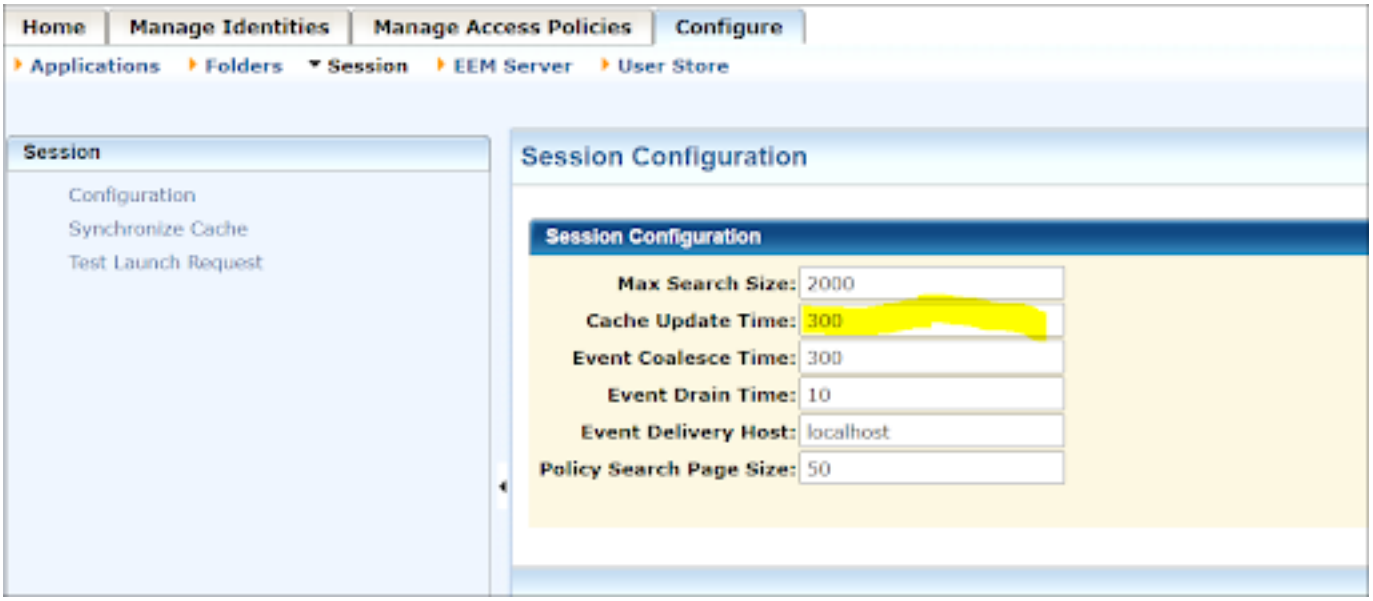
1. Login to SDM and navigate to Administration -> Security and Role Management-> Data partition-> Data Partition List.
2. In Data Partition List, create a constraint (O_COMMENTS with the condition id=0) and link it to Customer, Employee and ServiceDesk Analyst data partitions.
3. Run the following commands from NX_ROOT/bin from the command prompt/ terminal:
   ```
   pdm_options_mgr -c -s KT_COMMENT_USE_SUPERUSER -v 0 -a pdm_option.inst
   pdm_options_mgr -c -s KT_COMMENT_USE_SUPERUSER -v 0 -a pdm_option.inst -t
   ```
4. Restart CA SDM and XFlow applications.

(Optional) Manual Post-Installation Task for EEM Cache Timeout Configuration Update

You may face issues while logging in to EEM or SDM using the EEM user accounts due to a session cache timeout mismatch between EEM and SDM.

You can update the EEM cache timeout configuration by adding a new NX variable, "NX_EEM_CACHE_UPDATE_TIME" in SDM.

Keep the value the same as the EEM cache timeout configuration value mentioned in the following screenshot.



To set the EEM cache update time to 300 seconds, perform the following steps:

To add the variable in the **nx.env** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst
```

To add the variable in the **nxenv_nt.tpl** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst -t
```

(Optional)casc_impersnate_user

Once both CA SDM and CA Service Catalog are integrated, do the following:

1. Login to CA SDM with the Administrator user type.
2. Navigate to Administration >> Options Manager >> CA Service Catalog >> in the Options List, install casc_impersnate_user
3. Restart the SDM service.

(Optional)Incident/Request/Problem Solution should display Short Resolution

To display a short solution for tickets, do the following:

1. Login to CA SDM with the Administrator user type.
2. Navigate to Administration >> Options Manager >> Knowledge >> use_resolution_short_for_soln.
3. In the use_resolution_short_for_soln window displayed, click **Edit,** and click **Install**.
4. Restart CA SDM.

(Optional)Logging SOAP Calls

Service Point

After successful installation of Service Point, refresh index and data on the Search Server.

> **NOTE**
> In case of Linux Search Server environment, ensure to run the following commands on the terminal:
>
> ```
> sysctl -w vm.max_map_count=262144
> ulimit -Hn 65535
> ```

(Optional) Manual Post-Installation Task for Rebuild Index

If you have upgraded or are planning to install CA EEM 12.6 with Service Point and CA Service Catalog, ensure that you rebuild the CA Service Catalog index after completing the CA EEM upgrade/installation process. For more information, see Perform Initial Load and Refresh Index Data on Search Server.
Run the following command to rebuild the index:

```
$ES_HOME/search/bin/pdm_es_rebuild_index.bat index=catalog_index
```

CA Service Catalog

Perform the following CA Service Catalog post-installation steps as per your environment requirements:

- Post-Installation Tasks after installing CA Service Catalog 17.4.3.0 on CA Service Catalog 17.4 or later
- gMSA Authentication

Post-Install Tasks After Installing CA Service Catalog 17.4.3.0 on CA Service Catalog 17.4 or later

Review and perform the post-installation steps based on your deployment requirements.

> **NOTE**
> If any SSL certificate(s) chain was imported to the Oracle Java's JAVA_HOME\lib\security\cacerts Keystore, those certificate(s) chain again need to be exported explicitly and imported properly into the new AdoptOpenJDK's %USM_HOME%\embedded\jdk\lib\security\cacerts Keystore.
>
> **NOTE**
> While upgrading from 17.4.2.0 to 17.4.3.0, re-run the removal_axis.bat file, if already ran it on 17.4.2.0.

Follow these steps

- (Optional) If Service Catalog is accessed from applications like Service Point, SharePoint and so on, then you must perform this step. Open the "**web.xml**" file located in *USM_HOME%\view\webapps\usm\WEB-INF* and update the respective parameter values for the parameter name "cors.allowed.origins" and "cors.support.credentials". For example, if you are using Catalog Widgets in Service Point
- (Optional) Unable to approve any workflow requests through email. The email sent for approvals is corrupted with special characters. Follow the KB article steps.

(Optional) Approve workflow requests through email

The workflow email sent for approvals is corrupted with special characters. To approve any workflow requests through email, follow the KB article steps.

gMSA Authentication Post-Install Steps for CA Service Catalog (Automated steps)

CA Service Catalog provides automated post-installation steps for gMSA Authentication. Review the following before you proceed:

- Prerequisites and Considerations
- How to Automate gMSA Post-Installation Steps
- Unconfigure gMSA

Prerequisites and Considerations:

Ensure that you have taken care of the below considerations on all nodes of CA Service Catalog where you want to automate the gMSA post-installation procedure.

1. If the accounting service is available, perform the automated post-install steps by updating serviceName to ServiceAccounting and providing the appropriate gMSA account name in the **gmsa.properties** file.
2. If the filestore path is external to the server from where the **GMSAPostInstall.bat** is running, ensure to provide appropriate access to the server for setting the gMSA account permissions.
3. While preparing the properties file, check for any copy/paste issues, special characters, spaces, and so on.

How to Automate gMSA Post-Installation Steps

Perform the following steps to enable the batch file **GMSAPostInstall.bat** and to automate the gMSA post-installation steps:

1. Create a gMSA account for CA Service Catalog.
2. Ensure that appropriate DB (database) permission is given to the newly created account(s).
3. Enter the below values in the **gmsa.properties** file and save it.

   NOTE
   A template file (**gmsa.properties**) is added in the common installer in *"\filestore\SLCM\gMSAProperties"* . Optionally. use this file as a template file to fill gmsa properties. This can be copied to "USM_HOME/scripts/GMSA" where Catalog is installed.

   NOTE
   All values are mandatory. Make sure that there are no leading or trialing spaces while providing these values.

   a. **serviceName**: Application service name to enable GMSA. In the case of Catalog, it will be ServiceCatalog and for accounting, it will be ServiceAccounting.
   b. **folderName**: Absolute path of the application installation directory.
   c. **gMSAName**: Name of the GMSA service account (domain\accountName$).
   d. **filestorePath**: Shared network path (\\Server_name\filestore) or local filestore path
4. Navigate to %USM_HOME% and launch **usm.cmd**
5. Navigate to %USM_HOME%/scripts/GMSA/ folder and run the "GMSAPostInstall.bat" in command prompt.

NOTE
If the command executes without any error, "Post-installation steps for GMSA executed successfully" message is displayed with exit code 0.
If there are errors, the error message is shown with a non-zero exit code. Correct the error to continue and perform step **5** again.

Unconfigure gMSA for CA Service Catalog

To disable or unconfigure gMSA for CA Service Catalog, follow the steps below:

1. Replace **config.properties** file from*%USM_HOME%*\ folder.
2. Revert back all the folder permissions which were assigned to domain users.
3. Start Catalog and Accounting Services with the local system account.

NOTE
In the case of load balancers, repeat steps 1-3 to unconfigure gMSA on all Catalog servers.

CA Asset Portfolio Management

Manual Post-Installation Task for Removing jQuery, log4net, and log4j Vulnerabilities

After installing 17.4.4.0, delete jquery-1.3.2.min.js, jquery-ui-1.7.2.custom.min.js , log4net.dll, and log4j from their respective installed folders.

NOTE
Perform the below steps if you have not done this earlier.

Perform the following steps:

1. Log on to CA APM Web Server, navigate to the <ITAM_install_dir>\ITAM\Web Server\castylesr5\javascript.
   Example: C:\Program Files (x86)\CA\ITAM\Web Server\castylesr5\javascript
   Locate and delete the jquery-1.3.2.min.js and jquery-ui-1.7.2.custom.min.js files from the directory.
2. Navigate to the ITAM_install_dir\CA\ITAM\Data Importer.
   Locate and delete the log4net.dll file from the directory.
3. Navigate to the C:\Program Files (x86)\CA\SC\CORA.
   Locate and delete the log4j-1.2.15.jar, log4j.properties, and ojdbc6.jar files from the directory.
4. Navigate to the ITAM_install_dir\CA\ITAM.
   Backup the following folders and manually delete them:
   - CDB
   - MDB
   - jre8

   NOTE
   jre8 folder has to be deleted manually for every RU installation.
5. Perform an IISRESET and delete the browser's cache to load the upgraded files.

For ITAM installation on French Operating System

Following the successful installation of version 17.4.3.0 on a French Windows operating system with the browser set to French, we are unable to log in to ITAM.

An error message is displayed: ***Une valeur Request.Form potentiellement dangereuse a été détectée à partir du client (_UPLOAD_NON_TXT_CSV_DATA_FILE_="...ectionné n'est pas de type...").***

To overcome this, follow these steps:

1. Navigate to *C:\Program Files (x86)\CA\ITAM\Web Server, edit web.config*, and search for *<pages enableEventValidation="false" viewStateEncryptionMode="Never" controlRenderingCompatibilityVersion="4.0" clientIDMode="AutoID">* , then add *validateRequest="false"* to it.

2. After adding, it looks like this:
   <pages enableEventValidation="false" viewStateEncryptionMode="Never" controlRenderingCompatibilityVersion="4.0" clientIDMode="AutoID" validateRequest="false">
3. Do an IISRESET.
4. You can now successfully login to ITAM.

Post-Installation Tasks - CA SDM Advance Availability 17.4.4.0

The installer wizard provides an option in the installer to automate the post-install tasks. Perform the following post-installation tasks for CA SDM Advanced Availability Configuration. For the Advanced Availability configuration, the following host names are considered as an example:

- The background server hostname is sdmcahost1.
- The standby server hostname is sdmcahost2.
- The application server hostname is sdmcahost3.

Manual Steps required for Automated Post-Installation

If you have selected **Yes** for automating the post-install steps while running the installer wizard, then perform manually the following post-install tasks:

Manual Post-Install Tasks - After Installing CA Service Desk Manager (SDM) 17.4.4.0 on CA SDM 17.4.3.0

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Start the CA SDM service on the standby server (sdmcahost2). Open the command prompt and run the following command:
   ```
   pdm_server_control -b
   ```
3. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server.
4. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```
5. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_04_insert.dat
   pdm_load -u -f r17_4_04_update.dat
   ```
6. Clear the browser cache by the following command:
   ```
   pdm_webcache -H
   ```
7. Ensure that the patch is applied on all the application servers and the remaining standby servers if any. If you selected YES to automate the post-install steps, then there are no more manual post-install steps to be performed.

Manual Post-Install Tasks - After Installing CA Service Desk Manager (SDM) 17.4.4.0 on CA SDM 17.4.2.0

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Start the CA SDM service on the standby server (sdmcahost2). Open the command prompt and run the following command:
   ```
   pdm_server_control -b
   ```
3. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server.
4. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```
5. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_03_insert.dat
   pdm_load -u -f r17_4_03_update.dat                                                    pdm_load -
   f r17_4_04_insert.dat
   pdm_load -u -f r17_4_04_update.dat
   ```

6. Clear the browser cache by the following command:
   ```
   pdm_webcache -H
   ```
7. Ensure that the patch is applied on all the application servers and the remaining standby servers if any. If you selected YES to automate the post-install steps, then there are no more manual post-install steps to be performed.

Manual Post-Install Tasks - After Installing CA Service Desk Manager (SDM) 17.4.4.0 on CA SDM 17.4 GA or CA SDM 17.4.1.0

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Start the CA SDM service on the standby server (sdmcahost2). Open the command prompt and run the following command:
   ```
   pdm_server_control -b
   ```
3. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server.
4. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```
5. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_02_insert.dat
   pdm_load -u -f r17_4_02_update.dat
   pdm_load -f r17_4_03_insert.dat
   pdm_load -u -f r17_4_03_update.dat                                                    pdm_load -
   f r17_4_04_insert.dat
   pdm_load -u -f r17_4_04_update.dat
   ```
6. Clear the browser cache by the following command:
   ```
   pdm_webcache -H
   ```
7. Ensure that the patch is applied on all the application servers and the remaining standby servers if any. If you selected YES to automate the post-install steps, then there are no more manual post-install steps to be performed.

Manual Steps required for Non-Automated Post-Installation

If you have selected **No** for automating the post-install steps while running the CA Service Management installer wizard, perform manually the appropriate post-install after this task as per your deployment requirements.

Post-Install Tasks After Installing CA SDM 17.4.4.0 on CA SDM 17.4.3.0 for Advanced Availability Configuration

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0. Open a command prompt and run the following command:
   ```
   pdm_configure
   ```
   On the installer configuration page, before clicking Finish, uncheck the Start service when completed checkbox. Failure to uncheck the box may result in the loss of custom schema changes and configurations that were changed on the server.
3. Promote the standby server as the new background server by performing the following steps. To suppress version control on the standby server and later on the background server, run the following command on each server:
   ```
   pdm_server_control -v
   ```
4. Start SDM service on the standby server. Run the following command:
   ```
   pdm_server_control -b
   ```
5. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server, and perform the following steps. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
   ```
   pdm_configure
   ```
6. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:

```
pdm_server_control –b
```

7. Change the directory to NX_ROOT/data, and execute the following command:

```
pdm_load –f r17_4_04_insert.dat
pdm_load –u –f r17_4_04_update.dat
```

8. Ensure that the patch is applied on all application servers and the remaining standby server if any. Perform the following steps on all application servers and standby servers.
9. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:

```
pdm_configure
```

10. Clear the browser cache by running the following command:

```
pdm_webcache –H
```

Post-Install Tasks After Installing CA SDM 17.4.4.0 on CA SDM 17.4.2.0 for Advanced Availability Configuration

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0. Open a command prompt and run the following command:

```
pdm_configure
```

   On the installer configuration page, before clicking Finish, uncheck the Start service when completed checkbox. Failure to uncheck the box may result in the loss of custom schema changes and configurations that were changed on the server.
3. Promote the standby server as the new background server by performing the following steps. To suppress version control on the standby server and later on the background server, run the following command on each server:

```
pdm_server_control –v
```

4. Start SDM service on the standby server. Run the following command:

```
pdm_server_control –b
```

5. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server, and perform the following steps. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:

```
pdm_configure
```

6. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:

```
pdm_server_control –b
```

7. Change the directory to NX_ROOT/data, and execute the following command:

```
pdm_load –f r17_4_03_insert.dat
pdm_load –u –f r17_4_03_update.dat                                                    pdm_load –
f r17_4_04_insert.dat
pdm_load –u –f r17_4_04_update.dat
```

8. Ensure that the patch is applied on all application servers and the remaining standby server if any. Perform the following steps on all application servers and standby servers.
9. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:

```
pdm_configure
```

10. Clear the browser cache by running the following command:

```
pdm_webcache –H
```

Post-Install Tasks After Installing CA SDM 17.4.4.0 on CA SDM 17.4 GA or CA SDM 17.4.1.0 for Advanced Availability Configuration

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0. Open a command prompt and run the following command:

```
pdm_configure
```

On the installer configuration page, before clicking Finish, uncheck the Start service when completed checkbox. Failure to uncheck the box may result in the loss of custom schema changes and configurations that were changed on the server.

3. Promote the standby server as the new background server by performing the following steps. To suppress version control on the standby server and later on the background server, run the following command on each server:
   ```
   pdm_server_control -v
   ```
4. Start SDM service on the standby server. Run the following command:
   ```
   pdm_server_control -b
   ```
5. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server, and perform the following steps. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
   ```
   pdm_configure
   ```
6. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```
7. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_02_insert.dat
   pdm_load -u -f r17_4_02_update.dat
   pdm_load -f r17_4_03_insert.dat
   pdm_load -u -f r17_4_03_update.dat                                                    pdm_load -f
   r17_4_04_insert.dat
   pdm_load -u -f r17_4_04_update.dat
   ```
8. Notification message template is modified to update the existing email templates with the html_description field. **Tool location**: NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate This has to be executed one time when the SDM service is up and running. **For e.g:**
   Windows
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.bat
   ```
   Linux
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.sh
   ```
9. Ensure that the patch is applied on all application servers and the remaining standby server if any. Perform the following steps on all application servers and standby servers.
10. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
    ```
    pdm_configure
    ```
11. Clear the browser cache by running the following command:
    ```
    pdm_webcache -H
    ```
12. Execute the following command.
    ```
    pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst
    ```
13. Execute the following command to add the new NX variable to NX.env.tpl file.
    ```
    pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst -t
    ```
14. Restart CA SDM Services.

Uninstall CA Service Management 17.4.4.0

Review the following topics to uninstall CA Service Management 17.4.4.0:

- Uninstall CA SDM 17.4.4.0 (Windows)
- Uninstall CA SDM 17.4.4.0 (Linux)
- Restore Visualizer Files (Windows)
- Restore Visualizer Files (Linux)
- Uninstall the XFlow Interface 17.4.4.0 (Windows)
- Uninstall the XFlow Interface 17.4.4.0 (Linux)
- Uninstall CA Service Catalog 17.4.4.0
- Uninstall CA Asset Portfolio Management
- Run the Database Cleanup Utility for CA Service Management 17.4.4.0

NOTE

Uninstalling CA Service Management 17.4.4.0 will revert to CA Service Management 17.4 version on which 17.4.3.0 is installed.

Example: If CA Service Management 17.4.4.0 is installed on 17.4, uninstalling 17.4.4.0 will revert to CA Service Management 17.4.

Uninstall CA SDM 17.4.4.0 (Windows)

Perform the following steps to uninstall CA SDM 17.4.4.0 on Windows:

1. Log in to CA SDM as an Administrator.
2. Shut down CA Service Desk Manager Server.
3. Navigate to **<NX_ROOT>\bin** and run the following command to backup the unsupported jars from the older versions.
   ```
   pdm_perl pdm_restore_files.pl gol-310_tool ..\PATCHES\DELIVERABLES_REMOVAL.TXT
   ```
4. Navigate to the location *where you have extracted the CA Service Management Installer zip package*.
5. Locate *CASM-17.4.4.0-CommonPatchInstaller-Win\filestore\utils\ApplyPTF* folder.
6. Run the **APPLYPTF.exe**.
7. Select **Backout PTF** from the local or remote node option.
8. Click **Next**.
9. Enter the patch number for the language that is installed in your environment:
   - **Binary**: gol-310_cum_C
   - **Language Master**: gol-310_langcombo
   - **English**: gol-310_en_US
   - **French**: gol-310_fr_FR
   - **German**: gol-310_de_DE
   - **Spanish**: gol-310_es_ES
   - **Italian:** gol-310_it_IT
   - **Japanese**: gol-310_ja_JP
   - **Brazilian Portuguese**: gol-310_pt_BR
10. Click **Next** to continue backing out the specific patch.
11. Repeat steps 6-9, to back out the language English (gol-310_en_US) patch.
12. Repeat steps 6-9 to back out the language master (gol-310-langcombo) patch.
13. Repeat steps 6-9 to back out the binary (gol-310_cum_C) patch.
14. Reboot the server.
15. Repeat steps **6-12** on all standby, application, or secondary servers in your environment.
16. Restore the MDB database to the database backup that you took before applying the patch.
17. Take the backup of the 5.40.0 folder and then delete the folder from <NX_ROOT>\bin\perllib.
18. Copy the 5.36.0 folder from <NX_ROOT>\REPLACED\perl_old and place the folder to <NX_ROOT>\bin\perllib.

19. Navigate to ***<NX_ROOT>\site\config.properties*** file and ensure that the value of ***resources.created*** parameter is set to **0.** This must display as ***resources.created=0*** in the file.
20. Open a command prompt and run the following command:
    ```
    pdm_configure
    ```
21. If you roll-back from 17.4.4.0 to the base version, copyright text year will not change. To change the year:
    a. Add @NX_COPYRIGHT_YEAR=2023 into NX.env file located under CA SDM install directory (NX_ROOT).
    b. To preserve the value even after pdm_configure, update the value of "genfact.gx_copyright_year in install.properties under NX_ROOT\site genfact.gx_copyright_year=2023

Uninstall CA SDM 17.4.4.0 (Linux)

Perform the following steps to uninstall CA SDM 17.4.4.0 on Linux:

1. Login to CA SDM as the root user.
2. Shut down the CA Service Desk Manager Server.
3. Navigate to **<NX_ROOT>/bin** and run the following command to backup the unsupported jars from the older versions.
   ```
   ./pdm_perl pdm_restore_files.pl ../replaced/gol-310_tool ../patches/deliverables_removal.txt
   ```
4. Navigate to the location where you have extracted the CA SDM Installer **tar.gz** package.
5. Navigate to the *casm_dvd.x86Linux \filestore\utils\ApplyPTF* folder.
6. The following binary/language-independent and language-dependent patches are supported:
   - Binary name: gol-310_x86_64Linux_cum_C
   - Language Master: gol-310_langcombo
   - English: gol-310_x86_64Linux_en_US
   - German: gol-310_x86_64Linux_de_DE
   - French: gol-310_x86_64Linux_fr_FR
   - Japanese: gol-310_x86_64Linux_ja_JP
7. Execute the following command to remove the binaries or the language-dependent patches. Replace the <patch_name> with the binary or language-dependent patch name:
   ```
   ./applyptf.<platform> -b <patch_name>  -e <NX_ROOT>
   ```
   Here, <platform> specifies the operating system and <NX_ROOT> specifies the CA SDM installed location.
   For example:
   ```
   ./applyptf.linux -b gol-310_x86_64Linux_cum_C  -e /opt/CA/ServiceDeskManager
   ```
8. Repeat step 6 to back out the language English (gol-310_x86_64Linux_en_US) patch.
9. Repeat step 6 to back out the binary (gol-310_x86_64Linux_cum_C) patch.

   NOTE

   During uninstallation, if you get any errors, ensure that the following steps are performed:

   1. Copy the subfolders from <NX_ROOT>/replaced/<gol-310_x86_64Linux_cum_C> and place it into the <NX_ROOT> folder.
   2. Copy the subfolders from <NX_ROOT>/replaced/<gol-310_x86_64Linux_en_US> and place it into the <NX_ROOT> folder.
   3. Open **.his** file under the <NX_ROOT> folder and delete the entries related to the current patch and save it.
10. To restore MDB database, revert to database backup that you took before applying the patch.
11. Take the backup of the 5.40.0 folder and then delete the folder from <NX_ROOT>\bin\perllib.
12. Copy the 5.36.0 folder from <NX_ROOT>\REPLACED\perl_old and place the folder to <NX_ROOT>\bin\perllib.
13. Navigate to ***<NX_ROOT>/site/config.properties*** file and ensure that the value of ***resources.created*** parameter is set to **0.** This must display as ***resources.created=0*** in the file.

14. Open a command prompt and run the following command:
    ```
    pdm_configure
    ```

15. If you roll-back from 17.4.4.0 to the base version, copyright text year will not change. To change the year:
    a. Add @NX_COPYRIGHT_YEAR=2023 into NX.env file located under CA SDM install directory (NX_ROOT).
    b. To preserve the value even after pdm_configure, update the value of "genfact.gx_copyright_year in install.properties under NX_ROOT\site genfact.gx_copyright_year=2023

       NOTE
       If you are unable to access the SDM URL, follow these steps:

       1. Stop the SDM Services.
       2. Navigate to the <NX_ROOT>/bopcfg/www/CATALINA_BASE folder and delete the work folder.
       3. Start the SDM Services.

Restore Visualizer Files (Windows)

NOTE
After uninstalling CA SDM, CMDB Visualizer does not function automatically as the uninstallation process will not restore the previous visualizer files . To restore, perform the below steps manually:

1. Navigate to the where you have installed Visualizer: <NX_ROOT>\bopcfg\www\CATALINA_BASE_VIZ\webapps.
2. Remove the CMDBVisualizer folder.
3. Copy the **Visualizer_17.4.3.0.OLD** folder from<NX_ROOT>*REPLACED* and paste it in <NX_ROOT>*bopcfg**www* *CATALINA_BASE_VIZ**webapps*. Now, rename the **Visualizer_17.4.3.0.OLD** to **CMDBVisualizer**.
4. Start the SDM Service.

CMDB Visualizer works as expected.

Restore Visualizer Files (Linux)

NOTE
After uninstalling CA SDM, CMDB Visualizer does not function automatically as the uninstallation process will not restore the previous visualizer files. To restore the files, perform the below steps manually:

1. Navigate to the where you have installed Visualizer: <NX_ROOT>/bopcfg/www/CATALINA_BASE_VIZ/webapps.
2. Remove the CMDBVisualizer folder.
3. Copy the **Visualizer_17.4.4.0.OLD** folder from <NX_ROOT>*/replaced* and paste it into <NX_ROOT>*bopcfg**www* *CATALINA_BASE_VIZ**webapps*. Now, rename the **Visualizer_17.4.4.0.OLD** to **CMDBVisualizer**.
4. Start the SDM Service.

CMDB Visualizer works as expected.

Uninstall the XFlow Interface 17.4.4.0 (Windows)

Perform the following steps to uninstall the xFlow Interface (Windows):

1. Click **Start**, **Control Panel**, **Uninstall a Program** to uninstall xFlow Interface 17.4.4.0.
   Or
2. Navigate to where you have installed the xFlow Interface (for example, <NX_ROOT>*/CA/xFlow)*.
3. Select xFlow Interface **17.4.4.0_installation** and click **Change xFlow Interface 17.4.4.0 Installation.exe** to complete the uninstallation process.
4. If you are using Microsoft SQL Server Database, run the following query:
   ```
   update ofProperty set propValue='select c.access_token from casm_active_sessions c WHERE
    UPPER(CONVERT(varchar(32), c.contact_uuid ,2)) = UPPER(?)' where name = 'jdbcAuthProvider.passwordSQL'
   ```

5. If you are using Oracle Database, run the following query;

```
update ofProperty set propValue='select c.access_token from casm_active_sessions c WHERE
 UPPER(CAST( c.contact_uuid AS varchar2(32))) = UPPER(?)' where name = 'jdbcAuthProvider.passwordSQL'
```

Uninstall the XFlow Interface 17.4.4.0 (Linux)

To uninstall the XFlow Interface, complete the following steps on Linux :

1. Navigate to where you have installed the xFlow Interface server.
   For example:
   /opt/CA/_xFlow Interface_installation
2. Execute the following command to uninstall the xFlow Interface 17.4.4.0:

```
./Change xFlow_Interface 17.4.4.0 Installation
```

3. Click **Finish** to complete the uninstallation process.
4. Run the following query for Oracle Database:

```
update ofProperty set propValue='select c.access_token from casm_active_sessions c WHERE
 UPPER(CAST( c.contact_uuid AS varchar2(32))) = UPPER(?)' where name = 'jdbcAuthProvider.passwordSQL'
```

Uninstall CA Service Catalog 17.4.4.0

To uninstall CA Service Catalog, perform the following steps:

1. Click **Control Panel**, **Programs and Features**.
2. Double-click the CA Service Catalog 17.4.4.0 and follow the uninstallation wizard instructions.

Uninstall CA Asset Portfolio Management

CA Asset Portfolio Management (CA APM) architecture is highly database-driven and has no rollback mechanism for database-related changes. Hence, the uninstallation of the CA APM patch is not recommended and is not supported. The only way to revert the database changes, if required, is by restoring the database backup which can lead to data loss.

Run the Database Cleanup Utility for CA Service Management 17.4.4.0

During the installation of CA Service Management products, each of the installed product is registered in a common mdb table that is reflected in the deployment summary of the selected environment. These entries must be cleared in the following cases:

- Uninstallation of any CA Service Management products
- Moving Database or migration from one server to another

Run the CA Service Management database cleanup utility on a Windows or a Non-Windows environment. The database cleanup utility removes the entries and adds to the following tables:

Table Name from where the entries are removed	Table Name to which the removed entries are added
al_cdb_componentinstallstate	al_cdb_comp_installstate_bkup
al_cdb_configurationparameters	al_cdb_config_params_backup

NOTE

On a Non-Windows Platform, before running the **DBCleanup.sh** Utility, you must export the Oracle Environment variables on the terminal.

Perform the following steps:

1. Navigate to *DVD1\DBCleanupUtility* directory.
2. For Windows, double-click the **DBCleanupUtility.bat** to launch the DBCleanup utility.
   On a Non-Windows platform, you must run the **DBCleanupUtility.sh** file.
3. Enter the following details:
   a. **Database Server Type**: Specify the type of server. Microsoft SQL or Oracle.
   b. **Database Host Name:** Enter the host name of the server where the database resides.
   c. **Database User Name**:
      - **SQL Server**: Enter the privileged user detail that has permission to create and modify the schema.
      - **Oracle**: Provide the mdbadmin user name.
   d. **Database Password**:
      - **MS SQL**: Enter the database password used by the database admin user.
      - **Oracle**: Enter the password for the Oracle database user.
   e. **Database Port Number:** Provide the database port number.
   f. **Enter Database Name :** Enter the database name. For example, mdb.
   g. **System Name**: Provide the hostname of the system from where you want to uninstall the CA Service Management product or products.
4. Select any of the following products that you want to uninstall:
   – **SDM -** CA Service Desk Manager
   – **SLCM-** CA Service Catalog
   – **xFlow Interface**
   – **ProductUpdates**
   Enter product name to be uninstalled: **ProductUpdates**
5. Enter product to clean up updates: (sdm,slcm,xflow,chatserver,searchserver): For example, slcm
6. Enter update version to clean up: 17.4.4.0
7. The utility cleans up the database and removes the entries of the selected product installed on the server.

Installing CA Service Management 17.4.3.0

Based on your CA Service Management implementation, review the following topics for a successful installation.

- Install Prerequisites for CA Service Management 17.4.3.0
- Install CA Service Management 17.4.3.0 (Windows)
- Install CA Service Management 17.4.3.0 (Linux)
- Validate CA Service Management 17.4.3.0 Installation
- Post-Installation Tasks for CA Service Management 17.4.3.0
- Post-Installation Tasks - CA SDM Advance Availability 17.4.3.0
- Uninstall CA Service Management 17.4.3.0

Install Prerequisites for CA Service Management 17.4.3.0

Installation prerequisites for patch 17.4.3.0

Verify and complete the following prerequisites before installation of CA Service Desk Manager, CA Service Catalog, and CA Asset Portfolio Management.

Installation Prerequisites for CA Service Desk Manager

Review the following CA SDM installation prerequisites before you start the installation process:

> **NOTE**
> We recommend that you place the installer in a shorter directory path, containing no spaces, to avoid errors when running the setup file. For example, unzip the file at: C:\WIN_17_4_3_0.

- Ensure that CA SDM and xFlow Interface 17.4 is installed in your environment.
- Back up the system database (MDB). The system database (MDB) backup is mandatory and ensures safe data recovery if any problems are encountered during the installation process.
- Do not invoke ApplyPTF manually in your system while running the CA Service Management CA Service Management 17.4.3.0 installer.
- In the case of, CA Service Desk Manager - Conventional configuration, if the Primary and Secondary servers are on separate nodes, and the xFlow Interface is deployed on the Secondary Server, then the Database Client (Microsoft SQL Native Client or Oracle Database Client) is required on the Secondary Server for a successful installation.
- CA SDM must be installed for deploying the xFlow Interface. In the conventional configuration, it's recommended to install the xFlow Interface server on a secondary server. Advanced Availability users can install the xFlow Interface only on an application server.

> **IMPORTANT**
> After upgrading to CA Service Management 17.4.3.0, Maileater does not work with POP3 on a non-TLS port with Clear Text as a security level. This is a limitation of the upgraded Javamail (Third-party jar) used in connection to the mail server. Users are required to use **LOGIN** as a security level.

(Optional) Backing up the CR_Stored_Queries Table

CA Service Management provides a fix to address a vulnerability related to SQL injection. As part of this fix, **Cr_Stored_Queries** table is updated. If you have made customizations or changes to the Out-of-the-box (OOTB) stored queries, perform the below steps to take a backup before applying the CA Service Management patch:

1. Open a command prompt window and navigate to *$NX_ROOT\patches.* and create a folder **Backup_Data**. Navigate to this folder and take the Backup of the Cr_Stored_Queries table by running the following command:
   ```
   pdm_extract Cr_Stored_Queries> Cr_Stored_Queries_17.4_backup.dat
   ```
2. Verify that the **Cr_Stored_Queries_17.4_backup.dat** file is successfully created.
3. Now, proceed with CA Service Management 17.4.3.0 installation.
4. After applying the patch, if required based on your customizations, you can update the database files created in **Step 1**. Navigate to $NX_ROOT\patches\Backup_Data location and run the following command:

   ```
   pdm_load -f Cr_Stored_Queries_17.4_backup.dat
   ```

Installation Prerequisites for CA Service Catalog

Verify and complete the following prerequisites before proceeding with CA Service Catalog 17.4.3.0 installation:

- Ensure that CA Service Catalog 17.4 is installed in your environment.
- Back up the system database (MDB). The system database (MDB) backup is mandatory and ensures safe data recovery if any problems are encountered during the installation process.

Installation Prerequisites for CA Asset Portfolio Management

Consider the following prerequisites:

WARNING
CA Asset Portfolio Management (CA APM) architecture is highly database-driven and has no rollback mechanism for database-related changes. Hence, the uninstallation of the CA APM patch is not recommended and is not supported. The only way to revert the database changes, if required, is by restoring the database backup which can lead to data loss.

Verify and complete the following prerequisites before proceeding with the CA Asset Portfolio Management installation:

- Ensure that CA Asset Portfolio Management 17.4 is installed in your environment.
- Back up the system database (MDB). The system database (MDB) backup is mandatory and ensures safe data recovery if any problems are encountered during the installation process.

Install CA Service Management 17.4.3.0 (Windows)

Installing CASM 17.4.3.0 on Windows

CA Service Management 17.4.3.0 installer performs prerequisite checks and mandatory validations before installing the patch.

NOTE

- We recommend that you place the installer in a **shorter directory path**, containing no spaces, to avoid errors when running the setup file. For example, unzip the file at: C:\WIN_17_4_3_0
- In CA Service Desk Manager (SDM) Availability mode, the xFlow Interface can only be installed on an Application Server. If you have converted CA SDM conventional configuration to advanced availability configuration, and you have a previously installed version of xFlow Interface, ensure that you uninstall xFlow Interface and then, reinstall xFlow Interface on an Application server.

Perform the following steps to install 17.4.3.0 on Windows:

1. From the Broadcom Support portal, download the **WIN_17_4_3_0.zip** file.
   NOTE
   CA Service Management 17.4.3.0 (Windows) installer is provided as a ZIP file and not as a CAZ file. Extract/unzip the content of the ZIP file to your local folder.
2. Copy **WIN_17_4_3_0.zip** on all servers in your environment.
3. Extract the contents of the ZIP file to the folder named WIN_17_4_3_0.
   NOTE
   In the extracted folder, CA EEM 12.6.7.0 installer is available in *CASM_17.4.3.0\filestore\EEM\Windows\12.6.7.0*
4. In CA SDM conventional configuration, stop CA SDM services in your environment. In the case of Advance Availability configuration, stop CA SDM services only on the server on which CA SDM will be installed.
5. Log in to the server as a local admin to install the patch.
6. Navigate to **WIN_17_4_3_0** folder. Right-click **Setup.exe** and select **run as administrator**.
7. Review and Accept the License Agreement Information.
8. Select **Microsoft SQL Server** or **Oracle** as the database type in the Database Configuration screen.
   For Windows Microsoft SQL Server database, *mdbadmin* is the default user.
9. Provide the database details:
   a. If you have selected **Microsoft SQL Server**, provide the following field information:
      - **Database Server**: Enter the hostname of the database server.
        If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
      - **Database Name**: Enter the Database name (MDB), the name of the target DBMS. The default value is mdb.
      - **Database Port**: Provide the port identifier for the target DBMS.
      - **Database Server Instance**: The database instance name. Leave this field blank if you are using the default instance.

If you are using the SQL named instance and a non-default port number, the **Database Server Instance** field should look like this: ***<database instance name>,<non-default port number>***
For example:

```
sdmcasmqa,10555
```

- **Database Admin User**: For Microsoft SQL Server, **sa** is the default value.
  The **sa** user is the admin user with permission to create a user and schema.
- **Database Admin Password**: Enter the database password of the user specified by a database admin user. No default value is assigned to this parameter.
- **mdbadmin Password**: Enter the password for the user specified for the MDB admin user.
- **Confirm mdbadmin Password**: Confirm the password.

b. If you have selected **Oracle** database, provide the following field information:
- **Database Server**: Enter the hostname of the Oracle database server.
  If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
- **Oracle Service Name**: Enter the Oracle Service name.
- **Listener Port**: Enter the listener port for the database.
- **Net Service Name**: Enter the Net Service Name of the Oracle database where the database resides.
  If the database is remote, use the Net Service Name that is defined in the Oracle client on the local computer. CA SDM has access to the database through a local installation of the Oracle client.
- **DBA User Name**: Provide the name of an Oracle user with DBA access. The default value for DBA user is **SYS**.
- **DBA User Password**: Enter the password for the DBA user.
- **mdbadmin Password**: Enter the password for MDB admin.
- **Confirm mdbadmin Password**: Confirm the password.
- **Tablespace Path (on DB Server)**: Provide the tablespace path on the database server.
- **Oracle Home Path**: Provide the Oracle Client 64-bit path. Provide Oracle client 32-bit path. For CA Asset Portfolio Management, provide the Oracle 32-bit client.
- **Data Tablespace Name**: Provide the name of the data tablespace. The default value is MDB_DATA.
- **Index Tablespace Name**: Provide the index Tablespace Name: The default value is MDB_INDEX.

Click **Next**.

10. On **Select the Products and Integrations** page, patch list is displayed. Select the required patch to install.

    WARNING

    CA Service Management does not support the uninstallation of the CA Asset Portfolio Management patch. Uninstalling the CA APM patch does not uninstall the MDB patch, as there is no rollback mechanism for the same. For more information, see Installation Prerequisites.

a. **CA Service Desk Manager**
- CA Service Desk Manager Patch
- xFlow Interface Patch
- Search Server Patch: Search server patch must be installed on a Server where an earlier version from previous releases is already installed.

b. **CA Service Catalog**
- CA Service Catalog Patch

c. **CA Asset Portfolio Management**
- CA Asset Portfolio Management Patch

Click **Next**.

11. Review the Pre-Installation Configuration page. Click **Next**.

NOTE

Select the option to perform the automated post-install steps for CA SDM. If this option is not selected, you must manually perform the post-install steps for CA SDM. Note that for CA Service Catalog and CA Asset Portfolio Management, you need to perform the manual post-install steps.

12. Click **Install** to proceed with the installation. Review the Installation progress and when the installation is complete, click **Next.**
13. To review the installation logs, navigate to the path shown in the installation summary screen.
14. Click **Finish** to exit the installer
15. After installing the Key definition for "xflow" not found in the DITA map., start the **CA Key definition for "xflow" not found in the DITA map.** service from Windows Services (**Control Panel, Administrative Tools, Services**) as this service is not started by default.
16. After installing the search server, if the **Search Server** service is not started by default or if it stops, start the services from **Control Panel, Administrative Tools, Services.**
17. Validate the Installation.
18. Complete the Post-Installation Tasks.

After installing the Key definition for "xflow" not found in the DITA map. and Search Server, you can configure the cache size for Key definition for "xflow" not found in the DITA map.. For more information about how to configure cache size, see Perform initial load and refresh index on the search server.

Install CA Service Management 17.4.3.0 (Linux)

Installing CASM 17.4.3.0 on Linux

NOTE
In CA SDM Advanced Availability mode, the xFlow Interface can only be installed on an Application Server. If you have converted CA SDM conventional configuration to advanced availability configuration, and you have a previously installed version of xFlow Interface, ensure that you uninstall xFlow Interface and then reinstall xFlow Interface on an Application server.

CA Service Management 17.4.3.0 installer performs prerequisite checks and mandatory validations before installing the patch.

IMPORTANT
We recommend that you place the installer in a **shorter directory path**, containing no spaces, to avoid errors when running the setup file. For example, unzip the file at: *<root>\casm_dvd.x86Linux\<installer_file>*

Perform the following steps to install 17.4.3.0 on Linux:

1. From the Broadcom Support portal, download the **LNX_17_4_3_0.tar.gz** file.
2. To extract the LNX_17_4_3_0.tar.gz, run the following command:
   ```
   tar -xvzf LNX_17_4_3_0.tar.gz
   ```
   NOTE
   In the extracted folder, CA EEM 12.6.7.0 installer is available in CASM_17.4.3.0\filestore\EEM\Linux\12.6.7.0
3. Set CA SDM bin and library environment variables.
   For example:
   ```
   export PATH=/opt/CA/ServiceDeskManager/bin:$PATH
   export LD_LIBRARY_PATH=/opt/CA/ServiceDeskManager/lib:$LD_LIBRARY_PATH
   ```
4. In SDM conventional configuration, stop CA SDM services in your environment. In the case of Advance Availability, stop CA SDM services only on the server on which CA SDM will be installed.
5. Run the setup file from the casm_dvd.x86Linux folder to install the application(s) using the following command:
   ```
   ./setup.sh
   ```
6. Review and accept the license agreement terms.

7. In Database Configuration, select **Oracle** as the database type. Ensure that you have installed the Oracle database client in your system.
   Provide the following Oracle database configuration details:
   - **Database Server**: Enter the hostname of the Oracle database server.
     If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
   - **Oracle Service Name**: Enter the Oracle Service name.
   - **Listener Port**: Enter the listener port for the database.
   - **Net Service Name**: Enter the Net Service Name of the Oracle database where the database resides.
     If the database is remote, use the Net Service Name that is defined in the Oracle client on the local computer. CA SDM has access to the database through a local installation of the Oracle client.
   - **DBA User Name**: Authenticates the name of the user with DBA access. The default value for DBA user is **SYS.**
   - **DBA User Password**: Authenticates the password for DBA user.
   - **mdbadmin Password**: Specify the password for the mdbadmin user.
   - **Confirm mdbadmin Password**: Confirm the mdbadmin user password.
   - **Tablespace Path (on DB Server)**: Enter the tablespace directory on the database server.
   - **Oracle Home Path**: Provide the Oracle Client 64-bit path. Provide Oracle client 32-bit path for fresh installation
   - **Data Tablespace Name**: Enter the tablespace name. The default value is **MDB_DATA**.
   - **Index Tablespace Name**: Enter the index tablespace name: The default value is **MDB_INDEX**.
8. Click **Next.**
9. On **Select the Products and Integrations** page, patch list is displayed. Select the required patch to install.
   a. **CA Service Desk Manager**
      - CA Service Desk Manager Patch
      - xFlow Interface Patch
      - Search Server Patch: A search server patch must be installed on a server where an earlier version from previous releases is already installed.
10. Click **Next**.
11. Review the Pre-Installation Summary page. Click **Next.**
12. Click **Install** to proceed with the installation. Review the Installation progress and when the installation is complete, click **Next.**
13. To review the installation logs, navigate to the path shown in the installation summary screen.
14. Click **Finish** to exit the installer.
    If patch installation fails for a product, retry the installation by using the **Retry** option in the wizard where it fails. If retry does not work, exit the installer and relaunch it. If the problem persists, contact Broadcom Support.
15. After installing XFlow Interface, start the**CA XFlow Interface**service. Navigate to /opt/CA/xFlow/Apps and run the following command from a command prompt window:
    ```
    ./xFlowServer start
    ```
    To stop the server, run the following command:
    ```
    ./xFlowServer Stop.sh
    ```
16. Navigate to /opt/CA/SC/SearchServer/elasticsearch/bin and run the following command from the command prompt window:
    ```
    ./ServicesStart.sh
    ```
    NOTE
    If you get the below message while restarting the search server service, ignore it and continue:
    ```
    groupadd: group 'elasticsearch' already exists
    useradd: user 'elasticuser' already exists
    runuser: warning: cannot change directory to /home/elasticuser: No such file or directory
    OpenJDK 64-Bit Server VM warning: Option UseConcMarkSweepGC was deprecated in version 9.0 and
     will likely be removed in a future release.
    ```

To stop the server, run the following command:

```
/ServicesStop.sh
```

17. Validate the Installation.
18. Complete the Post-Installation Tasks.

After installing the XFlow Interface and Search Server, you can configure the cache size for XFlow Interface. For more information about how to configure cache size, see Perform initial load and refresh index on the search server.

Validate CA Service Management 17.4.3.0 Installation

Validate CASM 17.4.3.0 installation

To verify the successful installation of CA Service Management 17.4.3.0, validate the installation for the corresponding applications in the following location. You can also navigate to the root folder where you have locally installed the application and select the "env.info" file for validating your installation:

- **CA ITAM:**

  NOTE
  CORA version: 17.4.0.57

  – C:\Program Files (x86)\CA\ITAM\env.info
  – product=ITAM
  – major=17
  – minor=4
  – cum=3
  – roll-up=0

  NOTE
  CA CASM uses Apache Tomcat version 9.0.86.
- **CA Service Catalog:**
  – %USM_HOME%*CA_SLCM_<hostname>.his* file
  – %USM_HOME%\view\webapps\usm\locale\about_info.xml
  – %USM_HOME%\build.xml
  – %USM_HOME%\env.info
- **xFlow Interface:**
  – *C:\Program Files\CA\xFlow\<hostname>.his* file
  – NX_ROOT\env.info
  – NX_ROOT\version.xml
- **CA SDM:**
  – *C:\Program Files (x86)\CA\Service Desk Manager\<hostname>.his* file
  – NX_ROOT\pdmconf\version
  – NX_ROOT\env.info
  – HELP, ABOUT

  NOTE
  For CA SDM, the Binary and Language Combo patches are installed irrespective of the environment. The language patch for the corresponding CA SDM environment is installed.

  To validate the successful installation of the patches, follow these steps:

  1. Open the <sdm_installation>\<hostname>.his file in a text editor.

2. Search for the instances of "gol-230". In a successful install, you should come across these three entries (Binary, Language Combo, and corresponding language patch).

Application	Windows	Linux
CA Asset Portfolio Management	T6EE1743	NA
CA Service Catalog	T6D91743	NA
CA Service Desk Manager - Binary	gol-230_cum_C	gol-230_x86_64Linux_cum_C
CA Service Desk Manager - Languag504e Comb	gol-230_langcombo	gol-230_langcombo
CA Service Desk Manager - English	gol-230_en_US	gol-230_x86_64Linux_en_US
CA Service Desk Manager - German	gol-230_de_DE	gol-230_x86_64Linux_en_DE
CA Service Desk Manager - Spanish	gol-230_es_ES	NA
CA Service Desk Manager - French	gol-230_fr_FR	gol-230_x86_64Linux_fr_FR
CA Service Desk Manager - Italian	gol-230_it_IT	NA
CA Service Desk Manager - Japanese	gol-230_ja_JP	gol-230_x86_64Linux_ja_JP
CA Service Desk Manager - Brazilian Portuguese	gol-230_pt_BR	NA

Post-Installation Tasks for CA Service Management 17.4.3.0

After successful installation of CA Service Management 17.4.3.0, perform the following post-installation steps.

CA Service Desk Manager

Manual Steps required for Automated Post-Installation

CA Service Management provides an option in the installer to automate the post-install tasks.
If you have selected **Yes** to automate the post-installation steps while running the CA Service Management Installer Wizard, you are not required to perform any manual post-install steps for CA Service Desk Manager (SDM).

If you have selected **No** for automating the post-install steps while running the CA Service Management Installer wizard, perform the below manual post-install tasks as per your deployment requirements:

Manual Post-Install Tasks After Installing CA Service Desk Manager (SDM) 17.4.3.0 on CA SDM 17.4.2.0
Perform the following steps:

1. Stop CA SDM services.
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0.
3. Open a command prompt window and run the following command:
   ```
   pdm_configure
   ```
4. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_03_insert.dat
   pdm_load -u -f r17_4_03_update.dat
   ```
5. Notification message template is modified to update the existing email templates with the html_description field.
   Tool location: NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate This has to be executed one time when the SDM service is up and running. **For e.g:**
   Windows
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.bat
   ```
   Linux

```
NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.sh
```

Manual Post-Install Tasks After Installing CA Service Desk Manager (SDM) 17.4.3.0 on CA SDM 17.4 or CA SDM 17.4.1.0

Perform the following steps:

1. Stop CA SDM services.
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0.
3. Open a command prompt window and run the following command:
   ```
   pdm_configure
   ```
4. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_02_insert.dat
   pdm_load -u -f r17_4_02_update.dat
   pdm_load -f r17_4_03_insert.dat
   pdm_load -u -f r17_4_03_update.dat
   ```
5. Notification message template is modified to update the existing email templates with the html_description field.
   Tool location: NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate This has to be executed one time when the SDM service is up and running. **For e.g:**
   Windows
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.bat
   ```
   Linux
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.sh
   ```

Manual Post-Installation Task for CMDB

> **NOTE**
> For Oracle 21c, set the **isClusterEnv** setting to "true" in the application.conf file of CMDB viz and restart the CA SDM Services.:
>
> - For Windows: <SDM installed location>\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\WEB-INF\classes\application.conf)
> - For LINUX: <SDM installed location>/bopcfg/www/CATALINA_BASE_VIZ/webapps/CMDBVisualizer/WEB-INF/classes/application.conf).

If you are applying CA Service Management 17.4.3.0 in an SSL environment, CA SDM Administrators must perform the following post-installation steps for CMDB Visualizer Interface to open using the **HTTPS** protocol when launching from CA SDM UI:

1. In the NX.env file, change the **NX_CMDB_VISUALIZER** entry to provide **HTTPS protocol** and **HTTPS port**.
   For example: @NX_CMDB_VISUALIZER=https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer
2. In **casm.conf.js** file from NX_root\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\conf, provide the HTTPS as protocol and HTTP port.
   For example:

```
sdm: {

      vizserver: 'https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer/viz',
      server: 'https:// sdmurl:<SDM_SSL_PORT>/CAisd/pdmweb.exe'
    }
```

Manual Post-Installation Task for OAuth 2.0 (Maileater/ Mailer)

After generating an access token for OAuth 2.0 to connect to a mail server, perform the mandatory post-install tasks on all servers in your environment:

1. Create a backup of web.xml from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF in your local directory to perform backout if required.
2. Open the web.xml file from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF.
3. Search for the following statement:
   ```
   <!--USD servlet mappings start-->
   ```
4. Copy and paste the following configuration under the statement:

   ```
   <servlet>
     <servlet-name>OAuthProcessor</servlet-name>
     <servlet-class>com.broadcom.oAuth.OAuthProcessor</servlet-class>
    </servlet>
   ```
5. Copy and paste the following configuration after any <servlet-mapping>:

   ```
    <servlet-mapping>
      <servlet-name>OAuthProcessor</servlet-name>
      <url-pattern>/OAuthProcessor</url-pattern>
    </servlet-mapping>
   ```
6. Save and close the web.xml file.
7. For OAuth servlet preservation in a primary and secondary environment, the secondary server must also have the OAuth processor entries in the following file: *<NX_ROOT>\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF\web.xml.last*.
8. Repeat Steps 1-8 to update **web.xml.tpl** file in the NX_ROOT\samples\pdmconf\web.xml.tpl folder to avoid losing these changes when pdm_configure is run at a later point of time.
9. Run the following command:
   ```
   pdm_tomcat_nxd -c STOP command
   ```
10. Wait for some time (a minute or so) and run the following command:
    ```
    pdm_tomcat_nxd -c START command
    ```

For more information, see Define a mailbox.

Manual Post-Installation Task for Excel file (.xlsx) export*

1. Execute the following command:
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst
   ```
2. Execute the following command to add the new NX variable to NX.env.tpl file.
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst -t
   ```
3. Restart CA SDM services.

HTML Tags Sanitization

The sanitization tool can be run any number of times. Administrators can choose if they wish to run the tool for each ticket type separately or for all ticket types at one go.

Before running the Sanitization tool ensure you take a backup of all the ticket objects using the following commands or by taking a backup of the sdm databse :

```
"Call_Req": pdm_extract Call_Req > cr_backup.txt
"Issue": pdm_extract Issue > Issue_backup.txt
"Change Orders": pdm_extract Change_Request >change_orders_backup.txt.
```

Both active and archived tickets can be sanitized.
For Windows:

```
NX_ROOT\samples\sdk\SanitizationTool\sanitizeTool.bat
```

For Linux:

```
NX_ROOT\samples\sdk\SanitizationTool\sanitizeTool.sh
```

NOTE
We recommend executing the sanitization during slack time on any Service Desk Manager server.

After executing the sanitization tool, execute the following command to update domsrvr memory cache:

```
pdm_cache_refresh -t Call_Req
pdm_cache_refresh -t Issue
pdm_cache_refresh -t Change_Request
```

(Optional) Enable HTML content for Ticket Description in the Activity log

NOTE

- A new Field Update activity association (**HTML Description** has been introduced)
- Either **HTML Description** or **Description** activity association has to be enbaled. Do not enable both the options.
- Out of the box Description Field Update activity association is enabled in the activity log for all ticket types (Incident,Request,Problem,ChangeOrder and Issue).

Enabling HTML content for Description in the Activity Log:

1. Login into CA SDM with Administrator access type.
2. Administration --> Notifications --> Activity Associations.
3. Click **Show Filter**.
4. Enter "HTML Description" in the Symbol text box in the Activity Association List page.
5. Click **Search**
6. For every **HTML Description Field Update association**, click **Edit** and check the **Log Me** checkbox, then save the changes made.

Disabling plain text for Description in the Activity log:

1. Click **Show Filter** in the Activity Association List page.
2. Enter "Description" in the Symbol text box and Select Object Type as "Request/Incident/Problem" and click **Search**.
3. Now see the one row and click the Description link in the Activity Association List page.
4. Unselect the **Log Me** check box and save the page.
5. For Change Order and Issue repeat the steps from 1 to 4.

(Optional) Manual Post-Installation Task for Knowledge Document export with German fields

If you are exporting Knowledge Document with German Fields (included in the export field list), then perform the below post-install steps for successful export of knowledge documents:

1. Take the backup of NX_ROOT\NX.env file (NX_ROOT is the installation directory for CA Service Desk Manager).
2. Open NX_ROOT\pdmconf\NX.env_nt.tpl file using any text editor and take the value of NX variable @NX_KEIT_AVAILABLE_FIELDS which is COPIED_DATA and update Nx.env in NX_ROOT by executing below command:

   ```
   pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA>  -a  pdm_option.inst
   ```

   Where COPIED_DATA is the value copied from NX.env_nt.tpl (step 2).
   For example, " pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <provide the exact string here> -a pdm_option.inst "

Note to avoid losing the updated changes when you run pdm_configure, execute the above command with the '-t' flag as follows:

```
pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA> -a pdm_option.inst -t
```

(Optional) Handle HTML Mails that contains Tables in MailEater

Java based Maileater already has the ability to process the emails that were sent in plain format or HTML format, to support the processing of emails that contain Tables in the HTML body. But the format in which the email was sent is lost when received by Maileater.

1. Run the following commands from NX_ROOT/bin from the command prompt/ terminal:

```
pdm_options_mgr -c -s NX_ENFORCE_HTML_EMAIL_PARSING -v "Yes" -a pdm_option.inst
pdm_options_mgr -c -s NX_ENFORCE_HTML_EMAIL_PARSING -v "Yes" -a pdm_option.inst -t
```

2. Restart CA SDM and XFlow applications.

(Optional) Comments are not allowed when a Data Partition (O_COMMENTS) is created with Customer, Employee and ServiceDesk Analyst

To disable comments:

1. Login to SDM and navigate to Administration -> Security and Role Management-> Data partition-> Data Partition List.
2. In Data Partition List, create a constraint (O_COMMENTS with the condition id=0) and link it to Customer, Employee and ServiceDesk Analyst data partitions.
3. Run the following commands from NX_ROOT/bin from the command prompt/ terminal:

```
pdm_options_mgr -c -s KT_COMMENT_USE_SUPERUSER -v 0 -a pdm_option.inst
pdm_options_mgr -c -s KT_COMMENT_USE_SUPERUSER -v 0 -a pdm_option.inst -t
```

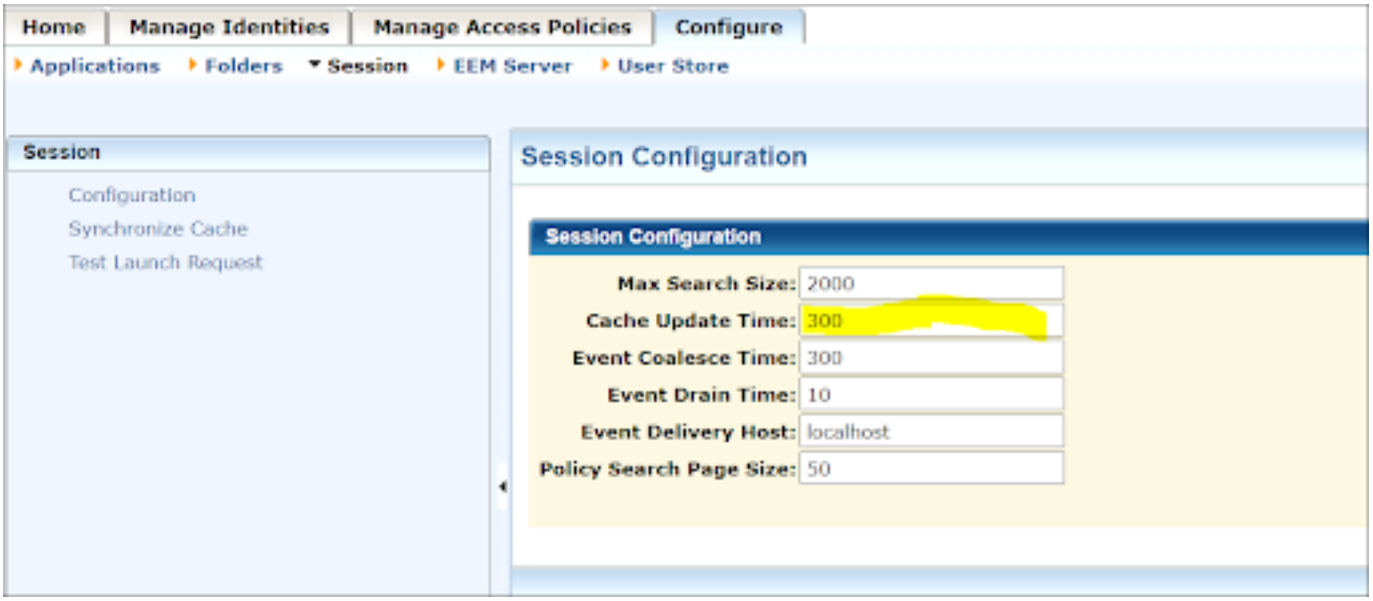
4. Restart CA SDM and XFlow applications.

(Optional) Manual Post-Installation Task for EEM Cache Timeout Configuration Update

You may face issues while logging in to EEM or SDM using the EEM user accounts due to a session cache timeout mismatch between EEM and SDM.

You can update the EEM cache timeout configuration by adding a new NX variable, "NX_EEM_CACHE_UPDATE_TIME" in SDM.

Keep the value the same as the EEM cache timeout configuration value mentioned in the following screenshot.



To set the EEM cache update time to 300 seconds, perform the following steps:

To add the variable in the **nx.env** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst
```

To add the variable in the **nxenv_nt.tpl** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst -t
```

(Optional)casc_impersnate_user

Once both CA SDM and CA Service Catalog are integrated, do the following:

1. Login to CA SDM with the Administrator user type.
2. Navigate to Administration >> Options Manager >> CA Service Catalog >> in the Options List, install casc_impersnate_user
3. Restart the SDM service.

Service Point

After successful installation of Service Point, refresh index and data on the Search Server.

> **NOTE**
> In case of Linux Search Server environment, ensure to run the following commands on the terminal:
>
> ```
> sysctl -w vm.max_map_count=262144
> ulimit -Hn 65535
> ```

(Optional) Manual Post-Installation Task for Rebuild Index

If you have upgraded or are planning to install CA EEM 12.6 with Service Point and CA Service Catalog, ensure that you rebuild the CA Service Catalog index after completing the CA EEM upgrade/installation process. For more information, see Perform Initial Load and Refresh Index Data on Search Server.
Run the following command to rebuild the index:

```
$ES_HOME/search/bin/pdm_es_rebuild_index.bat index=catalog_index
```

CA Service Catalog

Perform the following CA Service Catalog post-installation steps as per your environment requirements:

- Post-Installation Tasks after installing CA Service Catalog 17.4.3.0 on CA Service Catalog 17.4 or later
- gMSA Authentication

Post-Install Tasks After Installing CA Service Catalog 17.4.3.0 on CA Service Catalog 17.4 or later

Review and perform the post-installation steps based on your deployment requirements.

> **NOTE**
> If any SSL certificate(s) chain was imported to the Oracle Java's JAVA_HOME\lib\security\cacerts Keystore, those certificate(s) chain again need to be exported explicitly and imported properly into the new AdoptOpenJDK's %USM_HOME%\embedded\jdk\lib\security\cacerts Keystore.

> **NOTE**
> While upgrading from 17.4.2.0 to 17.4.3.0, re-run the removal_axis.bat file, if already ran it on 17.4.2.0.

Follow these steps

- (Optional) If Service Catalog is accessed from applications like Service Point, SharePoint and so on, then you must perform this step. Open the "**web.xml**" file located in *USM_HOME%\view\webapps\usm\WEB-INF* and update the

respective parameter values for the parameter name "cors.allowed.origins" and "cors.support.credentials". For example, if you are using Catalog Widgets in Service Point

- (Optional) Unable to approve any workflow requests through email. The email sent for approvals is corrupted with special characters. Follow the KB article steps.

(Optional) Approve workflow requests through email

The workflow email sent for approvals is corrupted with special characters. To approve any workflow requests through email, follow the KB article steps.

gMSA Authentication Post-Install Steps for CA Service Catalog (Automated steps)

CA Service Catalog provides automated post-installation steps for gMSA Authentication. Review the following before you proceed:

- Prerequisites and Considerations
- How to Automate gMSA Post-Installation Steps
- Unconfigure gMSA

Prerequisites and Considerations:

Ensure that you have taken care of the below considerations on all nodes of CA Service Catalog where you want to automate the gMSA post-installation procedure.

1. If the accounting service is available, perform the automated post-install steps by updating serviceName to ServiceAccounting and providing the appropriate gMSA account name in the **gmsa.properties** file.
2. If the filestore path is external to the server from where the **GMSAPostInstall.bat** is running, ensure to provide appropriate access to the server for setting the gMSA account permissions.
3. While preparing the properties file, check for any copy/paste issues, special characters, spaces, and so on.

How to Automate gMSA Post-Installation Steps

Perform the following steps to enable the batch file **GMSAPostInstall.bat** and to automate the gMSA post-installation steps:

1. Create a gMSA account for CA Service Catalog.
2. Ensure that appropriate DB (database) permission is given to the newly created account(s).
3. Enter the below values in the **gmsa.properties** file and save it.

   NOTE
   A template file (**gmsa.properties**) is added in the common installer in *"\filestore\SLCM\gMSAProperties"* . Optionally. use this file as a template file to fill gmsa properties. This can be copied to "USM_HOME/scripts/GMSA" where Catalog is installed.

   NOTE
   All values are mandatory. Make sure that there are no leading or trialing spaces while providing these values.

   a. **serviceName**: Application service name to enable GMSA. In the case of Catalog, it will be ServiceCatalog and for accounting, it will be ServiceAccounting.
   b. **folderName**: Absolute path of the application installation directory.
   c. **gMSAName**: Name of the GMSA service account (domain\accountName$).
   d. **filestorePath**: Shared network path (\\Server_name\filestore) or local filestore path
4. Navigate to %USM_HOME% and launch **usm.cmd**
5. Navigate to %USM_HOME%/scripts/GMSA/ folder and run the "GMSAPostInstall.bat" in command prompt.

   NOTE
   If the command executes without any error, "Post-installation steps for GMSA executed successfully" message is displayed with exit code 0.

If there are errors, the error message is shown with a non-zero exit code. Correct the error to continue and perform step **5** again.

Unconfigure gMSA for CA Service Catalog

To disable or unconfigure gMSA for CA Service Catalog, follow the steps below:

1. Replace **config.properties** file from*%USM_HOME%*\ folder.
2. Revert back all the folder permissions which were assigned to domain users.
3. Start Catalog and Accounting Services with the local system account.

NOTE
In the case of load balancers, repeat steps 1-3 to unconfigure gMSA on all Catalog servers.

CA Asset Portfolio Management

Manual Post-Installation Task for Removing jQuery, log4net, and log4j Vulnerabilities

After installing 17.4.3.0, delete jquery-1.3.2.min.js, jquery-ui-1.7.2.custom.min.js , log4net.dll, and log4j from their respective installed folders.

NOTE
Perform the below steps if you have not done this earlier.

Perform the following steps:

1. Log on to CA APM Web Server, navigate to the <ITAM_install_dir>\ITAM\Web Server\castylesr5\javascript.
   Example: C:\Program Files (x86)\CA\ITAM\Web Server\castylesr5\javascript
   Locate and delete the jquery-1.3.2.min.js and jquery-ui-1.7.2.custom.min.js files from the directory.
2. Navigate to the ITAM_install_dir\CA\ITAM\Data Importer.
   Locate and delete the log4net.dll file from the directory.
3. Navigate to the C:\Program Files (x86)\CA\SC\CORA.
   Locate and delete the log4j-1.2.15.jar, log4j.properties, and ojdbc6.jar files from the directory.
4. Navigate to the ITAM_install_dir\CA\ITAM.
   Backup the following folders and manually delete them:
   - CDB
   - MDB
   - jre8

   NOTE
   jre8 folder has to be deleted manually for every RU installation.
5. Perform an IISRESET and delete the browser's cache to load the upgraded files.

For ITAM installation on French Operating System

Following the successful installation of version 17.4.3.0 on a French Windows operating system with the browser set to French, we are unable to log in to ITAM.

An error message is displayed: ***Une valeur Request.Form potentiellement dangereuse a été détectée à partir du client (_UPLOAD_NON_TXT_CSV_DATA_FILE_="...ectionné n'est pas de type...").***

To overcome this, follow these steps:

1. Navigate to *C:\Program Files (x86)\CA\ITAM\Web Server, edit web.config*, and search for *<pages enableEventValidation="false" viewStateEncryptionMode="Never" controlRenderingCompatibilityVersion="4.0" clientIDMode="AutoID">* , then add *validateRequest="false"* to it.
2. After adding, it looks like this:
   <pages enableEventValidation="false" viewStateEncryptionMode="Never" controlRenderingCompatibilityVersion="4.0" clientIDMode="AutoID" validateRequest="false">

3. Do an IISRESET.
4. You can now successfully login to ITAM.

Post-Installation Tasks - CA SDM Advance Availability 17.4.3.0

The installer wizard provides an option in the installer to automate the post-install tasks. Perform the following post-installation tasks for CA SDM Advanced Availability Configuration. For the Advanced Availability configuration, the following host names are considered as an example:

- The background server hostname is sdmcahost1.
- The standby server hostname is sdmcahost2.
- The application server hostname is sdmcahost3.

Manual Steps required for Automated Post-Installation

If you have selected **Yes** for automating the post-install steps while running the installer wizard, then perform manually the following post-install tasks:

Manual Post-Install Tasks - After Installing CA Service Desk Manager (SDM) 17.4.3.0 on CA SDM 17.4.2.0

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Start the CA SDM service on the standby server (sdmcahost2). Open the command prompt and run the following command:
   ```
   pdm_server_control -b
   ```
3. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server.
4. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```
5. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_03_insert.dat
   pdm_load -u -f r17_4_03_update.dat
   ```
6. Clear the browser cache by the following command:
   ```
   pdm_webcache -H
   ```
7. Ensure that the patch is applied on all the application servers and the remaining standby servers if any. If you selected YES to automate the post-install steps, then there are no more manual post-install steps to be performed.

Manual Post-Install Tasks - After Installing CA Service Desk Manager (SDM) 17.4.3.0 on CA SDM 17.4 GA or CA SDM 17.4.1.0

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Start the CA SDM service on the standby server (sdmcahost2). Open the command prompt and run the following command:
   ```
   pdm_server_control -b
   ```
3. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server.
4. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```
5. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_02_insert.dat
   pdm_load -u -f r17_4_02_update.dat
   pdm_load -f r17_4_03_insert.dat
   pdm_load -u -f r17_4_03_update.dat
   ```
6. Clear the browser cache by the following command:

```
pdm_webcache -H
```

7. Ensure that the patch is applied on all the application servers and the remaining standby servers if any. If you selected YES to automate the post-install steps, then there are no more manual post-install steps to be performed.

Manual Steps required for Non-Automated Post-Installation

If you have selected **No** for automating the post-install steps while running the CA Service Management installer wizard, perform manually the appropriate post-install after this task as per your deployment requirements.

Post-Install Tasks After Installing CA SDM 17.4.3.0 on CA SDM 17.4.2.0 for Advanced Availability Configuration

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0. Open a command prompt and run the following command:
   ```
   pdm_configure
   ```
   On the installer configuration page, before clicking Finish, uncheck the Start service when completed checkbox. Failure to uncheck the box may result in the loss of custom schema changes and configurations that were changed on the server.
3. Promote the standby server as the new background server by performing the following steps. To suppress version control on the standby server and later on the background server, run the following command on each server:
   ```
   pdm_server_control -v
   ```
4. Start SDM service on the standby server. Run the following command:
   ```
   pdm_server_control -b
   ```
5. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server, and perform the following steps. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
   ```
   pdm_configure
   ```
6. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```
7. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_03_insert.dat
   pdm_load -u -f r17_4_03_update.dat
   ```
8. Ensure that the patch is applied on all application servers and the remaining standby server if any. Perform the following steps on all application servers and standby servers.
9. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
   ```
   pdm_configure
   ```
10. Clear the browser cache by running the following command:
    ```
    pdm_webcache -H
    ```

Post-Install Tasks After Installing CA SDM 17.4.3.0 on CA SDM 17.4 GA or CA SDM 17.4.1.0 for Advanced Availability Configuration

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0. Open a command prompt and run the following command:
   ```
   pdm_configure
   ```
   On the installer configuration page, before clicking Finish, uncheck the Start service when completed checkbox. Failure to uncheck the box may result in the loss of custom schema changes and configurations that were changed on the server.

3. Promote the standby server as the new background server by performing the following steps. To suppress version control on the standby server and later on the background server, run the following command on each server:
   ```
   pdm_server_control -v
   ```
4. Start SDM service on the standby server. Run the following command:
   ```
   pdm_server_control -b
   ```
5. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server, and perform the following steps. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
   ```
   pdm_configure
   ```
6. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```
7. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_02_insert.dat
   pdm_load -u -f r17_4_02_update.dat
   pdm_load -f r17_4_03_insert.dat
   pdm_load -u -f r17_4_03_update.dat
   ```
8. Notification message template is modified to update the existing email templates with the html_description field. **Tool location**: NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate This has to be executed one time when the SDM service is up and running. **For e.g:**
   Windows
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.bat
   ```
   Linux
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.sh
   ```
9. Ensure that the patch is applied on all application servers and the remaining standby server if any. Perform the following steps on all application servers and standby servers.
10. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
    ```
    pdm_configure
    ```
11. Clear the browser cache by running the following command:
    ```
    pdm_webcache -H
    ```
12. Execute the following command.
    ```
    pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst
    ```
13. Execute the following command to add the new NX variable to NX.env.tpl file.
    ```
    pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst -t
    ```
14. Restart CA SDM Services.

Uninstall CA Service Management 17.4.3.0

Review the following topics to uninstall CA Service Management 17.4.3.0:

- Uninstall CA SDM 17.4.3.0 (Windows)
- Uninstall CA SDM 17.4.3.0 (Linux)
- Restore Visualizer Files (Windows)
- Restore Visualizer Files (Linux)
- Uninstall the XFlow Interface 17.4.3.0 (Windows)
- Uninstall the XFlow Interface 17.4.3.0 (Linux)
- Uninstall CA Service Catalog 17.4.3.0
- Uninstall CA Asset Portfolio Management
- Run the Database Cleanup Utility for CA Service Management 17.4.3.0

NOTE

Uninstalling CA Service Management 17.4.3.0 will revert to CA Service Management 17.4 version on which 17.4.3.0 is installed.

Example: If CA Service Management 17.4.3.0 is installed on 17.4, uninstalling 17.4.3.0 will revert to CA Service Management 17.4.

Uninstall CA SDM 17.4.3.0 (Windows)

Perform the following steps to uninstall CA SDM 17.4.3.0 on Windows:

1. Log in to CA SDM as an Administrator.
2. Shut down CA Service Desk Manager Server.
3. Navigate to **<NX_ROOT>\bin** and run the following command to backup the unsupported jars from the older versions.
   ```
   pdm_perl pdm_restore_files.pl gol-230_tool ..\PATCHES\DELIVERABLES_REMOVAL.TXT
   ```
4. Navigate to the location *where you have extracted the CA Service Management Installer zip package*.
5. Locate *CASM-17.4.3.0-CommonPatchInstaller-Win\filestore\utils\ApplyPTF* folder.
6. Run the **APPLYPTF.exe**.
7. Select **Backout PTF** from the local or remote node option.
8. Click **Next**.
9. Enter the patch number for the language that is installed in your environment:
   - **Binary**: gol-230_cum_C
   - **Language Master**: gol-230_langcombo
   - **English**: gol-230_en_US
   - **French**: gol-230_fr_FR
   - **German**: gol-230_de_DE
   - **Spanish**: gol-230_es_ES
   - **Italian:** gol-230_it_IT
   - **Japanese**: gol-230_ja_JP
   - **Brazilian Portuguese**: gol-230_pt_BR
10. Click **Next** to continue backing out the specific patch.
11. Repeat steps 6-9, to back out the language English (gol-230_en_US) patch.
12. Repeat steps 6-9 to back out the language master (gol-230-langcombo) patch.
13. Repeat steps 6-9 to back out the binary (gol-230_cum_C) patch.
14. Reboot the server.
15. Repeat steps **6-12** on all standby, application, or secondary servers in your environment.
16. Restore the MDB database to the database backup that you took before applying the patch.
17. Take the backup of the 5.40.0 folder and then delete the folder from <NX_ROOT>\bin\perllib.
18. Copy the 5.36.0 folder from <NX_ROOT>\REPLACED\perl_old and place the folder to <NX_ROOT>\bin\perllib.

19. Navigate to ***<NX_ROOT>\site\config.properties*** file and ensure that the value of ***resources.created*** parameter is set to **0.** This must display as ***resources.created=0*** in the file.
20. Open a command prompt and run the following command:
    ```
    pdm_configure
    ```
21. If you roll-back from 17.4.3.0 to the base version, copyright text year will not change. To change the year:
    a. Add @NX_COPYRIGHT_YEAR=2023 into NX.env file located under CA SDM install directory (NX_ROOT).
    b. To preserve the value even after pdm_configure, update the value of "genfact.gx_copyright_year in install.properties under NX_ROOT\site genfact.gx_copyright_year=2023

Uninstall CA SDM 17.4.3.0 (Linux)

Perform the following steps to uninstall CA SDM 17.4.3.0 on Linux:

1. Login to CA SDM as the root user.
2. Shut down the CA Service Desk Manager Server.
3. Navigate to **<NX_ROOT>/bin** and run the following command to backup the unsupported jars from the older versions.
   ```
   ./pdm_perl pdm_restore_files.pl ../replaced/gol-230_tool ../patches/deliverables_removal.txt
   ```
4. Navigate to the location where you have extracted the CA SDM Installer **tar.gz** package.
5. Navigate to the *casm_dvd.x86Linux \filestore\utils\ApplyPTF* folder.
6. The following binary/language-independent and language-dependent patches are supported:
   - Binary name: gol-230_x86_64Linux_cum_C
   - Language Master: gol-230_langcombo
   - English: gol-230_x86_64Linux_en_US
   - German: gol-230_x86_64Linux_de_DE
   - French: gol-230_x86_64Linux_fr_FR
   - Japanese: gol-230_x86_64Linux_ja_JP
7. Execute the following command to remove the binaries or the language-dependent patches. Replace the <patch_name> with the binary or language-dependent patch name:
   ```
   ./applyptf.<platform> -b <patch_name>  -e <NX_ROOT>
   ```
   Here, <platform> specifies the operating system and <NX_ROOT> specifies the CA SDM installed location.
   For example:
   ```
   ./applyptf.linux -b gol-230_x86_64Linux_cum_C  -e /opt/CA/ServiceDeskManager
   ```
8. Repeat step 6 to back out the language English (gol-230_x86_64Linux_en_US) patch.
9. Repeat step 6 to back out the binary (gol-230_x86_64Linux_cum_C) patch.

   NOTE
   During uninstallation, if you get any errors, ensure that the following steps are performed:

   1. Copy the subfolders from <NX_ROOT>/replaced/<gol-230_x86_64Linux_cum_C> and place it into the <NX_ROOT> folder.
   2. Copy the subfolders from <NX_ROOT>/replaced/<gol-230_x86_64Linux_en_US> and place it into the <NX_ROOT> folder.
   3. Open **.his** file under the <NX_ROOT> folder and delete the entries related to the current patch and save it.
10. To restore MDB database, revert to database backup that you took before applying the patch.
11. Take the backup of the 5.40.0 folder and then delete the folder from <NX_ROOT>\bin\perllib.
12. Copy the 5.36.0 folder from <NX_ROOT>\REPLACED\perl_old and place the folder to <NX_ROOT>\bin\perllib.
13. Navigate to ***<NX_ROOT>/site/config.properties*** file and ensure that the value of ***resources.created*** parameter is set to **0.** This must display as ***resources.created=0*** in the file.
14. Open a command prompt and run the following command:
    ```
    pdm_configure
    ```

15. If you roll-back from 17.4.3.0 to the base version, copyright text year will not change. To change the year:
    a. Add @NX_COPYRIGHT_YEAR=2023 into NX.env file located under CA SDM install directory (NX_ROOT).
    b. To preserve the value even after pdm_configure, update the value of "genfact.gx_copyright_year in install.properties under NX_ROOT\site genfact.gx_copyright_year=2023

NOTE
If you are unable to access the SDM URL, follow these steps:

1. Stop the SDM Services.
2. Navigate to the <NX_ROOT>/bopcfg/www/CATALINA_BASE folder and delete the work folder.
3. Start the SDM Services.

Restore Visualizer Files (Windows)

NOTE
After uninstalling CA SDM, CMDB Visualizer does not function automatically as the uninstallation process will not restore the previous visualizer files . To restore, perform the below steps manually:

1. Navigate to the where you have installed Visualizer: <NX_ROOT>\bopcfg\www\CATALINA_BASE_VIZ\webapps.
2. Remove the CMDBVisualizer folder.
3. Copy the **Visualizer_17.4.3.0.OLD** folder from<NX_ROOT>*REPLACED* and paste it in <NX_ROOT>*bopcfg**www* *CATALINA_BASE_VIZ**webapps*. Now, rename the **Visualizer_17.4.3.0.OLD** to **CMDBVisualizer**.
4. Start the SDM Service.

CMDB Visualizer works as expected.

Restore Visualizer Files (Linux)

NOTE
After uninstalling CA SDM, CMDB Visualizer does not function automatically as the uninstallation process will not restore the previous visualizer files. To restore the files, perform the below steps manually:

1. Navigate to the where you have installed Visualizer: <NX_ROOT>/bopcfg/www/CATALINA_BASE_VIZ/webapps.
2. Remove the CMDBVisualizer folder.
3. Copy the **Visualizer_17.4.3.0.OLD** folder from <NX_ROOT>*/replaced* and paste it into <NX_ROOT>*bopcfg**www* *CATALINA_BASE_VIZ**webapps*. Now, rename the **Visualizer_17.4.3.0.OLD** to **CMDBVisualizer**.
4. Start the SDM Service.

CMDB Visualizer works as expected.

Uninstall the XFlow Interface 17.4.3.0 (Windows)

Perform the following steps to uninstall the xFlow Interface (Windows):

1. Click **Start**, **Control Panel**, **Uninstall a Program** to uninstall xFlow Interface 17.4.3.0.
   Or
2. Navigate to where you have installed the xFlow Interface (for example, <NX_ROOT>*/CA/xFlow)*.
3. Select xFlow Interface **17.4.3.0_installation** and click **Change xFlow Interface 17.4.3.0 Installation.exe** to complete the uninstallation process.
4. If you are using Microsoft SQL Server Database, run the following query:

```
update ofProperty set propValue='select c.access_token from casm_active_sessions c WHERE
 UPPER(CONVERT(varchar(32), c.contact_uuid ,2)) = UPPER(?)' where name = 'jdbcAuthProvider.passwordSQL'
```

5. If you are using Oracle Database, run the following query;

```
update ofProperty set propValue='select c.access_token from casm_active_sessions c WHERE
 UPPER(CAST( c.contact_uuid AS varchar2(32))) = UPPER(?)' where name = 'jdbcAuthProvider.passwordSQL'
```

Uninstall the XFlow Interface 17.4.3.0 (Linux)

To uninstall the XFlow Interface, complete the following steps on Linux :

1. Navigate to where you have installed the xFlow Interface server.
   For example:
   /opt/CA/_xFlow Interface_installation
2. Execute the following command to uninstall the xFlow Interface 17.4.3.0:
   ```
   ./Change xFlow_Interface 17.4.3.0 Installation
   ```
3. Click **Finish** to complete the uninstallation process.
4. Run the following query for Oracle Database:
   ```
   update ofProperty set propValue='select c.access_token from casm_active_sessions c WHERE
    UPPER(CAST( c.contact_uuid AS varchar2(32))) = UPPER(?)' where name = 'jdbcAuthProvider.passwordSQL'
   ```

Uninstall CA Service Catalog 17.4.3.0

To uninstall CA Service Catalog, perform the following steps:

1. Click **Control Panel**, **Programs and Features**.
2. Double-click the CA Service Catalog 17.4.3.0 and follow the uninstallation wizard instructions.

Uninstall CA Asset Portfolio Management

CA Asset Portfolio Management (CA APM) architecture is highly database-driven and has no rollback mechanism for database-related changes. Hence, the uninstallation of the CA APM patch is not recommended and is not supported. The only way to revert the database changes, if required, is by restoring the database backup which can lead to data loss.

Run the Database Cleanup Utility for CA Service Management 17.4.3.0

During the installation of CA Service Management products, each of the installed product is registered in a common mdb table that is reflected in the deployment summary of the selected environment. These entries must be cleared in the following cases:

- Uninstallation of any CA Service Management products
- Moving Database or migration from one server to another

Run the CA Service Management database cleanup utility on a Windows or a Non-Windows environment. The database cleanup utility removes the entries and adds to the following tables:

Table Name from where the entries are removed	Table Name to which the removed entries are added
al_cdb_componentinstallstate	al_cdb_comp_installstate_bkup
al_cdb_configurationparameters	al_cdb_config_params_backup

NOTE

On a Non-Windows Platform, before running the **DBCleanup.sh** Utility, you must export the Oracle Environment variables on the terminal.

Perform the following steps:

1. Navigate to *DVD1\DBCleanupUtility* directory.
2. For Windows, double-click the **DBCleanupUtility.bat** to launch the DBCleanup utility.
   On a Non-Windows platform, you must run the **DBCleanupUtility.sh** file.
3. Enter the following details:
   a. **Database Server Type**: Specify the type of server. Microsoft SQL or Oracle.

b. **Database Host Name:** Enter the host name of the server where the database resides.
c. **Database User Name**:
   - **SQL Server**: Enter the privileged user detail that has permission to create and modify the schema.
   - **Oracle**: Provide the mdbadmin user name.
d. **Database Password**:
   - **MS SQL**: Enter the database password used by the database admin user.
   - **Oracle**: Enter the password for the Oracle database user.
e. **Database Port Number:** Provide the database port number.
f. **Enter Database Name :** Enter the database name. For example, mdb.
g. **System Name**: Provide the hostname of the system from where you want to uninstall the CA Service Management product or products.

4. Select any of the following products that you want to uninstall:
   – **SDM -** CA Service Desk Manager
   – **SLCM-** CA Service Catalog
   – **xFlow Interface**
   – **ProductUpdates**
   Enter product name to be uninstalled: **ProductUpdates**
5. Enter product to clean up updates: (sdm,slcm,xflow,chatserver,searchserver): For example, slcm
6. Enter update version to clean up: 17.4.3.0
7. The utility cleans up the database and removes the entries of the selected product installed on the server.

Installing CA Service Management 17.4.2.0

Based on your CA Service Management implementation, review the following topics for a successful installation.

- Install Prerequisites for CA Service Management 17.4.2.0
- Install CA Service Management 17.4.2.0 (Windows)
- Install CA Service Management 17.4.2.0 (Linux)
- Validate CA Service Management 17.4.2.0 Installation
- Post-Installation Tasks for CA Service Management 17.4.2.0
- Post-Installation Tasks - CA SDM Advance Availability 17.4.2.0
- Uninstall CA Service Management 17.4.2.0

Install Prerequisites for CA Service Management 17.4.2.0

Installation prerequisites for patch 17.4.2.0

Verify and complete the following prerequisites before installation of CA Service Desk Manager, CA Service Catalog, and CA Asset Portfolio Management.

Installation Prerequisites for CA Service Desk Manager
Review the following CA SDM installation prerequisites before you start the installation process:

> **NOTE**
> We recommend that you place the installer in a shorter directory path, containing no spaces, to avoid errors when running the setup file. For example, unzip the file at: C:\WIN_17_4_2_0.

- Ensure that CA SDM and xFlow Interface 17.4 is installed in your environment.
- Back up the system database (MDB). The system database (MDB) backup is mandatory and ensures safe data recovery if any problems are encountered during the installation process.
- Do not invoke ApplyPTF manually in your system while running the CA Service Management CA Service Management 17.4.2.0 installer.
- In the case of, CA Service Desk Manager - Conventional configuration, if the Primary and Secondary servers are on separate nodes, and the xFlow Interface is deployed on the Secondary Server, then the Database Client (Microsoft SQL Native Client or Oracle Database Client) is required on the Secondary Server for a successful installation.
- CA SDM must be installed for deploying the xFlow Interface. In the conventional configuration, it's recommended to install the xFlow Interface server on a secondary server. Advanced Availability users can install the xFlow Interface only on an application server.

  IMPORTANT
  After upgrading to CA Service Management 17.4.2.0, Maileater does not work with POP3 on a non-TLS port with Clear Text as a security level. This is a limitation of the upgraded Javamail (Third-party jar) used in connection to the mail server. Users are required to use **LOGIN** as a security level.

(Optional) Backing up the CR_Stored_Queries Table

CA Service Management provides a fix to address a vulnerability related to SQL injection. As part of this fix, **Cr_Stored_Queries** table is updated. If you have made customizations or changes to the Out-of-the-box (OOTB) stored queries, perform the below steps to take a backup before applying the CA Service Management patch:

1. Open a command prompt window and navigate to *$NX_ROOT\patches.* and create a folder **Backup_Data**. Navigate to this folder and take the Backup of the Cr_Stored_Queries table by running the following command:
   ```
   pdm_extract Cr_Stored_Queries> Cr_Stored_Queries_17.4_backup.dat
   ```
2. Verify that the **Cr_Stored_Queries_17.4_backup.dat** file is successfully created.
3. Now, proceed with CA Service Management 17.4.2.0 installation.
4. After applying the patch, if required based on your customizations, you can update the database files created in **Step 1**. Navigate to $NX_ROOT\patches\Backup_Data location and run the following command:

   ```
   pdm_load -f Cr_Stored_Queries_17.4_backup.dat
   ```

Installation Prerequisites for CA Service Catalog

Verify and complete the following prerequisites before proceeding with CA Service Catalog 17.4.2.0 installation:

- Ensure that CA Service Catalog 17.4 is installed in your environment.
- Back up the system database (MDB). The system database (MDB) backup is mandatory and ensures safe data recovery if any problems are encountered during the installation process.

Installation Prerequisites for CA Asset Portfolio Management

Consider the following prerequisites:

WARNING
CA Asset Portfolio Management (CA APM) architecture is highly database-driven and has no rollback mechanism for database-related changes. Hence, the uninstallation of the CA APM patch is not recommended and is not supported. The only way to revert the database changes, if required, is by restoring the database backup which can lead to data loss.

Verify and complete the following prerequisites before proceeding with the CA Asset Portfolio Management installation:

- Ensure that CA Asset Portfolio Management 17.4 is installed in your environment.
- Back up the system database (MDB). The system database (MDB) backup is mandatory and ensures safe data recovery if any problems are encountered during the installation process.

Install CA Service Management 17.4.2.0 (Windows)

Installing CASM 17.4.2.0 on Windows

CA Service Management 17.4.2.0 installer performs prerequisite checks and mandatory validations before installing the patch.

NOTE

- We recommend that you place the installer in a **shorter directory path**, containing no spaces, to avoid errors when running the setup file. For example, unzip the file at: C:\WIN_17_4_2_0
- In CA Service Desk Manager (SDM) Availability mode, the xFlow Interface can only be installed on an Application Server. If you have converted CA SDM conventional configuration to advanced availability configuration, and you have a previously installed version of xFlow Interface, ensure that you uninstall xFlow Interface and then, reinstall xFlow Interface on an Application server.

Perform the following steps to install 17.4.2.0 on Windows:

1. From the Broadcom Support portal, download the **WIN_17_4_2_0.zip** file.
   NOTE
   CA Service Management 17.4.2.0 (Windows) installer is provided as a ZIP file and not as a CAZ file. Extract/unzip the content of the ZIP file to your local folder.
2. Copy **WIN_17_4_2_0.zip** on all servers in your environment.
3. Extract the contents of the ZIP file to the folder named WIN_17_4_2_0.
   NOTE
   In the extracted folder, CA EEM 12.6.7.0 installer is available in *CASM_17.4.2.0\filestore\EEM\Windows\12.6.7.0*
4. In CA SDM conventional configuration, stop CA SDM services in your environment. In the case of Advance Availability configuration, stop CA SDM services only on the server on which CA SDM will be installed.
5. Log in to the server as a local admin to install the patch.
6. Navigate to **WIN_17_4_2_0** folder. Right-click **Setup.exe** and select **run as administrator**.
7. Review and Accept the License Agreement Information.
8. Select **Microsoft SQL Server** or **Oracle** as the database type in the Database Configuration screen.
   For Windows Microsoft SQL Server database, *mdbadmin* is the default user.
9. Provide the database details:
   a. If you have selected **Microsoft SQL Server**, provide the following field information:
      - **Database Server**: Enter the hostname of the database server.
        If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
      - **Database Name**: Enter the Database name (MDB), the name of the target DBMS. The default value is mdb.
      - **Database Port**: Provide the port identifier for the target DBMS.
      - **Database Server Instance**: The database instance name. Leave this field blank if you are using the default instance.
        If you are using the SQL named instance and a non-default port number, the **Database Server Instance** field should look like this: ***<database instance name>,<non-default port number>***
        For example:
        ```
        sdmcasmqa,10555
        ```
      - **Database Admin User**: For Microsoft SQL Server, **sa** is the default value.

The **sa** user is the admin user with permission to create a user and schema.

- **Database Admin Password**: Enter the database password of the user specified by a database admin user. No default value is assigned to this parameter.
- **mdbadmin Password**: Enter the password for the user specified for the MDB admin user.
- **Confirm mdbadmin Password**: Confirm the password.

b. If you have selected **Oracle** database, provide the following field information:

- **Database Server**: Enter the hostname of the Oracle database server.
  If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
- **Oracle Service Name**: Enter the Oracle Service name.
- **Listener Port**: Enter the listener port for the database.
- **Net Service Name**: Enter the Net Service Name of the Oracle database where the database resides.
  If the database is remote, use the Net Service Name that is defined in the Oracle client on the local computer. CA SDM has access to the database through a local installation of the Oracle client.
- **DBA User Name**: Provide the name of an Oracle user with DBA access. The default value for DBA user is **SYS**.
- **DBA User Password**: Enter the password for the DBA user.
- **mdbadmin Password**: Enter the password for MDB admin.
- **Confirm mdbadmin Password**: Confirm the password.
- **Tablespace Path (on DB Server)**: Provide the tablespace path on the database server.
- **Oracle Home Path**: Provide the Oracle Client 64-bit path. Provide Oracle client 32-bit path. For CA Asset Portfolio Management, provide the Oracle 32-bit client.
- **Data Tablespace Name**: Provide the name of the data tablespace. The default value is MDB_DATA.
- **Index Tablespace Name**: Provide the index Tablespace Name: The default value is MDB_INDEX.

Click **Next**.

10. On **Select the Products and Integrations** page, patch list is displayed. Select the required patch to install.

    WARNING

    CA Service Management does not support the uninstallation of the CA Asset Portfolio Management patch. Uninstalling the CA APM patch does not uninstall the MDB patch, as there is no rollback mechanism for the same. For more information, see Installation Prerequisites.

    a. **CA Service Desk Manager**
    - CA Service Desk Manager Patch
    - xFlow Interface Patch
    - Search Server Patch: Search server patch must be installed on a Server where an earlier version from previous releases is already installed.

    b. **CA Service Catalog**
    - CA Service Catalog Patch

    c. **CA Asset Portfolio Management**
    - CA Asset Portfolio Management Patch

    Click **Next**.

11. Review the Pre-Installation Configuration page. Click **Next**.

    NOTE

    Select the option to perform the automated post-install steps for CA SDM. If this option is not selected, you must manually perform the post-install steps for CA SDM. Note that for CA Service Catalog and CA Asset Portfolio Management, you need to perform the manual post-install steps.

12. Click **Install** to proceed with the installation. Review the Installation progress and when the installation is complete, click **Next.**
13. To review the installation logs, navigate to the path shown in the installation summary screen.
14. Click **Finish** to exit the installer

15. After installing the Key definition for "xflow" not found in the DITA map., start the **CA Key definition for "xflow" not found in the DITA map.** service from Windows Services (**Control Panel, Administrative Tools, Services**) as this service is not started by default.
16. After installing the search server, if the **Search Server** service is not started by default or if it stops, start the services from **Control Panel, Administrative Tools, Services.**
17. Validate the Installation.
18. Complete the Post-Installation Tasks.

After installing the Key definition for "xflow" not found in the DITA map. and Search Server, you can configure the cache size for Key definition for "xflow" not found in the DITA map.. For more information about how to configure cache size, see Perform initial load and refresh index on the search server.

Install CA Service Management 17.4.2.0 (Linux)

Installing CASM 17.4.2.0 on Linux

> **NOTE**
> In CA SDM Advanced Availability mode, the xFlow Interface can only be installed on an Application Server. If you have converted CA SDM conventional configuration to advanced availability configuration, and you have a previously installed version of xFlow Interface, ensure that you uninstall xFlow Interface and then reinstall xFlow Interface on an Application server.

CA Service Management 17.4.2.0 installer performs prerequisite checks and mandatory validations before installing the patch.

> **IMPORTANT**
> We recommend that you place the installer in a **shorter directory path**, containing no spaces, to avoid errors when running the setup file. For example, unzip the file at: *<root>\casm_dvd.x86Linux\<installer_file>*

Perform the following steps to install 17.4.2.0 on Linux:

1. From the Broadcom Support portal, download the **LNX_17_4_2_0.tar.gz** file.
2. To extract the LNX_17_4_2_0.tar.gz, run the following command:
   ```
   tar -xvzf LNX_17_4_2_0.tar.gz
   ```
   > **NOTE**
   > In the extracted folder, CA EEM 12.6.7.0 installer is available in CASM_17.4.2.0\filestore\EEM\Linux\12.6.7.0
3. Set CA SDM bin and library environment variables.
   For example:
   ```
   export PATH=/opt/CA/ServiceDeskManager/bin:$PATH
   export LD_LIBRARY_PATH=/opt/CA/ServiceDeskManager/lib:$LD_LIBRARY_PATH
   ```
4. In SDM conventional configuration, stop CA SDM services in your environment. In the case of Advance Availability, stop CA SDM services only on the server on which CA SDM will be installed.
5. Run the setup file from the casm_dvd.x86Linux folder to install the application(s) using the following command:
   ```
   ./setup.sh
   ```
6. Review and accept the license agreement terms.
7. In Database Configuration, select **Oracle** as the database type. Ensure that you have installed the Oracle database client in your system.
   Provide the following Oracle database configuration details:
   - **Database Server**: Enter the hostname of the Oracle database server.
     If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
   - **Oracle Service Name**: Enter the Oracle Service name.
   - **Listener Port**: Enter the listener port for the database.
   - **Net Service Name**: Enter the Net Service Name of the Oracle database where the database resides.

If the database is remote, use the Net Service Name that is defined in the Oracle client on the local computer. CA SDM has access to the database through a local installation of the Oracle client.
- **DBA User Name**: Authenticates the name of the user with DBA access. The default value for DBA user is **SYS.**
- **DBA User Password**: Authenticates the password for DBA user.
- **mdbadmin Password**: Specify the password for the mdbadmin user.
- **Confirm mdbadmin Password**: Confirm the mdbadmin user password.
- **Tablespace Path (on DB Server)**: Enter the tablespace directory on the database server.
- **Oracle Home Path**: Provide the Oracle Client 64-bit path. Provide Oracle client 32-bit path for fresh installation
- **Data Tablespace Name**: Enter the tablespace name. The default value is **MDB_DATA**.
- **Index Tablespace Name**: Enter the index tablespace name: The default value is **MDB_INDEX**.

8. Click **Next.**
9. On **Select the Products and Integrations** page, patch list is displayed. Select the required patch to install.
   a. **CA Service Desk Manager**
      - CA Service Desk Manager Patch
      - xFlow Interface Patch
      - Search Server Patch: A search server patch must be installed on a server where an earlier version from previous releases is already installed.
10. Click **Next**.
11. Review the Pre-Installation Summary page. Click **Next.**
12. Click **Install** to proceed with the installation. Review the Installation progress and when the installation is complete, click **Next.**
13. To review the installation logs, navigate to the path shown in the installation summary screen.
14. Click **Finish** to exit the installer.
    If patch installation fails for a product, retry the installation by using the **Retry** option in the wizard where it fails. If retry does not work, exit the installer and relaunch it. If the problem persists, contact Broadcom Support.
15. After installing XFlow Interface, start the**CA XFlow Interface**service. Navigate to /opt/CA/xFlow/Apps and run the following command from a command prompt window:
    ```
    ./xFlowServer start
    ```
    To stop the server, run the following command:
    ```
    ./xFlowServer Stop.sh
    ```
16. Navigate to /opt/CA/SC/SearchServer/elasticsearch/bin and run the following command from the command prompt window:
    ```
    ./ServicesStart.sh
    ```
    NOTE
    If you get the below message while restarting the search server service, ignore it and continue:
    ```
    groupadd: group 'elasticsearch' already exists
    useradd: user 'elasticuser' already exists
    runuser: warning: cannot change directory to /home/elasticuser: No such file or directory
    OpenJDK 64-Bit Server VM warning: Option UseConcMarkSweepGC was deprecated in version 9.0 and
     will likely be removed in a future release.
    ```
    To stop the server, run the following command:
    ```
    /ServicesStop.sh
    ```
17. Validate the Installation.
18. Complete the Post-Installation Tasks.

After installing the XFlow Interface and Search Server, you can configure the cache size for XFlow Interface. For more information about how to configure cache size, see Perform initial load and refresh index on the search server.

Validate CA Service Management 17.4.2.0 Installation

Validate CASM 17.4.2.0 installation

To verify the successful installation of CA Service Management 17.4.2.0, validate the installation for the corresponding applications in the following location. You can also navigate to the root folder where you have locally installed the application and select the "env.info" file for validating your installation:

- **CA ITAM:**

  NOTE
  CORA version: 17.4.0.57

  – C:\Program Files (x86)\CA\ITAM\env.info
  – product=ITAM
  – major=17
  – minor=4
  – cum=2
  – roll-up=0

  NOTE
  CA CASM uses Apache Tomcat version 9.0.86.
- **CA Service Catalog:**
  – %USM_HOME%*CA_SLCM_<hostname>.his* file
  – %USM_HOME%\view\webapps\usm\locale\about_info.xml
  – %USM_HOME%\build.xml
  – %USM_HOME%\env.info
- **xFlow Interface:**
  – *C:\Program Files\CA\xFlow\<hostname>.his* file
  – NX_ROOT\env.info
  – NX_ROOT\version.xml
- **CA SDM:**
  – *C:\Program Files (x86)\CA\Service Desk Manager\<hostname>.his* file
  – NX_ROOT\pdmconf\version
  – NX_ROOT\env.info
  – HELP, ABOUT

  NOTE
  For CA SDM, the Binary and Language Combo patches are installed irrespective of the environment. The language patch for the corresponding CA SDM environment is installed.

  To validate the successful installation of the patches, follow these steps:

  1. Open the <sdm_installation>\<hostname>.his file in a text editor.
  2. Search for the instances of "gol-168". In a successful install, you should come across these three entries (Binary, Language Combo, and corresponding language patch).

Application	Windows	Linux
CA Asset Portfolio Management	T6EE1742	NA
CA Service Catalog	T6D91742	NA
CA Service Desk Manager - Binary	gol-168_cum_C	gol-168_x86_64Linux_cum_C
CA Service Desk Manager - Languag504e Comb	gol-168_langcombo	gol-168_langcombo

Application	Windows	Linux
CA Service Desk Manager - English	gol-168_en_US	gol-168_x86_64Linux_en_US
CA Service Desk Manager - German	gol-168_de_DE	gol-168_x86_64Linux_en_DE
CA Service Desk Manager - Spanish	gol-168_es_ES	NA
CA Service Desk Manager - French	gol-168_fr_FR	gol-168_x86_64Linux_fr_FR
CA Service Desk Manager - Italian	gol-168_it_IT	NA
CA Service Desk Manager - Japanese	gol-168_ja_JP	gol-168_x86_64Linux_ja_JP
CA Service Desk Manager - Brazilian Portuguese	gol-168_pt_BR	NA

Post-Installation Tasks for CA Service Management 17.4.2.0

After successful installation of CA Service Management 17.4.2.0, perform the following post-installation steps.

CA Service Desk Manager

Manual Steps required for Automated Post-Installation

CA Service Management provides an option in the installer to automate the post-install tasks.
If you have selected **Yes** to automate the post-installation steps while running the CA Service Management Installer Wizard, you are not required to perform any manual post-install steps for CA Service Desk Manager (SDM).

If you have selected **No** for automating the post-install steps while running the CA Service Management Installer wizard, perform the below manual post-install tasks as per your deployment requirements:

Manual Post-Install Tasks After Installing CA Service Desk Manager (SDM) 17.4.2.0 on CA SDM 17.4 or CA SDM 17.4.1.0

Perform the following steps:

1. Stop CA SDM services.
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0.
3. Open a command prompt window and run the following command:
   ```
   pdm_configure
   ```
4. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_02_insert.dat
   pdm_load -u -f r17_4_02_update.dat
   ```
5. Notification message template is modified to update the existing email templates with the html_description field.
   Tool location: NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate This has to be executed one time when the SDM service is up and running. **For e.g:**
   Windows
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.bat
   ```
   Linux
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.sh
   ```

Manual Post-Installation Task for CMDB

> **NOTE**
> For Oracle 21c, set the **isClusterEnv** setting to "true" in the application.conf file of CMDB viz and restart the CA SDM Services.:

- For Windows: <SDM installed location>\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\WEB-INF\classes\application.conf)
- For LINUX: <SDM installed location>/bopcfg/www/CATALINA_BASE_VIZ/webapps/CMDBVisualizer/WEB-INF/classes/application.conf).

If you are applying CA Service Management 17.4.2.0 in an SSL environment, CA SDM Administrators must perform the following post-installation steps for CMDB Visualizer Interface to open using the **HTTPS** protocol when launching from CA SDM UI:

1. In the NX.env file, change the **NX_CMDB_VISUALIZER** entry to provide **HTTPS protocol** and **HTTPS port**.
   For example: @NX_CMDB_VISUALIZER=https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer
2. In **casm.conf.js** file from NX_root\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\conf, provide the HTTPS as protocol and HTTP port.
   For example:

```
sdm: {

      vizserver: 'https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer/viz',
      server: 'https:// sdmurl:<SDM_SSL_PORT>/CAisd/pdmweb.exe'
    }
```

Manual Post-Installation Task for OAuth 2.0 (Maileater/ Mailer)

After generating an access token for OAuth 2.0 to connect to a mail server, perform the mandatory post-install tasks on all servers in your environment:

1. Create a backup of web.xml from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF in your local directory to perform backout if required.
2. Open the web.xml file from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF.
3. Search for the following statement:
   ```
   <!--USD servlet mappings start-->
   ```
4. Copy and paste the following configuration under the statement:

```
<servlet>
  <servlet-name>OAuthProcessor</servlet-name>
  <servlet-class>com.broadcom.oAuth.OAuthProcessor</servlet-class>
 </servlet>
```

5. Copy and paste the following configuration after any <servlet-mapping>:

```
<servlet-mapping>
  <servlet-name>OAuthProcessor</servlet-name>
  <url-pattern>/OAuthProcessor</url-pattern>
</servlet-mapping>
```

6. Save and close the web.xml file.
7. For OAuth servlet preservation in a primary and secondary environment, the secondary server must also have the OAuth processor entries in the following file: *<NX_ROOT>\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF\web.xml.last*.
8. Repeat Steps 1-8 to update **web.xml.tpl** file in the NX_ROOT\samples\pdmconf\web.xml.tpl folder to avoid losing these changes when pdm_configure is run at a later point of time.
9. Run the following command:
   ```
   pdm_tomcat_nxd -c STOP command
   ```
10. Wait for some time (a minute or so) and run the following command:

```
pdm_tomcat_nxd -c START command
```

For more information, see Define a mailbox.

Manual Post-Installation Task for Excel file (.xlsx) export*

1. Execute the following command:
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst
   ```
2. Execute the following command to add the new NX variable to NX.env.tpl file.
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst -t
   ```
3. Restart CA SDM services.

HTML Tags Sanitization

The sanitization tool can be run any number of times. Administrators can choose if they wish to run the tool for each ticket type separately or for all ticket types at one go.

Before running the Sanitization tool ensure you take a backup of all the ticket objects using the following commands or by taking a backup of the sdm databse :

```
"Call_Req": pdm_extract Call_Req > cr_backup.txt
"Issue": pdm_extract Issue > Issue_backup.txt
"Change Orders": pdm_extract Change_Request >change_orders_backup.txt.
```

Both active and archived tickets can be sanitized.
For Windows:

```
NX_ROOT\samples\sdk\SanitizationTool\sanitizeTool.bat
```

For Linux:

```
NX_ROOT\samples\sdk\SanitizationTool\sanitizeTool.sh
```

> **NOTE**
> We recommend executing the sanitization during slack time on any Service Desk Manager server.

After executing the sanitization tool, execute the following command to update domsrvr memory cache:

```
pdm_cache_refresh -t Call_Req
pdm_cache_refresh -t Issue
pdm_cache_refresh -t Change_Request
```

(Optional) Manual Post-Installation Task for Knowledge Document export with German fields

If you are exporting Knowledge Document with German Fields (included in the export field list), then perform the below post-install steps for successful export of knowledge documents:

1. Take the backup of NX_ROOT\NX.env file (NX_ROOT is the installation directory for CA Service Desk Manager).
2. Open NX_ROOT\pdmconf\NX.env_nt.tpl file using any text editor and take the value of NX variable @NX_KEIT_AVAILABLE_FIELDS which is COPIED_DATA and update Nx.env in NX_ROOT by executing below command:
   ```
   pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA>  -a  pdm_option.inst
   ```
   Where COPIED_DATA is the value copied from NX.env_nt.tpl (step 2).
   For example, " pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <provide the exact string here> -a pdm_option.inst "
   Note to avoid losing the updated changes when you run pdm_configure, execute the above command with the '-t' flag as follows:
   ```
   pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA> -a pdm_option.inst -t
   ```

(Optional) Handle HTML Mails that contains Tables in MailEater

Java based Maileater already has the ability to process the emails that were sent in plain format or HTML format, to support the processing of emails that contain Tables in the HTML body. But the format in which the email was sent is lost when received by Maileater.

1. Run the following commands from NX_ROOT/bin from the command prompt/ terminal:
   ```
   pdm_options_mgr -c -s NX_ENFORCE_HTML_EMAIL_PARSING -v "Yes" -a pdm_option.inst
   pdm_options_mgr -c -s NX_ENFORCE_HTML_EMAIL_PARSING -v "Yes" -a pdm_option.inst -t
   ```
2. Restart CA SDM and XFlow applications.

(Optional) Comments are not allowed when a Data Partition (O_COMMENTS) is created with Customer, Employee and ServiceDesk Analyst

To disable comments:

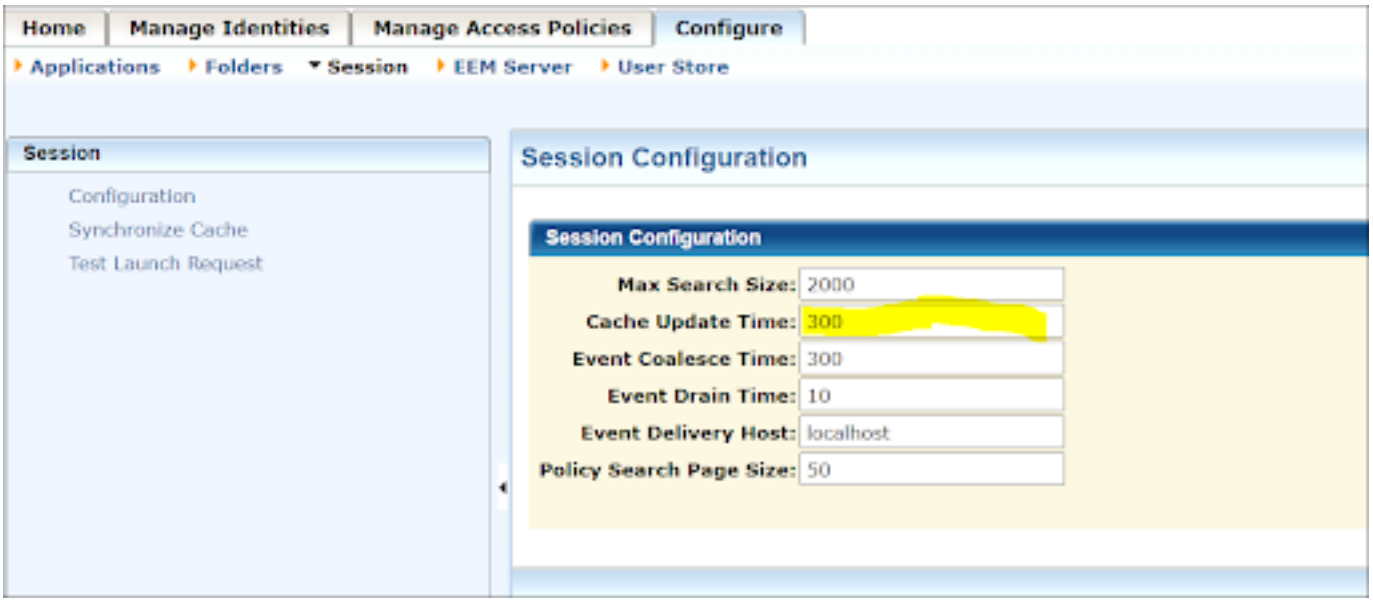
1. Login to SDM and navigate to Administration -> Security and Role Management-> Data partition-> Data Partition List.
2. In Data Partition List, create a constraint (O_COMMENTS with the condition id=0) and link it to Customer, Employee and ServiceDesk Analyst data partitions.
3. Run the following commands from NX_ROOT/bin from the command prompt/ terminal:
   ```
   pdm_options_mgr -c -s KT_COMMENT_USE_SUPERUSER -v 0 -a pdm_option.inst
   pdm_options_mgr -c -s KT_COMMENT_USE_SUPERUSER -v 0 -a pdm_option.inst -t
   ```
4. Restart CA SDM and XFlow applications.

(Optional) Manual Post-Installation Task for EEM Cache Timeout Configuration Update

You may face issues while logging in to EEM or SDM using the EEM user accounts due to a session cache timeout mismatch between EEM and SDM.

You can update the EEM cache timeout configuration by adding a new NX variable, "NX_EEM_CACHE_UPDATE_TIME" in SDM.

Keep the value the same as the EEM cache timeout configuration value mentioned in the following screenshot.



To set the EEM cache update time to 300 seconds, perform the following steps:

To add the variable in the **nx.env** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst
```

To add the variable in the **nxenv_nt.tpl** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst -t
```

Service Point

After successful installation of Service Point, refresh index and data on the Search Server.

> **NOTE**
> In case of Linux Search Server environment, ensure to run the following commands on the terminal:
>
> ```
> sysctl -w vm.max_map_count=262144
> ulimit -Hn 65535
> ```

(Optional) Manual Post-Installation Task for Rebuild Index

If you have upgraded or are planning to install CA EEM 12.6 with Service Point and CA Service Catalog, ensure that you rebuild the CA Service Catalog index after completing the CA EEM upgrade/installation process. For more information, see Perform Initial Load and Refresh Index Data on Search Server.
Run the following command to rebuild the index:

```
$ES_HOME/search/bin/pdm_es_rebuild_index.bat index=catalog_index
```

CA Service Catalog

Perform the following CA Service Catalog post-installation steps as per your environment requirements:

- Post-Installation Tasks after installing CA Service Catalog 17.4.2.0 on CA Service Catalog 17.4
- gMSA Authentication

Post-Install Tasks After Installing CA Service Catalog 17.4.2.0 on CA Service Catalog 17.4 or later versions

Review and perform the post-installation steps based on your deployment requirements.

> **NOTE**
> If any SSL certificate(s) chain was imported to the Oracle Java's JAVA_HOME\lib\security\cacerts Keystore, those certificate(s) chain again need to be exported explicitly and imported properly into the new AdoptOpenJDK's %USM_HOME%\embedded\jdk\lib\security\cacerts Keystore.

Follow these steps

- (Optional) If Service Catalog is accessed from applications like Service Point, SharePoint and so on, then you must perform this step. Open the "**web.xml**" file located in *USM_HOME%\view\webapps\usm\WEB-INF* and update the respective parameter values for the parameter name "cors.allowed.origins" and "cors.support.credentials". For example, if you are using Catalog Widgets in Service Point
- (Optional) Unable to approve any workflow requests through email. The email sent for approvals is corrupted with special characters. Follow the KB article steps.

(Optional) Approve workflow requests through email

The workflow email sent for approvals is corrupted with special characters. To approve any workflow requests through email, follow the KB article steps.

gMSA Authentication Post-Install Steps for CA Service Catalog (Automated steps)

CA Service Catalog provides automated post-installation steps for gMSA Authentication. Review the following before you proceed:

- Prerequisites and Considerations
- How to Automate gMSA Post-Installation Steps
- Unconfigure gMSA

Prerequisites and Considerations:

Ensure that you have taken care of the below considerations on all nodes of CA Service Catalog where you want to automate the gMSA post-installation procedure.

1. If the accounting service is available, perform the automated post-install steps by updating serviceName to ServiceAccounting and providing the appropriate gMSA account name in the **gmsa.properties** file.
2. If the filestore path is external to the server from where the **GMSAPostInstall.bat** is running, ensure to provide appropriate access to the server for setting the gMSA account permissions.
3. While preparing the properties file, check for any copy/paste issues, special characters, spaces, and so on.

How to Automate gMSA Post-Installation Steps

Perform the following steps to enable the batch file **GMSAPostInstall.bat** and to automate the gMSA post-installation steps:

1. Create a gMSA account for CA Service Catalog.
2. Ensure that appropriate DB (database) permission is given to the newly created account(s).
3. Enter the below values in the **gmsa.properties** file and save it.

   NOTE
   A template file (**gmsa.properties**) is added in the common installer in *"\filestore\SLCM\gMSAProperties"* . Optionally. use this file as a template file to fill gmsa properties. This can be copied to "USM_HOME/scripts/GMSA" where Catalog is installed.

   NOTE
   All values are mandatory. Make sure that there are no leading or trialing spaces while providing these values.

   a. **serviceName**: Application service name to enable GMSA. In the case of Catalog, it will be ServiceCatalog and for accounting, it will be ServiceAccounting.
   b. **folderName**: Absolute path of the application installation directory.
   c. **gMSAName**: Name of the GMSA service account (domain\accountName$).
   d. **filestorePath**: Shared network path (\\Server_name\filestore) or local filestore path
4. Navigate to %USM_HOME% and launch **usm.cmd**
5. Navigate to %USM_HOME%/scripts/GMSA/ folder and run the "GMSAPostInstall.bat" in command prompt.

   NOTE
   If the command executes without any error, "Post-installation steps for GMSA executed successfully" message is displayed with exit code 0.
   If there are errors, the error message is shown with a non-zero exit code. Correct the error to continue and perform step **5** again.

Unconfigure gMSA for CA Service Catalog

To disable or unconfigure gMSA for CA Service Catalog, follow the steps below:

1. Replace **config.properties** file from*%USM_HOME%*\ folder.
2. Revert back all the folder permissions which were assigned to domain users.
3. Start Catalog and Accounting Services with the local system account.

   NOTE
   In the case of load balancers, repeat steps 1-3 to unconfigure gMSA on all Catalog servers.

CA Asset Portfolio Management

Manual Post-Installation Task for Removing jQuery, log4net, and log4j Vulnerabilities

After installing 17.4.2.0, delete jquery-1.3.2.min.js, jquery-ui-1.7.2.custom.min.js , log4net.dll, and log4j from their respective installed folders.

NOTE
Perform the below steps if you have not done this earlier.

Perform the following steps:

1. Log on to CA APM Web Server, navigate to the <ITAM_install_dir>\ITAM\Web Server\castylesr5\javascript.
   Example: C:\Program Files (x86)\CA\ITAM\Web Server\castylesr5\javascript
   Locate and delete the jquery-1.3.2.min.js and jquery-ui-1.7.2.custom.min.js files from the directory.
2. Navigate to the ITAM_install_dir\CA\ITAM\Data Importer.
   Locate and delete the log4net.dll file from the directory.
3. Navigate to the C:\Program Files (x86)\CA\SC\CORA.
   Locate and delete the log4j-1.2.15.jar, log4j.properties, and ojdbc6.jar files from the directory.
4. Navigate to the ITAM_install_dir\CA\ITAM.
   Backup the following folders and manually delete them:
   - CDB
   - MDB
   - jre8

     NOTE
     jre8 folder has to be deleted manually for every RU installation.
5. Perform an IISRESET and delete the browser's cache to load the upgraded files.

Post-Installation Tasks - CA SDM Advance Availability 17.4.2.0

The installer wizard provides an option in the installer to automate the post-install tasks. Perform the following post-installation tasks for CA SDM Advanced Availability Configuration.

For the Advanced Availability configuration, the following host names are considered as an example:

- The background server hostname is sdmcahost1.
- The standby server hostname is sdmcahost2.
- The application server hostname is sdmcahost3.

Manual Steps required for Automated Post-Installation

If you have selected **Yes** for automating the post-install steps while running the installer wizard, then perform manually the following post-install tasks:

Manual Post-Install Tasks - After Installing CA Service Desk Manager (SDM) 17.4.2.0 on CA SDM 17.4 GA or CA SDM 17.4.1.0

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Start the CA SDM service on the standby server (sdmcahost2). Open the command prompt and run the following command:
   ```
   pdm_server_control -b
   ```
3. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server.
4. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```
5. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_02_insert.dat
   pdm_load -u -f r17_4_02_update.dat
   ```
6. Clear the browser cache by the following command:
   ```
   pdm_webcache -H
   ```

7. Ensure that the patch is applied on all the application servers and the remaining standby servers if any. If you selected YES to automate the post-install steps, then there are no more manual post-install steps to be performed.

Manual Steps required for Non-Automated Post-Installation

If you have selected **No** for automating the post-install steps while running the CA Service Management installer wizard, perform manually the below appropriate post-install task as per your deployment requirements.

Post-Install Tasks After Installing CA SDM 17.4.2.0 on CA SDM 17.4 GA or CA SDM 17.4.1.0 for Advanced Availability Configuration

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0. Open a command prompt and run the following command:
   ```
   pdm_configure
   ```
   On the installer configuration page, before clicking Finish, uncheck the Start service when completed checkbox. Failure to uncheck the box may result in the loss of custom schema changes and configurations that were changed on the server.
3. Promote the standby server as the new background server by performing the following steps. To suppress version control on the standby server and later on the background server, run the following command on each server:
   ```
   pdm_server_control -v
   ```
4. Start SDM service on the standby server. Run the following command:
   ```
   pdm_server_control -b
   ```
5. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server, and perform the following steps. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
   ```
   pdm_configure
   ```
6. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control –b
   ```
7. Change the directory to NX_ROOT/data, and execute the following command:
   ```
   pdm_load -f r17_4_02_insert.dat
   pdm_load -u -f r17_4_02_update.dat
   ```
8. Notification message template is modified to update the existing email templates with the html_description field. **Tool location**: NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate This has to be executed one time when the SDM service is up and running. **For e.g:**
   Windows
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.bat
   ```
   Linux
   ```
   NX_ROOT\samples\sdk\SanitizationTool\SDMNotifUpdate.sh
   ```
9. Ensure that the patch is applied on all application servers and the remaining standby server if any. Perform the following steps on all application servers and standby servers.
10. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
    ```
    pdm_configure
    ```
11. Clear the browser cache by running the following command:
    ```
    pdm_webcache -H
    ```
12. Execute the following command.
    ```
    pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst
    ```
13. Execute the following command to add the new NX variable to NX.env.tpl file.
    ```
    pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst -t
    ```
14. Restart CA SDM Services.

NOTE
Ensure that

Uninstall CA Service Management 17.4.2.0

Review the following topics to uninstall CA Service Management 17.4.2.0.

NOTE

Uninstalling CA Service Management 17.4.2.0 will revert to CA Service Management 17.4 version on which 17.4.2.0 is installed.

Example: If CA Service Management 17.4.2.0 is installed on 17.4, uninstalling 17.4.2.0 will revert to CA Service Management 17.4.

Uninstall CA SDM 17.4.2.0 (Windows)
Perform the following steps to uninstall CA SDM 17.4.2.0 on Windows:

1. Log in to CA SDM as an Administrator.
2. Shut down CA Service Desk Manager Server.
3. Navigate to **<NX_ROOT>\bin** and run the following command to backup the unsupported jars from the older versions.
   ```
   pdm_perl pdm_restore_files.pl gol-168_tool ..\PATCHES\DELIVERABLES_REMOVAL.TXT
   ```
4. Navigate to the location *where you have extracted the CA Service Management Installer zip package*.
5. Locate *CASM-17.4.2.0-CommonPatchInstaller-Win\filestore\utils\ApplyPTF* folder.
6. Run the **APPLYPTF.exe**.
7. Select **Backout PTF** from the local or remote node option.
8. Click **Next**.
9. Enter the patch number for the language that is installed in your environment:
   - **Binary**: gol-168_cum_C
   - **Language Master**: gol-168_langcombo
   - **English**: gol-168_en_US
   - **French**: gol-168_fr_FR
   - **German**: gol-168_de_DE
   - **Spanish**: gol-168_es_ES
   - **Italian:** gol-168_it_IT
   - **Japanese**: gol-168_ja_JP
   - **Brazilian Portuguese**: gol-168_pt_BR
10. Click **Next** to continue backing out the specific patch.
11. Repeat steps 6-9, to back out the language English (gol-168_en_US) patch.
12. Repeat steps 6-9 to back out the language master (gol-168-langcombo) patch.
13. Repeat steps 6-9 to back out the binary (gol-168_cum_C) patch.
14. Reboot the server.
15. Repeat steps **6-12** on all standby, application, or secondary servers in your environment.
16. Restore the MDB database to the database backup that you took before applying the patch.
17. Navigate to ***<NX_ROOT>\site\config.properties*** file and ensure that the value of ***resources.created*** parameter is set to **0.** This must display as ***resources.created=0*** in the file.
18. Open a command prompt and run the following command:
    ```
    pdm_configure
    ```
19. If you roll-back from 17.4.2.0 to the base version, copyright text year will not change. To change the year:
    a. Add @NX_COPYRIGHT_YEAR=2023 into NX.env file located under CA SDM install directory (NX_ROOT).

b. To preserve the value even after pdm_configure, update the value of "genfact.gx_copyright_year in install.properties under NX_ROOT\site genfact.gx_copyright_year=2023

Uninstall CA SDM 17.4.2.0 (Linux)

Perform the following steps to uninstall CA SDM 17.4.2.0 on Linux:

1. Login to CA SDM as the root user.
2. Shut down the CA Service Desk Manager Server.
3. Navigate to **<NX_ROOT>/bin** and run the following command to backup the unsupported jars from the older versions.
   ```
   ./pdm_perl pdm_restore_files.pl ../replaced/gol-168_tool ../patches/deliverables_removal.txt
   ```
4. Navigate to the location where you have extracted the CA SDM Installer **tar.gz** package.
5. Navigate to the *casm_dvd.x86Linux \filestore\utils\ApplyPTF* folder.
6. The following binary/language-independent and language-dependent patches are supported:
   - Binary name: gol-168_x86_64Linux_cum_C
   - Language Master: gol-168_langcombo
   - English: gol-168_x86_64Linux_en_US
   - German: gol-168_x86_64Linux_de_DE
   - French: gol-168_x86_64Linux_fr_FR
   - Japanese: gol-168_x86_64Linux_ja_JP
7. Execute the following command to remove the binaries or the language-dependent patches. Replace the <patch_name> with the binary or language-dependent patch name:
   ```
   ./applyptf.<platform> -b <patch_name>  -e <NX_ROOT>
   ```
   Here, <platform> specifies the operating system and <NX_ROOT> specifies the CA SDM installed location.
   For example:
   ```
   ./applyptf.linux -b gol-168_x86_64Linux_cum_C  -e /opt/CA/ServiceDeskManager
   ```
8. Repeat step 6 to back out the language English (gol-168_x86_64Linux_en_US) patch.
9. Repeat step 6 to back out the binary (gol-168_x86_64Linux_cum_C) patch.

   NOTE
   During uninstallation, if you get any errors, ensure that the following steps are performed:

   1. Copy the subfolders from <NX_ROOT>/replaced/<gol-168_x86_64Linux_cum_C> and place it into the <NX_ROOT> folder.
   2. Copy the subfolders from <NX_ROOT>/replaced/<gol-168_x86_64Linux_en_US> and place it into the <NX_ROOT> folder.
   3. Open **.his** file under the <NX_ROOT> folder and delete the entries related to the current patch and save it.
10. To restore MDB database, revert to database backup that you took before applying the patch.
11. Navigate to ***<NX_ROOT>/site/config.properties*** file and ensure that the value of ***resources.created*** parameter is set to **0.** This must display as ***resources.created=0*** in the file.
12. Open a command prompt and run the following command:
    ```
    pdm_configure
    ```
13. If you roll-back from 17.4.2.0 to the base version, copyright text year will not change. To change the year:
    a. Add @NX_COPYRIGHT_YEAR=2023 into NX.env file located under CA SDM install directory (NX_ROOT).
    b. To preserve the value even after pdm_configure, update the value of "genfact.gx_copyright_year in install.properties under NX_ROOT\site genfact.gx_copyright_year=2023

    NOTE
    If you are unable to access the SDM URL, follow these steps:

    1. Stop the SDM Services.

2. Navigate to the <NX_ROOT>/bopcfg/www/CATALINA_BASE folder and delete the work folder.
3. Start the SDM Services.

Restore Visualizer Files (Windows)

NOTE
After uninstalling CA SDM, CMDB Visualizer does not function automatically as the uninstallation process will not restore the previous visualizer files . To restore, perform the below steps manually:

1. Navigate to the where you have installed Visualizer: <NX_ROOT>\bopcfg\www\CATALINA_BASE_VIZ\webapps.
2. Remove the CMDBVisualizer folder.
3. Copy the **Visualizer_17.4.2.0.OLD** folder from<NX_ROOT>*REPLACED* and paste it in <NX_ROOT>*bopcfg**www* *CATALINA_BASE_VIZ**webapps*. Now, rename the **Visualizer_17.4.2.0.OLD** to **CMDBVisualizer**.
4. Start the SDM Service.

CMDB Visualizer works as expected.

Restore Visualizer Files (Linux)

NOTE
After uninstalling CA SDM, CMDB Visualizer does not function automatically as the uninstallation process will not restore the previous visualizer files. To restore the files, perform the below steps manually:

1. Navigate to the where you have installed Visualizer: <NX_ROOT>/bopcfg/www/CATALINA_BASE_VIZ/webapps.
2. Remove the CMDBVisualizer folder.
3. Copy the **Visualizer_17.4.2.0.OLD** folder from <NX_ROOT>*/replaced* and paste it into <NX_ROOT>*bopcfg**www* *CATALINA_BASE_VIZ**webapps*. Now, rename the **Visualizer_17.4.2.0.OLD** to **CMDBVisualizer**.
4. Start the SDM Service.

CMDB Visualizer works as expected.

Uninstall the xFlow Interface 17.4.2.0 (Windows)
Perform the following steps to uninstall the xFlow Interface (Windows):

1. Click **Start**, **Control Panel**, **Uninstall a Program** to uninstall xFlow Interface 17.4.2.0.
   Or
2. Navigate to where you have installed the xFlow Interface (for example, <NX_ROOT>*/CA/xFlow)*.
3. Select xFlow Interface **17.4.2.0_installation** and click **Change xFlow Interface 17.4.2.0 Installation.exe** to complete the uninstallation process.
4. If you are using Microsoft SQL Server Database, run the following query:
```
update ofProperty set propValue='select c.access_token from casm_active_sessions c WHERE
 UPPER(CONVERT(varchar(32), c.contact_uuid ,2)) = UPPER(?)' where name = 'jdbcAuthProvider.passwordSQL'
```
5. If you are using Oracle Database, run the following query;
```
update ofProperty set propValue='select c.access_token from casm_active_sessions c WHERE
 UPPER(CAST( c.contact_uuid AS varchar2(32))) = UPPER(?)' where name = 'jdbcAuthProvider.passwordSQL'
```

Uninstall the xFlow Interface 17.4.2.0 (Linux)
To uninstall the xFlow Interface, complete the following steps on Linux :

1. Navigate to where you have installed the xFlow Interface server.
   For example:
   /opt/CA/_xFlow Interface_installation
2. Execute the following command to uninstall the xFlow Interface 17.4.2.0:
```
./Change xFlow_Interface 17.4.2.0 Installation
```

3. Click **Finish** to complete the uninstallation process.
4. Run the following query for Oracle Database:

```
update ofProperty set propValue='select c.access_token from casm_active_sessions c WHERE
 UPPER(CAST( c.contact_uuid AS varchar2(32))) = UPPER(?)' where name = 'jdbcAuthProvider.passwordSQL'
```

Uninstall CA Service Catalog 17.4.2.0

To uninstall CA Service Catalog, perform the following steps:

1. Click **Control Panel**, **Programs and Features**.
2. Double-click the CA Service Catalog 17.4.2.0 and follow the uninstallation wizard instructions.

Uninstall CA Asset Portfolio Management

CA Asset Portfolio Management (CA APM) architecture is highly database-driven and has no rollback mechanism for database-related changes. Hence, the uninstallation of the CA APM patch is not recommended and is not supported. The only way to revert the database changes, if required, is by restoring the database backup which can lead to data loss.

Run the Database Cleanup Utility for CA Service Management 17.4.2.0

During the installation of CA Service Management products, each of the installed product is registered in a common mdb table that is reflected in the deployment summary of the selected environment. These entries must be cleared in the following cases:

- Uninstallation of any CA Service Management products
- Moving Database or migration from one server to another

Run the CA Service Management database cleanup utility on a Windows or a Non-Windows environment. The database cleanup utility removes the entries and adds to the following tables:

Table Name from where the entries are removed	Table Name to which the removed entries are added
al_cdb_componentinstallstate	al_cdb_comp_installstate_bkup
al_cdb_configurationparameters	al_cdb_config_params_backup

NOTE

On a Non-Windows Platform, before running the **DBCleanup.sh** Utility, you must export the Oracle Environment variables on the terminal.

Perform the following steps:

1. Navigate to *DVD1\DBCleanupUtility* directory.
2. For Windows, double-click the **DBCleanupUtility.bat** to launch the DBCleanup utility.
   On a Non-Windows platform, you must run the **DBCleanupUtility.sh** file.
3. Enter the following details:
   a. **Database Server Type**: Specify the type of server. Microsoft SQL or Oracle.
   b. **Database Host Name:** Enter the host name of the server where the database resides.
   c. **Database User Name**:
      - **SQL Server**: Enter the privileged user detail that has permission to create and modify the schema.
      - **Oracle**: Provide the mdbadmin user name.
   d. **Database Password**:
      - **MS SQL**: Enter the database password used by the database admin user.
      - **Oracle**: Enter the password for the Oracle database user.
   e. **Database Port Number:** Provide the database port number.
   f. **Enter Database Name :** Enter the database name. For example, mdb.

g. **System Name**: Provide the hostname of the system from where you want to uninstall the CA Service Management product or products.

4. Select any of the following products that you want to uninstall:
   - **SDM -** CA Service Desk Manager
   - **SLCM-** CA Service Catalog
   - **xFlow Interface**
   - **ProductUpdates**
     Enter product name to be uninstalled: **ProductUpdates**
5. Enter product to clean up updates: (sdm,slcm,xflow,chatserver,searchserver): For example, slcm
6. Enter update version to clean up: 17.4.2.0
7. The utility cleans up the database and removes the entries of the selected product installed on the server.

Installing CA Service Management 17.4.1.0

Install CASM patch 17.4.1.0

Based on your CA Service Management implementation, review the following topics for a successful installation.

- Install Prerequisites for CA Service Management 17.4.1.0
- Install CA Service Management 17.4.1.0 (Windows)
- Install CA Service Management 17.4.1.0 (Linux)
- Validate CA Service Management 17.4.1.0 Installation
- Post-Installation Tasks for CA Service Management 17.4.1.0
- Post-Installation Tasks - CA SDM Advance Availability 17.4.1.0
- Uninstall CA Service Management 17.4.1.0

Install Prerequisites for CA Service Management 17.4.1.0

Installation prerequisites for patch 17.4.1.0

Verify and complete the following prerequisites before installation of CA Service Desk Manager, CA Service Catalog, and CA Asset Portfolio Management.

Installation Prerequisites for CA Service Desk Manager

Review the following CA SDM installation prerequisites before you start the installation process:

> **NOTE**
> We recommend that you place the installer in a shorter directory path, containing no spaces, to avoid errors when running the setup file. For example, unzip the file at: C:\WIN_17_4_1_0.

- Ensure that CA SDM and xFlow Interface 17.4 is installed in your environment.
- Back up the system database (MDB). The system database (MDB) backup is mandatory and ensures safe data recovery if any problems are encountered during the installation process.
- Do not invoke ApplyPTF manually in your system while running the CA Service Management CA Service Management 17.4.1.0 installer.
- In the case of, CA Service Desk Manager - Conventional configuration, if the Primary and Secondary servers are on separate nodes, and the xFlow Interface is deployed on the Secondary Server, then the Database Client (Microsoft SQL Native Client or Oracle Database Client) is required on the Secondary Server for a successful installation.
- CA SDM must be installed for deploying the xFlow Interface. In the conventional configuration, it's recommended to install the xFlow Interface server on a secondary server. Advanced Availability users can install the xFlow Interface only on an application server.

IMPORTANT
After upgrading to CA Service Management 17.4.1.0, Maileater does not work with POP3 on a non-TLS port with Clear Text as a security level. This is a limitation of the upgraded Javamail (Third-party jar) used in connection to the mail server. Users are required to use **LOGIN** as a security level.

(Optional) Backing up the CR_Stored_Queries Table

CA Service Management provides a fix to address a vulnerability related to SQL injection. As part of this fix, **Cr_Stored_Queries** table is updated. If you have made customizations or changes to the Out-of-the-box (OOTB) stored queries, perform the below steps to take a backup before applying the CA Service Management patch:

1. Open a command prompt window and navigate to *$NX_ROOT\patches.* and create a folder **Backup_Data**. Navigate to this folder and take the Backup of the Cr_Stored_Queries table by running the following command:
   ```
   pdm_extract Cr_Stored_Queries> Cr_Stored_Queries_17.4_backup.dat
   ```
2. Verify that the **Cr_Stored_Queries_17.4_backup.dat** file is successfully created.
3. Now, proceed with CA Service Management 17.4.1.0 installation.
4. After applying the patch, if required based on your customizations, you can update the database files created in **Step 1**. Navigate to $NX_ROOT\patches\Backup_Data location and run the following command:

   ```
   pdm_load -f Cr_Stored_Queries_17.4_backup.dat
   ```

Installation Prerequisites for CA Service Catalog
Verify and complete the following prerequisites before proceeding with CA Service Catalog 17.4.1.0 installation:

- Ensure that CA Service Catalog 17.4 is installed in your environment.
- Back up the system database (MDB). The system database (MDB) backup is mandatory and ensures safe data recovery if any problems are encountered during the installation process.

Installation Prerequisites for CA Asset Portfolio Management
Consider the following prerequisites:

WARNING
CA Asset Portfolio Management (CA APM) architecture is highly database-driven and has no rollback mechanism for database-related changes. Hence, the uninstallation of the CA APM patch is not recommended and is not supported. The only way to revert the database changes, if required, is by restoring the database backup which can lead to data loss.

Verify and complete the following prerequisites before proceeding with the CA Asset Portfolio Management installation:

- Ensure that CA Asset Portfolio Management 17.4 is installed in your environment.
- Back up the system database (MDB). The system database (MDB) backup is mandatory and ensures safe data recovery if any problems are encountered during the installation process.

Install CA Service Management 17.4.1.0 (Windows)

Installing CASM 17.4.0.1 for windows.

CA Service Management 17.4.1.0 installer performs prerequisite checks and mandatory validations before installing the patch.

NOTE

- We recommend that you place the installer in a **shorter directory path**, containing no spaces, to avoid errors when running the setup file. For example, unzip the file at: C:\WIN_17_4_1_0
- In CA Service Desk Manager (SDM) Availability mode, the xFlow Interface can only be installed on an Application Server. If you have converted CA SDM conventional configuration to advanced availability configuration, and you have a previously installed version of xFlow Interface, ensure that you uninstall xFlow Interface and then, reinstall xFlow Interface on an Application server.

Perform the following steps to install 17.4.1.0 on Windows:

1. From the Broadcom Support portal, download the **WIN_17_4_1_0.zip** file.

   NOTE
   CA Service Management 17.4.1.0 (Windows) installer is provided as a ZIP file and not as a CAZ file. Extract/unzip the content of the ZIP file to your local folder.
2. Copy **WIN_17_4_1_0.zip** on all servers in your environment.
3. Extract the contents of the ZIP file to the folder named WIN_17_4_1_0.
4. In CA SDM conventional configuration, stop CA SDM services in your environment. In the case of Advance Availability configuration, stop CA SDM services only on the server on which CA SDM will be installed.
5. Log in to the server as a local admin to install the patch.
6. Navigate to **WIN_17_4_1_0** folder. Right-click **Setup.exe** and select **run as administrator**.
7. Review and Accept the License Agreement Information.
8. Select **Microsoft SQL Server** or **Oracle** as the database type in the Database Configuration screen.
   For Windows Microsoft SQL Server database, *mdbadmin* is the default user.
9. Provide the database details:
   a. If you have selected **Microsoft SQL Server**, provide the following field information:
      - **Database Server**: Enter the hostname of the database server.
        If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
      - **Database Name**: Enter the Database name (MDB), the name of the target DBMS. The default value is mdb.
      - **Database Port**: Provide the port identifier for the target DBMS.
      - **Database Server Instance**: The database instance name. Leave this field blank if you are using the default instance.
        If you are using the SQL named instance and a non-default port number, the **Database Server Instance** field should look like this: ***<database instance name>,<non-default port number>***
        For example:
        ```
        sdmcasmqa,10555
        ```
      - **Database Admin User**: For Microsoft SQL Server, **sa** is the default value.
        The **sa** user is the admin user with permission to create a user and schema.
      - **Database Admin Password**: Enter the database password of the user specified by a database admin user. No default value is assigned to this parameter.
      - **mdbadmin Password**: Enter the password for the user specified for the MDB admin user.
      - **Confirm mdbadmin Password**: Confirm the password.
   b. If you have selected **Oracle** database, provide the following field information:
      - **Database Server**: Enter the hostname of the Oracle database server.
        If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
      - **Oracle Service Name**: Enter the Oracle Service name.
      - **Listener Port**: Enter the listener port for the database.
      - **Net Service Name**: Enter the Net Service Name of the Oracle database where the database resides.

If the database is remote, use the Net Service Name that is defined in the Oracle client on the local computer. CA SDM has access to the database through a local installation of the Oracle client.

- **DBA User Name**: Provide the name of an Oracle user with DBA access. The default value for DBA user is **SYS**.
- **DBA User Password**: Enter the password for the DBA user.
- **mdbadmin Password**: Enter the password for MDB admin.
- **Confirm mdbadmin Password**: Confirm the password.
- **Tablespace Path (on DB Server)**: Provide the tablespace path on the database server.
- **Oracle Home Path**: Provide the Oracle Client 64-bit path. Provide Oracle client 32-bit path. For CA Asset Portfolio Management, provide the Oracle 32-bit client.
- **Data Tablespace Name**: Provide the name of the data tablespace. The default value is MDB_DATA.
- **Index Tablespace Name**: Provide the index Tablespace Name: The default value is MDB_INDEX.

Click **Next**.

10. On **Select the Products and Integrations** page, patch list is displayed. Select the required patch to install.

    WARNING

    CA Service Management does not support the uninstallation of the CA Asset Portfolio Management patch. Uninstalling the CA APM patch does not uninstall the MDB patch, as there is no rollback mechanism for the same. For more information, see Installation Prerequisites.

    a. **CA Service Desk Manager**
       - CA Service Desk Manager Patch
       - xFlow Interface Patch
       - Search Server Patch: Search server patch must be installed on a Server where an earlier version from previous releases is already installed.

    b. **CA Service Catalog**
       - CA Service Catalog Patch

    c. **CA Asset Portfolio Management**
       - CA Asset Portfolio Management Patch

    Click **Next**.

11. Review the Pre-Installation Configuration page. Click **Next**.

    NOTE

    Select the option to perform the automated post-install steps for CA SDM. If this option is not selected, you must manually perform the post-install steps for CA SDM. Note that for CA Service Catalog and CA Asset Portfolio Management, you need to perform the manual post-install steps.

12. Click **Install** to proceed with the installation. Review the Installation progress and when the installation is complete, click **Next.**
13. To review the installation logs, navigate to the path shown in the installation summary screen.
14. Click **Finish** to exit the installer
15. After installing the Key definition for "xflow" not found in the DITA map., start the **CA Key definition for "xflow" not found in the DITA map.** service from Windows Services (**Control Panel, Administrative Tools, Services**) as this service is not started by default.
16. After installing the search server, if the **Search Server** service is not started by default or if it stops, start the services from **Control Panel, Administrative Tools, Services.**
17. Validate the Installation.
18. Complete the Post-Installation Tasks.

For more information about how to configure cache size, see Perform initial load and refresh index on the search server.

Install CA Service Management 17.4.1.0 (Linux)

NOTE
In CA SDM Advanced Availability mode, the xFlow Interface can only be installed on an Application Server. If you have converted CA SDM conventional configuration to advanced availability configuration, and you have a previously installed version of xFlow Interface, ensure that you uninstall xFlow Interface and then reinstall xFlow Interface on an Application server.

CA Service Management 17.4.1.0 installer performs prerequisite checks and mandatory validations before installing the patch.

IMPORTANT
We recommend that you place the installer in a **shorter directory path**, containing no spaces, to avoid errors when running the setup file. For example, unzip the file at: *<root>\casm_dvd.x86Linux\<installer_file>*

Perform the following steps to install 17.4.1.0 on Linux:

1. From the Broadcom Support portal, download the **LNX_17_4_1_0.zip** file.
2. a. Unzip the LNX_17_4_1_0.zip file. After you extract the zip file, ensure the **CASM_17.4.1.0.tar** file is available. Run the following command to untar the **CASM_17.4.1.0.tar** file:
   ```
   $ tar -xvf CASM_17.4.1.0.tar
   ```
   OR
   b. Run the following command:
   ```
   tar -xvzf LNX_17_4_1_0.zip
   ```
3. Set CA SDM bin and library environment variables.
   For example:
   ```
   export PATH=/opt/CA/ServiceDeskManager/bin:$PATH
   export LD_LIBRARY_PATH=/opt/CA/ServiceDeskManager/lib:$LD_LIBRARY_PATH
   ```
4. In SDM conventional configuration, stop CA SDM services in your environment. In the case of Advance Availability, stop CA SDM services only on the server on which CA SDM will be installed.
5. Run the setup file from the casm_dvd.x86Linux folder to install the application(s) using the following command:
   ```
   ./setup.sh
   ```
6. Review and accept the license agreement terms.
7. In Database Configuration, select **Oracle** as the database type. Ensure that you have installed the Oracle database client in your system.
   Provide the following Oracle database configuration details:
   - **Database Server**: Enter the hostname of the Oracle database server.
     If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
   - **Oracle Service Name**: Enter the Oracle Service name.
   - **Listener Port**: Enter the listener port for the database.
   - **Net Service Name**: Enter the Net Service Name of the Oracle database where the database resides.
     If the database is remote, use the Net Service Name that is defined in the Oracle client on the local computer. CA SDM has access to the database through a local installation of the Oracle client.
   - **DBA User Name**: Authenticates the name of the user with DBA access. The default value for DBA user is **SYS.**
   - **DBA User Password**: Authenticates the password for DBA user.
   - **mdbadmin Password**: Specify the password for the mdbadmin user.
   - **Confirm mdbadmin Password**: Confirm the mdbadmin user password.
   - **Tablespace Path (on DB Server)**: Enter the tablespace directory on the database server.
   - **Oracle Home Path**: Provide the Oracle Client 64-bit path. Provide Oracle client 32-bit path for fresh installation
   - **Data Tablespace Name**: Enter the tablespace name. The default value is **MDB_DATA**.
   - **Index Tablespace Name**: Enter the index tablespace name: The default value is **MDB_INDEX**.
8. Click **Next.**

9. On **Select the Products and Integrations** page, patch list is displayed. Select the required patch to install.
   a. **CA Service Desk Manager**
      - CA Service Desk Manager Patch
      - xFlow Interface Patch
      - Search Server Patch: A search server patch must be installed on a server where an earlier version from previous releases is already installed.
10. Click **Next**.
11. Review the Pre-Installation Summary page. Click **Next.**
12. Click **Install** to proceed with the installation. Review the Installation progress and when the installation is complete, click **Next.**
13. To review the installation logs, navigate to the path shown in the installation summary screen.
14. Click **Finish** to exit the installer.
    If patch installation fails for a product, retry the installation by using the **Retry** option in the wizard where it fails. If retry does not work, exit the installer and relaunch it. If the problem persists, contact Broadcom Support.
15. After installing XFlow Interface, start the**CA XFlow Interface**service. Navigate to /opt/CA/xFlow/Apps and run the following command from a command prompt window:
    ```
    ./xFlowServer start
    ```
    To stop the server, run the following command:
    ```
    ./xFlowServer Stop.sh
    ```
16. Navigate to /opt/CA/SC/SearchServer/elasticsearch/bin and run the following command from the command prompt window:
    ```
    ./ServicesStart.sh
    ```
    NOTE
    If you get the below message while restarting the search server service, ignore it and continue:
    ```
    groupadd: group 'elasticsearch' already exists
    useradd: user 'elasticuser' already exists
    runuser: warning: cannot change directory to /home/elasticuser: No such file or directory
    OpenJDK 64-Bit Server VM warning: Option UseConcMarkSweepGC was deprecated in version 9.0 and
     will likely be removed in a future release.
    ```
    To stop the server, run the following command:
    ```
    /ServicesStop.sh
    ```
17. Validate the Installation.
18. Complete the Post-Installation Tasks.

After installing the XFlow Interface and Search Server, you can configure the cache size for XFlow Interface. For more information about how to configure cache size, see Perform initial load and refresh index on the search server.

Validate CA Service Management 17.4.1.0 Installation

Validate CASM 17.4.1.0. insta.llation

To verify the successful installation of CA Service Management 17.4.1.0, validate the installation for the corresponding applications in the following location. You can also navigate to the root folder where you have locally installed the application and select the "env.info" file for validating your installation:

- **CA ITAM:**
  NOTE
  CORA version: 17.4.0.57

  - C:\Program Files (x86)\CA\ITAM\env.info
  - product=ITAM
  - major=17
  - minor=4
  - cum=1
  - roll-up=0

    NOTE
    CA CASM uses Apache Tomcat version 9.0.71.
- **CA Service Catalog:**
  - %USM_HOME%*CA_SLCM_<hostname>.his* file
  - %USM_HOME%\view\webapps\usm\locale\about_info.xml
  - %USM_HOME%\build.xml
  - %USM_HOME%\env.info
- **xFlow Interface:**
  - *C:\Program Files\CA\xFlow\<hostname>.his* file
  - NX_ROOT\env.info
  - NX_ROOT\version.xml
- **CA SDM:**
  - *C:\Program Files (x86)\CA\Service Desk Manager\<hostname>.his* file
  - NX_ROOT\pdmconf\version
  - NX_ROOT\env.info
  - HELP, ABOUT

    NOTE
    For CA SDM, the Binary and Language Combo patches are installed irrespective of the environment. The language patch for the corresponding CA SDM environment is installed.

    To validate the successful installation of the patches, follow these steps:

    1. Open the <sdm_installation>\<hostname>.his file in a text editor.
    2. Search for the instances of "gol-94". In a successful install, you should come across these three entries (Binary, Language Combo, and corresponding language patch).

Application	Windows	Linux
CA Asset Portfolio Management	T6EE815	NA
CA Service Catalog	T6D91741	NA
CA Service Desk Manager - Binary	gol-94_cum_C	gol-94_x86_64Linux_cum_C
CA Service Desk Manager - Languag504e Comb	gol-94_langcombo	gol-94_langcombo
CA Service Desk Manager - English	gol-94_en_US	gol-94_x86_64Linux_en_US
CA Service Desk Manager - German	gol-94_de_DE	gol-94_x86_64Linux_en_DE
CA Service Desk Manager - Spanish	gol-94_es_ES	NA
CA Service Desk Manager - French	gol-94_fr_FR	gol-94_x86_64Linux_fr_FR
CA Service Desk Manager - Italian	gol-94_it_IT	NA
CA Service Desk Manager - Japanese	gol-94_ja_JP	gol-94_x86_64Linux_ja_JP
CA Service Desk Manager - Brazilian Portuguese	gol-94_pt_BR	NA

Post-Installation Tasks for CA Service Management 17.4.1.0

After successful installation of CA Service Management 17.4.1.0, perform the following post-installation steps.

CA Service Desk Manager

Manual Steps required for Automated Post-Installation

CA Service Management provides an option in the installer to automate the post-install tasks.
If you have selected **Yes** to automate the post-installation steps while running the CA Service Management Installer Wizard, you are not required to perform any manual post-install steps for CA Service Desk Manager (SDM).

If you have selected **No** for automating the post-install steps while running the CA Service Management Installer wizard, perform the below manual post-install tasks as per your deployment requirements:

Manual Post-Install Tasks After Installing CA Service Desk Manager (SDM) 17.4.1.0 on CA SDM 17.4.

Perform the following steps:

1. Stop CA SDM services.
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0.
3. Open a command prompt window and run the following command:
   ```
   pdm_configure
   ```

Manual Post-Installation Task for CMDB

> **NOTE**
> For Oracle 21c, set the **isClusterEnv** setting to "true" in the application.conf file of CMDB viz and restart the CA SDM Services.:
>
> - For Windows: <SDM installed location>\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\WEB-INF\classes\application.conf)
> - For LINUX: <SDM installed location>/bopcfg/www/CATALINA_BASE_VIZ/webapps/CMDBVisualizer/WEB-INF/classes/application.conf).

If you are applying CA Service Management 17.4.1.0 in an SSL environment, CA SDM Administrators must perform the following post-installation steps for CMDB Visualizer Interface to open using the **HTTPS** protocol when launching from CA SDM UI:

1. In the NX.env file, change the **NX_CMDB_VISUALIZER** entry to provide **HTTPS protocol** and **HTTPS port**.
   For example: @NX_CMDB_VISUALIZER=https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer
2. In **casm.conf.js** file from NX_root\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\conf, provide the HTTPS as protocol and HTTP port.
   For example:

```
sdm: {

      vizserver: 'https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer/viz',
      server: 'https:// sdmurl:<SDM_SSL_PORT>/CAisd/pdmweb.exe'
    }
```

Manual Post-Installation Task for OAuth 2.0 (Maileater/ Mailer)

After generating an access token for OAuth 2.0 to connect to a mail server, perform the mandatory post-install tasks on all servers in your environment:

1. Create a backup of web.xml from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF in your local directory to perform backout if required.
2. Open the web.xml file from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF.
3. Search for the following statement:
   ```
   <!--USD servlet mappings start-->
   ```
4. Copy and paste the following configuration under the statement:

   ```
   <servlet>
     <servlet-name>OAuthProcessor</servlet-name>
     <servlet-class>com.broadcom.oAuth.OAuthProcessor</servlet-class>
    </servlet>
   ```
5. Copy and paste the following configuration after any <servlet-mapping>:

   ```
    <servlet-mapping>
      <servlet-name>OAuthProcessor</servlet-name>
      <url-pattern>/OAuthProcessor</url-pattern>
    </servlet-mapping>
   ```
6. Save and close the web.xml file.
7. For OAuth servlet preservation in a primary and secondary environment, the secondary server must also have the OAuth processor entries in the following file: *<NX_ROOT>\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF\web.xml.last*.
8. Repeat Steps 1-8 to update **web.xml.tpl** file in the NX_ROOT\samples\pdmconf\web.xml.tpl folder to avoid losing these changes when pdm_configure is run at a later point of time.
9. Run the following command:
   ```
   pdm_tomcat_nxd -c STOP command
   ```
10. Wait for some time (a minute or so) and run the following command:
    ```
    pdm_tomcat_nxd -c START command
    ```

For more information, see Define a mailbox.

Manual Post-Installation Task for Excel file (.xlsx) export*

1. Execute the following command:
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst
   ```
2. Execute the following command to add the new NX variable to NX.env.tpl file.
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst -t
   ```
3. Restart CA SDM services.

(Optional) Manual Post-Installation Task for Knowledge Document export with German fields

If you are exporting Knowledge Document with German Fields (included in the export field list), then perform the below post-install steps for successful export of knowledge documents:

1. Take the backup of NX_ROOT\NX.env file (NX_ROOT is the installation directory for CA Service Desk Manager).
2. Open NX_ROOT\pdmconf\NX.env_nt.tpl file using any text editor and take the value of NX variable @NX_KEIT_AVAILABLE_FIELDS which is COPIED_DATA and update Nx.env in NX_ROOT by executing below command:
   ```
   pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA>  -a  pdm_option.inst
   ```
   Where COPIED_DATA is the value copied from NX.env_nt.tpl (step 2).

For example, " pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <provide the exact string here> -a pdm_option.inst "
Note to avoid losing the updated changes when you run pdm_configure, execute the above command with the '-t' flag as follows:

```
pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA> -a pdm_option.inst -t
```

(Optional) Handle HTML Mails that contains Tables in MailEater

Java based Maileater already has the ability to process the emails that were sent in plain format or HTML format, to support the processing of emails that contain Tables in the HTML body. But the format in which the email was sent is lost when received by Maileater.

1. Run the following commands from NX_ROOT/bin from the command prompt/ terminal:

```
pdm_options_mgr -c -s NX_ENFORCE_HTML_EMAIL_PARSING -v "Yes" -a pdm_option.inst
pdm_options_mgr -c -s NX_ENFORCE_HTML_EMAIL_PARSING -v "Yes" -a pdm_option.inst -t
```

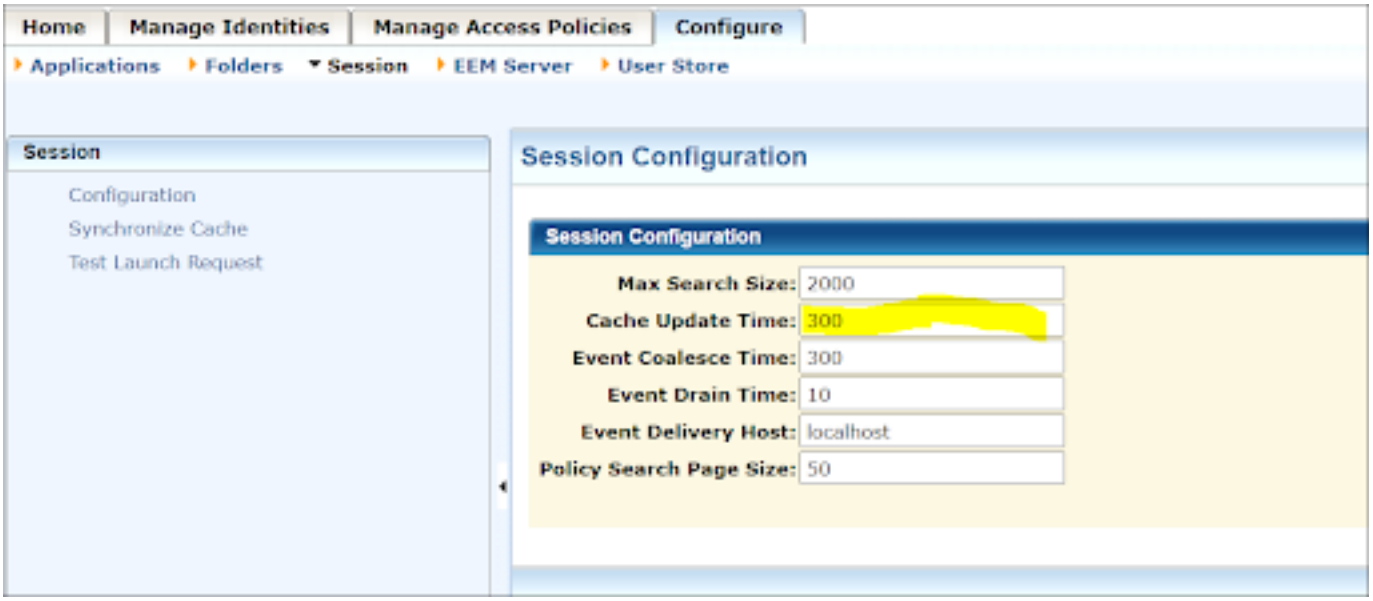
2. Restart CA SDM and XFlow applications.

(Optional) Manual Post-Installation Task for EEM Cache Timeout Configuration Update

You may face issues while logging in to EEM or SDM using the EEM user accounts due to a session cache timeout mismatch between EEM and SDM.

You can update the EEM cache timeout configuration by adding a new NX variable, "NX_EEM_CACHE_UPDATE_TIME" in SDM.

Keep the value the same as the EEM cache timeout configuration value mentioned in the following screenshot.



To set the EEM cache update time to 300 seconds, perform the following steps:

To add the variable in the **nx.env** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst
```

To add the variable in the **nxenv_nt.tpl** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst -t
```

Service Point

After successful installation of Service Point, refresh index and data on the Search Server.

NOTE
In case of Linux Search Server environment, ensure to run the following commands on the terminal:

```
sysctl -w vm.max_map_count=262144
ulimit -Hn 65535
```

(Optional) Manual Post-Installation Task for Rebuild Index

If you have upgraded or are planning to install CA EEM 12.6 with Service Point and CA Service Catalog, ensure that you rebuild the CA Service Catalog index after completing the CA EEM upgrade/installation process. For more information, see Perform Initial Load and Refresh Index Data on Search Server.
Run the following command to rebuild the index:

```
$ES_HOME/search/bin/pdm_es_rebuild_index.bat index=catalog_index
```

CA Service Catalog

Perform the following CA Service Catalog post-installation steps as per your environment requirements:

- Post-Installation Tasks after installing CA Service Catalog 17.4.1.0 on CA Service Catalog 17.4
- gMSA Authentication

Post-Install Tasks After Installing CA Service Catalog 17.4.1.0 on CA Service Catalog 17.4

Review and perform the post-installation steps based on your deployment requirements.

NOTE
If any SSL certificate(s) chain was imported to the Oracle Java's JAVA_HOME\lib\security\cacerts Keystore, those certificate(s) chain again need to be exported explicitly and imported properly into the new AdoptOpenJDK's %USM_HOME%\embedded\jdk\lib\security\cacerts Keystore.

Follow these steps

- (Optional) If Service Catalog is accessed from applications like Service Point, SharePoint and so on, then you must perform this step. Open the "**web.xml**" file located in *USM_HOME%\view\webapps\usm\WEB-INF* and update the respective parameter values for the parameter name "cors.allowed.origins" and "cors.support.credentials". For example, if you are using Catalog Widgets in Service Point
- (Optional) Unable to approve any workflow requests through email. The email sent for approvals is corrupted with special characters. Follow the KB article steps.

(Optional) Approve workflow requests through email

The workflow email sent for approvals is corrupted with special characters. To approve any workflow requests through email, follow the KB article steps.

gMSA Authentication Post-Install Steps for CA Service Catalog (Automated steps)

CA Service Catalog provides automated post-installation steps for gMSA Authentication. Review the following before you proceed:

- Prerequisites and Considerations
- How to Automate gMSA Post-Installation Steps
- Unconfigure gMSA

Prerequisites and Considerations:

Ensure that you have taken care of the below considerations on all nodes of CA Service Catalog where you want to automate the gMSA post-installation procedure.

1. If the accounting service is available, perform the automated post-install steps by updating serviceName to ServiceAccounting and providing the appropriate gMSA account name in the **gmsa.properties** file.

2. If the filestore path is external to the server from where the **GMSAPostInstall.bat** is running, ensure to provide appropriate access to the server for setting the gMSA account permissions.
3. While preparing the properties file, check for any copy/paste issues, special characters, spaces, and so on.

How to Automate gMSA Post-Installation Steps

Perform the following steps to enable the batch file **GMSAPostInstall.bat** and to automate the gMSA post-installation steps:

1. Create a gMSA account for CA Service Catalog.
2. Ensure that appropriate DB (database) permission is given to the newly created account(s).
3. Enter the below values in the **gmsa.properties** file and save it.

   NOTE
   A template file (**gmsa.properties**) is added in the common installer in *"\filestore\SLCM\gMSAProperties"* . Optionally. use this file as a template file to fill gmsa properties. This can be copied to "USM_HOME/scripts/GMSA" where Catalog is installed.

   NOTE
   All values are mandatory. Make sure that there are no leading or trialing spaces while providing these values.

   a. **serviceName**: Application service name to enable GMSA. In the case of Catalog, it will be ServiceCatalog and for accounting, it will be ServiceAccounting.
   b. **folderName**: Absolute path of the application installation directory.
   c. **gMSAName**: Name of the GMSA service account (domain\accountName$).
   d. **filestorePath**: Shared network path (\\Server_name\filestore) or local filestore path
4. Navigate to %USM_HOME% and launch **usm.cmd**
5. Navigate to %USM_HOME%/scripts/GMSA/ folder and run the "GMSAPostInstall.bat" in command prompt.

   NOTE
   If the command executes without any error, "Post-installation steps for GMSA executed successfully" message is displayed with exit code 0.
   If there are errors, the error message is shown with a non-zero exit code. Correct the error to continue and perform step **5** again.

Unconfigure gMSA for CA Service Catalog

To disable or unconfigure gMSA for CA Service Catalog, follow the steps below:

1. Replace **config.properties** file from*%USM_HOME%*\ folder.
2. Revert back all the folder permissions which were assigned to domain users.
3. Start Catalog and Accounting Services with the local system account.

   NOTE
   In the case of load balancers, repeat steps 1-3 to unconfigure gMSA on all Catalog servers.

CA Asset Portfolio Management

Manual Post-Installation Task for Removing jQuery, log4net, and log4j Vulnerabilities

After installing 17.4.1.0, delete jquery-1.3.2.min.js, jquery-ui-1.7.2.custom.min.js , log4net.dll, and log4j from their respective installed folders.

NOTE
Perform the below steps if you have not done this earlier.

Perform the following steps:

1. Log on to CA APM Web Server, navigate to the <ITAM_install_dir>\ITAM\Web Server\castylesr5\javascript.
   Example: C:\Program Files (x86)\CA\ITAM\Web Server\castylesr5\javascript
   Locate and delete the jquery-1.3.2.min.js and jquery-ui-1.7.2.custom.min.js files from the directory.

2. Navigate to the ITAM_install_dir\CA\ITAM\Data Importer.
   Locate and delete the log4net.dll file from the directory.
3. Navigate to the C:\Program Files (x86)\CA\SC\CORA.
   Locate and delete the log4j-1.2.15.jar, log4j.properties, and ojdbc6.jar files from the directory.
4. Navigate to the ITAM_install_dir\CA\ITAM.
   Backup the following folders and manually delete them:
   - CDB
   - MDB
   - jre8

     NOTE
     jre8 folder has to be deleted manually for every RU installation.
5. Perform an IISRESET and delete the browser's cache to load the upgraded files.

Post-Installation Tasks - CA SDM Advance Availability 17.4.1.0

post install tasks for casm ru1

The installer wizard provides an option in the installer to automate the post-install tasks. Perform the following post-installation tasks for CA SDM Advanced Availability Configuration.

For the Advanced Availability configuration, the following host names are considered as an example:

- The background server hostname is sdmcahost1.
- The standby server hostname is sdmcahost2.
- The application server hostname is sdmcahost3.

Manual Steps required for Automated Post-Installation

If you have selected **Yes** for automating the post-install steps while running the installer wizard, then perform manually the following post-install tasks:

Manual Post-Install Tasks - After Installing CA Service Desk Manager (SDM) 17.4.1.0 on CA SDM 17.4 GA

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Start the CA SDM service on the standby server (sdmcahost2). Open the command prompt and run the following command:
   ```
   pdm_server_control -b
   ```
3. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server.
4. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control -b
   ```
5. Clear the browser cache by the following command:
   ```
   pdm_webcache -H
   ```
6. Ensure that the patch is applied on all the application servers and the remaining standby servers if any. If you selected YES to automate the post-install steps, then there are no more manual post-install steps to be performed.

Manual Steps required for Non-Automated Post-Installation

If you have selected **No** for automating the post-install steps while running the CA Service Management installer wizard, perform manually the below appropriate post-install task as per your deployment requirements.

Post-Install Tasks After Installing CA SDM 17.4.1.0 on CA SDM 17.4 GA for Advanced Availability Configuration

1. Ensure that the patch is applied on the standby server (sdmcahost2).
2. Navigate to NX_ROOT/site/config.properties file and ensure the value of resources.created parameter is set to 0.
   Open a command prompt and run the following command:

```
pdm-configure
```

On the installer configuration page, before clicking Finish, uncheck the Start service when completed checkbox. Failure to uncheck the box may result in the loss of custom schema changes and configurations that were changed on the server.

3. Promote the standby server as the new background server by performing the following steps. To suppress version control on the standby server and later on the background server, run the following command on each server:
   ```
   pdm_server_control -v
   ```
4. Start SDM service on the standby server. Run the following command:
   ```
   pdm_server_control -b
   ```
5. Ensure that the patch is applied on the original background server (sdmcahost1) which is now a standby server, and perform the following steps. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
   ```
   pdm_configure
   ```
6. Start the CA SDM service on the standby server (sdmcahost1). Run the following command:
   ```
   pdm_server_control –b
   ```
7. Ensure that the patch is applied on all application servers and the remaining standby server if any. Perform the following steps on all application servers and standby servers.
8. Navigate to NX_ROOT/site/config.properties file and ensure that the value of resources.created parameter is set to 0. Run the following command:
   ```
   pdm_configure
   ```
9. Clear the browser cache by running the following command:
   ```
   pdm_webcache -H
   ```
10. Execute the following command.
    ```
    pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst
    ```
11. Execute the following command to add the new NX variable to NX.env.tpl file.
    ```
    pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst -t
    ```
12. Restart CA SDM Services.

Uninstall CA Service Management 17.4.1.0

uninstall CASM 17.4.1.0

Review the following topics to uninstall CA Service Management 17.4.1.0.

> **NOTE**
>
> Uninstalling CA Service Management 17.4.1.0 will revert to CA Service Management 17.4 version on which 17.4.1.0 is installed.
>
> **Example**: If CA Service Management 17.4.1.0 is installed on 17.4, uninstalling 17.4.1.0 will revert to CA Service Management 17.4.

Uninstall CA SDM 17.4.1.0 (Windows)

Perform the following steps to uninstall CA SDM 17.4.1.0 on Windows:

1. Log in to CA SDM as an Administrator.
2. Shut down CA Service Desk Manager Server.
3. Navigate to **<NX_ROOT>\bin** and run the following command to backup the unsupported jars from the older versions.
   ```
   pdm_perl pdm_restore_files.pl gol-94_tool ..\PATCHES\DELIVERABLES_REMOVAL.TXT
   ```
4. Navigate to the location *where you have extracted the CA Service Management Installer zip package*.
5. Locate *CASM-17.4.1.0-CommonPatchInstaller-Win\filestore\utils\ApplyPTF* folder.
6. Run the **APPLYPTF.exe**.

7. Select **Backout PTF** from the local or remote node option.
8. Click **Next**.
9. Enter the patch number for the language that is installed in your environment:
   - **Binary**: gol-94_cum_C
   - **Language Master**: gol-94_langcombo
   - **English**: gol-94_en_US
   - **French**: gol-94_fr_FR
   - **German**: gol-94_de_DE
   - **Spanish**: gol-94_es_ES
   - **Italian:** gol-94_it_IT
   - **Japanese**: gol-94_ja_JP
   - **Brazilian Portuguese**: gol-94_pt_BR
10. Click **Next** to continue backing out the specific patch.
11. Repeat steps 6-9, to back out the language English (gol-94_en_US) patch.
12. Repeat steps 6-9 to back out the language master (gol-94-langcombo) patch.
13. Repeat steps 6-9 to back out the binary (gol-94_cum_C) patch.
14. Reboot the server.
15. Repeat steps **6-12** on all standby, application, or secondary servers in your environment.
16. Restore the MDB database to the database backup that you took before applying the patch.
17. Navigate to ***<NX_ROOT>\site\config.properties*** file and ensure that the value of ***resources.created*** parameter is set to **0.** This must display as ***resources.created=0*** in the file.
18. Open a command prompt and run the following command:
    ```
    pdm_configure
    ```

Uninstall CA SDM 17.4.1.0 (Linux)

Perform the following steps to uninstall CA SDM 17.4.1.0 on Linux:

1. Login to CA SDM as the root user.
2. Shut down the CA Service Desk Manager Server.
3. Navigate to **<NX_ROOT>/bin** and run the following command to backup the unsupported jars from the older versions.
   ```
   ./pdm_perl pdm_restore_files.pl gol-94_tool deliverables_removal.txt
   ```
4. Navigate to the location where you have extracted the CA SDM Installer **tar.gz** package.
5. Navigate to the *casm_dvd.x86Linux \filestore\utils\ApplyPTF* folder.
6. The following binary/language-independent and language-dependent patches are supported:
   - Binary name: gol-94_x86_64Linux_cum_C
   - Language Master: gol-94_langcombo
   - English: gol-94_x86_64Linux_en_US
   - German: gol-94_x86_64Linux_de_DE
   - French: gol-94_x86_64Linux_fr_FR
   - Japanese: gol-94_x86_64Linux_ja_JP
7. Execute the following command to remove the binaries or the language-dependent patches. Replace the <patch_name> with the binary or language-dependent patch name:
   ```
   ./applyptf.<platform> -b <patch_name>  -e <NX_ROOT>
   ```
   Here, <platform> specifies the operating system and <NX_ROOT> specifies the CA SDM installed location.
   For example:
   ```
   ./applyptf.linux -b gol-94_x86_64Linux_cum_C  -e /opt/CA/ServiceDeskManager
   ```
8. Repeat step 6 to back out the language English (gol-94_x86_64Linux_en_US) patch.

9. Repeat step 6 to back out the language master (gol-94_langcombo) patch.
10. Repeat step 6 to back out the binary (gol-94_x86_64Linux_cum_C) patch.

    NOTE
    During uninstallation, if you get any errors, ensure that the following steps are performed:

    1. Copy the subfolders from <NX_ROOT>/replaced/<gol-94_x86_64Linux_cum_C> and place it into the <NX_ROOT> folder.
    2. Copy the subfolders from <NX_ROOT>/replaced/<gol-94_x86_64Linux_en_US> and place it into the <NX_ROOT> folder.
    3. Open **.his** file under the <NX_ROOT> folder and delete the entries related to the current patch and save it.
11. To restore MDB database, revert to database backup that you took before applying the patch.
12. Navigate to ***<NX_ROOT>/site/config.properties*** file and ensure that the value of ***resources.created*** parameter is set to **0.** This must display as ***resources.created=0*** in the file.
13. Open a command prompt and run the following command:

    ```
    pdm_configure
    ```

    NOTE
    If you are unable to access the SDM URL, follow these steps:

    1. Stop the SDM Services.
    2. Navigate to the <NX_ROOT>/bopcfg/www/CATALINA_BASE folder and delete the work folder.
    3. Start the SDM Services.

Restore Visualizer Files (Windows)

NOTE
After uninstalling CA SDM, CMDB Visualizer does not function automatically as the uninstallation process will not restore the previous visualizer files . To restore, perform the below steps manually:

1. Navigate to the where you have installed Visualizer: <NX_ROOT>\bopcfg\www\CATALINA_BASE_VIZ\webapps.
2. Remove the CMDBVisualizer folder.
3. Copy the **Visualizer_17.4.1.0.OLD** folder from<NX_ROOT>*REPLACED* and paste it in <NX_ROOT>*bopcfg**www* *CATALINA_BASE_VIZ**webapps*. Now, rename the **Visualizer_17.4.1.0.OLD** to **CMDBVisualizer**.
4. Start the SDM Service.

CMDB Visualizer works as expected.

Restore Visualizer Files (Linux)

NOTE
After uninstalling CA SDM, CMDB Visualizer does not function automatically as the uninstallation process will not restore the previous visualizer files. To restore the files, perform the below steps manually:

1. Navigate to the where you have installed Visualizer: <NX_ROOT>/bopcfg/www/CATALINA_BASE_VIZ/webapps.
2. Remove the CMDBVisualizer folder.
3. Copy the **Visualizer_17.4.1.0.OLD** folder from <NX_ROOT>*/replaced* and paste it into <NX_ROOT>*bopcfg**www* *CATALINA_BASE_VIZ**webapps*. Now, rename the **Visualizer_17.4.1.0.OLD** to **CMDBVisualizer**.
4. Start the SDM Service.

CMDB Visualizer works as expected.

Uninstall the xFlow Interface 17.4.1.0 (Windows)

Perform the following steps to uninstall the xFlow Interface (Windows):

1. Click **Start**, **Control Panel**, **Uninstall a Program** to uninstall xFlow Interface 17.4.1.0.
   Or
2. Navigate to where you have installed the xFlow Interface (for example, <NX_ROOT>*/CA/xFlow)*.
3. Select xFlow Interface **17.4.1.0_installation** and click **Change xFlow Interface 17.4.1.0 Installation.exe** to complete the uninstallation process.
4. If you are using Microsoft SQL Server Database, run the following query:
   ```
   update ofProperty set propValue='select c.access_token from casm_active_sessions c WHERE
    UPPER(CONVERT(varchar(32), c.contact_uuid ,2)) = UPPER(?)' where name = 'jdbcAuthProvider.passwordSQL'
   ```
5. If you are using Oracle Database, run the following query;
   ```
   update ofProperty set propValue='select c.access_token from casm_active_sessions c WHERE
    UPPER(CAST( c.contact_uuid AS varchar2(32))) = UPPER(?)' where name = 'jdbcAuthProvider.passwordSQL'
   ```

Uninstall the xFlow Interface 17.4.1.0 (Linux)

To uninstall the xFlow Interface, complete the following steps on Linux :

1. Navigate to where you have installed the xFlow Interface server.
   For example:
   /opt/CA/_xFlow Interface_installation
2. Execute the following command to uninstall the xFlow Interface 17.4.1.0:
   ```
   ./Change xFlow_Interface 17.4.1.0 Installation
   ```
3. Click **Finish** to complete the uninstallation process.
4. Run the following query for Oracle Database:
   ```
   update ofProperty set propValue='select c.access_token from casm_active_sessions c WHERE
    UPPER(CAST( c.contact_uuid AS varchar2(32))) = UPPER(?)' where name = 'jdbcAuthProvider.passwordSQL'
   ```

Uninstall CA Service Catalog 17.4.1.0

To uninstall CA Service Catalog, perform the following steps:

1. Click **Control Panel**, **Programs and Features**.
2. Double-click the CA Service Catalog 17.4.1.0 and follow the uninstallation wizard instructions.

Uninstall CA Asset Portfolio Management

CA Asset Portfolio Management (CA APM) architecture is highly database-driven and has no rollback mechanism for database-related changes. Hence, the uninstallation of the CA APM patch is not recommended and is not supported. The only way to revert the database changes, if required, is by restoring the database backup which can lead to data loss.

Run the Database Cleanup Utility for CA Service Management 17.4.1.0

During the installation of CA Service Management products, each of the installed product is registered in a common mdb table that is reflected in the deployment summary of the selected environment. These entries must be cleared in the following cases:

- Uninstallation of any CA Service Management products
- Moving Database or migration from one server to another

Run the CA Service Management database cleanup utility on a Windows or a Non-Windows environment. The database cleanup utility removes the entries and adds to the following tables:

Table Name from where the entries are removed	Table Name to which the removed entries are added
al_cdb_componentinstallstate	al_cdb_comp_installstate_bkup
al_cdb_configurationparameters	al_cdb_config_params_backup

NOTE

On a Non-Windows Platform, before running the **DBCleanup.sh** Utility, you must export the Oracle Environment variables on the terminal.

Perform the following steps:

1. Navigate to *DVD1\DBCleanupUtility* directory.
2. For Windows, double-click the **DBCleanupUtility.bat** to launch the DBCleanup utility.
   On a Non-Windows platform, you must run the **DBCleanupUtility.sh** file.
3. Enter the following details:
   a. **Database Server Type**: Specify the type of server. Microsoft SQL or Oracle.
   b. **Database Host Name:** Enter the host name of the server where the database resides.
   c. **Database User Name**:
      - **SQL Server**: Enter the privileged user detail that has permission to create and modify the schema.
      - **Oracle**: Provide the mdbadmin user name.
   d. **Database Password**:
      - **MS SQL**: Enter the database password used by the database admin user.
      - **Oracle**: Enter the password for the Oracle database user.
   e. **Database Port Number:** Provide the database port number.
   f. **Enter Database Name :** Enter the database name. For example, mdb.
   g. **System Name**: Provide the hostname of the system from where you want to uninstall the CA Service Management product or products.
4. Select any of the following products that you want to uninstall:
   – **SDM -** CA Service Desk Manager
   – **SLCM-** CA Service Catalog
   – **xFlow Interface**
   – **ProductUpdates**
   Enter product name to be uninstalled: **ProductUpdates**
5. Enter product to clean up updates: (sdm,slcm,xflow,chatserver,searchserver): For example, slcm
6. Enter update version to clean up: 17.4.1.0
7. The utility cleans up the database and removes the entries of the selected product installed on the server.

CA Service Management 17.4 Solution Architecture

CA Service Management is a seamless integration of CA Service Desk Manager, CA Service Catalog, and CA IT Asset Manager products. The solution consists of a collaborative platform and offers self-service and customization capabilities. CA Service Management delivers a modern, collaborative user experience interface to access knowledge, collaborate, resolve issues, request services, and manage IT assets.

The following diagram illustrates high-level CA Service Management solution architecture and the capabilities offered by the solution:

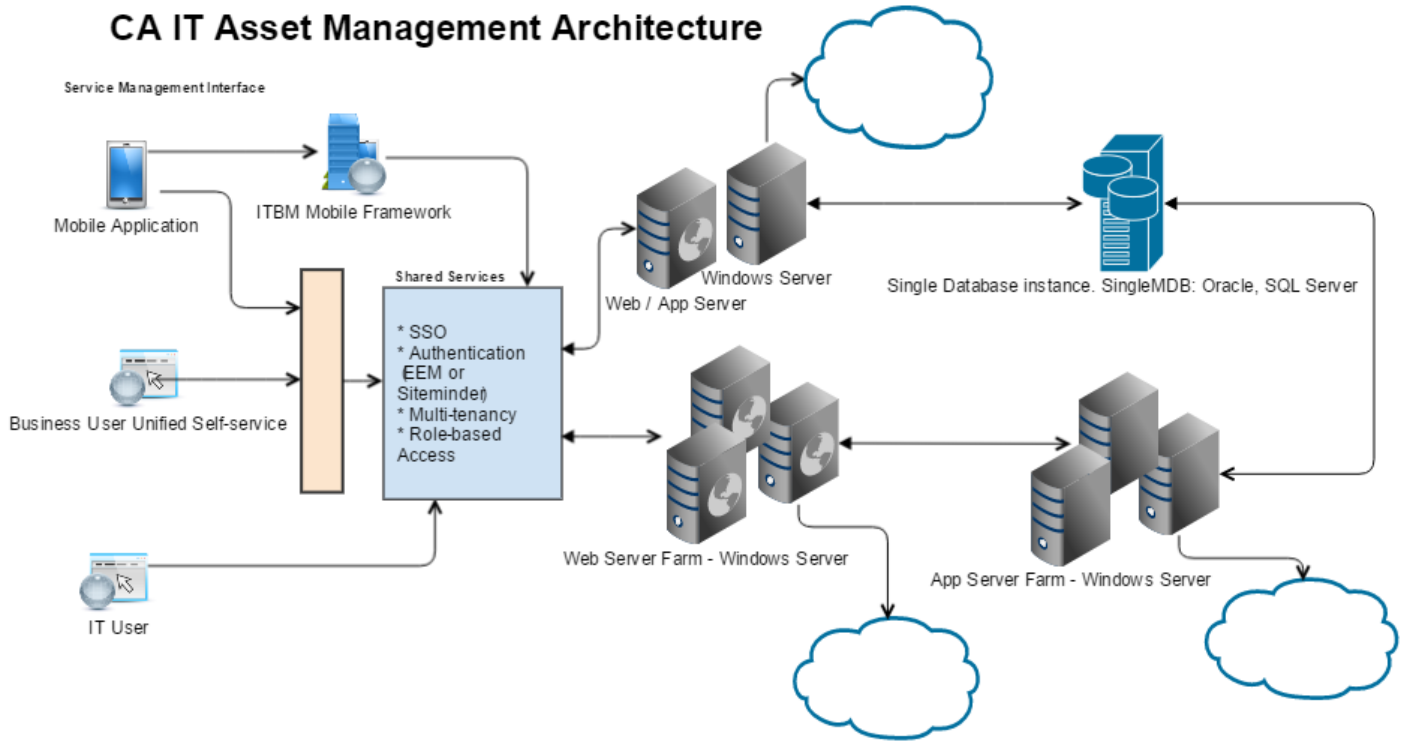
CA Asset Portfolio Management Product Architecture

CA IT Asset Management supports two types of architectures.

NOTE
CA Asset Portfolio Management(formerly 'ITAM') must be deployed on a separate server as part of Service Management Solution Deployment. Service Management supports 64-bit architecture where as Asset Portfolio Management is supported on 32-bit architecture.

CA Asset Portfolio Management supports two types of architectures:

- When only one server hosts the Web UI and services.
- When different servers host the Web UI and services.



CA Service Catalog Product Architecture

The number of *catalog requests per minute* is the key criterion for determining the size of the system architecture as small, medium and large. This criterion is much more important than the number of users in the system or other criteria. The higher the number of catalog requests per minute, the larger the recommended architecture. The following are the three types of architecture sizes that are typically followed in the CA Service Catalog implementation.

Availability refers to the ability of an organization to deliver consistent, predictable access to applications and data.

In addition to small, medium, and large architectures, CA Service Catalog also supports a standalone deployment. In a standalone deployment, all the components, including the database server, are installed on a single physical computer. This deployment is for demonstration (demo) purposes only and is *not* suitable for a production environment.

WARNING

Verify that the DBMS is tuned for maximum performance. This directive applies to all system architecture (large, medium, and small).

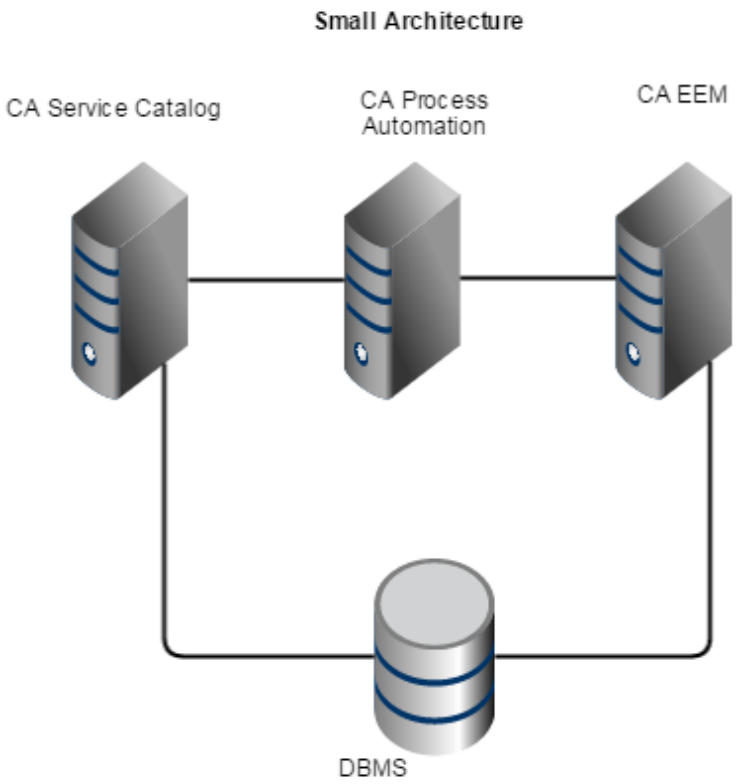
Review the following sections, to identify the CA Service Catalog architecture that suits your organization requirements:

Small Architecture

CA Service Catalog small architecture can include the following servers:

- Server 1 hosts CA Service Catalog, including Service Accounting Component.
- Server 2 hosts CA Process Automation.
- Server 3 hosts the DBMS.
- Server 4 hosts CA EEM. You use configuration parameters to connect CA Service Catalog and CA Process Automation with CA EEM.

Figure 1: CA Service Catalog System Architecture _Small



This model applies to systems with an expected average load of 1 to 10 catalog requests per minute. Clustering does *not* apply to small architecture.

Also consider, any performance-related information and other stress test data that you have from implementing previous releases of CA Service Catalog

Medium Architecture

If you expect more than a small percentage of the requests to be complex, consider moving to medium architecture. Complex requests include one or more of the following items:

- Multiple (two or more) services or service options.
- Attachments.
- Long forms or multiple forms.
- Long web service calls or multiple web service calls.
- Multiple CA Process Automation processes.

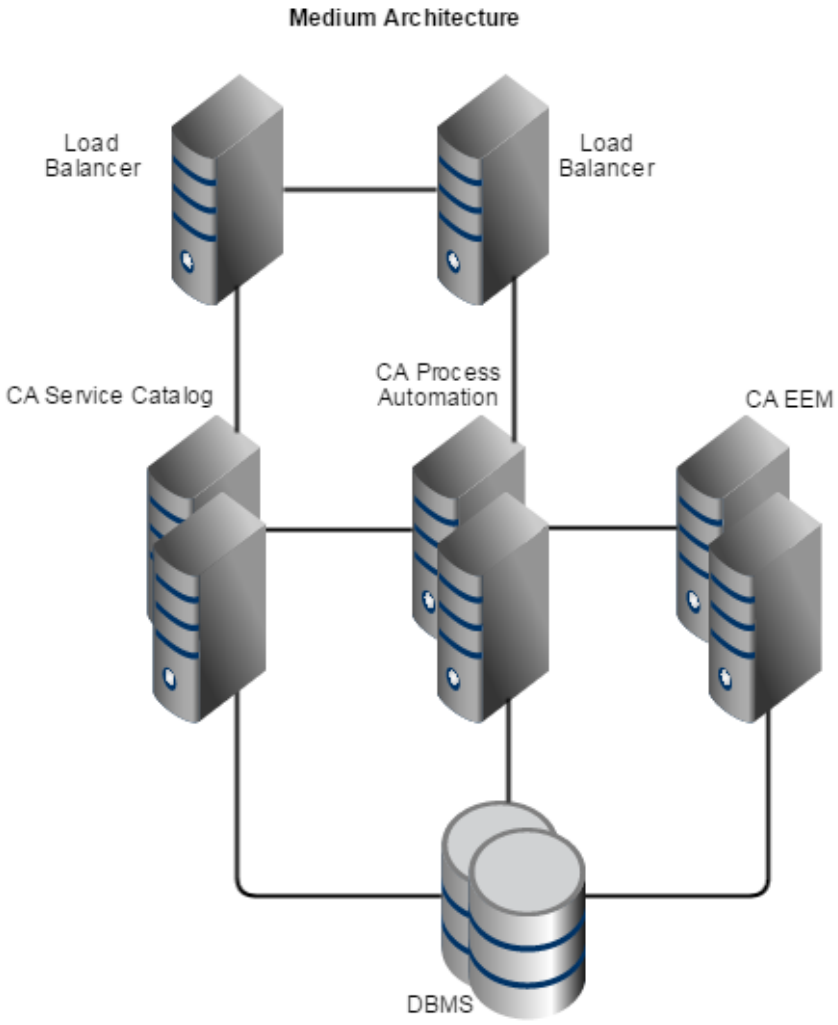
Similarly, consider moving to a medium architecture if both of the following conditions exist:

- Your implementation includes more than 5,000 users.
- You expect more than a small percentage of these users to access the catalog simultaneously.

The medium architecture can efficiently process a load of 11-50 catalog requests per minute. The medium architecture expands the small architecture by implementing clustering and adding load balancers.

The following diagram illustrates an example of medium architecture:

Figure 2: CA Service Catalog System Architecture _Medium



The medium architecture setup is as follows: *Two clustered servers* refers to two virtual computers that are clustered.

- Two clustered servers host CA Service Catalog.
- Two clustered servers host CA Process Automation. Install and cluster CA Process Automation according to the specifications in the CA Process Automation documentation.
- One or preferably two load balancers are the front ends to the clusters for CA Process Automation and CA Service Catalog, respectively. Both the CA Process Automation and CA Service Catalog clusters require a load balancer, which can be an existing instance. CA Process Automation and CA Service Catalog can share a load balancer.
- Two clustered servers host CA EEM. CA EEM is clustered but is not used with a load balancer.

  NOTE

  We recommend clustering CA EEM to provide failover protection. You can use the configuration parameters to connect CA Service Catalog and CA Process Automation with the CA EEM cluster. You can increase CA EEM performance by pointing each CA Process Automation domain server to different CA EEM servers. Install and cluster CA EEM according to the specifications in the CA EEM documentation.
- Two clustered servers host the DBMS. You cluster the DBMS to improve throughput and responsiveness. Verify that you cluster your DBMS using a method that improves performance. For example, using an **active-active** database cluster can reduce performance, especially on write operations. For more information about clustering your DBMS, see its documentation.
- Some virtual machine software uses scheduling algorithms that result in decreased performance when the number of CPUs increases. This performance loss is most noticeable when the virtual machines are sharing systems with other virtual machines that also have a lower number of CPUs allocated. Typically, in such cases, increasing the number of virtual machines in the cluster improves performance more than increasing the number of CPUs per virtual machine.

Large Architecture

Consider the factors for moving from a small architecture to a medium architecture. The same factors in greater magnitude can cause you to add computers and clusters to a large architecture. The large architecture can efficiently process an average load of 50 or more catalog requests per minute.

The large architecture includes multiple web server clusters and a database cluster.

The following diagram illustrates an example of large architecture:

Figure 3: CA Service Catalog System Architecture _Large

The large architecture setup is as follows:

- Two or more clustered servers host CA Service Catalog.
- Two or more clustered servers host CA Process Automation.
- You install and cluster CA Process Automation according to the specifications in the CA Process Automation documentation. Alternatively, you install CA Process Automation on three physical computers and place them in a separate cluster.
- Two or more load balancers are the front ends to the clusters for CA Process Automation and the Catalog Component of CA Service Catalog, respectively. Both the CA Process Automation and CA Service Catalog clusters require a load balancer, which can be an existing instance.
- Two or more clustered servers host CA EEM. CA EEM does not *use* a load balancer.

  NOTE

  We recommend clustering CA EEM to provide failover protection and high availability. You use configuration parameters to connect CA Service Catalog and CA Process Automation with the CA EEM cluster. You can increase CA EEM performance by pointing each CA Process Automation domain server to different CA EEM servers.
- Two or more clustered servers host the DBMS.
- Some virtual machine software uses scheduling algorithms that result in decreased performance when the number of CPUs increases. This performance loss is most noticeable when the virtual machines are sharing systems with other

virtual machines that also have a lower number of CPUs allocated. Typically, in such cases, increasing the number of virtual machines in the cluster improves performance more than increasing the number of CPUs per virtual machine.

- A filestore serves as the single location for all shared files.

Example Business Scenario

This example illustrates, how to architect the CA Service Catalog system to meet the requirements for the following scenario: **I have 60,000 end users who are distributed across the country. My system is located in a centralized data center that the service provider owns. We want to integrate with CA Service Desk Manager, CA Asset Portfolio Management, and other products. Our expected load is about 100 requests per day with an average of 1000 concurrent connections. We are planning to define CA Process Automation processes to meet our requirements.** This scenario is common. Because of the variability around usage patterns and architecture there is no simple calculator that can generate sizing estimates. However, the CA Service Catalog components are dynamically scalable. The CA Service Catalog architecture is n-tier, stateless, and web-based. Users make HTTP requests. These HTTP requests are passed to one or more application servers. The application server performs the processing logic using a pool of connections to the database. The web server can be co-located with application server instances. Also, several computational engines run independently from any user interactions; examples include the billing and SLA correlation engines. Other web servers, application servers, and computational engines can be added, as required. The database, where state is maintained, is the potential performance bottleneck. The scalability of the database platform is important. Therefore, the sizing and architecture of the MDB is the most critical factor. **Metrics** First, consider the concurrent connection metric: Because HTTP is connectionless, there is no "concurrent user" metric. Next, consider the metric of 100 requests per day: This metric does not provide the peak usage, a critical factor. If the 100 requests are evenly distributed through the day, system impact would be minimal. However if the 100 requests are all made at the same time of day, then the system could be challenged. Also, the type of request has a direct impact on the resource utilization. **Variations** Variations in usage and architecture are important. Consider them carefully and tune your system. Adjust your tuning over time as changes in demand or system load occur, if necessary. **Business Requirements and Updates** When you are implementing CA Service Catalog, it is critical to capture accurate business requirements. Use the business requirements to formulate efficient technical architectures. Therefore, you start an implementation with a discovery phase per module to capture the business requirements. Next, you study the limitations and capabilities of the client operational infrastructure, and you formulate an efficient implementation architecture. Once you obtain all the information, you can formulate an initial sizing estimate. Based on the possible variations in usage and architecture, organizations must start with two instances of each component. One component for fail-over and one component for load balancing. Because the components can be co-located, start with at least two server class machines in addition to the database servers. If computational engines are heavily used during peak activity times, they require their own server or servers. The usage-patterns typically change over time as the clients grow to understand the features and use them more heavily. As you discover usage patterns over time, you can dynamically add and distribute components as needed.

CA Service Desk Manager Product Architecture

For a successful CA SDM implementation, you need to understand the CA SDM deployment model and the different installation types that are available. Review the following sections, to identify the CA Service Desk Manager architecture that suits your organization requirements:

CA SDM Deployment Model

You can deploy CA SDM using one of the following deployment models that best suits your requirement:

- **Centralized** – Installs and configures all product components on one server. This is the default installation. You can implement multiple Object Managers and web engines for load balancing and failover, however your business may outgrow this implementation.
- **Distributed** – Installs and configures product components on servers that are closer to the clients receiving the service. For example, a business with many subnets can have analysts that are using the Web Client. In this type of deployment, you place an additional server at the branch location. This server performs caching and reduces the network traffic and response time between the branch location and the main server location. This type of implementation supports the implementation of multiple Object Managers and web engines for load balancing and failover.
- **Global** – Having two or more centralized or distributed implementations that are known as regions. The main server of a region replicates minimal information to and from a master region. Hence, a single region can have all necessary information about all other regions. An analyst is aware of tickets from all regions and can connect to a region when required. This type of implementation is useful when network bandwidth is too limited for a distributed implementation. For example, business locations in different countries with a slow link between them.

CA SDM deployment model consists of the following categories:

1. Small
2. Medium
3. Large

CA SDM Advanced Availability configuration is recommended since it provides better performance and availability. Customers with less demanding needs can opt for CA SDM conventional configuration. Use clustered DBMS configurations to support large data volumes. Each server in advanced availability mode requires DBMS connectivity.

NOTE

It is difficult to give exact recommendations for deployment models since customers use CA SDM differently based on environment and organization requirements. Deployment model showcased in this article is based on feedback received from CA SDM customers, in general.

NOTE

If you are using a load balancer for Advanced Availability, ensure that the load balancer is configured to use sticky sessions.

This table describes CA SDM environment category and concurrent Analyst sessions for CA SDM, xFlow Interface, Insights, Search Server, and Collaboration Server:

Category	Analyst Sessions	CA Service Desk Manager + xFlow Interface	MDB Database	Datamart / Aggregate Reporting Database	Transactional Reporting Database	Search Server
Tiny* (*Not depicted in the diagram)	1-10	(recommended 16-24 GB RAM) Conventional Configuration: • 1 Primary Server (along with **xFlow Interface**)	On CA SDM Server	On CA SDM Server	On CA SDM Server	On CA SDM Server

Small	1-200	(recommended 16-24 GB RAM) Advanced Availability Configuration: • 1 Background Server • 1 Standby Server • 2 Application Servers (along with **xFlow Interface**)	Dedicated Server	Dedicated Server	Dedicated DBMS replica with shadow SDM	Dedicated Server
Medium	200-5000	(recommended 16-32 GB RAM) Advanced Availability Configuration: • 1 Background Server • 1 Standby Server • 5 Application Servers (along with **xFlow Interface**)	Clustered	Dedicated Server	Dedicated DBMS replica with shadow SDM with multiple domsrvrs	Cluster of 3
Large	5000+	(recommended 16-32 GB RAM) Advanced Availability Configuration: • 1 Background Server • 1 Standby Server • 5+ Application Servers (along with **xFlow Interface**)	Clustered	Dedicated Server	Dedicated DBMS replica cluster with shadow sdm and multiple domsrvrs	Cluster > 3 Solid-State Drive (SSD) disks

NOTE

For Aggregate(Datamart) JasperReports database, we recommend that you use HDD data storage for small architecture, and SAN or RAID for medium and large architectures.

RASA Server Requirements

Consider the following RASA server requirements for environment category and concurrent Employee sessions for CA SDM and xFlow Interface

Table 2: RASA Server Requirements

Category	Employee Sessions	RASA Server	Considerations
Small	1-180	One machine (recommended 16-24 GB RAM and 8 CPUs)	1. Each RASA server node can scale up to 60 users. 2. Ability to configure upto 8 Rasa Server nodes with load balancer enabled.
Medium	180-5000	6 - 8 machines (recommended 32-64 GB RAM and 16 CPUs)	1. Each RASA server node can scale up to 60 users.. 2. Ability to configure upto 16 Rasa Server nodes on each machine with load balancer enabled.

CA SDM Architecture

CA SDM can be deployed using one of the architectures from Conventional Configuration and Advanced Availability using the centralized and distributed models.

CA SDM Architecture for Conventional Configuration

The conventional configuration includes a primary server, a database, and one or more secondary servers. The secondary servers are optional; configure them to support a large number of concurrent users. Only the primary server can access the database directly. The secondary servers access the database through a central process that is available on the primary server. A secondary server can be removed from the configuration without disrupting other servers. Only the users connected to the specific secondary server are affected.

When to Opt for Conventional Configuration

Conventional is the default configuration and is suitable for smaller deployments. Choose the conventional configuration if you do not require any high availability. Be aware that with conventional configuration, the application is not available during maintenance or when a server crashes.

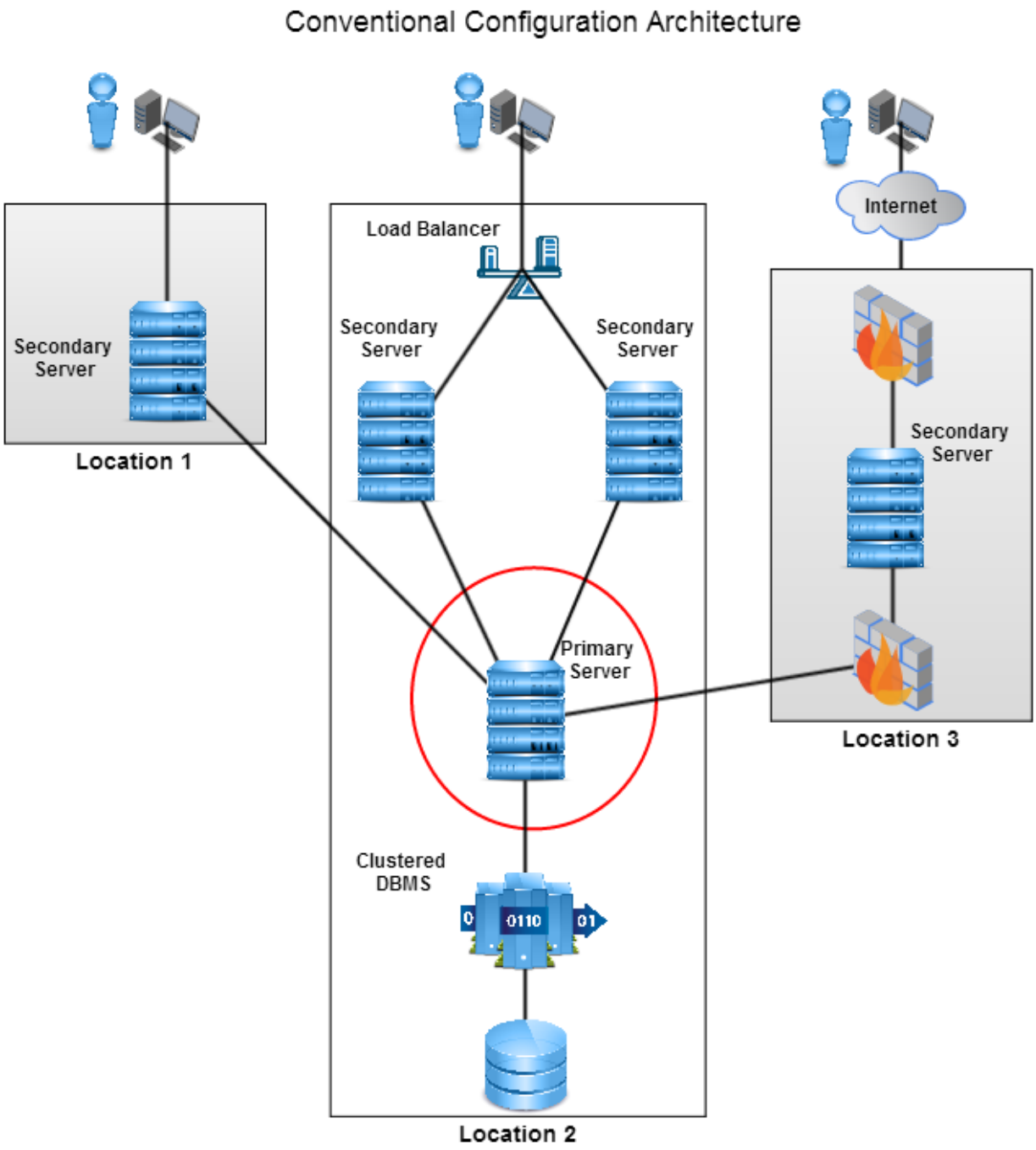
Primary Server

The Primary Server is the main server in a conventional configuration. In a single server CA SDM configuration, the server is the Primary Server. When CA SDM is distributed across two or more servers, one server is designated as the Primary Server. This server performs roles in the configuration such as accessing the database. There is always only one Primary Server.

Secondary Server

A Secondary Server is configured when there are more than one CA SDM server in a configuration. In this scenario, all servers other than the Primary Server are configured as Secondary Servers. Secondary servers perform only a subset of the roles of the Primary Server. Secondary Servers are added to a configuration to scale CA SDM to handle more users.

The following diagram provides an example of the conventional configuration architecture:



CA SDM Architecture for Advanced Availability

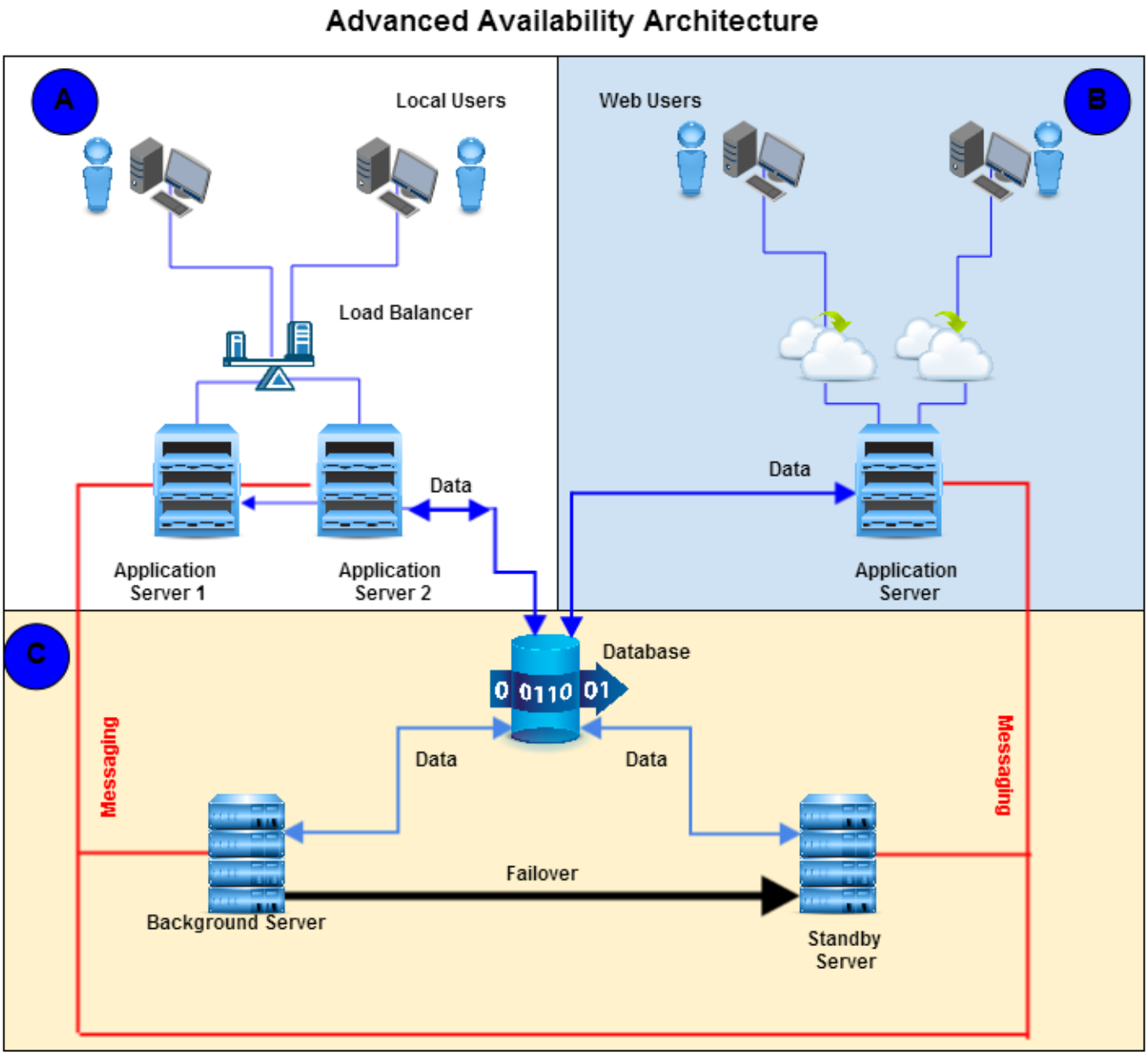
The advanced availability configuration includes a background server, one or more standby servers, and one or more application servers. To reduce the single point of failure, each of these servers has a direct connection with the database. All the components of the architecture communicate using an internal CA protocol.

When to Opt for Advanced Availability Configuration

The advanced availability configuration provides higher availability, reduces down times, and supports rolling maintenance. Consider the advanced availability configuration when any of the following factors is true:

- You require a high degree of CA SDM availability.
- You require the CA SDM servers to be more independent and more resilient to failures.
- You require an ability to remove and return the CA SDM servers to service, without bringing down the entire CA SDM installation.
- You require near zero downtime during rolling maintenance.

The following diagram provides an example of advanced availability architecture:



The architecture is spread across three different locations A, B, and C. Location A has two application servers, serving users through a load balancer. The location B has an application server directly serving the users and the location C has the background server, the standby server, and the database. Each of the servers has direct connection to the database. The blue lines mark the flow to and from the database. The red lines identify the internal communication between the components.

Background Server

The background server is the core of the advanced availability architecture. This server provides ancillary services to other servers and runs all singleton processes of CA SDM. A process can be is designated as singleton when only single copy of it can be active in any SDM installation. Only the users with the Administrator Access Type have access to the background server. To increase the availability, the standby server shadows the background server. You can switch the standby server as the background server in case of any failure or when performing rolling maintenance.

Standby Server

The primary function of the standby server is to act as a warm standby for the background server. The standby server has the same hardware and OS platform as the background server. The standby server can run all processes that run on the background server. The standby server stays idle during normal working of the system. However, it listens to the internal CA SDM system messages for database changes and updates the critical caches continuously. If the background server fails or requires rolling maintenance, you can promote the standby server to background server. When you promote the server, the application servers as well as the end users have minimal disruption. You can invoke a utility to perform this switchover.

The standby server is only running a small subset of the processes that normally run on the background server. You cannot log on to the web interface on the standby server.

Application Server

The application server has all the CA SDM components necessary to serve the end users through various interfaces like web, SOAP, and RESTful web services. The application servers are independent of each other and resilient to the background server outages for short periods of time. Users access the application servers. You can individually remove the application servers and return to service by using the new Quiesce facility. The quiesce facility allows current users to complete their work and then sign in to an alternate application server.

Installation Prerequisites

CA Service Management installer checks for deployment prerequisites and performs mandatory validations before installing the products. Review the following installation prerequisites before you proceed with the installation:

- Operating System Requirements
- Server and Hardware Requirements
- Software Requirements
- Database Requirements
- Supported Ports and Port Ranges
- Supported Browser and Network Settings

Operating System Requirements

This article describes the following topics:

Review the following OS requirements for CA Service Management 17.4. For more information, see Supportability Matrix.

Red Hat Enterprise Linux Operating Systems

> **NOTE**
> Ensure that you have glibc 2.14 or later in your environment.

Red Hat Enterprise Linux 9.4

To install CA SDM on Red Hat Enterprise Linux 9.4, verify that you have the following RPM packages and their dependencies:

Required RPM	Dependencies
libXp-1.0.3-3.el9.i686_64.rpm libXp-1.0.3-3.el9.i686.rpm	libXau-1.0.9-8.el9.x86_64 libXau-1.0.9-8.el9.i686 libX11-1.7.0-9.el9.x86_64 libX11-1.7.0-9.el9.i686 libxcb-1.13.1-9.el9.x86_64 libxcb-1.13.1-9.el9.i686 libXext-1.3.4-8.el9.x86_64 libXext-1.3.4-8.el9.i686
libXtst-1.2.3-16.el9.x86_64 libXtst-1.2.3-16.el9.i686	libXi-1.7.10-8.el9.x86_64 libXi-1.7.10-8.el9.i686
openssl-libs-3.0.7-27.el9.x86_64 openssl-libs-3.0.7-27.el9.i686	zlib-1.2.11-40.el9.x86_64 Zlib-1.2.11-40.el9.i686 libselinux-3.6-1.el9.x86_64 Libselinux-3.6-1.el9.i686 keyutils-libs-1.6.3-1.el9.x86_64 keyutils-libs-1.6.3-1.el9.i686 krb5-libs-1.21.1-1.el9.x86_64 Krb5-libs-1.21.1-1.el9.i686
openldap-2.6.6-3.el9.x86_64 openldap-2.6.6-3.el9.i686	audit-libs-3.1.2-2.el9.x86_64 cracklib-2.9.6-27.el9.x86_64 cyrus-sasl-2.1.27-21.el9.x86_64 cyrus-sasl-lib-2.1.27-21.el9.x86_64 cyrus-sasl-lib-2.1.27-21.el9.i686 libcap-ng-0.8.2-7.el9.x86_64 libdb-5.3.28-53.el9.x86_64 pam-1.5.1-19.el9.x86_64

Verify that you have the following *pcre* and *libuuid* packages:

- pcre-8.44-3.el9.3.x86_64 *and* pcre-8.44-3.el9.3.i686
- libuuid-2.37.4-18.el9.x86_64 *and* libuuid-2.37.4-18.el9.i686

> **NOTE**
> These 32-bit packages (pcre-8 and libuuid-2) are required on both 32-bit and 64-bit systems.

Red Hat Enterprise Linux 8.6

To install CA SDM on Red Hat Enterprise Linux 8.6, verify that you have the following RPM packages and their dependencies:

Required RPM	Dependencies
libXp-1.0.3-3.el8.i686.rpm	libXau-1.0.8-3.el8.i686.rpm libX11-1.6.8-5.el8.i686.rpm libxcb-1.13.1-1.el8.i686.rpm libXext-1.3.4-1.el8.i686.rpm
libXtst-1.2.3-7.el8.i686.rpm	libXi-1.7.10-1.el8.i686.rpm
openssl-libs-1.1.1k-6.el8_5.i686.rpm	zlib-1.2.11-18.el8_5.i686.rpm libselinux-2.9-5.el8.i686.rpm keyutils-libs-1.5.10-9.el8.i686.rpm krb5-libs-1.18.2-14.el8.i686.rpm

openldap-2.4.46-18.el8.i686.rpm	openldap-devel-2.4.46-18.el8.i686.rpm audit-libs-3.0.7-2.el8.2.i686.rpm cracklib-2.9.6-15.el8.i686.rpm cyrus-sasl-2.1.27-6.el8_5.i686.rpm cyrus-sasl-devel-2.1.27-6.el8_5.x86_64.rpm cyrus-sasl-devel-2.1.27-6.el8_5.i686.rpm cyrus-sasl-lib-2.1.27-6.el8_5.i686.rpm libcap-ng-0.7.11-1.el8.i686.rpm libdb-5.3.28-42.el8_4.i686.rpm pam-1.3.1-16.el8.i686.rpm

Verify that you have the following *pcre* and *libuuid* packages:

- pcre-8.42-4.el8.i686.rpm
- libuuid-2.32.1-17.el8.i686.rpm

NOTE
These 32-bit packages (pcre-8 and libuuid-2) are required on both 32-bit and 64-bit systems.

Red Hat Enterprise Linux 7.2

To install CA SDM on Red Hat Enterprise Linux 7.2, verify that you have the following RPM packages and their dependencies:

Required RPM	Dependencies
libXp-1.0.2-2.1.el7.i686.rpm	libXau-1.0.8-2.1.el7.i686.rpm libxcb-1.11-4.el7.i686.rpm libXext-1.3.3-3.el7.i686.rpm libX11-1.6.3-2.el7.i686.rpm
libXtst-1.2.2-2.1.el7.i686.rpm	libXi-1.7.4-2.el7.i686.rpm
openssl-1.0.1e-42.el7_1.10.i686.rpm	zlib-1.2.7-15.el7.i686.rpm libselinux-2.2.2-6.el7.i686.rpm keyutils-libs-1.5.8-3.el7.i686.rpm krb5-libs-1.13.2-10.el7.i686.rpm
openldap-2.4.40-8.el7.i686.rpm	cyrus-sasl-lib-2.1.26-19.2.el7.i686.rpm

Verify that you have the following *pcre* and *libuuid* packages:

- pcre-8.42-6.el8.i686.rpm
- libuuid-2.32.1-35.el8.i686.rpm

```
These 32-bit packages (pcre-8 and libuuid-2) are required on both 32-bit and 64-bit
systems.
```

Suse Linux Enterprise Server 15.6

To install CA SDM on Suse Linux Enterprise Server 15.6, verify that you have the following RPM packages and their dependencies:

Required RPM	Dependencies
glibc-2.38-150600.14.5.1.x86_64 glibc-32bit-2.38-150600.14.5.1.x86_64	glibc-devel-2.38-150600.14.5.1.x86_64 glibc-devel-32bit-2.38-150600.14.5.1.x86_64

libXp6-1.0.3-1.24.x86_64 libXp6-32bit-1.0.3-1.24.x86_64	libXau6-1.0.8-1.26.x86_64 libXau6-32bit-1.0.8-1.26.x86_64 libX11-6-1.8.7-150600.1.2.x86_64 libX11-6-32bit-1.8.7-150600.1.2.x86_64 libxcb1-1.13-150000.3.9.1.x86_64 libxcb1-32bit-1.13-150000.3.9.1.x86_64 libXext6-1.3.3-1.30.x86_64 libXext6-32bit-1.3.3-1.30.x86_64 libXext-devel-1.3.3-1.30.x86_64 libXext-devel-32bit-1.3.3-1.30.x86_64
libXtst6-1.2.3-1.24.x86_64 libXtst6-32bit-1.2.3-1.24.x86_64 libXtst-devel-1.2.3-1.24.x86_64 libXtst-devel-32bit-1.2.3-1.24.x86_64	libXi6-1.7.9-3.2.1.x86_64 libXi6-32bit-1.7.9-3.2.1.x86_64 libXi-devel-1.7.9-3.2.1.x86_64 libXi-devel-32bit-1.7.9-3.2.1.x86_64
libopenssl3-3.1.4-150600.3.6.x86_64 libopenssl3-32bit-3.1.4-150600.5.7.1.x86_64	zlib-devel-1.2.13-150500.4.3.1.x86_64 zlib-devel-32bit-1.2.13-150500.4.3.1.x86_64 libselinux1-3.5-150600.1.46.x86_64 libselinux1-32bit-3.5-150600.1.46.x86_64 libkeyutils1-1.6.3-5.6.1.x86_64 libkeyutils1-32bit-1.6.3-5.6.1.x86_64 krb5-1.20.1-150600.9.2.x86_64 krb5-32bit-1.20.1-150600.11.3.1.x86_64
openldap2-2.4.46-150600.23.21.x86_64	openldap2-back-perl-2.4.46-150600.23.21.x86_64 openldap2-client-2.4.46-150600.23.21.x86_64 openldap2-devel-2.4.46-150600.23.21.x86_64 openldap2-devel-32bit-2.4.46-150600.23.21.x86_64 cracklib-2.9.11-150600.1.90.x86_64 cyrus-sasl-2.1.28-150600.5.3.x86_64 cyrus-sasl-32bit-2.1.28-150600.5.3.x86_64 cyrus-sasl-bdb-devel-2.1.28-150600.5.3.x86_64 cyrus-sasl-crammd5-2.1.28-150600.5.3.x86_64 cyrus-sasl-digestmd5-2.1.28-150600.5.3.x86_64 cyrus-sasl-gssapi-2.1.28-150600.5.3.x86_64 cyrus-sasl-plain-2.1.28-150600.5.3.x86_64 libcap-ng0-0.7.9-4.37.x86_64 libcap-ng0-32bit-0.7.9-4.37.x86_64 libcap-ng-devel-0.7.9-4.37.x86_64 libcap-ng-utils-0.7.9-4.37.x86_64 libcap-progs-2.63-150400.3.3.1.x86_64 libdb-4_8-4.8.30-150000.7.9.1.x86_64 pam-1.3.0-150000.6.66.1.x86_64 pam-config-1.1-150600.14.3.x86_64 pam-doc-1.3.0-150000.6.66.1.noarch
pcre2-devel-10.42-150600.1.26.x86_64 pcre2-doc-10.42-150600.1.26.noarch pcre2-tools-10.42-150600.1.26.x86_64	libpcre1-32bit-8.45-150000.20.13.1.x86_64 libpcre16-0-32bit-8.45-150000.20.13.1.x86_64 libpcre16-0-8.45-150000.20.13.1.x86_64 libpcre1-8.45-150000.20.13.1.x86_64 libpcre2-16-0-10.42-150600.1.26.x86_64 libpcre2-32-0-10.42-150600.1.26.x86_64 libpcre2-8-0-10.42-150600.1.26.x86_64 libpcre2-8-0-32bit-10.42-150600.1.26.x86_64 libpcre2-posix3-10.42-150600.1.26.x86_64

libuuid1-2.39.3-150600.2.1.x86_64 libuuid1-32bit-2.39.3-150600.4.6.2.x86_64 libuuid-devel-2.39.3-150600.4.6.2.x86_64 libuuid-devel-32bit-2.39.3-150600.4.6.2.x86_64 libldap-32bit	

Operating System Requirements for CA Asset Portfolio Manager (ITAM)

Complete the following prerequisites before proceeding with CA APM installation:

- Install .NET Framework on Windows Server
- Add ASP.NET in IIS on Windows Server

Install .NET 4.6 Framework on Windows Server 2016, .NET 4.7 Framework on Windows Server 2019, .NET 4.8 Framework on Windows Server 2022

For more information see, .NET Framework System Requirements.

Install .NET 4.6 Framework on Windows Server 2016

To install the .NET 4.6 Framework on Windows Server 2016, complete the following steps:

1. Launch the Server Manager.
2. Click **Add roles and features**. Click **Next**.
3. Select **Role-based or feature-based installation**.
4. In the **Server** section, select your server and click **Next**.
5. Scroll down to **Web Server (IIS)** to set up the following required features.
6. Expand **Application Development** and select, **.Net Extensibility 3.5**, **.Net Extensibility 4.6**, **Application Initialization**, **ASP.NET 3.5,** and **ASP.NET 4.6**. Click Next.
7. Navigate to **Features** > **.NET framework 4.6** > **WCF Services**. Select **HTTP Activation**.
8. Follow on-screen instructions. Click **Next**.
9. Review the summary screen.
10. Click **Finish** to exit the installer. The .NET 4.6 framework is now installed.

Install .NET 4.7 Framework on Windows Server 2019

To install the .NET 4.7 framework on Windows Server 2019, complete the following steps:

1. Launch the Server Manager.
2. Click **Add roles and features**. Click **Next**.
3. Select **Role-based or feature-based installation**.
4. In the **Server** section, select your server and click **Next**.
5. Scroll down to **Web Server (IIS)** to set up the following required features.
6. Expand **Application Development** and select, **.Net Extensibility 3.5**, **.Net Extensibility 4.7**, **Application Initialization**, **ASP.NET 3.5,** and **ASP.NET 4.7**. Click Next.
7. Navigate to **Features** > **.NET framework 4.7** > **WCF Services**. Select **HTTP Activation**.
8. Follow on-screen instructions. Click **Next**.
9. Review the summary screen.
10. Click **Finish** to exit the installer. The .NET 4.7 framework is now installed.

Install .NET 4.8 Framework on Windows Server 2022

To install the .NET 4.8 framework on Windows Server 2022, complete the following steps:

1. Launch the Server Manager.
2. Click **Add roles and features**. Click **Next**.
3. Select **Role-based or feature-based installation**.
4. In the **Server** section, select your server and click **Next**.
5. Scroll down to **Web Server (IIS)** to set up the following required features.
6. Expand **Common HTTP Feeatures** and select **Static Content.**
7. Expand **Performance** and select **Static Content Compression.**
8. Expand **Application Development** and select, **.Net Extensibility 3.5**, **.Net Extensibility 4.8**, **Application Initialization**, **ASP.NET 3.5,** and **ASP.NET 4.8**. Click **Next**.
9. Navigate to **Features** > **.NET framework 4.8** > **WCF Services**. Select **HTTP Activation**.
10. Follow on-screen instructions. Click **Next**.
11. Review the summary screen.
12. Click **Finish** to exit the installer. The .NET 4.8 framework is now installed.

Add ASP.NET 4.6 on Windows Server 2016, and ASP.NET 4.7 on Windows Server 2019, and ASP.NET 4.8 on Windows Server 2022

For more information, see Install ASP.NET in IIS Web Server.

To add ASP.NET in IIS on Windows Server, perform the following steps:

1. Open Server Manager.
2. Navigate to **Manage**, **Add Roles and Features**.
3. Follow on-screen instructions. Select installation type and destination server.
4. Select **Roles** from the select server roles pane, expand **Web Server (IIS)**, **Application Development (IIS)**.
5. Select **ASP.NET 3.5** and **ASP.NET 4.5**. or **ASP.NET 4.6** or **ASP.NET 4.7** or **ASP.NET 4.8** as per your deployment.
6. Click **Next**.
7. Follow on-screen instructions and complete the installation.
   ASP.Net is now added in IIS.

For more information on supported operating system, databases, web browsers, and so on, see Supportability Matrix.

Operating System Requirements for CA Service Catalog

Install CA Service Catalog products and components on supported Windows Server operating systems only.

CA Service Catalog is certified on several operating system languages. Language certification implies that you can install CA Service Catalog for a supported operating system in a supported language. Similarly, Service Catalog can store and retrieve data using the character set of the supported language. Web application server for CA Service Catalog is Apache Tomcat. When you install the base files for CA Service Catalog using the CA Service Management installation media, Apache Tomcat and related files are installed.

```
Ensure that all server operating systems use the same language.
```

Server and Hardware Requirements

Recommendation is that your environment must have at least 4 GB of temporary disk space for CA Service Management installer to extract required product installation files. If the temporary available disk space is less than 4 GB, use the **IATEMPDIR** variable to redirect this temporary space of the installer to another folder.

This article contains the following topics:

Server Requirements for CA Service Management Solution Products

The following table lists the CA Service Management Products, Common Components, and the minimum number of servers that you require for installation:

Component	Number of Servers (Minimum)	Description
CA MDB	1	Service Management Master Database
CA Service Desk Manager (Advanced Availability Configuration)	3	One Background Server, one Standby Server, one or more Application Servers **Recommended:** 4 servers [One Background Server, one Standby Server, two Application Servers]
CA Service Desk Manager (Conventional Configuration)	1	One Primary Server
CA Service Catalog	1	CA Service Catalog and the accounting components are installed on the same server.
CA IT Asset Manager	1	Web and Application Server can be installed on the same server. **Recommended:** 2 servers; one each for Web and Application servers.
CA Embedded Entitlements Manager	Cluster: 2 Standalone: 1	
CA Business Intelligence JasperReports Server	1	
CA Process Automation	1	

CA Service Desk Manager Hardware Requirements

Based on the size of your CA SDM environment, the following requirements must be met or exceeded for CA SDM to install and run properly:

Database Size	Hardware	Requirements
Small—Used for installing CA SDM in a test environment.	• CPU • RAM • Disk Space	• Minimum Dual Processor 2.0 GHz • Minimum 6 GB • 8 GB minimum. This will increase over time to accommodate database growth.
Medium—The CA SDM default. The recommended setting for most CA SDM installations.	• CPU • RAM • Disk Space	• Minimum Dual Processor 2.0 GHz • Minimum 12GB • 16 GB minimum. This will increase over time to accommodate database growth.
Large—Used for large CA SDM installations.	• CPU • RAM • Disk Space	• Quad Processor 2.0 GHz • Minimum 16 GB • 24 GB minimum. This will increase over time to accommodate database growth.

NOTE

The domsrvr or DOM server is the internal daemon which manages the CA SDM objects in the memory. The webengine is the internal daemon which generates the HTML pages and sends it to the client. A single webengine and domsrvr are deployed in pairs. Each pair requires one dedicated processor core and at least 1 GB of RAM. Each webengine or domsrvr pair can serve 200-250 users. If you want to add more than one pair for scaling up the system, ensure that you have the necessary processor and the RAM available.

NOTE
The data files directory of the database server for the MDB requires at least 2 GB of space.

When CA SDM is up and running, it is recommended to check the process memory. For optimal performance, we recommend the following method:

1. Set a notification when the process memory exceeds 1.25 GB and check the processes that are running.
2. Set a warning notification when the process memory exceeds 1.5 GB and take corrective actions to check the memory usage.

CA Service Management Virtual Analyst Requirements

Review the following hardware requirements for Virtual Analyst Implementation in CA SDM.

Hardware	Requirements
CPU	4-8 Core CPU CPU must support AVX Instruction set. In general, CPUs manufactured on or after year 2011 support AVX. For more information, see your CPU manual instructions.
RAM	Minimum 16 GB. Recommended 32 GB
Disk Space	40 GB

Requirements

Review the following hardware requirements for xFlow Interface installation.

Hardware	Requirements
CPU	Minimum Dual Processor 2.0 GHz Preferred Quad Core Processor 2.0 GHz
RAM	Minimum 16 GB Recommended 32 GB
Disk Space	Minimum 12 GB

Insights (CABI JasperReports Server) Requirements

Review the following hardware requirements for Insights (CABI JasperReports Server) installation:

NOTE

The rendering time for Insights App varies with the CABI JasperReports server response time and network lag (if any) when the Insights App is launched from CA SDM or xFlow Interface.

Hardware	Requirements
CPU	Minimum Dual Processor 2.0 GHz Preferred Quad Core Processor 2.0 GHz
RAM	Minimum 16 GB Recommended 32 GB

Disk Space	Minimum 12 GB

Search Server Requirements

Review the following hardware requirements for, Search Server.

Hardware	Requirements
CPU	4-8 Core CPU
RAM	Minimum 16 GB Recommended 32 GB
Disk Space	Minimum 12 GB

CA SDM Advanced Availability Requirements

NOTE
Use only one *DOM server or web engine pair for each CPU and allocate 1GB RAM per pair.

The following hardware requirements must be met or exceeded for a CA SDM Advanced Availability server:

Server Type	CPU Requirements	RAM Requirements	Disk Space Requirements
Standby or Background server	Minimum Dual Processor 2 GHz Preferred Quad Core Processor 2 GHz	Minimum 4 GB Recommended 8 GB	4 GB (for product installation and for minimal application log growth.)
Application server (minimum recommended configuration for each of the application servers must be the same)	Minimum Dual Processor 2 GHz Preferred Quad Core Processor 2 GHz	Minimum 4 GB Recommended 8 GB (Can be increased based on the number of *domsrvr or webengine pairs)	4 GB
Remote MDB	Minimum Dual Processor 2 GHz Preferred Quad Core Processor 2 GHz	Minimum 4 GB Recommended 8 GB	20 GB (recommended minimum disk space) You should also allow for incremental growth to accommodate both new MDB table entries and to provide enough space for extra Service Desk-related document
Attachments We do not recommend having the attachment repository location as part of the standby or background or application server to avoid any link breakup during failover. We recommend keeping it on an independent file server, preferably on a shared storage location. For advanced availability configuration, you may use SAN with RAID 5 configured. Best performance is achieved by using multiple repositories on multiple application servers.			Based on your organization requirements.

CA SDM Conventional Configuration Requirements

NOTE
Use only one *DOM server or web engine pair for each CPU and allocate 1GB RAM per pair.

The following hardware requirements must be met or exceeded for a CA SDM conventional configuration:

Server Type	CPU Requirements	RAM Requirements	Disk Space Requirements
Primary server	Minimum Dual Processor 2 GHz	Minimum 4 GB Recommended 6 GB (Can be increased based on the number of domsrvr or webengine pairs)	4 GB
Secondary server	Minimum Dual Processor 2 GHz	Minimum 4 GB Recommended 6 GB (Can be increased based on the number of *domsrvr or webengine pairs)	4 GB
Remote MDB	Minimum Dual Processor 2 GHz Recommended: Quad Core Processor 2 GHz	Minimum 4 GB Recommended 8 GB	20 GB (recommended minimum disk space) You should also allow for incremental growth to accommodate both new MDB table entries and to provide enough space for extra Service Desk-related documents.

CA SDM Web Client Requirements

The following hardware requirements must be met or exceeded for a CA SDM Web Client computer to access CA SDM:

Hardware	Requirements
CPU	Dual Processor 2.0 GHz preferred
RAM	Minimum 2 GB
Disk Space	4 GB

CA Service Catalog Hardware Requirements

Ensure that your system meets the following hardware requirements before proceeding with the CA Service Catalog installation:

Hardware	Requirements
CPU	Minimum Intel Platform 3.0 GHz Processor Recommended: Multi-Processor
RAM	Minimum 4 GB Recommended 6 GB
Disk Space	80 GB with at least 4 GB free space

WARNING

For production, it is recommended to install each required application on separate servers. For example, you install Catalog Component on server 1, CA Process Automation on server 2, Accounting Component on server 3, and the DBMS server software on server 4.

CA IT Asset Management Hardware Requirements

The following hardware requirements assume that you have 80 -100 concurrent users. For assistance with deployment architectures exceeding 100 concurrent users, contact Broadcom Support

Component	Web Server and Application Server
Processor	3 GHz (dual core processor)
RAM	8 GB
Disk Space	5 GB

Component	Database Server
Processor	3 GHz (dual core processor)
RAM	8 GB
Disk Space	4 GB

Software Requirements

CA Service Management installer performs prerequisite checks and mandatory validations before installing CA Service Management solution products. Ensure that you have considered the following prerequisites for CA Service Desk Manager (CA SDM), CA Service Catalog, and CA Asset Portfolio Management (CA APM) before proceeding with the installation process:

CA Service Desk Manager Requirements

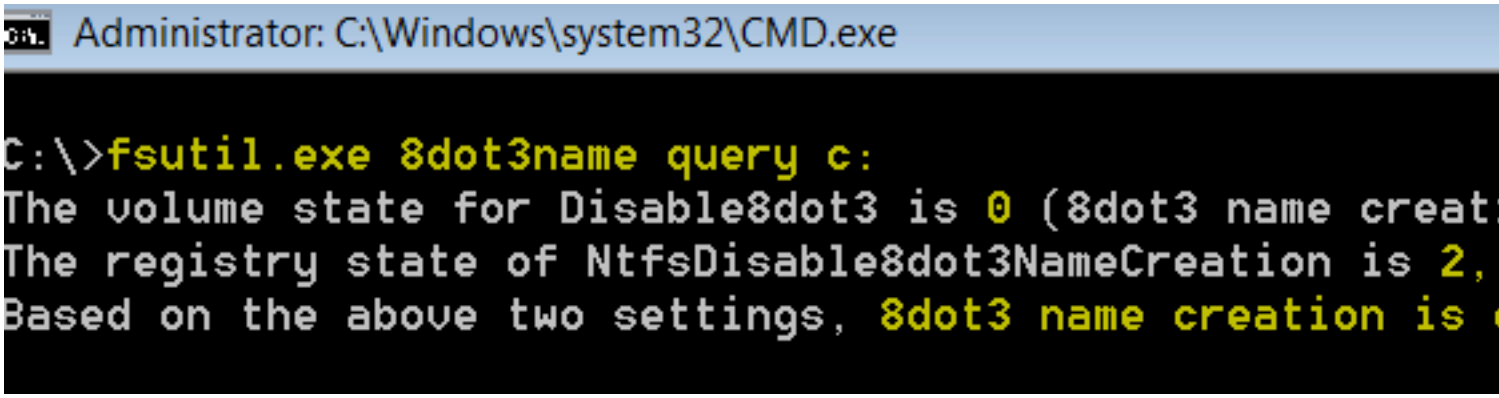
The following software applications are required for CA Service Desk Manager:

8dot3 name creation

Required for CA Service Desk Manager (CA SDM). You must enable the **8.3 name creation** before installing CA Service Desk Manager on the Windows operating system.

Follow these steps:

1. Execute the following command on the drive where you want to install CA SDM:
   <installation drive>:\Users\Administrator>FSUTIL.EXE 8dot3name query <installation drive>:
   This example screenshot displays the status of 8dot3 name at volume and registry level:



2. Changes the file system behavior for 8dot3 name in the following instances:
   When *DefaultValue* is specified, the registry key,**HKLM\System\CurrentControlSet\Control\FileSystem \NtfsDisable8dot3NameCreation**, is set to the *DefaultValue*.
   The *DefaultValue* can have the following values:
   - **0**: Enables 8dot3 name creation for all volumes on the system.
   - **1**: Disables 8dot3 name creation for all volumes on the system.
   - **2**: Sets 8dot3 name creation on a per volume basis.
   - **3**: Disables 8dot3 name creation for all volumes except the system volume.
3. To enable 8dot3 name, run the following command:
   <installation drive>:\Users\Administrator>FSUTIL.EXE 8dot3name set <installation drive>: 0

TIP

1. For more information about how to enable 8dot3 name, see Microsoft Website and search for Fsutil 8dot3name.
2. Ensure 8dot3name is enabled while creating a directory, otherwise the folder is created without a valid short name.

(Optional) PsTools Suite

The PsTools Suite from Microsoft is optional for CA SDM. The PsTools Suite helps implement the CA SDM Interval Logging utility. This utility is used for collecting application performance and system statistics. Extract **pstools.zip** to any directory of your choice and add the directory to the **Windows %PATH%** environment variable. If PsTools is not available on your local environment, download it from Microsoft.

JointJS

Required for xFlow interface. Jointjs is a JavaScript library which helps to visualize the SDM classic workflow as a graphical representation. The xFlow user interface uses Jointjs and requires certain files for successful installation. You must download the required JointJS files before starting the installation.

Perform the following steps:

1. Download the **CA SERVICE MANAGEMENT 17.4 3RD PARTY COMPONENTS.iso** from Broadcom Support.
2. Extract the Jointjs folder from the Jointjs.zip file.
3. Verify that Jointjs folder includes the following files:
   joint.min.js
   joint.min..css

WARNING

Note the path where you placed the Jointjs folder that includes the files. When you run the installer, on the screens corresponding to the xFlow installation you must provide the path for the Jointjs folder you saved.

CA Asset Portfolio Management Requirements

The following software applications are required for CA Asset Portfolio Management:

CA Embedded Entitlements Manager

The CA Embedded Entitlements Manager is required for installing CA Asset Portfolio Management Requirements.

Microsoft .NET Framework

Ensure that you have any of these supported version of Microsoft .NET Framework. This is mandatory for CA Asset Portfolio Management installation:

- Microsoft .NET Framework 2.0
- Microsoft .NET Framework 3.5
- Microsoft .NET Framework 4.0
- Microsoft .NET Framework 4.5.2/ 4.6/ 4.7/ 4.8

Windows Installer 5.0

Microsoft Windows Installer is a component of the Windows operating system. Windows Installer provides a standard foundation for installing and uninstalling the software.

Internet Information Services (IIS)

Required for CA APM and supports IIS version 7.5 or above. IIS is a flexible, general-purpose web server from Microsoft that runs on Windows systems to serve requested HTML pages or files.

Database Requirements

Consider the following database requirements (Oracle or Microsoft SQL Server) for installing CA Service Management.:

NOTE
The SDM privileged user must have read access to **tnsnames.ora** as well as **sqlnet.ora** files.

NOTE
Add the mdbadmin user on all nodes of Microsoft SQL Server AlwaysOn Environment.
For example:

```
CREATE LOGIN mdbadmin WITH PASSWORD = 'xxxx@123', SID = 0x9033FDA09F4C49438D0A4B094C243CDA; GO
```

NOTE
For Oracle database in CA Service Management Solution deployment, ensure that the Oracle-32 bit path is set on top of the Environment path variable list for CA APM.

CA Service Desk Manager (SDM)

NOTE
Interactive ODBC SQL (isql) tool is no longer supported for CA SDM.

Consider the following if you are installing CA SDM with Microsoft SQL Server or Oracle Database:

SDM SQL Server (2016, 2017, 2019 or 2022) Database Requirements

If you are using a remote MDB, the database client tools for Microsoft SQL Server must be installed on the same system or computer where you install CA Service Desk Manager (SDM).

SDM Advanced Availability Configuration: Database Client tools for Microsoft SQL Server are required on SDM standby server, background server, and application servers.

SDM Conventional Configuration: Database Client Tools for Microsoft SQL Server are required on the Primary server.

NOTE
Use client tools and management tools for command line utilities when installing or configuring the SDM server (**pdm_configure**). In case, if you do not want to install these components for the command line, use the Microsoft ODBC driver for Microsoft SQL Server.

Download the following from Microsoft Download Center:

- Microsoft ODBC Driver for Microsoft SQL Server. For more information and compatibility, see Microsoft ODBC Driver compatibility.
- Microsoft Command Line Utilities for Microsoft SQL Server. For more information, see Microsoft Command Line Utilities.
- Microsoft SQL Server Management Studio (SSMS). For more information, see SQL Server Management Studio.

NOTE
Prerequisite for MS SQL Server 2022: Microsoft has removed the SQL Native Client 11 for out of the box SQL 2022. CA Service Management uses SQL Native Client 11, you must manualy install it from the Microsoft website. Download the 64-bit SQL Native Client 11", named as Microsoft® SQL Server® 2012 Native Client from: https:// www.microsoft.com/en-us/download/confirmation.aspx?id=50402

SDM Oracle Database Requirements

- If you are doing a fresh installation of 17.4 or upgrading from an earlier version, ensure that the Oracle PATH environment variable refers to the 32-bit Oracle client first, and then, to the 64-bit Oracle client. To change the sequence, navigate to **Environment Variables**, **System Variables**, **Path**, **Edit**.
- For Linux Oracle implementations, set the Oracle environment variables before you install or migrate CA SDM.
- (For Oracle database only) 32-bit and 64-bit ORACLE client with administrator must be installed on all CA SDM servers.

CA Service Catalog

- **Oracle Dtabase**
  - If you are planning to use an Oracle database, ensure that you identify the system identifier (SID) and verify that the ORACLE_HOME and ORACLE_SID variables are set correctly.
  - In a distributed implementation, any computer on which you install CA Service Catalog must have either your DBMS server or DBMS client installed. This requirement applies to both Microsoft SQL Server and Oracle.
- **Microsoft SQL Server Database**
  Use a user ID that has **SA (System Administrator)** privileges to install or upgrade CA Service Catalog.

CA Asset Portfolio Management (APM)

Consider the following database requirements for CA APM or ITAM:

NOTE
CA APM uses Oracle 32-bit client for supported Oracle databases. For more information, see Supportability Matrix.

NOTE
For installing CA APM on Oracle 21c database, add a new system environment variable "TNS_ADMIN" on the CA APM Web / Application servers. This variable value must point to the Oracle 21c 32-bit client installed location. Set the variable value, for example <oracle32bit_client_installed_location>\product \21.0.0\client32bit_1\NETWORK\ADMIN

NOTE

If CA APM is installed on a SQL Server database, install the 64-bit SQL Native Client 11.0 on the APM Application server to ensure CORA functionality works, as CORA uses the SQLNCLI11 provider to connect to

the database. Download the 64-bit SQL Native Client 11", named as Microsoft® SQL Server® 2012 Native Client from: https://www.microsoft.com/en-us/download/confirmation.aspx?id=50402

Supported Ports and Port Ranges

CA Service Managementinstallation requires various ports and port ranges to be opened for configuration and integration with CA products and other third-party products. This article describes the following:

Ports and Port Ranges for CA Service Desk Manager

By default, CA SDM selects a port based on availability. The system reserves ports less than 1024, but can request for a port as high as 65335. The ports that you require to open on a firewall depends on the settings defined in the NX.env file.

The following NX.env variables set the starting port (2100) and the incremental increase (plus 1):

- *NX_SLUMP_FIXED_SOCKETS=1*
- *NX_SLUMP_SECONDARY_SOCKET=2100*

The following list displays default and recommended ports (and port ranges) for a typical CA SDM installation:

Port Name	Port Number
xFlow Interface/Service Point	• Web Server Port: 9002 • Incident Service Port: 9002 • Search Service Port: 9006 • Notification Search Service Port (requires Web Sockets Protocol): 9008
RASA NLU Server	5005 (Core) 5055 (Action)
Insights Services Port	• 9016
Search Server	• HTTP Port: 9012 • TCP Port: 9300
CA SDM Tomcat	• Tomcat Port: 8080 • Tomcat Shutdown Port: 8085
Database	• Oracle: 1521 • SQL Server: 1433

CA SDM	• FTP: 21 • SMTP: 25 • IIS: 80 • HTTP: 8080 • HTTPS: 8443 • HTTPS (secondary): 8081 • POP3: 110 • IMAP: 143 • LDAP: 389 • WebEx: 1270 • mstsc: 1389 • Slump Socket: 2100 • qserver: 2234 • Proctor Socket: 2300 • Communications: 2365 • Apache Tomcat: 8080 • Apache Tomcat Shutdown: 8085 • SSL on Apache Tomcat: 8443
CA CMDB	• Visualizer: 9080 • Visualizer Apache Tomcat Shutdown: 9085
CA EEM	• Administration Port: 5250
REST	• Apache Tomcat: 8050 • Apache Tomcat Shutdown: 8055
CA Business Intelligence	• Apache Tomcat: 8080 **Recommended:** 8070 • Apache Tomcat Redirect: 8443 • Apache Tomcat Shutdown: 8075
EBR Search	• Base Port: 1300
Portal Server	• Apache Tomcat: 8080 • Apache Tomcat Shutdown: 8085 • SSL Functionality: 8443 • Portal_Safe_List: 8444
Visualizer	• Apache Tomcat: 9080 • Apache Tomcat Shutdown: 9085

Ports and Port Ranges for CA Service Catalog

Specify the startup and shutdown port numbers, if you do not want to use the default ports **8080** and **8085**.

> **WARNING**
> We recommend that you do *not* use port **7777** as the startup or shutdown port. Port **7777** is reserved for Java Messaging Service (JMS). If you must use port 7777 for the startup or shutdown port, reset the JMS port number after you have finished running the setup utility. Otherwise, port conflicts occur, and the product does not function correctly.

On each system where you plan to install CA Service Catalog, specify the port number for CA Service Catalog to communicate with the web server. You can change the default values during installation. If these ports are not available, have other non-conflicting port numbers available. You require port numbers **8080** and **8085** for Catalog Component service communication.

NOTE

CA EEM by default uses port number **5250** (for iTechnology iGateway service). You cannot change it during installation.

CA Service Catalog requires the following ports for communication with products and components. If you have disabled these ports, enable them before proceeding with CA Service Catalog Installation. If you are not using the default ports, verify that they are open before using them. You do not need to open the shutdown ports for the products and components listed in this table:

Product or Component	Default Port
CA Service Catalog startup port (includes access for CA Service Catalog and Service Accounting Component)	8080
CA Process Automation, CA Service Desk Manager, CA Business Intelligence/CABI Jaspersoft Server Reports and other CA products that integrate with CA Service Catalog	8080
CA Service Catalog shut down port	8085
Visualizer port for CA Service Desk Manager	9080
SQL Server server	1433
Oracle server	1521
CA EEM	5250
IMQ port of CA EEM	7676

If you plan to implement clustering for CA Service Catalog, open all Apache JServ Protocol (AJP) and load balancing ports.

Ports and Port Ranges for CA Asset Portfolio Manager

The following list displays default ports for a typical CA APM installation:

Product or Component	Default Port
ITAM default	99
SQL Server server	1433
Oracle server	1521
CA EEM	5250
ITAM default https	443

Supported Browser and Network Settings

For the detailed list of CA Service Management supported web browsers, see Supportability Matrix. Review the following related to browser and network settings before you start the installation:

IPV 6 Settings

- **IPV6**: If you are installing CA SDM in an IPV6 environment, the installation may fail. Ensure that the **NX_PROTOCOL_ONLY** variable is set to IPV6 and click **Retry Install.**
  You can change the default TCP Internet Protocol (TCP/IP) setting on one or more servers. This setting cannot be forced on the client if it is not supported on the server.

The TCP/IP setting is controlled by using the **NX.env** file, which is found in the **$NX_ROOT** directory. Use a text editor such as WordPad to edit this file. The following option controls the TCP/IP setting:
NX_PROTOCOL_ONLY=mode
For example: *NX_PROTOCOL_ONLY=ipv4*

- For Windows installation, install the **pslist.exe** tool and add its directory path variable to the system path variable.
- IIS 8.0 with support for ASP.NET 3.5 and ASP.NET 4.5 (Applicable only for ITAM).

Pre Installation Considerations

Before you proceed with CA Service Management installation, review the following installation considerations:

- CA Service Management Default User Information
- CA Service Desk Manager Considerations
- CA Service Catalog Considerations
- CA Asset Portfolio Manager Considerations
- Oracle Real Application Clusters (RAC) Considerations

CA Service Management Default User Information

The following table lists default user information for CA Service Management implementation:

CA Service Management Products	DBMS Users	Roles
• CA Service Desk Manager • CA Service Catalog • CA Asset Portfolio Manager		Needs user with DBMS Administrator privileges for installation.
CA Service Management	MDBADMIN	• **ams_group** • **amsgroup** • **db_ddladmin** • **db_owner** • **db_securityadmin** • **regadmin** • **service_desk_admin_group** • **service_desk_ro_group** • **usmgroup** • **workflow_admin_group**
CA Service Desk Manager	mdbadmin	• **ams_group, amsgroup** • **db_ddladmin** • **dbsecurityadmin** • **regadmin** • **service_desk_admin_group** • **service_desk_ro_group** • **workflow_admin_group**
CA Asset Portfolio Manager (ITAM)	uapmadmin	• **uapmadmin_group** • **Db_owner**
CA Asset Portfolio Manager	uapmbatch	**uapmbatch_group**
CA Asset Portfolio Manager	uapmreporting	**uapmreporting_group**

CA Process Automation (formerly known as IT PAM)	Defined during Installation	Users of this type require the DB owner privileges when starting CA Process Automation server for the first time. After that, permissions to Select, Insert, Update, and/or Delete the workflow processes are sufficed.
CA Service Management Products	**OS/CA EEM Users**	**Rights**
CA Asset Portfolio Manager (ITAM)	casmadmin	OS – Admin full rights
CA Asset Portfolio Manager	uapmadmin	CA APM internal application use only
CA Service Desk Manager	ServiceDesk	OS - Admin Full Rights
CA Service Catalog	spadmin	CA EEM Only
CA Process Automation	pamadmin	CA EEM Only.

For more information, see CA Service Desk Manager Default User Information

CA Service Desk Manager Considerations

Review the following for implementing CA Service Desk Manager (CA SDM):

NOTE

When planning for CA SDM deployment, for Scalability purposes, ensure that the upper limit of 64 processes is adhered to for a Primary Server. If the number of processes exceeds 64 on a Primary Server, plan to have more processes on CA SDM secondary server. CA SDM installation can ideally have up to 40 Processes on a single server. Adding a new web engine domserver to your existing environment results in adding 3 more processes.

NOTE

Ensure to restart xFlow interface services whenever you restart CA SDM Services. This is applicable for an environment that CA SDM and xFlow Interface Installation.

NOTE
For SUSE 15.6, before you start the 17.4GA SDM Installation, ensure you create a soft link using the command mentioned below:

```
ln -s /usr/bin/ifconfig /sbin/ifconfig
```

SUSE Linux Enterprise Server 15.6 supports 17.4.3.0 and above.

NOTE
For RHEL 9.4, before you start the 17.4GA SDM Installation, ensure you create soft links using the commands mentioned below:

```
yum install libnsl
 yum install libnsl2
 ln -s /usr/bin/ifconfig /sbin/ifconfig
 sudo ln -s /lib64/libnsl.so.3 /lib64/libnsl.so.2
```

RHEL 9.4 supports **17.4.5.0** and above.

Prerequisites for SDM 17.4.0.0 on SUSE Enterprise Linux 15.6 and RHEL 9.4

Follow these steps:

1. Download the latest applyPTF tool for SUSE with APAR No. 99112268 from Broadcom Support, and extract the T5UG572.tar.Z file.

2. Download the CA Service Desk Manager 17.4 installer DVD from Broadcom Support, and extract "**CA SERVICE DESK MANAGER 17.4 FOR LINUX.iso**"
3. Paste the file **applyptf.linux** in the path:
   ```
   <extracted "CA SERVICE DESK MANAGER 17.4 FOR LINUX.iso">\filestore\utils\ApplyPTF\
   ```
4. Before you start the 17.4GA SDM Installation, ensure you create a soft link using the command mentioned below:

   NOTE
   For SUSE 15.6, before you start the 17.4GA SDM Installation, ensure you create a soft link using the command mentioned below:
   ```
   ln -s /usr/bin/ifconfig /sbin/ifconfig
   ```
   SUSE Linux Enterprise Server 15.6 supports 17.4.3.0 and above.

   NOTE
   For RHEL 9.4, before you start the 17.4GA SDM Installation, ensure you create soft links using the commands mentioned below:

   ```
   yum install libnsl
    yum install libnsl2
    ln -s /usr/bin/ifconfig /sbin/ifconfig
    sudo ln -s /lib64/libnsl.so.3 /lib64/libnsl.so.2
   ```

   RHEL 9.4 supports **17.4.5.0** and above.
5. SUSE Linux Enterprise Edition 15.6 and RHEL 9.4:
   a. Ensure that you have exported Oracle variables on the server where you want to install CA SDM.
   b. Navigate to the root directory extracted from step 2, and start the installation by running the following command:
      ```
      ./setup.sh
      ```
      To install CA Service Management from the non-root user on Unix or Linux, you must have the permissions to execute the sudo command.

      NOTE
      To install CA Service Management from the non-root user on Unix or Linux, you must have the permissions to execute the sudo command. Follow these steps:

      1. Export the appropriate oracle variables and library paths. Navigate to where you have downloaded the CA Service Management installer DVD and execute the below command:
      2. **sudo -E PATH=$PATH ./setup.sh**

For the rest of the installation follow from **Step 5** from Installing CA Service Desk Manager.

Search Server

Consider dedicating a server to install Search Server. The Search server has a built-in clustering feature and can be installed on a standalone setup. For large setups, we recommend installing the search server on three or more servers and manually configuring it to a fault-tolerant cluster.

Default User Information

The following table lists default user information for a typical CA SDM implementation:

OS	User	Default Username	OS Level	How it is Created?
Windows	CA SDM	ServiceDesk	Yes	Automatically
	CA EEM	EiamAdmin	No	Created while configuring CA EEM.

	Microsoft SQL Server	mdbadmin	No	Created in Microsoft SQL Server during configuration.
	Oracle	mdbadmin	No	Created in Oracle database during configuration.
	CA Service Management Admin User	CASMAdmin	No	Manually created.
	Oracle	mdbadmin	No	Created in the MDB during configuration.
	CA Service Management Admin User	CASMAdmin	No	Manually created.
Linux	CA SDM	srvcdesk	Yes	Manually created.
	Oracle	mdbadmin	No	Created in the database during configuration.
	CA Service Management Admin User	CASMAdmin	No	Manually created.

Password Policy

User authentication does not work if the system is using shadow files and there is an **x** in the password field of the /etc/passwd file. Users cannot log in to CA SDM. The password that you have specified for the CA SDM privileged user must conform to the password policy imposed by the network domain. If password confirms to the policy, only then CA SDM configuration works.

Shared Components

- When installing CA SDM, do not install the CA Shared Components in the same directory as the CA SDM installation directory (**NX_ROOT**).
- References to NX_ROOT pertain to the environmental variable containing the installation path of CA SDM. The NX_ROOT variable is set in the **NX.env** configuration file that is used to set environment variables for CA SDM. **Example:** NX_ROOT Definition @NX_ROOT=C:\Program Files(x86)\CA\Service Desk Manager
- Do not specify spaces in the installation media path and folder name to avoid installation failure.

Options Manager

When you are installing CA Service Desk Manager for the first time, the Options Manager **copy_inactive** web option is not installed by default. The links to inactive objects are not copied. Consider the following information when you do not install the **copy_inactive** option:

- CA SDM copies SRELs that point to inactive objects when SREL is required.
  For example, Organization org1 has an inactive Location. When you copy org1, the new organization does not have a location. However, if SREL is required, CA SDM ignores this rule and copies the location. CA SDM ignores the rule and you can copy a CI with an inactive class.
- CA SDM does not copy LREL (many-to-many) relationships to inactive objects. For example, the CI named **CITest1** has a relationship to the inactive Organization org2. When you copy **CITest1**, org2 does not link to the new CI.
- These rules also apply when you copy a ticket, even when you create a ticket from a template. Even if you have installed copy_inactive, the exception to the rule is for Inactive Areas and Categories, which are not copied from existing tickets or populated from the templates.

Database Considerations

- If CA SDM has been configured with one database, and then the configuration is run a second time with a different database type, the configuration does not work. For example, you initially configure the Microsoft SQL Server and then, you configure it again for an Oracle database. The workaround is to restart the computer before the second configuration is run.
- Database connection information, if different, is not accepted in subsequent configurations. If an extra configuration is needed as a result of a change in the database connection information, delete the $NX_ROOT\NX.env file before proceeding.
- Suppose that you are using an Oracle database and you want to use existing tablespaces. Create a Data tablespace that is at least 400 MB, and an Index tablespace that is at least 100 MB before configuring CA SDM.
- Set the Oracle environment variables before you install or migrate CA SDM, as follows:
  - Verify that the ORACLE_HOME variable is set correctly. You must export TWO_TASK variable when Oracle 32-bit client variables are exported on non-windows.
  - Include the 32-bit and 64-bit Oracle libraries in the library path variable LD_LIBRARY_PATH. It is applicable only on LINUX servers.

Web Server Considerations

- Tomcat is set as the default CA SDM web server during the product installation. You can also use the following web servers based on your requirement:
  - **Internet Information Services (IIS)**
    Applicable only on Windows. If you want to use IIS as the CA SDM web server, you must configure the IIS before installing CA SDM. For more information about how to configure IIS, see Configuring IIS.
  - **Apache**
    Applicable only on Linux and Solaris. If you want to use Apache as the CA SDM web server, you must configure the Apache before installing CA SDM.
- If Tomcat is configured with the external authentication on the primary server, set up a secondary server with a webengine and repository daemon. This setup allows users that are not authenticated to use attachments. The Tomcat installation on the secondary server cannot use external authentication.
- The CA SDM installation uses the Tomcat port **8080**. Other CA products, such as CA Asset Portfolio Management or the Service Delivery Suite of products also uses Tomcat port to 8080. If you are installing multiple CA products on the same server, select a port number other than 8080 for subsequent CA product installations. Selecting a different port number helps the products function properly together. To change the Tomcat port number to something other than 8080 for CA SDM, install the product and if the product is already installed, re-run the product configuration and specify an available port number for Tomcat when prompted.
- After CA SDM restart, the CA SDM Tomcat process may not start properly. To start Tomcat properly, stop and restart Tomcat using the following commands:

```
pdm_tomcat_nxd -c STOP
pdm_tomcat_nxd -c START
```

Configuring Internet Information Services (IIS)

Follow these steps:

1. Launch the **Server Manager** on the IIS installed server.
   Server Manager **Dashboard** screen opens.
2. Navigate to **Local Server** and go to the **ROLES AND FEATURES** section.
3. Select the **Add Roles and Features** option from the **TASKS** drop-down list.
   The Add Roles and Features wizard opens.
4. Click **Next** on the **Before You Begin** screen.

The Select installation type screen opens.

5. Select the **Role-based or feature-based installation** radio button and click Next.
   The Select destination server screen opens.
6. Select the **Select a server from the server pool** radio button and ensure that the local server is listed in the Server Pool section. Click **Next**.
7. The **Select server roles** screen opens.
8. Select **WEB SERVER (IIS)** from the Roles list.
   The Add features that are required for Web Server (IIS)? window opens.
9. Click **Add Features** and then click **Next**.
   The Select features page opens.
10. Select the additional IIS features as required and click **Next**.
    The Web Server Role (IIS) screen opens.
11. Click **Next**.
    The Role Services page opens.
12. Select the following from the **Role Services** and click **Next.** You can optionally select the other role services as required.
    - Web Server
    - Common HTTP Features
    - Default Document
    - Directory Browsing
    - HTTP Errors
    - Static Content
    - Health and Diagnostics
    - HTTP Logging
    - Performance
    - Static Content Compression
    - Security
    - Request Filtering
    - Management Tools
    - IIS Management Console
    - Application Development
    - CGI
    - Server Side Includes

    NOTE

    If you do select CGI, then only the web browser launches CA Service Desk Manager. The web browser attempts to open the PDMWEB.EXE file.

    The Confirm installation section screen opens.
13. Review the roles and features that you selected to install, and click **Install**.
14. Click **Close** when the installation is completed.
15. Launch the **Command Line Interface** and run the following command:
    pdm_configure

    NOTE

    To run the command pdm_configure, you must have the privileged user access or root access on the CA SDM server.

Internationalization

- **Multi-byte Characters -** You cannot use multi-byte characters for the user you are logged in as or for the CA SDM privileged username. This applies when you install on multi-byte operating systems such as Simplified Chinese and Japanese. Knowledge searches containing multi-byte Japanese characters works properly with SQL Server only when SQL Server is installed with Windows collation. Ensure to specify the Collation option for your data during the SQL Server installation.

  NOTE

  Do not specify multi-byte characters in the file path names during installation and configuration. Doing so causes failure.

- **UTF-8** - CA SDM must run on UTF-8 locale on Linux and UNIX platforms.
- **Timespan** Symbol names that are provided with the default CA SDM installation (**Administration** tab, **Service Desk**, **Application Data**, **Codes**, **Timespans**) are in English. For example, TODAY, YESTERDAY, THIS MONTH**.** For localized versions of the product, administrators may want to define new localized Timespans as required. Do not delete or modify the default Timespans.
- **Date formats-** in CA SDM do not support international specifiers such as, localized date-picture specifiers. For example, "jj/MM/AAAA" for French. The syntax is limited to generic specifiers such as "DD/MM/YYYY". However, many international short date-time patterns can be constructed from these generic specifiers (for example, "YYYY.MM.DD" would supply a common Japanese short date format).
  International users may want to adjust the **DateFormat** property in **web.cfg** to use the date and date-time formats.
- For outbound plain-text email notifications, the **NX_SMTP_HEADER_CHARSET** and **NX_SMTP_BODY_CHARSET** options may have to be adjusted in the NX.env file. Adjusting the options helps to tag the email message correctly with the character encoding used by the international operating environment. The default values for these options are set to UTF-8 on all platforms.
- International users may want to change the default spell check locale in lexicon to a regional locale. Use the **LEX_LANG** option in the Options Manager.
- **Short File Names** - If you have disabled short file names on your Windows operating system enable these before installing CA SDM. Set both the **TEMP** and **TMP** environment variables to a short file name. For example, **c:\temp**, after enabling short file names. For more information, see Microsoft Knowledge Base Article 121007.
- **Web Interface and Internet Information Services (IIS)**: To configure the web interface with IIS 7.0 on Windows 2008, install the CGI and Metabase Compatibility components of IIS 7.0. Add these components using the Roles section of the Server Manager, after installing the IIS Management Compatibility modules.
- Localized releases of CA SDM are supported only on the matching localized Windows server operating environment. In all cases, configure the **Language for non-Unicode programs** for your system to support the target certified language. This setting is available in the **Regional and Language Options** window in the Control Panel.
  For more information about the localized releases of Windows Server operating systems, see the Microsoft Global Development and Computing Portal.

CA SDM Server Components

CA SDM includes components that work together and run on different servers, depending on the CA SDM configuration. Before you begin your implementation, ensure that you have a basic understanding of the following components:

SDM Server Components Name	Description
Daemon Manager (pdm_d_mgr)	Starts the process sets as defined in the startup file, pdm_startup. By default, the daemon manager tries to start a failed component up to 10 times. To check the status of all CA SDM components, use the **pdm_status** utility. The **pdm_d_refresh** utility instructs the daemon manager to start a new cycle of 10 attempts to start any process marked as previously failed. The daemon manager runs on all CA SDM servers.
Message Dispatcher (sslump_nxd)	Acts as a common bus or message passing system. Components that require to communicate with each other first register with the Message Dispatcher. When a component sends a message, the Message Dispatcher delivers it to those components that have registered to receive that type of message. If two components communicate so much that it would be inefficient to pass the messages through the Message Dispatcher, they create a fast channel between them. You can view a list of registered components using theslstatutility. The message dispatcher runs on the following servers, depending on your CA SDM configuration: • Conventional: **Primary server** • Advanced Availability: **All servers**
Database Agent (platform_agent)	Performs Microsoft SQL Server queries on the database. Database agents adhere to the logical schema of CA SDM and translate the SQL at this level to the physical database platform SQL. **Note:** The database agent detects momentary disconnection and failed queries, and attempts to reconnect and communicate with the database. This is only meant for short outages, such as for a brief network outage and momentary disconnection. It is not meant for long outages such as taking down a database service for maintenance and so forth. The agent only retries the connection for a defined number of times (the default is 3 times), and only for a short time period of a few minutes interval. If the outage is longer than a few minutes, the agent stops trying to connect. CA SDM must be recycled after the database is made available again. The database agent runs on the following servers based on the CA SDM configuration: • Conventional: **Primary server** • Advanced Availability: **All servers**

Virtual Database (bpvirtdb_srvr)	Enables the operation of multiple Object Managers. All Object Managers running on primary or secondary servers connect to the Virtual Database, which arbitrates their access to database agents. For example, when retrieving a new range of ticket reference numbers, the Virtual Database ensures that only one Object Manager at a time accesses the table containing the reference numbers. The Virtual Database also performs caching of database information for Object Managers. The Virtual Database runs on the following servers: • Conventional: **Primary server** • Advanced Availability: **All servers**
Continuous Archive and Purge (arcpur_srvr)	Runs the archive and purge rules as configured by the CA SDM administrator. The Continuous Archive and Purge runs on the following servers: • Conventional: **Primary server** • Advanced Availability: **Background server**
Database Monitor (dbmonitor_nxd)	Monitors changes to common tables in CA MDB, for example, ca_contact. The Database Monitor runs on the following servers: • Conventional: **Primary server** • Advanced Availability: **Background server**
Mail Daemon (pdm_mail_nxd)	Sends outbound email notifications. The Mail Daemon runs on the following servers: • Conventional: **Primary server** • Advanced Availability: **All servers**
Mail Eater (pdm_maileater_nxd)	Accepts inbound email for ticket creation and updates. The Mail Eater daemon runs on the following servers: • Conventional: **Primary server** • Advanced Availability: **Background server**
Notification Manager (bpnotify_nxd)	Manages notifications in a Windows environment. The Notification Manager runs on the following servers: • Conventional: **Primary server** • Advanced Availability: **Application, background, and standby servers**
Spell Checker (lexagent_nxd)	Performs spell checking as requested by users. The Spell Checker runs on all CA SDM servers.
Text API Daemon (pdm_text_nxd)	Creates and updates tickets by external interfaces, such as the command line and email. The Text API daemon runs on all CA SDM servers.
Timed Event (animator_nxd)	Runs the delay times of events. If an implementation has many service types or contracts, the Timed Event engine tracks the active events. You must dedicate the primary server Object Manager entirely to the Timed Event engine. You can configure other Object Managers on the primary or secondary servers for product access as appropriate. The Timed Event daemon runs on the following servers: • Conventional: **Primary server** • Advanced Availability: **Background server**

Time-To-Violation (ttv_nxd)	Calculates projected violation times for service types. The Time-To-Violation daemon runs on the following servers: • Conventional: **Primary server** • Advanced Availability: **Background server**
Proctor Daemon (pdm_proctor_nxd)	(For Windows only) Starts and restarts the CA SDM components, as instructed by the Daemon Manager, on primary and secondary servers. When you install a secondary server, the **pdm_proctor_nxd** process is installed as the CA SDM Remote Daemon Proctor service. When the primary server starts, the Daemon Manager instructs the Remote Daemon Proctor to connect to the Message Dispatcher. The Daemon Manager then instructs the Remote Daemon Proctor to start components on the secondary server. The process for starting the components is defined by the Process Sets in the startup file **pdm_startup**. The Proctor daemon runs on all the CA SDM servers in advanced availability configuration.
Object Manager (domsrvr)	Acts as the server process of CA SDM. When you install a primary server, by default, two Object Managers are installed: one for connections to the product, and one dedicated to the Web Screen Painter. Having multiple Object Managers helps you test your modifications without affecting the production environment. When you install a secondary server, you can configure more Object Managers. There must always be a default Object Manager running on the primary server to which clients such as the Timed Event engine can connect. The Object Manager also caches various records and tables for clients. If you use **pdm_userload** to manipulate these records, you can also use the **pdm_cache_refresh** utility to make the Object Manager retrieve the new data. The Object Manager runs on all the CA SDM servers in advanced availability configuration. The dedicated object managers have been added for heat weather daemon and reminder daemon.
Method Engine (spel_srvr)	Runs SPEL code, event, macros, and so forth, for an Object Manager. We recommend that you run every Object Manager with its own method engine. The Method Engine runs on all CA SDM servers.
Login Server (boplgin)	Manages authenticated user sessions. The Login Server runs on the following servers: • Conventional: **Primary server** • Advanced Availability: **All servers**
LDAP Virtual Database (ldap_virtdb)	Interfaces with an LDAP directory. The Method Engine runs on the following servers: • Conventional: **Primary or secondary server** • Advanced Availability: **Background or application server**

Knowledge Management Search Daemon (bpebr_nxd)	Performs knowledge base searches. Upon CA SDM startup, the **bpebr_nxd** daemon caches Knowledge Document data in its memory from the database. With a large document base, you might have memory resource issues. The **bpebr_nxd** daemon has the following size requirements: Knowledge Management Search • 100,000 documents • Memory size = 332,000 KB The Knowledge Management daemon runs on the following servers: • Conventional: **Primary or secondary server** • Advanced Availability: **Background server**
Knowledge Management/Keyword Indexing Daemon (bpeid_nxd)	Indexes the knowledge base The Keyword Index daemon runs on the following servers: • Conventional: **Primary or secondary server** • Advanced Availability: **Background server**
Knowledge Management FAQ Ratings Daemon (bu_daemon)	The bu_daemon runs on the following servers: • — Conventional: **Primary or secondary server** — Advanced Availability: **Background server**
Knowledge Report Card Daemon (krc_daemon)	Performs the calculations for the Knowledge Management Knowledge Report Card (KRC) feature. This feature enables analysts and managers to display different matrix views of their knowledge contributions and provide feedback about which documents are most effective. The information that is provided can be used to improve the processes of creating knowledge documents and providing the best support to customers. The Knowledge Report Card daemon runs on the following servers: • Conventional: **Primary or secondary server** • Advanced Availability: **Background server**
Knowledge Management Daemon (kt_daemon)	Manages knowledge base administration and knowledge management logic. It also manages notifications and the document approval process. The Knowledge Management daemon runs on all CA SDM servers.
Repository Daemon (rep_daemon)	Manages the attachment repositories for CA SDM and the Knowledge Management/Keyword Search Daemon. The Repository Daemon runs on the following servers: • Conventional: **Primary or secondary servers** • Advanced Availability: **All servers**
Version Control Daemon (pdm_ver_nxd)	Synchronizes the schema files between a primary and secondary server to ensure that they are using the same schema. The Version Control Daemon runs on the following servers: • Conventional: **Primary server** • Advanced Availability: **Background server**
Apache Tomcat Web Server (javaw)	Enables certain features to be implemented, regardless of whether Microsoft Internet Information Server (IIS) is used as the web server for access to CA SDM. These features include Graph Items, Attachments, and Web Services. The Apache Tomcat web server can be administered with the Apache Tomcat controller (***pdm_tomcat_nxd***). The Apache Tomcat Web Server runs on all CA SDM servers.

Web Engine (webengine)	Connects to web browsers through apdmwebcgirunning on a Microsoft IIS or Apache Tomcat web server. There must be at least one web engine for WSP on the following servers. • Conventional: **Primary server** • Advanced availability: **Application, standby, and background servers** This process ensures that Web Screen Painter Schema Designer can write schema files. Web engines are the true client of an Object Manager, which the web browser uses to access the product. Web engine uses cache.htmplweb forms for connected users. You can manipulate the cache using the **pdm_webcache** utility and can see connection statistics using the **pdm_webstat** utility. The Web Engine runs on all CA SDM servers.
RF Broker (pdm_rfbroker_nxd)	(Applicable for advanced availability configuration only). Manage the roles of the servers and controls them across the configuration. This daemon runs on all servers in advanced availability configuration and performs the following tasks: • Acquire information about background and standby servers. • Update information (such as slump ID, node name, server type) in the server status monitor class. • Receive broadcast messages of server status changes. • Quiesce the requests. Register for **SLUMP_NODE_GONE** messages that are forwarded to ServerStatusMonitor objects when the failing node is the background server.
Login User Authentication (bopauth_nxd)	This daemon performs the operating system user account validation. To match a user with an access type, the contact record lookups using the System Login field. If your business provides CA SDM for other client businesses, place the Login server on a secondary server at a single client location. The external authentication is then enabled in access types. Based on the CA SDM configuration, the daemon runs on the following servers: • Conventional: **Primary or secondary (if configured) server** • Advanced Availability: **Background or application (if configured) server**
Interval Logger (pdm_intrvlog_nxd)	The interval logger daemon gathers the debugging information for debugging the system. Runs on all CA SDM servers.
QRY KPI Daemon (kpi_qry_daemon)	Executes the SQL queries for updating the Key Performance Indicators (KPIs) in the database. The QRY KPI daemon runs on the following servers: • Conventional: **Primary server** • Advanced Availability: **Background server**
SYS KPI Daemon (kpi_sys_daemon)	Daemon to collect system type key performance indicators and write to the database. The SYS KPI daemon runs on the following servers: • Conventional: **Primary server** • Advanced Availability: **All servers**
Confirm Database (confirm_db)	A utility to verify database access that runs on all CA SDM servers.
Data Dictionary (ddictbuild)	A utility to build the data dictionary and runs on all CA SDM servers.

Set LogFile (pdm_logfile)	A utility to display or set the log file size limits. The **pdm_logfile** daemon runs on all CA SDM servers.
Report Manager (pcrpt_nxd)	A utility for PC reporting that runs on all CA SDM servers.
RPC Server (rpc_srvr)	Used for making outbound SOAP web service calls and runs on all CA SDM servers.
CA SA Tomcat (sa_tomcat)	CA SATomcat to run support automation and can be configured on any CA SDM server.
Visualizer Tomcat (viz_tomcat)	The Tomcat instance to run visualizer and can be configured on the following servers: • Conventional: **All servers** • Advanced Availability: **Application server**
Knowledge Management Indexing Daemon (bpeid_nxd)	Responsible for indexing the knowledge documents and runs on the following servers: • Conventional: **Primary server** • Advanced Availability: **Background server**
Registration Server (mdb_registration_nxd)	An agent for handling MDB registration requests. The Registration Server runs on the following servers: • Conventional: **Primary server** • Advanced Availability: **All servers**
Search Deamon (es_ebl)	Responsible for the CA SDM data sync with the elastic search server.
Catalog_sync	Deamon syncs new log comments & attachments with Ca. **Note:** Sync only happens for Incidents/Change Order that is created by Catalog.
SDM Telemetry Deamon (sdmtelemetry)	A new singleton process which reads kpi data and push into the segment for Telemetry.
Heat Weather Deamon (pdm_hw_nxd)	Calculates the heat of tickets and weather for analyst **Note:** The Demon runs only when the xFlow is enabled.
Reminder Deamon (reminder_nxd)	Responsible for triggering the follow-up reminders for tick **Note:** The Demon runs only when the xFlow is enabled.

Considerations for Conventional Configuration

Review the following topics before you install CA Service Desk Manager (CA SDM) in Conventional Configuration:

Login Permissions

For CA SDM conventional configuration, perform the following installation tasks:

1. Install Primary Server
2. Install Secondary Servers
3. Install xFlow Interface
4. Install Search Server

Complete the following steps:

- (Windows) Log in as an Administrator and have full Administrative permissions.
- (Linux) Log in as the root user and have the correct permissions to the root account.
- Create the **CA Service Desk Manager Server Privileged** user before you begin the installation.

Installation Home Directory

Complete the following steps:

- Determine the home directory in which you want to install the product. The default home directory for CA SDM (on Windows) is *\Program Files (x86)\CA\Service Desk Manager*.
- Determine the home directory in which you want to install the shared components that the product uses (for example, the Java Runtime Environment (JRE) and Apache Tomcat). The default home directory for the shared components is *\Program Files (x86)\CA\SC*.

Database

CA Service Management supports both Microsoft SQL Server and Oracle databases.

Microsoft SQL Server

Microsoft SQL Server is only natively supported on the Windows platform. For example, if your implementation consists of servers with heterogeneous operating systems like Windows and Linux, select Oracle as the DBMS as Microsoft SQL Server is not supported on Linux.

Obtain the following information:

- – The named instance of the server that is running Microsoft SQL Server
  – The Microsoft SQL Server database user name and password
  – The Microsoft SQL Server database port number

Oracle Server

To ensure that you can configure the product and components on Oracle, obtain the following information:

- – Whether the Oracle database is local or remote
  – Whether you are required to create tablespaces
  – The Net service name
  – The DBA user name and password
  – The data and index tablespace name
  – The complete path for the tablespace
  – JDBC connection information, including the system identifier (SID) and listener port

For configuring Oracle 19c/ 21c database

Install the Oracle 32-Bit/ 64-Bit Client with adminstrator option. Also, create a net service name on the Oracle client to point to the Oracle database server instance.

If you are using an Oracle database and you want to use an existing tablespace:

- Create a Data tablespace that is at least 400 MB, and an Index tablespace that is at least 100 MB before configuring CA SDM.
- Increase the hardware configuration of the DBMS server. Each of the servers is directly connected to the database, leading to an increased resource contention at the DBMS level.

Primary, Secondary Server, and User Configuration

NOTE

If in the Conventional configuration the Primary and Secondary are on separate nodes, and the **xFlow Interface** is deployed on the Secondary server, it is observed that a Database Client is needed on the Secondary Server for the installation to complete successfully.

To ensure that you can configure both the primary and optional secondary servers at the end of the product installation, have the following information available:

- The Domain Name System (DNS) of the primary and secondary servers
- The name or alias of the server
- The name or aliases of the object manager to which you want to establish a connection
- The name or IP address of the secondary server
- The slump socket port to be used for communication by all of CA SDM
- The proctor socket port to be used by the secondary server to listen for messages from the primary server
- To ensure that you can configure account information for product administration at the end of the product installation, have the **CA SDM privileged user name** and password ready
  (Linux) Manually create your privileged user. On Windows, this user is automatically created during the product configuration, but on Linux, you must manually create this user.

Web Interface Configuration

To ensure that you can configure the web interface at the end of the product installation, have the following information available:

- The web host name and web server
- (Windows NT only) The website to be used by the CA SDM server
- The Tomcat port number and socket port number for shutdown requests
- Browsers and Antivirus Software—Complete the following steps:
  - (Firefox) Optionally enable Windows authentication (enterprise single sign-on).
  - Temporarily disable any antivirus software, as it speed up the installation. After the installation, enable your antivirus software again.

For More Information, see IIS Prerequisites.

Web Screen Painter

- Web Screen Painter automatically installs on the primary server when you install CA SDM. You can install Web Screen Painter on separate computers using the installation media.
- File Name Creation: CA SDM requires 8.3 File Name Creation. See your operating system Help and Support documentation for more information.

Considerations for Advanced Availability

CA SDM advanced availability configuration provides higher availability, reduces down times, and supports rolling maintenance. Consider opting for the advanced availability configuration when any of the following factors in your CA SDM environment are true:

- You require a high degree of CA SDM availability.
- You require the CA SDM servers to be more independent and more resilient to failures.
- You require an ability to remove and return the CA SDM servers to service, without bringing down the entire CA SDM installation.

- You require near zero downtime during rolling maintenance.

The biggest difference between conventional (**Primary**/**Secondary**), and Advanced Availability (**Background**, **Standby**, **App servers**), is that in a conventional configuration, only the primary server talks to the database. However, in an Advanced Availability environment, each server independently talks to the database and has its own virtual database process and database agents for better performance.

CA SDM advanced availability configuration offers more resilience to server outages, higher application availability, and supports rolling maintenance with minimal end-user disruption. This configuration requires the following type of servers:

- **Database server for the MDB**
- **One Background Server**
  The Background server takes the place of the Primary server in the conventional mode. It handles all background transactions, for example, triggering events and notifications, checking for conditions, and so on.
- **One or more Standby Servers**
  The Standby server is meant for standby purposes. During server failures and disruptions, the standby server is promoted as the background server when the original background server is revived and repaired.
- **One or more Application servers**
  End-user traffic only hits the Application servers, on which you use an F5 load balancer to balance the load between two (or more) application servers. End-user traffic load does not affect the Background or Standby servers. Use the background server and log in to the CA SDM console to configure the standby and app servers.

When you opt for CA SDM Advanced Availability mode, you perform the following installation tasks:

1. Install Background Server
2. Install Application Servers
3. Install xFlow Interface
4. Install Search Server

Ensure that you have considered the following planning consideration before installing CA SDM Advanced availability configuration:

Login Permissions

Complete the following steps:

- (Windows) Log in as an Administrator and have full Administrative permissions.
- (Non-Windows) Log in as the root user and have the correct permissions to the root account.
- Create the **CA Service Desk Manager Server Privileged** user before you begin the installation.

Installation Home Directory

Complete the following steps:

- Determine the home directory in which you want to install the product. The default home directory for the product is *C:\Program Files (x86)\CA\Service Desk Manager*.
- Determine the home directory in which you want to install the shared components that the product uses (for example, the Java Runtime Environment (JRE) and Apache Tomcat). The default home directory for the shared components is *C:\Program Files (x86)\CA\SC*.

Database Considerations

CA Service Management supports both Microsoft SQL Server and Oracle databases. Select a database as per your environment requirement.

- Microsoft SQL Server is only natively supported on the Windows platform. For example, if your implementation consists of servers with heterogeneous operating systems like Windows and Linux, select Oracle as the DBMS as Microsoft SQL Server is not supported on Linux. Obtain the following information:
  - The named instance of the server that is running Microsoft SQL Server.
  - The Microsoft SQL Server database user name and password.
  - The Microsoft SQL Server database port number.
- To help ensure that you can configure the product and components on Oracle, obtain the following information:
  - Whether the Oracle database is local or remote.
  - Whether you are required to create tablespaces.
  - The Net service name.
  - The DBA user name and password.
  - The data and index tablespace name.
  - The complete path for the tablespace.
  - JDBC connection information, including the system identifier (SID) and listener port.
- To configure a 32-bit/ 64-bit Oracle 19c/ 21c database, do the following:
  - Install the Oracle 32-bit/ 64-bit client with adminstrator option on the server. Also, create a net service name on the Oracle client to point to the Oracle database server instance.
- If you are using an Oracle database and you want to use an existing tablespaces, do the following:
  - Create a Data tablespace that is at least 400 MB, and an Index tablespace that is at least 100 MB before configuring CA SDM.
  - Increase the hardware configuration of the DBMS server. Each of the servers is directly connected to the database, leading to an increased resource contention at the DBMS level.

Advanced Availability Configuration Considerations

Planning Consideration	Checklist
General Advanced Availability Consideration	• Additional hardware costs are expected as you require one background, at least one standby, and one or more application servers. The configuration of standby and background server must be identical. • You require a remote database server and a server to share knowledge tools index files, knowledge tools import/export files, archive purge output files, and attachment repositories. To allow the background and the standby server to access these files, a shared location is required. Linux installations can use NFS mounts. UNC support has been added for Windows installations. • The CA SDM performance is expected to remain the same even with the additional servers for background and standby operations. If you deploy more application servers, the performance can possibly improve. • Each of the servers is directly connected to the database, leading to an increased resource contention at the DBMS level. We recommend increasing the hardware configuration of the DBMS server. • Converting a conventional configuration to an advanced availability configuration is a manual effort. The larger implementations are more complex and you can engage CA Services for assistance. • Install the background and the standby server on the same network subnet to make ping times and latencies from different application servers similar. • Consider locating the background and the standby servers in a central location with a good network connectivity to all your users. The application servers can be located either centrally or they can be distributed across the globe. • An advanced availability configuration must always have **one background server** and at least **one standby server**. • (Recommended) Ensure that both background server and all other standby servers have similar configuration. During a failover when a standby server becomes the new background server, it functions exactly like the old background server. • Any number of standby servers can be configured. To increase the CA SDM availability, consider placing one standby server in your backup data center or the disaster recovery site. • The minimum advanced availability implementation requires one application server. We recommend you to have two application servers to increase the availability and a load balancer to direct web traffic. • Except for CA SDM administrators, no other users are allowed to log in to the background server. Also, no users are allowed to log in to the standby servers. • You can send email notifications from all CA SDM servers. There is no way to limit or configure this option. Every server in the advanced availability configuration must have a connection to the mail server. • An email notification resulting from an end user interaction is sent from the application server to which the user is connected.to. • An email notification resulting from any background processes (such as Animator processing an Attached Event) is send by the **pdm_mail_nxd** utility running on the background server. • During a background server failure, the queued emails are sent when the background server comes up as a standby server.

Failover Consideration	During failover of the background server to the standby server, consider the following: • New users cannot log in during a failover process. • For those users who are already connected, the following actions do not work during a failover. These actions must be repeated by the user after the failover is complete: • — Creating the tickets with attachments. — Downloading the attachments. — Searching Knowledge documents. — Indexing the new knowledge documents. — Inbound emails. — The SLA events that are not triggered until the failover is complete. **Warning!** If you have configured a third-party tool to enable auto-failover of CA SDM servers, you must disable the tool before performing a rolling maintenance.
Database Considerations	• A direct connection exists from each of the servers to the other as well as to the database. If CA SDM server is within DMZ, you are required to open the firewall ports or implement a tunneling proxy technology for this connectivity. Also consider licensing arrangements with your DBMS vendor. • Ensure that you install the database client on all the CA SDM servers. • In advanced availability configuration, all the servers still connect to a single database. As the database can be a single point of failure, consider taking the advantages of database clustering to increase the availability of DBMS. • Microsoft SQL Server is only natively supported on the Windows platform. For example, if your implementation consists of servers with heterogeneous operating systems like Windows and Linux, select Oracle as the DBMS as Microsoft SQL Server is not supported on Linux.

System Configuration, Administration, and Operation	• SOAP Web Services and RESTful Web Services are only supported on the application servers. Web directors can be configured on all CA SDM servers. • The application servers are independent of each other,, therefore, web directors can only service web engines running on the same application server. Web directors cannot service web engines across application servers. • Because the servers in the advanced availability configuration have a higher degree of independence, most command-line utilities function only on the local server. For example, **pdm_status** only shows you the CA SDM processes running on the server on which you are executing the command. The **pdm_webcache** utility only refreshes the form caches on the server on which it was issued. • Unlike conventional configuration, where you start and stop CA SDM processes by **pdm_d_mgr** running on the primary server, in advanced availability, processes on each server are controlled independently. • We recommend you to use the new pdm_server_control instead of pdm_halt command to shut down application servers. Before the shutdown, you can ask the active users to move to another application server. Notify the users using the **quiesce** option. • The pdm_edit utility has been replaced with a new graphical user interface, eliminating many manual configuration file changes required earlier. • Unlike conventional, in the advanced availability configuration, you can use the bopauth_host option from the Options Manager of the background server web UI to specify the authentication server details. You no longer make this configuration change in pdm_edit for the advanced availability configuration. Users cannot log in when the authentication server is unavailable. **Note:** In the conventional configuration, you can use a secondary server to integrate CA SDM with an authentication system running on a different system or even on a different hardware platform. • To prevent rogue servers from joining the advanced availability configuration, all servers must be defined from the background server web UI, before they can be configured. • The role and other information for a server in the advanced availability configuration can be changed from the Administration tab. Stop the CA SDM services before attempting to change a server definition. The server reconfiguration is required for the changes to take effect. • You can change a server between the conventional configuration and the advanced availability configuration by running the configuration utility. Be sure to change all servers in the implementation. Your data remains unaffected, but the manual updates are required to change the settings. • Use the latest **pdm_startup** files when migrating to the latest release version. Do not use the files from previous versions of CA SDM. For example, files that are generated by the **pdm_edit** utility **Warning!** Do not change the **Nx.env** variable manually unless you are instructed to do so. • A new facility generates ticket numbers and numeric record keys. To avoid possible database corruption, never try to load or manually alter the **Key_Control** table. • You cannot move the Knowledge Tools daemons to another server in the advanced availability configuration. The **kt_daemon** runs on all servers. All other Knowledge Tools daemons run as singletons on the background server. • Knowledge Tools now support UNC file paths on Windows

General Web User Interface Considerations	• A web server is required on all the servers of the advanced availability configuration. • When the background server is not available due to failover, a Delayed Server Response form is presented to the web users. Users are allowed to resume their work when the standby server is promoted to the background server. • The value of the web_cgi_url option must point to: • The load balancer if you have more than one application servers. • The application server, if you have only one application server.
Attachment Considerations	• You can increase the availability of attachments by configuring multiple document repository processes to access a shared file repository.
Web Screen Painter Considerations	You can use the Web Screen Painter (WSP) only on the background server. • Follow the recommended procedure to publish WSP form changes so that the updated forms are distributed to all the servers in the installation. For more information, see How to Modify Schema Using Web Screen Painter. • The virtual database layer daemons run on all servers. Install the CA SDM database customizations and object definitions on all the servers.
Reporting Considerations	CABI Jaspersoft can automatically retrieve data from alternate application servers. You can configure this feature to increase the availability of CA SDM reporting. For more information, see Integrate JasperReports Server 8.1.1 with CA Service Management.
Options Manager Considerations	• You can install or uninstall options through Options Manager only from the background server web UI. Use the rolling maintenance procedure to propagate the changes to all servers in the configuration. For more details on performing the rolling maintenance, see the Performing Rolling Maintenance on the CA SDM Server scenario.
Web Services Considerations	• You can configure web services only on the application servers. • The webservices_domsrvr option no longer exists in the Options Manager. You can configure NX_WEBSERVICES_DOMSRVR variable independently on each application server by modifying NX.env.

General Integration Considerations	• The URL to the CA SDM web user interface must point to a properly configured application server. The ca_application_registration contains a URL to your CA SDM installation that is used by other CA products. This URL points to the CA SDM server that is configured first, which is most commonly the background server. Only the CA SDM administrators can change the value through the administration facility. If you are using a load balancer, point this URL to the load balancer instead of a single application server. • Most end-users interaction and integrations with other software products are done at the application server level. There is no failover for the application servers. If the application server is unavailable, the web services for the application server are also unavailable. To increase the application server availability, you can deploy a load balancer to route requests among different application servers • CA Process Automation can be installed on any of the application servers. If the application server goes down, workflow becomes unavailable..
Federated Search Considerations	• To enable and use the federated search feature, select the federated search option while configuring the application servers.

CA SDM Configuration Conversion Considerations

Consider the following when converting the CA SDM server configuration mode:

- You can convert the background server to a primary server only.
- You can convert the primary server to a background server only.
- You can convert the secondary server to a standby server or application server only.
- You can convert the standby or an application server to the secondary server only.

Advanced Availability Additional Considerations

All the planning considerations that are applicable for CA SDM conventional configuration are valid for the advanced availability configuration too. Consider the following before you decide to implement an advanced availability configuration:

General Considerations

Review and implement the following considerations if you are opting for CA SDM Advanced Availability configuration:

- An advanced availability configuration must always have one background server and at least one standby server. (Recommended) Ensure that both background server and all other standby servers have similar configuration. This process ensures that during a failover when a standby server becomes the new background server, it can function exactly like the old background server. Except for CA SDM administrators, no other users are allowed to log in to the background server. Also, no users are allowed to log in to the standby servers. Any number of standby servers can be

configured. To increase the CA SDM availability, consider placing one standby server in your backup data center or the disaster recover site.

- The CA SDM performance is expected to remain the same even with the additional servers for background and standby operations. If you deploy more application servers, the performance can possibly improve.
- The minimum advanced availability implementation requires one application server. We recommend you to have two application servers to increase the availability and a load balancer to direct web traffic.
- Additional hardware costs are expected as you require one background, at least one standby, and one or more application servers. The configuration of standby and background server must be identical.
- You require a remote database server and a server to share knowledge tools index files, knowledge tools import/export files, archive purge output files, and attachment repositories. To allow the background and the standby server to access these files, a shared location is required.
- Linux installations can use NFS mounts. UNC is supported for Windows installations.
- We can send email notifications from all CA SDM servers. Every server in the advanced availability configuration must have a connection to the mail server.
- Install the background and the standby server on the same network subnet to make ping times and latencies from different application servers similar.
- Consider locating the background and the standby servers in a central location with a good network connectivity to all your users. The application servers can be located either centrally or they can be distributed across the globe.
- An email notification resulting from an end user interaction is sent from the application server to which the user is connected.
- An email notification resulting from any background processes (such as Animator processing an Attached Event) are sent by the pdm_mail_nxd utility running on the background server.
- During a background server failure, the queued emails are sent when the background server comes up as a standby server.

System Configuration, Administration, and Operation Considerations

Consider the following before you implement CA SDM Advanced Availability Configuration:

- SOAP Web Services and REST Web Services are only supported on the application servers. Web directors can be configured on all CA SDM servers. As the application servers are independent of each other, web directors can only service web engines running on the same application server. Web directors cannot service web engines across application servers.
- Since the servers in the advanced availability configuration have a higher degree of independence, most command-line utilities function only on the local server. For example, pdm_status only shows you the CA SDM processes running on the server on which you are executing the command. The pdm_webcache utility only refreshes the form caches on the server on which it was issued.
- In advanced availability, processes on each server are controlled independently. This is unlike conventional configuration, where you have to run **pdm_d_mgr** on the primary server to start and stop CA SDM processes.

  NOTE

  We recommend that you use the new **pdm_server_control** instead of **pdm_halt** command to shutdown application servers. Before the shutdown, you can move active users to another application server. This can be done by notifying users that are using the **quiesce** option.
- Unlike conventional, in the advanced availability configuration, you can use the **bopauth_host** option from the **Options Manager** of the background server Web UI to specify the authentication server details. Users cannot log in when the authentication server is unavailable.

NOTE

The pdm_edit utility is replaced with a new graphical user interface, eliminating many manual configuration file changes required earlier.

- Unlike conventional, in the advanced availability configuration, you can use the **bopauth_host** option from the Options Manager of the background server Web UI to specify the authentication server details. You no longer make this configuration change in **pdm_edit** for the advanced availability configuration. Users cannot log in when the authentication server is unavailable.
- In the conventional configuration, you can use a secondary server to integrate CA SDM with an authentication system running on a different system or even on a different hardware platform.
- To prevent rogue servers from joining the advanced availability configuration, all servers must be defined from the background server Web UI, before they are configured.
- The role and other information for an advanced availability server can be changed from the **CA SDM Console, Administration** tab . Stop the CA SDM services before attempting to change a server definition. The server reconfiguration is required for the changes to take effect.
- You can change a server between the conventional configuration and the advanced availability configuration by running the configuration utility. Be sure to change all servers in the implementation. Your data remains unaffected, but manual updates are required to change the settings.
- New environment variables have been added to NX.env to support the advanced availability and the system automatically maintains the variable values.

WARNING

Do not change NX.env manually unless instructed to do so.

- Use the latest **pdm_startup** files while migrating/upgrading CA SDM. Do not use the files from previous versions of CA SDM. For example, files that are generated by the pdm_edit utility.
- A facility generates ticket numbers and numeric record keys. To avoid possible database corruption, never try to load or manually alter the **Key_Control** table.

NOTE

In Advanced Availability configuration, ticket numbers and numeric record keys, may be out of order. This is not an error. Other fields can be used for sorting based on timestamp such as Open Date for tickets.

- You cannot move the Knowledge Tools daemons to another server in the advanced availability configuration. The **kt_daemon** runs on all servers. All other Knowledge Tools daemons run as singletons on the background server.
- Knowledge Tools support UNC file paths on Windows for the location of the EBR index files and input/output files that are used by the Knowledge Export Import facility. This feature is available for both the advanced availability and the conventional configurations.

WARNING

EBR Index Files path & KEIT Files path must refer to the same UNC credentials and the path must reside on a same server to support this.

- Version control distributes the files (for example, htmpl, .maj, .mod, and .sch) that are configured in the server_secondary_custom.ver file from the background server. Upon startup, the standby or the application server runs the version control client to pull updated files from the background server.
- Archive/Purge runs on the background server and supports UNC file paths on Windows for output files. This feature is available in the advanced availability and the conventional configurations.
- Attachments on incoming emails are now stored by pdm_maileater when the repository is on a remote server.
- Ensure that you allow the daemon manager to modify the procsets and do not run the pdm_dmnmode command for this action.

CA SDM Configuration Conversion

Consider the following before you start converting a conventional configuration to the advanced availability or vice-versa:

- You can convert the background server to a primary server only.
- You can convert the primary server to a background server only.
- You can convert the secondary server to a standby server or application server only.
- You can convert the standby or an application server to the secondary server only.

Failover Considerations

During a failover of the background server to the standby server, consider the following:

- New users must not log in. For users who are already connected, the following actions will not work during the failover process and must be attempted again (retried) by the user after failover is complete:
  - Creating tickets with attachments.
  - Downloading attachments.
  - Searching Knowledge documents.
  - Indexing new knowledge documents.
  - Inbound email.
  - The SLA events that are not triggered until the failover process is complete. .

    NOTE

    If you are using a third-party tool to auto-failover CA SDM servers, you must disable it before starting rolling maintenance.

Web User Interface and Web Server Considerations

Web User Interface

A web server is required on all servers having an advanced availability configuration. When background server is not available due to failover, a delayed server response form is presented to the web users. Users are allowed to resume their work when the standby server is promoted to the background server. The value of the **web_cgi_url** option must point to:

- The load balancer if you have more than one application servers.
- The application server, if you have only one application server.

Web Services

You can configure web services only on the application servers. The **webservices_domsrvr** option no longer exists in the **Options Manager**. You can configure the **NX_WEBSERVICES_DOMSRVR** variable independently on each application server by modifying the **NX.env** file.

Integration Considerations

- The URL to the CA SDM web user interface must point to a properly configured application server. The *ca_application_registration* contains a URL to your CA SDM installation that is used by other CA products. This URL points to the CA SDM server that is configured first, which is most commonly the background server. Only the CA SDM administrators can change the value through the administration facility. If you are using a load balancer, point this URL to the load balancer instead of a single application server.
- Most end-users interaction and integrations with other software products are done at the application server level. There is no failover for the application servers. If the application server is unavailable, the web services for the application server are also unavailable. To increase the application server availability, you can deploy a load balancer to route requests through different application servers

Database Considerations

A direct connection exists from each of the servers to the other as well as to the database. If the CA SDM server is within DMZ, you are required to open the firewall ports or implement a tunneling proxy technology for this connectivity. Also consider licensing arrangements with your DBMS vendor.

- Ensure that you install the database client on all the CA SDM servers.
- In advanced availability configuration, all servers connect to a single database. As the database can be a single point of failure, consider taking the advantages of database clustering to increase the availability of DBMS.
- Microsoft SQL Server is only natively supported on the Windows platform. For example, if your implementation consists of servers with heterogeneous operating systems like Windows and Linux, select Oracle as the DBMS since Microsoft SQL Server is not supported on Linux.

Options Manager Considerations

You can install or uninstall options through Options Manager only from the background server web UI. Use the rolling maintenance procedure to propagate the changes to all servers in the configuration. For more details on performing the rolling maintenance, *see the Performing Rolling Maintenance on the CA SDM Server scenario.*

Web Screen Painter Considerations

- You can use the Web Screen Painter (WSP) only on the background server.
- Follow the recommended procedure to publish WSP form changes so that the updated forms are distributed to all the servers in the installation. For more information, see *Web Interface Modifications*.
- The virtual database layer daemons run on all servers. Install the CA SDM database customizations and object definitions on all the servers.

xFlow Interface Considerations

xFlow Interface uses Jointjs and requires certain files for successful installation. Jointjs is a JavaScript library which helps to visualize the SDM classic workflow as a graphical representation. The xFlow Interface uses Jointjs and requires certain files for successful installation. You must download the required Jointjs files before starting the installation.

Follow these steps:

1. Download the CA Service Management **CA SERVICE MANAGEMENT 17.4 3RD PARTY COMPONENTS.iso** from Broadcom Support.
2. Extract the **Jointjs** folder from the **Jointjs.zip** file available in the DVD.
3. Verify that Jointjs folder includes the following files:
   a. **joint.min.js**
   b. **joint.min..css**
4. Note the path where you placed the Jointjs folder that includes the files.

> **NOTE**
>
> When you run the CA Service Management 17.4 installer, you are required to browse and provide the Jointjs folder path from where you have saved the files in the xFlow Interface configuration details page.

CABI JasperReports Server (Insights) Considerations

This article contains the following topics:

> **NOTE**
>
> CA Service Management no longer supports role switching for Insights App. The default OOTB role for Insights app is available with CA Service Management 17.4, and it is similar to the Out-of-the-Box (OOTB) Reporting

role of an Access Type for a contact. In case, if you change the default role of the app to any other role than the Reporting role, ensure that you consider the following:

- Aggregated Reporting reflects data as per the new default role the app
- Transactional Reporting honors only reporting role of the contact

Ensure that the reporting role of an access type and default role of an access type for Insights App is same.

NOTE

ODBC is deprecated for CA Service Management 17.4 and for JasperReports Server - Transactional Reports.

CABI JasperReports Server 17.4 uses the JDBC driver client which only supports one instance of CA SDM. If you setup JasperReports Server in a clustered setup, each JasperSoft Server instance can point to a different CA SDM server. This helps in scalability and better performance. We recommend that you use a replicated database and a shadow/passive CA SDM for Transactional reporting. If you configure Transactional reporting on an online or active CA SDM server, the performance of the online/active system may get impacted. For more information, see Replicated Database for Offline Reporting.

You can add multiple domsrvrs for better performance of Transactional reporting through the JDBC driver. For more information, see Configure Multiple Domsrvr for Transactional Domain Reports.

NOTE

CA Service Management Installer automatically sets the CA SDM data source for CABI JasperReports Server. However, if you are manually editing the CA SDM data source through CABI JasperReports Server, you must restart the CABI services.

NOTE

If you have installed CA SDM with Oracle database on a Server (Example: Server **A**) and installed CABI JasperSoft on another server (Example: Server **B**), in order to integrate CABI with CA Service Management Install & Integrate CABI JasperReports server with CA Service Management), you must install the Oracle Client on the CABI JasperSoft Server and establish an Oracle -Server-Client connection.

Upgrade Considerations

NOTE
As a prerequisite, upgrade to JasperReports Server 8.1.1.

Refer Upgrade JasperReports Server from 7.9 to 8.1.1 for upgrading to JasperServer Reports 8.1.1

Enable KPI Option "kpi_ticket_data_table"

The KPI option "kpi_ticket_data_table", enables the KPI daemon to collect data whenever CA SDM tickets are opened, closed, or certain fields are modified. Ticket data is written into the usp_kpi_ticket_data database table and is available for generating web-based reports. For more information about enabling the kpi_ticket_data_table in CA Service Desk Manager, see Options Manager.

- The "kpi_ticket_data_table" option must be enabled before generating the report. Not enabling the “kpi_ticket_data_table” leads to inconsistent data in the report.
- To ensure that the data is consistent and correct in the report, configure the Trend Start Date to a date later than that of CA Service Management implementation.
- The "kpi_ticket_data_table" option is enhanced to capture additional information while creating tickets. The tickets that are created prior to CA Service Management 17.4 implementation/upgrade, do not include this additional information.

So, the reports which include these ticket details will show the incorrect data, if Trend Start Date is configured to a date prior than that of your CA Service Management 17.4 implementation. The following reports are impacted:

- Incidents Cost for Top Categories (USD)
- Incidents Cost Trend for Top Categories (USD)
- MTTR for Top Categories (In Hours) Report
- MTTR Quarterly Trend for Top Categories (In Hours) Report
- MTTR Weekly Trend for Top Categories (In Hours) Report
- MTTR Yearly Trend for Top Categories (In Hours) Report

CA Service Catalog Considerations

The CA Service Catalog products and components are an integrated set of business applications that a Service Delivery Manager uses to manage services. To optimize your use of CA Service Catalog and the system resources that it uses, prepare your environment for a successful implementation.

As a best practice, verify that all CA Service Catalog systems are *geographically collocated.*

Review the following considerations before you start installing CA Service Catalog:

Network Share

When you install CA Service Catalog products and components on a network share, map a drive letter to this share. Alternatively, copy the installation image to a local folder with a short path name, for example, **C:** or **C:\Temp**. Run the installation programs locally from this folder.

NOTE

Avoid long path names because they can cause problems during the installation process. The maximum length of a path name in Windows is 260 characters. If the path names exceed this limit, the installation (or upgrade) can fail. Verify that the path name where you install CA Service Catalog does not exceed this limit.

Service Provider Business Unit ID

During the CA Service Catalog installation, you can specify the **service provider** business unit ID. This ID specifies the top level (root) business unit. All other business units are structured under the root business unit. As a best practice, specify your company domain name or a short version of your company name for this ID.

NOTE

After you have installed CA Service Catalog, you cannot change the business unit ID of the service provider. But, you can change the business unit login ID and the business unit name service provider.

Determine the Authentication Methods to Use

Administrators can authenticate users in CA Service Catalog by any one of the following methods:

- CA EEM alone
- CA EEM with an external directory. For example: an LDAP directory, such as Microsoft Active Directory
- Windows NTLM authentication

You can specify whether to use CA EEM as standalone or with an external directory. When you use CA EEM standalone, CA EEM stores the user information and passwords in its data store. When you use CA EEM with an external directory, such as Microsoft Active Directory, the external directory stores the users and passwords in its data store. If the client system uses a Windows domain, you can configure CA Service Catalog to use NTLM authentication. NTLM authentication enables a user who has been authenticated to the domain to skip the login page.

Failover Mechanisms

As part of your implementation, you can perform the following tasks to provide failover mechanisms:

- Deploy CA Service Catalog in a cluster.
- If you use CA Process Automation, deploy clustering in CA Process Automation.
- Use *database* clusters. CA Service Catalog components are also compatible with database clusters which are recommended for environments experiencing heavy load.
- For information about database clustering, see your DBMS documentation.
- Use a load balancer. Clustering requires a load balancer. To maximize the uptime of CA Service Catalog, include the load balancer and cluster computers in your failover plan.
- Implement clustering on the load balancer computer.
- Move user-generated files from the local file system to a filestore (a central location) on a high-availability network shared file system. CA Service Catalog maintains several user-generated files on the local file system. These files include forms and XSL and XML customizations.
- If you are using CA EEM for authentication, cluster CA EEM, using operating system level clustering.
- For information about clustering and implementing failover with CA EEM, see CA EEM Documentation.
- Improve the robustness of the hardware used in your environment, by using all applicable means, including RAID hard disks, redundant power supply, and network cards. For more information, see your hardware documentation.

Session Management

Session management is very important for performance. Use a process automation tool such as CA Process Automation to handle session management effectively.

Effective use of web service sessions is crucial for maintaining throughput as your environment scales up. Creating a web service session is a task which requires multiple invocations of authentication and authorization logic and accounts for the majority of time and effort spent when using web services. Design CA Process Automation processes and other web service clients to avoid creating extra web service sessions. Exception handling should create a session only when the exception is known to be the result of an invalid or expired session.

CA Service Catalog Integration with CA Process Automation

If you want to integrate CA Service Catalog with CA Process Automation, ensure that:

- The CA Service Catalog computer name must not begin with a number.
- If you are using a load balancer for CA Service Catalog or CA Process Automation, the computer name of the load balancer must not begin with a number.

Failure to meet these requirements can result in a web service call error. For example:

```
Caused by: com.sun.xml.messaging.saaj.util.JaxmURI$MalformedURIException: Host is not a well formed address!
```

CA Asset Portfolio Manager Considerations

NOTE
We do not recommend the installation of CA Asset Portfolio Management components, except the Management Database, on a 64-bit computer that hosts a 64-bit Oracle database server.

NOTE
For Oracle database in CA Service Management Solution deployment, ensure that the Oracle-32 bit path is set on top of the Environment path variable list for CA APM.

WARNING

CA Asset Portfolio Management (APM) architecture is highly database driven and has no rollback mechanism for database-related changes. Hence, CA Service Management, currently does not provide support for uninstallation of a CA APM patch. The only mechanism to revert database changes is by restoring database backup, which may result in data loss.

Perform the following pre-installation tasks as part of CA APM installation and configuration:

WARNING

CA APM has time-sensitive processes. Verify that all servers are set to their correct date and time with their respective time zones.

NOTE

If Microsoft Visual C++ Redistributable 2015 32-bit is a pre-requisite for CORA functionality. If not installed, download and install Microsoft Visual C++ Redistributable 2015 32-bit on the Application server of CA APM.

NOTE

If CA APM is installed on a SQL Server database, install the 64-bit SQL Native Client 11.0 on the APM Application server to ensure CORA functionality works, as CORA uses the SQLNCLI11 provider to connect to the database. Download the 64-bit SQL Native Client 11", named as Microsoft® SQL Server® 2012 Native Client from: https://www.microsoft.com/en-us/download/confirmation.aspx?id=50402

Verify the Internet Information Services Installation

Before you install CA APM, Install IIS and ASP.NET on all applications and web servers. If the service is not on a server, add the service before you begin the installation.

Perform the following steps:

1. Navigate to Windows Server Manager.
2. From **Manage** menu, select **Add Roles and Features**.
3. Follow on-screen instructions and provide the installation type and destination server details.
4. In the Select server roles pane, under **Roles**, expand **Application Server**, and select the appropriate ASP.NET version.
   Click **Next**.Follow the on-screen instructions and complete the installation.
   ASP.NET is now registered with IIS.
5. Refer Operating System Requirements for CA Asset Portfolio Manager ITAM.

Oracle Real Application Clusters (RAC) Considerations

Oracle RAC provides high availability to applications by removing the single server as a single point of failure. You can deploy one or a combination of CA Service Management solution products as required. This section describes how to deploy the CA Service Management Products (CA Service Desk Manager, CA Service Catalog, and CA IT Asset Manager) with Oracle Real Application Clusters (RAC).

NOTE

For more information about installing Oracle RAC, see the Oracle documentation.

Ensure to verify and complete the following Oracle RAC installation prerequisites:

- Download Oracle Database (Oracle Grid infrastructure and database) software from Oracle Downloads. Create the Oracle database (SID).
- The Single Client Access Name (SCAN) must be defined in the Domain Name Service (DNS) or Oracle Grid Naming Service (GNS).
- Create at least one single DNS or GNS name that resolves to three IP addresses using a round-robin algorithm. For the installation to succeed, the SCAN must resolve to at least one IP address.
- SCAN Virtual IP (VIP) addresses must be on the same subnet as the virtual IP addresses and public IP addresses.
- The virtual machines can be limited to 2 GB of swap but this results in a prerequisite install check failure. The best practices define a 3+ GB of swap.
- Ensure that you have Windows Server 2016/ 2019/ 2022 with Oracle 19c.
- Install the Oracle Grid Infrastructure following the Oracle Documentation installation procedures.

How to Install Oracle RAC With CA Service Management

To install Oracle RAC with CA Service Management (CA Service Desk Manager, CA Service Catalog, and CA IT Asset Manager), complete the following steps:

Follow these steps:

1. Install Oracle 19c 32-bit/ 64-bit client with administrator option on system or systems where you plan to install the CA Service Management products.
2. Create a TNS entry on the system or systems.
3. Enable the client-side connect-time failover by setting FAILOVER=ON in the corresponding client-side TNS entry:

```
TESTRAC =
(DESCRIPTION =
(ADDRESS_LIST=
(FAILOVER = ON)
(ADDRESS = (PROTOCOL = TCP)(HOST = <scan_host_name>)(PORT
= <1521>))
)(CONNECT_DATA = (SERVICE_NAME = <netservice name>))
```

Install the CA Service Management Products with Oracle RAC

For Installing the CA Service Management Product or products with Oracle RAC, perform the following steps:

Follow these steps:

1. Launch the CA Service Management installer.
   For more information, see Installing CA Service Management 17.4.
2. Select **CA Service Management** from Select the Required Installer screen.
3. Review and accept the license agreement information.
4. Select **Oracle** as the database type in the Database Configuration Screen and enter the following Oracle database details
   a. **Database Server**: Enter the Oracle database host name <SCAN_HOST_NAME>.
   b. **Tablespace Path (on DB Server)**: Enter +DATA.
      Complete all fields on the Oracle Database Configuration Screen.
5. Select the required CA Service Management product or products.
6. Follow the on-screen instructions on the installer wizard and complete the CA Service Management installation.

Installing CA Service Management 17.4

CA Service Management installer installs, upgrades, configures and integrates CA Service Desk Manager, CA Service Catalog, and CA Asset Portfolio Management. CA Service Management common installer performs prerequisite

installation checks and mandatory validations before installing the solution products. CA Service Desk Manager (SDM) installation includes xFlow Interface, Insights, Service Point, and Search Server.

WARNING

If you are using CA EEM configured with Active Directory, ensure to create **CasmAdmin** user before you install CA Service Management.

NOTE

mdbadmin is the default user across all CA Service Management products.

Follow these articles:

- Installing Common Components (CA EEM, Jaspersoft, and CA Process Automation)
- Enable and Configure Telemetry Services
- Installing CA Service Desk Manager
- Installing Web Screen Painter
- Installing CA Service Catalog
- Installing CA Asset Portfolio Management

Installing Common Components (CA EEM, Jaspersoft, and CA Process Automation)

The common components are software components that are used by different CA Technologies products. This section provides an overview of the CA components used by CA Service Management solution describes how to install and configure them.

Install the following CA Common Components and configure them with CA Service Management solution.

- Jaspersoft for CA Service Management
- Install CA Process Automation for CA Service Management
- Install CA EEM for CA Service Management

Jaspersoft for CA Service Management

Jaspersoft Studio allows you to create and customize reports. Once you have Jasper Studio installed, configure the data source, and connect it to your JasperReports Server so that you can create, edit and publish reports to JasperReports Server. JasperReports Server is a high-performance standalone and embedded Business Intelligence (BI) platform that provides rich reporting and analysis capabilities.

This article contains the following topics:

- Jaspersoft Studio
- JasperReports Server

Jaspersoft Studio

Jasper Studio allows you to create and customize the reports. Once you have Jasper Studio installed, configure the data source, and connect it to your Jasper Reports Server so that you can edit or create reports, and publish them to JasperReports Server.

TIP

We recommend that you save the customized reports with a different name. This prevents the customized reports from being overwritten when you import the same reports again.

Visit the Jaspersoft Community for related tutorials, tips and tricks, and other helpful information.

Integrate CA Service Management using one of these Jaspersoft Studio versions.

- Integrate Jaspersoft Studio 8.1.0 with CA Service Management
- Integrate Jaspersoft Studio 9.0.3 with CA Service Management
- Integrate Jaspersoft Studio 9.0.0 with CA Service Management

Integrate Jaspersoft Studio 8.1.0 with CA Service Management

CA Service Management 17.4 and later provide the ability to integrate Jaspersoft Studio 8.1.0. This integration enables reporting users to design and run report templates, write complex expressions, and layout visual components. The connection between CA Service Management's data access layer and Jaspersoft Studio is established via the JDBC driver. All the existing Out-of-the-box reports built for Jaspersoft Server work with Jaspersoft Studio as well.

> **NOTE**
> ODBC Driver is not supported on CA Service Management 17.4.

Download and Install Jaspersoft Studio Professional 8.1.0

To download and install Jaspersoft Studio 8.1.0 in your environment, perform the following steps:

1. Download the **CA Business Intelligence JasperSoft Studio 8.1.0 for CA Service Management.iso** installer from Broadcom Support.
2. Right-click the ISO and select **Mount**.
3. Right-click the **TIB_js-jss_8.1.0_windows_x86_64** application and run as **Administrator**.
4. Follow the Installation Wizard instructions to install Jaspersoft Studio 8.1.0 in a location of your choice.
5. After installing Jaspersoft Studio, the out-of-the-box installation package (For example, C:\Program Files\TIBCO \Jaspersoft StudioProfessional-8.1.0) displays the Evaluation License Version.

Update Jaspersoft Studio License (Evaluation License to PerpetualLicense)

To change the Evaluation License (default) provided with Jaspersoft Studio 8.1.0 to Perpetual or Full License, perform the following steps:

1. Navigate to where you have downloaded and extracted CA Service Management 17.4.
2. Navigate to **Filestore**, **Jasper**, **8.1.1**, **License**.

   > **NOTE**
   > Jaspersoft Reports 8.1.1 Install folder contains the Jaspersoft Studio Professional 8.1.0 License Information.
3. Now, launch Jaspersoft Studio 8.1.0.
4. Navigate to **Help**, **License Manager**, **Install New License**.
5. Browse and click open the License folder which is available in step 2 and select **jasperserver.license**.
6. Click **OK**.

Create Data Adapter

Perform the following steps to create a Data Adapter:

1. Open Jaspersoft Studio.
2. Right-click a project in Project Explorer and select **New**, **Data Adapter**.
3. In the Data Adapter File window, select a project where you want to save the data adapter file. The selected project must contain the report(s) that you want to use with your data adapter. The Data Adapter Wizard is displayed.
4. Enter a name for your adapter in the Data Adapter Wizard and click **Next**
   For example, Database JDBC Connection.

5. Provide the CA Service Desk Manager (SDM) connection details.
6. In the Database Location tab, enter the properties required by the data adapter type. For example, for database JDBC connection, select the JDBC driver, set the URL, and provide the CA SDM database username and password:
   a. JDBC Driver: com.ca.sdm.dal.sql.jdbc.driver.DalJdbcDriver
   b. JDBC URL: jdbc:sdm:bop:@<SDM Host Name>:2100;<UserUuid>=<ServiceDeskUUID>
   c. UserName: ServiceDesk
   d. Password: Encrypted Password must be used. Perform the following steps to encrypt the password:
      a. Navigate to CASM 17.4_DVD\filestore\Jasper\Utils folder.

         NOTE
         Ensure that java is added to PATH env. variable to run the below command.

      b. Run the following command:

         ```
         gen_passwd.bat <servide desk username> <service desk password>
         ```

         For example: Username: servicedesk. Password: servicedesk password (example: see the below image in Linux environment)

7. In the Driver Classpath tab, select the list of jars.
8. Provide the library location for the jars, the same jars as the Jaspersoft Reports 8.1.0 `DVD \filestore\Jasper \8.1.1\ExternalAuthJars`
9. a. BOPIntegration.jar
   b. domsrvr_utils.jar
   c. log4j-api-2.17.1.jar
   d. log4j-core-2.17.1.jar
   e. DurationFormatter.jar
   f. sd-utils.jar
   g. slump.jar
10. To test the connection, click the **Test button**.
11. Click **Finish** to create the Adapter.

Create a Report and Publish

To create a report in Jaspersoft Studio and publish it to your existing JasperReports Server (8.1.0), see TIBCO Documentation.

Create Subreports

To create subreports, perform the following steps:

Pass string parameters to subreports

Testing subreport in Jaspersoft Studio

Publish the Subreport in Jaspersoft Studio

Pass string parameters to subreports

Consider the following example scenarios:

Jaspersoft Studio Reports - Example 1

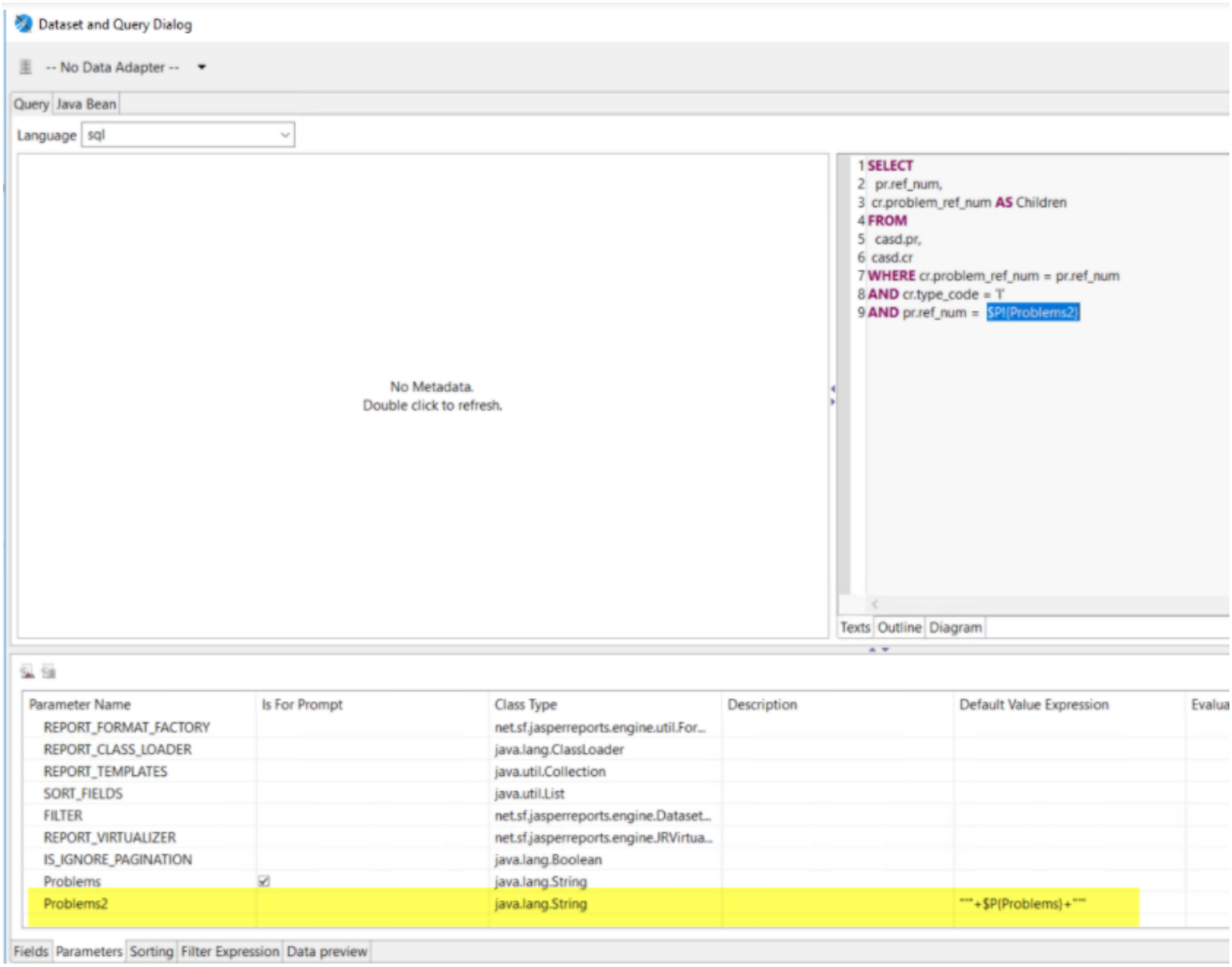
Jaspersoft Studio Reports - Example 2

Jaspersoft Studio Reports - Example 1

In this example report, a list of problem tickets is presented with a subreport to show the child Incidents related to each problem. The following query shows a parameter, defined as **java.util.Collection**, to prompt the user to enter or select a list of problem tickets. The parameter is flagged for **Is For Prompt**.

```
SELECT casd.pr.id,
casd.pr.ref_num,
casd.pr.summary,
casd.pr.group_name,
casd.pr.impact_symbol,
casd.pr.priority_symbol,
casd.pr.urgency_symbol,
casd.pr.status_symbol,
casd.pr.rootcause_symbol,
casd.pr.open_date,
casd.pr.description,
casd.pr.change_summary,
casd.pr.change_chg_ref_num,
casd.pr.caused_by_change_order_summary,
casd.pr.caused_by_change_order_chg_ref_num,
casd.alg.type,
casd.alg.type_symbol,
casd.alg.description,
casd.alg.last_mod_dt
FROM casd.alg,
casd.pr
WHERE $X{IN,pr.ref_num, Problems} AND
pr.persistent_id = alg.call_req_id
```

NOTE
Ensure that your subreport string parameter has a default expression to hard code single quotes. Also, this parameter must not be set for prompting the user to enter or select a list of problem tickets.



Jaspersoft Studio Reports - Example 2

In this example, the main report has data about one or more change orders. The subreport returns the correct properties for the change orders as a list.

Perform the following steps:

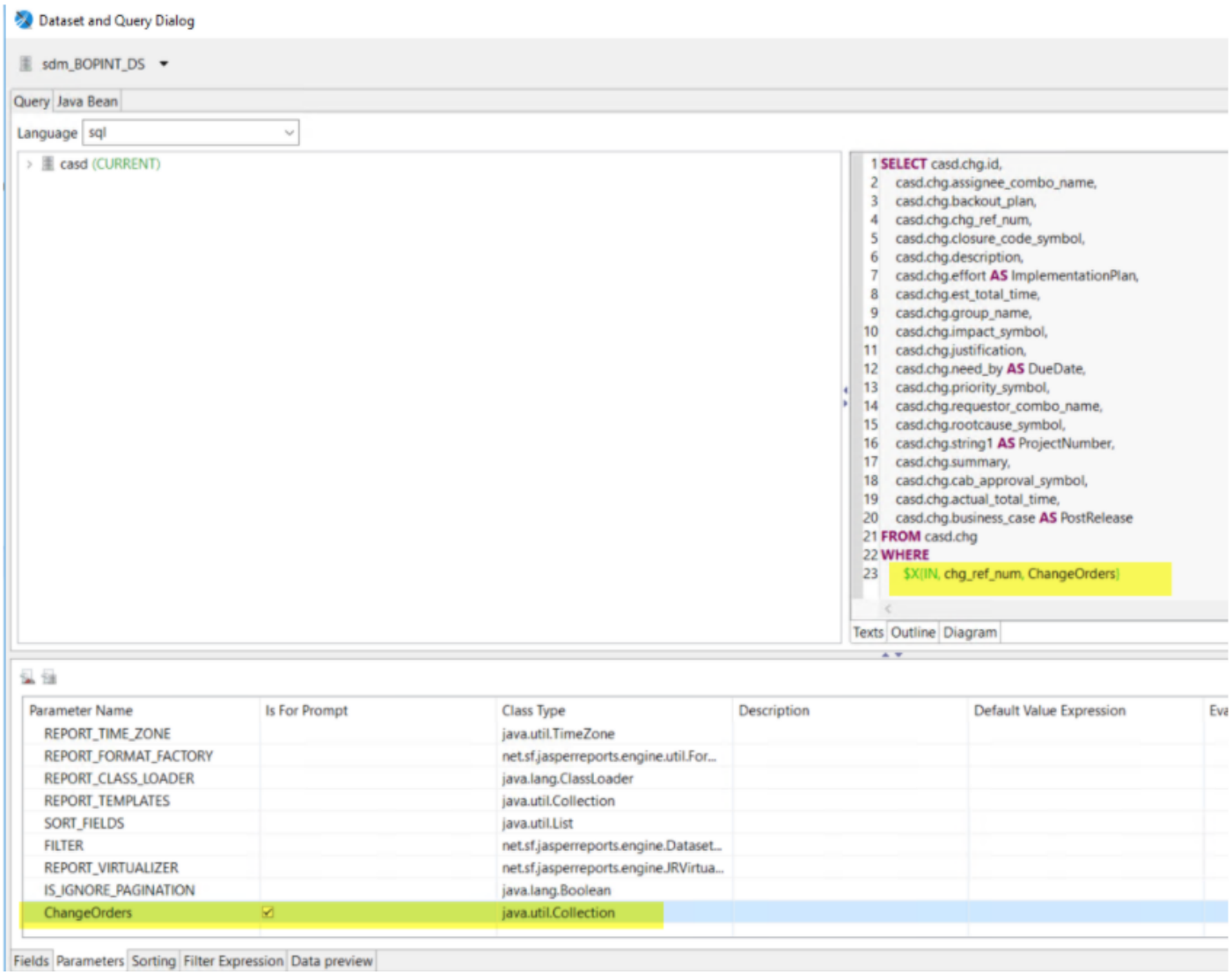
1. Run the following example query for the main report:

```
SELECT casd.chg.id,
casd.chg.assignee_combo_name,
casd.chg.backout_plan,
casd.chg.chg_ref_num,
casd.chg.closure_code_symbol,
casd.chg.description,
casd.chg.effort AS ImplementationPlan,
casd.chg.est_total_time,
```

```
casd.chg.group_name,
casd.chg.impact_symbol,
casd.chg.justification,
casd.chg.need_by AS DueDate,
casd.chg.priority_symbol,
casd.chg.requestor_combo_name,
casd.chg.rootcause_symbol,
casd.chg.string1 AS ProjectNumber,
casd.chg.summary,
casd.chg.cab_approval_symbol,
casd.chg.actual_total_time,
casd.chg.business_case AS PostRelease
FROM casd.chg
WHERE
$X{IN, chg_ref_num, ChangeOrders}
```

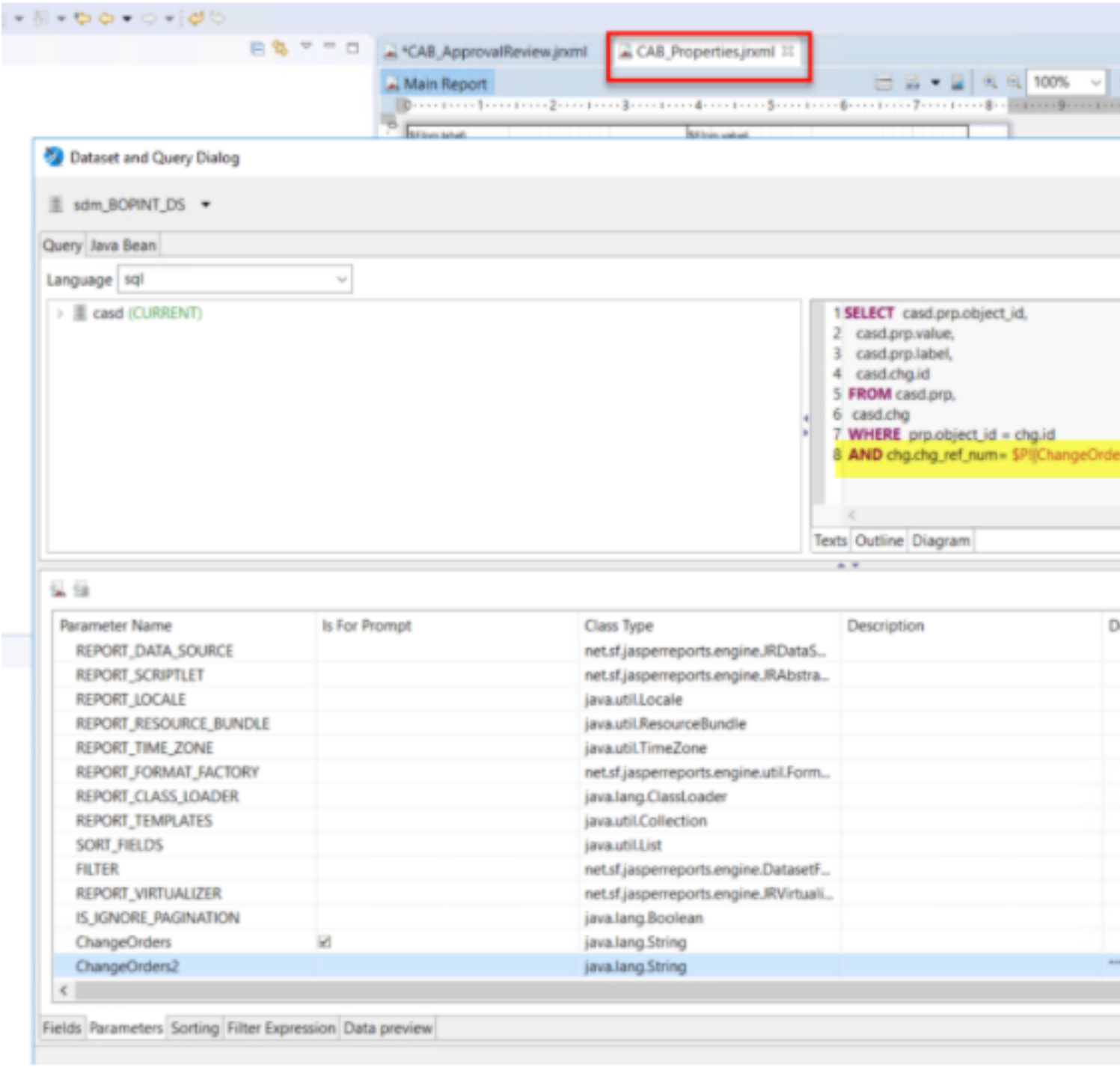
2. Run the following query to pass the list of values:

```
SELECT casd.chg.id,
casd.chg.assignee_combo_name,
casd.chg.backout_plan,
casd.chg.chg_ref_num,
casd.chg.closure_code_symbol,
casd.chg.description,
casd.chg.effort AS ImplementationPlan,
casd.chg.est_total_time,
casd.chg.group_name,
casd.chg.impact_symbol,
casd.chg.justification,
casd.chg.need_by AS DueDate,
casd.chg.priority_symbol,
casd.chg.requestor_combo_name,
casd.chg.rootcause_symbol,
casd.chg.string1 AS ProjectNumber,
casd.chg.summary,
casd.chg.cab_approval_symbol,
casd.chg.actual_total_time,
casd.chg.business_case AS PostRelease
FROM casd.chg
WHERE
$X(IN, chg_ref_num, ChangeOrders)
```



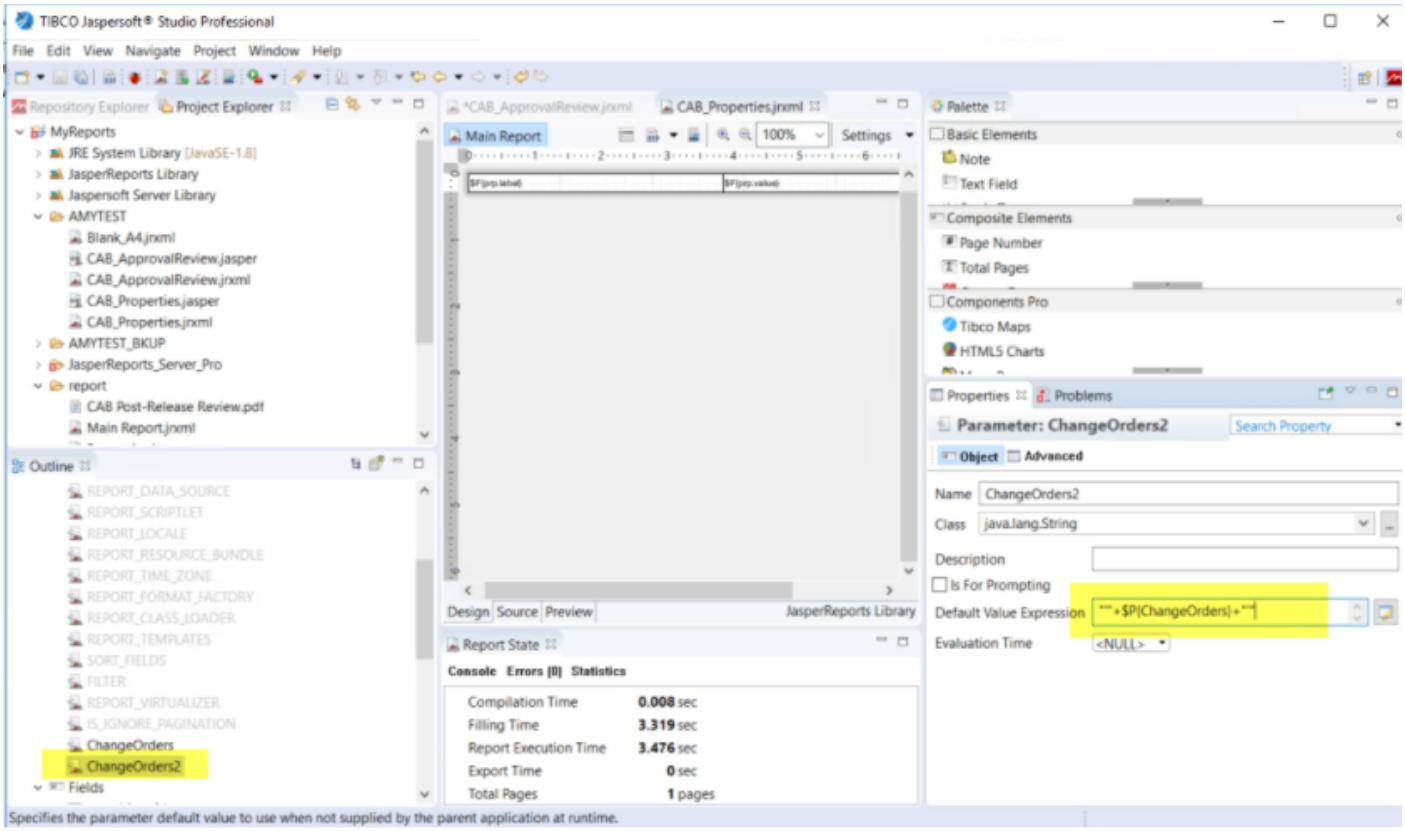
3. Run the following example query for subreports:

```
SELECT casd.prp.object_id,
casd.prp.value,
casd.prp.label,
casd.chg.id
FROM casd.prp,
casd.chg
WHERE prp.object_id = chg.id
AND chg.chg_ref_num= $P!{ChangeOrders2}
```

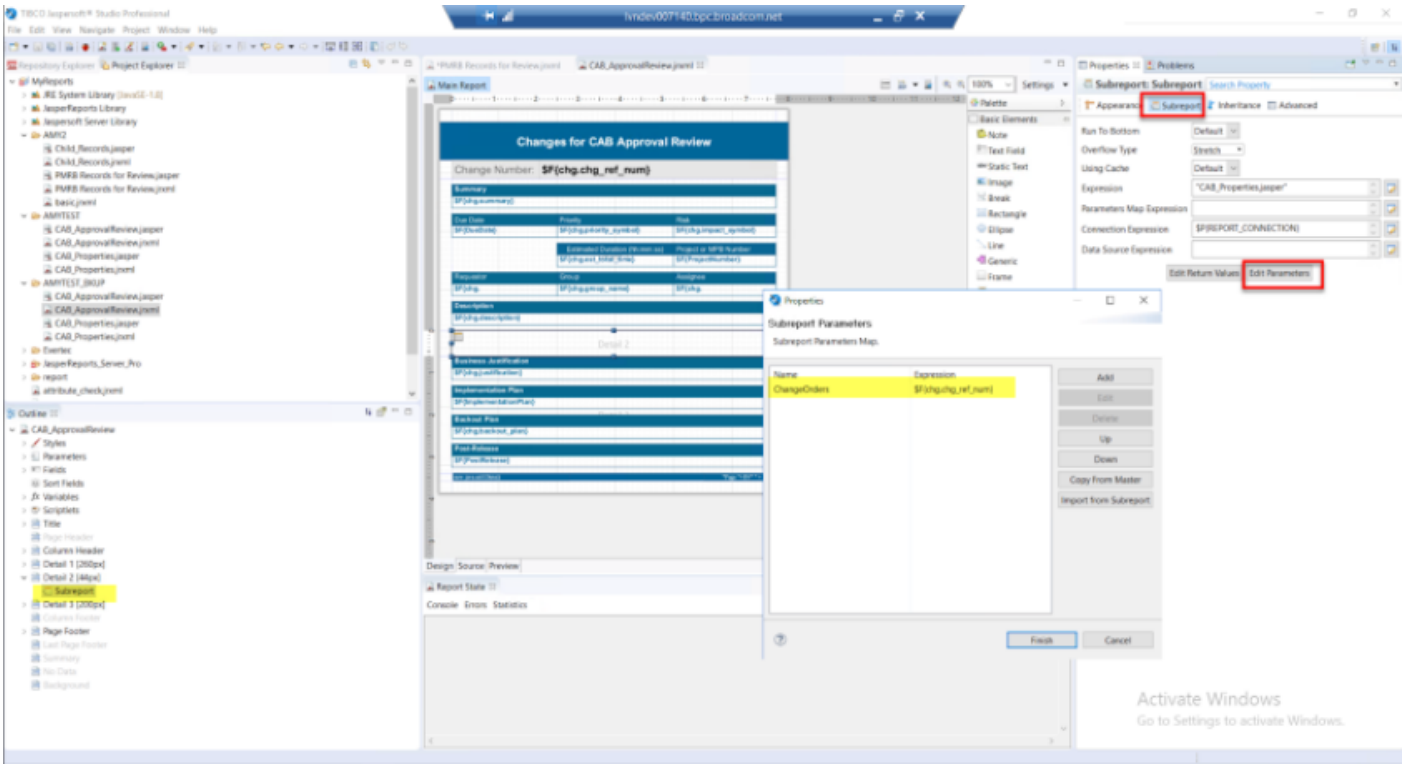


4. Add a new parameter to your subreport as shown below.

```
ChangeOrders2
Class: java.lang.String
Default Value Expression: "'"+$P{ChangeOrders}+"'"
```



5. Validate that the parameter to pass to the subreport is set as follows.



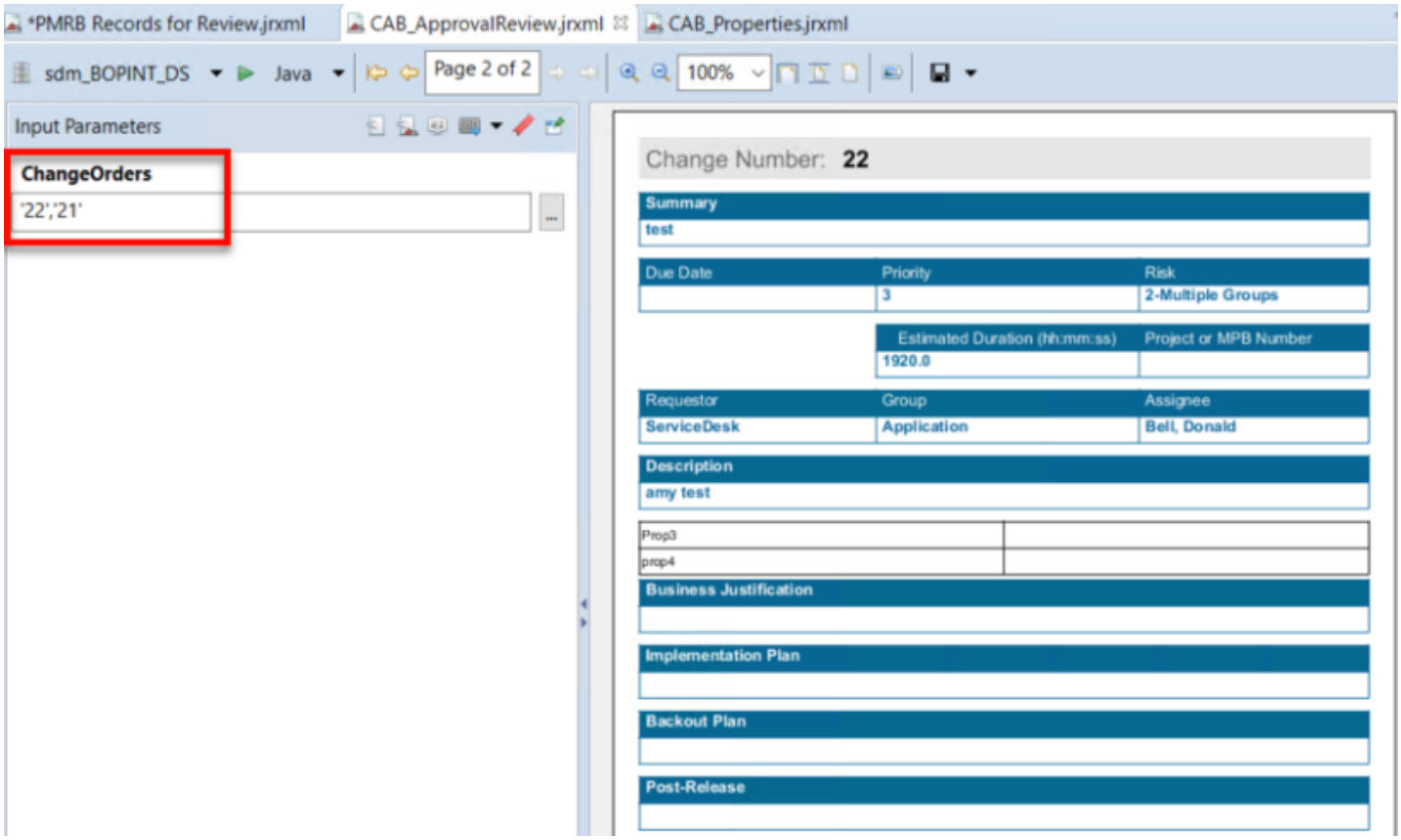
6. Compile the subreport after it is saved.

Testing Subreports in Jaspersoft Studio

Validate the subreports that you have created from Jaspersoft Studio. Ensure to list the change order numbers with single quotes, separated by a comma.

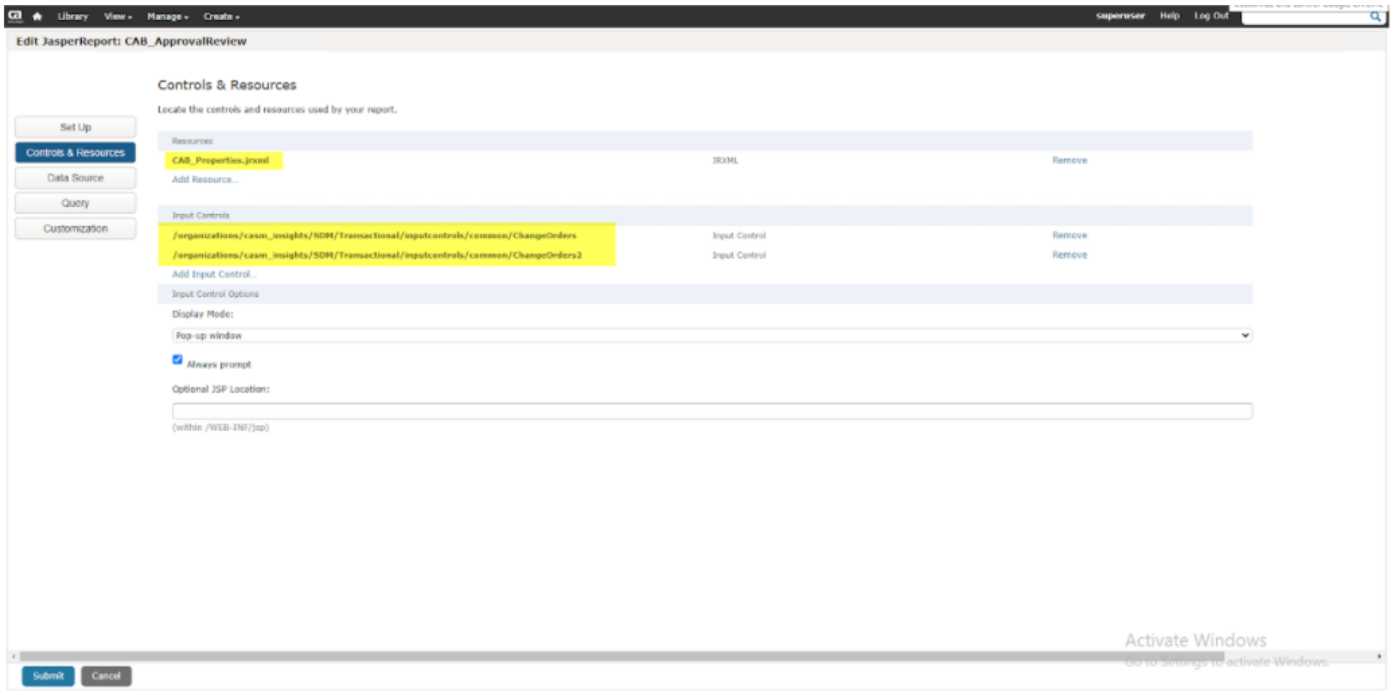
Perform the following steps:

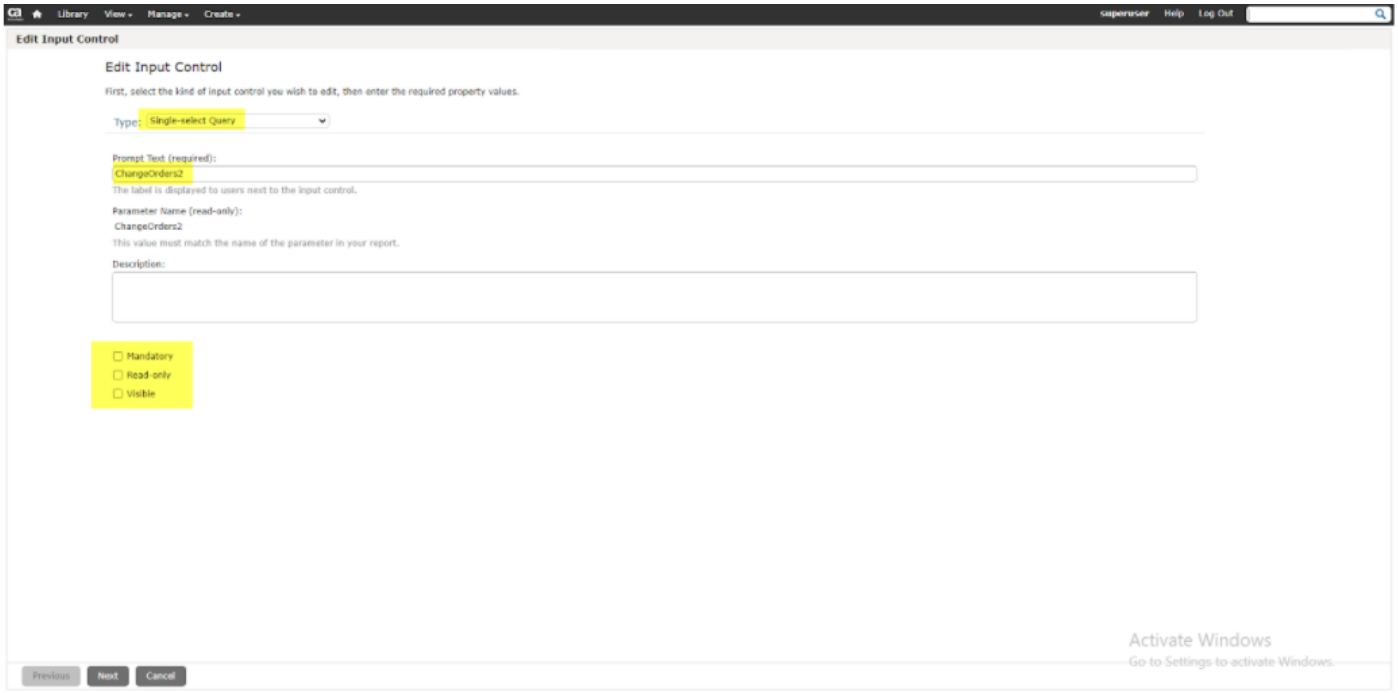
- Run the subreports and enter the change order numbers as shown below to test the subreport:



Publish the Subreport in Jaspersoft Studio

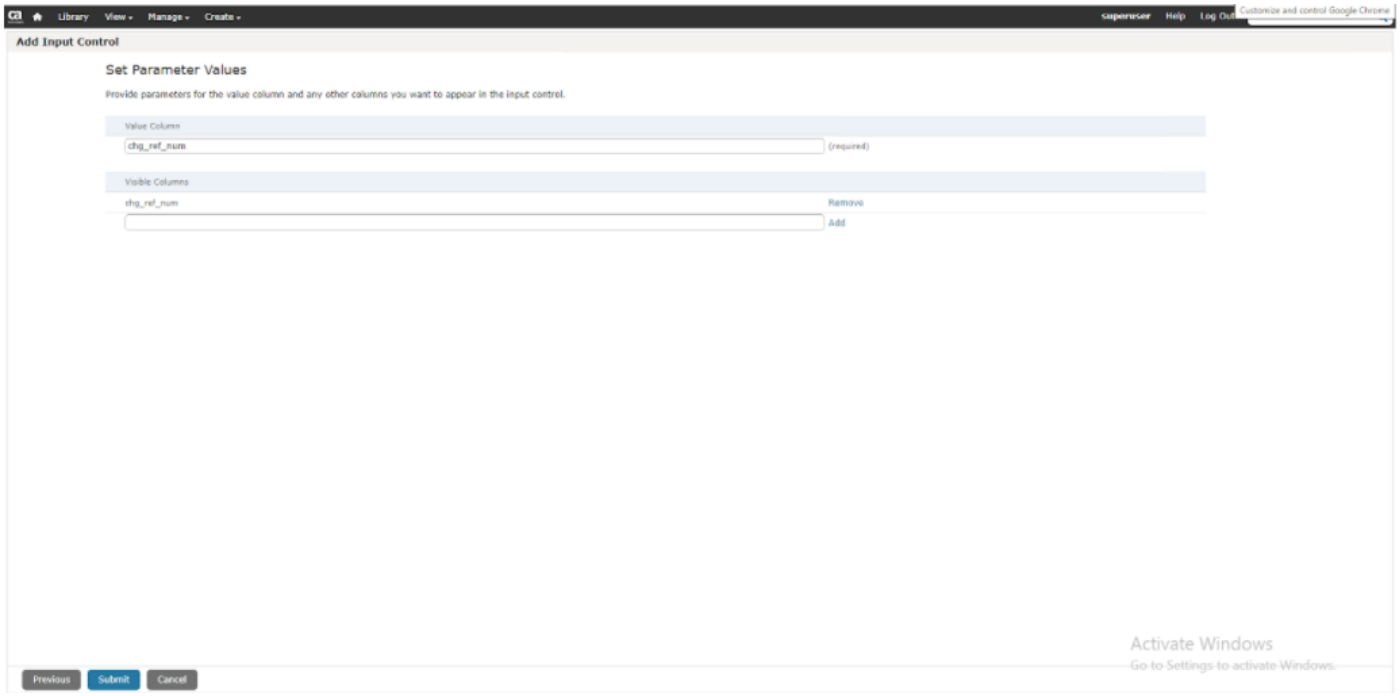
1. Add the **ChangeOrders2** input control as well. Ensure that the respective jrxml file is associated as a resource



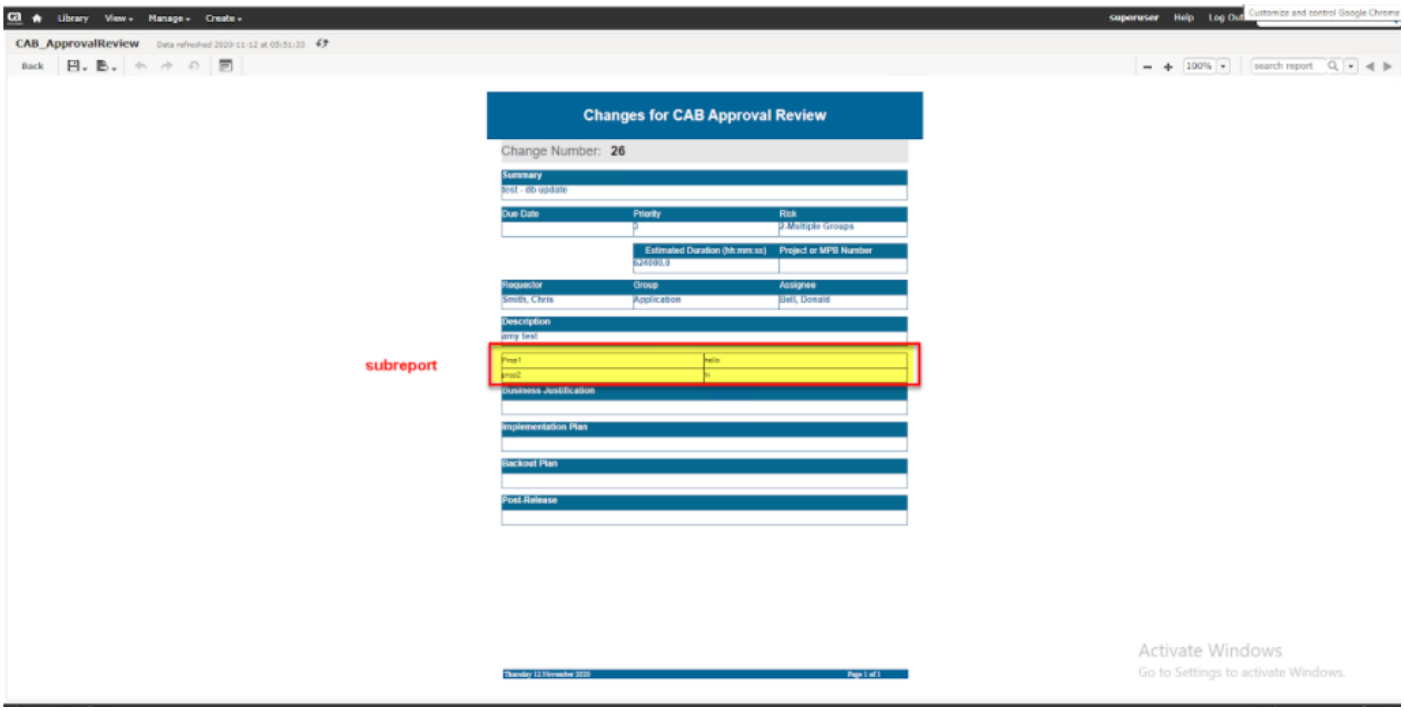


2. Link it to a query, in this example **Chg_Single** , pointing to the data source, with a query string of:

```
select ''''+chg_ref_num+'''' as chg_ref_num FROM chg where active =1
```



3. The subreport displays as follows.

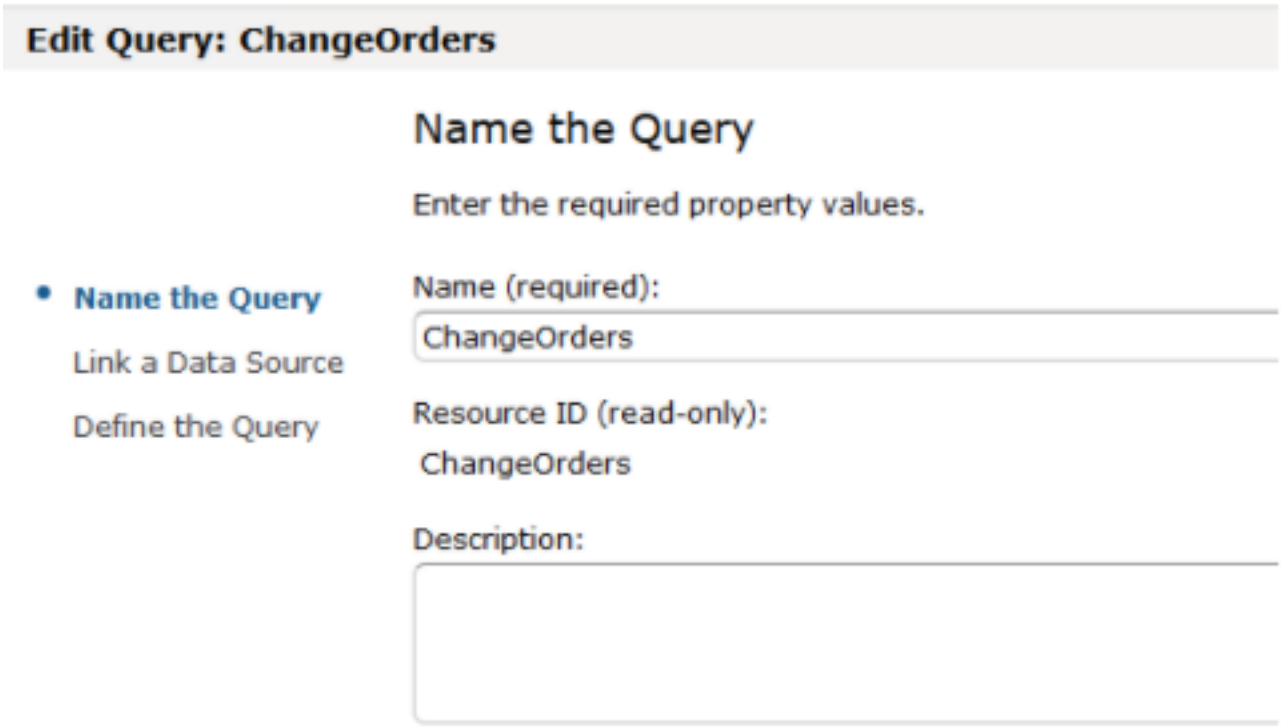


Modify the Input Parameters

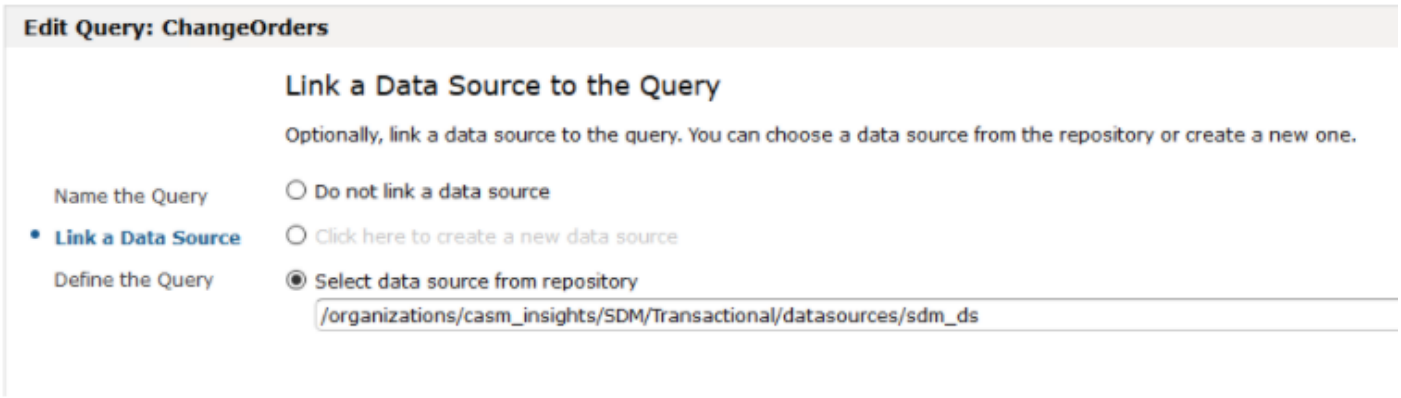
Input parameters/controls are used to prompt users for input or to pass the input to the report. Based on the input, the server modifiesthe WHERE filter clauses. Note Use single quotes to input a collection into a report, such as a list of Change Orders to return.

Perform the following steps:

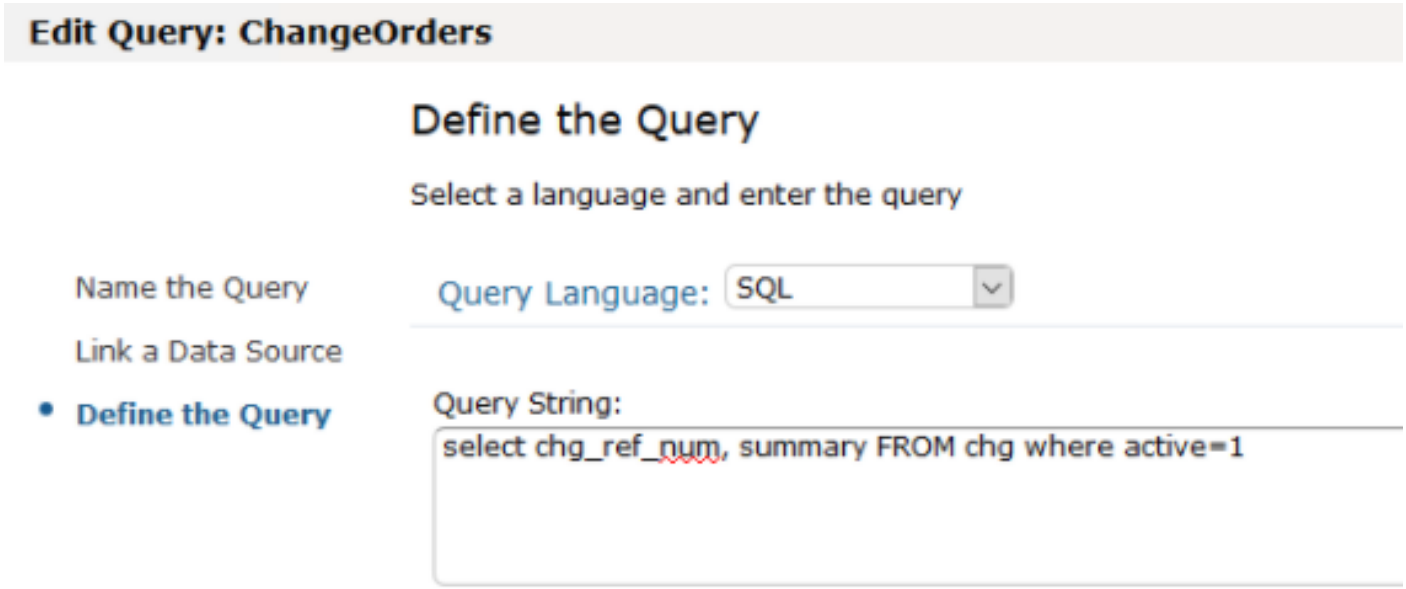
1. Create a query that the input parameter will use:
   a. View the repository, and drill down to: `\Organizations\SDM\Transactional\inputcontrols\queries`
   b. Right-click Queries, select Add Resource, Query.
   c. Provide a name as shown in the image below



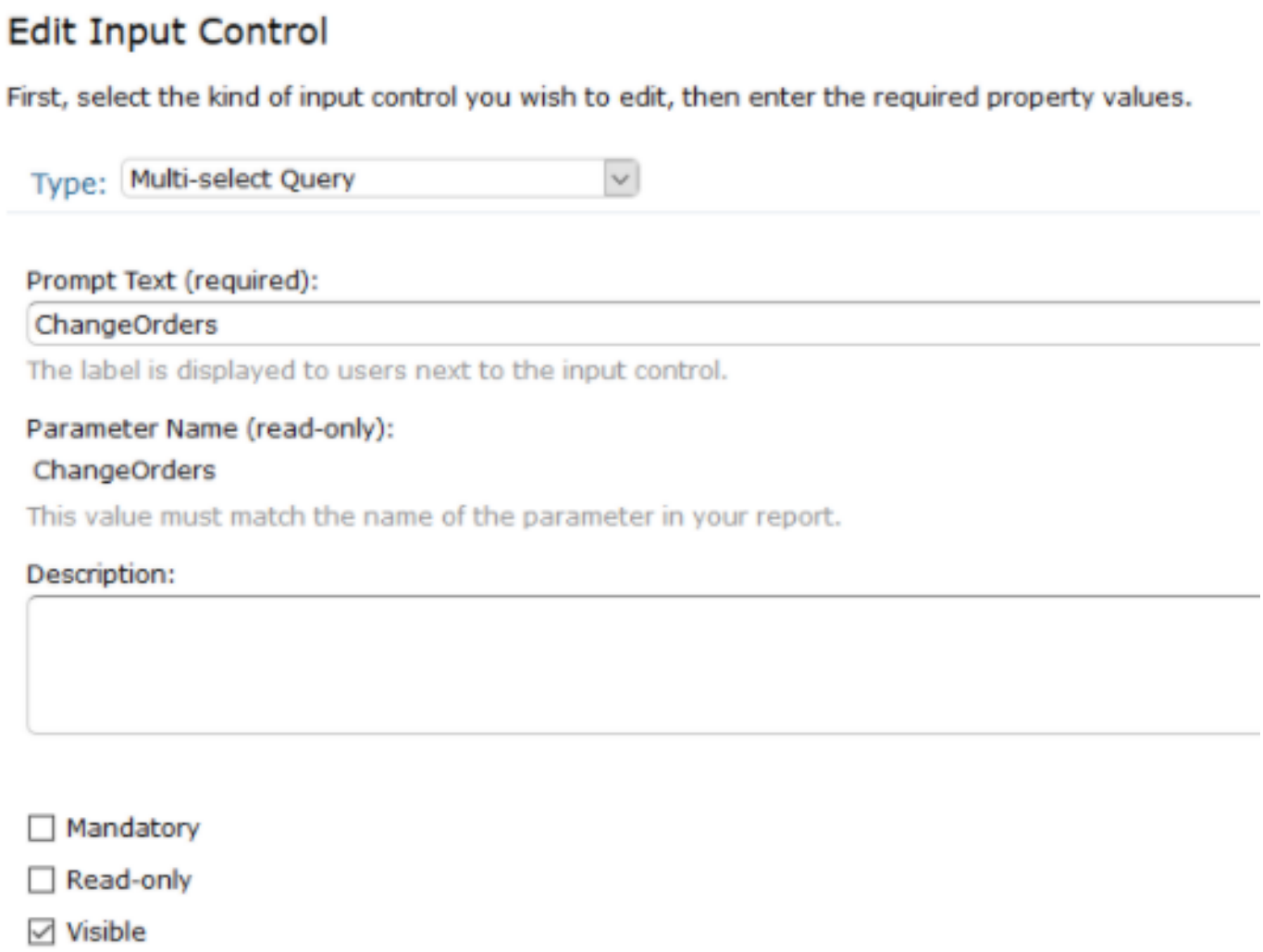
2. Select the data source



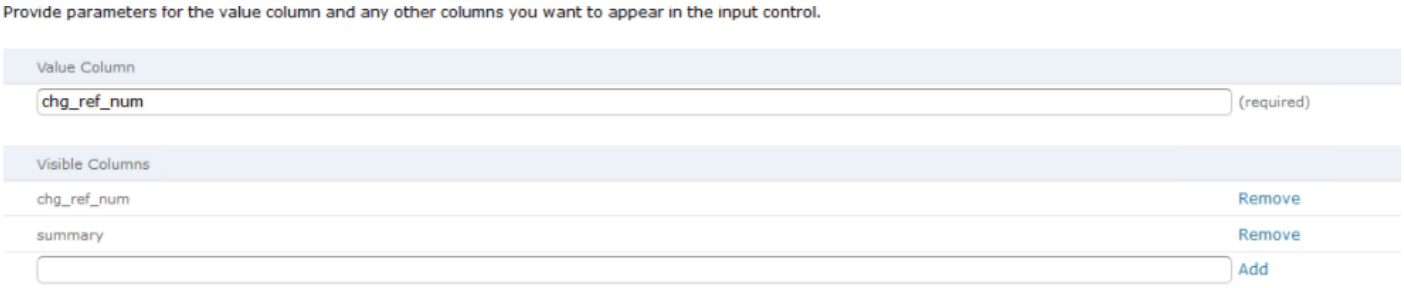
3. Define the query string with object names. For example: select chg_ref_num, summary FROM chg where active=1



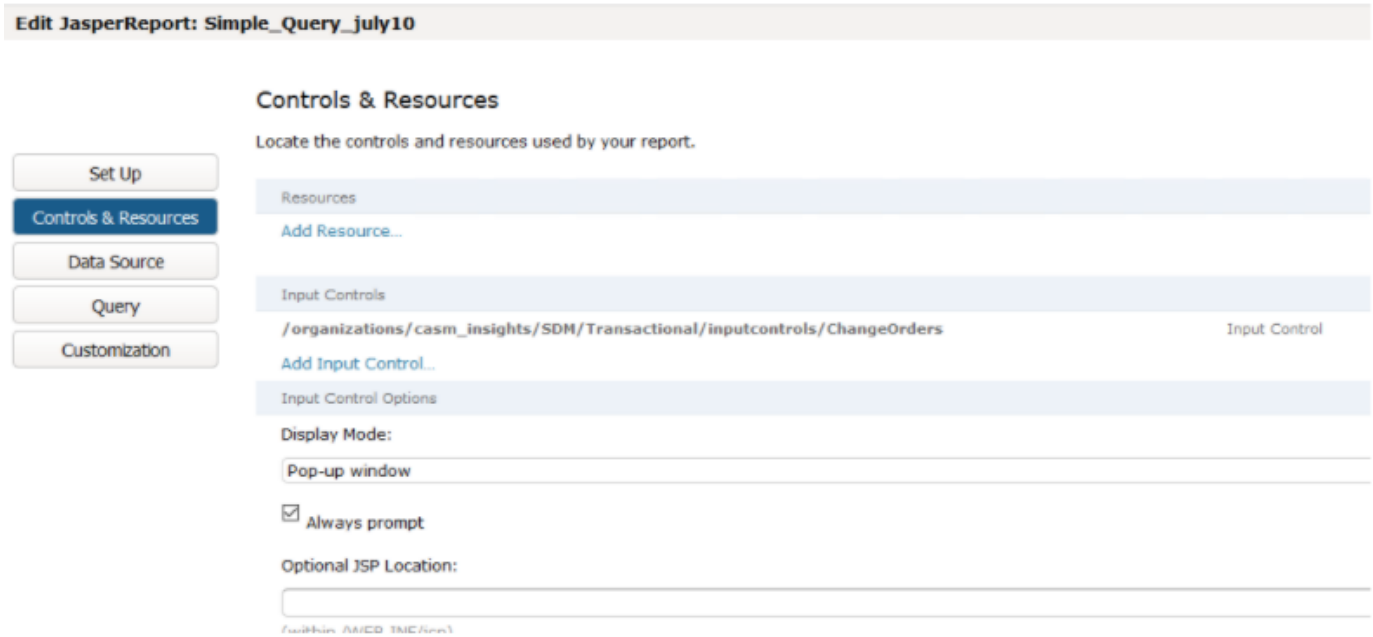
4. Right-click input controls, and select Add Resource, Input Control.
   a. Select the Type from the drop-down list. Example: Select Multi-select Query and select Visible.



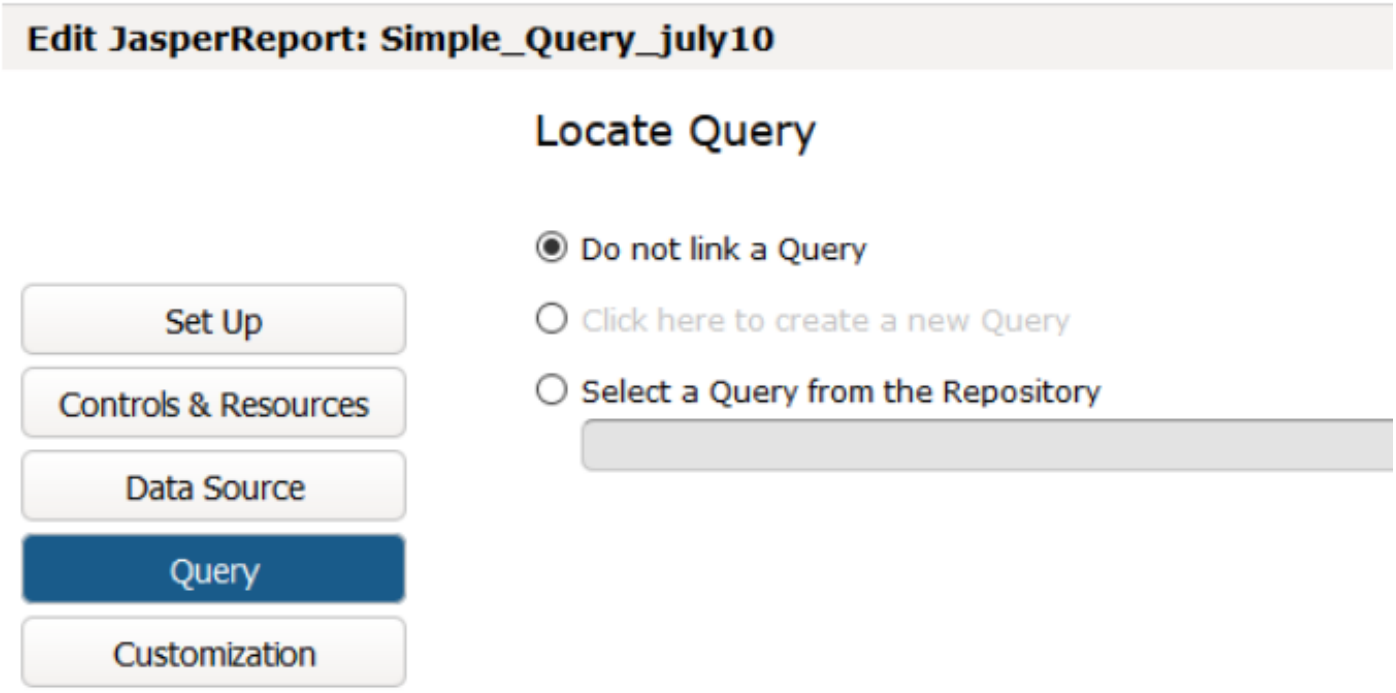
5. Select the query that you created in the first step:



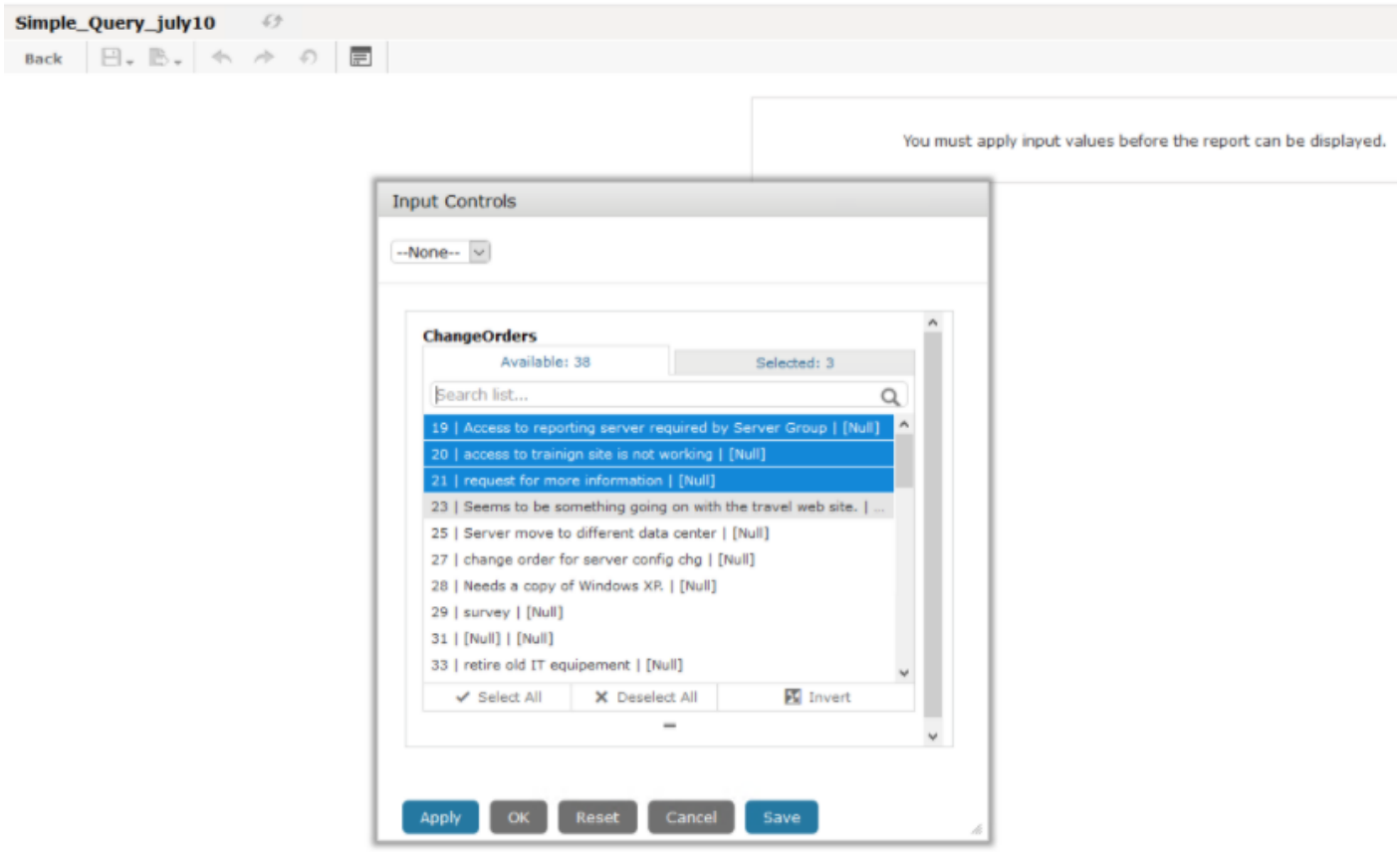
6. Select the column name that is queried, and create the columns that to display when the parameter is shown:



7. Browse to your report in the repository. Right-click to edit it.
8. Under Controls & Resources, browse to the Input Control you created in step 2.



9. Run your report to test the input control.

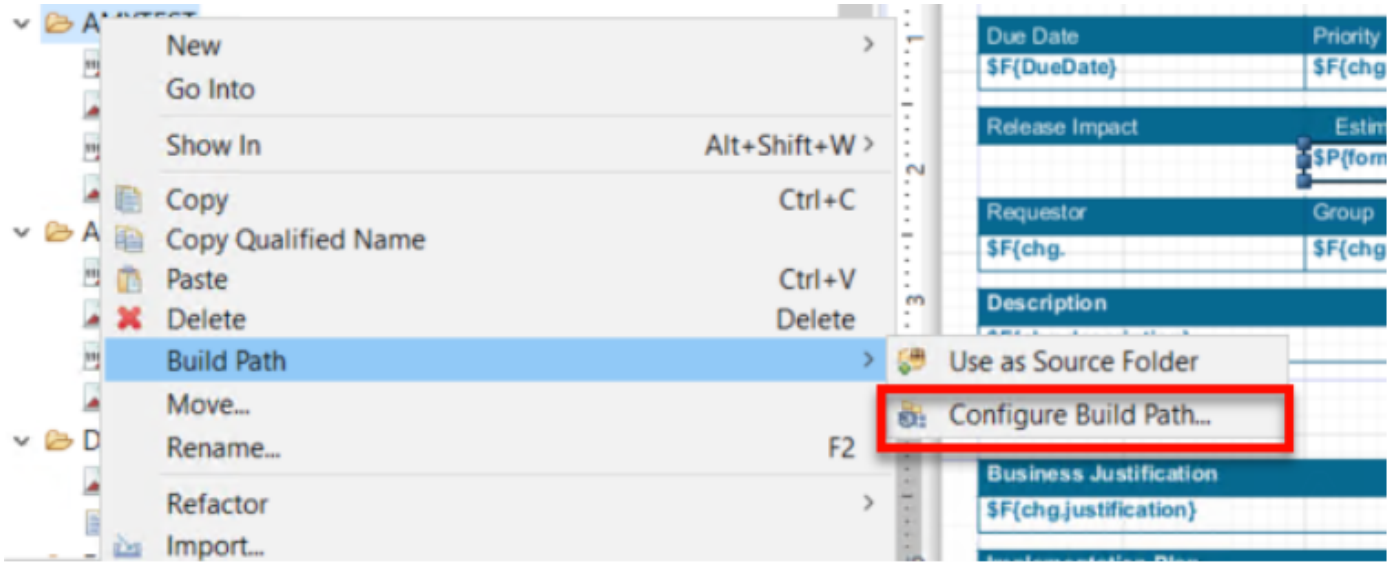


Add Duration Fields

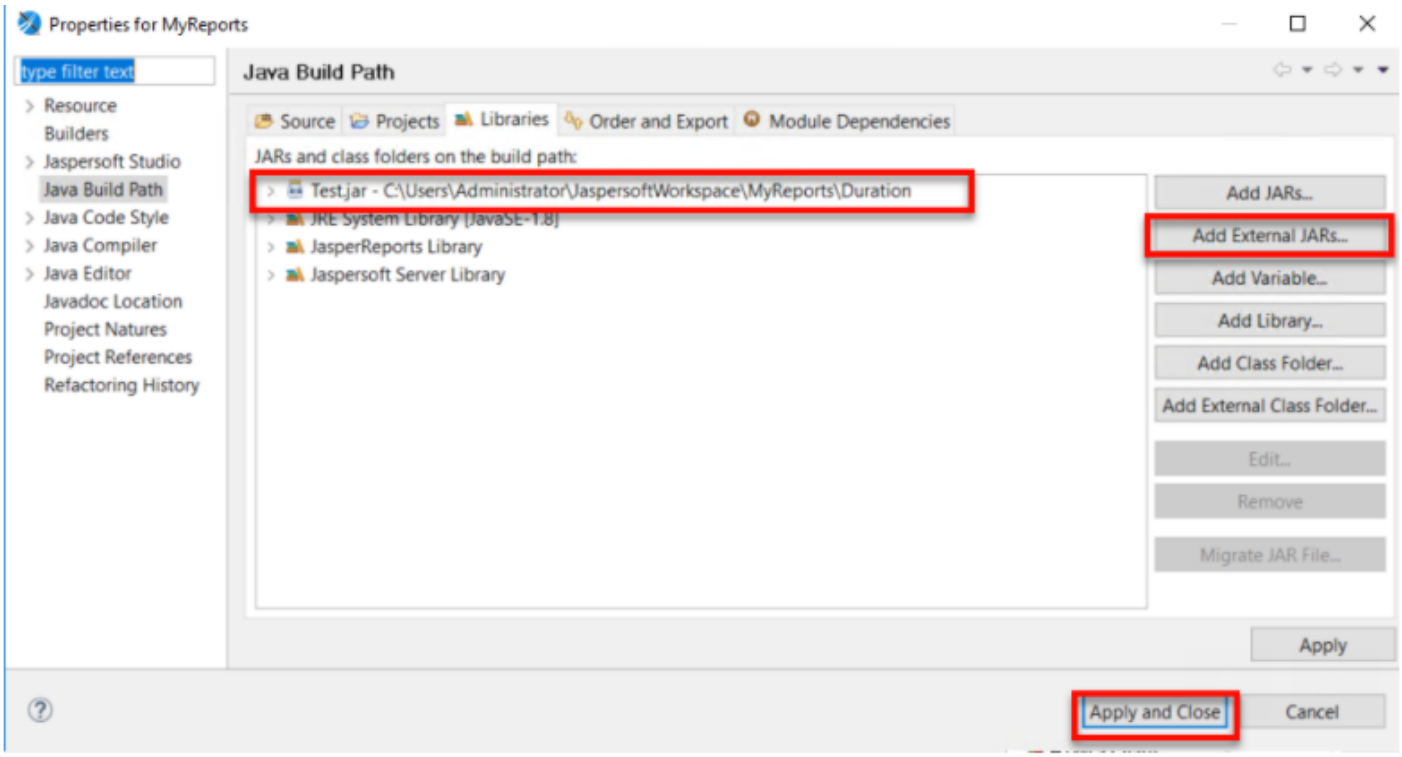
Add the duration fields to the build path of your project in the libraries. Then, import the class: com.duration.converter.

Perform the following steps:

1. Right-click a project folder, select **Build Path**, and **Configure Build Path**.



2. Copy the Jar from the DVD. CASM_17.4\filestore\Jasper\8.1.1\ExternalAuthJars and, copy it into the jasperserver tomcat lib folder. For eg: *C:\apache-tomcat-9.0.65\webapps\jasperserver-pro\WEB-INF\lib*.
3. Navigate to the **Libraries** tab, click **Add External JARs**, and browse for the above **DurationFormatter.jar** that you copied in Step 1.

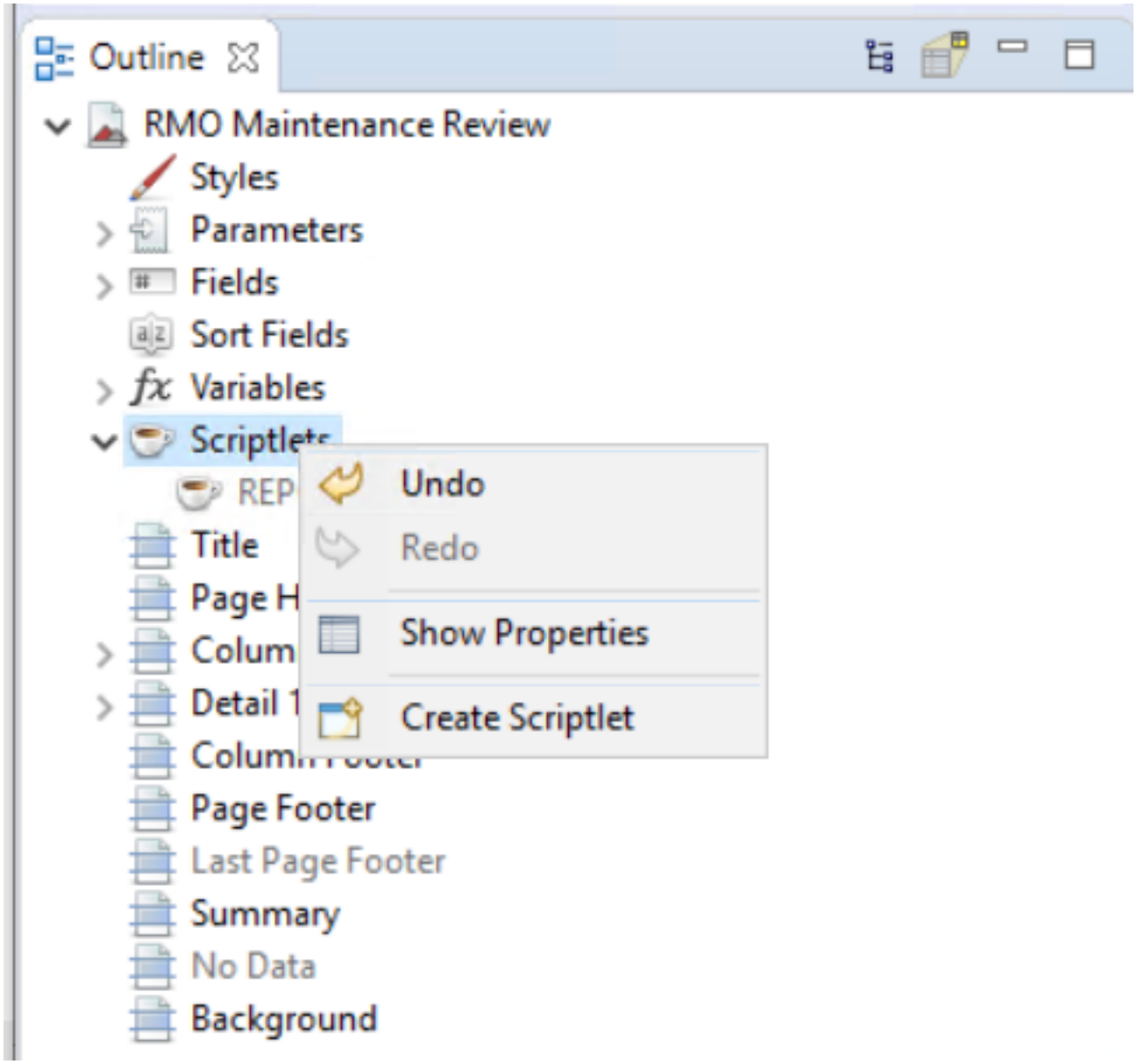


4. Click **Apply and Close**.

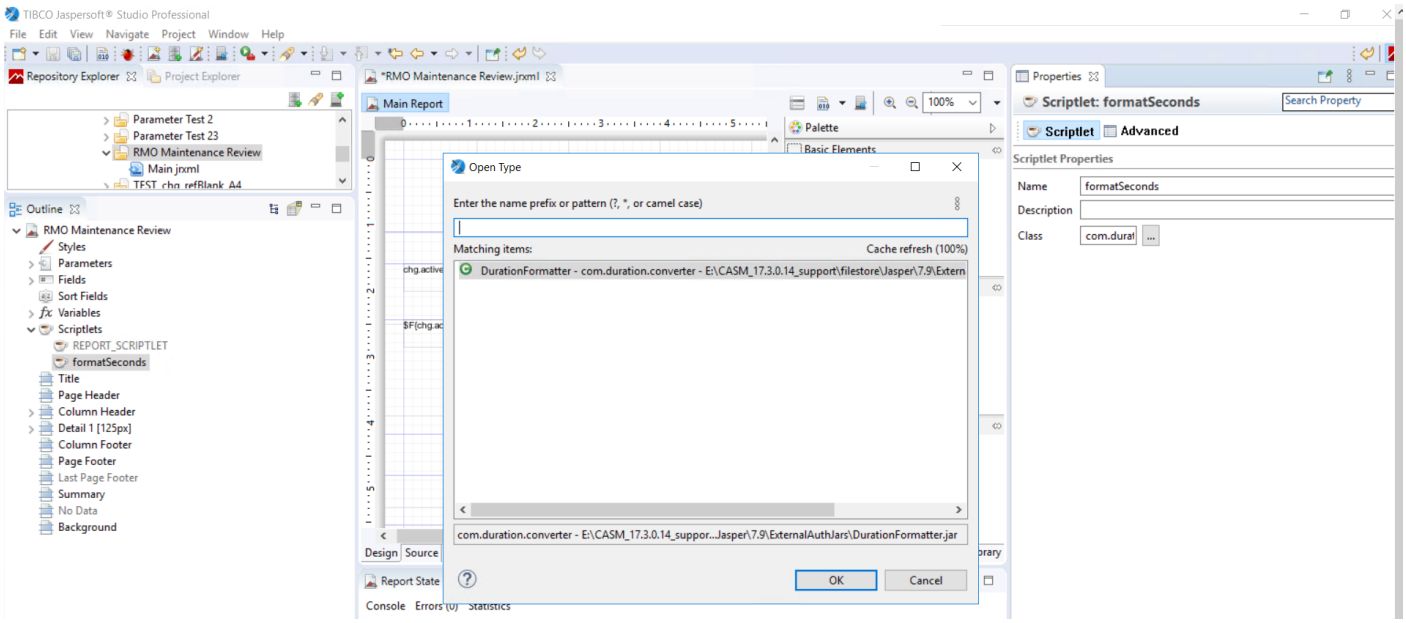
Add Parameter for Scriptlet

Perform the following steps:

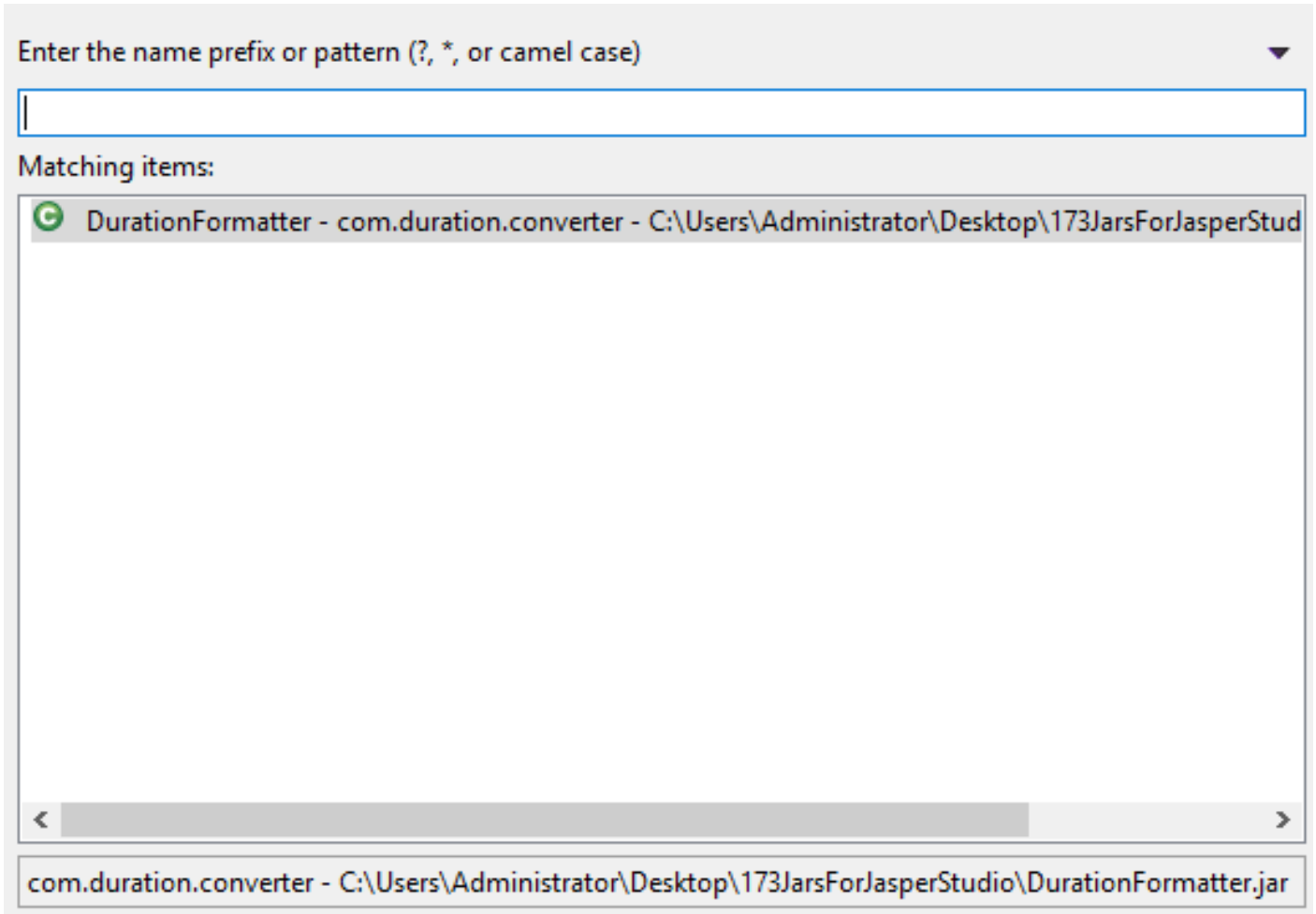
1. Right-click Scriptlets (under the report outline tab), and Click **Create Scriptlet** option.

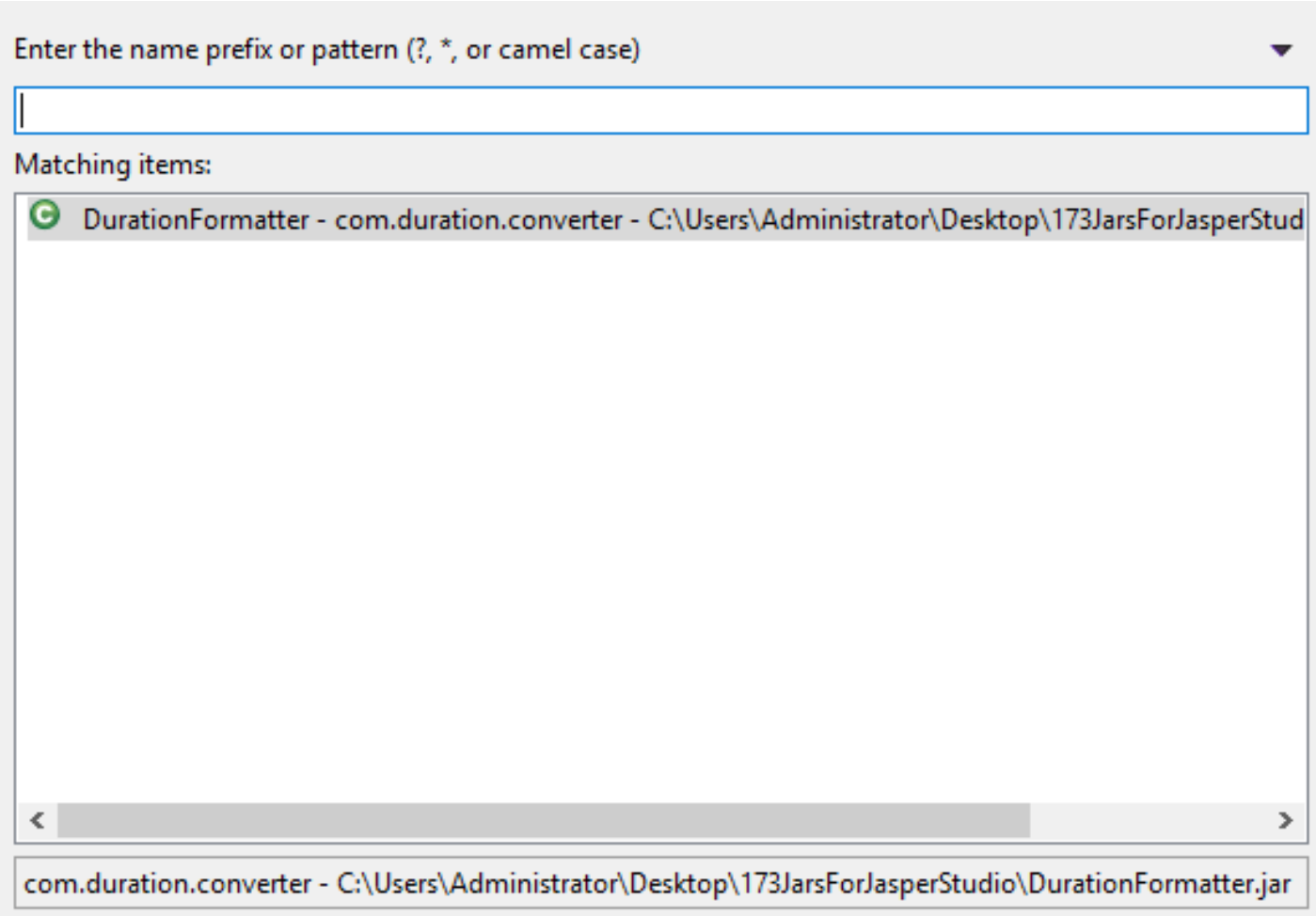


2. Under the **Properties** tab, give the scriptlet a name: **formatSeconds**
3. Next to the Class field, select the ellipsis (...)

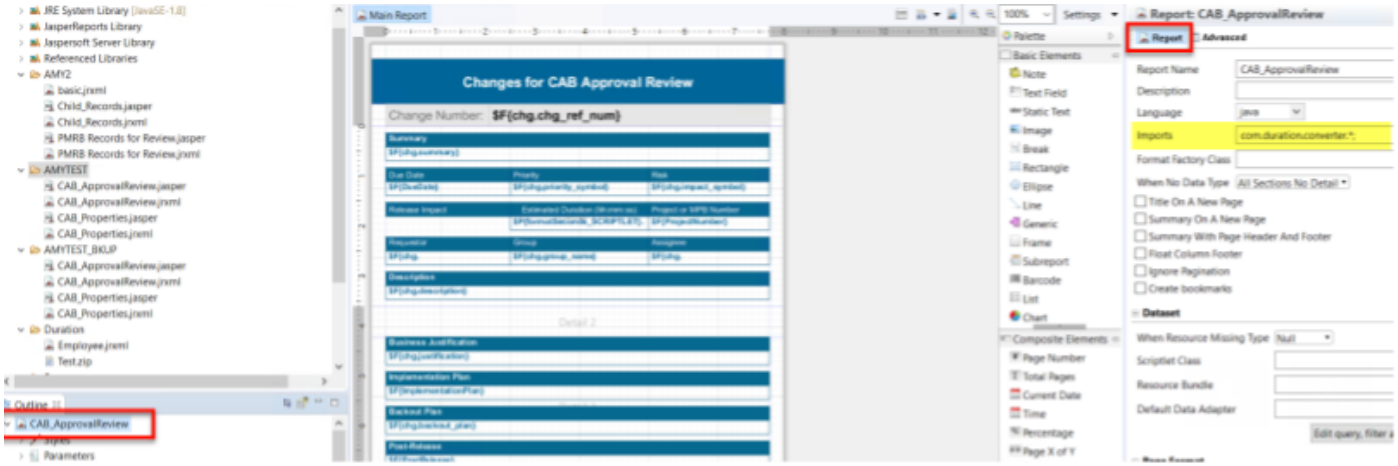


4. Select the **DurationFormatter**.

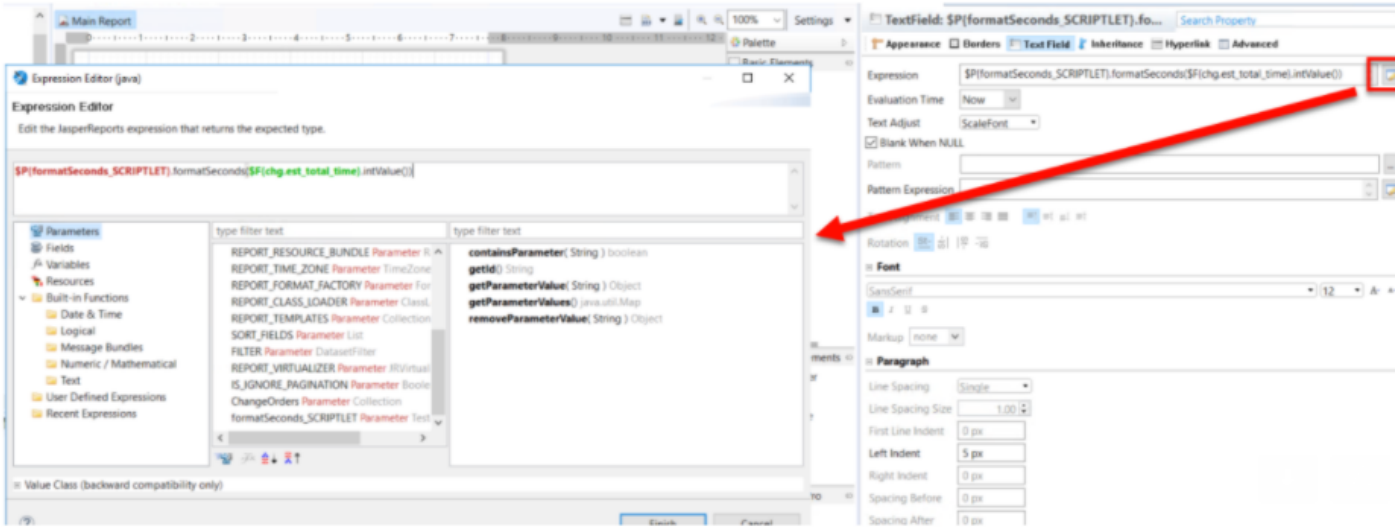




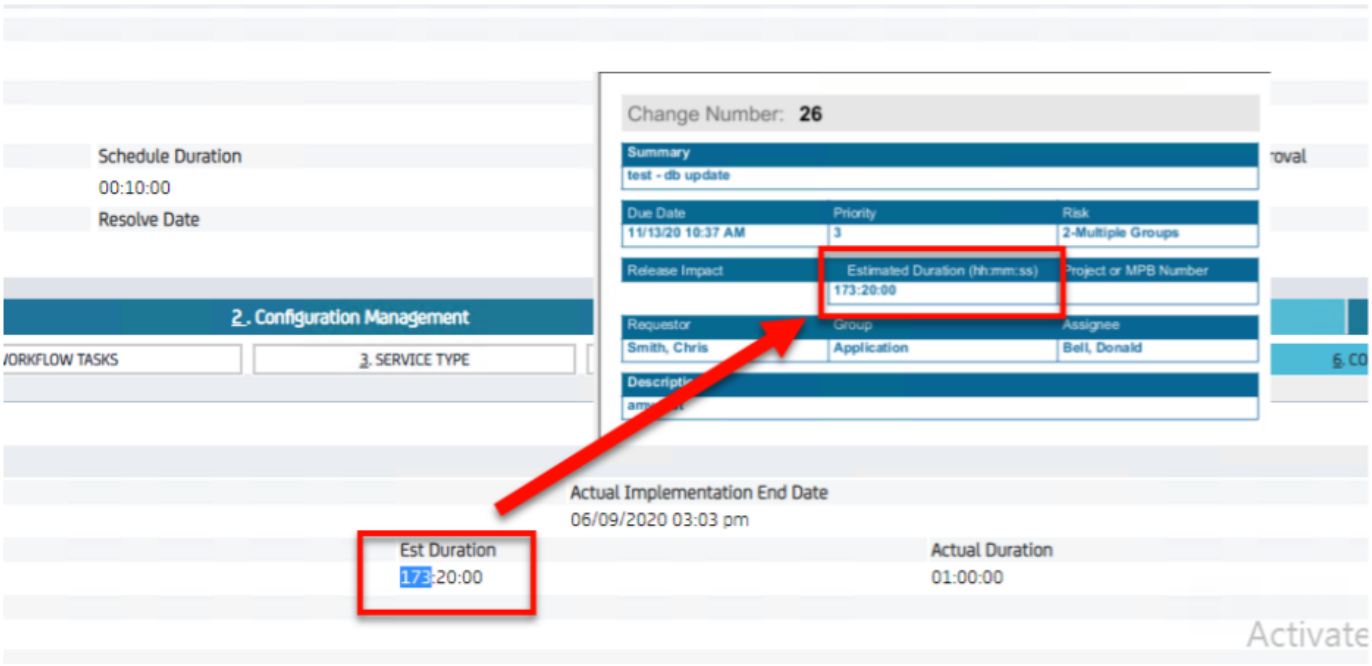
5. Click the Duration field in the report.
6. Click to add an expression for the Estimated Duration field:
   ```
   $P{formatSeconds_SCRIPTLET}.formatSeconds($F{chg.est_total_time}.intValue())
   ```
7. Click the report name under Outline Properties and click the dots next to Imports.
8. Select **Add Package**.
9. Select **com.duration.converter**.



10. Click **OK**.



11. Run the Report. It should match as shown in the following screenshot.



NOTE
If you are unable to run the report, ensure that the <import value> and <scriptlet name> parameters in the source tab of the report have the following values

```
<import value="com.duration.converter.*"/> <scriptlet name="formatSeconds"
 class="com.duration.converter.DurationFormatter"/>
```

12. Publish this report in the Jasper Server from the Jasper Studio.
13. Log in to the Jasper Server URL in which the report is published.
14. From the service console, stop the service: *jasperreportsTomcat.*
15. Copy the *DurationFormatter.jar* from CASM_17.4.2.0\filestore\Jasper\9.0.0\ExternalAuthJars and, copy it into the jasperserver tomcat lib folder. For eg: *C:\apache-tomcat-9.0.65\webapps\jasperserver-pro\WEB-INF\lib*.

16. Start the service: *jasperreportsTomcat.*
17. Run the report.
    The duration should match the report and SDM application as mentioned in Step 11.

SQL Queries with Strings

If you are using SQL queries with a string type parameter, ensure that the parameters are enclosed in single quotes. See the following example.

```
For example: select * from cr where cr.ref_num='$P{refNumParam}' Where refNumParam is a string.
```

When using FLOAT/ DOUBLE ATTRIBUTES in JasperStudio

If double type fields are in use, they convert to a String by default in JasperStudio. This string must be manually changed to double type.
For e.g. if field class is java.lang.String, then manually change it to java.lang.double

Integrate Jaspersoft Studio 9.0.0 with CA Service Management

CA Service Management 17.4 RU2 and later provide the ability to integrate Jaspersoft Studio 9.0.0. This integration enables reporting users to design and run report templates, write complex expressions, and layout visual components. The connection between CA Service Management's data access layer and Jaspersoft Studio is established via the JDBC driver. All the existing Out-of-the-box reports built for Jaspersoft Server work with Jaspersoft Studio as well.

> **NOTE**
> ODBC Driver is not supported on CA Service Management 17.4.

Download and Install Jaspersoft Studio Professional 9.0.0

To download and install Jaspersoft Studio 9.0.0 in your environment, perform the following steps:

1. Download the **CA Business Intelligence JasperSoft Studio 9.0.0 for CA Service Management.iso** installer from Broadcom Support.
2. Right-click the ISO and select **Mount**.
3. Right-click the **TIB_js-jss_9.0.0_windows_x86_64** application and run as **Administrator**.
4. Follow the Installation Wizard instructions to install Jaspersoft Studio 9.0.0 in a location of your choice.

   > **NOTE**
   > You must join the Jaspersoft Community to use Jasper Studio, refer **Logging in to the Jaspersoft Community** section in TIBCO documentation.
   >
   > **NOTE**
   >
   > - If you have trouble logging-in with your community credentials, apply the Hotfix 99112369.
   > - For more information refer this Jasper Community page.
5. After installing Jaspersoft Studio, the out-of-the-box installation package (For example, C:\Program Files\TIBCO \Jaspersoft StudioProfessional-9.0.0) displays the Evaluation License Version.

Update Jaspersoft Studio License (Evaluation License to PerpetualLicense)

To change the Evaluation License (default) provided with Jaspersoft Studio 9.0.0 to Perpetual or Full License, perform the following steps:

1. Navigate to where you have downloaded and extracted CA Service Management 17.4. RU2.
2. Navigate to **Filestore**, **Jasper**, **9.0.0**, **License**.

NOTE
Jaspersoft Reports 9.0.0 Install folder contains the Jaspersoft Studio Professional 9.0.0 License Information.

3. Now, launch Jaspersoft Studio 9.0.0.
4. Navigate to **Help**> **License Manager**> **Install New License**.
5. Browse and click open the License folder which is available in step 2 and select **jasperserver.license**.
6. Click **OK.**
7. If you see Edge related errors on your screen, follow the steps mentioned in **Using Edge in Jaspersoft Studio Window Builds** in TIBCO documentation.

Create Data Adapter

Perform the following steps to create a Data Adapter:

1. Open Jaspersoft Studio.
2. Right-click a project in Project Explorer and select **New**, **Data Adapter**.
3. In the Data Adapter File window, select a project where you want to save the data adapter file. The selected project must contain the report(s) that you want to use with your data adapter. The Data Adapter Wizard is displayed.
4. Enter a name for your adapter in the Data Adapter Wizard and click **Next**
   For example, Database JDBC Connection.
5. Provide the CA Service Desk Manager (SDM) connection details.
6. In the Database Location tab, enter the properties required by the data adapter type. For example, for database JDBC connection, select the JDBC driver, set the URL, and provide the CA SDM database username and password:
   a. JDBC Driver: com.ca.sdm.dal.sql.jdbc.driver.DalJdbcDriver
   b. JDBC URL: jdbc:sdm:bop:@<SDM Host Name>:2100;<UserUuid>=<ServiceDeskUUID>
   c. UserName: ServiceDesk
   d. Password: Encrypted Password must be used. Perform the following steps to encrypt the password:
      a. Navigate to CASM 17.4RU2_DVD\filestore\Jasper\Utils folder.

         NOTE
         Ensure that java is added to PATH env. variable to run the below command.

      b. Run the following command:

         ```
         gen_passwd.bat <servide desk username> <service desk password>
         ```

         For example: Username: servicedesk. Password: servicedesk password (example: see the below image in Linux environment)

7. In the Driver Classpath tab, select the list of jars.
8. Provide the library location for the jars, the same jars as the Jaspersoft Reports 9.0.0 `DVD \filestore\Jasper \9.0.0\ExternalAuthJars`
9. a. BOPIntegration.jar
   b. domsrvr_utils.jar
   c. log4j-api-2.17.1.jar
   d. log4j-core-2.17.1.jar
   e. DurationFormatter.jar
   f. sd-utils.jar
   g. slump.jar

10. To test the connection, click the **Test button**.
11. Click **Finish** to create the Adapter.

Create a Report and Publish

To create a report in Jaspersoft Studio and publish it to your existing JasperReports Server (9.0.0), see TIBCO Documentation.

Create Subreports

To create subreports, perform the following steps:

Pass string parameters to subreports

Testing subreport in Jaspersoft Studio

Publish the Subreport in Jaspersoft Studio

Pass string parameters to subreports

Consider the following example scenarios:

Jaspersoft Studio Reports - Example 1

Jaspersoft Studio Reports - Example 2

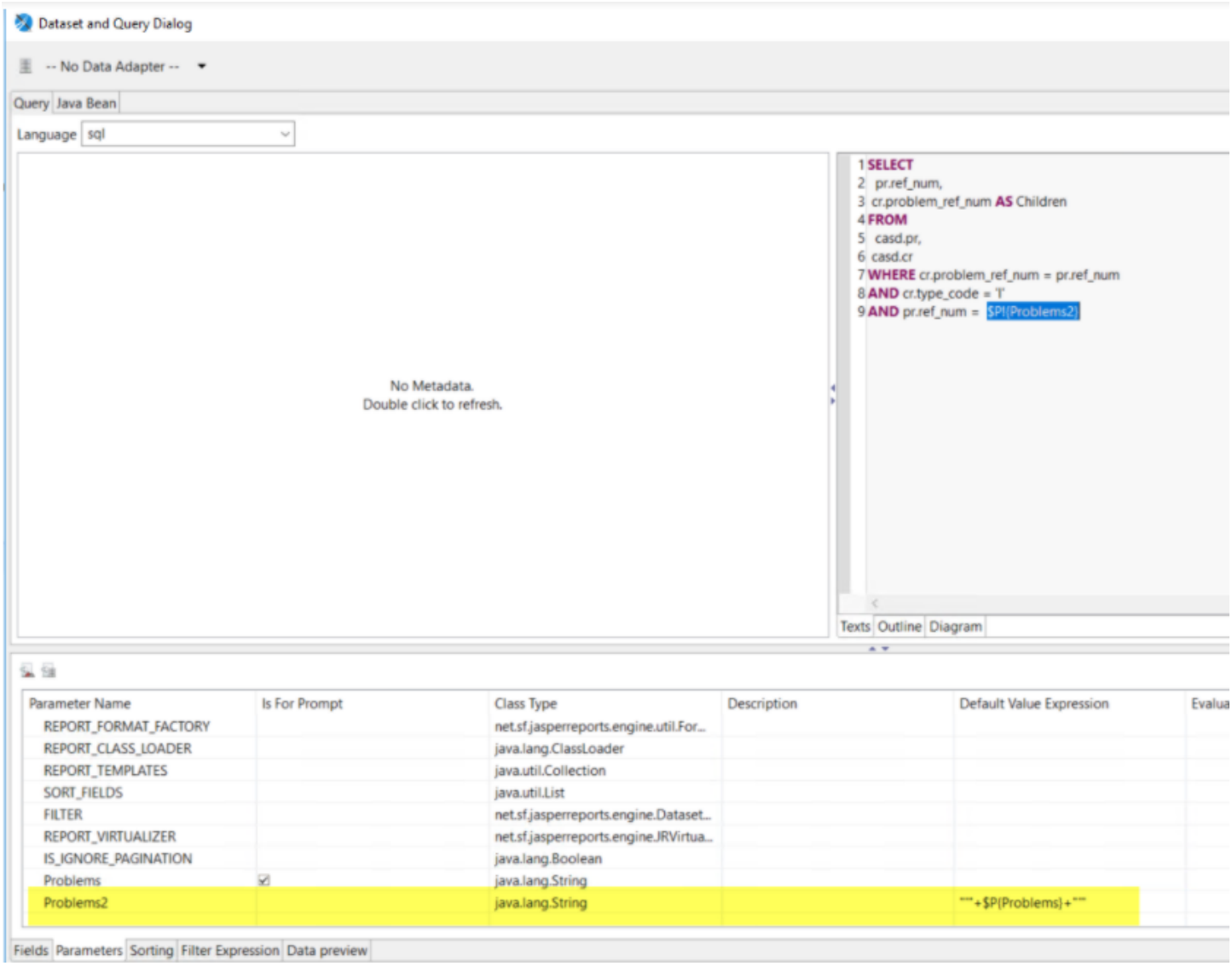
Jaspersoft Studio Reports - Example 1

In this example report, a list of problem tickets is presented with a subreport to show the child Incidents related to each problem. The following query shows a parameter, defined as **java.util.Collection**, to prompt the user to enter or select a list of problem tickets. The parameter is flagged for **Is For Prompt**.

```
SELECT casd.pr.id,
casd.pr.ref_num,
casd.pr.summary,
casd.pr.group_name,
casd.pr.impact_symbol,
casd.pr.priority_symbol,
casd.pr.urgency_symbol,
casd.pr.status_symbol,
casd.pr.rootcause_symbol,
casd.pr.open_date,
casd.pr.description,
casd.pr.change_summary,
casd.pr.change_chg_ref_num,
casd.pr.caused_by_change_order_summary,
casd.pr.caused_by_change_order_chg_ref_num,
casd.alg.type,
casd.alg.type_symbol,
casd.alg.description,
casd.alg.last_mod_dt
FROM casd.alg,
casd.pr
WHERE $X{IN,pr.ref_num, Problems} AND
```

```
pr.persistent_id = alg.call_req_id
```

NOTE
Ensure that your subreport string parameter has a default expression to hard code single quotes. Also, this parameter must not be set for prompting the user to enter or select a list of problem tickets.



Jaspersoft Studio Reports - Example 2

In this example, the main report has data about one or more change orders. The subreport returns the correct properties for the change orders as a list.

Perform the following steps:

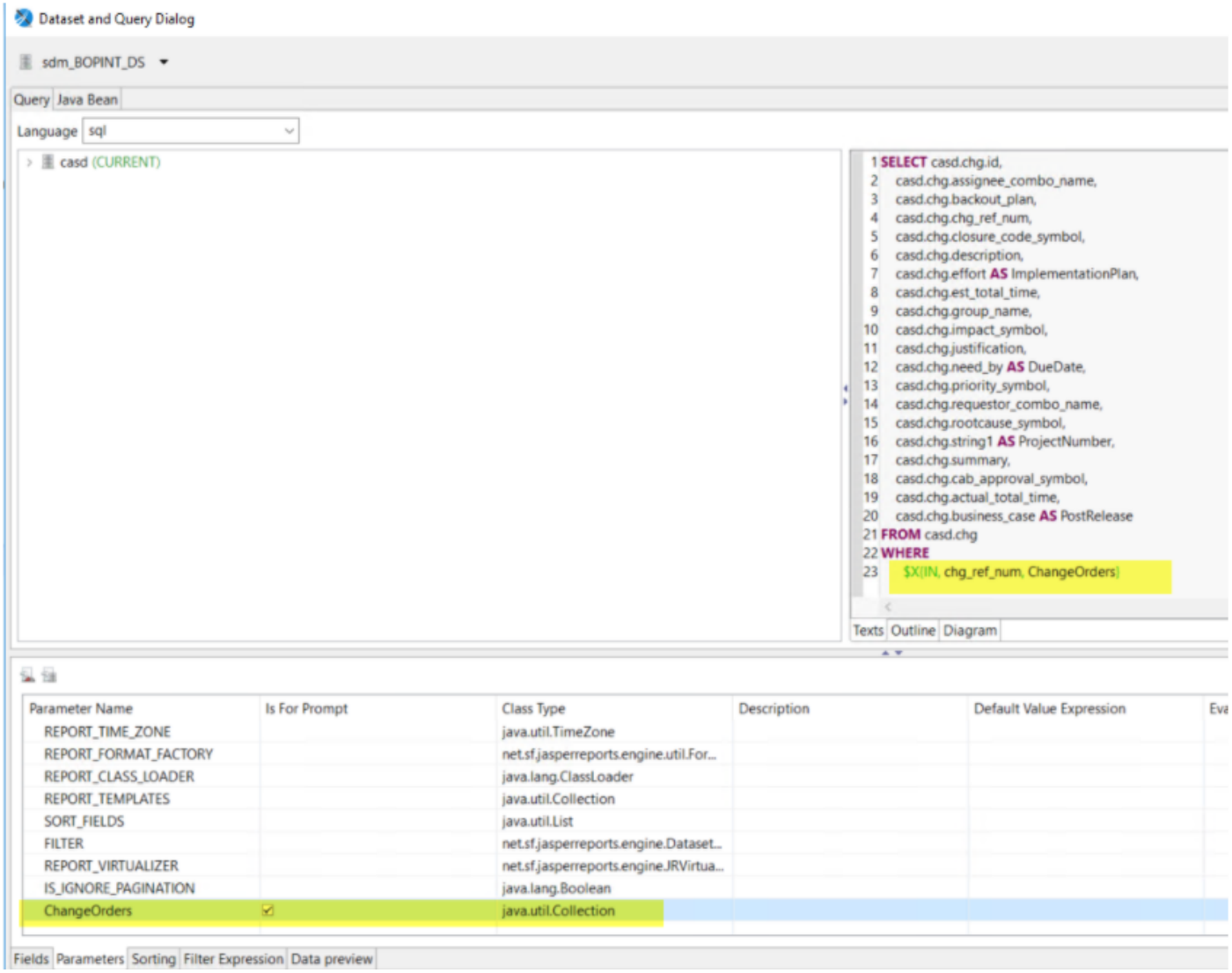
1. Run the following example query for the main report:

```
SELECT casd.chg.id,
casd.chg.assignee_combo_name,
casd.chg.backout_plan,
```

```
casd.chg.chg_ref_num,
casd.chg.closure_code_symbol,
casd.chg.description,
casd.chg.effort AS ImplementationPlan,
casd.chg.est_total_time,
casd.chg.group_name,
casd.chg.impact_symbol,
casd.chg.justification,
casd.chg.need_by AS DueDate,
casd.chg.priority_symbol,
casd.chg.requestor_combo_name,
casd.chg.rootcause_symbol,
casd.chg.string1 AS ProjectNumber,
casd.chg.summary,
casd.chg.cab_approval_symbol,
casd.chg.actual_total_time,
casd.chg.business_case AS PostRelease
FROM casd.chg
WHERE
$X{IN, chg_ref_num, ChangeOrders}
```

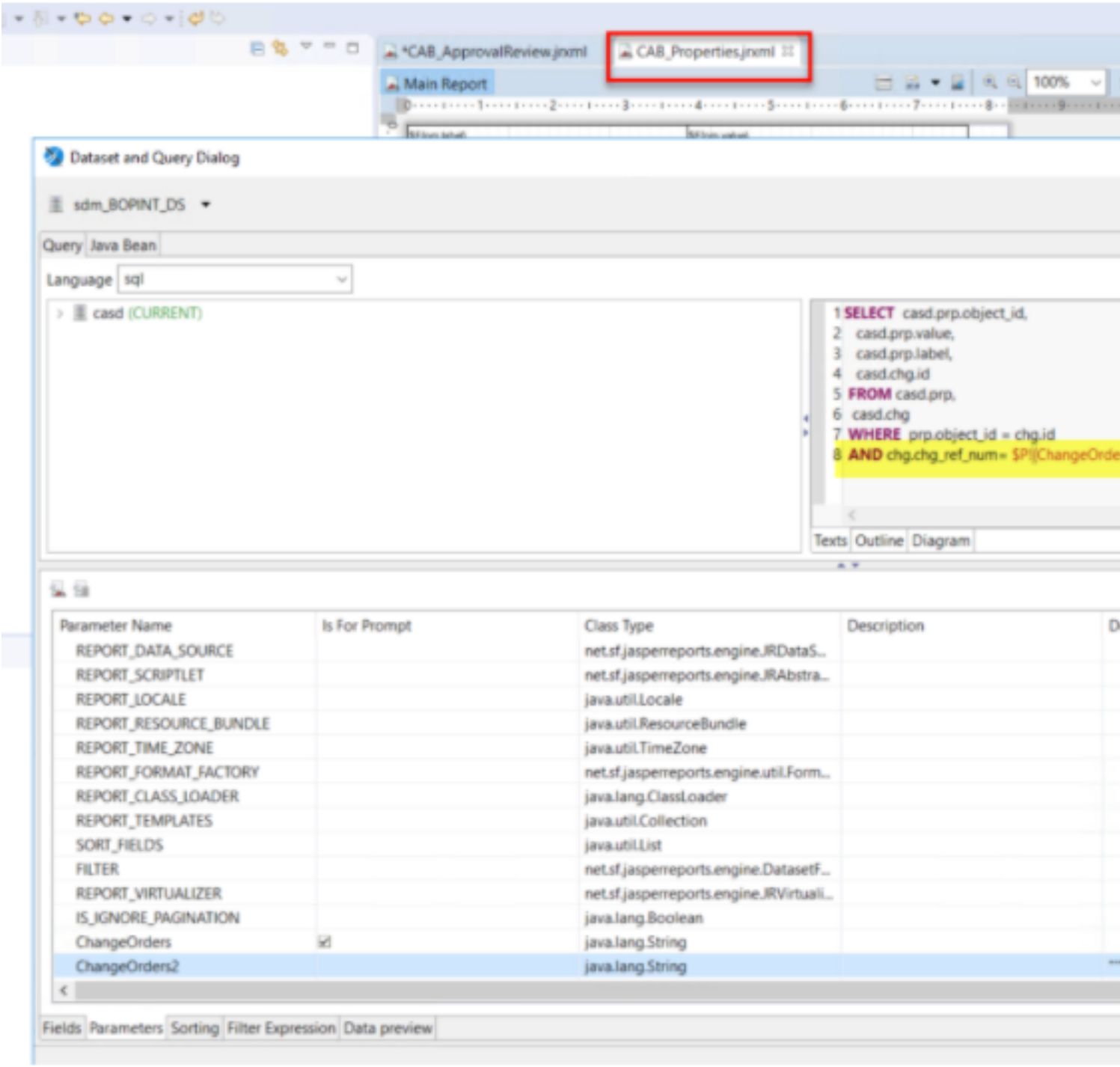
2. Run the following query to pass the list of values:

```
SELECT casd.chg.id,
casd.chg.assignee_combo_name,
casd.chg.backout_plan,
casd.chg.chg_ref_num,
casd.chg.closure_code_symbol,
casd.chg.description,
casd.chg.effort AS ImplementationPlan,
casd.chg.est_total_time,
casd.chg.group_name,
casd.chg.impact_symbol,
casd.chg.justification,
casd.chg.need_by AS DueDate,
casd.chg.priority_symbol,
casd.chg.requestor_combo_name,
casd.chg.rootcause_symbol,
casd.chg.string1 AS ProjectNumber,
casd.chg.summary,
casd.chg.cab_approval_symbol,
casd.chg.actual_total_time,
casd.chg.business_case AS PostRelease
FROM casd.chg
WHERE
$X(IN, chg_ref_num, ChangeOrders)
```



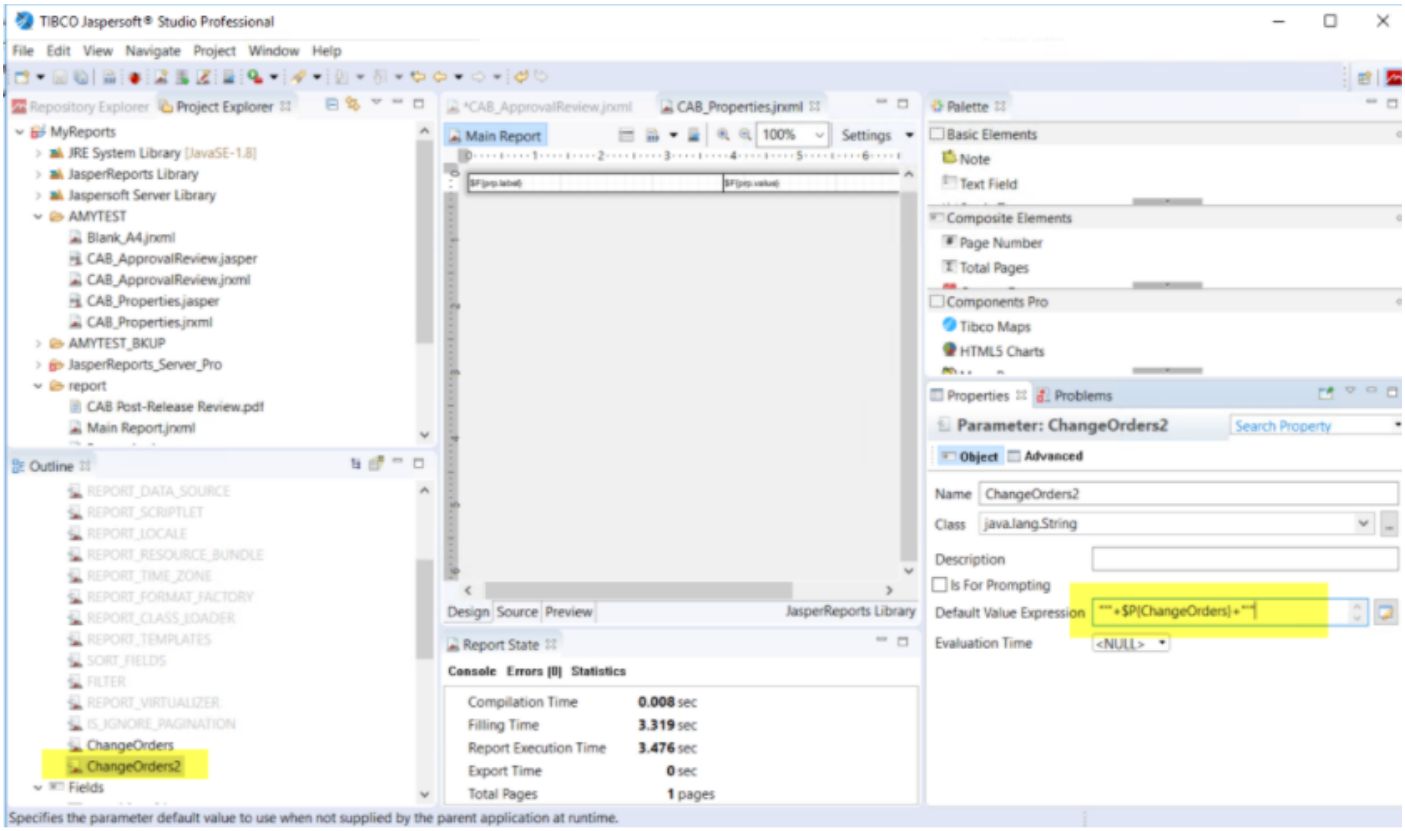
3. Run the following example query for subreports:

```
SELECT casd.prp.object_id,
casd.prp.value,
casd.prp.label,
casd.chg.id
FROM casd.prp,
casd.chg
WHERE prp.object_id = chg.id
AND chg.chg_ref_num= $P!{ChangeOrders2}
```

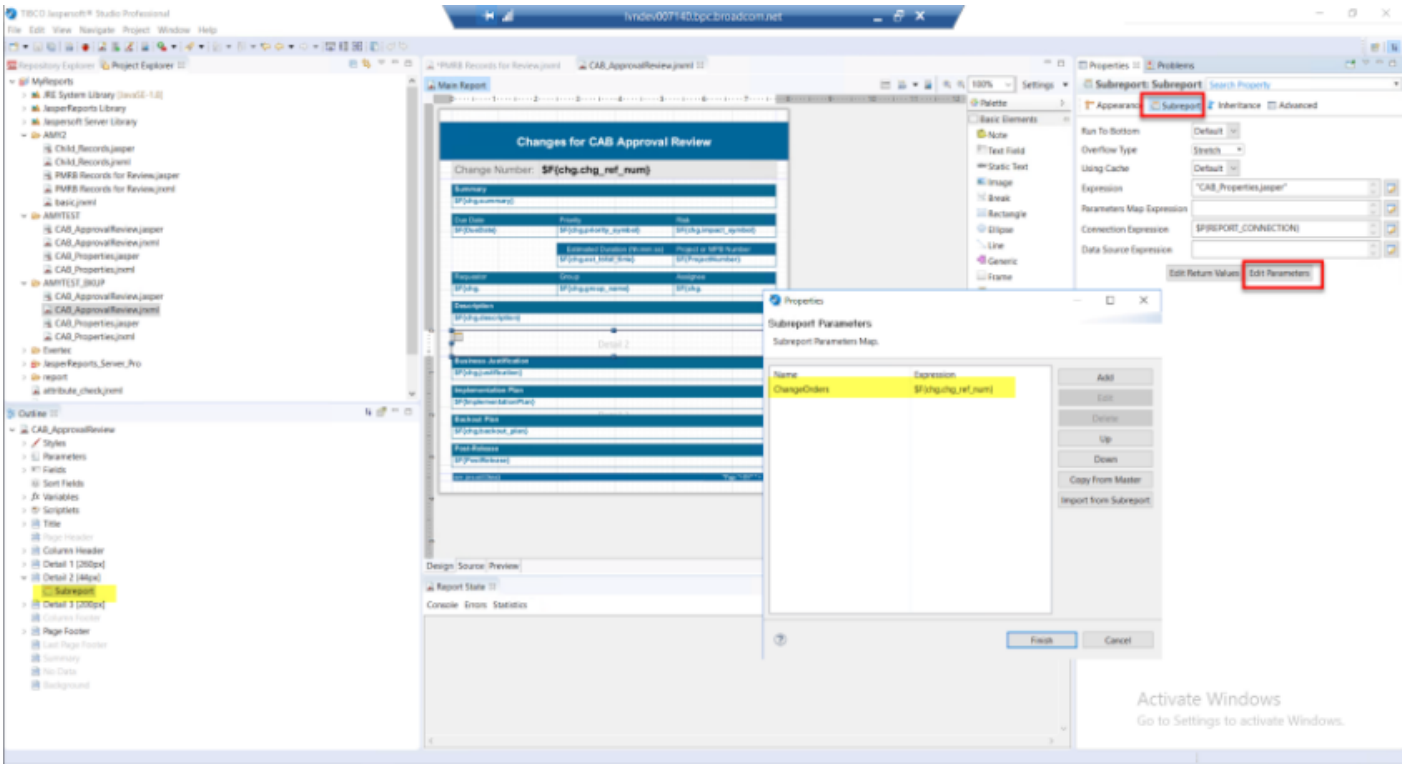


4. Add a new parameter to your subreport as shown below.

```
ChangeOrders2
Class: java.lang.String
Default Value Expression: "'"+$P{ChangeOrders}+"'"
```



5. Validate that the parameter to pass to the subreport is set as follows.



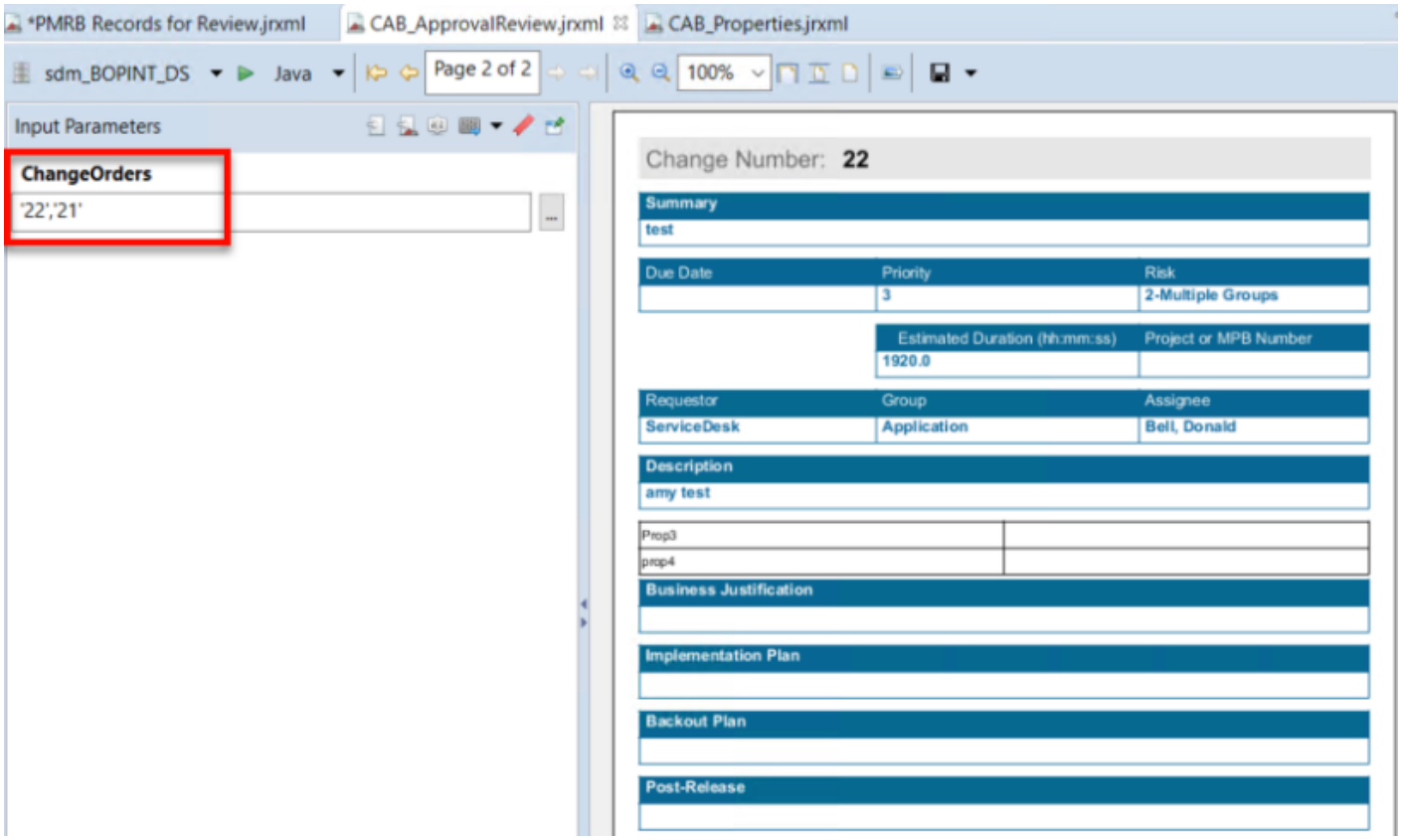
6. Compile the subreport after it is saved.

Testing Subreports in Jaspersoft Studio

Validate the subreports that you have created from Jaspersoft Studio. Ensure to list the change order numbers with single quotes, separated by a comma.

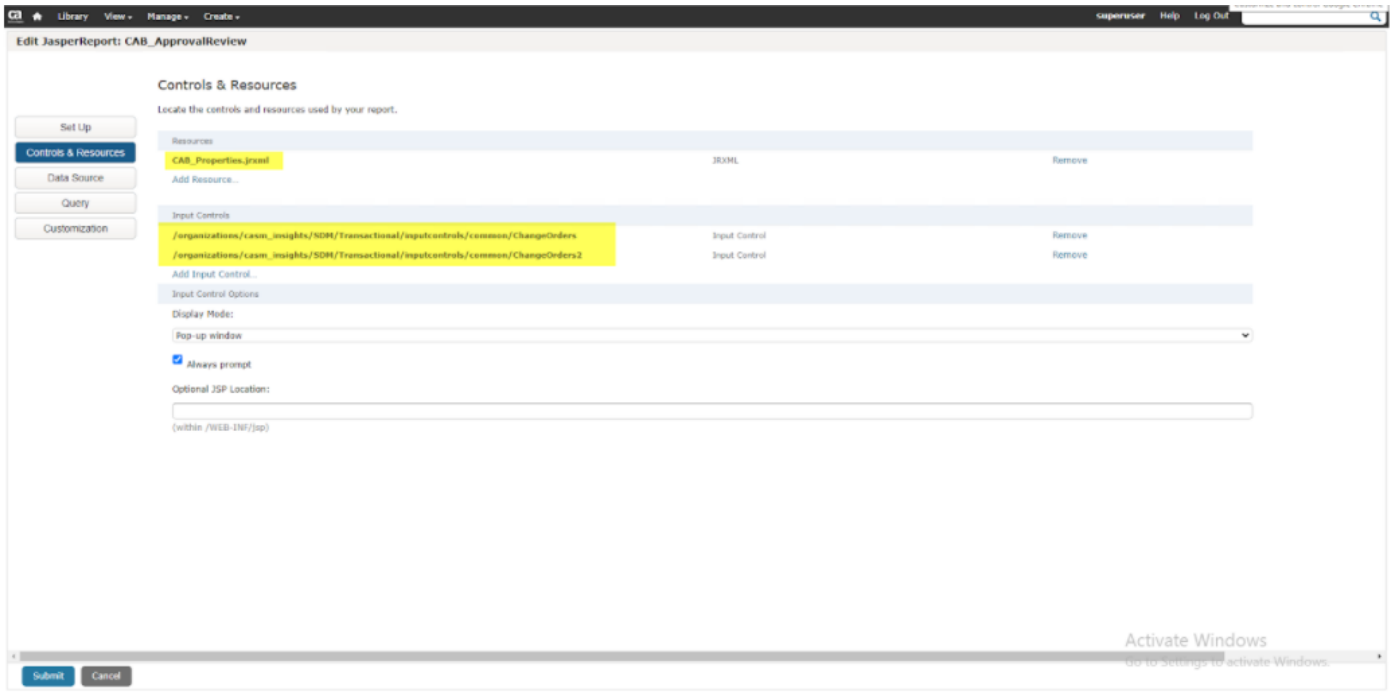
Perform the following steps:

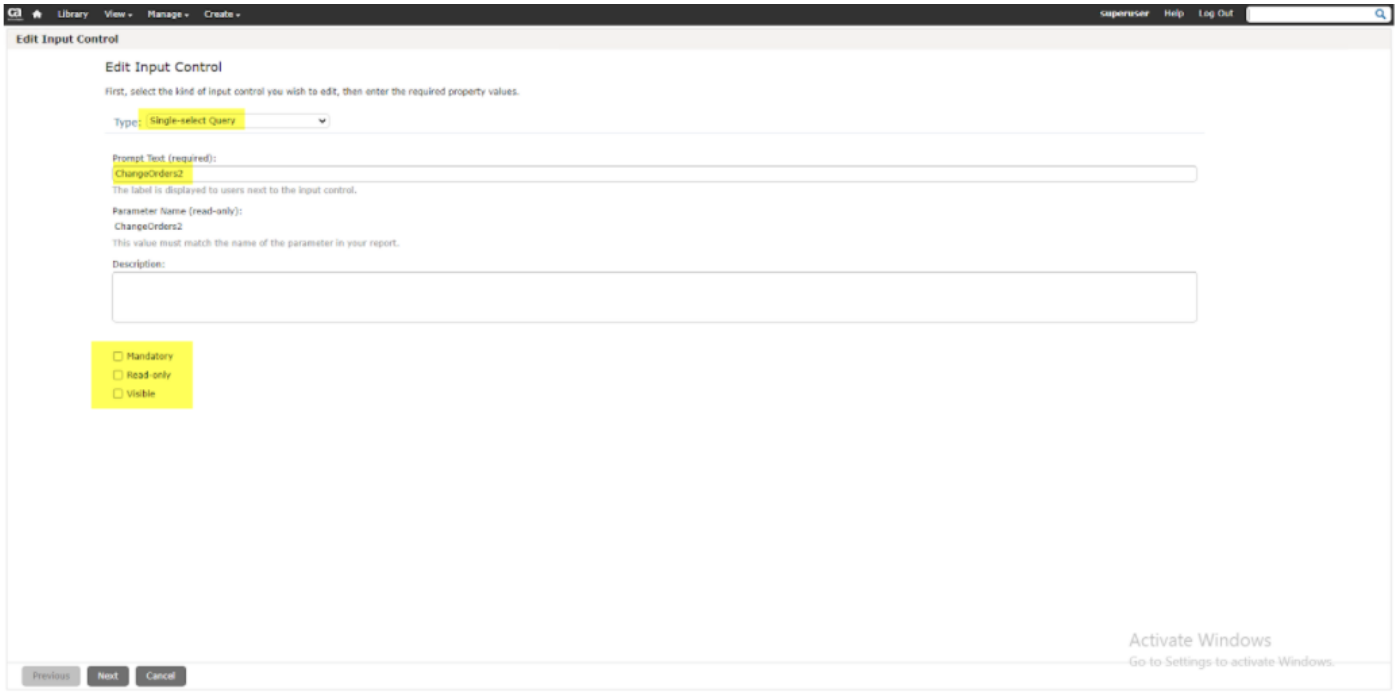
- Run the subreports and enter the change order numbers as shown below to test the subreport:



Publish the Subreport in Jaspersoft Studio

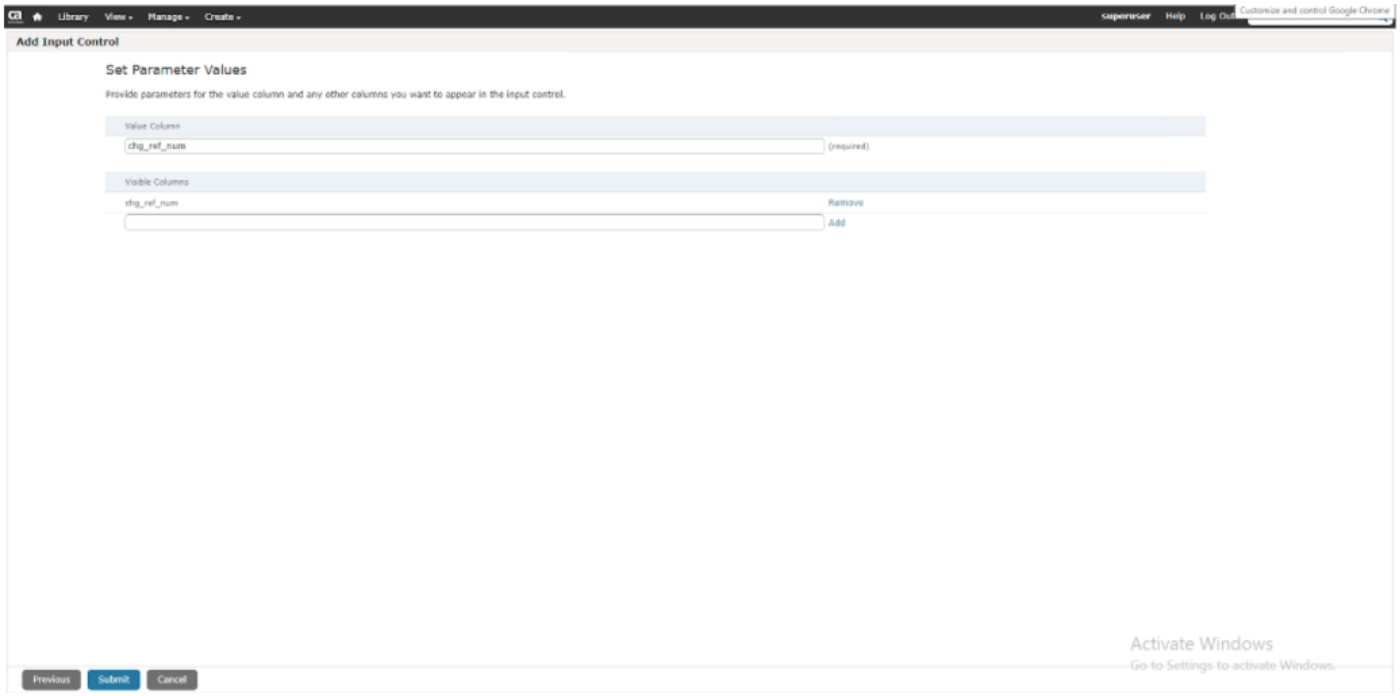
1. Add the **ChangeOrders2** input control as well. Ensure that the respective jrxml file is associated as a resource



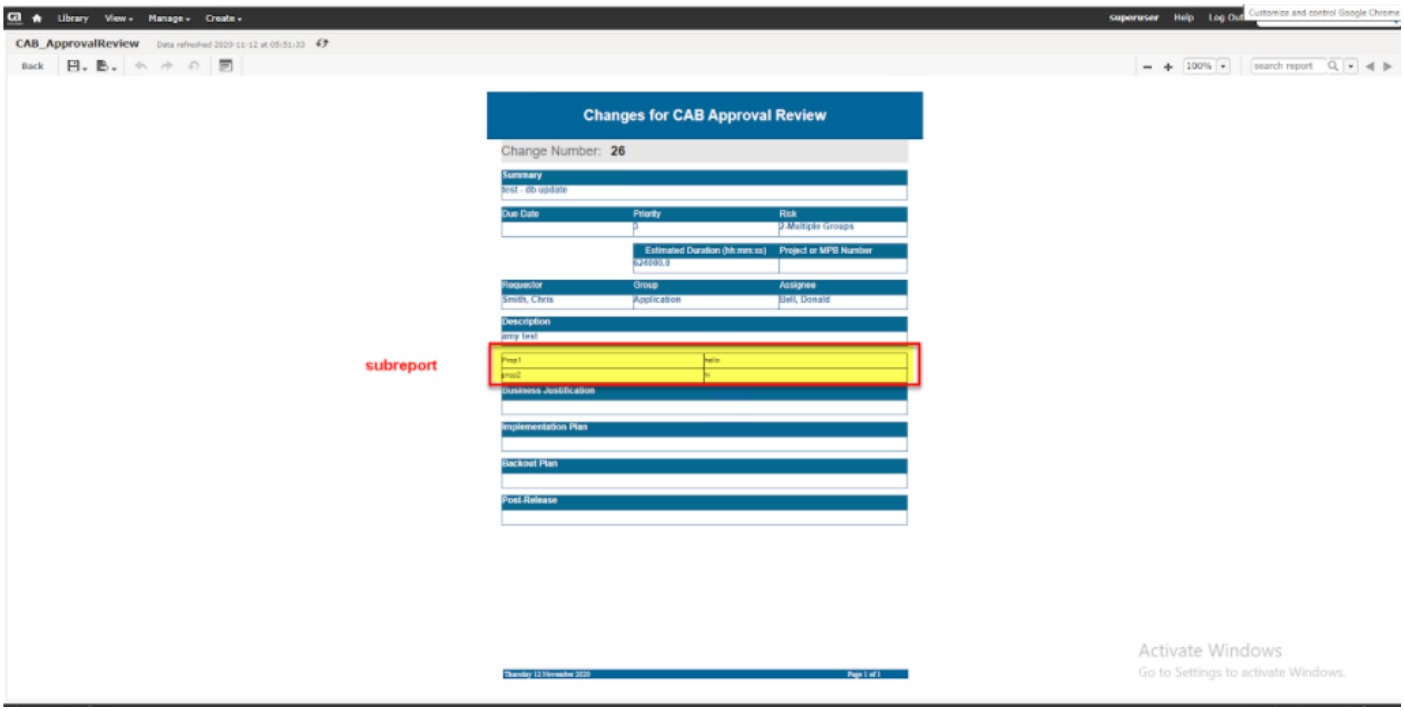


2. Link it to a query, in this example **Chg_Single** , pointing to the data source, with a query string of:

```
select ''''+chg_ref_num+'''' as chg_ref_num FROM chg where active =1
```



3. The subreport displays as follows.

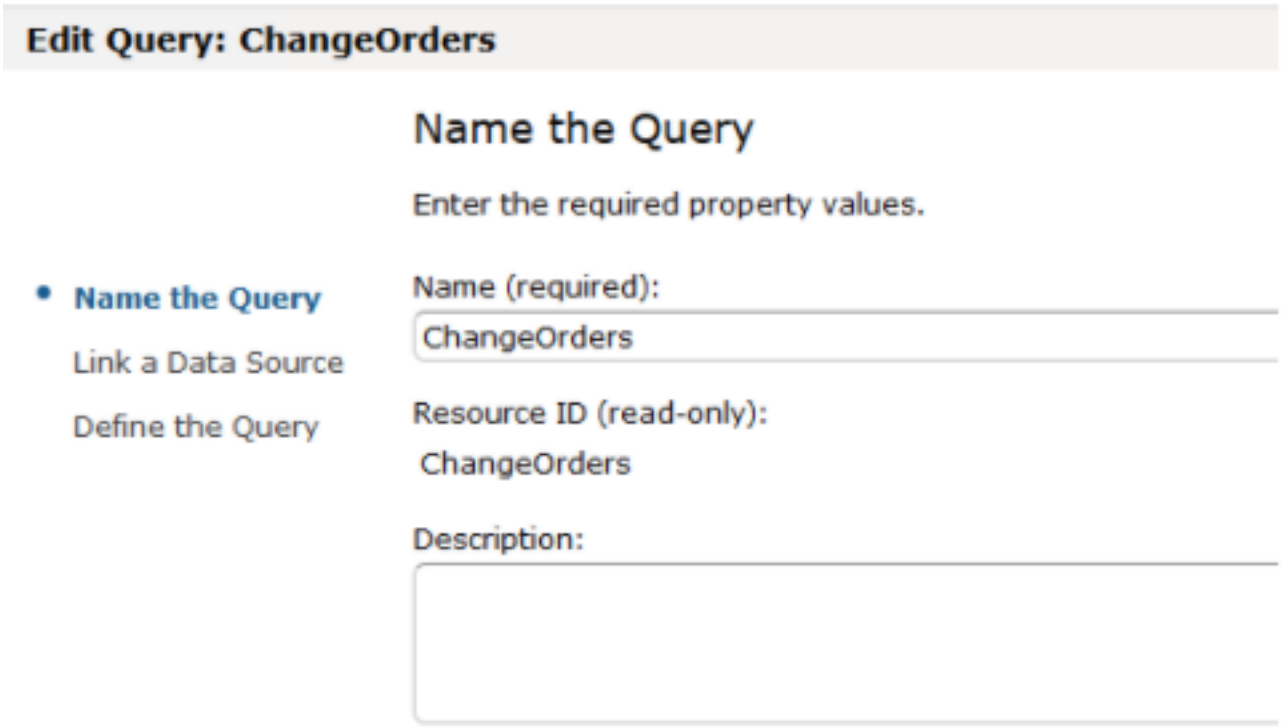


Modify the Input Parameters

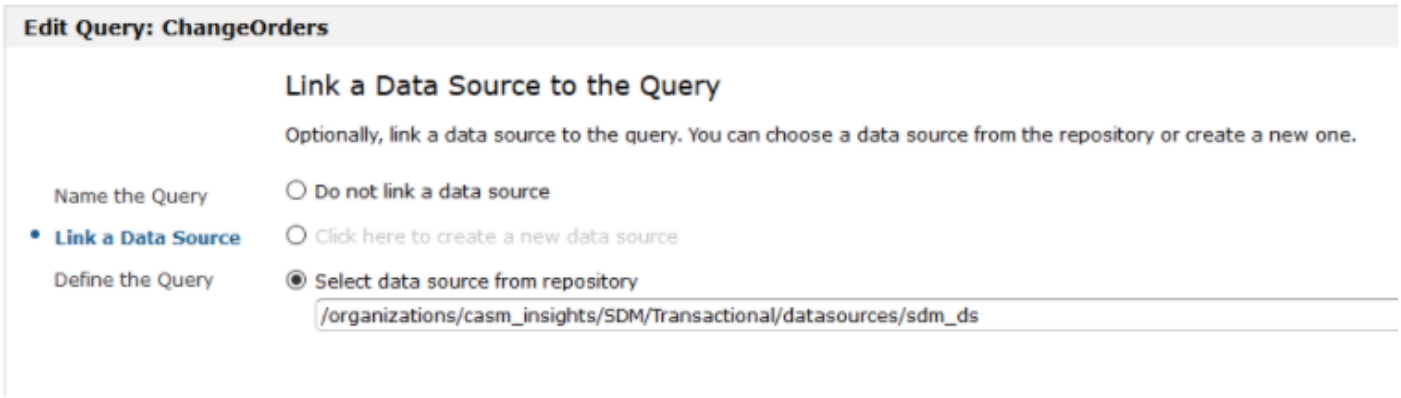
Input parameters/controls are used to prompt users for input or to pass the input to the report. Based on the input, the server modifiesthe WHERE filter clauses. Note Use single quotes to input a collection into a report, such as a list of Change Orders to return.

Perform the following steps:

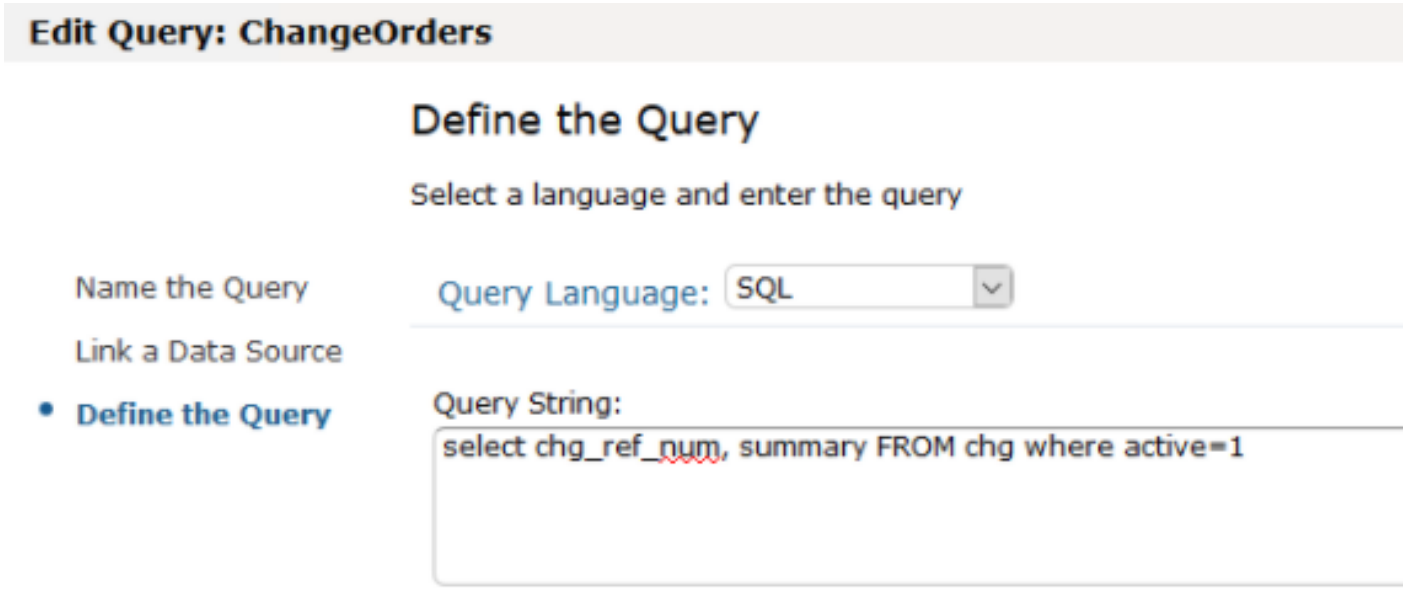
1. Create a query that the input parameter will use:
   a. View the repository, and drill down to: `\Organizations\SDM\Transactional\inputcontrols\queries`
   b. Right-click Queries, select Add Resource, Query.
   c. Provide a name as shown in the image below



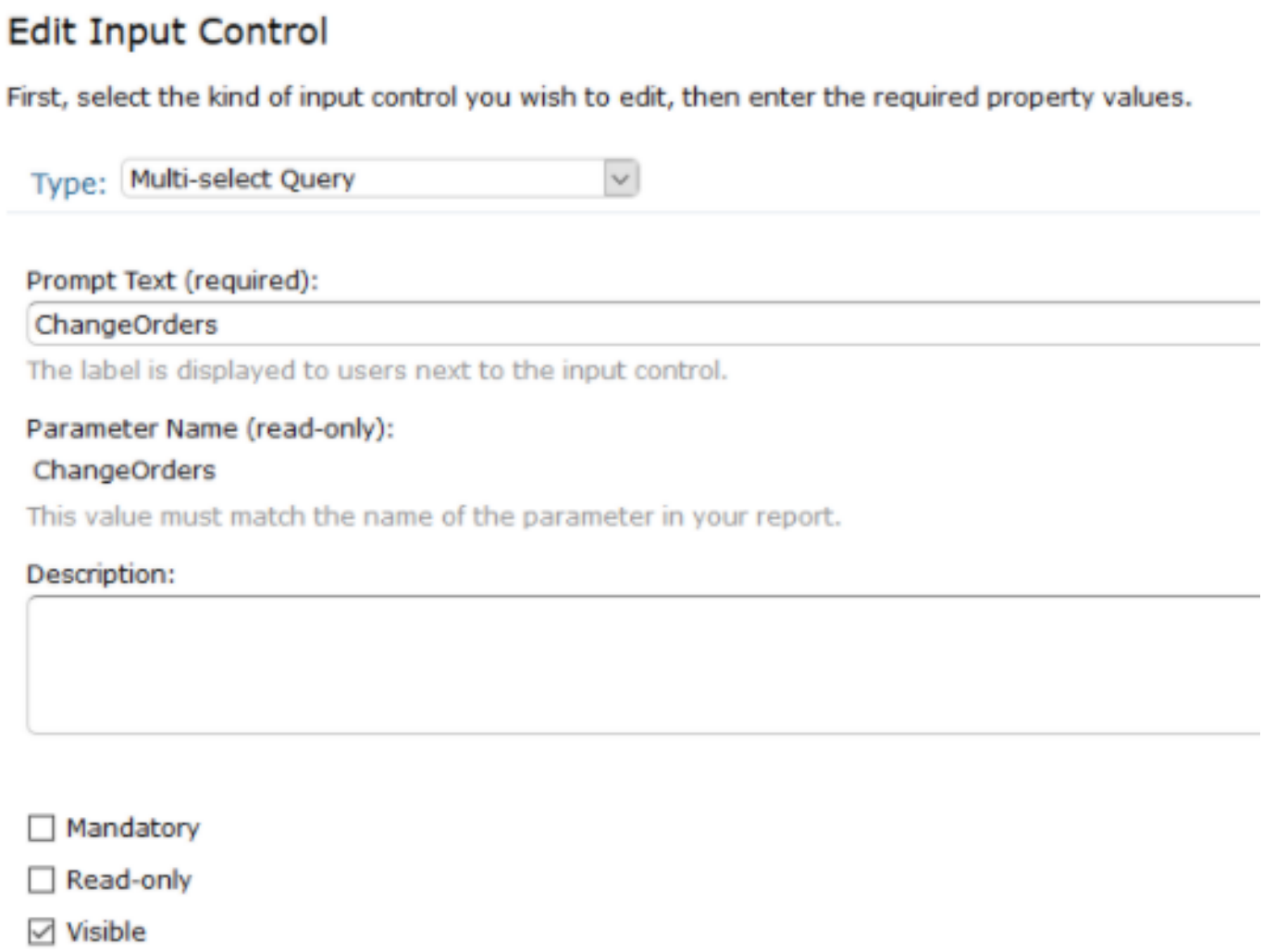
2. Select the data source



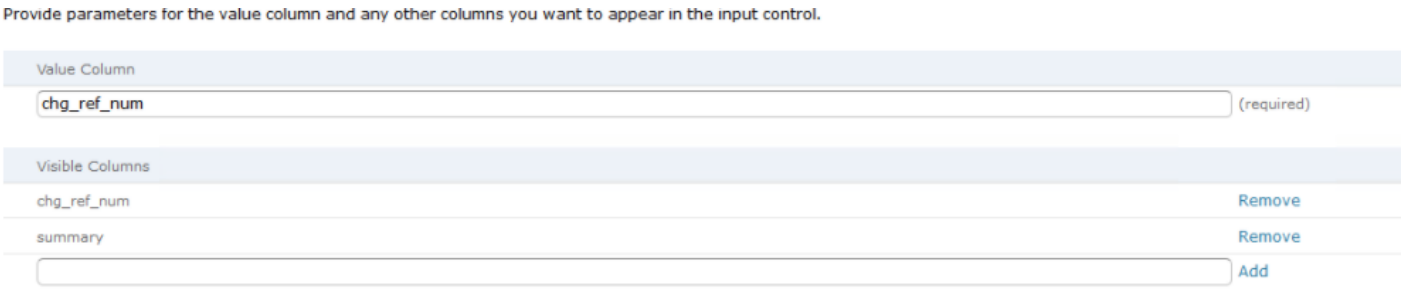
3. Define the query string with object names. For example: select chg_ref_num, summary FROM chg where active=1



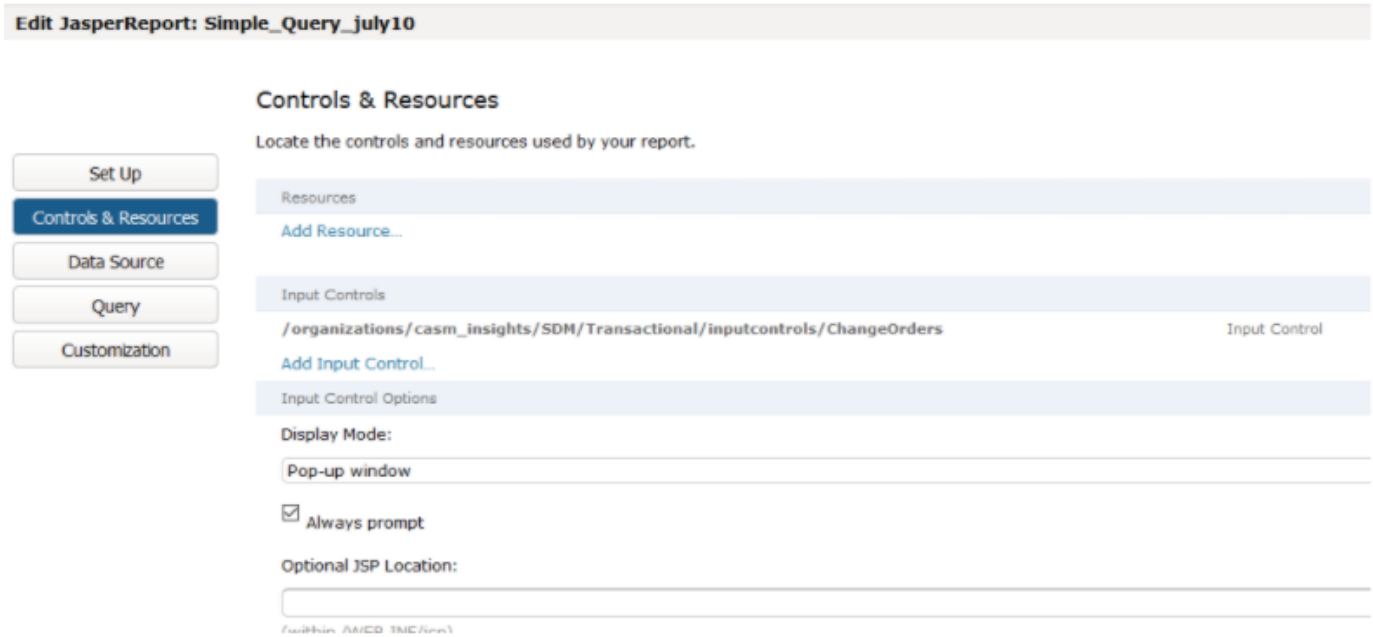
4. Right-click input controls, and select Add Resource, Input Control.
   a. Select the Type from the drop-down list. Example: Select Multi-select Query and select Visible.



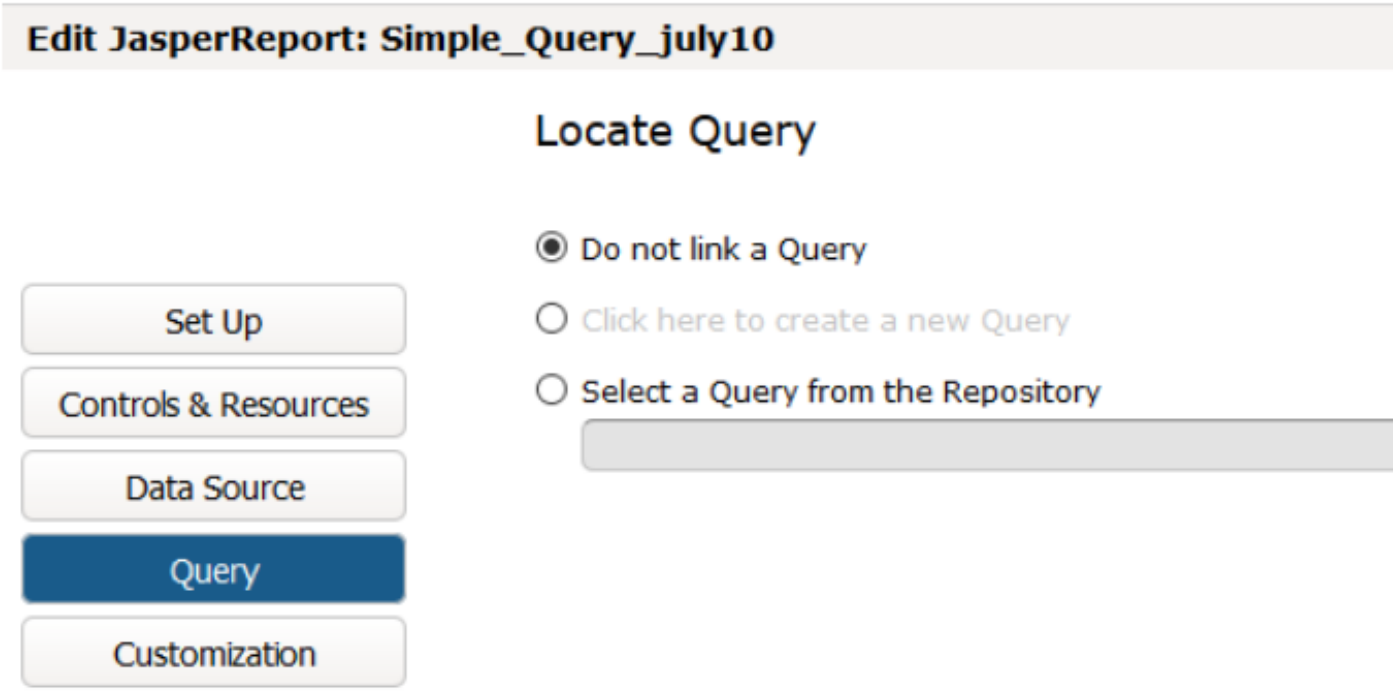
5. Select the query that you created in the first step:



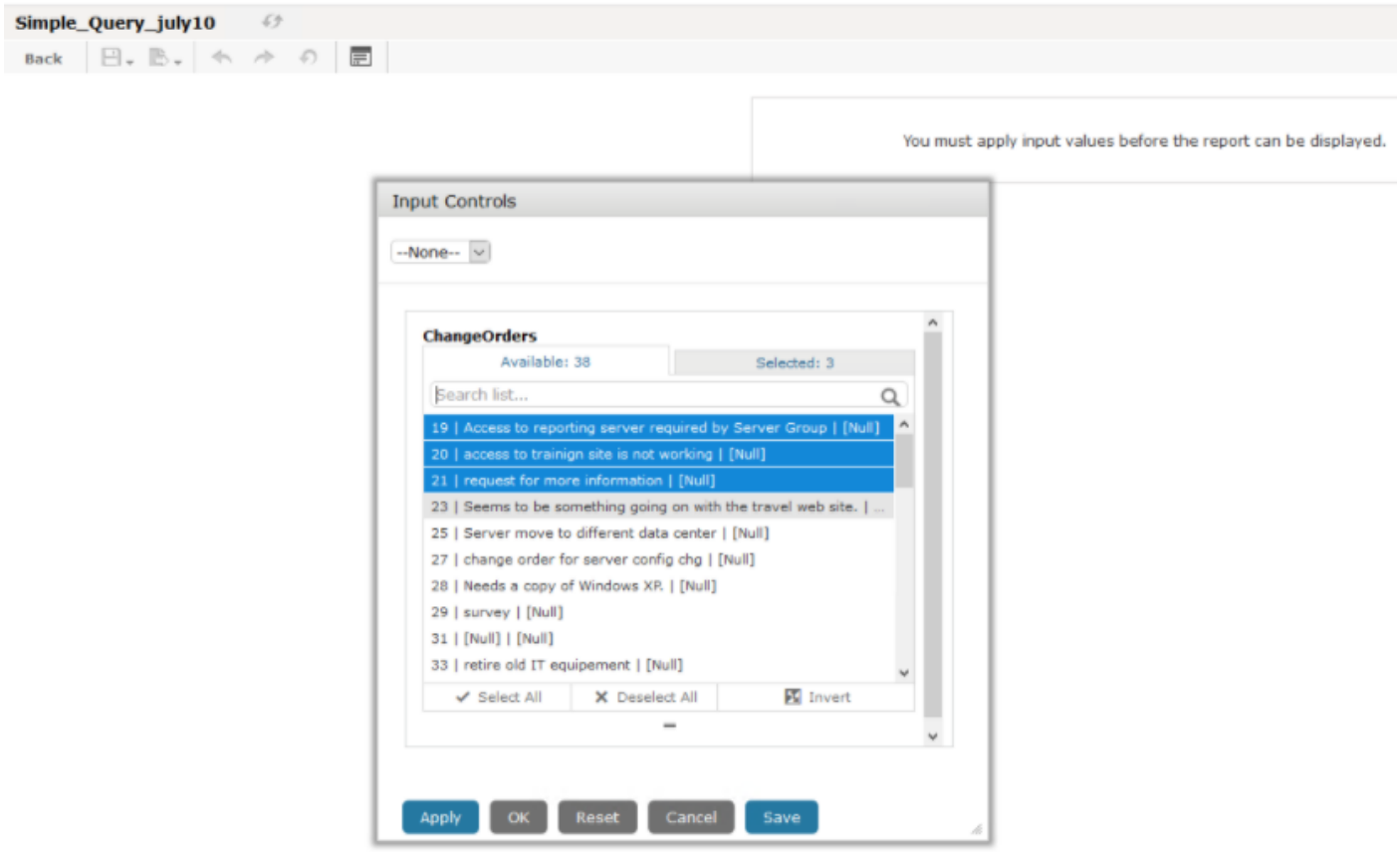
6. Select the column name that is queried, and create the columns that to display when the parameter is shown:



7. Browse to your report in the repository. Right-click to edit it.
8. Under Controls & Resources, browse to the Input Control you created in step 2.



9. Run your report to test the input control.

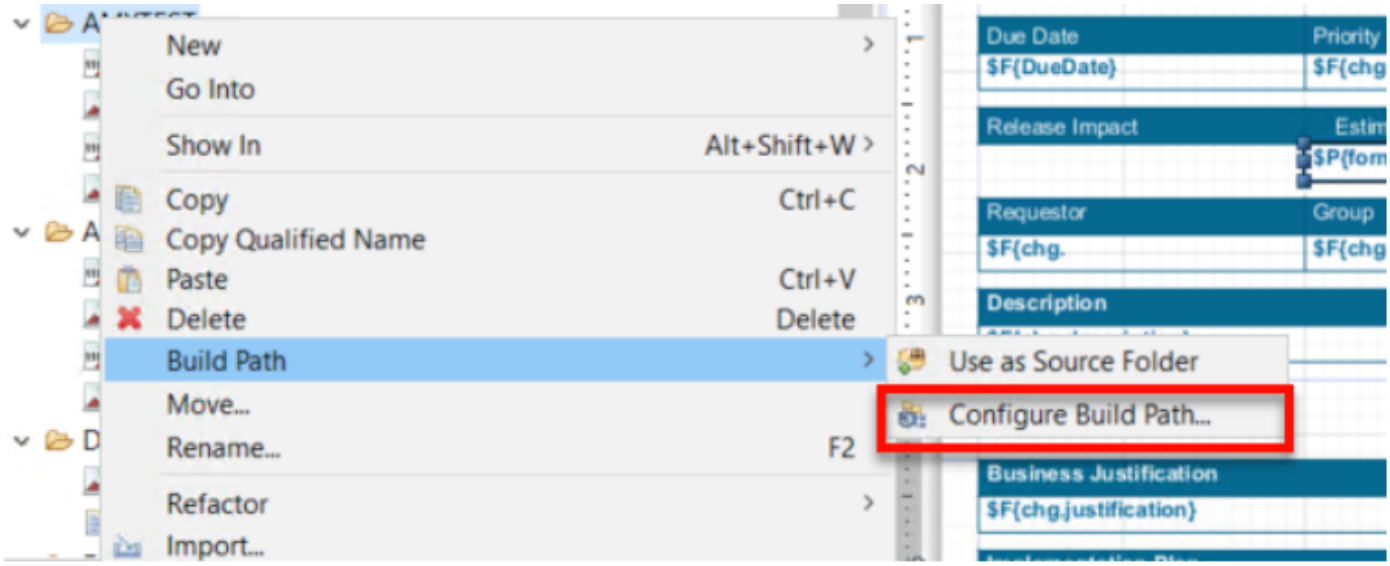


Add Duration Fields

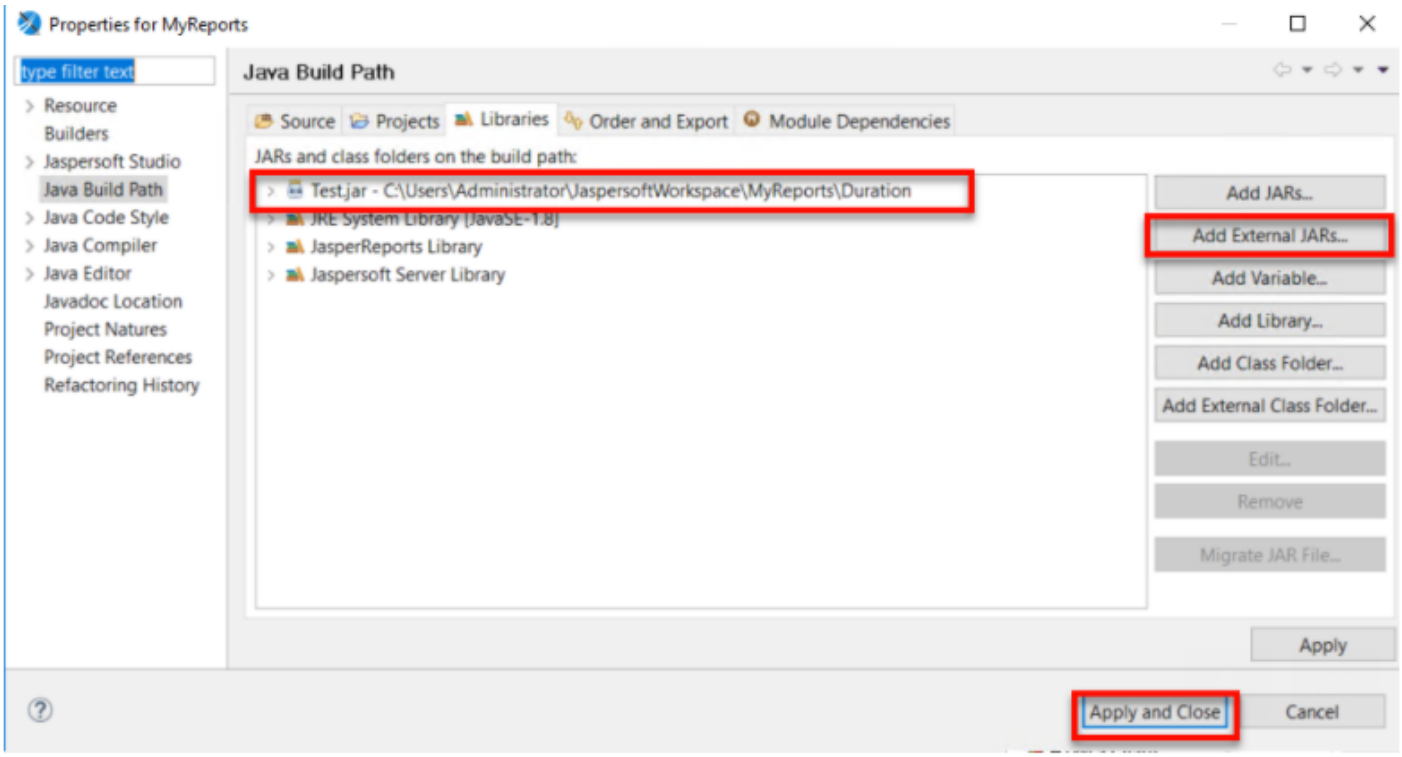
Add the duration fields to the build path of your project in the libraries. Then, import the class: com.duration.converter.

Perform the following steps:

1. Right-click a project folder, select **Build Path**, and **Configure Build Path**.



2. Copy the Jar from the DVD. CASM_17.4.2.0\filestore\Jasper\9.0.0\ExternalAuthJars and, copy it into the jasperserver tomcat lib folder. For eg: *C:\apache-tomcat-9.0.65\webapps\jasperserver-pro\WEB-INF\lib*.
3. Navigate to the **Libraries** tab, click **Add External JARs**, and browse for the above **DurationFormatter.jar** that you copied in Step 1.

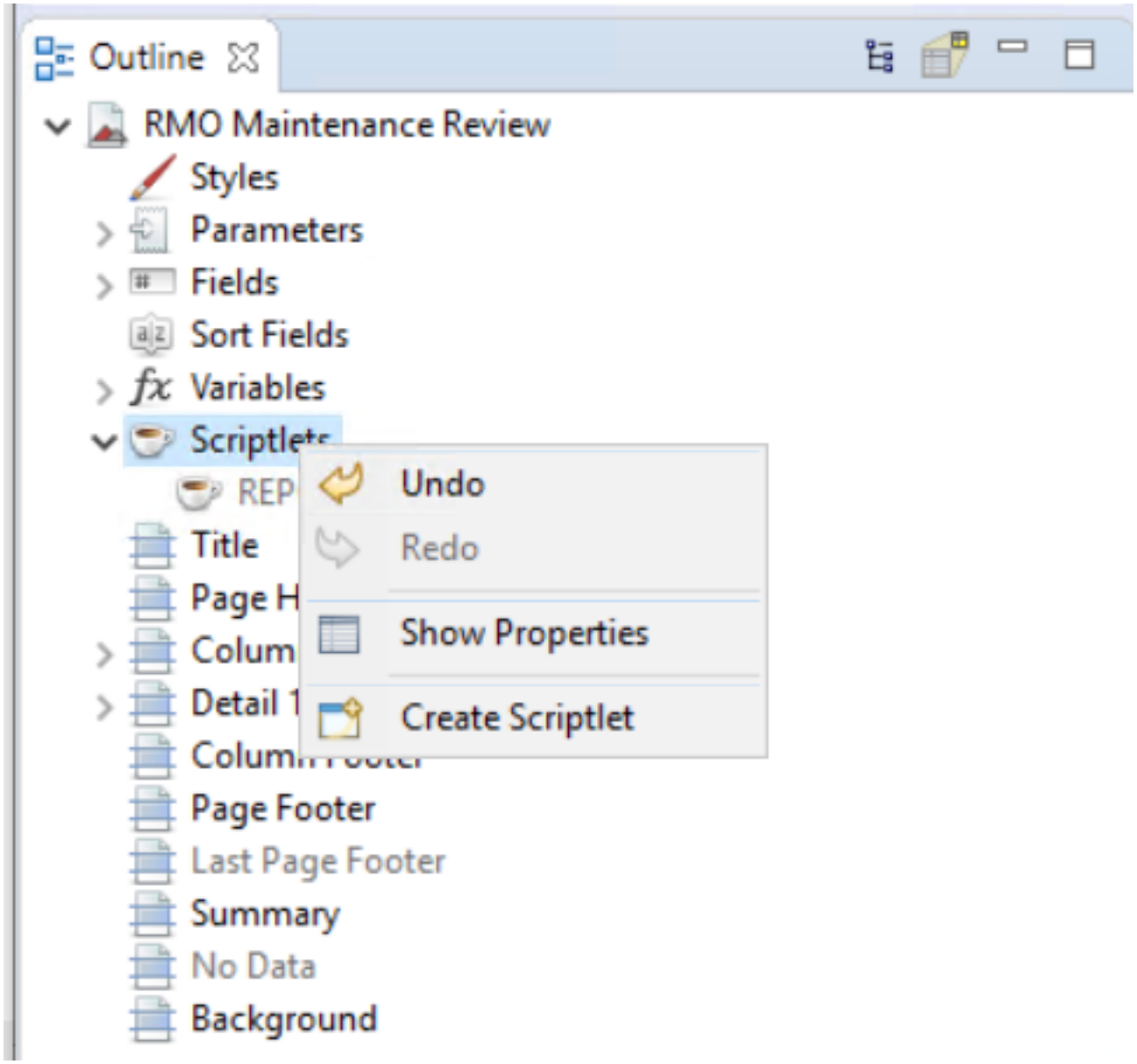


4. Click **Apply and Close**.

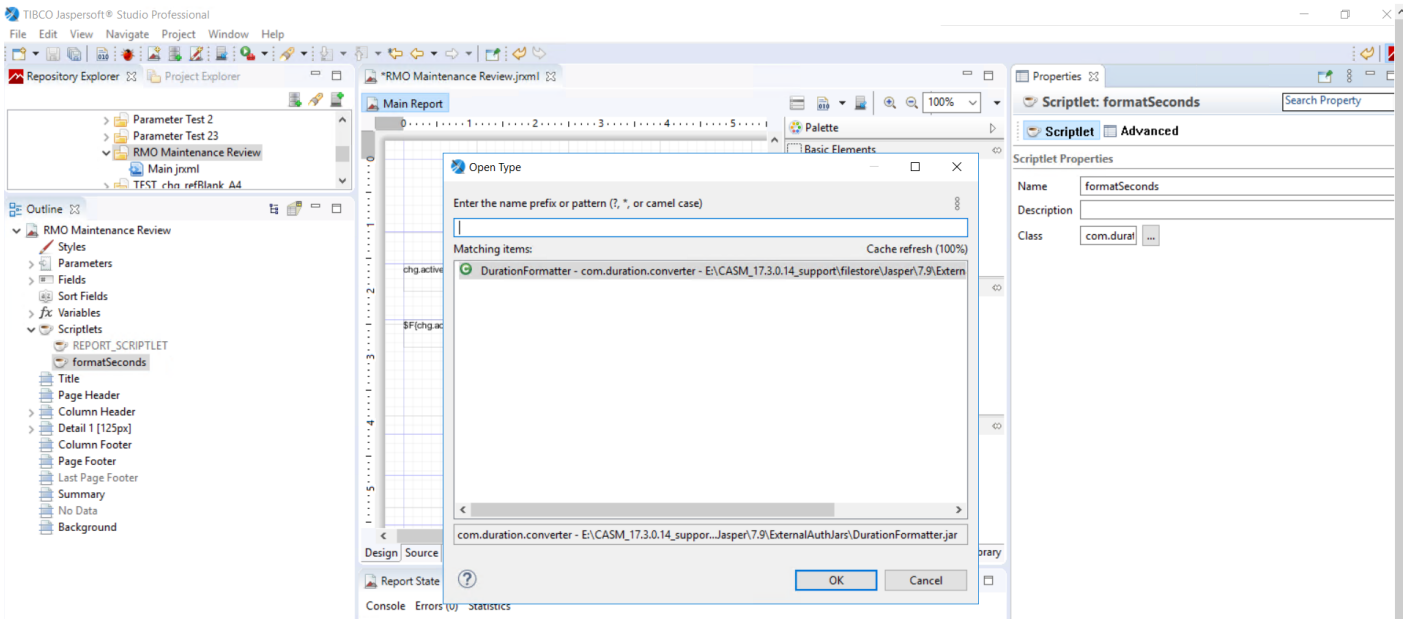
Add Parameter for Scriptlet

Perform the following steps:

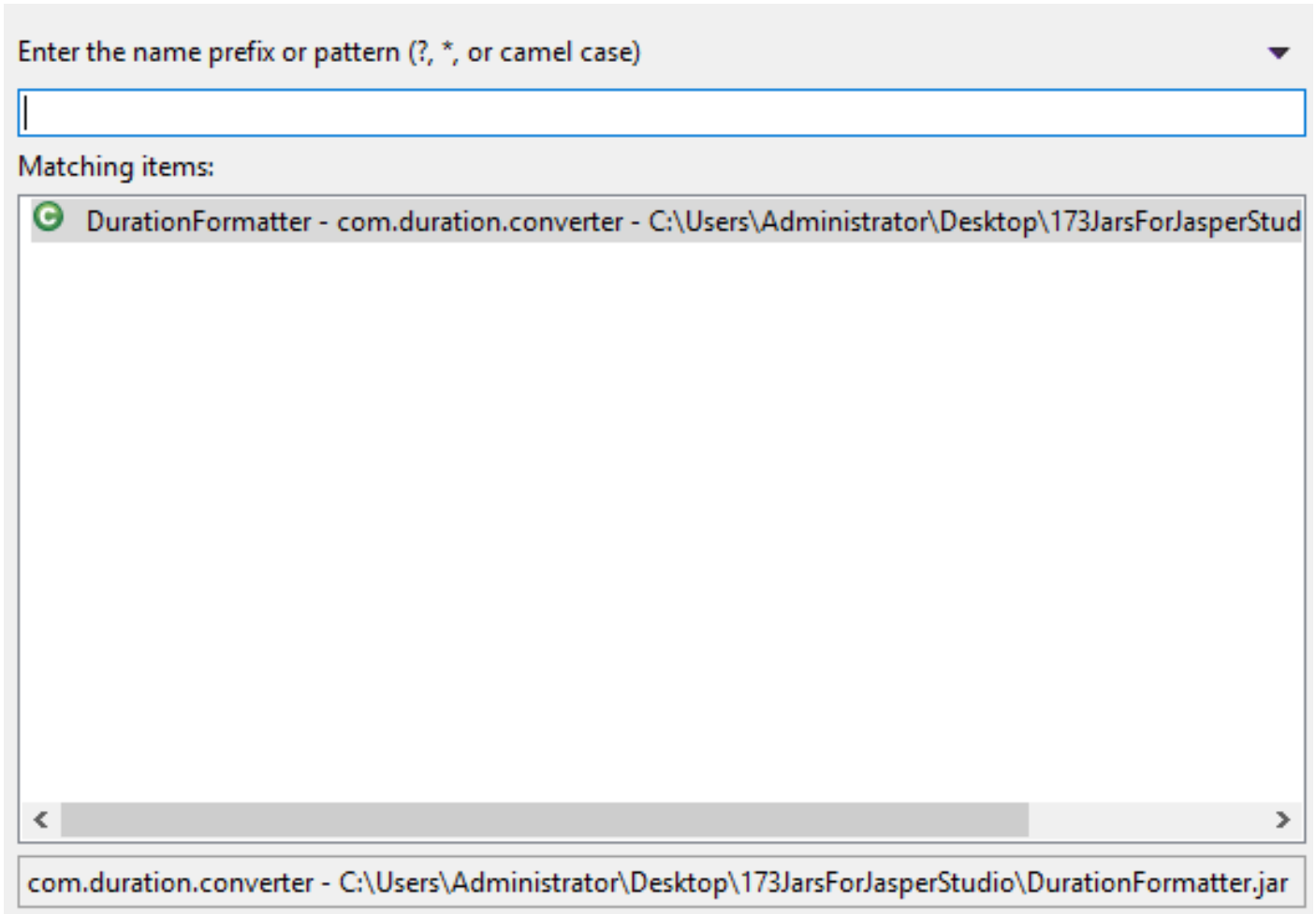
1. Right-click Scriptlets (under the report outline tab), and Click **Create Scriptlet** option.



2. Under the **Properties** tab, give the scriptlet a name: **formatSeconds**
3. Next to the Class field, select the ellipsis (...)



4. Select the **DurationFormatter**.

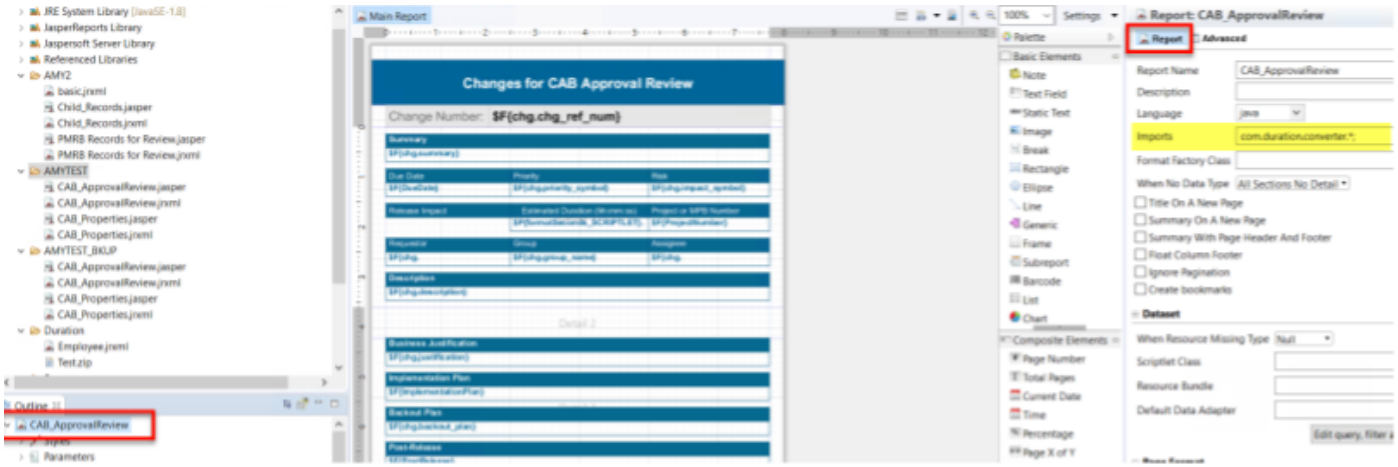


5. Click the Duration field in the report.

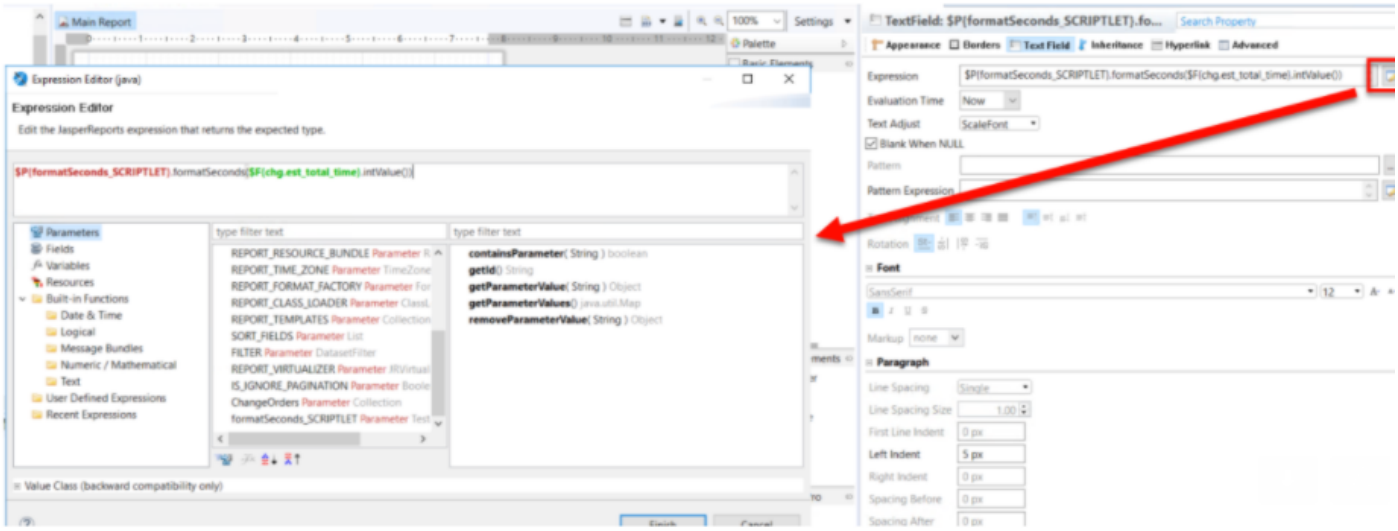
6. Click to add an expression for the Estimated Duration field:

   ```
   $P{formatSeconds_SCRIPTLET}.formatSeconds($F{chg.est_total_time}.intValue())
   ```

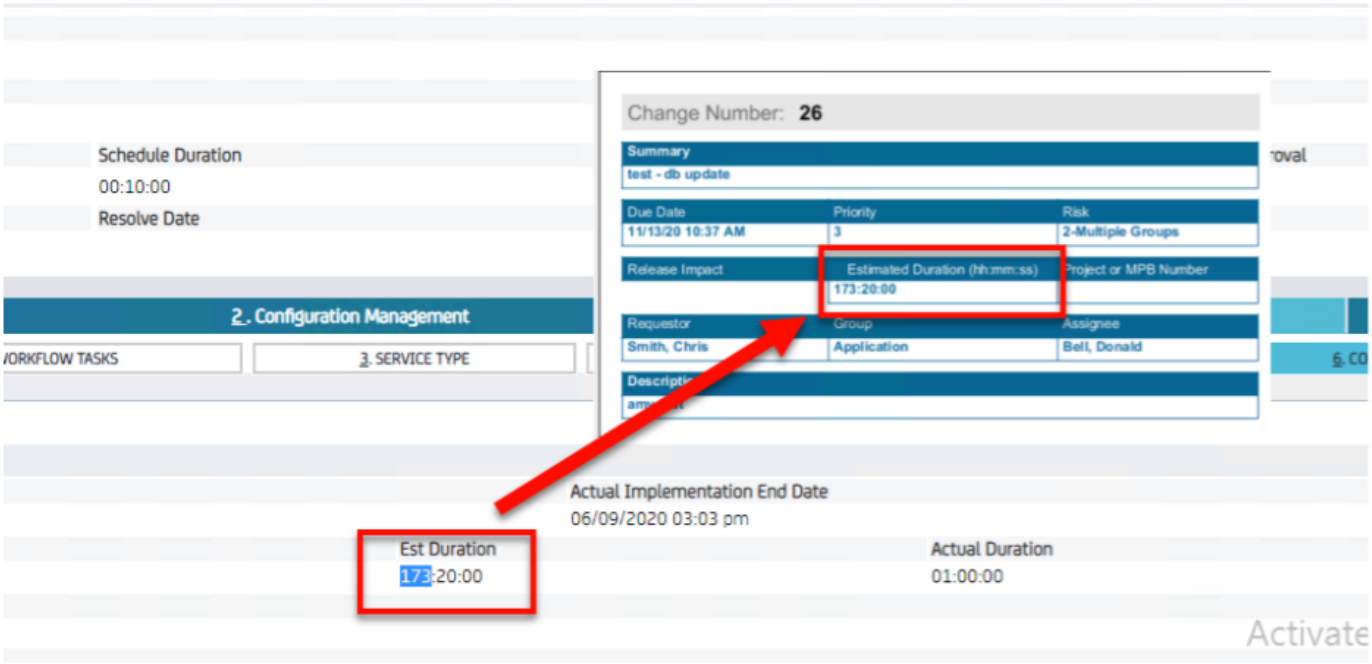
7. Click the report name under Outline Properties and click the dots next to Imports.
8. Select **Add Package**.
9. Select **com.duration.converter**.



10. Click **OK**.



11. Run the Report. It should match as shown in the following screenshot.



NOTE
If you are unable to run the report, ensure that the <import value> and <scriptlet name> parameters in the source tab of the report have the following values

```
<import value="com.duration.converter.*"/> <scriptlet name="formatSeconds"
 class="com.duration.converter.DurationFormatter"/>
```

12. Publish this report in the Jasper Server from the Jasper Studio.
13. Log in to the Jasper Server URL in which the report is published.
14. From the service console, stop the service: *jasperreportsTomcat.*
15. Copy the *DurationFormatter.jar* from CASM_17.4.2.0\filestore\Jasper\9.0.0\ExternalAuthJars and, copy it into the jasperserver tomcat lib folder. For eg: *C:\apache-tomcat-9.0.65\webapps\jasperserver-pro\WEB-INF\lib*.
16. Start the service: *jasperreportsTomcat.*
17. Run the report.
    The duration should match the report and SDM application as mentioned in Step 11.

SQL Queries with Strings

If you are using SQL queries with a string type parameter, ensure that the parameters are enclosed in single quotes. See the following example.

```
For example: select * from cr where cr.ref_num='$P{refNumParam}' Where refNumParam is a string.
```

When using FLOAT/ DOUBLE ATTRIBUTES in JasperStudio

If double type fields are in use, they convert to a String by default in JasperStudio. This string must be manually changed to double type.
For e.g. if field class is java.lang.String, then manually change it to java.lang.double

JasperReports Server

This topic provides complete information about the following releases of JasperReports Server.

- 8.1.1
- 9.0.0

JasperReports Server 9.0.0

- Tutorial for JasperReports Server 9.0.0
- Install JasperReports Server 9.0.0
- Upgrade JasperReports Server to 9.0.0
- Integrate JasperReports Server 9.0.0 with CA Service Management
- Configure JasperReports Server 9.0.0 for High Availability
- Configure JasperReports Server 9.0.0 to Use SSL

Tutorial for JasperReports Server 9.0.0

Learn about JasperReports Server, TIBCO Jaspersoft Online Learning Portal, training, documentation, and the Jaspersoft Community.

TIBCO Jaspersoft Online Learning Portal

Broadcom Inc., in partnership with TIBCO Software, Inc., has authorized its partners and licensed customers to have free access to the TIBCO Jaspersoft Online Learning Portal.

Recommended Jaspersoft Training - TIBCO Software, Inc

Jaspersoft training is available from TIBCO Software, Inc. The following recommended Jaspersoft training courses will help you in your Jaspersoft implementation.

Recommended Training for Analysts and Managers

The following online training and resources are recommended for end users such as Analysts and Managers:

Training Topic	Type	Description	Duration	Video	TIBCO Docs
Ad Hoc Views	Video	This video helps you learn how to create and configure Ad Hoc views using the Ad Hoc Editor.	8 Minutes	Ad Hoc Views Tutorial	Adhoc Views
Reports	Video	This video helps you learn how to create and format reports from existing Ad Hoc views.	9 Minutes	Ad Hoc Reports Tutorial	Reports
Dashboards	Video	This video helps you learn how to create dashboards and dashlets.	9 Minutes	Dashboards Tutorial	Dashboards

Recommended Training for Administrators.

All the training materials covered in online training resources in the following table are recommended for administrators such as SDM administrators:

Training Topic	Type	Description	Duration	Video	Registration Required?	TIBCO Docs
Domains	Video	This video helps you learn how to create domains.	8 Minutes	Domains Tutorial	No	NA
Jasper Reports User Guide	User Guide	Introduction to JasperReports Server and its capabilities.		NA	No	User Guide
JasperReports Administration Guide	Administrative Guide	Overview of JasperReports Server Administration.		NA	No	Admin Guide

Jaspersoft Community

Visit the Jaspersoft Community for related tutorials, tips and tricks, and other helpful information.

Install JasperReports Server 9.0.0

The installation of JasperReports 9.0.0 is supported on Microsoft SQL Server, Oracle, and PostgreSQL databases. The installation is available in English (United States) and is supported only on Windows Operating System.

We recommend using the WAR file distribution for installation or upgrade to JasperReports Server 9.0.0, as the WAR distribution includes customizations needed for SDM integration.

> **NOTE**
> Ensure the supported Java is installed, refer TIBCO Platform Support for JasperReports 9.0.0 Documentation.
> To use TIB_js-jrs_9.0.0_win_x86_64.exe to install JasperReports Server 9.0.0 (**for evaluation purpose only**) and to import an existing repository, refer the KB article.

Download a Vendor JDBC Driver

Pick the following database versions for installation:

Jasper Database	Name of jar to be copied to manually before installation	Sample Location
SQL	mssql-jdbc-12.4.1.jre8	..\js-jrs_9.0.0_bin\jasperreports-server-pro-9.0.0-b server\jdbc
Oracle	Rename ojdbc8.jar with ojdbc8-23.2.0.0.jar	..\js-jrs_9.0.0_bin\jasperreports-server-pro-9.0.0-b cle\jdbc
PostgreSQL	Not required	TIBCO provides a driver by default.

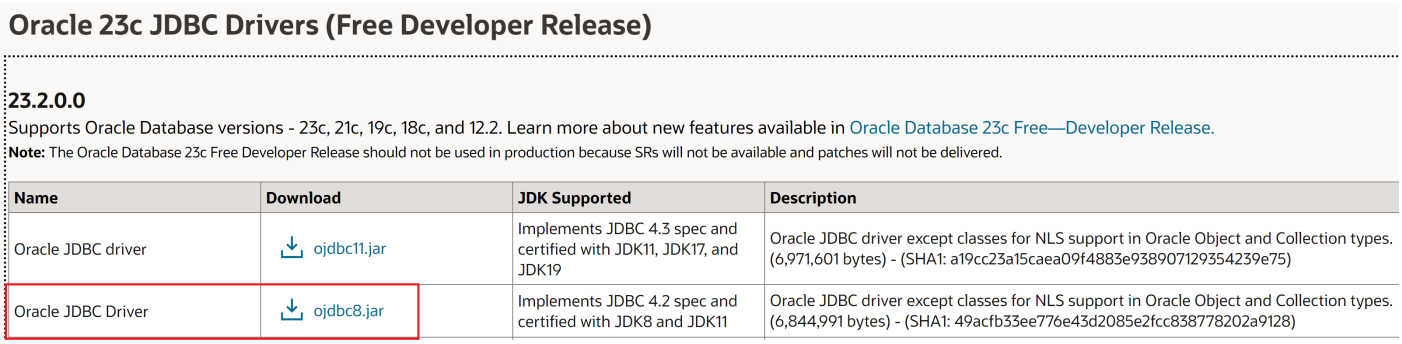
For SQL Installation

1. Go to the Microsoft SQL page search and extract the file: "*Download Microsoft JDBC Driver 12.4.1 for SQL Server (zip)".*

2. Once you unzip the downloaded file, go to the extracted zip file location **>sqljdbc_12.4.1.0_enu> sqljdbc_12.4> enu> jars** and look for the *mssql-jdbc-12.4.1.jre8* file.

For Oracle Installation

1. Go to the Oracle page, and look for "Oracle 23c JDBC Drivers (Free Developer Release)" . Expand the **23.2.0.0** section and download the **ojdbc8.jar.**



Oracle 23c JDBC Drivers (Free Developer Release)

23.2.0.0

Supports Oracle Database versions - 23c, 21c, 19c, 18c, and 12.2. Learn more about new features available in Oracle Database 23c Free—Developer Release.

Note: The Oracle Database 23c Free Developer Release should not be used in production because SRs will not be available and patches will not be delivered.

Name	Download	JDK Supported	Description
Oracle JDBC driver	ojdbc11.jar	Implements JDBC 4.3 spec and certified with JDK11, JDK17, and JDK19	Oracle JDBC driver except classes for NLS support in Oracle Object and Collection types. (6,971,601 bytes) - (SHA1: a19cc23a15caea09f4883e938907129354239e75)
Oracle JDBC Driver	ojdbc8.jar	Implements JDBC 4.2 spec and certified with JDK8 and JDK11	Oracle JDBC driver except classes for NLS support in Oracle Object and Collection types. (6,844,991 bytes) - (SHA1: 49acfb33ee776e43d2085e2fcc838778202a9128)

2. Once the jar is downloaded, rename **ojdbc8.jar** as ***ojdbc8-23.2.0.0.jar***.

For more information, refer "Download a Vendor JDBC Driver" section in TIBCO documentation.

Install Jaspersoft Server 9.0.0 on PostgreSQL, Microsoft SQL Server, or Oracle database using WAR distribution

Perform the following steps to Install JasperReports Server 9.0.0 on PostgreSQL, Microsoft SQL Server or Oracle database.

> **NOTE**
> Before installing, review the TIBCO Platform Support for JasperReports 9.0.0 Documentation, for the supported versions of Apache Tomcat/ Java/ Databases/ Operating systems.

1. From Broadcom support portal, download the JasperReports Server 9.0.0 installer file, search with the name "**CA Business Intelligence JasperReports Server 9.0.0 for CA Service Management.iso**".
2. Mount the JasperReports Server 9.0.0 installer after downloading it to your environment and, ensure you unzip the installer contents in your local system.
3. Now Download a Vendor JDBC Driver for JasperServer.
4. For installation instructions, refer the "*Installing the WAR File for Production"* section in TIBCO documentation.
5. Configure Tomcat as an installed service.

   > **NOTE**
   > The paths shown in the below procedure are given as examples only. The paths may differ based on your environment and installation requirements.

   a. Install a supported Java version. For more information about a supported Java version, see TIBCO Platform Support for JasperReports 9.0.0 Documentation.
   b. Set CATALINA_HOME to the tomcat folder.

c. Set JAVA_HOME to the java folder.

d. From `CATALINA_HOME\bin` , execute the following command to install JasperReports Service.

```
tomcat9.exe //IS//jasperreportsTomcat --DisplayName="Apache Tomcat 9.0 jasperreportsTomcat"
 --Install="C:\JRS9_0_0\tomcat9570\bin\tomcat9.exe" --Jvm=auto --StartMode=jvm --
StopMode=jvm --StartClass=org.apache.catalina.startup.Bootstrap --StartParams=start --
StopClass=org.apache.catalina.startup.Bootstrap --StopParams=stop
```

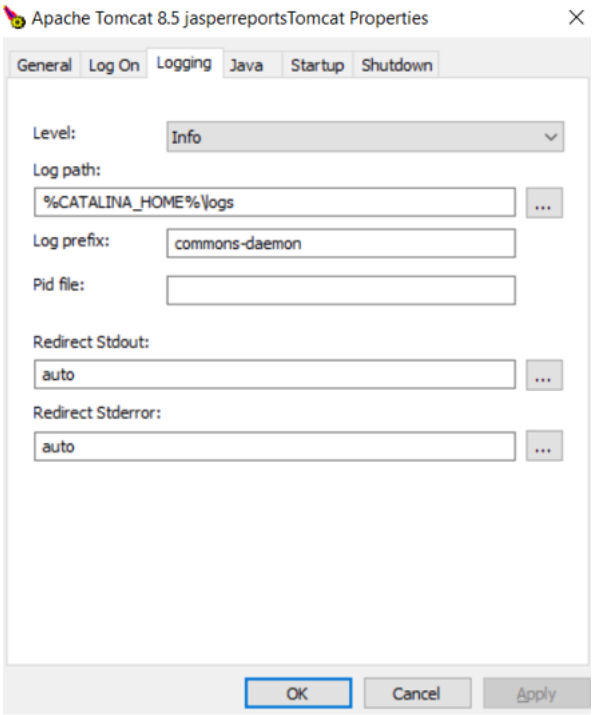
NOTE

- Use the service name **jasperreportsTomcat,** identified by the common installer.
- See the Troubleshooting CA Service Management section, if the following error message is displayed: Failed to grant service user 'NT AUTHORITY\LocalService' write permissions to log path 'C:\Windows \system32\LogFiles\Apache' due to error '19: The media is write protected.'

e. From `CATALINA_HOME\bin` , execute the following command to configure JasperReports Service.

```
tomcat9w.exe //ES//jasperreportsTomcat
```

f. On the UI Window, select the **Logging** tab and set the below settings:



g. On the Java tab, set the below options:

```
JVM------
%JAVA_HOME%\jre\bin\server\jvm.dll
Java Classpath---------
%JAVA_HOME%\lib\tools.jar;%CATALINA_HOME%\bin\bootstrap.jar;%CATALINA_HOME%\bin\tomcat-juli.jar

Java Options-----------
-Djs.license.directory=C:\Users\Administrator
-Djs.license.directory=C:\Users\Administrator
-Dcatalina.home=C:\JRS9_0_0\tomcat9570
-Dcatalina.base=C:\JRS9_0_0\tomcat9570
-Djava.io.tmpdir=C:\JRS9_0_0\tomcat9570\temp
-Djs.license.directory=C:\Users\Administrator
-Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager
```

```
-Djava.util.logging.config.file=C:\JRS9_0_0\tomcat9570\conf\logging.properties


Java  9 options-----------
--add-opens=java.base/java.lang=ALL-UNNAMED
--add-opens=java.base/java.io=ALL-UNNAMED
--add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED
-Xdebug
-Xrunjdwp:transport=dt_socket,address=9999,server=y,suspend=n
```

h. Click **Apply** and **OK**.
i. Start Apache Tomcat 9.0 jasperreportsTomcat services from Windows services console.

6. Launch JasperReports Server 9.0.0.
7. If an invalid license error message pops up, follow the steps from 8-11, mentioned below.
8. Stop Tomcat 9.0 jasperreportsTomcat services.
9. Navigate to `CASM_DVD\filestore\Jasper\9.0.0\License` folder and copy the jasperserver.license file.
10. Navigate to the path displayed in the error message and paste the copied file.
11. Start Tomcat 9.0 jasperreportsTomcat services.

Next Step:

- Integrate JasperReports Server 9.0.0 with CA Service Management

Upgrade JasperReports Server to 9.0.0

Review the supported upgrade paths for JasperReports Server and upgrade by following the recommended upgrade procedure. You can upgrade directly to JasperReports Server 9.0.0 from JasperReports Server 8.1.1.

Download a Vendor JDBC Driver
This section provides instructions to upgrade from JasperReports Server 8.1.1 to JasperReports Server 9.0.0.

Download a Vendor JDBC Driver

Pick the following database versions for installation:

Jasper Database	Name of jar to be copied to manually before upgrade	Sample Location
SQL	mssql-jdbc-12.4.1.jre8	..\js-jrs_9.0.0_bin\jasperreports-server-pro-9.0.0-b server\jdbc
Oracle	Rename ojdbc8.jar with ojdbc8-23.2.0.0.jar	..\js-jrs_9.0.0_bin\jasperreports-server-pro-9.0.0-b cle\jdbc
PostgreSQL	Not required	TIBCO provides a driver by default.

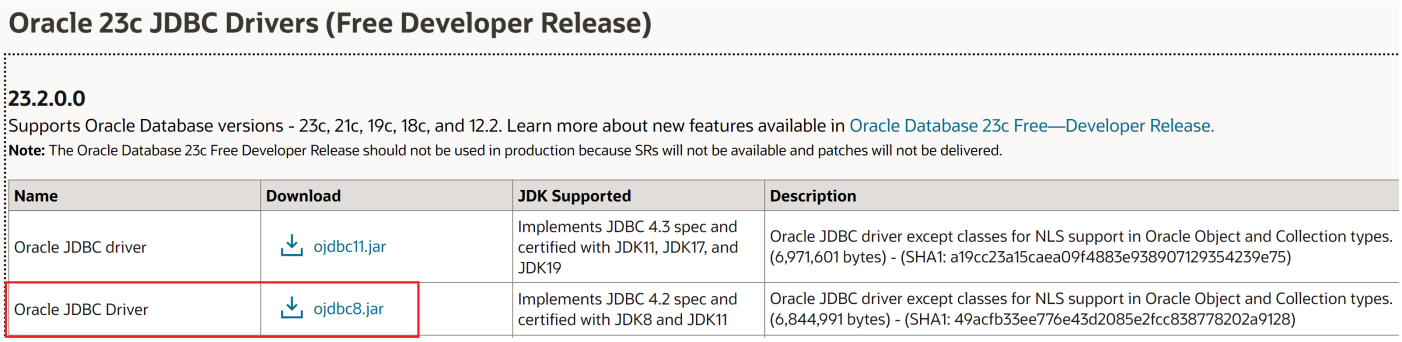
For SQL Installation

1. Go to SQL page search, extract the file: "Download Microsoft JDBC Driver 12.4.1 for SQL Server (zip)".

2. Once you unzip the downloaded file, go to the extracted zip file location **>sqljdbc_12.4.1.0_enu> sqljdbc_12.4> enu> jars** and look for the *mssql-jdbc-12.4.1.jre8* file.

For Oracle Installation

1. Go to the Oracle page, and look for "Oracle 23c JDBC Drivers (Free Developer Release)" . Expand the **23.2.0.0** section and download the **ojdbc8.jar.**



Oracle 23c JDBC Drivers (Free Developer Release)

23.2.0.0

Supports Oracle Database versions - 23c, 21c, 19c, 18c, and 12.2. Learn more about new features available in Oracle Database 23c Free—Developer Release.

Note: The Oracle Database 23c Free Developer Release should not be used in production because SRs will not be available and patches will not be delivered.

Name	Download	JDK Supported	Description
Oracle JDBC driver	ojdbc11.jar	Implements JDBC 4.3 spec and certified with JDK11, JDK17, and JDK19	Oracle JDBC driver except classes for NLS support in Oracle Object and Collection types. (6,971,601 bytes) - (SHA1: a19cc23a15caea09f4883e938907129354239e75)
Oracle JDBC Driver	ojdbc8.jar	Implements JDBC 4.2 spec and certified with JDK8 and JDK11	Oracle JDBC driver except classes for NLS support in Oracle Object and Collection types. (6,844,991 bytes) - (SHA1: 49acfb33ee776e43d2085e2fcc838778202a9128)

2. Once the jar is downloaded, rename **ojdbc8.jar** as ***ojdbc8-23.2.0.0.jar***.

For more information, refer "Download a Vendor JDBC Driver" section in TIBCO documentation.

Upgrade JasperReports Server 8.1.1 to JasperReports Server 9.0.0

This section provides instructions to upgrade from JasperReports Server 8.1.1 to JasperReports Server 9.0.0.
Follow these steps:

1. From Broadcom support portal , download the JasperReports Server 9.0.0 installer file, search with the name **"CA Business Intelligence JasperReports Server 9.0.0 for CA Service Management.iso".**
2. Ensure you Download a Vendor JDBC Driver.
3. Upgrade to JasperReports Server 9.0.0 by performing the tasks listed in the *Upgrading from 8.0.x - 8.1.x to 9.0* section in the TIBCO JasperReports Server 9.0.0 upgrade guide.

   NOTE
   While performing the Jasperreports Server upgrade, if you get an error CATALINA_HOME and CATALINA_BASE are both NOT set, then edit the default_master.properties file for your database; set the CATALINA_HOME and CATALINA_BASE values to Tomcat app server root directory and save the file. Run the Upgrade command again.
4. Database driver prerequisite for JasperReports Integration.

   NOTE
   For this section, use the drivers you have downloaded previously. For more information refer the Download a Vendor JDBC Driver.

Jasper Database	SDM Database	Name of Jar to be copied into jasperserver webapps lib folder before integration	
PostgreSQL	SQL	\jaspertomcatdir\webapps\jasperserver-pro\WEB-INF\lib\mssql-jdbc-12.4.1.jre8.jar\mssql-jdbc-12.4.1.jre8.jar	Jasp
PostgreSQL	Oracle	Not required	
SQL	SQL	Manual copy is not required, Jasper installation on SQL would place the driver.	
SQL	Oracle	Not required	

Jasper Database	SDM Database	Name of Jar to be copied into jasperserver webapps lib folder before integration	
Oracle	SQL	\jaspertomcatdir\webapps\jasperserver-pro\WEB-INF\lib\mssql-jdbc-12.4.1.jre8.jar\mssql-jdbc-12.4.1.jre8.jar	Jasp
Oracle	Oracle	Not required	

NOTE
If Jasperserver service is not restarted here, the integration of Jasperserver with CA SDM will fail.

5. To **Integrate JasperReports Server 9.0.0 with CA Service Management**, launch the CA Service Management common installer on the server where you installed JasperReports Server 9.0.0.
   a. On the **Installer selection** window, select **Integrate CABI JasperReports Server with CA Service Management**.
   b. On the **License Agreement** window, review the license information, accept the agreement, and click **Next**.
   c. On the **Database Configuration** window, in the **Database Type** field, select a database.
   d. Enter the details for CA Service Desk Manager (CA SDM) and click **Next**

Field	Description
Database Server	The location where you've installed the CA SDM database server.
Database Name	The database name on the CA SDM database name.
Database Port	The port number that CA SDM uses.
Database Server Instance	The instance of the database server.
Database Admin User	The database admin user name with permissions to create users and schema. The default user for Microsoft SQL Server.
Database Admin Password	The password to authenticate the database administrator user.
mdbadmin Password	The password for the CA SDM database user.
Confirm mdbadmin Password	Reconfirm the password for the CA SDM database use

   e. On the **CABI JasperReports Server Configuration** window, enter the details for configuring JasperReports Server 9.0.0, and click **Next.**

Field	Description
Server Name	The name of the server where JasperReports Server 9.0.0 is installed.
Organization	The organization name created in JasperReports Server 9.0.0
Port	The port number that JasperReports Server 9.0.0 uses.
Insights Admin User Name	The user name with complete access to all organizations and tenants.
Admin User	The default admin user is superuser.
Insights Admin Password	The admin password with complete access to all organizations and tenants.
Admin Password	The password to authenticate the database administrator user.

Field	Description
Confirm Insights Admin Password	Reconfirm the admin password with complete access to all organizations and tenants.
Server Protocol	The server protocol used during JasperReports Server 9.0.0 installation.
Webapp Directory	The location within the tomcat folder where JasperReports Server 9.0.0 is deployed. Example, <JasperReports Server 9.0.0 installed directory>\apache-tomcat\webapps\jasperserver-pr

f. (Optional) To enable aggregate reports, select the **Aggregate Reports** checkbox.
On the **Aggregate Reports** window, verify the datamart database server details, and click **Next.**

g. (Optional) To enable aggregate reports, select the **Aggregate Reports** checkbox.
On the **Aggregate Reports** window, verify the datamart database server details, and click **Next.**

h. (Optional) To enable transactional reports, select the **Transactional Reports** checkbox.
On the **Transactional Reports** window, verify the transactional database server details, and click **Next.**

i. In the Jasper Database Configuration window, select the required database against Jasper Database Type to automatically increase the password field length of the jijdbcdatasource table on jasperserverdb. If your JasperReports server is deployed on any of the following databases, perform the following steps:
   a. Postgre SQL Database
   b. Microsoft SQL Server Database
   c. Oracle Database

NOTE

- Ensure that you have the required database client installed based on the Jasper database server type.
- The Jasper Database Admin User is the privileged user who can create and update the Jasper database entries.

1. **PostgreSQL Database**

Prerequisites

For successful integration, verify the following.

- Open the pg_hba.conf file under the PostgreSQL directory.
  <jasper-report-server-path>\postgresql\data
  Under # IPv4 local connections, add the following new entry:
  host all all <—IP address-->/32 md5
  IP address of the installed Jasper Server.
- Open the postgresql.conf file under the PostgreSQL directory.
  <jasper-report-server-path>\postgresql\data
  Update the listen_addresses entry from 127.0.0.1 to * as follows and save.
  Example: listen_addresses = '*'

Restart the PostgreSQL database service.

a. Complete the following mandatory fields:
   - Jasper Database Hostname
     Specifies the Jasper Database Hostname.
   - Jasper Database Name
     Specifies the name of the Jasper Database.
     Default: jasperserver
   - Jasper Database Admin User

Specifies the Jasper Database Admin User name.
Default: postgres
- Jasper Database Admin Password
Specifies the Jasper Database Admin Password.
Note: On the PosrgreSQL database page the password is pre populated.
- Jasper Database Port
Specifies the Jasper Database Port number.
Default: 5432
- PostgreSQL Home Path
Specifies the Jasper directory where the Postgres SQL database is installed.

b. Click Next to proceed.

Microsoft SQL Server Database

a. Complete the following mandatory fields:
- Jasper Database Hostname
Specifies the Jasper Database Hostname.
- Jasper Database Name
Specifies the name of the Jasper Database.
Default: jasperserver
- Jasper Database Port
Specifies the Jasper Database Port number.
Default: 1433
- Jasper Database Admin User
Specifies the Jasper Database Admin User name.
Default: sa
- Jasper Database Admin Password
Specifies the Jasper Database Admin Password.

b. Complete the following optional field:
- Jasper Database Server Instance
Specifies the Jasper Database Server Instance.

c. Click Next to proceed.

Oracle Database

a. Complete the following mandatory fields:
- Jasper Database Hostname
Specifies the Jasper Database Hostname.
- Jasper Database Port
Specifies the Jasper Database Port number.
Default: 1521
- Jasper Database Net Service Name
Specifies the Jasper Database Net Service Name.
- Jasper Database Admin User
Specifies the Jasper Database Admin User name.
Default: system
- Jasper Database Admin Password
Specifies the Jasper Database Admin Password
- Jasper Database - SID
Specifies the Jasper Database - SID.

b. Click Next to proceed.

2. Review the information on the **Configuration** window, and click **Next.**

3. Click **Install** to start the integration process.

What's next After integrating JasperReports Server 9.0.0 with CA Service Management, complete the following tasks:

- Post-installation tasks.
- Additional tasks.

Integrate JasperReports Server 9.0.0 with CA Service Management

This article provides an overview of the integration between JasperReports Server 9.0.0 and CA Service Management.

Overview

JasperReports Server is a stand-alone and embeddable reporting software that CA Service Management uses for reporting and analytics purposes. CA Service Management integration with JasperReports Server provides enterprise IT management critical information on a real-time or scheduled basis. The integration offers pre-configured reports, dashboard views, and the ability to create sophisticated layouts containing charts, images, subreports, crosstabs, and more that matter for your business objectives.

CA Service Management integration with JasperReports Server provides the following benefits:

- Visualize data quickly using pre-configured reports and interactive dashboard views.
- Generate reports, and customize them to your needs by going one step further than regular table-based reports: Drag-and-drop report designer supporting interactive tables, charts, crosstabs, and analytic views.
- Drill down into reports and analyze them to enable data-driven decision-making.
- Ability to modify underlying data cube on the fly as part of analysis capabilities.

JasperReports Server 9.0.0 installation is a distinct and separate activity within the overall CA Service Management installation process.

> **NOTE**
> If JasperReports Server 9.0.0 DVD is installed, then ***Allow.jar*** file is visible in the UI.
>
> To eliminate the CVE-2024-10271, apply the hotfix T5UG571.caz with APAR No : 99112250.

You can integrate the JasperReports Server 9.0.0 with CA Service Management using the following information:

Integrate 17.4 RU2 and Later

The following sections help you integrate the JasperReports Server 9.0.0 with CA Service Management 17.4 RU2 and later.

Prerequisites

For successful integration, verify the following.

- Review the JasperReports Server 9.0.0 supportability matrix from TIBCO Software.
- Ensure that JasperReports Server 9.0.0 is installed.

Enable authentication files for communication

Enable authentication files for communication between JasperReports Server 9.0.0 and CA Service Desk Manager (CA SDM).

Follow these steps:

1. Add contactUUID for superuser.
   a. Log in to JasperReports Server 9.0.0 as a superuser.

b. Navigate to **Manage** > **Users**.
c. From the left navigation, select **root**.
d. Select the **superuser**.
e. From the **Properties** pane, click **Edit**.
f. From the **Attributes** tab, click **Add new attribute**.
g. In the Name field, enter `contactUUID`.
h. Click **OK** and **Save**.

NOTE
For this section, use the drivers you have downloaded previously. For more information refer the Download a Vendor JDBC Driver.

Database driver prerequisite for JasperReports Integration

Jasper Database	SDM Database	Name of Jar to be copied into jasperserver webapps lib folder before integration	Notes
PostgreSQL	SQL	\jaspertomcatdir\webapps\jasperserver-pro\WEB-INF\lib\mssql-jdbc-12.4.1.jre8.jar	Jasper server service must be restarted.
PostgreSQL	Oracle	Not required	
SQL	SQL	manual copy is not required, Jasper installation on SQL would place the driver.	
SQL	Oracle	Not required	
Oracle	SQL	\jaspertomcatdir\webapps\jasperserver-pro\WEB-INF\lib\mssql-jdbc-12.4.1.jre8.jar\mssql-jdbc-12.4.1.jre8.jar	Jasper server service must be restarted.
Oracle	Oracle	Not required	

NOTE
If Jasperserver service is not restarted here, the integration of Jasperserver with CA SDM fails.

Integrate using the common installer

To enable integration between JasperReports Server 9.0.0 with CA Service Management, launch the CA Service Management common installer on the server where you installed JasperReports Server 9.0.0.

Follow these steps:

1. On the **Installer selection** window, select **Integrate CABI JasperReports Server with CA Service Management**.
2. On the **License Agreement** window, review the license information, accept the agreement, and click **Next**.
3. On the **Database Configuration** window, in the **Database Type** field, select a database.
4. Enter the details for CA Service Desk Manager (CA SDM) and click **Next**.

Field	Description
Database Server	The location where you've installed the CA SDM database server.
Database Name	The database name on the CA SDM database name.

Field	Description
Database Port	The port number that CA SDM uses.
Database Server Instance	The instance of the database server.
Database Admin User	The database admin user name with permissions to create users and schema. The default user for Microsoft SQL Server is sa.
Database Admin Password	The password to authenticate the database administrator user.
mdbadmin Password	The password for the CA SDM database user.
Confirm mdbadmin Password	Reconfirm the password for the CA SDM database user.

5. On the **CABI JasperReports Server Configuration** window, enter the details for configuring JasperReports Server 9.0.0, and click **Next.**

Field	Description
Server Name	The name of the server where JasperReports Server 9.0.0 is installed.
Organization	The organization name created in JasperReports Server 9.0.0
Port	The port number that JasperReports Server 9.0.0 uses.
Insights Admin User Name	The user name with complete access to all organizations and tenants.
Admin User	The default admin user is superuser.
Insights Admin Password	The admin password with complete access to all organizations and tenants.
Admin Password	The password to authenticate the database administrator user.
Confirm Insights Admin Password	Reconfirm the admin password with complete access to all organizations and tenants.
Server Protocol	The server protocol used during JasperReports Server 9.0.0 installation.
Webapp Directory	The location within the tomcat folder where JasperReports Server 9.0.0 is deployed. Example, <JasperReports Server 9.0.0 installed directory>\apache-tomcat\webapps\jasperserver-pro

6. (Optional) To enable aggregate reports, select the **Aggregate Reports** checkbox.
   On the **Aggregate Reports** window, verify the datamart database server details, and click **Next.**
7. (Optional) To enable transactional reports, select the **Transactional Reports** checkbox.
   On the **Transactional Reports** window, verify the transactional database server details, and click **Next.**
8. In the Jasper Database Configuration window, select the required database against Jasper Database Type to automatically increase the password field length of the jijdbcdatasource table on jasperserverdb.
   If your JasperReports server is deployed on any of the following databases, perform the following steps:
   a. PostgreSQL Database
   b. Microsoft SQL Server Database
   c. Oracle Database

NOTE

- Ensure that you have the required database client installed based on the Jasper database server type.
- The Jasper Database Admin User is the privileged user who can create and update the Jasper database entries.

PostgreSQL Database

Prerequisites

For successful integration, verify the following.

- Open the pg_hba.conf file under the PostgreSQL directory.
  <jasper-report-server-path>\postgresql\data
  Under # IPv4 local connections, add the following new entry:
  host all <—IP address-->/32 md5
  IP address of the installed Jasper Server.
- Open the postgresql.conf file under the PostgreSQL directory.
  <jasper-report-server-path>\postgresql\data
  Update the listen_addresses entry from 127.0.0.1 to * as follows and save.
  Example: listen_addresses = '*'

Restart the PostgreSQL database service.

a. Complete the following mandatory fields:
   - Jasper Database Hostname
     Specifies the Jasper Database Hostname.
   - Jasper Database Name
     Specifies the name of the Jasper Database.
     Default: jasperserver
   - Jasper Database Admin User
     Specifies the Jasper Database Admin User name.
     Default: postgres
   - Jasper Database Admin Password
     Specifies the Jasper Database Admin Password.
     Note: On the PosrgreSQL database page the password is pre- populated.
   - Jasper Database Port
     Specifies the Jasper Database Port number.
     Default: 5432
   - PostgreSQL Home Path
     Specifies the Jasper directory where the Postgres SQL database is installed.
b. Click Next to proceed.

Microsoft SQL Server Database

a. Complete the following mandatory fields:
   - Jasper Database Hostname
     Specifies the Jasper Database Hostname.
   - Jasper Database Name
     Specifies the name of the Jasper Database.
     Default: jasperserver
   - Jasper Database Port
     Specifies the Jasper Database Port number.

Default: 1433

- Jasper Database Admin User
  Specifies the Jasper Database Admin User name.
  Default: sa
- Jasper Database Admin Password
  Specifies the Jasper Database Admin Password.

b. Complete the following optional field:
- Jasper Database Server Instance
  Specifies the Jasper Database Server Instance.

c. Click Next to proceed.

Oracle Database

a. Complete the following mandatory fields:
- Jasper Database Hostname
  Specifies the Jasper Database Hostname.
- Jasper Database Port
  Specifies the Jasper Database Port number.
  Default: 1521
- Jasper Database Net Service Name
  Specifies the Jasper Database Net Service Name.
- Jasper Database Admin User
  Specifies the Jasper Database Admin User name.
  Default: system
- Jasper Database Admin Password
  Specifies the Jasper Database Admin Password
- Jasper Database - SID
  Specifies the Jasper Database - SID.

b. Click Next to proceed.

9. Review the information on the **Configuration** window, and click **Next.**
10. Click **Install** to start the integration process.

NOTE
The required JARS and XML files are copied automatically. On the progress page, you may view the following password increment and copy tasks.

- Password Field Length Increment On JasperReport Server Database
- Copying Custom Files to JasperReport Server

NOTE
JasperReports Server Directory/Path Traversal Vulnerability: A directory traversal vulnerability has been identified that potentially allows unauthenticated users to access sensitive information on the host system. An attacker could send a specially-crafted URL request to view files on the server.
This fix should be applied on the 17.4 RU4 and above, integrated with JasperReports 9.0.0 environment.
To eliminate the CVE-2024-3325, apply the hotfix T5UG584.caz with APAR No : 99112426.

NOTE
For 17.4 RU2 and later, integrated with the JasperReports 9.0.0 environment, apply Hotfix T5UG596.caz with APAR No: 99112519 to resolve a performance issue in ad-hoc views when adding additional fields to crosstab rows.

Post-installation tasks

After integrating JasperReports Server 9.0.0 with CA Service Management, complete the following post-integration tasks to have Insights up and running.

- Enable Insights from the xFlow Interface
- Enable Insights from CA SDM

Enable Insights from the xFlow Interface

An administrator must enable Insights app for it to appear on the xFlow Interface.

Follow these steps:

1. Restart the JasperReports Tomcat Server.
2. Login to CA SDM as an administrator.
3. Navigate to the **Administration** tab, select **xFlow Interface**, **App Configuration**, and **Insights**.
4. Click **Edit**.
5. From the drop-down list, select **Active**, and click **Save**.
6. Restart the xFlow Interface services based on your configuration.
   a. For Advanced Availability: Restart services on the application server.
   b. For Conventional: Restart services on the primary and secondary servers.

Result: Insights app is enabled and available from the xFlow Interface.

Enable Insights from CA SDM

If you have not installed the xFlow Interface or Insights app, you can access Insights directly from CA SDM.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to **Administration**, **Security and Role Management**, **Role Management**, and **Role List.**
3. Select a role.
4. Select the Resources tab and click **Update Tabs** to select the Insights tab.
5. Assign the insights app to the selected role.
6. Log in to CA SDM with the selected role.
7. From the Menu bar, select **Insights** and click **Launch Insights**.

Result: Insights app is enabled and available for use from CA SDM.

Additional tasks

After successfully integrating JasperReports Server 9.0.0 with CA Service Management, review the following additional tasks. You can complete the tasks that you need for your environment.

- Configure the data aggregation date
- Match the duration format

Configure the data aggregation date

Get accurate and consistent reports by configuring the data aggregation date to a date later than that of CA Service Management 17.4 implementation. If the Data Aggregation Date is not configured, inaccurate information is displayed, and the following reports are impacted:

- Incidents Cost for Top Categories (USD)
- Incidents Cost Trend for Top Categories (USD)
- MTTR for Top Categories (In Hours) Report
- MTTR Quarterly Trend for Top Categories (In Hours) Report
- MTTR Weekly Trend for Top Categories (In Hours) Report
- MTTR Yearly Trend for Top Categories (In Hours) Report

Follow these steps:

1. Navigate to the `<datamart installed folder>\config\config.properties` file.
2. Search for the string **trend_start_date='yyyy-mm-dd'**.
3. Replace the **trend_start_date** value with the data aggregation date.
   During the next ETL job, the data older than the data aggregation date is deleted.

   NOTE
   For the vulnerability of ***Log4j1 Removal and adding the log4j2 jars and related dependencies in Datamart Installed location*** apply the patch with APAR No : 99112251

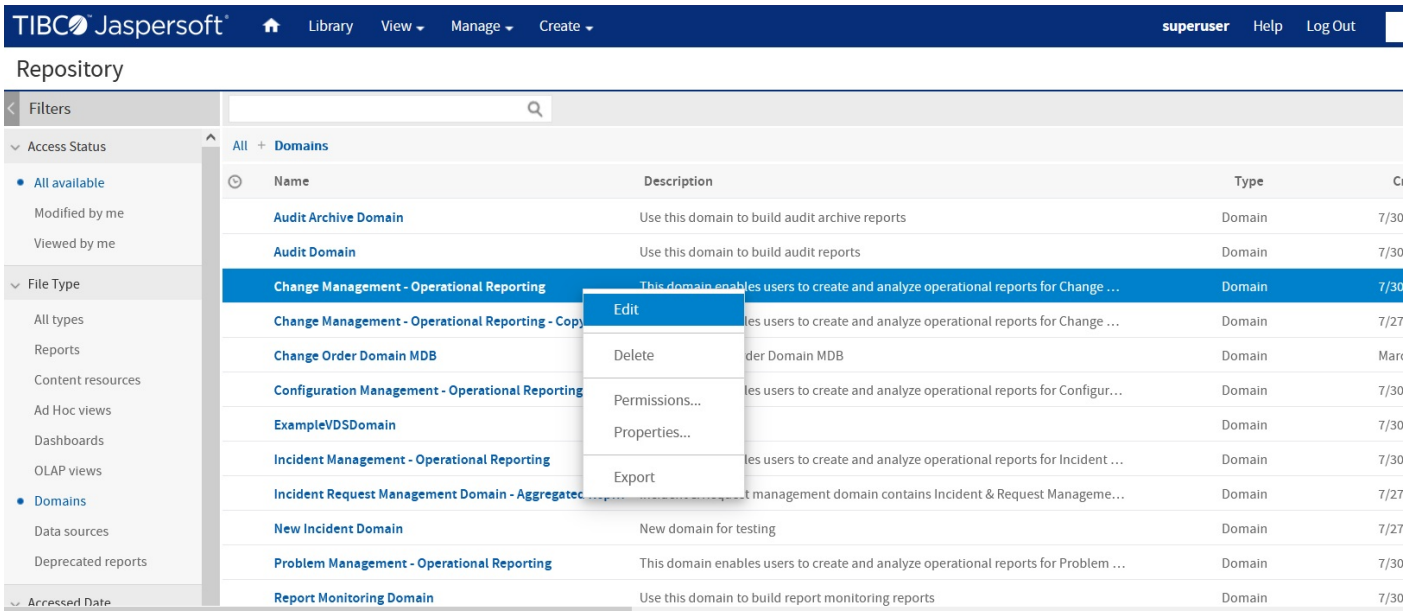
Match the duration format

The duration format is represented differently in CA SDM and Insights. You can match the duration format to appear in HH:MM:SS format in both CA SDM and Insights.

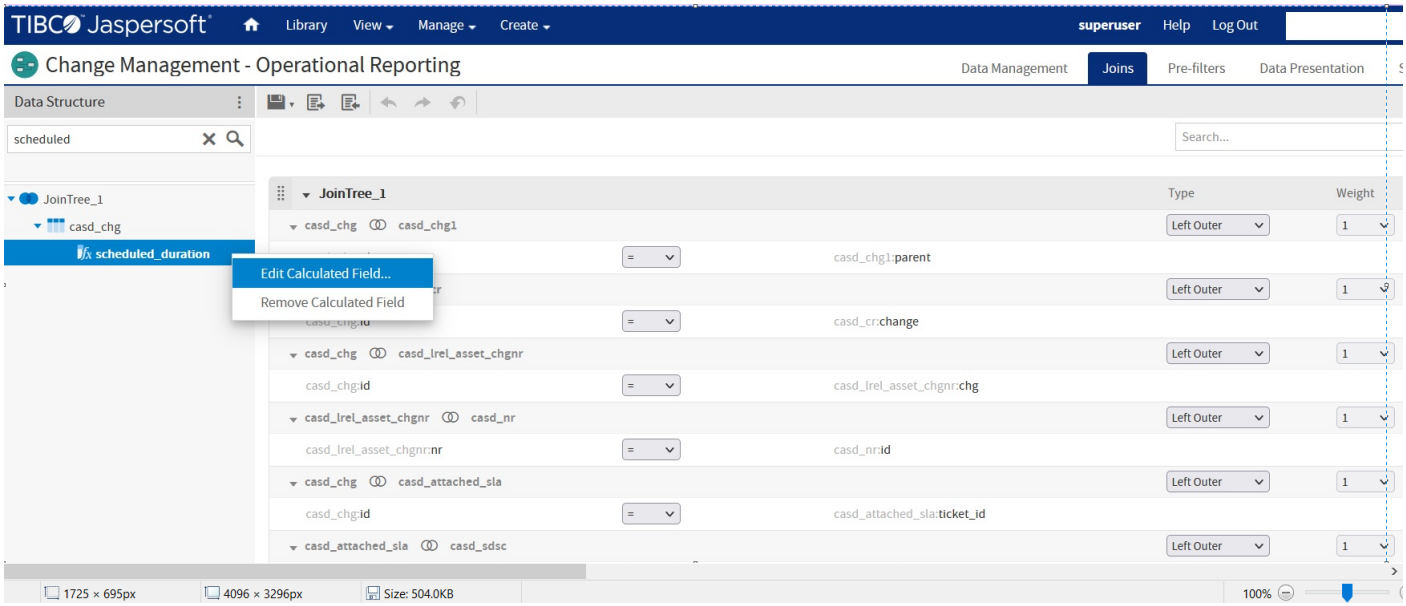
Follow these steps

1. Connect to the Jasper Reports server machine.
2. Copy the *DurationFormatter.jar* from CASM_17.4.2.0\filestore\Jasper\9.0.0\ExternalAuthJars and, copy it into the jasperserver tomcat lib folder. For eg: *C:\apache-tomcat-9.0.65\webapps\jasperserver-pro\WEB-INF\lib*.
3. From the service console, stop the service:*jasperreportsTomcat.*
4. From the folder where you installed JasperReports Server 9.0.0, navigate to the `apache-tomcat\webapps\jasperserver-pro\WEB-INF` folder and back up applicationContext-el-operators.xml file.
5. From the folder where you installed JasperReports Server 9.0.0, navigate to the `apache-tomcat\webapps\jasperserver-pro\WEB-INF\bundles` folder and back up the following files.
   - adhoc_messages.properties
   - adhoc_messages_es.properties
   - adhoc_messages_fr.properties
   - adhoc_messages_ja.properties
   - adhoc_messages_pt_BR.properties
6. Navigate to the DVD `filestore\Jasper\9.0.0\ExternalAuthXMLs` folder and copy the applicationContext-el-operators.xml file.
7. From the folder where you installed JasperReports Server 9.0.0, navigate to the `apache-tomcat\webapps\jasperserver-pro\WEB-INF` folder and paste the copied file.
8. Navigate to the DVD `filestore\Jasper\9.0.0\Bundles` folder and copy the following files.
   - adhoc_messages.properties
   - adhoc_messages_es.properties
   - adhoc_messages_fr.properties
   - adhoc_messages_ja.properties
   - adhoc_messages_pt_BR.properties
9. From the folder where you installed JasperReports Server 9.0.0, navigate to the `apache-tomcat\webapps\jasperserver-pro\WEB-INF\bundles` folder and paste the copied files.
10. Navigate to the DVD `filestore\Jasper\9.0.0\Groovy` folder and copy the BaseGroovyColumn.groovy file.

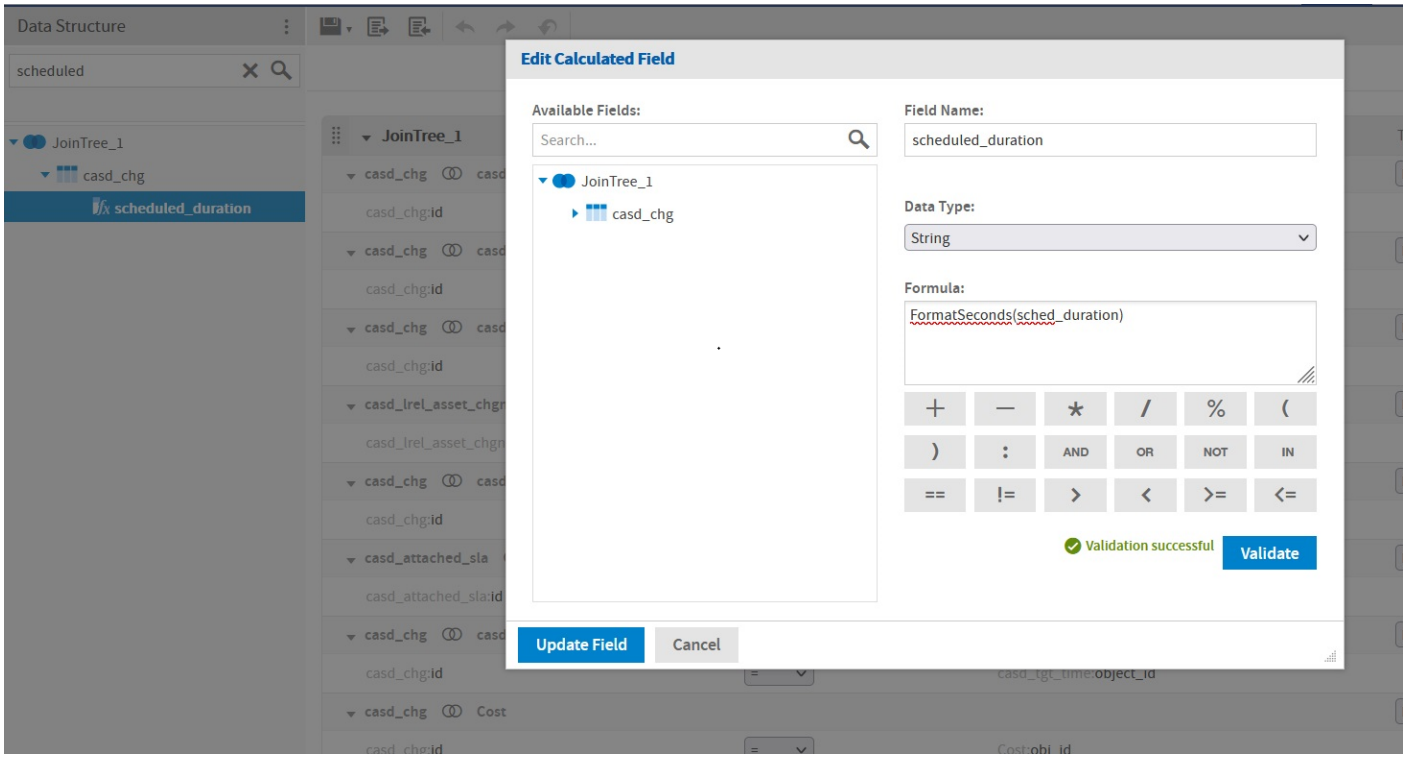
11. From the folder where you installed JasperReports Server 9.0.0, navigate to the `apache-tomcat\webapps\jasperserver-pro\WEB-INF\groovy\groovy_column` folder and paste the copied file.
12. From the folder where you installed JasperReports Server 9.0.0, navigate to the `apache-tomcat\temp` folder and empty the folder.
13. From the folder where you installed JasperReports Server 9.0.0, navigate to the `apache-tomcat\work` folder and empty the folder.
14. From the service console, start the service: *jasperreportsTomcat.*
15. Clear the browser cache.
16. Log in to Jasper UI in the browser, and edit the required domain. This image shows how to edit the Change Management - Operational Reporting domain.



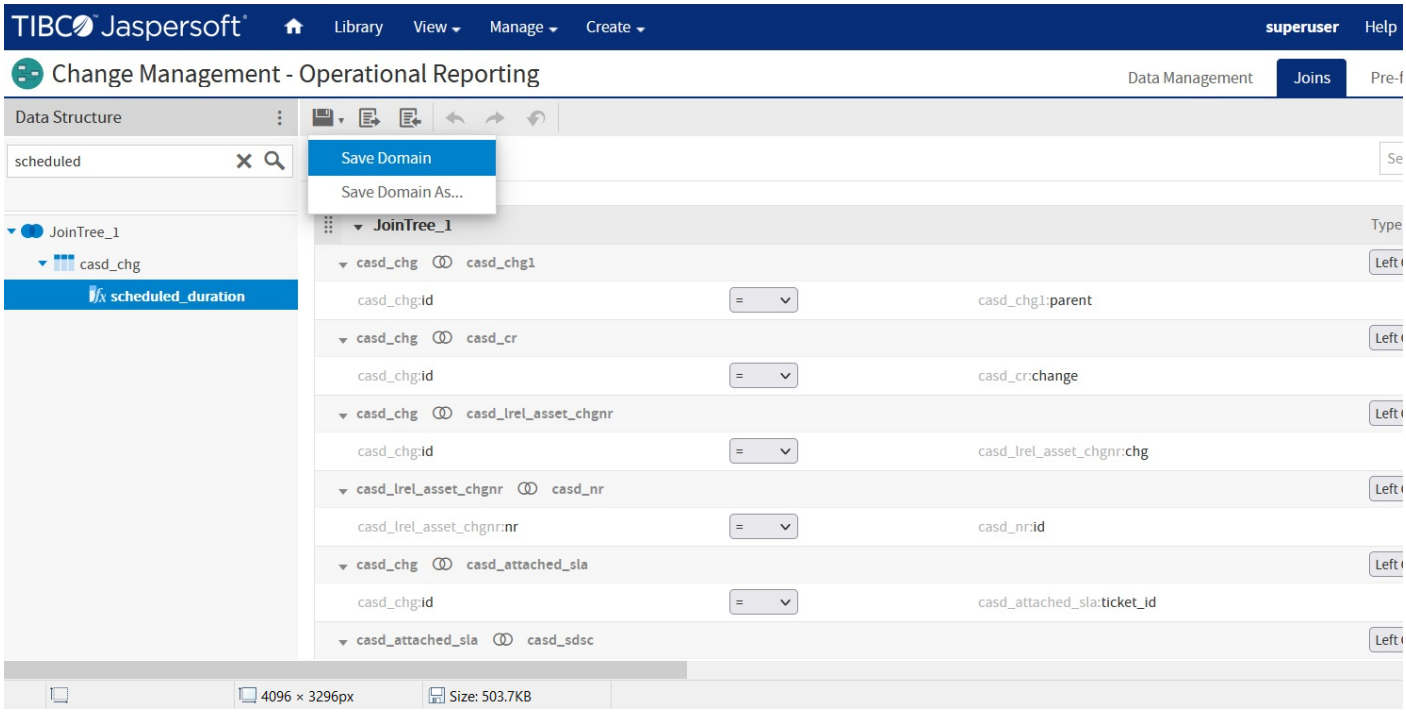
17. Under the Joins tab, search for the required calculated field and right-click the field to edit it. This image shows how to edit the scheduled_duration field.



18. Under Formula, edit the formula for the calculated field. This image shows the formula for the scheduled_duration field.



19. Change the data type to String.
20. Click **Validate**.
21. Click **Update Field**.
22. To save the domain, click the **Save** icon.



23. To verify that the duration value is displayed in HH: MM: SS format, use this field in your reports.

Duration Measure error for default users

Follow these steps:

1. Login to jasperserver application.
2. Click **Domains**.
3. Right click on the required Domain (e.g.: Incident Management- Operational Reporting) to modify and click **Edit.**
4. Click **Joins** and search for the Calculated field (e.g.: Resolution_Time_Hours) that are used for measures.
5. Right click on calculated field and click **Edit Calculated Field**.
6. Remove the SUM from the formula then click **Validate** and **Update Field**.
7. Save the Domain.
8. Restart the Jasperreports Tomcat service.

If CA SDM is on Oracle follow these additional steps

1. Navigate to the location where Jasper server is installed. (e.g.: C:\apache-tomcat-9.0.65\webapps\jasperserver-pro)
2. Under the folder WEB-INF, select and edit *applicationContext-semanticLayer.xml*
3. Search for the tag
   ```
   <bean id="sqlserverGenerator" parent="defaultSQLGenerator" scope="prototype">
   ```
4. Find the entry <entry key ="divide"> under it.
5. For removing the data type *casting of money*, change
   "cast ($top as money) / (case when (($bottom) = 0) then null else ($bottom) end)" as String
   to *"($top) / (case when (($bottom) = 0) then null else ($bottom) end)"* as String.
6. Save the applicationContext-semanticLayer.xml
7. Restart the Jasper server tomcat service.

Supported Languages

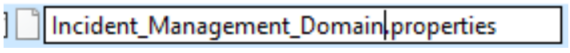
JasperReports supports (fr_FR (French), de_DE (German), pt_BR (Brazil- Portuguese), es_ES (spanish), languages, and the reports work as expected.

For supported languages:

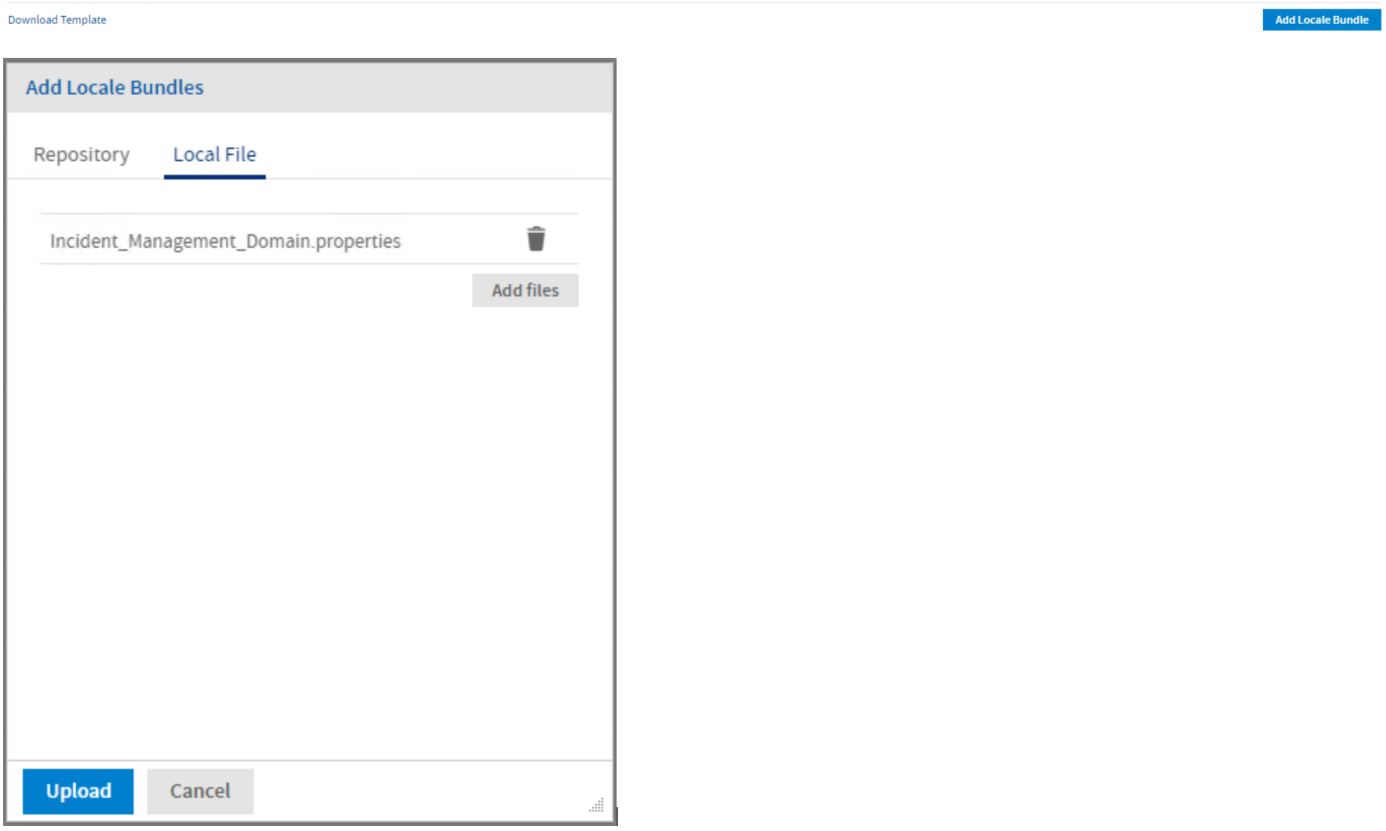
1. Go to **Browser Settings**> **Languages**, choose a supported language, and clean the browser cache.
2. Restart JasperReports.
3. The Reports/ad hoc views are displayed in the selected language.

For unsupported languages:

1. Login to Jasper UI. Right click on a domain (**for example**: *Incident Management - Operational Reporting*) and choose **Edit** and navigate to **Locales** tab.
2. Download the Incident_Management_Domain_en_US.properties file of the domain.
3. Make a copy of the same locally and rename the file to *Incident_Management_Domain.properties*



4. Go to **Add Locale Bundle**, and click **Local File.** Now **Upload** the Incident_Management_Domain.properties file.



5. Now **Save** the domain.
6. Delete the browser cache and login to Jaspersoft web page and run the report(s).

JasperReports Server 9.0.0 Out-of-the Box Transactional Reports Error.

Problem: In case of SDM installed with Oracle database and integrated with Jasper 9.0.0, you may see an issue while running some of the reports which has bigint datatype.

Solution: Perform the following steps:

1. To resolve this issue, update and modify the applicationcontext-semanticLayer.xml file. Open this file in Jasper installed location: (for example: C:\Jaspersoft\jasperreports-server-7.9.0\apache-tomcat\webapps\jasperserver-pro \WEB-INF) and search for the following:

```
switch (args[0]?.javaType)
{
case 'java.lang.Integer':
case 'java.lang.Short':
return String.format("SUM(cast(%s as bigint))", sqlArgs[0])
default:
return String.format("SUM(%s)", sqlArgs[0])
            }
```

2. Comment the following lines from the above step:

```
<!--case 'java.lang.Integer':
case 'java.lang.Short':
return String.format("SUM(cast(%s as bigint))", sqlArgs[0])-->
```

3. Save the file and restart JasperReports Tomcat services.

Configure JasperReports Server 9.0.0 for High Availability Set Up Cluster

Prerequisites

This document provides updates on how to Configure JasperReports Server 9.0.0 High Availability Set Up Cluster.

Prerequisites

The following are the prerequisites:

- Complete the installation of Jasper 9.0.0 with supported databases. (Call it Node 1)
- Verify the Jasper URL.
- Install ActiveMQ on any one server in the cluster.
- Install JDK (OpenJDK11) and Apache Tomcat (9) on all the nodes
- Integrate SDM or xFlow to the JasperReports Server 9.0.0

EhCache Replication

The repository cache in JasperReports Server is implemented internally through the Ehcache component. Edit the Ehcache configuration files as described below to replicate the repository cache among all instances that share a single repository. You do not have to configure a cluster for cache replication.JMS – Java Message Services can provide cache distribution for nodes in a cloud where IP addresses may change. Jaspersoft provides a configuration for using the Apache ActiveMQ JMS Server. You must first install and configure ActiveMQ on one of the computers in your cluster.

Configure a High Availability Environment on Node 1

Perform the following steps on JasperReports Server, Node 1:**Follow these steps:**

1. Stop the JasperReports Server Apache Tomcat service on Node1.
2. Take a backup of the following files on each node:
   Open the file <web-app>/WEB-INF/classes/ehcache_hibernate.xml and <web-app>/WEB-INF/ehcache.xml on each node.
   Example:C:\Jaspersoft\jasperreports-server-9.0.0\apache-tomcat\webapps\jasperserver-pro
3. In both ehcache_hibernate.xml and ehcache.xml files, comment out or remove the section of the file in between "NO CLUSTERING" and "END of NO CLUSTERING" comments.

```
<!-- ********************************************************************** NO CLUSTERING **********
<!-- START
<cache name="aclCache"
                maxElementsInMemory="100000"
                statistics="false"
                eternal="true"
                overflowToDisk="false"
                timeToIdleSeconds="36000"
                timeToLiveSeconds="180000"
                diskExpiryThreadIntervalSeconds="120"
                diskPersistent="false"/>
END -->
<!-- ********************************************************************** END of NO CLUSTERING *******
```

4. Uncomment the JMS section in <web-app>/WEB-INF/classes/ehcache_hibernate.xml and <web-app>/WEB-INF/ehcache.xml on each node.

Example:

5. Set the providerURL properties in both files to the address of your JMS server.
   It is shown in the following code example:

```
<!-- START
        <cacheManagerPeerProviderFactory
                class="net.sf.ehcache.distribution.jms.JMSCacheManagerPeerProviderFactory"

properties="initialContextFactoryName=com.jaspersoft.jasperserver.api.engine.replication.JRSActiveMQInitialContextF
                providerURL=tcp://ActivemQ_host:61616,
                replicationTopicConnectionFactoryBindingName=topicConnectionFactory,
                replicationTopicBindingName=ehcacheAcl,
                getQueueConnectionFactoryBindingName=queueConnectionFactory,
                getQueueBindingName=ehcacheQueueAcl,
                topicConnectionFactoryBindingName=topicConnectionFactory,
                topicBindingName=ehcacheAcl"
                propertySeparator=","/>
```

6. On each node, edit the file <web-app>/META-INF/context.xml. Locate the Manager pathname near the end, and comment it out as follows:
   C:\Jaspersoft\jasperreports-server-9.0.0\apache-tomcat\webapps\jasperserver-pro

```
<!-- Manager pathname="" -->
```

Configure a High Availability Environment on Node 2

Perform the following steps on JasperReports Server, Node 2:
Follow these steps:

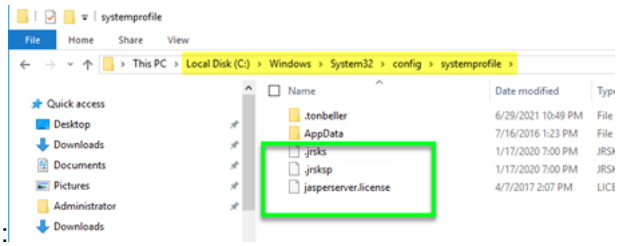
1. Use the same installer that is used in Node 1 (Use original installer).
2. Copy the default_master.properties file from Node 1 to Node 2.
3. Stop Apache Tomcat services.
4. Ensure that Apache Tomcat and Database details are pointing to the correct database.
   Example:
   # Tomcat app server root dir
   appServerDir = C:\\Program Files\\Apache Software Foundation\\Tomcat 9.0
   # appServerDir = /home/devuser/apache-tomcat-9.0
   If it is a Linux package managed Tomcat instance, set the following two properties, for e.g:
   – CATALINA_HOME = C:\\Program Files\\Apache Software Foundation\\Tomcat 9.0
   – CATALINA_BASE = C:\\Program Files\\Apache Software Foundation\\Tomcat 9.0
5. Install only webapp in Node 2.

Navigate to the C:\TIB_js-jrs_9.0.0_bin\jasperreports-server-pro-9.0.0-bin\buildomatic folder and run the following command

```
js-ant deploy-webapp-pro
```

6. Ensure you have a non default port under:
   C:\Program Files\Apache Software Foundation\Tomcat 9.0\webapps\jasperserver-pro\WEB-INF
   diagnostic.port = 10990
   Change the port number.
7. Repeat the steps 3-6 from the above Configure a High Availability Environment on Node 1 section.
8. Copy the following files from NODE1's system user directory from when you installed Jasper 9.0.0, to NODE2's system user directory. For example, C:\Users\Administrator.
   Give full permissions to the following files for Local Service:
   - .jrsks
   - .jrslic
   - .jrsksp
   - Jasperserver.license

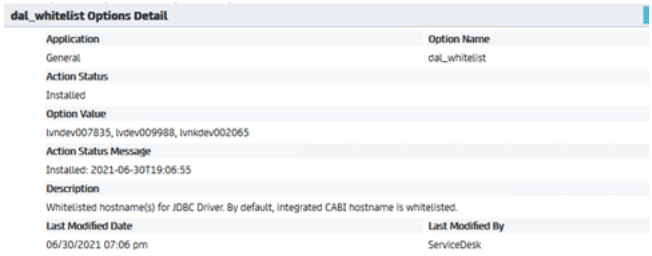
   In some cases, you may need to copy the files to the systemprofile location:



Integrate with CA Service Management Server

To integrate Jasper High Availability servers with CA SDM:

1. Access both Node-1 and Node-2 url's after successful installation.
2. Integrate CASM solution from Node-1 and complete the integration.
3. Configure CA SDM to accept communication from the new node by updating the dal_whitelist Option.
4. Under Options manager> General:



5. Copy the ServiceManagemnt Specific custom libraries /xmls from Node-1 Jasper Installation location to Node-2 location. For e.g: C:\apache-tomcat-9.0.65\webapps\jasperserver-pro\WEB-INF\lib

```
BOPIntegration.jar
cabi-jasperreportsserver-commonutil.jar
cabi-jasperreportsserver-dynamicjdbc.jar
cabi-jasperreportsserver-rolebasedflow.jar
cabi-jasperreportsserver-sso.jar
cabi-jasperreportsserver-userlocale.jar
cabi-jasperreportsserver-versionapi.jar
```

```
domsrvr_utils.jar
sd-utils.jar
slump.jar
```

C:\apache-tomcat-9.0.65\webapps\jasperserver-pro\WEB-INF

```
applicationContext.xml
applicationContext-el-operators.xml
applicationContext-externalAuth-template-mt.xml
applicationContext-jdbc-metadata.xml
applicationContext-rest-services.xml
applicationContext-security.xml
applicationContext-security-web.xml
applicationContext-semanticLayer.xml
```

6. Restart the Tomcat Service's of Node-2 and Node-1
7. Make sure you see your reports from both the URl's. Create a new ad hoc view on NODE2, then log into Jasper-Node1 on NODE1 and make sure you can see and run the report.

Configure JasperReports Server 9.0.0 to Use SSL

You can configure JasperReports Server to use Secure Socket Layer (SSL) and communicate with CA Service Management. SSL establishes an encrypted link between a server and a client. For example, a web server and a browser; or a mail server and a mail client. This link helps ensure that all data passed between the server and client remain private and integral. When you configure JasperReports server to use SSL, you change its communication method from HTTP to HTTPS while integrating with CA Service Management.
Follow these procedures:

Create a Keystore File
A keystore file is required to enable SSL. Create a keystore file if you do not have one already for another CA product that integrates with CA Service Management. You can use a keystore file for a single product or for multiple products. If you must create individual keystores for each product, you can optionally merge your keystore files.**Follow these steps:**

1. Open a command window on the JasperReports server. Navigate to the location where JasperServer is installed.
   Example: C:\Jaspersoft\jasperreports-server-9.0.0\java\bin>
2. Execute the following command:
   ```
   keytool -genkey -alias tomcat -keyalg RSA -keystore <Selected Path>/<filename>.keystore
   ```
   Enter the required information when prompted.
3. Enter the password at the “Enter keystore password” prompt. To make configuring Tomcat easier, you can use “changeit” as the password.
4. Record your password for reference.

Add the Certificate to the Keystore
After creating the keystore file, you must add a CA certificate to the keystore.**Follow these steps:**

1. Execute the following command to add the certificate into the keystore.
   ```
   keytool -importkeystore -srckeystore <Selected Path>/<filename>.keystore -destkeystore <Selected Path>/<filename>.keystore -deststoretype pkcs12
   keytool -export -storepass PASSWORD -alias tomcat -keystore <Selected Path>/<filename>.keystore -file <Selected Path>/<filename>/Jasperssl.crt
   ```
   The certificate is exported to keystore.

Edit the Server.xml File to Support SSL

As part of configuring the JasperReports server to use Secure Socket Layer (SSL), edit the server.xml file to support SSL.**Follow these steps:**

1. Navigate to the JasperServer installed location.
2. Edit the <*Jaspersever Installed Location*>\apache-tomcat\conf**server.xml** file.
3. Search for the <Service name="Catalina"> section.
4. Updated the Connector related information under the <Service name="Catalina"> section:

```
<Connector SSLEnabled="true" maxThreads="200" port="8443"
 protocol="org.apache.coyote.http11.Http11NioProtocol"
 sslImplementationName="org.apache.tomcat.util.net.jsse.JSSEImplementation">
      <SSLHostConfig
 ciphers="TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384,TLS_DHE_RSA_WITH_AES_128_GCM_S
 TLS_DHE_RSA_WITH_AES_256_GCM_SHA384,TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256,TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA,
 TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384,TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA,TLS_DHE_RSA_WITH_AES_128_CBC_SHA256,
 TLS_DHE_RSA_WITH_AES_128_CBC_SHA,TLS_DHE_RSA_WITH_AES_256_CBC_SHA256,TLS_DHE_RSA_WITH_AES_256_CBC_SHA,
 TLS_RSA_WITH_AES_128_GCM_SHA256,TLS_RSA_WITH_AES_256_GCM_SHA384,TLS_RSA_WITH_AES_128_CBC_SHA256,
 TLS_RSA_WITH_AES_256_CBC_SHA256,TLS_RSA_WITH_AES_128_CBC_SHA,TLS_RSA_WITH_AES_256_CBC_SHA,
 SSL_RSA_WITH_3DES_EDE_CBC_SHA" honorCipherOrder="true" protocols="TLSv1.2" sslProtocol="TLSv1.2">
        <Certificate certificateKeyAlias="<Alias Name>" certificateKeystoreFile="<Selected
 Path>/<filename>.keystore" certificateKeystorePassword="PASSWORD" certificateKeystoreType="PKCS12"/>
      </SSLHostConfig>
    </Connector>
```

   – Default value before making this change.

```
<Connector port="8080" URIEncoding="UTF-8" protocol="HTTP/1.1" connectionTimeout="20000"
 redirectPort="8443"/>
```

5. Save and close the server.xml file.
6. Restart the Tomcat server.
7. Now you can launch the Jaspersoft url using https://HOST_NAME:SSL_PORT/jasperserver-pro/flow.html?_flowId=homeFlow
   Ensure that Jasper URL is accessible.

Integrate CA SDM with the SSL-enabled JasperReports Server

Configure the SSL-enabled JasperReports Server to integrate with CA SDM using the CA Service Management Common Installer.**Follow these steps:**

1. Download the common installer to the required location in the JasperServer.
   Example: C:\Users\Administrator\Downloads\<*RUXXXXX*>
2. Extract the installer and go to the Java folder
   Example: \java\java11
3. To integrate HTTPS enabled Jasper Server with Common Installer, the certificate you exported in the *Add the Certificate to the Keystore* section should be imported by executing the following command:

```
keytool -importcert -alias <aliasname>-file <Selected Path>/<filename>.crt -keystore  ..\lib\security
\cacerts
```

4. Launch the CA Service Management Common Installer.
5. Provide the following mandatory information in the "CABI JasperReports Server Configuration" screen.
   Enter the port number that you have provided in the server.xml file.
   Select HTTPS from the Server Protocol drop-down.
6. Proceed with the next steps in the wizard to complete the integration.

   NOTE
   After the integration is successful, log in to the classic CA SDM, Administration, xFlow Interface, General, and confirm the value *insights.jasper.domain* is populated in the General Configuration List.

Enable JasperServer SSL URL from CA SDM Classic UI

1. Navigate to <jasper_installation_path>\jasperserver-pro\WEB-INF\config and copy the following files.
   - casm_insights.jks
   - casm_insights.properties
2. Paste the copied files from **step 1** to CA SDM Server.**For example**: NX_ROOT\site folder.
3. Restart CA Service Desk Manager Server and JRS services.

NOTE
Integration of CA Service Management Release 17.4 with Jasper Reports Server 9.0.0 is automated.

Add JasperReports Server SSL Certificate to the CA SDM

To launch HTTPS enabled JasperReports server from CA SDM Classic UI, the certificate you exported in the *Add the Certificate to the Keystore* section should be imported.Follow these steps:

1. Navigate to the JRE bin folder in the CA SDM server.
   For example, C:\Program Files (x86)\CA\SC\JRE\11.0.12\bin
2. Import the certificate or any trusted certificate.
   ```
   keytool -importcert -alias <aliasname>-file <Selected Path>/<filename>.crt -keystore  ..\lib\security
   \cacert
   ```
3. You can now launch Insights from CA SDM classic UI.

Adding JasperReports Certificate to Insights Trust Store

For Insights to connect to HTTPS enabled JasperReports server, the certificate you exported in the *Add the Certificate to the Keystore* section should be imported into the Insights trust store.

1. Navigate to <xFlowInstallation Home Dir>\jre\bin
   Example: C:\Program Files\CA\SC\JRE\11.0.3\bin
2. Import the certificate or any trusted certificate.
   ```
   keytool -importcert -alias <aliasname>-file <Selected Path>/<filename>.crt -keystore  ..\lib\security
   \cacert
   ```
3. Restart the CA xFlow Analyst Interface services.

JasperReports Server 8.1.1

This topic provides complete information about JasperReports Server Release 8.1.1.

- Tutorial for JasperReports Server
- Install JasperReports Server 8.1.1
- Upgrade JasperReports Server from 7.9 to 8.1.1
- Integrate JasperReports Server 8.1.1 with CA Service Management
- Configure JasperReports Server 8.1.1 for High Availability
- Configure JasperReports Server 8.1.1 to Use SSL

Tutorial for JasperReports Server

Learn about JasperReports Server, TIBCO Jaspersoft Online Learning Portal, training, documentation, and the Jaspersoft Community.

TIBCO Jaspersoft Online Learning Portal

Broadcom Inc., in partnership with TIBCO Software, Inc., has authorized its partners and licensed customers to have free access to theTIBCO Jaspersoft Online Learning Portal..

Recommended Jaspersoft Training - TIBCO Software, Inc

Jaspersoft training is available from TIBCO Software, Inc. The following recommended Jaspersoft training courses will help you in your Jaspersoft implementation.

Recommended Training for Analysts and Managers

The following online training and resources are recommended for end users such as Analysts and Managers:

Training Topic	Type	Description	Duration	Video	TIBCO Docs
Ad Hoc Views	Video	This video helps you learn how to create and configure Ad Hoc views using the Ad Hoc Editor.	8 Minutes	Ad Hoc Views Tutorial	Adhoc Views
Reports	Video	This video helps you learn how to create and format reports from existing Ad Hoc views.	9 Minutes	Ad Hoc Reports Tutorial	Reports
Dashboards	Video	This video helps you learn how to create dashboards and dashlets.	9 Minutes	Dashboards Tutorial	Dashboards

Recommended Training for Administrators.

All the training materials covered in online training resources in the following table are recommended for administrators such as SDM administrators:

Training Topic	Type	Description	Duration	Video	Registration Required?	TIBCO Docs
Domains	Video	This video helps you learn how to create domains.	8 Minutes	Domains Tutorial	No	
Jasper Reports User Guide	User Guide	Introduction to JasperReports Server and its capabilities.		NA	No	User Guide
JasperReports Administration Guide	Administrative Guide	Overview of JasperReports Server Administration.		NA	No	Admin Guide

Jaspersoft Community

Visit the Jaspersoft Community for related tutorials, tips and tricks, and other helpful information.

Install JasperReports Server 8.1.1

The installation of JasperReports 8.1.1 is supported on Microsoft SQL Server, Oracle, and PostgreSQL databases. The installation is available in English (United States) and is supported only on Windows Operating System.

We recommend using the WAR file distribution for installation or upgrade to JasperReports Server 8.1.1, as the WAR distribution includes customizations needed for SDM integration.

NOTE
To use TIB_js-jrs_8.1.1_win_x86_64.exe to install JasperReports Server 8.1.1 (**for evaluation purpose only**) and to import an existing repository, refer the KB article https://knowledge.broadcom.com/external/article?articleId=263988

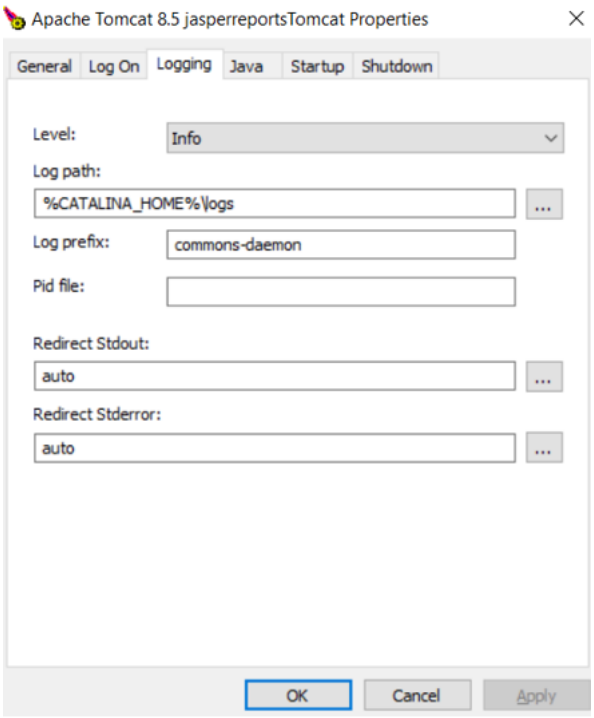
Install Jaspersoft Server 8.1.1 on PostgreSQL, Microsoft SQL Server, or Oracle database using WAR distribution

Perform the following steps to Install JasperReports Server 8.1.1 on PostgreSQL, Microsoft SQL Server or Oracle database.

NOTE
Before installing, review the TIBCO Platform Support for JasperReports 8.0. Documentation.

1. From Broadcom support portal, download the JasperReports Server 8.1.1 installer file **CA Business Intelligence JasperReports Server 8.1.1 for CA Service Management.iso**.
2. Mount the JasperReports Server 8.1.1 installer after downloading it to your environment. Or ensure to unzip the installer contents in your local system.
   NOTE
   TIBCO Release 8.0.0 documentation is applicable for Release 8.1.1 too. For example, though the documentation refers to TIB_js-jrs_8.0.0_bin.zip, you can consider it as TIB_js-jrs_8.1.1_bin.zip.
3. For installation instructions, see the chapter "*Installing the WAR File for Production"* in TIBCO documentation.
4. Configure Tomcat as an installed service.
   NOTE
   The paths shown in the below procedure are given as examples only. The paths may differ based on your environment and installation requirements.
   a. Install a supported Java version. For more information about a supported Java version, see TIBCO Platform Support for JasperReports 8.0 Documentation.
   b. Set CATALINA_HOME to the tomcat folder.
   c. Set JAVA_HOME to the java folder.
   d. From `CATALINA_HOME\bin` , execute the following command to install JasperReports Service.
   ```
   tomcat8.exe //IS//jasperreportsTomcat --DisplayName="Apache Tomcat 8.5 jasperreportsTomcat"
    --Install="C:\JRS8_1_1\tomcat8570\bin\tomcat8.exe" --Jvm=auto --StartMode=jvm --
   StopMode=jvm --StartClass=org.apache.catalina.startup.Bootstrap --StartParams=start --
   StopClass=org.apache.catalina.startup.Bootstrap --StopParams=stop
   ```
   NOTE
   - Use the service name **jasperreportsTomcat,** identified by the common installer.
   - See the Troubleshooting CA Service Management section, if the following error message is displayed: Failed to grant service user 'NT AUTHORITY\LocalService' write permissions to log path 'C:\Windows \system32\LogFiles\Apache' due to error '19: The media is write protected.'

   e. From `CATALINA_HOME\bin` , execute the following command to configure JasperReports Service.
   ```
   tomcat8w.exe //ES//jasperreportsTomcat
   ```
   f. On the UI Window, select the **Logging** tab and set the below settings:



g. On the Java tab, set the below options:

```
JVM------
%JAVA_HOME%\jre\bin\server\jvm.dll
Java Classpath---------
%JAVA_HOME%\lib\tools.jar;%CATALINA_HOME%\bin\bootstrap.jar;%CATALINA_HOME%\bin\tomcat-juli.jar

Java Options-----------
-Djs.license.directory=C:\Users\Administrator
-Djs.license.directory=C:\Users\Administrator
-Dcatalina.home=C:\JRS8_1_1\tomcat8570
-Dcatalina.base=C:\JRS8_1_1\tomcat8570
-Djava.io.tmpdir=C:\JRS8_1_1\tomcat8570\temp
-Djs.license.directory=C:\Users\Administrator
-Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager
-Djava.util.logging.config.file=C:\JRS8_1_1\tomcat8570\conf\logging.properties

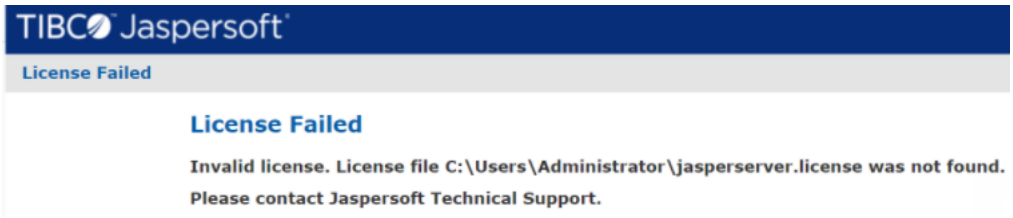

Java  9 options-----------
--add-opens=java.base/java.lang=ALL-UNNAMED
--add-opens=java.base/java.io=ALL-UNNAMED
--add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED
-Xdebug
-Xrunjdwp:transport=dt_socket,address=9999,server=y,suspend=n
```

h. Click **Apply** and **OK**.

i. Start Apache Tomcat 8.5 jasperreportsTomcat services from Windows services console.

5. Launch JasperReports Server 8.1.1.

If the error message License Failed is displayed, copy the jasperserver.license file and paste it into the folder as follows:



6. Stop Tomcat 8.5 jasperreportsTomcat services.
7. Navigate to `CASM_DVD\filestore\Jasper\8.1.1\License` folder and copy the jasperserver.license file.
8. Navigate to the path displayed in the error message and paste the copied file.
9. Start Tomcat 8.5 jasperreportsTomcat services.

Next Step:

- Integrate JasperReports Server 8.1.1 with CA Service Management

Upgrade JasperReports Server from 7.9 to 8.1.1

Upgrade JasperReports Server from 7.9 to 8.1.1 for CA SDM

Review the supported upgrade paths for JasperReports Server and upgrade by following the recommended upgrade procedure.

You can upgrade directly to JasperReports Server 8.1.1 from JasperReports Server 7.9.

Upgrade JasperReports Server 7.9 to JasperReports Server 8.1.1

This section provides instructions to upgrade from JasperReports Server 7.9 to JasperReports Server 8.1.1.

Follow these steps:

1. From Broadcom support portal, download the JasperReports Server 8.1.1 installer file **CA Business Intelligence JasperReports Server 8.1.1 for CA Service Management.iso**.
2. Upgrade to JasperReports Server 8.1.1 by performing the tasks listed in the *CHAPTER 3 UPGRADING FROM 7.9 TO 8.0 .* section in the TIBCO JasperReports Server 8.0.0 upgrade guide

   NOTE
   TIBCO Release 8.0.0 documentation is applicable for Release 8.1.1 too. For example, though the documentation refers to TIB_js-jrs_8.0.0_bin.zip, you can consider it as TIB_js-jrs_8.1.1_bin.zip.

   NOTE
   While performing the Jasperreports Server upgrade, if you get an error CATALINA_HOME and CATALINA_BASE are both NOT set, then edit the default_master.properties file for your database; set the CATALINA_HOME and CATALINA_BASE values to Tomcat app server root directory and save the file. Run the Upgrade command again.
3. To integrate JasperReports Server 8.1.1 with CA Service Management, launch the CA Service Management common installer on the server where you installed JasperReports Server 8.1.1.
   a. On the **Installer selection** window, select **Integrate CABI JasperReports Server with CA Service Management**.
   b. On the **License Agreement** window, review the license information, accept the agreement, and click **Next**.
   c. On the **Database Configuration** window, in the **Database Type** field, select a database.

d. Enter the details for CA Service Desk Manager (CA SDM) and click **Next**.

Field	Description
Database Server	The location where you've installed the CA SDM database server.
Database Name	The database name on the CA SDM database name.
Database Port	The port number that CA SDM uses.
Database Server Instance	The instance of the database server.
Database Admin User	The database admin user name with permissions to create users and schema. The default user for Microsoft SQL Server is sa.
Database Admin Password	The password to authenticate the database administrator user.
mdbadmin Password	The password for the CA SDM database user.
Confirm mdbadmin Password	Reconfirm the password for the CA SDM database user.

e. On the **CABI JasperReports Server Configuration** window, enter the details for configuring JasperReports Server 8.1.1, and click **Next.**

Field	Description
Server Name	The name of the server where JasperReports Server 8.1.1 is installed.
Organization	The organization name created in JasperReports Server 8.1.1
Port	The port number that JasperReports Server 8.1.1 uses.
Insights Admin User Name	The user name with complete access to all organizations and tenants.
Admin User	The default admin user is superuser.
Insights Admin Password	The admin password with complete access to all organizations and tenants.
Admin Password	The password to authenticate the database administrator user.
Confirm Insights Admin Password	Reconfirm the admin password with complete access to all organizations and tenants.
Server Protocol	The server protocol used during JasperReports Server 8.1.1 installation.
Webapp Directory	The location within the tomcat folder where JasperReports Server 8.1.1 is deployed. Example, <JasperReports Server 8.1.1 installed directory>\apache-tomcat\webapps\jasperserver-pro

f. (Optional) To enable aggregate reports, select the **Aggregate Reports** checkbox.
On the **Aggregate Reports** window, verify the datamart database server details, and click **Next.**

g. (Optional) To enable transactional reports, select the **Transactional Reports** checkbox.
On the **Transactional Reports** window, verify the transactional database server details, and click **Next.**

h. In the Jasper Database Configuration window, select the required database against Jasper Database Type to automatically increase the password field length of the jijdbcdatasource table on jasperserverdb.
If your JasperReports server is deployed on any of the following databases, perform the following steps:
   a. PostgreSQL Database
   b. Microsoft SQL Server Database
   c. Oracle Database

NOTE

- Ensure that you have the required database client installed based on the Jasper database server type.
- The Jasper Database Admin User is the privileged user who can create and update the Jasper database entries.

PostgreSQL Database

Prerequisites

For successful integration, verify the following.

- Open the pg_hba.conf file under the PostgreSQL directory.
  <jasper-report-server-path>\postgresql\data
  Under # IPv4 local connections, add the following new entry:
  host all all <—IP address-->/32 md5
  IP address of the installed Jasper Server.
- Open the postgresql.conf file under the PostgreSQL directory.
  <jasper-report-server-path>\postgresql\data
  Update the listen_addresses entry from 127.0.0.1 to * as follows and save.
  Example: listen_addresses = '*'

Restart the PostgreSQL database service.

a. Complete the following mandatory fields:
   - Jasper Database Hostname
     Specifies the Jasper Database Hostname.
   - Jasper Database Name
     Specifies the name of the Jasper Database.
     Default: jasperserver
   - Jasper Database Admin User
     Specifies the Jasper Database Admin User name.
     Default: postgres
   - Jasper Database Admin Password
     Specifies the Jasper Database Admin Password.
     Note: On the PosrgreSQL database page the password is pre populated.
   - Jasper Database Port
     Specifies the Jasper Database Port number.
     Default: 5432
   - PostgreSQL Home Path
     Specifies the Jasper directory where the Postgres SQL database is installed.

b. Click Next to proceed.

Microsoft SQL Server Database

a. Complete the following mandatory fields:
   - Jasper Database Hostname

Specifies the Jasper Database Hostname.

- Jasper Database Name
  Specifies the name of the Jasper Database.
  Default: jasperserver
- Jasper Database Port
  Specifies the Jasper Database Port number.
  Default: 1433
- Jasper Database Admin User
  Specifies the Jasper Database Admin User name.
  Default: sa
- Jasper Database Admin Password
  Specifies the Jasper Database Admin Password.

b. Complete the following optional field:
- Jasper Database Server Instance
  Specifies the Jasper Database Server Instance.

c. Click Next to proceed.

Oracle Database

a. Complete the following mandatory fields:
- Jasper Database Hostname
  Specifies the Jasper Database Hostname.
- Jasper Database Port
  Specifies the Jasper Database Port number.
  Default: 1521
- Jasper Database Net Service Name
  Specifies the Jasper Database Net Service Name.
- Jasper Database Admin User
  Specifies the Jasper Database Admin User name.
  Default: system
- Jasper Database Admin Password
  Specifies the Jasper Database Admin Password
- Jasper Database - SID
  Specifies the Jasper Database - SID.

b. Click Next to proceed.

i. Review the information on the **Configuration** window, and click **Next.**

j. Click **Install** to start the integration process.

What's next

After integrating JasperReports Server 8.1.1 with CA Service Management, complete the following tasks:

- Post-installation tasks.
- Additional tasks.

Integrate JasperReports Server 8.1.1 with CA Service Management

Overview of the integration between JasperReports Server 8.1.1 and CA Service Management.

This article provides an overview of the integration between JasperReports Server 8.1.1 and CA Service Management.

Overview

JasperReports Server is a stand-alone and embeddable reporting software that CA Service Management uses for reporting and analytics purposes. CA Service Management integration with JasperReports Server provides enterprise IT management critical information on a real-time or scheduled basis. The integration offers pre-configured reports, dashboard views, and the ability to create sophisticated layouts containing charts, images, subreports, crosstabs, and more that matter for your business objectives.

CA Service Management integration with JasperReports Server provides the following benefits:

- Visualize data quickly using pre-configured reports and interactive dashboard views.
- Generate reports, and customize them to your needs by going one step further than regular table-based reports: Drag-and-drop report designer supporting interactive tables, charts, crosstabs, and analytic views.
- Drill down into reports and analyze them to enable data-driven decision-making.
- Ability to modify underlying data cube on the fly as part of analysis capabilities.

JasperReports Server 8.1.1 installation is a distinct and separate activity within the overall CA Service Management installation process.

You can integrate the JasperReports Server 8.1.1 with CA Service Management using the following information:

Integrate 17.4 and Later

The following sections help you integrate the JasperReports Server 8.1.1 with CA Service Management 17.4 and later.

Prerequisites

For successful integration, verify the following.

- Review the JasperReports Server 8.0.0 supportability matrix from TIBCO Software.

  NOTE
  TIBCO Release 8.0.0 documentation is applicable for Release 8.1.1 too. For example, though the documentation refers to TIB_js-jrs_8.0.0_bin.zip, you can consider it as TIB_js-jrs_8.1.1_bin.zip.
- Ensure that JasperReports Server 8.1.1 is installed.

Enable authentication files for communication

Enable authentication files for communication between JasperReports Server 8.1.1 and CA Service Desk Manager (CA SDM).

Follow these steps:

1. Add contactUUID for superuser.
   a. Log in to JasperReports Server 8.1.1 as a superuser.
   b. Navigate to **Manage** > **Users**.
   c. From the left navigation, select **root**.
   d. Select the **superuser**.
   e. From the **Properties** pane, click **Edit**.
   f. From the **Attributes** tab, click **Add new attribute**.
   g. In the Name field, enter `contactUUID`.
   h. Click **OK** and **Save**.

Integrate using the common installer

To enable integration between JasperReports Server 8.1.1 with CA Service Management, launch the CA Service Management common installer on the server where you installed JasperReports Server 8.1.1.

Follow these steps:

1. On the **Installer selection** window, select **Integrate CABI JasperReports Server with CA Service Management**.
2. On the **License Agreement** window, review the license information, accept the agreement, and click **Next**.
3. On the **Database Configuration** window, in the **Database Type** field, select a database.
4. Enter the details for CA Service Desk Manager (CA SDM) and click **Next**.

Field	Description
Database Server	The location where you've installed the CA SDM database server.
Database Name	The database name on the CA SDM database name.
Database Port	The port number that CA SDM uses.
Database Server Instance	The instance of the database server.
Database Admin User	The database admin user name with permissions to create users and schema. The default user for Microsoft SQL Server is sa.
Database Admin Password	The password to authenticate the database administrator user.
mdbadmin Password	The password for the CA SDM database user.
Confirm mdbadmin Password	Reconfirm the password for the CA SDM database user.

5. On the **CABI JasperReports Server Configuration** window, enter the details for configuring JasperReports Server 8.1.1, and click **Next.**

Field	Description
Server Name	The name of the server where JasperReports Server 8.1.1 is installed.
Organization	The organization name created in JasperReports Server 8.1.1
Port	The port number that JasperReports Server 8.1.1 uses.
Insights Admin User Name	The user name with complete access to all organizations and tenants.
Admin User	The default admin user is superuser.
Insights Admin Password	The admin password with complete access to all organizations and tenants.
Admin Password	The password to authenticate the database administrator user.
Confirm Insights Admin Password	Reconfirm the admin password with complete access to all organizations and tenants.
Server Protocol	The server protocol used during JasperReports Server 8.1.1 installation.
Webapp Directory	The location within the tomcat folder where JasperReports Server 8.1.1 is deployed. Example, <JasperReports Server 8.1.1 installed directory>\apache-tomcat\webapps\jasperserver-pro

6. (Optional) To enable aggregate reports, select the **Aggregate Reports** checkbox.
   On the **Aggregate Reports** window, verify the datamart database server details, and click **Next.**
7. (Optional) To enable transactional reports, select the **Transactional Reports** checkbox.

On the **Transactional Reports** window, verify the transactional database server details, and click **Next.**

8. In the Jasper Database Configuration window, select the required database against Jasper Database Type to automatically increase the password field length of the jijdbcdatasource table on jasperserverdb.
   If your JasperReports server is deployed on any of the following databases, perform the following steps:
   a. PostgreSQL Database
   b. Microsoft SQL Server Database
   c. Oracle Database

      NOTE

      - Ensure that you have the required database client installed based on the Jasper database server type.
      - The Jasper Database Admin User is the privileged user who can create and update the Jasper database entries.

PostgreSQL Database

Prerequisites

For successful integration, verify the following.

- Open the pg_hba.conf file under the PostgreSQL directory.
  <jasper-report-server-path>\postgresql\data
  Under # IPv4 local connections, add the following new entry:
  host all all <—IP address-->/32 md5
  IP address of the installed Jasper Server.
- Open the postgresql.conf file under the PostgreSQL directory.
  <jasper-report-server-path>\postgresql\data
  Update the listen_addresses entry from 127.0.0.1 to * as follows and save.
  Example: listen_addresses = '*'

Restart the PostgreSQL database service.

a. Complete the following mandatory fields:
   - Jasper Database Hostname
     Specifies the Jasper Database Hostname.
   - Jasper Database Name
     Specifies the name of the Jasper Database.
     Default: jasperserver
   - Jasper Database Admin User
     Specifies the Jasper Database Admin User name.
     Default: postgres
   - Jasper Database Admin Password
     Specifies the Jasper Database Admin Password.
     Note: On the PosrgreSQL database page the password is pre populated.
   - Jasper Database Port
     Specifies the Jasper Database Port number.
     Default: 5432
   - PostgreSQL Home Path
     Specifies the Jasper directory where the Postgres SQL database is installed.

b. Click Next to proceed.

Microsoft SQL Server Database

a. Complete the following mandatory fields:
   - Jasper Database Hostname

Specifies the Jasper Database Hostname.

- Jasper Database Name
  Specifies the name of the Jasper Database.
  Default: jasperserver
- Jasper Database Port
  Specifies the Jasper Database Port number.
  Default: 1433
- Jasper Database Admin User
  Specifies the Jasper Database Admin User name.
  Default: sa
- Jasper Database Admin Password
  Specifies the Jasper Database Admin Password.

b. Complete the following optional field:

- Jasper Database Server Instance
  Specifies the Jasper Database Server Instance.

c. Click Next to proceed.

Oracle Database

a. Complete the following mandatory fields:

- Jasper Database Hostname
  Specifies the Jasper Database Hostname.
- Jasper Database Port
  Specifies the Jasper Database Port number.
  Default: 1521
- Jasper Database Net Service Name
  Specifies the Jasper Database Net Service Name.
- Jasper Database Admin User
  Specifies the Jasper Database Admin User name.
  Default: system
- Jasper Database Admin Password
  Specifies the Jasper Database Admin Password
- Jasper Database - SID
  Specifies the Jasper Database - SID.

b. Click Next to proceed.

9. Review the information on the **Configuration** window, and click **Next.**
10. Click **Install** to start the integration process.

NOTE
The required JARS and XML files are copied automatically. On the progress page, you may view the following password increment and copy tasks.

- Password Field Length Increment On JasperReport Server Database
- Copying Custom Files to JasperReport Server

Post-installation tasks

After integrating JasperReports Server 8.1.1 with CA Service Management, complete the following post-integration tasks to have Insights up and running.

- Enable Insights from the xFlow Interface
- Enable Insights from CA SDM

Enable Insights from the xFlow Interface

An administrator must enable Insights app for it to appear on the xFlow Interface.

Follow these steps:

1. Restart the JasperReports Tomcat Server.
2. Login to CA SDM as an administrator.
3. Navigate to the **Administration** tab, select **xFlow Interface**, **App Configuration**, and **Insights**.
4. Click **Edit**.
5. From the drop-down list, select **Active**, and click **Save**.
6. Restart the xFlow Interface services based on your configuration.
   a. For Advanced Availability: Restart services on the application server.
   b. For Conventional: Restart services on the primary and secondary servers.

Result: Insights app is enabled and available from the xFlow Interface.

Enable Insights from CA SDM

If you have not installed the xFlow Interface or Insights app, you can access Insights directly from CA SDM.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to **Administration**, **Security and Role Management**, **Role Management**, and **Role List.**
3. Select a role.
4. Select the Resources tab and click **Update Tabs** to select the Insights tab.
5. Assign the insights app to the selected role.
6. Log in to CA SDM with the selected role.
7. From the Menu bar, select **Insights** and click **Launch Insights**.

Result: Insights app is enabled and available for use from CA SDM.

Additional tasks

After successfully integrating JasperReports Server 8.1.1 with CA Service Management, review the following additional tasks. You can complete the tasks that you need for your environment.

- Configure the data aggregation date
- Match the duration format

Configure the data aggregation date

Get accurate and consistent reports by configuring the data aggregation date to a date later than that of CA Service Management 17.4 implementation. If the Data Aggregation Date is not configured, inaccurate information is displayed, and the following reports are impacted:

- Incidents Cost for Top Categories (USD)
- Incidents Cost Trend for Top Categories (USD)
- MTTR for Top Categories (In Hours) Report
- MTTR Quarterly Trend for Top Categories (In Hours) Report
- MTTR Weekly Trend for Top Categories (In Hours) Report
- MTTR Yearly Trend for Top Categories (In Hours) Report

Follow these steps:

1. Navigate to the `<datamart installed folder>\config\config.properties` file.

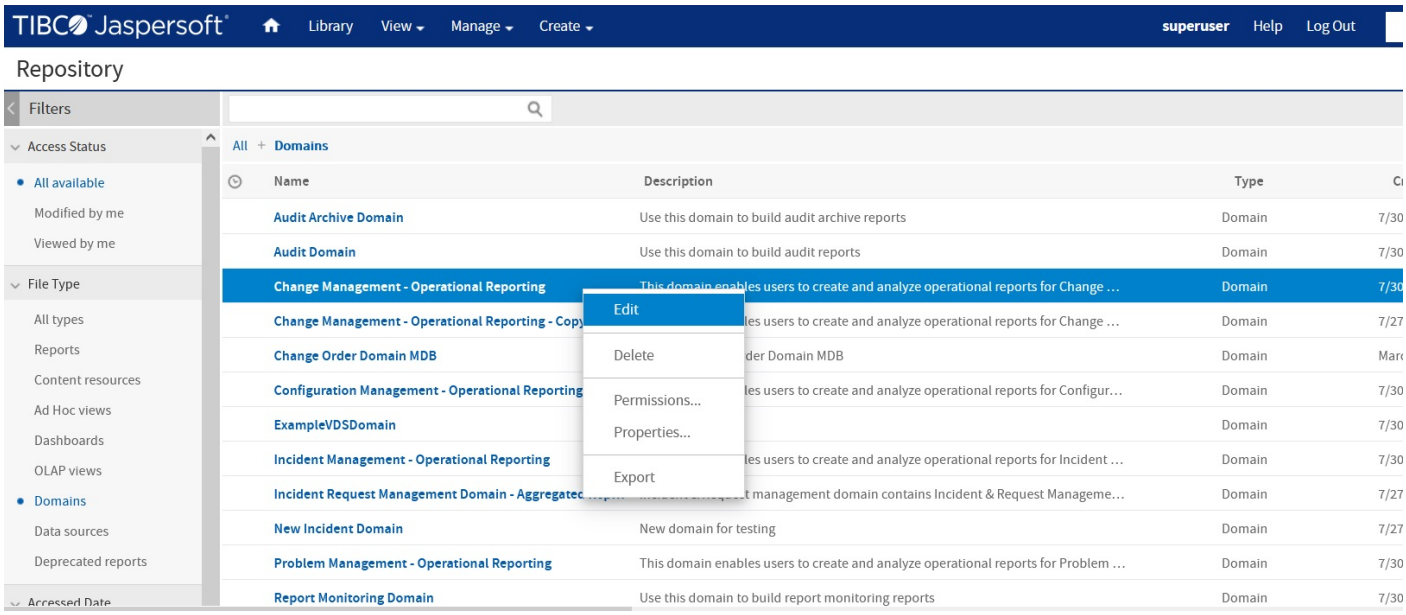
2. Search for the string **trend_start_date='yyyy-mm-dd'**.
3. Replace the **trend_start_date** value with the data aggregation date.
   During the next ETL job, the data older than the data aggregation date is deleted.

Match the duration format

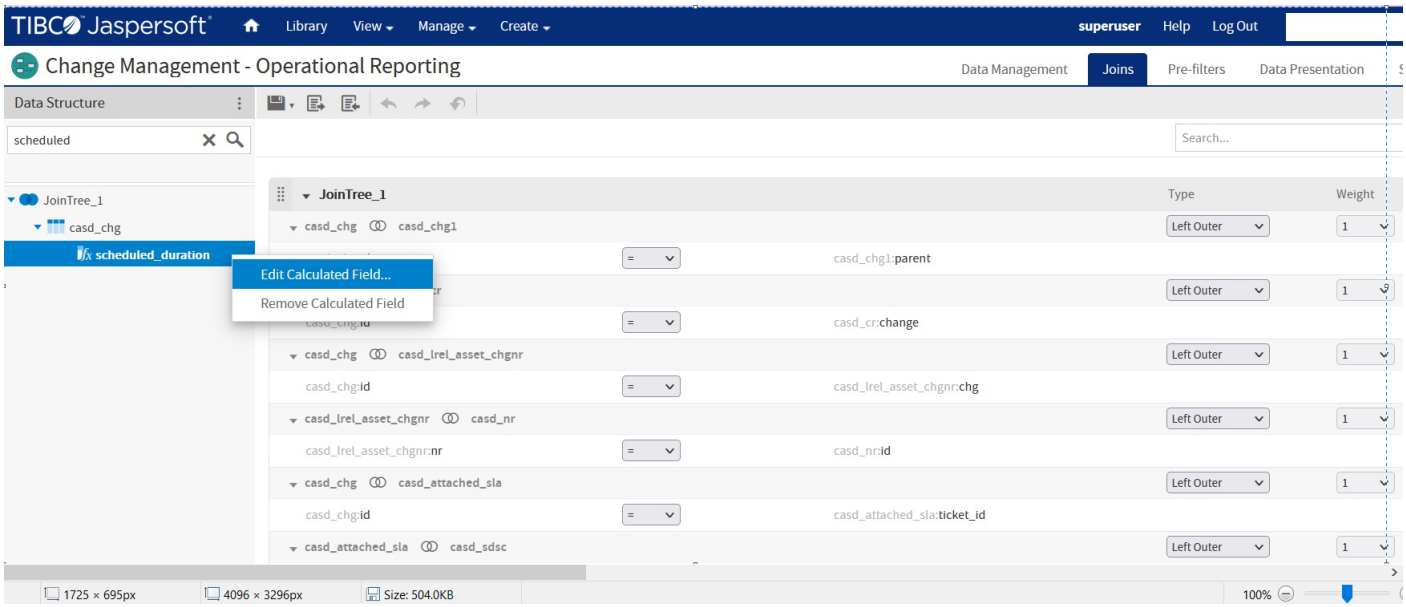
The duration format is represented differently in CA SDM and Insights. You can match the duration format to appear in HH:MM:SS format in both CA SDM and Insights.

Follow these steps

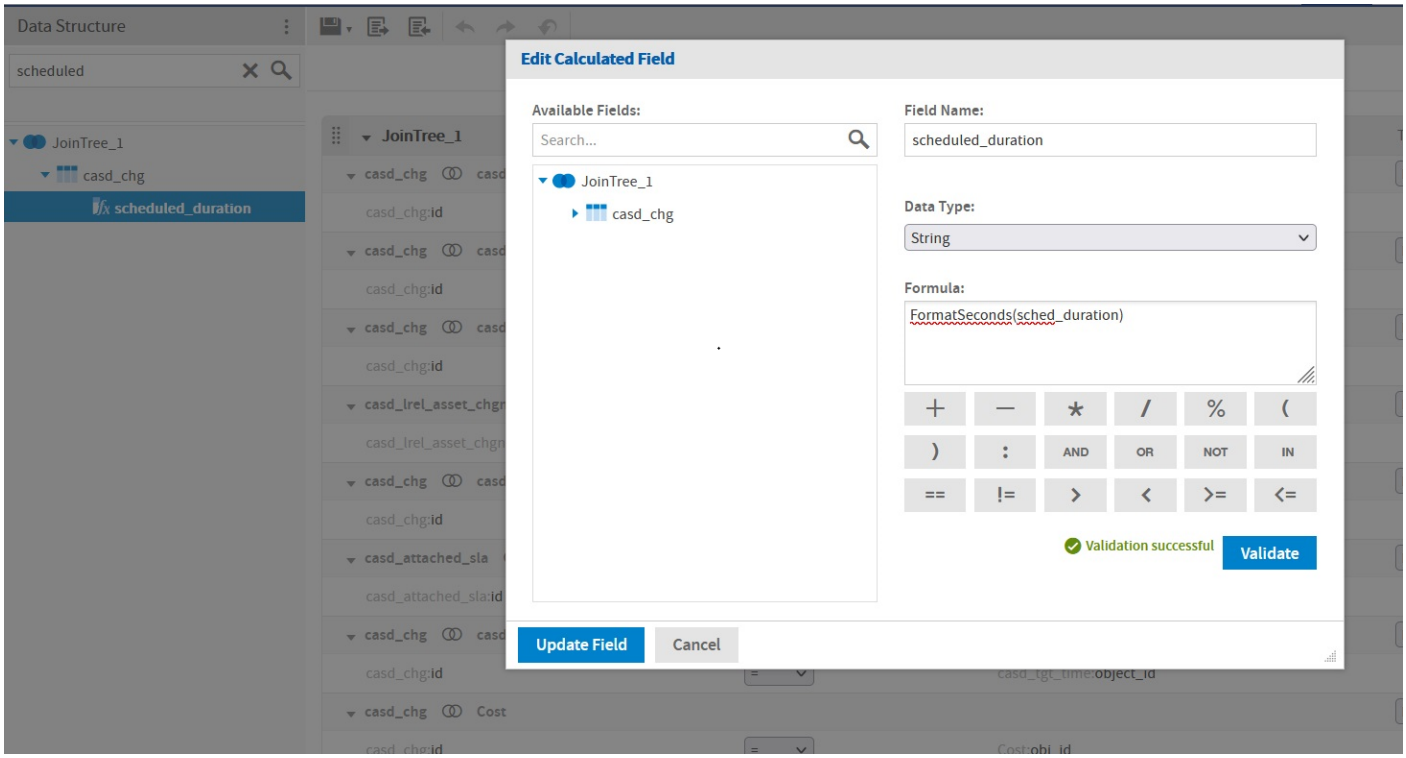
1. Connect to the Jasper Reports server machine.
2. Copy the *DurationFormatter.jar* from CASM_17.4.\filestore\Jasper\8.1.1\ExternalAuthJars and, copy it into the jasperserver tomcat lib folder. For eg: *C:\apache-tomcat-9.0.65\webapps\jasperserver-pro\WEB-INF\lib*.
3. From the service console, stop the service: jasperreportsTomcat.
4. From the folder where you installed JasperReports Server 8.1.1, navigate to the `apache-tomcat\webapps\jasperserver-pro\WEB-INF` folder and back up applicationContext-el-operators.xml file.
5. From the folder where you installed JasperReports Server 8.1.1, navigate to the `apache-tomcat\webapps\jasperserver-pro\WEB-INF\bundles` folder and back up the following files.
   - adhoc_messages.properties
   - adhoc_messages_es.properties
   - adhoc_messages_fr.properties
   - adhoc_messages_ja.properties
   - adhoc_messages_pt_BR.properties
6. Navigate to the DVD `filestore\Jasper\8.1.1\ExternalAuthXMLs` folder and copy the applicationContext-el-operators.xml file.
7. From the folder where you installed JasperReports Server 8.1.1, navigate to the `apache-tomcat\webapps\jasperserver-pro\WEB-INF` folder and paste the copied file.
8. Navigate to the DVD `filestore\Jasper\8.1.1\Bundles` folder and copy the following files.
   - adhoc_messages.properties
   - adhoc_messages_es.properties
   - adhoc_messages_fr.properties
   - adhoc_messages_ja.properties
   - adhoc_messages_pt_BR.properties
9. From the folder where you installed JasperReports Server 8.1.1, navigate to the `apache-tomcat\webapps\jasperserver-pro\WEB-INF\bundles` folder and paste the copied files.
10. Navigate to the DVD `filestore\Jasper\8.1.1\Groovy` folder and copy the BaseGroovyColumn.groovy file.
11. From the folder where you installed JasperReports Server 8.1.1, navigate to the `apache-tomcat\webapps\jasperserver-pro\WEB-INF\groovy\groovy_column` folder and paste the copied file.
12. From the folder where you installed JasperReports Server 8.1.1, navigate to the `apache-tomcat\temp` folder and empty the folder.
13. From the folder where you installed JasperReports Server 8.1.1, navigate to the `apache-tomcat\work` folder and empty the folder.
14. From the service console, start the service: jasperreportsTomcat.
15. Clear the browser cache.
16. Log in to Jasper UI in the browser, and edit the required domain. This image shows how to edit the Change Management - Operational Reporting domain.



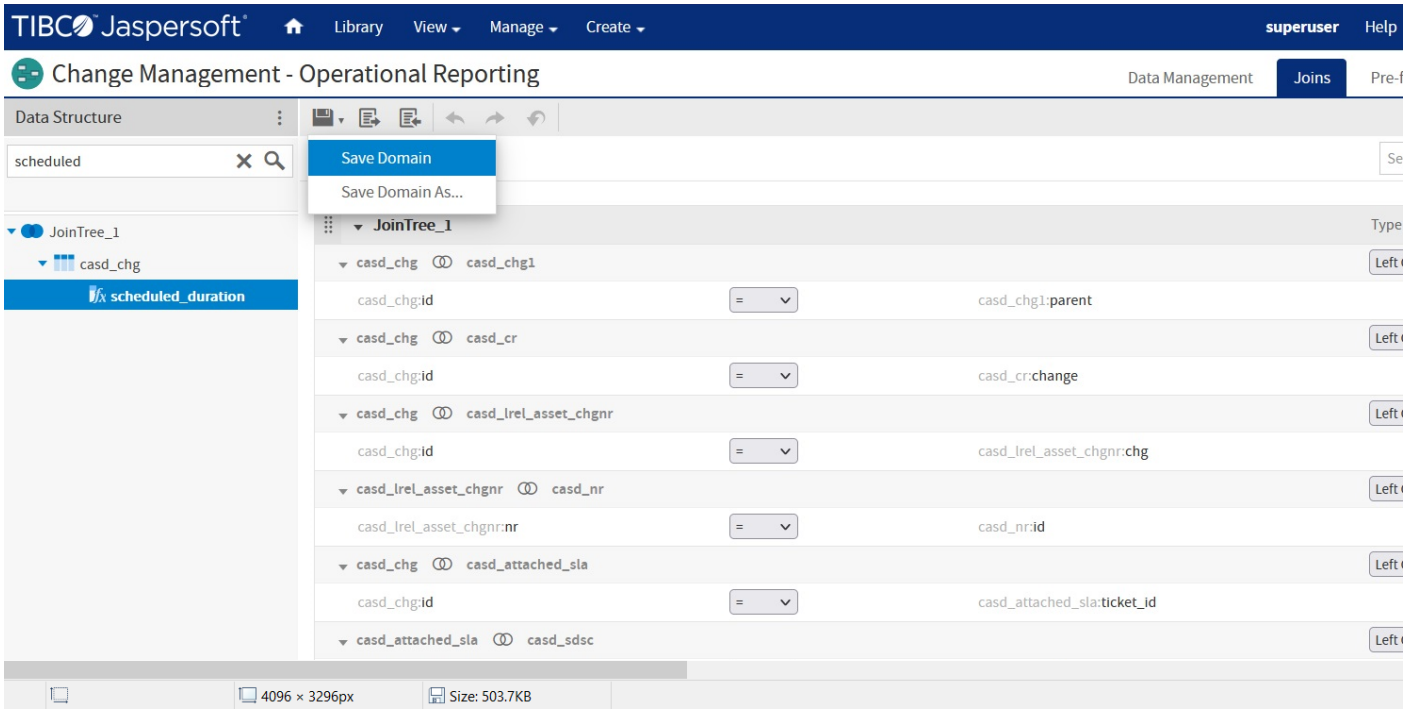
17. Under the Joins tab, search for the required calculated field and right-click the field to edit it. This image shows how to edit the scheduled_duration field.



18. Under Formula, edit the formula for the calculated field. This image shows the formula for the scheduled_duration field.



19. Change the data type to String.
20. Click **Validate**.
21. Click **Update Field**.
22. To save the domain, click the **Save** icon.



23. To verify that the duration value is displayed in HH: MM: SS format, use this field in your reports.

Duration Measure error for default users

Follow these steps:

1. Login to jasperserver application.
2. Click **Domains**.
3. Right click on the required Domain (e.g.: Incident Management- Operational Reporting) to modify and click **Edit.**
4. Click **Joins** and search for the Calculated field (e.g.: Resolution_Time_Hours) that are used for measures.
5. Right click on calculated field and click **Edit Calculated Field**.
6. Remove the SUM from the formula then click **Validate** and **Update Field**.
7. Save the Domain.
8. Restart the Jasperreports Tomcat service.

If CA SDM is on Oracle follow these additional steps

1. Navigate to the location where Jasper server is installed. (e.g.: C:\apache-tomcat-9.0.65\webapps\jasperserver-pro)
2. Under the folder WEB-INF, select and edit *applicationContext-semanticLayer.xml*
3. Search for the tag
   ```
   <bean id="sqlserverGenerator" parent="defaultSQLGenerator" scope="prototype">
   ```
4. Find the entry <entry key ="divide"> under it.
5. For removing the data type *casting of money*, change
   "cast ($top as money) / (case when (($bottom) = 0) then null else ($bottom) end)" as String
   to *"($top) / (case when (($bottom) = 0) then null else ($bottom) end)"* as String.
6. Save the applicationContext-semanticLayer.xml
7. Restart the Jasper server tomcat service.

Configure JasperReports Server 8.1.1 High Availability Set Up Cluster

This document provides updates on how to Configure JasperReports Server 8.1.1 High Availability Set Up Cluster.

Prerequisites

The following are the prerequisites:

- Complete the installation of Jasper 8.1.1 with supported databases. (Call it Node 1)
- Verify the Jasper URL.
- Install ActiveMQ on any one server in the cluster.
- Install JDK (OpenJDK11) and Apache Tomcat (9) on all the nodes
- Integrate SDM or xFlow to the JasperReports Server 8.1.1

EhCache Replication

The repository cache in JasperReports Server is implemented internally through the Ehcache component. Edit the Ehcache configuration files as described below to replicate the repository cache among all instances that share a single repository. You do not have to configure a cluster for cache replication.

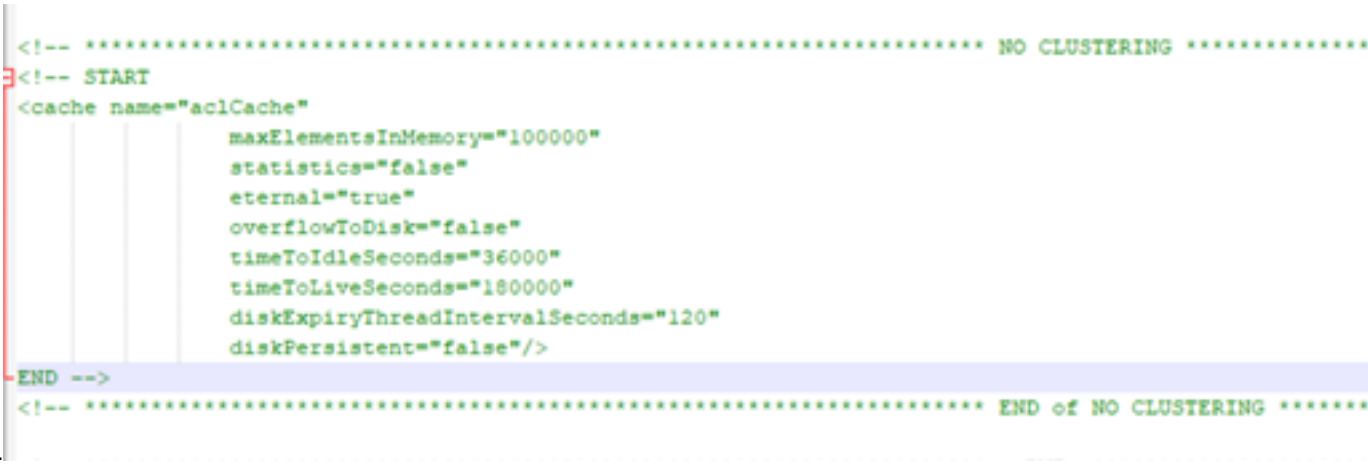
JMS – Java Message Services can provide cache distribution for nodes in a cloud where IP addresses may change. Jaspersoft provides a configuration for using the Apache ActiveMQ JMS Server. You must first install and configure ActiveMQ on one of the computers in your cluster.

Configure a High Availability Environment on Node 1

Perform the following steps on JasperReports Server, Node 1:

Follow these steps:

1. Stop the JasperReports Server Apache Tomcat service on Node1.
2. Take a backup of the following files on each node:
   Open the file <web-app>/WEB-INF/classes/ehcache_hibernate.xml and <web-app>/WEB-INF/ehcache.xml on each node.
   Example:C:\Jaspersoft\jasperreports-server-8.1.1\apache-tomcat\webapps\jasperserver-pro
3. In both ehcache_hibernate.xml and ehcache.xml files, comment out or remove the section of the file in between "NO CLUSTERING" and "END of NO CLUSTERING" comments.



4. Uncomment the JMS section in <web-app>/WEB-INF/classes/ehcache_hibernate.xml and <web-app>/WEB-INF/ehcache.xml on each node.
   Example:

5. Set the providerURL properties in both files to the address of your JMS server.
   For example, there are several providerURL properties to set in each element; only the first one is shown in the following code example:

```
<!-- START
        <cacheManagerPeerProviderFactory
                class="net.sf.ehcache.distribution.jms.JMSCacheManagerPeerProviderFactory"

properties="initialContextFactoryName=com.jaspersoft.jasperserver.api.engine.replication.JRSActiveMQInitialContextF
                providerURL=tcp://ActivemQ_host:61616,
                replicationTopicConnectionFactoryBindingName=topicConnectionFactory,
                replicationTopicBindingName=ehcacheAcl,
                getQueueConnectionFactoryBindingName=queueConnectionFactory,
```

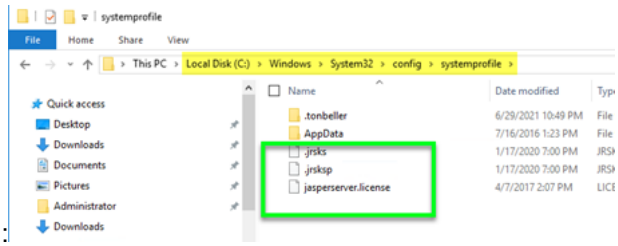
```
getQueueBindingName=ehcacheQueueAcl,
topicConnectionFactoryBindingName=topicConnectionFactory,
topicBindingName=ehcacheAcl"
propertySeparator=","/>
```

6. On each node, edit the file <web-app>/META-INF/context.xml. Locate the Manager pathname near the end, and comment it out as follows:
   C:\Jaspersoft\jasperreports-server-8.1.1\apache-tomcat\webapps\jasperserver-pro
   ```
   <!-- Manager pathname="" -->
   ```

Configure a High Availability Environment on Node 2

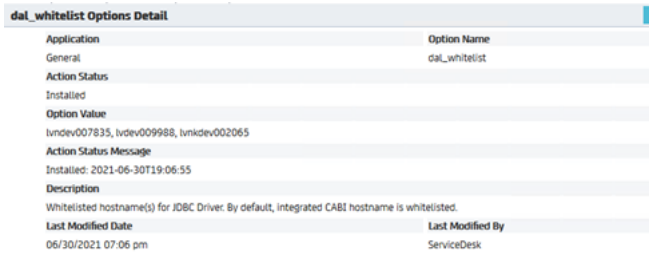
Perform the following steps on JasperReports Server, Node 2:

Follow these steps:

1. Use the same installer that is used in Node 1 (Use original installer).
2. Copy the default_master.properties file from Node 1 to Node 2.
3. Stop Apache Tomcat services.
4. Ensure that Apache Tomcat and Database details are pointing to the correct database.
   Example:
   # Tomcat app server root dir
   appServerDir = C:\\Program Files\\Apache Software Foundation\\Tomcat 9.0
   # appServerDir = /home/devuser/apache-tomcat-9.0
   If it is a Linux package managed Tomcat instance, set the following two properties:
   – CATALINA_HOME = C:\\Program Files\\Apache Software Foundation\\Tomcat 9.0
   – CATALINA_BASE = C:\\Program Files\\Apache Software Foundation\\Tomcat 9.0
5. Install only webapp in Node 2.
   Navigate to the C:\TIB_js-jrs_8.1.1_bin\jasperreports-server-pro-8.1.1-bin\buildomatic folder and run the following command
   ```
   js-ant deploy-webapp-pro
   ```
6. Ensure you have a non default port under:
   C:\Program Files\Apache Software Foundation\Tomcat 9.0\webapps\jasperserver-pro\WEB-INF
   diagnostic.port = 10990
   Change the port number.
7. Repeat the steps 3-6 from the above Configure a High Availability Environment on Node 1 section.
8. Copy the following files from NODE1's system user directory from when you installed Jasper 8.1.1, to NODE2's system user directory. For example, C:\Users\Administrator.
   – .jrsks
   – .jrslic
   – .jrsksp
   – Jasperserver.license
   In some cases, you may need to copy the files to the systemprofile location:
   

9. Start Apache Tomcat services.

10. Configure SDM to accept communication from the new Node by updating the dal_whitelist option.



11. Make sure you can view your reports. Create a new adhoc view on NODE2; log in to JasperReports Server, Node1 and verify you can view and run the report from Node 1.

Configure JasperReports Server 8.1.1 to Use SSL

You can configure JasperReports Server to use Secure Socket Layer (SSL) and communicate with CA Service Management. SSL establishes an encrypted link between a server and a client. For example, a web server and a browser; or a mail server and a mail client. This link helps ensure that all data passed between the server and client remain private and integral. When you configure JasperReports server to use SSL, you change its communication method from HTTP to HTTPS while integrating with CA Service Management.

Follow these procedures:

Create a Keystore File

A keystore file is required to enable SSL. Create a keystore file if you do not have one already for another CA product that integrates with CA Service Management. You can use a keystore file for a single product or for multiple products. If you must create individual keystores for each product, you can optionally merge your keystore files.

Follow these steps:

1. Open a command window on the JasperReports server. Navigate to the location where JasperServer is installed. Example: C:\Jaspersoft\jasperreports-server-8.1.1\java\bin>
2. Execute the following command:

   ```
   keytool -genkey -alias tomcat -keyalg RSA -keystore <Selected Path>/<filename>.keystore
   ```

   Enter the required information when prompted.
3. Enter the password at the “Enter keystore password” prompt. To make configuring Tomcat easier, you can use “changeit” as the password.
4. Record your password for reference.

Add the Certificate to the Keystore

After creating the keystore file, you must add a CA certificate to the keystore.

Follow these steps:

1. Execute the following command to add the certificate into the keystore.

   ```
   keytool -importkeystore -srckeystore <Selected Path>/<filename>.keystore -destkeystore <Selected Path>/<filename>.keystore -deststoretype pkcs12
   keytool -export -storepass PASSWORD -alias tomcat -keystore <Selected Path>/<filename>.keystore -file <Selected Path>/<filename>/Jasperssl.crt
   ```

   The certificate is exported to keystore.

Edit the Server.xml File to Support SSL

As part of configuring the JasperReports server to use Secure Socket Layer (SSL), edit the server.xml file to support SSL.

Follow these steps:

1. Navigate to the JasperServer installed location.
2. Edit the <*Jaspersever Installed Location*>\apache-tomcat\conf**server.xml** file.
3. Search for the <Service name="Catalina"> section.
4. Updated the Connector related information under the <Service name="Catalina"> section:

```
<Connector SSLEnabled="true" maxThreads="200" port="8443"
 protocol="org.apache.coyote.http11.Http11NioProtocol"
 sslImplementationName="org.apache.tomcat.util.net.jsse.JSSEImplementation">
      <SSLHostConfig
 ciphers="TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384,TLS_DHE_RSA_WITH_AES_128_GCM_S
 TLS_DHE_RSA_WITH_AES_256_GCM_SHA384,TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256,TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA,
 TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384,TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA,TLS_DHE_RSA_WITH_AES_128_CBC_SHA256,
 TLS_DHE_RSA_WITH_AES_128_CBC_SHA,TLS_DHE_RSA_WITH_AES_256_CBC_SHA256,TLS_DHE_RSA_WITH_AES_256_CBC_SHA,
 TLS_RSA_WITH_AES_128_GCM_SHA256,TLS_RSA_WITH_AES_256_GCM_SHA384,TLS_RSA_WITH_AES_128_CBC_SHA256,
 TLS_RSA_WITH_AES_256_CBC_SHA256,TLS_RSA_WITH_AES_128_CBC_SHA,TLS_RSA_WITH_AES_256_CBC_SHA,
 SSL_RSA_WITH_3DES_EDE_CBC_SHA" honorCipherOrder="true" protocols="TLSv1.2" sslProtocol="TLSv1.2">
        <Certificate certificateKeyAlias="<Alias Name>" certificateKeystoreFile="<Selected
 Path>/<filename>.keystore" certificateKeystorePassword="PASSWORD" certificateKeystoreType="PKCS12"/>
      </SSLHostConfig>
   </Connector>
```

   – Default value before making this change.

```
<Connector port="8080" URIEncoding="UTF-8" protocol="HTTP/1.1" connectionTimeout="20000"
 redirectPort="8443"/>
```

5. Save and close the server.xml file.
6. Restart the Tomcat server.
7. Now you can launch the Jaspersoft url using https://HOST_NAME:SSL_PORT/jasperserver-pro/flow.html?_flowId=homeFlow
   Ensure that Jasper URL is accessible.

Integrate CA SDM with the SSL-enabled JasperReports Server

Configure the SSL-enabled JasperReports Server to integrate with CA SDM using the CA Service Management Common Installer.

Follow these steps:

1. Download the common installer to the required location in the JasperServer.
   Example: C:\Users\Administrator\Downloads\<*RUXXXXX*>
2. Extract the installer and go to the Java folder
   Example: \java\java11
3. To integrate HTTPS enabled Jasper Server with Common Installer, the certificate you exported in the *Add the Certificate to the Keystore* section should be imported by executing the following command:

```
keytool -importcert -alias <aliasname>-file <Selected Path>/<filename>.crt -keystore  ..\lib\security
\cacerts
```

4. Launch the CA Service Management Common Installer.
5. Provide the following mandatory information in the "CABI JasperReports Server Configuration" screen.
   Enter the port number that you have provided in the server.xml file.
   Select HTTPS from the Server Protocol drop-down.
6. Proceed with the next steps in the wizard to complete the integration.

NOTE
After the integration is successful, log in to the classic CA SDM, Administration, xFlow Interface, General, and confirm the value *insights.jasper.domain* is populated in the General Configuration List.

Enable JasperServer SSL URL from CA SDM Classic UI

1. Navigate to <jasper_installation_path>\jasperserver-pro\WEB-INF\config and copy the following files.
   – casm_insights.jks
   – casm_insights.properties
2. Paste the copied files from **step 1** to CA SDM Server.**For example**: NX_ROOT\site folder.
3. Restart CA Service Desk Manager Server and JRS services.

NOTE
Integration of CA Service Management Release 17.4 with Jasper Reports Server 8.1.1 is automated.

Add JasperReports Server SSL Certificate to the CA SDM

To launch HTTPS enabled JasperReports server from CA SDM Classic UI, the certificate you exported in the *Add the Certificate to the Keystore* section should be imported.

Follow these steps:

1. Navigate to the JRE bin folder in the CA SDM server.
   For example, C:\Program Files (x86)\CA\SC\JRE\11.0.12\bin
2. Import the certificate or any trusted certificate.
   ```
   keytool -importcert -alias <aliasname>-file <Selected Path>/<filename>.crt -keystore  ..\lib\security
   \cacert
   ```
3. You can now launch Insights from CA SDM classic UI.

Adding JasperReports Certificate to Insights Trust Store

For Insights to connect to HTTPS enabled JasperReports server, the certificate you exported in the *Add the Certificate to the Keystore* section should be imported into the Insights trust store.

1. Navigate to <xFlowInstallation Home Dir>\jre\bin
   Example: C:\Program Files\CA\SC\JRE\11.0.3\bin
2. Import the certificate or any trusted certificate.
   ```
   keytool -importcert -alias <aliasname>-file <Selected Path>/<filename>.crt -keystore  ..\lib\security
   \cacert
   ```
3. Restart the CA xFlow Analyst Interface services.

Install CA Process Automation for CA Service Management

The CA Process Automation integrates with the CA Service Management products to automate the workflow business processes. For more information, see CA Process Automation Documentation.

WARNING

If you use CA Process Automation, we recommend that you do *not* install the CA Process Automation domain orchestrator and CA Process Automation components on the same computer.

CA Process Automation Installation in an HTTPS Environment

Before starting the installation or upgrade process, if your CA Process Automation instance is SSL enabled (configured with https), ensure that you add a CA Process Automation SSL certificate to the Java Run time Environment (JRE) keystore. Without this certificate, JRE, under which the installer is running, does not connect to the CA Process Automation SSL URL and the following SSL validation errors may appear during the installation:

```
2014/11/11 19.04.39.728 DEBUG [AWT-EventQueue-0] [PAMWebserviceManager] Get ITPAM version
2014/11/11 19.04.39.752 ERROR [AWT-EventQueue-0] [PAMValidator] CA PAM validation failed
com.ca.smsi.installcore.webservices.WebserviceException: ; nested exception is:
javax.net.ssl.SSLHandshakeException: sun.security.validator.ValidatorException: PKIX path building failed:
 sun.security.provider.certpath.SunCertPathBuilderException: unable to find valid certification path to
 requested target
at com.ca.smsi.installcore.webservices.PAMWebserviceManager.getITPAMVersion(PAMWebserviceManager.java:265)
```

Complete the following steps to add a CA Process Automation SSL certificate to the JREkeystore:

1. Download the SSL Certificate by accessing the CA Process Automation URL.
   Copy this certificate file to some location. For example, c:\ItpamCert.cer
2. Extract the DVD to a local drive.
3. Import the Process Automation SSL Certificate to a keystore file using the following command:
   ```
   ..\java\jre\bin\keytool -importcert -file c:\ItpamCert.cer -alias somealias -keystore c:/keystore_file -
   storepass somepass\
   ```
4. To enter the keystore and password values to the JRE, open the **setup.lax** file in WordPad from where you have installed CA Service Management or a similar text editor.
   `To avoid any formatting issues, it is recommended that you do not use the Notepad.`
5. Identify the "**lax.nl.java.option.additional**" property.
6. Append the reference to the JAVA keystore and keystorepass entries as follows:
   ```
   -Djavax.net.ssl.trustStore=c:/keystore_file -Djavax.net.ssl.trustStorePassword=somepass
   ```
   After the modification, the line would look like:
   ```
   lax.nl.java.option.additional=-Djava.library.path="\\java\\bin; -DGET_SD_PRESETS=true -
   Djavax.net.ssl.trustStore=c:/keystore_file -Djavax.net.ssl.trustStorePassword=somepass
   ```
7. Save the file and restart the installer by executing a **setup.exe** command.

CA Process Automation Integration with CA Embedded Entitlements Manager (EEM)

If you have installed EEM and Process Automation and have integrated one or more CA Service Management products with Process Automation and EEM, ensure that you provide the same EEM information that you provided while installing Process Automation for the integration to be successful.

You can configure Process Automation to use EEM as an authentication server.

If you have installed CA EEM and CA Process Automation and integrate one or more CA Service Management products with these common components, ensure that you provide the same EEM information that you provided while installing Process Automation for auto-integration to be successful

The Process Automation uses EEM for user authentication and authorization. **CA EEM is a required prerequisite**.

- If you use EEM with CA Service Management, verify that it is a version that CA Process Automation supports. See Platform Support and Requirements.
- If you do not have EEM or your EEM instance is of an earlier version, then download and install EEM.
- Plan how to authenticate CA Process Automation users.
- Gather EEM-related information for the Domain Orchestrator installation.
- To create two EEM instances during the installation (one to use and another for failover). You can complete this optional procedure later.

Installing CA Process Automation

Follow the steps:

1. Launch the CA Service Management Installer by right-click **Setup.exe** to select the **run as administrator** option.
2. Select the language and click **Next**

3. Follow the installation wizard to install or upgrade CA Process Automation.

Generating CASM Reports with APIs

Reporting APIs are supported from CA SDM 17.4.4.0.

This page talks about:

- Overview of Reporting API's
- Deployment
- Getting started with the API's
- Logging
- Limitations

Overview

Reporting is a crucial feature of CA SDM, enabling administrators, service owners, and service delivery managers to analyze service desk performance, ticket management, and workload trends. These reports are key for assessing performance and identifying areas for improvement. Previously, we integrated JasperSoft transactional Reporting to provide visualizations, drill-down capabilities, and custom dashboards. However, as JasperSoft will no longer be supported, we are transitioning to a more flexible and extensible reporting solution based on APIs.

> **NOTE**
>
> The current reporting APIs do not support or expose endpoints for Aggregated Reports. Hence out-of-the-box Aggregated reports will also not be available.
>
> **Custom Reports**
>
> To create custom reports, connect your reporting tool to ***casm_dm*** (datamart) using the appropriate Microsoft SQL driver.

API-Based Transactional Reporting Solution:

The new reporting APIs offer enhanced flexibility by allowing users to select and integrate their preferred Business Intelligence (BI) tools, as long as they are compatible with API integration. This shift provides the ability to create custom dashboards and reports tailored to specific business needs, thus extending the overall reporting functionality of CA SDM.

List of APIs

1. **API Overview**
   - Title: CASM Reports
   - Description: The API allows users to interact with CASM Reports, including accessing reports, designing custom reports, and more.
2. **Available Operations (Endpoints)**
   - **Access Token Operations:**
     - POST /v1/access: Retrieve an access token for the specified user. You must use basic authentication for this.
     - DELETE /v1/access: Delete the access token for the user. Requires the token to be passed in the X-AccessToken header.
   - **Report Operations:**

- GET /v1/getReportCodes: Retrieve a list of available report codes with descriptions.
- GET /v1/getReport: Fetch a predefined report based on the provided report code.
- GET /v1/{factoryName}/: Retrieve a custom list-type report. You can specify the fields, filters, and sorting in headers.
- POST /v1/{factoryName}/createReport: Create a new report based on a specified factory name (in, cr, pr, etc.). The request body should contain the report data in JSON format.
- DELETE /v1/{factoryName}/deleteReport: Delete a report identified by a report code, which must be passed in the header.
- PUT /v1/{factoryName}/updateReport: Updates the existing reports.
- GET /v1/getReportMetaData: Fetch metadata for a report, requiring the report code.

NOTE

- The API "v1/getReport" endpoint gives the report data for a given report code and provides the "createReportMetaData" payload. This payload can be used to customize the OOTB report or creating new report using the "v1/{factoryName}/createReport" API.
- The "/v1/{factoryName}/deleteReport" and "/v1/{factoryName}/updateReport" API endpoints are applicable only for Custom reports. OOTB reports cannot be deleted or updated.

– **ITIL Reports:**
  - GET /v1/itilreport: Retrieve ITIL reports based on various filter options, including report code, ticket type, service, and other parameters.

3. **Parameters and Authentication:**
   – Path Parameters: For example, factoryName is required in many endpoints to specify the type of report or factory (e.g., in, cr).
   – Query Parameters: For e.g.: Use top (pagination) and skip to control the number of results returned.
   – Headers: Headers like X-Report-Code, X-Select, and X-Filter are used to define which report or data is returned and can apply filters and sorting.
4. **Request Body Example for Creating/Updating Reports:**
   – When creating or updating a report, the request body must be in the format of a CASMReport object, which includes:
     - id: A unique identifier for the report.
     - r_name: The name of the report.
     - attributes: A collection of key-value pairs representing the attributes of the report.

   Example of API Usage:
   – **To Create a Report**
     - Endpoint: POST /v1/{factoryName}/createReport
     - Required Header: X-AccessToken (your authentication token)
     - Body: A CASMReport object specifying the report’s details.
   – **To Fetch a Predefined Report by Code:**
     - Endpoint: GET /v1/getReport
     - Required Header: X-Report-Code (the code for the report you are fetching)
5. **Security:**
   – Basic Authentication: Required for accessing the POST /v1/access endpoint.
   – API Key (X-AccessToken): Required for many other endpoints to authorize requests. This is passed in the request header.

Deployment

Enable reporting API services by executing pdm_configure wizard.

1. After the successful installation of 17.4 RU4, perform pdm_configure on the SDM Server (Primary server or APP server or Secondary Server).
2. In the pdm_configure wizard, you can see a new screen **Reporting Services** for enabling the reporting services, with the required ports.

   NOTE

   For Secondary Server:

   1. To run the reporting services on the Secondary server, update the existing configuration under **Administration>> System>> Configurations**.
      In the screen displayed, update the configuration with Current Configuration status as **Yes.** Now click **Additional Processes** tab>>click **Add Process**. In the screen displayed under **Process** drop down, select **Reporting Service Tomcat Server** option.
      Update the **Host Name** with a secondary server(s) hostname.
   2. Run *pdm_configure* on Secondary server(s), followed by Primary server.
3. Click **Configure Reporting Services** check box, and you can see the **Reporting Service Tomcat** port and the **Shutdown** port details. The tomcat ports that are displayed can be modified as per your requirement.
4. Click **Next**, and complete the *pdm_configure*.
5. After the successful completion of pdm_configure, launch the Reporting Services url to view the list of APIs.
   For e.g.: http://HOST_NAME:REPORTING_PORT/CASMReports/api/
6. In the "<NX_ROOT>\bopcfg\CATALINA_BASE_RPT\webapps\CASMReports" location, ensure CASMReports.war is successfully deployed.
7. In case of Advanced Availability configuration, the Reporting Service option is available only in the Application servers, and is not available for both Standby and Background servers.
8. Base URL for Reporting API endpoints: http://HOST_NAME:REPORTING_PORT/CASMReports/reports/<Endpoint>
   For e.g.:
   For access endpoint: http://HOST_NAME:8070/CASMReports/reports/v1/access
   For createReport: http://reportingservername:reportingport/CASMReports/reports/v1/in/createReport

How to Use CASM Reporting APIs with Swagger UI

1. **Security and Authentication:**
   a. All API endpoints are secured and require authentication.
   b. Basic Authentication is only required for the /access endpoint, where you generate an access token.
   c. Once the token is generated, use it to authenticate and try out other APIs.
2. **Swagger UI Usage:**
   a. Swagger UI provides an Authorize button, where you can input your credentials (Basic Authentication and Access Token).
   b. Once authenticated, Swagger automatically includes the provided credentials in all subsequent API requests.
3. **Token Expiration:**
   a. The generated token will expire after one hour (by default).
   b. Once expired, you must generate a new token to continue making requests.
4. **Executing APIs in Other Tools (e.g., Swagger):**
   a. When using tools like Postman, you must explicitly pass the required credentials (Basic Authentication and the Access Token) with each request.
5. Delete Report & Update Report API methods are applicable only for Custom reports. OOTB reports cannot be deleted or updated.

6. **ITIL Reports:**
   a. Swagger UI allows you to dynamically input parameters for ITIL reports.
   b. Depending on the selected report code, you will be prompted to provide the relevant parameters to retrieve the desired data.

Get started with Report API Endpoints

1. The base URL to access the reporting APIs is: *http://<SDM_Reporting_Servername>:ReportingServiceTomacatport/CASMReports/api/.* This page lists out all the supported API endpoints to get, create, update, and delete the reports.
2. In the page displayed, click **Authorize**.
3. In the **Available authorizations** window, enter your Username and Password (the user must have access to Reports), and click **Authorize.** Now close the window.
4. In the **POST /access** method, click **Try it out**>> **Execute.**
5. In the Response body, copy the value of the ***accessToken***, and click **Authorize**.
6. In the Available authorizations window displayed, in **X-AccessToken**, paste the *accessToken* in **Value**, close the window. Now click **Authorize** and close the window.

NOTE

- X-AccessToken is generated based on the Reporting Role of the contact, and insights_reports functional access.
- If you execute the **DELETE /access endpoint**, the authorized access token is deleted.
- If the session is idle for 60 minutes, the access token auto-expires.
- If the session expires when you choose to delete the token, you have to **Authorize** again.


```
{
  "openapi" : "3.0.1",
  "info" : {
    "title" : "CASM Reports.",
    "description" : "CASM Reports - Try it out."
  },
  "servers" : [ {
    "url" : "/CASMReports/reports"
  } ],
  "tags" : [ {
    "name" : "Access Token Operations.",
    "description" : "Operations related to Access Tokens."
  }, {
    "name" : "Get Available Report Codes with Description.",
    "description" : "Operations related to get the CASM Report codes with Description."
  }, {
    "name" : "Get Predefined Report By Report Code.",
    "description" : "Operations related to get the Predefined Report By Report Code."
  }, {
    "name" : "Design your Report.",
    "description" : "Operations related to Design your Report - Customise the required select fields, filters
 and conditions as needed."
  }, {
    "name" : "ITIL Reports.",
    "description" : "Operations related to ITIL Reports - Get ITIL Reports by ITIL Report Code."
  } ],
  "paths" : {
    "/v1/access" : {
```

```
    "post" : {
      "tags" : [ "Access Token Operations." ],
      "summary" : "Get The Access Token.",
      "description" : "Retrieves the Access Token for the specified user.",
      "operationId" : "createAccess",
      "responses" : {
        "200" : {
          "description" : "Successful operation."
        },
        "401" : {
          "description" : "Unauthorized."
        }
      },
      "security" : [ {
        "basicAuth" : [ ]
      } ]
    },
    "delete" : {
      "tags" : [ "Access Token Operations." ],
      "summary" : "Delete The Access Token.",
      "description" : "Delete the Access Token for the specified user.",
      "operationId" : "deleteAccess",
      "responses" : {
        "200" : {
          "description" : "Successful operation."
        },
        "401" : {
          "description" : "Unauthorized."
        }
      },
      "security" : [ {
        "X-AccessToken" : [ ]
      } ]
    }
  },
  "/v1/{factoryName}/createReport" : {
    "post" : {
      "tags" : [ "Design your Report." ],
      "summary" : "Create Report.",
      "description" : "Operation to Create a Report.",
      "operationId" : "createReport",
      "parameters" : [ {
        "name" : "factoryName",
        "in" : "path",
        "description" : "Provide the factoryName, valid factoryName are in,cr,pr,nr,chg and iss.",
        "required" : true,
        "schema" : {
          "type" : "string"
        }
      } ],
      "requestBody" : {
        "content" : {
          "application/json" : {
```

```
            "schema" : {
              "$ref" : "#/components/schemas/CASMReport"
            }
          }
        }
      },
      "responses" : {
        "200" : {
          "description" : "Successful operation."
        },
        "401" : {
          "description" : "Unauthorized."
        }
      },
      "security" : [ {
        "X-AccessToken" : [ ]
      } ]
    }
  },
  "/v1/{factoryName}/deleteReport" : {
    "delete" : {
      "tags" : [ "Design your Report." ],
      "summary" : "Delete Report.",
      "description" : "Operation to Delete a Report.",
      "operationId" : "deleteReport",
      "parameters" : [ {
        "name" : "factoryName",
        "in" : "path",
        "description" : "Provide the factoryName, valid factoryName are in,cr,pr,nr,chg and iss.",
        "required" : true,
        "schema" : {
          "type" : "string"
        }
      }, {
        "name" : "X-Report-Code",
        "in" : "header",
        "description" : "Provide the valid reportCode, which can be retrieved from getReportCodes API.",
        "required" : true,
        "schema" : {
          "type" : "string"
        }
      } ],
      "responses" : {
        "200" : {
          "description" : "Successful operation."
        },
        "401" : {
          "description" : "Unauthorized."
        }
      },
      "security" : [ {
        "X-AccessToken" : [ ]
      } ]
```

```
      }
    },
    "/v1/itilreport" : {
      "get" : {
        "tags" : [ "ITIL Reports." ],
        "summary" : "ITIL Reports.",
        "description" : "Operations related to ITIL Reports - Get ITIL Reports by ITIL Report Code.",
        "operationId" : "getITILObjects",
        "parameters" : [ {
          "name" : "X-Report-Code",
          "in" : "header",
          "description" : "Get ITIL Report By Code.",
          "required" : true,
          "schema" : {
            "type" : "string",
            "enum" : [ "Agreements_Contracts_Expiring_in_Next_Days_for_all_tickets_files",
 "Business_Impact_Change_Orders_and_Releases_files", "BusinessImpact_Main_files",
 "Closed_Categories_vs_Initial_Categories___Changes_and_Releases_files",
 "Closed_Categories_vs_Initial_Categories_files", "Configuration_Items_Cards_files",
 "SLA__and_Enterprise_Services_files", "SLA_Management_By_Enterprise_Services_Charts_and_Summary_files",
 "SLA_MANAGEMENT_TICKETS_files", "Survey_files", "Surveys_By_Survey_Template_files", "Top_Categories_files",
 "Top_Configurations_Items_Classes_files", "Top_Configurations_Items_Families_files",
 "Top_Configurations_Items_files", "Top_Customer_Organizations_files" ]
          }
        }, {
          "name" : "P_days",
          "in" : "header",
          "description" : "Number of Days before Contract Violation.",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "P_all_type",
          "in" : "header",
          "description" : "Ticket Type.(Numeric value) 1: Incidents,2: Requests,3: Problems,4: Incidents/
Requests/Problems,5: Changes/Releases,6: Changes,7: Releases,9: All Tickets",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "P_Int_SLM_type",
          "in" : "header",
          "description" : "SLM Type.(Numeric value) 0: All SLM Types, 1: OLAs, 2: SLAs,3: UCs",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "P_SLM",
          "in" : "header",
          "description" : "Provide Agreements/Contracts.",
          "schema" : {
            "type" : "string"
          }
```

```
        }, {
          "name" : "P_Category",
          "in" : "header",
          "description" : "Provide Categories.",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "P_chg_Category",
          "in" : "header",
          "description" : "Provide Change/Release Category.",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "P_Service_Name",
          "in" : "header",
          "description" : "Provide Enterprice Service.",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "p_finalstartdate",
          "in" : "header",
          "description" : "Provide Start Date.",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "p_finalenddate",
          "in" : "header",
          "description" : "Provide End Date.",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "P_chg_type",
          "in" : "header",
          "description" : "Provide Change/Release Type. (Numeric value)1: Standard Change Orders, 2: Normal
Change Orders, 3: Emergency Change Orders, 4: Major Releases, 5: Minor Releases, 6: Emergency Releases, 7:
All Types of Change Order, 8: All Types of Release, 9: All Types of Change Order and Release",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "p_chg_status_symbol",
          "in" : "header",
          "description" : "Provide Change Status Symbol.",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "p_status_symbol",
```

```
          "in" : "header",
          "description" : "Provide Status.",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "P_Status",
          "in" : "header",
          "description" : "Provide the Status Mode.(Numeric value) 1: All Statuses, 2: All Statuses except
Resolved and Closed, 3: No Status Selected",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "P_type",
          "in" : "header",
          "description" : "Provide Ticket Type.(Numeric value) 1: Incidents, 2: Requests3: Problems, 4:
Requests/Incidents, 5: All Ticket Type",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "P_Family",
          "in" : "header",
          "description" : "Provide the Configuration Item Family.",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "p_earliest_open_date",
          "in" : "header",
          "description" : "Provide the Earliest Date.",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "p_latest_open_date",
          "in" : "header",
          "description" : "Provide the Latest Date.",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "P_Survey",
          "in" : "header",
          "description" : "Provide the Survey.",
          "schema" : {
            "type" : "string"
          }
        }, {
          "name" : "p_topn",
          "in" : "header",
          "description" : "Provide the TOPN value.",
```

```
    "schema" : {
      "type" : "string"
    }
  }, {
    "name" : "P_CiClass",
    "in" : "header",
    "description" : "Provide the CI CLASS.",
    "schema" : {
      "type" : "string"
    }
  }, {
    "name" : "P_CiFam",
    "in" : "header",
    "description" : "Provide the CI FAM.",
    "schema" : {
      "type" : "string"
    }
  }, {
    "name" : "P_Ci",
    "in" : "header",
    "description" : "Provide the CI.",
    "schema" : {
      "type" : "string"
    }
  }, {
    "name" : "P_CustOrg",
    "in" : "header",
    "description" : "Provide the Customer Organization.",
    "schema" : {
      "type" : "string"
    }
  }, {
    "name" : "top",
    "in" : "query",
    "description" : "Provide the top value for the pagination. Default value is 100.",
    "schema" : {
      "maximum" : 200000,
      "minimum" : 10,
      "type" : "integer",
      "format" : "int64",
      "default" : 100
    }
  }, {
    "name" : "skip",
    "in" : "query",
    "description" : "Provide the skip value for the pagination. Default value is 0.",
    "schema" : {
      "maximum" : 500,
      "minimum" : 0,
      "type" : "integer",
      "format" : "int64",
      "default" : 0
    }
```

```
    } ],
    "responses" : {
      "200" : {
        "description" : "Successful operation."
      },
      "401" : {
        "description" : "Unauthorized."
      }
    },
    "security" : [ {
      "X-AccessToken" : [ ]
    } ]
  }
},
"/v1/{factoryName}" : {
  "get" : {
    "tags" : [ "Design your Report." ],
    "summary" : "Design Custom List type Report.",
    "description" : "Operation to Design your custom List type Report.",
    "operationId" : "getObjects",
    "parameters" : [ {
      "name" : "factoryName",
      "in" : "path",
      "description" : "Provide the factoryName, valid factoryName are in,cr,pr,nr,chg and iss.",
      "required" : true,
      "schema" : {
        "type" : "string"
      },
      "example" : "cr"
    }, {
      "name" : "X-Select",
      "in" : "header",
      "description" : "Provide the list of fields to be selected/retrieved.",
      "required" : true,
      "schema" : {
        "type" : "string"
      },
      "example" : "id"
    }, {
      "name" : "X-Filter",
      "in" : "header",
      "description" : "Provide the custom filter if required.",
      "schema" : {
        "type" : "string"
      },
      "example" : "type='R' and status.sym!='Closed'"
    }, {
      "name" : "X-Expand",
      "in" : "header",
      "description" : "Provide the expand field name for resolving the child object information.",
      "schema" : {
        "type" : "string"
      },
```

```
        "example" : "assignee.organization($select=name)"
      }, {
        "name" : "X-OrderBy",
        "in" : "header",
        "description" : "Provide the orderBy field name for sorting.",
        "schema" : {
          "type" : "string"
        },
        "example" : "ref_num ASC"
      }, {
        "name" : "top",
        "in" : "query",
        "description" : "Provide the top value for the pagination. Default value is 100.",
        "schema" : {
          "maximum" : 200000,
          "minimum" : 1,
          "type" : "integer",
          "format" : "int64",
          "default" : 100
        }
      }, {
        "name" : "skip",
        "in" : "query",
        "description" : "Provide the skip value for the pagination. Default value is 0.",
        "schema" : {
          "maximum" : 500,
          "minimum" : 0,
          "type" : "integer",
          "format" : "int64",
          "default" : 0
        }
      } ],
      "responses" : {
        "200" : {
          "description" : "Successful operation."
        },
        "401" : {
          "description" : "Unauthorized."
        }
      },
      "security" : [ {
        "X-AccessToken" : [ ]
      } ]
    }
  },
  "/v1/getReport" : {
    "get" : {
      "tags" : [ "Get Predefined Report By Report Code." ],
      "summary" : "Get Predefined Report By Report Code.",
      "description" : "Operations related to get the Predefined Report By Report Code.",
      "operationId" : "getReportByCode",
      "parameters" : [ {
        "name" : "X-Report-Code",
```

```
    "in" : "header",
    "description" : "Provide the valid reportCode, which can be retrieved from getReportCodes API.",
    "required" : true,
    "schema" : {
      "type" : "string"
    },
    "example" : "Incidents_Average_Time_to_Resolve_by_Priority"
  }, {
    "name" : "X-Filter",
    "in" : "header",
    "description" : "Provide the custom filter if required.",
    "schema" : {
      "type" : "string"
    },
    "example" : "type='R' and status.sym!='Closed'"
  }, {
    "name" : "X-OrderBy",
    "in" : "header",
    "description" : "Provide the orderBy field name for sorting.",
    "schema" : {
      "type" : "string"
    },
    "example" : "ref_num ASC"
  }, {
    "name" : "top",
    "in" : "query",
    "description" : "Provide the top value for the pagination. Default value is 100.",
    "schema" : {
      "maximum" : 200000,
      "minimum" : 1,
      "type" : "integer",
      "format" : "int64",
      "default" : 100
    }
  }, {
    "name" : "skip",
    "in" : "query",
    "description" : "Provide the skip value for the pagination. Default value is 0.",
    "schema" : {
      "maximum" : 500,
      "minimum" : 0,
      "type" : "integer",
      "format" : "int64",
      "default" : 0
    }
  } ],
  "responses" : {
    "200" : {
      "description" : "Successful operation."
    },
    "401" : {
      "description" : "Unauthorized."
    }
```

```
      },
      "security" : [ {
        "X-AccessToken" : [ ]
      } ]
    }
  },
  "/v1/getReportCodes" : {
    "get" : {
      "tags" : [ "Get Available Report Codes with Description." ],
      "summary" : "Get Available Report Codes with Description.",
      "description" : "Operations related to get the CASM Report codes with Description. The response
contains isCustomReport to identify whether it's custom report or pre-defined report.",
      "operationId" : "getReportCodesWithNames",
      "responses" : {
        "200" : {
          "description" : "Successful operation."
        },
        "401" : {
          "description" : "Unauthorized."
        }
      },
      "security" : [ {
        "X-AccessToken" : [ ]
      } ]
    }
  },
  "/v1/getReportMetaData" : {
    "get" : {
      "tags" : [ "Design your Report." ],
      "summary" : "Get Report Meta Data.",
      "description" : "Operation to Get the Report Meta Data.",
      "operationId" : "getReportMetaData",
      "parameters" : [ {
        "name" : "X-Report-Code",
        "in" : "header",
        "description" : "Provide the valid reportCode, which can be retrieved from getReportCodes API.",
        "schema" : {
          "type" : "string"
        }
      } ],
      "responses" : {
        "200" : {
          "description" : "Successful operation."
        },
        "401" : {
          "description" : "Unauthorized."
        }
      },
      "security" : [ {
        "X-AccessToken" : [ ]
      } ]
    }
  },
```

```
  "/v1/{factoryName}/updateReport" : {
    "put" : {
      "tags" : [ "Design your Report." ],
      "summary" : "Update Report.",
      "description" : "Operation to Update a Report.",
      "operationId" : "updateReport",
      "parameters" : [ {
        "name" : "factoryName",
        "in" : "path",
        "description" : "Provide the factoryName, valid factoryName are in,cr,pr,nr,chg and iss.",
        "required" : true,
        "schema" : {
          "type" : "string"
        }
      } ],
      "requestBody" : {
        "content" : {
          "application/json" : {
            "schema" : {
              "$ref" : "#/components/schemas/CASMReport"
            }
          }
        }
      },
      "responses" : {
        "200" : {
          "description" : "Successful operation."
        },
        "401" : {
          "description" : "Unauthorized."
        }
      },
      "security" : [ {
        "X-AccessToken" : [ ]
      } ]
    }
  }
},
"components" : {
  "schemas" : {
    "CASMReport" : {
      "type" : "object",
      "properties" : {
        "id" : {
          "type" : "string"
        },
        "r_name" : {
          "type" : "string"
        },
        "attributes" : {
          "type" : "object",
          "additionalProperties" : {
            "type" : "string"
```

```
            }
          }
        }
      }
    },
    "securitySchemes" : {
      "basicAuth" : {
        "type" : "http",
        "scheme" : "basic"
      },
      "X-AccessToken" : {
        "type" : "apiKey",
        "name" : "X-AccessToken",
        "in" : "header"
      }
    }
  }
}
```

Sample code snippet to create an incident report:

```
{
    "r_name": "Demo_ListType_ReportName",
    "attributes": {
      "r_code": "Demo_ListType_ReportCode",
      "r_orderby": "ref_num ASC",
      "r_expand": "assignee.organization($select=name)",
      "r_filter": "category.sym eq 'Software'",
      "r_select": "id,ref_num,assignee,description,category",
      "r_type": "List"
    }
}
```

To update a custom report, get the report metadata using /v1/getReportMetaData endpoint, and update it using /v1/{factoryName}/updateReport endpoint.

Logging

1. Check the log information in *reportMS.log* located in NX_ROOT\log.
2. To enable debug logs for Reporting, update the file *logback.properties* present in NX_ROOT\bopcfg\www\CATALINA_BASE_RPT\webapps\CASMReports\WEB-INF\classes to update the value of rootLogger.level. For e,g.: INFO to DEBUG
3. Restart the reporting_server Tomcat_service, with the following commands:

   ```
   pdm_tomcat_nxd -c STOP -t RPT

   pdm_tomcat_nxd -c START -t RPT
   ```

4. Alternatively, to enable the debug logging by using the following command from the command line:

   ```
   nxcd bin

   pdm_log4j_config -f RPT -a -l DEBUG
   ```

5. To check the status of the log level from the cmd prompt, run the following commands:

```
pdm_log4j_config -f RPT -d
```

Limitations

1. ITIL reports are supported only on CA SDM installed using SQL server.
2. Reports created using Reporting APIs are not reflected in Insights App (for both App Launcher and SDM classic UI)

PDM Functions Usage

PdmFunctions are specialized functions within the domsrvr process that can be used to query objects. Currently, 10 functions are supported by domsrvr. This document provides an overview of how these functions can be utilized through the Reporting Service, using version 1 (v1) APIs. Each section in the document includes:

- A description of the function.
- Steps to create a report using the function.
- Instructions to render or retrieve the report.

  NOTE
  Queries must be written in Majic query format and not in standard SQL syntax.

- PdmAddDays
  PdmAddDays(date,count)

```
    When used with one argument, adds the number of days in its argument to today's date and returns the
 result.

When used with two arguments, adds the number of days in its second argument to the value of the date
 column specified in its first argument and returns the result.

This function may be used anywhere in the query.
```

Create

```
POST http://hostname:port/CASMReports/reports/v1/in/createReport
{
"r_name": "PdmAddDays",
"attributes": {
"r_query": "select id, PdmAddDays(2) as aftertwodays, PdmAddDays(last_mod_dt,2) as lastmoddateaftertwodays
 from in where id >= 400001 and id <= 400010",
"r_code": "PdmAddDays",
"r_type": "GroupBy"
}
}
```

Response

```
GET http://hostname:port/CASMReports/reports/v1/getReport
{
"results": [
{
"id": 400001,
"aftertwodays": 1757757527
```

```
},
{
"id": 400002,
"aftertwodays": 1757757527
},
{
"id": 400003,
"aftertwodays": 1757757527
},
{
"id": 400004,
"aftertwodays": 1757757527
}
]
}
```

- PdmAddMonths

PdmAddMonths(date,count)

When used with one argument, adds the number of months in its argument to today’s date and returns the result.

When used with two arguments, adds the number of months in its second argument to the value of the date column specified in it first argument and returns the result.

The single argument form can be used anywhere in the query.

The two-argument form can be used only in the selection list.

Create

```
POST http://hostname:port/CASMReports/reports/v1/in/createReport
{
"r_name": "PdmAddMonths",
"attributes": {
"r_query": "select id, PdmAddMonths(2) as aftertwomonths, PdmAddMonths(last_mod_dt,2) as
 lastmoddateaftertwomonths from in where id >= 400001 and id <= 400010",
"r_code": "PdmAddMonths",
"r_type": "GroupBy"
}
}
```

Response

```
GET http://hostname:port/CASMReports/reports/v1/getReport
{
"results": [
{
```

```
"id": 400001,
"aftertwomonths": 1762757266,
"lastmoddateaftertwomonths": 1762707600
},
{
"id": 400002,
"aftertwomonths": 1762757266,
"lastmoddateaftertwomonths": 1744943008
},
{
"id": 400003,
"aftertwomonths": 1762757266,
"lastmoddateaftertwomonths": 1762706706
},
{
"id": 400004,
"aftertwomonths": 1762757266,
"lastmoddateaftertwomonths": 1744943007
}
]
}
```

- PdmDay

PdmDay(date)

When used with no arguments, returns the current day as an integer.

When used with one argument, returns the day associated with the value of the date column specified in its argument.

The zero argument form can be used anywhere in the query. The one-argument form can be used only in the selection list.

Create

```
POST http://hostname:port/CASMReports/reports/v1/in/createReport
{
"r_name": "PdmDay",
"attributes": {
"r_query": "select id, PdmDay() as currentday, PdmDay(last_mod_dt) as lastmodsday from in where id >=
 400001 and id <= 400010",
"r_code": "PdmDay",
"r_type": "GroupBy"
}
}
```

Response

```
GET http://hostname:port/CASMReports/reports/v1/getReport
```

```
{
"results": [
{
"id": 400001,
"currentday": 10,
"lastmodsday": 9
},
{
"id": 400002,
"currentday": 10,
"lastmodsday": 18
},
{
"id": 400003,
"currentday": 10,
"lastmodsday": 9
},
{
"id": 400004,
"currentday": 10,
"lastmodsday": 18
}
]
}
```

- PdmMonth

PdmMonth(date)

When used with no arguments, returns the current month as an integer from 1 to 12.

When used with one argument, returns the month associated with the value of the date column specified in its argument.

The zero argument form can be used anywhere in the query.

The one-argument form can be used only in the selection list.

Create

```
POST http://hostname:port/CASMReports/reports/v1/in/createReport                          {
                                                                                  "r_name": "PdmMonth",
                                                                "attributes": {"r_query": "select id, PdmMonth() as
 currentmonth, PdmMonth(last_mod_dt) as lastmodsmonth from in where id >= 400001 and id <= 400010",
                            "r_code": "PdmMonth",
      "r_type": "GroupBy"                                                                 }
                                                                          }
```

Response

```
GET http://hostname:port/CASMReports/reports/v1/getReport
{
"results": [
{
```

```
"id": 400001,
"currentmonth": 9,
"lastmodsmonth": 9
},
{
"id": 400002,
"currentmonth": 9,
"lastmodsmonth": 2
},
{
"id": 400003,
"currentmonth": 9,
"lastmodsmonth": 9
},
{
"id": 400004,
"currentmonth": 9,
"lastmodsmonth": 2
}
]
}
```

- PdmMonthName
  PdmMonthName(date)

When used with no arguments, returns the localized name of the current month (“January”, “February”, etc) .

When used with one argument, returns the localized name of the value of the date column specified in its argument.

The zero argument form can be used anywhere in the query.

The one-argument form can be used only in the selection list.

Create

```
POST http://hostname:port/CASMReports/reports/v1/in/createReport
{
"r_name": "PdmMonthName",
"attributes": {
"r_query": "select id, PdmMonthName() as currentmonthname, PdmMonthName(last_mod_dt) as lastmodsmonthname
 from in where id > 400001 and id < 00010",
"r_code": "PdmMonthName",
"r_type": "GroupBy"
}
}
```

Response

```
GET http://hostname:port/CASMReports/reports/v1/getReport
{
"results": [
{
"id": 400001,
"currentmonthname": "September",
"lastmodsmonthname": "September"
},
{
"id": 400002,
"currentmonthname": "September",
"lastmodsmonthname": "Febrary"
},
{
"id": 400003,
"currentmonthname": "September",
"lastmodsmonthname": "September"
},
{
"id": 400004,
"currentmonthname": "September",
"lastmodsmonthname": "Febrary"
}
]
}
```

- PdmSeconds

PdmSeconds(date)

Returns the value of the date column specified in its argument in its raw form as the number of seconds from midnight January 1, 1970.

This function can be used only in the selection list.

Note that the argument is required.

Create

```
POST http://hostname:port/CASMReports/reports/v1/in/createReport
{
"r_name": "PdmSeconds",
"attributes": {"r_query": "select id, PdmSeconds(last_mod_dt) as utcseconds from in where id >= 400001 and id <= 400010","r_code": "PdmSeconds",
"r_type": "GroupBy"
}
}
```

Response

```
GET http://hostname:port/CASMReports/reports/v1/getReport
```

```
{
"results": [
{
"id": 400001,
"utcseconds": 1757522726
},
{
"id": 400002,
"utcseconds": 1739845408
},
{
"id": 400003,
"utcseconds": 1757522706
},
{
"id": 400004,
"utcseconds": 1739845407
}
]
}
```

- PdmString
  PdmString(column)

```
Returns the string equivalent of value of the column specified in its argument.

This function can be used with UUID or string columns.

It can be used only in the selection list.
```

Create

```
POST http://hostname:port/CASMReports/reports/v1/in/createReport
{
"r_name": "PdmString",
"attributes": {
 "r_query"
:
"select id, PdmString(ref_num) as strrefnum, PdmString(last_mod_by) as strlastmodby
 from in where id >= 400001 and id <= 400010"
,

"r_code": "PdmString",
"r_type": "GroupBy"
}
}
```

Response

```
GET http://hostname:port/CASMReports/reports/v1/getReport
{
"results": [
{
"id": 400001,
"strrefnum": "22",
"strlastmodby": "73AC2369BB0C9145BC1A84975FD1AE65"
},
{
"id": 400002,
"strrefnum": "23",
"strlastmodby": "73AC2369BB0C9145BC1A84975FD1AE65"
},
{
"id": 400003,
"strrefnum": "24",
"strlastmodby": "73AC2369BB0C9145BC1A84975FD1AE65"
},
{
"id": 400004,
"strrefnum": "25",
"strlastmodby": "73AC2369BB0C9145BC1A84975FD1AE65"
}
]
}
```

- PdmToday
  PdmToday()

  Returns the current date.

  This function may be used anywhere in the query.

  Create

```
POST http://hostname:port/CASMReports/reports/v1/in/createReport
{
"r_name": "PdmToday",
"attributes": {

"r_query"
:
"select id, PdmToday() as today from in where id >= 400001 and id <= 400010"
,

"r_code": "PdmToday",
"r_type": "GroupBy"
```

```
}
}
```

Response

```
GET http://hostname:port/CASMReports/reports/v1/getReport
{
"results": [
{
"id": 400001,
"today": 1757573981
},
{
"id": 400002,
"today": 1757573981
},
{
"id": 400003,
"today": 1757573981
},
{
"id": 400004,
"today": 1757573981
}
]
}
```

- PdmYear

PdmYear(date)

When used with one argument, returns the year associated with the value of the date column specified in its argument.

The zero argument form can be used anywhere in the query.

The one-argument form can be used only in the selection list.

Create

```
  POST http://hostname:port/CASMReports/reports/v1/in/createReport
{
"r_name": "PdmYear",
"attributes": {
 "r_query"
:
"select id, PdmYear() as currentyear, PdmYear(last_mod_dt) as lastmoddtyear from in
 where id >= 400001 and id <= 400010"
```

```
,

 "r_code": "PdmYear",
"r_type": "GroupBy"
}
}
```

Response

```
GET http://hostname:port/CASMReports/reports/v1/getReport
{
"results": [
{
"id": 400001,
"currentyear": 2025,
"lastmoddtyear": 2025
},
{
"id": 400002,
"currentyear": 2025,
"lastmoddtyear": 2025
},
{
"id": 400003,
"currentyear": 2025,
"lastmoddtyear": 2025
},
{
"id": 400004,
"currentyear": 2025,
"lastmoddtyear": 2025
}
]
}
```

- PdmDownTime
  PdmDownTime(slaName, workshift, startDate, endDate)

```
    Calculates the downtime between two dates under the specified SLA and workshift.

This function can be used only in the selection list.
```

Create

```
POST http://hostname:port/CASMReports/reports/v1/in/createReport
{
"r_name": "PdmDownTime",
"attributes": {"r_query": "select id, PdmDownTime('TestSLA', 'Mon - Fri { 9:00 am - 5:00 pm }',
 open_date,resolve_date) as downtime from in where id >= 400001 and id <= 400010","r_code": "PdmDownTime",
"r_type": "GroupBy"
```

```
}
}
```

Response

```
    GET http://hostname:port/CASMReports/reports/v1/getReport
{
"results": [
{
"id": 400001,
"downtime": "26289"
},
{
"id": 400002,
"downtime": "24387"
},
{
"id": 400003,
"downtime": "22134"
},
{
"id": 400004,
"downtime": "19359"
}
]
}
```

Install CA EEM for CA Service Management

CA Service ManagementEmbedded Entitlements Manager (CA EEM) enables you to secure objects such as jobs, calendars, cycles, global variables, machines, and resources, and delegate role-based access to the users in your enterprise. EEM provides authentication and authorization capability. EEM eliminates plain text user names and passwords from being passed for authentication purposes.

Required for CA Service Catalog and CA Asset Portfolio Management. Install CA EEM 12.6.6 using the CA Service Management Installation Media or upgrade an earlier installed version of EEM to EEM 12.6.6. Note the details of the server where you have installed EEM. EEM authenticates users during login and authorizes users to access CA Service Catalog.

NOTE

- If you plan to implement CA SDM and EEM on the same server, you must install CA EEM 12.6.6 version.
- The CA Service Catalog and all products that integrate with it must share CA EEM.

You can install CA EEM using CA Service Management installation media. Review the following information:

1. The CA Service Management installer launches the CA EEM installer based on the system architecture. The CA Service Management installer launches the 64-bit CA EEM installer.
2. For better performance, we recommend that you install CA EEM on a different server than the DBMS server.
   Make a note of your CA EEM administration password that you provide during the CA EEM installation. You have to use this password while integrating CA Service Management products with CA EEM.

If you have a clustered environment, ensure that all the nodes in the CA EEM cluster are up and running before you install any of the CA Service Management products.

NOTE

If you are installing EEM on Linux, ensure to launch the EEM installer from Linux Terminal using the install path for Linux: SMSDMLINUXP990*/product/EEM*

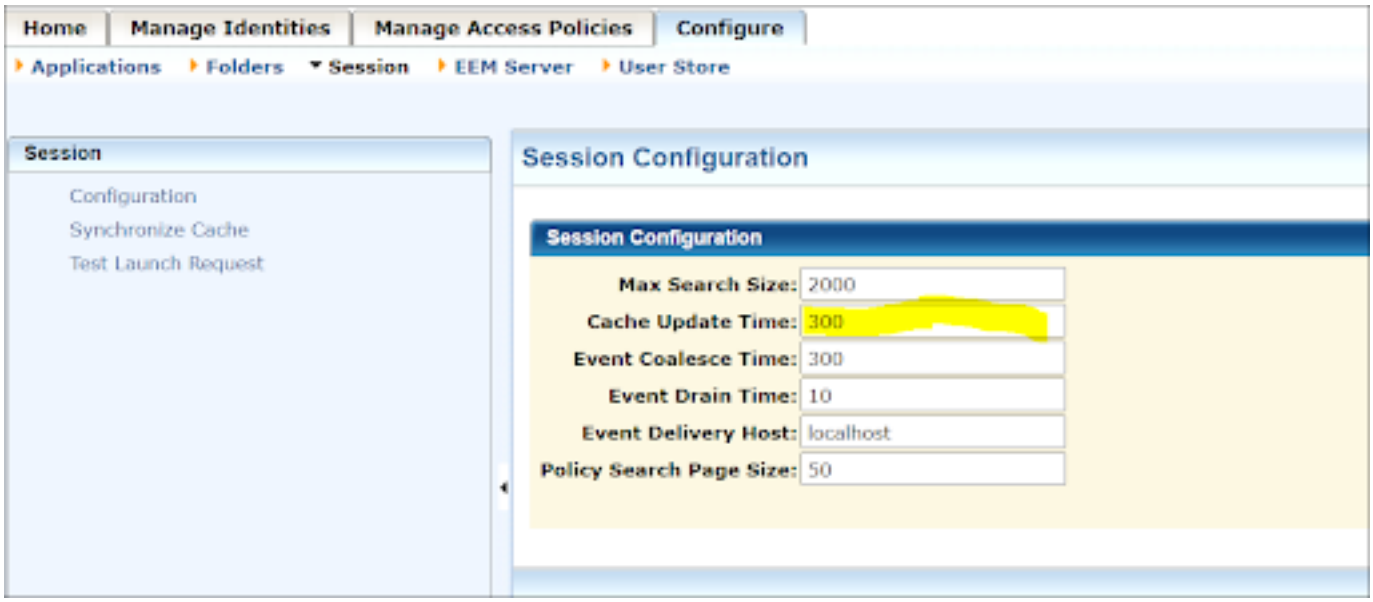
On Linux, do not launch CA EEM using the CA Service Management Common Installer wizard.

Perform the following steps to install CA EEM:

1. Download the CA Service Management installer 17.4 DVD from Broadcom Suopport.
2. Copy and extract the files from the installation media on the servers.
3. Launch the CA Service Management 17.4 installer by right-click **Setup.exe** to select the **run as administrator** option.
4. Select the language and click **Next**.
5. Select **CA Embedded Entitlements Manager (CA EEM)**.
   The wizard guides you through the installation or upgrade process.
   `The CA EEM and CA Service Catalog each installs and uses its own JRE. Therefore, the JRE version of each product can be different.`
   CA EEM is installed in your environment.

   NOTE
   While installing EEM, a popup screen appears asking, **Would you like to enable Cyberark support for EiamAdmin account?** Choose "**No**."
6. Navigate to **Start**, **All Programs**, **CA**, **Embedded Entitlements Manager**, **Admin UI** on the EEM server. Log in to the application to ensure that the CA EEM installation was successful.
   `If CA EEM is configured to use an external LDAP data store, the privileged user must be created in the LDAP Directory.`
   You can optionally install one or more instances of CA EEM. To obtain high availability and failover protection for EEM, cluster all EEM servers. For more information about EEM Failover configuration, see the CA EEM installation documentation.
7. (Optional) EEM Cache Timeout Configuration Update.

   NOTE
   If you are using EEM and SDM, set the EEM cache timeout same as SDM cache timeout.
   You can optionally set the EEM cache timeout configuration as mentioned in the following screenshot.



You may face issues while logging in to EEM or SDM using the EEM user accounts due to session cache timeout mismatch between EEM and SDM. Starting from 17.4, you can update the EEM cache timeout configuration by adding a new NX variable, "NX_EEM_CACHE_UPDATE_TIME" in SDM.

To set the EEM cache update time to 300 seconds, perform the following steps:

To add the variable in the nx.env file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst
```

To add the variable in the nxenv_nt.tpl file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst -t
```

Enable and Configure Telemetry Services

As per the new Portfolio License Agreement (PLA) model initiated by Broadcom Inc, Telemetry Services will collect and report CA Service Management product(s)-specific monthly consumption. The product consumption metrics captured for CA Service Management product(s) will vary depending on the number of installation/upgrade instances, number of users, number of authentications, and so on.

NOTE

PLA Customers are required to upgrade product versions within a time frame as specified in their PLA agreement. PLA and Non-PLA Customers are expected to collect the product telemetry metrics and report it quarterly to Broadcom via Cloud Services.

Perform the following steps to enable and configure Telemetry Services

1. Launch the CA Service Management Installer to initiate and configure Telemetry services. Monthly usage calculation is handled in-product.
2. To enable Telemetry Service Settings in your environment, select “**Configure Telemetry Service**” from the Product and Integrations Selection Page as shown in the figure below:

3. Click **Next** to provide Telemetry Service Setting values for selected CA Service Management Products as shown below:

Field	Description
Enterprise Site ID	This is a mandatory field. Your Enterprise Site ID is listed on your PLA program and the Broadcom Support (Login to Support Portal. Navigate to **My Account**, **Profile**, **Site ID**).
Environment Type	This is a mandatory field. Specify whether you are deploying the selected CA Service Management product(s) on a Production or a Non-Production Environment.
Internal Identifier	Indicate the department or cost center used for internal tracking of product consumption metrics.
Company Domain	Specify the last part of your company or organization's email address.
Proxy Server URL	If your network is secured by a firewall, or if you do not have direct access to internet, specify the proxy server details for connecting to Telemetry Services.
Proxy User	Enter the Proxy Server User name.
Proxy Password	Enter the proxy server password.

4. Click **Save & Close** to enable Telemetry Service Settings for selected CA Service Management product(s).

Installing CA Service Desk Manager

Ensure that you have reviewed the Pre-Installation Considerations before proceeding with the installation.

NOTE
Before you install CA Service Desk Manager 17.4 on Oracle Database, ensure that the Oracle PATH environment variable refers to 32-bit oracle client first, and then, to 64-bit Oracle client. To change the sequence, navigate to Environment Variables, System Variables, Path, Edit.

The CA Service Management installer performs prerequisite checks and mandatory validations before installing CA Service Desk Manager.

If you are upgrading from CA SDM 17.3 and later version of CA SDM:

- Stop the Search service, xFlow Interface Service from Windows **Control Panel, Administrative Tools, Services**.
- Back up the system database (MDB). The system database (MDB) backup is mandatory and ensures safe data recovery if any problems are encountered during the installation process.

NOTE

In Advanced Availability mode, the xFlow Interface can only be installed on an Application Server. If you have converted CA SDM conventional to advanced availability configuration, and you have a previously installed version of xFlow Interface, ensure that you uninstall the xFlow Interface and reinstall the xFlow Interface on an Application server.

NOTE

If in a case of a Conventional configuration if the Primary and Secondary servers are on separate nodes, and the xFlow Interface is deployed on the Secondary server, it is observed that a Database Client is needed on the Secondary Server for the installation to complete successfully.

NOTE

Whenever you restart the CA Service Desk Manager Services due to environment/other requirements, ensure that you restart the xFlow Analyst Interface Services too.

NOTE
SUSE Linux Enterprise Server 15.6 supports 17.4 GA, and 17.4.3.0 and above.

Follow these pre-requisite steps for SUSE Linux Enterprise 15.6 and RHEL 9.4:

1. Download the latest applyPTF tool for SUSE with APAR No. 99112268 from Broadcom Support, and extract the T5UG572.tar.Z file.
2. Download the CA Service Desk Manager 17.4 installer DVD from Broadcom Support, and extract "**CA SERVICE DESK MANAGER 17.4 FOR LINUX.iso**"
3. Paste the file **applyptf.linux** in the path:
   ```
   <extracted "CA SERVICE DESK MANAGER 17.4 FOR LINUX.iso">\filestore\utils\ApplyPTF\
   ```
4. Before you start the 17.4GA SDM Installation, ensure you create a soft link using the command mentioned below:

   NOTE
   For SUSE 15.6, before you start the 17.4GA SDM Installation, ensure you create a soft link using the command mentioned below:
   ```
   ln -s /usr/bin/ifconfig /sbin/ifconfig
   ```
   SUSE Linux Enterprise Server 15.6 supports 17.4.3.0 and above.

   NOTE
   For RHEL 9.4, before you start the 17.4GA SDM Installation, ensure you create soft links using the commands mentioned below:

   ```
   yum install libnsl
    yum install libnsl2
    ln -s /usr/bin/ifconfig /sbin/ifconfig
    sudo ln -s /lib64/libnsl.so.3 /lib64/libnsl.so.2
   ```

   RHEL 9.4 supports **17.4.5.0** and above.
5. For the remaining process, follow from Step 2 mentioned in the below section.

Follow these steps:

1. Download the CA Service Desk Manager 17.4 installer DVD from Broadcom Support.
   - **CA SERVICE MANAGEMENT SDM PACKAGE 17.4 FOR WINDOWS (WITH CATALOG).iso**
   - **CA SERVICE DESK MANAGER 17.4 FOR WINDOWS (WITHOUT CATALOG).iso**
   - **CA SERVICE DESK MANAGER 17.4 FOR LINUX.iso**
2. The installer installs the following products in your environment:
   - Server Installation
   - xFlow Interface
   - Search Server
   - Datamart (Aggregate Data for JasperReports Server)
3. Mount or extract the files from the installation media on the server.
4. Based on the operating system you are using, do the following:
   a. **Windows**
      a. Navigate to the folder where you have copied the installation files and *r*ight-click on the **Setup.exe** file and select **run as administrator**.
         ```
         You must be logged in to the server as a local admin to install CA SDM.
         ```
   b. **Linux**
      a. Ensure that you have exported Oracle variables on the server where you want to install CA SDM.
      b. Navigate to the root directory of the installation media and start the installation by running the following command:
         ```
         ./setup.sh
         ```

To install CA Service Management from the **non-root** user on Unix or Linux, you must have the permissions to execute the sudo command.

WARNING

To install CA Service Management from the **non-root** user on Unix or Linux, you must have the permissions to execute the sudo command.

Follow these steps:

1. Export the appropriate oracle variables and library paths.
2. Navigate to where you have downloaded the CA Service Management installer DVD and execute the below command:
   sudo -E PATH=$PATH ./setup.sh

5. Select the language and click **Next**.
6. Select **CA Service Management** .
7. Review and Accept the License Agreement Information and click **Next**.
8. Select **Microsoft SQL Server** or **Oracle** as the database type in the Database Configuration screen.
   For the Windows Microsoft SQL Server database, **mdbadmin** is the default user.
   For the Oracle database, verify that you are providing a valid tablespace path. If this path is invalid, the database installation fails. The following path is an example of a valid Oracle tablespace path:
   ```
   C:\app\Administrator\oradata\Oracle_Service_Name.
   ```
9. Provide the database details:
   a. If you have **a Microsoft SQL Server** database, provide the following field information:
      - **Database Server**: Enter the hostname of the database server.
        If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
      - **Database Name**: Enter the Database name (MDB), the name of the target DBMS. The default value ismdb.
      - **Database Port**: Provide the port identifier for the target DBMS.
      - **Database Server Instance**: The database instance name. Leave this field blank if you are using the *default instance* and *default port*.
        - If you are using the *SQL default instance* and *non-default port number*, the Database Server Instance field should look like this:
          <database default instance name>,<non-default port number>
          For example:
          ```
          mssqlserver,10555
          ```
        - If you are using the *SQL named instance* and *default port number*, the Database Server Instance field should look like this: <database named instance name>,<default port number>
          For example:
          ```
          sdmcasmqa,1433
          ```
        - If you are using the *SQL named instance* and *non-default port number*, the Database Server Instance field should look like this: <database named instance name>,<non-default port number>
          For example:
          ```
          sdmcasmqa,10555
          ```
      - **Database Admin User**: For Microsoft SQL Server, **sa** is the default value.
        The **sa** user is the admin user with permission to create a user and schema.
      - **Database Admin Password**: Enter the database password of the user specified by a database admin user. No default value is assigned to this parameter.
      - **mdbadmin Password**: Enter the password for the user specified for the MDB admin user.
      - **Confirm mdbadmin Password**: Confirm the password.
   b. If you have the **Oracle** database, provide the following field information:

- **Database Server**: Enter the host name of the Oracle database server.
  If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
- **Oracle Service Name**: Enter the Oracle Service name.
- **Listener Port**: Enter the listener port for the database.
- **Net Service Name**: Enter the Net Service Name of Oracle database where the database resides.
  If the database is remote, use the Net Service Name that is defined in the Oracle client on the local computer. CA SDM has access to the database through a local installation of the Oracle client.
- **DBA User Name**: Provide the name of an Oracle user with DBA access. The default value for DBA user is **SYS**.
- **DBA User Password**: Enter the password for the DBA user.
- **mdbadmin Password**: Enter the password for MDB admin.
- **Confirm mdbadmin Password**: Confirm the password.
- **Tablespace Path (on DB Server)**: Provide the tablespace path on the database server.
- **Oracle Home Path**:
  - Provide the Oracle Client 64-bit path.

    NOTE
    After successful installation, CA Service Desk Manager 17.4 will use Oracle Client 64-bit architecture.
  - Oracle client 32-bit path for fresh installation.
- **Data Tablespace Name**: Provide the name of the data tablespace. The default value is MDB_DATA.
- **Index Tablespace Name**: Provide the index Tablespace Name: The default value is MDB_INDEX.

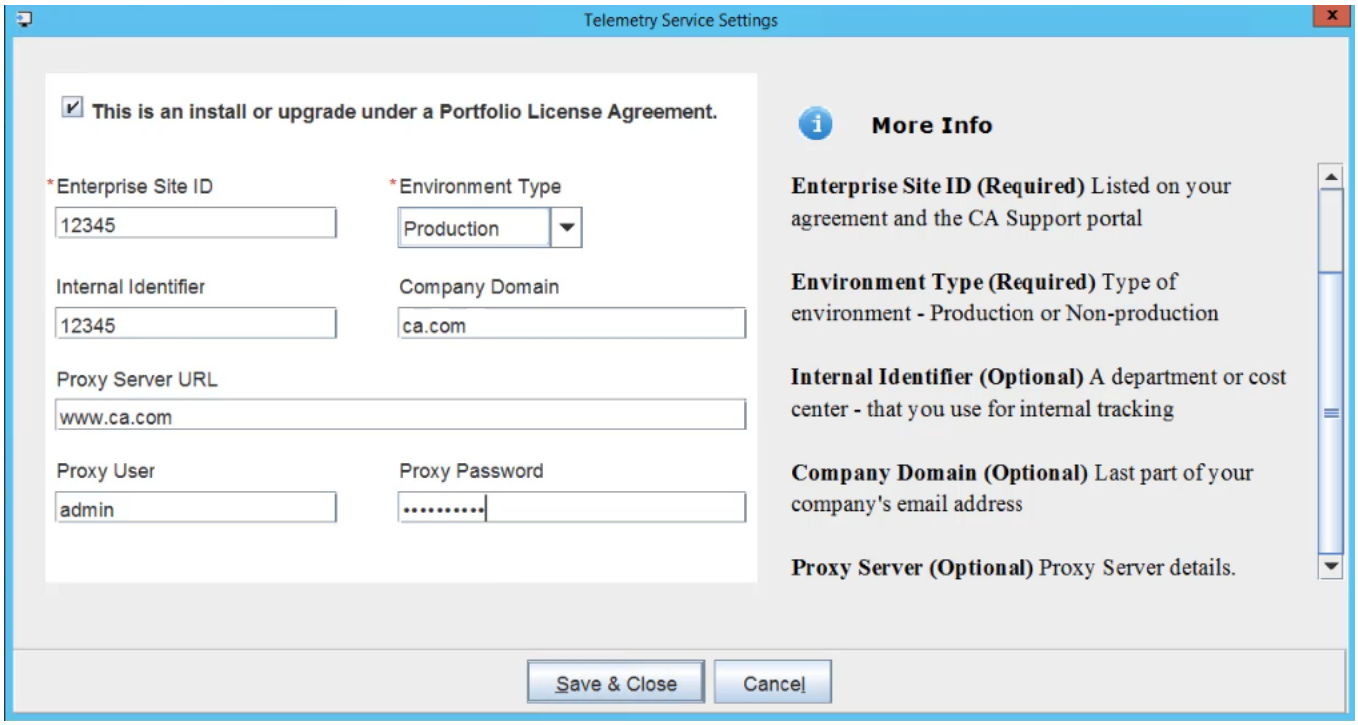
Click **Next**.

10. Select **CA Service Desk Manager** from Select the Products and Integrations screen. Select the following product options as per your environment and integration requirement:
    a. **Server Installation**
    b. **CA xFlow Interface**
    The xFlow Interface provides analysts the ability to understand the IT environment, prioritize work and make decisions.Data driven intelligence improves the productivity of the analyst and enables them to understand the workload for the day and the list of activities to be completed.
    The CA xFlow Interface server is deployed only if you have installed CA Service Desk Manager in your environment.
    For **conventional configuration**, it is recommended that you install the xFlow Interface server on a secondary server.
    Advanced Availability users can only install the xFlow Interface server on an application server.
    c. **Search Server**
    Search Server facilitates CA SDM Analysts to search from the xFlow Interface for users, tickets (including incidents and requests), configuration items (CIs), and knowledge base articles.
    - Search Server can be deployed as a standalone setup in your environment. Search server currently supports the following server:
      - Microsoft Windows Server 2016/2019/2022 (64-bit)
      - Red Hat Linux 7.2/8.1(64-bit)
    - If you install Search Server after installing the xFlow Interface, you must restart the xFlow Interface services again to perform searches in the xFlow Interface. The restart is needed because a table - **usp_es_servers** needs to have the CA Search Server information populated in it before the Search functionality can be used. Without this table, the Search functionality will not work as the initial data load into the Search server does not recognize the table contents and will quit the process.

    d. **Datamart**
    The Datamart server stores the data that is required by the Advanced Reporting engine to render the report data on the xFlow Interface.
    Click **Next.**

11. Select **This is an install or upgrade under a Portfolio License Agreement** on Telemetry Service Settings page. Provide the following field information:
    - **Enter Site ID :** Listed on your agreement and the CA Support portal
    - **Environment type:** Type of Environment Production or Non-Production
    - **Internal Identifier (optional):** A department or cost center- that you use for internal tracking
    - **Company Domain (optional):** Last part of your company email address
    - **Proxy Server URL (optional):** Proxy server details
    - **Proxy User:** Proxy server username
    - **Proxy Password:** Proxy server password

    Click **Save & Close**.



12. Review the Installation Prerequisites page and click **Next.**
13. The **CA Service Desk Manager Product Configuration** screen shows the server configuration details options as per the server component opted by you in **step 7**. The installer wizard guides you through the following configuration details screen. As per the selected component by you in **step 7**, complete the following configuration details screens:
    a. Advanced Availability Configuration
    b. Conventional Configuration Details
    c. Search Server Configuration Details
    d. xFlow Interface Configuration Details
    e. Install and Integrate CABI JasperReports Server with CA Service Management.
14. Click **Next**.
15. Review the Pre-Installation Configuration page. Click **Next**.
16. Select **YES** to automatically execute the CA SDM mandatory post-install steps or select **NO**.
17. Click **Install**. Review the Installation progress and when the installation is complete, click **Next.**
18. Review the Post Deployment summary page.
19. Select **Finish** to exit the installer.
    If the installation fails for a product, retry the installation by using the **Retry** option in the wizard where it fails. If retry does not work, exit the installer and relaunch it. If the problem persists, contact Broadcom Support.

20. Complete the Post-Instalation Tasks.
21. After installing the xFlow Interface, you must start the **CA** xFlow Interface **Server** service as this is not started by default. Perform the following steps to start the xFlow Interface Server services:
    a. **Windows**
    Start the **CA xFlow Interface service**, **Collaboration Server service**, and **CA Service Management Search Server service** from Windows Services (**Control Panel, Administrative Tools, Services**).
    b. **Linux**

    NOTE
    In case of Linux Search Server environment, ensure to run the following commands on the terminal:

    ```
    sysctl -w vm.max_map_count=262144
    ulimit -Hn 65535
    ```

    c. a. Navigate to */opt/CA/xFlow/Apps* and run the below command from command prompt window:

    ```
    ./xFlowServer start
    ```

    If you want to stop the server, run the below command:

    ```
    ./xFlowServer Stop.sh
    ```

    b. Navigate to */opt/CA/SC/SearchServer/elasticsearch-7.17.6/bin*and run the below command from command prompt window:

    ```
    ./ServicesStart.sh
    ```

    If you want to stop the server, run the below command:

    ```
    ./ServicesStop.sh
    ```

After installing the xFlow Interface and Search Server, you can configure the cache size for the xFlow Interface. For more information about how to configure cache size, see Configure the xFlow Interface Server Services.

Complete the CA SDM Post Installation Tasks.

CA SDM Advanced Availability Server Configuration Details

If you have opted for CA SDM Advanced Availability configuration, complete the following options as displayed on the installer:

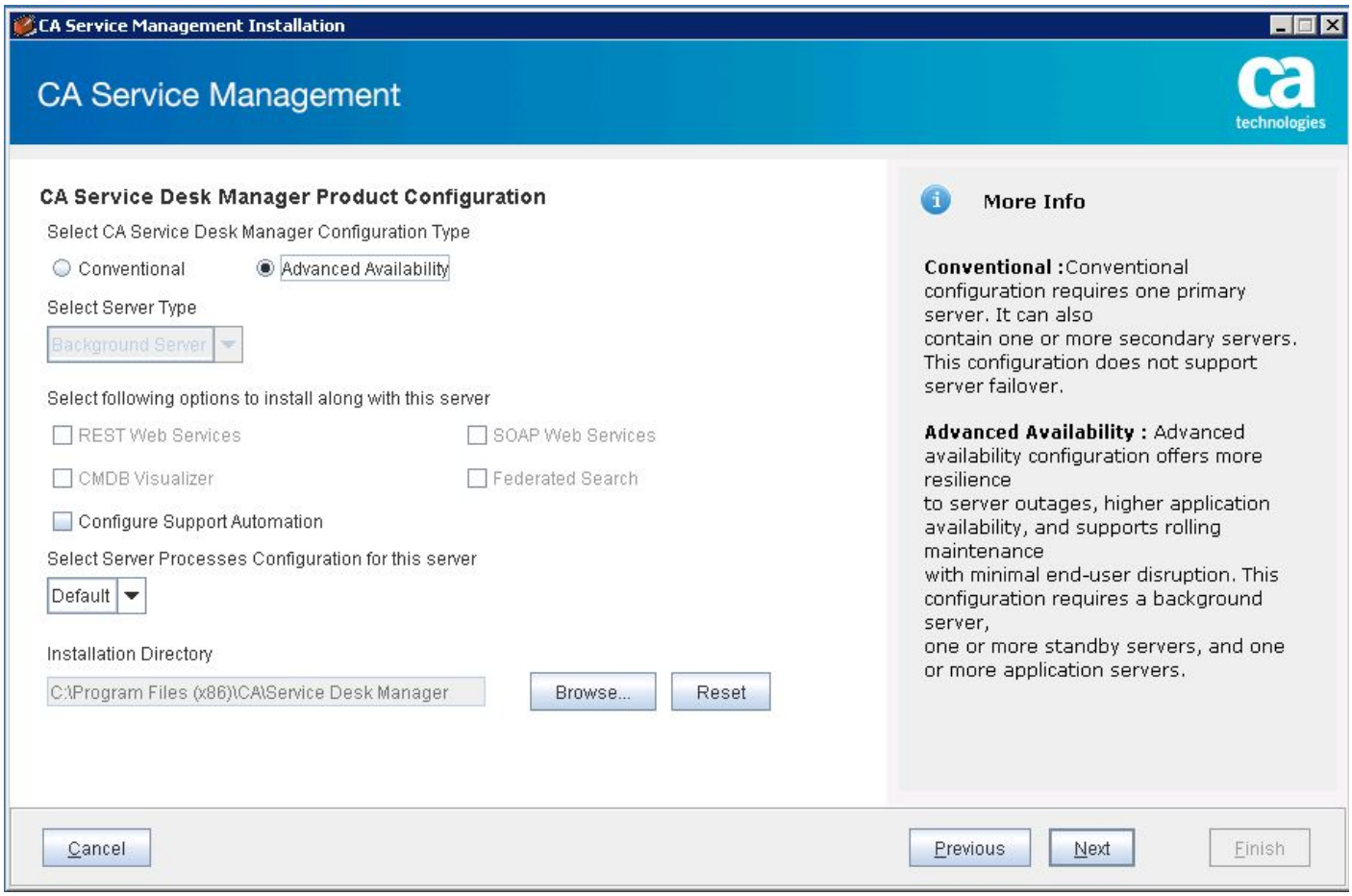
NOTE

Create the **CA Service Desk Manager Server Privileged** user before you begin the CA SDM installation.

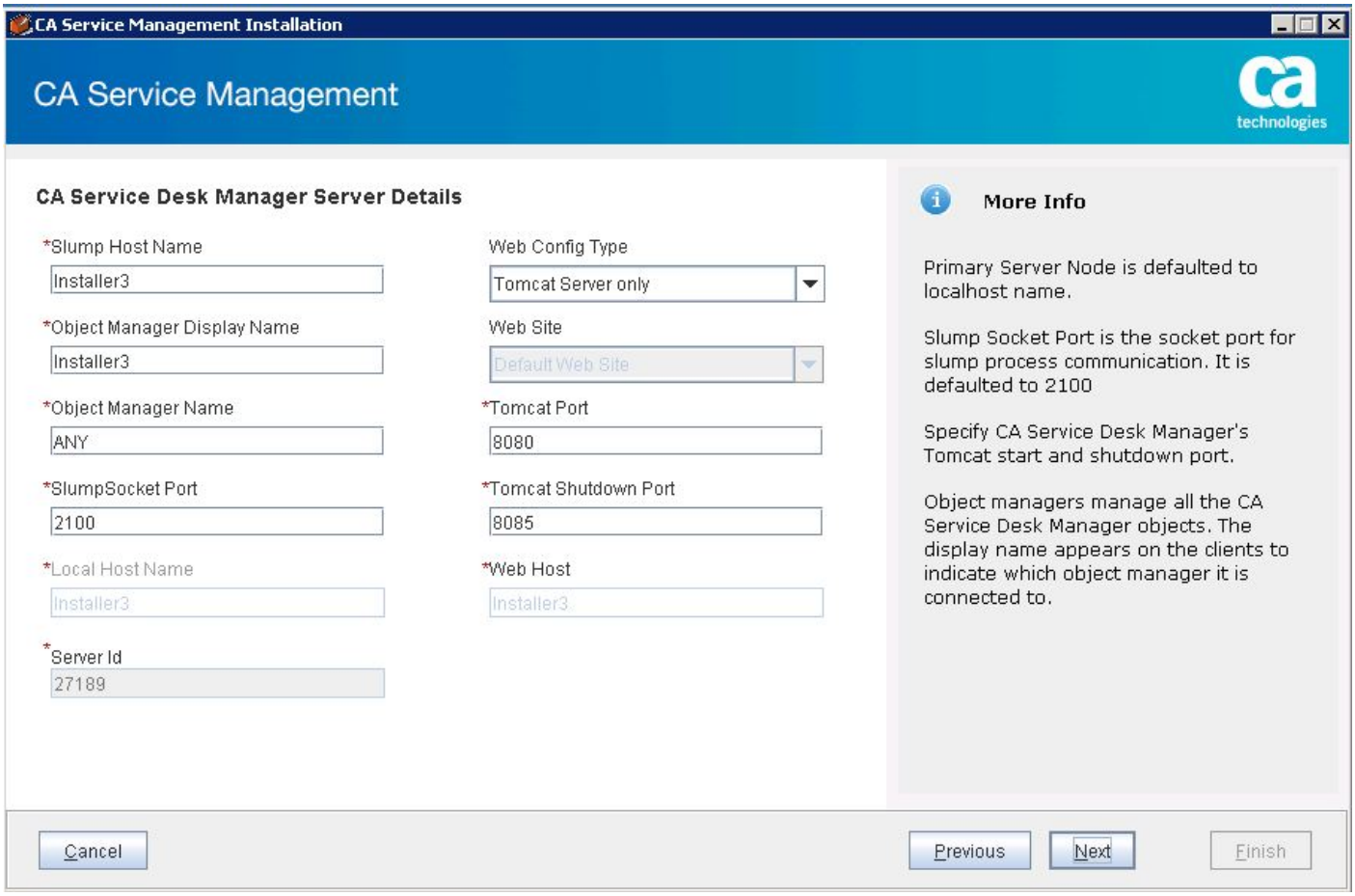
Advanced Availability Server

Perform the following steps:

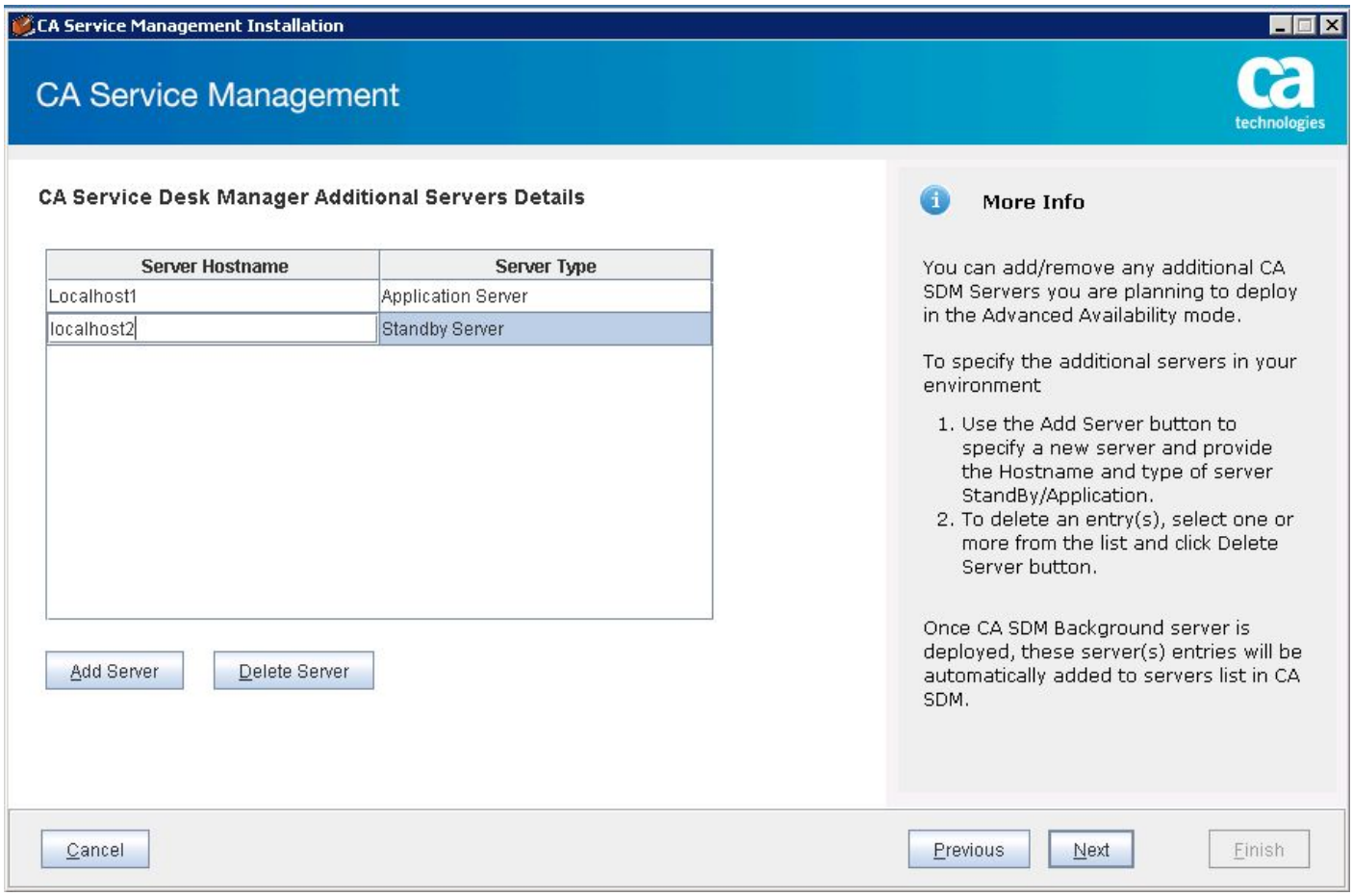
1. Navigate to **CA Service Desk Manager Product Configuration** screen.
    a. Select the **Advanced Availability** radio button.
    The background server option is selected by default.
    b. Keep the **Default** option selected.
    c. Click **Browse** to select the installation directory and click **Next**.
    d. Select **Advanced Availability** from the CA SDM Product Configuration Screen:



2. Enter the following information in the **CA Service Desk Manager Server Details** screen:



3. On the **CA Service Desk Manager Server Privileged User Details** screen, select the **Load Default Data** check box to load on the predefined data and enter the privileged user details.
4. On the **CA Service Desk Manager Additional Server Details** screen, enter the following information of the servers that you want to add:



CA SDM Conventional Server Configuration Details

If you have opted for CA SDM conventional configuration, complete the following options that are displayed on the CA SDM Installer:

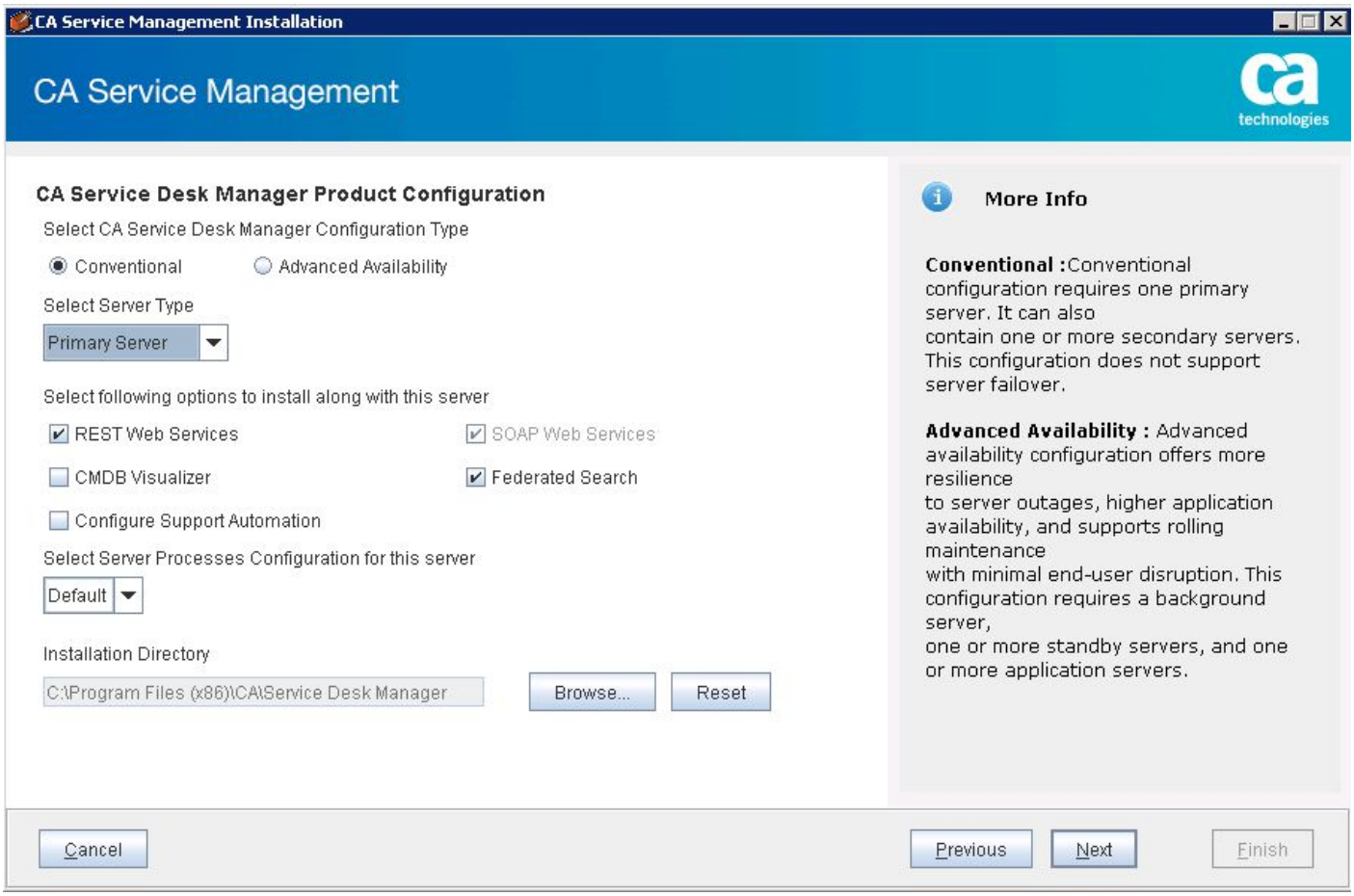
NOTE

Create the **CA Service Desk Manager Server Privileged** user before you begin the installation.

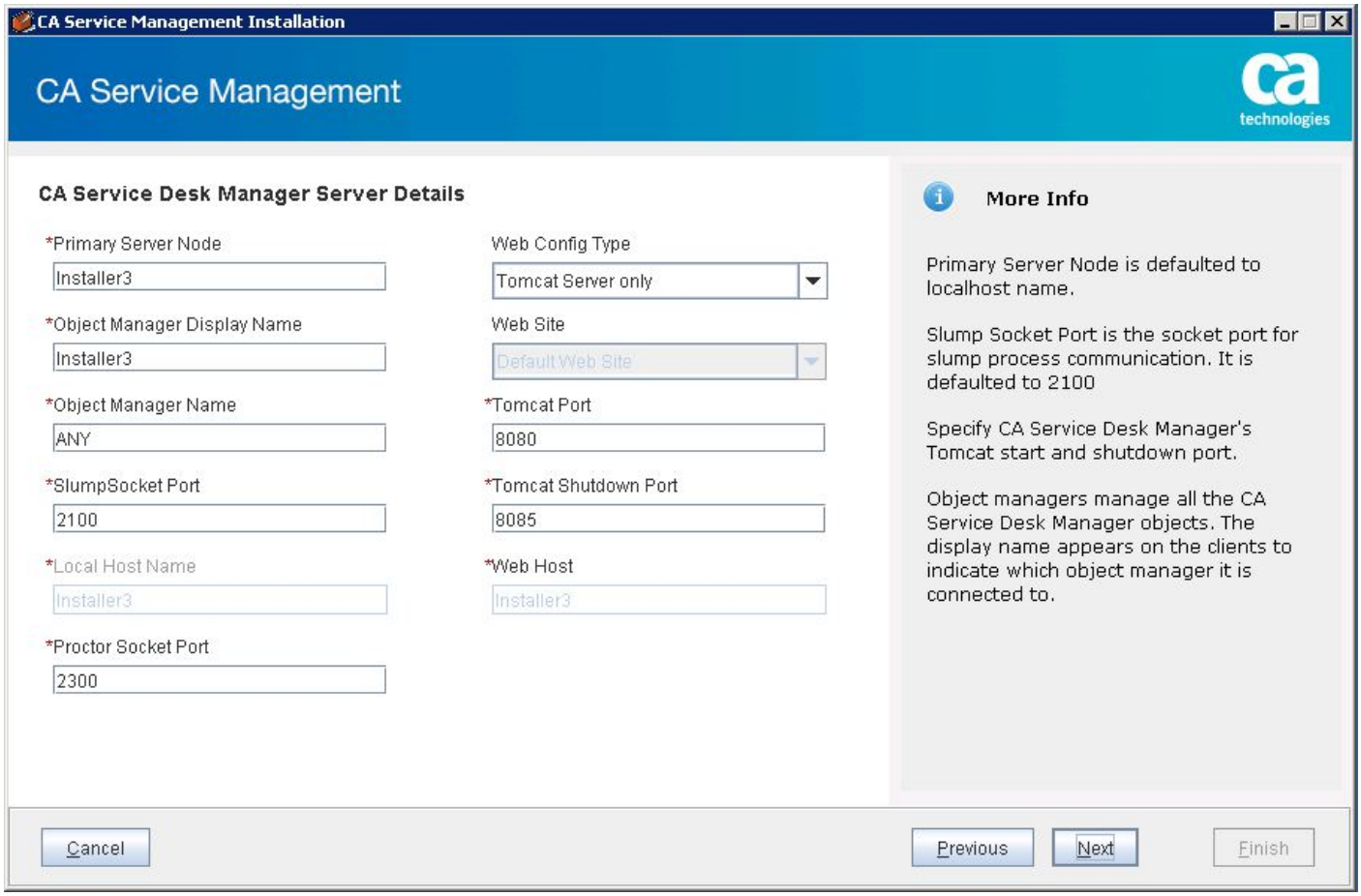
Conventional Server

Primary Server Configuration

1. Navigate to **CA Service Desk Manager Product Configuration** screen.
   a. Select the **Conventional** option**.**
   b. Select **Primary Server** from the **Select Server Type** drop-down.
   c. Select the other components that you want to install on this server. For example, Federated Search
   d. If you have already created server process configuration, then select the configuration or else keep the **Default** option selected.
   e. Click **Browse** to select the installation directory and click **Next**. The following screenshot corresponds to the installation on Windows.



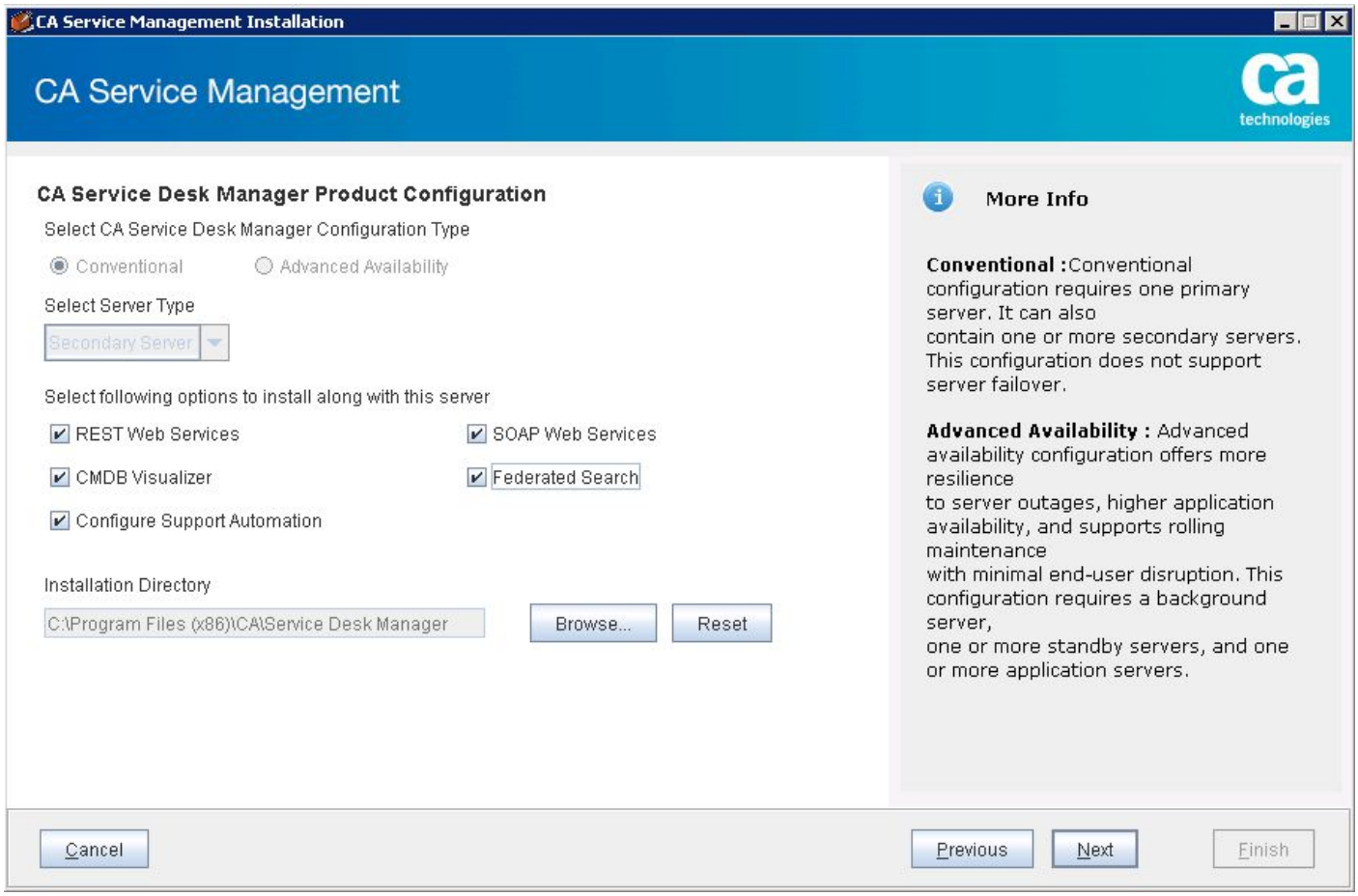
2. Enter the following information in the **CA Service Desk Manager Server Details** screen:
   a. Enter the primary server host name in the **Primary Server Node** field (defaulted to the localhost name).



3. On the **CA Service Desk Manager Server Privileged User Details** screen, select the **Load Default Data** check box to load on the predefined date and enter the privileged user details.
4. On the **CA Service Desk Manager Optional Component Details** screen, enter the port numbers to install the components that you have selected.
   CA

Secondary Server Configuration

1. After installing CA SDM on the primary server, create the server process configuration for the secondary server.
2. Log in to the secondary server and start the CA Service Management installation.
   You can skip the Database Configuration screen to go to the **CA Service Desk Manager Product Configuration** screen.
   - If you skip the Database Configuration screen (recommended in DMZ installations where you need SDM only), you can install SDM secondary server but you cannot add xFlow Interface.
   - If you want to install SDM secondary server and the xFlow component, enter the database server details to which the primary server is pointing to.



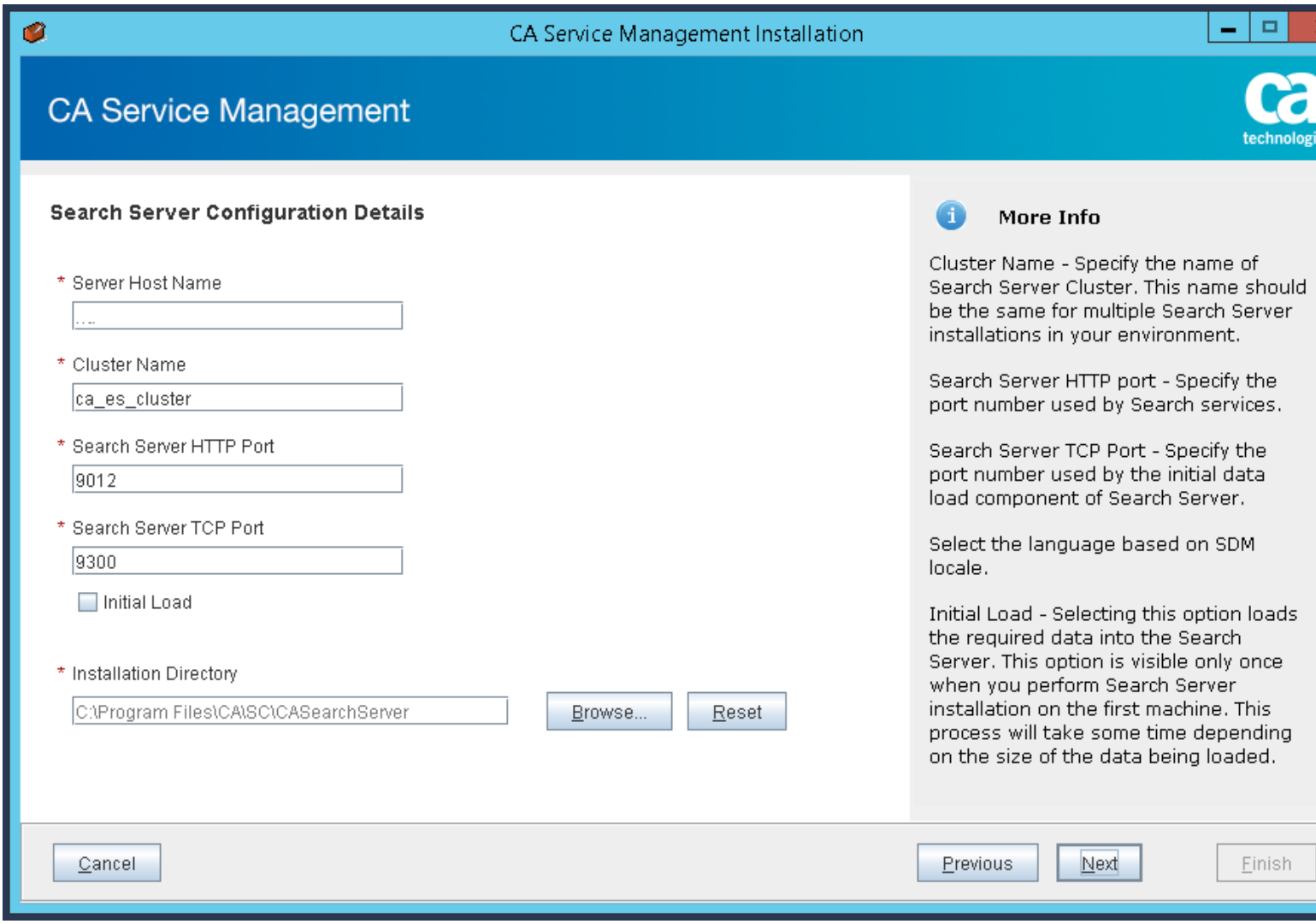
3. Do one of the following on the Database Configuration screen.
4. After you install CA SDM (or migrate to CA SDM from a CMDB standalone system), run the *cmdb_update_ambiguity* utility. Use the -h command to view the mandatory options.

NOTE

If configuration fails during the Validate Extension Tables step, database connectivity can be an issue. Retry the installation and verify that you provided the correct database connectivity information.

Search Server Configuration Details

If you opted to install Search server while installing CA SDM, complete the following search server configuration details:



Field Options	Description
Server Host Name	Provide the system name or IP address where you have installed search server.
Search Server HTTP Port	Provide the search server port details. The default value is 9012.
(Optional) Initial Data	Selecting this option loads the required data into the Search Server. This option is visible only when you perform the search server installation for the first time in your implementation. This process takes some time to complete depending on the size of the data being loaded and may slow down the performance during peak hours. We recommend that you perform the initial load task during off peak hours as it affects the CA SDM server performance. When initial load is checked, the pdm_es_rebuild will run by default. when initial load is not checked, you need manually execute the pdm_es_rebuild command. For more information, see the Perform Initial Load and Refresh Index and Data on Search Server.

Installation Directory	By default, the search server is installed in *C:\Program Files\CA \SC\SearchServer*. Click **Browse** to change the default install location.

Click **Next**.

xFlow Interface Configuration Details

If you have opted to install the xFlow Interface, complete the xFlow Interface Configuration installation screen options and click **Next**. Ensure that the ports are open before configuring the port information.

NOTE

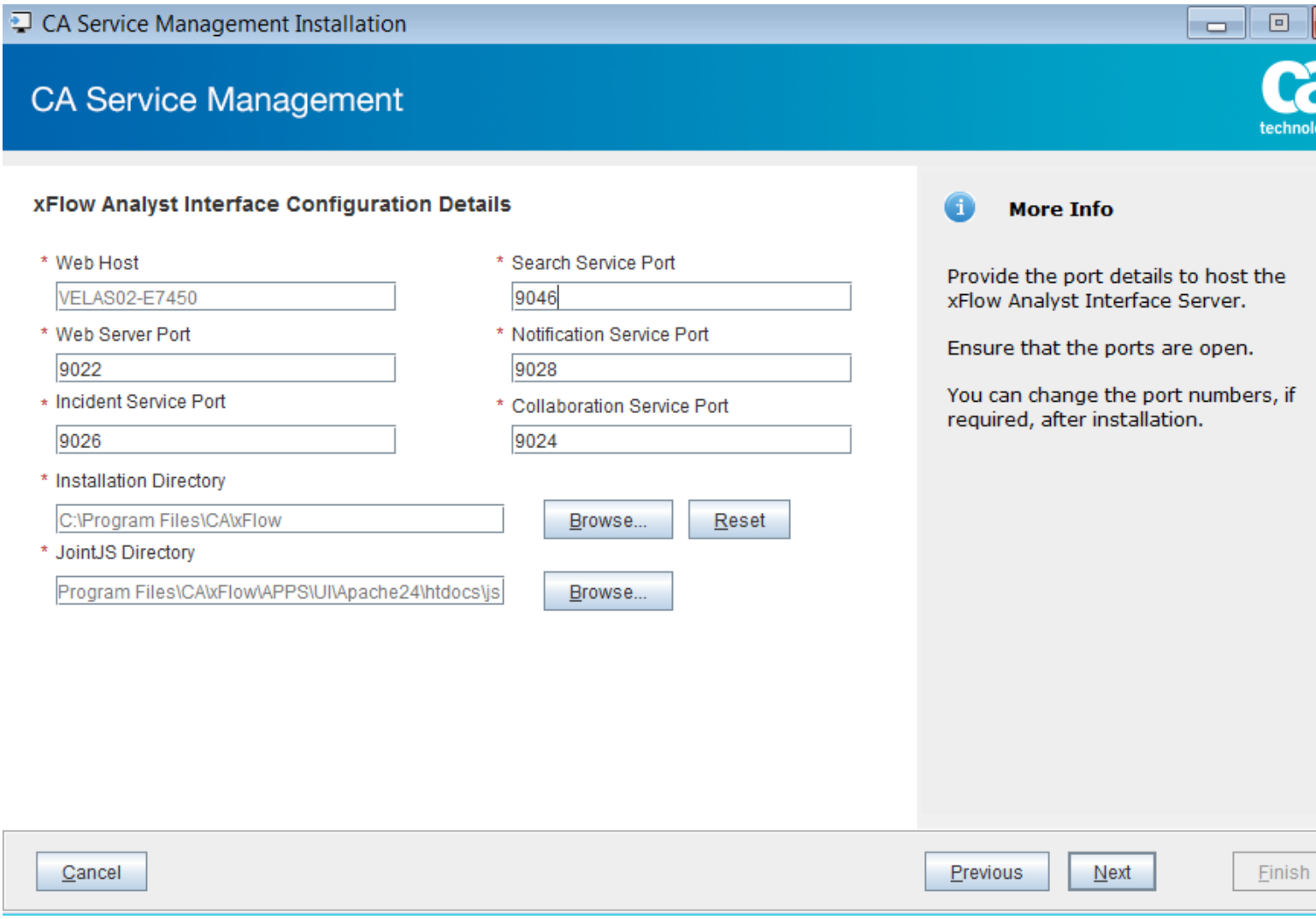
Service Point App is available with the xFlow Interface installation and can be launched from the xFlow Interface App Launcher.

WARNING

Jointjs is a JavaScript library which helps to visualize the SDM classic workflow as a graphical representation. The xFlow Interface uses Jointjs and requires certain files for successful installation. You must download the required Jointjs files before starting the installation.

Follow these steps:

1. Download the **CA SERVICE MANAGEMENT 17.4 3RD PARTY COMPONENTS.iso** from Broadcom Support.
2. Extract the Jointjs folder from the Jointjs.zip file.
3. Verify that Jointjs folder includes the following files:
   a. **joint.min.js**
   b. joint.min.css
4. Note the path where you placed the Jointjs folder that includes the files.
   When you run the installer, on the screens corresponding to the xFlow Interface installation you must provide the path for the Jointjs folder you saved.



Field Options	Description
Web Host	Specify the system that hosts the web server. By default, your web host is your local system.
Search Service Port	Specify the port number of the Search Service. The default value is 9046.
Web Server Port	Specify the port number of the web server. The default value is 9002.
Notification Service Port	Specify the port number for the Notification Services. The default value is 9008.
Collaboration Service Port	Specify the port number for the Collaboration Server Service. The default value is 9016.
Installation Directory	By Default, the xFlow Interface Interface is installed in *C:\Program Files\CA\xFlow*. Click **Browse** if you want to change the default install location.

Workflow JointJS Directory	Click Browse and select the path of the folder which contains the Jointjs files. The specified folder must include the following files: • – joint.min.js – joint.min..css

NOTE

After completing the installation, the xFlow Interface UI files are available under the path *C:\Program Files\CA\xFlow\Apps\Services\incidentmicroservice<version>\public*. If you are upgrading from an earlier version to CA Service Management 17.4, the xFlow Interface UI files are moved to the preceding path from the formerly installed (Apache/IIS) location.

Verify CA Service Desk Manager Installation

After you complete the installation process for CA SDM, perform the following to verify that the installation was successful:

1. Verify that a system environment variable for the path is set for the product installation directory that you specified. The default install home directory is ***\Program Files\CA\Service Desk Manager*** . For a 64-bit Windows operating system, the path is ***\Program Files (x86)\CA\Service Desk Manager*** . For a non-Windows operating system, the path is **/opt/CAisd/**.
2. Verify the following information:

1. a. In the Control Panel (Add or Remove Programs), verify that an entry appears for CA SDM.
   b. From the Start menu, verify that the following options appear:
      - Start the Configuration Wizard.
      - Start the Web Interface
      - Start the Web Screen Painter
      - Contact Technical Support.
      - Uninstall CA Service Desk Manager.
2. From the Windows Start menu, click Start, CA, Service Desk Manager, Service Desk Manager Web Client. If the Web Interface starts properly, then your installation is successful. For more information about the default user accounts created at the time of installation, see the CA SDM Default User List topic. If you cannot verify this information, the product has not been installed correctly. In this case, start the CA SDM installation again to modify the installation.
3. When CA SDM is up and running, it is recommended to check the process memory. For optimal performance, we recommend the following method:
   a. Set a notification when the process memory exceeds 1.25 GB and begin a check on the processes that are running.
   b. Set a warning notification when the process memory exceeds 1.5 GB and take corrective actions to check the memory usage.
4. The default tenant Home Page is displayed.
5. Check the Status of the Required Ports.

Securing CA SDM from Cross-Site Scripting Vulnerabilities

The CA SDM installation is susceptible to reflected cross-site scripting vulnerabilities, which might result in the infected URL being reflected back to the user. To secure CA SDM from such vulnerabilities, validation parameters exist in the web.cfg file. These parameters perform a white list validation in the webengine. Also, install the NX option on the primary and secondary servers to secure CA SDM.

Points to consider before you proceed with securing CA SDM:

- The SDM URL parameters that are defined in the web.cfg file are validated for securing CA SDM.

- You can add SDM URL parameters in the web.cfg file with required validation pattern. You can also add validation patterns, if necessary.

Secure CA SDM from Cross-Site Scripting Vulnerabilities in Conventional and Advanced Availability Mode

Follow these steps:

1. Stop the CA SDM services.
2. On the primary server, execute the following command to install the NX option.
   ```
   pdm_options_mgr -c -a pdm_option.inst -s VALIDATE_REQUEST_PARAMETER -v 1
   ```

   NOTE

   For each secondary server, manually add or update the NX option in the *NX.env* file that is located in *$NX_ROOT* directory.
3. (Optional) To avoid losing the changes when you run the *pdm_configure* command with the *-t* flag.
   ```
   pdm_options_mgr -c -a pdm_option.inst -s VALIDATE_REQUEST_PARAMETER -v 1 -t
   ```
4. Restart the CA SDM services.
5. (Optional) In Advanced Availability mode, perform rolling maintenance to apply the NX option on all servers.

How to Map a Parameter to an Existing White List Pattern

When you identify a parameter to be vulnerable, in the *web.cfg* file, you can map the parameter to a white list pattern against which the parameter is validated. For example, *CONTACT_ID* is the parameter that is vulnerable and you want to map the *AlphaNumericOnly* white list pattern to the parameter, which means the CONTACT_ID parameter can accept only alphanumeric characters.

Follow the steps:

1. Edit the *web.cfg* file, and in the *XSS Vulnerability* section, add the parameter that you want to validate against a pattern in the following way:
   ```
   SecureParameter.CONTACT_ID AlphaNumericOnly
   ```
2. *Edit the web.cfg.tpl file and update the same parameter.*
3. Restart the CA SDM services.

How to Create a White List Pattern

Apart from the out-of-the-box patterns, you can add or edit existing patterns to the web.cfg file.

Edit an Existing Pattern

For example, edit the following pattern to add colon as the allowable character: *AlphaNumericOnly ^[A-Za-z0-9]*$* to *AlphaNumericOnly ^[A-Za-z0-9:]*$*

Non-Windows

Edit the *web.cfg* file, locate **Patterns for Non-Windows** section and add the allowable character in the following way:

```
Non_windows_SecureValidator.AlphaNumericColonOnly ^[A-Za-z0-9:]*$
```

Windows

Edit the *web.cfg* file, locate **Patterns for Windows** section and add the allowable character in the following way:

```
Windows_SecureValidator.AlphaNumericColonOnly ^[A-Za-z0-9:]*$
```

After you add the patterns in the *web.cfg* file, perform the following steps:

1. *Edit the web.cfg.tpl file and update the same pattern.*
2. Restart the CA SDM services.

Installing Web Screen Painter

Web Screen Painter helps you to modify the CA Service Desk Manager schema and web interface. We recommend you to install the Web Screen Painter on a different server from where you have installed CA Service Desk Manager.

Perform the following steps:

1. Log in to the server. The following CA SDM configuration applies:
   – Primary or secondary server or any remote computer, if you have installed CA SDM with Conventional Configuration.
   – Background server, if you have installed CA SDM with Advanced availability.

   WARNING
   Web Screen Painter can only run on the background server. The Web Screen Painter installer does not stop you from installing it on any other server in the Advanced Availability configuration. For example, if you install the Web Screen Painter on a server other than the background server, it prevents you from running on a standalone, or on the standby server and application server. The Web Screen Painter shortcuts appear on the Start menu and but on clicking the shortcut, an error message is displayed.
2. Launch the installer.
3. Select CA Web Screen Painter for Service Desk Manager.
4. Follow the on-screen instructions to complete the Web Screen Painter installation.
   Web Screen Painter is installed.
   In a Web Screen Painter preview session in test mode, search filters are ignored on new tables that are not published.

Installing CA Service Catalog

Ensure that you have reviewed the Pre-Installation Considerations before proceeding with the installation.

Ensure that you have reviewed the Integrate the CA Service Management Products.

NOTE

- By default, the Service Catalog Content Pack is skipped as the parameter `slcm.skip.import.contentpack` is set to 1 in the **common_config.properties** file.

Perform the following steps:

1. Download the CA Service Management installer 17.4 from Broadcom Support.
   – **CA SERVICE MANAGEMENT SDM PACKAGE 17.4 FOR WINDOWS (WITH CATALOG).iso**
   – **CA SERVICE CATALOG 17.4.iso**
   – **CA SERVICE MANAGEMENT APM PACKAGE 17.4 FOR WINDOWS (WITH CATALOG).iso**
2. Copy and extract the files from the installation media on all servers in your environment.
3. Navigate to the folder where you have copied the installation files and right-click **Setup.exe** to select the **run as administrator** option.
4. Select the language and click **Next**.
5. Select **CA Service Management** and click **Next**.
6. Review and Accept the License Agreement Information and click **Next**.
7. Select **Microsoft SQL Server** or **Oracle** as the database type in the Database Configuration screen.

For the Windows Microsoft SQL Server database, **mdbadmin** is the default user.

For the Oracle database, verify that you are providing a valid tablespace path. If this path is invalid the database installation fails. The following path is an example of a valid Oracle tablespace path:

```
C:\app\Administrator\oradata\Oracle_Service_Name.
```

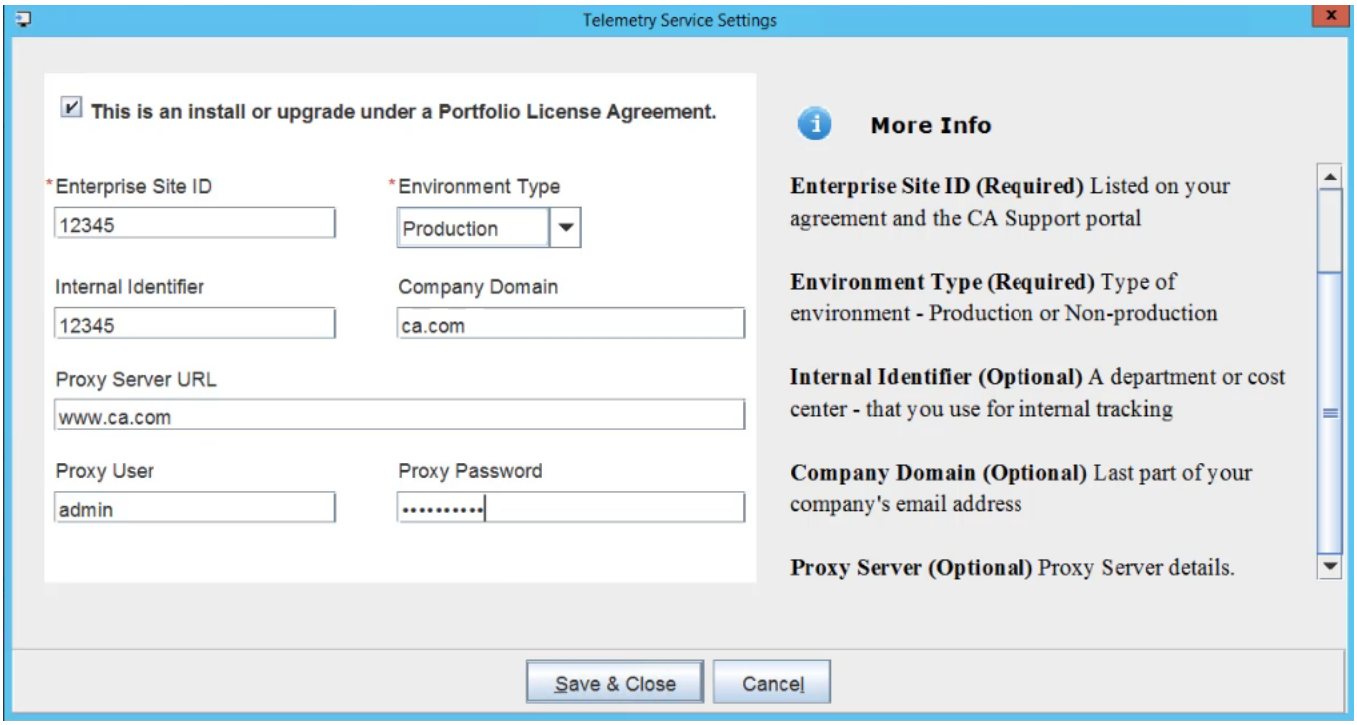
a. If you have **the Microsoft SQL Server** database, provide the following field information:
   - **Database Server**: Enter the hostname of the database server.
     If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
   - **Database Name**: Enter the Database name (MDB), the name of the target DBMS. The default value ismdb.
   - **Database Port**: Provide the port identifier for the target DBMS.
   - **Database Server Instance**: The database instance name. Leave this field blank if you are using the default instance.
   - **Database Admin User**: For Microsoft SQL Server, **sa** is the default value.
     The **sa** user is the admin user with permission to create a user and schema.
   - **Database Admin Password**: Enter the database password of the user specified by a database admin user. No default value is assigned to this parameter.
   - **mdbadmin Password**: Enter the password for the user specified for the MDB admin user.
   - **Confirm mdbadmin Password**: Confirm the password.

b. If you have the **Oracle** database, provide the following field information:

– • **Database Server**: Enter the host name of the Oracle database server.
  If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
  - **Oracle Service Name**: Enter the Oracle Service name.
  - **Listener Port**: Enter the listener port for the database.
  - **Net Service Name**: Enter the Net Service Name of Oracle database where the database resides.
    If the database is remote, use the Net Service Name that is defined in the Oracle client on the local computer. CA SDM has access to the database through a local installation of the Oracle client.
  - **DBA User Name**: Provide the name of an Oracle user with DBA access. The default value for DBA user is **SYS**.
  - **DBA User Password**: Enter the password for the DBA user.
  - **mdbadmin Password**: Enter the password for MDB admin.
  - **Confirm mdbadmin Password**: Confirm the password.
  - **Tablespace Path (on DB Server)**: Provide the tablespace path on the database server.
  - **Oracle Home Path**: Provide the Oracle client 32-bit path.
  - **Data Tablespace Name**: Provide the name of the data tablespace. The default value is MDB_DATA.
  - **Index Tablespace Name**: Provide the index Tablespace Name: The default value is MDB_INDEX.

  Click **Next**.

8. Select **CA Service Catalog** from Select the Products and Integrations screen and click **Next**.
9. Select **This is an install or upgrade under a Portfolio License Agreement** on Telemetry Service Settings page. Provide the following field information:
   – **Enter Site ID :** Listed on your agreement and the CA Support portal.
   – **Environment type:** Type of Environment Production or Non-Production.
   – **Internal Identifier (optional):** A department or cost center- that you use for internal tracking.
   – **Company Domain (optional):** Last part of your company email address.
   – **Proxy Server URL (optional):** Proxy server details.
   – **Proxy User:** Proxy server username.
   – **Proxy Password:** Proxy server password.
   Click **Save & Close**.



10. Review the Installation Prerequisites page and click **Next**.
11. Enter the **CA Service Management Administrator (CASMAdmin)** credentials for CA Service Management Click **Next**.
12. Select the services to configure in the **CA Service Catalog Configuration Details** page and click **Next**.
13. Enter the **CA Embedded Entitlements Manager (CA EEM)** details and click **Next**.
14. Review the **Pre-Installation Configuration Summary** page and click **Next**.
15. Click **Install**. Review the Installation progress and when the installation is complete, click **Next**.
16. Review the Post Deployment summary page and click **Finish** to exit the installer.
17. Review the log files **stdout.txt and stderr.txt** in the C:\ directory or the location where you have installed CA Service Catalog if the installation failed. Fix the errors and click **Retry Install**.
18. After the installation is complete, verify the CA Service Catalog installation by logging into the following URL as a Catalog administrator to view the CA Service Catalog User Interface:
    `http://<CA_Service_Catalog_Server_Name>:<Port_Number>/usm/`
19. Complete the CA Service Catalog Post Installation Tasks.

Installing CA Asset Portfolio Management

CA Service Management installer performs prerequisite checks and mandatory validations before installing CA Asset Portfolio Management. Ensure that you have reviewed the Pre-Installation Considerations topics before proceeding with the installation.

Follow these procedures:

Install CA Asset Portfolio Management

NOTE

CA Asset Portfolio Management App Server uses the reserved port 99 and CA Service Catalog uses the reserved port 7777. Reserved ports are set internally and are not configurable during CA Service Management installation. Do not configure any other services on these reserved ports.

NOTE
For installing CA APM on Oracle 21c database, add a new system environment variable "TNS_ADMIN" on the CA APM Web / Application servers. This variable value must point to the Oracle 21c 32-bit client installed location. Set the variable value, for example <oracle32bit_client_installed_location>\product \21.0.0\client32bit_1\NETWORK\ADMIN

- Refer Operating System Requirements for CA Asset Portfolio Manager ITAM

Follow these steps:

1. Download the CA Service Management 17.4 DVD from Broadcom Support.
   – CA ASSET PORTFOLIO MANAGEMENT 17.4 FOR WINDOWS DVD (WITHOUT CATALOG).ISO
   – CA SERVICE MANAGEMENT APM PACKAGE 17.4 FOR WINDOWS (WITH CATALOG).ISO
2. Copy and extract the files from the installation media on the servers in your environment.
3. Right-click **Setup.exe** to select the **run as administrator** option.
4. Select the language and click **Next**.
5. Select CA Service Management.
6. Review and Accept the License Agreement Information and click **Next**.
7. Select **Microsoft SQL Server** or **Oracle** as the database type in the Database Configuration screen:
   For Windows Microsoft SQL Server database, **mdbadmin** is the default user.
   Verify that you are providing a valid tablespace path if you have an Oracle database. If this path is invalid the database installation fails. The following path is an example of a valid Oracle tablespace path:
   `C:\app\Administrator\oradata\Oracle_Service_Name.`
8. Provide the database details:
   a. If you have **a Microsoft SQL Server** database, provide the following field information:
      - **Database Server**: Enter the hostname of the database server.
        If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
      - **Database Name**: Enter the Database name (MDB), the name of the target DBMS. The default value is mdb.
      - **Database Port**: Provide the port identifier for the target DBMS.
      - **Database Server Instance**: The database instance name. Leave this field blank if you are using the default instance.
      - **Database Admin User**: For Microsoft SQL Server, **sa** is the default value.
        The **sa** user is the admin user with permission to create a user and schema.
      - **Database Admin Password**: Enter the database password of the user specified by a database admin user. No default value is assigned to this parameter.
      - **mdbadmin Password**: Enter the password for the user specified for the MDB admin user.
      - **Confirm mdbadmin Password**: Confirm the password.
   b. If you have the **Oracle** database, provide the following field information:
      NOTE
      For Oracle database in CA Service Management Solution deployment, ensure that the Oracle-32 bit path is set on top of the Environment path variable list for CA APM. CA APM installation does not require the 64-bit path, ignore this field while performing CA APM installation.
      - **Database Server**: Enter the host name of the Oracle database server.
        If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
      - **Oracle Service Name**: Enter the Oracle Service name.
      - **Listener Port**: Enter the listener port for the database.
      - **Net Service Name**: Enter the Net Service Name of Oracle database where the database resides.

If the database is remote, use the Net Service Name that is defined in the Oracle client on the local computer. CA SDM has access to the database through a local installation of the Oracle client.

- **DBA User Name**: Provide the name of an Oracle user with DBA access. The default value for DBA user is **SYS**.
- **DBA User Password**: Enter the password for the DBA user.
- **mdbadmin Password**: Enter the password for MDB admin.
- **Confirm mdbadmin Password**: Confirm the password.
- **Tablespace Path (on DB Server)**: Provide the tablespace path on the database server.
- **Oracle Home Path**: Provide the Oracle client 32-bit path.
- **Data Tablespace Name**: Provide the name of the data tablespace. The default value is MDB_DATA.
- **Index Tablespace Name**: Provide the index Tablespace Name: The default value is MDB_INDEX.
  Click **Next**

WARNING

CA Service Management does not support the un-installation of the CA Asset Portfolio Management. Uninstalling CA APM does not uninstall CA MDB, as there is no rollback mechanism for the same. For more information, see CA Asset Portfolio Manager Considerations.

9. Select **CA Asset Portfolio Management** and click **Next**.
10. Select **This is an install or upgrade under a Portfolio License Agreement** on Telemetry Service Settings page. Provide the following field information:
    - **Enter Site ID :** Listed on your agreement and the CA Support portal
    - **Environment type:** Type of Environment Production or Non-Production
    - **Internal Identifier (optional):** A department or cost center- that you use for internal tracking
    - **Company Domain (optional):** Last part of your company email address
    - **Proxy Server URL (optional):** Proxy server details
    - **Proxy User:** Proxy server username
    - **Proxy Password:** Proxy server password
    - Click **Save & Close**.
11. Review the Installation Prerequisites page and click **Next.**
12. Enter the **CA Service Management Administrator (CASMAdmin)** credentials for CA Service Management Click **Next.**
13. Enter the **CA Asset Portfolio Management Server Details** and click **Next**.
14. Enter the **CA Embedded Entitlements Manager (CA EEM)** details and click **Next.**
15. Review the **Pre-Installation Configuration Summary** page and click **Next**.
16. Click **Install**. Review the Installation progress and when the installation is complete, click **Next**.
17. Review the Post Deployment summary page and click **Finish** to exit the installer.
18. Complete the CA Asset Portfolio Management Post Installation Tasks

Start the CA APM Services

After the installation is complete, start all services. Perform the following steps:

1. Navigate to **Start**, **Settings**, **Control Panel**.
2. Double-click **Administrative** Tools.
3. Double-click **Services**.
4. Locate and start each of the following services:

- – Apache Tomcat 9.0 AMS Service
  – CA Asset Portfolio Management - Data Importer Engine
  – CA Asset Portfolio Management - Event Service
  – CA Asset Portfolio Management - Export Service
  – CA Asset Portfolio Management - Registration Service
  – CA Asset Portfolio Management - HW Reconciliation Engine
  – CA Asset Portfolio Management - LDAP Import Service
  – CA CASM

    `For performance reasons, it is recommended that you do not start the CA CASM service when you are not using multi-tenancy.`
  – CA iTechnology iGateway 4.7

Start the Web Interface for CA APM

After installing CA APM, start the web interface to verify that CA APM is ready to use. Provide the login credentials and web interface URL to users. As an Administrator you can configure security, user interface, set up hardware, and, if necessary, configure the product components.

1. Start the web interface using one of the following methods:
   a. Open a supported Web browser and enter the appropriate URL in the following format:

      `http://servername:port/itam`
   b. Replace server name and port with the name of the server and the port that are hosting the CA APM web servers.
   c. A Start menu shortcut is created on your web server that references the URL location. Click **Start**, **Programs**, **CA**, **Asset Portfolio Management**.
   d. To log in to CA APM, enter the credentials provided during the installation.

      NOTE

      If you changed the password during the installation, use the password that you created.

   A browser error or a user name error can appear in certain cases. You can resolve these errors by following the troubleshooting instructions that are given in the error message. For more information, see: Troubleshooting CA Asset Portfolio Management

Verify the CA APM Installation

After you have completed all installation procedures, you can verify that CA APM was installed successfully. Perform the following steps:

1. Log in to the servers where you installed CA APM.
2. Click **Start**, **Control Panel**, **Programs**.
3. Verify that the following component is available on all applicable servers:
   CA Asset Portfolio Management
   You have completed the installation verification.

   NOTE

   If you backed up the Storage folder contents before installing, restore the contents now. Use the Storage folder contents that you copied and paste them into the following location:

   [ITAM Root Path]/Storage/

   If you receive a prompt about folders that already exist, merge the folders.

   NOTE

   After installing CA APM 17.4, the release version is shown as **17.0.0.1034** on the application user interface.

The env.info file for CA APM 17.4 in the CA APM installed folder includes the following information:

- product=ITAM
- major=17
- minor=4
- cum=0
- roll-up=0

Other Products Installed with CA APM

The following products are installed when you install CA APM:

- Common Asset Viewer
- CORA 17.4.0.57
- Common Administration for Service Management (CASM)

NOTE
The only supported ITAM WCF service is the WSDL, which can be downloaded from the ITAM Web UI by navigating to the Administration tab >> Web Services tab >> WSDL Download.

Post Installation Tasks

Post installation tasks for CASM 17.4.

After successful installation, CA Service Management products (CA SDM, CA Service Catalog, and CA Asset Portfolio Management require specific configurations. Review the following to understand the mandatory and optional post-installation tasks.

- CA Service Desk Manager Post Installation Tasks
- CA Service Catalog Post Installation Tasks
- CA Asset Portfolio Management Post Installation Tasks

CA Service Desk Manager Post Installation Tasks

The CA Service Management provides an option in the installer to automate the post-install tasks.

CA Service Management provides an option in the installer to automate the post-install tasks. If you have selected **Yes** to automate the post-installation steps while running the CA Service Management Installer Wizard, you are not required to perform any manual post-install steps for CA Service Desk Manager (SDM).

If you have selected **No** for automating the post-install steps while running the installer wizard, perform manually the following post-install tasks as per your deployment requirement.

- Post-Installation Tasks After Installing CA SDM 17.4
- Post-Installation Tasks After Installing CA SDM 17.4 on CA SDM 17.1
- Post-Installation Tasks After Installing CA SDM 17.4 on CA SDM 17.2
- Post-Installation Tasks After Installing CA SDM 17.4 on CA SDM 17.3

Post-Installation Tasks After Installing CA SDM 17.4

CA Service Desk Manager

If you have selected **No** for automating the post-install steps while running the installer wizard, perform manually the following post-install tasks as per your deployment requirement.

NOTE

For **Advance Availability Mode**: Perform the following steps sequentially on Background, Standby, and Application servers.
For **Conventional Mode**: Perform the following steps sequentially on Primary and Secondary Servers.

Update the Minor Version in Nx.Env file

The help option (Help Option on this Window) from the CA SDM User Interface provides context-sensitive product documentation. Perform the following steps to view the Help Window:

Perform the following steps:

1. Navigate to the `$NX_ROOT\site` folder.
   a. Open the `install.properties` in a text editor.
   b. Find the parameter: `genfact.gx_minor_ver`
   c. Modify the value to **4**
   d. genfact.gx_copyright_year=2023
2. Navigate to *NX_ROOT/site/config.properties* file and ensure that the value of ***resources.created parameter*** is set to 0.
3. Navigate to the NX_ROOT/bin and open the command prompt and run the below commands:

```
pdm_options_mgr -c -s attach_force_save_dialog -v 1 -a pdm_option.inst -t
pdm_options_mgr -c -s doc_url -v "https://docops.ca.com/rest/ca/product/latest/topic" -a pdm_option.ins -
tt
pdm_options_mgr -c -s doc_default_hid -v "HID_Service_Management_Home_Page" -a pdm_option.inst
pdm_options_mgr -c -s doc_space_prefix -v "CASM" -a pdm_option.inst
```

4. For conventional mode and all application servers and stand-by servers for advanced availability, run:

```
pdm_configure
```

   Advanced availability on the BG Server run:

```
pdm_configure -S
```

5. Change the directory to **NX_ROOT/data**, and execute the following command:

   NOTE

   Apply the *pdm_load* command once on Primary Server in Conventional mode or Background Server in Advance Availability mode.

```
pdm_load -f r17_1_insert.dat
pdm_load -u -f r17_1_update.dat
pdm_load -r -f r17_1_delete.dat
pdm_load -f r17_2_insert.dat
pdm_load -u -f r17_2_update.dat
pdm_load -f r17_3_insert.dat
pdm_load -u -f r17_3_update.dat
pdm_load -f r17_4_insert.dat
pdm_load -u -f r17_4_update.dat
```

6. Clear the browser cache by running the following command:

```
pdm_webcache -H
pdm_webcache -b
```

7. Restart the CA SDM Services.

Remote Web Screen Painter (WSP) and Java Update

If you have installed Remote Web Screen Painter in your environment, perform the following steps:

1. Log in to the server where CA SDM is installed:
   a. Navigate to the `$NX_ROOT/java/lib` folder.
   b. Copy the following jar files:
      1. WebScreenPainter.jar
      2. sd-utils.jar
      3. domsrvr_utils.jar
      4. slump.jar
      5. log4j-api-2.17.1.jar
      6. log4j-core-2.17.1.jar
      7. log4j-jcl-2.17.1.jar
      8. bc-fips-1.0.2.1.jar
      9. bctls-fips-1.0.12.2.jar
2. Log in to the server where remote WSP is installed:
   a. Navigate to the `$NX_ROOT/java/lib` folder and paste the jar files that you copied in Step 1.
3. Log in to the server where CA SDM is installed:
   a. Navigate to the shared JRE folder `ex: C:\Program Files (x86)\CA\SC\JRE` folder.
   b. Copy the `11.0.18` folder.
4. Log in to the server where remote WSP is installed:
   a. Navigate to the shared JRE folder `ex: C:\Program Files (x86)\CA\SC\JRE` folder.
   b. Paste the `11.0.18` folder.
   c. Navigate to the `$NX_ROOT` folder.
   d. Open the `NX.env` file in a text editor.
   e. Find the parameter `NX_JRE_INSTALL_DIR=`
   f. Modify the parameter value from 1.8.0_112 to **11.0.18**
   g. On the next line, add the following:
   `@NX_LOG=C:/PROGRA~2/CA/SERVIC~1/log`
   a. Save the `NX.env` file.
   Remote WSP is upgraded to AdoptOpenJDK 11.0.18
5. Log in to the server where CA SDM is installed and copy the following files to the same folder location in the remote WSP server:
   a. $NX_ROOT/bin/pdm_wsp.exe
   b. $NX_ROOT/bin/bop_sinfo.frg
   c. $NX_ROOT/site/ddict.sch
   d. $NX_ROOT/site/cfg/log4j2.properties
   e. $NX_ROOT/site/cfg/log4j2.properties.tpl
6. Launch remote WSP to verify that you are able to design and customize CA SDM forms.

Manual Post-Installation Task for CMDB

To run CA Service Management 17.4 in SSL environment, CA SDM Administrators must perform the following post-installation steps for CMDB Visualizer Interface to open using the **HTTPS** protocol when launching from CA SDM UI:

1. In the NX.env file, change the **NX_CMDB_VISUALIZER** entry to provide **HTTPS protocol** and **HTTPS port**.
   For example: @NX_CMDB_VISUALIZER=https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer
2. In **casm.conf.js** file from NX_root\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\conf, provide the HTTPS as protocol and HTTP port.
   For example:

```
sdm: {

      vizserver: 'https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer/viz',
      server: 'https:// sdmurl:<SDM_SSL_PORT>/CAisd/pdmweb.exe'
    }
```

Manual Post-Installation Task for CMDB

Manual Post-Installation Task for OAuth 2.0 (Maileater/Mailer)

After generating an access token for OAuth 2.0 to connect to a mail server, perform the mandatory post-install tasks on all servers in your environment:

1. Create a backup of web.xml from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF in your local directory to perform backout if required.
2. Open the web.xml file from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF.
3. Search for the following statement:
   ```
   <!--USD servlet mappings start-->
   ```
4. Copy and paste the following configuration under the statement:

   ```
   <servlet>
     <servlet-name>OAuthProcessor</servlet-name>
     <servlet-class>com.broadcom.oAuth.OAuthProcessor</servlet-class>
    </servlet>
   ```
5. Copy and paste the following configuration after any <servlet-mapping>:

   ```
   <servlet-mapping>
     <servlet-name>OAuthProcessor</servlet-name>
     <url-pattern>/OAuthProcessor</url-pattern>
   </servlet-mapping>
   ```
6. Save and close the web.xml file.
7. For OAuth servlet preservation in a primary and secondary environment, the secondary server must also have the OAuth processor entries in the following file: *<NX_ROOT>\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF\web.xml.last*.
8. Repeat Steps 1-8 to update **web.xml.tpl** file in the NX_ROOT\samples\pdmconf\web.xml.tpl folder to avoid losing these changes when pdm_configure is run at a later point of time.
9. Run the following command:
   ```
   pdm_tomcat_nxd -c STOP command
   ```
10. Wait for some time (a minute or so) and run the following command:
    ```
    pdm_tomcat_nxd -c START command
    ```

For more information, see Define a mailbox.

Manual Post-Installation Task for Excel file (.xlsx) export*

1. Execute the following command:
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst
   ```
2. Execute the following command to add the new NX variable to NX.env.tpl file.
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst -t
   ```
3. Restart CA SDM services.

(Optional) Manual Post-Installation Task for Knowledge Document export with German fields

If you are exporting Knowledge Document with German Fields (included in the export field list), then perform the below post-install steps for successful export of knowledge documents:

1. Take the backup of NX_ROOT\NX.env file (NX_ROOT is the installation directory for CA Service Desk Manager).
2. Open NX_ROOT\pdmconf\NX.env_nt.tpl file using any text editor and take the value of NX variable @NX_KEIT_AVAILABLE_FIELDS which is COPIED_DATA and update Nx.env in NX_ROOT by executing below command:

   ```
   pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA>  -a  pdm_option.inst
   ```

   Where COPIED_DATA is the value copied from NX.env_nt.tpl (step 2).
   For example, " pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <provide the exact string here> -a pdm_option.inst "
   Note to avoid losing the updated changes when you run pdm_configure, execute the above command with the '-t' flag as follows:

   ```
   pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA> -a pdm_option.inst -t
   ```

(Optional)Manually added NX variables need to be merged, after upgrading to 17.4.0.0

Manually updated NX variables are lost when a GA release or RU patch delivers a new `NX.env_nt.tpl` file (located in `NX_ROOT/pdmconf`). This necessitates an additional manual step to merge the custom values from the backup file of `NX.env_nt.tpl` created by `pdm_configure`.

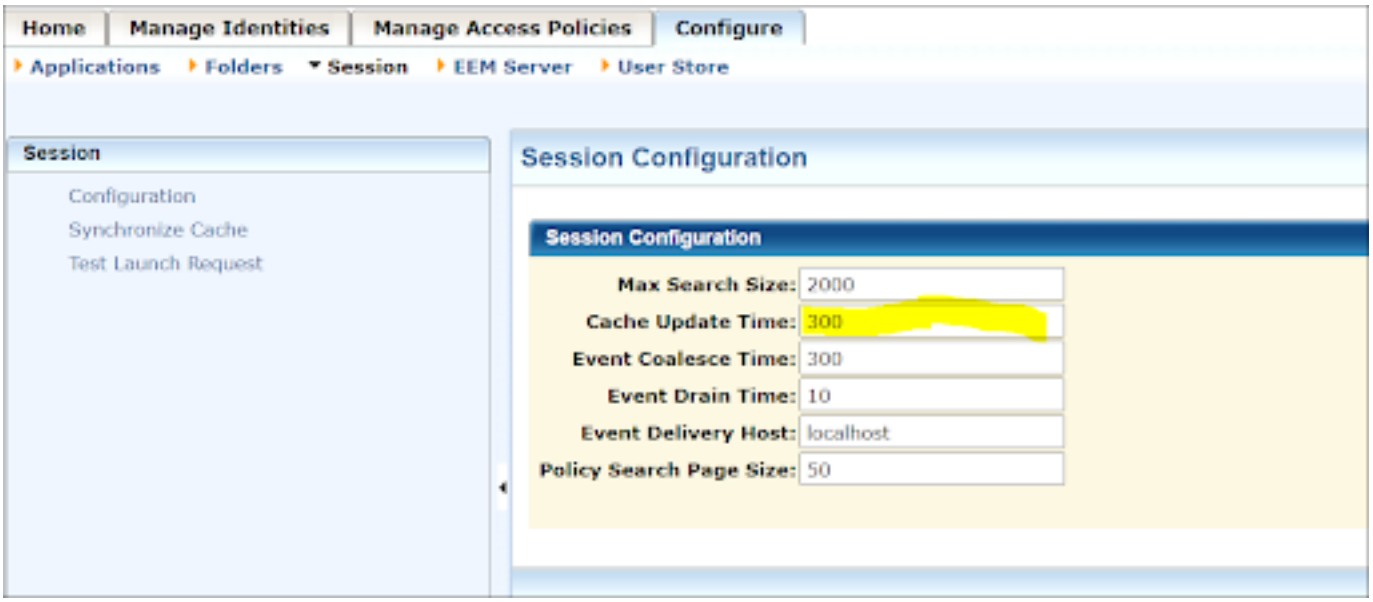
- Manually merge the required changes into the NX.env_nt.tpl file located in NX_ROOT\pdmconf from the backup file at NX_ROOT\REPLACED\GOL-78_EN_US.OLD\pdmconf\NX.env_nt.tpl.
- Restart the SDM services.
- **For Conventional Mode**: Restart the SDM services on all secondary servers.
- **For Advanced Availability Mode**: Restart the SDM services on all standby servers and application servers.

(Optional) Manual Post-Installation Task for EEM Cache Timeout Configuration Update

You may face issues while logging in to EEM or SDM using the EEM user accounts due to a session cache timeout mismatch between EEM and SDM.

In 17.4, you can update the EEM cache timeout configuration by adding a new NX variable, "NX_EEM_CACHE_UPDATE_TIME" in SDM.

Keep the value the same as the EEM cache timeout configuration value mentioned in the following screenshot.



To set the EEM cache update time to 300 seconds, perform the following steps:

To add the variable in the **nx.env** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst
```

To add the variable in the **nxenv_nt.tpl** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst -t
```

If 17.4 xFlow/ Service Point is installed

After succesful installation of 17.4 xFlow/ Service Point refresh index and data on the Search Server, then perform the following steps:

Run the PDM_ES_REBUILD_INDEX Command Utility

To view the Service Catalog data (Service request and Service offering) in the Service Point, execute one of the files based on your operating system:

- $ES_HOME/search/bin/pdm_es_rebuild_index.bat (Windows)
- $ES_HOME/search/bin/pdm_es_rebuild_index.sh (Linux)

NOTE
In case of Linux Search Server environment, ensure to run the following commands on the terminal:

```
sysctl -w vm.max_map_count=262144
ulimit -Hn 65535
```

WARNING
We recommend that you execute the pdm_es_rebuild_index command during off-peak hours as it affects the CA Service Catalog server performance.

(Optional) Manual Post-Installation Task for Rebuild Index

If you have upgraded or are planning to install CA EEM 12.6 with Service Point and CA Service Catalog, ensure that you rebuild the CA Service Catalog index after completing the CA EEM upgrade/installation process. For more information, see Perform Initial Load and Refresh Index Data on Search Server.
Run the following command to rebuild the index:

```
$ES_HOME/search/bin/pdm_es_rebuild_index.bat index=catalog_index
```

Display CA Service Catalog Featured Offerings in Service Point

After successful installation or after upgrading to CA Service Management 17.4, perform the following mandatory steps if you have an integrated environment of CA SDM and CA Service Catalog and you want to display Service Catalog Featured Offerings in Service Point. Before performing the following steps, ensure that you execute the "*pdm_es_rebuild_index.bat*" from where you have installed Search Server <*SearchServer_Installed_location>/search/bin*.

There are two ways to perform these mandatory steps, you can select one of these approaches as per your environmental requirements:

Approach 1

1. Navigate to the where you have installed xFlow Interface <*xFlow_Installed_location>/APPS/Services/searchmicroservice-<xx.x.xxx>/conf*> and edit the ***eiam.config*** file.
2. Search for *Logger Configuration* and comment it.
3. After commenting *Logger Configuration*, add the following step right after *Logger Configuration*:
   ```
   <LoggerConfiguration file="<xFlow_Installed_Location>\APPS\Services\searchmicroservice-.<xx.x.xxx>\conf
   \eiam.log4j.config"/>
   ```
4. Save the file.
5. Restart the xFlow Interface Services.

OR

Approach 2

1. After successful installation or after upgrading to CA Service Management 17.4, relaunch the CA Service Management 17.4 Installer wizard.
2. Select and select "**Integrate pre-installed solution components**" and continue with the installation process to complete integrating the solution components.

Update the Federated Search Configuration File

These steps are mandatory if you have configured Federated Search for CA Service Management 17.4. These steps must be performed for Advanced Availability and Conventional Configurations.

1. Shut down the Federated Search Tomcat server if CA SDM Services are up and running by executing the following command:
   ```
   pdm_tomcat_nxd -c stop -t FS
   ```
2. Update the *catalina.properties* file (*bopcfg\www\CATALINA_BASE_FS\conf\catalina.properties)*
   Search for *"common.loader=*" and replace it with the following:
   ```
   common.loader="${catalina.base}/lib/axis2/*.jar","${catalina.base}/lib","${catalina.base}/lib/
   *.jar","${catalina.home}/lib","${catalina.home}/lib/*.jar","${catalina.base}/../../../java/lib/CXF/*.jar"
   ```
3. Start the Federated Search Tomcat Server by executing the following command:
   ```
   pdm_tomcat_nxd -c start -t FS
   ```

Post-Installation Tasks After Installing CA SDM 17.4 on CA SDM 17.1

Post-Installation Tasks After Installing CA SDM 17.4 on CA SDM 17.1

CA Service Desk Manager

If you have selected **No** for automating the post-install steps while running the installer wizard, perform manually the following post-install tasks as per your deployment requirement.

> **NOTE**
>
> For **Advance Availability Mode**: Perform the following steps sequentially on Background, Standby, and Application servers.
> For **Conventional Mode**: Perform the following steps sequentially on Primary and Secondary Servers.

Update the Minor Version in Nx.Env file

The help option (Help Option on this Window) from the CA SDM User Interface provides context-sensitive product documentation. Perform the following steps to view the Help Window:

Perform the following steps:

1. Navigate to the `$NX_ROOT\site` folder.
   a. Open the `install.properties` in a text editor.
   b. Find the parameter: `genfact.gx_minor_ver`
   c. Modify the value to **4**
   d. genfact.gx_copyright_year=2023
2. Navigate to *NX_ROOT/site/config.properties* file and ensure that the value of ***resources.created parameter*** is set to 0.
3. Navigate to the NX_ROOT/bin and open the command prompt and run the below commands:

```
pdm_options_mgr -c -s attach_force_save_dialog -v 1 -a pdm_option.inst -t
pdm_options_mgr -c -s doc_url -v "https://docops.ca.com/rest/ca/product/latest/topic" -a pdm_option.ins -
tt
pdm_options_mgr -c -s doc_default_hid -v "HID_Service_Management_Home_Page" -a pdm_option.inst
pdm_options_mgr -c -s doc_space_prefix -v "CASM" -a pdm_option.inst
```

4. For conventional mode and all application servers and stand-by servers for advanced availability, run:

```
pdm_configure
```

   Advanced availability on the BG Server run:

```
pdm_configure -S
```

5. Change the directory to **NX_ROOT/data**, and execute the following command:

> **NOTE**
>
> Apply the *pdm_load* command once on Primary Server in Conventional mode or Background Server in Advance Availability mode.

```
pdm_load -f r17_2_insert.dat
pdm_load -u -f r17_2_update.dat
pdm_load -f r17_3_insert.dat
pdm_load -u -f r17_3_update.dat
pdm_load -f r17_4_insert.dat
pdm_load -u -f r17_4_update.dat
```

6. Clear the browser cache by running the following command:

```
pdm_webcache -H
pdm_webcache -b
```

7. Restart the CA SDM Services.

Remote Web Screen Painter (WSP) and Java Update

If you have installed Remote Web Screen Painter in your environment, perform the following steps:

1. Log in to the server where CA SDM is installed:
   a. Navigate to `$NX_ROOT/java/lib` folder.
   b. Copy the following jar files:
      1. WebScreenPainter.jar
      2. sd-utils.jar
      3. domsrvr_utils.jar
      4. slump.jar
      5. log4j-api-2.17.1.jar
      6. log4j-core-2.17.1.jar
      7. log4j-jcl-2.17.1.jar
2. Log in to the server where remote WSP is installed:
   a. Navigate to `$NX_ROOT/java/lib` folder and paste the jar files that you copied in Step 1.
3. Log in to the server where CA SDM is installed:
   a. Navigate to `$NX_ROOT/SC/JRE` folder.
   b. Copy the `11.0.18` folder.
4. Log in to the server where remote WSP is installed:
   a. Navigate to `$NX_ROOT/SC/JRE` folder.
   b. Paste the `11.0.18` folder.
   c. Navigate to the `$NX_ROOT` folder.
   d. Open the `NX.env` file in a text editor.
   e. Find the parameter `NX_JRE_INSTALL_DIR=`
   f. Modify the parameter value from 1.8.0_112 to **11.0.18**
   a. Save the `NX.env` file.
   Remote WSP is upgraded to AdoptOpenJDK 11.0.18
5. Launch remote WSP to verify that you are able to design and customize CA SDM forms.

Manual Post-Installation Task for CMDB

To run CA Service Management 17.4 in SSL environment, CA SDM Administrators must perform the following post-installation steps for CMDB Visualizer Interface to open using the **HTTPS** protocol when launching from CA SDM UI:

1. In the NX.env file, change the **NX_CMDB_VISUALIZER** entry to provide **HTTPS protocol** and **HTTPS port**.
   For example: @NX_CMDB_VISUALIZER=https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer
2. In **casm.conf.js** file from NX_root\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\conf, provide the HTTPS as protocol and HTTP port.
   For example:

```
sdm: {

      vizserver: 'https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer/viz',
      server: 'https:// sdmurl:<SDM_SSL_PORT>/CAisd/pdmweb.exe'
    }
```

Manual Post-Installation Task for OAuth 2.0 (Maileater/Mailer)

After generating an access token for OAuth 2.0 to connect to a mail server, perform the mandatory post-install tasks on all servers in your environment:

1. Create a backup of web.xml from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF in your local directory to perform backout if required.
2. Open the web.xml file from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF.
3. Search for the following statement:
   ```
   <!--USD servlet mappings start-->
   ```
4. Copy and paste the following configuration under the statement:

   ```
   <servlet>
     <servlet-name>OAuthProcessor</servlet-name>
     <servlet-class>com.broadcom.oAuth.OAuthProcessor</servlet-class>
    </servlet>
   ```
5. Copy and paste the following configuration after any <servlet-mapping>:

   ```
    <servlet-mapping>
      <servlet-name>OAuthProcessor</servlet-name>
      <url-pattern>/OAuthProcessor</url-pattern>
    </servlet-mapping>
   ```
6. Save and close the web.xml file.
7. For OAuth servlet preservation in a primary and secondary environment, the secondary server must also have the OAuth processor entries in the following file: *<NX_ROOT>\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF\web.xml.last*.
8. Repeat Steps 1-8 to update **web.xml.tpl** file in the NX_ROOT\samples\pdmconf\web.xml.tpl folder to avoid losing these changes when pdm_configure is run at a later point of time.
9. Run the following command:
   ```
   pdm_tomcat_nxd -c STOP command
   ```
10. Wait for some time (a minute or so) and run the following command:
    ```
    pdm_tomcat_nxd -c START command
    ```

For more information, see Define a mailbox.

Manual Post-Installation Task for Excel file (.xlsx) export*

1. Execute the following command:
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst
   ```
2. Execute the following command to add the new NX variable to NX.env.tpl file.
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst -t
   ```
3. Restart CA SDM services.

(Optional) Manual Post-Installation Task for Knowledge Document export with German fields

If you are exporting Knowledge Document with German Fields (included in the export field list), then perform the below post-install steps for successful export of knowledge documents:

1. Take the backup of NX_ROOT\NX.env file (NX_ROOT is the installation directory for CA Service Desk Manager).
2. Open NX_ROOT\pdmconf\NX.env_nt.tpl file using any text editor and take the value of NX variable @NX_KEIT_AVAILABLE_FIELDS which is COPIED_DATA and update Nx.env in NX_ROOT by executing below command:
   ```
   pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA>  -a  pdm_option.inst
   ```
   Where COPIED_DATA is the value copied from NX.env_nt.tpl (step 2).

For example, " pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <provide the exact string here> -a pdm_option.inst "

Note to avoid losing the updated changes when you run pdm_configure, execute the above command with the '-t' flag as follows:

```
pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA> -a pdm_option.inst -t
```

(Optional)Manually added NX variables need to be merged, after upgrading to 17.4.0.0

Manually updated NX variables are lost when a GA release or RU patch delivers a new `NX.env_nt.tpl` file (located in `NX_ROOT/pdmconf`). This necessitates an additional manual step to merge the custom values from the backup file of `NX.env_nt.tpl` created by `pdm_configure`.

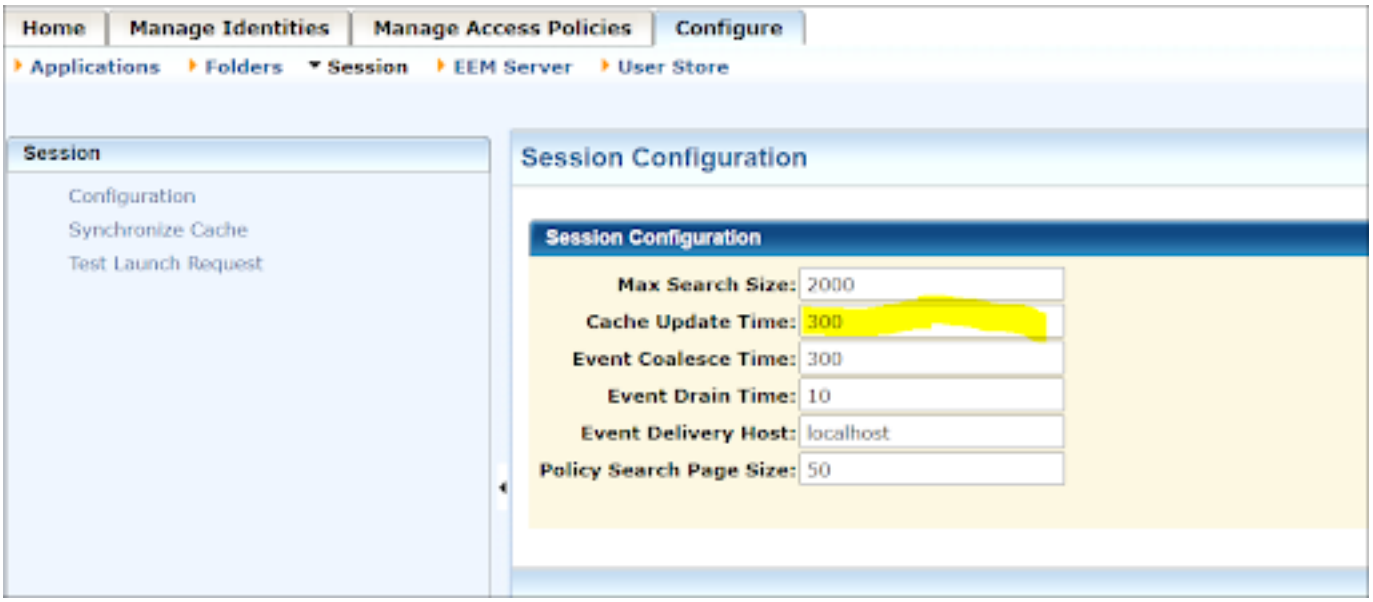
- Manually merge the required changes into the NX.env_nt.tpl file located in NX_ROOT\pdmconf from the backup file at NX_ROOT\REPLACED\GOL-78_EN_US.OLD\pdmconf\NX.env_nt.tpl.
- Restart the SDM services.
- **For Conventional Mode**: Restart the SDM services on all secondary servers.
- **For Advanced Availability Mode**: Restart the SDM services on all standby servers and application servers.

(Optional) Manual Post-Installation Task for EEM Cache Timeout Configuration Update

You may face issues while logging in to EEM or SDM using the EEM user accounts due to a session cache timeout mismatch between EEM and SDM.

In 17.4, you can update the EEM cache timeout configuration by adding a new NX variable, "NX_EEM_CACHE_UPDATE_TIME" in SDM.

Keep the value the same as the EEM cache timeout configuration value mentioned in the following screenshot.



To set the EEM cache update time to 300 seconds, perform the following steps:

To add the variable in the **nx.env** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst
```

To add the variable in the **nxenv_nt.tpl** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst -t
```

If 17.4 xFlow/ Service Point is installed

After succesful installation of 17.4 xFlow/ Service Point refresh index and data on the Search Server, then perform the following steps:

Run the PDM_ES_REBUILD_INDEX Command Utility

To view the Service Catalog data (Service request and Service offering) in the Service Point, execute one of the files based on your operating system:

- $ES_HOME/search/bin/pdm_es_rebuild_index.bat (Windows)
- $ES_HOME/search/bin/pdm_es_rebuild_index.sh (Linux)

NOTE
In case of Linux Search Server environment, ensure to run the following commands on the terminal:

```
sysctl -w vm.max_map_count=262144
ulimit -Hn 65535
```

WARNING
We recommend that you execute the pdm_es_rebuild_index command during off-peak hours as it affects the CA Service Catalog server performance.

(Optional) Manual Post-Installation Task for Rebuild Index

If you have upgraded or are planning to install CA EEM 12.6 with Service Point and CA Service Catalog, ensure that you rebuild the CA Service Catalog index after completing the CA EEM upgrade/installation process. For more information, see Perform Initial Load and Refresh Index Data on Search Server.
Run the following command to rebuild the index:

```
$ES_HOME/search/bin/pdm_es_rebuild_index.bat index=catalog_index
```

Display CA Service Catalog Featured Offerings in Service Point

After successful installation or after upgrading to CA Service Management 17.4, perform the following mandatory steps if you have an integrated environment of CA SDM and CA Service Catalog and you want to display Service Catalog Featured Offerings in Service Point. Before performing the following steps, ensure that you execute the "*pdm_es_rebuild_index.bat*" from where you have installed Search Server <*SearchServer_Installed_location*>*/search/bin*.

There are two ways to perform these mandatory steps, you can select one of these approaches as per your environmental requirements:

Approach 1

1. Navigate to the where you have installed xFlow Interface <*xFlow_Installed_location*>*/APPS/Services/searchmicroservice-*<*xx.x.xxx*>*/conf*> and edit the ***eiam.config*** file.
2. Search for *Logger Configuration* and comment it.
3. After commenting *Logger Configuration*, add the following step right after *Logger Configuration*:

```
<LoggerConfiguration file="<xFlow_Installed_Location>\APPS\Services\searchmicroservice-.<xx.x.xxx>\conf
\eiam.log4j.config"/>
```

4. Save the file.
5. Restart the xFlow Interface Services.

OR

Approach 2

1. After successful installation or after upgrading to CA Service Management 17.4, relaunch the CA Service Management 17.4 Installer wizard.
2. Select and select "**Integrate pre-installed solution components**" and continue with the installation process to complete integrating the solution components.

Update the Federated Search Configuration File

These steps are mandatory if you have configured Federated Search for CA Service Management 17.4. These steps must be performed for Advanced Availability and Conventional Configurations.

1. Shut down the Federated Search Tomcat server if CA SDM Services are up and running by executing the following command:
   ```
   pdm_tomcat_nxd -c stop -t FS
   ```
2. Update the *catalina.properties* file (*bopcfg\www\CATALINA_BASE_FS\conf\catalina.properties)*
   Search for *"common.loader=*" and replace it with the following:
   ```
   common.loader="${catalina.base}/lib/axis2/*.jar","${catalina.base}/lib","${catalina.base}/lib/
   *.jar","${catalina.home}/lib","${catalina.home}/lib/*.jar","${catalina.base}/../../../java/lib/CXF/*.jar"
   ```
3. Start the Federated Search Tomcat Server by executing the following command:
   ```
   pdm_tomcat_nxd -c start -t FS
   ```

Post-Installation Tasks After Installing CA SDM 17.4 on CA SDM 17.2

Post-Installation Tasks After Installing CA SDM 17.4 on CA SDM 17.2

CA Service Desk Manager

If you have selected **No** for automating the post-install steps while running the installer wizard, perform manually the following post-install tasks as per your deployment requirement.

> **NOTE**
>
> For **Advance Availability Mode**: Perform the following steps sequentially on Background, Standby, and Application servers.
> For **Conventional Mode**: Perform the following steps sequentially on Primary and Secondary Servers.

Update the Minor Version in Nx.Env file

The help option (Help Option on this Window) from the CA SDM User Interface provides context-sensitive product documentation. Perform the following steps to view the Help Window:

Perform the following steps:

1. Navigate to the `$NX_ROOT\site` folder.
   a. Open the `install.properties` in a text editor.
   b. Find the parameter: `genfact.gx_minor_ver`
   c. Modify the value to **4**
   d. genfact.gx_copyright_year=2023
2. Navigate to *NX_ROOT/site/config.properties* file and ensure that the value of ***resources.created parameter*** is set to 0.
3. Navigate to the NX_ROOT/bin and open the command prompt and run the below commands:
   ```
   pdm_options_mgr -c -s attach_force_save_dialog -v 1 -a pdm_option.inst -t
   pdm_options_mgr -c -s doc_url -v "https://docops.ca.com/rest/ca/product/latest/topic" -a pdm_option.ins -
   tt
   pdm_options_mgr -c -s doc_default_hid -v "HID_Service_Management_Home_Page" -a pdm_option.inst
   pdm_options_mgr -c -s doc_space_prefix -v "CASM" -a pdm_option.inst
   ```
4. For conventional mode and all application servers and stand-by servers for advanced availability, run:
   ```
   pdm_configure
   ```
   Advanced availability on the BG Server run:
   ```
   pdm_configure -S
   ```

5. Change the directory to **NX_ROOT/data**, and execute the following command:

   NOTE

   Apply the *pdm_load* command once on Primary Server in Conventional mode or Background Server in Advance Availability mode.

```
pdm_load -f r17_3_insert.dat
pdm_load -u -f r17_3_update.dat
pdm_load -f r17_4_insert.dat
pdm_load -u -f r17_4_update.dat
```

6. Clear the browser cache by running the following command:

```
pdm_webcache -H
pdm_webcache -b
```

7. Restart the CA SDM Services.

Remote Web Screen Painter (WSP) and Java Update

If you have installed Remote Web Screen Painter in your environment, perform the following steps:

1. Log in to the server where CA SDM is installed:
   a. Navigate to `$NX_ROOT/java/lib` folder.
   b. Copy the following jar files:
      1. WebScreenPainter.jar
      2. sd-utils.jar
      3. domsrvr_utils.jar
      4. slump.jar
      5. log4j-api-2.17.1.jar
      6. log4j-core-2.17.1.jar
      7. log4j-jcl-2.17.1.jar
2. Log in to the server where remote WSP is installed:
   a. Navigate to `$NX_ROOT/java/lib` folder and paste the jar files that you copied in Step 1.
3. Log in to the server where CA SDM is installed:
   a. Navigate to `$NX_ROOT/SC/JRE` folder.
   b. Copy the `11.0.18` folder.
4. Log in to the server where remote WSP is installed:
   a. Navigate to `$NX_ROOT/SC/JRE` folder.
   b. Paste the `11.0.18` folder.
   c. Navigate to the `$NX_ROOT` folder.
   d. Open the `NX.env` file in a text editor.
   e. Find the parameter `NX_JRE_INSTALL_DIR=`
   f. Modify the parameter value from 1.8.0_112 to **11.0.18**
   a. Save the `NX.env` file.
   Remote WSP is upgraded to AdoptOpenJDK 11.0.18
5. Launch remote WSP to verify that you are able to design and customize CA SDM forms.

Manual Post-Installation Task for CMDB

To run CA Service Management 17.4 in SSL environment, CA SDM Administrators must perform the following post-installation steps for CMDB Visualizer Interface to open using the **HTTPS** protocol when launching from CA SDM UI:

1. In the NX.env file, change the **NX_CMDB_VISUALIZER** entry to provide **HTTPS protocol** and **HTTPS port**.

For example: @NX_CMDB_VISUALIZER=https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer

2. In **casm.conf.js** file from NX_root\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\conf, provide the HTTPS as protocol and HTTP port.
   For example:

```
sdm: {

      vizserver: 'https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer/viz',
      server: 'https:// sdmurl:<SDM_SSL_PORT>/CAisd/pdmweb.exe'
    }
```

Manual Post-Installation Task for OAuth 2.0 (Maileater/Mailer)

After generating an access token for OAuth 2.0 to connect to a mail server, perform the mandatory post-install tasks on all servers in your environment:

1. Create a backup of web.xml from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF in your local directory to perform backout if required.
2. Open the web.xml file from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF.
3. Search for the following statement:
   ```
   <!--USD servlet mappings start-->
   ```
4. Copy and paste the following configuration under the statement:

```
<servlet>
  <servlet-name>OAuthProcessor</servlet-name>
  <servlet-class>com.broadcom.oAuth.OAuthProcessor</servlet-class>
 </servlet>
```

5. Copy and paste the following configuration after any <servlet-mapping>:

```
<servlet-mapping>
  <servlet-name>OAuthProcessor</servlet-name>
  <url-pattern>/OAuthProcessor</url-pattern>
</servlet-mapping>
```

6. Save and close the web.xml file.
7. For OAuth servlet preservation in a primary and secondary environment, the secondary server must also have the OAuth processor entries in the following file: *<NX_ROOT>\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF\web.xml.last*.
8. Repeat Steps 1-8 to update **web.xml.tpl** file in the NX_ROOT\samples\pdmconf\web.xml.tpl folder to avoid losing these changes when pdm_configure is run at a later point of time.
9. Run the following command:
   ```
   pdm_tomcat_nxd -c STOP command
   ```
10. Wait for some time (a minute or so) and run the following command:
    ```
    pdm_tomcat_nxd -c START command
    ```

For more information, see Define a mailbox.

Manual Post-Installation Task for Excel file (.xlsx) export*

1. Execute the following command:
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst
   ```
2. Execute the following command to add the new NX variable to NX.env.tpl file.

```
pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst -t
```

3. Restart CA SDM services.

(Optional) Manual Post-Installation Task for Knowledge Document export with German fields

If you are exporting Knowledge Document with German Fields (included in the export field list), then perform the below post-install steps for successful export of knowledge documents:

1. Take the backup of NX_ROOT\NX.env file (NX_ROOT is the installation directory for CA Service Desk Manager).
2. Open NX_ROOT\pdmconf\NX.env_nt.tpl file using any text editor and take the value of NX variable @NX_KEIT_AVAILABLE_FIELDS which is COPIED_DATA and update Nx.env in NX_ROOT by executing below command:

   ```
   pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA>  -a  pdm_option.inst
   ```

   Where COPIED_DATA is the value copied from NX.env_nt.tpl (step 2).
   For example, " pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <provide the exact string here> -a pdm_option.inst "
   Note to avoid losing the updated changes when you run pdm_configure, execute the above command with the '-t' flag as follows:

   ```
   pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA> -a pdm_option.inst -t
   ```

(Optional)Manually added NX variables need to be merged, after upgrading to 17.4.0.0

Manually updated NX variables are lost when a GA release or RU patch delivers a new `NX.env_nt.tpl` file (located in `NX_ROOT/pdmconf`). This necessitates an additional manual step to merge the custom values from the backup file of `NX.env_nt.tpl` created by `pdm_configure`.

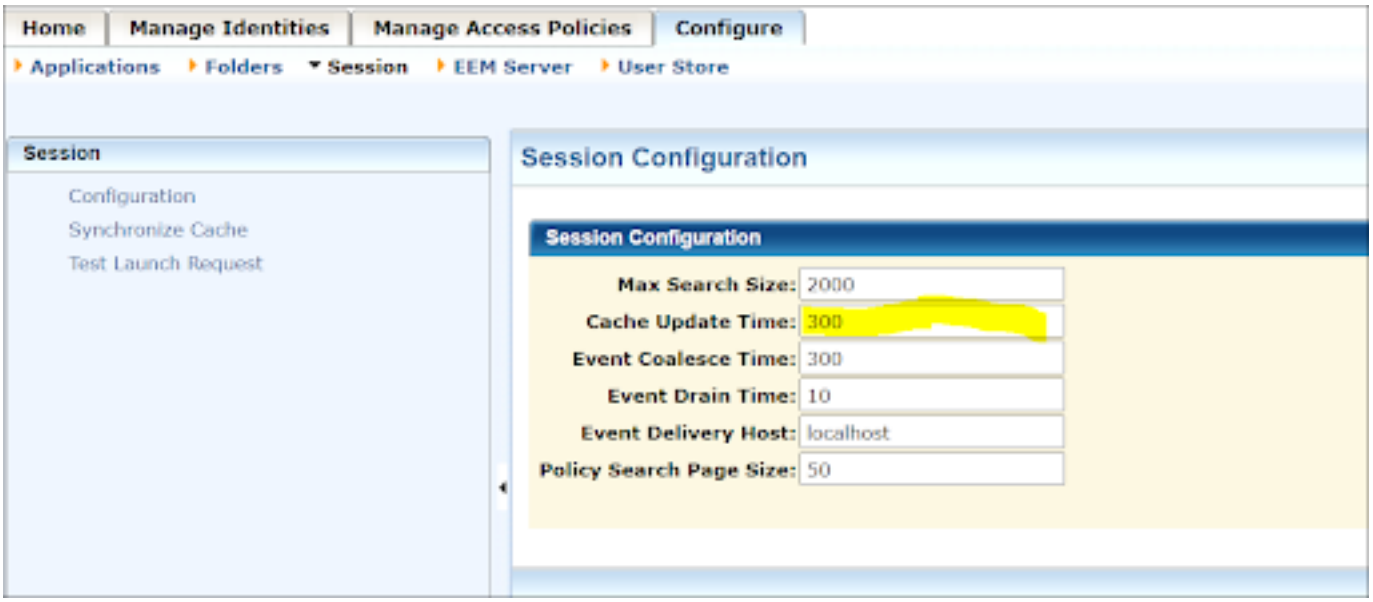
- Manually merge the required changes into the NX.env_nt.tpl file located in NX_ROOT\pdmconf from the backup file at NX_ROOT\REPLACED\GOL-78_EN_US.OLD\pdmconf\NX.env_nt.tpl.
- Restart the SDM services.
- **For Conventional Mode**: Restart the SDM services on all secondary servers.
- **For Advanced Availability Mode**: Restart the SDM services on all standby servers and application servers.

(Optional) Manual Post-Installation Task for EEM Cache Timeout Configuration Update

You may face issues while logging in to EEM or SDM using the EEM user accounts due to a session cache timeout mismatch between EEM and SDM.

In 17.4, you can update the EEM cache timeout configuration by adding a new NX variable, "NX_EEM_CACHE_UPDATE_TIME" in SDM.

Keep the value the same as the EEM cache timeout configuration value mentioned in the following screenshot.



To set the EEM cache update time to 300 seconds, perform the following steps:

To add the variable in the **nx.env** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst
```

To add the variable in the **nxenv_nt.tpl** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst -t
```

If 17.4 xFlow/ Service Point is installed

After succesful installation of 17.4 xFlow/ Service Point refresh index and data on the Search Server, then perform the following steps:

Run the PDM_ES_REBUILD_INDEX Command Utility

To view the Service Catalog data (Service request and Service offering) in the Service Point, execute one of the files based on your operating system:

- $ES_HOME/search/bin/pdm_es_rebuild_index.bat (Windows)
- $ES_HOME/search/bin/pdm_es_rebuild_index.sh (Linux)

NOTE
In case of Linux Search Server environment, ensure to run the following commands on the terminal:

```
sysctl -w vm.max_map_count=262144
ulimit -Hn 65535
```

WARNING
We recommend that you execute the pdm_es_rebuild_index command during off-peak hours as it affects the CA Service Catalog server performance.

(Optional) Manual Post-Installation Task for Rebuild Index

If you have upgraded or are planning to install CA EEM 12.6 with Service Point and CA Service Catalog, ensure that you rebuild the CA Service Catalog index after completing the CA EEM upgrade/installation process. For more information, see Perform Initial Load and Refresh Index Data on Search Server.
Run the following command to rebuild the index:

```
$ES_HOME/search/bin/pdm_es_rebuild_index.bat index=catalog_index
```

Display CA Service Catalog Featured Offerings in Service Point

After successful installation or after upgrading to CA Service Management 17.4, perform the following mandatory steps if you have an integrated environment of CA SDM and CA Service Catalog and you want to display Service Catalog Featured Offerings in Service Point. Before performing the following steps, ensure that you execute the "*pdm_es_rebuild_index.bat*" from where you have installed Search Server <*SearchServer_Installed_location>/search/bin*.

There are two ways to perform these mandatory steps, you can select one of these approaches as per your environmental requirements:

Approach 1

1. Navigate to the where you have installed xFlow Interface <*xFlow_Installed_location>/APPS/Services/searchmicroservice-<xx.x.xxx>/conf*> and edit the ***eiam.config*** file.
2. Search for *Logger Configuration* and comment it.
3. After commenting *Logger Configuration*, add the following step right after *Logger Configuration*:
   ```
   <LoggerConfiguration file="<xFlow_Installed_Location>\APPS\Services\searchmicroservice-.<xx.x.xxx>\conf
   \eiam.log4j.config"/>
   ```
4. Save the file.
5. Restart the xFlow Interface Services.

OR

Approach 2

1. After successful installation or after upgrading to CA Service Management 17.4, relaunch the CA Service Management 17.4 Installer wizard.
2. Select and select "**Integrate pre-installed solution components**" and continue with the installation process to complete integrating the solution components.

Update the Federated Search Configuration File

These steps are mandatory if you have configured Federated Search for CA Service Management 17.4. These steps must be performed for Advanced Availability and Conventional Configurations.

1. Shut down the Federated Search Tomcat server if CA SDM Services are up and running by executing the following command:
   ```
   pdm_tomcat_nxd -c stop -t FS
   ```
2. Update the *catalina.properties* file (*bopcfg\www\CATALINA_BASE_FS\conf\catalina.properties)*
   Search for *"common.loader=*" and replace it with the following:
   ```
   common.loader="${catalina.base}/lib/axis2/*.jar","${catalina.base}/lib","${catalina.base}/lib/
   *.jar","${catalina.home}/lib","${catalina.home}/lib/*.jar","${catalina.base}/../../../java/lib/CXF/*.jar"
   ```
3. Start the Federated Search Tomcat Server by executing the following command:
   ```
   pdm_tomcat_nxd -c start -t FS
   ```

Post-Installation Tasks After Installing CA SDM 17.4 on CA SDM 17.3

CA Service Desk Manager

If you have selected **No** for automating the post-install steps while running the installer wizard, perform manually the following post-install tasks as per your deployment requirement.

> **NOTE**
>
> For **Advance Availability Mode**: Perform the following steps sequentially on Background, Standby, and Application servers.

For **Conventional Mode**: Perform the following steps sequentially on Primary and Secondary Servers.

Update the Minor Version in Nx.Env file

The help option (Help Option on this Window) from the CA SDM User Interface provides context-sensitive product documentation. Perform the following steps to view the Help Window:

Perform the following steps:

1. Navigate to the `$NX_ROOT\site` folder.
   a. Open the `install.properties` in a text editor.
   b. Find the parameter: `genfact.gx_minor_ver`
   c. Modify the value to **4**
   d. genfact.gx_copyright_year=2023
2. Navigate to *NX_ROOT/site/config.properties* file and ensure that the value of ***resources.created parameter*** is set to 0.
3. Navigate to the NX_ROOT/bin and open the command prompt and run the below commands:

```
pdm_options_mgr -c -s attach_force_save_dialog -v 1 -a pdm_option.inst -t
pdm_options_mgr -c -s doc_url -v "https://docops.ca.com/rest/ca/product/latest/topic" -a pdm_option.ins -
tt
pdm_options_mgr -c -s doc_default_hid -v "HID_Service_Management_Home_Page" -a pdm_option.inst
pdm_options_mgr -c -s doc_space_prefix -v "CASM" -a pdm_option.inst
```

4. For conventional mode and all application servers and stand-by servers for advanced availability, run:

```
pdm_configure
```

   Advanced availability on the BG Server run:

```
pdm_configure -S
```

5. Change the directory to **NX_ROOT/data**, and execute the following command:

   NOTE

   Apply the *pdm_load* command once on Primary Server in Conventional mode or Background Server in Advance Availability mode.

```
pdm_load -f r17_4_insert.dat
pdm_load -u -f r17_4_update.dat
```

6. Clear the browser cache by running the following command:

```
pdm_webcache -H
pdm_webcache -b
```

7. Restart the CA SDM Services.

Remote Web Screen Painter (WSP) and Java Update

If you have installed Remote Web Screen Painter in your environment, perform the following steps:

1. Log in to the server where CA SDM is installed:
   a. Navigate to `$NX_ROOT/java/lib` folder.
   b. Copy the following jar files:
      1. WebScreenPainter.jar
      2. sd-utils.jar
      3. domsrvr_utils.jar
      4. slump.jar
      5. log4j-api-2.17.1.jar
      6. log4j-core-2.17.1.jar

7. log4j-jcl-2.17.1.jar

2. Log in to the server where remote WSP is installed:
   a. Navigate to `$NX_ROOT/java/lib` folder and paste the jar files that you copied in Step 1.
3. Log in to the server where CA SDM is installed:
   a. Navigate to `$NX_ROOT/SC/JRE` folder.
   b. Copy the `11.0.18` folder.
4. Log in to the server where remote WSP is installed:
   a. Navigate to `$NX_ROOT/SC/JRE` folder.
   b. Paste the `11.0.18` folder.
   c. Navigate to the `$NX_ROOT` folder.
   d. Open the `NX.env` file in a text editor.
   e. Find the parameter `NX_JRE_INSTALL_DIR=`
   f. Modify the parameter value from **11.0.3** to **11.0.18**
   a. Save the `NX.env` file.
   Remote WSP is upgraded to AdoptOpenJDK 11.0.18
5. Launch remote WSP to verify that you are able to design and customize CA SDM forms.

Manual Post-Installation Task for CMDB

To run CA Service Management 17.4 in SSL environment, CA SDM Administrators must perform the following post-installation steps for CMDB Visualizer Interface to open using the **HTTPS** protocol when launching from CA SDM UI:

1. In the NX.env file, change the **NX_CMDB_VISUALIZER** entry to provide **HTTPS protocol** and **HTTPS port**.
   For example: @NX_CMDB_VISUALIZER=https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer
2. In **casm.conf.js** file from NX_root\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\conf, provide the HTTPS as protocol and HTTP port.
   For example:

```
sdm: {

      vizserver: 'https://sdmurl:<VIZ_SSL_PORT>/CMDBVisualizer/viz',
      server: 'https:// sdmurl:<SDM_SSL_PORT>/CAisd/pdmweb.exe'
    }
```

Manual Post-Installation Task for OAuth 2.0 (Maileater/Mailer)

After generating an access token for OAuth 2.0 to connect to a mail server, perform the mandatory post-install tasks on all servers in your environment:

1. Create a backup of web.xml from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF in your local directory to perform backout if required.
2. Open the web.xml file from NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF.
3. Search for the following statement:

```
<!--USD servlet mappings start-->
```

4. Copy and paste the following configuration under the statement:

```
<servlet>
  <servlet-name>OAuthProcessor</servlet-name>
  <servlet-class>com.broadcom.oAuth.OAuthProcessor</servlet-class>
 </servlet>
```

5. Copy and paste the following configuration after any <servlet-mapping>:

```
<servlet-mapping>
  <servlet-name>OAuthProcessor</servlet-name>
  <url-pattern>/OAuthProcessor</url-pattern>
</servlet-mapping>
```

6. Save and close the web.xml file.
7. For OAuth servlet preservation in a primary and secondary environment, the secondary server must also have the OAuth processor entries in the following file: *<NX_ROOT>\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF\web.xml.last*.
8. Repeat Steps 1-8 to update **web.xml.tpl** file in the NX_ROOT\samples\pdmconf\web.xml.tpl folder to avoid losing these changes when pdm_configure is run at a later point of time.
9. Run the following command:
   ```
   pdm_tomcat_nxd -c STOP command
   ```
10. Wait for some time (a minute or so) and run the following command:
    ```
    pdm_tomcat_nxd -c START command
    ```

For more information, see Define a mailbox.

Manual Post-Installation Task for Excel file (.xlsx) export*

1. Execute the following command:
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst
   ```
2. Execute the following command to add the new NX variable to NX.env.tpl file.
   ```
   pdm_options_mgr -c -s use_legacy_export -v "No" -a pdm_option.inst -t
   ```
3. Restart CA SDM services.

(Optional) Manual Post-Installation Task for Knowledge Document export with German fields

If you are exporting Knowledge Document with German Fields (included in the export field list), then perform the below post-install steps for successful export of knowledge documents:

1. Take the backup of NX_ROOT\NX.env file (NX_ROOT is the installation directory for CA Service Desk Manager).
2. Open NX_ROOT\pdmconf\NX.env_nt.tpl file using any text editor and take the value of NX variable @NX_KEIT_AVAILABLE_FIELDS which is COPIED_DATA and update Nx.env in NX_ROOT by executing below command:
   ```
   pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA>  -a  pdm_option.inst
   ```
   Where COPIED_DATA is the value copied from NX.env_nt.tpl (step 2).
   For example, " pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <provide the exact string here> -a pdm_option.inst "
   Note to avoid losing the updated changes when you run pdm_configure, execute the above command with the '-t' flag as follows:
   ```
   pdm_options_mgr -c -s KEIT_AVAILABLE_FIELDS -v <COPIED_DATA> -a pdm_option.inst -t
   ```

(Optional)Manually added NX variables need to be merged, after upgrading to 17.4.0.0

Manually updated NX variables are lost when a GA release or RU patch delivers a new `NX.env_nt.tpl` file (located in `NX_ROOT/pdmconf`). This necessitates an additional manual step to merge the custom values from the backup file of `NX.env_nt.tpl` created by `pdm_configure`.

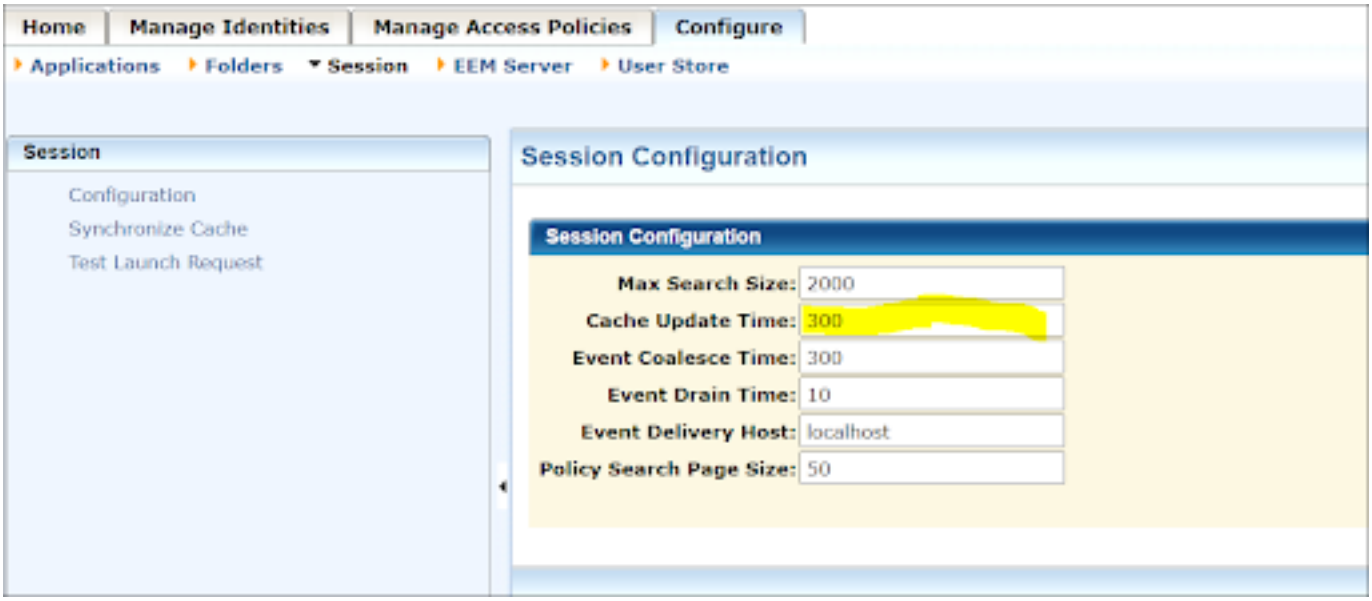
- Manually merge the required changes into the NX.env_nt.tpl file located in NX_ROOT\pdmconf from the backup file at NX_ROOT\REPLACED\GOL-78_EN_US.OLD\pdmconf\NX.env_nt.tpl.
- Restart the SDM services.
- **For Conventional Mode**: Restart the SDM services on all secondary servers.
- **For Advanced Availability Mode**: Restart the SDM services on all standby servers and application servers.

(Optional) Manual Post-Installation Task for EEM Cache Timeout Configuration Update

You may face issues while logging in to EEM or SDM using the EEM user accounts due to a session cache timeout mismatch between EEM and SDM.

In 17.4, you can update the EEM cache timeout configuration by adding a new NX variable, "NX_EEM_CACHE_UPDATE_TIME" in SDM.

Keep the value the same as the EEM cache timeout configuration value mentioned in the following screenshot.



To set the EEM cache update time to 300 seconds, perform the following steps:

To add the variable in the **nx.env** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst
```

To add the variable in the **nxenv_nt.tpl** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst -t
```

If 17.4 xFlow/ Service Point is installed

After succesful installation of 17.4 xFlow/ Service Point refresh index and data on the Search Server, then perform the following steps:

Run the PDM_ES_REBUILD_INDEX Command Utility

To view the Service Catalog data (Service request and Service offering) in the Service Point, execute one of the files based on your operating system:

- $ES_HOME/search/bin/pdm_es_rebuild_index.bat (Windows)
- $ES_HOME/search/bin/pdm_es_rebuild_index.sh (Linux)

NOTE
In case of Linux Search Server environment, ensure to run the following commands on the terminal:

```
sysctl -w vm.max_map_count=262144
ulimit -Hn 65535
```

WARNING
We recommend that you execute the pdm_es_rebuild_index command during off-peak hours as it affects the CA Service Catalog server performance.

(Optional) Manual Post-Installation Task for Rebuild Index

If you have upgraded or are planning to install CA EEM 12.6 with Service Point and CA Service Catalog, ensure that you rebuild the CA Service Catalog index after completing the CA EEM upgrade/installation process. For more information, see Perform Initial Load and Refresh Index Data on Search Server.
Run the following command to rebuild the index:

```
$ES_HOME/search/bin/pdm_es_rebuild_index.bat index=catalog_index
```

Display CA Service Catalog Featured Offerings in Service Point

After successful installation or after upgrading to CA Service Management 17.4, perform the following mandatory steps if you have an integrated environment of CA SDM and CA Service Catalog and you want to display Service Catalog Featured Offerings in Service Point. Before performing the following steps, ensure that you execute the "*pdm_es_rebuild_index.bat*" from where you have installed Search Server <*SearchServer_Installed_location*>*/search/bin*.

There are two ways to perform these mandatory steps, you can select one of these approaches as per your environmental requirements:

Approach 1

1. Navigate to the where you have installed xFlow Interface <*xFlow_Installed_location*>*/APPS/Services/searchmicroservice-*<*xx.x.xxx*>*/conf*> and edit the ***eiam.config*** file.
2. Search for *Logger Configuration* and comment it.
3. After commenting *Logger Configuration*, add the following step right after *Logger Configuration*:

   ```
   <LoggerConfiguration file="<xFlow_Installed_Location>\APPS\Services\searchmicroservice-.<xx.x.xxx>\conf
   \eiam.log4j.config"/>
   ```
4. Save the file.
5. Restart the xFlow Interface Services.

OR

Approach 2

1. After successful installation or after upgrading to CA Service Management 17.4, relaunch the CA Service Management 17.4 Installer wizard.
2. Select and select "**Integrate pre-installed solution components**" and continue with the installation process to complete integrating the solution components.

Update the Federated Search Configuration File

These steps are mandatory if you have configured Federated Search for CA Service Management 17.4. These steps must be performed for Advanced Availability and Conventional Configurations.

1. Shut down the Federated Search Tomcat server if CA SDM Services are up and running by executing the following command:

   ```
   pdm_tomcat_nxd -c stop -t FS
   ```
2. Update the *catalina.properties* file (*bopcfg\www\CATALINA_BASE_FS\conf\catalina.properties)*
   Search for *"common.loader=*" and replace it with the following:

   ```
   common.loader="${catalina.base}/lib/axis2/*.jar","${catalina.base}/lib","${catalina.base}/lib/
   *.jar","${catalina.home}/lib","${catalina.home}/lib/*.jar","${catalina.base}/../../../java/lib/CXF/*.jar"
   ```
3. Start the Federated Search Tomcat Server by executing the following command:

   ```
   pdm_tomcat_nxd -c start -t FS
   ```

CA Service Catalog Post Installation Tasks

WARNING

If the CA Service Catalog is installed after installing CA SDM and xFlow Interface, then run 'Integrate Pre-installed Solution Components.'

NOTE
If you install CA Service Catalog 17.4, follow these post- installation tasks.

As part of implementing CA Service Catalog, review the following manual post-installation tasks and perform the tasks that apply:

- Post installation tasks after installing CA Service Catalog 17.4 on 17.3.0.1 or later versions
- After Upgrading to CA Service Catalog 17.4 from 17.1/17.2/17.3

After Installing CA Service Catalog 17.4 on 17.3.0.1 or later versions

Post installation tasks after installing CA Service Catalog 17.4 on 17.3.0.1 or later versions.

Review and perform the post-installation steps based on your deployment requirements.

NOTE
If any SSL certificate(s) chain was imported to the Oracle Java's JAVA_HOME\lib\security\cacerts Keystore, those certificate(s) chain again need to be exported explicitly and imported properly into the new AdoptOpenJDK's %USM_HOME%\embedded\jdk\lib\security\cacerts Keystore.

Follow these steps

- (Optional) If Service Catalog is accessed from applications like Service Point, SharePoint and so on, then you must perform this step. Open the "**web.xml**" file located at *USM_HOME%\view\webapps\usm\WEB-INF* and update the respective parameter values for the parameter name "cors.allowed.origins" and "cors.support.credentials".
  For example, if you are using Catalog Widgets in Service Point:
  - Search the parameter name "**cors.allowed.origins**" and update the respective parameter value as shown here.

    ```
    <param-name>cors.allowed.origins</param-name>
    <param-value>http://servicepointhostname:servicepointport</param-value>
    ```

    NOTE
    If you have to add multiple entries, use "," (comma) as a separator.
  - Search the parameter name "**cors.support.credentials**" and update the respective parameter value to true as shown here.

    ```
    <param-name>cors.support.credentials</param-name>
    <param-value>true</param-value>
    ```

    Retain the default configuration in web XML, if no other application is accessing catalog:

    ```
    <param-name>cors.allowed.origins</param-name>
    <param-value>*</param-value>
    <param-name>cors.support.credentials</param-name>
    <param-value>false</param-value>
    ```
- Restart the CA Service Catalog services.
- (Optional) Unable to approve any workflow requests through email. The email sent for approvals is corrupted with special characters. Follow the KB article steps.

After Upgrading to CA Service Catalog 17.4 from 17.1/17.2/17.3

Post Installation tasks after installing CA Service catalog 17.4

Review and perform the post-installation steps based on your deployment requirements:

Update configuration files

After upgrading to CA Service Catalog 17.4, perform the following post-install steps:

1. (Mandatory) Update the **server.xml** file in USM_HOME\view\conf location. Add the below attributes to **HTTP** and **HTTPS** connector tags to allow below characters which are mandatory for the application to work.

   ```
   relaxedPathChars="[]|{}^&#x5c;&#x60;&quot;&lt;&gt;" relaxedQueryChars="[]|{}^&#x5c;&#x60;&quot;&lt;&gt;"
   ```

   For example:

   ```
   <!-- Define a HTTP/1.1 Connector on port XXXX -->
   <Connector port="XXXX" disableUploadTimeout="true" URIEncoding="UTF-8"
   ..
   ..
   relaxedPathChars="[]|{}^&#x5c;&#x60;&quot;&lt;&gt;" relaxedQueryChars="[]|{}^&#x5c;&#x60;&quot;&lt;&gt;"
   />

   <!-- Define a SSL HTTP/1.1 Connector on port XXXX -->
   <Connector port="XXXX" disableUploadTimeout="true" URIEncoding="UTF-8"
   ..
   ..
   relaxedPathChars="[]|{}^&#x5c;&#x60;&quot;&lt;&gt;" relaxedQueryChars="[]|{}^&#x5c;&#x60;&quot;&lt;&gt;"
   />
   ```

   Tomcat **server.xml** allows special characters in Hex code format in **relaxedPathChars** and **relaxedQueryChars**. For more information, see the special characters and their respective Hex codes:

Special Character Name	Special Character	Hex Code
Backslash	\	\
Grave accent	`	`
Double quotes	"	"
Less than	<	<
Greater than	>	>

2. Open USM_HOME\view\conf\viewService.conf file to edit as an Administrator and search for: **wrapper.java.additional.23=-Xverify:none**
   a. Add the following entries after **wrapper.java.additional.23=-Xverify:none**

      ```
      #trust these packages for serialize for upgraded activemq
      wrapper.java.additional.24=-
      Dorg.apache.activemq.SERIALIZABLE_PACKAGES=java.lang,javax.security,java.util,org.apache.activemq,org.fusesource
      ```

      NOTE
      While adding the entry "wrapper.java.additional.**24**", maintain the step sequence.

   b. Save the file.
   c. Restart the CA Service Catalog services.
3. (Optional) If Service Catalog is accessed from applications like ServicePoint, SharePoint, and so on, you must perform this step. Open the "**web.xml**" file that is located at *USM_HOME%\view\webapps\usm\WEB-INF* and update the respective parameter values for the parameter name "cors.allowed.origins" and "cors.support.credentials". For example, if you are using Catalog Widgets in ServicePoint:
   – Search the parameter name "**cors.allowed.origins**" and update the respective parameter value as shown here.

      NOTE
      Hostname in the below URL address must be provided in lower case.

      ```
      <param-name>cors.allowed.origins</param-name>
      ```

```
<param-value>http://<servicepointhostname>:<servicepointport></param-value>
```

– Search the parameter name "**cors.support.credentials**" and update the respective parameter value to true as shown here.

```
<param-name>cors.support.credentials</param-name>
<param-value>true</param-value>
```

Retain the default configuration in web XML, if no other application is accessing catalog:

```
<param-name>cors.allowed.origins</param-name>
<param-value>*</param-value>

<param-name>cors.support.credentials</param-name>
<param-value>false</param-value>
```

NOTE
If CA Service Catalog and CA Service Point are installed in two different machines (cross domain), then it is mandatory to enable SSL and setting Service Catalog Cookie to crossdomain. Follow the steps mentioned in this page.

Update the Login Logo file for Shared Filestore

After successful installation of CA Service Catalog 17.4, if you have a shared filestore, you must merge the **logon.css** file from USM_HOME\ filestore\themes\common\css\ to your filestore location. Changes for logo are found in **.loginlogoimage class**. This is required to avoid multiple overlaps of login logo files.

NOTE
In case, if you notice overlapping of login logo files after making the above changes, try clearing your browser cache.

Change Your Password

Changing the password is especially recommended for the user named spadmin (the Service Delivery administrator). In addition, you can also change it at any time, for various security-related reasons.

Follow these steps:

1. Log in to CA Service Catalog with your current user name and password.
2. Click **Profile**.
3. Click the **Change Password** button at the top right of the page.
4. Enter your old and new passwords in the fields provided.
5. Click **OK**.

You have changed your password.

Assign the Service Delivery Administrator Role to a User

Typically, the CA Service Catalog installation creates a user who is named *spadmin* and assigns it the Service Delivery administrator role. This user has complete control of the Catalog system. By default, the user name and the password are the same.

However, if *all* of the following conditions exist, then the installation *cannot* create this user.

- You have installed CA Service Catalog for the first time (*not* upgraded).
- CA EEM is already installed.
- CA EEM is already configured to use an external store, such as Microsoft Active Directory.

In this case, assign the Service Delivery Administrator role to another user. Doing so enables this user to log in to CA Service Catalog using the Service Delivery Administrator role. You can also assign the Service Delivery Administrator role to other users. Doing so is optional but is beneficial if redundancy is important in your organization.

Follow these steps:

1. Open the CA Service Catalog command prompt by clicking Start, Programs, CA, Service Catalog, Service Catalog Command Prompt.
2. Enter the following command at the CA Service Catalog command prompt:
   ```
   ant add-spadmin-user
   ```
   NOTE
   For a list of ant commands and their descriptions, open a command window and enter **ant -p**.
3. Follow the prompts to add the spadmin administrator role to a specific user, using the following information:
   - If CA EEM *is* configured to use an external directory (such as Microsoft Active Directory), specify an existing user name.
   - The command utility creates the user in the CA Service Catalog user database if both of the following conditions exist:
     - CA EEM is *not* configured to use an external directory.
     - The user name that you specify is new.
   - The command utility does *not* prompt you for the password of the new or updated user.
   - If CA EEM *is* configured to use an external directory, the password is defined and stored in the external directory.
   - The new password is the same as the user name, if both of the following conditions exist:
     - CA EEM is *not* configured to use an external directory.
     - The user name that you specify is new.
4. If necessary, cancel and rerun the ant add-spadmin-user command to correct any errors.
5. Verify that the new or updated user can log in to CA Service Catalog as a Service Delivery Administrator and can perform the spadmin functions.
6. (Optional) Instruct the new user to change the password.

You have assigned the Service Delivery Administrator role to a user.

(Optional) Apache Tomcat Server – AJP Connector Configuration

Perform the following steps if you have configured CA Service Catalog with Apache Tomcat Load balancer:

1. Navigate to where Apache Tomcat is installed and open the **worker.properties** file.
2. Add secret to each worker node in the **worker.properties** file as shown below:
   ```
   worker.<JVMRoute>.secret= <Secret_value>
   ```
   For example: **worker.COMPUTER2_USMView.secret= <xxxx>**
   NOTE
   This attribute must be specified with a **non-null**, **non-zero length** value. For more details, refer to APJ Connector.
3. Restart Apache Tomcat server.

Update MaxKeepAliveRequests

If you installed CA EEM 12.6.6 in a clustered setup and have updated the MaxKeepAliveRequests property in the Apache proxy.conf file to meet the CA EEM installation requirements. Now, reset the MaxKeepAliveRequests property to its original value.

NOTE

The installation or upgrade program backs up your existing CA EEM data to the following location:

***USM_HOME*\conf-backup\upgrade-eem-backup.xml**

Integrate Pre-Installed Solution Components

This article is applicable only if you have installed the CA Service Management 17.4 for the first time and followed the sequence of the product installation as shown below:

1. CA Service Desk Manager
2. xFlow Interface
3. CA Service Catalog
4. Other Components

If you have installed CA Service Management 17.4 products in above sequence, run the "**Integrate Pre-Installed Solution Components**" on the server in which **CA SDM/xFlow Interface** is installed.

Follow these steps:

1. Launch the CA Service Management 17.4 installer on the server where you have installed CA Service Desk Manager. Right-click the **Setup.exe** file to select the **run as administrator** option.
2. Select the language and click **Next**.
3. Select **Integrate Pre-installed solution components**.
4. Review and Accept the License Agreement Information and click **Next**.
5. Select **Microsoft SQL Server** or **Oracle** as the database type in the Database Configuration screen.
   For the Windows Microsoft SQL Server database, **mdadmin** is the default user.
   For the Oracle database, verify that you are providing a valid tablespace path. If this path is invalid, then the database installation fails. The following path is an example of a valid Oracle tablespace path: `C:\app\Administrator\oradata\Oracle_Service_Name.`
6. Provide the database details:
   a. If you have **a Microsoft SQL Server** database, provide the following field information:
      - **Database Server**: Enter the hostname of the database server.
        If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
      - **Database Name**: Enter the Database name (MDB), the name of the target DBMS. The default value is mdb.
      - **Database Port**: Provide the port identifier for the target DBMS.
      - **Database Server Instance**: The database instance name, Leave this field blank if you are using the default instance.
      - **Database Admin User**: For Microsoft SQL Server, **sa** is the default value.
        The **sa** user is the admin user with permission to create a user and schema.
      - **Database Admin Password**: Enter the database password of the user specified by a database admin user. No default value is assigned to this parameter.
      - **mdbadmin Password**: Enter the password for the user specified for the MDB admin user.
      - **Confirm mdbadmin Password**: Confirm the password.
   b. If you have the **Oracle** database, provide the following field information:
      - **Database Server**: Enter the host name of the Oracle database server.
        If the target instance is part of a clustered instance, the virtual hostname of the cluster must be used.
      - **Oracle Service Name**: Enter the Oracle Service name.
      - **Listener Port**: Enter the listener port for the database.
      - **Net Service Name**: Enter the Net Service Name of Oracle database where the database resides.

If the database is remote, use the Net Service Name that is defined in the Oracle client on the local computer. CA SDM has access to the database through a local installation of the Oracle client.

- **DBA User Name**: Provide the name of an Oracle user with DBA access. The default value for DBA user is **SYS**.
- **DBA User Password**: Enter the password for the DBA user.
- **mdbadmin Password**: Enter the password for MDB admin.
- **Confirm mdbadmin Password**: Confirm the password.
- **Tablespace Path (on DB Server)**: Provide the tablespace path on the database server.
- **Oracle Home Path**: Provide the Oracle client 32-bit path.
- **Data Tablespace Name**: Provide the name of the data tablespace. The default value is MDB_DATA.
- **Index Tablespace Name**: Provide the index Tablespace Name: The default value is MDB_INDEX.

Click **Next**.

7. On ''Select the Products and Integrations'' page, configure common components is already selected. Click **Next**.
8. Review Product Endpoint and click **Next**.
9. Provide the CA EEM and click **Next**.
10. Provide the CA Process Automation Details and click **Next**.
11. Review the Pre-Installation Configuration page. Click **Next**.
12. Click **Install**. Review the integration progress and when complete, click **Next**.
13. Click **Finish** to exit the installer.

CA Asset Portfolio Management Post Installation Tasks

Post installation taks for CA Asset Portfolio Management.

After installing CA Asset Portfolio Management, perform the following post-installation tasks:

- Post-Installation Tasks After Installing CA APM on SQL Database
- Post-Installation Tasks After Installing CA APM on Oracle Database
- Additional Components Configuration
- Apache Tomcat Configuration
- (Optional) Configure Secure Network Communication
- (Optional) Config File Changes in WCF Service to Address Security vulnerabilities

Post-Installation Tasks After Installing CA APM on SQL Database

Manual Post-Installation Task for Removing jQuery, log4net, and log4j Vulnerabilities

After installing 17.4, delete jquery-1.3.2.min.js, jquery-ui-1.7.2.custom.min.js , log4net.dll, and log4j from their respective installed folders.
Perform the following steps:

1. Log on to CA APM Web Server, navigate to the <ITAM_install_dir>\ITAM\Web Server\castylesr5\javascript.
   Example: C:\Program Files (x86)\CA\ITAM\Web Server\castylesr5\javascript
   Locate and delete the jquery-1.3.2.min.js and jquery-ui-1.7.2.custom.min.js files from the directory.
2. Navigate to the ITAM_install_dir\CA\ITAM\Data Importer.
   Locate and delete the log4net.dll file from the directory.
3. Navigate to the C:\Program Files (x86)\CA\SC\CORA.
   Locate and delete the log4j-1.2.15.jar, log4j.properties, and ojdbc6.jar files from the directory.
4. Navigate to the ITAM_install_dir\CA\ITAM.
   Backup the following folders and manually delete them:

- CDB
- MDB
- jre8

5. Delete the *roslyn* folder from *<ITAM_intsalled_dir>\ITAM RestApi Service\bin* on the CA APM Application Server host, and Web Server host.
6. Perform an IISRESET and delete the browser's cache to load the upgraded files. Restart all CA APM services, AMS, and CA CASM services.

Post-Installation Tasks After Installing CA APM on Oracle Database

Perform the following manual post-install steps if you installed CA APM 17.4 on any Oracle database.

1. Open the **<<INSTALL_DIR>>\ITAM\Storage Manager Service\web.config** file and add the following entries inside the <configuration> element.

```
<system.data>
 <DbProviderFactories>
<!-- Remove in case this is already defined in the machine.config -->
<remove invariant="Oracle.ManagedDataAccess.Client" />
<add name="Oracle Data Provider for .NET" invariant="Oracle.ManagedDataAccess.Client" description="Oracle
 Data Provider for .NET, Managed Driver" type="Oracle.ManagedDataAccess.Client.OracleClientFactory,
 Oracle.ManagedDataAccess, Version=4.122.19.1, Culture=neutral, PublicKeyToken=89b483f429c47342" />
</DbProviderFactories>
</system.data>
```

   For example:

```
<configuration>
-----------------
<system.data>
<DbProviderFactories>
<!-- Remove in case this is already defined in the machine.config -->
<remove invariant="Oracle.ManagedDataAccess.Client" />
<add name="Oracle Data Provider for .NET" invariant="Oracle.ManagedDataAccess.Client" description="Oracle
 Data Provider for .NET, Managed Driver" type="Oracle.ManagedDataAccess.Client.OracleClientFactory,
 Oracle.ManagedDataAccess, Version=4.122.19.1, Culture=neutral, PublicKeyToken=89b483f429c47342" />
</DbProviderFactories>
</system.data>
----------------
</configuration>
```

2. Search the entry

```
<add assembly="System.Data.OracleClient, Version=4.0.0.0, Culture=neutral,
 PublicKeyToken=B77A5C561934E089"/>
```

   and replace with the following entry. Save the file.

```
<!--<add assembly="System.Data.OracleClient, Version=4.0.0.0, Culture=neutral,
 PublicKeyToken=B77A5C561934E089"/>-->
```

3. Open the **<<INSTALL_DIR>>\ITAM\WCF Service\web.config** file and add the following entries inside the <configuration> element.

```
<system.data>
<DbProviderFactories>
<!-- Remove in case this is already defined in the machine.config -->
<remove invariant="Oracle.ManagedDataAccess.Client" />
```

```
<add name="Oracle Data Provider for .NET" invariant="Oracle.ManagedDataAccess.Client" description="Oracle
 Data Provider for .NET, Managed Driver" type="Oracle.ManagedDataAccess.Client.OracleClientFactory,
 Oracle.ManagedDataAccess, Version=4.122.19.1, Culture=neutral, PublicKeyToken=89b483f429c47342" />

</DbProviderFactories>
</system.data>
```

For example:

```
<configuration>
-----------------
<system.data>
<DbProviderFactories>
<!-- Remove in case this is already defined in the machine.config -->
<remove invariant="Oracle.ManagedDataAccess.Client" />

<add name="Oracle Data Provider for .NET" invariant="Oracle.ManagedDataAccess.Client" description="Oracle
 Data Provider for .NET, Managed Driver" type="Oracle.ManagedDataAccess.Client.OracleClientFactory,
 Oracle.ManagedDataAccess, Version=4.122.19.1, Culture=neutral, PublicKeyToken=89b483f429c47342" />

</DbProviderFactories>
</system.data>
----------------
</configuration>
```

4. Search for the entry **<compilation debug="true" targetFramework="4.5"/>** in **<system.web>** element. Replace with the following:

```
<compilation debug="true" targetFramework="4.5">
<assemblies>
<add assembly="CA.Common.Transactions, Version=12.9.0.0, Culture=neutral, PublicKeyToken=null"/>
</assemblies>
</compilation>
```

5. Open the **<<INSTALL_DIR>>\ITAM\Import Processor\ImportProcessor.exe.config** file and add the following entries inside the configuration element.

```
<system.data>
<DbProviderFactories>
<!-- Remove in case this is already defined in the machine.config -->
<remove invariant="Oracle.ManagedDataAccess.Client" />

<add name="Oracle Data Provider for .NET" invariant="Oracle.ManagedDataAccess.Client" description="Oracle
 Data Provider for .NET, Managed Driver" type="Oracle.ManagedDataAccess.Client.OracleClientFactory,
 Oracle.ManagedDataAccess, Version=4.122.19.1, Culture=neutral, PublicKeyToken=89b483f429c47342" />

</DbProviderFactories>
</system.data>
```

For example:

```
<configuration>
-----------------
<system.data>
<DbProviderFactories>
<!-- Remove in case this is already defined in the machine.config -->
```

```
<remove invariant="Oracle.ManagedDataAccess.Client" />
<add name="Oracle Data Provider for .NET" invariant="Oracle.ManagedDataAccess.Client" description="Oracle
 Data Provider for .NET, Managed Driver" type="Oracle.ManagedDataAccess.Client.OracleClientFactory,
 Oracle.ManagedDataAccess, Version=4.122.19.1, Culture=neutral, PublicKeyToken=89b483f429c47342" />
</DbProviderFactories>
</system.data>
----------------
</configuration>
```

Manual Post-Installation Task for Removing jQuery, log4net, and log4j Vulnerabilities

After installing 17.4, delete jquery-1.3.2.min.js, jquery-ui-1.7.2.custom.min.js , log4net.dll, and log4j from their respective installed folders..

Perform the following steps:

1. Log on to CA APM Web Server, navigate to the <ITAM_install_dir>\ITAM\Web Server\castylesr5\javascript.
   Example: C:\Program Files (x86)\CA\ITAM\Web Server\castylesr5\javascript
   Locate and delete the jquery-1.3.2.min.js and jquery-ui-1.7.2.custom.min.js files from the directory.
2. Log on to CA APM Application Server and navigate to the ITAM_install_dir\CA\ITAM\Data Importer.
   Locate and delete the log4net.dll file from the directory.
3. Log on to CA APM Application Server and navigate to the C:\Program Files (x86)\CA\SC\CORA.
   Locate and delete the log4j-1.2.15.jar, log4j.properties, and ojdbc6.jar files from the directory.
4. Navigate to the ITAM_install_dir\CA\ITAM.
   Backup the following folders and manually delete them:
   - CDB
   - MDB
   - jre8
5. Delete the *roslyn* folder from *<ITAM_intsalled_dir>\ITAM RestApi Service\bin* on the CAAPM Application Server host, and Web Server host.
6. Perform an IISRESET and delete the browser's cache to load the upgraded files. Restart all CA APM services, AMS, and CA CASM services.

Additional Components Configuration

After installing CA APM, you can change the component configurations and can configure additional product components.

Follow these steps:

1. Log in to CA APM as an administrator.
2. Click **Administration**, **System Configuration**.
3. Click the product component that you want to configure.
4. Configure the settings. Click **Save**.
5. Do the following steps to recycle the settings in the application pool:
   a. Click **Start** menu, **Control Panel**.
   b. Double-click **Administrative Tools.**
   c. Double-click **Internet Information Services (IIS) Manager**.
   d. In the Connections pane, expand the server name, click **Application Pools,** select ITAM.
   e. In the Actions Pane, click **Stop** and then, click **Start**.
6. Restart the services.
7. For more information, see **Start the Services**.

NOTE
You cannot configure the database server from the System Configuration page. Update the corresponding configuration files for any database server configuration settings.

For more information about changing the component configurations and adding servers, see Manage Product Components.

Following components are configurable in the CA APM product:

Database Server

The database server is a product component that hosts the Oracle or SQL Server database management system for CA APM. The CA MDB is installed on the database server. The application server, Hardware Reconciliation Engine, and other product components retrieve data from and store data in the CA MDB.

The following fields require explanation:

- **MS SQL Server Instance**
  Defines the name of the MS SQL Server instance that is being configured. Enter the instance name only when multiple SQL Servers named instances exist. Leave the field blank if there is only one (default) instance.

  NOTE

  You can change the component configurations and can configure additional components for your enterprise after you install the product. For more information about changing the component configurations and adding servers, see Managing Product Components.

Web Server

The web server is the main server that hosts the web application and builds the CA APM user interface. This server communicates with the user, the application server, and the database server. The following fields require explanation:

- **Web Server or Load Balancer IP/Host**
  The CA APM installation, by default, sets this field to the web server host name. In a single web server environment, you can enter the web server host name, or the web server IP address. In a multiple web server environment, you can enter either the web server host name, or the IP address of the Load Balancer.

  NOTE
  The web server can be registered with a different name in the Domain Name System (DNS) than what is registered as the web server host name. In this situation, specify the different name in this field.

  You can configure additional web server components after you install the product.

Application Server

The application server is the server that connects the database server and the web server for CA APM. The business and data access logic reside on the application server. To allow for scalability, the application server and web server are on two distinct servers.

You can have more than one application server. The Export Service component and the Storage Management Service component must be installed on one of the application servers, but not necessarily on the same server.

The following fields require explanation:

- **Application Server or Load Balancer IP/Host**
  The CA APM installation, by default, sets this field to the application server host name. In a single application server environment, you can enter the application server host name, or the application server IP address. In a multiple application server environment, you can enter either the application server host name, or the IP address of the Load Balancer.

NOTE
The application server can be registered with a different name in the Domain Name System (DNS) than what is registered as the application server host name. In this situation, specify the different name in this field.

You can configure more application server components after you install the product.

Hardware Reconciliation Engine

The Hardware Reconciliation Engine is the service that matches discovered assets to their corresponding owned assets from different logical repositories. You can manage the assets based on your business practices. The Hardware Reconciliation Engine retrieves data from and stores the results in the CA MDB. You can install the Hardware Reconciliation Engine on one or more servers.

You can configure more Hardware Reconciliation Engine components after you install the product.

CA EEM

CA APM uses CA EEM for authentication. Other products that need CA EEM for authentication can use the same CA EEM server that CA APM uses.

- To manage security centrally for multiple CA Technologies products, specify the name, location, and login credentials for the existing CA EEM server.
- To manage CA APM security independently from other CA Technologies products, install CA EEM on any single application or web server other than the one where the existing CA EEM is installed.

CA Business Intelligence JasperSoft Reports

CA Business Intelligence JasperSoft administers, monitors, and configures the reporting environment. CA APM uses CA Business Intelligence to integrate, analyze, and present information required for effective enterprise IT management.

For more information, see **Install CABI JasperReports Server.**

CA Process Automation Manager

CA Process Automation can be integrated with CA Asset Portfolio Management to automate workflow business processes. For more information, see Install CA Process Automation.

Export Service

The Export Service exports data from CA APM and saves the results in formats such as a comma-separated value (CSV) file. To accomplish this task, the Export Service interacts with the Storage Manager Service so that you can specify where the exported files are stored.

Data Importer Engine Service

The Data Importer Engine Service imports bulk product information into the CA MDB through column and field mapping.

Import Driver

The Import Driver processes discovered hardware data exports from CA SAM. CA APM uses the discovered hardware data to link ownership and discovery data. CA APM exports ownership data back to CA SAM.

LDAP Data Import and Sync Service

The LDAP Data Import and Sync Service imports data into CA APM from CA EEM or external data sources (LDAP or CA SiteMinder). Install the LDAP Data Import and Sync Service on one of the Data Importer servers.

Storage Manager Service

The Storage Manager Service stores exported files, attachment files, data import data and map files, and log files for data import and mass change.

CA APM Registration Service

The CA APM Registration Service consolidates individual CA APM CORA services into one main service. You can have installations of other CA Technologies products that also use the CORA API. The changes that you make to the CORA API in your CA APM environment do not affect the use of the CORA API by other CA Technologies products.

Common Administration for Service Management (CASM)

The Common Administration for Service Management (CASM) provides administrative functionality, such as multi-tenancy administration, to CA APM. Multi-tenancy is the ability for multiple independent tenants (and their users) to share a single implementation of CA APM.

NOTE
For more information about implementing multi-tenancy, see How to Implement Multi-Tenancy.

Event Service

The Event Service manages the events and notifications process in CA APM. Events are important activities or data changes that you want to track and that you define in CA APM. After a defined event has occurred, notifications are sent to alert appropriate users and administrators about the event.

To perform the notification function, the Event Service interacts with a workflow provider (for example, CA Process Automation) using the Web Service. A workflow provider manages automated processes. If your workflow provider is CA Process Automation, you can specify the existing instance of CA Process Automation during the installation. You can also share CA Process Automation with CA Service Desk Manager and CA Service Catalog.

Common Asset Viewer

The Common Asset Viewer lets you view discovered and owned data for an asset that has been linked through reconciliation, including system configuration, operating system, system devices, and file systems. You can view this data on the Asset Details page by clicking the Owned Information or Discovered Information link.

The Common Asset Viewer requires the following components to install and execute successfully:

- Apache Tomcat server, which is included with the CA APM installation. The default value for the Apache Tomcat server port is 9080. You can change this value after the installation. You first update the port in the Apache Tomcat configuration file. Then, you change the port in the product (Administration tab, System Configuration, Common Asset Viewer).

After you install the Common Asset Viewer, the component is configured for non-secure network communication (http). You can configure the component for secure network communication (https) by first configuring the Apache Tomcat server (where the Common Asset Viewer is installed) to support the Secure Socket Layer (SSL) protocol. Then you need to change a setting for the Common Asset Viewer component in the web configuration file.

WARNING

The Tomcat port number for CA APM defaults to 9080. If another product that is integrated with CA APM uses this port number, change the port number in CA APM so that you do not have a conflict.

WCF Service

The Windows Communications Foundation (WCF) service implements the web services in CA APM. The web services let you use a standards-based interface to build client applications that integrate with CA APM.

The web services let you create, search, update, copy, and delete CA APM objects from your external client application. Your assigned user role determines whether you have permission to access the web services in CA APM. Your role also restricts the objects and data (classes and attributes) that you can view or modify.

Specify the server name for the WCF Service component. You can modify the WCF Service protocol setting. You can change the configuration of the WCF Service component after you install the product.

NOTE

You can change the component configurations and can configure additional components for your enterprise after you install the product. For more information about changing the component configurations and adding servers, see Managing Product Components.

CA Service Desk Manager

CA Service Desk Manager component is used for integrating CA APM and CA SDM. Refer to Integrating CA APM and CA Service Desk Manager.

Apache Tomcat Configuration

This article explains how to update the port in the Apache Tomcat configuration file.

This is an optional step and is required only when you want to change the AMS port. The Common Asset Viewer lets you view discovered and owned data for an asset that has been linked through reconciliation. This data includes system configuration, operating system, system devices, and file systems. The Common Asset Viewer requires the Apache Tomcat server, which is included with the CA APM installation. You first update the port in the Apache Tomcat configuration file. Then, you change the port in the product (**Administration** tab, **System Configuration**, **Common Asset Viewer**).

NOTE

The Tomcat port number for CA APM defaults to 9080. If another product that is integrated with CA APM uses this port number, change the port number in CA APM so that you do not have a conflict.

Perform the following steps:

1. On the application server where the Common Asset Viewer is installed, navigate to one of the following folders, depending on your server:
   a. C:\Program Files\CA\SC\AMS\TomCat\conf (for 32-bit operating systems)
   b. C:\Program Files (x86)\CA\SharedComponents\AMS\TomCat\conf (for 64-bit operating systems)
2. Select and open the **server.xml** file.
3. Navigate to the following section of the server.xml file:

```
<Connector port="9080" protocol="HTTP/1.1"connectionTimeout="20000" redirectPort="8443" />
```

4. Update the Tomcat port number with the same number that CA APM uses (**Administration** tab, **System Configuration**, **Common Asset Viewer**).
5. Save the server.xml file.

(Optional) Configure Secure Network Communication

After the installation is complete, configure CA APM for non-secure or secure network communication. For secure network communication (https), first configure IIS on the product servers to support the Secure Socket Layer (SSL) protocol. Then, set the CA APM configuration parameters for secure network communication.

Configure IIS for Secure Network Communication

Configure IIS on the product servers to support the Secure Socket Layer (SSL) protocol. Perform the following steps:

1. Launch the Internet Information Services (IIS) Manager on the CA APM web server.
2. Select **Server Certificates**.
3. Click **Create Self-Signed Certificate** and specify a certificate name.
4. Select the website (on the left) where CA APM is installed.
5. Click **Bindings** under **Actions** on the right.
6. Click **Add**.
7. Select **https** for the Type.
8. Specify the port and the SSL certificate name.
9. Perform these (1-8) steps on the CA APM application server.

Configure CA APM for Secure Network Communication

Configure CA APM on the product servers to support the Secure Socket Layer (SSL) protocol. Perform the following steps:

1. Log in to CA APM and navigate to **Administration**, **System Configuration**.
2. Click **Web Server** on the left.
3. Change the server protocol to **https** and click **Save**.
4. Click **WCF Service** on the left.
5. Change the server protocol to **https** and click **Save**.
6. Click **Application Server** on the left and select the **Show Advanced Options** checkbox to see all configuration parameters.
7. Change the server protocol to **https**.
8. Change the **server port** and the **component server port** to the port of the https protocol (by default, **443**) and click **Save**.
9. Execute the following query on your database tp update the webserver SSL port (by default, **443**).
   ```
   update al_cdb_configurationparameters set configvalue=443 where configkey = 'ServerPort' and
    componentkey='Web_Server'
   ```
10. Reset IIS on the web server and the application server.
    You can now start the product web interface using secure network communication. Open a supported web browser and enter the following URL:
    ```
    https://servername/ITAM/Pages/UserLogin.aspx
    ```
    Replace *server name* with the name of the server that is hosting the CA APM web server.

Configure AMS for Secure Network Communication

Configure AMS on the product servers to support the Secure Socket Layer (SSL) protocol. Perform the following steps:

Create a Keystore File

A keystore file is required to enable SSL. Create a keystore file if you do not have one already for another CA product that integrates with CA Service Management. **Follow these steps:**

1. Create “certificates" folder in the C drive of AMS server.
2. Open a command window.
   Example: C:\Program Files (x86)\CA\SharedComponents\AMS\OpenJdk11\bin> <Installed location of AMS>
3. Enter the following command:
   ```
   keytool -genkey -alias tomcat -keyalg RSA -keystore c:/certificates/AMS.keystore
   ```
   alias_name

Specifies the logical name for the certificate that you are using for CA Service Catalog and possibly for other products. Record this alias name for reference.

4. Enter the password at the “Enter keystore password” prompt. To make configuring Tomcat easier, you can use “changeit” as the password.
5. Record your password for reference.
6. Enter your password at the prompt.
7. Run the following command to add the certificate into the keystore.

```
keytool -importkeystore -srckeystore c:/certificates/AMS.keystore -destkeystore c:/certificates/
AMS.keystore -deststoretype pkcs12
```

Edit the Server.xml File to Support SSL

As part of configuring AMS server to use Secure Socket Layer (SSL), edit the server.xml file to support SSL. **Follow these steps:**

1. Open the C:\Program Files (x86)\CA\SharedComponents\AMS\TomCat\conf file.
2. Search for the <Service name="Catalina"> section.
3. Updated the Connector related information under the <Service name="Catalina"> section:

```
<Connector SSLEnabled="true" maxThreads="200" port="8443"
 protocol="org.apache.coyote.http11.Http11NioProtocol"
 sslImplementationName="org.apache.tomcat.util.net.jsse.JSSEImplementation">
      <SSLHostConfig
 ciphers="TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384,TLS_DHE_RSA_WITH_AES_128_GCM_S
 TLS_DHE_RSA_WITH_AES_256_GCM_SHA384,TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256,TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA,
 TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384,TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA,TLS_DHE_RSA_WITH_AES_128_CBC_SHA256,
 TLS_DHE_RSA_WITH_AES_128_CBC_SHA,TLS_DHE_RSA_WITH_AES_256_CBC_SHA256,TLS_DHE_RSA_WITH_AES_256_CBC_SHA,
 TLS_RSA_WITH_AES_128_GCM_SHA256,TLS_RSA_WITH_AES_256_GCM_SHA384,TLS_RSA_WITH_AES_128_CBC_SHA256,
 TLS_RSA_WITH_AES_256_CBC_SHA256,TLS_RSA_WITH_AES_128_CBC_SHA,TLS_RSA_WITH_AES_256_CBC_SHA,
 SSL_RSA_WITH_3DES_EDE_CBC_SHA" honorCipherOrder="true" protocols="TLSv1.2" sslProtocol="TLSv1.2">
        <Certificate certificateKeyAlias="tomcat" certificateKeystoreFile="C:/certificates/AMS.keystore"
 certificateKeystorePassword="N0tallowed" certificateKeystoreType="PKCS12"/>
      </SSLHostConfig>
   </Connector>
```

   – Default value before making this change.

```
<Connector port="8080" URIEncoding="UTF-8" protocol="HTTP/1.1" connectionTimeout="20000"
 redirectPort="8443"/>
```

4. Save and close the server.xml file.

Enable SSL for AMS

To enable SSL for AMS, **follow these steps:**

1. Login to CA APM (ITAM).
2. Navigate to **Administration**, **System**, **Configuration**, **Common Asset Viewer**.
3. From the Common Asset View page, provide the AMS Web Server URL as (use HTTPS) and the port as 8443. For example: `https://<hostname>:8443/AMS/login.do`
4. Restart the AMS service.
5. Restart IIS.

Configure CA CASM for Secure Network Connection

1. Stop CA CASM Services
2. Connect to ITAM database
3. Run the following query:

```
select * from ca_application_registration where product_code=2070 and product_name='CACASM'
Update ca_application_registration set product_specific_data='https://ApplicationServerHostName:9060/itsm/
itsm.html'
where product_code=2070 and product_name='CACASM'
```

NOTE
Make sure to replace ApplicationServerHostName with the correct value of the ITAM Application Server HostName

4. Navigate to C:\Program Files (x86)\CA\SharedComponents\CASM\apache-tomcat\conf' and take a backup of the existing Server.xml file. Open Server.xml file in Edit mode.
5. Search for Connector port as mentioned and delete it:

```
<Connector port="9060" protocol="HTTP/1.1"   connectionTimeout="20000"    redirectPort="8443" />
```

Use the following tags instead:

```
<Connector SSLEnabled="true" maxThreads="200" port="9060" protocol="HTTP/1.1"
 sslImplementationName="org.apache.tomcat.util.net.jsse.JSSEImplementation">

<SSLHostConfig
 ciphers="TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384,TLS_DHE_RSA_WITH_AES_128_GCM_S
 TLS_DHE_RSA_WITH_AES_256_GCM_SHA384,TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256,TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA,
 TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384,TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA,TLS_DHE_RSA_WITH_AES_128_CBC_SHA256,
 TLS_DHE_RSA_WITH_AES_128_CBC_SHA,TLS_DHE_RSA_WITH_AES_256_CBC_SHA256,TLS_DHE_RSA_WITH_AES_256_CBC_SHA,
 TLS_RSA_WITH_AES_128_GCM_SHA256,TLS_RSA_WITH_AES_256_GCM_SHA384,TLS_RSA_WITH_AES_128_CBC_SHA256,
 TLS_RSA_WITH_AES_256_CBC_SHA256,TLS_RSA_WITH_AES_128_CBC_SHA,TLS_RSA_WITH_AES_256_CBC_SHA,
 SSL_RSA_WITH_3DES_EDE_CBC_SHA" honorCipherOrder="true" protocols="TLSv1.2"
 sslProtocol="TLSv1.2"><Certificate certificateKeyAlias="<cert_Alias>"
 certificateKeystoreFile="<cert_location>" certificateKeystorePassword="<cert_password>"
 certificateKeystoreType="PKCS12"/> </SSLHostConfig>

 </Connector>
```

6. Start the CA CASM services.
7. Reset IIS.

Verify the Installation

After you have completed all installation procedures, you can verify that CA APM was installed successfully. To start the web interface, open a web browser and enter the following URL:

```
https://servername/itam
```

Update the SSL ports for patch installations

In the integrated environment of CA APM and SDM, SSL is enabled on both the products. Follow these steps when any patch of 17.4 is installed:

1. product_specific_data column value is reset to default in ca_application_registration table. So you must update the value to https manually, along with its port by executing the below query against the mdb

```
UPDATE ca_application_registration SET product_specific_data='https://<SDMHost>:<port>/CAisd/pdmweb.exe'
 where product_code = 2003 and major_version = 17 and minor_version = 4
```

2. Open the AMS.properties file under AMS directory (For example C:\Program Files (x86)\CA\SharedComponents\AMS \TomCat\webapps\AMS\WEB-INF\classes) and update the value of the entry 'appreg.2003.product_specific_data', it should refer to https and SSL port in the SDM URL.

```
https://<hostname>:<port>/CAisd/pdmweb.exe
```

3. On the CA APM application server, restart the AMS Service and perform the iisreset.
4. Perform iisreset on CA APM Web Server

NOTE
When Secure Socket Layer (SSL) is configured on the CA APM, additional configuration is required for the Windows Communication Foundation services (WCF services).
Follow the steps mentioned in the KB Article 36958.

Secure Cookies

The RequireSSL property enables or disables output of the secure cookie attribute as described in RFC 2109. When used by compliant browsers, the cookie will only be sent back over a connection using SSL/TLS.

To secure the cookies, it is recommended to follow these steps:

- The default value is false.

On ITAM web server, navigate to the web server folder, for example:

```
<ITAM_installed_drive>\CA\ITAM\Web Server
```

- Edit web.config file, and you will see the following entry:

```
<httpCookies requireSSL="false" />
```

- Manually change the value to **true**:

```
<httpCookies requireSSL="true" />
```

- Perform IIS Reset.

(Optional) Config File Changes in WCF Service to Address Security vulnerabilities

WCF service response returns full stack trace incase of a service fault exception.

Perform the following steps to turn off exception details in any fault response:

1. Open your CA APM installed directory and navigate to the WCF service folder, for example C:\Program Files (x86)\CA \ITAM\WCF Service.
2. Open web.config file and search for **serviceDebug** element and set the **includeExceptionDetailInFaults** attribute to false, for e.g:

```
<serviceBehaviors>
    <behavior name="CA.Applications.Web.WCF.IServiceBehavior">
      <!-- To avoid disclosing metadata information,
      set the value below to false and remove the metadata endpoint above before deployment -->
      <serviceMetadata httpGetEnabled="false"/>
      <!-- To receive exception details in faults for debugging purposes,
      set the value below to true.  Set to false before deployment
      to avoid disclosing exception information -->
      <serviceDebug includeExceptionDetailInFaults="false"/>
      <ITAMServiceBehavior/>
    </behavior>
</serviceBehaviors>
```

3. Perform IISReset.

CA Service Management Solution Integration

If you want to integrate one or more CA Service Management Products (CA SDM, CA Service Catalog, and CA APM) with CA Common Components (CA EEM, CA Process Automation, and CABI Jaspersoft Reports), consider the following integration requirements that may require manual intervention.

NOTE

CA Service Management Installer automates the integration process. The procedures that are described in this article must be performed manually only if the automation of the integration process fails or does not work for some unforeseen reason.

For more information about the CA Common component release versions that are supported for CA Service Management, see Supportability Matrix.

Integrate CA Service Management with CA Common Components

Consider the following items, if you are integrating the following products:

- **CA EEM and CA Process Automation Considerations:** If you have installed CA EEM and CA Process Automation and integrated one or more CA Service Management products with CA EEM and CA Process Automation, ensure that you provide the same CA EEM information that you provided while installing CA Process Automation for auto-integration to be successful.
- **CASM_Policy File**: Do not modify or edit the **CASM_Policy** file from **CA SDM**, **Administration**, **SOAP Web Services Policy**, **Policies**, **CASM_Policy File**. If you modify the file, generate a new **CASM_policy** file and manually copy this file from the CA SDM installation directory to the installation directory of all CA Service Management products. For more information, see Use the Web Service Access Policy File.

Integrate CA Service Management Product(s)

You can integrate the following CA Service Management Products or Products in these combinations:

- Integrate CA Asset Portfolio Management with CA Service Catalog
- Integrate CA Service Desk Manager with CA Asset Portfolio Management
- Integrate CA Service Catalog with CA Service Desk Manager

For more information about how to troubleshoot an integration, see Troubleshooting the Integration.

CA Service Desk Manager and Common Components

Perform the following after installation is complete to integrate CA Service Management products with CA Common Components (CA EEM, CA Process Automation, and CABI JasperSoft Reports Server).

Integrate CA Service Desk Manager with CABI JasperSoft

WARNING

Customers purchasing CA Service Management products on or after April 1, 2016 are only eligible to use CABI JasperReports® Server r6.4.3 or later. We will continue to support CABI 4.x for customers who purchased CA Service Management, CA IT Asset Manager, or CA Service Catalog products prior to April 1, 2016.

CABI JasperReports® Server r.6.4.3 requires that you import the CA Service Management CABI content, to deliver the CA Service Management reports. Also, you need to configure the data source and user permissions. For more information, see CABI JasperReports® Server r6.2.0 for CA Service Management.

Integrate CA Service Desk Manager with CA EEM (Solaris)

NOTE

If you plan to implement CA SDM and CA EEM on the same server, you must install the **CA EEM 12.51 CR05** version.

If you are planning to auto-integrate CA SDM with CA EEM on Solaris, install the CA SDM Secondary or Application Server on Windows or any other Operating System other than Solaris. If you are planning to upgrade CA SDM on Solaris to CA Service Management, upgrade the CA SDM Primary or Background server. Move the bopuath_nxd authentication module daemon to the Secondary Server. Complete the following based on your CA SDM configuration:

Move the Authentication Module Daemon for CA SDM Conventional Configuration

On the CA SDM Conventional Setup, complete the following steps:

1. Upgrade CA SDM Primary Server to CA Service Management.*Or*
   Install CA SDM Secondary on Windows.
   You can perform a fresh CA SDM Installation or can upgrade an earlier CA SDM release to CA Service Management.
2. Log in to the CA SDM web user interface as an administrator on the Primary server with **srvcdesk** credentials of the local host.
3. Navigate to System, Configurations on the Administration tab.
4. Click Create New to add the primary server details.
5. In the Host Name field, specify the primary host name for the configuration. Click Save.
6. On the Configuration Detail page, select the Additional Processes tab.
7. Select and edit the Login User Authentication and in the Update Process page provide the secondary server name.
8. Launch the ***pdm_configure*** on the Primary Server.
   The Config Options page drop-down has the last configuration that you created. Select the configuration and click Finish.
   After successful configuration, user login is successfully authenticated by CA EEM.

Move the Authentication Module Daemon for CA SDM Advanced Availability Configuration

For CA SDM Advanced Availability, move the bopauth_nxd authentication daemon from the background server to any other server (such as application server). Perform the following steps:

1. Log in to the CA SDM background server web interface as an administrator.
2. Select the Administration tab. Expand Options Manager, Security.
3. Click *the bopauth_nxd_host* entry in the Options List. Click Edit.
4. Select the host name of the target server from the Option Value drop-down list. Click Save.
5. Restart the background server after changing the default value of *bopauth_host*.
6. Start version control as a client on the application server using the following command:
   pdm_ver_nxd -c
7. Restart the application server.
   You have successfully configured the redirection of authentications requests to an external server.**Note:** By default, the authentication module runs on the background server.

Integrate CA SDM with CA EEM Manually

If you are not able to integrate CA SDM and CA EEM through the CA Service Management installer, perform the following steps:

1. Ensure that you have installed CA EEM.
2. Depending on the CA SDM configuration, log in to the following server:
   - Conventional: **Primary server**
   - Advanced availability: **Background server**
3. Install the following options:
   - **use_eiam_authentication**
     Allows use of EEM authentication when users log in to CA SDM / Knowledge Management and the access type authentication is set to the operating system.
   - **use_eiam_artifact**

Allows EEM artifact to be used when users log in to CA SDM / Knowledge Management using URL.

– **eiam_hostname**
Specifies the hostname of the server where EEM is running. Required for CA EEM user name and password or artifact authentication.

4. Restart CA SDM services and add the CA SDM privileged user in EEM.

You have successfully integrated CA EEM with CA SDM.

Integrate CA SDM with CA Process Automation Manually

CA Process Automation is a stand-alone CA product with features for automating and tracking hardware and software administration tasks in enterprise IT environments. CA Process Automation automates tasks and manages user interactions, such as approvals and notifications for compliance and accuracy within production environments.

When you integrate CA SDM and CA Process Automation, you can use/leverage the benefits of CA Process Automation workflow capabilities from key points within CA SDM. An effective integration between CA SDM and CA Process Automation requires you to understand both products.

CA Process Automation Components

CA Process Automation offers multiple capabilities and structures that facilitate a wide range of activities as part of CA Process Automation process management. For the integration with CA SDM, however, only the following CA Process Automation components are critical for CA Process Automation integration:

- **Process Definition --** Identifies a collective series of tasks, steps, and conditions that are structured in a specific order to be initiated, and completed by various individuals or parties. This component is the central building block of all CA Process Automation content.
- **Start Request Form --** An object containing descriptive information for end users. The Start Request Form presents a process definition to users while hiding the technical details of the process definition.
- **Keywords --** A list of pre-defined words or phrases to attach to Start Request Forms.
- **Automation Library --** An area within CA Process Automation that stores and shows Process Definitions and Start Request Forms.
- **Library Path or Reference Path --** A folder structure that organizes and describes Process Definitions and Start Request Forms within the automation library.
- **Process Instance --** An active entity that executes the rules that are defined in a process definition. The process instance progresses until the process definition state is complete.
- **Process Instance Log Messages --** A configurable, running record that details the progression of activities of the process instance. Log message categories are useful to the CA SDM integration with CA Process Automation.

Note: The scope of the definitions of CA Process Automation components is limited to the usage within CA SDM. For information about CA Process Automation components and capabilities, see the CA Process Automation documentation.

Complete the following steps only if you are not able to integrate CA Process Automation and CA SDM through the CA Service Management installer:

Verify the Integration Requirements

You can integrate CA Process Automation and CA SDM to coexist on a single server when the server architecture supports both products. When CA Process Automation or CA SDM components cannot integrate on the same server, consider installing each product on separate servers.

WARNING
If CA Process Automation is configured in FIPS mode, you must also configure CA EEM server and CA EEM SDK in FIPS mode. For more information about configuring the EEM SDK in FIPS mode, see the CA EEM Documentation.

Before you configure CA Process Automation and CA SDM, confirm that both products are installed and are working independently. Perform the following steps:

1. Open a browser on the server that hosts CA SDM and verify that a CA Process Automation user can log in to CA Process Automation. Change the place holders to match the target CA Process Automation installation.
   `http(s)://<server>:CA Portal/itpam`
2. Enter the following URL. Change the place holders to match the target CA Process Automation installation.
   `http(s)://<server>:CA Portal/itpam/JNLPRequestProcessor?processType=startUI`

   The CA Process Automation product is accessible from the CA SDM host.

Install the CA Process Automation Workflow Options

When you install the CA Process Automation Workflow options, you specify connectivity between CA SDM and CA Process Automation. If you are using CA EEM for authentication, you also specify the CA EEM host name. Perform the following steps:

1. On the Administration Tab, select Options Manager, CA Process Automation Workflow.
   The Option List appears.
2. Right-click the name of each option and select Edit from the context menu. Install the following options:
   - **caextwf_eem_hostname**
     Specifies the name of the CA EEM server. For example, *<wf_hostname>* identifies the authentication host. You install caextwf_eem_hostname only if you configured CA Process Automation to use CA EEM as an authentication server. CA SDM uses this value to transform a user name and password into a CA EEM token. Then, the user name and password do not pass in plain text over HTTP.

     NOTE
     If the CA Process Automation installation is not using CA EEM, do not place a value in the caextwf_eem_hostname option, and do not install the **caextwf_eem_hostname**. Placing a false value or installing caextwf_eem_hostname when it is not necessary causes the integration to fail.
   - **caextwf_endpoint**
     Specifies the URL that points to the CA Process Automation web services by including the CA Process Automation host name, port, and the mandatory /itpam/soap path. For example, http://<wf_hostname>:<wf_tomcat_port>/itpam/soap identifies the endpoint. If your implementation uses CA EEM, installing the **caextwf_eem_hostname** option is required for the integration between CA Process Automation and CA SDM to operate properly.
   - **caextwf_log_categories**
     Specifies a comma-separated list of CA Process Automation process instance log category names to appear on the CA SDM Request, Change Order, and Issue Workflow Tasks tab. For example, *Operator,Response,MyOwnCategory* supplies three-log categories.
     You install caextwf_log_categories based on business decisions from the CA SDM and CA Process Automation process design personnel. This option adjusts the default data that appears on the Workflow Tasks tab for requests, change orders, and issues.
     When you install the caextwf_log_categories option, all CA Process Automation process instance log messages from the Process category and the categories that you specify appear on the Workflow Tasks tab. When you do not install caextwf_log_categories, only the CA Process Automation process instance log messages from the Process category appear on the Workflow Tasks tab.
   - **caextwf_processdisplay_url**
     Specifies how to launch a graphical snapshot of a CA Process Automation process instance by supplying the host name and the mandatory */itpam/Web.jsp?page=runtimeeditor&ROID* path. For example, *http://<wf_hostname>:<wf_tomcat_port>/itpam/Web.jsp?page=runtimeeditor&ROID=* launches a snapshot of a process instance. On the Workflow Tasks tab of a request, change order or issue, the user selects View Process to see the snapshot.
     Installing the caextwf_processdisplay_url option is required for the integration between CA Process Automation and CA SDM to operate appropriately.
   - **caextwf_worklist_url**

Specifies the process instance path by supplying the host name and the mandatory */itpam?Page=tasklist* path. For example, *http://<wf_hostname>:<wf_tomcat_port>/itpam?page=tasklist* enables CA SDM users to see a list of CA Process Automation process instances that require attention. The list appears in CA Process Automation when the CA SDM user selects a link associated with any listed task in the request, change order, or issue Workflow Tasks tab.
Installing the caextwf_worklist_url option is required for the integration between CA Process Automation and CA SDM to operate properly.

- **caextwf_ws_password**
  Specifies the administrative password associated with the CA Process Automation user name from the caextwf_ws_user option. CA SDM uses the user name and password to access the CA Process Automation web service functions to perform integration activities such as selecting start request forms, process definition information, and process instance information.
  Installing the caextwf_ws_password option is required for the integration between CA Process Automation and CA SDM. The password and user name that you specify requires the appropriate access to CA Process Automation. However, it is not necessary the CA Process Automation user name and password to exist within the CA SDM contact records.
- **caextwf_ws_user**
  Specifies the CA Process Automation administrative user name associated with the CA Process Automation user name from the caextwf_ws_password option. CA SDM uses the user name and password to access the CA Process Automation web service functions. These services perform integration activities such as selecting start request forms, selecting process definition information, selecting process instance information, or launching process instances.
  Installing the caextwf_ws_user option is required for the integration between CA Process Automation and CA SDM to operate. The user name and password that you specify requires the appropriate access to CA Process Automation. However, it is not necessary the CA Process Automation user name and password to exist within the CA SDM contact records.
- **caextwf_retry_count**
  Specifies the number of times an event is triggered if CA Process Automation is unavailable. CA Process Automation Workflows can be attached to CA SDM tickets (for example,Change Orders) through CA SDM Events and Macros. The attached events are triggered by CA SDM at the specified retry interval duration. During retry, if the CA Process Automation is unavailable, the attached event is marked as Unknown and the process is not executed. CA SDM retry mechanism automatically re-triggers the attached Unknown events when CA Process Automation is unavailable. Default value is 3 and can be set in the range [1 – 20].
- **caextwf_retry_interval**
  Specifies time interval after which the event is again triggered in case CA Process Automation is unavailable.

3. Click Install.
4. Restart the CA SDM service.
   The CA SDM and CA Process Automation can communicate even though there is no process instance data. CA SDM and CA Process Automation are ready for you to create CA Process Automation process definitions and CA Process Automation start request forms.

Set Up SSL Communications with CA Process Automation

For security reasons, CA Process Automation implementers can install or reconfigure CA Process Automation to require SSL communications. If CA Process Automation is configured to require SSL communications, integrated applications such as CA SDM require a certificate from the CA Process Automation keystore for communication. Perform the following steps:

1. Configure CA SDM options to use the CA Process Automation HTTPS address.
2. Export the CA Process Automation keystore certificate to a file and copy the file to CA SDM.
3. Load the certificate file into CA SDM using the CA SDM pdm_keystore_mgr utility.
4. If applicable to your CA SDM architecture, update the version control files to deliver the CA SDM keystore to all secondary servers.

5. Restart CA SDM.

Enable Communications When CA Process Automation is SSL Enabled

When CA Process Automation communicates with SSL, configure the CA SDM servers to communicate with CA Process Automation.

To enable communications when CA Process Automation is SSL enabled, do the following tasks:

1. Verify that you can use CA Process Automation in a browser, without launching CA SDM. Record the CA Process Automation URL and use it for reference when you configure the CA Process Automation Workflow options in Options Manager.
2. Log in to CA SDM and install or modify the CA Process Automation Workflow options in Options Manager. For each of the following options, use the syntax *https://server:8443/* instead of http://server:8080/ for reaching the SSL enabled CA Process Automation application. If the CA Process Automation installation uses another port instead of the 8443 SSL port, specify the appropriate port number.
   - caextwf_endpoint
   - caextwf_processdisplay_url
   - caextwf_worklist_url

     NOTE
     If the values do not match the actual CA Process Automation installation values, CA SDM cannot communicate with CA Process Automation. A runtime error occurs. Verify that the values match the actual CA Process Automation installation values, because the CA Process Automation installer might have selected a different port instead of port 8443.
3. On the CA Process Automation server, locate the KEYSTOREID and itpam.web.keystorealias entries in the following file:

   ```
   C:\Progra~1\ITPAM\server\c2o\.config\OasisConfig.properties
   ```

4. Copy the KEYSTOREID. Be prepared to paste the KEYSTOREID value as the password after you issue the keytool command.
5. On the CA Process Automation server, issue the following keytool command as one line on the command line:

   ```
   C:\Progra~1\ca\sc\jre\1.6.0_00\bin\keytool.exe -keystore C:\Progra~1\ITPAM\server\c2o\.config\c2okeystore
    -export -alias <keystorealias> -file itpam.cer
   ```

   The keytool utility prompts you for a password.
6. Paste or type the KEYSTOREID value as the password.
   The keytool utility uses the final parameter (-file itpam.cer) to create a file that is named *itpam.cer*. The *itapm.cer* file contains the necessary certificate information for communications with CA SDM.
7. Move the *itpam.cer* file to one of the following locations on the CA SDM server:
   - (Windows) %NX_ROOT%\bin
   - (UNIX) $NX_ROOT/bin
8. Import the CA Process Automation certificate information into CA SDM by entering the following command:

   ```
   (Windows) pdm_perl %NX_ROOT%\bin\pdm_keystore_mgr.pl -import %NX_ROOT%\bin\itpam.cer
   (UNIX) pdm_perl $NX_ROOT/bin/pdm_keystore_mgr.pl -import $NX_ROOT/bin itpam.cer
   ```

   The *pdm_keystore_mgr.pl* script generates the keystore file in the following locations:
   - (Windows) %NX_ROOT%\pdmconf\nx.keystore
   - (UNIX) $NX_ROOT/pdmconf/nx.keystore
9. The *nx.keystore* must be delivered to the following CA SDM servers, depending on the CA SDM configuration:
   - Conventional: **Secondary server**
   - Advanced Availability: **Application and Standby server**

   Open the *server_secondary.ver* file from one of the following locations:

- (Windows): *%NX_ROOT%\site\server_secondary.ver*
- (UNIX): *$NX_ROOT/site/server_secondary.ver*

10. Modify the *server_secondary.ver* for version control by adding the following information:

```
[SSL_Keystore]
filename = "nx.keystore"
directory = "$NX_ROOT/pdmconf"
component_type = "file"
O_mode = "RW"
g_mode = "RW"
w_mode = "RW"
file_ctl
```

11. Restart CA SDM.
    The CA SDM server can communicate with the SSL enabled CA Process Automation application.

Configure the Integration (CA SDM and CA Process Automation)

Both CA Process Automation and CA SDM, as stand-alone products, have individual requirements for authentication and authorization. To support a unified Service Oriented Architecture (SOA) strategy, you can configure both products to use CA EEM for authentication.

When you install CA Process Automation with CA EEM as the authentication server, the installer creates several policies and eight essential entities by default:

- Four Application Users: pamadmin, pamuser, pamproduser, and pamdesigner.
- Four application groups: PAMAdmins, PAMUsers, Designers, and Production Users.

CA SDM users who also use CA Process Automation can be divided between PAMAdmins and PAMUsers as follows:

- CA SDM analysts must be members of PAMUsers when their duties entail:
  - Approving, rejecting, or otherwise responding to CA Process Automation Interaction Request Forms.
  - Listing CA Process Automation process instances assigned to the user.
  - View the graphical display, click the **View Process** button of CA Process Automation process status screen. The CA Process Automation PAMUsers group requires an additional CA Process Automation policy to grant access the graphic.
- CA SDM analysts are members of PAMAdmins when their duties entail:
  - Creating and checking in CA Process Automation process definitions and/or start request forms.
  - Terminating process instances directly within CA Process Automation. Terminating process instances are an administrative exception to expected integration procedures.
  - Delegating CA Process Automation process instance tasks.
  - If the user is the user name defined in CA SDM Options Manager.
- CA SDM users require no access to CA Process Automation when their duties entail:
  - Creating requests, change orders, and issues that launch CA Process Automation instances.
  - Reviewing the Workflow tab which shows CA Process Automation process instance status and task information.
  - Changing the status of a request, change order, or issue which causes the termination of a CA Process Automation process (such as canceling a change order).
  - Selecting a CA Process Automation process definition on a CA SDM request area, change category, issue category.

Finalize the Integration (CA SDM and CA Process Automation)

If you have CA SDM and CA Process Automation integration, set up the Single Sign-On (SSO). Perform the following steps:

1. Verify that the following requirements have been met:

- CA SDM and CA Process Automation are configured to use the same CA EEM installation.
- The user that logs in to CA SDM is also a user in CA Process Automation.
- When CA EEM uses the internal database as a user store, the users must have either global permissions or must belong to the same folder. If CA EEM references an external user store like an external directory or CA SiteMinder, the users must be of the same store to access Single Sign-On.

2. Install CA EEM or use any existing CA EEM installation (for example, CA EEM for CA Process Automation).
3. On the CA SDM Administration tab, install the following options from under the Options Manager, Security folder:
   - eiam_hostname
   - use_eiam_authentication

   NOTE
   You do not need to install the option caextwf_eem_hostname under Options Manager, CA Process Automation folder. But if you do install it, the value must be the same as eiam_hostname option.
4. Restart CA SDM.
5. To create a user in CA Process Automation, perform the following steps:
   a. Log in to CA EEM using the CA Process Automation application context using the **EiamAdmin** userid or any other administration user.
   b. Select the Manage Identities tab and click the icon next to the Users folder.
   c. On the New User page, the Name field at the top is the userid that must match the userid in the CA SDM contact table.
   d. Click the Add Application User Details button and complete the following steps:
      - Add any of the groups that are listed. Add at least one of these groups to be able to log in to CA Process Automation.
      - Complete the New User fields, such as the First Name, Last Name, Display, and Password.
6. Create a user in the CA SDM contact table with the same userid. Verify that the Access Type Validation Type field for the user is set to CA EEM.
   You can log in to CA EEM and CA SDM with this user and the password specified in CA EEM.

CA Process Automation Integration with CA SDM at Run Time

When you enable the integration, CA SDM users experience the following:

- On a new Request, Change Order, or Issue, a CA Process Automation process instance initiates based on the ticket category or area. Summary information immediately appears on the Workflow Tasks tab.
- When a Request Area, Change Category, or Issue Category changes, an attached CA Process Automation process instance terminates and a new process instance initiates.
- When a CA SDM user attempts to close a Request, Change Order, or Issue where the CA Process Automation process instance is not yet complete, the user cannot close the ticket. Instead, the user must first cancel the ticket. The Cancel status terminates the CA Process Automation process instance before the ticket closes.
- When a user wants to understand the state of the process instance without navigating away from the ticket, the user can click the ticket Workflow Tasks tab. The Workflow Tasks tab shows the process instance start date, end date, current state, and a current audit trail of messages indicating the path of the process instance.
- When a user wants to see the current path of the process instance relative to the entire process, the user selects the View Process button on the Workflow Tasks tab. The View Process button launches a graphical snapshot of the entire process instance, and shows the current path.
- When a user wants to see CA Process Automation interaction request forms that are waiting for user action, the user can select any entry in the Workflow Tasks tab. The Workflow Tasks tab contains an audit trail of process instance messages that appear on the CA Process Automation task list.

NOTE
When a user selects the CA SDM View Process button or CA Process Automation process instance messages, the system prompts for a CA Process Automation user name and password for a single browser session. After the initial prompt, the system does not prompt the user again until the CA SDM browser closes.

How to Create a Process Definition

When you create the process definition, you populate CA Process Automation with content to appear in CA SDM. You use the CA Process Automation graphical process designer to create, test, and check in a process definition. From CA SDM, you can also create macros to initiate CA Process Automation processes. To create a process definition in CA Process Automation, perform the following steps:

1. Log in to the CA Process Automation client as an administrative user.
2. Use the CA Process Automation graphical process designer to create, test, and check in a process definition. When you work with the process definition, use the instructions in the CA Process Automation user documentation.

   NOTE
   If you fail to check in the process definition before attempting to use it, the workflow cannot operate properly in CA SDM.

   The following process definition items are available in CA SDM:
   - **Process Name**
     Appears on the Request Area, Change Category, and Issue Detail page. The process name also appears on the Workflow Tasks tab of a ticket. The process name describes the process definition to CA SDM to Analysts and other users who manage tickets.
   - **Process Reference Path**
     Appears on the Request Area, Change Category, and Issue Detail pages. The Process Reference Path also appears on the ticket Workflow Tasks tab. The Process Reference Path can be useful to describe the purpose of the process to end users. For example, "/Processes/Approval" is not helpful. Instead, a reference path like "/Office Supplies/Approvals/Over-200-CA SDM" describes the workflow to manage orders that exceed $200 US dollars.
   - **Process Log Messages**
     Appears on the CA SDM ticket Workflow Tasks tab. By default, a record of activities stores with the process instance. Process log messages have the Process category. A process designer can create custom messages to appear on the Workflow Tasks tab of a CA SDM ticket.

Create a Start Request Form

When you create a Start Request Form, you associate it with the process definition and check it in. You include the appropriate keywords in the Start Request Form properties. If the appropriate keyword is missing from the Start Request Form properties, the Start Request Form and its associated process definition fail to appear in CA SDM. Perform the following steps:

1. Log in to the CA Process Automation client as an administrative user.
2. Open the CA Process Automation Library and navigate to the path Start Request Form.
   The Start Request Form appears in the right pane of the CA Process Automation library.
3. Select the Start Request Form from the list.
   A shortcut menu appears.
4. Select Properties.
   The Library Object Properties page appears.
5. (Optional) Click the General tab and modify the description of the Start Request Form. Add a description that identifies the proper usage of the Start Request Form and the associated Process Definition to the CA SDM Administrator.
6. Click the Keywords tab.
   The Keywords tab is active.
7. Click the ab+ icon.
   A row adds to the empty list.

8. Click the row.
   A blinking cursor highlights the row and indicates the row is ready for typing.
   a. Enter one of the following values to associate a keyword to the appropriate ticket area or category.
      For example, to make a Start Request Form available for a CA SDM request area, enter the **pcat** keyword.
   b. Add a row to the list for each applicable keyword.
      For example, to make the Start Request Form appear on both request areas and change categories, add one row for the **chgcat** keyword and another row for the **pcat** keyword.
   c. Click **OK**.
      CA Process Automation saves the keywords and description and closes the Library Object Properties dialog.
   d. Check in the Start Request Form.

Ticket	Use Keyword
Request Area	pcat
Change Category	chgcat
Issue Category	isscat

NOTE

If you fail to check in the Start Request Form, the form fails to appear in CA SDM.

The CA Process Automation Start Request Form information appears on the CA SDM Start Request Form List. The CA SDM administrator can associate the CA SDM Start Request Form with the Process Definition, on a Request Area, Change Category, or Issue Category Detail page.
The following Start Request Form items appear in CA SDM:

1. – **Start Request Form Name**
   Appears on the Request Areas, Change Categories, and Issue Categories list pages.
   – **Start Request Form Reference Path**
   Appears on the Request Areas, Change Categories, and Issue Categories list pages.
   – **Start Request Form Description**
   Appears on the Request Areas, Change Categories, and Issue Categories list pages. The text in this field describes how the Start Request Form is appropriate for selection on a particular Request Area, Change Category, or Issue Category.

CA Service Catalog and Common Components

After you install CA Service Catalog, perform the following to complete the integration:

NOTE

Note: CA EEM is a mandatory component for CA Service Catalog. Hence, CA EEM is automatically integrated with CA Service Catalog. After the CA Process Automation processes are loaded and configured, update the CA Service Catalog login credentials in the CA Process Automation Dataset, to complete the integration between the two products. For more information, see the Integrate CA Service Desk Manager with CA Service Catalog topic.

CA Asset Portfolio Management and Common Components

The integration of CA Asset Portfolio Management with CA EEM, CA Process Automation, and/or CABI Jaspersoft is a completely automated process that is performed by the CA Service Management Installer.

You are not required to perform any manual steps to finalize the integration. However, if you are migrating CA Asset Portfolio Manager from CA Business Intelligence 3.3 to 4.1 SP3 , you must manually update the custom universe as follows:

WARNING

This step applies only if you are migrating from CA Business Intelligence 3.3 to CA Business Intelligence 4.1 SP3. For more information, see CABI Documentation.

If you have Custom Universe based on the OOTB Universe, refresh the link with CA Service Management Universe.

1. Launch Universe Design Tool from the machine where CA Business Intelligence Release 4.1 SP3 client tools are installed.
2. Copy the backup Custom Universe to this machine.
3. Click **File**, **Open**.
4. Click **File**, **Parameters**.
5. Click the **Links** Tab
6. From the Name column, click the universe of your product,
7. Click the **Change Source** button.
8. Select the location of the universe file.

   NOTE

   The .unv file is typically located in the CA Universes folder.
9. Click Open, OK.
   The universe link is updated.
10. Export the custom universe.

CA Service Desk Manager and CA Service Catalog

Ensure that you have selected both CA SDM and CA Service Catalog during the CA Service Management installation. After the successful installation of CA Service Management, perform the following manual steps to ensure that CA SDM and CA Service Catalog are integrated successfully.

Update Login Credentials in CA Process Automation Dataset

WARNING

Ensure that you have CA SDM Connector installed before you proceed with the following procedure.

Follow these steps:

1. Log in to CA Process Automation as an Administrator.
2. Click Library from the Home page.
3. Select the CA SDM folder and double click SDM_GlobalDataset.
4. Click Check out and enter the CA Service Catalog user credentials.
5. Click Save.
6. Click Check in and Close.
7. From the CA SDM SRF folder, double click HWSW_FilledFormInventory.
8. Click Check Out from the Common menu.
9. On the properties tab, enter the details of Tags field as chgcat and save.
10. Click Check in and Close.
11. From the CA Service Catalog folder, double click CA SLCM.
12. Click Check out and enter the CA Service Catalog user credentials
13. Click Save.

14. Click on Check in and close.

    WARNING

    Ensure that you have CA SDM connector installed to complete steps 15 and 16.
15. On the **Configuration** tab, click **Domain**, **Modules**.
16. Click **Lock** from the title bar of the configuration panel to lock the domain. Double-click **CA ServiceDesk**.
17. Give the CA SDM Host name In the Default ServiceDesk WebService URL.
18. Enter the password for Default User.
19. Save and close.
20. Click **Unlock**.

Create Groups in CA Service Desk Manager

Follow these steps:

1. Log in to CA SDM as an Administrator.
2. From the Administrator tab select Security and Role Management, Groups.
3. Click Create New and enter the group name for CA Service Catalog
4. Click Members, Service contracts, Auto Assignment.
5. Click Update Members.
6. Search and update the members.
7. Click OK and Save.
8. Associate this group with the following categories, and add CA SDM administrator as the assign of these categories:
   - SLCM.HWFFI
   - SLCM.SWFFI

   WARNING

   Ensure that you restart CA SDM and CA Service Catalog services after performing all the steps.

At the end of this topic, you have successfully integrated CA Service Catalog with CA Service Desk Manager.

In the rare event that the integration between the two products fail, see integrate CA Service Catalog with CA Service Desk Manager manually to troubleshoot the integration.

How to Handle Enabled Rules Defined in the Event-Rule-Action Page

When you auto-integrate CA Service Management components, some of the rules that are defined in the Event-Rule-Action page of CA Service Catalog are enabled. The rules that are enabled are required for the common administration components and for the integration to work properly. The rules are enabled in the following scenarios:

- When you install CA Service Management for the first time and auto-integrate the components
- When you upgrade from a standalone CA Service Catalog to the integrated environment
- When you upgrade from a standalone CA SDM to the integrated environment

  NOTE

  When you upgrade to the integrated environment, some of the rules that you had disabled may also be enabled. We recommend that you do not disable such rules, as disabling them may cause the common administration and the integration to not work properly.

The following table gives the list of rules that are enabled. Rules with filter that are triggered once for each hardware request item when status goes from approval range to Pending Fulfillment:

Rule	Description
Common Administration and Content Pack Related Rules	
CASM Auto Integration Rule	Rule with filter that fires once for CASM Auto Integration service
CASM Common Configuration Rule	Rule with filter that fires once for CASM Common Configuration service
CASM Contact Sync Rule	Rule with filter that fires once for each CASM Contact Sync service
CASM Create LDAP Rule	Rule with filter that fires once for each CASM add LDAP service
CASM Create Role Rule	Rule with filter that fires once for each CASM Custom Role service
CASM Custom Role Mapping Rule	Rule with filter that fires once for each CASM Custom Role Mapping service
CASM Custom Tenant Mapping Rule	Rule with filter that fires once for each CASM Custom Tenant Mapping service
CASM Import LDAP Users Rule	Rule with filter that fires once for each CASM import LDAP users service
CASM Manage LDAP Rule	Rule with filter that fires once for each CASM manage LDAP service
CASM Tenant Mapping Rule	Rule with filter that fires once for each CASM tenant onboarding service
CASM Tenant Onboarding Rule	Rule with filter that fires once for each CASM tenant onboarding service
CASM Update MT Options	Enable Multi Tenancy Options for Common Administration
CASM Update Role Rule	Rule with filter that fires once for each CASM Create Role Service
CASM Update Tenant	Rule with filter that fires once for each CASM tenant update service
CASM Update User Rule	Rule with filter that fires once for each CASM update user service
CASM User Onboarding Rule	Rule with filter that fires once for each CASM user onboarding service
When Category is Service Management Content and Status is Canceled	Rule with filter that fires once for each service option of Service Management Content category when status updates to Canceled
When Category is Service Management Content and Status is Completed	Rule with filter that fires once for each service option of Service Management Content category when status updates to Completed
When Category is Service Management Content and Status is Pending Fulfillment	Rule with filter that fires once for each service option of Service Management Content category when status updates to Pending Fulfillment
Out-of-The-Box rules The out-of-the-box rules include multiple actions and only the actions mentioned here are enabled.	
When Category is Hardware and Status is Filled from Inventory Action: Launch HWSWFilledFromInv_SDM_SYNC SRF	Rule with filter that fires once for each HW request item when status goes to Filled From Inventory (WFHWFulfillment)

When Category is Hardware and Status is Pending Fulfillment Actions: Launch CheckAvailability SRF for Hardware Auto assign the selected asset to the user	Rule with filter that fires once for each HW request item when status goes from approval range to Pending Fulfillment (WFHWCheckAvailability)
When Category is Software and Status is Filled from Inventory Action: Launch HWSWFilledFromInv_SDM_SYNC SRF for Software	Rule with filter that fires once for each SW request item when status goes to Filled From Inventory (WFSWFulfillment)
When Category is Software and Status is Pending Fulfillment Action: Launch CheckAvailability SRF for Software availability	Rule with filter that fires once for each SW request item when status goes from approval range to Pending Fulfillment (WFSWCheckAvailability)
When Status is Submitted and Approval Process is driven by Workflow Action: Launch Approval SRF	Rule with filter that fires once for each service when a request is submitted for approval (WFAllSubmit)
Notes and Attachment Synchronization Rules	
When notes is added to Service Catalog request	When notes are added to CA Service Catalog request, add them to the corresponding CA Service Desk ticket
When attachment is added to Service Catalog request	When attachment is added to CA Service Catalog request, add it as a link to the corresponding CA Service Desk ticket

CA Service Desk Manager and CA Asset Portfolio Management

The integration between CA Service Desk Manager and CA Asset Portfolio Management is completely automated. You do not have to perform any manual steps to finalize this integration.

In the rare event that the integration between the two products fail, see integrate CA Service Desk Manager with CA Asset Portfolio Management manually to troubleshoot the integration manually:

To manually integrate CA APM and CA SDM, perform the following steps:

1. Enable SSO from CA APM to CA SDM
2. Verify Single Sign-On
3. Launch the CA APM in context from CA SDM

WARNING

Ensure that CA APM and CA Service Desk Manager are installed on the same database.

NOTE

CA APM uses the fallback mechanism during the Single-Sign On process. If both the mechanisms are configured, the Policy File mechanism is used. If the Policy file does not work, then CA Service Desk Manager Administrator credentials are used. If neither of these are successful, then the CA Service Desk Manager login screen is displayed on clicking the respective hyperlink in CA APM. If the CA APM user does not have valid access rights (role/group) in CA Service Desk Manager, the Single Sign-On login screen is displayed on clicking the CA Service Desk Manager links on the CA APM page.

Enable SSO from CA APM to CA SDM

CA Service Management provides the following ways to enable SSO from CA Asset Portfolio Management (CA APM) to CA Service Desk Manager.

Use the Web Server Access Policy File

You create the Access Policy file in CA Service Desk Manager and then use this file in CA APM. The Access Policy file contains encrypted policy details that are assigned to a CA Service Desk Manager user who has access to various capabilities in the CA SDM application.

Follow these steps:

1. Log in to the CA Service Desk Manager as an Administrator. Use the Administration tab to perform the following tasks:
   a. Create the Web Services Access Policy using the SOAP Web Services Policy option. Complete the following:
      a. Assign a name. The default name and code are CASM_POLICY; however, you can use any name as per your requirements.
      b. Assign a proxy user to this policy.

         NOTE

         This proxy user must have administrative access to CA Service Desk Manager.
      c. Verify that the Allow Impersonate property of CA Service Desk Manager is set to Yes.
      CA Service Desk Manager checks the access type of the CA APM user to be impersonated against the access type of the proxy user of the policy. If the access level of the CA APM user access type is less than or equal to the grant level of the proxy user access type, CA SDM replaces the CA APM user with the proxy user.
   b. Generate the policy key file with the help of pdm_pki tool for the Web Services policy that you created.
   In the Service Desk environment, open the Command Prompt and enter:
   ```
   C:>pdm_pki -p CASM_POLICY -f
   ```
   The policy file CASM_POLICY.p12 is generated in the same folder.
   The default/recommended name is CASM_POLICY.p12, but you can use any name as per your requirement.
   c. Verify that the HashKey property is set to Yes in the newly created Web Services access policy. The system typically configures this setting automatically when you generate the policy key file.
   d. Copy and save the policy key file to the CA APM default installation directory of the CA APM Web Server.
   ```
   C:\Program Files (x86)\CA\ITAM
   ```

      NOTE
      If you are using application clustering, copy the policy key file to the CA APM installation directory of every CA APM Webserver Component cluster node. If you re-configure the Policy in CA SDM, then you must re-generate the policy file and copy it to CA APM Web Server installed directory.
2. Log in to CA APM as an Administrator.
   Default: *uapmadmin*.
3. Click Administration, System Configuration, Service Desk Manager.
4. Enter the complete URL for CA Service Desk Manager SOAP Web Services in the following format:
   ```
   http://<ServiceDeskHostName>:<PortNum>/axis/services/USD_R11_WebService.
   ```
5. Copy the CA Service Desk Manager SOAP Policy file from CA Service Desk Manager Server to the CA APM web server installation folder. Default installation directory :
   ```
   C:\Program Files (x86)\CA\ITAM
   ```
6. Enter the file name of CA Service Desk Manager policy file that is copied on the CA APM Web server component installation folder.
7. Enter the CA Service Desk Manager policy code that is the policy name as configured in the CA Service Desk Manager Administration SOAP Web Services policy configuration page.
8. Click Save.

   NOTE
   Do not perform IISReset on the CA APM Web Server or CA APM Application Server.

Both the mechanisms internally use the CA Service Desk Manager SOAP Web Services to achieve the SSO from CA APM to CA SDM. You can configure either one or both the mechanisms to achieve SSO from CA APM to CA SDM.

WARNING
Ensure that CA APM and CA Service Desk Manager are installed on the same database.

NOTE
CA APM uses the fallback mechanism during the Single-Sign On process. If both the mechanisms are configured, the Policy File mechanism is used. If the Policy file does not work, then CA Service Desk Manager Administrator credentials are used. If neither of these are successful, then the CA Service Desk Manager login screen is displayed on clicking the respective hyperlink in CA APM. If the CA APM user does not have valid access rights (role/group) in CA Service Desk Manager, the Single Sign-On login screen is displayed on clicking the CA Service Desk Manager links on the CA APM page

Use the CA Service Desk Manager Administrator Credentials

The CA Service Desk Manager admin credentials are used from CA APM to login to CA Service Desk Manager through the CA Service Desk Manager Web Services:
Perform the following steps:

1. Log in to CA APM as an Administrator. Default user name is *uapmadmin*.
2. Click Administration, System Configuration, Service Desk Manager.
3. Enter the complete URL for CA Service Desk Manager SOAP Web Services in the following format:
   `http://<ServiceDeskHostName>:<PortNum>/axis/services/USD_R11_WebService`
4. Enter the CA Service Desk Manager Administrator login user name and password.
5. Click Save.

NOTE
Do not perform IISReset on the CA APM Web Server or CA APM Application Server.

Verify Single Sign-On

To verify that Single Sign-On from CA APM to CA Service Desk Manager has been enabled, click on any *one* of the following CA Service Desk Manager hyperlinks in the CA APM page:

- Log in to CA APM, Asset Details page and click on the CA Service Desk Manager link on the top
- Click Asset Relationships (menu on the left), CA Service Desk Manager Tickets links. A Criteria and Results page with Service Desk Ticket ID Column values in hyperlink is displayed in the results section. The hyperlinks are the entry point to CA SDM and display ticket details after you have successfully logged into CA SDM.
- Click on the Launch CA Service Desk Manager link in the Asset Viewer page. This launches the CA Service Desk Manager application.

View the Contact Update Details

You can view the contact update details for existing or new assets (known as configuration items in CA Service Desk Manager) in the following ways:

- Contact Update Details for existing Contacts in CA APM
  Ensure that you have enabled Single Sign-On from CA APM to CA Service Desk Manager as shown above for existing contact details to be updated and reflected in CA Service Desk Manager Contact's Quick Profile.
- Contact Update Details for new Assets in CA APM
  In this release, if you have integrated CA APM and CA Service Desk Manager, new Assets that are created have the Contact details automatically saved and reflected in CA Service Desk Manager Contact's Quick Profile.

Launch the CA APM in context from CA SDM

To launch CA APM in context from CA SDM, you need to add a Management Data Repository (MDR) in CA SDM to point to the proper CA APM URL.

Perform the following steps:

1. Create an MDR in CA SDM with the following properties:
   a. Button Name: CA APM
   b. MDR Name: CA APM
   c. MDR Class: GLOBAL
   d. Hostname: CA APM Server Name
   e. Path: ITAM/Pages/Asset.aspx
   f. Parameters: assetid={federated_asset_id}
2. Click the CA APM button.
   The corresponding asset record opens in CA APM.

CA Service Catalog and CA Asset Portfolio Management

The integration between CA Asset Portfolio Management and CA Service Catalog is completely automated. You do not have to perform any manual steps to finalize this integration.

In the rare event that the integration between the two products fail, see integrate CA Asset Portfolio Management with CA Service Catalog manually to troubleshoot the integration.

> **WARNING**
>
> Review the following section only if you want to manually integrate CA Asset Portfolio Management with CA Service Catalog when the automatic integration between the two has failed.

CA Service Catalog integrates with CA Asset Portfolio Management in several areas, including these areas:

- Requested items can be associated with CA Asset Portfolio Management Assets. Asset managers can associate CA Asset Portfolio Management assets with items requested from the catalog during request fulfillment.
- Catalog entries can be associated with CA Asset Portfolio Management Models. CA Service Catalog catalog administrators can associate one or more CA Asset Portfolio Management models with a service option. This association can assist with automated asset creation and association during request fulfillment using a workflow process and the CA Service Catalog web services.

> **WARNING**
> CA Service Catalog and CA Asset Portfolio Management must share the same MDB and the same CA EEM for this integration to work properly.

Requirements for the integration:

- You must have a role defined in CA Asset Portfolio Management.
- Verify that CA Asset Portfolio Management supports your internet browser.
- Check whether the CA Asset Portfolio Management application and web servers reside on the same computer or different computers. If they reside on different computers, complete the CA Asset Portfolio Management web services section in the administration configuration settings on the CA Service Catalog GUI.
- Authorized CA Service Catalog users use the CA Asset Portfolio Management Model Assignments window to assign eligible assets during the fulfillment of a request.

You can assign the assets through CA Service Catalog services if the following requirements are met:

- The assets must be active.
- The assets must not be assigned to any user.

You can associate one or more CA APM models with a service option, if the following requirements are met.

CA Service Catalog and CA Asset Portfolio Management administrators work together to use one of the following methods to authorize CA Service Catalog users (typically request managers) to assign assets:

1. Assign to the user the Fulfiller role in CA Asset Portfolio Management.
2. Assign to the user another role in CA Asset Portfolio Management that can update the Asset Fulfillment object. If you use another role, it must have the same Asset Fulfillment security settings as the CA Asset Portfolio Management Fulfiller role.
3. Specify a proxy user for CA Asset Portfolio Management in CA Service Catalog. To specify this proxy user, select **Administration**, **CA APM Web Services**, and complete the fields. Specify this CA Asset Portfolio Management user in the **User ID** field.
   In this case, CA Service Catalog users inherit the authority of the proxy user to assign assets.

   NOTE

   For information about fulfilling requests from inventory, see How to Fulfill Requests from Inventory.

Perform the following steps:

Step 1 - Create Users in CA Service Catalog

Step 2 - Create or Update Administrator in CA EEM

Step 3 - Update CA APM Web Services in CA Service Catalog Configuration

Step 4 - (Optional) Create New Rules and Actions

Step 5 - Create a Service and Request

Step 6 - Configure for Single Sign-On

Verify the CA Asset Portfolio Management - CA Service Catalog Integration

Uninstalling CA Service Management

How to Uninstall the xFlow Interface

Uninstall the xFlow Interface (Windows)

Perform the following steps to uninstall the xFlow Interface (Windows):

1. Click **Start**, **Control Panel**, **Uninstall a Program** to uninstall xFlow Interface.
   Or
2. Navigate to where you have installed the xFlow Interface (for example, *C:/Program Files/CA/xFlow)* and do the following:
   – Select **xFlow Interface_installation** and click **Change xFlow Interface Installation.exe** to complete the uninstallation process.

Uninstall the xFlow Interface (Non-Windows)

To uninstall the xFlow Interface, complete the following steps on Linux or Solaris as per your implementation:

1. Navigate to where you have installed the xFlow Interface server.
2. For example: */opt/CA/xFlow/_xFlow Interface 17.4.0.0_installation*
3. Execute the following command to uninstall:
   ```
   ./ Change xFlow Interface 17.4.0.0 Installation
   ```
4. Click **Finish** to complete the uninstallation process.

How to Uninstall the Search Server

Uninstall the Search Server

- Perform the following steps to uninstall the search server (Windows):
  a. Stop the CA Service Management Search Server service.

     NOTE
     Back up the search server folder. For example, back up the search server folder at C:/Program Files/CA/SC/CASearchServer
  b. Click **Start**, **Control Panel**, **Uninstall a Program and** right-click **SearchServer** to uninstall the Search Server.
     Or
  c. Navigate to where you have installed Search Server (for example, *C:/Program Files/CA/SC).*
  d. *Open* the **SearchServer** folder and select **SearchServer_Installation.** Click **Change SearchServer Installation.exe** to complete the uninstallation process.
- Perform the following steps to uninstall the search server (Linux):
  a. Stop the CA Service Management Search Server service.
  b. Navigate to /opt/CA/SC/SearchServer/elasticsearch-x.xx.x/bin on terminal using the following command
     ```
     ./ServiceStop.sh
     ```

     NOTE
     Back up the search server folder. For example, back up the search server folder at /opt/CA/SC/SearchServer
  c. Navigate to where you have installed Search Server (for example, /opt/CA/SC/SearchServer_Search Server 17.4.0.xx_installation) on the terminal and run the following command:
     ```
     ./Change\ Search\ Server\ 17.4.0.xx\ Installation
     ```
     Example:
     ```
     ./Change\ Search\ Server\ 17.4.0.0\ Installation
     ```
     The Uninstall Search Server window opens. Follow the on screen instructions.

Unable to run pdm_rebuild_index in Search Server after rolling back to previous 17.3 version from current RU version.

Perform the following steps to run pdm_rebuild_index:

1. Check the Java version present in <Search Server Install location>/search/bin/java_home.txt.
2. Check the Java version is available in the following locations:
   – **Windows:** C:\Program Files\CA\SC\JRE
   – **Linux:** /opt/CA/SC/JRE
3. Copy the specified JRE from jre_backup and place it in the above mentioned location in Step 2.
4. Run pdm_rebuild_index.

Uninstall DataMart

Perform the following steps to uninstall DataMart:

1. Navigate to the location where you have installed DataMart.
   For example,*C:\Program Files\CA\Datamart_CA Datamart_installation*
2. Run **Uninstall CA Datamart.exe**
3. Click **Next**
4. Click **Finish** to exit the wizard.
5. Navigate to *DVD1\DBCleanupUtility* directory.
6. Double-click the **DBCleanupUtility.bat** to launch the DBCleanup utility.
7. Enter the following details:
   a. **Database Server Type**: Specify the type of server, MS SQL or Oracle.

b. **Database Host Name:** Enter the host name of the server where the database resides.
c. **Database User Name**:
   • **SQL Server**: Enter the privileged user detail that has permission to create and modify schema.
d. **Database Password**:
   • **MS SQL**: Enter the database password used by the database admin user.
e. **Database Port Number:** Provide the database port number.
f. **Enter Database Name :** Enter the database name. For example, mdb.
g. **System Name**: Provide the host name of the system to cleanup the database.

8. On the product selection screen, select **DataMart**.
9. Launch MS SQL Server Management Studio.
10. As an administrator connect to the MS SQL Server where datamart DB is created.
11. Delete the DataMart schema.

Uninstall CA Service Desk Manager

To uninstall CA Service Desk Manager, perform the following steps:

1. Depending upon the operating system,
   – **Windows**:
      • Copy SDUninstall folder from NX_ROOT\REPLACED\sduninstall_backup folder to NX_ROOT.
      • Copy JDK folder 1.8.0_112 from SC\REPLACED\jre_old to CA\SC\JRE. Example: C:\Program Files(x86)\CA\SC \REPLACED\jre_old to C:\Program Files (x86)\CA\SC\JRE.
      • Stop the ServiceDesk Manager Services.
      • Run SDUninstall.exe from NX_ROOT\SDUninstall folder.
   – **Linux:**
      • Copy SDUninstall folder folder from NX_ROOT/REPLACED/sduninstall_backup folder to NX_ROOT.
      • Copy JDK folder 1.8.0_112 from SC/REPLACED/jre_old to CA/SC/JRE. Example: From /opt/CA/SC/ REPLACED/jre_old to /opt/CA/SC/JRE.
      • Stop the ServiceDesk Manager Services.
      • Run SDUninstall from NXROOT/SDUnistall folder.

   A message window appears.
2. Select the option to restart the system (recommended for Windows). Click **Done**.
   You have successfully uninstalled CA SDM.

Uninstall CA Service Catalog

You can uninstall CA Service Catalog and CA EEM from one or more computers, if you want to at some point of time. Perform the following steps:

1. Stop all CA Service Catalog Windows services on all computers.
2. Open the Windows Control Panel, select **CA Service Catalog**, and click **Uninstall**.

   NOTE

   The CA Service Catalog data in MDB is saved for any future CA Service Catalog installations in your enterprise. If you delete MDB data, you cannot recover it.
3. If no other CA Technologies products in your enterprise use CA EEM, use the Windows Control Panel to uninstall it.

   WARNING
   Before you uninstall CA EEM, unregister all registered applications.
4. (Optional) Review the uninstallation log files in the Windows Temp directory:

- **CA_Service_Catalog_view.log**
- **CA_Service_Catalog_Uninstall.log**

NOTE

Tomcat and the JRE are installed automatically during CA Service Catalog installation and are uninstalled automatically when you uninstall CA Service Catalog. After Uninstalling Service Catalog, you must uninstall CA Service Management Administration.

NOTE

If the uninstallation fails for any reason, perform the following actions and uninstall again:

- Install JRE 1.7.51 or later.
- Set JAVA_HOME to the JRE path and add JAVA_HOME in the Environment PATH variable.

Run the Database Cleanup Utility

During the installation of CA Service Management products, each of the installed product is registered in a common mdb table that is reflected in the deployment summary of the selected environment. These entries must be cleared in the following cases:

- Uninstallation of any CA Service Management products
- Moving Database or migration from one server to another

Run the CA Service Management database cleanup utility on a Windows or a Non-Windows environment. The database cleanup utility removes the entries and adds these to the following tables:

Table Name from where the entries are removed	Table Name to which the removed entries are added
al_cdb_componentinstallstate	al_cdb_comp_installstate_bkup
al_cdb_configurationparameters	al_cdb_config_params_backup

NOTE

On a Non-Windows Platform, before running the **DBCleanup.sh** Utility, you must export the Oracle Environment variables on the terminal.

Follow these steps:

1. Navigate to *DVD1\DBCleanupUtility* directory.
2. For Windows, double-click the **DBCleanupUtility.bat** to launch the DBCleanup utility.
   On a Non-Windows platform, you must run the **DBCleanupUtility.sh** file.
3. Enter the following details:
   a. **Database Server Type**: Specify the type of server, MS SQL or Oracle.
   b. **Database Host Name:** Enter the host name of the server where the database resides.
   c. **Database User Name**:
      - **SQL Server**: Enter the privileged user detail that has permission to create and modify schema.
      - **Oracle**: Provide the mdbadmin user name.
   d. **Database Password**:
      - **MS SQL**: Enter the database password used by the database admin user.
      - **Oracle**: Enter the password for the Oracle database user.
   e. **Database Port Number:** Provide the database port number.
   f. **Enter Database Name :** Enter the database name. For example, mdb.

g. **System Name**: Provide the host name of the system from where you want to uninstall the CA Service Management product or products.

4. Select any of the following products that you want to uninstall:
   - **SDM -** CA Service Desk Manager.
   - **SLCM-** CA Service Catalog.
   - The utility cleans up the database and removes the entries of the selected product installed on the server.

Post-Backout Tasks

This article explains the steps to perform after CA Service Management is uninstalled, and then roll back to a previous version.

Roll Back Java Update

CA Service Management 17.3 is shipped with Java 11. To revert to an older version of Java (1.8.0_112), follow the steps:

CA SDM

For CA SDM, perform the following steps:

1. Stop the `CA Service Desk Manager` service.
2. Navigate to the `$NX_ROOT` folder.
   a. Open the `NX.env` file in a text editor.
   b. Find the parameter `NX_JRE_INSTALL_DIR=C:/Program Files (x86)/CA/SC/JRE/11.0.3`
   c. Modify the value from **11.0.3** to **1.8.0_112**
   d. Save the `NX.env` file.
3. Restart the `CA Service Desk Manager` service.

For remote Web Screen Painter, perform the following steps:

1. On the system where CA SDM is installed, navigate to `$NX_ROOT/java/lib` folder.
2. Copy the `sd-utils.jar` file.
3. On the remote WSP server, paste the .jar file on `$NX_ROOT/java/lib` folder.
4. Execute the jar file.
5. Launch the remote WSP.
6. Copy Java 11 folder from the above path (java) on the remote WSP
7. Open the `NX.env` file in a text editor.
8. Modify the value from **11.0.3** to **1.8.0_112.**

Upgrading

As you prepare to upgrade to CA Service Management 17.4, consider the value of the new features added in the latest release. Based on these assessments, you can plan and make effective decisions toward a successful upgrade.

Review the following before you upgrade to CA Service Management 17.4:

Getting Started

- Review the CA Service Management 17.4 Release Notes.
- Review the Supportability Matrix.

Upgrade Planning

For more information, see Pre-Planning for an Upgrade.

Prerequisites and Considerations

Review and verify the upgrade prerequisites and planning consideration for the following CA Service Management products:

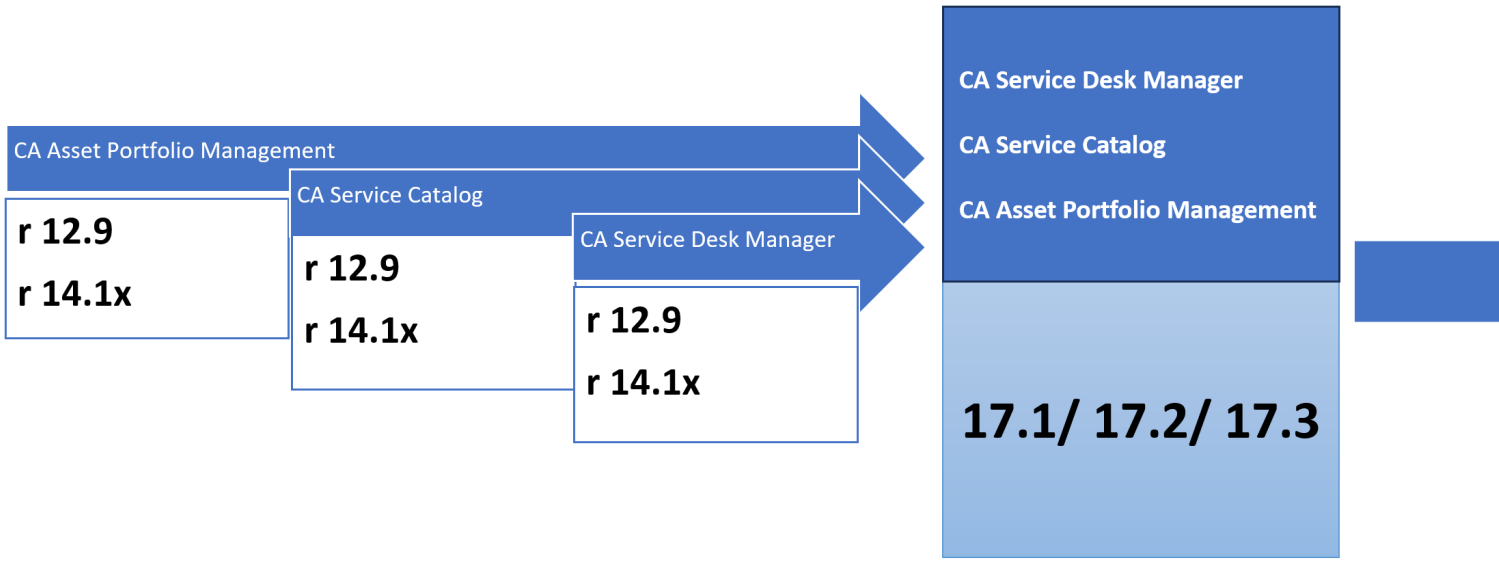
1. Upgrading to CA Service Desk Manager 17.4
2. Upgrading to CA Service Catalog 17.4
3. Upgrading to CA Asset Portfolio Management 17.4

NOTE

If you are planning to upgrade a product that is integrated and configured with Active Directory, ensure to create the **CasmAdmin** user and **OpenSpaceAdminGroup** before upgrading to CA Service Management 17.4. Associate the **CasmAdmin** to **OpenSpaceAdminGroup** in the Active Directory.

NOTE

Insights - Aggregate Reports are currently supported only on Microsoft SQL Server Database and not on Oracle Database.



Supported Upgrade Paths

The following table lists the earliest version from which a CA Service Management product can be migrated to the current version of CA Service Management:

CA Service Management Products	Supported Upgrade Paths
CA Service Desk Manager	Upgrade from **17.1/17.2/17.3** release is supported. CA Service Desk Manager provides the xFlow Interface, App Launcher with Service Point App/Insights App for CABI JasperReports Server, and search server capabilities.
CA Service Catalog	Upgrade from CA Service Catalog **17.1/17.2/17.3** release is supported.
CA Asset Portfolio Management	Upgrade from **CA Asset Portfolio Management (CA ITAM) 17.1/17.2/17.3** release is supported. **Note:** If you are upgrading CA Asset Portfolio Management from an earlier release to 17.4, the changes you made to configuration files from an earlier release are not automatically merged with CA Service Management 17.4. Refer to the backup files located in the ***%Temp%\ITAM*** directory and manually merge these changes with the currently installed version.

How to Upgrade

Launch the CA Service Management installer on the systems/servers that you want to upgrade. The installer automatically detects the installed CA Service Management version in your environment and upgrades it to 17.4.

For more information about how the upgrade process works, see Upgrade to CA Service Management 17.4.

Integrations

Integrations After completing the upgrade process, perform integration to integrate one or more CA Service Management products in your environment. For example, you can integrate CA Service Desk Manager with CA Service Catalog. For more information see, Integrate CA Service Management Solution.

Why Upgrade to CA Service Management 17.4

Timely upgrades help improve ROI both quantitatively and qualitatively. It ensures business continuity and avoids unnecessary failures. An older product version may lack important security updates, new features, and the ability to integrate with current technology trends and advancements. Unsupported software also poses serious security risks for an Enterprise. Timely upgrades verify that you are using an improved and enhanced product to increase productivity and usability. Being on a current release also helps to protect your systems and applications against virus, spyware, and/or other malicious software attack.

Some upgrades can be easily handled in-house with the right resources, while others require external support. As you complete your assessment and build your strategy, consider your options for assistance and map out your upgrade requirements and timing. CA Services and Support help with your upgrade plan and provide resources, options, and expertise to ensure that you get the most from your software investment. You can access resources that are provided by CA Support for a hassle-free upgrade experience from start to finish. For more information, contact Broadcom Support.

Benefits of Upgrading

Upgrading to the latest release version of CA Service Management gives you the following benefits:

- Improved application with new features and enhancements
- Mobility
- Out-of-the box services
- Availability of new fixes and enhancements
- Improved solution architecture for speed and minimal downtime
- Better administrative capabilities and improved service delivery efficiency
- Easy future upgrades and on-boarding
- Better alignment with business objectives

New Features and Enhancements

Upgrading to the latest version helps you avail the new features and enhancements that are provided in CA Service Management. It also provides operational efficiency to conduct your daily IT business. For more information about new features and enhancements, see CA Service Management 17.4 Release Notes.

End of Service (EOS) for Release Versions

EOS Release dates have been announced for older versions of CA Service Management. Ensure that you are up-to-date and using the latest version of CA Service Management to have continued support. Supportability options on an EOS release are minimal and restricts your access to published patches and new code creation. You have limited ability to receive standard support for products that you are using. Staying on an Older release version also increases vulnerability and security concerns. For more information, see Release and Support Life Cycle Matrix.

Pre-Planning for an Upgrade

Planning Considerations

Planning well ensures upgrade readiness before starting the actual upgrade process. Before proceeding with the actual upgrade process, review, consider, and plan for the following tasks:

Task	Additional Information
Download CA Service Management 17.4 Installer from CA Support.	For more information, see Broadcom Support.
Review the Release Note	For more information, see CA Service Management 17.4 Release Notes.
Operating Systems and Database Compatability	For more information, see Supportability Matrix.
Ensure that your environment meets all minimum version requirements. Review Supported Upgrade Paths for CA Service Management 17.4.	For more information, see Upgrade to CA Service Management 17.4
Backing up the Operating Systems and database	For more information, see Database Backup and Database Restore.
Backing up all customizations.	• Are there any customizations you have performed in your environment? • Do you need these customizations after upgrading? For more information, see Pre-Planning for an Upgrade.
Server and Hardware requirements	For more information, see Server and Hardware Requirements.
Environment Scalability	• Whether your Current Environment is meeting your business needs? • Is it scalable?
New Features/Enhancements from Latest Release	• Which new features you need to enable/install? For more information, see CA Service Management 17.4 Release Notes. • Will these new features require more servers?
Which Applications must be upgraded first?	• CA Process Automation and CA Embedded Entitlements Manager must be upgraded first as this allows the CA Service Management installer to automatically integrate these products. • Begin the upgrade with the 17.4 CA Service Catalog, and subsequently proceed to upgrade CA SDM and its sub-components. If you have CA Asset Portfolio Management in your environment, upgrade it in an order of your preference.
Integration	• If the upgrade environment is integrated with CA EEM: Enable active directory only after CA Service Management is successfully upgraded. • If the upgrade environment is configured with a load balancer, SAML, or NTLM: Reconfigure the settings after CA Service Management is successfully upgraded. • Are there any CA Service Management Integrated Products in your environment? Will they need to be upgraded? For more information, see Integrating.

Other Considerations

- When upgrading to Release 17.4 and later, older versions of JRE, MDB, and Tomcat which are not used by Service Desk Manager are backed up in the jre_old, mdb_backup, and tomcat_old folders.
  Example:

- **Windows:** Older version/s of Tomcat is moved from the C:Program Files (x86)/CA/SC/tomcat folder to C:Program Files (x86)/CA/SC/REPLACED/tomcat_old.
- **Linux:** Older version/s of Tomcat is moved from the /opt/CA/SC/tomcat folder to /opt/CA/SC/REPLACED/tomcat_old.

  NOTE
  If other applications are using these older versions of JRE and Tomcat, copy them back to the original location.

- When upgrading to Release 17.4 and later, older versions of jquery and perl are backed up in the jquery and perl_old folders.
  Example:
  - **Windows:** Older version/s of perl is moved from the NX_ROOT/bin/perllib folder to NX_ROOT/REPLACED/perl_old.
  - **Linux:** Older version/s of perl is moved from the NX_ROOT\bin\perllib folder to NX_ROOT\REPLACED\perl_old.

    NOTE
    If your update is successful and other integrations are intact, you can delete the "REPLACED" folders as it may contain older versions of vulnerable jars.

Create a Backup Plan

Plan to take a full backup of your database and ensure it is validated. Plan to take multiple snapshots or a backup of the entire installation folder during the course of the upgrade procedure. This saves time, in case, you need to revert to a certain point while upgrading. Temporarily allocate additional disk space for backups.

For more information, see Database Backup and Database Restore.

Decide the Upgrade Method to Use

- Decide the type of architecture you want to use for CA SDM: **Conventional** or **Advanced Availability**.
- Identify the product components on all systems in your environment.
- Synchronize production data over to the development environment (**Production** > **Development**).
- Prepare a checklist of all tasks that needs to be performed during the upgrade/migration process.
- If time permits, plan a test upgrade to capture actual migration time, validate tasks, and record any possible issues and provide resolutions.
- Ensure that you have the necessary administrative privileges for performing the upgrade procedures.

For more information, see Upgrade Tools.

Plan for Environments

Plan for at least two environments: **Development** and **Production** environments. If a test or development environment is not available, consider Swing Box as the development environment. For more information about Swing Box Migration Method, see Swing-Box Methods to Upgrade to CA Service Management 17.4.

Scaling the Environment

In case of CA SDM Conventional Configuration, limit the number of components installed on a Primary Server for a better upgrade experience, scalability, and better application performance.

Upgrade Tools

This article describes the different ways you can upgrade and migrate from an earlier release version of CA Service Management to the latest.

CA Service Management Common Installer

CA Service Management Common Installer performs the following tasks on launch:

- Detects the version of the CA Service Management products (CA SDM, CA Service Catalog, and CA APM) that are currently installed in your environment and performs the appropriate Upgrade tasks for a given release version.
- Remotely connects to database server (no need to run installer on a remote database).
- Performs automated integration tasks for CA Service Management products (CA SDM, CA Service Catalog, and CA APM), if you had

CA Service Management Swing-Box Upgrade

The Swing-Box Upgrade method is used for the following:

- Applicable for CA SDM
- When a new hardware/Operating System (OS)/database is upgraded
- When MDB (database) is not shared or is shared with other applications that are either already updated or are updated. For more information, see Swing-Box Methods to Upgrade to CA Service Management 17.4.

Pros	Cons
• Allows the use of new/latest hardware with current OS/ database versions. • The remains of the legacy customizations and previous upgrade are cleaned up. • Less risk as current production environment is touched.	More cost of new hardware (need temporary hardware for SWING environment).

CA SDM Customization Upgrade Utility

CA Service Desk Manager allows customers to extend the design forms as required. The Customization Upgrade utility facilitates smooth upgrade and merging of the customized forms into the latest CA SDM version. For more information, see

How does the Customization Upgrade Utility Works?

CA SDM Customization Upgrade Utility does the following tasks:

- Performs comparisons on HTMPL and CSS customizations.
- Attempts to merge the customizations into newer forms.
- Semi-automatic in nature.

For the utility to run successfully, the following are required:

- Out-of-the-Box (OOB) files from an earlier CA SDM Release that you are upgrading from.
- Customized Files from earlier CA SDM Release that you are upgrading from.
- Out of the Box files from latest CA SDM Release (In this case, CA SDM 17.3).
- Target Merged files as a Result. The utility generates a log file containing the status of each file: **Merged**, **Unable to Merge**, and **Copied**.

  NOTE

  Any files that are not merged successfully through the utility must be merged manually.

Upgrade to CA Service Management 17.4

To upgrade CA Service Management, review the following upgrade tasks.

NOTE

If you are planning to upgrade a product that is integrated and configured with Active Directory, ensure to create the **CasmAdmin** user before upgrading to CA Service Management 17.4. Associate the **CasmAdmin** in the Active Directory.

NOTE

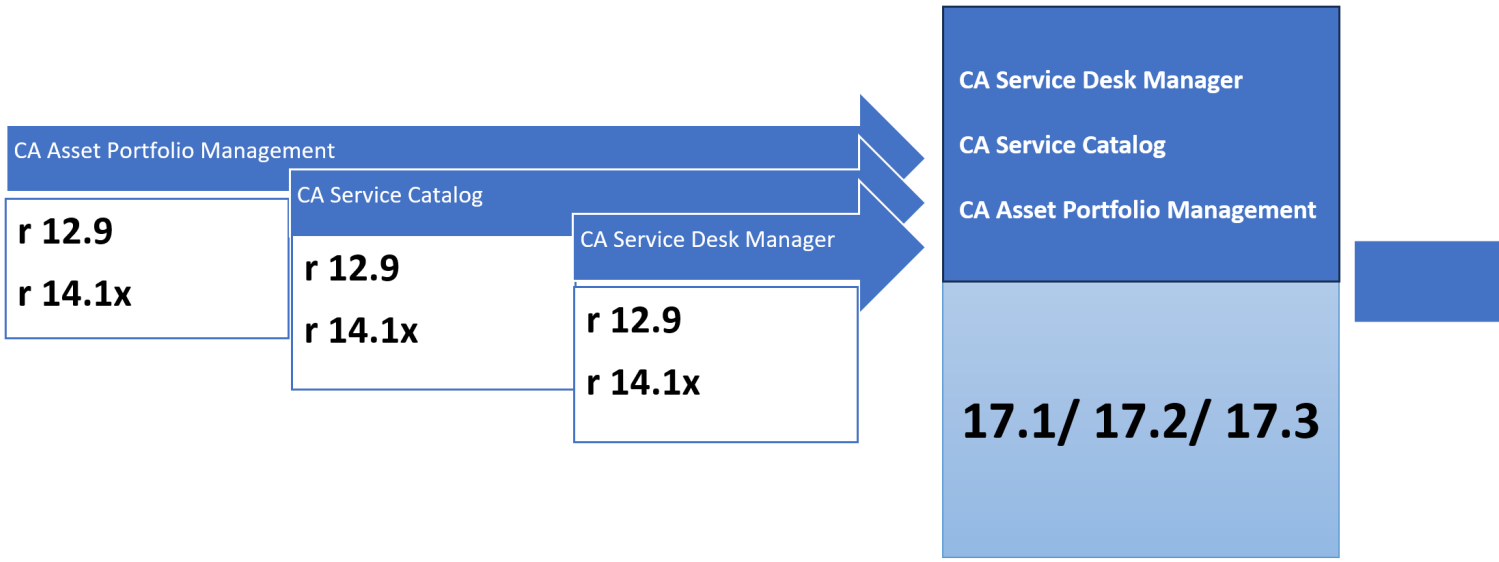
If you are upgrading from CA Service Management 17.3 to CA Service Management 17.4, ensure to rename the *CUM1* folder in *<Nx_ROOT>/patches/* to update the latest MDB changes.

To upgrade CA Service Management products, review the following upgrade tasks:

1. Review and verify the upgrade prerequisites and planning consideration for the following CA Service Management products:
   a. Upgrade CA Service Desk Manager
   b. Upgrade CA Service Catalog
   c. Upgrade CA Asset Portfolio Management
2. Launch the CA Service Management installer on the system/server that you want to upgrade.
   The installer automatically detects the installed CA Service Management release version and upgrades it to 17.4.
3. Complete the upgrade process. Now, you can integrate one or more CA Service Management products in your environment. Perform more steps to finalize the integration.

Supported Upgrade Paths

The following supported products and release versions can be upgraded to CA Service Management 17.4:



CA Service Management Products	Supported Upgrade Paths
CA Service Desk Manager	Upgrade from **17.3** release is supported. CA Service Desk Manager provides the xFlow Interface, App Launcher with Service Point App/Insights App for CABI JasperReports Server, and search server capabilities.
CA Service Catalog	Upgrade from CA Service Catalog **17.3** release is supported.
CA Asset Portfolio Management	Upgrade from **CA Asset Portfolio Management (CA ITAM) 17.3** release is supported. **Note:** If you are upgrading CA Asset Portfolio Management from an earlier release to 17.4, the changes you made to configuration files from an earlier release are not automatically merged with CA Service Management 17.4. Refer to the backup files located in the ***%Temp%\ITAM*** directory and manually merge these changes with the currently installed version.

Upgrading to CA Service Desk Manager 17.4

Ensure to review the upgrade tasks before upgrading on a production or test system.

NOTE
Before you start upgrading to CA Service Desk Manager 17.4, for Oracle Database, ensure that the Oracle PATH environment variable refers to 32-bit oracle client first, and then, to 64-bit Oracle client. To change the sequence, navigate to Environment Variables, System Variables, Path, Edit.

For more information, see the following topics:

- Upgrade Prerequisites - CA SDM
- Pre-Upgrade Tasks
- Upgrade CA Service Desk Manager 17.4
- How to Upgrade Oracle Database Client
- Perform the CA SDM Post-Upgrade Tasks

Upgrade Prerequisites - CA SDM

This article describes the upgrade prerequisites for a successful CA SDM upgrade process. Ensure to complete these prerequisites before you start the upgrade process on a production or test system.

CA SDM 17.4 supports upgrade from 17.1/17.2/17.3 for all supported platforms. For more information, see the Supportability Matrix.

NOTE

If you are on an older release version CA SDM, prior to 17.1 it is recommended that you first upgrade to CA SDM 17.1/17.2/17.3 or above and then, follow the upgrade process for CA SDM 17.4

- Windows users can upgrade directly to the latest version of CA Service Management using the CA Service Management Common Installer.
- Linux users must run the automated upgrade script for CA SDM and must move CA SDM to a supported platform and database before running the automated upgrade script for the release of CA Service Management 17.4
- If you have CA SDM modifications/customization, you must apply these customizations once after running the final automated upgrade script to the release of CA SDM 17.4

Prerequisites

- If you have manually added a variable in the NX.env file or updated a variable value and want to retain the variable value after migration, ensure to add the variables to the NX.env template file before performing the migration process.
- CA SDM supports only ITIL. If you are upgrading from a non-ITIL system, the CA Service Management installation upgrades your environment to an ITIL environment.
- Before migrating from an earlier supported version of CA SDM to CA SDM 17.4, validate that the servlet path URL (in each Repository Server) is in the correct format.
  To validate, complete the following steps:
  a. a. Log into CA SDM.
     b. Navigate to **Administration**, **Attachment Library**, **Repository**.
     c. Edit each repository servlet path URL. Remove the Fully Qualified Domain Name (FQDN) from the servlet path and replace it with the hostname in the servlet URL.
     For example, change (http://myhostname.ca.com:8080/CAisd/UploadServlet) to (https://myhostname:8080/CAisd/UploadServlet)
     Ensure that the repository server functionality is intact and working.
- UTF-8 locale must be installed on Linux platforms.

  NOTE

  On Linux, CA SDM no longer uses the **smtp_mail** script to process outgoing mail notifications.
- If you are using **smtp_mail**, and you upgrade to CA SDM 17.4, your CA SDM administrator must configure the appropriate mail options using the **Default Mailbox Detail** page to enable the CA SDM mail notification feature.

  WARNING

  Migration from an advanced availability environment is not possible using the CA SDM rolling maintenance process. You must shutdown the CA SDM services on all the application and standby servers before starting the migration activity. You need to first migrate the background server and then, verify whether the

background server processes are running. Migrate other servers only after ensuring that the background server processes are up and running.

- If an earlier supported version of CA SDM uses ***domsrvr:01*** , rename the server ID before starting the upgrade. Execute the ***pdm_edit*** command and update the domsrvr ID from **01** to another ID value.

  NOTE

  The ***pdm_edit*** command provides an easier interface to configure the servers. The ***domsrvr:01*** name for object manager is reserved and cannot be used by a custom configuration.

Pre-Upgrade Tasks

Perform the following pre-upgrade tasks as per your environment requirements.

Upgrade your Database

Consider the following before upgrading your CA SDM database:

- If you are using CA SDM 17.3 and want to upgrade to CA SDM 17.4, run the following commands to a supported database: Backup your existing database using the database backup procedures:

```
update mdb_schema_information set FileTimestamp='2012-04-23T09:19:09+0000'
  where FileName like 'doc_rep.xml'
```

- (Applicable for all non-windows machines) If the supported upgrade release versions of CA SDM is configured, install Oracle 19c/ 21c client before you upgrade. For more information, see Supportability Matrix.

  WARNING
  Before upgrading, change the Oracle home path to Oracle 19c/ 21c client path in the *$NXROOT/NX.env* file.

- Archive the installation directory ($NX_ROOT) using your typical archive procedures.
  This action lowers amount of data movement and saves disk space.

  IMPORTANT
  You must take the backup of the installation directory ($NX_ROOT), which helps add any missing files after the upgrade.
- Upgrade CA SDM 17.3 or later to 17.4 on a supported database (Microsoft SQL Server or Oracle).

Run the following script from a command prompt to identify any duplicate records on the Microsoft SQL Server or Oracle database:

NOTE
Run this script on the secondary servers. If you want to run this script using Microsoft SQL Query Analyzer, edit the **SQLCheckr12UniqueIndexes.sql** script and remove the **EXIT** argument before you run the command.

WARNING
These scripts identify the duplicate records. Delete identified duplicate records before you proceed with the upgrade process.

- (Oracle) Run **OracleCheckr12UniqueIndexes.sql**, located in the **..\products\SDM\Migrate** directory on the installation media.
- (Microsoft SQL Server): Perform the following steps:

  NOTE

  The database is named as **use mdb** by default. Before you run the script, update the database name manually to **use <other_database_name>** using the **SQLCheckr12UniqueIndexes.sql** file.

  a. Open a Command Prompt window and run **SQLCheckr12UniqueIndexes.sql** as follows:

```
cd $NX_ROOT\samples\views\SQLServer
```

  b. Run the following command:

```
Sqlcmd  - E  - e < SQLCheckr12UniqueIndexes.sql
```

NOTE
After upgrade, you can find the files at ***$NX_ROOT/samples/views/SQLServer*** or *$NX_ROOT/samples/views/Oracle* on the server.

- If you are using an earlier version of CA SDM on a non-supported operating system and database, move CA SDM to a supported operating environment and database before you upgrade.

  NOTE

  Special Windows characters, such as a long hyphen are not properly stored in the database on a non-Windows system in CA SDM or Knowledge Management.
- **Oracle** -- Oracle does not support case-insensitive indexes for Configuration Item registration. Before you start the migration on Oracle, verify SQLPlus and Oracle DB are able to communicate using the host name. If communication fails, verify that the Oracle database is configured with a loopback adaptor.

  NOTE

  When you migrate a double-byte character environment with an Oracle database, increase the maximum open cursor limit to at least **500**. For more information, see Oracle documents **ORA-01000: maximum open cursor exceeded**.
- **Microsoft SQL Server** -- To upgrade the Microsoft SQL Server database to the current release of CA SDM 17.4, the default configured database must be MDB, if not the upgrade fails and displays the following message:
  The acctyp_v2 table does not exist on the MDB .
- **Table Updates** -- Consider the following table updates that happen during the upgrade process:
  - **Status Tables** -- These tables are also updated with the appropriate status records when the same code values do not exist in your database.
    For example, ***Cr_Status*** is updated with the code ***AEUR*** (Awaiting End User Response).
  - **Functional Areas** -- For each role, the upgrade process automatically adds a row for each ***usp_functional_access*** record. Migration sets the access level to the same level for each CA SDM functional area that the *usp_role* table includes. New functional areas are mapped using a reference field.
- **Foreign Keys** -- Consider the following information:
  - Foreign keys (SRELs) that reference tables, in which the primary key is a **UUID**, change from integer type to **UUID type** (or **BYTE 16**).
  - If you dropped the foreign key constraints in the earlier CA SDM release version that you were using to mass load data, recreate the foreign key constraints before you run the upgrade. The scripts that drop the constraints are found in the following locations:
    - **Oracle**
      $NX_ROOT/samples/views/Oracle/OracleDropConstraints.sql
    - **Microsoft SQL Server**
      $NX_ROOT/samples/views/SQLServer/SQLDropConstraints.sql

      NOTE
      Reapply the dropped constraints by running the appropriate script ***OracleAddConstraints.sql*** or ***SQLServer/SQLAddConstraints.sql*** . These scripts are found in the same directory as the drop constraints and contains instructions.
- **MDB** -- MDB provides a consistent database schema for various IT management data. Data elements from earlier CA SDM environment are incorporated into this schema during upgrade. The size of the data elements can increase and consequently, increases the overall database size.

  NOTE

  When standard data elements extend beyond the column width defined for MDB, the update process can truncate data in these elements. Messages alert you to any truncation that occurs during the upgrade.

- **Distributed Setup** -- We recommend you upgrade your servers in the following order, depending on your CA SDM configuration:
  a. **Conventional**
     - Primary server
     - One or more secondary servers
  b. **Advanced Availability**
     - Background server
     - One or more standby servers
     - One or more application servers
- **Remote Database** -- If you are using Microsoft SQL Server MDB database, **sqlcmd** must be on the client computer before connecting to the remote MDB.

Knowledge Management Considerations

For **EBR** search engine, navigate to CA SDM **Options Manager**, **Search Engine** and edit **ebr_version** to specify KT Search Engine.

LREL Migration Considerations

A *List Relationship* (LREL) represents an association between two objects. An LREL has a left-hand side (lhs) and right-hand side (rhs) relationship. Each side of the relationship is an attribute of the majic object that contains the data relationship.

> **NOTE**
>
> In earlier releases of CA SDM, the **.maj** LREL statements and objects described many-to-many DBMS data relationships. Now, the any-to-many relationships no longer use the LREL majic statement. Instead, individual tables store both sides of the relationship. Objects access the relationship with a standard BREL statement. For example, you can see the relationship between change orders and CIs by reviewing the new **usp_lrel_asset_chgnr** table and in the corresponding **lrel_asset_chgnr** object.

The LREL changes eliminate the need to store attribute names in the database. The two sides of the relationship are foreign key single relationships (SREL) that are easy to join and index. If necessary, the relationship can contain additional relational attributes.

During an upgrade, the following activities happen when the LREL table data migrates to CA SDM 17.4:

- Automatically migrates tables and objects with LREL relationships.
- Names new tables using the ***usp_lrel_lhsName_rhsName*** format.
  For example, the ***usp_lrel_asset_chgnr*** table has a left-hand relationship to assets and a right-hand relationship to change orders.
- Names the corresponding objects using the ***lrel_lhsName_rhsName*** .

For example, the ***lrel_asset_chgnr*** object corresponds to the ***usp_lrel_asset_chgnr*** table.

- Due to database limitation, some names are abbreviated.
- Your data is migrated from the old tables and CA SDM code is modified to use CA SDM 17.4.
- Old LREL database tables are no longer used, such as ***bmlrel*** . However, for reference purposes, the old tables retain the data.
- A backward relation (BREL) attribute to the new object replaces the original LREL attribute in each related majic object definition.
- If you are using a supported API, such as the ***CreateLrelRelationship()*** web service method, the code works as expected.
- If you have included any custom LREL style relationships, CA SDM migrates/upgrades the Custom LREL to the tables of the updated release.
- Any site-defined code or reports that directly access the old LREL tables operate on old data because the system no longer uses those tables. We recommend that you update the code to tables and reports of the updated version of the code.

WARNING
If your code directly accesses legacy LREL objects or tables, the code fails after upgrade. We recommend that you upgrade the code before upgrading. For example, if your code uses **majic** statements to establish LREL relationships, use the ***createLrelRelationships()*** method instead of directly populating a table.

NOTE
We recommend that you verify site-defined code or reports that directly access the database or address the legacy LREL majic objects such as the **lrel1** object to verify that they operate properly. You can update your code to use a supported interface, such as Web Services. You can also update the necessary table names. For reports, you can also update the queries with the new DBMS table references.

How to Upgrade CA EEM

NOTE
The CA SDM17.4 supports CA EEM version 12.6.6.0

To upgrade CA EEM, perform the following tasks:

To upgrade from an earlier supported version of CA EEM, see CA EEM Documentation.

1. Verify that you have upgraded to the current version of CA SDM 17.4.
2. Install/ upgrade CA EEM version 12.6.6.0
3. After you install CA EEM, manually set the appropriate options in CA SDM Options Manager (***eiam_hostname*** , ***use_eiam_artifact*** , and ***use_eiam_authentication***). Review the upgraded options carefully. The default Process Manager URL is no longer **pmService**.

NOTE

The ***cawf_pm_url*** option has changed to default: ***http://<wf_hostname>:<wf_tomcat_port>/pm/service/pmService2*** . You *must* manually change the **pmService** to **pmService**2 for CAWF communication to remain functional.

WARNING
After you upgrade CA EEM, set the ***eiam_hostname*** , ***use_eiam_artifact*** , and ***use_eiam_authentication*** options in CA SDM **Options Manager**, **Security** if you were earlier using **eIAM** CA SDM user authentication.

CA Process Automation Integration

To ensure that CA SDM and CA Process Automation integration works properly, add the **ServiceDesk** user in CA EEM (if not already added) during the upgrade process.

Retain your Customization

Before you upgrade, review the following information to see how you can retain CA SDM customization/modification performed in your environment :

WARNING

Any **modifications** or **adaptions** or c**onfigurations** that are done administratively through the interface (web browser, command-line, Web Screen Painter) are supported, meaning CA Support can assist with basic suggestions and troubleshooting. CA Support will not perform any modifications for the customer. The customer is responsible for the changes and customization that are carried out in an environment.

For example, adding a field to a table and putting the field on a form through Web Screen Painter is a fully supported “modification”. Similarly, installing or uninstalling a feature through the CA SDM Options Manager administration is a fully supported “configuration”. Anything to do with SPEL code, Java scripting (or any language scripting), or a customer-specific change to the underlying base code-line (done by CA Services or a Partner), is not supported by CA Support. The customer can perform these actions, but is responsible for the support, maintenance, and troubleshooting when an issue occurs. If such “customization” affect the expected out-of-the-box behavior, CA Support will ask the customer to remove the customization and see if the behavior persists. For more information, see Customization Upgrade Utility.

- **Modified Forms** -- Upgrade retains the modification of the forms from an earlier release of CA SDM. However, you cannot view the CA SDM Release 17.4 functionality on the modified forms after you upgrade.
- **Modified Form Buttons** -- After upgrade completes, buttons on the modified forms in ***/site/mods/htmpl*** that did not have quotes around the **msgtxt(n)** part of the code show an error message, instead of the button name.
  For example, in the **detail_cr.htmpl** form, modify **msgtxt(441)** with quotes as follows to display the correct button name:

```
ImgBtnCreate("btnchg", "msgtext(441)", "detailSave('NEW_CHANGE')",
true, 0, msgtext(440)); // Save and Create Change Order
```

- **Retaining Modifications**--. When any form is modified, the customer changes are overwritten and has to be recreated. Upgrade process attempts to detect and identify the differences in the forms, and report it to the user. However, it is highly recommended to document your customization before you start the upgrade process. The custom JavaScript file must be customized again according to the CA SDM base release JavaScript file.

  NOTE
  If you modify the ***acctypedtl.rpt*** and ***acctypesum.rpt*** reports, the return data in CA SDM Release 17.4 is obsolete.
- **Notification Rules** -- If you have removed the default activity notifications **Contact**, **Object Contacts**, and **Contact Types** from an earlier installation of CA SDM and want to retain this functionality, then make a note of the default contacts that were removed before migration. After you upgrade to the new version, remove the default Notification contacts again.
- **Role-based Functionality** -- Upgrade may cause issues with the role-based functionality. If you modified any of the following forms, they are considered as read-only by the Web Screen Painter in CA SDM Release 17.4 and they include an **xxx_site.htmpl** version where you can use custom code:
  - **ahdtop.htmpl**
  - **menu_frames.htmpl**
  - **reports.htmpl**
  - **std_body.htmplca**
  - **std_footer.htmpl**
  - **std_head.htmpl**
  - **styles.htmpl**
  - **msg_cat.js**
  - **menu_frames_role.htmpl**

NOTE

The script does not restore files from backup folders to the legacy or **site/mods** directories as in previous releases of CA SDM.

How to Upgrade Oracle Database Client

This article describes how to upgrade the Oracle Database Client:

For Windows

Perform the following steps:

1. Install the Oracle 32-bit client for Oracle 19c/ 21c with Administration Client.
2. Establish the connection with Oracle_Server_SID.
3. Create a Local Net Service Name and test the connection.
4. Edit the PATH variable to have oracle_client home and remove server home from PATH, as shown in the following example:
   C:\app\Administrator\product\<oracle_database_version>\client_32bit\bin
5. Execute pdm_configure with ORACLE_HOME set to client_home, as shown in the following example:
   C:\app\Administrator\product\<oracle_database_version>\client_32bit\
6. Install the Oracle 64-bit client for Oracle 19c/ 21c with Administration Client.
7. Establish the connection with Oracle_Server_SID.
8. Create a Local Net Service Name and test the connection.
9. Edit the PATH variable to have oracle_client home and remove server home from PATH, as shown in the following example:
   C:\app\Administrator\product\<oracle_database_version>\client_64bit\bin
10. Execute pdm_configure with ORACLE_HOME set to client_home, as shown in the following example:
    C:\app\Administrator\product\<oracle_database_version>\client_64bit\

    NOTE
    Before you start installing CA Service Desk Manager 17.4, for Oracle Database, ensure that the Oracle PATH environment variable refers to 32-bit oracle client first, and then, to 64-bit Oracle client. To change the sequence, navigate to Environment Variables, System Variables, Path, Edit.
11. Install CA SDM 17.4

For Linux

Perform the following steps:

1. Install the Oracle 32-bit client with Administration client.
2. Establish the connection with Oracle_Server_SID.
3. Create a Local Net Service Name and test the connection.
4. Export TWO_TASK variable.
5. Edit the ORACLE_HOME and PATH variables to have oracle_client home and remove server home, as shown in the following example:

   ```
   ORACLE_HOME= /opt/app/product/<oracle_database_version>/client_32bit
   PATH= /opt/app/product/<oracle_database_version>/client_32bit/bin
   ```
6. Install the Oracle 64-bit client with Administration client.
7. Establish the connection with Oracle_Server_SID.
8. Create a Local Net Service Name and test the connection.
9. Export TWO_TASK variable.

10. Edit the ORACLE_HOME and PATH variables to have oracle_client home and remove server home, as shown in the following example:

```
ORACLE_HOME= /opt/app/product/<oracle_database_version>/client_64bit
PATH= /opt/app/product/<oracle_database_version>/client_64bit/bin
```

NOTE
Before you start installing CA Service Desk Manager 17.4, for Oracle Database, ensure that the Oracle PATH environment variable refers to 32-bit oracle client first, and then to 64-bit Oracle client in the terminal.

11. Install CA SDM 17.4

Install and Configure Apache Tomcat

CA SDM provides out-of-the-box Apache Tomcat. This document provides steps to upgrade the Apache Tomcat version 8.x.xx or 9.x.xx to Apache Tomcat 9.x.xx.

First perform these steps in your Pre-prod environment; after successful internal validation, perform the same steps in production.

These are all the generic steps for Tomcat upgrade, you may have to perform some more additional steps based on the selected Tomcat version.

Follow these steps:

1. Shut down CA SDM Daemon Service or CA SDM Proctor Service, or both on the relevant CA SDM server (primary or secondary, or both).
2. Download Tomcat 9.x.xx (apache-tomcat-9.x.xx.zip).

NOTE
CA SDM supports service packs and point releases of Operating Systems, Databases, Web Servers, Web Browsers, Java, Servlets, etc., not necessarily noted on the certification matrix as long as the problem reported is reproducible with versions that are listed on the matrix. We reserve the right to refuse support of new point releases should the reported problem require a major SDM rework or redesign in order to function properly. Both Technical Support and Sustaining Engineering will do their best to resolve any issues that occur in a timely manner. If the resolution to the problem is determined to be outside the realm of their support responsibilities, they may ask that you escalate your request for certification to your local account team.

3. Unzip apache-tomcat-9.x.xx.zip and place the files at <drive>:\<install_directory>\CA\SC\tomcat\9.x.xx

NOTE
After unzipping, ensure that <drive>:\<install_directory>\CA\SC\tomcat\9.x.xx contains conf, bin, webapps, and other directories.

4. Make a backup copy of the NX.ENV located at the CA SDM install directory (NX_ROOT).
5. Modify NX.ENV located at the CA SDM install directory (NX_ROOT) as follows:

```
@NX_TOMCAT_INSTALL_DIR=C:\Program Files\CA\SC\tomcat\9.x.xx
```

6. Backup NX_ROOT\bopcfg\www\CATALINA_BASE\conf and copy all the files in the <drive>:\<install_directory>\CA\SC\tomcat\9.x.xx\conf
to
NX_ROOT\bopcfg\www\CATALINA_BASE\conf
7. Verify/update that the server.xml file located in directory NX_ROOT\bopcfg\www\CATALINA_BASE\conf has the same startup (connector) and shutdown ports as defined in the original server.xml file.

NOTE
If any other changes were made to the original server.xml file, then they would need to be made to the new server.xml file.

8. Edit the file NX_ROOT\bopcfg\www\CATALINA_BASE\conf\catalina.properties with a text editor.
Look for the following entry in the file:

shared.loader=
Modify it as follows:

```
shared.loader=${catalina.base}/shared/lib/*.jar
```

Save the file.

9. If CA Visualizer is installed, back up the directory NX_ROOT\bopcfg\www\CATALINA_BASE_VIZ\conf and copy all the files in the <drive>:\<install_directory>\CA\SC\tomcat\9.x.xx\conf
to
NX_ROOT\bopcfg\www\CATALINA_BASE_VIZ\conf
10. Verify that the server.xml file located in the directory NX_ROOT\bopcfg\www\CATALINA_BASE_VIZ\conf has the same startup (connector) and shutdown ports as defined in the original server.xml file.

    NOTE
    If any other changes were made to the original server.xml file, then they would need to be made to the new server.xml file.
11. Edit the file NX_ROOT\bopcfg\www\CATALINA_BASE_VIZ\conf\catalina.properties with a text editor.
Look for the following entry in the file:
shared.loader=
Modify it as follows:

```
shared.loader=${catalina.base}/shared/lib/*.jar
```

Save the file.

    NOTE
    If you encounter any issue related to accessing the latest Visualizer try the followiing steps:

    - Add the following entry in the Visualizer context.xml and Save the file.
      NX_ROOT\bopcfg\www\CATALINA_BASE_VIZ\conf
      <Valve className="org.apache.catalina.valves.rewrite.RewriteValve"/>
12. If CA REST Webservices are enabled, back up the directory NX_ROOT\bopcfg\www\CATALINA_BASE_REST\conf and copy all the files in the <drive>:\<install_directory>\CA\SC\tomcat\9.x.xx\conf
to
NX_ROOT\bopcfg\www\CATALINA_BASE_REST\conf
13. Verify that the server.xml file located in the directory NX_ROOT\bopcfg\www\CATALINA_BASE_REST\conf has the same startup (connector) and shutdown ports as defined in the original server.xml file.

    NOTE

    - If any other changes were made to the original server.xml file, then they would need to be made to the new server.xml file.
    - If you encounter any access related issue try the following steps:
      Update the file catalina.properties under NX_ROOT\bopcfg\www\CATALINA_BASE_REST\conf
      \catalina.properties with the following value:
      common.loader="${catalina.base}/lib","${catalina.base}/lib/*.jar","${catalina.home}/lib","${catalina.home}/lib/*.jar",${catalina.base}/../../../java/lib/CXF/*.jar
      tomcat.util.scan.StandardJarScanFilter.jarsToSkip=*.jar,\
14. Modify config.properties located in directory NX_ROOT\site as follows:
web.tomcat_home=<drive>\:\\<install_directory>\\CA\\SharedComponents\\tomcat\\9.x.xx
web.tomcat.version=9.x.xx \
15. If CA Federated Search components are installed, perform Steps 6 and 7 for CATALINA_BASE_FS folder.
16. Restart CA SDM service.

Tomcat Upgrade for CA Service Catalog using ANT

upgrading tomcat using ant cmd

CA SDM provides out-of-the-box Apache Tomcat. This document provides steps to upgrade the Apache Tomcat version 9.0.71 to Apache Tomcat 9.0.x.

These are all the generic steps for Tomcat upgrade, you may have to perform some more additional steps based on the selected Tomcat version.

1. Download Tomcat 9.0.x version from here.
   a. Choose the version to be upgrade (For Ex if we want to upgrade 9.0.80)
   b. Click on the version link (For Ex : v9.0.80/)
   c. Go to **bin** folder.
   d. Download the .zip file (Ex : apache-tomcat-9.0.80-windows-x64.zip)Note : Catalog product supports 64 bit tomcat prefer to download only 64 bit .zip file.
   e. Once downloaded, follow the steps metioned below.
      a. Launch cmd from %USM_HOME% location.
      b. Type command
         ```
         ant upgrade-tomcat
         ```
      c. Follow the instructions to upgrade Tomcat version.
2. Start the CA Service Catalog services again.

Upgrade Tomcat and JRE (CASM Admin)

The following procedures help you upgrade Tomcat and JRE for CA Service Management Administration.

> **NOTE**
> Perform the same steps when CA Service Catalog is installed on multiple nodes.

Tomcat

Download the Tomcat, which is the later version of 9.0.71 (32-bit version) and place it in the server machine where we are upgrading the Tomcat.

Follow the steps once the downloaded Tomcat.zip is available:

1. Open the *% CASMAdmin_HOME %\scripts\utilities\tomcat* folder.
2. Stop the CA Service Management Admin service.
3. Run the *Tomcat_Upgrade.bat* in Administrator mode.
4. Enter the location of tomcat.zip when prompted.
   Upgrade starts. Wait for the process to upgrade the Tomcat version.

JRE

Download the JRE, which is the later version of 1.8.0_362 (32-bit version) and place it in the server machine where we are upgrading the JRE.

Follow the steps once the downloaded JRE.zip is available:

1. Open the *% CASMAdmin_HOME %\scripts\utilities\jre* folder.
2. Stop the CA Service Management Admin service.
3. Run the *Jre_Upgrade.bat* in Administrator mode.
4. Enter the location of JRE.zip when prompted.
   Upgrade starts. Wait for the process to upgrade the JRE version.

> **NOTE**
> *%CASMAdmin_HOME%* refers the installed location of CA Service Management Administration.
> (**Example:** C:\Program Files\CA\Service Management Administration\).

Upgrade Tomcat and JRE (CAF ITAM)

The following procedures help you upgrade Tomcat and JRE for CAF CA Asset Portfolio Manager.

> **NOTE**
> Perform the same steps when CA Asset Portfolio Manager is installed on multiple nodes.

Tomcat

Download the Tomcat, which is the later version of 9.0.71 (32-bit version) and place it in the server machine where we are upgrading the Tomcat.

Follow the steps once the downloaded Tomcat.zip is available:

1. Open the *%CAF_ITAM_HOME%\scripts\utilities\tomcat* folder.
2. Stop the CA CASM service.
3. Run the *Tomcat_Upgrade.bat* in Administrator mode.
4. Enter the location of tomcat.zip when prompted.
   Upgrade starts. Wait for the process to upgrade the Tomcat version.

JRE

Download the JRE, which is the later version of 1.8.0_362 (32-bit version) and place it in the server machine where we are upgrading the JRE.

Follow the steps once the downloaded JRE.zip is available:

1. Open the *%CAF_ITAM_HOME%\scripts\utilities\jre* folder.
2. Stop the CA CASM service.
3. Run the *Jre_Upgrade.bat* in Administrator mode.
4. Enter the location of JRE.zip when prompted.
   Upgrade starts. Wait for the process to upgrade the JRE version.

> **NOTE**
> *%CAF_ITAM_HOME%*refers the installed location of CA Asset Portfolio Manger.
> (**Example:** *C:\Program Files (x86)\CA\SharedComponents\CASM*).

Perform the CA SDM Post-Upgrade Tasks

After a successful upgrade to CA SDM 17.4, perform the following post-upgrade tasks:

- Customization Upgrade Utility
- Configure CA SDM Servers after Upgrade
- Manually added NX variables need to be merged
- Configure your Tickets
- Enable Services in xFlow Interface
- Modify the New Functional Access Areas Post-Migration
- Review the CA SDM Email Mailbox Options
- Upgrade Knowledge Management
- Add Insights Functional Access to Existing Custom Roles
- Set Active Status for Default Area

After successful completion of the post-upgrade task, perform additional post-installation tasks.

Customization Upgrade Utility

Customization Upgrade for Patch Migration

CA Service Desk Manager allows customers to extend the design forms as required. The Customization Upgrade utility facilitates smooth upgrade and merging of the customized forms into the latest CA SDM version.

WARNING
If you are upgrading from 17.x, ensure to take backup of **bopcfg** (*$NX_ROOT/bopcfg/www/htmpl*) file and provide the same file path into

```
PREVIOUS_BASE_FOLDER
```

Prerequisites

- Create a backup of the *site/mods* folder. This folder is used by the utility to identify the customized files for processing.
- Upgrade to CA SDM 17.3.

TIP

It is recommended that you clean up the unused HTMPL and CSS files from the customization folder before merging the content.

How the Utility Works

For the utility to run successfully, the following folder locations are required:

- Out of the box (OOTB) files from the previous release
- Customized files
- Out of the box files from the latest release
- Target location of the merged files

The utility merges the customization in the OOTB files of the latest release, and then places the merged files in the target location. A log file is also generated, which indicates the status of each file:

- **Merged**: Indicates that the files merged correctly. These files can be placed in the *sites/mods* folder by the administrator.
- **Unable to merge**: The files that fail to merge correctly are created with '**_partial_merge**' in the file name. You can use these files to merge the customization manually.
- **Copied**: When the previous or the current releases do not have a corresponding OOTB file, the utility copies the file in the target location.

WARNING

We recommend that you use the utility to merge only the HTMPL and CSS files. We currently do not provide support to merge the failed files and such files must be merged manually.

Execute the Utility

Follow these steps:

1. Login to CA SDM as an administrator. Navigate to the location that contains the utility.

   WARNING

   When using **** as the file separator to define folder locations on Windows, use another **** as the escape character.
2. *Navigate to **$NX_ROOT/site/cfg/ folder, edit the cu_env.properties** file and define the following properties:*
   - **DO_SDM_CONV**

Set the value to 1, if the CA SDM version that is being upgraded is 12.5 or earlier.
Retain the default value, 0, if the CA SDM version that is being 12.6 and above.

- **PREVIOUS_BASE_FOLDER**
  Define the location of the OOTB files from the previous release.
  For example, *PREVIOUS_BASE_FOLDER=$NX_ROOT/migrate/tor17_2/prer17_2/bopcfg/www/htmpl*
  In case of a patch upgrade, point the variable to backup folder taken for the patch migration.
- **CUSTOMIZED_FOLDER**
  Define the location of the customized files that you saved as the backup.
  For example, *CUSTOMIZED_FOLDER=<Site-mods-backup-path>/www/htmpl*
  If you failed to create a backup of the *site/mods* folder, you can define the CUSTOMIZED_FOLDER path as $NX_ROOT/site/mods/www/htmpl folder. However, the migration process renames some HTMPL files to *_incompatible_for_r<major_release>_<minor_release>* (example, **_incompatible_r17_2**). Rename the files back to the original name before you execute the utility.
- **NEW_BASE_FOLDER**
  Define the location of the OOTB files from the latest release.
  For example, *NEW_BASE_FOLDER=$NX_ROOT/bopcfg/www/htmpl*

  WARNING

  This utility reads the folder recursively to find corresponding files to compare and merge, like any other compare tool. Therefore, the relative path to an htmpl file from PREVIOUS_BASE_FOLDER, CUSTOMIZED_FOLDER, and NEW_BASE_FOLDER folders must match.
  For example, if you want to merge *about.htmpl* file that is located in *$CUSTOMIZED_FOLDER/web/analyst*; ensure that the file exists in *$PREVIOUS_BASE_FOLDER/web/analyst* and *$NEW_BASE_FOLDER/web/analyst* folder locations.

  If the relative folder paths mismatch, the utility might throw errors.
- **MERGED_FOLDER**
  Define the location of the target folder that is used by the utility to create the merged files.

  TIP

  Do not define the MERGED_FOLDER path to any existing CA SDM folder, as the utility might rename the folder.
- **TEMP_LOC**
  Temporary location to contain the processing files.
  For example, TEMP_LOC=$NX_ROOT/temp

  NOTE

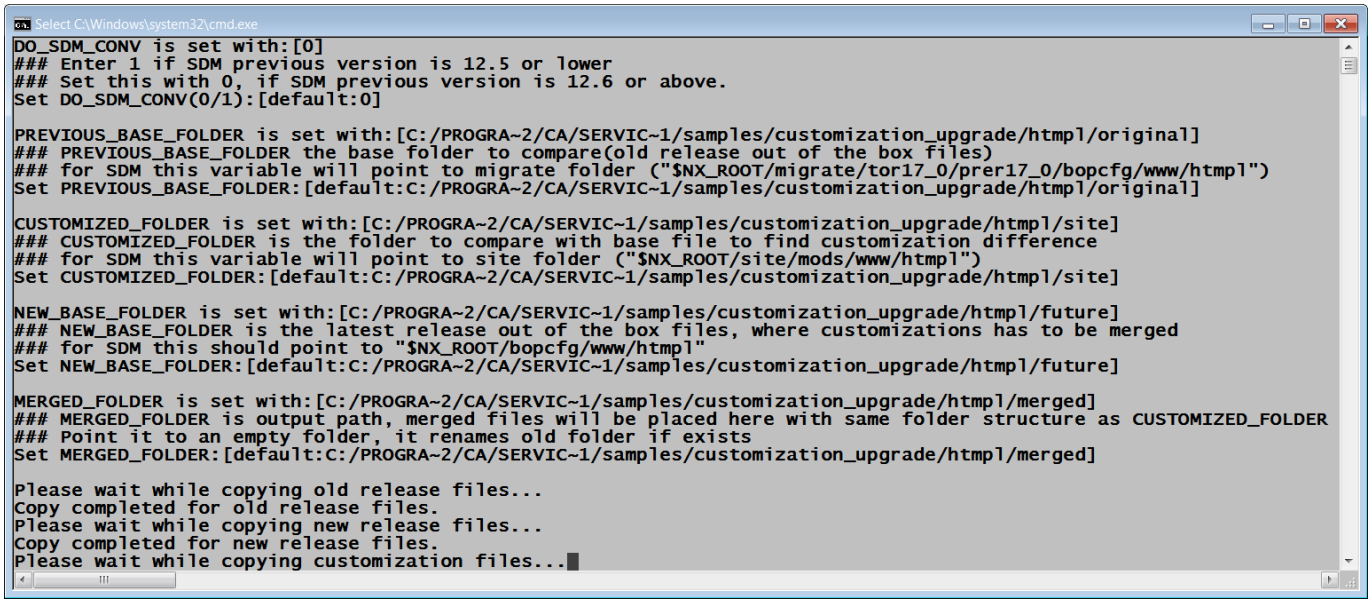
  Ensure enough space to contain the files in the PREVIOUS_BASE_FOLDER, CUSTOMIZED_FOLDER, and NEW_BASE_FOLDER locations.
- **LOG_LOC**
  Location that contains the log and differences files for debugging.
  For example, LOG_LOC=$NX_ROOT/log/UpgradeFactoryLogs

3. *Navigate to $NX_ROOT/bin folder and execute the following file to start the merge process:*
   - *Windows: customization_upgrade.bat*
   - Unix/Linux: *customization_upgrade.sh*

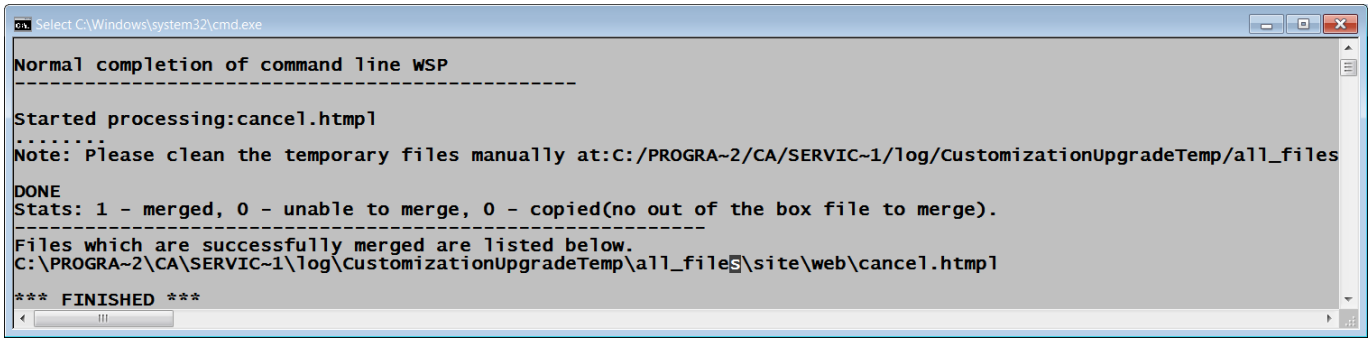
     NOTE

     When you execute the script, it prompts the variables that you defined in *cu_env.properties*, which you can update, if necessary.

   The merge process runs as indicated in this image:



After the merge process completes successfully, you can view the result as indicated in the image:



4. Verify the ***$NX_ROOT/log/customization_merging.log*** log file to ensure successful merging of customization.
   To manage the log level, edit the ***$NX_ROOT/site/cfg/log4j.properties*** *file.*
5. Manually merge the files that the utility failed to merge.
6. Copy the files from the merged folder location to ***$NX_ROOT/site/mods*** folder.
7. Execute the following command on the CA SDM server to clear the cache:
   ```
   pdm_webcache -bH
   ```

Understand the Files Created in the Output Folder

- Successfully merged files are placed in the **MERGED_FOLDER** with the same file extension. For example, <file_name>.htmpl; <file_name>.css
- Partially merged files are created with ***_partial_merge*** in the filename. For example, <file_name>.htmpl_partial_merge
  You can analyze the partially merged files and can perform the manual merge.
- Customizations that do not have a corresponding OOTB file in the previous release are considered as new and copied in the output folder with the same name.
- *Customizations that do not have a corresponding OOTB file in the latest release are considered as unmerged and copied in the output folder with the* ***_unmerged*** *in the file name.*
- The .htmpl file format of the copied files can change based on the value defined in DO_SDM_CONV property in the ***$NX_ROOT/site/cfg/cu_env.properties*** file.
- The log file contains the status of each file processed.

Manually Verify the Merged Files

You can use any diff tools to verify that the merging process was successful.

Verification method 1: Compare the differences

- Compare the files in the **CUSTOMIZED_FOLDER** and the **MERGED_FOLDER**, and
- Compare the files in the **NEW_BASE_FOLDER** and **PREVIOUS_BASE_FOLDER**.

In both the comparisons the differences must match, which indicates that the merge was successful. If not, then, manually merge the files.

Verification method 2: Compare the Customization

- Compare the **PREVIOUS_BASE_FOLDER** and **CUSTOMIZED_FOLDER** to identify the customization
- Compare the NEW_BASE_FOLDER and the MERGED_FOLDER to identify the customization

In both the comparisons, the customization must match. If not, then, manually merge the files.

SITEMODS.JS File

Lines of code added to the ***sitemods.js*** file of the previous version, called from an HTMPL page, must be merged into the current ***sitemods.js*** file before the code works.

Exceptions to Merging Customization

The merge utility fails when your customization falls under the following scenarios:

- When the number of customizations in the file is large.
- When the customized content location has changed only slightly, that is, immediately before or after the customized content, the utility fails to find the reference location and eventually fails.
- When the customized file is split into multiple Out-of-the-Box (OOTB) HTMPL and CSS files.

Configure CA SDM Servers after Upgrade

Inactive Servers Other Than Primary or Secondary

After you upgrade, all the servers that are not entered as primary server in the **usp_servers** table are converted to secondary server. Before upgrading, if you have added any server in the **usp_servers** table, which are not secondary or primary server, inactive them after you upgrade. You can inactive servers from the primary server Web User Interface. The local host name is stored in the **usp_servers** table in **local_host** column.

Clear the Webengine and Browser Cache

> **NOTE**
> After you upgrade CA SDM, run the **pdm_webcache** utility to clear the cache for the web engine and browser.

```
pdm_webcache -b -H
```

- **-b**
  Warns the user to clear their browser cache.
- **-H**
  Clears the webengine cache.

Manually added NX variables need to be merged

Manually updated NX variables are lost when a GA release or RU patch delivers a new `NX.env_nt.tpl` file (located in `NX_ROOT/pdmconf`). This necessitates an additional manual step to merge the custom values from the backup file of `NX.env_nt.tpl` created by `pdm_configure`.

- Manually merge the required changes into the NX.env_nt.tpl file located in NX_ROOT\pdmconf from the backup file at NX_ROOT\REPLACED\GOL-78_EN_US.OLD\pdmconf\NX.env_nt.tpl.
- Restart the SDM services.
- **For Conventional Mode**: Restart the SDM services on all secondary servers.
- **For Advanced Availability Mode**: Restart the SDM services on all standby servers and application servers.

Configure your Tickets

How to Add the Incident Priority Field to Incidents

This article describes how to configure your CA SDM Tickets

The *incident priority* is the sum of the Urgency and Impact values. The incident priority is only for the incident ticket type. The Incident Priority value appears on the incidents after you install the **use_incident_priority** option and add it to the incident Detail page form with Web Screen Painter.

To add the Incident Priority field to incident, do the following:

1. Install the **use_incident_priority** option from the **Options Manager**, **Request Mgr**.
2. Use Web Screen Painter to add the Incident Priority field to the Incident Detail pages.
   The Incident Priority value appears on saved Incident Detail page when the **use_incident_priority** option is installed. When the **use_incident_priority** option is not installed, the Incident Priority value is zero.

   NOTE
   The use_incident_priority option only manages the Incident Priority value. This option is not related to priority calculation.

Add the Urgency Field to Employee Tickets

By default the **Urgency** field does not appear on Employee incidents or requests. However, you can add the **Urgency** field by using the **urgency_on_employee** option.

NOTE
When you uninstall the **urgency_on_employee** option and disable priority calculation, the Priority field appears on the Request and Incident Detail pages for self-service Users.

To add the **Urgency** field to Employee tickets, install the **urgency_on_employee** option from the **Options Manager**, **Request Mgr**. The Urgency field appears on Employee incidents or requests. Self-service users can override the value on the incident.

Activate Status Transitions

After the upgrade, all predefined status transitions are inactive, so status transitions are not in effect. You can activate and modify these status transitions to accommodate the ticket status transition flow you want.

NOTE
All modified status code descriptions that appear on ticket forms are retained during the upgrade process.

To activate a status transition, perform the following steps:

1. On the Administration tab, expand the **Service Desk** node, and select one of the following ticket types:
   – **Change Orders**

    - Change Order Transitions
  - **Issues**
    - Issue Transitions
  - **Requests/Incidents/Problems**
    - Incident Transitions
    - Problem Transitions
    - Request Transitions

  The Transitions List appears.
2. Select **Show Filter** on the Transitions List page.
   The top portion of the page reveals additional search fields.
3. Select **Inactive** in the Record Status field and click **Search**.
   The Transitions List at the bottom of the page displays all inactive transitions.
4. Open the transition for editing.
5. Select **Active in the Record Status** drop-down list.
6. Click **Save** and close the window .
7. Click **Search**.
   The Transition List displays the active transition.

Enable Priority Calculation

Priority calculation is a set of values that automatically set Priority, Urgency, and Impact values on problems and incidents. For new CA SDM installations, the default priority calculation is enabled for problem and incident ticket types by default. However, if you are upgrading from a previous release, the default priority calculation is inactive.

If you create and activate a different priority calculation, the ticket values reflect the settings in the active priority calculation that is associated with an incident or a problem. When no priority calculation is active, users can manually set the Priority and other values on tickets.

> **NOTE**
>
> The modified forms on the Employee and Customer interface operate in the same manner as the previous versions. The Self-Service users can directly change the Priority regardless of the settings in Priority calculation.

To enable priority calculation after the migration, perform the following steps:

1. On the Administration tab, navigate to **Service Desk**, **Request/Incidents/Problems**, **Priority Calculation**.
2. Right-click the default priority calculation or another priority calculation and select **Edit** from the short-cut menu.
   The Update Priority Calculation page appears.
3. Set the status to **Active**.
4. Select one or more of the following ticket types:
   - **Incidents**
     Enables this priority calculation to manage incident tickets. Only one active priority calculation can manage incidents.
   - **Problems**
     Enables this priority calculation to manage problem tickets. Only one active priority calculation can manage problems.
5. Click **Save**.
   The values on the default priority calculation apply to new tickets unless you activated another priority calculation. On new tickets that use a priority calculation, the Priority field is read-only.

Activate Transition Types

By default, all predefined transition types delivered with the product are inactive, so status transition buttons are not in effect. You can activate and modify these transition types to accommodate the status transition flow you want.

To activate a transition type, perform the following steps:

1. Select **Show Filter** on the Transition Type List page.
   The top portion of the page reveals additional search fields.
2. Select **Inactive** in the Record Status field and click **Search**.
   The Transition Type List displays all inactive transition types.
3. Right-click the title of the transition type and select **Edit** from the menu.
4. Select **Active** in the Record Status drop-down list.
5. Click **Save** and close the window.
6. Click **Search**. The Transition Type List displays the active transition type.

Enable Services in xFlow Interface

Enable the xFlow Analyst Interface services after successful installation of the xFlow Interface.

> **NOTE**
>
> Failure to enable services, results in interruption of all the daemons in the xFlow Interface and does not allow the latest information to be displayed.

Perform the following steps to enable services in xFlow Interface:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **Options Manager**, and **General**.
3. From the list of options, select **xFlow_installed**.
4. Click **Edit** and set the **Optional Value** to **Yes.**
5. Click **Save**.
6. Restart the *CA Service Desk Manager* services.

Modify the New Functional Access Areas Post-Migration

This article describes how to modify the functional access areas after you have upgraded to CA SDM 17.3.

A *functional access area* is a group of objects that let you restrict user access. Previous versions of CA SDM included eight fixed functional access groups to restrict access to code components.

During migration, the functional access groups migrate to new functional access areas for each role. Migration automatically handles the Majic changes, the default reference data, and role mapping to the new functional access areas.

After migration, consider the following actions:

- Review how the objects map to existing and new functional access areas and role permissions to each area. Use Web Screen Painter to verify the functional access areas.
- Use CA SDM to remap or change permissions. Verify that users have the proper access to features and objects.

The following table maps Functional access areas to code components:

Functional access area	Code Component	New
Administration	admin	No
Incident/Problem/Request	call_mgr	No

Change Order	change_mgr	No
Inventory	inventory	No
Issue	issue_mgr	No
Knowledge Document	kd	No
Notification	notify	No
Reference	reference	No
Security	security	No
Announcement	announcement	Yes
Incident/Problem/Request Reference	call_mgr_reference	Yes
Incident/Problem/Request Template	call_mgr_template	Yes
Change Order Template	change_mgr_template	Yes
Change Order Reference	change_reference	Yes
Configuration Item	ci	Yes
Configuration Item Common	ci_common_ro	Yes
Configuration Item Reference	ci_reference	Yes
Contact	contact	Yes
Group	group	Yes
Issue Template	issue_mgr_template	Yes
Issue Reference	issue_reference	Yes
Location	location	Yes
Notification Reference	notification_reference	Yes
Organization	organization	Yes
Prioritization	prioritization	Yes
Service Level	service_level	Yes
Site	site	Yes
Stored Query	stored_queries	Yes
Survey	survey	Yes
Tenant Admin	tenant_admin	Yes
Timezone	timezone	Yes
Workflow Reference	workflow_reference	Yes
Workshift	workshifts	Yes

Post-Migration Access Level Changes

After migration, you can verify functional access levels for every role. Because the objects moved to another functional access areas, the user could have access to some screens in some situations that they were denied previously. They can also be denied access to forms to which they had access previously. Both situations can occur when a new functional access area manages permissions for two of the original functional access areas.

Edit Access Types

When you upgrade from CA SDM, the upgrade process automatically creates roles for all access types and correctly assigns access and permissions to the roles. If you want to take advantage of the new roles, you can create roles for access types.

To create roles for access types

1. Log in to the web interface as a user with the ability to access the Administration tab.
2. Click the Administration tab.
3. In the tree on the left, select Security and Role Management, Access Types.
   All available access types display.
4. Click an available access type.
5. Click the Roles tab.
6. Select a new role for the access type and click Update Roles.
   The new role is associated with the access type.

   NOTE
   You can also create a custom role and assign it to the access type.

Adjust Access Types

If you modified access types and data partitions in the previous release of CA SDM, you may have a problem with the Knowledge Management data partitions settings after upgrading. These customizations can cause a problem with the permission groups settings on categories and documents. For example, a user has access to restricted information.

NOTE
Even if you recreated a data partition or an access type after deleting it, verify your access type and data partition settings after the upgrade.

To adjust access types

1. Click the **Administration** tab from the classic CA SDM Interface.
2. Click **Security and Role Management**, **Role Management**, **Role List**.
3. Complete the following steps for each role:
   a. Right-click the role and select **Edit**.
   b. Review the Data Partition Name field under the **Authorization** tab.
      If this field is empty, there is no data partition associated with the selected access type, so the user has no restrictions and can access any document or category in the product, even if you set up permission groups. This action can be appropriate for administrators, but not for all roles. If there is no data partition associated with the role, you can create or modify one.

Review the CA SDM Email Mailbox Options

CA SDM replaces the Options Manager, Email inbound email options with a Mailbox (usp_mailbox table) that supplies corresponding options. The Email outbound email options are still present under the Options Manager. When you upgrade, CA SDM uses your existing email settings to configure a mailbox, instead of the default Mailbox settings that are supplied with the release. Each email option, except **EMAIL_ATTACHMENT_DIR** (which is no longer needed), is mapped to an option in the **usp_mailbox** table. Any option that is not set, is set as null in the table.

The following table lists options that are removed from the Email options, provided in the *usp_mailbox* table, and indicates their labels in the Mailbox Detail page:

Email Option	usp_mailbox Option	De
EMAIL_ALLOW_ANONYMOUS	allow_anonymous	All
EMAIL_ATTACHMENT_DIR	N/A	N/A **No** ma EM
EMAIL_ATTACHMENT_ REPOSITORY	attmnt_repository	Att

EMAIL_FORCE_ATTACHMENT_ SPLITOUT	split_out_attachment	Fo
EMAIL_IS_ATTACHMENT	attach_email	Att
EMAIL_SAVE_UNKNOWN_ EMAILS	save_unknown_emails	Sa
MAIL_EATER_IMAP_HOST_PORT	host_port	Po
MAILEATER_CHECK_MAIL_ INTERVAL	check_interval	Ch
MAILEATER_HOST	host_name	Ho
MAILEATER_LOGIN_PASSWORD	password	Pa
MAILEATER_LOGIN_USERID	userid	Us
MAILEATER_POP3_HOST_PORT	host_port	Po
MAILEATER_SECURITY_LEVEL	security_lvl	Se
MAILEATER_SERVER_TYPE	email_type	Em

WARNING
The Attachment Directory setting is deprecated after you upgrade, so you *must* specify an Attachment Repository before you continue polling the mailboxes.

Maileater.cfg Considerations

Information from your upgraded system that was included in the ***maileater.cfg*** file maps to the ***usp_mailbox_rule*** table after you upgrade. Consider the following information:

- The '**-i**' at the beginning of the line denotes case insensitivity and maps to the **filter_ignore_case** field.
- The search filter "**Subject: *…**" previously denoted a regular expression on which to filter. The "**Subject:**" is removed and is replaced with a "**^**" symbol, and the remaining value maps to the ***filter_string*** field. The **Filter_type** is set to "**Subject Contains**" type.
- The "**TEXT_API xxx**" denotes the object that is processed for the rule. The string "**TEXT_API** " is removed, and the remainder maps to the ***action_object*** field. The ***action_operation*** field is set to **Create/Update** Object.
- The reply to the user typically contains "**PDM_MAIL …**". If you have "**PDM_MAIL**" set, set **reply_method** to 1800 or else leave it set to null.
- If the **- s** parameter is set, remove the subject field from the text and set the ***reply_subject*** with this value.
- The functionality maintains the order of entries. A sequence number starting at 100 and incremented by 100 is set for each valid row.

The other fields in the ***usp_mailbox_rule*** are set as follows:

Field	Value
mailbox	Default
action_write_to_log	0
action_log_prefix	null
delete_flag	0
description	Migrated from pdm_maileater.cfg file
reply_failure_html	<leave blank to inherit default action>
reply_failure_text	<leave blank to inherit default action>
reply_success_html	<leave blank to inherit default action>
reply_success_text	<leave blank to inherit default action>
text_api_defaults	null

text_api_ignore_incoming	null
action_subject_handling	null
last_mod_dt	null
last_mod_by	null
inclusion_list	"*"
email_address_per_hour	-1
exclusion_list	null
log_policy_violation	1

Upgrade Knowledge Management

How to Upgrade Knowledge Management

When you upgrade, CA Service Management Installer automatically upgrades your Knowledge Management environment.

After successful upgrade, perform the following steps to upgrade your knowledge management component:

1. Map links created in a resolution of a document to the database to locate broken links.

   NOTE
   You use the default *Flag broken links* policy to locate broken links.
2. On the **Administration** tab, browse to **Knowledge**, **Automated Policies**, **Policies**, **Scheduling**.
   The Scheduling page appears.
3. Select the **Run Calculation** check box in the **Last Updated** field.
4. Enter a date in the Schedule text box or click the **Calendar** icon to select a date.
5. Select the time interval to perform the calculation and run the policies.
6. Click **Save**.
   The policies are processed at the specified date and time.
7. On the **Administration** tab, browse to **Knowledge**, **Systems**, **General Settings**.
   The General Settings page opens.
8. Complete the following information:
   Path for EBR Index Files
   Defines the location for storing EBR index files. CA SDM creates EBR index files when you save and publish any knowledge document. Depending on your configuration type, consider the following points while defining the EBR index file path:
   – Conventional: If you have not created a UNC path, CA SDM uses the default path to store EBR index files. If you are using a UNC shared drive, manually copy the **ebr/ebr_ADM**folders from the default location (**$NX_ROOT/site**/) to the UNC shared path.
   – Advanced availability: If you are upgrading to the advanced availability configuration, you must create the UNC shared path and use it to store EBR index files. The UNC credentials are not required for the default path. After you create the UNC path, manually copy the **ebr/ebr_ADM** folders from the default location (**$NX_ROOT/site/**) to the UNC shared path.

     WARNING
     EBR Index Files path & **KEIT Files path** must refer to the same UNC credentials and the path must reside on a same server to support this.

   Default: $NX_ROOT/site/ebr
   – **Path For Knowledge Import/Export Files**

Defines the location for storing KEIT import/export packages during an import/export operation. Depending on your configuration type, consider the following points while defining KEIT file path:

- Conventional: If you have not created a UNC path, CA SDM uses the default path to store the KEIT files. If you are using a UNC shared drive, manually copy the Import/Export package folders from the default location ($NX_ROOT/site/keit) to the UNC shared path.
- Advanced availability: If you are upgrading to the advanced availability configuration, you must create the UNC shared path and use it to store the KEIT files. The UNC credentials are not required for the default path. After you create the UNC path, manually copy the Import/Export package folders from the default location (**$NX_ROOT/site/keit**) to the UNC shared path.

  WARNING
  EBR Index Files path & **KEIT Files path** must refer to the same UNC credentials and the path must reside on a same server to support this.

Default: $NX_ROOT/site/keit

- **UNC Credentials**

You can use this option to create UNC credentials to access the network shared drive to access EBR indexing files and import/export packages. Use the **UNC Credentials** link to create the UNC credentials.

NOTE
UNC paths and UNC credentials are required in case of the advanced availability configuration. Restart the CA SDM service when you change any of the UNC details (UNC paths or UNC credentials).

9. (For Keyword Search implementations) Enter the following command at the command prompt:

```
pdm_k_reindex factory:all
```

WARNING
The *all* variable is case sensitive. If you use factory:ALL, errors appear in the log files.

The Knowledge Management environment is upgraded.

NOTE

After the upgrade, printing Knowledge Documents can result in a large space inserted after the Resolution section of the document. This space is inserted due to an issue with upgrading document templates from a previous release.

WARNING
After you upgrade, Knowledge Management notification data from previous releases of CA SDM uses the latest notification engine. For example, there are default activity notifications and notification rules for object types, such as the Knowledge Report Card.

Add Insights Functional Access to Existing Custom Roles

Add Insights functional access to existing custom roles after successful upgrade to 17.3.

Run the following command from **$NX_ROOT\bin** folder to add Insights Functional Access to custom roles:

```
pdm_perl migrate_custom_role_func_access.pl
```

Set Active Status for Default Area

As an administrator, ensure to set an active status of the default areas.

Perform the following steps to set the active status for the following default areas:

1. Log in to Classic CA SDM user interface as an administrator.
2. Navigate to **Administration** tab, select **xFlow Interface**, **ServicePoint**, and **Default Areas**.

3. In the Name column, click the desired tenant configuration name for which the default areas are set to active.
4. If the area status is **Inactive**, click inactive area name.
5. Click **Edit** in the top right corner.
6. In Record Status, set the drop-down value to **Active**.
7. Click **Save**.

Upgrading to CA Service Catalog 17.4

- CA Service Management 17.4 supports direct upgrades from CA Service Catalog 17.1/17.2/17.3, CA Service Management Common Installer detects installed product version in your environment and automatically upgrades it to CA Service Catalog 17.4.
- If you are on versions prior to 17.1, you must first upgrade to 17.1/17.2/17.3 and then, proceed with the 17.4 upgrade process.

  NOTE

  If you want to use custom port numbers, specify them in the fields provided. This requirement applies even if you have specified custom port numbers CA Service Catalog.

To upgrade successfully, review the following upgrade considerations:

- Review Supportability Matrix.
- You can upgrade directly from CA Service Catalog 17.1/17.2/17.3 using the Traditional and Migration Model Upgrades.

  NOTE

  Beginning with CA Service Catalog Release 12.8, **%USM_HOME%** is no longer a Windows environment variable. Instead, **USM_HOME** is the documentation convention that specifies the local CA Service Catalog installation directory. For 32-bit computers, the default path name is *C:\Program Files\CA\Service Catalog*. For 64-bit computers, the default path name is *C:\Program Files (x86)\CA\Service Catalog* for 32-bit installations or *C:\Program Files\CA\Service Catalog* for 64-bit installations.

- Decide if you need to perform a traditional or migration model upgrade.
- Verify that the systems on which you plan to install CA Service Catalog meet the system requirements. For best performance, we recommend that you do *not* install CA Service Catalog on the same system as your Microsoft SQL Server server or Oracle server.

Complete the CA Service Catalog Post Installation Tasks.

Upgrade Prerequisites and Planning Consideration

Preparing and planning for CA Service Catalog upgrade can be done using either a migration model or a traditional model.

Review and perform the following:

1. Find and list out all systems in your implementation that have one or more CA Service Catalog components installed. Use this list to verify and upgrade all required systems efficiently and in the correct sequence/order.
2. Stop all Microsoft Windows services for your existing CA Service Catalog setups.
3. Review whether you have shared any of the folders under CA Service Catalog and Shared Components. If the folders are shared, unshare them before performing the upgrade and restore the shared folders after the upgrade is finished.
4. Verify that your *user ID* is defined as an administrator with *elevated privileges* on each *applicable Windows system*, as follows:
   - ***user ID***
     Specifies your administrative user ID for a Microsoft Windows system (*not* your user ID for CA Service Catalog).
   - ***elevated privileges***

Specifies additional rights on a Microsoft Windows system that are available to administrative users only. To install or upgrade CA Service Catalog, administrators require the elevated privilege named **Log on as Service**.

– ***applicable Windows computer***
Specifies a Windows system on which you plan to install or upgrade CA Service Catalog and which supports elevated privileges for administrators.

5. Verify that your administrative user ID has **Log on as Service** rights on each applicable Windows computer, as follows:
   a. Click **Start**, **All Programs**, **Administrative Tools**, **Local Security Policy**, **Security Settings**, **Local Policies**, **User Rights Assignment**, **Log on as Service**.
   b. Verify that your administrative user ID is listed.
6. (For Migration Model Upgrade Only) If you have an existing filestore that you want to continue using after the upgrade, perform this step. Otherwise, skip this step.
   Record the computer name and the complete pathname of the filestore. Save them for reference after the upgrade. For traditional upgrades, the filestore is backed in the following location:
   C:\Program Files \CA\Service Catalog\SCA_Backup\previous_release\r17.n\Service Catalog
   r17.*n* depending on the version from which you are upgrading.

Upgrade Prerequisites

If you are upgrading to CA Service Catalog 17.4, ensure to perform the following steps:

1. Take a backup of the MDB before proceeding with the upgrade process.
2. The install script for CA Service Catalog relies on the **spadmin** User ID. Run the following command line query to check if the **spadmin** user is present and active. If you do not have the **spadmin** user, create the user:
   ```
   SELECT *FROM ca_contact WHERE userid = 'spadmin' and inactive = 0
   ```
3. Run the following query to get all the duplicate User IDs. . If the list contains **spadmin** and if this user has duplicates IDs, make the inactive User ID unique
   ```
   SELECT userid, COUNT(userid) AS NumOccurrences FROM ca_contact GROUP BY userid HAVING ( COUNT(userid) >
    1 )
   ```

   NOTE

   If the **spadmin** user ID has duplicates and if any one such ID is active, the upgrade fails.

   NOTE

   The **spadmin** duplicate User ID may create problems while upgrading CA Service Catalog. Other duplicate user IDs will not hamper the upgrade process in general, but may show inconsistent application behavior for users with duplicate user IDs.
4. Update the inactive User IDs to some unique value in the database by executing the update statement directly on the MDB:

   NOTE

   Disable the **usm_trigger_update_all_table** trigger in the ca_contact table before updating the user IDs. Enable **usm_trigger_update_all_table** trigger after updating the user IDs.

   NOTE

   This procedure must be followed to update any duplicate inactive user IDs in the **ca_contact** table.

Avoid Re-importing the Service Catalog Content Pack

To avoid importing the CA Service Catalog Content Pack again while upgrading from an earlier release of CA Service Management, perform the following steps before upgrading to CA Service Catalog:

1. Navigate to where you have downloaded the CA Service Management Common Installer files.

2. Locate the <*Common Installer Image File location*>/*Config* folder and open the **common_config.properties** file. Modify the ***slcm.skip.import.contentpack*** value to **1**.
   Changing the above value to **1** helps to skip re-importing the content pack while upgrading.

   NOTE

   The default value of slcm.skip.import.contentpack is '**0**', which reimports the content pack during an upgrade process.

Prepare the Catalog Database and the DBMS

Prepare the Catalog database and the DBMS for the upgrade. Perform the following steps:

1. Review the contents of the Catalog database:
   - The Management Database (MDB) tables that apply to CA Service Catalog; their names typically begin with **CA_ prefix**. The MDB is the common,shared database for CA Technologies products. The MDB provides the database schema for CA Service Catalog and other CA Technologies products.
   - The CA Service Catalog-specific tables; their names typically begin with a **USM_** prefix. These tables are not included in the MDB.
2. Back up your existing Catalog database on the computer where it currently resides.
3. Optimize the System Change Detail tables.
4. If applicable on your DBMS server, upgrade the Oracle or Microsoft SQL Server database to a supported version. For more information, see Supportability Matrix.
5. Apply all patches and other maintenance for your DBMS (Microsoft SQL Server or Oracle). For more information, see your SQL Server or Oracle documentation.
6. Remove replication in your DBMS.

   WARNING
   When replication has been added, the database tables are locked from any schema changes, even if replication is turned off. As a result, the upgrade cannot run successfully. For information, see your database documentation about how to remove replication and how to add it again, after you have completed the upgrade.
7. Verify that the DBMS client software is installed on the computer from which you plan to run the installation program.
8. Record any customizations that you made to XML, XSL, or other CA Service Catalog files, for example, **server.xml**. The upgrade program installs new versions of these files. After the upgrade, you can make the same customizations to the new versions of these files. Follow the same for any customizations that you make to third-party files such as **Tomcat files**.
9. If you have copied the installation media to a network share, map a drive letter to this share. This mapping is required because a batch file runs as part of the upgrade and batch files cannot run from a UNC path.

Optimize the System Change Detail Tables

The System Change Detail tables are important for auditing and database performance. You can optimize the System Change Detail tables by eliminating redundant or obsolete records. This procedure helps optimize the performance of the database and CA Service Catalog. Perform the following steps:

1. Determine the number of each type of record, as follows:
   a. Enter the following command to determine the number of useful records:
      ```
      select count(*) from usm_system_change_detail where old_value!=new_value
      ```
   b. Enter the following command to determine the number of obsolete records:
      ```
      select count(*) from usm_system_change_detail where old_value=new_value
      ```
   Complete the remaining steps if a significant number of obsolete records exist, especially If more obsolete records than useful records exist. Otherwise, skip the remaining steps and finish preparing the Catalog database and the DBMS.

2. Enter the following command to filter the useful records from **usm_system_change_detail** table into a temporary table:

```
Select * into usm_system_change_detail_temp from usm_system_change_detail where old_value!=new_value
```

3. Drop **usm_system_change_detail** table, as follows:
   a. Back up the indexes and database constraints for the following tables:
      - usm_system_change_detail
      - usm_system_change_detail_ext

      NOTE
      For more information about how to perform the backup, see your DBMS documentation. For a sample script that restores indexes and constraints, see the sample script for Microsoft SQL Server.
   b. Drop the foreign key constraints for the **usm_system_change_detail_ext** table.
   c. Drop the **usm_system_change_detail** table.
4. Rename the **usm_system_change_detail_temp** table to **usm_system_change_detail**.
5. Use the script that you created in Step 3a to restore the indexes and database constraints for the following tables.
   – **usm_system_change_detail**
   – **usm_system_change_detail_ext**

Sample Script

The following sample script applies to Microsoft SQL Server. This script backs up indexes and database constraints. You can use this script as a model for backing up the indexes and database constraints in **Step 3a**.

```
ALTER TABLE [dbo].[usm_system_change_detail]
ADD CONSTRAINT [XPKusm_system_change_detail] PRIMARY KEY CLUSTERED
(
[id] ASC,
[name] ASC
)
WITH
(
PAD_INDEX = OFF,
STATISTICS_NORECOMPUTE = OFF,
SORT_IN_TEMPDB = OFF,
IGNORE_DUP_KEY = OFF,
ONLINE = OFF,
ALLOW_ROW_LOCKS = ON,
ALLOW_PAGE_LOCKS = ON
)
ON [PRIMARY]
ALTER TABLE [dbo].[usm_system_change_detail]
WITH CHECK ADD CONSTRAINT [$usm_s_r00002c5700000000] FOREIGN KEY
(
[id]
)
REFERENCES [dbo].[usm_system_change] ([id])
ALTER TABLE [dbo].[usm_system_change_detail]
CHECK CONSTRAINT [$usm_s_r00002c5700000000]
ALTER TABLE [dbo].[usm_system_change_detail_ext]
WITH CHECK ADD CONSTRAINT [$usm_s_r00002c6100000000] FOREIGN KEY
(
[id],
[name]
```

```
)
REFERENCES [dbo].[usm_system_change_detail] ([id], [name])
ALTER TABLE [dbo].[usm_system_change_detail_ext]
CHECK CONSTRAINT [$usm_s_r00002c6100000000]
```

Back Up CA EEM Data

Backing up CA EEM data is a required task whether you upgrade using a migration model or a traditional model. For more information, see CA EEM Documentation.

WARNING
If CA EEM is configured with high availability, perform these steps on the Primary CA EEM server.

Follow these steps:

1. Back up your existing CA EEM on the computer where it currently resides, as follows:
   a. If other CA Service Catalog components exist on the same computer, shut down their Windows services.
   b. Log in to CA EEM; on the login page, select your current CA Service Catalog release as the application instance.
2. Click **Configure**, **EEM Server**, **Export Application**.
3. Perform the following steps:
   a. Select all check boxes under Object List, and verify that **Override the Max Search Size** is *not* selected.
   If CA EEM is configured with External Directory, clear these options: **Global Users**, **Global User Groups**, **Global Folders**, and **Global Settings**.
   b. Click **Export**.
   c. On the File Download prompt, select a location to save the ServiceCatalog.xml file. Record the location where you saved the file, for future reference.
4. Log out of CA EEM.
5. Back up the CA EEM data on the Catalog Component computer in your test system, as follows.

   NOTE

   If you have multiple Catalog Component computers, perform this step on the *first* (formerly *primary*) Catalog Component computer *only*. You run this command once, regardless of how many Catalog Component computers your system has.

   a. Open the command prompt and run the following command:

   ```
   ant backup-eem-app
   ```

   This action generates a file named **eem-backup.xml** in the **USM_HOME** directory.
   USM_HOME is the documentation convention that specifies the local CA Service Catalog installation directory. For 32-bit computers, the default path name is **C:\Program Files\CA\Service Catalog**. For 64-bit computers, the default path name is **C:\Program Files (x86)\CA\Service Catalog** for 32-bit installations or **C:\Program Files\CA\Service Catalog for 64-bit installations**.You have now backed up CA EEM data.

Traditional and Migration Model Upgrades

- You can upgrade directly to the current release of CA Service Catalog using *either* a traditional model or migration model upgrade.
- You *cannot* upgrade from one DBMS to another. For example, you cannot upgrade a CA Service Catalog installation running SQL Server to the current release of CA Service Catalog installation running Oracle.
- The installer upgrades CA Service Catalog in its existing installation directory, except in the following case: If you perform a migration model upgrade to the current release of CA Service Catalog on a 64-bit computer, CA Service Catalog is upgraded to a 64-bit application, and its default installation directory is updated accordingly.

Traditional and Migration Model Upgrades

You can upgrade CA Service Catalog to the current release using *either* a migration model or a traditional model.

- In a traditional model, you upgrade on existing CA Service Catalog computers.
- In a migration model, you install the current release of CA Service Catalog on *new* computers. When you install CA Service Catalog on the new computers, you use the Catalog database and CA EEM computers.

The migration model works as follows:

Consider this sample scenario: Your setup has Catalog Component and the Catalog database on existing Computer 1. Catalog Component and Service Accounting Component are on existing Computer 2, and CA Process Automation on existing Computer 3. Perform the following actions:

- Verify that the current release of CA Service Catalog Release supports the DBMS version on existing Computer 1.
- Verify that the DBMS client software is installed on the new computers on which you plan to run the upgrade.
- Install Catalog Component on new Computer 4, pointing to the existing Catalog database on existing Computer 1. Similarly, install Catalog Component and Service Accounting Component on new Computer 5. When you are prompted for information about the Catalog database or CA EEM, use the *existing* Catalog database and CA EEM specifications, such as the CA EEM host name and application name.
- Utilize CA Process Automation on Computer 3.

  NOTE

  Do *not* upgrade or delete CA Service Catalog on existing Computer 1 until all other computers are upgraded to the current release of CA Service Catalog and are running successfully.

Migrate the CA Service Catalog Data

- T*est system* refers to all computers *from* which you are migrating
- P*roduction system* refers to all computers *to* which you migrating

Thus, these terms describe a typical scenario where you migrate from a test system to a production system. If you are migrating for other reasons, the test system is the *source* system. The production system is the *target* or *destination* system.

This documentation assumes a multi-computer system architecture. For example, one computer for the DBMS, one computer for CA Service Catalog and Service Accounting Component and one computer for CA Process Automation. Unless indicated otherwise, *system* refers to *all* CA Service Catalog computers in your test or production, including the DBMS computer. If you are migrating one stand-alone CA Service Catalog system to another stand-alone system, ignore references to multiple computers.

Verify the Migration Prerequisites

Perform the following prerequisite tasks before you migrate CA Service Catalog. These tasks are required to help ensure that you can complete the migration process successfully.

1. Verify that the setup on the production system matches the setup on the test system. Verify that the production system has CA Service Catalog installed on the same number of computers as the test system. Verify that the same products and components are installed on each computer in the production system as its "matching" computer in the test system. You migrate from the old system to the new system incrementally in pairs.
   For example, suppose that the first test computer has installed CA Service Catalog, the Catalog Content, and Service Accounting Component. In that case, the first production computer must also have each of these components installed. Moreover, the production computer must not have any *additional* CA Service Catalog products and components installed.
2. Verify that the business unit ID of the service provider is *the same* in both systems.

WARNING
The service provider business unit ID for the new installation *must match* the previous installation.

3. Verify that all computers in both systems are at the same patch level for all important software.
4. Verify that the CA EEM Application Name for CA Service Catalog is the same on both the test and production systems. By default, the name is *Service Catalog*.
   If necessary, update the Application Name, on either or both systems.
5. Stop the CA Service Catalog services on all CA Service Catalog computers using the Windows Control Panel.
6. Back up the MDB of your *production* system and also your *test* system, on the computer where the MDB is installed. This requirement applies, regardless of whether the system is installed on one computer or multiple computers.
   You have now completed the prerequisite tasks for migrating CA Service Catalog.

How to Migrate

You can migrate CA Service Catalog products and components from one system to another. Perform the following steps:

1. Restore the MDB database backup of the test system into your production system. During this process, select the option for overwriting the existing MDB database.
2. Unregister the CA EEM application that the production system uses, as follows:
3. 1. Log in to the Global Application of CA EEM as the EiamAdmin user.
      If your systems use a different name than *Service Catalog* for the CA Service Catalog application name, use that name.
   2. Select Configure, Applications, Service Catalog application and click the UnRegister button.
4. Back up the CA EEM data on the CA Service Catalog computer in your test system, as follows.
   If you have multiple CA Service Catalog computers, perform this step on the *first* CA Service Catalog computer. You run this command once, regardless of how many CA Service Catalog computers your system has.
   a. Open the CA Service Catalog command prompt by clicking **Start**, **Programs**, **CA**, **Service Catalog**, Service Catalog Command Prompt.
   b. Run the following command:
      ```
      ant backup-eem-app
      ```
      This action generates a file that is named *eem-backup.xml* in the *USM_HOME* directory.
5. Copy the following files from the test CA Service Catalog computer to the production CA Service Catalog computer.
   If you have multiple CA Service Catalog computers, perform this step on the *first* CA Service Catalog computer in each system. You copy these files once, regardless of how many CA Service Catalog computers your system has.
   – *USM_HOME\seeddata.properties*
   – *USM_HOME\filestore folder*
     This folder can exist on a different computer than the current computer. To verify its location, select **Administration**, **Configuration**, **Filestore** Information. As a best practice, soon after the migration, set up a single location for shared files.
   – *USM_HOME\eem-backup.xml*
6. Run the following commands on all CA Service Catalog computers in the production system, one computer at a time.
   a. Update the Database Host, Password, Instance, Service Name, or Port
   b. Update the database host, password, instance, service name, or port for the database, as needed.

      WARNING
      The database host name that you specify must be different from the one used in the test system. Updating the other settings is optional.
   c. Run the *ant restore-eem-app* command.
      This command restores the CA EEM database from test to production, using the *eem-backup.xml* that you copied.
   d. Run the *ant update-usm-host* command.
      This command enables you to change the host name and port number for a computer running CA Service Catalog.

WARNING
All host names for production CA Service Catalog must be different from the ones in the test system. Updating the other settings is optional.

For more information about this command, see Update the Host Name and Port Number Using the ant Command section.

7. Start CA Service Catalog services on all computers in the production system.
8. Log in to CA Service Catalog in the production system. Select **Administration**, **Configuration**. Update the configuration sections for Filestore, CA Service Desk Manager Integration, and CA CMDB Integration.

   WARNING
   If your post-migration environment does *not* include any option in the list, update the configuration settings to *remove* the integration! Similarly, if your post-migration environment *does* include any of these options, perform the following step: Verify that the host names, port numbers, and other configuration data are correct for your post-migration environment. Remember that setting up a filestore location is a best practice.
9. Test by verifying that the same users, services, and requests, that existed in test also exist in production.

Post-Upgrade Tasks - CA Service Catalog

NOTE

In case, if you are upgrading from CA Service Management 17.3 to CA Service Management 17.4, the below steps are optional and are not required.

1. If you have upgraded to CA EEM in a clustered setup and have updated the **MaxKeepAliveRequests** property in the Apache **proxy.conf** file to meet the CA EEM installation requirements. After upgrading to CA Service Catalog 17.4, reset the **MaxKeepAliveRequests** property to its original value.
2. Reset the JMS port number only if you specified **7777** as the CA Service Catalog startup or shutdown port when you ran the installation program. We recommend that you do *not* use port **7777** as the startup or shutdown port. Port **7777** is reserved for Java Messaging Service (JMS). Perform the following steps:

   WARNING
   If you must use port 7777 for the startup or shutdown port, reset the JMS port number after you have finished running the setup utility. Otherwise, port conflicts occur, and the product does not function correctly.

   a. Open the file that is named **USM_HOME/config.properties** in a text editor.
   b. Update the value of the *jms.port* property to a new value.
   c. Restart the Windows service named CA Service Catalog.
      You have reset the JMS port number.
3. Log in to the root business unit of CA Service Catalog as a **Service Delivery Administrator**.
4. If you are performing a migration model upgrade, perform the following step. Otherwise, skip this step.
   a. Click **Catalog**, **Configuration**, **System Configuration**.
      The System Configuration page appear.
   b. Set the value of the **Use Service Provider Catalog Only** option to **Yes** or **No**. You decided the value when you System Configuration.

      NOTE
      The value that you set applies to the entire Catalog system.
5. Verify that you and other users can log in to CA Service Catalog on the new computers. Verify that the CA Service Catalog components are working properly.
6. If you have integrated CA Service Catalog with CA Service Desk Manager in the previous release, update the host computer names. Also reconfigure other connection details between these two products.

7. If you have enabled the following features in the previous release, to continue using them enable them for the current release:
   – Secure Socket Layer (SSL)
   – Windows NTLM authentication

   NOTE
   These options are disabled by default, for both new installations and upgrades. Verify that your configuration specifications were retained and reset them, if necessary.
8. Set up shared and customized files. This step can include re-applying customizations from the previous release and setting up the filestore, if applicable.
9. (For Migration Model Upgrades Only) Uninstall the previous versions of CA Service Catalog products and components from the "old" CA Service Catalog computers. If you installed a new version of CA EEM, uninstall the old version of CA EEM from the old CA Service Catalog computers.
   You have finalized the upgrade.

Set Up Shared and Customized Files

Setting up shared and customized files is a required task when you finalize the upgrade. For example, re-apply any customizations that you recorded for XML, XSL, or other CA Service Catalog files when you prepared for the upgrade. Perform the following steps:

1. (For Multiple Catalog Component Computers) Perform the following steps:
   a. If you have not already done so, set up the filestore (the single location for shared files). Otherwise, skip this step.
   b. Click **Administration**, **Configuration**, **Filestore** on any Catalog Component computer. Verify that the filestore location is correct. If necessary, correct the location.
   c. Access the filestore computer and verify that the filestore folder includes the themes folder. If necessary, copy the themes folder to the filestore folder.
2. To use the old filestore location, if applicable, proceed as follows. Otherwise, skip this step.
   a. Copy the themes folder from the local filestore to the old filestore.
   b. Click **Administration**, **Configuration**, **Filestore**.
   c. Specify the original filestore location. You recorded this location when you prepared for the upgrade.
3. Copy all filestore contents (files shared by all users in the environment) from the previous filestore location to the new location, as follows:
   – Documents: Copy from *USM_HOME\view\documents to USM_HOME\filestore\documents*
   – Images:
     • Copy from *USM_HOME\view\webapps\usm\images\offerings to USM_HOME\filestore\images\offerings*
     • Copy from *USM_HOME\view\webapps\usm\images\rateplans to USM_HOME\filestore\images\rateplans*
   – Forms: Copy from *USM_HOME\view\forms to USM_HOME\filestore\forms*
   – Custom directory: Copy any files that you want to update or replace from *USM_HOME\view\webapps\usm\custom to USM_HOME\filestore\custom*

   NOTE
   Your installation can have some or all of these folders, depending on the previous version of CA Service Catalog that was installed.
4. Re-apply any customizations that you recorded for XML, XSL, or other CA Service Catalog files, such as **category.xml** or **requestshared.xml**. Restore or verify your customizations, as follows:
   (For **Migration model** upgrade only)

   WARNING
   You can replace certain new files with your backup files, in certain cases. However, *before* doing so, contact Technical Support and verify each file that you want to replace.

- Compare each customized file that you backed up earlier with the files supplied in this release. Update the new file with your customizations from the backup file.
  Follow the same process for any customizations that you made to related third-party files such as Tomcat files.

(For Traditional upgrade only)

- Review any CA Service Catalog files that you customized, such as **custom.xml** or **requestshared.xml**. Verify that your customizations are intact and are applicable for this release.

The traditional upgrade automatically backs up customized files before the upgrade and restores them after the upgrade.

5. Perform this step if you customized the CA Service Repository Agent in the previous release and you want to continue using the same customizations for the current release:

   NOTE
   The CA Service Repository Agent is also named the Data Mediation Data Repository Agent. This repository agent automates the process of importing usage data in Delimiter Separated File format or Fixed Length File format.

   NOTE
   After upgrade, images for the newly created content pack on the previous CA Service Catalog version may get lost and the content pack offering images on the latest CA Service Catalog version are seen as broken on the application. To resolve this issue, copy the missing images from the backup location USM_HOME\SCA_Backup \filestore\images folder to USM_HOME\filestore\images.

After successful completion of the above tasks, perform additional post-installation tasks.

Upgrading to CA Asset Portfolio Management 17.4

CA Service Management 17.4 supports CA Asset Portfolio Management (CA APM, formerly CA ITAM) upgrades from CA APM release 17.1/17.2/17.3. To upgrade release versions prior to 17.1, it is recommended that you first upgrade to APM 17.1/17.2/17.3, and then proceed to upgrade to 17.4 version.

Prerequisites

Verify and consider the following prerequisites before upgrading to CA APM 17.4:

1. To upgrade from CA APM 17.4, ensure to set the database compatibility level to Microsoft SQL Server and Oracle Database Server. For more information, see Supportability Matrix.
2. Stop all the CA APM services.
3. Stop the scheduled tasks for the integrated CA Service Management products: CA Service Catalog, CA Service Desk Manager, CA Client Automation.
4. Back up the system database (MDB) This ensures safe data recovery, in case the upgrade is not complete.
5. Back up the configuration files.
6. Upgrade CA EEM from the CA APM 17.4 installer.

Upgrade to CA Asset Portfolio Management 17.4

Verify and complete the prerequisites before proceeding with the upgrade.

NOTE

In CA APM distributed environment, Service Management must be installed on all web and application servers. The database screen is shown only on the component server and you must install the MDB patch from the component server.

1. Download the CA Asset Portfolio Management installer from Broadcom Support.

2. The DVD contains the installer for CA Asset Portfolio Management 17.4. If you have integrated CA APM with CA Service Catalog and if you are upgrading or want to integrate CA APM with Catalog while doing a fresh or new installation, download the appropriate installer DVD.
   a. CA ASSET PORTFOLIO MANAGEMENT 17.4 FOR WINDOWS DVD (WITHOUT CATALOG).ISO
   b. CA SERVICE MANAGEMENT APM PACKAGE 17.4 FOR WINDOWS (WITH CATALOG).ISO
3. Copy and extract the installation ZIP files on all servers in your environment where you plan to install CA APM.
4. Right-click **Setup.exe** and launch the CA Service Management Installer.

   NOTE

   You must be logged in as a local admin.
5. Select a language from the **Language Selection** screen.
6. Click the CA Service Management link.
7. Accept the license agreement and click **Next.**
8. The installer automatically detects the CA APM release a version that is installed in your environment and proceeds with upgrading it to CA APM 17.4.
   Click **Next** to proceed.
9. Provide the database configuration details as per the selected database (Oracle or Microsoft SQL Server):
   a. **Microsoft SQL Server**: Complete the following if you have installed the Microsoft SQL Server database:
      - **Database Server**: The host name of the database server. If the target instance is part of a clustered instance, the virtual host name of the cluster must be used.
      - **Database Name**: Specifies the Database name (mdb), the name of the target DBMS. The default value is mdb.
      - **Database Port**: Specifies the port identifier for the target DBMS.
      - **Database Server Instance**: The database instance name. Leave this field blank if you are using the default instance.
      - **Database Admin User**: For SQL Server, **sa** is the default value. Provides permission to create users and schema.
      - **Database Admin Password**: Specifies the database password of users who are specified by the database admin user.
      - **mdbadmin Password**: Specify the password for the mdbadmin user.
      - **Confirm mdbadmin Password**: Confirm the mdbadmin user password.
   b. **Oracle**: Complete the following if you have installed Oracle database:
      - **Database Server**: Specifies the host name of Oracle database server.
        If the target instance is part of a clustered instance, the virtual host name of the cluster must be used.
      - **Oracle Service Name**: Specifies the Oracle Service name.
      - **Listener Port**: Specifies the listener port for database.
      - **Net Service Name**: Identifies the Net Service Name of Oracle database where the database resides.
        If database is remote, use the Net Service Name that is defined in Oracle client on the local computer. CA SDM accesses the database through a local installation of the Oracle client.
      - **DBA User Name**: Authenticates the name of the user with DBA access. The default value for DBA user is **SYS.**
      - **DBA User Password**: Authenticates the password for DBA user.
      - **mdbadmin Password**: Specify password for mdbadmin user.
      - **Confirm mdbadmin Password**: Confirm mdbadmin user password.
      - **Tablespace Path (on DB Server)**: Specifies the tablespace directory on the database server.
      - **Oracle Home Path**: Specifies the Oracle client 32-bit path.
      - **Data Tablespace Name**: Specifies tablespace name. The default value is **MDB_DATA**.
      - **Index Tablespace Name**: Specifies index tablespace name: Default value is **MDB_INDEX**.
10. Click **Next.**
11. The installer lists the CA Service Management products and CA Common Components that is installed on the selected MDB. Click **Next** to proceed.

12. Review the **Select the Products and Integrations** screen. The installer automatically selects the CA APM components that are already installed in your environment to upgrade it to the latest version.
13. 1. **Common Components**: Select this option to integrate CA Asset Portfolio Management with common components such as CA EEM, CA Process Automation.
14. Click **Next**.
15. Review the Pre-installation configuration summary and click **Next**.
16. Review the pre-populated values and provide the CA Asset Portfolio Management Server details. Click **Next**.
17. If CA Embedded Entitlements Manager (CA EEM) is installed in your environment, provide the CA Embedded Entitlements Manager (CA EEM) details:
    a. **CA EEM Server Name**: Specifies the CA EEM server name which is configured with CA Service Catalog.
    b. **CA EEM Admin User Name**: Authenticates the name of the user with Admin access. The default value for the Admin user is **EiamAdmin.**
    c. **CA EEM Admin Password**: Specify password for CA EEM admin user.
18. If CA Process Automation (CA PAM) is installed in your environment, provide the CA Process Automation details.
19. Review the installation progress. Click **Install** to upgrade the select product/products.
20. After the upgrade or installation is complete, click **Finish**.

NOTE

After installing CA APM 17.4, the release version is shown as **17.0.0.1034** on the application user interface

The env.info file for CA APM 17.4 in the CA APM installed folder includes the following information:

- product=ITAM
- major=17
- minor=4
- cum=0
- roll-up=0

21. Complete the CA APM Post Install Steps.

NOTE
The only supported ITAM WCF service is the WSDL, which can be downloaded from the ITAM Web UI by navigating to the Administration tab >> Web Services tab >> WSDL Download.

Post-Upgrade Tasks for CA APM 17.4

NOTE
After successfully upgrading to CA APM 17.4, perform the tasks mentioned in CA Asset Portfolio Management Post Installation Tasks.

Start all the Services

1. Open the **Control Panel** (click Start> Settings> Control Panel).
2. Double-click Administrative Tools.
3. Double-click Services.
4. Locate and start each of the following services:
    a. Apache Tomcat 9.0 AMS Service
    b. CA Asset Portfolio Management - Data Importer Engine
    c. CA Asset Portfolio Management - Event Service
    d. CA Asset Portfolio Management - Export Service
    e. CA Asset Portfolio Management - Registration Service
    f. CA Asset Portfolio Management - HW Reconciliation Engine
    g. CA Asset Portfolio Management - LDAP Import Service

h. CA CASM

NOTE

For performance reasons, we recommend that you do not start the CA CASM service when you do not use multi-tenancy.

Start the Web Interface

After the installation is complete, you can start the web interface to verify that CA APM is ready to use. After you verify that the web interface starts, provide all administrators with the URL and login credentials to log in and prepare the product for users. Administrators can then set up security, configure the user interface, set up hardware reconciliation, and, if necessary, configure the product components. After the administrators prepare the product, they can provide users with the URL and login credentials.

Follow these steps:

1. Start the web interface using one of the following methods:

- – Open a supported web browser and enter the appropriate URL in the following format:
  `http://servername:port/itam`
  – Replace servername and port with the name of the server and the port that are hosting the CA APM web servers.
- – To log in to CA APM, enter the following default credentials:
  - **User Name**
    uapmadmin
  - **Password**
    uapmadmin

    NOTE

    If you changed the password during the installation, use the password that you created.

  In some situations, a browser error or a user name error appears. You can resolve these errors by following the troubleshooting instructions in the message.

IIS Reset

Execute the following command to reset IIS on the web server and the application server:

```
iisreset
```

Replace *servername* with the name of the server that is hosting the CA APM web servers.

Upgrade Troubleshooting

System Resource Issues

Example:

- DEBUG [AWT-EventQueue-0] [DiskSpacePrereqValidator] Dir: /sd/Servicedesk1 Drive: / Usable Disk Space: 4976779264
- DEBUG [AWT-EventQueue-0] [DiskSpacePrereqValidator] Available Disk Space: 4746, Required Disk Space: 5120
- ERROR [AWT-EventQueue-0] [DiskSpacePrereqValidator] Required disk space is insufficient. Minimum diskspace error limit: 5120

Solution: Ensure systems are correctly allocated:

1. Move attachments from *NX_ROOT\site\attachments* to another drive.

2. Move archived backups.
3. Remove old logs, backup version folders.

Integration Issues

CA EEM

Example: CA Service Management Common Installer attempts to create Administrator accounts that which are failing. Applications are unable to integrate with CA EEM.

Solution: Verify that CA EEM is integrated with external LDAP. Create admin accounts manually in LDAP first. For more information,.

CA Process Automation

Example: CA Process Automation with SSL. For more information, see Install CA Process Automation for CA Service Management.

DEBUG [AWT-EventQueue-0] [PAMWebserviceManager] Get ITPAM version

Error Message:[AWT-EventQueue-0] [PAMValidator] CA PAM validation
failed com.ca.smsi.installcore.webservices.WebserviceException: ; nested exception is:

```
javax.net.ssl.SSLHandshakeException: sun.security.validator.ValidatorException: PKIX path building failed:
sun.security.provider.certpath.SunCertPathBuilderException: unable to find valid certification path to
 requested target
at com.ca.smsi.installcore.webservices.PAMWebserviceManager.getITPAMVersion(PAMWebserviceManager.java:265)
Solution:
```

CDB Tables

Invalid information in the CDB tables. For more information, see Run the Database Cleanup Utility.
Example: References to old or invalid installation information may cause failures during installation.

Solution: Run the Database Cleanup Utility to clean up references to old application installs.

> **NOTE**
>
> CDB tables only exist in CA Service Management 14.1 and later versions.

xFlow Interface

Confirm the web server type (Apache Tomcat vs IIS) prior to upgrading CA Service Management. If you are upgrading using a different web server, a full uninstall and reinstall may be required to resolve issues.

Example: Newer xFlow functionality is missing after upgrade

Solution:

- Clear the browser cache.
- Delete the contents from where you have installed xFlow Interface, for example, *c:\windows\temp\casmplayapi*
  In case of CA Service Management 17.3.0.4 Widows OS, navigate to \Program Files\CA\xFlow\APPS\temp instances.
  For CA Service Management 17.3.0.4 Linux OS, navigate to /opt/CA/xFlow/APPS/temp
- Restart the CA xFlow Analyst Interface Services.

Log List

Verify the following logs, in case of any installation issues:

Common Installer Logs

- %temp%\casm
- *<Install Drive>\Program Files (x86)\CA\SC\logs<timeStamp>*
- <Install Drive>*stdout.txt* and <Install Drive>*stderr.txt*
- Database logs (*stderr.log* for example)

CA Service Desk Manager

- <NX_ROOT>/log
- <Install Drive>\stdout.txt and <Install Drive>\stderr.txt
- Database logs (stderr.log for example)

CA Service Catalog logs

- <USM_HOME>\logs

CA Asset Portfolio Manager

- %TEMP%\CA_Asset_Portfolio_Management_xxxxx.log
- <Install Drive>\Program Files (x86)\CA\ITAM\Application Server*.log
- <Install Drive>\Program Files (x86)\CA\ITAM\Web Server*.log

Swing-Box Methods to Upgrade to CA Service Management 17.4

CA Service Management - Swing Box Migration Overview
This document describes the Swing Box Migration process to upgrade CA Service Management 17.4 for the following:

- CA Service Desk Manager
- CA Service Catalog

> **IMPORTANT**
> Unified Self-Service (USS) is not supported in CA Service Management 17.4. If you have USS in your production environment, it is recommended that you test and adapt Service Point (Self-Service) 17.4.

> **IMPORTANT**
> If your upgrade environment is having CA EEM integration, ensure that you do not enable active directory. Enable Active directory only after upgrading to CA Service Management to 17.4 on your swing-box environment.

> **IMPORTANT**
> If your environment is configured with a load balancer or SAML or NTLM, ensure that you do not enable these configurations during migration. Reconfigure these settings after upgrading to CA Service Management 17.4 on your swing-box environment.

Review the following topics:

- CA Service Desk Manager (SDM) - Swing Box Migration
- CA Service Catalog - Swing Box Migration
- Use Case 1 - Swing-Box Migration to Upgrade CA Service Management 17.4 and Above
- Use Case 2 – CA Service Management 17.3.0.3 Swing Box Move from the Production to the Target Server

CA Service Desk Manager (SDM) - Swing Box Migration

This article describes the Swing Box Migration process to upgrade to CA Service Desk Manager (CA SDM) 17.4.
Use any of the below two Swing-box migration methods to upgrade to CA SDM 17.4:

- Swing-Box Migration using Production Replica

> **IMPORTANT**
> In case of Production Replica, moving the production delta data can be tedious (delta here means the additional data or any data added over a period of time post the Swing Box migration). To move the delta

data in your production server, you must repeat the entire Swing box migration process on your complete production data and not just on the delta or supplementary data.

- Swing-Box Migration using Intermittent Server

  NOTE
  This method facilitates the move of the production delta data in an easier way as compared to the "**Swing-Box Migration Using Production replica**".

The above two methods leave your current production system fully intact in case of a disaster recovery issue or in case of a failed migration. With VMware environment as the swing system/intermittent servers, you have the advantage of using the snapshot functionality to test the migration process multiple times, and gather data with timestamp and required steps.

Approach 1 - Production Server		**Approach 2 - Replica of Production Server (Swing Box)**
Moving the delta is tedious and time consuming (if production and target server release levels are different). For example: CA Service Management 17.1/17.2/17.3 with CA Service Management 17.4 It's recommended to have similar release versions on the production and target servers (that is, CA Service Management 17.4 and above patches.)		Moving the delta data is an easier approach (Adding intermittent server to move the delta data). For example: CA Service Management 17.1/17.2/17.3 with CA Service Management 17.4 It is recommended to have different release versions for production and target server (For example: CA Service Management 17.1/17.2/17.3)
Number Of servers required to setup production replica is same as number of servers used in the production deployment. For more information, see OS and Database Requirements for Swing Box Migration Using Production Replica.		Number of Servers required for intermittent server swing-box migration is either one or two along with the same number of servers used in production deployment. In case of Conventional Deployment, number of servers can be **1** (one for Intermittent Server) along with the number of servers used in production deployment. In case of Advance Availability Deployment, number of servers can be **2** (**1** for background server and **1** for Application server) along with the number of servers used in production deployment. It's recommended that you have the Intermittent servers in a Virtual machine, as this facilitates taking snapshot and helps in reverting back to the stable state of intermittent server, while migrating the delta data. For more information, see OS and Database Requirements for Swing Box Migration using Intermittent Servers

CA SDM Recommended OS and Database Requirements

Review the following:

- OS and Database Requirements for Production Replica
- OS and Database Requirements for Intermittent Servers

  NOTE
  For more information, review the following Supportability or compatability Matrix:

  - CA Service Management 17.4
  - CA Service Management 17.3

CA SDM Recommended OS and Database Requirements using Production Replica

The below table showcases the operating system and database requirements for Swing-Box Migration methods discussed in this article:

IMPORTANT
Your target server must be a mirror of the products or applications installed in your production server.

Production Server	**Replica of Production Server (Swing Box - Target Server)**
Operating System: CA SDM 17.1/17.2/17.3 or above • Windows 2012/2016/ 2019 • RHEL 7.x	**Operating System**: CA SDM 17.1/17.2/17.3 or above (Migrate to CA SDM 17.4) • Windows 2016/ 2019/ 2022 • RHEL 7.x/ 8x
Database: • MS SQL server 2012/2016/2019 • Oracle 12c/19c	**Database**: • MS SQL server 2016/ 2019/ 2022 Oracle 19c/ 21c

OS and Database Requirements for Swing Box Migration using Intermittent Servers

Production Server	**Replica of Production Server (Swing Box)**	**Target Server 17.4 (Swing box)**
CA SDM 17.1/17.2/17.3 or above **Operating System**: • Windows 2012/ 2016/ 2019 • RHEL 7.x	**CA SDM 17.1/17.2/17.3 or above (Migrate to CA SDM 17.4)** **Operating System**: • Windows 2016/2019 • RHEL 7.x	**CA SDM: 17.4** **Operating System:** • Windows 2016/2019/ 2022 • RHEL 7.x or 8.x
Database: • MS SQL server 2012/ 2016/ 2019 • Oracle 12c/19c	**Database**: • MS SQL server 2016/ 2019 • Oracle 19c	**Database**: • MS SQL server 2016/ 2019/ 2022 • Oracle 19c/ 21c

Prerequisites and considerations for using CA SDM Swing Box Migration

Ensure that you consider the following prerequisites before using the swing-box migration procedures. This activity saves lot of time while upgrading:

- No form, or schema (tables and columns) customization, or changes must be made to any system once upgrade testing is started (CA SDM production, development, test, Swing box, and so on).
- If you are planning to add or modify additional tables or columns, it must be done in your production environment prior to upgrading. This can also be done after your production upgrade is complete and when you are planning for another outage or downtime using CA Service Desk 17.4 on a new production system.
- It is assumed that you have already done a test upgrade on a replica of your production system. Complete re-customizing any of your custom forms from earlier CA SDM versions that are incompatible with CA SDM 17.4.
- Now, you have a final customized and complete site\mods directory on CA SDM 17.4.

CA SDM Swing-Box Migration Using Production Replica High Level Steps

The below table showcases a high-level overview of your production environment along with supported CA SDM versions and deployment modes (Conventional/Advance Availability) and upgrade to the target server 17.4 using the swing-box migration. For more information, see Install CA Service Desk Manager.

IMPORTANT
Your target server must be a mirror of the products or applications installed in your production server.

S.No	Mode of Deployment	Production Environment	Target 17.4 (Swing Box)
1	**Conventional**	• **Release Version: CA SDM 17.4 and above** • **Mode of deployment: Conventional**	• Setup CA SDM 17.4 and above, similar your production deployment model. Example: Windows Server 2016/ 2019/ 2 or RHEL 7.x/ 8.x • Take a **snapshot-1.** • Move MDB from the production environm • Copy customizations from the productio environment. • Take a **snapshot-2.** • Upgrade to CA SDM 17.4 • Merge the customizations with CA SDM forms. • Take **snapshot 3** (CA SDM 17.4 with production data, customizations).
	• **Release Version: CA SDM 17.4** • **Mode of deployment: Conventional**	• Setup CA SDM 17.4 with supported OS with the same production mode of deployment. Example Windows Server 2016/2019/2022 or RHEL 7.x/8.x • Take a snapshot-1 • Move MDB from the production environment. • Copy customizations from the production environment. • Take snapshot 2 (CA SDM 17.4 with production data, customizations).	

2	**Advance Availability**	• Release Version: CA SDM 17.4 and above • Mode of deployment: Advance Availability	• Setup CA SDM 17.4 and above, same a your production deployment model. Example: Windows Server 2016/ 2019/ 2 or RHEL 7.x/ 8.x • Take a **snapshot-1.** • Move MDB from the production environm • Copy customizations from the productio environment to target environment with 1 mapping (Production to target Backgrou (BG) server, standby server, and all application servers). • Take a **snapshot-2.** • Upgrade to CA SDM 17.4 • Merge the customizations with **1-1** mapp (Production to target BG server, standby server, and all app servers) with CA SDM 17.4 forms • Take snapshot 3 (CA SDM 17.4 with production data, customizations).
	• **Release Version: CA SDM 17.4** • **Mode of deployment: Advance Availability**	• Setup CA SDM 17.4 with supported OS with the same production mode of deployment. Example: Windows Server 2016/2019/2022 or RHEL 7.x/8.x • Take a **snapshot-1.** • Move MDB from the production environment. • Merge file customization from the production environment to target environment with **1-1** mapping (production to target BG server, standby server, and all app servers). • Take a snapshot-2 (CA SDM 17.4 with production data, customizations)	

CA SDM Swing-Box Migration using Production Replica to Upgrade to CA SDM 17.4

This method of swing-box migration uses a separate server to replicate your current production environment to upgrade to CA SDM 17.4. Move the upgraded CA SDM data and customizations over to a fresh system (a system that was never before upgraded) and install on a new hardware. This method leaves your current production system fully intact in case of a disaster recovery issue or in event of a failed migration.
In this method of migration, you have to prepare a replica of your Production environment via Swing box with CA SDM 17.4.

NOTE
This process requires a Production System and a replica of your current Production System.

Review and perform the following:

1. Prepare a Production replica Server (Swing Box) with supported hardware and software for CA SDM.
   For more information, see Supported OS and Database.
   For more information Server Nomenclature.
2. Backup and restore database and customizations from your production environment.
   a. For more information, see Backup Files/Data from your Production Server.
   b. For more information, see Prepare a Swing Box from you production Servers.

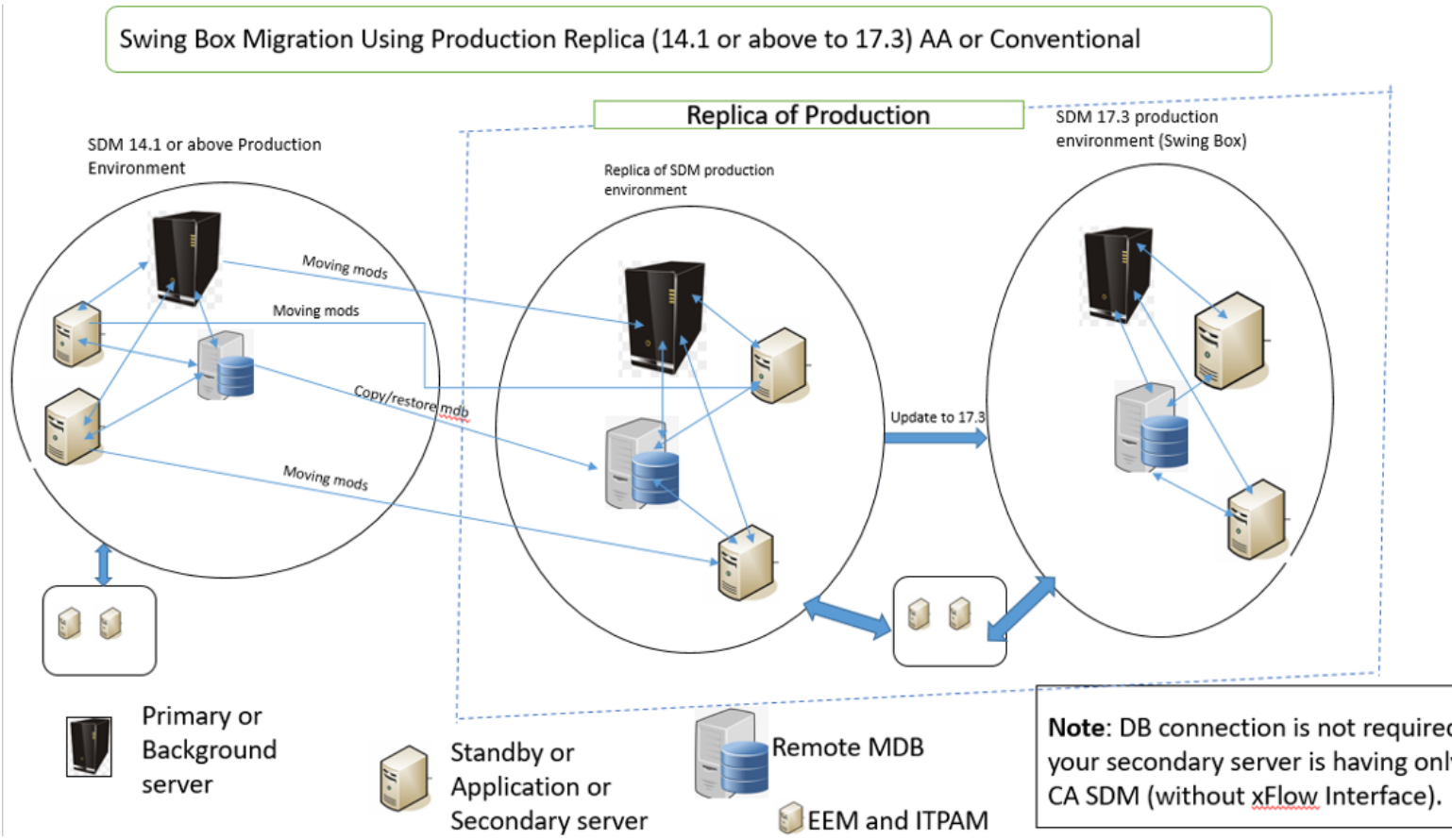
3. Prepare to Upgrade to CA SDM 17.4 using the Swing-Box Migration process on the Production Replica server.
   a. For more information, see Upgrade to CA SDM 17.4.

Server Nomenclature for Swing-Box Production Replica and Process Flow Diagram

The below example server names are used to describe the Swing-Box Production Replica procedure to update CA Service Desk Manager CA SDM 17.1/17.2/17.3 and above to CA SDM 17.4 or above.
Here, the following abbreviated terms have been used for Server names:

- **bg**: Background Server
- **mdb**: MDB Database for Conventional and Advance Availability
- **prim**: Primary Server
- **sec**: Secondary Server
- **app**: Application Server
- **sb**: Standby Server
- **new**: If server names are appended with new, they denote the target servers.

Production Servers		Swing Box/Production Repl		
		Replica of Your Production Servers > M		
CA SDM 17.1/17.2/17.3 or above (Advance Availability Configuration)	**CA SDM 17.1/17.2/17.3 or above (Conventional Configuration)**	**CA SDM 17.1/17.2/17.3 or above (Advance Availability Configuration)**	**CA SDM 17.1/17.2/17.3 or above (Conventional Configuration)**	**CA SDM Availabi**
testqa-mdb	testqa-mdb	testqa-mdb-new	testqa-mdb-new	testqa-m
testqa-bg	testqa-prim	testqa-bg-new	testqa-prim-new	testqa-bg
testqa-sb	testqa-sec	testqa-sb-new	testqa-sec-new	testqa-sb
testq-app	testqa-sec	testqa-app-new	testqa-sec-new	testqa-ap



Step 1: Backup Files/Data from your Production Servers

Perform the following steps:

1. Take a SQL Backup of MDB from the current production database server (**testqa-mdb**) and save it as **.bak** file (usually a normal SQL backup).
2. From the NX_ROOT\site\ directory, make a copy of the “mods\" directory and these two **tblobj.cfg** and **ddict.sch** files from the (**testqa-bg**, **testqa-sb**, **testqa-app**, or **testqa-prim**, **testqa-sec**) based on the current production deployment model (**Advance Availability** or **Conventional Configuration**).
   It is recommended that you compile this into a Zip format.

   IMPORTANT
   Copy the files from your application, standby, background, or primary and secondary servers based on your deployment model.
3. Make a copy of "NX_ROOT\site\attachments" from **testqa-bg** or **testqa-prim** based on your deployment model.

   NOTE
   In case, your attachment repositories are configured on different servers/locations, ensure to copy from this location.

   It is recommended that you compile this into a Zip format.
4. Copy the above files from **Step 1**, **Step 2**, and **Step 3** to any of your swing-box environments to a standby directory and do not place the files in their respective locations for now.

Step 2: Prepare a Swing Box (Production Replica) Environment

For reference purposes, we will refer to the database server here as "**testqa-mdb-new**" and the application server as "**testqa-bg-new**" (Advance Availability) or "**testq-prim-new**" (Conventional Configuration) on a swing box environment.

1. On **testqa-mdb-new**, verify that Microsoft SQL Server is already installed on this server.
2. On **testqa-bg-new** or **testq-prim-new** - install the SQL client and native client (workstation tools).
3. Install the same version of CA SDM as your current production environment on **testqa-bg-new** or **testq-prim-new**. Now, you have a basic CA SDM installation up and running without customizations.
4. Install Standby and number of Application servers (Advance Availability similar to your current production environment.)
   For conventional configuration, this will be replaced by Secondary servers, if any, similar to your current production environment.
5. **NOTE**
   Ensure to set the option to **OVERWRITE** the entire database when restoring.
   Also, follow the instructions in Move the CA MDB Data from the Production to the Target Systems based on your database type (MS SQL or Oracle).
   IMPORTANT
   Additionally, ensure that the below step **6** - step **13** are performed appropriately in your environment.
6. As a database administrator, execute the Microsoft SQL Restore command and restore the MDB from your current production (**testqa-mdb**) that you have backed up on the **testq-mdb-new** server as shown in Step 1: Backup Files/ Data from your Current Production System. For more information, see Server Nomenclature for Swing-Box Production Replica Method.
   a. Here, Production MDB server will be referred as **testqa-mdb** and target MDB server is **testqa-mdb-new**.
   b. For more information, see Server Nomenclature for Swing-Box Production Replica Method.
7. After you have completed the Microsoft SQL Server restore process on the target mdb (**testqa-mdb-new**) server, execute the stored procedure in SQL to fix the orphaned users which are created when restoring a database from one Microsoft SQL Server instance to another.
   For Example:
   ```
   sp_change_users_login 'AUTO_FIX','mdbadmin'
   ```
8. Copy the NX_ROOT/site/mods directory and the **tblobj.cfg ddict.sch** files from the production server to the Swing Box server:
   NOTE
   The below steps are applicable only if your production environment is customized. This is not applicable for the out-of-the-box (OOTB) content.
   a. **Advance Availability**:
      - Production background server (BG) (**testqa-bg**) to the target background server (BG) (**testqa-bg-new**).
      - Production Standby (**testqa-sb**) to the target Standby (**testqa-sb-new**).
      - Production Application Server (**testqa-app**) to the target Application Server (**testqa-app-new**).
   b. **Conventional Configuration**:
      - Production Primary (**testqa-prim**) to the target Primary server (**testqa-prim-new**).
      - Production Secondary (**testqa-sec**) to the target Secondary server (**testqa-sec-new**).

      NOTE
      Perform on all secondary servers in your environment.
      NOTE
      The above steps are applicable only if you have customizations in your environment. This is not applicable for the out-of-the-box (OOTB) content.
      NOTE
      Ensure that you are taking a backup of NX_ROOT\site\mods directory, NX_ROOT/**site/tblobj.cfg**, and NX_ROOT/**site/ddict.sch** files on the swing box/Production Replica. You will need to rebuild your form customizations using the CA SDM 17.4 form delivered with the installer as the earlier customized

CA SDM forms will not work. Once you have completed the rebuilding of your customizations, take a backup of your NX_ROOT\site\mods directory, NX_ROOT/**site/tblobj.cfg**, and NX_ROOT/**site/ddict.sch** files on the swing box/production Replica.

c. Copy the site\attachments directory (or where you have placed the attachments directory) that you backed up from the production server (**testqa-bg** or **testqa-prim**) and place it in the swing-box environment (**testqa-bg-new**or **testqa-prim-new**) based on the deployment model.

9. Execute the pdm-configure command:
   a. Run **pdm_configure -S** on the target BG Server (**testqa-bg-new**) for Advance Availability.
   It configures silently. Monitor the progress of **pdm_configure** using **NX_ROOT/logs/configure.log** file.
   b. Run **pdm_configure** on the target primary server (**testqa-prim-new**) for Conventional Configuration.
   c. Ensure that you do not select the **LOAD DEFAULT DATA** checkbox option.
10. In the configuration wizard, click **Next** on the database section.
   A message is displayed to show that the database was previously configured. This message indicates that you have restored the MDB from a different environment.
   Click **Yes**.
11. After **pdm_configure** is complete, start the CA SDM services (if the services are not started already).
12. Run **pdm_configure** on the Swing Box standby and all application servers (Advance Availability) and on all secondary servers (conventional configuration).
13. Verify and test the functionality to ensure that CA SDM is up and running with customizations seamlessly as a full replica of your production environment.

Step 3: Upgrade to CA SDM 17.4

Now you can upgrade to CA SDM 17.4 based on your deployment configuration. For more information, see Install CA Service Desk Manager 17.4.
Perform the following steps:

1. Mount the CA SDM 17.4 install media and then, run the **setup.exe**, run as **Administrator** from the local folder where you have copied the CA SDM installer.
   a. If you are extracting an ISO of the install media, it should be stored in a path and the folder should have no spaces in the name.

      NOTE
      Download the CA Service Desk Manager 17.4 installer DVD from Broadcom Support.

      - CA SERVICE MANAGEMENT SDM PACKAGE 17.4 FOR WINDOWS (WITH CATALOG).iso
      - CA SERVICE DESK MANAGER 17.4 FOR WINDOWS (WITHOUT CATALOG).iso
      - CA SERVICE DESK MANAGER 17.4 FOR LINUX.iso
2. Launch the CA Service Management 17.4 installer.
3. Select a language of your choice and click **Next**.
   For example, select **English**.
4. Select CA Service Management.
5. Follow the installation wizard to complete the installation process.
   This may take some time to complete depending upon the size of the CA SDM database that you are upgrading/migrating.
6. At this point, if you are doing a test run on the intermittent server, you must re-build your form customizations using the CA SDM 17.4 form delivered with the installer as the earlier (17.1/17.2/17.3 or above) customized CA SDM forms will not work. Once you have completed the rebuilding of your customizations, take a backup of your NX_ROOT**site\mods** directory, NX_ROOT/**site/tblobj.cfg** and NX_ROOT/**site/ddict.sch** files on the Intermittent Server and store it somewhere outside of this environment for later use.
7. Execute the **pdm-configure**command:
   a. Run **pdm_configure -S** on the target BG Server (t**estqa-bg-new**) for Advance Availability.
   It configures silently. Monitor the progress of **pdm_congure** using *NX_ROOT/logs/configure.log* file.

b. Run **pdm_configure** on the target primary server (**testqa-prim-new**) for Conventional Configuration.
c. Ensure that you do not select the **LOAD DEFAULT DATA** checkbox option.

8. Test the migrated and customized CA SDM 17.4 production replica server for integrity and functionality.
9. If all tests are successful, you can now start the process of moving your migrated data and customizations to the target or swing box 17.4 environment.
10. Upgrade Standby and number of Application servers (Advanced Availability similar to your current production environment.)
    For conventional configuration, this will be replaced by Secondary servers, if any, similar to your current production environment.

CA SDM Swing-Box Migration using Intermittent Server - High Level Steps

CA SDM Swing-Box Migration using Production Replica - High Level Steps

The below table showcases a high-level overview of your production environment along with supported CA SDM versions and deployment modes (Conventional/Advance Availability) and upgrade to the target server 17.4 using the swing-box migration. For more information, see Install CA Service Desk Manager.

S.No	Mode of Deployment	Production Environment	Intermittent Server Migra
1	**Conventional Configuration**	• Release Version: CA SDM 17.3 and above • Mode of deployment: Conventional	• Setup Intermittent 17.3 supported OS . Example: Windows 201 RHEL 7.x/ 8.x • Take a **snapshot-1.** • Move MDB from the pro • Take a **snapshot-2.** • Upgrade to CA SDM 17 • Take **snapshot-3**
		• Release Version: CA SDM 17.4 and above • Mode of deployment: Conventional	• Setup CA SDM 17.4 wi the same production m Example Windows Serv or RHEL 7.x/8.x • Take a **snapshot-1** • Move MDB from the pro • Copy customizations fr environment. • Take **snapshot 2** (CA S production data, custom
2	**Advance Availability Configuration**	• Release Version: CA SDM 17.3 • Mode of deployment: Advance Availability	• Setup Intermittent 17.3 Example: Windows Ser or RHEL 7.x • Take a **snapshot-1** • Move MDB from the pro • Take a **snapshot-2** • Upgrade to CA SDM 17 • Take **snapshot 3.**

		• Release Version: CA SDM 17.3 • Mode of deployment: Advance Availability	• Setup CA SDM 17.4 wi the same production m Example: Windows Ser RHEL 7.x/8.x • Take a **snapshot-1.** • Move MDB from the pr • Merge file customizatio environment to target e mapping (production to standby server, and all • Take a s**napshot-2** (CA production data, custom

Swing-Box Migration using Intermittent Server to Upgrade to CA SDM 17.4

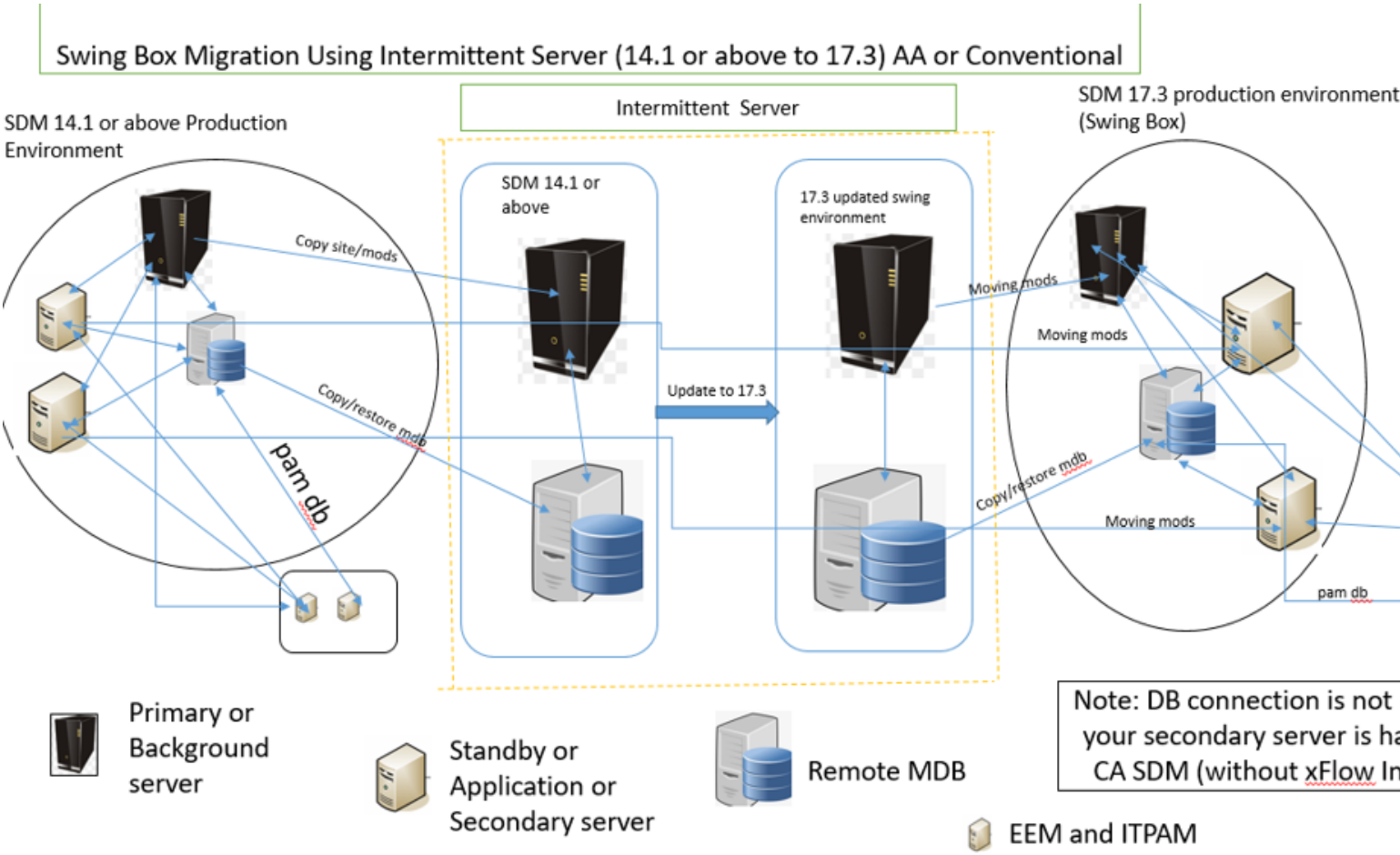
This method of swing-box migration is performed by using an intermittent server to migrate your current CA SDM production system data along with customizations, and then upgrade it to CA SDM 17.4. It involves an intermittent server (**Stand alone** in case of conventional configuration or two boxes, one for the background and one for the Application server in case of **Advance Availability**).
Review and Perform the following:

1. Prepare an Intermittent Server. Ensure that you have reviewed the OS and Database requirements
   For more information, see Supported OS and Database and follow the Server Nomenclature for Swing-Box Intermittent Server.
2. Take a database and file customization backup of your production environment.
   – For more information, see Step 1: Collect Files/Data from your current Production System.
   – For more information, see Step 2: Prepare the intermittent Server.
3. Install a fresh or new instance of CA SDM 17.4 server (a server that was never upgraded before) on the target server (swing box). This server should have the same deployment as your production server.
   For more information, see Install CA Service Desk Manager
4. Take a database backup of your intermittent server and customizations from application/secondary servers of the production environment and restore it to the target servers.
   – For more information, see Step 3: Upgrade to CA SDM 17.4 on the Intermittent Server.
   – For more information, see Step 4: Move the upgraded Intermittent Server install to a Swing-Box (Target) 17.4.

Server Nomenclature for Swing-Box Intermittent Server and process flow diagram
The below example server names are used to describe the Swing-Box Intermittent Server Migration to update CA Service Desk Manager (CA SDM) 17.1/17.2/17.3 or above to CA SDM 17.4 or above.
The following abbreviated terms have been used for Server names:

- **bg**: Background Server
- **mdb**: MDB Database for Conventional and Advance Availability
- **prim**: Primary Server
- **sec**: Secondary Server
- **app**: Application Server
- **sb**: Standby Server
- **new**: If server names are appended with new, they denote the target servers.

Production Server		Intermittent Server	Target S
CA SDM Release 17.1/17.2/17.3 or above with Advance Availability Configuration	**CA SDM Release 17.1/17.2/17.3 or above with Conventional Configuration**	**CA SDM 17.1/17.2/17.3 Intermittent Server**	**CA SDM 17.4 Advance Availability**
testqa-mdb	testqa-mdb	testqa-mdb-inter (bg server or primary server)	testqa-mdb-new
testqa-bg	testqa-prim	testqa-app-inter (bg server or primary server	testqa-bg-new
testqa-sb	testqa-sec		testqa-app-new
testqa-app	testqa-sec		testqa-app-new



Step 1: Collect Files/Data from your Current Production System

Perform the following steps to collect files or data from your production system :

1. Take a SQL Backup of MDB from the current production database server (**testqa-mdb**) and save it as **.bak** file (usually a normal SQL backup).
2. Make a copy of the “\Program Files\CA\Service Desk\site\mods\" directory, **tblobj.cfg**, and **ddict.sch** files from the (**testqa-bg**, **testqa-sb**, **testqa-app**, or **testqa-prim**,**testqa-sec**) based on the current production deployment model (Advance Availability or Conventional Configuration).
   It’s recommended that you compile this into a Zip format.

   NOTE
   Copy the files from all your application, standby, background, or primary and secondary servers based on your deployment model.
3. Make a copy of "Program Files\CA\Service Desk\site\attachments" from **testqa-bg** or **testqa-prim** based on your deployment model.

   NOTE
   In case, if your attachment repositories are configured on different servers/location, ensure to copy from this location.

   It’s recommended that you compile this into a Zip format.

Step 2: Prepare the Intermittent Server

NOTE
For reference purposes, in this section we will refer to the database server as “**testqa-mdb-inter**" and the application server as "**testqa-app-inter**". Here, **testqa-app-inter** is SDM Primary or SDM BG server based on your production environment deployment mode.

Perform the following steps:

1. It is assumed that you have already installed Microsoft SQL Server (**testqa-mdb-inter**).
2. On **testqa-app-inter** server, ensure to install the SQL client and native client (workstation tools).

   NOTE
   If your intermittent server has both Database and Application running on a single server you may skip the **Step 2** and also ensure to install the Microsoft SQL Server Client and native client (workstation tools) on **testqa-mdb-inter**.
3. Install the same release version of production server CA SDM on **testqa-app-inter**, so that you can have a basic installation of CA SDM up and running in your environment (without customizations).
4. As a database administrator, execute the Microsoft SQL Restore command and restore the MDB from your current production (**testqa-mdb**) that you have backed up on the **testqa-mdb-inter** server.

   NOTE
   Ensure to set the option to **OVERWRITE**the entire database when restoring.

   – Here, production mdb server will be referred as **testqa-mdb** and target mdb server is **testqa-mdb-inter**. For more information, see Server Nomenclature.
   – Also, follow the instructions in Move the CA MDB Data from the production to the Target Systems based on your database type (MS SQL or Oracle).

   IMPORTANT
   Additionally, ensure that the below step 6 - step 13 are performed appropriately in your environment.
5. After you have completed the Microsoft SQL Server restore process on the target mdb (**testqa-mdb-inter**) server, execute the stored procedure in SQL to fix the orphaned users which are created when restoring a database from one Microsoft SQL Server instance to another.
   For example:

```
sp_change_users_login 'AUTO_FIX','mdbadmin'
```

6. Copy the site/mods directory **tblobj.cfg** and **ddict.sch** files from the production server to the Swing Box server:
   a. **Advance Availability**: production BG (**testqa-bg**) to the target BG (**testqa-app-inter**).
   b. **Conventional Configuration**: production Primary (**testqa-prim**) to the target Primary server (**testqa-app-inter**)

      NOTE
      The above steps are applicable only if you have customizations in your environment. This is not applicable for the out-of-the-box (OOTB) content.
7. Copy the *site**attachments* directory (or where you have placed the attachments directory) that you backed up from the production server (**testqa-bg** or**testqa-prim**) and place it in the swing-box environment (**testqa-app-inter**) based on the deployment model.
8. Execute the pdm-configure command:
   a. Run **pdm_configure -S** on the target BG Server (**testqa-app-inter**) for Advance Availability.
      It configures silently. Monitor the progress of **pdm_congure** using NX_ROOT/logs/configure.log file.
   b. Run pdm_configure on the target primary server (**testqa-app-inter**) for Conventional Configuration.
   c. Ensure that you do not select the **LOAD DEFAULT DATA** checkbox option.
9. In the configuration wizard, click **Next** on the database section.
   A message is displayed to show that the database was previously configured. This message indicates that you have restored the MDB from a different environment.
   Click **Yes**.
10. After **pdm_configure** is complete, start the CA SDM services (if the services are not started already).
11. Verify and test the functionality to ensure that CA SDM is up and running with customizations seamlessly as a full replica of your production environment.

Step 3: Upgrade to CA SDM 17.4 on the Intermittent Server

Test the migration to become familiar with the processes and issues that you may encounter while migrating your production system. Ensure that you have considered the following:

- If your Intermittent Server is on a VMware environment, it's recommended that you take a snapshot. You can roll back to that snapshot when you take down your production system for maintenance or other purposes.
- Perform a Microsoft SQL Server backup on the production system. Restore it to the Intermittent Server and complete the below steps mentioned in this section.
  In case, if the migration to SDM 17.4 fails, you can revert to the CA SDM production system and start the services.
- Retest migration on the intermittent server again and schedule the migration for another time.

Perform the following steps:

1. Mount the CA SDM 17.4 install media and then, run the **setup.exe**, run as **Administrator** from the local folder where you have copied the CA SDM installer.
   a. If you are extracting an ISO of the install media, it must be stored in a path and the folder must have no spaces in the name.
   b. Run locally from the same drive volume where you plan to install CA SDM 17.4. **Do not run the installer from a network drive or mounted share as this may cause install issues and failures**.

      NOTE
      Download the CA Service Desk Manager 17.4 installer DVD from Broadcom Support.

      - CA SERVICE MANAGEMENT SDM PACKAGE 17.4 FOR WINDOWS (WITH CATALOG).iso
      - CA SERVICE DESK MANAGER 17.4 FOR WINDOWS (WITHOUT CATALOG).iso
      - CA SERVICE DESK MANAGER 17.4 FOR LINUX.iso
2. Launch the CA Service Management 17.4 installer.
3. Select a language of your choice and click **Next**.
4. Select **CA Service Management**.
5. Follow the installation wizard to complete the installation process.

This may take some time to complete depending upon the size of the CA SDM database that you are upgrading/ migrating.

6. At this point, if you are doing a test run on the intermittent server, you must re-build your form customizations using the CA SDM 17.4 form delivered with the installer as the earlier (17.4 or above) customized CA SDM forms will not work. Once you have completed the rebuilding of your customizations, take a backup of your site\mods directory, site/ tblobj.cfg and site/ddict.sch files on the Intermittent Server and store it somewhere outside of this environment for later use.
7. Execute the **pdm-configure** command:
   a. Run **pdm_configure -S** on the target BG Server (**testqa-app-inter**) for Advance Availability.
      It configures silently. Monitor the progress of pdm_congure using NX_ROOT/logs/configure.log file.
   b. Run **pdm_configure** on the target primary server (**testqa-app-inter**) for Conventional Configuration.
   c. Ensure that you do not select the **LOAD DEFAULT DATA** checkbox option.
8. Test the migrated and customized CA SDM 17.4 Intermittent Server for integrity and functionality.
9. If all tests are successfully completed, you can now start the process of moving your migrated data and customizations to the target or swing box 17.4 environment.

Step 4: Move Upgraded Intermittent Server Install to a Swing-Box (Target) 17.4

Your Swing-box (Target) environment must already be up and running with a clean or fresh installation of CA Service Desk Manager 17.4 same as your production deployment model.
Perform a Microsoft SQL Server backup on the intermittent server (**testqa-mdb-inter**) and have your database administrator perform a Microsoft SQL Server Restore onto the SQL database instance on your Swing-box or target server (**testqa-mdb-new**).
Perform the following steps:

1. Create a Microsoft SQL Server backup on the intermittent server (**testqa-mdb-inter**), and have your database administrator perform a SQL Restore on the SQL instance used by your Swing-box or target server (**testqa-mdb-new**).

   NOTE
   Ensure to set the option to **OVERWRITE** the entire database when restoring.

   – Here, production mdb server will be referred as **testqa-mdb-inter** and target mdb server is **testqa-mdb-new**.
   – Also, follow the instructions in Move the CA MDB Data from the production to the Target Systems based on your database type (MS SQL or Oracle).

   NOTE
   Additionally, ensure that the below step 2 - step 6 are performed appropriately in your environment.
2. After you have completed the Microsoft SQL Server restore process on the target mdb (**testqa-mdb-inter**) server, execute the stored procedure in SQL to fix the orphaned users which are created when restoring a database from one Microsoft SQL Server instance to another. For example:

   ```
   sp_change_users_login 'AUTO_FIX','mdbadmin'
   ```
3. **Advance Availability**:
   Perform the following steps:
   a. Ensure that you have installed the following components:
      a. (Oracle) database client is installed and the net service name is configured to connect with the new database server.
      b. (MS SQL) MS SQL database client is installed.
      c. Copy the site/mods directory, site/tblobj.cfg, and site/ddict.sch files from the production CA SDM server to the target CA SDM server:
         a. production BG (**testqa-bg**) to the target BG (**testqa-bg-new**).
         b. production Standby (**testqa-sb**) to the target Standby (**testqa-sb-new**).
         c. production Application Server (**testqa-app**) to the target Application Server (**testqa-app-new**) (Perform on all application servers in an environment).

NOTE
The above steps are applicable only if your production environment is customized. This is not applicable for the out-of-the-box (OOTB) content.

NOTE
Ensure that you are taking a backup of site\mods directory, **site/tblobj.cfg**, and**site/ddict.sch** files on the swing-box or target servers. You will need to rebuild your form customizations using the CA SDM 17.4 form delivered with the installer as the earlier customized CA SDM forms will not work. Once you have completed the rebuilding of your customizations, take a backup of your **site\mods** directory, **site/tblobj.cfg**, and **site/ddict.sch** files on the swing-box or target servers.

d. Run **pdm_configure -S** on BG Server (**testqa-bg-new**). It configures silently. Monitor the progress of **pdm_congure** using NX_ROOT/logs/configure.log file.
e. Run pdm_configure on Standby Server (**testqa-sb-new**).
f. Run the pdm_configure on all application servers (**testqa-app1-new**) and **testqa-app2-new**).
g. Execute the pdm_k_reindex command on the CA SDM target system of BG Server machine **testqa-bg-new**to index knowledge documents that are created and copied from the production system of BG Server machine **testqa-bg**.

4. **Conventional Configuration**: Perform the following steps:
   a. Copy the site/mods directory, **site/tblobj.cfg** and**site/ddict.sch** files from the production CA SDM server to the target CA SDM server.
   b. production Primary (**testqa-prim**) to the target Primary (**testqa-prim-new**).
   c. production Secondary (**testqa-sec**) to the target secondary (**testqa-sec-new**) (Perform on all secondary servers in this environment).

      NOTE
      The above steps are applicable only if your production environment is customized. This is not applicable for the out-of-the-box (OOTB) content.

      NOTE
      Ensure that you are taking a backup of *site**mods* directory, **site/tblobj.cfg**, and **site/ddict.sch** files on the swing-box or target servers. You will need to rebuild your form customizations using the CA SDM 17.4 form delivered with the installer as the earlier customized CA SDM forms will not work. Once you have completed the rebuilding of your customizations, take a backup of your **site\mods** directory, **site/tblobj.cfg**, and **site/ddict.sch** files on the swing-box or target servers.

   d. If conventional configuration, add **NX_SWING_BOX_MIGRATE=Yes** variable to NX.env file and run the **pdm_configure** command.
   e. Do not select **TO LOAD DEFAULT DATA**.
   f. In the configuration wizard, click **Next** in the database section.
      A message displays, stating that the database was previously configured. This message indicates that you have restored the MDB from a different environment.
   g. Click **Yes**.
   h. After the configuration is complete, perform the following tasks:
      a. Start the CA SDM services (if not started).
      b. Run the **pdm_configure** on all application servers (**testqa-sec1-new**) and **testqa-sec2-new**).
      c. Execute the **pdm_k_reindex** command on the CA SDM target system of Primary Server machine **testqa-prim-new** to index knowledge documents that are created and copied from the production system of Primary Server machine **testqa-prim**.

5. Test the system functionality to ensure the swing-box or target environment with CA Service Desk Manager 17.4 and above is up and running with data and customizations intact.

CA Service Catalog - Swing Box Migration

This document describes how to upgrade to CA Service Catalog 17.4 using swing box migration. Use any of the below two swing-box methods to upgrade to CA Service Catalog 17.4:

- Swing-Box Migration using Production Replica

  NOTE
  In case of Production Replica, moving the production delta or supplementary data can be tedious (supplementary here means the additional data or any data added over a period of time post the Swing Box migration). To move the supplementary data in your production server, you must repeat the entire Swing box migration process on your complete production data and not just on the delta or supplementary data.
- Swing-Box Migration using Intermittent Server

  NOTE
  This method facilitates the move of the production delta or supplementary data in an easier way as compared to “Swing-Box Migration Using Production replica”.

CA Service Catalog Recommended OS and Database Requirements

- Production Replica - OS and Database Requirements
- Intermittent Servers -OS and Database Requirements

The below table showcases the Operating system and database requirements for Swing-Box Migration methods discussed in this article:

OS and Database Requirements for Swing Box Migration using Production Replica

Production Server	**Replica of Production Server (Swing Box)**
CA Service Catalog: 17.1/17.2/17.3 or above	*CA Service Catalog: 17.1/17.2/17.3 or above (migrate to CA Service Catalog 17.4)*
Operating System: • Windows 2012/2016/2019	**Operating System**: • Windows 2016/ 2019
Database: • MS SQL server 2012/2016/2019 • Oracle 19c	**Database**: • MS SQL server 2016/ 2019 • Oracle 19c

OS and Database Requirements for Swing Box Migration using Intermittent Server

Production Server	**Replica of Production Server (Swing Box)**	**Target Server with 17.4 (Swing Box)**
CA Service Catalog: 17.1/17.2/17.3 or above	*CA Service Catalog: 17.1/17.2/17.3 or above (migrate to CA Service Catalog 17.4)*	CA Service Catalog: 17.4

Operating System: • Windows 2012/2016/ 2019	**Operating System**: • Windows 2016/ 2019	**Operating System**: • Windows 2016/ 2019/ 2022
Database: • MS SQL server 2012/2016/ 2019 • Oracle 12c/19c	**Database**: • MS SQL server 2016/2019 • Oracle 19c	**Database**: • MS SQL server 2016/2019/ 2022 • Oracle 19c/ 21c

Prerequisites and Considerations for using CA Service Catalog Swing Box Migration

Ensure that you consider the following prerequisites before using the swing-box method. This activity saves you a lot of time while upgrading:

- **Do not enable Active Directory on EEM Application** . Enable Active Directory only after installing CA Service Catalog 17.4.
- No form, or schema (tables and columns) customization, or changes must be made to any system once upgrade testing is started (CA Service Catalog production, development, test, swing box, and so on).
- If you are planning to add or modify additional tables or columns, it must be done in your production environment prior to upgrade. This can also be done after your production upgrade is complete and when you are planning for another outage or downtime using CA Service Catalog 17.4 on a new production system.
- You have already done a test upgrade on a replica of your production system. Complete re-customizing any of your custom forms from earlier CA Service Catalog versions that are incompatible with CA Service Catalog 17.4.
  Now, you have a final customized and complete filestore\custom directory on CA Service Catalog 17.4.

  NOTE
  If you have a shared filestore, copy from the shared filestore location. This is recommended to avoid multiple overlap of common files.
- Ensure to take a backup of CA EEM policies on the production CA Service Catalog server.

CA Service Catalog Swing-Box Production Replica - High Level Steps

The table below shows a high-level overview of your production environment with supported CA Service Catalog release versions and upgrade to target CA Service Catalog Server 17.4 using the swing-box migration.

NOTE
The tabular information is applicable for both standalone and multiple server instances of CA Service Catalog. For more information about how to upgrade to CA Service Catalog 17.4, see Install CA Service Catalog.

S.No	Intermittent Server	Target 17.4 (Swing Box)
1	*CA Service Catalog: 17.3 or above (migrate to CA Service Catalog 17.4)*	• Setup CA Service Catalog 17.3 similar to your production deployment model. Example: Windows 2016/ 2019 • Take a **snapshot-1** • Move MDB from the production environment. • Copy customizations from the production environment to target environment with **1-1** mapping (production to target of Catalog application Servers). If you have a shared filestore, you need to copy from the shared filestore location (example: filestore/custom). This is required to avoid multiple overlap of common files. • Take a **snapshot-2** • Upgrade to 17.4. • Merge the customizations with **1-1** mapping (production to target of Catalog application Servers) with CA Service Catalog 17.4 • Take snapshot 3 (CA Service Catalog 17.4 with production data, customizations).
2	Version: CA Service Catalog 17.4	• Setup CA Service Catalog 17.4 with supported OS with the same production mode of deployment. Example: Windows 2016/ 2019/ 2022 • Take a **snapshot-1** • Move MDB from the production environment. • Merge file customization from the production environment to target environment with **1-1** mapping (production to target of Service Catalog Application servers). If you have a shared filestore, you need to copy from the shared filestore location (example: filestore/custom). This is required to avoid multiple overlap of common files. • Take **snapshot 2** (CA Service Catalog 17.4 with production data, customizations).

Swing-Box Migration Using Production Replica to Upgrade to CA Service Catalog 17.4

This method uses the replica method on a separate server to replicate your current production environment to upgrade to CA Service Catalog 17.4. Move the upgraded CA Service Catalog data and customizations over to a fresh system (a system that was never upgraded before) and install on a new hardware. This method leaves your current production system fully intact in a disaster recovery issue or in event of a failed migration.

In this method of migration, you have to prepare a replica of your production environment using swing box with CA Service Catalog 17.4.

NOTE
This process requires a production system and a replica of your current production system.

Review and perform the following:

1. Prepare a Production replica Server (Swing Box) with supported hardware and software for CA Service Catalog.
   For more information, see Production Replica - OS and Database Requirements and Server Nomenclature for Swing-Box Production Replica.
2. Backup and restore database and customizations from your production environment.
   For more information, see Step:1 Collect Files/Data from your Current Production System.
3. Prepare to upgrade from CA Service Catalog 17.1/17.2/17.3 using the Swing-Box Migration process on the Production Replica server.
   For more information, see Prepare to Upgrade to CA Service Catalog 17.4.

   NOTE
   Moving the production delta or supplementary data can be tedious (supplementary here means the additional data or any data added over a period of time post the Swing Box migration). To move the supplementary data in your production server, you must repeat the entire Swing box migration process on your complete production data and not just on the delta or supplementary data.

Server Nomenclature for Swing-Box Production Replica Method and Process Flow Diagram

The following example server names are used to describe the swing-box production replica method to upgrade CA Service Catalog 17.1/17.2/17.3 or above to CA Service Catalog 17.4 or above release versions.

- **mdb**: MDB database
- **catalog1**: Catalog Production Server
- **catalog1-new**: Replica of Catalog Production Server
- **catalog2**: Catalog Production Server 2
- **catalog2-new**: Replica of Production Server 2
- **new**: If Server names are appended with new, they denote the target servers.

Production Servers	**Swing Box/Production Replica Server**	
	Replica of Your Production Servers > Migrated to 17.4	
CA Service Catalog 17.1/17.2/17.3 or above	*CA Service Catalog: 17.1/17.2/17.3 or above (migrate to CA Service Catalog 17.4)*	CA Service Catalog 17.4
testqa-mdb testqa-catalog1 testqa-catalog2	CA Service Catalog 17.1/17.2/17.3 or above testqa-mdb-new testqa-catalog1-new testqa-catalog2-new	CA Service Catalog 17.4 testqa-mdb-new testqa-catalog1-new testqa-catalog2-new

Step 1: Backup Files/Data from your Production Servers

Perform the following steps:

1. Take a SQL Backup of MDB from the current production database server (**testqa-mdb**) and save it as **.bak** file (usually a normal SQL backup).
2. Make a copy of the shared filestore directory of the (**testqa-catalog1**and **testqa-catalog2**) based on the current production deployment model.

   It is recommended that you compile this into a Zip format.

   NOTE

   If you have a shared filestore, you must copy from the shared filestore location. This is required to avoid multiple overlaps of common files.

   NOTE

   Make a copy of the "*Program Files**CA**Service Catalog**filestore**custom*" directory files from the **testqa-catalog1**if you have single catalog server.
3. Copy the above files from Step 1 and Step 2 to any of your swing-box environments to a standby directory and do not place the files in their respective locations for now.

Step 2: Prepare a Swing Box (Production Replica) Environment

For reference purposes, we refer to the database server here as "**testqa-mdb-new**" and the application server as **"testqa-catalog1-new**" on a swing box environment.

1. On **testqa-mdb-new**, verify that Microsoft SQL Server is already installed on this server.
2. On **testqa-catalog1-new**- install the SQL client and native client (workstation tools).
3. Install the same version of CA Service Catalog as your current production environment on **testqa-catalog1-new**. Now, you have a basic CA Service Catalog installation up and running without customizations.
4. Install number of Catalog Application servers (similar to your current production environment).

CA EEM Policies Backup and Restore on CA Service Catalog server

Perform the following steps to backup CA EEM data on production Catalog server (**testqa-catalog1**):

1. Open the CA Service Catalog command prompt. Click **Start**, **Programs**, **CA**, **Service Catalog**, usm command prompt (*%usm_home%**usm.cmd*) from Catalog Server 17.1/17.2/17.3 or above installed as example **testqa-catalog1** in this document.
2. Run the following command:

   ```
   ant backup-eem-app
   ```

   NOTE

   During execution, provide input parameters like EEM password . After successful completion, **"eem-backup.xml"** file is generated in the %USM_HOME% directory.

   NOTE

   If you have multiple CA Service Catalog computers, perform this step on the first CA Service Catalog computer. Run this command once, regardless of the number of Service Catalog servers in your environment.

Perform the following steps to restore CA EEM on the target Catalog server **(testqa-catalog1-new)**:

1. Copy file (**eem-backup.xml**) from **testqa-catalog1** to the **testqa-catalog1-new** in the *%USM_HOME%* folder.
2. Run the following command:

   ```
   ant restore-eem-app
   ```

   After successful completion-related entries are updated in the EEM.
3. Restart CA Service Catalog services.
4. As database administrator, execute the Microsoft SQL Restore command and restore MDB from your current production (**testqa-mdb**) that is backed up on the **testq-mdb-new** server.

NOTE
Ensure to set the option to OVERWRITE the entire database when restoring.

For more information, see Step 1: Backup Files/Data from your Production Servers.

5. Here, production mdb server is referred as testqa-mdb and target mdb server is testqa-mdb-new. For more information, see Server Nomenclature for Swing-Box Production Replica Method.
6. Also, follow the instructions in Move the CA MDB Data from the Production to the Target Systems based on your database type.

   NOTE
   Also, ensure that the below **step 6 - step 9** are performed appropriately in your environment.
7. After you have completed the Microsoft SQL Server restore process on the target mdb (testqa-mdb-new) server, execute the stored procedure in SQL to fix the orphaned users which are created when restoring a database from one Microsoft SQL Server instance to another.

   Example:
   sp_change_users_login 'AUTO_FIX','mdbadmin'
8. Copy shared filestore directory files from the production server (testqa-catalog1/testqa-catalog2) to the Swing Box server.

   NOTE
   The below steps are applicable only if your production environment is customized. This is not applicable for the out-of-the-box (OOTB) content.
9. If you have a shared filestore, copy from the shared filestore location. This is required to avoid multiple overlaps of common files.
10. Make a copy of the “<usm_home>\filestore\custom" directory files from the **testqa-catalog1**if you have single catalog server.
11. Ensure that you are taking a backup of ***filestore**custom*** directory files on the swing box Production Replica.
    You are required to rebuild your form customizations using CA Service Catalog 17.4 form that is delivered with the CA Service Management installer as earlier customizations for CA Service Catalog will not work.
12. Once you have completed the rebuilding of your customizations, take a backup of your filestore/custom directory files on the swing box/production Replica.
13. Start the CA Service Catalog services.
14. Verify and test the functionality to ensure that CA Service Catalog is up and running with customizations seamlessly as a full replica of your production environment.

Step 3. Upgrade to CA Service Catalog 17.4

Now, you are ready to upgrade to CA Service Catalog 17.4 based on your deployment configuration.

IMPORTANT
Test the migration to become familiar with the processes and issues that you may encounter while migrating your production system. Perform a Microsoft SQL Server backup on the production system. Restore it to the Swingbox/Replica Server and complete the below steps that are mentioned in this section. Ensure that you have considered the following:

Perform the following steps:

1. Mount the CA Service Catalog 17.4 install media and then, run the **setup.exe**, run as **Administrator** from the local folder where you have copied the CA Service Catalog installer.
   a. If you are extracting an ISO of the install media, it must be stored in a path and the folder must have no spaces in the folder name.

      NOTE

      - Download the CA Service Management installer 17.4 from Broadcom Support.

- **CA SERVICE MANAGEMENT SDM PACKAGE 17.4 FOR WINDOWS (WITH CATALOG).iso**
- **CA SERVICE CATALOG 17.4.iso**
- **CA SERVICE MANAGEMENT APM PACKAGE 17.4 FOR WINDOWS (WITH CATALOG).iso**

b. Run locally from the same drive volume where you plan to install CA Service Catalog 17.4.

IMPORTANT
Do not run the installer from a network drive or mounted share as this has been known to cause install issues and failures.

2. Launch the CA Service Management 17.4 installer.
3. Select a language of your choice and click Next.
   For example, select **English**
4. Select CA Service Management.
5. Follow the installation wizard to complete the installation process.
   This may take some time to complete depending upon the size of the CA Service Catalog database that you are upgrading/migrating.

CA Service Catalog Swing Box Migration Intermittent Server - High Level Steps

The below table shows a high-level overview of your production environment, intermittent Servers, along with supported CA Service Catalog versions and upgrade it to target servers 17.4 using the swing-box migration.

NOTE
The below information is applicable for both standalone and multiple server instances of CA Service Catalog.

S.No	Production Environment	Intermittent Server - Migration	Target 17.4 (Swing Box)
1	Version: CA Service Catalog 17.3	• Setup Intermittent server 17.3 with supported OS Example: Windows 2016/ 2019 • **Take snapshot -1** • Move MDB from Production • **Take snapshot -2** • Upgrade to 17.4 • **Take snapshot -3**	• Fresh Install 17.4 same as of deployment. Example: Windows 2016/ • **Take a snapshot - 1** • Move MDB from the Interm • Copy/Merge file customiza the production environmen environment with **1-1** mapp target of Catalog Applicatio have a shared filestore, yo from the shared filestore lo filestore/custom). This is re multiple overlap of commo • Take **snapshot 2** (CA Serv with production data, custo
2	Version: CA Service Catalog 17.4	• Setup Intermittent server 17.4 with supported Operating System. Example: Windows Server 2016/ 2019 • Take a**snapshot-1** • Move MDB and Customizations from the production environment. • Take **snapshot 2** (CA Service Catalog 17.4 with production data, customizations). • Update to latest rollup of 17.4 if required	• Fresh Install 17.4 same as of deployment. Example: Windows Server • Take a **snapshot-1** • Move MDB from the Interm • Copy/Merge file customiza the production environmen environment with **1-1** mapp target Catalog Application • Take **snapshot 2** (CA Serv with production data, custo

Use the Swing Box Intermittent Server to Upgrade to CA Service Catalog 17.4

This method of swing-box migration is performed using an intermittent server to migrate the current production system data along with customizations, and then upgrade it to CA Service Catalog 17.4.

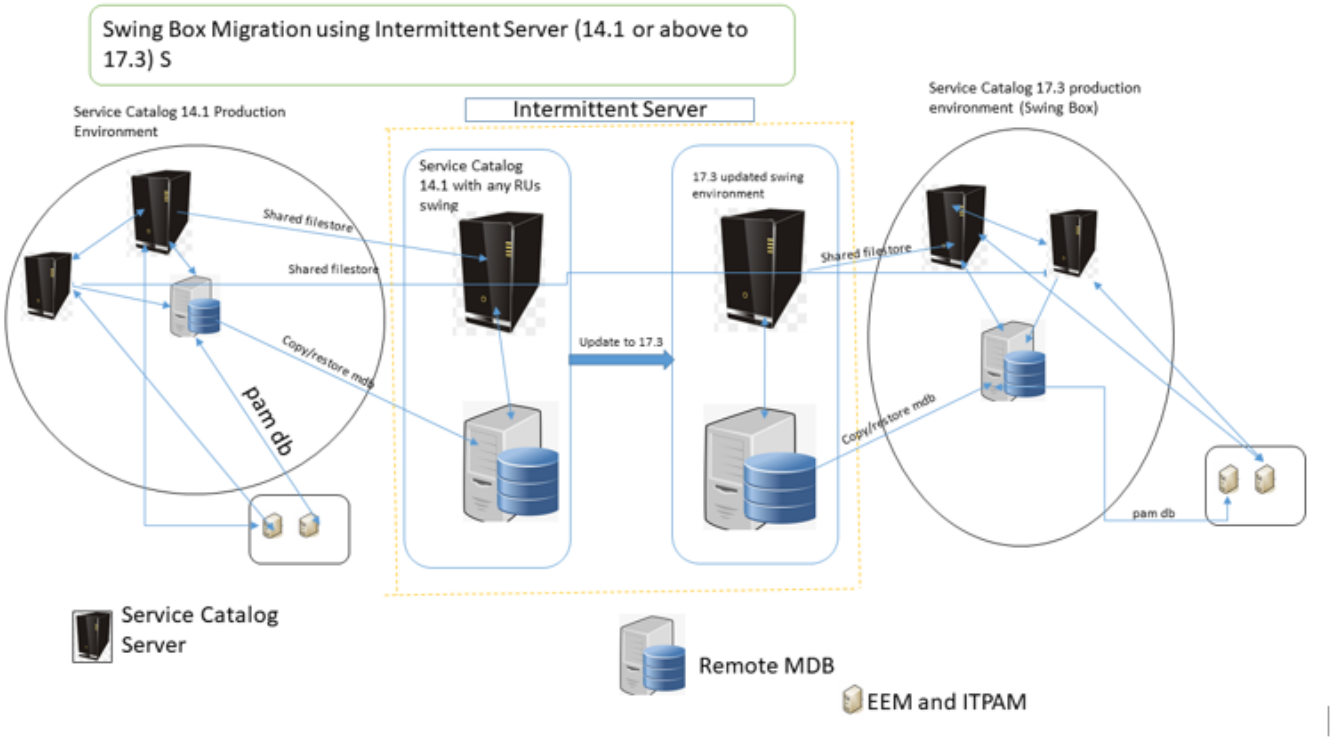
Review and Perform the following:

1. Prepare an Intermittent Server.
   For more information, see Recommended OS and Database Requirements and Server Nomenclature for Swing-Box Intermittent Server Migration.
2. Take a database and file customization backup of your production environment.
   For more information, see Step 1: Collect Files/Data from your Current Production System and Step 2: Prepare the Intermittent Server .
3. Upgrade to CA SLCM 17.4 on the Intermittent Server. For more information, see Step 3: Upgrade to CA SLCM 17.4 on the Intermittent Server.
4. Install a fresh or new instance of CA Service Catalog 17.4 server (a server that you have not upgraded before) on the target server (swing box) with the same configuration and deployment as your production server.
   For more information, see Install CA Service Catalog 17.4.
5. Take a database backup of your intermittent server and customizations from application or secondary servers of the production environment and restore it to the target servers.
   – For more information, see Step 4: Move Upgraded Intermittent Server Install to a Swing-Box (Target) 17.4 .
   – For CA EEM, see CA EEM Policies Backup and Restore on CA Service Catalog server.

Server Nomenclature for Swing-Box Production Replica Method and Process Flow Diagram

The below example server names are used to describe the swing-box production replica method to upgrade CA Service Catalog 17.1/17.2/17.3 or above to CA Service Catalog 17.4 or above release versions.

Production Server	Intermittent Server	Target Server
CA Service Catalog 17.1/17.2/17.3 or above	CA Service Catalog: 17.1/17.2/17.3 Intermittent server	CA Service Catalog 17.4
testqa-mdb	CA Service Catalog 17.1/17.2/17.3 or above testqa-mdb-inter	CA Service Catalog 17.4 testqa-mdb-new
testqa-catalog 1	testqa-app-int er	testqa-catalog1-new
testqa-catalog 2		testqa-catalog2-new



Step 1: Collect Files/Data from your Current Production System

Perform the following steps to collect files or data from your production system:

1. Take a SQL Backup of MDB from the current production database server (**testqa-mdb**) and save it as **.bak** file (usually a normal SQL backup).
2. Make a copy of the shared filestore directory of the (**testqa-catalog1**) and (**testqa-catalog2**) based on the current production deployment model.
   It's recommended that you compile this into a Zip format.

   NOTE
   If you have a shared filestore, you are required to copy from the shared filestore location. This is required to avoid multiple overlap of common files.

   NOTE
   Make a copy of the "<usm_home>\filestore\custom" directory files from **testqa-catalog1** if you have single catalog server.
3. Copy the above files from Step 1, Step 2 (only files copied from **testqa-catalog1**) to your intermittent app server (**testqa-app-inter**) to a standby directory and do not place the files in their respective locations for now.

Step 2: Prepare the Intermittent Server

Perform the following steps:

NOTE
For reference purposes, in this section we will refer to the database server as "**testqa-mdb-inter**" and the Service Catalog application server as "**testqa-app-inter**".

1. It is assumed that you have already installed Microsoft SQL server on **testqa-mdb-inter**.
2. On **testqa-app-inter** server, ensure to install the SQL client and native client (workstation tools).

3. Install the same release version of production server CA Service Catalog on **testqa-app-inter**and run configuration so that you have a basic installation of CA Service Catalog running in your environment (without customizations).
4. As database administrator, execute the Microsoft SQL Restore and restore the MDB from your current production that you have backed up, to the**testqa-mdb-inter** server.
5. As database administrator, execute the Microsoft SQL Restore command and restore the MDB from your current production (**testqa-mdb**) that you have backed up on the **testqa-mdb-inter** server.

   NOTE
   Ensure to set the option to **OVERWRITE** the entire database when restoring.

   – Here, source mdb server will be referred as **testqa-mdb** and target mdb server is **testqa-mdb-inter**. For more information, see Server Nomenclature for Swing-Box Production Replica Method.
   – Also, follow the instructions in Moving the CA MDB Database on Microsoft SQL Server based on your database type (MS SQL or Oracle).

   NOTE
   Additionally, ensure that the below **step 6 - step 9** are performed appropriately in your environment.
6. After you have completed the Microsoft SQL Server restore process on the target mdb (**testqa-mdb-inter**) server, execute the stored procedure in SQL to fix the orphaned users which are created when restoring a database from one Microsoft SQL Server instance to another.
   For Example:
   ```
   sp_change_users_login 'AUTO_FIX','mdbadmin'
   ```
7. Copy shared filestore directory files from the production server (**testqa-catalog1**/**testqa-catalog2**) to the swing box server.

   NOTE
   The below steps are applicable only if your production environment is customized. This is not applicable for the out-of-the-box (OOTB) content.
8. If you have a shared filestore, copy from a shared filestore location. This is recommended to avoid multiple overlap of common files.
9. Make a copy of the “Program Files\CA\Service Catalog\filestore\custom" directory files from the **testqa-catalog1** if you have a single catalog server.
10. Ensure that you are taking a backup of filestore\custom directory files on the swing box (production replica).
    You are required to rebuild your form customizations using CA Service Catalog 17.1/17.2/17.3 or above form packaged with the CA Service Management installer as the earlier customized CA Service Catalog forms will not work. Once you have completed the rebuilding of your customizations, take a backup of your filestore/custom directory files on the swing box/production Replica.
11. Start the CA Service Catalog services.
12. Verify and test the functionality to ensure that CA Service Catalog is up and running with customizations seamlessly as a full replica of your production environment.

Step 3: Upgrade to CA Service Catalog 17.4 on the Intermittent Server

Test the migration to become familiar with the processes and issues that you may encounter while migrating your production system.

Ensure that you have considered the following:

- If your Intermittent Server is on a VMware environment, it is recommended that you take a snapshot. You can roll back to that snapshot when you have a down time of your production system for maintenance or other purposes.
- Perform a Microsoft SQL Server backup on the production system. Restore it to the Intermittent Server and complete the below steps mentioned in this section.
  In case, if the migration to CA Service Catalog 17.4 fails, you can revert to the CA Service Catalog production system and start the services.
- Retest migration on the intermittent server again and schedule the migration for another time.

Perform the following steps to upgrade to CA Service Catalog 17.4 on the Intermittent Server:

1. Mount the CA Service Catalog 17.4 install media and then, run the **setup.exe**, run as **Administrator** from the local folder where you have copied the CA Service Catalog installer.
   a. If you are extracting an ISO of the install media, it must be stored in a path and the folder must have no spaces in the folder name, for example: catalog174Setup.

      NOTE

      Download the CA Service Management installer 17.4 from Broadcom Support.

      - **CA SERVICE MANAGEMENT SDM PACKAGE 17.4 FOR WINDOWS (WITH CATALOG).iso**
      - **CA SERVICE CATALOG 17.4.iso**
      - **CA SERVICE MANAGEMENT APM PACKAGE 17.4 FOR WINDOWS (WITH CATALOG).iso**

   b. Run locally from the same drive volume where you plan to install CA Service Catalog 17.4.

      NOTE
      Do not run the installer from a network drive or mounted share as this has been known to cause install issues and failures.
2. Launch the CA Service Management 17.4 installer.
3. Select a language of your choice and click **Next**.
   For example, select **English**
   .
4. Select **CA Service Management**.
5. Follow the installation wizard to complete the installation process.
   This may take some time to complete depending upon the size of the CA Service Catalog database that you are upgrading/migrating.

CA EEM Policies Backup and Restore on CA Service Catalog server

Perform the following steps to backup CA EEM data on CA Service Catalog server (**testqa-catalog1**):

1. Open the CA Service Catalog command prompt. Click **Start**, **Programs**, **CA**, and **Service Catalog**, **usm Command Prompt** (% usm_home% \usm.cmd) from where you have installed Catalog 17.1/17.2/17.3 and above.
   Example: **testqa-catalog1**
   a. Run the following command:
      ```
      ant backup-eem-app
      ```
   b. During execution, provide input parameters like EEM password and so on.
   c. After successful completion, "**eem-backup.xml**" file is generated in the *%USM_HOME%* directory.

      NOTE
      If you have multiple CA Service Catalog computers, perform this step on the first CA Service Catalog computer. Run this command once, regardless of the number of CA Service Catalog servers in your environment.
2. Perform the following steps to restore CA EEM on the target Catalog server (**testqa-app-inter**):
   a. Copy the file (**eem-backup.xml**) from **testqa-catalog1**to the **testqa-app-inter** in the *%USM_HOME%* folder.
   b. Run the following command:
      ```
      ant restore-eem-app
      ```
   c. After successful completion related entries are updated in the EEM.
   d. Restart CA Service Catalog services.
3. At this point, if you are doing a test run on the intermittent server, you must rebuild your customizations using CA Service Catalog 17.4 packaged with the installer as the earlier CA Service Catalog 17.1/17.2/17.3 or above customizations will not work.
4. Once you have completed the rebuilding of your customizations, take a backup of your shared filestore directory on the Intermittent Server and store it somewhere outside of this environment for later use.

5. Test the migrated and customized CA Service Catalog 17.4 Intermittent Server for integration and functionality.
6. After successful completion, start the process of moving your migrated data and customizations to a target or swing box 17.4 environment.

Step 4: Move Upgraded Intermittent Server Install to a Swing-Box (Target) 17.4

Your swing-box (target) environment must already be up and running with a clean or fresh installation of CA Service Catalog 17.4 same as your production deployment model.

CA EEM Policies Backup and Restore on CA Service Catalog server

Perform the following steps to backup CA EEM data on the source Catalog server (**testqa-app-inter**):

1. Open the CA Service Catalog command prompt. Click**Start**, **Programs**, **CA**, **Service Catalog**, usm Command Prompt (*%usm_home%\usm.cmd*) from the Catalog 17.1/17.2/17.3 or above installed source server:
   Example: **testqa-catalog1**
2. Run the following command:
   ```
   ant backup-eem-app
   ```
   NOTE
   During execution, provide input parameters like EEM password and so on. After successful completion, "**eem-backup.xml**" file is generated in the %USM_HOME% directory.
   NOTE
   If you have multiple CA Service Catalog computers, perform this step on the first CA Service Catalog computer. Run this command once, regardless of the number of CA Service Catalog servers in your environment.
3. Perform the following steps to restore CA EEM on the target Catalog server (t**estqa-catalog1-new**):
   a. Copy the file (eem-backup.xml) from **testqa-app-inter**to the **testqa-catalog1-new**in the %USM_HOME% folder.
   b. Run the following command:
      ```
      ant restore-eem-app
      ```
   c. After successful completion related entries are updated in CA EEM.
   d. Restart CA Service Catalog services.
4. Perform a Microsoft SQL Server backup on the intermittent server (**testqa-mdb-inter**) and have your database administrator perform a Microsoft SQL Server Restore onto the SQL database instance on your Swing-box or target server (**testqa-mdb-new**).
   NOTE
   Ensure to set the option to **OVERWRITE** the entire database when restoring.
5. Here, source mdb server will be referred as **testqa-mdb-inter** and target mdb server is **testqa-mdb-new**. For more information, see Server Nomenclature.
6. Also, follow the instructions in Move the CA MDB Data from the Source to the Target Systems based on your database type (MS SQL or Oracle).
   NOTE
   Additionally, ensure that the below step 6 - step 9 are performed appropriately in your environment.
7. After you have completed the Microsoft SQL Server restore process on the target mdb (**testqa-mdb-inter**) server, execute the stored procedure in SQL to fix the orphaned users which are created when restoring a database from one Microsoft SQL Server instance to another.
   For example:
   ```
   sp_change_users_login 'AUTO_FIX','mdbadmin'
   ```
8. Copy the shared filestore directory files from the production server (**testqa-catalog1**/**testqa-catalog2**) to the swing box server.

If you have a shared filestore, copy from a shared filestore location. This is recommended to avoid multiple overlap of common files.

NOTE
The below steps are applicable only if your production environment is customized. This is not applicable for the out-of-the-box (OOTB) content.

9. Make a copy of the "*Program Files**CA**Service Catalog**filestore**custom*" directory files from the **testqa-catalog1** if you have a single catalog server.
10. Ensure that you are taking a backup of filestore\custom directory files on the swing box (production replica).
    You are required to rebuild your form customizations using CA Service Catalog 17.4 form packaged with the CA Service Management installer as the earlier customized CA Service Catalog forms will not work. Once you have completed the rebuilding of your customizations, take a backup of your filestore/custom directory files on the swing box/production Replica.
11. Start the CA Service Catalog services.
12. Test the system functionality to ensure the swing-box or target environment with CA Service Catalog 17.4 is up and running with data and customizations intact.

Troubleshooting- Swing-Box Migration Issues

This article describes the following Swing-Box Migration issues and how to troubleshoot:

Unable to Submit Catalog Request from Service Point
Problem: Unable to submit CA Service Catalog request from service point as a result getting a below error message on Chrome browser Version V80 + only.
For more information, see Chrome Browser issues in Service Point.
Solution: Enable the following flags to disable this constraint:

- SameSite by default cookies
- Cookies without SameSite must be secure

Perform the following to disable Chrome flags:

1. Open Chrome browser.
2. Enter Chrome://flags.
3. Search with above flags and disable.

Reconfigure CA EEM
If you are using CA EEM for authentication on your CA SDM environment, and if the configured CA EEM is not accessible from CA SDM or if you want to configure a new CA EEM server, perform the following steps:

1. Edit the NX.env file and set **USE_EIAM_FOR_AUTH=No**.
2. Restart the CA Service Desk Manager services.
3. Login to CA Service Desk Manager as an administrator.
4. Navigate to Options Manager, and reconfigure all CA EEM options to point to the appropriate CA EEM server, along with proper credentials and configurations for that server.
5. Edit the NX.env file again to **USE_EIAM_FOR_AUTH=Yes**.
6. Restart CA Service Desk Manager Services again.

Use Case 1 - Swing-Box Migration to Upgrade CA Service Management 17.4

This article describes the Swing Box Migration use case scenario to upgrade to CA Service Management 17.4 and above. CA Service Management installation comprises CA Service Desk Manager and CA Service Catalog along with CA common components (CA EEM and CA Process Automation).

Perform the following:

- Upgrade from CA Service Management 17.1 (production) to CA Service Management 17.1 (Target)
- Upgrade to CA Service Management 17.3 update
- Update to CA Service Management 17.3.0.2
- Update to CA Service Management 17.4

> **NOTE**
> These steps are mentioned for a particular use case, and are not mandatory. You can upgrade from 17.1 to 17.4

Product Installation Sequence

Install the CA Service Management Solution products in the sequence shown below:

1. CA EEM Server

   > **NOTE**
   > Do not enable Active Directory on EEM Application. Enable Active Directory after installing CA Service Management 17.4.
2. CA Process Automation
3. CA Service Desk Manager
4. CA Service Catalog

Server Nomenclature

The following server/host names were used in-house for testing the procedures outlined in this article for installing, upgrading CA Service Management 17.1 to CA Service Management 17.3 to updating it to CA Service Management 17.3.0.2, and upgrade to CA Service Management 17.4.

Following server names are used for CA Service Catalog installation and upgrade scenarios wherever applicable for Production and Target servers.

- **mdb**: MDB database.
- **catalog1**: Catalog Production Server 1
- **catalog1-new**: Replica of Catalog Production Server 1
- **catalog 2**: Catalog Production Server 2
- **catalog2-new**: Replica of Catalog Production Server 2
- **new**: If server names are appended with new, they denote the target servers.

After installing CA Service Management 17.3, update it to CA Service Management 17.3.0.2, and upgrade to CA Service Management 17.4.

production System (CA Service Management 17.1)	**Target System (Swing Box move from production CA Service Management 17.1)**	**Upgrade Management 17.3.0.2, a**
Remote MDB: **testqa-mdb**	Remote MDB: **testqa-mdb-new**	Remote MDB: **testq**
CA EEM Server: **testqa-eem**	CA EEM Server: **testqa-eem-new**	CA EEM Server: **tes**
CA Process Automation: **testqa-pam**	CA Process Automation: **testqa-pam-new**	CA Process Automa
CA Service Desk Manager: **Advance Availability**		
Background Server (BG Server): **testqa-bg**	Background Server: **testqa-bg-new**	Background Server:
Standby Server: **testqa-sb**	Standby Server: **testqa-sb-new**	Standby Server: **tes**

Application Server 1: **testqa-app1** Application Server 2: **testqa-app1**	Application Server 1: **testqa-app1-new** Application Server 2: **testqa-app1-new**	Application Server 1 Application Server 2
CA Service Catalog Server: **testqa-sc**	CA Service Catalog Server: **testqa-sc-new**	Application Server 3 CA Search Server: **t** Virtual Analyst (RAS CA Service Catalog

NOTE
Assumption is that the production system is already installed with CA Service Management 17.1 in your environment. The below procedures are for replicating the production system to the target system using the Swing Box mechanism to CA Service Management 17.3, then update it to CA Service Management 17.3.0.2, and upgrade to CA Service Management 17.4.

Perform the following:

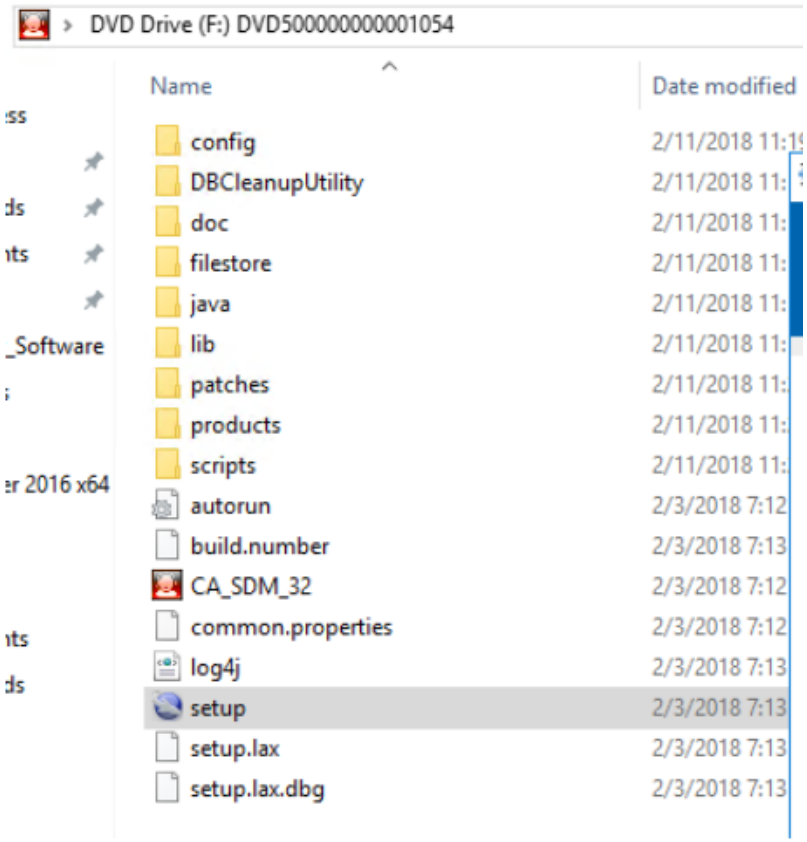
- Install CA EEM (testqa-eem-new)
- Install CA Process Automation 4.3.0.5 (testqa-pam-new)
- Install CA Service Management 17.1
- CA Service Management Solution Integration
- Upgrade CA Service Management 17.1 to CA Service Management 17.4
- Troubleshooting

Install CA EEM (testqa-eem-new)

WARNING
Do not enable Active Directory on CA EEM server (testqa-eem-new) for a successful install experience. After installing CA Service Management 17.3.0.2, enable Active Directory in your environment.

Perform the following steps:

1. Download CA Service Management Installer 17.1 from Broadcom Support.
2. Mount CA Service Management 17.1. DVD on **testqa-eem-new**.
3. Select **Setup.exe**, right-click Run as **Administrator**.
4. Select a language of your choice. For example: English
   Click **Next**.
5. Click CA Embedded Entitlements Manager (CA EEM) to run the CA EEM installation wizard.
6. CA EEM installation process begins.
7. Click **OK**.
8. Click **Next**.
9. Review and accept the License Agreement.
10. Enter the path of Shared Components and CA Directory.
11. Click **Next**.
12. Click **Next**.
    Provide the CA Directory Information (Port/DB size).
13. Enter Password and Confirm Password in the "EiamAdmin Password" Screen.
14. Click **Next**. Enter the EiamAdmin Password
15. Click **Next**. Enable FIPS information, if required.



16. Click **Install**.
    CA EEM Installation Process begins.
17. Click **Done** after successful installation.
18. Open and launch the CA EEM URL *https://<hostname>:<portnumber>/spin/eiam> For example: https://testqa-eem-new:5250/spin/eiam.*
19. Enter User Name and Password in the EEM login page.
20. Click Log In to login to CA EEM.
    CA EEM Home page is displayed.

This completes the CA EEM Installation process.

Install CA Process Automation 4.3.0.5 (testqa-pam-new)

This articles describes the CA Process Automation Installation process.

Prerequisites
Review the following supported JDK versions for CA Process Automation:

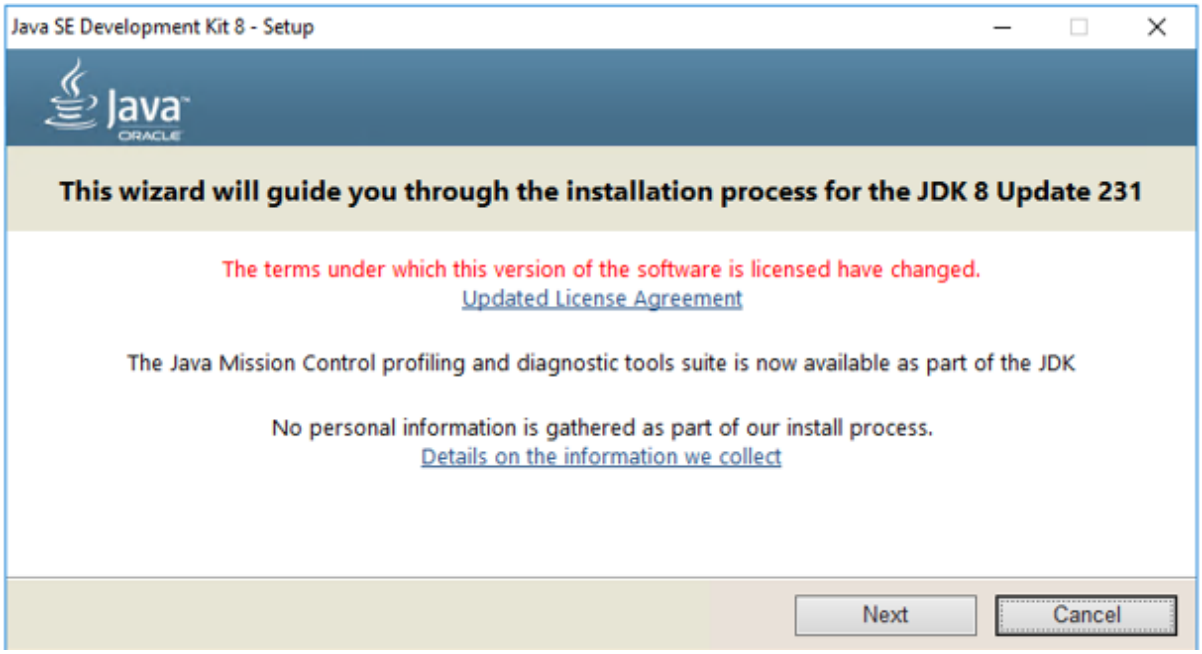
- Oracle Java Development Kit (JDK) 1.8.162 and above OR
- One of the following JDK requirements:
  - AdoptOpenJDK HotSpot jdk8u222-b10
  - AdoptOpenJDK OpenJ9 jdk8u222-b10_openj9-0.15.1 along with AdoptOpenJDK icedtea-web-1.8.3

For more information, see JDK Prerequisites.

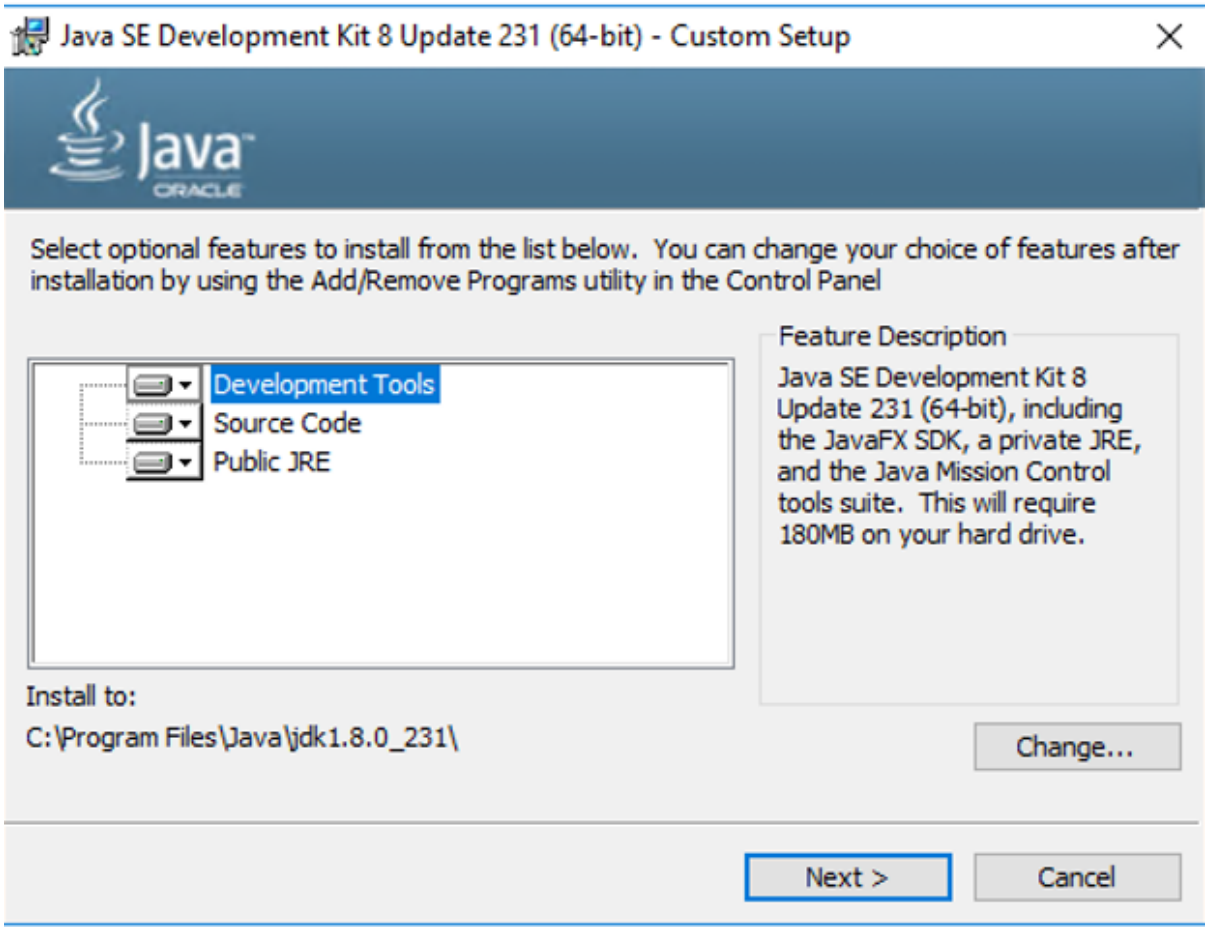
To install CA Process Automation Workflow, perform the following steps. For example: **testqa-pam-new**.

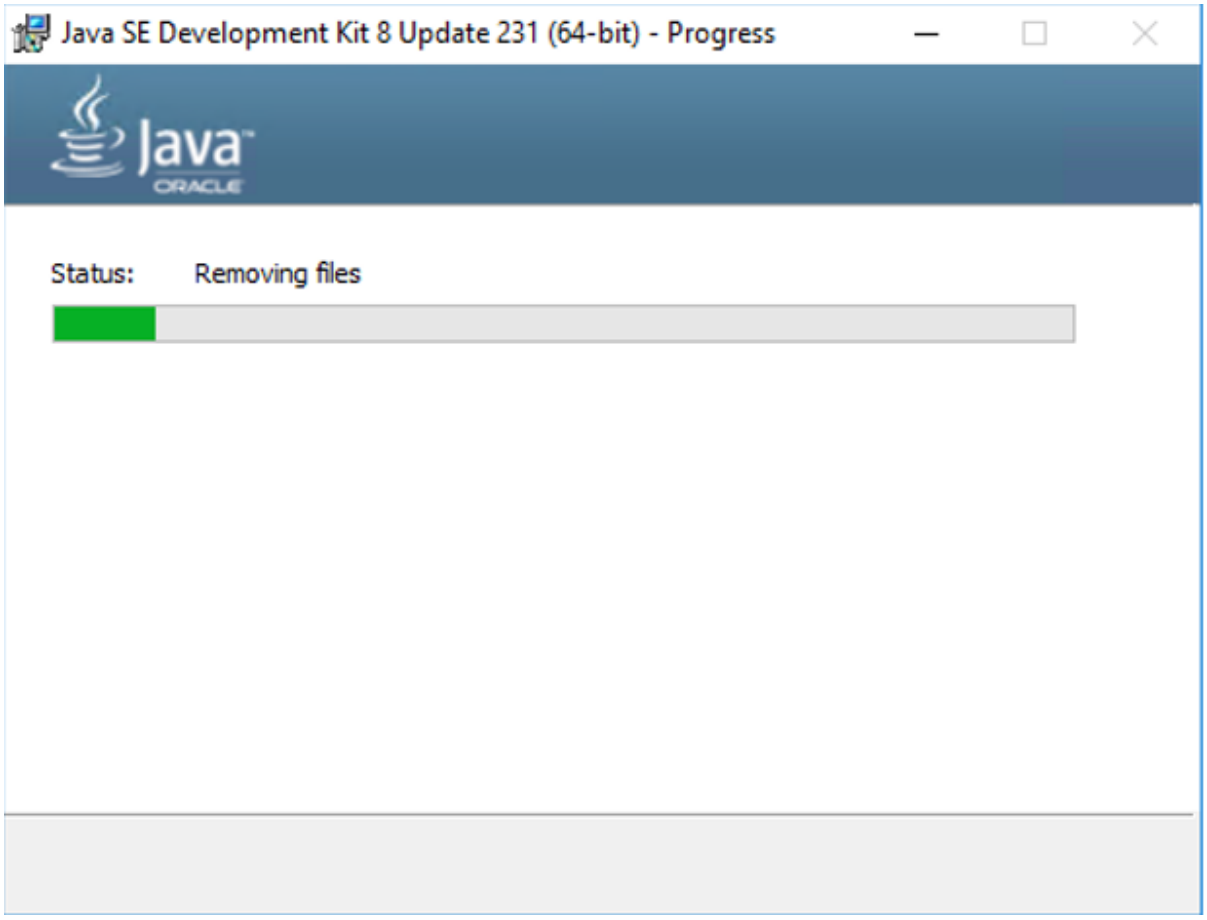
Install JDK

1. Download JDK. Click **setup.exe** and right click to select "**Run as Administrator**" option.
   Click **Next**.

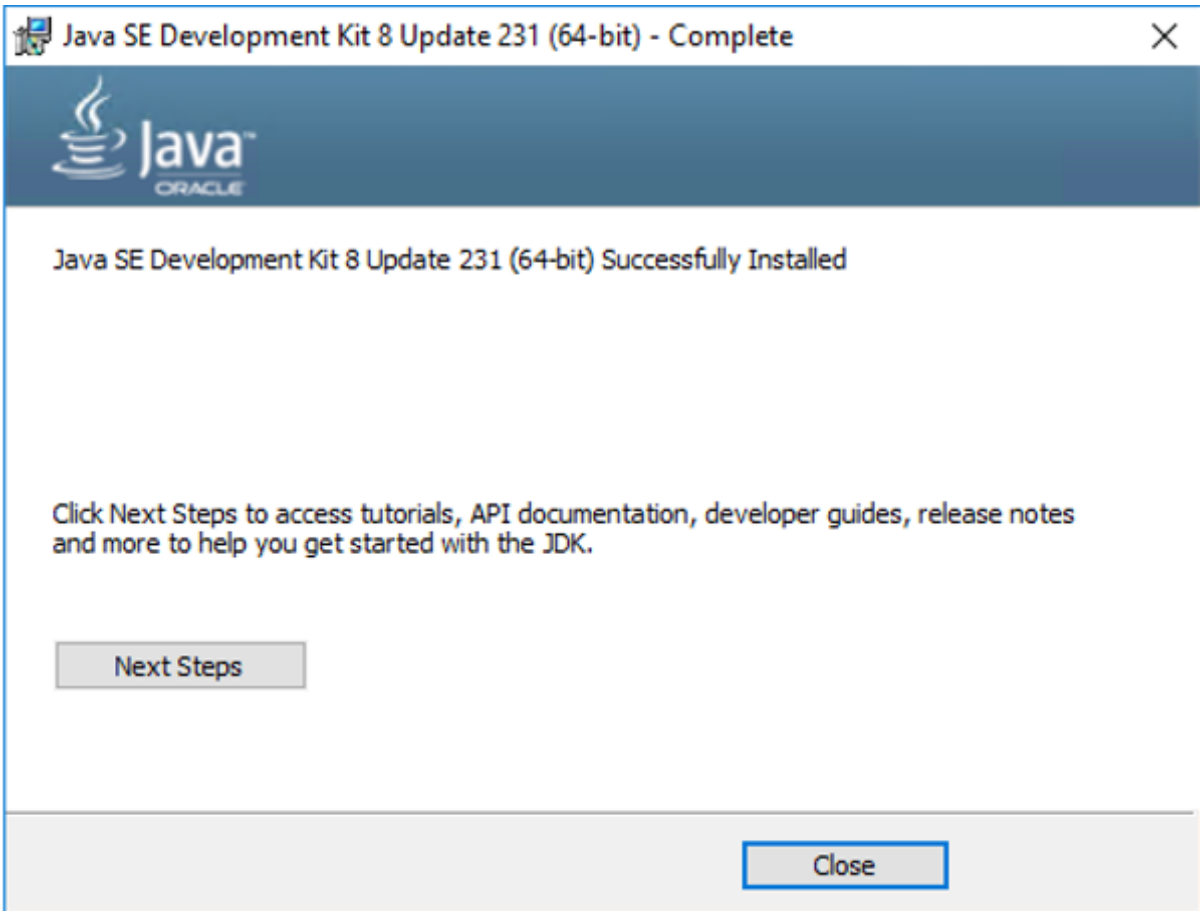


2. Click **Next**.


Java SE Development Kit 8 Update 231 (64-bit) - Custom Setup
Java
ORACLE
Select optional features to install from the list below. You can change your choice of features after installation by using the Add/Remove Programs utility in the Control Panel
Development Tools
Source Code
Public JRE
Feature Description
Java SE Development Kit 8 Update 231 (64-bit), including the JavaFX SDK, a private JRE, and the Java Mission Control tools suite. This will require 180MB on your hard drive.
Install to:
C:\Program Files\Java\jdk1.8.0_231\
Change...
Next >
Cancel



3. Follow the installation Wizard.

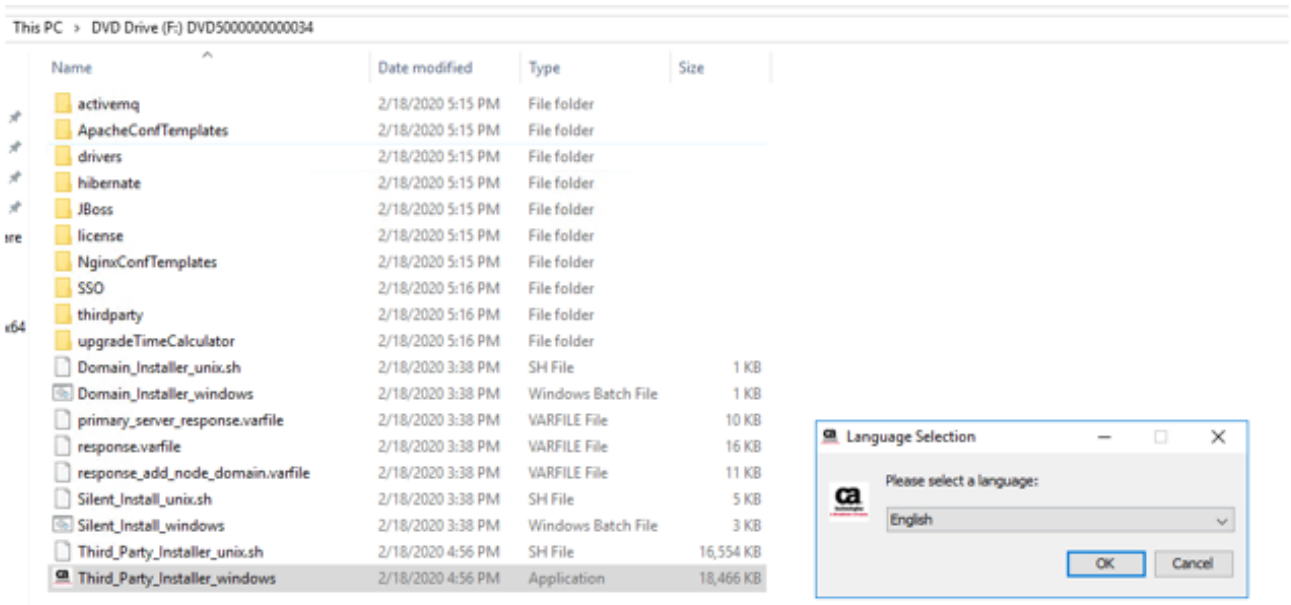


4. Click Close once the installation is successful.

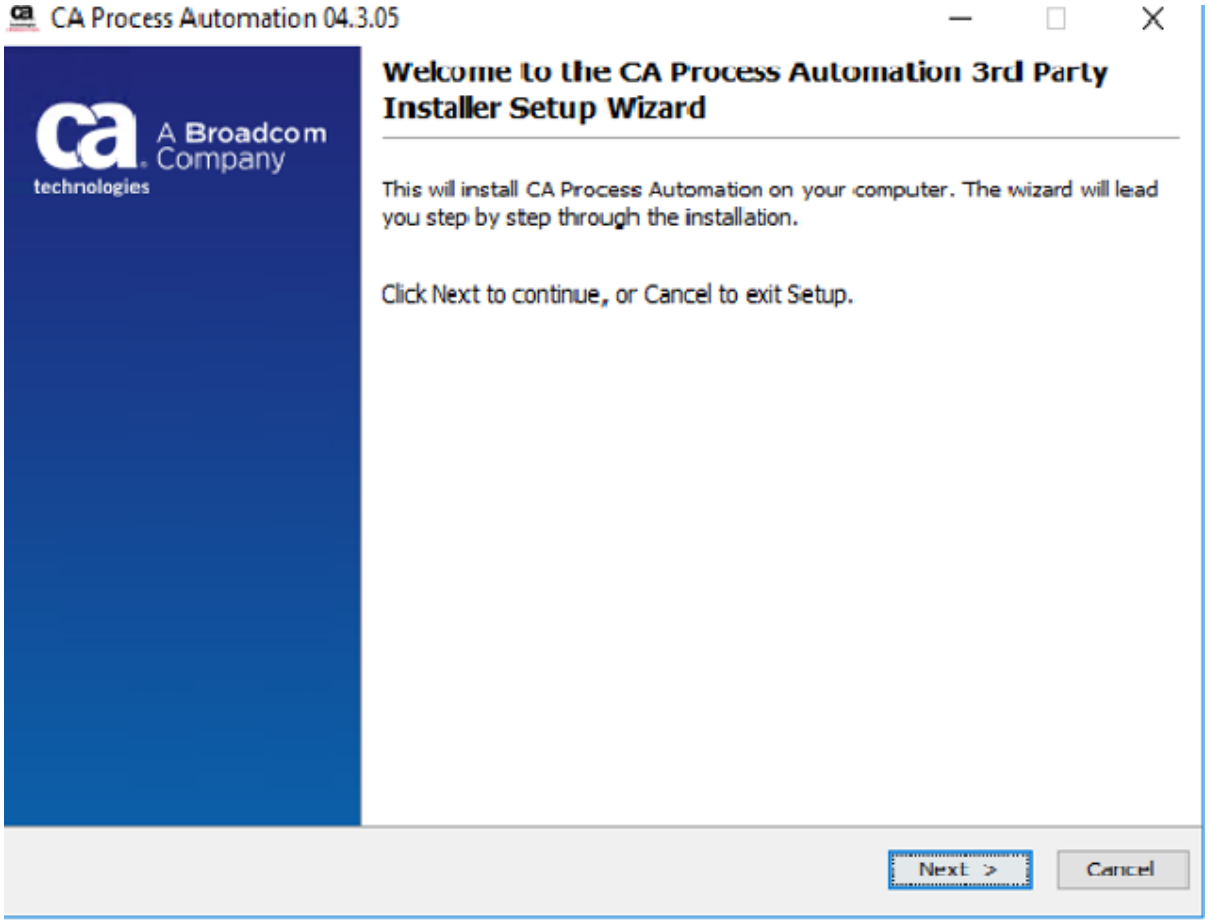
Install CA Process Automation

Perform the following steps:

1. Mount the CA Process Automation **DVD (DVD500000000003410)**on **testqa-pam-new** and Select "**Third_Party_Installer_Installer_windows**" and right-click to select "**Run as Administrator**".

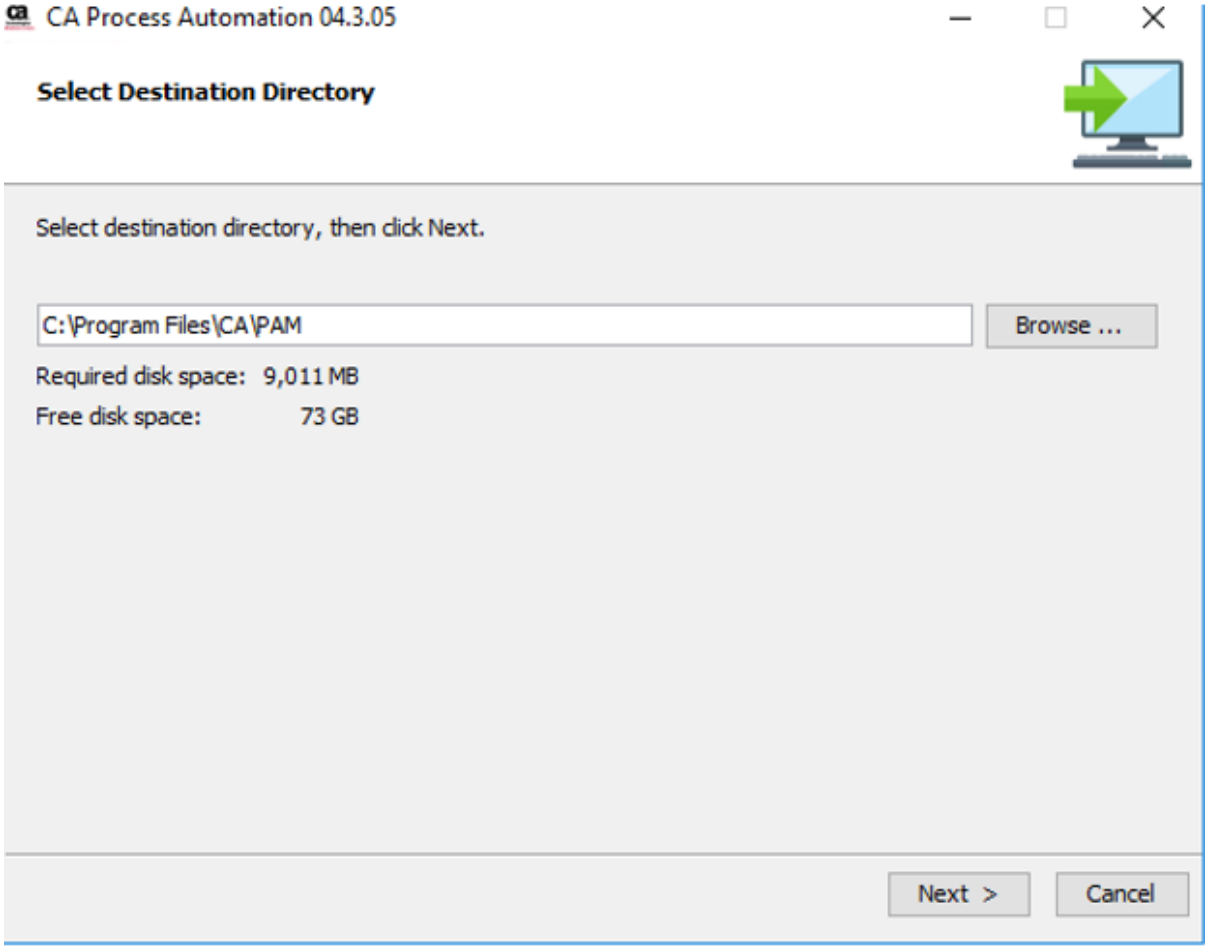


2. Click OK.
3. Click Next.

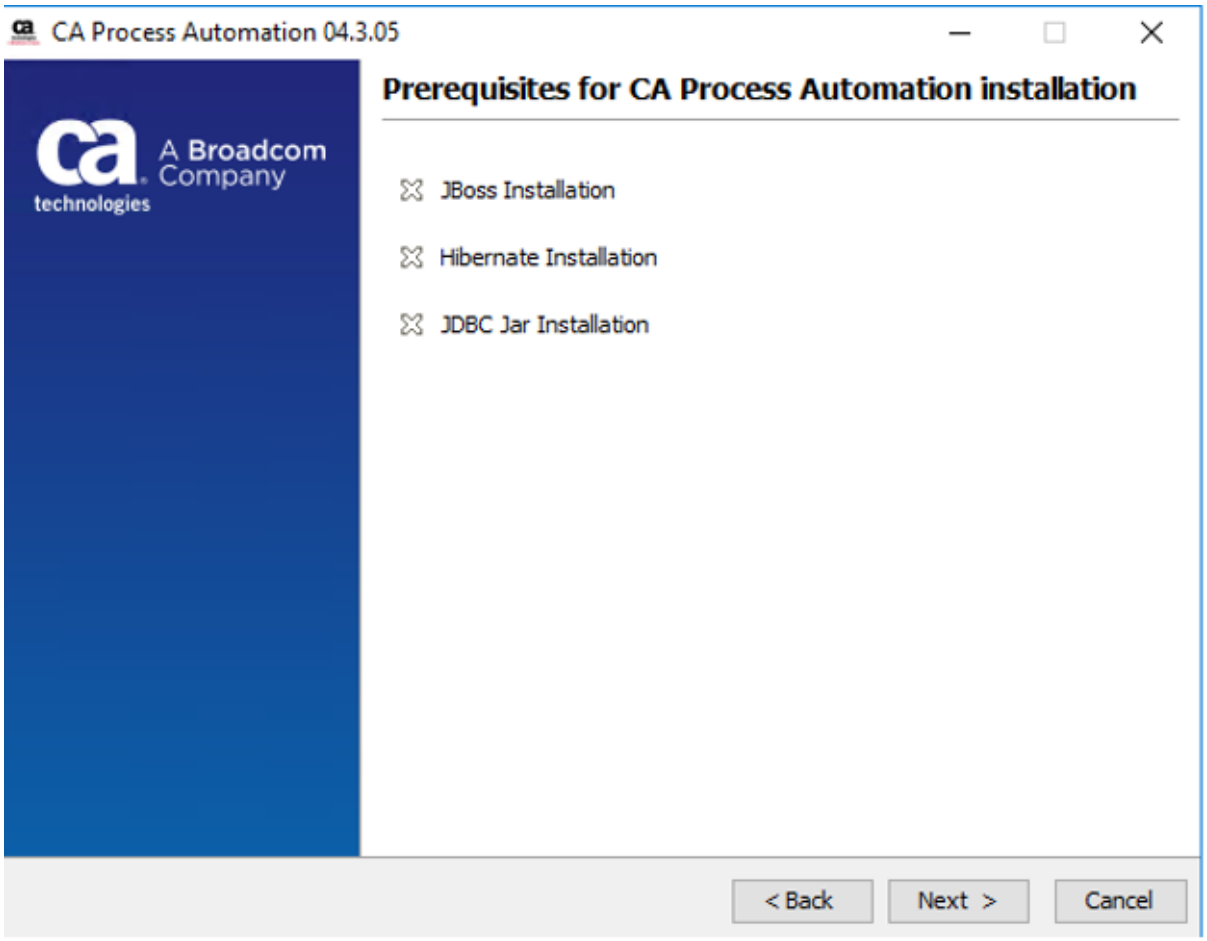


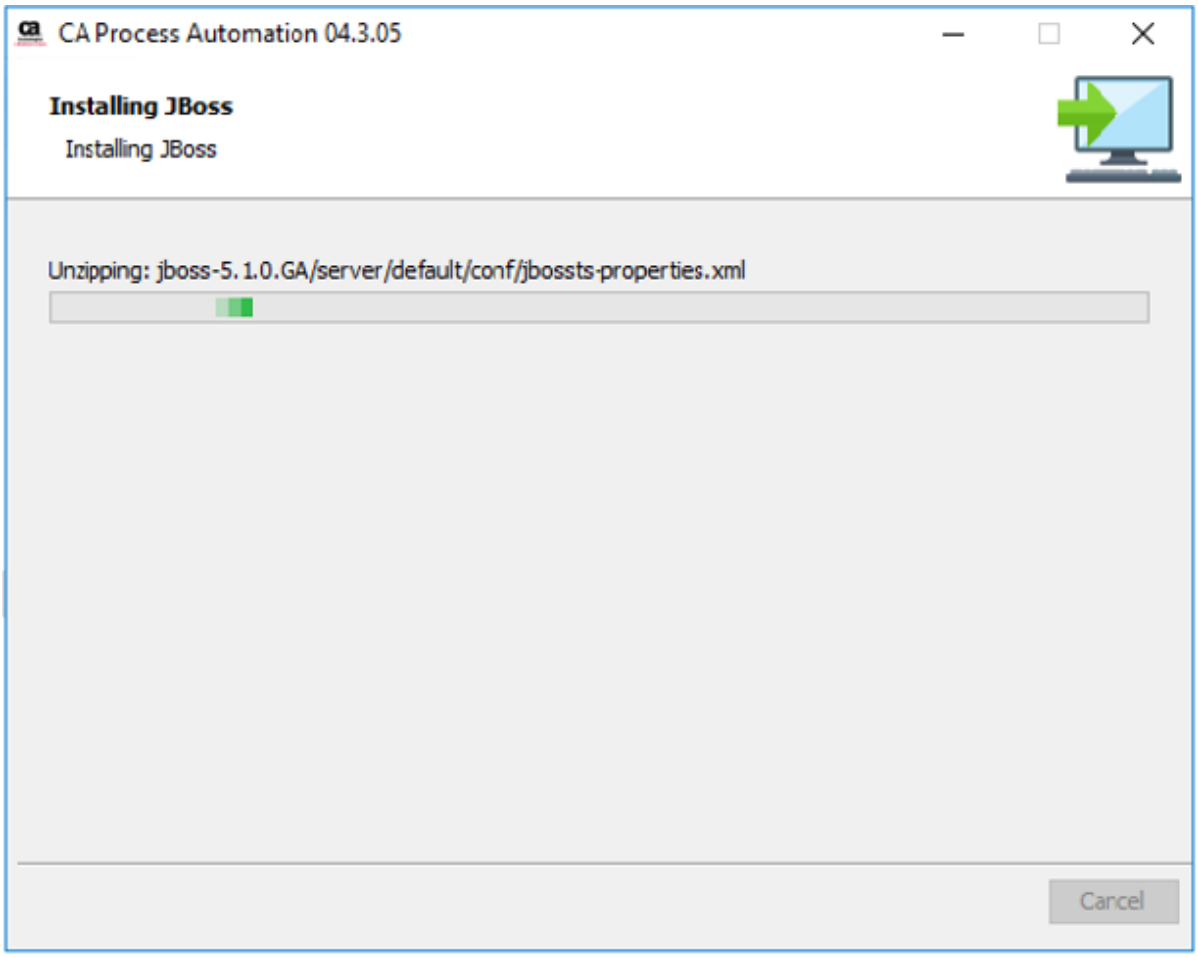
4. Review and accept the license agreement.

5. Click **Next** and enter the installation path by selecting the destination directory.

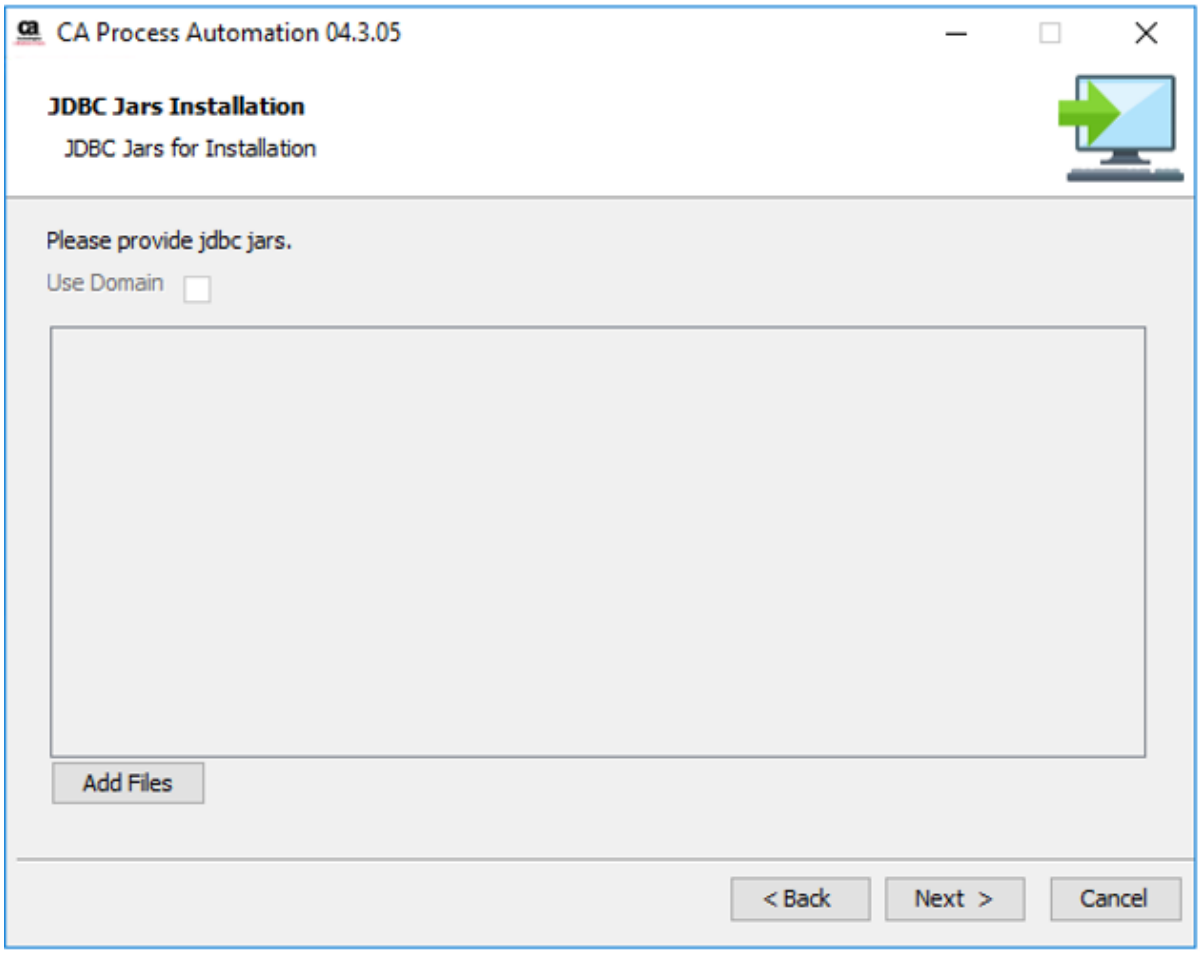


6. Click **Next**. Provide the **JDBC**jar location path.
7. Click **Next**.
   The Installation process begins.
8. Verify that you completed the prerequisites shown below.

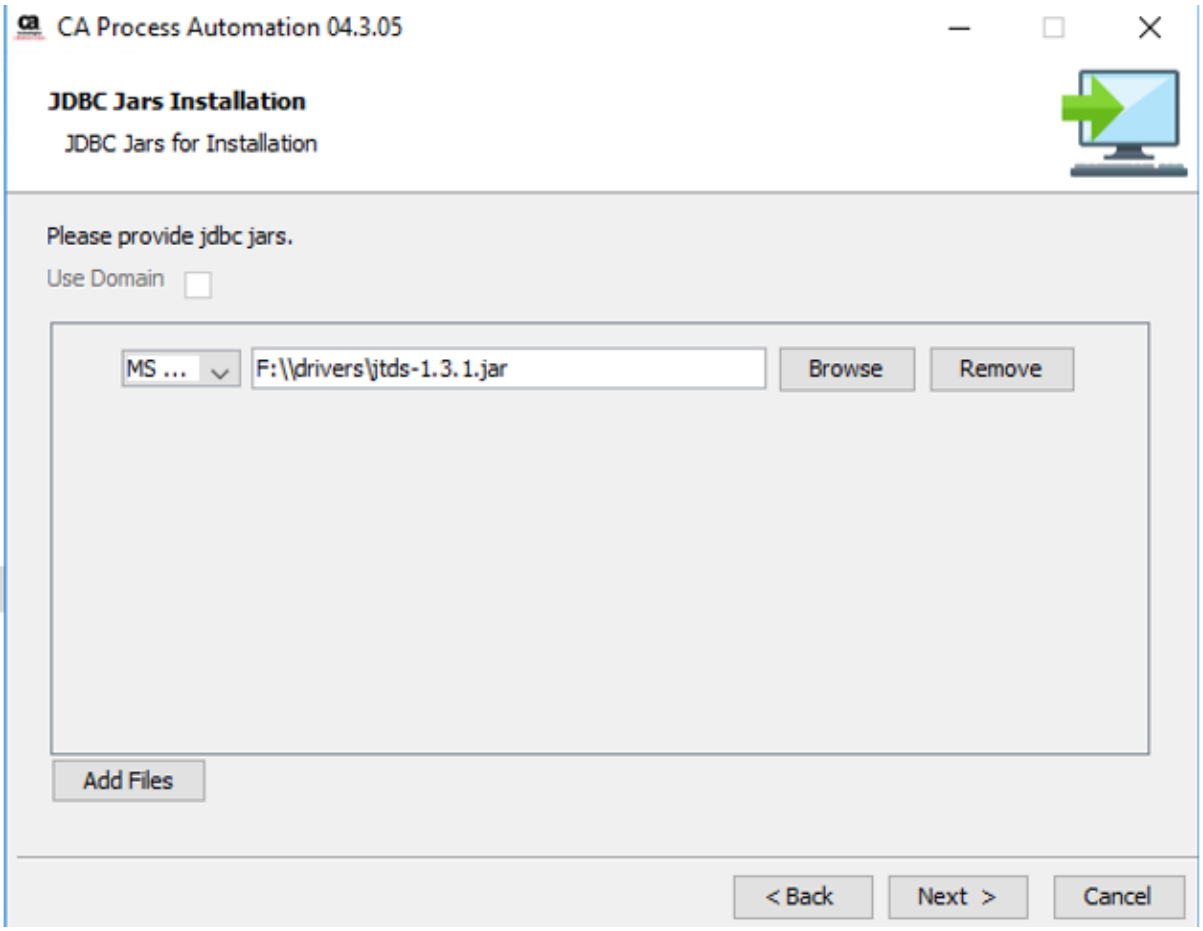

CA Process Automation 04.3.05
A Broadcom Company
technologies
Prerequisites for CA Process Automation installation
JBoss Installation
Hibernate Installation
JDBC Jar Installation
< Back
Next >
Cancel



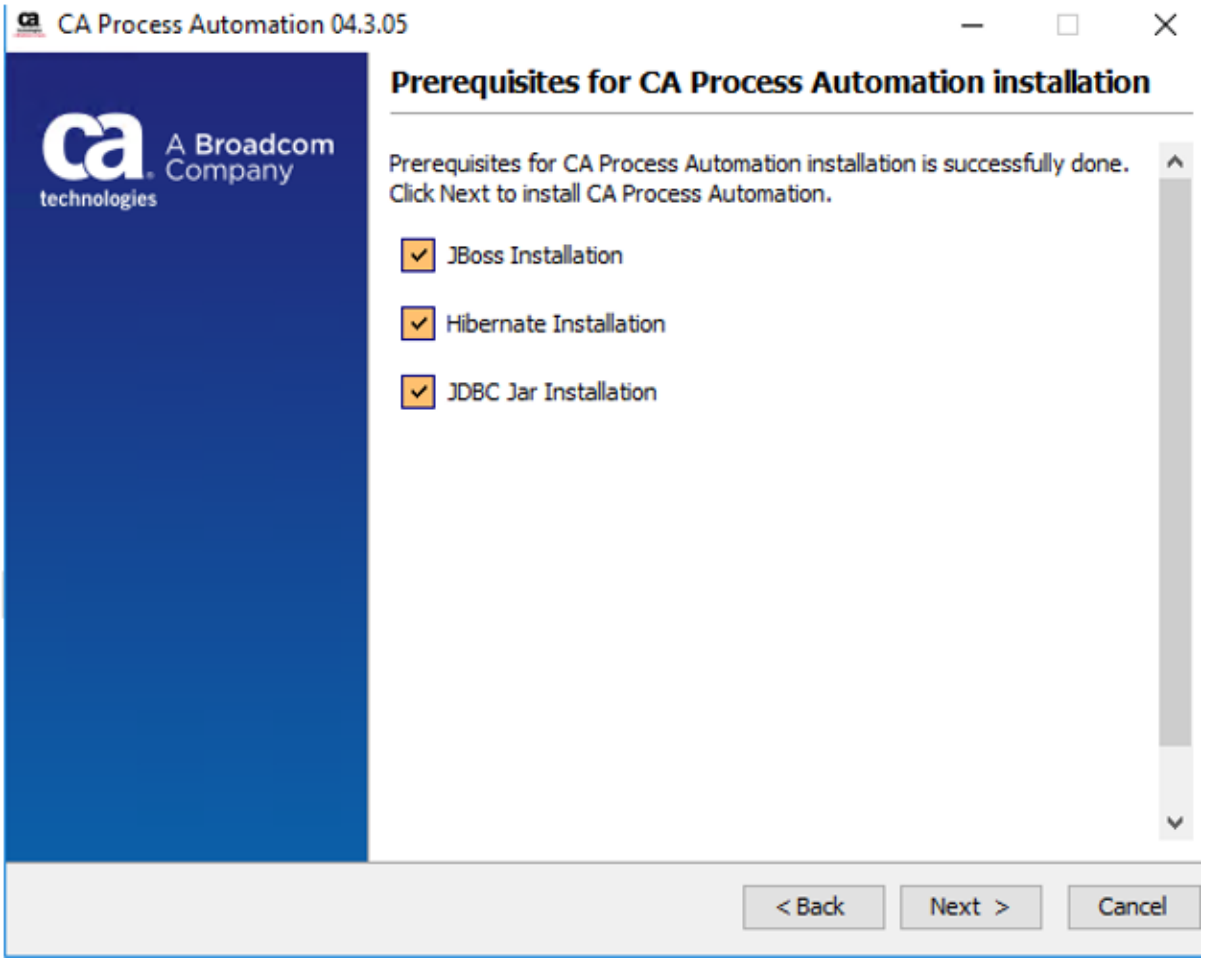
9. Click "**Add Files**" and Select the Microsoft SQL Server.



10. Click **Next**. Provide the JDBC jar location path.



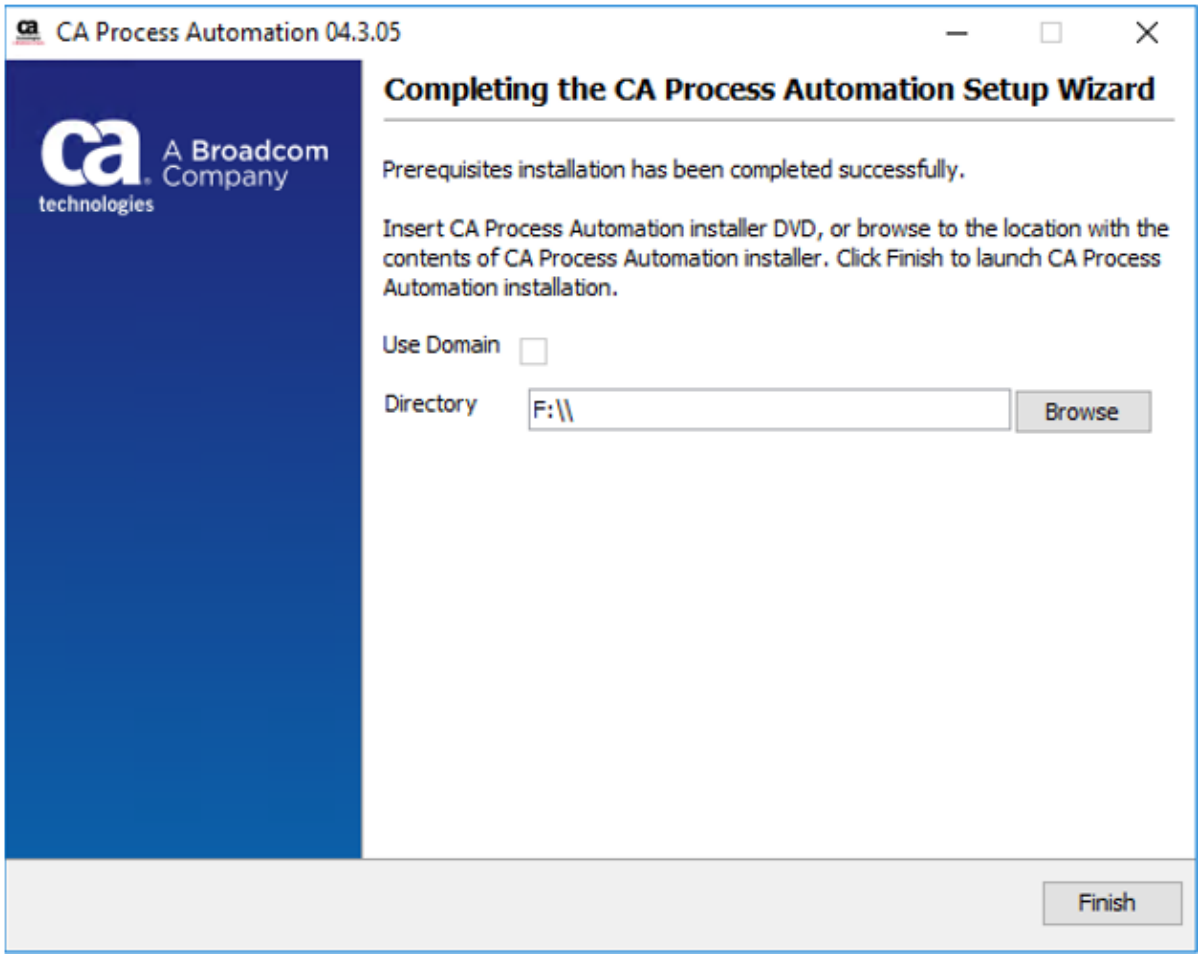
11. Click **Next**.
12. Review the installation prerequisites and proceed if these components are installed in your environment.



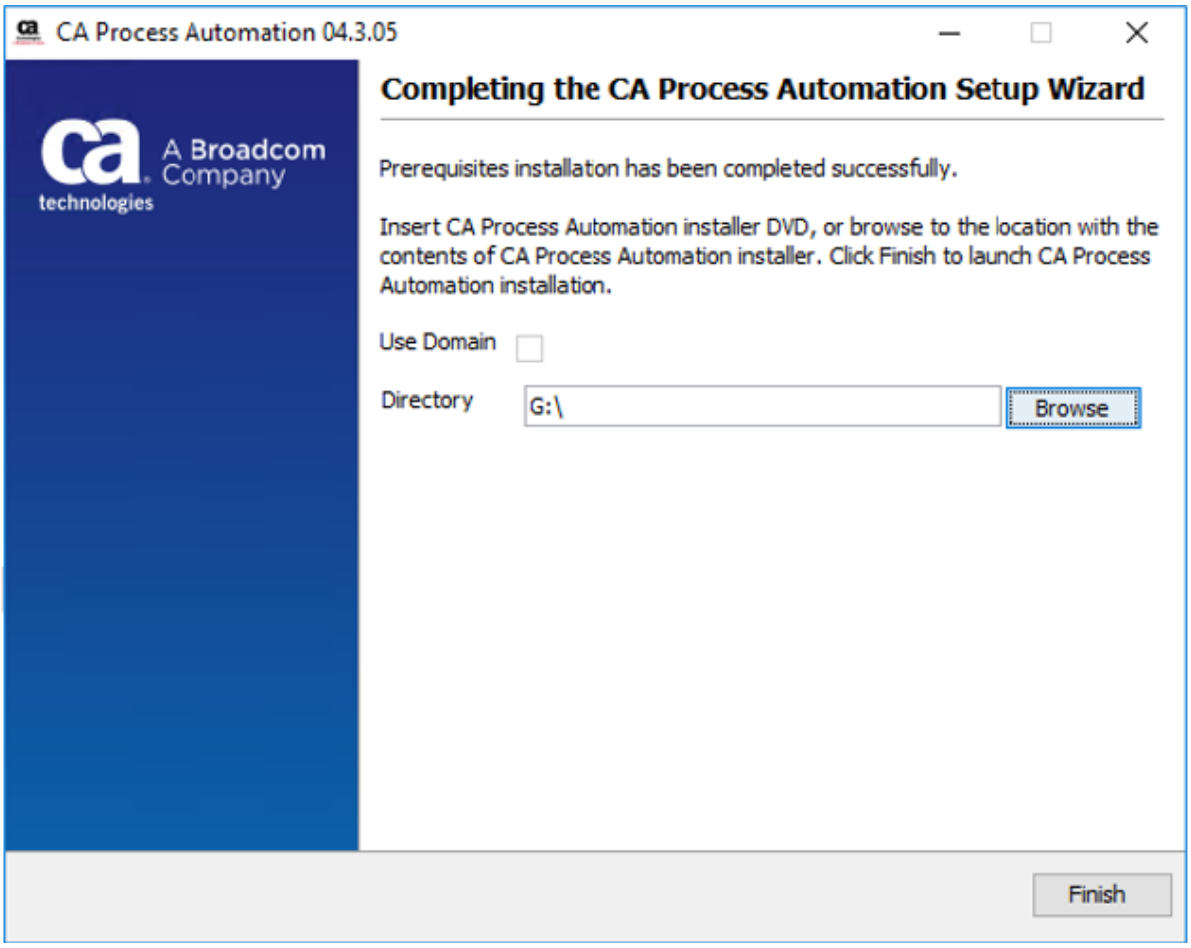
13. Mount the CA Process Automation second **DVD (DVD500000000003420)** on **testqa-pam-new**.

    NOTE

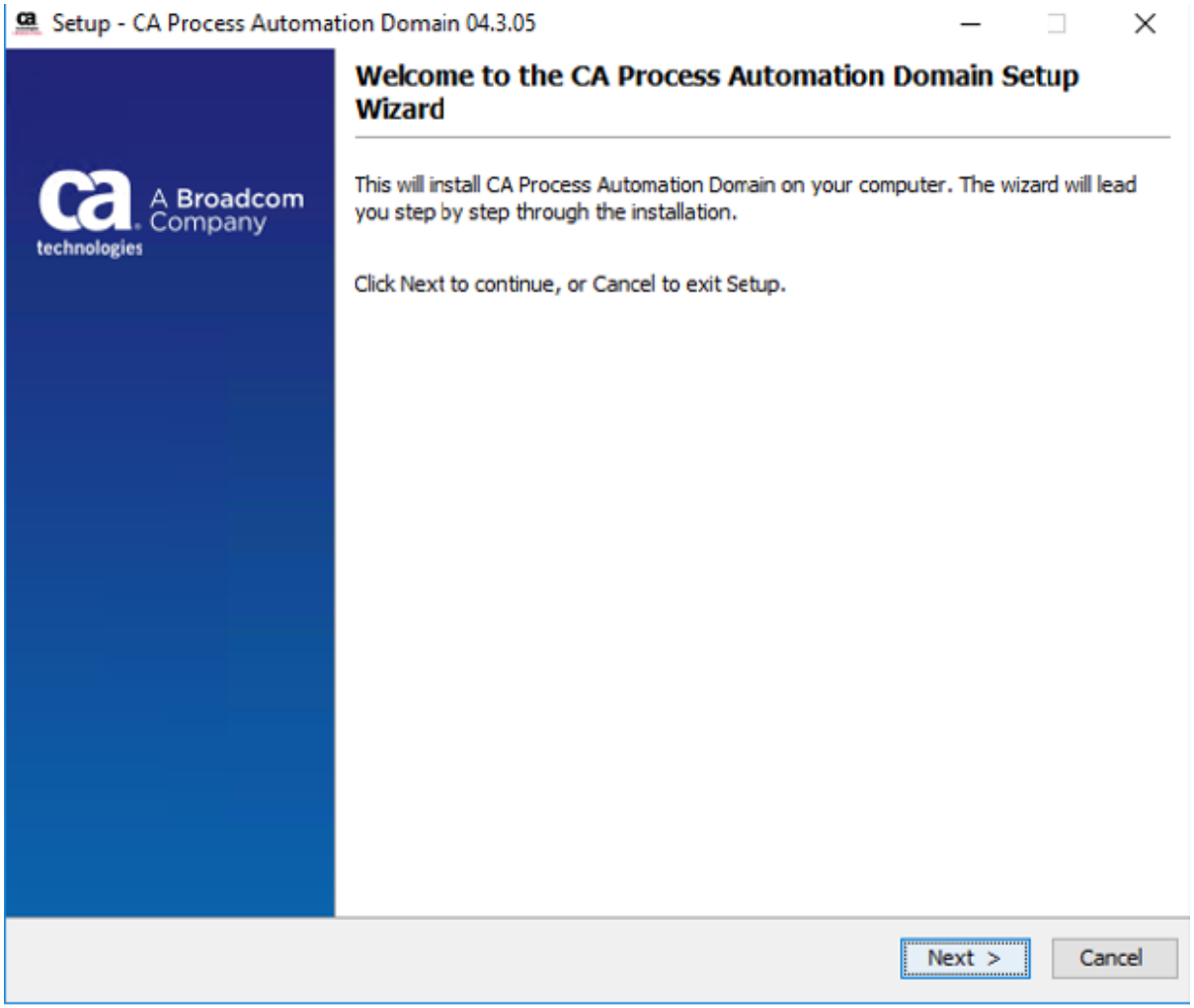
    For example: Mount**DVD500000000003420** on G: drive and change the directory path to G: drive. The first PAM DVD is mounted on F: drive as shown in the image.



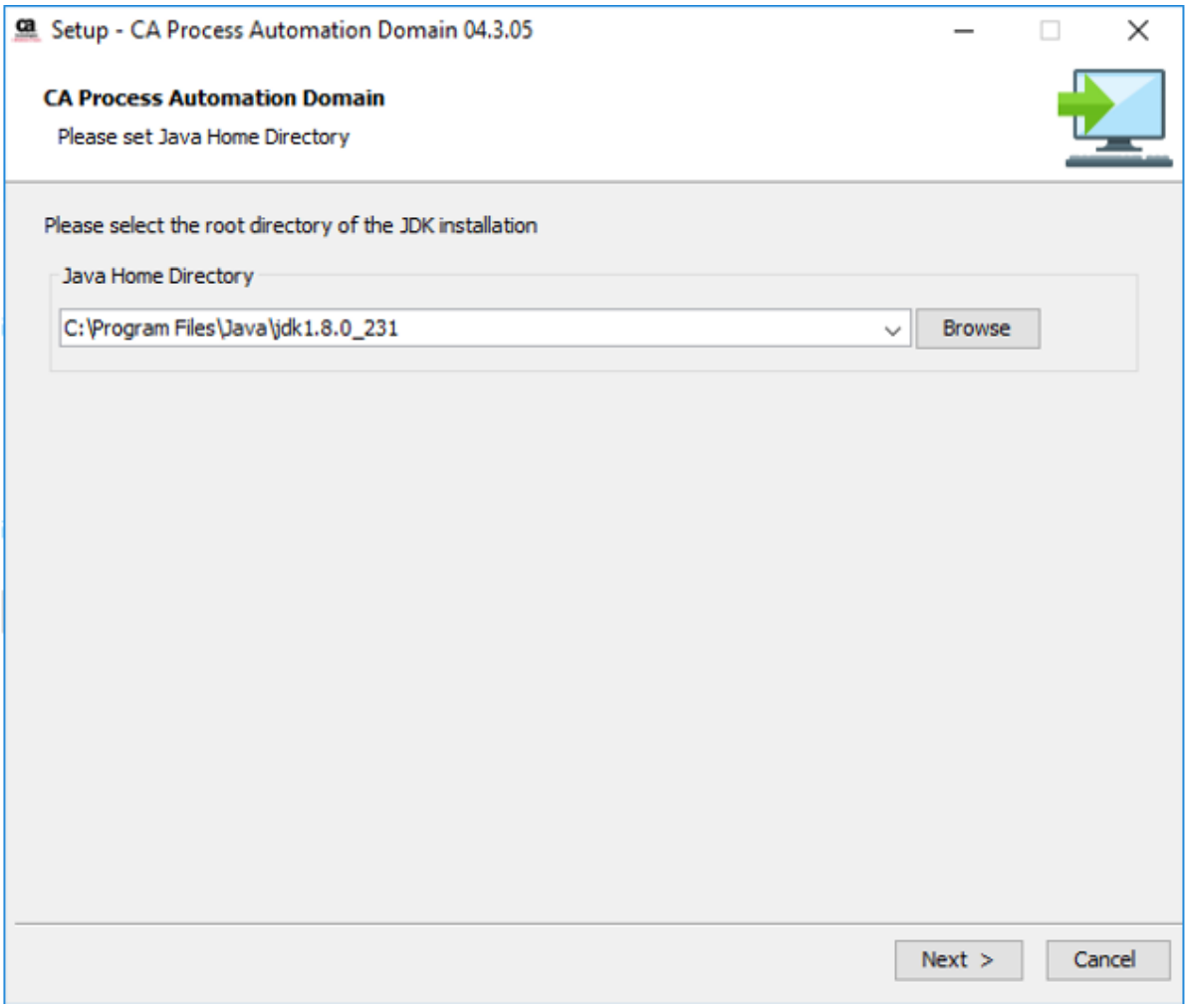
14. Click **Finish**.
    Installer takes some time to fetch details for the second DVD installation screen.



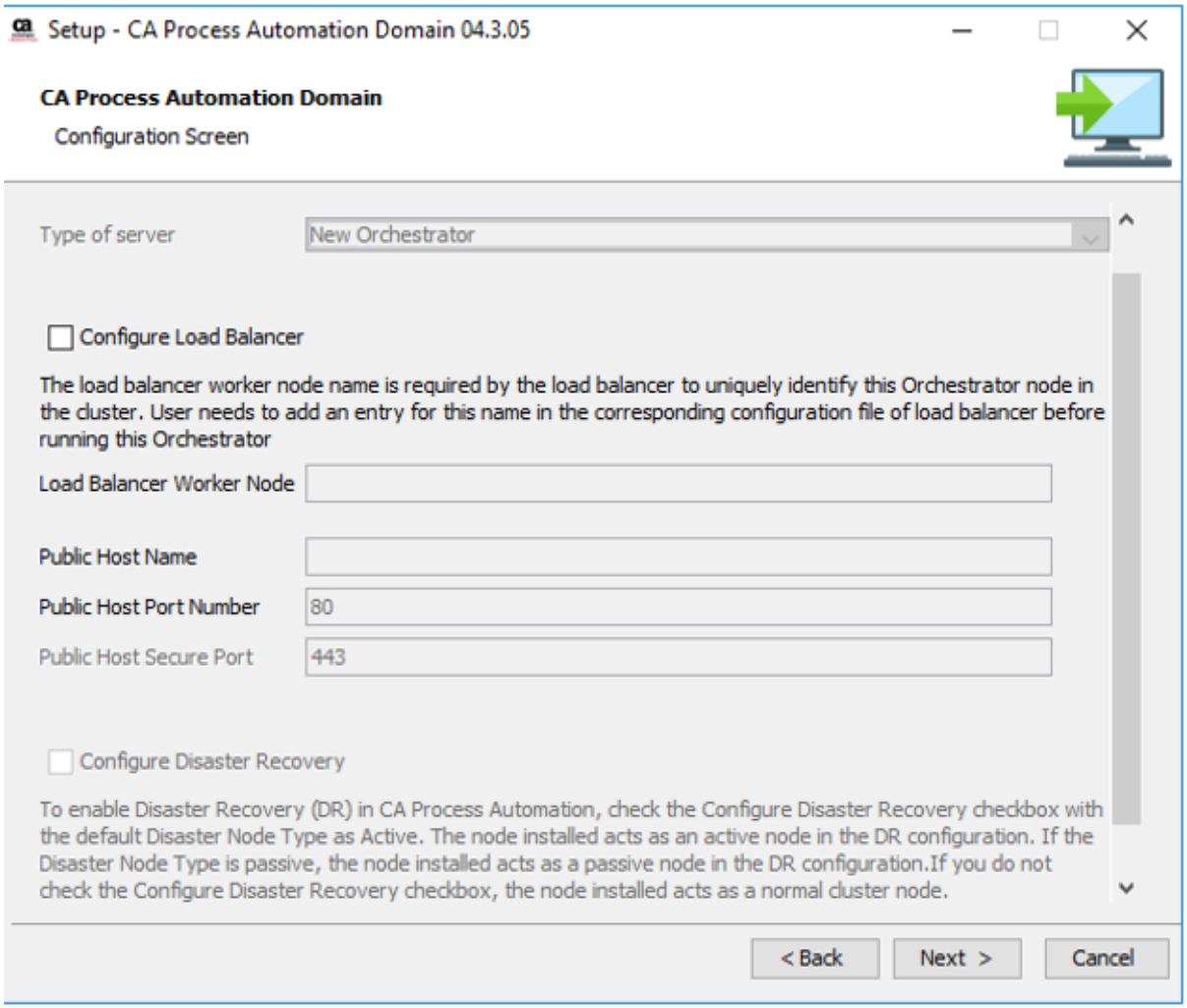
15. Click**Next**.



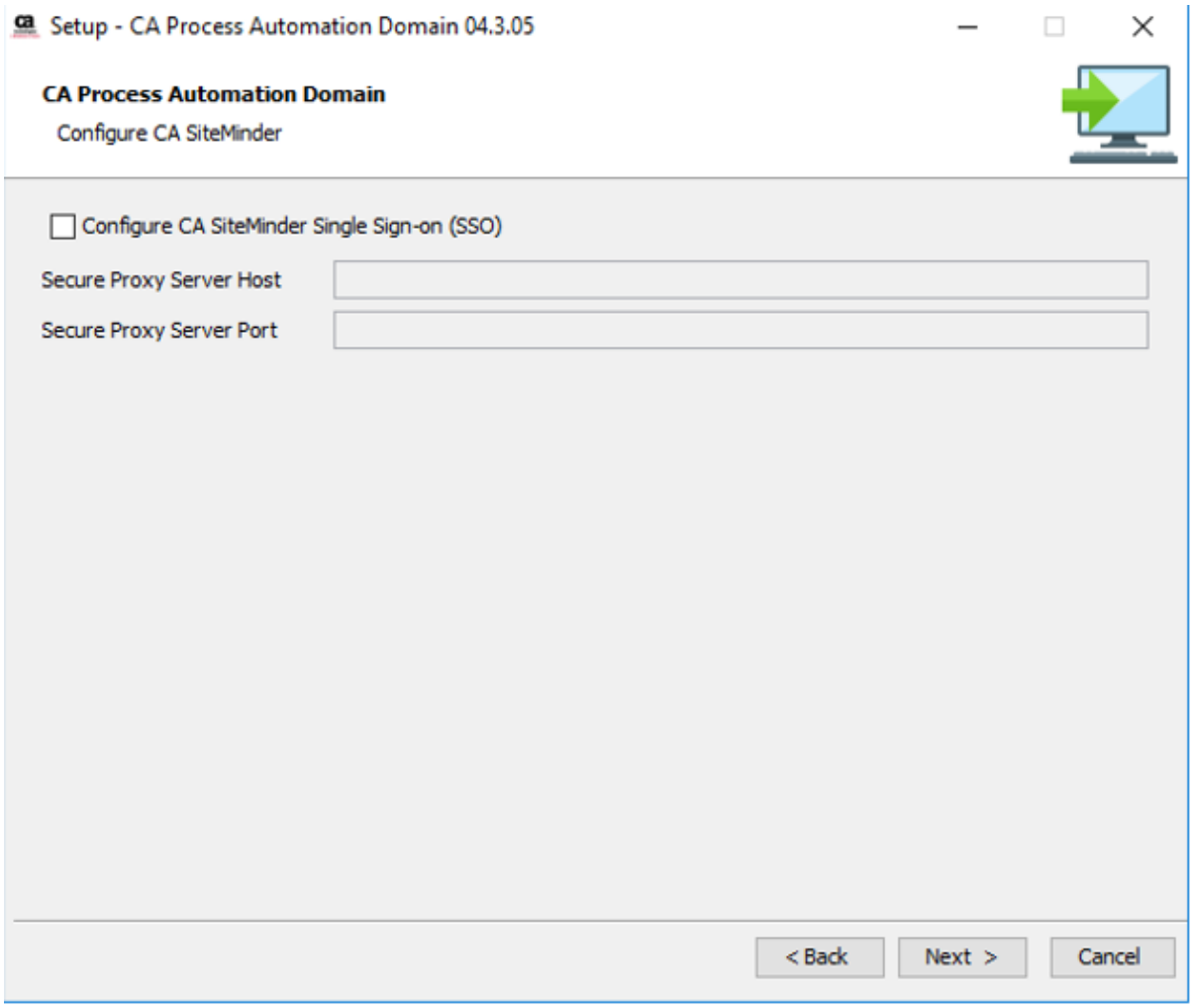
16. Review and accept the License Agreement.
17. Click **Next** and select the java home path (JDK installation location).



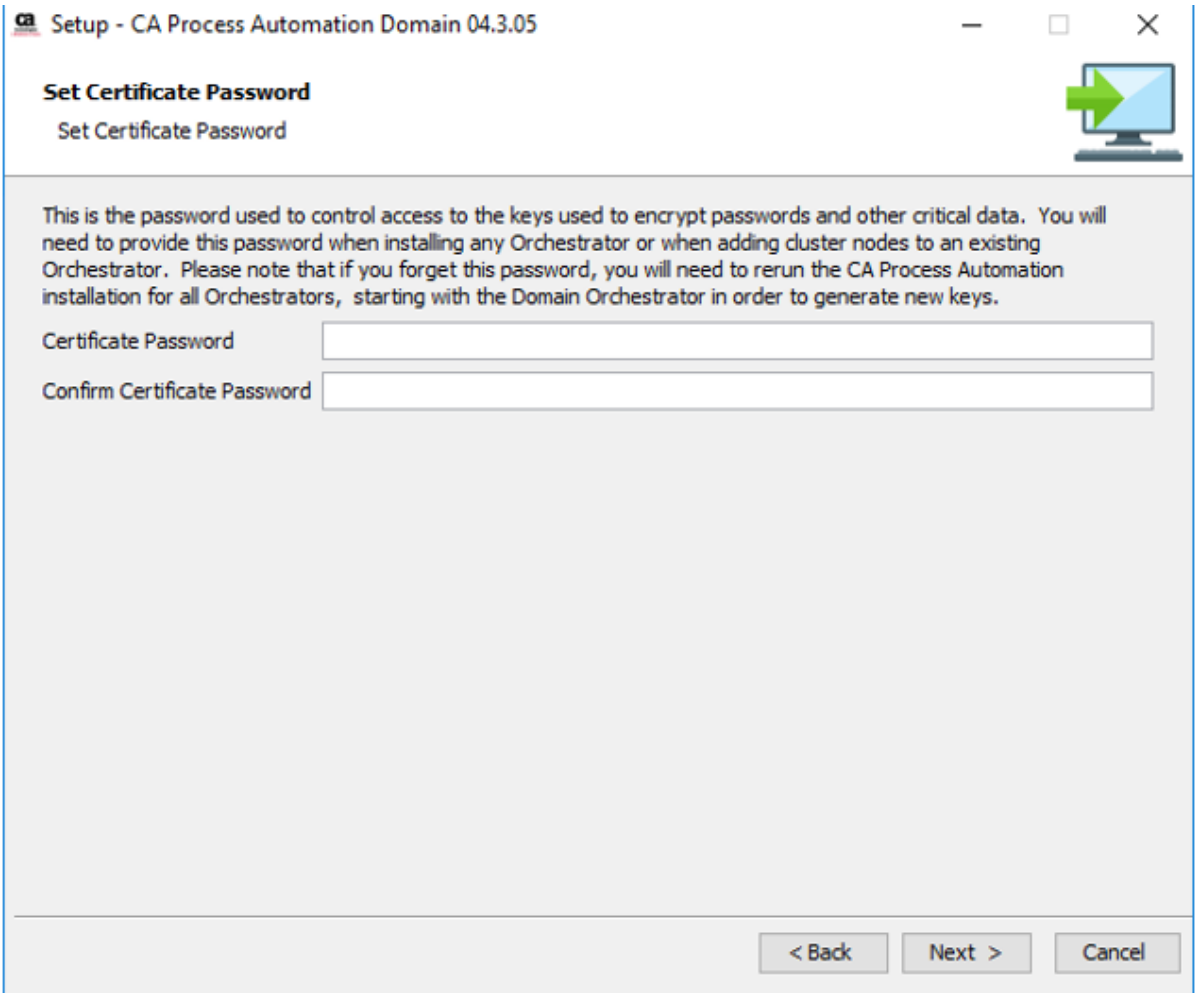
18. Click **Next** with default details as this configuration does not have a Load balancer.



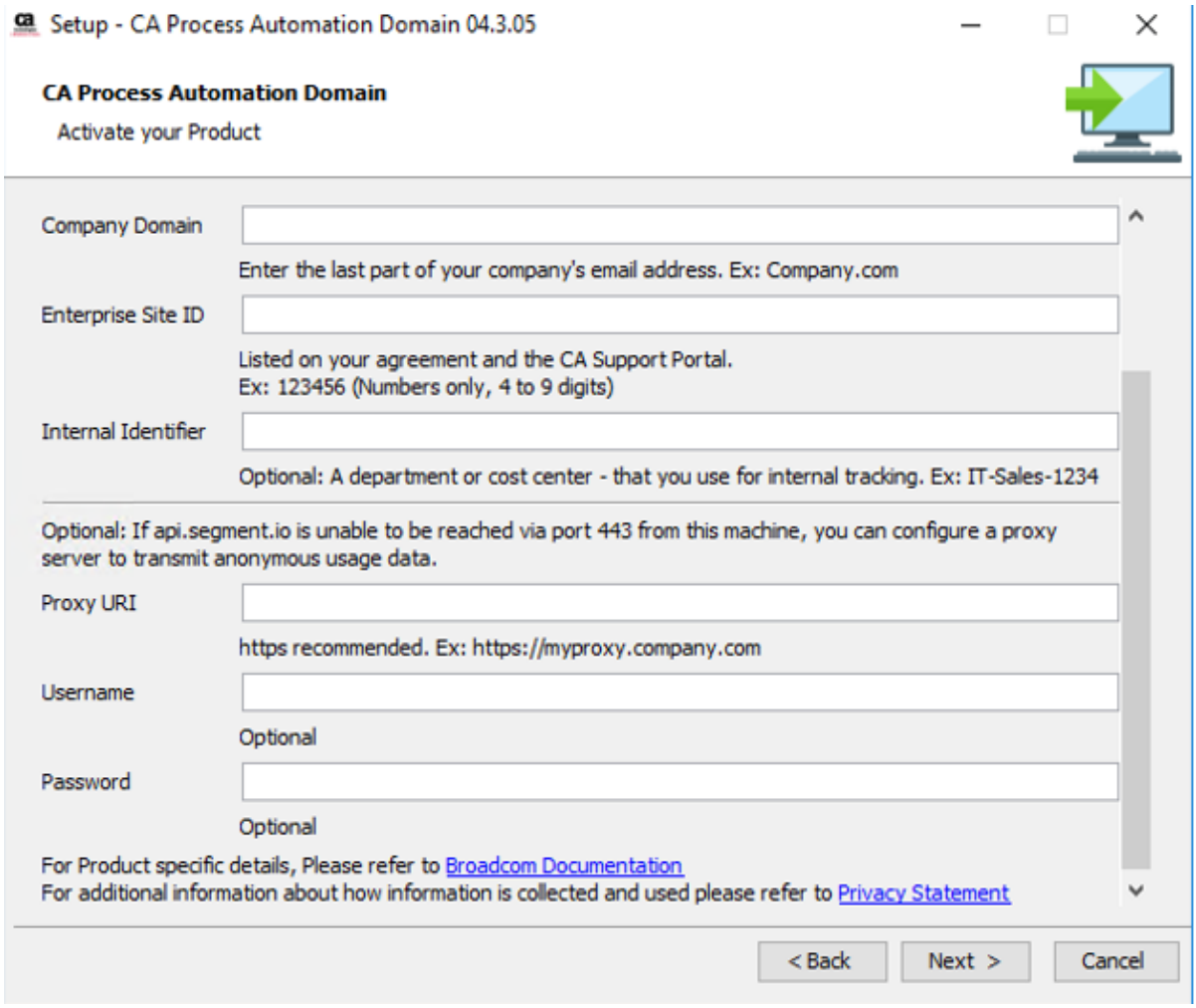
19. Click **Next**.

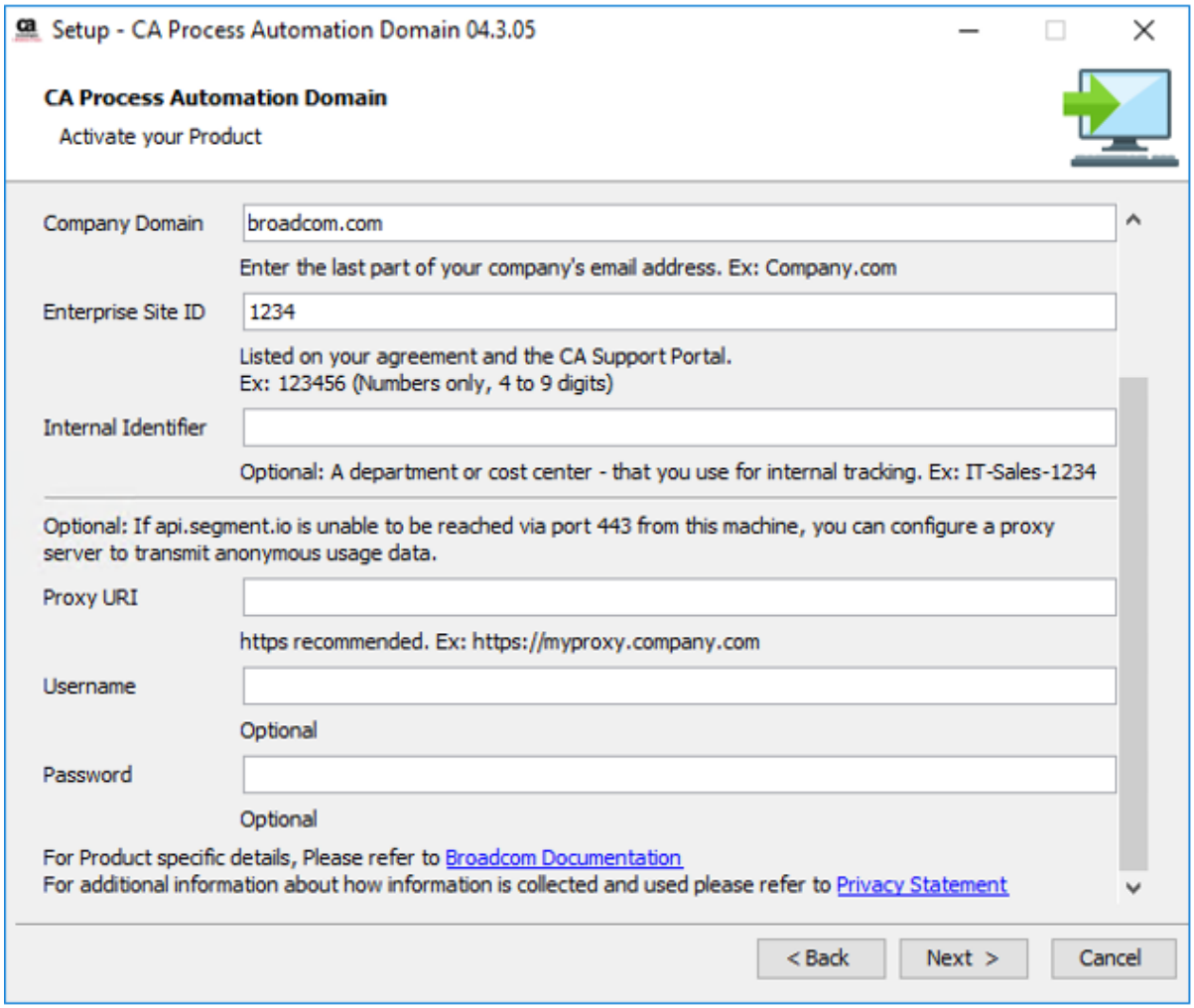


20. Enter certificate password and confirm Certificate Password.

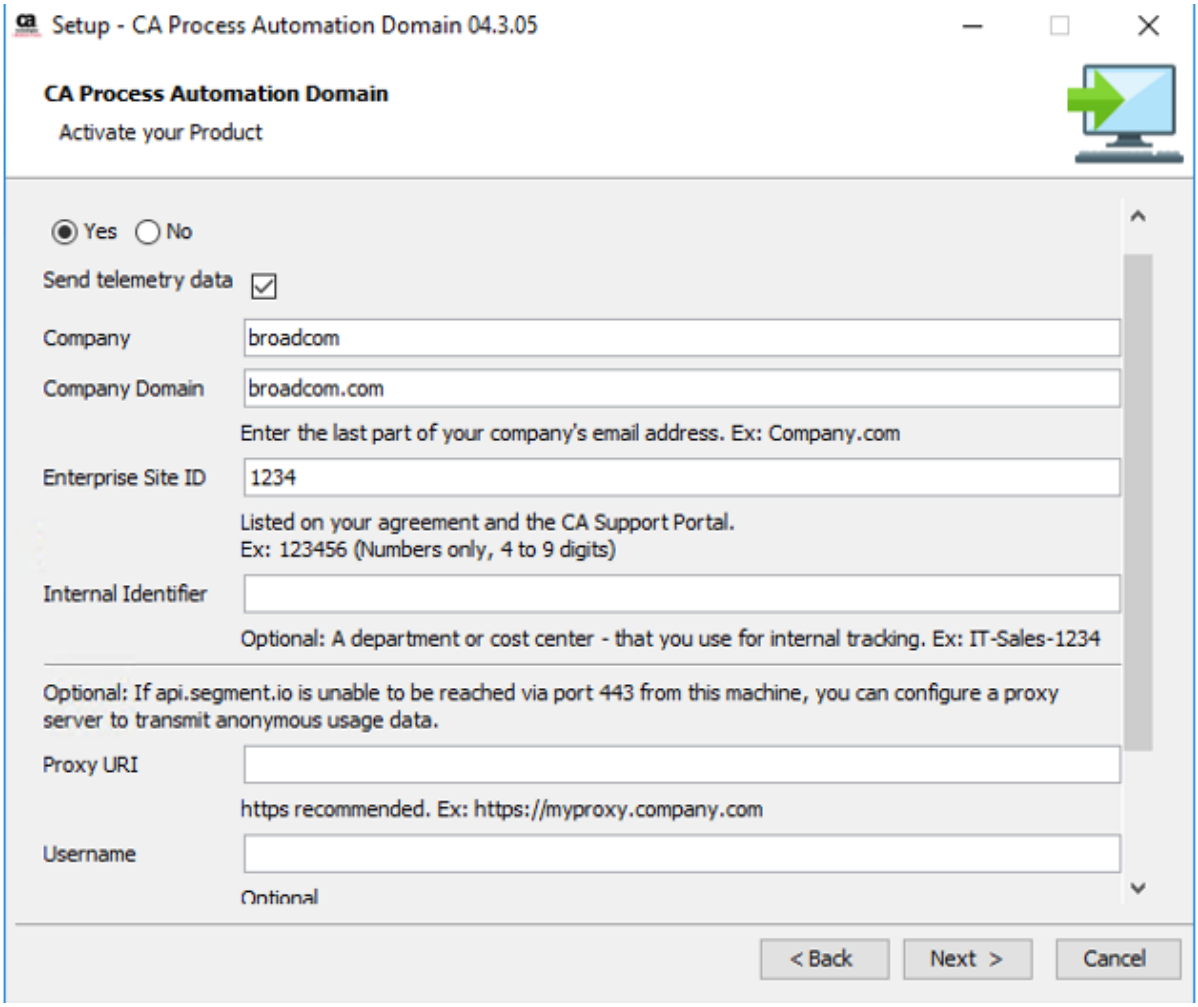


21. Enter **Enterprise Site ID**, **Company Domain** and **Internal Identifier**.

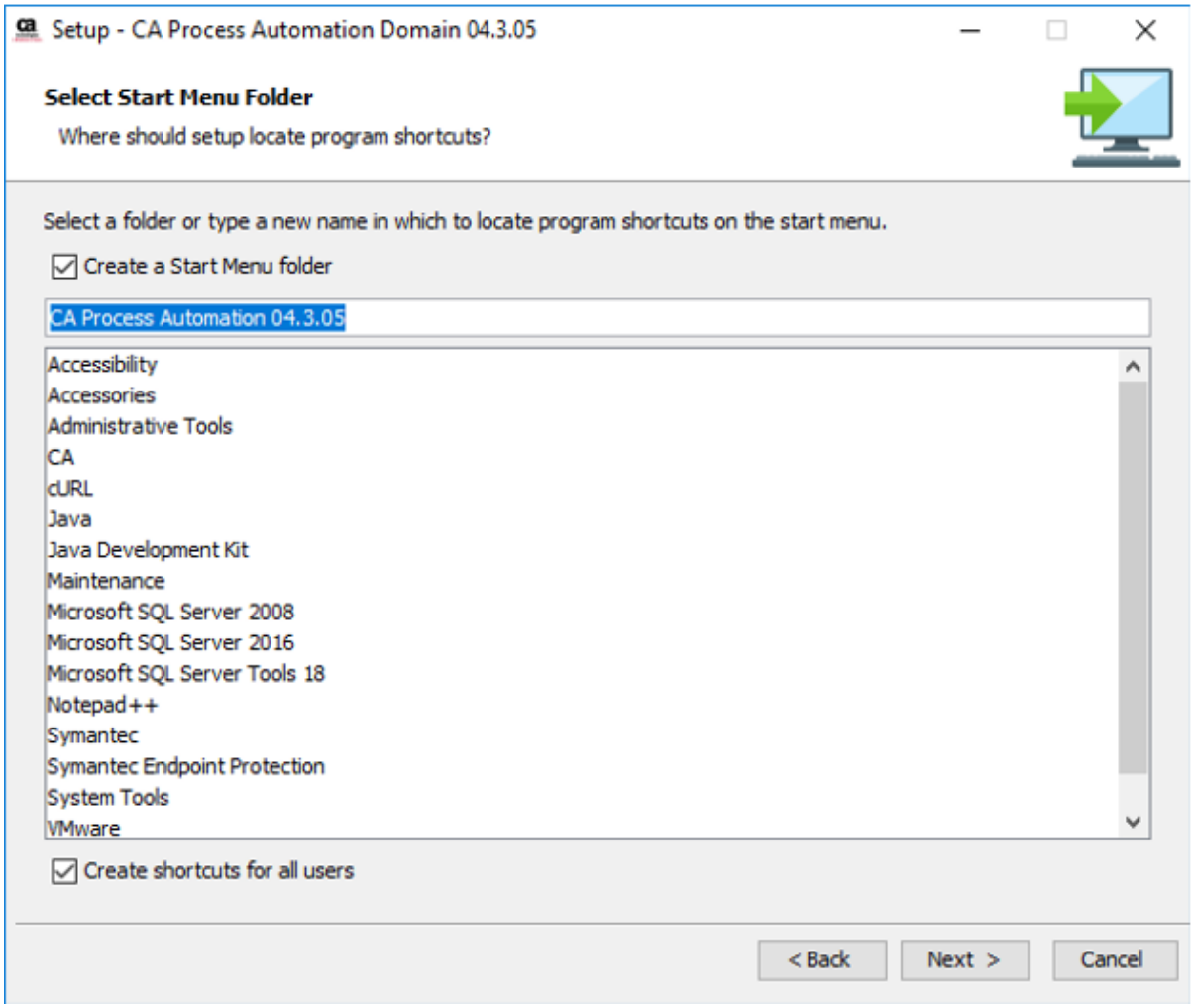

Setup - CA Process Automation Domain 04.3.05
CA Process Automation Domain
Activate your Product
Company Domain
Enter the last part of your company's email address. Ex: Company.com
Enterprise Site ID
Listed on your agreement and the CA Support Portal.
Ex: 123456 (Numbers only, 4 to 9 digits)
Internal Identifier
Optional: A department or cost center - that you use for internal tracking. Ex: IT-Sales-1234
Optional: If api.segment.io is unable to be reached via port 443 from this machine, you can configure a proxy server to transmit anonymous usage data.
Proxy URI
https recommended. Ex: https://myproxy.company.com
Username
Optional
Password
Optional
For Product specific details, Please refer to Broadcom Documentation
For additional information about how information is collected and used please refer to Privacy Statement
< Back
Next >
Cancel



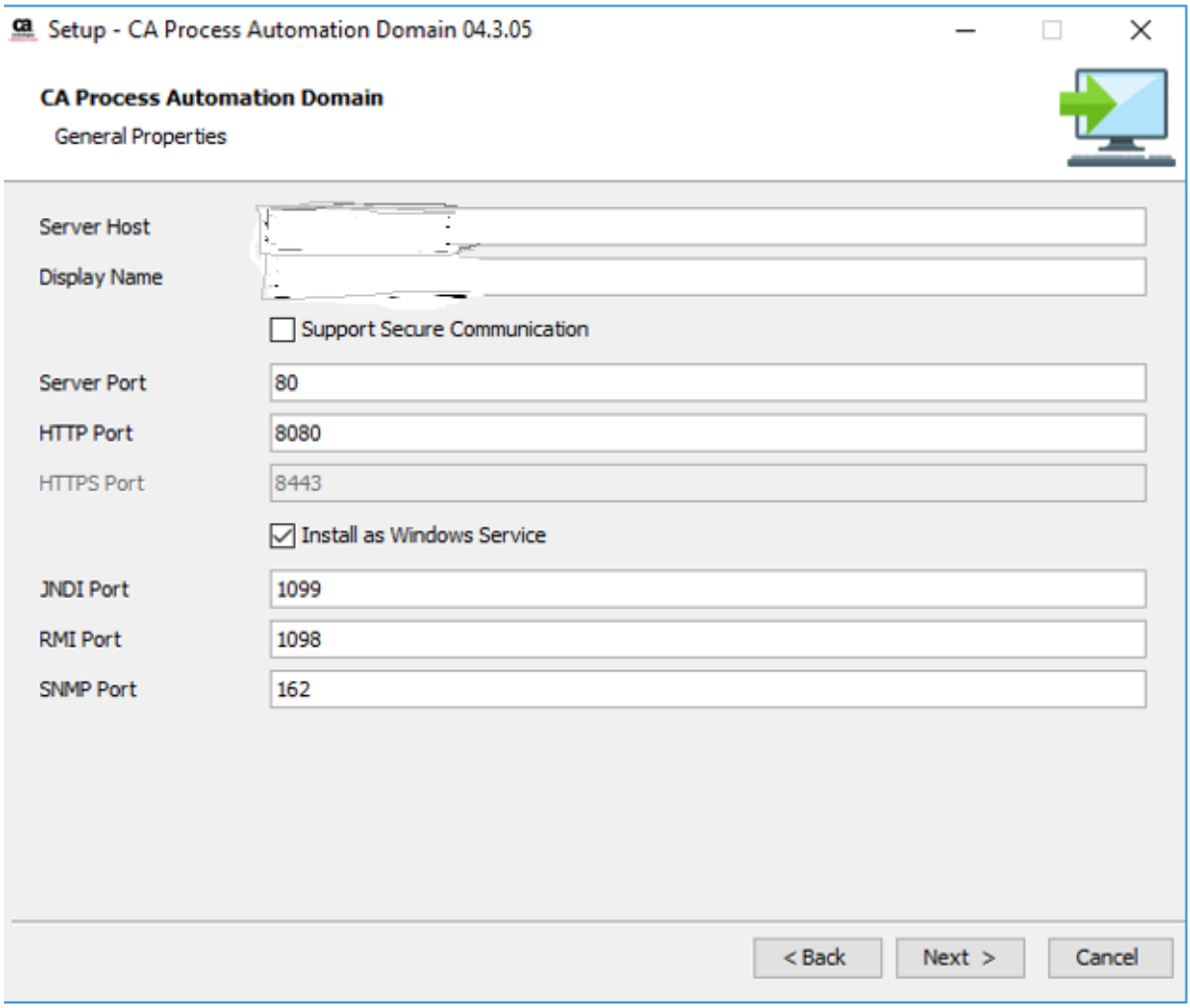
22. Enter Company and Select telemetry data check box.



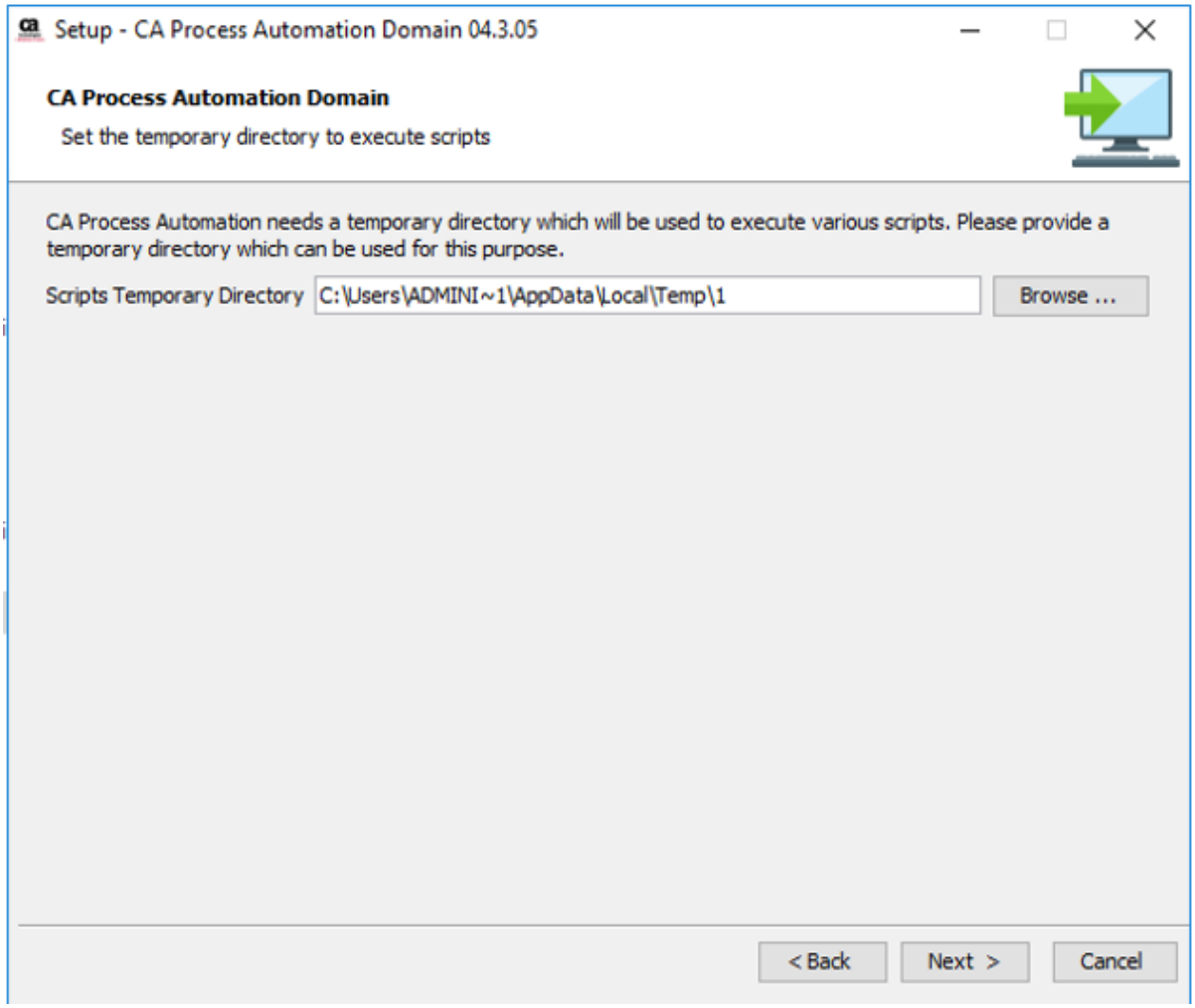
23. Click **Next**.



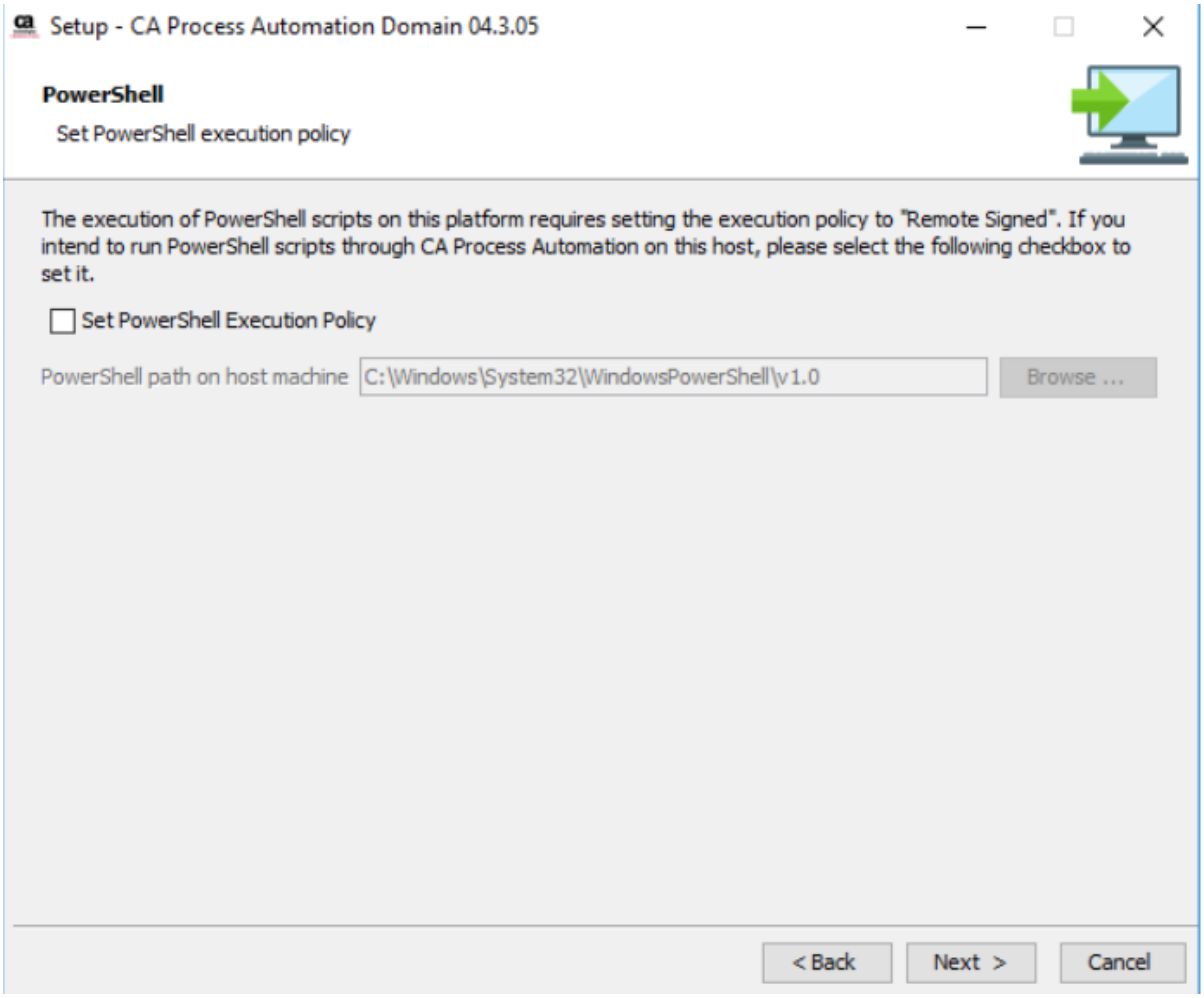
24. Click **Next**.



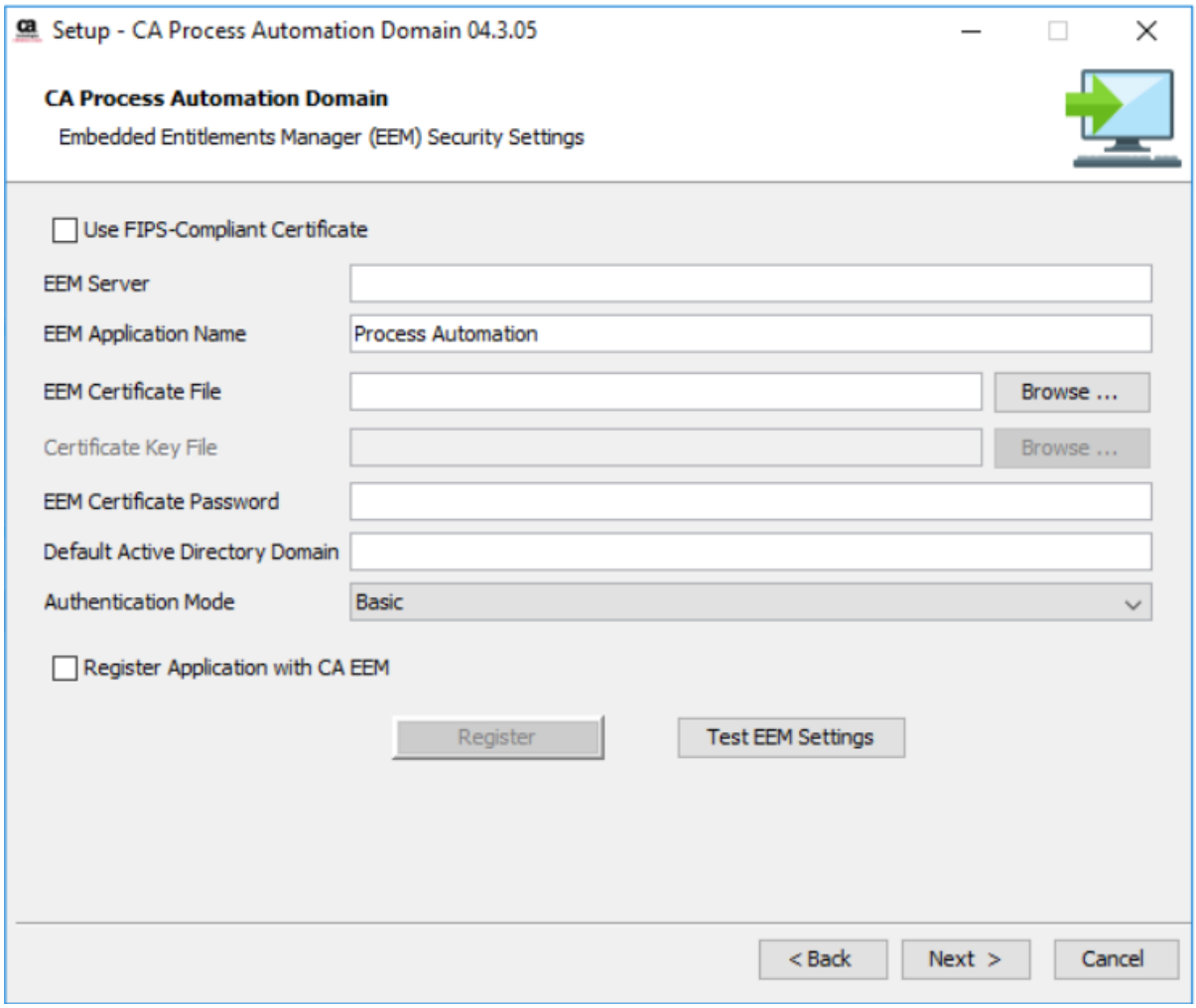
25. Click **Next**.



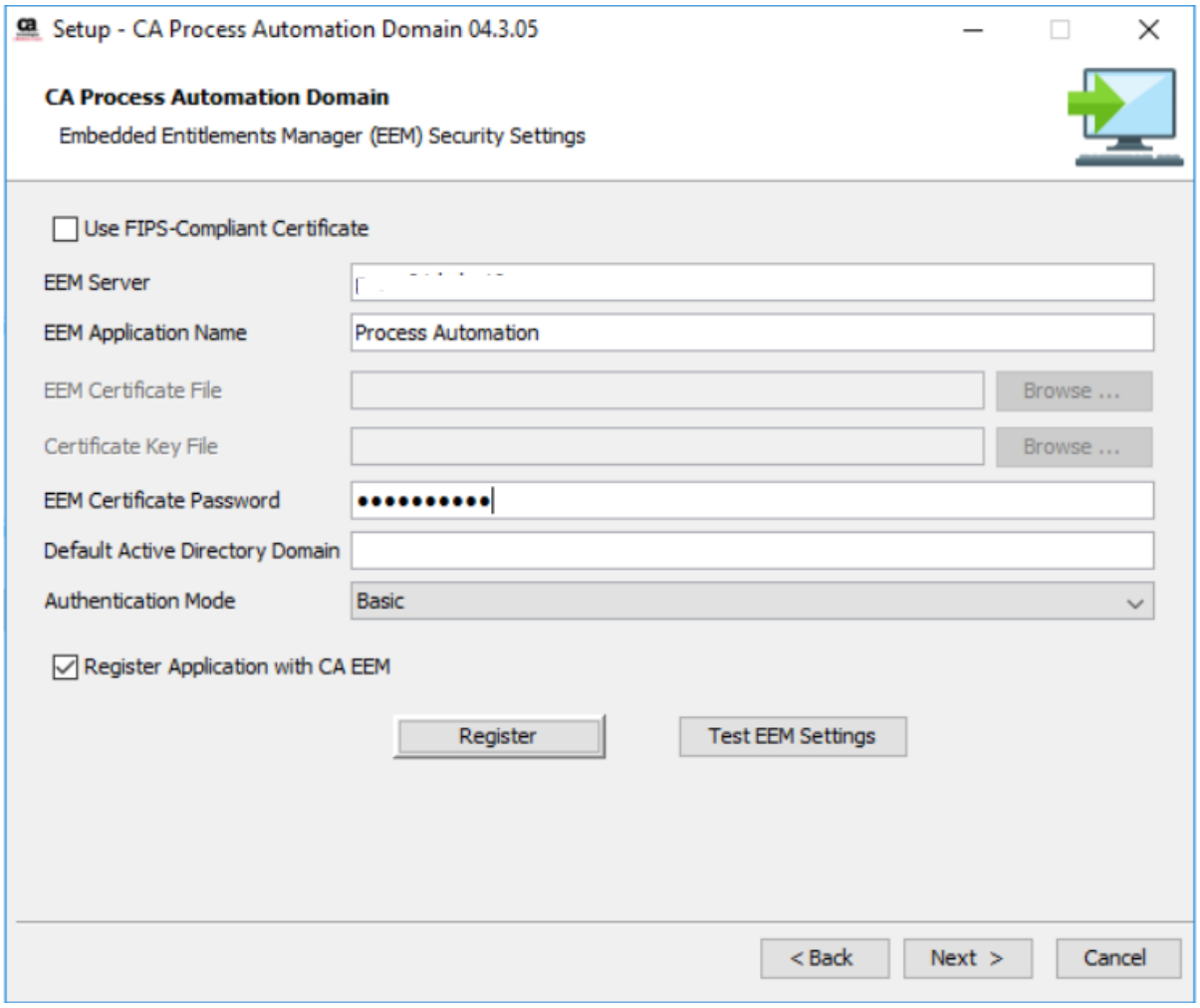
26. Click**Next**.



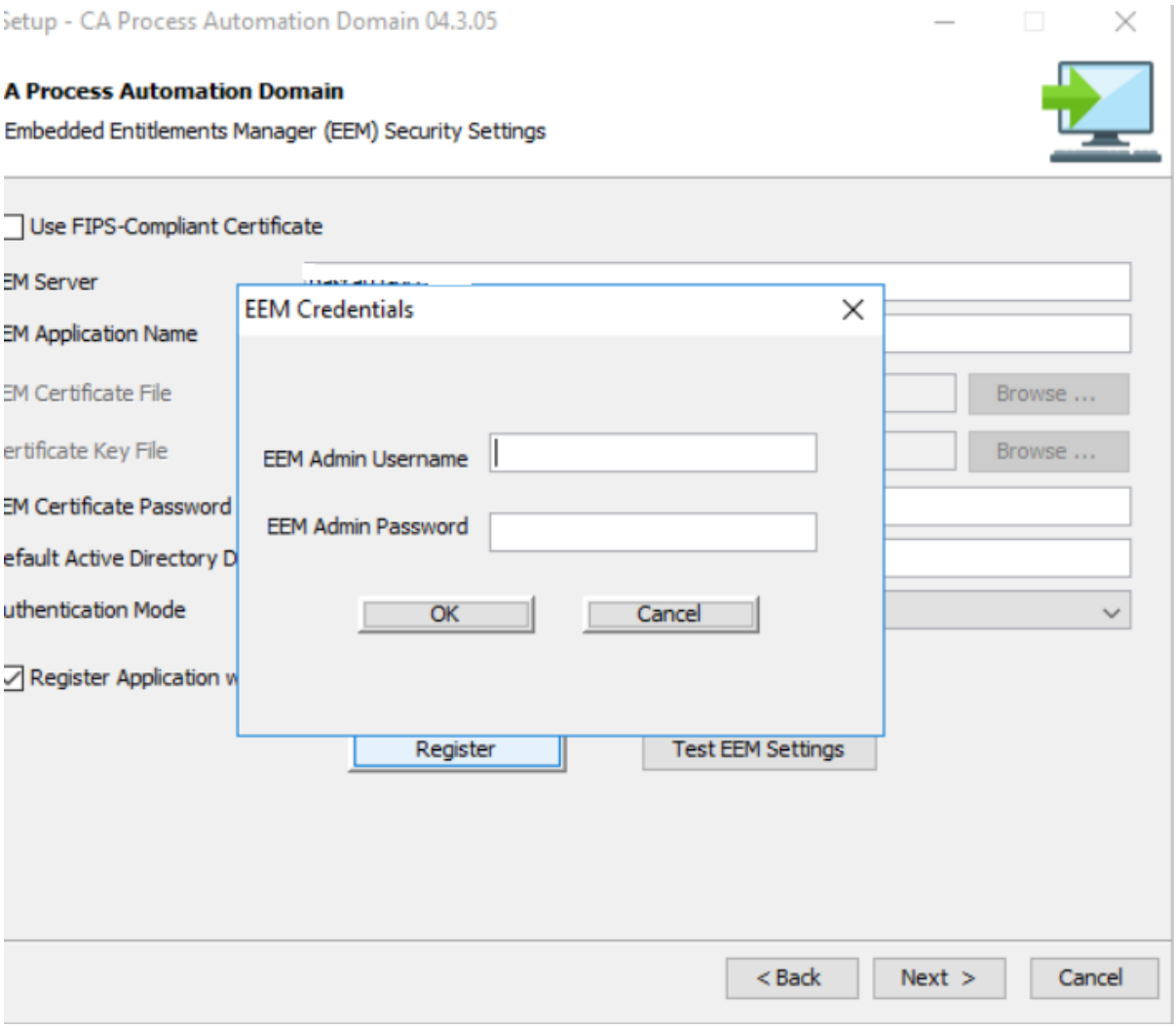
27. Enter EEM server Host name like **testqa-eem-new** and EEM Certificate Password. Here EEM application name can be given any name but we go with default value.

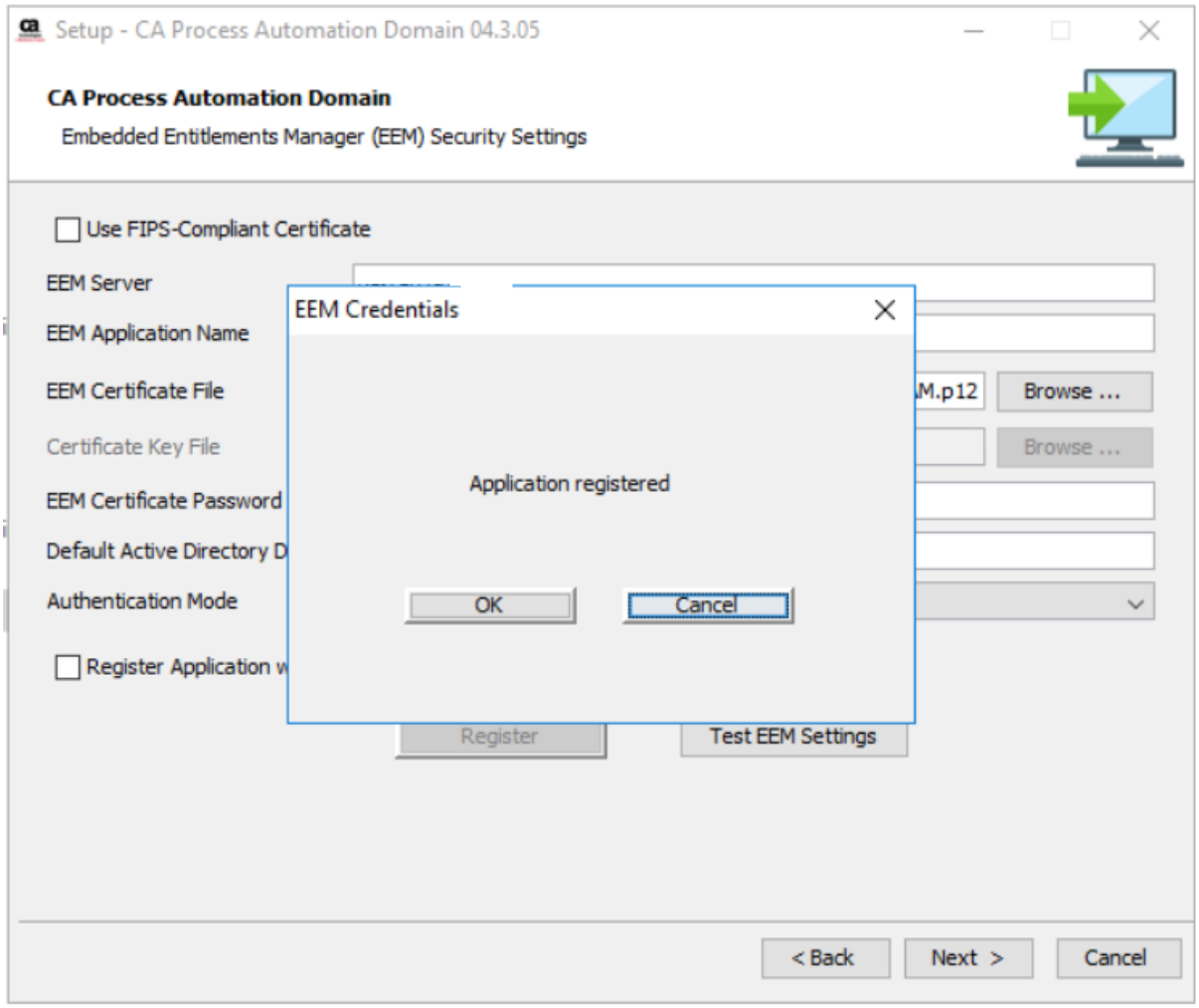


28. Click **Register**.

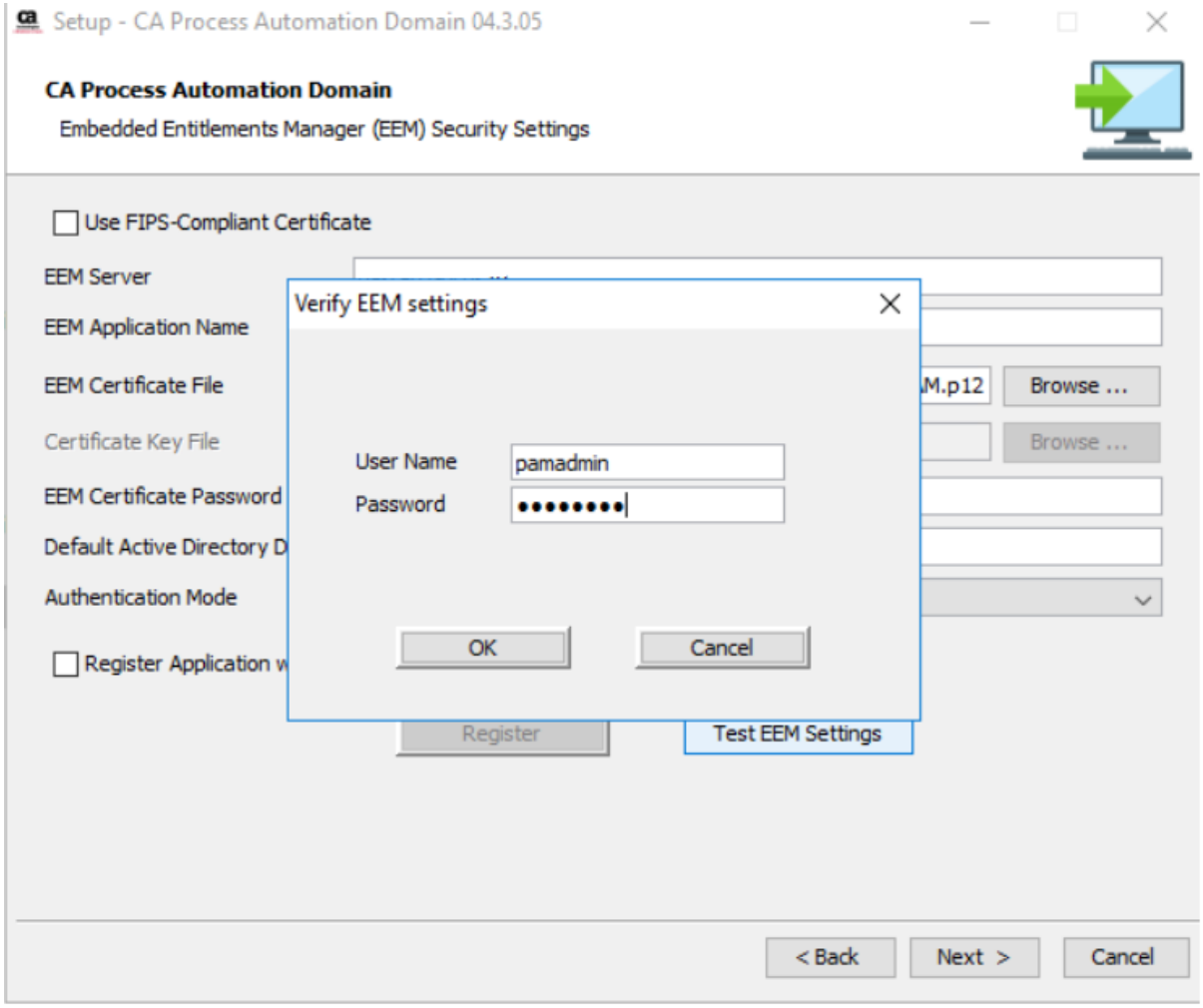


29. Enter EEM Admin user name and password.
30. Click **OK**.

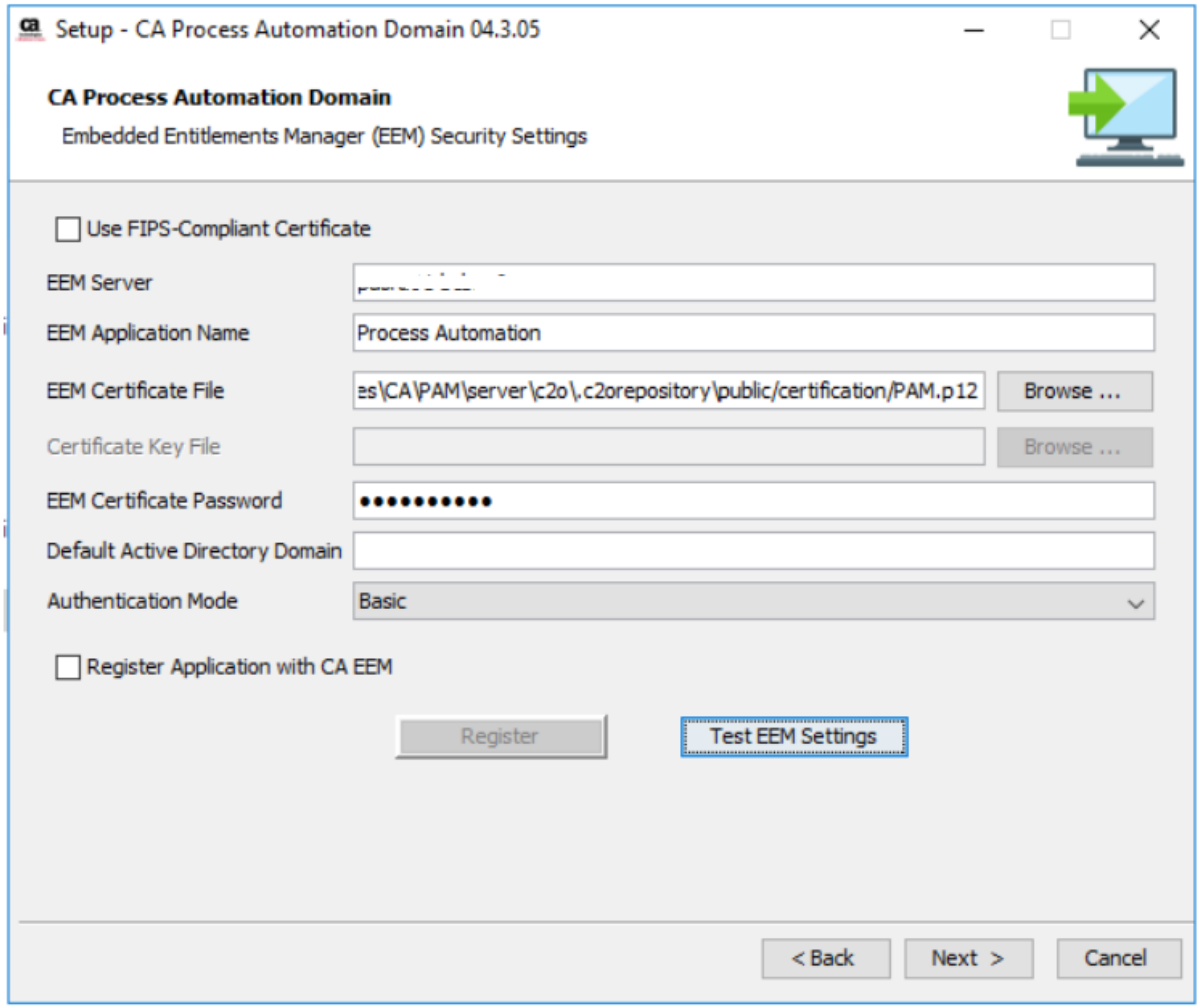

Setup - CA Process Automation Domain 04.3.05
A Process Automation Domain
Embedded Entitlements Manager (EEM) Security Settings
Use FIPS-Compliant Certificate
EM Server
EM Application Name
EM Certificate File
ertificate Key File
EM Certificate Password
efault Active Directory D
uthentication Mode
Register Application w
Browse ...
Browse ...
EEM Credentials
EEM Admin Username
EEM Admin Password
OK
Cancel
Register
Test EEM Settings
< Back
Next >
Cancel


Setup - CA Process Automation Domain 04.3.05
CA Process Automation Domain
Embedded Entitlements Manager (EEM) Security Settings
Use FIPS-Compliant Certificate
EEM Server
EEM Application Name
EEM Certificate File
Browse ...
Certificate Key File
Browse ...
EEM Certificate Password
Default Active Directory D
Authentication Mode
Register Application w
Register
Test EEM Settings
EEM Credentials
Application registered
OK
Cancel
< Back
Next >
Cancel

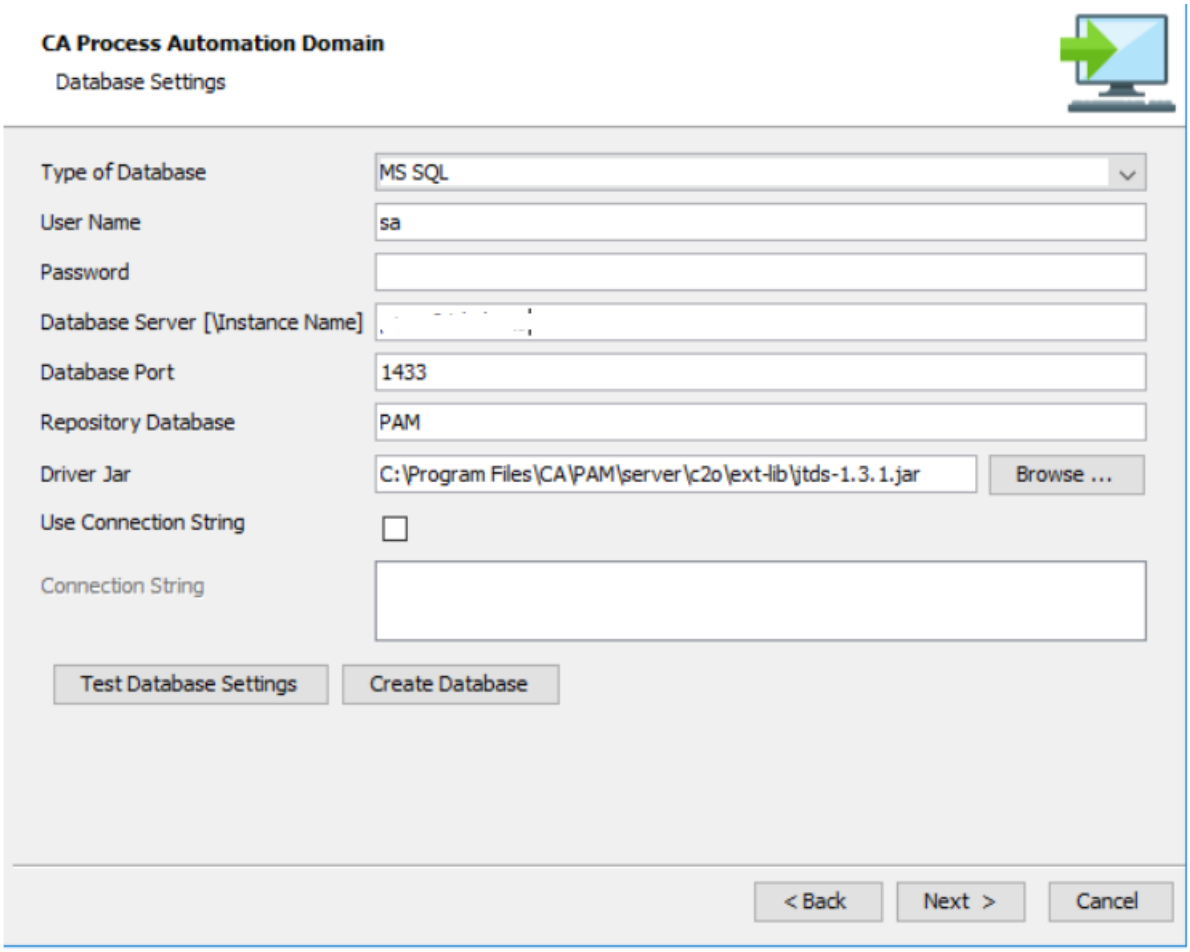
31. Click **Ok**.



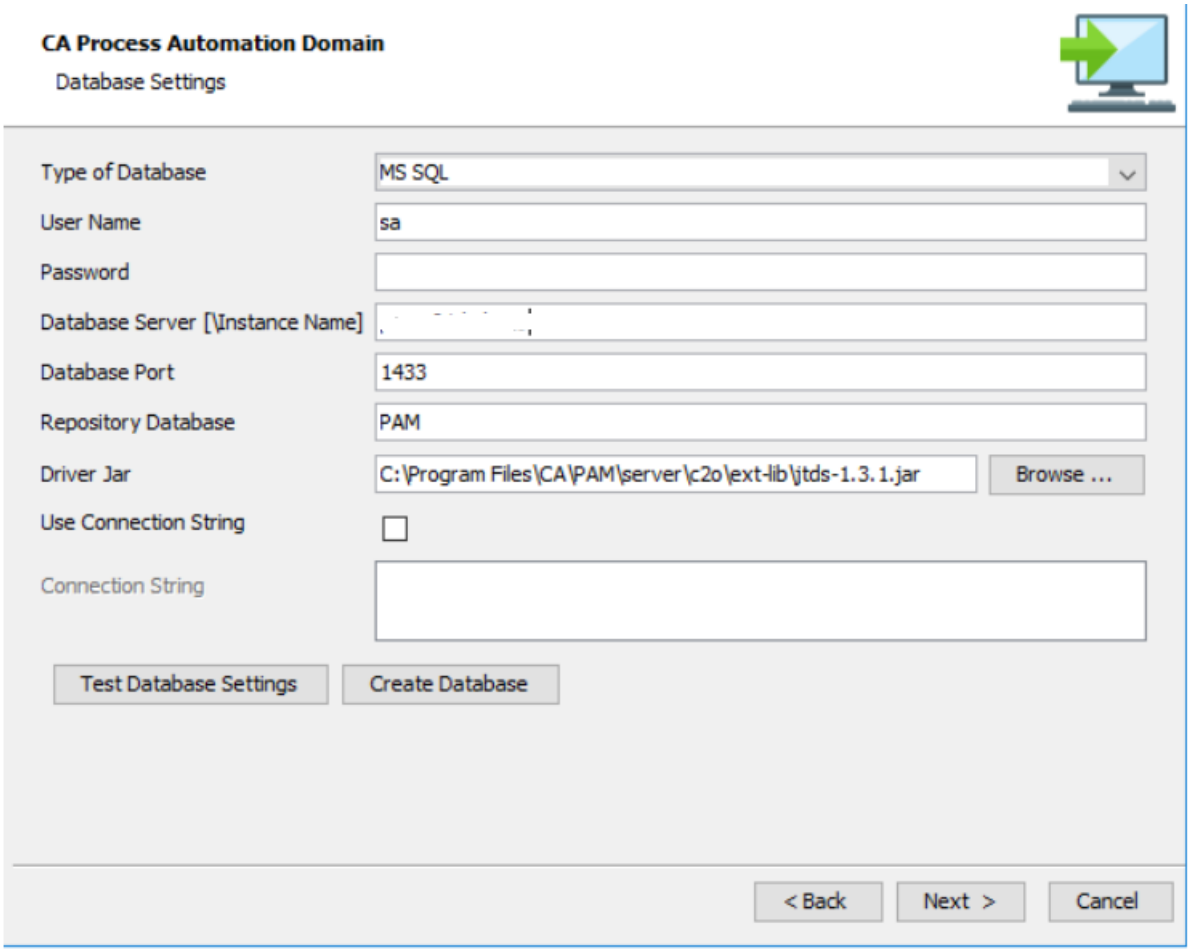
32. Click **Next**.



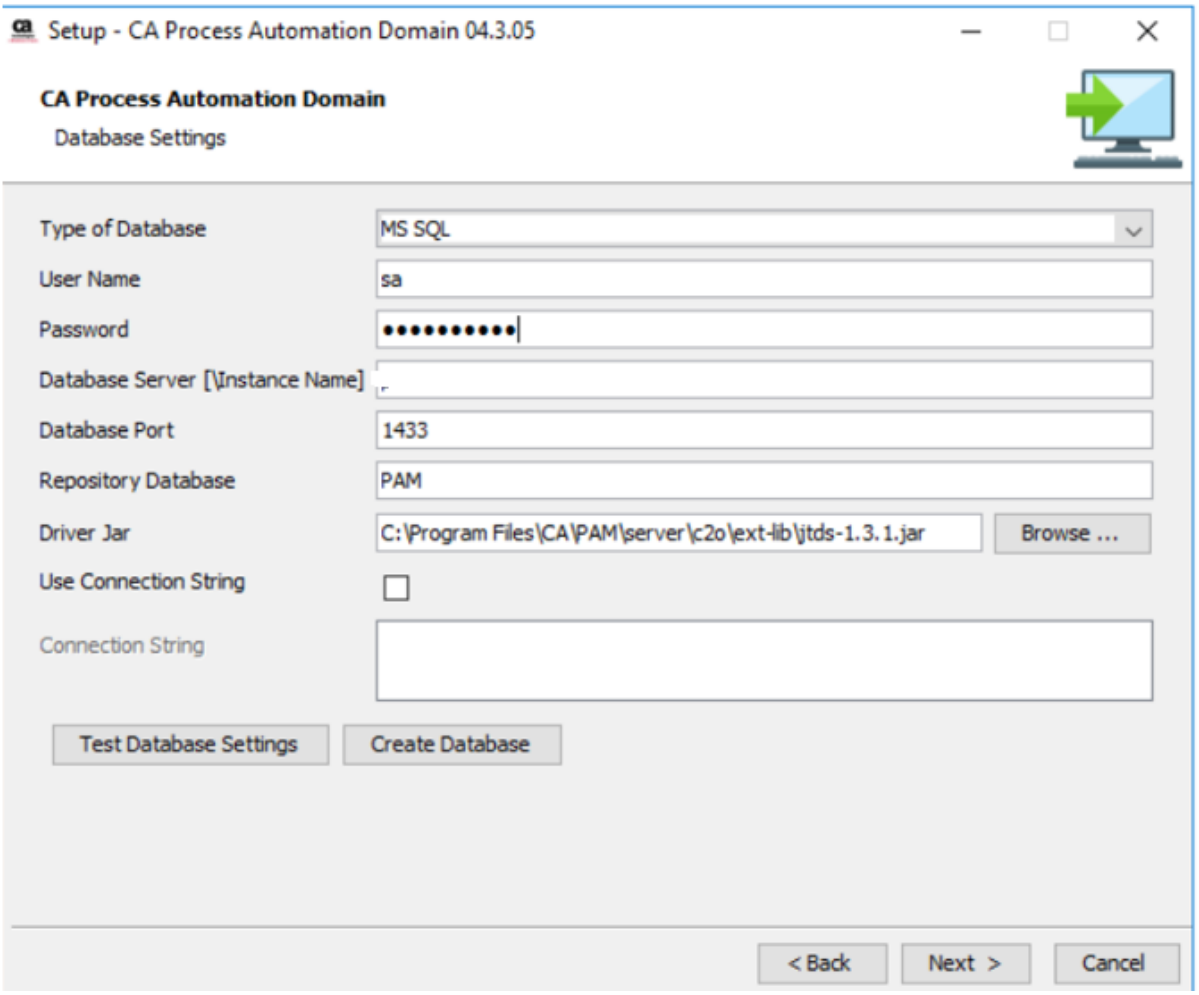
33. Click Test EEM Settings.
34. Select type of Database, user name, password and Host Name.



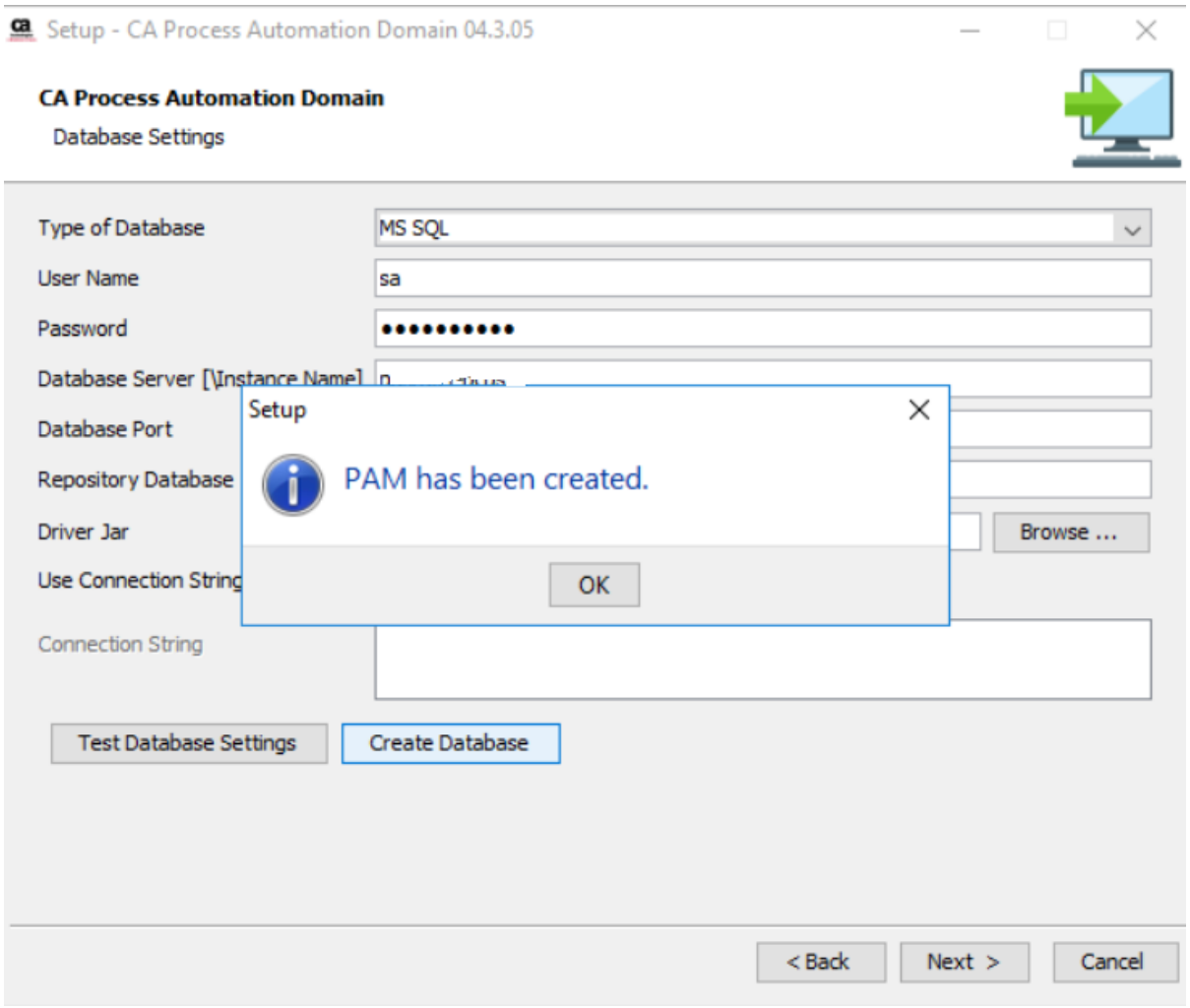
35. Click **Create Database**.


CA Process Automation Domain
Database Settings
Type of Database
MS SQL
User Name
sa
Password
Database Server [\Instance Name]
Database Port
1433
Repository Database
PAM
Driver Jar
C:\Program Files\CA\PAM\server\c2o\ext-lib\jtds-1.3.1.jar
Browse ...
Use Connection String
Connection String
Test Database Settings
Create Database
< Back
Next >
Cancel

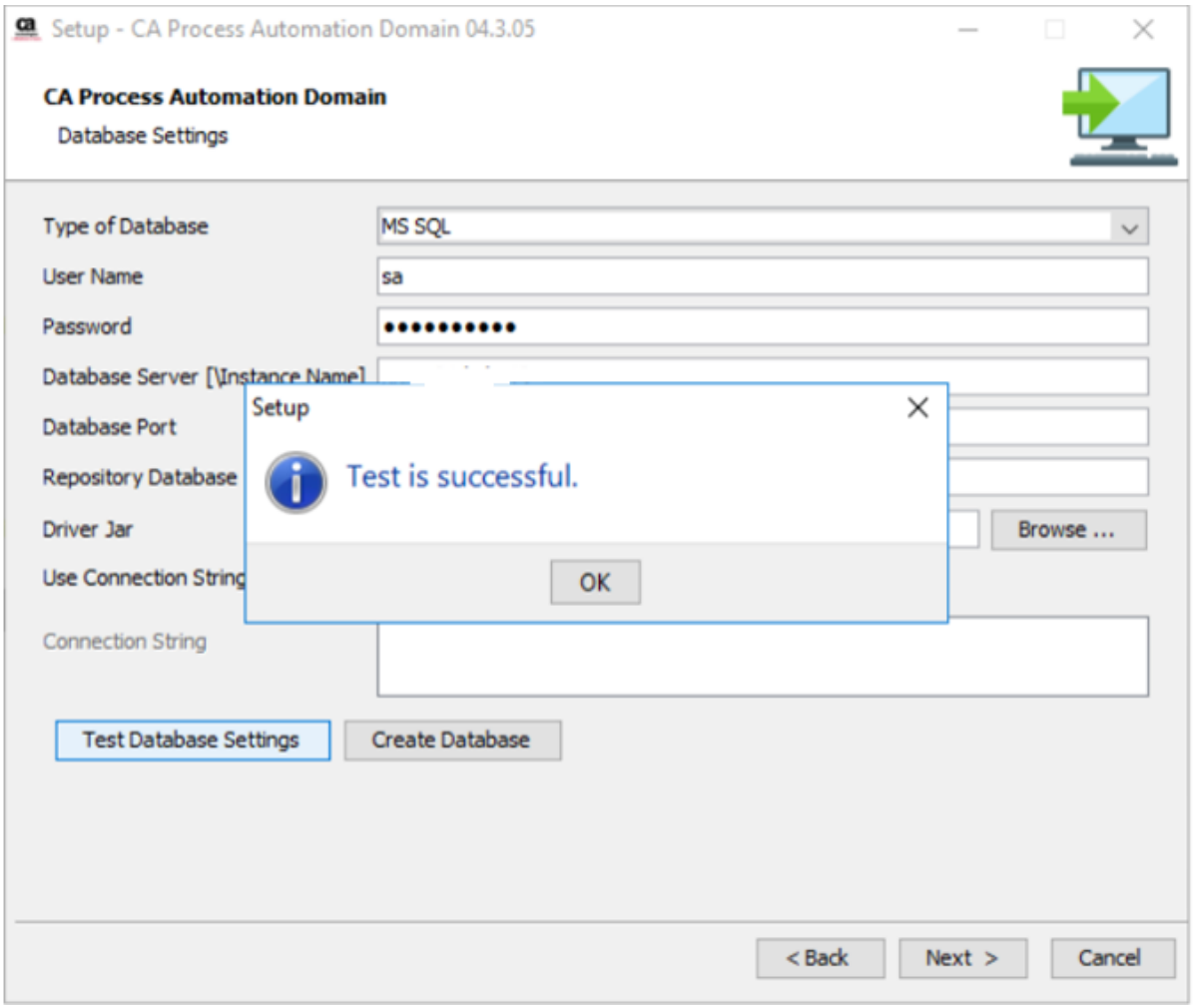
36. Click **OK**.



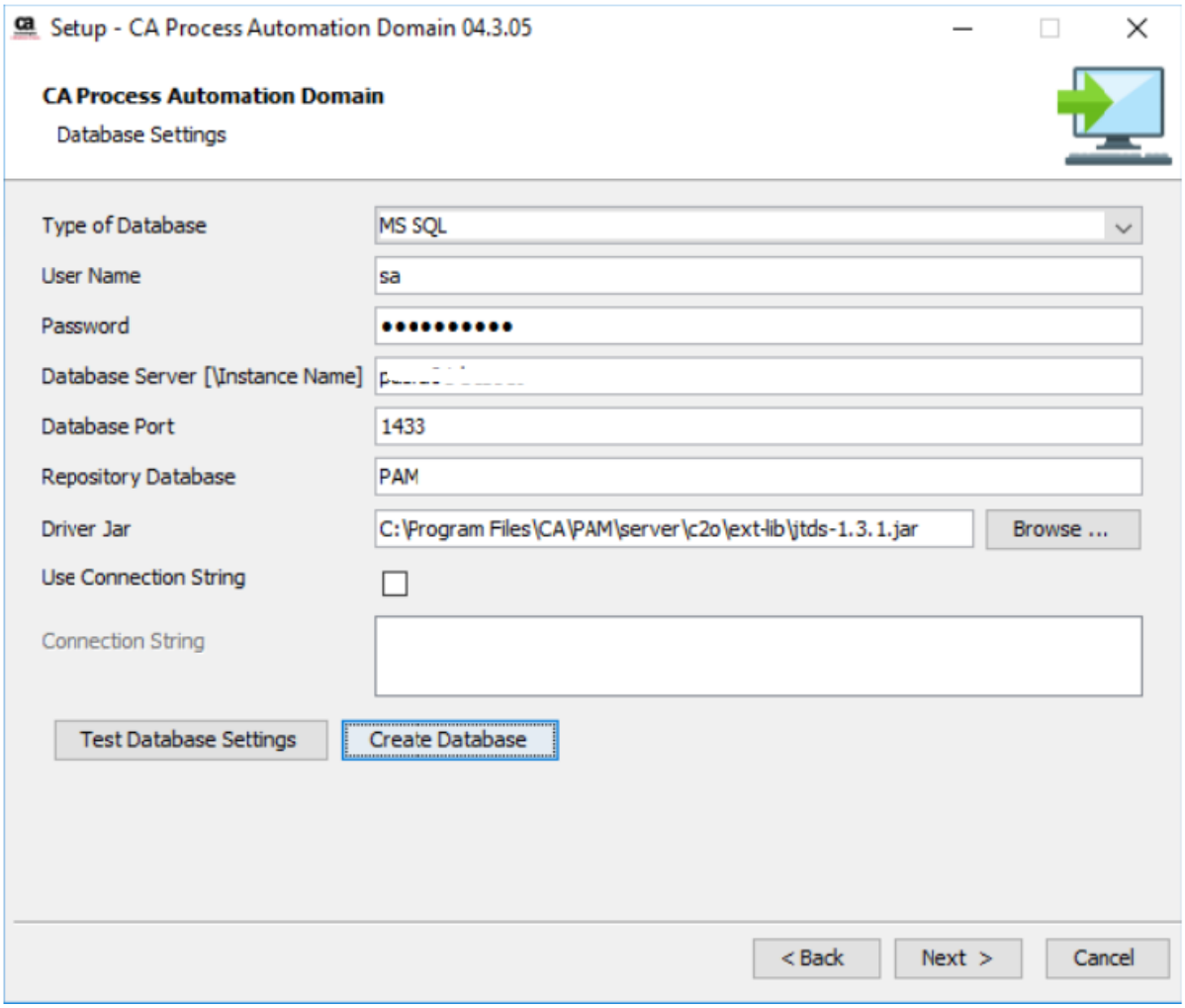
37. Click **Test Database Settings**.



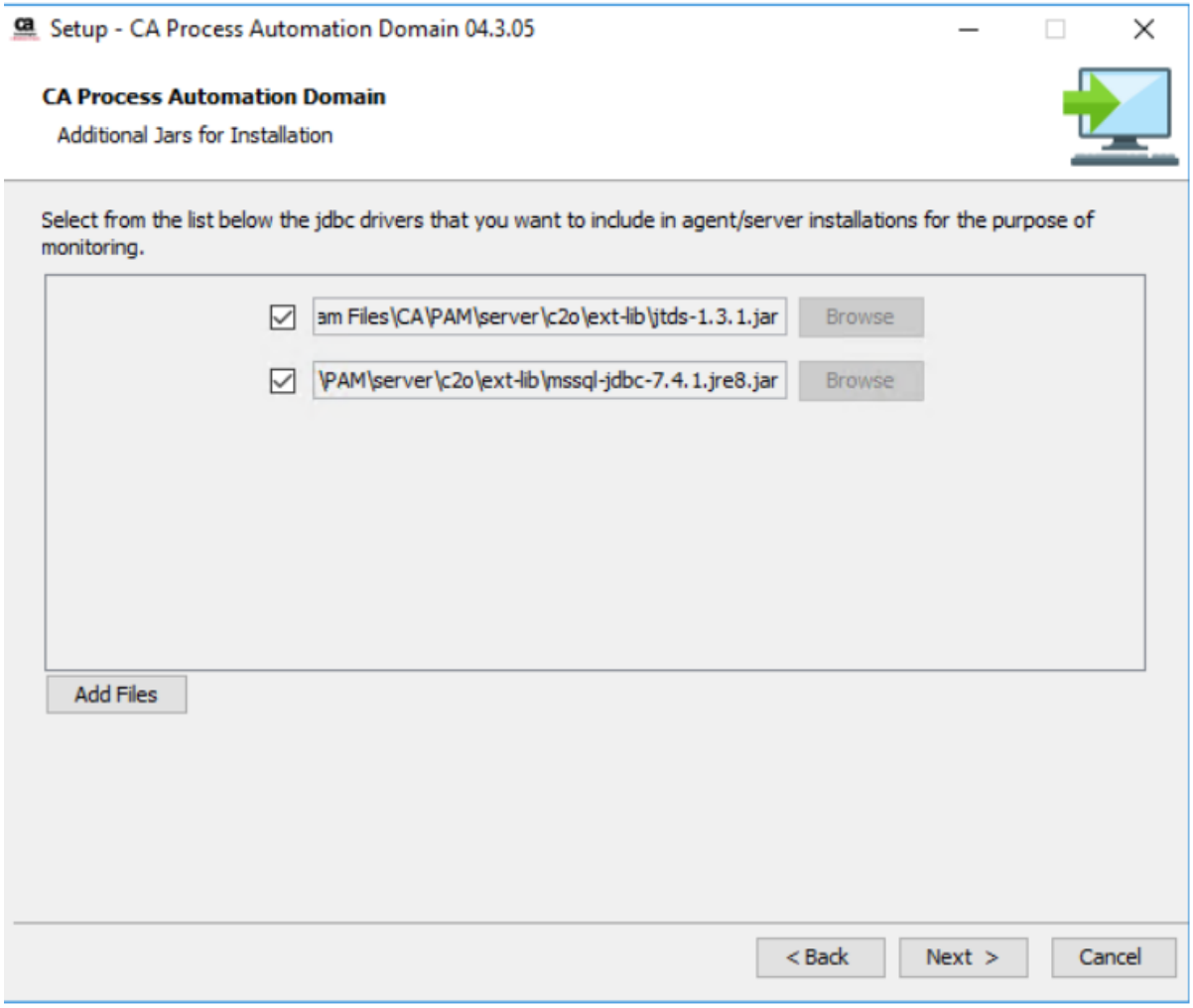
38. Select checkbox "**copy from main repository"** and enter Runtime Database name like "**PAM_runtime**".
39. Click Create Database.

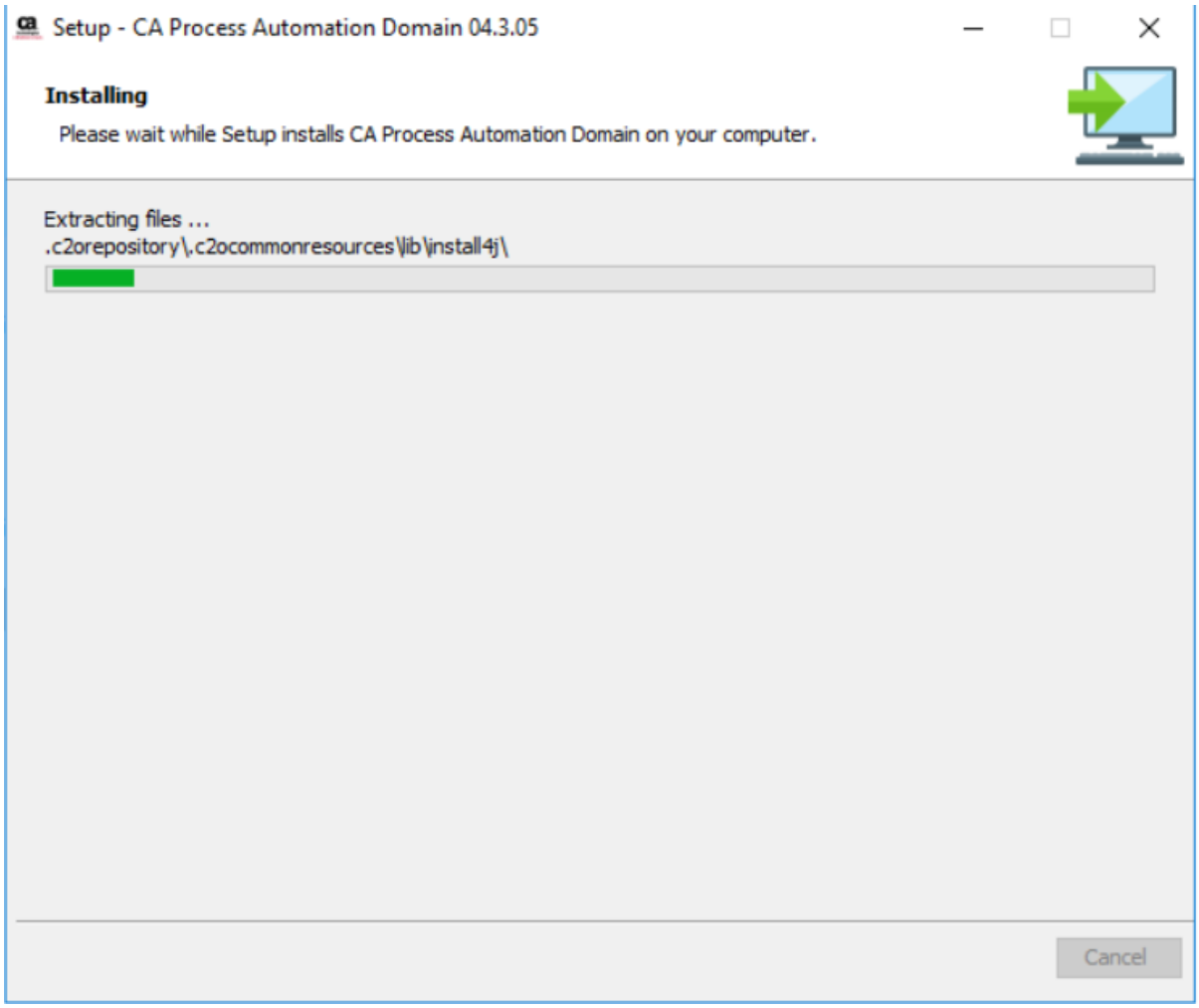


40. Click **Next**.

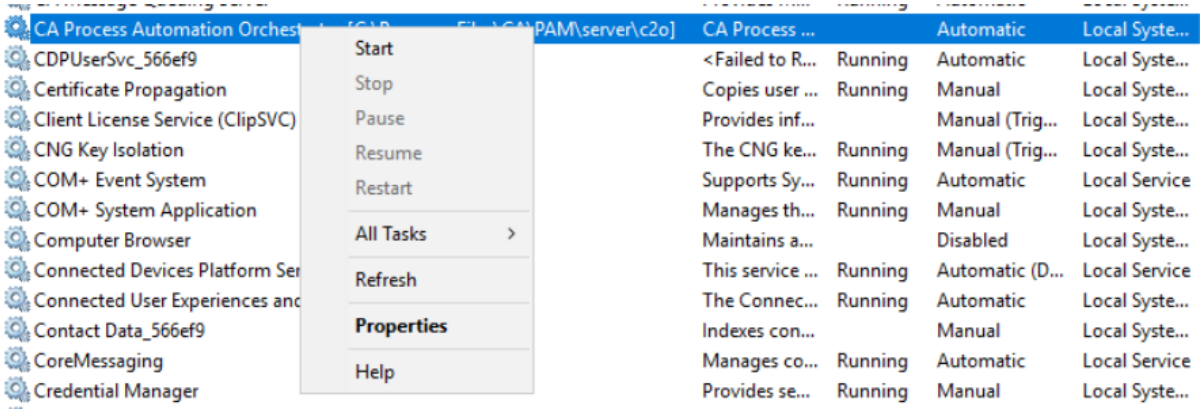


41. Click "**Test database Settings**" and Click OK.
42. Select checkbox and click **Next**.

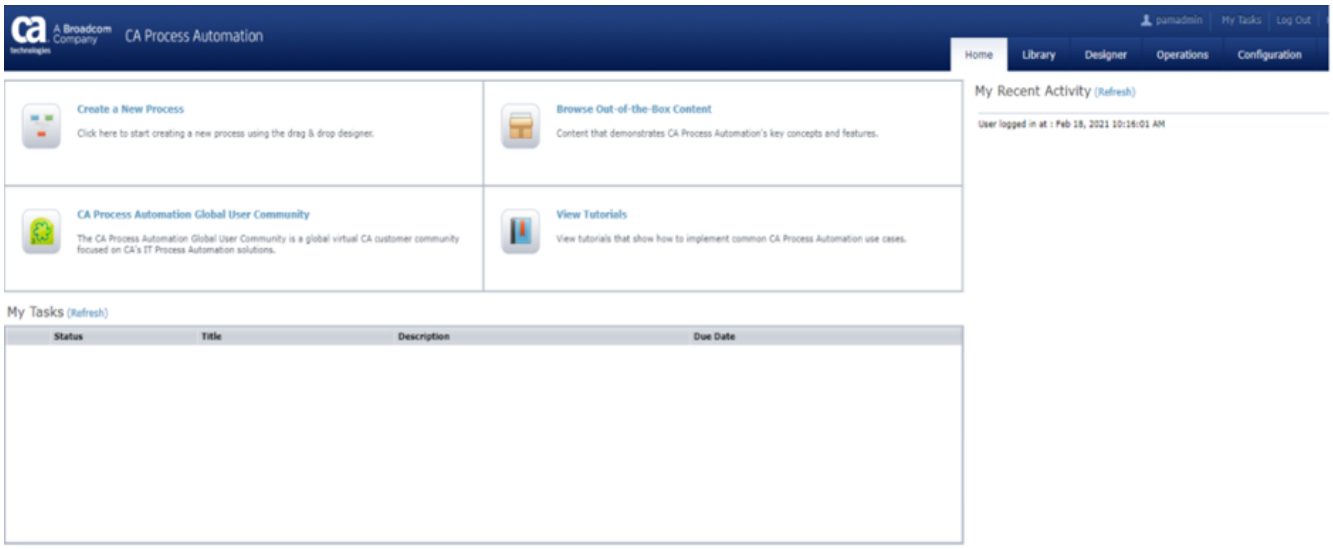

Setup - CA Process Automation Domain 04.3.05
CA Process Automation Domain
Additional Jars for Installation
Select from the list below the jdbc drivers that you want to include in agent/server installations for the purpose of monitoring.
am Files\CA\PAM\server\c2o\ext-lib\jtds-1.3.1.jar
Browse
\PAM\server\c2o\ext-lib\mssql-jdbc-7.4.1.jre8.jar
Browse
Add Files
< Back
Next >
Cancel



43. Click **Finish**.
44. Navigate to Windows Services and start the CA Process Automation Services.



45. Open the CA Process Automation Web URL: ***http://<hostname>:<portnumber>/itpam***
46. Enter the username and password.
47. CA Process Automation home page is displayed.



Install CA Service Management 17.4

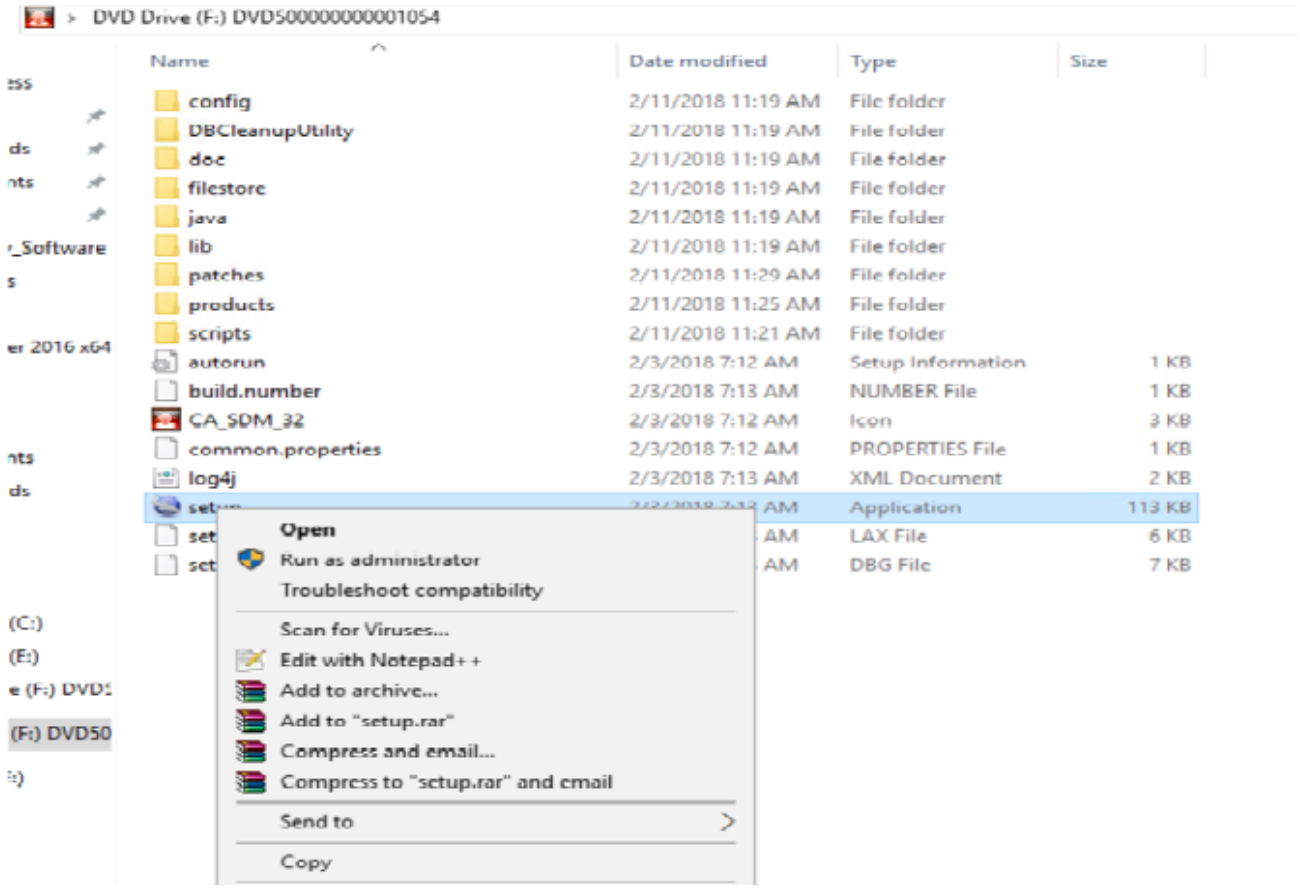
This section describes CA Service Management 17.4 installation procedures for CA Service Desk Manager (CMDB Visualizer), and CA Service Catalog with CA Common Components – CA EEM and CA Process Automation Workflow.

Install CA Service Desk Manager (SDM) 17.4
Perform the following steps to install CA Service Desk Manager 17.4 (Advanced Availability Configuration):

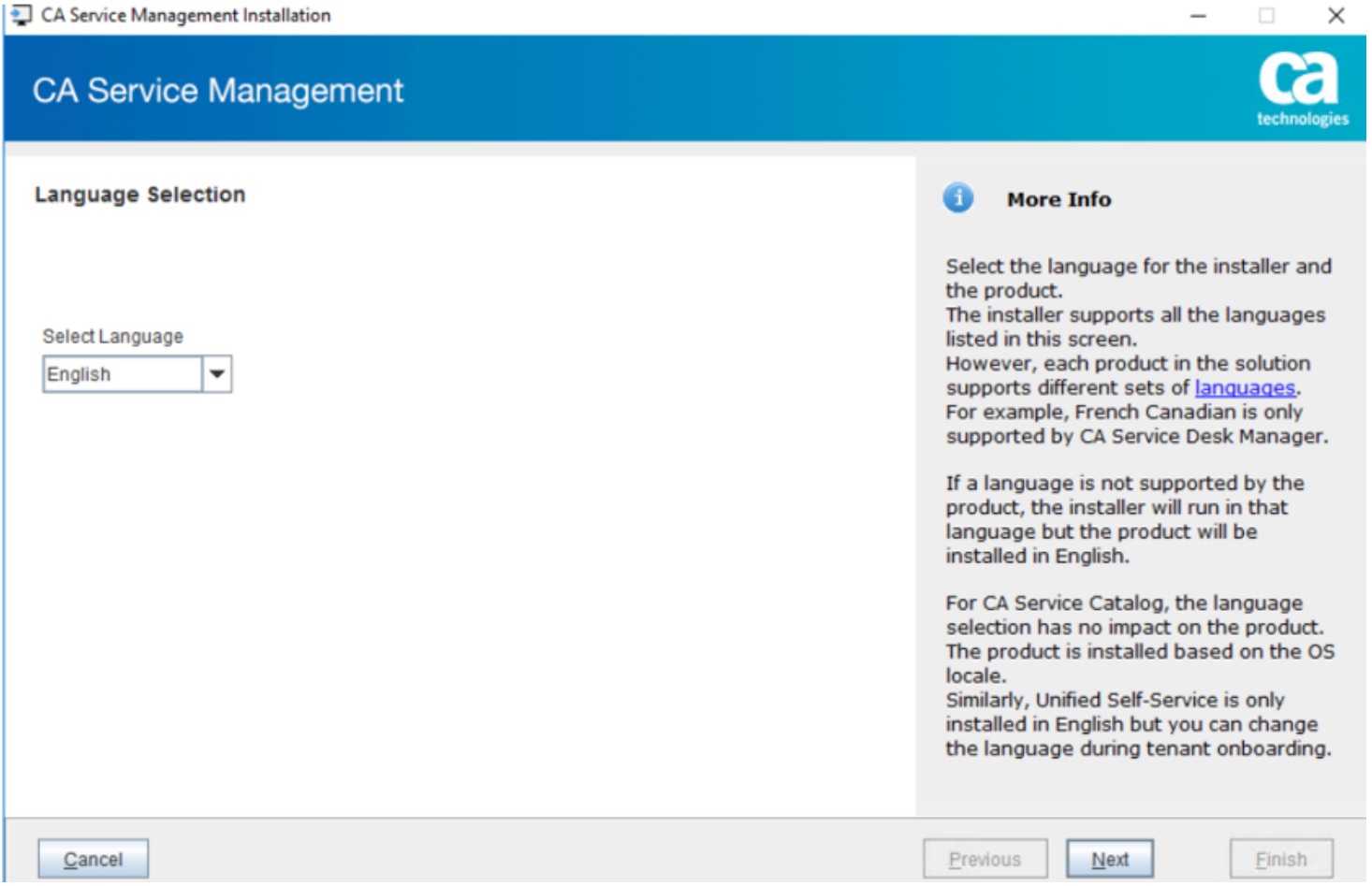
- CA SDM Installation 17.4 on Background server (**testqa-bg-new**)
- Install the Standby Server (**testqa-sb-new**)
- Install the First Application Server installation (**testqa-app1-new**)
- Install the Second SDM Application Server on (t**estqa-app2-new**)

Install CA SDM Background Server (testqa-bg-new)
Perform the following steps:

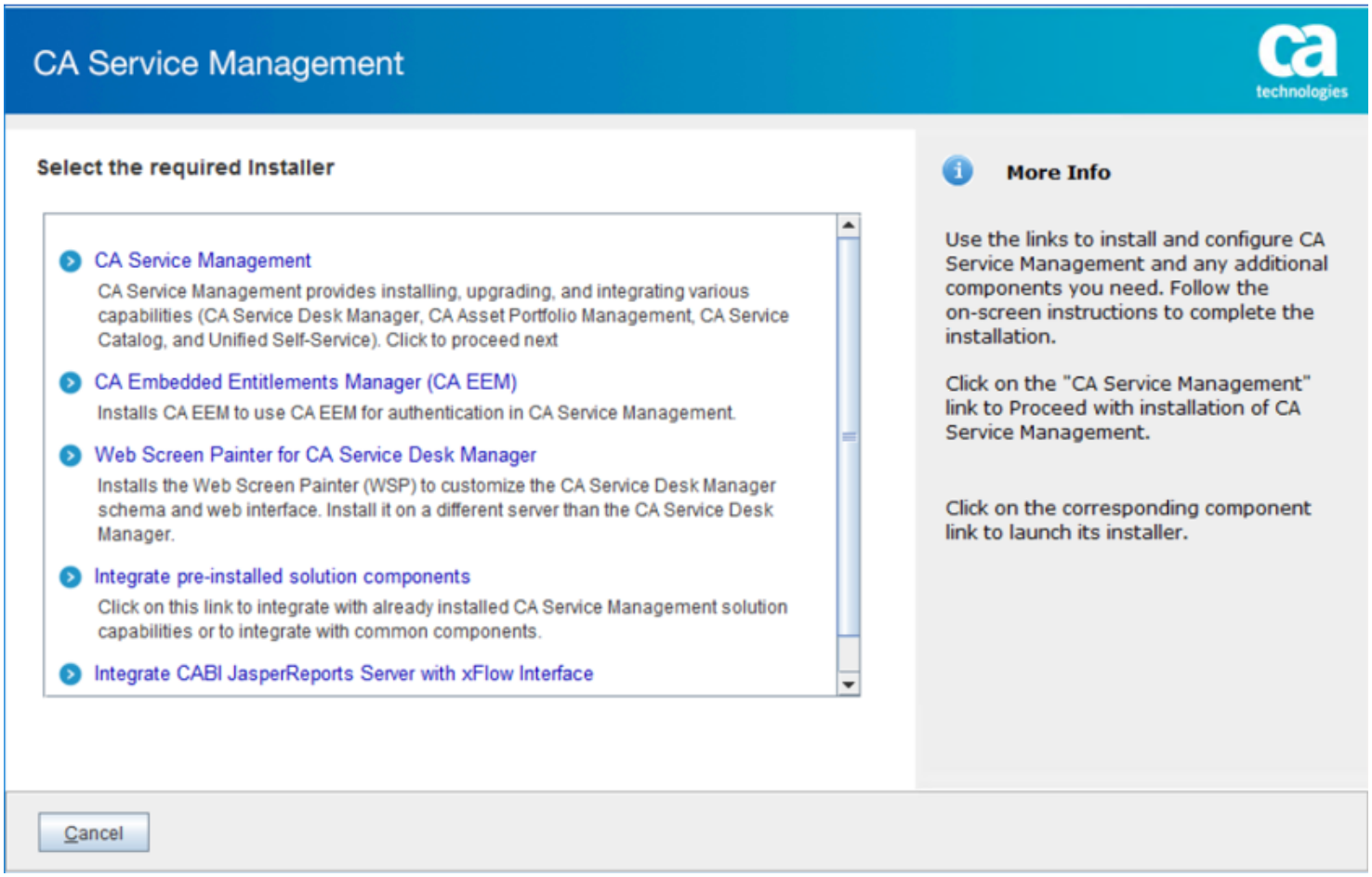
1. Mount the CA SDM DVD and click the **setup.exe**. Right-click and select **Run as administrator**.



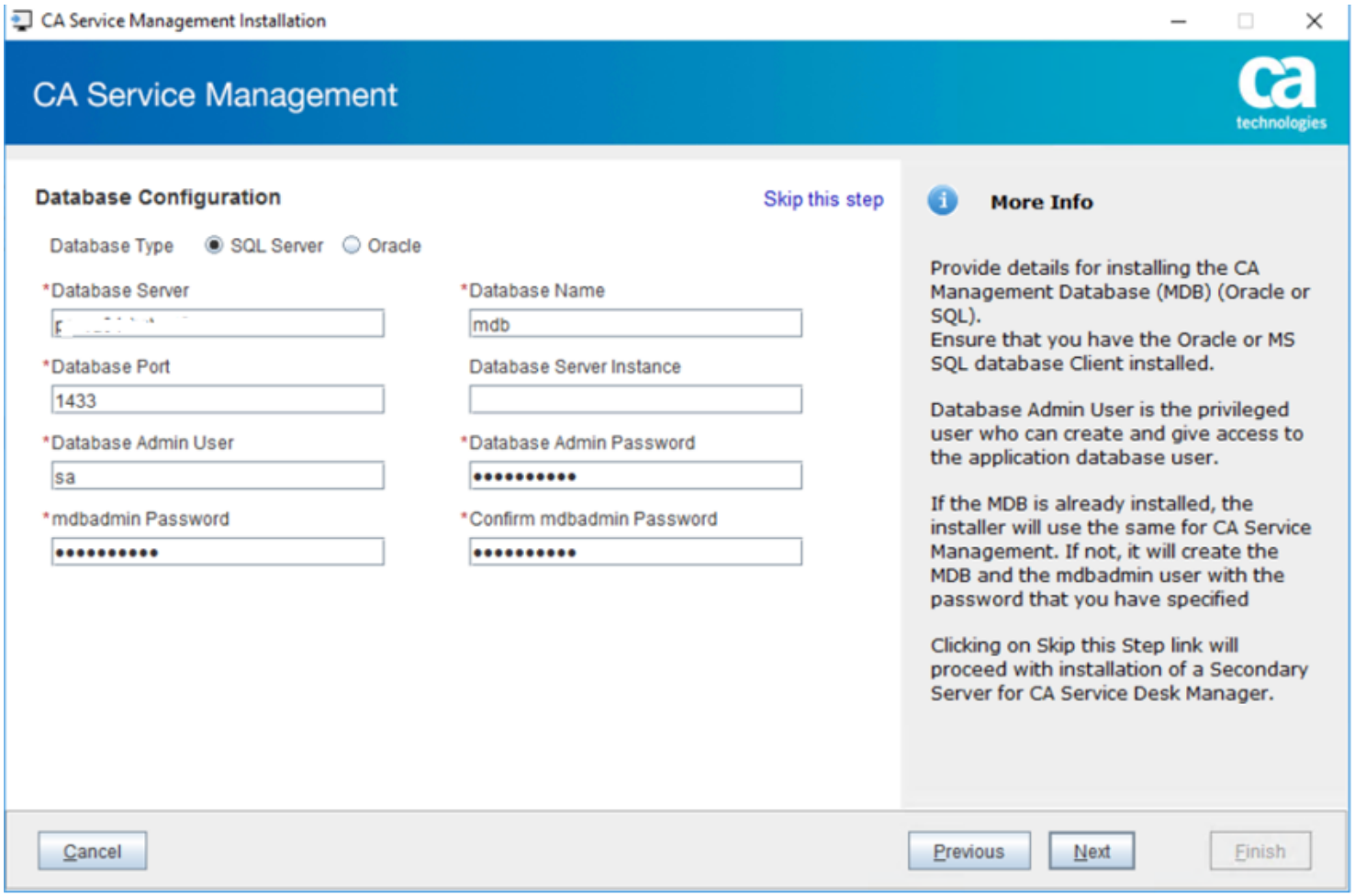
2. Click **Next** and select a language of your choice.



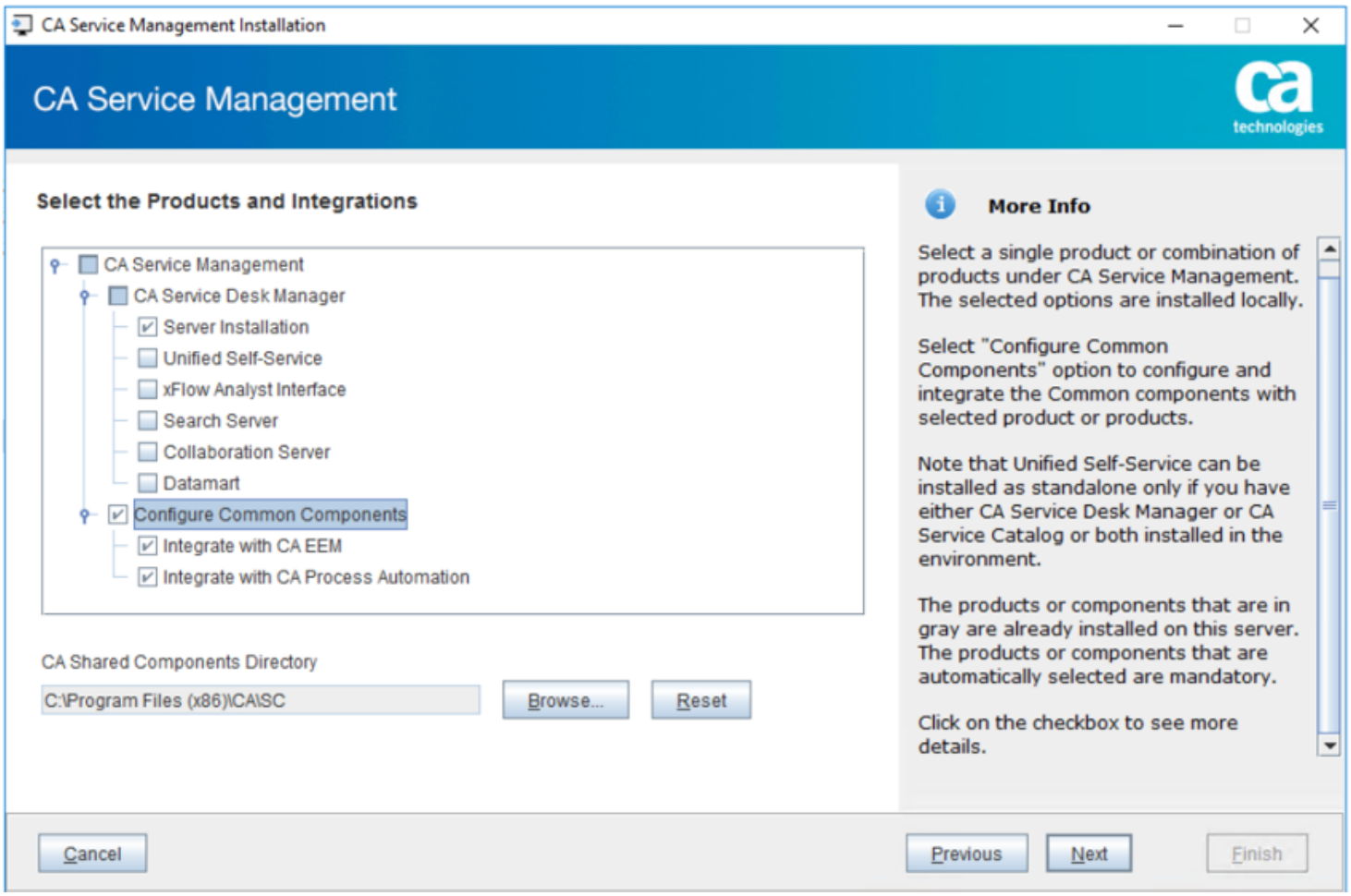
3. Select the "**CA Service Management**" link.



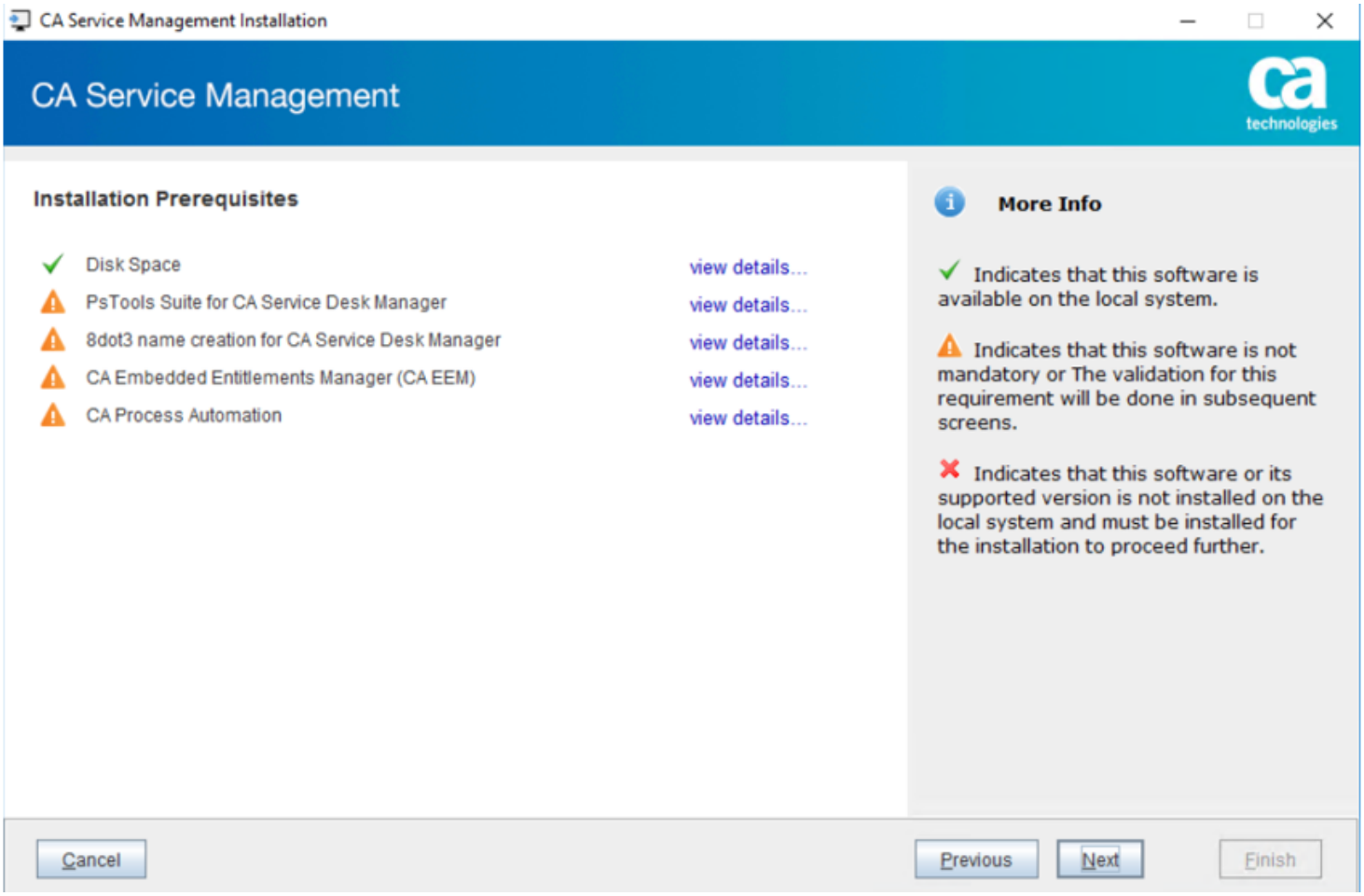
4. Review and accept the License Agreement.
5. Click **Next**.
   Select Database Type as **SQL Server**. Enter the database details.



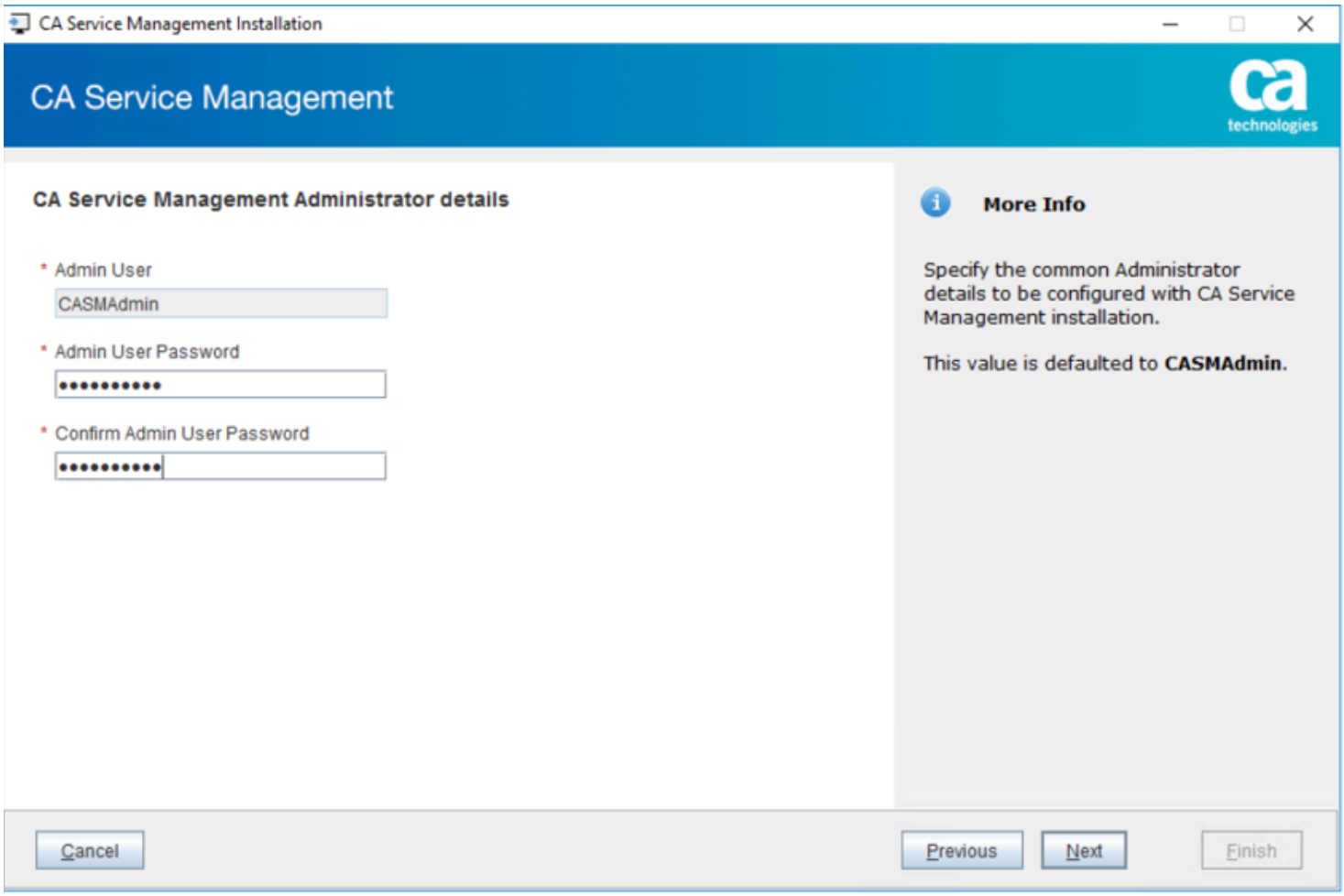
6. Click **Next** and configure the Common Components (CA EEM, CA Process Automation Workflow).


CA Service Management Installation
CA Service Management
ca technologies
Select the Products and Integrations
CA Service Management
CA Service Desk Manager
Server Installation
Unified Self-Service
xFlow Analyst Interface
Search Server
Collaboration Server
Datamart
Configure Common Components
Integrate with CA EEM
Integrate with CA Process Automation
CA Shared Components Directory
C:\Program Files (x86)\CA\SC
Browse...
Reset
More Info
Select a single product or combination of products under CA Service Management. The selected options are installed locally.
Select "Configure Common Components" option to configure and integrate the Common components with selected product or products.
Note that Unified Self-Service can be installed as standalone only if you have either CA Service Desk Manager or CA Service Catalog or both installed in the environment.
The products or components that are in gray are already installed on this server. The products or components that are automatically selected are mandatory.
Click on the checkbox to see more details.
Cancel
Previous
Next
Finish

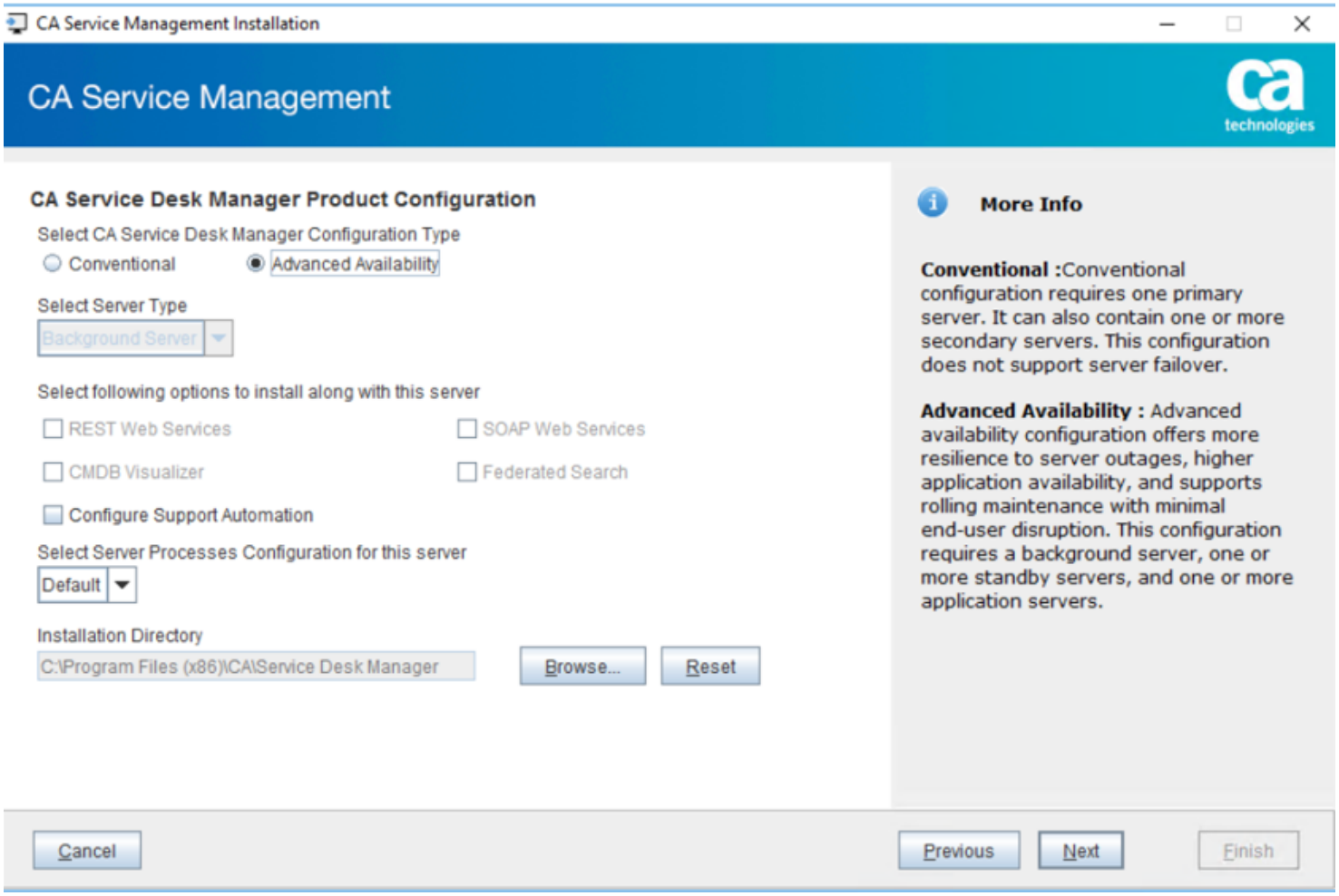
7. Click **Next**.



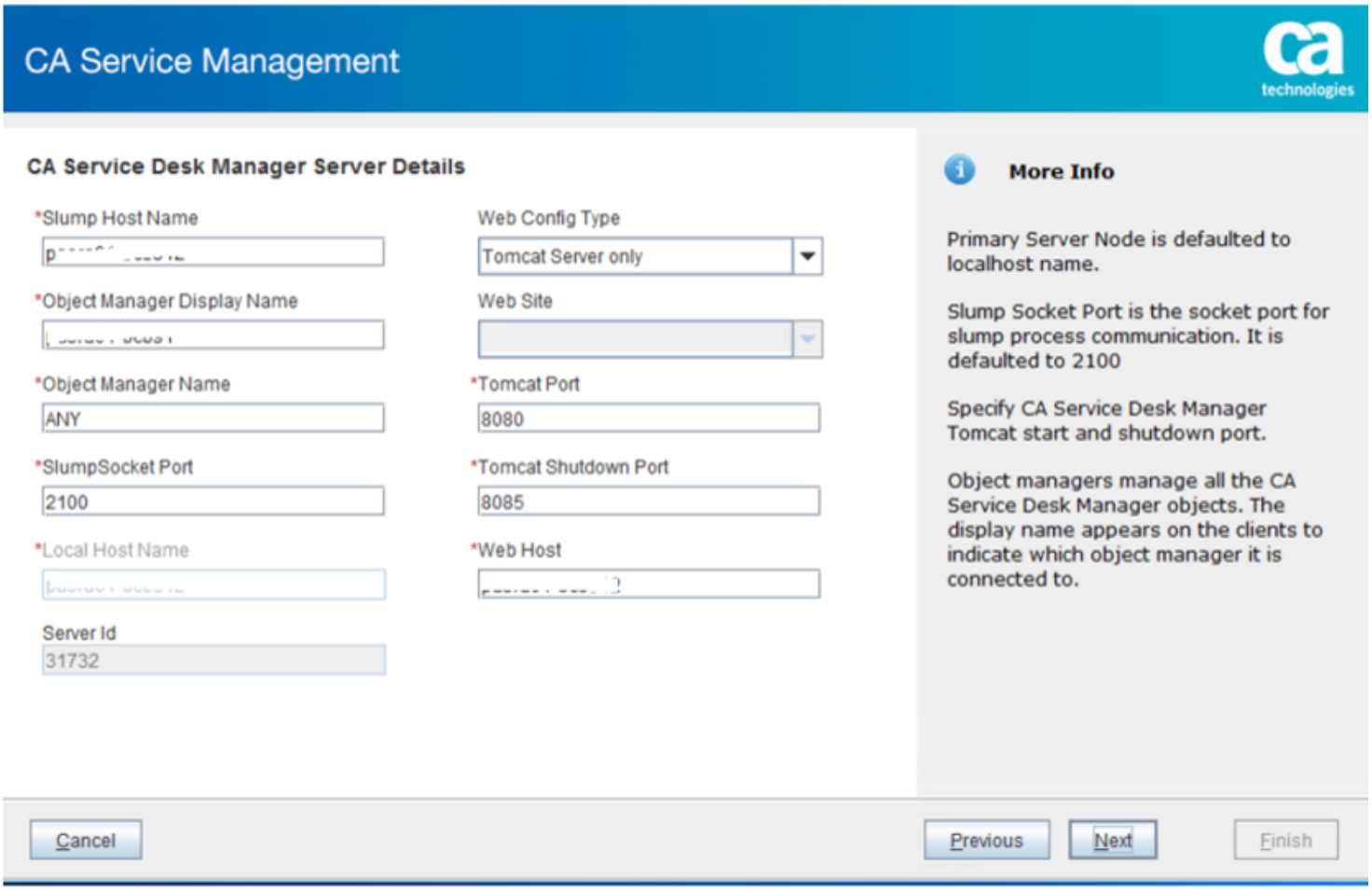
8. Enter CASM Admin User password.



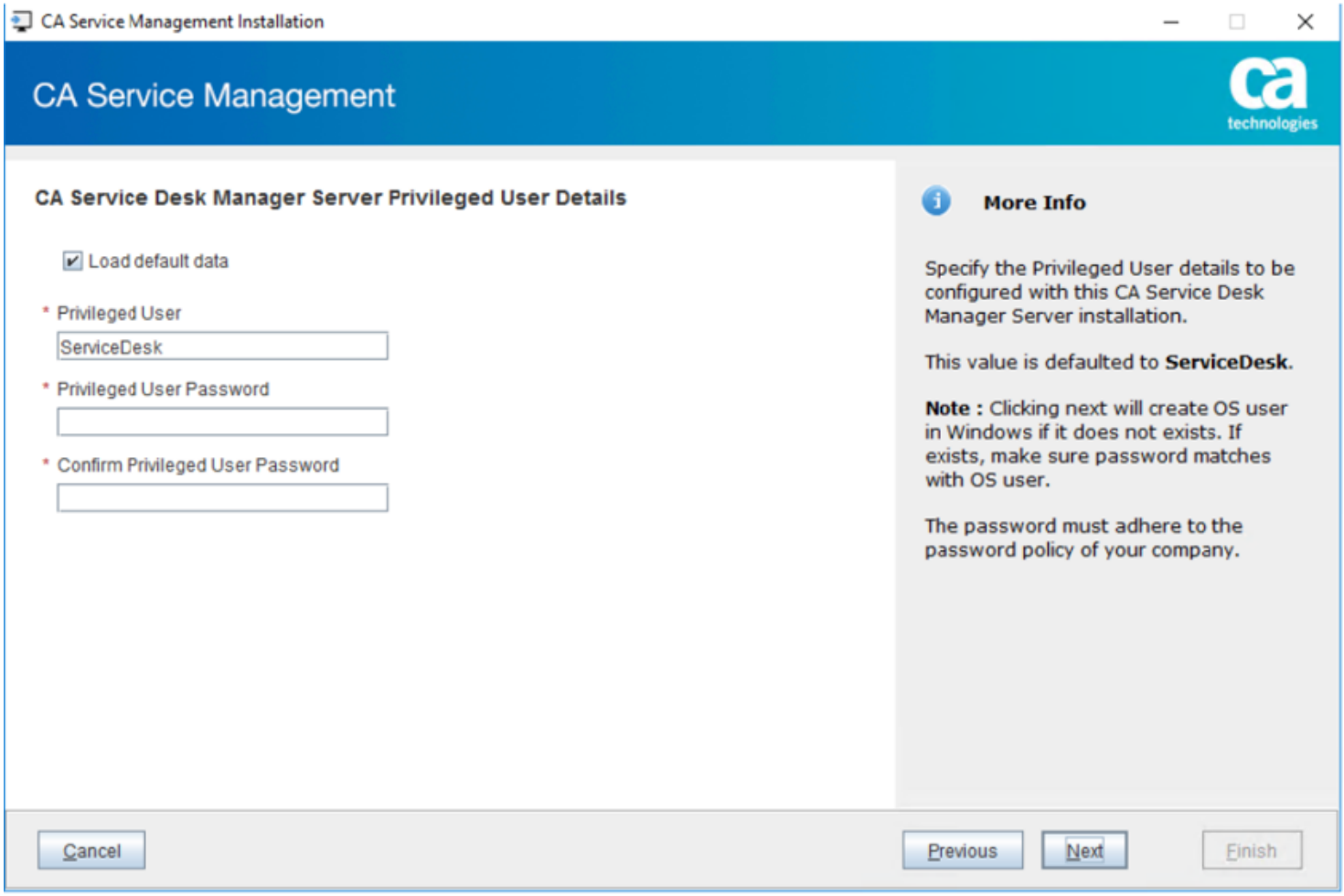
9. Click **Next**.
   Select **Advanced Availability** as the mode of SDM configuration and provide the required details.



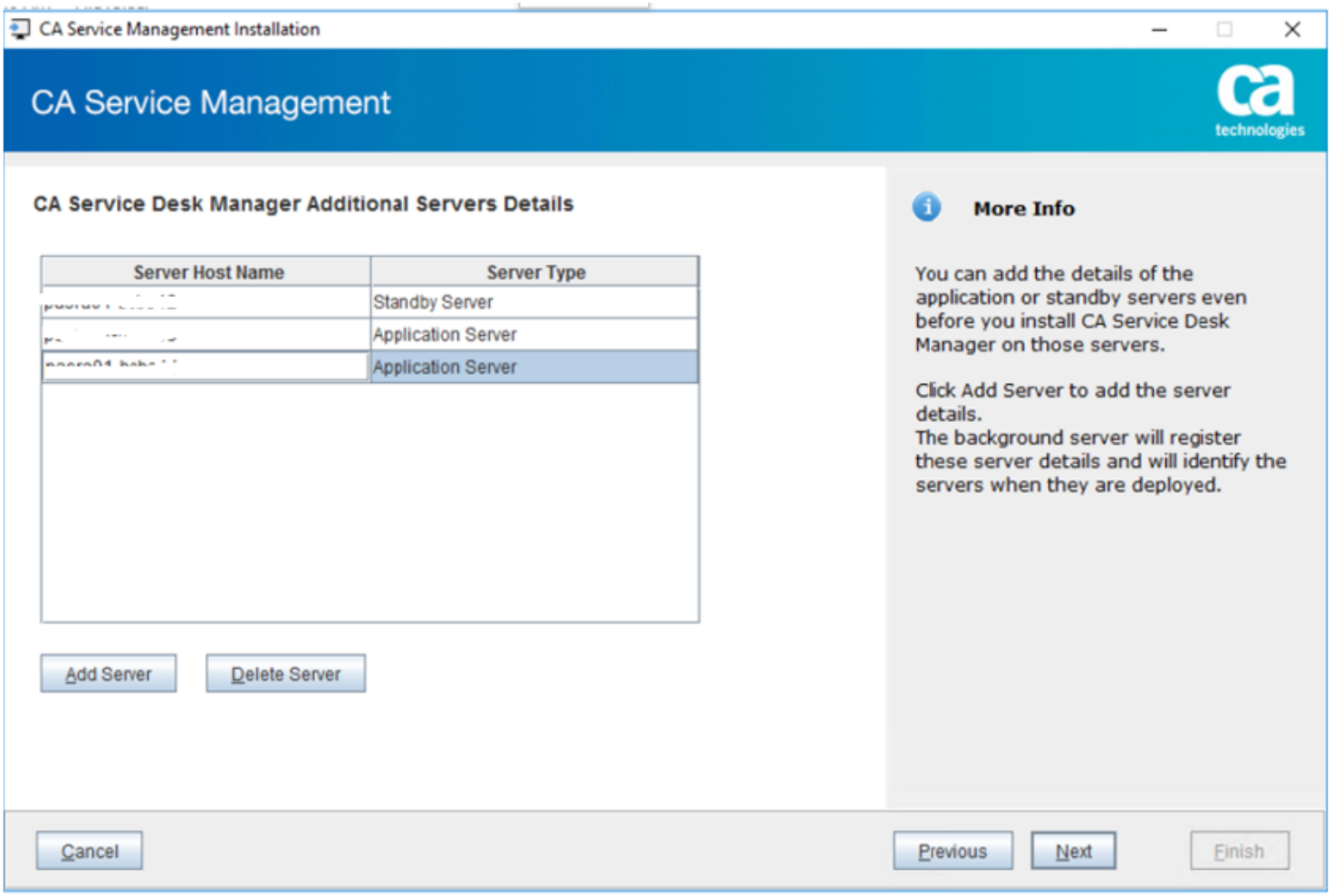
10. Click **Next**.
    Enter slump host name, slump port details, object manager name, and tomcat port and other details.



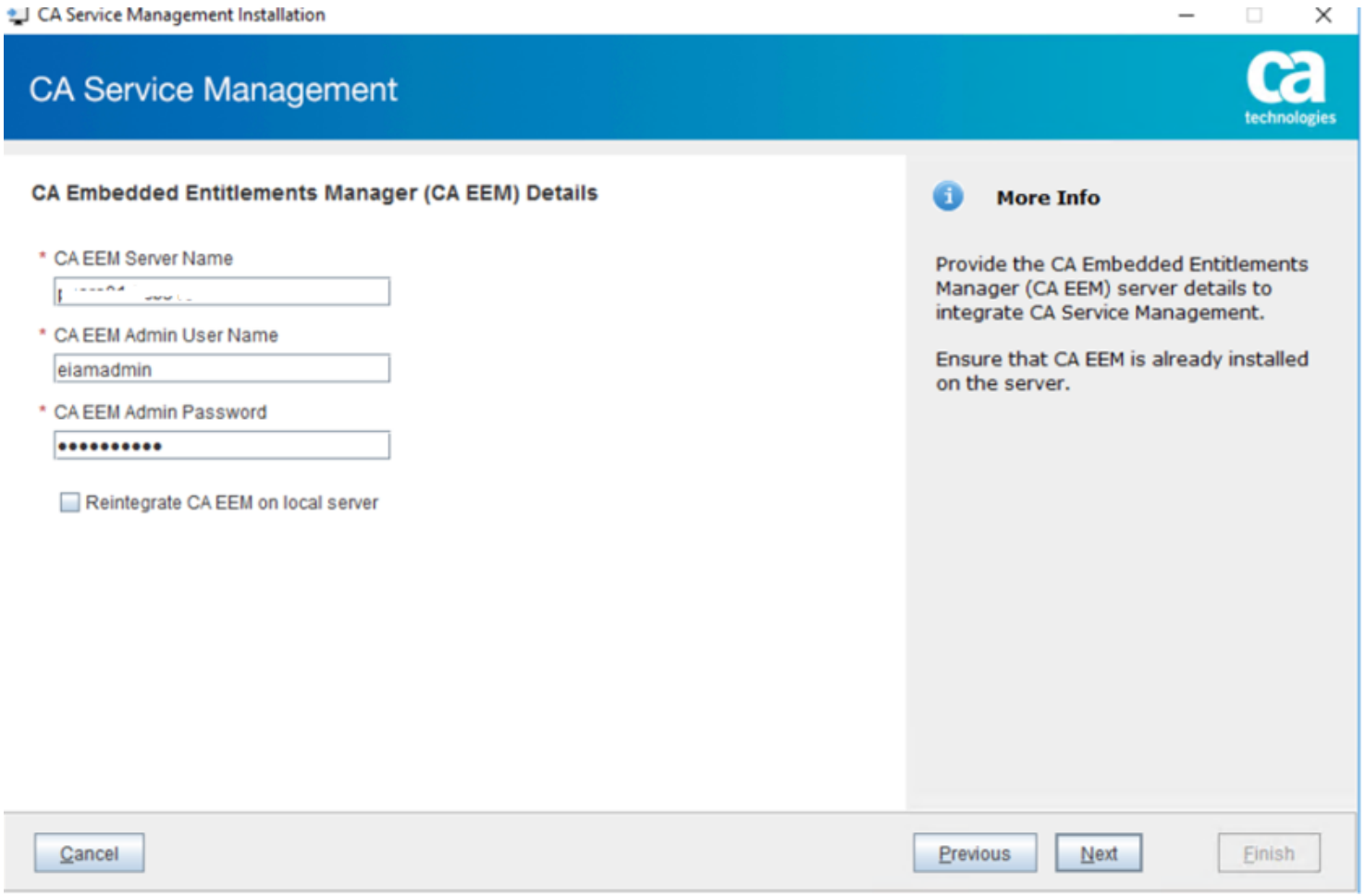
11. Click **Next**.
    Enter the Privileged User password and confirm password.



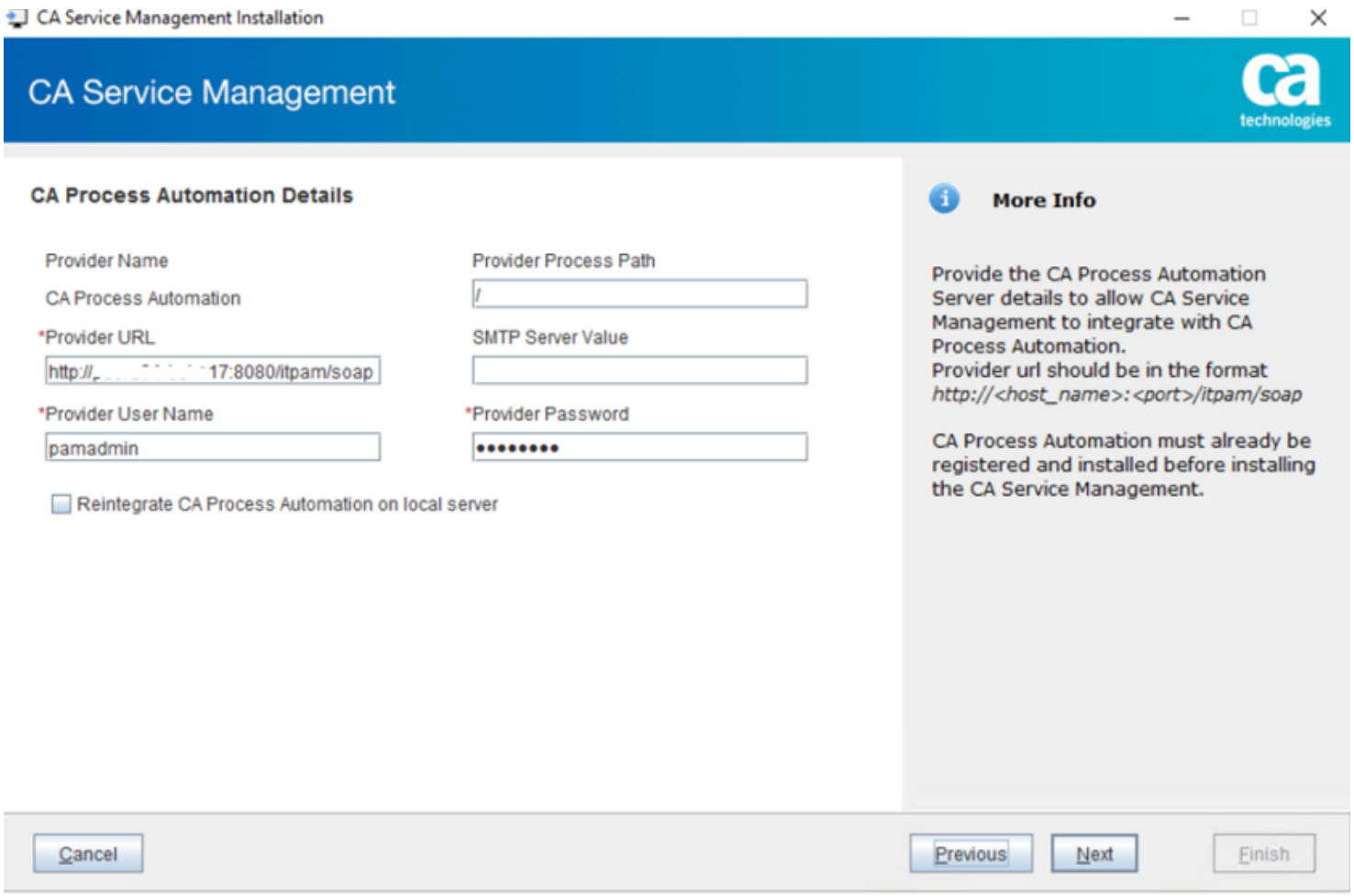
12. Click **Next**.
    Add the Standby Server **testqa-sb-new**, two Application Server - **testqa-app1-new** and **testqa-app2-new**.



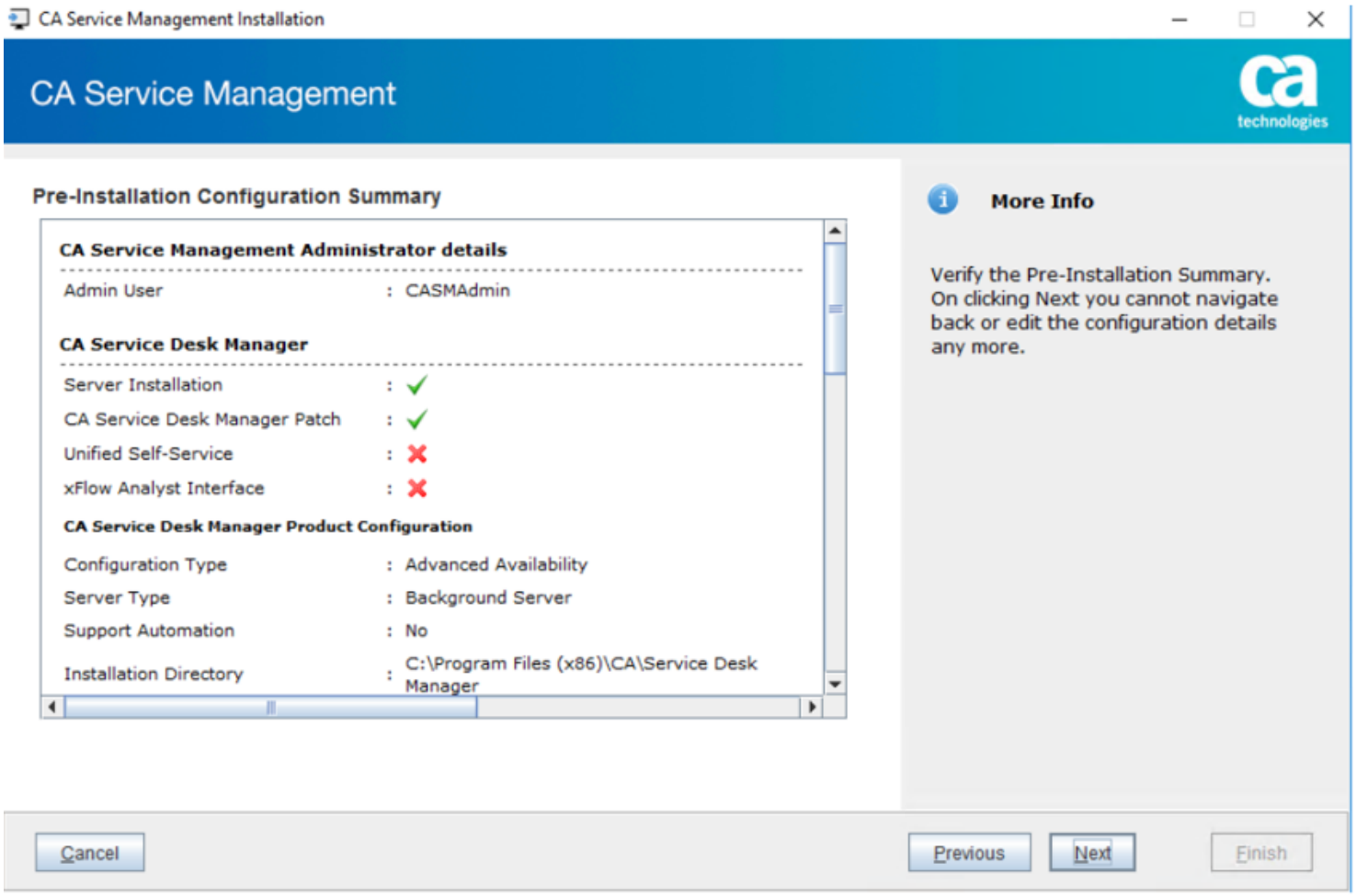
13. Enter EEM Host name (**testqa-eem-new**) and Password.



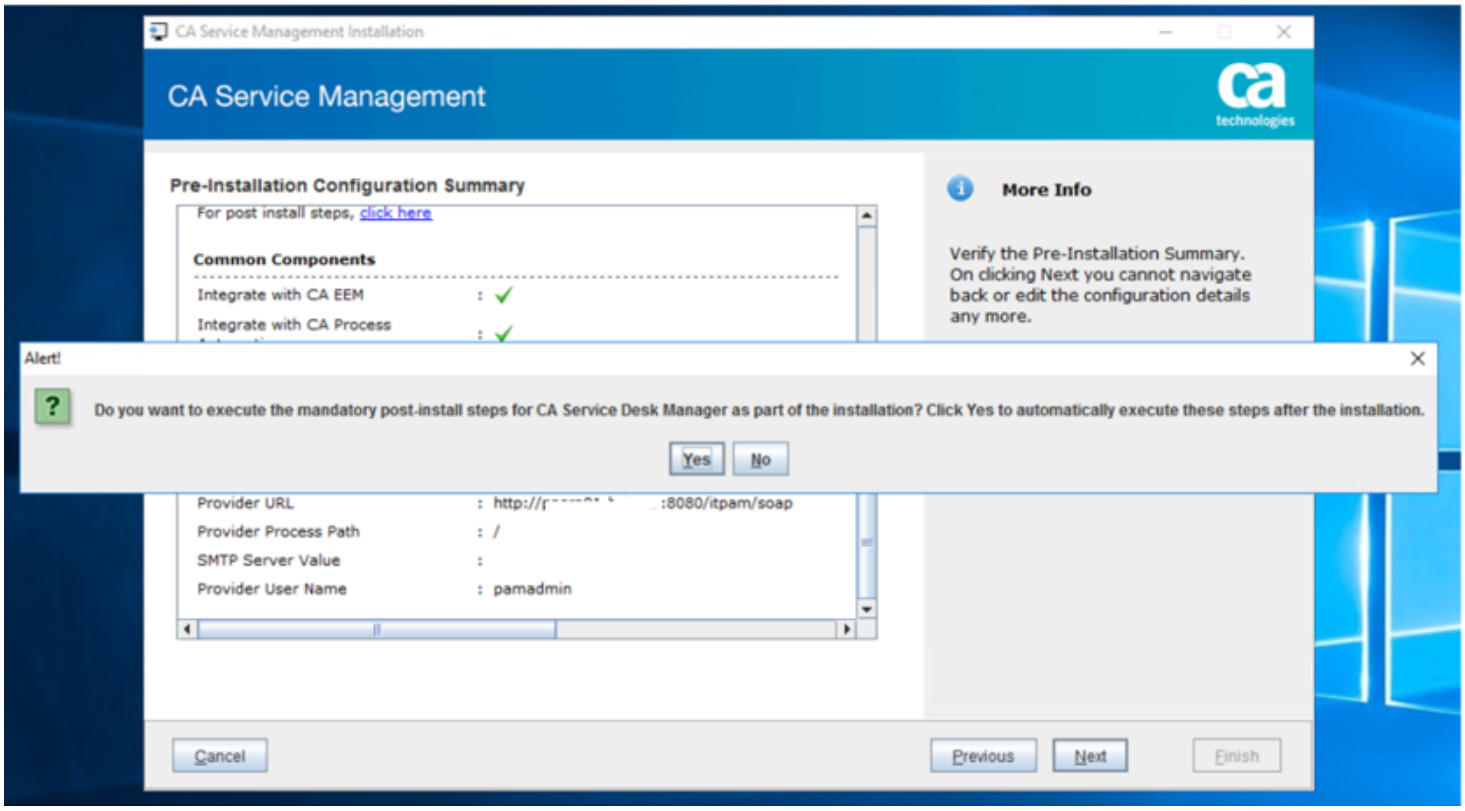
14. Click **Next**.
    Enter CA Process Automation URL: *http://testqa-pam-new:8080/itpam/soap*
15. Enter the user name and Password.



16. Click **Next**. Review the Pre-installation configuration summary.



17. Click **Next**. Follow the installation wizard.
18. Click **Yes** to automatically execute the post-installation steps for CA SDM. Click **No** if you want to perform the manual post-install steps once installation completes successfully.



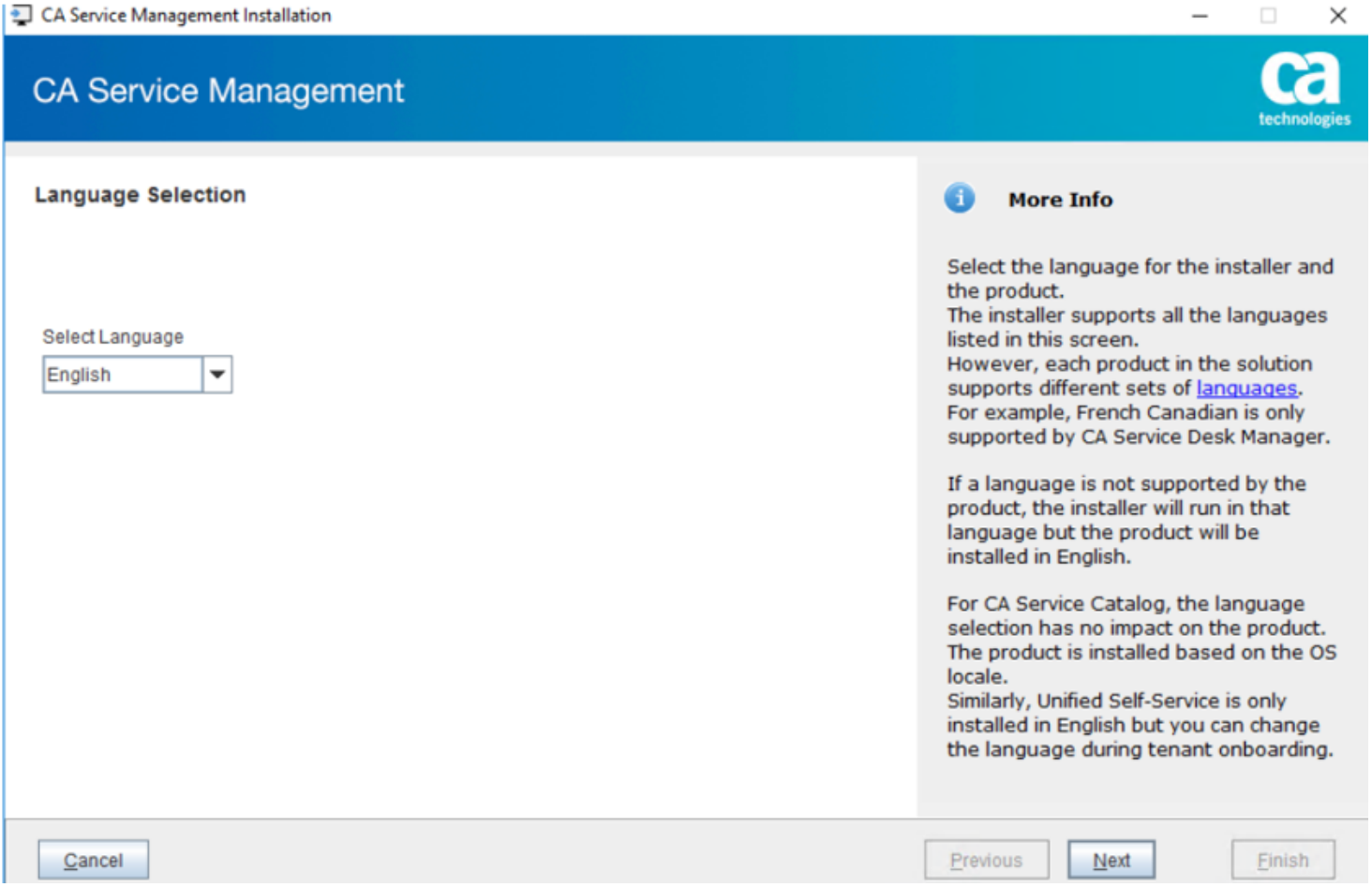
19. Click **Install**.
    The installation process begins.
20. Click **Finish** once installation is successful.
21. Now open the CA SDM Background server web interface: *http://testqa-bg-new:8080/CAisd/pdmweb.exe*
22. Click **Login**.

CA SDM background server is launched.

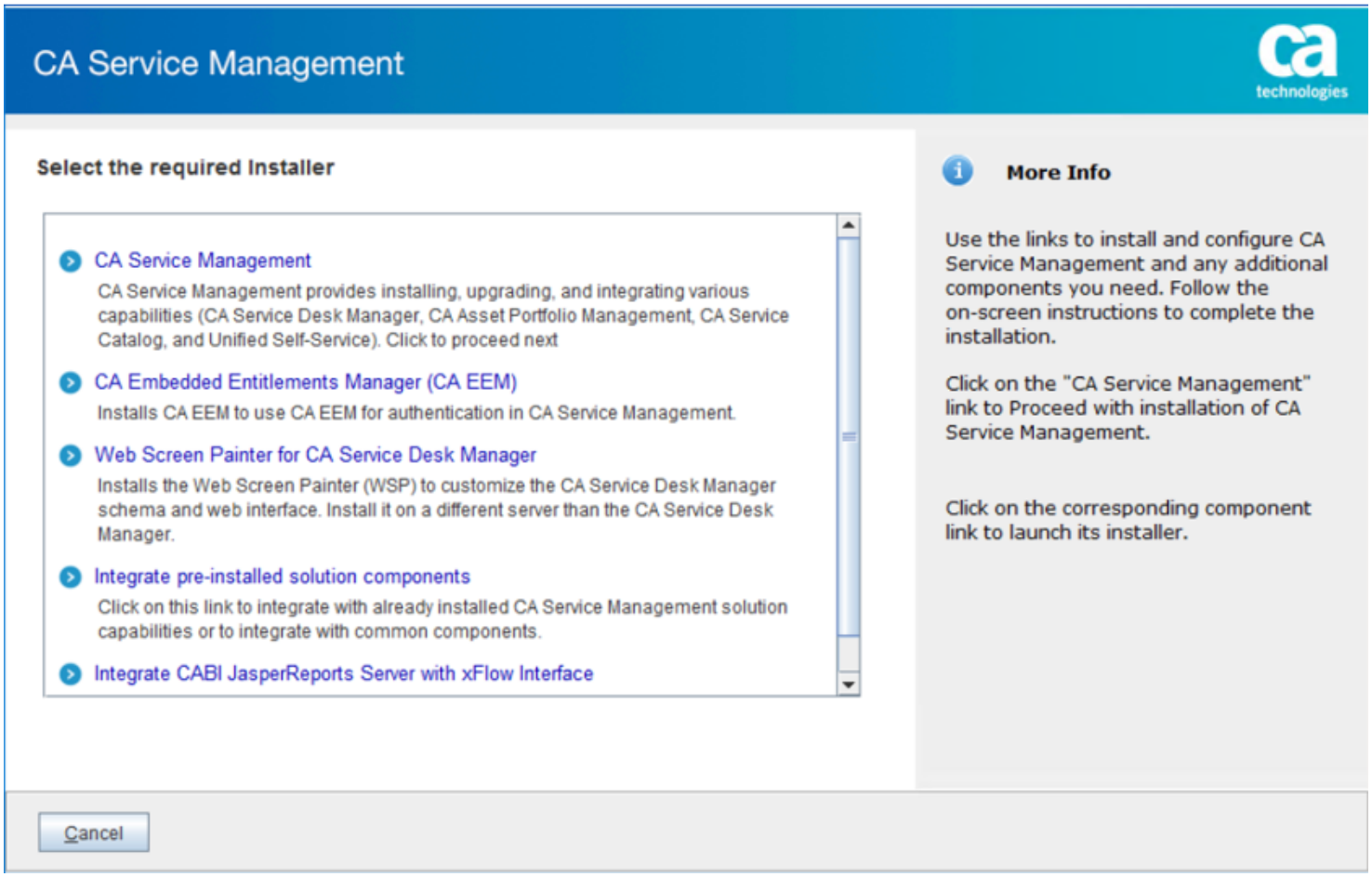
Install CA SDM Standby Server (testqa-sb-new)

Perform the following steps to install the standby server:

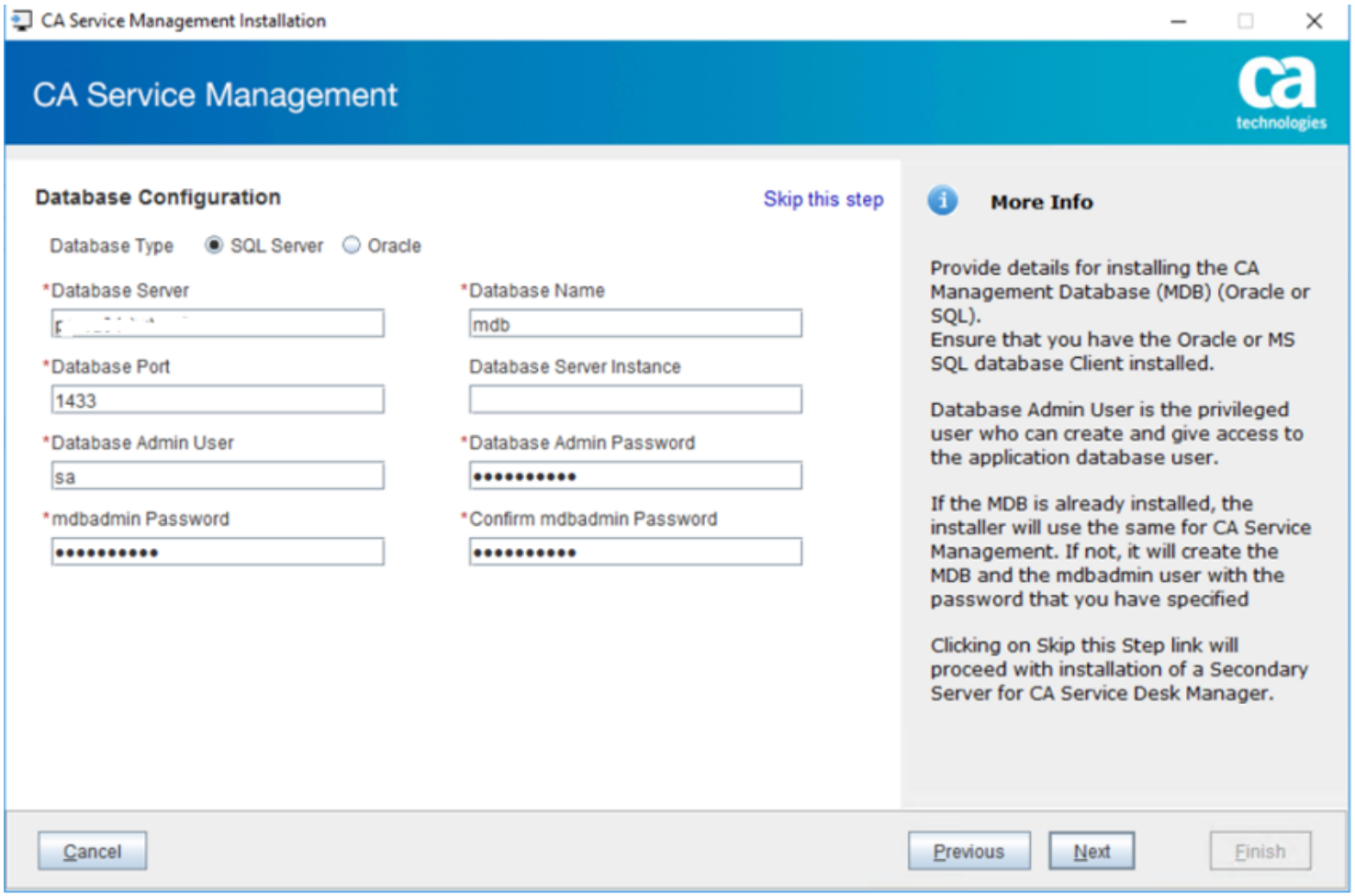
1. Login to Broadcom Support.
2. Mount CA Service Management 17.4 DVD and Launch the installer.
3. Select Language as **English**.



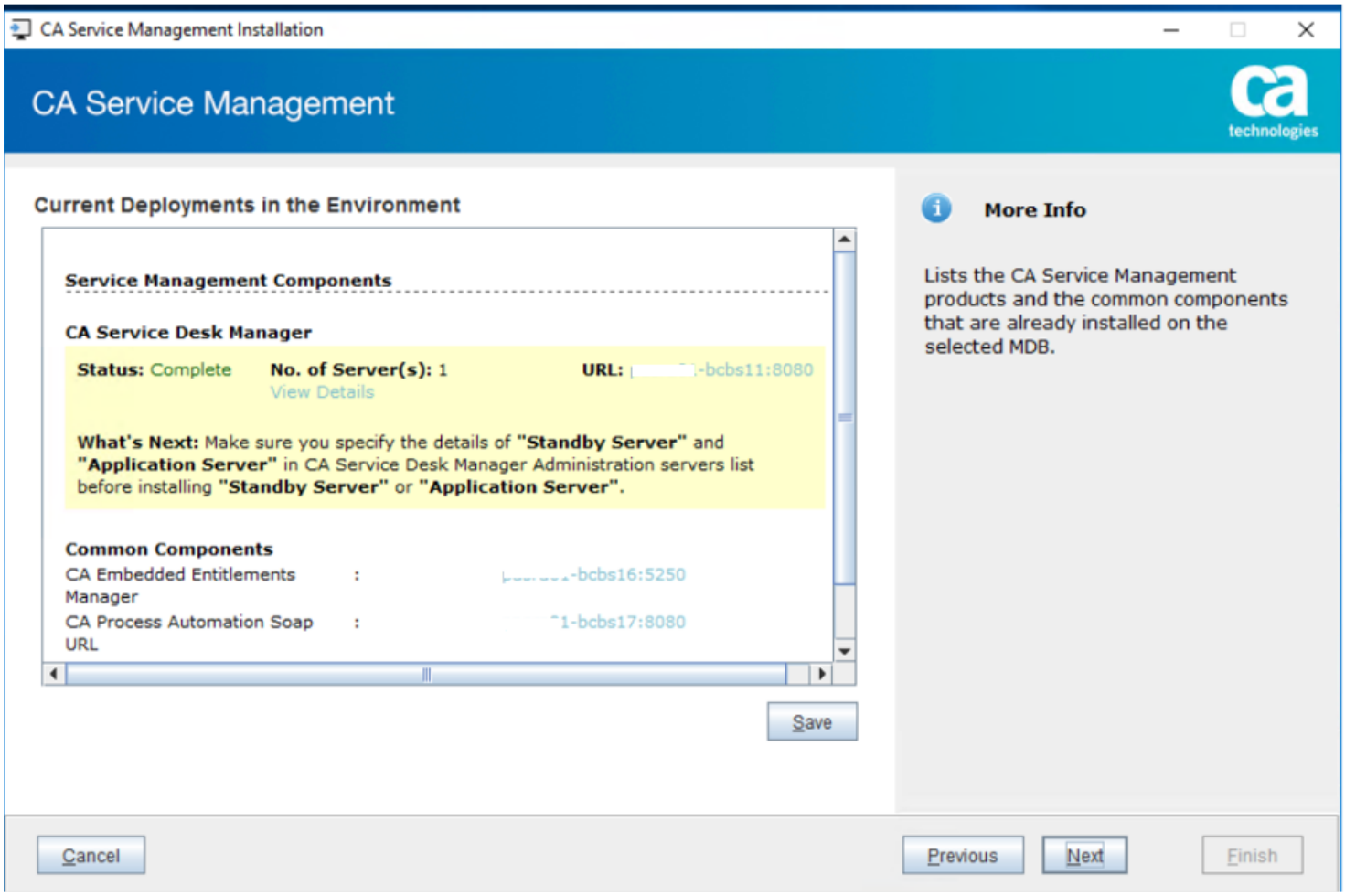
4. Click CA Service Management Link to run the CA Service Management installation wizard.



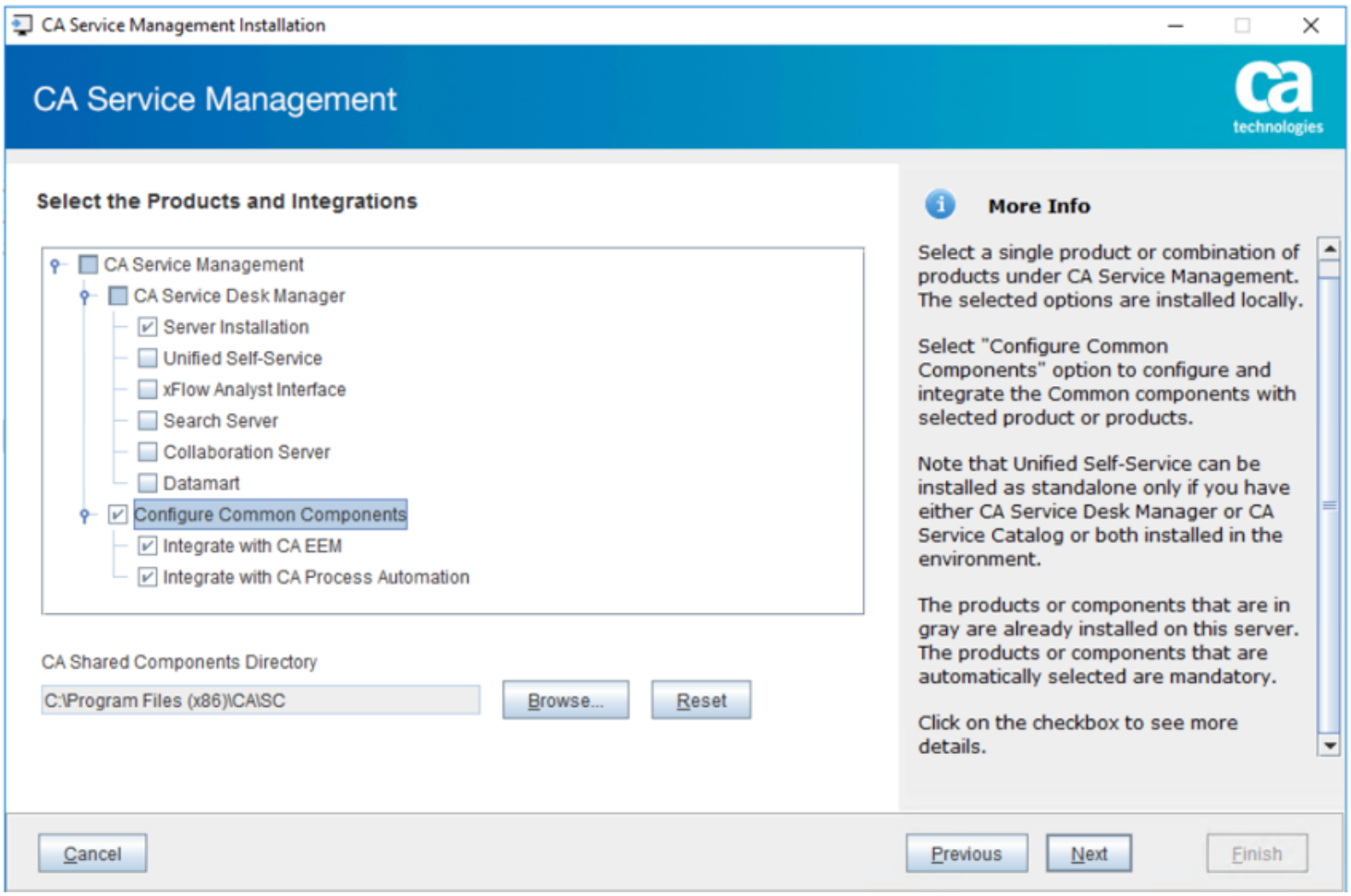
5. Review and accept the license agreement.
6. Select Database type as SQL Server. Enter the Database Server details (**testqa-mdb-new**) and provide the database password.


CA Service Management Installation
CA Service Management
technologies
Database Configuration
Skip this step
Database Type
SQL Server
Oracle
*Database Server
*Database Name
mdb
*Database Port
1433
Database Server Instance
*Database Admin User
sa
*Database Admin Password
*mdbadmin Password
*Confirm mdbadmin Password
More Info
Provide details for installing the CA Management Database (MDB) (Oracle or SQL).
Ensure that you have the Oracle or MS SQL database Client installed.
Database Admin User is the privileged user who can create and give access to the application database user.
If the MDB is already installed, the installer will use the same for CA Service Management. If not, it will create the MDB and the mdbadmin user with the password that you have specified
Clicking on Skip this Step link will proceed with installation of a Secondary Server for CA Service Desk Manager.
Cancel
Previous
Next
Finish

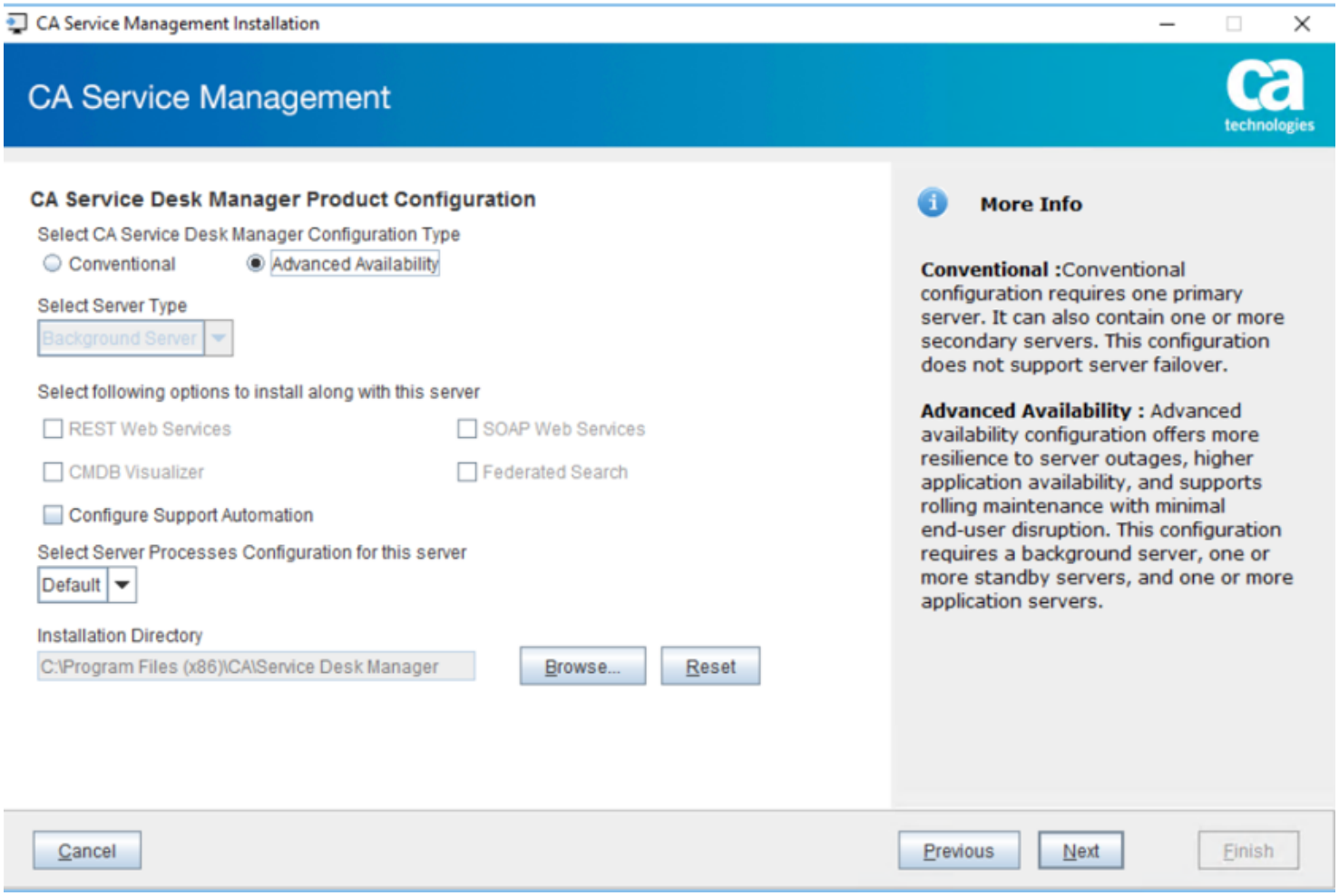
7. Click **Next**.



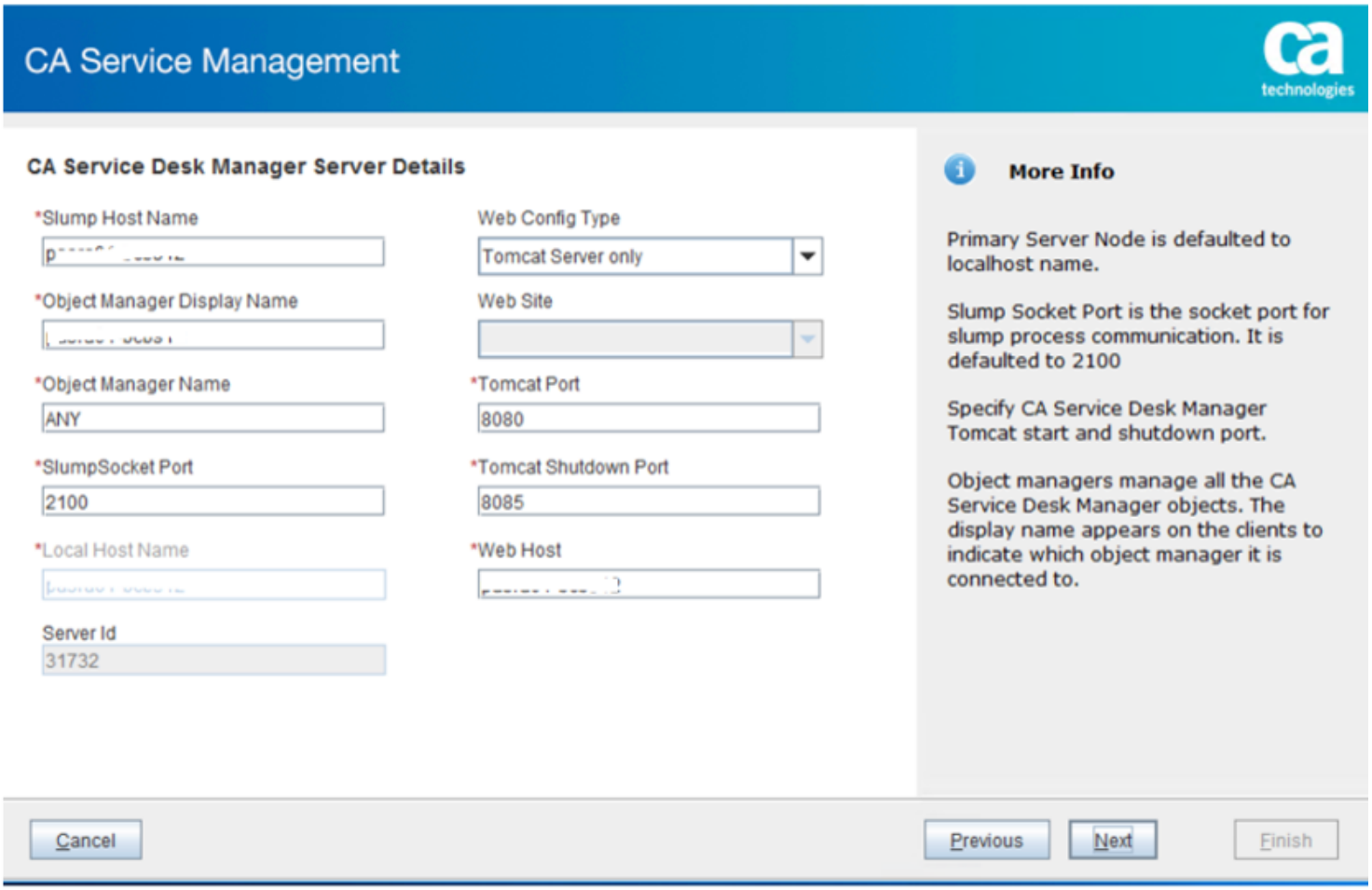
8. Click **Next**.
9. Select Configure Common Components (CA EEM and CA Process Automation).



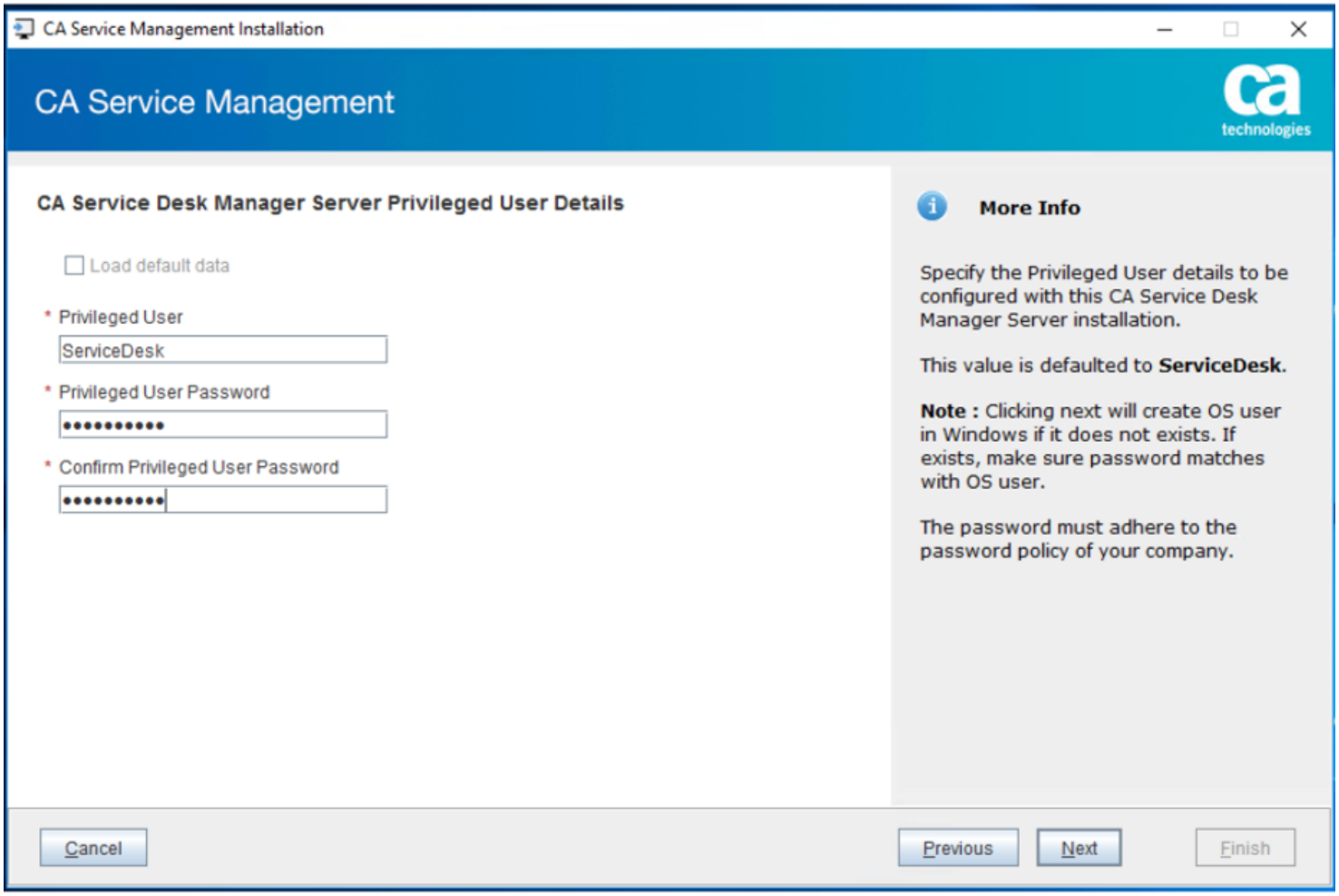
10. Review the installation prerequisites.
11. Click **Next**.
12. Click **Next**.
    Select Advanced Availability configuration. Enter the required details.



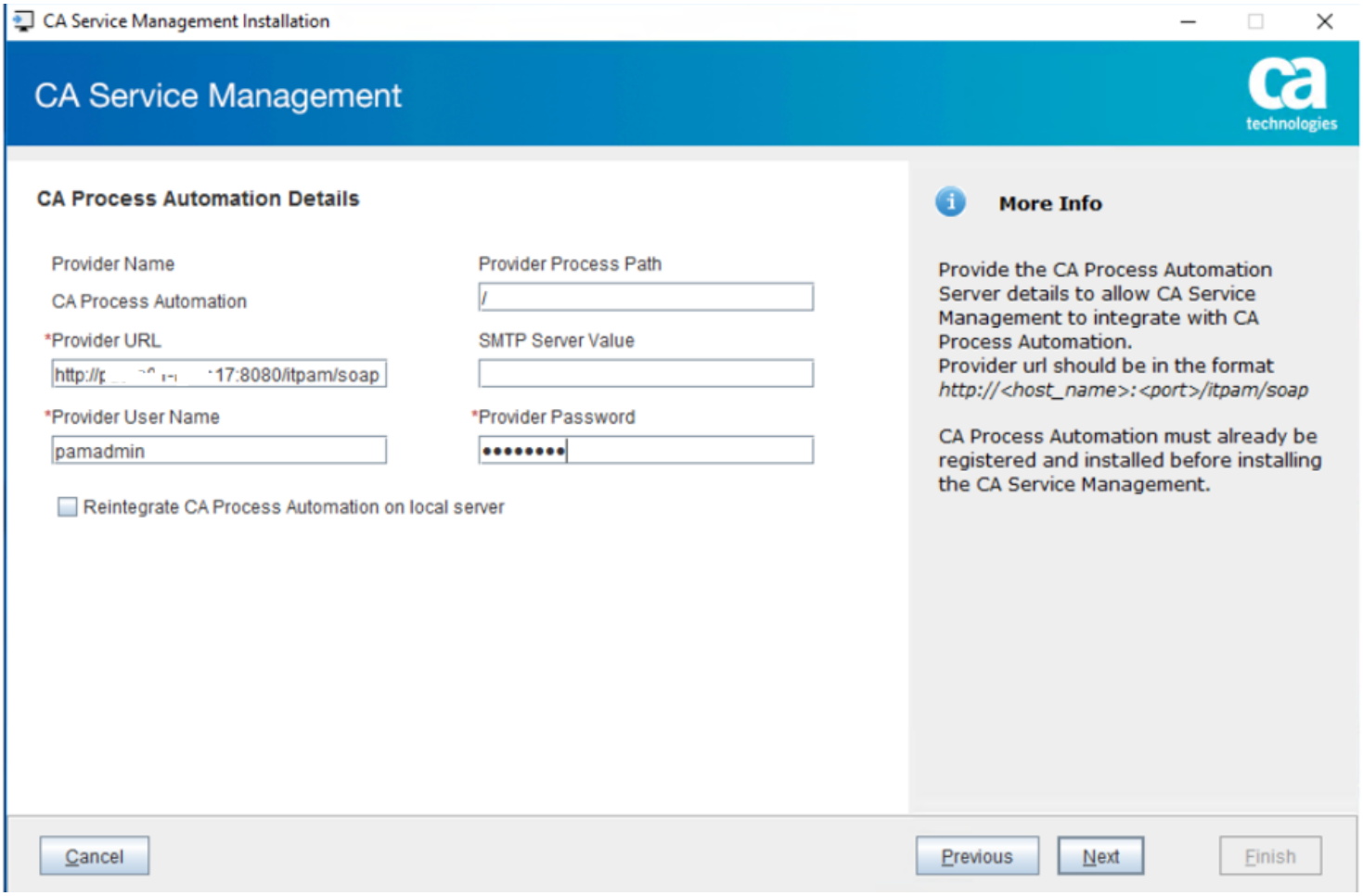
13. Click **Next**.
14. Provide Slump host name and other details (Tomcat port, slump host name, slump port, Object Manager name and web host details) as seen below.



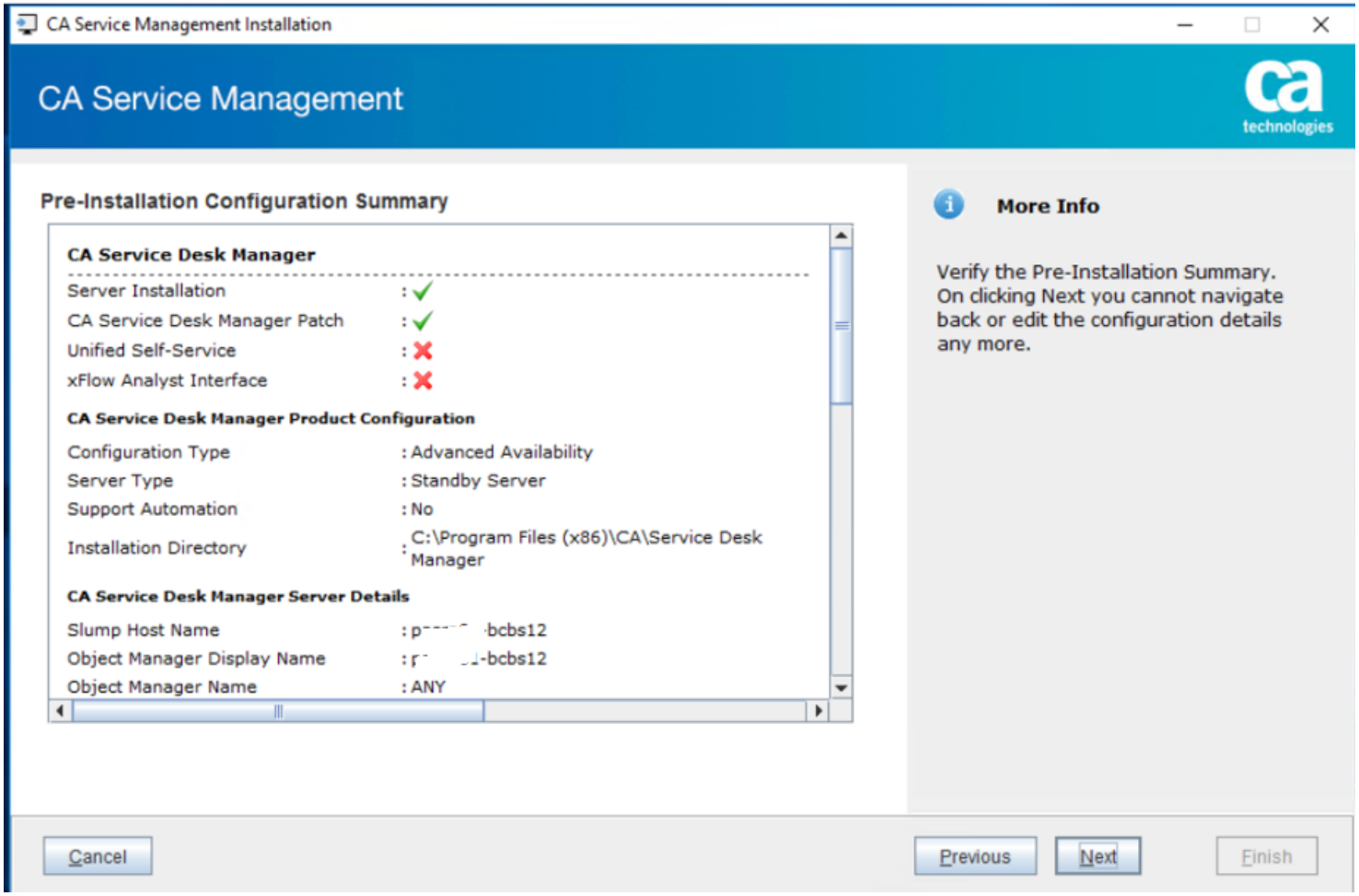
15. Click**Next**.
    Enter Privileged User Password and Confirm Privileged User Password



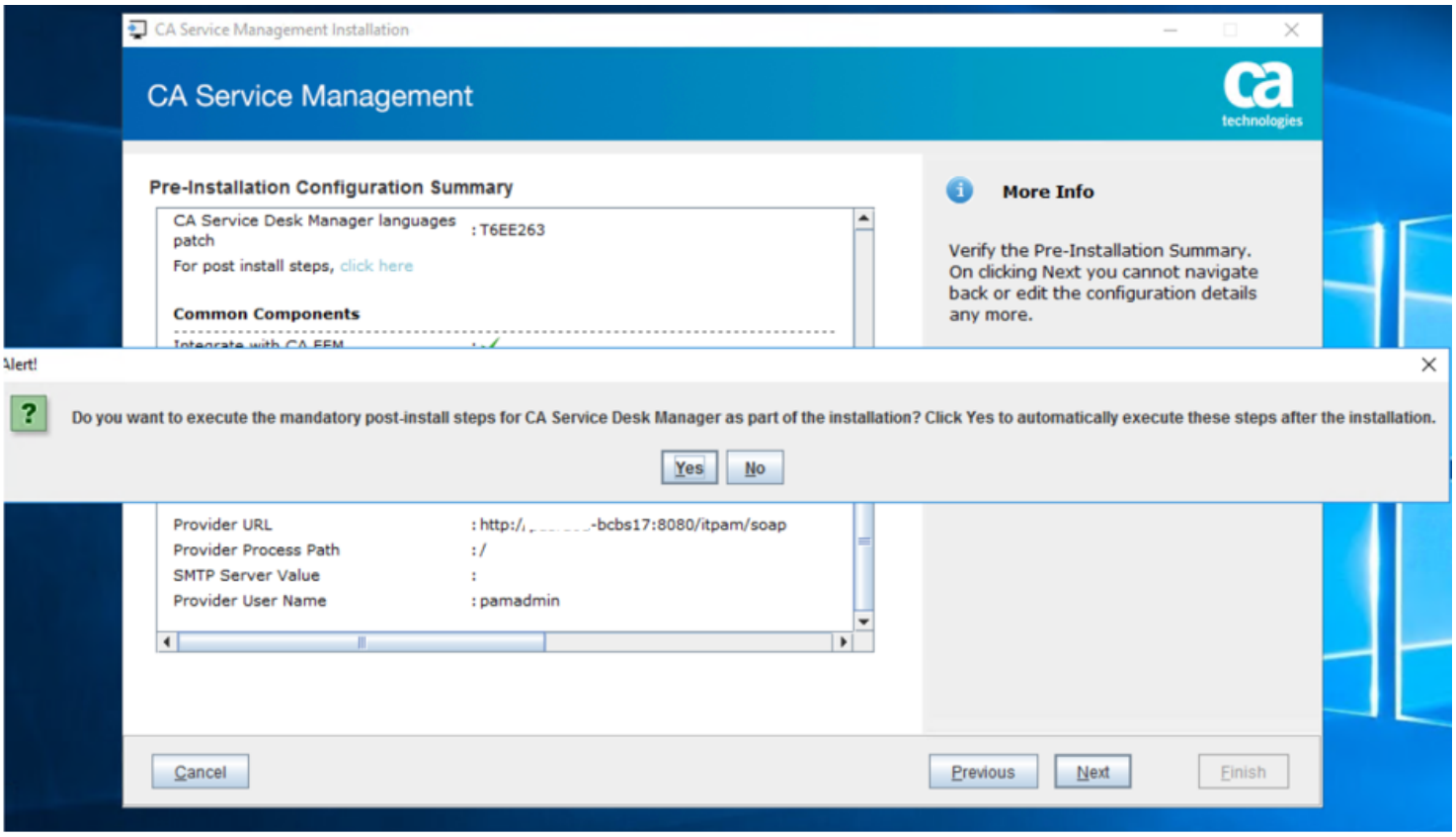
16. Click **Next**.
    Enter EEM Admin Password
17. Click **Next**.
    Enter CA Process Automation Provider Password



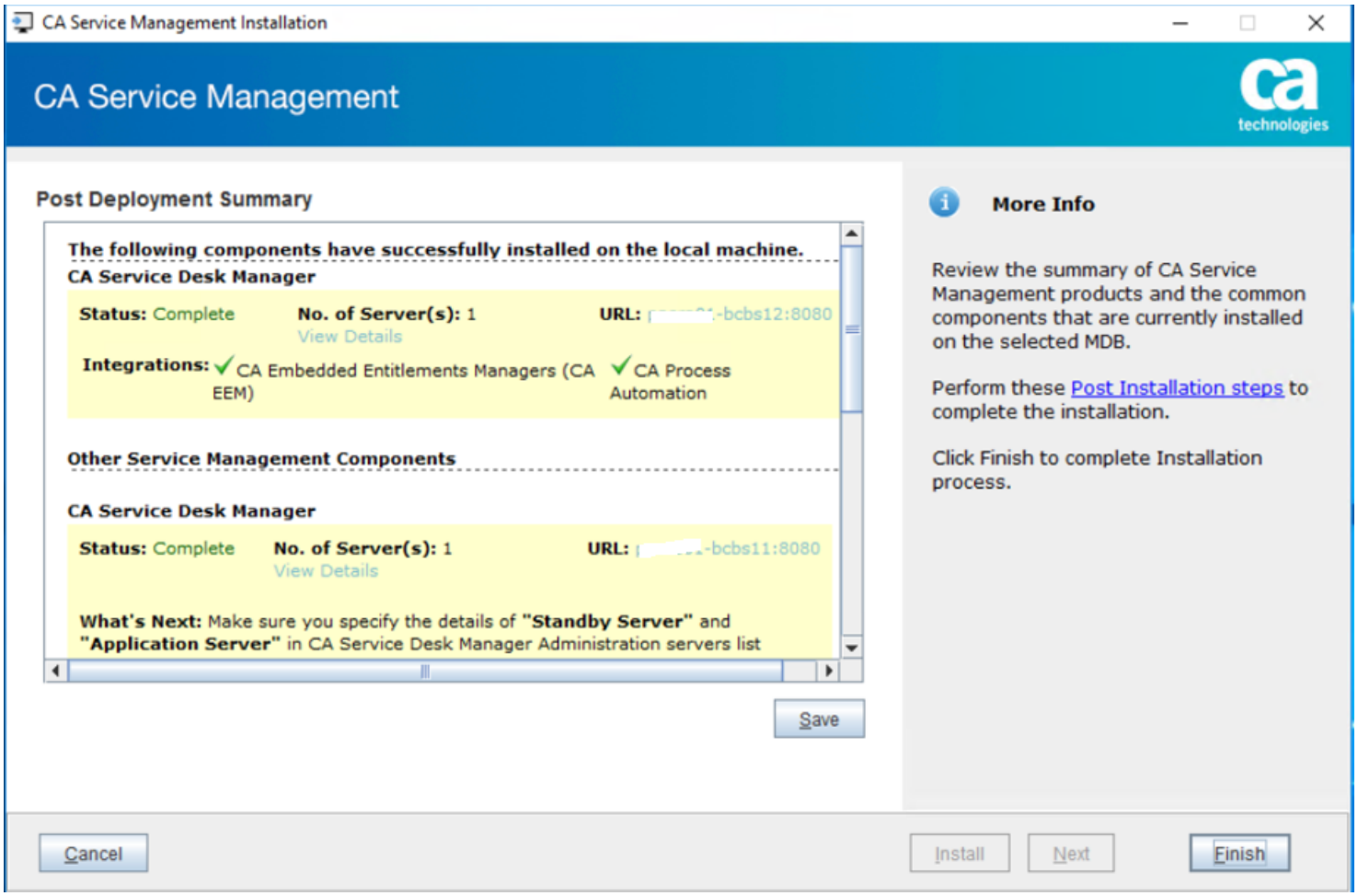
18. Click **Next**.
19. Review the Post-Installation Configuration Summary.
    It shows installation progress for selected components.



20. Click **Yes** to automatically execute the post-installation steps. Click **No** to perform manual post-installation steps.



21. Click **Install**.

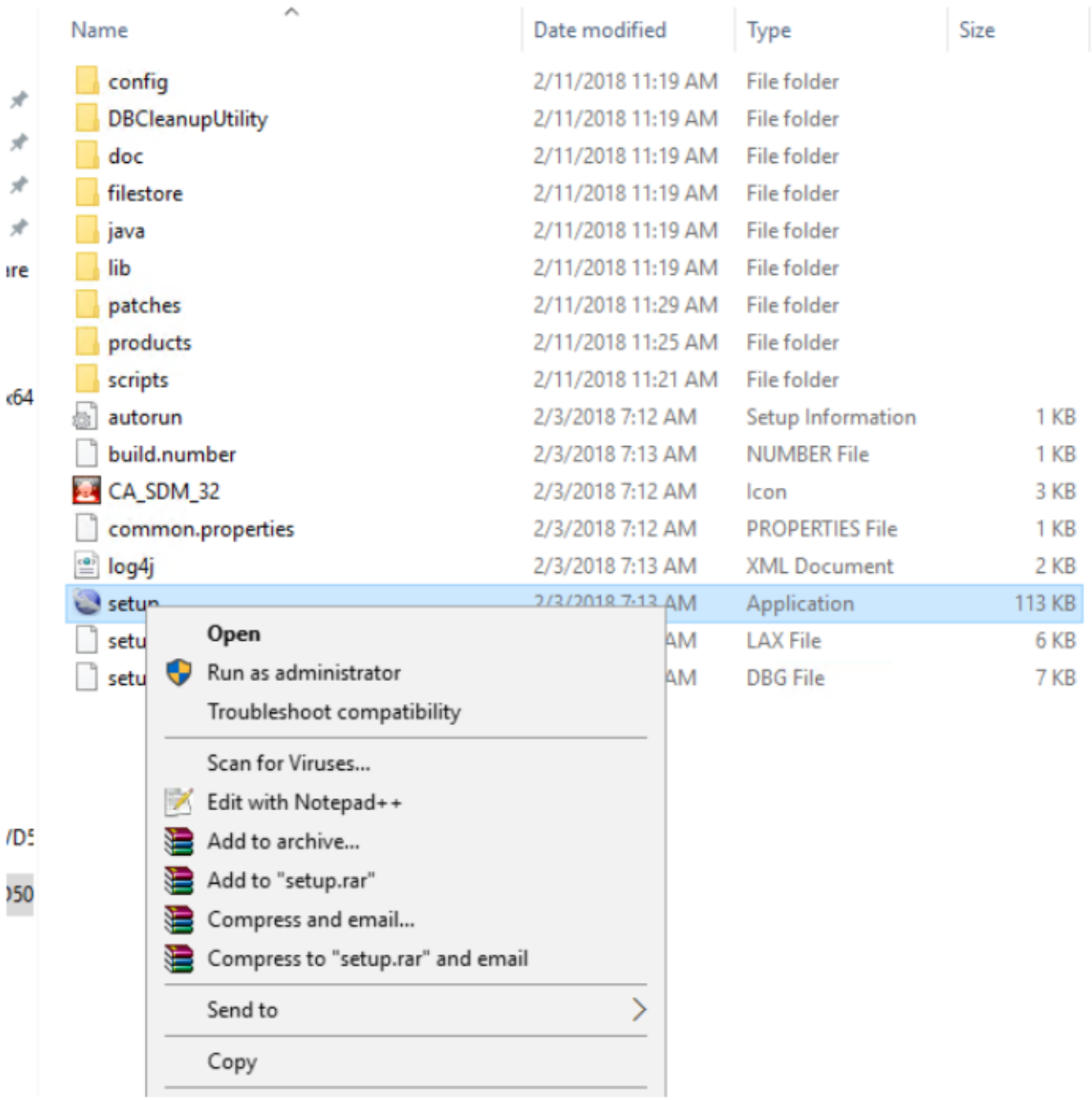


22. Click **Finish**.

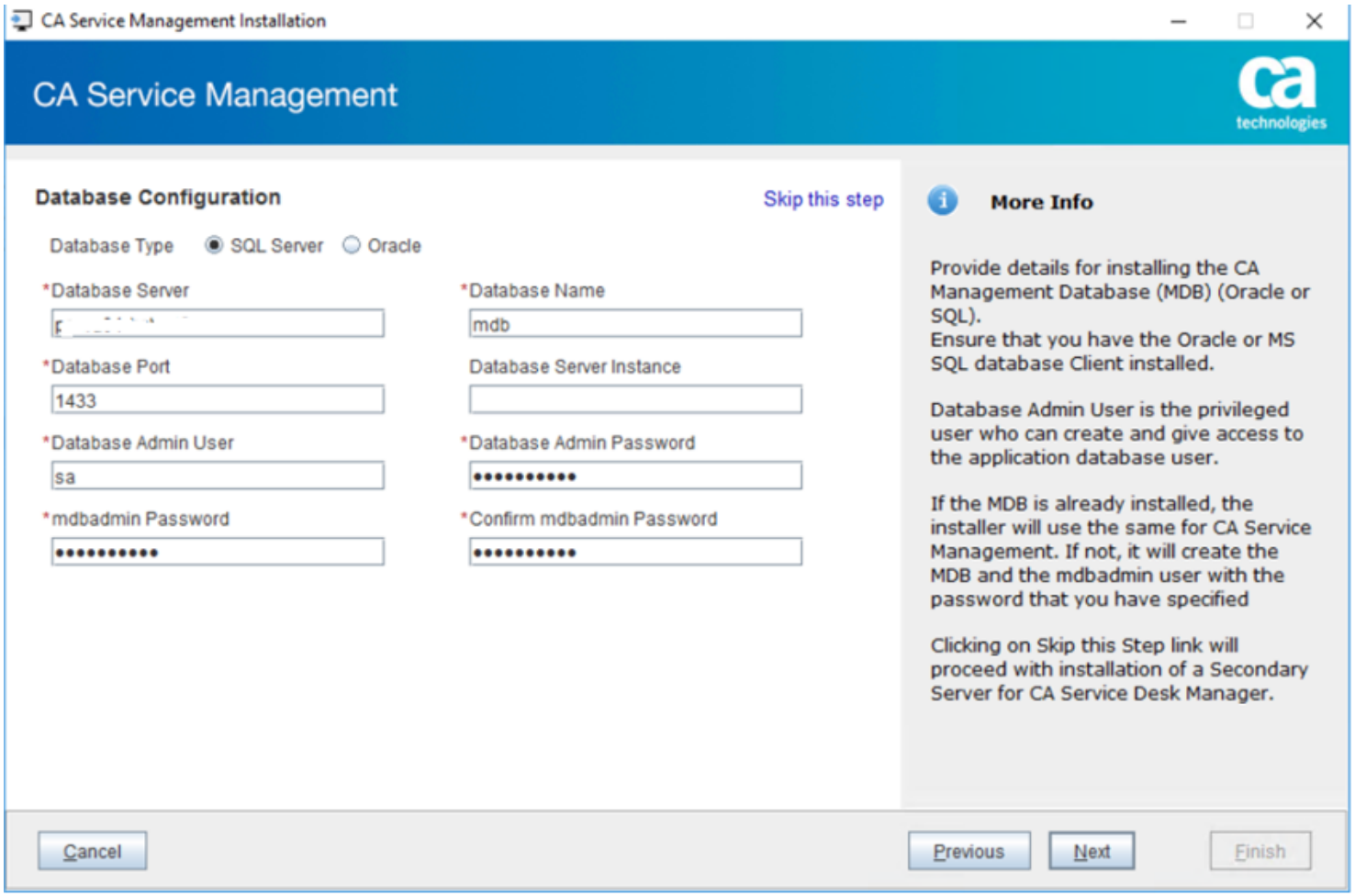
Install the First Application Server (testqa-app1-new)

Perform the following steps:

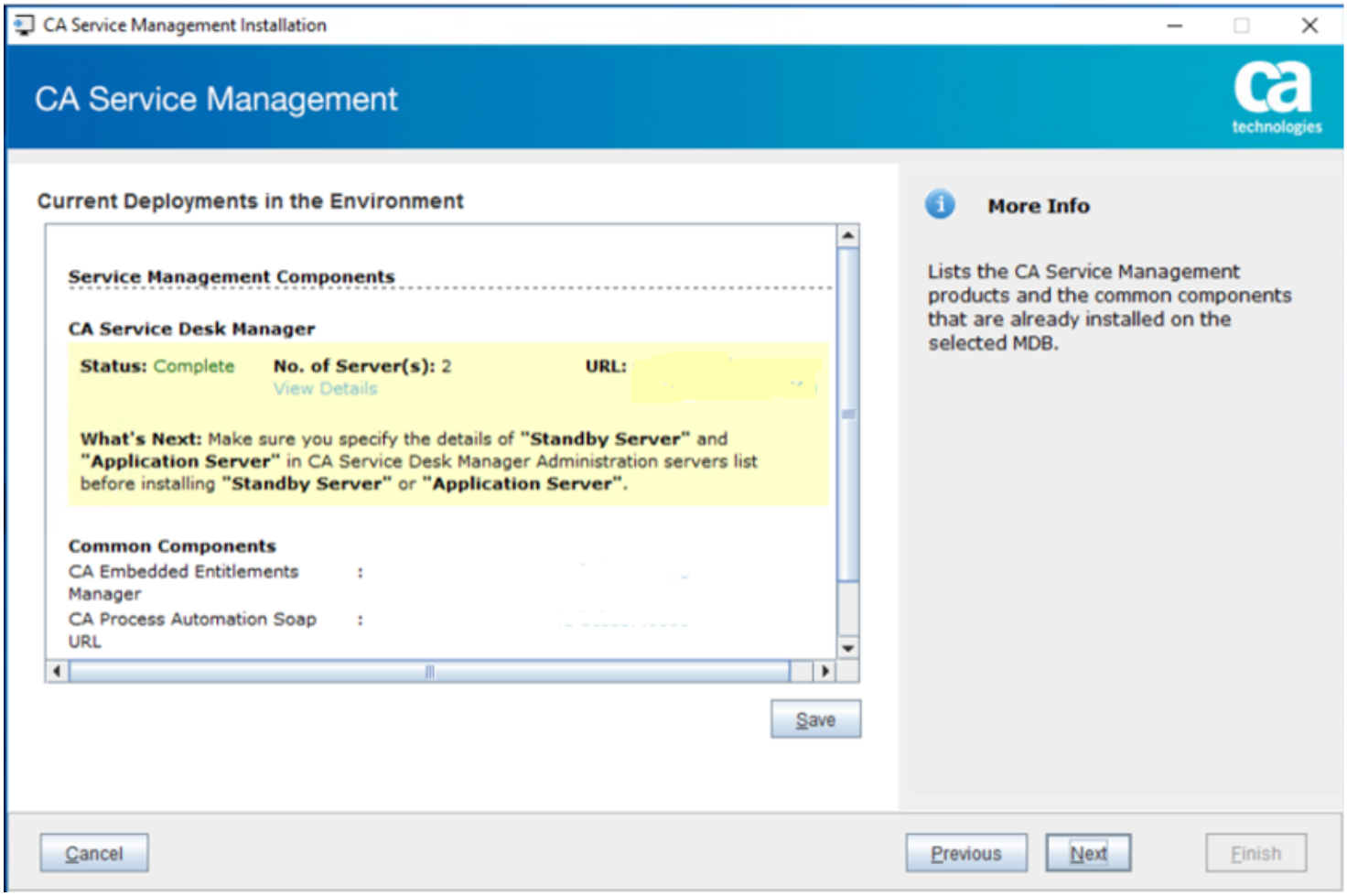
1. Mount the SDM DVD and select **setup.exe**, right click '**Run as Administrator**'.



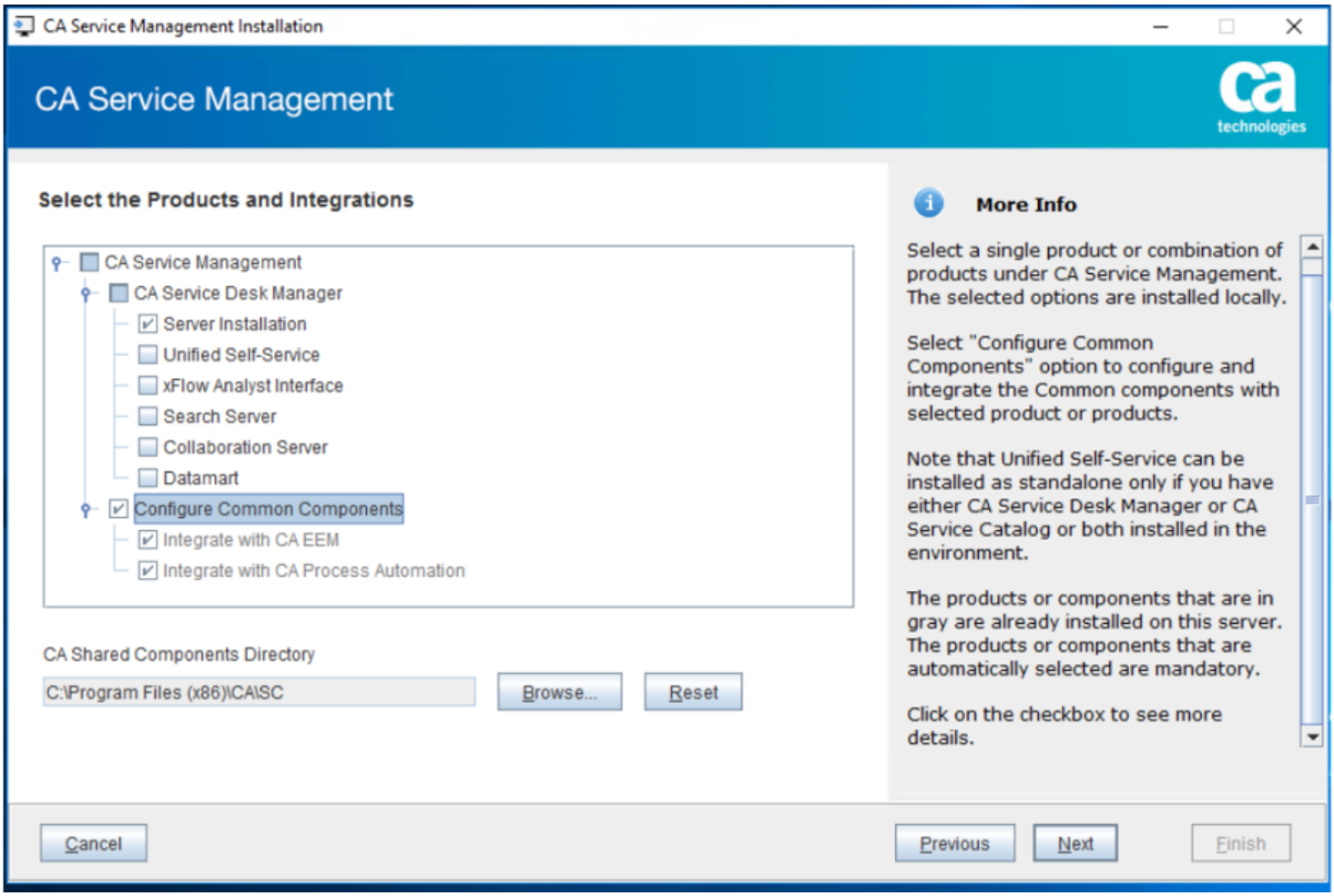
2. Click **Next**. Select Language “**English**”.
3. Click **CA Service Management** link to run the CA Service Management Installation wizard.
4. Review and accept the license agreement.
5. Click **Next**.
6. Enter Database Server as **testqa-mdb-new**and enter all passwords.


CA Service Management Installation
CA Service Management
ca technologies
Database Configuration
Skip this step
More Info
Database Type
SQL Server
Oracle
*Database Server
*Database Name
mdb
*Database Port
1433
Database Server Instance
*Database Admin User
sa
*Database Admin Password
*mdbadmin Password
*Confirm mdbadmin Password
Provide details for installing the CA Management Database (MDB) (Oracle or SQL).
Ensure that you have the Oracle or MS SQL database Client installed.
Database Admin User is the privileged user who can create and give access to the application database user.
If the MDB is already installed, the installer will use the same for CA Service Management. If not, it will create the MDB and the mdbadmin user with the password that you have specified
Clicking on Skip this Step link will proceed with installation of a Secondary Server for CA Service Desk Manager.
Cancel
Previous
Next
Finish

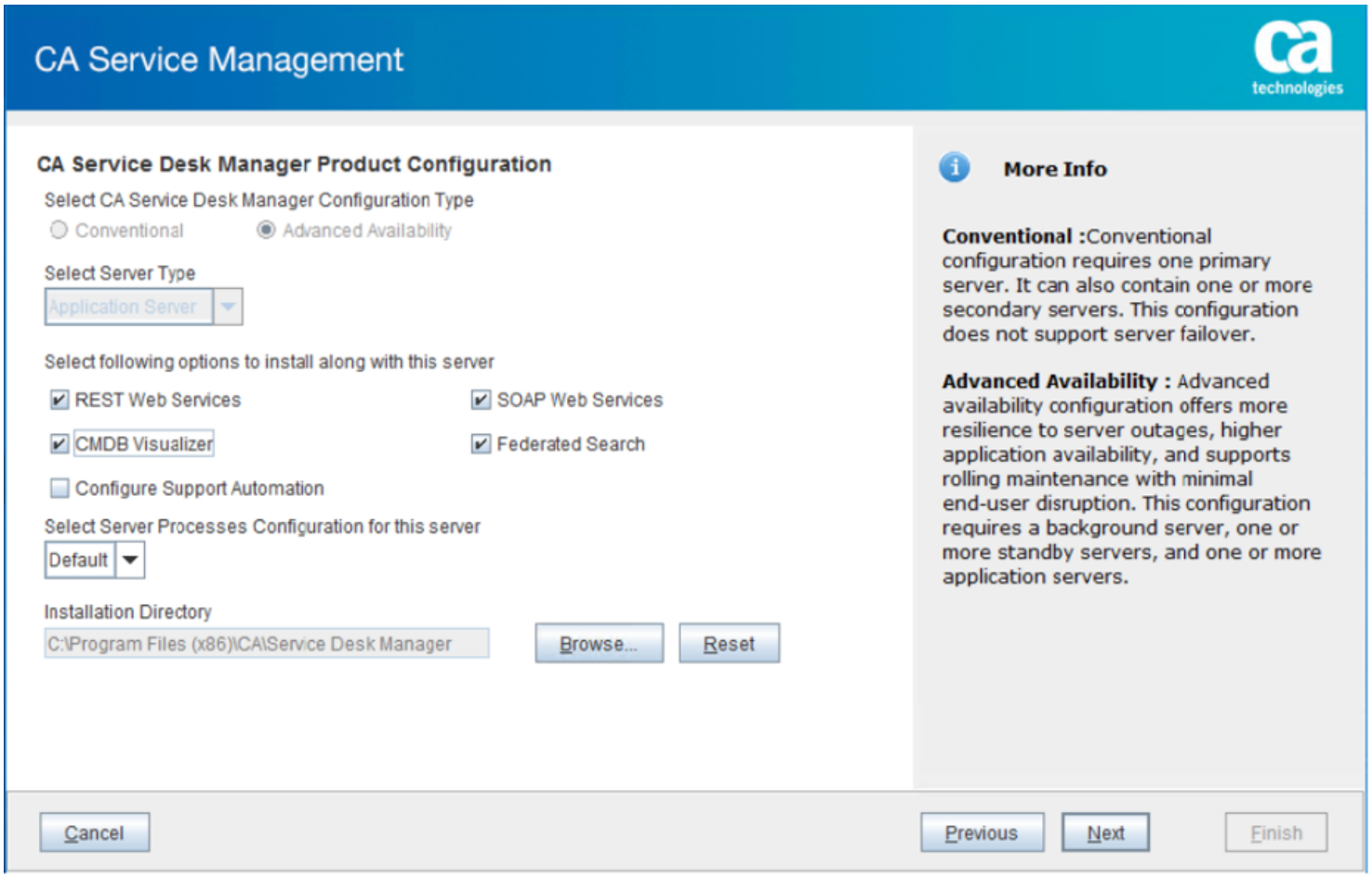
7. Click **Next**.



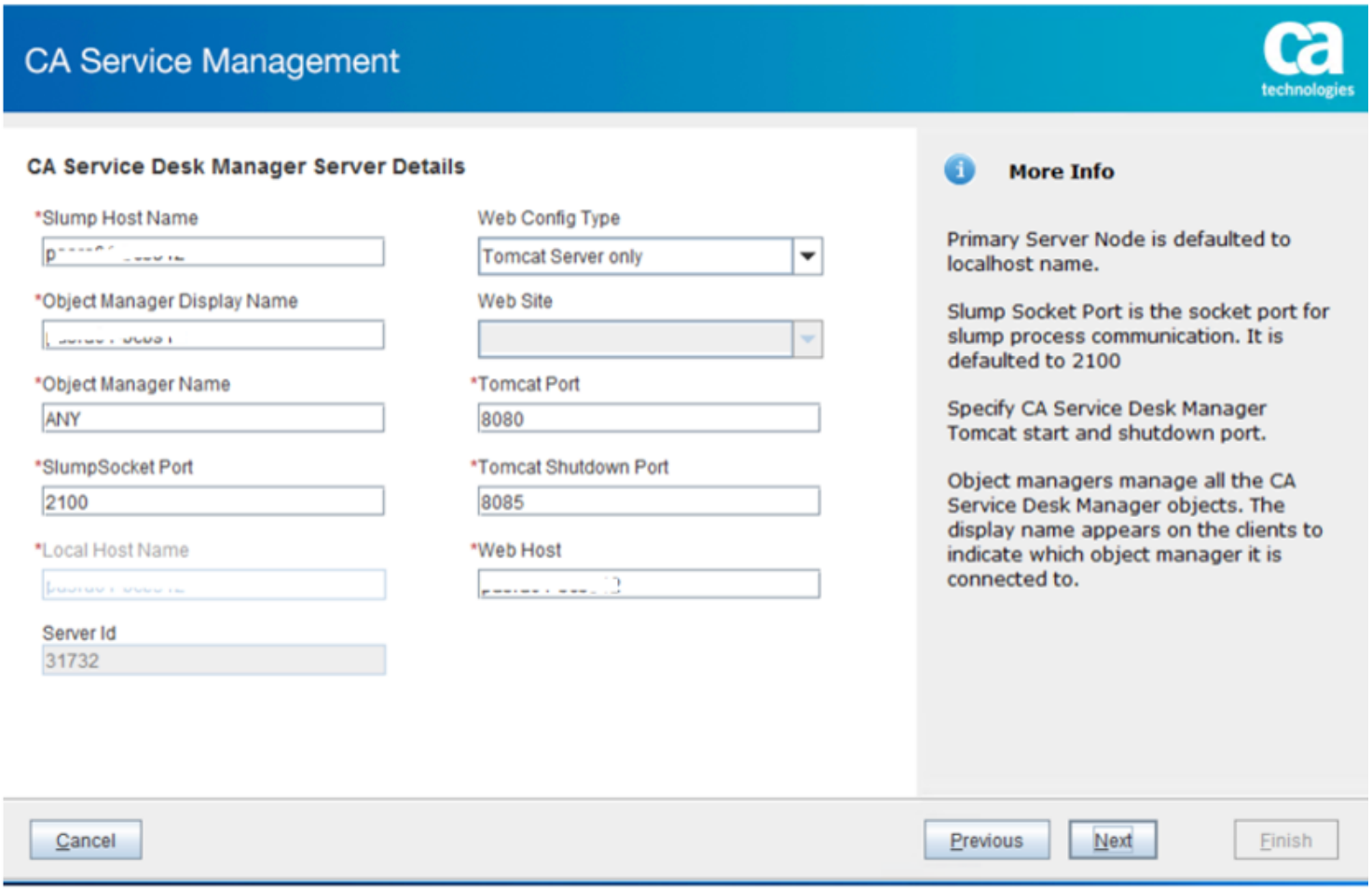
8. Click **Next**.
9. Check the '**Server Installation**' checkbox under the CA Service Desk manager and Select Configure Common Components check box (EEM and ITPAM)'.



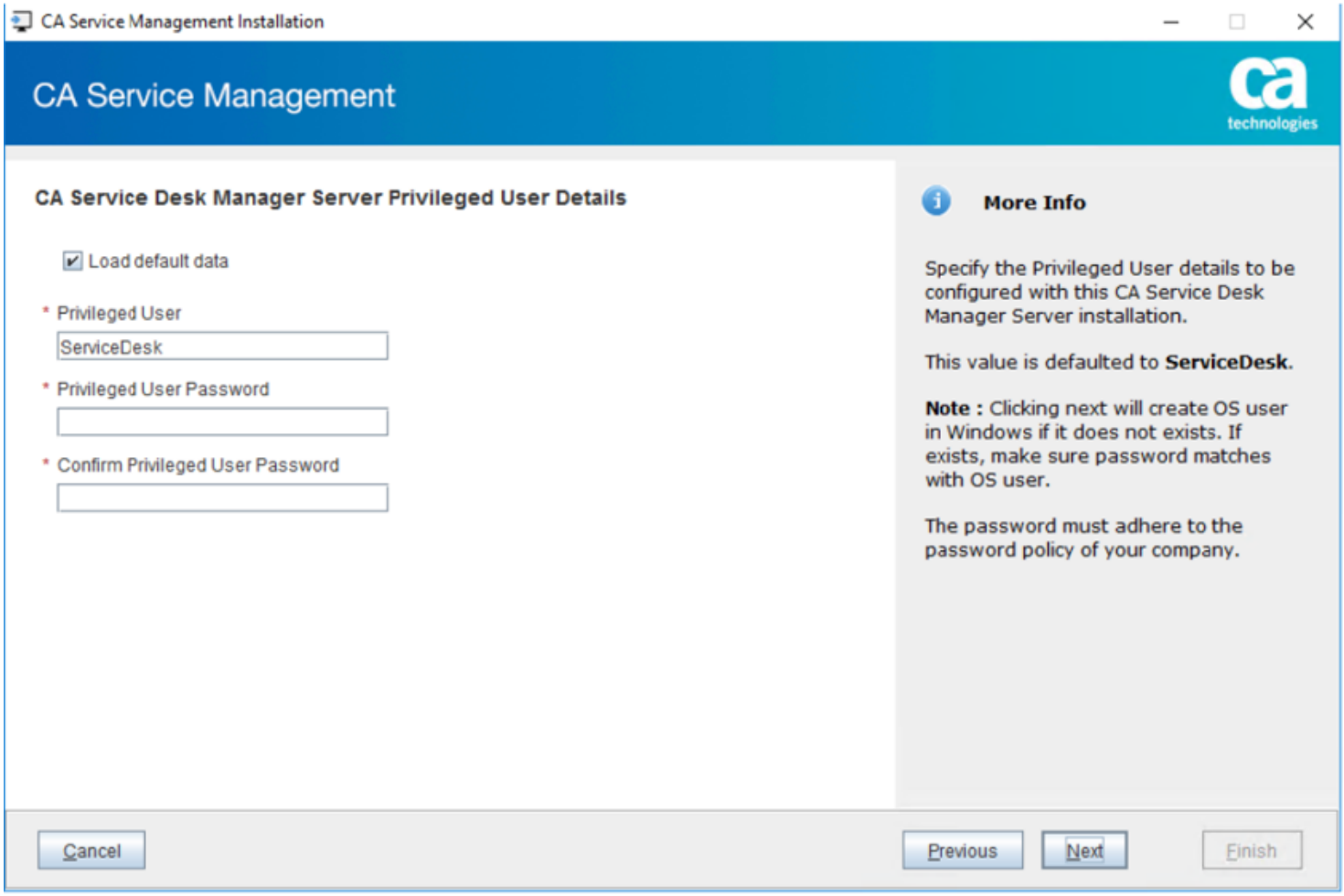
10. Click **Next** and Review the Installation Prerequisites.
11. Select the **CMDB Visualizer**checkbox.



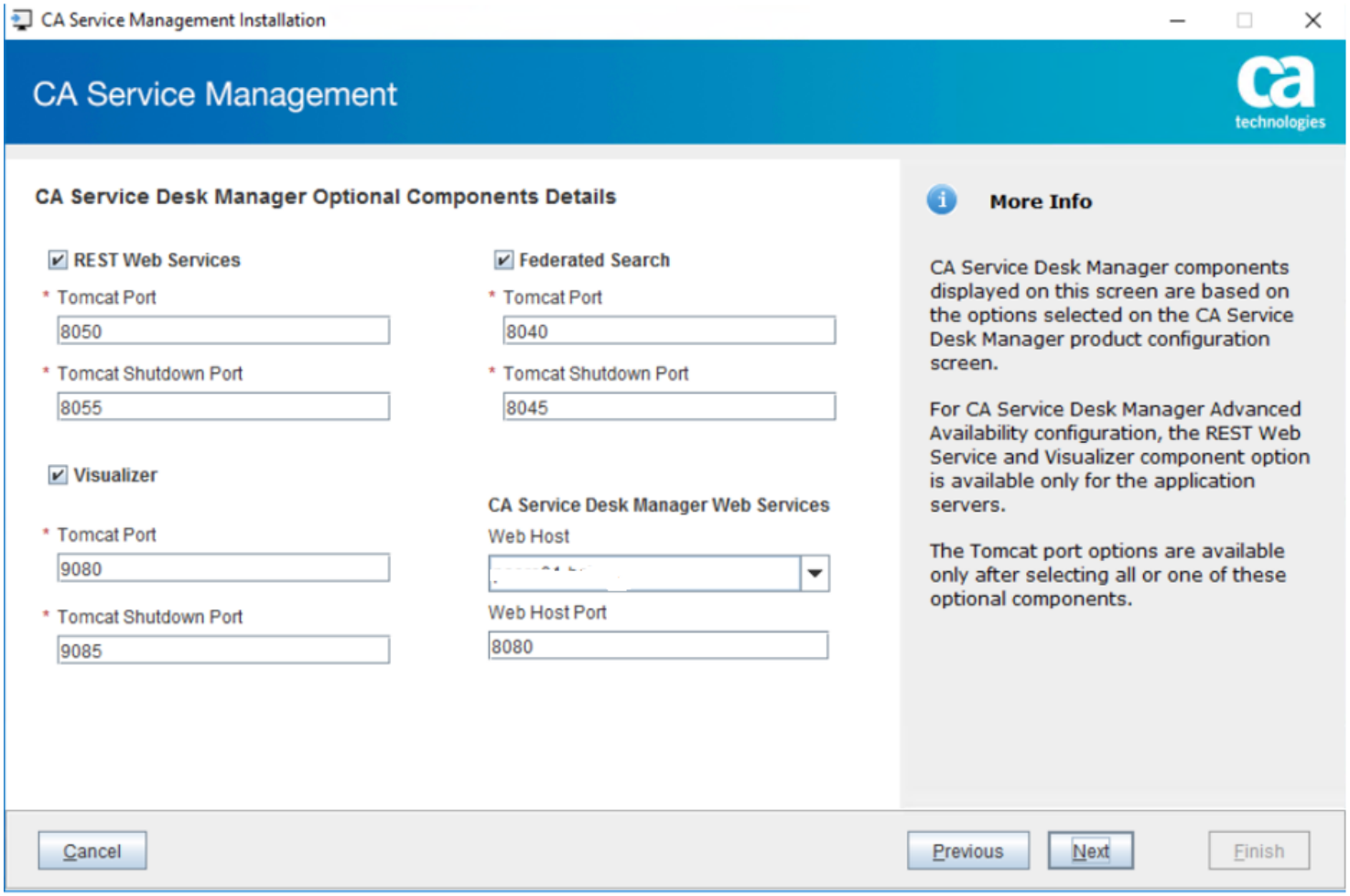
12. Click **Next**.
13. Click **Next**.
    Enter slum socket port details, Tomcat Port, Object Manager Name to manage CA SDM Objects.



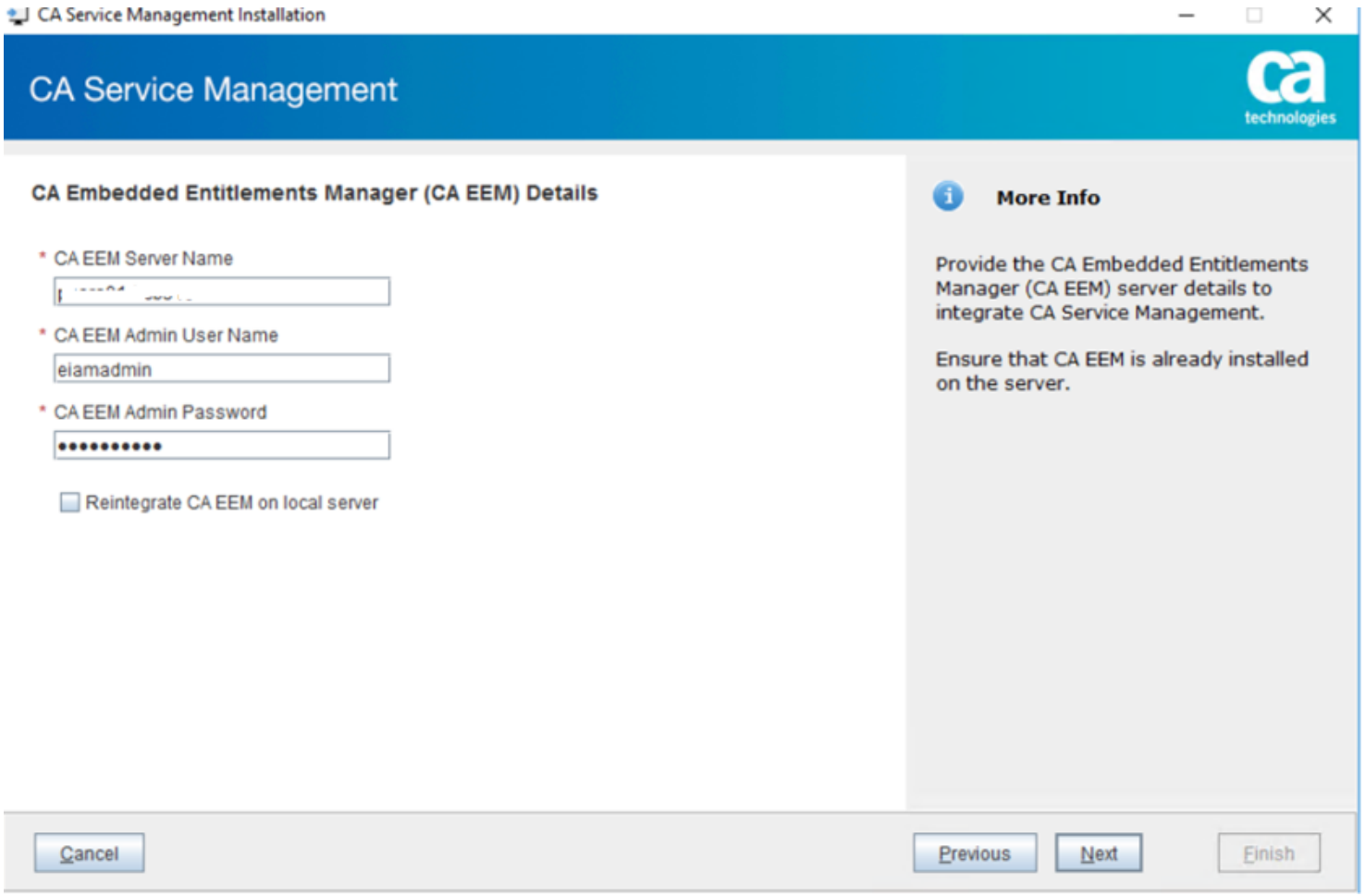
14. Enter Privileged User Password and Confirm Privileged User Password.



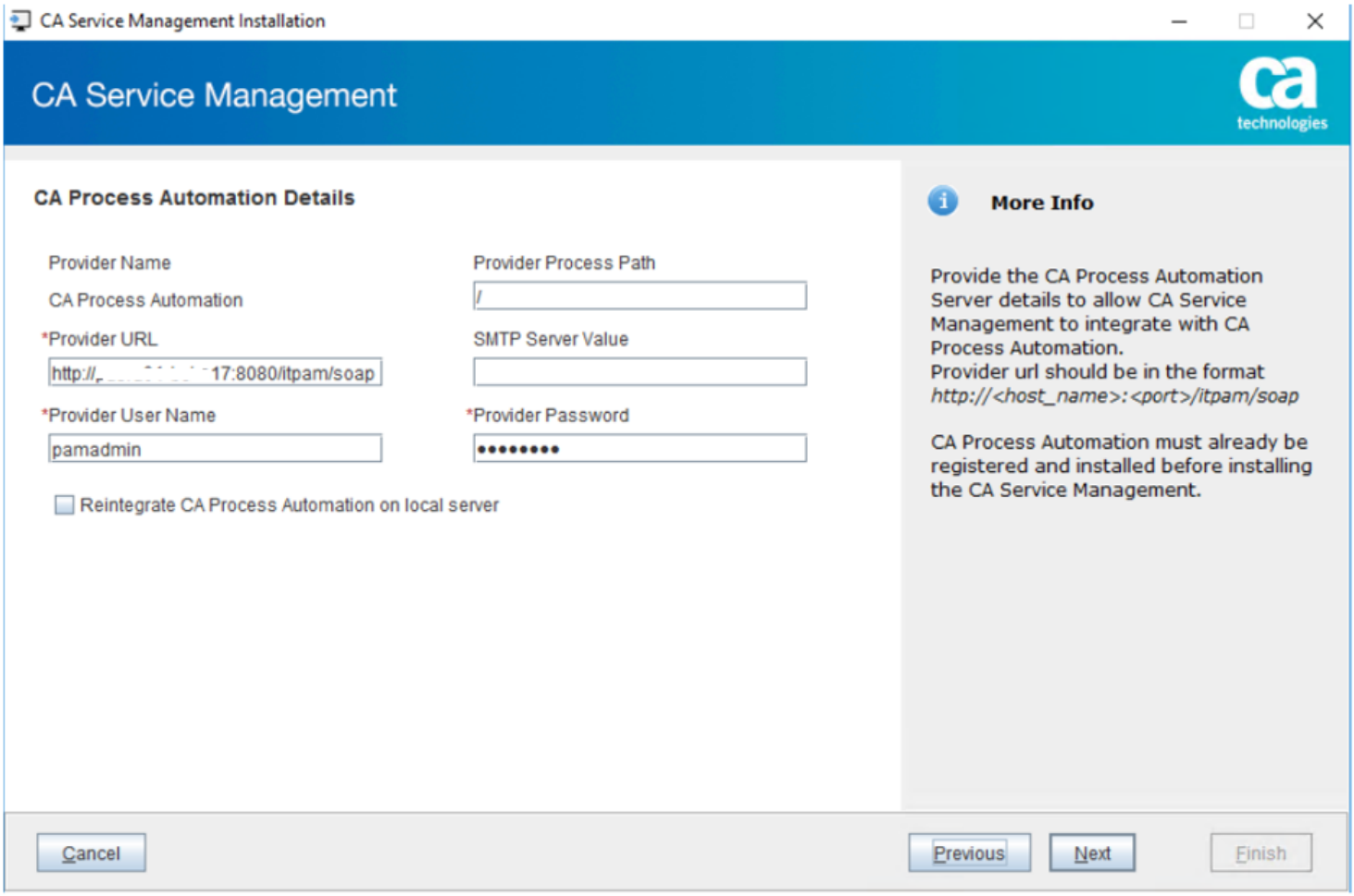
15. Click **Next**.
16. Enter the Optional Component Details.



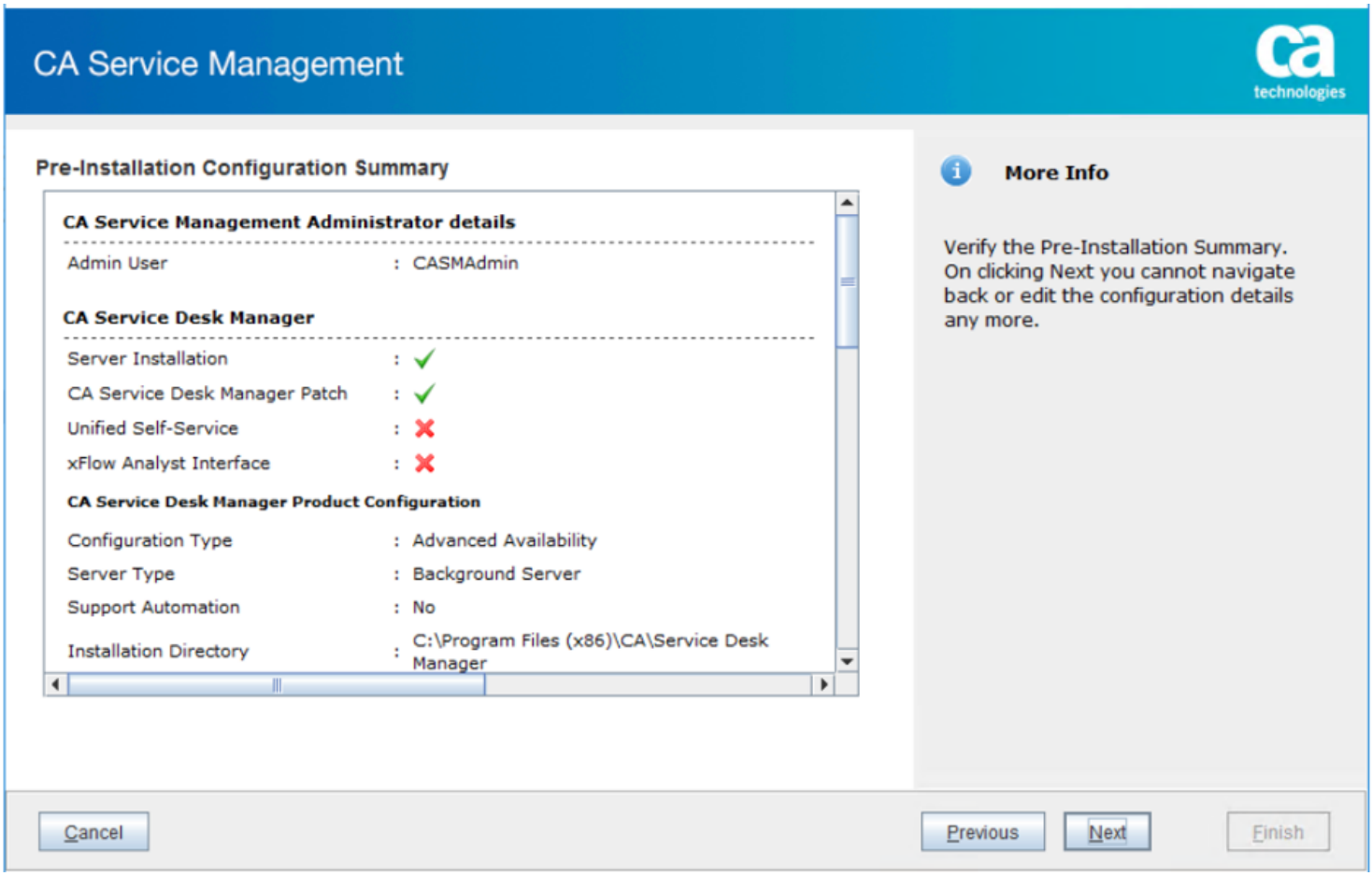
17. Click **Next**.
    Enter EEM Admin Password.



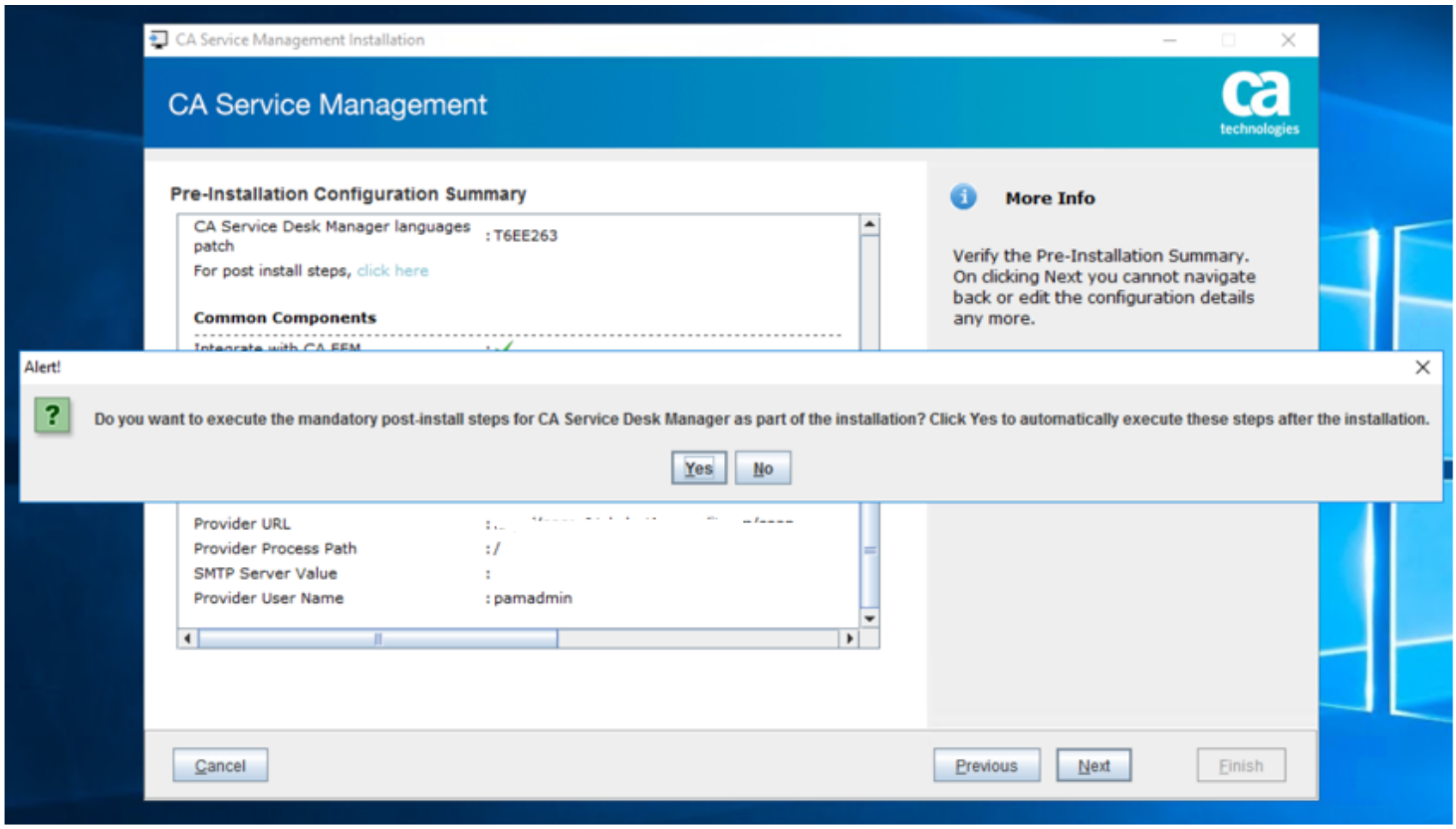
18. Click **Next**.
    Enter the Process Automation Provider password.



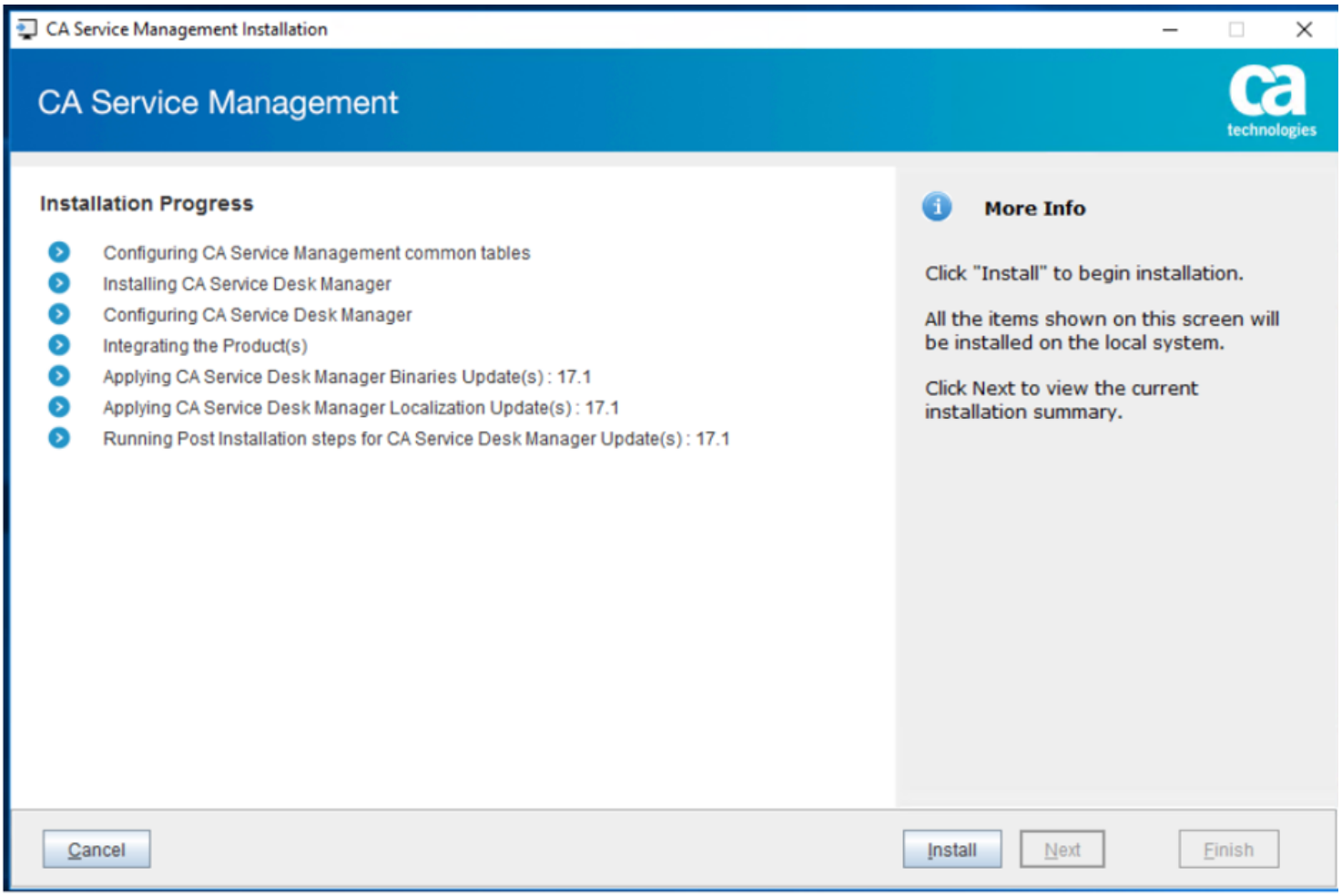
19. Click **Next**.
    Review the Pre-installation configuration summary.



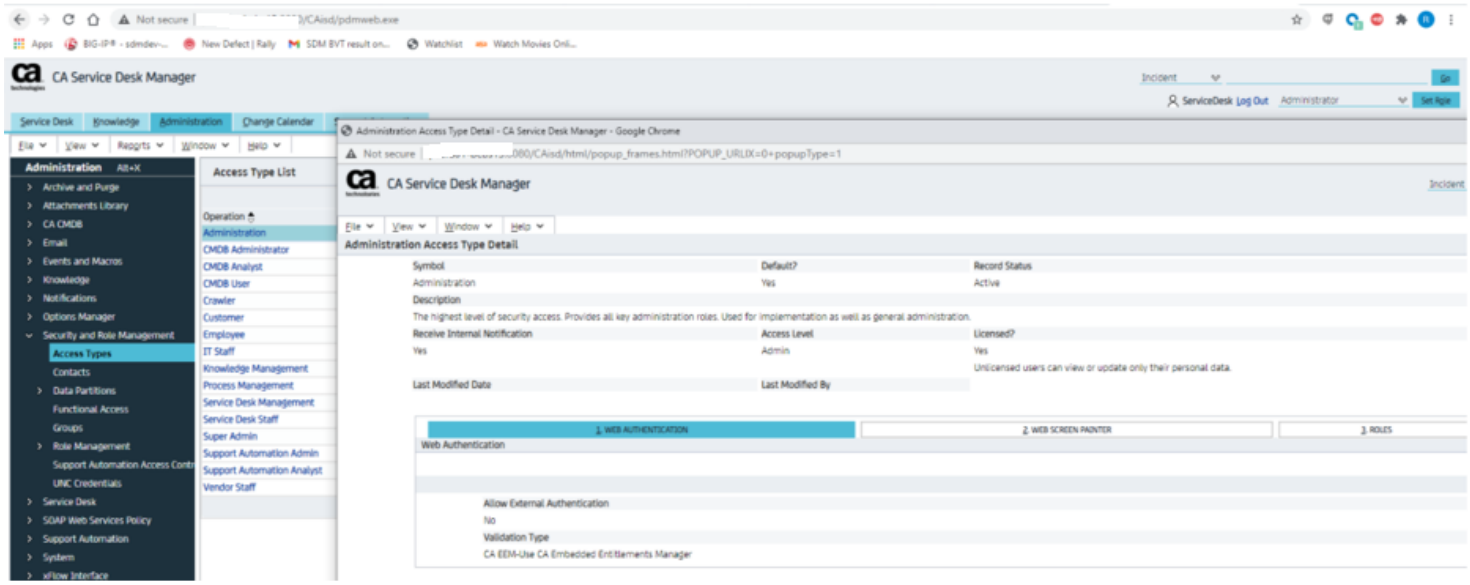
20. Click **Next**.
    Follow the Installation wizard.
21. Click **Next**.
22. Click **Yes** to automate the post-install steps. Click **No** to perform manual post-install steps.



23. Click **Install**.



24. Click **Finish** once installation completes.
25. Review the Post-Installation Summary.
26. Open the SDM Application Server URL: *http://testqa-app1-new:8080/CAisd/pdmweb.exe*
27. Enter Login or access credentials (Username and password).
28. Check the EEM option in the Administration Access type detail page using the Navigation path Administration tab, Security and Role Management, Access Type.

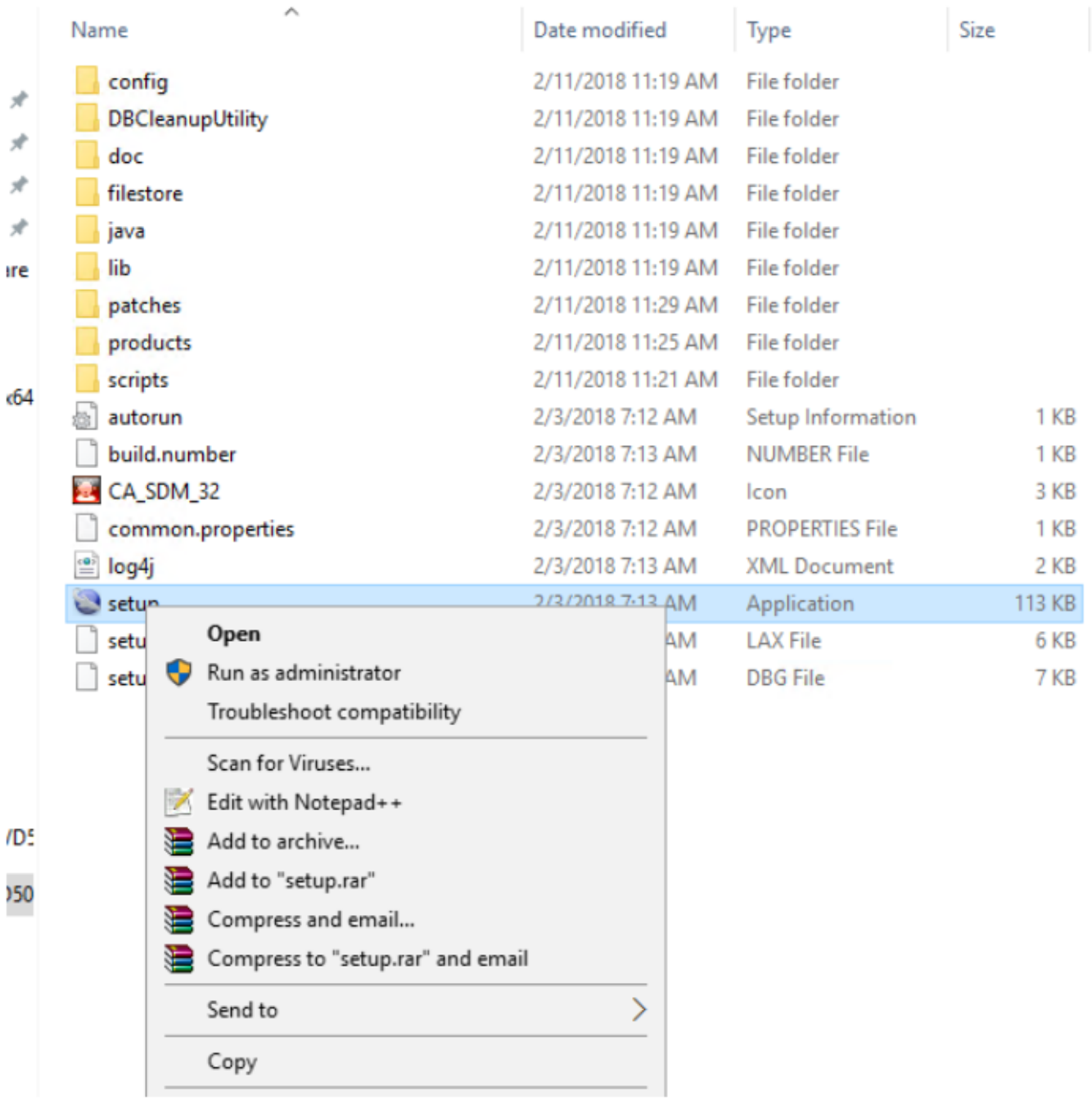


This completes the Application Server 1 (**testqa-app1-new**) installation.

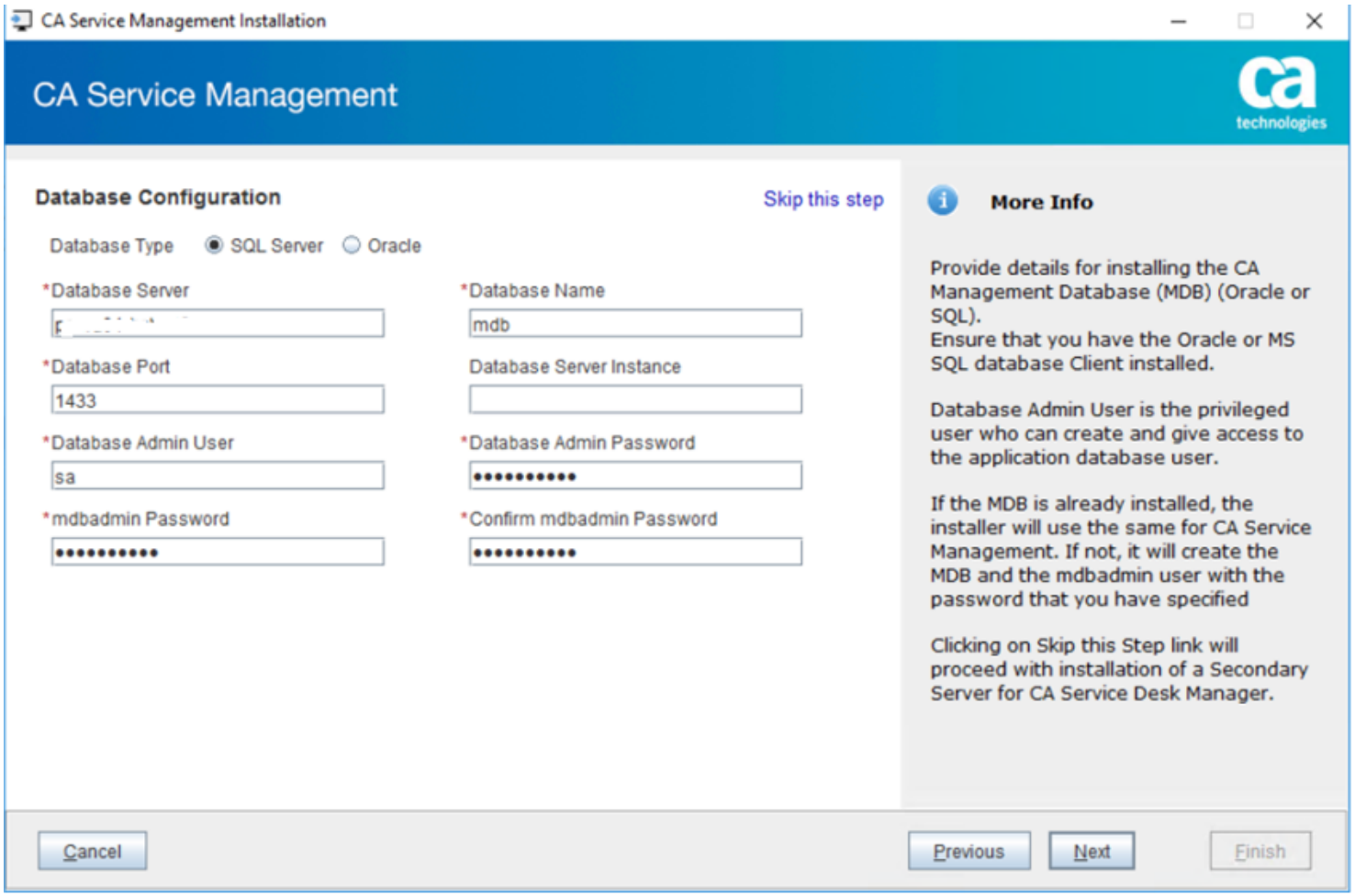
Install the Second Application Server (testqa-app2-new)

Perform the following steps:

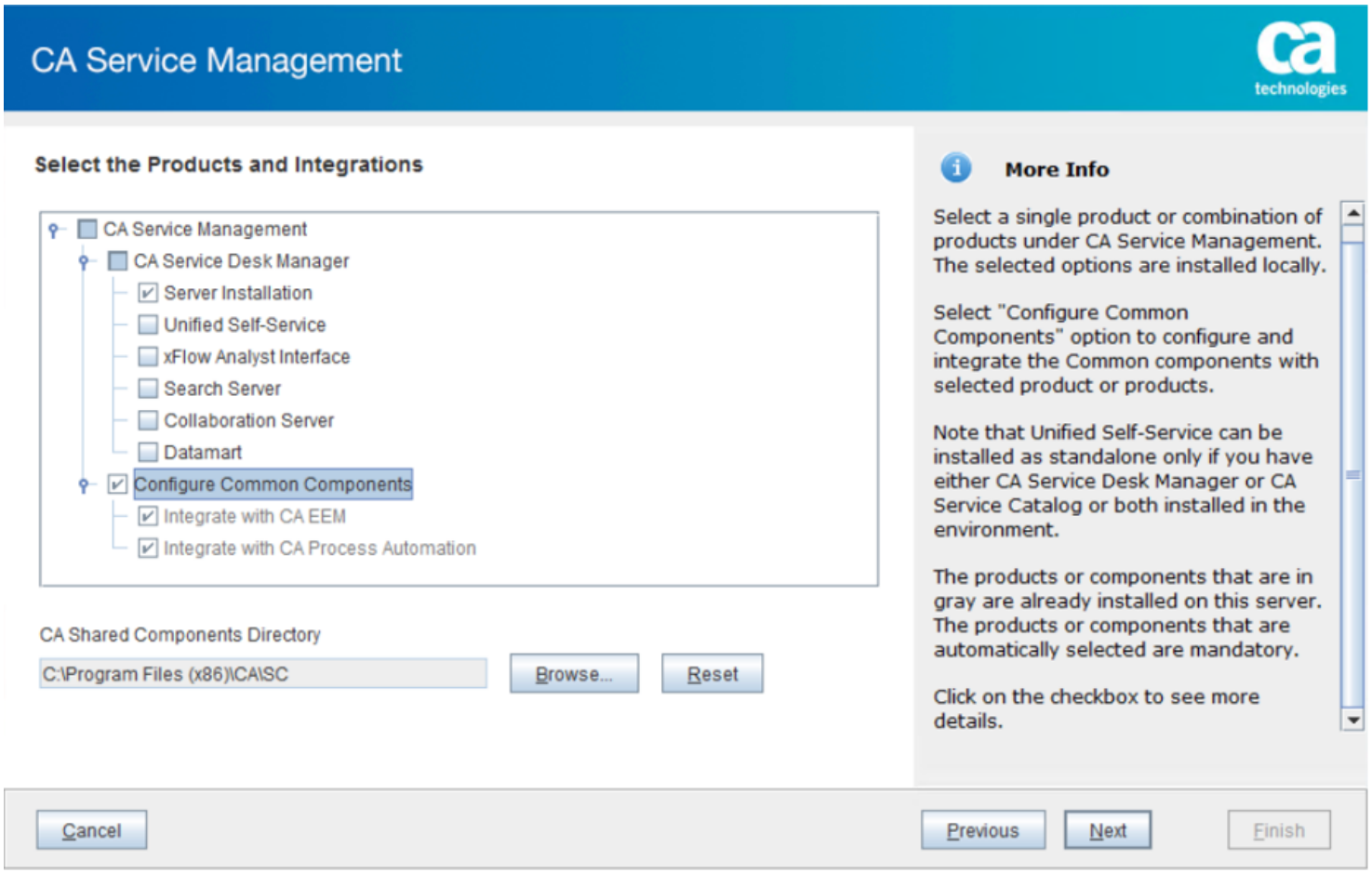
1. Mount the SDM DVD and Select the setup.exe, right click "**Run as Administrator**".



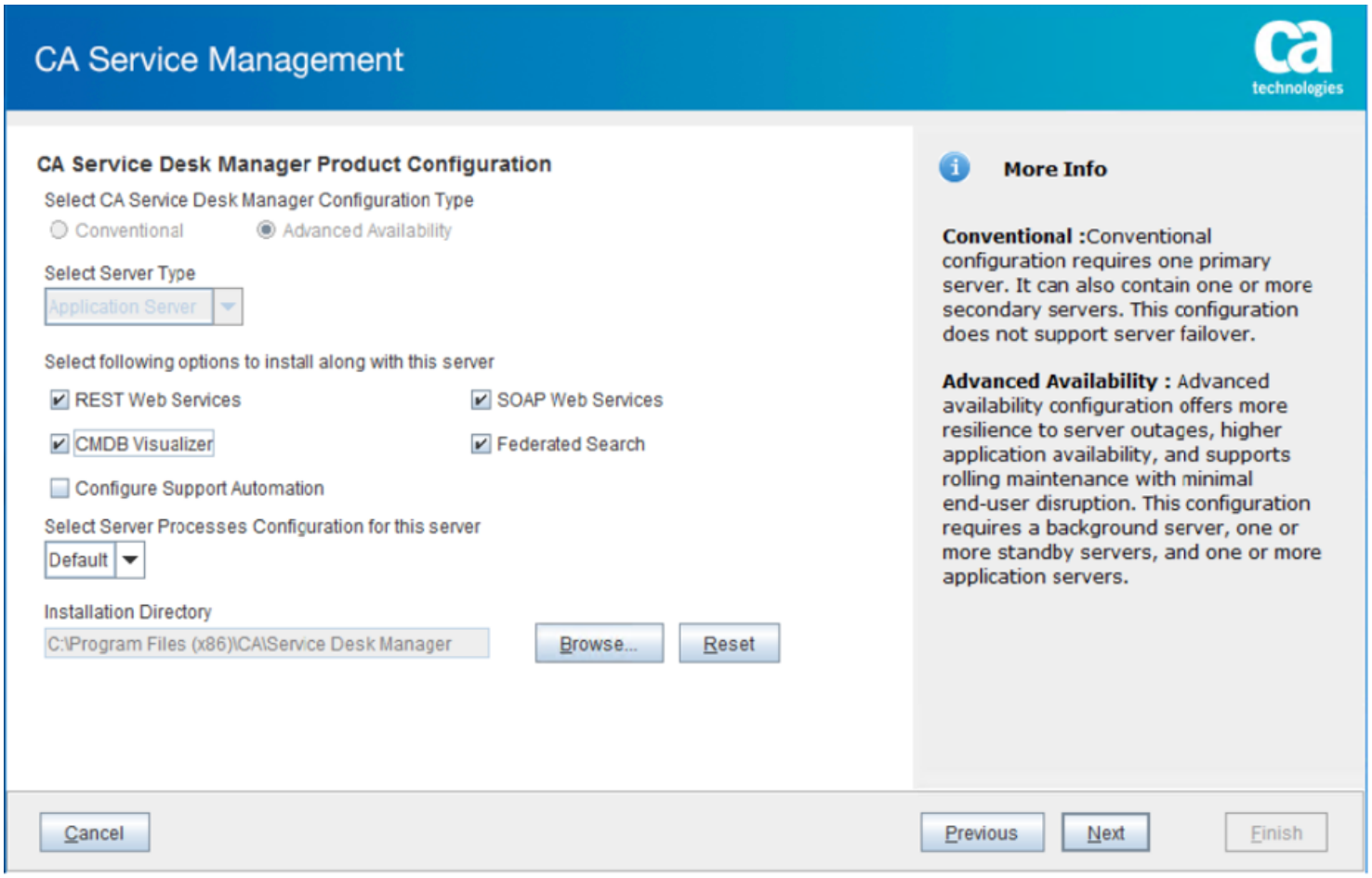
2. Click **Next**. Select **English** as Language.
3. Click "**CA Service Management**" link to run the CA Service Management Installation Wizard.
4. Review and accept the license agreement.
5. Enter the database details based on your environment requirements.



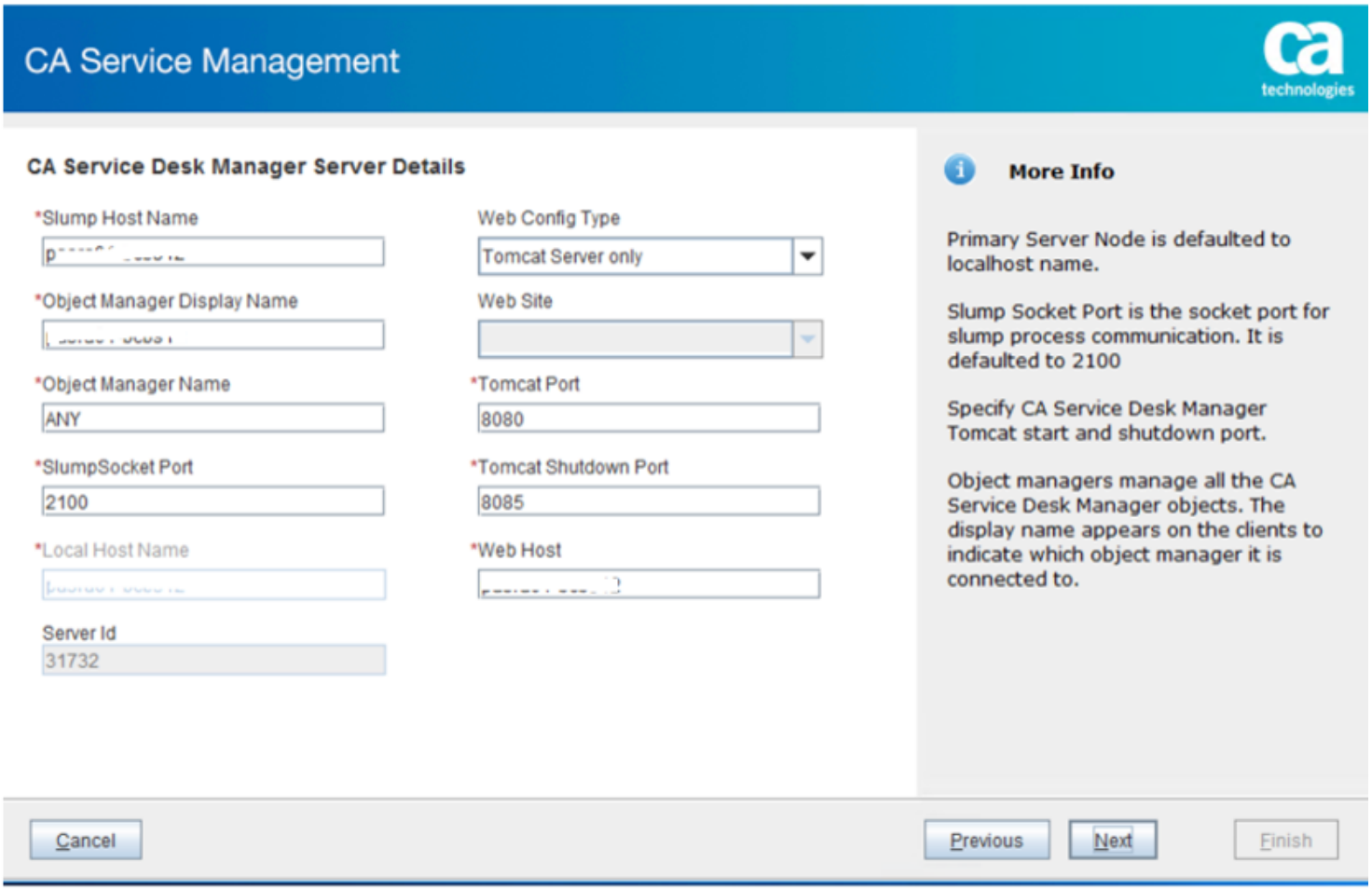
6. Click **Next**. Review the Current Deployment details in your environment.
7. Click **Next**. Check the "**Server Installation**" checkbox under CA Service Desk Manager and Select "**Configure Common Components**" check box (CA EEM and CA Process Automation).



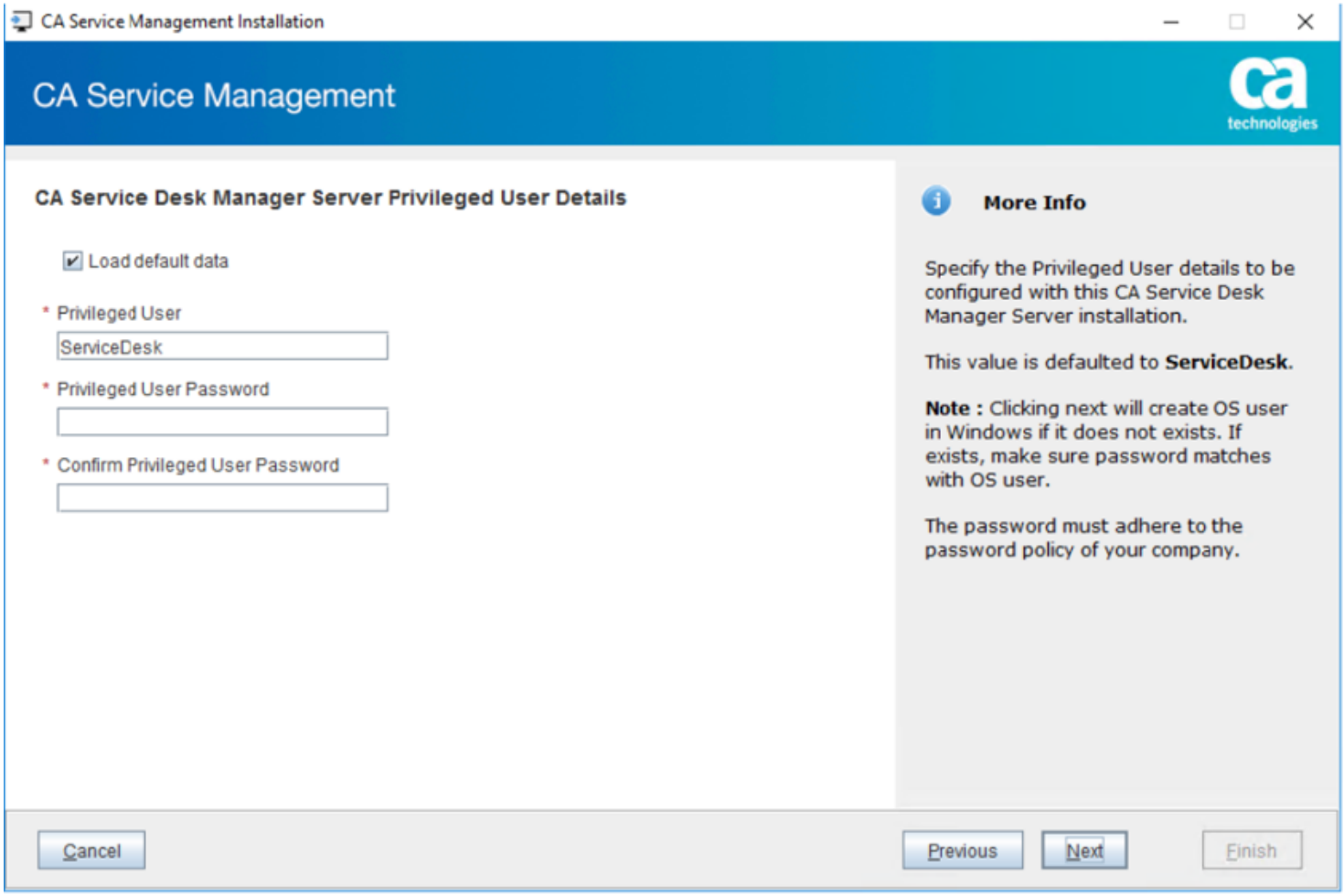
8. Click **Next**. Review the installation prerequisites.
9. Click **Next**. Select the Advanced Availability configuration.



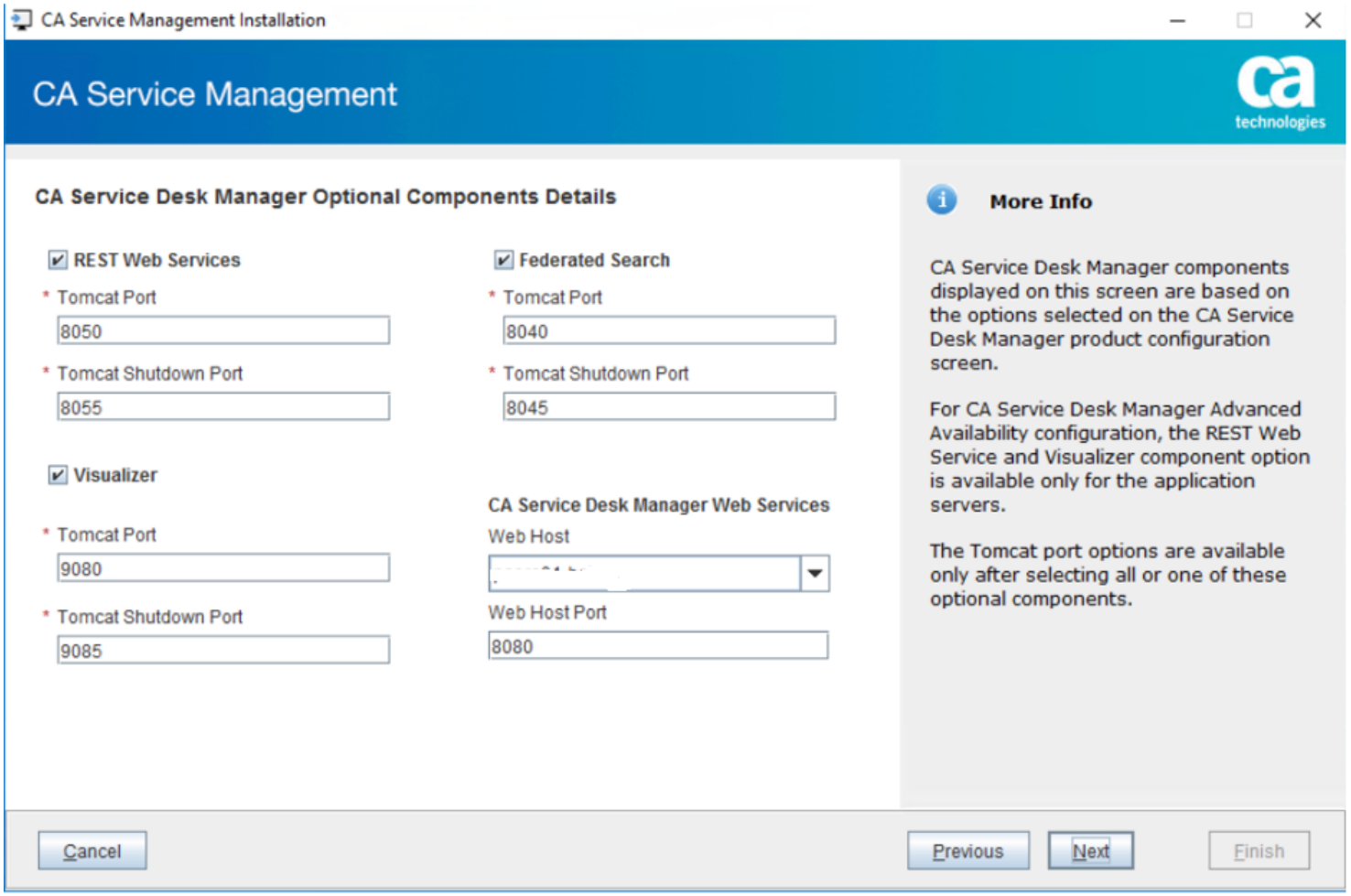
10. Provide the Slump host name, slum port details, Tomcat Port, and object manager name.



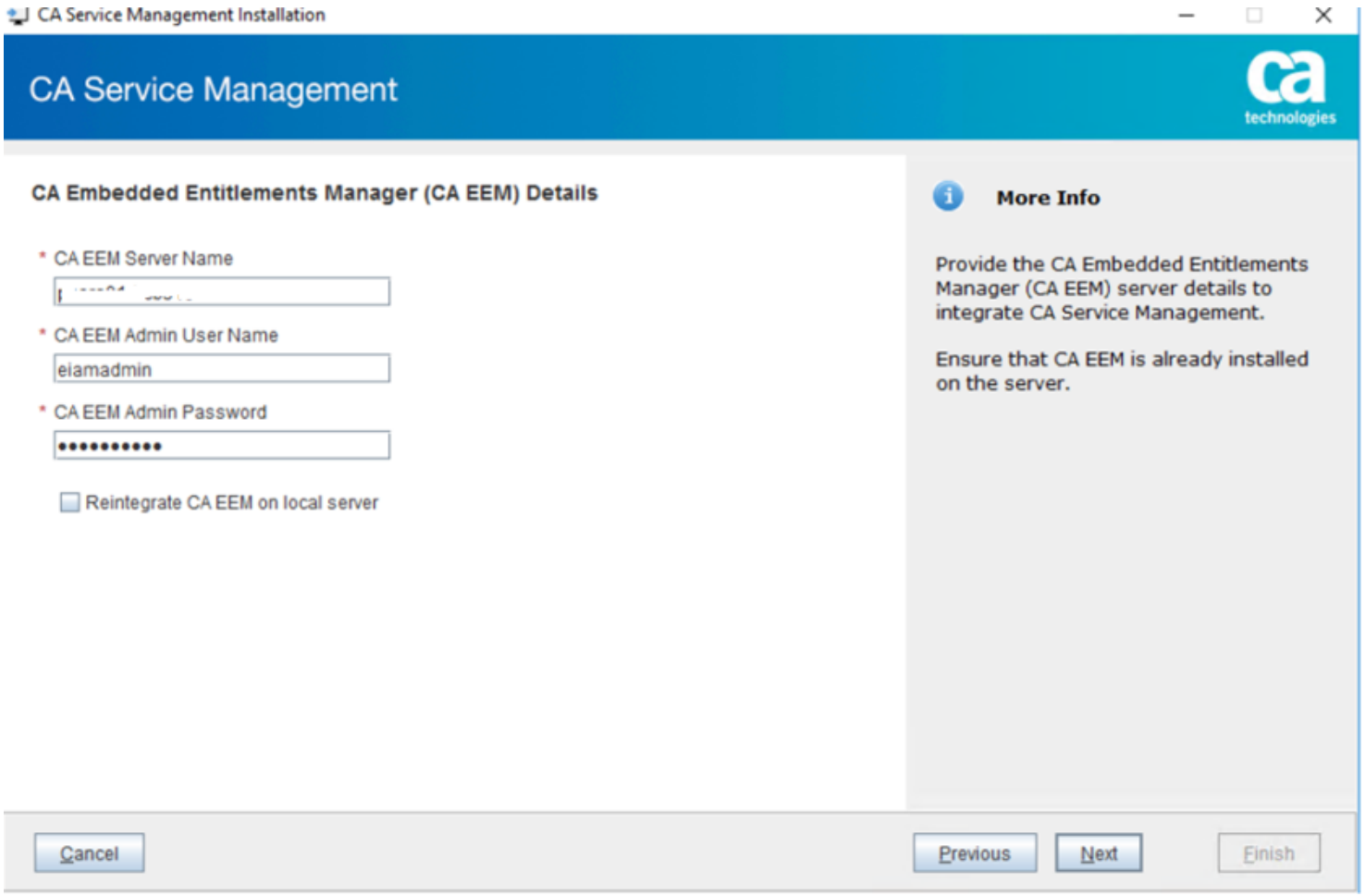
11. Enter the privileged User password and Confirm Privileged User Password.



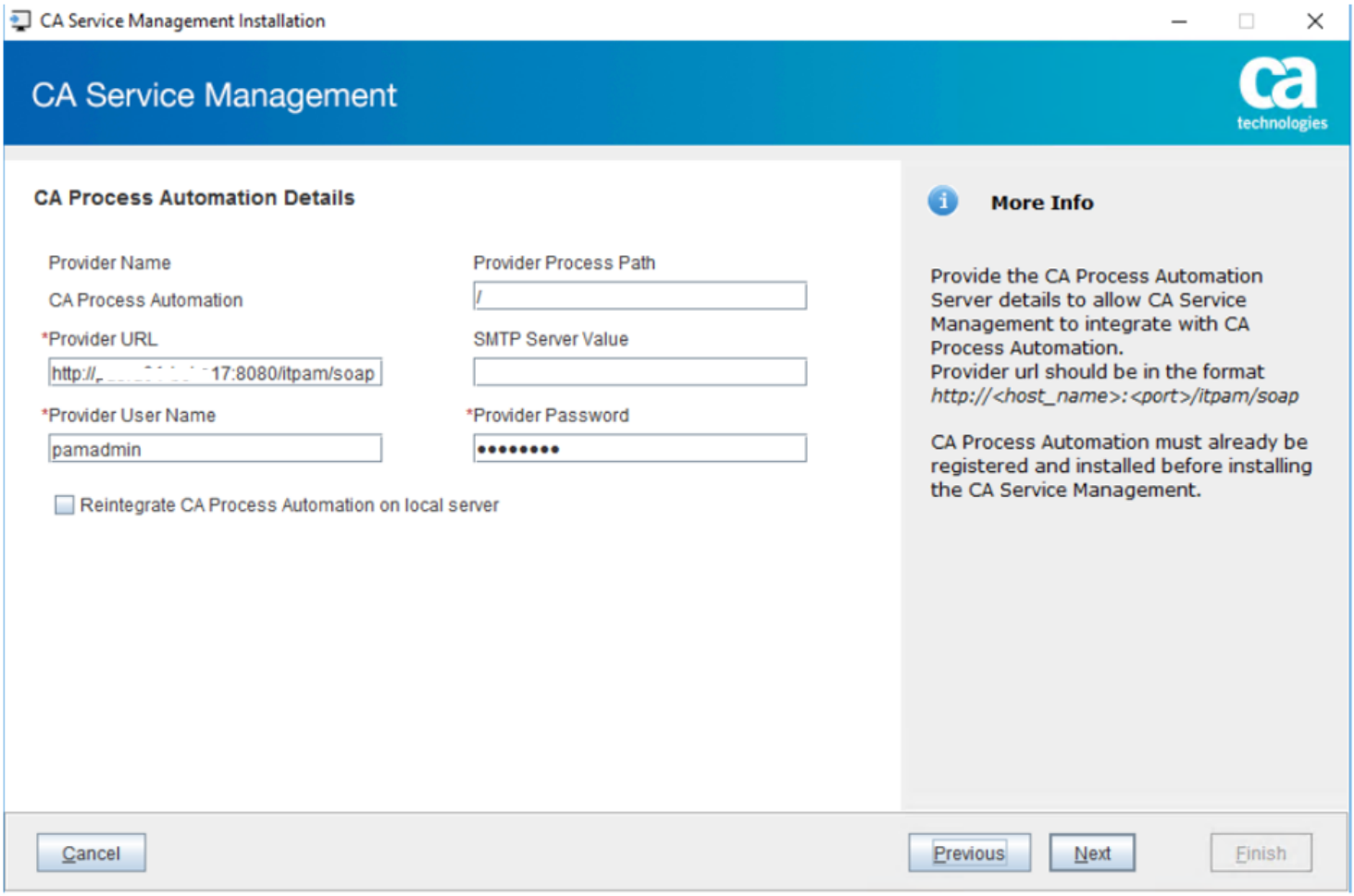
12. Enter the CA SDM Optional Component Details like Rest Web Services, Tomcat Shutdown port, Federated Search, and Visualizer details.



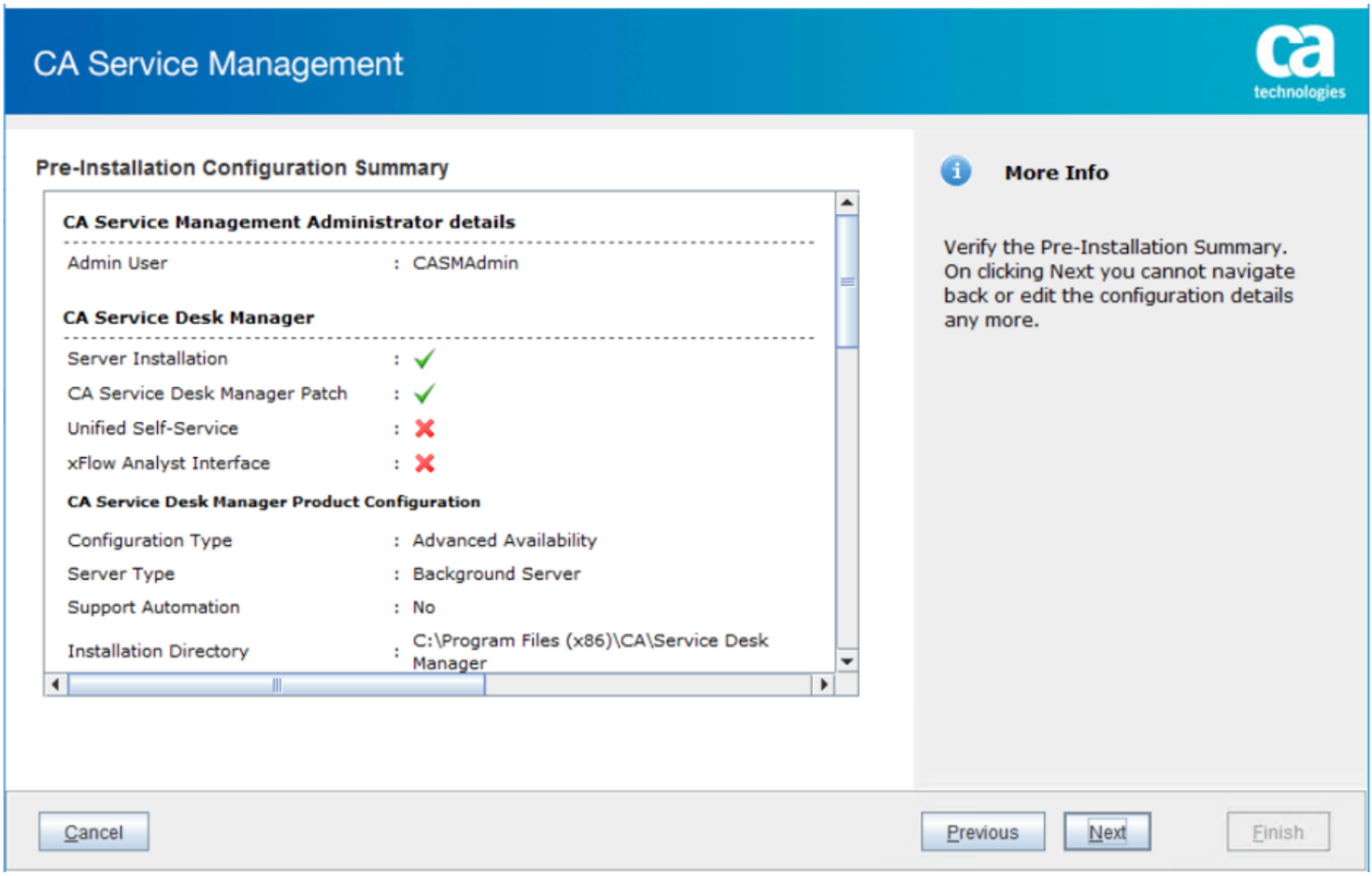
13. Enter CA EEM Admin Password.



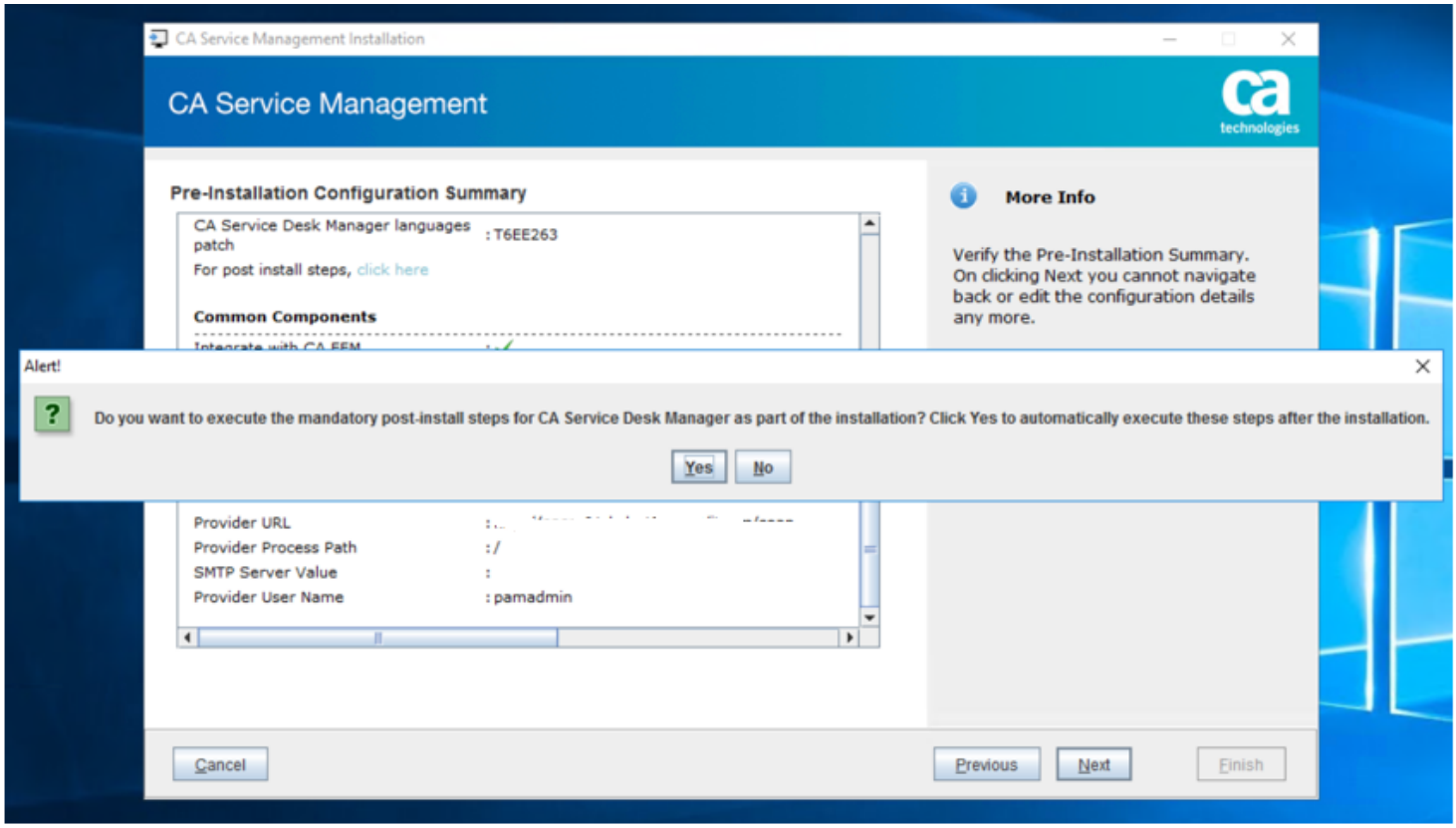
14. Enter CA Process Automation Workflow Provider name.



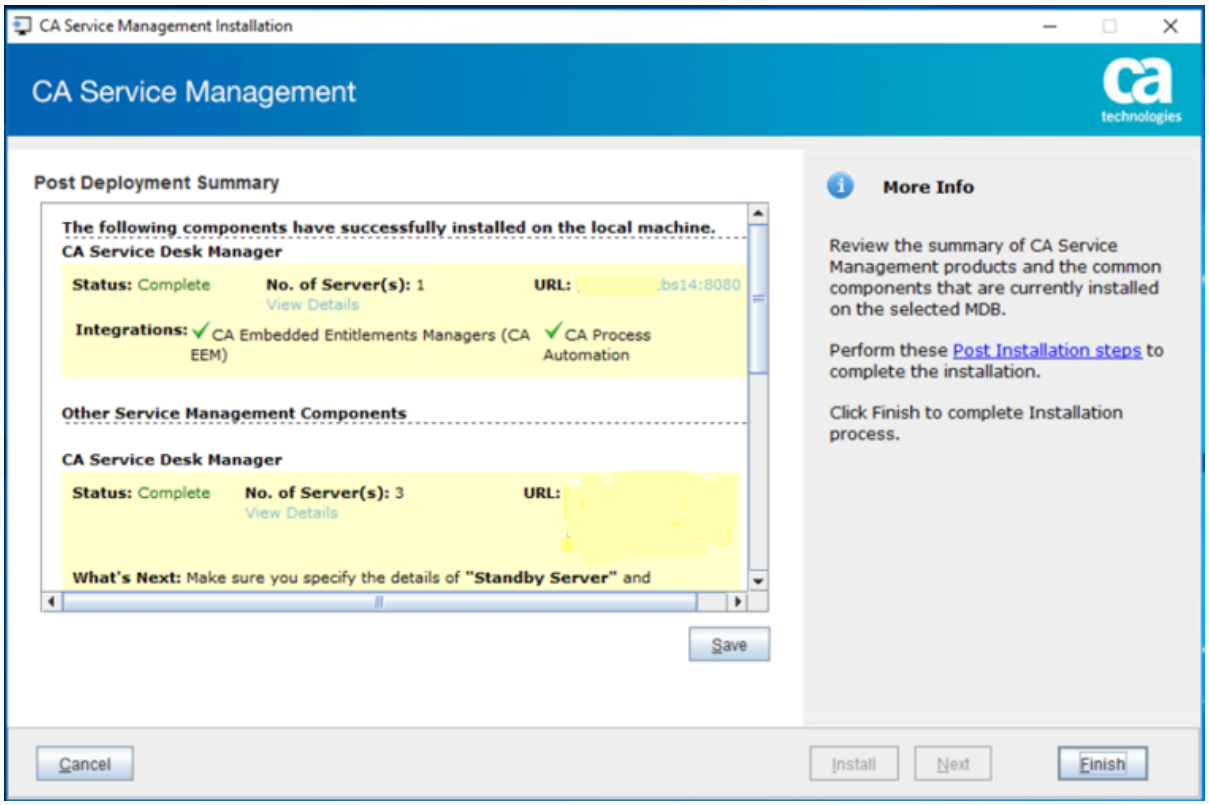
15. Review the pre-installation configuration summary.



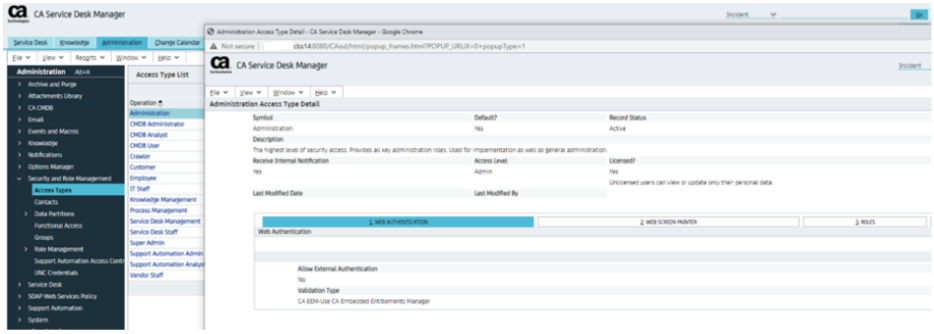
16. To perform manual post-installation steps, click **Yes**. Click **No** to perform manual post-installation steps.



17. Click **Install** to begin the installation process.
18. Click **Finish** to complete the installation.
19. Review the post-installation summary.



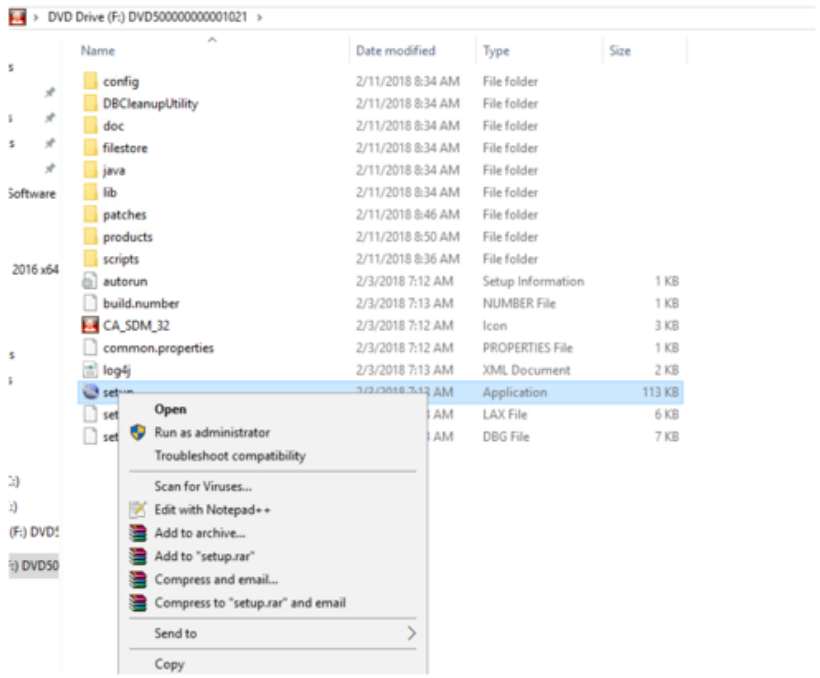
20. Launch the CA SDM Web Interface and enter login credentials.



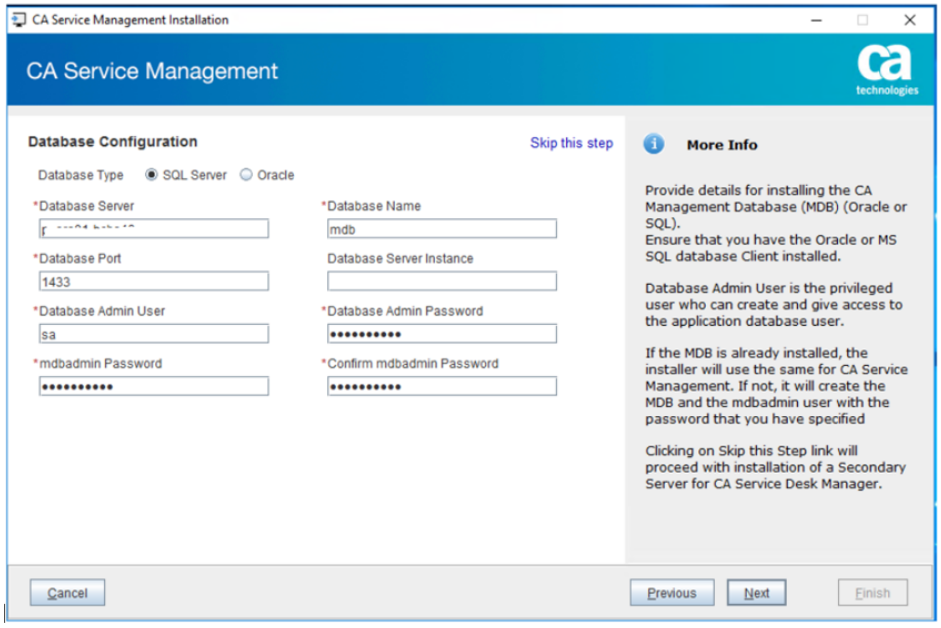
Install CA Service Catalog (testqa-sc-new)

Perform the following steps to install and configure CA Service Catalog:

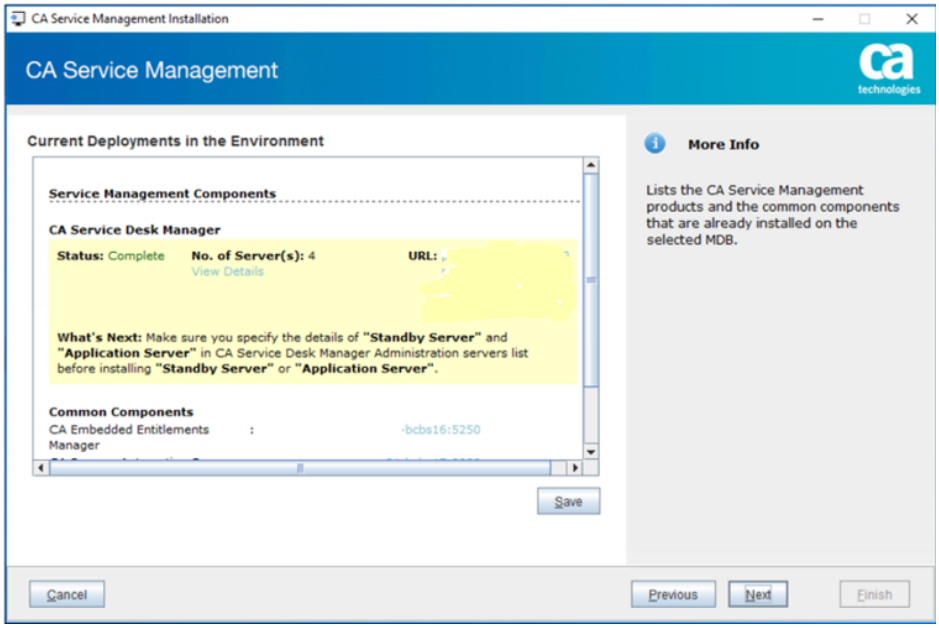
1. Login to Broadcom Support and download the CA Service Catalog 17.4 installer.



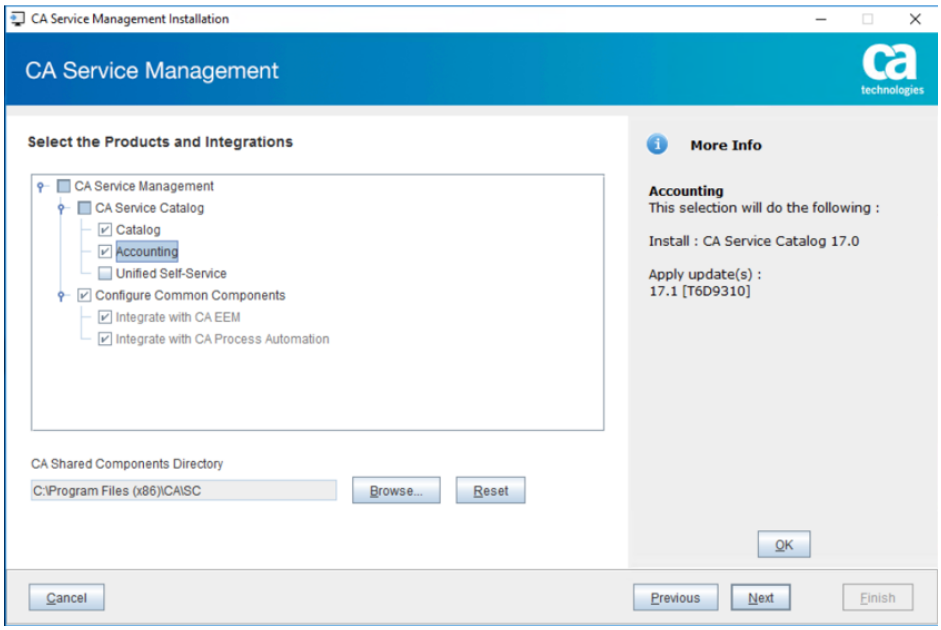
2. Right-click the **setup.exe** file and select "**Run as Administrator**".
3. Select **English** as Language.
4. Select **CA Service Management** to run the CA Service Management installer wizard.
5. Review and accept the license agreement.
6. Select Database Type as SQL Server. Enter the Database Details for Microsoft SQL Server database.



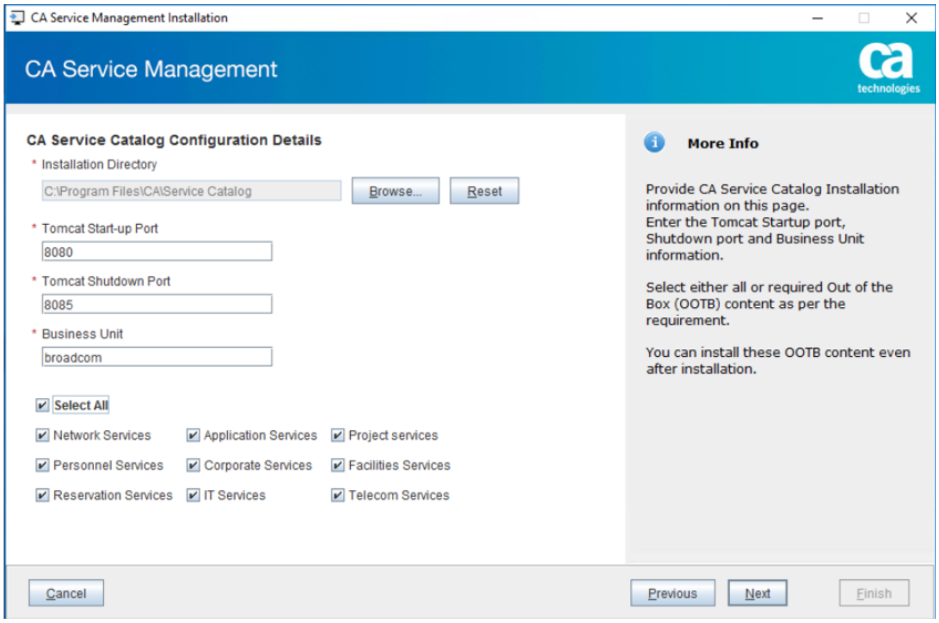
7. Review the Current Deployment details.



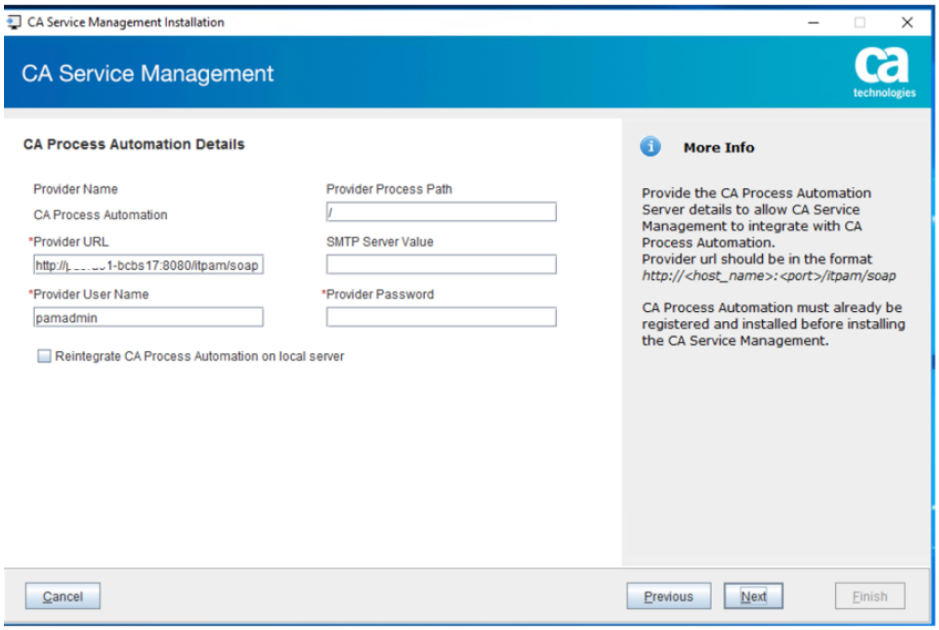
8. Select CA Service Catalog components. For example:**Accounting**.



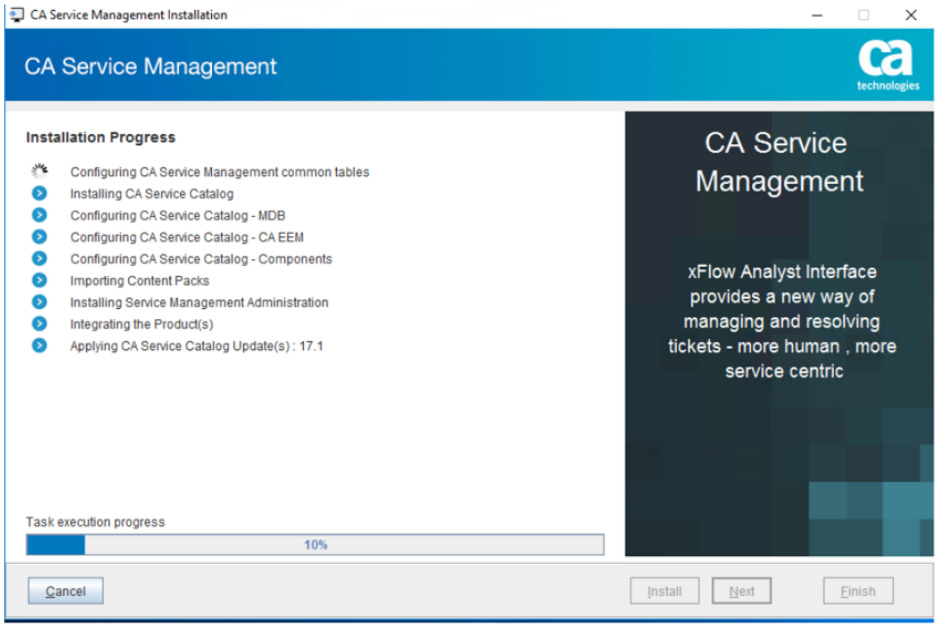
9. Review the installation prerequisites summary.
10. Provide CA Service Catalog Configuration Details.



11. Provide CA EEM details.
12. Provide the CA Process Automation Workflow provider details.



13. Review the pre-installation configuration summary.
    Click**Next**.
14. Installation progress begins.



15. Click **Next**. Follow the installation wizard.
16. Click **Finish** once installation completes.
17. Review the post- installation summary details.
18. Now, launch the CA Service Catalog web interface and login to CA Service Catalog.

CA Service Catalog Installation is complete.

CA Service Management Solution Integration

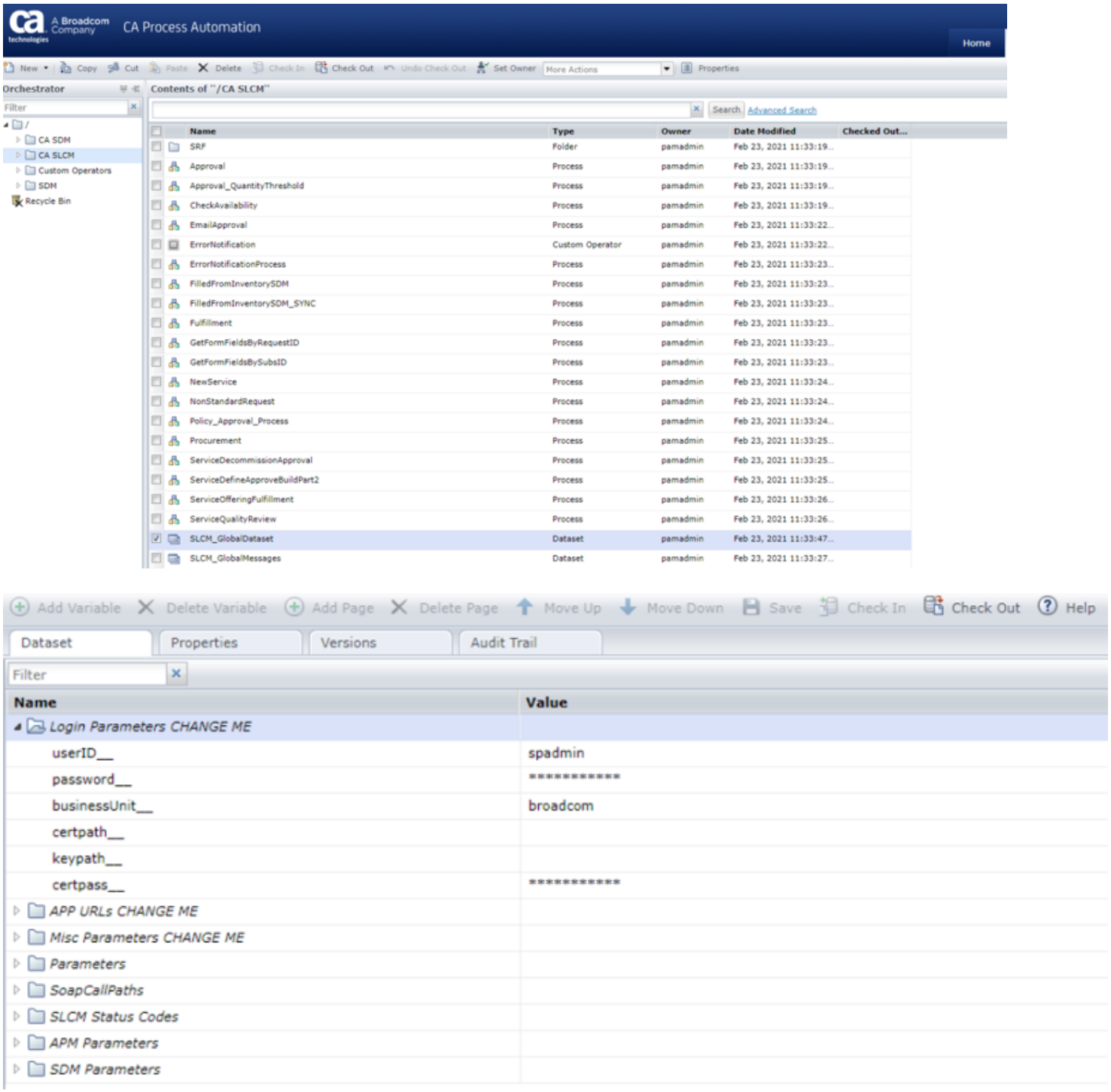
> **NOTE**
> If both CA Service Catalog and ITPAM are SSL enabled, to load the OOTB content of CA Service Catalog into ITPAM, the administrator has to import the certificate of PAM into the truststore of CA Service Catalog.

This article describes CA Service Management Solution integration with the following:

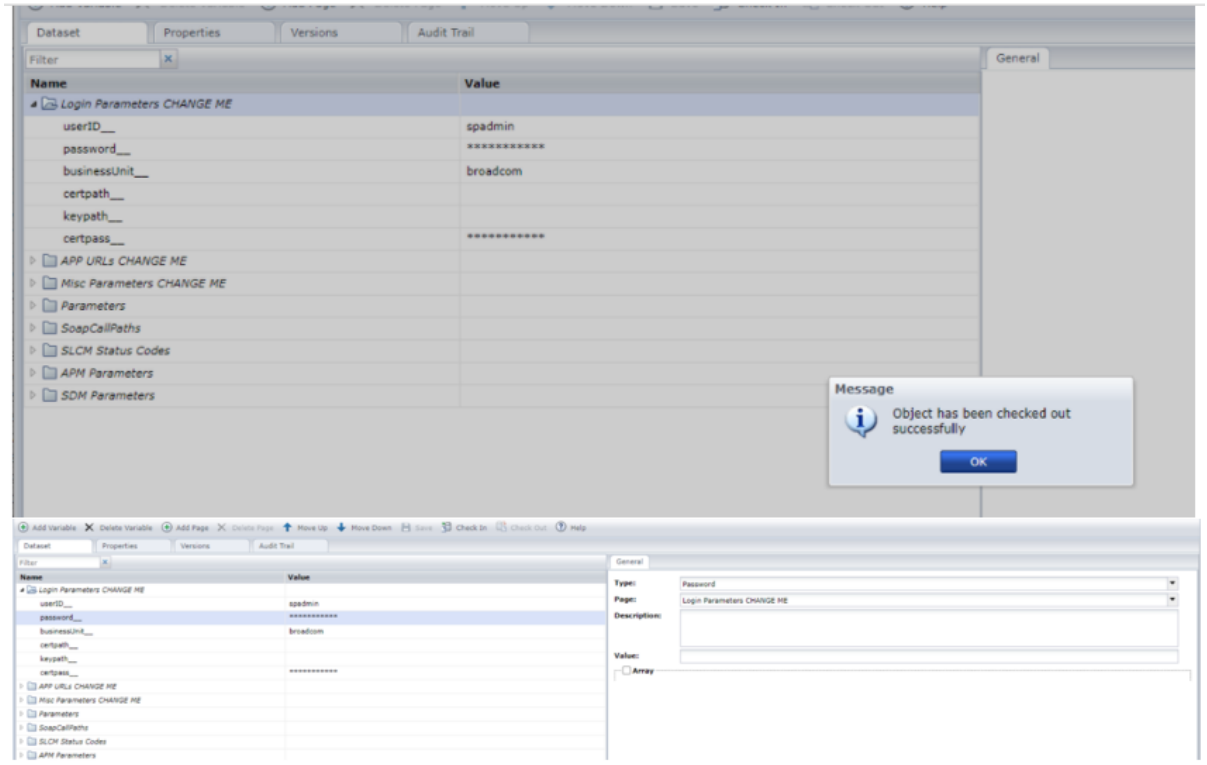
Updating the Service Catalog and SDM entries in Process Automation

Perform the following steps:

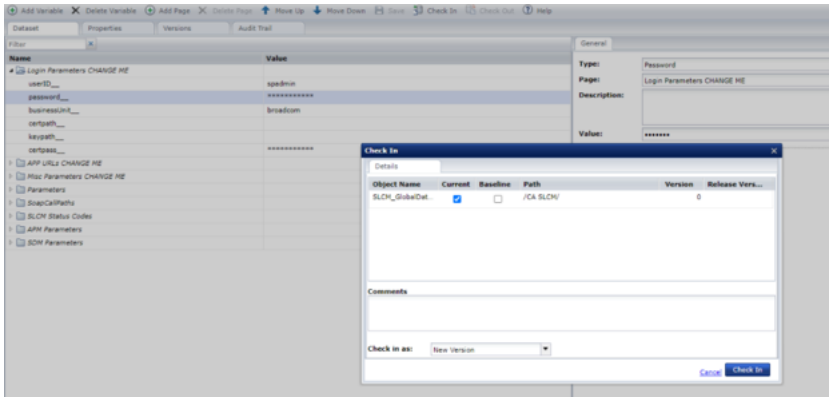
1. Login into CA Process Automation Interface: (*http://testqa-pam-new:8080/itpam*).
2. Navigate to Library, CA SLCM folder, Click **SLCM_GlobalDatasheet** dataset.



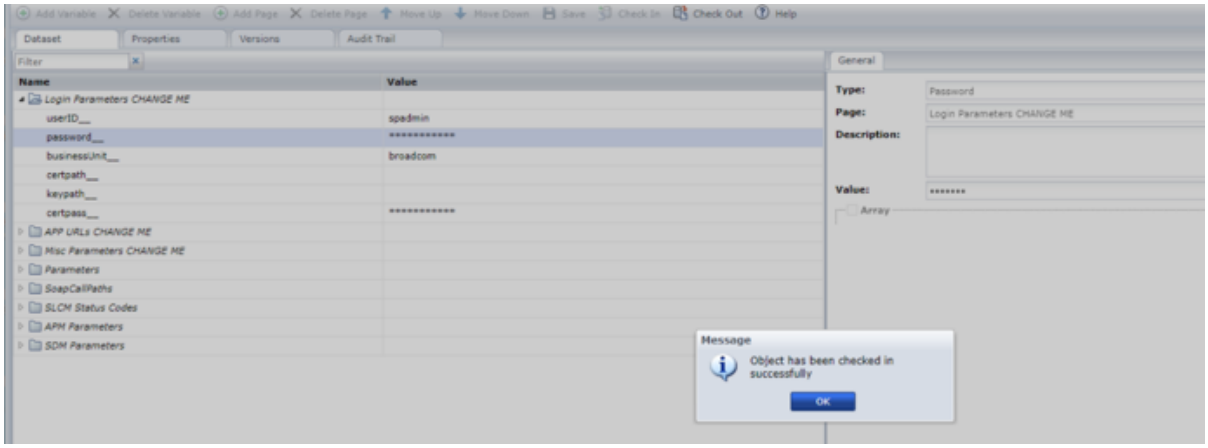
3. Click the **Checkout** link.



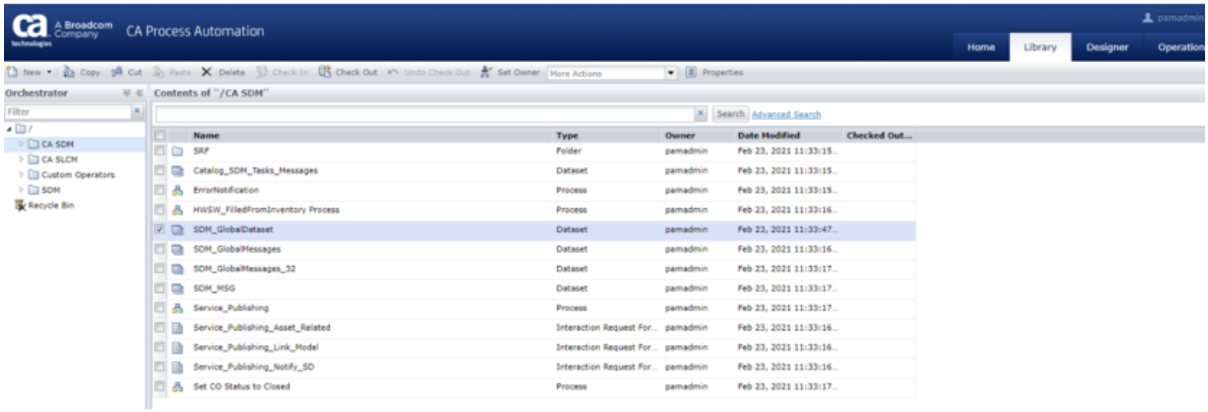
4. Update the entries **userID_**, **Password_**,**businessUnit_** parameters. Click **Save** and **Check in**.



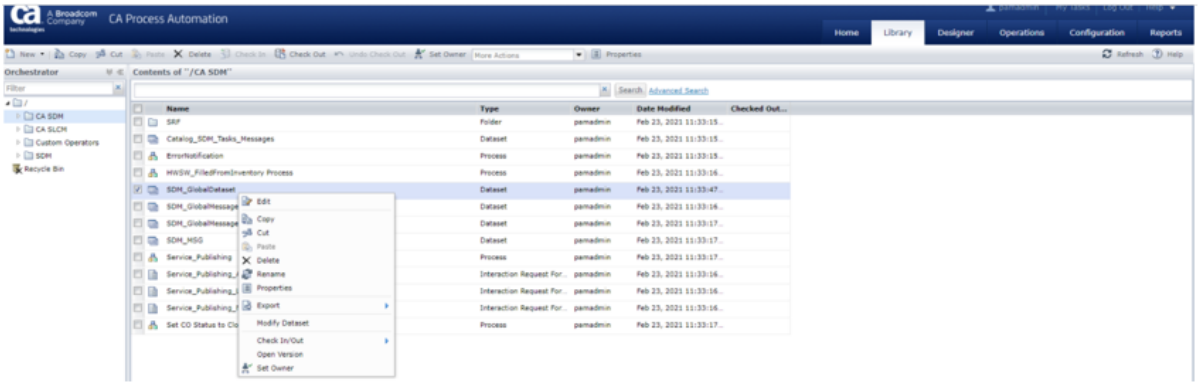
5. Click **OK** to update the above mentioned entries.

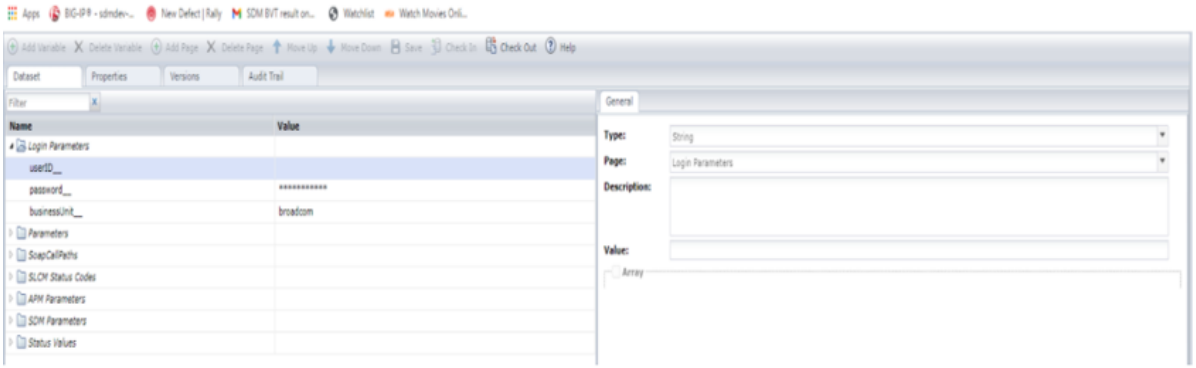


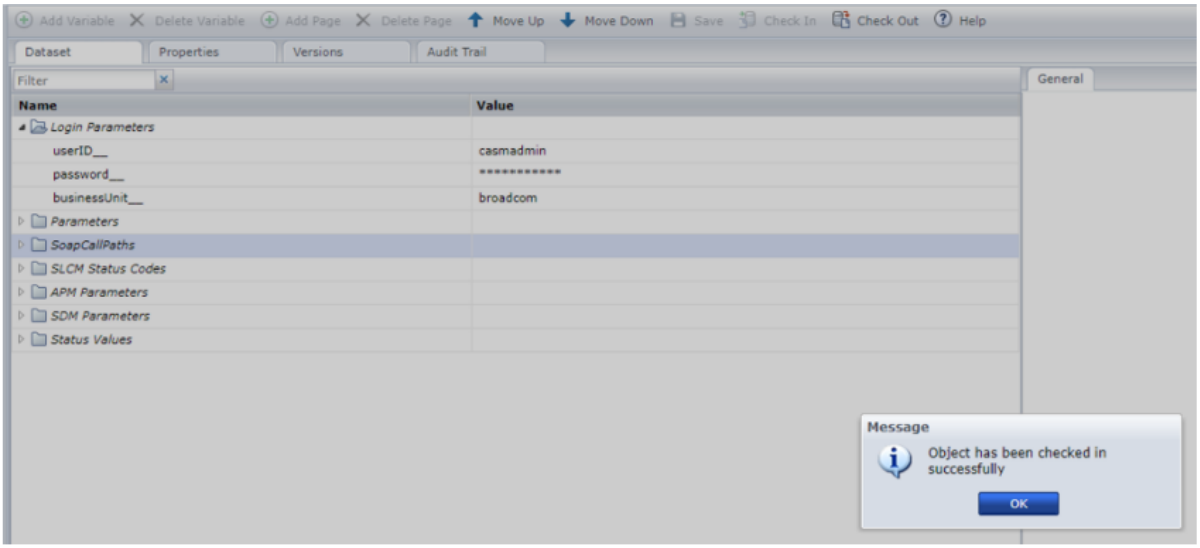
6. Navigate to **Library**, **CA SDM**, **SDM_GlobalDataset**.



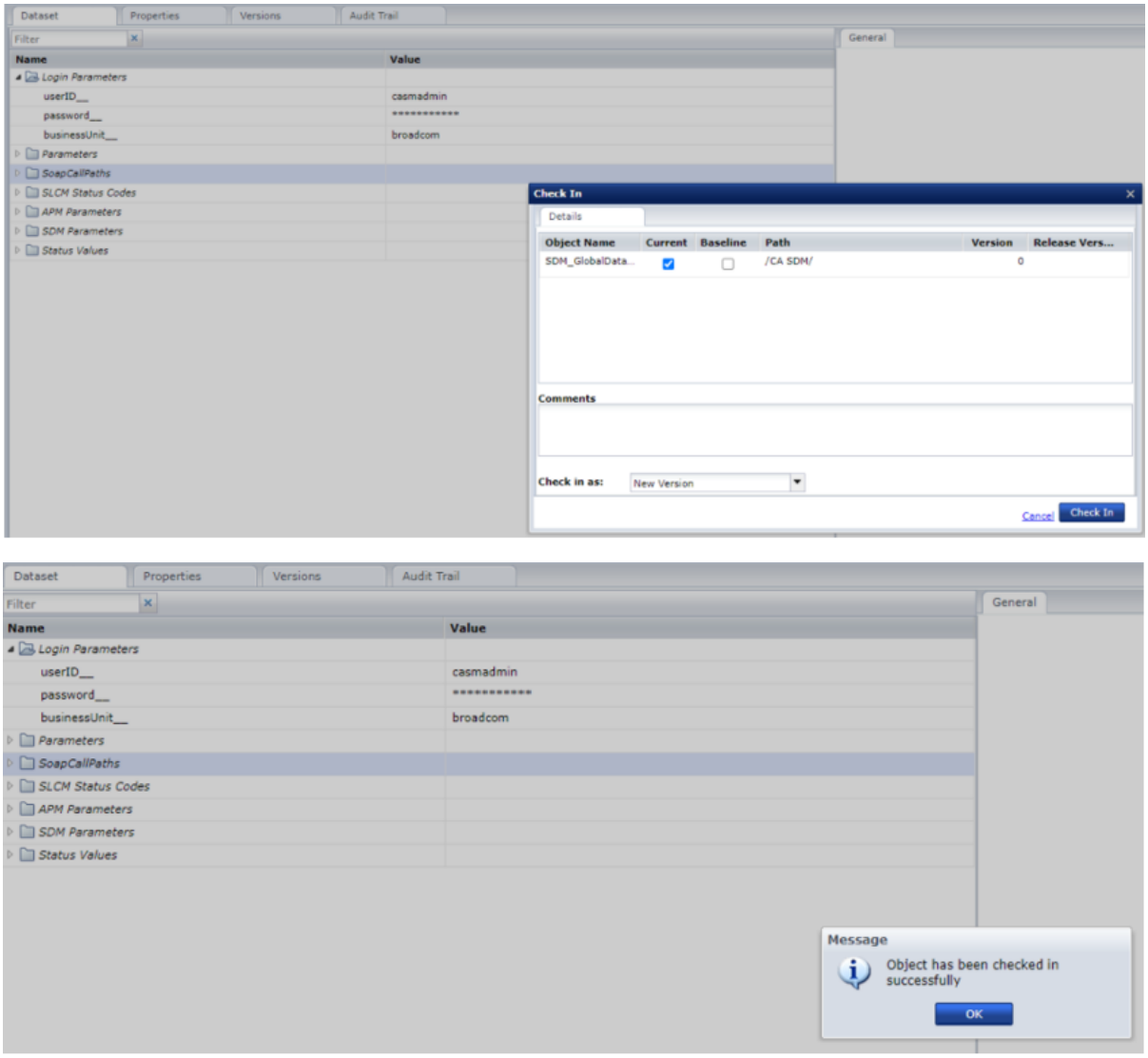
7. Click **Edit SDM_GlobalDataset**.







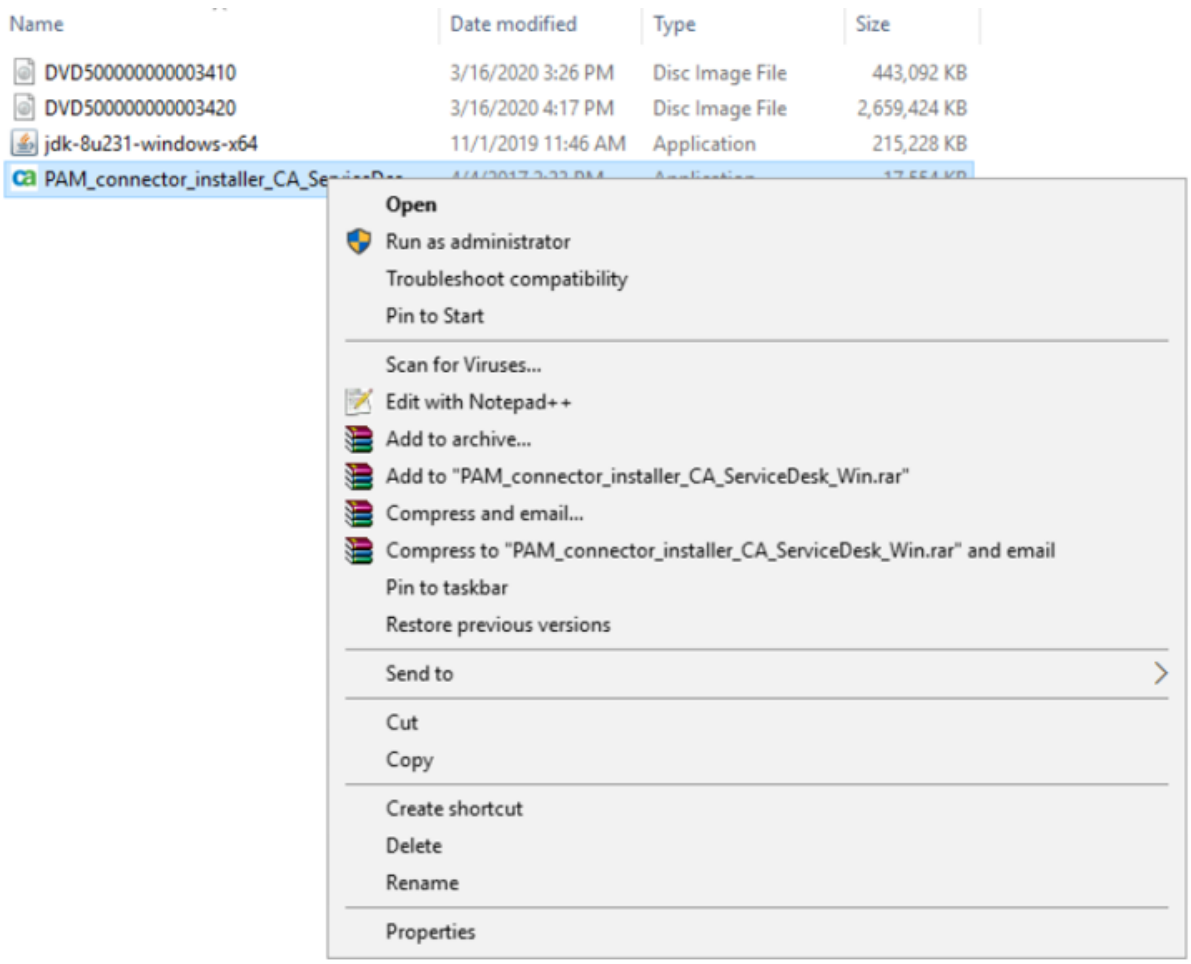
8. Update the entries of **UserID**, **Password**, and **businessUnit_values**. Click **Save** and **Check In**.



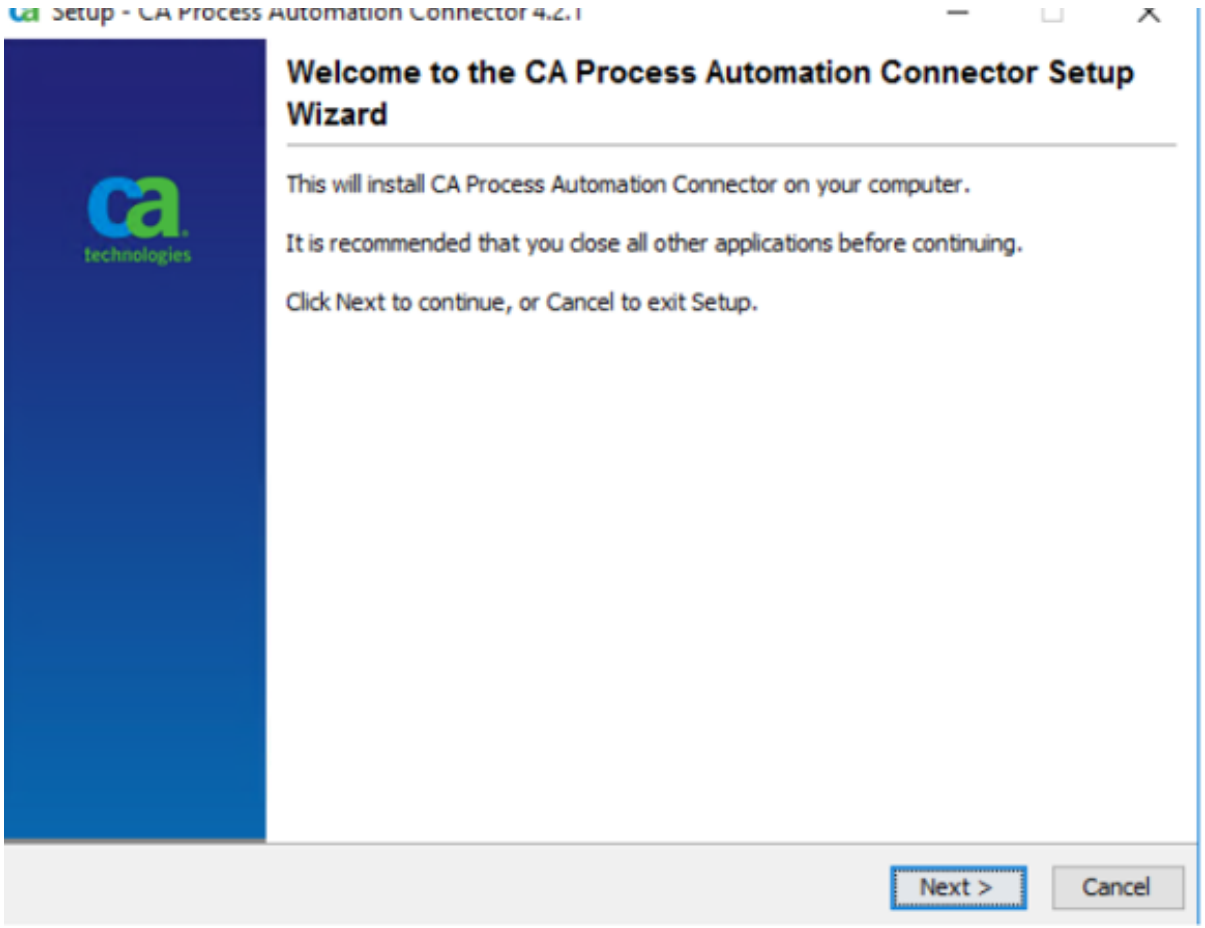
Install PAM Connector for SDM (testqa-pam-new)

Perform the following steps:

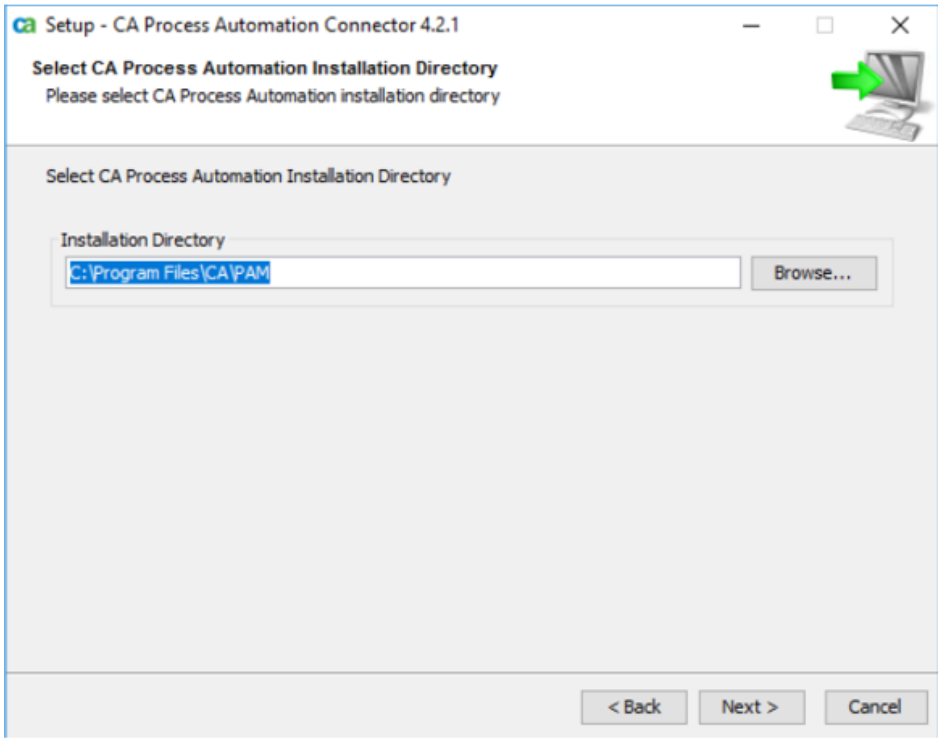
1. Stop CA Process Automation Service on **testqa-pam-new** and install the PAM connector for SDM.
2. Login to Broadcom Support and download **PAM_connector_installer_CA_ServiceDesk_Win.exe** on **testqa-pam-new**.



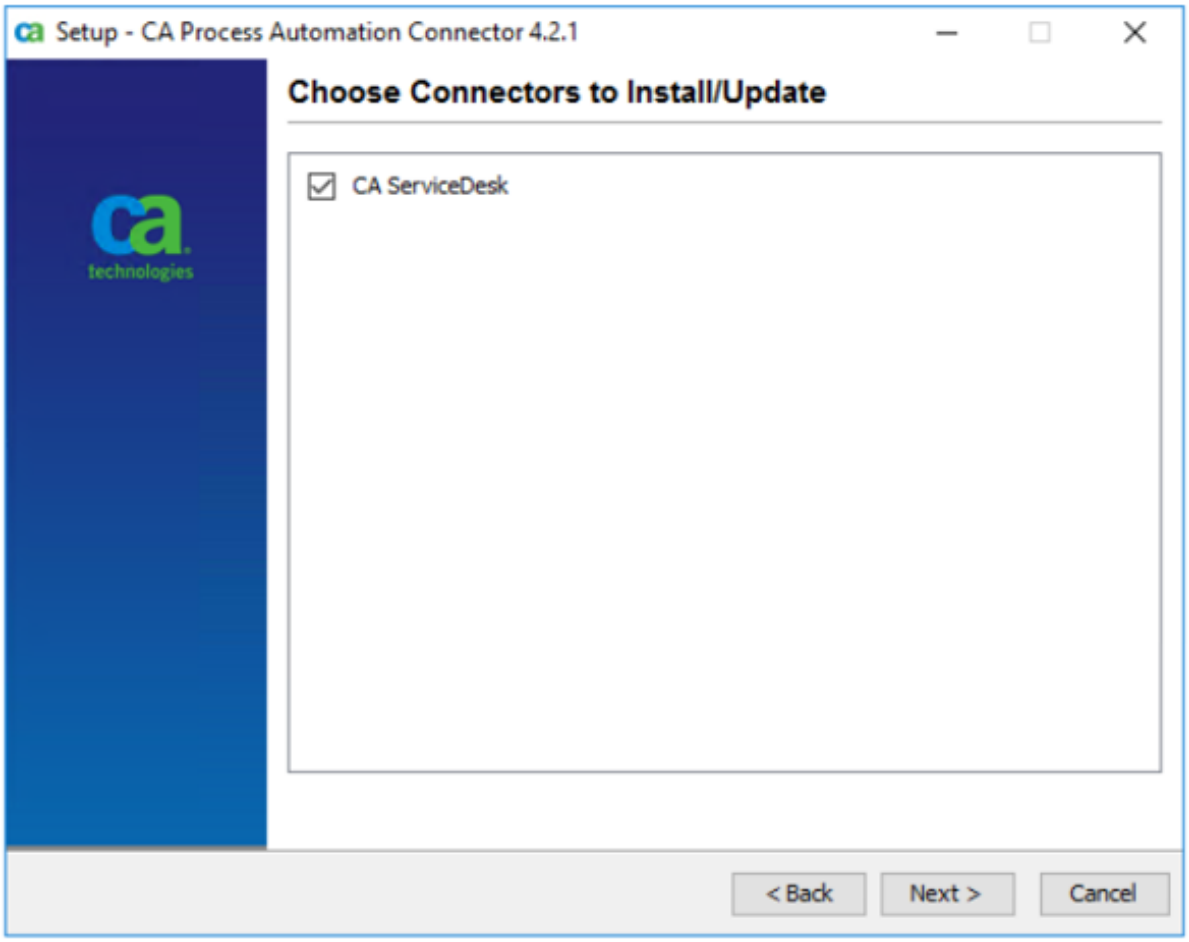
3. Right click the **setup.exe** and Select **Run as Administrator**.
4. CA Process Automation Connector 4.2.1 Installation process begins.



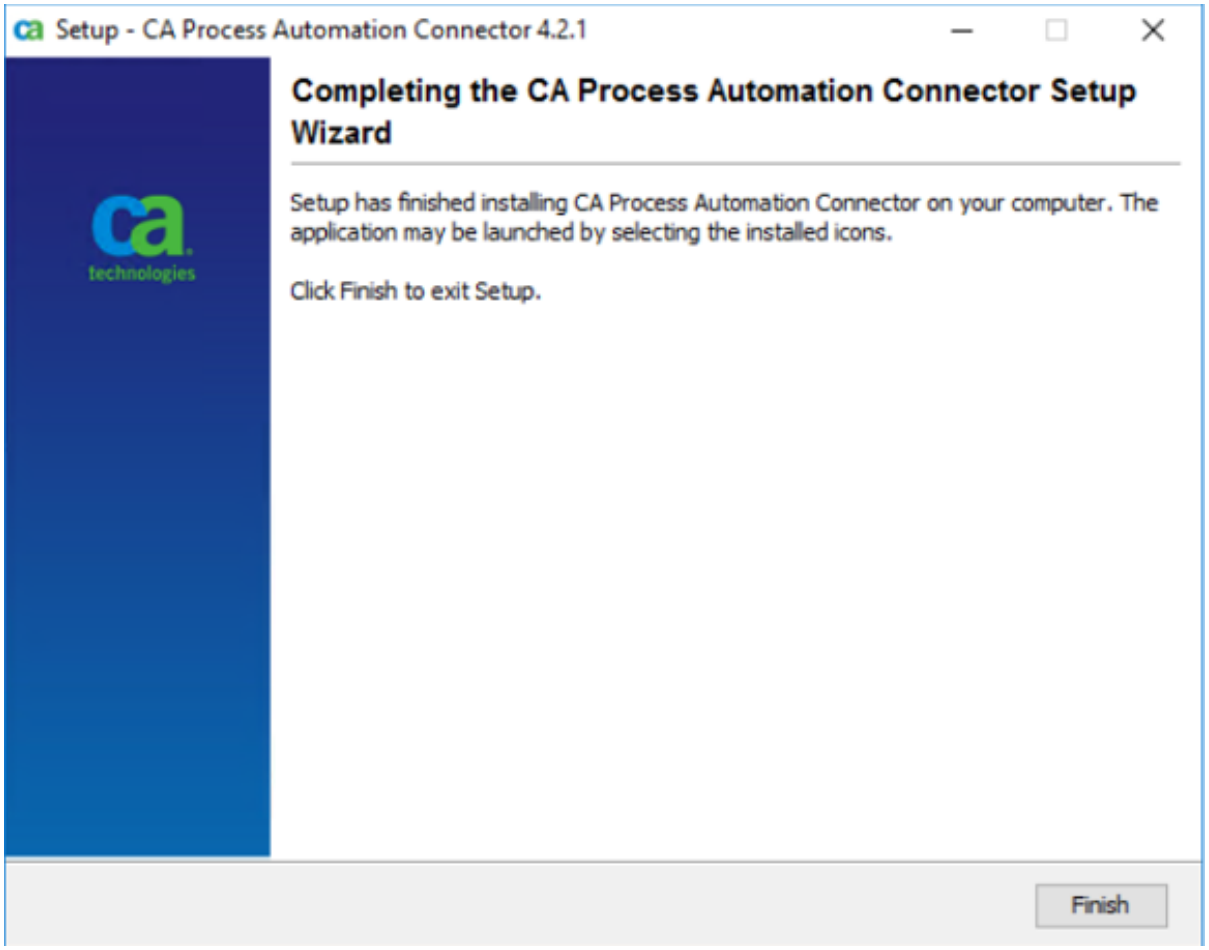
5. Review and accept the license agreement.
6. Provide the installation directory path.



7. Select CA Service Desk Manager.
   Click Next.



8. Click**Finish** to complete the Process Automation Connector Setup Wizard.



9. CA Process Automation Connector Installation is complete.
   Start the ITPAM Services on **testqa-pam-new** and then start the below services:
   1. Stop Catalog and Start the Catalog Services on **testqa-sc-new**.
   2. Stop and start the SDM Application services in the BG Server on **testqa-bg-new**.
   3. Stop and start the SDM Application services in the Standby Server on **testqa-sb-new**.
   4. Stop and start the SDM Application services in App1 Server on **testqa-app1-new**.
   5. Stop and start the SDM Application services in App2 Server on **testqa-app2-new**

Moving MDB Data from the Source (testqa-mdb) to the Target System (testqa-mdb-new)

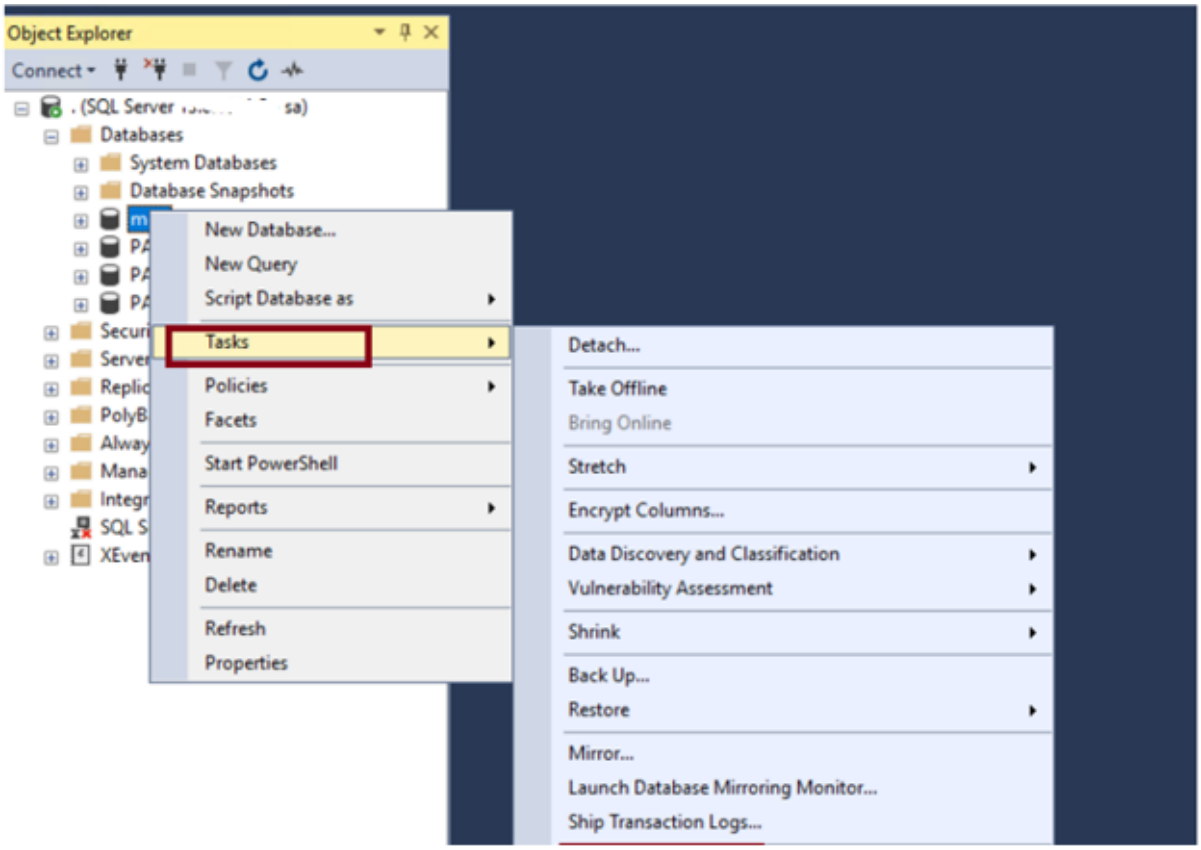
Perform the following steps:

1. Review the following CA Service Management Tables on **testqa-mdb-new** to ensure that these tables have system-specific configuration. For more information, see CA Service Management - Impacted Tables.

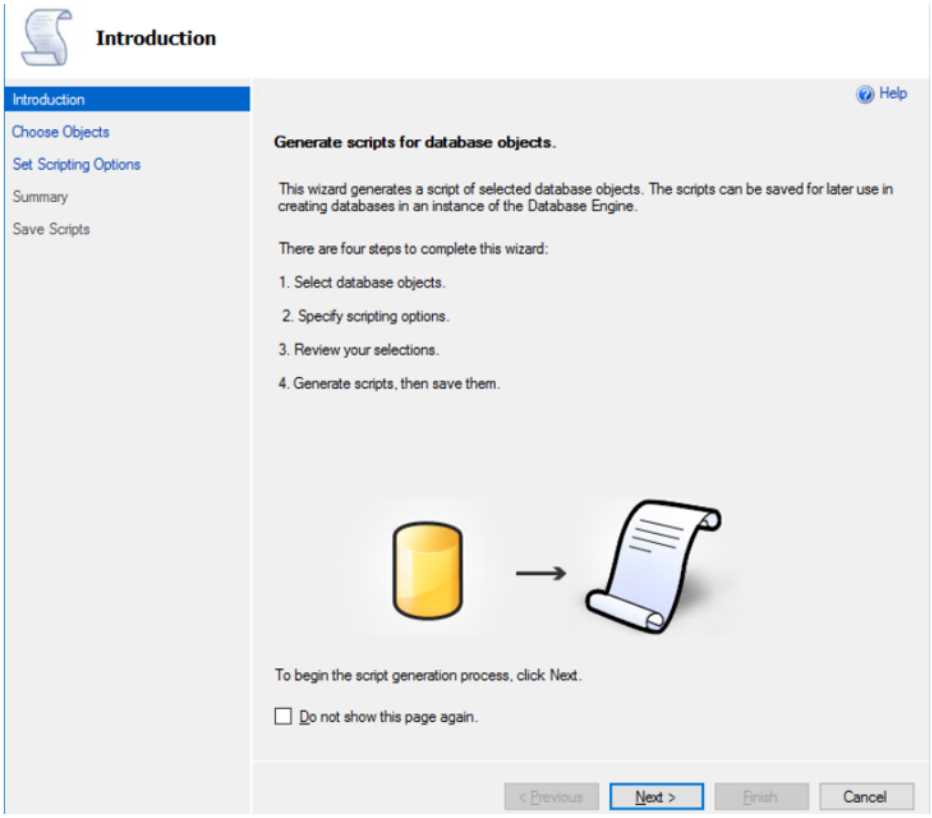
CA Service Management Tables:

Common Tables	CA SDM	CA Service Catalog
al_cdb_comp_installstate_bkup al_cdb_componentinstallstate al_cdb_config_params_backup al_cdb_configurationparameters al_cdb_files ca_application_registration	ci_mdr_provider options sapolicy usp_servers object_promotion promo_hist usp_configuration usp_webeng_domsrvr usp_webengine usp_webeng_alias usp_domsrvr	usm_configuration

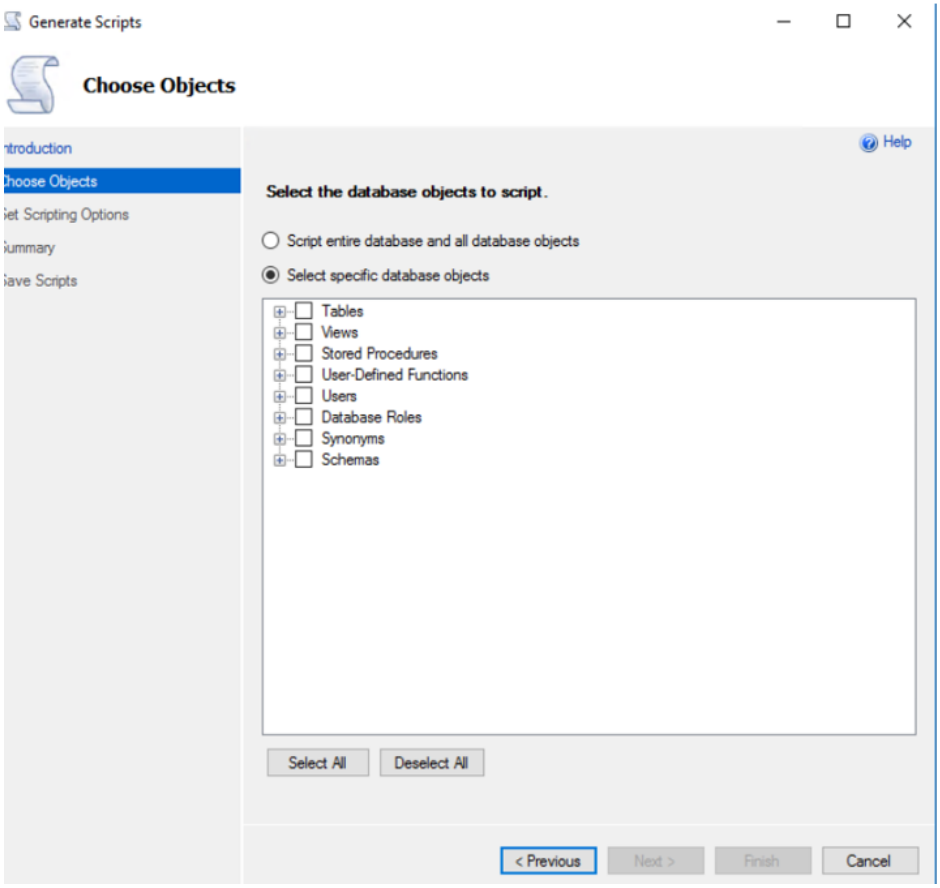
2. Stop the following CA Service Management services on both the source and target servers:
   a. **CA SDM**: CA Service Desk Manager Server (Background, Standby, and Application)
   b. **CA Service Catalog**: CA Service Accounting, CA Service Catalog and CA Service Management Admin (Catalog Server)
   c. **CA Process Automation**: CA Process Automation Orchestrator (CA Process Automation Server)
3. On the source system (**testqa-mdb**), backup MDB. Copy the MDB backup file to the target system (**testqa-mdb-new**).
4. On the target system (**testqa-mdb-new**), perform the following steps to backup the tables which are mentioned in the CA Service Management Tables in step 1.
5. Navigate to Microsoft SQL Management Studio, right-click MDB (or the appropriate MDB instance name applicable for your system), and select **Tasks**, **Generate Scripts**.



6. Click **Next**.



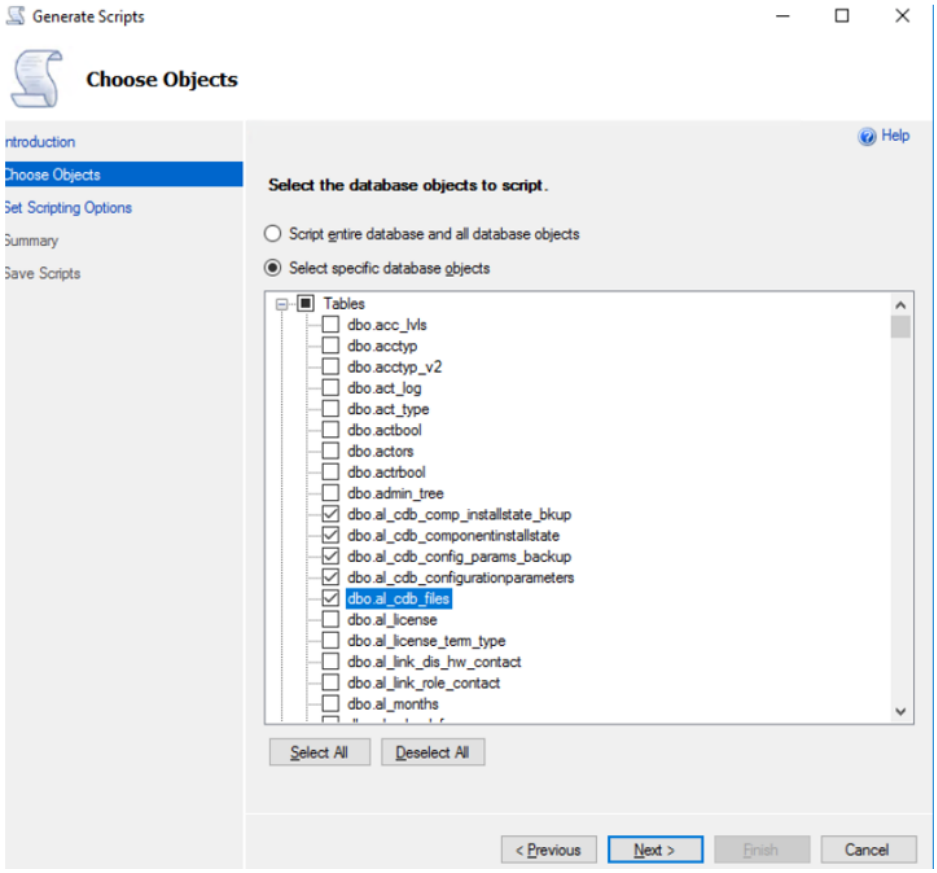
7. Click Select Specific database objects.



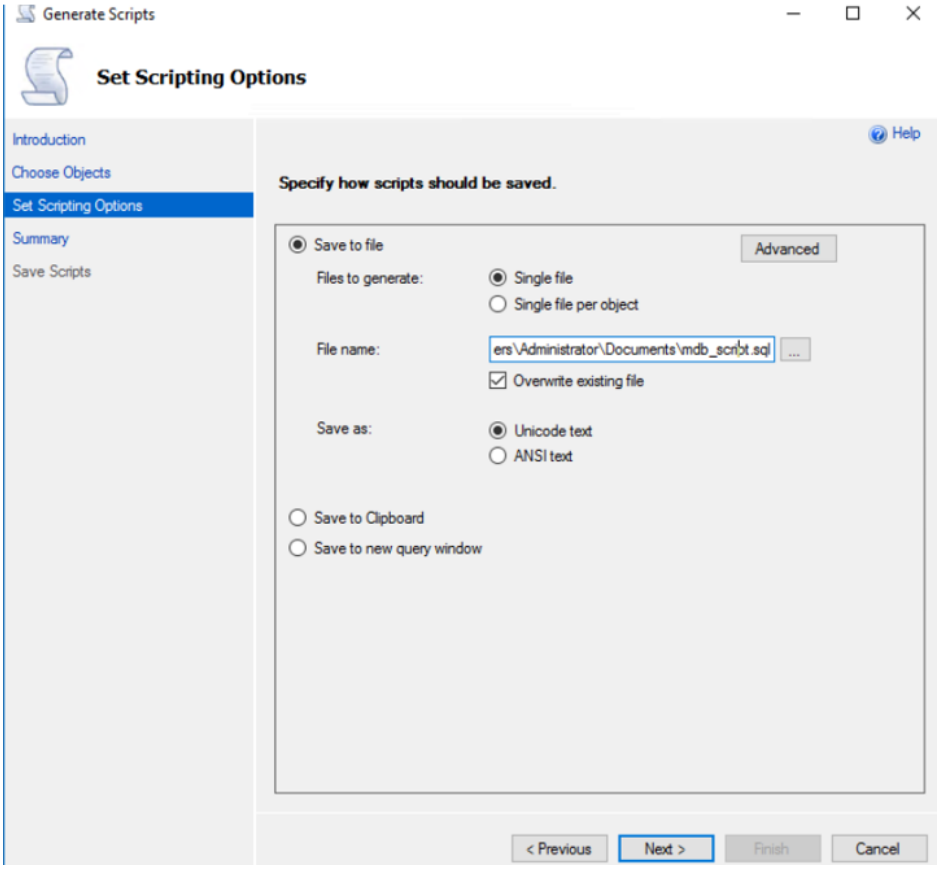
8. Below are the steps to take a backup of the CA Service Management Tables and then, generate scripts.
   a. Select the Tables option and choose the appropriate tables.
   b. Select list of tables from the CA Service Management Tables.

Common Tables	CA SDM	CA Service Catalog
al_cdb_comp_installstate_bkup al_cdb_componentinstallstate al_cdb_config_params_backup al_cdb_configurationparameters al_cdb_files ca_application_registration	ci_mdr_provider options sapolicy usp_servers object_promotion promo_hist usp_configuration usp_webeng_domsrvr usp_webengine usp_webeng_alias usp_domsrvr	usm_configuration

NOTE
Following example image shows few selected tables but you must select all the **18** tables (mentioned in step 8) to proceed further.

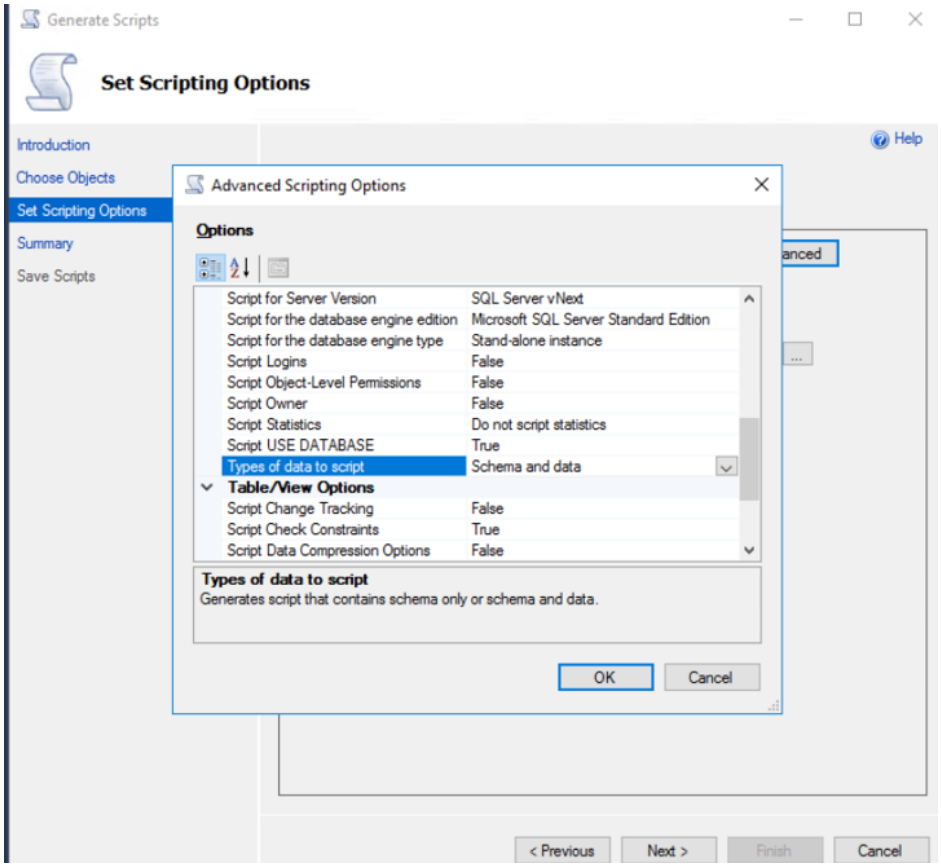


c. Click**Next**.
Select the Advanced tab on the upper right corner (image) and navigate to Types of Data.

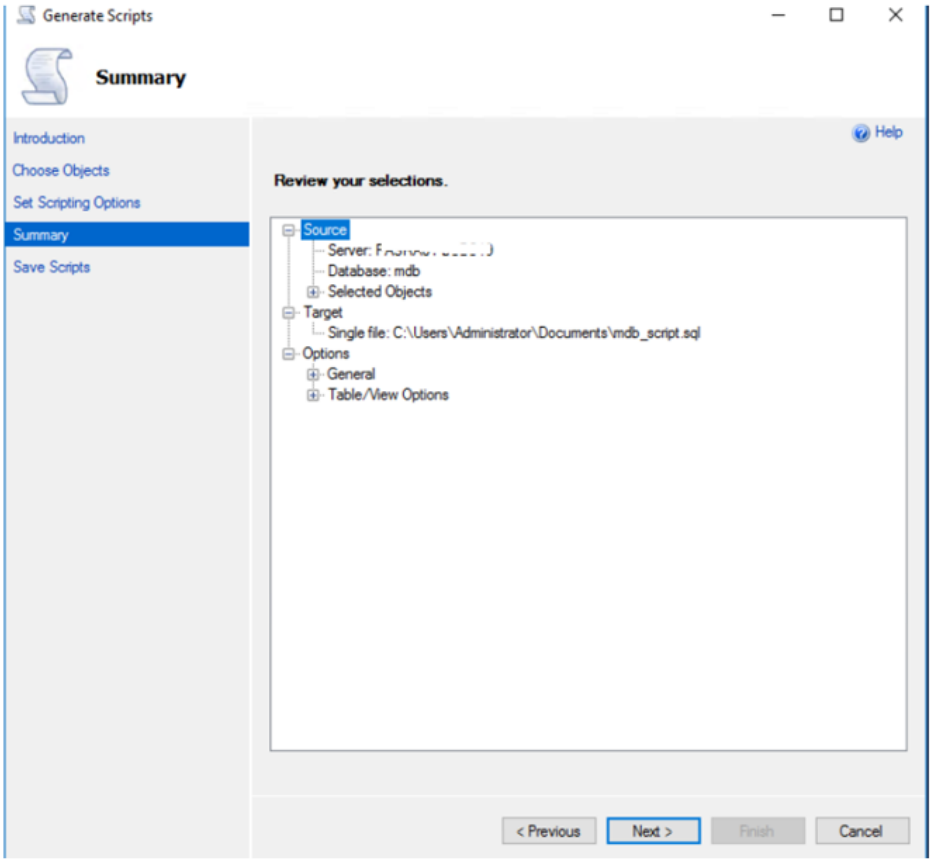


9. Change the Types of data to script option value from **Schema Only** to **Schema and Data**, and click **OK**. Your selection is now saved to a file *%user%\documents\mdb_script.sql*.

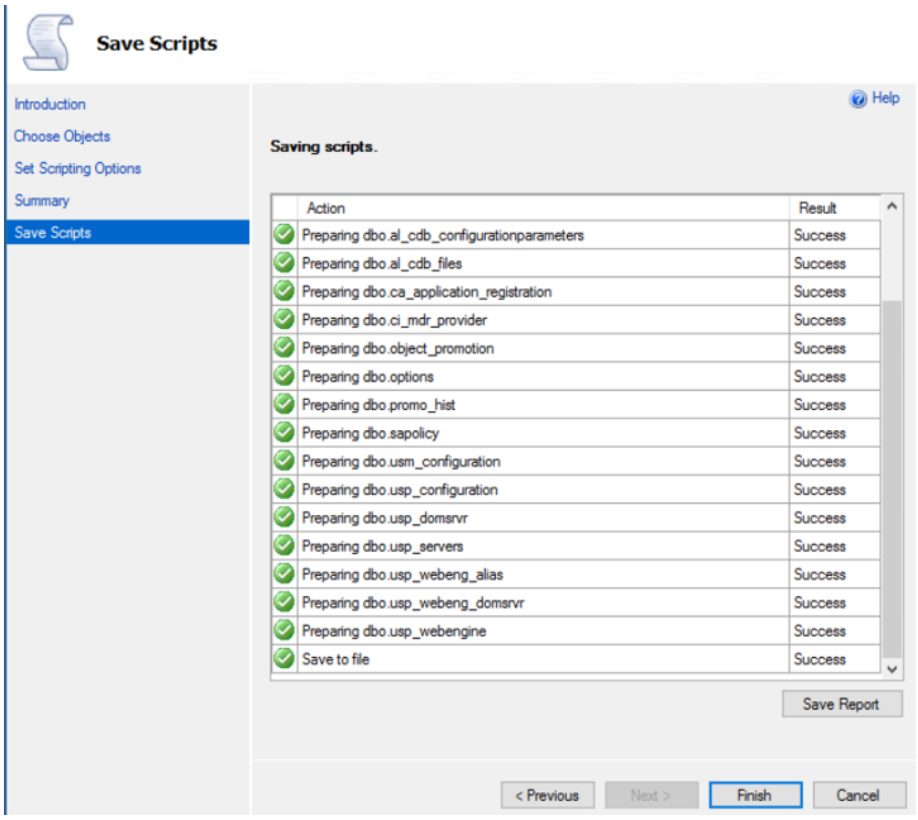
   NOTE
   It's recommended to use non-default file names. By default, the file name is **script.sql**.



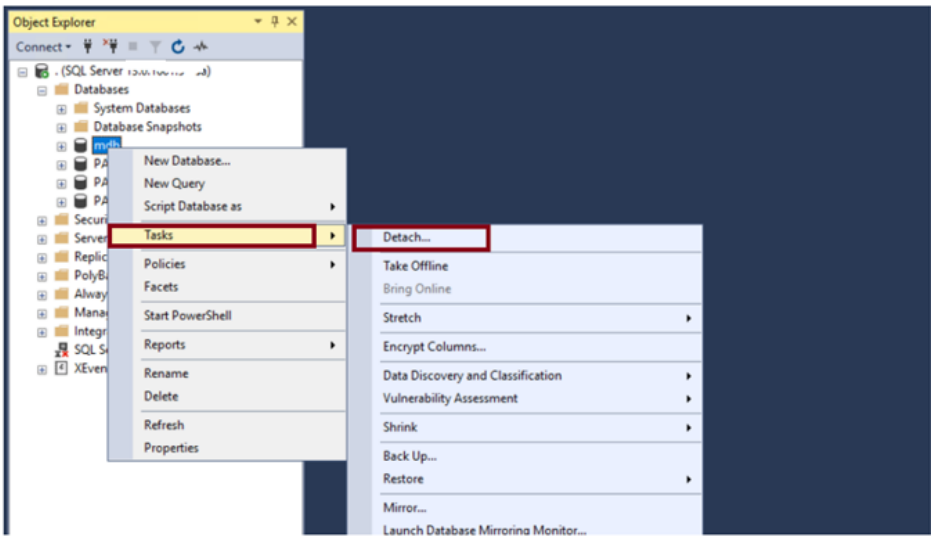
10. Click **Next**.



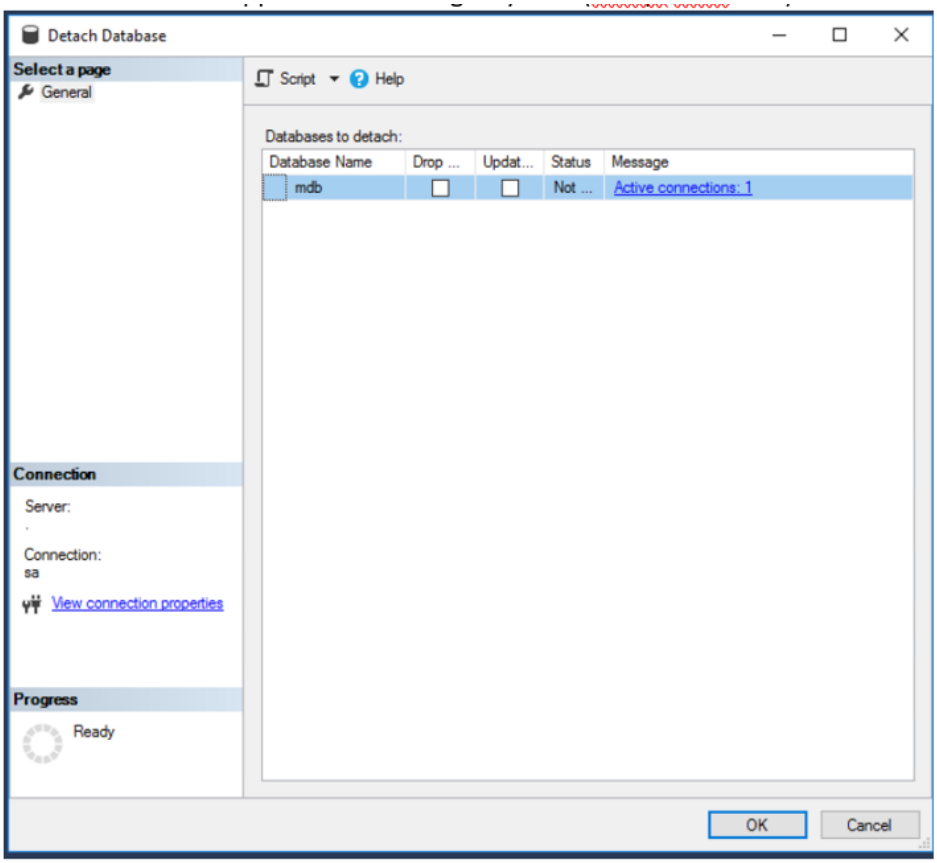
11. Click **Finish**.



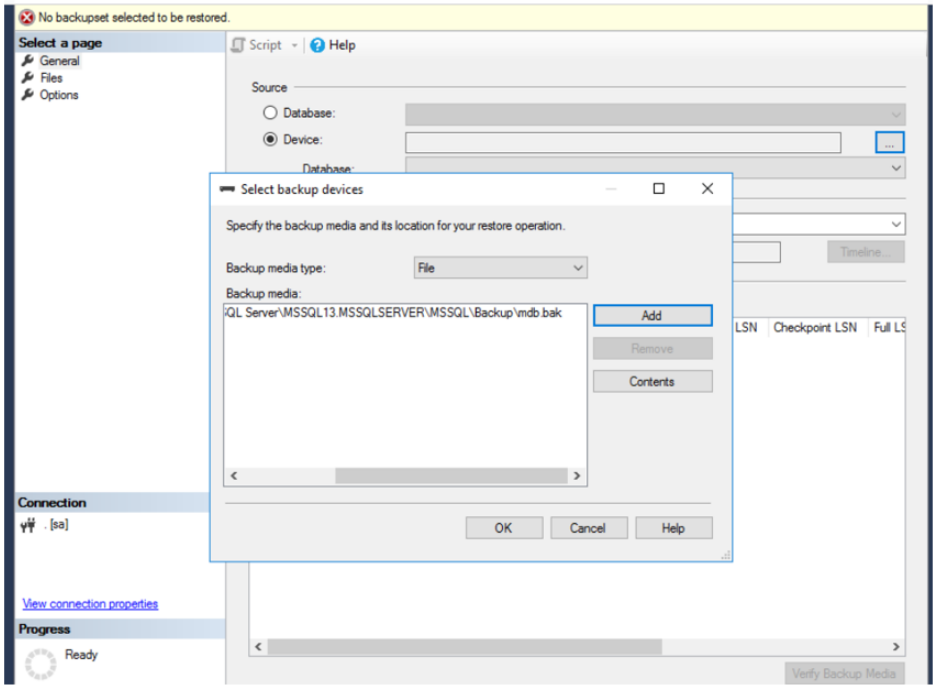
12. On the Detach Database screen, check the**Drop Connections** checkbox.



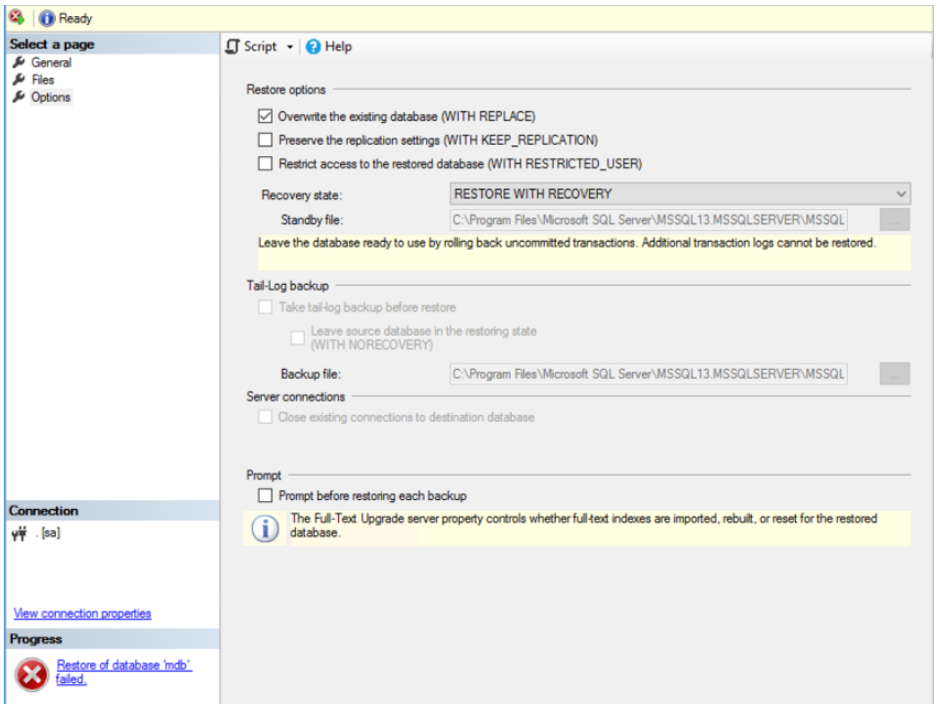
13. Check the **Drop** Checkbox and click OK.
    MDB instance is dropped from the target system (**testqa-mdb-new**).



14. To restore MDB backup into the target system (**testqa-mdb-new**) which was copied from the source system (**testqa-mdb**).
    a. Open the SQL Management studio with DB Admin credentials (**sa**).
    b. Navigate to Databases option, right-Click and select the **Restore Database** option.
    c. Select the **Device** Radio button in the Restore database screen.
    d. Click **Browse (...)**. Click **Add** and select the backup file where you have copied from source system.
    For example: \Program Files\Microsoft SQL Server\MSSQL13.MSSQLSERVER\MSSQL\Backup\mdb.bak)
    e. Click "**options**" on the Restore Database Screen.

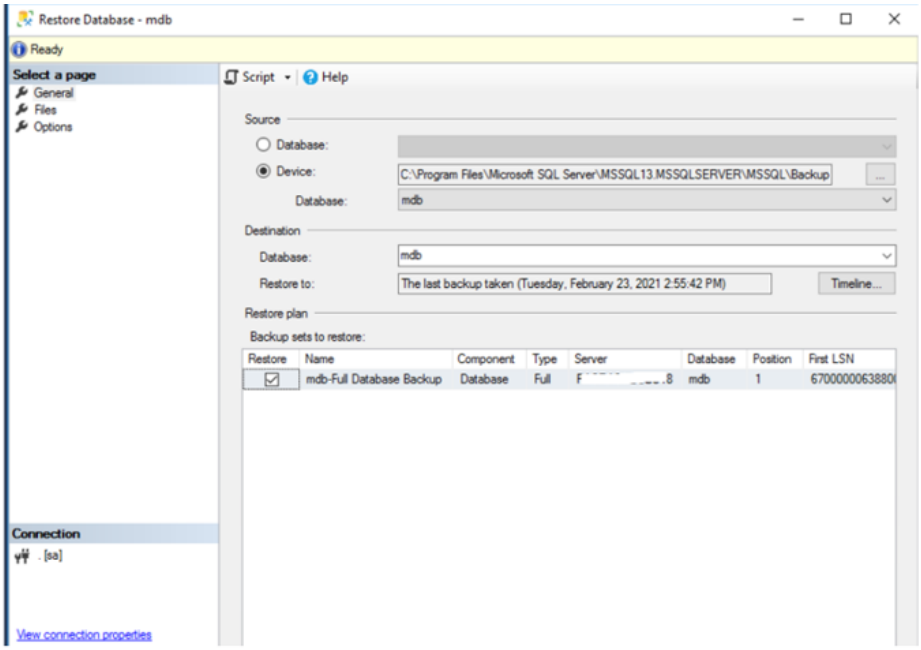


15. Check the checkbox of Overwrite the existing database (WITH REPLACE). Ensure that the database name on the target system is retained.



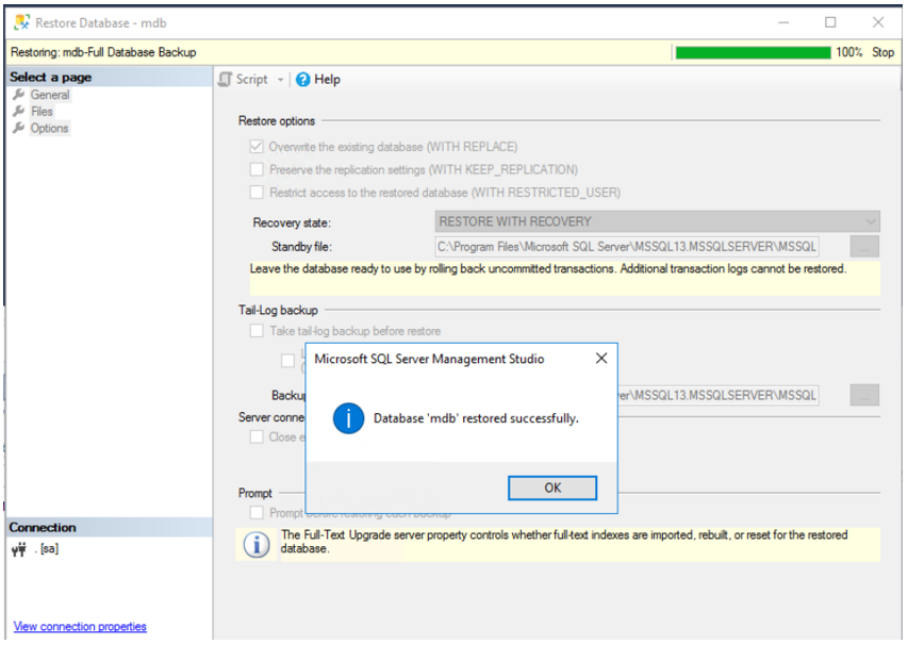
Click Ok.

16. The restore database name on the target system (**testq-mdb-new**) is retained and successful.



Click OK.

17. Click OK.



Database “mdb” is restored successfully.

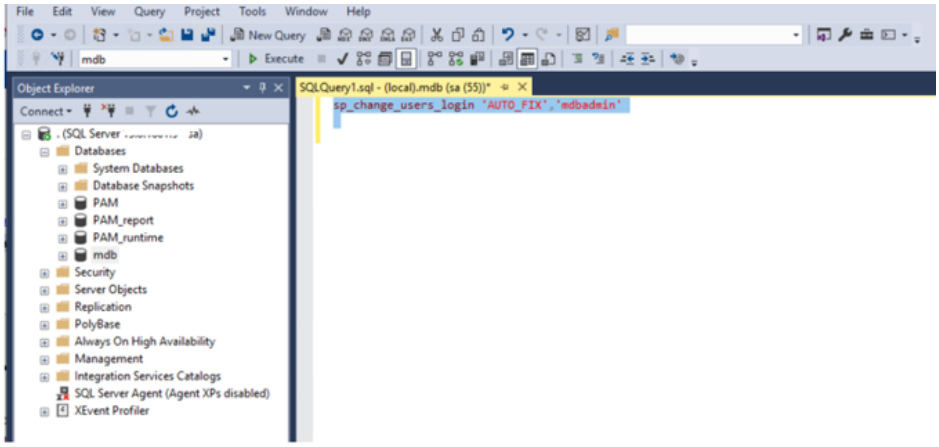
NOTE

Provide read and execute privileges to the mdb.bak file that you copied from the source database (testqa-mdb) to the SQLServerMSSQLUser$$MSSQLSERVER service account on the target MDB server (testqa-mdb-new). Failure to grant the read and execute privilege to mdb.bak may result in error “file folder cannot be located or it does not have the adequate permissions”.

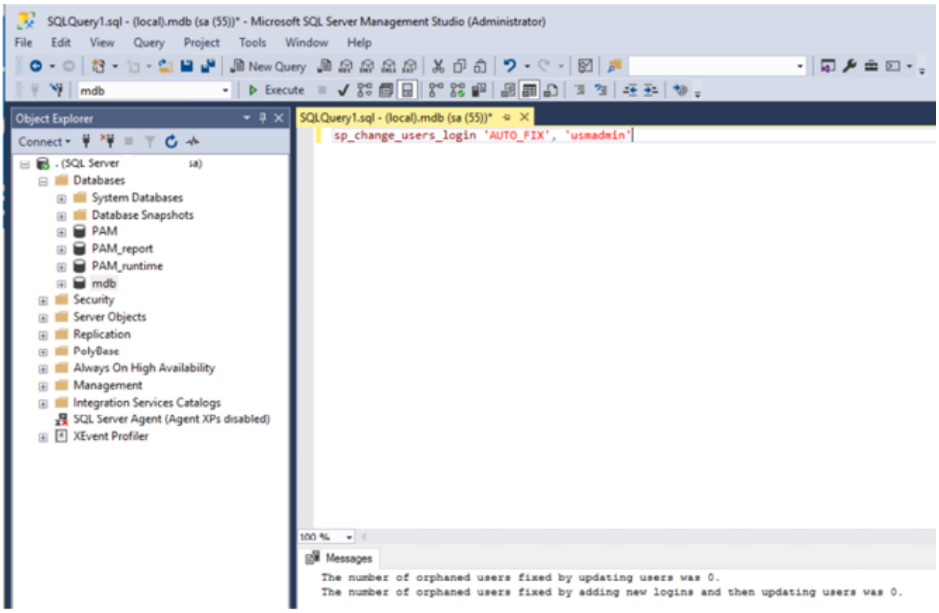
NOTE

Restore must complete without errors

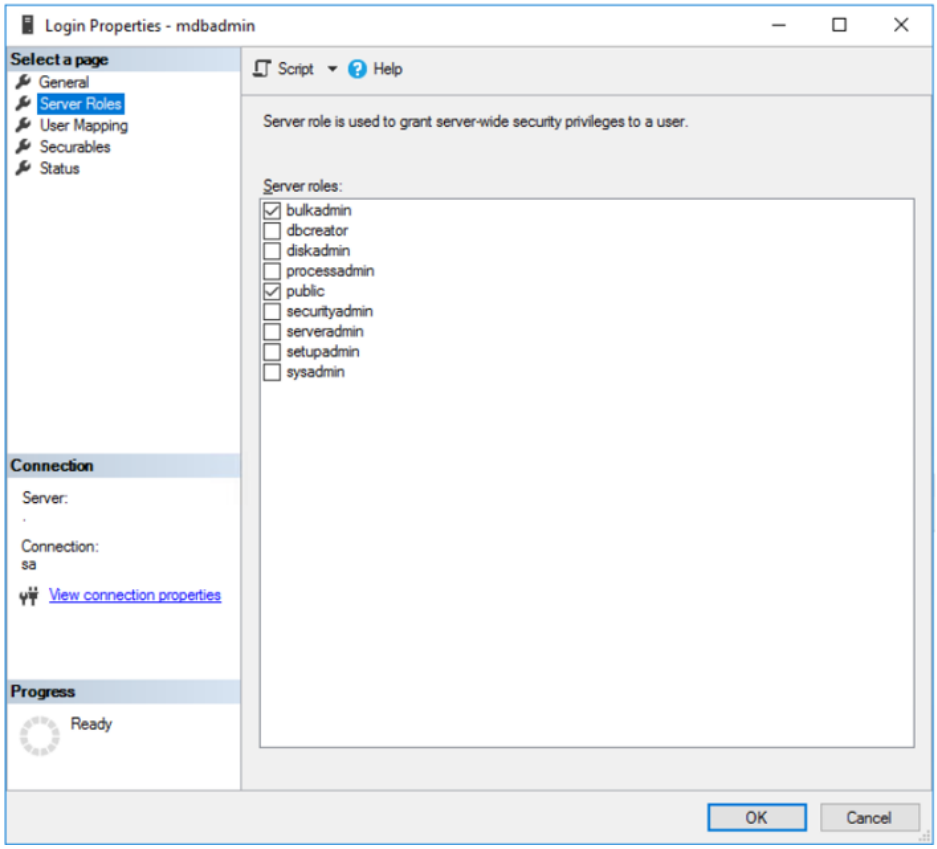
18. Run the following Microsoft SQL auto_fix_user script for the mdbadmin user: sp_change_users_login 'AUTO_FIX','mdbadmin'.



19. sp_change_users_login 'AUTO_FIX', 'usmadmin' (Applicable only if CA Service Catalog is installed and upgraded from previous release versions).



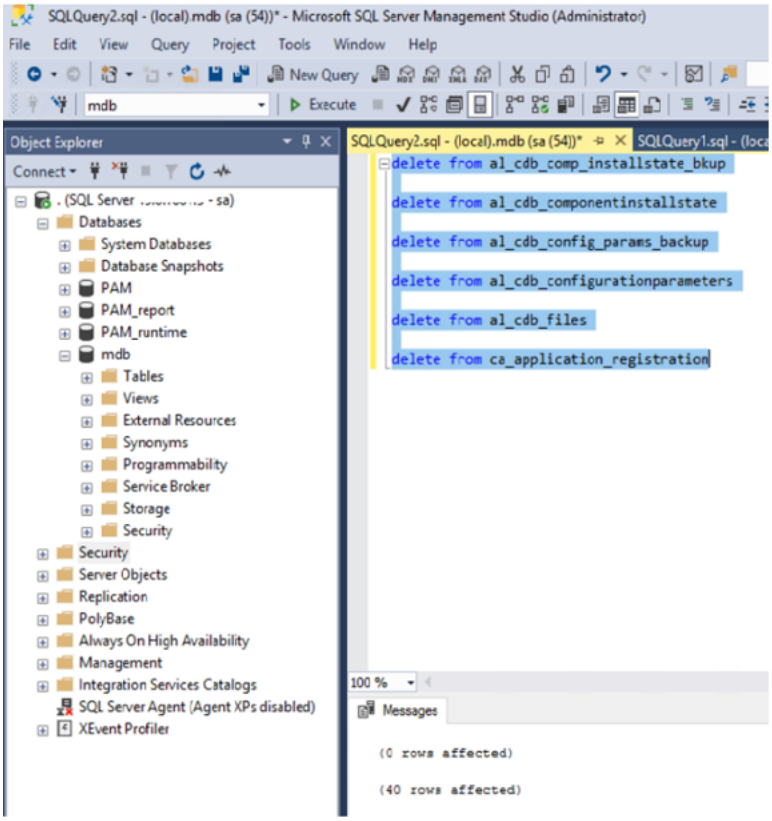
20. Verify that the mdbadmin is now mapped to mdb and have bulk admin, public roles.



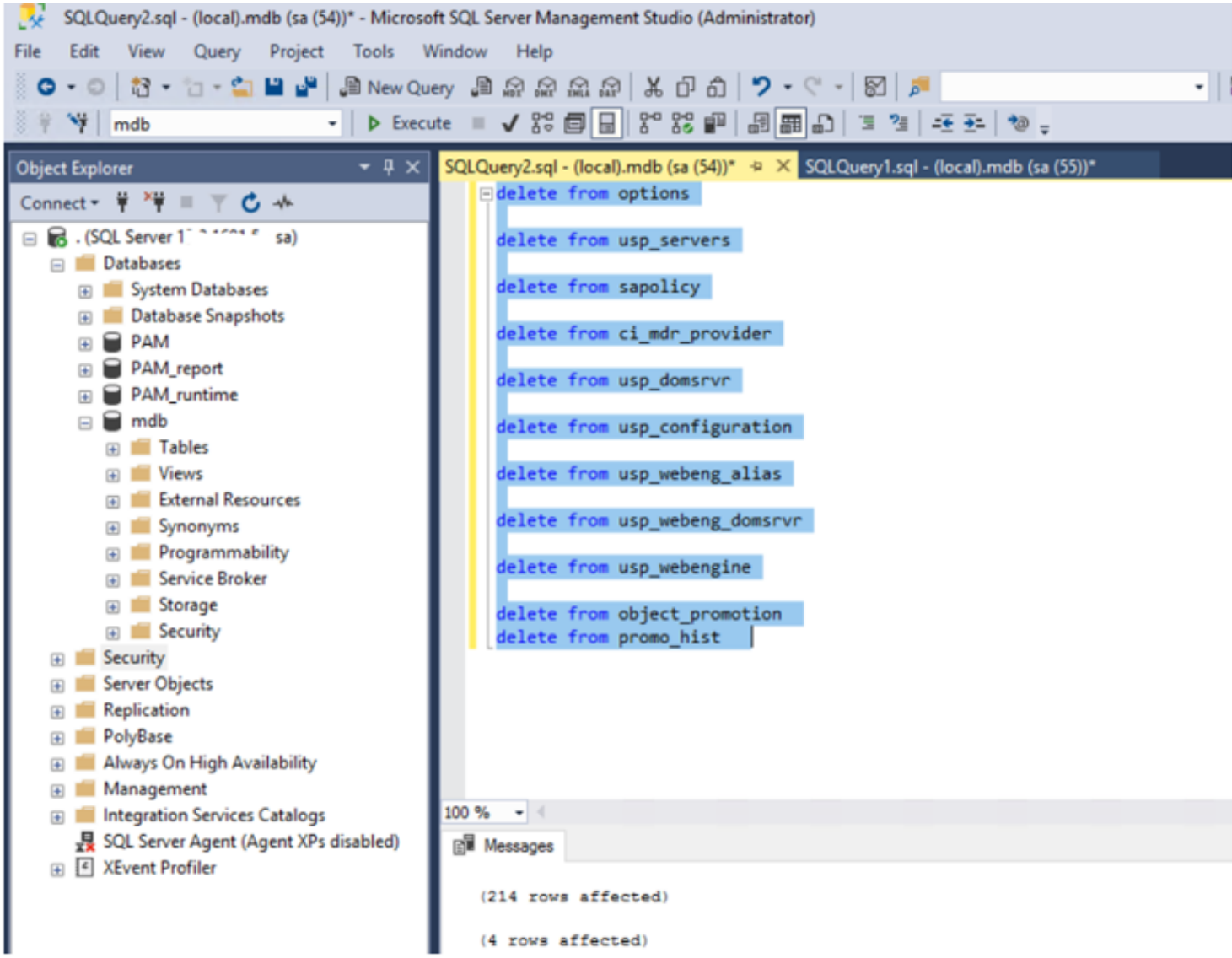
21. On the target system (**testqa-mdb-new**), delete data from the following tables and commit the changes:
    a. Use mdb and run the following queries to delete data from the required tables:

       Common Tables:

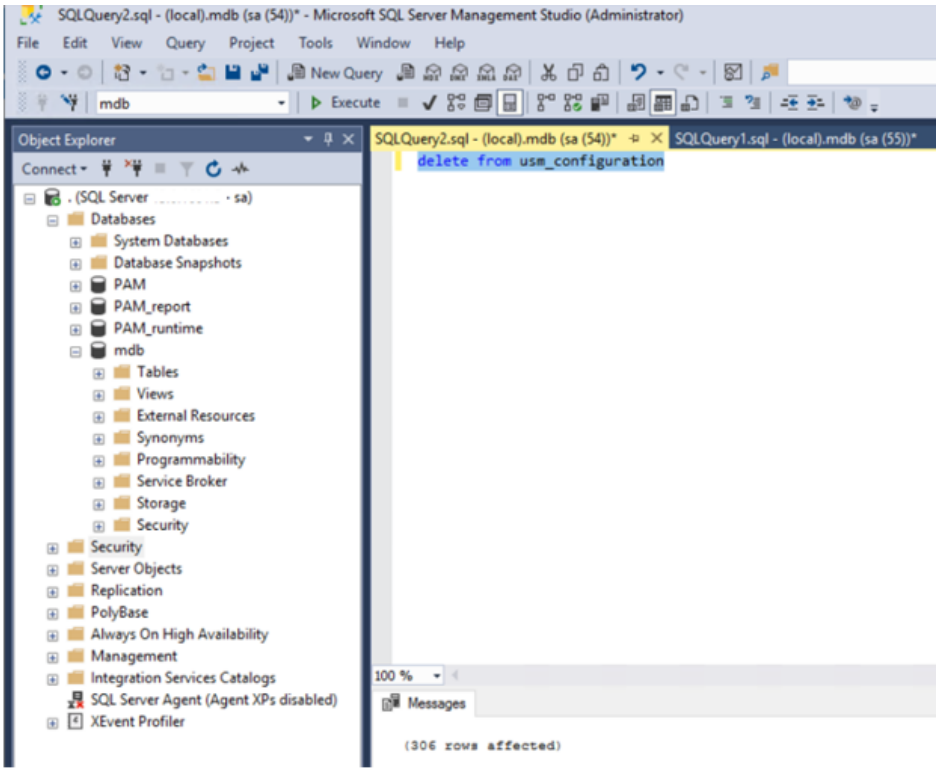
       - delete from al_cdb_comp_installstate_bkup
       - delete from al_cdb_componentinstallstate
       - delete from al_cdb_config_params_backup
       - delete from al_cdb_configurationparameters
       - delete from al_cdb_files
       - delete from ca_application_registration



22. CA SDM Tables:
    - delete from options
    - delete from usp_servers
    - delete from sapolicy
    - delete from ci_mdr_provider
    - delete from usp_domsrvr
    - delete from usp_configuration
    - delete from usp_webeng_alias
    - delete from usp_webeng_domsrvr
    - delete from usp_webengine
    - delete from object_promotion
      (applicable only if CA Service Management 14.1.02 is installed)
    - delete from promo_hist
      (applicable only if CA Service Management 14.1.02 is installed)



23. CA Service Catalog Tables: Delete from **usm_configuration**.



24. Execute the **mdb_script.sql** scripts that were created.

    IMPORTANT
    Ignore the warning message during execution “There is already an object named in the Database”.

CA Service Desk Manager (SDM) Customizations

Perform the following steps:

1. Copy the site/mods directory and tblobj.cfg and ddict.sch files from the source CA SDM server to the target CA SDM server:
   a. Source BG (**testqa-bg**) to the target BG (**testqa-bg-new**).
   b. Source Standby (**testqa-sb**) to the target Standby (**testqa-sb-new**).
   c. Source Application Server (**testqa-app1**) to the target Application Server (**testqa-app1-new**)

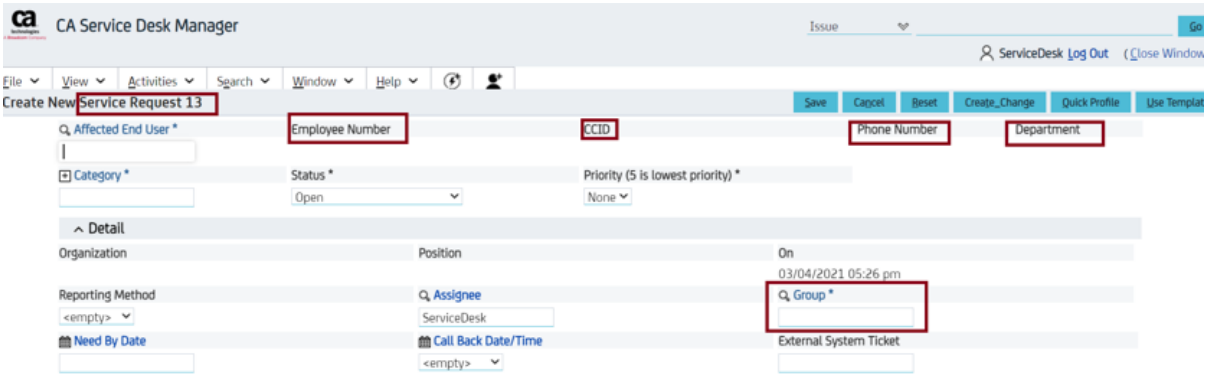
      NOTE
      Perform on all application servers in an environment.

      NOTE
      These steps are applicable only if your application environment is customized. This is not applicable for the out-of-the-box (OOTB) content.
2. Run **pdm_configure -S** on BG Server (**testqa-bg-new**). It configures silently. Monitor the progress of pdm_configure using NX_ROOT/logs/configure.log file. Perform the following steps:
   a. Launch the CA SDM BG URL as an Administrator (*http://testqa-bg-new:8080*).
   b. Navigate to **Administration** tab, **Security and Role Management**, Access Types link and click the **Administration** link.
   "CA EEM-Use CA Embedded Entitlements Manager" value should be displayed in the validation type list box.
   c. Verify that the new EEM server name (**testqa-eem-new**) is available in the NX_ROOT/NX.env file.
   d. Run **pdm_configure** on Standby Server (**testqa-sb-new**). Verify that the new EEM server name (**testqa-eem-new**) is available in the NX_ROOT/NX.env file in the Standby Server.

e. Run the **pdm_configure** on two application servers (**testqa-app1-new**) and (**testqa-app2-new**). Verify that the new EEM server name (**testqa-eem-new**) is available in the NX_ROOT/NX.env file for these two Application Servers.

f. Execute the **pdm_k_reindex** command on the CA SDM target system of BG Server **testqa-bg-new** to index knowledge documents that are created and copied from the source system of BG Server machine **testqa-bg**.

g. Login and ensure the target environment with CA Service Desk 17.1 is running with data and customizations from the source system.
For example, the below image shows customized items in the target system. Login to CA SDM, Navigate to **Service Desk**,**File**, **New Issue**.



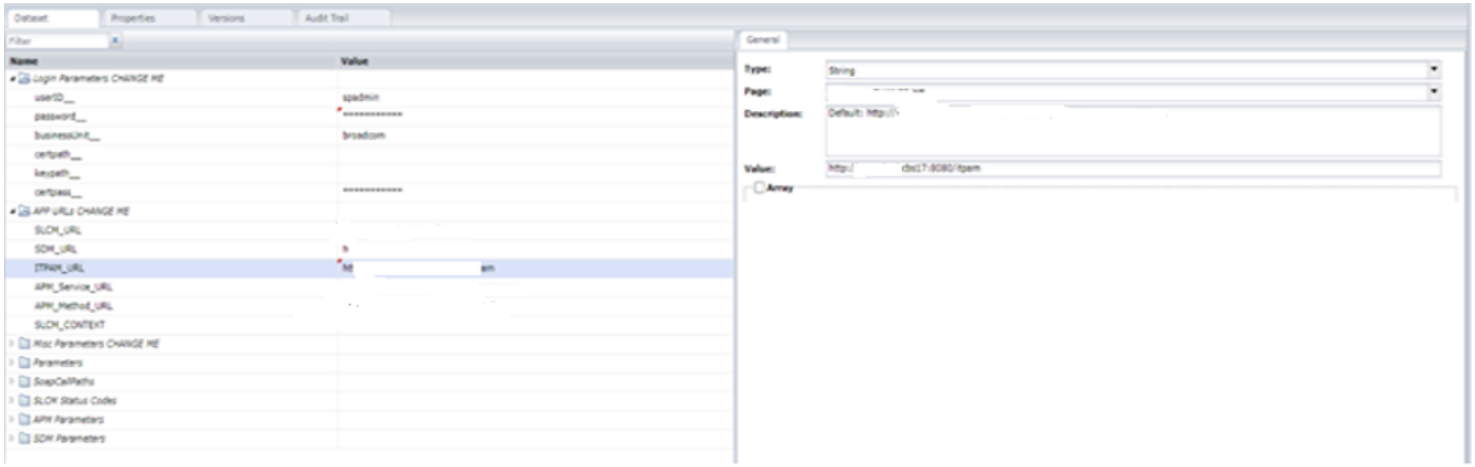
3. CA SDM Attachments (applicable only if you have NOT used the OOTB attachments path):
   a. Copy all the attachments using upload Path of all the repositories from a source server to the same path in a target server.
   For Example: Service Desk repository upload path is: <NX_ROOT>/site/attachments/default/servicedesk/file on source server. Copy the attachments from a source server to the target server.
4. CA SDM Repositories (applicable only if you have NOT used the OOTB repositories):
   a. Log in to CA SDM URL as an administrator.
   b. Navigate to **Administration**, **Attachments Library**, **Repositories**.
   c. Open the repositories one by one and update the target CA SDM server name in the Servlet Server and Repository Server fields.

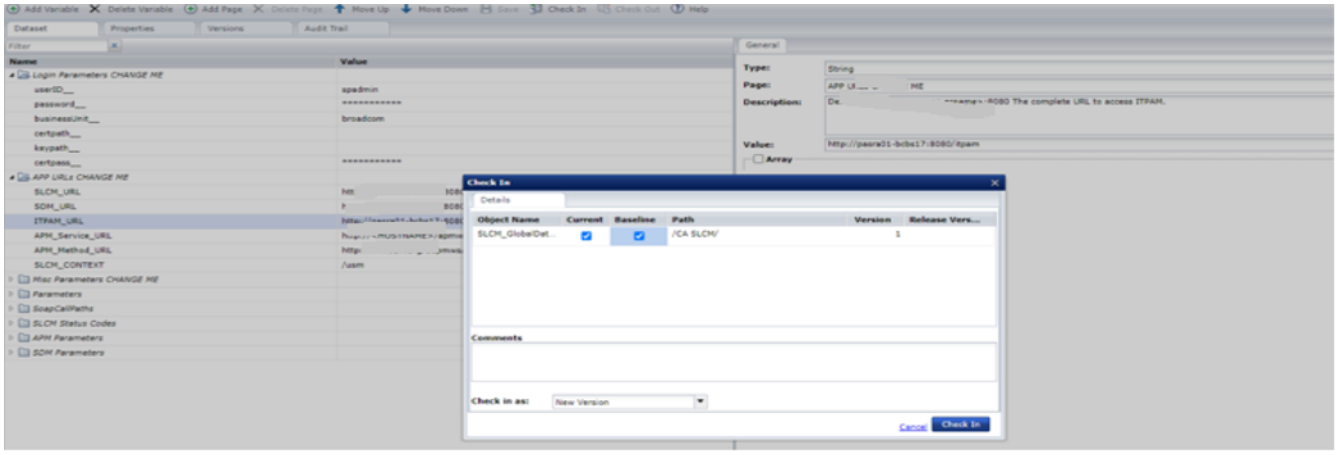
NOTE
Update the CA SDM server name in all the repositories (in the out-of-the-box and Created Repositories)

CA Process Automation

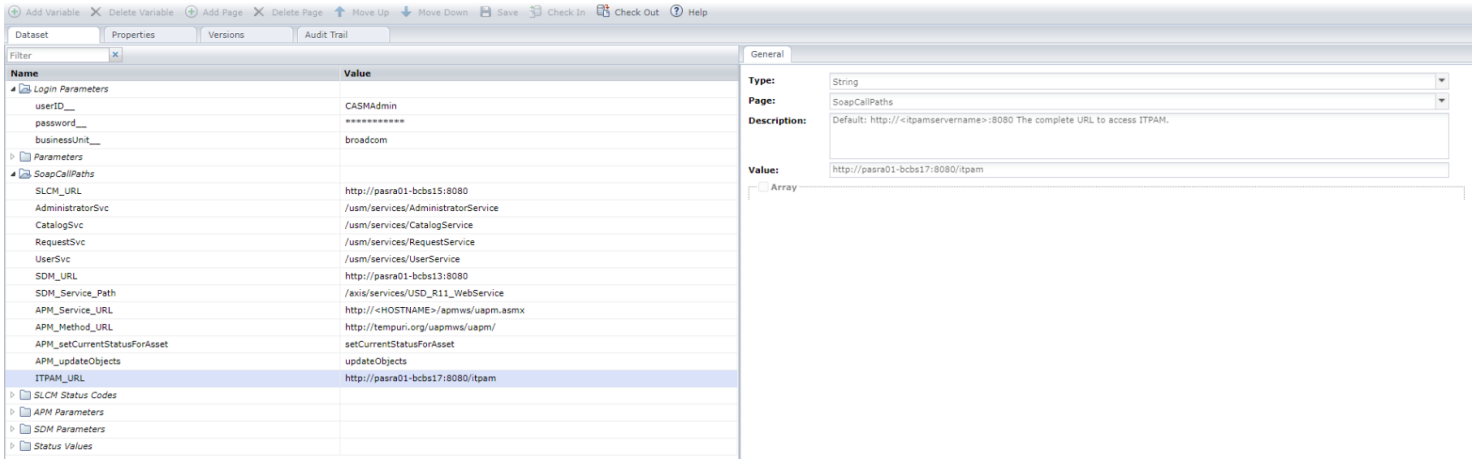
After successful installation, perform the following steps for CA Process Automation to ensure integration is seamless:

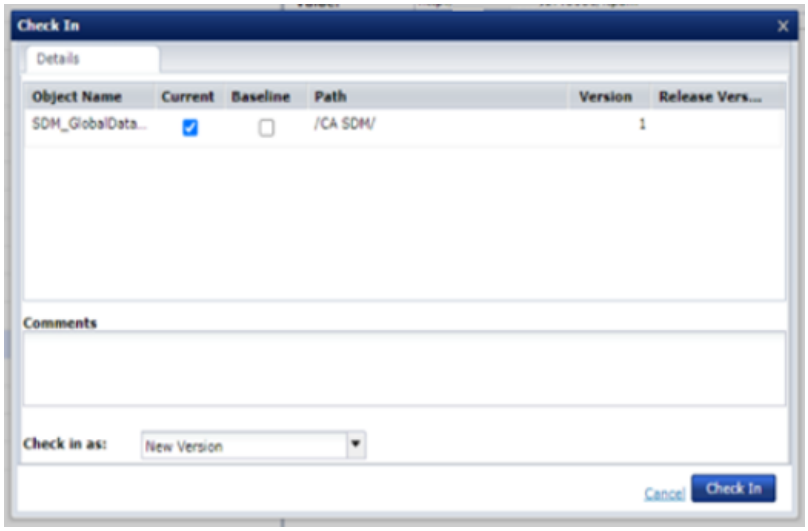
1. Start the CA Process Automation Service on **testqa-pam-new**.
2. Launch the CA Process Automation server (*http://testqa-pam-new:8080/itpam*) and access **SLCM_Globaldataset**.
3. Navigate to**Library**, **CA SLCM** folder, Click **SLCM_GlobalDatasheet** dataset.
4. Update Password, Catalog URL (**testqa-sc-new)**, SDM App URL (**testqa-app1-new**) and CA Process Automation (**testqa-pam-new**) URL of the target machines.





5. Access SDM_Globaldataset and Update password, Catalog URL, SDM URL and ITPAM URL of target machines: (*http://testqa-pam-new:8080/itpam*) and access **SDM_Globaldataset**.
6. Navigate to **Library**, CA SDM folder, Click **SDM_GlobalDatasheet** dataset.





Update Password, Catalog URL (**testqa-sc-new**), SDM App URL (**testqa-app1-new**) and CA Process Automation (**testqa-pam-new**) URL

7. Perform the following steps before you start the CA Service Catalog services on **testqa-sc-new**:
   a. If your ITPAM source system (t**estqa-pam**) has customized CA Process Automation content, export all relevant CA Process Automation content from the source CA Process Automation server (**testqa-pam**) and import to the target CA Process Automation server (**testqa-pam-new**). Complete the required configurations for CA Process Automation content.
   b. If your EEM source system (**testqa-eem**) has customized CA EEM policies, export all the relevant CA EEM policies from the source CA EEM server (**testqa-eem**) and import to the target CA EEM server (**testqa-eem-new**).

CA Service Catalog in-flight (waiting state) CA Process Automation Process

This section describes how to regenerate your waiting state CA Service catalog Process Automation workflow type requests from the source server to your target server. In-flight specifies running, waiting, suspended, blocked and so on. This document describes only the "**Waiting State**".

Approval Process

> **NOTE**
> In this example, the use case is for the "**Approval**" process instance in CA Process Automation, also known as Pending Approval request status in CA Service Catalog.

Perform the following steps:

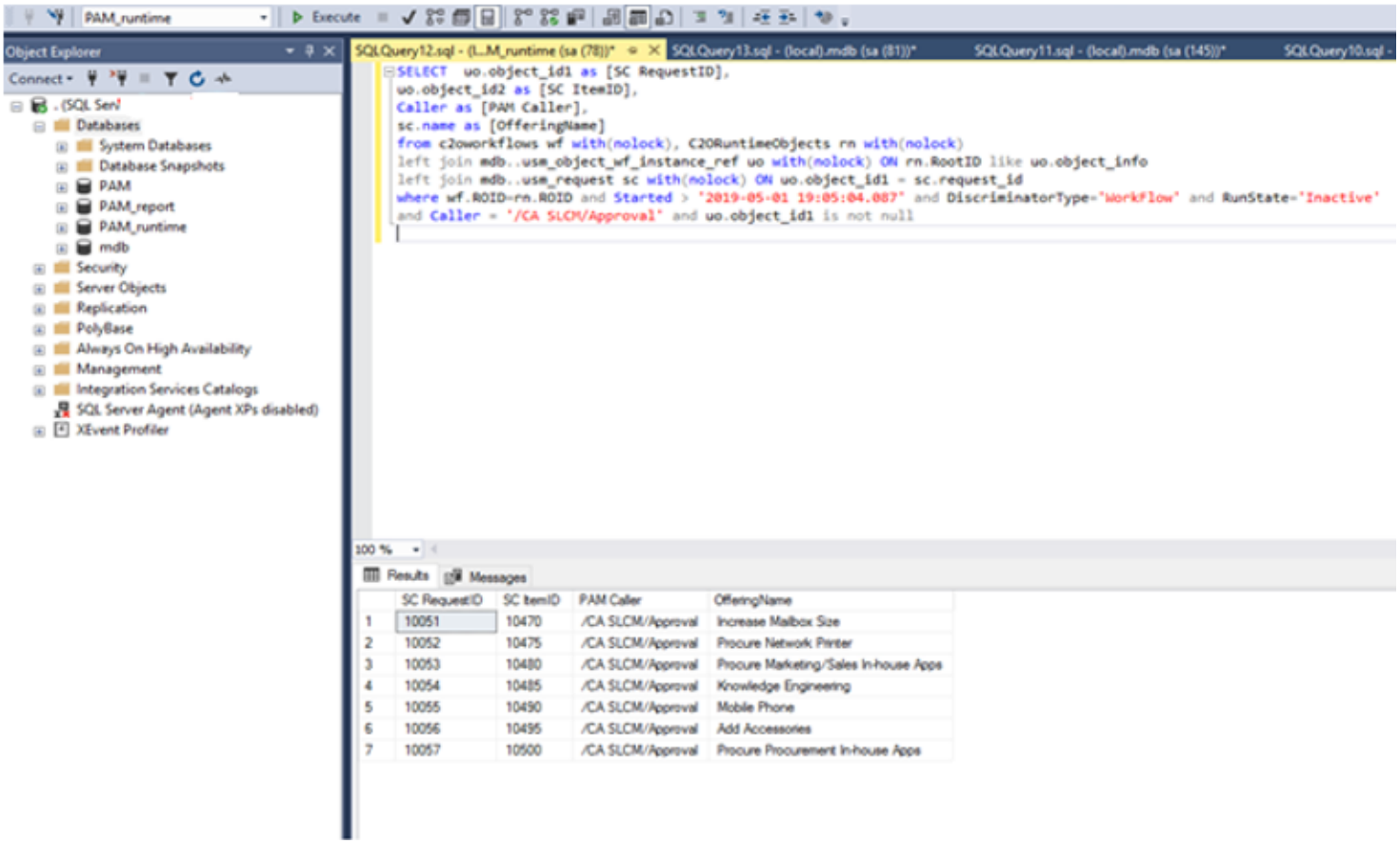
1. Ensure that the Service Catalog request is in **pending** status with the CA Process Automation Workflow instance on your source server (**testqa-mdb**).
2. Run the below database query on **testqa-mdb** to fetch pending CA Service Catalog request status by selecting the **PAM_runtime** database name:

```
DB query:
SELECT     uo.object_id1 as [SC RequestID]
uo.object_id2 as [SC ItemID],
Caller as [PAM Caller],
sc.name as [OfferingName]
from c2oworkflows wf with(nolock), C2ORuntimeObjects rn with(nolock)
left join mdb..usm_object_wf_instance_ref uo with(nolock) ON rn.RootID like uo.object_info
left join mdb..usm_request sc with(nolock) ON uo.object_id1 = sc.request_id
```

```
where wf.ROID=rn.ROID and Started > '2019-05-01 19:05:04.087' and DiscriminatorType='WorkFlow' and
 RunState='Inactive'
and Caller = '/CA SLCM/Approval' and uo.object_id1 is not null
```

NOTE
In the above query, the following parameters can be changed as required:

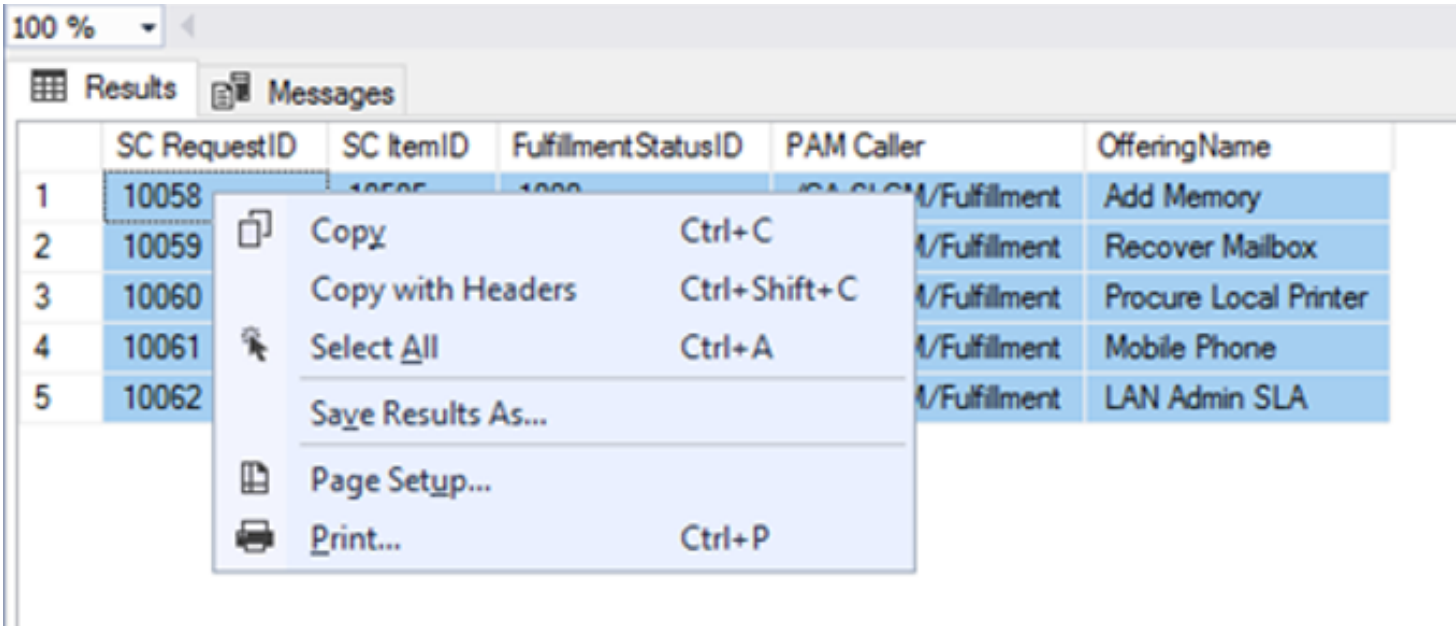
- Change the date and time as required.
  For example:
  Given **Started > '2019-05-01 19:05:04.087'**can be changed as required.
- Caller path value can be changed as per the CA Process Automation workflow process.
  For example:
  OOTB approval process : Given **Caller = '/CA SLCM/Approval'**.
- For the Out-of-the-Box (OOTB) Approval instance, input parameters are: **RequestID**, **ItemID,** and offering name. These input parameters can be changed based on a custom-defined process.



3. Copy the output values from step 2 (see image).
   Create the **catalog_approval.csv file as following:**
   a. Select the results data for the Approval or Fulfillment query and then select option as “**Copy**”.

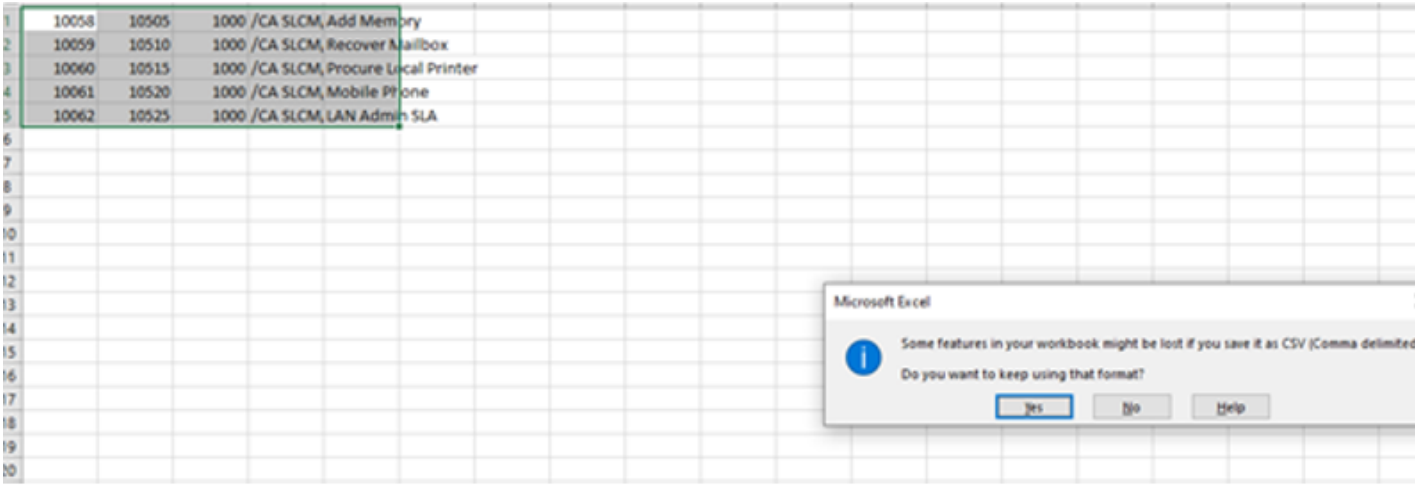
      NOTE
      Ensure that **step 2** is performed on a server where Microsoft Office (Excel) is available.

   b. Open a new excel file and paste the results in the excel file.



c. Click on "**Save as**" excel file and select:
   a. SLCM_CSV_Filepath # E:\\Catalog_Data
   b. SLCM_CSV_FileName # **catalog_approval.csv**
   Click **Save**.

   c. Click **Yes**.

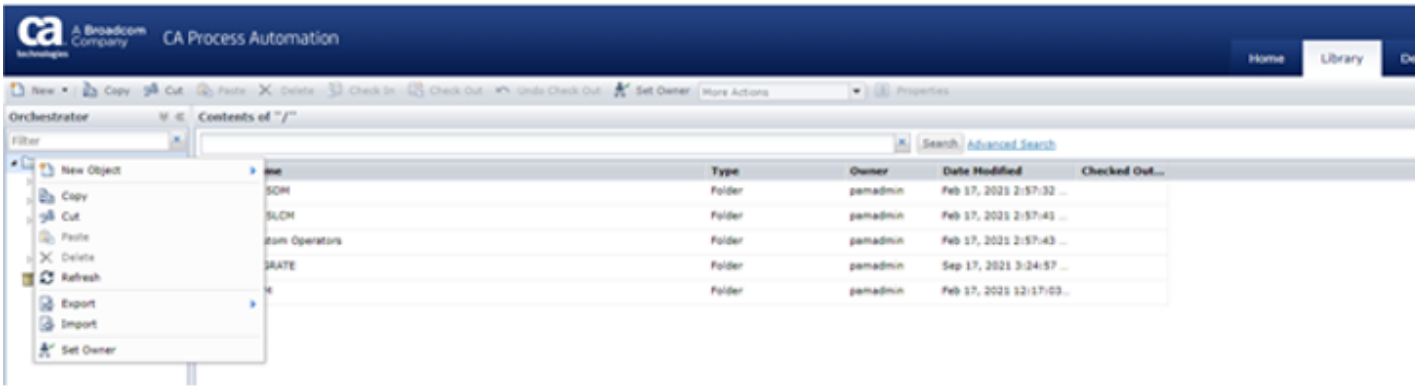


   d. Click **Save**.
   File is generated as a CSV file (**catalog_approval.csv**).
   Ensure this file is copied or available in CA Process Automation Target Server (**testqa-pam-new**).

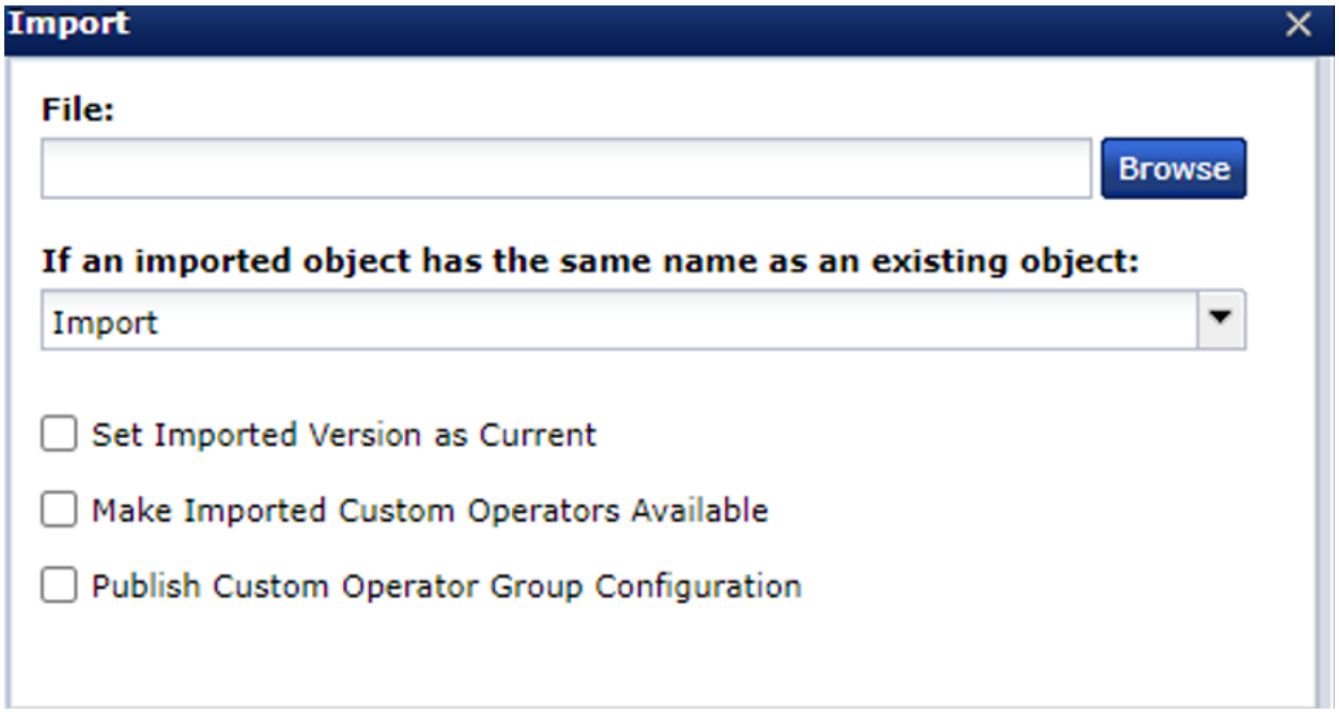
4. Import the **PAM-InflightProcess.xml**.

IMPORTANT
This file is provided with the Run book.

a. Perform the following Steps to import the file:
   a. Login to CA Process Automation target server (**testqa-pam-new**) with Admin privileges.
   Example: **pamadmin**
   b. Navigate to **Home**, **Library**.
   c. Right-click the root folder, select the **Import** option from the menu.

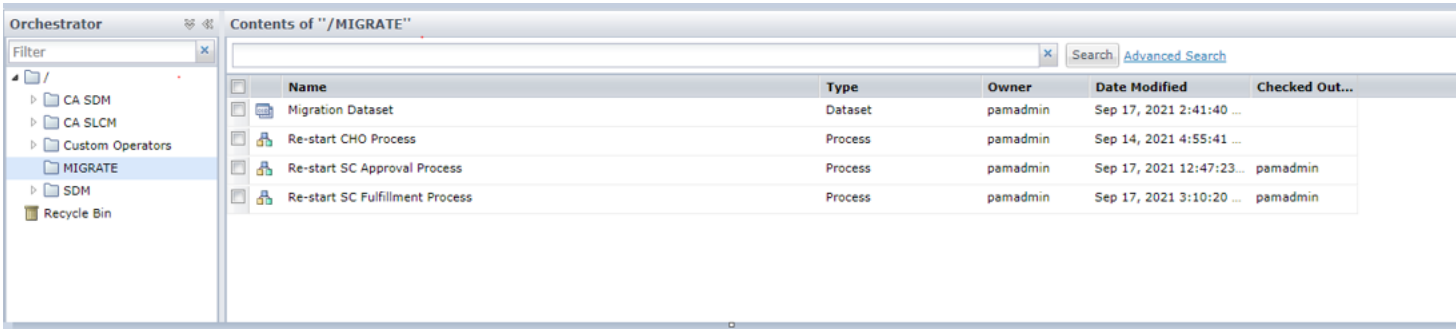


   d. Click **Import**.
   e. Click **Browse** to select the file (from the Import window, see the image in step 4 iii).

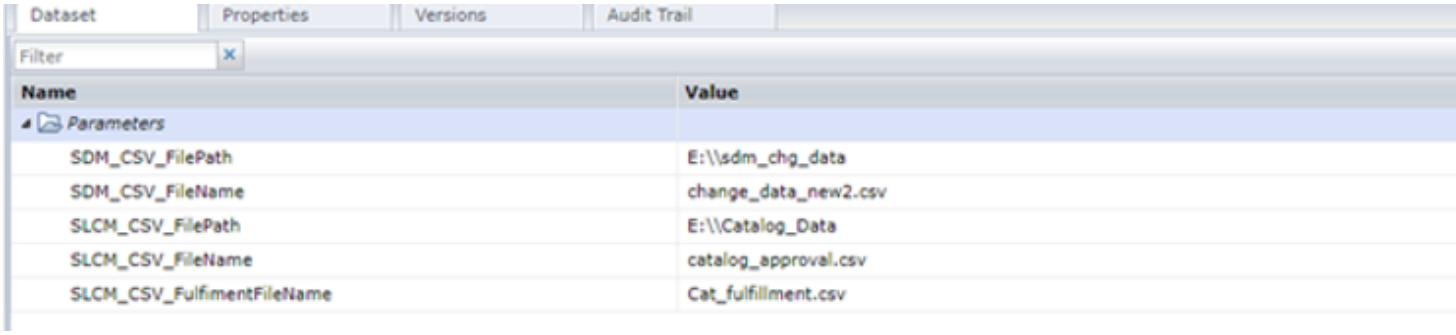


   f. Click **Submit**.

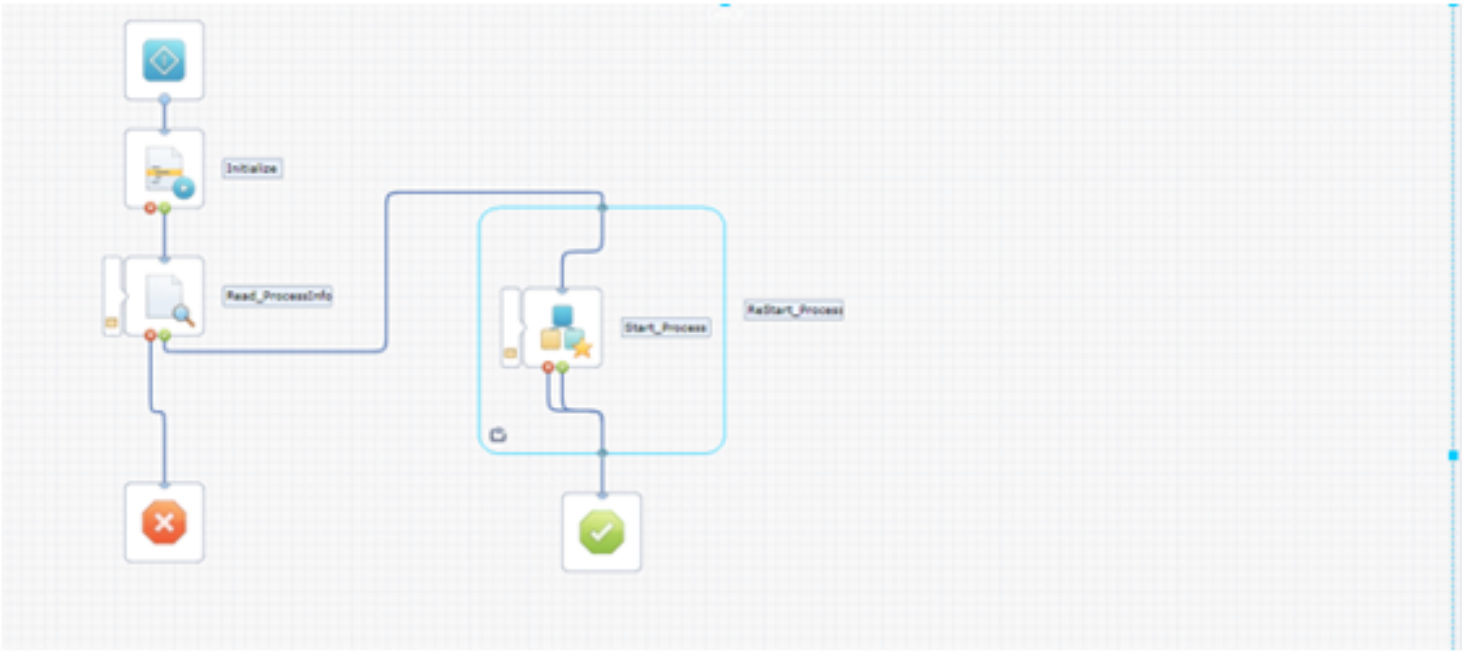
5. After successful import on the Process Automation target server (**testqa-pam-new**), the "**MIGRATE**" folder is added with three workflow processes in Process Automation:
   a. Migration Dataset
   b. Re-start CHO Process
   c. Re-start SC Approval Process
   d. Re-start SC Fulfillment Process



6. Edit the Migration Dataset and then update the below entries:
   a. SLCM_CSV_FilePath # E:\\catalog_data
   b. SLCM_CSV_FileName # **catalog_approval.csv**
   Click **Save**.



7. Edit "**Re-start SC Process**" process and click **Start**.
   It will now create the Process Automation workflow Process instances in "**waiting**" state.



8. Login to CA Service Catalog and verify the catalog request status on the target server (**testqa-sc-new)**. Approve the request to move the request status from one state to another. Example: pending approval to pending fulfillment.

Fulfillment Process

NOTE
In this example, the use case is for the “**Fulfillment**” process instance in CA Process Automation, also known as **Pending Fulfillment** request status in CA Service Catalog.

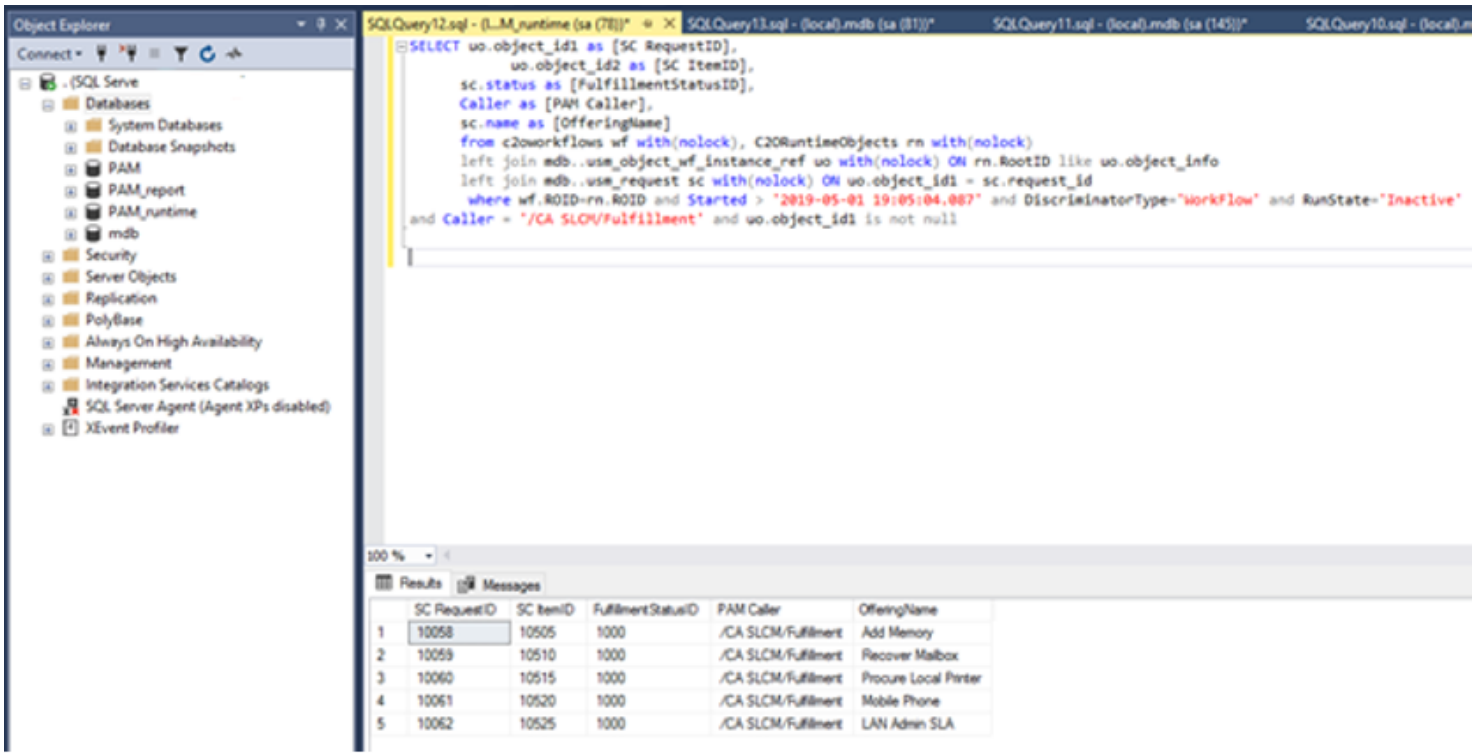
Perform the following steps:

1. Run the below query:

```
  DB Query:
----------------
    SELECT uo.object_id1 as [SC RequestID],
           uo.object_id2 as [SC ItemID],
      sc.status as [FulfillmentStatusID],
      Caller as [PAM Caller],
      sc.name as [OfferingName]
      from c2oworkflows wf with(nolock), C2ORuntimeObjects rn with(nolock)
      left join mdb..usm_object_wf_instance_ref uo with(nolock) ON rn.RootID like uo.object_info
      left join mdb..usm_request sc with(nolock) ON uo.object_id1 = sc.request_id
       where wf.ROID=rn.ROID and Started > '2019-05-01 19:05:04.087' and DiscriminatorType='WorkFlow' and
 RunState='Inactive'
and Caller = '/CA SLCM/Fulfillment' and uo.object_id1 is not null
```

NOTE
In this query, the following parameters can be changed as required.

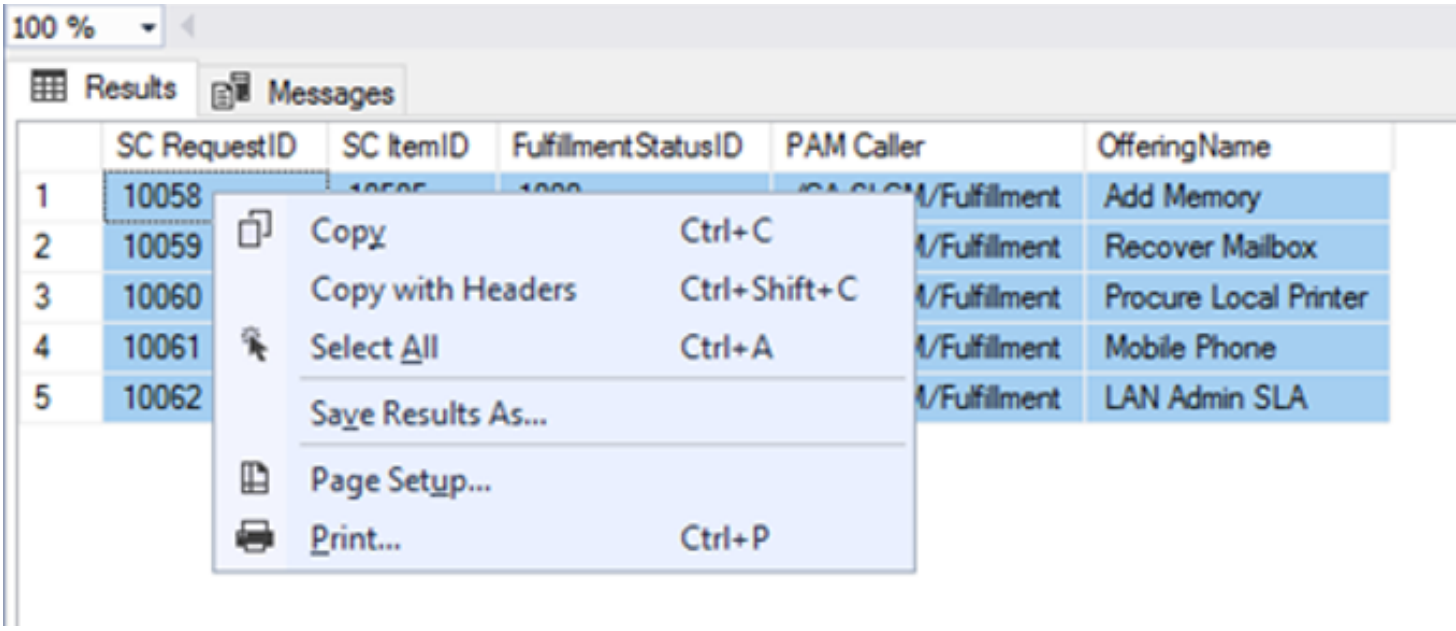
- Change the date and time based on requirement. For example:
  Given Started > '2019-05-01 19:05:04.087.
- Caller path value can be changed based on the PAM process. For example:
  Given Caller = '/CA SLCM/Fulfillment'
- For the Fulfillment instance, input parameters like RequestID, ItemID, FulfillmentStatusID and offering name. These input parameters can be changed based on a custom-defined process.



2. Copy the above output values and create it as **Cat_fulfillment.csv** on the CA Process Automation server target machine. Record the CSV file path. (ex: E:\\Catalog_Data).
   a. Create a **Cat_fulfillment.csv** file as shown below on the CA Process Automation target server (**testqa-pam-new**) and save it. For example: E:\\Catalog_Data
   Create a CSV File:
   b. Select the results data for the Approval or Fulfillment query and then select option as “**Copy**”.

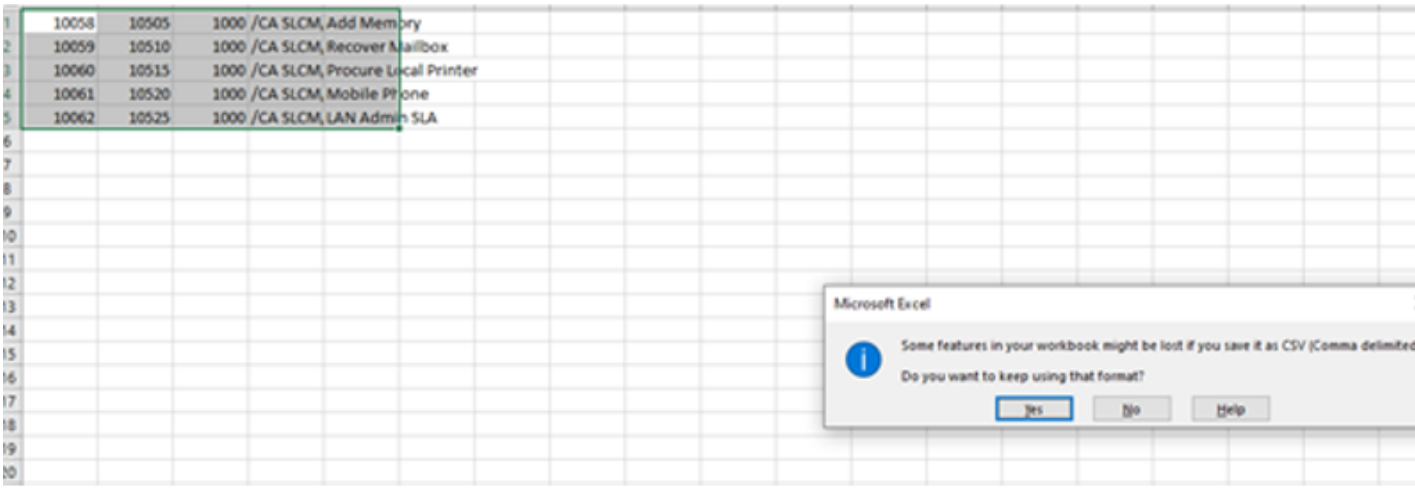
      NOTE
      Ensure that step 2 is performed on a server where Microsoft Office (Excel) is available.

   c. Open a new excel file and paste the results in the excel file.



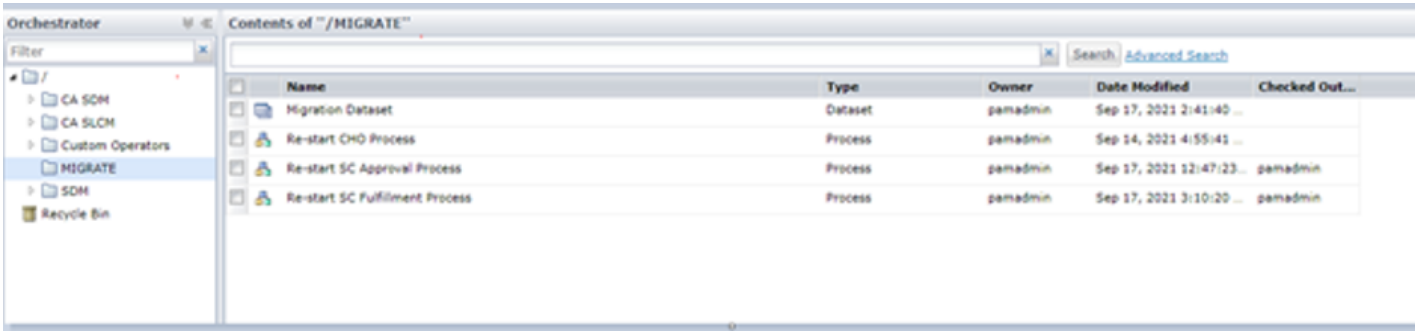
d. Click on "**Save as**" excel file and select:
   a. SLCM_CSV_FilePath # E:\\catalog_data
   b. SLCM_CSV_FulfillmentFileName # **cat_fulfillment.csv**
   c. Save as type " CSV(Comma delimited) (*.csv).
   Click **Save**.

   d. Click **Yes**.

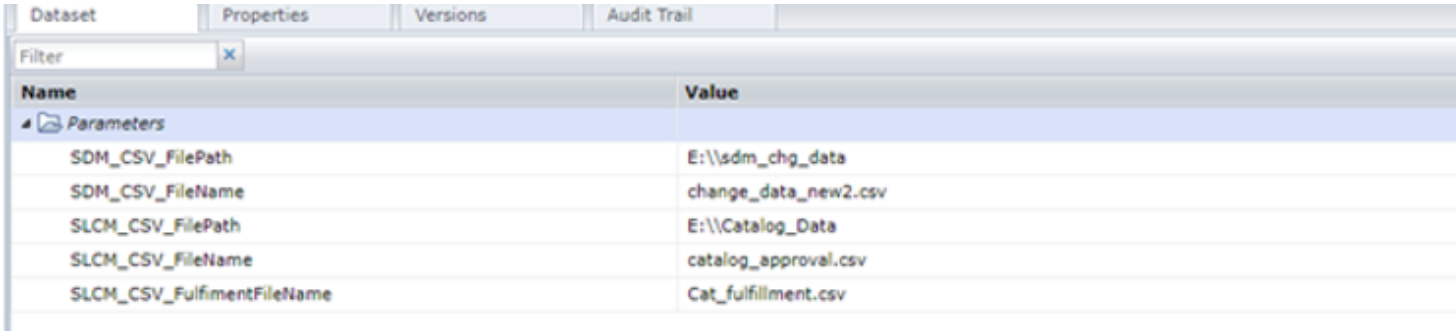


   e. Click **Save**.
   File is generated as a csv file (**catalog_approval.csv**).

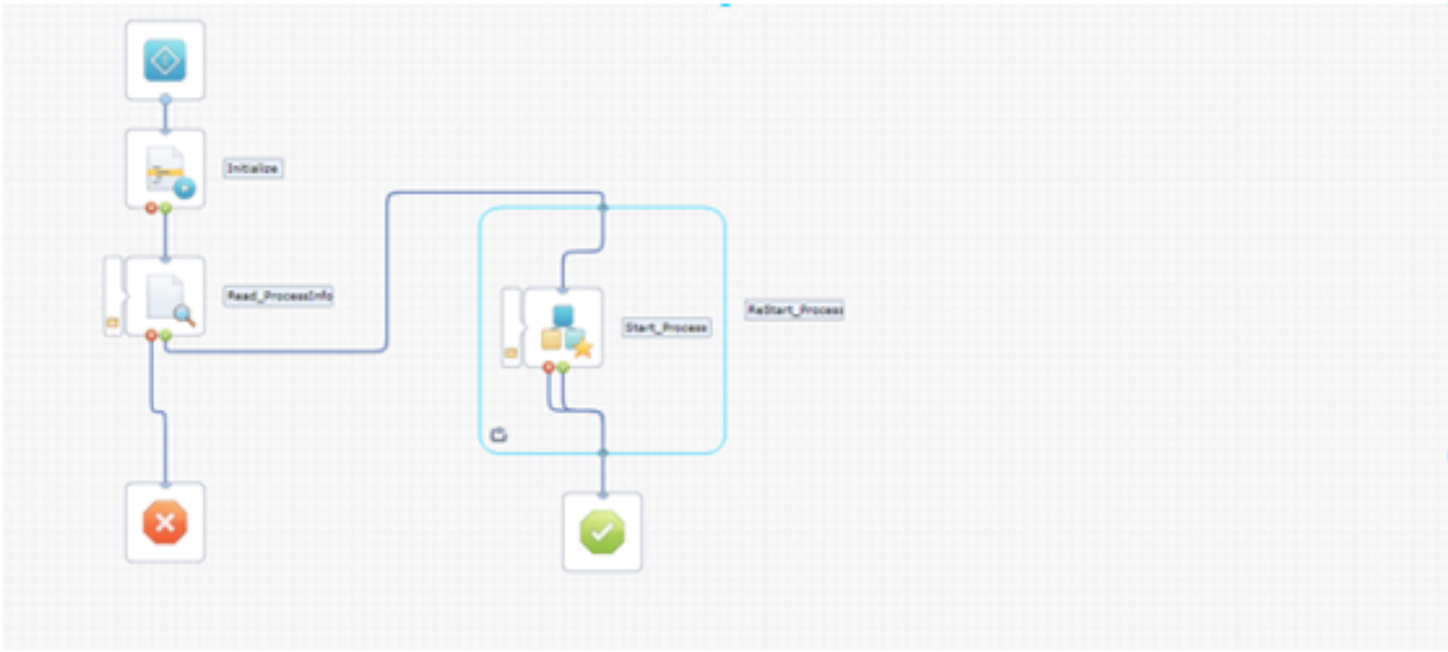
NOTE
Ensure this file is copied or available in CA Process Automation Target Server (**testqa-pam-new**)



3. Edit the Migration Dataset as seen in the above image. Update the below entries. Click **Save**.
   a. SDM_CSV_FilePath # E:\\catalog_data
   b. SLCM_CSV_FulfillmentFileName # **cat_fulfillment.csv**
   Click **Save**.



   c. Edit "**Re-start SC Fulfillment Process**" process and click **Start**.
   It will now create the Process Automation workflow Process instances instances in "**waiting**" state.



4. Login to CA Service Catalog and verify that the catalog request status is pending fulfillment on the target server (**testqa-sc-new**). Fulfill the request to move the request status from one state to another. Example: pending fulfilment to completed.

NOTE
CA Process Automation Process instances names shows as "**unknown**" in the CA service catalog request tracking tab when the instances are moved from Source CA service Catalog server (**testqa-sc**) to Target CA Service Catalog server (**testqa-sc-new**). However, as part of inflight process, after regenerating a new process instance with the same request ID on the CA Process Automation Target server (**testqa-pam-new**), the requests are able to move from one state to another and subsequent process instance names are displayed on the CA Service Catalog Request Tracking tab.

CA SDM In-flight CA Process Automation Process for Change Order

This section describes how to regenerate change orders with "waiting" state of theProcess Automation workflow of CA Service Desk Manager (SDM), from a source server (testqa-pam) to a target server (**testqa-pam-new**) to a target server (**testqa-pam-new**). In-flight here specifies running, waiting, suspended, blocked and so on. This document describes only the “**Waiting State**”.

Change Order Process

NOTE
In this example, the use case is for “Order PC” (Out-of-the box) process instance in CA Process Automation, also known as Change Order in CA SDM.

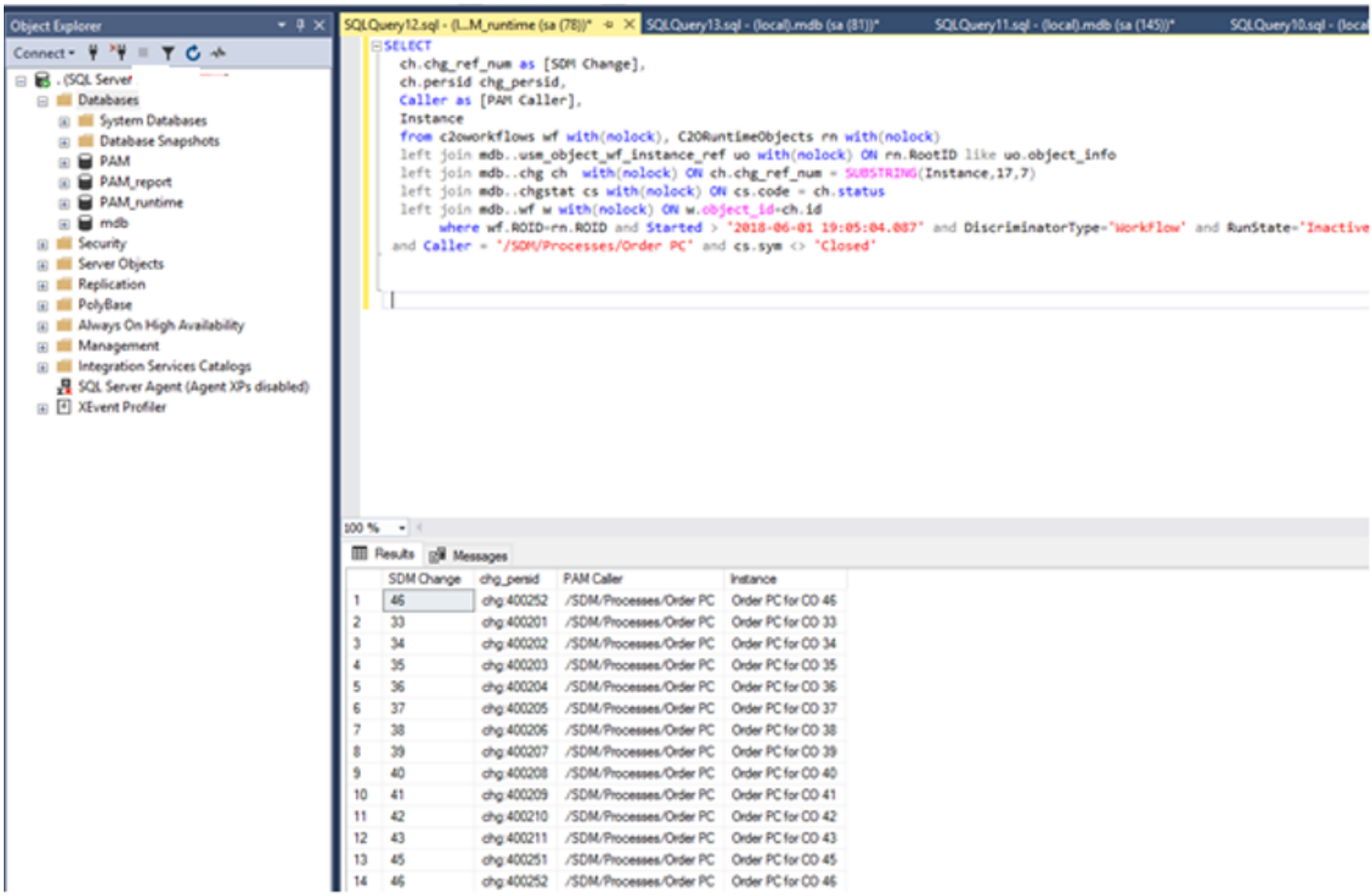
Perform the following steps:

1. Ensure that you have few Change Orders with categories in the Process Automation Workflow (**Add.IT.Other**) on any of the source SDM application servers (**testqa-app1**).
2. Run the below database query on the source database server (**testqa-mdb**) to get the CA Process Automation workflow process list for change order by selecting the **PAM_runtime** database name.

```
 SELECT
 ch.chg_ref_num as [SDM Change],
 ch.persid chg_persid,
 Caller as [PAM Caller],
 Instance
 from c2oworkflows wf with(nolock), C2ORuntimeObjects rn with(nolock)
 left join mdb..usm_object_wf_instance_ref uo with(nolock) ON rn.RootID like uo.object_info
 left join mdb..chg ch  with(nolock) ON ch.chg_ref_num = SUBSTRING(Instance,17,7)
 left join mdb..chgstat cs with(nolock) ON cs.code = ch.status
 left join mdb..wf w with(nolock) ON w.object_id=ch.id
      where wf.ROID=rn.ROID and Started > '2018-06-01 19:05:04.087' and DiscriminatorType='WorkFlow' and
RunState='Inactive'
and Caller = '/SDM/Processes/Order PC' and cs.sym <> 'Closed'
```

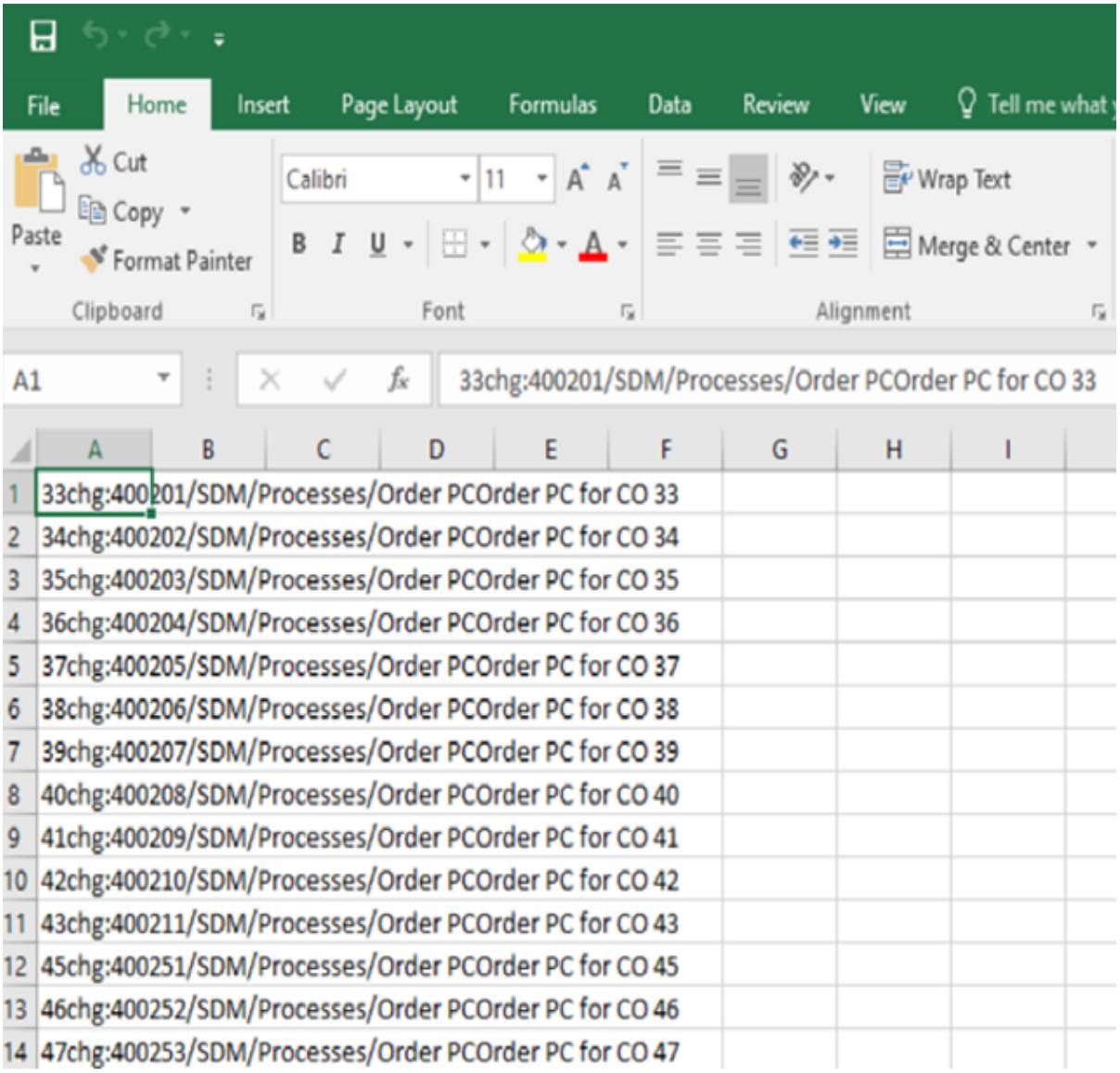
NOTE
In this query, the following parameters can be changed as required:

- Change the date and time based on requirement. For example:
  Given Started > '2018-06-01 19:05:04.087
  This can be changed based on the requirement.
- Caller path value can be changed based on the PAM process. For example:
  Given Caller = '/SDM/Processes/Order PC'
- For the OOTB Order PC instance, input parameters are: **chg_persid** and**chg_ref_num**. These input parameters can be changed based on a custom-defined process.

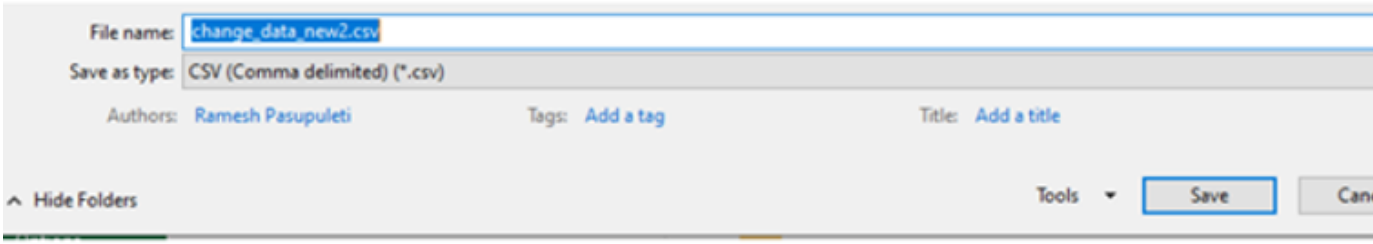


3. Copy the above output values and create it as **change_data_new2.csv** format on the target Process Automation server (**testqa-pam-new**). Record the csv file path
   (For example: E:\\s**dm_chg_data**)
   Example of CSV File
   .
   – 33,chg:400201,/SDM/Processes/Order PC,Order PC for CO 33
   – 34,chg:400202,/SDM/Processes/Order PC,Order PC for CO 34
   – 35,chg:400203,/SDM/Processes/Order PC,Order PC for CO 35
4. Copy the above output values as shown in the image above.
   a. Create a **change_data_new2.csv** file as shown below on the CA Process Automation target server (testqa-pam-new) and save it. For example: \ex: E:\\sdm_chg_data
      Create a CSV File as following:
      a. Select the results data for the change order query and then select option as “**Copy**”.

         NOTE
         Ensure that the below step is performed on a server where Microsoft Office (Excel) is available.
      b. Open a new excel file and paste the results in the excel file.



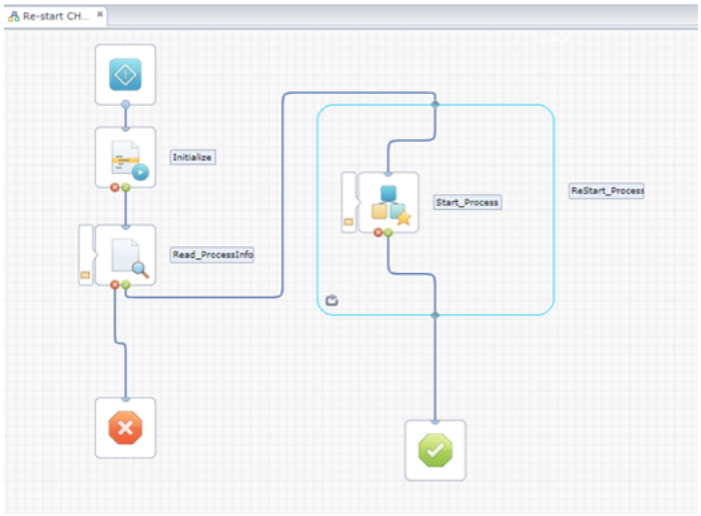
c. Click “**Save as**” excel file and select: For example: Filename as “**change_data_new2.csv**”
   a. Save as type “ CSV(Comma delimited) (*.csv)
   b. Save file.



   c. Click **Yes**.
   d. Click**Save**.
   File is generated as a csv file (**change_data_new2.csv**).

NOTE
Ensure that this file is copied or available in the CA Process Automation Target Server (testqa-pam-new).

5. Edit the Migration Dataset. Update the below entries.
   Click **Save**.
   a. SDM_CSV_FilePath # E:\\sdm_chg_data
   b. SDM_CSV_FileName # **change_data_new2.csv**
6. Edit "**Re-start CHO Process**" process and click **Start**.



Process Automation workflow Process instances are in the "**waiting**" state.

7. Update the list of **Instance_Id** in the table **usp_caextwf_instances** of MDB database with the latest ROID in the table C2ORuntimeObjects of **PAM_runtime** database on the target database server (**testqa-mdb-new)**.

```
Update uci
 set uci.instance_id=rn.ROID
  from c2oworkflows wf with(nolock), C2ORuntimeObjects rn with(nolock)
  left join mdb..usm_object_wf_instance_ref uo with(nolock) ON rn.RootID like uo.object_info
  left join mdb..chg ch  with(nolock) ON ch.chg_ref_num = SUBSTRING(Instance,17,7)
  left join mdb..usp_caextwf_instances uci with(nolock) ON uci.object_persid = ch.persid
  left join mdb..chgstat cs with(nolock) ON cs.code = ch.status
  left join mdb..wf w with(nolock) ON w.object_id=ch.id
       where wf.ROID=rn.ROID and Started > '2018-06-01 19:05:04.087' and DiscriminatorType='WorkFlow' and
 RunState='Inactive'
 and Caller = '/SDM/Processes/Order PC' and cs.sym <> 'Closed'
```

NOTE
In the above query, the following parameters can be changed as required. Change the date and time as required. Example: Given Started > '2018-06-01 19:05:04.087' can change based on requirement.
Caller path value can be changed based on the PAM process: Ex: Given Caller = '/SDM/Processes/Order PC'

8. Restart SDM services on all the target servers (**testqa-bg-new**, **testqa-sb-new**, **testqa-app1-new**, and**testqa-app2-new**).
9. Verify the CA SDM Change Order status.
   By default, the status is Open and it will be changed to another status based on the actions taken on the Process Automation target server (**testqa-pam-new**).

CA EEM Policies Backup and Restore on CA Service Catalog server

Perform the following steps to backup CA EEM data on source Catalog server (**testqa-sc**):

1. Open the CA Service Catalog command prompt by clicking Start, Programs, CA, Service Catalog, usm Command Prompt (that is, *%usm_home%\usm.cmd*) from the Catalog 17.1 installed source server **testqa-sc**.
2. Run the following command:

   ```
   ant backup-eem-app
   ```

   – During execution, provide input parameters like EEM password and so on.
   – After successful completion, "eem-backup.xml" file is generated in the *%USM_HOME%* directory.

   NOTE
   If you have multiple CA Service Catalog computers, perform this step on the first CA Service Catalog computer. Run this command once, regardless of how many CA Service Catalog system your environment may have.

   – Perform the following steps to restore CA EEM on the target Catalog server (**testqa-sc-new**):
   a. Copy the file (**eem-backup.xml**) from testqa-sc to the testqa-sc-new in the %USM_HOME% folder.
   b. Run the following command:

      ```
      ant restore-eem-app
      ```

   c. After successful completion related entries are updated in CA EEM.
   d. Restart CA Service Catalog services.

CA Service Catalog Customization

Perform the following steps:

1. Copy filestore, custom directory from the source CA Service Catalog server (**testqa-sc**) to the target CA Service Catalog server (**testqa-sc-new**).

   NOTE
   This is applicable only if customization is carried out in the application environment. This is not applicable for the OOTB content.
2. In the Catalog customizations, we have only one file, that is, **requestprofile.xsl** for customization in this environment.
3. Start the following CA Service Catalog services:
   a. CA Service Accounting
   b. CA Service Catalog

After completing the above, CA Service Management customizations are successfully moved.

Upgrade CA Service Management 17.1 to CA Service Management 17.4

After successful installation of CA Service Management 17.1, upgrade it to CA Service Management 17.3 first and then update it to CA Service Management 17.3.0.2, and upgrade to CA Service Management 17.4:

Prerequisites

1. Shutdown all CA Service Desk and Service Catalog services.
2. CA EEM Services must be up and running on **testqa-eem-new**.
3. CA ITPAM application services must be up and running on **testqa-pam-new**.
4. Download “Joint JS”.
   For more information about download and installation instructions, see JointJs.

   NOTE
   Take a backup of your customizations before upgrading from CA Service Management 17.3.0.2 to 17.4.

Upgrade from CA Service Management 17.1 to CA Service Management 17.3

Perform the following steps:

1. Start upgrade to CA SDM 17.3, then update to 17.3.0.2 in a sequence.
   a. Start the upgrade on the BG server (**testqa-bg-new**)
      Ensure that BG Server is up and running after successful installation.
   b. Start upgrade on the Standby Server (**testqa-sb-new**).
   c. Start upgrade of **Appserver1**, **Appserver2**and install **Appserver3**:
      a. Appserver1 (**testqa-app1-new**):
         Components: SDM Server with Visualizer
      b. Appserver2 (**testqa-app2-new**):
         Components: Only SDM Server
      c. Appserver3 (**testqa-app3-new**) (Newly added server as part of CA Service Management 17.3 and 17.3.0.2 installation): Components: SDM Server, xFlow Interface/Service Point

         NOTE
         For more information, see Installing CA Service Desk Manager.
2. Start install Search Server on **testqa-es-new**and during installation of search server, follow the steps to configure search server.
3. Start installing Virtual Analyst on **testqa-rasa-new**.
   For more information, see Install and Configure Virtual Analyst.
   a. Train the RASA NLU Sever.
      For more information, see How to train the NLU Server Train the NLU Server.
4. Configure Virtual Analyst in Service Desk Manager. For more information, see Train the NLU Server to configure Virtual Analyst.
5. Start the upgrade process on CA Service Catalog Server (**testqa-sc-new**).
   a. For more information, see Install CA Service Catalog .
   b. Perform CA Service Catalog Post-installation steps.

      NOTE
      Make sure to merge all customizations with CA Service Management 17.3, and 17.3.0.2.

      NOTE
      Ensure that all CA SDM and Service Catalog application Services are up and running after successful Upgrade with configurations and customizations.
6. Now, apply CA Service Management 17.4. Follow the below installation process sequence:

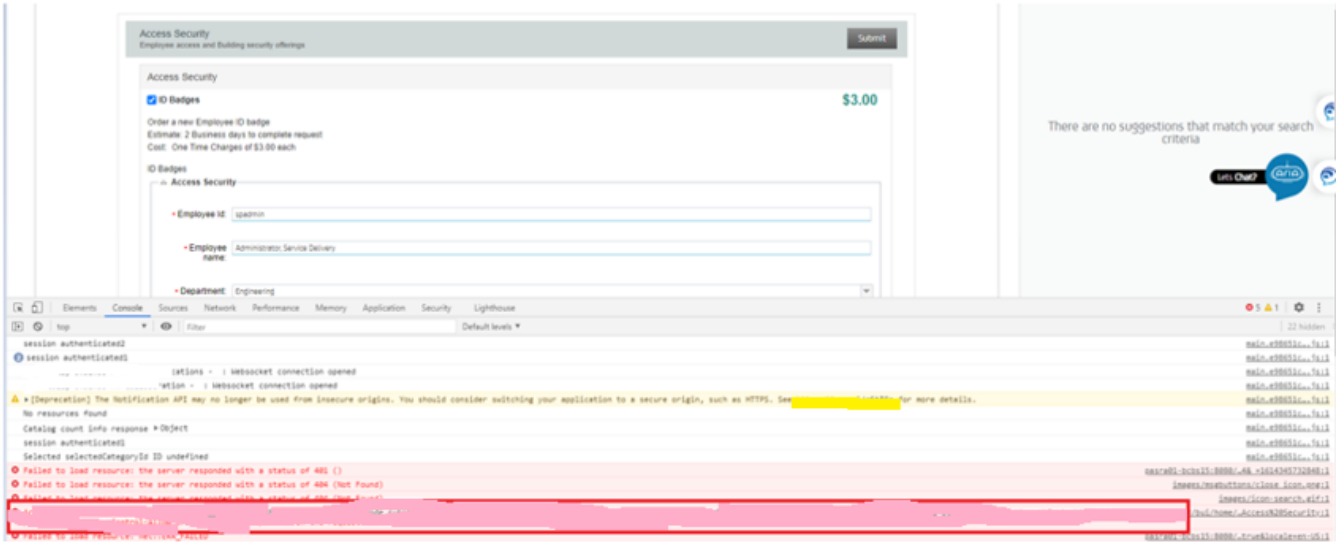
   NOTE
   Stop all CA SDM and Service Catalog Servers.
   a. Start updating CA Service Management 17.4 on BG Server (t**estqa-bg-new**).
      Ensure that BG server is up and running after successful installation.
   b. Start updates to CA Service Management 17.4 on Standby Server (**testqa-sb-new**).
7. Start updates to CA Service Management 17.4 on Appserver1, Appserver2 and Appserver3:
   a. Appserver1 (**testqa-app1-new)**:
      Components: SDM Server with Visualizer
   b. Appserver2 (**testqa-app2-new**)
      Components: Only SDM Server
   c. Appserver3 (**testqa-app3-new**)
      Components: SDM Server, xFlow Interface/Service Point
8. Start Update of CA Service Management 17.4 on Search Server (**testqa-es-new**).
9. Start update of CA Service Management 17.4 on Catalog Server (**testqa-sc-new**).
   For more information, see CA Service Catalog Post-installation steps.

   NOTE
   Ensure to merge all customizations with CA Service Management 17.4

10. After successful configuration, ensure that all applications are up and running.
11. Enable/Configure Active directory in CA EEM.
12. Ensure CA SDM, xFlow Interface, and Catalog server functionality is working fine with the Active directory integration.

Troubleshooting

Problem: Unable to submit CA Service Catalog request from service point as a result getting the following error message on Chrome browser Version V80 + only.

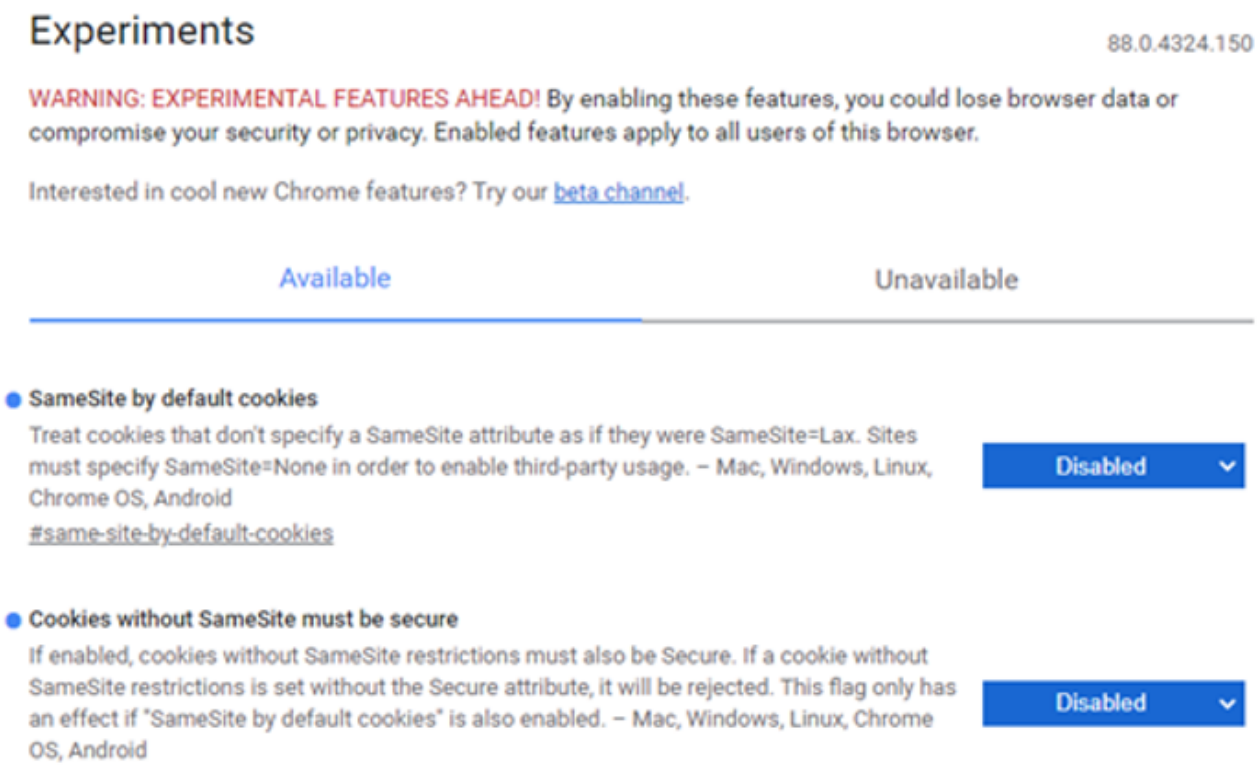


Solution: Chrome provides the following flags to disable this constraint:

- SameSite by default cookies.
- Cookies without SameSite must be secure.

Perform the following steps to disable the Chrome browser flags:

1. Open Chrome Browser.
2. Enter Chrome://flags.
3. Search with above flags and disable.



Use Case 2 – CA Service Management 17.4 Swing Box Move from the Production to the Target Server

This article describes the Swing Box Move procedure for CA Service Desk Manager (CA SDM) 17.4 from a Production to a Target (Test) Server with enhanced system configurations by adding CA Service Catalog also into the target server.

Server Terminology
In this example use case, the following terminology is used for the Production and Target servers:

- Production MDB database host name: **testqa-mdb**
- Target MDB database host name: **testqa-mdb-new**

Implement the Swing-Box method in your environment to move CA MDB data from the Production server (**testqa-mdb**) to a target server (**testqa-mdb-new**).

Production System (CA Service Management 17.4)	Target System (CA Service Management 17.4)
Remote MDB: **testqa-mdb**	Remote MDB: **testqa-mdb-new**
CA Service Desk Manager: **Advanced Availability with Multi-Tenancy and NTLM Authentication**	CA Service Desk Manager: **Advanced Availability with Multi-Tenancy**
Background Server (BG Server): **testqa-bg**	Background Server (BG Server): **testqa-bg-new**
Standby Server: **testqa-standby**	Standby Server: **testqa-standby-new**
Application Server 1: **testqa-app1**	Application Server 1: **testqa-app1-new**
Standby Server: **testqa-sb**	Standby Server: **testqa-sb-new**

Application Server 1: **testqa-app1**	Application Server 1: **testqa-app1-new** CA EEM Server: **testqa-eem-new**
CA Service Catalog Server: **testqa-sc**	CA Service Catalog Cluster with Automic Integration and F5 Load Balancer Configured: • testqa-catalog-new • testqa-catalog-new2 Reconfigure NTLM for CA Service Management (CA SDM, xFlow Interface, and CA Service Catalog)

Perform the following steps:

1. Maintain the similar number of Windows Servers in your Production as well as your target environment.
2. Install CA Service Desk Manager (SDM) 17.4 on Windows 2016/ 2019/ 2022 Server pointing to Microsoft SQL 2019/ 2022 Database Server in your target environment.

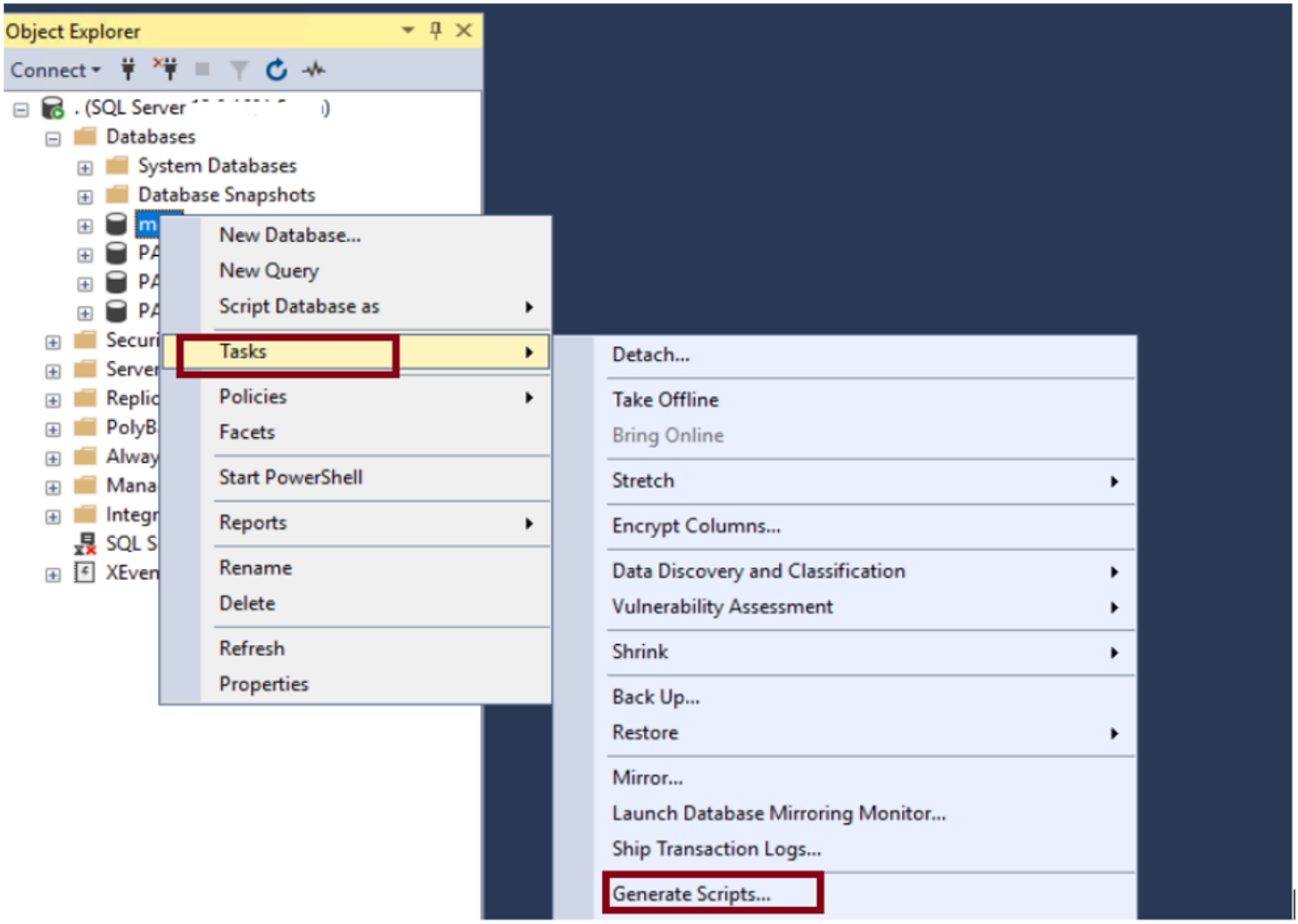
   NOTE
   Download the CA Service Desk Manager 17.4 installer DVD from Broadcom Support.

   - CA SERVICE MANAGEMENT SDM PACKAGE 17.4 FOR WINDOWS (WITH CATALOG).iso
   - CA SERVICE DESK MANAGER 17.4 FOR WINDOWS (WITHOUT CATALOG).iso

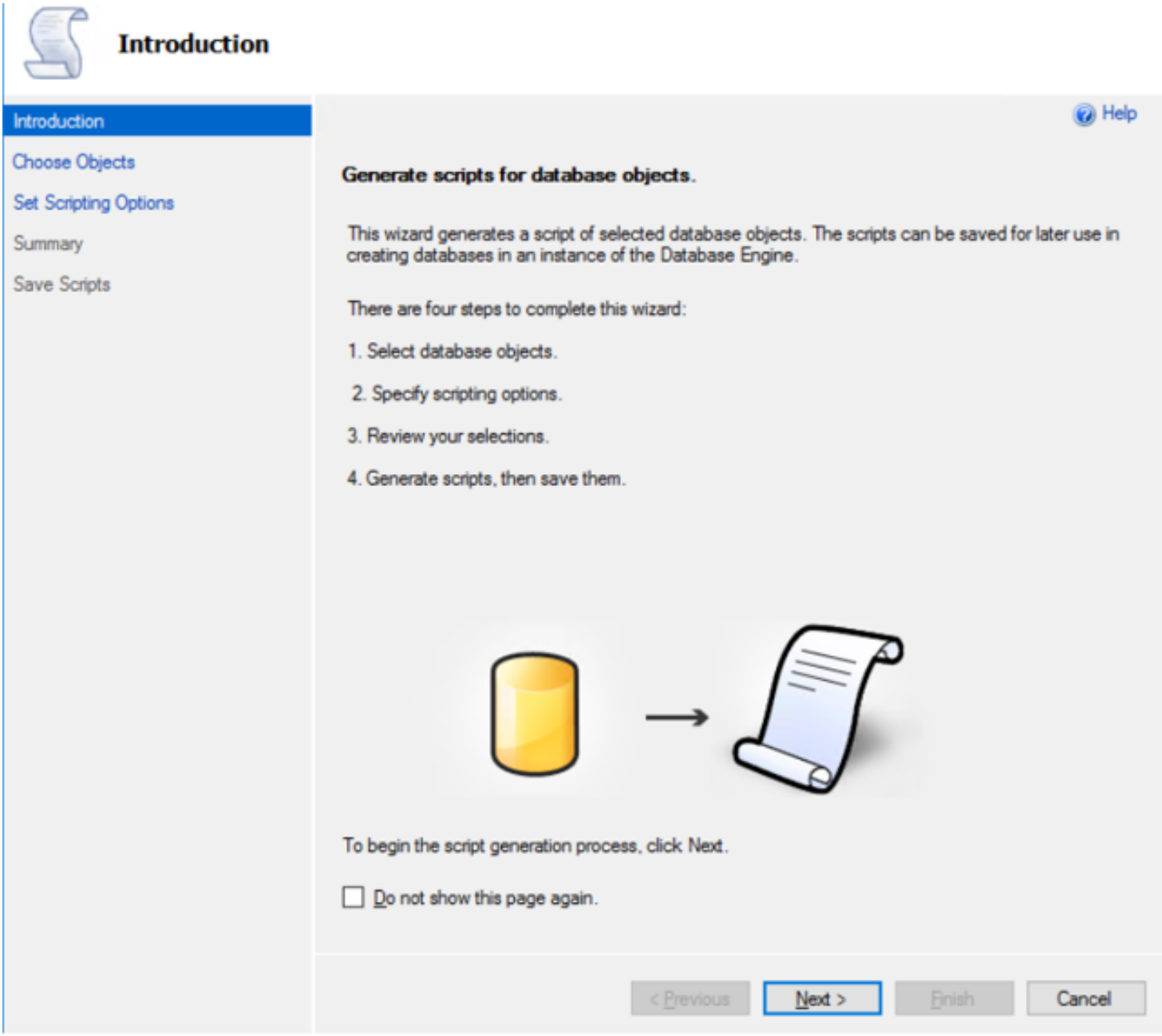
   Refer Installing CA Service Desk Manager and Installing CA Service Catalog.
3. After successful completion of step 1 and 2, enable multi-tenancy. Tenant name must be your production tenant name. Example: Tenant1
4. Review the following CA Service Desk Manager Tables on your test server (testqa-mdb-new) to ensure that these tables have system-specific configurations.

Common Table	CA SDM
al_cdb_comp_installstate_bkup	ci_mdr_provider
al_cdb_componentinstallstate	options
al_cdb_config_params_backup	sapolicy
al_cdb_configurationparameters	usp_servers
al_cdb_files	object_promotion
ca_application_registration	promo_hist
	usp_configuration
	usp_webeng_domsrvr
	usp_webengine
	usp_webeng_alias
	usp_domsrvr

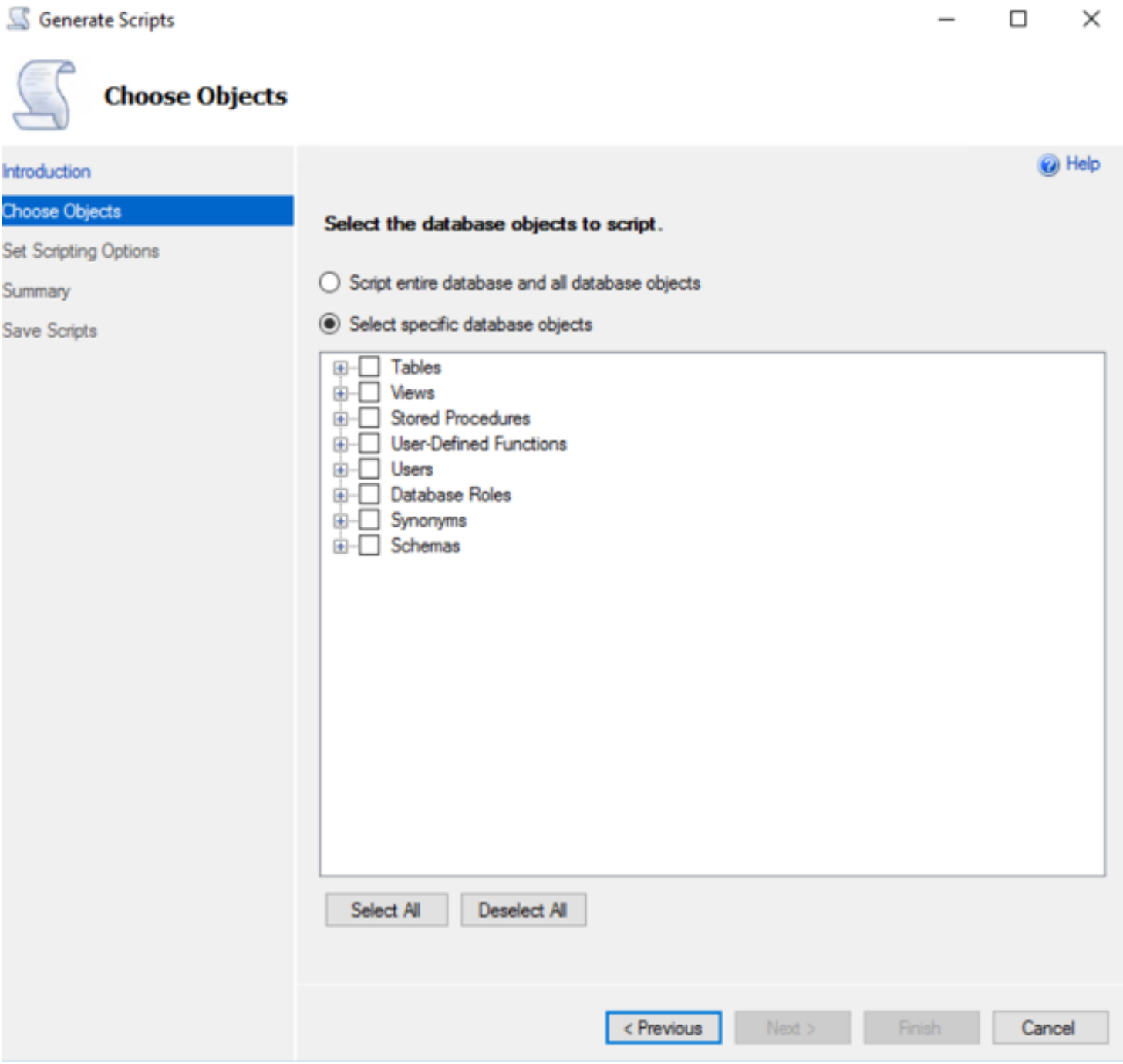
5. Stop the following CA Service Desk Manager services on both the Production and target servers:
   CA SDM: CA Service Desk Manager Servers (Background, Standby, and Application Servers (xFlow application).
6. On the Production system (**testqa-mdb**), backup your MDB database. Copy the MDB backup files to the target system (**testqa-mdb-new**).
7. On the target system (**testqa-mdb-new**), perform the following steps to backup CA Service Desk Manager Tables (shown in Step 4).
8. Navigate to Microsoft SQL Management Studio, right-click MDB (or the appropriate MDB instance name applicable for your system), and select **Tasks**, **Generate Scripts**.


Object Explorer
Connect
. (SQL Server
Databases
System Databases
Database Snapshots
New Database...
New Query
Script Database as
Tasks
Policies
Facets
Start PowerShell
Reports
Rename
Delete
Refresh
Properties
Detach...
Take Offline
Bring Online
Stretch
Encrypt Columns...
Data Discovery and Classification
Vulnerability Assessment
Shrink
Back Up...
Restore
Mirror...
Launch Database Mirroring Monitor...
Ship Transaction Logs...
Generate Scripts...

a. Click **Next**.



b. Select Specific database objects.



c. Take a backup of the CA Service Desk Manager Tables (see step d below) and then, generate the scripts.

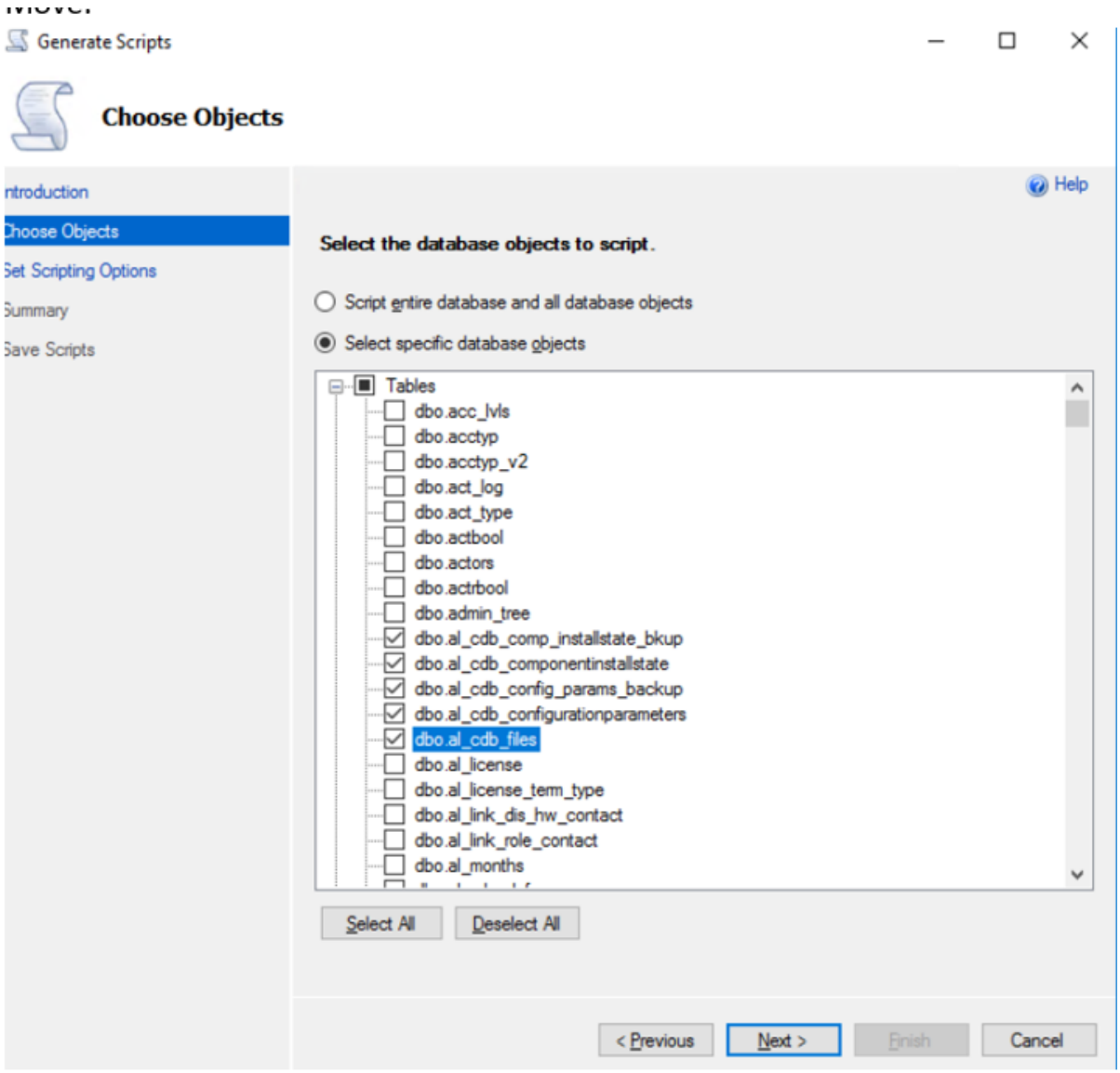
d. Select the list of tables from CA Service Desk Manager Tables:

Common Table	CA SDM	xFlow Interface
al_cdb_comp_installstate_bkup	ci_mdr_provider	usp_es_servers
al_cdb_componentinstallstate	object_promotion	l1admin_general_config
al_cdb_config_params_backup	>options	
al_cdb_configurationparameters	promo_hist	
al_cdb_files	sapolicy	
ca_application_registration	usp_servers	
	usp_configuration	
	>usp_domsrvr	
	usp_webeng_alias	
	usp_webeng_domsrvr	
	usp_webengine	

e. Select the Tables option and choose the appropriate tables. Select list of tables from above CA Service Desk Manager Tables.

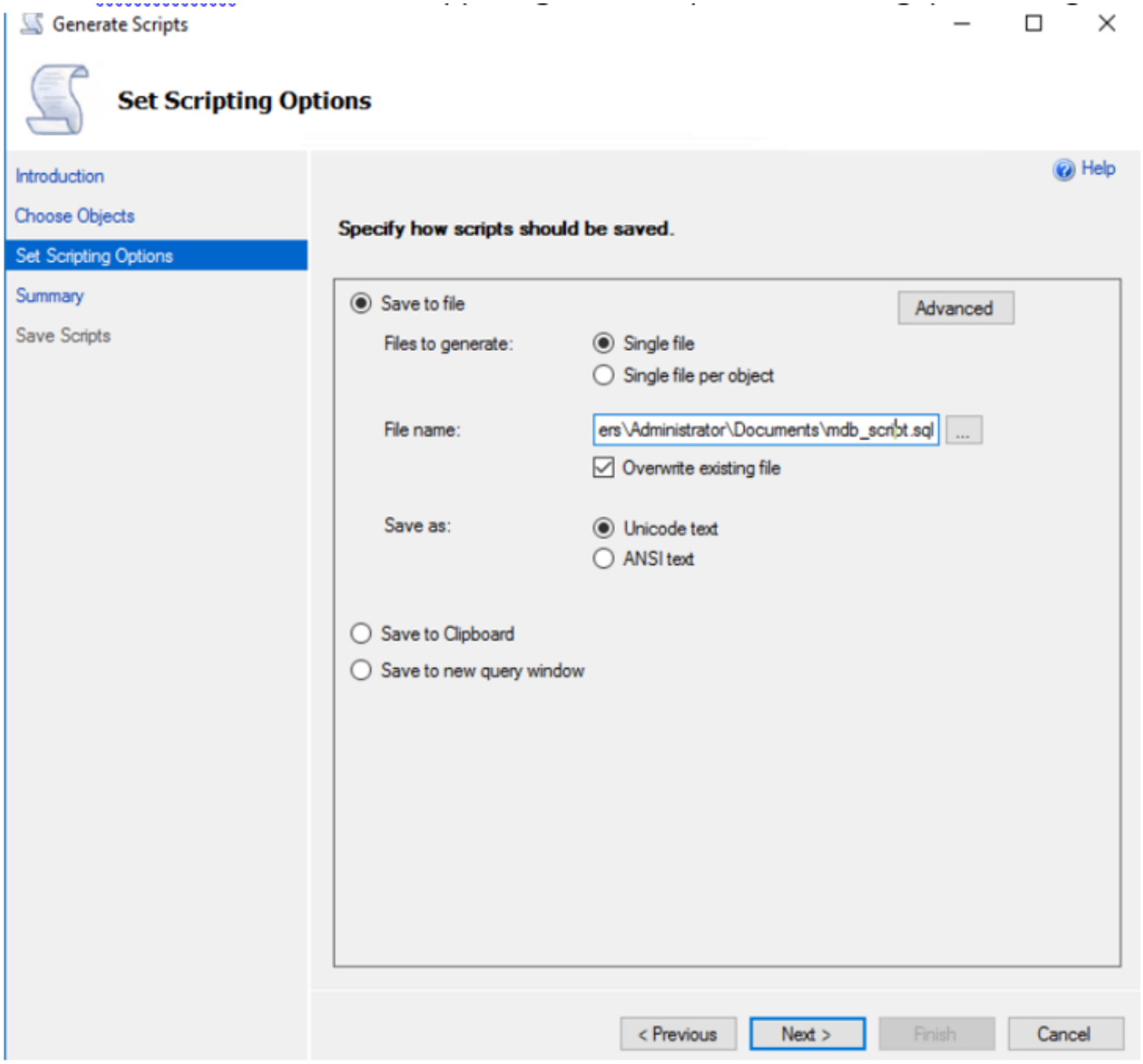
NOTE

The following screenshot shows few selected tables but ensure to select all the**19** tables (mentioned in **step d**) to continue with the Swing Box Move.



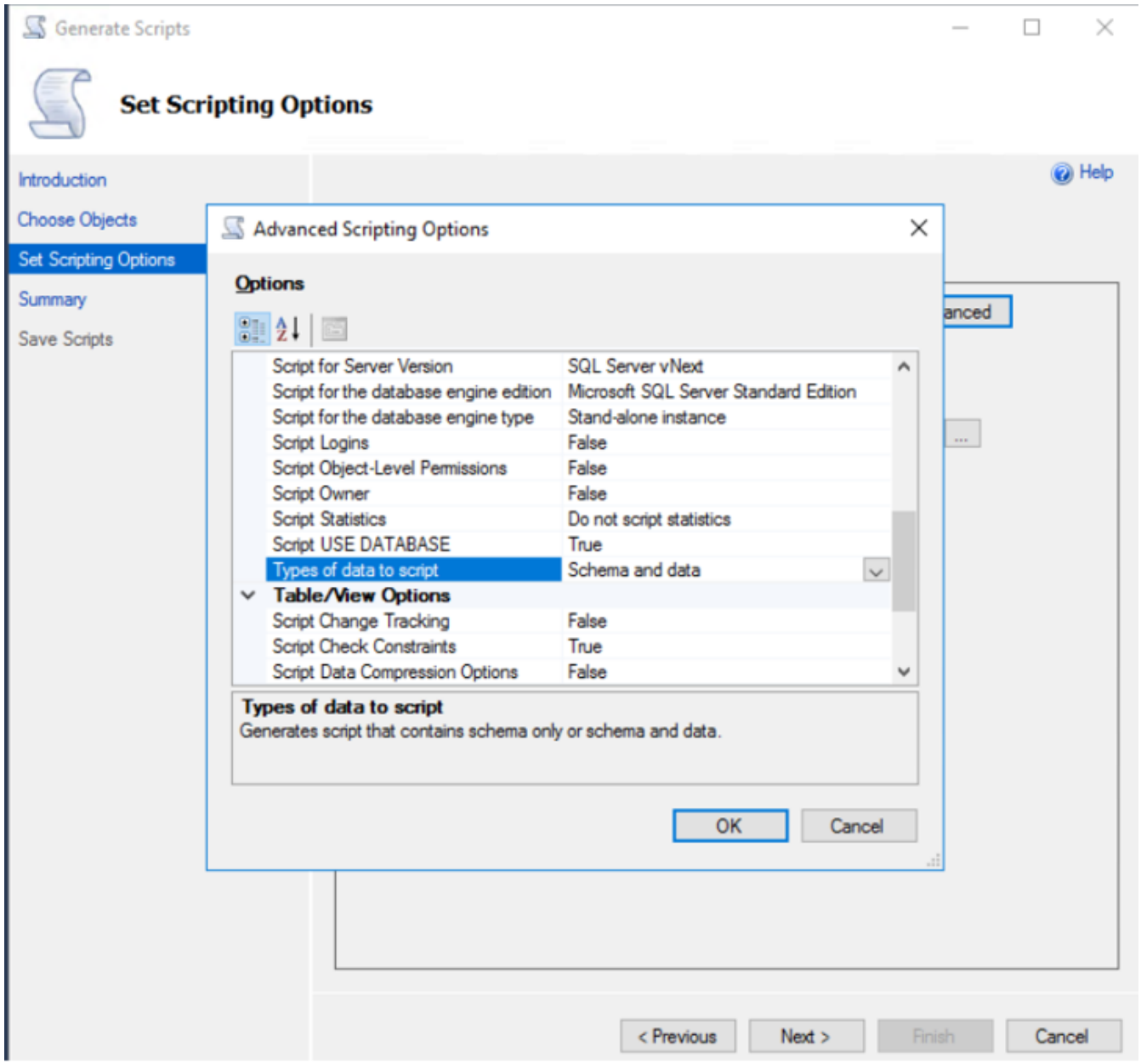
Click **Next**.

f. Select the **Advanced** tab from the upper right corner (see below image) and navigate to Types of Data.

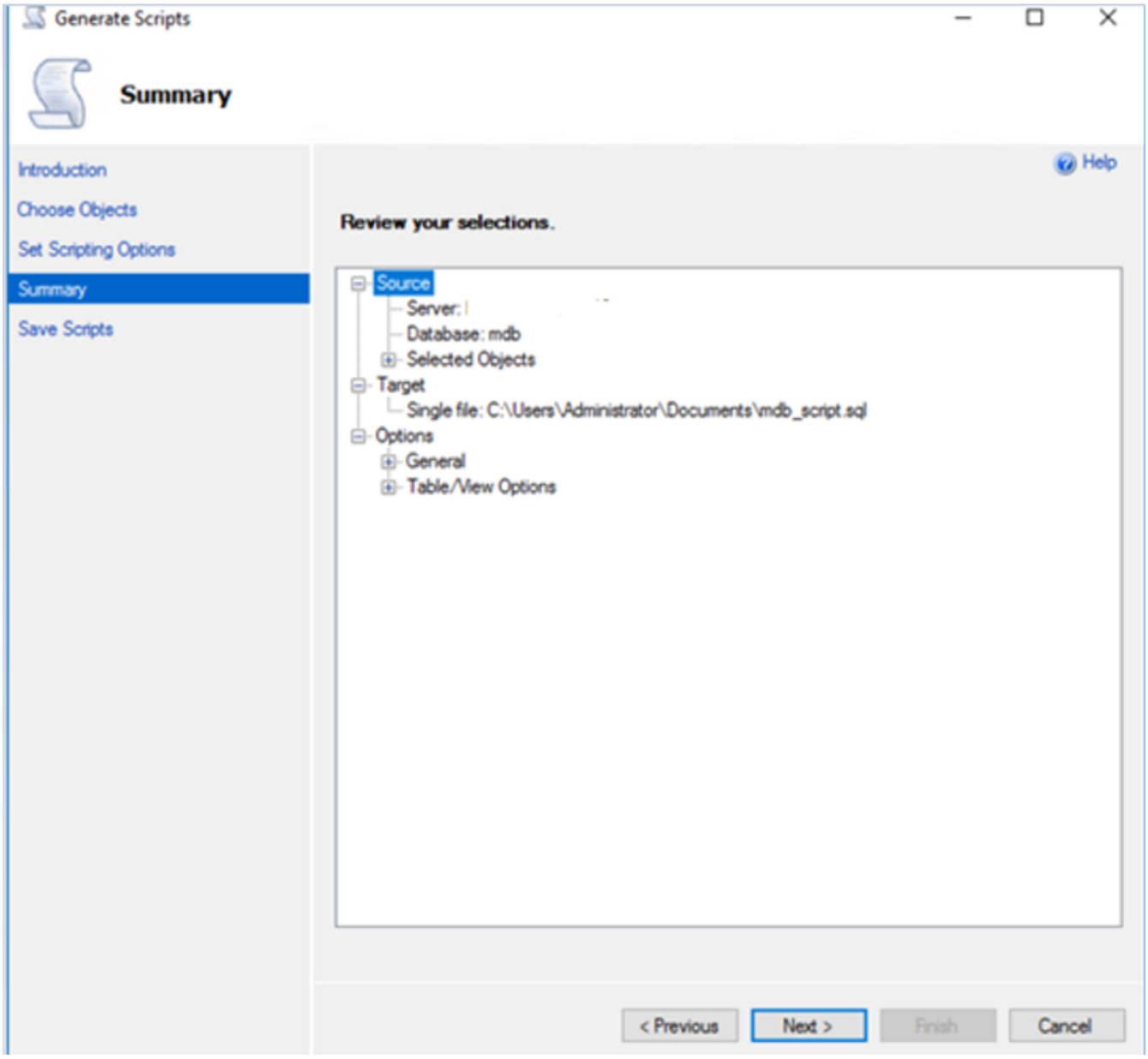


g. Change the Types of data to script option value from Schema Only to Schema and Data.
h. Click **OK**.
Your selection is now saved to a file %user%\documents\mdb_script.sql

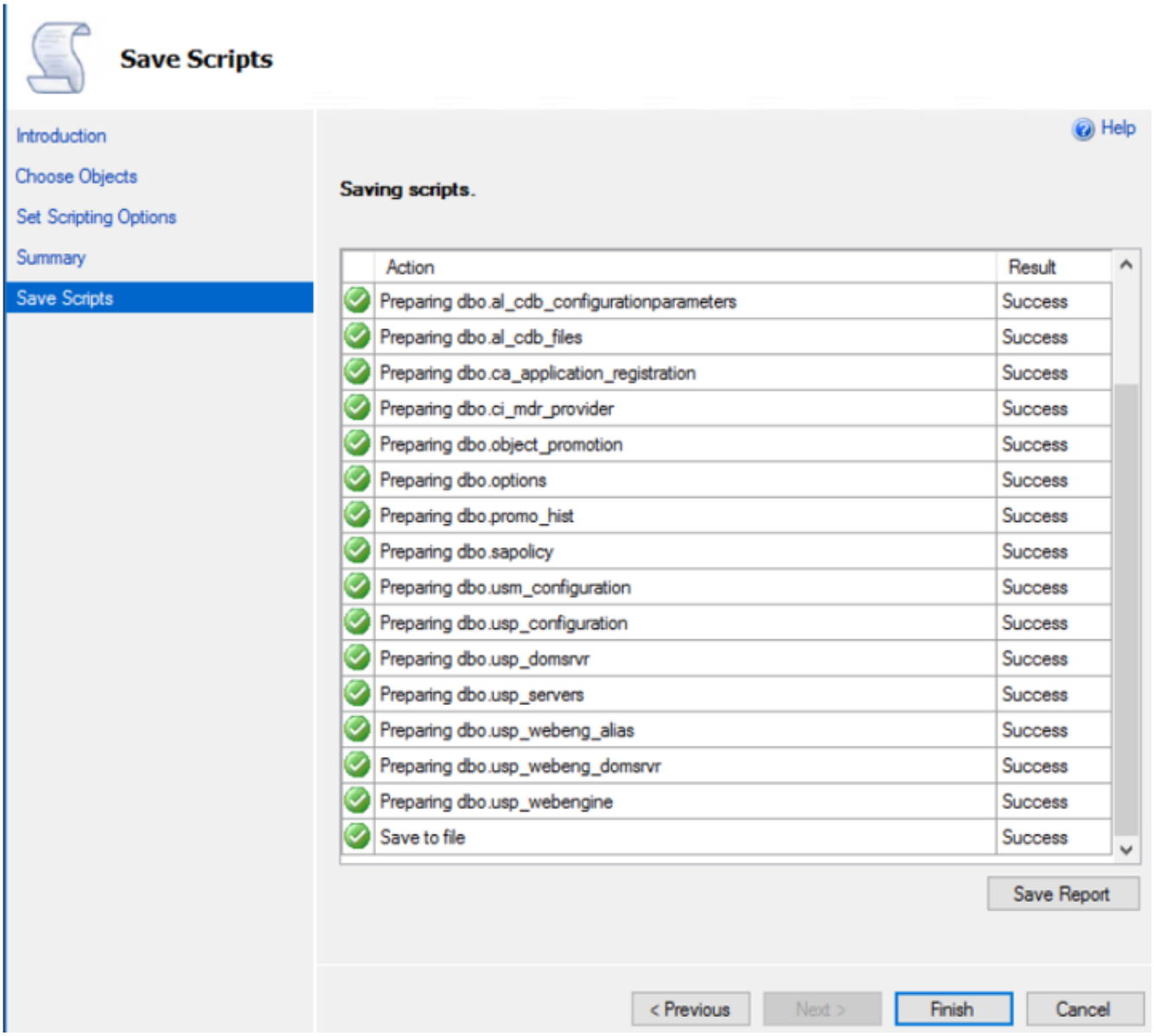
NOTE
It's recommended that you use non-default file names. By default, the file name is script.sql.


Generate Scripts
Set Scripting Options
Introduction
Choose Objects
Set Scripting Options
Summary
Save Scripts
Help
Advanced Scripting Options
Options
Script for Server Version
SQL Server vNext
Script for the database engine edition
Microsoft SQL Server Standard Edition
Script for the database engine type
Stand-alone instance
Script Logins
False
Script Object-Level Permissions
False
Script Owner
False
Script Statistics
Do not script statistics
Script USE DATABASE
True
Types of data to script
Schema and data
Table/View Options
Script Change Tracking
False
Script Check Constraints
True
Script Data Compression Options
False
Types of data to script
Generates script that contains schema only or schema and data.
OK
Cancel
< Previous
Next >
Finish
Cancel

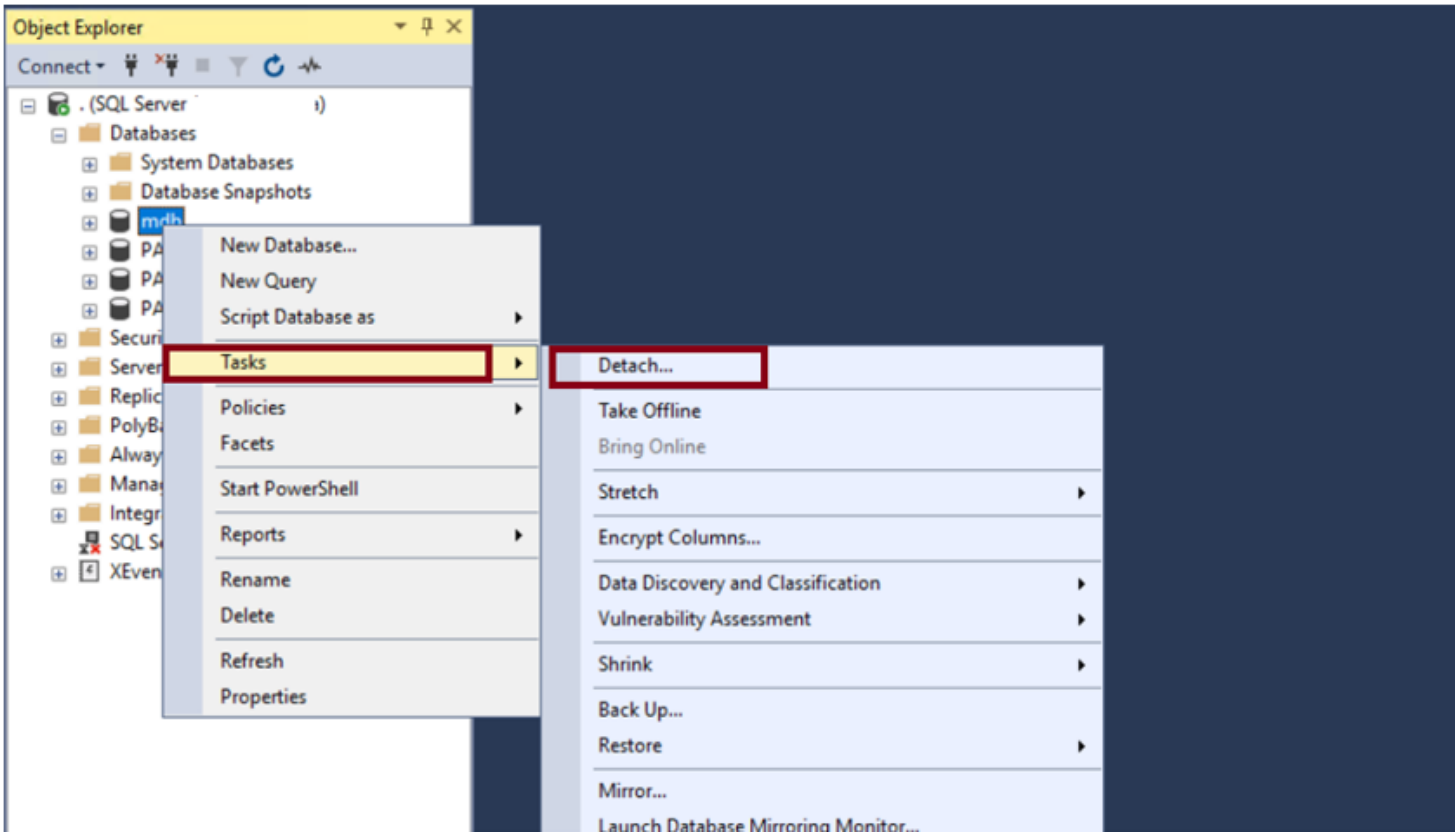
i. Click Next.



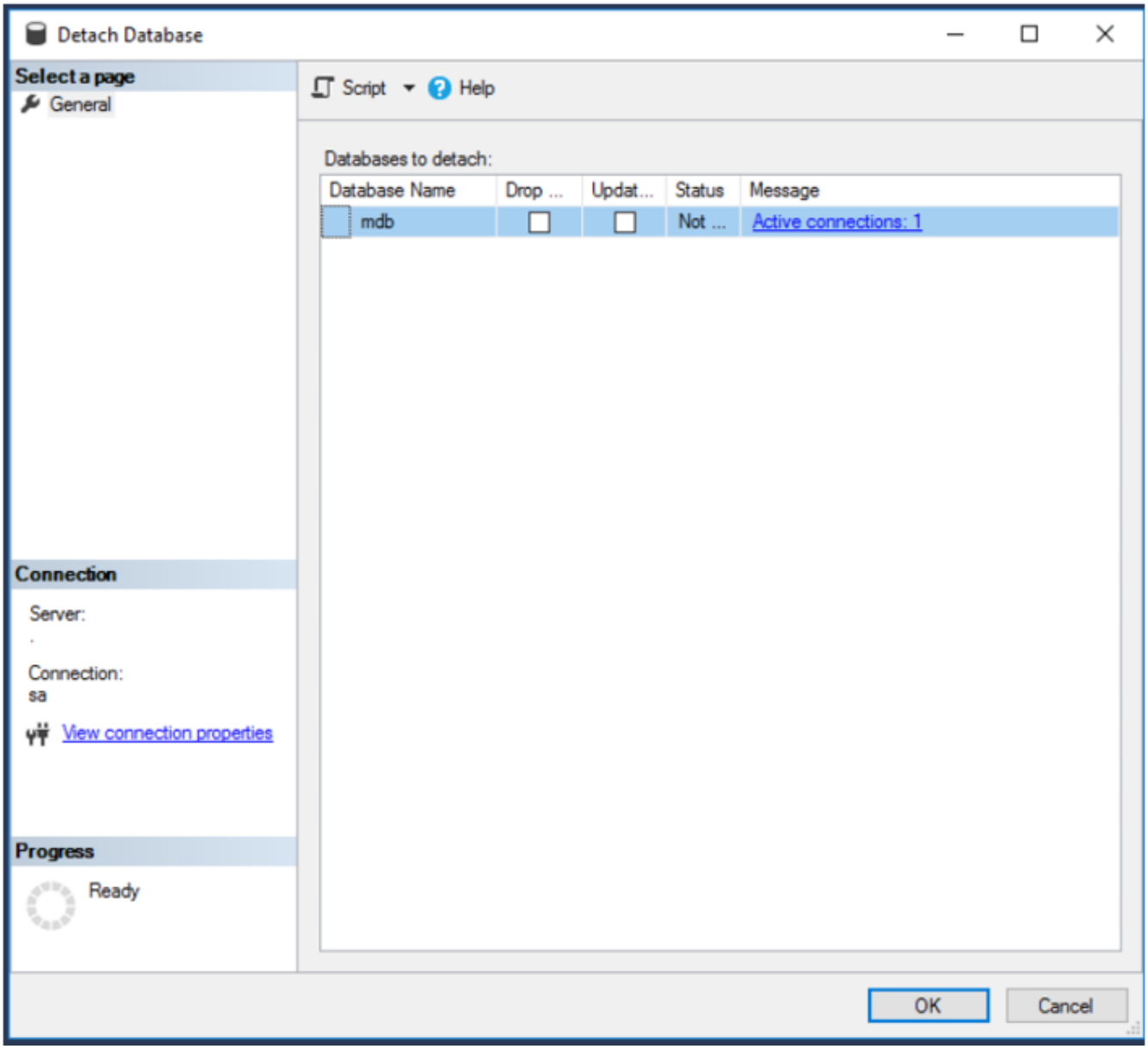
j. Click**Finish**.



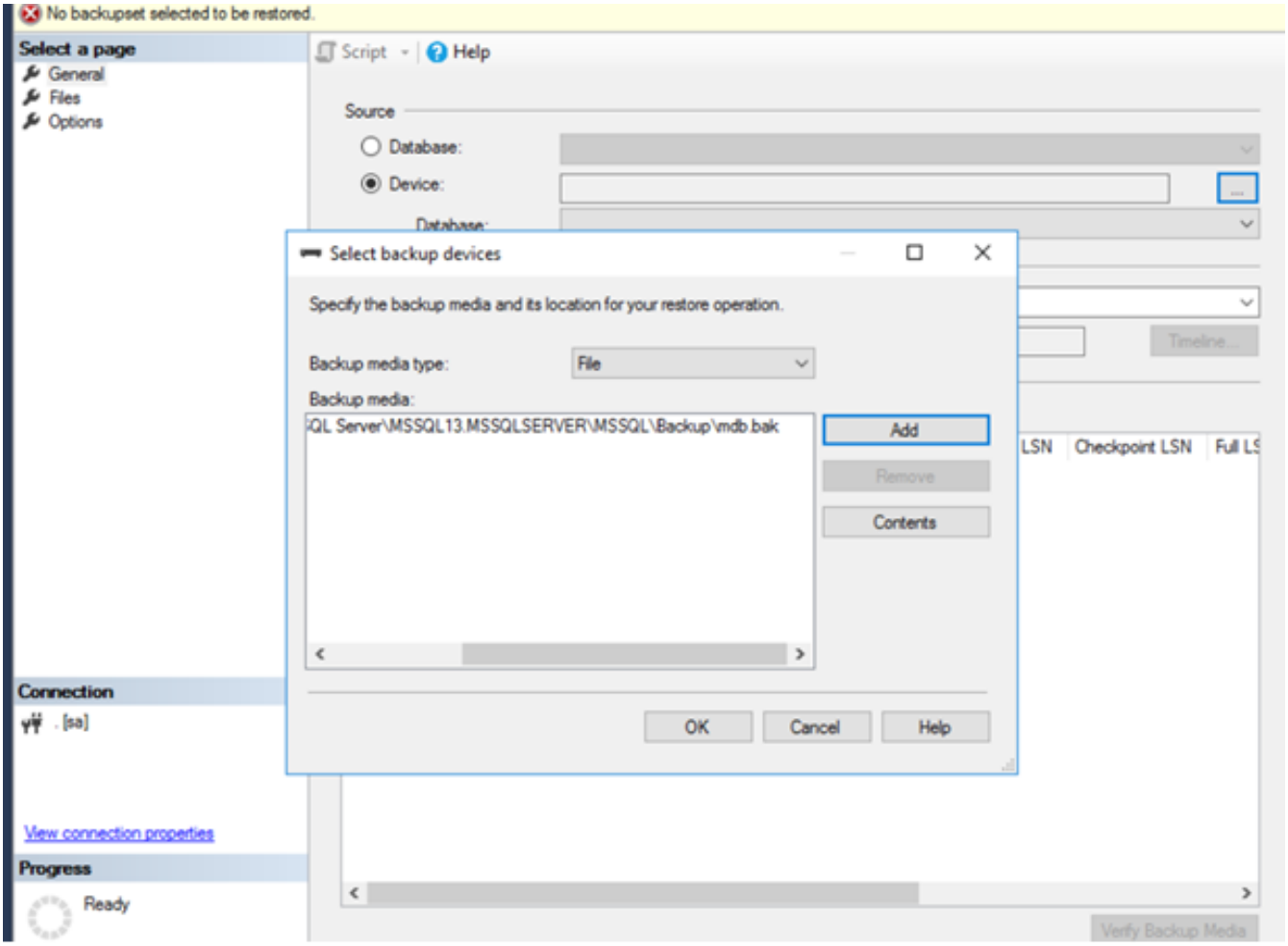
k. From the Detach Database screen, check the Drop Connections checkbox.



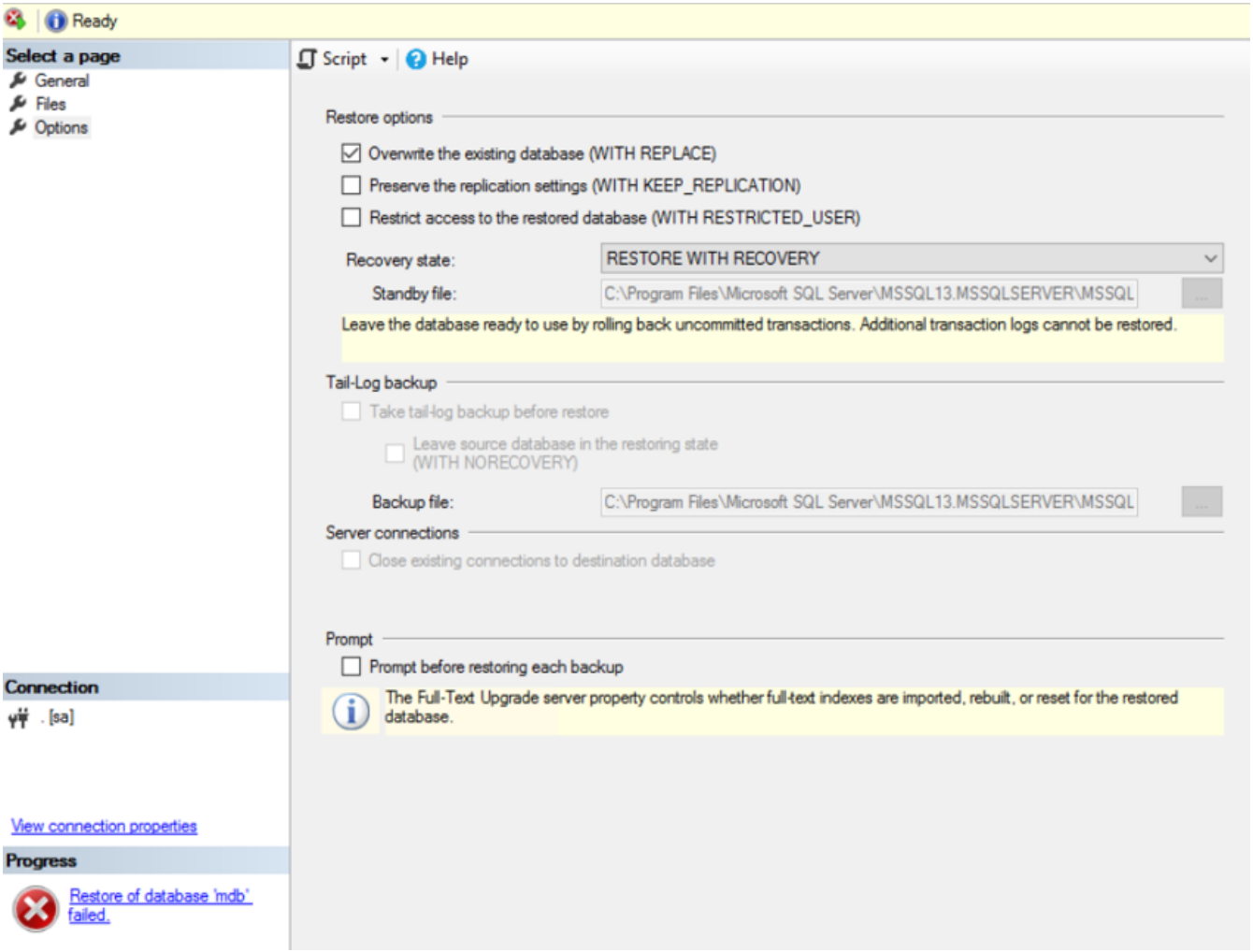
l. Select the **Drop Checkbox** and click **OK**.



9. MDB instance is dropped from the target system (testqa-mdb-new).
10. To restore MDB backup to the target system (testqa-mdb-new) which was copied from the Production system (testqa-mdb), perform the following steps: Important: Provide read and execute privileges to the mdb.bak file that you copied from the Production database (testqa-mdb) to the SQLServerMSSQLUser$$MSSQLSERVER service account on the target MDB server (testqa-mdb-new). Failure to grant the read and execute privilege to mdb.bak may result in error “file folder cannot be located or it does not have the adequate permissions”.
    a. Open the Microsoft SQL Management studio with Database Admin credentials (sa).
    b. Navigate to Databases option, right-click and select the Restore Database option.
    c. Select Device Radio button in the Restore database screen.
    d. Click Browse (...). Click Add and select the backup file where you copied from the Production system: For example: C:\Program Files\Microsoft SQL Server\MSSQL13.MSSQLSERVER\MSSQL\Backup\mdb.bak)
    e. Click "**Options**" on the Restore Database Screen.

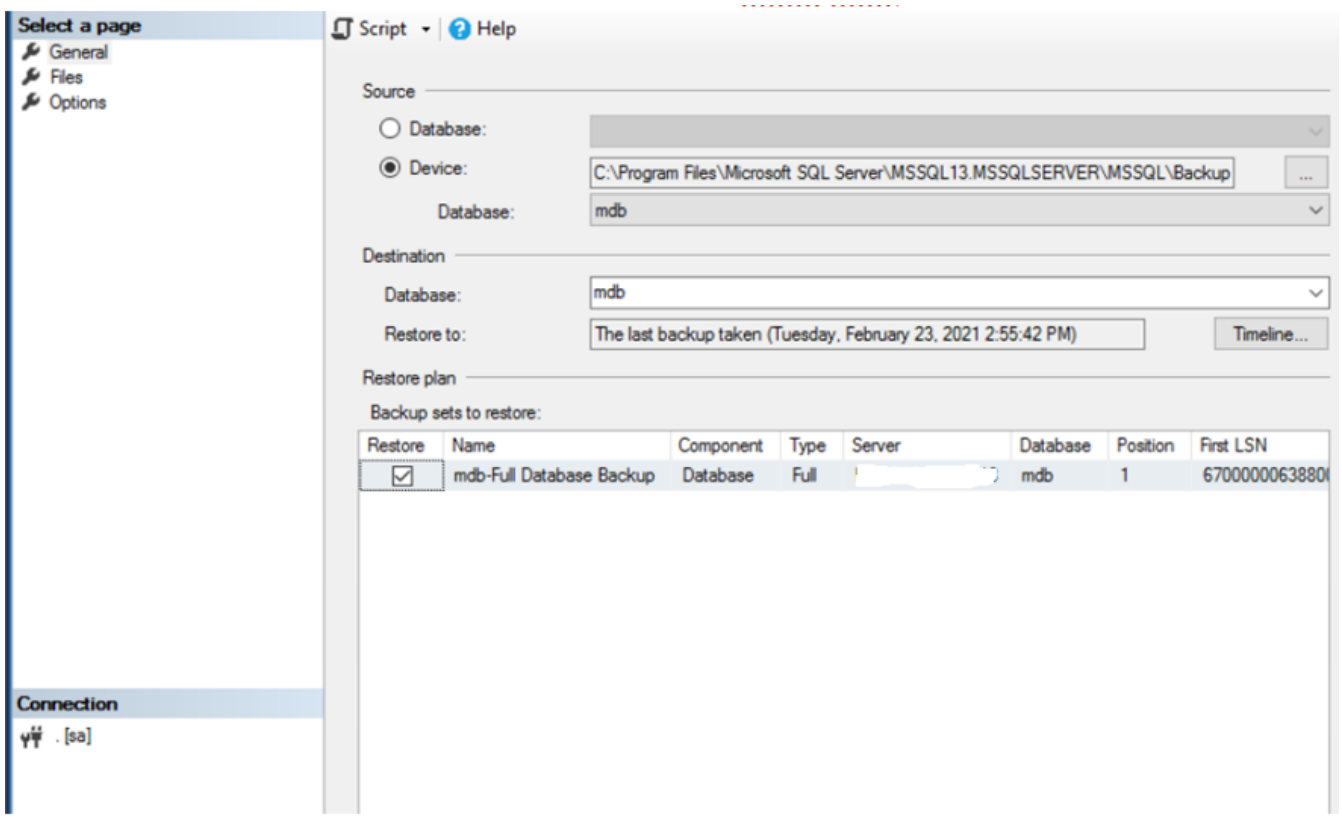


f. Select the checkbox Overwrite the existing database (WITH REPLACE).
Ensure that the database name on the target system is retained.

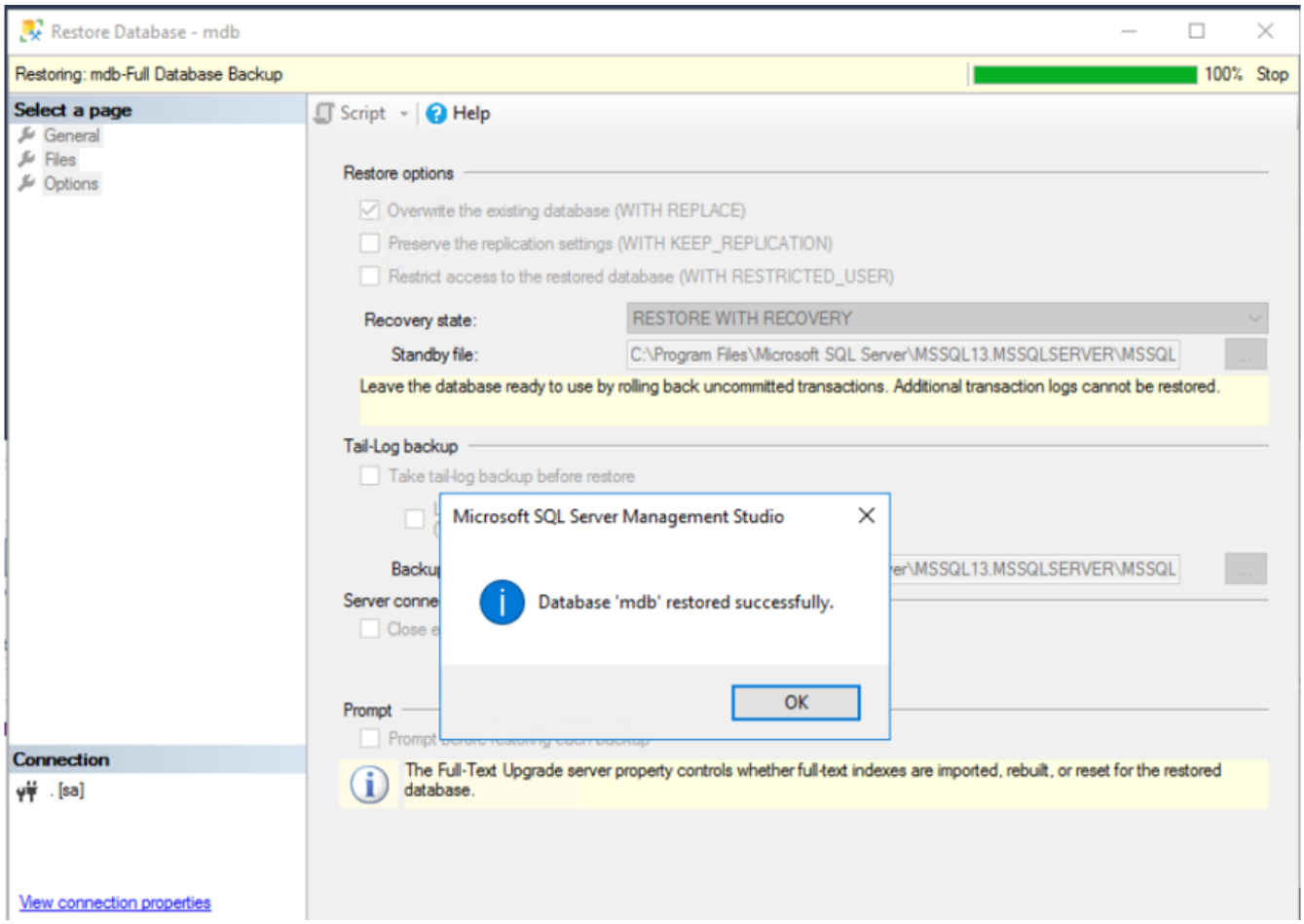


Click OK.

g. The restore database name on the target system (**testq-mdb-new**) is retained and successful.


Select a page
General
Files
Options
Connection
. [sa]
Script
Help
Source
Database:
Device:
C:\Program Files\Microsoft SQL Server\MSSQL13.MSSQLSERVER\MSSQL\Backup
Database:
mdb
Destination
Database:
mdb
Restore to:
The last backup taken (Tuesday, February 23, 2021 2:55:42 PM)
Timeline...
Restore plan
Backup sets to restore:
Restore
Name
Component
Type
Server
Database
Position
First LSN
mdb-Full Database Backup
Database
Full
mdb
1

h. Click OK.



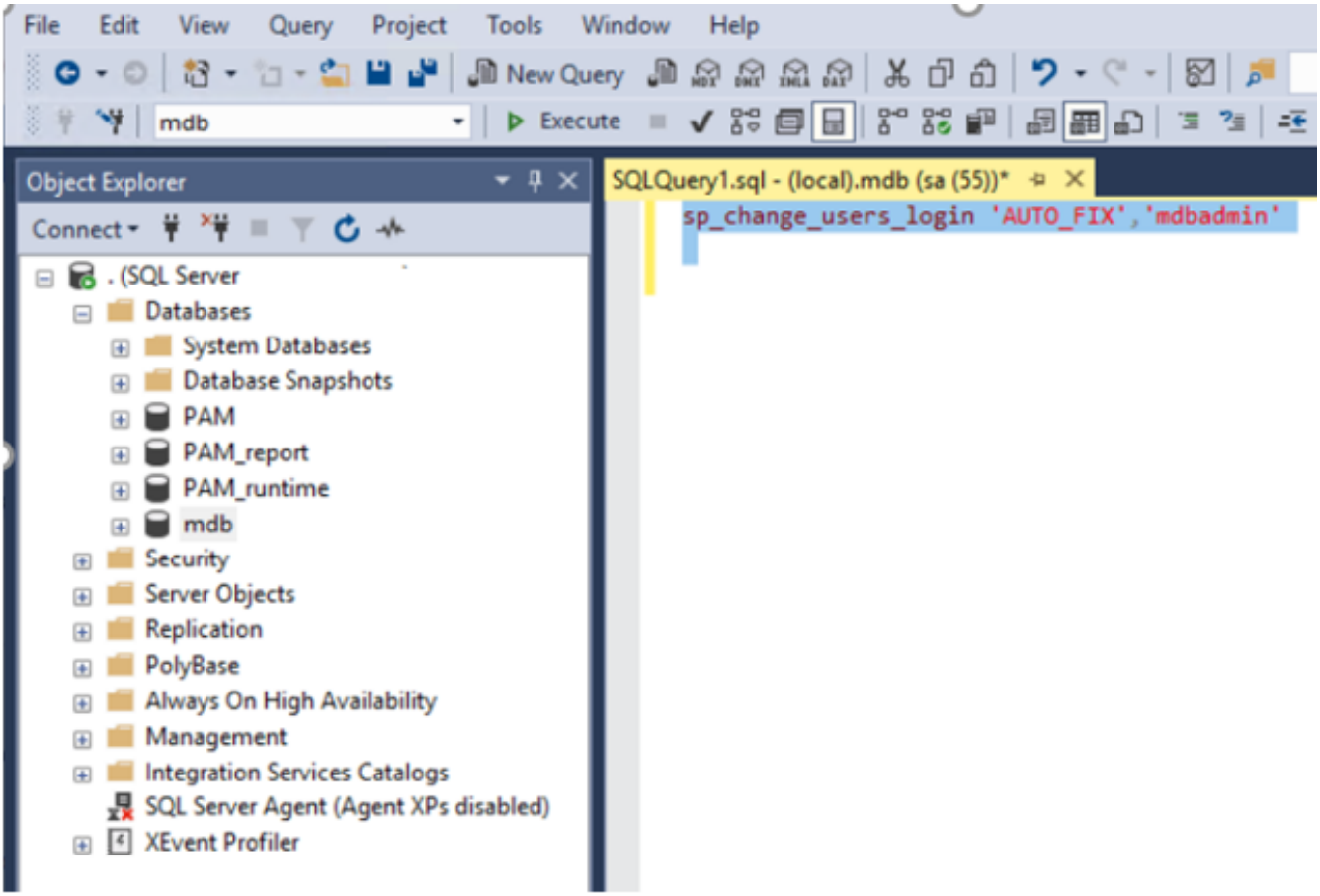
11. Database “mdb” is restored successfully. Important: Provide read and execute privileges to the mdb.bak file that you copied from the Production database (testqa-mdb) to the SQLServerMSSQLUser$$MSSQLSERVER service account on the target MDB server (testqa-mdb-new). Failure to grant the read and execute privilege to mdb.bak may result in error “file folder cannot be located or it does not have the adequate permissions”.

    NOTE

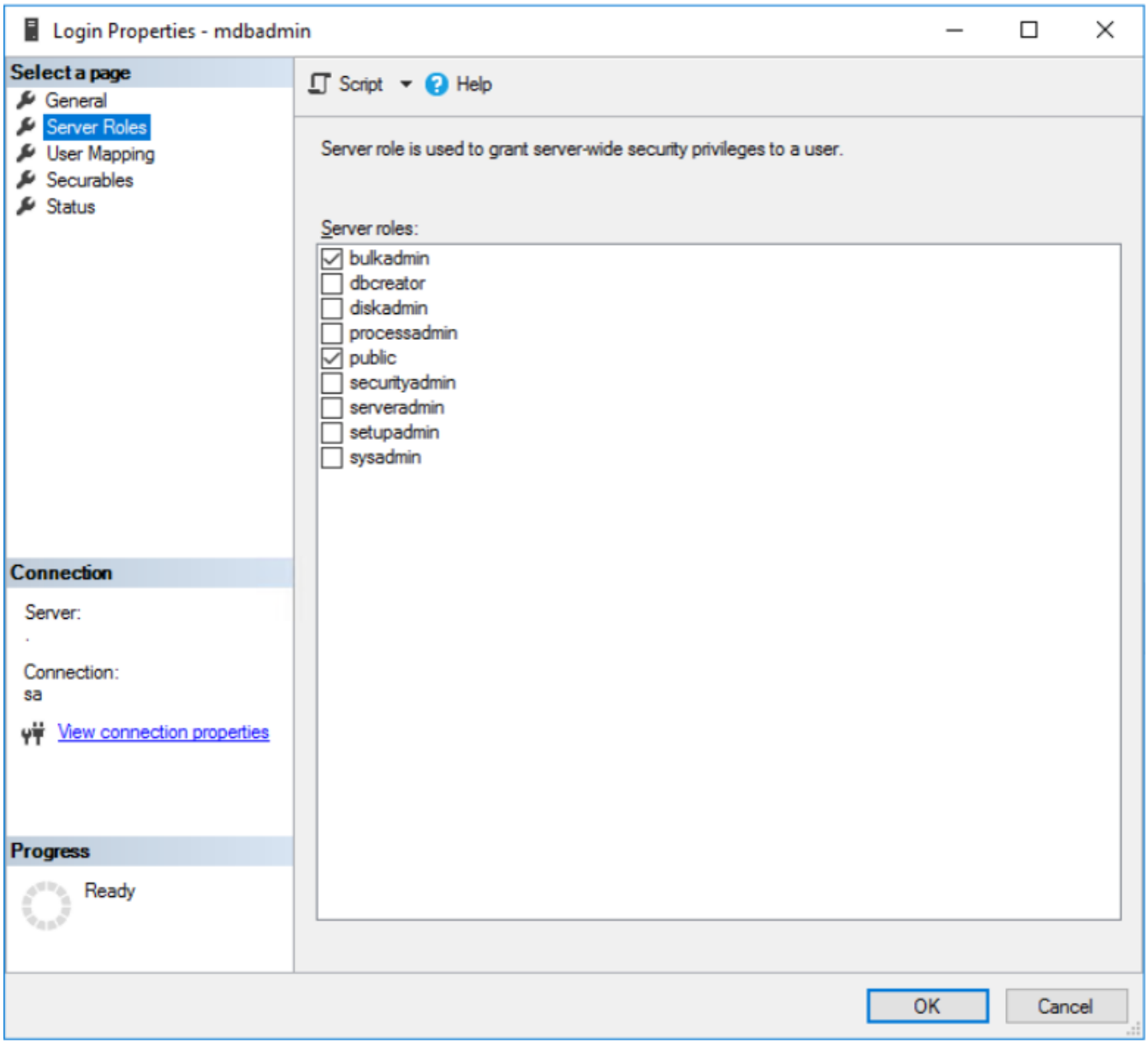
    Restore must complete without errors.

12. Run the following Microsoft SQL auto_fix_user script for the mdbadmin user:

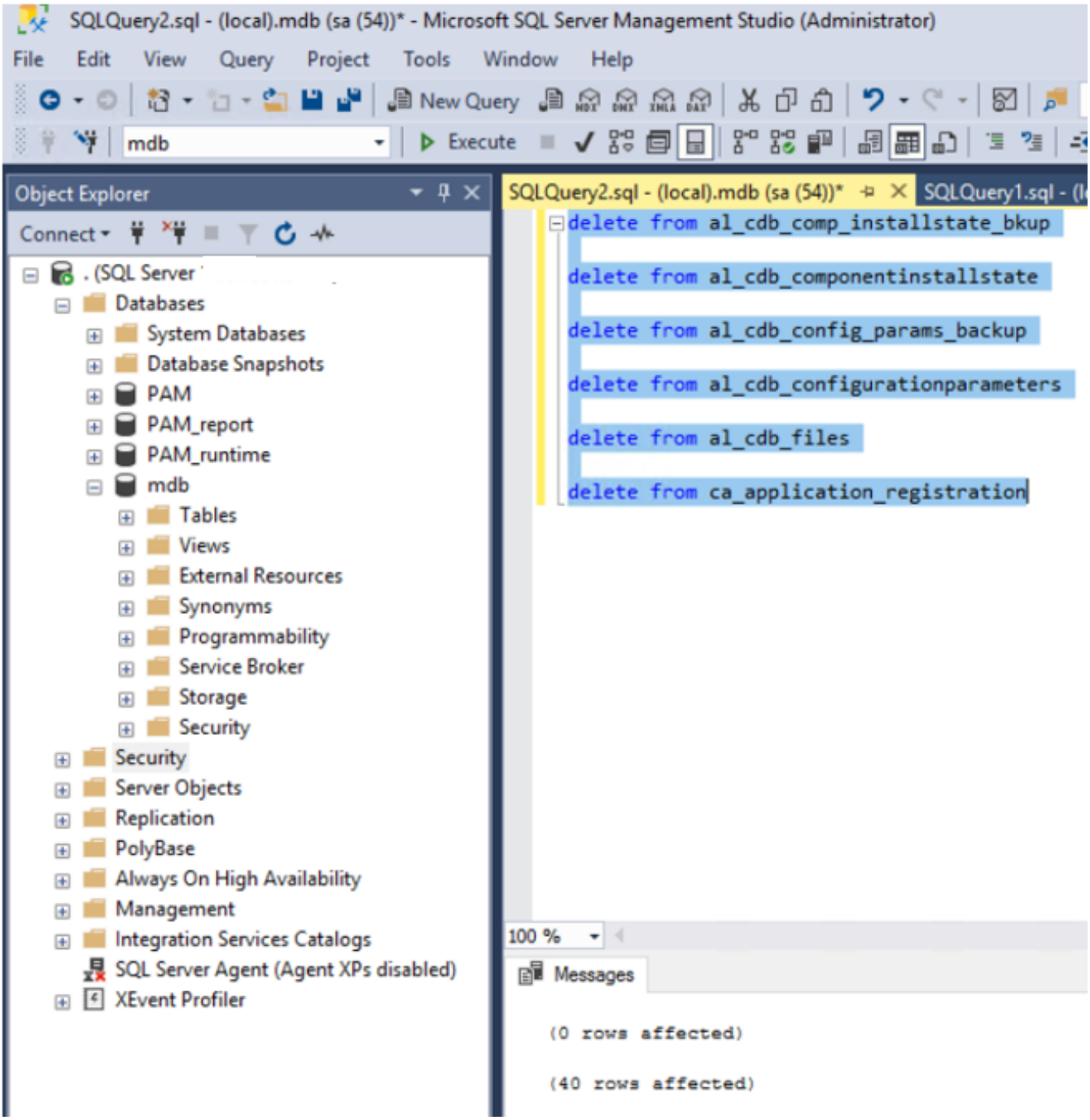
```
sp_change_users_login 'AUTO_FIX','mdbadmin'
```



13. Verify that the mdbadmin is now mapped to mdb and has the bulk admin and public roles.

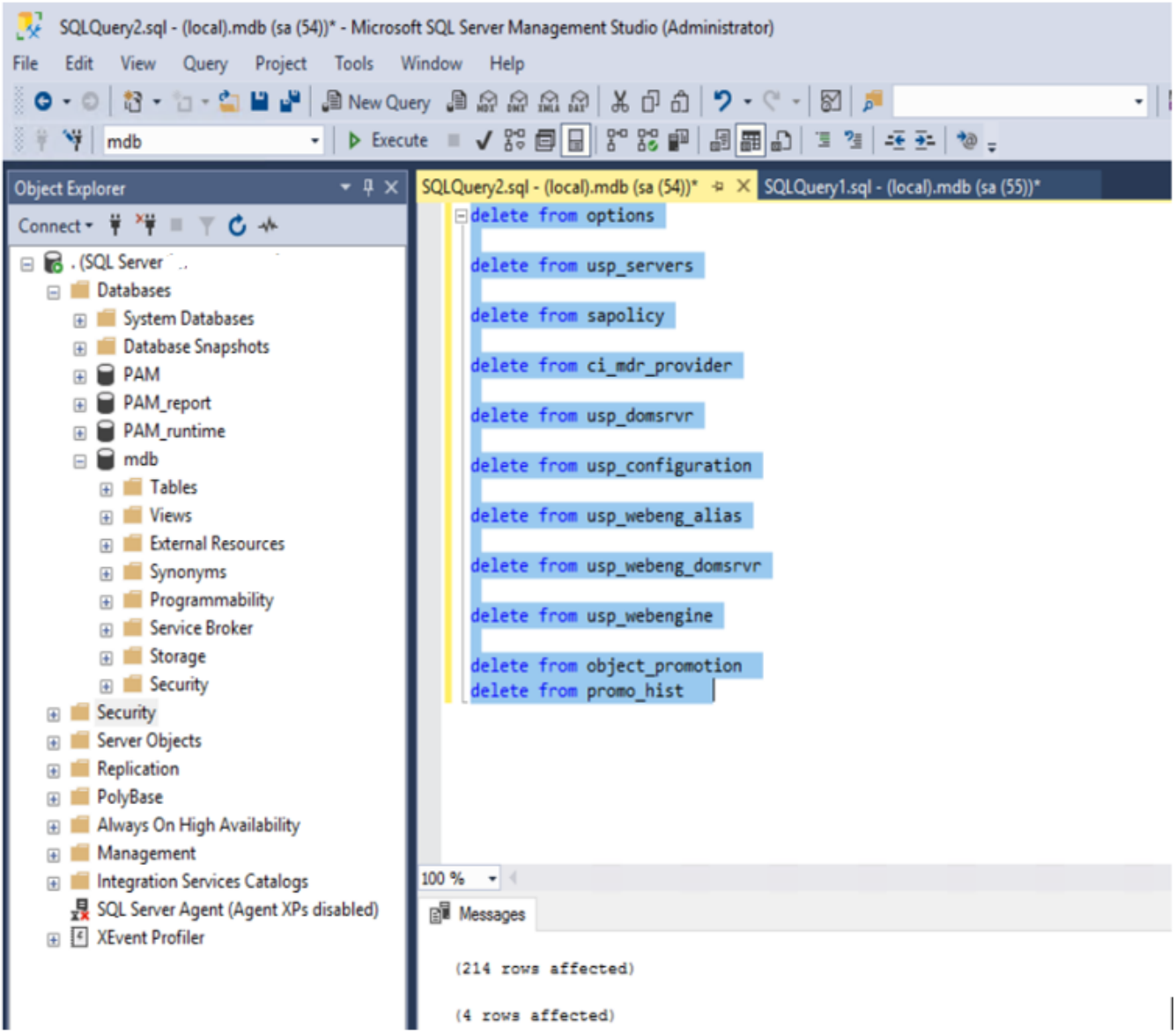


14. On the target system (testqa-mdb-new), delete data from the following tables and commit the changes:
    a. use mdb and run the following queries to delete data from the required tables:
    Common Tables
    - delete from al_cdb_comp_installstate_bkup
    - delete from al_cdb_componentinstallstate
    - delete from al_cdb_config_params_backup
    - delete from al_cdb_configurationparameters
    - delete from al_cdb_files
    - delete from ca_application_registration



b. CA SDM Tables:
   a. delete from options
   b. delete from usp_servers
   c. delete from sapolicy
   d. delete from ci_mdr_provider
   e. delete from usp_domsrvr
   f. delete from usp_configuration
   g. delete from usp_webeng_alias

h. delete from usp_webeng_domsrvr
i. delete from usp_webengine
j. delete from object_promotion
k. delete from promo_hist



c. **xFlow Interface Tables**:
   a. delete from usp_es_servers
   b. delete from l1admin_general_config
   c. From the File Menu, Select Open, Select File, now select mdb_script.sql (created in step 8, sub step i) and click the Execute button.

      NOTE
      Ignore the warning message during execution: “There is already an object named in the Database”.

Merge Customizations from CA SDM Production to Target Server

If you have customizations in your environment that needs to be moved, copy the site/mods directory and tblobj.cfg and ddict.sch files from the Production CA SDM server to the target CA SDM server perform the following steps:

> **IMPORTANT**
> These steps are applicable only if your application environment is customized. These are not applicable for the out-of-the-box (OOTB) content.

1. From the Production background server (testqa-bg) to the target background server (testqa-bg-new).
2. From the Production Standby (testqa-standby) to the target Standby (testqa-standby-new).
3. From the Production Application Server (**testqa-app1**) to the target Application Server (testqa-app1-new) (Perform on all application servers in an environment).

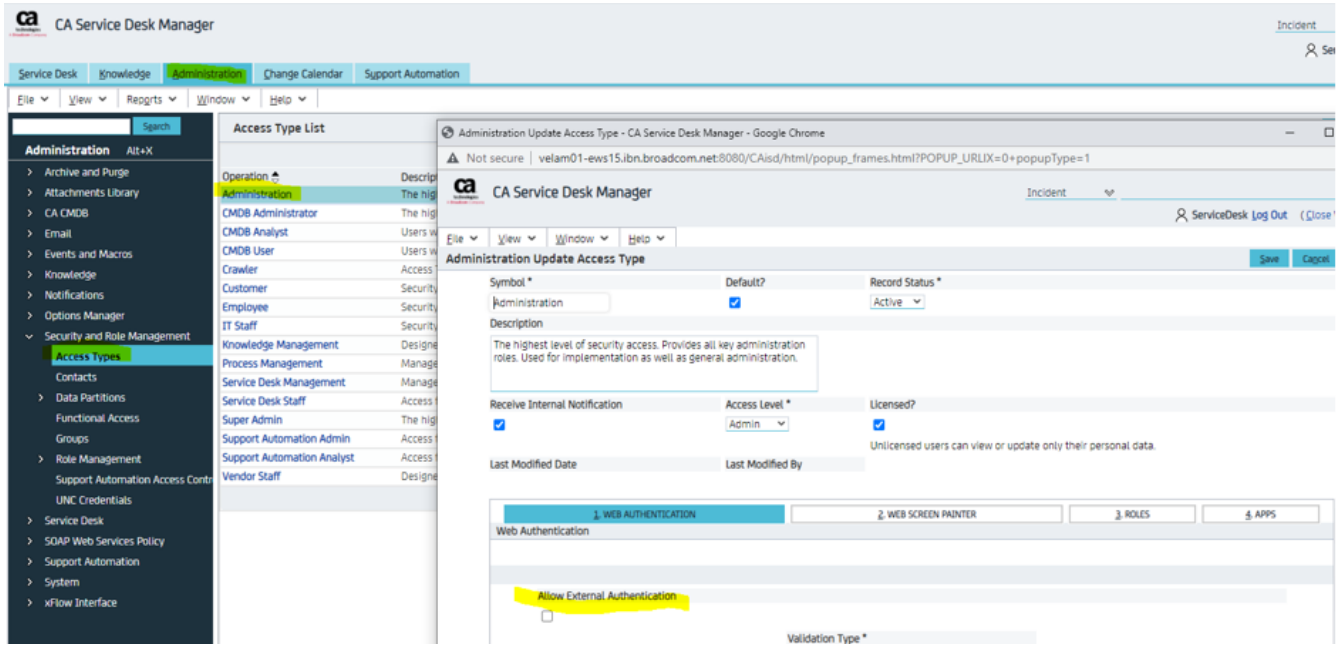
1. Run pdm_configure -S on the target background server (**testqa-bg-new**). It will configure silently:
   a. Monitor the progress of pdm_configure using NX_ROOT/logs/configure.log file.
   b. Launch the CA SDM Background URL as an Administrator (http://testqa-bg-new:8080/CAisd/pdmweb.exe).
   c. Run pdm_configure on the Standby Server (testqa-standby-new).
   d. Run pdm_configure on all application servers (testqa-app1-new and testqa-app2-new, and so on).
   e. Execute the pdm_k_reindex command on the target background server (testqa-bg-new) to index knowledge documents that are created and copied from the Production background server (testqa-bg).
2. CA SDM Repositories (applicable only if you have NOT used the OOTB repositories):
   a. Log in to CA SDM URL as an administrator.
   b. Navigate to Administration, Attachments Library, Repositories.
   c. Open the repositories one by one and update the target CA SDM server name in the Servlet Server and Repository Server fields.

      > **NOTE**
      > Update the CA SDM server name in all the repositories (in the out-of-the-box and Created Repositories).
3. CA SDM Attachments (applicable only if you have NOT used the OOTB attachments path):
   a. Copy all the attachments using upload Path of all the repositories from a Production server to the same path in a target server
      For Example: Service Desk repository upload path is: <NX_ROOT>/site/attachments/default/servicedesk/file on Production server. Copy the attachments from a Production server to the target server.
4. Customizations are now moved to the target server.

Disable NTLM on CA SDM servers

Perform the following steps:

1. Open BG SDM URL, Uncheck the “Allow External Authentication” and Click on Save Button.



2. Restart the services of all SDM,xFlow servers (BG, Standby and Application servers, xFlow application).

   IMPORTANT

   If you have plan to do archive and purge of the historical data later in the production, perform the steps in the pre-prod environment also.

   For more information, see How to Archive and Purge Historical Data .

   IMPORTANT

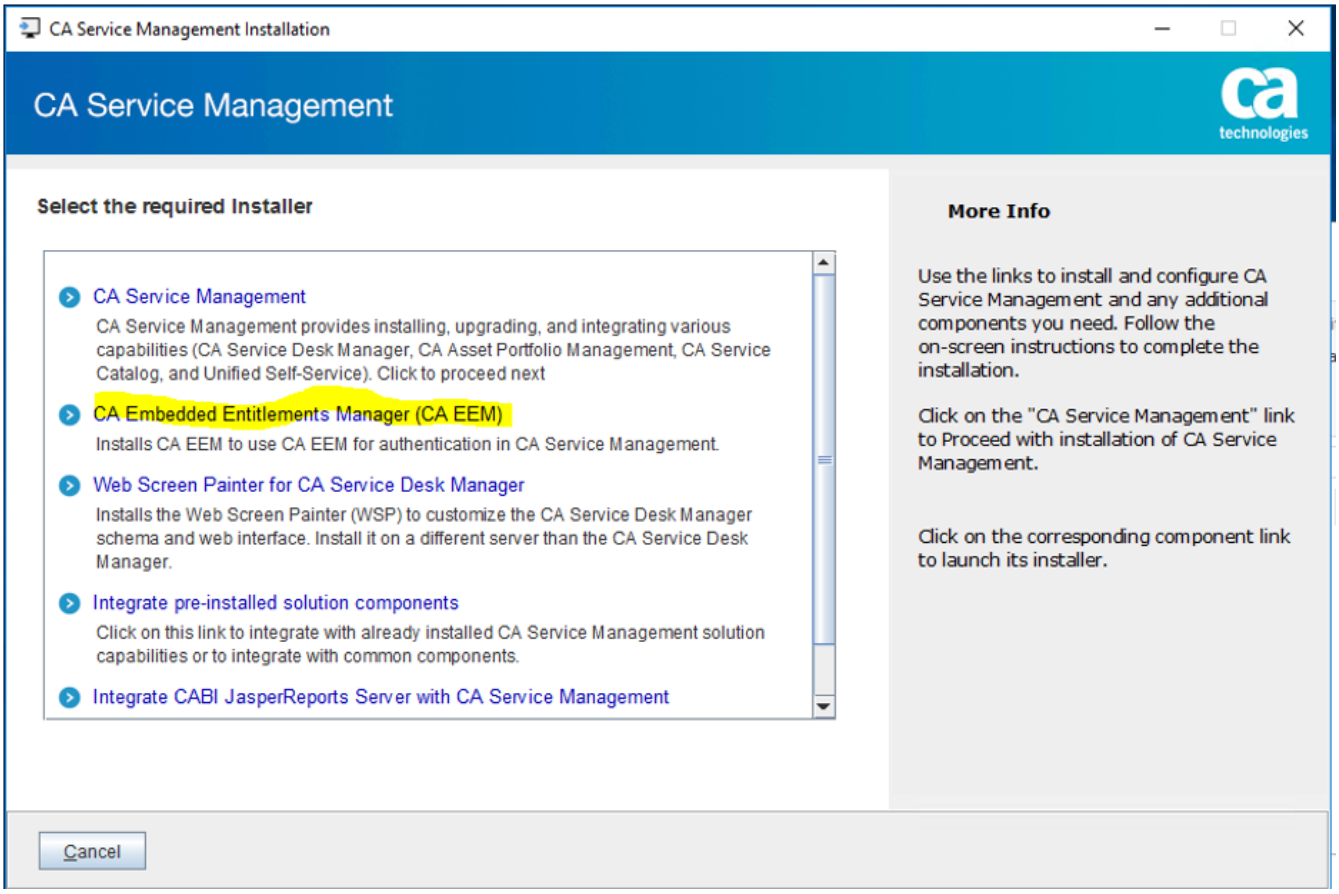
   Rebuild index execution time may vary based on the data in the MDB.
3. Run the rebuild index (pdm_es_rebuildindex.bat) from Search server machine.
4. Verify the installation of CA SDM,xFlow Interface and Service Point application.
5. Launch the CA SDM Background URL as an Administrator (http://testqa-bg-new:8080/CAisd/pdmweb.exe) and verify the ticket creation flow.
6. Launch the xFlow/ServicePoint URL : *http://testqa-app-new:9002/#/login*
   a. Login as an analyst to create a ticket from global search in the xFlow application.
   b. Login as an end user to create a ticket from service point application.
7. After successful SDM swing box migration, take snapshots of all above servers.

You have now completed the CA SDM Swing box migration successfully without CA EEM and LDAP integration.

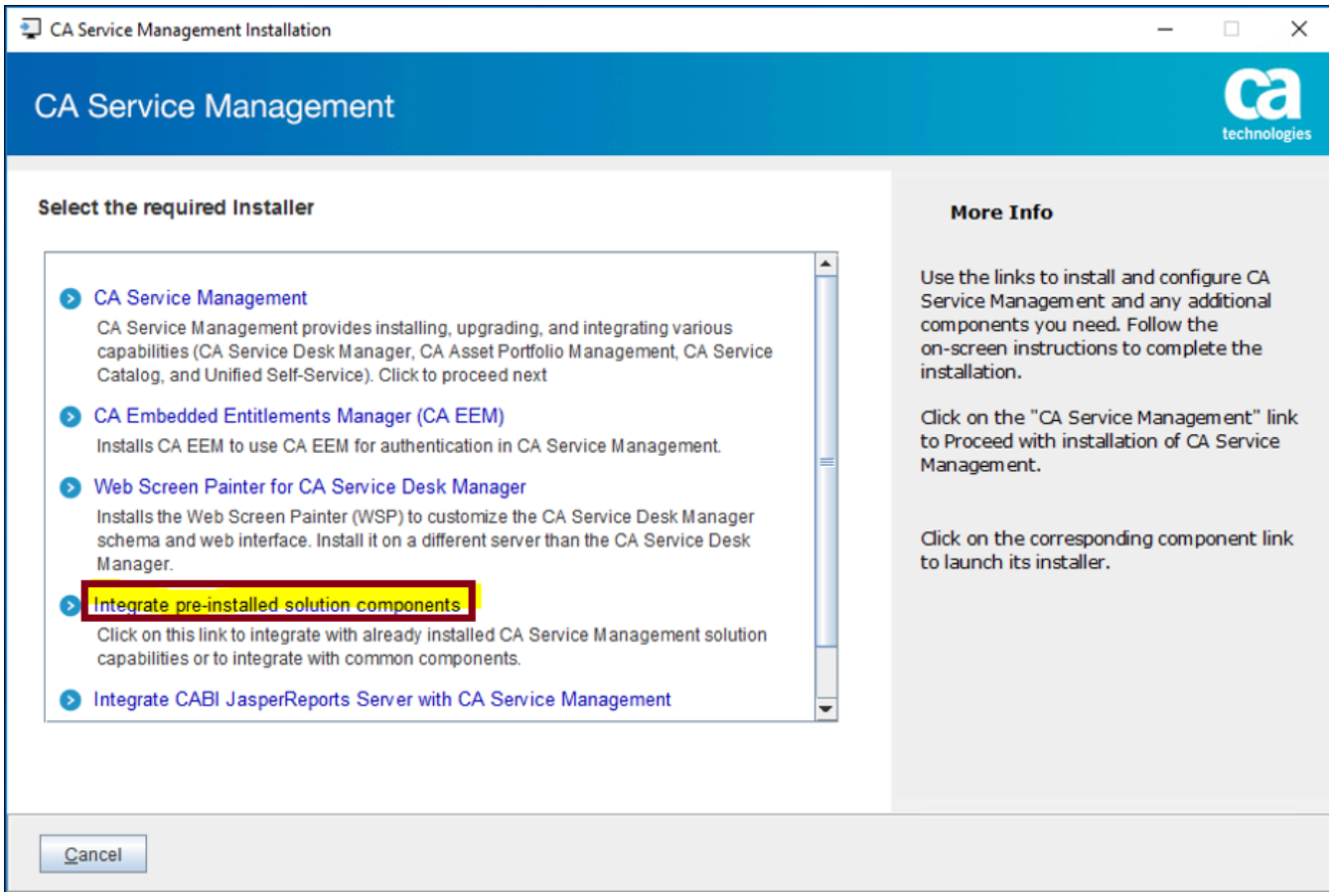
CA EEM installation and Integration with CA SDM

Perform the following steps:

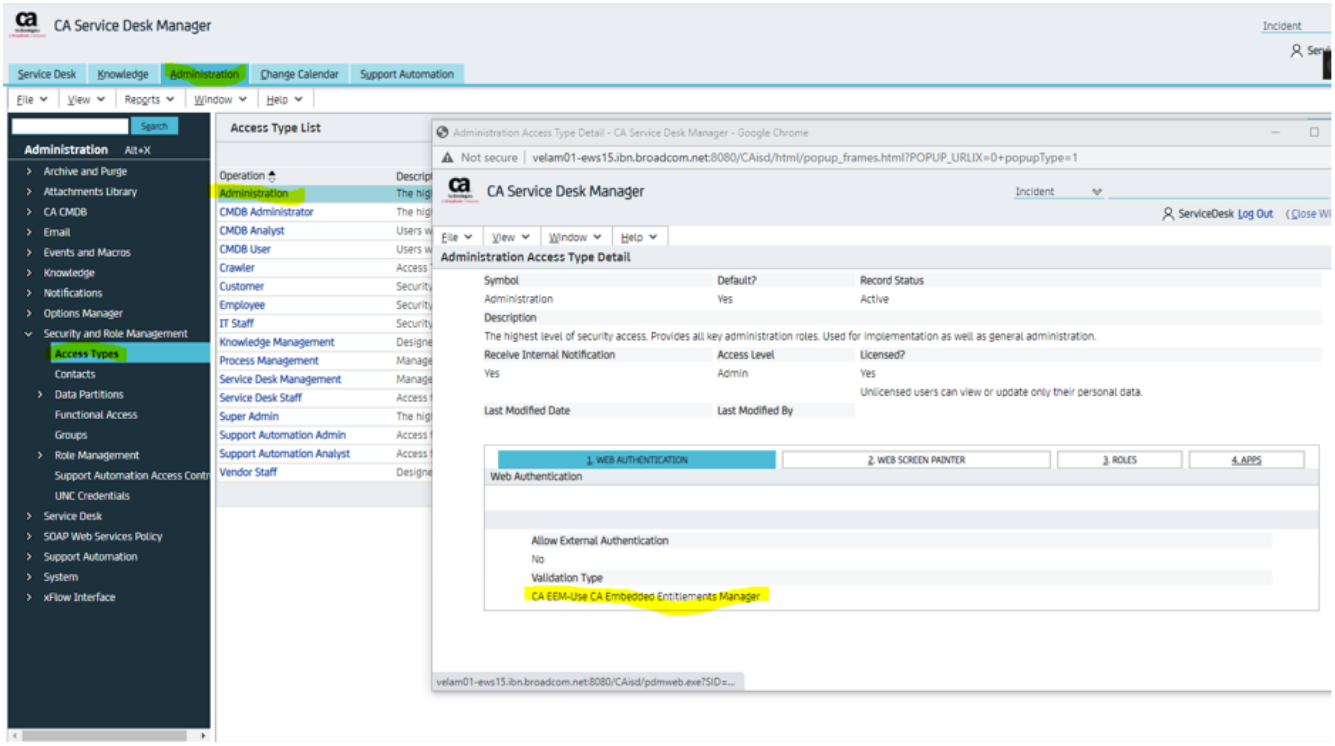
1. Install EEM application using 17.4 iso image on testqa-eem-new.



2. To Configure EEM application with BG server, launch the CA Service Management ISO image on BG server, that is, testqa- bg-new and Click "Integrate pre-installed solution components" and complete remaining steps.



3. To synchronize the EEM application information with other SDM servers, restart the services of SDM Standby and all application servers.
4. Validation check points:
   a. Ensure that Eiam details are updated in the NX.env file on SDM servers. For Example:
      a. @NX_EIAM_HOSTNAME= testqa-eem-new
      b. @NX_USE_EIAM_AUTHENTICATION=Yes
      c. @NX_EIAM_CERT_PASSWORD=aMgRLqnrDkGdtVKszPoC9d3Hcn5bzGYnYG8/dh4YAVM2+5Qh6nfTVv0G7/e5bP4T
      d. @NX_EIAM_APPINST_LABEL=
   b. Open the CA SDM Interface URL. Navigate to Administration, Security and Role management, Access types: Administration::: Validation type as: CA EEM-Use
   c. CA Embedded Entitlements Manager



d. Basic Sanity test after integration with EEM:
   a. Create tickets from SDM application.
   b. Create ticket from the global search in xFlow Interface.
   c. Create ticket from Service Point.

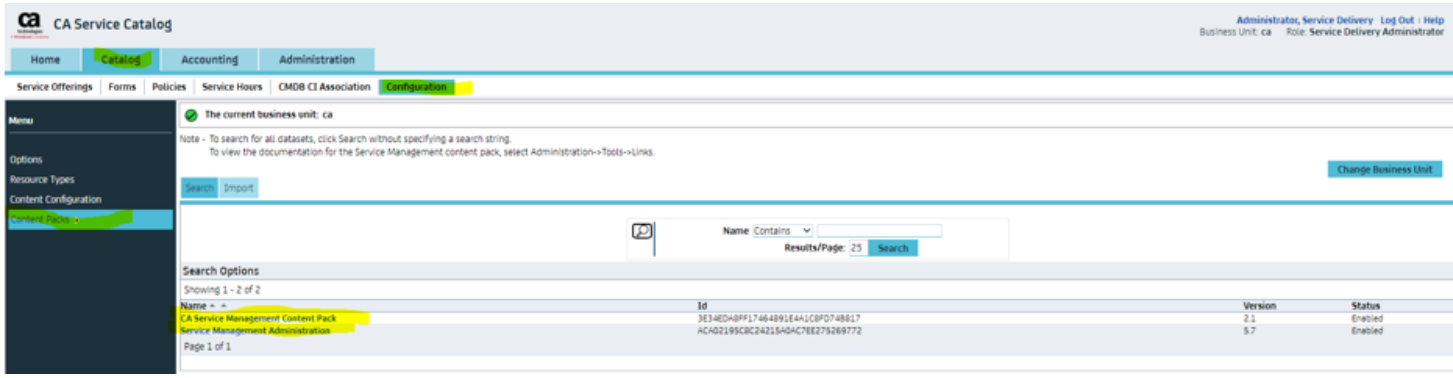
5. CA SDM is now integrated with EEM successfully.

NOTE
It's recommended that you take snapshots of all the above-mentioned servers.

Install CA Service Catalog

Perform the following steps:

1. Install CA Service Catalog 17.4 pointing to the same EEM server (testqa-eem-new) on new machine**testqa-catalog-new** and pointing to the same mdb on **testqa-mdb-new** machine.
   For more information, see Install CA Service Catalog.
2. Update CA Service Catalog post-installation steps for 17.4 on**testqa-catalog-new**.
3. Restart catalog services on testqa-catalog-new.
   a. CA Service Catalog
   b. CA Service Accounting
4. Import the content packs in the catalog application URL: Import the below content packs from the location %usm_home%/filestore/contentpacks on catalog installed machine (testqa-catalog-new).
   a. "CA Service Management Content Pack 2.0"
      "Service Management Admin Content Pack for SQL"
   b. Login to CA Service Catalog application with admin credentials (spadmin/spadmin).
   c. Navigate to Home, Catalog, Configurations.
   d. Click on content packs, Import.
   e. Select the content packs from step 4 of Install CA Service Catalog section and then import.
   f. Ensure that after successful import, the content pack names are displayed as mentioned in the following image:



5. Basic Sanity test on Catalog: Create a catalog request with admin credentials (For example: spadmin/spadmin)

Install Multiple Catalog Cluster Nodes

> **NOTE**
> If you are adding CA Service Catalog Nodes to an existing CA Service Catalog Cluster Node before Installing/ Upgrading, perform the following steps. For more information, see CA Service Catalog Clustering.

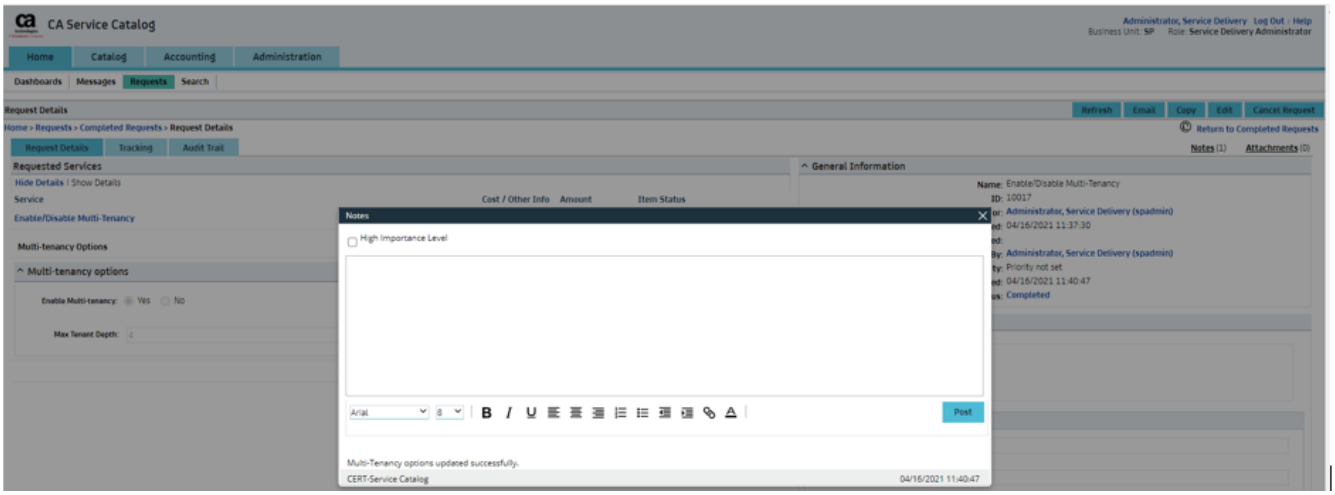
Perform the following steps:

1. Navigate to where you have downloaded and extracted CA Service Management ISO.
2. Install CA Service Catalog 17.4 pointing to the same EEM server (testqa-eem-new) on new machine **testqa-catalog-new2**and pointing the same MDB on **testqa-mdb-new** machine. For more information, CA Service Catalog Installation.
3. Restart catalog services on testqa-catalog-new2:
   a. CA Service Catalog
   b. CA Service Accounting
4. Add multiple catalog node(s) by following Step 1- Step 3(Section Name: “Install Multiple Catalog Cluster Nodes”).

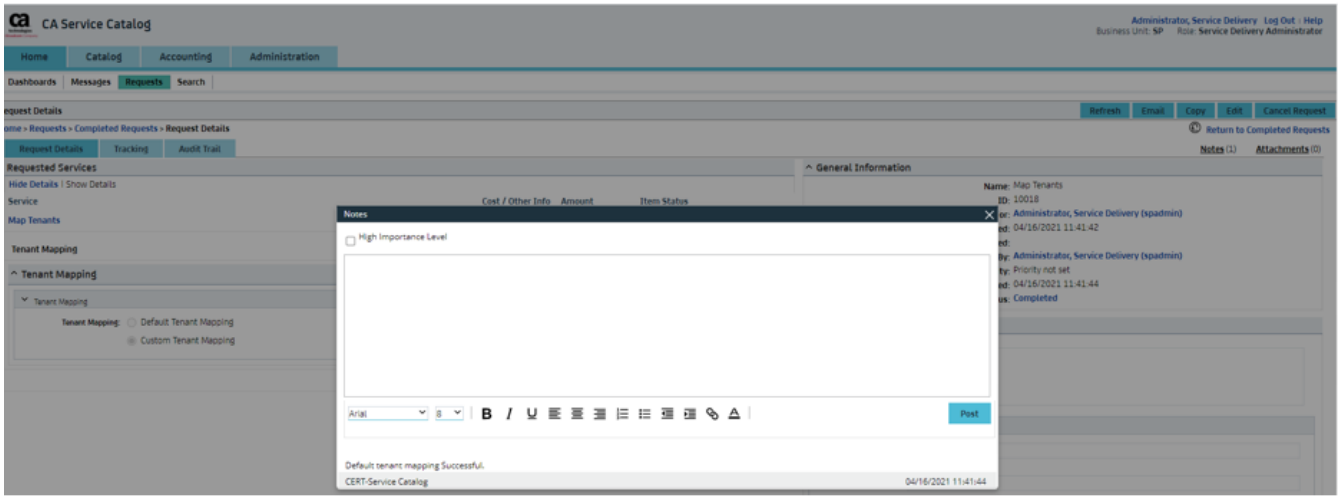
Enable Multi Tenancy in CA Service Catalog

Prequisite: Ensure that “CA Service Management Admin” service is up and running.
Perform the following steps to enable multi-tenancy in CA Service Catalog:

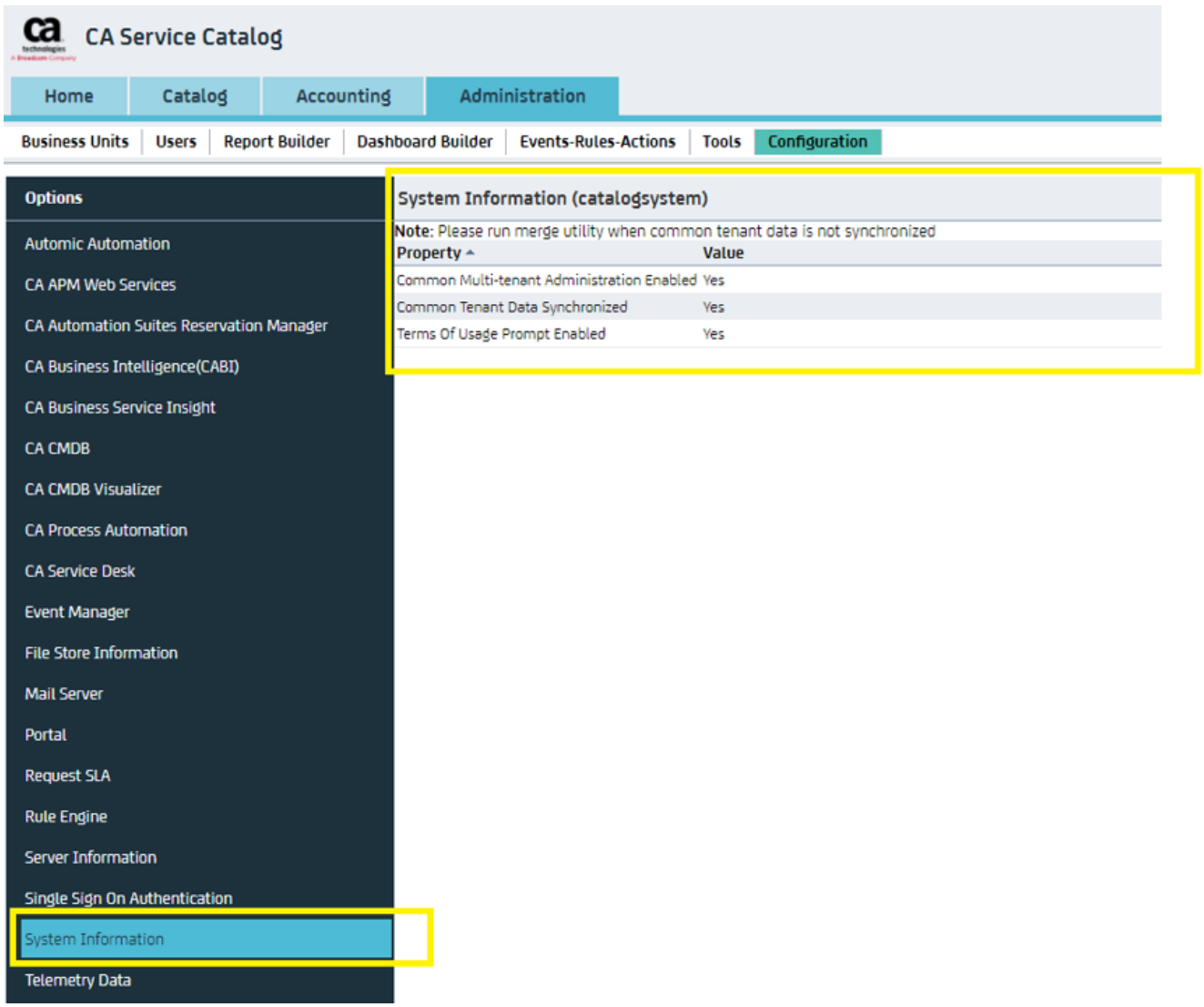
1. Login to CA Service Catalog with admin privileges (for example: spadmin/spadmin).
2. Navigate to Home, Requests, Click on Service Management Administration. Select offering as “Enable/Disable Multi tenancy” and then, submit request.



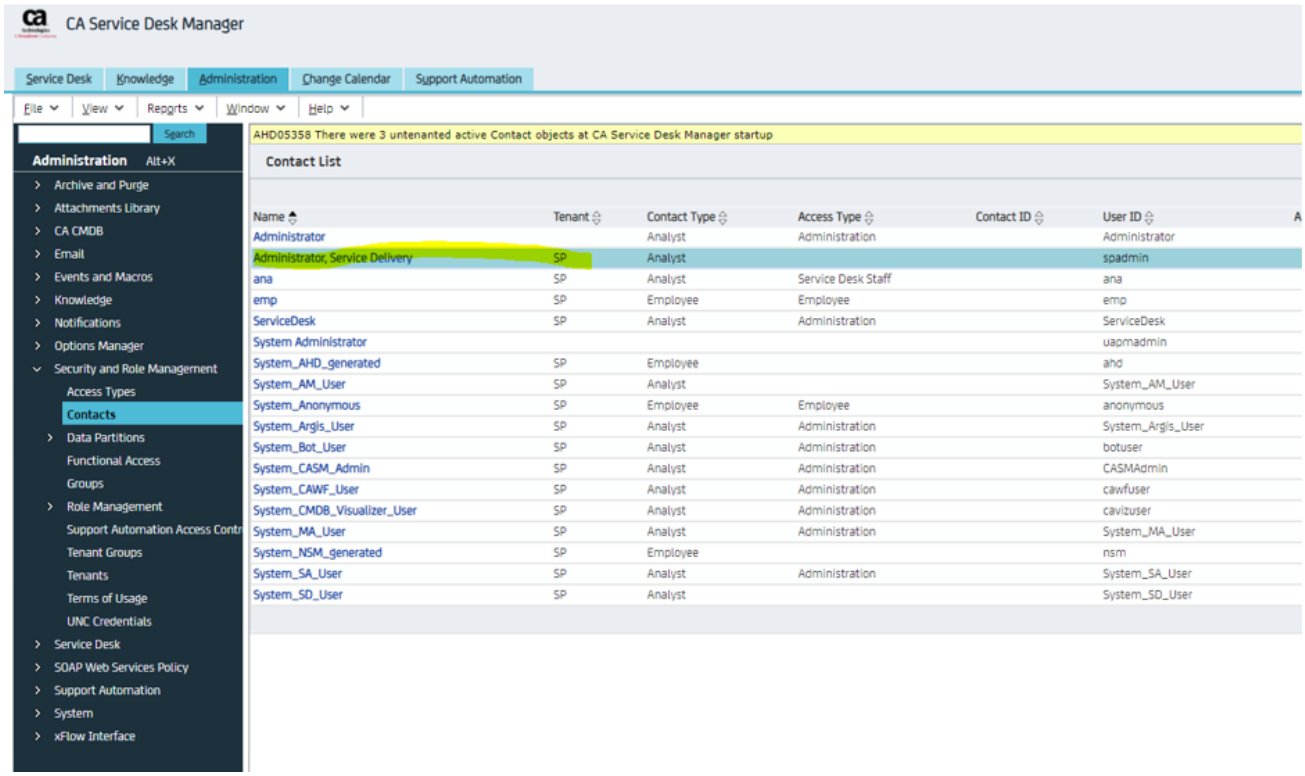
3. Navigate to Home, Requests. Click **Service Management Administration** and select offering as “**Map tenants**” and select custom tenant mapping option and then, submit the request.



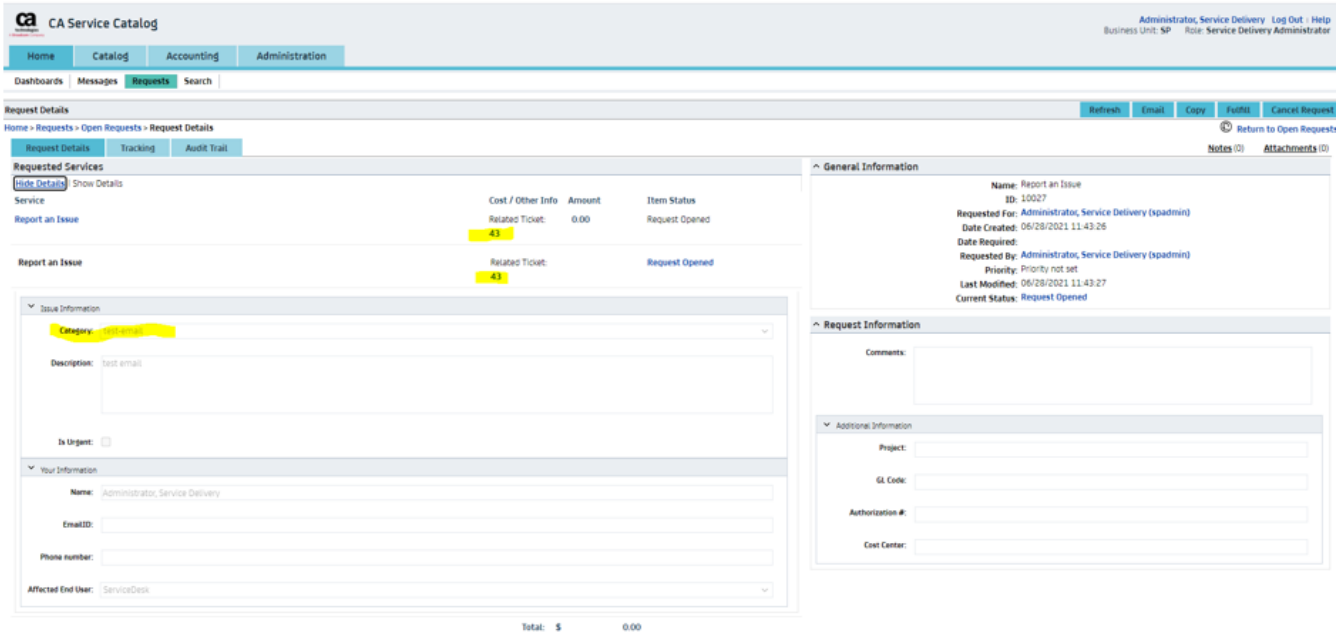
4. Restart the below CA Service Catalog services:
   a. CA Service Catalog
   b. CA Service Accounting
5. Validation Check: After successful configuration with multi-Tenancy. Below values are set as “**Yes**” in CA Service Catalog as mentioned in the image below:



6. Functional Sanity check with SDM. Pre-requisite: Please make sure that Service catalog default user (i.e., **spadmin**) should have tenant associated in SDM (see image):



7. Validation Check: Create a request with “report an issue” offering from catalog application and verify the incident number is available in the request details page. Same Incident is available in the SDM application as well.

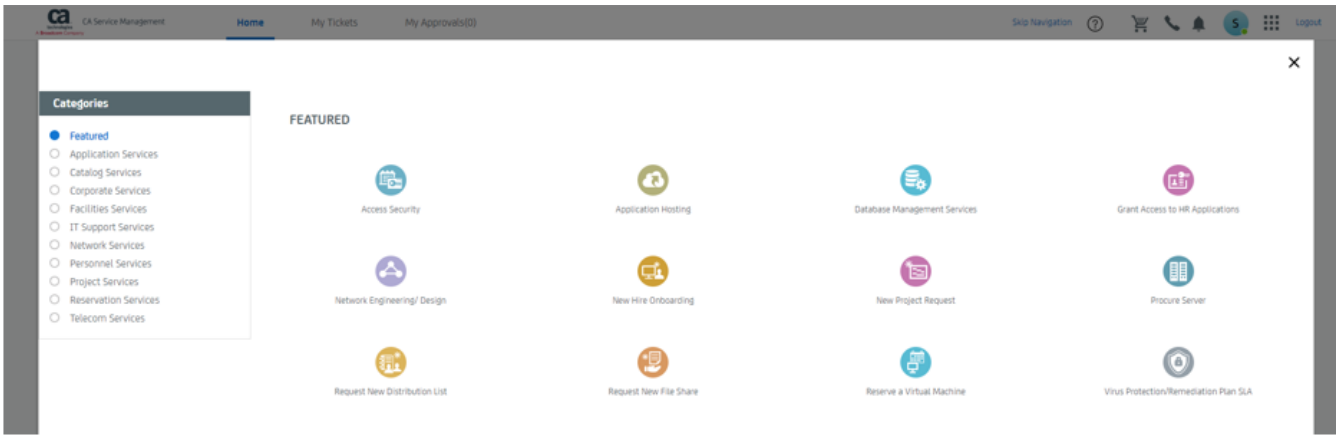


8. If you are unable to view the catalog offerings in the service point, perform the steps for the following sections from CA Service Catalog Troubleshooting:
   a. Validate CA Service Catalog Integration and Configurations
   b. Validate xFlow Interface and CA EEM Integration

NOTE
Rebuild index execution time will vary based on the data in the MDB database.

9. Run the pdm_es_rebuildindex.bat from command prompt on search server installed machine (testq-xflow-new).
10. Restart CA Service Desk Manager, xFlow Interface, and CA Service Catalog services.
11. Access the xFlow Interface application and validate the catalog offerings from service point application.

    NOTE
    In Service Point, catalog offerings are now with Policy driven approval as we don't have integration with Automic or Process Automation.
12. Follow the steps mentioned in Use Policies to Manage Requests to enable in service catalog offerings.



13. After successful Integration with CA SDM, xFlow Interface, and CA Service Catalog, take snapshots of all the servers.

CA Service Catalog Customizations

To maintain common file store location or setup custom shared location for shared files across all the catalog nodes, see Setup a Single Location for Shared Files.

- Set Up a Custom Location for Shared Files
- Set Up Single Location for Shared Files

Configure F5 Load Balancer for CA Service Catalog

To configure F5 Load Balancer with CA Service Catalog, see Configure F5 Load Balancer for CA Service Catalog Server. After enabling F5 load-balancer, update CA Service Catalog F5 Load balancer URL link in CA SDM. Perform the following steps:

1. Login to CA Service Desk Manager.
2. Navigate to **Administration**,**Options Manager**, CA Service Catalog, **casc_endpoint**.
3. Restart the CA SDM, xFlow Interface, and CA Catalog services.

Integrate CA Service Catalog with CA Automic

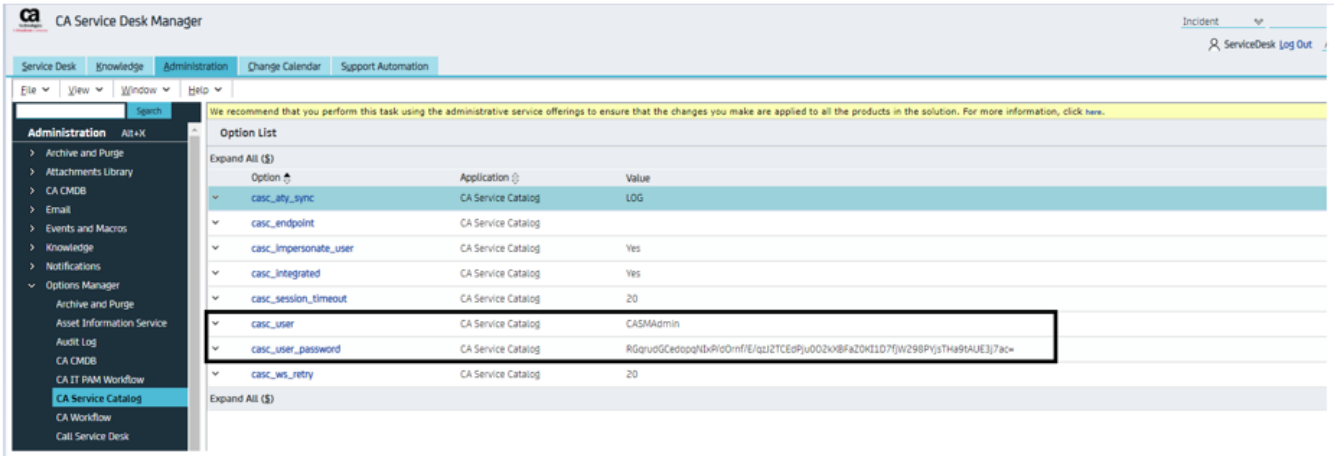
To integrate CA Service Catalog and CA Service Automic, see Automic Service Orchestration and refer the section "**Parameters**, **Automic Automation**".
After successful completion of the above steps, CA Service Catalog is integrated with Automic.

Reconfigure CA Service Catalog and CA SDM with NTLM and LDAP

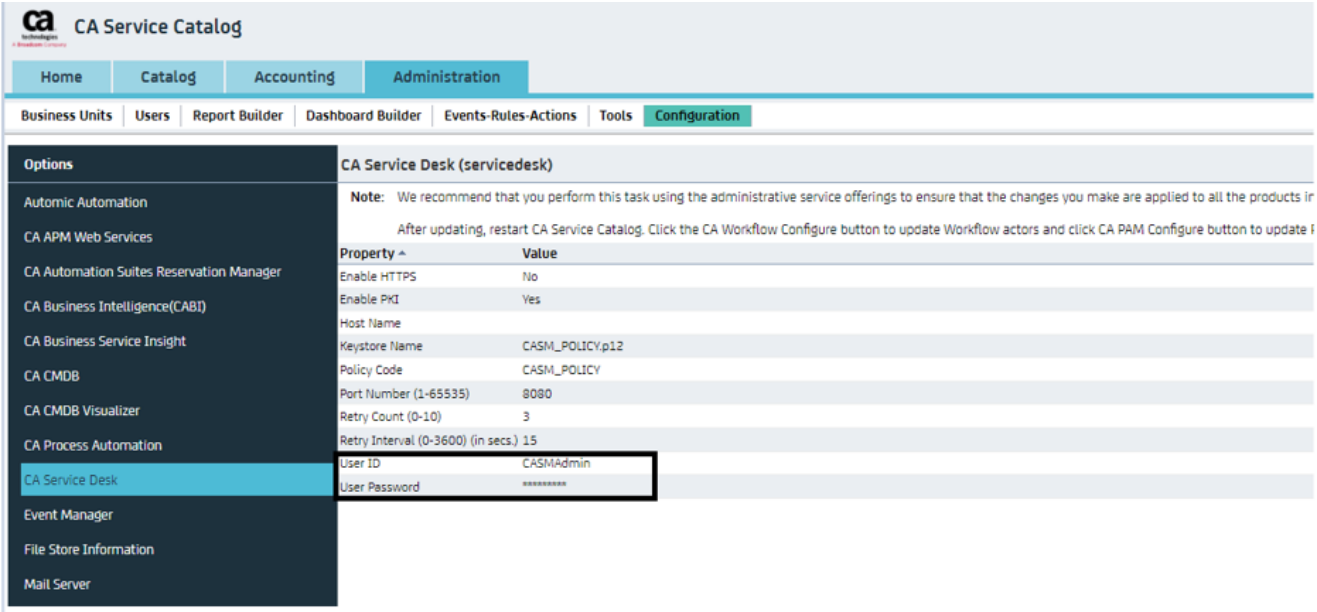
Perform the following steps:

1. Enable NTLM with SDM, xFlow Interface, and CA Service Catalog.
2. Enable LDAP in CA EEM.

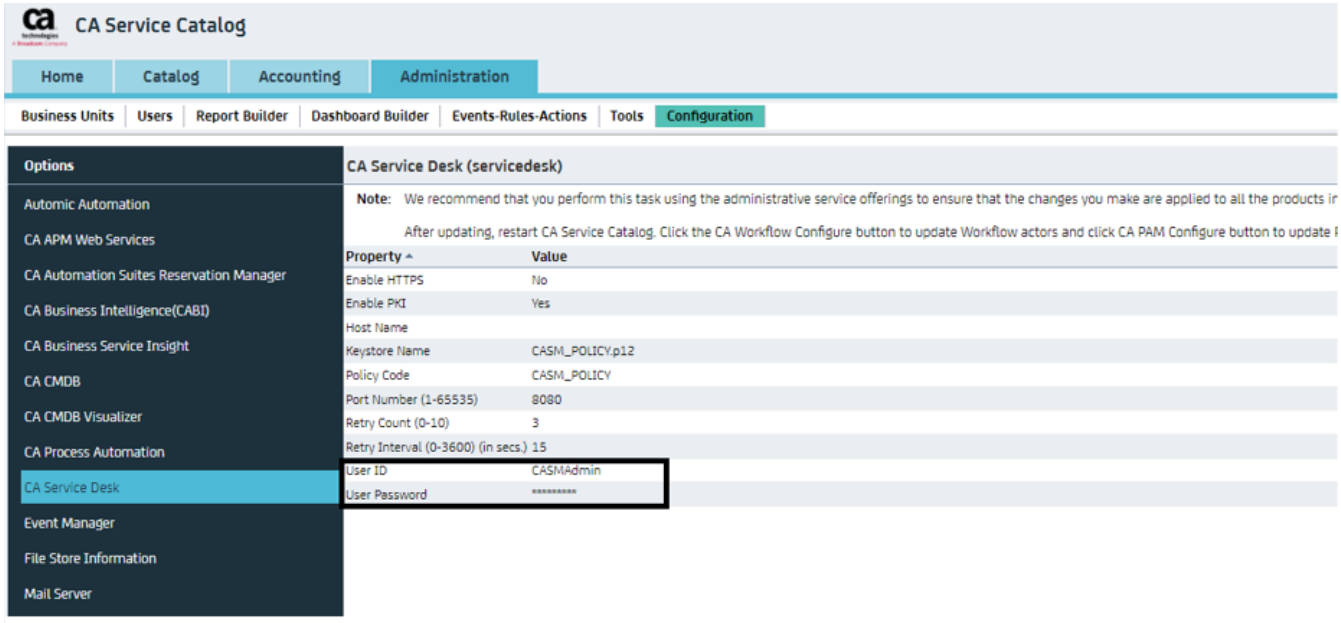
3. Update the Username and Password of LDAP account for Service Catalog under SDM Options Manager. Restart the SDM services in the all machines (BG, Standby and All application Server Servers).

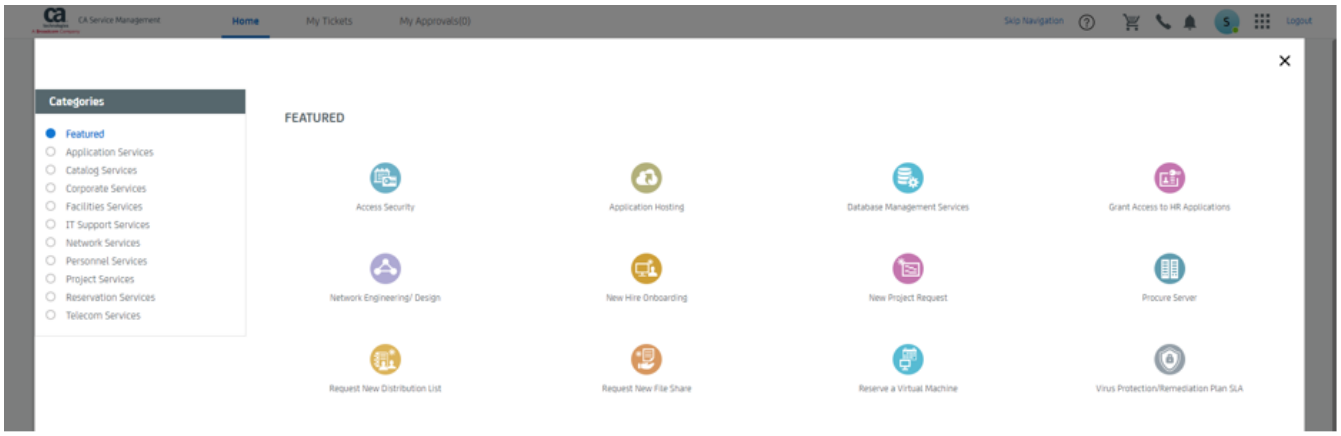


4. Restart xFlow Interface Analyst Services.
5. Update username and password for LDAP in CA SDM, xFlow Interface, and CA Service Catalog.



6. Run the **pdm_es_rebuildindex.bat** from the search server installed machine (**testqa-xflow-new**).
7. Try to access xFlow Interface and validate the catalog offerings from Service Point.





8. After successful Integration and configuration of NTLM and LDAP with CA SDM, xFlow Interface, and CA Service Catalog, take snapshots of all above servers.

Troubleshooting

Chrome Browser shows errors in Service Point with NTLM, Please refer below link. For more information, see Chrome Browser Errors.

Administering

Common Administration

Managing Tenants

Managing Roles

Managing Users

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CA Service Desk Manager

Managing Servers

Configure the Mailbox

Manage Multi-tenancy

More...

CA Service Catalog

Configuring Business Units

Enable External Authentication

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CA Asset Portfolio Management

Managing Roles and Users

Import Data

Configure the User Interface

More...

Administering CA Service Management

CA Service Management provides the ability to perform several administrative tasks that are common to the integrated products from a single interface. Earlier, it was required to manage tenants, users, roles, and other configurations such as CA EEM integration from the respective products' interfaces. However, with CA Service Management, you can manage all the common administrative tasks from the same interface.

Each administrative and configuration task is a service offering that you can access from CA Service Catalog. The configurations you make are applied to all the installed products of the solution.

As an Administrator, if you want to move MDB from the production to a pre-production environment, you can move the database from the source system and configure it to work on the target system. For more information, see Moving the CA MDB Data from the Source to the Target Systems

WARNING

Any “modifications” or “adaptions” or “configurations” that are done administratively through the interface (web browser, command-line, Web Screen Painter) are “supported”, meaning CA Support can assist with

basic suggestions and trouble-shooting. CA Support does not do the changes for the customer, they are the customer's responsibility. For example, adding a field to a table and putting the field on a form through Web Screen Painter is a fully supported "modification". Another example, installing or uninstalling a feature through the Options Manager administration is a fully supported "configuration".

Anything to do with SPEL code, Java scripting (or any language scripting), or a customer-specific change to the underlying base code-line (done by CA Services or a Partner), is not supported by CA Support. The customer can do these things, but the customer is responsible for the support, maintenance, and trouble-shooting when things go wrong. If such "customizations" affect expected out-of-the-box behavior, CA Support will ask the customer to remove the customizations and see if the behavior persists.

NOTE

The common administration service offerings are available only when you install CA Service Catalog as part of the solution.

You can perform the following tasks using the service offerings:

- Manage tenants
  Create a new tenant and map tenant structures across integrated products to specify a single tenant structure.
- Manage roles
  Create service management roles and map roles across integrated products.
- Manage users
  Add existing users to the solution; create new users and assign roles to the users.
- Configure LDAP server settings
  Add LDAP servers to the solution to import and synchronize contacts from LDAP user records.
- Configure common components
  Manage component settings that are required to integrate CA Service Management products. Components that you can manage include CA EEM, CA Business Intelligence, CA Process Automation, and mail server.
- Manage product integrations
  Manage the settings of integrated products in the solution.

To view a list of available administrative service offerings and learn how to use them, see Administrative Service Offerings.

When to Use the Solution Administration Capability

It is important to understand the scenarios in which you can leverage the solution administration capability:

Scenario 1: When you install a single product

If you install CA Asset Portfolio Management or CA Service Desk Manager, the common administration service offerings are not applicable. You must perform any administrative tasks through the respective product's administration page.

However, if you install CA Service Catalog only, you can use the administrative service offerings to perform administrative tasks of CA Service Catalog.

Scenario 2: When you install a combination of products

- CA Service Desk Manager and CA Asset Portfolio Management
  The common administration service offerings are available only when you install CA Service Catalog. In this scenario, you must use the respective product's administration pages to perform any administrative tasks. For example, to create a user you must create separate users in CA Service Desk Manager and CA Asset Portfolio Management.
- CA Service Desk Manager and CA Service Catalog (or) CA Asset Portfolio Management and CA Service Catalog
  You can use the administrative service offerings to perform administrative tasks.
- CA Service Desk Manager, CA Asset Portfolio Management, and CA Service Catalog
  You can use the administrative service offerings to perform administrative tasks.

Administrative Service Offerings

Administrative service offerings let you perform administrative tasks that are common to the products in the solution from a single location. Each service offering has a life cycle and has several phases. By default, the administrative service offerings life cycle has no approvals. To understand more about the request life cycle, see Request Life Cycle.

The table below lists the administrative service offerings and the tasks you can perform:

Service Offering	Task Description
Enable/Disable Multi-tenancy	Enable or disable tenancy management.
Add Tenant	Create a new tenant to access the capabilities of the solution.
Map Tenants	Map the existing tenant structures across integrated products to create a single logical tenant structure.
Manage Tenants	View and update tenants in your organization.
Add Role	Add a role that specifies permissions across products in the solution.
Manage Roles	Search and update CA Service Management roles to change what the user sees after logging in to the product. Map the CA Service Management roles to appropriate roles in integrated products.
Add User	Add new users to the solution and assign roles to the users.
Manage Users	Search and update details of users of the solution.
Add LDAP Server	Add an LDAP server to import a list of users from an external user store.
Synchronize Users	Automatically create users and assign roles to the users of a product in the solution, that earlier did not have any roles in other products
Manage LDAP Servers	View and update LDAP servers that you have integrated with solution.
Import Users from LDAP	Import a list of users from an external user store such as an active directory.
Configure Components	Manage settings of CA EEM, CA Business Intelligence, CA Process Automation, and mail server.
Manage Integrations	Manage the settings of integrated products in the solution.

Manage Tenants

This article contains the following topics:

Multi-tenancy is the ability for multiple independent tenants (and their users) to share a single implementation of CA Service Management. Through CA Service Management, you can create and manage the tenants of products in the solution from a single location. As part of tenancy management, you can perform the following tasks by using the administrative service offerings:

- Create new tenants.
- Map tenancy structures across integrated products.
- Search for a list of available tenants.
- Update tenant details.

Enable or Disable Multi-tenancy

To mange tenants in the solution, you must enable multi-tenancy. You can disable multi-tenancy when you have no tenants in the solution or when you do not want to manage tenants.

NOTE

When you disable multi-tenancy, tenant related data is available to everyone who has access to the application.

Create Tenant

You can define as many tenants as required to manage multiple separate enterprises that provide support to clients. When you create a new tenant, the tenant views the CA Service Management implementation as solely for its own use and cannot update or view another tenant's data.

View and Update Tenants

You can view a list of all the tenants that you manage in your organization. When required, you can make changes to tenant details. For example, you can change the terms of usage of a tenant to meet the new terms. If you no longer want to manage the tenant, you can inactivate it.

NOTE

You cannot update the parent details of the tenant using the administrative service offerings. To update the parent details, you must access the respective product's user interface.

Map Tenancy Structures across Products in the Solution

As a solution Administrator, if you installed CA Service Management with a combination of products, you must map tenancy structures across the products to create a single tenant structure. If you already have more than one product in your environment and have implemented multi-tenancy, you can leverage your existing tenancy data for performing tenant management from a single location. It is possible that you have named your tenants differently in different products. So, as a first step, you map the tenants in different products and specify how to match them.

For example, *Tenant1* in CA SDM corresponds to *BusinessUnit3* in CA Service Catalog and *Tenant4* in CA APM. Hence, you specify the mapping information to simplify your tenancy management tasks. Once you perform the mapping, you can manage tenants from a single location.

Follow these steps:

1. Step 1: Know When to Map Tenancy Structures.
2. Step 2: Review Your Existing Tenancy Structures.
3. Step 3: Understand Default and Customized Tenancy Mapping.
4. Step 4: Complete the Tenancy Mapping.

Know When to Map Tenancy Structures

It is important to map tenancy structures in the following scenarios:

- You upgraded older versions of your products to CA Service Management. For example, you upgraded CA SDM 12.9 and CA Service Catalog 12.9 to CA Service Management.
- You first installed CA Service Management, but not all the products. However, you choose to integrate another product. For example, you initially installed CA SDM and CA APM as part of the solution. At a later point of time, you decided to install CA Service Catalog.

Review Your Existing Tenancy Structures

Before you perform any of the tenancy management tasks, it is important to review your existing tenant structures in each of the products you installed.

Follow these steps:

1. To view the tenants and the tenant structure in CA SDM, complete the following steps:
   a. Log in to CA SDM as an Administrator
   b. On the Administration tab, select Security and Role Management.
   c. Click the tenant to view the tenant information
   For more information, click Multi-tenancy in CA Service Desk Manager.
2. To view the tenants and the tenant structure in CA APM, complete the following steps:
   a. Log in CA APM as an Administrator.
   b. Click Administration, Tenancy Management.
   The Multi-Tenancy Administration page appears.
   c. On the left, click Tenant.
   d. Click Go to view all the tenants.
   e. Click the tenant to view the tenant information.
   For more information, click Implementing Multi-tenancy.
3. To view the tenants and the tenant structure in CA Service Catalog, complete the following steps:
   a. Log in to CA Service Catalog as an Administrator.
   b. Select Administration, Business Units.
   c. Expand the business unit tree and find the business unit that you want.
   d. Click a business unit to view the business unit information.
   For more information, click Multi-tenancy in CA Service Catalog.

Understand Default and Custom Tenancy Mapping

When you map tenancy structures across the products in the solution, you can choose to automatically map tenants or map them manually. You must understand the different mapping processes and identify the one that better suits your requirements.

Default Tenancy Mapping

In the automatic tenant mapping process, you can choose to automatically map your tenancy structures. Tenant structures are mapped as follows:

- Service Providers of the products are first mapped to each other.
- Based on the names and database IDs of the tenants, each tenant is mapped to tenants in the other product.
- If the number of tenants at a level are different in the products, a new tenant is created. For example, at level 2, CA SDM has four tenants but CA Service Catalog has five tenants. A new tenant is created and is mapped to the fifth tenant in CA Service Catalog.
- Tenants are mapped only if they are at the same level in the products. For example, a first-level tenant in CA Service Desk Manager is mapped only to first-level business unit in CA Service Catalog.

WARNING

When you choose to map tenant structures automatically, the mapping is successful only if all the products you upgraded to CA Service Management have the same database. If the products are installed on different databases, tenant information on databases other than the CA Service Management database is not migrated. For example, you installed CA SDM 12.9 and CA Service Catalog 12.9 on *database_server1* and *database_server2* respectively. While upgrading, you installed the solution on *database_server1*. Hence, tenant information on *database_server2* is not considered.

Custom Tenancy Mapping

You can map tenancy structures manually to meet your business requirements. Map your tenancy structures using the following rules:

- Do not map tenants that are at different levels. For example, do not map a seventh-level business unit in CA Service Desk Manager to a third-level tenant in CA Service Catalog.
- Do not map the same tenant more than once.
- Do not map a child without mapping all its parents. Map the parents from the tenant level directly above the child through the level directly under the Service Provider tenant.
- Verify that tenant names are unique across all products. For example, if you have a tenant named AAA in CA Service Desk Manager and CA Service Catalog, rename at least one of them to a unique, meaningful name.
- Verify that CA Service Catalog has no duplicate tenant names within its own tenant structure.

Complete the Tenancy Mapping

The Map Tenants service offering lets you perform the tenancy mapping.

Follow these steps:

1. In the Home page, click Request a Service.
   NOTE
   You can access the service offerings from the CA Service Catalog user interface.
   The Request Page Appears.
2. Under Categories, click Service Management Administration.
3. Click Map Tenants.
4. To map tenancy structures automatically, complete the following steps:
   a. Select Default Tenant Mapping.
   b. Click Submit.
5. To map tenancy structures manually, complete the following topics:
   a. Select Custom Tenant Mapping.
   b. Perform the mapping to meet your requirements and click Update Mapped Tenants.
   c. Click Submit.

Manage Service Management Roles

This article contains the following topics:

User Role defines the tasks that a user typically performs. You define roles to control security and user interface navigation. Each role defines a focused view of the system by exposing only the functionality necessary for users in that role.

A user's default role determines the system view that is presented upon login. Users with multiple role assignments can switch from one role to another to see different views of the system without having to log out and log back in again.

CA Service Management lets you define service management roles that span across multiple products in the solution. A service management role is group of product roles that lets you assign multiple product roles to user at once. In the solution, you manage several service management roles. For example, System Administrator is a service management role. The System Administrator is responsible for managing configurations and administering roles and users of the individual products in the solution. Earlier, the role of an administrator was different in different products in the solution. Now, through the solution, you create a System Administrator service management role that defines the role of an administrator without having to create the roles separately in each of the products.

You can create and manage roles of integrated solutions from a single location. As part of role management, you can perform the following tasks by using the administrative service offerings:

- Define service management roles.
- Map the service management roles to appropriate roles in integrated products.
- Search for roles.
- Update role details.

Define a Role

You can create a service management role to meet your business requirements.

When you create a service management role, identify the corresponding roles in the integrated products. The integrated product role is either a predefined role or a role that you manually created. For example, a service management role you create must define the following:

- Name of the service management role: Business User
- Tasks performed by the role: Requests services, reports issues, and collaborates for self-help.
- Role in CA SDM: Employee, Customer
- Role in CA Service Catalog: End User
- Role in CA APM: Default User

NOTE

Every service management role you create must have a corresponding role in at least one of the integrated products.

The solution provides several predefined service management roles. For more information on predefined service management roles and the corresponding integrated product roles, see Predefined Service Management Roles.

To see predefined roles in the each of the installed products, see the following information:

- Predefined roles in CA SDM
- Predefined roles in CA Service Catalog
- Predefined roles in CA APM

Predefined Service Management Roles

The solution provides several predefined service management roles. When you add a new role from predefined service management roles, the solution automatically identifies the corresponding roles in the integrated products.

The table details the predefined service management roles:

Service Management Role	Description	Roles in other products
System Administrator	System Administrators are responsible for managing configurations and administering roles and users of the individual products in the solution.	• CA SDM role: System Administrator • CA Service Catalog role: Administrator • CA APM role: System Administrator
Business User	Business users request services, report issues, and collaborate for self-help for issues related to IT.	• CA SDM role: Employee, Customer • CA Service Catalog role: End User • CA APM role: Default User
Approver	Approvers manage the change order process, but typically not the analysts who work on change order tickets.	• CA SDM role: Process Manager • CA Service Catalog role: Request Manager • CA APM role: NA

Service Catalog Administrator	Service Catalog Administrators create, define, and manage services for a specific tenant or business unit.	• CA SDM role: NA • CA Service Catalog role: Catalog Administrator • CA APM role: NA
Service Desk Administrator	Service Desk Administrators administer CA Service Desk Manager.	• CA SDM role: Administrator • CA Service Catalog role: NA • CA APM role: NA
L1 Analyst	L1 Analysts provide first-line support within your organization.	• CA SDM role: Service Desk Staff • CA Service Catalog role: Request Manager • CA APM role: Default User
L2 Analyst	L2 Analysts provide second-line support within your organization, which requires more advanced subject matter expertise.	• CA SDM role: IT Staff • CA Service Catalog role: Request Manager • CA APM role: Default User
Configuration Analyst	Configuration Analysts perform tasks within the configuration item life cycle process and second-line CMDB support within your organization.	• CA SDM role: CMDB Analyst • CA Service Catalog role: Request Manager • CA APM role: Asset Technician
Asset Manager	Asset Managers manage asset life cycle and asset fulfillment.	• CA SDM role: NA • CA Service Catalog role: Request Manager • CA APM role: Asset Fulfiller
Asset Management Administrator	Asset Management Administrators administer CA IT Asset Manager.	• CA SDM role: NA • CA Service Catalog role: NA • CA APM role: System Administrator
Service Owner	Service Owners are responsible for managing a specific IT service.	• CA SDM role: CMDB Administrator • CA Service Catalog role: Catalog Administrator • CA APM role: NA
Executive User	Executive users are decision makers, who would be primarily interested in metrics to understand performance and efficiency. The users must have view-only access to reports and dashboards.	• CA SDM role: Employee • CA Service Catalog role: End User • CA APM role: Default User

View and Update Roles

You can search for the available roles in the solution. When required, you can make changes to the role details. For example, you can change the name of the service management role.

You can also map the service management roles to appropriate roles in integrated products. For example, you first created a service management role with the following role definitions:

- Role in CA SDM: Employee, Customer
- Role in CA Service Catalog: End User
- Role in CA APM: Default User

However, at a later of point of time you decided to change the role in CA APM to Asset Technician.

Follow these steps:

1. You can access the service offerings from the CA Service Catalog user interface.

2. In the Home page, click Request a Service.
   The Request Page Appears.
3. Under Categories, click Service Management Administration.
4. Click Manage Roles.
5. Search for the list of available roles.
6. Select the role that you want to update and click Update.
7. Update the role details and click Update.
8. Click Submit.

Manage Users

This article contains the following topics:

An important part of CA Service Management administration is to define the users accessing the solution. You must establish user security when you add new users to the product and assign a user ID and password. CA Service Management lets you create and manage users of integrated products from a single location. As part of user management, you can perform the following tasks by using the various administrative service offerings:

- Define users.
- Import user details from LDAP.
- Assign roles to users.
- Update user details.

Create a User

You can add a new user and define solution access rights to the user. When you create a user, the user details are available across the integrated products. For example, if you installed CA Service Desk Manager and CA Service Catalog, you can create a user and the user information is available in both the products.

When you add a new user, assign a role to the user. You can choose any predefined service management role or a custom role service management that you created.

If you have configured LDAP servers, you can also import the list of users. For more information, see Import Users Details from LDAP.

Import User Details from LDAP

You can import a list of users from an external user store such as an active directory. Importing users helps you save time when defining your users and makes it easy to synchronize contacts with network user data.

You can import user details from only those LDAP servers that you configure with the solution. Therefore, before you import user details from an LDAP server, ensure that you add the LDAP server to the solution. For more information, see Add LDAP Servers.

> **NOTE**
>
> To ensure user authentication, the LDAP server from which you import user details must be integrated with CA EEM.

When you add or create user details in an active directory, ensure that the you review the user name policies (such as special characters, mandatory parameters) of all the integrated products in the solution and specify the appropriate details.

Synchronize User Details across Integrated Products

You can synchronize users in the following scenarios:

- You upgraded older versions of your products to CA Service Management. For example, you upgraded CA SDM 12.9 and CA Service Catalog 12.9 to CA Service Management.
- You first installed CA Service Management, but not all the products. However, you choose to integrate another product. For example, you initially installed CA SDM and CA APM as part of the solution. At a later point of time, you decided to install CA Service Catalog.

In both the scenarios, the users of one product do not have a role assigned in other products of the solution. To ensure, that the every user has a role in the solution, you must synchronize the users.

When you use the synchronize contacts service offering, users of a product in the solution, that earlier did not have a role in other products, are automatically assigned a role. To assign a role, the service offering performs the following tasks in the order mentioned below:

- Identify the current role of the user in the product. For example, the role of User1 in CA SDM is Employee.
- Determine the service management role that the user's role corresponds to. For example, Employee in CA SDM corresponds to Business User service management role.
- Identify the corresponding role in the other products of the solution. For example, a Business User has the role of End User in CA Service Catalog, and Default User in CA APM.
- Search for the user details in the other integrated products. If the user does not exist, create the user. For example, if User1 details are not available in CA APM, User1 is created.
- Assign the respective product roles to the user.

If a user doesn't have any role assigned, you must specify the default role to be assigned to the user. For example, you imported user details from an LDAP server, but did not assign roles to the users. You specify the default role to be assigned to the user and then synchronize the users.

NOTE

You can assign only a service management role as a default role.

View and Update User Details

You can search for the list of users and update the details, when required. For example, you initially imported users from an LDAP server and later want to change the email id of a few users. You can also inactivate a user when you do not manage the user or the user is no longer part of your organization.

Follow these steps:

1. You can access the service offerings from the CA Service Catalog user interface.
2. In the Home page, click Request a Service.
   The Request Page Appears.
3. Under Categories, click Service Management Administration.
4. Click Manage Users.
5. Search for the list of available users.
6. Select the user that you want to update and click Update.
7. Update the user details and click Submit.

Manage LDAP Servers

You can integrate multiple LDAP servers with CA Service Management. The administrative service offerings let you perform the following tasks:

- Add LDAP servers to the solution
- View and update LDAP server details, when required.

This article contains the following topics:

Add LDAP Server

CA Service Management allows you to import user details from an LDAP server. To import users from LDAP servers, you must first add the LDAP server to the solution. You can import users from only those LDAP servers that you added to the solution.

Follow these steps:

1. You can access the service offerings from the CA Service Catalog user interface.
2. In the Home page, click Request a Service.
   The Request Page Appears.
3. Under Categories, click Service Management Administration.
4. Click Add LDAP Server.
5. Specify the details of the LDAP server and click Submit

Manage LDAP Servers

You can view and modify the LDAP servers integrated with the solution. When you no longer want an LDAP server integrated with the solution, you can inactivate it.

Follow these steps:

1. You can access the service offerings from the CA Service Catalog user interface.
2. In the Home page, click Request a Service.
   The Request Page Appears.
3. Under Categories, click Service Management Administration.
4. Click Manage LDAP Servers.
5. Search for the list of available LDAP servers.
6. Select the LDAP server that you want to update and click Update.
7. Update the LDAP server details and click Submit.

For more information on how to manage LDAP servers using CA SDM, see How to integrate CA SDM with LDAP.

Configure Common Components

This article contains the following topics:

CA Common Components are the components that are shared by all the products in the solution and have specific functions to perform. For example, you can configure CA Business Intelligence to generate reports. You can install the components through the CA Service Management solution or install them separately when required.

You can manage the settings of the following common components:

- CA Process Automation
- CA Business Intelligence
- CA EEM

When to configure the Common Components

You configure the CA common components in the following scenarios:

- You upgraded older versions of your products to CA Service Management and the older versions of the products already use the components.
  In this scenario, you must specify the details of the installed common components. If you integrated each product with different instances of common components, you must identify the instance that you want to integrate with the solution. For example, you integrated CA SDM with CA EEM on *server1* but integrated CA APM with CA EEM on *server2*. You must determine the CA EEM server instance that you want to integrate with the solution.
- You installed CA Service Management for the first time and installed the common components separately.
  In this scenario, you must specify the details of the installed common components to integrate with the solution.
- You already specified the common component details but want to change the settings.

For example, you installed CA Process Automation on *server1* and it is scheduled for maintenance soon. To avoid any downtime, you install CA Process Automation on *server2* and integrate the solution with *server2*.

WARNING

After you configure the common components, restart the CA SDM and CA Service Catalog services. If you have installed CA APM, you must recycle the Application Pool.

How to Configure the Common Components

Follow these steps:

1. You can access the service offerings from the CA Service Catalog user interface.
2. In the Home page, click Request a Service.
   The Request Page Appears.
3. Under Categories, click Service Management Administration.
4. Click Configure Components.
5. To configure the settings of a component, select the respective tab.
   For example, to update CA EEM settings, click the CA EEM tab.
6. Specify the details and click Submit.
   Note: When you change CA EEM settings, you may experience issues with logging in to the products of the solution. To resolve these issues, you must manually integrate CA EEM with the individual products.

Manage Product Integrations

You can manage the settings of integrated products in the solution. After you integrated products in the solution, you can change several configuration settings of the integrated products. Based on the new settings, the products are automatically integrated. For example, you changed the host name of the server on which you installed CA SDM. You specify the new host name using the service offering and integration is performed automatically.

Follow these steps:

1. You can access the service offerings from the CA Service Catalog user interface.
2. In the Home page, click Request a Service.
   The Request Page Appears.
3. Under Categories, click Service Management Administration.
4. Click Product Integrations.
5. To configure the settings of a product, select the respective tab.
   For example, to update CA SDM settings, click the CA Service Desk Manager tab.
6. Specify the settings and click Submit.

7. (Optional) Any changes you make the CA SDM details are not updated in the *ca_registration* table in the MDB. If necessary, you must update the details in the table manually.
8. (Optional) If CA Process Automation is integrated with CA Service Catalog, any changes you make to the CA Service Catalog details are not updated in the CA SDM and CA Service Catalog datasets in CA Process Automation. You manually update the changes in CA Process Automation.

Business Value Dashboards

A Business value dashboard is a real-time graphical representation of an organization's Key Performance Indicators (KPI) that enables decision makers such as IT managers to make informed decisions. The dashboards represent detailed information about the following:

- Service support metrics such as the categories of incidents or requests, the associated volume, cost, and time spent for incident resolution.
- Operational and functional efficiency information such as, how support groups are occupied, information on ticket transfer from one group to another and the impact on time and cost.

Each dashboard is a service offering that is exposed through CA Service Catalog and accessible through the CA Business Intelligence interface.

As an Administrator, you are responsible for the following:

- Configuring the business value dashboard reports
- Defining the users and user access permissions to the dashboards
- Troubleshooting any issues related to business value dashboards

Complete the following steps to implement business value dashboards:

1. Know When the Business Value Dashboards are Available
2. Review the Business Value Dashboards Prerequisites
3. Determine the Business Value Dashboards User Permissions
4. Configure the Business Value Dashboard Reports
5. (Optional) Learn How to Schedule the Business Value Dashboard Reports
6. (Optional) Understand the Important Metrics

Know When the Business Value Dashboards are Available

It is important to understand the various scenarios when you can access the business value dashboards.

Installation Scenarios

The type of installation determines the type and availability of the dashboards.

Type of Installation	Dashboards Availability
New installation For example, as part of CA Service Management, you installed CA Service Catalog and CA SDM for the first time.	Dashboards are available automatically. No configuration is required.
Upgrade For example, you upgraded older releases of CA Service Catalog or CA SDM or both to release 14.1.	Dashboards are available automatically. No configuration is required.

You first installed CA Service Catalog and at a later point installed CA SDM.	Import the the CASM.biar file manually. The file is available at the following location: [ISO_ROOT\filestore\BOXI\biconfig] **Note:** Verify that CA Service Catalog and CA SDM BIAR files are already imported.

Products Installed

The dashboards and the comprising reports that are available to you are based on the products you installed as part of the CA Service Management solution.

Products Installed	Available Dashboards
CA Service Catalog only	Service Demand - Requests
CA Service Desk Manager only	Service Demand - Incidents Operational Effectiveness
CA Service Catalog and CA Service Desk Manager	Service Demand - Requests Service Demand - Incidents Service Demand Operational Effectiveness

Review the Business Value Dashboards Prerequisites

Before you use the dashboard reports, ensure that you considered the following prerequisites:

- Both CA Service Desk Manager and CA Service Catalog must use the same CA Business Intelligence server.
- Run the following queries on an Oracle MDB:

```
create or replace function MDBADMIN.left(str1 varchar2 :=NULL,num1 NUMBER)
RETURN VARCHAR2
as
ascii_chr varchar2(32767);
begin
ascii_chr:= substr(str1,1, num1);
return ascii_chr;
end;
Update report_labels set language_code ='en' where language_code='en ';
```

- Install the Arial Unicode MS font on the server where the BIAR file will be imported. Additionally, the same font must be available on the machine from where the reports will be viewed. The font files must be copied to the Windows font folder directory.
  Following code must be added to the fontalias.xml located in below location:
  <BusinessObjects Installation Path>\ SAP BusinessObjects Enterprise XI 4.0\win64_x64\fonts

```
<FONT NAME="Arial Unicode MS">
<FONTFAMILY PLATFORM="ttf" NAME="Arial Unicode MS">
<FONTATTRIBUTE BOLD="false" ITALIC="false" LOGICAL="Arial Unicode MS" PHYSICAL="ARIALUNI.TTF"/>
</FONTFAMILY>
<FONTFAMILY PLATFORM="win" NAME="Arial Unicode MS"/>
<FONTFAMILY PLATFORM="java" NAME=" 'Arial Unicode MS', 'Arial Unicode MS'"/>
<FONTFAMILY PLATFORM="html" NAME=" 'Arial Unicode MS', 'Arial Unicode MS'"/>
</FONT>
```

- Enable the Key Performance Indicators (KPIs) for Operational Effectiveness Dashboard.
  Enabling KPI for Operational Effectives Business Objects Dashboard is mandatory to retrieve the Group information when CA SDM incidents are transferred to multiple Groups for ticket resolution. Consider the following:

To show Support Groups Productivity and Functional Escalations in Operational Effectiveness dashboard, navigate to the CA SDM Administration tab, Options Manager, KPI and ensure that the status of kpi_ticket_data_table is installed.

- Configure the NX_KPI_TICKET_DATA_TABLE_DELAY variable to specify how frequently the incident data (for example, incident transfer) is updated in the database. The recommended value is 900 seconds.
  To configure the variable, complete the following steps in the order mentioned below:
  – Navigate to [SDM Root folder].
  – Open the NX.env file in notepad.
  – Search for @NX_KPI_TICKET_DATA_TABLE_DELAY and specify the value.
  For example, @NX_KPI_TICKET_DATA_TABLE_DELAY = 900.
- Make sure that every analyst is associated with a support group. Data related to an analyst with no support group is not displayed on business dashboard.
- Every time an analyst moves from one group to another, make sure to note down the analyst metrics in the previous group. The business dashboards display any metrics related to the older group as belonging to the new group.

Determine the Business Value Dashboards User Permissions

The role and user permissions determine the users accessing the dashboard and the users configuring the dashboards. To provide permissions to any user to configure the reports or access the dashboard, you must add the user to the CA Service Management group.

Important! You must add the *CASMAdmin* user to the CA Service Management group. The CASMAdmin user has the privileges to access the solution dashboard and also run the reports.

Follow these steps:

1. Log in Central Management Console as an administrator.
   http://cabi-hostname:Port_number/BOE/CMC
2. Under Organize, click Users and Groups.
3. Click User List.
4. Right-click the user you want to add to the CA Service Management group and click Join Group.
5. Select CA ServiceManagement and move it from Available Group(s) to Destination Group(s).
6. Click Ok and save the changes.

Configure the Business Value Dashboard Reports

This article contains the following topics:

Scheduled and On-demand Reports

CA Service Management provides you the flexibility of generating reports, when required. The report generation is classified as Scheduled reporting and On-demand reporting.

Scheduled Reporting

You can use the scheduled reports when your organization requires reports to be generated at frequent intervals. Also, when you have a large data to generate reports from, you can use the scheduled reports. For example, a report is to be generated on a weekly basis. So, you use the scheduled reports and configure them to generate reports once in a week.

Note: We recommend that you use scheduled reporting when the data size is more than 10,000 records.

On-demand Reporting

You can use the On-demand reports to generate the business value dashboards when required or to process small data. For example, your organization does not require reports to be generated at frequent intervals. However, you need to generate a report whenever an executive user requires you to generate it.

Note: We recommend that you use scheduled reporting when the data size less than 10,000 records.

To know the data size, identify the count of the following tables in your MDB:

- *usm_subscription_detail* for the number of requests
- *call_req* for the number of incidents

Dashboards and the Associated Reports

Each dashboard has several reports and sub-reports associated that you can view and configure to meet your requirements. The reports are accessible through the BusinessObjects Central Management Console.

Follow these steps to access the reports associated with the dashboards:

1. Log in Central Management Console as an administrator.
   http://cabi-hostname:Port_number/BOE/CMC
2. Navigate to Folders, All Folders, CA Reports, CA Service Management.
3. Do one of the following:
   - Click View on Demand Dashboard to view the reports that you can generate on-demand.
   - Click Scheduled Dashboard to view or schedule reports as per your requirement.
   - Click Sub Reports to view the sub reports of Service Demand and Operational effectiveness dashboards.

Learn How to Schedule the Business Value Dashboard Reports

This articles explains how to schedule dashboard reports and tasks you must perform before and after scheduling the reports.

Note: Perform these tasks only if you or the end user access the dashboards from the Unified Self-Service user interface.

Complete the following:

Configure the Business Value Dashboard Service Offerings

By default, the dashboard service offerings represent data generated by on-demand reports. To display scheduled reports, you must perform several configurations on service offerings.

Complete the following steps:

- Identify the Report ID
- Configure the Service Offering

Identify the Report ID

Follow these steps:

1. Log in to Central Management Console as an administrator or a user with permissions to schedule report generation.
   http://cabi-hostname:Port number/BOE/CMC
2. Navigate to Folders, All Folders, CA Reports, CA Service Management, Scheduled Dashboard.
3. Right click the report that you want to schedule and select Properties.
4. From the ID, CUID field, make a note of the CUID of the report.
   For example, in **ID, CUID:17810, ATKoKrurIjlCqTvnya20W1s**, the CUID is ATKoKrurIjlCqTvnya20W1s.

Configure the Service Offering

Follow these steps:

1. Log in to CA Service Catalog as an Administrator.

2. Navigate to Catalog, Service Offerings, Offerings.
3. Expand Service Management Dashboards.
4. Select the service offering that represents the dashboard for which you want to schedule a report.
   For example, Operational Effectiveness.
5. Click the copy icon.
6. Create a copy of the service offering in the same folder and specify the name of the new service offering.
   For example, Operational Effectiveness - Scheduled Report.
7. Click the new service offering copy.
   In this example, click Operational Effectiveness - Scheduled Report.
8. In the Details tab, set the Availability date.
9. In the Permissions tab, enable permissions for the appropriate roles.
10. In the Definition tab, click Edit Service Option.
11. From the Form field, make a note of the Form name.
12. Navigate to Catalog, Forms.
13. Expand Forms, Service Management Dashboards Forms.
14. Click the form that you earlier noted and create a copy of the form as you created a copy of the service offering.
15. Click the new form and the click Script.
16. Search for the following statement. In this example, the form is the Operational Effectiveness form.

```
cabiURL = 'http://' + ca_fd.js.cabiHostName + ':' + ca_fd.js.cabiPort + '/BOE/OpenDocument/opendoc/
openDocument.jsp?iDocID=AbEmxu8mNftBjHQIMRu2uLs&sIDType=CUID&sReportMode=weblayout&token=' +
 rows[0].value;
```

17. Replace the iDocID variable with the appropriate CUID that you earlier noted.
    For example, replace AbEmxu8mNftBjHQIMRu2uLs with ATKoKrurIjlCqTvnya20W1s.
18. Click Save.
19. In the new service offering you created, update the Form name in the Definition tab to the new form name.
20. Save the changes.

Schedule the Business Value Dashboard Reports

Schedule a report when the date size is more than 10,000 records. It is important to configure both the reports and the associated sub-reports.

To know the data size, identify the count of the following tables in your MDB:

- *usm_subscription_detail* for the number of requests
- *call_req* for the number of incidents

Follow these steps:

1. Log in to Central Management Console as an administrator or a user with permissions to schedule report generation.
   http://cabi-hostname:Port_number/BOE/CMC
2. **To schedule the Service Demand-Incidents dashboard, complete the following steps:**
   a. Navigate to Folders, All Folders, CA Reports, CA Service Management, Sub Reports.
   b. Right-click Incident Cost Details, Incident Effort Details, Incident Volume Details and select Schedule.
   c. Configure the appropriate parameters and click Schedule.
   d. Navigate to Folders, CA Reports, CA Service Management, Scheduled Dashboard, Service Demand – Incidents.
   e. Right-click Service Demand- Incidents and click Schedule.
   f. Configure the appropriate parameters and click Schedule.
      Note: Ensure you that parameter values you configure for reports and their sub-reports are the same.
3. **To schedule the Service Demand-Request dashboard, complete the following steps:**
   a. Navigate to Folders, All Folders, CA Reports, CA Service Management, Sub Reports.

b. Right-click Request Volume/Cost for Services and select Schedule.
c. Configure the appropriate parameters and click Schedule.
d. Navigate to Folders, CA Reports, CA Service Management, Scheduled Dashboard, Service Demand – Requests.
e. Right-click Service Demand- Requests and select Schedule.
f. Configure the appropriate parameters and click Schedule.
Note: Ensure you that parameter values you configure for reports and their sub-reports are the same.

4. **To schedule the Service Demand dashboard, complete the following steps:**
   a. Navigate to Folders, All Folders, CA Reports, CA Service Management, Sub Reports, Service Demand Scheduling.
   b. Right-click on each of the reports and select Schedule.
   c. Configure the appropriate parameters and click Schedule.
   d. Navigate to Folders, CA Reports, CA Service Management, Scheduled Dashboard, Service Demand.
   e. Right-click Service Demand and select Schedule.
   f. Configure the appropriate parameters and click Schedule.
   Note: Ensure you that parameter values you configure for reports and their sub-reports are the same.
5. **To schedule the Operational Effectiveness dashboard, complete the following steps:**
   a. Navigate to Folders, All Folders, CA Reports, CA Service Management, Sub Reports.
   b. Right-click FCR For Incident Categories, Groups By Effort Incurred On Resolution Functions, MTTR for Incident Categories and select Schedule.
   c. Configure the appropriate parameters and click Schedule.
   d. Navigate to Folders, CA Reports, CA Service Management, Scheduled Dashboard, Operational Effectiveness.
   e. Right-click Operational Effectiveness and select Schedule.
   f. Configure the appropriate parameters and click Schedule.
   Note: Ensure you that parameter values you configure for reports and their sub-reports are the same.

Verify the Scheduled Dashboards

After you schedule the report generation, you must verify if the dashboards are created correctly. Each dashboard is a service offering that is exposed through CA Service Catalog.

Understand the Important Metrics

The three parameters related to request and incidents/categories that are significant to decision makers are cost, effort, and volume.

It is important to understand how these parameters are calculated:

Effort

Effort is the time total time spent on an incident resolution. The time spent may be by one user or multiple users in resolving an incident. The effort is calculated only when the users appropriately log in the time spent on an incident. It is important that your organization recommends time logging.

The **Time Spent** field in the incident details determines the time spent on the incident.

Volume

Volume is the total number of requests/incidents in a specific category. It is important that each incident/request has a specific category associated with it. For example, Hardware, Networking, Applications are categories.

Cost

Cost is a function of effort spent and the resource rate. It is also the cost of a service.

Each incident/request has a category associated and the total cost of a category is the sum of all costs of incidents/requests in a specific category.

- For an incident, cost is calculated as follows:
  Cost of an incident = Total time spent (in hours) by a resource * Rate (per hour) of the resource
  The following example describes how the cost parameter is calculated for an incident involving multiple resources.

Data	Analyst_1	Analyst_2
Resource Rate (per hour)	$ 100	$ 75
Time spent (in hours)	10	15

  Total cost = ($100 * 10) + ($75*15) = $ 2125
  Note: The **Cost** field in the **Contact Details** page determines the resource rate.
- For a request, the cost is the cost field that you associate for a service offering.

Offline Reporting through Microsoft SQL Server Database Replication

CA Service Desk Manager lets you generate reports that help you analyze critical information. CA SDM accesses the database to fetch information whenever you create a report. When your database contains large volumes of data, the data retrieval process impacts the performance of your application. An effective way to reduce performance issues is to replicate or create an offline version of your database. While users continue to access the application to perform day-to-day tasks, the application accesses the replicated or offline database for any report generation activity.

NOTE

The database replication process is applicable only if you use Microsoft SQL Server in your environment.

This article contains the following topics:

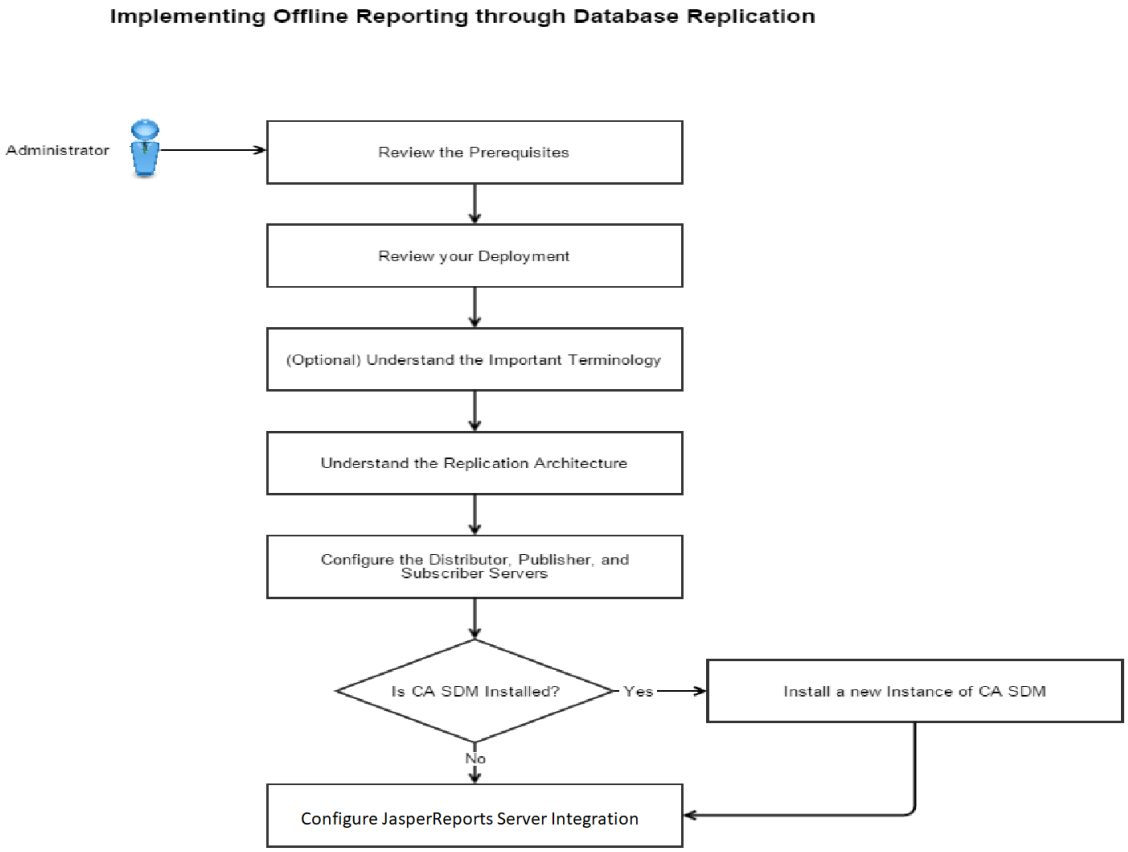
The Environment

You can implement offline reporting in the following environment:

- **Products installed**: CA Service Desk Manager 17.3.
- **Database:** Supported versions of Microsoft SQL Server. For more information on the supported versions, see Supportability Matrix.

The following diagram illustrates how a solution administrator implements database replication for offline reporting:

Figure 4: Implementing Offline Reporting through Database Replication

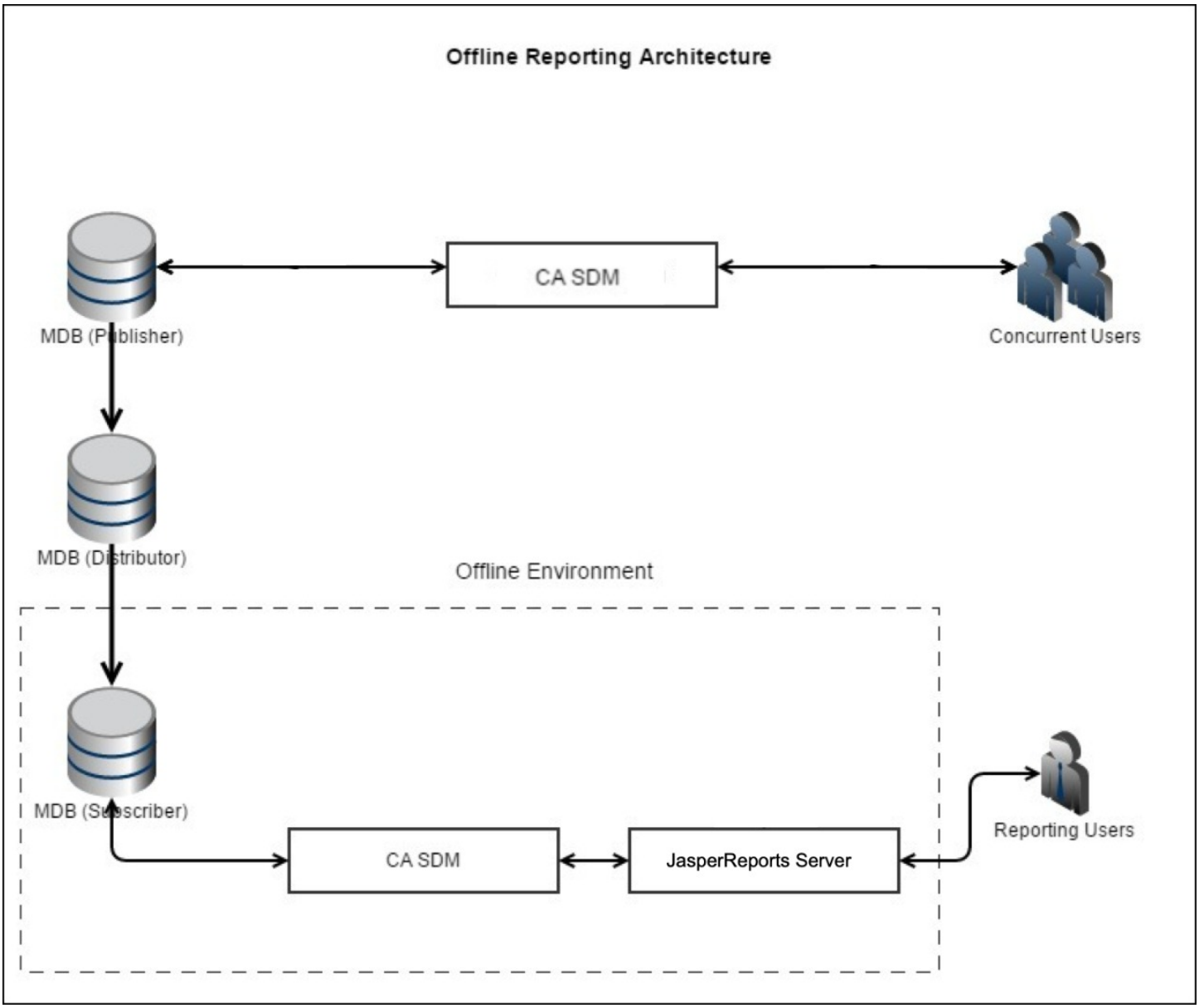


For more information about installing the Replication feature, see Microsoft SQL Server documentation.

The Architecture

The following diagram illustrates the architecture of a replicated database environment for offline reporting:

Figure 5: Offline Reporting Architecture



Benefits

Replicating the database through transactional replication has the following benefits:

- Incremental changes are propagated to the Subscribers as they occur.
- Report generation does not require any changes to be made by the Subscriber; thus, data need not be propagated back to the Publisher. In transactional replication, Subscribers are read-only and therefore do not affect the performance of your environment.
- Low latency between the time changes is made at the Publisher, and the changes arrive at the Subscriber.
- Effective management of high insert, update, and delete activity compared to other replication types.

NOTE

With support for transactional replication in CA Service Desk Manager, support for the merge replication process is deprecated. If you already implemented merge replication on CA Service Desk Manager, we recommend you disable it and implement transactional replication.

Best Practice

Scenario 1: New Installation of CA Service Desk Manager

If your environment follows the database replication process for the first time, you must install CA Service Desk Manager and enable transactional replication.

Scenario 2: Upgrade CA Service Desk Manager

If your environment follows the database replication process, you must disable the replication process (merge or transactional), upgrade the CA Service Desk Manager, and enable the transactional replication.

Scenario 3: Integration of CA Service Desk Manager Products

If your environment follows the database replication process, you must disable the replication process (merge or transactional), Integrate the CA Service Management products, and enable the transactional replication.

Glossary

The following terms are used frequently in the configuration information. Understand the meaning of the terms before you start configuring offline reporting:

Production Database

The production database refers to the database server that CA Service Desk Manager uses. This is the database from where you replicate another database for offline reporting.

A production database is typically the database server in a production environment.

Reporting Database

Reporting database refers to the database that you replicated from the production database. The reporting database is used only for offline reporting.

Publisher

The Publisher is a server that makes data available for replication to other servers. In addition to being the server where you specify the data to be replicated, the Publisher also detects the data that has changed and maintains information about all publications at that site.

The publisher is typically the production database server in the offline reporting scenario.

Distributor

The Distributor is a server that contains the distribution database and stores metadata, history data, and/or transactions. The Distributor can be a separate server from the Publisher (remote Distributor) or the same server as the Publisher (local Distributor).

Subscriber

A subscriber is a server that receives replicated data. Subscribers subscribe to publications, not individual articles within a publication, and they subscribe only to the publications they need, not necessarily all of the publications available from a Publisher.

Verify the Prerequisites

We recommend that you complete the checklist before you proceed.

1. Verify that CA Service Desk Manager 17.3 and above version is installed on the production and offline server.
2. You must have DBA privileges.
3. The following Windows services must be running. We recommend that you set the start mode to **Automatic**.
   - SQL Server
   - SQL Server Agent
   - SQL Server Browser
4. The SQL Server Agents must run on the publisher, distributor, and subscriber servers.
5. You can perform either of the following steps:
   a. **Use a domain account for every agent.** Add the domain account to the SQL Server logins and assign the sysadmin role. Set **Master** as the default database.
   b. **For each agent, create a local Windows account on the respective servers.** Add the Windows user account that you created to the SQL Server logins and assign the sysadmin role. Set **Master** as the default database.
6. Create a Snapshot Folder. Ensure that the Snapshot Folder is a network folder and not a local folder.
7. To implement replication, Microsoft mandates that every table is associated with a primary key. Hence, ensure any new table that you want to replicate has a primary key that is associated with it.
8. To connect to the database, the replication utility requires SQL server authentication-based login credentials. We recommend the default system admin account be used.

Back up and restore the existing MDB instance

Review the 3-step procedure to back up and restore the existing instance of MDB.

Step 1: Back up and Restore

Back up the production server database and restore it to the offline server. See information specific to CA Service Desk Manager on Moving the CA MDB Database on Microsoft SQL Server.

Step 2: Implement Replication

An effective way to reduce application performance issues is by replicating or creating an offline version of your database. The users can continue to access the application to perform day-to-day tasks, and the application accesses the replicated or offline database for any report generation activity. For more information, see How To Implement Replication.

Step 3: Configure CA SDM with offline MDB

Follow these steps:

1. Using a text editor, open the NX_ROOT\pdmconf\NX.env_nt.tpl file and perform the following steps:
   a. Use the following sample code to verify that auditing is commented out:

```
# Create audit log entry for field update.
! @NX_AUDIT_UPD=
# Create audit log entry for deletion.
! @NX_AUDIT_DEL=
```

   b. Add the following code and parameters at the end of the file:

```
@NX_EVENT_LOG_EXCLUDE=ALL
@NX_SESSION_LOG_EXCLUDE=ALL
#this is just a required comment
```

2. Using a text editor, open the NX_ROOT\samples\pdmconf\pdm_startup.tpl file.

3. Edit the file by changing the line *[default procset MAIN_PROCSET_REPLACE]* to *[default procset OFFLINE_REPORTING]*.
4. Save and close the file.
5. Configure CA SDM with replicated db and do not select “load-default’ data check box. To configure CA SDM run the following command on the command prompt.
   ```
   pdm_configure
   ```
6. Complete the following steps in Microsoft SQL Server Management Studio:
   a. Expand **Databases, MDB, Security, Users**.
   b. Right-click **mdbadmin** and select **Properties**.
   c. In the **Database User** window, next to **Default schema**, click the search option.
   d. In the **Select Schema** window, enter *[dbo]* and click **Browse**.
   e. In the **Browse for Objects** window, select the **mdbadmin** object and click **OK**.
   f. In the **Database User** window, select the following entries under **Role Members** and click **OK.**
   – • ams_group
   • db_ddladmin
   • db_owner
   • db_securityadmin
   • regadmin
   • service_desk_admin_group
   • service_desk_ro_group
   • usmgroup
   • workflow_admin_group
7. In the Windows command prompt, run the following command to view the status of the services:
   ```
   pdm_status
   ```
8. Configure JasperReports Server and integrate with the offline server (Offline MDB). JasperReports Server that you configured to access the production database must now access the reporting database. JasperReports Server does not access the production database directly. Configure JasperReports Server with CA SDM offline server to access the reporting database. For more information, see Integrate JasperReports Server 8.1.1 with CA Service Management.

   NOTE
   Access CA SDM reports from JasperReports Server.
9. To access the CA SDM reports, log in to JasperReports Server, create a user and add the contactUUID attribute.
   a. Log in to JasperReports Server.
      ```
      http://<machine_name>:< jaspereportsserver_port>/jasperserver-pro/login.html
      ```
   b. From the JasperReports Server home page, click **Manage** and select **Users**.
   c. From the left navigation select **casm_insights.**
   d. Click the **Add User** option.
   e. On the **Add User** form, fill in the fields.
   f. Click **Add User to casm_insights**.
   g. Select the newly created User.
   h. From the **Properties** pane, click **Edit**.
   i. From the **Attributes** tab, click **Add new attribute**.
   j. In the Name field, enter contactUUID.
   k. Click **OK** and **Save**.

   NOTE

   If multi-tenancy is enabled on the production instance of CA SDM, then manually install the same on the offline instance of CA SDM. For more information about allowing multi-tenancy, see How Multi-Tenancy Works.

Implement Replication

This article describes the process to implement replication.

> **TIP**
>
> Initiate the replication process during a planned maintenance window as it involves creating a snapshot of the database. While this process is in progress, access to your database is locked. The time period depends on the volume of data in the production database. For instance, time period for a new database is 5-6 minutes, whereas for a 112-GB database, it is 6 hours and 20 minutes.

Configure the Distributor, Publisher, and Subscriber Servers

You can configure the servers in one of the following ways:

- Run the Configuration Utility
- Manual Configuration

Perform the Validation Check

Perform the checks in Distributor/Publisher based on your configurations.

Follow these steps:

1. Navigate to Microsoft SQL Server Management Studio.
2. Expand **Databases,** and verify the new database "**Distribution**".
3. Expand **Security**, **Logins**, and verify the new login "**Distributor_admin**".
4. Expand **Server Objects**, **Linked Servers**, and verify the new linked server "**repl_distributor**".
5. Expand **SQL Server Agent, Jobs,** and verify the 6 new jobs that are created automatically.
6. Expand **Replication**, **Local Publications,** and verify the newly created publication.
7. Expand the **New Publication,** and verify the newly created subscription.
8. To confirm snapshot creation, right-click on the **Publication** and select **View Snapshot Agent Status.**

> **WARNING**
>
> Failure in the above steps indicates that the replication process is unsuccessful. In this case, the user must navigate to the log file created as part of replication and check for errors.

Run the Configuration Utility

This article explains the steps to download and run the configuration utility.

Select a Mode

You can either specify the details in the utility or perform a silent replication.

- Ensure you review the 'ReplicationInputData.csv' file for the interactive mode process to comprehend the required input values.
- For the silent replication process, ensure that you specify all the details in the 'ReplicationInputData.*csv'* file before you run the utility.

Download and run the configuration utility

>From the Broadcom Support portal, download the Configuration Utility.

Follow these steps:

1. Unzip the configuration utility file.
2. Run the *CreateReplication.bat* file.
3. Specify if you want to perform a silent replication.
   The replication process starts.
4. If you choose to specify the details in the utility, follow the on-screen instructions.
5. After you specify all the details, the replication process starts.
   The progress is displayed.
6. To understand whether the replication process was successful, view the *Replication.log* file.

Include/Exclude Tables for Replication

During the replication process, the configuration utility replicates only the tables that are part of CA Service Management. Tables that are related to other products are not replicated. For example, if you integrated CA SDM with JasperReports Server, the configuration utility replicates only CA SDM tables and not JasperReports Server tables. However, you can manually include these tables for replication.

Include a Table for Replication

Follow these steps:

1. In the configuration utility folder, open the **Creation Script** folder.
2. Open the *CreatePublication.sql* file using a text editor.
3. Copy the following code:

```
IF (EXISTS (SELECT * FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_NAME = 'Table_Name'))
BEGIN
exec sp_addarticle @publication = N'publication_mdb', @article = N'Table_Name', @source_owner = N'dbo',
 @source_object = N'Table_Name', @type = N'logbased',
@description = null, @creation_script = null, @pre_creation_cmd = N'drop', @schema_option =
 0x000000000803509F, @identityrangemanagementoption = N'manual',
@destination_table = N'Table_Name', @destination_owner = N'dbo', @vertical_partition = N'false', @ins_cmd
 = N'CALL sp_MSins_dboxent_map', @del_cmd = N'CALL
sp_MSdel_dboxent_map', @upd_cmd = N'SCALL sp_MSupd_dboxent_map'
END
GO
```

4. Paste the code before the following entry in the file:

```
PRINT 'Publication created'
```

5. Replace 'Table_Name' in the code with the name of the table you want to include.

Exclude a Table for Replication

Follow these steps:

1. In the configuration utility folder, open the **Creation Script** folder.
2. Open the *CreatePublication.sql* file using a text editor.
3. Search for the name of the table and find the following entry. For example, if the name of the table is **xent_map**, you find the following code in the file:

```
IF (EXISTS (SELECT * FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_NAME = 'xent_map'))
BEGIN
exec sp_addarticle @publication = N'publication_mdb', @article = N'xent_map', @source_owner = N'dbo',
 @source_object = N'xent_map', @type = N'logbased',
```

```
@description = null, @creation_script = null, @pre_creation_cmd = N'drop', @schema_option =
 0x000000000803509F, @identityrangemanagementoption = N'manual',
@destination_table = N'xent_map', @destination_owner = N'dbo', @vertical_partition = N'false', @ins_cmd =
 N'CALL sp_MSins_dboxent_map', @del_cmd = N'CALL
sp_MSdel_dboxent_map', @upd_cmd = N'SCALL sp_MSupd_dboxent_map'
END
GO
```

4. Delete the code and save the file.

Manual Configuration

You can manually implement database replication if you do not want to use the utility.

Complete the following steps:

1. Configure Distribution on the Production Database Server.
2. Create a Publication on the Production Database using Transactional Replication.
3. Update the Publication Properties
4. Configure Subscription on the Reporting Database.

Configure Distribution on the Production Database Server

For database replication, you must first configure the distributor. The distributor is the same server as the production database server.

Follow these steps:

1. In Microsoft SQL Server Management Studio, connect to the server that will be the Distributor (in many cases, the Publisher and Distributor are the same server), and then expand the server node.
2. Right-click the **Replication** folder, and then click **Configure Distribution**.
3. In the Distributor pane, select **'<ServerName>' will act as its own Distributor; SQL Server will create a distribution database and log** and click Next.
4. In the **SQL Server Agent Start** pane, select **Yes, configure SQL Server Agent service to start automatically** and click Next.
5. In the **Snapshot Folder** pane, specify a root snapshot folder (for a local Distributor), and click **Next**.
6. In the **Distribution Database** pane, review and edit the information, if necessary, and click **Next**.
7. In the **Publishers** pane, optionally enable other Publishers to use the Distributor and click **Next**.
8. In the **Wizard Action** pane, select an appropriate action, and click **Finish**.

Create a Publication on the Production Database using Transactional Replication

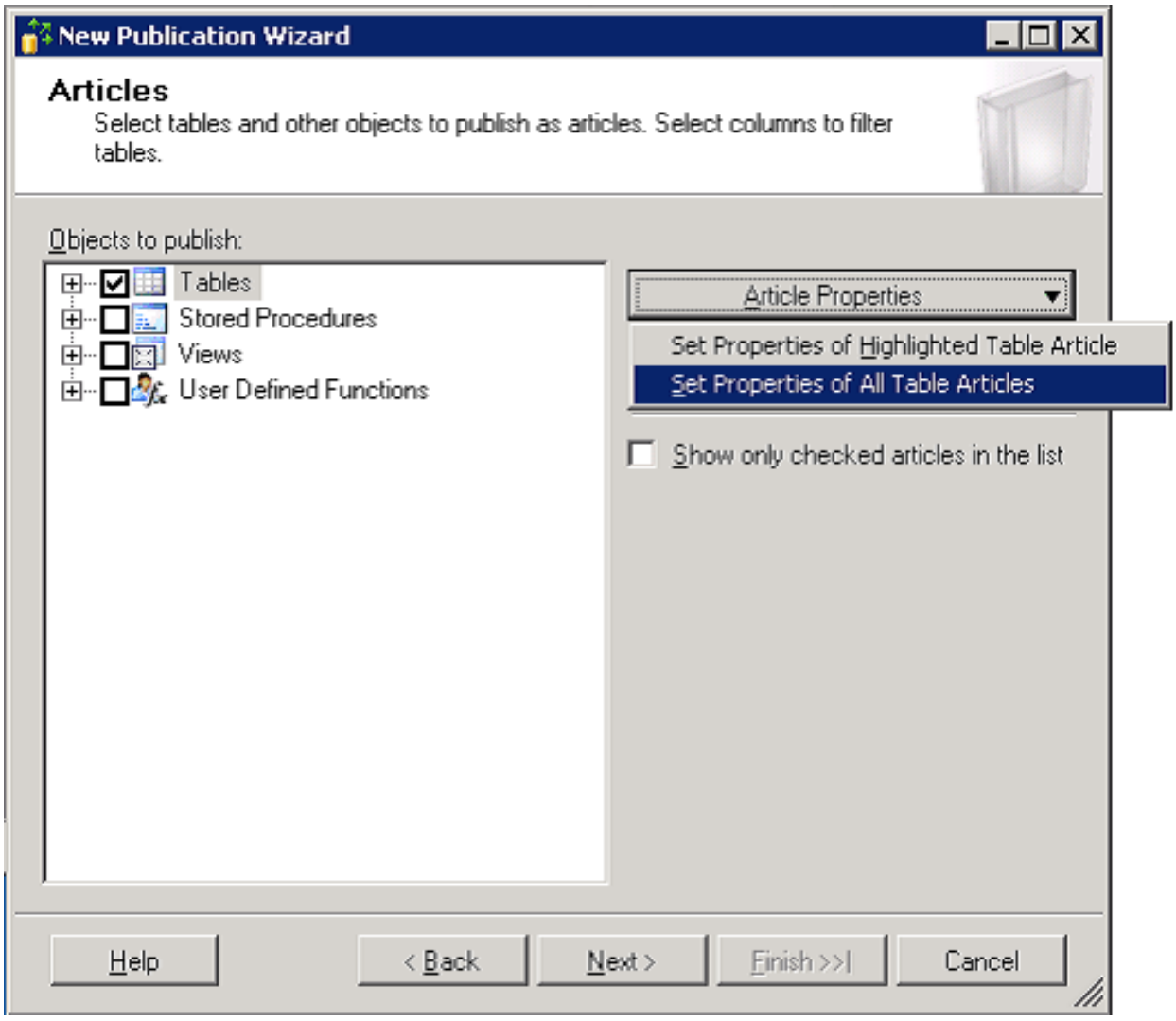
After you configure the distributor, you must create a publication on the publication database. The subscriber or the replication database shall subscribe to the publication for successful data replication.

Follow these steps:

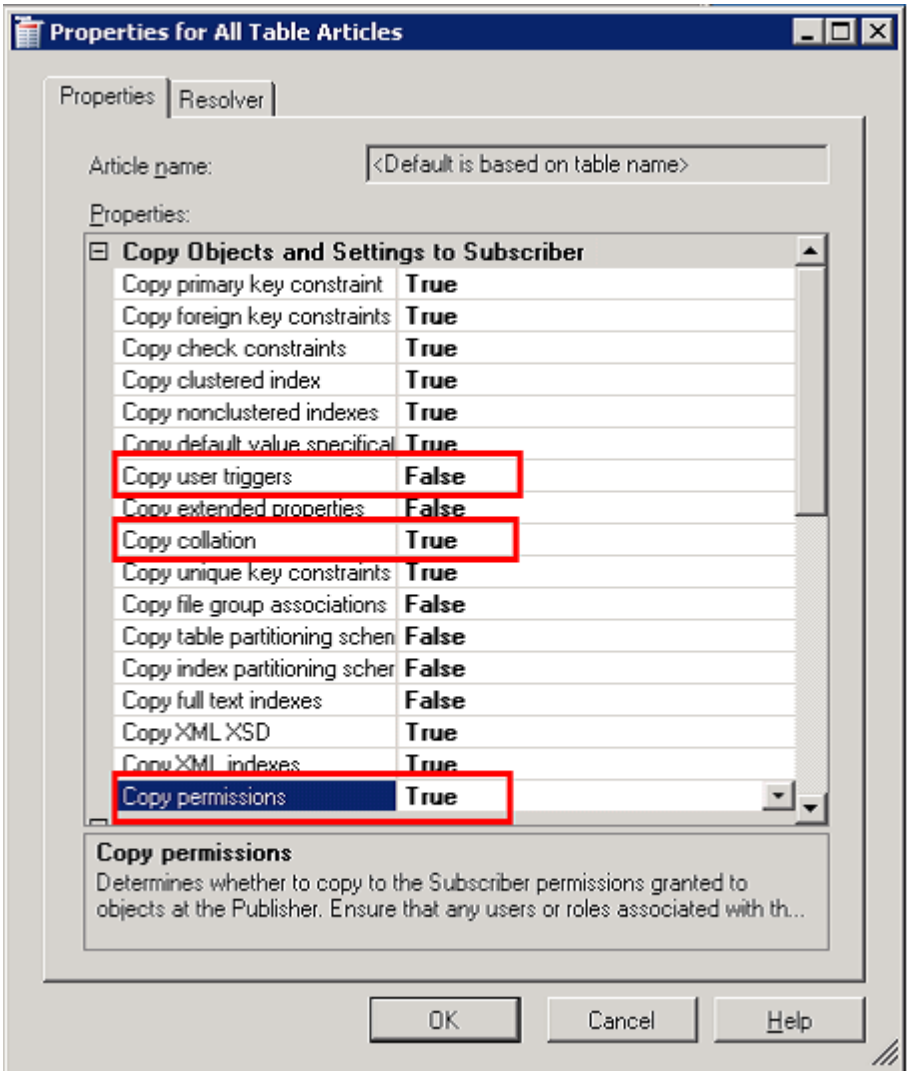
1. In the Microsoft SQL Server Management Studio, connect to the production database server.
2. Expand the **Replication** folder, right-click **Local Publications** and select **New Publication**.
3. In the **Publication Database** pane, select the database you want to replicate, and click **Next**.
4. In the **Publication Type** pane, under **Publication Type** select **Transactional publication** and click **Next**.
5. In the **Articles** pane, select **Tables**, expand **Tables** and *exclude* the following entries:

- ci_twa_ci
- dlgtsrv
- ebr_fulltext
- ebr_fulltext_adm
- ebr_patterns
- ebr_properties
- ebr_synonyms
- ebr_synonyms_adm
- options
- server_types
- usp_alias
- usp_bopsid
- usp_config_daemon_type
- usp_config_other_daemons
- usp_configuration
- usp_domsrvr
- usp_interval_log
- usp_mailbox
- usp_record_lock
- usp_servers
- usp_session_ext
- usp_webeng_alias
- usp_webeng_domsrvr
- usp_webengine

6. Click **Article Properties,** select **Set Properties for All Table Articles**.



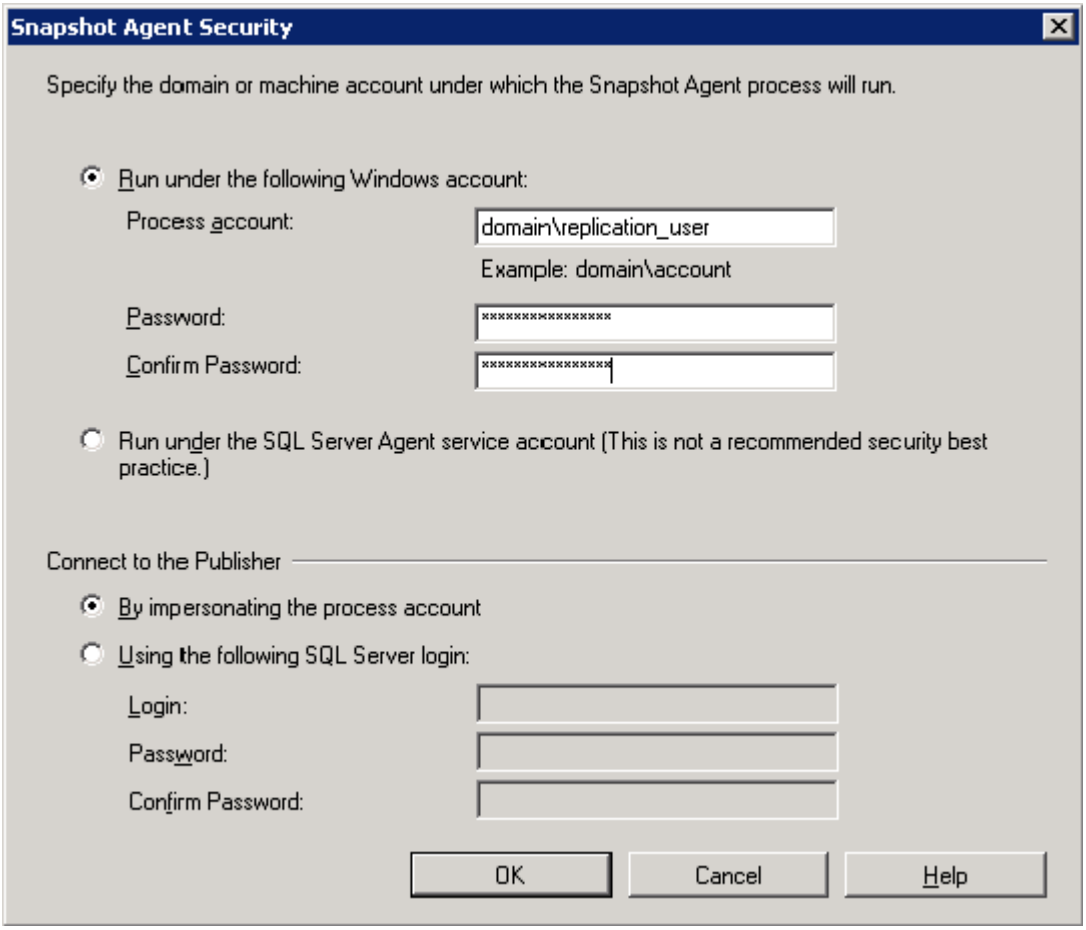
7. In the **Properties for All Table Articles** window, expand **Copy Objects and Settings to Subscriber**.



8. Perform the following steps:
   a. Set **Copy user triggers** to **False**.
   b. Set **Copy collation** to **False**.
   c. Set **Copy permissions** to **True**.
9. Click **OK** and then click **Next**.
10. In the **Filter Table Rows** ensure that you do not add any filters and click **Next**.
11. In the Snapshot Agent page, select **Create a snapshot immediately and keep the snapshot available to initialize subscriptions** and click **Next**.
12. In the **Agent Security** pane, click **Security Settings**.
    The **Snapshot Agent Security** window appears.
13. Select **Run under the following Windows account**, specify the details and click **OK.**

    NOTE

    For testing purposes, you can use a local administrator account that has access to the replication snapshot shared folder. Do not change the **Connect to the Publisher** option and retain the default value to impersonate the process account.



14. In the **Agent Security** pane, click **Next**.
15. In the **Wizard Action** pane, select **Create the Publication** and click **Next**.
16. In the Complete the Wizard pane, specify a name for the publication and click **Finish**.
    The publishing process begins.
17. Update the publication properties.

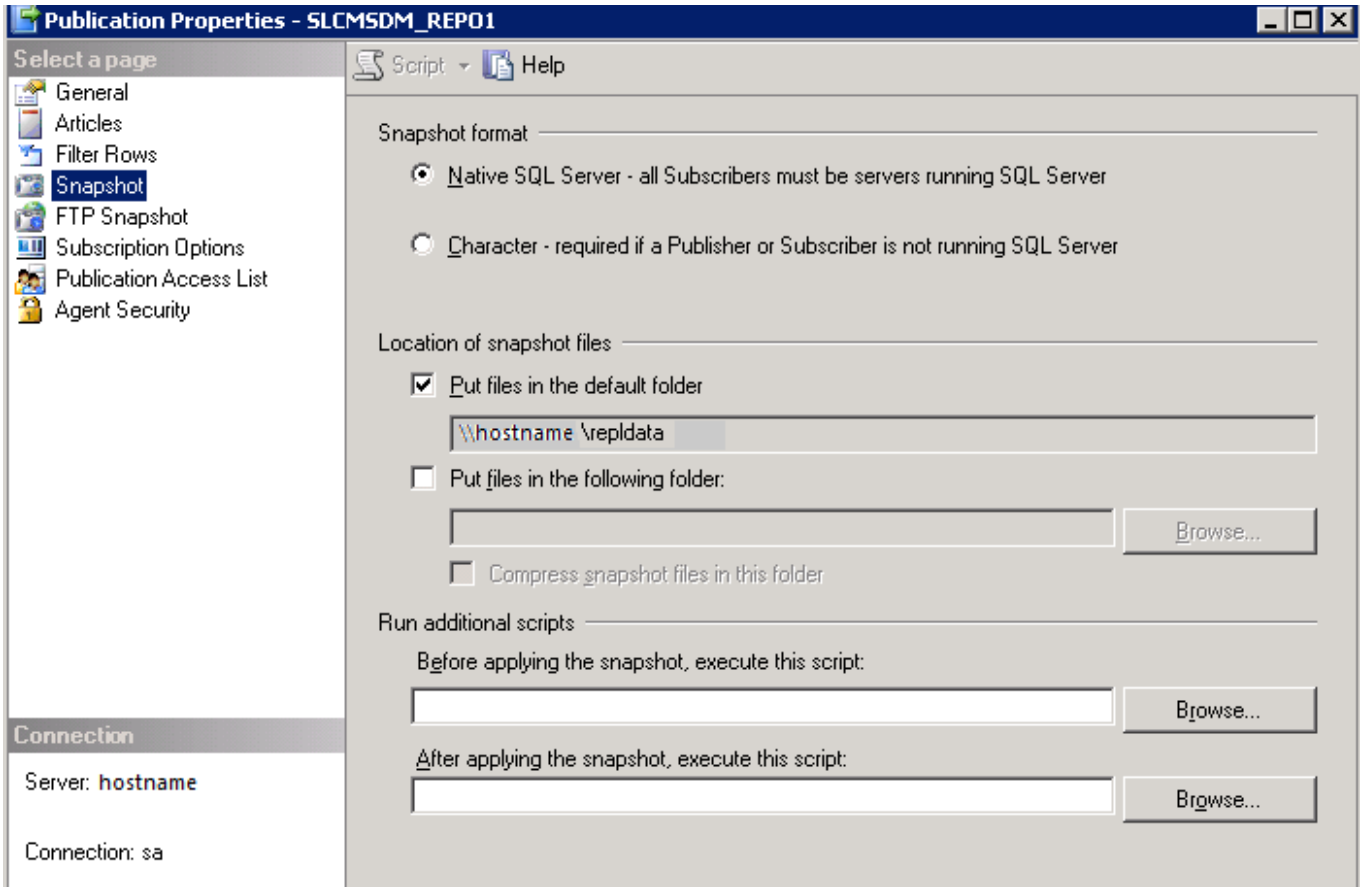
Update the Publication Properties

Follow these steps:

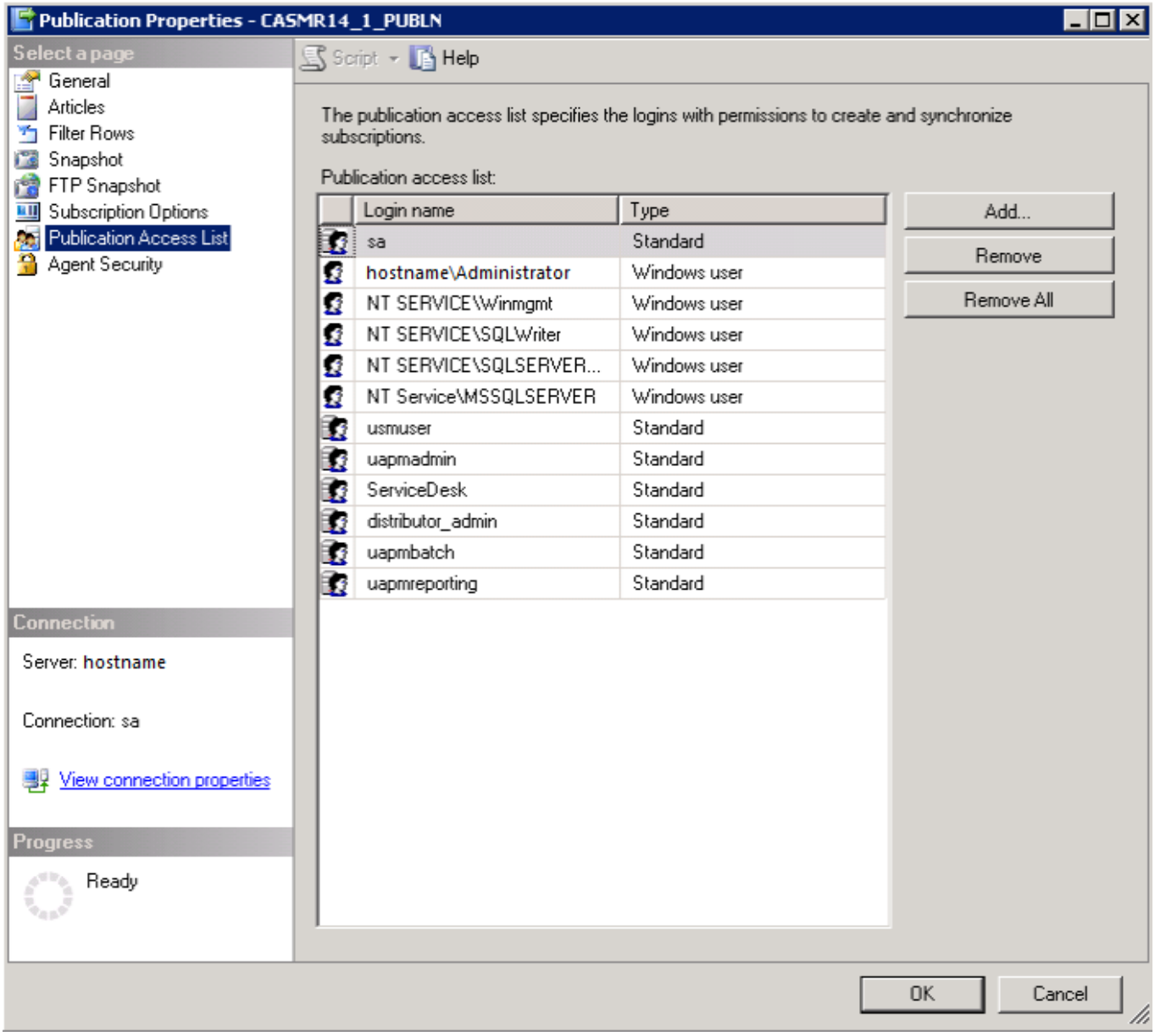
1. In the Microsoft SQL Server Management Studio, connect to the production database server.
2. Expand **Replication, Local Publications**.
3. Right-click the publication that you created and select **Properties.**

   NOTE
   If you change a property that requires the regeneration of the snapshot, you are prompted to regenerate the snapshot.

4. In the **Publication Properties** window, complete the following steps:
   a. In the tree on the left, click **Snapshot**.



b. Clear the **Put files in the default folder** check box.
c. Select the **Put files in the following folder** check box.
d. Specify the path to the file share.
e. In the tree on the left, click **Publication Access List**.



f. Click **Add**.
g. In the **Add Publication Access** window add *ServiceDesk, uapmadmin, uapmreporting, usmuser* (and *mdbadmin,* if available) users
h. Click **OK** and then click **Yes.**

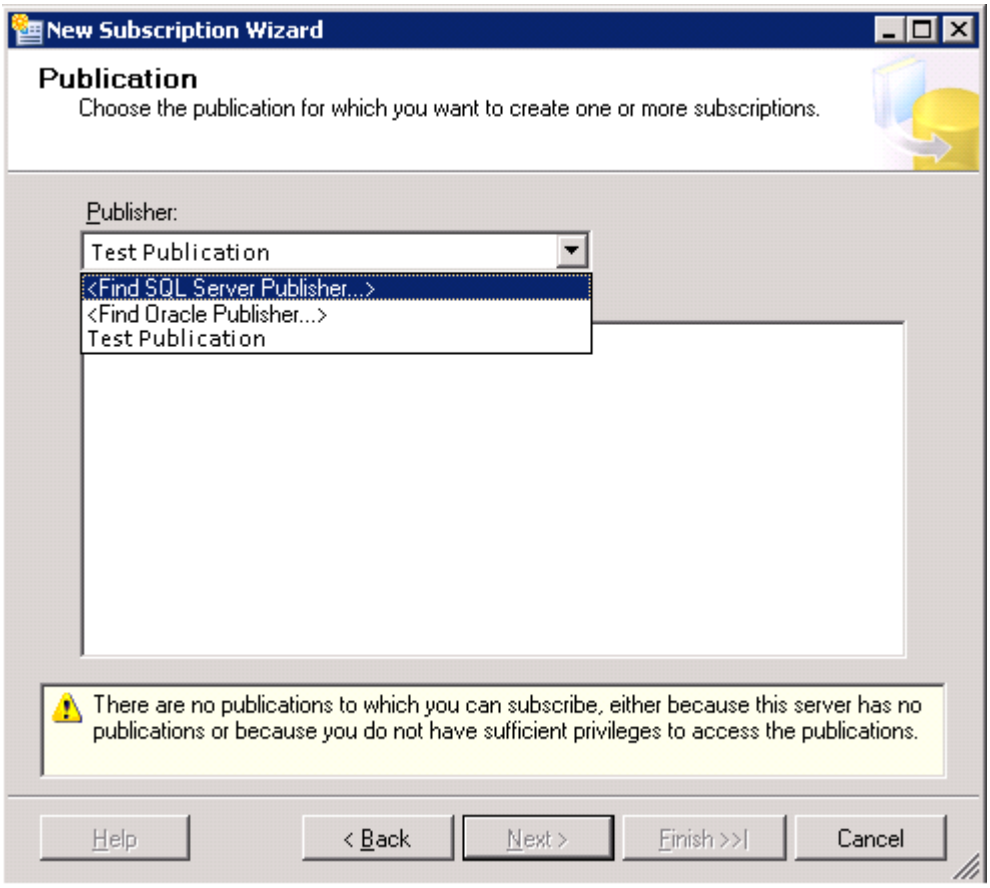
Configure Subscription on the Reporting Database

To replicate the database, the reporting database (subscriber) must subscribe to the specific publication of the publisher database.

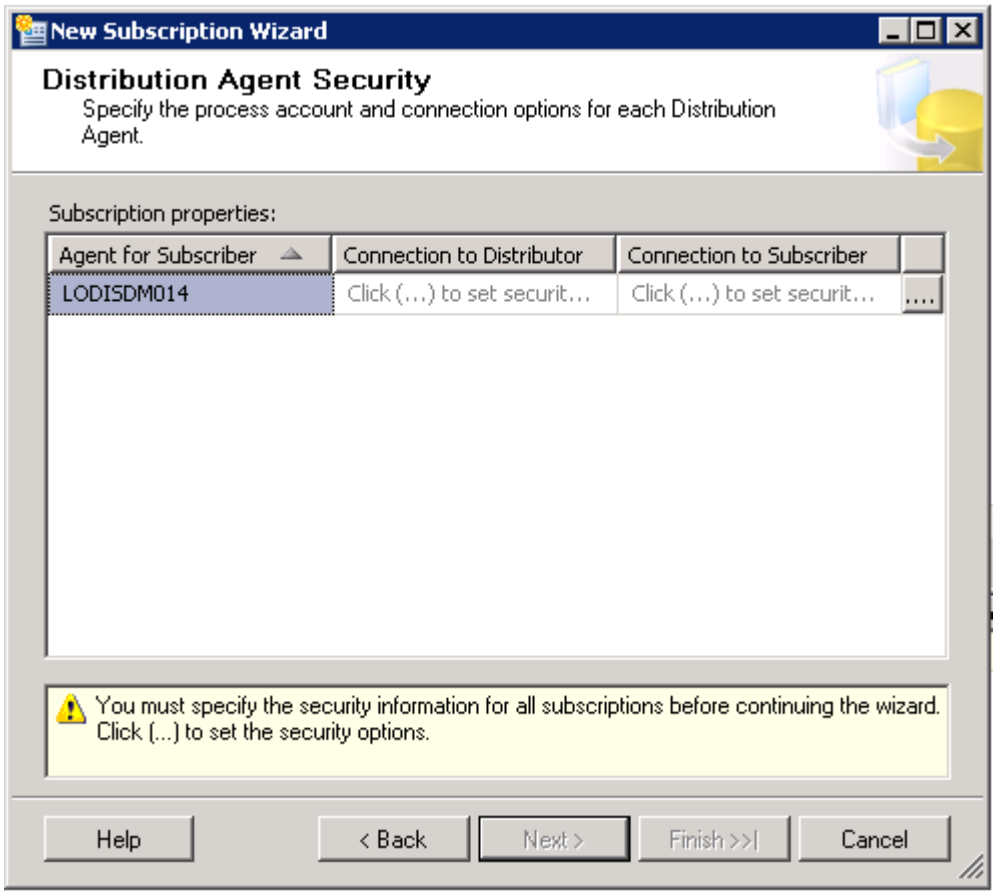
Follow these steps:

1. In the Microsoft SQL Server Management Studio, connect to the reporting database server.
2. Expand the **Replication** folder, right-click **Local Subscription** and select **New Subscription.**
3. In the **New Subscription Wizard** window, click **Next**.

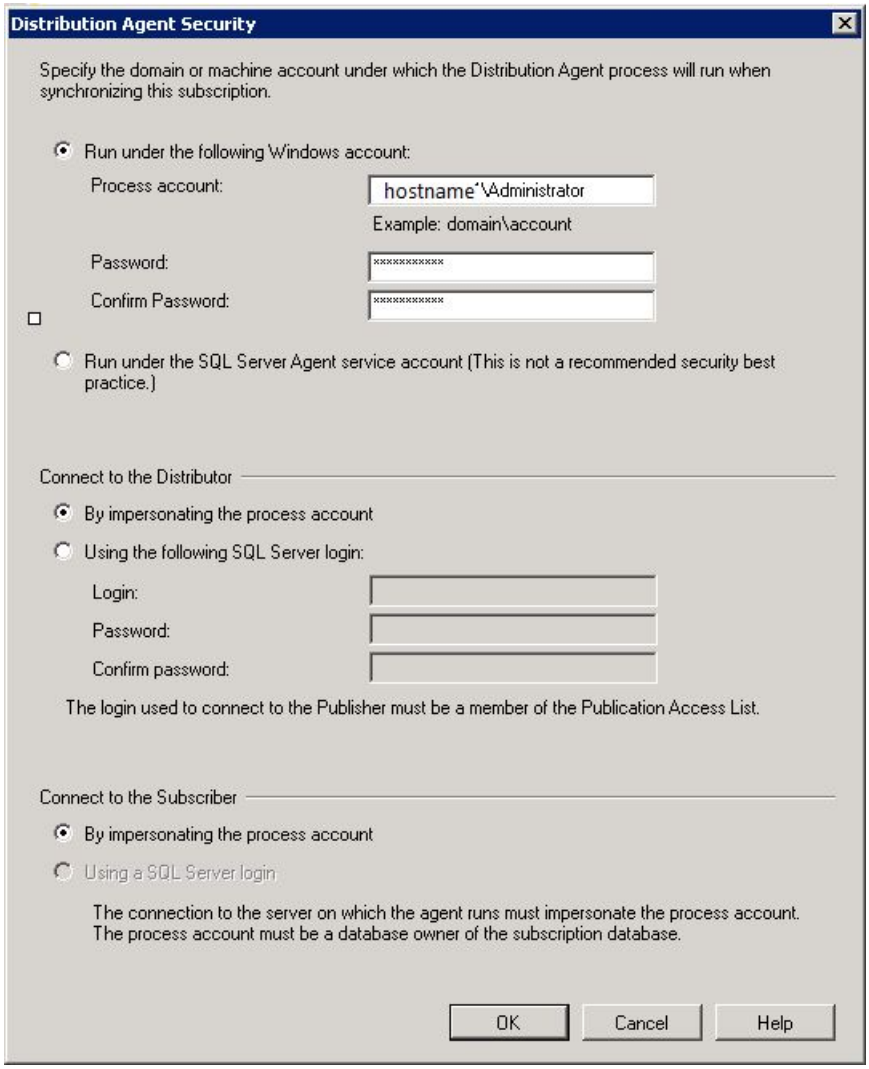
4. In the **Publications** pane, click **<Find SQL Server Publisher>** in the **Publisher** drop-down list.



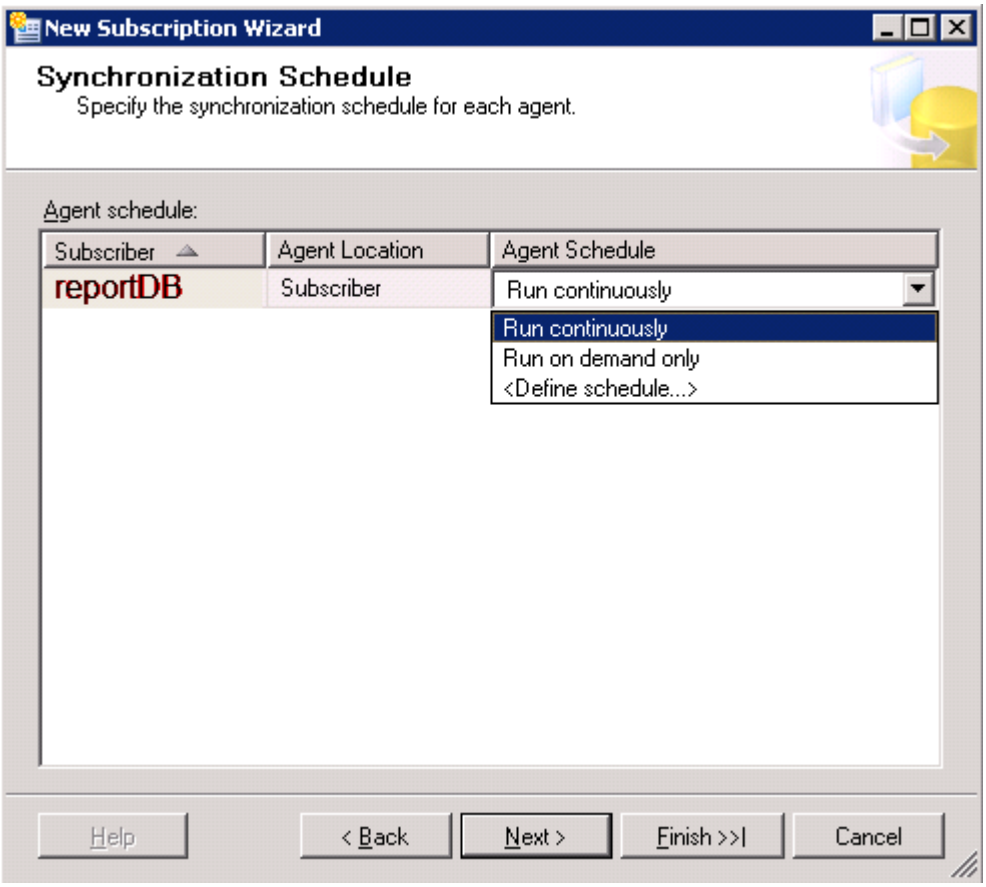
5. Specify the production database and the appropriate details and click **Connect.**
   After successful connection, the **Publication** pane displays all the available publications in the database server.
6. Select the publication that you previously configured and click **Next**.
7. In the **Distribution Agent Location** pane, select **Run each agent as its Subscriber (pull subscriptions)** and click **Next**.
8. In the **Subscribers** pane, ensure that **Subscription Database** is **mdb** and click **Next.**
9. In the **Distribution Agent Security** pane, click the button next to **Connection to the Subscriber**.



10. Specify the security credentials for the Subscriber Agent, click **OK,** and click **Next**.



11. In the **Synchronization Schedule** pane, under **Agent Schedule** select **Run continuously**.
    To schedule the replication process, select **Define Schedule**.



12. In the **Initialize Subscriptions** pane, click **Next**.
13. In the **Wizard Action** pane, select **Create the subscription** and click **Next**.
14. In the **Complete the Wizard** pane, click **Finish**.
15. After the subscription creation process is complete, click **Close**.

Troubleshoot offline reporting

The following topics explain how to troubleshoot the most frequent issues.

Replication Process Fails

When you run the configuration utility for the database replication, the replication process may fail for several reasons. To know why the replication process failed, you can view the *Replication.log* file available in the **Configuration Utility** folder.

For information about the error messages, refer to Microsoft SQL Server documentation for the respective error message.

After you understand the log messages, run the **DisableReplication.bat** file before you start the replication process again.

If database replication using the configuration utility fails repeatedly, perform the replication process manually. For more information see, Manual Configuration.

Unable to Copy Data to the Replicated Database

During replication, some information may not be copied to the subscriber server. Since data is not copied to the replication database, reports are not available.

This is a known issue. See the following Microsoft SQL Server documentation for more information.

Convert a PFX Keystore to a Java Keystore

This article describes how to convert a .pfx keystore to a Java keystore when configuring SSL in your CA Service Management environment.

Follow these steps:

1. Copy the .pfx file to the product root folder.
   For example, copy to <USM_HOME>
2. Open command prompt and execute the following command:
   ```
   keytool -importkeystore -deststorepass <pfx_password> -destkeystore "<$HOME>\.keystore" -srckeystore
   <pfx_file> -srcstoretype PKCS12 -srcstorepass <pfx_password>
   ```
   Replace the following parameters with the actual values: <pfx_password>, <pfx_file>, <$HOME>
3. Navigate to <$HOME>\viewService.conf and update the keystore password in the file.
   ```
   wrapper.java.additional.XX=-Djavax.net.ssl.trustStore="<$HOME>\.keystore"
   wrapper.java.additional.xx=-Djavax.net.ssl.trustPass="<password_for_.keystore>"
   ```
   Replace the <password_for_.keystore> and <$HOME> with the actual values. The vale for *xx* in the parameter is the sequence number for the parameter. For example, wrapper.java.additional.24=-Djavax.net.ssl.trustStore="<$HOME>\.keystore"
4. Navigate to the following folder and add the following content to the file: <$HOME>\view\conf\server.xml
   ```
   <Connector port="443" maxHttpHeaderSize="8192" maxThreads="150" minSpareThreads="25" maxSpareThreads="75"
   enableLookups="false" disableUploadTimeout="true" acceptCount="100" scheme="https" secure="true"
  keystorePass="<pfx_password>" clientAuth="false" sslProtocol="TLS" keystoreFile="<certificate_location>
  \.keystore"/>
   ```
   Replace the following parameters with the actual content: <pfx_password>, <certificate_location>
5. Save and close the file.
6. Restart the services.

Transparent Data Encryption for CA Service Management

This article describes the Transparent Data Encryption (TDE) procedures for CA Service Management Databases (Microsoft SQL Server and Oracle). The TDE procedures described in this article are in accordance with the GDPR and other compliance requirements mandated by the European Union (EU) law on data protection and privacy within the European Union (EU) and European Economic Area (EEA).

This article contains the following topics:

- Enabling Transparent Data Encryption for Microsoft SQL Server Database
- Enabling Transparent Data Encryption for Oracle Database

Enabling Transparent Data Encryption for Microsoft SQL Server Database

This article describes the Transparent Data Encryption (TDE) procedures for Microsoft SQL Server (2019, 2016, 2014, 2012, 2008, 2008 R2) database. TDE provides the ability to encrypt the CA Service Management MDB database schema. The procedure outlined in this article can be performed by Administrators/database administrators while performing a fresh installation or while upgrading an existing CA Service Management database.

This article contains the following topics:

How Transparent Data Encryption Works

TDE uses the Database Encryption Key (DEK) for encrypting. TDE protects data at the physical storage layer, that is, data at rest – data and log files. It performs real-time encryption of the database, associated backups, and transaction log files without requiring changes to the application. Since, protection is at the level of the physical device, it prevents risk of compromising the data. It provides the ability to protect data at the storage level as required by GDPR regulations and other compliance. In Microsoft SQL Server, keys and certificates are first created, and then, encryption must be set to **ON** in the database for the data to be encrypted. Data is encrypted using the AES and 3DES encryption algorithms without making any changes to existing applications. The data in the unencrypted data files can be read by restoring the files to another server.

NOTE

It is recommended that you review Microsoft Documentation for various TDE considerations and restrictions before implementing it on your CA Service Management production environment.

For example, if file groups associated with the database are set as read-only, TDE encryption fails. TDE does not protect data in memory or data transmitted between an application and the SQL Server.

Encrypt CA Service Management Database

Login to Microsoft SQL Server Database as a System Administrator (SA) and connect to the CA Service Management master database. Perform the following steps to encrypt the CA Service Management Database (for example, MDB). CA Service Management Solution Products (CA Service Desk Manager, CA Service Catalog, and CA Asset Portfolio Manager (CA ITAM).

1. Create a Master Key
2. Create a Certificate in the Master Database
3. Create a Database Encryption Key or Symmetric Key in the User Database
4. Enable Database for Encryption
5. Restore the Microsoft SQL Server Database on the Target or Destination Server

NOTE

Before enabling the TDE procedures, ensure that you have taken a backup of the CA Service Management Database (for example, MDB).

Create a Master Key

In Microsoft SQL Server 2012, 2014, 2016, 2019 the master key is encrypted using AES 256 with a user specified password. For Microsoft SQL Server 2008 and SQL Server 2008 R2, Triple DES algorithm is used.

Run the following command to create a Master key:

```
USE master;
GO

CREATE MASTER KEY
ENCRYPTION BY PASSWORD = '<Specifypassword>';
```

In the above command, <*Specifypassword*> specifies the password for the master key.

Verify the Data Master Key and Service Master Key

To verify that the the Data Master Key (DMK) is created, you can query the *sys.symmetric_keys catalog* view. The query results display both Data Master Key (DMK) and Service Master Key (SMK). Service Master Key is unique to each instance of database based on how it is generated. Microsoft SQL Server creates the Service Master Key automatically in the master database. Both DMK and SMK are based on the 256-bit AES encryption algorithm.

Run the following command to query the *sys.symmetric_keys catalog view* and verify that DMK is created:

```
SELECT name KeyName,
 symmetric_key_id KeyID,
 key_length KeyLength,
 algorithm_desc KeyAlgorithm
FROM sys.symmetric_keys;
```

Create the Certificate in the Master Database

NOTE

In case of Microsoft SQL Server database, even if the TDE Certificate is out of date, you can still export and import TDE enabled databases without any apparent issues on the destination or target server. It is only in the log files a message is logged that the certificate is out of date and expired. Hence, it is recommended that you validate the security of your data in all stages of its life cycle.

You can create and use a self-signed or external certificate as per your requirement. A self-signed certificate is an Asymmetric key created by the SQL Server database engine. A certificate protects the TDE database or encrypted Microsoft SQL Server backup. Create a self-signed certificate as shown below. Based on your environment requirements, you can also use any external certificate authority to generate a certificate.

Run the following command to generate a certificate:

```
CREATE CERTIFICATE <Cert Name>
WITH SUBJECT = 'Name of certificate or issuer name'
```

Verify the Certifcate

Run the following command to verify the certificate is generated:

```
SELECT name CertName,
  certificate_id CertID,
  pvt_key_encryption_type_desc EncryptType,
  issuer_name Issuer
FROM sys.certificates
WHERE issuer_name = 'Name of certificate or issuer name';
```

NOTE

For external certificate, database administrators have to ensure that the external certificates are copied to *sys.certificates* database. All certificates and associated private keys must be backed up.

Create the Database Encryption Key or Symmetric Key in the User Database

Create the Database Encryption Key (DEK) in the CA Service Management Database. Run the following query:

NOTE

You will receive a warning when the below query is executed. The warning is regarding taking a backup of certificates and private keys. Ensure to take a backup of certificates and private keys.

```
USE '<Product_Database>';
GO
```

```
CREATE DATABASE ENCRYPTION KEY
WITH ALGORITHM = AES_256
ENCRYPTION BY SERVER CERTIFICATE <Cert Name>;
```

In the above command, <Product_Database> specifies the name of your MDB database.

Verify the Database Encryption Key

Query the *sys.dm_database_encryption_keys dynamic management view* to verify that the Database Encryption Key (DEK) is created. Run the following command to verify DEK is created.

```
SELECT DB_NAME(database_id) DbName,
encryption_state EncryptState,
key_algorithm KeyAlgorithm,
key_length KeyLength,
encryptor_type EncryptType
FROM sys.dm_database_encryption_keys;
```

NOTE

If you are using Microsoft SQL Server 2008/2008R2, you must remove the ***encryptor_type*** *(Encrypt Type)* from the above query since ***encryptor_type*** is not supported by Microsoft SQL Server 2008/2008 R2 databases.

The above query will list the database where the values for "Encrypt State" are referencing the following:

```
0 : No database encryption key present, no encryption
1 : Unencrypted
2 : Encryption in progress
3 : Encrypted
4 : Key change in progress
5 : Decryption in progress
6 : The certificate or asymmetric key encrypting the DEK is being changed
```

Enable Database for Encryption

Turn on the database for encryption. Run the following query to enable database encryption:

```
ALTER DATABASE <Product_Database>
SET ENCRYPTION ON;
```

In the above command, **<Product_Database>** specifies the name of your MDB database or any custom name that you want to provide for your database as per your requirement.

Run the following command:

```
SELECT DB_NAME(database_id) DbName,
  encryption_state EncryptState,
  key_algorithm KeyAlgorithm,
  key_length KeyLength,
  encryptor_type EncryptType
FROM sys.dm_database_encryption_keys;
```

The above query shows the addition of a row on the *tempdb* database. When TDE is implemented on any user table, Microsoft SQL Server encrypts the *tempdb* database as well.The database contains items like temporary user objects, internal objects, and row versions (all of which may contain sensitive data).

Take a Backup of the Master Key and Certificate

Take a backup of the certificate and keys, preferably, right after creating the certificate and keys. This helps you to have access to the TDE protected database even after migrating to a different server. Take a backup of your Service Master Key (SMK) also. Specify a backup folder location for the backed up private keys.

```
Use master;
GO

BACKUP SERVICE MASTER KEY
TO FILE = 'C:\Backuplocation\SvcMasterKey.key'
ENCRYPTION BY PASSWORD = '<Specifypassword>'
```

Backup the Data Master Key

Take a backup of the Data Master Key (DMK). Run the following query to take a backup:

```
BACKUP MASTER KEY
TO FILE = 'C:\Backuplocation\DbMasterKey.key'
ENCRYPTION BY PASSWORD = '<Specifypassword>'
```

Backup the Certificate

Taking a backup of a certificate is little different from taking a backup of SMK and DMK. In case of certificate backups, you may want to take an explicit backup of the private key along with the certificate. To achieve this, it is recommended that you use a Backup Certificate statement that includes the private key clause. Run the following command to take a backup:

```
BACKUP CERTIFICATE <Cert Name>
TO FILE = 'C:\Backuplocation\ProductDBNameCert.cer'
WITH PRIVATE KEY(
 FILE = 'C:\Backuplocation\ProductDBNameCert.prvk',
 ENCRYPTION BY PASSWORD = 'Specifypassword'
);
```

NOTE

: In case of JasperSoft Aggegate/DataMart reports, you must perform the above steps separately on the Aggregate/DataMart database <database name> (For example: casm_dm).

Restore the Microsoft SQL Server on the Target or Destination Server

Perform the below steps to restore the Microsoft SQL Server database on the target server:

- Create a Master Key on the Target or Destination Server
- Create the Certificate Using the Backed Up Certificate Files from the Source Server

Create a Master Key on the Target or Destination Server

Run the following command to create a master key on the target server.

NOTE

Master key must be created by using the master database only.

```
USE master
GO
CREATE MASTER KEY ENCRYPTION BY PASSWORD = '<Specifypassword>'
```

Create the Certificate Using the Backed Up Certificate Files from the Source Server

For more information, see Create the Certificate in the Master Database. Run the following command to create a certificate using the backed up certificate files:

```
CREATE CERTIFICATE <certificateName>
FROM FILE = 'C:\backup\ ProductDBNameCert.cer'
WITH PRIVATE KEY(
FILE = 'C:\backup\ ProductDBNameCert.prvk',
DECRYPTION BY PASSWORD = '<Specifypassword>'
)
```

NOTE

In case of JasperSoft Aggegate/DataMart reports, you must perform the above steps separately on the Aggregate/DataMart database <database name> (For example: casm_dm).

Enabling Transparent Data Encryption for Oracle Database

This article contains the following topics:

Overview

Oracle Transparent Data Encryption (TDE) secures sensitive data within an Oracle database. It encrypts oracle database columns and manages encryption keys for CA Service Management Solution Products Database (CA Service Desk Manager, CA Service Catalog, and CA Asset Portfolio Management (CA ITAM)). Transparent Data Encryption is certified with Oracle Database 11gR2 and Oraycle 12c.

NOTE

TDE procedure documented in this article is applicable only for fresh installation of CCA Service Management Oracle database, say, CA Service Management 17.3 or 17.1.

How Transparent Data Encryption Works

NOTE

For more information about Transparent Data Encryption considerations and restrictions, see Oracle Documentation before you start implementing it on your production environment.

Transparent data encryption is a key-based access control system. The encrypted data cannot be understood until authorized decryption occurs. Authorized decryption is automatic for authorized users accessing the database table. To use transparent data encryption, you must have the ALTER SYSTEM privileges and a valid password to the Oracle wallet. If an Oracle wallet does not exist, create a new wallet. For more information, see Create a Wallet and Encrypted TableSpace.

Enabling Transparent Data Encryption for Oracle Database (Windows and Non-Windows)

To enable Transparent Data Encryption for Oracle Database for CA Service Management, perform the following steps using Command Line:

Create a Wallet and Encrypted TableSpace

Login into the Oracle 11gR2, 12c, 19c as database administrator. Create a wallet and update the WALLET_LOCATION parameter in the *sqlnet.ora* file as outlined in the procedure below:

> **NOTE**
>
> For Non-Windows, ensure that the *ORACLE_BASE* variable has been set. TDE best practices recommend using a wallet location outside of $Oracle_Base.

For **Windows**, perform the following steps:

1. Create a directory <*WALLET*> manually anywhere in your system:
   Windows: For example:
   ```
   C:\TDE\MDB\WALLET
   ```
   Non-Windows:
   ```
   /root/TDE/MDB/WALLET
   ```
2. Add the following content to *sqlnet.ora* file to create the encryption wallet keystore file in the specified directory.
   Navigate to *sqlnet.ora file location.*
   For example, on Windows: C:\app\Administrator\product\<Oracle_database_version>\dbhome_1\NETWORK\ADMIN\sqlnet.ora
   ```
   ENCRYPTION_WALLET_LOCATION
   (SOURCE =
   (METHOD = FILE)
   (METHOD_DATA =
   (DIRECTORY = C:\TDE\MDB\WALLET)
   )SQLNET.AUTHENTICATION_SERVICES = (NTS)
   ```
   For example, on Non-Windows: /opt/oracle/app/product/<*Oracle_database_version*>/dbhome_1/network/admin/sqlnet.ora
   ```
   ENCRYPTION_WALLET_LOCATION
   (SOURCE =
   (METHOD = FILE)
   (METHOD_DATA =
   (DIRECTORY = /root/TDE/MDB/WALLET
   )SQLNET.AUTHENTICATION_SERVICES = (NTS)
   ```
3. Create the wallet, along with a secure password, preferably containing a mix of alphanumeric characters.

   > **NOTE**
   >
   > The password is case-sensitive and must contain a mix of alphanumeric characters.
4. Navigate to Oracle Home/BIN. For example, on **Windows**: C:\app\Administrator\product\<*Oracle_database_version*>\dbhome_1\BIN
   For example, on **Non-Windows**: /opt/oracle/app/product/<*Oracle_database_version*>/dbhome_1/bin
   a. For **Windows**, run the following command:
      ```
      SQL> alter system set encryption key identified by <password>;
      ```

b. For **Non-Windows**, open shell/terminal, login as ORACLEUSER and export Oracle DB variables and enter the following command to connect to Oracle database:

```
$ sqlplus sys as sysdba
```

Provide the password and hit Enter. Now, you should be able to connect to database.

5. Verify the status of the Wallet that you created by running the following command:
   a. For **Windows**, run the following command:

```
SQL> select * from v$encryption_wallet;
```

   b. For **Non-Windows**, run the following command:

```
SQL > select * from v$encryption_wallet;
```

6. If the wallet status is CLOSE, open it by running the following command:
   a. For **Windows**:

```
SQL> alter system set wallet open identified by <password>;
System altered
```

   b. For **Non-Windows**:

```
SQL> alter system set wallet open identified by <password>;
System altered
```

7. Now, create an encrypted tablespace ***MDB_DATA_TEST DATAFILE*** by running the following command:
   For **Windows**:

```
create tablespace <tablespacename>

For example:
create tablespace MDB_DATA_TEST DATAFILE 'C:\app\Administrator\oradata\mdb\MDB_DATA_TEST.DBF' SIZE 100M
 autoextend on next 100M ENCRYPTION USING 'AES256' DEFAULT STORAGE (ENCRYPT);
```

   For Non-Windows:

```
create tablespace <tablespacename>

For example:
create tablespace MDB_DATA_TEST DATAFILE opt/oracle/app/product/<Oracle_database_version>/dbhome_1/bin
\MDB_DATA_TEST.DBF' SIZE 100M autoextend on next 100M ENCRYPTION USING 'AES256' DEFAULT STORAGE (ENCRYPT);
```

8. Similarly, create an encrypted tablespace index ***MDB_INDEX_TEST DATAFILE*** by running the following command (**Windows and Non-Windows**):
   For **Windows**:

```
create tablespace <indexname>


 For example:
create tablespace MDB_INDEX_TEST DATAFILE 'C:\app\Administrator\oradata\MDB\MDB_INDEX_TEST.DBF' SIZE 100M
 autoextend on next 100M ENCRYPTION USING 'AES256' DEFAULT STORAGE (ENCRYPT);
```

   For **Non-Windows**:

```
create tablespace <indexname>
```

```
For example:
create tablespace MDB_INDEX_TEST DATAFILE '/opt/oracle/app/oradata/MDB/MDB_INDEX_TEST.DBF' SIZE 100M
autoextend on next 100M ENCRYPTION USING 'AES256' DEFAULT STORAGE (ENCRYPT);
```

9. Now, create a user for the tablespace name that you created in step 7 For Windows/Non-Windows as:
   For **Windows/Non-Windows**:

```
create user for <tablespacename>


For example:


create user mdbadmin identified by <password> default tablespace MDB_DATA_TEST;
```

10. Similarly, alter a user for tablespace index that you created in step 8:

```
alter user for <indexname>


For example:


alter user mdbadmin identified by <password> default tablespace MDB_INDEX_TEST;
```

11. Alter the users with unlimited quota for tablespace.
    For example:

```
alter user mdbadmin quota unlimited on <MDB_DATA_TEST>;
```

12. Alter the users with unlimited quota for index.
    For example:

```
alter user mdbadmin quota unlimited on <MDB_INDEX_TEST>;
```

13. Now, grant required privileges to the user. Run the following command:

```
grant CREATE PROCEDURE to mdbadmin
grant CREATE SEQUENCE to mdbadmin
grant CREATE SESSION to mdbadmin
grant CREATE TABLE to mdbadmin
grant CREATE TRIGGER to mdbadmin
grant CREATE VIEW to mdbadmin
grant CONNECT to mdbadmin
grant RESOURCE to mdbadmin
```

14. Commit the changes. Run the following command:

```
Commit;
```

Verify Transparent Data Encryption is Enabled

Ensure that your tablespace is TDE enabled. Run the below command to confirm and verify whether your tablespace is TDE enabled or not encryption status of the database:

```
SELECT WRL_PARAMETER FROM V$ENCRYPTION_WALLET;
SELECT * FROM V$ENCRYPTED_TABLESPACES;
```

The output shows the list you the list of encryption table spaces where you can confirm whether the table space created by you is encrypted or not.

Configure Service Point

Service Point provides self-service capabilities with embedded assistive intelligence for business-users. Service Point uses Natural Language Processing (NLP) that provides business users with context-based solutions and suggestions.

Business users can perform the following actions using Service Point:

- Access knowledge
- Create, view, and modify tickets
- Contextually collaborate on tickets using live chat
- View allocated resources
- Manage ticket notifications

With a business-centric focus, Service Point helps an organization achieve the following:

- Broad adoption of services
- Context-based solutions
- Improved productivity
- Reduced costs
- Improved collaboration and knowledge sharing
- Elevated perception of IT organization

This article explains how to configure and manage Service Point from the CA SDM administrator interface. Perform the following steps to set up Service Point for the first time:

1. Define Permissions to set Service Point as the default app for a particular access type.
2. Define the User Context to display the most relevant information to the user in Service Point.
3. Define the Default Categories that appear on the Service Point.
4. Manage Ticket Attributes to be displayed while viewing a ticket.
5. Manage Features to customize the related features.
6. Set Up My Resources to list all the assets assigned to the user, along with asset information.
7. Manage CA Service Catalog Featured Offerings to appear on Service Point, after integrating CA Service Desk Manager and CA Service Catalog.
8. Manage Smart Search to optimize the search relevance.
9. (Optional) Enable the Guest Login
10. Add Additional Information

Implementing CA Service Management Virtual Analyst

This article describes the following topics:

- Overview
- Supported Virtual Analyst Services
- Use Case
- Additional Information
- Hardware Requirements

Overview

CA Service Management now provides the Virtual Analyst "ARIA" (Automated Response with Intelligent Automation). It provides the following capabilities:

- Interacts with business users through a chat Interface to resolve common IT support issues. The chat interface is available via Service Point.
- Ability to understand Natural Language using Machine Learning techniques.
- Provides real-time assistance to business users to resolve routine IT issues.
- Provides ticket update and can handle queries for CA Service Desk Manager (SDM) tickets (requests and incidents)

Supported Virtual Analyst Services

Business users can connect with the Virtual Analyst for assistance and interact with the service bot over a chat window. The service bot provides optimal transfer of issues to a Service Desk Manager agent to assist with queries and issues in the following supported scenarios:

- **Contextual scenario**, where the business user has already logged a ticket with IT support for a said issue and is looking for information pertaining to ticket status, details, resolution, and so on. When additional assistance or intervention is required, Virtual Analyst provides the business users with “**Connect to an Analyst**” option in the chat interface.
- **Non-Contextual scenario**, where there is no open ticket for an issue and a new ticket is opened by IT for a reported issue. In this case, a business user can request the bot to connect to a Service Desk Manager agent (when agent intervention is required).

A business user gets assistance for the following Services from the bot on Service Point:

- Get Ticket Status and Details
- Create a Ticket
- Self-Service through KB Solution
- Chat with an Analyst/Hand over the issue to an Analyst
- Provide Feedback to the Virtual Analyst.
- Reset Password
- Update Ticket
  - Close Ticket
  - Urgency
  - Ticket Description
  - Add Comment to a Ticket

NOTE
Starting from CA Service Management 17.4 or higher versions, you can use either RASA 3.0.6 or RASA 3.6.19. You can choose the version of you choice to use for the Virtual Analyst. For installing Virtual Analyst, refer to Automated Installation of Virtual Analyst for Windows.

Use Case

Business users can leverage the Virtual Analyst on Service Point for the following services:

- **Get Ticket Status and Details:**
  Users can inquire about the current status and specific details of any previously logged ticket.
- **Create a Ticket**:
  The Virtual Analyst provides the capability to create new tickets directly through the chat interface. When a user initiates this request, the Virtual Analyst will prompt for a description of the issue and any information corresponding to mandatory attributes configured for ticket creation.

Additionally, the system employs a smart search mechanism (identical to the one used in Service Point) to analyze the issue description. If the description matches a pre-configured system category, the ticket is automatically mapped accordingly. The Virtual Analyst will then request any additional information required by that category's specific attributes.

For example:

A user is unable to connect to Wi-Fi and needs to create a ticket. The organization requires an employee ID for all new tickets. This issue is mapped to the "Network" category, which requires the laptop's OS and firmware version.

- **User:** I want to create a new ticket.
- **Bot:** Can you provide the description for this new ticket?
- **User:** I'm unable to connect to wifi from my laptop.
- **Bot:** Please provide your employee id.
- **User:** <employee_id>
- **Bot:** What OS are you using?
- **User:** Linux (OpenSUSE)
- **Bot:** What is your laptop's firmware version?
- **User:** <firmware_details>
- **Bot:** Do you want to create the ticket? Please confirm Yes/No.
- **User:** Yes
- **Bot:** A ticket has been created successfully.

Self-Service through KB Solution

The Virtual Analyst's default workflow is to facilitate self-service by searching for relevant knowledge base documents. When a user describes an issue, the bot searches the system for existing knowledge documents that may provide a solution. For this functionality to operate as intended, it is recommended to train the bot with knowledge documents under the `search_kd` intent. For detailed instructions, please refer to: Virtual Analyst Intents.

Chat with an Analyst / Handover to an Analyst

If the automated solutions do not meet the user's needs, they can request a transfer to a live analyst. This process utilizes the standard Service Point transfer mechanism, which places the ticket in an analyst's queue to be manually accepted by the next available agent. For more information, refer to the Live Chat documentation.

Provide Feedback to the Virtual Analyst

Following the conclusion of a chat session with an analyst, users have the option to submit feedback regarding their experience directly within the Virtual Analyst interface.

Update Actions on a Ticket

The Virtual Analyst supports several update actions on existing tickets. To perform any of these actions, a ticket ID is required. If the ID is not available in the current chat context, the user will be prompted to provide one. The user will also be prompted for any additional information needed to complete the action. As these are write/update operations, the system will require user confirmation before executing the change.

Supported Update Actions:

- Close Ticket
- Update Urgency
- Update Ticket Description
- Add Comment to a Ticket

Hardware Requirements

> **NOTE**
> Recommendation is to install the Virtual Analyst on a separate server or on a server where Search Server is installed with adequate hardware requirements. For more information, see Search Server Installation Requirements.

- **CPU**: 4-8 Core CPU
  > **NOTE**
  > CPU must support AVX Instruction set. In general, CPUs manufactured on or after 2011 support AVX. For more information, see your CPU manual instructions.
- **RAM**: Minimum 16 GB. Recommended 32 GB
- Disk Space: 40 GB

Install Virtual Analyst

Virtual Analyst can be installed in two ways:

- Automated Installation of Virtual Analyst for Windows
- Manual Installation of Virtual Analyst for Non- Windows

Automated Installation of Virtual Analyst for Windows

Installation of RASA Server using installer script for Windows Environment.

Installation for 17.4 RU3

Create a folder in any drive, and copy the casmBot.zip file present in <17.4 RU3 DVD>/filestore\Virtual_Analyst and paste it in the created folder.
For rest of the process follow from step 2 in the section below.

Installation for 17.4 RU2

1. Create a folder in any drive, and copy the casmBot.zip file present in <17.4 RU2 DVD>/filestore\Virtual_Analyst and paste it in the created folder.
   > **NOTE**
   > For 17.4 GA the support site URL Virtual Analyst Installer (Automated) for CA Service Management.
2. Extract the casmBot.zip file, and follow these steps:
   a. Open requirements.txt file and copy the components at the end of *requirements.txt* file:
   ```
   setuptools==69.2.0
   pip==23.0.1
   wheel==0.43.0
   ```
   b. Save the file.
3. Run the installer.bat present in the casm-bot folder.
4. Select the locale. Refer the following example mentioned below:
   ```
   casm-bot>installer.bat
   This script installs all the dependencies for Virtual Analyst.
   Select a Language Option for Virtual Analyst.
   English (en)
   German (de)
   Spanish (es)
   Portuguese (pt)
   French (fr)
   Italian (it)
   ```

```
Enter your choice(en, de, es, pt, fr, it):_
```

5. Once the installation is done, if the locale is a language other than English follow these steps:
   a. Navigate to casm-bot folder -> config.yml file, open the file, and enter the chosen language in *language* and *model*, and save the file. For example:

```
recipe: default.v1
language: fr

pipeline:
  - name: "SpacyNLP"
    model: "fr_core_news_md"
    case_sensitive: False
```

   List of languages :

```
model[en]=en_core_web_md"
model[de]=de_core_news_md"
model[es]=es_core_news_md"
model[pt]=pt_core_news_md"
model[fr]=fr_core_news_md"
model[it]=it_core_news_md"
```

   b. Navigate to *casm-bot-> locale*, open the required language folder, and copy the *data* and *domain.yml* files, and replace them in casm-bot.
6. Now, run the following command to activate the virtual environment:

```
botenv\Scripts\activate.bat
```

7. Run the following command to Navigate to the casm-bot folder and train the RASA Model:

```
rasa_train.bat
```

8. Run the RASA Server:

```
rasa_run.bat
```

Manual Installation of Virtual Analyst for Non- Windows

To install and configure the Virtual Analyst Service Bot, perform the following steps:

> **NOTE**
> If you do not have Internet connectivity, you can opt to install the Virtual Analyst in an offline mode. For more information, see Offline Installation - CA Service Management Virtual Analyst.

Download and Install CA Service Management Virtual Analyst Using RASA Server 3.0.6

Perform the following steps to install RASA Server 3.0.6:

1. Download and Install Python 3.8.0. Make sure you enable the **ADD TO PATH** option during installation.
2. Install Rust Toolchain for the dependency compilations.
3. Locate the **casmbot.zip** (CA Service Management *DVD:\filestore\Virtual_Analyst\casm-bot.zip*).
4. Choose a directory to install Virtual Analyst and create a folder, for example, **VA**. Extract the contents of **casmbot.zip** to the casm-bot folder inside the VA directory.
   This directory structure will contain the virutal analyst scripts and the associated Rasa virtual environment that the virtual analyst will use.
5. Create a virtual environment for VA in the same folder with the following command:

```
python -m venv ./botenv
```

6. Run the following command to activate the virtual environment.

```
./botenv/Scripts/activate
```

7. Upgrade pip inside the virtual environment by running the command:
   ```
   python -m pip install --upgrade pip
   ```
8. Run the following command to install RASA v3.0.6:
   ```
   pip install rasa==3.0.6
   ```
9. Run the following command to install Spacy:
   ```
   pip install rasa[spacy]
   pip install websockets==10.0
   pip install protobuf==3.2.0
   ```
10. Download the spacy language model. Refer to the Virtual Analyst Supported Locale to download language models for other languages. (Ignore the step to link the model as linking is deprecated in latest spacy versions).
    ```
    python -m spacy download en_core_web_md
    ```
11. Navigate to casm-bot folder and train the RASA Model, by running the following command:
    ```
    cd casm-bot
    rasa_train.sh
    ```
12. Run the RASA Server.
    ```
    rasa_run.sh
    ```

Download and Install CA Service Management Virtual Analyst Using RASA Server 3.6.19 for 17.4.3.0

Perform the following steps to install RASA Server 3.6.19:

1. Download and Install Python 3.10.11, and make sure you enable the **ADD TO PATH** option during installation.
2. Install Rust Toolchain for the dependency compilations.
3. Locate the **casmbot.zip** (CA Service Management *17.4.3.0:\filestore\Virtual_Analyst\casm-bot.zip*).
4. Choose a directory to install Virtual Analyst and create a folder, for example, **VA**. Extract the contents of **casmbot.zip** to the casm-bot folder inside the VA directory.
   This directory structure will contain the virutal analyst scripts and the associated Rasa virtual environment that the virtual analyst will use.
5. Create a virtual environment for VA in the same folder with the following command:
   ```
   python -m venv ./botenv
   ```
6. Run the following command to activate the virtual environment.
   ```
   ./botenv/Scripts/activate
   ```
7. Upgrade pip inside the virtual environment by running the command:
   ```
   python -m pip install --upgrade pip
   ```
8. Run the following command to install RASA v3.6.19:
   ```
   pip install rasa==3.6.19
   ```
9. Run the following command to install Spacy:
   ```
   pip install rasa[spacy]
   pip install websockets==10.4
   pip install protobuf==4.23.3
   ```
10. Download the spacy language model. Refer to the Virtual Analyst Supported Locale to download language models for other languages. (Ignore the step to link the model as linking is deprecated in latest spacy versions).
    ```
    python -m spacy download en_core_web_md
    ```
11. Navigate to casm-bot folder and train the RASA Model, by running the following command:
    ```
    cd casm-bot
    rasa_train.sh
    ```

12. Run the RASA Server.

```
rasa_run.sh
```

Virtual Analyst Supported Locales

NOTE
Japanese is currently not supported on Virtual Analyst.

Table 3: Virtual Analyst Supported Locales

Supported Locales	Supported Commands
English (en)	Run the following command to download English: `python -m spacy download en_core_web_md`
German (de)	Run the following command to download German: `python -m spacy download de_core_news_md`
Spanish (es)	Run the following command to download Spanish: `python -m spacy download es_core_news_md`

French (fr)	Run the following command to download French:
	`python -m spacy download fr_core_news_md`
	`python -m spacy link fr_core_news_md fr`

Italian (it)	Run the following command to download Italian. **""md""** stands for medium. `python -m spacy download it_core_news_md`
Portuguese (pt)	Run the following command to download Portuguese. **""md""** stands for medium. `python -m spacy download pt_core_news_md`

Offline Installation - CA Service Management Virtual Analyst

Offline Installation of Virtual Analyst requires two systems or servers:

1. One system with Internet connectivity.
2. The other, an offline system with no Internet connectivity.

Perform the following steps to download all the required packages and libraries for Installation on a system that has Internet connectivity:

Using RASA Server 3.0.6

1. Create a directory named **VA** under the root directory.
2. Download Python 3.8.10. Make sure you enable the **ADD TO PATH** option during installation.
3. Follow steps 1-4 from Automated Installation of Virtual Analyst for Windows.
4. Zip the content of *casm-bot folder* to casm-bot.zip.

Perform the following steps for offline installation:

1. Transfer Python 3.8.10 installation file which was downloaded in the other machine (system with internet connectivity) and install it.

2. Create a directory named VA under root directory(maintain the same path that you gave for the system with internet connectivity). In this case, C drive is used as root directory and hence the VA home is C:/VA.
3. Navigate to VA home directory from the Command Prompt(Administrator) and copy the casm-bot.zip file and extract the content to casm-bot folder.
4. Install Python 3.8.10 on the offline system to the same path used in online system. By default, this path is *%ProgramFiles%\Python38.*
5. Navigate to the casm-bot folder from the command prompt and run the following command to activate the virtual environment in the offline machine:
   ```
   .\botenv\Scripts\activate
   ```
6. Run the **rasa_train.bat** to train the VA. After training the model, run the **rasa_run.bat** to run the VA.
7. Type *http://<localhost>:5005* in a browser to see a "Hello" message from RASA.

Using RASA Server 3.6.19 for 17.4.3.0

1. Create a directory named **VA** under the root directory.
2. Download Python 3.10.11. Make sure you enable the **ADD TO PATH** option during installation.
3. Follow steps 1-4 from Automated Installation of Virtual Analyst for Windows for 17.4.3.0.
4. Zip the content of *casm-bot folder* to casm-bot.zip.

Perform the following steps for offline installation:

1. Transfer Python 3.10.11 installation file which was downloaded in the other machine (system with internet connectivity) and install it.
2. Create a directory named VA under root directory(maintain the same path that you gave for the system with internet connectivity). In this case, C drive is used as root directory and hence the VA home is C:/VA.
3. Navigate to VA home directory from the Command Prompt(Administrator) and copy the casm-bot.zip file and extract the content to casm-bot folder.
4. Install Python 3.10.11 on the offline system to the same path used in online system. By default, this path is *%ProgramFiles%\Python310.*
5. Navigate to the casm-bot folder from the command prompt and run the following command to activate the virtual environment in the offline machine:
   ```
   .\botenv\Scripts\activate
   ```
6. Run the **rasa_train.bat** to train the VA. After training the model, run the **rasa_run.bat** to run the VA.
7. Type *http://<localhost>:5005* in a browser to see a "Hello" message from RASA.

Train the NLU Server

After successful installation of Virtual Analyst, training the RASA NLU server is mandatory for Virtual Analyst to work in your environment. You must also train the NLU server whenever you make any changes in Rasa (adding utterances, modifying or customizing intents, and so on).

Perform the following steps to train the NLU server:

1. If you have installed a virtual environment for installing Rasa. ensure that you activate the virtual environment before training the NLU server. For more information, see Install Virtual Analyst.
2. Navigate to the directory VA which was created during installation and run the following command to activate the virtual environment:
   (Windows)
   ```
   .\botenv\Scripts\activate
   ```
   (Non-Windows)
   ```
   ./botenv/Scripts/activate
   ```
3. Change the directory to casm-bot. Run the following command:

```
cd casm-bot
```

4. Train the Rasa model.
   a. **(Windows)**
      Run the following command:
      ```
      rasa_train.bat
      ```
   b. **(Non-Windows)**
      Run the following command:
      ```
      rasa_train.sh
      ```

Start/Stop the NLU Services

Execute the following commands from Anaconda Prompt to start and stop the Rasa NLU services:

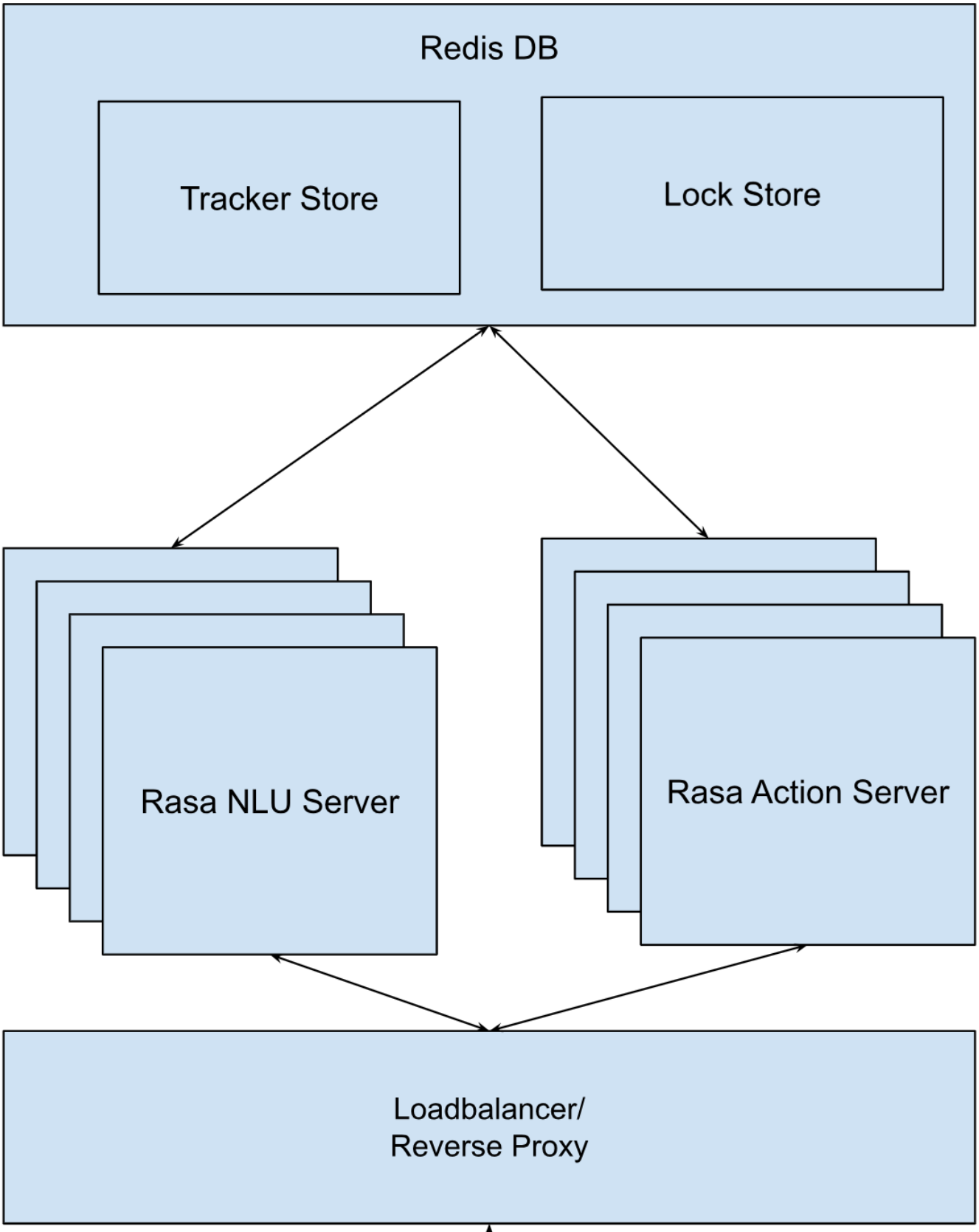
1. **(Windows)**
   Run the following command:
   ```
   rasa_run.bat
   ```
2. **(Non-Windows)**
   Run the following command:
   ```
   rasa_run.sh
   ```
3. Verify if Rasa services are up and running. Open the Rasa URL browser window (*http://<localhost>:5005*) and check for **"Hello from Rasa"**message.
4. To stop Rasa NLU services, run the following commands:
   a. **(Windows)**
      Run the following command to stop the services:
      ```
      rasa_halt.bat
      ```
   b. **(Non-Windows)**
      Run the following command to stop the services:
      ```
      rasa_halt.sh
      ```

RASA Clustering

RASA clustering with and without Dockers

RASA Clustering using Dockers

Rasa server can be scaled as shown:


Redis DB
Tracker Store
Lock Store
Rasa NLU Server
Rasa Action Server
Loadbalancer/
Reverse Proxy

RASA cluster contains the following components:

- **Load Balancer**:
  A load balancer, for eg, Nginx listens to RASA NLU requests and sends them to one of the many available Rasa NLU Servers. This is also used to balance the requests that are sent by Rasa NLU servers to one of the many available RASA Action Servers.
- **RASA NLU Servers**:
  The NLU server interprets the user's inputs, this helps to understand the context of the user's, and MS Teams Bot responds accordingly. The NLU server can handle tasks like tokenization, named entity recognition, and intent classification.
- **RASA Action Servers**:
  The Action server is where the responses to the user's inputs are handled. Actions in RASA can be anything from simple responses to executing complex tasks like calling an API, querying a database, or any other custom action defined in the RASA framework. Actions can also influence the flow of the conversation by determining which question or statement the chatbot should deliver next based on the current state of the conversation.
- **Redis DB:**
  By default RASA keeps the session information and the locking mechanism in memory, which is not a good practice in multi node configuration. In a multi node configuration of RASA the locking and session information must be kept in a shared location. RASA offsets support for Redis DB to keep this information. For more details, check “Trackler Stores” and “Lock Stores” in RASA documentation. It is highly recommended to configure the cluster using Docker Compose technology. See below a few samples of the key configuration files.

It is recommended to configure the cluster using Docker Compose technology.

Refer the following samples of the key configuration files:

docker-compose.yaml

```
version: '3.7'
services:
  core:
    ports:
      - 5006-5016:5005
    volumes:
      - ./casm-bot:/app
    image: rasa/rasa:3.0.6-full
    entrypoint: ["rasa", "run", "-m", "models", "-p", "5005", "--endpoints", "endpoints.yml", "--enable-api"]
    restart: unless-stopped
  action:
    ports:
      - 5056-5066:5055
    volumes:
      - ./casm-bot:/app
    image: rasa/rasa:3.0.6-full
    entrypoint: ["rasa", "run", "actions", "--actions", "actions", "-p", "5055"]
    restart: unless-stopped
  nginx:
    container_name: rasa-nginx
    image: nginx
    volumes:
     - ./nginx/nginx.conf:/etc/nginx/nginx.conf:ro
     - ./nginx/logs:/logs
ports:
     - "5005:5005"
     - "5055:5055"
```

```
  redis:
    container_name: rasa-redis
    image: redis:latest
    ports:
      - "6379:6379"
```

nginx.conf

```
upstream rasa_server {
        server EXTERNAL_LB_IP:5006;
        server EXTERNAL_LB_IP:5007;
        server EXTERNAL_LB_IP:5008;
        server EXTERNAL_LB_IP:5009;
        server EXTERNAL_LB_IP:5010;
        server EXTERNAL_LB_IP:5011;
        server EXTERNAL_LB_IP:5012;
        server EXTERNAL_LB_IP:5013;
        server EXTERNAL_LB_IP:5014;
        server EXTERNAL_LB_IP:5015;
        server EXTERNAL_LB_IP:5016;
    }

    upstream actions_server {
        server EXTERNAL_LB_IP:5056;
        server EXTERNAL_LB_IP:5057;
        server EXTERNAL_LB_IP:5058;
        server EXTERNAL_LB_IP:5059;
        server EXTERNAL_LB_IP:5060;
        server EXTERNAL_LB_IP:5061;
        server EXTERNAL_LB_IP:5062;
        server EXTERNAL_LB_IP:5063;
        server EXTERNAL_LB_IP:5064;
        server EXTERNAL_LB_IP:5065;
        server EXTERNAL_LB_IP:5066;
    }

    server {
listen       5005;
        server_name  localhost;

        keepalive_timeout   30;
        client_max_body_size 800M;

        location / {
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $remote_addr;
            proxy_set_header X-Forwarded-Proto $scheme;
            proxy_set_header Host $host;
            proxy_pass http://rasa_server;
        }

        location /socket.io {
```

```
            proxy_http_version 1.1;
            proxy_buffering off;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $remote_addr;
            proxy_set_header X-Forwarded-Proto $scheme;
            proxy_set_header Upgrade $http_upgrade;
            proxy_set_header Connection "Upgrade";
            proxy_pass http://rasa_server;
        }

        location /robots.txt {
            return 200 "User-agent: *\nDisallow: /\n";
        }
    }

    server {
        listen       5055;
        server_name  localhost;
        location / {
            root   html;
            index  index.html index.htm;
            proxy_pass http://actions_server;
        }
    }
```

endpoints.yml

```
  action_endpoint:
  url: "http://rasa-nginx:5055/webhook"

tracker_store:
  type: "redis"
  url: "rasa-redis"

lock_store:
  type: "redis"
  url: "rasa-redis"
```

Run this command to scale RASA servers.

```
"docker-compose up -d –scale core=X action=Y"
```

This would start X core containers and Y action containers.

RASA Clustering without Docker

1. Install RASA using the steps from Install Virtual Analyst.
2. Install Redis Database Server from https://redis.io/docs/getting-started/installation/.
3. Refer to comments in rasa_run.bat to spawn multiple RASA instances. Replicate existing endpoint.yml file for each instance that you want to create and update the content of endpoints.yml file as shown:
   action_endpoint

   ```
   url: "http://localhost:5055/webhook"
   ```

tracker_store

```
type: redis
    url: <url of the redis instance, e.g. localhost>
    port: <port of your redis instance, usually 6379>
    db: <number of your database within redis, e.g. 0>
    key_prefix: <alphanumeric value to prepend to tracker store keys>
    password: <password used for authentication>
    use_ssl: <whether or not the communication is encrypted, default `false`
```

lock_store

```
type: redis
    url: <url of the redis instance, e.g. localhost>
    port: <port of your redis instance, usually 6379>
    db: <number of your database within redis, e.g. 0>
    key_prefix: <alphanumeric value to prepend to tracker store keys>
    password: <password used for authentication>
    use_ssl: <whether or not the communication is encrypted, default `false`>
```

Configure Virtual Analyst in Service Desk Manager

This article describes the following topics. Perform the following tasks to configure Virtual Analyst in CA Service Desk Manager:

NOTE
If **va_chat_sessions** table is already populated before enabling multi-tenancy, it's recommended that you run the script after enabling multi tenancy. For more information, see Manage Multi-Tenancy. For example: SREL Type Syntax

```
<Object name="va_chat_sessions">
   <TenantRule type="SREL">last_mod_by</TenantRule>
   <Where>tenant is null</Where>
</Object>
```

NOTE
For a multi-tenanted environment, ensure that **System_Bot_User** (Service Bot User) in SDM is added to the Service Provider tenant.

Enable Virtual Analyst

Perform the following steps to enable the Virtual Analyst Bot Services:

1. Login to CA Service Desk Manager.
2. Navigate to **Administration**, **xFlow Interface**, **General**, Open the **va.bot.nlu.server.url** configuration.
3. Update the value "**configuration value"** to rasa service (*http://<RASASERVER>:5005)*. Default port is **5005**.
4. You can enable Virtual Analyst Chat Engine in Service Point Configuration. For more information about how to add these configurations in Virtual Analyst, see Manage Features.

Configure the Collaboration Microservices

NOTE
For Linux, add absolute path for **local.dir** and **leveldb.dir** in application.conf file (*<XFLOW HOME>/Apps/Services/collabmicroservice-version>/conf*).

For example:

```
local.dir = "/opt/CA/xFlow/APPS/Services/collabmicroservice-17.0.479/chatlog-persistence/snapshots
leveldb.dir="/opt/CA/xFlow/APPS/Services/collabmicroservice-17.0.479/chatlog-persistence/journal
```

Perform the following steps to configure Collaboration Microservices:

1. Navigate to *<XFLOW HOME>/Apps/Services/collabmicroservice-version>/conf*.
2. Take a backup of **application.conf** file.
3. Open the **application.conf** file, verify the property **search.ms.url**, ensure that the value is pointing to **search microservice**. Example: "*search.ms.url="http://<LOCAL HOST>:9006"*
4. Restart the xFlow Interface services.

Data Configuration

If you have configured your ticket numbers with any prefix or suffix in the CA Service Desk, perform the below steps.

Example Use Case: Consider a use case where ticket prefix is **abcd** and suffix is **12**:

1. Navigate to **Administration**, **Service Desk**, and **Sequence Numbers**.
2. Open **Request/Incident/Problem Number**. Verify the prefix and suffix value.
3. Navigate to the installed location of Rasa, open the **data**/**nlu.md** file.
4. Under the **inform** intent, you can find some training examples for getting ticket details. Consider the following statement
   a. Please show ticket **[40](tkt_id)**
      40 is the ticket ID.
      Please show ticket [40](tkt_id) is modified to "**Please show ticket [abcd4012](tkt_id)**"
      abcd = ticket prefix and **12=** ticket suffix.
5. Perform these replacements for at least **5** statements in the inform intent section.
6. Add at least **5** more ticket IDs under the inform intent.
7. Ensure that you use different ticket IDs for proper Rasa training.
8. Navigate to **regex:tkt_id** and add a regex which will describe ticket IDs as per the CA SDM configuration. For this example, regex would be **abcd[0-9]{2}**.
9. Now, start training the NLU server, Run the command in Anaconda Prompt:
   ```
   rasa_train.bat
   ```
   Restart Rasa services.
10. Navigate to Collaboration Microservices location (***<xFLOW HOME>/Apps/Services/collabmicroservice-<version>/conf***).
11. Open the **entity_mapper.cfg** file. Check the '**where**' clause under the entity name **tkt_id**. Ensure that it is inline with the ticket format configured in CA SDM.

Urgency

Urgency for a ticket can be configured in SDM. When some changes are made to urgency configuration, users must update the same configuration in Rasa also in order to work with the virtual analyst.

Out of the box, urgency values are **1,2,3,4,5**. If you want to add a value, say **6**, perform the following steps to configure Urgency for a ticket:

1. Navigate to **cfg/config.json** and under the urgency field, add new urgency values to the list of urgency values.
2. If some training is required for Rasa to pick up this urgency, you may have to add some training samples to the **nlu.md** file under the i**nform_urgency** intent.

Chat Session Details

Administrators can run a query on Oracle or Microsoft SQL database server to get additional details of user-initiated chat session with the Virtual Analyst. The query will provide the following information:

- Details of a user initiated chat session with the service bot for a specific issue.
  For example: User Name, chat text, chat room name, and chat session ID, user provided feedback, chat session rating, and comment details.

```
select A.sender,A.msg_text, B.rating, B.comments from chat_log A INNER JOIN va_feedback B ON A.chat_room_name
 = B.va_chat_session_id
```

User Chat Logs

User chat log provides information about a user chat utterance and confidence of the Service Bot in classifying that utterance to an intent. This log is generated as comma separated values. The log can be opened with any CSV reader for better readability purposes.

Perform the following to view the user chat log file:

1. To locate the log, navigate to **xFlow Home>/APPS/logs** and change the extension of the file from **user_chat.log** to **user_chat.csv**.
2. User chat log provides details for the following values:
   a. **Time**: The time at which the user chat was initiated.
   b. **Session ID**: Each user utterance in the bot is tied to a chat session and the log can be filtered based on this value to identify the user utterances in a particular session.
   c. **Utterance**: Utterance pertains to the user chat text that is send to the Service Bot. Actions are special kind of utterances which are always classified as a particular intent with confidence level **1**. Actions start with character **"/"** followed by the intent name. For example: Action "**/get_tkt_status**" will get the status of a ticket.
   d. **Intent**: Specifies an intent for a given user utterance. This is classified by the Service Bot.
   e. **Main Intent**: There are scenarios where the bot can request information from a user to fulfil a request. One example of this is adding comments to a ticket. In this case, the bot needs both ticket ID and added comment to fulfil a request. When the user provides an information, these utterances are also classified into intents. The Main Intent here is "**adding a comment**" and "**sub-intent**" (known as intent in general) can be "**ticket id**" and "**providing comment**".
   f. **Confidence**: Confidence is a metric which shows how confident the service bot is with intent prediction. The threshold value for confidence metric is **0.7**. If the confidence level is below**0.7**, service bot is unable to comprehend the issue described by a . The user utterance can be a good sample data for training the bot.

Service bot must be trained with sample log data to help with accurate intent prediction. The user log files provide details for utterances that has low confidence threshold values. The utterances with confidence value less than 0.8 can be used to train the bot to improve intent prediction. Administrators must review these utterances before training.

Virtual Analyst Intents

CA Service Management supports the following intents for Virtual Analyst:

Task Name	Supported Bot Intents	Description	Example
My Open Tickets	**my_open_tickets**	Retrieves the list of open tickets.	• Can you show my ticket list? • Ticket Status.
Ticket Status	**get_tkt_status**	Displays the status of a specific ticket.	• Can you tell me the status of ticket 50?

Search Knowledge Base	**search_kd**	This intent helps the virtual analyst to search the CA SDM knowledge Base and get relevant KB articles for a said problem/issue.	• Can you suggest me a solution? • I have a problem with my laptop?
Create a Ticket	create_ticket	This intent helps the bot to create a ticket for a said problem/issue.	• Can you create an incident for me? • Help me create a ticket.
Close Ticket	**close_tkt0**	Ticket is closed	• Can you please close the ticket. • I want to close ticket 23
Update Urgency	**upd_tkt_urgency**	Updates Ticket Urgency.	• Please change the urgency of ticket 23. • Update Urgency
Add comment to a Ticket	**add_tkt_comment**	Adds a comment to a ticket.	• Add comment to ticket 23.
Transfer the chat to an Analyst	**transfer_to_analyst**	This intent connects the business user to a human analyst through a chat queue.	• Please connect me to an Analyst. • I want to talk to a human analyst.
Forgot password Administrators can set the password reset url in forgot password option: <**CollborationMicroservicePath**> **\conf\intent_mapper.cfg** >	**forgot_password**	Shows the Password Reset URL link (set by Administrators)	• Hey I Forgot password. • Reset my password
Reset the chat **NOTE** '**Reset**' is a powerful command and helps in clearing or revoking an existing conversation to start a fresh one. You can use this command when the bot is not responding well to queries or is stuck while performing a task.	**Reset**	This intent lets the virtual analyst or bot stop an ongoing conversation and start a fresh conversation.	• Begin • Reset • Restart • You can't help me

How to train the Service Bot with additional data or when it is unable to understand the given intents?

This scenario may arise when a logged in business user tries to make a statement, and the service bot responds with "**I am unable to understand**".

Verify the **collabMS.log** in ***<XFLOW Home>/logs/collabMS.log*** location. This log file contains chat text entered by a business user along with confidence-level. If confidence level is less than the suggested threshold leve**l (0.6>)**, train the intent with new utterances.

Sample example: When the Training Threshold is not Enough-

Training needed : The confidence is **0.3912886982016276** for **intent:affirm**

For example, "**Delegate the conversation to an analyst"** may not be understood by the service bot as compared to the intent "**transfer to an analyst**"".

- Navigate to where you have installed Rasa Server.
- Open the **nlu.md** file.
- Search the said intent (**transfer_to_analyst**).
- Add the utterance which is failing in the model data format.
- Open the Anaconda prompt window in the Rasa home directory.
- Stop Rasa Services.
- Train the model by using command 'rasa train'.
- Start the service using '**run**'.

> **NOTE**
> Ensure the length of the new utterances are similar to the existing ones for an intent. Do not train with rarely used utterances or statements. Always use frequently used utterances.

Few sample examples below showcase recommended and complex utterances:

Recommended Utterances	Complex Utterances
• Show my open tickets • What are my open tickets?	• Awesome, for displaying the complete list. But, want to see my open tickets.
• Want to see my ticket details • Show my ticket details	• In a jiffy, can you show me my ticket details?
• What is my ticket status? • Tell my ticket status	• Are my open tickets assigned to an analyst or are they still in waiting state?

It is recommended that you use simple utterances to train the bot. Complex sentences complicate the initial training data. Try to use brief and simple utterances or sentences.

Customize the Virtual Analyst Responses

Customizing the Configuration Files

Customize the Virtual Analyst responses by making changes to the **Entity Mapper** and **Intent Mapper** Configuration Files.

Perform the following steps to customize the responses:

1. **Intent Mapper:** This file is used to map an intent coming from NLU with an action executed in Service Management and the response. Location of this file: ***<xFLOW HOME>/APPS/Services/collabmicroservice-<version>/conf/intent_mapper.cfg***
2. **Entity Mapper**: This file is used to map an entity in the NLU server with system parameter. For example, a ticket number will be translated to persistent ID. Location of this file: ***<xFLOW HOME>/APPS/Services/collabmicroservice-<version>/conf/entity_mapper.cfg***

> **NOTE**
> You can update the response string by adding new message code which is defined in the **messages.language** file. Response strings can be edited by users. it is recommended that configuration file must always be edited by Administrators.

It is recommended that the configuration file must be edited by Administrators.

To customize or modify the service bot responses, retrieve the bot responses as shown below. Virtual Analyst response are kept in two locations, RASA server and collaboration microservices.

1. **Rasa Services**:
   - Open the **domain.yml** file.
   - Navigate to the templates section, search for the Utterance that you want to update. Change it and train the NLU server using the “**rasa train**” command. In case, if you want to add a new utterance refer to the existing text format and replicate the format.
   - The response comes from RASA server, when a query is initiated to fulfill an intent or when a query does not require a Service Management action. This response is derived from the **domain.yml**. The response may contain text and buttons. The response is defined under the templates node in the **domain.yml** file. RASA responds or provides different text for the same conversation flow at different points of time.

     Example: "*Provide me the ticket number*." OR "*Did you forgot to provide the ticket number, please give the details*."

     Such responses can be defined in an array node, so that RASA can pick one randomly. The response contains text and buttons. In case you want to remove a button, remove the title and the payload pair together.
   - After making the above changes, ensure to stop the RASA server, train it, and start it again.
     For more information, see RASA documentation.
2. **Collaboration Microservice**s
   - Navigate to the configuration file: ***<xFLOW HOME>/APPS/Services/collabmicroservice-<version>/conf***.
   - To update the response text, open the **messages.en-US** file or a supported language-specific file.
   - Search for the message string that you want to edit, update the message, and restart the xFlow Analyst Interface Services.
   - All bot-related messages start with **BOT, BCF,** and **BWP.** **BCF** messages are used in the intent mapper and entity mapper file. **BWP** are confirmation prompt messages.
   - When the query requires action from CA SDM, the response is built at the microservice layer. To update or change the response of any specific flow, first figure out the intent. Open the **intent_mapper.cfg**, and go to the specific intent.
   - The response node contains array of message code starting with **BOT**, **BCF**, and **BWP**. BCF messages are used in the intent mapper and entity mapper file. BWP are confirmation prompt messages:
     - Add additional responses, in the JSON array with a comma-separated value, so that virtual analyst can pick these responses randomly.

       NOTE
       Ensure that the newly added message code do not start with **BOT**,**BCF**, or **BWP**.
       For example, you can add "TEST0001", and the string can be added in the **messages.en-US** file with the same code.
     - To update the label of a button, pick the message code from the buttons array node in the footer, and search for the code in the **messages.en_US** file. You can edit the text as required.

       NOTE
       CA Service Management supports only a single message code.
     - To remove the buttons from the response, remove the complete JSON node containing label and action.

Customize the Virtual Analyst Help Text

When a user enters "Help"/"help" in the Virtual Analyst chat interface or instructs the Virtual Analyst to perform certain tasks, it shows a help content with few pre-defined user queries. For example: "I want to know the status of my ticket", "Printer is not working" and so on.

You can customize the help content as required. To customize the Help text for Virtual Analyst, perform the following steps:

1. Navigate to where you have installed collaboration micro services.
2. Locate the folder <**CollaborationMicroServiceHome>/conf/vafiles**

3. In the language specific folder, locate the configured language folder.
   For example if you have **English** as the default language, the folder location is <*CollaborationMicroServiceHome*>/*conf/vafiles/en_US.*
4. Open the file **help.txt** and edit as required.
   This file supports HTML formatting. Ensure not to add rich html content and validate the HTML formatting before saving the file.
5. From the Service Point interface, open the Virtual Analyst chat interface, and enter help/Help. Verify your changes.

Custom Attributes for Ticket Creation

If you are using custom attributes for ticket creation and want to add these attributes as part of ticket creation, review the following two options on how you can add custom attributes in Virtual Analyst as part of ticket creation process;

1. **Updating the Virtual Analyst (VA) Template**: After the VA template is updated in Service Point configuration via Service Desk Manager, the custom attributes values are picked from the VA template. Custom attribute values are similar for all the tickets created using the Virtual Analyst. In case of a tenanted environment, the Service Point configurations will differ for each tenant if you have not applied the default configuration.
2. **Service Bot prompts user for custom attribute field inputs** The Virtual analyst Service Bot understands the user string with the help of the NLU engine (Rasa engine). It is recommended that you update the collaboration microservices for actions and Rasa engine for the conversation flow.

The **entity mapper** file in Virtual Analyst takes care of querying the business user when data is not available or confusing.

Example: Let's create a ticket and add two attributes that will be queried to a business user while creating this ticket.

- Take '**impact**' as one of the attributes. **"Impact**" is a relation or SREL attribute (stores data in another table).
- Second attribute will be 'ticket avoided', which is a plain attribute.

Add Custom Attributes to an Intent

Perform the following steps to add custom attributes to an intent:

Step 1

Open the **intent_mapper.cfg**. Add the attributes in entity list under **create_ticket** intent.

Add the following code under '**entities**' node:

Example: Single RELation (SREL) Attribute

```
  {
 "name" : "my_impact",
 "required" : "false",
 "default" : "_SKIP_NLU_"
         },
```

Example: Plain Attribute

```
 {
 "name" : "tkt_avoided",
 "required" : "false",
 "default" : "_SKIP_NLU_"
 }
```

NOTE
_SKIP_NLU_ is required if you want to skip training the NLU engine to manage the conversation. Also "**required**" is kept as false, so that the xFlow Interface does not expect inputs from NLU server.

Step 2

Update the intent **create_ticket** for the action, this is required to map the entity with system attribute (majic layer attributes).

1. Add the following code under the **action' > 'Parameters** node:

Example: Single RELation (SREL) Attribute

```
{
"name" : "impact",
"data_type" : "String",
"from_entity" : "my_impact"
},
```

Example: Plain Attribute

```
{
 "name" : "ticket_avoided",
 "data_type" : "Integer",
 "from_entity" : "tkt_avoided"
}
```

In parameter definition, ensure '**name**' field matches with **cr** object attributes (**impact** and **ticket_avoided**). Data type can be taken from the majic layer or from xFlow Interface java pojo source file. The **'from_entity'** must match the name defined in **Step 1**. After updating **Step 1** and **Step 2**, ensure that JSON format is not corrupted.

Step 3

After defining entities and system parameter in the intent mapper, update the entity mapper file. This file is used to map the user provided data with system data. For example, user **ServiceDesk** must be mapped to the UUID value. This helps to define a set of input expected from user when the data is a plan attribute and user is free to enter anything as input text.

1. Open the **entity_mapper.cfg** and add the following code:

Example: Single RELation (SREL) Attribute

```
{
"name" : "my_impact",
"ui_name" : "Ticket Impact",
"description" : "This entity is ticket impact to be mapped to imp enum",
"system" : "sdm",
"object" : "imp",
"id_attribute" : "enum1",
"display_attribute" : "sym",
"where" : "sym like '%@@my_impact@@%'"
},
```

Example: Plain Attribute

```
    {
  "name" : "tkt_avoided",
  "ui_name" : "Avoid Ticket",
  "description" : "This entity is avoid ticket flag, no mapping needed",
  "system" : "none",
  "object" : "",
  "id_attribute" : "",
  "display_attribute" : "0,1",
  "where" : ""
 }
```

The first entity mapping is done to map user input for impact attribute with SDM impact object:

> **NOTE**
> After a newly added JSON is formatted with customer attributes, it is better to cross-check changes in third-party tools to ensure that JSON format is properly defined. Once **intent_mapper.cfg** and **entity_maper.cfg** are updated, verify whether the JSON formatting is good by using any JSON formatter tool.

- **name**: This value must match with entity name defined in **Step 1**.
- **ui_name**: This string is used to querying the user. For example bot may ask, “**please provide 'Ticket Impact'"**. This can be put into message. LANG (LANG can be **en_US**) file as key value pairs and provide the key like 'USERMSG0001' here.
- **description**: This is purely for reference/remarks purposes. Bot does not use it.
- **system**: The name of the system from where we map the data.
  > **NOTE**
  > Currently, only CA SDM is supported.
- **object**: Provide the majic factory name. For '**impact'** the factory name is **imp**.
- **id_attribute**: Specify the relational attribute(relAttr) of the factory mentioned in object field. In the above example, 'enum' is the relation attribute of 'imp' factory, however enum1 is a special name given in xFlow Interface layer only for enum attribute. Rest all attribute are same as majic layer attribute name.
- **display_attribute**: Attribute used to display user name, in case, if multiple record exists, and user has to choose one, which is common name attribute in case of Service Desk.
- **where**: This is the clause used to filter the data based on users input. You can use the user entered data, which is entity name (**my_impact**) in the form of **@@my_impact@@**. This will get replaced with users input. Note that flexibility can be added by adding **%**, so that the user does not have to submit the exact string. The **where** clause can be made rigid or flexible, depending on the use case.

Define Permissions

As an administrator, control the permissions for all the apps within the xFlow Interface. Set the Service Point as a default app for a particular access type:

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Interface**, and **App Configuration**.
3. From the list, select **Service Point**.
4. Select an access type. (For example, Administration)
5. From the Access Type Detail page, navigate to the **APPS** tab.

6. Select **Service Point** as the default app.
7. Click **Save.**Service Point is set as the default app.

Define the User Context

To display the most relevant information to the user in Service Point, you must define the user context. The most important pre-defined aspects of user context are: location, organization, and department of the user.

> **NOTE**
>
> You can create a user context at a group level or at a tenant level for each object type.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Interface**, **Service Point**, and **User Context**.
3. In the Name column, select the desired tenant configuration name for which the user context must be defined.
4. Click **Edit**.
   The predefined (User Context) attributes are displayed.
5. From the list of attributes, you can modify only the attribute weight.
6. Click **Edit in List**.
7. To modify the attribute weight, select the attribute from the list.
8. Specify the weight.
9. Click **Save**.

Predefined User Context

For a default tenant, the following are predefined attributes for User Context:

Attribute	Weight
Location	10
Organization	8
Department	9

> **NOTE**
>
> Based on organizational requirements, an administrator can turn-off user context. To turn-off user context, set the attribute weights to 0.

Customizing Service Point appearance

An administrator can alter the look and feel of the Service Point interface by modifying CSS. This is the starting point for branding your Service Point instance and changing basic system defaults, such as the background image.

It is recommended that you do not make extensive CSS customizations that can alter the product's UI.

Follow these steps:

1. On the server where xFlow Interface is installed, navigate to the `xFlow Interface root directory\APPS \Services\incidentmicroservice-17.0.479/public` folder and open the `casm.html` file in a text editor.
2. Ensure that the following line is uncommented.

```
<link rel="stylesheet" href="custom.css">
```

3. In the case of CA Service Management 17.3.0.16 or a below version, add the following line inside the head element.

```
<link rel="stylesheet" href="custom.css">
```

4. Save the file.
5. On the server where xFlow Interface is installed, navigate to the `xFlow Interface root directory\APPS\Services\incidentmicroservice-17.0.479/public` folder and create a new file `custom.css`.
6. Open `custom.css` file in a text editor.
7. Create your own CSS class to override the system's default CSS class.
   a) For example: Copy and paste the following content in `custom.css` file

```
.searchSection{
background-image:url(/assets/images/icons/BG4_SP.jpg);
background-repeat:no-repeat;
background-size:cover
}
```

   b) Replace **BG4_SP.jpg** with your background image.
8. Save the file.
9. Refresh the browser cache.

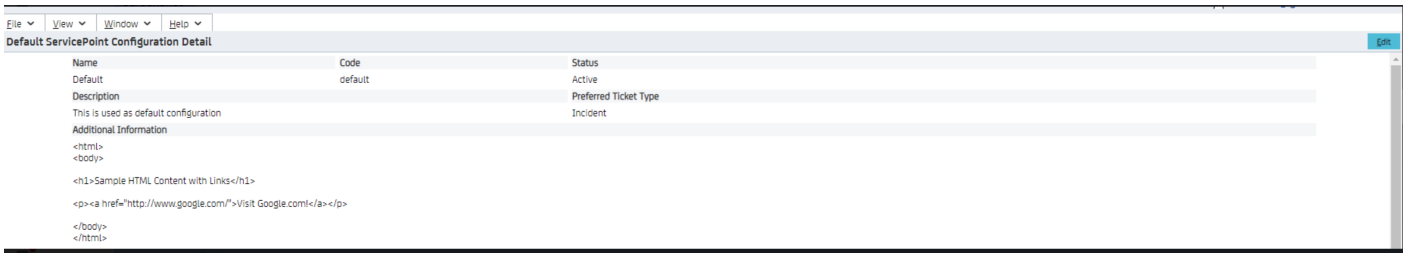
Add Additional Information

This article describes how to add additional information in CA Service Desk Manager (SDM). This information is displayed on the Service Point Home page to provide important information to end users. Additional information is displayed in the Call icon available on the Service Point Home page (top right corner of the interface). Click the Call icon to view additional Information along with contact details.

> **NOTE**
> You must enable the **call_service_desk_mobile_enable** option for the call or contact Help Desk icon to appear on the Service Point Home Page. By default the **@NX_CALL_SERVICE_DESK_MOBILE_ENABLE** option is set to **Yes**. To configure this option, you can edit the CA SDM Options Manager.

Perform the following steps to add additional information:

1. Login to CA Service Desk Interface.
2. Navigate to **Administration**, **xFlow Interface**, **Service Point**, **Configuration**, **Default**.
3. Click **Edit** to open the configuration in edit mode.



4. You can either enter HTML content or plain text in the additional information text area (text size limit is 1500 characters).
5. Save the Configuration in SDM and refresh the Home page in Service Point to reflect the changes.

Define the Default Categories

As an administrator, you can define the default **five** categories, that appear on Service Point. When a category is assigned to a ticket, the properties associated with that category are automatically added to the ticket.

NOTE

Service Point displays categories only when the following fields are selected, **Active**, **Self-Service Include** and **Self-Service Symbol**. When you change the **Incident Dependent Attributes** control (**CA SDM Administration**, **Service Desk**, **Request/Incident/Problem Areas**, **Status**), restart the xFlow Interface services.

Follow these steps to define default categories:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Interface**, **Service Point**, and **Default Areas**.
3. In the Name column, select the desired tenant configuration name for which the default categories or areas are to be defined.

   TIP

   In a multi-tenant environment, the global default category is displayed on Service Point if none of the tenants have a category assigned to them.
4. Click **Update Default Areas**.
5. Click **Search**.
6. Select from the list of categories or areas.
7. Click **OK**.
   The selected categories now appear on Service Point.

Manage Ticket Attributes

As an administrator, define the set of attributes to be displayed during ticket creation and manage the attributes that appear when viewing a ticket.

NOTE

To restrict Urgency data in Service Point, it is recommended that CA SDM Administrators create Data Partition restrictions for respective employee roles.

NOTE

The Preferred Ticket Type attribute specifies the default type of a ticket created by Service Point when the logged in user does not select a category/Area or when the selected category is designed for both incidents and requests.

NOTE

Service Point honors all transition types supported by CA Service Desk Manager employee interface. If you are changing the Status Transition type for a ticket in CA SDM, ensure to restart the xFlow Analyst Interface Services.

Create New Ticket Attribute

An administrator can configure the attributes that are displayed during ticket creation in Service Point.

NOTE

We recommend not to add the Local attributes of an object as a ticket attribute, as they are temporal and do not have a database storage.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Analyst Interface**, **Service Point**, and **Configurations.**

3. Select a configuration from the configuration list.
   The Service Point Configuration Detail page appears.
4. Click **Create New**.
   Administrators can create a tenant-specific configuration.
5. Complete the fields as appropriate:
   a. **Configuration**: The configuration to which the new ticket attribute belongs to.
   b. **Active**: Only active attributes are displayed while creating a ticket.
   c. **Ticket Attribute**: Identifies the ticket attribute name. To add a custom attribute, see Objects and Attributes.

      NOTE

      We recommend not to add the following system level attributes to be displayed while creating a ticket. This includes and is not limited to the following attributes: **Active**, **Call Back Flag**, **Closed On**, **Customer, Description**, **Heat**, **Id**, **Last Modified**, **Modified By**, **Opened On**, **Resolved On**, **Reported by**, **SLA Violation**, **Ticket Number**, **Type**, **Persistent id**, **Predicted SLA Violation**, and **Priority** (When APC is enabled).
      The following attributes will not be shown in the Service Point new ticket creation page: **Category**, **Problem**, **Status**, and **Team**.
   d. **Default Value**: A pre-filled value in the field that is displayed while creating a ticket. It displays the Majic attributes of the New Ticket attribute that is selected.
      For example, for a request object (cr), the attribute value can be *assignee* or *assignee.organization* or *summary*.
6. Click **Save**.

Manage Ticket Summary Attributes

An administrator can configure the attributes that are displayed while viewing a ticket in Service Point.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Analyst Interface**, **Service Point**, and **Configurations.**
3. Select a configuration from the configuration list.
   The Service Point Configuration Detail page appears.
4. Click **Add Ticket Attribute**.
5. Click Ticket Summary Attribute.
6. Complete the fields as appropriate:
   – **Sequence:** Represents the order in which the ticket attributes appear in the ticket summary section of the Ticket Details page.
   – **Attribute Name:** Click the magnifier icon to search for an attribute from the **Object Attribute List**. To add a custom attribute, see Objects and Attributes.
7. Click **Save.**

ServicePoint Request Approval Process

Service Point now provides the ability to approve your Service Catalog requests, provided you have an integrated environment of CA SDM and CA Service Catalog. Any logged in user in Service Point can approve, reject, Transfer, Delegate, and Take/Return CA Service Catalog Requests.

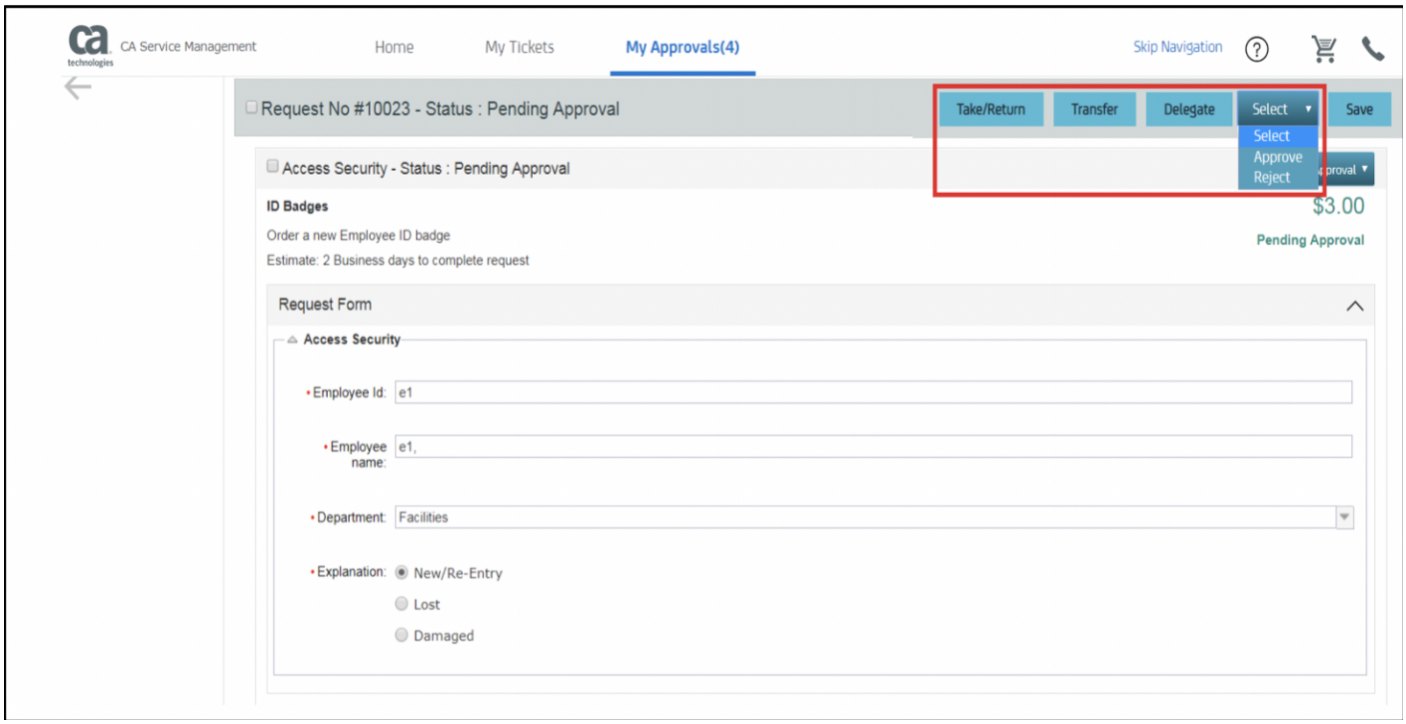
Service Point will fetch pending approval requests from CA Service Catalog and will approve the requests through Catalog Request wizards. Requests that are in Pending Approval state can be viewed by the logged in user or user groups. In case, when there are no requests assigned to a user or group, Service Point will display appropriate message. In Service Point, My Approval Tab displays all pending approval requests.

Perform the following steps:

1. Login to CA Service Point and Click on the My Approvals tab from the Home Page.
2. Select a request that is in Pending Approval state, and from the Request Page take appropriate action(s).
3. **Select**: Provides drop-down options to either approve or reject a request.

   NOTE
   Rejecting a service rejects the entire request. The request is returned to the user who submitted it and the user decides whether to update the request and resubmit it or accept the rejection as final. Once all services in the request are approved, the request wizard moves the request to the next phase of the request life cycle.

Figure 6: Figure - Request Approvals in Service point



4. **Take/Return**: Logged in users have the option to either assign a request to another analyst, user or group, or return the assigned requests in a queue back to the request assignee. Take or return ownership of requests pending action in a queue. Users can take ownership from a group queue or the queue of another user in the same business unit or sub-business unit. Similarly, you can return ownership of requests pending action from your queue to the queue of your group. You can take or return ownership of requests pending action that is assigned to your group.

   NOTE
   Take and Return works only if you are logged in as a user of the group to which the request pending action is assigned. Only CA Service Catalog administrators and other users with the required access control setting can do so.
5. **Transfer**: Request managers can transfer the requests pending actions that are assigned to themselves or other request managers. You can transfer a request pending action that is assigned to you or another user.
   For example, when a user is unavailable to perform or approve an assigned task.
6. **Delegate**: Request managers can delegate the request pending actions. Request managers can auto-delegate requests pending actions that are assigned to themselves or other request managers.
   For example, when users are unavailable to perform or approve an assigned task. Auto-delegation is useful for managers who monitor requests but who do not address requests directly. Thus, auto-delegation helps ensure that requests are addressed in a timely manner. A request pending action can be delegated only once.

NOTE
Only the owner of the request pending action can delegate, if the owner has the required access control setting.

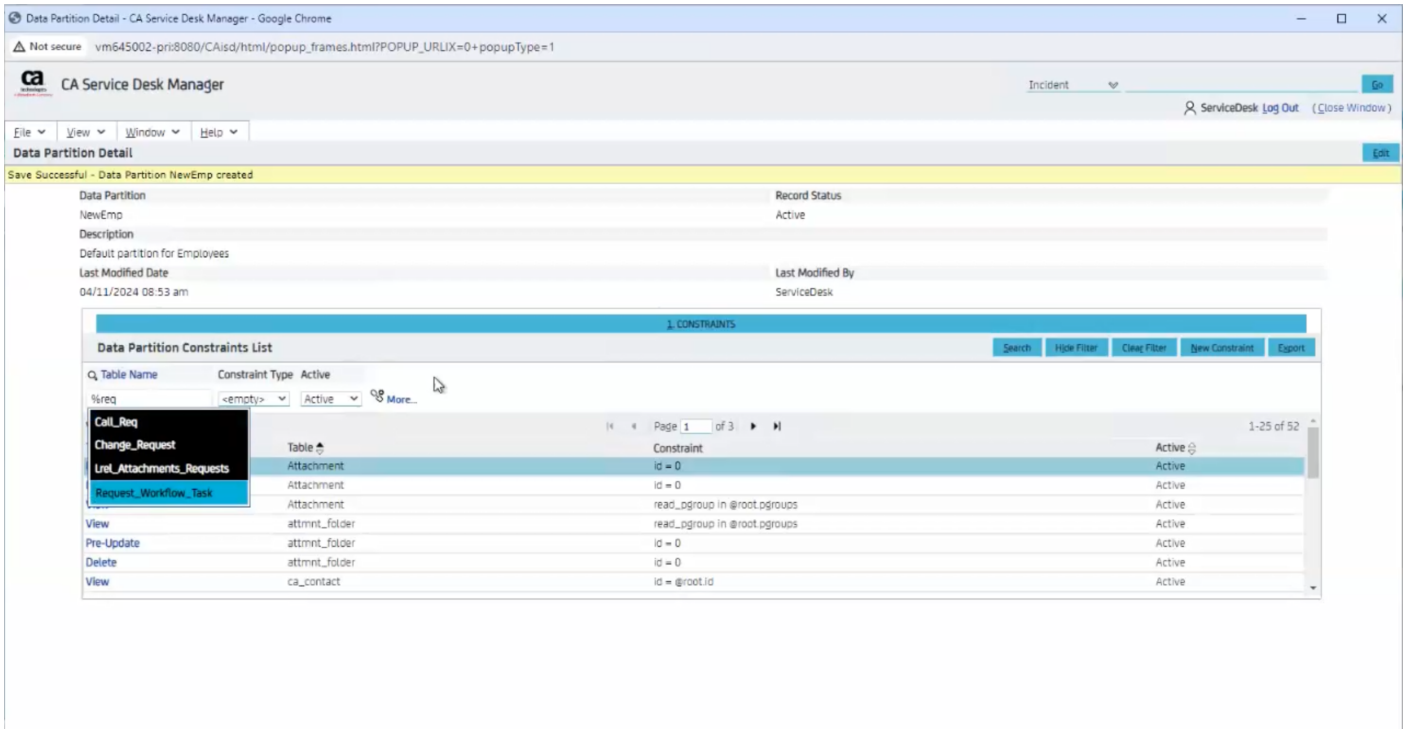
ServicePoint Task Approval Process for Classic workflow

Service Point now provides the ability to approve your customer's pending Classic CA SDM workflow tasks. In Service Point, **My Approval** displays all the pending approval requests. Service Point user can now perform actions like approve, reject or any custom task configured in the CA SDM classic workflow.

Managing Workflow tasks for Service Point

For **Employee** data partition, disable the *create* condition for the Request workflow task. Follow these steps to disable the condition:

1. In CA SDM Classic workflow, go to **Administration**> **Security and Role Management** > **Data Partition constraints** > **Data Partition** > **Data partition List**.
2. In the **Data Partition List** page that's displayed, right click on the **Employee**, and click **Copy**.
3. A page, **Create New Data Partition** is displayed, enter the **Data Partition** name as NewEmp and **Save** it.



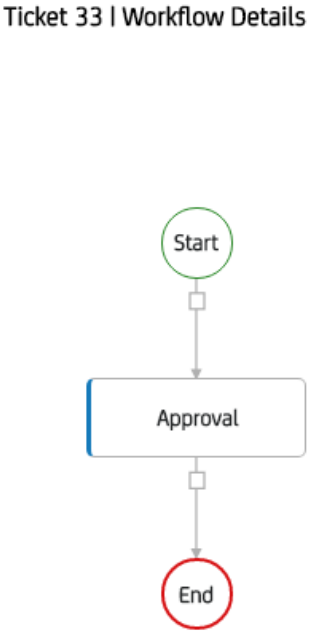
4. In the newly created Data Partition List, search for **Request_Workflow_Task.**
5. In the List displayed, click **Create** > **Edit**.
6. In the screen displayed, set the **Create** data partition constraint to **Inactive**, and click **Save**.
7. Go to Administration > Security and Role Management> Role Management> Employee Role List.
8. In the **Employee Role Detail**, update the **Data Partition Name** created in the earlier steps and set the **Override Contact Data Partition** to **Yes.**

Predefined Workflow Tasks

The following workflow tasks are available by default:

Pending Tasks

- When the end user logs in to Service Point, a new My Approval tab is displayed with a list of pending tasks that need an action by the user.
- Only the the tasks with status set to **Pending,** are assigned to the Employee user are displayed in this list. You can see the number of tasks and the graphical representation of the tasks.
- User can view the graphical representation of the workflows by clicking **View Workflow.**



Legend:
Green color indicates that the task is complete.
Blue color indicates that the task is pending.
Orange color indicates it is a future task

- Click on the required task to view the workflow details.
- Perform the required operations, and click **Save** to save the changes made.

Customize workflow fields in Service Point

- The fields displayed in the the Service Point workflow task detail page have to be configured in CA SDM (Classic interface) .
- In CA SDM, navigate to **Administration** > **XFlow Interface** > **View Configuration** > **Detail View** > **Request/Incident/Problem Workflow Task.**
- In the **Detail View Attribute List** window displayed, choose the fields you want displayed in the **Service Point Workflow Details** screen.

Manage Features

As an administrator, you can enable (show) or disable (hide) the My Resources features in Service Point:

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to **Administration** tab, select **xFlow Interface**, **Service Point**, and **Configurations**.
3. In the Name column, select the desired tenant name to enable or disable the features.

4. Click **Edit**.
5. From the **My Resources** drop-down list, select **YES** to show or **NO** to hide.
6. Click **Save**.

Add Features in Service Point

CA SDM Administrators can now add, view, and modify configurations or features for business users in Service Point. It helps administrators easily manage various capabilities of Service point and Virtual Analyst as described in this article.

To add feature configurations, perform the following steps:

1. Login to the Classic SDM UI.
2. Navigate to **Administrator**, **xFlow Interface**, **Service Point**, **Configurations**, **Default**. Click the **Feature Configurations** tab, and click the **Add Feature** button. Click **Look Up** (Search icon) on the Name Field. Click **Search**. The feature list page is displayed.
3. **Table 4: Add Features in Service Point**

Feature	Description
Chat Engine	An Option to choose between Virtual Analyst and Live Chat Support.
Virtual Analyst Greeting	Option to choose the Greeting format or welcome message in Virtual Analyst bot as L,M or F,M or L,F or L or F or M e.t.c. Last Name(L), Middle Name(M), First Name(F). Regex being used internally for validation: ([LlMmFf])([,]*)([LlMmFf]?)([,]*)([LlMmFf]?) For example: When specified as **L**, welcome message is Hi <**LastName**>. When specified as **L M** welcome message is Hi <**LastName**> <**MiddleName**>. . When specified as **L**, **M** welcome message is Hi <**LastName**>, <**MiddleName**>. In case of an invalid format, the welcome message will be something like: Hi <**Lastname**>, <**Firstname**> <**Middle Name**>
Chat with Analyst	Provides an ability to enable/disable chat with an analyst in Service Point.
Mask Analyst Name	Specify a preferred name to mask the Analyst name in Service Point.
Chat Icon	Upload and save an icon of your choice. This icon will be used to launch the Chat interface. Recommended image dimensions in the range of 126x140 pixels. Prohibited File Type formats defined by SDM Administrators will not be used. To view the Prohibited File types, navigate to SDM **Administration** tab, **Attachments Library**, **Repositories**, **ServiceDesk**, right-click and select **View**.
Virtual Analyst Launch Text	Add a text of your choice for the Chat Interface. This text appears adjacent to the Chat Icon.
Virtual Analyst Name	Name of the Virtual Analyst Service Bot. The default name provided with Service Point Virtual Analyst can be changed as per your requirement.

Feature	Description
Virtual Analyst Incident Template	Template used while creating an incident through Virtual Analyst.
Virtual Analyst Request Template	Request Template used while creating a request using Virtual Analyst.
Virtual Analyst Auto Categorization	Enable/Disable Auto-Categorization while creating an incident or a request. When disabled, template category is used while creating a ticket. If enabled, Virtual Analyst uses smart search to identify ticket category.
Allow Ticket Edit	Controls the end-users' ability to edit the ticket in Service Point. When set to **No**, ticket attributes cannot be modified in Service Point.
Notification Redirect This option is only available with CA Service Catalog and CA SDM integration.	When this option is selected from the Create New Feature page, the notification redirect is enabled for either: • CA Service Catalog Service Request details page. OR • The corresponding CA SDM Ticket details page. After adding Notification Redirect option using the Manage Feature/Create New Feature configuration, the **Service Request** option for Service Point is added. For Notification Redirect feature, select **Service Request** from the Create New Feature page. Default option is CA SDM Ticket. If this value is set to **Service Request**, notifications for Incidents/Requests which are created via CA Service Catalog requests will display notifications with CA Service Catalog request ID Appended to it. Click on the notification from Service Point to redirect to Catalog request. If the value is set to **SDM Ticket**, all the notifications of Incidents/Requests which are created through Catalog requests will be redirected to SDM Incident/Request. If this configuration is not set in the feature configuration list, then the behavior will remain same as before which is similar to '**SDM Ticket**'.

4. Select any feature that you want to add from the displayed Feature List. Select or give a value to a feature. To add Chat Engine, select **Virtual Analyst** as the value. For example, feature **"Chat with Analyst"** can be configured as "**Yes**" "**No**". When set to **Yes**, it provides an ability to enable chat with an Analyst in Service Point.
5. Click **Save** on the Create New Feature page.

Set Up My Resources

The vertical segment, **My Resources**, lists all the assets and Configuration Items (CI) assigned to the user. As an administrator, you must configure My Resources, for it to be displayed in Service Point.

The following predefined attributes are available for every asset and CI: `Asset Tag, Name, DNS Name, Host Name, Mac Address, and Serial Number`. Based on the organizational requirements, an administrator can create custom attributes for an asset and CI.

How to create custom object attribute

In case the predefined object attributes are not sufficient or do not match your environment, you can create custom object attributes.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Interface**, and **Object Attributes**.
3. Click **Create New**.
4. Complete the fields as appropriate:
   a. **Object Type**Set the type as **Configuration Item**
   b. **Attribute Name**
   Name of the attribute.
   c. **Status**
   Current state of the attribute, such as active or inactive.
   d. **Attribute Value**The majic level object attribute.
   e. (Optional) **Show in Streams**
   Select the **Show in Streams** check box to display the attribute in work streams rule list.
   f. (Optional) **Show in Attributes**
   Select the **Show in Attributes** check box to display the attribute on the Ticket Details page.
5. Click **Save**.

How to display attributes

Two (2) attributes are displayed under **My Resources** section for every asset and CI assigned to the business user. The following steps explain how you can select those two attributes.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Analyst Interface**, **Service Point**, and **Configurations.**
3. Select a configuration from the configuration list.
4. Navigate to the **My Resources Attributes** tab.
5. Click **Create New**.
6. Complete the fields as appropriate:
   a. **Active**
   Set the state of the attribute, such as active or inactive.
   b. **Configuration Item Attribute**
   Click the magnifier icon to search for an attribute from the Object Attribute List.
   c. **First Field?**
   Represents the order in which the attribute appears for the CI in Service Point.
7. Click **Save.**

Manage CA Service Catalog Featured Offerings

You can now manage CA Service Catalog featured offerings and requests in Service Point. CA Service Catalog offerings are tenant specific.

NOTE
It is recommended to avoid using page size attribute for the Select component in CA Service Catalog Form designer, when the same form needs to be rendered in Service Point.

NOTE

Ensure that you have taken into consideration the mapping between Business Unit(s) and Tenants in your environment. Common tenant administration or multi-tenant administration allows administrators to use a

single administration tool to simultaneously create and maintain business units (tenants) for multiple products, including CA Service Catalog and CA Service Desk Manager.

Perform the following steps:

1. Log in to CA SDM as an administrator.
2. Navigate to **Administration**, select **xFlow Interface**, **Service Point**, and **Configurations**.
3. Select a configuration from the configuration list, and click **Edit.**
4. From the **Catalog** drop-down list, select **YES** to show or **NO** to disable/enable the catalog featured offerings and requests in Service Point.
5. Click **Save**.

CA Service Catalog Clustered Environment Considerations

If CA Service Catalog is clustered in your environment, and you want to use the load balancer to communicate to a Catalog instance in Service Point.

NOTE

Service Point depends on CA SDM Options Manager (**Administration, CA Service Catalog, casc_endpoint**) for displaying Catalog Featured Offerings in Service Point. But in case, CA Service Catalog is clustered and you want Service Point to point to any other Catalog server or load balancer, you can override this option using the CA SDM **Administration**, **xFlow Interface**, **General, casc_endpoint** key

Follow these steps:

1. Login to CA SDM as an administrator.
2. Navigate to **Administration**, select **xFlow Interface**, **General.**
3. Search for the Configuration Key **casc_endpoint.** This configuration key is set to Inactive state by default.
4. Select the configuration key, and click **Edit**.
5. Modify the configuration Value and point to the Catalog Cluster URL and set to **Active** state.

After configuring Service Point, if you encounter some issues, for example, if you are not able to search for Catalog Featured Offerings in Service Point and other such issues, see Troubleshooting Service Point.

View Knowledge Base Articles from Service Point

Service Point provides the ability for business users to view and use the Knowledge Base (KB) documents as a solution to their immediate problem, rate and provide feedback for a KB article.

Perform the following self-service options in Service Point for a given knowledge base document:

View Knowledge Base Documents from Search Results

Service Point uses Natural Language Search (NLS) and provides relevant KB solutions based on your problem description. Or Click on any categories displayed on the Service Point Home page to see relevant KB solutions. Suggested Solutions display the list of published knowledge base articles. It provides a short synopsis of a solution for a reported issue. Click the KB article to review. You can also Open the KB solution in a new browser tab Or simply, click the Copy to Clipboard icon to copy the KB article.
When you attach a KB solution to a ticket it is displayed On Service Point Home Page, Timeline view.

View a Knowledge Base Documents from Ticket Timeline

When a KB solution is attached to a ticket, the Knowledge Base document gets linked to that ticket and is logged in the ticket timeline on the Service Point ticket detail page. In Service Point, navigate to the ticket timeline, look for or click the Solutions Used Hyperlink in a said ticket and view the KB article that is linked to the ticket. If a KB document addresses a reported issue, click **Use as a Solution** option. The suggested solution gets appended to the Ticket Timeline as a link.

Provide Comments and Feedback for a KB Solution

After viewing the complete KB article, you can provide your comments or feedback about a KB article/solution provided to you in the comment box. Click Submit. You can view any other existing comments on a selected KB document.

Rate a Knowledge Base Document

You can rate a KB article and give a rating on a scale of 1-5.. If a KB article is not rated, "Be the first to rate this Document" is displayed so that user can rate it. if it is rated, the rating is displayed on a scale of 1-5 star representation. Additionally, if a KB article is accepted as a solution for a ticket, that information is available as Number of times it has been accepted as a solution.
The rating on a scale of 1-5 represents the following:

- 1 Star & 1.5 star - Poor
- 2 star & 2.5 star - Fair
- 3 star & 3.5 star - Average
- 4 star &4.5 star - Good
- 5 Star - Excellent

Manage Smart Search

This article introduces you to Smart Search and related concepts.

> **TIP**
>
> This article is applicable for an CA Service Desk Manager and CA Service Catalog integrated environment. Before you follow this article, follow the instructions on Manage CA Service Catalog Featured Offerings.

Introduction to Smart Search

Business-users can ask a question or describe an issue using the Smart Search bar. The Smart Search displays categories, service offerings, and 20 most relevant Suggested Solutions that match the search query. The Smart Search results are based on the following parameters: `search terms` , user context, minimum weight, and maximum weight.

Based on the organizational requirement and to optimize the search relevance, an administrator can customize smart search.

Customize Smart Search

Based on the organizational requirement and to optimize the search relevance, an administrator can customize smart search.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Interface**, **Service Point**, and **Configuration**.
3. In the name column, select the desired tenant configuration name.
4. From the list of components, select **Smart Search**.
   The smart search details for categories and service offerings are displayed.
5. To modify the smart search details, click **Edit** at the top right of the screen.

   > **WARNING**
   >
   > It is recommended you do not modify the minimum and maximum weights. However, you may `swap` the maximum values for Categories and Service Offerings.
6. Edit the following fields for Categories:

- **Minimum Weight**
- **Maximum Weight**
- **Search Term**

7. Edit the following fields for Service Offerings:
   - **Minimum Weight**
   - **Maximum Weight**
   - **Search Term**
8. Click **Save**.
9. If you modify the **Search Term**, verify the Elasticsearch cluster node is working.
10. Manually execute the **pdm_es_initial_load** command:
    `$ES_HOME/search/bin/pdm_es_initial_load.bat index=sp_index.autocat_search`

Smart Search with Fuzziness

Fuzziness is an elasticsearch concept which returns a list of results even though the search strings or spellings may not exactly match. Fuzziness is a way for smart search to handle small typos in search strings and still return the right results. For example, matching the string, `network printer not functional` with `network printre not functional` requires some form of fuzzy matching.

Fuzziness is disabled for Service Point as it may have a negative impact on the search relevance. However, based on an organisation's requirements, an administrator can enable fuzziness.

Follow these steps:

1. Navigate to the default location, <`xFlow_Home>\APPS\Services\searchmicroservice-17.0.479\conf` and open **application.conf** file in a text editor.
2. Update the `searchQueryWithContext` and `searchQueryWithContextAndTenant` as follows:
   Current:

   ```
   "minimum_should_match": "%s",
   "tie_breaker": 0.3
   }
   ```

   Update:

   ```
   "minimum_should_match": "%s",
    "tie_breaker": 0.3,
     "fuzziness": "AUTO",
     "prefix_length": "3"
   }
   ```

3. Restart the xFlow Interface services.

Configuration Keys for Smart Search

The following pre-defined configuration keys are used by Smart Search:

- esearch.default.slcm.context.range.time
- esearch.default.slcm.context.scheduler.interval
- smartsearch.query.minmatch
- smartsearch.tickets.resolved.time

> **NOTE**
>
> **Information:** To lookup for the configuration keys, navigate to the **Administration** tab, select **xFlow Interface**, and **General**.

Recommendations:

- Use the predefined configuration **key:values**.
- Restart the xFlow Services if you modify the pre-defined configuration key values.

Enable the Guest Login

Guest accounts allow temporary login to Service Point without requiring to provide credentials. The guest login option is available on the Service Point login page.

While creating a ticket in Service Point, a guest user must provide an email address or phone number to complete the ticket creation process. Guest users cannot view the ticket they created if they logout and log back in. This article explains how to enable guest login for Service Point.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to **Administration**, **Options Manager**, **Security,** use the **GUEST_USER_NAME** option in SDM to Install/ Uninstall Guest Login.
   a. Value of **GUEST_USER_NAME** is the user ID of contact which is used for guest login.
   b. If **GUEST_USER_NAME** option is uninstalled or the value is empty, it will disable the guest login.
   c. Guest login works only when Authentication type is set to **Basic Authentication** type (Navigate to **Administration**, **xFlow Interface**, **General**, **General Configuration**, and update the authentication type to **BASIC**.

Configure the xFlow Interface

The xFlow Interface provides analysts the ability to understand the IT environment, prioritize work and make decisions. Data-driven intelligence improves the productivity of the analyst and enables them to understand the workload for the day and the list of activities to be completed.

This article contains the following topics:

- Configure Security Plugin to Access the Search Server
- Enable Services for xFlow Interface
- Configure Search Server Cluster
- Configure Search Servers in CA SDM
- Configure Event-Based Load
- Perform Initial Load and Refresh Index and Data on the Search Server
- View Configuration
- Command Bar
- Configure the xFlow Interface Server Services
- Heat
- Ticket Attributes
- Weather
- Work Stream
- Push Notification Service
- Enable the Research Tool Extension
- Configure the xFlow Interface Cache Size
- Enable SAML Authentication for xFlow Interface
- Configuration Changes for xFlow Interface
- App Launcher - Apps Configuration and Role Switching
- Manage Chat

Configure Security Plugin to Access the Search Server

Configure the CA Security Plugin to authorize the hosts that access the search server based on your environment requirements.

NOTE

The search server uses two ports, an HTTP port and a TCP port. The CA security plugin authorizes only the HTTP port.

Elastic Search Server 7.17.6

Perform the following steps if you have Elastic Search Server 7.17.6 in your environment:

1. Navigate to where you have installed Elastic Search Server 7.17.6 and edit the elasticsearch.yml file:
   For example:
   ```
   <install_home>\CASearchServer\elasticsearch-7.17.6\config
   ```
   In case, if you do not have elasticsearch-7.17.6 full license, add the below in the elasticsearch.yml file:
   ```
   xpack.security.enabled: false
   ```
2. Configure the IP address or the host names of the servers that can access the search servers.
   ```
   ca-es-security-plugin:
         enable: true
         enable-x-forwarded-for: true
         allowed-list-of-hosts: [host1, host2, ...]
   ```
   NOTE

   By default, the security is disabled for the search servers.
3. Save and close the file.
4. Restart the *CA Search Server* service.
   The security is enabled and only the hosts that are specified in the *allowed-list-of-hosts* parameter can access the search server.

Enable Services for xFlow Interface

Enable the xFlow Interface services and Search Server services if the xFlow Interface is installed.

The failure to do so, results in the following:

- xFlow Interface will not load to the login screen.
- Interruption of all the daemons of the xFlow Interface and does not allow the latest information to be displayed.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **Options Manager**, and **General**.
3. From the list of options, select **xFlow_installed**.
4. Click **Edit** and set the **Optional Value** to **Yes.**
5. Click **Save**.
6. Restart the *CA Service Desk Manager* service.
7. Restart the *CA xFlow Interface* service.
8. *Restart the CA Service Management Search Server service.*

Configure Search Server Cluster

A CA Service Desk Manager administrator can optionally create a cluster of search servers to provide improved performance and fail over protection.

Elastic Search Server 7.17.6

Perform the following steps if you have Elastic Search Server 7.17.6 in your environment:

- Create a Cluster for Search Server 7.17.6
- Manage Search Server 7.17.6

Create a Cluster for Search Server 7.17.6

Perform the below steps:

1. As an administrator, log in to the search server where you want to add the cluster.
2. Navigate to the location where you have installed Elastic Search Server 7.17.6: <elastic-search-installation>\elasticsearch-7.17.6\config folder and edit the *elasticsearch.yml* file.
3. Navigate to the *Cluster* section and add the following properties after the *cluster.name* property:
   a. ***node.master****: true*
      Define the value as *true* to ensure that the node is eligible as a master node.
   b. ***node.data****: true*
      Define the value as *true* to ensure that the node also services the data requests.
   c. **discovery.seed_hosts**: ["<search server 1>:9300", "<search server 2>:9300","<search server 3>:9300"] and so on for other search servers in your environment.
      Edit the *elasticsearch.yml* file in *search server 1,* and define the *search server 2*, and *search server 3* host names in the file. Similarly, repeat the configuration in other hosts.
   d. **cluster.initial_master_nodes**: Host name of your first search server in a cluster. For example: <search server 1>.
      NOTE
      For all search servers in a cluster, provide the host name that you have provided for the first search server <search server 1>. For example, in case of search server 2, provide the host name in **cluster.initial_master_nodes** as **<search server 1>**.
4. Save and close the file.
5. Restart the *CA Search Server* service.
6. Restart the *CA **xFlow Interface*** service.
   After the services start, the search cluster is created.
7. Repeat steps 1 - 6 on all the search servers.
8. Manage Search Server 7.17.6

Manage Search Server 7.17.6

As an administrator, you can add or edit the details of the servers in the cluster. For example, if you add an eligible master search server to the cluster, you can edit the minimum number of eligible master nodes property.

Follow these steps:

1. Navigate to the <elastic-search-installation>\config*elasticsearch.yml* file and edit these properties, if necessary, complete the following information:
   a. *http.port:* <port-number-for-the-search-server>
      Update the port number when the default port is changed. For example, if the port number is already in use, define another port number for the search server.
2. Update the cluster details in the CA SDM, **Administration**, **xFlow Interface** , **Search Servers** page.
   For more information about configuring search servers, see Configure Search Servers in CA SDM.
3. Restart the CA Search Server service.

4. Restart the *CA Service Management xFlow Server* service.
5. Verify the configuration:
   a. Ensure that the search servers and the **xFlow Interface** services are running.
      To troubleshoot Search Server, refer the search server logs.
   b. Log in to the **xFlow Interface** and perform the search.

NOTE

After you configure and manage the cluster, you can configure the CA Security plugin to secure the search server. For more information, see Configure Security Plugin to Access the Search Server.

Configure Search Servers in CA SDM

The **Search Servers** page allows you to view and update the search server details. The configuration is internally referred to by the search processes. For example: Loading the CA SDM data into Elasticsearch server processes. When you update the server details on the search server and when you configure the search server cluster, update the same configuration on the CA SDM server. Perform the following steps to configure search servers in CA SDM:

1. Log in to CA SDM as an Administrator.
2. Navigate to **Administration**, **xFlow Interface** , **Search Servers**.
   The list of search servers that are configured during installation is displayed.
3. Search for the required server by entering the server name.
4. Click **Edit** to edit the applicable server details, such as the HTTP or TCP port and save.
5. Restart the CA SDM services.
6. Restart the *CA **xFlow Interface*** service.
   The search services reference the updated server details.

Configure Event-Based Load

Event-based load (EBL) is a process that runs in the background and periodically updates the search server with the changes in CA SDM. EBL is a preconfigured process and takes the default values.

Configure EBL on the following servers:

- Primary server in conventional configuration
- Background server in advanced availability configuration

NOTE

We recommend that you configure the EBL variables only in scenarios, such as implementing SSL or when you encounter issues that are related to connectivity or database.

Follow these steps:

1. Log in to the CA SDM server as an administrator.
2. Execute the following command to configure the required EBL variable:

```
pdm_options_mgr -c -a pdm_option.inst -s <parameter_name> -v <parameter_new_value> -t
```

Where,
-c, updates the changes to the secondary servers
-a, updates the NX.env file
-t, updates the parameter in the template file
For example, if you want to increase the wait time of the connection before timeout to 20 seconds, configure the parameter as follows:

```
pdm_options_mgr -c -a pdm_option.inst -s ES_HTTP_CONNECTION_WAIT_TIMEOUT -v 20 -t
```

3. Repeat step 2 to configure more variables.
   For more information about the default EBL parameters that you can configure, see Default EBL Variables.
4. Restart the *CA Service Desk Manager* service.
5. (Optional) In Advanced Availability, perform rolling maintenance to apply the NX options on all application servers.
6. Navigate to the *$NX_ROOT* folder and verify that the variables are set to the updated values in the *NX.env* file.

Default EBL Variables

Variable Name	Description	Default Value
ES_HTTP_CONNECTION_WAIT_TIMEOUT	Max wait time in seconds to get HTTP connection.	10 (in seconds)
ES_HTTP_CONNECTION_TIMEOUT_AFTER	Timeout connection after x seconds.	10 (in seconds)
ES_HTTP_CM_CONNECTION_KEEP_ALIVE_TIME	Time to keep the HTTP connection alive (seconds).	300 (in seconds)
ES_HTTP_IDLE_CONNECTION_CLOSE_TIME	The idle time of connections to be closed (seconds).	300 (in seconds)
ES_HTTP_WAIT_TIME_BEFORE_CLOSE	The idle time of connections to be closed (seconds).	240 (in seconds)
ES_PUSH_THREAD_SLEEP_TIME	Thread sleep time for search Data Push (seconds).	30 (in seconds)
ES_DB_SUPPORT	*If value is set to On, it uses database to store data, before sending to search.*	on
ES_ORA_MIN_LIMIT	Oracle database minimum connection pool.	2
ES_ORA_INITIAL_LIMIT	Oracle database initial connection pool.	2
ES_ORA_MAX_LIMIT	Oracle database maximum connection pool.	5
ES_DB_BATCH_SIZE	Oracle database batch size.	100
ES_HTTPS_CERT_PATH	For the HTTPS protocol, define the certificate path.	<certificate_path>\server.keystore
ES_HTTPS_CERT_PWD	For the HTTPS protocol, define the encrypted password.	<encrypted password>

NOTE

If you configure HTTPS to connect to the search server, the ES_HTTPS_CERT_PATH and ES_HTTPS_CERT_PWD variables are mandatory.

Perform Initial Load and Refresh Index and Data on the Search Server

This article contains the following topics:

Configure the Indexes

Before you perform initial load or refresh index and data on the search server, you can configure the indexes that you want to load or refresh. Search currently supports the following indexes:

- sdm_index_cnt (users)
- sdm_index_kd (knowledge document)
- sdm_index_nr (configuration items)
- sdm_index_cr (incidents and requests)
- sdm_index_kdlinks (suggested solutions)
- sdm_index_pcat (category)
- catalog_index_service_offering (offerings)
- catalog_index_service_request (requests)

NOTE

By default, the **dataload_limit** is set to one year. You can use this parameter to limit the initial load data for number of years.

By default, all these indexes are configured.

WARNING

You cannot add new or custom indexes other than what **Search** server currently supports.

Follow these steps:

1. As an administrator, log in to the search server and navigate to the *<es_install>\config* folder and edit the *config.yml* file.
2. In the config.yml file, remove the index you don't want to configure from the **objects** and **index** section.
   For example:
   ```
   sdm_index_kd (knowledge document)
   catalog_index_service_request (requests)
   ```
3. By default, the **dataload_limit** is set to one year. You can use this parameter to limit the initial load data for number of years.

   ```
   dataload_limit: 1y
   ```
4. Save and close the file and perform initial load.

Data Verification

Execute the query against the MDB database to verify data:

- **Oracle**:

```
select * from usm_offering where (date_available <= to_date('1901-12-13','yyyy-mm-dd')
or date_available >= to_date('2038-01-19','yyyy-mm-dd'))
or (date_unavailable <= to_date('1901-12-13','yyyy-mm-dd')
or date_unavailable >= to_date('2038-01-19','yyyy-mm-dd'))
or (date_cancelled <= to_date('1901-12-13','yyyy-mm-dd')
or date_cancelled >= to_date('2038-01-19','yyyy-mm-dd'))
```

- **SQL**:

```
select * from usm_offering where (date_available <= CONVERT(DATETIME, '1901-12-13', 102)
or date_available >= CONVERT(DATETIME, '2038-01-19', 102))
or (date_unavailable <= CONVERT(DATETIME, '1901-12-13', 102)
or date_unavailable >= CONVERT(DATETIME, '2038-01-19', 102))
or (date_cancelled <= CONVERT(DATETIME, '1901-12-13', 102)
or date_cancelled >= CONVERT(DATETIME, '2038-01-19', 102))
```

WARNING

We recommend you update the data and run the rebuild index, if the query returns any results.

Perform Initial Load on the Search Server

Initial load is an activity that is typically performed when you install search server using the installer. Initial load prepares the search server with the index and data that are imported from the CA SDM and CA Service Catalog database that is used by search in xFlow Interface.

WARNING

We recommend that you perform the initial load task during off peak hours as it affects the CA SDM server performance.

If you perform the initial load action after installation, execute one of the files based on your operating system:

- *$ES_HOME/search/bin/pdm_es_initial_load.bat (Windows)*
- *$ES_HOME/search/bin/pdm_es_initial_load.sh (Linux)*

This file creates an index and loads the data into the search server.

When you execute the file from the command prompt, you can use the following options:

- *$ES_HOME/search/bin/pdm_es_initial_load.bat [v, V] [h, help] (Windows)*
- *$ES_HOME/search/bin/pdm_es_initial_load.sh [v, V] [h, help] (Linux)*

Where,

v, V: This option displays the data load version. You can use any option with the batch file.

h, help: This option displays the help on the batch file. You can use any option with the batch file.

NOTE

To perform initial load on a specific index on Windows, execute the command as *$ES_HOME/search/bin/pdm_es_initial_load.bat index=<index_name>.*

For example, to perform initial load on *sdm_index*, execute the command as *$ES_HOME/search/bin/pdm_es_initial_load.bat index=sdm_index*

Refresh Index and Data on the Search Server

Typical scenarios to refresh the index and data:

- When you execute the *pdm_load* command on the users, assets, knowledge base, or tickets (incidents and requests) data.
- To refresh the changes that are done to the dependent indexes, such as changes to the location or asset class name.
- When you enable or disable tenancy.

If you selected to run initial load while configuring the Search Server details from the common installer, pdm_es_rebuild is run by default. when initial load is not checked while configuring the Search Server Details, you must manually execute the *pdm_es_rebuild* command.

- *$ES_HOME/search/bin/pdm_es_rebuild_index.bat (Windows)*
- *$ES_HOME/search/bin/pdm_es_rebuild_index.sh (Linux)*

When you execute the batch file from the command prompt, you can use the following options:

- *$ES_HOME/search/bin/pdm_es_rebuild_index.bat [v, V] [h, help] (Windows)*
- *$ES_HOME/search/bin/pdm_es_rebuild_index.sh [v, V] [h, help] (Linux)*

Where,

v, V: You can use any option with the file. This option displays the data load version.

h, help: You can use any option with the file. This option displays the help on the file.

NOTE

To refresh a specific index on Windows, execute the command as *$ES_HOME/search/bin/ pdm_es_rebuild_index.bat index =<index_name>.*

For example, to refresh the *catalog_index* index, execute the command as *$ES_HOME/search/bin/ pdm_es_rebuild_index.bat index=catalog_index*

NOTE

To load the local specific service offerings, execute the command as: *$ES_HOME/search/bin/ pdm_es_rebuild_index locale=local_name*

For example, to load the service offering data in Japanese, execute the command as *$ES_HOME/search/bin/ pdm_es_rebuild_index locale=ja_JP*

Refresh the Smart Words for Command Bar in the Search Server

Smart Words that are created in CA SDM are not automatically updated in the Search Server. Therefore, when you type a new smart word in the Command Bar, the associated categories and templates do not display because the Search Server is not updated with the latest Smart Words.

TIP

: We recommend that you schedule a reindex at specific intervals to maintain the latest smart words and associated categories and templates in the Search Server.

Schedule a Refresh on Windows

The example in this procedure schedules the refresh once every week.

Follow these steps:

1. From the **Start** menu on the *CA Search Server*, open the **Windows Task Scheduler**.
2. For example, on Windows Server 2016/ 2019/ 2022, go to **Control Panel**, **System and Security**, **Administrative Tools**, **Task Scheduler**.
3. Click **Create Task**.
4. On the **General** tab, type a name for the task.
   - Select the check box for **Run** whether user is logged in or not.
5. Navigate to the **Actions** tab and click **New**.
   - In the **Action** field, select **Start** a program.
   - In the **Program/script** field, type the execution script name as *pdm_es_rebuild_index.bat index=sdm_autocat_index*.
   - In the **Start in (Optional)** field, set the root path of *pdm_es_rebuild_index.bat index=sdm_autocat_index* script and click **OK**.
     For Example: C:\es\bin\
6. Navigate to the **Triggers** tab and click **New**.
   - In the **Settings** field, select *Weekly*.
   - In the **Start** field, select *12:00:00 AM*.
   - In **Recur every** field, type *1*, and select a day from the list, and click **OK**.
7. On the **Create Task** dialog, click **OK**.
   You have completed scheduling the Windows task to refresh the auto-categorization index.

Schedule a Reindex on Non-Windows

Define a cron job to schedule a reindex on Linux.

Prerequisite

Enable the execute permission for the user executing the reindex script.

Follow these steps:

1. As an administrator, log in to the search server.
2. Navigate to the *<search_server_install>/search/bin* folder.
3. Execute the following command to open the cron job file for editing.

```
crontab –e
```

4. Define the cron job schedule.

Crontab job scheduling pattern:

```
# Minute  Hour         Day of Month       Month                   Day of Week          Command
# (0-59)  (0-23)       (1-31)             (1-12 or Jan-Dec)       (0-6 or Sun-Sat)     path_to_script
```

Cron job command examples:

```
# execute every minute
* * * * *  cd /root/<search_server>/search/bin && ./pdm_es_rebuild_index.sh index=sdm_autocat_index

# execute every 5 minutes
*/5 * * * *  cd /root/<search_server>/search/bin && ./pdm_es_rebuild_index.sh index=sdm_autocat_index

# execute every Friday 1AM
0 1 * * 5  cd /root/<search_server>/search/bin && ./pdm_es_rebuild_index.sh index=sdm_autocat_index

# execute on workdays 1AM
0 1 * * 1-5 cd /root/<search_server>/search/bin && ./pdm_es_rebuild_index.sh index=sdm_autocat_index

# execute 10 past after every hour on the 1st of every month
10 * 1 * * cd /root/<search_server>/search/bin && ./pdm_es_rebuild_index.sh index=sdm_autocat_index
```

You can define a cron job that is based on the scheduling pattern or can enable the cron job from the examples.

5. Save and close the file.

NOTE

To view the cron job, execute the following command: *$ crontab -l*

select * from usm_offering where (date_available <= to_date('1901-12-13','yyyy-mm-dd') or date_available >= to_date('2038-01-19','yyyy-mm-dd')) or (date_unavailable <= to_date('1901-12-13','yyyy-mm-dd') or date_unavailable >= to_date('2038-01-19','yyyy-mm-dd')) or (date_cancelled <= to_date('1901-12-13','yyyy-mm-dd') or date_cancelled >= to_date('2038-01-19','yyyy-mm-dd'))

View Configuration

This article contains the following topics:

Card and List View

An administrator can configure the attributes displayed in the card and list view.

A total of 10 predefined attributes are set to be displayed on the card and list view. The first four attributes are fixed and cannot be customized. Rest of the attributes (6) can be customized.

Follow these steps:

1. Log in to CA SDM as a system administrator.
2. Navigate to the **Administration**, **xFlow Interface** , **View Configuration**, **Card and List View**.
3. Select the **Ticket type**.
   The List and Card View Configuration Detail page appears.
4. Click **Edit**.
5. From the list of customizable attributes, select an attribute that you wish to modify.
6. Click **Edit**.
7. Enter the **Attribute Name** or click the magnifier icon to search for an attribute name.
   Search results display OOTB attributes available in the Object Attribute List (**Administration**, **xFlow Interface**, **Object Attributes**).
   Filter the attributes based on the object type: **Incident**, **Request**, and **Problem**.
8. Select the attribute and click **Save**.

   NOTE

   The analysts cannot sort tickets by description, due to limitation in the database for NTEXT or NCLOB data types.

   The attributes configured here will be used by the analyst to filter and sort tickets on the interface. For more information on sorting, see Enable Sort for New Attributes.

   Notify the analysts to re-login for the change to take effect.

Enable Sort for New Attributes

After configuring a new attribute to be displayed in the card and list views, perform the following steps to make the attribute available for sorting on the interface.

Follow these steps:

1. Log in to the following CA SDM server where WSP is installed, depending on your CA SDM configuration:
   - Conventional: Primary server
   - Advanced availability: Background server
2. Start WSP using any of the following actions, depending on the operating system that is installed on the CA SDM server:
   - (Windows) From the Start menu, select Program Files, CA, CA SDM, Web Screen Painter.
   - (UNIX) Enter the command pdm_wsp with $NX_ROOT/bin in your path.

   The Web Screen Painter login window opens.
3. Enter your login credentials.
4. Select Tools, and Schema Designer.
5. For ticket type as Incident:
   a. Select *in (Incident)* table and click the Domset tab.
   b. Right click on *in_list_l1_web* and click Properties.
   c. Add the new Attribute Value in the **sort columns** field.
   d. Select **Save**.
   e. Right click on *in_list_l1_web_static,* click Properties, and follow steps 5c - 5d.

f. Select File, Save and Publish.

6. For ticket type as Request:
   a. Select *cr (Request)* table and click the Domset tab.
   b. Right click on *cr_list_l1_web and* click Properties.
   c. Add the new Attribute Value in the **sort columns** field.
   d. Select **Save**.
   e. Right click on cr_list_l1_web_static, click Properties, and follow steps 6c - 6d.
   f. Select File, Save and Publish.
7. Stop the *CA Service Desk Manager Server* service.
8. Run the following command:

```
pdm_publish
```

9. Restart the *CA Service Desk Manager Server* service.

Detail View

An administrator can configure the attributes that are displayed in the Workflow Tasks details.

The first three attributes are fixed and cannot be customized. Those attributes are: **Task**, **Sequence**, and **Description**.

Follow these steps:

1. Log in to CA SDM as a system administrator.
2. Navigate to the **Administration** tab, select **System**, **xFlow Interface** , **View Configuration**, and **Detail View**.
3. Select the **Object type**.
   The Detail View Attributes page appears.
4. Click **Detailed View Attribute**.
5. Enter the **Sequence** of the attribute.
6. Enter the **Attribute Name** or click the magnifier icon to search for an attribute name.
   Search results display the attributes available in the Object Attribute List.
7. Set the **Attribute Status** as **Active.**
8. Select the **Read Only** checkbox if the attribute value cannot be modified.
9. Select the **Required** checkbox if the attribute is mandatory and requires an input.
10. Click **Save**.

Command Bar

You can use the Command Bar to manage text commands, add additional ticket attributes for new tickets, and configure smart words.

> **NOTE**
> Copy and Paste operations are not supported for Command Bar in xFlow Interface.

This article contains the following topics:

Commands

All the commands are in the *Active* state by default.

Disable a Command

As an administrator, you can only disable command from the command bar. You cannot create or add a new command.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Interface** , **Command Bar,** and **Commands**.
3. Select the command you want to disable.
   The command detail page appears.
4. Click **Edit**.
5. Set the **Status** to **Inactive**, and click **Save**.

Modify the Command Alias

As an administrator, you can modify the command alias. An Alias is optional and can be used in place of the command.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Interface** , **Command Bar,** and **Commands**.
3. Select the command for which you want to modify the alias.
   The command detail page appears.
4. Click **Edit**.
5. Modify the **Alias 1**, and click **Save**.
   The command Alias is saved.

Exclude Attributes for the Set command

An administrator must exclude system level attributes for the Set command.

Follow these steps:

1. Access the **casm.conf.do-not-change.js** file located in C:*Program Files\CA\xFlow\APPS\UI\Apache24\htdocs\conf* folder.
2. Locate the key: *customattribExclusions.*
3. To the existing list, add the system level attribute that you want to exclude for the Set command.
4. Click **Save**.
5. Restart the **CA Service Management xFlow Server** Service.

> **NOTE**
>
> We recommend to exclude any system level attributes for the Set command. This includes and is not limited to the following attributes: *Call Back Flag, Closed On, Heat, Id, Last Modified, Modified By, Opened On, Resolved On, Reported by, SLA Violation, Ticket Number, Type, Persistent id, Predicted Sla Violation*, and *Priority* (When APC is enabled)

New Ticket

Default Ticket Type

A ticket in the **xFlow Interface** can be an incident or a request. An administrator can set a default ticket type for the tickets created by the analysts.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **System**, **xFlow Interface** , and **General**.
3. From the General Configuration List, select **TicketType**.
4. Click **Edit** and enter one of the following values in the **Configuration Value** field:
   0: The ticket type is Incident.
   1: The ticket type is Request.

2: The ticket types are Incident and Request. However, the default ticket type is Incident.
3: The ticket types are Incident and Request. However, the default ticket type is Request.

5. Click **Save**
   The default ticket type is saved.

Attributes Displayed While Creating a Ticket

To capture additional information for a ticket, an administrator can mandate attributes to be displayed when a ticket is created.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Interface** , **Command Bar**, and **New Ticket**.
3. Click **Create New**.
   The Create a New Ticket Attribute page appears.
4. Complete the fields as appropriate:
   StatusIndicate if the attribute is active or inactive.
   Ticket Attribute
   Click the magnifier icon to search for an attribute to be displayed while creating a ticket.Search results display attributes available in the Ticket Attribute list.

   NOTE

   We recommend not adding any system level attributes to be displayed while creating a ticket. This includes and is not limited to the following attributes: *Active, Call Back Flag, Closed On, Customer, Description, Heat, Id, Last Modified, Modified By, Opened On, Resolved On, Reported by, SLA Violation, Ticket Number, Type, Persistent id, Predicted Sla Violation*, and *Priority* (When APC is enabled)

   Default Value(Optional) Click the magnifier icon to search and assign a default value to the attribute.
5. Click **Save**.

Smart Words

Smart Words are suggestions with which you can auto-categorize and apply template to a ticket.

Command bar identifies the smart words when an analyst enters the description of the reported issue. Each smart word has one or more associated categories and templates.

For example, you can create a mapping for smart words like *outlook*, *express*, *lotusnotes* and *macmail*, and assign a Category (For example, *Email*) and a Template (For example, *ITIL Email* and *ITIL Software*).

When you upgrade to CA Service Management 14.1.03, a list of smart words are generated based on the ticket information stored within the Search Server or CA SDM Server. However, smart words defined by the administrator have higher preference than the server-generated smart words.

Create a Smart Word

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Interface** , **Command Bar**, and **Smart Words**.
3. Click **Create New**.
   The Create New Smart Word page appears.
4. Complete the following fields as appropriate:
   a. **Category**: Click the magnifier icon to search for a category.
   b. **Keyword**: Provide the list of keywords (separated by space) to be associated with the selected category.
   c. **Status**: Indicates if the attribute is active or inactive.

5. Click **Save**.
6. After creating a smart word, you can add or attach a template to it. Select a Smart word and from the Smart Word Detail page, select **Add Template**.
   You can add a template based on your ticket type (Incident or Request).

   TIP

   Ensure that you have configured a schedule to refresh the smart words in the search server. The smart words that you define in CA SDM are added to the search server only when you refresh the auto categorization index. For more information about refreshing the auto categorization index, see *Refresh the Smart Words for Command Bar in the Search Server* topic in Perform Initial Load and Refresh Index and Data on the Search Server.

Configure the Smart Words

As an administrator, you can configure the number of categories and templates that you can view in the command bar.

Follow these steps:

1. Log into CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Interface** , **General**.
3. Edit the *esearch.default.categoryNtemplate.suggestions* property and define a configuration value.
   For example, set the value as *esearch.default.categoryNtemplate.suggestions=5.*
4. Click **Save**.
5. Restart the *CA Service Management xFlow Server* service.
   The command bar displays the defined number of categories and templates that match the smart word. For example, if your smart word is *Outlook* and the parameter value is 5, the command bar displays the first five categories and templates that match the smart word. In this example, it displays 10 suggestions, five each of the matching category and template.

Attach a Service Type Event or Update a SLA

You can also attach a Service Type Event to a ticket or Update a SLA using the **xFlow Interface** Command Bar Option. For more information see, Manage Service Type Events (SLA) - xFlow Interface.

Configure the xFlow Interface Server Services

CA SDM Administrators can configure xFlow Interface Server services from the CA SDM console **Administration**, **xFlow Interface** , **General** tab. The variables that you define on this page are used by the incident, search, and notification services.

NOTE

After updating the variables in the **General** configuration page, restart the xFlow Interface services. The xFlow Interface services read only the active records or configurations.

If you want to override these variables from any specific service, define the variables in the **application.conf** file of the corresponding service that is located in the conf directory. Always restart the xFlow Interface Server services after defining the variables in the **application.conf** file. Variables that start with cache are related to caching. The format of cache variable is ***cache.<cache-name>.<cache-parameter>*** . The description of each configuration variable can be found in the respective configuration details page.

WARNING

Define the values for the variables in the specified format to avoid erroneous values for the xFlow Interface.

For more information about configuring the servers, see the following topics:

- Configure xFlow Interface Cache Size
- Configure Suggested Experts Search
- Configure Suggested Solution Search
- Configure Multi-faceted Search

Configure Suggested Experts Search

As an administrator, you can configure the search scope for suggested experts. The suggested experts are listed based on the maximum number of resolved tickets that match the defined scope.

NOTE

To make the search more efficient and relevant for the analyst, you can configure the noise words. After you configure the noise words, refresh the index and data on the search server to include the latest noise words in your searches.

Configure Suggested Experts from CA SDM

Follow these steps:

1. As an administrator, log in to CA SDM and navigate to **Administration**, **xFlow Interface** , **General**.
2. Edit the following parameters:
   – suggested.experts.numberOfSuggestedExperts
   Indicates that number of suggested experts to display in the ticket details page. For example, if you configure the number as 15, the ticket details page, displays the first five and upon scrolling you can view the next five results, until the limit is reached.
   – suggested.experts.searchDurationInMonths
   Indicates that the search is performed on records for the defined period (in months). For example, if you define the duration as one year, the search is conducted on records of the past one year.
3. Restart the *CA Service Management xFlow Interface Server* service.

Configure Suggested Solution Search

As an administrator, you can configure the search scope for suggested solutions. The suggested solutions are listed based on the maximum number of matching keywords in the published articles and resolved tickets that match the defined scope.

NOTE

To make the search more efficient and relevant for the analyst, you can configure the noise words. After you configure the noise words, refresh the index and data on the search server to include the latest noise words in your searches.

Configure Suggested Solution from CA SDM

Follow these steps:

1. As an administrator, log in to CA SDM and navigate to **Administration**, **xFlow Interface** , **General**.
2. Edit the following parameter:
   – esearch.default.numberOf.suggestedsolutions
   Indicates that number of suggested solutions to display in the ticket details page. For example, if you configure the number as 15, the ticket details page, displays the first five and upon scrolling you can view the next five results, until the limit is reached.
3. Restart the *CA Service Management xFlow Interface Server* service.

Configure Multi-faceted Search

As an administrator, you can configure the multi-faceted search scope on specific fields that are related to facets; for example, you can set the search scope for users who are based on their email and user ID.

NOTE

To make the search more efficient and relevant for the analyst, you can configure the noise words. After you configure the noise words, refresh the index and data on the search server to include the latest noise words in your searches.

Configure Multi-faceted Search from CA SDM

Follow these steps:

1. As an administrator, log in to CA SDM and navigate to **Administration**, **xFlow Interface** , **General**.
2. Edit the following parameters:
   - esearch.searchable.objects: Enable or disable the facets in the Multi-faceted search.
     Search currently supports four facets: users (cnt); knowledge document (kd); configuration items (nr); incidents and requests (cr). This parameter allows you to enable or disable the facets in the Multi-faceted search drop-down.
   - esearch.cnt.searchable.columns: Define the searchable fields for the User facet.
     You can define from the following fields: cnt: first_name; last_name; middle_name; userid; email_address; tenant.name; location.name; phone_number; alt_phone; mobile_phone
   - esearch.kd.searchable.columns: Define the searchable fields for knowledge base facet.
     You can define from the following fields: persid; title; problem; kd_summary; resolution; tenant.name

     NOTE

     We currently support only 32 KB of characters in *resolution* field. The search scope is limited to the number of characters you define in the *resolution* field.
   - esearch.nr.searchable.columns: Define the searchable fields for the Assets facet.
     You can define from the following fields: resource_name; class.name; family.name; mac_address; dns_name; serial_number; location.name; tenant.name
   - esearch.cr.searchable.columns: Define the searchable fields for the incidents and requests.
     You can define from the following fields: inc_id; summary; description; tenant.name; ref_num
3. Restart the *CA xFlow Interface Server* service.

Heat

Heat is a composite representation of various attributes and conditions that define the criticality of a ticket.

Heat provides the analysts with a multi-dimensional scoring model to prioritize tickets and understand the amount of attention a ticket must receive.

NOTE

You can create a heat configuration at a group level or at a tenant level for each object type (Incident and Request).

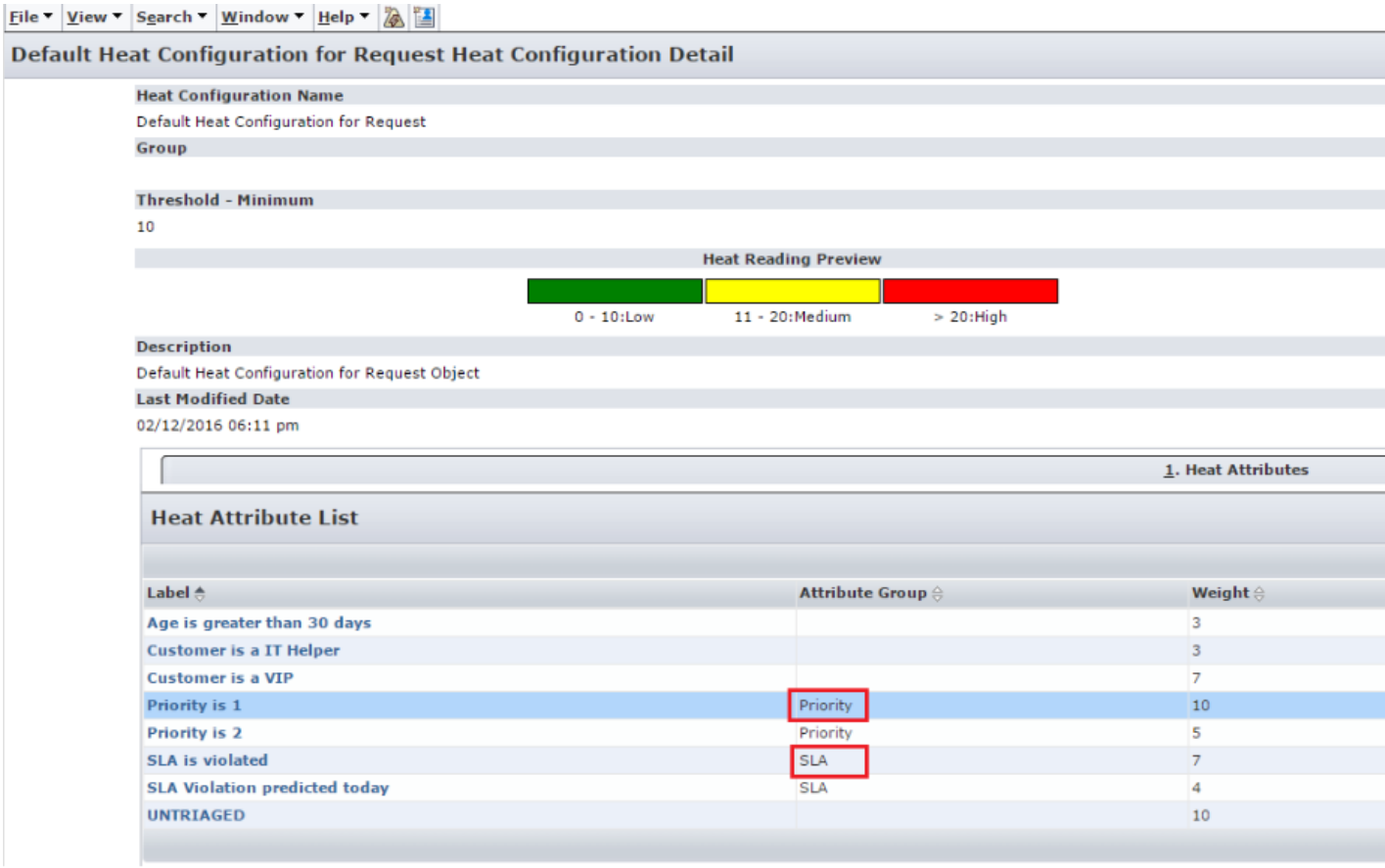
This article contains the following topics:

Configure Heat

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Interface** , and **Heat**.

3. Click **Create New**.
   The Create New Heat Configuration page appears.
4. Complete the following fields as appropriate:
   - **Heat Configuration Name**Specify the heat configuration name.
   - **Code**Specify the internal code for the heat configuration.
   - **Status**Indicate if the heat configuration is active or inactive.
   - **Group**Select the group to calculate heat for the tickets belonging to that group.
   - **Object Type**Indicate if the heat configuration is incident or request.
   - **Threshold - Minimum**Specify the minimum value to calculate range for the heat configuration.
   - **Threshold - Maximum**Specify the maximum value to calculate range for the heat configuration.
   - **Maximum Weight**Specify the maximum possible heat associated with a ticket.
     This is a non-editable field and the value is calculated automatically based on the sum of the weight for the individual heat attributes and the highest weight for each attribute group (in case an attribute is grouped under a attribute group)
     For example, Maximum Weight for the following heat configuration is 40, which is the sum of the weight for the individual heat attributes (3,3,7,and 10) and the highest weight for each attribute group (Priority (10), and SLA (7)).



   - **Description**Describe the heat configuration in detail.
5. Click **Save**.
   The heat configuration is saved.
6. On the Heat Attribute List, click **Create New**.
   The Create New Heat Attribute page appears.
7. Complete the following fields as appropriate:

- **Label**
  Specify the label assigned to the heat attribute.
- **Code**
  Specify the internal code for the heat attribute.
- **Status**
  Indicate if the heat attribute is active or inactive.
- **Weight**
  Specify the weight for the heat attribute.
- **Description**Describe the heat attribute in detail.
- **Condition**Specify the condition for the heat attribute.
- **(Optional) Attribute Group**
  Group the mutually exclusive heat attributes.
  For example, the following attributes can be grouped under the SLA attribute group, as they are mutually exclusive and a ticket can only have one SLA at any point in time.

Attribute	Weight
SLA violated	30
SLA violation predicted for today	20

8. Click **Save**.
   The heat attribute is saved. Repeat the steps 6-8 to add another heat attribute.

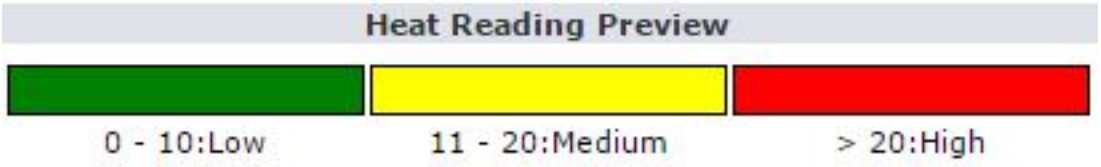
Predefined Heat Configurations

For a default tenant, following are the predefined attributes for the heat configuration:

Heat Attribute	Weight
Untriaged	10
Priority 1	10
Priority 2	5
SLA Critical	7
SLA Warning	4
Age	3
VIP User	7
IT Helper	3
Major Incident	20

Predefined Heat Range

For a default tenant, following is the predefined heat range:



On-Access Heat Calculation

The real time heat of the ticket is displayed when the ticket is accessed from the xFlow Interface

The on-access heat calculation functionality is disabled by default. An administrator can enable this functionality, however this can affect the overall performance of the application.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **Options Manager**, and **Heat-Weather**.
3. From the list of options, select **on_access_heat_calculation**.
4. Click **Edit** and **Install**.
   The on-access heat calculation is enabled.
5. Restart the CA *Service Desk Manager* Server service.

Heat Reindexing

An administrator can define the time interval (in hours) to calculate heat of the ticket.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **Options Manager**, and **Heat-Weather**.
3. From the list of options, select **Heat_Reindexing_Interval**.
4. Click **Edit** and specify the number of hours for each heat reindexing interval in the **Option Value** field.The default time is 6 hours. The minimum value can be 3 hours and the maximum value can be 72 hours.
5. Click **Save**.
6. Restart the CA *Service Desk Manager* Server service.

Enable or Disable Heat

An administrator can enable or disable heat.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **Options Manager**, and **Heat-Weather**.
3. From the list of options, select **Heat_Weather**.
4. Click **Edit** and set the **Optional Value**.
   a. **Off**
      Indicates that heat and weather are disabled**.**
   b. **Heat Only**Indicates that the heat is enabled and the weather is disabled.
   c. **Both Heat and Weather**
      Indicates that heat and weather are enabled.
5. Click **Save**.
6. Restart the CA *Service Desk Manager* Server service.

Ticket Attributes

The Ticket Attribute List page allows administrators to create, edit, export, and search for tickets.

NOTE

We recommend you not to add the LOCAL attributes of an object as a ticket attribute, as they are temporal and do not have a database storage.

Follow these steps:

1. Log in to CA SDM Interface as an administrator.
2. Navigate to the **Administration**, **xFlow Interface**, **Command Bar**, **New Ticket**.
3. Click **Create New** Attribute to add an attribute for tickets.
   Or
   Click an attribute and then click **Edit** to modify the attribute details.
4. Define the fields as appropriate.
   - **Attribute Name**
     Identifies the attribute name.
   - **Status**
     Specifies whether the attribute is active or inactive.
   - **Attribute Value**
     Specifies the majic level object attribute.
     For example, for a request object (cr), the attribute value can be *assignee* or *assignee.organization* or *summary*.
   - (Optional) **Show in Streams**
     Select the **Show in Streams** check box to display the attribute in work streams rule list.
   - (Optional) **Show in Attributes**
     Select the **Show in Attributes** check box to display the attribute on the Ticket Details page.
     By default, the attributes do not display.
5. Click **Save**.

WARNING

When you disable an attribute, the corresponding attribute rule is grayed out for an existing work stream.

Weather

Weather enables an analyst to view the workload forecast for the day, based on the tickets assigned to the analyst, analyst groups, and the planned environmental changes.

NOTE

If Multi-tenancy is enabled, at any point in time only one weather configuration can be active per tenant.

The weather icon is not displayed when a new contact accesses the interface for the first time. It is displayed after the weather is calculated.

This article contains the following topics:

Configure Weather

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Interface** , and **Weather**.
3. Click **Create New**.
   The Create New Weather Configuration page opens.
4. Complete the fields as appropriate:

Weather Configuration Name Specify the weather configuration name.**Status**Indicate if the weather configuration is active or inactive.**Description**Describe the weather configuration in detail.**Show Suggestions**Select to enable suggestions for the analyst.**Threshold - Minimum**Specify the minimum value to calculate the range for the weather configuration.**Threshold - Maximum**Specify the maximum value to calculate the range for the weather configuration.

5. Click **Save**.
   The weather configuration is saved and the Weather Criteria List is displayed.
6. The weather criteria list displays the following predefined criterion:

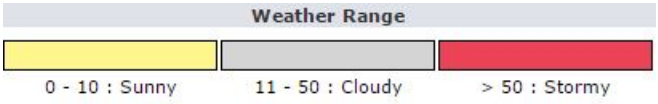
   NOTE

   The administrator cannot add or remove criteria.

   PersonalThese criteria is determined by calculating the weight of the high heat tickets assigned to the logged-in analyst. The default weight per high heat ticket is 10 and this value can be customized by the administrator.**Team**These criteria is determined by calculating the weight of the untriaged tickets and high heat tickets assigned to the logged-in analyst groups. The default weight for an untriaged ticket is 4 and for a high heat ticket is 2. These values can be customized by the administrator.**Environmental**This criteria is used to determine the weather for a scheduled environmental change (Change Order). The default weight per ticket is 6. The administrator can customize all the attributes for this criteria.
7. Click **Save**.

Predefined Weather Range

For a default tenant, following is the predefined weather range:



Weather Reindexing

An administrator can define the time interval (in minutes) to calculate weather.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **Options Manager**, and **Heat-Weather**.
3. From the list of options, select **Weather_Reindexing_Interval**.
4. Click **Edit** and specify the number of minutes for each weather reindexing interval in the **Option Value** field.
   The default time is 60 minutes. The minimum value can be 30 minutes and the maximum value can be 720 minutes.
5. Click **Save**.
6. Restart the CA *Service Desk Manager* Server service.

Enable or Disable Weather

An administrator can enable or disable weather.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **Options Manager**, and **Heat-Weather**.
3. From the list of options, select **Heat_Weather**.
4. Click **Edit** and set the **Optional Value**.
   a. **Off**
      Indicates that heat and weather are disabled**.**
   b. **Heat Only**Indicates that the heat is enabled and the weather is disabled.

c. **Both Heat and Weather**
Indicates that heat and weather are enabled.

5. Click **Save**.
6. Restart the CA *Service Desk Manager* Server service.

Work Stream

Work streams provide logical grouping of tickets based on the specific search criteria.

This article contains the following topics:

Create a Work Stream

An administrator can create default work streams that are available for all the analysts.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Interface** , and **Work Streams.**
3. Click **Create New**.
The Create New Stream page opens.
4. Complete the fields as appropriate.
LabelSpecify the label assigned to the work stream.
StatusIndicate if the work stream is active or inactive.**Stored Query**Specify the stored query associated with the work stream.**Description**Describe the work stream in detail.
5. Click **Save** to save the default work stream.
The work stream is displayed on the navigation pane.

Disable a Work Stream

An administrator can disable a default work stream for all the analysts.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Interface** , and **Work Streams.**
3. From the list, select the work stream you want to disable.
The work stream detail page appears.
4. Click **Edit**.
5. Set the **Status** to **Inactive**, and click **Save**.

Push Notification Service

The push notification services are configured to enable communication between the CA SDM server and the xFlow Interface.

This service is enabled when you install the CA Service Management 14.1.03 cumulative patch and the xFlow Interface.

Push notifications ensure the logged-in analyst receives real-time change notifications when a ticket is updated or modified by another analyst.

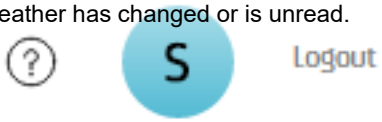
The xFlow Interface receives real-time push notifications only during an active session. This service is configured for the following features:

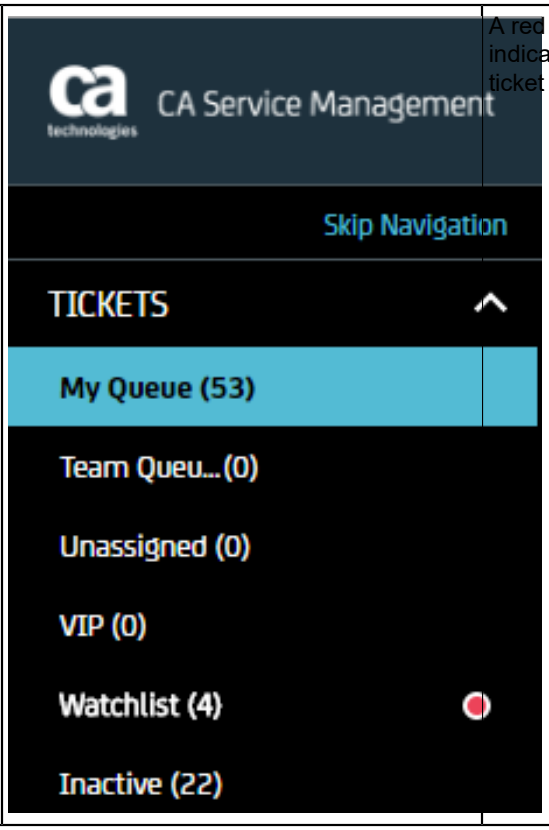
- Notifications
- Team Messages
- Timeline
- Weather
- Work Streams

NOTE

To improve the overall performance of the application, an administrator can choose to enable the polling mechanism to receive periodical updates on work streams. For more information, see Polling for Work Streams.

Following is a representation of the real-time push notifications:

Feature	Representation of the push notification	Description
Notification		Number of unread notifications are displayed.
Team Messages		A red dot on the icon indicates a new message.
Weather		A red circle on the icon indicates if the weather has changed or is unread. 

Work Streams		A red dot icon on a work stream label indicates a ticket is added or an active ticket is updated in the work stream.

Polling for Work Stream

An administrator can enable the polling mechanism to receive periodical updates on work streams.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **xFlow Interface,** and **General.**
3. From the list of options, select **enableStreamPoll.**
4. Click **Edit**.
5. Set the **Configuration Value** to **True**.
6. Click **Save.**

Set the Polling Interval Value

An administrator can define the time interval (in seconds) to receive periodical updates on the work streams from the polling server.

Follow these steps:

1. Log in to CA SDM as a system administrator.
2. Navigate to the **Administration** tab, select **xFlow Interface,** and **General.**
3. From the list of options select **StreamPollInterval.**
4. Click **Edit** and modify the **Configuration Value**.
   The default time is set as 60 seconds.
5. Click **Save**The polling interval is saved.
6. Restart the *CA Service Management xFlow Interface Server* service.

Enable the Research Tool Extension

Research Tool is available as an extension in the following supported browsers:

- CA Service Management Extension for Chrome
- CA Service Management Extension for Firefox

> **NOTE**
>
> Research Tool extension for Microsoft Internet Explorer and Safari is not supported.

To enable the Research Tool for your browser, follow these steps:

Step 1: Configure the Download Links

As an administrator, configure the extension download URLs.

Follow these steps:

1. Stop the required application server.
2. Based on the application server, edit the following file:
   For Apache, navigate to **<xflow_Install_location>\APPS\Services\incidentmicroservice-17.0.479\public\conf** and edit the **casm.conf.do-not-change.js** file.
3. Locate the *var* ***casm_admin*** section and add the following details:

```
 researchTool:{
             url:"http://www.google.com",
             storeURLs: {
             firefox: 'https://support.broadcom.com/download-center/solution-detail.html?
aparNo=99111302&os=WINDOWS-ALL',
             chrome:  'https://support.broadcom.com/download-center/solution-detail.html?
aparNo=99111301&os=WINDOWS-ALL';
   }
 }
```

4. Save and close the file and restart the application *server*

Step 2: Enable the Extension

Enable Extension for Chrome

Click on the Research Tool icon from the xFlow Interface. This will take you to the Broadcom Support page and download the extension under the **Solution Downloads** section.

> **NOTE**
> To install the extension, perform the following steps:
>
> - Visit chrome://extensions (via omnibox or menu > More tools > Extensions).
> - Enable Developer mode by ticking the checkbox in the upper-right corner.
> - Click on the "Load unpacked extension..." button.
> - Select the directory containing your unpacked extension

After you enable the extension, refresh the **xFlow Interface** .

Enable Extension for Firefox

Follow these steps:

1. Click on the Research Tool icon from the xFlow Interface. This will take you to the Broadcom Support page and download the extension under the **Solutions Download** section.

2. Verify that the Download folder contains the *cart.xpi file.*
3. Launch Firefox and from the download, drag and drop the extension to any Firefox window.
4. Click a Firefox window to initiate the installation. Click **Install**.
5. Confirm the installation.
6. Open a Firefox tab and from the toolbar, click the CA Service Management Extension for Firefox extension.
7. Accept the End User License Agreement and click **Next** to complete the installation.
8. Refresh the **xFlow Interface** .

NOTE
This plugin works for firefox ESR Version lesser than 46.0.

Configure the xFlow Interface Cache Size

Cache Considerations

After installing the xFlow Interface as described in Install CA Service Desk Manager, you can configure the cache size for xFlow Interface server.

Consider the following:

- Ensure to review the cache size of your environment before starting the xFlow Interface and Search Server services, review the cache size with respect to your environment.
- Your environment may already have some CA SDM factory records. It is recommended that you increase the record count for such factories in the CA SDM console (**Administration , xFlow Interface** , **General)**.

NOTE

We recommend that you cache all the active records of CA SDM factories that are mentioned in this article. Set the max record limit to more than the number of active records available in your environment.

The following cache and corresponding CA SDM factory names are given with out-of-the-box (OOTB) CA SDM max record count:

Cache Name	CA SDM Factory	Default value for Max Cache Record	Configuration Key in CA SDM Console xFlow Interface Administration
role_access	func_access_role	10000 records	cache.role_access.maxEntriesLocalHeap
role_auth	role	2000 records	cache.role_auth.maxEntriesLocalHeap
dcon	dcon	50000 records	cache.dcon.maxEntriesLocalHeap
acctyp_role	acctyp_role	2000 records	cache.acctyp_role.maxEntriesLocalHeap
tenant	tenant	2000 records	cache.tenant.maxEntriesLocalHeap
tgm_members	tgm_members	2000 records	cache.tgm_members.maxEntriesLocalHeap

Enable SAML Authentication for xFlow Interface

Prerequisites

1. Ensure that SSL is enabled for CA XFlow. For more information see, Enable SSL Authentication for CA SM 17.4.
2. **For ADFS**: If you want to use ADFS as the identity provider, you must install ADFS on Windows Server 2012 R2 and above. After completing the installation of ADFS and CA Service Desk Manager, you must configure the ADFS with following the CA XFlow related information:
   a. While configuring Add Relying Party Trust on ADFS specify the following on the **"Configure URL"** screen:
      - Select the **"Enable support for the WS-Federation Passive protocol"** check-box.
      - Enter the value in the **"Relying party WS-Federation Passive protocol URL:"** as below: ***https://<xflow_url>:<port_number>/samllogin***
      - Select the **"Enable support for the SAML 2.0 WebSSO protocol"** check-box.
      - Enter the value in the **"Relying party SAML 2.0 SSO service URL:"** as below: ***https://<xflow_url>:<port_number>/samllogin***
   b. While adding claim rules, specify the following in the **"Configure Claim Rule"** step:
      - **Claim rule name**
        Specify a name for the claim rule. For example, XFlow Claim Rule.
      - **Attribute store**
        Select the datasource from the drop-down list. For example, Active Directory.
      - **LDAP Attribute**
        Select "**SAM-Account-Name**" from the drop-down list.
      - **Outgoing Claim Type**
        Select "**Name**" from the drop-down list.
   c. Verify that the certificate you want to use to configure the Identity Provider is SHA 256 signed.
3. **For CA Siteminder**: If you want to use CA Single Sign-On (formerly known as CA Siteminder), as your Identify Provider, do the following:
   a. Launch CA Single Sign-on as Administrator.
   b. Go to the Federation Partnership Partnership Federation and click Partnerships.
      Federation Partnership List appears.
   c. Click the Federation Partnership name which you want to use.
      Partnerships properties page opens.
   d. Expand the IP Restrictions section and verify the Assertion Attributes properties.
   e. The value of the Namespace field must be the URI for the respective Claim. If the value is specified as "Unspecified" the SAML authentication will not work.
   f. Verify that the certificate you want to use to configure the Identity Provider is SHA 256 signed.
4. **For Azure Identity Provider (IdP):** If you want to use Azure Identity Provider (IdP), as your Identify Provider, do the following:
   a. Login to Microsoft Azure Portal.
   b. Click on the "**Azure Active Directory**" link under Azure Services.
   c. Click on the "**Enterprise Applications**" link from left side window.
   d. Click "**New Application**" link and again click "**Create your own application**" link.
   e. Enter name example "**CASM tomcat**" in the "**What's the name of your App**?" text box and Click **Create** on the right-pane.
   f. Click on the "2. Set up single sign on" link and Click on the "SAML" link.
   g. Click on the Edit button on section 1 i.e "Basic SAML Configuration".
   h. Enter SDM SSL tomcat/iis URL like ***https://<xflow_url>:<port_number>/samllogin*** for all below text fields:
      a. "**Identifier (Entity ID)**"

b. "**Reply URL (Assertion Consumer Service URL**"
c. "**Sign on URL**"
d. "**Relay State**"
e. "**Logout Url**"

i. Click **Save**.

SAML authentication for xFlow Interface

Security Assertion Markup Language (SAML) is a standard for logging users into applications based on their sessions in another context. SAML SSO works by transferring the user's identity from one place (the identity provider) to another (the service provider). This is done through an exchange of digitally signed XML documents.

This article explain how to enable SAML authentication for xFlow Interface.

Follow these steps:

1. Launch CA Service Desk Manager and login as Administrator.
2. Go to the Administration tab and navigate to xFlow Interface, General in the left hand navigation.
   The General Configurations List page opens.
3. Find the following from the Configuration Key column and specify the values as specified below:
4. **federation.audienceuris**: Specifies the URI of the application from where to accept the tokens.
   Example: *https://<xflow_url>:<port_number>/samllogin|https://<xflow_url>:<port_number>/samllogin*
5. **federation.enableManualRedirect**: Specifies whether you want to enable manual redirection of the token or not. True indicates that you want to enable the manual redirection and False indicates that you do not want to enable the manual redirection.
   Default: false
6. **federation.realm**: Specifies the location from where the tokens are sent.
   Example: *https://<xflow_url>:<port_number>/samllogin*
7. **federation.reply**: Specifies the URL of the location that receives responses.
   Example: *https://<xflow_url>:<port_number>/samllogin*
8. **federation.trustedissuers.friendlyname**:
   ADFS Idp (Identity Provider)
   Specifies a common name for the Identity Provider.
   Example: *ADFS Signing - <trusted_issuer_URL>.*
   Microsoft Azure Idp:
   Perform the following:
   a. Login to Microsoft Azure portal, navigate to **Enterprise Application**, <**CASM Application Name**>, Single-Sign-On, Block/Section 4, Azure ID Identifier (use the copy option to copy the value).
      For example: *https://sts.windows.net/<tenant_id>/*
   b. Replace **saml2** with **wsfed** in the URL.
      For example: *https://login.microsoftonline.com/<microsoft_tenentid>/wsfed*
9. **federation.trustedissuers.issuer**:
   ADFS Idp (Identity Provider):
   Specifies the URL of the ADFS Identity Provider.
   Example: https://<trusted_issuer_URL>/<identity_provider>/ls/idpinitiatedsignon.aspx.
   For Microsoft Azure Idp (Identity Provider):
   Perform the following:
   a. Login to Microsoft Azure portal, navigate to **Enterprise Application**, <**CASM Application Name**>, Single-Sign-On, Block/Section 4, Login URL. Copy the URL (use the copy option adjacent to the URL).
      For example: *https://login.microsoftonline.com/<tenant_ID>/saml2*
   b. Replace **saml2** with **wsfed** in the URL.

For example: *https://login.microsoftonline.com/<microsoft_tenentid>/wsfed*

10. **federation.trustedissuers.thumbprint**:
    ADFS Identity Provider
    Specifies the value of the certificate thumbprint provided by the Identity Provider.
    Example: *0214c3035d002505b9e5e672a117d9bf5c5d4d02*

    NOTE

    The certificate you want to use to configure the Identity Provider must be **SHA-256** signed.

    The thumbprint you use from the SHA-256 certificate must be the **SHA-1** thumbprint.

    Azure Idp: federation.trustedissuers.thumbprint
    Perform the following:
    a. Login to Microsoft Azure portal, navigate to **Enterprise Application**, <**CASM Application Name**>, Single-Sign-On, Block/Section 3, Thumbprint. Copy the value.
       For example: 7844FB57B8EB854C10F2CBD8BE78B0BF574B729D
11. **authenticationtype**: Specifies the authentication mechanism you want to enable for single sign-on. The available options are SAML, SSO, NTLM and BASIC.
    Enter SAML as the configuration value.
12. Restart the CA xFlow Interface service.
    You have successfully enabled SAML based single sign-on authentication for CA Service Management.

Configuration Changes for xFlow Interface

The main components of the xFlow Interface are a set of services (Incident, search, and notification services). These services share the default domsrvr out-of-the-box. For better performance, dedicate your domsrvrs for these services. From the CA SDM console, navigate to **Administration**, xFlow Interface, **General** tab and update the following domsrvr variables:

- sdm.domsrvr.inms (for Incident services)
- sdm.domsrvr.srms (for search services)
- sdm.domsrvr.psms (for notification services)
- sdm.domsrvr.coms (for collaboration services)

You can provide multiple domsrvrs separated by a comma (,) for these services. After updating these variables, restart the xFlow Interface Server services. For more information, see Configure the xFlow Interface Server Services.

The xFlow Interface requires additional hardware resources and scales better horizontally. xFlow Interface services scaling needs to be implemented based on the performance needs at a particular installation site. Scaling of xFlow Interface services involves essentially three steps. Each step should give a performance benefit over the previous one. Follow the below order of steps to scale the xFlow Interface micro-services better.

1. Attempt to increase the number of domsrvrs that each micro-service connects.
   – For more information about adding new domsrvrs to an installation, see How to Configure Processes for CA SDM Servers.
   – Review the documentation on this page to configure the micro-services to use these new domsrvrs.
2. If the above does not result in satisfactory performance, next step is to scale the micro-service instances. The micro-services (and the UI servers) can be scaled by installing xFlow Interface on different servers and using a load balancer to point to the new instances of the UI servers.

App Launcher - Apps Configuration and Role Switching

This article contains the following topics:

NOTE

If you have upgraded to xFlow Interface 17.4, and have users with custom access types, these users may not be able to access xFlow Interface directly. To be able to access the xFlow Interfacet App, update the custom access type to xFlow Interface app from App Configuration. For more information about how to add multiple access types in an App, see Adding Apps in Bulk To an Access Type.

xFlow Interface expands beyond Analysts in this release. The App driven approach delivers target user experience for different types of users. It provides the capability to switch between Apps and switch between Roles.

In the classic CA SDM UI, functional access and data access is controlled using roles. A contact or user in CA SDM has Access Types assigned to it.The access type in turn is mapped to assigned roles, and one of these is the default role.

The xFlow Interface provides the ability to configure and launch Apps as per user authorization (Access Types and Roles) and is based on the following considerations:

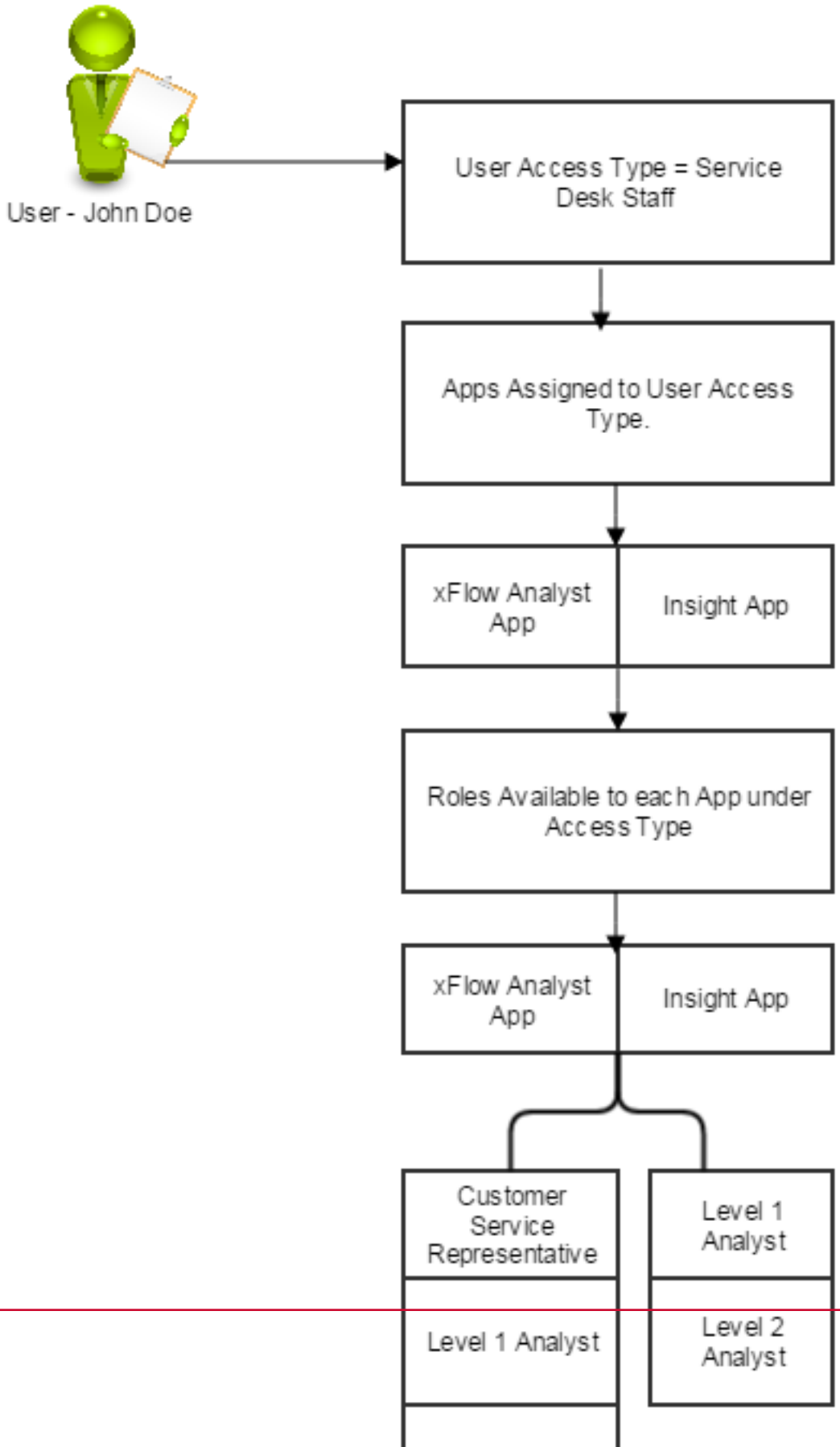
- Access to Apps is given based on the App configured in the Access Type of the user.
- When you login to the xFlow Interface App, you will be logged in with the default App of access type attached to a user.
- The App Launcher will have Out of the box (OOTB) Apps. You can update or change the App assigned to an Access Type.

  NOTE

  You can enable or disable an App but you cannot delete an app. You can enable or disable the app through the Classic CA SDM Interface by making the App Active or Inactive.

  Each App will have a list of roles assigned to it. When you launch the App Launcher for the first time, the current role will be the default role of the App . You can update or change the default role assigned to an app. For more information about APP to Role Mapping, see How to Update Roles for an Access Type.

Figure 7: Manage Appp Authorization in App Launcher



Managing Apps configuration in the xFlow Interface is a two-step process:

- Configure Apps Authorization (Access Types and Roles) - Classic CA SDM Interface
- Accessing the App Launcher - xFlow Interface

Configure Apps Authorization (Access Types and Roles) - Classic CA SDM Interface

To update Access Types and Roles for a user as per the Authorization level, perform the following from the Classic CA SDM UI:

- Update Access Types for an App
- Set the Default App for an access Type
- How to Update Roles for an Access Type
- Adding Multiple Apps in Bulk To an Access Type

Update Apps for an Access Type

To update an app for a given access type, perform the following steps from the Classic CA SDM UI:

1. Click **Administration**, **Security and Role Management**, **Access Types.**
2. From the Access Type List, select the desired Access Type.
   For example, select the access type as **Service Desk Staff.**
   The Access Type Details page is displayed.
   The Apps assigned to an Access Type is displayed on the Access Type Details page.
3. Select the **Apps** tab. Click **Update Apps**, if you want to modify or update the App for an access type.
   For example, **xFlow Interface or Insights.**
4. Click **Search** on the App Configuration List page.
5. Select the desired apps and move it from Available list to the Assigned list.
   Click **OK**.
   The App is updated or modified for the Access Type.

Set the Default App for an Access Type

Log into the Classic CA SDM Interface and perform the following steps:

1. Click **Administration**, **Security and Role Manag.ement**, **Access Types.**
2. Select and Open the required access type**.** For example: **Service Desk Management**
   The Service Desk Management Access Type Detail page is displayed.
   On the **Apps** tab, select the App that you want to set as the default App and click **Set Default App**.
   The selected app is now set as the default app for the user having above access type (Service Desk Management).

Adding Apps in Bulk To an Access Type

> **NOTE**
>
> The topic describes an alternate way of updating apps for an access type other than what has been described in the preceding section. For more information, see Update Apps for an Access Type.

A new node **App Configuration** is added to the classic CA SDM UI. Login to the Classic CA SDM UI and click **Administration**, **xFlow Interface** , **App Configuration** to view, modify or update the Apps assigned to an Access Type and add multiple apps at once.

To add multiple Apps to an Access Type at a go, perform the following steps on the Classic CA SDM UI:

1. Click **Administration**, **xFlow Interface** , **App Configuration**.
2. Select an **App**.
   For example, xFlow Interface App.

The Access Types assigned to an App is displayed on the App Details page.

3. Click **Update Access Types**, if you want to update the access type.
   For example, **Service Desk Management**.
4. Click Search on the Access Type Search page.
5. Select the access type that you want to update from the list.
   Click forward arrow keys to move your selection to the **Access Type Assigned** list.
6. Click **OK**.
   The Access Type is added for an App.

How to Update Roles for an App in Access Type

You can map roles to an Access Type by performing the following:

NOTE

When assigning Roles for an App, ensure that the role is designed for the App's persona.

Perform the following steps to update roles for an App in Access Type:

1. Login to the classic CA Service Desk interface as an Administrator.
2. Navigate to **Administration**, **Security and Role Management**, **Access Types**. For example: **Administration**
   The Administration Access Type Detail Page is displayed.
3. Click on the **Apps** tab.
   The list of Apps assigned to this access type (in this case, Administration) is displayed.
4. Click the App hyperlink for which you want to update the Role.
   This will display the Detail page of the App under a particular Access Type (Example: Administration).
5. Click **Update Roles**,
   The Role Search Page is displayed. Click **Search**.
   Only Roles assigned to an Access Type will be displayed.
6. Select the desired role and move it from Roles Available to Roles Assigned.
   Click **Ok**.
   Now you can also update the Default Role of an App using the **Set Default Role** button.

NOTE

When no roles are assigned to an app, it takes the default role of an access type.

Accessing the App Launcher - xFlow Interface

App launcher is the new UI element in the xFlow Interface on the top right pane. You can view and access the following OOTB Apps.

- xFlow Interface App: Provides the ability to create and manage tickets, manage the service type events/SLA, and view the Knowledge base articles.
- **Insight App**: View the JasperSoft reports and dashboard in your environment. (Insight App is only supported on Microsoft SQL Server Database).

NOTE

If only one app with a single role is assigned to a user, the App Launcher Icon on the xFlow Interface is not displayed.

Perform the following steps to launch the App Launcher:

1. Login to the xFlow Interface.
2. Click the **App Launcher** icon on the top right pane. By default, an App will be selected as per the user Access Type. For example: **xFlow Interface App.**

For more information, see Set the Default App for an Access Type.

3. Click the drop-down list from the selected App to view the role mapped to an app. Select the desired role.For example: In case of the xFlow Interface App, the drop-down list will show the roles available to an analyst user - **Administration**, **Incident Manager**, **Vendor Analyst**, **Level 1 Analyst**, **Level 2 Analyst**.

   NOTE

   This role drop-down list will not appear if the app has a single role assigned to it.

Manage Chat

As an administrator, you can define how chat requests are managed by analysts.

Set Up Access Level Permission

Administrator must ensure users have the required access level permissions.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **Security and Role Management**, and **Support Automation Access List**.
3. The following access levels are available:
   a. **Analyst** Defines the rights for users who provide live chat assistance to end users.
   b. **End User**
   Defines the rights for users that request live chat assistance from analysts, such as employees.

Create a Queue

An administrator can add or define a uniquely named queue.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **Support Automation**, **Queues**, and **Queue List**.
3. Click **Create New**.
4. Complete the following fields:
   - **Tenant**Specifies the tenant for the queue.
   - **Queue Name**Specifies the queue name.
   - **End User Display Name**Specifies the name that displays for the end user.
   - **Default**Specifies if the queue is the default queue.
   - **Status**Specifies the status of the queue.
   - **Default Chat Preset**Specifies the default chat preset for the queue.
   - **Queue Hours**Specifies the queue hours.
   - **Issue Category**Specifies the Issue Category associated with this queue.
   - **Incident/Request Area**Specifies the Inciedent/request Area associated with this queue.
5. Click **Save**.
   The queue is created.

Assign a Default Queue

You can route live chat to a default queue. You can configure only one default queue per tenant.

Follow these steps:

1. Log in to CA SDM as an administrator.

2. Navigate to the **Administration** tab, select **Support Automation**, **Queues**, and **Queue List**.
3. Click a queue.
4. Click **Edit** to modify to the settings.
5. Select the **Default** check box and click **Save**.
   The queue is set as default.

Set Queue Hours of Operation

You can set the hours of operation for each queue. The queues determine multi-tenancy, which are tenant optional.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **Support Automation**, **Queues**, and **Queue List**.
3. Click a queue.
4. Click **Edit** to modify to the settings.
5. Click **Queue Hours**.
   The Workshift Search page appears.
6. Complete the information and click **Save**.
   The hours of operation are saved.

Restrict the Live Chat

You cannot disable the Live Chat. However, you can stop the Live Chat operation by assigning the Workshift which have no work hours set, to the default queue (support). To set a Workshift with no work hours, enter a past date in the schedule. For more information, see Workshift Setup.

> **NOTE**
>
> If you are using Support Automation and using Queues then don't Restrict the Live Chat as it impacts the Support Automation behavior.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **Support Automation**, **Queues**, and **Queue List**.
3. Click a queue.
4. Click **Edit** to modify to the settings.
5. In **Queue Hours**, enter the Workshift and click **Save**.
   The Live Chat is stopped and "*Live Chat not available*" message will be displayed to the user trying to initiate the chat.

Establish an Auto Transfer or Escalate Process

You can establish an auto transfer or escalation process for each queue, when a live chat is not addressed within a specified time interval.

Follow these steps:

1. Log in to CA SDM as an administrator.
2. Navigate to the **Administration** tab, select **Support Automation**, **Queues**, and **Queue List**.
3. Click a queue.
4. Click **Edit** to modify to the settings.
5. Complete the following fields:**Auto Transfer Users**
   Select the check box to automatically transfer or escalate a live chat request to another queue.
   Timeout
   Specifies the timeout value (in seconds)

Target Queue
Specifies the target queue where the live chat request will be transferred or escalated.

6. Click **Save**.
   The auto transfer or escalate process is saved for the queue.

Configuration keys for Live Chat

The following pre-defined configuration keys are used by Live Chat:

- **chat.cleanUpThreadInterval**
  The interval of time (in minutes) to calculate session timeout.
- **chat.escalateThreadInterval**
  The interval of time (in minutes) to calculate escalation timeout for a live chat (when a live chat request is not addressed within a specified time, it is then escalated to a different queue). The **chat.escalateThreadIntervall** value must be less than the **chat.session Timeout** value
- **chat.sessionTimeout**
  Number of minutes to wait without any activity, before ending the live chat session automatically. Making the intervals longer can lead to the unnecessary maintenance of inactive sessions in memory.

> **NOTE**
>
> **Information:** To lookup for the configuration keys, navigate to the **Administration** tab, select **xFlow Interface**, and **General**.

Recommendations:

- Use the predefined configuration **key:values**.
- Higher values can consume significant chunk of memory and may affect performance.
- Restart the xFlow Services if you have modified the pre-defined configuration **key:value**.

Administer the xFlow Interface

The xFlow Interface provides analysts the ability to understand the IT environment, prioritize work and make decisions. Data-driven intelligence improves the productivity of the analyst and enables them to understand the workload for the day and the list of activities to be completed.

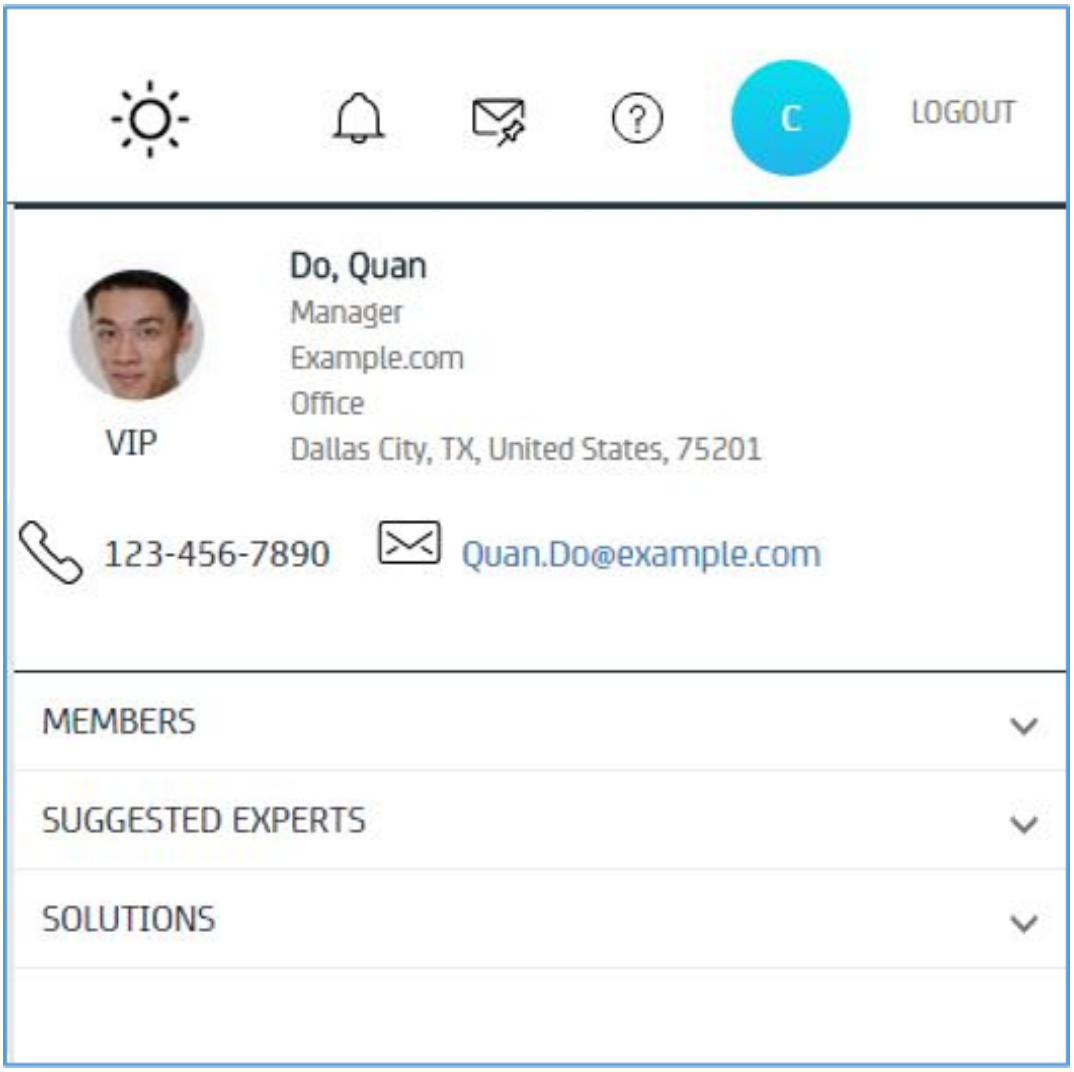
This article contains the following topics:

- Display the User Information
- Customize the Branding
- Modify Date and Time Format
- Manage Attachments
- Team Messages
- Configure External Redirect URL on Logout
- Configure SSO for xFlow Interface/Service Point
- Rate Limiting in xFlow

Display the User Information

The user information (contact widget) is displayed when an analyst creates a new ticket and when a ticket is accessed in either the card or grid view.

The contact widget displays the name, designation, organization name, address, email address, VIP status and profile picture of the user, as shown in the example.



For more information about user accounts, see Configuring User Accounts.

Customize the Branding

As an administrator you can customize the brand logo, branding text, and the favicon.

Brand Logo

The default logo at the global and tenant level for xFlow Interface, Insights, and Service Point is the **CA logo.**

Recommendations for the logo:

- Optimal logo size: 166 x 40 (width x height)
- Supported logo file formats: .png,.jpg, .jpeg, .gif, .bmp, and .tiff;
  Service Point also supports .svg format.

Customize Brand Logo Using pdm_branding Utility

To change the default logo for xFlow Interface, Insights, and Service Point, execute the pdm_branding utility command on the primary server of your CA SDM environment. The default logo path is `C:\Program Files (x86)\CA\Service Desk Manager\bopcfg\www\wwwroot\img`.

NOTE

When multi-tenancy is enabled, you cannot have separate logos for xFlow Interface, Insights, and Service Point.

Follow these steps:

1. Execute the following command: `pdm_branding [-h] | [-m] | [-g] | [-a] | [-d]`
   - **pdm_branding -h**
     Prints the command line help
   - **pdm_branding -m**
     Migrates the default as well as the tenanted logos to database from the filesystem.
   - **pdm_branding -m -g**
     Migrates all the tenant-specific logos and the logo present in the given path as the default/global logo.
   - **pdm_branding -a -t**
     Adds a logo to a particular tenant.
     For example: `pdm_branding -a -t <tenant:persistent ID>` To obtain the tenant persistent ID, execute the following command: `bop_odump domsrvr tenant "name='TenantName'"`
   - **pdm_branding -d -t**
     Deletes a logo for a tenant.
     For example: `pdm_branding -d -t <tenant:persistent ID>` To obtain the tenant persistent ID, execute the following command: `bop_odump domsrvr tenant "name='TenantName'"`
2. Restart the xFlow Interface services.

Customize Brand Logo Without the Utility

When multi-tenancy is not enabled, you can have separate logos for xFlow Interface, Insights, and Service Point.

To replace the CA logo with yours for Service Point, perform the following steps:

1. Navigate to the default location, `C:\Program Files\CA\xFlow\APPS\Services \incidentmicroservice-17.0.479\public\assets\images\logos` and add the new logo file "CA-logo-mod.svg".
   The reference to"CA-logo-mod.svg" will apply for the logo for Service Point.
2. Navigate to the default location, `C:\Program Files\CA\xFlow\APPS\Services \incidentmicroservice-17.0.479\public\conf` and open the **casm.conf.js** file in a text editor.
3. Add the following new key:value, after the **locale** key:

```
bui_branding : {
      builogo : 'assets/images/logos//CA-logo-mod.svg',
  },
```

4. Save and close the **casm.conf.js** file.
5. Restart the xFlow Interface services.

To replace the CA logo with yours for xFlow Interface and Insights, perform the following steps:

1. Navigate to the default location, `C:\Program Files\CA\xFlow\APPS\Services \incidentmicroservice-17.0.479\resources\images\icons` and add the new logo file CA-SM_logo-white-mod.svg".
   The reference to "CA-SM_logo-white-mod.svg" will apply to the logo for the main xFlow login screen and the xFlow logo.
2. Navigate to the default location, `C:\Program Files\CA\xFlow\APPS\Services \incidentmicroservice-17.0.479\public\conf and` open **casm.conf.js** file in a text editor.
3. Add the following new key:value, after the **locale** key:

```
logo : 'resources/images/icons/CA-SM_logo-white-mod.svg'
```

4. Save and close the **casm.conf.js** file.
5. Restart the xFlow Interface services.

Customize the Favicon

The favicon appears in the bookmarks and browser address bar for the platform.

To update the favicon, perform the following steps:

1. Navigate to the default location, `C:\Program Files\Files\CA\xFlow\APPS\Services \incidentmicroservice-17.0.479\public\resources\images\icons`
2. Replace the **favicon.ico** file.
3. Restart the xFlow Interface services.

Customize Brand Name

The brand name is displayed in the top left hand corner next to the logo. The default brand name at the global and tenant level for xFlow Interface, Insights, and Service Point is **CA Service Management** .

Currently, you can replace the CA brand name only for Service Point, and not for xFlow Interface and Insights.

To replace the CA brand name for Service Point, perform the following steps:

1. Navigate to the default location, `C:\Program Files\CA\xFlow\APPS\Services \incidentmicroservice-17.0.479\public\conf` and open **casm.conf.js** file in a text editor.
2. Add the following new key:value, after the **locale** key:

```
logo : 'resources/images/icons/CA-SM_logo-white.svg',
bui_branding : {
            builogo : 'assets/images/logos/CA-SM_logo.svg',
            brandname : 'CA Service Management',
        }
```

3. Override the default CA brand name with your brand name.
4. Save and close the **casm.conf.js** file.
5. Restart the xFlow Interface services.

To replace the CA brand name at a tenant level for Service Point, perform the following steps:

1. Navigate to the default location, `C:\Program Files\CA\xFlow\APPS\Services \incidentmicroservice-17.0.479\public\conf` and open **casm.conf.js** file in a text editor.
2. Add the following new key:value, after the **locale** key:

```
logo : 'resources/images/icons/CA-SM_logo-white.svg',
bui_branding : {
            builogo : 'assets/images/logos/CA-SM_logo.svg',
            brandname : 'CA Service Management',
        }
```

3. Remove **CA Service Management** as the brand name, and let the value be empty.
4. Save and close the **casm.conf.js** file.
5. Restart the xFlow Interface services.
   Tenant name is displayed as the brand name.

Modify Date and Time Format

By default the xFlow Interface displays date and time in dd/MM/YYYY and hh:mm a format.

12 Hour vs. 24 Hour Time

Capital letters denote 24 hour time and lower case 12 hour time.

Example

To use a US date format, perform the following steps:

1. Access the **casm.conf.do-not-change.js** file from \Program Files\CA\xFlow\APPS\Services \incidentmicroservice-17.0.479\public\conf folder.
2. Modify the default date format as shown:
   Current
   ```
   "defaultDate": "dd/MMM/yyyy"
   ```
   New
   ```
   "defaultDate": "MM/dd/yyyy"
   ```
   IMPORTANT
   CA Service Management 17.3.0.8 and above release versions support the **dd/MM/yyyy HH:mm:ss** date format. Modify the date format as shown below:
   Current
   ```
   dateTimeFormat_default": dd/MM/yyyy HH:mm:ss
   ```
   New
   ```
   dateTimeFormat_default":  MM/dd/yyyy HH:mm:ss
   ```
3. Save the file
4. Restart the **CA xFlow Analyst Interface** Service.

Manage Attachments

This article contains the following topics:

Enable Attachments on xFlow Interface

Perform the

1. Log in to CA SDM as an administrator.
2. Navigate to **Administration** tab, **xFlow Interface**, **General**, select **attachments.allowed-content-types**. click **Edit**.

Supported File Formats

The following file formats can be attached to a ticket from xFlow Interface.

File Type	File Format
Word Processing Format	.doc, .docx, .txt
Spreadsheet Formats	.xls, .xlsx
Presentation Formats	.ppt, .pptx
Graphic Formats	.pdf, .jpg, .jpeg, .png

Configurable File Formats

The following file formats are excluded due to security vulnerabilities: **zip**, .**xml**, .**exe**, and .**mht.** However, based on the environment, an administrator can configure these file formats to be attached to a ticket.

For example, to add **.zip** file format, perform the following steps:

1. Log in to CA SDM as an administrator.
2. Navigate to **Administration** tab, **xFlow Interface** , and **General**.
3. From the list of options, select **attachments.allowed-content-types** and click **Edit**.
4. Add the following value toward the end of the existing **Configuration Value**:
   ```
   , .zip:application/zip
   ```
5. Click **Save**.
6. Restart the CA Service Management xFlow Server Service.

Team Messages

This article contains the following topics to manage the Team Messages feature:

Archive and Purge Team Messages

To keep the database at a manageable size for optimum performance, periodically remove historical team messages. A configuration Object Name *team message* has been added to the Archive and Purge Rule. The *team message* object is disabled by default.

> **NOTE**
>
> For more information about creating rules or activating the predefined team message rule, refer, How to Archive and Purge Historical Data.

Disable Team Messages

An administrator can disable the Team Messages feature. The Team Messages feature is enabled by default.

Follow these steps:

1. Log in to CA SDM as a system administrator.
2. Navigate to **Administration** tab, **xFlow Interface** , **General**.
3. From the list of options select **TeamMessagesEnabled**.
4. Click **Edit** and modify the **Configuration Value** to **false**.
5. Click **Save**.

The Team Message feature is disabled. Verify the update by viewing the xFlow Interface, the Team Message icon does not appear on the interface.

Configure External Redirect URL on Logout

After logging out from xFlow Interface or Service Point, you can redirect the browser URL to any configured location by making changes to the casm.conf.js file from the xFlow Interface Installed location:

Perform the following steps:

1. Open the casm.conf.do-not-change.js file where you have installed the xFlow Interface.
   <xFlow Interface Installed Directory>\APPS\Services\incidentmicroservice-17.0.479\public\conf
   The above location has two files: **casm.conf.do-not-change.js** and **casm.conf.js**
2. Open the **casm.conf.do-not-change.js** file and copy the following strings:

```
 onLogout: {
url:'/login', //Custom placeholder for url redirection when clicked on Logout
isExternal : false //Default:false change this to true if you want to keep it o external url like
 www.google.com
},
```

3. Now, paste the above strings in the **casm.conf.js** file.
4. To redirect to an external URL, put your custom URL location in "url:'/login' string and change '*isExternal'* to **True** in the above command (step2).
5. Refresh the browser cache.

Configure SSO for xFlow Interface/Service Point

To configure SSO for xFlow Interface/Service Point, navigate to Administration, xFlow Interface, General, perform the following step

1. **authenticationType**: Specifies the authentication mechanism. The configuration value must be set to SSO. This is mandatory to enable SSO for xFlow or Service Point.
2. **SSOArtifactType**. For SSO authentication, the configuration value must be set to**COOKIE**, **HEADER**, and **PARAMETER.**
3. **SSOArtifactName**: **sm_user** is the fixed value. Users must use this default value.
4. **sso.security.cookies**: By default this value is set to False. If you need to have secured cookies, set this value to True. If set to True, ensure that CA Service Desk Manager (SDM) and xFlow interface is SSL enabled.

Users must be present in CA SDM with access policy set to **Allow external authentication**.
If a user is not present in CA SDM, Error 401 (unauthorized error) is shown.

Rate Limiting in xFlow

> **NOTE**
> This is applicable from CASM 17.4 RU5 and above.

You can now control the number of API requests allowed within a specific time frame using the **Rate Limiter Filter**.
The Rate Limiter Filter is a comprehensive API protection mechanism that safeguards against API abuse by enforcing configurable request limits.

It enforces configurable limits on API calls and supports both token-based and IP-based identification to ensure fair and secure access.

By default, rate limiter filter is disabled (ratelimit.enabled = false). When set to true, the remaining parameters become applicable.

To enable or disable the rate- limiting functionality:

- Locate the file ***application.conf*** at the default location. *(Default location: **C:\Program Files\CA\xFlow\APPS\Services \incidentmicroservice-17.0.479\conf**).*
- Edit the below configurations as per your requirements.

  > **NOTE**
  > This is applicable only for the end-points present in ***incident microservice*** (for both xFlow and ServicePoint)
- Restart xFlow service.

Rate Limitation Configuration

Configuration for application.conf	Description	
ratelimit.enabled = false	Enables or disables rate limiting for API endpoints.	Set
ratelimit.defaultPerUserLimit = 50	Defines the number of requests allowed per API endpoint per X-Access Token within the configured time window.	For ***= 1*** min
ratelimit.timeWindowMinutes = 1	Duration of the rate-limiting time window (in minutes). Users can make the configured number of requests within this window.	Use allo
ratelimit.customIpHeader = ""	Allows specifying a custom header to extract the client IP address for rate limiting. When configured, this header is checked before the standard ones (X-Forwarded-For, X-Real-IP, Remote-Addr).	Exa Cus with

Administer Insights

As a CA Service Management administrator, you can create users and map roles/access types , create domains, add custom attributes, and so on as described in this article for Insights (Aggregate Reports and Transactional Reports for JasperReports):

This article contains the following topics:

Assign Roles and Functional Access

NOTE

CA Service Management no longer supports role switching for Insights App. The default OOTB role for Insights app is available with CA Service Management 17.4, and it is similar to the Out-of-the-Box (OOTB) Reporting role of an Access Type for a contact. In case, if you change the default role of the app to any other role than the Reporting role, ensure that you consider the following:

- Aggregated Reporting reflects data as per the new default role the app
- Transactional Reporting honors only reporting role of the contact

Ensure that the reporting role of an access type and default role of an access type for Insights App is same.

The below table lists out the CA SDM Roles and Functional Access. Ensure that you have assigned the appropriate roles and access for access to reporting domains.

The List of Roles, and Functional Access

Access Type	Roles List	Default Role
Administration	Administrator Level 1 Analyst Level 2 Analyst Service Desk Administrator Service Desk Manager Super Administrator	Default
Process Management	Incident Manager Level 2 Analyst Service Desk Manager Level 1 Analyst	Default

Service Desk Management	Customer Service Manager Level 1 Analyst, Level 2 Analyst Service Desk Manager	Default
Service Desk Staff	Customer Service Representative Level 1 Analyst Level 2 Analyst	Default
Super Admin	AdministratorCustomer Service Manager Customer Service Representative Incident Manager Level 1 Analyst Level 2 Analyst Service Desk Administrator Service Desk Manager	Default

(Optional) Create and Configure Users

Create Roles

You can create roles and can assign roles to the users. The users can access the reports and can view the data in the reports based on the roles assigned to them.

Follow these steps:

1. Log in to JasperReports® Server 8.1.1, as *superuser*, using the following credentials:
   - User name: *superuser*
   - Password: *superuser*
2. Navigate to **Manage**, **Roles**, **casm_insights**.
3. Click **Add Role** in the **Roles** section.
4. Define the role in the following format: *ROLE_<role_name>.*
5. Click **Add Role to servicemanagement** to create and save the role.

Create Users

You can create users and can assign roles to the users.

Follow these steps:

1. Log in to JasperReports® Server 8.1.1, as *superuser*, using the following credentials:
   - User name: *superuser*
   - Password: *superuser*
2. Navigate to **Manage**, **Users**, **casm_insights**.
3. Click **Add User** in the **Users** section.
4. Define the user details for the *servicemanagement* organization.
5. Click **Add User** to **casm_insights** to create and save the user to the *servicemanagement* organization.
6. Associate a role to the user:
   - Click the required user from the list and click **Edit** in the **Properties** pane on the right.
   - From the list in **Available Roles**, move the required role to **Roles Assigned** and then click **Save**.
     The user can now access the reports data based on the assigned role.

(Optional) Configuring Insights with CABI JasperReports Server Load Balancer

If you have configured CABI JasperReports Server with high availability, then you must manually configure Insights with CABI JasperReports Server Load Balancer.

Follow these steps:

1. Verify that Insights is integrated with one of the high availability environment CABI JasperReports Servers.
2. Copy the following content from < *CA Business Intelligence Installed Directory>\apache-tomcat\webapps\jasperserver-pro\WEB-INF* folder:
   a. *applicationContext-externalAuth-template-mt.xml* file
   b. *actionModel-navigation.xml* file
   c. `Config` folder
3. On another high availability environment CABI JasperReports Server, navigate to < *CA Business Intelligence Installed Directory>\apache-tomcat\webapps\jasperserver-pro\WEB-INF* folder.
   a. Replace the existing content with the copied content.
   b. Restart the *CABI JasperReports Tomcat service*.
4. Login to CA SDM as an administrator. Navigate to the **Administration** tab, select **xFlow Interface** , and **General**.
5. Search for the configuration key, **insights.jasper.domain,** and click **Edit.**
   a. Replace the **Configuration Value** with the domain name of the load balancing server. For example, **<hostname:8080>**
   b. Click **Save**.
6. Search for the configuration key, **insights.jasper.hostname,** and click **Edit.**
   a. Replace the **Configuration Value** with the host name of the load balancing server. For example, **<hostname>**
   b. Click **Save**.
7. Restart the *xFlow Interface* service based on your configuration:
   a. **Advanced Availability**: Restart services on the application server.
   b. **Conventional**: Restart services on the primary and secondary servers.

Enable the KPI Option

The KPI option enables the KPI daemon to collect data when a ticket is opened, closed, or modified. The KPI option is enhanced to capture additional information during the ticket creation process. However, for the tickets that were created prior to the implementation of CA Service Management 17.3, this additional information is not captured.

Perform the following steps to enable the KPI options:

1. Login to CA SDM as an Administrator.
2. Navigate to Administration, Options Manager, KPI and select the **kpi_ticket_data_table** and **kpi_ticket_data_enable_workshift** options. Click **Edit** and **Install**.
3. Restart the CA SDM Services.

Insights Architecture Diagram

This article consists of Architecture Diagrams for Transactional and Aggregate Reports for CABI JasperReports Server:

Transactional Reports

The following deployment diagram displays Transactional Reports with offline CA SDM Reporting Components:

Figure 8: Figure: Insights - Transactional Reports 17.3

Aggregate Reports

The following deployment diagram shows various components for Aggregate Reports:

Figure 9: Figure: Insights - Aggregate Components

Transactional Reports

Create a Domain

Pre-requisites to create a new domain

1. Login to CABI as jasperadmin of casm_insights organization or as a superuser.
2. Navigate to **Manage**, **Users**.
3. Click jasperadmin/superuser and click **Edit** on the right panel
4. Navigate to the **Attributes** tab.
5. Click **Add new attribute**. Enter '**contactUUID**' in the Name column. Leave Value column as empty.
6. Click **Ok**. Click **Save**.

How to Create a New Domain?

This video demonstrates how to create a domain in CABI JasperReports. For more information, see the articles that follow the video.

> **TIP**
>
> To play the video in full screen, click the YouTube logo in the right of Settings at the bottom of the video.
>
> To change the video quality, navigate to **Settings**, and set **Quality** to **720p**.

Perform the following steps to create a new Domain:

1. Click Domains, Create from JasperReports Server Home Page.
2. Enter the **Name** and ID of the organization.
   Ensure that the ID is prefixed with letter '**z**' to avoid overwriting of the domain on importing latest releases.
3. For **superuser**:
   a. Select Save Location as: /organizations/casm_insights/SDM/Transactional/domains
   b. Select Data Source as: /organizations/casm_insights/SDM/Transactional/datasources/sdm_d
4. For **jasperadmin**:
   a. Select Save Location as: /organizations/casm_insights/SDM/Transactional/domains
   b. Select Data Source as: /organizations/casm_insights/SDM/Transactional/datasources/sdm_ds
5. Click **Create with Domain Designer**.
6. Select '**casd**' as Database Schema.
   Click **OK**.
7. Under **Tables** tab, pull the required magic objects from **Data Source** column to **Selected Tables** column.
   You can create derived tables from the **Derived Tables** tab.
   Ensure that you write the query in the following syntax:
   ```
   SELECT <columns> from <object_name> Example: select * from chg and NOT LIKE select * from casd_chg
   ```
8. Click **Run** Query.
   If the query is successful, you will see a list of columns. Enter a unique Query ID and select required columns from the list.
   Click **Save Table**.
9. Define joins to this derived table under the **Joins** tab.
   You can create calcuated fields under the **Calculated Fields** tab.
   Syntax to use attributes is as follows:
   ```
   <schema>_<object_name>.<alias_name>
   ```

For example, **casd_chg.assignee**

10. Under the **Joins** tab, create joins between tables based on the primary keys.
    CA Service Management supports **Left outer**, **Right outer** and **Inner joins** only.
    Ensure that all tables are joined under one Join Tree.
11. Under the **Display** tab, pull in the attributes from the **Resources** column to **Set and Items**.
12. (If you donot find the required alias, please create attribute alias in SDM and refresh the CABI browser.)
    Edit their **Properties** as required:
13. Click **OK**. Click **Submit**.
14. The Domain saved dialog appears.
    Now you can use this domain to create your ad-hoc views and reports.

Add Custom Attributes to Reporting Domains

This article is intended for CA Service Management Administrators and describes how to add attributes to a modified MDB schema for a reporting domain. It provides step by step instructions on how you can update the Out-of-the-Box (OOTB) domains provided with CA Service Management to include the additional attributes for reporting. Once SDM and Jaspersoft are integrated and you have imported the default reporting domains, you can start building your own custom reports. If you have modified the MDB schema and need to incorporate the new fields into your report building, you can update the out of the box domains to include them.

The use case used in the example showcases how an attribute is added to the OOTB I**ncident Management** Domain. The newly added attribute will be used to identify incidents that are incorrectly categorized.

CA SDM Web Screen painter was used to add an attribute (**zcatergory_init**) to CA MDB. The attribute that we are going to add is **zcategory_init** and it references the **pcat** object (SREL Table).

Before you Begin

Ensure to consider the following best practices before you start adding a custom attribute to your reporting domains:

1. It is recommended that you take a backup of the domains and save it under a different unique resource ID to perform your customizations.
2. If you are planning to modify the Out-of-the-Box domains, you are only eligible to add custom attributes which exist in tables that you have pulled into your reporting domains.
3. These custom attributes must be added to the empty set that is provided for custom attributes in **Display** Tab. To access the Display tab from JasperReports Server, click **Domains**, **View List**, Select a **Domain** (for example: Incident Management), Click **Edit**, access the **Display** tab.

> **NOTE**
>
> If you plan to modify the Out-of-the-Box domains, do not add new tables or derived tables into an existing domain schema or modify existing joins. Do not modify the existing Out-of-the-Box fields that are provided.

Create an Attribute Alias in CA SDM for the Additional Attributes

This video demonstrates how to add attributes to a modified MDB schema for a reporting domain in CA Service Management 17.3. For more information, see the articles that follow the video. After integrating CA SDM and Insights (CABI JasperReports), and after importing the default reporting domains, you can start building your own custom reports.To use this attribute for reporting, a corresponding attribute alias has to be created, if it is not already present in your reporting domain.

> **TIP**
>
> To play the video in full screen, click the YouTube logo in the right of Settings at the bottom of the video.
>
> To change the video quality, navigate to **Settings**, and set **Quality** to **720p**.

Perform the following steps to create an attribute alias:

1. Login to CA SDM as an Administrator.
2. Click **Administration, Service Desk**, **Application Data**, **Codes**, **Attribute Aliases**. Click **Create New**.

   NOTE

   Attribute Alias codes provide a flattened view of the database and allows you to include columns from related tables. Attribute alias name is same as the majic join name, but underscores are used in place of the dots when defining the alias name.
3. Complete the following fields on the Create New Attribute Alias page:
   a. **Object Name**: Select the Object Name as **IN** for Incident on the Create New Attribute Alias page.
   b. **Alias Name**: Use underscores to create alias name as **zcategory_init_sym**
   c. **Alias Value**: Create Alias Value as **zcategory_init.sym**
   d. **Description**: (Optional) Provide an appropriate description for the new attribute alias.
4. Click **Save**.

   NOTE

   After creating a new attribute alias for the custom attribute, you need to add this to a domain and use this for reporting purposes. Also, in order to Create an Adhoc View for the Custom Attribute using the custom attribute, the attribute alias must be included in the reporting domain first.

Add Attributes to a Non-Tenanted Environment

The Out-of-the-Box domains in JasperReports server are built for tenanted environments. Hence, to add attributes to a domain for a non-tenanted environment, you are required to perform few additional steps as shown below. These steps must be performed before adding any custom attributes to a domain and required only the first time the domain is being modified.

Perform the following steps:

1. Login to JasperReports Server as **Superuser**.
2. Navigate to **Domains**. Click **View list**.
3. Select the domain (in this case, for example, select the **Request Management** Domain) and right-click to **Edit** the domain.
4. Select the '**Edit with Domain Designer**' option under '**Edit Domain**' section.
5. Select the **Display** Tab, **Sets and Items** selection box, and expand the domain (**Request Management** Domain).
6. Expand the selected domain and scroll down to select tenant related fields. In this case, select **Tenant Name** for Request Management Domain and delete it.
7. Select the **Tables** Tab.
   A popup message is displayed.
8. Click **Ok** to delete the tenanted resources and related information from the tenanted schema.
9. Click **Ok** on the Domain Designer. Click **Submit** to update the domain.
   Your domain is now ready to be customized.

Add Custom Attributes to Domains

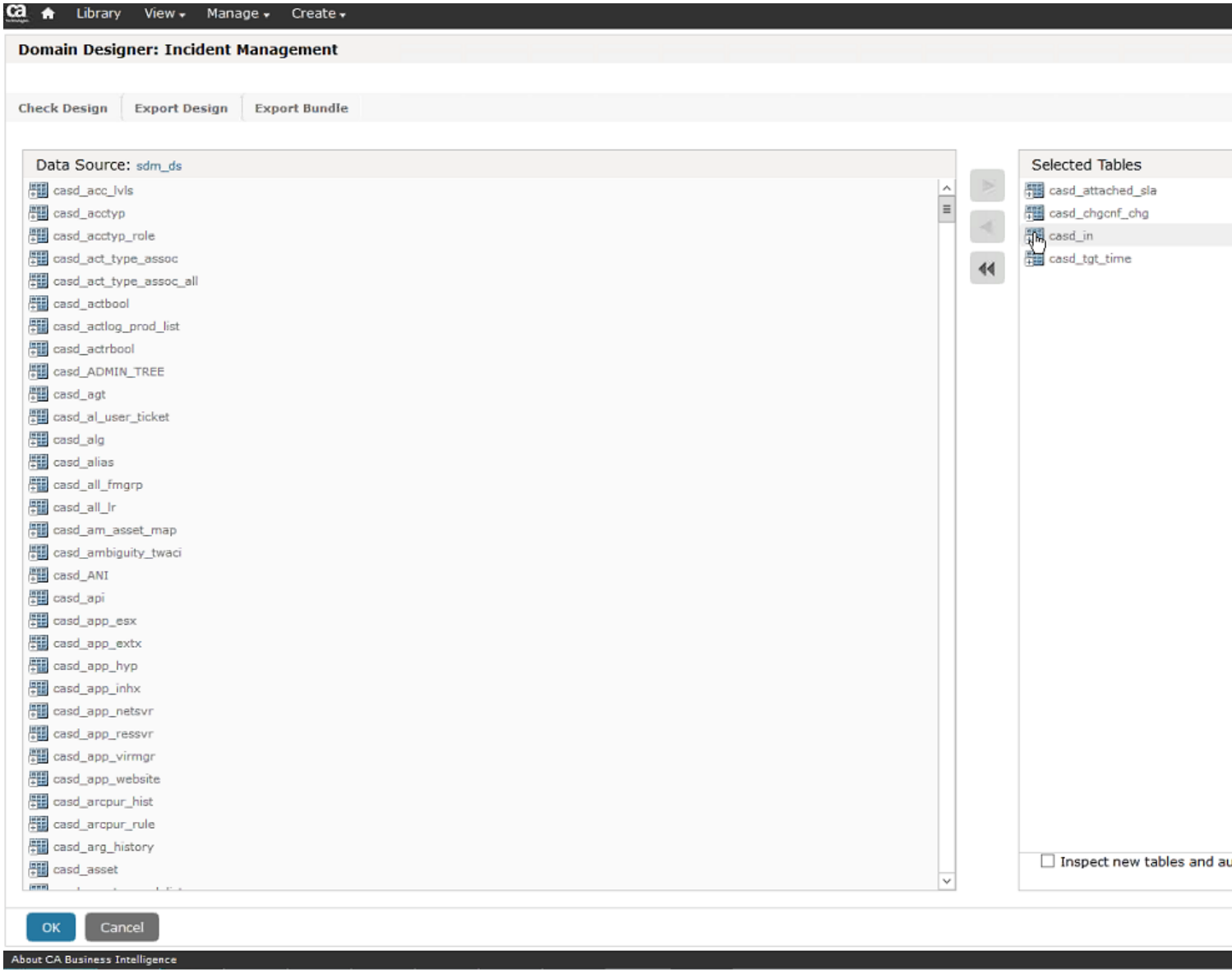
Add the custom attributes (for which attribute aliases were created) to the Out-of-the-Box domain by performing the following steps:

1. Login to CA Business Intelligence JasperReports Server as a **Superuser**.
2. Navigate to **Domains**, click **View List** to see the available domains.
3. Right-click the domain and select **Edit**. In this case, select the **Incident Management** domain.
4. Select the **Edit with Domain Designer** option under the **Edit Domain** section.
   By Default, the **Display Tab** is displayed.

5. Expand the Domain (In this case, **Incident Management**) from the **Sets and Items** section to view all the fields in a domain.
6. Scroll down to verify if the new field (**zcategory_init_symbo**l added in step Create an Attribute Alias) is part of the list or not.

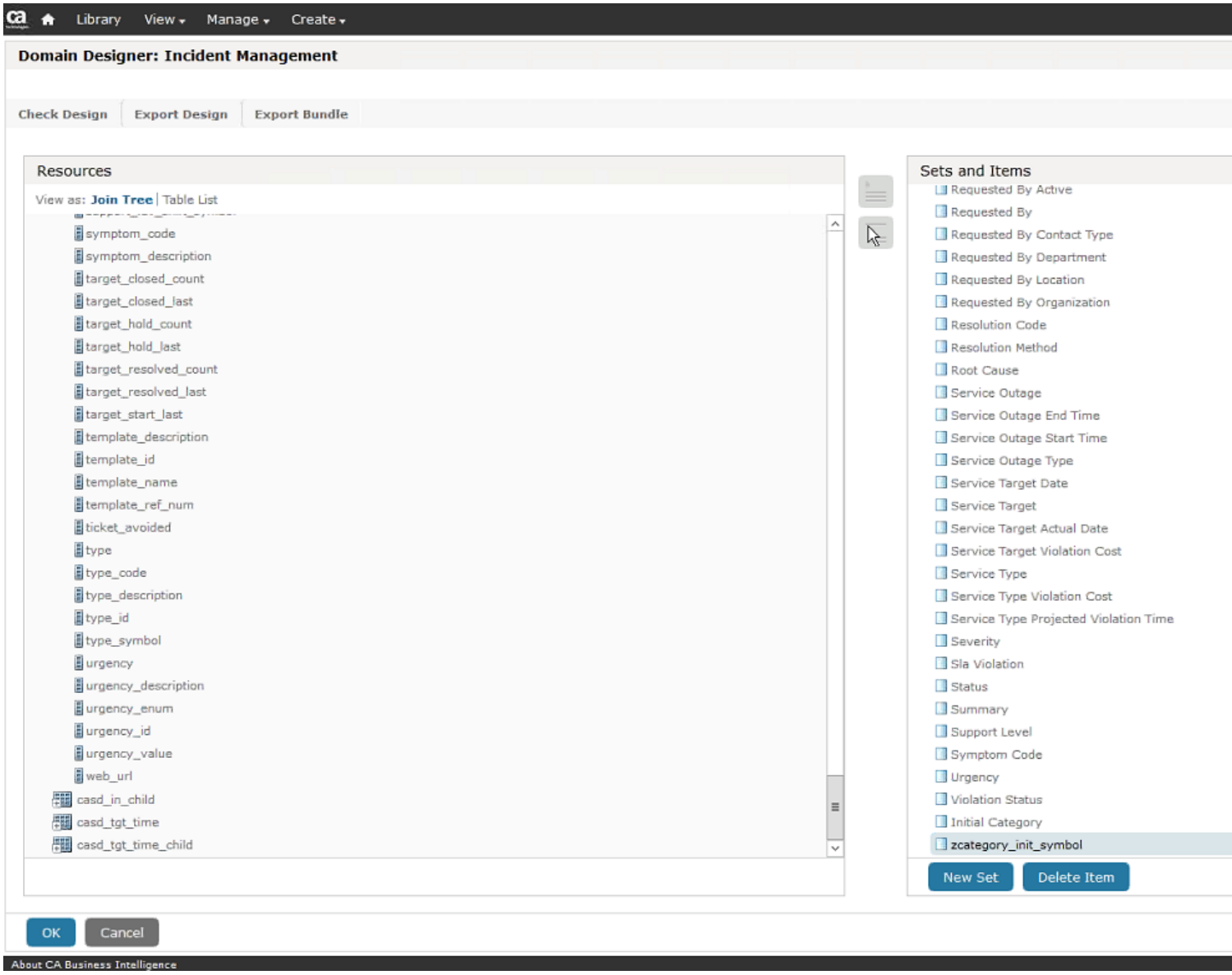
   NOTE

   In this case, the field will be not be listed and must be added manually.
7. Click the Tables Tab. Click **casd_in** table to view where the incident data is located.
   The right-hand side section, that is the, **Selected Tables** contains fields that the schema knows about and this is used to create ad hoc views.



8. Expand **casd_in** and scroll down the list.
9. The attribute alias **zcategory_init_symbol** that was created in step Create an Attribute Alias is visible on the list. This confirms that the newly added attribute alias is present in the schema.
10. Pull this attribute into the domain. Navigate back to the **Display** tab and from the **Resources** section, expand the **casd_in** node under the **Resources** section.

11. Scroll down and select the **zcategory_init_symbol** and click **Add to Selected Set** to add it to the existing domain.
12. Scroll down and verify if **zcategory_init_symbol** is present.
13. Now, modify, if required, the label name of the newly added attribute. Click **Edit** on the extreme right.



14. After editing is complete, click **Save**.
15. Click **Ok** on the Domain Designer.
16. Click **Submit** to update the domain.

(Optional) Create an Adhoc View for the Custom Attribute

Additionally, you can also create a new adhoc view for the custom attribute that is added to the domain.

Perform the following steps:

1. Select **Create** from the menu, select **Ad Hoc Views** to create an Ad Hoc View.
2. Select the **Incident Management** domain from **Select Data**. Click the **Choose Data…** option.
3. From the Source column, expand the **Incident Management** domain and select the fields that you want to generate a report for.

Click **OK**.

4. On the new Ad Hoc View page, click the **Crosstab** drop-down list and select the **Table Type** to Table.
5. Expand the Incident Management fields, drag and drop or double-click the fields that you would like to add to the table Columns and Groups.
   You can group by Category and can even add a Filter.
   Additional Attributes can be added in the same way.

Editing Reporting Domains for Untenanted CA SDM

While customizing JasperReports Server domains, if you have an untenanted CA SDM environment, perform the following steps:

NOTE

These steps must be performed for all domains in your reporting environment before saving a domain:

1. Login to JasperReports Server as an Administrator.
2. Navigate to Domains. Click **View list**.
3. Select the domain of your choice and right-click to **Edit** the domain.
4. On the Edit Domain page, navigate to **Domain Design, Edit with Domain Designer...** option.
5. Select the **Display** Tab, **Sets and Items** selection box, and expand the domain of your choice, for example: **Request Management** domain.
6. Select any tenant related fields, for example Tenant Name and double-click to remove it.
7. Select the **Tables** Tab. A popup message is displayed.
   Click **Ok** to delete the tenanted resources and related information from the tenanted schema.
8. Click **Ok** on the Domain Designer. Click **Submit** to update a domain.
   Your domain is now ready to be customized.

Configure JasperReports Whitelist for Transactional Reporting

For security purposes, perform the following steps to configure JasperReports Hostname whitelist for Transactional Domain reports:

1. Login to Classic CA SDM Interface as an Administrator.
2. Configure the hostname whitelist in CA SDM Options Manager for Transactional Domain Reporting. Navigate to **Administration**, **Options Manager**, **General**, **dal_whitelist**. Click **Edit**.
3. In the Option Value box, enter the host names by comma separated values without quotes. For example:

```
hostname_01<.domain.com>, hostname:02<.domain.com>
```

4. Click **Install** to configure multiple whitelists for transactional domain reports.
5. Restart CA SDM Services.

Configure Multiple Domsrvr for Transactional Domain Reports

Perform the following steps to configure multiple domsrvrs for transactional domain reports for scalability purposes:

1. Login to Classic CA SDM Interface as an Administrator.
2. If you want to add additional object managers or domsrvrs on a CA SDM server for scalability or load balancing, see Add Object Managers.
3. Now, configure the domsrvrs in CA SDM Options Manager for Transactional Domain Reporting. Navigate to **Administration**, **Options Manager**, **General**, **dal_domsrvr**. Click **Edit**.

4. In the Option Value box, enter the domsrvr names by comma separated values without quotes:
   For example, **domsrvr**, **domsrvr:01**, **domsrvr:02**
5. Click **Install** to configure multiple domsrvr for transactional domain reports.

   NOTE

   Ensure that you have provide the **Communication Server Name** for domsrvrs.
6. Restart CA SDM Services.

Aggregate Reports

To get the Aggregate Reports up and running, perform the following as an Administrator:

CA SDM Data Warehouse

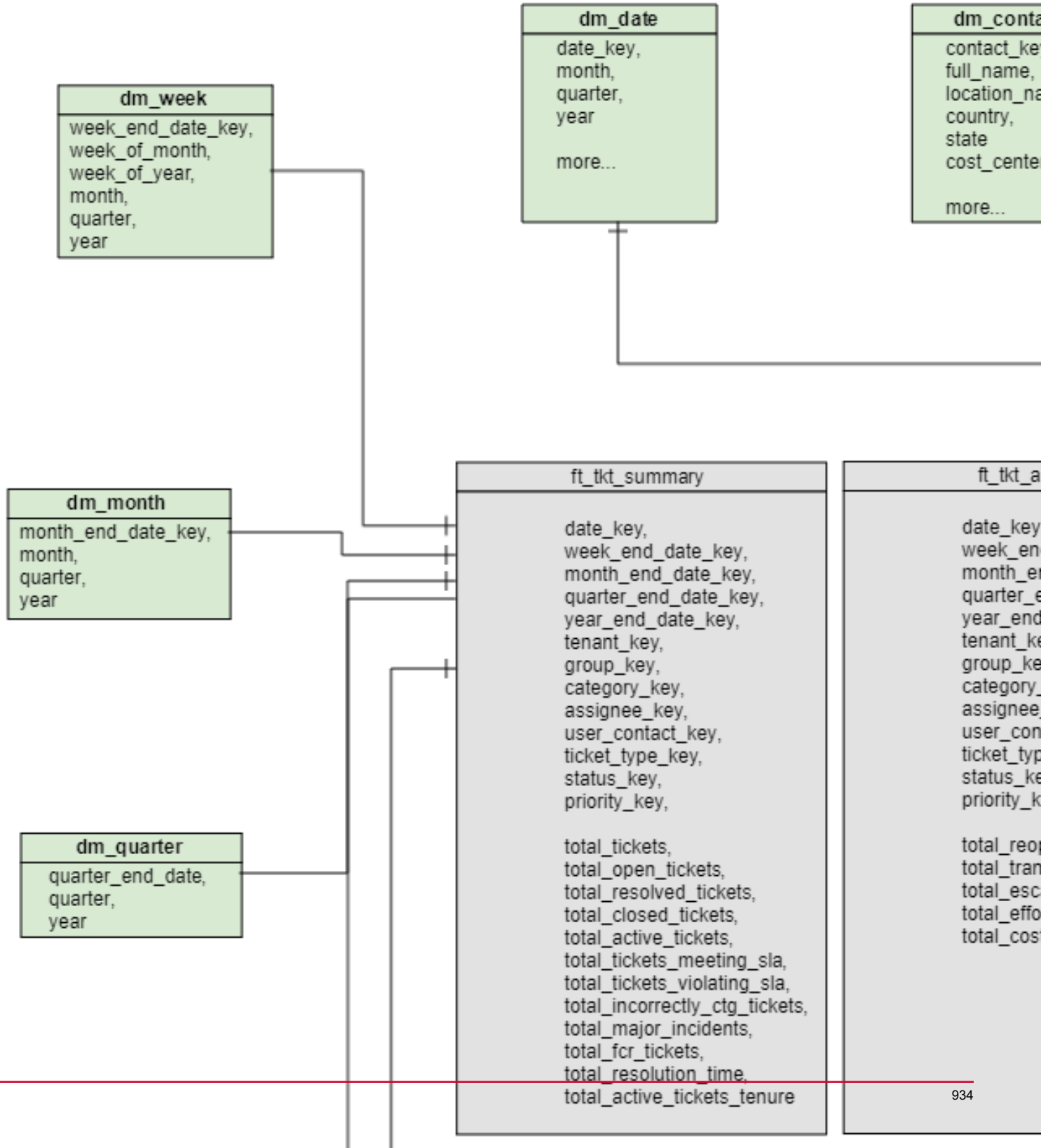
CA SDM data warehouse is the primary data source for Insights and has the following characteristics:

- Optimized for reporting purposes.
- Includes the incident and request management data.
- Can include custom attributes.

Aggregated (Datamart) Schema

The Datamart schema and entity relationship.

Datamart Entity Relationship Diagram (ERD)



Add Custom Attributes to the Dimensions

In the Ad Hoc Designer View, you see dimensions under the fields tab on the left side panel. Expanding each dimension lets you see the exposed attributes associated with that dimension. If you need to add additional attributes, first edit the domain and see the list of attributes associated with that dimension. If the additional attribute that you need is not available in the list, add a new attribute.

Perform the following procedures:

- Add Custom Attributes to the Aggregate (Datamart) Schema
- Expose the Attributes in the Domain and Topic

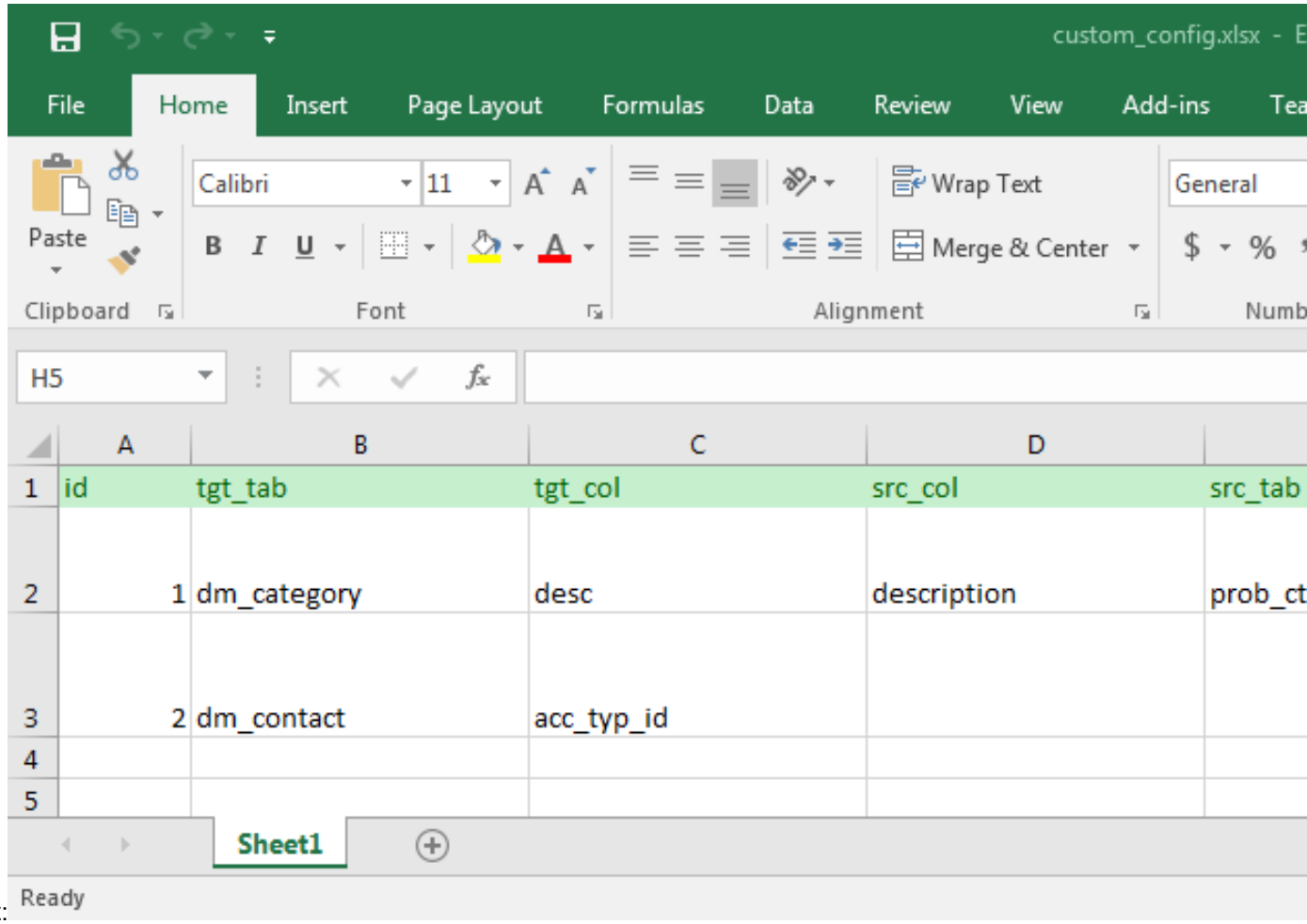
Add Custom Attributes to the Aggregate (Datamart) Schema

1. Navigate to where you have installed Aggregate Reports and open the **custom_config.xlsx** file under the config directory.
2. The file includes the following column. Fill the values under each column as explained below:
   a. **tgt_tab**
   Specifies the target dimension table in which you want to add the new column/attribute.
   b. **tgt_col**
   Specifies the new column that you want to add to the target dimension table which corresponds to the custom attribute.
   c. **src_tab**
   The **base table** in the **MDB** which is the source for the **dimension table** in the **Datamart**. If the new attribute/ column that you want to add is present in the base table, then specify the base table name here. If not, skip this and jump to filling the sql_query field. The below table shows the mapping between dimension tables in the datamart and their corresponding base tables in the MDB:

Dimension	tgt_tab	base table
Analyst	dm_contact	ca_contact
Business User	dm_contact	ca_contact
Group	dm_contact	ca_contact
Category	dm_category	prob_ctg
Priority	dm_priority	pri
Status	dm_status	cr_stat
Tenant	dm_tenant	ca_tenant
Ticket Type	dm_ticket_type	crt

   d. **src_col**
   The column name in the base table of the attribute that you are trying to add. If the attribute that you are trying to add is not present in the base table, then skip filling this field and proceed on to fill the **sql_query** field.
   e. **sql_query**
   SQL query is required to add a column to the target table when the respective column is present in a table other than the base table. This query extracts the required column from the other table along with the **src_lkp** field, which is the primary key of the base table. So, the output of the query must be just two columns, which is, **(src_lkp, src_ col)**.

Example: If you want to add **description** and **c_acctyp_id** attributes/columns to **category** and **analyst** dimensions respectively, this is how you fill the excel sheet:



Adding **c_acctyp_id** to **dm_contact** requires specifying the SQL query. In this case, the base table for **dm_contact** is **ca_contact**. The required column **c_acctyp_id** is not present in **ca_contact** but is present in another table named **usp_contact**. The SQL query joins **ca_contact** with **usp_contact** and extracts the **c_acctyp_id** column.

In this query **ca_contact.contact_uuid** is the **src_lkp** field and **usp_contact.c_acctyp_id** is the **src_col** field.

select ca_contact.contact_uuid,usp_contact.c_acctyp_id from ca_contact left join usp_contact on ca_contact.contact_uuid=usp_contact.contact_uuid

NOTE

You must choose the SQL query option to add attributes to the following two tables even if the attributes are present in the base table. This is because of the data type mismatch between the primary keys of the datamart tables and their corresponding MDB tables.

- – dm_contact
  Spcify the SQL query in the below format:
  select convert(nvarchar(34),contact_uuid,2) as contact_uuid, <src_col> from ca_contact <joins and other conditions if required>
  – dm_tenant
  Spcify the SQL query in the below format:

select convert(nvarchar(34),id,2) as tenant_uuid,<*src_col*> from ca_tenant <*joins and other conditions if required*>

3. Run the ETL job for the changes to take effect in the datamart schema.
   You have successfully added the attributes to the datamart schema.

Expose the Attributes in the Domain and Topic

1. Login into **JasperReports Server** as **Administrator**.
2. Click **View List** under **Domains**.
   Repository page opens with the list of available Domains.
3. Click the **Incident Request Management Domain** under the **Name** column.
   The Edit Domain page opens.
4. Click the **Edit with Domain Designer** option under Domain Design.
   The **Domain Designer** page opens. The **Display** tab shows Resources, Sets and Items, and Properties panels.
5. Expand the JoinTree_1 node and navigate to the dimension to which you have added the new attribute.
6. Select the attribute from the **Resources** panel and select the group from the **Sets and Items** panel under which you want to show the attribute on UI.
7. Click the **Add to Selected Set** button.
   The attribute is added to the selected group under the Sets and Items panel
8. Select the newly added attribute under the Sets and Items panel, and then click **Edit** in the **Properties** panel.
9. Go to **Field or Measure** drop-down list and select **Dimension**.
10. Click **Save** and click **OK.** Click **Submit** on the Edit Domain page.
11. Go to the home page and navigate to **View**, **Repository**.
12. From the left hand panel, expand the **casm_insights**, **capability**, and select **topics**.
13. Select **Incident Request Management Topic** and click **Edit**.
14. Select the attribute under the **Source** panel, select the category under the **Selected Fields** panel, and click the **Add** button.
15. Click **Save**.
    You have successfully added the custom attributes to the existing dimension.

Purging Data from the FACT Tables in Datamart

FACT tables are the tables which contain the aggregated data. The required data for reporting is available in the FACT tables within the datamart. The data is available from the date set as the data aggregation date. Any data that is prior to the data integration date will be automatically purged. Data purging is based on the data aggregation date parameter in the config.properties file. For example, if data was available from 2017-01-01 initially and then the data aggregation date was changed to 2017-06-01, all of the data prior to the date 2017-06-01 will be deleted from the FACT tables.

For more information, see Configure the Data Aggregation Date.

NOTE
For removing the vulnerability of ***Log4j1 Removal and adding the log4j2 jars and related dependencies in Datamart Installed location,*** apply the patch with APAR No : 99112251

Run the Extract Transfer and Load (ETL) Job

Run the ETL job to load the initial data to Aggregate (Datamart) Reports.

Follow these steps:

1. Navigate to `<Datamart Installed Directory>\bin` folder.
2. Run the `std-load` batch file, to load the initial data to Datamart.
   The time taken to complete the ETL job is based on the data size and the server configuration.

NOTE

We recommend to run the ETL job once every day during the non-peak hours.

3. (Optional) Once the ETL job is complete, if the SQL server (MSSQLSERVER) service consumes high memory, then restart the SQL Server service.

The following is an observation when you run an ETL job for different volumes of data on a server with the below configuration:

Test Results Observation for the ETL Job

- **Server:** Physical
- **Operating System:** Windows Server 2016 (64bit)
- **RAM:** 32 GB
- **Processor:** 2.10 GHz (2 processors)
- **Disk Space:** Total available - 1 TB; MDB Consumption - 40 GB
- **Database Server:** MS SQL 2014
- **MDB and Datamart Installation:** On the same server
- **MS SQL Memory Consumption:** Less than 50%

Total Number of Records in MDB	Data Aggregation Period	Initial Load Results	Delta Load Results
Number of Requests = 789,593 Number of Activities = 11,167,780	3 months	Number of Requests Loaded = 26,527 Number of Activities Loaded = 360,063 Time Taken = 20 minutes Disk space consumed = 3 GB	Number of Requests Loaded = 4,810 Number of Activities Loaded = 57,984 Time Taken = 2 minutes
	6 months	Number of Requests Loaded = 78,484 Number of Activities Loaded = 830,574 Time Taken = 3 hours 20 minutes Disk space consumed = 7 GB	Number of Requests Loaded = 4,810 Number of Activities Loaded = 57,984 Time Taken = 30 minutes
	12 months	Number of Requests Loaded = 174,463 Number of Activities Loaded = 1,439,205 Time Taken = 7 hours 40 minutes Disk space consumed = 58 GB	Number of Requests Loaded = 4,810 Number of Activities Loaded = 57,984 Time Taken = 1 hour 2 minutes
	24 months	Number of Requests Loaded = 385,622 Number of Activities Loaded = 5,833,017 Time Taken = 26 hours 19 minutes Disk space consumed = 196 GB	N/A

(Optional) Modify Multi Tenancy

To modify the multi tenancy settings in datamart, and for the changes to reflect instantly in Insights, navigate to the `<Datamart Installed Directory>\bin` folder using the command line interface, and run the `role-tenant-map.bat` file**.**

Schedule the Extract Transfer and Load (ETL) Job

You can use a task scheduling tool to schedule the ETL job at regular intervals or when a certain event occurs. It is recommended to backup the Datamart database before you run an ETL job.

For example, follow these steps to schedule an ETL job using Windows Scheduler:

1. Enter **sched** in the Start search box to open Task Scheduler.
2. In the Actions pane, click **Create Basic Task**.
3. Enter a name and description for the task.
4. Click **Next** to choose a trigger for the task.
5. Choose the task (ETL job) frequency as **Daily**.
6. From the **Start** drop-down, select a date and time when you want the job to begin.
7. Click **OK**.
8. Click **Start a Program** from the **Action** drop-down.
9. Browse the `std-load.bat` file from the `< Datamart Installed Directory>\bin` folder.
10. Click **OK,** and **Finish**.

NOTE

For more information about how to use the Transactional and Aggregate reports, see Using Insights.

For information on how to troubleshoot issues with Insights, see Troubleshooting Insights (CABI JasperReports).

Configure CA Service Desk Manager

Navigate to the required topics for configuring CA SDM.

This section contains the following articles:

- Configuring CA SDM Servers
- How to Configure Surveys
- How to Configure Notifications
- Configure CA SDM and the Components
- Configure the CA SDM Environment
- Managing Servers
- Managing Your Database
- Setting Up Multi-Tenancy
- Handling the Open Redirect Vulnerability Issue in \"LINK WITH BOPSID URL\" Parameter
- Setting Up Security
- Establishing Support Structure
- Auto Assignment
- Manage Roles
- Configuring User Accounts
- Configuration File Modification
- Creating the Business Structure
- CA SDM User Authentication
- Encrypt Session IDs to Address Vulnerability Issues
- Retry Mechanism for CA SDM and CA Process Automation Workflow Options
- Apache HTTP Load Balancer for CA SDM
- How to Configure the F5 Load Balancer for CA Service Desk Manager
- How to Configure the Mailbox to Handle Inbound Emails
- How to Set Up the Data Partition
- How to Perform Proactive Monitoring for CA SDM
- How to Configure Multiple CA SDM Web Engines on Application Servers?

WARNING

Any “modifications” or “adaptions” or “configurations” that are done administratively through the interface (web browser, command-line, Web Screen Painter) are “supported”, meaning CA Support can assist with basic suggestions and trouble-shooting. CA Support does not do the changes for the customer, they are the customer’s responsibility. For example, adding a field to a table and putting the field on a form through Web Screen Painter is a fully supported “modification”. Another example, installing or uninstalling a feature through the Options Manager administration is a fully supported “configuration”.

Anything to do with SPEL code, Java scripting (or any language scripting), or a customer-specific change to the underlying base code-line (done by CA Services or a Partner), is not supported by CA Support. The customer can do these things, but the customer is responsible for the support, maintenance, and trouble-shooting when things go wrong. If such “customizations” affect expected out-of-the-box behavior, CA Support will ask the customer to remove the customizations and see if the behavior persists.

WARNING

Any general configuration that you define for CA Service Desk Manager must be in the valid format to avoid erroneous values for the parameters.

Configuring CA SDM Servers

As a system administrator, configure CA SDM servers. Review the following:

NOTE

You must configure all the CA SDM servers in your environment before you start using the CA SDM application. If the servers are not configured, you will not be able to access the CA SDM application.

- Convert CA SDM Conventional to Advanced Availability Configuration.
- Convert from CA SDM Advanced Availability to Conventional Configuration.

Changing the Server Configuration from Conventional to Advanced Availability

Before proceeding with changing the configuration, ensure that you have upgrade to CA Service Desk Manager 17.4 from earlier an version of CA Service Management or installed CA Service Desk Manager 17.4 on all the servers in your environment.

Follow these procedures:

1. Plan for the Advanced Availability Configuration.
   For more information see, How to Plan for CA SDM Advanced Availability Installation.
2. Stop all services on the secondary servers.
3. Inactivate All the Secondary Servers.
4. Configure the Background Server.
5. Change Secondary Server Type.
6. Configure the Standby and Application Servers.
7. Verify Server Details.

Inactivate All the Secondary Servers

Before you configure the servers for the advanced availability configuration, ensure that you inactivate all the secondary servers. Perform the following steps:

1. Log in to the web interface of the primary server.
2. Select **System**, **Servers** on the **Administration** tab.
   The **Server List** page opens.
3. Click the host name of the secondary server.
   The **Server Detail** page opens.
4. Click **Edit**.
5. Change the **Record** status to **Inactive**.

   WARNING

   If you have inactivated any server, it is recommended not to start CA SDM services on that server. This action may result in unexpected behaviour.
6. Click **Save**.
   The secondary server is set to inactive.
7. Perform steps 3-6 for all the other secondary servers.
   All secondary servers are set to inactive.

Configure the Background Server

Configure the background server before configuring the application or standby servers. Perform the following steps:

1. Log in to the primary server.
2. Use the pdm_configure command to start the configuration.
   The **Select Server Configuration** screen opens.
3. Select **Advanced Availability** as the configuration type and follow the prompts to complete the configuration. For more information, see the Server Configuration Utility.

The background server is configured.

Change Secondary Server Type

After you configure the background server, change the server type for all the secondary servers to application and standby servers and activate them. Perform the following steps:

1. Make sure that the background server is running.
2. Log in to the web interface of the background server.
3. Select **System**, **Servers** on the **Administration** tab.
   The **Server List** page opens.
4. Set the Record Status for all secondary servers as Active.
5. Change the server type of all the secondary servers to application or standby servers, as required.
   The server type of all the secondary servers is changed.

Configure the Standby and Application Servers

You configure each standby and application server after configuring the background server. This configuration is required to establish communication with the background server. Perform the following steps:

> **WARNING**
> (Recommended) Ensure that both background server and all other standby servers have similar configuration. This process ensures that during a failover when a standby server becomes the new background server, it can function exactly like the old background server.

1. Log in to a secondary server.
2. Use the pdm_configure command to start the configuration.
   The **Select Server Configuration** screen opens.
3. Select **Advanced Availability** as the configuration type and follow the prompts to complete the configuration. For more information, see the Server Configuration Utility.
4. Repeat steps 1-3 on each of the servers that you want to configure as the standby and application server.
   You have configured the standby and application servers.
5. If you want to use a load balancer to track the requests to and from the different application servers, configure the load balancer.

   > **NOTE**
   > To use a load balancer URL or an application server URL for the links in the CA SDM notifications, complete the following steps:
   >
   > 1. Log in to the background server.
   > 2. Select **Options Manager**, **Notifications** on the **Administration** tab.
   > 3. Change the value of the web_cgi_url option to point to:
   >   – The load balancer if you have more than one application servers.
   >   – The application server, if you have only one application server.

6. Verify the Server Details.

Verify Server Details

After you have configured all the servers, ensure that each one is properly configured and available. Perform the following steps:

1. Make sure that all background, standby, and application servers are running.
2. Log in to the background server Web UI.
3. Select **System**, **Servers** on the **Administration** tab.
   The **Server List** page opens.
4. Verify that the **Configured** field displays Yes for the standby and application server records.

5. If **Record Status** is set to inactive, edit the server and set to Active. If you change the record status of any server, reconfigure it.
   You have successfully configured servers for the advanced availability configuration.

   NOTE
   You can click any of the Local Host entries on the Server Details page to view the details of the server.

Changing the Server Configuration from Advanced Availability to Conventional Configuration

You can convert the CA SDM conventional configuration to advanced availability. Please see the following:

Convert from Advanced Availability to Conventional

Follow these steps:

1. Verify the Prerequisites
2. Stop all services on the application and standby servers.
3. Inactivate all Application and Standby Servers.
4. Configure the Primary Server.
5. Change the Application and Standby Server Types.
6. Configure the Secondary Servers.
7. Verify Server Details

Verify the Prerequisites

Complete planning for the conventional configuration. For more information, see Plan the CA SDM Installation.

NOTE

Note: You can configure only the background server as the primary server. You can configure only the standby and application servers as the secondary servers.

Inactivate All Application and Standby Servers

Before you configure the servers for the conventional configuration, ensure that you inactivate all the application and standby servers. Perform the following steps:

1. Log in to the web interface of the background server.
2. Select **System**, **Servers** on the **Administration** tab.
   The **Server List** page opens.
3. Click the host name of the application server.
   The **Server Detail** page opens.
4. Click **Edit**.
5. Change the **Record** status to **Inactive**.

   WARNING
   If you have inactivated any server, it is recommended not to start CA SDM services on that server. This action may result in unexpected behaviour.
6. Click **Save**.
   The application server is set to inactive.
7. Perform steps 3-6 for all the other application and standby servers.
   All application and standby servers are set to inactive.

Configure the Primary Server

Primary server must be configured before you configure any secondary servers. Perform the following steps:

1. Log in to the background server web interface and inactivate it.
2. Use the pdm_configure command and start the configuration.
   The **Select Server Configuration** screen opens.
3. Select **Conventional** as the configuration type and follow the prompts to complete the configuration. For more information about the configuration, see the Server Configuration Utility.
   The primary server is now configured.

Change the Application and Standby Server Types

After you configure the primary server, change the server type for all the application and standby servers to secondary servers and then activate them. Perform the following steps:

1. Make sure that the primary server is running.
2. Log in to the primary server web interface.
3. Select **System**, **Servers** on the **Administration** tab.
   The **Server List** page opens.
4. Set the **Record Status** for all application and standby servers as **Active**.
5. Change the server type of all the application and standby servers to secondary servers.
   The server type for all application and standby servers are changed.

Configure the Secondary Servers

You configure each secondary server after configuring the primary server. This configuration is required to establish communication with the primary server. Perform the following steps:

1. Log in in to the application or standby server that you want to configure as the new secondary server.
2. Use the pdm_configure command to start the configuration.
   The **Select Server Configuration** screen opens.
3. Select **Conventional** as the configuration type and follow the prompts to complete the configuration.
   The secondary server is now configured.
4. Repeat steps 1-3 on each of servers that you want to configure as the secondary server.

Verify Server Details

After you have configured all the servers, ensure that each one is properly configured and available. Perform the following steps:

1. Make sure that primary and all secondary servers are running.
2. Log in to the primary server Web UI.
3. Select **System**, **Servers** on the **Administration** tab.
   The **Server List** page opens.
4. If **Record Status** is set to inactive, edit the server and set to **Active**. If you change the record status, ensure that you reconfigure that server again.
   You have successfully configured the servers.

How to Configure Surveys

This article contains the following topics:

Customer surveys let CA SDM administrators systematically collect and analyze customer feedback about service desk performance. You can modify surveys to suit the needs of your site.

Configure Your System for Surveys

Before you can use surveys, you need to configure your system properly, which involves two steps:

1. Install and configure the CA SDM web interface. When a user accesses the URL for a survey, the web interface formats the survey and populates the survey information.
2. Using the Options Manager, configure and install the web_cgi_url option to specify the location of the CA SDM web engine. For more information, see Install/Uninstall Options Manager Options.

Prepare a Survey

You prepare surveys using the Customer Survey List, which is a typical list window. For example, you can use this window to view all surveys or a subset based on search criteria that you enter; you can create new surveys; you can view details on a particular survey; and you can report on the surveys currently listed.

Each survey has the following features that you can define:

- A name that you can use for searching and reporting purposes
- An introduction that you can use to explain the purpose of the survey to customers
- An ordered list of questions for the customer to answer, each of which includes a set of possible answers
- An optional area where the user can enter free-form comments
- A completion message to display after the user submits the survey

Define Survey Notifications

You can use the Survey tab on the Update Activity Notification page to define a survey notification for an activity notification. When the selected activity notification is triggered, the contact who initiated the activity receives the survey notification. An activity log is generated both when a survey notification is sent and when one is received back from a customer.

Follow these steps:

1. On the Administration tab, browse to Notifications, Activity Notifications.
   The Activity Notification List appears.
2. Select the desired activity notification.
   The detail page displays.
3. Click the Edit button.
   The Update Activity Notification page displays.
4. Edit the fields as appropriate.
5. Select the appropriate object type from the drop-down list.
6. Click the Survey tab.
   This tab contains the following fields:
   - **Send Survey**
     This check box allows you to activate or deactivate the survey. If selected, the survey is sent to the contact when the selected activity notification is triggered.
   - **Default Survey**
     Specify a default survey using the search icon or specify your own in the text box.
   - **Notification Method**
     Choose *one* of the following notification methods:
     - Email
     - Notification
     - Pager_Email
   - **Survey Message Title**

Enter the title for the survey.

– **Survey Message Body**
Enter a message for the contact. When a contact receives notification of a survey, the message body automatically includes a URL that they can access from their web browser to find and fill out the survey form.

7. Save the activity notification.
When the selected activity notification is triggered, the contact who initiated the activity receives the survey notification.

Survey Reporting

Within CA SDM you can report on surveys from within the administration tab of the web client in all of the usual ways. For example, from the Customer Survey List window, you can choose Reports from the File menu, and then choose a Summary or Detail report. You can also choose Print Form from the various detail windows to print the form data for your surveys, questions, and answers.

You can also fashion your own reports based on the survey data maintained in the CA SDM database.

Create a Managed Survey

Contents

Create a Managed Survey

The Managed Survey lets the CA SDM administrator select a desired survey sample population and match it to a specific survey. The administrator can then distribute targeted requests for respondents to take the survey at a specific time. This gives the administrator the flexibility of creating open survey periods, while maintaining the ability to have activity-based and category-based surveys related to Requests, Change Orders, and Issues.

The purpose of Managed Surveys is to provide a mechanism for managing surveys. This function can be useful when for survey forms that need to be monitored from time to time (for example, surveys only used during a short period every year or surveys that have been offline too long).

> **WARNING**
> If you want to send the survey to a large number of contacts, set the default value of NX_SURVEY_ILIMIT in NX.env to a higher limit, such as 1073741824.

You can create a managed survey from the Service Desk node on the Administration tab.

Follow these steps:

1. Click the Administration tab.
The Administration page appears.
2. Click Service Desk, Surveys.
Click Managed Surveys, Managed Survey List.
The Managed Survey List appears.
3. Click Create New.
The Create New Managed Survey page appears.
4. Complete the fields as appropriate.
See Managed Survey Fields for field descriptions.
5. Use the controls available on the tabs at the bottom of this page to configure the managed survey as appropriate.
See Managed Survey Tabs for more information.
6. Click Save.
The managed survey definition is saved and the Managed Survey Detail page appears.

The following buttons are available:

- **Event History**

Opens the Event History window, which lists the status, time loaded, fire time, and condition for each event associated with the managed survey.

- **Attach Event**
  Opens the Attach Managed Survey Events window, which allows you to attach events with an object type of Managed Survey.

Managed Survey Fields

The following fields require explanation:

- **Name**
  Identifies the title of the managed survey.
- **Managed Survey Status**
  Select *one* of the following options:
- (Default) Survey Period Open
  - Configuration in Progress
  - Survey Cancelled
  - Survey on Hold
  - Survey Period Closed

  You can create a managed survey status from the Service Desk node on the Administration tab.
- **Active**
  Specifies whether the managed survey is active or inactive.
- **Assignee**
  Specifies the contact that is assigned as the survey owner.
- **Group**
  Specifies the group that is associated with the survey.
- **Survey**
  Allows you to search for existing surveys.

Managed Survey Tabs

The following tabs are available on the Create Managed Survey, Managed Survey Detail and Update Managed Survey pages:

- **Activities**
  Allows you to view the Managed Survey Activity Log List. To display only the log entries of interest, you can filter the list by specifying:
  - Earliest Log Date
  - Latest Log Date
  - Activity Type
  - Created By
- **Objects**
  Allows you to specify assignee (managed survey owner) or associated groups as notification recipients for the managed survey.
- **Contacts**
  Allows you to specify one or more contacts as notification recipients for the managed survey.
- **Types**
  Allows you to specify one or more contact types as notification recipients for the managed survey (for example, Analyst, Customer, Manager, or Vendor).
- **Initial Message**
  Allows you to view or edit the text for the initial message title and body. Also allows you to specify the notification method for the initial message (for example, e-mail, CA SDM notification, FAXserve, or pager e-mail).
- **Reminder Message**

Allows you to view or edit the text for the reminder message title and body. Also allows you to specify the notification method for the reminder message (for example, e-mail, CA SDM notification, FAXserve, or pager e-mail).

View or Edit a Managed Survey Activity

You can view or edit an entry on the managed survey activity log.

Follow these steps:

1. Select Service Desk, Surveys, Managed Surveys, Managed Survey List on the Administration tab.
   The Managed Survey List appears.
2. Click a survey in the Managed Survey Name column.
   The Managed Survey Detail page appears.
3. Select the Activities tab.
   The Managed Survey Activity Log Search fields appear.
4. (Optional) Complete any of the following search filter fields to list only the activity log entries of interest:
   a. **Tenant**Select the appropriate tenant, if you are operating in a multi-tenancy environment.
   b. **Public Data**
      Select whether you want to include, exclude, or list only public data activities.
   c. **Activity Type**
      Search for an activity type if you want to filter the list by activity notification type.
5. Click Search.
   The Managed Survey Activity Log List displays the log entries that match your search criteria.
6. Click Search.
7. The Managed Survey Activity Log List displays the log entries that match your search criteria.
8. Right-click the log entry of interest and select View.
   The Managed Survey Activity Detail page appears.
9. (Optional) Click Edit. The following editable fields require explanation:
   a. **Internal**
      Specifies whether the log entry is public or private.
   b. **User Description**
      By default, this field contains a copy of the System Description field, which specifies the details of the activity (for example, definition of an initial message for the survey. You can edit the User Description to provide an easy way of identifying the activity.
10. Click Save, if you have chosen to edit the log entry.
    Your edits to the log entry are saved.

Managed Survey Status

You can view the default survey status options by navigating to Service Desk, Surveys, Managed Survey Status from the Administration tab.

From this page, you can perform the following tasks:

- Create a new status.
- Edit an existing status.
- Search the Status List.
- Enable or disable a status.

Survey Status symbols, along with their codes, descriptions, and record status appear in the list. The survey symbol is a field that defines the record status of a managed survey.

The following are default Managed Survey Status List symbols:

- Configuration in progress
- Survey Cancelled
- Survey on Hold
- Survey Period Closed
- Survey Period Open

Search Managed Surveys

The Managed Survey List page has the following options:

Show Filter Displays a search filter based on the following criteria: Name Description Active Related Survey Assignee Status Clear Filter Clears the search filter.

> **NOTE**
>
> If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

Search Managed Survey Status

The Managed Survey Status List page has the following options:

Show Filter

Displays a search filter based on the following criteria:

- Symbol
- Code
- Active
- Description

Clear Filter Clears the search filter.

Create a Survey Template

You can create a survey template from the Service Desk node on the Administration tab.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. Click the Administration tab.
2. Click Service Desk, Surveys, Survey Templates.
   The Customer Survey List appears.
3. Click Create New.
   The Create New Survey page appears.
4. Complete the appropriate fields.
   See Survey Template Fields for field definitions.

   > **NOTE**
   > The Survey Name field and the Active? dropdown are required.
5. Click Save.
   The template is saved and questions can be added.
6. Click Add Question to specify the question's sequence, comment label, status and question text. Or click Add Answer to open the Survey Answer Template Detail page.

NOTE

If multi-tenancy is installed, the list page displays a tenant column and a tenant drop-down list in the search filter. Specifying <empty> in the search filter searches for public objects. On detail pages, select the appropriate tenant from the drop-down list. If you select <empty>, the object is public.

Survey Template Fields

The following fields require explanation:

- **Comment Label**
  Indicates the label for the comment text box.

  NOTE
  Use this field when you indicate that you want to include comments in the survey.
- **Active**
  Sets the template to active or inactive.
- **Submit Cycle**
  Indicates to send the survey every 'nth' time.
- **Survey Introduction**
  Shows the introduction text of the selected survey.
- **Survey Completion Message**
  Shows the text when the survey is completed.
- **Include Comments**
  Allows a customer to enter free-form comments by adding a text box to the end of the survey form.
- **Use Stricter Rules?**
  Prevents a user from submitting the same survey more than once.

Search Survey Templates

The Customer Survey List page contains the following options:

Show Filter Displays a search filter based on the following criteria: Active Survey Name Survey Introduction Clear Filter Clears the search filter.

NOTE

If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

Search Survey Templates

The Customer Survey List page contains the following options:

Show Filter
Displays a search filter based on the following criteria:

- Active
- Survey Name
- Survey Introduction

Clear Filter
Clears the search filter.

NOTE

If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

Survey Question List

The Survey Question List appears when you create or open a survey template.

NOTE

If multi-tenancy is installed, the list page displays a tenant column and a tenant drop-down list in the search filter. Specifying <empty> in the search filter searches for public objects. On detail pages, select the appropriate tenant from the drop-down list. If you select <empty>, the object is public

The page contains the following fields:

Search
Allows you to search for all questions added to your survey.

Show Filter Displays a search filter.

Clear Filter Clears the search filter.

Add Question Opens the Create New Survey Question window, allowing you to specify the question's sequence, comment label, status and question text.

Sequence
(Required) Indicates a number (greater than zero) for the order in which this question appears on the survey question list. The questions appear in the order in which they are created. Numbers do not have to be sequential.

Question Displays the question text.

Status(Required) Indicates if the question is active or inactive.

Survey Question Detail Page

The Survey Question Template Detail page appears when you create or modify a survey question.

This page displays the following:

Tenant
Specifies the tenant associated with this question. If this field says public (shared), then the question is public.

NOTE

This field only appears in multi-tenancy installations.

Sequence Defines the order in which the question appears in the survey. For example, if the value is 1, then this is the first question in the survey.

Include Comments Allows a customer to enter free-form comments by adding a text box to the end of the survey form.

Comment Label Indicates the label for the comment text box.

NOTE

Use this field when you indicate that you want to include comments in the question.

Active? Specifies if the question is active or inactive.

Response Required?
Indicates that the user is required to answer this question.

Multiple Response Question? Indicates that this question has multiple answers.

Question text
Specifies the question text that appears in the survey.

Last Modified Date Specifies the last date the question was modified.

Last Modified By Specifies the last user who modified the question.

This page also displays the Survey Answer List.

Survey Answer List

The Survey Answer List appears when you create or open a survey question template.

> **NOTE**
>
> If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

The page contains the following fields:

Search
Allows you to search for all answers added to your question.

Show Filter
Displays a search filter.

Clear Filter
Clears the search filter.

Add Answer Opens the Create New Survey Answer window, allowing you to specify the answer's sequence, comment label, status and answer text.

Sequence (Required)
Indicates a number (greater than zero) for the order in which this answer appears on the survey answer list. The answers appear in the order in which they are created. Numbers do not have to be sequential.

Answer
Displays the answer text. Status Indicates if the answer is active or inactive.

Survey Answer Detail Page

The Survey Answer Template Detail page appears when you create or modify an answer.

This page displays the following:

Tenant
Specifies the tenant associated with this answer. If this field says public (shared), then the answer is public. Note: This field only appears in multi-tenancy installations.

Sequence
Defines the order in which the answer appears in the survey. For example, if the value is 1, then this is the first answer to the question.

Active?
Specifies if the answer is active or inactive.

Answer Text
Specifies the answer text that appears in the question.

How to Configure Notifications

This article contains the following topics:

With CA SDM, you can automatically notify key personnel about ticket activities (researching, escalating) and events (opening a ticket, for example). You can also notify key personnel about Knowledge Report Card (KRC) and Support Automation Assistance Sessions. When a significant activity or event occurs, CA Service Desk Manager creates a notification message that does the following:

- Identifies the ticket activity or the notification event
- References the ticket
- Includes other optional information
- Can identify potential contacts

You can view a notification message for a ticket because of a system action. A system action includes opening, closing, or modifying a ticket through its history information.

Setting up automatic notifications involves the following tasks:

- Defining activity notifications that determine the types of activities that generate notifications.
- Defining object contact notifications that determine the object contacts that can be used to send notifications in an activity notification.
- Identifying the methods used to send messages.

Create Object Contact Notifications

Object contact notifications let you notify the recipients based on the current value of a field on the ticket. Instead of identifying a person to notify, as in a notification method, you identify an object. For example, you can identify the To field to ensure that notification goes to the person currently identified in the To field, even if the value has changed since the ticket was defined.

Follow these steps:

1. Select Notifications, Object Contact Notifications on the Administration tab.
   The Object Contact Notification List page opens.
2. Click Create New.
   The Create New Object Contact Notification page opens.
3. Complete the following fields:
   - **Symbol**
     Defines a unique identifier for the object contact notification.
   - **Status**
     Specifies if the object contact notification is active or inactive.
   - **Object Type**
     Displays the name of the object to which the attribute applies.
   - **Object Attribute Name**
     Provides the name of the object contact notification (in the Symbol field) in Majic, which is internal CA code. The attribute name depends on the Object Type selection:
     - If the Object Type is Issue or Workflow Task, the attribute name is assignee, requester, or group; these are the attribute names in the chg objects and they map to fields in the Change_Request tables.
     - If the Object Type is an Issue Activity log, the attribute name must start with the attribute name in the activity log object that links it to a specific instantiation of the chg object. The attribute name could be change_id.group.
   - **Description**
     Describes the object contact notification.
4. Click Save.
   The new object contact notification is displayed in the Object Contact Notification List.

Previous Assignee Notifications

You can define Previous Assignee or Group values for an activity notification that detects changes to key fields when a ticket is saved. Previous values let you notify a previous assignee when a ticket is transferred, or notify both the current and previous groups when the priority of a ticket is escalated.

The Previous value fields of a ticket are local fields that exist only in memory and not in the database. The fields are populated during the save operation of the ticket only when respective attributes are changed and cleared at the completion of the notification processing. A previous value field is associated with a particular activity type through an activity association.

You can define Previous values that detect changes to the following key fields of a ticket:

Field	Requests, Incidents, Problems	Change Orders	Issues
Status	Yes	Yes	Yes
Active	Yes	Yes	Yes
Assignee	Yes	Yes	Yes
Request Area/Category	Yes	Yes	Yes
Group	Yes	Yes	Yes
Impact	Yes	Yes	Yes
Priority	Yes	Yes	Yes
Urgency	Yes	No	No
Severity	Yes	No	No

There are several contacts that you can specify for each object type (request, incident, problem, change order, or issue), which notify the current and previous contacts when an activity occurs.

- **Assignee**
  Person assigned to handle the ticket.
- **Assignee Previous**
  Person previously assigned to handle the ticket.
- **Group**
  Group assigned to handle the ticket.
- **Group Previous**Group previously assigned to handle the ticket.

After the notification rule is saved, the Assignee Previous and Group Previous fields display on the Object Contact Notifications List page.

Example: Configure Current and Previous Values for Key Fields

The following usage example describes how an administrator configures current and previous values for key fields to help ensure that the previous support representative is notified when a request is transferred away from them.

1. **Situation**A support representative is frustrated because a ticket was transferred away from them and they were never notified.
2. **Task** The administrator adds the Assignee and Assignee Previous object contacts to the notification rule for the Transfer activity notification. They attach a message template and specify the current and previous assignees to notify on the request form.
3. **Action** When the request is saved, the Assignee and Assignee Previous fields of the request are populated. When the activity occurs (ticket is transferred), the condition for the rule is evaluated.

4. **Result** If the condition is met, a notification message that describes the ticket activity is sent to the current assignee and the previous assignee.

Notify Contacts When a Ticket is Transferred Example

You can notify both the current and previous contacts when a CA SDM ticket is transferred.

Example: Notify both the current and previous contacts when a ticket is transferred

1. On the Administration tab, browse to Notifications, Activity Notifications.
   The Activity Notifications List page appears.
2. Select the Transfer activity notification.
   The Transfer Activity Notification Detail page appears.
3. In the Object Type field, select Requests/Incidents/Problems.
4. On the Notification Rules tab, under Symbol, select the Default Transfer Notification Rule.
   The Default Transfer Notification Rule page appears.
5. On the Object Contacts tab, click Update Object Contacts.
6. Click Search.
   The Notification Rule Update Recipients page appears.
7. From the Object Contacts list, select Assignee and Assignee Previous from the list on the left, and click the contact selection button (>>).
   The selected item is added to the list on the right.

   NOTE

   : Use the CTRL or SHIFT keys plus the left mouse button to select multiple object contacts.
8. Click OK.
9. Save the notification rule.
   The Object Contacts list displays the selected object contact.
10. On the Default Transfer Notification Rule page, click Message Template. Select a template and ensure that the Auto Notification option is enabled.
11. Create a request, specify an Assignee, and click Save.
12. On the Request Detail page, select Activities, and Transfer from the File menu.
13. Specify a new Assignee, and click Save.
    The notification is sent to the current and previous assignees when the transfer activity occurs.

Configuration Item Notifications

A configuration item (CI) notification lets you define an activity notification that is associated with a specific CI that is associated with a specific CA SDM ticket. This feature lets you track information about the users, organizations, and vendors of a CI. You can specify the CI object contacts on the Notification Rules Update Recipients page, such as CI Maint Org, CI Primary Contact, and so on.

Notify the Primary Contact of a Configuration Item for an Issue Example

You can define an activity notification for a primary contact that are sent for a specific CI for a specific CA SDM ticket.

Example: Notify the primary contact of a configuration item for an issue

1. On the Administration tab, browse to Notifications, Activity Notifications.
   The Activity Notification List page appears.
2. Select the Initial Activity Notification from the list.
   The Initial Activity Notification Detail page appears.
3. Select the object type you want to use.
4. On the Notification Rules tab, select the Default Notification Rule link.
   The Default Notification Rule page appears.

5. Select the Default Message Template link and ensure that the Auto Notification option is enabled.
6. Select the Object Contacts tab, and click Update Object Contacts.
   The Object Contact Notification Search page appears.
7. Click Search. A list of object contacts appears.
8. Select CI's Primary Contact from the list on the left, and click the contact selection button (>>).
   The selected item is added to the list on the right.
   You can use the CTRL or SHIFT keys plus the left mouse button to select multiple object contacts. You can add one object for a request and multiple objects for a change order or issue.
   The object contact is in the list on the right.
9. Click OK.
10. Save the notification rule.
    The Object Contacts list displays the selected object contact.
11. Complete the following tasks:
    – On the Service Desk tab, create or update an existing CI.
    – Add the primary contact listed on the Object Contacts tab. The selected object contact appears on the Configuration Items Detail page.
    – Add the CI to the Issue.

When an activity event occurs, the rule is implemented and the condition is evaluated. If the criteria for the condition is met, a notification message that describes the ticket activity is sent to all contacts of the CI associated with this notification rule.

Notification Log Reader

The Notification Log Reader displays the notifications received for the logged-in user by date, urgency, and status. With the Notification Log Reader, you can do the following:

- Change the sort order and set menu options to have the Notification Log Reader appear automatically when new messages are received.
- Double-click a notification message to request that CA SDM display the detail page for the ticket associated with the notification.
- Monitor notification messages by entering specific selection criteria to query the database for analysis or for selection of notification messages based on data entered in the fields. For example, you can list only those notification messages that have not been cleared by changing the Message Status field to Not Cleared.
- Clear notification messages to keep your list of notifications to a manageable size. Cleared notifications are not displayed when you first access the Notification Log Reader, although you can display them, if needed.

Set Notification Log Reader Options

You can set options for the Notification Log Reader to define how you are notified when new messages are received for an issue.

Follow these steps:

1. On the ServiceDesk tab, browse to View, Log Reader.
   The Notification Log Reader page appears.
2. Use the check box to the left of each notification to set the following options. You can select items to perform operations such as Clear Selected or Delete Selected.
   – **Header**
     Displays the header of the message, which usually contains the number of the ticket and the activity type.
   – **Start Date**
     Displays the date and time the notification was sent to your Log Reader window.
   – **Status**

Displays the status of the notification.

- **Urgency**
  Defines the level of urgency for the notification (low, normal, high, or emergency), which indicates the relative importance of different activities. Urgency levels are predefined; however, the system administrator is responsible for setting up other aspects of notification, such as notification methods and activity associations. The system administrator also defines the method of notification used for contacts and groups for each urgency level.
- **Message Text**
  The full message text for the notification.

The Log Reader displays any changes.

3. Click Close.
   The Notification Log Reader page closes and the options are set.

Personalized Responses

You can create personalized responses and attach them to requests, issues, and change order records when adding activities to the record. For example, you can append a personalized response on the Status Change or Log Comment windows available from the Activities menu.

Create a Personalized Response

You can create a personalized response to append to requests, issues, and change order records.

Follow these steps:

1. From the Administration tab, navigate to Service Desk, Personal Responses.
   The Personal Response list page displays.
2. Click Create New.
   The Create New Personalized Response page displays.
3. Complete the fields on the page:
   - **Response Owner**
     Specifies the contact who owns the response. If this field is left blank, the response is available to all analysts.
   - **Response**
     Specifies the text delivered to all those who receive this response. This field can be up to 1000 characters long. You can use variables in this field, for example:

     ```
     Ticket ref_num: @{call_req_id.ref_num}
     Assignee: @{call_req_id.assignee.combo_name}
     Customer: @{call_req_id.customer.combo_name}
     Description: @{call_req_id.description}
     ```
4. Select the type of records for which you want this response available. Click Save.
   A personalized response is created.

Personalized Response Variable Substitution

Variables can be embedded in the text of a Personal Response. These variables allow information to be substituted from the corresponding Request, Change Order, Issue, Problem or Incident. The syntax of the variables is the same as is used elsewhere in the CA SDM product, such as in the Activity Notification Message Templates and the Manual Notify Activity Message Text. The information can only be substituted from the corresponding Request, Change Order, Issue, Problem or Incident. Activity Notification Message Templates and the Manual Notify Activity Message Text allow information from the Activity Log Record to be included as well.

Check boxes for each object type (Requests, Change Orders, Issues, Incidents, and Problems) allow Responses to be filtered during selection. If the object type is not checked, the Response is not available for that object. For example, if only the Request box is checked, the Response is only presented in Activities for a Request.

A single Response can be used for all object types (Requests, Change Orders, Issues, Problems or Incidents). Because each object has different attributes, information that does not apply to the object is not substituted (for example, attempting to substitute the Request Number in a Response for an Issue).

A Response text example and the variable substitutions that occur for each object type follows:

```
This is Request # '@{call_req_id.ref_num}'
This is Change Order #' '@{change_id.chg_ref_num}'
This is Issue # '@{issue_id.ref_num}'
```

For a *Request*, the following substitution occurs:

```
This is Request # 'cr_demo:11'
This is Change Order # "
This is Issue # "
```

For a *Change Order*, the following substitution occurs:

```
This is Request # "
This is Change Order # 'chg_demo:3'
This is Issue # "
```

For an *Issue*, the following substitution occurs:

```
This is Request # "
This is Change Order # "
This is Issue # 'iss_demo:6'
```

By using the "Display this Response for" check boxes, you can create different versions of a Response with the appropriate substitution variables for the corresponding object (Requests, Change Orders, Issues, Problems or Incidents).

The format of the substitution variables for the different objects is as follows.

Object	Variable Format
Request / Incident / Problem	@{call_req_id.attr}
Change Order	@{change_id.attr}
Issue	@{issue_id.attr}

The substitution occurs when the Response is copied to the User Description field. The Response is copied after it is selected from the Personalized Response drop-down list and the drop-down list loses focus.

Internal Logs

You can define whether a particular access type is qualified to view internal logs. If allowed to view internal logs, contacts see a check box labeled Internal on each of the Log Activity windows, which they can select to mark the activity as internal. When activities are marked as internal, only contacts with an access type that is qualified to view internal logs sees the activity or is notified of it.

View the Log Reader

The Log Reader lists the notifications that are received for the logged-in user. You can filter the list by clicking Show Filter and specifying search criteria in the search fields. From the menu select View, Log Reader.

> **NOTE**
> If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

View Notification History

You can see the notification history for the currently logged in user (from the main CA SDM page or a list page, select View, Notification History) , or for the currently open ticket record (select View, Notification History on the menu bar)

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

List All Recipients of a Manual Notification

You can list all recipients for a manual notification by viewing the Notification List.

Follow these steps:

1. Locate text (similar to the following example) in the manual notification and click the URL link:
2. Click on the following URL to view the Notification List:

   ```
   http://<hostname>/CAisd/pdmweb.exe?OP=SEARCH+FACTORY=cr+SKIPLIST=1+QBE.EQ.id=400001+HTMPL=cr_lr.htmpl
   +INSTANCE=400027
   ```

   The Notification List that applies to the ticket appears and lists the notification recipients.

   > **NOTE**
   >
   > Only analysts can view all recipients of the notification. When other roles click the URL, a message informs them that they do not have sufficient security settings.

View Response Time Statistics

You can monitor the response time of your system to evaluate performance issues. Choose View, Response Time Statistics.

Create a Notification Method

A notification method lets you inform people when an activity that affects a ticket takes place. You can assign different notification methods for different notification levels (low, medium, high, or emergency).

Follow these steps:

1. Select Notifications, Notification Methods on the Administration tab.
   The Notification Method List page opens.
2. Click Create New.
   The Create New Notification Method page opens. Complete the following fields, as appropriate:
   - **Symbol**Defines a unique identifier for this record.
   - **Write to File**Specifies if the context information of the notification method is written to a file. If you select this option, the TEMP (Windows) or TMP (UNIX) environment variable determines the directory used for creating notification files. The environment used depends on who started the CA SDM services. On Windows, for example, configure the System Variables environment and specify TEMP if it is not already defined. If the system cannot find the environment variable, notification files are written to the root directory of the default drive.
   - **Supports SMTP**Specifies that SMTP addresses are supported for email recipients.
   - **Record Status**Specifies if the database record is active or inactive.
   - **Description**Gives a description of the notification method.
   - **Notification Method**
     Shows the name (including the path if it is required) of the program or executable script that runs, and any applicable switches. For example: Launchit c:\\backup\\applog.bat or pdm_mail -p.

     > **NOTE**
     > Because the notification method runs from the CA SDM server, put the notification method script in a directory that can be accessed from the path on the server. Alternatively, specify the full path to the

script. On UNIX, depending on the shell you are running, you can check the path by executing which pathname_to_script command.

Out of the box, the following Notification Methods are provided:

- Email
- Notification
- Pager_Email
- xMatters/Email
- xMatters/Notification
- xMatters/Pager_Email

NOTE

CA SDM supports only one notification method at a time. If you are using Email, then you cannot use Notification at the same time. This applies to all out of the box notification methods like Email, Notification, Pager_Email, xMatters/Email, xMatters/Notification, and xMatters/Pager_Email. When xMatters integration is disabled, the Notification Method must be manually reset to either Email, Notification, or Pager_Email.

Select one of the notification method from the Low, Normal, High, or Emergency drop-down list. When xMatters is integrated, select Notification Method as **xMatters/Email**. It indicates that the notification will be sent to the xMatters agent for incidents, requests, or problems. CA SDM objects like Change Orders, Issues, or Knowledge Documents are processed using the Email notification method.

3. Click Save.
   The notification method is defined and the Notification Method Detail page appears. If there appears to be a problem with the notification methods, examine the logs. The logs are in the $NX_ROOT/log directory on UNIX or $NX_ROOT \log on Windows.

Create a Notification Rule

This article contains the following topics:

Create the notification rule to specify the contacts to be notified and under what circumstances.

NOTE
If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

NOTE

The "Document about to expire" activity notification is activated seven days before the actual expiration date that displays in the knowledge document.

Follow these steps:

1. Select Notifications, Notification Rules from the Administration tab.
   The Notification Rule List page opens.
2. Click Create New.
   The Create New Notification Rule page opens.
3. Complete the following fields:
   - **Symbol**
     Defines a unique identifier for this notification rule. For example, enter Ticket Description Update.
   - **Object Type**
     Defines the object to which the rule applies. For example, select Request/ Incident/ Problem for an activity related to a ticket.
4. Click Save & Continue.

5. Click Condition to select the macro you can use to define a condition for this rule. Do one of the following actions to define the condition:

   NOTE

   A notification rule without a condition notifies all contacts every time the activity occurs.

   – Search for the macro from the list and select it.
   – Click Create New to create a macro.
6. Click Message Template to add a message template that you have created for this rule. Do one of the following:
   – Search for the template from the list and select it.
   – Click Create New to create a message template.
7. Choose the appropriate contacts to notify from the following tabs:

   NOTE

   Use the Update Contacts button that appears on each tab to search for and select more contacts to notify.

   – **Object Contacts**
   Displays the available organizations, vendors, and configuration items for the selected Object type that receive notification about tickets. For example, you can select Affected End User or Affected End User's Admin Org to notify.
   – **Contacts**
   Displays the individuals who are added to the notification rule, regardless of their association with the ticket.
   – **Contact Types**
   Displays the users who are defined within the notification rule with the same classification, such as analyst or customer.
8. Click Save.
   The notification rule is created.

Example Notify the Current and Previous Assignees

You can configure previous values for an activity notification that detects changes to key fields when a ticket is saved. Using previous values, you can notify both the current and previous assignee when a ticket is transferred, and much more. **Follow these steps:**

1. Open the Transfer activity notification.
   The Transfer Activity Notification Detail page opens.
2. Verify the following fields are set to the following values:
   – **Object**: Requests/Incidents/Problems.
   – **Notification Rules**: Default Transfer Notification Rule for request/incident/problem
3. Click Default Transfer Notification Rule for request/incident/problem rule.
   The Default Transfer Notification Rule for request/incident/problem Notification Rule Detail page opens.
4. Verify that the Auto Notification option is enabled on the Default Transfer message template for request/incident/problem template.

   NOTE
   The Auto Notification check box sends the notification associated with this template automatically when the activity occurs. For example, you set up an initial notification, set up the objects to notify, and set up the message template, but you are not ready to turn on the notifications. In this case, you do not select Auto Notification. When you are ready to start automatic notifications, you select the check box. The notification becomes active and occurs as defined.
5. On the Object Contacts tab, click Update Object Contacts.
   The Object Contact Notification Search page opens.
6. Click Search.

The Notification Recipients - Update page opens.

7. From the Object Contacts list, select Assignee and Assignee Previous and click the move button (>>).
   The selected items are added to the list on the right.

   NOTE

   : Use the CTRL or SHIFT keys plus the left mouse button to select multiple object contacts.
8. When the object contacts are in the list on the right, click OK.
9. Click Close Window.
   The Object Contacts list displays the selected object contacts.
10. Complete the following steps to test this notification:
    a. Create a request, specify an assignee, and save the request.
    b. On the Request Detail page, select Activities, Transfer. Specify a new assignee and save the request.

    The notification is sent to both the current and previous assignees when the ticket is transferred.

Example Notify the Primary Contact of a CI/Asset

You can define a notification rule for an activity notification that is sent for a specific configuration item associated with a service desk ticket. These notifications let you notify the person responsible for maintaining the configuration item when a request, problem, incident, change order, or issue is created.
Follow these steps:

1. Open the Initian activity notification.
   The Initial Activity Notification Detail page opens.
2. Verify the following fields are set to the following values:
   – **Object**: Requests/Incidents/Problems.
   – **Notification Rule**: Default initial Notification Rule for request/incident/problem
3. Click Default initial Notification Rule for request/incident/problem.
   The Default initial Notification Rule for request/incident/problem Notification Rule Detail page opens.
4. Verify that the Auto Notification option is enabled on the Default initial message template for request/incident/problem template.

   NOTE
   The Auto Notification check box sends the notification associated with this template automatically when the activity occurs. For example, you set up an initial notification, set up the objects to notify, and set up the message template, but you are not ready to turn on the notifications. In this case, you do not select Auto Notification. When you are ready to start automatic notifications, you select the check box. The notification becomes active and occurs as defined.
5. On the Object Contacts tab, click Update Object Contacts.
   The Object Contact Notification Search page opens.
6. Click Search.
   The Notification Recipients - Update page opens.
7. Select the Primary Contact of the CI from the list on the left and click the move button {>>).
   The selected items are added to the list on the right.

   NOTE

   : Use the CTRL or SHIFT keys plus the left mouse button to select multiple object contacts.
8. When the object contacts are in the list on the right, click OK.
9. Click Close Window.
   The Object Contacts list displays the selected object contacts.
10. Complete the following steps to test this notification:

a. On the Service Desk tab, create or update an existing CI.
b. Add the primary contact listed on the Object Contacts tab. The selected object contact appears on the Configuration Items Detail page.
c. Add the CI to the Issue.

When the activity occurs, a notification message that describes the ticket activity is sent to the contact responsible for the CI.

Create Message Templates

Create a message template that contains the values to use for the notification message. When you send multiple notification messages, you can use the message templates to simplify your workload.

Follow these steps:

1. Select Notifications, Message Templates from the Administration tab.
   The Message Template List page opens.
2. Click Create New.
   The Create New Message Template page opens.
3. Complete the fields as appropriate.
   - **Symbol**
     Defines a unique identifier for this message template.
   - **Object Type**
     Specifies the object type associated with this template. For example, select Request/ Incident/ Problem for any notification related to a ticket.
   - **Record Status**
     Specifies the status of the template as either active or inactive. Set the status to Active to use the message template.
   - **Auto Notification**
     Specifies to send the notification associated with this template automatically, when the activity occurs. For example, you set up an initial notification, set up the objects to notify, and set up the message template, but you are not ready to turn on the notifications. In this case, you do not select Auto Notification. When you are ready to start automatic notifications, you select the check box. The notification becomes active and occurs as defined.
   - **Notify Level**
     Indicates the relative importance of sending this notification. For example, select Emergency if you want to send the email notification to the contact immediately when the associated activity occurs.
   - **Notification Message Title**
     Specifies the summary title of the message. You can use variables to insert the incident number in the message title. For example, @{call_req_id.type.sym} @{call_req_id.ref_num} @{type.sym}.
   - **Notification Message Body**
     Specifies the content of the message. You can use variables to insert the analyst name, end-user name, and description into the message. For example

```
@{call_req_id.type.sym} @{call_req_id.ref_num} @{type.sym}.
Assigned to: @{call_req_id.assignee.combo_name}
Customer: @{call_req_id.customer.combo_name}
Description: @{call_req_id.description}

Click on the following URL to view:
@{call_req_id.web_url}
```

     You can add a survey URL to a message template. The survey url uses the following parameters:

```
#{SURVEY_ID,CNT_ID,CONTEXT_PERSID}
```

If the parameters **CNT_ID** and **CONTEXT_PERSID** are preceded by **@**, then the value must be expanded, otherwise the value is replaced in the survey URL.

```
http://NX_WEB_CGI_URL?OP=DO_SURVEY
+SVY_ID=SURVEY_ID+CNT_ID=CNT_ID+CNTXT_PERSID=CONTEXT_PERSID+MSG_DIGEST=message_digest_of_the_url
```

Replace the current string with the new string to update the survey URL link in the message template:

Current String

```
<a href="https://servicedeskhost/CAisd/pdmweb1.exe?OP=DO_SURVEY
+SVY_ID=400001+CNT_ID=@{call_req_id.customer.id}+CNTXT_PERSID=@{call_req_id.persistent_id}"></a>
```

New String

```
<a href="#{400001,@call_req_id.customer.id,@call_req_id.persistent_id}"></a>
```

You can use the ARTIFACT keyword to specify how artifacts are handled in outbound messages, message templates, notifications, and auto-replies. The ARTIFACT keyword uses the following values:

- **NONE** -- Specifies no validation of artifacts. This value is the same as not using the keyword.
- **PROTECTED** -- Validates a ticket against the hash for confirmation. If confirmation fails, the artifact is considered invalid and filtering fails when filtering searching for an artifact ({{object_id}}).
- **SECURE** -- Decrypts the ticket number. If the value is not a valid password, the artifact is considered invalid and filtering fails when filtering is searching for an artifact ({{object_id}}).

– **HTML Message**
Specifies the HTML message that is displayed to the recipient. If the recipient receives the message on an external device, such as a cell phone or PDA, the message displays in plain text only. Click Edit HTML Message to open the HTML Editor.

NOTE
If the URL variables do not appear as active links in email templates. Update the template to enclose the URL variable in <a> tags with the href attribute.

– **Quick View**
Displays the message as it appears to the recipient.

– **HTML Source**
Displays the message in the HTML source code.

4. Click Save.
The message template is created.

Create a Manual Notification

You can define a manual activity notification that is sent for a specific activity. An activity is an action that someone performs, such as resolving a service desk ticket or sending a managed survey. Even daily activities such as returning a call, canceling or closing a record, increasing priority, or updating status can result in a notification being sent.

Manual notifications let you do the following activities:

- Use temporary email addresses
- Add default users to manual notifications
- Show in an email notification all other recipients of the notification
- Process the outgoing email queue after a server error that leaves emails in the queue
- Notify the attached Object Contacts, Contacts, and Contact Types

NOTE
If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. On the Service Desk tab, select one of the following:
   - Incident from the Incident List page
   - Problem from the Problem List page
   - Request from the Request List page
   - Change order from the Change Order List page
   - Issue from the Issue List page
2. Select Activities, Manual Notify on the menu bar.
   The Manual Notification page appears.
3. Add the recipients of the notification in the To or Cc fields. The following fields enable you to add the recipients:

   WARNING

   You cannot add the same recipient in both To and Cc fields.

   NOTE

   The option to add recipients in the To and Cc list can be enabled if you install the **mail_show_to_cc_list** option from the Options Manager. If you disable the option, you can add recipients in the To list only**.** This option shows all the recipients of the email notification. For manual email notification, both To and Cc list recipients are displayed and for automatic email notification, only To list recipients are displayed. Pager email notification method does not support the listing of all the recipients in To or CC list.

   - **Available Recipients:** Lists relevant contacts and Contact Objects that you can add to the Recipients list, as specified in the Manual Notify settings in Activity Notifications options. When you select a contact object name, only the contact names that are associated with the object are added to the Recipients list. After selecting the recipient(s), click **Add To Recipients** (required) or click **Add Cc Recipients** (optional). The selected recipients appear on the respective Selected Recipient list.
   - **Contact:** Enter characters into the contact field of the ticket. The field displays results as you type and generates a list of suggested results. After selecting the contacts, click **Add To Contact** (required) or click **Add Cc Contact** (optional). The selected contact appears on the respective Selected Recipient list.
   - **Email address:** Accepts an SMTP address for an email recipient. You can specify any valid email address including addresses that are not associated with a contact record. Separate multiple addresses with a semi-colon. This field appears when the option notification_allow_temp_address is installed using Options Manager. After selecting the email address, click **Add To Email** (required) or click **Add Cc Email** (optional). The selected email address appears on the respective Selected Recipient list.
4. Complete the following fields, as appropriate:
   - **Urgency:** Specifies the relative importance of the notification (by default: low, normal, high, or emergency). The system administrator defines the Urgency levels.
   - **Preferred Method:** Specifies the notification method to use, such as Email or Notification.

NOTE

When you specify an email address, set the method to one that supports SMTP, otherwise, remove email addresses from the Selected Recipients list.

- **Internal:** Makes the notification visible to internal contacts only.
- **Message Title**: Specifies a title for the notification message. This title is a short description of the message, similar to an email subject line. The default title is the ticket type and number, and the words "Manual Notify" (for example, "Incident 32 Manual Notify").
- **Message Text:** Specifies the entire text of the notification. You can click Spelling to check the spelling of the message text.
- **Personalized Response:** Specifies a personalized response such as a generic or administrator signature. The response is added to the end of the message text.

NOTE

The administrator must set up a personalized response before you can use this option. For more information about personalized responses, see Personalized Response

5. You can send documents or URLs to the selected recipients, if you have selected the **Email** method and if you have installed the **mail_allow_attmnts** option.
   Ensure that you read the following considerations before using this feature:
   - Attachment repository that is used for storing manual notification attachments should be configured to match the sender and receiver email server restrictions in terms of file size and extensions. There is no separate repository to store the attachments. As an Administrator, you may need to add extra validations to match Exchange limitations.
   - Mailer daemon needs to be explicitly configured to use SSL (use the options manager) if encryption is needed.
   - The SOAP web services method **notifyContact** does not support this functionality.
   - Based on the size of the attachments, emails with many attachments may slightly degrade the performance of sending emails.
   - No changes are made to the mobile interface as Manual Notify is not supported from the mobile interface.
   - CA SDM does not handle bounce emails and emails that are failed to be delivered by the mail server.
   - Email attachments are not encrypted.
   - CA SDM is not responsible for language differences in the email content. An email in Portuguese can be sent to the English supported device.
   - CA SDM does not have the mechanism to detect if huge (number/size) emails are sent. Ensure that you check the email size at the firewall level.
   - The attachments can be sent to CA SDM contacts, and to the contacts that are not created in CA SDM (using email addresses directly). CA SDM will not take responsibility of the following:
     - attachments being forwarded to other contacts.
     - attachments may contain virus or malware content in the files.
     - attachments may contain sensitive information (such as personal information)
   - If an Analyst sends the same attachment as part of two different manuals notify activities, then, there would be duplicate attachments created and linked to the cr object. Ensure that you check before uploading any attachment.

   To upload the documents or URLs, click **Attach Documents** or **Attach URLs** from the **Attachments** tab and follow the on-screen instructions. After you upload, the URL or the document is listed on the tab.
6. Click Notify.
   The notification is sent and the activity is recorded on the Activities tab.

How to Set Up Notification for an Activity

Setting up Notifications for activities in CASM.

This article contains the following topics:

You can define a notification that is sent for a specific activity. An activity is an action that someone performs, such as resolving a ticket, sending a managed survey, running the Knowledge Report Card. Even daily activities such as returning a call, canceling or closing a record, increasing priority, or updating status are activities that can result in a notification being sent.

Follow these steps:

1. Open the CA SDM Web UI.
2. Verify the Prerequisites.
3. If you do not want to use a predefined message template, Create a Message Template.
4. If you do not want to use a predefined notification rule, Create a Notification Rule.
5. If you do not want to use a predefined activity association, Create an Activity Association.
6. You can Search for an Activity Association.
7. If you do not want to use a predefined activity notification, Create an Activity Notification.
8. You can Search for an Activity Notification.

Open CA SDM Web UI

Log in to the web UI from the following servers, depending on your CA SDM configuration:

- Conventional: Primary or secondary servers
- Advanced availability: Application or background servers

Verify the Prerequisites

Verify the following prerequisites, before you being the setup:

- Installed the option to add URL in the notification.
- Verified the notification method for the recipient.
- If you want to send a manual notification to an email address that is not associated with a contact, allow temporary email address.

Install the Option to Add URL Hyperlinks to Notifications

The web_url field in the Change_Request and Workflow_Task tables holds a URL value that allows a user to access a particular change order or workflow task through the web interface. When used in email notifications, a user can click the URL and can go to the web interface without any further querying.

Before you can implement the URL hyperlinks in notifications, configure the system as follows:

1. Install and configure the CA SDM web interface.
2. Using the Options Manager, configure and install the web_cgi_url option to specify the location of the CA SDM web engine. For example, http://hostname/scripts/pdmcgi.exe. For advanced availability, the host name must point to the application server or the load balancer.

Verify the Notification Method for the Recipient

Ensure that the contact to whom you want to send the notification, is assigned to that particular notification method.

Follow these steps:

1. Select Security and Role Management, Contacts on the Administration tab.
   The Contact Search page opens.
2. Search for the contact using the filter and select the contact that you want to notify from the search result.
   The contact detail page opens.
3. Select Notification on the Contact Details tab.
4. Verify the notification method. Based on your notification priority, select the required option. For example, you want to send an email notification for any emergency notification. Select the Email option in the Emergency field.

5. To change the notification method, click Edit, change the option, and click Save.
   The notification method for the contact is verified.

Allow Temporary Email Addresses

A temporary email address is an address that is not associated with a contact in the system. Temporary email addresses are useful in circumstances such as the following one:

- An end user is out of the office or is having difficulty accessing their standard email account.
- The analyst wants to use email to track interactions with the user.
- The analyst sends a manual notification to a *temporary* email address for the user.
- The analyst can view the activity log, which is updated with the manual notification.

Recipients cannot reply to temporary addresses when their email address is not associated with a contact record or does not have permission to update the ticket.

> **NOTE**
> Temporary email addresses are always SMTP email addresses, and are supported only when the Preferred Method supports SMTP. For information about how to set up temporary email addresses, see the *Online Help*.

Follow these steps:

1. Select Options Manager, Notifications on the Administration tab.
   The Option List page opens.
2. Click notification_allow_temp_address.
   The notification_allow_temp_address Options Detail page opens.
3. Click Edit.
4. Click Install.
   The notification_allow_temp_address Options Detail page opens.
5. Click Close Window.
6. Restart the CA SDM servers.Temporary email addresses are now allowed for manual notifications.

Create Message Templates

Create a message template that contains the values to use for the notification message. When you send multiple notification messages, you can use the message templates to simplify your workload.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. Select Notifications, Message Templates from the Administration tab.
   The Message Template List page opens.
2. Click Create New.
   The Create New Message Template page opens.
3. Complete the fields as appropriate.
   - **Symbol**
     Defines a unique identifier for this message template.
   - **Object Type**
     Specifies the object type associated with this template. For example, select Request/ Incident/ Problem for any notification related to a ticket.
   - **Record Status**

Specifies the status of the template as either active or inactive. Set the status to Active to use the message template.

- **Auto Notification**
  Specifies to send the notification associated with this template automatically, when the activity occurs. For example, you set up an initial notification, set up the objects to notify, and set up the message template, but you are not ready to turn on the notifications. In this case, you do not select Auto Notification. When you are ready to start automatic notifications, you select the check box. The notification becomes active and occurs as defined.
- **Notify Level**
  Indicates the relative importance of sending this notification. For example, select Emergency if you want to send the email notification to the contact immediately when the associated activity occurs.
- **Notification Message Title**
  Specifies the summary title of the message. You can use variables to insert the incident number in the message title. For example, @{call_req_id.type.sym} @{call_req_id.ref_num} @{type.sym}.
- **Notification Message Body**
  Specifies the content of the message. You can use variables to insert the analyst name, end-user name, and description into the message. For example,

```
@{call_req_id.type.sym} @{call_req_id.ref_num} @{type.sym}.
Assigned to: @{call_req_id.assignee.combo_name}
Customer: @{call_req_id.customer.combo_name}
Description: @{call_req_id.description}

Click on the following URL to view:
@{call_req_id.web_url}
```

  You can use the ARTIFACT keyword to specify how artifacts are handled in outbound messages, message templates, notifications, and auto-replies. The ARTIFACT keyword uses the following values:
  - **NONE** -- Specifies no validation of artifacts. This value is the same as not using the keyword.
  - **PROTECTED** -- Validates a ticket against the hash for confirmation. If confirmation fails, the artifact is considered invalid and filtering fails when filtering searching for an artifact ({{object_id}}).
  - **SECURE** -- Decrypts the ticket number. If the value is not a valid password, the artifact is considered invalid and filtering fails when filtering is searching for an artifact ({{object_id}}).
- **HTML Message**
  Specifies the HTML message that is displayed to the recipient. If the recipient receives the message on an external device, such as a cell phone or PDA, the message displays in plain text only. Click Edit HTML Message to open the HTML Editor.
- **Quick View**
  Displays the message as it appears to the recipient.
- **HTML Source**
  Displays the message in the HTML source code.

4. Click Save.
   The message template is created.

Create a Notification Rule

Create the notification rule to specify the contacts to be notified and under what circumstances.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. Select Notifications, Notification Rules from the Administration tab.
   The Notification Rule List page opens.

2. Click Create New.
   The Create New Notification Rule page opens.
3. Complete the following fields:
   - **Symbol**
     Defines a unique identifier for this notification rule. For example, enter Ticket Description Update.
   - **Object Type**
     Defines the object to which the rule applies. For example, select Request/ Incident/ Problem for an activity related to a ticket.
4. Click Save & Continue.
5. Click Condition to select the macro you can use to define a condition for this rule. Do one of the following actions to define the condition:

   NOTE

   A notification rule without a condition notifies all contacts every time the activity occurs.

   - Search for the macro from the list and select it.
   - Click Create New to create a macro.
6. Click Message Template to add a message template that you have created for this rule. Do one of the following:
   - Search for the template from the list and select it.
   - Click Create New to create a message template.
7. Select the appropriate contacts to notify from the following tabs:

   NOTE

   Use the Update Contacts button that appears on each tab to search for and select more contacts to notify.

   - **Object Contacts**
     Displays the available organizations, vendors, and configuration items for the selected Object type that receive notification about tickets. For example, you can select Affected End User or Affected End User Admin Org to notify.
   - **Contacts**
     Displays the individuals who are added to the notification rule, regardless of their association with the ticket.
   - **Contact Types**
     Displays the users who are defined within the notification rule with the same classification, such as analyst or customer.
8. Click Save.
   The notification rule is created.

Create an Activity Association

Associate an activity with the object attribute to track the changes to the related object attribute. For example, associate Field Update activity with the object attribute of the Description field of an incident record. This enables you to send a notification whenever the description of the incident is updated. An object attribute can have only one activity notification.

NOTE
If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. Select Notifications, Activity Associations on the Administration tab.
   The Activity Association List page opens.
2. Click Create New.
   The Create New Activity Association Type page opens.
3. Complete the following fields:
   - **Symbol**

Defines a unique identifier for this association.

- **Code**
  Defines an internal code for the activity association.
- **Object Type**
  Specifies the name of the object to which the attribute applies. For example, select Request/ Incident/ Problem.
- **Object Type Attribute**
  Defines the object attribute to which the activity type is associated. For example, enter the object attribute of the Description field of an incident record.
- **Activity Type**
  Indicates the type of activity. For example, select Field Update to see when the selected object attribute is updated.
- **Log Me**
  Determines if this activity association creates a log entry in the Audit Log.

4. Click Save.
5. Click Close window.

Search for an Activity Association

You can use the search function to filter the Activity Association List to show only the items you want to see. You can also locate a specific activity association to view or update.

> **NOTE**
>
> If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

Follow these steps:

1. On the Administration Tab, browse to Notifications, Activity Associations.
   The Activity Association List appears.
2. Click Show Filter.
3. Complete one or more of the following search fields:
   Symbol Defines a unique identifier for this record. Object Type Specifies the name of the object to which the attribute applies. Object Type Attribute Defines the field to which the activity type is associated. Activity Type Indicates the type of event.
   Click the More icon to display additional search fields.
4. Click Search.
   The Activity Association List populates with all activity associations that match your search criteria.

Create an Activity Notification

Associate the activity with a notification. When the activity takes place, the associated notification is sent to the selected contacts.

> **NOTE**
>
> If multi-tenancy is installed, select the appropriate tenant from the drop-
> down list. The public (shared) option creates the object for all tenants. You
> cannot maintain different Manual Notify activity notifications per tenant, or cannot copy a Manual Notify activity notification.

> **NOTE**
>
> Note: If select Object Type" is "**Knowledge Documents**" a new field "**Notification Recipients Type**" is displayed for the list of recipients that are notified. By default it is set to the following:

- "**Document Subscribers**", ONLY the emails listed in the **Emails** Tab will be notified.
- "**KD Approval Process Subscribers**", ONLY the contacts listed in the **Notification** Tab of the forum will be notified.
- "**Everyone**", both of the above listed options together will be notified.

Follow these steps:

1. Select Notifications, Activity Notifications from the Administration tab.
   The Activity Notification List page opens.
2. Click Create New.
   The Create New Activity Notification page opens.
3. Complete the following fields:
   - **Symbol**
     Defines a unique identifier for this activity notification.
   - **Code**
     Defines an internal code for the activity notification.
   - **Activity Valid for**
     Specifies the object to which this activity applies.
4. Click Save & Continue.
   The Update Activity Notification page opens.
5. Complete the following fields:
   - **Internal**
     Specifies if the activity notification is available only to internal employees or visible to end users.
   - **Record Status**
     Defines if the activity notification is active or inactive. Set the status to active to use the activity association.
   - **Related Ticket Activity**
     Specifies if the activity log on a ticket is propagated to all related tickets. The activity log propagation is only valid for related incidents, problems, and change orders. If you use multi-tenancy, perform the following actions:
     - Specify the appropriate tenant type from the Related Ticket Activity drop-down list.
     - Enter the name of the tenant in the tenant field, or click the search icon to search for a tenant.
   - **Object Type**
     Specifies the name of the object to which the activity applies.
6. Select the Notification Rules tab and click Update Notification Rules. Do one of the following actions:
   - Enter the search criteria, click Search, and select the notification rule from the search result.
   - Click Create New to create a notification rule.
7. (If you want to send a survey to the recipient of the notification) Select the Survey tab and define a survey notification. Surveys let you collect and analyze the customer feedback. An activity log is generated when a survey notification is sent and when one is received back from a customer.

   NOTE
   The Survey tab applies to all object types except Knowledge Documents, Knowledge Document Comments, and Knowledge Report Card. When specified from the Object Type list, the Emails tab appears instead of the Survey tab on the Update Activity Notification page. From the Emails tab, you can search for one or more email messages to associate with this notification or define a new one.
8. (If you want to trigger an event after the notification is sent) Select the Events tab. Events are procedures that an issue management system follows after a certain amount of time has elapsed. When the activity notification is triggered, the selected events occur. For example, update the status of the incident to Close. Search for the event and click Update Events button to add the event to an activity notification.
9. Click Save.
   The Activity Notification Detail page opens.
10. Click Close Window.

The activity notification is created. If an error occurs on the outgoing mail server, email notifications are not sent and queue in the $NX_ROOT/site/mail_queue directory. When the mail server becomes active again, after an interval it processes and sends email. You can set the email retry interval variable to recycle the email that was queued when the mail server was busy.
Notification email messages that the outgoing mail server fails to send are resubmitted until you do one of the following actions:

– Stop the Mail Daemon (pdm_mail_nxd) that handles outbound email notifications.
– Manually delete the messages from the $NX_ROOT/site/mail_queue directory.

Search for an Activity Notification

You can use the search function to filter the Activity Notification List to show only the items you want to see. You can also locate a specific activity notification to view or update.

NOTE

If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

Follow these steps:

1. On the Administration Tab, browse to Notifications, Activity Notifications.
   The Activity Notification List appears.
2. Click Show Filter.
3. Complete one or more of the following search fields:
   Symbol Defines a unique identifier for the activity notification. Code Defines an internal code for the activity notification. Record Status Specifies if the activity notification is currently active or inactive. Description Describes the activity notification.
4. (Optional) Click the More icon to display additional search fields.
5. Click Search.
   The Activity Notification List populates with all activity notifications that match your search criteria.

Define Survey Notifications

Complete the following fields, as appropriate:

- **Send Survey**
  Indicates whether to activate or deactivate the survey. If selected, the survey is sent to the contact when the selected activity notification is triggered.
- **Default Survey**
  Specify a default survey using the search icon or specify your own in the text box.
- **Survey Message Title**
  Enter the title for the survey.
- **Survey Message Body**
  Enter a message for the contact. When a contact receives notification of a survey, the message includes a URL that they can access from their web browser to find and fill out the survey form.

Set the Email Retry Interval Variable

You can define the time interval (in seconds) to retry failed attempts to send outgoing email to the mail server.

NOTE
CA SDM does not retry sending messages that the outgoing mail server accepts, but cannot be delivered. For these messages, the outgoing mail server retry capabilities and policies, if any, are in effect.

Retries are on a per-message basis. If the mail server is unavailable for a period, each message is retried when its own timer expires, rather than all the messages being sent at once. However, if you restart the outgoing mail daemon, all unsent messages attempt to be sent at that time, and if they all fail to be sent, their retry timers are all reset simultaneously.

The setting (NX_EMAIL_RETRY_INTERVAL) in the NX.env file controls the retry interval. You can change the default retry interval setting on one or more servers.

Follow these steps:

1. Log in to the following server, depending on your CA SDM configuration:
   - Conventional: Primary server
   - Advanced availability: Background server.
2. Navigate to the $NX_ROOT directory
3. Use a text editor such as WordPad to open the NX.env file.
4. Modify the value of to the NX_EMAIL_RETRY_INTERVAL interval you want as follows:
   ```
   NX_EMAIL_RETRY_INTERVAL=number_of_seconds
   ```
   - **NX_EMAIL_RETRY_INTERVAL**
     Defines the time interval (in seconds) to retry failed email attempts.
   - ***number_of_seconds***
     Specifies the number of seconds for each email retry interval. The default time is 600 seconds (10 minutes). The minimum value that you can use is 20 seconds. If you set a value that is less than the minimum of 20 seconds or more than 2000000 seconds, the default value of 10 minutes is used.
5. Save and close the file.
   Restart the CA SDM servers.
   The change takes effect.

Configure CA SDM and the Components

After you install CA SDM and any additional products you select, complete the following steps:

- Configure the CA SDM components (servers, the database, the web interface)
- Configure the web interface if the web server and primary server are on different computers
- Implement Knowledge Document Life Cycle Reports for Automated Policies
- Configure CA Business Intelligence
- Implement multi-tenancy

This article contains the following topics:

Understand the CA SDM Components

CA SDM components include CMDB, Visualizer, Web Screen Painter.

Configuration Management Database (CMDB)
CMDB consolidates and reconciles disparate sources of IT-related data in the context of business services. It lets you view the configuration item (CI) information such as resource attributes, relationships, and dependencies. You can use CMDB to display the relationships that are associated with a contemplated change, and to understand the impact that is associated with such a change from a business service perspective (impact analysis). CMDB is installed automatically when you install CA SDM.

Visualizer

Visualizer provides a graphical depiction of a CI and its related CIs including the types of the relationships. You can double-click a CI in the Visualizer to follow relationship trails and understand the impact that one CI has on the other

CIs. You can configure Visualizer on the Primary Server or a Secondary Server when you install CA SDM on that server. However, if you want to configure Visualizer after installing CA SDM, you launch CA SDM, then use the Configuration Wizard Visualizer.

Web Screen Painter
You can use Web Screen Painter to modify the database schema of CA SDM to meet your needs. Web Screen Painter also allows you to test your schema changes on your own web forms before updating the physical DBMS schema or affecting other users. In the conventional configuration, you can install Web Screen Painter on the Primary Server or any Secondary Server. However, in the advanced availability configuration, install the Web Screen Painter only on the Background Server. If you install it on a standby server or an application server, you get an error message upon launching it.

Configure the CA SDM Server and Database

The CA SDM installation, initializes, and configures the server and database. For more information about configuring the server and database, see How to Configure Servers for Advanced Availability.

Configure CMDB

Set Up the CMDB Audit Log

The object and trigger definitions, attributes, and HTML forms that are used by CMDB audit has changed.

Follow these steps:

1. Remove the *cmdb_write_audit_log_site* trigger if you have created site/mods/*extension*.mod (*extension* specifies the extension name).
   In this release of the product, auditing is automatically created and enabled.
2. Add 'UI_INFO("AUDIT_LOG")' to each attribute that you want to log.
3. To migrate your existing HTML forms, use the new CA SDM templates.

Configure Single Point of Entry

Configuring single point of entry in CA SDM allows Single Sign-On (SSO) to CA Service Catalog.

Define CA Catalog Details in CA SDM

Defining the CA Catalog details allows the CA SDM server to identify the location of CA Catalog installation.

Follow these steps:

1. Log in to CA SDM and navigate to **Administration** tab, **Options Manager**, **General**, and select **catalog_server**.
2. Configure the required CA Catalog details.
3. Restart CA SDM server.

Modify the System Environment

You can customize the system environment by using the Options Manager (recommended) or editing the environment template file.

Use Options Manager to Modify the System Environment

After you log into CA SDM, navigate to the **Administration** tab, **Options Manager**, and perform the following actions to modify the system environment:

- Obtain a list of all the available options.
- View a summary of each option such as the application with which it is associated, a brief description, and its status.
- View the details of any specific option.

When you view the detailed information for an option, you can view any special actions that you are required to take when changing the option. For example, after installing or uninstalling some options, ensure that you restart the CA SDM server for the option to take effect.

- Review the status of all options at the summary level.
- Uninstall any of the available options (applicable only on the background server).
  Many of the options are preconfigured and installed during the CA SDM installation. Using the Options Manager to install or uninstall options can alter some or all of the default settings.
- Install any defined option (applicable only on the background server).

Edit the Environment Template to Modify the System Environment

CA SDM uses environment variables that are specified in the environment template file (NX.env.tpl) to determine certain behaviors. You can use environment variables to modify some system behaviors. You typically use the Options Manager to control system behavior, but at times CA Technical Support instructs you to modify a particular environment variable directly.

Points to consider when editing the environment template file:

- Environment variables set in this file can be overridden by setting the environment variable in the process space in which a process runs. Although convenient in some limited cases, this setup is not wanted. Preceding a variable setting with an at symbol (@) prevents variables in the process space overriding the variable. Unless there is a specific reason for allowing an override, the @ symbol always precedes the variable name in the template file.
- The comment characters for this file are pound (#) and exclamation point (!). The exclamation point character is also used to disable an option.

  WARNING
  Modify the template file (NX.env.tpl) and allow the configuration process to apply the changes to the environment file. Never modify the environment file (NX.env) directly.

Follow these steps:

1. Back up the environment template file (.tpl) that corresponds to your system environment:
   – UNIX -- $NX_ROOT/pdmconf/NX.env.tpl
   – Windows -- installation-directory\pdmconf\NX.env_nt.tpl
2. Edit the environment template file on the following server, depending on your CA SDM configuration:
   – Conventional: Primary server
   – Advanced availability: One of the standby servers

   NOTE
   You can view and modify this file using any text editor (Windows users use WordPad).
3. Make the required updates and save the template.
4. To apply the changes you made to the environment template file to the actual environment file, run the configuration utility on the primary or standby server installation.
5. Depending on your CA SDM configuration, complete one of the following actions:
   – Conventional: For the changes that are made to the environment file to take effect, restart the primary server.
   – Advanced availability: Restart all the CA SDM servers.

   NOTE
   To avoid shutting down your system, your support technician can instruct you to restart only certain processes rather than recycling your entire CA SDM server.

How to Move the Authentication Module to an External Server

This article contains the following topics:

The authentication module in CA SDM is a singleton daemon that is responsible for the user authentication. The authentication module runs on the primary server for the conventional configuration and on the background server for the advanced availability configuration. You can move the authentication module to an external server for the following scenarios:

- **Conventional configuration:**
  - The primary server is on a Linux platform and you require a Windows authentication, you move the authentication module to a Windows secondary server.
  - The primary server is on a Solaris platform and you require the CA EEM authentication, you move the authentication module to the secondary server as CA SDM does not support the CA EEM integration on Solaris.
- **Advanced availability configuration:**
  - The background server is on a Linux platform and you require a Windows authentication, you move the authentication module to a Windows application server.
  - The background server is on a Solaris platform and you require the CA EEM integration, you move authentication module to an application server as CA SDM does not support the EEM integration on Solaris.

The following diagram illustrates how to move the authentication module:

NOTE
The selected external server must have CA SDM installed. If you move the authentication module to an external server such as an application server, it becomes a single point of failure. If the application server is down, users cannot log in.

Verify the Prerequisites

The analysis of the CA SDM server platforms helps you to decide how to move the authentication module to an external server.

Follow these steps:

- Identify the operating system of the following server, which is based on the configuration type:
  - **Conventional:** Primary server.
  - **Advanced Availability**: Application server.
- Identify the type of required authentication from the following models:
  - Windows authentication for the CA SDM implementation on Linux.
  - CA EEM authentication for the CA SDM implementation on Solaris.
- Make sure that CA SDM is installed on the selected external server.

Configure CA SDM for Using External Authentication

To redirect the authentication requests to an external server, specify the hostname of the target server where the authentication module is present. The process of configuring CA SDM for moving the authentication module is different for the conventional and advanced availability configuration of CA SDM.

Conventional Configuration

Follow these steps:

1. Log in to the primary server as an administrator.
2. From the command prompt, change the directory to samples\pdmconf under NX_ROOT and run pdm_perlpdm_edit.pl.
3. From the pdm_edit.pl top menu select U, and press Enter.
   The User Validation server submenu appears.
4. Enter E to specify the hostname of the external server, and press Enter.
5. Enter primary for the primary server or the hostname of the secondary server, and press Enter.

You have configured the redirection of authentication requests to an external server.

Advanced Availability Configuration

Follow these steps:

1. Log in to the background server web interface as an administrator.
2. Select the Administration tab.
3. Expand Options Manager, Security.
4. Click the bopauth_nxd_host entry in the Option List.
   The bopauth_nxd_host Options Details page is displayed.
5. Click Edit.
6. Select the hostname of the target server from the Option Value drop-down list.
7. Click Save.

   NOTE
   By default, the authentication module runs on the background server.

You have configured the redirection of authentication requests to an external server.

How to Restart the CA SDM Servers

Depending on your CA SDM configuration, perform the following processes:

- Restart the CA SDM servers in conventional configuration.
- Restart the CA SDM servers in advanced availability configuration.

Restart the CA SDM Servers in Conventional Configuration

For the conventional configuration, you restart the servers in the following order:

NOTE

To restart a server click Start, Settings, Control Panel, Administrative Tools, Services. Right-click the CA SDM Server and select Start.

1. Restart the secondary server.
2. Restart the primary server.

Restart the CA SDM Servers in Advanced Availability Configuration

For the advanced availability configuration, we recommend that you restart the CA SDM servers in the following order:

NOTE

To restart a server click Start, Settings, Control Panel, Administrative Tools, Services. Right-click the CA SDM Server and select Start.

1. Restart all Standby Servers.
2. Promote the Standby Server as the New Background Server.
3. Start the Old Background Server.
   When you start the background server, it becomes a standby server.
4. Choose the Less Active Application Server.
5. Restart the Less Active Application Server.
6. Stop the Other Application Server.
7. Start the Application Server.
8. Perform the steps 6 and 7 for the other application servers.

Promote the Standby Server as the New Background Server

Before you stop the background server, promote the standby server (that you have upgraded) as the new background server. If Support Automation is installed with CA SDM, notify the active Support Automation users about the background server shutdown.

Follow these steps:

1. Execute the following command on the background server to notify all active users using Support Automation to save their work:
   ```
   sa_server_notifier [-h] | [-q seconds] | [-c]
   ```
   - **-h**
     Displays the help page.
   - **-q seconds**
     This option notifies a local server (background) to quiesce in a specified time interval. This interval is the number of seconds before the server goes offline. This option cannot be used for a standby server or application server.
   - **-c**
     This option cancels a previously sent quiesce request.

   A pop-up message is displayed to all the active users using Support Automation. This message notifies the users about the server shutdown and the time that is left for the shutdown. The users must save their work and logout within that scheduled time.
2. Execute the following command on the standby server that you wish to promote as the new background server:
   ```
   pdm_server_control  -b
   ```
   - **-b**
     Notifies a local standby server to become the background server. The standby server must already be running. If the server is not running, it is started but no failover is performed; to start a failover, run the command again.

   The background server shuts down automatically and the standby server is promoted as the new background server. This change does not affect the end-user sessions. The in-progress updates (if any) are stored and delayed, until the new background server comes online.

Choose the Less Active Application Server

You choose an application server with the least user activity. Run the following command on each application server to choose the one with no or minimal active sessions:

```
pdm_webstat
```

> **NOTE**
>
> This command does not capture the SOAP or REST Web Service sessions.

Stop the Other Application Server

You inform all the active users on an application server to move to the less active application server before you stop it. Ensure that you have restarted the less active application server before moving all the users to it.

Follow these steps:

1. (Recommended) Inform all active Support Automation analysts on the application server which you want to stop, to create a ticket in CA SDM with their session information. This process ensures that the session information is not lost. For example, the Support Automation analyst is in a session with a customer to resolve a hardware issue. In such a case, the Support Automation analyst can create an issue in CA SDM with the session information before the application server shuts down.
2. Send a notification (for example, an email notification) to all the active users on the application server to move to the less active application server that you just restarted. This notification can include the details of the updated application server.
3. Execute the following command on the application server:

```
pdm_server_control [-h] -q interval -s server_name
```

- **-h**
  Displays the help page.
- **-q interval -s server_name**
  Notifies a local or remote application server to quiesce in a specified time interval. This interval is the number of seconds before the server goes offline. When using this option without a server_name, the local server is notified to quiesce. This option cannot be used for a background or a standby server.

A pop-up message is displayed to all the active users on the application server to notify them about the server shutdown and the time left for the shutdown. The users must save their work and logout within that time. The application server stops after the specified time. The users log on to the other application server to resume their work. The Support Automation analyst can refer to the ticket and resume their work.
The application server is stopped successfully.

Verify the Authentication

Verify the authentication with a user ID that has a corresponding contact record in Windows or CA EEM, based on the authentication type. A successful login indicates that you have successfully moved the authentication module.

Follow these steps:

1. Launch the browser and enter the CA SDM URL.
2. Log in to CA SDM with a user name having a corresponding contact record in Windows or CA EEM, based on the authentication type.

If the module has been moved successfully, the Service Desk Home Page opens.

Server Configuration Utility

The CA Service Desk Manager Server Configuration utility (pdm_configure) initializes and configures the CA Service Desk Manager server, database, and web environment. Configuration runs automatically during CA Service Desk Manager installation. The system administrator also can run Configuration on demand. The system administrator can configure the Administrative Client on Linux.

Follow these steps

Select Server Configuration

You can select the CA Service Desk Manager installation configuration type and the type of database to configure.

Select Server Configuration

Specifies the configuration type for your CA Service Desk Manager installation. The options in the subsequent screens automatically updates to correspond to the configuration type you select. You can select the Advanced Availability option or Conventional option.

General Settings

Configure the General Settings page with the following options:

- **Configuration Type**
  The server type to configure.

  - Advanced Availability: If you are executing the Server Configuration utility for the first time, background server is automatically selected. Configure the background server and then add the application or stand-by servers to support advanced availability and failover.
  - Conventional: If you are executing the Server Configuration for the first time, primary server is automatically selected. After the primary server is configured, you can add secondary servers to manage high traffic volume and improve performance.
    For example, you can add more Object Managers and web engines to another server. There can be zero or more secondary servers in a CA Service Desk Manager enterprise system.
- **Slump Host Name**
  Available only for the Advanced Availability configuration. Slump host name specifies the system on which the server is installed.
- **Server Id**
  Available only for the Advanced Availability configuration. This field is automatically populated.
- **Primary Server Node**
  Available only for the Conventional configuration.
  Specifies the host name of the primary server. This field is case-sensitive. The Primary Server Node value is the same for both primary and secondary servers in the CA Service Desk Manager installation.
- **Object Manager Display Name**
  Specifies the server connecting to the server object managers (domsrvr). All object managers that start on this computer use this name for display purposes only. You can change this value when the object manager is started.
- **Object Manager Name**
  Specifies the names or aliases of the object managers to which you want to establish a connection. You can enter a single name or a list of names. The default of ANY connects to any running object manager, and is acceptable in most situations.

    WARNING
    Verify the names of the object manager or group of object managers with your system administrator.
- **Local Host Name**
  Specifies the name of the server. This field is case-sensitive.
  - Conventional: You can enter the value only for secondary servers. This field is case-sensitive.
  - Advanced Availability: The value is automatically populated from your background server configuration details.
- **Slump Socket Port**
  Specifies the name of the main communications socket port. CA Service Desk Manager uses this port for all servers that are configured in your environment. The default value is 2100. The primary server always opens a listen port on the slump socket port. Every daemon (service) that communicates across the slump port connects to the primary server using this port.
- **Proctor Socket Port**
  Available only for the Conventional configuration. Applies only to systems that use secondary servers (proctors). Each secondary server listens on this port for messages from the primary server. The secondary servers open this port only when they cannot connect to the slump port on the primary server. For example, the port opens when the primary server is not running.
  Default: 2300

System Accounts

Configure the System Accounts page with the following options:

- **Privileged User Name**
  Identifies an operating system user ID with administration privileges. On Windows, if the name you specify does not exist, the system creates a local use with domain rights to different places, depending on how your local computer is defined:

- If the local computer is not defined as a primary or backup domain controller, the privilegeduser is added to the local computer as a local user.
- If the local computer is defined as a primary domain controller, the privilegeduser is added as a domain user.
- If the local computer is defined as a backup domain controller, the privilegeduser is added to the primary domain controller as a domain user.

WARNING
If you have modified the user and domain user rights, contact the system administrator to set up the privileged user.

- **Privileged User Password (Windows)**
  Specifies a password for the privileged user account. If the privileged user name exists on the local computer, and the local computer is not defined as a primary or backup domain controller, the local user is checked. If the local computer is defined as a primary domain controller or a backup domain controller, the domain user is checked.
- **Privileged User Password (Linux)**
  Specifies a privileged user password for integrations with AMS and Workflow.
- **Restricted User Name (Windows only)**
  Specifies an operating system user ID that is used on behalf of clients as they access functions. The user name is restricted because it has limited user rights. The default is rhd. The rhd user is added in the same manner as the privileged user.
- **Restricted User Password (Windows only)**
  Specifies a required password for the restricted user name is added. The configuration stores the value.

Load Default Data

Load default data is available only for the Advanced Availability configuration.

- **Load default data**
  Specifies whether to load the CA Service Desk Manager default system data into the database. If you have modified any system default values, selecting this option replaces the data values.

NOTE
You can select the option only during the background server configuration.

Select Database

Specifies the type of database to configure for your CA Service Desk Manager installation. The available options on the left automatically update to correspond the database type you select. For example, if the Database Type is SQL, the options update with MS SQL Database Config.

- Advanced Availability: The database type that you select must match the background server database.
- Conventional: The database type that you select must match the primary server database.

MS SQL Database Configuration

If you are using MS SQL Database, specify the following details:

- **Load Default Data**
  Applicable only for Conventional configuration.
  Load Default Data whether to load the CA Service Desk Manager default system data into the database. If you have modified any system default values, selecting this option replaces the data values.
- **Database Server Node**
  The name of the MS SQL instance that CA Service Desk Manager uses for configuration. The service instance is in the format hostname\instance_name. If the default instance is used, you only have to specify the hostname.
  If you are using the SQL named instance and a non-default port number, the **Database Server Node** field should look like this:
  <databaseHostname>\<database instance name>,<non-default port number>

For example:

```
testsqlserver\sdmcasmqa,10555
```

- Advanced Availability: The database server node must match the background server database node.
- Conventional: The database server node must match the primary server database node.

- **Database Name**
  Indicates the name of the database.
  - Advanced Availability: The database name must match the background server database.
  - Conventional: The database name must match the primary server database.
- **Database Admin User**
  Specifies the admin user who has permission to create user and schema.
- **Database Admin Password**
  Specifies the database password of user specified by database admin user.
- **Database Userid**
  The database user ID and password. The SQL user ID is added while creating the database.

  NOTE
  The special at-sign character (@) cannot be used in database IDs or passwords.
- **Database Password**
  The password for the SQL database.
- **SQL Listening Port**
  The listening port of the defined SQL server instance.
  Default: 1433

Oracle Database Configuration

If you using the Oracle database, specify the following details:

CA Service Desk Manager requires a Net Service Name specifying the Oracle database where the MDB resides. CA Service Desk Manager also requires a system identifier (SID) for the database. CA Service Desk Manager requires the Net Service name and SID information to access the database with both Oracle client technology and JDBC technology. For information about Service Names and SIDs, refer to Oracle documentation.

- **Load Default Data**
  Applicable only for Conventional configuration and indicates the CA Service Desk Manager default system data into the database. If you have modified any system default values, selecting this option replaces the data values.
- **Remote Database**
  To indicate whether the Oracle MDB is on a remote server (selected) or the local server. If the Oracle server is local, the MDB is created during the configuration. If it is remote, the MDB must already be configured with the remote installer.

  NOTE
  For Advanced Availability configuration, we recommend that you configure the Oracle MDB on a remote server.
- **Create Tablespaces**
  Specifies whether to create tablespaces for the MDB database or use existing tablespaces.
  - If this check box is selected, the script creates tablespaces with the default names MDB_DATA and MDB_INDEX.

    NOTE
    The Data Tablespace Name and Index Tablespace Name fields are disabled.
  - If this check box is not selected, provide names of the existing tablespaces in the Data Tablespace Name and Index Tablespace Name.

WARNING
To use the existing tablespaces, configure the database to use at least 500 MB of tablespace, and an index tablespace of at least 150 MB.

- **Net Service Name**
  Identifies the Net Service Name of the Oracle database where the MDB resides. If the database is remote, provide the Net Service Name that is defined within the Oracle client on the local computer.
- **mdbadmin User Password**
  Specifies the password of the mdbadmin user for accessing the mdb.
- **DBA User Name**
  Identifies the name of an Oracle user with DBA access.
- **DBA Password**
  Identifies the password for the DBA user.

  NOTE
  The special at-sign character (@) cannot be used in database IDs or passwords.
- **Data Tablespace Name**
  Specifies the name for the MDB data tablespace you want to create. Used only if the Oracle server is on the local computer.
- **Index Tablespace Name**
  Specifies the name for the MDB index tablespace you want to create. Used only if the Oracle server is on the local computer.
- **Tablespace Path in DB Server**
  Specifies the directory path to the physical tablespace location.
- **Oracle Home Path**
  Specifies the Oracle Home path.
- **JDBC Connectivity**
  Specifies whether to use JDBC Connectivity. Several components of CA Service Desk Manager use JDBC technology to access the database and require specific information about the Oracle server.
  - **Database Host Name**
    Identifies the name of the computer where the Oracle server is installed.
  - **SID**
    Specifies the system identifier (SID) for the database as defined on the Oracle server computer. The SID value must be the SID for the MDB, and is not the same as the Net Service Name.
  - **Listener Port**
    Specifies the listener port for the database.

    WARNING
    For Windows installation, configure the Oracle MDB to use UTF-8 as the database character encoding to support multilingual search.

Web Interface

Configure the Web Interface with the following options:

- **Web Host**
  Specifies the system that hosts the web server. This value is used to build URLs for the system. By default, the web host is your local system.
  - Advanced Availability: You can configure web services only on application server.
  - Conventional: Available only if you are on a primary server, and defaults to the primary server name. You can change this value if your default web server is on another computer. For example, if you move your web server to

a secondary server, you can change this value to the name of your secondary server. This value is used to build URLs for the system.

- **Config Type**
  Specifies the web server for your installation:

  NOTE
  The Tomcat port is always used to access upload, the Asset Maintenance System, and pdmgraph, regardless of this setting.

  Select the Manual option only when using an alternate http or servlet server.

  – On Windows, you can select Tomcat or IIS.
- **Web Site**
  NT only. Identifies the web site for the CA Service Desk Manager Server. The names appearing identify the available web sites. The IIS port is used to specify the port that is associated with the site.

  NOTE
  You can install any mail system client on your server that complies with Microsoft MAPI 1.0 standard.
- **Tomcat Port**
  Specifies the Tomcat port number. Defaults to 8080, but you can set this port value. This value appears in the URL that is constructed to access the web site, for example,*http://bobcat:8080/CAisd/UploadServlet*.
- **Tomcat Shutdown Port**
  Specifies the socket port for Tomcat to monitor for shutdown requests.
- **IIS/Apache Port**
  Specifies the IIS or Apache server port number. If you configure with IIS or Apache, and are not using port 80, enter an alternate port number. If you are using port 80, leave this field blank.
- Deploy SOAP Web Services
  – Advanced Availability: Applicable only for the application server. Enabling this option notifies the active users on the application server about the server shutdown details. For example, during rolling maintenance of the application server, a form is displayed to the active users stating that the application server shuts down within a specified time.
  – Conventional: This check box is only available if you are configuring a secondary server. This option lets you deploy the web services on the secondary server. By default, web services are installed on the primary server.

REST Web Services

Configure the REST Web Services page with the following options:

- **REST Tomcat Port**
  Specifies the REST Tomcat Port.
  Default: 8050
- **Tomcat Shutdown Port**
  Specifies the REST Tomcat Shutdown Port.
  Default: 8055

  NOTE
  For Advanced Availability configuration, this option is available only for application servers.

Federated Search

Configure the Federated Search Tomcat with the following options:

- **Configure Federated Search**
  Specifies the option to select Federated Search. The Tomcat options are available only after selecting this option.
- **Tomcat Port**
  Specifies the Federated Search Tomcat Port.

Default: 8040

- **Tomcat Shutdown Port**
  Specifies the Federated Search Tomcat Shutdown Port.
  Default: 8045

Visualizer

Configure the Visualizer page with the following options:

- Service Desk Manager Web Services
  - **Web Host**
    Displays the web host address.
  - **Web Host Port**
    Displays the Visualizer web host port.
- Visualizer Tomcat
  - **Tomcat Port**
    Specifies the Visualizer Tomcat Port.
    Default: 9080
  - **Tomcat Shutdown Port**
    Specifies the Visualizer Tomcat Shutdown Port.
    Default: 9085

    NOTE
    For Advanced Availability configuration, this option is available only for application servers.

Support Automation

On the Support Automation page, select the appropriate Configuration Type:

- **None**
  Configures CA Service Desk Manager without Support Automation functionality.

  WARNING
  If you want to configure Support Automation in a multi-tenancy environment, we recommend that you separately migrate the CA Support Automation r6.0 SR1 eFix5 divisions to Release 12.9 tenants before enabling Support Automation on a CA Service Desk Manager server.
- **Main Server**
  Configures the Support Automation server in main server (standalone) mode.
  Note: If you select the Main Server option, and are planning to configure one or more socket proxy servers, set the Socket Server host name and external port to the socket proxy host. For multiple socket proxies, you set the Socket Server to the host and external port of the load balancer server.

  NOTE
  For Advanced Availability, the main server resides on background and stand-by servers.

  - HTTP
    - **Host Name or IP** -- Specifies the address of your server.
    - **External Port** -- Specifies the external port of your server.
      Default: 8070
  - Socket Server
    - **Host Name or IP** -- Specifies the address of your socket server.
    - **External Port** -- Specifies the external port of your socket server.
      Default: 10443
    - **Internal Port** -- Specifies the internal port of your socket server.

Default: 7005
- **Bind to IP** -- Specifies the IP where you want to bind the server.

- **Socket Proxy Server**
  Configures the Support Automation server in socket proxy mode. Use a Socket Proxy Server to off-load some of the CPU-intensive operations of Support Automation, such as encryption/decryption from the main server.
  Advanced Availability: This option is available only for the application server.
  Conventional: This option is available only for the secondary server.
  – Socket Configuration
    - **Main Server Host Name or IP** -- Specifies the address of the main server.
    - **Main Server Internal Port** -- Specifies the internal port of the main server.
      Default: 7005
    - **Main Server HTTP Port** -- Specifies the HTTP port of the main server. This field is available only for conventional configuration.
    - **External Port** -- Specifies the external port of the server.
      Default: 10444
    - **Bind to IP** -- Specifies the IP where you want to bind the server.
- **Message Routing Server**
  Configures the Support Automation server in message routing server mode. Use Message Routing Servers (MRS) to manage multiple Remote Control sessions that are based on the geographical location of the local server. Using MRS helps improve performance during assistance sessions.
  Advanced Availability: This option is available only for the application server.
  Conventional: This option is available only for the secondary server.
  – Socket Configuration
    - **External Port** -- Specifies the external port of the socket.
      Default: 10444
    - **Bind to IP** -- Specifies the IP where you want to bind the server.
- Tomcat Configuration
  a. **Tomcat Port**
    - Specifies the Support Automation Tomcat port.
      Default: 8070
  b. **Tomcat Shutdown Port**
    - Specifies the Support Automation Tomcat Shutdown port.
      Default: 8075

      NOTE
      When you change the main server Tomcat port, also change the port references in the server.properties file with tomcat server.xml.

Configuration Options

Configure the Config Options page with the following options:

- **Server Configuration**
  Select the configuration that you want to apply for this server from the drop-down list. Select Default to apply default CA Service Desk Manager configurations.
- Windows
  – **Start service when completed**
    Select this check box for the server to start when the configuration is complete.
  – **Start service when completed**
    Select this check box if you want the server to start when the configuration is complete.
  – **Put CA Service Desk Manager links in /usr/bin**

This option only works if you are at root. Selecting this option puts important CA Service Desk Manager links into /usr/bin.

– **Start Event converter daemons**
Select this option to enable the event converter daemons. This option allows you to create the CA Service Desk Manager tickets from either CA NSM World View or CA NSM Event Monitor.

Character Encoding (Linux)

The ICU converter provides the character encoding mapping that CA Service Desk Manager uses when transcoding character data between Unicode and the operating system character encoding.

NOTE
This configuration page is only available on Linux system.

Configure the Character Encoding page with the following options:

- **Use detected converter**
Auto-detection has determined the converter available for your Linux system. Leave selected if you want to use this converter.
- **Specify a converter**
The ICU converter name field provides feedback of the converter CA Service Desk Manager has determined best matches the operating environment. The converter also provides a means of overriding the auto-detected converter by specifying a converter name. The converter names are not case-sensitive.
Give special attention to two converter names, if these converters appear as auto-detected:
  – **US-ASCII**
  Appears when the following conditions occur:
    - The auto-detection could not determine a matching converter and defaulted to US-ASCII. In this case, the ICU Converter Explorer or an equivalent tool must be used to determine a converter best matching the character encoding of the operating system locale. Enter its name into the ICU converter name field.
    - The character set of the operating system locale only supports a limited 7-bit character encoding such as C, POSIX, US-ASCII, ISO646-US. In this case, we recommend that you select a different operating system locale which utilizes a character set encoding that supports a wider selection of characters. See your operating system documentation for details for changing the locale environment for the user or shell that is used to invoke the CA Service Desk Manager configuration program. A matching locale environment must be used for all subsequent accounts that are used to launch and operate CA Service Desk Manager.
- **Default HTTP character set**
Specifies the character set that is used on web pages that are sent from CA Service Desk Manager web servers. This value is specified in the web page HTTP character set declaration as the charset parameter in the HTTP Content-Type header, and is described in RFC 2616 Hypertext Transfer Protocol - HTTP/1.1.
The configuration application attempts to determine whether this value is an alias of the ICU Unicode converter that is specified in ICU Unicode converter name. If it is not, you can override the value.

Configure the CA SDM Environment

This section contains the following topics:

- How to Configure the Employee and Guest Interface
- How to Configure the Web Interface
- How to Configure Integrated Windows Authentication for CA SDM
- Search Engine Configuration
- How to Enable Auto-Failover
- Screen Reader Usage
- Deploy the Health Servlet on the Application Server

How to Configure the Employee and Guest Interface

This article contains the following topics:

You can use CA SDM to configure separate interfaces for employees and guests. You configure these separate interfaces through the Options Manager in the Administration tab. The following values control these interfaces:

- **employee_intf_incident_support**
  Displays the following values:
  – Request only
  – Incident only
  – Both Incident and Request
- **guest_intf_incident_support**
  Displays the following values:
  – Request only
  – Incident only
  – Both Incident and Request

  WARNING
  For a new installation, ITIL is configured by default value set to *Incident only*. If you are migrating from a previous non-ITIL configuration, though the options are installed, the values are set to *Request only*.

Configure the Employee Interface

You can configure the employee interface to display incidents, requests, or both.

Follow these steps:

1. Click the Administration tab.
   The Administration console appears.
2. Click Options Manager, Request Mgr.
   The Option List appears.
3. Click employee_intf_incident_support.
   The Options Detail page appears.
4. Change the Option Value field to one of the following values:
   – **Incident Only**
   (ITIL Default) Displays only Incident ticket types on the employee interface.
   – **Request Only**
   Displays only Request ticket types on the employee interface.
   – **Both Incident and Request**
   Displays both Incident and Request ticket types on the employee interface.
   Click Save.
5. Click Refresh to confirm your selections.
   The Options Detail is updated.
6. Close the Options Detail.

Configure the Guest Interface

You can configure the guest interface to display incidents, requests, or both.

Follow these steps:

1. Click the Administration tab.
2. Click Options Manager, Request Mgr.
3. Click guest_intf_incident_support.

4. Change the Option Value field to one of the following values:
   - **Incident Only**
     (Default) Displays only Incident ticket types on the guest interface.
   - **Request Only**
     Displays only Request ticket types on the guest interface.
   - **Both Incident and Request**
     Displays both Incident and Request ticket types on the guest interface.

   Click Save.
5. Click Refresh to confirm your selections.
   The Options Detail is updated.
6. Close the Options Detail.

How to Configure the Web Interface

This article contains the following topics:

When you install CA SDM, the *web interface* (commonly referred to as the *browser interface*) is automatically installed.

Select the web server depending on the CA SDM configuration. Consider the following use cases:

- Conventional: No additional action is required when the web server and the primary server are on the same computer. If they are installed on a different computer, install and configure both primary and secondary server.
- Advanced Availability: No additional action is required when the web server and the background server are on the same computer. But if they are on a different computer, install and configure both background and application server. The web interface for end users is never hosted from the background server. The web interface must be hosted only on the application servers.

Select the computer on which the web server resides depending on your CA SDM configuration:

- Conventional: Secondary server. Install this server after you install the primary server.
- Advanced Availability: Application server. Install this server after you install the background server.

> **WARNING**
> By default, Tomcat is the default web server. If you want to use IIS as your web server, manually configure the product and select IIS. For information, see the *Server Configuration Online Help*.

To configure the web interface, complete the following steps:

1. (Required) Enable the web engine on the secondary server on Windows or Non-Wndows .
2. (Required) Configure the web interface.
   You can then start the web interface.

Web Engines or Web Directors

Web engines help prepare web pages for the web client. All systems have one or more web engines. Each web engine connects to an object manager for processing all requests to CA SDM objects. Depending on your configuration type, CA SDM installs a default web engine on the following servers:

- Conventional: Primary server.
- Advanced Availability: All servers.

> **NOTE**
>
> You can modify the default web engine to suit your requirements.

Web directors are optional, and are used when there are two or more web engines installed on a single server. Web Director receives connection requests from users, selects a web engine to handle the request, and redirects the request to that web engine.

Add Web Engines or Web Directors

Web engines connect to an object manager for processing all requests to CA SDM objects. Web directors are optional, and are used when two or more web engines are installed on a single server. You can configure web directors on any server. Depending on the CA SDM configuration, CA SDM installs a default web engine on the following servers:

- Conventional: Primary server.
- Advanced Availability: All servers.

Follow these steps:

1. Select Systems, Configurations on the Administration tab.
   The Configurations List page opens.
2. Select the configuration to which you want to add the web engine or web director.
   The Configuration Detail page opens.

   NOTE

   :f you are changing the configuration for the first time, then create a configuration first. When you want to make a configuration change, always create or copy an existing one. This process allows you to revert to the previous configuration, if needed.
3. Select the Web Engines/Web Directors tab.
   The Web Engine/Web Directors List page displays the web engines and web directors that are configured for the server.
   – Conventional: A web engine exists by default on the primary server. You can add web directors to any server.
   – Advanced availability: A web engine exists by default on all servers. You can add more web directors on any CA SDM server.
4. Click Add Web Engine/Web Director.
   The Create New Web Engine/Web Director page opens.
5. Complete the following fields:

   WARNING

   Enter only English characters for all the input fields for any localized language.

- **Host Name**
  Specifies the host name for the web engine or web director. You can click Search to look up for the servers.
  For an advanced availability configuration type, the host name is read-only and is automatically populated based on the host name you specified while creating the configuration.
- **Type**
  Specifies if you are configuring a web engine or web director. Based on the option that is selected, the relevant fields are automatically populated.
  – Select Web Engine if you want to configure a web engine.
  – Select Web Director if you want to configure a web director.

  NOTE

  Ensure that you have selected the appropriate option. You cannot edit the process type after you have saved the configuration.
- **Web Director**
  Specifies the web director that is assigned to the web engine. You can click Search to look up for the web directors added to the server.

NOTE

When implementing any web engine load-balancing scheme, SSL-Login, or both, at least two web engines must be assigned to the same web director.

- **CGI Name**
  Specifies the unique CGI name for the web engine. It is the name of an actual CGI executable when IIS or Apache is used as the HTTP server; it is a servlet parameter when Tomcat is used as the HTTP server.
  Examples: (web engines) pdmweb1, pdmweb2, (web directors) pdmweb_d1, and pdmweb_d2.
  Default: pdmweb.exe (The CGI name must be unique).
- **CGI Port Number**
  Specifies the port on which CA SDM web clients can connect. The CGI port number is the same port on which the tomcat server is running.
  Default: 8080
- **Protocol**
  Specifies the protocol for accessing the web engine.
  - Select HTTPS if the web engine is configured to handle all CA SDM web-client user authentication requests.
  - Select HTTP if the web engine is configured to handle all web client non-user authentication requests (after user is authenticated through the secure login web engine).
- **Record Status**
  Specifies whether the web engine or web director is active or inactive.

  NOTE

  Before setting the record status of a web director to inactive, remove the link between the web director and the associated web engines.
- **Object Manager**
  Specifies the object manager that you want to assign to the web engine.
  - **Default**
    Specifies that the default object manager is assigned to the web engine.
  - **ANY**
    Specifies that the web engine can connect to any available object manager with more willingness value. Willingness value is the availability of the server to accept new clients. A willingness value of zero means that the web engine does not accept any sessions.
  - **Choose**
    Allows you to specify an object manager for the web engine. Selecting this option provides you the option to add multiple object managers or aliases to the configuration.

Click Save.

The web engine or web director that you added appears in the Web Engines/Web Directors List.

Web Director List

Web directors are used when there are two or more web engines installed on a single server. Web Director receives connection requests from users, selects a web engine to handle the request, and redirects the request to that web engine.
All

The Web directors List page opens that displays all the web directors that are configured for the server.

Configure the Web Interface

If the default web interface configuration specified does not meet the CA SDM installation requirements, modify the *web.cfg* file. The *web.cfg* is located in the *installation-directory* \bopcfg\www. Edit the file using a text editor, such as Notepad or WordPad.

Each entry in the file consists of a single line containing a property name, optionally followed by a value. Lines beginning with a pound sign (#) are treated as comments and are ignored.

Start the Web Interface

Verify that the Daemon Server services and the database server are started before starting the web interface. If the secondary server is configured, the Remote Daemon Proctor service must be started before starting the primary server service. To start the services on Windows and Linux OS, perform the following steps:

- (Windows): Select Start, Control Panel, Administrative Tools, Services). Right-click the CA SDM Remote Proctor and select Start.
- (Linux): Run the *pdm_init* command to start the primary server and *pdm_proctor_init* to start the secondary server.

Verify that the services are started and then start the web interface. You can also start the web interfacefroman internal web site. If you are on the primary or secondary server system and are using the Internet Information Services (IIS), perform the following steps:

- Select Start, All Programs, CA, Service Desk Manager, Service Desk Manager Web Client.
- On a system that is not a primary or a secondary server, open a web browser and enter the following URL address:

  ```
  http://servername:8080/CAisd/pdmweb.exe
  ```

  In this URL, *servername* is the name of the computer that is hosting the CA SDM web server.
- On a System that is not a primary or secondary server with IIS as the web server, open a web browser and enter the following URL address:

  ```
  http://servername/CAisd/pdmweb.exe
  ```
- To start the web interface from an internal website, add */pdmweb.exe* to the URL for your web pages. Use the following sample HTML code as a guide:

  ```
  A HREF=http://<server-name>:<port-no>/CAisd/pdmweb.exe
  ```

  In this URL, *server-name* identifies your computer and *port-no* is the port on which your web server is listening.

  NOTE
  If the Internet Explorer browser security is set to high, a warning message appears after starting the web interface. To avoid this message, add the website to your trusted sites or lower your security settings.
  In advance availability, the default web interface is readily available with the default installation and configuration.

Record Locking Behavior in the Web Interface

Editing a database record using the web interface, the user is given an exclusive lock of two minutes. You can change the default time by using the ExclLockSeconds property in the web.cfg file.

The following conditions apply when a database record is updated:

- If a user *can* edit and submit changes within an allotted time, the changes are included the database.
  When the database record is locked, other users (both web and non-web users) can view the record, but cannot edit the record. If they try to edit the record when it is locked, an error message appears.
- If a user *cannot* edit and submit changes within an allotted time, the record lock automatically drops and other users can edit the record.
  When the user submits the updates, time stamps are checked to ensure that nobody else has changed the record:
  - If the record has not been changed since the exclusive lock was dropped, the user updates are saved to the database.
  - If another user has edited the record after the lock expires, the user receives an error response and the changes are not saved. The user must restart the edit process and must reapply the changes.

Enabling the CAPA Help

CA Productivity Accelerator (CAPA) provides context-sensitive in-application performance support for the CA SDM analyst web forms. The CAPA player package directly integrates with the Context Help menus of CA SDM analyst web pages. The integration of CAPA Help content with CA SDM is achieved through the help menu scripts. These scripts launch the CAPA recorded Help content from the CA SDM Help menu drop-downs and right-click context help.

NOTE

This section does not include information about the CAPA recording tool, obtaining and configuring the CAPA Help content, and installing the Smart Help.

CA SDM provides a unique identifier to recognize each web form uniquely. If an identification element is present in the top frame of the CA SDM, the CAPA recording tool recognizes the web application form. Once the top frame is recognized, all sub frames are considered to be part of this application. The sub frames that are loaded from a different domain must also be identified. Otherwise, these sub frames are not considered a part of CA SDM. To enable CAPA help, prepare the help content and configure the options to launch the CAPA help.

NOTE

To modify the HTML files, use the invoke methods from the ias_helper.js files for the CAPA recorder to recognize the html forms.

CAPA Help is not supported on the CA SDM Employee, Customer, or PDA interfaces.

Follow these steps:

1. Open the capa.properties file from the following CA SDM directory:
   `%NX_ROOT/bopcfg/www/wwwroot`
2. Change the following values in the file:
   - **Enable_Alerts** = 0. This option verifies and displays the javascript alert messages in a browser console when the CA SDM web page loads.
3. Save the file.
4. Go to the web.cfg file in the CA SDM directory, add DebugScript=1, and restart the web engine.
   The recorded content is prepared, published, and deployed on the CAPA server.
5. Open the capa.properties file and change the following values:
   - **Enable_For_Recording** = 0
   - **Show_Learn_Links** = 1. This value displays the CAPA Help option in the CA SDM Help Menu and in the right-click Help context menu.
   - **Server_Name** = name of the server where CAPA published content is deployed.
   - **Server_Port =** CAPA server port where the CAPA published content is deployed. For example, the TOMCAT port.
   - **Virtual_Dir** = virtual directory where the CAPA published content is deployed on the CAPA server.
   - **Namespace** = application context namespace. For example, app.SDM.Project123;en.
   - **Enable_Alerts** = 0
6. Save the file.
   You are ready to launch the CAPA Help from the CA SDM analyst web forms.

How to Configure Integrated Windows Authentication for CA SDM

As a system administrator, you can authenticate CA SDM users through an existing external authentication method. The following scenario describes configuring Integrated Windows Authentication (IWA) to authenticate CA SDM users through Microsoft Active Directory (AD). Configure CA SDM access types like administrators to use IWA. The users of the selected access types are directly authenticated through Active Directory.

Follow these steps:

1. Verify the Prerequisites
2. Configure Internet Information Services (IIS) for IWA.
3. Configure CA SDM for IWA.
4. Test IWA

Verify the Prerequisites

Verify the following requirements before you configure the external authentication for CA SDM:

- You have configured servers for CA SDM.
- You have installed IIS 7.0 on the following server depending on your configuration:
  - Conventional configuration: primary server.
  - Advanced Availability: application server.
- You have configured CA SDM to use IIS 7.0.
- You have the list of access types for which the external authentication is required. For example, CMDB Administrator, Employee, and CMDB User.

Configure IIS for IWA

You configure the IIS server to enable IWA. The configuration changes the authentication mode of IIS from Anonymous to Windows Authentication. After the configuration, the server starts redirecting authentication requests to the external source.

Follow these steps:

1. Log in as administrator to the following server depending on your configuration:
   - Conventional: Primary or secondary server.
   - Advanced Availability: Background server.
2. Open the Administrative Tools, Internet Information Services Manager.
3. Expand the Server Name node, where Server Name is the name of the CA SDM server.
4. Expand Sites, Default Web Site, CAisd node.
   The CAisd Home page opens.
5. Double-click the Authentication icon.
   The Authentication Settings page opens.
6. Disable the Anonymous Authentication by clicking the existing Status value.
7. Enable the Windows Authentication by clicking the existing Status value.
8. Restart IIS to apply the changes.

You have configured IIS for IWA.

Configure CA SDM for IWA

You configure the IWA for each of the selected access types. After an access type is configured for the external authentication, CA SDM externally authenticates the contacts of that access type.

Follow these steps:

1. Log in to CA SDM as a system administrator.
2. Select the Administration tab, Security and Role Management, Access Types.
   The Access Type List page opens.
3. Click the access type for which the external authentication is required.
   The Update Access Type form opens.
4. Select the Web Authentication tab and select the Allow External Authentication check box.

5. Select an appropriate value from the Validation Type drop-down list. The following values require description:
   - **No Access**Specifies that the selected access type is not allowed to access.
   - **Open**Specifies that the access is open to all.
   - **OS**Specifies that the CA SDM server operating system credentials are required to access CA SDM. If you have configured EEM, the login request would be redirected to the EEM server.
   - **PIN**Specifies that the value of selected CA SDM contact record field is required for the authentication. For example, if you select Pin as Validation Type and select Contact_Number from the PIN Field drop-down, the contact number of the user is required for authentication.

     NOTE
     The Validation Type is not used for the IWA validation, but it is used when the user logs in through the CA SDM login form. The CA SDM Login page appears only when the user clicks the Logout link or when the credentials do not match with IIS credentials.
6. Click Save.
   To configure the external authentication for other access types, repeat steps 1 through 6.

Test IWA

You test the IWA with a user id having the corresponding contact record in CA SDM. A successful login indicates that you have configured the IWA successfully.

Follow these steps:

1. Log in to the following server depending on your configuration, with a user id having the corresponding contact record in CA SDM:
   - Conventional: Primary or secondary server.
   - Advanced Availability: Application server.

     NOTE
     If the LDAP options have been installed in the Options Manager, you can also create new contact in CA SDM from any Active Directory contact.
2. Launch the browser and enter the CA SDM URL.
   The Service Desk Home page opens.

Search Engine Configuration

This article contains the following topics:

The EBR Search Engine installs with CA SDM by default.

Knowledge Reindex Utility

The Knowledge reindex utility, pdm_k_reindex.exe, is located under the Knowledge Management installation directory.

NOTE
Reindexing the documents in the knowledge base can be a time-consuming operation based on your database size. It is recommended that you run the Knowledge Reindex utility after all the changes have been added. For advanced availability configuration, you cannot execute the knowledge reindex utility during the failover process.

WARNING
On Non-Windows, the LIBPATH must be set before running several CA SDM utilities. Use pdm_task to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

Follow these steps:

1. Open the command prompt.

2. To run the knowledge reindex, enter the following command:

For example:

```
pdm_k_reindex
```

The following options are available with this command.

- **-D**
  Defines the debug mode, such as printing to the command window.
- **-v**
  Defines the verbose mode, such as printing to the stdlog file.
- **-i**
  Does not create table indexes in the reindex table after reindexing.

  NOTE
  Parameters with a dash as a prefix, such as " - D", must precede other parameters that do not have this prefix.

The other option is as follow:

- ***File*:reindex.txt**
  Documents are reindexed to the specified file.
- **+i**
  Creates the indexes of the reindexed table only, which is the search table after reindexing. The old indexes are dropped before reindexing.
- **+t**
  Switches the names of search and reindex tables only.

  NOTE
  A “+” prefix denotes only this parameter applies.

- **sdtout**
  Defines the frequency of statistic appearing in the command window. By default the knowledge reindex utility provides statistics into the command window for every 1000 documents processed. However, sometimes statistics are required to be provided more often. Use the following parameter:

  ```
  pdm_k_reindex -i sdtout:10
  ```

  In this case, statistics display in the command window for every ten documents.

The documents are reindexed in the knowledge base.

New Tags to Configure a Domserver for the Crawler Surface

You must have a dedicated server for the Crawler Surface depending on your CA SDM configuration and data size for indexing. The dedicated server minimizes the impact on the production environment:

- Conventional: Primary or secondary server
- Advanced Availability: Application server
  Note: For more information on how to configure CA SDM server, see Server Configuration Utility.

The following two new tags are introduced in the crawler_surface_config.xml file to take care of the indexing:

- A new tag has been introduced in crawler_surface_config.xml file root\general_configuration\sdm_domsrvr_name to let the customer mention the name of a dedicated domserver instance for the Crawler Surface.
  Default: domsrvr
  This value is a default domserver (object manager) instance name.
- Another new tag has been introduced in the crawler_surface_config.xml file *sharepoint_properties_file* at the *root \general_configuration\.*
  Default: sharepoint

This value is the name of *sharepoint.properties* file available by default at NX_ROOT\CATALINA_BASE_FS\lib.

How to Enable Auto-Failover

Server health monitoring observes how a server reacts to the operating load and tracks its responsiveness to the client requests. The purpose of this monitoring is to prevent server failures by ensuring that the server always retains sufficient capacity to conduct the required tasks. Most of the standard health monitoring tools support remote server monitoring through the HTTP and HTTPS protocols.

CA SDM automatic failover feature exposes the following HTTP-based standard interfaces:

- Health Monitoring Interface: HTTP(S) interface to monitor the background server health. It also makes reliable failover decisions to initiate the failover on a chosen standby server in case of any disruptions to the background server's availability or ability to conduct required tasks.
- Failover Initiating Interface: HTTP(S) interface to promote the chosen standby server as the new background server without causing any service disruption.

Follow these steps:

1. Install Apache Tomcat on the background and all standby servers.

   NOTE
   Ensure that Tomcat is using JRE version same as that of CA SDM. Also, verify that Tomcat is not using a port number that is configured for CA SDM components.
2. (Optional) Configure SSL on the Tomcat servers that you have installed. For more information about configuring SSL, see How to Configure SSL Authentication.
3. Log in to the background server.
4. Deploy the health servlet. Complete the following steps:
   a. Copy the HealthServlet.war file from the $NX_ROOT/samples/HealthServlet folder to the *TOMCAT_HOME*/webapps folder.
   b. Restart Tomcat.

   The HealthServlet.war file is deployed in the webapps folder. To confirm the deployment, verify that the HealthServlet folder is created in the same webapps folder.
   After the successful deployment, the health servlet is ready to perform the health checks. It includes checking the status of the SLUMP and health of the CA SDM processes that are defined in the health.xml file. Find the health.xml file in the following location:
   ```
   TOMCAT_HOME/webapps/HealthServlet/WEB-INF/classes
   ```
5. (Optional) Customize the health.xml based on your organization needs. For example, you want to monitor the webengine process. Add the process in the health.xml file with the correct tagname, as defined in CA SDM. Complete the following steps to find the tagname:
   a. Open the pdm_startup.i and pdm_startup files from the $NX_ROOT/pdmconf directory.
   b. Look for the process that you want to monitor in both the files.
   c. Find the corresponding tagname by matching the variables in both the files.
      For example, webengine process is defined in the pdm_startup.i file as follows:
      ```
      #define WEBENGINE(_TAG,_HOST,_SLUMP_NAME,_DOMSRVR, _CFG, _WEBDIRECTOR, _RPC_NAME)
      ```
      The webengine process is defined in the pdm_startup file as follows:
      ```
      WEBENGINE(webengine, $NX_LOCAL_HOST, web:local, domsrvr, $NX_ROOT/bopcfg/www/web.cfg, "", "rpc_srvr:
      %h")
      ```
      From the example, we can find out that the tagname for webengine process is webengine.

      WARNING
      For creating a new process, the existing process is commented out in the pdm_startup file and new entries are added. Ensure that you look for the tagname in the new process entries.

WARNING
If you modify health.xml, ensure that the XML does not have any errors and that you restart Tomcat to reflect the changes that are made to the XML.

6. Perform steps 4 and 5 for all standby servers.
7. Configure the chosen third-party tool to monitor the health of the background server at regular intervals. To monitor the health, use the following HTTP URL:

   ```
   http(s)://Background_server_name:port_number/HealthServlet/GetHealth
   ```

8. Configure the chosen third-party tool to initiate a failover logic when the background server health degrades. We recommend that you to configure the failover logic so as to promote the standby server as the new background server. Use the following failover servlet in the failover logic:

   WARNING
   We recommended you to configure the failover servlet on SSL with the access privileges given only to predefined users. Follow this recommendation for configuring third-party tools to initiate failover.

   ```
   http(s)://Standby_server_name:port_number/HealthServlet/FailoverServlet
   ```

   You have enabled auto-failover.
9. After the successful configuration, the third-party tool starts monitoring the background server health using the health servlet URL.
   - Each server type has its own set of processes. If the SLUMP and all the CA SDM processes are working properly, the third-party tool receives an HTTP 200 response from the background server with a predefined payload:

     ```
     AA-Server-Status: All OK!
     AA-Server-Role: BG
     ```

   - If a SLUMP or any of the CA SDM processes (listed in health.xml) stop working and cannot resume, the third-party tool receives an HTTP 503 response from the background server with a predefined payload as:

     ```
     AA-Server-Status: NOT OK!
     AA-Server-Role: BG
     ```

10. If the HTTP 503 response is received, the third-party tool automatically initiates the failover logic.

Screen Reader Usage

You can modify system behavior for optimal use with a screen reader for visually impaired users.

To enable screen reader usage:

1. Enable screen reader usage on the main menu from View, Preferences, and select the Using Screen Reader option.

   WARNING

   You must log off and log back on for this change to take effect.
2. From the Help menu, select Screen Reader Usage for an overview of using CA SDM with a screen reader.

Deploy the Health Servlet on the Application Server

Server health monitoring prevents server failures and downtimes by ensuring that the server always retains sufficient capacity to conduct the required tasks.

NOTE

Ensure that your health servlet is installed on a separate instance of Tomcat server than CA SDM.

To deploy health servlet on the CA SDM Application Server, complete the following:

Follow these steps:

1. Install Apache Tomcat on the CA SDM Application Servers.

   NOTE

   Ensure that Tomcat is using JRE version same as that of CA Service Desk Manager. Also, verify that Tomcat is not using a port number that is configured for CA SDM components.
2. (Optional) Configure SSL on the Tomcat servers that you have installed.
   For more information about configuring SSL, see How to Configure SSL Authentication.
3. Log in to the application server and deploy the health servlet by completing the following steps:
   a. Copy the HealthServlet.war file from the $NX_ROOT\samples\HealthServlet\HealthServlet.WAR to the *Apache Tomcat Webapps directory (TOMCAT_HOME*/webapps).
   b. Restart Tomcat.
      The HealthServlet.war file is deployed in the webapps folder. To confirm the deployment, verify that the HealthServlet folder is created in the same webapps folder.
      After the successful deployment, the health servlet is ready to perform the health checks. It includes checking the status of the SLUMP and health of the CA SDM processes that are defined in the health.xml file. Find the health.xml file in the following location:

      ```
      TOMCAT_HOME/webapps/HealthServlet/WEB-INF/classes
      ```
4. (Optional) Customize the health.xml based on your organization needs. For example, you want to monitor the webengine process. Add the process in the health.xml file with the correct tag name, as defined in CA SDM. Complete the following steps to find the tag name:
   a. Open the pdm_startup.i and pdm_startup files from the $NX_ROOT/pdmconf directory.
   b. Look for the process that you want to monitor in both the files.
   c. Find the corresponding tag name by matching the variables in both the files.
      For example, webengine process is defined in the **pdm_startup.i** file as follows:

      ```
      #define WEBENGINE(_TAG,_HOST,_SLUMP_NAME,_DOMSRVR, _CFG, _WEBDIRECTOR, _RPC_NAME)
      ```
   d. The webengine process is defined in the **pdm_startup** file as follows:

      ```
      WEBENGINE(webengine, $NX_LOCAL_HOST, web:local, domsrvr, $NX_ROOT/bopcfg/www/web.cfg, "", "rpc_srvr:
      %h")
      ```

      WARNING

      For creating a new process, the existing process is commented out in the pdm_startup file and new entries are added. Ensure that you look for the tag name in the new process entries.

      WARNING

      If you modify health.xml, ensure that the XML does not have any errors and that you restart Tomcat to reflect the changes that are made to the XML.
5. Configure the chosen third-party tool to monitor the health of the application server at regular intervals. To monitor the health of the server, use the following HTTP URL:

   ```
   http(s)://application_server_name:port_number/HealthServlet/GetHealth
   ```
6. After the successful configuration, the third-party tool starts monitoring the CA SDM application server health using the health servlet URL.
   For more information on how to deploy health servlet on the background and standby servers, see How to Enable Auto-Failover.
7. Each server type has its own set of processes. If the SLUMP and all the CA SDM processes are working properly, the third-party tool will receive an HTTP 200 response from the Application server with a predefined payload, as follows:

```
AA-Server-Status: All OK!
AA-Server-Role: AP
```

If a SLUMP or any of the CA SDM process (listed in health.xml) stops working and cannot resume, the third-party tool receives an HTTP 503 response from the application server with a predefined payload, as follows:

```
AA-Server-Status: NOT OK!
AA-Server-Role: AP
```

You have successfully deployed the health servlet for the CA SDM Application Server.

Managing Servers

This article talks about CASDM server components.

This article contains the following topics:

The CA SDM installation consists of one or more server components that an administrator can manage. The number of servers depends on the CA SDM configuration:

- Conventional: One primary server and one or more secondary servers.
- Advanced Availability: One background server, one or more standby servers, and one or more application servers.

After you install CA SDM, configure each computer that runs the CA SDM components. You can run the server configuration as part of the installation process, or you can run it later. The CA SDM services must be restarted after you change the server configuration.

How to Change a Server Configuration

As an administrator, you can configure the servers for CA SDM installation. The number and type of servers depend on the CA SDM configuration. The initial configuration occurs as part of the CA SDM installation process.

> **NOTE**
>
> A change in the system environment can require changes to the server configuration. For example, the database management system or integration changes with a web server (EEM server or Tomcat)

Follow these steps:

1. Log in to the server you want to configure again.
2. From the Windows Start menu, select Programs, CA, CA SDM, Configuration.
   The CA SDM Configuration utility opens.
3. Complete the utility fields, and click Next.
   The right panel changes to show the appropriate fields for the link that is highlighted in the navigation pane on the left.
4. Continue following the on-screen instructions and click Finish.
   The server configuration is changed.

How to Configure TCP/IP

You can change the default TCP Internet Protocol (TCP/IP) setting on one or more servers. This setting cannot be forced on the client if it is not supported on the server.

The TCP/IP setting is controlled by using the NX.env file, which is found in the $NX_ROOT directory. Use a text editor such as WordPad for editing this file. The following option controls the TCP/IP setting:

```
NX_PROTOCOL_ONLY=mode
```

where *mode* can be one of the following values:

- **IPv4**
  In IPv4 mode, the system opens up sockets for slump processes that listen to IPv4 traffic.
- **IPv6**
  In IPv6 mode, the system opens up sockets for slump processes that listen to IPv6 traffic.
- **Mixed**
  In mixed mode, the system opens up sockets for slump processes that listen to both IPv4 and IPv6 traffic. Depending upon your CA SDM configuration, you can configure mixed mode in the following circumstances:
  – Conventional: The Secondary servers that use a different Internet Protocol from the primary server or each other.
  – Advanced Availability: The Application servers that use a different Internet Protocol from the background server or each other.

    NOTE
    If IPv4 and IPv6 hosts coexist on the network, ensure that the appropriate transition strategies, tools, and mechanisms to support these technologies are in place before you change the server configuration.

Example

```
NX_PROTOCOL_ONLY=ipv4
```

Activity Log Security

The Activity Log security option prevents end users from updating fields on an activity log. You can select the internal option to prevent a customer from seeing the log.

Activity Log security affects activity logs from the following ticket types:

- Request
- Change order
- Issue
- Incident
- Problem

Enable Activity Log Security

You can enable Activity Log Security from the Options Manager in the Administration tab.

Follow these steps:

1. Click the Administration tab.
2. Click Options Manager, Request-Change-Issue.
3. Click activity_log_security.
4. Click Edit, and select one of the following Option Values:
   – **Editable**
   (Default) Allows all fields on the activity log to be editable through the web interface or web services.
   – **Write Protected**
   Displays the activity log as read-only. If you select the internal option, only internal users can edit the log which cannot be viewed by the customer.

     NOTE

     If the security option is enabled and you try to edit the log using the web interface or external web services, an error message displays that the activity log is read-only

   Click Save.
5. Click Refresh to confirm your selections. Close Window.
   Activity log security is enabled.

WARNING
The activity_log_security option cannot be uninstalled. You can only change the value of the option to Editable or Write Protected in Options Manager, Request-Change-Issue.

Impact on Web Screen Painter

The Activity Log Security feature, $NX_ACTIVITY_LOG_ SECURITY, includes the following attributes (time_spent, time_stamp, and description) for the alg, chgalg, issalg objects in majic.

Example: $NX_ACTIVITY_LOG_SECURITY for Object alg in cm.maj

In this example, for object alg in cm.maj, $NX_ACTIVITY_LOG_SECURITY appears on the three attributes:

```
time_spent DURATION $NX_ACTIVITY_LOG_SECURITY {ON_POST_VAL update_cr_timespent( call_req_id ) 50 ;
 } ;
time_stamp DATE $NX_ACTIVITY_LOG_SECURITY { ON_NEW DEFAULT NOW ; } ;
description STRING  $NX_ACTIVITY_LOG_SECURITY;
```

In Web Screen Painter, *Updatable only for new record* field is disabled when the value of the keyword evaluates to WRITE_NEW.

Improve Performance With Browser Caching

The CA SDM web interface uses many JavaScript, style sheets, and image files, which can be fairly large and can affect performance.

To improve performance of the web interface, set up your HTTP server so that the user browser caches these files and loads only once a day.

The web interface performance improves.

NOTE
The default installation automatically configures caching for Apache and IIS; however, you can configure it manually.

Configure the Microsoft Internet Information Server

You can configure Microsoft Internet Information Server (IIS) to notify the browser that files loaded from the CA SDM directory expire one day after loading. The browser queries the server for these files only once a day, regardless of how many times they are used.

Follow these steps:

1. Launch the Internet Services Manager application (for Windows 2000 and XP, select Start Menu, Windows Administrative Tools, Internet Services Manager).
2. Navigate to the CA SDM file folder, which is typically CAisd:
   a. Click the plus sign that is adjacent to the server running the CA SDM web interface.
   b. Click the plus sign that is adjacent to Default Web Site.
   c. Scroll down to CAisd.
3. Right-click the CAisd folder, and select Properties.
   The Properties page appears.
4. Click the HTTP Headers tab.
5. Select the Enable Content Expiration check box.
6. Select the Expire After option, enter 1 into the text field, and select a day from the drop-down list.
7. Click OK.
   The properties are saved and the changes take effect immediately.

Configure Apache

You can configure Apache to notify the browser that files loaded from the CA SDM directory expire one day after loading. This configuration means that the browser queries the server about these files only once per day, regardless of how many times they are used.

You configure Apache by updating a text configuration file. The default installation modifies your active configuration file in the apache conf directory (typically httpd.conf) to contain the statement:

```
Include installation-directory/bopcfg/www/CAisd_apache.conf
```

Installation-directory must be replaced with a full path. On Windows, this path is typically c:\Program Files\CA\CA SDM. On UNIX, replace *installation-directory* with the value of $NX_ROOT.

The file CAisd_apache.conf, which is referenced by the Include statement, contains the following text. Again, *installation-directory* is replaced with the full path as it was in the Include statement.

```
<IfModule mod_alias.c>
    Alias /CAisd installation-directory/bopcfg/www/wwwroot/
    <IfModule mod_expires.c>
           <Directory installation-directory/bopcfg/www/wwwroot>
              ExpiresActive   On
              ExpiresDefault  "access plus 1 day"
           </Directory>
    </IfModule>
</IfModule>
```

To configure Apache manually for browser caching of the CA SDM files, include statements similar to those in CAisd_apache.conf in your Apache configuration file. You can either add them directly to the file, or add an Include statement referencing a separate file, like the default installation.

Changes to Apache configuration files take effect only after you recycle Apache.

Clear the Cache

If you change a JavaScript, image, style sheet, HTML, or help file loaded by the HTTP server itself, you must instruct the users to clear their browser cache.

> **NOTE**
> For changes to HTMPL files to take effect, you must either recycle the web engine or use the pdm_webcache utility. In a development environment, you can avoid this task by specifying the configuration file property SuppressHtmplCache.

To clear the browser cache for Internet Explorer, perform the following:

Follow these steps:

1. Select Tools, Internet Options.
   An Internet Options dialog appears.
2. Click Delete Files.
   A confirmation window appears.
3. Click OK.
   The browser cache is cleared

To clear the browser cache for Firefox, perform the following:

1. Select Tools, Clear Private Area.
2. Click the Clear Private Data Now button.
   The browser cache is cleared.

Add a CA SDM Server

If you want to install a new server in your CA SDM deployment, you must first add the corresponding server record before you configure it.

Follow these steps:

1. Log in to the following server, depending on your CA SDM configuration:
   – Conventional: Primary server
   – Advanced availability: Background server
2. Select System, Servers from the Administration tab.
3. Click Create New to add a server record for the following server, depending on CA SDM configuration:
   – Conventional: Secondary server
   – Advanced availability: Application or standby server
4. Complete the fields as appropriate for the server.
5. Click Save to add the server detail.

Create Server Fields

The following fields appear when you create or update a server:

- **Host Name**
  Specifies the local host name of the server. The local host name is stored in the usp_servers table in local_host column.

  WARNING
  Ensure that host name is entered as case-sensitive in the usp_servers table.
- **Attachment Servlet Path**
  You must specify the fully qualified domain name of a server using this field:
  http://<host>:<port>/CAisd/UploadServlet
  Where <host> is the fully qualified domain name of a server.
  We recommend that you configure this field.
- **Time Zone**
  Specifies the time zone where the server is located. This time zone value is used to trigger events in the application. This value is used only if the Use End User Time Zone option is not selected, or if no time zone is specified for the service type.
- **Record Status**
  Indicates the state of the server. Active status indicates that the server is a part of the CA SDM deployment.

  WARNING
  If you have inactivated any server, it is recommended not to start CA SDM services on that server. This action may result in unexpected behaviour.
- **Server Type**
  Specifies the type of server that you want to configure. Following server types can be selected, depending on your CA SDM configuration:
  – Advanced Availability: Application or standby server
  – Conventional: Secondary server
- **Configured**
  Available only for advanced availability configuration. This field indicates the state of the configured server. The default value of this field is No. The value is updated to Yes after you successfully run pdm_configure on that server. If you edit any of the automatically entered field values of a server record, the Configured field turns to No.

How to Start the CA SDM Servers

This article contains the following topics:

Depending on your CA SDM configuration, complete the following actions:

- Start the CA SDM servers in conventional configuration
- Start the CA SDM servers in advanced availability configuration.

The following table describes the processes that start automatically after you start the CA SDM servers:

NOTE
In this table, the Notification Manager process applies only to the Windows environment and the Default DB process applies only to the Non-Windows environment.

Process	Description
Daemon Agent (pdm_proctor_nxd)	The daemon agent responsible for starting the managed daemons
Daemon Monitor (pdm_d_mgr)	Monitors daemon processes
BOP Virtual DB (bpvirtdb_srvr)	BOP virtual database server
Data Dictionary (ddictbuild)	Rebuilds data dictionary each time the system is started -- this runs and then goes away
KPI Daemon (kpi_daemon)	Manages the collection, organization, and storage of KPI data.
Oracle agent (orcl_agent)	Agent for Oracle database -- many instances, depending upon load
Oracle DB (orcl_prov_nxd)	Oracle database provider
SQL agent (sql_agent) (runs only if you are using MS SQL)	Agent for SQL Server database -- many instances, depending upon load
SQL DB (sql_prov_nxd) (runs only if you are using MS SQL)	Microsoft SQL Server database provider
Message Dispatcher (sslump_nxd)	Dispatches messages
Notification Manager (bpnotify_nxd)	Manages notifications
Object Engine (domsrvr)	CA SDM Object Manager
Report Manager (pcrpt_nxd)	PC reporting
Software Version Control (pdm_ver_nxd)	Manages versions of specified system components
Method Engine (spel_srvr)	Spell code interpretation server
Text API (pdm_text_nxd)	Text API daemon for email and CA NSM interfaces
Timed Events/Notifications (animator_nxd)	Timed events and notifications
User Validation (boplgin)	User account validation
User Authentication (bopauth)	User account authentication
Web Engine (webengine)	Runs the engine for the web client
Archive Purge Daemon (arcpur_srvr)	Manages the background Archive and Purge processing
BU Daemon (bu_daemon)	Handles FAQ Rating calculation for Knowledge Documents
DB Monitor (dbmonitor_nxd)	Monitors CA common tables for changes
EBR Daemon (bpebr_nxd)	Handles knowledge search requests
EBR Idx Daemon (bpeid_nxd)	Handles EBR keyword indexing/re-indexing
KRC Daemon (krc_daemon)	Manages statistical calculations and notifications for the Knowledge Report Card
KT Daemon (kt_daemon)	Manages Knowledge Documents (KD approval process, permissions, notifications, and so on)
LDAP virtdb (ldap_virtdb)	Agent for communication with LDAP Servers
Mail Daemon (pdm_mail_nxd)	Handles outbound email notifications

Mail Eater (pdm_maileater_nxd)	Handles inbound email notifications
MDB Registration Server (mdb_registration_nxd)	Agent for handling MDB registration requests
PDM RPC (PDM_RPC)	Manages the Web Services requests
Repository Daemon (rep_daemon)	Handles attachment repositories
Spell checker (lexagent_nxd)	Handles spell checking requests
Time-to-Violation (ttv_nxd)	SLA Violation forecaster
tomcat controller (pdm_tomcat_nxd)	Manages Tomcat services

Start the CA SDM Servers in Conventional Configuration

Start the servers in the following order:

1. Start the secondary server.
2. Start the primary server.

Start the Secondary Server

If your installation includes one or more secondary servers, you must start the secondary servers prior to starting the primary server.

Follow these steps:

- (Windows) Select Services from the Control Panel, select the CA SDM Remote Proctor service and click Start.
- (Non-Windows) Use pdm_init from the command line.

Start the Primary Server

Every CA SDM installation has one primary server that handles basic product functionality.

> **WARNING**
> If your installation includes one or more secondary servers, you must start the secondary servers before starting the primary server.

Follow these steps:

- (Windows) Select Control Panel, CA SDM Server service and click Start. You can start the service manually each time you need it, or you can configure it to start automatically like any other Windows service.
- (Non-Windows) Use pdm_init from the command line.

Start the CA SDM Servers in Advanced Availability Configuration

Start the servers in the following order:

> **NOTE**
>
> (Windows) Select Control Panel, CA SDM Server service and click Start. You can start the service manually each time you need it, or you can configure it to start automatically like any other Windows service. (Non-Windows) Use pdm_init from the command line.

1. Start the background server.
2. Start the standby and application servers (in any order).

How to Stop the CA SDM Servers

This article contains the following topics:

Depending on your CA SDM configuration, do one of the following actions:

- Stop the CA SDM servers in conventional configuration.
- Stop the CA SDM servers in advanced availability configuration.

Stop the CA SDM Servers in Conventional Configuration

You stop the following servers:

- Stop the primary server.
- Stop the secondary server.

(Conventional Configuration) Stop the Primary Server

You can stop the primary server in the Windows or UNIX environment.

(Windows), follow these steps:

1. Select Services from the Control Panel.
2. Select the CA Service Desk Manager Server service and click Stop.
   The primary server is stopped. The processes or daemons on the secondary servers are also stopped.

(UNIX), follow these steps:

- Enter the following command in the command prompt:
  ```
  pdm_halt
  ```
  The primary server is stopped. The processes or daemons on the secondary servers are also stopped.

(Conventional Configuration) Stop the Secondary Server

You can stop the secondary server in the Windows or UNIX environment.

(Windows) Follow these steps:

1. Select Services from the Control Panel.
2. Select the CA Service Desk Manager Remote Proctor service and click Stop.
   The secondary server is stopped.

(Non-Windows) Follow these steps:

- Enter the following command in the command prompt:
  ```
  pdm_halt
  ```
  The secondary server is stopped.

Stop the CA SDM Servers in Advanced Availability Configuration

You stop the following servers:

- Stop the standby server.
- Stop the background server.
- Stop the application server.

(Advanced Availability Configuration) Stop the Standby Server

You can stop the standby server in the Windows or UNIX environment.

(Windows), follow these steps:

1. Select Services from the Control Panel.
2. Select the CA Service Desk Manager Server service and click Stop.
   The standby server is stopped.

(UNIX), follow these steps:

- Enter the following command in the command prompt:

  ```
  pdm_halt
  ```

  The standby server is stopped.

(Advanced Availability Configuration) Stop the Background Server

Before you stop the background server, promote the standby server (that you have upgraded) as the new background server.

Follow these steps:

1. Execute the following command on the standby server that you wish to promote as the new background server:

   ```
   pdm_server_control  -b
   ```

   – **-b**
   Notifies a local standby server to become the background server. The standby server must already be running. If the server is not running, it is started but no failover is performed; to start a failover, run the command again.

   The background server shuts down automatically and the standby server is promoted as the new background server. This change does not affect the end-user sessions. The in-progress updates (if any) are stored and delayed, until the new background server comes online.

(Advanced Availability Configuration) Stop the Application Server

Before you stop an application server, notify all active users to move to another application server.

Follow these steps:

1. Send a notification to all the active users on the application server for stopping servers. This notification can include the details of other application servers.
2. Execute the following command on the application server that you want to stop:

   ```
   pdm_server_control -q interval -s server_name
   ```

   – **-q interval -s server_name**
   Notifies a local or remote application server to quiesce in a specified time interval. This interval is the number of seconds before the server goes offline. When using this option without a server_name, the local server is notified to quiesce. This option cannot be used for a background or a standby server.

   A pop-up message is displayed to all the active users on the application server. This message notifies the users about the server shutdown and the time left for the shutdown. The users must save their work and logout within that time. The application server stops after the specified time. The users log on to the other application server to resume their work.

How to Configure Processes for CA SDM Servers

As an administrator, you can add or modify processes and daemons to enhance the performance of the CA SDM servers. You can configure object managers, web engines, web directors, and other processes for the CA SDM servers in your environment. You can add the processes across several servers to increase volume, performance, and enhance security.

Follow these steps:

1. Add a Server
2. Create a Process Configuration
3. Add CA SDM Server Processes
4. Configure the Server
5. Verify the Configuration

Add a Server

If you want to install a new server in your CA SDM deployment, you must first add the corresponding server record before you configure it.

Follow these steps:

1. Log in to the following server, depending on your CA SDM configuration:
   - Conventional: Primary server
   - Advanced availability: Background server
2. Select System, Servers from the Administration tab.
3. Click Create New to add a server record for the following server, depending on CA SDM configuration:
   - Conventional: Secondary server
   - Advanced availability: Application or standby server
4. Complete the server fields as appropriate.
5. Click Save to add the server detail.

Server Fields

The following fields appear when you create or update a server:

- **Host Name**
  Specifies the local host name of the server. The local host name is stored in the usp_servers table in local_host column.

  WARNING
  Ensure that host name is entered as case-sensitive in the usp_servers table.
- **Attachment Servlet Path**
  You must specify the fully qualified domain name of a server using this field:
  http://<host>:<port>/CAisd/Upload/Servlet
  Where <host> is the fully qualified domain name of a server.
  We recommend that you configure this field.
- **Time Zone**
  Specifies the time zone where the server is located. This time zone value is used to trigger events in the application. This value is used only if the Use End User Time Zone option is not selected, or if no time zone is specified for the service type.
- **Record Status**
  Indicates the state of the server. Active status indicates that the server is a part of the CA SDM deployment.

  WARNING
  If you have inactivated any server, it is recommended not to start CA SDM services on that server. This action can result in an unexpected behavior.
- **Server Type**
  Specifies the type of server that you want to configure. Following server types can be selected, depending on your CA SDM configuration:
  - Advanced Availability: Application or standby server
  - Conventional: Secondary server
- **Configured**
  Available only for advanced availability configuration. This field indicates the state of the configured server. The default value of this field is No. The value is updated to Yes after you successfully run **pdm_configure** on that server. If you edit any of the automatically entered field values of a server record, the Configured field turns to No.

Create a Process Configuration

WARNING

(Recommended) Ensure that both background server and all other standby servers have similar configuration. The standby server becomes the new background after failover and function likes the old background server.

You can create a process configuration for your CA SDM installation, or can modify an existing configuration to suit your requirement.

Consider the following points while creating or modifying a process configuration:

- If you want to modify a process configuration, we recommend you to create a new one or copy the existing configuration and then modify it. This process allows you to revert to the previous configuration, if necessary.
- In conventional configuration, you create configuration only for a primary server and not for secondary servers. All secondary server configuration details must be included within the configuration created for the primary server. For example, to add a web engine with the secondary server host name details, open the configuration for the primary server and add the necessary details.
- For an advanced availability configuration type, you can create configurations only for the specific server. Ensure that you create the same configuration for the background server and standby server.

Follow these steps:

1. Select Systems, Configurations on the Administration tab.
2. Click Create New to add a server configuration.
3. Complete the following fields and click Save:

WARNING
Enter only English characters for all the input fields for any localized language.

- **Configuration Name**
  Specifies the name that you want to assign to the configuration you create.
- **Advanced Availability**
  Indicates whether the configuration is created for conventional configuration or for the advanced availability configuration type. Select **Yes** if the configuration is valid for advanced availability.

NOTE
Configurations that are created for one configuration type (advanced availability or conventional) cannot be implemented for the other configuration type.

Host Name
Specifies the host name for the configuration. The host name is taken from the servers records configured under the Servers page. You can click the Search button to look up for the servers added to your CA SDM installation.

- **Record Status**
  Specifies if the server configuration record is active.
- **Current Configuration**
  Indicates that the configuration is applied on the selected CA SDM server. This field is read-only and is updated based on the configuration you select while executing server configuration utility (**pdm_configure**).

The configuration is saved. New tabs are enabled on the page to add object managers, web engines, web directors, and other processes.

NOTE
The host name and the advanced availability fields become read-only once you have saved the configuration.

Click **Edit in List** on the Configuration List page to modify the record status of a process configuration.

NOTE
You cannot modify the record status if the current configuration of a selected server is in use.

Add CA SDM Server Processes

You can configure object managers, web engines, web directors, and other processes for the CA SDM servers in your environment. We recommend that you become familiar with the following information before configuring the CA SDM processes:

WARNING
Reconfigure the specific server by running the pdm_configure utility on the server after adding the CA SDM processes.

- **Object Manager**
  Object managers manage all the CA SDM objects. Each object manager has an associated name, which it uses to communicate with other objects. Enterprise systems with multiprocessor servers can add object managers to spread out the processing load. Depending on your configuration type, CA SDM installs a default object manager on the following servers:
  - Conventional: Primary server
  - Advanced Availability: All servers

  NOTE
  You cannot modify the default object manager, but can add extra object managers to any server.
- **Web Engines**
  Web engines help prepare web pages for the web client. All systems have one or more web engines. Each web engine connects to an object manager for processing all requests to CA SDM objects. Each web engine can run and be accessed directly. In direct access, every Web browser enters the specific CGI interface for a specific web engine. With this approach, all clients can connect to one web engine and can overburden the web engine, while the other web engines go unused. To handle such heavy traffic, you can assign two or more web engines to a single web director. All requests that go to web engines are directed to the web director. The web director then redirects the client to the most available web engine.
  Depending on your configuration type, CA SDM installs a default web engine on the following servers:
  - Conventional: Primary server
  - Advanced Availability: All servers
- **Web Directors**
  Web directors are optional, and are used when two or more web engines are installed on a single server. Web Director receives connection requests from users, selects a web engine to handle the request, and redirects the request to that web engine. This process is transparent to the end user who always accesses CA SDM with the same URL, regardless of the number of web engines configured. Web directors can be used for the following purposes:
  - To configure CA SDM for efficient use of secure sockets (SSL).
    The HTTPS protocol allows web transactions to be encrypted, providing maximum security for sensitive data, especially passwords. However, pages using SSL are ineligible for browser caching, which can have a negative impact on the performance.
  - To direct logins to a specific web engine (or web engines).
    After a user is authenticated, Web Director can move the session to a different web engine, which can be on a different HTTP server. setup lets you configure SSL for a web engine, providing protection for your passwords while using HTTP protocol for transactions.
  - To have multiple Web Directors, each handling a different group of web engines.
    This setup can be useful in geographically dispersed organizations that want to locate groups of web engines physically closer to their End Users.
- **Aliases**

Aliases are easily identifiable names you can create for object manager Slump names. You can create aliases for a specific object manager or a group of object managers. These aliases can be used in place of an object manager name while configuring CA SDM processes. Aliases are optional in CA SDM.

- **Knowledge Daemons**
  Applicable only to a conventional configuration type. Knowledge daemons provide the knowledge base for CA SDM (knowledge documents, approval process, permissions, notifications, and so on). CA SDM configures the knowledge daemon by default on the primary server. You can move the daemon to a secondary server by changing the host name.
- **Login User Authentication**
  Applicable only to a conventional configuration type. CA SDM configures the login user authentication daemon by default on the primary server. You can move the daemon to a secondary server by changing the host name.
- **LDAP Virtual DB**
  Applicable only to a conventional configuration type. CA SDM configures the LDAP virtual database daemon by default on the primary server. You can move the daemon to a secondary server by changing the host name.
- **Repository Daemons**
  Applicable only to a conventional configuration type. The Repository daemon enables you to locate the attachment records saved in the repository directory. Using the CA SDM web UI, you can add the processes only for secondary servers. After you have added the process through the web UI, enable the specific option during the CA SDM configuration. For more information about enabling the processes, see the Visualizer options.
- **Visualizer**
  Applicable only to a conventional configuration type. You can configure visualizer on a CA SDM server to use web services. Using the CA SDM web UI, you can add the processes only for secondary servers. After you have added the process through the web UI, enable the specific option during the CA SDM configuration.
- **REST Web Services for Tomcat Server**
  Applicable only to a conventional configuration type. REST Web Services enable you to configure the web UI for CA SDM to be able to communicate with external world. Using the CA SDM web UI, you can add the processes only for secondary servers. After you have added the process through the web UI, enable the specific option during the CA SDM configuration.
- – **TNG Converter**
  You can add a TNG converter to any one of the servers running on the Windows platform. You can add the TNG converter to the following servers:
  - Conventional: Primary server, secondary server
  - Advanced Availability: Application server

  NOTE
  If the Daemon Manager manages the TNG converter, it starts and stops with the other daemons. If you need the TNG converter to catch events after the CA SDM daemons are shut down, start and stop the TNG converter as a service.

Add Object Managers

Adding object managers to CA SDM servers improves the overall system performance. CA SDM installs a default object manager on all servers. You cannot modify the default object manager, but you can add more object managers on any CA SDM server to increase the server performance. Depending on your configuration type, CA SDM installs a default object manager on the following servers:

- Conventional: **Primary server**.
- Advanced Availability: **All servers**.

NOTE
You cannot modify the default object manager, but can add additional object managers to any server.

Follow these steps:

1. Select **Systems**, **Configurations** under the **Administration** tab.
   The **Configurations List** page opens.
2. Select the configuration to which you want to add the object manager.
   The **Configuration Detail** page opens.
3. Select the **Object Managers** tab.
   The **Object Manager List** page opens displaying the object managers that are configured for the server. The default object manager (if any) is also displayed in the list.
4. Click **Add Object Manager**.
   The **Create New Object Manager** page opens.
5. Complete the following fields and click **Save**.

   WARNING
   Enter only English characters for all the input fields for any localized language.

   Host Name
   Specifies the host name to which you want to add the object manager. You can click the Search button to look up for the servers added to your CA SDM installation.
   For an advanced availability configuration type, the host name is read-only and is automatically populated based on the host name you specified while creating the configuration.
   - **Display Name**
     Specifies a display name for the object manager. The display name appears on the clients to indicate which object manager it is connected to.
   - **Object Manager Group**
     Specifies the group name to which the object manager is added.

     WARNING
     Enter only English characters for the object group name for any localized language.
   - You can group object managers so that the groups can be assigned to provide service to specific groups of web engines. Users that require this feature often have web engines that are geographically separated from the primary server and want to collocate an object manager and the webengine. The user groups the object managers that are assigned to the local webengines, and then assign the webengines to this group of object managers.
   - **Accept Mask**
     Specifies the mask for the object manager. The Accept mask feature tells the object manager from which clients it accepts connections.
     For example, a web engine attempts to connect to an object manager with names such as web:seattle:1, web:seattle:2 or web:texas:1. You can specify an Accept mask like web:seattle.* to accept all Seattle connections and reject others. You also specify a mask like web:.* to accept connections from the webengines and reject connections from clients.
   - **Record Status**
     Specifies whether the object manager is active or inactive.

     NOTE
     Before setting the record status of an object manager to inactive, you must remove the link between the object manager and associated webengines.

   The object manager that you added appears in the **Object Managers** list.

Add Web Engines or Web Directors

Web directors are optional, and are used when two or more web engines are installed on a single server. The Web Director receives connection requests from users and selects a web engine to handle the request. The request is then redirected to that web engine.

Web engines connect to an object manager for processing all requests to CA SDM objects. Web directors are optional, and are used when two or more web engines are installed on a single server. You can configure web directors on any server. Depending on the CA SDM configuration, CA SDM installs a default web engine on the following servers:

- Conventional: Primary server.
- Advanced Availability: All servers.

Follow these steps:

1. Select **Systems**, **Configurations** on the **Administration** tab.
   The **Configurations List** page opens.
2. Select the configuration to which you want to add the web engine or web director.
   The **Configuration Detail** page opens.

   NOTE
   If you are changing the configuration for the first time, then create a configuration first. When you want to make a configuration change, always create or copy an existing one. This process allows you to revert to the previous configuration, if needed.
3. Select the **Web Engines/Web Directors** tab.
   The **Web Engine/Web Directors List** page opens displays the web engines and web directors that are configured for the server.
   – Conventional: A web engine exists by default on the primary server. You can add web directors to any server.
   – Advanced availability: A web engine exists by default on all servers. You can add more web directors on any CA SDM server.
4. Click **Add Web Engine/Web Director**.
   The **Create New Web Engine/Web Director** page opens.
5. Complete the following fields and click **Save**.

   WARNING

   Enter only English characters for all the input fields for any localized language.

- **Host Name**
  Specifies the host name for the web engine or web director. You can click Search to look up for the servers.
  For an advanced availability configuration type, the host name is read-only and is automatically populated based on the host name you specified while creating the configuration.
- **Type**
  Specifies if you are configuring a web engine or web director. Based on the option that is selected, the relevant fields are automatically populated.
  – Select Web Engine if you want to configure a web engine.
  – Select Web Director if you want to configure a web director.

  NOTE
  Ensure that you have selected the appropriate option. You cannot edit the process type after you have saved the configuration.
- **Web Director**
  Specifies the web director that is assigned to the web engine. You can click Search to look up for the web directors added to the server.

  NOTE
  When implementing any web engine load-balancing scheme, SSL-Login, or both, at least two web engines must be assigned to the same web director.
- **CGI Name**
  Specifies the unique CGI name for the web engine. It is the name of an actual CGI executable when IIS or Apache is used as the HTTP server; it is a servlet parameter when Tomcat is used as the HTTP server.

Examples: (web engines) **pdmweb1**, **pdmweb2**, (web directors) **pdmweb_d1**, and **pdmweb_d2**.
Default: pdmweb.exe (The CGI name must be unique).

- **CGI Port Number**
  Specifies the port on which CA SDM web clients can connect. The CGI port number is the same port on which the tomcat server is running.
  Default: 8080
- **Protocol**
  Specifies the protocol for accessing the web engine.
  – Select HTTPS if the web engine is configured to handle all CA SDM web-client user authentication requests.
  – Select HTTP if the web engine is configured to handle all web client non-user authentication requests (after user is authenticated through the secure login web engine).
- **Record Status**
  Specifies whether the web engine or web director is active or inactive.

  NOTE
  Before setting the record status of a web director to inactive, remove the link between the web director and the associated web engines.
- **Object Manager**
  Specifies the object manager that you want to assign to the web engine.
  – **Default**
  Specifies that the default object manager is assigned to the web engine.
  – **ANY**
  Specifies that the web engine can connect to any available object manager with more willingness value. Willingness value is the availability of the server to accept new clients. A willingness value of zero means that the web engine does not accept any sessions.
  – **Choose**
  Allows you to specify an object manager for the web engine. Selecting this option provides you the option to add multiple object managers or aliases to the configuration.

The web engine or web director that you added appears in the web Engines/Web Directors List.

Add Aliases

Clients and other daemons can be connected to a specific Object Server using aliases. Aliases are easily identifiable names you can create for object manager Slump names. You can create aliases for a specific object manager or a group of object managers. These aliases can be used in place of an object manager name while configuring the CA SDM processes. Aliases are optional in CA SDM. Before you create aliases, perform the following activities:

- Define object managers and add some to groups.
  For example, domsrvr:group1:11, domsrvr:seattle:12, domsrvr:seattle:13, domsrvr:Tacoma:11.
- Enter a regular expression that matches a group of object managers.
  For example, you want the Java clients that are located in Washington to connect to object managers located in Seattle. You connect them to /domsrvr: seattle.*. Or, you can define an alias that is named SEATTLE and can assign it the value /domsrvr:seattle.*.
- Define web engines and assign aliases to the web engines.

  WARNING
  If you have created aliases for an object manager group, ensure that you modify the respective aliases after modifying the object manager group.

Follow these steps:

1. Select **Systems**, **Configurations** under the **Administration** tab.
2. Select the configuration to which you want to add the alias.
3. Select the **Aliases** tab.

4. Click **Add Alias**.
5. Complete the following fields and click Save:

   WARNING

   Enter only English characters for all the input fields for any localized language.

- – **Name:** Specifies the name for the alias you created.
  – **Definition:** Specifies the regular expression that is associated with the alias. The following format is used for an alias definition:

    ```
    domsrvr:[[set the product group or family]:][<host_id><id>][.*]
    ```

    Default: /domsrvr:.*
  – **Record Status:** Specifies whether the alias is active or inactive.
- The alias that you added appears in the Aliases List.

Add Additional Processes

CA SDM enables you to configure visualizer, LDAP, KT, TNG/UNI, BopLogin repository daemons, and REST web services process to a server. For more information about the processes, see System Configurations.

Follow these steps:

1. Select **Systems**, **Configurations** on the **Administration** tab.
2. Select the configuration to which you want to add the additional processes.
3. Select the **Additional Processes** tab.
   The **Additional Process List** displays the list of processes that are configured for the server with the default processes, if any.
4. Click **Add Process**.
5. Complete the following fields as appropriate and click **Save**:

   WARNING
   Enter only English characters for all the input fields for any localized language.

- **Process**Indicates the CA SDM server process that you want to add.
- **Repository Daemon**Specifies the repository daemon that is configured for the server. Repository daemons support attachment repositories. CA SDM configures a repository daemon by default on the primary server. You can move the daemon to a secondary server by changing the host name.
- **REST Web Services for Tomcat Server**
  Applicable only to a conventional configuration type.
  Specifies the REST Web Services configures for the server. REST Web Services enable you to configure the web UI for CA SDM to be able to communicate with external world. Using the CA SDM web UI, you can add the processes only for secondary servers. After you have added the process through the web UI, enable the specific option during the CA SDM configuration.
- **TNG Converter**
  You can add a TNG converter to any one of the servers running on the Windows platform. You can add the TNG converter to the following servers:
- **Conventional:** Primary server, secondary server.
- **Advanced Availability:** Application server.

  NOTE

  If the Daemon Manager manages the TNG converter, it is started and stopped with the other daemons. If you want the TNG converter to catch events after the CA SDM daemons are shut down, start and stop the TNG converter service.

- **UNI Converter**

Specifies the UNI converter that is configured to respond to UNIX events. These events can be filtered and configured to create tickets and trigger other work in the Service Desk. You can add UNI Converter to any one of the servers running on the UNIX platform. You can add the UNI converter to the following servers:
Conventional: Primary server, secondary server.
Advanced Availability: Application server.

- **Visualizer Tomcat Server**
  Applicable only to a conventional configuration type. Specifies the visualizer that is configured for the server.
  You can configure visualizer on a CA SDM server to use web services. Using the CA SDM web UI, you can add the processes only for secondary servers. After you have added the process through the web UI, enable the specific option during the CA SDM configuration. For more information about enabling the processes, see the *Server Configuration Online Help*.

  NOTE

  You can configure only one Visualizer Tomcat Server on a Secondary Server.

- **Hostname**Specifies the server that hosts the process you added. You can click the Search button to look up for the servers.
- **Record Status**Specifies whether the process you added is active or inactive.

The process that you added are displayed in the Additional Process List page.

Configure the Server

Use the Server Configuration utility (pdm_configure) to initialize and configure the CA Service Desk Manager server, database, and web environment. For more information, see Server Configuration Utility

Verify the Configuration

1. After the configuration is completed and the services are properly started, execute the pdm_status or task manager command to verify that the new daemons added.
2. Verify that the current configuration is updated with the configuration you selected during pdm_configure.

How to Perform Rolling Maintenance on CA SDM Servers

WARNING

Migration from an advanced availability environment is not possible using rolling maintenance. You must shutdown the CA SDM services in all the application and standby servers before starting the migration activity. For more information, see How to Upgrade CA SDM.

This article contains the following topics:

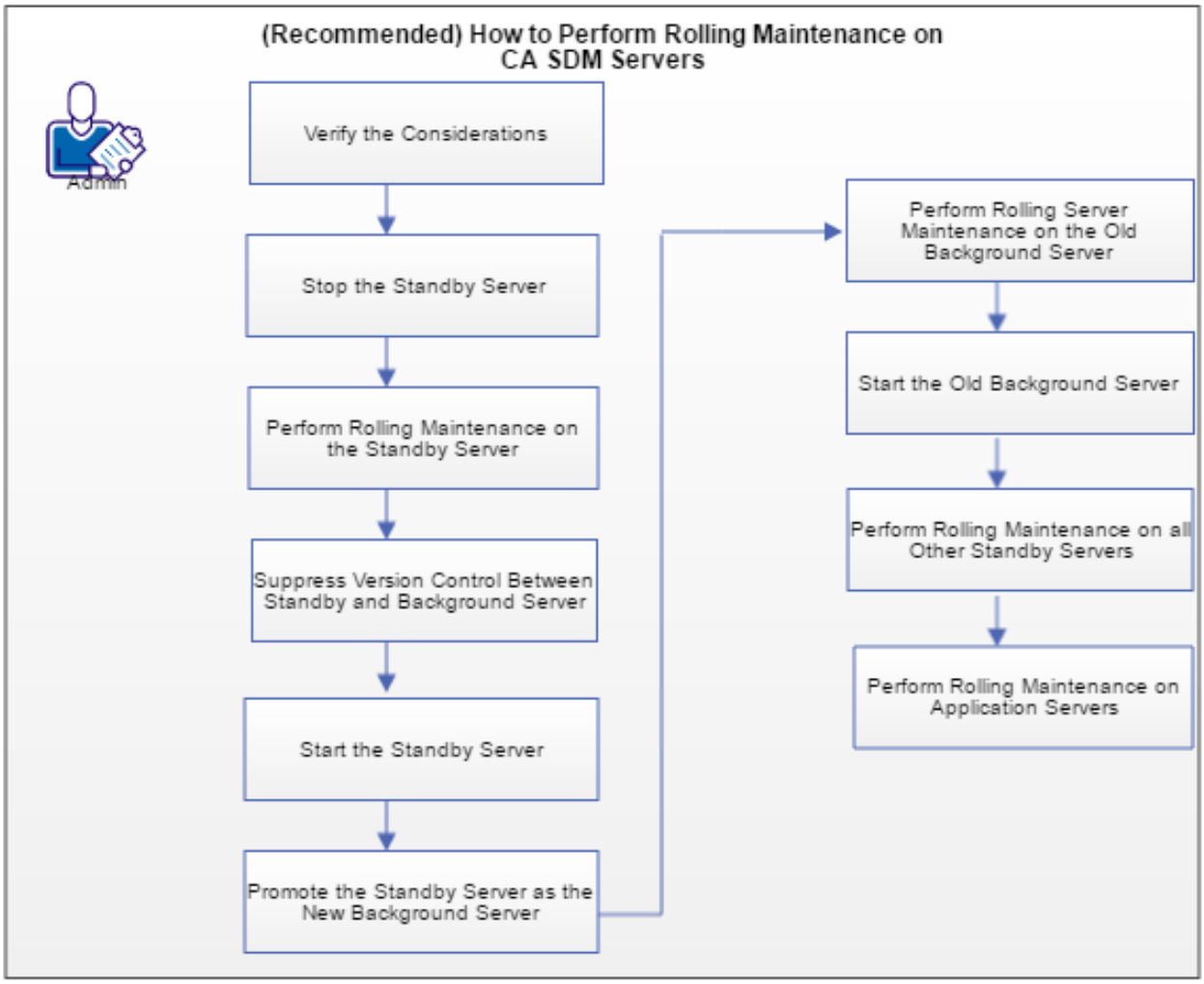
As a system administrator, you perform the rolling server maintenance on the CA SDM servers. This maintenance can be performed to apply patches or to perform a general maintenance on the servers. We recommend you to perform the maintenance on all the servers in a specific order. This process ensures all the servers are updated with the similar changes causing minimal or no disruption to end users. For any server-specific changes, you do not need to update all the other servers.

WARNING
Before applying a common MDB patch or an OS or security patch, shut down all the CA SDM servers. In such cases, the user task is disrupted with no access to CA SDM until all servers are up and running. We recommend the system administrator to plan the patch application accordingly.

The following diagram shows the recommended process to perform a rolling maintenance on the CA SDM servers:

NOTE
Depending upon your organization standards, the rolling maintenance process in your organization may differ from the recommended process.

Figure 10: How to Perform Rolling Maintenance on CA SDM Servers



Follow these steps:

1. Verify the Considerations.
2. Stop the Standby Server (that you wish to promote as the new background).
3. Perform Rolling Maintenance on the Standby Server.
4. Suppress Version Control between the Standby and Background Server.
5. Start the Standby Server.
6. Promote the Standby Server as the New Background Server.
7. Perform Rolling Maintenance on the Old Background Server.
8. Start the Old Background Server.
   When you start the background server, it becomes a standby server.
9. Perform Rolling Maintenance on all Standby Servers.

NOTE

Stop the standby server, perform a rolling maintenance, and start the server.

10. Perform Rolling Maintenance on Application Servers.

Verify the Considerations

During a failover of the background server to the standby server, consider the following points:

- The new users cannot log in.
- For the users that are already connected, some actions do not work during the failover. The users must try the actions after the failover. The following actions do not work:
  - Creating the tickets with attachments
  - Downloading the attachments
  - Searching Knowledge documents
  - Indexing the new knowledge documents
  - Inbound email
  - The SLA events that are not triggered until the failover has completed

WARNING
If you have configured your third-party tool to enable the auto-failover of the CA SDM servers, you must disable it before starting the rolling maintenance.

Suppress Version Control between the Standby and Background Server

CA SDM version control helps you to manage the system modifications across all CA SDM servers. Ensure that you suppress the version control on the standby server before starting it. This process ensures, that the standby server is not upgraded to the system modifications of the background server. To suppress the version control, run the following command on the standby server that you upgraded.

NOTE

When you run the following command, version control should not be disabled.

```
pdm_server_control  -v
```

Promote the Standby Server as the New Background Server

Before you stop the background server, promote the standby server (that you have upgraded) as the new background server. If Support Automation is installed with CA SDM, notify the active Support Automation users about the background server shutdown.

Follow these steps:

1. Execute the following command on the background server to notify all active users using Support Automation to save their work:

   ```
   sa_server_notifier [-h] | [-q seconds] | [-c]
   ```

   - **-h**
     Displays the help page.
   - **-q seconds**
     This option notifies a local server (background) to quiesce in a specified time interval. This interval is the number of seconds before the server goes offline. This option cannot be used for a standby server or application server.
   - **-c**
     This option cancels a previously sent quiesce request.

A pop-up message is displayed to all the active users using Support Automation. This message notifies the users about the server shutdown and the time that is left for the shutdown. The users must save their work and logout within that scheduled time.

2. Execute the following command on the standby server that you wish to promote as the new background server:

```
pdm_server_control  -b
```

– **-b**
Notifies a local standby server to become the background server. The standby server must already be running. If the server is not running, it is started but no failover is performed; to start a failover, run the command again.

The background server shuts down automatically and the standby server is promoted as the new background server. This change does not affect the end-user sessions. The in-progress updates (if any) are stored and delayed, until the new background server comes online.

Perform Rolling Maintenance on Application Servers

As a best practice, you can first perform the rolling maintenance on an application server that has the minimal or no users connected to it. Then, inform all users active on other application servers to log in to the updated server. Finally, you can perform the maintenance on the other application servers. This process ensures that the users are not moved between application servers multiple times.

Follow these steps:

1. Choose the less active application server.
2. Stop the less active application server, perform the rolling maintenance, and start it.
   The application server is updated with all the changes.
3. Stop the other application server.
4. Perform the rolling maintenance on the application server and start it.
   The application server is updated with all the changes.
5. Perform steps 3 and 4 for the other application servers.
   All the application servers are updated with all the changes.

Choose the Less Active Application Server

You choose an application server with the least user activity. Run the following command on each application server to choose the one with no or minimal active sessions.

```
pdm_webstat
```

NOTE
This command does not capture the SOAP or REST Web Service sessions.

Stop the Other Application Server

You inform all the active users on an application server to move to the less active application server before you stop it. Ensure that you have restarted the less active application server before moving all the users to it.

Follow these steps:

1. (Recommended) Inform all active Support Automation analysts on the application server which you want to stop, to create a ticket in CA SDM with their session information. This process ensures that the session information is not lost. For example, the Support Automation analyst is in a session with a customer to resolve a hardware issue. In such a case, the Support Automation analyst can create an issue in CA SDM with the session information before the application server shuts down.
2. Send a notification (for example, an email notification) to all the active users on the application server to move to the less active application server that you just restarted. This notification can include the details of the updated application server.
3. Execute the following command on the application server:

```
pdm_server_control [-h] -q interval -s server_name
```

– **-h**
  Displays the help page.
– **-q interval -s server_name**
  Notifies a local or remote application server to quiesce in a specified time interval. This interval is the number of seconds before the server goes offline. When using this option without a server_name, the local server is notified to quiesce. This option cannot be used for a background or a standby server.

A pop-up message is displayed to all the active users on the application server to notify them about the server shutdown and the time left for the shutdown. The users must save their work and logout within that time. The application server stops after the specified time. The users log on to the other application server to resume their work. The Support Automation analyst can refer to the ticket and resume their work.
The application server is stopped successfully.

How to Configure SSL Authentication

As a system administrator, you can configure the web director direct login requests to a specific web engine using the Secure Socket Layer (SSL) protocol. Configuring the SSL authentication provides enhanced login security. For more information on how to configure SSL for the xFlow Interface and CA Search Server, see Enable Secure Socket Layer (SSL) for xFlow Interface.

This article contains the following topics:

Verify the Prerequisites (Setting up SSL)

Verify the following prerequisites before you begin the SSL configuration:

- Installed and configured CA SDM on the server where you want to implement SSL.
- Configured and assigned at least two web engines configured and assigned to a web director. For more information about how to configure web engines and web directors, see the How to Configure Processes for CA SDM Servers.

Set the Web Engine Capability by Web Director Parameters

After you configure the web engine to use a web director, the web engine can handle the web client requests. The web engine can service login requests (redirect nonlogins elsewhere) or nonlogin requests (redirect logins elsewhere) or both (‘general purpose’ web engine). The WebDirector parameter found in the <Host_Name>-web[#].cfg file determines how webengine can service login requests.

The following table shows the relationship between the web engine role and web director parameter setting:

Web Engine Capability	‘<Host_Name>-web[#].cfg’ ‘webdirector’ Parameter Settings
Service login requests	‘UseDirector AfterLogin’; ‘Willingness 0’
Service non-login activity	‘UseDirector BeforeLogin’; ‘Willingness [1 thru 10]’
General purpose	‘UseDirector Yes’; Willingness [1-10]’

Choose the SSL Login Environment

You can use the web director for a targeted SSL login in a mixed SSL/non-SSL web environment to redirect every web-login request to the specific SSL web engines. All other requests can be redirected to and serviced by the non-SSL web engines.

Choose from the following SSL Login environment:

Non-SSL Environment with Basic Load-Balancing

You can use web director in a non-SSL environment with basic load-balancing. Web director balances load across all web engines according to each web engine willingness value. Each web engine can service login requests and nonlogin requests. The HTTP protocol is used for communication between the web client and the web server.

For each web engine under web director control, set the web director parameters in the web engine '<*Host_Name*>-web[#].cfg' as follows:

- UseDirector: Yes
- WebDirectorSlumpName: (do not change this value)
- WillingnessValue: [1 through 10]
- RedirectingURL: (the preappended protocol value can be either missing or 'http')

Global SSL Environment with Basic Load-Balancing

You can use web director in a global SSL environment with basic load-balancing. Web director balances load across all web engines according to each web engine willingness value. Each web engine can service login and nonlogin requests. The HTTPS protocol must be used for all communications between web clients and the web server.

For each web engine under web director control, set the web director parameters in the web engine '<*Host_Name*>-web[#].cfg' as follows:

- UseDirector: Yes
- WebDirectorSlumpName: (do not change this value)
- WillingnessValue: [1 through 10]
- RedirectingURL: (the preappended protocol value must be 'https')

Targeted Login in a Non-SSL Environment with Optional Load-Balancing

You can use web director for a targeted login in a non-SSL environment with optional load-balancing. The login-only web engine services only the login requests. The remaining web engines under web director control, service all other requests. This configuration puts the entire burden of servicing login requests on the specified login-only web engines. The HTTP protocol is used for communications between the web client and the web server.

For the login-only web engines, set the web director parameters in the web engine '<*Host_Name*>-web[#].cfg' as follows:

- UseDirector: AfterLogin
- WebDirectorSlumpName: (do not change this value)
- WillingnessValue: 0
- RedirectingURL: (the preappended protocol value can either be missing or 'http')

For the nonlogin web engines, set the web director parameters in the web engine '<*Host_Name*>-web[#].cfg' as follows:

- UseDirector: Before Login
- WebDirectorSlumpName: (do not change this value)
- WillingnessValue: [1 through 10]
- RedirectingURL: (the preappended protocol value can be either missing or 'http')

Targeted SSL Login in a Mixed Environment with Optional Load-Balancing

You can use web director for a targeted SSL login in a mixed SSL/non-SSL web environment with optional load-balancing. Every web-login request is redirected and serviced by the SSL web engines, while other requests being serviced by the non-SSL web engines. The HTTPS protocol must be used for all communications between the web client and the SSL web engines.

Set Up SSL Login Environment

Setting up SSL allows web transactions to be encrypted, providing maximum security for sensitive data, especially passwords. Depending on your configuration type, you can implement an SSL login environment on the configured CA SDM servers.

Follow these steps:

1. Log in to the following server, depending on your CA SDM configuration:
   – Advanced Availability: Application server
   – Conventional: Primary or secondary server
2. Verify that the server has successfully imported an SSL certificate.
3. Create a copy (including subdirectories) of the directory '$NX_ROOT/bopcfg/www/wwwroot', and assign it the following name:
   '$NX_ROOT/bopcfg/www/wwwrootsec'
4. Add a new virtual directory for the web server named CAisdsec.
5. Point this virtual directory to the following physical directory:
   '$NX_ROOT/bopcfg/www/wwwrootsec'
6. Verify that the virtual directory permissions for CAisdsec match the CAisd virtual directory permissions for the script execution. Enforce SSL for the CAisdsec virtual directory.

   NOTE
   In this example, CAisdsec is user-defined and can be renamed.
7. Save the changes.

   NOTE
   Webdirectors do not use a '<Host_Name>-web[#].cfg' file. However, web engines require a unique '<Host_Name>-web[#].cfg' file. Sample web.cfg files are automatically generated while running the configuration. You can import the modifications in the original web.cfg to the new web configuration files by specifying the original web.cfg as the template file you want to use.
8. Copy and save the following files, because a backup of these files is useful whenever you decide to restore the original environment:
   – $NX_ROOT/pdmconf/pdm_startup.tpl
   – $NX_ROOT/pdmconf/pdm_startup
   – $NX_ROOT/bopcfg/www/web.cfg file
   – Any existing primary-web[#].cfg files
   – $NX_ROOT/bopcfg/www/CATALINA_BASE/webapps/CAisd/WEB-INF/web.xml and web.xml.tpl
   – For a secondary server configuration, save backup copies of any existing $NX_ROOT/bopcfg/www/web.cfg or <Secondary_Server_Host_Name>-web[#].cfg files and $NX_ROOT/bopcfg/www/CATALINA_BASE/webapps/CAisd/WEB-INF/web.xml*
9. For each web engine assigned to a webdirector, ensure that the parameters (<Host_Name>-web[#].cfg 'webdirector') of the web engine are set correctly by examining the file in a text editor. If necessary, modify the 'webdirector' parameter values to reflect the web engine role you want. Then copy them to the directory: $NX_ROOT/bopcfg/www.
10. Move all $NX_ROOT/samples/pdmconf/primary-web[#].cfg files to the $NX_ROOT/bopcfg/www directory.
    For the secondary server configuration, Move all $NX_ROOT/samples/pdmconf/'secondary_server_name-web[#].cfg' from the primary server to the secondary server $NX_ROOT/bopcfg/www directory
11. For the servlet server like Tomcat, CA SDM creates web.xml files that can replace the web.xml file on each server hosting a webengine. These files are name primary-web.xml. Rename the files and copy them to the directory: $NX_ROOT/bopcfg/www/CATALINA_BASE/webapps/CAisd/WEB-INF directory.
    For the HTTP server like IIS or Apache, create copies of 'pdmweb.exe' in the $NX_ROOT/bopcfg/www/wwwroot directory, a 'pdmweb[#].exe' for each web engine, and a 'pdmweb_d[#].exe' for each webdirector that has been configured. Verify that the 'pdmweb[#].exe' and 'pdmweb_d[#].exe' are named according to the correct CGI I/F values (for example: 'pdmweb1.exe', 'pdmweb2.exe', pdmweb_d1.exe', and so on).
12. If you are using IIS and want to add Server Extensions for each CGI interface, you can take the file primary-site.dat file and copy it to your $NX_ROOT/bopcfg/www directory as site.dat. When the system is reconfigured these sites is added to IIS.
13. Reconfigure the primary server without reinitializing the database and start services.

14. After reconfiguring verify that the current settings are valid. Start the CA SDM daemons. Verify that there are no errors in the stdlog files. Use pdm_status to view the daemons and their status. Use http://localhost:8080/CAisd/pdmweb.exe to access the system.
15. For Knowledge Management to CA SDM integration, if SSL has been enforced for CA SDM, the CA SDM URL protocol value must be changed.
    a. From the Knowledge Management Tool Settings Manager, General, Integration, change the CA SDM URL protocol value from http to https.
    b. Save and exit.
16. Open a web browser to the CA SDM login page and verify that a user can log in and that the expected redirect/login behavior is observed.

Implement SSL Login Environment

To implement the SSL login you have set up, make changes to the web director parameter values.

Follow these steps:

1. For Secure Login Web engines, edit the <Host_Name>-web[#].cfg as follows:
    a. Change the CAisd parameter value from /CAisd to /CAisdsec.
    b. Change the UseDirector parameter value from Yes to AfterLogin if the web director uses pass through an authentication.
    c. Change the Willingness parameter value from 5 to 0.
    d. Verify that the RedirectingURL value protocol is listed as https.
    e. Change the Redirecting URL <cgi directory> value from CAisd to CAisdsec.
    f. Save the changes.
2. For non-secure web engines handling all other activity, edit the <Host_Name>-web[#].cfg files as follows:
    a. Verify that the CAisd parameter value is /CAisd.
    b. Change the UseDirector parameter value from Yes to BeforeLogin.
    c. Maintain the Willingness value of 5 or set it to any integer value from 1 to 10, depending on the particular loading weight desired.
    d. Verify that the RedirectingURL value protocol is listed as http.
    e. Verify that the RedirectingURL <cgi directory> value is CAisd.
    f. Save the changes.
    After the configuration, restart Service Desk. After the service restarts, test the login by accessing the non-ssl web engine using HTTP. Verify if it automatically redirects you to the HTTPS secure webengine for the login. Once you are logged in, it automatically redirects you back to the non-ssl HTTP webengine for the normal Service Desk activity.

Verify SSL Login Environment

You can verify the SSL login environment for web engines.

Follow these steps:

- The secure-login web engines must reside in the physical directory that is mapped to the SSL-enforced virtual directory (CAisdsec in this example).
  For secure-login web engines, create instances of pdmweb.exe in the $NX_ROOT/bopcfg/www/ wwwrootsec directory with the name of pdmweb[#].exe. The executable name must match the CGI I/F value for each secure-login web engine.
  Example: If you have assigned the CGI I/F value of the secure-login web engine to pdmweb2, create a physical copy of pdmweb.exe, and rename it pdmweb2.exe.
- The non-secure web engines and web directors must reside in the physical directory that is mapped to the non-SSL virtual directory CAisd.
  For non-secure web engines and web directors, create instances of pdmweb.exe in the $NX_ROOT/bopcfg/www/ wwwroot directory. A copy of pdmweb.exe must exist for each non-secure web engine and webdirector configured.

Rename the copies so that the new names of the executables match the CGI I/F values that have been defined for the web engines and web directors.
Example: If you have assigned the CGI I/F value of pdmweb3 to the non-secure web engine and the value of pdmweb_d1 to the web director, then create two copies of pdmweb.exe. Rename the first copy to pdmweb3.exe, and then rename the second copy to pdmweb_d1.exe.

Set Up SSL for Tomcat Server

Configure SSL on Tomcat Servers in your CA SDM environment.

Follow these steps:

1. To create a key store on each CA SDM server that requires an SSL certificate, perform the following steps:

   NOTE
   A keystore is a store or storage unit for certificates, in which the certificates are imported to, and then Tomcat points to use that keystore and certificates for SSL.

   a. Create a directory under the C: drive (or the local drive you want) with the name, certificates.
   b. Using the command line, navigate to the JRE bin directory (for the JRE installed with Service Desk - usually /SC/JRE)
   c. Run the command "keytool -genkey -alias tomcat -keyalg RSA -keystore c:/certificates/.keystore".
   d. Fill in the fields as appropriate (make sure to note what you filled in each filed as you may need this information later).
      A .keystore file is created in the C:\certificates\ directory.
2. Generate the Certificate Request for each server. Perform the following steps to generate the certificate request:
   a. Using the command line, navigate to the JRE bin directory (for the JRE installed with Service Desk - usually /SC/JRE)
   b. Run the command "keytool -certreq -alias tomcat -keystore c:/certificates/keystore.jks -file servername-certreq.csr"
      A .csr file is created in the c:/certificates directory on each server where you generated the certificate request.
   c. Send the .csr files to the vendor of your choice who will then generate the appropriate certificates you need based on the certificate request, for each server.

      NOTE
      The certificate you receive from each is different. Some vendors will send you multiple certificated possibly including a root certificate, an intermediate certificate, and a certificate of authority. Each vendor has different instructions on which certificates they provide need to be imported into the keystore. So the key here is to ask the specific vendor that you used to generate the certificates for you, for specific instructions on how to import their certificates into a tomcat keystore.

   Once you received the specific instructions from the vendor, you can follow those to import the appropriate certificates into the keystore on each server. Once that is complete, you can now configure Tomcat on the Service Desk side of things to point it to that keystore where the certificates have been imported.
3. Open the \bopcfg\www\CATALINA_BASE\conf\server.xml file using a text editor and locate the following:

```
 <!--<Connector port="8443" protocol="org.apache.coyote.http11.Http11NioProtocol"
 maxThreads="150" SSLEnabled="true">
<SSLHostConfig>
<Certificate certificateKeystoreFile="conf/localhost-rsa.jks"
type="RSA" />
</SSLHostConfig>
</Connector>-->
```

4. Change the code to the following:

```
 <Connector port="8443" protocol="org.apache.coyote.http11.Http11NioProtocol"
maxThreads="150" SSLEnabled="true"
```

```
 keystoreFile="C:\certs\keystore.jks" keystorePass="password">
<!--<SSLHostConfig>
<Certificate certificateKeystoreFile="conf/localhost-rsa.jks"
type="RSA" />
</SSLHostConfig>-->
</Connector>
```

NOTE
Be sure to remove the <-- and --> tags that currently comment out the HTTPS/SSL connector for Tomcat, and set the appropriate path and password for your keystore that you generated in the beginning.

5. Save the server.xml file.
6. Restart Tomcat by using the following commands:

```
pdm_tomcat_nxd  - c stop
pdm_tomcat_nxd  - c start
```

NOTE
We recommend you to restart all the CA SDM servers to ensure that the Tomcat is restarted.

7. Test your tomcat SSL connection by opening a browser and navigating to the Service Desk URL, using the HTTPS protocol, and the tomcat port. For example, use the following URL:

```
https://servername:8443/CAisd/pdmweb.exe
```

The Service Desk Login Screen should open. You have successfully configured SSL on Tomcat.

Set Up SSL on IIS

For more information about how to set up authentication on IIS, see Microsoft Documentation.

System Configurations

This article contains the following topics:

To enhance the performance of CA SDM Servers, add or modify the processes and daemons. Configure object managers, web engines, web directors, and other processes for the CA SDM servers in your environment. You can add the processes across several servers to increase volume, performance, and enhance security.

Add Additional Processes

CA SDM enables you to configure visualizer, LDAP, KT, TNG/UNI, BopLogin repository daemons, and REST web services process to a server.

Follow these steps:

1. Select Systems, Configurations on the Administration tab.
   The Configurations List page opens.
2. Select the configuration to which you want to add the additional processes.
   The Configuration Detail page opens.
3. Select the Additional Processes tab.
   The Additional Process List displays the list of processes that are configured for the server along with the default processes, if any.
4. Click Add Process.
   The Create New Process page opens.
5. Complete the following fields as appropriate:
   ProcessIndicates the CA SDM server process that you want to add.
   Repository Daemon Applicable only to a conventional configuration type.Specifies the repository daemon that is configured for the server. Repository daemons support attachment repositories. CA SDM configures a repository

daemon by default on the primary server. You can move the daemon to a secondary server by changing the host name.

REST Web Services for Tomcat Server

Applicable only to a conventional configuration type.

Specifies the REST Web Services configures for the server. REST Web Services enables you to configure the web UI for CA SDM to be able to communicate with external world. Using the CA SDM web UI, you can add the processes only for secondary servers. After you have added the process through the web UI, enable the specific option during the CA SDM configuration. For more information about enabling the processes, see the Server Configuration Online Help.

TNG Converter

You can add a TNG converter to any one of the servers running on the Windows platform. You can add the TNG converter to the following servers:

Conventional: Primary server, secondary server.

Advanced Availability: Application server.

NOTE

If the Daemon Manager manages the TNG converter, it is started and stopped with the other daemons. If you want the TNG converter to catch events after the CA SDM daemons are shut down, start and stop the TNG converter service.

Visualizer Tomcat Server

Applicable only to a conventional configuration type.

Specifies the visualizer that is configured for the server.

You can configure visualizer on a CA SDM server to use web services. Using the CA SDM web UI, you can add the processes only for secondary servers. After you have added the process through the web UI, enable the specific option during the CA SDM configuration. For more information about enabling the processes, see the Server Configuration Online Help.

NOTE

You can configure only one Visualizer Tomcat Server on a Secondary Server.

Hostname

Specifies the server that hosts the process you added. You can click the Search button to look up for the servers.

Record Status

Specifies whether the process you added is active or inactive.

6. Click Save.
7. The process that you added are displayed in the Additional Process List page.

Additional Processes

Depending on your configuration type, you can configure the following extra processes for the CA SDM servers:

Processes	Configuration details
Knowledge Daemons Login User Authentication LDAP Virtual DB	Applicable only to the conventional configuration. CA SDM configures these daemons by default on the primary server. You cannot delete the daemons but can move the daemons to a secondary server by changing the host name.
Repository daemons REST Web Services for Tomcat Server Visualizer	Applicable only to the conventional configuration. Using the CA SDM web UI, you can add these processes only for secondary servers. After you have added the process through the web UI, enable the specific option during the CA SDM configuration.

TNG converter	Depending on the operating system, you can add the processes to the following servers: Conventional: Primary server, secondary server. Advanced Availability: Application server.

Current Locks

Current Locks let you view and release locks on any object. During normal operations, the system locks objects while users work with data. If there is a system failure, you can free objects for future updates. Because the initial locks default to 5 (five) minutes, you unlock the objects when another user needs the same data.

View Current Object Locks

You can view a snapshot of currently locked objects.

Follow these steps:

1. On the Administration tab, select System, Current Locks.
   The Current Locks page lists the following information about object locks:
   - **Lock**
     Process that requested the lock.
   - **Name**
     The lock name. If the lock is related to a ticket, the lock name can reflect the ticket number.
   - **Table Name**
     Location of the locked object.
   - **Object Name**
     The Majic object.
   - **Lock Owner**
     The table user.
   - **Locked Since**
     Date and time of the object lock.
   - **Process**
     Process that initiated the lock.
2. Refresh the browser.
   The Current Locks page shows the most recent object locks.

Unlock Objects

You can review and release object locks to allow other users to access the object.

Follow these steps:

1. On the Administration tab, select System, Current Locks.
   The Current Locks page shows a list of object locks.
2. Refresh the browser.
   The Current Locks page shows the most recent object locks.
3. Right click the row.
   A Delete Lock menu appears.
4. Click Delete Lock.
   A confirmation message appears.
5. Click OK.
   The Current Locks page updates and a message notifies you that the object is free.

Configure a Dedicated Domsrvr for SOAP Web Service - Advanced Availability

In CA SDM Advanced Availability, there is no option for **webservice_domsrvr**. This option **webservice_domsrvr** is only present in the CA SDM Conventional Installation.

As a result, all SOAP web service calls use the domsrvr on the CA SDM Background Server. This can impact performance in both the web service application and the CA SDM application itself. This article explains how a dedicated DOMSRVR can be configured to handle the SOAP web service calls.

1. Create an additional DOMSRVR (for example: **domsrvr:03**).
   This DOMSRVR could be on any CA SDM server, depending where the SOAP web service calls. For more information on how to add a domsrvr, see How to Configure Processes for CA SDM Servers.
2. On the CA SDM server that the newly created DOMSRVR (**domsrvr:03**) resides, edit the **NX.env** file with the *PDM_OPTIONS_MGR* command line utility to add or change content within the NX.ENV configuration file. As a best practice use it to avoid this file from becoming corrupted by a text editor.

   ```
   pdm_options_mgr -c -s webservice_domsrvr -v domsrvr:03 -a pdm_option.inst -t
   pdm_options_mgr -c -s webservice_domsrvr -v domsrvr:03 -a pdm_option.inst
   ```

3. Restart the CA SDM service.
   If this is not your Background server, only the CA SDM service on this particular server needs to be restarted.
   If it is the CA SDM Background server, promote the CA SDM Standby server to be the new CA SDM Background server, recycle the CA SDM service and promote it back to the CA SDM Background server.

Managing Your Database

This page talks about the ways to manage databases in CA SDM.

This article contains the following topics:

- Database Backup
- Database Restore
- Database Table Replacement
- Data Dereferencing
- Use the Dbadmin Mode
- How to Use pdm_deref Example
- Database Loader
- Drop and Restore Constraints
- How to Create and Use an Input File
- How to Archive and Purge Historical Data

You can run utilities to manage your database while CA SDM is shut down. If you are running database management utilities and using a database other than the CA SDM default repository, the database environment variables must be set.

> **NOTE**
> For information about setting environment variables, see the database documentation you are using.

Database Backup

You can back up the contents of a single database table, multiple database tables, or your entire CA SDM database, using the database backup utility, **pdm_backup**. The output of the backup utility is an ASCII file that the **pdm_restore** utility can use.

As part of its processing, **pdm_backup** first shuts down the daemons (UNIX) or services (Windows). pdm_backup stops CA SDM and then writes one or more tables from a CA SDM database to an ASCII file. You can use this output

file as the input file to pdm_restore. In addition to the contents of the database, pdm_backup backs up application configuration files.

If you have operating environment-specific backup tools, we recommend that you use them instead of pdm_backup. Because pdm_backup is a generic tool, it can be slow on some operating environment combinations.

> **NOTE**
> As part of its processing, pdm_backup first shuts down the daemons (UNIX) or services (Windows).

Syntax

This command has the following format:

```
pdm_backup [-d] [-g] [-v] -f filename [ALL | table1...tableN]
```

-d

Specifies to back up the database data only.

-g

Specifies that only non-database data be backed up. This means only windows (forms) and other non-database data is backed up.

-v

Specifies verbose mode, which writes comments about command progress to stdout.

-f *filename*

Specifies an output file.

ALL|table1...tableN

Specifies ALL files or the table or tables to write. If more than one table is specified, separate each with spaces.

- You can find table names in the CA SDM database schema file, ddict.sch, located in $NX_ROOT/site (UNIX) or *installation-directory*\site (Windows). $NX_ROOT or *installation-directory* is the directory where you installed CA SDM.
- If no table is specified, the entire CA SDM database is written, including window groups and menu registration files.

Restrictions

pdm_backup cannot be run while CA SDM is active.

> **WARNING**
> On UNIX, the LIBPATH must be set before running several CA SDM utilities. Use *pdm_task* to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

Database Restore

The database restore utility, pdm_restore, loads an output file from pdm_backup to the CA SDM database. The pdm_restore utility first shuts down the daemons (UNIX) or services (Windows). Then it restores, clears, and replaces all existing database records. Use the formatted ASCII file created by pdm_backup as input for the pdm_restore utility.

You can also use the pdm_restore and pdm_userload utilities to gain access to the CA SDM application if there is a catastrophic database corruption. If your database is damaged to the point where you cannot gain any access to the application and if all other measures have failed, rerun configuration and reinitialize the database to rebuild your database and populate reference data and system tables.

This procedure initializes your database in the same way that you did during the original installation. You can now access CA SDM. The pdm_restore utility can be used to restore the last backed up copy of your database.

For more information, see pdm_restore--Restore a Database.

pdm_restore--Restore a Database

pdm_restore stops CA SDM and then deletes all records from a CA SDM database and replaces them with records from a file you specify with the -f option. The data from the input file is the only data that will be in the CA SDM database after running pdm_restore.

The input file must be created using pdm_extract or pdm_backup, or otherwise formatted for pdm_restore. pdm_backup can back up non-database data, and pdm_restore can restore this data also. pdm_backup and pdm_restore are not recommended when other backup and restoration tools are available.

> **NOTE**
> As part of its processing, pdm_restore first shuts down the daemons (UNIX) or services (Windows).

Syntax

This command has the following format:

```
pdm_restore [-d] [-g] [-n] [-w] [-v]  -f filename
```

Restrictions

pdm_restore can be run only on a CA SDM server. Only the privileged user or root can run pdm_restore. The following restrictions are applicable if you are using the advanced availability configuration:

- If you are restoring the database, run the pdm_restore command only on the background server.
- Ensure that you have stopped all servers (application, background, and standby) before you run the pdm_restore command.

> **WARNING**
> Use pdm_restore only with a full database backup created by pdm_backup, because your current database is deleted and replaced by the backup file. Do not run pdm_restore when users are logged in to CA SDM.

-d
Specifies that only database data is restored.

-g
Specifies that only non-database data be restored. Only windows (forms) and other non-database data are restored.

-n
Specifies that NX.env is restored if restoring non-database data. By default, NX.env is not restored. We recommend that the NX.env file not be restored unless the restore is to the same server the backup came from. Restoring an incorrect NX.env can affect unintended databases.

-w
Specifies that web.cfg is restored if restoring non-database data. By default, web.cfg is not restored.

-v
Specifies verbose mode.

-f *filename*

Specifies an input file that contains a full backup created by pdm_backup.

> **WARNING**
>
> On UNIX, the LIBPATH must be set before running several CA SDM utilities. Use *pdm_task* to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

Database Table Replacement

This article contains the following topics:

The pdm_replace utility is a quick and easy way to replace the entire contents of a CA SDM database table with new information. This utility can be useful for massive revisions of look-up tables.

NOTE
pdm_replace takes the same input file format as pdm_userload. You can create an input file for pdm_replace using pdm_extract; however, you cannot use the output of pdm_backup as input to pdm_replace.

NOTE
For more information, see How to Create and Use an Input File.

pdm_replace--Replace a Database Table

pdm_replace deletes a table in a CA SDM database and replaces it with a table from a temporary file you specify with the -f option; the data from the input file is the only data that is in that table after running pdm_replace. Back up your table before running pdm_replace.

NOTE
As part of its processing, pdm_replace first shuts down the daemons (UNIX) or services (Windows).

pdm_replace accepts a text file as input, which is the same file format used by pdm_userload. You can create an input file for pdm_replace using pdm_extract; however, you cannot use the output of pdm_backup as input to pdm_replace.

WARNING
Be sure to name your input file with a name different from the table name you are attempting to replace. For example, if you are replacing a table named ca_contacts and you name the input file ca_contacts.dat, after you execute the pdm_replace command to point to the input file (ca_contacts.dat), it deletes the file after execution because it has the same name as the table.

Restrictions

- pdm_replace can be run only on the following servers, depending on your CA SDM configuration:
  - Conventional: Primary server
  - Advanced availability: Background server

    WARNING
    Ensure that you have stopped all application and standby servers before running this command on the background server.
- Only the privileged user or root can run pdm_replace.
- Do not run pdm_replace when users are logged in to CA SDM.

Syntax

This command has the following format:

```
pdm_replace [-v] -f filename
```

-v

Specifies verbose mode.

-f *filename*

Specifies an ASCII file with the following format:

```
TABLE table_name
fieldname1fieldname2 . . . . fieldnameN
{ "value11", "value12", . . . "value1N" }
{ "value21", "value22", . . . "value2N" }
.
.
```

```
.
{ "valueN1", "valueN2", . . . "valueNN" }
```

This format is the same file format used by pdm_userload. You can create an input file for pdm_replace using pdm_extract; however, you cannot use the output of pdm_backup as input to pdm_replace.

Data Extraction

The pdm_extract utility extracts data from the CA SDM database and produces output in various formats. You can further process this data or enter it into other applications, such as a spreadsheet or another database.

With the pdm_extract utility, you can do the following:

- Dump the entire CA SDM database
- Dump one or more database tables
- Extract specific information from the database and produce output in one of the following three formats:
  - Output compatible with pdm_userload
  - Comma-separated value (CSV) output
  - Informal report-style output

For more information, see pdm_extract--Extract Data from Database.

pdm_extract--Extract Data from Database

The pdm_extract command extracts data from specified CA SDM database tables or the entire CA SDM database, and outputs it as ASCII-formatted text.

Syntax

This command has the following format:

```
pdm_extract [-c|-e|-r] [-d] [-h] [-u] [-v] [-C] --B] [-f formatstring| ALL | table1 . . . TableN]
```

-c

Produces comma-separated value (CSV) output, such as:

```
"field1","field2","field3"
```

The -c, -e, and -r output format options are mutually exclusive.

-e

Produces comma-separated value (CSV) output with embedded double quotes escaped by another double quote. For example:

```
"Text with a "quoted string" in it"
```

The -c, -e, and -r output format options are mutually exclusive.

-r

Produces left-justified output in the formats if the column labels are not supplied in the input file:

```
"label":"value"
```

or

```
"value"
```

This option is intended for use with line printers, for example:

```
Field_Name: Field Value
```

The -c, -e, and -r output format options are mutually exclusive.

-d

Uses the date format found in the file $NX_ROOT/fig/english/cfg/dataent.fmt (UNIX) or *installation-directory*\fig\english\cfg\dataent.fmt (Windows), which you can edit to suit your requirements.

-h

Displays help and usage information.

-u

Produces output without headers.

-v

Specifies verbose mode, which writes comments about command progress to stdout.

-C
Changes encoding from UTF-8 to another charset. The default output is UTF-8.
Example: To convert the output to JIS, you would run "-C iso-2022-jp"
Example: To encode to the operating system's native charset, use "DEFAULT" or "NATIVE".

-B
Suppresses the Byte Order Mark if the variable NX_ADD_UTF8_BYTE_ORDER_MARK is set.
The NX_ADD_UTF8_BYTE_ORDER_MARK option is a signature into a file. It allows editors that support UTF-8 to maintain the UTF-8 integrity of the file.
Note: This is only needed for non-ASCII data. If this is not installed, the default behavior omits the Byte Order Mark (BOM). If installed, set it to "1" or "Yes".

-f *formatstring*

Extracts specific records and fields according to *formatstring*, which is an SQL subset statement.

For a date after a period, use the following syntax:

pdm_extract -v -f "select id, ref_num from Call_Req where open_date >= DATE '2005-02-24'" > daterange1.txt

For a date range, use the following syntax:

pdm_extract -v -f "select id, ref_num from Call_Req where open_date >= DATE '2004-01-20' and open_date < DATE '2004-02-25'" > daterange2.txt

> **NOTE**
> Use single quotes around the date in the YYYY-MM-DD format.

The syntax for DATE is as follows:

```
DATE 'yyyy-mm-dd'
```

yyyy = integer representing year (between 1970 and 2038)

mm = integer representing month

dd = integer representing day

Examples:

```
DATE '2005-01-18'
DATE '1999-12-25'
```

The syntax for TIMESTAMP is:

```
TIMESTAMP 'yyyy-mm-dd hh.mm.ss[.nnnnnn][[+|-][hh.mm]]
```

yyyy = integer representing year (between 1970 and 2038)

mm = integer representing month

dd = integer representing day

hh = integer representing hour

mm = integer representing minutes

ss = integer representing seconds

nnnn = optional integer representing fractions of sec.

[+|-][hh.mm] = optional time zone interval.

Examples:

```
TIMESTAMP '1998-04-28 12:00:00.000000'
TIMESTAMP '2004-10-17 18:30:45'
TIMESTAMP '2005-03-21 12:00:12+08:00'
TIMESTAMP '1999-05-10 09:12:23.005-03:30'
```

NOTE
The -d option is not needed, as it only affects the format of the output.

A command usage example follows:

```
pdm_extract -f "select * from Call_Req where open_date > TIMESTAMP '2004-01-12 12:00:00'"
```

In this example, all columns are being extracted from the Call_Req table where the open_date is after midnight 1/12/2004.

ALL

Extracts output from all tables in the database.

```
table1. . . tableN
```

Extracts output from the specified tables. Table names must be separated by spaces.

The default format, if none is specified, is an ASCII file compatible with pdm_userload.

Data Selection for Extraction

To select the data for extraction, the data extractor uses an integral SQL subset with the following rules:

- The following SQL functions are specifically supported in the subset:
  - IS NULL, IS NOT NULL, LIKE
  - SELECT statements and WHERE clauses
- The following SQL functionality is not supported in the subset:
  - Joins
  - Embedded tabs and new-line characters
  - Clauses other than SELECT, FROM, and WHERE
  - Asterisks (*) in SELECT statements
  - Nested SELECTs
  - Aggregate functions
- All SQL reserved words must be in uppercase characters
- Every token in a WHERE clause must be surrounded by white space
- All date/time specifications must be in one of three formats:
  - Elapsed seconds from 12:00:00 a.m. 1/1/1970 Greenwich Mean Time (GMT):

```
start_date< or >174182431500
```

– SQL DATE format:

```
start_date< or >DATE '2001-11-28'
```

– SQL TIMESTAMP format:

```
start_date< or >TIMESTAMP '2001-07-04 15:45:00'
```

NOTE
TIMESTAMP format uses GMT. You can adjust time zones by adding or subtracting the appropriate number of hours, such as: 2001-03-23 14:00:00+02:00 or 2001-06-06 04:45:00-09:00.

Use the Data Extractor on UNIX

Before you use the CA SDM data extractor on UNIX, you must do the following:

1. Set the value of the $NX_ROOT environment variable to the full path name of the CA SDM installation directory that you defined during installation.
2. Append $NX_ROOT/bin to your PATH environment variable.
3. Set umask to 000.

Data Dereferencing

This article contains the following topics:

The pdm_deref utility is a dereferencing tool that converts data from various sources into a format suitable for loading into the CA SDM database. Dereferencing extracts internal IDs for cross-referenced fields. The utility can also be used to calculate down time and SLA down time values.

The pdm_deref utility converts data into one of the following formats:

- Output compatible with pdm_userload, suitable for loading into the CA SDM database
- Comma-separated value (CSV) output
- Informal report-style output

NOTE

For more information, see pdm_deref--Dereference ASCII Data.

pdm_deref--Dereference ASCII Data

pdm_deref processes ASCII-formatted input to exchange data found in one database table for data found in another database table. It can be used to create files compatible with pdm_userload from a non-CA SDM database or spreadsheet. It can also be used to create reports or output files for a non-CA SDM database or spreadsheet.

WARNING
Do not use pdm_deref if you are unfamiliar with SQL. The directories "export" and "import" in $NX_ROOT/samples (UNIX) or installation-directory\samples (Windows) contain a standard set of specfiles for viewing.

Syntax

This command has the following format:

```
name pdm_deref  - s specfile [-c|-e|-r] [-d] [-h] [-n] [-u] [-v] <filename>
```

-s specfile

(Required) Specifies a file that defines which data is exchanged and the conditions under which it changes.

Specfile contains a list of SQL commands in the following format (note that "att" means attribute and "atts" means attributes):

```
Deref
{
input = <list of "from" atts from input file>
output = <list of "to" atts for output file>
rule = "SELECT <atts used to fill output atts> \
  FROM <table name>  \
  WHERE <att from table name to match input 1> =?\
  AND <att from table name to match input 2> = ? \
  OR <att from table name to match input 3> =?"
}
```

-c

Produces comma-separated value (CSV) output, such as:

```
"field1","field2","field3"
```

The -c, -e, and -r output format options are mutually exclusive.

-e

Produces comma-separated value (CSV) output with embedded double quotes escaped by another double quote. For example:

```
"Text with a "quoted string" in it".
```

The -c, -e, and -r output format options are mutually exclusive.

-r

Produces left-justified output in the formats if the column labels are not supplied in the input file:

```
"label":"value"
```

or

```
"value"
```

This option is intended for use with line printers, for example:

```
Field_Name: Field Value
```

The -c, -e, and -r output format options are mutually exclusive.

-d

Produces diagnostic information.

-h

Displays help and usage information.

-n

Specifies that you do not want to treat 0 valued foreign keys as NULL. This argument should be used only under special circumstances when recovering a damaged database.

-u

Produces output without headers.

-v

Specifies verbose mode.

filename

(Optional) Specifies the ASCII-formatted input file to be processed. If omitted, stdin is used.

Restriction -- Valid on UNIX only

Before using pdm_deref on UNIX, the $NX_ROOT environment variable must be set to the path of the CA SDM installation directory, and the PATH environment variable must include $NX_ROOT/bin.

Example: Using pdm_deref to Set Up Data for Input

This example shows how you can use pdm_deref to set up data for input.

Given the following data in the ca_location table:

```
id    location_name_name
86873FA40BA4234A8CF7A418D7C8B2DB    "Boulder NCC"
```

the following statement in the specfile:

```
Deref
{
input = place
output = location_uuid
rule = "SELECT id FROM ca_location WHERE location_name=?"
}
```

would change the following input:

```
last_name, first_name, place
{"Potts", "Elmore", "Boulder NCC"}
```

to the following output, which can be loaded into the ca.contact table with pdm_userload:

```
last_name, first_name, location_uuid
{"Potts", "Elmore", "86873FA40BA4234A8CF7A418D7C8B2DB"}
```

Example: Using pdm_deref to Set Up Data for Output

This example shows how you can use pdm_deref to set up data for output.

Given the following data in the ca_contact table:

```
id  last_name first_name
"69499D5A2424884887E62EC9823F5E47"    "Potts"    "Elmore"
```

the following statement in the specfile:

```
Deref
{
input = primary_contact_uuid
output = firstname, lastname
rule = "SELECT first_name, last_name FROM ca_contact
WHERE id=?"
}
```

would change the following input:

```
location_name, primary_contact_uuid
{"Boulder NCC", "69499D5A2424884887E62EC9823F5E47"}
```

to the following output, which can be printed or sent to a spreadsheet:

```
location_name, firstname, lastname
{"Boulder NCC", "Elmore", "Potts"}
```

Use the Dbadmin Mode

The dbadmin mode is a utility starts or triggers the data manipulation layer of the CA SDM system without starting the object layer. The utility provides the ability to lock the entire database to perform low-level data maintenance and data integrity.

For example, use pdm_extract, pdm_load, pdm_deref, and pdm_replace for performing batch data updates on the system. By using the dbadmin command, the administrator is essentially placing a database lock on the entire system. The backup (pdm_backup) and restore (pdm_restore) utilities, both automatically place the system in dbadmin mode to guarantee a consistent backup and restore.

The dbadmin mode is also useful to configure the system without starting the object layer until the data is modified. For example, an attribute “required” in majic on an existing system can confuse the animator if an update to an object that has the required attribute null. You can put the system into dbadmin mode and update the objects using pdm_load, and then start the system as usual.

Follow these steps:

1. Halt CA SDM either from Windows Service Manager or by running pdm_halt from the command line.

   NOTE
   It is a good practice to send an announcement to alert users and to check for logged in users before halting the system.
2. From the command line, enter the following command using the same case as shown:

   ```
   pdm_d_mgr  - s DBADMIN
   ```

   NOTE
   There is no return message, but a pause occurs before the command prompt returns. If there is no pause, check your spelling to verify that you entered data correctly.
3. Run pdm_status to verify that the system is in dbadmin mode.

   NOTE
   When the system is in dbadmin mode, the system returns the following status, which indicates that it is safe to work on the system:

   ```
   C:\>pdm_status
   The Daemons are not running.
   ```
4. When all work is complete, run pdm_halt to shut down dbadmin mode.
5. Restart the system by following your usual procedures.

How to Use pdm_deref Example

The following example shows how to use the pdm_deref utility in a CA SDM ticket tracking system.

Assume that an existing ticket tracking system that you implemented on a spreadsheet has columns labeled Trouble Description, Technician First and Last Name, and Entry Date. These columns correspond to the description, assignee, and open_date fields in the CA SDM Change_Request table. The Trouble Description field contains the same data type as the description field. But the assignee field is a numeric field, and the Technicianfield on the spreadsheet is in a “last name, first name” format.

Follow these steps:

1. Load the technicians’ names into the Contact table.
2. Prepare a pdm_deref input file with the existing information.

3. Build a specifications file to map the new contact names to assignee values.
4. Prepare a pdm_userload input file to be used to update the Change_Request table.

This process is described in more detail in the following steps:

1. Prepare a pdm_userload input file, location.dat, for the Location table as follows:

```
TABLE ca_location
location_name address_2 address_2
{"Boulder NCC - NQ", "716 Main
    Street","Boulder, CO 84302"}
{"Colorado Springs NCC", "2765 Spring Street",
    "Colorado Springs, CO 84303"}
{"Denver NCC", "3765 Stoneridge Way", "Denver,
    CO 80254"}
```

2. Load the data as follows:

```
pdm_load -f location.dat
```

3. Prepare an input file, contact.dat, with the original information as follows:

```
TABLE ca_contact
last_name first_name middle_name location pri_phone_number
{"Harrison", "Frank," "Harold", "NCC - HQ", "303-555-2333"}
{"Hertzog", "William", "I.", "Colorado Springs NCC", "303-966-1987"}
{"Lyman", "Jeanie", "L.", "Denver NCC", "303-966-5301"}
```

4. Prepare a dereferencing tool specifications file, contact.spec as follows:

```
Deref
{
input = c_location
output = location_uuid
rule = "SELECT id FROM ca_location WHERE location_name=?"
}
```

   WARNING
   Do not place a blank space in front of the SELECT keyword. Deref uses the new contacts' first and last names to obtain the appropriate numeric ID fields for loading the Change_Request table. In addition, the "hooks" represented by question marks (?), correspond to the specified input fields. You must have the same number of hooks as input fields, and they must be in the same order.

5. Run pdm_deref as follows:

```
pdm_deref -s contact.spec < contact.dat > contact.out
```

   The output file, contact.out, looks like the following:

```
TABLE ca_contact
last_name first_name middle_name location.uuid pri_phone_number
{"Harrison", "Frank", "Harold", "69499D5A2424884887E62EC9823F5E47", "303-555-2333"}
{"Hertzog", "William", "I.", "86873FA40BA4234A8CF7A418D7C8B2DB", "303-966-1987"}
{"Lyman", "Jeanie", "L.", "58AA42789957734E8BEE146D07F7AD49", "303-966-5301"}
```

6. Load the contact.out file into the CA SDM database as follows:

```
pdm_load -i -f contact.out
```

   NOTE
   You must use the pdm_load command to use the - i option.

7. (UNIX only, optional) Write a script, Convert_Ticket, as shown in the following:

```
#!/bin/sh
pdm_load -i $1
cat $2 | pdm_deref -s $3 | pdm_load -i
```

You can run this script, as shown by the following:

```
Convert Ticket location.dat contact.dat contact.spec
```

In this example, pdm_load with the -i flag is used to speed the process. If you are making these updates on a regular basis, you can drop the -i flag so that pdm_load checks for duplicate records.

The following are additional examples of dereferencing tool specification files:

```
Deref
{
    input = first_name, last_name, middle_name
    output = assignee
    rule = " SELECT id from ca_contact \
           WHERE first_name=? \
           AND last_name=? \
           AND middle_name=? "
}
```

This rule converts three fields labeled first_name, last_name, and middle_name to the appropriate contact UUID. If all three input fields are not present, the rule is not applied. No match produces an error message and processing continues. For multiple matches, the first value is used; an error message is produced, and processing continues.

Database Loader

This article contains the following topic:

The database loader utility, pdm_userload, adds, updates, and deletes CA SDM database records. You create a formatted ASCII input file for the pdm_userload utility and select the tables to load and the optional fields to add.

> **WARNING**
> The pdm_userload utility accesses the CA SDM database and does not directly interact with application software processes. The inventory items added to the database with pdm_userload do not update the helper/selection lists until the application has been halted and restarted.

Although the pdm_userload utility can run with the application active, system performance is degraded. For best results, ensure that the background server is running but that no users are using the client interface before you run pdm_userload.

For more information, see pdm_userload--Add, Update, and Delete Database Records.

> **NOTE**
> To add records with cross-referenced fields, use the pdm_deref utility pdm_deref--Dereference ASCII Data.

pdm_userload--Add, Update, and Delete Database Records

The pdm_userload utility updates a CA SDM database using an input file you specify.

> **WARNING**
> You should always backup your database before you perform a pdm_userload.

Whenever you upload tickets (such as issues or requests), your ticket number should include a unique prefix or suffix in its string. CA SDM views this number as a string of characters not as a sequential number, and does not guarantee that it assigns a unique number to the uploaded tickets. As long as you assign a unique prefix or suffix using awk or another text processor, however, you can upload tickets without CA SDM writing over previously assigned numbers.

Syntax

This command has the following format:

```
pdm_userload [-a] [-c] [-h] [-r] [-v] [-u] [-m]  - f filename
```

Input File Format

The input file entries follow this format:

```
TABLE table_name
fieldname1fieldname2 . . . . fieldnameN
{ "value11", "value12", . . . "value1N" }
{ "value21", "value22", . . . "value2N" }
.
.
.
{ "valueN1", "valueN2", . . . "valueNN" }
```

table_name is the name of the table to be loaded, as listed in the CA SDM database schema file, which is located in $NX_ROOT/site/sch/schema.sch (UNIX) or *installation-directory*\site\sch\schema.sch (Windows), where $NX_ROOT or *installation-directory* is the directory where you installed CA SDM.

-f filename

Specifies an input ASCII file.

-a

Updates all existing records, regardless of whether more than one existing record matches a single input record. Without this option, input records that match more than one existing record are rejected.

> **WARNING**
> Use this option carefully.

-c

Checks the input file against the database and reports on the updates that would be made, but does not make the updates.

-r

Removes database records that match input records. The -a option can be used with the -r option.

> **NOTE**
> Make a backup copy of the database before running pdm_userload with this option. old database records are removed, you must restore the CA SDM database with this backup copy if you wish to recover any deleted records.

-v

Specifies verbose mode.

-u

Updates existing records, but does not insert new records into the database.

-m

Mass update. Specify when you are using pdm_userload to add or delete a large number of records. This option suppresses all client notifications of updates and sends a cache refresh message for a table when pdm_userload finishes processing the table.

-x
Uses locale sensitive numeric input formats.

-t
Specifies the name or UUID of tenant to associate all loaded data with the specified tenant. This argument is valid only when multi-tenancy is installed.

Pdm_userload supports new arguments on the TABLE statement, "Truncate" and "NoNewID". These arguments are specified in an optional parenthesized option after the table name. For example:

```
TABLE Call_Req (TRUNCATE, NONEWID)
```

- **Truncate**
  Causes pdm_userload to issue a database-specific TRUNCATE command for the table prior to loading any data. In addition, it forces pdm_userload logic to use insert-only logic regardless of command-line arguments, as all records are new.
- **NoNewID**
  Causes pdm_userload to use the id value from its input control file for new rows in the table, rather than generating a new ID for inserted data.

Restrictions

You can run pdm_userload while CA SDM is active, but performance can become very slow. It is best to run pdm_userload when no one is using CA SDM.

Drop and Restore Constraints

Some of the mdb tables that start with ca_ (such as ca_contact_) have referential constraints that may impact mass loading of data using pdm_load, pdm_userload, and pdm_restore tools. If mass loading these tables is required, you may need to drop the referential constraints prior to mass loading of the data.

Two SQL scripts are provided for each DBMS type to drop and restore constraints. Prior to mass loading of data affecting ca_* tables, run the Drop version of the script. After the data load has completed, run the Add version of the script.

For SQL Server the scripts are located in *installation-directory*\samples\views\SQLServer directory. Run the following command to drop constraints:

```
osql -E -e < SQLDropConstraints.sql'
```

Run the following command to add back constraints:

```
osql -E -e < SQLAddConstraints.sql
```

For Oracle, the scripts are located in *installation-directory*\samples\views\Oracle directory.

Run the following command to drop constraints:

```
sqlplus mdbadmin/ <password> < OracleDropConstraints.sql
```

Run the following command to add back constraints:

```
sqlplus mdbadmin/ <password> < OracleAddConstraints.sql
```

How to Create and Use an Input File

You can use an input file and the pdm_userload utility to populate database tables.

The input file format is as follows:

- Surround field values with double quotes (“value”) and separate values with a comma and a space (“value1”, “value2”).
- Precede double quotes with a backslash (\) to embed them in text strings. To set a date/time field to the current date and time, use @NOW@ for the input value.
- Surround each record with curly braces separated by spaces, as follows:

```
( { record field values } )
```

- If properly delimited, input records can span more than one line in the input file as long as individual fields stay on one line. For a multi-line field such as comments, values can include a new line character (\n) to force a new line when the database field is displayed.
- Explicit new line characters are needed only for special formatting. Ordinary running text is automatically displayed with appropriate line breaks, as shown by the following example:

```
"Record status is \"COMPLETE\""
```

To create an input file for the pdm_userload utility, perform the following steps:

Follow these steps:

1. Determine which table you want to load, and the fields you want to populate in that table.
   You must populate the Name or Symbol key field for each record that you load.
2. Make a copy of the appropriate *filename*.dat file for the table that you are loading.
3. Edit your newly created copy of the *filename*.dat file, as follows:
   a. Add an entry for each record to be loaded.
   b. From under the TABLE line (see the following example), remove fields that you do not want to populate.

```
TABLE table_name
fieldname1 fieldname4 . . . fieldnameN
```

4. Save your file and exit from the editor.
5. Run the pdm_userload utility and specify the file, as shown by the following example. In this example, the input file name is myData.dat:

```
pdm_userload -f myData.dat
```

   The database table is populated.

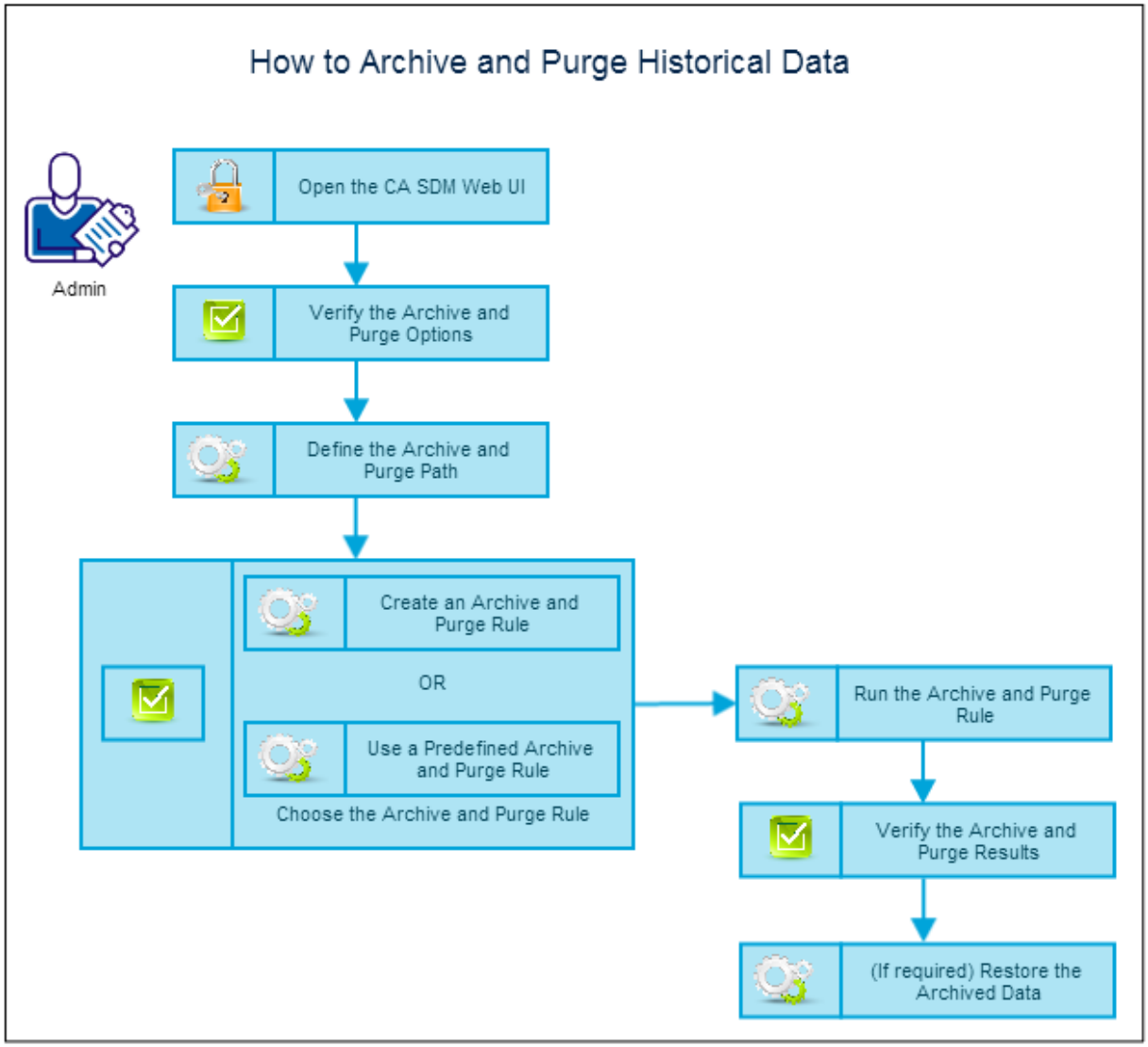
How to Archive and Purge Historical Data

Periodically remove the historical records from the system to keep the database at a manageable size for optimum performance. You create rules or activate the predefined rules to archive the historical records and purge them from the database.

Perform the following steps to archive and purge historical data:

The following diagram shows how to archive and purge the historical data:

Figure 11: Archive and Purge



Open CA SDM Web UI

Log in to the web UI from the following servers, depending on your CA SDM configuration:

- Conventional: Primary or secondary servers
- Advanced availability: Application or background servers

Verify the Archive and Purge Options

Before you set up the archive and purge rule, verify the archive and purge options that you may require to install or uninstall, depending on your organization needs. For example, the default_schedule option specifies the default schedule (workshift entry) used by the archive and purge rules. If the option value is set to an invalid workshift entry

name or empty string, the rule does not execute the rule with the schedule value. The option value field sets the NX_DEFAULT_SCHEDULE variable, which is located in the NX.env file.

Select Options Manager, Archive and Purge on the Administration tab. Install or uninstall the option that you require.

Define the Archive and Purge Path

You can define the location where you want to store the archived data. This path can be the root directory of a remote server or a UNC path. Depending on your CA SDM configuration, the following servers must run on Windows to access the UNC path:

- Conventional: Primary server
- Advanced Availability: Background server

Follow these steps:

1. Log in to web UI from the following CA SDM servers, depending on your CA SDM configuration:
   - Conventional: Primary or secondary servers
   - Advanced Availability: Application or background servers
2. To archive and purge attachments, complete the following steps:
   a. Select Attachments Library, Repositories on the Administration tab.
   The Repositories page opens.
   b. Right-click on a repository (such as Service Desk) and click Edit.
   The repository details page opens.
   c. Edit the following fields:
      - **Archive Type**
      Specifies the archive and purge action to be taken on the contents of the repository. Following values are valid:
      None: No archive and purge process is performed.
      Archive and Purge:The historic records are written to the file specified in the archive field and purged from the database.
      Purge Only:The historic records are purged from the database, but are not written to the archive file.
      - **Archive Path**
      Specifies the directory path or the UNC path to which files in the repository are moved during the archive process.
      - **UNC Credentials**
      Specifies the credentials to access the UNC path. Click UNC Credentials to open the Credentials Search page.
      - If you have already created the credentials to access the specified UNC path, search using the fields and select the credentials.
      - If you want to create the credentials, click Create New. For more information about creating credentials, see the Create UNC Credentials topic.
   d. Click Save.
3. If you want to archive and purge data other than attachments, complete the following steps:
   a. Select Archive and Purge, Archive and Purge Settings on the Administration tab.
   b. Enter the path where you want to store the archived and purged data as the Archive Purge Rule Path.
   c. If you are using a UNC path to store the archived data, click UNC Credentials.
   The Credentials Search page opens.
      - If you have already created the credentials to access the specified UNC path, search using the fields and select the UNC credentials.
      - If you want to create the credentials, click Create New. For more information about creating credentials, see the Create UNC Credentials topic.
   d. Click Save.

   The archive and purge path is defined.

Choose the Archive and Purge Rule

You are required to select the appropriate archive and purge rule. Consider the following possibilities:

- Create an archive and purge rule. For example, you create an archive and purge rule for Knowledge Management forums.
- Use a predefined archive and purge rule. For example, you use the predefined rules for archiving and purging KPI data.

Create an Archive and Purge Rule

Before you can perform an archive or purge, you create a rule. The rule defines what you want to archive and when.

Example: You create an archive and purge rule to remove deleted forums from the database.

Follow these steps:

1. Select Archive and Purge, Archive and Purge Rules on the Administration tab.
   The Archive and Purge Rule List page opens.
2. Click Create New.
   The Create New Archive and Purge Rule page opens.
3. Complete the archive and purge rule fields, as appropriate.
   - Select knowledge document as the Config. Object Name.
   - Add KS_TYPE=20 as the Additional Query.
4. Click Save.
   The new rule is displayed on the Archive and Purge Rule List page.

Example: You create an optional configuration rule inside the arcpur_cfg.xml and itil_arcpur_cfg.xml files to archive and purge the KPI_Ticket_Data node.

Follow these steps:

1. Log in to the following server, depending on your CA SDM configuration:
   - Conventional: Primary server
   - Advanced availability: Background server
2. Find the corresponding file in the $NX_ROOT/site/cfg/ directory based on the NX_OTB_MARKET value:
   - "itil" -> itil_arcpur_cfg.xml
   - else -> arcpur_cfg.xml
3. Copy the XML file into the $NX_ROOT/site/mods/cfg/ directory. This file will overwrite the out-of-box file.
4. Edit the files to define the end_time (last_mod_dt) in KPI_Ticket_Data node. It specifies the criteria to select the records for archive and purge.
5. Link the records in the KPI_Ticket_Data table to the records in the Ticket table (such as cr, chg, or iss). This ensures that all ticket-related records in the KPI_Ticket_Data table are archived and purged.
6. KPI_Ticket_Data node does not have a SREL relationship with any Ticket node and it relies on two fields, obj_name, and obj_id, to link with a ticket. The obj_name value can be cr, chg, or iss and the obj_id value is the ticket id. Define a main_obj for each ticket object.
   The following is a sample main_object definition for the ticket object, cr:

```
<!-- KPI Ticket Data -->
<main_obj>
<name>KPI Ticket Data</name>
<internal_name>KPI Ticket Data</internal_name>
<factory>ktd</factory>
<default_query>obj_name='cr'</default_query>
<date_field>end_time</date_field>
<ref_by value="obj_id">cr.id</ref_by>
```

```
</main_obj>
```

NOTE
The configuration rule can only select records for cr. The ref_by tag can match the value of obj_id in KPI Ticket Data to the value of id in cr. If a match is found, it means that a KPI Ticket Data record is referenced by a cr record, so the KPI Ticket Data record is not archived and purged.

7. After adding the configuration rules for all ticket objects, depending on your CA SDM configuration, perform the following steps:
   - Restart the CA SDM servers for conventional.
   - Restart the CA SDM servers for advanced availability.

   These configuration rules become selectable configuration object names in the Archive and Purge Rule Detail form.

Archive and Purge Rule Fields

You can use the following fields to define or edit rule definitions.

- **Rule Name**
  Specifies a unique identifier for the rule.
- **Status**
  Indicates whether this rule is active. The inactive rule runs only once, even if it is scheduled for a recurrent process.
- **Schedule**
  Specifies a workshift during which the rule should be in effect.
- **Recurrence Interval**
  Specifies how often this rule will run.
- **Archive File Name**
  Specifies the name of the file where you want to store the historic records. Enter the file name that you have mentioned while defining the archive and purge path. For more information, see the Define the Archive and Purge Path topic.
- **Operation Type**
  Specifies one of the following types of operation that the rule must execute:
  - **Archive/Purge**
    Archives the historic records to a file and purges the archived records from the database.
  - **Purge Only**
    Purges historic records from the database, but they are not written to the archive file.
  - **Archive Only (Test Run)**
    Writes historic records to the archive file without purging them from the database. Use this option to test a newly created or edited archive and purge rule.
- **Config. Object Name**
  Specifies the name of the database object this rule can archive and purge. The Object Name field is automatically populated according to your selection in the Config. Object Name field.
- **Days Inactive**
  Specifies the number of days a record is inactive to be eligible for the archive and purge from the database.
- **Additional Query**
  Archives and purges specific inactive records among the existing inactive records. Use this field when you want to create different rules for archiving and purging the subsets of expired records for the same object. Use the same syntax as you use for stored queries.
  The following query archives and purges only assigned inactive request records with a priority of 1:

  ```
  priority = 1 AND (assignee IS NOT NULL OR group IS NOT NULL) and active = 0
  ```

  The following query format archives and purges records that are based on time-span:

  ```
  close_date < EndAtTime(\'LAST_YEAR\')
  ```

Use a Predefined Rule

You can use the predefined rules in CA SDM to archive and purge historical data. These predefined rules are set to Inactive, by default. Set it active to use the rule.

Example: Use the predefined rule to archive and purge the KPI data.

Follow these steps:

1. Select Archive and Purge, Archive and Purge Rules on the Administration tab.
2. Search for any of the following predefined rules for KPI data:
   - KPI Ticket Data
   - KPI Data (System)
   - KPI Data (Stored Query)
   - KPI Data (SQL)
3. Select the rule from the search result.
4. Click **Edit** to modify the archive and purge rule fields.

   WARNING
   Ensure that you select the **Active** option from the **Status** field.
5. Click Save.
   The predefined rule is ready for use.

Verify the Archive and Purge Results

You check the results of the archive and purge rule that you have scheduled. Do one or both of the following actions:

- View the Archive and Purge History.
- View the Archive and Purge Log File

View the Archive and Purge History

You can view the history for each archive and purge rule. For example, view the objects that are purged by a rule.

Follow these steps:

1. Select Archive and Purge, Archive and Purge History on the Administration tab.
   The Archive and Purge History List page opens.
   Note: To display the Additional Search Arguments field, click the spigot icon. This field is intended only for expert users who understand SQL and Majic and can use it to specify search arguments that are not available in the standard search filter fields. To specify an additional search argument, enter a SQL WHERE clause in this field.
2. Click Show Filter and specify the filter criteria. For example, enter the earliest start date to show only entries from the specified time frame.

The list of matching rules are displayed.

- Click the rule name for which you want to review the rule configuration.
  The Archive and Purge Rule Detail page opens.

View the Archive and Purge Log File

You can check the arcpur.log file from the $NX_ROOT/log/ directory to find the errors that have occurred while executing the scheduled archive and purge rule.

NOTE
The size limitation for arcpur.log files is defined in $NX_ROOT/NX.env as

```
# The size limit for the Archive and Purge log file and data file.
```

```
@NX_ARCPUR_FILESIZE=2000000000
```

Archive and purge creates arcpur.log.0, arcpur.log.1 though arcpur.log.9 after reaching the file limit for each log files.

(If necessary) Restore the Archived Data

You can restore the archived data when you need them in the database again.

Follow these steps:

1. Start the daemons in dbadmin mode. Dbadmin mode allows limited access, so you can safely run pdm_load to restore the archived data. Run the pdm_d_mgr -s DBADMIN command on the following servers, depending on your CA SDM configuration:
   - Conventional: Primary server
   - Advanced Availability: Background server
2. Go to the root directory or the UNC location where you have stored the archived data.
3. Locate the archived data file (.dat file).
4. Copy the file to the following servers, depending on your CA SDM configuration:
   - Conventional: Primary server
   - Advanced Availability: Background server
5. Run pdm_load against the data file. For example:
   ```
   pdm_load -a -f 2004611T1726_Call_Request.dat
   ```
6. If there is a problem with the pdm_load command, complete the following steps:
   a. Check the command line and $NX_ROOT/log/arcpur.log for errors.
   b. Restart the CA SDM servers.
7. To prevent the record from being archived and purged at the next cycle, complete the following steps:
   a. Update the record to make it active again.
   b. Inactivate the associated archive and purge rule.

Additional Logging for Archive and Purge Rules (Conventional Configuration)

Set the NX_ARCPUR_RULE_LOGGING variable value to YES to specify additional logging output for Archive and Purge rules.

Perform the following steps:

1. Install this option by executing the following command on the primary server:
   ```
   pdm_options_mgr -c -s ARCPUR_RULE_LOGGING -v Yes -a pdm_option.inst
   ```

   NOTE

   For each secondary server that you have configured, ensure to manually add or update the above NX variable in each secondary server environment. The NX.env file is located in *$NX_ROOT* directory.
2. To avoid losing changes to *pdm_configure* command, execute this command with the *-t* option:
   ```
   pdm_options_mgr -c -s ARCPUR_RULE_LOGGING -v Yes -a pdm_option.inst -t
   ```
3. Restart CA SDM for the NX variable values to take effect.

Setting Up Multi-Tenancy

This article contains the following topics:

- How to Implement Multi-Tenancy
- Manage Multi-Tenancy
- How to Export and Import Tenant Data
- Utilities Used for Multi-Tenancy
- Setting Up Terms of Usage

How to Implement Multi-Tenancy

This scenario describes how an administrator, as the CA SDM privileged user, implements multi-tenancy for the first time. From start to finish, the CA SDM implementation changes as follows:

1. A single client uses a single implementation.
2. Multiple independent clients (tenants) and their users *share* a single implementation. Each tenant experiences the implementation as solely for its own use.

As the administrator, you use the CA SDM Administration interface to perform these steps:

1. Step 1: Install and enable multi-tenancy in setup mode
2. Step 2: Create the service provider tenant
3. Step 3: Create additional tenants
4. Step 4: Assign tenant access for a role
5. Step 5: Create subtenants
6. Step 6: Create tenant groups
7. Step 7: Change multi-tenancy to on mode and restart services
8. Step 8: Review the implementation and correct any problems

Step 1: Install and Enable Multi-Tenancy

You activate multi-tenancy by installing a multi-tenancy option in the product, and then enabling the setup mode. The setup mode specifies that multi-tenancy features are in effect for administrators. This mode allows the administrators to view and edit tenant-related objects and attributes. However, the product does not enforce tenancy restrictions, and the end users see no changes. This mode lets you prepare multi-tenancy by performing tasks such as defining tenants or assigning tenants to roles without impacting normal use of the product.

WARNING
When multi-tenancy is in setup mode, web interface changes are active for the service provider administrators. This behavior lets you view and edit tenancy-related objects and data on the web interface. However, tenancy restrictions are not enforced, and users other than service provider administrators do not see any product interface changes. Therefore, you can continue to use the product while implementing multi-tenancy.

NOTE

Create a backup of Domain_Constraint and usp_role tables before you proceed with implementing multi-tenancy. If multi-tenancy fails, you can use the back up to restore the tables.

Follow these steps:

1. Log in to CA SDM as an administrator and click the Administration tab.
2. In the tree on the left, click Options Manager, Multi-Tenancy.
   The Option List page appears.
3. Click multi_tenancy.
   The multi_tenancy Options Detail page appears.
4. Click Edit.
   The Update Options page appears.

5. Select setup from the Option Value drop-down list.
6. Click Install.
   The multi_tenancy option is installed.
7. Click Refresh.
   The page displays your changes.
8. Close the window.
   The Option List page reappears.
9. Restart services.
   Multi-tenancy is ready for you to implement in setup mode.

Step 2: Create the Service Provider Tenant

You use the product to create the service provider tenant. When you create the first tenant, the following occurs:

1. The first tenant always becomes the service provider.

   WARNING
   You cannot change this designation -- on the Create Tenant page, the Service Provider check box and Record Status field are read-only.
2. The product associates the privileged user (typically ServiceDesk on Windows, or srvcdesk on Linux/UNIX) to the service provider tenant. The product sets all system contacts (such as System_AHD_Generated) to belong to the new service provider tenant.
   Note: Windows provides an Administrator system user. The privileged user must assign a tenant to the Administrator user manually.

Follow these steps:

1. Select Security and Role Management, Tenants on the Administration tab.

   NOTE
   The Security and Role Management, Tenants option is available only when multi-tenancy is installed and in either setup or on mode.
2. Click Create New.
   The Create New Tenant page appears.
3. Complete the following fields:
   - **Name**
     Displays the tenant name.
   - **Service Provider**
     Identifies that this tenant is the service provider.
   - **Tenant Number**
     (Information Only) Displays the tenant number. CA SDM does not use this option.
   - **Record Status**
     Sets the tenant to Active or Inactive.
   - **Parent Tenant**
     Specifies another tenant above this tenant, making this tenant a subtenant in a tenant hierarchy.
   - **Subtenants Allowed**
     Allows this tenant to have subtenants. The tenant cannot modify the setting.
   - **Tenant Depth**
     (Information Only) Indicates the tenant depth of this tenant.
   - **Supertenant Group**
     (Information Only) Identifies the system-maintained tenant group that contains this tenant and all tenants above it in the tenant hierarchy.
   - **Subtenant Group**

(Information Only) Identifies the system-maintained tenant group that contains this tenant and all tenants below it in the tenant hierarchy.

- **Foreign Key Group**
  (Information Only) Identifies the system-maintained tenant group that contains tenants that can be referenced from an SREL in data that belongs to this tenant. The foreign key group is the same as the supertenant group.
- **Related Tenant Group**
  (Information Only) Identifies the system-maintained tenant group consisting of both the supertenant and subtenant groups for this tenant.
- **Terms of Usage**
  Specifies the Terms of Usage statement for the tenant. For more information about creating a terms of usage statement, see Setting Up Terms of Usage.
- **Logo**
  Specifies the URL for the tenant logo file, which can be any web image type.
- **Location**
  Displays the Location lookup page, which lets you specify a location.
- **Contact**
  Displays the Contact lookup page, which lets you specify a contact.

  NOTE
  If no contact is associated with the respective tenant, the Email Address and Pager Email Address fields are inactive.

4. Click Save.
   The product creates the service provider tenant.
5. Close the window.
6. Right-click the Tenant list and click Refresh.
   The Tenant List is updated and displays the service provider tenant.
7. Log out of CA SDM.

Step 3: Create Tenants

You use the product to create additional tenants. You can create as many tenants as required to manage multiple separate enterprises that provide support to clients.

Follow these steps:

1. Log in to the Administrator interface as a member of the service provider. An easy way to do this login is to log in as the privileged user (for example, ServiceDesk). This user automatically belongs to the service provider tenant.
2. Select Security and Role Management, Tenants on the Administration tab.

   NOTE
   The Security and Role Management, Tenants option is available only when multi-tenancy is installed and in either setup or on mode.
3. Click Create New.
   The Create New Tenant page appears.
4. Complete the fields in this page. Some of the fields are self-explanatory. Following fields require explanation:
   - **Service Provider**
     Identifies whether a tenant is the service provider. The first created tenant is always the service provider, afterward, this check box is read-only.
   - **Tenant Number**
     (Information Only) Displays the tenant number. CA SDM does not use this option.
   - **Subtenants Allowed**

Allows this tenant to have subtenants. The tenant cannot modify the setting.

- **Tenant Depth**
  (Information Only) Indicates the tenant depth of this tenant.
- **Supertenant Group**
  (Information Only) Identifies the system-maintained tenant group that contains this tenant and all tenants above it in the tenant hierarchy.
- **Subtenant Group**
  (Information Only) Identifies the system-maintained tenant group that contains this tenant and all tenants below it in the tenant hierarchy.
- **Foreign Key Group**
  (Information Only) Identifies the system-maintained tenant group that contains tenants that can be referenced from an SREL in data that belongs to this tenant. The foreign key group is the same as the supertenant group.
- **Related Tenant Group**
  (Information Only) Identifies the system-maintained tenant group consisting of both the supertenant and subtenant groups for this tenant.
- **Terms of Usage**
  Specifies the Terms of Usage statement for the tenant. For more information about creating a terms of usage statement, see Setting Up Terms of Usage.
- **Logo**
  Specifies the URL for the tenant logo file, which can be any web image type.

NOTE
If no contact is associated with the respective tenant, the Email Address and Pager Email Address fields are inactive.

5. Click Save and close the window.
   The product creates the tenant.
6. (Optional) Click Tenant Groups to add this tenant to a new group or to an existing one.

Step 4: Assign Tenant Access for a Role

The role of a CA SDM user governs both access authorization and the user interface. The roles available to users depend on their access type. Multi-tenancy lets you control the tenant or tenant group that a user can access within the role. When multi-tenancy is installed, the Role Detail page includes additional options that let you assign or edit tenant access.

NOTE
You can grant tenant users access to data other than their own. Non-service provider tenant analysts only have access to their own tenant and subtenants. However, you can you update their function access to include the tenant of the analyst. For example, you can define a role to set separate read and write access to certain tenant groups for users within that role.

Follow these steps:

1. Navigate to Security and Role Management, Role Management, Role List.
   The Role List appears.
2. Click the role for which you want to assign tenant access.
   The Role Detail page appears and provides Tenant Access and Tenant Write Access drop-down lists on its Authorization tab. Tenant Access is view-only, and Tenant Write Access allows create and update also.
3. Click Edit.
   The Update Role page appears.
4. Select options for Tenant Access and Tenant Write Access:
   - **Same As Tenant Access**

Sets the access to be the same as the Tenant Access setting. This value is the default for the Tenant Write Access drop-down list and is only available for the Tenant Write Access option.

- **All Tenants**
  Removes tenant restrictions. A user in a role with this access can do the following:
  - View any object in the database (read access).
  - Create and update (write access) any tenanted object in the database.

  When a user with All Tenants access creates an object, the user must select the tenant of the new object.
- **Single Tenant**
  Sets tenant access for a role to a named tenant. When you select this option, another field appears that lets you select a specific tenant. A user in this role can access only those objects associated with the named tenant.
- **Tenant Group**
  Sets tenant access for a role to a user-defined or system-maintained tenant group. After you select this option, another field appears that lets you select a specific tenant group. A user in this role can access only those objects associated with one of the tenants in this group. When a user with tenant group access creates an object, the user must select the tenant for the new object.
- **Contact's Tenant**
  Sets tenant access for the role to the tenant of the contact using it. A user in this role can access only those objects associated with their own tenant.
- **Contact's Tenant Group**
  Sets role access for an analyst role to the tenant group that the analyst works with, as specified on the contact record for the analyst. If the user with the role is not an analyst, this selection has the same effect as Contact's Tenant. This option is only available for analysts.
- **Contact's Subtenant Group**
  Sets tenant access for the role to the subtenant group of the contact using it. A user in this role can access only those objects associated with their own subtenant group.
- **Contact's Supertenant Group**
  Sets tenant access for the role to the supertenant group of the contact using it. A user in this role can access only those objects associated with their own supertenant group.
- **Contact's Related Tenant Group**
  Sets tenant access for the role to the Related Tenants Group of the contact using it. A user in this role can access only those objects associated with their own related-tenant group.
- **Update Public**
  Controls whether a service provider user in the role has the authorization to create or update public data. All users can view public data, regardless of access rights for the current role. Tenant users (users belonging to a tenant other than the service provider) cannot update public data, regardless of their role.
- **Click Save**
  Tenant access is assigned for the role. When a user queries the database, the product restricts the results to objects belonging to tenants associated with the role of the user.

Step 5: Create Subtenants

Subtenancy lets you define and modify tenant hierarchies for organizational and data-sharing purposes. To place a tenant into a tenant hierarchy, you assign the tenant a parent tenant.

Follow these steps:

1. On the Administration tab, select Security and Role Management, Tenants.
   The Tenant List appears.

   NOTE
   The Security and Role Management, Tenants option is available only when multi-tenancy is enabled.
2. Click an existing tenant to Edit, or click Create New.

The Tenant Detail page appears, which lets you enter any required data or changes.

3. Select a Parent Tenant.

   NOTE
   The Parent Tenant drop-down list only displays tenants that are allowed to have subtenants.

4. Click Save.
   The tenant is a subtenant of the parent tenant.
   Note: When a tenant is a subtenant, it belongs to the subtenant group of the parent tenant. The parent tenant joins the supertenant group of the subtenant. Each tenant joins the Related Tenants group of the other.

Step 6: Create Tenant Groups

A tenant group is a collection of tenants that share access to CA SDM objects. Tenant groups let you classify, manage, and control access to tenants. You can assign a role to a tenant or tenant group. When multi-tenancy is active, the product associates each role with: all tenants (public), a single tenant, or a single tenant group. Use tenant groups whenever a role needs access to more than one tenant. For example, you can assign analysts to a tenant group containing tenants belonging to a particular geographic location.

The product generates and maintains three tenant groups automatically for each tenant in a tenant hierarchy (*tenant* is the tenant name):

- *tenant*_subtenants (tenant, its child tenants, and their lower subtenants)
- *tenant*_supertenants (tenant, its parent tenant and its higher supertenants)
- *tenant*_relatedtenants (entire single hierarchy)

You use the system-maintained tenant groups like user-defined tenant groups. However, you can only change the system-maintained tenant group names and descriptions.

Example: Role A Needs Access to Tenant A, Tenant B, and Tenant J

Instead of assigning the role to each tenant separately, you can do the following:

1. Create a tenant group, and add Tenant A, Tenant B, and Tenant J to the group.
2. Assign Role A to this tenant group.
   Users (contacts) assigned to Role A can access the tenant group, which is comprised of Tenants A, B, and J.

Follow these steps:

1. Log in as the service provider, click the Administration tab, and select Security and Role Management.
2. Click Tenant Groups.

   NOTE
   The Security and Role Management, Tenant Groups option is available only when multi-tenancy is installed (either on or setup).

3. Click Create New.
   The Create New Tenant Group page appears.
4. Complete the tenant group fields and click Save.
   The tenant group is created.
5. Close the window.
   The Tenant Group List appears.
6. Right-click the Tenant List and select Refresh.
   The Tenant Group List is updated.
7. Click Update Tenants on the Tenant Group Detail page and add tenant members to the group.
8. (Optional) Repeat Steps 3 through 6 for each tenant group you want to create.

Step 7: Change Multi-Tenancy to On Mode

You change changing the Multi-Tenancy option to on mode to make the multi-tenancy implementation function fully. Each tenant then views the implementation as solely for its own use. Each tenant cannot update or view the data of another tenant.

Follow these steps:

1. Log in to CA SDM as an administrator, and click the Administration tab.
2. In the tree on the left, click Options Manager, Multi-Tenancy.
   The Option List page appears.
3. Click multi_tenancy.
   The multi_tenancy Options Detail page appears.
4. Click Edit.
   The Update Options page appears.
5. Select setup from the Option Value drop-down list.
6. Click Edit.
   The Update Options page appears.
7. Select on (the default) from the following values in the Option Value drop-down list:
   - **on**
     (Default) Disallows a check-in to a tenant-required table when the tenant is null and an SREL to a table with a tenant is not available.
   - **on (warn)**
     Writes an error to the log but allows the check-in to proceed when a tenant-required object with a null tenant is created or updated.
   - **on (allow)**
     Writes a warning to the log but allows the check-in to proceed when a tenant-required object with a null tenant is created or updated.
8. Click Save, and then Refresh.
   The page displays your changes.
9. Close the window.
   The Option List page reappears.
10. Restart services.
    Multi-tenancy is fully functional.

Step 8: Review the Implementation and Correct

Review the multi-tenancy implementation and correct any problems.

Follow these steps:

1. Log in to CA SDM using the privileged username (typically ServiceDesk).
2. Click the Administration tab and browse to the Tenant List.
   The Provider shows as Yes for the privileged user in the Tenant Name.
3. Verify that your multi-tenancy restrictions are enforced by browsing to a Contact List.
   If tenant-required tables incorrectly include untenanted data in a multi-tenancy system, the following message appears in the Contact List:

```
AHD05358 There were nn untenanted active Contact objects at CA Service Desk Manager startup.
```

WARNING
If untenanted data is in the database, you can set the multi-tenancy option mode to on (warn) or on (allow). These modes let you update tenant-required tables with a null tenant. This method prevents data loss when a service level agreement (SLA) or attached event executes for a ticket that does not include a tenant.

4. (Optional) Disable multi-tenancy if problems occur and complete the following steps:
   a. Restore the Domain_Constraint and usp_role tables.
   b. Set the Multi-Tenancy option to the setup mode.
   c. Recycle the system.
   The site can resume previous operations while you correct whatever issues required the reversion.

Manage Multi-Tenancy

This article contains the following topics:

How to Initialize a New Tenant

As the service provider, you may want to build a standard set of data for a new tenant, such as categories, data partitions, ticket templates, and so on. This task can be done by using pdm_extract or pdm_tenant_extract to build a pdm_userload input file containing the data you want.

If necessary, you can edit this file with any text editor. It can then be loaded into the database using pdm_userload with the -t argument to set the tenant column to the new tenant.

The following process describes how to initialize a new tenant:

1. Create the tenant in the ca_tenant table. Use the online Create Tenant page.
2. Load the standard data as previously described.
   Use pdm_userload -t to set the tenant.
3. Create contact records for the new tenant.
   Load outside data or use pdm_userload -t.

How to Convert an Existing Tenant Implementation to the Tenant Object

You may have used data partitions and another CA SDM object to achieve some of the functionality now provided by multi-tenancy. If you want to convert an implementation to multi-tenancy, the first step is to map the data in the previously-used object to the new tenant object. The previously-used object is called the *pre-tenant* object. For most sites with these requirements, the org (organization) object is the pre-tenant object, but the following approach can be used for any pre-tenant object.

1. If the pre-tenant object is not org, verify that its Majic object definition specifies TENANT_REQUIRED.
2. Verify the attribute mappings from the pre-tenant object to the new tenant object in the buildtenant.xml file in the following location:
   $NX_ROOT/samples/multi_tenancy

   NOTE
   You must copy buildtenant.xml to the $NX_ROOT/site/cfg directory. In addition, buildtenant.xsd must be in the same directory as buildtenant.xml, or you will receive an error. When you install the product, buildtenant.xsd is located in $NX_ROOT/site/cfg, so you do not have to copy this file.

   The default settings are based on org. If the pre-tenant object is not org, you must edit the file.
3. Run pdm_buildtenant -f.
   A new tenant is created for each pre-tenant object, and sets the tenant attribute in the pre-tenant object to reference the new tenant.
4. Log in to CA SDM, and review both the tenant object and the pre-tenant object.

   NOTE
   In some situations, you want to map multiple pre-tenant objects to a single tenant object. To do this, manually update the pre-tenant objects affected, and then delete or inactivate the unused tenants.

How to Populate the Tenant Attributes in Your Tables

To populate the tenant attribute in all or a subset of a table, use the *pdm_settenant* utility. This utility uses a configuration file to select the objects to be tenanted and to specify where to obtain the tenant for the objects. You can specify an explicit tenant, or specify that the tenant should be derived from an SREL reference in the object to be tenanted.

To populate the tenant attributes in your tables using pdm_settenant, complete the following steps:

1. Create or edit a configuration file.
   The configuration file selects the rows that will have their tenant attribute set and specifies a source for the tenant attribute's value. The product provides a sample settenant.xml file in the following location:
   $NX_ROOT/samples/multi_tenancy

   NOTE
   You can modify the sample settenant.xml file, or create a file and copy it to the $NX_ROOT/site/cfg directory. In addition, settenant.xsd must be in the same directory as settenant.xml, or you will receive an error. When you install the product, settenant.xsd is located in $NX_ROOT/site/cfg, so you do not have to copy this file.
2. Run pdm_settenant -f [configuration file] -r
   The pdm_settenant utility reads its configuration file and processes each rule it defines in sequence.
   We recommend that you use this utility first to populate the tenant attribute in the cnt (contact) object, and then use the cnt object as a source for populating tenant into other objects.
   After the cnt object is properly tenanted, it can be used as a base for setting tenant in other tables by completing the following steps:
   a. Specify a TenantRule with type="SREL" in the configuration file for an attribute referencing the cnt object to set tenant in other tables.
   b. (Optional) Specify a TenantRule with type="Name" < tenantname > to set tenant explicitly in some of the tables.
3. Run pdm_settenant with a new configuration file.
4. Rerun pdm_settenant as required.
   After you have populated the tenant column in an object, you can use SRELs to that object as the basis of an SREL TenantRule for setting tenant in other objects.

Example: SREL Type Syntax

SREL type syntaxt checks for cnt objects that do not have a tenant value specified and uses the tenant value from the linked organization object:

```
<Object name="cnt">
<TenantRule type="SREL">organization</TenantRule>
<Where>tenant is null</Where>
</Object>
```

Example: Name Type Syntax

Name type syntax checks for org objects that do not have a tenant value specified and sets their tenant field to the name of an actual Tenant object:

```
<Object name="org">
<TenantRule type="Name">Tenant A</TenantRule>
<Where>tenant is null</Where>
</Object>
```

Tenant Hierarchies

A *tenant hierarchy* is a structured tenant group that is system-created or modified when you assign a *parent tenant* to a tenant. The tenant becomes a *subtenant* of the parent and higher tenants (if any) in that hierarchy.

NOTE
The service provider can create multiple unrelated hierarchies, or none. Even in a system with tenant hierarchies, you can define standalone tenants.

A subtenant typically represents a subdivision within its *supertenants*. A subtenant can have its own business rules and data, and supertenant data is "pushed" to the subtenant automatically on a read-only basis.

CA SDM supports a tenant hierarchy of unlimited depth. However, the *service provider* can specify a limit on the total number of tenants and the depth of tenant hierarchies (default is four levels). The service provider also determines whether individual tenants can have subtenants.

NOTE
The service provider can participate in tenant hierarchies, but this is not required. The service provider cannot have a parent tenant.

Create a Subtenant

Subtenancy allows you to build and modify tenant hierarchies for organizational and data-sharing purposes. To place a tenant into a tenant hierarchy, you assign it a parent tenant.

Follow these steps:

1. On the Administration tab, select Security and Role Management, Tenants.

   NOTE
   The Security and Role Management, Tenants option is available only when multi-tenancy is enabled.
2. Click an existing tenant to Edit, or click Create New.
   Enter any required data or changes.
3. Select a Parent Tenant.

   NOTE
   The Parent Tenant drop-down only displays tenants that are allowed to have subtenants.
4. Click Save.
   The tenant is a subtenant of the parent tenant.

   NOTE

   When a tenant is a subtenant, it belongs to the Subtenant group of the parent tenant, as do the subtenants (if any) of that subtenant, and so on. The parent tenant joins the Supertenant group of the subtenant, as do the supertenants (if any) of that supertenant, and so on. Each joins the Related Tenants group of the other.

System-Maintained Tenant Groups

CA SDM generates and maintains three tenant groups automatically for each tenant in a tenant hierarchy (*tenant* is the tenant name):

- *tenant*_subtenants (tenant, its child tenants, and their lower subtenants)
- *tenant*_supertenants (tenant, its parent tenant and its higher supertenants)
- *tenant*_relatedtenants (entire single hierarchy)

System-maintained tenant groups can be used like user-defined tenant groups. However, only their names and descriptions can be modified.

Tenant Data Assignments

CA SDM displays the tenant in the same format in both the View and Edit versions of a detail page for an existing object, because the tenant for an existing object cannot be changed from the web interface.

When you edit a tenanted object, drop-down lists on the edit page are automatically restricted to values that are public, owned by the same tenant as the base object or any tenants above it in the tenant hierarchy, or owned by the service provider (if the drop-down list applies to a SERVICE_PROVIDER_ELIGIBLE attribute).

There are no changes on the detail page for lookup fields associated with a tenanted object. If a user with access to multiple tenants clicks a lookup link to a tenanted table, the web engine automatically restricts the lookup to values appropriate for the attribute, and displays a banner message on the pop-up search or list page.

NOTE
Tenant restrictions are not displayed in the search filter, and they cannot be changed by the user.

Tenant becomes a selector (either a lookup or a drop-down list) when you ask to create a tenant-required object.

If the tenant field is empty, you can specify a tenant value directly by filling in the field, or indirectly by specifying a value for a tenant-implying attribute (such as Affected End User). The interface displays the following suffixes:

- **(T)**
  Indicates an attribute that is tenant-implying; that is a lookup to a tenant-required table.
- **(TO)**
  Indicates an attribute that is optionally tenant implying that is, a lookup to a tenant-optional table.

web.cfg properties control the text of these indicators.

Except for the tenant attribute itself, tenant-implying attributes are always shown as lookups, even if created with a dtlDropdown macro.

CA SDM automatically sets the tenant when you look up or autofill a tenanted value into any tenant-implying field (except that filling a SERVICE_PROVIDER_ELIGIBLE field with a reference to a service provider object does not set the tenant). After the tenant is set, lookups for tenant-implying fields are restricted in the same way as existing tenanted objects.

NOTE
Until you save the object, the tenant field remains editable, and you can change the tenant by directly updating it. When you change the tenant, CA SDM automatically clears any tenant-implying fields containing references to objects belonging to the previous tenant.

CA SDM typically initializes the Tenant selector to empty. You can change this behavior in several ways:

- Open the Create New page from a page such as the Quick Profile, which pre-populates a tenant-implying field
- Set the Retain Tenant user preference
  This is a new user preference that initializes the tenant for new objects to the same tenant as the last detail page viewed or updated, or in the last list page search filter restriction.
- Open the page with a URL that explicitly specifies a tenant
  This is not provided in any predefined URL, but is available to allow sites to create menu items or buttons that specify a tenant.

NOTE

If you create configuration items from another CA Technologies product (such as CA APM) or the command-line interface, then the object is Public.

Create a Tenanted Object

The service provider can add tenant-specific data to objects such as issues, requests, change orders, and so on. You can add a tenant to a ticket (such as an incident) that is created from a Scoreboard tab.

Follow these steps:

1. Click File, New Incident.
2. Complete any of the following steps:
   a. Select the tenant from the Tenant drop-down.

b. Click Affected End User (or any other tenant-implying field).
The Contact Search page appears. Search for a user; you can filter the search by tenant.
c. Enter a name into the Affected End User field.
The tenant data completes automatically.
3. Continue to create the incident.

Activity Notifications

Activity Notifications control both the contents of notifications and which contacts receive notifications for various events in the history of a ticket.

In a multi-tenancy environment, the notification rule is a tenant-optional object. Public notification rules apply to all tickets; tenanted rules apply only to tickets with the same tenant as the rule, or to tenants in its subtenant hierarchy. The tenanting restriction is applied in addition to any condition specified with the rule itself.

Default Notification Rules are stored as public objects. If multi-tenancy is installed, you must create a copy of the Notification Rule for each of the tenants, otherwise the Update Contacts option is restricted.

Repositories

The repository (doc_rep) object is tenant-optional. Tenants can define their own repositories, and it is possible to define public repositories for objects such as attachments to public knowledge documents. Each tenant can have its own default repository, and you can specify a default public repository.

All attachments are either public or associated with a single tenant. If a tenant does not have its own default repository, the public repository is displayed as the default for its tenanted objects.

How to Export and Import Tenant Data

This article contains the following topics:

The service provider can extract tenant data from an existing multi-tenancy implementation and import it into a new system.

> **NOTE**
> Depending on the volume of your data, the extraction process can take several hours. You may have to perform the extract and import in multiple phases, as follows:

- **Initial**
  Extracts a base line and creates a control file used in subsequent phases .
- **Update**
  Uses the control file to extract only data that has changed since the previous run.
- **Final**
  Performs the same steps as Update, except that it also extracts animations. Animations are omitted from both the Initial and Update phases.

To extract data from one database and import it into another, complete the following steps:

1. Run an initial phase of pdm_tenant_extract to extract base-line data. This builds the control file used by subsequent phases.
2. Prepare a new, clean MDB for the extracted data.

> **WARNING**
> The output from the initial phase *must* be loaded into a database that has never been used for the product or for any other product. Each table loaded from initial phase data is truncated prior to the load, which could cause loss of data if the database is already in use.

3. To avoid duplicate privileged contacts appearing on the new system, you must inactivate the privileged contacts. Log onto CA SDM and change the Status of these contacts to "inactive" before loading the extracted data.
4. To avoid referential problems during the data load, run the appropriate drop constraints script:
   – (Oracle) Run $NX_ROOT/samples/views/Oracle/OracleDropConstraints.sql
   – (SQL Server) Run $NX_ROOT/samples/views/SQLServer/SQLDropConstraints.sql
5. Use pdm_userload to load the data from the initial phase into the clean MDB prepared in steps 2 and 3.
6. Run an update or final phase of pdm_tenant_extract to extract additional data created or modified since the previous phase. Pdm_tenant_extract uses the control file created in step 1 to determine the data already processed by the previous phase.
7. Use pdm_userload to load the data extracted in step 5 into the same MDB containing data loaded from the previous phases.

   NOTE
   For more information about this utility, see pdm_userload.
8. Repeat steps 5 and 6 as required until all data has been imported into the new database. The last run should be the final phase.
9. To protect the integrity of the new database, restore the constraints dropped in step 3 by running the appropriate add constraints script:
   – (Oracle) Run $NX_ROOT/samples/views/Oracle/OracleAddConstraints.sql
   – (SQL Server) Run $NX_ROOT/samples/views/SQLServer/SQLAddConstraints.sql
10. Use pdm_tenant_delete to delete the extracted data from the original database.
11. Ensure that all repositories associated with extracted tenants are copied to the target settings.

How to Handle Attachments and Repositories

Attachments are stored in repositories. You must copy all repositories that are associated with extracted tenants to the target system, including public repositories. This process is primarily a manual operation, with the following steps:

1. Redefine location-specific information for all repositories, after completion of the initial load of the data into the target system. This task includes changing the following values:
   – Server Name
   – Upload Path
   – Servlet Path
   – Archive Path
2. Manually create all required directories and folders.
3. Copy all attachment files from the previous location to the new repository location after (or during) the load of the data from the Final phase.

After you complete these steps, all references for attachments in the target system should be successful. However, copies of the attachments remain on the source system. Use the pdm_clean_attachments.pl utility to clean these redundant attachments.

Utilities Used for Multi-Tenancy

This article contains the following topics:

This section describes utilities that are used to manage a multi-tenancy environment.

Note: Required parameters are enclosed within "{ }", while optional parameters are in "[ ]".

pdm_buildtenant -- Creating Tenants from Another Object

The *pdm_buildtenant* utility is used to create tenants from another object. You may have used data partitions and another CA SDM object to achieve some of the functionality now provided by multi-tenancy. If you want to convert an implementation to multi-tenancy, the first step is to use pdm_buildtenant to map the data in the previously-used object to the new tenant object.

> **WARNING**
> Before you run pdm_buildtenant, you *must* configure the service provider.

In this section, the object used to hold tenant-like information is named the pre-tenant object. For most sites with these requirements, the org (organization) object is the pre-tenant object, but the following approach can be used for any pre-tenant object.

The pdm_buildtenant utility builds the tenant objects from pre-tenant objects. This application creates a tenant for each pre-tenant object, and sets the tenant attribute in the pre-tenant object to reference the new tenant. This utility has the following syntax:

```
pdm_buildtenant [-h] | [-f [configuration_file]
```

- **-f configuration_file**
  (Optional) Specifies the location of a configuration file specifying the rules for creating tenants from the pre-tenant object. If this argument is not included, pdm_buildtenant uses the configuration file from the $NX_ROOT/site/cfg directory. This file assumes the pre-tenant object is org; if this is not the case, you *must* edit the configuration file before using pdm_buildtenant.

  > **NOTE**
  > You *must* copy buildtenant.xml to the $NX_ROOT/site/cfg directory. In addition, buildtenant.xsd must be in the same directory as buildtenant.xml, or you receive an error. When you install the product, buildtenant.xsd is located in $NX_ROOT/site/cfg, so you do not have to copy this file.
- **-h**
  Displays usage information for pdm_buildtenant.

The following is the format of the configuration file:

```
<?xml version="1.0" encoding="utf-8" ?>
<BuildTenant>
   <Object from="MajicObjectName">
    <Attribute from="sourceAttribute1" to="tenantAttribute1" />
    <Attribute from="sourceAttribute2" to="tenantAttribute2" />
   </Object>
</BuildTenant>
```

The *from* attribute of the Object tag identifies the pre-tenant object. Each Attribute tag identifies an attribute to be copied from the pre-tenant object to an attribute of the new tenant.

> **WARNING**
> For UNIX implementations of multi-tenancy, you *must* run pdm_task to export LIBPATH before executing the pdm_settenant and pdm_buildtenant utilities. If you do not run pdm_task before executing these utilities, you receive system errors. Use ../pdm_task to run the command.

pdm_clean_attachments -- Delete Redundent Attachments After Importing Tenant Data

After importing tenant data, you should delete redundant attachments. This utility has the following syntax:

```
pdm_perl pdm_clean_attachments.pl [-h] | [-n repository_name] | [-S|-K]
```

- **-h**

Specifies to display command line help.

- **-n *repository_name***
  Specifies the name of the repository to process. If not specified, all repositories are processed.
- **-S**
  Specifies that only CA SDM repositories are processed.
- **-K**
  Specifies that only Knowledge Management and Embedded Images repositories are processed.

NOTE
Running the pdm_clean_attachments.pl command without any arguments processes all repositories.

WARNING

On UNIX, the LIBPATH must be set before running several CA SDM utilities. Use *pdm_task* to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

pdm_settenant -- Assigning Tenants to Objects

After you define tenants, you must use the *pdm_settenant* utility to set the tenant column in other objects. This utility has the following syntax:

```
pdm_settentant [-h] | {-f [configuration_file] | -r} [-d domsrvr]
```

- **-d domsrvr**
  (Optional) Specifies a domsrvr to use. If this argument is not specified, pdm_settenant uses the default domsrvr.
- **-f *configuration_file***
  (Optional) Specifies the location of a configuration file specifying the data that will be updated and the rules for updating the file. If this argument is not specified, pdm_settenant uses the configuration file from the $NX_ROOT/site/cfg directory (after the configuration file is copied to the $NX_ROOT/site/cfg folder).
  Note: You can modify the sample settenant.xml file, or create a file and copy it to the $NX_ROOT/site/cfg directory. In addition, settenant.xsd must be in the same directory as settenant.xml, or you will receive an error. When you install the product, settenant.xsd is located in $NX_ROOT/site/cfg, so you do not have to copy this file.

The following sample XML code describes the format of this file:

```
<?xml version="1.0" encoding="utf-8" ?>
<SetTenant>
   <Object name="MajicObjectName">
    <TenantRule type="SREL">MajicColumName</TenantRule>
    <Where>tenant is null</Where>
   </Object>
   <Object name="MajicObjectName">
   <TenantRule type="Name">TenantName</TenantRule>
   <Where>tenant is null</Where>
   </Object>
</SetTenant>
```

Each Object tag specifies a CA SDM object to be tenanted. The TenantRule tag specifies how pdm_settenant should determine the tenant, and the Where tag selects the objects to be tenanted. There are two types of TenantRule tags:

- type="Name"
  Specifies an explicit tenant by name.
- type="SREL"
  Specifies an SREL attribute in the object. Pdm_settenant copies the tenant of the object referenced by the SREL.
- **-h**

Displays usage information for pdm_settenant.

- **-r**
  Outputs a report displaying the total number of rows in each tenant-required table, and how many have a null tenant column.

  NOTE
  If both the -f and -r arguments are specified, pdm_settenant outputs a report after completing its update. If you only specify the -r argument, pdm_settenant outputs a report, but does not update any data.

Running pdm_settenant without any arguments displays usage information. To run pdm_settenant using the default configuration file, specify the -f option without the configuration_file argument. The pdm_settenant utility reads its configuration file and processes each rule it defines in sequence. It writes output to the pdm_settenant.log file in the $NX_ROOT/log directory.

You can run pdm_settenant as many times as needed. The first pass may take a significant time (possibly several hours at a large site). Subsequent passes run faster, as they only need to process rows that have not been updated. This prepares the database prior to installing the multi-tenancy option.

WARNING
On UNIX implementations of multi-tenancy, you must run pdm_task to export LIBPATH before executing the pdm_settenant and pdm_buildtenant utilities. If you do not run pdm_task before executing these utilities, you will receive system errors. Use ../pdm_task to run the command.

Assign Tenants to Objects Considerations

After you define tenants, you can use the pdm_settenant (assign tenants to objects) utility to set the tenant column in other objects. When you change the tenant for an object, you must consider whether to change the tenancy on related tenanted objects in order to maintain data integrity. Failure to keep these objects synchronized can cause data to appear missing from CIs, relationships, MDRs, versioning, and so on. The following CA CMDB objects are tenanted:

- nr -- CI definitions
- nr_com -- Log entries associated with a CI
- bmhier -- Relationships associated with CIs
- mdr_idmap -- MDR provider definitions
- ci_mdr_idmap -- CI/MDR federated mappings

For each CI, do the following to synchronize data when you use pdm_settenant to change tenancy:

- Specify nr for the CI object name.
- Change the log entries associated with the CI in nr_com so that you can view the log entries for the new tenant.

Example: XML to Change the Tenant and Log

The following XML changes the tenant for a CI named CITest to T2 and also changes the corresponding log entries in nr_com:

```
<Object name="nr">
<TenantRule type="Name">T2</TenantRule>
<Where>name = 'CITest'</Where>
</Object>
<Object name="nr_com">
<TenantRule type="Name">T2</TenantRule>
<Where>asset_id.name = 'CITest'</Where>
</Object>
```

pdm_tenant_delete -- Deleting Tenant Data from a Database

The *pdm_tenant_delete* utility removes all data for a specified tenant from the database.

WARNING
The referential constraints on the ca_ tables must be dropped before running pdm_tenant_delete and restored afterwards.

This utility has the following syntax:

```
pdm_tenant_delete -h|-t tenant_name [-C|-R] [-Q]
```

- **-h**
  Displays the usage information for pdm_tenant_delete.
- **-t *tenant_name***
  Specifies the name of the tenant of the data to be deleted.

  NOTE
  The tenant must be marked inactive before you can use this utility to delete the data.
- **-C**
  Specifies that all contacts for a tenant will be marked inactive. Since contacts can be shared between products, default logic should not mass delete or mass inactivate contacts unless explicitly requested.

  NOTE
  This option is ignored if the -R option is specified.
- **-R**
  Specifies that all rows in all tenanted tables marked CA_COMMON in ddict.sch will be deleted, including the tenant object itself.

  WARNING
  These tables are shared between multiple products, so use this option with caution.
- **-Q**
  Specifies quick query processing to execute database queries as fast as possible. If this argument is not specified, the utility uses background query processing so that queries run only when the system is otherwise idle. This argument improves running time at the expense of a greater impact on an active system.

  WARNING

  On UNIX, the LIBPATH must be set before running several CA SDM utilities. Use *pdm_task* to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

pdm_tenant_extract -- Extracting Tenant Data

The *pdm_tenant_extract* utility extracts all data for a specified tenant from the database. It extracts the data in pdm_userload format so that it can easily be loaded into another database. This utility has the following syntax:

```
pdm_tenant_extract -h | -c control_file [-d domsrvr] [-g yes|no] [-o output_file]    -p phase  [[-t
 tenant_name]...] [-Q] [table1 [table2...]]
```

- **-h**
  Displays the usage information for pdm_tenant_extract.
- **-c *control_file***
  Specifies the location of the control file for this tenant extract. For the Initial phase, the file is created in the specified location (and must not already exist). The file must exist for the Update and Final phases.
- **-d domsrvr**
  (Optional) Specifies a domsrvr to use.
- **-g yes|no**
  (Optional) Specifies whether or not public data is included in the output file. If this argument is not specified, public data from all tables is included.
- **-o *output_file***

(Optional) Specifies the location of the output file. If this argument is not specified, output is directed to stdout.

- **-p *phase***
  Specifies the phase of the extract. Use one of the following values:
  I -- Initial
  U -- Update
  F -- Final
- **-t *tenant_name***
  Specifies the name of a tenant to be extracted. This argument is required for the Initial phase and can be repeated for multiple tenants. It is not valid on the Update and Final phase.
- **-Q**
  Specifies quick query processing to execute database queries as fast as possible. If this argument is not specified, the utility uses background query processing so that queries run only when the system is otherwise idle
- ***table1 [table2...]***
  (Optional) Specifies the tables to extract. If omitted, all tables are extracted.

  WARNING
  The output from the initial phase must be loaded into a database that has never been used for CA SDM or for any other product. Each table loaded from initial phase data is truncated prior to the load, which could cause loss of data if the database is already in use.

  NOTE
  To avoid referential problems during the data load, run the appropriate drop constraints script ($NX_ROOT/samples/views/Oracle/OracleDropConstraints.sql or $NX_ROOT/samples/views/SQLServer/SQLDropConstraints.sql). After the loads complete, re-apply the constraints with the appropriate xxxAddConstraints.sql script found in the same directory.

pdm_userload -- Load Tenant Data

The *pdm_userload* utility is used to load data into a CA SDM database. This utility is available even if multi-tenancy is not installed. Multi-tenancy adds support for one additional argument (-t) that specifies the name of a tenant whose id must be inserted into the tenant column of all rows inserted or updated into a tenanted table. `The specified tenant must already be in the database.

When extracting data, complete the following steps to avoid errors in stdlog:

1. Before you start loading data, shut down CA SDM and restart the product in DBADMIN mode as follows:
   – **Windows**
     Run pdm_d_mgr -s DBADMIN
   – **UNIX**
     Run pdm_init -s DBADMIN
2. After the data loads, shut down CA SDM using the pdm_halt command.
3. Restart CA SDM in normal mode.

Setting Up Terms of Usage

For each tenant, you can configure a terms of usage statement which presents the end user with an initial statement when they log in to CA SDM. The terms of usage statement reminds the end user about the proper use of the product that they must agree to before they can continue to log in to CA SDM. A log is generated after the user Accepts or Rejects the statement.

In CA SDM, you can configure a terms of usage statement from the Security and Role Management node on the Administration tab. You can set the Service Provider tenant or subtenants to display the same or different terms of usage statements. You select the terms of usage statement in the Terms of Usage field on the Tenant detail form.

NOTE

You must enable multi-tenancy and configure one or more tenants before you can associate a terms of usage statement with a tenant.

Create a Terms of Usage Statement

End users are presented with a terms of usage statement when they log in to CA SDM that they must agree to before they can continue to log in. You can create a terms of usage statement and select it to display for a tenant or an analyst.

Follow these steps:

1. Select Security and Role Management, Terms of Usage on the Administration tab.
   The Terms of Usage List page appears.
2. Click Create New.
   The Create New Terms of Usage page appears.
3. Complete the following fields and click Save:
   Name
   Displays the name of the terms of usage statement.
   Status
   Sets the terms of usage statement as active or inactive.
   Description
   Displays a description of the terms of usage statement.
   Text
   Specifies the text of the terms of usage statement. Click the Edit Terms of Usage button to edit the text in an HTML editor.
   The Terms of Usage statement is saved. You can now associate the terms of usage with your tenants.

Update Terms of Usage Statement of a Tenant

You can configure a tenant to display a terms of usage statement upon login. The end user must accept the terms of usage statement before they can continue to log in.

Follow these steps:

1. Select Security and Role Management, Tenants on the Administration tab.
   The Tenant List page appears.
2. Select the Tenant link you want to update.
   The Tenant Detail page appears.
3. Click Edit.
   The Update Tenant page appears.
4. Select the terms of usage statement from the Terms of Usage drop-down and click Save.

   NOTE

   : If you select <empty> in the Terms of Usage drop-down, CA SDM displays the terms of usage statement for the tenants parent, grandparent, and so on until a terms of usage statement is found. If a terms of usage statement is not found at any level, CA SDM proceeds with the login. If you create a terms of usage statement without text, CA SDM does not display a terms of usage statement and lets the end user log in to CA SDM.

   The tenant is updated.

Handling the Open Redirect Vulnerability Issue in "LINK WITH BOPSID URL" Parameter

Invalid URL redirects and forwards happen when a web application accepts input URL requests from untrusted sources. Such untrusted URL inputs can be used to create a malicious site to launch a phishing attack. The pdmweb.exe "LINK_WITH_BOPSID+URL" parameter is the key CA SDM component in this vulnerability where the given variable can be exploited to create an untrusted and unauthorised URL input.

Resolution

Use the CA SDM NX Variable: *@NX_VALID_EXT_URLS* command to avoid this issue. This variable can be set to a semicolon separated list of valid URLs to validate the URL parameter against a whitelist. If an URL is found to be invalid, the redirect request is blocked.

This option can be installed by running the following **pdm_configure** command on a CA SDM Primary server machine:

```
pdm_options_mgr -c -s VALID_EXT_URLS -v URL1; URL2; URL3 -a pdm_option.inst
```

To avoid losing the changes while executing the above command, ensure to run this command with the '**-t**' option. For example:

```
pdm_options_mgr -c -s VALID_EXT_URLS -v http://www.domain1.com; http://www.domain2.com; http://www.domain3.com
 -a pdm_option.inst-t
```

After making the above changes, restart the CA SDM Services.

Setting Up Security

This article contains the following topics:

Before you allow people to use CA SDM, it is important that you set up security to determine the following:

- Which users can access the system
- What level or levels of access users can have
- How users are authenticated when they log in

CA EEM User Base Configurations

CA EEM is a central repository of user information (identities). CA EEM defines user authentication and access to other applications. If you have several CA Technologies products installed, some of them can use CA EEM to store identities and access policies. CA SDM only uses CA EEM for authentication. CA EEM is not a CA SDM configuration option and must be installed separately.

The CA EEM repository of user records is *either* of the following sources:

- An external LDAP directory
- Its own internal tables in the MDB

CA EEM has an LDAP interface for use when it is configured to use the MDB.

> **NOTE**
> The MDB tables used by CA EEM are different from the ones used by CA SDM.

If your organization uses a directory server, such as Active Directory or eTrust Directory, consider configuring CA EEM to use the directory for its user base. This configuration makes the users in your directory accessible by any other application that uses CA EEM. Because CA EEM centralizes access management, it is typically installed on a single server.

CA SDM

CA SDM stores contact information in MDB tables. These tables have no relationship to CA EEM. CA SDM does not use CA EEM for access or identity management. CA SDM manages its own access and security with Access Types and Data Partitions.

CA SDM uses CA EEM only for authentication. If you want to use CA EEM to authenticate users in CA SDM, install CA EEM. If you integrate CA SDM with CA EEM, it replaces the CA SDM operating system authentication with CA EEM authentication.

NOTE
To integrate CA EEM and CA SDM, you must set the *eiam_hostname*, *use_eiam_artifact*, and *use_eiam_authentication* options in Options Manager, Security.

To summarize:

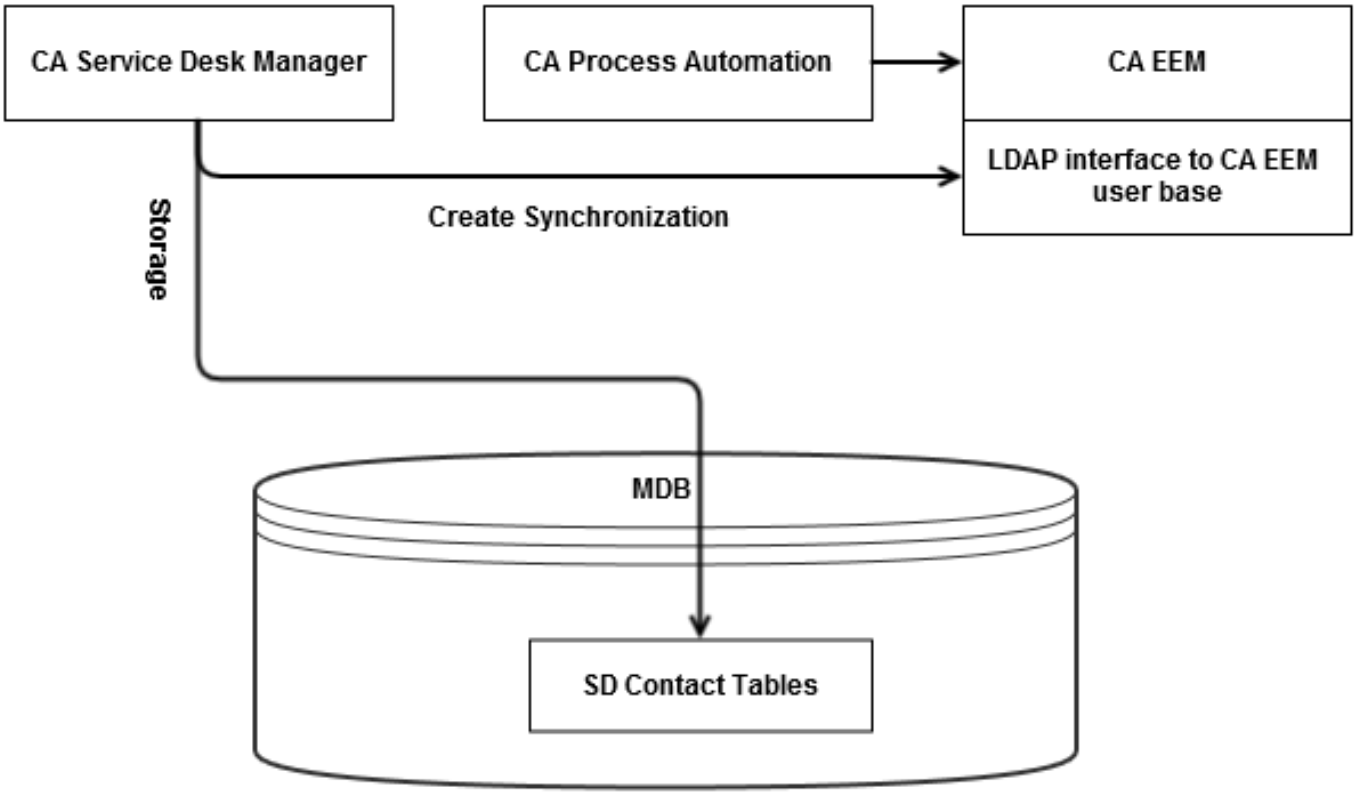
- The CA SDM user base is separate from that of CA EEM.
- CA SDM uses the MDB to store Contact information. CA SDM also features an LDAP integration, which allows it to create new Contacts from an LDAP server and synchronize existing contacts with the directory.
- CA EEM is CA's solution to centralized user management. If you have several CA products installed, they all may be using CA EEM to store identities and access policies.
- CA EEM may be configured to either point to an external (LDAP) directory or use the MDB to store user information. CA EEM itself has an LDAP interface for use when it is configured to use the MDB.

  NOTE
  The tables used by CA EEM in the MDB are different from the ones used by CA SDM.

CA EEM as LDAP Configuration

When CA EEM is configured to use MDB rather than an external directory to store user information, CA EEM exposes the user directory using an LDAP interface. If your site does not use an external LDAP server, you can still get the advantages of external LDAP configuration by configuring CA SDM to use CA EEM as an LDAP source. This configuration can be useful if your site does not use an LDAP server but you want to consolidate user management in CA EEM. Other CA products also use CA EEM, which can greatly simplify user management.

Figure 12: Diagram depicting CA EEM as LDAP configuration



NOTE
This configuration is applicable only when CA EEM is configured to use the MDB. If CA EEM is configured to an external LDAP server, configure CA SDM to point to the same LDAP server, *not* CA EEM. For more information, see How to integrate CA SDM with LDAP.

Configure the CA EEM r8.4 SP4 CR05 User Store

You can configure CA EEM r8.4 SP4 CR05 to store user records in an external LDAP directory or in its own internal MDB tables. When CA EEM uses an external LDAP directory, it is a read-only interface; you cannot add or modify users through the CA EEM interface.

Follow these steps:

1. Click Start, Programs, CA, Embedded Entitlements Manager, EEM UI.
   The CA EEM user interface appears.
2. Click the Configure tab.
3. Click the EEM Server sub-tab.
4. On the left-hand pane, click the Global Users / Global Groups link.
5. On the right-hand pane, select one of the following options:
   - Store in internal datastore
   - Reference from an external directory
   - Reference from CA SiteMinder

NOTE

If you select the Reference from an external directory option, you are prompted for the LDAP server details.

6. Click Save.
   The user store configuration for CA EEM is complete.

Configure the CA EEM r12 CR02 User Store

You can configure CA EEM r12 CR02 to store user records in an external LDAP directory or in its own internal MDB tables. When CA EEM uses an external LDAP directory, it is a read-only interface; you cannot add or modify users through the CA EEM interface.

Follow these steps:

1. Click Start, Programs, CA, Embedded Entitlements Manager, Admin UI.
   The CA EEM user interface appears.
2. Click the Configure Tab.
3. Click on User Store subtab.
4. On the left-hand pane, click the User Store link.
5. On the right-hand pane, select one of the following options:
   - Store in internal user store
   - Reference from an external LDAP Directory.
   - Reference from CA SiteMinder

   NOTE
   If you select the Reference from an external directory option, you are prompted for the LDAP server details.

6. Click Save.
The user store configuration for CA EEM is complete.

Add Users and Groups

If CA EEM is configured to reference an external directory, you cannot add users using the CA EEM user interface. CA EEM is a read-only interface to the LDAP server. You must use whatever interface is provided with your particular LDAP server product to update user records.

Follow these steps:

1. Click Start, Programs, CA, Embedded Entitlements Manager, Admin UI/EEM UI.
2. Log in using the CA EEM administrator user name and password. These are specified during the CA EEM installation. CA EEM must be installed separately and is not a configuration option for CA SDM.
3. Click the Manage Identities tab.
4. On the left-hand pane, click the Users tab to search for and update existing user records.

   NOTE
   To manage the CA EEM groups, click the Groups tab.

5. Click the icon to the left of the Users folder.
   The form for creating a user record appears.
6. Complete the form and click Save.
   The new CA EEM user record is saved in the MDB.

   NOTE
   The steps to edit an existing user record and maintain group records are similar to these steps.

Security Considerations

When you first install CA SDM, the system is set up to allow maximum access to any contact that does not have an explicit access type that is defined in the contact record. Perform the following steps before using the application:

1. Review the predefined access types to determine a reasonable default for your system.
   Administrator is set as the default access type, which is not a good choice for most sites. For example, some sites offer read-only CA access to most members of the IT organization. If you set CMDB User as the default access type, you do not have to set the access type of new users unless they need additional privileges. Similarly, if most users require the privilege to write configuration information, you can select CMDB Analyst as the default access type.
2. Assign the access types of remaining contacts explicitly.
   For example, if you select CMDB User as the default access type, modify the contact records for your analyst contacts to assign an access type of analyst.

CA EEM Authentication for CA Process Automation

CA SDM and CA Process Automation communicate using a web services exchange over HTTP. Although every measure is made to pass minimal amounts of sensitive information between the products, a malicious entity can access user names, passwords, and proprietary information. You can take deliberate steps to secure server communication.

For CA Process Automation authentication, consider the following recommendations:

- As an option, you can configure CA Process Automation to use CA EEM as an authentication server. CA Process Automation implements default groups and policies within CA EEM. You can modify the default groups and policies to meet the needs of your organization.
- Using CA EEM eliminates the need to pass plain text user names and passwords for authentication purposes. If you are using multi-tenancy, CA EEM is required for enabling multi-tenancy within CA Process Automation.

  NOTE
  To achieve authentication security in this integration, it is not necessary to have CA SDM configured to use CA EEM. However, CA EEM is required for CA Process Automation multi-tenancy implementation.
- Configure CA Process Automation to communicate using secure communications over HTTPS. HTTPS URLs use SSL/TLS to eliminate plain text exchanges while protecting proprietary and other sensitive data from accidental or malicious disclosure.

User Authentication

This article contains the following topics:

CA SDM provides a user authentication solution that you can modify as part of the access type. The same authentication is used by all CA SDM interfaces and by other CA products.

Authentication is flexible, allowing you to take advantage of external authentication mechanisms, such as Windows, HTTPD user validation or LDAP authentication. You can also select from a variety of internal authentication options, including operating system password, PIN, guest user access, or no access at all.

How CA SDM Authenticates Users

CA SDM authenticates users based on the user ID defined in their contact record. The product also does the following when a user requests access to the system:

1. If an external user ID is available (from HTTPD or Windows validation), CA SDM looks up the contact by login ID. If the contact is found and has an access type that permits external authentication, the user is allowed into the product.
2. If there was no successful external authentication, CA SDM prompts the user for a user ID and password. The product looks up a contact record for the user ID, obtains the access type, and then authenticates the user as specified by the access type.

Many installations find the predefined access types define authentication that is reasonable for that type of user; however, in some cases you may need to modify the authentication information for a predefined access type or define a new access type to handle a different authentication method for some of your users. You should review the authentication settings for the predefined access types to determine if they meet your needs, or if you need to modify them, or define additional types.

External Authentication

CA SDM permits users to access the system without supplying a user ID if all of the following conditions are met:

- External authentication is set for the user.
- The user's externally authenticated user ID is associated with a contact in your contact table.
- The contact record has an access type whose authentication definition permits external authentication.

External authentication does not permit users to access the system in the following cases:

- A user attempts access through a non-secure server.
- A user attempts access but is assigned to an access type that does not allow external authentication.

None of the predefined access types use external authentication. If you want to use external authentication for users, consider modifying the employee, analyst, and administrator access types to set external authentication. Your individual site requirements and different types of users determine whether to allow external authentication. When external authentication is used, the server configuration controls the access to files and directories. When you define authentication for an access type, you can decide the usage as follows:

- Do not use any external authentication that is already implemented, such as the user login on Windows or validation by the HTTPD server.
- Use the authentication that is implemented and allow or deny access based on it.

  NOTE
  If external authentication is not allowed, the user is authenticated based on the validation type that you specify.

Following are some examples of external authentication:

- If a user who has administrator access logs into a Windows computer, the user can perform administrative tasks without re-entering any login information.
- If a user who has HTTPD server validation, the user can access the web interface without re-entering any login information. Because the administrator access type specifies the analyst web user type, the appropriate web interface for the analyst is presented automatically.

Validation Types

Validation types authenticate users only under the following circumstances:

- The user access type does not permit external authentication.
- The user access type permits external authentication, but the user has not been validated externally (for example, the user has attempted access through a nonsecure server).

CA SDM provides you with the following validation options:

- **No Access** -- Users of this type have no access unless external authentication is allowed and is valid.
- **Open** -- Users of this type have access, with no additional authentication required.
  - **OS** -- Users of this type enter their operating system password for access. The operating system used for validation is the one running User Validation Host. This option is the default validation type for the administrator, analyst, and employee access types.
  - **PIN** -- Users of this type gain access by entering the correct value for the PIN field in their contact record as their password. You define the PIN field by entering the field attribute name when you select PIN as the validation

type. PIN is the default validation type for the customer access type, which uses the value in the customer ID (contact_num) field as the PIN.

Logged In User Counts and Session Counts

The following KPIs count the number of unique licensed users that are logged in to the system (for example, CA SDM Web UI, SOAP Web Services, REST Web Services, and so on), regardless of how many sessions each user has opened:

NOTE
For a licensed user, ensure that the Licensed? check box is selected from the Access type page of the contact (navigate to Security and Role Management, Access Type on the Administration tab and search for the contact).

- webConcurrentLicenseCt
- webConcurrentSOAPLicenseCt
- webConcurrentRESTLicenseCt
- webConcurrentTotalLicenseCt

The following KPIs count the numbers of unique unlicensed users that are logged in to the system, regardless of how many sessions each user has opened:

- webConcurrentNonLicenseCt
- webConcurrentSOAPNonLicenseCt
- webConcurrentRESTNonLicenseCt
- webConcurrentTotalNonLicenseCt

The following KPIs count the number of unique sessions that started during the interval:

- webSessionCt
- webSOAPSessionCt
- webRESTSessionCt

For more information about the KPI description, see the KPI detail page (navigate to Service Desk, KPIs on the Administration tab and search for the KPI). For more information about how these KPIs count from different session types, see the How KPIs Count from Different Session Types topic.

How KPIs Count from Different Session Types

There are different session types that are defined in the system. The following table shows how KPIs count these sessions:

Note: All predefined KPIs are installed as Inactive. For a KPI to begin functioning in your system, it must be set to Active. Navigate to Service Desk, KPIs on the Administration tab and search for the inactive KPI. Open the KPI and click Activate.

WARNING
Multiple versions of a KPI with the same name cannot be active at the same time.

Session Type	Session Type Description	Counted by KPIs
Web Client	Web browser session	webSessionCt webConcurrentLicenseCt webConcurrentNonLicenseCt webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt
Java Client	Java client session	webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt

Web Services	SOAP Web services session	webSOAPSessionCt webConcurrentSOAPLicenseCt webConcurrentSOAPNonLicenseCt
Utility	Server utility session	webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt
Portal	Portal session	webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt
Knowledge Chat	Knowledge Chat session	webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt
Mail Server	Mail Server session	webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt
Custom Application	Custom Application session	webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt
PDA Client	PDA Client session	webSessionCt webConcurrentLicenseCt webConcurrentNonLicenseCt webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt
REST Client	REST Web Services Session	webRESTSessionCt webConcurrentRESTLicenseCt webConcurrentRESTNonLicenseCt webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt

Example: KPIs calculating user counts

One licensed (that have the Licensed? check box selected) and two unlicensed end users are logged into the web self-service interface and reviewing some announcements.

At the same time five licensed analysts (that have the Licensed? check box selected) are logged into the analyst interface and working on incidents. One of the analysts also logs in to the SOAP Web Services interface.

- The webConcurrentLicenseCt KPI shows a count of six, meaning that six licenses are currently being used, irrespective of the number of interfaces each user is using.
- The webConcurrentNonLicenseCt KPI shows the count of two, which means that two unlicensed users are logged on to the system, irrespective of the number of interfaces each user is using.
- The webSessionCt KPI shows a count of eight, meaning that eight total users are logged in to the CA SDM Web UI.
- The webSOAPSessionCt KPI shows a count of one, meaning that one user is logged in to the SOAP Web Services interface.

(Applicable for advanced availability configuration only) Example: KPIs calculating user counts from different nodes

A licensed analyst logs in to the analyst interface from the background server and works on incidents. The same analyst logs in to the analyst interface from the application server. The webConcurrentLicenseCt KPI shows a count of one, meaning that one license is currently being used, irrespective of the number of nodes or servers the user has logged in from.

Establishing Support Structure

This article contains the following topics:

Setting Up Your CA SDM System

This section contains the following topics:

- Application Data
- Shared Codes
- Status Codes
- Define Code for Application Data Components
- Create Personal Responses
- Edit a Sequence Number Format
- Define Closure Codes
- Define Change Type Codes
- Define Change Order Status Transition
- Define Workflow Task Status Code
- Define Change Status Codes
- Risk Level
- Define Issue Status
- Define a Priority Calculation
- Manage Request/Incident/Problem Status
- Categories

Application Data

You can manage the following data components:

- Codes for various data types used in the application
- Data related to configuration items
- Remote references

Codes

You can define codes for your application data components.

Attribute Aliases

CA SDM includes an ODBC driver that allows an external application, such as Web Intelligence, to submit SQL SELECT queries directly to the CA SDM object manager (domsrvr). The driver depicts CA SDM Majic object names and attributes as tables and columns of a virtual database. In addition, the driver allows an unlimited number of attribute alias columns to be added to each object. Each attribute alias defines a pseudo-column corresponding to a column of a joined table. For example, the attribute alias assignee_organization_name corresponds to the Majic dotted join assignee.organization.name.
Attribute alias codes provide a flattened view of the database. They allow you to include columns from related tables. By convention, an attribute alias name is the same as the Majic join name, with the dots replaced by underscores. For example, if a Call Request report needs to show the assignee's organization name, and this column was not provided out of the box, you could define an alias called assignee_organization_name that equates to assignee.organization.name.

Auto Close Settings

You can use a configurable setting to allow automatic closure of CA SDM tickets. When a ticket is set to a Resolved status, the ticket is automatically closed in the number of business hours specified. The Auto Close activity notification sent to the end user displays the number of business hours before the ticket is closed. You can configure business hours for each tenant. If the status is changed before the configurable number of hours ends, the ticket closure is canceled.

As a system administrator, you can perform the following actions:

- Define an Auto Close ticket setting to control the number of business hours, for the end user, before the ticket is automatically closed.
- Set up an Auto Close activity notification to notify the appropriate contacts when automatic closure is scheduled for a ticket.

Define Auto Close Settings

You can define an auto close setting to allow automatic closure of tickets. When an analyst changes the status of a ticket to a Resolved status, the ticket is automatically closed in the number of business hours specified. After the ticket is closed, an activity log is added to the ticket. If the analyst changes the status of the ticket before the hours expire, automatic closure is canceled.

For each ticket type and tenant, you can define a different number of business hours. If business hours are not defined for a particular tenant, the public setting is used instead.

To define an auto close setting, **follow these steps:**

1. On the Administration tab, select Service Desk, Application Data, Codes, Auto Close.
   The Auto Close List page appears.
2. Click Create.
   The Create New Auto Close Settings page appears.
3. Complete the following fields as appropriate.
   Symbol Defines the setting name. Record Status Indicates if the item is active or inactive. Request/Incident/Problem/Change Order/Issue Defines the number of business hours, after the ticket is set to a Resolved status, before the ticket is closed. If the status is changed before the number of hours ends, the ticket closure is canceled. 0 (zero) hours indicates that automatic closure is not implemented for the ticket. Description Provides a brief description of the item. The auto close setting is defined.
4. Click Save, Close window.
   The new auto close setting appears on the Auto Close List page when you redisplay the list.

Contact Types

Contact type codes are classifications for the people associated with your network, such as vendors, analysts, or customers.

Contact type codes let you sort and select information based on the kind of person involved with a ticket. For example, you can report on how many tickets were created by an outside vendor contact. You can also use contact types to restrict certain functions to certain types of users. For example, only users with the contact type of analyst can resolve tickets.

Cost Centers

A cost center is a group or unit that uses purchased assets. Cost center codes are used to roll up groups of purchased assets for budgeting and reporting purposes. A cost center could be an internal department, a subsidiary, or any group that needs to have its assets tracked separately.

Countries

You can associate country codes with your organization's location codes, as part of the address information.

Departments

You can define department codes to reflect the departments within your organization. Departments can then be associated with records such as contacts and groups.

iCalendar Event Templates

iCalendar event templates control the information that is exported to iCalendar format.

The following predefined templates are installed with CA SDM:

- Change Schedule
- KnowledgeScheduleCreation
- KnowledgeScheduleExpired
- KnowledgeScheduleReview
- KnowledgeScheduleStart

NOTE
You can edit the predefined iCalendar event template codes, but you cannot delete them or create new ones.

WARNING

The SchedExpMaximum variable in web.cfg controls the maximum events allowed for an export. Increasing the default (1000) could cause system instability. If you attempt to export more than the value specified in SchedExpMaximum, a message appears refusing your exporting request.

Impacts

Impact codes indicate how much a ticket affects work being performed. You can use impact codes to select and sort records, and organize them for reports.

The predefined impact codes are:

1-Entire organization

2-Multiple Groups

3-Single Group

4-Small Group

5-One person

None = No Impact is assigned

You can change the symbols for impact codes. For example, your site might decide not to use numbers at all and assign codes of low, medium, high, and so on.

NOTE
You can edit the predefined impact codes, but you cannot delete them or create new ones.

Locations

Location codes precisely identify a physical place. You can use location codes to identify where inventory components reside, such as contacts and resources like printers. Location codes can identify any type of place, for example, a city, building, or a specific floor of a building.

Outage Types

Outage types represent a single general type of outage for an incident that a user reports, or to specify other types of outages such as a scheduled or test outage. When a user reports an outage incident, you can specify an outage type on

the ticket to help categorize and track the incident. For example, you can use an outage type of Facilities on a ticket to indicate that an outage that affected the end-user facilities.

You can do the following outage type actions:

- Use or edit the predefined outage types.
- Create outage types. For example, you can create an Intermittent Outage type to use for incidents concerning intermittent service outages.
- List outage types.
- Search for incidents by outage type. For example, you can search for all incidents that have a Network outage type, and then view each incident for information to help resolve an incident.

Create an Outage Type

You can create outage types to identify a single general outage type for services. For example, you can create an Intermittent Outage type to use for incidents concerning intermittent service outages.

To create an outage type, **follow these steps:**

1. On the Administration tab, select Service Desk, Application Data, Codes, Outage Types.
   The Outage Types List page appears.
2. Click Create New.
   The Create New Outage type page appears.
3. Complete the fields as appropriate, and click Save.
   The new type definition is saved, and you can use it for tickets to identify outage types.

List Outage Types

You can list outage types to see all the types that are defined in your system.

To list outage types, **follow these steps:**

1. On the Administration tab, select Service Desk, Application Data, Codes, Outage Types.
   The Outage Types List page appears and lists outage types and corresponding information.
2. (Optional) Click a symbol link.
   Details about the outage type appear.

Resolution Codes

Incident management focuses on getting the end user up and running as fast as possible. Analysts can indicate what they did to resolve an incident by using a resolution code. Resolution codes specify the category code of the ticket resolution, for example, an Applied Patch resolution code indicates that the ticket was addressed by applying a patch or fix to the software.

You can do the following resolution code actions:

- Create resolution codes. For example, you can create a resolution code to use for requests and that details what resolution was used.
- Edit resolution codes. For example, after the administrator loads the sample resolution codes, you can edit the Provided Documentation code to a more precise name such as Provided Implementation Documentation.
- List resolution codes.
- Search for incidents or requests by resolution code. For example, you can search for all incidents that have a Patch resolution code, and then view each incident for information to help resolve another incident.

List Resolution Codes

You can list resolution codes to see all the resolution codes that are defined in your system.

To list resolution codes, **follow these steps:**

1. On the Administration tab, browse to Service Desk, Application Data, Codes, Resolution Codes.
   The Resolution Codes List page appears, and lists resolution codes and corresponding information.
2. (Optional) Click a code link.
   Details about the resolution code appears.

Create a Resolution Code

You can create resolution codes to identify a single general resolution for an incident or request. For example, a Restart resolution code represents a resolution that involved restarting the end-user computer.

To create a resolution code, **follow these steps:**

1. On the Administration tab, browse to Service Desk, Application Data, Codes, Resolution Codes.
   The Resolution Code List page appears.
2. Click Create New.
   The Create New Resolution Code page appears.
3. Complete the fields as appropriate, and click Save.
   The new resolution code definition is saved, and analysts can use it for incidents and requests to identify resolution methods.

Resolution Methods

Incident management focuses on getting the end user up and running as fast as possible. Analysts can indicate how they resolved an incident by using a resolution method. Resolution methods define the ways the service desk can resolve tickets. For example, an analyst can resolve an incident or request through a chat session or an on-site visit.

You can do the following resolution method actions:

- Create resolution methods. For example, you can create a File Transfer resolution method to use for incidents that a file transfer addressed.
- Edit resolution methods. For example, after the administrator loads the sample resolution methods, you can edit the Remote Control Tool method to a more precise name such as Seamless Window.
- List resolution methods.
- Search for incidents or requests by resolution method. For example, you can search for all incidents that have an Electronic Software Delivery method code, and then view each incident for information to help resolve another incident.

List Resolution Methods

You can list resolution methods to see all the resolution methods that are defined in your system.

To list resolution methods, **follow these steps:**

1. On the Administration tab, browse to Service Desk, Application Data, Codes, Resolution Methods.
   The Resolution Methods List page appears, and lists resolution methods and corresponding information.
2. (Optional) Click a method link.
   Details about the resolution method appears.

Create a Resolution Method

You can create a resolution method to identify a single general method for how an incident or request was resolved. For example, a Seamless Window resolution method represents a resolution that involved using a remote tool function on a client computer.

To create a resolution method, **follow these steps:**

1. On the Administration tab, browse to Service Desk, Application Data, Codes, Resolution Methods.
   The Resolution Methods List page appears.

2. Click Create New.
   The Create New Resolution Method page appears.
3. Complete the fields as appropriate, and click Save.
   The new resolution method definition is saved, and analysts can use it for incidents and requests to identify resolution methods.

Symptom Codes

Symptom codes represent a single general symptom for an incident that a user reports. When a user reports an incident, you can specify a symptom code on the ticket to help categorize and track the incident. For example, you can use a symptom code of Slow Response on a ticket for an application that is not running as fast as a user expects.

You can do the following symptom code actions:

- Use or edit the predefined symptom codes.
- Create symptom codes. For example, you can create a Cannot Activate Device code to use for incidents concerning activating wireless devices.
- List symptom codes.
- Search for incidents by symptom code. For example, you can search for all incidents that have an Application Not Responding symptom code, and then view each incident for information to help resolve an incident.

Create a Symptom Code

You can create symptom codes to identify a single general symptom for an incident. For example, the Unable to Log in symptom code represents a problem for a user who is unable to log in to an application.

To create a symptom code, **follow these steps:**

1. On the Administration tab, select Service Desk, Application Data, Codes, Symptom Codes.
   The Symptom Code List page appears.
2. Click Create New.
   The Create New Symptom page appears.
3. Complete the fields as appropriate, and click Save.
   The new code definition is saved, and you can use it for tickets to identify symptoms.

List Symptom Codes

You can list symptom codes that you use in tickets to reference symptoms of problems.

To list symptom codes

1. On the Administration tab, select Service Desk, Application Data, Codes, Symptom Codes.
   The Symptom Code page appears and lists symptom codes.
2. (Optional) Click a symbol link.
   Details about the symptom code appear.

Timezones

Timezone codes are specified for users in their contact record. A timezone code displays the local time for users, regardless of their locations around the world, in the Time field on the Quick Profile, thus allowing you to see if it is currently during business hours for them.

You can add new timezone codes, and modify the predefined ones. Timezone codes indicate the time difference difference between a particular time zone and Greenwich Mean Time (GMT). You can also specify when to start and end Daylight Savings Time, if applicable.

Sites

Site codes represent general groupings, such as cities or regions. Sites can include several locations, which are physical places like buildings or floors. Site records are referenced in location records. You should create site records before location records so the sites are available when you define the locations.

Web Macro

CA SDM includes a number of predefined macros. Most of these macros insert JavaScript text to create an element on a web form. Use Web Screen Painter to create and modify forms using these macros.

NOTE

For more information about these web macros, see the Implementation Guide.

Create a Web Macro

Create a web macro in the web interface that you can use with WSP.

Follow these steps:

1. On the Administration tab, select Service Desk, Application Data, Codes, Web Macro List.
2. Click Create New.
3. Enter a PDM_MACRO name.
4. Select a macro type from the drop-down list.
5. Select a control type from the drop-down list.
6. Enter a description for the macro.
7. Enter the macro text.
8. Click Save.

Configuration Item - Application Data

Configuration items include all the hardware, software, and services associated with your system. Configuration item records uniquely identify each item and its exact location. These records are part of the inventory components that make up your database. Users can reference these records in incident and problem tickets.

Remote References

You can configure remote references to external programs that you want to execute from within the CA SDM application.

For more information, see Create a Remote Reference.

Shared Codes

This article contains the following topics:

In CA SDM, different types of tickets share certain underlying codes, such as priority, severity, impact, and urgency codes. Requests and change orders share some codes, and all types of tickets share other codes.

Consider the following information about shared codes:

- By default, numeric values rank them.
- You can modify the codes.
- You cannot add or delete shared codes.
- You can use impact, priority, severity, and urgency codes.

Based on your service desk model, set up the following codes:

Shared Codes	Description
Priority	Must be set up for all service desk models.
Severity	Must be set up for internal and combined service desk models.
Impact	Must be set up for internal and combined service desk models.
Urgency	Must be set up for internal and combined service desk models.

Priority Codes

Priority codesindicate a ranking order by which the service desk should respond to tickets (that is, they specify the level of attention a ticket should receive). Priority codes are referenced in requests, change orders, and issues; therefore, they apply to all service desk models.

You can use priorities to escalate tickets manually or automatically by monitoring events. In many service desk installations, priority codes are used on the scoreboard to provide analysts with a real-time status of their requests and change orders.

You can assign a service type to a priority code, which is then automatically assigned to tickets when the priority code is specified. This lets you associate a specific level of service to a ticket based on the assigned priority. For example, the system-defined service type, 4-hour resolution, is automatically associated with priority 1. Tickets that are assigned a priority of 1, therefore, are automatically assigned this service type, including all the service type events that are associated with the 4-hour resolution service type.

Severity Codes

Severity codesidentify the extent of the damage to equipment affected by a request. Severity codes are referenced in requests only; therefore, they apply only to internal and combined service desk models.

> **NOTE**
> Severity is often used as a synonym for priority. Some sites use priority only, ignoring severity altogether. If you want to distinguish between how serious a problem is on a technical level (severity) and how quickly you want it handled (priority), you can use severity and priority codes.

Impact Code values helps to calculate the Incident Priority.

Impact Codes

Impact codesmeasure the significance of a ticket on the functioning of the system. For example, if a change order could affect the functioning of the entire system, it would be assigned a high impact. Impact codes are referenced in requests and change orders only; therefore, they apply only to internal and combined service desk models.

> **NOTE**
> Impact and urgency codes are similar, but they have distinct purposes.

Urgency Codes

Urgency codesmeasure the significance of the request for users of the system (that is, they indicate the importance of the request to the overall production environment). For example, if a request could jeopardize the mission of the enterprise, the Urgency code can be a value of 5-Immediate. Urgency codes are referenced in requests only; therefore, they apply only to internal and combined service desk models.

Urgency and impact codes serve distinct purposes, but are often confused because they coincide. For example, a request to report a fire in a critical data center can have a 3-Single Group Impact and 5-Immediate Urgency. These codes apply

because the fire impacts more than one group but not necessarily the entire organization. Because the data center is critical for operations, the urgency requires immediate attention.

Status Codes

This article contains the following topics:

Status codes are used to track the status of an item. Separate status codes track requests, change orders, issues, and workflow tasks. In each case, there are predefined status codes that you can use if they suit your needs. Otherwise, you can modify the predefined status codes or define new ones that are specific to your site. Depending on your service desk model, you set up the following codes:

Status Codes	Description
Request/ Incident/ Problem	Must be set up for internal and combined service desk models.
Change Order	Must be set up for internal and combined service desk models.
Issue	Must be set up for external and combined service desk models.
Task	Must be set up for all service desk models.

Status codes let analysts sort and select information based on the status so that they can carefully track their progress. How carefully you define the status codes determines how accurate the analysts can be in describing the actual status of an item.

You can mark any status code as active or inactive. When you mark a status code as inactive, it is no longer available for analysts to use, but it remains available for future use (that is, it is not deleted from the database). If you decide later to use the status code, you need only to go back to it and mark it as active.

> **NOTE**
> The same area definitions are available for request, incident, and problem tickets. On the Administration tab, these areas are referred to as request/incident/problem areas. For brevity, they are referred to here simply as request areas.

Request Status Codes

The following table describes the predefined status codes for request tickets.

Request Status Code	Description
Acknowledged	The receipt of a request has been acknowledged.
Closed	A request has been completely resolved.
Closed -- Unresolved	A request has been closed but still must be resolved.
Fix in Progress	A request is pending a fix.
Hold	The service type events for the request are on hold.
Open	A request has been defined and is being used to monitor and manage its completion.
Problem Closed	A problem request is completely closed.
Problem Fixed	A problem request is fixed, but not closed.
Problem Open	A request has been identified as a problem.
Researching	A request is open pending additional research and analysis.
Work in Progress	Work is being done to fix a request.

If your site uses other terminology for identifying the request status, you should define status codes that suit your needs and ignore the predefined status codes, or change the definitions to match your use. For example, you may want to define additional request status codes, such as the following codes:

Request Status Code	Description
Duplicate	Requests that have been opened but may be a duplicate of an existing request for another user.
Emergency	Critical requests that must be addressed immediately.
Report	Requests that are resolved and closed, but should be reported on at a management level.
Test	Requests that are resolved, but should be tested for one week before closing.

Problem Status Codes

The following table describes the predefined status codes for problem tickets.

Problem Status Code	Description
Acknowledged	The receipt of a problem has been acknowledged.
Closed	A problem has been completely resolved.
Closed -- Unresolved	A problem has been closed but still must be resolved.
Fix in Progress	A problem is pending a fix.
Hold	The service type events for the problem are on hold.
Open	A problem has been defined and is being used to monitor and manage its completion.
Problem Closed	A problem is completely closed.
Problem Fixed	A problem is fixed, but not closed.
Problem Open	A request has been identified as a problem.
Researching	A problem is open pending additional research and analysis.
Work in Progress	Work is being done to fix a problem.

If your site uses other terminology for identifying the problem status, you should define status codes that suit your needs and ignore the predefined status codes, or change the definitions to match your use. For example, you may want to define additional problem status codes, such as the following codes:

Problem Status Code	Description
Duplicate	Problems that have been opened but may be a duplicate of an existing problem for another user.
Emergency	Critical problems that must be addressed immediately.
Report	Problems that are resolved and closed, but should be reported on at a management level.
Test	Problems that are resolved, but should be tested for one week before closing.

Incident Status Codes

The following table describes the predefined status codes for incident tickets.

Request Status Code	Description
Acknowledged	The receipt of an incident has been acknowledged.
Closed	An incident has been completely resolved.
Closed -- Unresolved	An incident has been closed but still must be resolved.
Fix in Progress	An incident is pending a fix.
Hold	The service type events for the incident are on hold.
Open	An incident has been defined and is being used to monitor and manage its completion.
Incident Closed	An incident is completely closed.
Incident Fixed	An incident is fixed, but not closed.
Incident Open	An incident has been identified as a problem.
Researching	An incident is open pending additional research and analysis.
Work in Progress	Work is being done to fix an incident.

If your site uses other terminology for identifying the incident status, you should define status codes that suit your needs and ignore the predefined status codes, or change the definitions to match your use. For example, you may want to define additional incident status codes, such as the following codes:

Request Status Code	Description
Duplicate	Incidents that have been opened but may be a duplicate of an existing request for another user.
Emergency	Critical incidents that must be addressed immediately.
Report	Incidents that are resolved and closed, but should be reported on at a management level.
Test	Incidents that are resolved, but should be tested for one week before closing.

Change Order Status Codes

The following table describes the predefined status codes for change order tickets.

Change Order Status Code	Description
Approval in Progress	A change order is open, pending approval.
Approved	A change order has been approved.
Cancelled	A change order has been cancelled.
Closed	A change order has been completed.
Hold	The service type events for the change order are on hold.
Implementation in Progress	A change order is being implemented.
Open	A service order has been defined in a change order and the change order is being used to monitor and manage its completion.

Rejected	A change order has been rejected.
Resolved	A change order has been resolved.
RFC	A request for change has been submitted.
Suspended	Stops workflow tasks on a change order.
Verification in Progress	A change order is being verified.
Backed Out	Implemented change order was backed out.
Implemented	Change was implemented.

If your site uses other terminology for identifying the status of a change order, you should define status codes that suit your needs and ignore the predefined status codes, or change the definitions to match your use. For example, you may want to define additional change order status codes, such as those listed in the following table:

Change Order Status Code	Description
Duplicate	Change orders that have been opened but may be a duplicate of an existing change order for another user.
Emergency	Critical change orders that must be addressed immediately.
Report	Change orders that are completed and closed, but should be reported on at a management level.

Closure Codes

Use closure codes to define the final outcome of change orders, such as successful or unsuccessful. Set closure codes manually or as part of the Update Status activity on a change order when the status is closed, finished or resolved.

Issue Status Codes

The following table describes the predefined status codes for issue tickets.

Issue Status Code	Description
Approval in Progress	An issue is open, pending approval.
Cancelled	An issue has been cancelled.
Closed	An issue has been completed.
Hold	The service type events for the issue are on hold.
Implementation in Progress	An issue is being implemented.
Open	An issue has been defined and open so that it can be monitored and managed until it is resolved.
Suspended	Stops workflow tasks on an issue.
Transaction in Progress	A transaction with a customer concerning this issue is in progress.
Verification in Progress	An issue is being verified.

If your site uses other terminology for identifying the status of an issue, you should define status codes that suit your needs and ignore the predefined status codes, or change the definitions to match your use. For example, you may want to define additional issue status codes, such as the following:

Issue Status Code	Description
Duplicate	Issues that have been opened but may be a duplicate of an existing issue for another user.
Emergency	Critical issues that must be addressed immediately.
Report	Issues that are completed and closed, but should be reported on at a management level.

Task Status Codes

Task status codes describe the different possible states of a workflow task. Each task in an issue or change order or request/ incident/ problem workflow has its own status, separate from the ticket status. Workflow tasks allow analysts to keep track of how long it takes to complete individual tasks within a ticket.

The following table describes the predefined status codes for workflow tasks.

Task Status Code	Description
Approve	Task approved.
Cancelled	Task is cancelled and no further updates are possible.
Complete	Task is complete.
Pending	Task is started.
Reject	Task is rejected.
Reopen	Task has been re-opened.
Reopen - Wait	A prior task was reopened.
Skip	Skip task.
Wait	Task has not started.

If your site uses other terminology for identifying the status of a workflow task, you should define status codes that suit your needs and ignore the predefined status codes, or change the definitions to match your use.

For each task status code, you can assign a type of behavior that occurs when the task reaches this state, which provides much more information about the progress toward completing the task. You can also use the accumulate function to track time and cost involved in completing the ticket.

Task Types

Task types help determine the behavior of specific workflow tasks and task status codes. To produce defining characteristics for each type of task, you can identify the task status codes or specific states that can be used.

Because change orders, issues, and request/ incident/ problem use workflow tasks, set up task types for all service desk models. The task status codes identify the behaviors associated with each task. For example, you can set the Approval task type to allow Approve, Reject, Pending, and Wait as available states. When the Approval task type enters the Pending state, you can send notification to a specific manager, analyst, and so forth.

You can mark any task type as active or inactive. When you mark a task type as inactive, it is no longer available for analysts to use, but it remains available for future use (it is not deleted from the database). If you decide later to use the task type, mark it as active again.

NOTE
You can view the predefined task types in the Task Type List page of the administrative function of the web interface.

Example: Task Status Codes

The following table contains some example status codes:

Task Status Code	Description
Approval	Approve or reject ticket
Group End Task	End of group tasks
Group Start Task	Start of group tasks
Start Approval	Approval to start the ticket

In this example, the Group Start Task and Group End Task types define a group of tasks in an issue or change category that must be completed. Tasks in the group can be executed in any order. After the Group Start Task is in the Pending state (started), all tasks in the group are also placed in this state.

Define Code for Application Data Components

You can define codes for your application data components. To create a code, select Service Desk, Application Data, Codes, <code_that_you_want_to_create> from the Administration tab.

NOTE
If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants. If you are the Service Provider, select the appropriate tenant from the dropdown menu. The public (shared) option creates the object for all tenants.

Code	Description
Attribute Alias	CA SDM includes an ODBC driver that allows an external application, such as Web Intelligence, to submit SQL SELECT queries directly to the CA SDM object manager (domsrvr). The driver depicts CA SDM Majic object names and attributes as tables and columns of a virtual database. In addition, the driver allows an unlimited number of attribute alias columns to be added to each object. Each attribute alias defines a pseudo-column corresponding to a column of a joined table. For example, the attribute alias assignee_organization_name corresponds to the Majic dotted join assignee.organization.name. Attribute alias codes provide a flattened view of the database. They allow you to include columns from related tables. By convention, an attribute alias name is the same as the Majic join name, with the dots replaced by underscores. For example, if a Call Request report needs to show the assignee's organization name, and this column was not provided out of the box, you could define an alias called assignee_organization_name that equates to assignee.organization.name. To create this code, the object name is the majic object name associated with the attribute alias. For more information about objects, see the Technical Reference topic.
Auto Close Settings	Auto Assignment

Contact Type	Contact type codes are classifications for the people associated with your network, such as vendors, analysts, or customers. Contact type codes let you sort and select information based on the kind of person involved with a ticket. For example, you can report on how many tickets were created by an outside vendor contact. You can also use contact types to restrict certain functions to certain types of users. For example, only users with the contact type of analyst can resolve tickets.
Cost Centers	A cost center is a group or unit that uses purchased assets. Cost center codes are used to roll up groups of purchased assets for budgeting and reporting purposes. A cost center could be an internal department, a subsidiary, or any group that needs to have its assets tracked separately.
Countries	You can associate country codes with your organization's location codes, as part of the address information.
Departments	You can define department codes to reflect the departments within your organization. Departments can then be associated with records such as contacts and groups.
End User Roles	End user role codes categorize the various people who contact the service desk. For example, a user might be an existing customer, a potential customer, or a sales manager.
iCalender Event Templates	iCalendar event templates control the information that is exported to iCalendar format. The following predefined templates are installed with CA SDM: • Change Schedule • KnowledgeScheduleCreation • KnowledgeScheduleExpired • KnowledgeScheduleReview • KnowledgeScheduleStart You can edit the predefined iCalendar event template codes, but you cannot delete or create them. The SchedExpMaximum variable in web.cfg controls the maximum events allowed for an export. Increasing the default (1000) could cause system instability. If you attempt to export more than the value specified in SchedExpMaximum, a message appears refusing your exporting request. You can set a display alarm (such as 1 Day before or 1 Hour before) for each event that creates a calendar.

Impact	Impact codes indicate how much a ticket affects work being performed. You can use impact codes to select and sort records, and organize them for reports. The predefined impact codes are: • 1-Entire organization • 2-Multiple Groups • 3-Single Group • 4-Small Group • 5-One person • None = No Impact is assigned You can change the symbols for impact codes. For example, your site might decide not to use numbers at all and assign codes of low, medium, high, and so on. You can edit the predefined impact codes, but you cannot delete or create them.
Location	Location codes precisely identify a physical place. You can use location codes to identify where inventory components reside, such as contacts and resources like printers. Location codes can identify any type of place, for example, a city, building, or a specific floor of a building. You can configure automatic ticket assignment for the location, based on request areas, change categories, issue categories, and groups. For example, to auto assign issue tickets in a certain category to analysts associated with this location, click the Update Issue Categories button on the Auto Assignment tab and select the category from the list. For more information about configuring Auto Assignment, see Auto Assignment.
Organization	Organization codes can define the companies, divisions, or departments in your enterprise. You can use organization codes to indicate the working and reporting relationships of your service desk users. This information is defined on a user's contact records and displayed on their tickets. Organizations can also be assigned to the resources in an asset class, or the issues assigned to a specific ticket area or category.
Outage Type	Outage types represent a single general type of outage for an incident that a user reports, or to specify other types of outages such as a scheduled or test outage. When a user reports an outage incident, you can specify an outage type on the ticket to help categorize and track the incident. For example, you can use an outage type of Facilities on a ticket to indicate that an outage that affected the end-user facilities.
Position	Position codes can indicate the job title of a person who opens a ticket.
Priority	Priority codes indicate the amount of attention a ticket should receive. Priority codes are used when defining automatic notification and external processing for specific types of situations. They are also used as a label in the scoreboard to prioritize tickets. You can modify a priority code's symbol. For example, your site might decide not to use numbers at all and assign priority codes of low, medium, high, and so on. You can modify a priority code's associated service type. For example, you could change the service type for priority code 1 from the default value of 04hr resolution to a custom service type of 02hr resolution.

Product	Product codes identify the products supported by your service desk.
Reason	Reason codes define the purposes for opening tickets. For example, a reason for opening an issue might be an inquiry, a complaint, or a suggestion.
Reporting Methods	Reporting method codes define the ways a ticket can be reported to the service desk. For example, a customer might create an issue over the phone or via e-mail.
Resolution	Incident management focuses on getting the end user up and running as fast as possible. Analysts can indicate what they did to resolve an incident by using a resolution code. Resolution codes specify the category code of the ticket resolution, for example, an Applied Patch resolution code indicates that the ticket was addressed by applying a patch or fix to the software.
Resolution Method	Resolution methods define the ways the service desk can resolve tickets. For example, an analyst can resolve an incident or request through a chat session or an on-site visit.
Root Cause	Root cause codes identify the source of a problem reported in your service desk. Root cause codes comprise a set of nodes, which are separated by periods to indicate where the root cause fits into a hierarchy of root causes. For example, the root cause name Network.Cable.Install indicates the Install root cause as part of the Cable root cause, which in turn is part of the Network root cause. By default, the Root Cause Selection window displays as a hierarchical selection list. If you want to display the Root Cause Selection window as a normal list, you can install the no_hier_list option through the Options Manager.
Severity	Severity codes indicate the effect a ticket has on people. For example, a printer problem might affect a larger number of people than the failure of user system or computer. Severity codes are used to select and sort tickets and organize them for reports.
State/Province	State/province codes identify the states and provinces where tickets might originate in your service desk.
Symptom	Symptom codes represent a single general symptom for an incident that a user reports. When a user reports an incident, you can specify a symptom code on the ticket to help categorize and track the incident. For example, you can use a symptom code of Slow Response on a ticket for an application that is not running as fast as a user expects.
Timer	Timers act as a stopwatch with various thresholds that give the analyst indications of elapsed time. You can define the amount of time the timer remains at each threshold, and have the timer change color, beep, or display a reminder as it reaches each threshold. A service desk analyst cannot control the stopwatch; only the administrator can control it. Requests are the only type of ticket that uses timers; therefore, set up timers for internal and combined service desk models. For more information, see Timer Setup.

Time Zone	You can set up specific time zones for servers, service types, contacts, and locations in your CA SDM system. You can set up local service types that apply to a specific time zone and global service types that apply across the entire enterprise. This setup eliminates the need for the administrator to know the time zone of a server and manually adjust the work shift times to fit different time zones. For more information, see Time Zone Setup.
Urgency	Urgency codes indicate the importance of the user tasks affected by a ticket. You assign urgency codes to tickets based on how it affects user tasks. Urgency codes can be used to select and sort information and to organize it for reports.
Web Macro	CA SDM includes a number of predefined macros. Most of these macros insert JavaScript text to create an element on a web form. Use Web Screen Painter to create and modify forms using these macros.
Workshift	Workshift codes specify the dates, days, and times when the monitoring of an event is in effect. You can add new workshifts or update existing workshifts. Workshifts often coincide with the times your service desk is operating. By defining workshifts, you can set up events and monitor them precisely. For more information, see Workshift Setup.
Timespan	Timespan codes allow you to define time spans that can be applied to stored queries. You can create time spans that control the start time, end time, and trigger time of stored queries.
Site	Site codes represent general groupings, such as cities or regions. Sites can include several locations, which are physical places like buildings or floors. Site records are referenced in location records. You should create site records before location records so the sites are available when you define the locations.

Timer Setup

The following threshold values are predefined for the timer:

Threshold Duration	Color
00:00:00	Green
00:01:00	Yellow
00:05:00	Red

The predefined values start the timer at green. After one minute, the timer changes to yellow. After five minutes, the timer changes to red. The elapsed time is displayed when the analyst views the request in detail and is reset each time a new request is selected. You can add steps to this process or change existing steps.

You can mark any timer as active or inactive. When you mark a timer as inactive, it is no longer part of the process, but it remains available for future use (that is, it is not deleted from the database). If you decide later to use the timer, back to it and mark it as active.

Timer codes control the reminders to analysts that they have been working on a request longer than the recommended amount of time. This way of reminding analysts has the recommended amount of time defined as a threshold, such as 5 minutes. The timer then uses various methods to indicate that the threshold has been reached. These methods include sounding beeps, changing the background color of the Timer field on the Request Detail window, or displaying messages telling the analyst to resolve the request as quickly as possible.

The Timer field that tracks the time spent on a request is reset each time the request is opened, and service desk analysts cannot control it or change it.

Time Zone Setup

This article contains the following topics:

Timezone codes are specified for users in their contact record. A timezone code displays the local time for users, regardless of their locations around the world, in the Time field on the Quick Profile, thus allowing you to see if it is currently during business hours for them.

You can add new timezone codes, and modify the predefined ones. Timezone codes indicate the time difference difference between a particular time zone and Greenwich Mean Time (GMT). You can also specify when to start and end Daylight Savings Time, if applicable.

Service Type Event Triggers

Service types define the events that are triggered after a specified delay time.

Example: Event Triggers by Work Shift Schedule

In this example, a work shift schedule associated with the service type constrains the actual time an event is triggered, as follows:

- Work shift is 8:00 a.m. to 5:00 p.m.
- Event delay is three hours
- Current server time is 3:00 p.m.

The event is triggered tomorrow at 9:00 a.m. server time because of the following:

- According to the current server time, two hours remain in the work shift for today (Current time is 3:00 p.m. Work shift ends at 5:00 p.m.)
- The event delay is three hours. Two hours of this delay are spent in the current day's work shift (3:00 p.m. to 5:00 p.m.). One additional delay hour is carried over to the next day's work shift (8:00 a.m. to 9:00 a.m.)

Consequently, the event begins at 9:00 a.m. the following day.

Time Zone Event Triggers

A time zone associated with the service type can constrain event trigger times.

Example: Event Triggers by Time Zone

In this example, a time zone associated with the service type constrains the actual time an event is triggered as follows:

- Work shift is 8:00 a.m. to 5:00 p.m.
- Event delay is three hours
- Current server time is 3:00 p.m.
- Current time in time zone is 12 noon

The event is triggered today at 6:00 p.m. server time because of the following:

- According to the current server time, two hours remain in the work shift for today (Current time is 3:00 p.m. Work shift ends at 5:00 p.m.).
- According to the current time zone time, five hours are left in the work shift for today (Current time is 12 noon Work shift ends at 5:00 p.m.).
- The event delay is three hours (12 noon to 3:00 p.m. time zone time. 3:00 p.m. to 6:00 p.m. server time).

Consequently, the event begins at 6:00 p.m. server time or 3:00 p.m. time zone time.

Time Zone Rules

You can specify a time zone for a server, for a location, and for a service type. You can also tell CA SDM to use the time zone of the Affected End User of a ticket. CA SDM uses the following rules to determine which time zone triggers an event:

- **Affected End User time zone** -- CA SDM uses this rule when the following conditions exist:
  - The Use End User Time Zone option is selected.
  - A time zone is specified for the affected end user of the ticket.
- **Affected End User location time zone** -- CA SDM uses this rule when the following conditions exist:
  - The Use End User Time Zone option is selected.
  - No time zone is specified for the affected end user of the ticket.
  - A time zone is specified for the affected end-user location.
- **Service type time zone** -- CA SDM uses this rule when the following conditions exist:
  - A time zone is specified for the service type.
  - The Use End User Time Zone option is not selected.
- **Server time zone** -- CA SDM uses this rule when the following conditions exist:
  - A time zone is specified for the server.
  - The Use End User Time Zone option is not selected.
  - No time zone is specified for the service type.
- **No time zone support** -- CA SDM uses this rule when the following conditions exist:
  - No time zone is specified for the service type.
  - The Use End User Time Zone option is not selected.
  - No server records exist.

Workshift Setup

Workshift codes specify the dates, days, and times when the monitoring of an event is in effect. You can add new workshifts or update existing workshifts.

Workshifts often coincide with the times your service desk is operating. By defining workshifts, you can set up events and monitor them precisely.

You can create and edit workshifts.

You can create new workshift codes.

Follow these steps:

1. Select Service Desk, Application Data, Codes, Workshifts on the Administration tab.
   The Workshift List page appears.
2. Click Create New.
   The Create New Workshift page appears.
3. Complete the fields as appropriate. The following tabs are available on the Create Workshift, Workshift Detail, and Update Workshift pages:
   - **Schedule**
     Specifies the days, date, and time of the workshift. You can specify days, or dates, or both days and dates. Specifying a time is optional. Use the following syntax:

     ```
     day_range date_range [{time}]
     ```

     - **day_range**
       Specifies a day or range of days in the following format:

       ```
       week_day | week_day - week_day
       ```

Valid week_day values are Sun, Mon, Tue, Wed, Thu, Fri, and Sat.

- **date_range**
  Specifies a date or range of dates in the following format:

  ```
  date | date - date
  ```

  Enter date as mm/dd/yy, where mm is the month, dd is the day, and yy is the year.
- **{time}**
  Specifies a time or range of times in the following format:

  ```
  {hh:mm[:ss] [am or pm]} | {hh:mm[:ss] [am or pm]} - {hh:mm[:ss] [am or pm]}
  ```

  Time values must be enclosed in braces. Enter time in hours, minutes, and optionally seconds, where hh represents hours, mm represents minutes, and ss represents seconds. You may optionally specify am to identify a time from midnight to noon; this is the default and need only be specified for CA. You must specify pm to set a time interval from noon to midnight. You may also omit am and pm and specify hours from noon to midnight in 24-hour format, for example 13:30. If am and pm are used, hour values greater than 12 are invalid.

  NOTE
  If the server and clients are in different time zones, events take place according to the time zone of the server.

Examples:

```
Mon - Fri {8:00 am - 5:00 pm}
Sun {9:00 - 12:00 2:00 pm - 4:00 pm}
Sat 12/24/08 - 1/1/05 {8:00 - 12:00 14:00 - 4:00 pm}
7/4/09
```

NOTE
Sat 12/24/08 - 1/1/09 indicates every day from December 24, 2008 through January 1, 2009. It does not mean just Saturdays within the specified date range. 7/4/09 means no work hours on July 4, 2009.

– **Auto Assignment**
Allows you to configure automatic ticket assignment for the workshift, based on request areas, change categories, and issue categories. These fields are optional. Editable. For more information about configuring Auto Assignment, see the Auto Assignment topic.

NOTE
You can click the More icon to display the Additional Search Arguments field. This field is intended only for expert users who understand SQL and Majic. You can enter a SQL WHERE clause in this field to specify an additional search argument.

4. (Optional) Perform the following steps to configure automatic assignment of request, incident, or problem tickets for this workshift:
   a. Click Update Request Areas.
      The Request/Incident/Problem Area Search page appears.
   b. (Optional) Complete the filter fields to display only the area records of interest and click Search.
      The Request/Incident/Problem Area Update page appears.
   c. Select the areas you want to assign to this workshift from the Request Areas Available list. To select multiple areas, hold down the CTRL key while clicking the left mouse button, or hold down the Shift key to select a range of areas.
   d. Click the double right-arrow button.
      The selected areas move to the Request Areas Assigned list.
   e. Click OK.
      The Workshift Detail page displays the selected request areas on the Auto Assignment tab.
5. (Optional) Perform the following steps to configure automatic assignment of change orders for this workshift:
   a. Click Update Change Categories.
      The Change Category Search page appears.
   b. (Optional) Complete the filter fields to display only the category records of interest and click Search.
      The Change Categories Assigned Update page appears.

c. Select the categories you want to assign to this workshift from the Change Categories Available list. To select multiple categories, hold down the CTRL key while clicking the left mouse button, or hold down the Shift key to select a range of categories.
d. Click double right-arrow button.
The selected categories move to the Change Categories Assigned list.
e. Click OK.
The Workshift Detail page displays the selected change categories on the Auto Assignment tab.

6. (Optional) Perform the following steps to configure automatic assignment of issue tickets for this workshift:
a. Click Update Issue Categories.
The Issue Category Search page appears.
b. (Optional) Complete the filter fields to display only the category records of interest and click Search.
c. The Issue Categories Assigned Update page appears.
d. Select the categories you want to assign to this workshift from the Issue Categories Available list. To select multiple categories, hold down the CTRL key while clicking the left mouse button, or hold down the Shift key to select a range of categories.
e. Click double right-arrow button.
The selected categories move to the Issue Categories Assigned list.
f. Click OK.
The Workshift Detail page displays the selected issue categories on the Auto Assignment tab.
7. Click Save.
The new code definition is saved.

Create Personal Responses

You can create personalized responses and attach them to requests, issues, and change order records when adding activities to the record. For example, you can append a personalized response on the Status Change or Log Comment windows available from the Activities menu.

Follow these steps:

1. From the Administration tab, navigate to Service Desk, Personal Responses.
The Personal Response list page appears.
2. Click Create New.
The Create New Personalized Response page appears.
3. Fill in the fields on this page including:
– **Response Owner**
The contact who owns the response. If this field is left blank, the response is available to all analysts.
– **Response**
The text delivered to all those who receive this response. This field can be up to 1000 characters long.
You can use variables in this field, for example:
Ticket ref_num: @{call_req_id.ref_num}
Assignee: @{call_req_id.assignee.combo_name}
Customer: @{call_req_id.customer.combo_name}
Description: @{call_req_id.description}
4. Select the type of records for which you want this response available. By default, all record types are selected.
5. Click Save.

Edit a Sequence Number Format

Sequence Numbers

Sequence numbers define the format for your ticket numbering scheme. You can modify the predefined sequence number formats, but you cannot create new ones.

> **NOTE**
> CA SDM treats the number format as a character string, not as a sequence of numbers. Therefore, when you load tickets into CA SDM from another system, assign a unique prefix or suffix so that existing tickets are not inadvertently overwritten.

When a ticket is opened, it is automatically assigned the next available sequential number. For example, if the last request opened is 5, the next one is assigned the number 6.

> **WARNING**
> After you install a new version of CA SDM, the internal record ID for all tickets is reset to 1. To help ensure that duplicate record IDs are not created, do not create tickets before restoring any backed-up data.

You can configure how requests, change orders, and issues are numbered by including a unique prefix or suffix in the numbering scheme for each one. For example, if you want to track requests by month, you can add a month identifier as a prefix or suffix to the request numbering scheme.

Because you define a separate numbering scheme for each type of ticket, set up numbering schemes for all service desk models. You can control the format of the numbering of requests, change orders, and issues by changing the Sequence Number settings. By default, new tickets are numbered using consecutive integers. Because the number field of a ticket is actually a string field and not numeric, you assign additional string values to use as prefixes or suffixes when the ticket number is generated for a new ticket. For example, you can specify r:, c:, and i: for prefixes for requests, change orders, and incidents. This setup lets users easily differentiate between the various ticket types and prevents confusion.

Incidents and problems share numbering schemes with requests, because incidents and problems are internally different types of requests.

> **NOTE**
>
> In Advanced Availability configuration, the ticket numbers are not in sequential order (may be out of order).

Edit a Sequence Number Format

You can edit an existing sequence number format, but you cannot create new ones.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants. A tenant dropdown appears in the search filter. If you select <empty> in this dropdown, the search is public. A tenant column also appears on the list page.

Follow these steps:

1. Select Service Desk, Sequence Numbers on the Administration tab.
   The Sequence Number Control List page appears.
2. Select the symbol for the sequence number format to edit.
   The Sequence Number Control Detail page appears.
3. Click Edit.
   The Update Sequence Number Control page appears.
4. Edit the fields as needed:
   - **Symbol**
     The name identifying the sequence number format.
   - **Prefix**

The character string that forms the first part of the ticket number.

- **Code**
  The internal code name for the number format. Cannot be changed.
- **Suffix**
  The character string that forms the last part of the ticket number.

5. Click Save.
   The format definition is saved and the Sequence Number Format Detail page appears.

Define Closure Codes

You use closure codes to indicate the final outcome of a change order. You set the field manually or when updating status activity on a change order. CA SDM provides the following default closure codes:

- **Successful**
  Indicates the change order was successful.
- **Unsuccessful**
  Indicates the change order was unsuccessful.
- **Successful with Errors**
  Indicates the change order was successful, but directly or indirectly caused errors.

You can define closure codes to use in managing change orders.

Follow these steps:

1. On the Administration tab, select Service Desk, Change Orders, Closure Code.
   The Closure Code List appears.
2. Click Create New.
   The Create New Closure Code page appears.
3. Complete the required fields.
4. Save and close the window.

Define Change Type Codes

Change type codes identify how change orders fit within the operating practices of the organization. Predefined codes are provided for the following ITIL change types:

- **Standard**
  A change to a service infrastructure that is pre-authorized by change management and has an accepted and established procedure to provide a specific change requirement
- **Normal**
  A change that is raised by request from the initiator (an individual or group) that requires the change
- **Emergency**A change that is accelerated outside of normal processing, which sometimes adds a higher risk or impact than if handled normally

  NOTE
  If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. Select Service Desk, Change Orders, Change Type on the Administration tab.
   The Change Type List appears.
2. Click Create New.
   The Create New Change Type page appears.

3. Complete the required fields:
   - **Symbol**
     Defines an identifier for the change type code. This identifier appears in the Type field on a ticket.
   - **Status**
     Specifies whether this database record is active or inactive. Select a value from the drop-down list.
4. Click Save.
   The change type record is saved and the Change Type Detail page appears.

Define Change Order Status Transition

This article contains the following topics:

Change order status transitions control the movement of a change order ticket from one discrete state to another (for example, from Open to Closed). With transitions, you can define a subset of the full status list and determine the next status of the ticket as it continues through its lifecycle. You can use the predefined transitions (listed on the Change Order Transitions List page), modify the transitions, or create transitions

> **NOTE**
> You can specify how strictly the system enforces Status policies by configuring the Status Policy Violations option in Options Manager (General Options). This option only applies to automated system processes, such as integrations and macros.

You can define a subset of the full status list and determine the next status of the ticket as it continues through its lifecycle.

> **NOTE**
> Because status transitions can be shared between integrations such as CA Workflow and Events and Macros, do not inactivate predefined status transitions unless explicitly requested.

Follow these steps:

1. On the Administration tab, select Service Desk, Change Orders.
   The Change Order Transition List page appears.
2. Complete the fields as appropriate. The following fields require explanation:
   - **From Status**
     Defines the current status of the ticket, for example, Open.
   - **To Status**
     Specifies a valid next status of the ticket, for example, Assigned.
   - **Default Transition**
     Specifies the default status transition. CA SDM uses the default transition when a user clicks the default transition button on the ticket detail page, or when a user (including a web services user) updates the status to a <d> value. You can only configure one default transition for each status (or one for each tenant in a multi-tenanted system).
   - **Must Comment**
     Indicates that a comment for the transition must be supplied. Specifying this option indicates that an analyst must supply an activity log comment when changing the status on a request.
   - **Condition**
     Specifies the condition by which the transition is allowed. For example, when the condition "nonprty 1&2 assigned req" is associated with the Request Transition of Acknowledged to Hold, the condition verifies if the transition to move the status from Acknowledged to Hold is allowed.
   - **Condition Error Message**
     Specifies the message returned to the user if the transition is rejected.
   - **Description**
     Specifies the message returned to the user if the transition is rejected.

   The transition is defined.

3. Click Save, Close Window.
   The new transition appears in the Change Order Transition List when you refresh the page.

Update Initial Status Transitions

You can update the list of statuses that are valid at initial creation of the ticket.

Follow these steps:

1. On the Transitions List page, click the Initial Transitions button.
   The Initial Transitions page appears and displays all available status codes and descriptions.
2. Fill in the following check boxes as appropriate:
   - **Allowed**
     Specifies a valid transition for the status list. Use this option to restrict status workflows.
   - **Default**
     Specifies the default status transition. CA SDM applies the default transition when a user clicks the default transition button on the ticket detail page, or when a user (including a web services user) updates the status to a <d> value. There is only one default transition for each status (or one for each tenant in a multi-tenanted system).
   - **Must Comment**
     Specifies that an activity log comment for the transition is required when changing the status on a ticket. An error message appears on the ticket form when the analyst attempts to close the ticket without adding a comment.
3. (Optional) Select a status code in the Name column to update its details.
4. Click Save.
   The updated status transition appears on the Transitions list.

Define Workflow Task Status Code

Workflow task status codes can be attached to change orders, issues or requests/ incidents/ problems. For example, this code can indicate how close the change order is to being completed. Each workflow task has its own status, which remains separate from the status for the entire change order. Workflow task status codes allow you to locate, categorize, and track change orders based on their workflow status.

> **NOTE**
>
> Out-of-the-box Workflow task status codes **Wait**, **Pending**, **Reopen-Wait**, and **Cancelled** will not appear in the status drop-down list while updating the Workflow task. This is because these tasks have special usage internally and marked as internal. To resolve, this add a new behavior. For example: **Cancelled1**.

Follow these steps:

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

1. On the Administration tab, browse to Service Desk and select any of the following:
   - Change Orders
   - Issues
   - Requests/ Incidents/ Problems
2. Select Workflow Task Status Codes.
   The Task Status Code List appears.
3. Click Create New.
   The Create New Task Status page appears.
4. Complete the fields as appropriate.

Symbol - Defines an identifier for the task status code. This identifier appears in the Status field next to the tasks listed on the Workflow Tasks notebook page of the issue or change order. This is a required field.

Code - Displays the internal code for the task status. This is a required field.

Record Status - Identifies whether the database record is active or inactive. This is a required field. Select a value from the drop-down list.

Allow Task Update - When checked, indicates that an analyst can change the status of the task. This allows a qualified person to override a status that is entered by default, such as Pending.

Task Completed - When checked, indicates that the task is complete when it reaches this status. This option cannot be selected if Allow Task Update is selected.

Allow Accumulate - When checked, includes tasks with this status in calculations of estimated cost and duration for the original issue or change order. If not selected, tasks with this status are not included in estimates made by choosing the Accumulate feature on the Issue Detail or Change Order Detail window.

Continue to Next Tasks - When checked, changes the status of the next task to Pending when the current task reaches this status. This option can only be selected if Task Completed is selected.

Stop Service Type Events - When checked, automatically delays service type events when a task is changed to this status. For example, the predefined Hold status has this option checked, so that when you change the status of a task to Hold, all service type events are stopped.

Cannot Update Message - Displays the contents of this field if Allow Task Update is not selected and someone tries to change a task with this status.

Last Modified Date - Displays when the record was last modified, in the time zone of the server. This field is read-only, and is filled automatically each time the record is updated.

Last Modified By - Displays the name of the contact who last updated this record. This field is read-only, and is filled automatically each time the record is updated.

5. Click Save.
   The task status code definition is saved.

Define Change Status Codes

Change order status codes indicate how close a change order is to being completed. The status code appears in the Status field on a change order. Once change orders are assigned status codes, you can locate, categorize, and track them based on their condition.

Follow these steps:

1. On the Administration tab, select Service Desk, Change Orders, Status.
   The Change Order Status List appears.
2. Click Create New.
   The Create New Change Order Status page appears.
3. Complete the fields as appropriate. The following fields require explanation:
   - **Make Change Order Active**
     Activates an inactive change order, request, or issue when assigned to a ticket.
   - **Stop Service Type Events and Targets**
     Automatically delays service type and events and targets for a ticket. For example, the predefined Hold status has this option checked, so when you change the status of a ticket to Hold, all service type events and targets are stopped.
   - **Make Change Order Resolved**
     Indicates that a change order is resolved.
4. (Optional) Use the controls available on the tabs at the bottom of the page to configure the following:
   - **Change Order Transitions**

Controls how users select available statuses on the ticket form. With transition controls, you can control how a ticket transitions through different statuses by limiting the status list. You can use predefined transitions, modify the transitions, or create transitions.

– **Change Order Dependent Attribute Controls**
Controls how attributes are designated as required (must supply) or locked (cannot update) depending on ticket status. With dependent attribute controls, you can determine which fields are shown, or required for the status. You can use predefined attributes, modify the attributes, or create dependent attributes.

5. Click Save.
The status record is saved and the Change Order Status Detail page appears.

Risk Level

Risk levels determine the amount of risk involved in a change order. Risk levels are calculated after submitting risk surveys for a change order.

- **Level**
Specifies the name of the risk level.
- **Value**
Specifies the value of the risk level.
- **Description**
Specifies the description of the risk level.
- **Status**
Specifies if the risk level is active or inactive.

Define Issue Status

Issue status codes indicate how close an issue is to being completed. The status code appears in the Status field on an issue. Once issues are assigned to status codes, you can locate, categorize, and track them based on their condition.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. On the Administration tab, browse to Service Desk, Issues, Status.
The Issue Status List displays.
2. Click Create New.
The Create New Issue Status page displays.
3. Complete the fields as appropriate.

Record Status

Shows whether this database record is active or inactive. This is a required field. Select a value from the drop-down list.

Code

Identifies the internal code for the status. This is a required field.

Make Issue Active

When checked, activates an inactive change order, request, or issue when you assign this status to the ticket.

Stop Service Type Events

When checked, automatically delays service type events when a ticket is changed to this status. For example, the predefined Hold status has this option checked, so that when you change the status of a ticket to Hold, all service type events are stopped.

Make Issue Resolved
When checked, indicates that an issue is resolved when it reaches this status.

4. Click Save.
   The Issue Status Detail page displays.
5. Click Close Window.
   The new status appears in the Issue Status List when you redisplay the list.

Update Initial Status Transitions

You can update the list of statuses that are valid at initial creation of the ticket.

To update an initial status transition

1. On the Transitions List page, click the Initial Transitions button.
   The Initial Transitions page appears and displays all available status codes and descriptions.
2. Fill in the following check boxes as appropriate:
   Allowed
   Specifies a valid transition for the status list. Use this option to restrict status workflows.
   Default
   Specifies the default status transition. CA SDM applies the default transition when a user clicks the default transition button on the ticket detail page, or when a user (including a web services user) updates the status to a <d> value. There is only one default transition for each status (or one for each tenant in a multi-tenanted system).
   Must Comment
   Specifies that an activity log comment for the transition is required when changing the status on a ticket. An error message appears on the ticket form when the analyst attempts to close the ticket without adding a comment.
3. (Optional) Select a status code in the Name column to update its details.
4. Click Save.
   The updated status transition appears on the Transitions list.

Update Initial Status Transitions
You can update the list of statuses that are valid at initial creation of the ticket.
To update an initial status transition
On the Transitions List page, click the Initial Transitions button.
The Initial Transitions page appears and displays all available status codes and descriptions.
Fill in the following check boxes as appropriate:
Allowed
Specifies a valid transition for the status list. Use this option to restrict status workflows.
Default
Specifies the default status transition. CA SDM applies the default transition when a user clicks the default transition button on the ticket detail page, or when a user (including a web services user) updates the status to a <d> value. There is only one default transition for each status (or one for each tenant in a multi-tenanted system).
Must Comment
Specifies that an activity log comment for the transition is required when changing the status on a ticket. An error message appears on the ticket form when the analyst attempts to close the ticket without adding a comment.
(Optional) Select a status code in the Name column to update its details.
Click Save.
The updated status transition appears on the Transitions list.

Define a Priority Calculation

This article contains the following topics:

Priority calculation is a predefined set of values that automatically set Priority, Urgency, and Impact values on problems and incident tickets.

NOTE
If you migrated from a previous release, the priority calculation values must be enabled after the migration. For more information about enabling priority calculation, see the Configure Ticket Management After Upgrade topic.

You can define a priority calculation to handle, incidents, problems, or both. However because two priority calculations cannot handle the same ticket type, you can review and update existing priority calculations before you create ones. For example, if you want to add a priority calculation to manage problem tickets, first clear the Problem field on the active priority calculation. Then, create a priority calculation to manage problem tickets.

If multi-tenancy is installed, you can use one public priority calculation that applies to all tenants or several tenant-specific priority calculations.

Follow these steps:

1. On the Administration tab, select Service Desk, Request/Incidents/Problems, Priority Calculation.
   The Priority Calculation List page appears.
2. Click Create New.
   The Create Priority Calculation page appears.
3. Complete the following fields as appropriate:
   - **Name**
     Identifies the priority calculation.
   - **Status**
     Specifies the ticket status for this priority calculation.
   - **Incidents**
     Specifies whether this priority calculation applies to incident ticket types.
   - **Problems**
     Specifies whether this priority calculation applies to problem ticket types.
   - **Spelling**
     Verifies the spelling of text in the Description field.
   - **Description**
     Specifies the purpose of the priority calculation.
4. Complete the priority calculation by setting the Priority value based on Impact and Urgency. For example, if the Urgency is 5-Immediate and the Impact affects the 1-Entire Organization, you can select the Priority value of 1 from the drop-down list.
5. Set the following Priority Calculation Options as appropriate:
   - **Impact Default**
     Specifies the initial Impact value to display when the user creates a ticket.
   - **Override Impact**
     Works with the Impact Increment field. Specifies whether the user can override the ticket Impact value with the Impact level specified in the attached Affected Service.
   - **Impact Increment**
     Works with the Override Impact field. Increments the Impact by the specified amount when the user overrides the Impact value. When the ticket uses a blackout window, the Impact value increments only when the open date is within the blackout window time frame. The Impact value can only raise to a maximum of 1-High. For example, if you set the value to 2 and the default Impact is 3-Medium, the user can only override the Impact on a ticket to 1-High.
   - **Urgency Default**
     Specifies the Urgency value to display when the user creates a ticket.
   - **Override Urgency**
     Specifies whether the user can override the ticket Urgency value with the Area Urgency level specified in Incident or Problem area.
   - **Urgency Increment**

Works with the Override Urgency field. Increments the Urgency level by the specified amount when the affected end user sets the Escalate Urgency flag.

- **Capture Reason**
  Specifies whether the user is required to enter the reason for changing the Urgency or Impact values. If the user changes the Urgency or Impact values, the Escalate Detail page appears while saving the ticket. The user is required to describe the reason for changing the Urgency or Impact values.
- **Enable for Templates**
  Specifies whether to use the priority calculation for tickets that use templates.

6. Click Reset Matrix if necessary to restore the settings to the defaults.
   The default values reset.
7. Click Save.
   The Update Priority Detail page appears. On the next new or updated ticket or Knowledge Document, the fields update according to the values in the active priority calculation. If no priority calculation is active for a ticket type, the system clears the Priority, Urgency, and Impact fields.

Priority Calculation Search Fields

The following fields are available to search for active and inactive priority calculations:

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The Public Data option specifies whether to include public and tenant priority calculations, exclude public priority calculations, or only show public priority calculations.

- **Name**
  Identifies the priority calculation.
- **Description**
  Specifies the purpose of the priority calculation.
- **Incident**
  Specifies whether to search for priority calculations that manage incidents.
- **Problem**
  Specifies whether to search for priority calculations that manage problems.
- **Status**
  Specifies whether the priority calculation is active or inactive.

> **NOTE**
> You can click the More icon to display the Additional Search Arguments field. This field is intended only for expert users who understand SQL and Majic and can use it to specify search arguments that are not available in the standard search filter fields. You can enter a SQL WHERE clause in this field to specify an additional search argument.

Use Ticket Templates to Calculate Priority Values

If you want ticket templates to calculate priority, you configure the priority calculation with the *Enable for Templates* option. If you enable this option the Urgency and Impact values come from the template, but the priority field comes from priority calculation record. The priority value displays as read-only.

If you do not enable this option, the Urgency, Impact and Priority fields come from the template. You can edit the priority field and no priority calculation is done for the ticket until it is saved.

To use ticket templates to calculate priority values

1. On the Administration tab, select Service Desk, Requests/Incidents/Problems, Priority Calculation.
   The Priority Calculation List appears.
2. Select the priority calculation that you want to use for calculating priority with templates.

You can also create a priority calculation to use ticket templates to calculate priority values.
3. Select *Enable for Templates* from the Priority Calculation Options list.
4. Click Save.
The option is enabled.

Manage Request/Incident/Problem Status

This topic contains the following topics:

Define a Request/Incident/Problem Status

Request/incident/problem status codes indicate how close a ticket is to resolution. Status codes allow you to track the progress of tickets, and categorize and locate tickets based on their condition. As a system administrator, you can use predefined status codes, modify the predefined status codes, or create new status codes.

You can mark any request/incident/problem status as active or inactive. When you mark a status code as inactive, it is no longer available for analysts to use, but it remains available in the database for future use. If you decide to use the request/incident/problem status in the future, you can go back and mark it as active.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. On the Administration Tab, select Service Desk, Requests/Incidents/Problems, Status.
The Request/Incident/Problem Status List page appears.
2. Click Create New.
The Create New Request/Incident/Problem Status page appears.
3. Complete the Request/Incident/ProblemStatus Fields as appropriate.
4. Click Save.
The Request/Incident/Problem Status Detail page appears.
5. Click Close Window.
The new request/incident/problem status appears in the Request/Incident/Problem Status List when you redisplay the list.

Request/Incident/Problem Status Fields

The following fields are required to create or update the request/incident/problem status:

- **Symbol** -- Defines a unique identifier for the record. This identifier appears in the Status field on a ticket.
- **Record Status** -- Shows whether the database record is active or inactive.
- **Code** -- Defines the internal code for the status.
- **Make Active** -- If checked, specifies that an inactive change order, request, or issue is active when you assign this status to it.
- **Stop Service Type Events and Targets** -- If checked, specifies that service type events are delayed automatically when a ticket is changed to this status. For example, the Hold status that is predefined for requests, issues, and change orders has this option checked. When you change the status of a ticket to Hold, all service type events are stopped.
- **Make Resolved** -- If checked, causes the request, change order, or issue to have the Resolve Date/Time automatically set when it is changed to this status.
- **Description** -- Gives a detailed description of the record.
- **Status valid for** -- Defines what ticket types the record is valid for. The default is checked for all ticket types (Requests, Incidents, and Problems).

NOTE
If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

Update a Request/Incident/Problem Status

As a system administrator, you can modify the predefined status codes or status codes you have created. Request/incident/problem status codes indicate how close a ticket is to being resolved. Status codes let you carefully track the progress of tickets, categorize and locate tickets based on their condition.

You can mark any request/incident/problem status as active or inactive. When you mark a status code as inactive, it is no longer available for analysts to use, but it remains available in the database for future use. If you decide to use the request/incident/problem status in the future, you can go back and mark it as active.

Follow these steps:

1. On the Administration Tab, browse to Service Desk, Requests/Incidents/Problems, Status.
   The Request/Incident/Problem Status List appears.
2. Select the request/incident/problem status.
   The Request/Incident/Problem Status Detail page appears.
3. Click Edit.
   The Update Request/Incident/Problem Status page appears.
4. Update one or more of the fields as needed.
5. (Optional) Use the controls available on the tabs at the bottom of the page to configure the following:
   - **Request/Incident/Problem Transitions**
     Controls how users select available statuses on the ticket form. With transition controls, you can control how a ticket transitions through different statuses by limiting the status list. You can use predefined transitions, modify the transitions, or create transitions.
   - **Request/Incident/Problem Dependent Attribute Controls**
     Controls how attributes are designated as required (must supply) or locked (cannot update) depending on ticket status. With dependent attribute controls, you can determine which fields are shown, or required for the status. You can use predefined attributes, modify the attributes, or create dependent attributes.
6. Click Save.
   The Request/Incident/Problem Status page appears.
7. Click Close Window.
   The modified request/incident/problem status appears in the Request/Incident/Problem Status List when you redisplay the list.

Define Dependent Attribute Controls

You can designate certain attributes as required or locked depending on the status of the CA SDM ticket (change order, issue, incident/problem/request). For example, a manager can prevent an analyst from editing the summary of a request after approval. For each ticket type, you can use predefined attribute controls, edit the attribute controls, or define attribute controls.

NOTE

: When defining "required" dependent attribute controls for a particular status (all ticket types), be aware that the Edit in List option available on the ticket's list page cannot present the dependent attributes field values that are required. If the required attribute field value is not already part of the saved ticket, and if the required attribute field is not presented in the edit in list format to display its values, then the end user cannot save the ticket. Consequently, the ticket’s detail page must be used instead of the Edit in List option to edit the dependent attribute fields values that are required.

Follow these steps:

1. On the ticket's detail page for the status, select the Request/Incident/Problem Dependent Attribute Control tab at the bottom of the page.
   The Attribute Control List appears.
2. Click Create.
   The Update Request Status Acknowledged Dependent Attribute Control page appears
3. Complete the following fields as appropriate:
   - **Tenant**
     Specifies the name of the tenant.
   - **Attribute**
     Specifies the name of the attribute control, for example, Summary.
   - **Locked**
     Specifies the attribute as locked (read only) while the ticket is in the status for which the dependent attribute is defined.
   - **Required**
     Specifies the attribute as required (must be supplied).
4. Click Save.
   The attribute control for the status appears on the list page.

Search for a Request/Incident/Problem Status

You can use the search function to filter the Request/Incident/Problem Status List to show only the items you want to see. You can also locate a specific request/incident/problem status to view or update.

> **NOTE**
>
> If you are using multi-tenancy, a tenant drop-down list appears in the search filter. If you select <empty> in this drop-down list, the search is public.

To search for a request/incident/problem status

1. Select Service Desk, Requests/Incidents/Problems, Status on the Administration tab.
   The Request/Incident/Problem Status List appears.
2. Click Show Filter.
3. Complete one or more of the following search fields:
   Active
   Shows whether the database record is active or inactive.
   Name
   Defines a unique identifier for the record. This identifier appears in the Status field on a ticket.
   Description
   Gives a detailed description of the record.
   Request
   Indicates whether the status is valid for requests.
   Incident
   Indicates whether the status is valid for incidents.
   Problem
   Indicates whether the status is valid for problems.

   > **NOTE**
   >
   > You can click the More icon to display the Additional Search Arguments field. This field is intended only for expert users who understand SQL and Majic and can use it to specify search arguments that are not available in the standard search filter fields. You can enter a SQL WHERE clause in this field to specify an additional search argument.

4. Click Search.
   The Request/Incident/Problem Status List populates with all items that match your search criteria.

Categories

Change categories allow you to define default values that are entered automatically on change order tickets assigned to the category.

Define Change Categories

You can assign change orders to categories so that default values defined for a category automatically appear in fields on new change orders. Defining categories lets you create change orders quickly.

> **NOTE**
>
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

To define a change category, **follow these steps:**

1. On the Administration tab, browse to CA SDM, Change Orders, Categories.
   The Change Category List appears.
2. Click Create New.
   The Create New Change Category page appears.
3. Complete the fields as appropriate.
4. (Optional) Use the controls available on the tabs at the bottom of the page to configure the following features:
   Properties Defines custom properties to be applied to change orders, when assigned to this category. Workflow Attaches a workflow to specify a standardized sequence of tasks to be performed to resolve change orders assigned to this category. Note: To use the process viewer, you must associate a CA Workflow process to the change category. Auto Assignment Specifies how change orders in this category are auto assigned.
5. Click Save.
   The Change Category Detail page appears.
6. Click Close Window.
   The new change order category appears in the Change Category List when you refresh the list.

Change Category Fields

The following fields appear on the Create Change Category, Change Category Detail, and Update Change Category pages:

Symbol

(Required) Defines an identifier for the category. This is a required field.
Code

(Required) Defines the internal code for the category. This is a required field.
Type

Defines the level at which change orders are implemented within an organization.
Record Status

(Required) Indicates whether this database record is active or inactive. You can select a value from the drop-down list.
CAB

Specifies the group that is responsible for reviewing Requests for Changes (RFCs). The CAB provides multiple perspectives necessary to ensure proper decision making about implementing changes. The CAB can include members

from the application team, development manager, component owner, QA, support, and any additional parties deemed necessary. You can enter the value directly or click the search icon to search for the group.

Organization

Identifies the company, division, or department that is associated with the change category. Enter a value directly or click the search icon to search for an organization.

Group

Identifies the group that is responsible for the record. Your system administrator defines groups of contacts that are responsible for different types of issues, requests, incidents, change orders, and so on. Any individual contact assigned to the group can handle the task once it is assigned to the group. Enter a value directly or click the search icon to search for a group.

Assignee

Identifies the person assigned to the record. Enter a value directly or click the search icon to search for a contact.

Service Type

Defines the level of support service received by the contact affected by the change order. For example, some users may have contracted for 24-hour support, while others might receive on-site training. Enter a value directly or click the search icon to select a defined service type.

Survey

Identifies the defined survey associated with this record. Enter a value directly or click the search icon to search for a survey.

Risk Survey

Indicates the defined risk for change orders associated with change categories.

Children Allowed

Allows change orders assigned to this category to have subordinate issues. Organizes issues into a hierarchy of parent-child relationships to divide large issues into smaller, more manageable issues.

Self-Service Include

Displays this category in the self-service interface.

Self-Service Symbol

Defines a unique identifier for this category in the self-service interface.

Description

Gives a detailed description of the record.

Last Modified Date

Displays when the record was last modified, in the time zone of the server. This field is read-only, and is filled automatically each time the record is updated.

Last Modified By

Displays the name of the contact who last updated this record. This field is read-only, and is filled automatically each time the record is updated.

Service Contract

Identifies the service contract name associated with this record.

Setting Up Category or Area

This topic contains the following information:

- Change Order and Issue Categories
- Create Workflow Task Types
- Request/Incident/Problem Areas
- Define a Category or Area
- Define Behavior Template

Change Order and Issue Categories

This article contains the following topics:

Change categories and issue categories define the logical groupings into which change orders and issues can be organized.

NOTE

Unlike request/incident/problem areas, change order categories and issues categories are managed separately:

- Set up change categories for internal and combined CA SDM models.
- Set up issue categories for external and combined CA SDM models.

You can use categories to specify default values for certain fields in tickets. You can automatically associate a level of service to tickets by assigning a default service type to categories. You can also associate a survey with a category.

For each category, you can specify attributes or qualities to be associated with the ticket and create a workflow that identifies all the individual tasks required to fulfill the ticket. By defining behaviors that are associated with the workflow tasks, you can notify key personnel when the status of the task changes or as activities close the ticket.

Whenever an analyst assigns a ticket to a category, all the information you have associated with the category is automatically entered on the ticket. For example, if you indicate a service type, it becomes associated with the ticket, and its associated service type events.

Predefined Change Categories

The following sets of predefined change categories are installed with CA SDM:

- Add
- Change
- Move
- Retire

NOTE

All these category sets are subdivided into more specific categories. For example, the Change category set includes categories for changing servers and workstations.

Predefined Issue Categories

The following predefined issue categories are installed with CA SDM:

- Hardware.pc.install
- Software.pc.install

You can set the status of any category to active or inactive. When you make a category inactive, it is no longer available for analysts to use, but it is not deleted from the database. If you decide at a later time to use the category, change the status back to active.

Rules for Changing Categories on a Ticket

The following rules affect only workflow tasks.

- If the previous category used Classic Workflow and the new category uses CA Process Automation, the CA Process Automation process definition links to a workflow on a CA Process Automation server.
- If both the old and new categories use CA SDM workflow, the rules from previous releases apply.
- If the new category uses CA Process Automation and the old uses CA SDM workflow, the following occurs:
  - All incomplete and pending (those tasks that can be updated) workflow tasks of the CA SDM 6.0 style, are set to Cancelled status, regardless of the KEEP_TASKS option. Any completed workflow tasks remain, but they cannot be reopened.
  - All incomplete and nonactive tasks (such as tasks in Wait status) are deleted.
  - The CA Process Automation definition instantiates.
- If the new category uses CA SDM workflow and the old category uses CA Process Automation, the following occurs:
  - The CA Process Automation instance is forcibly set to a status of Terminated.
  - The new category workflow tasks are added as normal.

In a CA SDM workflow, a ticket with a running CA Process Automation instance cannot be closed.

Create Workflow Task Types

Workflow task types specify lists of allowed status codes for workflow tasks assigned to those types. For example, Approval tasks may only allow Approve, Reject, Pending, and Wait status codes, while Install Software tasks may allow Pending, Wait, and Completed status codes.

> **NOTE**
>
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. On the Administration tab, browse to Service Desk and select any of the following:
   - Change Orders
   - Issues
   - Requests/ Incidents/ Problems
2. Select Workflow Task Types.
   The Task Type List displays.
3. Click Create New.
   The Create New Task Type page displays.
4. Complete the following fields as appropriate.
5. (Optional) Select Update Status Codes on the Status Codes tab and select the codes that you want to assign to the task type.
6. Click Save.
   The workflow task type is created.

Request/Incident/Problem Areas

This article contains the following topics:

Request areas define the logical groupings into which you can organize request, incident, and problem tickets. For example, tickets pertaining to an application can be assigned to the predefined Applications area. Whenever an analyst assigns a ticket to a request area, all the information you have associated with the request area is automatically entered

on the ticket. For example, if you indicate a service type, it becomes associated with the ticket and all its associated service type events.

NOTE

The same area definitions are available for request, incident, and problem tickets. On the CA SDM Web Interface Administration tab, these areas are referred to as request/incident/problem areas. For brevity, they are referred to here simply as request areas.

You can set the status of any request area to active or inactive. When you make a request area inactive, it is no longer available for analysts to use, but it is not deleted from the database. If you decide later to use the request area, you need only change the status back to active.

You can use request areas to do the following:

- Specify default values for the group and assignee fields on tickets.
- Automatically associate a level of service to tickets by assigning a default service type to the request area.
- Associate a survey with a request area.
- Select and report on tickets by area by defining your own custom request areas. Eventually study request trends and analyze problem causes. Focusing your view to specific request areas can help make these studies more significant and revealing.

The following predefined request areas are installed with CA SDM:

- Applications
- Email
- Hardware
- Networks
- Printer
- Software
  Software is subdivided into several request areas.

Request/Incident/Problem Area Properties

You can use properties to add custom attributes or qualities to a specific request area. If you have added properties to a request area, when an analyst assigns a ticket to that area the associated properties automatically appear on the ticket's properties tab. For example, you can add properties to the predefined email request area to specify the email server or mailbox size.

As you define properties, you can specify whether a value is required or optional. For properties with a required value, users must enter a value (or accept the default) before they can save the ticket.

The keep_tasks option determines what happens when you assign an existing ticket to a different request area:

- If keep_tasks is not installed, existing properties (and workflow tasks) are removed from the ticket, and any properties or tasks associated with the new request area are added.
- If keep_tasks is installed, existing properties and tasks are retained on the ticket, and any properties or tasks associated with the new request area are added.

Property Validation Rules

You can define validation rules to restrict the property values users can enter to a predefined set of selectable values. For example, if you have defined a property named Operating System, you can define the validation rule as a drop-down list containing the options Windows, UNIX, and Linux.

Properties defined without validation rules are presented to the user as free form text boxes, which allow any text string to be entered. Validation rules make reporting on area and category values less complex and less error prone.

NOTE

Property validation rules are reusable. They are not specific to a particular property. You can apply any existing validation rule to properties defined for change categories, issue categories, or request/incident/problem areas.

Depending on which type of validation rule you have configured for a property, when the user assigns a ticket to the category or area to which that property is attached, one of the following controls appears on the ticket properties tab:

- Text Edit Box—No validation rule has been defined and no default value can be specified. Users are expected to enter values that follow the examples you provide.
- Check Box—A two-state check box appears on the Properties tab. By default, the check box does not contain a checkmark. The user can accept the default, or select the check box. Boxes that contain a checkmark are displayed on the ticket detail page with Yes in the value column. Cleared boxes are displayed with No in the value column.
- Drop-down List—A drop-down list containing a predefined set of options appears on the Properties tab. If you have defined a default value, it is selected when the drop-down list is first displayed. The option the user selects is shown in the value column of the ticket detail page.

Create Request/Incident/Problem Areas

Request/incident/problem areas define the general areas into which request, incident, and problem tickets can be organized. Categorizing these ticket types lets you generate reports, study trends, and analyze problem causes.

To define a request/incident/problem area

1. On the Administration tab, browse to Service Desk, Requests/Incidents/Problems, Areas.
   The Request/Incident/Problem Area List appears.
2. Click Create New.
   The Request/Incident/Problem Area Detail page appears.
3. Complete the fields as appropriate.
4. (Optional) Use the controls available on the tabs at the bottom of the page to configure the following features:
   - **Properties**
     Defines custom properties to apply to tickets assigned to this area.
   - **Auto Assignment**
     Specifies how tickets in this area are auto assigned.
5. Click Save.
   The new request/incident/problem area is created.

Attach a CA Workflow Process Definition

Note: Unlike change categories and issue categories, which support both CA Workflow process definitions and CA SDM classic workflows, request/incident/problem areas support only CA Workflow process definitions. For more information, see the *Administration Guide*. For information about integrating CA Workflow with your CA SDM installation, see the *Implementation Guide*.

The Workflow Tab allows you to select a CA Workflow process definition for the request/incident/problem area. A workflow specifies tasks that must be completed on tickets associated with that area.

To allow flexibility, you can define workflow tasks as optional, which allows the analyst to delete them from the task list.

All mandatory workflow tasks must be complete before the ticket can be closed. The only exception is that if the ticket status is set to Cancelled, the workflow process terminates automatically.

If a ticket is removed from an area with an attached workflow, or assigned to a different area, the workflow process terminates automatically. If the new area has an attached workflow, its process starts automatically.

To attach a workflow process definition to a request/incident/problem area

1. Select Service Desk, Requests/Incidents/Problems, Areas on the Administration tab.

The Request/Incident/Problem Area List appears.
2. Select the area to edit.
   The Area Detail page appears.
3. Click Edit.
   The Area Update page appears.
   4. Select the Workflow tab and click Add Classic Workflow.
   Note: The button is only present on systems where CA Workflow is installed and configured to work with CA SDM. For more information about workflows, see the *Implementation Guide*.
   The Workflow Definition List appears.
4. Click Save.
   The Area Detail page displays a Save Successful message.

Define Request/Incident/Problem Areas for Self-Service

The Self-Service Include option lets you define which request/incident/problem areas to include on tickets for self-service. You can also define different self-service symbols than those seen by the analyst. When the ticket is saved, the self-service symbol displays in the Request/Incident/Problem Area field. If the ticket displays in the analyst interface, the normal symbol for the area appears. The self-service user is not allowed to edit the symbol; it is read-only.

Follow these steps:

1. On the Administration tab, select Service Desk, Requests/Incidents/Problems, Areas.
   The Requests/Incidents/Problems Area List appears.
2. Select Edit In List.
   The top portion of the page displays the editable fields.
3. Open the desired incident/problem/request area for editing on the Symbol list (Hardware.pc.install, for example).
4. Complete the following fields:
   - **Self-Service Include**
     Specifies whether this request/incident/problem area is shown in the self-service interface.
   - **Self-Service Symbol**
     Specifies a unique identifier for this request/incident/problem area in the self-service interface.
   - **Active**
     Specifies whether the request/incident/problem area is active or inactive.

   The request/incident/problem area is defined for self-service.
5. Click Save.
   The updated request/incident/problem area appears in the Request/Incident/Problem Area List when you redisplay the list.

Define a Category or Area

Issues or change orders can be assigned to categories so that default values defined for a category automatically appear in the fields while creating the issues or change orders.

Request/incident/problem areas define the general areas into which request, incident, and problem tickets can be organized. Categorizing these ticket types lets you generate reports, study trends, and analyze problem causes.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. On the Administration tab, browse to Service Desk, and select any of the following:

- Change Orders, Categories
- Issues, Categories
- Requests/Incidents/Problems, Areas

The corresponding Area or Category List appears.

2. Click Create New and complete the fields, as appropriate:
   - For information about request/incident/problem areas, see Request/Incident/Problem Area Fields.
   - For information about issue category fields, see Issue Category Fields.
   - For information about change category fields, see Change Category Fields
3. (Optional) Add Properties to Requests/Incidents/Problems Areas from the **Properties** tab.
4. (Optional) Attach a classic workflow or attach a CA Process Automation workflow (if you have integrated CA SDM with CA Process Automation).

   NOTE

   You can attach classic workflow to each ticket type area only if you have applied a patch. Find the patch and download details from CA Support Online.
5. (Optional) Enable Auto Assignments from the **Auto Assignment** tab.
6. Click Save and close the window.
   The category or area is defined.

Request/Incident/Problem Area Fields

The following fields appear on the Create, Detail, and the Update pages for Request/Incident/Problem Areas:

- **Symbol**
  Defines a unique identifier for the record.
- **Organization**
  Specifies the company, division, or department that is associated with the request/incident/problem area. You can enter a value directly or click the magnifier to search for an organization.
- **Record Status**
  Shows whether the database record is active or inactive.
- **Group**
  Identifies the group that is responsible for this record.
- **Assignee**
  Shows the name of the person assigned to the record.
- **Service Type**
  Shows the level of support service received by the contact affected by the record.

  NOTE
  Service types are automatically attached to issues based on the service type defined for its issue category. If different service types have been defined for other values on the issue (priority and contact, for example) the service type used is the one with the best (lowest) ranking. For example, if an issue is opened and the From contact has a 12-hour resolution service type, which is ranked as 2, while the priority code has a 4-hour resolution service type, which is ranked as 1, the service type for the issue is a 4-hour resolution.
- **Survey**
  Identifies the defined survey associated with this record. You can enter a value directly or click the magnifier to search for a survey.
- **Area Urgency**
  Specifies how critical the [Request, Incident, Problem] is to the business area.
- **Self-Service Include**
  Displays the request/incident/problem area in the self-service interface.
- **Self-Service Symbol**

Defines a unique identifier for this request/incident/problem area in the self-service interface.

- **Description**
  Provides a detailed description of the record.
- **This category is valid for the following**
  Defines what ticket types the record is valid for. The default is enabled for all three ticket types (Requests, Incidents, and Problems).

Issue Category Fields

The following fields appear on the Create Issue Category, Issue Category Detail, and Update Issue Category pages:

- **Symbol**
  (Required) Defines an identifier for the category.
- **Code**
  (Required) Defines the internal code for the category.
- **Record Status**
  (Required) Indicates whether this database record is active or inactive. You can select a value from the drop-down list.
- **Group**
  Identifies the group that is responsible for the record. Your system administrator defines groups of contacts that are responsible for different types of issues, requests, incidents, change orders, and so on. Any individual contact assigned to the group can handle the task once it is assigned to the group. Enter a value directly or click the search icon to search for a group.
- **Assignee**
  Identifies the person assigned to the record. Enter a value directly or click the search icon to search for a contact.
- **Service Contract**
  Identifies the service contract name associated with this record. This field is read-only.
- **Service Type**
  Defines the level of support service received by the contact affected by the change order. For example, some users may have contracted for 24-hour support, while others might receive on-site training. Enter a value directly or click the search icon to select a defined service type.
- **Survey**
  Identifies the defined survey associated with this record. Enter a value directly or click the search icon to search for a survey.
- **Children Allowed**
  Allows issues assigned to this category to have subordinate issues. Organizes issues into a hierarchy of parent-child relationships to divide large issues into smaller, more manageable issues.
- **Self-Service Include**
  Displays this category in the self-service interface.
- **Self-Service Symbol**
  Defines a unique identifier for this category in the self-service interface.
- **Description**
  Gives a detailed description of the record.

Change Category Fields

The following fields appear on the Create Change Category, Change Category Detail, and Update Change Category pages:

- **Code**
  (Required) Defines the internal code for the category.
- **Type**

Defines the level at which change orders are implemented within an organization.

- **Record Status**
  (Required) Indicates whether this database record is active or inactive. You can select a value from the drop-down list.
- **CAB**
  Specifies the group that is responsible for reviewing Requests for Changes (RFCs). The CAB provides multiple perspectives necessary to ensure proper decision making about implementing changes. The CAB can include members from the application team, development manager, component owner, QA, support, and any additional parties deemed necessary. You can enter the value directly or click the search icon to search for the group.
- **Organization**
  Identifies the company, division, or department that is associated with the change category. Enter a value directly or click the search icon to search for an organization.
- **Group**
  Identifies the group that is responsible for the record. Your system administrator defines groups of contacts that are responsible for different types of issues, requests, incidents, change orders, and so on. Any individual contact assigned to the group can handle the task once it is assigned to the group. Enter a value directly or click the search icon to search for a group.
- **Assignee**
  Identifies the person assigned to the record. Enter a value directly or click the search icon to search for a contact.
- **Service Type**
  Defines the level of support service received by the contact affected by the change order. For example, some users may have contracted for 24-hour support, while others might receive on-site training. Enter a value directly or click the search icon to select a defined service type.
- **Survey**
  Identifies the defined survey associated with this record. Enter a value directly or click the search icon to search for a survey.
- **Risk Survey**
  Indicates the defined risk for change orders associated with change categories.
- **Children Allowed**
  Allows change orders assigned to this category to have subordinate issues. Organizes issues into a hierarchy of parent-child relationships to divide large issues into smaller, more manageable issues.
- **Self-Service Include**
  Displays this category in the self-service interface.
- **Self-Service Symbol**
  Defines a unique identifier for this category in the self-service interface.
- **Service Contract**
  Identifies the service contract name associated with this record.

Add Properties

Properties are characteristics or attributes of tickets that are affected by the area or category the ticket is assigned to. For example, when you assign a request to a request area, the properties associated with that area are automatically assigned to the request.

Follow these steps:

1. On the Properties tab, click Add Property.
   The Create New Property Template page appears.
2. Complete the fields as appropriate. To limit the property values to selections from a predefined set, you can specify validation rules.
   The following values on the Properties pages are not self-explanatory:
   – **Sequence**

Defines the order in which the properties appear on the Properties tab of the tickets assigned to this Area or Category. The property with the lowest sequence number is listed first.

- **Active**
  Specifies whether the property is active or inactive.
- **Value Required**
  Indicates whether a value must be entered for this property on tickets assigned to this Area or Category. If selected, the word Yes appears in the Required column when this property is listed on the Properties tab of a ticket.
- **Validation Rule**
  Indicates the name of a rule that has been defined to validate user-entered values for this property. Enter the validation rule name directly or click the search button to search for a validation rule.
- **Label**
  The name of the property as it appears on the Properties tab.
- **Examples**
  Examples of appropriate values for the property. For example, if a freeform text field is intended to contain information about a network type, this field might include LAN, WAN, and SAN as example values. These examples appear on the ticket to provide guidance for entering the actual property value.

3. Click Save.
   The detail page lists the new property on the Properties tab.
   The keep_tasks option determines what happens when you assign an existing ticket to a different category:
   - If keep_tasks is not installed, existing properties and workflow tasks are removed from the ticket, and any properties or tasks associated with the new category are added.
   - If keep_tasks is installed, existing properties and tasks are retained on the ticket, and any properties or tasks associated with the new category are added.

Create Property Validation Rules

You can create validation rules to ensure that only valid values are assigned to custom properties. For example, if you have defined a property named Operating System, you can define the validation rule as a drop-down list containing the options Windows, UNIX, and Linux.

Properties that are defined without validation rules are presented to the user as freeform text boxes, which allow any text string to be entered. Validation rules make reporting on area and category values less complex and less error prone.

> **NOTE**
> Property validation rules are reusable. They are not specific to a particular property. You can apply any existing validation rule to properties defined for change categories, issue categories, or request/incident/problem areas.

Depending on which type of validation rule you have configured for a property, when the user assigns a ticket to the category or area to which that property is attached, one of the following controls appears on the ticket properties tab:

- Text Edit Box
- Check Box
- Drop-down List.

Follow these steps:

1. Access the create, detail, or update page for the custom property you want to assign a property validation rule. For example:
   a. Select Service Desk, Change Orders, Categories on the Administration tab.
      The Change Category List page appears.
   b. Click the Add.IT.Server link.
      The Add.IT.Server Change Category Detail page appears.
   c. Select the Properties tab and click the Add Property button.
      The Create New Property Template page appears.
2. Click the Validation Rule link.

The Property Validation Rule Search page appears.

3. Specify any filter criteria and then click Search.
   The rules that match your filter criteria are displayed.

   NOTE
   Validation rules are not specific to a particular category or area type. You can apply any existing validation rule to properties for change categories, issue categories, or request/incident/problem areas. Although you can create multiple checkbox validation rules, it is not necessary to do so. You can reuse an existing checkbox rule as many times times as you want by applying a different label. For example, if you have an existing checkbox rule named CB, and you later want to apply a checkbox validation rule to a different property, you can select CB from the Property Validation Rule List and then enter a different label on the Create New Property Template page.
4. Click Create New if there is no existing validation rule that meets your needs.
   The Create New Property Validation Rule page appears.
5. Complete the fields as appropriate.
6. For Dropdown validation rules only, the Property Value List tab appears on the Create New Property Validation Rule page. To specify the list of options that appears in the dropdown menu:
   a. Click Add Property Value.
      The Create New Property Value page appears.
   b. Complete the fields as appropriate and click Accept.

      NOTE
      The Accept button accepts the changes, closes the window, and applies them to the current record, but you must still click the Save button on the main record window to make the changes permanent.

      The Create New Property Validation Rule page appears with a highlighted status indicator showing the number of property values that are pending.
   c. Repeat the above steps until you have added all the property values you want the dropdown list to contain.
7. Click Save on the Create New Property Value page.
   The Create New Property Template page appears, with the validation rule name filled in.

Attach a CA Process Automation Workflow

A workflow *process definition* identifies a collective series of tasks, steps, and conditions that are structured in a specific order for individuals or parties to initiate and complete. An administrator creates the workflow process definition and stores it on the CA Process Automation server.

From CA SDM, you attach a workflow process definition to a ticket area or category. Later, when a user creates a ticket with the area or category, the system automatically launches a workflow process instance. The Workflow Tasks tab tracks the workflow progress to completion.

NOTE

You can attach this workflow for each ticket type (request or problem or incident).

Follow these steps:

1. Click Use ITPAM from the Workflow tab.
   The CA Start Request Form is displayed with the following fields:
   - **Name**
     Specifies a name of CA Process Automation workflow process definition.
   - **Path**
     Specifies the location where the CA Process Automation process is stored on the CA Process Automation server. For example: ServiceDesk/Process/.
   - **Description**
     Describes the workflow.

2. Select the process definition.
   The Start Request Form closes and the process definition appears on the Workflow tab.
3. Click Save.
   The system saves the process settings. The next ticket that a user creates in the specified ticket category or area automatically attaches the workflow and creates a process instance. When the ticket area or category has an attached CA Process Automation process definition, the Workflow Tab contains the following fields:
   - **CA IT PAM Start Request Form**
     Searches for CA IT PAM Start Request forms.
   - **CA IT PAM Process Name**
     Specifies a CA Process Automation workflow process name. The process name you specify must match one of the processes that you specified in the Options Manager *caextwf_log_categories* parameter.
   - **CA IT PAM Process Path**
     Specifies the location where the process is stored on the CA Process Automation server. For example: ServiceDesk/Process/.

Attach a Classic Workflow

Classic workflow is provided as a standard CA SDM component. To allow flexibility, you can define workflow tasks as optional, which allows the analyst to delete them from the task list. All mandatory workflow tasks must be complete before the ticket can be closed. The only exception is that if the ticket status is set to Cancelled, the workflow process terminates automatically. If a ticket is removed from a category with an attached workflow, or assigned to a different category, the workflow process terminates automatically. If the new category has an attached workflow, its process starts automatically.

Ensure that you apply the patch before attaching this workflow to the request/incident/problem area. If you have customized the following files, ensure that you save the customization, apply the patch and then reapply the changes to these files:

- detail_pcat.htmpl
- pcat_wftpl_tab.htmpl
- detail_in.htmpl
- detail_cr.htmpl
- detail_pr.htmpl
- detail_pr.htmpl
- detail_pcat.htmpl
- list_wftpl.htmpl
- detail_wftpl.htmpl
- detail_bhvtpl.htmpl
- detail_cr.htmpl
- detail_in.htmpl
- list_ntfl.htmpl
- list_ntfm.htmpl
- detail_act_type_assoc.htmpl
- detail_evt.htmpl
- detail_macro.htmpl
- list_act_type_assoc.htmpl
- detail_ntfm.htmpl
- list_evt.htmpl
- detail_ntfl.htmpl
- list_slatpl.htmpl
- std_head.htmpl
- detail_sdsc.htmpl
- list_macro.htmpl
- xx_wf_tab.htmpl

Follow these steps:

1. Click Add Classic Workflow from the Workflow tab.

   NOTE

   For each ticket type (request or problem or incident), you can attach a classic workflow. For example, select the Request tab and click Add Classic Workflow. Similarly click Incident or Problem tab to attach the classic workflow, if required.

   The Workflow Definition List appears.
2. Complete the following fields as appropriate:
   - **Task**
     (Required) Defines the CA SDM workflow task type. Click this link to select a task from the Task Type List. For more information about task types, see Define Workflow Task Types.
   - **Sequence**
     The order in which this task appears on the ticket's Workflow Tasks tab. The task with the lowest sequence number is listed first. Numbers do not need to be sequential.
   - **Record Status**
     (Required) Identifies whether the database record is active or inactive. Select a value from the drop-down list.
   - **Assignee**
     The name of the person assigned to the ticket. Enter a value directly or click the search icon to search for a contact.
   - **Group**

The group that is responsible for the ticket. Any individual contact assigned to the group can handle the task once it has been assigned to the group. Enter a value directly or click the search icon to search for a group.

- **Deleteable**
  Determines whether the user can delete this task from tickets.
- **Service Type**
  The level of support received by the organization. For example, some users may have contracted for 24-hour support, and others might receive on-site training. Enter the service type directly into the field, or click the search icon to select the desired service type.
- **Estimated Duration**
  Indicates the estimated date when the task should be finished. The date and time display in mm/dd/yyyy hh:mm am | pm format. Enter the date and time directly or click the calendar icon to select a date.
- **Estimated Cost**
  Estimates how much it costs to complete the task. Enter the amount as an integer without commas or periods. It is best to enter the amount in dollars only.

3. Click Accept. Repeat steps 1 and 2 to add more workflows.
   The workflow tasks appear on the Workflow tab.
4. Use the controls on the following tabs to configure the workflow tasks:
   - **Behaviors**
     Specifies behavior macros to perform actions based on conditions associated with the issue workflow status. For details about working with this tab, see Create a Behavior.
   - **Auto Assignment**
     (Optional) Specifies how workflow tasks are automatically assigned to groups. For details about working with this tab, see Enable Workflow Auto Assignment.

Create a Behavior

You can configure behavior macros to perform specified actions automatically, based on conditions associated with the issue workflow status.

Follow these steps:

1. Click Add Behavior.
   The Create New Behavior Template page appears.
2. Select a workflow task status and active or inactive from the drop-down lists and enter a description of what activities the behavior is configured to perform, then click Save.

   NOTE
   You cannot define the actual functionality of the behavior until it has been saved.
3. Click the behavior's workflow status code on the Behavior tab of the Workflow Template page.
   The Behavior Template Detail page appears.
4. Click the To Do tab.
   Any conditions and macros that have been defined for the behavior appear in the Macro List.
5. Click Edit to update the Macro List.
   Any conditions and macros that have been defined for the behavior appear in the Macro List.
6. Click the Condition link.
   Macro List page appears, displaying the available condition macros.
7. Select a macro from the list.
   The macro symbol appears in the Condition field of the To Do tab on the Update Behavior Template page. For more information about macros, see create a macro from the How to Configure SLAs topic.
8. Click Update Actions On True.
   The Macro List page appears.
9. Select Action in the Type drop-down list and click Search.

The Actions On True Update page appears.

10. Select the actions you want to apply when the condition is true and click the arrow button to move them to the selected list, then click OK.
    The Update Behavior Template page appears, displaying your action selections.
11. Click Update Actions On False.
    The Macro List page appears.
12. Select Action in the Type drop-down list and click Search.
    The Actions On False Update page appears.
13. Select the actions you want to apply when the condition is false and click the arrow button to move them to the selected list, then click OK.
    The Update Behavior Template page appears, displaying your action selections.

Enable Auto Assignment

You can specify how CA SDM automatically assigns tickets in an area. You can select location-based auto assignment or CI-based auto assignment.

> **NOTE**
>
> The Options Manager autoasg_override option controls the circumstances under which auto assignment takes place. If the value of this option is set to 1 (default), CA SDM ignores any existing assignee and group settings and auto assigns tickets in all cases. If you want CA SDM to auto assign tickets only if they are not already assigned, set the option value to 0.

Follow these steps:

Complete the following fields from the Auto Assignment tab and click Save.

- **Auto Assignment Mode**
  Specifies how auto-assignment occurs. You use the Configuration Item Based option to base the auto assignment on the CI Assignable Attribute value.
  - **Disabled**
    Bases the auto-assignment on the Area Defaults option when it is installed.
  - **Configuration Item Based**
    Bases the auto-assignment on the CI Assignable Attribute value.
  - **Location Based**
    Bases the auto-assignment on the location value.
    The automatic assignment process begins with the workshifts defined for the area, proceeds next to groups, and finally to locations. If at any point relationships are defined that fail to satisfy the requirements of the auto assignment process, the process attempts to assign the Default Assignee and Default Group of the area. If no defaults have been defined, the ticket assignment is not changed.

    > **NOTE**
    > For details about the auto assignment processing logic, see the Auto Assignment topic.
- **Assignable CI Attribute**
  Specifies the configuration item attribute to use for the group assignment. You can enter a value directly or click the magnifier to search for an attribute.

Assign Groups

To configure auto assignment for a request/incident/problem area, you must at a minimum define the relationships between analyst groups and the area.

Assignees are chosen from groups that meet all of your specified auto assignment criteria. If no additional constraints are defined, tickets are auto assigned to the group member with the fewest active tickets.

If no groups are associated with the area, the default group and assignee are assigned. If these defaults are not defined, the ticket is left for manual assignment.

Follow these steps:

1. On the Auto Assignment tab of one of the pages, click Update Groups.
   – Change Orders Detail
   – Issues Detail
   – Requests/Incidents/Problems, Areas Detail

   The Group Search window opens.
2. To filter the search, complete one or more of the search fields, or leave all fields blank to search for all groups, then click Search.
   The Groups Assigned Update window opens.
3. Use the arrow buttons to move available groups to or from the Groups Assigned list.
4. Click OK.

Assign Locations

If you assign a location to a request/incident/problem area, the tickets in that area are auto assigned only if a matching location is found. For example, a request ticket is auto assigned if there is an eligible analyst at:

- The affected asset's location
- The affected customer's location

If the affected asset or customer has no specified location, the request is assigned to the default group and assignee. If these defaults are not defined, the request is left for manual assignment.

Follow these steps:

1. On the Auto Assignment tab of one of the following pages, click Update Locations.
   – Change Orders Detail
   – Issues Detail
   – Requests/Incidents/Problems, Areas Detail

   The Location Search window opens.
2. To filter the search, complete one or more of the search fields, or leave all fields blank to search for all locations, then click Search.
   The Locations Assigned Update window opens.
3. Use the arrow buttons to move available locations to or from the Locations Assigned list.
4. Click OK.

Assign Workshifts

You can constrain auto assignment by relating the request/incident/problem area to a workshift. The workshift determines the timeframe within which tickets are eligible for auto assignment. Tickets opened outside the hours of the workshift are assigned to the default group and analyst or left for manual assignment.

Follow these steps:

1. On the Auto Assignment tab of one of the following pages, click Update Workshifts.
   – Change Orders Detail
   – Issues Detail
   – Requests/Incidents/Problems, Areas Detail

   The Workshift Search window opens.
2. To filter the search, complete one or more of the search fields, or leave all fields blank to search for all workshifts, then click Search.
   The Workshifts Assigned Update window opens.

3. Use the arrow buttons to move available workshifts to or from the Workshifts Assigned list.
4. Click OK.

Define Issue Categories

Issues can be assigned to categories so that default values defined for a category automatically appear in fields on new issues. Defining categories lets you quickly create issues.

NOTE

If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

To define an issue category

1. On the Administration tab, browse to CA SDM, Issues, Categories.
   The Issue Category List appears.
2. Click Create New.
   The Create New Issue Category page appears.
3. Complete the fields as appropriate.
4. (Optional) Use the controls available on the tabs at the bottom of the page to configure the following features:
   Properties
   Defines custom properties to apply to issues assigned to this category.
   Workflow
   Attaches a workflow to specify a standardized sequence of tasks to resolve issues assigned to this category.
   Auto Assignment
   Specifies how issues in this category are auto assigned.
5. Click Save.
   The Issue Category Detail page appears.
6. Click Close Window.
   The new issue category appears in the Issue Category List when you refresh the list.

Enable Auto Assignment

You can specify how CA SDM automatically assigns tickets in an area. You can base auto assignment on groups, locations, and workshifts.

The automatic assignment process begins with the workshifts defined for the issue category, proceeds next to groups, and finally to locations. If at any point relationships are defined that fail to satisfy the requirements of the auto assignment process, the process attempts to assign the Default Assignee and Default Group of the category. If no defaults have been defined, the ticket assignment is not changed.

NOTE

For details about the auto assignment processing logic, see the Administrator Guide.

NOTE

The Options Manager autoasg_override option controls the circumstances under which auto assignment takes place. If the value of this option is set to 1 (default), CA SDM ignores any existing assignee and group settings and auto assigns tickets in all cases. If you want CA SDM to auto assign tickets only if they are not already assigned, set the option value to 0.

To enable auto assignment

1. On the Administration tab, browse to Service Desk, Issues, Categories.
   The Issue Category List appears.
2. Select the category to edit.
   The Category Detail page appears.

3. Click Edit.
   The Update Issue page appears.
4. Select the Auto Assignment tab, and select the Auto Assignment Enabled check box.
5. Click Save.
   Auto assignment is enabled.
6. Click Save.
   Auto assignment is enabled.

Define Behavior Template

A behavior template defines an action when a task changes status. For example, an email notification can be sent to a manager when the status of an approval task changes to Pending.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

The Create New Behavior Template page contains the following fields:

- **Status**
  Specifies the status of the template, such as Approve or Pending.
- **Active**
  Specifies if the template is active or inactive.
- **Description**
  Specifies a description of the template.

This page also contains the following tabs:

- **To Do**
  Identifies selected macros to run when the status change identified in the Status field takes place.
- **Transition Information**
  Identifies conditions that prevent the task from reaching the specified status, prior to the macros on the To Do tab run.

> **NOTE**
> All behaviors defined for a specific issue or change category are listed on the Workflow Task Template Detail window for that category and become part of any issue or change order that is assigned to that category.

Install Incident Tracking

This article contains the following topics:

Incident tracking lets analysts track an incident by selecting one or more flags for the incident. The information that analysts specify provides your organization with metrics about incidents for reports. For example, analysts can indicate that an incident was assigned incorrectly. When a large percentage of incorrectly assigned tickets appears in a report, your organization is aware that assignments must be adjusted.

For example, analysts can specify information to help your organization do the following:

- Improve SLA responsiveness and closure at lower levels within the support organization.
- Identify tickets that are incorrectly assigned.
- Indicate that a remote control tool was used to resolve the ticket.

You install the efficiency_tracking Options Manager option, so that analysts can use tracking options that appear on the Efficiency Tracking tab of incident detail pages.

Install Incident Tracking

You can install an option to let analysts track an incident by setting particular flags for the incident. The information that analysts specify provides your organization with metrics about incidents for reports. After you install the option, the tracking flags appear on the Efficiency Tracking tab of the Incident Detail pages.

Follow these steps:

1. On the Administration tab, browse to Options Manager, Request Mgr.
   The Options List appears.
2. Click efficiency_tracking.
   The efficiency_tracking Options Detail page appears with default values set.
3. Click Edit.
   The efficiency_tracking Options Update page appears with default values set, and you can edit the Description.
4. Click Install.
   The efficiency_tracking Options Detail page appears.
5. Click Close Window.
   Incident tracking is installed; however, the Efficiency Tracking tab does not appear on incident detail pages.
6. Restart CA SDM.
   The Efficiency Tracking tab appears on incident detail pages.

Related Ticket Activities

This article contains the following topics:

When an activity is generated for a CA SDM ticket, you can propagate the activity to one or more related tickets. For example, a Problem record created from an Incident can update the Incident record when the Problem is resolved. When the activity occurs, an activity log is generated for the related ticket that includes the following information:

- Related ticket activity type, for example, Update Status
- Contact name
- Parent ticket type and its reference number
- Activity log description, for example, status updated from Work in Progress to Open

Activity logs are propagated to related tickets based on the properties set within each activity notification. The attributes of the related tickets are not modified. The following relationships are propagated:

Problems propagate to all active Incidents.

- Change Orders propagate to all active Problems and Incidents.

As a system administrator, you can perform the following actions:

- Set up activity notifications to propagate related ticket activities.
- Configure a Related Ticket activity notification to notify the appropriate contacts when the activity is propagated to related tickets.

How to Define Activity Notifications for Related Tickets

You can propagate activity logs to related tickets based on the properties set in each activity notification. The attributes of the related tickets are not modified.

To define an activity notification for related ticket activities, do the following:

1. On the Administration tab, select Notifications, Activity Notifications.
2. Open the appropriate activity notification for editing.
3. On the detail page, click the Related Ticket Activity check box to mark it as active.

4. If you have multi-tenancy, perform the following:
   – Specify the appropriate tenant type from the Related Ticket Activity drop-down list.
   – Enter the name of the tenant in the tenant field, or click the search icon to search for a tenant.
5. (Optional) Update the default Notification Rule and specify contacts to receive notification.
6. Click Save, Close Window.
   The updated activity notification appears in the Activity Notification List when you redisplay the list.

How to Define Related Ticket Activity Notifications

You can change the default settings in the Related Ticket activity notification to notify the appropriate contacts when the activity logs are propagated to related tickets (requests/incidents/problems, change orders, and issues).

To define a related ticket activity notification, do the following:

1. Open the Related Tickets activity notification on the Activity Notification List page.
2. Click Edit and change one or more of the fields as appropriate on the detail page.
3. (Optional) Update the default Notification Rule for and specify contacts to receive notification.
4. Click Save, Close Window.
   The updated Related Ticket activity notification appears in the Activity Notification List when you redisplay the list.

Audit Log Use

CA SDM creates an audit log that records the following information:

- All changes to an issue (Issue)
- All changes to a request (Call_Req)
- All changes to a change order (Change_Request)
- All changes to a data partition (Domain) tables
- The login ID of the person associated with the change and an associated day/date/time stamp
- The before and after values of the update, insert, or delete
- The creation of contacts
- Creation and update activities to the nr object

> **NOTE**
> The audit log captures modifications to only the contact data partition field, but no other contact record updates.

The audit log feature is automatically installed, and you can enable it by installing two Audit Log options with the Options Manager: audit_ins and audit_upd. After you have installed these options, you can access the audit log on the Administration Tab by selecting Service Desk, Audit Log List. You can search the audit log list with a built-in search tool, and use it to facilitate report generation.

Priority Calculation

This article contains the following topics:

Priority calculation is a predefined set of values that automatically set Priority, Urgency, and Impact fields on problems and incidents. Priority calculation helps you manage incidents and problems for your business needs and IT capabilities. ITIL recommends that you prioritize tickets by using a data calculation that is based on Urgency and Impact values. Support organizations define this calculation based on their unique processes, how this calculation determines Service Level Agreements (SLAs), and other key events in the system. This calculation can also include the criticality of the CI that is linked to the incident and problem. Prioritizing tickets effectively helps you accomplish the following:

- Allocate IT resources for tickets
- Better serve customers
- Reduce costs

The CA SDM solution for Priority Calculation includes the following components:

- A Priority Calculation Matrix based on the Urgency and Impact values
- Default values for Urgency and Impact
- Urgency and Impact adjustment options based on Affected Service, Open Date, Affected End User, and Incident or Problem Area
- A Capture Reason option for manually modifying Urgency or Impact
- An Enable for Templates option for creating a ticket from a Template

When you install CA SDM, a Default priority calculation automatically manages ticket values. You can modify the Default priority calculation settings, or create additional priority calculations to manage incidents or problems. In the priority calculation, you define the outcome based on business scenarios to make the level of importance and ticket handling more consistent. Users can override some settings, but they cannot set the Priority on the ticket because this value is data-driven. For multi-tenancy, you or the tenants can create additional priority calculations with specific settings for each tenant.

When an analyst opens an incident or problem, the system automatically uses an active priority calculation and ticket values to generate Priority, Urgency, and Impact settings. The settings are based on one or more of the following fields:

- Urgency
- Impact
- Affected End User
- Incident or problem Area
- Open Date
- Affected Service

Analysts can override Urgency and Impact values as necessary. Depending on how you configure Options Manager, employees can only override incident Urgency values when the *urgency_on_employee* option is installed. When the Capture Reason flag is enabled and users override Urgency or Impact values and click Save, the Escalate Detail page appears to let users describe a reason for the change.

All ticket priority calculations, manual overrides, and reason information appear in the New Activity Log. If no priority calculation adjustments occur, the system does not create an activity log entry.

> **NOTE**
> If you migrated from a previous version, priority calculation is disabled by default.

Priorities

Priority codes indicate the amount of attention a ticket should receive. Priority codes are used when defining automatic notification and external processing for specific types of situations. They are also used as a label in the scoreboard to prioritize tickets.

The predefined priority codes are:

1 = High priority

2 = Medium-high priority

3 = Medium priority

4 = Medium-low priority

5 = Low priority

None = No priority is assigned.

You can modify a priority code's symbol. For example, your site might decide not to use numbers at all and assign priority codes of low, medium, high, and so on.

You can modify a priority code's associated service type. For example, you could change the service type for priority code 1 from the default value of 04hr resolution to a custom service type of 02hr resolution.

NOTE
You can edit the predefined priority codes, but you cannot delete them or create new ones.

How Priority Calculation Manages Ticket Values

The system adjusts problem and incident values based on active priority calculation settings to assist Analysts in handling tickets more effectively. The following table summarizes how priority calculation changes fields based on the priority calculation and user actions for problems, incidents, web service, email, and the Text API:

Action	Automatic Field Changes	Description
User changes Affected Service	Impact Priority	The system evaluates the Service Impact value on the CI of type Service to calculate the new Impact value. The CIs of type Service are defined as CIs with their class defined in the Enterprise Service family. If the open date of the ticket is within the blackout window time frame, the system increments a new Impact value based on the Impact Increment field. The system only replaces the Impact value when the new value is greater than the initial Impact value.
User changes Incident Area	Urgency	The Urgency value changes only when the new value is greater than the default value.
User changes Incident Area *and* Affected End User	Urgency Priority	If the user sets the Incident Area field first, the Urgency value changes after the user sets the Affected End User. The priority calculation sets the Priority.

User changes Urgency *and* Impact	Priority Impact	The system evaluates the Service Impact value on the CI of type Service to calculate the new Impact value. If the open date of the ticket is within a time defined by a blackout window, the system increments a new Impact value based on the Impact Increment field. The system only replaces the Impact value when the new value is greater than the initial Impact value. If the Administrator sets Capture Reason, the user must provide a reason for the modification. If the user changes the Urgency or Impact values, these values remain the same throughout entire ticket creation or ticket update unless the user modifies them again. However, the system can update the overridden values for Urgency or Impact the next time the user updates the ticket. After the system adjusts Urgency and Impact, the priority calculation sets the Priority value.
User selects New Incident based on the Knowledge Document and system has overrides for Knowledge Documents	Impact Urgency	The system always uses the Knowledge Document or knowledge solution values irrespective of whether the values are greater than or lesser than the default priority calculation values. For example, if a priority calculation has an Impact value of 3-Single Group and Urgency value of 3-Quickly, and Knowledge Documents have an Impact value of 2-Multiple Groups and Urgency value of 4-Very Quickly, the system applies the values from the Knowledge Document to the incident. The priority value always derives from the priority calculation.
User accepts Knowledge Document as solution to problem or incident	Impact Urgency	The system uses the values from the Knowledge Document for Impact and Urgency. The system also uses the priority calculation to set the Priority value.
User derives incident from Knowledge Document by selecting New Incident	Impact Urgency Priority	The system uses the priority calculation values.
User derives incident from Knowledge Document without system overrides for Knowledge documents	Impact Urgency Priority	The system uses the priority calculation values regardless of how the user created the incident for the Knowledge Document or knowledge solution.

Ticket accepts Knowledge Document as solution to problem or incident and the system does not override for knowledge documents	Impact Urgency Priority	The system uses values from problem or incident. The Priority value originates from the priority calculation.
Ticket that is in Read Only Mode accepts Knowledge Document as solution to the problem or incident and system overrides for Knowledge Documents	Impact Urgency Priority	The system uses the Impact and Urgency values from the Knowledge Document when the values are not empty. If the Impact/Urgency value in the Knowledge Document is empty the system uses values from the problem or incident. The Priority value originates from the priority calculation.
Ticket that is in Edit Mode accepts Knowledge Document as solution to the problem or incident and system overrides for knowledge documents.	Impact Urgency Priority	The system uses values from the problem or incident. The Priority value originates from the priority calculation.

Priority Calculation Generates Urgency Value After Saving Self-Service Tickets

By design, priority calculation generates Urgency values only after self-service users save incidents. Self-service users, such as VIP employees, employees, and anonymous users, can view the generated value after saving a ticket.

For self-service users, priority calculation uses the following settings and values to generate Urgency values:

- Urgency_On_Employee is set to Yes in Options Manager
- Override Urgency value is enabled in the active priority calculation for incidents
- *Web.cfg* Urgency settings such as AnonymousUrg for anonymous users, ESCEmpUrg for VIP employees, and EmpUrg for all other employees
- Area Urgency values
- Manual user overrides

The following table summarizes how priority calculation sets the Urgency value for self-service incidents:

Self-Service User Action	Urgency Value
The user saves an incident with the default Urgency and an empty Incident Area.	The ticket shows the default Urgency value from the *web.cfg*.
The user saves an incident after overriding the Urgency value.	Regardless of the Area Urgency, *web.cfg*, or priority calculation settings, the ticket shows the Urgency value that the user selected.
The user saves an incident after selecting an Incident Area. The Incident Area does not have a predefined Area Urgency value.	The ticket shows the default Urgency value from the *web.cfg*.
The user saves an incident after selecting an Incident Area that has a predefined Area Urgency value. The Override Urgency option is also enabled on the active priority calculation for incidents.	If the Area Urgency value is greater than the Urgency in the *web.cfg*, the ticket shows the Area Urgency value. However, the updated Urgency field is not visible while the user is creating or editing the ticket. When the user saves and reopens the incident, the updated Urgency value appears on the incident.
The user saves an incident after selecting an Incident Area that has a predefined Area Urgency value. However, the Override Urgency option is disabled on the active priority calculation for incidents.	The ticket shows an Urgency value from the *web.cfg*.

The user edits an existing incident that has an Incident Area with a predefined Area Urgency value.	The Urgency drop-down list shows the Area Urgency value and all applicable *web.cfg* values.

How to Set Priority Calculation

This article contains the following topics:

By default, ticket values such as priority are based on a priority calculation. You can find and adjust the initial values for Priority and Urgency in the web.cfg. The web.cfg has separate settings for various users such as guest, VIP-user, and employee.

> **NOTE**
> If you migrated from a previous version, priority calculation is disabled by default. Modified Incident and Problem Detail pages require additional configuration to operate properly.

To set priority calculation, do the following:

1. On the Administration tab, select Service Desk, Request/Incidents/Problems, Priority Calculation.
   The Priority Calculation List appears.
2. Right-click the default priority calculation and select Edit from the short-cut menu.
   The default Priority Calculation Detail page shows default settings for incident and problem tickets.
3. Review default priority calculation and adjust the values accordingly. When you set priority calculation values, consider the following issues for your working environment:
   - **Issue escalation** -- When tickets require escalation to a particular VIP, you can increase the value for Urgency.
   - **Critical CIs** -- For critical CIs, you can configure Service Impact for each CI.
   - **Critical Service Uptime** -- When CIs require high availability, add a blackout window.
   - **Blackout window** -- When CI-related tickets use a particular blackout window, you can increase the Service Impact value on the priority calculation.
4. Use the Manual Override setting to allow users to change tickets settings as necessary.
5. If you want a separate priority calculation to manage, problems or incidents, configure the ticket type.
6. Click Save.
   On the next new or updated ticket or Knowledge Document, the fields update according to the values in the active priority calculation.
7. Consider creating additional priority calculations for each ticket type. For multi-tenancy, create and activate additional priority calculations to manage tickets for each tenant.

Define a Priority Calculation

Priority calculation is a predefined set of values that automatically set Priority, Urgency, and Impact values on problems and incident tickets.

> **NOTE**
> If you migrated from a previous release, the priority calculation values must be enabled after the migration.

You can define a priority calculation to handle, incidents, problems, or both. However because two priority calculations cannot handle the same ticket type, you can review and update existing priority calculations before you create ones. For example, if you want to add a priority calculation to manage problem tickets, first clear the Problem field on the active priority calculation. Then, create a priority calculation to manage problem tickets.

If multi-tenancy is installed, you can use one public priority calculation that applies to all tenants or several tenant-specific priority calculations.

Follow these steps:

1. On the Administration tab, select Service Desk, Request/Incidents/Problems, Priority Calculation.

The Priority Calculation List page appears.

2. Click Create New.
   The Create Priority Calculation page appears.
3. Complete the following fields as appropriate:
   - **Name**
     Identifies the priority calculation.
   - **Status**
     Specifies the ticket status for this priority calculation.
   - **Incidents**
     Specifies whether this priority calculation applies to incident ticket types.
   - **Problems**
     Specifies whether this priority calculation applies to problem ticket types.
   - **Spelling**
     Verifies the spelling of text in the Description field.
   - **Description**
     Specifies the purpose of the priority calculation.
4. Complete the priority calculation by setting the Priority value based on Impact and Urgency. For example, if the Urgency is 5-Immediate and the Impact affects the 1-Entire Organization, you can select the Priority value of 1 from the drop-down list.
5. Set the following Priority Calculation Options as appropriate:
   - **Impact Default**
     Specifies the initial Impact value to display when the user creates a ticket.
   - **Override Impact**
     Works with the Impact Increment field. Specifies whether the user can override the ticket Impact value with the Impact level specified in the attached Affected Service.
   - **Impact Increment**
     Works with the Override Impact field. Increments the Impact by the specified amount when the user overrides the Impact value. When the ticket uses a blackout window, the Impact value increments only when the open date is within the blackout window time frame. The Impact value can only raise to a maximum of 1-High. For example, if you set the value to 2 and the default Impact is 3-Medium, the user can only override the Impact on a ticket to 1-High.
   - **Urgency Default**
     Specifies the Urgency value to display when the user creates a ticket.
   - **Override Urgency**
     Specifies whether the user can override the ticket Urgency value with the Area Urgency level specified in Incident or Problem area.
   - **Urgency Increment**
     Works with the Override Urgency field. Increments the Urgency level by the specified amount when the affected end user sets the Escalate Urgency flag.
   - **Capture Reason**
     Specifies whether the user is required to enter the reason for changing the Urgency or Impact values. If the user changes the Urgency or Impact values, the Escalate Detail page appears while saving the ticket. The user is required to describe the reason for changing the Urgency or Impact values.
   - **Enable for Templates**
     Specifies whether to use the priority calculation for tickets that use templates.
6. Click Reset Matrix if necessary to restore the settings to the defaults.
   The default values reset.
7. Click Save.

The Update Priority Detail page appears. On the next new or updated ticket or Knowledge Document, the fields update according to the values in the active priority calculation. If no priority calculation is active for a ticket type, the system clears the Priority, Urgency, and Impact fields.

Use Ticket Templates to Calculate Priority Values

If you want ticket templates to calculate priority, you configure the priority calculation with the *Enable for Templates* option. If you enable this option the Urgency and Impact values come from the template, but the priority field comes from priority calculation record. The priority value displays as read-only.

If you do not enable this option, the Urgency, Impact and Priority fields come from the template. You can edit the priority field and no priority calculation is done for the ticket until it is saved.

To use ticket templates to calculate priority values

1. On the Administration tab, select Service Desk, Requests/Incidents/Problems, Priority Calculation.
   The Priority Calculation List appears.
2. Select the priority calculation that you want to use for calculating priority with templates.
   You can also create a priority calculation to use ticket templates to calculate priority values.
3. Select *Enable for Templates* from the Priority Calculation Options list.
4. Click Save.
   The option is enabled.

Multiple Priority Calculations

You can set up more than one priority calculation. However, only one active priority calculation handles problems, or incidents, or both. For example, you can have an active priority calculation for problems, and another for incidents. You can also save several inactive priority calculations for future use.

Priority Calculation Assignment for Multi-Tenancy

You or your tenants can create tenant-specific priority calculations to manage incidents and problems. When you assign priority calculations for multi-tenancy, consider the following:

- When a priority calculation has no assigned tenant, it is considered public. The status of a public priority calculation is either Active or Inactive. A priority calculation is no longer considered public when it is assigned to a tenant.
- If a tenant has no priority calculation assignment, the Default priority calculation or another active public priority calculation automatically manages problems and incidents.
- A priority calculation manages problems and incidents for one tenant. However, a separate tenant-specific priority calculation can handle each ticket type. For example, Company X has one priority calculation to handle incidents and another to manage problems.
- When tenants create their own priority calculations while public priority calculations are active, the tenant-specific priority calculation applies only to tickets for the respective tenant.
  For example, if the Default priority calculation is active, Company X tenant can create a tenant-specific priority calculation named new_priority_calculation. The new_priority_calculation settings and configurations apply only to Company X incidents and problems.
- If the tenant inactivates a priority calculation, the system uses an active public priority calculation to manage tenant problems and incidents.
  For example, Company X inactivates the tenant-specific priority calculation while there is still an active Default priority calculation. Priority calculation remains enabled for Company X because the system uses the Default priority calculation to manage incidents and problems for Company X.

NOTE
Because tenants can delete their own priority calculation records, we recommend that you inactivate the public priority calculations that manage incidents and problems. Instead, you or the tenants can create tenant-specific priority calculations.

- When you disable multi-tenancy and there is more than one active priority calculation that manages tenants, leave *only one* priority calculation to manage incidents and problems. For example, you can inactivate all priority calculations except one to manage incidents and another to handle problems.

How to Assign a Tenant to a Priority Calculation

For multi-tenancy, you can assign a tenant to a priority calculation. First, you inactivate public priority calculations. Then, you assign the tenant to a priority calculation and activate it.

To assign a tenant to a priority calculation, do the following:

1. On the Administration tab, select Service Desk, Request/Incidents/Problems, Priority Calculation.
   The Priority Calculation List appears.
2. Edit each public priority calculation, such as Default. Set the status to Inactive and click Save.
   The system disables the public priority calculations.
3. For each tenant, create or edit a priority calculation with tenant-specific settings for Impact, Urgency, and Priority.
   The Create Priority Calculation or Update Priority Calculation page appears.
4. In the Name field, specify the tenant.
5. In the Status field, select Active.
6. Click Save.
   The system applies tenant-specific values for Impact, Urgency, and Priority on new incidents and problems.

Ticket Type Considerations for Priority Calculation

When you configure ticket types for a priority calculation, consider the following:

- The default priority calculation lets you manage both incident and problem ticket types.
- If you are migrating from a previous release, you enable the default priority calculation or create a priority calculation to manage problems and incidents.
- Although you can have many priority calculations, only one active priority calculation can handle a particular ticket type. For example, one active priority calculation can manage problems and another can manage incidents.
- If you want to create a priority calculation and an active priority calculation already handles a particular ticket type, you first disable the ticket type on the active priority calculation. For example, if you want a priority calculation to manage problems, you disable the problem ticket type on the active priority calculation and create an active priority calculation that manages problems.

Configure Ticket Types for a Priority Calculation

You can specify the ticket types that the priority calculation manages. When the priority calculation is active, it manages Priority, Impact, and Urgency values on new tickets.

To configure ticket types for a priority calculation

1. On the Administration tab, select Service Desk, Request/Incidents/Problems, Priority Calculation.
   The Priority Calculation List appears.
2. Right-click a priority calculation and select Edit.
   The Update Priority Calculation page appears.
3. Select or clear one or more of the following options:
   - **Incidents**
     Enables or disables the priority calculation to manage new incidents.
   - **Problems**

Enables or disables this priority calculation to manage new problem tickets.

4. Click Save.
   CA SDM uses the settings in the priority calculation to manage ticket values for new incidents, problems or both.

Use Ticket Templates to Calculate Priority Values

If you want ticket templates to calculate priority, you configure the priority calculation with the *Enable for Templates* option. If you enable this option the Urgency and Impact values come from the template, but the priority field comes from priority calculation record. The priority value displays as read-only.

If you do not enable this option, the Urgency, Impact and Priority fields come from the template. You can edit the priority field and no priority calculation is done for the ticket until it is saved.

To use ticket templates to calculate priority values

1. On the Administration tab, select Service Desk, Requests/Incidents/Problems, Priority Calculation.
   The Priority Calculation List appears.
2. Select the priority calculation that you want to use for calculating priority with templates.
   You can also create a priority calculation to use ticket templates to calculate priority values.
3. Select *Enable for Templates* from the Priority Calculation Options list.
4. Click Save.
   The option is enabled.

Use Knowledge Documents to Calculate Priority Values

If you want knowledge documents to calculate priority, you can update the field mapping. After you configure the field mapping, the analyst can create incident or problem tickets from knowledge documents. The knowledge document calculates Impact and Urgency values on the tickets. If the Impact or Urgency value is missing, the values originate from the priority calculation.

> **WARNING**
> If you modify and save a ticket that already contains Impact and Urgency values calculated by a knowledge document, the priority calculation overrides the values set by Knowledge Management. The audit log displays these activities.

To use knowledge documents to calculate priority values

1. On the Administration tab, select Knowledge, Service Desk Integration.
2. Select Field Mapping.
   The Field Mapping page appears.
3. For Impact and Urgency, select the following check boxes as appropriate:
   - **Populate Empty Service Desk values**
     Specifies whether to use information from Knowledge Management to populate fields in issues or requests.
   - **Overwrite Service Desk values**
     Identifies the fields in issues or requests that correspond to fields listed in the Knowledge Management column.

     > **NOTE**
     > When the Override Service Desk values field is not enabled but Populate Empty Service Desk values field is enabled for Impact and Urgency, the knowledge values for Impact and Urgency override the Incident values.
4. Click Save.
   Incidents and problems are created, using the Impact and Urgency values from the knowledge document to calculate the Priority value. If the values are missing, the ticket obtains the values from an active priority calculation. If no priority calculation is active for the ticket type, the system clears the Priority, Urgency, and Impact fields.

Manually Override the Impact Value

When you manually override the Impact value on a problem or incident, the active priority calculation that manages the ticket type automatically adjusts the Priority value.

To manually override the Impact value

1. Open the details page for the problem or incident you want to change.
2. Change the Impact value.
   If there is an active priority calculation that manages the ticket type, the Priority value automatically changes based on the settings in the priority calculation.
3. Save the incident.
   The Activity Log on the Incident Detail page reflects the Impact values changes.

Example: Manually override the Impact value on a new incident

1. Create an incident.
   By default, the Urgency value is 3-Quickly. The Impact value is 3-Single Group. The Priority value is 3.
2. Override the Impact value to 1-Entire Organization.
   The Priority value automatically changes based on the values in the active priority calculation that manages incidents.
3. Save the incident.
   The Activity Log on the Incident Detail page reflects the Impact value changes.

Manually Override the Urgency Value

When you manually override the Urgency value on a ticket, the active priority calculation that manages the ticket type automatically adjusts the Priority value.

To manually override the Urgency value

1. Open the details page for the incident you want to change.
2. Change the Urgency value.
   If there is an active priority calculation that manages the ticket type, the Priority value automatically changes based on the settings in the priority calculation.
3. Save the incident.
   The Activity Log on the Incident Detail page reflects the Urgency values changes.

Example: Manually Override the Urgency Value on a New Incident

1. Create an incident.
   By default, the Urgency value is 3-Quickly. The Impact value is 3-Single Group. The Priority value is 3.
2. Override the Urgency value to 5-Immediate.
   The Priority value automatically changes based on the values in the active priority calculation that manages incidents.
3. Save the incident.
   The Activity Log on the Incident Detail page reflects the change in the Urgency value.

Automatically Adjust Impact for a Problem or Incident

For Configuration Items defined with a family of Enterprise Service, you can automatically adjust the Impact value for problems or incidents. When you select the Problem or Incident Area and select an Affected Service, the impact adjusts according to the CI Service Impact settings for Enterprise Service CI's and the priority calculation.

To automatically adjust Impact for a Problem of Incident

1. Create a problem or incident for an Enterprise Service type CI.
2. Select an Affected Service.
3. Select a Problem or Incident Area.

If there is an active priority calculation that manages the ticket type, the Impact value changes based on the Increment Impact value (used for Blackout Window impact assessment) in the priority calculation and the Service Impact value from the affected service.
If you are using the default priority calculation, with a Service Impact for the Enterprise Service CI set to 1-Entire Organization, and the Problem or Incident is not opened within a Blackout Window the Impact value in the Problem or Incident is set to 1 and the Priority value on the ticket raises to a 2.

4. Save the ticket.
   The Activity Log on the Incident Detail page reflects the Impact value's changes.

Adjust Impact for a Problem or Incident Example

The following example shows you how to adjust impact for a Problem or Incident.

1. Create an Enterprise Service CI named CI-APC and set the class as one that comes under the family Enterprise Service.
   For example, you can set the class as Other Service, Business Services, or Infrastructure Service.
2. In the Service tab within the CI Detail page, set the Service Impact field to 2-Multiple Groups.
3. On the Service Desk tab, create an incident and set the Affected Service to CI-APC.
4. Save the ticket.
   The Impact field in the incident reflects the value from the Service Impact of the selected affected service (CI-APC). In this case, the Impact value is set to 2 -Multiple Groups.
   Note: If you are using the default priority calculation and the current ticket is created during a Blackout Window period then the Impact value increments by 1 and in the case above the Impact value is then set to 1-Entire organization.

Automatically Adjust Urgency for a Problem or Incident

For problems and incidents, you can automatically adjust Urgency and Priority by specifying an affected end user that requires Special Handling or by specifying a Problem/Incident Area that has an Area Urgency value.

When you assign special handling, with the Escalate Urgency turned on, for a contact or set an Area Urgency value for a Problem/Incident area, the Urgency value within the Problem/Incident automatically adjusts according to the values in the priority calculation and the Area Urgency value for the affected end user.

Follow these steps:

1. Create a problem or incident.
2. Select an Affected End user. For an elevated urgency, select an Affected End user that requires Special Handling that has the Escalate Urgency on.
   If there is an active priority calculation that manages the ticket type, the Urgency value changes based on the Urgency Increment value in the active priority calculation.
3. Select a Problem or Incident Area.
   If there is an active priority calculation that manages the ticket type, the Urgency value changes based on the Area Urgency value in the Problem/Incident Area definition.
4. Save the ticket.
   A confirmation message reminds you that the ticket requires special handling. The Activity Log on the Problem/Incident Detail page reflects the changes of the Urgency value.

Adjust Urgency for a Problem or Incident Example

The following example shows you how to adjust urgency for a Problem or Incident:

1. On the Administrator tab, create a contact named Non-VIP.
2. Create a special handling contact named VIP and set the Escalate Urgency value on.
3. Create an area named Test Area and specify Area Urgency as 2-Very Quickly.
4. On the Service Desk tab, create an incident.
5. For the Affected End User, select Non-VIP.

6. In the Incident Area, select Test Area and save the ticket.
   The Urgency field reflects the Area Urgency value from the Incident Area definition. In this case, the Urgency is set to 2-Very Quickly.
7. Change the Affected End User to VIP and save the ticket.
   If the default Priority Calculation matrix is being used, the Urgency value is incremented by 1 and is set to 1-Immediate. A confirmation message appears that reminds you that the ticket requires special handling. The Activity Log on the Incident Detail page reflects the Urgency value's changes.

How to Set Up the Attachments Library

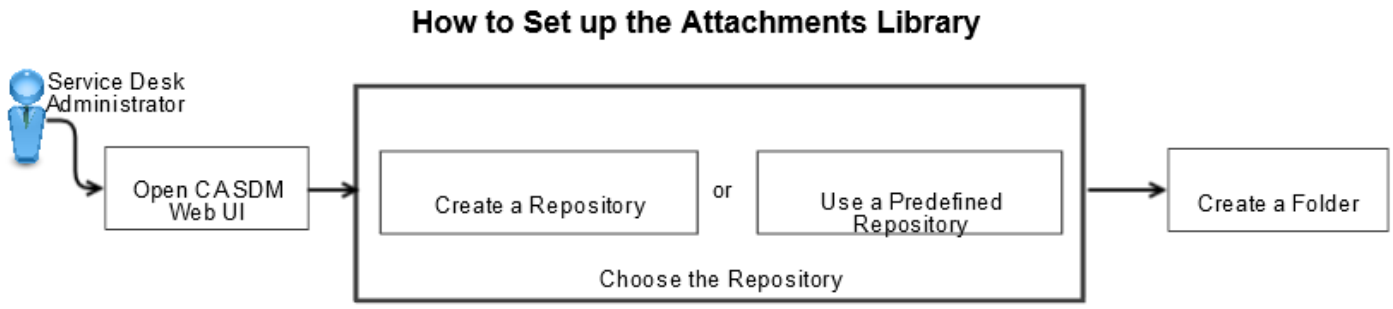
You can add different types of attachments to the various CA SDM entities. For example, a customer can attach a snapshot of an error to the incident. As an administrator, you set up the attachments library from where users can upload or download attachments.

Attachments can be classified, as follows:

- **Stored**: The web server uses the HTTP protocol to upload and store the stored attachments in a repository. When an analyst reviews a stored attachment, the file is retrieved from the repository and the file is displayed locally. Utilizing a web server for file storage allows storage and retrieval from the user interface.
- **Linked**: Stores only a link to the file in the database.

The following diagram shows how to set up an attachments library:

Figure 13: How to Set up the Attachments Library



Follow these steps:

Open CA SDM Web UI

Log in to the web UI from the following servers, depending on your CA SDM configuration:

- Conventional: Primary or secondary servers
- Advanced availability: Application or background servers

Choose the Repository

The attachments are stored in repositories and you required to set up the repositories before users can work with attachments. You can add as many repositories to best suit your organization needs. For example, you might add separate repositories to store the file attachments and images. You can also add folders to your repositories to further organize your files, then upload files to the appropriate locations.

All client interfaces can access existing repositories for the upload and download of file attachments except as follows:

- Shared file repositories can be accessed only if the repository daemon is running on a computer that can access the shared file. The server name on the repository record (detail form) must be a Windows computer that has access to the share. A CA SDM repository daemon must also be running on the computer.
- Download of zip files is based on when they were uploaded. Attachments from earlier releases are downloaded without unzipping them. Unzip the file on the client computer; otherwise, attachments that are uploaded from a client interface, are also downloaded without unzipping them. That is, the server unzips the file before returning it, during a client download request.

The distributed architecture lets a site configure their repositories according to their needs. The servlet for a repository does not have to reside on the same server as the attached files. Sites can have a central servlet to access all their distributed repositories or have a dedicated servlet for each of their repository servers.

Consider the following setups when configuring repositories:

- **Repository server on a protected CA SDM server** -- Sites that designate a repository server on a protected CA SDM server behind the firewall do not need to expose the servlet on that computer. Rather, specify a servlet running on another CA SDM server or an unrestricted CA SDM server and still upload and download successfully to that repository. Depending on the network between the repository server and servlet server, some performance impact on the upload and download can occur. A good way to set up a remote repository is to install and configure another server with the repository on the remote server, and set the upload path to a local path. For more information, see the Setup a Repository on a Remote Computer topic.
- **Servlet on the same server as the repository server** -- Sites that want the optimal performance for the attachment upload and download, and if the network performance is an issue and if very large files are being attached, should consider this setup. This approach requires exposing the Tomcat port (typically 8080) on the server and should be noted if the computer is behind a firewall.

CA SDM provides predefined repositories and also lets you create your own repositories to best suit your organization needs. Choose from the following options:

- Create a repository.
- Use a predefined repository.

Setup a Repository on a Remote Computer

By default, the repositories are located on the following server, depending on your CA SDM configuration:

- Conventional: Primary server
- Advanced availability: Background server

The default repositories use both the servlet and the repository daemon (rep_daemon) from these servers. To create a repository on a remote computer you must install and configure the following servers, depending on your CA SDM configuration:

- Conventional: Secondary server. The servlet runs on the primary server and the rep_daemon runs on the secondary server.
- Advanced availability: Background server. The servlet runs on the application server and the rep_daemon runs on the background server.

In the advanced availability configuration, the rep_daemon runs on all the servers, by default.

In the conventional configuration, the rep_daemon runs on the primary server, by default and you are required to verify that the rep_daemon runs on the secondary server for this setup.

(Conventional configuration only) Follow these steps:

1. Select **System**, **Configurations** on the **Administration** tab.
   The **Configuration List** page opens.
2. Select the configuration for the secondary server.
   The **Configuration Detail** page opens.

3. Click **Additional Processes**.
   The **Additional Process List** is displayed.
4. Click **Add Process**.
5. Select **Repository Deamon** as the **Process**.
6. Click **Save**.
   The repository is set up on the secondary server.

Create a Repository

Depending upon your organization needs, you can have one large repository or several small ones. Moving or combining repositories is simple because all the attributes about a repository are defined in the repository record.

Follow these steps:

1. Select **Attachments Library**, **Repositories** on the **Administration** tab.
   The **Repositories** page opens.
2. Click **Create New**.
   The **Create New Repository** page opens.

   NOTE
   If you are the service provider, select the appropriate tenant from the **Create New Repository** page. The public (shared) option makes the repository available for all tenants.
3. Complete the Repository Fields as appropriate.
4. Click **Save**.
   The repository is created.

   NOTE

   To delete a repository, select Attachments Library, Repositories from the Administration tab, right-click on the repository, and select Delete. When you delete a repository, all files and folders within a repository are deleted as well.

Use a Predefined Repository

You can use a predefined repository (Service Desk or Knowledge or Images).You can edit a repository to best suit your organization needs.

Follow these steps:

1. Select **Attachments Library**, **Repositories** on the **Administration** tab.
   The **Repositories List** page opens.
2. Right-click the repository that you want to edit and select **Edit**.
   The **Update Repository** page opens.
3. Edit the fields as appropriate. For more information, see the Repository Fields topic.
4. Click **Save**.
   The repository definition is saved.

Repository Fields

The following fields are used to edit or create a repository.

Name

Specifies the name to uniquely identify the repository. For example, Incident Images repository can store all the images that are related to an incident.

- **Repository Type**
  Indicates the type of content that is stored in the repository. For example, to store image attachments, select **Images**.
- **Default**

Indicates whether this is the default repository for the specified repository type. For example, when the user is creating the incident and user wants to attach an attachment to the incident, the default repository is displayed for the selection. You can only set one repository to default.

- **File Limit Size (KB)**
  Specifies the maximum size of file, in kilobytes, that a user can upload to the repository.
- **Upload Path**
  Specifies the full root directory path or the UNC path where files uploaded to the repository reside.
- **UNC Credentials**
  Specifies the credentials to access the UNC path specified in the **Upload Path** field. Click **UNC Credentials** to open the **Credentials Search** page.
  - If you have already created the credentials to access the specified UNC path, search using the fields and select the credentials.
  - If you want to create the credentials, click **Create New**. For more information about creating credentials, see the Create UNC Credentials topic.
- **Background Services**

  NOTE

  : The options for background server services appears only in Advanced Availability configuration.

  Specifies the background server services for the servlet path and rep_daemon.
  - **None**
    Indicates that the background server is not used for the servlet path or for the rep_daemon. If you select this option, enter the values for **Servlet Server** and the **Repository Server** fields.
  - **Servlet Only**
    Indicates that servlet is hosted on the background server. If you select this option, the **Servlet Server** field is auto-populated with **Background Server** value. Enter the value for the **Repository Server** field. If the background server shuts down and if the standby server is promoted as the new background server, the **Servlet Server** field is populated with the new background server value.
  - **Daemon Only**
    Indicates that the rep_daemon is running on the background server. If you select this option, the **Repository Server** field is auto-populated with **Background Server** value. Enter the value for the **Servlet Server** field. If the background server shuts down and if the standby server is promoted as the new background server, the **Repository Server** field is populated with the new background server value.
  - **Servlet and Daemon**
    Indicates that background server is used for the servlet path and the rep_daemon. If you select this option, **Servlet Server** and **Repository Server** fields are auto-populated with the **Background Server** value. If the background server shuts down and if the standby server is promoted as the new background server, these fields are populated with the new background server value.
- **Servlet Server**
  Specifies the server where the servlet is running.
- **Repository Server**
  Specifies the server where the rep_daemon is up and running.
- **Archive Type**
  The archive and purge action to be taken on the contents of the repository.
  - **None**
    No archive and purge process is performed.
  - **Archive and Purge**
    The historic records are written to the file specified in the archive field and purged from the database.
  - **Purge Only**
    The historic records are purged from the database, but are not written to the archive file.
- **Archive Path**

Specifies the directory path or the UNC path to which files in the repository are moved during the archive process.

- **UNC Credentials**
  Specifies the credentials to access the UNC path. Click **UNC Credentials** to open the **Credentials Search** page.
  – If you have already created the credentials to access the specified UNC path, search using the fields and select the credentials.
  – If you want to create the credentials, click **Create New**. For more information, see Create UNC Credentials.
- **Prohibited File Types**
  The file extensions that users may not upload to the repository.

  NOTE
  If the value in this field begins with an exclamation point (!), these file types are allowed in the current repository. For example, a value of jpg,gif in the list denotes that files with .jpg and .gif extensions are prohibited in the repository. However, a value of !jpg,gif denotes that only files with .jpg and .gif extensions are allowed in the repository.

Create UNC Credentials

You create the UNC credentials to allow users to access shared resources from the CA SDM servers using UNC path.

WARNING
The UNC component does not work when the CA SDM server is in the domain and the shared location is in the WORKGROUP. The UNC credentials that you use, must exist on the CA SDM server.

Follow these steps:

1. Click the UNC Credentials in the General Setting page or select Security and Role Management, UNC Credentials on the Administration tab.
   The Credentials List page opens.
2. Click Create New.
   The Create New Credentials page opens.
3. Complete the following fields as appropriate:
   – **Symbol**
     Specifies the unique identifier to identify the credentials during a search easily.
   – **Userid**
     Specifies the username to access the UNC path. The user can be a local or domain Windows user having access to the Service Desk server.
   – **Password**
     Specifies the password to access the UNC path.
   – **Active**
     Specifies if the UNC credentials are active or inactive. The inactive credentials cannot be used.
4. Click Save.
   The UNC credentials are created.

Create a Folder

Folders are used to organize the documents in repositories. For example, you can create a folder Error Images under the Images repository. This folder can contain all the snapshots of errors messages that the user has encountered.You cannot create a folder for the Service Desk Attachments repository type.

Follow these steps:

1. Select **Attachments Library**, **Repositories** on the **Administration** tab.
   The **Repositories List** page opens.
2. Right-click the repository where you want to create the folder and select **Add Folder**.
   The **Create New Folder** page opens.

3. Enter the name of the folder and a description of its contents.
4. Select the **Permissions** tab and specify the appropriate access rights.
5. Click **Save**.
   The folder is created.

Access Rights

You can add the following access rights to the folder in the repository:

- **Inherit from Parent**
  Indicates that this folder has the same permission settings as its parent folder. This option is only displayed for sub-folders.
- **Control by Group**
  Indicated the read or write access on this folder for specified groups. This option appears for all folders and sub-folders.
  - **Grant Write Permission to Everyone**
    Indicates that all users have write access to the folder.
  - **Grant Read Permission to Everyone**
    Indicates that all users have read access to the folder. Read permission indicates that you can view the folder, but you cannot edit, delete, or store files in it. Users with administrative rights can edit a folder even if their associated permission group cannot. If a user belongs to multiple permission groups with varying levels of access to the document, the user gets the highest available access level (for example, if one group has read-only access and the other write access, the user gets write access).

    NOTE
    The **Grant Read Permission to Everyone** check box is automatically selected if you select the **Grant Write Permission to Everyone** check box.
  - **Available Groups**
    Displays all the groups. You can choose the groups from this list. For example, select a group and click > for **Groups with Write Permission** to provide read and write access to this folder for all the users in that group. Use **Show Filter** to specify criteria and filter the groups.

Add a File to a Folder or Repository

A knowledge administrator or a knowledge analyst can add a file to a folder or a repository. Ensure that you have an attachment library in place before proceeding.

Follow these steps:

1. Select **Attachments Library**, **Repositories** on the **Administration** tab.
   The **Repositories List** page opens.
2. Expand the repository, right-click the folder where you want to add the file, and select **Add File**.
   The **Add File** page opens.
3. Click **Browse** to navigate to the desired file and select it.
4. Enter a name for identifying the file, and a description, if necessary.
5. Click **Upload**.
   The file is uploaded to the folder of the repository.

Service Level Agreements (SLA)

A service level agreement (SLA) or service type is an agreement between a service desk and its customers and usually describes the level of service to be provided by the service desk. If this level of service is not provided, the service desk can be penalized. For example, a service desk that operates on a “pay per service” basis may not receive full payment for

service that does not meet an agreed upon level of service. Thus, most service desks view service level agreements very seriously and make every effort to meet the type of service specified in these agreements.

In addition, most service desks keep meticulous records of when service level agreements are met or violated. The service types defined with CA SDM are designed to help the service desk personnel meet their service level agreements and keep the records they need to verify that their service level agreements have been met.

For more information about SLA, see the following topics.

- How to Configure SLAs
- How to Create Service Targets
- Service Contracts

How to Configure SLAs

Action Information Fields

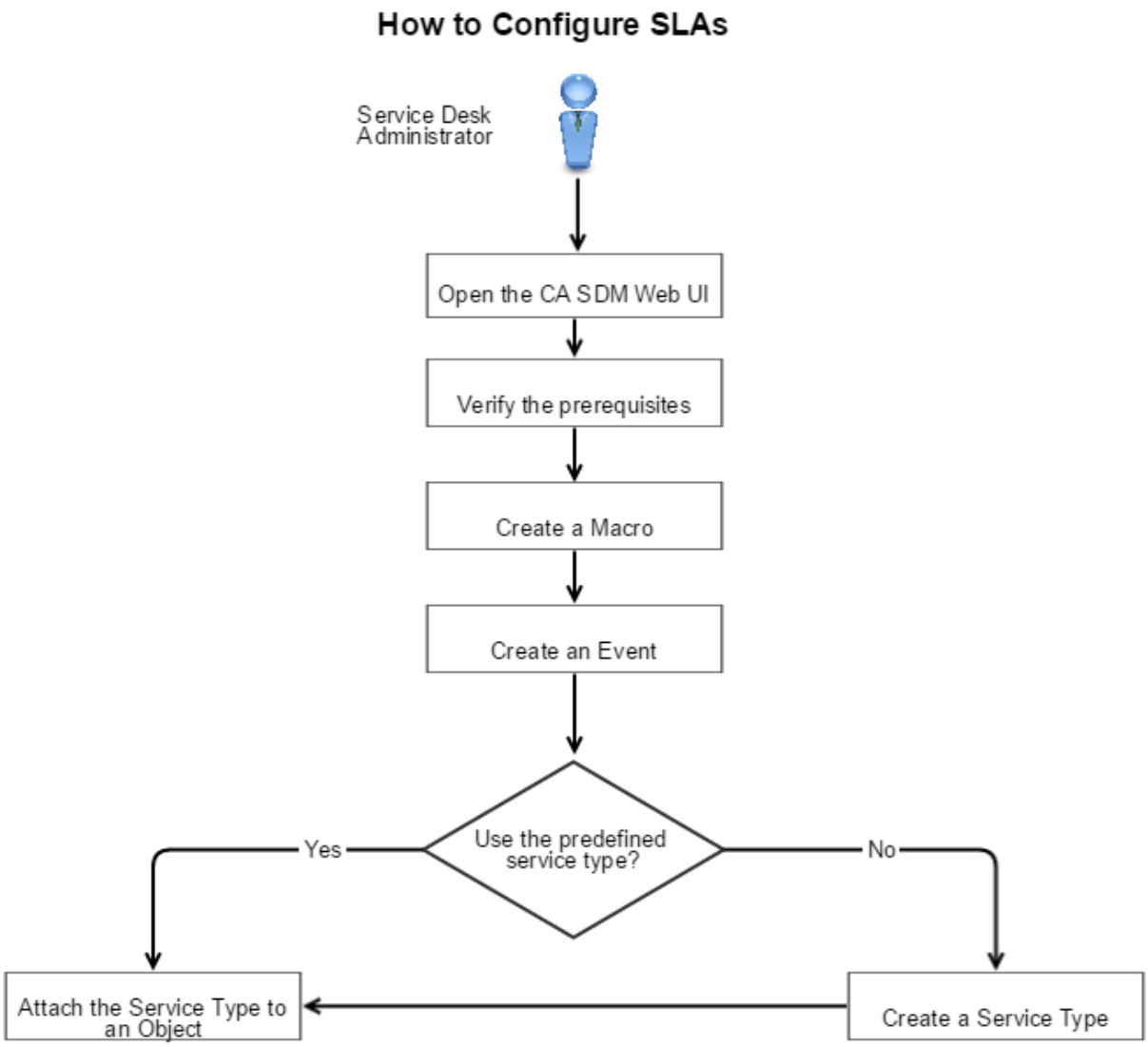
In CA SDM, the SLA or service type describes the level of service that the service desk analyst provides to the customer. To track your enterprise commitments and schedules (as they relate to specific tickets), events are attached to service types. Events are used to define the condition under which the service type is violated and the actions to be taken after the violation. Each event has three generic behavior characteristics: conditions, actions on true, and actions on false.

- Condition identifies the measurable state of a ticket.
- Action identifies the processes that occur automatically when the condition is true or false after a specified amount of time.

Example: As a system administrator, you want to set up an SLA of 24-hours for a hardware request, failing which an email notification is sent to the Customer Support Manager and the analyst. This example is used throughout the scenario to explain how the email notification is configured for an SLA.

The following diagram showcases how you can set up an SLA of 24-hours for a hardware request:

Figure 14: How to Configure SLAs for CA SDM



Follow these steps:

1. Open the CA SDM Web UI.
2. Verify the Prerequisites.
3. Create a Macro.
4. Create an Event.
5. Create a Service Type.
6. Attach the Service Type to an Object.

Open CA SDM Web UI

Log in to the web UI from the following servers, depending on your CA SDM configuration:

- Conventional: Primary or secondary servers
- Advanced availability: Application or background servers

Verify the Prerequisites

Before you configure the service level agreement, ensure that you have completed the following steps:

- Installed the SLA options which are necessary for your organization needs.
- Verified the Notification Method for the Recipient
- Created a Message Template, if you want send an email notification upon an SLA violation and if you do not want to use a predefined message.

SLA Options

Depending on your organization needs, install the SLA options that you require to set up the SLA.

For example, in "classic" SLA processing (enabled if the classic_sla_processing option is installed in Options Manager) only one service type can apply to a ticket at any given time. When different attributes on a ticket have different service types associated with them, the higher ranked service type is used. The rank of a service type is defined when the service type is created, with the highest ranking being 1, the next being 2, and so on. For example, assume that the issue has a service type of 12-hour resolution (ranking 2), was assigned a priority code of 1, which has a service type of 4-hour resolution (ranking 1). The higher ranked service type determines the service behavior for the associated issue. In this example, 4-hour resolution is ranked higher than 12-hour resolution, so the 4-hour resolution service type is applied to the issue.

The following options can be installed from the Options Manager:

Option	Description
change_allow_sla_downgrade	Alters the behavior of the chg_sla option by allowing the system to automatically downgrade a change order's Service Type. The chg_sla option selects the best Service Type from among several change order attributes, but cannot replace the change order's current Service Type with one of lesser rank. If this option is installed, the Service Types for all affected attributes are evaluated whenever one of the attributes changes. The change order's Service Type is set to highest ranked Type found, even if the new Service Type is lesser in rank to the change order's current Service Type. The Service Type with the smallest Rank value is considered the best service. If all the Service Types considered are equal in Rank (including Service Types with empty Rank values), the Service Type created first in the database is selected. The chg_sla option must be installed for this option to function correctly. You can install similar option for issue, and request.

ttv_enabled	Runs the Time to Violation daemon, which monitors the SLAs for all open tickets and tasks. This process does not set the SLA violation, but records the date the ticket or task is violated in its current state. This projection is updated when the ticket or task is updated. This option must be installed in order for the other Time to Violation options to function correctly. **Important!** This option does not require you to install the classic_sla_processing option.
set_sla_evt_open_date	Uses the open date/time value of a change order, issue, or request as the start date/time of attached events. The attached events are triggered as soon as the ticket is saved.

Verify the Notification Method for the Recipient

Ensure that the contact to whom you want to send the notification, is assigned to that particular notification method.

Follow these steps:

1. Select Security and Role Management, Contacts on the Administration tab.
   The Contact Search page opens.
2. Search for the contact using the filter and select the contact that you want to notify from the search result.
   The contact detail page opens.
3. Select Notification on the Contact Details tab.
4. Verify the notification method. Based on your notification priority, choose the required option. For example, you want to send an email notification for any emergency notification. Select the Email option in the Emergency field.
5. To change the notification method, click Edit, change the option, and click Save.
   The notification method for the contact is verified.

Create Message Templates

Create a message template that contains the values to use for the notification message. When you send multiple notification messages, you can use the message templates to simplify your workload.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. Select Notifications, Message Templates from the Administration tab.
   The Message Template List page opens.
2. Click Create New.
   The Create New Message Template page opens.
3. Complete the fields as appropriate.
   - **Symbol**
     Defines a unique identifier for this message template.
   - **Object Type**
     Specifies the object type associated with this template. For example, select Request/ Incident/ Problem for any notification related to a ticket.
   - **Record Status**
     Specifies the status of the template as either active or inactive. Set the status to Active to use the message template.
   - **Auto Notification**
     Specifies to send the notification associated with this template automatically, when the activity occurs. For example, you set up an initial notification, set up the objects to notify, and set up the message template, but you are not ready

to turn on the notifications. In this case, you do not select Auto Notification. When you are ready to start automatic notifications, you select the check box. The notification becomes active and occurs as defined.

- **Notify Level**
  Indicates the relative importance of sending this notification. For example, select Emergency if you want to send the email notification to the contact immediately when the associated activity occurs.
- **Notification Message Title**
  Specifies the summary title of the message. You can use variables to insert the incident number in the message title. For example, @{call_req_id.type.sym} @{call_req_id.ref_num} @{type.sym}.
- **Notification Message Body**
  Specifies the content of the message. You can use variables to insert the analyst name, end-user name, and description into the message. For example,

  ```
  @{call_req_id.type.sym} @{call_req_id.ref_num} @{type.sym}.Assigned to:
   @{call_req_id.assignee.combo_name}Customer: @{call_req_id.customer.combo_name}Description:
   @{call_req_id.description}Click on the following URL to view:@{call_req_id.web_url}
  ```

  You can use the ARTIFACT keyword to specify how artifacts are handled in outbound messages, message templates, notifications, and auto-replies. The ARTIFACT keyword uses the following values:
  - **NONE** -- Specifies no validation of artifacts. This value is the same as not using the keyword.
  - **PROTECTED** -- Validates a ticket against the hash for confirmation. If confirmation fails, the artifact is considered invalid and filtering fails when filtering searching for an artifact ({{object_id}}).
  - **SECURE** -- Decrypts the ticket number. If the value is not a valid password, the artifact is considered invalid and filtering fails when filtering is searching for an artifact ({{object_id}}).
- **HTML Message**
  Specifies the HTML message that is displayed to the recipient. If the recipient receives the message on an external device, such as a cell phone or PDA, the message displays in plain text only. Click Edit HTML Message to open the HTML Editor.
- **Quick View**
  Displays the message as it appears to the recipient.
- **HTML Source**
  Displays the message in the HTML source code.

4. Click Save.
   The message template is created.

Create a Macro

Macros are small scripts that define either conditions or actions. When Events or Behaviors execute, they can execute one or more Action macros. Before the macros are executed, you can use a conditional macro to determine which set of Action macros to execute.

You can use macros in the following areas:

- Events
- Behavior templates
- Activity Notifications

CA SDM includes several macros. Users can create their own macros too.

> **NOTE**
> Customers cannot add Action or Condition macros but can create simple macros with site-defined conditions. The Site-defined conditions are noncomplex macros that are created from GUI dialogs; they are not replacements for condition-type macros.

For each macro, you specify the object type that you want the macro to use. If you create a site-defined conditional to verify a Requests value, you set the type to Request. CA SDM displays only those object type macros that match the ones on the Events or Behaviors.

Create a macro. Use this macro to add actions to objects, check for certain characteristics, or conditions.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. Select **Events and Macros**, **Macros** on the **Administration** tab.
2. Click **Create New**.
3. Complete the fields as follows:
   - **Symbol**
     Enter a descriptive identifier for the macro.
   - **Macro Type**
     Select the type for this macro. The selected macro type controls the remaining data that is supplied.

     > **NOTE**
     > The **Execute CA IT PAM Action** selection is only available when CA Process Automation Workflow is configured with CA SDM.
   - **Object Type**
     Select the type of object on which the macro can be run.
4. Click **Continue**.
   The page fills in with the remaining data.

Attach Event Macro

This macro type attaches an event object to a change order, issue, or request.

Example: Use Macros in the Behavior of Category or Area

You can also use macros in the Behavior for Issue and Change categories, and also for the Request/ Incident/ Problem areas. On a Category/ Area detail, select the Workflow tab, Workflow status, Workflow detail. Click Behavior and specify a Conditional-type macro. The macro is executed when a user tries to change the task status value to a new or different value. If the condition is set to false, updating the task status is not allowed.

> **NOTE**
> Use an Attach Event macro to attach an event to a workflow task.

Follow these steps:

1. Type the event name directly in the **Event** field, or click the search icon to select the desired event.
2. Click **Save**.
   The attach event macro is created.

Execute Remote Reference Macro

This type of macro launches an external program on the server.

Follow these steps:

1. Complete the following fields:
   - **Remote Reference**
     The name of the remote reference to the external program you want the macro to launch. Type the remote reference name directly into this field, or click the search icon to select the remote reference.
   - **Parameters to Command**
     Additional parameters required for running the command.
2. Click **Save**.
   The execute remote reference macro is created.

Create Multiple Notification Macro

This macro type lets you send a notification to one or more contacts. You can specify the message to send, the recipients of the message, and the urgency level.

Multiple Notification

The message body of the multiple notification points directly to the table of the object selected. Therefore, when referencing fields on the object table, (CR (Call_Req) for example,) you can reference the field attribute name. If you want to specify the description information of the Call Request ticket, enter the following command:

```
Description: @{description}
```

A QREL (Query Relation) is a relation which contains a list of objects. The list of objects are by an SQL type WHERE clause. You can add the QREL to the description of a multiple notification macro. To use the QREL in the act_log QREL <-- alg, enter the following command in a notification macro:

```
@{act_log.0.description}
```

You can use replacement variables to make notification messages more relevant and dynamic. These replacement variables take the form of @{attribute_path_here} where attribute_path_here is an attribute of some CA SDM object. When the notification is sent, the variable is replaced with the attribute value specified.
A notification (from an activity log or a multiple notification) always has some base context (a ticket or a workflow task). On multiple notification, the Notification macro type field specifies the base object. You use the @{} syntax to reference any attribute on that object.

Example: A Multiple Notification of type Request references any attribute in the 'cr' (Call_Req) object. To include the Request description, specify @{description} (dot-notation is used to follow references to other objects). For example, to include the Request assignee last name: @{assignee.last_name}.

Follow these steps:

1. Type the template name directly in the **Message Template** field, or click the search icon to select the desired template. This template is used to create the notification message.
2. Click **Save**.
3. Choose the appropriate contacts to notify from the following tabs:

   NOTE
   Use the **Update Contacts** button that appears on each tab to search for and select more contacts to notify.

   - **Object Contacts**
     The available organizations, vendors, and configuration items for the selected Object type. For example, you can select Affected End User or Affected End User's Org to notify.
   - **Contacts**
     The individuals who are added to the notification macro, regardless of their association with the ticket.
   - **Contact Types**
     The users who are defined within the notification macro with the same classification, such as analyst or customer.
4. Click **Save**.
   The multiple notification macro is created.

Create a Site-Defined Condition Macro

This type of macro evaluates true and false conditional statements, and can be created by administrators to provide specific behaviors to individual sites.

Site-Defined Condition
The Conditional macro is made up of one or more *atomic* conditions. Each atom tests the value of a single attribute. The Conditional macro uses two atoms, one for Assignee and one for Category. The individual atoms in the Conditional Macro event are connected with a Boolean operator 'AND' or 'OR'.

The Conditional Macro Detail shows a list of Atoms, which is read from left to right as follows:

Attribute	Operator	Value	Logical	Translation
Assignee	Equals	Jones	AND	The assignee is Jones and…
Cost	Less than	600	OR	Cost is an integer value less than 600 or…
Group	Empty/NULL		AND	The Group field is blank.

NOTE

The Logical connectors (AND, OR) connect two atoms. When there are two atoms, Atom A and Atom B, the atom A connector connects the atom. Atom B connector is not used.

Follow these steps:

1. Complete the following fields:
   - **If All Conditions Succeed Return**
     Select the value you want to return if all conditions succeed. Valid values are True or False.
   - **Record Status**
     Select whether the macro is Active or Inactive.
2. Click **Save**.
3. Click **Add Condition** on the **Conditions** tab.
   The **Create New Atomic Condition** page opens.
4. Fill in the following fields as appropriate:
   - **Sequence**
     The order in which the condition is evaluated within the site-defined condition.
   - **Select an Attribute**
     The attribute to be evaluated. Enter the attribute name directly in this field or click the search icon to select the desired attribute.
   - **Choose Operator**
     Select the conditional operator to use in the evaluation. Valid values are: Equals, Does Not Equal, Greater Than, Less Than, Empty/Null, Not Empty/Null.
   - **Select Attribute or Data Value**
     Choose either Attribute or Data Value to be evaluated.
   - **Attribute Value/Data Value**
     The value to be evaluated. Enter the attribute or data value directly into the field, or click the search icon to select the desired value.
5. Click **Save**.
6. Repeat this process until all the desired conditions have been created.
   The site-defined macro is created.

Execute CA IT PAM Action Macro

You can create a macro to initiate a CA Process Automation process. This process definition can be associated with any CA SDM ticket type. The change category, issues category, or request/incident/problem area the ticket is assigned to determines the process definition that is applied to the individual tickets. If a CA Process Automation workflow has been attached to the selected category, the workflow is automatically attached and initiated when the ticket is saved.

Complete the macro description, record status, and the **CA IT PAM** start form and click **Save**.

Create an Event

You can configure events that are attached to objects to execute configured actions. Events are procedures that execute after a certain amount of time has elapsed. For example, an event sends a message to a service desk analyst if a "priority 1" issue is not received within an hour. Other parts of the system use events, for example, Service Types.

You can define events for Requests, incidents, problems, change orders, issues, contacts, configuration items, and global and specific tenants. CA SDM schedules the events execution time that is based on the delay time and workshift.

NOTE

If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Create an event and attach a macro to this event. This event is executed after certain time is elapsed. If any macro is attached, an action is performed.

1. Select Events and Macros, Events on the Administration tab.
2. Click Create New.
3. Complete the event fields and the configuration information fields.
4. Click Save.
5. To add the macro to this event, complete the action information fields.
   The new event is saved.

Event Fields

- **Object Type**
  Indicates if the event is attached to an issue, request, change order, workflow task, knowledge document, knowledge report card, assistance session, or managed survey.
  This field can only be edited when creating an event. This field is read-only when you want to update the event.
- **Record Status**
  Indicates whether the event is active or inactive. Only active events can be used.

Configuration Information Fields

Complete the following configuration information fields:

- **Delay Time**
  The time after which the event is triggered.
- **Repeat Delay Time**
  The interval of time after which you want the event to be triggered again.
- **Allow Time Resetting**
  Indicates that the desk analyst can change the **Delay Time**. Select this option to use an event as a Service Type event.
- **Work Shift**
  The dates, days, and hours when the service type is in effect.
- **On Done Event Flag**
  The action that is taken once the event is complete.
  - **Repeat Event**
    The event at the specified time interval until the issue is closed.
  - **Save History**
    Records the history of activities that are taken on the event.
  - **No History**
    Do not record any history of the event. The event does not appear in the **Event History** window.
- **Condition**

Displays the macro (if associated with the event) indicating the condition that is checked for by the event.

- **Text**
  Defines the event configuration. For some action macros, this field is used for a specific purpose. For example, the **Transfer to Event Contact** action macro contains the User ID of the person to whom the ticket can be transferred.

Select the actions to be associated with the event as follows:

1. Click the **Action Information** tab.
2. Select one or both of the Set SLA Violation for Actions on True/False Macro check boxes. Selecting these check boxes logs a Service Level Agreement (SLA) violation when a true or false condition is encountered for the event.

   NOTE
   Specify appropriate macros for true or false condition under the action list to log the SLA violations.
3. Click **Update Actions on True**.
   The **Macro Search** page opens.
4. Search for the macros to be performed if the event condition is true.
5. Select the desired macros from the list on the left, and click More (>>).
   The selected macros are added to the list on the right.
6. When all desired macros are in the list on the right, click **OK**.
   The selected macros appear in the **Actions on True Macro List**.
7. Click **Update Actions on False**, and repeat the previous procedure to select the macros to be performed if the event condition is false.

Create a Service Type

You can create a service type that suits your requirements. You can also modify a predefined service type.

Follow these steps:

1. Select **Service Desk**, **Service Types** on the **Administration** tab.
   The **Service Type List** page opens.
2. Click **Create New**.
   The **Create New Service Type** page opens.
3. If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.
4. Complete the following fields, as appropriate:
   - **Symbol**
     Defines a unique identifier for the service type. In this example, you assign the symbol as 24_hr_resolution.
   - **Ranking**
     Defines a ranking for the service type. In this example, the hardware request ticket may be associated with multiple service types. The ranking value determines the applicable service type. The service type with low ranking has the highest priority.
     Enter 1.
   - **Workshift**
     Specifies the dates, days, and hours when the service type is in effect. The following rules apply to workshifts:
     - If you apply a workshift to a service type, stop and restart the service for the workshift to take effect immediately.
     - If a workshift for a service type has been specified, but not for an event, the service type workshift is in effect.
     - If a workshift for an event has been specified, but not for a service type, the service type workshift is ignored.
     - If a workshift for the event and service type have been specified, the service type workshift is ignored.
   - **Timezone**
     Specifies the time zone for the service type. This time zone is used for triggering events in the system if the **Use End User's Time Zone** option is not selected.
   - **Use End User's Timezone**

Specifies the timezone of the affected end user to trigger the events.

– **Violation Cost**
Specifies the cost that is incurred if the service type time limit is violated.

5. Click **Save**.
The service type is saved.
6. To attach a service type event, select the appropriate tab (Requests, Change Orders, Change Order Tasks, Issues, or Issue Tasks) and click **Add Service Type Event**.
The **Create New Service Type Event** page opens.
7. Click **Event**.
The **Event List page** opens.
8. Select one of the existing events from the list.
The selected or created event is displayed in the **Event** field.
9. Click **Continue**.
The service type detail page is displayed with the attached event.

Attach the Service Type to an Object

Service types can be associated with various objects such as contacts, organizations, categories, priority codes. According to the example, you attach the service type to the Incident Area, which is Hardware.

Follow these steps:

1. Open the related ticket for which you want to assign an SLA.
2. Click **Hardware** from the **Incident Area** field.
The **Hardware Update Request/Incident/Problem Area** page opens.
3. Click **Edit**.
4. Click **Service Type**.
The **Service Type List** page opens.
5. Search for 24_hr_resolution service type that you have created.
6. Click 24_hr_resolution from the search result.
The **Hardware Update Request/ Incident/ Problem Area** page opens with the updated service type.
7. Click **Save**.
The service type is attached to the Hardware incident area.

How to Create Service Targets

This article contains the following topics:

To minimize the SLA violations, you can create a set of service target templates to measure each stage of ticket resolution. Like Service Types, each service target contains a condition and estimate of completion time. However, service targets do not provide an action mechanism.

During the ticket creation, a Service Type assigns one or more service targets to track each stage of the ticket resolution. Each time the ticket changes, the active service targets evaluate the condition. If the condition is met, the ticket and activity log show the actual completion time. When the time exceeds the estimated time, the ticket displays the amount of time by which the target was missed.

Service targets let you do the following actions:

- Verify that tickets of the same Service Type follow the same service targets.
- Monitor whether tickets are closed within the required time frames.
- View information such as the remaining number of minutes before a service target completes.

When creating service target templates, consider the following points:

- Consider the required outcome for meeting the service target. Use an existing Condition or Site-Defined Condition Macro to evaluate ticket data. If necessary, modify or write a new macro to manage the service target.
- Calculate the projected violation and penalty costs based on the SLA agreements.
- Assign at least one service target template to a Service Type.

Follow these steps:

1. Create a service target template to manage requests, incidents, problems, change orders, or issues.
2. Link the service target template to a Service Type.
3. Assign the service target template detail to a ticket category such as request, incident, problem, change order, or issue tickets.
   At the time of ticket creation, the appropriate template automatically attaches to the ticket based on the Service Type.
   Each time a user creates a ticket, the status of the service target automatically displays in the Service Type tab.

Create a Service Target Template

You can create a service target template to measure service targets for requests, incidents, problems, issues, or change orders. After you create the template, you can link it to a service type.

> **NOTE**
> If multi-tenancy is installed, specify the tenant in the Tenant field.

Follow these steps:

1. Select **Service Desk**, **Service Target Templates** on the **Administration** tab.
   The **Service Target Template List** page opens.
2. Click **Create New**.
   The **Create New Service Target Template** page opens.
3. Select a ticket type from the **Object Type** field, and click **Continue**.
4. Complete the following fields as appropriate:
   - **Name**
     Defines a descriptive identifier for the service target.
   - **Object Type**
     Identifies the ticket type for this service target.
   - **Target Duration**
     Specifies the maximum amount of time for service target completion in the hh:mm:ss format.
   - **Workshift**
     Specifies the workshift which contains a range of working hours that are used in time calculations for a service target, for example, M-F 08:00-17:00.
   - **Cost**
     Defines information such as the estimated cost associated with missing the service target.
   - **Record Status**
     Indicates whether this target is active or inactive.
   - **Condition**
     Specifies the name of the condition macro or site-defined condition that evaluates the ticket data and determines whether the service target is met.
   - **Required Outcome**
     Specifies a True or False value that indicates whether the service target is complete.
   - **Allow Set Actual**
     Specifies whether to allow users to set the date and time for a service target.
   - **Allow Reset Actual**
     Specifies whether to allow users to reset the date and time for a service target.
5. Click **Save**.

The service target template is created.

Link a Service Target Template to a Service Type

When you link a service target template to a service type, the targets are automatically attached to a ticket when the service type is assigned. You can only assign a service target template once to a service type. Many different service targets can be linked to a service type.

Follow these steps:

1. Select **Service Desk**, **Service Types** on the **Administration** tab.
   The **Service Type List** page opens.
2. Select a service type.
   The **Service Type Detail** page opens.
3. Select the target tab for the selected object type. For example, if you have created the service target for request, select **Request Targets** tab to manage problem, incident, and request tickets.
4. Click **Link Service Target Template**.
   The **Create New Linked Service Target Template** page opens.
5. You can enter a value directly or click the magnifying glass to search for a service type.
6. Click **Continue**.
7. Make changes to the service target template, if necessary.
8. Click **Save**.
   The service target template is linked to the service type.

View Ticket Counters and Timers for Service Targets

If an analyst assigned a set of service targets to a ticket, you can view the status and deadlines for completing each target.

Follow these steps:

1. On the Service Desk tab, display a list of Incidents, Problems, Requests, Change Orders, or Issues.
   The ticket detail page appears.
2. Click the Service Type tab.
   Additional Service Type information appears at the bottom of the ticket.
3. In the Target column, click the Service Target for additional information.
   The Ticket Counters and Timers section appears near the bottom of the Assigned Service Target Detail page. The Assigned Service Target Detail page displays the following fields:
   - **Name**
     Displays the name of the service target.
   - **Target Duration**
     Displays the amount of allotted time to perform the service target. You can only override this value by editing the ticket.
   - **Workshift**
     Displays the schedule used for time calculations for the service target.
   - **Condition**
     Displays the condition or site-defined condition macro that evaluates the ticket data to determine whether the work can complete within the target time frame.
   - **Required Outcome**
     Displays the required result of the condition or site-defined condition Macro.
   - **Cost**
     Displays the penalty that incurs for missing the target. This information also displays on the ticket.
   - **Target Date/Time**

Displays the deadline for completing the target. If the ticket is in a Hold status or the service target has been met, this field is blank.

- **Actual Date/Time**
  Specifies the date and time that the condition was satisfied or the user clicked Set Actual.
- **Time Left**
  Displays the amount of remaining time for the service target. A negative value indicates the amount of time that exceeded the target time frame.
- **Allow Set Actual**
  Displays whether you can set the actual time. Yes indicates that you can set the Actual Date/Time of a Service Target. No indicates that you cannot override the Actual Date/Time.
- **Allow Reset Actual**
  Displays whether you can restart the time. Yes indicates that you can reset the Actual Date/Time of a Service Target. No indicates that you cannot reset the Actual Date/Time.
- **Last Modified Date/Time**
  Displays the date that this ticket was last modified.
- **Last Modified By**
  Displays the name of the last person who edited the ticket.
- **Service Type**
  Displays the name of the service type that attached this service target.
- **Service Target Template**
  Displays the name of the service target template that was linked to the service type that was used to create this Service Target.
- **Lock Target Date/Time From Hold Recalculations**
  Locks the Target Date/Time from being automatically updated when the ticket goes on hold or is delayed.
- **Last Start Date/Time**
  Displays the last time the service target timer was started.
- **Ticket Status**
  Displays the value of the Status field of the ticket.
- **Hold Status**
  Displays whether the ticket status has placed the ticket on hold.
- **Last Hold Date/Time**
  Displays the last time the ticket was placed on hold.
- **Hold Count**
  Displays the number of times the ticket was placed on hold.
- **Last Resolved Date/Time**
  Displays the last time the ticket transitioned to a resolved status.
- **Resolved Count**
  Displays the number of times that the ticket changed to resolved status.
- **Last Closed Date/Time**
  Displays the last time the ticket was changed to a closed status.
- **Closed Count**
  Displays the number of times the ticket changed to a closed status.
- **Ticket Open Date/Time**
  Displays the date and time the ticket opened.
- **Ticket Resolved Date/Time**
  Displays the date and time the ticket resolved.
- **Ticket Closed Date/Time**
  Displays the date and time the ticket closed.

View Service Target Status

On an open ticket, you can view the status for each service target. Status information such as Time Left and Violation Cost help you prioritize your work.

Follow these steps:

1. On the Service Desk tab, display a list of Incidents, Problems, Requests, Change Orders, or Issues.
   The respective ticket list displays with the following Service Target information:
   - **Service Target**
     Displays the time that the next service target is due.
   - **Projected Violation**
     Displays the incurred cost when the service type time limit is violated.
2. Select the ticket you want from the list page.
   The ticket detail page appears.
3. Select the Service Type tab.

   NOTE
   Service targets appear on tickets that meet the conditions that the administrator sets up. Priority calculation can be a factor in how target information calculates and displays.

   If the ticket meets pre-defined target conditions, the Service Targets List the following information about service targets:
   - **Action**
     Sets or resets the Actual Date/Time to the current date and time.
   - **Target**
     Specifies the current service target for the ticket.
   - **Target Date/Time**
     Specifies date and time when this service target is due. If the ticket is in a Hold status, this value is blank.
   - **Actual Date/Time**
     Specifies the time when the target condition was met. If no value appears, the target condition has not been met.
   - **Time Left**
     Specifies the amount of remaining time for the service target when the ticket is on hold. If the service target has been met, the Time Left field shows the unused time. A negative value indicates the amount of time that elapsed since the missed target date.
   - **Violation Cost**
     Displays the penalty that incurs for missing the target. This information also displays on the ticket.

Service Contracts

This article contains the following topics:

Introduction

The SLA model includes the Service Contract. The Service Contract defines the SLA for a particular organization, including its Service Types, Request areas, and Issue or Change Categories. These definitions are referred to as *private* Categories and *private* Service Types.

NOTE
The *private* Categories and Service Types can only be used on tickets where the Service Contract is used.

The Service Contract applicable to a ticket is determined by the ticket's *affected organization*, which is always the Organization of the Affected End User on the ticket (this is the Organization field on a Contact record). Only the Areas or Categories that are listed on the Service Contract can be selected for the ticket. In addition, the only Service Types that

can be applied are the *private* ones that are listed on the Contract. This ensures that the SLAs for one Organization are not accidentally mixed with another's.

A Service Contract also maps Service Types to common reference fields on a ticket, such as Priority and Asset. This mapping associates Service Types with attributes of a ticket. For example, an Organization's contract can assign Service Types to each of the five Priority objects. When a ticket is created with a certain priority, the mapped Service Type is applied.

Categories and Service Types can be defined outside of a Contract and are considered *public*. The public definitions are used when a ticket has no Service Contract. The lack of a Service Contract can occur if the end user has no organization, or the organization's Contract is inactive. A public definition is a helpful backup or default mechanism. Public Service Types are set directly on categories and other reference field objects.

All applicable Service Types are assigned to the ticket. This ensures that all aspects of an SLA are visible and enforced, for example:

- A Printer may have a Service Type that requires a technician to be dispatched within two days.
- A priority object's Service Type may require a callback within one hour.

With both Service Types applied, these required actions are enforced.

Tracking multiple Service Types also helps prioritize tickets. For example, a ticket concerning a broken keyboard is assigned a low priority Service Type. However, if the affected end user is in urgent need of the keyboard, the service priority can be increased.

> **NOTE**
> The SLA model is enforced by default. Past versions of CA SDM applied only a single Service Type to a ticket. The Service Type selection that is involved finding the highest *ranked* Type among all the possible Service Types. A model using the ranking scheme can still be used by installing the Option 'classic_sla_processing'.

Service Contracts Example Use Case

A company, **ABC Inc**, has configured CA SDM environment in a way that SLAs are associated to Service Types, and these Service Types are linked to priority
The SLAs are acceptable in most circumstances, however a group of VIPs require a faster resolution time than what the normal SLAs provide. In such a case, Service Contract can be created to accommodate this.
A Service Contract can be configured for special private Service Types which can contain different SLAs than those used by the default Service Types. These private Service Types can be mapped to a Priority or Urgency. You are then able to associate specific Users, Groups, or Configuration Items to this Service Contract that it applies to.
For example:

1. CEO of **ABC Inc** is associated to a specific Service Contract, that would have special SLAs.
2. A specific server at ABC Inc requires high availability which can be associated to a Service Contract so that if a ticket were created for that Configuration Item it would be associated to the special SLAs.

You may link a Service Contract to an Area\Category, so that if they were specified in the ticket they would be associated to the special SLAs too.
A Service Contract is not a replacement for SLAs, it provides a method to associate unique SLAs to certain situations to help meet the business requirements.

Service Contracts Migration

If CA SDM is installed as a migration, the Option classic_sla_processing is turned ON by default, so your SLA processing can continue as before the migration. This continuation gives you time to create appropriate Service Contracts and eventually deactivate classic_sla_processing.

When building Service Contracts, you do not have to create Service Types, Request Areas, or Categories. You can use the Copy button found on the Service Contract detail to copy existing objects to the Contract.

If the previous installation marked the support_lev field as required for any ticket type, this restriction must be removed. The support_lev field still exists, but is not set in the current model so a required field error results with new tickets. This affects the objects Request (cr), Issue (iss), or Change Order (chg).

Time to Violation

When the SLA model is in use, CA SDM's Time to Violation (TTV) system can help you track and prioritize tickets according to their projected violation time. This system monitors all active tickets and sets the projected violation time for each Service Type. You can report and sort tickets based on their violation time and cost, helping you resolve the most urgent and costly issues first.

The TTV system monitors all active tickets and evaluates their SLA events silently, to determine which events set the SLA violation flag. The events are not executed, but the system looks at the outcome of each event based on the current state of the ticket. If the evaluation results in an action that sets the SLA violation flag, the ticket's attached Service Type is updated with a Time to Violation value; this value is the date/time when the event fires that sets the SLA violation.

Evaluation occurs whenever a ticket is inserted or updated. Because tickets are often updated in rapid succession, the evaluation is delayed for a short period. The delay interval is controlled by the ttv_evaluation_delay Option. After the delay expires, the TTV system evaluates all the SLA events that might set the SLA violation flag.

Each Event has an optional condition and a set of actions (Macros) that start based on the outcome of the condition. To ensure adequate performance, Event template information is cached by the TTV system and is refreshed periodically. Projections that are made by TTV involving recently updated Event templates may be inaccurate for several minutes.

> **NOTE**
> The TTV projections appear on the Service Type tab of each ticket. The TTV system is activated with the ttv_enabled Option.

Time Zones and Workshifts

To address the complex business demands of automated application execution, CA SDM lets you define as many time zones and workshifts as you want, and to record them for easy reference.

- Time zones identify the time zone where the user, CI, and so forth, are located.
- Workshifts define the period during which event monitoring or the work hours of a service type or SLA are in effect.

Being able to determine a course of action to perform based on when an event occurs can be critical to the proper handling of the event. The time zones and workshifts you define are available for use by any of the CA SDM functions.

Time Zones Setup

Time zone codesdefine the time zone from which a user usually accesses the system (that is, the user's local time zone) or the time zone in which a CI is located. Business hours are always entered in the time zone of the CA SDM server. This means that business hours can always be compared uniformly.

Time zones are used to manage service types, escalations, and overall response to affected end users based on the ability of CA SDM to present the correct time across multiple time zones. CA SDM automatically adjusts the offset between the time zone where a user is logging in or a CI is located, and the time zone where the server is running. Time zones use the Greenwich Mean Time (GMT) format.

How to Manage Multiple Time Zones for Service Types

CA SDM servers and users can be located in different time zones. The time difference can cause users to miss Service Type expiration dates and times.

To correct the time difference, you can configure CA SDM to display Service Type expiration times in the end-user time zone. If two users in different time zones view the same ticket, each user sees Expiration Time values based on the local computer time zone. However, the Start Time values always reflect the server time zone.

To configure for the end-user time zone, do the following:

1. Create a server code that identifies the server name and time zone.
2. Create or update a Service Type. Set the Use End User Time zone field to Yes.
   A value of Yes causes the Expiration Time to display and update according to each user time zone.

Example: Show Service Type Expiration Dates in Any User Time Zone

In this example, you configure CA SDM to show Service Type expiration dates in any user time zone. The server and user are on separate computers and in different time zones.

To create a server code that identifies the server name and time zone

1. On the Administration Tab of the host server, select Service Desk, Application Code.
2. Click Codes, Servers.
   The Server List displays
3. Click the Local Host server.
   The Server Name Detail page appears.
4. Set the Time Zone. For example, set the time zone to GMT (EU). The local host name must match the NX_LOCAL_HOST value that is stored in NX.env for the server
5. Click Save.
   The Host Server uses the new time zone. When the user views a ticket, the Start Time reflects the server time zone.

To create a service type

1. On the Administration Tab, select Service Types.
   The list of Service Types appears.
2. Click Priority 1 Resolution or another Service Type that manages Priority 1 Requests.
   The Update Service Type page displays
3. Select the Use End User Time Zone check box.
4. Click Save.
   The Service Type record is updated

To view the time zones on the ticket

1. On another computer, open a Request ticket, and set the Priority to 1.

   NOTE
   If you are using Service Targets, set the values in the ticket to cause the target to use Priority 1 Resolution.
2. View the ticket and click the Service Type tab.
   The Start Time reflects the server time zone. The Expiration Time reflects the time on the end-user local computer.
3. Close the page that displays the Request ticket.
4. Change the time zone on the end-user local computer.
5. View the ticket Service Type on the end-user local computer to see the corresponding Expiration Time values based on the user time zone.
   The Expiration Time reflects the new time zone.

Workshifts Setup

Workshiftsidentify the days, dates, and times when an event or schedule is in effect. You can specify days or dates, or days and dates. Specifying a time is optional.

When you are monitoring events for tickets, workshifts define when the event is monitored or, in other words, when the clock is running. For example, using the predefined event P1 Active Issue Notify assignee, if a priority 1 issue is opened at 4:45 PM and the workshift schedule is 9:00 AM to 5:00 PM, the monitored event automatically sends notification to the issue assignee at 9:45 AM the next day.

NOTE
Workshifts are also used for the purpose of automatically assigning tickets to contacts.

Create a Service Contract

This article contains the following topics:

You can create service contracts and map them to contacts, configuration items, and priorities.

NOTE

If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. Select Service Desk, Service Contracts on the Administration tab.
   The Service Contract List page appears.
2. Click Create New.
   The Create New Service Contract page appears.
3. Complete the Service Contract fields as appropriate.
4. Configure the service contract mapping as appropriate.
5. Click Save.
   The service contract definition is saved and the Service Contract Detail page appears.

Service Contract Fields

The following fields require explanation to create or update a service contract:

- **Client Advocate**
  The person within the company that runs the help desk. This person is the point of contact between the “Customer” company and the “Help Desk” company. This person is a member of the organization that would get special notifications
- **Client Contact**
  This is the contact person that would get notified on tickets/service types/etc.
- **Assigned to Organization(s)**
  Displays the name of an organization assigned to the service contract. If an organization has not been assigned, None appears.
  Add an organization to a service contract by first creating the new service contract. Then, access the Scoreboard tab, select File, Create New Organization and while creating the organization, select the contract previously saved. When you access the service contract, the Assigned to Organization(s) field is populated with the updated organization name.
- **Organization's Service Type**
  If set, this service type is ALWAYS applied to any ticket opened for the organization. This is useful for adding baseline events, notifications or SLA parameters that need to be on every ticket regardless of priority, category, etc. For example, the Client Advocate gets a notification for every ticket opened/closed for the Organization, so they have a general idea of how their customer is doing.

  NOTE
  If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

Service Contract Tabs

The following tabs are available on the Service Contract Detail and Update Service Contract pages:

- **Contacts**
  Allows you to map one or more contacts to the service contract. Also allows you to update a contact's service type. Click Map Single Contact.
- **Groups**
  Allows you to map one or more groups to the service contract. Also allows you to update a group's service type. Click Map Single Group.
- **Config. Items**
  Allows you to map one or more configuration items to the service contract. Also allows you to update a configuration item's service type. Click Map Single CI.
- **Priorities**
  Allows you to map a priority to the service contract. Click Map Priority.
- **Urgencies**
  Allows you to map an urgency to the service contract. Click Map Urgency.

The following tabs appear on the Service Contract Detail page after a new service contract record has been saved:

- **Private_Service_Types**
  Allows you to create a new service type to map to this service contract. Click Add Private Service Type.
  Request Areas
- Allows you to create a new request area or copy an existing request area to map to the service contract. Click Add New Request Area.
  Change Categories
- Allows you to create a new change category or copy an existing change category to map to the service contract. Click Add New Change Category.
  Issue Categories
- Allows you to create a new issue category or copy an existing issue category to map to the service contract. Click Add New Issue Category.

Add an Issue Category to a Service Contract

You can create one or more new issue categories to associate with this service contract.

To create a new issue category for the service contract

1. Select the Issue Categories tab on the Service Contract Detail or Update Service Contract page.
2. Click Add New Issue Category.
   The Create New Issue Category page appears.
3. Fill in the fields as appropriate to define the area.
4. Click Save.
   The issue category definition is saved and the Issue Category Detail page appears, with the name of the associated contract in the Service Contract field.
5. Close the Issue Category Detail page and return to the Service Contract Detail page.
   The new issue category is listed on the Issue Categories tab.

To copy an existing issue category for the service contract

1. Select the Issue Categories tab on the Service Contract Detail or Update Service Contract page.
2. Click Copy Existing Issue Category.
   The Issue Category Search page appears.
3. (Optional) Complete one or more search fields to filter your search.
4. Click Search.
   The Issue Category List page lists the issue categories that match your search criteria.
5. Select the issue category you want to copy.
   The Create New Issue Category page appears with the fields of the copied issue category filled in.

6. (Optional) Update the fields as appropriate for the new issue category.
7. Click Save.
   The issue category definition is saved and the Issue Category Detail page appears, with the name of the associated contract in the Service Contract field.
8. Close the Issue Category Detail page and return to the Service Contract Detail page.
   The new issue category is listed on the Issue Categories tab.

Automatic Closure of Tickets

You can use a configurable setting to allow automatic closure of tickets (requests/incidents/problems, change orders, or issues). When a ticket is set to a Resolved status, the ticket is automatically closed in the number of business hours specified. The Auto Close activity notification sent to the end user displays the number of business hours before the ticket is closed. The number of hours is configurable and tenant-specific. If the status is changed before the configurable number of hours ends, the ticket closure is canceled.

Administrators, can perform the following actions:

- Define an Auto Close ticket setting to control the number of business hours, for the end user, before the ticket is automatically closed.
- Set up an Auto Close activity notification to notify the appropriate contacts when automatic closure is scheduled for a ticket.

If you use multi-tenancy, consider the following:

- The system uses the default public Auto Close setting when a tenant-specific Auto Close setting is not found.
- There is one Auto Close setting for each tenant.

NOTE

If the end user and requester of a ticket are different, the workshift of the requester is given priority and considered for auto close calculations.

How to Define Auto Close Ticket Settings

You can define the number of business hours before a ticket is closed (all ticket types) as follows:

1. On the Administration tab, select Service Desk, Application Data, Codes, Auto Close.
2. Click Create New on the list page.
3. Complete the following fields on the detail page:
   - **Symbol**
     Defines the system setting name.
   - **Request/Incident/Problem/Change Order/Issue**
     Defines the number of business hours, after the ticket is set to Resolved status, before the ticket is closed. If the status is changed before the number of hours ends, the ticket closure is canceled. 0 (zero) hours indicates that automatic closure is not implemented for the ticket type.
   - **Description**
     Provides a brief description of the record.
   - **Status**
     Indicates if the record is active or inactive.

   The auto close setting is defined.
4. Click Save, Close Window.
   The new setting appears on the Auto Close List page when you redisplay the list.

How to Define an Auto Close Activity Notification

You can change the settings in the default Auto Close activity notification to notify the appropriate contacts when automatic closure is scheduled for a ticket. The activity is valid for all CA SDM ticket types, and include a default notification rule for requests/incidents/problems, change orders and issues.

To define an auto close activity notification, do the following:

1. On the Administration tab, select Notifications, Activity Notifications.
2. Select the Auto Close activity notification on the list page to open it.
3. Update the default Auto Close Notification Rule and specify contacts to receive notification.
4. Click Save, Close window.
   The updated Auto Close activity notification appears in the Activity Notification List when you redisplay the list.

Search Attachments

You can enter search criteria to filter the Attachments List to display only the attachments you want to see. You can also search for individual attachments to view or edit.

Follow these steps:

1. From the Administration tab, select Attachments Library, Attachments.
   The Attachment Search page opens.
2. Complete the search fields. The following fields require explanation:

   NOTE

   All search fields that allow text entry support use of the % wildcard character.

   – **Repository** -- The repository where the file resides.
   – **Status** -- Choose from one of the following values:
   - **Installed** - Attachment is available and in place.
   - **Link Only** - Attachment is a URL link.
   - **Not Installed** - Attachment is not in place. Request has not completed successfully.
   - **Archived** - Attachment is archived.
   - **Not Available** - The attachment could not be found after an archive and purge.
   - **Knowledge File** - Attachment is a text file.
3. Click Search.
   The Attachment List page displays the attachments that match your filter criteria. You can select an attachment to view or edit it.

   NOTE

   If you are using multi-tenancy, a tenant drop-down list appears in the search filter. If you select <empty> in this drop-down list, the search is public.

Create an Announcement

You can use CA SDM to post announcements to users. Announcements help decrease the number of incoming calls and promote increased productivity through proactive resolution of tickets and communication of important information to all affected users. Users can scroll through stored multiple announcements.

NOTE
Announcements apply to all service desk models.

You can add new announcements and update existing ones. Announcements are part of the reference data function of CA SDM, so by using the access type you can control which contacts can create announcements.

An announcement can specify either or both of the following:

- **Location**
  Specifies a physical location, for example, a city, building, or floor.
- **Organization**
  Specifies an organization ID. When Organization is set for an Announcement, only individuals assigned to that Organization can see the Announcement.

When Location or Organization is set for an announcement, only contacts at that location or organization receive it. Any contact can still see all announcements that are not restricted by location or organization. For example, if neither is set, contacts in every location and organization can see an announcement.

> **NOTE**
> You can specify announcements using the administrative function of the web interface.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. On the Scoreboard, select File, New Announcement.
   The Create New Announcement window opens.
2. Fill in the fields as appropriate. See Announcement Fields for field definitions.
3. To insert a link to a Knowledge Document in the announcement, click Knowledge Doc.
   The Knowledge Document search page displays.
4. Search for the document you want to include. See Search Knowledge Documents for more information.
5. In the Link Text field, type the text to be used to link to the document.
6. Click Insert Document.
   The HTML code linking to the Knowledge Document is added to the Text field.
7. Click Save to save the announcement.
   The new announcement is added to the right pane of the main window.

Announcement Fields

- **Text**
  Enter the text of the announcement. Click Spelling to perform a spell check of the text you entered. This field is required.
  If you insert a link to a Knowledge Document, the HTML code for that link is displayed in this field.
- **Location**
  The location where the announcement is displayed. Enter the name of the location directly into the field, or click the search icon to search for the location.
- **Internal**
  Select this option if this announcement should only be shown to internal employees.
- **Organization**
  The organization where the announcement is displayed. Enter the name of the organization directly into the field, or click the search icon to search for the organization.
- **Announcement Type**
  Determines the urgency of the announcement and the color of the announcement text displayed in the window. Asterisks are displayed next to the posted date for color blind or visually impaired users.
  The possible values are:
  - **Routine**
    Black text. No asterisks.
  - **Advisory**

Orange text. One asterisk, after the date.
  - **Emergency**
    Red text. Two asterisks, after the date.
- **Status**
  Indicates whether the announcement is active (displayed) or inactive (not displayed).
- **Close Date/Time**
  The date and time to stop displaying the announcement. Enter the date and time in the format mm/dd/yyyy hh:mm AM | PM, or click the calendar icon to select the date and time.

Spell Check Text Entry Fields

The Spell Checker utility is available for checking your spelling in text entry fields throughout the user interface.

To Use the Spell Checker

1. Enter text into the text entry field.
2. Click the Spelling button above the text entry field.
   - If the Spell Checker does not find any errors, a Spell Check Complete message appears at the top of the page.
   - If the Spell Checker does find errors, the Spell Check Results window opens with the first error highlighted.
3. Evaluate each error and decide whether to ignore or change it.
   When spell checking is complete, the Spell Check Results window closes and any corrections you have made appear in the text entry field.

File Types

You can define acceptable file types to be used as attachments, and assign an icon to each file type to make it easy for users to identify.

Create a File Type

CA SDM provides predefined file types that can support the attachments. For example, the predefined file type, Adobe Acrobat supports the pdf file formats as attachments. You can also create file types. For example, create JPEG file type for users to upload snapshot for an incident.

Follow these steps:

1. Log in to web UI from the following CA SDM servers, depending on your CA SDM configuration:
   - Conventional: Primary or secondary servers
   - Advanced availability: Application or background servers
2. Select **Administration**, **Attachments Library**, Prohibited Files Types of Attachments library.
3. Click **Create New**.
   The **Create New File Type** page opens.
4. Complete the following fields:
   Type Specifies the file type or identifies the associated software application, for example Word document, Image, Executable. Extension Specifies the characters appended to the base filename that identify the file type. For example, the extension doc identifies a Microsoft Word document. Icon Name Specifies the name of the graphic file that contains the icon to be used to identify this file type. Status Indicates whether the record is active in the database and available for use.
   Click Save.
   The file type is created.

Auto Assignment

This article contains the following topics:

CA SDM *Auto Assignment* reduces the time that is required to manage tickets by automating the process of assigning them to analysts. This automation can make the tasks that are associated with balancing workloads and coordinating work schedules much more efficient.

You can configure auto assignment to assign tickets that are based on the following factors:

- Which analyst groups work on which tickets or tasks.
- When the work must be done.
- The locations catering the affected customers.
- The workload and availability of each analyst.
- The value of an attribute of a configuration item that is associated with the ticket.

> **NOTE**
> Configuration item-based auto assignment lets you create group-specific assignments for Request/Incident/Problem tickets only. You do not have to implement auto assignment all at once. You can develop a strategy for phasing it in gradually. For example, you can begin by enabling it only for selected ticket types, analyst groups, or locations.

Auto Assignment Relationships

The auto assignment process can involve many CA SDM elements. The element relationships are as follows:

- Areas and categories relate to groups, locations, and workshifts.
- Groups relate to locations and workshifts.

To make the auto assignment easy to administer, all relationships can be maintained from either related element. For example, when relating an analyst group to a change category, you can maintain the association from either the Change Category Detail page or the Group Detail page.

Auto Assignment Methods

The following basic methods let you auto assign tickets:

- **Location-based auto assignment**
  Creates assignments for all ticket types that are based on the following criteria:
  - Areas and categories that are related to groups, locations, and workshifts
  - Groups that are related to locations and workshifts

  > **NOTE**
  > The Location-based auto assignment is referred to simply as auto assignment.
- **Configuration item-based auto assignment**
  Creates group assignments for request, problem, and incident ticket types that are based on the following criteria:
  - Areas that are related to tickets associated with configuration items
  - Configuration item attributes used to record contact/group information

Location-based auto assignment and configuration item-based auto assignment are exclusive options. This is because you can select only one algorithm for use on a given Request/Incident/Problem Area. Location-based auto assignment and configuration item-based methods both serve to assign tickets when. However, Configuration item-based auto assignment is distinct, because it also reevaluates the assignments for a ticket upon changing the Area or configuration item of a Request/Incident/Problem ticket. If a configuration item-based auto assignment is specified, without successfully generating a group assignment for a ticket, the Area_Defaults option is consulted to determine whether the default Group and Assignee values on the Area should be used to assign the ticket.

Default Group and Assignee

When the auto assignment logic is unable to identify a suitable group or assignee, the ticket is assigned to the default group and assignee. You can specify these defaults in the Group and Assignee fields of the following web interface pages:

- Request/Incident/Problem Area Detail
- Change Category Detail
- Issue Detail

If auto assignment cannot identify a suitable group or assignee and the defaults are not specified, the ticket is left for manual assignment.

Auto Assignment Enablement

Options and controls let you configure auto assignment. You can control whether auto assignment processing occurs as follows:

- For requests, incidents, and problems assigned to that area, use the Auto Assignment Enabled tab on the Request/Incident/Problem Area Detail page.
- For change orders and issues, select the Auto Assignment tab and Auto Assignment Enabled check box on the Change Category Detail page, Issue Category Detail page.

Example: Enable Auto Assignment for a Request/Incident/Problem Area

This example demonstrates how to enable auto assignment for a Request/Incident/Problem area.

1. On the Administration tab, browse to Service Desk, Requests/Incidents/Problems, Areas.
   The Requests/Incidents/Problems Area List page appears.
2. Click the area for which you want to configure auto assignment.
   The Requests/Incidents/Problems Area Detail page appears.
3. Click Edit.
   The Update Requests/Incidents/Problems Area page appears.
4. Select the Auto Assignment tab, and complete the following fields as follows:
   - **Auto Assignment Mode**
     Specifies how auto-assignment occurs. You use the Configuration Item Based option to base the auto assignment on the CI Assignable Attribute value.
     - **Disabled** -- Bases the auto-assignment on the Area Defaults option when it is installed.
     - **Configuration Item Based** -- Bases the auto-assignment on the CI Assignable Attribute value.
     - **Location Based** -- Bases the auto-assignment on the location value.
   - **Assignable CI Attribute**
     Specifies the configuration item attribute to use for the group assignment. You can enter a value directly or click the magnifier to search for an attribute.
5. Click Save.
   The Request/Incident/Problem area is enabled to use default values that are entered automatically on the tickets that are assigned to the area.

Auto Assignment Override

You can use the Auto Assignment Override (autoasg_override) option in the Options Manager to determine if the auto assignment overrides an analyst or a group that is set when a ticket is created.

Other CA SDM features may have set the assignee and/or group before auto assignment obtains control. You can configure your system by setting this option to one of the following values:

- **0**

Honors the existing assignee and/or group. If the assignee and/or group was set during the creation of the ticket, auto assignment processing does not occur.

- **1**
  Ignores the existing assignee and/or group. Auto assignment processing continues and attempts to find an assignee and/or group.

You can set the Assignee and/or Group in various ways:

- Manually by the Analyst
- Area Defaults and the Assignee set options
- Request Templates

NOTE
Uninstalling the Auto Assignment Override option causes it to operate in its default mode, which is 0.

Configure Auto Assignment by Location

The Auto Assignment tab allows you to establish a relationship between this location and request areas, change categories, issues categories, and groups. This allows you to assign tickets to eligible members of the group within the location automatically.

WARNING

You must enable automatic assignment for each request area and category individually. Listing them on the Location Auto Assignment tab does not enable this function.

Follow these steps:

1. On the Location Detail page, click Edit and select the Auto Assignment tab.
2. Click one of the following buttons:
   - Update Request Areas to select request areas for auto assignment.
   - Update Change Categories to select change categories for auto assignment.
   - Update Issue Categories to select issue categories for auto assignment.
   - Update Groups to select groups for auto assignment.

   The Search page for the selected item appears.
3. Enter the search criteria to display the desired items and click Search.
   The Update page appears, listing the items that matched the search criteria.
4. From the list on the left, select the desired items. To select multiple items, hold down the CTRL key while clicking the left mouse button.
5. When you have selected all the items that you want, click the double right-arrow button.
   The selected items move to the Assigned list on the right.
6. Click OK.
   The Location Detail page displays, with the selected items listed on the Auto Assignment tab.

Assignment Controls

Options and controls let you configure the auto assignment. The CA SDM features can affect the assignee, the group fields, or both on a ticket before auto assignment processing occurs. We recommend that you review assignment controls before implementing auto assignment.

Manual Assignment

The analyst can manually select the assignee and/or group while creating a ticket.

Assignee_set Option

By default, CA SDM automatically sets the logged in analyst user as the assignee of the request. An Options Manager option, Assignee_set, lets you control this behavior. This option is typically installed by default.

Iss assignee_set

The Iss assignee_set option automatically sets the logged in analyst user as the assignee of the Issue. It is similar to Assignee_set, except it is for Issues instead of Requests.

Area_Defaults

The Area_Defaults option determines what happens when the request area is specified on a request detail page. This option lets you set the assignee and group whenever the request area changes. The default assignee and group from the request area are used, and this processing occurs before auto assignment processing.

This option is not installed by default.

> **NOTE**
> The Category_Defaults option provides similar functionality for change orders. The Iss_Category_Defaults option provides similar functionality for issues.

Require Assignee and Group Options

Several options are available in Options Manager for controlling the creation of unassigned tickets. These options are global in scope. They affect the creation of all of the indicated ticket types, regardless of whether auto assignment is in effect. If the indicated condition is not satisfied, the new or updated ticket cannot be saved.

The following options control the creation of tickets without an assignee specified:

- **require_change_assignee**
  Application: Change Order Mgr
  Description: Makes the Assignee field of a change order ticket required
- **require_issue_assignee**
  Application: Issue Mgr
  Description: Makes the Assignee field of an issue ticket required
- **require_incident_assignee**
  Application: Request Mgr
  Description: Makes the Assignee field of an incident ticket required
- **require_problem_assignee**
  Application: Request Mgr
  Description: Makes the Assignee field of a problem ticket required
- **require_request_assignee**
  Application: Request Mgr
  Description: Makes the Assignee field of a request ticket required

The following options control the creation of tickets without a group specified:

- **require_change_group**
  Application: Change Order Mgr
  Description: Makes the Group field of a change order ticket required
- **require_issue_group**
  Application: Issue Mgr
  Description: Makes the Group field of an issue ticket required
- **require_incident_group**
  Application: Request Mgr
  Description: Makes the Group field of an incident ticket required
- **require_request_group**
  Application: Request Mgr

Description: Makes the Group field of a request ticket required

- **require_problem_group**
  Application: Request Mgr
  Description: Makes the Group field of a problem ticket required

Templates

You can use templates to set the values on a new request, change order, or issue. Templates can affect assignee and group fields.

How Auto Assignment Assigns Tickets

This article contains the following topics:

How Tickets are Assigned

Auto-assignment assigns tickets as follows:

1. The initial save operation of a new ticket invokes auto assignment.
   If an area or category is not configured for auto assignment, processing stops.
2. Auto assignment determines whether **Autoasg_override** is installed.
   If it is not installed and the ticket has an assignee or group, processing stops.
3. If work shifts are related to the ticket, the open date is evaluated to determine the inclusion of a ticket.
   If a work shift does not include the ticket, processing stops. Auto assignment attempts to assign the default group and assignee.
4. Auto assignment determines whether any groups are related to the ticket.
   If no groups are related, processing stops, and auto-assignment attempts to assign the default group and assignee.
5. Auto assignment filters any groups with work shift, where the open date is outside the time frame of the work shift.
   Groups without a work shift bypass filtering.
6. The following processing occurs for locations that are related to the ticket:
   – If this ticket is a request, and it has a configuration item:
     The location of the configuration item is matched against the locations that are related to the request area. If no match occurs, processing stops, and auto assignment attempts to assign the default group and assignee. Otherwise, the customer location is matched against the locations that are related to the area or category. If no match occurs, processing stops, and auto assignment attempts to assign the default group and assignee.
   – If locations are associated with the request area or category:
     Suppose that locations are associated with the request area and the configuration item (during request auto assignment processing), or the customer has no location. Then auto assignment stops processing.
7. Auto assignment filters the groups that have related locations that do not match the configuration item location (for requests only) or customer location. If no groups remain, processing stops, and auto assignment attempts to assign the default group and assignee.
8. Auto assignment creates a list of analysts from each of the remaining eligible groups.
9. Unavailable analysts are filtered.
10. The remaining analysts are checked for Work Schedules. Available analysts that have Work Schedules are filtered when the open date is not within their work schedules.
11. If no analysts remain, processing stops and auto assignment attempts to assign the default group and assignee.
12. All the remaining analysts are ranked according to the number of active tickets that are assigned to them.
13. The analyst (and associated group) with the least number of active tickets is assigned to the ticket. If a tie occurs, the first analyst that occurs in the group is selected.
    For more information, see Ticket Process Flow Diagram.

Ticket Process Flow Diagram

The following flow diagrams show how auto-assignment assigns ticket:

Figure 15: Diagram depicting ticket workflow Page 1 v1

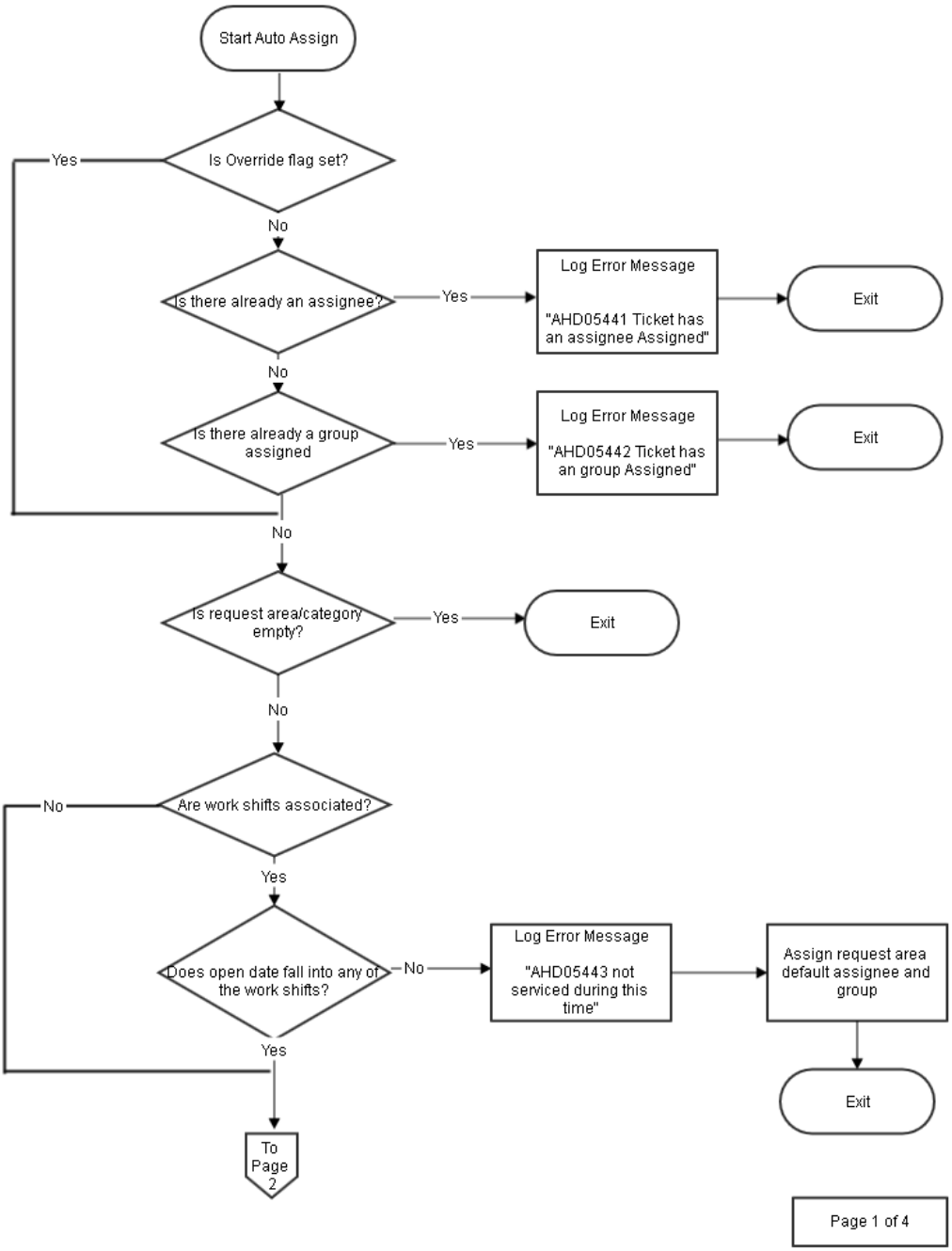


Figure 16: Diagram Showing Ticket Workflow for Page 2

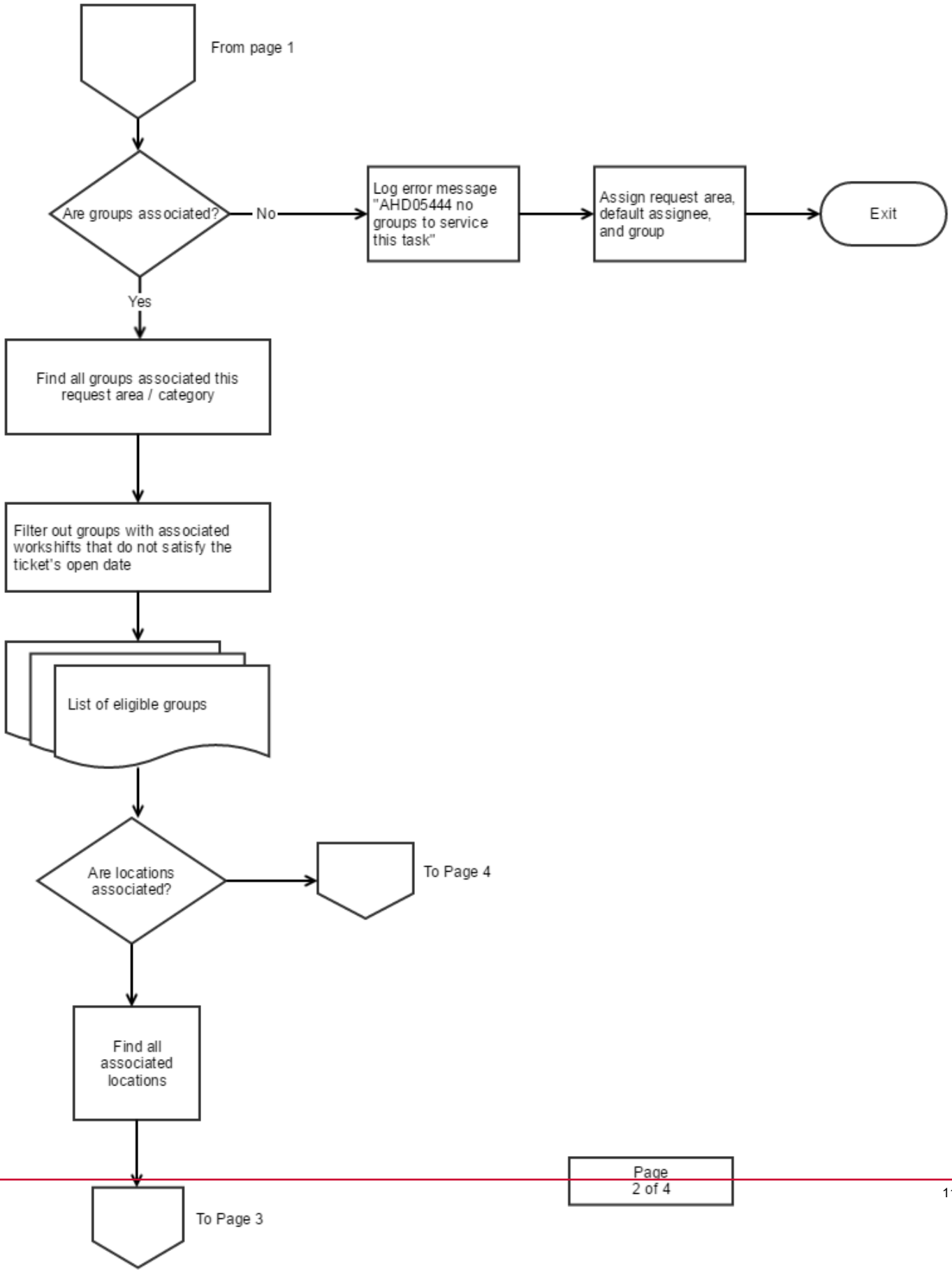


Figure 17: Diagram for depicting ticket workflow Page 3

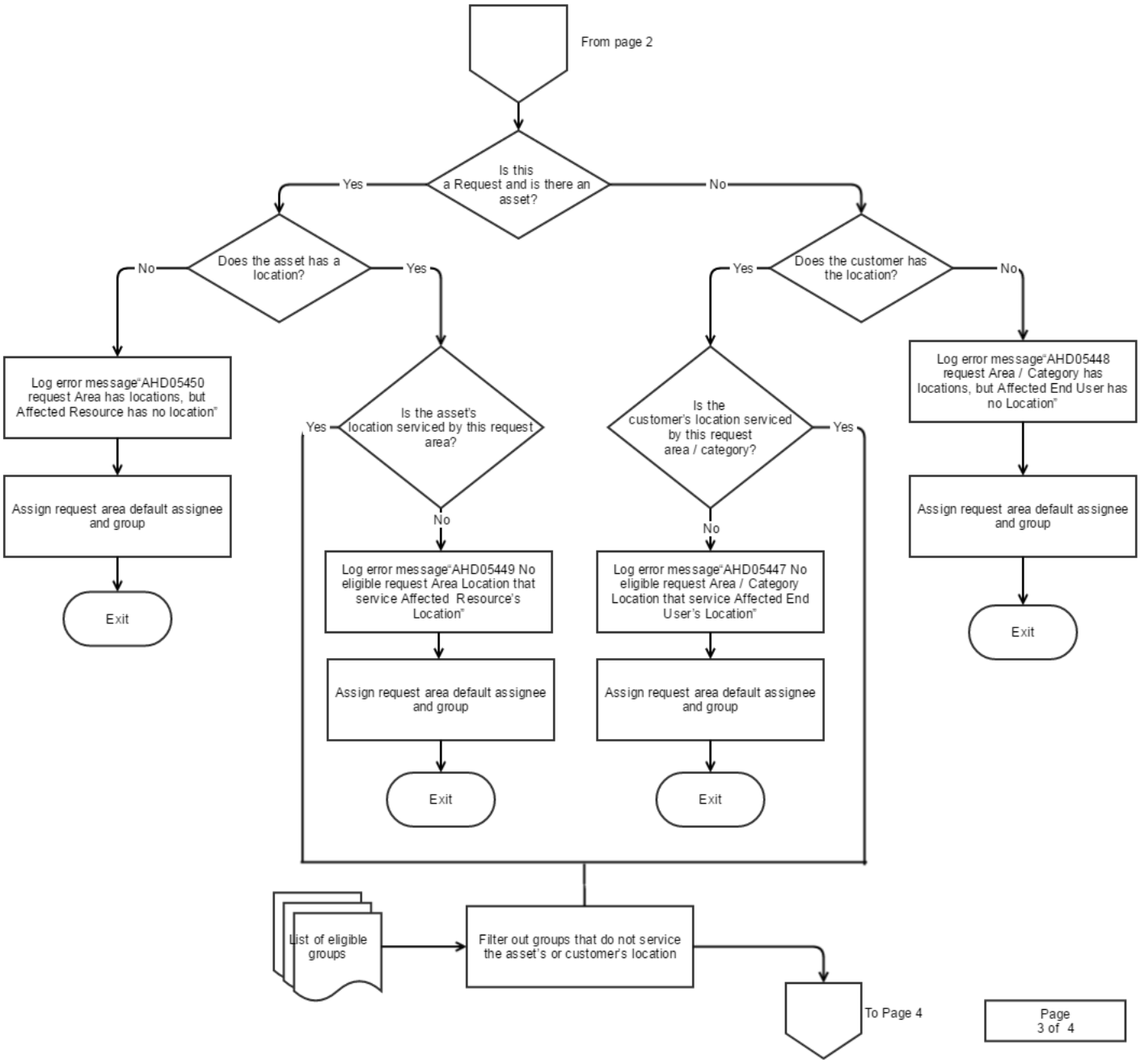
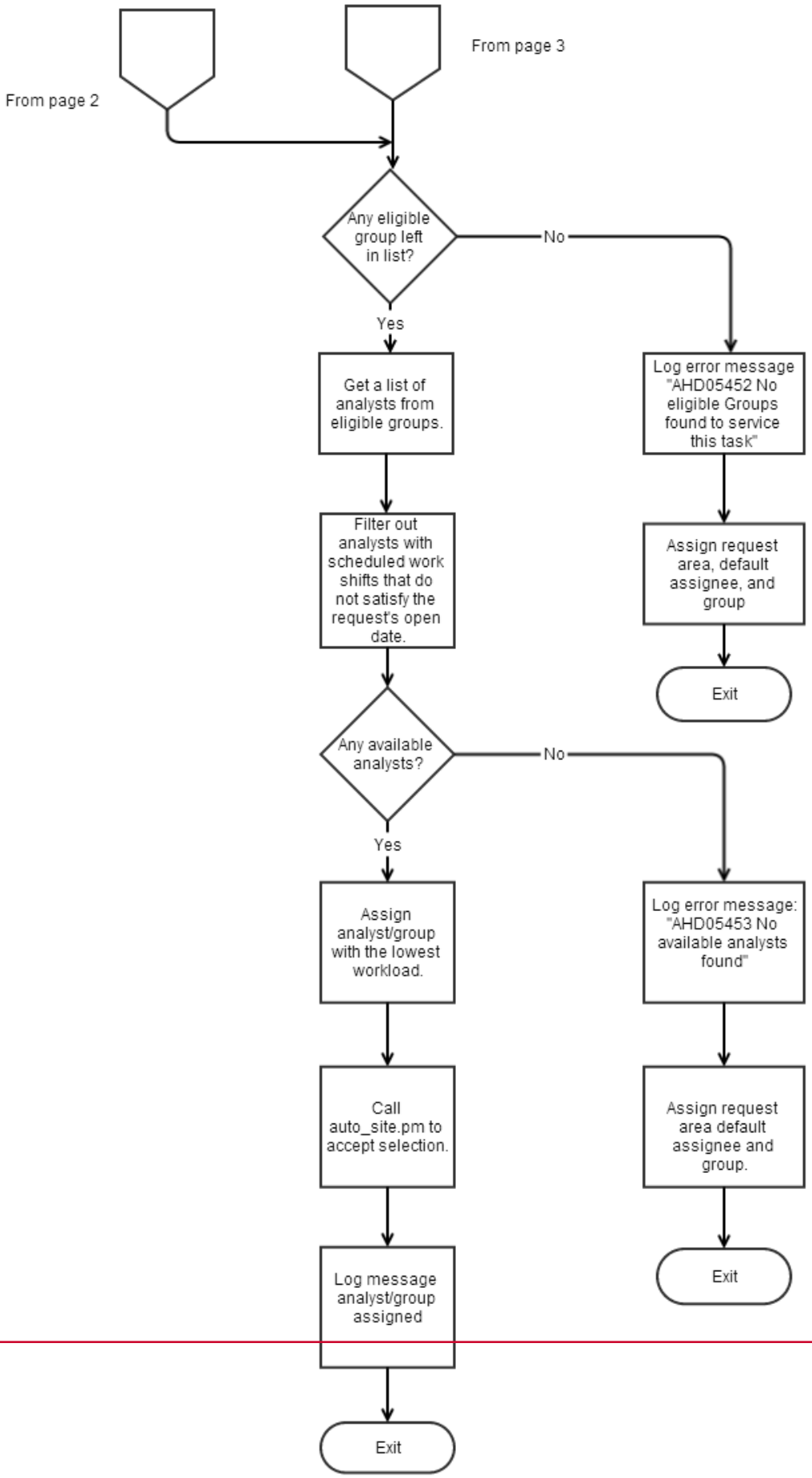


Figure 18: Diagram depicting ticket workflow Page 4



How Workflow Tasks are Assigned

Auto Assignment assigns workflow tasks as follows:

1. Auto assignment is invoked when the status of a workflow task changes to pending. If the workflow template that the workflow task was created from is not enabled for auto assignment, processing stops. If the parent change order category or Issue category is not enabled for auto assignment, processing stops.
2. Auto assignment checks to determine if Autoasg_override is installed. If not installed and the task has an assignee or group, processing stops.
3. The workflow template that the workflow task was created from is checked to see if any contacts are associated with it. If there are no contacts, processing stops.
4. Auto assignment builds a list of contacts that are members of the groups that are currently associated to the workflow template. The workflow template must be the template from which the workflow task was created. Any contacts in this list that are groups are filtered out.
5. All of the remaining contacts are ranked according to the number of active change order tasks or the corresponding issue tasks.
6. The contact and associated group with the least number of active tasks is assigned to the task.
   For more information, see Work Process Flow Diagram.

Work Process Flow Diagram

Figure 19: Diagram Showing Workflow Process Page 1

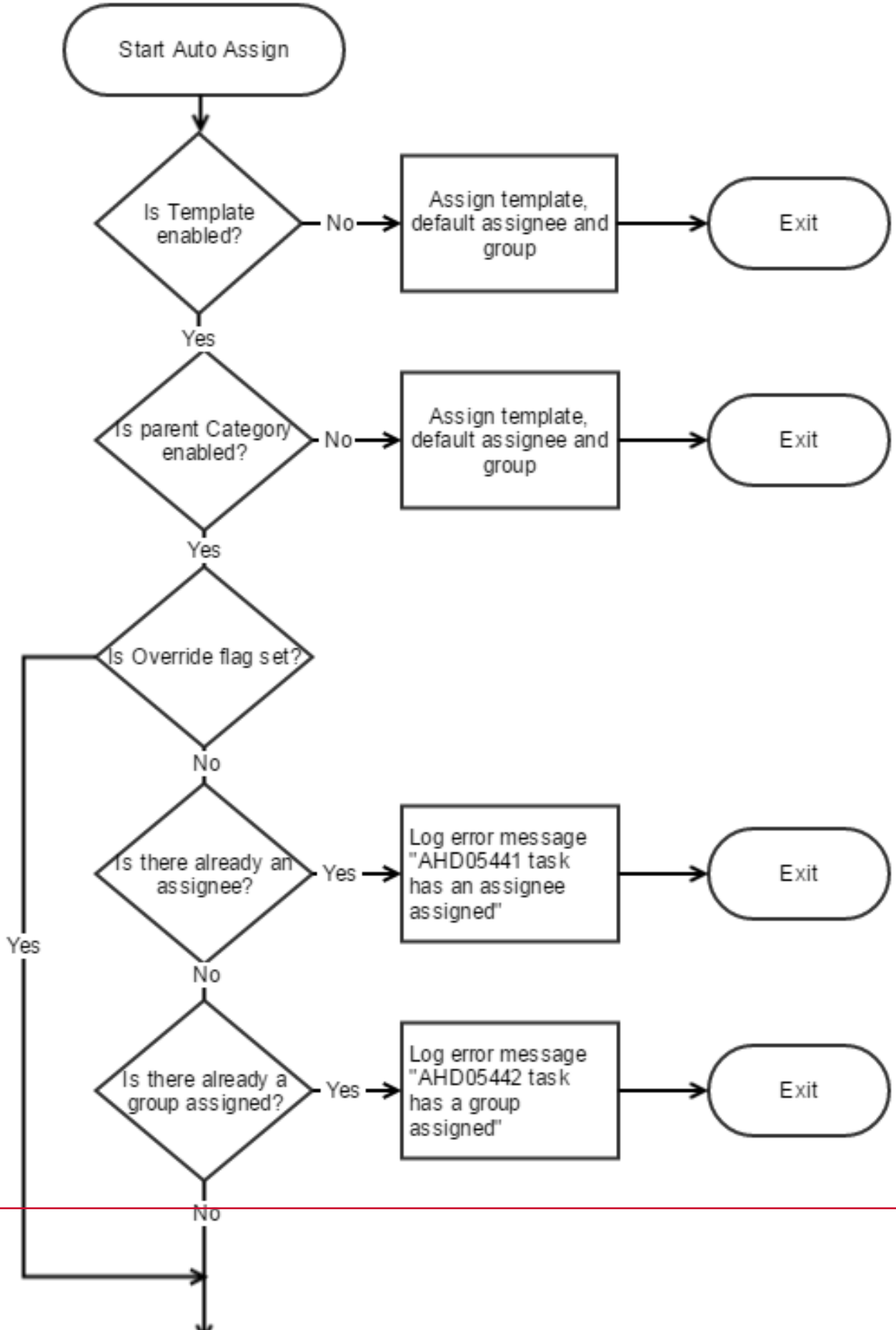
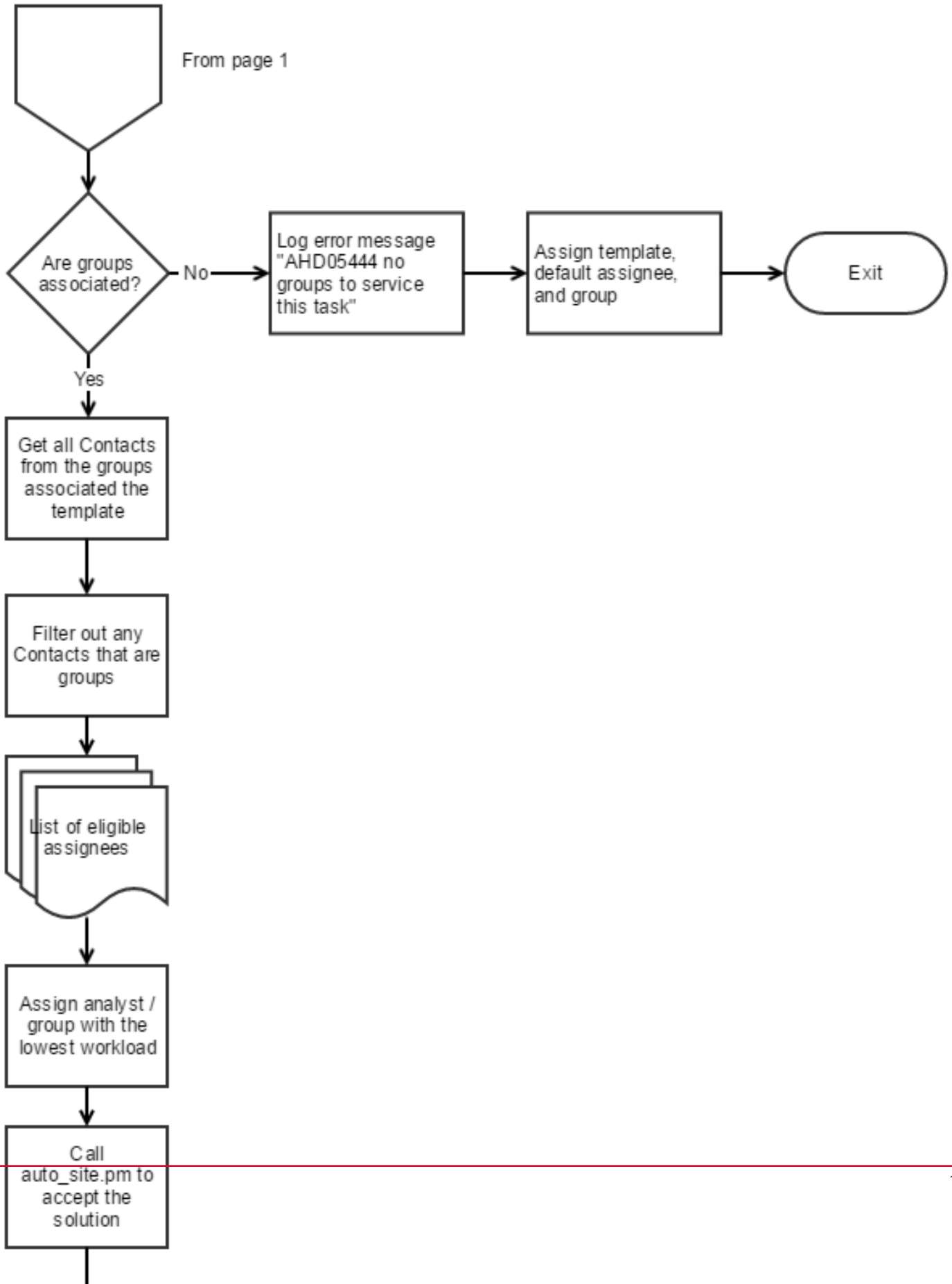


Figure 20: Diagram Showing Workflow Process Page 2



How Auto Assignment Assigns Tickets - Flow Diagram

Figure 21: Diagram depicting Workflow process page 1

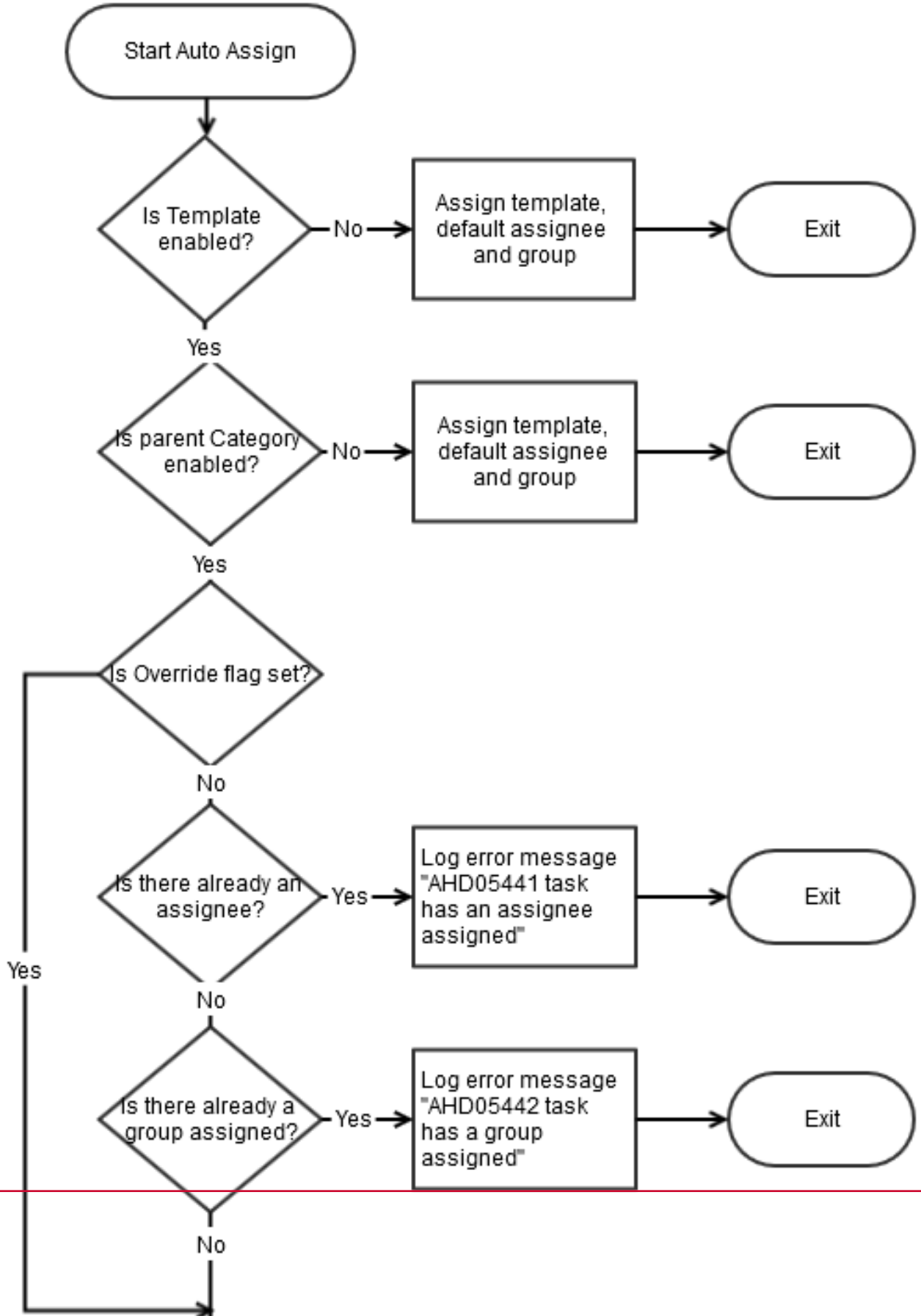


Figure 22: Diagram depicting workflow process page 2

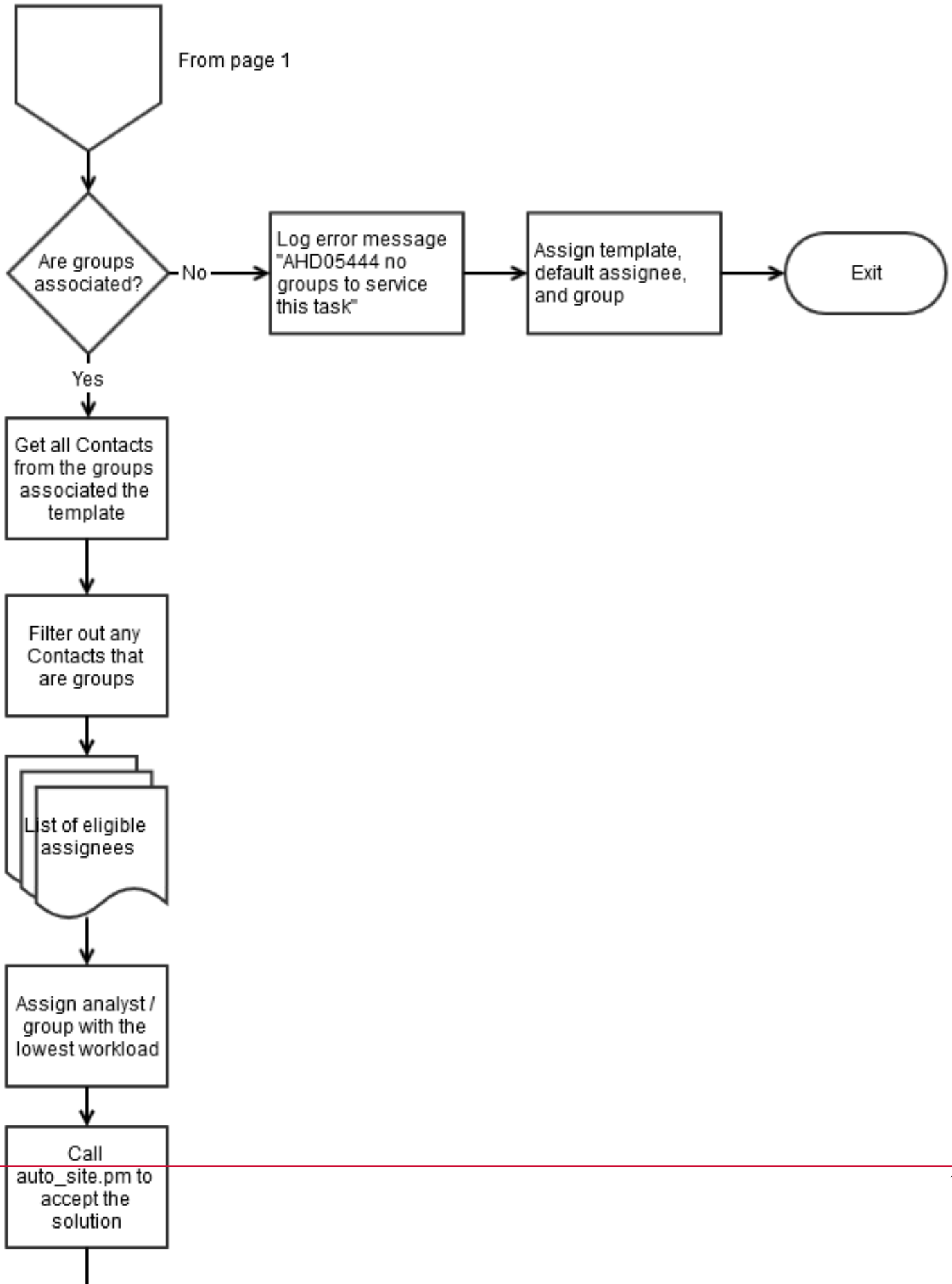


Figure 23: Diagram depicting ticket workflow Page 2

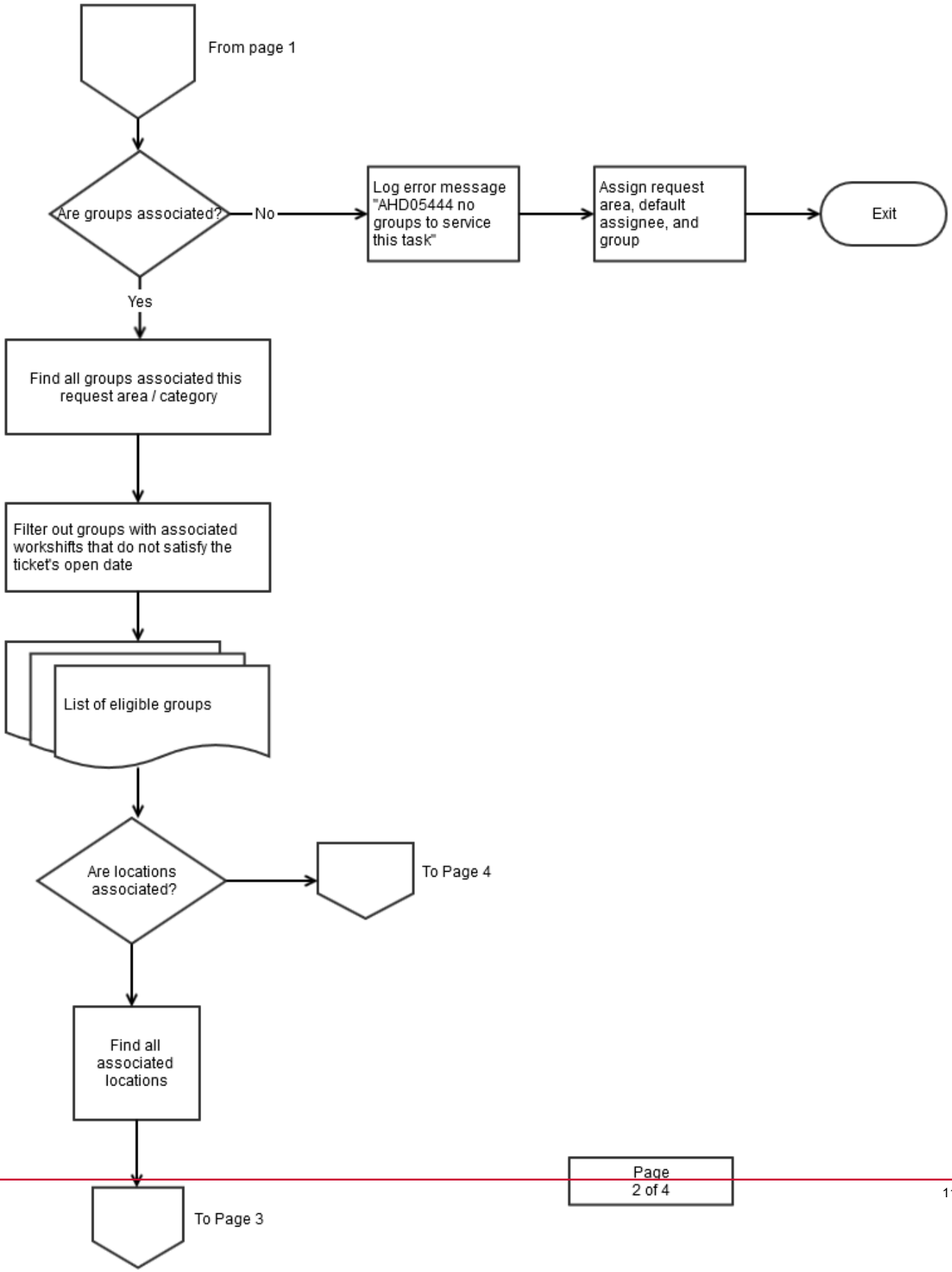
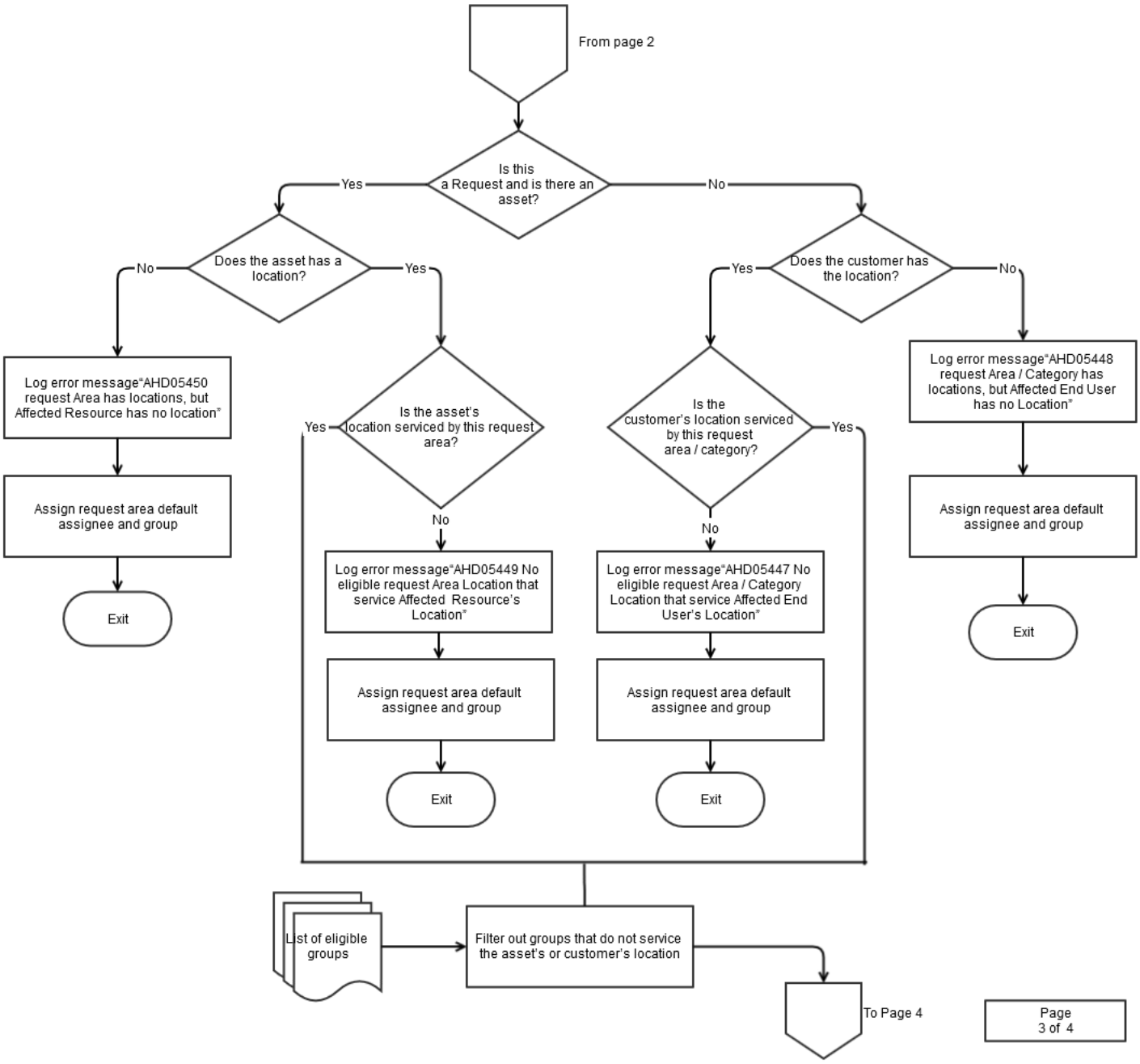
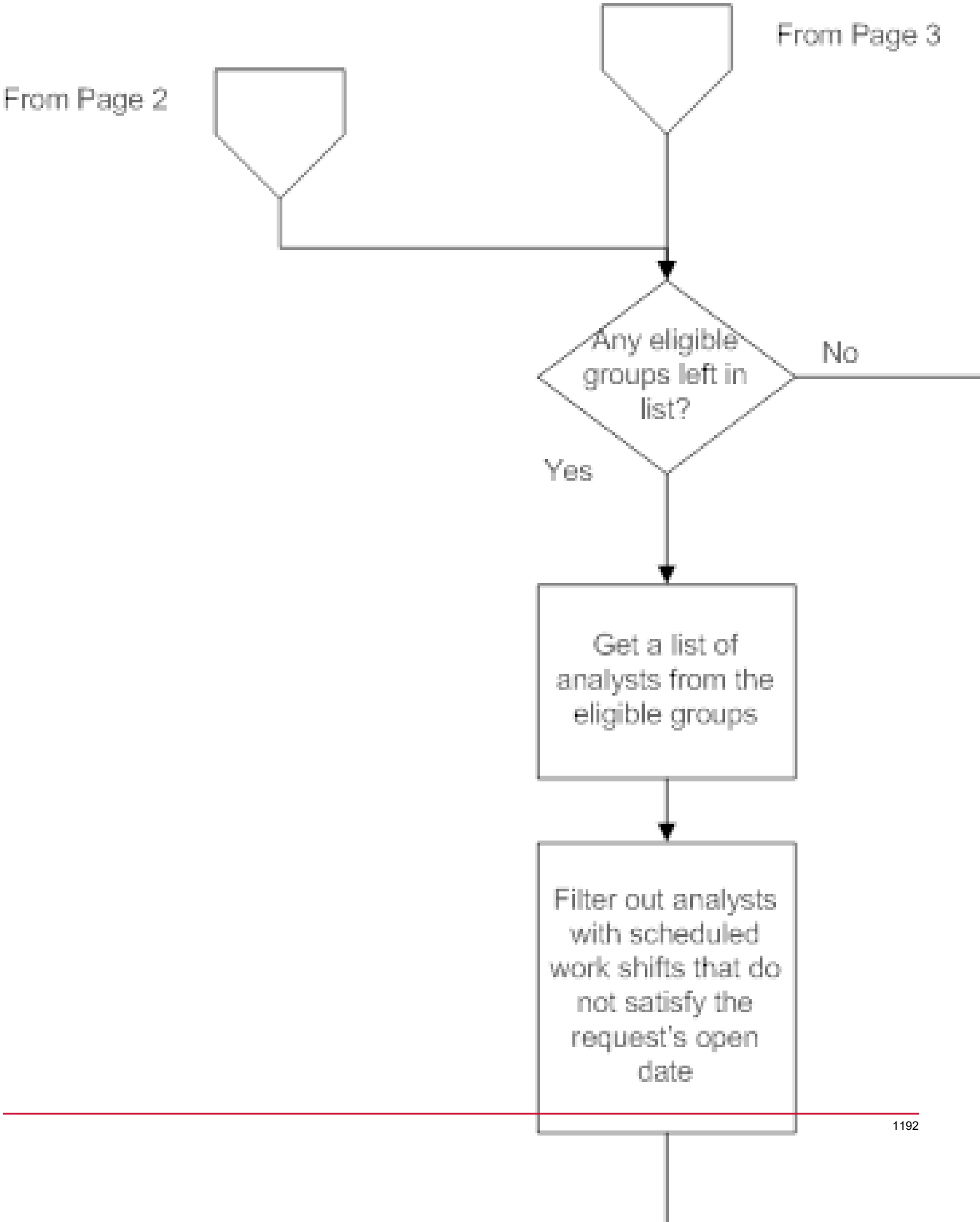


Figure 24: Diagram depicting ticket workflow page 3




From Page 3
From Page 2
Any eligible groups left in list?
No
Yes
Get a list of analysts from the eligible groups
Filter out analysts with scheduled work shifts that do not satisfy the request's open date

How to Begin Implementing Auto Assignment

This article contains the following topics:

Follow these guidelines to begin implementing auto assignment for selected analyst groups and analysts:

1. Identify one or more areas or categories for which you want to enable auto assignment.

   NOTE
   To review your site area and category configurations, examine the settings in the CA SDM web interface.
2. By default, auto assignment is disabled. Enable it only for the areas and categories where you want to use it.
3. Build the relationships between an identified area or category and the analyst groups that can be assigned tickets through that area or category.
4. Mark individual members of the analyst groups as available.

Areas and Categories

NOTE
When you configure areas and categories, consider setting default groups and assignees.

To configure the auto assignment of request, incident, and problem tickets, use the following options in the Auto Assignment tab of Request/Incident/Problem Area Detail page:

- Update Groups
- Update Locations
- Update Workshifts

NOTE
The check box for enabling auto assignment is also on the Auto Assignment tab of each of these pages. It is visible only while the page is in edit mode.

The following pages on the interface provide the same controls for configuring auto assignment of change orders and issues:

- The Change Category Detail
- The Issue Category Detail

Analyst Groups

To configure auto assignment you must, at a minimum, define the relationships between analyst groups and areas or categories. Assignees are chosen from groups that meet *all* of your specified auto assignment criteria. If no additional constraints are defined, tickets are auto assigned to the group member with the fewest active tickets.

If no groups are associated with an area or category, the default group and assignee are assigned. If these defaults are not defined, the ticket is left for manual assignment.

You can maintain the relationships between groups and areas or categories on the area or category detail page.

Alternatively, use the following options to maintain the same relationships on the Auto Assignment tab of the Group Detail page:

- Update Request Areas
- Update Change Categories
- Update Issue Categories
- Update Locations

Analysts

Fields on the Analyst Detail page help determine whether the analyst is eligible for auto assignment.

Analysts are eligible for auto assignment only if they are marked as available.

The Analyst Detail page provides an Available check box that enables auto assignment.

Consider allowing analysts to control their own auto assignment availability. You can monitor analyst availability by using stored queries.

NOTE
The Available check box is not considered during auto assignment of workflow tasks.

The Work Schedule field allows analysts to have tickets auto-assigned only during their scheduled workshift. Analysts that are marked as available and are without work schedule are eligible for auto assignment at any time. However, no other constraints must result in an ineligible status.

Stored Queries

Two stored queries are provided for monitoring the availability of analysts. The CA SDM managers can add the following queries to their Scoreboard:

- **Available Analysts** -- Analysts that are marked as available for auto assignment
- **Unavailable Analysts** -- Analysts that are marked as unavailable for auto assignment

How to Auto Assign Tickets to a Group and Not to Contacts Within the Group

Auto assignment assigns tickets to contacts that have the Available option selected in the contact record. However, you can auto assign incidents/issues/change orders/requests to a group and not to contacts within the group. The NX_AUTOASG_GROUP_ONLY option controls the auto assignment behavior for groups. You install this option to auto assign the tickets to the group instead of individual contacts. NX_AUTOASG_GROUP_ONLY is not available on the web interface, you install it from the command prompt.

Follow these steps:

1. Verify that your CA SDM installation is at a minimum level of Release 12.9 Cumulative 2 patch.
2. Open the command prompt on any CA SDM server.
3. Run the following command:
   ```
   pdm_options_mgr -c -s AUTOASG_GROUP_ONLY -v 1 -a pdm_option.inst
   ```
   By default, new options that you add to the nx.env file are not saved after you run pdm_configure. You can save the changes permanently by specifying the -t option as follows:
   ```
   pdm_options_mgr -c -s AUTOASG_GROUP_ONLY -v 1 -a pdm_option.inst -t
   ```
   The commands update the following files in CA SDM with the new option:
   - Windows -- NX_ROOT/NX.env and NX_ROOT\pdmconf\nx.env.nt.tpl
   - UNIX/Linux -- NX_ROOT/pdmconf/NX.env.tpl
4. Open the NX.env file in the CA SDM installation folder and search for the variable @NX_AUTOASG_GROUP_ONLY=1. Typically, the variable is found at the end of the file.
5. Open the file nx.env.nt.tpl present under NX_ROOT/pdmconf and search for the @NX_AUTOASG_GROUP_ONLY=1 option.
6. Restart the CA SDM Daemon Server service.
   CA SDM auto assigns the tickets to the group only.
7. Restart the CA SDM servers.

Auto Assignment by Location

If your service area is large and it consists of many locations that service different customer communities. You can use location as a factor in your auto assignment configuration as follows:

- **Location Assigned to an Area or Category** -- If you assign a location to an area or a category, tickets are auto assigned only when a matching location is found. For example, a request ticket is auto assigned if there is an eligible analyst at the following locations:
  a. The affected asset location.
  b. The affected customer location.

  If the affected asset or customer has no specified location, the request is assigned to the default group and assignee. If these defaults are not defined, the request is left for manual assignment. You can use the area or category detail pages to maintain relationships between locations and areas or categories.
- **Location Assigned to a Group** -- When a location is associated with a group, only members of that group are eligible for auto assignment of tickets. Further, only the tickets that pertain to the location of the member can be assigned. You can use group detail pages to maintain relationships between groups and locations.

To maintain location relationships with areas, categories, or groups, use the following options on the Auto Assignment tab of Location Detail page:

- Update Request Areas
- Update Change Categories
- Update Issue Categories
- Update Groups

Examples: Use Location in Auto Assignment Configuration

The following are examples of how you can use location in your auto assignment configuration:

- **Auto assign tickets only at a specified location** -- Tickets from other locations receive the default group and assignee, or are left for manual assignment. For example, you can have many users at your company headquarters, and smaller groups of users at regional offices. An analyst group located at the headquarters services their local users, while mobile analyst groups visit the regional offices. You can configure auto assignment of tickets to the headquarters analysts only, and manually assign tickets to the mobile analysts.
- **Auto assign tickets by user or asset location** -- You can restrict auto assignment eligibility to analyst groups at locations that match the affected user location, or the affected asset location. For example, your organization can have many offices, but only the groups that are located in a particular office handle tickets for the office. You can relate each group to appropriate areas or categories *and* to the appropriate location. The auto assignment logic selects eligible analysts only from groups at the correct location.

Auto Assignment by Workshift

You can constrain auto assignment by relating an area or category to a workshift. The workshift determines the timeframe within which tickets are eligible for auto assignment. Tickets opened outside the workshift are assigned to the default group and analyst, or left for manual assignment.

You can also use workshifts to control group and analyst eligibility for assignment:

- If you assign a work schedule to a group. Analysts in that group are eligible for auto assignment only to tickets opened during their work schedule.

  NOTE
  The workshift for a group is specified in the Work Schedule field of the Group Detail page.
- If you assign a work schedule to an individual analyst. The analyst is eligible for auto assignment only to tickets opened during that work schedule.

  NOTE
  The workshift for an analyst is specified in the Work Schedule field of the Analyst Detail page.

When a ticket is created, the group that meets the assignment eligibility criteria is identified. Within the group, the analyst with the fewest active tickets is identified. If an appropriate analyst is not identified, the ticket is assigned to the default group and assignee. If these defaults are not defined, the ticket is left for manual assignment.

For workflow tasks associated with change orders or issues, auto assignment uses a simpler selection strategy. Assignees are selected from the groups that are associated with workflow templates. Workflow task assignees can be of any contact type except group. When a task changes to pending status, the contact that has the fewest change order or issue workflow tasks is selected. To prevent unwanted results, if the parent change order category or issue category is disabled for auto assignment, tasks are not assigned automatically. Workflow tasks may encompass individuals external to service desk. So, relying on them to reflect their availability with the available flag is not recommended.

You can use the area or category detail pages to relate a workshift to an area or category. Or, you can use the following controls on the Workshift Detail page:

- Update Request Areas
- Update Change Categories
- Update Issue Categories

Examples: Use Workshifts in Auto Assignment Configuration

The following examples show how you can use workshifts in your auto assignment configuration:

- **Auto assign tickets to the shift on duty** -- Suppose that your service desk operates 24 hours per day and you have tickets about network outage problems. You may want to configure auto assignment of such tickets to the shift on duty when the ticket is opened.
- **Allow auto assignment only during a specified shift** -- You may want to restrict further auto assignment of tickets in some areas or categories. For example, if analysts are on duty only during day shift, you may auto assign application problems only during the day shift.

Configuration Item-Based Auto Assignments

Configuration item-based auto assignment lets you create group-specific assignments that apply to specific scenarios. You can specify that for a particular Request/Incident/Problem area, the value of a configuration item attribute controls the assignment of the ticket.

Configuration item-based and location-based auto assignments are exclusive options because you can select only one algorithm for use. Configuration item-based and location-based auto assignment modes both serve to assign tickets. However, configuration item-based auto assignment is distinct, because it also reassigns tickets after changing the Request/Incident/Problem Area.

Example: Request/Incident/Problem Area Assigns Tickets to a Group

Suppose that you configure the Network Area to assign configuration item-based auto assignments using the network_contact_uuid attribute as the Assignable CI Attribute value. Tickets that are opened for the Network Area are assigned automatically to the group that is specified for Network Operations in the corresponding CI. If no CI is associated with the ticket, the Network Operations value of the CI is blank, or a group is not specified, the ticket is not assigned. In such cases, the system acts in accordance with the Option Manager option, Area Defaults, and assigns the ticket using the Group and Assignee fields of the Category.

How Configuration Item-Based Auto Assignment Works

Request/Incident/Problem tickets must specify the following for configuration item-based auto assignment to occur:

- A configuration item and an Area.
- The Area must have the Auto Assignment option set to Configuration Item Based.

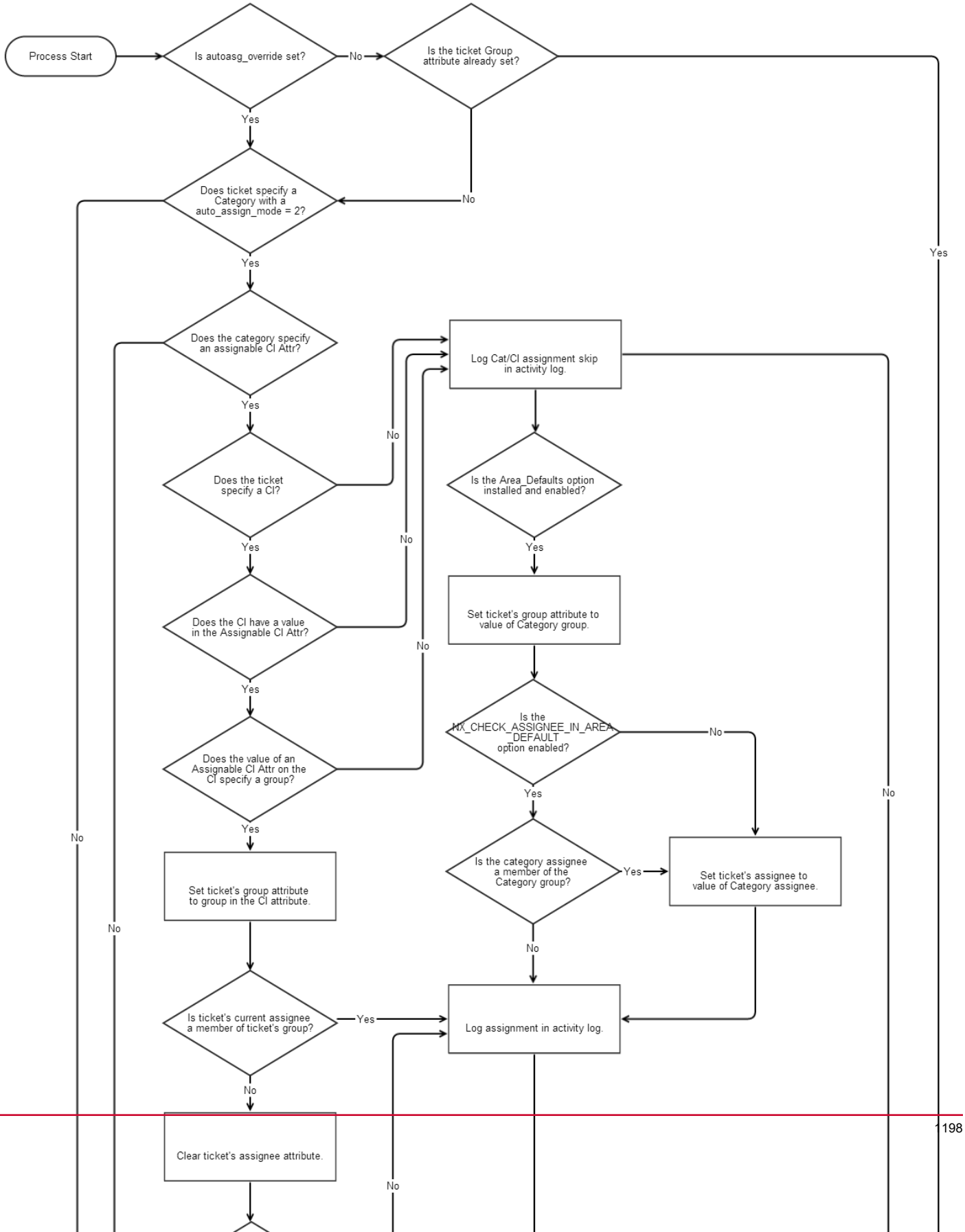
When an analyst creates and assigns a ticket to this Area, or changes the Area on an existing ticket, CA SDM examines the Assignable CI Attribute field of the Area. CA SDM uses the value of Assignable CI Attribute as the name of an attribute and then attempts to find an identically named attribute on the configuration item that is associated with the ticket. If the attribute on the configuration item includes a group, the ticket is assigned to that group.

The following diagram describes the process for configuration item-based auto assignment in more detail:

1. CA SDM checks for the following:
   a. Is the autoasg_override option set?
   b. Does a ticket specify an Area with its Auto Assignment auto_assign_mode=2?
   c. Does the category specify an assignable CI attribute?
   d. Does the ticket specify a configuration item?
   e. Does the configuration item have an assignable CI attribute value?
   f. Does the value specify a group in the CI attribute?
   g. Is the assignee of the ticket a member of the ticket group?
2. When the answers are positive for all questions in the previous step, CA SDM sets the group attribute of the ticket to the group attribute of the CI, and logs the assignment in the activity log.

   NOTE
   The diagram shows how Configuration Item-Based Auto Assignment uses the NX_CHECK_ASSIGNEE_IN_AREA_DEFAULTS variable to determine if the Area option is enabled. NX_CHECK_ASSIGNEE_IN_AREA_DEFAULTS is a variable in the NX.env file, which is located in the $NX_ROOT\ directory.
3. CA SDM assigns the ticket to the group.

Figure 25: Configuration Item - Based Auto Configuration



Enable Configuration Item-Based Auto Assignments

Configuration item-based auto assignment lets you create group-specific assignments that apply to specific scenarios. You can specify that for Request/Incident/Problem tickets opened for a particular Area, the value of an attribute of the configuration item associated with the ticket controls its assignment. Configuration item-based auto assignment reassigns tickets whenever the Request/Incident/Problem Area of a ticket or configuration item changes.

> **NOTE**
> The Options Manager autoasg_override option controls the circumstances under which auto assignment takes place. When you set this option to 1, CA SDM ignores any existing assignee and group settings and auto assigns tickets in all cases. If you want CA SDM to auto assign tickets only if they are not already assigned, set the option value to 0.

Follow these steps:

1. On the Administration tab, browse to Service Desk, Request/Incident/Problems, Areas.
   The Request/Incident/Problem Area List appears.
2. Click the area to edit.
   The Area Detail page appears.
3. Click Edit.
   The Update Area page appears.
4. Select the Auto Assignment tab, and complete the fields as follows:
   - **Auto Assignment Mode**
     Specifies how auto-assignment occurs. You use the Configuration Item Based option to base the auto assignment on the CI Assignable Attribute value.
   - **Assignable CI Attribute**
     Specifies the configuration item attribute to use for the group assignment. You can enter a value directly or click the magnifier to search for an attribute.

   Click Save.
   Auto assignment is enabled. CA SDM performs configuration item-based auto assignments using the attribute that you specified in the Assignable CI Attribute field.

Activity Logging

Auto assignment logs information as events in the ticket's activity log. The success or failure of auto assignment is logged, and any anomalies that may have occurred during processing.

Enable Activity Logging for Additional Attributes

Activity logs display changes attribute values of objects. Objects such as all ticket types, classic workflow tasks, Surveys, Contacts, and Configuration Items feature activity logging. You can enable activity logging for additional attributes In WSP.

Note: You can view activity log simple attributes, including strings, integers, date, duration, and SREL types. You cannot view activity logs for list types, such as QRELs and BRELs.

Follow these steps:

1. Open WSP and start the Schema Designer.
2. Select the table that you want to modify, and add additional attributes.
3. For each attribute, add +AUDITLOG() to the site-defined UI_INFO field.

   > **WARNING**
   > When you add the AUDITLOG flag, you *must* also remove the val_fieldupdate_site() function to prevent duplicate activity logs.
4. Save and publish the schema
5. Open WSP again and add the new attributes to the appropriate detail forms.

6. Open CA SDM and define an Activity Association for the additional attributes.
7. Test your changes.
   For example, open a previously-saved instance of the object and modify the affected attributes to verify that the appropriate activity logs appear.

Auto Assignment Tracing

In a complex auto assignment implementation, auto assignment may not make the decisions that were expected. Tracing can be enabled to help you understand the processing flow. Tracing is normally turned off but when turned on, numerous messages are written to the stdlog files in $NX_ROOT\log that describe the various decisions made by auto assignment.

Use caution when implementing this in a high-volume installation, as the number of messages generated could cause the log files to grow and eventually wrap. On very active systems, performance degradation may occur. Tracing is controlled with the pdm_logstat utility. The parameters used by this utility are case-sensitive. Be sure to enter them as shown.

To turn tracing on, run the following command on each CA SDM the server:

```
pdm_logstat  -f auto.pm milestone
```

To turn off tracing, run the following command on each CA SDM the server:

```
pdm_logstat  -f auto.pm
```

Create a Stored Query

CA SDM supports use of stored queries to produce two kinds of data:

- Scoreboard Queries let you configure the web interface scoreboard by adding counter fields for items of interest.
- KPI Queries let you retrieve historical information for a specified time period on the value of a counter for use in web-based reporting.

Stored queries apply to all service desk models. Stored queries are not intended to bring users to a new page.

Administrators define the stored queries that are available to users. You can use the predefined stored queries that are installed with CA SDM, or define your own. You can define stored queries by using the administrative function of the web interface.

Consider the following information about stored query definitions:

- A valid stored query clause is integrated into an appropriate SELECT statement and passed to the underlying DBMS engine for processing. To develop the SELECT statements, see the object definition files in the following directories:
  - Windows: installation-directory\bopcfg\majic directory
  - UNIX: $NX_ROOT/bopcfg/majic directory
- Stored queries use the object and attributes statements for building the SQL WHERE clause instead of the field names at the database level.
- CA SDM does not support Cartesian product joins for stored queries. To help ensure that your stored query does not produce a Cartesian product join, enter the following command appropriate for your system.
  - Windows: bop_cmd -f *installation-directory*\bopcfg\interp\bop_diag.frg "check_queries()"
  - UNIX: bop_cmd -f $NX_ROOT/bopcfg/interp/bop_diag.frg "check_queries()"

  Update any queries that are flagged as a result to avoid Cartesian product joins.
- The URL stored query is an option offered while creating stored queries. The URL stored query runs a query and returns numerical results that only work with Knowledge Management.

Stored queries are supported for the following:

- **Scoreboard Queries** -- Allow you to configure the Web Interface scoreboard by adding counter fields to monitor items of interest.
- **KPI Queries** -- Allow you to retrieve historical information for a specified time period on the value of a counter for use in web-based reporting.
- **Automated Policy Queries** -- Allow you to retrieve information about the documents that are flagged for correction and promoted to publication or retirement during the various stages of the document lifecycle.

You can create stored queries for use in configuring the Scoreboard, or defining KPIs, or both.

> **NOTE**
> When you save a stored query, validation testing is performed on parameters. If you attempt to save a stored query with an invalid parameter (for example, datespan @cnt,@root) the save fails and an error message appears.

Follow these steps:

1. Click CA SDM, Application Data, Stored Queries on the Administration tab.
   The Stored Query List appears.
2. Click Create New.
   The Create New Stored Query page appears.
3. Complete the stored query fields as appropriate.
4. Click Save.
   The Create Stored Query Detail page displays your entries.

Stored Query Fields

You can use the fields on the Stored Query pages to define and edit query definitions.

> **NOTE**
>
> If you are using multi-tenancy, a tenant column appears on the list page.

The following fields require explanation:

- **Code**
  Defines a unique identifier for the stored query.
- **Scoreboard Usage**
  Specifies whether the query is available for configuring the Scoreboard.
- **KPI Usage**
  Specifies whether the query is available for configuring a KPI
- **Record Status**
  Specifies whether the stored query is active or inactive.
- **Type**
  Specifies the type of CA SDM object for which the stored query produces data (Contact, Request, Change Order, and so on).
- **Label**
  Defines the text string to be displayed on the Stored Query List page.
- **Where Clause**
  The SQL WHERE clause for the query. A WHERE clause includes a comparison predicate, which restricts the number of rows returned by the query.

Manage Roles

Contents

Roles are the primary records that control CA SDM security and user interface navigation. Each role defines a focused view of the system by exposing only the functionality necessary for users to perform the tasks typically assigned to the role they perform within their business organization.

A user's default role determines the system view that is presented upon login. Users with multiple role assignments can switch from one role to another to see different views of the system without having to log out and log back in again.

Predefined Roles

You can use the predefined roles in their default configuration, modify them to meet your business requirements, or create new roles.

The following table describes the predefined roles installed with CA SDM. These roles are designed to align with ITIL v3 best practices, and thereby reduce the amount of site-specific modifications required to bring your IT organization into ITIL compliance.

CA SDM only supports ITIL, and the CA SDM documentation is ITIL-oriented. For more information, see ITIL Configuration.

Role Type	Role Name	Description
End Users	Configuration Viewer	Performs basic CI viewing and research tasks from inside your organization.
	Customer	Performs basic self-service tasks from outside your organization.
	Employee	Performs basic self-service tasks from inside your organization.
Analysts	Configuration Analyst	Performs tasks within the configuration item life cycle process and second-line CMDB support within your organization.
	Customer Service Representative	Supports users external to your organization, most often customers.
	Knowledge Analyst	Performs tasks within the knowledge management life cycle process.
	Level 1 Analyst	Provides first-line support within your organization.
	Level 2 Analyst	Provides second-line support within your organization, which requires more advanced subject matter expertise.
	Support Automation Analyst	Provides first-line support within your live assistance environment.
	Vendor Analyst	Supports a limited segment of your IT environment from outside your organization, such as vendor-specific hardware.
Managers	Change Manager	Manages the change order process, but typically not the analysts who work on change order tickets.
	Customer Service Manager	Manages Customer Service Representatives and the external support process.

	Incident Manager	Manages the incident process, but typically not the analysts who work on incident tickets.
	Knowledge Manager	Supervises Knowledge Analysts, knowledge document reassignments and escalations, and day-to-day knowledge administration.
	Problem Manager	Manages the problem process, but typically not the analysts who work on problem tickets.
	Service Desk Manager	Handles escalations and supervises Level 1 Analysts. Also may manage overall service desk operations.
Administrators	Administrator	Performs administrative tasks throughout your CA SDM and Knowledge Management implementation. This role typically installs, configures, and integrates the products.
	Configuration Administrator	Performs administrative tasks related to your CA CMDB implementation. This role typically administers CMDB and configuration item infrastructure and data structures.
	Knowledge Management Administrator	Configures and monitors knowledge management settings.
	Service Desk Administrator	Performs administrative tasks on data and processes, such as creating and updating categories, contacts, service types, root causes, and so on.
	Support Automation Administrator	Performs administrative tasks related to your Support Automation environment, such as configuring queues and analyst tool permissions.
	System Administrator	Performs administrative tasks related to your CA SDM implementation, configuration and adaptation, such as setting options, configuring integrations and modifying web forms.
	Tenant Administrator	Performs multi-tenancy administrative tasks specific to a particular tenant or supporting organization.

Create a Role

Roles are the primary records that control security and user interface navigation. Each role defines a focused view of the system by exposing only the functionality necessary for users to perform the tasks typically assigned to the role they perform within their business organization.

Predefined roles are provided that are designed to align with ITIL v3 Best Practices and thereby reduce the amount of site-specific customization required to bring your IT organization into ITIL compliance. You can use the predefined roles in their default configuration, modify them to meet your business requirements, or create new roles.

Administrators can create roles to meet site-specific business requirements.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. Select Security and Role Management, Role Management, Role List on the Administration tab.
2. Click Create New.
3. Complete the following fields:
   - **Code**
     Specifies the code that identifies the role to the system.

     > **NOTE**
     > After you save the record, this field value cannot be changed.
   - **Record Status**
     Indicates whether the role is Active or Inactive.
   - **Default?**
     Indicates whether this role is the default role.
   - **Customization Form Group**
     Specifies a predefined or custom form group.
   - **Preferred Document**
     Specifies the document used by this role for entering tickets into the system.

   Click Save.
   The role definition is saved and the Role Detail page appears.
4. Update the information in the role tabs.

Role Tabs

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

The following tabs are available on the Role Detail and Update Role pages:

- **Authorization**
  Allows you to define the authorization level assigned to the role. Complete the following fields as appropriate:
  - **Grant Level**
    Specifies the access permission that users assigned to this role can grant to others. The grant level is used to determine which access types a user can grant to another user. You can assign an access type to the contact record of another user only if the access level of the access type you are attempting to assign is ranked the same as or lower than the grant level for your own access type. These levels are ranked as follows:
    - Admin (highest)
    - Analyst
    - Cust/Emp
    - None (lowest)
  - **View Internal Logs?**
    Allows users assigned to this role to view internal log files.
  - **Data Partition Name**

The name of the data partition assigned to this role. Data partitions are subsets of the database with restricted access to data records, based on their content. You restrict that access by defining a set of constraints for each data partition.
Enter the data partition name directly into the field, or click the search icon to search for a data partition name.

– **Override Contact Data Partition?**
Select this option if you want the data partition defined for the access type to override the data partition defined on the contact record. This option can help prevent conflicts from arising between data partitions specified on the contact records and data partitions specified on the role record.

– **Multi-Tenancy Settings**
The following options apply to systems where multi-tenancy is enabled:
Update Public (Service Provider only)
Select this option if you want users that are assigned to this role to update data for all tenants and non-tenanted data.
Tenant Access
Select the tenant or tenant group that you want users assigned to this role to be able to read. If you select Single Tenant, you can enter the name of the tenant that you want this role to read. You can assign the following associations to roles:

- **Same As Tenant Access (Tenant Write Access Only)**
  Sets Tenant Write Access to be the same as the Tenant Access setting. Default for Tenant Write Access and only valid for Tenant Write Access.
- **All Tenants**
  Removes tenant restrictions. CA SDM allows a user in a role with this access to view any object in the database (read access) or create and update (write access) any tenanted object in the database. When users with All Tenant access create an object, CA SDM requires that they select the tenant of the new object.
- **Single Tenant**
  Sets a role's tenant access to a named tenant. When this option is selected, a second field appears on the web UI that allows selection of a specific tenant. CA SDM restricts a user in a role with this access to view (read access) or create and update (write access) only those objects associated with the named tenant. This selection is valid for either Tenant Access or Tenant Write Access.
- **Tenant Group**
  Sets a role's tenant access to a user-defined or system-maintained tenant group. When the Tenant Group option is selected, a second field appears on the web UI that allows selection of a specific tenant group. CA SDM restricts a user with the role to view (read access) or create and update (write access) only those objects associated with one of the tenants in the group. When a user with tenant group access creates an object, CA SDM requires that they select the tenant for the new object. This selection is valid for either Tenant Access or Tenant Write Access.
- **Contact's Tenant**
  Sets a role's tenant access to the tenant of the contact using it. CA SDM restricts a user in a role with this access to view (read access) or create and update (write access) only those objects associated with their own tenant. This selection is valid for either Tenant Access or Tenant Write Access.
- **Contact's Tenant Group (Analyst Only)**
  Sets an analyst's role access to the tenant group that the analyst works with, as specified on the analyst's contact record. If the user with the role is not an analyst, this selection has the same effect as Contact's Tenant. It is valid for either Tenant Access or Tenant Write Access.
- **Contact's Subtenant Group**
  Sets a role's tenant access to the Subtenant group of the contact using it. CA SDM restricts a user in a role with this access to view (read access) or create and update (write access) only those objects associated with their own Subtenant group. This selection is valid for either Tenant Access or Tenant Write Access.
- **Contact's Supertenant Group**

Sets a role's tenant access to the Supertenant group of the contact using it. CA SDM restricts a user in a role with this access to view (read access) or create and update (write access) only those objects associated with their own Supertenant group. This selection is valid for either Tenant Access or Tenant Write Access.

- **Contact's Related Tenant Group**
  Sets a role's tenant access to the Related Tenants Group of the contact using it. CA SDM restricts a user in a role with this access to view (read access) or create and update (write access) only those objects associated with their own Related Tenants Group. This selection is valid for either Tenant Access or Tenant Write Access.

All users can view public data, regardless of their current role's access rights. The Update Public check box controls whether a service provider user in the role has the authorization to create or update public data. Tenant users (users belonging to a tenant other than the service provider) cannot update public data, regardless of their role.

– **Tenant Write Access**
Select the tenant or tenant group that you want users assigned to this role to be able to create and update. If you select Single Tenant, an additional field displays where you can enter the name of the tenant you want this role to access.

NOTE
Either the Tenant Access or Tenant Write Access fields can be set to Contact's Tenant Group to reference the Analyst's Tenant Group on the Contact Detail page. If a user that is not an Analyst, or an Analyst with no Analyst's Tenant Group defined, uses a role with this access, their access is Contact's Tenant.

– **Support Automation Access**
Defines the appropriate Support Automation access for this role.

- **Function Access**
  Allows you to define the role's access to each functional area.
- **Web Interface**
  Allows you to configure the web interface for the role by defining the web pages and online help content the users can access. Complete the following fields as appropriate:
  – **Web User Interface Type**
  The kind of Web Interface that used to present the product features. Most of the predefined roles use the Analyst interface. The Customer and Employee roles are assigned a restricted interface type because they are not allowed access to analyst, management, and administrative functionality.
  – **Web Initial Form**
  The initial web form that appears for this role.

WARNING
This must be set to menu_frames_role.htmpl in order for role-based functionality to be active. Changing the name of this form to anything else negates the role-based functionality.

– **Help View**
The name of the help set that appears for this role. Enter the name of the help set directly or click the search icon to select the help set from a list.

NOTE
The list of available help sets is based on the web user interface type selected in the role. Only active help sets belonging to that interface type are available for selection. If you do not select the web user interface type before selecting the help view, no help sets are available for selection.

- **Knowledge Management**
  Allows you to specify the Knowledge Management privileges for the role. Fill in the following fields as appropriate:
  – **Open Issue/Request**
  Allows the role to open an issue and request.
  – **Open Issue/Request based on Document**
  Allows the role to open an issue and request based on a Knowledge Document.
  – **Bypass Approval Process**
  Allows the role to bypass the knowledge approval process.
  – **Change Approval Process Template**

Allows the role to change the approval process template.

- **Add, Edit, Copy and Paste Categories**
  Allows the role to manage Knowledge Categories.
- **Delete Multiple Categories and Documents**
  Allows the role to delete parents and child categories and documents.
- **Delete Category**
  Allows the role to delete a Knowledge Category.
- **Create Document**
  Allows the role to Create a Knowledge Document.
- **Create Document with Attachments**
  Allows the role to create a document with attachments.

  NOTE
  If this option is disabled, the Attach File and Attach File from Library buttons do not appear for that role.
- **Delete Knowledge**
  Allows the role to delete knowledge documents, forums and files.
- **View Forum**
  Allows the role to view forums.
- **Create Forum**
  Allows the role to create a forum.
- **Reply Forum**
  Allows the role to reply to a forum.
- **Edit Forum**
  Allows the role to edit a forum.
- **View File**
  Allows the role to view files.
- **Edit File**
  Allows the role to edit files.
- **View External Repository**
  Allows the role to view the external repository.
- **View Related Tickets**
  Allows the role to view related support tickets.
- **Add Bookmark**
  Allows the role to add bookmark to My Bookmarks.
- **Use Preferences**
  Allows the role to use preferences.
- **Allow Export**
  Allows the role to export knowledge packages.
- **Allow Import**
  Allows the role to import knowledge packages.

- **KT Document Visibility**
  Allows you to specify which document statuses the role is allowed to view (for example, draft, retired, and published).
- **Tabs**
  Allows you to define the tabs that appear when a user assigned to this role is logged in.

  WARNING
  Include *only* tabs that contain forms that are in the form group assigned to the role you are creating or editing. For example, assigning the Customer tab or Employee tab to the Administrator role causes an error when users attempt to access that tab. Including more tabs than your browser window can display causes some tabs to be inaccessible to the user.

NOTE
The role's form group is specified in the Customization Form Group field on the Role Detail page, and is also displayed in the Form Group column on the Role List page.

- **Report Web Forms**
  Allows you to define the report web forms that are available to this role.
- **Go Resources**
  Users can search for items in by a number, name, or ID. In the upper right-hand corner of the main window, there is a drop down list containing the searchable record types. These searchable record types are referred to as "go" resources. Allows you to specify which record types appear in the Go drop-down list for the role.

How to Implement a Custom Role

For many sites, the predefined roles are sufficient. There may be situations, however, when you want to create a custom role and tailor it to meet site-specific business needs within your organization.

The following process outlines the tasks required in implementing a new role. The example shown here describes how you might implement a role for a small group of analysts tasked with reviewing and authorizing change order tickets.

Follow these steps:

1. Create a new role record using the following field values:
   - **Role Name**
     Change Analyst
   - **Code**
     chg_anal
   - **Customization Form Group**
     Analyst
   - **Preferred Document**
     Incident
2. Select Service Desk Analyst in the Data Partition field on the Authorization tab.
3. Select the Modify in the Change Orders field on the Function Access tab.
4. Enter the following values on the Web Interface tab:
   - **Web User Interface Type**
     Analyst
   - **Help View**
     Change Analyst
5. Select the following tabs:
   - Reports tab - Change Analyst
   - Service Desk tab - Change Analyst
   - Change Calendar tab
6. Select the following reports on the Report Web Forms tab:
   - Active Change Orders Aging by Priority for Status
   - Active Change Orders at Weeks End
   - Change Orders by Failed Service Type for Change Categories
7. Add the Change Order resource on the Go Resources tab.
8. Create a custom help set named Change Analyst that includes all content appropriate for the new role.
   For more information, see Create and Publish a Help Set.
9. Create the following custom tabs using features appropriate for the new role:
   - Reports tab - Change Analyst
   - Service Desk tab - Change Analyst
10. Create a custom menu tree that includes all nodes appropriate for the new role.

For more information see How to Implement a Custom Menu Tree.

Switch Roles

Roles define the system functionality each user can access. Depending on your assigned role, a specific set of menus, tabs, and toolbar controls are presented when you log in in the Web Interface client. For example, administrative roles have access to the Administration tab, while analyst and manager roles typically do not.

Some users are allowed to access multiple roles, enabling them to switch from one view of the system to another. If you are assigned multiple roles, you can switch between them at any time without having to log out and back in again.

Follow these steps:

1. Select the desired role from the Role drop down list in the upper right corner of the main page of the Web Interface.
2. Click Set Role.
   The Web Interface and available functionality change to match the new role setting.

Create Help Sets

This article describes the following:

Help sets are the collections of online help topics available to users depending on their role assignments and current role setting. If you log in using the Administrator role, for example, you can view the online help topics included in the Administrator help set. If you switch to the Employee role, you can view the Employee help set.

Each predefined role has a corresponding predefined help set. You can create custom help sets for any custom roles you might define.

You can create custom help sets for any custom roles you might define.

Follow these steps:

1. Select Security and Role Management, Role Management, Help Sets on the Administration tab.
2. Click Create New.
3. Complete the following fields:
   - **Help Set Name**
     The unique name of this help set.
   - **Interface Type**
     The interface type of the help set (such as Analyst, Employee, or Customer).
   - **Record Status**
     Indicates whether the help set is Active or Inactive.
   - **File Name Prefix**
     The prefix you want to attach to the help files generated for this help set. Do not include spaces in the name.

     NOTE
     Assign a prefix that allows you to identify the files belonging to this help set. For example, you may want to use ANAL for an analyst's help set prefix.
   - **Internal**
     This is automatically set to NO for user-defined help sets. The value in this field cannot be changed.
4. Click Save.
5. Click Define Content and select the content you want to include in the help set.

   WARNING
   Some topics are required, and are included in your new help set regardless of whether you select them. For example, the CA SDM home page and other front matter topics are always included. Also, nested topics are dependent on their container topics. Container topics are included automatically if you include any of their

nested topics. For example, if you select the "Use the Scoreboard" topic, the container topic "Navigate CA SDM" is included when you publish the help set. If you edit the contents of an existing help set by adding or removing topics, you may not see your content changes reflected in the help set's table of contents until you clear your browser cache. See your browser documentation for information on how to clear the cache

6. Click OK.
   The Selected Help Update window closes and the content is listed on the Contents tab.
7. Click Publish.
   This generates the help set by packaging the selected topics into a help system you can display in a web browser.
8. Wait a few moments for the publishing process to complete; then select View, Refresh on the menu bar.
   The View Help button becomes active.
9. Click View Help.
   The custom help set appears in your default web browser.

Define Help Set Contents

You can choose the topics that appear in the Table of Contents of a help set.

To define the help set contents, **follow these steps:**

1. Click Define Content on the Contents tab of the Help Set Detail page.
   The Selected Help Update page appears.
2. Select the topics you want to include in the help set from the Available Help list. To select multiple topics, hold down the CTRL key while clicking the left mouse button.
3. Move the selected topics to the help list on the right.
4. Click OK.
   The Help Set Detail page appears, with the selected topics listed on the Contents tab.

How to Manage Roles Using Menu Trees

This article describes the following:

You can manage roles using the menu trees, implementing a custom menu tree, and copying a menu tree.

Menu Trees

Menu trees are the hierarchical listings of nodes (menu tree resources) that are displayed in the navigation pane on the left-hand side of the main web interface window. Menu tree resources define the items users can access from the menu tree. A menu tree resource consists of a name, description, and a URL fragment or HTMPL filename used by the web engine that controls the web page displayed.

A role can have a menu tree, which provides nodes for access to many functional areas of the system. For example, the predefined Administrator role has a menu tree that includes nodes to the System and Role Management administration features, Service Desk administration features, and many others.

For roles that include a menu tree, the menu tree provides access to a specified set of resources that provide access to functional areas of the system.

CA SDM provides predefined menu trees for the following roles:

- Administrator (admin_tree)
- CA CMDB Administrator (cmdb_adm_tree)
- Knowledge Management Administrator (kt_adm_tree)
- Knowledge Manager (kt_mgr_tree)
- Support Automation Administrator (sa_admin_tree)
- Service Desk Administrator (sd_adm_tree)
- System Administrator (sys_adm_tree)
- Tenant Administrator (tn_admin_tree)

You can edit the Name, Record Status, and Description fields of the predefined menu tree records, but you cannot modify them by adding or removing their menu tree resources.

To produce a custom a menu tree, you can create new menu tree record or copy and modify one of the predefined menu trees.

> **NOTE**
> The non-modifiable Internal field on each menu tree record indicates whether the menu tree can be modified. A value of YES in the Internal field indicates a predefined menu tree, which cannot be modified. A value of NO indicates a site-defined menu tree, which can be modified. The Customize Menu button appears only on menu tree detail records with an Internal field value of NO.

When you attach a menu tree to a tab, it becomes available for all roles that have access to that tab.

How to Implement a Custom Menu Tree

For many sites, the predefined menu trees are sufficient. There can be situations, however, when you want to configure a role by implementing a custom menu tree for it.

In most cases, it is easier to start with a copy of a predefined menu tree and then add, remove, or reorganize nodes within the hierarchy. Alternatively, you can create a menu tree and construct an all new hierarchy of nodes.

You can use either of the following methods to make custom menu tree available to a role:

- Replace the menu tree in the web form (Start Page) for the tab that shows the original admin_tree.
- Create a web form and attach the new web form with the new menu tree to a tab.

Follow these steps:

1. Copy one of the predefined menu trees.

   > **NOTE**
   > Make a note of value you enter in the Code field.
2. Create a web form using the following field values:
   - **Type:** HTMPL
   - **Resource:**

     ```
     $cgi?SID=$SESSION.SID+FID=123+OP=DISPLAY_FORM+HTMPL=admin_main_role.htmpl+KEEP.tree_code=menu_tree_code
     ```

     > **NOTE**
     > Specify the value of the code for the menu tree you created in Step 1 for menu_tree_code. The admin_main_role.htmpl code uses the value of the KEEP.tree_code variable as its menu tree.
3. Create a tab record using the following field values:
   - **Starting Page:** The web form you created in Step 2
   - **Menu Bar:** Administration

     > **NOTE**
     > Administration is a generic menu bar used by many roles; it is not role-specific.
4. Assign the tab you created in Step 3 to the role you want to have access to the custom menu tree.

5. Log out of CA SDM and log back in again.
   The Administration tab displays your custom menu tree.

Copy a Menu Tree

You can copy an existing menu tree to use as a starting point for a modified menu tree.

Follow these steps:

1. Select Security and Role Management, Role Management, Menu Trees on the Administration tab.
   The Menu Tree List page appears.
2. Click the Menu Tree to copy.
   The Menu Tree Detail page appears.
3. Click File, Copy.
   The Create New Menu Tree page appears.
4. Complete the following fields:
   - **Menu Tree Name**
     (Required) Specifies the name you assign that identifies this menu tree.
   - **Code**
     (Required) Specifies the code that identifies the menu tree to the system. After the code is defined, it cannot be changed.
   - **Record Status**
     Indicates whether this menu tree is active or inactive.
   - **Description**
     Describes the menu tree. The description can be used to give additional details about the menu tree and the roles that use it.

   Click Save.
   The Menu Tree Detail page for the new menu tree appears.
5. Click Customize Menu.
   A copy of the original menu tree appears.
6. Modify the menu tree as you want.

Create and Format a Menu Tree

You can create and format menu trees, based on one of the default menu trees provided.

Follow these steps:

1. Select Security and Role Management, Role Management, Menu Trees on the Administration tab.
2. Click Create New. The following fields require explanation:
   - **Menu Tree Name**
     (Required) The name you assign that identifies this menu tree.
   - **Code**
     (Required) The code that identifies the menu tree to the system. Once the code is defined, it cannot be changed.
   - **Record Status**
     Indicates whether this menu tree is active or inactive.
3. Click Save.
4. Click Customize Menu.
   A form appears, allowing you to modify the menu tree. At this point, the menu tree contains only a top node with the text you entered as the menu tree name.
5. Right-click the node in the menu tree and select Create New Node.
   The following fields require explanation:

- **Node Name**
  Enter a name for the node. This is the name that appears in the menu tree.
- **Description**
  Enter a description for the node. This description can be used to further define the purpose of the node.
- **Resource**
  Enter the resource name directly in the field, or click the search icon to select the resource from a list. The menu tree resource determines the action to perform when the user selects the node from the menu tree.

6. Repeat steps 4 and 5 as many times as necessary to create the set of nodes you want to appear in the menu tree.
7. Click Save.
   The menu tree definition is saved and the Menu Tree Detail page appears.

Create a Menu Tree Resource

Menu tree resources are the items users can access from the left pane of one of the tabs. A menu tree resource consists of a name, its description, and a URL fragment or HTMPL filename used by the web engine that controls the web page displayed.

You can create your own menu tree resources to configure the user access points available from the left pane of a tab.

Follow these steps:

1. Select Security and Role Management, Role Management, Menu Tree Resources on the Administration tab.
2. Click Create New.
3. Fill in the following fields and click Save:

Name

The name displayed in the menu tree in the left pane of the tab. This is a required field.

Status

Indicate whether the resource is active or inactive. This is a required field.

Description

The description of the resource. The description can be used to further define the purpose of the resource.

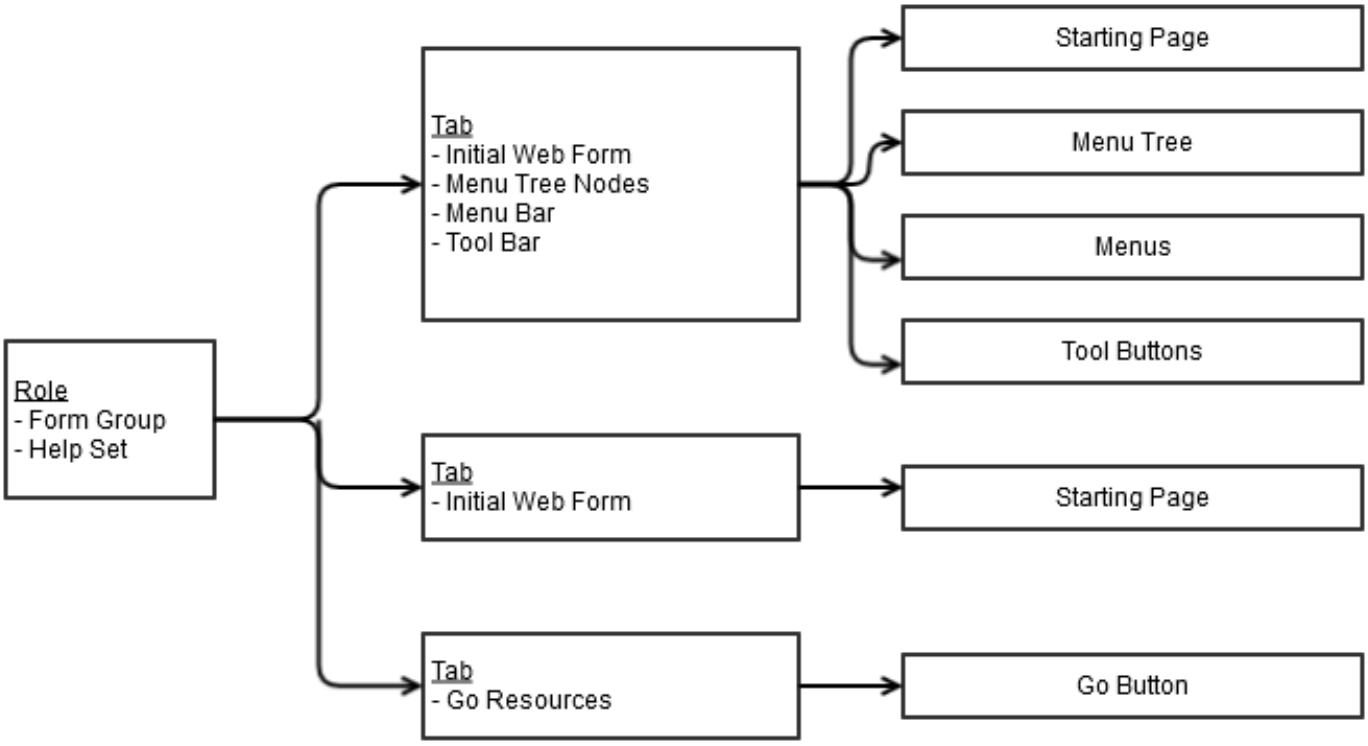
Resource

The action performed when the user selects the resource from the menu tree. This can be the name of the form that is displayed or the search operation that is performed.

How Role-Based Navigation Works

Each user's view of the web interface is defined by a role. Users with multiple role assignments can switch to multiple web interface views. The following diagram shows how roles interrelate with other objects to produce a role-based presentation of the user interface:

Figure 26: Diagram depicting how role-based navigation works



Create a Tab

A tab is a graphical display entity that links to a role in order to present features to the users of that role. When a user logs in to the system, the main window displays the tabs assigned to the user default role.

Tabs define major subdivisions in the web interface main window. Each tab is configured to expose an appropriate set of user interface features to the role or roles that use it.

All roles must have at least one tab. You can associate one or more tabs with a role. Each tab has a sequence number that controls its display order in the main window. If only one tab is associated with a role, the tab starting page is displayed and not the tab.

You can configure tabs to include the following kinds of display features:

- A starting page (default web form) that is displayed when a user selects the tab. The starting page is a required element of all tabs. You can assign only one starting page to each tab.

  NOTE
  You can configure a starting page to display graphical reports from Business Object, which enables users to generate reports at run time. For more information, see CA Business Intelligence Reporting documentation.
- A menu bar, which presents drop-down lists of commands, such as File, View, and Search commands. The menu bar is optional. You can assign only one menu bar to each tab.
- A toolbar, which presents tool buttons for easy access to frequently used menu commands. The toolbar is optional.

CA SDM provides several predefined tabs. You can assign the predefined tabs to a role, modify the predefined tabs, and create custom tabs.

WARNING
Do not assign more tabs than your browser window can display; doing so causes tabs with higher sequence numbers to extend beyond the window viewable display area and make them inaccessible to the user.

WARNING
Include only tabs that contain forms that are included in the form group that is assigned to the role you are creating or editing. For example, do not assign the Customer tab or the Employee tab to the Administrator role; doing so causes an error when users attempt to access that tab. The role form group is specified in the Customization Form Group field on the Role Detail page, and is also displayed in the Form Group column on the Role List page. For a list of the web forms in each form group, see Form Groups.

You can create your own tabs to display on the main page. You can then define the tabs that display when users of different roles log into the system.

Follow these steps:

1. Select Security and Role Management, Role Management, Tabs on the Administration tab.
2. Click Create New.
3. Fill in the tab fields as appropriate and click Save.

Tab Fields

Code

The code that identifies this tab to the system. After you save the tab record, the code cannot be changed.

Record Status

Indicates whether the tab is active or inactive.

Display Name

The name of the tab that is displayed to users.

Starting Page

The initial web page that appears in the main window when a user selects this tab. The starting page and menu bar must belong to the same form group. Defining a tab with a starting page and a menu bar that belongs to different form groups causes an error when users access the tab.

Menu Bar

The menu bar that appears in the main window when a user selects this tab.

Predefined Tabs

The following table shows the predefined tabs that are assigned to each role. The tabs are listed in sequence number order, indicating their left-to-right position in the window.

NOTE
In many cases, there are multiple versions of tabs with the same display name. For example, the Service Desk tab for the Administrator role provides full access to CA SDM functionality, while the Service Desk tab for the Change Manager role is more focused on change orders.

Role	Tabs
Administrator	Service Desk tab with full menu and scoreboard Knowledge tab Administration tab with full menu tree Reports tab - Administrator Change Calendar tab CA CMDB tab Support Automation tab
Change Manager	Reports tab - Change Manager Service Desk tab - Change Manager Change Calendar tab
Configuration Administrator	CA CMDB tab Administration tab - Configuration Administrator
Configuration Analyst	CA CMDB Scoreboard - Configuration Analyst
Configuration Viewer	CA CMDB Scoreboard - Configuration Viewer
Customer	Customer tab
Customer Service Manager	Service Desk tab - Customer Service Manager Knowledge tab Reports tab - Customer Service Manager
Customer Service Representative	Service Desk tab - Customer Service Rep Quick Profile tab Knowledge tab
Employee	Employee tab
Incident Manager	Reports tab - Incident Manager Service Desk tab - Incident Manager Knowledge tab
Knowledge Analyst	Service Desk tab - Knowledge Analyst Knowledge tab Knowledge Management Schedule Knowledge Report Card tab Reports tab - Knowledge Analyst
Knowledge Manager	Service Desk tab - Knowledge Manager Knowledge tab Knowledge Management Schedule Administration tab - Knowledge Manager Knowledge Report Card tab Reports tab - Knowledge Manager
Knowledge Management Administrator	Administration tab - Knowledge Administrator

Level 1 Analyst	Service Desk tab - Level 1 Analyst Quick Profile tab Knowledge tab
Level 2 Analyst	Service Desk tab - Level 2 Analyst Knowledge tab Change Calendar tab
Problem Manager	Reports tab - Problem Manager Service Desk tab - Problem Manager Knowledge tab
Service Desk Administrator	Administration tab - Service Desk Administrator
Service Desk Manager	Reports tab - Service Desk Manager Service Desk tab - Service Desk Manager Knowledge tab Change Calendar tab
Support Automation Administrator	Service Desk tab - Level 1 Analyst Support Automation tab Support Automation Admin Quick Profile tab Knowledge tab
Support Automation Analyst	Service Desk tab - Level 1 Analyst Support Automation tab Quick Profile tab Knowledge tab
System Administrator	Administration tab - System Administrator
Tenant Administrator	Administration tab - Tenant Administrator
Vendor Analyst	Service Desk tab - Vendor Analyst

Create a Web Form

Web forms define the pages that appear in the CA SDM web interface.

There are four web form types:

- Business Object Report URL
- HTMPL page
- GO resource
- Custom (for example, a URL for a third-party web page)

Administrators can create web forms to be the starting pages for tabs, reports to display on tabs, "go button" resources, or another URL.

Follow these steps:

1. From the Administration tab, navigate to Security and Role Management, Role Management, Web Forms.
   The Web Forms List page displays.
2. Click Create New.
   The Create New Web Form page displays.
3. Complete the following fields:

- **Web Form Name**
  (Required) Specifies the name that identifies the web form.
- **Record Status**
  Indicates whether this form is active or inactive.
- **Code**
  (Required) Specifies the code that identifies the web form to the system. After the code is defined, it cannot be changed.

  NOTE
  This field specifies the web_form_name on the Properties tab for a multiframe form in Web Screen Painter.
- **Type**
  Specifies one of the following types of web form that you are creating:
  - **HTMPL page** -- Displays a web page to use as the starting page for one of the custom tabs you create.
  - **Repor**t -- Specifies a CA SDM report that displays on any tab.
  - **Go Resource** -- Specifies a "Go Button" resource.
  - **Other** -- Accesses any other external Web page through URL.
- **Description**
  Describes the web form. Use this description to further identify this web form, where it displays, and its purpose.
- **Resource**
  Specifies the code that calls the web form. This code can be command line code or a URL.
  Example: Open a simple htmpl form "menu_tab_dflt.htmpl":

```
$cgi?SID=$SESSION.SID+FID=123+OP=DISPLAY_FORM+HTMPL=menu_tab_dflt.htmpl
```

Click Save.

Form Groups

Form groups define the sets of CA SDM web interface pages that are available to a role. Each role has one form group. Users can display only the web pages that are included in the form group that is assigned to their role.

Each interface type has an associated form group, a set of HTMPL files that define the pages displayed to the users.

CA SDM provides the following predefined form groups:

- Analyst
- Customer
- Employee
- ITIL

You can use the predefined form groups in their default configuration, modify the predefined form groups, and create new form groups using Web Screen Painter. You can view a listing of the HTMPL filenames included in each predefined form group.

Create a Menu Bar

A menu bar is a user interface element that displays a horizontal list of menus in the web interface main window. Each menu contains a drop-down list of options or commands. You can define custom menu bars for any custom roles you might create. Menu bar records specify the HTMPL form that controls the menu items that the menu bar can access.

NOTE

To define the functionality of the menu bar, you must use the Web Screen Painter application. For information about configuring the functionality of a predefined or custom menu bar, see the *Web Screen Painter Online Help*.

The following table lists the predefined menu bars and identifies the predefined tabs that use them.

Menu Bar	Associated Tabs
Administration	Administration tab - Configuration Administrator Administration tab - Knowledge Administrator Administration tab - Knowledge Manager Administration tab - Service Desk Administrator Administration tab - System Administrator Administration tab - Tenant Administrator Administration tab with full menu tree
CA CMDB	CA CMDB tab with full menu and scoreboard
Change Calendar	Change Calendar tab
Knowledge	Knowledge tab Knowledge Management Schedule
Service Desk	Service Desk tab with full menu and scoreboard
Service Desk-Change Manager	Service Desk tab - Change Manager
Service Desk-Cust Service Mgr	Service Desk tab - Customer Service Manager
Service Desk-Cust Service Rep	Service Desk tab - Customer Service Rep
Service Desk-Incident Manager	Service Desk tab - Incident Manager
Service Desk-Knowledge Analyst	Service Desk tab - Knowledge Analyst
Service Desk-Knowledge Manager	Service Desk tab - Knowledge Manager
Service Desk-Level 1 Analyst	Service Desk tab - Level 1 Analyst
Service Desk-Level 2 Analyst	Service Desk tab - Level 2 Analyst
Service Desk-Problem Manager	Service Desk tab - Problem Manager
Service Desk-Service Desk Mgr	Service Desk tab - Service Desk Manager
Service Desk-Vendor Analyst	Service Desk tab - Vendor Analyst
Support Automation-Analyst	Support Automation tab - Support Automation Analyst

You can create your own menu bars to control the access to system functionality for your user-defined roles.

Follow these steps:

1. From the Administration tab, navigate to Security and Role Management, Role Management, Menu Bars.
2. Click Create New.
3. Fill in the menu bar fields as appropriate.
4. Click Save.

Menu Bar Fields

Code

The code that identifies this menu bar to the system. Once the code is defined, it cannot be changed. This is a required field.

Record Status

Indicates whether the menu bar is Active or Inactive.

HTMLP Name

The name of the HTMPL form that contains the menu bar definition. The menu bar is actually designed using the Web Screen Painter.

Go Resources

The *Go* button provides an easy means of locating a particular record.

Go resources are a type of web form. If a role has associated Go resources, when a user logs in with that role the Go button appears in the upper-right corner of the main CA SDM window and in all popup windows. The Go button has two associated fields in the user interface:

- A drop-down list for selecting the type of record to search for (for example, Change Order)
- A text box for entering a value to identify a particular record (for example, 135 to locate Change Order 135)

By assigning Go resources to a role, you can specify the kinds of records users in that role can search for. For example, the predefined Administrator role has the following Go resources:

- Change Order
- Document by ID
- Incident
- Issue
- Knowledge
- Problem
- Request
- User by ID
- User by Name
- User by Phone

Toolbars

Toolbars extend the functionality of menu bars by adding the capability to display one or more tool buttons to the right of the menus.

Tool buttons appear as icons on the toolbar. Clicking a tool button gives the user easy access to frequently used menu options or commands.

> **NOTE**
>
> You can use the Web Screen Painter application to define the functionality of the toolbar. For more information, see Using the Web Screen Painter (WSP).

Search Menu Bars

You can enter search criteria to filter the Menu Bar List to display only the menu bars you want to see. You can also search for individual menu bars that you want to view or edit.

To search for a menu bar:

1. From the Administration tab, navigate to Security and Role Management, Role Management, Menu Bars.
   The Menu Bar List page displays.
2. Click Show Filter.
   The available search fields display at the top of the list page.
3. Complete one or more of the search fields, and click Search.
   The Menu Bar List page displays the menu bars that match your search criteria. You can select a menu bar to view or edit.

Search Tabs

You can enter search criteria to filter the Tab List to display only the tabs you want to see. You can also search for individual tabs that you want to view or edit.

To search for a tab:

1. From the Administration tab, navigate to Security and Role Management, Role Management, Tabs.
   The Tab List page displays.
2. Click Show Filter.
   The available search fields display at the top of the list page.
3. Complete one or more of the following fields:
   Tab Name
   The name that identifies the tab.
   Display Name
   The name of the tab that is displayed to users.
   Code
   The code that identifies this tab to the system.
   Record Status
   Indicates whether the tab is active or inactive.
   Menu Bar
   The menu bar that displays when a user selects this tab.
   Starting Page
   The page that is displayed when a user selects this tab.
   Description
   A description of this tab.

   NOTE

   You can click the More icon to display the Additional Search Arguments field. This field is intended only for expert users who understand SQL and Majic and can use it to specify search arguments that are not available in the standard search filter fields. You can enter a SQL WHERE clause in this field to specify an additional search argument.
4. Click Search.
   The Tab List page opens to display the roles that match your search criteria. You can select a tab to view or edit.

Create a Functional Access Area

This article contains the following topics:

Functional access areas define the role level access to ticket records and other system components. The usp_functional_access_type table defines the area and the usp_functional_access_level tables describe user access.

Name	Code Name	New
Administration	admin	No
Incident/Problem/Request	call_mgr	No
Change Order	change_mgr	No
Inventory	inventory	No
Issue	issue_mgr	No
Knowledge Document	kd	No

Notification	notify	No
Reference	reference	No
Security	security	No
Announcement	announcement	Yes
Incident/Problem/Request Reference	call_mgr_reference	Yes
Incident/Problem/Request Template	call_mgr_template	Yes
Change Order Template	change_mgr_template	Yes
Change Order Reference	change_reference	Yes
Configuration Item	ci	Yes
Configuration Item Common Readonly	ci_common_ro	Yes
Configuration Item Reference	ci_reference	Yes
Contact	contact	Yes
Group	group	Yes
Issue Template	issue_mgr_template	Yes
Issue Reference	issue_reference	Yes
Location	location	Yes
Multisite Administration	multisite_admin	Yes
Multisite Reference	multisite_reference	Yes
Notification Reference	notification_reference	Yes
Organization	organization	Yes
Prioritization	prioritization	Yes
Service Level	service_level	Yes
Site	site	Yes
Stored Query	stored_queries	Yes
Support Automation	sa	Yes
Survey	survey	Yes
Tenant Admin	tenant_admin	Yes
Timezone	timezone	Yes
Workflow Reference	workflow_reference	Yes
Workshift	workshifts	Yes

When you add a functional access area, the existing roles automatically have Modify access. You can review and change the access levels to grant the appropriate authority.

Follow these steps:

1. On the Administration tab, select Security and Role Management, Functional Access.
2. Click Create New.
   The Create New Functional Access List page appears.
3. Complete the functional access area fields as appropriate. The Code field defines the CA SDM or site-defined object name. Objects can only map to one functional access area. However, one functional access area can manage multiple objects.
4. Click Save.
   The Functional Access Detail page appears.
5. Apply access levels to one or more roles.

Apply Access Levels to Many Roles

For a functional access area, you can set the access levels for every role to save time. Instead of using Role Management, you update the functional access area with the appropriate role access levels.

Follow these steps:

1. On the Administration tab, select Security and Role Management, Functional Access.
2. Click the name of the function area.
3. On the Roles Tab, click Edit in List.
4. Review and update each role as appropriate. The following access levels are available:
   - **None**
     Denies the role access to the function object.
   - **View**
     Grants read-only capability to the function object.
   - **Modify**
     Grants read/write access to the function object.
5. Continue selecting roles and choosing access levels.
6. (Optional) Click Change All.
7. Click Save.
   The changes for the roles apply immediately.

Example: Change the Access Levels for the Announcement

This example shows you can role access levels for the Announcement functional access area.

1. On the Administration tab, select Security and Role Management, Functional Access.
   The Functional Access Detail page appears.
2. Click Announcement.
3. On the Roles Tab, click Edit in List.
4. Select Level 2 Analyst.
5. Select View from the Access Level.
   The Access Level for the Level 2 Analyst role highlights with the value of View.
6. Select Configuration Analyst.
7. Select Modify from the Access Level.
   The Access Level for the Configuration Analyst highlights with the value of Modify.
8. Click Save.
   A message confirms the change.
9. Log in as a Level 2 Analyst role.
10. Select View Announcements.
    The Announcements page appears.
11. Click an announcement.
    A message reminds you that as a Level 2 Analyst, you can only view announcements.
12. Set the Role to Configuration Analyst.
13. Select View Announcements.
    The Announcements page appears.
14. Click an announcement.
    The Update Announcement page appears. As a Configuration Analyst, you can modify announcements.

Apply an Access Level to a Role

You can use Role Management to change the way users access the user interface. When you change the access levels for a role, the user interface shows only objects, pages, and menu items based on the access level. For example, if a role can no longer create contacts, the File menu omits New Contacts.

Follow these steps:

1. On the Administration tab, select the Security and Role Management, Role Management, Role List.
2. On the Role List, right-click the role name and select Edit from the short-cut menu.
3. Click Edit in List on the Function Access tab.
4. Click a function name.
   The row highlights.
5. Update the functional access areas with the following access levels as appropriate:
   - **None**
     Denies the role access to the function object.
   - **View**
     Grants read-only capability to the function object.
   - **Modify**
     Grants read/write access to the function object.
6. Click Save.
   A message confirms the change. The role can immediately use the functional access area at the specified access level.
7. Verify the access level, by logging in as the role and checking menus, page options, and buttons.

Example: Grant the Level 2 Analyst Role Modify Access to Tickets

This example shows how the user interface changes when you grant a Level 2 Analyst access to modify tickets.

1. On the Administration tab, select the Security and Role Management, Role Management, Role List.
   The Role List appears.
2. Right-click Level 2 Analyst and select Edit from the short-cut menu.
3. Click Edit in List on the Function Access tab.
4. Select Incident/Problem/Request Reference.
5. Select Modify on the Access Level.
   The access level updates to Modify.
6. Click Save and log out.
   A message confirms the change.
7. Log in as a Level 2 Analyst role.
8. Select Search, Incidents.
9. Click Search and open an incident.
   The Incident Detail page includes an Edit in List button. As a Level 2 Analyst, you can modify the ticket.

Role-Based Security

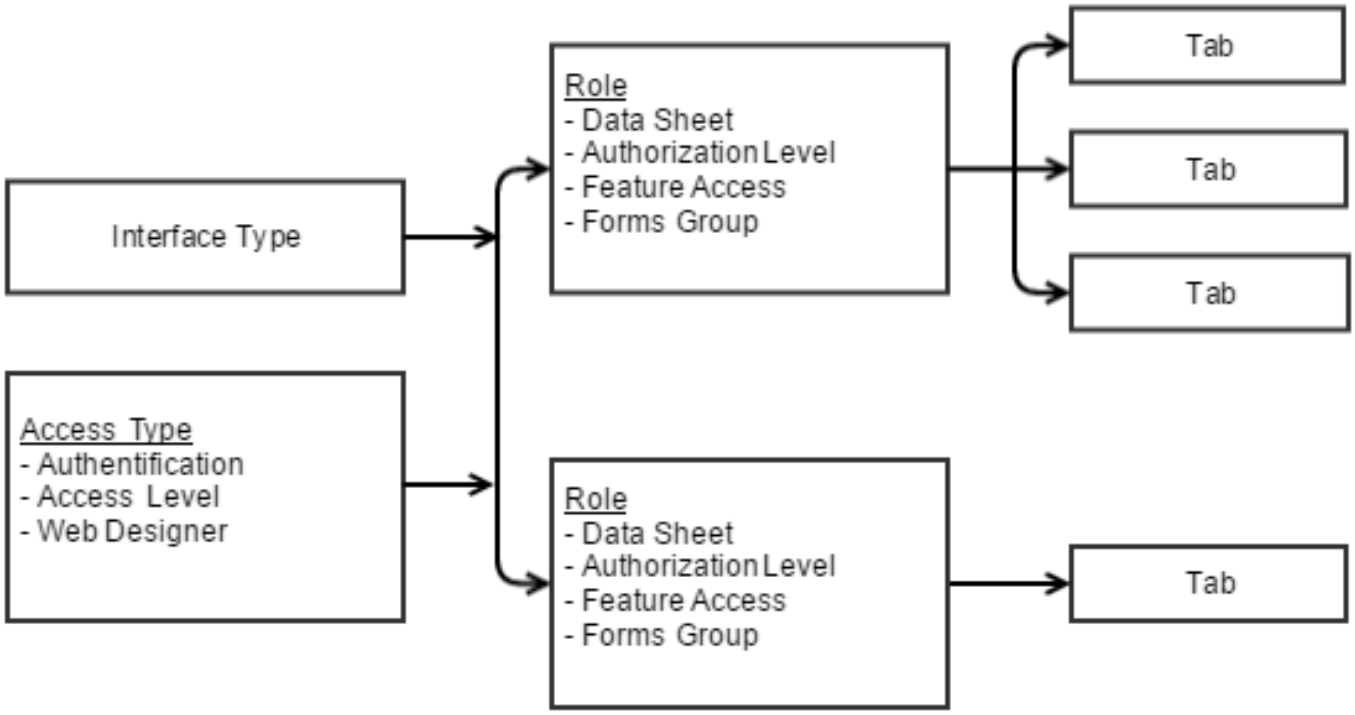
This article contains the following topics:

Access types and roles are the primary components you use to control CA SDM security.

> **NOTE**
> For more information about other aspects security, see Security.

The following diagram shows an overview of how roles interrelate with other system objects to provide role-based security.

Figure 27: Providing Role-based security



How Access Types Work

Each access type for a user controls the following aspects of system behavior:

- How CA SDM performs web authentication when the user logs in
- The access level for the user
- Whether the user can modify web forms or the database schema using Web Screen Painter
- Which roles are available to the user

You can associate an access type with a contact by selecting the access type while creating or updating the contact record.

NOTE
The following table lists the predefined access types, identifies their linked roles, and gives a brief description.

Access Type	Linked Roles	Description
Administration	Administrator (default) Configuration Administrator Employee Level 2 Analyst Service Desk Administrator System Administrator Tenant Administrator	Provides the highest level of security access to all key administration roles. Used during implementation and ongoing administration. **Note:** The Administration access type is preconfigured to allow administrators to switch to any of the linked roles. For example, to see a different view of the system, administrators can switch to the Employee role without having to log out and log in again.
Customer	Customer	Provides highly restricted access to *external* customers who use the self-service view.
Employee	Employee	Provides highly restricted access to *internal* employees who use the self-service view. Used to create new incident and update incident pages.
IT Staff	Configuration Analyst Employee Level 2 Analyst (default) Knowledge Analyst Knowledge Management Administrator Knowledge Manager	Provides analyst-oriented access to users who work within your IT organization but are not actual members of the support team. This access is designed specifically for users who need access to Knowledge Management.
Knowledge Management	Configuration Administrator Configuration Analyst Configuration Viewer Employee Knowledge Analyst Knowledge Management Administrator (default) Knowledge Manager Level 2 Analyst	Provides administrative access tailored to users who administer Knowledge Management features.
Process Management	Change Manager Configuration Analyst Employee Incident Manager (default) Level 2 Analyst Problem Manager Service Desk Manager	Provides access tailored to users who perform key process management roles.

Service Desk Management	Customer Service Manager Configuration Analyst Employee Level 1 Analyst Level 2 Analyst Service Desk Manager (default)	Provides access tailored to users who manage IT support or external customer support functions (typically front-line support supervisors).
Service Desk Staff	Configuration Analyst Configuration Viewer Customer Service Representative Employee Level 1 Analyst (default) Level 2 Analyst	Provides access tailored to users who perform support tasks. Access is focused on those users that perform frontline support.
Support Automation Admin	Support Automation Administrator	Provides access to users that perform Support Automation administration.
Support Automation Analyst	Support Automation Analyst	Provides access to users that provide live assistance to end users.
Vendor Staff	Vendor Analyst	Provides highly restricted access to external vendors, who work only on items directly related to their product (for example, a particular brand of hardware).

Role Records

You can assign roles to an access type, or directly to a user contact record. If a role assignment conflict occurs, the contact role assignments take precedence.

Each role record must be configured with the following components:

- One form group
- One user interface type
- Function access settings
- One or more tabs
- One help set

The following optional components can also contribute to each role definition:

- Menu trees
- Scoreboards
- Menu bars
- Toolbars
- One data partition
- Knowledge Management access
- Support Automation access levels
- Report web forms
- Go resources

Data Partitions

Data partitions are subsets of the CA SDM database that enable you to control access at the record level. You can associate a data partition to a role to control access to tickets and other records accessible through the web interface.

For information about working with data partitions, see How to Set Up the Data Partition.

Configuring User Accounts

Configure the user accounts to set up contacts, contact types, groups, special handling types, LDAP data authentication, and communication.

Contacts

An important part of establishing a working service desk is defining the users who are going to access it. In CA SDM, users are named *contacts*, and you can perform several tasks to set up and manage them:

- Set up contacts manually.
- The Organize contacts into groups that define areas of responsibility.
- Create contact types to organize your CA SDM contacts into logical groupings.
- Import LDAP user information into a CA SDM contact record.
- Assign a contact to a role to define the accessible system functionality.
- Assign a special handling type, such as Very Important Person (VIP), to a contact.

Contact Definitions

Everyone who uses CA SDM must be defined as a contact. The contact record of a user defines the user information that the system needs as follows:

- **Basic Identification**
  Defines basic identification, such as the name of a user and contact type. The Contact name is used as the primary identifier when selecting a contact or filling in contact information in other contexts.
- **Login**
  Defines login information, such as the user ID and in some cases, a PIN field to use as a password that verifies the user upon login. The user ID is used to identify the user in the contact table for authentication purposes and for determining the access types assigned to the user. Depending on your security configuration, another field such as the contact ID, can be specified as the PIN field. The user can use the PIN as login password.
- **Security**
  Defines the access type that is assigned in their contact record or by a default access type, depending on how you set up the security for your system. In addition, the access type of a user can be assigned based on their membership in an LDAP Directory group.
  The user access type determines all aspects of security, including authentication, web interface they can view, and what product functionality they can access.
  Security management is a feature of the web interface.
- **Service Type**
  Determines the level of service a user receives. A service type defines the level of service a user can receive. The Service Level Agreements (SLAs) are negotiated with CA SDM customers. Service types serve as the mechanism for CA SDM to implement SLAs. As an Administrator, you can use service type to set up SLAs using the web interface.
- **Automatic Assignment**
  Defines automatic assignment information, such as work shift and availability (used for analyst contact types only). You can set up analyst contacts to determine if they are eligible for automatic assignment. The Automatic assignment is valid only for requests, and is defined as part of the request area definition. Automatic Assignment is also linked to the groups to which the analyst belongs.
- **How to Send Users Notification Messages**

Defines the notification information of a contact for the following:

- – Various email addresses and telephone numbers to be used for notification
  – Method to be used for the notifications with different urgency levels
  – work shifts during which notifications are received
  The notification delay calculation takes the Contact Time Zone into account. If the Contact Time Zone is not set, the server time zone is used, instead. Using the server time zone may result in notifications firing at times, perceived to be outside the Work shift settings.
  Organizational information (such as location, organization, and department) lets you group contacts based on the organization to which they belong. For example, associating a contact with a location links the contact to a physical address and also helps in determining automatic assignment. The organization can be assigned a service type, making it easier to manage SLAs by organization rather than by individual contacts.
- **Groups to which a User Belongs**
  Organizes contacts into groups that represent specific areas of responsibility within your service desk. You can set up and can define contacts using the web interface.

Groups

A group is a collection of contacts that share a common area of responsibility. In CA SDM, groups are implemented using the predefined group contact type, making a group just a special type of contact. A group has the same basic information as a contact, with the important additional feature that groups are one of the keys to automatically assigning requests. You can associate request areas, locations, and a work shift with a group. These attributes are used to determine if and when the contacts in the group can accept the automatic assignment of a request.

User Profile Picture Synchronization from LDAP Server

You can import and synchronize user profile pictures in the new **xFlow Interface** through CA SDM LDAP integration. The profile pictures can be loaded in bulk from an LDAP server (Active Directory) using existing CA SDM utilities such as **pdm_ldap_sync** and **pdm_ldap_import**.

> **NOTE**
>
> CA SDM must be integrated with an Active Directory server to display user profile pictures in the **xFlow Interface** .

If CA SDM is already integrated with an LDAP Server, system administrators can use the **pdm_ldap_sync** utility to update existing CA SDM user records from the LDAP server. Any subsequent new users added to the LDAP server can be copied over to CA SDM using the **pdm_ldap_import** command utility to display the profile picture. If a CA SDM user does not have a profile picture, the **xFlow Interface** displays a colored circle with user initials appended to it. For more information, see Importing and synchronizing CA SDM records from the LDAP Server.

How to Disable User Profile Picture Synchronization from the LDAP Server

To disable a user's profile pictures from the LDAP server, the CA SDM ldap object must be modified as follows:

1. Navigate to $NX_ROOT/site/mods/majic directory.
2. Create a new **.mod** file or edit an existing one.
3. Add the following code in a separate line:

```
MODIFY ldap profile_pic dummy ;
```

4. Save the mod file and restart the CA SDM Server.

Working with Special Handling Types

This article contains the following topics:

You can define special handling types that identify contacts who require special attention. You can use the special handling types that CA Service Desk Manager provides or create your own types. You can view and locate tickets that specify an affected end user who requires special attention. For example, analysts can browse the V.I.P. folder on the Scoreboard to identify tickets that specify a VIP as the affected end user.

The following examples are contacts that special handling types can identify:

- Very Important Persons (VIPs) such as executives
- Customers with support renewal in process
- Customers with disabilities requiring special handling or equipment
- Visitors
- Contacts suspected of misusing or abusing system resources

When one or more special handling types are assigned to a contact, tickets that specify the contact in the Affected End User field show an alert banner, icon, or both. You can use ticket fields and special handling types to track tickets, and distinguish between two related but possibly distinct contact types. For example, a VIP (Affected End User) has an assistant (Requestor) acting on their behalf. When the Affected End User is a contact assigned to a VIP special handling type, an analyst can prioritize tickets more accurately.

How to Configure Special Handling Contacts

To configure special handling contacts, do the following steps:

1. Create special handling types.
2. Associate a contact to any number of special handling types. Similarly, a special handling type can have many contacts.
   A contact that is associated with one or more Special Handling types is visually distinguished in the Contact Detail form and the Quick Profile browser using a banner at the top of each page. This banner displays the alert icon and alert text for each Special Handling type that is assigned to the contact.
   Additionally, any tickets that identify the contact as the Affected End User are indicated as follows:
   – Alert Icons and Alert text appear in a banner at the top of the ticket detail form.
   – Alert Icons appear in the ticket list.
   – The Scoreboard includes a V.I.P. folder and subfolders for each ticket type. The V.I.P. subfolders include tickets for affected end users who are VIP special handling contacts.

   NOTE

   The V.I.P. Scoreboard folder displays for analyst roles.

   : When you launch the Knowledge Document search form via Quick Profile browser, the configured Federated Search checkboxes are not displayed. As a result, users are unable to perform federated searches on Knowledge Search forms displayed via the Quick Profile browser.

Associate a Contact to a Special Handling Type

You can assign a special handling type to a contact to alert analysts about tickets that affect end users with special requirements, such as for a visually impaired person or a contact that poses a security risk.

To associate a contact to a special handling type, perform the following:

Follow these steps:

1. On the Contact Detail page, select the Special Handling tab.

The Associated Special Handling List tab lists special handling types that are associated with the contact.

2. Click the Update Contact's Special Handlings button.
   The Search filter appears.
3. Search for the special handling type that you want to associate to the contact.
   The Special Handlings Update page appears.
4. Select one or more handling types from the left column and use the move button (>>) to move the types to the right column. Click OK.

   NOTE
   You can remove an association from a contact by using the move button (<<) to move the type from the right column to the left column. You can click the search icon to search for the value you want.

   The contact is associated to a handling type.
   CA SDM displays the following depending on the handling type, when a ticket specifies the contact in the Affected End-User field:
   - An alert banner appears on the Contact Detail for the affected end user on a ticket.
   - Alert text appears as a banner at the top of the ticket detail page and in the Quick Profile.
   - Ticket lists highlight the contact row and show an alert flag.
   - A V.I.P. folder appears in the Scoreboard for analyst roles. The folder contains all tickets that are associated with contacts (Affected End Users) that have a VIP special handling type.

Special Handling Types Options

This article contains the following:

Create a Special Handling Type

You can associate contacts to special handling types. This association lets you easily view and locate contacts that require special attention, for example, to identify tickets in a list that specify Very Important Person (VIP), vision that is impaired, or visitor as affected end users. You can create special handling types and can associate them to contacts.

To create a special handling type, perform the following:

Follow these steps:

1. On the Administration tab, select Service Desk, Application Data, Codes, Special Handling Types.
   The Special Handling List page appears.
2. Click Create New.
   The Create New Special Handling page appears.
3. Complete the fields as appropriate, and click Save.
4. Click Update Special Handling Members.
   The Contact Search page appears.
5. Complete the search fields as appropriate and click Search.
6. Select one or more contacts from the Contacts column and use the move button (>>) to move the contacts to the Special Handling column. Click OK.

   NOTE
   You can remove a contact from an association by using the move button (<<). Move the contact from the right column to the left column.

   The special handling type is created and it is associated with contacts.
7. Click Close Window.

Edit a Special Handling Type

You can edit special handling types that you can use to associate to contacts. For example, you can add contacts (members) to a special handling type.

To edit special handling types, perform the following:

Follow these steps:

1. On the Administration tab, select Service Desk, Application Data, Codes, Special Handling Types.
   The Special Handling List page appears and lists special handling types.
2. Click a name link.
   The Special Handling Detail page appears.
3. Click Edit.
   The Update Special Handling page appears.
4. Edit the fields as appropriate.
5. Select the Special Handling tab, and click Update Special Handling Membership.
   The Special Handling Search page appears.
6. Complete the page fields as appropriate, and click Search.
   The Special Handling Update page appears.
7. Use the move controls (>> and <<) to add or remove special handling type associations. Click OK.
   The special handling type is updated.
8. Click Save.
   The special handling type is updated.

Special Handling Types Fields

The following fields require explanation:

- Name
  Specifies the name of the special handling type, for example, VIP, Visually Impaired, Visitor, and so on.
- **Icon**
  Specifies the URL for the icon graphic that displays with the alert banner, for example, icon\visitor_icon.jpg, to display an alert banner. If you do not specify an icon URL, no alert icon appears on the ticket detail page. The format of the URL is http[s]://*server_name*:*server_port*/CAisd/img/icon_file. For example:
  ```
  http://server_name:8080/CAisd/img/glasses.gif
  ```
- **Escalate Urgency**
  Uses the special handling type to escalate the corresponding ticket priority.
- **Auto-Display Notes**
  Indicates whether the contact notes (ca_contact.comments) automatically display in an alert banner following the alert text banner when the affected user of the ticket is associated with the special handling type.
- **Alert Text**
  Specifies the text to display in an alert banner, for example, Visitor. If you do not specify text, no alert text or highlighting appears on the ticket detail page.
- **Description**
  Describes the special handling type, such as Very Important Person.

Add an Icon to a Special Handling Object

You can add an icon to a special handling object to identify the object. The icon file is located in the NX_ROOT/bopfg/www/wwwroot/img folder.

To add an icon to a special handling object, perform the following:

Follow these steps:

1. On the Administration tab, select Service Desk, Application Data, Codes, Special Handling Types.
   The Special Handling List page appears and lists special handling types.
2. Click a name link.
   The Special Handling Detail page appears.
3. Click Edit.
   The Update Special Handling page appears.
4. Specify the URL in the Icon field and click Save. The format of the URL is http[s]://*server_name*:*server_port*/CAisd/img/*icon_file*. For example:

   ```
   http://server_name:8080/CAisd/img/glasses.gif
   ```

   NOTE
   The filename can be case-sensitive depending on the web server used. If the case is incorrect, the icon is not displayed.

   The special handling icon is added.

How to Identify Tickets for Special Handling Contacts

Special handling contacts are contacts that require special attention, for example, security-risk monitoring, a visitor badge, or requiring special equipment. You can identify tickets for which the affected end user is a contact that requires special attention. For example, you can identify tickets in a list that specify Very Important Person (V.I.P.), vision that is impaired, or visitor as affected end users.

To identify tickets for which the affected end user is a special handling contact, perform the following:

- Navigate the Scoreboard to ticket lists.
  Special handling contact rows are highlighted and show an alert flag.
- As an analyst, navigate the Scoreboard to the V.I.P. folder.
  The V.I.P. folder includes subfolders for each ticket type. The subfolders include tickets for affected end users who are VIP special handling contacts.
- Navigate the Scoreboard, open ticket Details, and click Quick Profile for the affected end user on a ticket.
  An alert banner appears on the page to identify contacts with special handling types.

  NOTE
  An alert icon also appears if an alert icon is associated with the special handling type.
- Browse the Administration tab and open Contact Details.
  An alert banner appears on the page to identify contacts with special handling types.
  Note: An alert icon also appears if an alert icon is associated with the special handling type.

List Special Handling Types

You can list special handling types that you can use to associate to contacts.

To list special handling types

1. On the Administration tab, select Service Desk, Application Data, Codes, Special Handling Types.
   The Special Handling List page appears, lists special handling types, and corresponding information.
2. (Optional) Click a name link.
   Details about the special handling type appear.

Contact Types

Contact types are used to categorize CA SDM users into logical groupings based on how they use the system. For example, some of the many contact types that are predefined by the system are analyst, customer, and group. These

predefined contact types meet the needs of most CA SDM implementations; however, if your circumstances require it, you can modify the predefined contact types and create contact types. When you define users as contacts, you can associate a contact type with each one. You can base notification on contact type, which allows you to send a notification message to all contacts of a particular type.

You can select users based on contact type in various contexts. For example, most list and selection windows that display contacts have a search field where you can select a contact type as a search criterion.

Determine Behavior Based on Contact Type

The contact *type* determines which contacts display (and have permission) in different situations. For example, when you manually assign any type of ticket, such as a request or an issue, the field to specify the assignee requires that the person you specify have a contact type of analyst. If you choose to select a contact from a selection list for this field, only contacts with the type of analyst display in the selection list. Entering a contact with a different type displays the analyst search screen only.

> **NOTE**
> An important feature of the contact type is in the implementation of groups of contacts through the predefined group contact type.

How to integrate CA SDM with LDAP

Configure LDAP Options

You can configure CA SDM to access LDAP directory data.

Follow these steps:

1. Manually install LDAP options using the Web Interface Options Manager.

   > **NOTE**
   > The options necessary for basic LDAP integration are identified as required in the Description column in the following table. Options identified as optional are features you can add only if all the required options are installed. The values you specify when installing these options are written to the $NX_ROOT/NX.env file.
2. Restart the CA SDM service.
   The changes take effect.

Manage LDAP Servers Using the LDAP Configuration Utility

You can use the LDAP Server utility to manage multiple LDAP servers. You can perform the following tasks by using the utility:

- Add a new LDAP server.
- Delete an LDAP server that you no longer want to use.
- View LDAP server details.
- Restart LDAP virtual database.

> **WARNING**
>
> In an advance availability configuration, run the utility on the background server. After you add a new LDAP server, restart the CA SDM services on all the standby servers.

Follow these steps:

1. (Optional) To specify the domain name of the default LDAP server, execute the following command:

   ```
   pdm_options_mgr -c -a pdm_option.inst -s LDAP_DOMAIN -v <Default_LDAP_DomainName>
   pdm_options_mgr -c -a pdm_option.inst -s LDAP_DOMAIN -v <Default_LDAP_DomainName> -t
   ```

Configure the default LDAP server domain name in the following cases:
- CA EEM server is configured with multiple Microsoft Active Directory domains.
- Default LDAP server and any other LDAP server configured with CA SDM have the same user details.

After the configuration is complete, the default LDAP users must log in to CA SDM in the format domain_name\userid.

2. Open the windows command and navigate to the location $NX_ROOT/bin.
3. Run the following command:
   ```
   pdm_perl pdm_ldap_config.pl
   ```
4. Based on the task that you want to perform, select the appropriate option.

Option	Default Value	Description
ldap_domain		Required for configuring mulitple LDAP servers. Specifies the domain name of the LDAP server.
default_ldap_tenant		Required for multi-tenancy installation. Specifies the default tenant assignment for contacts imported from LDAP. You must use the tenant UUID when setting the Option Value field. You can get the tenant UUID from a database query. For example, "SELECT * FROM ca_tenant".
ldap_enable	Yes	Required. Enables LDAP integration with CA SDM.
ldap_host		Required. Specifies the LDAP database server host name or IP address.
ldap_port	389	Required. Specifies the LDAP server port number. If LDAP is enabled with SSL/ TLS, provide the valid LDAPS port number.
ldap_dn		Required. Specifies the LDAP server logon distinguishedName. **Example**: CN=Joe, CN=Users, DC=KLAND, DC=AD, DC=com If the LDAP server supports anonymous binds, this value can be empty.
ldap_pwd		Required. Specifies the password for LDAP server logon distinguishedName. If the LDAP server supports anonymous binds, this value can be empty.
ldap_search_base		Required. Specifies the starting point for searches in the LDAP schema tree: (UNIX) You must specify a starting container. For example: CN=Users, DC=KLAND, DC=AD, DC=com (Windows) You do not have to specify a container. You may start at the top of the schema tree. For example: DC=KLAND, DC=AD, DC=com

ldap_filter_prefix	(&(objectClass= user)	Specifies the prefix applied to an automatically generated filter when searching for LDAP users. This variable has been superseded by the ldap_user_object_class option. It is not available in Options Manager, but can be set manually in the NX.env file.
ldap_filter_suffix	)	Specifies the suffix applied to an automatically generated filter when searching for LDAP users. This variable has been superseded by the ldap_user_object_class option. It is not available in Options Manager, but can be set manually in the NX.env file.
ldap_user_object_class	person	Required. Specifies the value of the LDAP objectClass attribute applied to an automatically generated filter when searching for LDAP users.
ldap_enable_group	Yes	Optional. Enables CA SDM access type assignment based on LDAP group membership.
ldap_group_object_class	group	Required only if the ldap_enable_group is installed. Specifies the object name applied to an automatically generated filter when searching for groups.
ldap_group_filter_prefix	(&(objectClass= group)	Specifies the prefix applied to an automatically generated filter when searching for LDAP groups. This variable has been superseded by the ldap_group_object_class option. It is not available in Options Manager, but can be set manually in the NX.env file.
ldap_group_filter_suffix	)	Specifies the suffix applied to an automatically generated filter when searching for LDAP groups. This variable has been superseded by the ldap_group_object_class option. It is not available in Options Manager, but can be set manually in the NX.env file.
ldap_enable_auto	Yes	Optional. Enables auto generation of contact records from LDAP data.
ldap_sync_on_null	Yes	Optional. Overwrites existing CA SDM contact attributes with null data if the corresponding LDAP user attribute contains a null value.

ldap_service_type	Active Directory	Optional. Use this option if the CA SDM operating environment is Windows and the LDAP directory is *not* Active Directory (for example, eTrust or Novell). On UNIX operating environment, "Non AD" functionality is used only if this option is *not* installed. If it is installed, the service type is set to Active Directory.
ldap_enable_tls	No	Optional. Specifies whether Transport Layer Security (TLS) is enabled during LDAP processing.

Verify LDAP Integration

After you have installed the necessary LDAP options, CA SDM users can import LDAP data on a case-by-case basis, eliminating the need to fill in all the contact attribute fields manually.

To verify that you can search for and import LDAP records

1. Select File, New Contact from LDAP on the Service Desk tab.
   The LDAP Directory Search window appears.
2. Specify filter criteria, and then click Search. For example, you could enter b% in the Last Name field to retrieve a list of the LDAP user entries with last names that begin with the letter B.

   NOTE
   If your LDAP directory contains thousands of entries and you do not filter your search, your request attempts to retrieve *all* of the LDAP user records. This can cause the request to time-out and return zero records.

   Search results matching your filter criteria are displayed.
3. Select an entry.
   The Create New Contact window appears, populated with imported LDAP attribute values.
4. Click Save.
   The contact record is created.

To verify that you can update a contact using LDAP data

NOTE
Before performing this procedure, for test purposes you may want to use whatever LDAP editing tool you have available to change one or more attribute values in the entry you used for the previous procedure. You can verify that the contact is updated with the latest LDAP data.

1. Select Search, Contacts on the Service Desk tab.
   The Contact Search window appears.
2. Specify filter criteria to search for a contact that has a corresponding LDAP user entry. For example, you could search for the contact you created in the previous procedure.
   Search results matching your filter criteria are displayed.
3. Select the contact you want to update with LDAP data.
   The Contact Detail page appears, populated with the CA SDM contact information.
4. Click Edit.
   The Contact Update page appears.
5. Click Merge LDAP.
   The LDAP Entry List page displays a list of any LDAP user entries that correspond with the selected CA SDM contact. To search the LDAP directory for other entries, you can click Show Filter, specify filter criteria, and then click Search.

NOTE
If your LDAP directory contains thousands of entries and you do not filter your search, your request attempts to retrieve *all* of the LDAP user records. This may cause the request to time-out and return zero records.

6. Click the LDAP entry of interest.
   The LDAP Detail page displays the attribute values for the selected entry. Verify that you have selected the correct entry for the contact you want to update, then click Close Window.
7. On the LDAP Entry List page, right-click the entry that best matches the contact you want to update, and then select Merge into Contact.
   The Contact Update page reappears, populated with the current LDAP attribute values. If the LDAP data has changed since you created or last updated the contact, the changes are reflected in the contact attribute fields.

   NOTE
   If you have installed the ldap_sync_on_null option, and the LDAP entry contains null values for any attribute fields that correspond to contact attributes that currently contain values, the values in the contact record are overwritten with null values when you save the contact data.
8. Click Save on the Contact Update page.
   The contact is updated with the corresponding LDAP data.

Create a Contact

A contact is a person who uses your system regularly, such as an analyst or customer. After you have created the business structure and groups, you create contacts and map them to their respective location and organization.

You can create contacts using the following ways:

Create a Contact Using Data From LDAP

If your installation is configured to access a Lightweight Directory Access Protocol (LDAP) server such as Microsoft Windows Active Directory and has the necessary options installed, you can create and update contacts using data from the LDAP database. This method makes it easy to synchronize contacts with network user data.

NOTE
Administrators can configure automated synchronization of contacts with LDAP data.

Follow these steps:

1. Select File, New Contact from LDAP from the menu bar of the Service Desk tab.
   The LDAP Directory Search page appears.
2. (Optional) Complete one or more of the following filter fields to limit the LDAP Entry list to the records of interest:
   - **Last Name**
     Specifies the last name of the user as it appears in the LDAP directory. For example, you could enter b% in the Last Name field to retrieve a list of the LDAP user entries with last names that begin with the letter B.
   - **First Name**
     Specifies the first name of the user as it appears in the LDAP directory.
   - **Middle Name**
     Specifies the middle name of the user as it appears in the LDAP directory.
   - **User ID**
     Specifies the user name for logging in to the system.
3. Click Search.
   The LDAP Entry List page displays the records that match your search criteria.

   NOTE
   To see the information contained in an LDAP record without creating a contact, right-click the record of interest and select View. The LDAP Entry Detail page appears.

All fields on the LDAP Entry Detail page are self-explanatory except for the following:

- **User ID**
  Specifies the ID the user enters to log in to the system.
- **Distinguished Name**
  Specifies the fully qualified LDAP login name. For example, CN=Joe, CN=Users, DC=KLAND, DC=AD, DC=com.

4. Click the LDAP entry to create a contact.
   The Create New Contact page appears and is partially populated with LDAP information.
5. Enter additional information as necessary.
6. Click Save.
   The contact record is saved and the Contact Detail page appears. The following buttons are now available for configuring the contact:
   - **Update Environment** -- Displays the Configuration Item/Asset Search window for the contact or organization, where you can specify search criteria for the assets you want to consider. When you click Search, the Environment Update window is displayed, where you can add and remove assets for this contact or organization.
     Update Groups -- Displays the Group Search window, where you can specify search criteria for the groups you want to consider for this contact. When you click Search, the Groups Update window is displayed, where you can add and remove groups for this contact.

Create a Contact Automatically

You can configure CA SDM to create a contact automatically from a corresponding LDAP user record whenever a new user logs in to CA SDM.

To enable this feature, install all of the required LDAP options plus the ldap_enable_auto option.

The contact record is automatically created as follows:

1. If a user logging in to CA SDM does not yet have a contact record, but the user's login name exists in an LDAP record, the LDAP data is automatically imported and a contact record is created.
2. The automatically created contact record inherits the defaultaccess type security settings.
3. The contact can then be assigned an access type explicitly, or the access type can be assigned based on the user's membership in an LDAP Group.

This process is completely transparent to the user, appearing as any other login session.

Create Contacts Manually

If you do not want to use an active directory such as LDAP for your contacts information, you can create the contacts manually in CA SDM.

> **NOTE**
>
> If multi-tenancy is enabled, select the appropriate tenant from the drop-down list.

Follow these steps:

1. Select File, New Contact from the menu bar on the Scoreboard.
   The Create New Contact window opens.
2. Complete the contact fields .
3. Click Save.
   The contact information is saved.

Contact Fields

Tenant

Specifies the tenant that is associated with the contact (for multi-tenancy installations).

Contact ID

Specifies a unique identifier for the contact. If the default user authentication is being used, the value in this field is used as the password when the user logs in.

User ID

Specifies the user name of the contact. The contact uses this value to log in to the system.

Service Type

Specifies the level of support that is received by the contact.

Data Partition

Specifies the data partition for this contact. This value determines the records that this contact can access.

Access Type

Specifies the access type. The access type determines the system functions the contact can access.

Available

Indicates whether the contact is available for ticket assignments.

Confirm Self-Service Save

Indicates whether the contact receives a confirmation when saving a record from the self-service interface.

Analyst's Tenant Group

(Analyst Contact Type Only) Specifies the tenant group that the analyst is responsible for.
To configure the contact, use the following controls available on the tabs.

Notification

Defines the contact information and method for notifying the contact.

- – Select the notification method from the drop-down list (Email, Notification, Pager_Email, xMatters/Email, xMatters/Notification, and xMatters/Pager_Email) that you want to use for each message urgency level for this contact.

  NOTE

  CA SDM supports only one notification method at a time. If you are using Email, then you cannot use Notification at the same time. This applies to all out of the box notification methods like Email, Notification, Pager_Email, xMatters/Email, xMatters/Notification, and xMatters/Pager_Email.

  NOTE

  CA SDM Administrators must update the notification method manually in the contact details page if the xMatters and CA SDM integration is disabled. For more information, see Options Manager, xMatters.
- – Select the workshift that is valid for each notification urgency level.

For example, you may assign a Regular workshift (five-day week, eight-hours a day) to the normal level notification, but a 24 hour workshift to the emergency level notification.

Address

Specifies the location of the contact.

Organizational Info

Specifies the functional or administrative organization, department, cost center, or vendor information of the contact.

Environment

Specifies the environment of the contact, such as equipment, software, and services.

Groups

Assigns a contact to a group (a collection of contacts with a common area of responsibility).

Roles

Assigns the contact to one or more roles.

Service Contracts

Displays any service contracts that have been associated with the contact.

Special Handling

Lists the special handing contacts and lets you search for and associate a contact to a special handling type, such as a visitor or security risk type.

Event Log

Lists events that are associated with the contact, such as self service and knowledge activities.

Activities

Lists the activity log for the contact.

Merge Contacts Using LDAP

You can synchronize existing contacts with the current LDAP data.

Follow these steps:

1. Select Search, Contacts on the Scoreboard.
   The Contact Search page appears.
2. Fill in the filter fields as desired (or leave all filter fields blank to see a listing of all contacts), then click Search.
   The Contact List page appears.
3. Click the contact you want to edit.
   The contact's Detail page appears.
4. Click Edit.
   The contact's Update page appears.
5. Click Merge LDAP.
   The LDAP Entry List page appears. If the contact you are editing has a corresponding LDAP record, it appears on this page.
6. Click the LDAP entry.
   The LDAP Detail page appears.
7. Click Close Window after you have verified that the LDAP Detail page contains the data for the correct user.
8. Right-click the entry on the LDAP Entry List page for the contact you are updating and select Merge Into Contact.
9. Click Save on the contact's Update page.

Assign Access Type Using LDAP Groups

Assign Access Types values to contacts automatically with a Lightweight Directory Access Protocol (LDAP) server.

> **NOTE**
> To enable this feature, install the ldap_enable_group and ldap_group_object_class options.

Follow these steps:

1. Select Security and Role Management, Access Types on the Administrator tab.
2. Select the Access Type you want to associate with an LDAP Group. For example, select Administration.
   If the ldap_enable_group option is installed, the LDAP Access Group field appears on the Web Authentication tab.

NOTE
If an LDAP Group is already associated with the selected Access Type, a link to the LDAP Group Detail appears. Click the link for a read-only description of the LDAP Group and a listing of its members.

3. Click Edit on the Access Type Detail page to associate an Access Type with an LDAP Group.
4. Click the LDAP Access Group link.
5. (Optional) Enter filter criteria to limit the search to the LDAP groups of interest.
6. Select the LDAP Group that you want to associate with this Access Type.
7. Click Save.
   Association of the selected LDAP Group with the Access Type is complete.

Attribute Mapping

CA SDM contact record attribute values are synchronized with LDAP user attribute values based on the attribute mapping definitions in the $NX_ROOT/bopcfg/majic/ldap.maj file.

The following excerpt from ldap.maj illustrates mapping. Attribute names in the left column (id) are the CA SDM contact attribute names. The center column (distinguishedName) contains the corresponding LDAP attribute names.

```
     id                distinguishedName               STRING 512;
 last_name           sn,pzLastName                     STRING ;
 first_name          givenName,pzFirstName             STRING ;
 middle_name      initials,pzMiddleName             STRING ;
 userid                uid,sAMAccountName,pzUserName     STRING ;
 phone_number     telephoneNumber,pzWorkPhoneNumber STRING ;
```

If an SREL (a single relationship or foreign key in another database table) exists on CA SDM, the contact attribute value is synchronized with the corresponding LDAP value. If the SREL does not exist, it is not created automatically during LDAP synchronization processing.

NOTE
By default, attribute mapping is configured for the Microsoft Active Directory LDAP schema. If necessary, you can change the mapping by using a mod file.

How to Modify Attribute Mapping

You can change the default attribute mapping.

To change the default attribute mapping, do the following steps:

1. Navigate to $NX_ROOT/site/mods/majic and open the mod file.
2. Use MODIFY statements in the mod file as follows.
   - MODIFY statements must always be stated first in the file.
   - Following the MODIFY statements, any additional fields that are not in the ldap.maj file should be stated using the syntax shown in the following example.
   - If you define a field that contains a hyphen character in the attribute name, you must enclose the name in single quotes; otherwise, when you build the mod file, the attribute fails with a syntax error. For example, the following attribute name must be enclosed in single quotes:

     ```
     c_nx_string1 'swsd-secret-question' STRING ;
     ```
3. Save and close the mod file.
4. Restart the CA SDM service.

   WARNING
   The web engine does not start if there is a discrepancy in the syntax or case.

   Your changes take effect.

Example: Use MODIFY Statements

The following example shows how to modify two existing fields and add a new field.

```
//
// Map CA SDM userid attribute to ADAM Userid
//
MODIFY ldap userid cn ;
MODIFY ldap middle_name middleName ;
OBJECT ldap LDAP {
ATTRIBUTES LDAP_Entry{
contact_num employeeNumber STRING ;
};
} ;
```

How CA SDM Uses LDAP Data to Communicate

Lightweight Directory Access Protocol (LDAP) is a network communications protocol for querying and modifying directory services running over a TCP/IP network. An LDAP directory is a tree structure that contains entries for managing users, groups, computers, printers, and other entities on a network.

CA SDM can be configured to access an LDAP directory, which allows you to use LDAP data in several ways:

- Synchronize contacts with LDAP user records. Synchronization can occur in the following ways:
  - **At login** -- When a user logs in to the product, if an LDAP record exists for that user but a corresponding contact record does not exist, a contact record is automatically created based on the LDAP information.
  - **New Contact** -- When you manually create a contact record, you can select an LDAP record and can merge its attribute values with their corresponding fields in the new contact record.
  - **Batch Update** -- You can run batch jobs to automate the processes of importing and updating contact records with information from the corresponding LDAP records.

    NOTE
    Synchronization with LDAP is a one-way process. The LDAP data can be used to create and update contacts, but the product does not support updates to the LDAP directory.
- Assign CA SDM access types that are based on LDAP group membership.
- Implement an alternative method of performing CA SDM authentication.

The ldap_virtb component provides LDAP integration functionality on the following servers depending on your CA SDM configuration, regardless of operating system type:

- Conventional: Primary or secondary server.
- Advanced Availability: Background or application server.

NOTE
The $NX_ROOT/bopcfg/majic/ldap.maj file specifies the mapping between LDAP attributes and contact record attributes.

WARNING
CA SDM requires that LDAP records have an entry in the last name field in order to facilitate search, view, and import the LDAP data.

WARNING
CA SDM supports *paged searching*, which searches all records in your LDAP directory. Paged searching also enables you to import new contact records or sync existing contact records from any number of LDAP records. These capabilities are limited, however, if you are using Sun Java System Directory Server or Novell eDirectory because these LDAP servers do not support paged searching. In that case, you can only search, import, and

sync with the number of LDAP records specified by NX_LDAP_MAX_FETCH. For more information about paged searching, see NX.env File.

Access Type Assignments From LDAP Groups

You can configure CA SDM to assign access type values to contacts automatically, based on LDAP group membership. With automatic access type assignment enabled, if an LDAP user record that was used to create a contact belongs to an LDAP group that is associated with one of the CA SDM access types, then the contact is automatically assigned that access type. Otherwise, the contact inherits the default access type.

To enable automatic access type assignment, you must install the ldap_enable_group and ldap_group_object_class options.

For more information, see Configure LDAP Options.

LDAP Authentication

You can use LDAP to authenticate users logging in to CA SDM. The LDAP authentication is available when the CA EEM authentication component is integrated with CA SDM, which replaces the default validation that is performed by the host operating system. The LDAP authentication is only applicable when CA EEM is configured to use an external LDAP directory and you have selected OS authentication for an user validation type in an access type record.

When a CA EEM feature is activated, login requests are checked with the CA EEM server. A log in request is granted only if the following occurs:

- The specified user ID matches a CA SDM contact record.
- The user ID matches a user profile in CA EEM.
- The user ID and password combination is successfully validated by CA EEM.

NOTE

For more information about using CA EEM for authentication and to move authentication module to external server, see How to Move the Authentication Module to an External Server. Also, see Assign Access Type Using LDAP Groups.

Transport Layer Security

You can configure CA SDM to use Transport Layer Security (TLS) during LDAP processing. TLS, a secure communications protocol, is the successor of Secure Socket Layer (SSL v3) security. You install the ldap_enable_tls option to enable TLS.

WARNING
If this feature is enabled, all communications between CA SDM and the LDAP server are encrypted. If this feature is *not* enabled, all data communications (including the administrative login and password that is used to access the LDAP server) are sent in clear text.

NOTE
Ensure the corresponding LDAP server specific certificate is installed on the system where CA SDM is installed.

NOTE
For information about configuring TLS, refer to your LDAP server and operating system documentation. Manually install the LDAP options using the Web Interface Options Manager. For more information, see Configure LDAP Options.

Create Contacts in Batch Mode Using LDAP Data

This article contains the following topics:

Batch Import Contacts Using LDAP Data

You can run the pdm_ldap_import command-line utility to create CA SDM contacts in batch mode using LDAP data.

> **NOTE**
> In addition to creating contacts, pdm_ldap_import updates existing contacts if they are not synchronized with their corresponding LDAP entries. You can use the pdm_ldap_sync batch process to update existing contacts, but not create ones.

When you import contacts using LDAP data, restart the CA SDM services if you encounter the following error:

pdm_ldap_import: Method got_record in Ldap_Group_Catcher failed (LDAP agent not found).

pdm_ldap_import has the following syntax:

```
pdm_ldap_import -n "domain_name" -l "ldap_where_clause" [-c "contact_where_clause"] [-u "userid"]
```

- **-n *"domain_name"***
  Specifies the LDAP directory domain name from where you want to import contacts to CA SDM. If you do not specify the domain name, CA SDM retrieves the data using the default LDAP domain name.
- **-l "*ldap_where_clause*"**
  Specifies the userids of LDAP records to be searched. Replacement variables are indicated with the '?' character. For example, for *userid* = *?*. The default value is *userid* = *?*. In this special case, id is mapped to the contact attribute ldap_dn.
  Note: Use the keywords as defined in the *ldap.maj* file. You can also search by using the memberOf = 'group_dn' syntax.
- **-c "*contact_where_clause*"**
  (Optional) Specifies how to determine whether the contact record already exists. If the contact record does not exist, a new contact record is inserted. If the contact record does exist and is not synchronized with the current LDAP data, the contact record is updated.
- **-u "*userid*"**
  (Optional) Specifies the login name under which the pdm_ldap_import program runs.

> **NOTE**
> You can use wildcards with pdm_ldap_import to specify multiple records.

Examples: Batch Imports Using LDAP Data

This example imports a single LDAP record for userid jsmith11 from the LDAP directory with the domain name example.com:

```
pdm_ldap_import -n "example.com" -l "userid = 'jsmith11'"
```

This example imports all LDAP records with a userid that begins with the letter C from the LDAP directory with the domain name example.com:

```
pdm_ldap_import -n "example.com" -l "userid = 'c%'"
```

This example imports all LDAP User records in the directory from the LDAP directory with the domain name example.com:

```
pdm_ldap_import -n "example.com" -l "userid = '%'"
```

Batch Import Contacts by Date and Time

You can configure the pdm_ldap_import utility to import LDAP records that were created before or after a specified date and time. To enable this functionality, create an ldap.mod file with the following content:

```
OBJECT ldap {
  ATTRIBUTES LDAP_Entry {
    whenCreated whenCreated STRING ;
```

```
  };
};
```

This adds the *whenCreated* attribute to the LDAP object.

The rules for filtering records using the whenCreated attribute are as follows:

- Use only the >= or <= operator.
- Specify *all* characters for the date/time value, including the Z. Place a 0 in any location you do not wish to explicitly state (for example, the time of day).
- Place the date/time specification at the beginning of the filter; do not use leading 0s at the beginning of the string.
- Do not include the leading century. For example, to specify the year 2008, use 08.

NOTE
Single quotation marks must surround the date/time value.

Example: Using the whenCreated Attribute to Import LDAP Entries

The following example uses the whenCreated attribute to import LDAP entries created after 3/11/2008.

```
Pdm_ldap_import -l "whenCreated >= '080312000000Z'"
```

Example: Using the whenCreated Attribute to Search for LDAP Records

The following example uses the whenCreated attribute with pdm_ldap_test to search for LDAP records created after 3/11/2008.

```
pdm_ldap_test.exe -f "whenCreated>=080312000000Z" -a whenCreated
Starting ldap_test.exe...
LDAP Directory Type    : active directory
Service Desk Platform    : windows
Search Base    : DC=kirklandsd,DC=ca,DC=com
Search Filter    : (&(objectClass=person)(whenCreated>=080312000000Z))
Administrator Username    : CN=Administrator,CN=Users,DC=kirklandsd,DC=ca,DC=com
Administrator Password    : **********
LDAP Host    : gecko.kirklandsd.ca.com
LDAP Port    : 389
LDAP API Version    : 3
DN: CN=aixmail,CN=Users,DC=kirklandsd,DC=ca,DC=com
        whenCreated(17)(0): 20080312035327.0Z
DN: CN=hpmail,CN=Users,DC=kirklandsd,DC=ca,DC=com
        whenCreated(17)(0): 20080312035425.0Z
DN: CN=sunmail,CN=Users,DC=kirklandsd,DC=ca,DC=com
        whenCreated(17)(0): 20080312035726.0Z
3 Total LDAP records found...
```

Batch Import Summary and Log Data

The pdm_ldap_import command maintains a detailed log of all activity of each run. The ldap_logging.0-n log file is located in the $NX_ROOT/log directory.

The following is an example of the summary data pdm_ldap_import returns at the command line:

```
pdm_ldap_import Starting...
pdm_ldap_import Summary:  Processed(21) Updated(1) No Matches(7) New Contacts(11) Multiple Matches(0) Empty
 Filter(2) Errors(0)
pdm_ldap_import Complete...
```

The following table describes the summary data.

Status	Count	Description
Processed	21	The number of CA SDM contacts with corresponding LDAP entries.
Updated	1	The number of contact records that were updated because the corresponding LDAP entry contained different information
No Matches	7	The number of CA SDM contact records with no corresponding LDAP entries.
New Contacts	11	The number of new contact records that were created based on corresponding LDAP entries
Multiple Matches	0	The number of LDAP entries with multiple matching contact records, as defined by the ldap_search_base option
Empty Filter	2	The number of LDAP entries that cannot be used to generate a valid search filter
Errors	0	The number of LDAP entries that encountered an error during processing. For example, LDAP records that do not contain a value in a field required by CA SDM (such as Last Name) are counted as failures and cannot be imported.

Batch Update Contacts Using LDAP Data

You run the pdm_ldap_sync utility to update contact records in batch mode using LDAP data.

WARNING
This utility overwrites the existing tenant of the LDAP contact defined in CA SDM. If you want to retain the tenant value, you must modify NX.env by adding the NX_RETAIN_TENANT_VALUE variable manually, and set it to "yes". If this variable is set to "no", missing, or not set properly, the utility overwrites the tenant information.

NOTE
The pdm_ldap_sync utility synchronizes existing contacts with corresponding LDAP entries, but does not create contacts. You can use the pdm_ldap_import batch process to create contacts.

pdm_ldap_sync has the following syntax:

```
pdm_ldap_sync -n "example.com" -l "ldap_where_clause" [-c "contact_where_clause"] [-u "userid"]
```

- **-n *"domain_name"***
  Specifies the LDAP directory domain name from where you want to import contacts to CA SDM. If you do not specify the domain name, CA SDM retrieves the data using the default LDAP domain name.
- **-l "ldap_where_clause"**
  Determines how to search for matching LDAP records. Replacement variables are indicated with the '?' character. For example, for *userid* = *?*, the default value is *id* = *?*. In this special case, id is mapped to the Contact attribute ldap_dn.
- **-c "contact_where_clause"**
  (Optional) Determines which contacts are used when searching for matching LDAP records.
  Default: "ldap_dn IS NOT NULL"
- **-u "userid"**

(Optional) Specifies the userid under which pdm_ldap_sync runs.

NOTE
You can use wildcards with pdm_ldap_sync to specify multiple records.

Examples:

This example establishes a baseline of contact records that have a corresponding LDAP record:

```
pdm_ldap_sync -n "example.com" -l "userid = ?" -c ""
```

This example uses the default parameters to update all contacts that have an LDAP distinguishedName:

```
pdm_ldap_sync -n "example.com"
```

This example updates a single contact:

```
pdm_ldap_sync -n "example.com" -l "userid = ?" -c "userid = 'jsmith11'"
```

Batch Update Summary and Log Data

The pdm_ldap_sync command maintains a detailed log of all activity for every run. The ldap_logging.0-n file is located in the $NX_ROOT/log directory.

The following is an example of the summary data pdm_ldap_sync returns at the command line:

```
pdm_ldap_sync Starting...
pdm_ldap_sync Summary:  Processed(21) Updated(1) No Matches(7) No Changes(11) Multiple Matches(0) Empty
 Filter(2) Errors(0)
pdm_ldap_sync Complete...
```

The following table describes the summary data:

Status	Count	Description
Processed	21	The number of CA SDM contacts with corresponding LDAP entries
Updated	1	The number of LDAP entries with information different from their corresponding CA SDM contact record
No Matches	7	The number of CA SDM contact records with no corresponding LDAP entries.
No Changes	11	The number of LDAP entries with information identical to their corresponding CA SDM contact record
Multiple Matches	0	The number of LDAP entries with multiple matching contact records in CA SDM, as defined by the ldap_search_base option
Empty Filter	2	The number of LDAP entries that cannot be used to generate a valid search filter
Errors	0	The number of LDAP entries that encountered an error during processing

Test Connections to LDAP Directories

This article contains the following topics:

Use the pdm_ldap_test command-line utility to test the connection to an LDAP directory, ensure that the search options are correctly configured, and test the TLS configuration.

By default, pdm_ldap_test uses the parameter settings that are entered in the $NX_ROOT/NX.env file when you install, edit, or uninstall LDAP options. To override the defaults, you can specify parameters at the pdm_ldap_test command line.

To see the available parameters for this command, enter the following command:

```
pdm_ldap_test -h
```

WARNING

On UNIX, the LIBPATH must be set before running several CA SDM utilities. Use *pdm_task* to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

Verify Connection to LDAP Server

To verify the connection to the LDAP server, run pdm_ldap_test without parameters:

```
pdm_ldap_test
```

Successful Connection to LDAP Server

If the connection is successful, you receive output similar to the following:

```
Starting pdm_ldap_test...
LDAP service type=active directory
Service Desk platform=windows
Using search base=DC=mycontroller,DC=xyz,DC=com
Using filter=(&(objectCategory=person))
ldap_init(myserver.mycontroller.xyz.com,389): (Success)
ldap_bind_s(Administrator) (Success)
LDAP API Verion 3
```

View Search Parameters

To verify that the search parameters are correctly configured, run pdm_ldap_test without parameters:

```
pdm_ldap_test
```

Successful Search

When your search is successful, you see output similar to the following:

```
DN: CN=John A. Smith,CN=Users,DC=COMPUTERTEST
       c(2)(0): US
       displayName(14)(0): John A. Smith
       mail(14)(0): account02@mycompany.com
       givenName(4)(0): John
       initials(1)(0): a
       distinguishedName(38)(0): CN=John a.
Smith,CN=Users,DC=COMPUTERTEST
       objectGUID(3)(0): 314738
       pager(12)(0): ###-111-1111
       postalCode(5)(0): 11111
       SAMAccountName(7)(0): account02
       sn(6)(0): Smith
       telephoneNumber(12)(0): ###-342-6265
```

```
       userPrincipalName(16)(0): account02@COMPUTERTEST
DN: CN=Mike Johnson,CN=Users,DC=COMPUTERTEST
       displayName(10)(0): Mike Johnson
       givenName(4)(0): Mike
       distinguishedName(34)(0): CN=Mike
Johnson,CN=Users,DC=COMPUTERTEST
       objectGUID(12)(0): 312328
       SAMAccountName(7)(0): account03
       sn(5)(0): Johnson
       userPrincipalName(16)(0): account03@COMPUTERTEST
```

Determine which Attribute Names have Values

Use the -a “*” parameter and the -f parameter with the pdm_ldap_test command to determine which attributes are defined for LDAP User or Group records. This test is useful for seeing if there are LDAP attributes that you want to map to Contact attributes, and to verify that a particular attribute has a value and should be available when creating or updating Contact records.

The following example shows output from an iPlanet directory:

```
pdm_ldap_test -a "*" -f sn=Account_1000001
2 LDAP records found...
DN: cn=Account_1000001,ou=200K_Plus,o=SmartLabs
   sn(15)(0): Account_1000001
   objectClass(13)(0): inetOrgPerson
   objectClass(20)(1): organizationalPerson
   objectClass(6)(2): Person
   objectClass(18)(3): ndsLoginProperties
   objectClass(3)(4): Top
DN: cn=Account_1000001,ou=2_Plus,o=SmartLabs
   mail(28)(0): ThisIsTheMailingAddressField
   uid(13)(0): Login_1000001
   givenName(17)(0): GivenNameOfPerson
   sn(15)(0): Account_1000001
   objectClass(13)(0): inetOrgPerson
   objectClass(20)(1): organizationalPerson
   objectClass(6)(2): Person
   objectClass(18)(3): ndsLoginProperties
   objectClass(3)(4): Top
```

The following example shows output from Active Directory:

```
Ldap_test  - a "*"  - f (&(sn=Brown)(initials=A))"
1 LDAP records found...
DN: CN=John A. Smith,CN=Users,DC=mycontroller,DC=xyz,DC=com
   objectClass(3)(0): top
   objectClass(6)(1): person
   objectClass(20)(2): organizationalPerson
   objectClass(4)(3): user
   cn(16)(0): John A. Smith
   sn(5)(0): Brown
   givenName(7)(0): John
   initials(1)(0): A
   distinguishedName(55)(0): CN=John A. Smith,CN=Users,DC=mycontroller,DC=xyz,DC=com
   displayName(16)(0): John A. Smith
```

```
memberOf(52)(0): CN=Domain Admins,CN=Users,DC=mycontroller,DC=xyz,DC=com
sAMAccountName(7)(0): smijo04
userPrincipalName(25)(0): smijo04@mydomain.xyz.com
objectCategory(63)(0): CN=Person,CN=Schema,CN=Configuration,DC=mycontroller,DC=xyz,DC=com
```

Narrow Your Search

Use the -f parameter with the pdm_ldap_test command to specify a filter to be added to the base filter for narrowing the search criteria. You must use appropriate LDAP syntax and LDAP schema attribute names in your filter. Surround your filter with double quotation marks and use parenthesis to clarify the order of operator precedence.

For example, use the following command to search for all records where sn=Account_10001:

```
pdm_ldap_test  - f "(sn=Account_10001)"
```

The pdm_ldap_test utility supports the following equality operators:

Equality Operator	Description
=	equal to
<=	less than or equal to
>=	greater than or equal to
~=	like

The pdm_ldap_test utility supports the following Boolean operators:

Boolean Operator	Description
&	AND
|	OR
!	NOT

The AND and OR operators affect each set of parenthesis () in the search filter. The NOT only affects the first set of parenthesis. Always place these operators *before* the search filters to be operated on, rather than between them. They can be applied to any number of filters, as shown in the following examples:

```
"(&(sn=Brown)(initials=A)) "
"(|(sn=Brown)(sn=Smith))"
"(!sn=Brown)"
```

Error Messages for Failed Connections to LDAP Directories

This article contains the following topics:

Failed Connection Server Down or Incorrect Name or Port

If the connection fails because the server is down or you specified an incorrect LDAP server name or port, you receive output similar to the following:

```
Starting pdm_ldap_test...
LDAP service type=active directory
Service Desk platform=windows
Using search base=DC=mycontroller,DC=xyz,DC=com
Using filter=(&(objectCategory=person))
```

```
ldap_init(junk,389): (Success)
ldap_bind_s(Administrator) (Server Down)
```

Failed Connection Invalid LDAP_DN or LDAP_PWD

If the connection fails because you specified an incorrect LDAP_DN or LDAP_PWD, you receive output similar to the following:

```
Starting pdm_ldap_test...
LDAP service type=active directory
Service Desk platform=windows
Using search base=DC=mycontroller,DC=xyz,DC=com
Using filter=(&(objectCategory=person))
ldap_init(myserver.mycontroller.xyz.com,389): (Success)
ldap_bind_s(junk) (No Such Object or Invalid Credentials)
```

Failed Search Invalid SEARCH_BASE

If the search fails because you have an invalid SEARCH_BASE, you receive output similar to the following:

```
Starting pdm_ldap_test...
LDAP service type=edirectory
Service Desk platform=windows
Using search base=o=SmartLabsx
Using filter=(&(objectClass=InetOrgPerson)
ldap_init(155.35.173.110,15389): (Success)

ldap_bind_s() (Success)
LDAP API Verion 3
ldap_search_st() (No Such Object or Referral)
```

Failed Search SIZELIMIT_EXCEEDED, TIMEOUT

The search can fail with a SIZELIMIT_EXCEEDED or TIMEOUT message if you specify a filter that does not sufficiently narrow the search. Most LDAP Servers limit the size of the result set returned from a search request. If you exceed this limit, you receive a SIZELIMIT_EXCEEDED message. If your search request takes longer than the default timeout of 20 seconds, the LDAP server times out your request and you receive a TIMEOUT error message similar to the following:

```
Starting pdm_ldap_test...
LDAP service type=edirectory
Service Desk platform=windows
Using search base=o=SmartLabsx
Using filter=(&(objectClass=InetOrgPerson)
ldap_init(155.35.173.110,15389): (Success)
ldap_bind_s() (Success)
LDAP API Verion 3
ldap_search_st() (TIMEOUT or SIZELIMIT_EXCEEDED)
```

Failed Search 0 Records Returned

The search may fail because your default filter is not correct. If pdm_ldap_test returns zero (0) records, always check the Using filter line, which is the base filter generated by the LDAP_FILTER_PREFIX and LDAP_FILTER_SUFFIX, or the LDAP_OBJECT_CLASS options:

```
Starting pdm_ldap_test...
```

```
LDAP service type=edirectory
Service Desk platform=windows
Using search base=o=SmartLabs
Using filter=(&(objectClass=InetOrgPerson)
ldap_init(155.35.173.110,15389): (Success)
ldap_bind_s() (Success)
LDAP API Verion 3
ldap_search_st() 0 records
```

Configuration File Modification

When you install the CA SDM web interface, a sample web engine configuration file (*web.cfg*) is installed that you can modify to suit your needs. The *web.cfg* file itself contains helpful comments that you can read by viewing the file. You can open the *web.cfg* file from the appropriate directory:

- (Windows) %NX_ROOT%\bopcfg\www\
- (UNIX) $NX_ROOT/bopcfg/www/

> **NOTE**
> Some additional configuration variables, such as charset, are also available in Options Manager. These are accessible using the Administration tab in the web interface.

AllowInactiveSrelEntry
Specifies whether a record can be saved when it references inactive records in a reference table.

- When this property is omitted or set to zero, inactive reference table entries (such as request status or change category) are not included in drop-down selects, and cannot be specified for lookup or hierarchical search fields.
- When this property is set to 1, the inactive flag is ignored in reference table entries.

Regardless of the setting of this flag, records already containing a reference to an inactive reference table entry can be saved without changing the reference; the flag affects only new field values.

AnnouncementLength
Specifies the maximum number of announcements to display on the opening screen for both customer and analyst interfaces. CA SDM begins the display with the most recent announcement, continuing for the number of announcements specified by this parameter. Users of the analyst interface can view additional announcements by selecting Announcements from the Search menu.
Default: 10, meaning that the ten most recent announcements display.

AnonymousPrio
Specifies valid priorities for tickets created by guest users. Such users can specify only one of the priorities in the AnonymousPrio list for their tickets. Entries in the list of priorities are separated by spaces. Each entry must be either a number between 1 and 5 or the word “none” (without quotes). The default priority for tickets created by guest users should be specified first, and can be repeated in the list.
The values valid for AnonymousPrio correspond to the symbolic names of the priorities as distributed. You can use the java client to modify these symbolic names; however, this does not affect the specification for AnonymousPrio, which must continue to reference priorities by their default names, where 1 corresponds to the highest priority.
Default: None, meaning that all requests created by a guest user have a priority of none.

Autofill
Specifies that the web interface should auto-fill lookup fields when a user keys data into them and presses Tab to exit the field. When a user does this and the Autofill option is selected, the browser asks the server to confirm that the update is correct. This results either in the full name filling in the field (if the user provided a partial name), or a pop-up search window appearing (if the user’s selection was incorrect or ambiguous).

This property is optional. Autofill is enabled by default so if this property ID is omitted or set to Yes, tabbing out of a lookup field automatically searches the database. If this property is set to No, no auto-fill occurs, and lookup fields are not verified until the record is saved.

CAisd
Specifies the path (including a leading slash) to the alias or virtual directory in your HTTP server that contains the files needed by the CA SDM web server. This property typically has a value of /CAisd in both UNIX and Windows installations. For Apache servers, it should be defined in an Alias statement in a configuration file. For IIS, it should match an Alias field in the Directory Properties Window.

CGI
Specifies the name of the CGI executable program supplied with the web interface (without the .exe suffix).
Default: pdmweb

> **NOTE**
>
> If you rename this program, you must update this property.

CgiReport

Specifies the name of the CGI executable program for web reports supplied with the web interface (without the .exe suffix).
Default: pdm_cgireport

> **NOTE**
>
> If you rename this program, you must update this property.

ContactAutoDesc

Specifies whether the contact's name should be inserted into the description of new issues and requests created in the customer and employee interfaces. If this property is omitted or specified as 0, no automatic information is added to the description of new issues and requests. If this property is specified as 1, the contact's name is automatically inserted into the description of issues and requests created in the customer and employee interfaces. This property has no effect on the analyst interface.

ContactAutoDescWithIP
Specifies whether the contact's IP address should be inserted into the description of new issues and requests created in the customer and employee interfaces. If this property is omitted or specified as 0, no IP address information is added to the description of new issues and requests. If this property and the ContactAutoDesc property are both specified as 1, the contact's name and IP address are automatically inserted into the description of issues and requests created in the customer and employee interfaces. This property has no effect on the analyst interface. It is ignored unless ContactAutoDesc is 1.

CstPrio
Valid priorities for issues created with the customer web interface. Users of the customer interface can only specify one of the priorities in the CstPrio list for their issues and cannot update the priority of an issue if an analyst has altered it to a value that is not in the list.
Entries in the list of priorities are separated by spaces. Each entry must be either a number between 1 and 5 or the word "none" (without quotes). The default priority for issues created with the customer interface should be specified first (and can be repeated in the list).
Default: none, 3, 4, 5
The values valid for CstPrio correspond to the symbolic names of the priorities as distributed. You can use the java client to modify these symbolic names; however, this does not affect the specification for CstPrio, which must continue to reference priorities by their default names, where 1 corresponds to the highest priority.

DateFormat
Defines the order of elements in dates.

Default: MM/DD/YYYY hh:mm a(am,pm)

Symbol	Description
M	Print 1 or 2 digits of month
MM	Print 2 digits of month
D	Print 1 or 2 digits of date
DD	Print 2 digits of date
YY	Print 2 digits of year
YYYY	Print 4 digits of year
H	Print 1 or 2 digits of hours on 24 hour clock
HH	Print 2 digits of hours on 24 hour clock
H	Print 1 or 2 digits of hours on 12 hour clock
hh	Print 2 digits of hours on 12 hour clock
M	Print 1 or 2 digits of minutes
mm	Print 2 digits of minutes
S	Print 1 or 2 digits of seconds
ss	Print 2 digits of seconds
a(am, pm)	Print am and pm as a string

DateFormatNoTime

Specifies the same definition as DateFormat, but without specifying the time portion.

DebugSource
Enables the standard browser right-click menu on CA SDM forms. When this property is not set, you can right-click a form to display a CA SDM menu. You should use caution when setting this property, because some of the options on the standard browser right-click menu can cause execution errors (this is the reason why it is usually disabled). On Internet Explorer, you can display the standard browser right-click menu although the DebugSource property is not set by pressing the Ctrl key when you right-click.

DebugTrace
Causes the web engine to write trace information to the stdlog file.

> **WARNING**
> This property should *not* be set for typical use. It should only be used when requested by CA Support.

EmpPrio
Valid priorities for requests created with the employee web interface. Users of the employee interface can only specify one of the priorities in the EmpPrio list for their requests and cannot update the priority of a request if an analyst has altered it to a value not in the list.
Entries in the list of priorities are separated by spaces. Each entry must be either a number between 1 and 5 or the word “none” (without quotes). The default priority for requests created with the employee interface should be specified first (and can be repeated in the list).
Default: none, 3, 4, 5
The values valid for EmpPrio correspond to the symbolic names of the priorities as distributed. You can use the java client to modify these symbolic names; however, this does not affect the specification for EmpPrio, which must continue to reference priorities by their default names, where 1 corresponds to the highest priority.

ExclLockSeconds
Specifies the maximum number of seconds that a user is given an exclusive lock on a record after clicking Edit. After this period elapses, the web engine releases the lock, allowing other users to update the record. The web engine attempts to

retake the lock if a user asks to save after ExclLockSeconds has expired. This attempt succeeds only if no other user has updated the record while the lock was available. If the attempt to retake the lock fails, the user must re-enter the updates.
Default: 120 (two minutes)
This argument is optional. If omitted, the default value is assumed.

> **NOTE**
> The ExclLockSeconds setting must be shorter than the Timeout setting. ExclLockSeconds is specified in seconds and Timeout is specified in minutes.

FormCacheMax
Specifies the maximum number of forms to be retained in web engine memory for each user. The web engine always retains the last FormCacheMax forms used by each user. Forms beyond this number are eligible to be timed out. Timed out forms cannot be accessed by the Back or Forward buttons on the main page, and can no longer be submitted on a pop-up form.
Default: 10
Timed out forms save memory in the web engine, but they occasionally require users to manually refresh them. You can set FormCacheMax to - 1 to disable the FormTimeout feature.

FormTimeout
Specifies the minimum number of seconds that a form is retained in the web engine before it is eligible for removal from the cache. Users always have at least the number of seconds specified in this parameter to work on a form before submitting it. In addition, the web engine always retains the most recently used FormCacheMax forms for each user. You can use the StayCacheList property to prevent specified forms from timing out.
Default: 180 (3 minutes)

FormTitle
Specifies a string to be included in the title bar of a web browser displaying a CA SDM web form. The value of FormTitle supplements the title of the specific form displayed.
Default: CA SDM
For example, if the default value is retained, and Microsoft Internet Explorer is used to display the Announcement Detail form, the title bar displays the following:
Announcement Detail -- CA SDM -- Microsoft Internet Explorer
This property is optional. If omitted, the analyst web interface does not use a constant value in the title. The customer and PDA web interfaces revert to the default value.

HitTrackFile
Specifies the full path to a file that receives a log of all web pages used. One line is written to this file each time a user requests a page. The file can grow indefinitely, so be cautious when specifying this property.

> **NOTE**
> Records containing a time stamp, user ID, database record ID, and HTMPL form name are appended to this file. The format of the records may change. You must periodically maintain this file so that it does not get too large.

This property is optional. If omitted, no hit tracking file is written.

HtmplCacheSize
Specifies the size of the HTMPL cache. When this size is exceeded, the least used form is removed from the cache.
Default: 1000.

ListAllMaximum
Specifies the maximum number of records that can be displayed in a list before a request to display the entire list produces a pop-up warning message advising the user that the request adversely impacts performance and is not allowed.
Default: 2500

ListAllWarn

Specifies the maximum number of records that can be displayed in a list before a request to display the entire list produces a pop-up warning message advising the user that the request may adversely impact performance, and asking for confirmation.
Default: 1000

ListBottomMaximum
Specifies the maximum number of records that can be displayed in a list before a request to scroll to the bottom produces a pop-up warning message advising the user that the request adversely impacts performance and is not allowed.
Default: 2500

ListBottomWarn
Specifies the maximum number of records that can be displayed in a list before a request to scroll to the bottom produces a pop-up warning message advising the user that the request may adversely impact performance, and asking for confirmation.
Default: 1000

ListPageLength
Specifies the maximum number of found records to be shown on a list page after performing a search.
Default: 10

LogoutURL
Specifies the complete URL of a web page to be displayed after a user logs out of CA SDM. This property is optional. If it is not specified, logging out returns the login form.

Lr_Refresh
Specifies the log reader refresh interval in seconds. If this property is non-zero, the Notification Log Reader automatically refreshes itself at the specified interval (with a minimum of thirty seconds).
This property is optional. If omitted, the log reader refreshes itself every 5 minutes (a default value of 300 seconds). If this property is specified as zero, the log reader does not automatically refresh at all.

MacroPath
Specifies a list of directory paths that the web engine searches to find files requested by the PDM_MACRO tag. You can specify multiple directories separated by spaces. You can include environment variables in the directory names by prefixing them with a dollar sign (for example, $NX_ROOT). For both Windows and UNIX, separate path components with a forward slash (/), not a backslash (\). This property is required. It is typically set as follows:

```
$NX_ROOT/site/mods/macro $NX_ROOT/bopcfg/www/macro
```

MatchesFound
Specifies the text of the message to display under a field when a user's key for a lookup field is ambiguous and the edit form must be redisplayed with a drop-down select list. This property is optional; if omitted, it defaults to Multiple Matches.

MaxRecordsAutoSuggest

Specifies the number of records that autosuggest displays when search as you type or autosuggest displays suggestions below a lookup.
Default: 25

MaxSelectList
Specifies the maximum number of matches to display in the drop-down select list shown when a user's key for a lookup field is ambiguous and the edit form must be redisplayed. If more than this many matches are found, the message specified for Too Many Matches is displayed.

MinCharsAutoSuggest
Specifies the minimum number of characters to enter in the lookup fields before search as you type or autosuggest displays suggestions.
Default: 3

MouseoverPreviewDelayTime

Specifies the delay time (in milliseconds) between hovering the mouse pointer over a link and the display of the mouseover preview.
If the mouse moves away from the link before the delay time expires, the preview does not appear.
Default: 1000

NoMatchesFound
Specifies the text of the message to display under a field when a user's key for a lookup field is incorrect and the edit form must be redisplayed. This property is optional; if omitted, it defaults to No Matches Found.

PreLogin Timeout
Specifies the maximum number of minutes the web engine keeps a session active before login. The web engine automatically starts a session when a user requests a login form, in anticipation of the user completing the login. If the user does not login within the time period specified, the web engine destroys the session. If the user subsequently logs in, the web engine creates a session that is transparent to the user.
This property has no end-user impact. Its sole purpose is performance -- balancing web engine memory usage versus the overhead of destroying and recreating a session. This property is optional; if omitted, it defaults to one minute.

RedirectingURL
Specifies the URL the WebDirector should use to send requests to this web engine. This property specifies the full URL of the web engine, including http. This property is required if you are using WebDirector. It is ignored if you are not.

SchedExpMaximum

Specifies the limit of the number of schedule events that can be exported at a time.
Default: 1000

> **WARNING**
> The default is the maximum exports that CA SDM can handle at a time. Increasing this default could cause system instability. If you attempt to export more than the value specified in SchedExpMaximum, a message appears refusing your exporting request.

SelListCacheExclude
Specifies the names of the factories (objects) to be excluded from caching for <PDM_SELECT> lists. To improve performance, the web engine usually caches in its own memory of the contents of small tables used in <PDM_SELECT> (drop-down) lists and hierarchical search lists. You may want to suppress caching for a table if you are using data partition constraints to specify that different users should receive different views of the table. In addition, including tables in the value of this property eliminates the need for the web engine to query their record count at startup, slightly improving startup performance. This property is optional. If specified, it should contain one or more object names separated by spaces.

SelListCacheMax
Defines the maximum number of records in a table that can be cached in the web engine. The web engine keeps the entire contents of tables at or below its cache size in its own memory, improving its performance in building <PDM_SELECT> lists using these tables. Specifying a higher value for this property improves performance at the expense of memory usage.
Default: 10
SelListCacheMax is ignored for tables used in hierarchical search lists, such as category on requests, issues, and change orders. The web engine always stores the entire contents of tables used in hierarchical search lists in its own memory. If you have a large number of values in any of these tables, you may want to specify the SelListCachePreload property.

SelListCachePreload

Specifies one or more tables to be loaded into the web engine's select cache at startup time. Tables not specified in this property are loaded the first time they are used. If SelListCacheMax is large, or if you have a large number of records in a hierarchical search list (such as category), you may want to specify the table in SelListCachePreload. This avoids a response time delay the first time a user accesses a form using the table.
The specification for the SelListCachePreload property is a blank-separated list of object names. Each object name can be followed by an optional list of attribute names in parentheses. The attributes specified in the list are loaded into the web

engine. If no attributes are specified, only the common name and rel attr value of the object are loaded. This is sufficient for drop-down selects, but may not be sufficient for hierarchical searches. If you modify the hierarchical search forms (hiersel_xx.htmpl, where xx is an object name), be sure that the SelListCachePreload property specifies every attribute used in the form. If you omit an attribute, the cache is reloaded when the form is used.
The SelListCachePreload property is optional. If it is omitted, nothing is loaded into the select cache until a user requests a form using a drop-down select or a hierarchical search.

```
chgcat(description owning_contract) chgstat crs isscat(description owning_contract) issstat pcat(description
 cr_flag in_flag pr_flag owning_contract) pri tskstat urg pcat_cr(description cr_flag in_flag pr_flag
 owning_contract) pcat_pr(description cr_flag in_flag pr_flag owning_contract) pcat_in(description cr_flag
 in_flag pr_flag owning_contract)
```

StayCacheList

Specifies the names of forms that are never removed from the forms cache, regardless of the length of time they have been displayed. This property ensures that the fixed frames on a frame display remain for the lifetime of a session. It can be used with caution to cause other forms to be permanently cached. The default is as follows:

```
scoreboard.htmpl top_splash.htmpl buttons.htmp hiersel_admin_tree.htmpl
```

SuppressHtmplCache

Specifies that the web engine should reread all files defining the contents of a page each time the page is requested. Parsing an HTMPL file takes a significant amount of web engine processing time, and usually involves reading several physical files (because most pages use PDM_INCLUDE tags). The web engine normally saves parsed files in its own memory so that future requests for the same page can be satisfied immediately. This markedly improves performance, but can be inconvenient for users in the process of developing new or updated pages, as the web engine must be recycled for changes to take effect.
This property is optional and requires no value. If it is specified, the web engine does not cache parsed files, and changes to HTMPL files take effect immediately. Because of its impact on performance, this property should not be specified in a production environment.

SuppressLoginAndLogoutMsg
Specifies that the web engine should not log a message to the CA SDM log file each time a user logs in or logs out of the web interface.
This property is optional. If it is not specified, the web engine logs a message each time a user logs in or logs out.

SuppressMacroCache
Specifies that the web engine should discard all saved macros each time a new page is requested. The web engine normally saves parsed macros in its own memory so that future requests for the macro can be satisfied immediately. This improves performance, but can be inconvenient for users in the process of developing new or updated macros, as the web engine must be recycled for changes to take effect.
This property is optional. If it is specified, the web engine does not cache parsed macros, and changes to macros take effect immediately. Because of its impact on performance, this property should not be specified in a production environment.

Timeout
Specifies the number of minutes that a user's session can be idle before it is automatically terminated, freeing up all server resources.

> **NOTE**
> The Timeout setting must be longer than the ExclLockSeconds setting. ExclLockSeconds is specified in seconds and Timeout is specified in minutes.

TooManyMatches
Specifies the text of the message to display under a field when a user's key for a lookup field is ambiguous and the number of matches for the key exceeds the value of MaxSelectList. This property is optional; if omitted, it defaults to Too Many Matches.

UpdatedAnnouncementsPopup
The interval that browser checks for a new announcement. When a new announcement is found, it automatically shows the announcement in a popup window. The interval value is in minutes. To reduce the impact to the browser performance, it's recommended to set this variable to the value greater than 5 (minutes).

UseDirector
Specifies when the WebDirector is controlling this web engine. The following table defines possible values:

Value	Description
No	The web engine is independent of the WebDirector. This is the default value.
Yes	The WebDirector must initiate all sessions, including the login form. If a user attempts to make a direct connection to the web engine, the web engine asks the WebDirector for a referral.
AfterLogin	The web engine refers a session to the WebDirector after authenticating a user. A web engine configured with UseDirector AfterLogin is responsible solely for authentication, and is thereby a candidate for the use of secure sockets (SSL) for maximum security.
BeforeLogin	The web engine refers a session to the WebDirector before authenticating a user. A web engine configured with UseDirectory BeforeLogin never displays a login page, and never accepts a login password.

This property is optional. If omitted, the web engine does not use the WebDirector.

UseNestedTabs
Specifies whether to display the nested tab control on detail forms.
Default: Enabled

WebDirectorSlumpName
Specifies the name of the WebDirector servicing this web engine. This property is needed only if you are running more than one WebDirector, or if you have configured your WebDirector to use a slump name other than its default of web:director.
This property is optional if you are using WebDirector. It is ignored if you are not.

WillingnessValue
Specifies the willingness of this web engine to accept sessions, based on a scale of 0 to 10. This property is used only if you are using WebDirector. This value is meaningful only in comparison with the willingness of other web engines associated with the same WebDirector. The WebDirector transfers sessions to web engines in proportion to their willingness values. A web engine with a willingness value of twice that of another web engine's, on average, services twice the number of sessions.
A WillingnessValue of zero means that the web engine does not accept any sessions. This value can be useful when UseDirector is AfterLogin.
This property is optional if you are using WebDirector. It is ignored if you are not. If omitted, the web engine sets its willingness to 5.

WorkFrameTimeout
Specifies the maximum number of seconds the web engine waits for a response to an internal server request before concluding the request has failed. The workframe is used for CA SDM web interface features requiring server data other than normal web pages. This includes such features as autofill, loading category properties, and updating scoreboard counts. Workframe requests to the CA SDM are unlikely to fail. However, workframe requests to other servers (such as integrated products, such as Knowledge Management) may fail if the targeted server is not running, or if a network problem prevents access to it. The WorkFrameTimeout property specifies a length of time before the request is considered to have failed, and the workframe is available for other requests.

NOTE
WorkFrameTimeout is not checked unless a workframe is needed and all workframes are in use. Therefore, it is quite likely that a remote server is going to have more time than that specified for WorkFrameTimeout to respond. The value of WorkFrameTimeout is a minimum.

This property is optional. If omitted, the web engine uses a 30 second work frame timeout.

Creating the Business Structure

Contents

A business structure is a logical representation of your enterprise in a CA Service Desk environment. This scenario describes how an administrator can create a business structure for an enterprise to manage and track contacts, groups, and assets that are spread across distributed locations. After you create a business structure, you can generate reports to analyze requests by site, organization, or group.

Example

The fictional company Forward, Inc. uses a request processing system for multiple offices that are spread across geographical locations. The company plans to implement CA Service Desk to analyze the number and types of requests that are generated from its various business segments.

To facilitate effective tracking and decision making, the organization must track the following elements:

- Contacts, contact groups, and locations
- Assets at various locations

Creating a business structure allows the management team at Forward, Inc. to perform the following actions:

- Generate reports to analyze requests by site, organization, and group
- Increase overall efficiency by implementing corrective measures for any gaps in the request process

Create a Contact

You must set the appropriate Validation Type for Web Authentication when creating a contact. For more information, see Configure Web Authentication for an Access Type.

NOTE

If multi-tenancy is installed, select the appropriate tenant from the drop-down list. Contacts require a tenant.

There are three methods you can use to create a contact:

Create a Contact from the File Menu

On the Scoreboard, you can create a new contact from the File menu.

NOTE

If multi-tenancy is installed, select the appropriate tenant from the drop-down list or lookup table.

Follow these steps:

1. Select File, New Contact from the menu bar on the Scoreboard.
   The Create New Contact window opens.

   NOTE

   If your installation is configured to access an LDAP server, and has the necessary options installed, the Merge LDAP button appears in the upper right corner of the Create New Contact page. For information

about basing the contact record on an LDAP entry, see From the File Menu Using LDAP. Also see the *Administration Guide* for additional information.

2. Complete the fields as appropriate for the contact.
   See Contact Fields for field definitions.
3. (Optional) Use the controls available on the tabs at the bottom of this page to configure the contact.
   See Contact Tabs for more information.
4. Click Save.
   The contact record is saved and the Contact Detail page appears. The following buttons are now available for configuring the contact:
   - **Update Environment**
     Displays the Configuration Item/Asset Search window for the contact or organization, where you can specify search criteria for the assets you want to consider. When you click Search, the Environment Update window is displayed, where you can add and remove assets for this contact or organization.
   - **Update Groups**
     Displays the Group Search window, where you can specify search criteria for the groups you want to consider for this contact. When you click Search, the Groups Update window is displayed, where you can add and remove groups for this contact.

Create a Contact Using Data From LDAP

If your installation is configured to access a Lightweight Directory Access Protocol (LDAP) server such as Microsoft Windows Active Directory and has the necessary options installed, you can create and update contacts using data from the LDAP database. This method makes it easy to synchronize contacts with network user data.

NOTE

Administrators can configure automated synchronization of contacts with LDAP data. For more information about configuration, see the *Administration Guide*.

To create a contact using data from LDAP, **follow these steps:**

1. Select File, New Contact from LDAP from the menu bar of the Service Desk tab.
   The LDAP Directory Search page appears.
2. (Optional) Complete one or more of the following filter fields to limit the LDAP Entry list to the records of interest:
   Last Name Specifies the last name of the user as it appears in the LDAP directory. For example, you could enter b% in the Last Name field to retrieve a list of the LDAP user entries with last names that begin with the letter B. First Name Specifies the first name of the user as it appears in the LDAP directory. Middle Name Specifies the middle name of the user as it appears in the LDAP directory. User ID Specifies the user name for logging in to the system.
3. Click Search.
   The LDAP Entry List page displays the records that match your search criteria.

   NOTE

   : To see the information contained in an LDAP record without creating a contact, right-click the record of interest and select View. The LDAP Entry Detail page appears.

   All fields on the LDAP Entry Detail page are self-explanatory except for the following:
   User ID Specifies the ID the user enters to log in to the system. Distinguished Name Specifies the fully qualified LDAP login name. For example, CN=Joe, CN=Users, DC=KLAND, DC=AD, DC=com.
4. Click the LDAP entry to create a contact.
   The Create New Contact page appears and is partially populated with LDAP information.
5. Enter additional information as necessary.
6. Click Save.

The contact record is saved and the Contact Detail page appears. The following buttons are now available for configuring the contact:

- **Update Environment**
  Displays the Configuration Item/Asset Search window for the contact or organization, where you can specify search criteria for the assets you want to consider. When you click Search, the Environment Update window is displayed, where you can add and remove assets for this contact or organization.
- **Update Groups**Displays the Group Search window, where you can specify search criteria for the groups you want to consider for this contact. When you click Search, the Groups Update window is displayed, where you can add and remove groups for this contact.

Create a Contact Using the Quick Profile

You can create a new contact using the Quick Profile.

To create a contact from the Quick Profile, **follow these steps:**

1. Select View, Quick Profile from the menu bar on the Service Desk tab.
   The Quick Profile Contact Search page appears.
2. Click Create New.
   The Create New Contact page appears.
3. Complete the fields as appropriate for the contact.
   See Contact Fields for field definitions.
4. Click Save.
   The contact record is saved and the Contact Detail page appears. The following buttons are now available for configuring the contact:
   - **Update Environment**
     Displays the Configuration Item/Asset Search window for the contact or organization, where you can specify search criteria for the assets you want to consider. When you click Search, the Environment Update window is displayed, where you can add and remove assets for this contact or organization.
   - **Update Groups**
     Displays the Group Search window, where you can specify search criteria for the groups you want to consider for this contact. When you click Search, the Groups Update window is displayed, where you can add and remove groups for this contact.

Contact Fields

The following fields appear on the Create Contact, Contact Detail, and Update Contact pages.

> **NOTE**
>
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Alert Banner Indicates that the contact requires special handling, such as a visitor contact.

TenantDisplays the tenant associated with the contact.

> **NOTE**
>
> This field appears for privileged users in multi-tenancy installations. Tenant activities also appear in the Event Log.

Last Name/First Name/Middle Name Specifies the last, first, and middle names of the contact. Only the Last Name field is required.

StatusIndicates whether the contact is Active or Inactive.

Contact ID Specifies a unique identifier for the contact. If the default user authentication is being used, the value in this field is used as the password when the user logs in.

User ID Specifies the user name of the contact. The contact uses this value to log into the system.

Contact Type Specifies the contact type.

Service Type Specifies the level of support received by the contact. For example, some users have 24-hour support, and others receive on-site training.You can enter a value directly or click the magnifying glass to search for the service type.

Time Zone Specifies the time zone where the contact is located.

> **NOTE**
>
> Regardless of the time zone specified for a user, the time zone where the server is located determines the time stamps on records.

Job Title Specifies the job title of the contact within the organization.

Data Partition Specifies the data partition for this contact. This value determines the records this contact can access. Data partitions are subsets of the database with restricted access to data records, based on their content. You restrict that access by defining a set of constraints for each data partition.

Access Type Specifies the access type. The access type determines the system functions the contact can access.

Work Schedule Indicates whether the contact is assigned to a specific work schedule.

AvailableIndicates whether the contact is available for ticket assignments.

Confirm Self-Service Save Indicates whether the contact receives a confirmation when saving a record from the self-service interface.

AliasSpecifies an alias can be assigned to the contact record, if appropriate.

Last Modified Date Indicates the last time that this record was modified.

Last Modified By Indicates the last user that modified this record.

Analyst's Tenant Group(Analyst Contact Type Only) Specifies the tenant group that the analyst is responsible for.

Associate a Contact to a Special Handling Type

You can assign a special handling type to a contact to alert analysts about tickets that affect end users with special requirements, such as for a visually impaired person, a contact that poses a security risk, and so on.

To associate a contact to a special handling type, **follow these steps:**

1. On the Contact Detail page, select the Special Handling tab.
   The Associated Special Handling List tab lists special handling types that are associated with the contact.
2. Click the Update Contact's Special Handlings button.
   The Search filter appears.
3. Search for the special handling type that you want to associate to the contact.
   The Special Handlings Update page appears.
4. Select one or more handling types from the left column and use the move button (>>) to move the types to the right column. Click OK.

   > **NOTE**
   >
   > You can remove an association from a contact by using the move button (<<) to move the type from the right column to the left column. You can click the search icon to search for the value you want.
5. The contact is associated to a handling type.

CA SDM displays any of the following, depending on the handling type, when a ticket specifies the contact in the Affected End User field:

- An alert banner appears on the Contact Detail for the affected end user on a ticket.
- Alert text appears as a banner at the top of the ticket detail page and in the Quick Profile.
- Ticket lists highlight the contact row and show an alert flag.
- A V.I.P. folder appears in the Scoreboard for analyst roles. The folder contains all tickets that are associated with contacts (Affected End Users) that have a VIP special handling type.

Create a Site

A site is a geographical region where your organization has one or more locations.For example, North America can be a site, with locations (offices) in New York, California, and Texas.

NOTE

If multi-tenancy is enabled, select the appropriate tenant from the drop-down list.

Follow these steps:

1. Select File, New Site from the menu bar on the CA Service Desk Scoreboard.
2. Enter the information as appropriate specific to your site.
3. Click Save.
   The site record is created and saved.

Create Locations

A location is a physical place such as an office address. For example, the addresses for New York, California, and Texas offices can be locations under the site North America.

Creating locations helps you manage contacts and resources in that location. After you create a location, you can assign it to a site.

Follow these steps:

1. Select File, New Location from the menu bar on the Scoreboard.
2. Enter the information specific to your location, then configure the location using the following controls on the tabs:
   - **Address**
     Specifies the physical address of the location.
   - **Auto Assignment**
     Automatically assigns the tickets (requests, change orders, and issues) to members in this location.
     - **Update Request Areas**
       Select the request areas that you want to auto-assign to members of the location.
     - **Update Change Categories**
       Select the change categories that you want to assign to members of the location.
     - **Update Issue Categories**
       Select the issue categories that you want to assign to members of the location.
     - **Update Groups**
       Select the groups that you want to update for auto assigning tickets.

       WARNING
       After updating the request areas and categories, enable the automatic assignment for each request area and category individually.
3. Click Save.
   The location details are saved.
4. (Optional) Repeat steps 1-3 to add more locations.

Configure Auto Assignment by Location

The Auto Assignment tab allows you to establish a relationship between this location and one or more request areas, change categories, issues categories, and groups. This allows you to automatically assign tickets (requests, change orders, and issues) that are defined using the request areas or category to eligible members of the group within the location.

NOTE

You must enable automatic assignment for each request area and category individually. Listing them on the Location Auto Assignment tab does not enable this function.

To add auto assignments to a location

1. On the Location Detail page, click Edit and select the Auto Assignment tab.
2. Click one of the following buttons:
   - Update Request Areas to select request areas for auto assignment.
   - Update Change Categories to select change categories for auto assignment.
   - Update Issue Categories to select issue categories for auto assignment.
   - Update Groups to select groups for auto assignment.

   The Search page for the selected item appears.
3. Enter the search criteria to display the desired items and click Search.
   The Update page appears, listing the items that matched the search criteria.
4. From the list on the left, choose the desired items. To choose multiple items, hold down the CTRL key while clicking the left mouse button.
5. When you have selected all the items you want, click Select button.
   The selected items move to the Assigned list on the right.
6. Click OK.
   The Location Detail page displays, with the selected items listed on the Auto Assignment tab.

Create Organizations

An organization refers to an internal department or division or an external company. You can assign organizations to tickets, Configuration Item (CI) classes, and contacts.

For example, you can define CIs for organizations to specify the hardware, software, and services that the organization uses.

NOTE

If multi-tenancy is enabled, select the appropriate tenant from the drop-down list.

Follow these steps:

1. Select File, New Organization from the menu bar on the Scoreboard.
2. Enter the information specific to your organization, then specify organization details in the following fields:
   - **Address**
     Displays the address of the location to which you associate the organization. The fields are automatically populated when you assign location to the organization.
   - **Environment**
     Displays the Configuration items (for example, equipment, software, and services) that the organization uses. You can associate one or more configuration items with the organization. Associating the CI items to the organization helps administrators to keep track of the resources used by the organizations in various locations.
3. Click Save.
   The organization details are saved.
4. (Optional) Repeat Steps 1-3 to add more organizations.

Set Up an Organization's Environment

An organization's environment consists of the equipment, software, and services they use.

Follow these steps:

1. Select the Environment tab on the Organization Detail page
2. Click Update Environment.
   The Configuration Item Search page appears.
3. Enter the search criteria to display the configuration items of interest and click Search.
   The Organization Environment Update page displays the configuration items that matched the search criteria.
4. From the list on the left, choose the configuration items you want to add to this organization's environment. To choose multiple items, hold down the CTRL key while clicking the left mouse button.
5. When you have selected all the configuration items you want, click the double right-arrow button.
   The selected configuration items move to the Organization Environment list on the right.
6. Click OK.
   The Organization Detail page displays the selected items listed on the Environment tab.

Create Groups

A group is a collection of contacts that represent a specific area of responsibility. Defining groups lets you assign the responsibility for resolving a ticket when that responsibility is shared among several individuals. For example, a Dallas Human Resources group is responsible for dealing with the personnel issues in the Dallas office of your organization. When an employee in that office has a problem, you can assign the problem to the Dallas Human Resources group for resolution.

> **NOTE**
>
> The public (shared) option creates the object for all tenants.
>
> **NOTE**
>
> If multi-tenancy is enabled, select the appropriate tenant from the drop-down list

Follow these steps:

1. Select File, New Group from the menu bar on the Scoreboard.
2. Enter the information specific to this group and your organization by specifying the group details in the following fields:
   - **Notification**
     Defines the contact information and method for notifying the group.
   - **Address**
     Assigns the group to a location.
   - **Organizational Info**
     Specifies the functional or administrative organization, department, cost center, or vendor information.
   - **Environment**
     Specifies the environment (for example, equipment, software, and services).
   - **Members**
     Adds or removes contacts.
   - **Service Contracts**
     Lists service contracts that have been associated with the group.
   - **Auto Assignment**
     Lists auto assignments of tickets that are based on the group membership.
   - **Remarks and Special Handling**
     Lists remarks and special handling types, such as VIP or security risk types. You can click Update Contact's Special Handling to search for special handling members.

3. Click Save.
   The group record is saved and the group detail page opens. The following buttons are now available for configuring the group:
   - **Update Environment**
     On the Contact Details, Environment tab, click this button to display the Configuration Item Search window for the group. You can specify search criteria for the assets you want to consider on this page. You can create new configuration item and search assets using the Update window of Contacts respectively.
   - **Update Members**
     On the Members, Service Contracts, Auto Assignment, Members tab, this button displays the contacts. You can add and remove contacts for this group.
4. (Optional) Repeat Steps 1-3 to add more groups as needed.

CA SDM User Authentication

This article contains the following topics:

CA SDM provides a user authentication solution that you can modify as part of the access type. The same authentication is used by all CA SDM interfaces and by other CA products.

Authentication is flexible, allowing you to take advantage of external authentication mechanisms, such as Windows, HTTPD user validation or LDAP authentication. You can also select from a variety of internal authentication options, including operating system password, PIN, guest user access, or no access at all.

How CA SDM Authenticates Users

CA SDM authenticates users based on the user ID defined in their contact record. The product also does the following when a user requests access to the system:

1. If an external user ID is available (from HTTPD or Windows validation), CA SDM looks up the contact by login ID. If the contact is found and has an access type that permits external authentication, the user is allowed into the product.
2. If there was no successful external authentication, CA SDM prompts the user for a user ID and password. The product looks up a contact record for the user ID, obtains the access type, and then authenticates the user as specified by the access type.

Many installations find the predefined access types define authentication that is reasonable for that type of user; however, in some cases you may need to modify the authentication information for a predefined access type or define a new access type to handle a different authentication method for some of your users. You should review the authentication settings for the predefined access types to determine if they meet your needs, or if you need to modify them, or define additional types.

External Authentication

CA SDM permits users to access the system without supplying a user ID if all of the following conditions are met:

- External authentication is set for the user.
- The user's externally authenticated user ID is associated with a contact in your contact table.
- The contact record has an access type whose authentication definition permits external authentication.

External authentication does not permit users to access the system in the following cases:

- A user attempts access through a non-secure server.
- A user attempts access but is assigned to an access type that does not allow external authentication.

None of the predefined access types use external authentication. If you want to use external authentication for users, consider modifying the employee, analyst, and administrator access types to set external authentication. Your individual site requirements and different types of users determine whether to allow external authentication. When

external authentication is used, the server configuration controls the access to files and directories. When you define authentication for an access type, you can decide the usage as follows:

- Do not use any external authentication that is already implemented, such as the user login on Windows or validation by the HTTPD server.
- Use the authentication that is implemented and allow or deny access based on it.

  NOTE
  If external authentication is not allowed, the user is authenticated based on the validation type that you specify.

Following are some examples of external authentication:

- If a user who has administrator access logs into a Windows computer, the user can perform administrative tasks without re-entering any login information.
- If a user who has HTTPD server validation, the user can access the web interface without re-entering any login information. Because the administrator access type specifies the analyst web user type, the appropriate web interface for the analyst is presented automatically.

Validation Types

Validation types authenticate users only under the following circumstances:

- The user access type does not permit external authentication.
- The user access type permits external authentication, but the user has not been validated externally (for example, the user has attempted access through a nonsecure server).

CA SDM provides you with the following validation options:

- **No Access** -- Users of this type have no access unless external authentication is allowed and is valid.
- **Open** -- Users of this type have access, with no additional authentication required.
  - **OS** -- Users of this type enter their operating system password for access. The operating system used for validation is the one running User Validation Host. This option is the default validation type for the administrator, analyst, and employee access types.
  - **PIN** -- Users of this type gain access by entering the correct value for the PIN field in their contact record as their password. You define the PIN field by entering the field attribute name when you select PIN as the validation type. PIN is the default validation type for the customer access type, which uses the value in the customer ID (contact_num) field as the PIN.

Logged In User Counts and Session Counts

The following KPIs count the number of unique licensed users that are logged in to the system (for example, CA SDM Web UI, SOAP Web Services, REST Web Services, and so on), regardless of how many sessions each user has opened:

NOTE
For a licensed user, ensure that the Licensed? check box is selected from the Access type page of the contact (navigate to Security and Role Management, Access Type on the Administration tab and search for the contact).

- webConcurrentLicenseCt
- webConcurrentSOAPLicenseCt
- webConcurrentRESTLicenseCt
- webConcurrentTotalLicenseCt

The following KPIs count the numbers of unique unlicensed users that are logged in to the system, regardless of how many sessions each user has opened:

- webConcurrentNonLicenseCt
- webConcurrentSOAPNonLicenseCt
- webConcurrentRESTNonLicenseCt
- webConcurrentTotalNonLicenseCt

The following KPIs count the number of unique sessions that started during the interval:

- webSessionCt
- webSOAPSessionCt
- webRESTSessionCt

For more information about the KPI description, see the KPI detail page (navigate to Service Desk, KPIs on the Administration tab and search for the KPI). For more information about how these KPIs count from different session types, see the How KPIs Count from Different Session Types topic.

How KPIs Count from Different Session Types

There are different session types that are defined in the system. The following table shows how KPIs count these sessions:

Note: All predefined KPIs are installed as Inactive. For a KPI to begin functioning in your system, it must be set to Active. Navigate to Service Desk, KPIs on the Administration tab and search for the inactive KPI. Open the KPI and click Activate.

> **WARNING**
> Multiple versions of a KPI with the same name cannot be active at the same time.

Session Type	Session Type Description	Counted by KPIs
Web Client	Web browser session	webSessionCt webConcurrentLicenseCt webConcurrentNonLicenseCt webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt
Java Client	Java client session	webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt
Web Services	SOAP Web services session	webSOAPSessionCt webConcurrentSOAPLicenseCt webConcurrentSOAPNonLicenseCt
Utility	Server utility session	webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt
Portal	Portal session	webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt
Knowledge Chat	Knowledge Chat session	webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt
Mail Server	Mail Server session	webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt
Custom Application	Custom Application session	webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt

PDA Client	PDA Client session	webSessionCt webConcurrentLicenseCt webConcurrentNonLicenseCt webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt
REST Client	REST Web Services Session	webRESTSessionCt webConcurrentRESTLicenseCt webConcurrentRESTNonLicenseCt webConcurrentTotalLicenseCt webConcurrentTotalNonLicenseCt

Example: KPIs calculating user counts

One licensed (that have the Licensed? check box selected) and two unlicensed end users are logged into the web self-service interface and reviewing some announcements.

At the same time five licensed analysts (that have the Licensed? check box selected) are logged into the analyst interface and working on incidents. One of the analysts also logs in to the SOAP Web Services interface.

- The webConcurrentLicenseCt KPI shows a count of six, meaning that six licenses are currently being used, irrespective of the number of interfaces each user is using.
- The webConcurrentNonLicenseCt KPI shows the count of two, which means that two unlicensed users are logged on to the system, irrespective of the number of interfaces each user is using.
- The webSessionCt KPI shows a count of eight, meaning that eight total users are logged in to the CA SDM Web UI.
- The webSOAPSessionCt KPI shows a count of one, meaning that one user is logged in to the SOAP Web Services interface.

(Applicable for advanced availability configuration only) Example: KPIs calculating user counts from different nodes

A licensed analyst logs in to the analyst interface from the background server and works on incidents. The same analyst logs in to the analyst interface from the application server. The webConcurrentLicenseCt KPI shows a count of one, meaning that one license is currently being used, irrespective of the number of nodes or servers the user has logged in from.

Encrypt Session IDs to Address Vulnerability Issues

CA Service Desk Manager (SDM) uses the Session ID for authenticating each request from the user. This Session ID is sent back and forth through the web browser. An attacker can auto-generate the Session ID and can gain unauthorized access to CA SDM, if it matches any of active SIDs in SDM. An attacker can sniff the SDM web URL using man-in-the-middle attack and can replay the URL to gain unauthorized access to SDM. Using encrypted Session ID and cookie for authenticating user requests may have some minimal performance impact on SDM.

The following attributes are added in Options Manager to support encrypted Session IDs:

- **use_encrypted_sid_and_cookie (optional)**
  Use the encrypted Session ID and cookie to prevent spoofing and Man-in-the-middle attack. By default, this attribute is disabled. If you want to have enhanced CA SDM security, this attribute can be enabled (Yes).
- **force_browser_to_send_cookie_only_in_ssl_connection** (optional)
  Force the browser to send the Session ID (SID) cookie only if there is an SSL connection. This attribute is applicable only if you have enabled the **use_encrypted_sid_and_cookie** to (Yes). By default, this is turned off. If this flag is enabled, CA SDM can only be accessed through an SSL connection.
  For more information, see Options Manager, Security Options.

Retry Mechanism for CA SDM and CA Process Automation Workflow Options

CA Process Automation Workflows can be attached to CA SDM tickets (for example,Change Orders) through CA SDM Events and Macros. The attached events are triggered by CA SDM at the specified retry interval duration. During retry, if the CA Process Automation is unavailable, the attached event is marked as **Unknown** and the process is not executed. CA SDM retry mechanism automatically re-triggers the attached **Unknown** events when CA Process Automation is available.

CA SDM administrators can enable the retry mechanism by installing additional options in the CA Process Automation Workflow Options Manager. For more information, see Options Manager, CA Process Automation Workflow Options.

- **caextwf_retry_count**: Default value is 3 and can be set in the range [1 – 20].
- **caextwf_retry_interval**: Default minimum value is set as 10 minutes in the range [10-999].

This feature can be disabled by uninstalling these options. Restart CA SDM services after installing or uninstalling these options.

How the CA SDM Retry Mechanism Works

Install the Options Manager options (caextwf_retry_count and caextwf_retry_interval) and restart the CA SDM services. The retry mechanism for a failed event checks if the CA Process Automation services are up and running during the specified time interval in the Options Manager value (default value of 10 minutes). If CA Process Automation Manager services are available, the CA SDM retry mechanism checks for any attached events that are in **Unknown** status. CA SDM triggers such events at the specified time.

The retry mechanism checks for **Unknown** status events for a maximum count value which is equal to Options Manager **caextwf_retry_count** value. This event is not re-triggered once it reaches the maximum retry count value.

For more information, see Options Manager, CA Process Automation Manager Workflow Options.

Apache HTTP Load Balancer for CA SDM

Consider the following example scenario to load balance your existing CA SDM Servers using Apache HTTP Server.

Perform the following steps:

Install Apache Web Server Load Balancer

Perform the following steps:

1. Downland and install Apache Web Server load balancer for Windows version 2.2.4. You can download it from the Apache web site, apache.org.

Configure Apache HTTP Server

Configuring Apache HTTP Server is a required task when you set up load balancing. Perform the following steps:

1. Access the Apache website.
2. Download Apache HTTP server 2.2.4 and install it.

   NOTE
   This example scenario was tested with HTTP Server 2.2.4.
3. Download Mod JK Tomcat connector.
4. Download the binary, click on JK 1.2 Binary Releases, win32, jk-1.2.21, mod_jk-apache-2.2.4.so.
5. Open the Apache Server Monitor and start the web server if it's not already running.
6. Point your browser to*http://localhost/* to verify that Apache is running on port 80.
   By default, Apache server uses port 80.

7. Stop the Apache Server.
8. Copy the “**mod_jk-apache-2.2.4.so**” to the modules directory in your Apache installation. Rename it to **mod_jk.so**.

(Optional) Disable Web Server Features

Disabling the web server features of Apache HTTP Server improves performance and reduces potential security risks. Perform these steps:

1. Delete the following folders under C:\Program Files\Apache Software Foundation\Apache2.2:
   a. \cgi-bin
   b. \errror
   c. \htdocs
   d. \icons
   e. \manual

   NOTE
   The "C:\Program Files\" portion of the path names reflect the typical location that is specified at installation time. This portion of your path names can vary depending on the installation directory that is specified for the \Apache Software Foundation folder at installation time
2. Rename **httpd.conf** to **_httpd.conf.bak** in the C:\Program Files\Apache Software Foundation \Apache2.2\confdirectory.
3. Find the **httpd.minimal.conf** file.
   This minimal configuration file is included in the Utilities/Apache Webserver directory of the CA Service Catalog installation media.
4. Copy the **httpd.minimal.conf** file to C:\Program Files\Apache Software Foundation\Apache2.2\conf.
5. Rename the **httpd.minimal.conf** file to **httpd.conf** in that directory.
6. Delete all files in that directory except the following files:
   a. **httpd.conf**
   b. **_httpd.conf.bak**
7. Change to the *C:\Program Files\Apache* Software Foundation\Apache2.2\modules directory.
8. Delete all files in that directory except the following files:
   a. **mod_jk.so**
   b. **mod_log_config.so**
   c. **mod_setenvif.so**

You have disabled the web server features.

Create and Configure the workers.properties File

Creating and configuring the worker.properties file is a required task when you set up load balancing.
Perform the following steps as shown in this example. Here, we have two SDM servers:

1. Edit the **workers.properties** in the *conf* directory where you have installed Apache HTTP Server. Add the following lines to this file:

```
worker.list=balancer
worker.worker1.port=8009
worker.worker1.host=<hostname> (First CA SDM server)
worker.worker1.type=ajp13
worker.worker1.lbfactor=1
worker.worker2.port=8010
worker.worker2.host=<hostname> (Second CA SDM server)
worker.worker2.type=ajp13
worker.worker2.lbfactor=10
worker.worker2.lbfactor=1
```

NOTE
Note: For all SDM servers in your environment, add additional set of worker nodes as shown above

2. In the same file as shown in step 1, add the following for all the worker nodes:

```
worker.balancer.type=lb
worker.balancer.balance_workers=worker1,worker2
worker.balancer.method=Busyness
worker.balancer.sticky_session=1
```

Update the httpd.conf File

Updating the **httpd.conf** file is a required task when you set up load balancing.

1. Open the "**httpd.conf**" file in conf directory of your Apache installation in a text edit, and add the following lines at the end of the set of LoadModule statements:

```
LoadModule jk_module modules/mod_jk.so
# Path to workers.properties
JkWorkersFile c:/Apache2/conf/workers.properties
# Path to jk logs
JkLogFile c:/Apache2/mod_jk.log
# Jk log level [debug/error/info]
JkLogLevel debug
# Jk log format
JkLogStampFormat "[%a %b %d %H:%M:%S %Y] "
#JkOptions     +ForwardKeySize +ForwardURICompat -ForwardDirectories
JkMount / balancer
JkMount /* balancer
```

(Optional) Disable HTTP Ports

To enhance security, disable the HTTP ports of the cluster computers when you update the **httpd.conf** file. Doing so routes users directly to the load balancer computer rather than directly to a cluster.
Perform the following steps:

1. Comment the connector tag that includes HTTP port information, in the server.xml files of each cluster computer. This port is the startup node that is provided during installation, typically 8080.
2. Recycle the CA Service Catalog Windows services of the clusters whose server.xml files were changed. The Windows services are named CA Service Accounting and CA Service Catalog.

Configuration on CA SDM Servers

Perform the following steps on each CA SDM server in your environment:

1. As shown in the examle use case above, in your first CA SDM server, open the **server.xml** file in edit mode for Tomcat (File path: *NX_ROOT*\ *bopcfg**www**CATALINA_BASE**conf*).
2. Replace the following:

```
<Connector enableLookups="false" port="8080" protocol="AJP/1.3" redirectPort="8443"/>
```

With

```
<Connector enableLookups="false" port="8009" protocol="AJP/1.3" redirectPort="8443"/>
```

Also, replace the following:

```
<Engine defaultHost="localhost" name="Catalina">
```

With

```
<Engine name="Standalone" defaultHost="localhost" jvmRoute="worker1" debug="0">
```

Verify the Load Balancer

1. Restart CA SDM services on both machines.
2. Now, navigate back to the server where Apache server is installed and start the Apache web server.
3. Open and point your browser window to *http://localhost*
   The CA SDM login page is displayed.

How to Configure the F5 Load Balancer for CA Service Desk Manager

F5 Load balancers help to enhance the reliability, availability, and scalability of CA Service Desk Manager (CA SDM) servers and improve the overall performance. An F5 load balancer monitors the CA SDM Health Servlet by sending requests to servers and applications that can respond in a timely manner.

To configure an F5 load balancer in your environment, complete the following steps:

- Prerequisites for configuring the F5 Load Balancer
- Create a Custom Health Monitor
- Create an F5 Pool for the CA Service Desk Manager Application Server
- Create an F5 Virtual Server or Node for CA Service Desk Manager
- Verify the F5 Load Balancer Configuration

> **NOTE**
>
> For more information on configuration tasks other than the ones discussed in this article, refer to your F5 Load Balancer documentation.

Prerequisites for configuring the F5 Load Balancer

Before you configure an F5 load balancer for CA SDM, complete and keep handy the following information:

- Identify the CA SDM hosts where you need the load balancing capability.
- Credentials to log in to the F5 interface.
- Before you deploy an F5 load balancer, ensure that you have configured the CA SDM Health Servlet on application servers. For more information on how to deploy the health servlet on a CA SDM Application Server, see Deploy the Health Servlet on the Application Server.

Configure the following F5 elements so that these elements can function with the CA SDM Application Servers:

1. Create a Custom Health Monitor
2. Create an F5 Pool for the CA Service Desk Manager Application Server
3. Create an F5 Virtual Server or Node for CA Service Desk Manager
4. Verify the F5 Load Balancer Configuration

Create a Custom Health Monitor

You can create a custom monitor when the values that you have defined for a pre-configured monitor does not meet your requirements, or when a pre-configured monitor is not available for the type of monitor you are creating. You can give a unique name to your monitor, specify the monitor type, and, if a monitor of that type already exists, import settings and values from the existing monitor. You can change the values of any imported settings.

To create a custom health monitor, complete the following steps:

Follow these steps:

1. Navigate to **Local Traffic**, **Monitors**, and Click **Create**. Complete the following field information:

a. **Name**: Enter the name of the health monitor.
   For example: **SDM_health**
b. **Type**: Select **HTTP** from the drop-down list.
   You must base each custom monitor on a monitor type. To specify a monitor type, simply choose the one that corresponds to the service you want to check.
c. **Send String**: Enter **GET /HealthServlet/GetHealth\r\n**.

   NOTE

   When you create a new HTTP monitor type, you must include **\r\n** at the end of a non-empty Send String. If you do not include **\r\n** at the end of the Send String, the HTTP monitor fails.
d. **Receive String**: Enter **AA-Server-Status: All OK!**
   The above string is displayed when the CA SDM server is healthy. When the server is down, the status displayed is **AA-Server-Status: NOT OK!**
e. **Receive Disable String**: Enter **Quiesce time remaining**.

2. Click **Finished**.
   You have successfully created a custom health monitor.
   When a user accesses the F5 load balancer URL, CA SDM health check is performed, and the user is routed to a server that is functional and not disabled or temporarily inactive (quiesced).

.

Create an F5 Pool for the CA Service Desk Manager Application Server

Create an F5 pool for the CA SDM server where you want to enable the F5 load balancing capability.

Follow these steps:

1. Select **Main**, **Local Traffic**, and click **Pools.**

   NOTE

   The Pool List is empty if you are setting up pools for the first time. The Pool List displays the details for each pool.
2. Click **Create**.
3. Complete the Configuration section on the New Pool page:
   a. Select **Basic** from the drop-down list.
   b. Enter a name and description for the new pool. .
   c. Select an already pre-configured health monitor from the available health monitors list. Select **HTTP** and move it to the active list.
   d. If a pre-configured health monitor is not available, create a custom health monitor. For more information on how to create a custom health monitor, see Create a Custom Health Monitor.

      NOTE

      Before you deploy the F5 Load balancer, ensure that you have configured the CA SDM Health Servlet on the application servers to verify the CA SDM application server status. Health Monitors monitor the health of your CA SDM application servers. For more information about CA SDM health servlets, see Deploy the Health Servlet on the CA SDM Application Server.
4. Select **Round Robin** from the Load Balancing Method drop-down list. Default option is Round Robin.
5. Select **Disabled** from the Priority Group Activation drop-down list.
6. Add each node or server to the new F5 pool as follows:
   a. Select **Node List** if you are adding a node that is already defined.
   b. If the nodes are not defined, select the **New Node** option to create a node.

c. Select the CA SDM Application Server IP address (host name) from the **Address** drop-down list that identifies the CA SDM node to add to this F5 pool.
If the applicant server is no longer valid or no longer in use, select the pool member and click **Disable**.

7. Click **Add**.
The details that you added for this node appear in the New Members list.
8. Click **Finished**.
The new pool is added to the F5 Pool List.

NOTE

Before quiescing the CA SDM Application Server server for performing a rolling maintenance, ensure that you have deployed the CA SDM application health servlet, or else, remove this server from the F5 Node List. For more information, see Deploy the Health Servlet on the CA SDM Application Server.

Create an F5 Virtual Server or Node for CA Service Desk Manager

You can create an F5 Virtual Server for CA SDM application servers to enable the F5 load balancing capability. Complete the following steps to create an F5 Virtual Server:

Follow these steps:

1. Log in to F5. Select **Main**, **Local Traffic**, **Virtual Servers**, and **Virtual Server List**.
2. Click **Create**.
3. Complete the following General Properties on the New Virtual Server page.
   a. **Name**: Specifies the name of the virtual server, for example, casdm.
   b. **Destination Type**: Specifies Host for a single IP address.
   This IP address should be available in the network and must not be assigned to any other device.
   For more information, see the F5 Load Balancer documentation.
   c. **Destination Address**: Specifies the IP address of the virtual server.
   d. **Service Port**: Specifies the associated port for the virtual server.
   For example: 80 for HTTP.
   e. **State:** Select **Enabled**.
   Specifies whether the virtual server is available for load balancing.
4. Complete the Configuration section. Accept all defaults, except for the **HTTP Profile**.
   a. **HTTP Profile**: Select **http**.
   Specifies the HTTP profile for managing HTTP traffic.
   Complete the Resources section:
   a. **iRules**: Specifies the iRules to enable for this virtual server.
   b. **Default Pool**: Specify the CA SDM pool which was created in **step 1** as the default pool. The virtual server routes traffic to this pool.
   c. **Default Persistence Profile**: Specifies the persistence profile for this virtual server. Select **cookie**.
   d. **Fallback Persistence Profile**: Specifies the persistence profile that this virtual server uses when the default persistence profile cannot be used.
   For example: **source_addr**.
5. Click **Finished**.
For more information about the F5 Load Balancer persistence profile, see the F5 documentation.
6. Navigate to **Local Traffic**, **Profiles**, **Persistence** to modify the properties of a persistence profile. Select the persistence profile that you want to modify.
For example, click **source_addr** and change the **Timeout** option to **600** seconds.
Click **Update**.

NOTE

For CA SDM, it is recommended that you set the **Timeout** option to a value of **600**.

7. Click **Finished**.

Verify the F5 Load Balancer Configuration

To verify the F5 load balancer configuration in your environment, complete the following steps:

Follow these steps:

1. Open any web browser, and enter the F5 Virtual Server destination IP address that you created in step 3 earlier in this article while creating the F5 virtual server.
   You must be able to view the CA SDM login page.
2. Enter the CA SDM login credentials and click Login.
   Login must be successful.
3. Repeat **step 1** and **2** on all CA SDM application servers and verify that you are routed to these servers by checking the CA SDM logs in the CA SDM install directory.

How to Configure the Mailbox to Handle Inbound Emails

Email lets you communicate with end users, such as employees or customers. The mailbox in CA SDM handles inbound emails from users and provides action according to the email. For example, the user sends an email to CA SDM to create an incident. Mailbox checks the email, creates an incident, and sends a notification back to the user.

NOTE

Maileater now supports configuring mail servers with wild card SSL certificates and/or where the mail server name is in the Certificate's subject alternate names (and not in the issues To Part).

CA SDM provides a default mailbox (named Default) that is active and ready for use within your organization. You can modify the default mailbox, create more mailboxes, or both. After creating a mailbox or modifying the Default mailbox, define the mailbox rules. The Mailbox rules let you configure any actions, replies, or both.

NOTE

The rules that are applicable for one mailbox cannot be associated with another mailbox. To reuse the same rules for a different mailbox, recreate them for the other mailbox. You can also copy the existing mailbox.
We recommend that you set the associated mailbox to inactive before configuring a mailbox rule. Else, any messages that are retrieved between your first change and last change are processed with rules that are in effect.

Follow these steps:

1. Log in to the web UI from the following servers, depending on your CA SDM configuration:
   – Conventional: Primary or secondary servers
   – Advanced availability: Application or background servers
2. Choose a Notification Phrase
3. Define a Mailbox

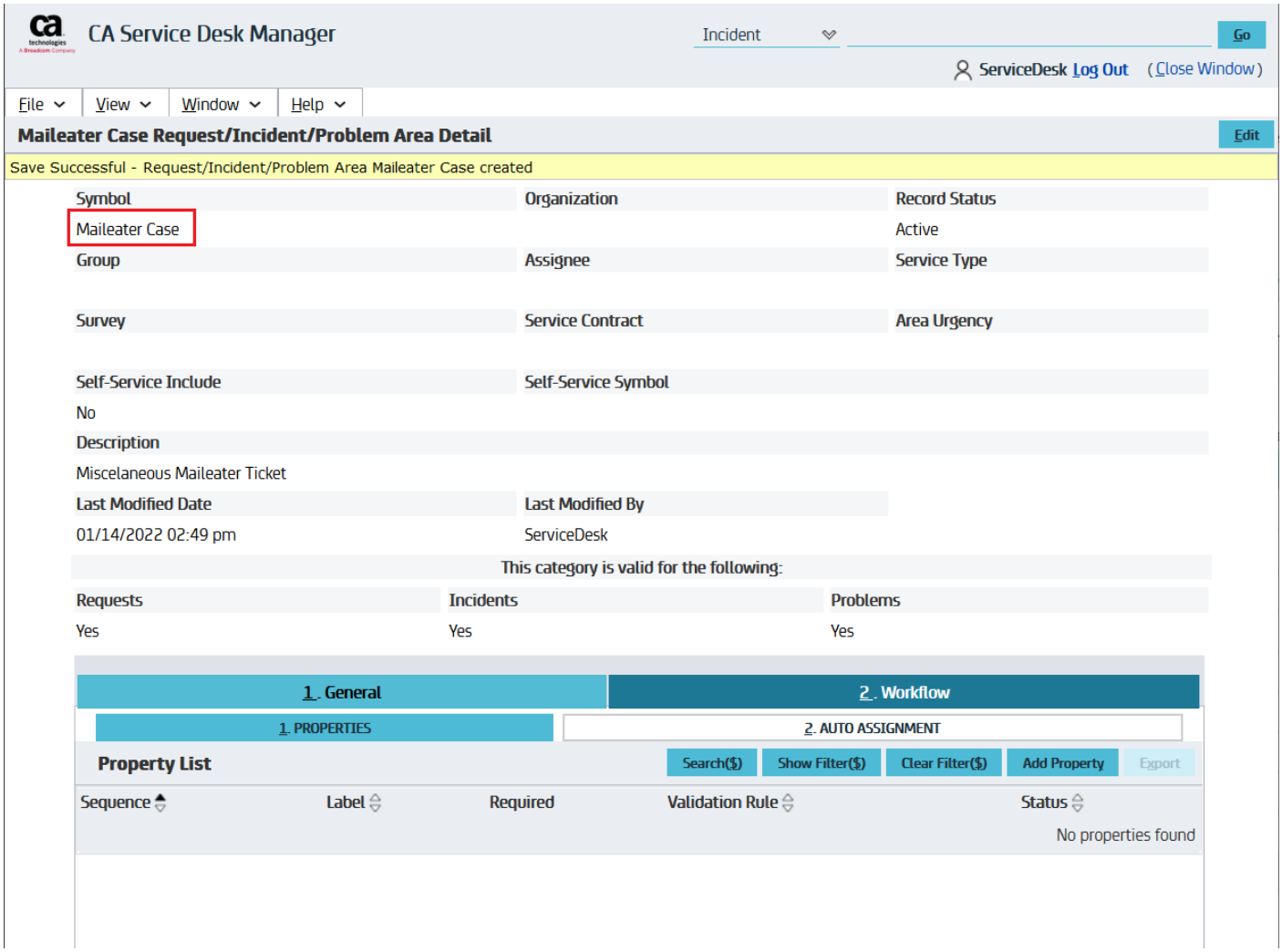
Real World Examples

Scenario 1

Create tickets with Category "Maileater Case" and Priority "4" when any other rule requirements do not meet.

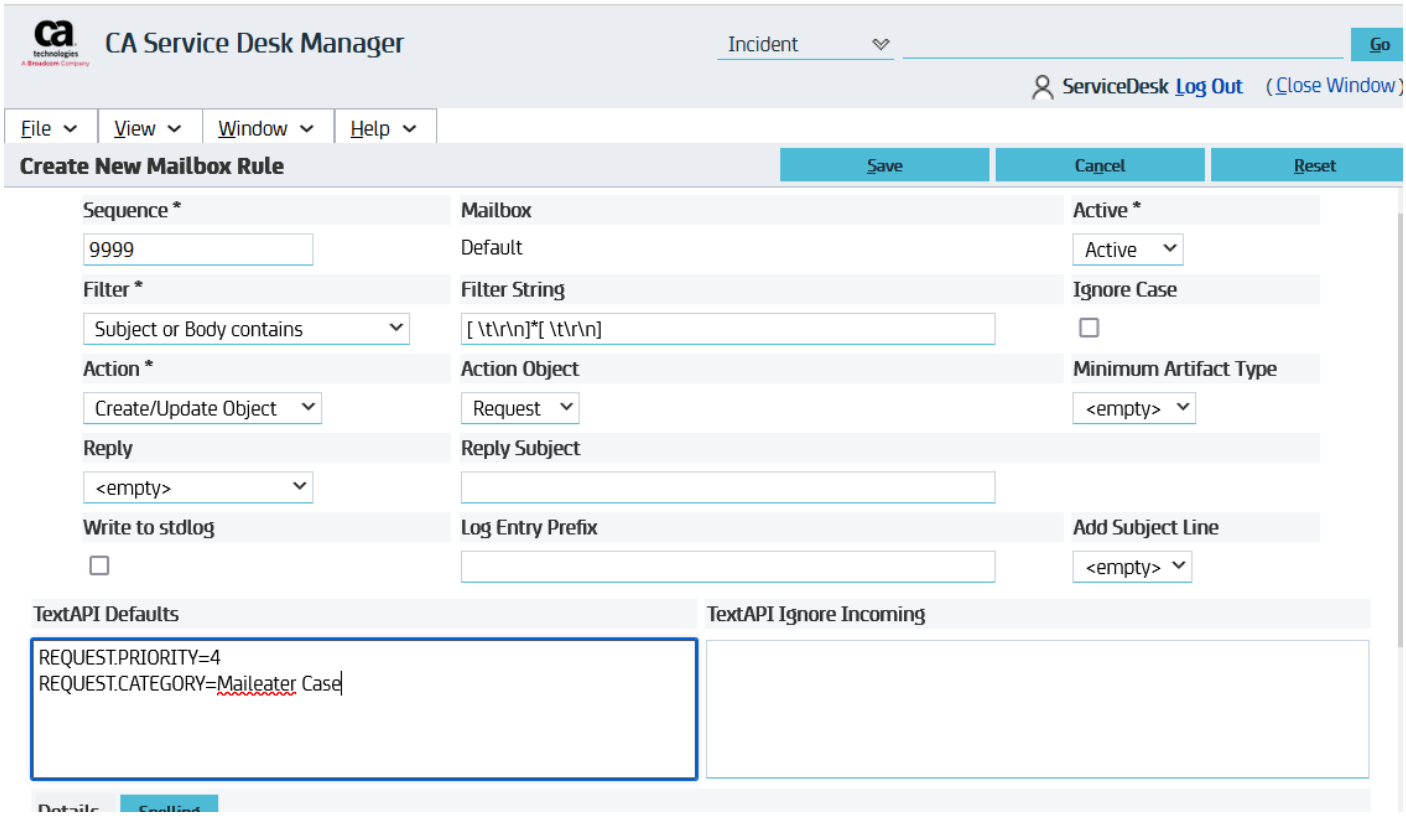
Follow these steps:

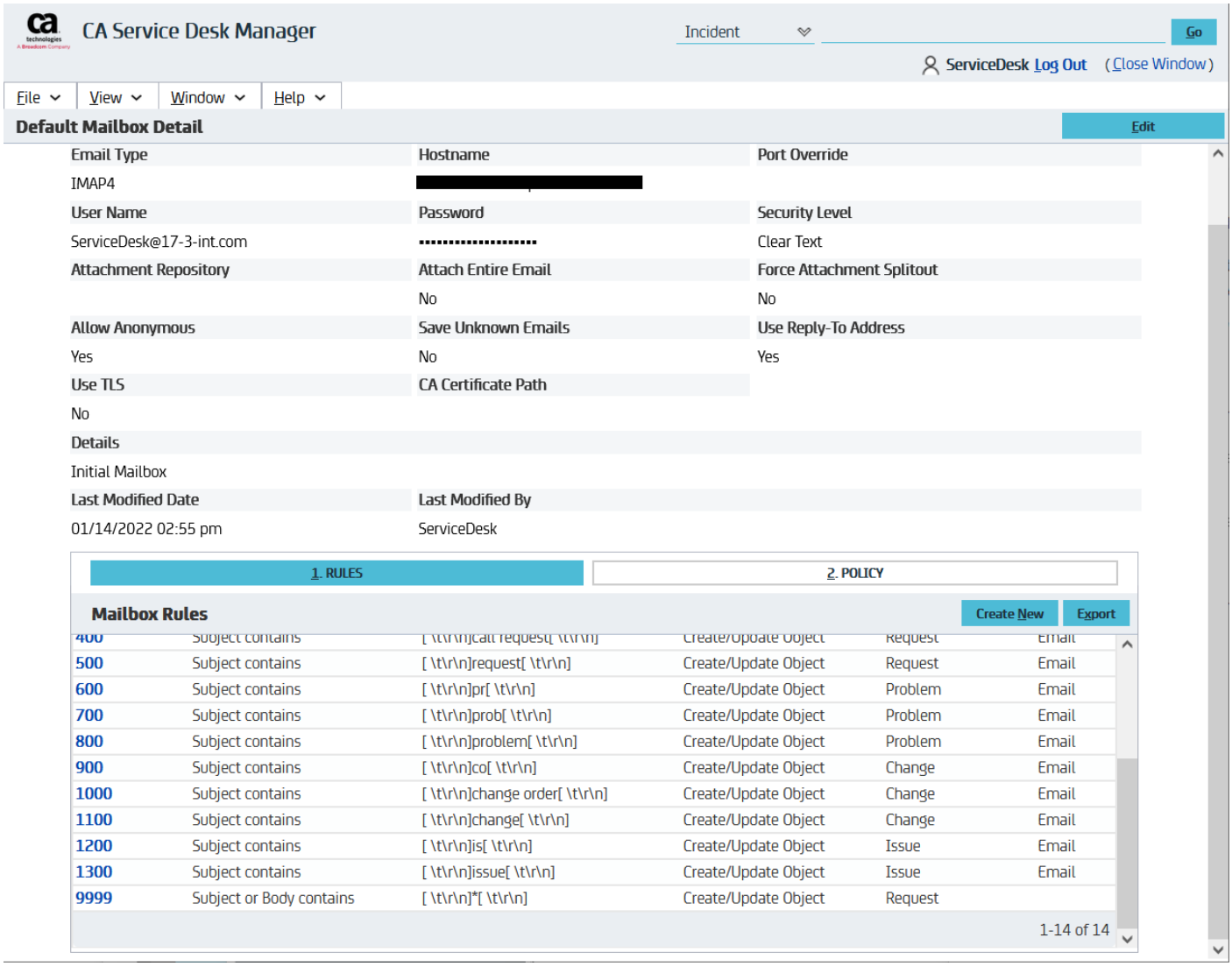
1. Create a new Request Area, "Maileater Case."



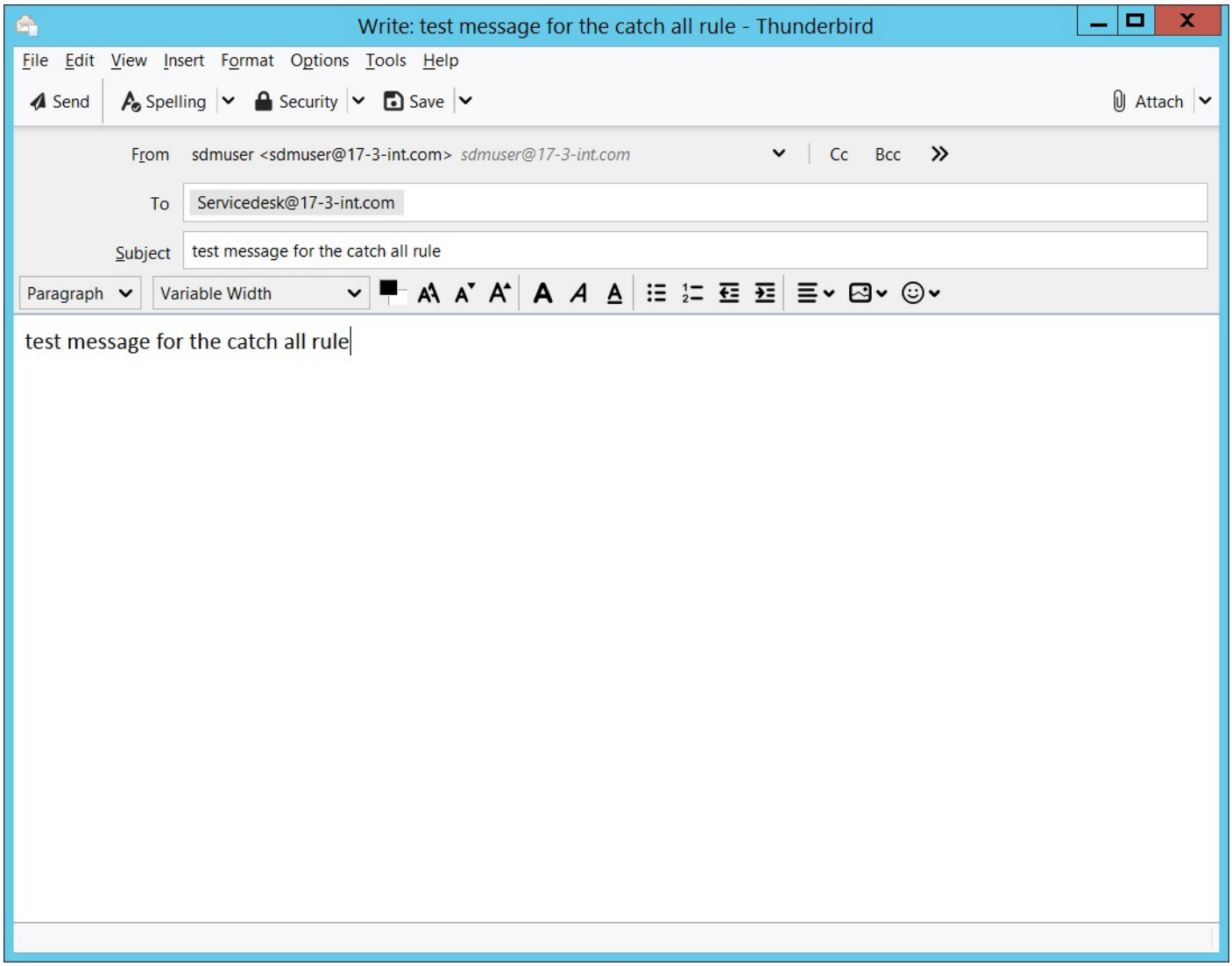
2. Create a new rule that serves as a "catch-all" rule in your mailbox. The following screenshots show the rule definition.
3. Under the "TextAPI Defaults," add this text:

```
REQUEST.PRIORITY=4
REQUEST.CATEGORY=Maileater Case
```

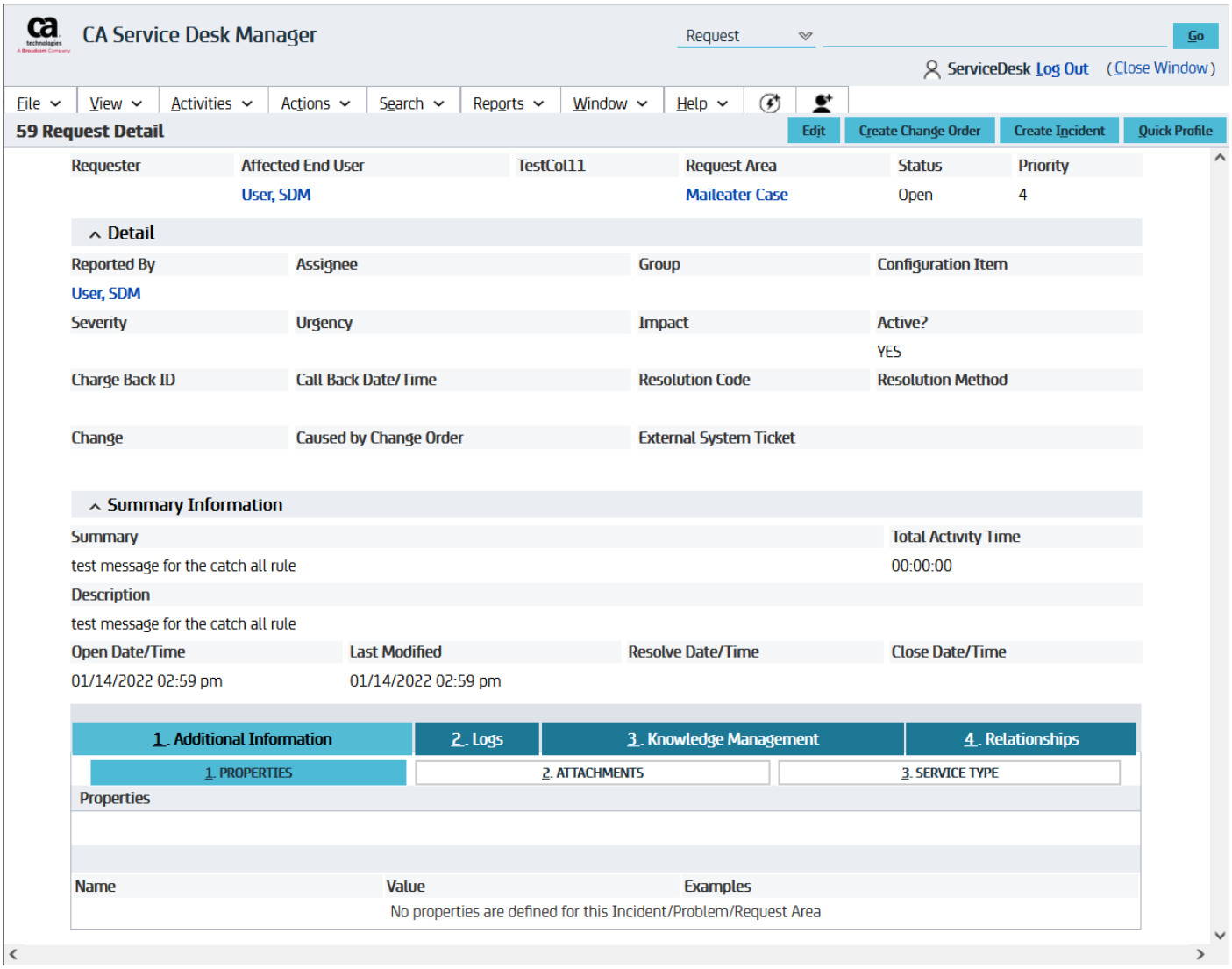

CA Service Desk Manager
Incident
Go
ServiceDesk Log Out
(Close Window)
File
View
Window
Help
Create New Mailbox Rule
Save
Cancel
Reset
Sequence *
9999
Mailbox
Default
Active *
Active
Filter *
Subject or Body contains
Filter String
[\t\r\n]*[\t\r\n]
Ignore Case
Action *
Create/Update Object
Action Object
Request
Minimum Artifact Type
<empty>
Reply
<empty>
Reply Subject
Write to stdlog
Log Entry Prefix
Add Subject Line
<empty>
TextAPI Defaults
REQUEST.PRIORITY=4
REQUEST.CATEGORY=Maileater Case
TextAPI Ignore Incoming



4. Create a test mail with content that does not match any prior rules.



5. After the mail has been sent, it processes the catching on this rule and creates a ticket with the specified default values.



Considerations:

- The rule that was devised was intentionally placed at the bottom of the rule list by giving a very high sequence number. This is because maileater rules are processed sequentially. Once a given rule has matched requirements, it acts on that rule, and no further rules are processed.
- Maileater does not act on multiple rules. It always work on the first rule that matches the criteria.
- The Filter string being used, [\t\r\n]*[\t\r\n] relies on REGEX, or Regular Expressions, as mentioned in the documentation.
- The rule relies on the TextAPI Defaults to populate the given request.
- The mailbox itself must have "Allow Anonymous" enabled to allow any sender to create a request.

Choose a Notification Phrase

The notification phrase is sent to the sender of the email. For example, to confirm that an incident has been created successfully. CA SDM provides predefined notification phrases which are set to inactive by default. You can activate and modify the phrases. You can also create notification phrases to best suit your organization needs.

Choose from the following options:

- Activate a predefined notification phrase.
- Create a notification phrase

Activate a Predefined Notification Phrase

By default, predefined notification phrases are set to inactive. You activate a notification phrase so that the notifications can use the phrase. Select **Administration**, **Notifications**, **Notification Phrases** and search for the notification phrase. Edit the notification phrase and select **Active** from the Active drop-down list.

Create a Notification Phrase

You can create notification phrase that contains standardized text and macros. This notification phrase is sent as a response to emails from users.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. Select Notifications, Notification Phrases on the Administration tab.
   The Notification Phrase List page opens.
2. Click Create New.
   The Create New Notification Phrase page opens.
3. Complete the Notification Phrase Fields, as appropriate.
4. Click Save.
   The notification phrase is created.

Notification Phrase Fields

Complete the following fields to edit or create a notification phrase:

- **Symbol**
  Defines a unique identifier for this record.
- **Code**
  Specifies a unique value to use for the notification phrase, in the usp_notification_phrase table. The usp_notification_phrase table lists common phrases that notification message templates can use. For information about the usp_notification_phrase table, see the *Technical Reference Guide*.
- **Phrase**
  Specifies the phrase for the notification. You can specify both plain-text and HTML versions. HTML consolidates most of the whitespace into single spaces, so you must specify line breaks or paragraph breaks with tags.
  For example, the following plain-text phrase is used in the Confidential Notice notification phrase:

  ```
  This e-mail and any files transmitted with it are for the sole use of the intended recipient(s) and
   contain information that may be privileged and confidential. Any unauthorized review, use, disclosure or
   distribution is prohibited. If you are not the intended recipient of this e-mail, please delete this e-
  mail and any files transmitted with it and notify the sender immediately.
  ```

  The following HTML phrase can be used to ask the user to view the notification list:

  ```
  Click on the following URL to view Notification List:
  @{call_req_id.web_url}+HTMPL=cr_lr.htmpl+INSTANCE=@{id}
  ```

  For more information about phrases, see the Notification Codes and Phrases topic.

Notification Codes and Phrases

Notification phrases let you add a standardized piece of information or text to a number of different notification messages, without having to enter and maintain the text or formulae separately in each notification template. For example:

Reply to this notification to add additional information to the ticket

Phrases standardize text for use in multiple message templates. For example, you
can maintain a common phrase such as a confidential notice in a single record and use it in multiple message templates. Notification phrases are also useful for message replies, such as a Reply Notice, or a web URL link. CA SDM provides phrases and you can create your own phrases. You can set a phrase as active or inactive for use in a message template globally. (Notification phrases are inactive by default.) When a phrase is inactive, the phrase is suppressed in all message templates that use the phrase.

Notification phrases can also be used in the automatic responses to incoming email messages. The processing context for this type of message is different; omit the prefix (change_id., issue_id., call_req_id.) used in certain macros such as ref_num and web_url for phrases that the message uses. As a result, you cannot share notification phrases between notification templates and email automatic responses.

For example, some of the phrases that CA SDM provides are as follows:

Symbol	Code	Phrase
Notify History - Change	notify_history_chg	Click the following URL to view the Notification List: @{change_id.web_url}+HTMPL=chg_lr.htmpl +INSTANCE=@{id}
Notify History - Issue	notify_history_iss	Click the following URL to view the Notification List: @{issue_id.web_url}+HTMPL=iss_lr.htmpl +INSTANCE=@{id}
Notify History - Request/Incident/Problem	notify_history_cr	Click the following URL to view the Notification List: @{call_req_id.web_url}+HTMPL=cr_lr.htmpl +INSTANCE=@{id}

Example: New Phrases

The following phrases are examples of phrases that you can create:

Symbol	Code	Phrase
Notify Confidential	ConfidentialNotice	This email and any files transmitted with it are for the sole use of the intended recipient(s) and contain information that may be privileged and confidential. Any unauthorized review, use, disclosure or distribution is prohibited. If you are not the intended recipient of this email, please delete this email and any files transmitted with it and notify the sender immediately.
Notify Incident Reply	IncidentReply	To add a comment to your Incident, just reply to this email or include the line below (on a line by itself). %incident_id=@{call_req_id.ref_num}
Notify Incident URL	IncidentURL	Click the following URL to view: @{call_req_id.web_url}

NOTE
Use separate phrases for the plain-text and HTML versions of message templates or email auto-replies. HTML consolidates the whitespaces into single spaces, so line- or paragraph-breaks must be specified with tags. HTML tags included in plain-text versions of messages are displayed rather than processed.

Notification Phrase Syntax

You insert notification phrases in message templates and email reply messages using the following macro syntax:

```
@{notification_phrase[code].phrase}
```

- ***code***
  Specifies the unique Code value, such as ConfidentialNotice, of the Message Phrases (usp_notification_phrase) table.

  NOTE
  The usp_notification_phrase table lists common phrases that notification message templates can use.

For example:

```
@{notification_phrase[IncidentURL1].phrase}
@{notification_phrase[RequestReply].phrase}
```

When CA SDM locates the macro, the phrase text from the usp_notification_phrase table replaces the syntax. If no matching code exists (or if it is inactive), an empty string replaces the macro. No errors are written to the standard log (STDLOG), instead a warning message is logged to help you resolve potential problems.

NOTE
Embedding phrases within other phrases is limited to a maximum depth value that you configure by setting the NX_MAX_EXPAND_DEPTH environment variable. This limitation prevents problems which can occur when processing phrases that are accidentally self-referenced (embed themselves) or circular-referenced (when a phrase embeds a phrase into which it is embedded).

Example How Notification Phrases Appear in a Message

This example demonstrates how notification phrases appear in a notification message. The example uses the following codes and phrases:

Code	Phrase
ConfidentialNotice	This e-mail and any files transmitted with it are for the sole use of the intended recipient(s) and contain information that may be privileged and confidential. Any unauthorized review, use, disclosure or distribution is prohibited. If you are not the intended recipient of this e-mail, please delete this e-mail and any files transmitted with it and notify the sender immediately.
IncidentReply	In order to add a comment to your Incident, just reply to this email or include the line below (on a line by itself). %incident_id=@{call_req_id.ref_num}
IncidentURL	Click the following URL to view: @{call_req_id.web_url}

The following message template includes the notification phrases:

```
@{call_req_id.type.sym} @{call_req_id.ref_num} Closed.
Assigned to: @{call_req_id.assignee.combo_name}
Customer: @{call_req_id.customer.combo_name}
Description: @{call_req_id.description}
```

```
@{notification_phrase[IncidentURL].phrase}
@{notification_phrase[IncidentReply].phrase}

@{notification_phrase[ConfidentialNotice].phrase}
```

The phrases appear in a message as follows:

```
Incident 1234 Closed.
Assigned to: Analyst, Joe
Customer: Doe, John
Description: This is a description of my problem

Click on the following URL to view:
http://helpdesk/CAisd/pdmweb.exe?OP=SEARCH+FACTORY=chg+SKIPLIST=1+QBE.EQ.id=400723

In order to add a comment to your Incident, just reply to this email or include the line below (on a line by
 itself).
%incident_id=1234

This e-mail and any files transmitted with it are for the sole use of the intended recipient(s) and contain
 information that may be privileged and confidential. Any unauthorized review, use, disclosure or distribution
 is prohibited. If you are not the intended recipient of this e-mail, please delete this e-mail and any files
 transmitted with it and notify the sender immediately.
```

Define a Mailbox

This topic contains the following information:

CA SDM provides a default mailbox to connect to the mail server. You can configure this default mailbox to change the password, user name, hostname, and so on. You can also create extra mailboxes to suit your organizational needs. Each mailbox can have its own definition, instead of using a single global set of definitions. You can define multiple mailboxes and can use different templates or default values for each mailbox. Multiple definitions allow individual tenants or organization use separate mailboxes with different settings.

Use the Default Mailbox or Create a Mailbox

CA SDM provides an active default mailbox. You can edit the mailbox for incoming mail delivery according to your requirement or create a mailbox that connects to the mail server. Configure the mailbox to set values for host, user, password, and so on.

Follow these steps:

1. Select Email, Mailboxes from the Administration tab.
   The Mailbox List page opens.
2. Click Default from the Name column and click Edit to edit a default mailbox or click Create New to create a mailbox.
3. Complete or update the other fields as appropriate:
   - **Check Interval**
     Specifies the time after which the mail server is polled for new emails.
   - **Active**
     Indicates the mailbox status.
   - **Email Type**
     Specifies the protocol that the mail server uses. CA SDM supports both POP3 and IMAP4. If you select IMAP4, CA SDM polls only the Inbox folder from the mailbox.

NOTE
For MS GRAPH API, set Email Type as GRAPH.

NOTE

For the Security Level, if Email Type selected is POP3, then, MD5 authentication is not a supported security Level.

NOTE
For the Security Level, if **Email Type** is selected as **POP3**, OAUTH 2.0 authentication is not a supported for it.

- **Hostname**
  Specifies the hostname of the email server.
- **Port Override**
  Specifies the port number when the default port number is overridden.
- **User Name**
  Specifies the user ID on the mail server.
- **Password**
  Specifies the password on the mail server.
- **Security Level**
  Specifies the the POP3/IMAP4/GRAPH security level.
  NOTE
  For GRAPH Email Type, set OAUTH 2.0 as the security level.
- **Attachment Repository**
  Specifies the repository where the email attachments are stored.
- **Attach Entire Email**
  Specifies whether to allow entire email as an attachment.
  - If this checkbox is enabled, it saves the entire mail as either .eml or .msg file.
  - If the checkbox is disabled, Email is not saved, but any embedded images and explicit files attached to the inbound email are stored as individual files and are found under the **Attachments** tab.
- **Force Attachment Splitout**
  Specifies whether to split all attachments in the email when an entire email is added as an attachment. The email and its attachments are split into separate files and attached to the tickets. Only applicable when the **Attach Entire Email** option is selected.
- **Allow Anonymous**
  Specifies whether tickets can be created from anonymous mails.
- **Save Unknown Emails**
  Specifies whether to save the emails that the rules defined in the mailbox did not process. These emails are stored in $NX_ROOT/site/mail_unknown.
- **Use Reply-To Address**
  Specifies whether to use the alternate email address for replies.
- **Use TLS**
  Specifies whether to use Transport Layer Security support in emails.
- **CA Certificate Path**
  Specifies the path where the trusted certificate has been deployed.

  NOTE
  For the advanced availability configuration, ensure that you deploy the trusted certificate on the same location for both background and standby servers. CA SDM supports only Base-64 encoded (PEM) format for CA Certificates.

NOTE
Microsoft Exchange OAuth 2.0 requires 2 certificates. Provide with a space separated value (for example: outlook.cer login_microsoft.cer)
Office 365 certificate
login_microsoft certificate
To generate login_microsoft.cer, open https://login.microsoftonline.com/ in a browser. Click the padlock symbol and export the certificate in Base64 cer format.

4. Click Create New from the Rules tab to create a mailbox rule.
   The Create New Mailbox Rule page opens.
5. Complete the Mailbox Rule Fields as appropriate and click **Save**.
6. Select the **Policy** tab to define the mailbox policies to protect your organization against certain types of email abuse. Complete the Mailbox Policy Fields as appropriate and click **Save**.
   The changes to the default mailbox are saved and applied or a new mailbox is created. The first poll occurs after one second.

Mailbox Rule Fields

WARNING

We recommend that you set the associated mailbox to Inactive before you configure a mailbox rule. Else, any messages that are retrieved between the first and the last change are processed with rules in effect.

The table (usp_mailbox_rule) maintains the rules that are used to connect to each mail server account (usp_mailbox). CA SDM provides predefined rules which you can use to modify or create mailbox rules.

Complete the following mailbox rule fields, as appropriate:

- **Sequence**
  Specifies the sequence number of the rule. Messages are checked against rules in sequence number order from lowest to highest.
- **Mailbox**
  Specifies the mailbox to which this rule belongs.
- **Active**
  Sets the rule to active or inactive.
- **Filter**
  Specifies what part of the email to search for the filter pattern, for example, Subject contains. For more information, see the Pattern Matching in Mailbox Rules topic.
- **Filter String**
  Specifies a regular expression string to match with the email content. For example, [\t\r\n]request[\t\r\n]. The placeholder {{object_id}} lets you specify the artifact value for the Text API to use for associating the message with a specific ticket. For more information, see the Filter String Object Identifier Restrictions and Special Characters in Regular Expressions topics.
- **Ignore Case**
  Specifies whether to consider letter case when matching patterns.
- **Action**
  Specifies the action to take when the filter criteria matches, for example, Create/Update Object.
- **Action Object**
  Displays the type of ticket object to which message actions apply, for example, Request.
- **Minimum Artifact Type**
  Sets the minimum type of artifact checking that you want to allow:
  - **NONE**
    Specifies no validation of artifacts. This value is the same as not adding the keyword to the input file. Also accepts Text API ticket ID commands.
  - **PROTECTED**

Validates a ticket against the hash for confirmation. If confirmation fails, the artifact is considered invalid and the email fails filtering where filtering is looking for an artifact ({{object_id}}).

– **SECURE**
Validates the ticket number from an encrypted data block. If the value is not a valid encrypted ticket number, the artifact is considered invalid and the email fails filtering where filtering is looking for an artifact ({{object_id}}).

NOTE
Types that are more secure than what is set are allowed. In other words, if you set the minimum type to **PROTECTED**, then both **PROTECTED** and **SECURE** are allowed, but **NONE** is not. Also, if **PROTECTED** or **SECURE** are selected, Text API ticket ID commands are not accepted. For more information about the artifacts, see the Artifacts Use Considerations topic.

- **Reply**
Specifies a notification method to send an automatic response. For example, Email. If you do not set this option, no response is returned. The response indicates success or failure of the actions performed for the message, and is separate from any notifications. If you are using multiple mailboxes, we recommend you to use the notification method to configure email replies.
- **Reply Subject**
Specifies a subject line for the reply, for example, Service Desk Response.
- **Write to stdlog**
Write email text to the standard log (stdlog) if the filter matches.
- **Log Entry Prefix**
Specifies a prefix to add when writing email text to the standard log entries. Use this option to enable matching log entries to rules.
- **Add Subject Line**
Adds the subject line from the original message to the message body before processing. You can append, prepend, or not add a subject line. The subject line is attached to either the ticket Description or a Log Comment, depending on the actions for the message.
- **Text API Defaults**
Specifies additional default commands for the Text API when the filter matches. Combines with the contents of the [EMAIL_DEFAULTS] section of the text_api.cfg file. The TextAPI Defaults field includes TextAPI keyword commands that are applied to a ticket when it is created from an email that matches a mailbox rule. The commands are not applied when the message affects an existing ticket.
To specify TextAPI Defaults commands, place each command on a separate line in the TextAPI Defaults field. Format the commands as follows:

```
OBJECT.FIELD=value
```

NOTE
Do not include a leading percentage symbol (%), which is required only for corresponding commands that are embedded in the body of the email.

For example, format the commands as follows:

```
REQUEST.PRIORITY=3
PROBLEM.CATEGORY=Facilities
INCIDENT.GROUP=Plumbing
```

- **Text API Ignore Incoming**
Specifies additional Ignore Details for the Text API when the filter matches. Combines with the contents of the [EMAIL_IGNORE_INCOMING] section of the text_api.cfg file.
The TextAPI Ignore Incoming field lists TextAPI keyword commands that are not permitted to be used in the text of the incoming email message. Any commands that are listed in this field are ignored when they are found in an incoming email message.
To specify TextAPI Ignore Incoming commands, do the following steps:
a. Place each command on a separate line in the TextAPI Ignore Incoming field.
b. Format the commands as follows:

```
OBJECT.FIELD
```

NOTE
Do not include a leading percentage symbol (%), which is required only for the corresponding commands that are embedded in the body of the email.

For example, format the commands as follows:

```
CHANGE.ASSIGNEE
PROBLEM.GROUP
REQUEST.EFFORT
```

c. Define all commands used in either field in the [KEYWORDS] section of the text_api.cfg file. This file is located in the "site" subdirectory of the CA SDM installation directory.

- **Details**
  Specifies information about the rule.
- **Success Text**
  Specifies the plain-text contents of a reply message when the message is processed successfully. For example:

  ```
  Thank you for submitting an update to your request. A support technician will contact you within the next
   24 hours.
  ```

  You can also enter a notification phrase, if already created. For example, you can use a notification phrase for email auto-replies.

  ```
  Thank you for submitting your request.
  @{notification_phrase[notification phrase code].phrase}
  ```

- **Success HTML**
  Specifies HTML contents of a reply message when the message processes successfully. The following options let you edit and preview HTML text:
  - **Edit Success HTML**
    Opens the HTML Editor which you can use to format the HTML.
  - **Quick View**
    Previews the HTML.
  - **HTML Source**
    Shows the HTML source.

  You can also use a notification phrase, for example,

  ```
  Thank you for submitting your request.</p>
  @{notification_phrase[notification phrase code].phrase}</p>
  ```

- **Failure Text**
  Specifies the plain-text contents of a reply message when the message does not process successfully. You can also enter a notification phrase, if already created. For example, you can use the following text:

  ```
  Thank you for submitting your request.
  @{notification_phrase[notification phrase code].phrase}
  ```

- **Failure HTML**
  Specifies HTML contents of a reply message when the message does not process successfully.

Mailbox Rule Actions

Mailbox rules let you perform any of the following email actions:

Ignore Email -- Does not process the email and does not reply.
This action is useful for system-level messages such as Out of Office or Mail Delivery errors.

Ignore Email and Reply -- Does not process the email, and replies to the sender. Response emails use the reply success messages and the reply failure messages are ignored.

Update Object -- Uses the filter string to determine the object identifier (for example, %Incident:{{object_id}}% in the email), and sends an update request to the Text API. If the object identifier is not found, the Text API does nothing.

This action typically handles email replies where the object identifier is embedded in the email. If no object identifier is present, the failure response message is typically sent.

Create/Update Object -- Uses the filter string to determine the object identifier (for example, %Incident:{{object_id}}% in the email), and sends an update request to the Text API. If the object identifier is found, the Text API updates a ticket. If the object identifier is not found, the Text API creates a ticket.
This action is the standard behavior of the mail daemon (Mail Eater) in which the email does or does not contain Text API keywords.

Pattern Matching in Mailbox Rules

The Mailbox rules use regular expressions for pattern matching. Consider the following whitespace characters that apply to regular expressions in mailbox rules:

- **\t**
  Used for representing a horizontal tab
- **\r**
  Used for representing a carriage return character
- **\n**
  Used for representing a line feed or new line character

The characters that represent line breaks in text can vary with the operating system, mail server, and mail client. For example:

- UNIX uses a \n.
- Microsoft uses \r\n
- Macintosh uses \r
- MacOS X uses \n

CA SDM mail processing elements exchange substitute line break characters to identify text elements like message and attached parameters. Use a line or paragraph break for building filters to match \r or\n. Line breaks between two keywords identify a sequence of one or more \r and \n characters.

Line wrapping can result in unexpected line breaks. Line breaks appear in the middle of the text. A space can change to a carriage return or line feed or both. The carriage return or line feed or both can be inserted after a space. Include spaces in the middle of a filter string through the Regular Expression Block:

[\t\n]+(open-bracket, space, backslash, t, backslash, r, backslash, n close-bracket, plus sign).

Example: Use [\t\r\n] to Match the exact keyword:

The sample example shows how to match the exact keyword "request" and ignore other possible similar keywords:

```
requester
requesting
requested
orequestra
```

To match the exact keyword "request", enter the keyword request by one or more white space characters as:

```
[ \t\r\n]request[ \t\r\n]
```

The Mail Eater performs a keyword-based search for "request and ignores other possible keywords like 'requester', requesting.

Filter String Object Identifier Restrictions

Filter strings determine the object identifier (for example, %Incident:{{object_id}}%) in an email. Text that surrounds an object identifier ({{object_id}}) must be unambiguous in both content and length. The text must clearly define the beginning and end of the ticket ID artifact value between the text.

The following restrictions apply to how the Mail Eater interprets the *start* of the ticket ID artifact value:

1. The {{object_id}} placeholder must not be the first element in the filter string. At least, one character is required. Generally, a distinctive keyword or a sequence of letters, numbers, and symbols must precede the object ID keyword.
2. The character immediately preceding the {{object_id}} placeholder must not be a repeatable or optional character having a plus sign (+), an asterisk (*), or a question mark (?). Whitespace characters (space, tab, carriage return, line feed) must not be part of a ticket ID artifact value.
3. The character immediately preceding the {{object_id}} placeholder must not be a repeatable or optional bracketed-set of characters.

The following restrictions apply to how the Mail Eater interprets the *length* of the ticket ID artifact value:

- The {{object_id}} placeholder must not be the last element in the Filter String. One or more characters must follow the {{object_id}} placeholder.
- The character immediately following the {{object_id}} placeholder must not be a repeatable or optional character with a plus sign (+), an asterisk (*), or a question mark (?). Whitespace characters (space, tab, carriage return, line feed) must not be part of a ticket ID artifact value.
- The first character after the {{object_id}} placeholder must not be a character.
- Avoid the following characters immediately before and after the {{object_id}} placeholder:
  - All upper or lower-case letters
  - Numerals
  - The plus sign (+)
  - The slash (/)
  - The comma (,)
  - The period (.), as it can represent any character in this list except for a line break.

  Any of these characters can exist within the ticket ID artifact value. If a bracketed set (several characters between brackets) precedes or follows the {{object_id}} placeholder, the bracketed set must not contain any of these listed characters.

Special Characters in Regular Expressions

Pattern matching for the filters in mailbox rules follows the rules for ASCII Regular Expressions with C-style special characters.

> **WARNING**
> We recommend that you are familiar with Regex syntax to use special characters in regular expressions.

For example, consider the following special characters for Regex patterns that apply to regular expressions in mailbox rules:

- **\t**
  The mailbox filter string \t specifies a horizontal tab. This string matches the beginning and end of text (and tabs), and is specific to MailEater.
- **\r**
  This string specifies a carriage return (return to the beginning of the current line).
  Note: Do not use \r for searching message subjects or sender addresses.
- **\n**
  This string specifies a newline (combination of line feed and carriage return).

  > **NOTE**
  > Do not use \n for searching message subjects or sender addresses.

\t, \r, and \n are the special characters that occur most often in regular expressions for mailbox rules.

Example: \t, \r, and \n Use

```
[ \t]request[ \t]
```

The filter string performs a keyword search for 'request'. The brackets match any one character from the set, including the space, so [\t] matches a space or a tab.

```
[\r\n]+critical[ \t\r\n]
```

The filter string performs a keyword search for 'critical'. This string searches at the start of a line, the start of the message, or the entire line. The brackets match any one character from the set. The + (plus sign) matches one or more of the previous, so [\r\n]+ matches one or more carriage returns and newlines.

Sample Text for Notification Phrases in a Mailbox Rule

This example shows sample text that you can use to include notification phrases in a mailbox rule. Define separate versions of a notification phrase for plain text and for HTML when the phrase contains any line breaks or other formatting.

Use the following text in Success Text, Success HTML, or both fields on the Update Mailbox Rule page, Reply Success tab:

Success Text

```
Thank you for submitting your request.
@{notification_phrase[IncidentURL1].phrase}
```

Success HTML

```
Thank you for submitting your request.</p>
@{notification_phrase[IncidentURL1H].phrase}</p>
```

Artifacts

An email *artifact* refers to something that arises from the mail process, for example, an email address that is included in a forwarded email. The Text API uses artifacts that contain a ticket ID (such as a reference number) for reply support. When the ticket ID is found, an existing Text API keyword (such as %INCIDENT_ID) is set and added to the input for the Text API. The Mail Eater identifies that a reply is associated with a particular ticket by finding the artifact in the message.

The Mailbox rules let you specify the artifact and the value that the Text API uses. For example, you can define a rule for incidents as Incident:**{{object_id}}%**. When a rule finds Incident:1234, the Text API uses **%INCIDENT_ID=1234** (1234 is the ref_num for the Incident). Because the artifact must be unique in an email and easy to find, you can make the artifact more distinctive such as **%Incident:{{object_id}}%**. A percentage sign (**%**), whitespace, or some other character that cannot appear in an artifact value must follow **{{object_id}}**. Uppercase and lowercase letters, numbers, forward slashes, commas, and plus symbols are potentially part of a value. The secure artifacts are Base64-encoded after encryption. If you do not use the Secure artifacts, the characters that follow the artifact must not be contained in the ticket ID suffix, if any, which has been configured for that type of ticket.

When using the filter string of the mailbox rules to identify the ticket ID Artifact, the keyword {{object_id}} represents the position in the filter string where the ticket ID artifact is expected. This keyword is case-sensitive, even if the mailbox rule is not.

Example: Email Artifact Use

The following example shows an ARTIFACT format for use in a mailbox rule for a change request ticket.

- **Usage**: %REQUEST=@{call_req_id.ref_num}%
  Example: %REQUEST=1234%

Artifacts Use Considerations

When you construct the filter string of the mailbox rule, consider the following conditions:

- A clear boundary must exist between the ticket ID artifact and the keywords which precede and follow it. We strongly recommended that you include whitespace text in this boundary text.
- Do not end the portion of the filter string that precedes the **{{object_id}}** keyword in a repeatable or optional pattern that can match the beginning of the ticket ID Artifact, and do not end a pattern whose length is ambiguous. For

example, the filter string must not contain the request(er|ed|ing)?{{object_id}}, because this construction causes an ambiguity whether a trailing **er**, **ed**, or **ing** is the end of the leading text or part of the prefix of an unprotected ticket ID.

- The portion of the filter string that follows the **{{object_id}}** keyword must not begin in a repeatable or optional pattern that may match the end of the ticket ID artifact, must not begin with a pattern whose length is ambiguous, and must contain at least one element of whitespace. For example:
  - The filter string must not contain **{{object_id}}[A-Z]?**, because the **[A-Z]?** may match the last character of the ticket ID artifact.
  - The filter string must not end with **{{object_id}}**Item, because it is possible for Item to appear in the ticket ID artifact, either as the suffix of a ticket ID in a plain-text or protected artifact, or as characters within a secure artifact.
  - **{{object_id}}** Item is acceptable, because the space cannot be part of a ticket ID artifact, and is not part of a protected or plain ticket ID artifact. However, **{{object_id}}**[**\t\r\n**]+**Item** (**open-bracket**, **space**, **backslash**, **t**, **backslash**, **r**, **backslash**, **n**, c**lose-bracket**, **plus sign**, +I**tem**) is better, because the [**\t\r\n**]+ compensates for the mail program inserting a line break after the **{{object_id}}**.
- When you construct the filter strings for different mailbox rules, avoid using a filter string that completely includes another mailbox rule filter string, or in which the portion before or after a **{{object_id}}** keyword completely includes that portion of another mailbox rule filter string. Depending on the order in which these filters are checked, a message match intended for one filter can match with another, with a portion of the ticket ID artifact matching the additional text that distinguishes between the two filter strings.

Artifacts Protection and Security

You can use artifacts to protect the mail delivery to a mailbox in the following ways:

Use the **ARTIFACT** keyword in notifications and notification phrases.

> **NOTE**
>
> The artifact security level must match the mailbox rule Minimum Artifact Type setting for the mailbox to which the email is sent. Types that are more secure than what is set are allowed. In other words, if you set **ARTIFACT=PROTECTED**, then both **PROTECTED** and **SECURE** are allowed, but **NONE** is not allowed.

Set the Minimum Artifact Type option in the Mailbox Rule Detail.
You can embed an artifact in a notification phrase and specify the security level for the artifact, for example:

```
%REQUEST=@{ARTIFACT=PROTECTED:call_req_id.ref_num}%
```

> **NOTE**
>
> For more information about the tables for the Minimum Artifact Type option (**filter_min_artifact_type**) and the Mailbox Rule Detail (**usp_mailbox_rule table**), see USP Mailbox.

Example: No Validation of Artifacts

The following example shows no ARTIFACT format. No validation of artifacts can be performed.

- **Usage**: %REQUEST=@{call_req_id.ref_num}%
  Example: %REQUEST=1234%
  Example: No Validation of Artifacts

The following example shows the ARTIFACT=NONE format. ARTIFACT=NONE is the same as not adding the keyword; no validation of artifacts can be performed.

- **Usage**: %REQUEST=@{ARTIFACT=NONE:call_req_id.ref_num}%
  Example: %REQUEST=1234%
  Example: Validate a Ticket Number Against the Hash for Confirmation

The following example shows the **ARTIFACT=PROTECTED** format. The **PROTECTED** format validates a ticket against the hash for confirmation. The letter "**A**" denotes the type of formatting. Commas separate "**A**", the hash code, and the ticket number.

- **Usage**: %REQUEST=@{ARTIFACT=PROTECTED:call_req_id.ref_num}%
  Example: %REQUEST=A,12345678,1234%
  Example: Encrypt the Ticket Number

The following example shows the ARTIFACT=SECURE format. The SECURE format encrypts the ref_num, and then encodes the result in Base64 encoding so it can safely be included in the message text. The letter "B" denotes the type of formatting. Commas separate "B" and a password encrypted ticket number.

- **Usage**: %REQUEST=@{ARTIFACT=SECURE:call_req_id.ref_num}%
  Example: %REQUEST=B,da1jhr+9U5GVfd2VGH4dsnx2+PaSvygDS2e3IqjpjtyNSDW2u/KNPX61nopDu/KB%

Example: How to Create a Mailbox Rule That Matches Every Inbound Message

You can create a mailbox rule that matches every inbound message that another mailbox rule does not filter. To create this type of rule, set the filter as Subject Contains and the filter string as a period and an asterisk **(".*")**.

- A period matches any character except the line break.
- An asterisk matches zero or more occurrences of the symbol immediately before it.
- As a result, this combination matches zero or more characters that are not line breaks.

Example: A "Catch-All" Mailbox Rule

This example demonstrates how you can use a **".*"** combination to match every inbound message:

```
Filter = "Subject contains"
Filter String = ".*"
```

Example: How to Use the Mailbox Rules TextAPI Defaults and TextAPI Ignore Incoming Settings

The TextAPI Defaults and TextAPI Ignore Incoming fields let you specify default values for incoming mailbox rules, and specify TextAPI commands that should not be accepted in incoming emails. These fields work with the default values that are set in the [EMAIL_DEFAULTS] section and with the forbidden-commands list in the [EMAIL_IGNORE_INCOMING] section of the text_api.cfg file. When a conflict occurs between the definition in a mailbox rule and the definition in the text_api.cfg file, the value set in the mailbox rule applies.

The TextAPI Defaults field includes TextAPI keyword commands that are applied to a ticket when it is created from an email that matches a mailbox rule. The commands are not applied when the message affects an existing ticket.

Follow these steps:

1. Place each command on a separate line in the TextAPI Defaults field.
2. Format the commands as follows:

   ```
   OBJECT.FIELD=value
   ```

   NOTE
   Do not include a leading percentage symbol (%), used only for corresponding comma embedded in the body of the email.

   For example, format the commands as follows:

   ```
   REQUEST.PRIORITY=3
   PROBLEM.CATEGORY=Facilities
   INCIDENT.GROUP=Plumbing
   ```

The TextAPI Ignore Incoming field lists TextAPI keyword commands that are not permitted to be used in the text of the incoming email message. Any commands that are listed in this field are ignored when they are found in an incoming email message.

To specify TextAPI Ignore Incoming commands, follow these steps:

1. Place each command on a separate line in the TextAPI Ignore Incoming field.
2. Format the commands as follows:

```
OBJECT.FIELD
```

NOTE
Do not include a leading percentage symbol (%), used only for the corresponding commands embedded in the body of the email.

For example, format the commands as follows:

```
CHANGE.ASSIGNEE
PROBLEM.GROUP
REQUEST.EFFORT
```

3. Define all commands used in either field in the [KEYWORDS] section of the text_api.cfg file. This file is located in the "site" subdirectory of the CA SDM installation directory.

Mailbox Policy Fields

Email Address/Hour
Specifies the maximum number of emails per email address per hour. You can specify the following values:

- -1 -- No limit (default)
- 0 -- No emails allowed.
- 1 – Maximum number of emails allowed.

Log Violation
Logs the violation to the standard log. You can specify the following values:

- Do not log
- First violation only (default)
- All violations

NOTE
The First violation only option keeps a list of email addresses associated with messages that violate mailbox policies and uses the list for the log to avoid duplicate log entries. The list is cleared when the Mail Eater daemon is restarted. However, if you change the setting from First violation only to one of the other options and back, the list of email addresses which were logged under this setting is not cleared. If a mailbox logs numerous violations while using this setting, we recommend that you restart the Mail Eater daemon periodically to clear the list, or use the Do not log option.

Inclusion List
Specifies email addresses or domains that are allowed to process emails -- only emails matching the list are allowed. You can specify multiple addresses or domains by delimiting them with a comma, semicolon, space character, or line break. An entry of an asterisk ('*') by itself is the "World Domain," and matches all domains that are not in the Exclusion List.

Exclusion List

Specifies email addresses or domains that are not allowed to process emails. You can specify multiple addresses or domains by delimiting them with a comma, semicolon, space, or line break.

NOTE
Addresses in the Exclusion List override any values in the Inclusion List. Addresses in the Inclusion List override domains in the Exclusion List, and can provide specific exemptions for specific senders in an otherwise-banned domain. Domains in the Exclusion List override the World Domain in the Inclusion List. The World Domain is not valid in the Exclusion List.

Multiple Mailboxes

CA SDM can process and manage multiple mailboxes. Each mailbox can have its own definition, instead of using a single global set of definitions. You can define multiple mailboxes and use different templates or default values for each mailbox. Multiple definitions let individual tenants use separate mailboxes, or let an individual tenant or organization use different

mailboxes, and have different behaviors for each mailbox. You can set up multiple mailboxes by using the Administration interface. Each mailbox uses the following tables:

- usp_mailbox -- Defines the mailbox.
- usp_mailbox_rule -- Specifies a set of rules for each mailbox.
  Because mailbox rules supply Text API defaults, you can establish email interfaces with other software and parameters (such as category, assignee, and so on) that are configured specifically for the interface.

  NOTE
  IMAP servers support multiple mailboxes for a single account, but alternate mailboxes are not supported; only the default inbox is supported.

How Multiple Mailboxes Use Rules

The Mail Eater (pdm_maileater_nxd) component on the primary server uses mailbox connections and rules to read and process messages from one or more accounts on one or more mail servers. The Mail Eater processes mailboxes serially (only one mailbox is processed at a time), and processes rules in sequence number order.

Multiple mailboxes use rules as follows:

1. Upon primary server startup, the Mail Eater reads the following tables:
   - **usp_mailbox**
     Represents a connection to a mail server.
   - **usp_mailbox_rules**
     Represents the rules that apply to the connection (usp_mailbox).
   - **Contact_Method**
     Represents the Contact Methods used for automatic replies.
   - **Document_Repository**
     Represents the Document Repositories for storing attachments.

   The Mail Eater automatically detects changes to the objects in any of these tables, including the addition of additional objects. If a change is made to usp_mailbox or usp_mailbox_rule, the polling interval for the affected mailbox is rescheduled to one second after the change is applied.
2. At the interval defined by each mailbox, the Mail Eater retrieves each email in the inbox for the associated account, and processes the email as:
   a. Checks the email address for policy violations. When the Mail Eater finds a violation, processing stops, and the standard log is affected according to the mailbox definition.
   b. Compares the email to each rule (mailbox_rule) that belongs to that mailbox.
   c. If a matching rule is found, the Mail Eater submits the message to the Text API for posting, and replies to the user as appropriate based on the specified action for the rule.
      For reply emails, the filter string identifies the object and uses the Text API for processing. After processing is complete, the response goes either to the Reply to or the From address.
   d. When a matching rule is located, no other rules are checked, and the Mail Eater processes the next email in the inbox.
   e. If no matching rule is found, the message is discarded.
3. After the Mail Eater processes all emails for an inbox, the processed and discarded messages are purged from the mail server, and the next processing interval is scheduled.

Set the Security Level as OAuth 2.0

CA Service Management supports Maileater OAuth2.0. (inbound IMAP and outbound SMTP). OAuth2.0 is only for IMAP and SMTP as POP3 is not supported by JavaMail API. The process to authorize the Maileater/Mailer daemon to connect to a Mailserver via OAuth 2.0 is a single step.

Set the Security Level as OAuth 2.0 - IMAP

Perform the following steps:

1. Login to CA SDM as Administrator.
2. Navigate to **Administration** Tab, **Email**, **Mailboxes** and specify the mailbox connection settings to a mail server for inbound processing. For more information, see Create Mailbox.
3. Select the **OAuth 2.0** security level in the mailbox detail page and click **Save**.
   A new OAuth Tab is displayed.
4. Click **Create New** to open the detail page for OAuth setting. Enter the following details:
   a. **Provider**:Select the respective provider as per the preferred mailbox (Microsoft Exchange or Google Mail).
   b. **Client ID**: Client ID is generated while app registration is done. For more information, about how to do an app registration, see App Registration.
   c. **Client Secret**: Client secret is generated while app registration is done.
   d. **Redirect Url**: Provide the **redirect_url** similar to the one provided in the app registration.
   For example: (https://<SDM_HOSTNAME_with_FQDN:SDM_PORT_NUMBER>/CAisd/OAuthProcessor)

      NOTE
      If the **redirect_url** is different to the one provided for Mail App registration, access token generation will fail due to a mismatch in the URL.

   e. **Scope**: Scope of access required to generate access token for IMAP (Maileater) is as follows:
      a. For IMAP (Maileater), perform the following steps:
         a. For Microsoft Exchange, scope will be offline_access https://outlook.office.com/IMAP.AccessAsUser.All
         b. For Google Mail, scope will be https://mail.google.com
         c. For Maileater, scope of access required: offline_access https://outlook.office.com/IMAP.AccessAsUser.All
   f. **Admin Consent Required?**: Optional field visible only in case the Provider is Microsoft Exchange and is unchecked by default. Select the checkbox, if the access token requests needs approval from Azure Admin.
   g. **Tenant ID**: Optional field visible only in case the Admin Consent Required? is set to Checked. Provide the Azure Tenant ID.
   From RU 21, optional field is visible and required in case the Admin Consent Required? is set to Checked or an APP is registered as a single tenant application in the Azure portal. Provide the Azure Tenant ID.
   h. Click **Save.**
   i. Before Generating access token, perform the mandatory post-install steps. For more information, see OAuth 2.0 Post-Installation Task.
   j. Click **Generate Access Token** which opens up the respective mail server page.
   Select **Next** to generate the access token.

Set the Security Level as OAuth 2.0 - GRAPH

Perform the following steps:

1. Login to CA SDM as Administrator.
2. Navigate to Administration Tab, Email, Mailboxes and specify the mailbox connection settings to a mail server for inbound processing. For more information, see Create Mailbox.
3. Select the Email Type as GRAPH.
4. Select the OAuth 2.0 security level in the mailbox detail page and click Save. A new OAuth tab is displayed.
5. Click Create New to open the detail page for OAuth setting. Enter the following details:
   a. **Provider**: Select the provider as Microsoft Exchange.
   b. **Client ID**: Client ID is generated while app registration is done. For more information, about how to do an app registration, see App Registration.
   c. **Client Secret**: Client secret is generated while app registration is done.
   d. **Redirect URL**: Provide the **redirect_url** similar to the one provided in the app registration.
   For example: (https://<SDM_HOSTNAME_with_FQDN:SDM_PORT_NUMBER>/CAisd/OAuthProcessor)

NOTE
If the *redirect_url* is different to the one provided for Mail App registration, access token generation will fail due to a mismatch in the URL.

e. **Scope**: Scope of access required to generate access token for GRAPH (Maileater) is as follows:
   a. Set the scope as *offline_access Mail.ReadWrite*
f. **Admin Consent Required?**: Select the checkbox, if the access token requests needs approval from Azure Admin.
g. **Tenant ID**: Optional field but required in case the Admin Consent Required? is set to Checked or an APP is registered as a single tenant application in the Azure portal. Provide the Azure Tenant ID.
h. Click **Save**.
i. Before Generating access token, perform the mandatory post-install steps. For more information, see OAuth 2.0 Post-Installation Task.
j. Click **Generate Access Token** which opens up the respective mail server page. Input the credentials in mail server page opened to generate the access token.

IMPORTANT
For **Microsoft Exchange**, in instances where **Admin Consent Required?** field is set to **Yes**, clicking the '**Generate access token**' opens up the Admin consent Microsoft Login page where the Azure Administrator needs to login first and grant the required consent/access. Once access is granted, the page redirects to another Microsoft login page where the mailbox user needs to login so that an access token gets generated successfully.

For Microsoft Exchange, in the case where **Admin Consent Required?** is set to No, clicking the '**Generate access token**' opens up a microsoft login page where the mailbox user needs to login so that an access token gets generated successfully.

For Gmail, clicking on '**Generate access token**' opens up a Gmail login page where the mailbox user needs to login so that an access token gets generated successfully.

Set the Security Level as OAuth 2.0 - SMTP

Perform the following steps:

1. Click **Create New** to open the detail page for OAuth setting. Enter the following details:
   a. **Provider**:Select the respective provider as per the preferred mailbox (Microsoft Exchange or Google Mail).
   b. **Client ID**: Client ID is generated while app registration is done. For more information, about how to do an app registration, see App Registration .
   c. **Client Secret**: Client secret is generated while app registration is done.
   d. **Redirect Url**: Provide the **redirect_url** similar to the one provided in the app registration.
   For example: (https://<SDM_HOSTNAME_with_FQDN:SDM_PORT_NUMBER>/CAisd/OAuthProcessor)

   NOTE
   If the **redirect_url** is different to the one provided for Mail App registration, access token generation will fail due to a mismatch in the URL.

   e. **Scope**: Scope of access required to generate access token for IMAP (Maileater) and SMTP (Mailer) is as follows:
      a. For SMTP (Mailer), perform the following steps:
         a. For Google mail, set the scope to: https://mail.google.com
         b. For Microsoft Exchange, set the scope to: offline_access https://outlook.office.com/SMTP.Send
         c. Restart the CA SDM Services.
   f. **Admin Consent Required?**: Optional field visible only in case the Provider is Microsoft Exchange and is unchecked by default. Select the checkbox, if the access token requests needs approval from Azure Admin.
   g. **Tenant ID**: Optional field visible only in case the Admin Consent Required? is set to Checked. Provide the Azure Tenant ID.

From RU 21, optional field is visible and required in case the Admin Consent Required? is set to Checked or an APP is registered as a single tenant application in the Azure portal. Provide the Azure Tenant ID.

h. Click **Save.**

i. Before Generating access token, perform the mandatory post-install steps. For more information, see OAuth 2.0 Post-Installation Task

j. Click **Generate Access Token** which opens up the respective mail server page.
Select **Next** to generate the access token.

IMPORTANT
For **Microsoft Exchange**, in instances where **Admin Consent Required?** field is set to **Yes**, clicking the '**Generate access token**' opens up the Admin consent Microsoft Login page where the Azure Administrator needs to login first and grant the required consent/access. Once access is granted, the page redirects to another Microsoft login page where the mailbox user needs to login so that an access token gets generated successfully.

For Microsoft Exchange, in cases where **Admin Consent Required?** is set to **No**, clicking the '**Generate access token**' opens up a Microsoft login page where the mailbox user needs to login so that an access token gets generated successfully.

For Gmail, clicking on '**Generate access token**' opens up a Gmail login page where the mailbox user needs to login so that an access token gets generated successfully.

App Registration - configure the OAuth provider

Setup the OAuth 2.0 provider for Gmail and Microsoft Exchange as described below:

App Registration for configuring Google service as an OAuth provider

Perform the following steps for App Registration for Gmail:

1. Create a new GMAIL account that you intend to use as a Mailbox.
2. Navigate to Google Developer Console and login with your Gmail account created in Step 1.
3. After logging in, Click Select a project.
4. Click Select a project and enter a project name.
Click Create.
After Google creates the project, the project dashboard appears.
5. After successful creation of project, click the Menu Bar and click '**APIs** &**Services**'.
6. Give OAuth Consent to the App.
7. After the Consent is provided, create client credentials.
8. Choose ApplicationType as Web Application and set "Authorized redirect URI" as https://<SDM_HOSTNAME__with_FQDN:SDM_PORT_NUMBER>/CAisd/OAuthProcessor and click Save. The client credentials gets created.

App Registration for Configuring Microsoft Exchange (Office 365/Microsoft 365) as an OAuth Provider

NOTE
OAuth IMAP access is supported only for Microsoft365 accounts and not personal accounts like outlook.com. To test OAuth Configuration, you need a Office365 (renamed to Microsoft365) account. For more information, see Microsoft Documentation about how to set up an Office 365 account.

Perform the following steps to get your Microsoft/OneDrive account's Client ID and Secret/Password Key:

1. Login to Microsoft Azure.
If you have created an account with Microsoft Azure, enter your credentials to login, else create an account.
2. Navigate to Azure Active Directory.
3. Select App Registrations.
4. Click on Register an Application.

5. Enter the name, select your desired option in Supported account types, select web option under Redirect URL.
6. Enter this URL (https://<SDM_HOSTNAME__with_FQDN:SDM_PORT_NUMBER>/CAisd/OAuthProcessor) in Redirect URL field.
7. Once you are finished, click **Register**.
8. From Certifications & Secrets option on the left, copy your **Application (Client) ID** and go to **Certificates and Secrets**.
9. Click on **New Client Secret.**
10. Enter **description**, select **expiry time** and click **Add.**
11. From Certificates & Secrets option on the left, copy your **Client Secret Key**.

For more information about the Option Manager changes in order to set up OAuth 2.0 in Email Options > **mail_smtp_security_level**, see Options Manager.

Connecting Maileater to Office 365 Mail

To configure the maileater to connect to Office 365 Mail, perform the following steps:

1. Obtain a valid SSL Certificate for the Office 365 Mail Server.

   NOTE

   As of Dec 2016, the Office 365 Certificate is signed by the "**DigiCert Global Root CA**". Information on this certificate can be found here. If *outlook.com* gets signed by a different root certificate in the future, you must either obtain the new root certificate from Microsoft or find out the proper root certificate is and get it from the vendor.

   NOTE

   Maileater now supports configuring mail servers with wild card SSL certificates and/or where the mail server name is in the Certificate's subject alternate names (and not in the issues To Part).

   NOTE

   CA SDM Maileater now supports parsing of HTML and Rich-Text Format (RTF) emails for incoming mails.
2. After collecting the root certificate information, create a file using *Notepad* with the certificate's information.
3. Save the file and copy it to the primary server and make note of the copied path.
   a. Use the following settings in the mailbox details screen (making sure to populate your full email address and file path to the certificate that you saved):

Field Name	Description
Name	Default (or the name of your mailbox)
Check Interval (in Sec)	recommended value is 30
Email Type	IMAP4
Hostname	outlook.office365.com
Port Override	**Note:** CA SDM now supports both **STARTTLS** and **Direct SSL** port connections.
User Name	The User ID of the Office365's account. (If it does not work, try and use the full email address)
Password	Fill in the password for the account
Security Level	Clear Text
Attachment Repository	Default is "Service Desk"

Attach Entire Email	Default is "No"
Force Attachment Splitout	Default is "No"
Allow Anonymous	Default is "Yes"
Save Unknown Emails	Default is "Yes"
Use Reply To Emails	Default is "Yes"
Use TLS	Yes
CA Certificate Path	Path to the Certificate file you just created.

The maileater is now connected to Office 365 Mail Server.

How to Connect CA SDM to the Office365 Servers Using SSL

Complete the following steps if you are unable to connect to Office365 servers:

1. Log in to outlook.office365.com using the credentials for the account that you wish to use with maileater.
   You are redirected to the login.microsoftonline.com page.
2. Save the root certificate from the login.microsoftonline.com page as a base 64 encoded.cer file, and copy it to the CA SDM server. Complete the following steps to save certificate:
   a. As an Administrator, click on the lock in the address bar of the page from where you want to save the certificate.
   b. From the Certification Path tab, select the root certificate and click View Certificate.
   c. From the Details tab, click Copy to File, and choose Base 64 Encoded x.509 (.CER) as the format.
3. Log in to CA SDM and use the same settings in the Mailbox details screen (ensure that you add the full email address and file path to the certificate that you saved).
4. Connect CA SDM to the Office365 servers and verify.

How to Configure Email Replies

This article contains the following topics:

The email notifications that you use in mailboxes are specific to the replies that are sent to a contact in response to their emails. For example, you can configure email so that when a contact clicks a reply link in an email notification, the reply email is directed to a mailbox.

> **NOTE**
> This setup differs from the regular email notifications.

Follow these steps:

1. Open the CA SDM Web UI.
2. Verify that you have configured the mailbox for inbound email.
3. Configure the mail server.
4. (Optional) Specify a notification email address in the contact definition.

   > **NOTE**
   > Select Security and Role Management, Contacts on the Administration tab and select the contact to edit.
5. Modify the email notification method.

Open CA SDM Web UI

Log in to the web UI from the following servers, depending on your CA SDM configuration:

- Conventional: Primary or secondary servers
- Advanced availability: Application or background servers

Configure the Mail Server

Notifications for an event (automatic and manual notify) are sent using a single mail server definition.

Follow these steps:

1. Select Options Manager, Email from the Administration tab.
   The Option List page opens.
2. Click the email option that you want to install.
   The Options Detail page opens.
3. Click Edit, complete the fields as appropriate, and click Install.
   The mail server is configured to send notifications (outbound mail).
4. Repeat the procedure until all relevant Option List options are configured.

Email Options

The email interface sends email notifications and lets users create tickets from an email.

Option	Description
mail_from_address	Specifies the mail notification From: address. The address is in the format Displayname<user@company.com>.
mail_login_password	Specifies the SMTP server login password.
mail_login_userid	Specifies the SMTP server login userid.
mail_max_threads	Specifies the maximum number of concurrent SMTP connections that can attempt to communicate with the server.
mail_reply_to_address	Defines the reply to address for email notifications. This option is useful if emails are sent from one user account, but you want replies sent to another email address. The default value is the same as the from address.
mail_smtp_domain_name	Defines the domain name of the SMTP server. You can clear the domain name by setting the value to NONE.
mail_smtp_hosts	Specifies a space-separated list of SMTP server host names for email notifications.
mail_smtp_host_port	Specifies an SMTP port to override the default SMTP port.
mail_smtp_security_level	Specifies the SMTP security level. Valid settings are: 0=no security, 1=basic authentication, 2=NTLM, 3=MD5 and 4=LOGIN. If you set this option to 1, set the mail_login_password and mail_login_userid options. Most SMTP servers do not require authentication.
mail_smtp_use_tls	Specifies the Transport Layer Security (TLS) usage in the email. The valid settings are Yes= Use TLS, No=Do not use TLS.
mail_ca_cert_path	Specifies the path where the trusted certificate has been deployed. For the advanced availability configuration, you must deploy the trusted certificate on the same location in all the CA SDM servers. CA SDM supports only Base-64 encoded (PEM) format for CA Certificates.

mail_show_to_cc_list	Shows all the recipients of the email notification. For manual email notification, both “To” and “Cc” list recipients are displayed and for automatic email notification, only “To” list recipients are displayed. By default, this option is installed. **Note:** Pager email notification method does not support the listing of all recipients in To or CC list.

Modify the Email Notification Method

Modify the email notification method to configure the email replies.

Follow these steps:

1. Select Notifications, Notification Methods, Email on the Administration tab and click Edit.
2. On the Email Update Notification Method page, select the Supports SMTP checkbox to enable SMTP.
3. Enter pdm_mail [- F from_email_address] [- T reply_to_email_address] as the Notification Method.
   - **from_email_address**
     Specifies the address that is used as the From address of the message. This address overrides the sender address in the outbound mail server configuration.
   - **reply_to_email_address**
     Specifies the address to which replies are sent. This address overrides the reply-to address in the outbound mail server configuration.
     When a reply-to address is set in the outbound mail server configuration and no reply-to address is specified in the Notification Method, the reply-to address in the outbound mail server configuration is used with that Notification Method.

     NOTE
     If the keyword “$(REPLY_FROM)” is specified as either address, that address is constructed from the username and mail server hostname for the mailbox. This keyword is only valid when a mailbox rule uses the Notification Method; Notification Methods that use it must not be used for any other purpose. For example, user name=dev, server name=mail32.ca.com, $(REPLY_FROM)=dev@mail32.ca.com. Only use this keyword if your mail server is configured to accept the mail server name as equivalent to the email domain name. Use this keyword with caution: If the hostname is not fully domain-extended in the mailbox configuration (for example, mailserver1 instead of mailserver1.customer7.com), it is not extended automatically by the field interpreter.

     NOTE

     The from_email_address and reply_to_email_address are the addresses that appear in the From and Reply-To headers of the message when the user reads it. If the addresses are identical, you can specify only the from_address.
4. Click Save.
   The email notification method is modified. When the contact replies to the email notification, the reply is addressed by default to the specified mailbox.

Alternate Sender Address Identification

You can use a -m parameter in the subject of the message so that CA SDM identifies the sender of the message using a different email address from the one that originally sent it. The -m keyword, followed by a space and by the email address that CA SDM recognizes, must be the last elements of the subject line. Consider the following information when you use the -m parameter in the subject:

- Both the actual From address and the alternate address that is specified with -m are verified in the Inclusion and Exclusion lists.
- The email address that is specified as the alternate address must contain only the address, and not the accompanying display name.
- If more than one word follows the -m parameter in the subject line, the alternate address is not recognized.

Mailbox Polling

If an error occurs on the outgoing mail server, email notifications are not sent and queue in the %NX_ROOT%\site \mail_queue directory. When the mail server becomes active again, after an interval it processes and sends email. You can change the interval to recycle the email that was queued when the mail server was busy.

Notification email messages that the outgoing mail server fails to send are resubmitted until you do one of the following:

- Stop the Mail Daemon (pdm_mail_nxd) that handles outbound email notifications.
- Manually delete the messages from the %NX_ROOT%\site\mail_queue directory.

Set the Email Retry Interval Variable

You can define the time interval (in seconds) to retry failed attempts to send outgoing email to the mail server.

> **NOTE**
> CA SDM does not retry sending messages that the outgoing mail server accepts, but cannot be delivered. For these messages, the outgoing mail server retry capabilities and policies, if any, are in effect.

Retries are on a per-message basis. If the mail server is unavailable for a period, each message is retried when its own timer expires, rather than all the messages being sent at once. However, if you restart the outgoing mail daemon, all unsent messages attempt to be sent at that time, and if they all fail to be sent, their retry timers are all reset at the same time.

The setting (NX_EMAIL_RETRY_INTERVAL) in the NX.env file controls the retry interval. You can change the default retry interval setting on one or more servers.

Follow these steps:

1. Log in to the following server, depending on your CA SDM configuration:
   - Conventional: Primary server
   - Advanced availability: Background server.
2. Navigate to the $NX_ROOT directory
3. Use a text editor such as WordPad to open the NX.env file.
4. Modify the value of to the NX_EMAIL_RETRY_INTERVAL interval you want as follows:
   ```
   NX_EMAIL_RETRY_INTERVAL=number_of_seconds
   ```
   - **NX_EMAIL_RETRY_INTERVAL**
     Defines the time interval (in seconds) to retry failed email attempts.
   - ***number_of_seconds***
     Specifies the number of seconds for each email retry interval. The default time is 600 seconds (10 minutes). The minimum value that you can use is 20 seconds. If you set a value that is less than the minimum of 20 seconds or more than 2000000 seconds, the default value of 10 minutes is used.
5. Save and close the file.
6. Restart the CA SDM servers.
   The change takes effect.

How Does the Maileater Process Inbound Emails?

CA Service Desk Manager checks incoming emails based on the **Check Interval** setting, which is set to 30 seconds as a default. If multiple emails arrive within 30 seconds, which of these will be processed first? Will the maileater process in the order of **Sent Date and Time** of the emails?

Generally emails are processed based on the order in which emails arrive at the CA Service Desk Manager maileater process.

Consider the following example scenarios:

Example 1:

Two emails (**Email 1** and **Email 2**) are sent out at the same time, but a mail server takes time in sending out the **Email 1** (due to attachments or some other reasons). **Email 2** is sent first to the mail server where CA Service Desk Manager connects to and receives the email by the maileater process, in this scenario, the maileater will process **Email 2** first.

- Email 1 sent with some attachments.
- Email 2 sent with plain texts.

If there are multiple emails in the inbox, CA Service Desk Manager will process in order of the time it arrives at the maileater process.

Example 2:

The following three emails are received and stored in the mail box after the previous mail check:

- Email 1: Received date/time 10:31 18/Jul/2017
- Email 2: Received date/time 10:32 18/Jul/2017
- Email 3: Received date/time 10:33 18/Jul/2017

Here, in this case, Email 1 is processed first by the maileater, then, Email 2, and then, Email 3 is processed.

For more information, see Define a Mailbox.

How to Set Up the Data Partition

Data partitions can be assigned to individual contacts, but the preferred method is to assign data partitions based on access type. After associating data partitions with different access types, you can associate a contact to a particular access type, and define its data partition. The access type has a specific option to override the contact data partition.

To associate a data partition to an access type, you set up data partitions that are meaningful to your site and then select one of the data partitions when you define or modify an access type.

> **WARNING**
> Other CA SDM security settings can take precedence over data partition.

The data partition is a subset of CA SDM database that controls a user access to tickets and other data records based on their content.

For example, you can restrict a user view of the database to only those configuration items that are assigned to the user organization. You can also restrict a user to update only assigned tickets and allow read-only access to other configuration items.

A data partition consists of a set of constraints. The data partition constraints identify the table that is being controlled by the data partition and the constraint type. The constraint typesspecify what the user can do, such as create, delete, update, view, and so forth, within the data partition. After you assign a data partition to an access type, which you in turn assign to a user contact record, the constraints and constraint types are what control the user access to the records in the CA SDM database table.

You can also view constraints independent of the data partition using them. For example, you want to view all constraints for a specific table, regardless of the data partition.

Follow these steps:

1. Verify the prerequisites
2. Create a data partition
3. Create an access type
4. Configure Knowledge Management Data Partition Constraints for Role-Based Permissions

Verify the Prerequisites

Before you set up the data partition, verify the following prerequisites:

- Data Security Structure and Policy
- Data Partitions Setup
- Constraint Specifications

Data Security Structure and Policy

Planning data security involves enforcing restrictions to access the specific portion of the database. These restrictions can be enforced on individual contacts either through roles or access types:

- **Roles**
  Defines the functionality that users in the role are allowed to access. You can assign one or more roles to an individual contact record, or to an access type to define the functional access for all of the access types who are associated with contacts.
- **Access Type**
  Defines how contacts are authenticated when they log in to the web interface. For example, an access type decides whether the contacts can modify web forms or the database schema using Web Screen Painter and which roles are available for the contacts.

As a service desk administrator, you can modify the predefined access types and also create new ones. You can enforce the restriction to the individual users or a group of users through Roles.

Identify the following:

- The objects and the type of restrictions you want to enforce on these objects.
- The Users or Roles on whom the Data Partition are applied. Data Partitions can be applied to contacts directly, but the preferred method is to assign data partitions based on the role and assign this role to all the contacts directly or through the access type.

Data Partitions Setup Specifications

You can define an unlimited number of data partitions. Each data partition consists of a set of constraints and validations on each database table that is restricted by the data partition. For each table in a data partition, you can specify independent authorizations to view, update, create, or delete records using criteria that are specified in a format similar to an SQL WHERE clause. You can base the restriction on any attribute in the record being accessed, combined with any data in the user contact record. This allows considerable flexibility when defining data partitions. For example, using the Vendor field in the Contact table allows data partition restrictions to be placed on vendors that are permitted to access CA SDM directly.

For performance reasons, CA SDM does not allow a data partition constraint to contain a Cartesian join. A Cartesian join results from a constraint containing an “OR” that does fully restrict all joined tables on both sides of the OR. To ensure that your data partition constraint does not produce a Cartesian product join, enter the following command:

- **Windows**

```
bop_cmd -f $NX_ROOT\bopcfg\interp\bop_diag.frg "check_queries()"
```

- **UNIX**

```
bop_cmd -f $NX_ROOT/bopcfg/interp/bop_diag.frg "check_queries()"
```

WARNING
Any data partitions that are flagged by this program must be updated appropriately.

Constraint Specifications

You specify constraints and validation tests in Majic using the object definition metalanguage.

Constraints that are defined in Majic closely resemble an SQL WHERE clause, with the following exceptions:

- Attribute names in the constraint are object attribute names, not the database attribute name from the schema.
- You can refer to the value of an attribute in the contact record for the logged-in user with a name of the following form, where *att_name* is the Majic name of the desired attribute:

  ```
  @root.att_name
  ```

  For example, specifying @root.location refers to the location ID of the current contact.
  You specify joins with a specification of the following form, where the *foreign-key* is the Majic name of the SREL attribute in the table for which you are writing the data partition constraint, and *attribute-in-referenced-table* is the Majic name of the attribute in the table being joined:

  ```
  foreign-key.attribute-in-referenced-table
  ```

  For example, to refer to the maintenance vendor of the asset that is associated with an incident report, specify the following:

  ```
  resource.vendor_repair
  ```

  This specification is recursive. For example, you could refer to the name of the vendor with the following name:

  ```
  resource.vendor_repair.name
  ```

The following table contains examples of valid constraints to use for the Change_Request table, used to store change order information:

Constraint Type	Code and Description
View	assignee.organization = @root.organization Specifies the user can only view change orders where the assignee's organization is the same as the user's organization.
Pre-Update	requestor = @root.id Specifies the user can only update the change orders where he is the caller or requester.

However, you cannot write a constraint that uses joins on both sides of the expression, as shown in the following example:

```
assignee.org = requestor.org
```

Configure Knowledge Management Data Partition Constraints for Role-Based Permissions

Knowledge Management data partitions are enabled to let you use group and role permissions by default in CA SDM. If you are upgrading from a previous release, the migration tool updates your data partition constraints.

If you used custom data partition constraints to manage knowledge permissions in a previous release, update the constraints manually for the O_INDEXES and SKELETONS tables. You can view the default data partition constraints and apply the changes, as appropriate to your environment.

Follow these steps:

1. Select Security and Role Management, Data Partitions, Data Partition Constraints on the Administration tab.

The Data Partition Constraints List page opens.
2. Click Show Filter and search use the following search criteria:
   – **Service Desk Analyst** in the Data Partition search.
   – **O_INDEXES** in the Table search.
3. Edit the View constraint type by modifying the Constraint tab to replace "READ_PGROUP in @root.pgroups" as follows:
   ```
   READ_PGROUP in @root.pgroups OR
   READ_PGROUP.[pgroup]contained_roles.role IN @root.id
   ```
4. Save the constraint.
5. Edit the Delete and Pre-Update constraint types by modifying the Constraint tab to replace "WRITE_PGROUP in @root.pgroups"as follows:
   ```
   WRITE_PGROUP in @root.pgroups OR WRITE_PGROUP.[pgroup]contained_roles.role IN @root.role
   ```
6. Save the constraints.
7. Repeat the steps to update the View, Delete, and Pre-Update constraints in the SKELETONS table in your data partition.
   The data partition constraints are updated.

Create an Access Type

This article contains the following topics:

Access types define how contacts are authenticated when they log in to the web interface, whether the contacts can modify web forms or the database schema using Web Screen Painter, and which roles are available for the contacts.

You can modify the predefined access types and create new ones.

The access types define all aspects of security. Several predefined access types are included, and you can modify them or can define new ones. Each access type for a user controls the following aspects of system behavior:

- How CA SDM performs the web authentication when the user logs in.
- The access level for the user.
- Whether the user can modify web forms or the database schema using Web Screen Painter.
- What are the roles available to the user.

Follow these steps:

1. Select Security and Role Management, Access Types on the Administration tab.
   The Access Type List page opens.
   Default: 15
2. Click Create New and complete the access type fields, as appropriate, on the Create New Access Type page.
3. Use the tabs to complete the following tasks:
   – Configure Web Authentication for an Access Type
   – Assign Web Screen Painter Permissions to an Access Type
   – Assign Roles to an Access Type
4. Click Save.
   The access type is created.

Access Type Fields

The following fields appear on the Create Access Type, Access Type Detail, and Update Access Type pages.

- **Symbol**

Specifies a unique identifier for the access type.

- **Default?**
  Indicates whether this access type is the default that is associated with contacts.
- **Record Status**
  Specifies whether this access type is Active or Inactive.
- **Description**
  Describes the access type. Use this field to help identify the characteristics of the access type.
- **Receive Internal Notification**
  Determines whether the contacts associated with the access type receive internal notification of activities that are related to tickets.
- **Access Level**
  Determines which access types a user can grant to another user. A user can assign an access type to the contact record of another user only if the access level of the access type they are attempting to assign is ranked the same as or lower than the grant level for their own access type. The levels are ranked as follows:
  - Admin (highest)
  - Analyst
  - Cust/Emp
  - None (lowest)
- **Licensed?**
  Determines whether this contact is a licensed access type. Contacts assigned to unlicensed access types can only view or update their own personal data.

  NOTE
  KPIs count the concurrent usages of the users from the system (for example, CA SDM Web UI, SOAP Web Services, REST Web Services, and so on). For example, the webConcurrentLicenseCt KPI counts the maximum number of unique users (with the "Licensed?" option selected) logged in to the CA SDM Web UI during the interval.The following fields appear on the Create Access Type, Access Type Detail, and Update Access Type pages.

Configure Web Authentication for an Access Type

You can configure the web authentication and validation type to specify how roles assigned to this access type are authenticated when users attempt to access the CA products. Complete the following fields in the **Web Authentication** tab.

- **Allow External Authentication**
  Select this check box if you want to allow contacts to be authenticated externally, for example by the HTTPD server or the operating system. If you select this option, users with this access type are validated by the appropriate external method as configured during installation. Checks ensure that no external validation has taken place (for example, that the user has not attempted access through a non-secure server) and that the user is defined as a valid contact in the system using the login ID. Then, it uses the access type to determine the correct interface to use.
- **Validation Type**
  Defines how users are authenticated when an external authorization is either not permitted or fails (for example, if the user is attempting access through a non-secure server). The available options are:
  - **No Access**
    Denies access to CA products unless external authentication is allowed and is valid.
  - **Open**
    Access to the CA products are always allowed, with no additional authentication required.
  - **OS**
    Access to the CA products are allowed through operating system user name and password.
  - **PIN / PIN Number**

Users of this type can access only if they enter the correct value for the PIN field in their contact record. If you select the PIN option, you can choose which field in the contact record stores the PIN by entering the field attribute name in the PIN Field edit box.

- **CA EEM**
  Access to the CA products are allowed through CA EEM. This option is available only if CA SDM is integrated with EEM.

Assign Web Screen Painter Permissions to an Access Type

The Web Screen Painter (WSP) utility allows CA SDM users to build and publish web forms and schemas. The Web Screen Painter tab also controls the database access for Web Screen Painter preview sessions. For the details about WSP, see Using the Web Screen Painter (WSP).

Select the permissions that you want to allow for an access type in the Web Screen Painter tab.

- **Modify Forms**
  Allows the users to do the changes to existing forms without doing the changes available to all users.
- **Modify Schema**
  Allows the users to do the changes to an existing schema without doing the changes available to all users.
- **Publish Forms**
  Allows the users to make their modified forms available to all users.
- **Publish Schema**
  Allows the users to make their modified schema available to all users.
- **Preview Session Can Update Database**
  Allows the users to do the changes to the database during a preview session. By default, database changes are not allowed during a preview session.

Assign Roles to an Access Type

Assign the roles to an access type to limit contacts access to functional areas for assigned roles.

Follow these steps:

1. Select the following roles for this access type:
   - **Reporting Role**
     Defines the reporting access for this type.
   - **REST Web Service API Role**
     Defines the access to the REST web services for this type.
   - **SOAP Web Service API Role**
     Defines the access to the SOAP web services for this type.
   - **Command Line Utility Role**
     Defines the access to the command-line utilities for this type.
2. Click Update Roles.
   The Role Search page opens.
3. Enter any search criteria that you want to limit the list to the roles of interest, and then click Search.
   The Roles Assigned - Update page opens, listing the roles that matched the search criteria.
4. Select the roles that you want to assign. To select multiple items, hold down the CTRL key while clicking the left mouse button.
5. Click the double right-directional arrows, after you have selected all the roles that you want.
   The selected roles move to the Roles Assigned list on the right.
6. Click OK.
   The Access Type Detail page opens, with the assigned roles listed on the Roles tab.
7. Click Save.

The Access Type Detail page opens, with a confirmation message that your changes have been saved.

8. Select the role that you want to be the default for this access type upon login. Click Set Default Role.
   Your selection for the default role is saved.

- **Symbol**
  Specifies a unique identifier for the access type.
- **Default?**
  Indicates whether this access type is the default that is associated with contacts.
- **Record Status**
  Specifies whether this access type is Active or Inactive.
- **Description**
  Describes the access type. Use this field to help identify the characteristics of the access type.
- **Receive Internal Notification**
  Determines whether the contacts associated with the access type receive internal notification of activities that are related to tickets.
- **Access Level**
  Determines which access types a user can grant to another user. A user can assign an access type to the contact record of another user only if the access level of the access type they are attempting to assign is ranked the same as or lower than the grant level for their own access type. The levels are ranked as follows:
  – Admin (highest)
  – Analyst
  – Cust/Emp
  – None (lowest)
- **Licensed?**
  Determines whether this contact is a licensed access type. Contacts assigned to unlicensed access types can only view or update their own personal data.

  NOTE
  KPIs count the concurrent usages of the users from the system (for example, CA SDM Web UI, SOAP Web Services, REST Web Services, and so on). For example, the webConcurrentLicenseCt KPI counts the maximum number of unique users (with the "Licensed?" option selected) logged in to the CA SDM Web UI during the interval.

Data Partitions

Data partitions are subsets of the database with restricted access to data records. Restrict access by defining a set of constraints for each data partition. Assign these data partitions to access types. The data partition assignment determines the records that the contact can access.

Create a Data Partition Constraint

Data partition constraints restrict the database record access for users that are assigned to the data partition.

Follow these steps:

1. Select Security and Role Management, Data Partitions, Data Partition Constraints on the Administration tab.
   The Data Partition Constraints List page opens.
2. Click Create New.
   The Create New Data Partition Constraint page opens.
3. Complete the data partition constraint fields, as appropriate.
4. Complete the information in the tabs, as appropriate:
   – **Constraint**

Specifies the criteria that controls the records in the table and can be viewed, created, updated, or deleted by a user that is assigned to the data partition. For example, you can specify that users can only update issues that are assigned to them. When a user in the data partition requests a record that does not match the condition then the record is read-only.
Limit: 4000 bytes

– **SQL Translation**
Displays the constraint definition in SQL format. The condition that you entered on the Constraint tab is validated, and the underlying SQL WHERE clause is built. This translation is displayed on the SQL Translation tab for the verification.

5. Click Save.
The constraint is saved and added to the data partition.

Example: Create a Data Partition Constraint for CAB Assignments

You can create a data partition constraint that lets users update only change orders that are assigned to a CAB to which the logged in user belongs.

To create a data partition constraint for CAB change order user assignments, assign the following constraint values for a Change_Request controlled table in a data partition:

- Constraint type: Pre-Update
- Constraint specification: cab.[group]group_list.member IN (@root.id)

The logged in user can only update change orders that are assigned to a CAB to which the user belongs.

Constraint Definition

Specify the condition in Majic format (the metalanguage used to define CA SDM objects).

If the Constraint Type is View, the condition can include joins to other tables and references in the form @root.att_name to Majic attributes in the contact record for the logged-in user. Otherwise, it cannot include joins to other tables, but it can include references in the form @root.att_name to Majic attributes in the contact record for the logged-in user.

If the Constraint Type is Defaults, you may specify one or more assignment statements, separated by semicolons, which specify values to be assigned to empty fields in a new record at the time the record is stored. The syntax of each assignment statement is:

```
att_name=value
```

where att_name is the name of a Majic attribute from the record, and value can be an integer, a string enclosed in quotes, or a reference in the form @root.att_name to a Majic attribute in the contact record for the logged-in user. The way CA SDM uses default values depends on the table they affect.

For tables updated by CA SDM, such as Issues, default values are placed into the record at the time it is displayed, and are shown on the initial display of a new record. A default value can be assigned to a reference field (a Majic SREL) by coding it in the form of a persistent ID (a table name followed by a colon and an integer ID). For example, you might set a default value for category by including the following in the Defaults specification:

```
category='PCAT:12345'
```

where 'PCAT' is the target of the SREL, as shown in the Majic file, and 12345 is the ID number of the desired category. You can list persistent IDs for a table with a command of the form:

```
bop_odump domsrvr pcat "" sym
```

Data Partition Constraints Fields

Complete the following fields to add or modify the data partition constraint fields:

- **Data Partition Name**

Specifies the name of the data partition for which the constraint being defined.

- **Table Name**
  Specifies the database table that is controlled by the constraint.
- **Constraint Type**
  The type of constraint being defined. There are six constraint types for each table in a data partition.
  - **Create**

  Specifies the criteria that must be met before creating a record. When a user in the data partition attempts to create a record that does not match the create test condition, CA SDM displays the error message that is associated with the constraint and does not save the record.
  - **Defaults**

  Specifies one or more assignment statements, separated by semicolons, defining values to be assigned to empty fields in a new record at the time the record is stored. The syntax of each assignment statement is, where *att_name* is the name of a Majic attribute from the record, and *value* can be an integer, a string that is enclosed in quotes, or a reference in the form @root.*att_name* to a Majic attribute in the contact record for the current user:

  ```
  att_name=value
  ```

  For tables updated for tickets, default values are placed into the record at the time it is displayed and are shown on the initial display of a new record. You can assign a default value to a reference field (a Majic SREL) by coding it in the form of a persistent ID. A persistent ID is an object name followed by a colon and an integer ID. For example, you can set a default value for category by including the following in the defaults specification, where PCAT is the target of the SREL (as shown in the Majic file) and 12345 is the ID number of the desired category:

  ```
  category='PCAT:12345'
  ```

  You can list persistent IDs for an object using a command of the form:

  ```
  bop_odump domsrvr pcat "" sym
  ```
  - **Delete**

  Specifies the criteria that must be met to delete a record. When a user in the data partition attempts to delete a record that does not match the delete condition, CA SDM displays the error message that is associated with the constraint and does not delete the record.
  - **Pre-Update**

  Specifies the records in the controlled table that a user can update in the data partition. When a user in the data partition requests a record that does not match the pre-update condition, CA SDM makes the record read-only and displays the error message that is defined with the constraint.
  - **Update**

  Specifies the criteria that must be met when a record is saved. When a user in the data partition attempts to save a record that does not match the update condition, CA SDM displays the error message that is associated with the constraint and does not save the record.
  - **View**

  Specifies the records in the controlled table that a user can view in the data partition. This constraint is automatically applied to all lists selected by a user in this data partition, in addition to any selection criteria explicitly specified by the user.
  View can include joins to other tables and references in the form @root.*att_name* to Majic attributes in the contact record for the current or logged-in user. The valid examples are:

  ```
  requestor.organization = @root.organization
  requestor.organization.name = 'MIS'
  assignee = @root.id
  assignee.organization = @root.organization
  ```

  NOTE
  The Create, Delete, Pre-Update, and Update constraint types now support joins to other tables. They can also include references in the form @root.attribute to attributes in the contact record for the current user.
- **Record Status**

Indicates whether the constraint is active or inactive.

- **Error Message**
  Specifies the message returned to the user, if the constraint criteria is not met. For example, "You can only update issues assigned to you" or, "You can only create issues for your organization" or, "You can update your contact record but cannot change the data partition."

Create a Data Partition

A data partition is a subset of a CA SDM database. Data partition controls a user access to tickets and other data records based on their content.

Follow these steps:

1. Select Security and Role Management, Data Partitions, Data Partitions List on the Administration tab.
2. Click Create New.
3. Complete the fields as appropriate:
   - **Data Partition**
     A unique identifier for the data partition.
   - **Record Status**
     Record status whether the partition is active or inactive.
4. Click Save.
5. Click New Constraint and attach constraint definitions to the partition.
6. Click Save.
   The data partition is saved with the data partition constraint.

View Data Partitions Controlled Tables

Controlled tables are tables to which you can restrict user access through partitioning.

> **NOTE**
>
> You cannot add or delete controlled tables or change their object names.

Follow these steps:

1. Select Security and Role Management, Data Partitions, Data Partitions Controlled Tables on the Administration tab.
   The Controlled Tables list appears.
2. (Optional) Click Show Filter and complete one or more of the search fields.
3. Select the name of the table you want to view.
   The Controlled Table Detail page appears.

Data Partition Controlled Tables Search Fields

- **Table Name** -- The database table controlled by the constraint.
- **Object** -- The database object related to the controlled table.
- **Description** -- A text description of the controlled table.
- **Record Status** -- Indicates whether the constraint is active or inactive.

How to Perform Proactive Monitoring for CA SDM

The steps outlined in this article are the steps recommended for proactive monitoring actions for CA Service Desk Manager.

WARNING

Clean up the NX_ROOT\log directory so that the CA Support Diagnostics utility only sends what it has to. Do not retain old dump, trace or other files in the NX_ROOT/log directory. Do not create backup folders in NX_ROOT/log directory.

Perform the following to proactively monitor CA SDM Servers:

1. a. Install the Microsoft PSLIST utility on all CA SDM servers.
   When invoking PSLIST from within a batch file, please invoke it in the following manner to ensure that it generates an easy to read CPU and memory output:

   ```
   'pslist -m' and 'pslist -s 2'
   ```
2. Install the performance logging script from CA Support on all CA SDM servers and have it running at all times.
   For more information, see TEC473185 for further details
3. Using a performance monitoring utility, monitor the main CA SDM processes in real time and, if possible, also from a historical perspective.
   a. Monitor the CPU, Memory and Thread count metrics for the following processes:
      - Domsrvr
      - Javaw
      - Pdm_proctor_nxd
      - Pdm_tomcat_nxd
      - Spel_srvr
      - Webengine
      - Boplgin.exe
      - Bpvirtdb_srvr.exe

   b. If the monitoring utility can set baselines on the above metrics, allow it to do so.
      Use the baseline metrics to set threshold levels. When the thresholds are exceeded, have alarms triggered, appropriate staff notified, and if possible dynamically invoke a script that runs the CA Support Diagnostics tool which will gather the performance logging files.
4. Monitor the CA SDM service on all CA SDM servers.
   When the CA SDM service is down, dynamically invoke a script that runs the CA Support Diagnostics tool which will gather the performance logging files.
5. Set up SQL profiler trace logging on the SQL server where the MDB is hosted.
   For more information, see Microsoft Documentation for SQL Server Profiler.
6. Monitor the MDB for long running queries and any queries which are returning a large number of rows.
7. Create a script/batch file to copy the **STDLOGs** from the *NX_ROOT\log directory* to another location and schedule this script/batch file to run every day around midnight.
   Ensure that the script creates a unique name for the copied **STDLOGs** that contains the date they were copied on. This will ensure that if a problem occurs and CA Support asks for the STDLOGs for a certain time period, you can provide the copied STDLOGs.

   NOTE

   As far as database maintenance is concerned, database administrators follow their own methods to perform re-indexing and other maintenance tasks. Hence, it is recommended that when a large amount of data is loaded into the MDB (that is, pdm_load), you execute a re-index prior to loading data into the MDB. Ensure that there is enough disk space available for database growth. In addition, a database re-indexing can be helpful when SQL performance issues are seen on the MDB database. The drawback to database re-indexing is that it does have a tendency to cause deadlocked tables when it is executed while the CA SDM is up and active. Therefore, if you plan to run database re-indexing or other database maintenance tasks on

a scheduled basis, you must shutdown the CA SDM service on the CA SDM server(s) prior to performing these tasks.

How to Configure Multiple CA SDM Web Engines on Application Servers?

Overview

In CA SDM Advanced Availability architecture, you normally only one webengine/domsrvr pair on each Application server. However, in certain CA SDM environments, you may want to have more than one webengine/domsrvr pair for each of your Applicatiton servers, This will help in distributing the server load without the need to configure additional hardware or virtual machines.

F5 load balancers can only distribute the load across servers, but not across webengines. The only way to distribute a load across webengines is to use a web director within CA SDM itself. Typically in an advanced availability environment, if you want to use more than one webengine/domsrvr pair on your application servers behind an F5 Load Balancer , you will distribute the F5 load balancers across servers, and then have a web director on each application servers to distribute that load across the web engines on that specific server. To do this, you must configure the CA SDM systems as following:

Example Use Case:

Consider that your CA SDM environment has the following server Configuration:

- Background Server
- Standy Server
- 3 Application Servers (APP1, APP2 and APP3)
- F5 load balancer dividing the load among APP1, APP2 and APP3
- One domsrvr/webengine pair on each APP server

Given the above setup configuration, your goal is to distribute, let us say, 4-domsrvr/webengine pairs on each APP server. Navigate to Administration tab, System, Configurations, and edit the configuration that you are using by performing the following on each APP server:

1. Create a Web Director.
2. Edit the existing webengine (the original one) and set it to use the new web director on that specific server.
3. Add 2 more Object Managers.
4. Add 2 more Web Engines, setting each one to use its own appropriate domsrvr on that specific server. Also, configure the web engines to use the Web Director you created in step-1.
5. Save the configuration.
6. Run pdm_configure on the BG (you will need to fail over to the SB first), SB (you will need to fail back to the original BG first), APP1, APP2 and APP3 servers in that order.
7. Point the F5 Load Balancer to distribute the load between the 3 Web Directors on the APP servers.

The configuration will look like this visually:

APP1

- WebDirector1
- Domsrvr1
- Domsrvr2
- Domsrvr3
- Webengine1 (Behind WebDirector1) (Set to use Domsrvr1)
- Webengine2 (Behind WebDirector2) (Set to use Domsrvr2)
- Webengine3 (Behind WebDirector2) (Set to use Domsrvr3)

APP2

- WebDirector1
- Domsrvr1
- Domsrvr2
- Domsrvr3
- Webengine1 (Behind WebDirector1) (Set to use Domsrvr1)
- Webengine2 (Behind WebDirector2) (Set to use Domsrvr2)
- Webengine3 (Behind WebDirector2) (Set to use Domsrvr3)

APP3

- WebDirector1
- Domsrvr1
- Domsrvr2
- Domsrvr3
- Webengine1 (Behind WebDirector1) (Set to use Domsrvr1)
- Webengine2 (Behind WebDirector2) (Set to use Domsrvr2)
- Webengine3 (Behind WebDirector2) (Set to use Domsrvr3)

F5 Load Balancer configuration will be as follows:

- APP1 →WebDirector
  - Webengine1
  - Webengine2
  - Webengine2
- APP2→WebDirector
  - Webengine1
  - Webengine2
  - Webengine2
- APP3 →WebDirector
  - Webengine1
  - Webengine2
  - Webengine2

Administering CA Service Desk Manager

The Administrator role has full access to all product functionality. This role is used typically when implementing the product to help ensure that all users and roles are set up correctly. The Administrator role can also be used for a product environment that has a single person who performs all administration tasks.

- Options Manager
- Multi-Tenancy
- Create a Remote Reference
- Audit Log List
- Define Form Groups
- SOAP Web Services Policy
- Manage Service Type and Service Type Events
- Create a Service Target Template
- Create Log Interval Configuration
- How to Manage Contact Groups
- CA SDM Environment Promotion
- Web Services for CA SDM - Best Practices and Recommendations
- Macros and Events
- Tags Used with Maileater
- License Usage
- CA CMDB Administration
- System
- Enable SAML Authentication for CA SDM

Options Manager

Contents

The Options Manager lets you modify the functionality of your system. You can easily install and uninstall specific modified options.

> **NOTE**
> In advanced availability configuration, you can install or uninstall an option only from the background server. You can view the options and the descriptions from the application server.

The Options Manager pages list the applications that can be modified, the options that apply to those applications, and the values that are associated with the options, if any. They also provide descriptions of each option and indicate the status of each option (installed or uninstalled).

You can install these customization options when you install the rest of the product, or you can install any or all the options later. (Some options are automatically installed when you install the application that uses them.) You can also change the values of installed options at any time.

> **WARNING**
> Depending on your role, you may not have access to all the options

> **WARNING**
>
> Installing or uninstalling an option requires you to restart the CA SDM services and xFlow Interface service. For advanced availability configuration, not all options require you to restart all the servers.

Archive and Purge Options

Option	Description
busy_agent_threshold	Controls the threshold for processing the archive and purge background requests.For efficiency, requests in the background process queue should be processed during low system utilization ("non-busy" times). The Option Value field allows you to specify the number of the busy agents in a percentage. For example, if the Option Value is set to 70, the system starts processing requests in the background process queue when the percentage of busy agents is lower than 70 percent of the total agents in the system. The Option Value field sets the NX_BUSY_AGENT_THRESHOLD variable, which is located in the NX.env file.
default_schedule	The Option Value specifies the default schedule (workshift entry) used by the archive and purge rules. If the Option Value is set to an invalid workshift entry name or empty string, the rule does not execute the rule with the schedule value. The Option Value field sets the NX_DEFAULT_SCHEDULE variable, which is located in the NX.env file.
rule_history_length	The Option Value specifies the maximum number of rule history records saved in the database. The Option Value field is retrieved by archive and purge process at the startup. The default value for this variable is 50, which means 50 of the most recent rule history records are saved. The Option Value field sets the NX_RULE_HISTORY_LENGTH variable, which is located in the NX.env file.

Asset Information Service Options

Option	Description
allow_exceed	An asset/item contains an Inventory Count field. The number of parents an asset/item can have cannot exceed the inventory count. Installing this option allows the number of parents to exceed the inventory count.
allow_unrestricted_asset_upd	If installed, users can fully modify assets/configuration items owned by CA Asset Portfolio Management. If this option is not installed (the default), only a limited number of fields of the asset/item record can be updated.

Audit Log Options

These options control the logging of activities on the data partitions, change orders/issues, and requests. If these options are installed, an audit log entry is created for each field update and insert operation. The operation to be monitored is determined using the following options:

Option	Description
audit_ins	Monitors insertion.
audit_upd	Monitors field updates.

CA CMDB Options

Option	Description
allow_updates_to_inactive_ci	Specifies whether to permit inactive CIs to be updated. Default is NO (new installation) or YES (product upgrade). To uninstall this option, click Edit, Deinstall, Refresh. Then stop and restart the CA SDM server.
allow_updates_to_superseded_ci	Specifies whether to permit superseded CIs to be updated. Default is NO. To uninstall this option, click Edit, Deinstall, Refresh. Then stop and restart the CA SDM server.
ci_filter	Specifies whether to search for CIs in the CI List page. Default is YES. If the option is not installed, the CI search filter has no initial value. To uninstall this option, click Edit, Deinstall, Refresh. Then stop and restart the CA SDM server.
ci_hierarchy_levels	Specifies the number for levels (depth) to search for dependent services for CIs. Valid values are integers 0 (zero) through 9 (nine).
cmdb_model_class_inconsistency	Specifies to prevent, warn, or allow a CI class from being inconsistent with the class that a user specified in the optional CI model. • **Prevent** Stops the CI save from occurring and displays an error message. • **Allow** Allows the class and the class of the model to remain inconsistent. **Note:** Use this option for compatibility with earlier releases of the product, such as CA SDM r12.6. • **Warn** Updates the standard log (stdlog) with a warning and lets the save occur. For example, use this option if you want to transition from Allow to Prevent. **Default:** Prevent
cmdb_versioning_maxrows	Specifies the maximum number of audit log and Change Order history entries that displays in the Versioning tab. **Default:** 400
critical_service_relationship	Specifies a relationship type to identify dependent services for CIs. You can specify multiple relationships separated by a comma. For example, supports, is managed by.

CA Process Automation Workflow Options

Option	Description
caextwf_eem_hostname	Specifies the name of the CA EEM server. For example, *http://<workflow_hostname>* identifies the authentication host. You install caextwf_eem_hostname only if you configured CA Process Automation to use CA EEM as an authentication server. CA SDM uses this value to transform a user name and password into a CA EEM token. Then, the user name and password do not pass in plain text over HTTP. If the CA Process Automation installation is not using CA EEM, do not place a value in the caextwf_eem_hostname option, and do not install the caextwf_eem_hostname. Placing a false value or installing caextwf_eem_hostname when it is not necessary causes the integration to fail.
caextwf_endpoint	Specifies the URL that points to the CA Process Automation web services by including the CA Process Automation host name, port, and the mandatory /itpam/soap path. For example, *http://<workflow_hostname>:<workflow_tomcat_port>/itpam/soap* identifies the endpoint. If your implementation uses CA EEM, installing the caextwf_eem_hostname option is required for the integration between CA Process Automation and CA SDM to operate properly.
caextwf_log_categories	Specifies a comma separated list of CA Process Automation process instance log category names to appear on the CA SDM Request, Change Order, and Issue Workflow Tasks tab. For example, *Operator,Response,MyOwnCategory* supplies three log categories. You install caextwf_log_categories based on business decisions from the CA SDM and CA Process Automation process design personnel. This option adjusts the default data that appears on the Workflow Tasks tab for requests, change orders, and issues. When you install the caextwf_log_categories option, all CA Process Automation process instance log messages from the Process category and the categories that you specify appear on the Workflow Tasks tab. When you do not install caextwf_log_categories, only the CA Process Automation process instance log messages from the Process category appear on the Workflow Tasks tab. For information about the CA Process Automation predefined log message categories, and defining custom message categories, see the CA Process Automation reference documentation.

caextwf_processdisplay_url	Specifies how to launch a graphical snapshot of a CA Process Automation process instance by supplying the host name and the mandatory /itpam/Web.jsp?page=runtimeeditor&ROID= path. For example, *http://<workflow_hostname>:<workflow_tomcat_port>/itpam/Web.jsp?page=runtimeeditor&ROID=* launches a snapshot of a process instance. On the Workflow Tasks tab of a request, change order or issue, the user selects View Process to see the snapshot. Installing the caextwf_processdisplay_url option is required for the integration between CA Process Automation and CA SDM to operate appropriately.
caextwf_retry_count	Specifies the number of times the CA PAM attached Workflow Events can be re-triggered to retry the Unknown status events. Default retry count value is 3 and range [1-20].
caextwf_retry_interval	Specifies the retry interval duration in minutes to re-trigger each unknown CA PAM Workflow events.Default retry interval duration is 10 and range [10-999].
caextwf_worklist_url	Specifies the process instance path by supplying the host name and the mandatory /itpam?page=tasklist path. For example, *http://<workflow_hostname>:<workflow_tomcat_port>/itpam?page=tasklist* enables CA SDM users to see a list of CA Process Automation process instances that require attention. The list appears in CA Process Automation when the CA SDM user selects a link associated with any listed task in the request, change order, or issue Workflow Tasks tab. Installing the caextwf_worklist_url option is required for the integration between CA Process Automation and CA SDM to operate properly.
caextwf_ws_password	Specifies the administrative password associated with the CA Process Automation user name from the caextwf_ws_user option. CA SDM uses the user name and password to access the CA Process Automation web service functions to perform integration activities such as selecting start request forms, process definition information, and process instance information. Installing the caextwf_ws_password option is required for the integration between CA Process Automation and CA SDM. The password and user name that you specify requires the appropriate access to CA Process Automation. However, it is not necessary for CA SDM to use the user name and password for authentication.

caextwf_ws_user	Specifies the CA Process Automation administrative user name associated with the CA Process Automation user name from the caextwf_ws_password option. CA SDM uses the user name and password to access the CA Process Automation web service functions to perform integration activities. Integration activities include selecting start request forms, selecting process definition information, selecting process instance information, or launching process instances. Installing the caextwf_ws_user option is required for the integration between CA Process Automation and CA SDM to operate. The user name and password that you specify requires the appropriate access to CA Process Automation. However, it is not necessary for CA SDM to use the user name and password for authentication.

How to Configure SLAs

CA Service Catalog Options

To enable the self-service integration, install the following options:

Option	Description
casc_aty_sync	Specifies the type of activity logs to be synchronized from CA SDM to CA Service Catalog. If this field is left blank, Log Comment activity logs are synchronized. To synchronize other activity log types, (such as, Escalate, Update Status) enter the respective log type codes, separated by comma.
casc_endpoint	Specifies the CA Service Catalog Web Services URL. **Format:** http://<CA_Service_ Catalog_hostname>:<CA_Service_Catalog_portnum>/usm/ services
casc_session_timeout	(Optional) Specifies the time in minutes for which the CA Service Catalog web services session will be cached. **Range:** 20 to 60 mins.
casc_user	Specifies the CA Service Catalog administrator username that is responsible for making the web service calls to CA Service Catalog. This user must exist in CA EEM against which CA SDM is configured.
casc_user_password	Specifies the password of the CA Service Catalog user as entered in the casc_user option.

casc_ws_retry	(Optional) Specifies the number of retries to synchronize a CA SDM ticket update (activity logs/ attachments/status) with CA Service Catalog, if there is a failure. Enter -1 to try indefinitely, until the synchronization succeeds. As a system administrator you can define the time interval after which you want the failed ticket update to be retried again. Run the following command to define the time interval: pdm_options_mgr -c -a pdm_option.inst -a option.inst -s NX_CASC_RETRY_INTERVAL -v [time interval in minutes] By default this time interval is set to 15 minutes. If you do not set any value or provide a non-numeric value to this variable, the failed update is retried according to the default time interval (15 mins.). **Range:** -1 to 50
casc_integrated	Enables the self-service integration.

Call Service Desk Options

Option	Description
call_service_desk_default_phone_number	Install this option and specify the value as a phone number for the mobile users to call the Service Desk using this specified number from their mobile.
call_service_desk_mobile_enable	(Installed by default) Enables the mobile users to call the Service Desk number from their mobile. The Service Desk number can be specified using the **call_service_desk_default_phone_number** or **call_service_desk_tenant_phone_number_field** (in case of multi-tenancy) option.
call_service_desk_tenant_phone_number_field	(Applicable if multi-tenancy is enabled) Install this option and set the value as phone_number or alt_phone, as specified in the tenant detail page. This option enables the tenant mobile users to call Service Desk using this specified number from their mobile.

Change Order Mgr Options

Option	Description
asset_only_one_chg	Restricts Assets from being attached to more than one active change order.
Category_Defaults	Automatically fills certain fields on a change order detail page when the user selects a category. A category can have a default type, assignee, group, and organization/business. When this option is activated, the corresponding fields on the change order detail page are populated with category defaults if they have not been manually filled by the user. If the values are entered by the user, the default rule does not apply.

chg_allow_sla_downgrade	Alters the behavior of the chg_sla option by allowing the system to automatically downgrade a change order's Service Type. The chg_sla option selects the best Service Type from among several change order attributes, but cannot replace the change order's current Service Type with one of lesser rank. If this option is installed, the Service Types for all affected attributes are evaluated whenever one of the attributes changes. The change order's Service Type is set to highest ranked Type found, even if the new Service Type is lesser in rank to the change order's current Service Type. The Service Type with the smallest Rank value is considered the best service. If all the Service Types considered are equal in Rank (including Service Types with empty Rank values), the Service Type created first in the database is selected. The chg_sla option must be installed for this option to function correctly.
Auto Events	Automatically attaches events to newly created change orders. This attachment is specified using the following options: • **chg_auto_events** -- Installs or uninstalls the option itself. • **chg_auto_events_name** -- Specifies the format of a name for the attached events. • **chg_auto_events_count** -- Specifies the number of events attached. Event names to be attached are constructed by joining the chg_auto_events_name to a number ranging from 1 to chg_auto_events_count. For example: • chg_auto_events_name is "Auto Event" • chg_auto_events_count is "3" With these settings, installing the chg_auto_events option would cause the following events to be attached to all new change orders: • Auto Event 1 • Auto Event 2 • Auto Event 3 To complete the implementation of this option, you must define an event with these name.
chg_calendar_start_day_of_week	Specifies the start day of the week in the Change Calender. **Default:** Friday
chg_sla	Installs the rule for defaulting the Service Type field of a change order. The rule is as follows: • The best service among the priority, requester, affected contact, and category is applied to the change order. • The service type with the smallest rank value is considered the best service. If all the service types considered are equal in rank (including service types with empty rank values), the service type created first in the database is selected. This option is only available if the Classic SLA Processing Option is installed.
conflict_added	Generates Activity Log entries when conflicts are added.

conflict_updated	Generates Activity Log entries when conflicts are updated.
force_closure_code	Modifies the closure code value of children to be identical to the parent. The following values are valid: • **Empty** - (Default) Forces the closure code value of the parent to the children when the child does not have a value for the closure code. It does not overwrite an existing closure code. • **All** - Forces the closure code value of the parent to all children even if they already have a closure code value.
impact_explorer_exclude_hier	Used when building a list of child CIs for a CI in the Impact Explorer tree. If this option is installed and set to Yes, child CIs that are related through the CMDB Relationships tab of a CI Detail page are *not* displayed in the Impact Explorer tree or in the CI Descendants List. **Note:** Child CIs that are related through CA CMDB are displayed.
require_change_assignee	Prohibits change orders with no value in the Assignee field from being saved.
require_change_group	Prohibits change orders with no value in the Group field from being saved.
require_closure_code	Requires a closure code value prior to closing or resolving a change order.
nx_number_of_windows_to_show	Number of Blackout and Maintenance windows to be shown on the web interface. Default value: 10 Allowed Values: All (to display all the window names) or any number.
nx_max_bw_title_length	Maximum allowed length to display the Blackout or Maintenance window name and description. Default value: 256

Change-Issue Options

- **edit_completed_tasks**
  When this option is installed with a value of Yes, you are allowed to set both the "Allow Task Update" and the "Task Completed" flags on a Change Order or Issue Task Status detail. However, if you set both the "Allow Task Update" and the "Task Completed" flags, you can only update the Workflow Task Description. The default behavior is to not allow setting of both the "Allow Task Update" and the "Task Completed" flags on a Task Status detail.

Email Options

The email interface sends email notifications and lets users create tickets from an email.

Option	Description
mail_from_address	Specifies the mail notification From: address. The address is in the format Displayname<user@company.com>.
mail_login_password	Specifies the SMTP server login password.
mail_login_userid	Specifies the SMTP server login userid.
mail_max_threads	Specifies the maximum number of concurrent SMTP connections that can attempt to communicate with the server.

mail_reply_to_address	Defines the reply to address for email notifications. This option is useful if emails are sent from one user account, but you want replies sent to another email address. The default value is the same as the from address.
mail_smtp_domain_name	Defines the domain name of the SMTP server. You can clear the domain name by setting the value to NONE.
mail_smtp_hosts	Specifies a space-separated list of SMTP server host names for email notifications.
mail_smtp_host_port	Specifies an SMTP port to override the default SMTP port.
mail_smtp_security_level	Specifies the SMTP security level. Valid settings are: **0**=no security, **1**=basic authentication, **2**=NTLM, **3**=MD5 and **4**=LOGIN, **5**=OAuth 2.0 (Click **Save**, the OAUTH Tab is displayed at the bottom. For more information about how to set the Security Level to OAuth 2.0, see Set the Security Level to OAuth 2.0 for SMTP. If you set this option to **1**, set the mail_login_password and mail_login_userid options. Most SMTP servers do not require authentication.
mail_smtp_use_tls	Specifies the Transport Layer Security (TLS) usage in the email. The valid settings are Yes= Use TLS, No=Do not use TLS.
mail_ca_cert_path	Specifies the path where the trusted certificate has been deployed. For the advanced availability configuration, you must deploy the trusted certificate on the same location in all the CA SDM servers. CA SDM supports only Base-64 encoded (PEM) format for CA Certificates.
mail_show_to_cc_list	Shows all the recipients of the email notification. For manual email notification, both “To” and “Cc” list recipients are displayed and for automatic email notification, only “To” list recipients are displayed. By default, this option is installed. **Note:** Pager email notification method does not support the listing of all recipients in To or CC list.

General Options

Option	Description
add_utf8_byte_order_mark	Causes the default data to include an encoding signature (UTF-8 Byte Order Mark). Editors that support UTF-8 use this signature to display and maintain UTF-8 encoding.
catalog_server	Specifies the CA Service Catalog server location (for example: http[s]://*servername*:8090). If a load balancer is in use, verify that this option specifies the load balancer.
default_superuser_id	Specifies the default superuser ID for CA SDM, such as ServiceDesk.

Status_Policy_Violations	Specifies how strictly the system enforces **Status** policies. This option only applies to automated system processes, such as integrations and macros. The end user is unaffected by this option. Valid option values are: • **Warn** -- Allows the status policy violation to proceed, but a warning message is written to the log. • **Allow** -- Allows the status policy violation to proceed, but *no* warning message is written to the log. • **Reject** -- Rejects the status policy violation and an error message appears.
dal_domsrvr	Comma separated list of domservers to be used by CABI/Jasper JDBC Driver
dal_whitelist	Whitelisted hostname(s) for JDBC Driver. By default, the integrated CABI hostname is whitelisted when integration is done using the common installer.
xflow_installed	Enable this option if the xFlow Interface is installed.
singleton_notify_list	CA SDM sends the singleton daemon failure notification when it fails to restart a failed singleton daemon. This option controls the list of email recipients that will be notified when a singleton daemon fails to restart on the CA SDM server. Some fields in this page are self-explanatory. Following are the fields that need explanation: **Action Status** Indicates the installation status of the option. • **Option Value** Specifies a list of comma-separated email addresses for sending the singleton daemon failure email. • **Action Status Message** Specifies the message text to be included in the singleton daemon failure email.

Heat-Weather

Option	Description
heat_reindexing_interval	The Heat Reindexing time interval in hours. Sets with default value 6 hours,if not installed. Range: 3 - 72 hours.
heat_weather	This option is to Enable/Disable the Heat & Weather feature on xFlow Analyst Interface. off - Feature is Turned Off. Heat - Only Heat is available. Heat & Weather - Both Heat and Weather are available
heat_weather_domsrvr	Slump name of domsrvr used by Heat and Weather Daemon. Must be a string of the form domsrvr:xxx, where xxx can be any value. By default, Heat and Weather Daemon uses "domsrvr".
on_access_heat_calculation	This option is to enable or disable the On access Heat Calculation.
weather_reindexing_interval	The Weather Reindexing time interval in minutes. Sets with default value 60 minutes,if not installed. Range: 30 - 720 minutes.

Issue Mgr Options

Option	Description
iss_allow_sla_downgrade	Alters the behavior of the Iss SLA Option by allowing the system to automatically downgrade an issue's Service Type. The Iss SLA Option selects the best Service Type from among several issue attributes, but cannot replace the issue's current Service Type with a type of lesser rank. If this option is installed, the Service Types for all affected attributes are evaluated whenever one of the attributes changes. The issue's Service Type is set to the highest ranked type found, even if the new Service Type is lesser in rank than the issue's current Service Type. The Service Type with the smallest Rank value is considered the best service. If all the Service Types considered are equal in Rank (including Service Types with empty Rank values), the Service Type created first in the database is selected. The Iss SLA Option must be installed for this option to function correctly.
iss_assignee_set	Sets the assignee for issues to that of the logged in user, if the user is an analyst. If the option is installed, when an analyst creates a new issue, it is automatically assigned to that analyst.
Issue Auto Events	• Automatically attach events to newly created issues. This is specified using three options: — **iss_auto_events** -- Installs or uninstalls the option itself. — **iss_auto_events_count** -- Specifies the number of events attached. — **iss_auto_events_name** -- Specifies the format of a name for the attached events. Event names to be attached are constructed by joining the iss_auto_events_name to a number ranging from 1 to iss_auto_events_count. For example: • iss_auto_events_name is "Auto Event" — iss_auto_events_count is "3" With these settings, installing the iss_auto_events option would cause the following events to be attached to all new issues: • Auto Event 1 — Auto Event 2 — Auto Event 3 To complete the implementation of this option, you must define events with these names.
Iss_Category_Defaults	Determines what happens when the category field is specified on an issue detail page. A category can have a default assignee, group, and organization/ business. When this option is activated, the corresponding fields on the issue detail page are populated with the category defaults, if they have not been manually entered by the user. If these values are filled by the user, the defaulting rule does not apply. Make sure that the default assignee is actually a member of the default group.

iss_sla	Installs the rule for defaulting the Service Type field of an issue. The rule is as follows: The best service among the priority, requester, affected contact, and category is applied to the issue. The service type with the smallest rank value is considered the best service. If all the service types considered are equal in rank (including service types with empty rank values), the service type created first in the database is selected. This option is only available if the Classic SLA Processing Option is installed.
require_issue_assignee	Prohibits issues with no value in the Assignee field from being saved.
require_issue_group	Prohibits issues with no value in the Group field from being saved.

Knowledge Options

This option controls the Knowledge functionality:

Option	Description
analyst_preferred_ticket	Sets the link for opening a ticket from a Knowledge Document for the analyst user interface. The analyst has a default setting of Incident, and the available options are Incident, Request, IncidentandRequest, and Issue.
kt_disallow_forums	If this option is set to Yes, the Forum feature is disallowed (inactive), which means you cannot create or update forums or view any menu items.
kt_report_card_issue_status	Allow issue statistics to be calculated and displayed in the Reuse section of the Knowledge Report Card. The validity of the where clause is not checked. **Example:** status='CL'
kt_report_card_request_status	Allow request statistics to be calculated and displayed in the Reuse section of the Knowledge Report Card. The validity of the where clause is not checked. **Example:** status='CL'

KPI Options

These options control the operation of the KPI daemon:

Option	Description
kpi_ticket_data_table	Enables the KPI daemon to collect data whenever CA SDM tickets are opened, closed, or certain fields are modified. Ticket data is written into the usp_kpi_ticket_data database table, and is available for generating web-based reports. Enabling this feature may result in degraded CA SDM performance.

concurrent_lic_refresh_interval	Specifies the concurrent refresh time interval (in seconds) for all the concurrent KPIs (for example, webConcurrentLicenseCt KPI). If this option is installed, all the concurrent KPIs calculate the count only after the specified interval. If you do not install this option, refresh interval time is defaulted to 3600 seconds for all the concurrent KPIs.**Range:** 600 to 7200 seconds.

LDAP Options

You can set the domain name for LDAP Servers that was configured through Options Manager, by adding the domain name in the NX.env file variable and executing the following commands:

```
pdm_options_mgr -c -a pdm_option.inst -s NX_LDAP_DOMAIN -v <DEFAULT_LDAP_SERVER_DOMAIN_NAME> -t
pdm_options_mgr -c -a pdm_option.inst -s NX_LDAP_DOMAIN -v <DEFAULT_LDAP_SERVER_DOMAIN_NAME>
```

NOTE

Restart the services and login to CA SDM with domain\username credentials.

The following LDAP options are available for installation from Options Manager on the CA SDM Administration tab.

NOTE
The options identified as required must be installed together to enable integration with an LDAP directory. The options identified as optional are features you can add only if the required options are installed.

Option	Description
default_ldap_tenant (required)	Specifies the default tenant for contacts imported from an LDAP directory into an installation that is configured for a multi-tenancy environment. The option value must set to the UUID for that tenant. You can get the tenant UUID from a database query. For example, "SELECT * FROM ca_tenant". **Important!** This option is only required if multi-tenancy is enabled. This option only takes effect if Option "multi_tenancy" is set to "on". Before you run the pdm_buildtenant utility, and if you want to retain the tenant value, you *must* modify NX.env by adding the NX_RETAIN_TENANT_VALUE variable manua it to "yes". If multi-tenancy is set to "on" and the variable is set to "no", missing, or not set properly, then the utility overwrites the tenant information. The variable has no effect when multi-tenancy is in "setup" mode.
ldap_dn	Specifies the LDAP distinguishedName for logging in to the LDAP server. **Example**: CN=Joe, CN=Users, DC=KLAND, DC=AD, DC=com Depending on your site's network configuration, the userid *may* be used instead of a distinguishedName. **Note:** If the LDAP server supports anonymous binds, this value can be empty. **Note:** For example, if the LDAP directory is the CA EEM identity store, this option is not required because CA EEM allows anonymous access.

ldap_enable	Specifies whether LDAP integration is enabled. The default value is Yes. In addition to this option, you install all other LDAP options indicated as required.
ldap_enable_auto (optional)	Specifies whether automatic creation of contacts from LDAP information is enabled. The default value is Yes. If you install this option, a contact is automatically created from LDAP information whenever a new user logs in.
ldap_enable_groups (optional)	Specifies whether CA SDM assigns a contact's Access Type based on LDAP Group membership. The default value is Yes. To use this feature, you associate CA SDM Access Types with LDAP Groups. This option is only applicable for Microsoft Active Directory.
ldap_enable_tls (optional)	Specifies whether Transport Layer Security (TLS) is enabled during LDAP processing. The default option value is Yes.
ldap_group_object_class	Specifies the value of the LDAP objectClass attribute. The default value is *group*. This value is always included in the where clause of the automatically generated filters used to search for LDAP groups.
ldap_host	Specifies the LDAP server hostname or IP address. If the LDAP directory is the CA EEM identity store, use the hostname of the machine where Ingres is installed.
ldap_port	Specifies the LDAP server port number. If the LDAP directory is the CA EEM identity store, use 1684 for the port number.
ldap_pwd	Specifies the password for the ldap_dn for logging in to the LDAP server. **Note:** If the LDAP server supports anonymous binds, this value can be empty. **Note:** For example, if the LDAP directory is the CA EEM identity store, this option is not required because CA EEM allows for anonymous access.
ldap_search_base (required)	**Note:** If you want to search contacts for a default LDAP server, keep the domain name field in the LDAP Directory Search page as blank or provide the exact LDAP domain name configured in CA SDM to retrieve the contacts in the specified LDAP directory. Specifies the starting point for searches in the LDAP schema tree: • **UNIX** -- You *must* specify a starting container, such as: CN=Users, DC=KLAND, DC=AD, DC=com • **Windows** -- You do not have to specify a container and may start at the top of the schema tree, such as: DC=KLAND, DC=AD, DC=com If the LDAP directory is the CA EEM identity store, then use: cn=Users,cn=Entities,cn=iTechPoz This is only applicable when CA EEM is configured to use the MDB rather than an external directory.

ldap_service_type (optional)	Install this option to establish the LDAP service type. If the LDAP type is Active Directory, specify the string “Active Directory”. If the LDAP type is not Active Directory, specify any other string, for example, “eTrust” or “Novell”.
ldap_sync_on_null (optional)	Specifies whether existing contact attribute values are overwritten with null data if the corresponding LDAP user attribute contains a null value. The default value is Yes.
ldap_user_object_class (required)	Sets the value of the LDAP objectClass attribute. The default value is *person*. This value is always included in the where clause of the automatically generated filters used to search for LDAP users. If the LDAP directory is the CA EEM identity store, use pozObject. For many non-AD LDAP stores, the correct setting is inetOrgPerson.
num_ldap_agents	Specifies the number of LDAP agents. Install this option only if you have multiple LDAP agents. The default value is 2.

Multi-Tenancy Options

Option	Description
max_tenant_depth	Specifies a limit to the depth of the tenant hierarchy. Default is four (4) levels of tenants.
multi_tenancy	Controls the multi-tenancy features. Multi-tenancy allows organizations or groups to share a single installation. The following values are available: • **off** -- Multi-tenancy is not in use. • **setup** -- Multi-tenancy is not in use, however, administrators can view and edit the tenant-related objects and attributes on the UI. • **on** -- Multi-tenancy is fully operational. You can also add enforcement level options after the "on" setting, as follows: — **warn** -- Untenanted objects can be created or updated, but an error message are written to the log. — **allow** -- Untenanted objects can be created or updated, but a warning message is written to stdlog. **Note:** Setting an enforcement level is useful during migration. It allows SLA and attached event updates to legacy untenanted data.

Notifications Options

Option	Description
add_know_list	Adds the "in-the-know" notify list to all future notifications regarding a change order, issue, or request. When this option is installed, the contacts listed in the notify list are added to all future notifications.

log_all_notify	Logs all notifications, regardless of the method used. Logging a notification creates a notification log object that records the recipient, date/time, method of notification, and the notification text. The administrator uses the **Notification History** list to view and maintain notification log objects. The notification history can also be viewed from an individual ticket. If this option is not installed, the contact receives only the notification specified in the Contact record. No record is kept of the notification attempt.
notification_allow_temp_address	Allows manual notifications to be sent to manually entered SMTP email addresses, rather than to registered contacts only.
web_cgi_url	Allows the recipient of a notification email to click a URL and view the object they were notified about. This option is also used in survey notifications. To define a URL for a user to reach the CA SDM, a reference to the web interface is needed. The reference value is the web interface URL that is used to access the CA SDM web component. The URL has the following syntax. `http://hostname/CAisd/pdmweb.exe` • **hostname** Specifies the location of your web server. (Recommended) For Advanced Availability configuration, it is strongly recommended that this option points to one of the application servers and not the background or standby server. It can also point to a Web Director or a third-party load balancer. Change the value of this option to point to: — The load balancer if you have more than one application servers. — The application server, if you have only one application server.

Request Mgr Options

Option	Description
Area_Defaults	Determines what happens when the Request Area field is specified on a Request Detail page. A Request Area can have a default assignee and group. When this option is activated, the Request Detail assignee and group are populated with the Request Area defaults.
Assignee_set	Sets the assignee for requests to that of the logged in user, when the user is an analyst. If the option is installed, when an analyst creates a request, it is assigned automatically to that analyst.

autoasg_ci_assign_lowest_agt	Enables the CI-based auto-assignment to assign an analyst from group. The analyst must have active Status as Active and Available as Yes. The tickets assigned to all of the analysts that meet this criteria are counted but the first one encountered with the least number of tickets assigned to them is assigned to the ticket. You have to install this option to use it. After installing, this option restart the CA SDM servers.
autoasg_override	Controls how Auto Assignment handles a Request that is created with an Assignee, Group, or both. Configure your system by setting this option to one of the following values: • Uses the existing Assignee, Group, or both. Auto Assignment processing does not occur if the Assignee, Group, or both was set during the creation of the Request. • Ignores the existing Assignee, Group, or both. Auto Assignment processing occurs and attempts to find an Assignee, Group, or both. The Assignee, Group, or both can be set in various ways: • Manually by the Analyst • Area Defaults and the Assignee set options • Request Templates • Uninstalling this option causes it to operate in its default mode, which is 0.
Auto Events	• Automatically attach events to newly created requests. This is specified using three options: — **auto_events --** Installs or uninstalls the option itself. — **auto_events_count --** Specifies the number of events attached. — **auto_events_name --** Specifies the format of a name for the attached events. Event names to be attached are constructed by joining the auto_events_name to a number ranging from 1 to auto_events_count. For example: • auto_events_name is "Auto Event" — auto_events_count is "3" With these settings, installing the auto_events option causes the following events to be attached to all new requests: • Auto Event 1 — Auto Event 2 — Auto Event 3 To complete the implementation of this option, define an event with these names.
clear_scratch_pad	Clears the Scratch Pad text is after creating a request. The default behavior is to apply the Scratch Pad text to all new requests.

cr_allow_sla_downgrade	Alters the behavior of the CR SLA Option by allowing the system to downgrade a request Service Type automatically. The CR SLA Option selects the best Service Type from among several request attributes, but never replace the request current Service Type with a type of lesser rank. If this option is installed, the Service Types for all affected attributes are evaluated whenever one of the attributes changes. The request Service Type is set to the highest ranked type found, even if the new Service Type is lesser in rank than the request current Service Type. The Service Type with the smallest Rank value is considered the best service. If all the Service Types considered are equal in Rank (including Service Types with empty Rank values), the Service Type created first in the database is selected. The CR SLA Option must be installed for this option to function correctly.
cr_sla	Selects the best Service Type from among several request attributes, and never replace the request current Service Type with a type of lesser rank. The Service Type with the smallest Rank value is considered the best service. If all the Service Types considered are equal in Rank (including Service Types with empty Rank values), the Service Type created first in the database is selected.
efficiency_tracking	Specifies Efficiency Tracking, which lets analysts use options to track incidents. This option displays the Show Efficiency Tracking tab on the Incident Detail page.
employee_intf_incident_support	Displays the Incident, Request, or both ticket types for users logged in under the Employee interface type. You can select one of the following Option Values: • Incident only • Request only • Incident and Request The employee_intf_incident_support option is installed during CA SDM installation and cannot be uninstalled.
force_resolution_code	Specifies how the Resolution Code field value is handled for requests and incidents when an analyst closes a request or an incident. When this option is installed, select one of the following values: **Empty** -- The resolution code value of the parent is forced to the children only when the child does not have a value for the resolution code field. The option does not overwrite an existing resolution code value. This is the default value. **All** -- The resolution code value of the parent is forced to all children even if they already have a value for the resolution code field. The option overwrites the current value with the value of the parent. • If this option is not installed, the resolution code values of all children remain unchanged. • If the option is installed, the resolution code value of the parent is propagated to its children.

force_resolution_method	Specifies how the Resolution Method field value is handled for requests and incidents when an analyst closes a request or an incident. When this option is installed, select one of the following values: **Empty** -- The resolution method value of the parent is forced to the children only when the child does not have a value for the resolution method field. The option does not overwrite an existing resolution method value. This is the default value. **All --** The resolution method value of the parent is forced to all children even if they already have a value for the resolution method field. The option overwrites the current value with the value of the parent. • If this option is not installed, the resolution method values of all children remain unchanged. • If the option is installed, the resolution method value of the parent is propagated to its children.
guest_intf_incident_support	Displays the Incident, Request, or both ticket types for users logged in under the Guest interface type. You can select one of the following Option Values: • Incident only • Request only • Incident and Request The guest_intf_incident_support option is installed during CA SDM installation and cannot be uninstalled.
monitor_joins	Specifies whether to monitor joined tables for updates to dynamic lists. Updates to another table can affect the contents of a list. This effect often occurs when you use data partition view constraints containing a join. For example, a view constraint on change orders/issues where: `assignee.organization = @root.organization` results in a join from the Change_Request table to the Contact table in all change order, issue, or both lists. Keeping a change order or issue list containing this constraint up-to-date requires monitoring updates to the Contact table, and refreshing the list whenever a contact changes its organization/ business. The result can be that a large number of lists updating simultaneously after an update to a joined table, with a possible performance degradation while the update is occurring. The monitor joins option allows the list updating (and the monitoring of the joined table) to be suppressed at sites where the up-to-date accuracy of such lists is less important than the performance degradation.
Netres_pty	Escalates a request based on the associated asset. If you specify an asset with a Priority greater than the Priority on the request, the request Priority is set to match the asset. An escalation activity is not created. This option does not override a Priority that you manually specified on the Request Detail page during the same editing session.
require_incident_assignee	Prohibits incidents with no value in the Assignee field from being saved.

require_incident_group	Prohibits incidents with no value in the Group field from being saved.
require_problem_assignee	Prohibits problems with no value in the Assignee field from being saved.
require_problem_group	Prohibits problems with no value in the Group field from being saved.
require_request_assignee	Prohibits requests with no value in the Assignee field from being saved.
require_request_group	Prohibits requests with no value in the Group field from being saved.
urgency_on_employee	Specifies whether to show the Urgency field on Self Service incidents or requests. When the urgency_on_employee option is not installed, the Priority field appears on Self Service incidents or requests. When urgency_on employee option is installed, the Urgency field appears instead of the Priority field. The range of Urgency values are based on *web.cfg* settings.
use_incident_priority	Specifies whether to calculate the Incident Priority field on the Incident Detail page. When the administrator uses Web Screen Painter to add the Incident Priority field, the Incident Priority field appears on the Incident Detail page. When the use_incident_priority option is not installed, the Incident Priority value calculated is zero. When the use_incident_priority option is installed the Incident Priority calculates as the sum of the Urgency and Impact values.

Request-Change Options

Option	Description
edit_inactive	Prevents editing change orders, issues, or requests unless their status is Active. Default behavior is to allow editing of inactive tickets.
set_sla_evt_open_date	Uses the open date/time value of a change order, issue, or request as the start date/time of attached events. The attached events are triggered as soon as the ticket is saved.

Request-Change-Issue Options

Option	Description
activity_log_security	Allows or prevents the Activity Log fields from being edited. The only exception to this option's setting is the Internal field. This is always editable. The activity_log_security option is installed during installation and cannot be uninstalled.

Any_Contact	Some CA SDM objects restrict field values for Contacts based on Contact Type. This type of restriction is typically found on fields labeled, "Assignee" and only allow Contacts with a Contact Type Analyst. This option removes the contact type restrictions across the entire CA SDM application, allowing any type of contact to be used as an assignee.
classic_sla_processing	Enables the "classic" model of service types and SLA's. This model uses the Rank value for Service Types to determine the correct Type for a ticket. Only one Service Type can be applied to a ticket. The Chg SLA Option, CR SLA Option, and Iss SLA Option are available if the Classic SLA Processing Option is installed.
delete_null_properties	Determines what happens to the properties when the Change Category, Issue Category, Request Area, Incident Area or Problem Area change and you install the keep_tasks option. A saved ticket with a specified category can have a number of properties and tasks. If this option is not installed, the following occurs: • All existing properties are preserved. • Any new properties, if any, from the new category are added. If this option is installed, the following occurs: • All existing properties that contain non-empty values are preserved. • All existing properties that contain empty values are deleted. • New properties from the new category, if any, that do not have the same label value as any of the existing kept properties are added. When new properties are added to a ticket with existing properties, the system automatically adjusts (increases) the sequence numbers of the new properties to ensure they do not conflict with existing properties. The new properties always have greater sequence numbers than the existing ones.
filter_template_search	Filter the initial template search with end user's Organization.
force_rootcause	Specifies how the Root Cause field is handled for requests, change orders, and issues. • If this option is not installed, the root cause field values of all children are left unchanged. • If the option is installed, the root cause value of the parent are propagated to its children. When this option is installed, choose one of the following values: • **Empty** -- The root cause value of the parent are forced onto the children only when the child does not have a value for the root cause field. It does not overwrite an existing root cause value. This is the default. • **All** -- The root cause value of the parent is forced onto all children even if they already have a value for the root cause field. It overwrites the current value with the value of the parent.

init_list_search	Performs a search when opening the Request, Change Order, and Issue List pages to display the logged-in user's list of tickets. If this option is not installed, these list pages are blank on opening until the user performs a search.
keep_tasks	Specifies what happens if you move a ticket to a different category or area: • If keep_tasks is *not* installed, all existing properties on the ticket are removed. Any properties or tasks associated with the new category or area are added. Tasks with a status of Wait are removed. Tasks with any other status are retained. • If keep_tasks *is* installed, all existing properties and tasks on the ticket are retained. Any properties or tasks associated with the new category or area are added. When new properties or tasks are added to a ticket with existing properties or tasks, the sequence numbers are automatically adjusted to prevent conflict. New properties and tasks are always assigned greater sequence numbers than the existing ones.
leave_children_open	Keeps child incidents, requests, problems, changes, or issues, open when the parent is closed.
lex_lang	Allows you to select the language to be used for spell checking. Default is US-English. This option sets the NX_LEX_LANG variable located in the NX.env configuration file.
sla_workshift_override	Use this option to default the workshift for Service Type Events. • When this option is not installed, events for a Service Type always uses the workshift specified each event's template. • Installing this option defaults the workshift for Service Type Events to the workshift specified on the Service Type. The default occurs only when the event does not specify its own workshift. If an event template specifies a workshift, it is always used. Set the event's workshift to the supplied '24 hour' workshift if you want the event to always start, regardless of any Service Type workshift.

Security Options

Option	Description
bopauth_host	(Applicable only for advanced availability configuration) Specifies the server name where bopauth_nxd is running. bopauth_nxd authenticates users accessing CA SDM through different interfaces. Select default if bopauth_nxd has to run in background server. This option cannot be uninstalled.
ldap_virtdb_host	(Applicable only for advanced availability configuration) Specifies the server name where ldap_virtdb is running. Select default if ldap_virtdb has to run in background server. This option cannot be uninstalled.

eiam_hostname	Specifies the hostname of the server where EEM is running. Required for EEM user name and password or artifact authentication.
force_browser_to_send_cookie_only_in_ssl_connection	(optional) Force the browser to send the Session ID (SID) cookie only if there is an SSL connection. This attribute is applicable only if you have enabled the **use_encrypted_sid_and_cookie** to (Yes). By default, this is turned off. If this flag is enabled, CA SDM can only be accessed through an SSL connection.
force_unique_userid	Prevents using duplicate active system login names. We recommend that you keep the force_unique_userid option enabled at all times. If this option is uninstalled, and there are multiple contact records with the same login id, you may experience problems with data partitions, multi-tenancy, security, and other functionality.
guest_user_name	Specifies the Guest user login ID. Default is Anonymous. Must be a valid user ID.
ignore_security_case	If installed, the user ID is not case sensitive.

Portal_Safe_List	Stores a list of machine names and port numbers where Portal is installed.
	The machine name and the port are entered as follows
	`machine-name:8080`
	For multiple Portal installations, separate the machine name/port number with a comma. For example:
	`machine-name1:8080,machine-name2:8081,machine-name3:8080`

url_etrust_password_reset	The URL for the eTrust Admin web interface. This enables the password reset link on the web interface. For example: `http[s]://hostname/EAOWebi/EAOLogin`
use_eiam_artifact	Allows EEM artifact to be used when users log in to CA SDM / Knowledge Management using URL.
use_eiam_authentication	Allows use of EEM authentication when users log in to CA SDM / Knowledge Management and the access type authentication is set to OS.
use_encrypted_sid_and_cookie	(optional) Use encrypted Session ID (SID) and cookie to prevent SID spoofing and Man-in-the-middle attack. By default, this attribute is disabled. If you want to have enhanced CA SDM security, this attribute can be enabled (Yes). Enabling this attribute can have some performance impact on the CA SDM Application.

Support Automation Options

Option	Description
sa_domsrvr	Specifies that the Support Automation main server communicates with the specified Support Automation domsrvr. If this variable is not installed, the Support Automation main server communicates with the default primary CA SDM Domsrvr named "domsrvr".
sa_primary_domsrvr	Specifies whether to start the Support Automation primary domsrvr. This variable is used to determine if a dedicated object server is already configured. If this option is not installed, the domsrvr:sa runs on one of the secondary CA SDM servers and you are required to startup for domsrvr:sa.
supportautomation_url	Specifies the URL of the Support Automation web application, which CA SDM uses to communicate with the Support Automation main server. The host name and port number should correspond with the values set during configuration. We recommend that you configure the supportautomation_url with a host name because you may encounter problems when you configure the supportautomation_url with an IP address, if you use an IP address which has an existing host name mapping, the HTTP request processing resolves to this host name. The administrator installs this option after you configure the Support Automation main server on a primary or secondary CA SDM server to enable Support Automation usage in CA SDM. Do not install this option if Support Automation is not configured.

Time-to-Violation Options

These options control the process that monitors the service level agreements on all open tickets and tasks, and records the time the Service Type enters violation:

Option	Description
ttv_enabled	Runs the Time to Violation daemon, which monitors the SLAs for all open tickets and tasks. This process does not set the SLA violation, but records the date the ticket or task is violated in its current state. This projection is updated when the ticket or task is updated. This option must be installed in order for the other Time to Violation options to function correctly. This option requires that the classic_sla_processing option *not* be installed.
ttv_evaluation_delay	Sets the delay between a request to the time-to-violation (TTV) daemon and when it actually evaluates a ticket. Whenever a ticket is updated, a request is sent to the TTV daemon to update the time-to-violation projections. Because tickets are often updated several times in quick succession, a delay in evaluation helps prevent needless evaluations and improves performance. This option is dependent on the ttv_enabled option, which must also be installed. This value may never be set below 60 seconds.

ttv_highlight	Highlights a ticket in a list if the ticket has a Time to Violation value set to start by a certain time. The default highlight color is a pale yellow. The ticket number is highlighted only if one of its Time to Violation values is less than the date/time specified by the Timespan, "TTV_THRESHOLD". By default, this Timespan is set to show tickets with Time to Violation values less than midnight of the current day, for example, "today". You may modify this Timespan to any value. This option only affects ticket list forms on the web interface client. It applies to lists of Requests, Issues, Change Orders, Incidents, Problems and Tasks. It's an option because the Time to Violation value is stored in a database table separate from the ticket, and requires an extra database fetch *for each row displayed*. Given this extra DBMS activity, installing this option may increase the load on your database so that performance may be affected. This option is dependent on the ttv_enabled option, which must also be installed.

Ver Ctl Options

The ver_ctl option controls the actions that are taken by the Version Control Manager when a version discrepancy is detected. The Version Control Manager detects discrepancies by comparing version files across all CA SDM servers (client and servers). Depending upon the CA SDM configuration, the following client and servers are used:

- Conventional:
  - Client: Secondary server
  - Server: Primary server
- Advanced availability:
  - Client: Application and Standby servers
  - Server: Background server

Only components that are listed in the version files are controlled. When you restart the servers and if a discrepancy is detected, the client connects to the server to send a list of its controlled components. The server compares the list to its own master list. The affected components on the client are upgraded or terminated, depending on the ver_ctl option.

ver_ctl
Installing this option activates the Version Control Manager. Valid option values are:

- **Fail** -- A version discrepancy terminates the secondary server's connection to the primary server. The secondary server is prevented from connecting to the primary server until it is manually updated.
- **Notify** -- Default value. On Windows, a version discrepancy results in a message window asking if you want the secondary server to continue or quit. On UNIX, a version discrepancy is logged, but the secondary server always completes initialization.
- **Upgrade** -- When a version discrepancy is detected, an upgrade of the affected components is attempted on the secondary server. If the upgrade is successful, the secondary server connection continues; otherwise, the secondary server connection terminates.
- **Disable** -- Version discrepancies are ignored. All secondary servers are allowed to connect to the primary server.

Web Options

These options control the CA SDM web interface operation:

Option	Description
auto_suggest_status	Displays matching records for the text entered in lookup fields within the Analyst interface. For this option to work correctly, the Using Screen Reader user preference must be disabled. To specify user preferences, select View, Preferences on the menu bar. **Default:** On
block_opcodes_for_get	Conversion or simulation of a POST operation using a GET operation is prohibited or strongly discouraged due to the critical security and data integrity risks. OP codes not valid for GET Method. Must be a string of OP codes separated by comma. This is available from 17.4.5.0 and above. Default values are: `ACCUMULATE,ACCUMULATE_ALL,ATTACH_STYPE_EVENT,CLOSE_ALL_CHILD,K`
copy_inactive	Copies the inactive attributes of an object (such as the previous assignee) from the original record to the new record. **Note:** By default, this option is uninstalled so that links to inactive objects are not copied. If you install this option, or if you migrated from CA SDM r12.5, links to inactive objects are copied.
charset	Defines the default HTTP character set for the web interface. You cannot uninstall this option. The value of this option is set in the charset parameter of the HTTP content-type header and it matches the character encoding of the operating environment when CA SDM is configured. Under normal circumstances, this option does not need updating after initial configuration. Restarting the CA SDM server is *not* required.
export_max_fetch_rows	Determines the maximum number of rows that users can export for a list. **Default:** 5000
max_files_to_upload_simultaneously	Defines the maximum number of files that you can simultaneously upload during the same operation. **Valid Range:** Greater than or equal to 0. Zero means unlimited. **Default:** 10
max_records_auto_suggest	Specifies the number of suggestions to display in matching results. For example, increase the number of suggestions if analysts in your environment request 50 suggestions. **Default:** 25
min_chars_auto_suggest	Specifies the minimum number of characters that a user enters in a lookup field before the field starts displaying matching results. For example, increase the number of characters if your environment contains many users that share the first three characters in their user names. T **Default:** 3

mouseover_preview_delay_time	Specifies the delay time, in milliseconds, between hovering the mouse cursor over an object link and the display of the mouseover preview. The preview is not generated if the mouse is moved away from the link before the delay time expires. Enter a value from 100 to 5000 in the Option Value field. **Default:** 1000
pdmweb_cache_disable	All regular files from bopcfg/www are cached. If this option is installed, the PDMWEB servlet does not cache the files, which means all files are retrieved from the directory each time one is needed.
pdmweb_cache_sitemods	The PDMWEB Servlet, by default, does not cache files from the $NX_ROOT/site/mods folder. If this option is installed, files are cached from the sitemods folder.
prompt_for_knowledge	Enables a dialog that prompts users to search the knowledge base before creating issues from the customer web interface and requests from the employee web interface.
scoreboard_entry_limit	Limit for adding new scoreboard nodes and folders. When the system reaches the limit, existing folders and nodes continue to display information. However, the user must delete unused folders or nodes before creating new ones.
suppress_web_hier_search	Suppresses the hierarchical search used on the web for request, incident, and problem areas, category, and root cause. This option is useful for tables that are not organized hierarchically, or when one or more of the hierarchical levels contains a great many records. When hierarchical search is suppressed for a table, search requests on the table are processed as lookups. The value of this option is list of object names of the tables for which hierarchical search is suppressed. If more than one table name is specified, values are separated by spaces. Valid values are: • **chgcat** - Change order category • **isscat** - Issue category • **pcat_cr** - Request area • **pcat_in** - Incident area • **pcat_pr** - Problem area • **rc** - Request root cause To avoid errors when using autocomplete, we recommend that you use this option instead of modifying the dtlHier macro to dtlLookup in the HTMPL file.
use_nested_tabs	Displays nested tabs within the current form. The analyst can expand each tab to view key details. If you want to revert to the notebook styles used in CA SDM r12.5, you can disable the nested tabs control. For example, analysts in your environment prefer to click a tab that shows only the content of that tab.

web_auto_update	Requests the automatic update of scoreboard counts for analyst web users. When this option is installed, each web analyst session requests updated scoreboard counts at the interval specified in the option. The value of this option is the number of seconds between updates, with a minimum value of 60 seconds (one minute). Web scoreboard updating occurs in the background. The analyst end user can see periodic activity on the main page. In addition, updated count values are highlighted on the scoreboard. The highlighting remains until the user either clicks on the scoreboard link to display the associated list, or presses the Update Counts button. Activating this option affects the overall load of the system as follows: • Each web user can send additional requests to the server at the frequency specified in the option. • The web engine receives messages from the object engine each time a database update causes the count associated with a query to change. When the web auto update option is not installed, the web engine requests counts from the object engine only when a web user explicitly requests an update.
web_wildcard_search	Installing this option enables a wildcard search for all text fields when using Search Filters. To perform a wildcard search, you add the "%" symbol after a text string. For example, entering the string "option%" retrieves all records that begin with the word "option". Installing this option eliminates the need for the "%" symbol and conducts wildcard searches by default. Installing this option enables functionality equivalent to using the "%" symbol only at the *end* of the search string, not at the beginning. For example, with this option installed, searching the Options Manager option list for the string "ldap" finds all options with names beginning with "ldap" but does not find the "default_ldap_tenant" option.
wsp_domsrvr	Specifies the name of an object engine dedicated to the Web Screen Painter (WSP). The WSP webengine, designated by the companion option wsp_webengine, is the only client process allowed to connect to the object engine identified by wsp_domsrvr. You must install both the wsp_domsrvr option and wsp_webengine option in order to enable the WSP schema modification test feature. The value of this option must be a string beginning with the characters "domsrvr:". It is recommended that you use domsrvr:wsp as the name of the WSP object engine. Installing this option automatically runs the WSP object engine on the primary server.

wsp_webengine	Specifies the slump name of the webengine used by Web Screen Painter users. All WSP users connect to this webengine, and use it for preview sessions. All files saved by WSP users are stored in the site/mods directory on the server running this webengine, which must be the CA SDM primary server. Because WSP saves information on the server running its webengine, all WSP users must connect to the same webengine. If the wsp_wengine option is not installed, WSP users connect to the default webengine: web:local, sharing it with other users. The value of this option must be a string beginning with the characters: "web:". We recommend you use web:wsp as the name of the WSP webengine.

Web Report Options

These options control the CA Business Intelligence reporting integration with the BusinessObjects server:

Option	Description
bo_server_auth	Specify which type of authentication you want to use for reporting. You can specify the following types of authentication: • **secEnterprise**. (Default) Specify Enterprise Authentication as your authentication type if you prefer to create distinct accounts and groups in BusinessObjects for use with CA Business Intelligence, or if a user hierarchy has not been set up on an LDAP server or a Windows AD server. If you choose the secEnterprise option, you need to add your CA SDM report users to the BusinessObjects Central Management Console (CMC). The CMC allows you to control users' access to Business Intelligence Launch Pad and other BusinessObjects applications. In the CMC, you must enter the same user names and passwords configured in CA SDM. • **secLDAP**. Specify LDAP Authentication as your authentication type if you have already set up an LDAP directory server and want to use your LDAP user accounts and groups in BusinessObjects for use with CA Business Intelligence. • When you map LDAP accounts to BusinessObjects, users can access CA Business Intelligence with their LDAP user name and password. This eliminates the need to recreate individual user and group accounts within BusinessObjects. • **secWinAD**. Specify Windows AD Authentication as your authentication type if you are working in a Windows 2000 environment and want to use your existing AD user accounts and groups in BusinessObjects for use with CA Business Intelligence. • **secExternal**. Specify External Authentication as your authentication type if you integrate the BusinessObjects authentication solution with a third-party authentication solution (for example, using JCIFS with Tomcat). This authentication type requires setting up Trusted Authentication in BuisnessObjects to allow users to log on without providing their passwords. During the Trusted Authentication configuration, you must set the user name retrieval method to REMOTE_USER in the BusinessObjects web.xml file.
bo_server_cms	Specify the name of the Central Management Server (CMS) that is responsible for maintaining a database of information about your BusinessObjects that you use with CA Business Intelligence. For the bo_hostname, use the hostname of the machine where CA Business Intelligence is installed. The default bo_cms_port is 6400.

bo_server_location	Specify bo_hostname by using the hostname of the machine where CA Business Intelligence is installed. CA SDM uses this URL to put together report URLs for requesting reports from the BusinessObjects server. The CMS location is specified by hostname and port. **Warning!** The default bo_application_server_port is 8080 (for example, this can be the Tomcat port). If you install CA SDM and CA Business Intelligence on the same server, you must specify a different port. Use the port specified when starting Business Intelligence Launch Pad. **Warning!** For example, http://hostname:8180/businessobjects/enterprise115/desktoplaunch/businessintelligencelaunchpad/logon/logon.do displays that 8180 is the value to use for bo_tomcat_port.

Web Service Options

Option	Description
rest_webservice_access_duration	Specifies the number of hours that the REST Web Service Access Key remains active before expiration. The Access Key timeout is not based on inactivity time, but it is based on duration time since the Access Key creation. After the Access Key meets the specified duration, the Access Key ends regardless of whether it is being used. Optionally, the REST client can also provide the Access Key duration time for the specific Access Key during the Access Key request. To provide the duration value, set it directly on the expiration_date attribute of the rest_access resource, as part of the POST request payload. **Valid Range:** 1-8760 hours **Defaut:** 168
rest_webservice_disable_basic_auth	Disables Basic Authentication in REST Web Services. **Default:** No
rest_webservice_list_max_length	Specifies the maximum number of rows that a REST Web Service query returns. **Default:** 500
rest_webservice_list_page_length	Specifies the default number of rows that a REST Web Service query returns per page. **Valid Range:** 1-500 **Default:** 25

rest_webservice_resources_to_expose	Specifies the list of Majic factories (resources) that CA SDM exposes through REST Web Services. This option overrides the default behavior. By default, CA SDM exposes all factories through REST Web Services. If you do not enter values in this option, the default behavior exposes any Majic factory that does not have the REST_OPERATIONS property set to NONE. By default, no Majic fatory has this property set to NONE. Use the REST_OPERATIONS property to set the specific HTTP CRUD (CREATE, READ, UPDATE, DELETE) methods for CA SDM to expose on a given Majic factory. **Default:** rest_access **Example:** rest_access, cnt, grp, cr, crs, pri, alg, urg, imp, pcat, org
hmac_algorithm	Specifies the algorithm that you use to compute the signature for Custom/Secret Key Auhentication in REST Web Services. **Default:** HmacSHA1
string_to_sign_fields	Specifies the fields that you use to compute the signature for Custom/Secret Key Authentication in REST Web Services, in addition to the default REQUEST_METHOD, REQUEST_URI, and QUERY_STRING fields. **Default:** blank
webservice_domsrvr	Specifies the name of the object engine that SOAP web services use. If not installed, SOAP web services use "domsrvr". The value of the option must be a string beginning with the characters "domsrvr:"
webservice_session_timeout	Sets the timeout value (in minutes) for SOAP Web Service sessions. When the time between successive web method calls is greater than the value specified, the session ID is marked expired. The session is then no longer valid. To prevent sessions from expiring due to activity, set the value for this option to 0. Other methods, such as logoff routines, can still invalid sessions.

xMatters

xMatters is used for mass notifications and Alerts. When a CA SDM ticket (incident, request, or problem) is created or updated, the integration mechanism sends the notification along with ticket information to the xMatters agent. The notification works only with CA SDM CR objects (incident, request, or problem). The CA SDM notification feature sends notifications to users based on the rules set for notification. CA SDM sends the Notify Log Reader information automatically along with ticket information to the configured xMatters agent.

CA SDM administrators can enable or disable integration with xMatters through the Options Manager options. You must install these options as these options are not installed out of the box. Restart CA SDM services after installing or uninstalling these options. On successful installation of xMatters options, a new daemon the pdm_xmatters_sync is started and can be monitored through the task manager process list. This daemon is a singleton process and runs on the primary or background server.

The pdm_xmatters_sync daemon gathers notification records with status **Pending** or **Error** based on the **xmatters_retry_count** option value. It also maintains a list of newly created notification records. After the existing record list is run, the pdm_xmatters_sync runs the newly created record list. This is an ongoing process. In case of failure, when the xMatters agent is not available, the records wait for the time specified in the Options Manager

xmatters_retry_interval option and re-triggers the pdm_xmatters_sync daemon till all records attain a status of either **Success** or **Abort**.

The pdm_xmatters_sync daemon runs the records in the same order in which they are created. Records having **xMatters-Pending** or **xMatters-Error** status are processed first and then, the newly created records are processed.

Based upon the set notification rules, the notification history page will have information along with the status.

Rule	Information
xMatters – Error	Return error code status to CA SDM. Indicates that this notification message was sent to xMatters agent but was not received by xMatters agent. This can happen due to xMatters agent not running or other issues with the agent. Notifications having this status will retry based on the Options Manager *xmatters_retry_count* and *xmatters_retry_interval*.
xMatters – Abort	Return error code status to CA SDM. Indicates that CA SDM triggered the number of iterations mentioned in the Options Manager *xmatters_retry_count* and *xmatters_retry_interval* options, but the xMatters agent did not receive this message.
xMatters – Success	This status indicates that xMatters agent received this notification message successfully and returned the success status code. These records can be shown from xMatters system web interface.

Install the following xMatter Options Manager Options. These options control the CA SDM and xMatters integration provides a reliable and stable mechanism to integrate notifications.

Option	Description
xmatters_cr_attr	The valid CR majic attribute name. Specify the request attributes for xMatters notification. This can include custom attributes.
xmatters_retry_count	The default and minimum value is set as 3 in the range [1-20]. This value indicates the number of times the retry option can be triggered for failed transactions or notifications.
xmatters_retry_interval	The default and minimum value is set as 15 minutes in the range of [15-999]. This value indicates the time gap between each retry and specifies the retry interval duration in minutes for each failed xMatters notification.
xmatters_URL	Valid xMatters server agent and port number. Specifies the xMatters integration URL link. **Example URL:** http://<xmattersagentname>:<port>/http/caservicedesk15_caservicedesk15

Add HTTP Headers in CA SDM Options Manager

This article describes how to add the HTTP headers in CA Service Desk Manager (SDM). SDM Administrators can now define multiple HTTP headers that are added to HTTP response. You can also add HTTP Security Headers to HTTP responses that will instruct the browsers to apply the specified organizational security policies. For example, you can enable the following HTTP headers values:

- Content-Security-Policy
- Referrer-Policy
- Expect-CT

The new option **http_headers** is included in CA SDM Options Manager. The option is not installed by default. It takes **Name-Value pairs** of HTTP Headers and the corresponding values as input. The **Name-Value pairs** are separated by pipe symbol ('|'). Name and Value for each pair is separated by Tilde or twiddle symbol ('~').

Perform the following steps to add the HTTP headers in Options Manager:

- Login as CA SDM Administrator user.
- Navigate to **Administration**, **Options Manager**, **Security**, **http_headers** option. By default, the **http_headers** option is not installed.
- Update the values in the Option Value field on the http_headers Options Detail page. You can add multiple pairs separated with '|' (pipe symbol):

  ```
  Header1~Value1|Header2~Value2
  ```

  For example:

  ```
  Content-Security-Policy~ policy|Referrer-Policy~ unsafe-url|Expect-CT~ max-age=86400, enforce, report-uri=https://test.example/report
  ```

- Restart SDM.
- After you enter the URL address in a browser and click Enter, ensure that the http headers values are updated in the response headers.

Install/Uninstall Options Manager Options

You can install Options when installing CA Service Desk Manager (SDM) or can install them later. Some Options are automatically installed when you install the application that uses them.

Install an Option

You can install an option from CA SDM. If you are using advanced availability configuration, you can install an option only from the background server. From the application or standby server, you can view the option details after it is installed.

Follow these steps:

1. Log in to the following server, depending upon your CA SDM configuration:
   - Conventional: Primary or secondary server
   - Advanced Availability: Background server
2. On the Administration Tab, browse to Options Manager.
   The Option Search page opens.
3. Search for the option you want to install.
   The Option Detail window opens.
4. Click Edit.
   The Update Option window opens.
5. Click Install.
   The Option Detail window displays a refresh message.
6. Click Refresh.
   The Option Detail window displays the Action Status of the option as "Installed."
7. Click Close Window.
8. Depending on your CA SDM configuration,

   WARNING
   Changes in some options do not require you to restart all the servers. Ensure that you read the Server Restart List before restarting any server.

   - Restart the CA SDM servers in conventional configuration.
   - Restart the CA SDM servers in advanced availability configuration.

The Option List displays the updated status of the option when you restart the servers.

You can also install options from the command line using the following script:

```
$NX_ROOT/bin/pdm_options_mgr -c -b -a pdm_option.inst -t
```

For example:*pdm_options_mgr -c -s attach_force_save_dialog -v 1 -a pdm_option.inst -t*

NOTE
In case if SDM options are not listed in the web interface (Administration tab, Administration, Options Manager), use the command line interface "$NX_ROOT/bin/pdm_options_mgr" to install the specified options. This need to be executed on all SDM servers:

1. Secondary Server (in case of Conventional Configuration)
2. Standby and Application servers (in case of Advance Availability)

Uninstall an Option

You can edit an option to uninstall it. If you are using advanced availability configuration, you can edit an option only from the background server. From the application or standby server, you can view the option details.

Follow these steps:

1. Log in to the following server, depending upon your CA SDM configuration:
   - Conventional: Primary or secondary server
   - Advanced Availability: Background server
2. On the Administration Tab, browse to Options Manager.
   The Option Search page opens.
3. Search for the option you want to uninstall.
   The Option Detail window opens.
4. Click Edit.
   The Update Option window opens.
5. Click Deinstall.
   The Option Detail window displays a refresh message.
6. Click Refresh.
   The Option Detail window displays the Action Status of the option as "Not Installed."
7. Click Close Window.
8. Depending on your CA SDM configuration,

   WARNING
   Changes in some options do not require you to restart all the servers. Ensure that you read the Server Restart List before restarting any server.

   - Restart the CA SDM servers in conventional configuration.
   - Restart the CA SDM servers in advanced availability configuration.

   The Option List displays the updated status of the option when you restart the servers.

You can also uninstall options from the command line using the following script:

```
$NX_ROOT/bin/pdm_options_mgr -c -b -a pdm_option.deinst -t
```

For example:*pdm_options_mgr -c -s attach_force_save_dialog -v 1 -a pdm_option.deinst -t*

NOTE
In case if SDM options are not listed in the web interface (Administration tab, Administration, Options Manager), use the command line interface "$NX_ROOT/bin/pdm_options_mgr" to uninstall the specified options. This need to be executed on all SDM servers:

1. Secondary Server (in case of conventional Configuration)

2. Standby and Application servers (in case of Advance Availability)

Restart the CA SDM Servers in Conventional Configuration

For the conventional configuration, you restart the servers in the following order:

NOTE

To restart a server click Start, Settings, Control Panel, Administrative Tools, Services. Right-click the CA SDM Server and select Start.

1. Restart the secondary server.
2. Restart the primary server.

Restart the CA SDM Servers in Advanced Availability Configuration

For the advanced availability configuration, we recommend that you restart the CA SDM servers in the following order:

NOTE

To restart a server click Start, Settings, Control Panel, Administrative Tools, Services. Right-click the CA SDM Server and select Start.

1. Restart all Standby Servers.
2. Promote the Standby Server as the New Background Server.
3. Start the Old Background Server.
   When you start the background server, it becomes a standby server.
4. Choose the Less Active Application Server.
5. Restart the Less Active Application Server.
6. Stop the Other Application Server.
7. Start the Application Server.
8. Perform the steps 6 and 7 for the other application servers.

Promote the Standby Server as the New Background Server

Before you stop the background server, promote the standby server (that you have upgraded) as the new background server. If Support Automation is installed with CA SDM, notify the active Support Automation users about the background server shutdown.

Follow these steps:

1. Execute the following command on the background server to notify all active users using Support Automation to save their work:

   ```
   sa_server_notifier [-h] | [-q seconds] | [-c]
   ```

   – **-h**
   Displays the help page.
   – **-q seconds**
   This option notifies a local server (background) to quiesce in a specified time interval. This interval is the number of seconds before the server goes offline. This option cannot be used for a standby server or application server.
   – **-c**
   This option cancels a previously sent quiesce request.

   A pop-up message is displayed to all the active users using Support Automation. This message notifies the users about the server shutdown and the time that is left for the shutdown. The users must save their work and logout within that scheduled time.
2. Execute the following command on the standby server that you wish to promote as the new background server:

   ```
   pdm_server_control  -b
   ```

– **-b**
Notifies a local standby server to become the background server. The standby server must already be running. If the server is not running, it is started but no failover is performed; to start a failover, run the command again.

The background server shuts down automatically and the standby server is promoted as the new background server. This change does not affect the end-user sessions. The in-progress updates (if any) are stored and delayed, until the new background server comes online.

Choose the Less Active Application Server

You choose an application server with the least user activity. Run the following command on each application server to choose the one with no or minimal active sessions.

```
pdm_webstat
```

NOTE
This command does not capture the SOAP or REST Web Service sessions.

Stop the Other Application Server

You inform all the active users on an application server to move to the less active application server before you stop it. Ensure that you have restarted the less active application server before moving all the users to it.

Follow these steps:

1. (Recommended) Inform all active Support Automation analysts on the application server which you want to stop, to create a ticket in CA SDM with their session information. This process ensures that the session information is not lost. For example, the Support Automation analyst is in a session with a customer to resolve a hardware issue. In such a case, the Support Automation analyst can create an issue in CA SDM with the session information before the application server shuts down.
2. Send a notification (for example, an email notification) to all the active users on the application server to move to the less active application server that you just restarted. This notification can include the details of the updated application server.
3. Execute the following command on the application server:
   ```
   pdm_server_control [-h] -q interval -s server_name
   ```
   – **-h**
   Displays the help page.
   – **-q interval -s server_name**
   Notifies a local or remote application server to quiesce in a specified time interval. This interval is the number of seconds before the server goes offline. When using this option without a server_name, the local server is notified to quiesce. This option cannot be used for a background or a standby server.

   A pop-up message is displayed to all the active users on the application server to notify them about the server shutdown and the time left for the shutdown. The users must save their work and logout within that time. The application server stops after the specified time. The users log on to the other application server to resume their work. The Support Automation analyst can refer to the ticket and resume their work.
   The application server is stopped successfully.

Server Restart List

Depending upon the CA SDM configuration, restart the following servers:

- Conventional: Primary server, Secondary Server
- Advanced Availability: Application, Background, and standby servers. The following options do not require you to restart all the servers:

NOTE
Options manager can be installed through web interface or command line interface. If any option which is not available on web interface then use pdm_options_mgr command to install and uninstall the SDM options.

Options Type	Options	Action
Change Order	Category_Defaults, chg_auto_events, chg_sla	Restart all servers *quickly* to maintain functionality consistency. Change in these options contains Majic and/ or Spel changes
Email	mail_from_address, mail_login_password, mail_login_userid, mail_max_threads, mail_reply_to_address, mail_smtp_domain_name, mail_smtp_hosts, mail_smtp_host_port, mail_smtp_security_level	Restart all standby servers and then the current background server.
Issue Mgr	iss_auto_events, Iss_Category_Defaults, iss_sla	Restart all servers *quickly* to maintain functionality consistency. Change in these options contains Majic and/ or Spel changes.
LDAP	ldap_dn, ldap_enable_tls, ldap_group_object_class, ldap_host, ldap_port, ldap_pwd, ldap_search_base, ldap_service_type, ldap_user_object_class, num_ldap_agents	Restart all standby servers and then the current background server.
Request Mgr	Area_Defaults, auto_events, cr_sla, Netres_pty	Restart all servers *quickly* to maintain functionality consistency. Change in these options contains Majic and/ or Spel changes.
Request-Change-Issue	Any_Contact	Restart all servers *quickly* to maintain functionality consistency. Change in these options contains Majic and/ or Spel changes.

Search Engine	ebr_cr_where_clause, ebr_index_queue_timeout, ebr_iss_where_clause, ebr_max_qps, ebr_qps_timeout, ebr_search_engine_baseport, ebr_search_engine_host, ebr_version	Restart all standby servers and then the current background server.
Security	force_unique_userid	Restart all servers *quickly* to maintain functionality consistency. Change in these options contains Majic and/ or Spel changes.
Support Automation	sa_domsrvr, sa_primary_domsrvr	Restart all standby servers and then the current background server.
Time-to-Violation	ttv_evaluation_delay	Restart all standby servers and then the current background server.
Web	export_max_fetch_rows	No action required. Changes to this option takes effect when the user exports the list.
Web	web_wildcard_search	Restart all standby servers and then the current background server.
Web Service	hmac_algorithm, rest_webservice_access_duration, rest_webservice_disable_basic_auth, rest_webservice_list_max_length, rest_webservice_list_page_length, rest_webservice_resources_to_expose, string_to_sign_fields	Restart all application servers where RESTful Web Services are configured.

Multi-Tenancy

This section contains the following articles:

- Service Provider
- How Multi-Tenancy Works
- User Interface Impact
- Knowledge Management Impact

Multi-tenancy enables multiple independent tenants (and their users) to share a single implementation of CA SDM. All Tenant users interact with each other in defined ways as per assigned roles and tenant hierarchies. Typically, each tenant views the CA SDM implementation as per the assigned access right. Tenants cannot update or view data that are related to other tenants unless specified.

Multi-tenancy allows tenants to share hardware and application support resources and reduce costs.

Service Provider

This article contains the following topic:

The service provider is the primary tenant (owner) in a CA SDM multi-tenancy installation. The first tenant added to a CA SDM installation is always the service provider tenant. The service provider tenant cannot have a parent tenant.

CA SDM associates the privileged user (typically ServiceDesk on Windows, or srvcdesk on Linux/UNIX) with the service provider tenant.

Only the service provider tenant can perform any of the following CA SDM tasks:

- Set the CA SDM options
- Set the Knowledge Management options
- Set the Support Automation options
- Create the tables or columns
- Create, edit, or delete the tenants
- Allow the tenants to have subtenants
- Update the public data

NOTE
An administrator can grant tenant users access to data other than their own. The Non-Service Provider tenant analysts have access to only their tenant and sub-tenants. For example, a role definition can set separate read and write access to certain tenant groups for users within that role.

WARNING
When a tenant is created and set as the service provider, the service provider check box and record status fields are read-only.

Service Provider Administration

The service provider can permit tenants to administer their own settings. The Tenant administrators have access to a subset of administration tasks that is the same for all tenants. The default Tenant Administrator role defines the administration functions that are available to tenant administrators. To designate a user as a tenant administrator, select the Tenant Administrator for that user role.

How Multi-Tenancy Works

This article contains the following topics:

After enabling CA SDM multi-tenancy, you can grant contacts access to all tenants (public), a single tenant, or a tenant group (user-defined or product-maintained). A contacts role controls access and specifies the read-write access independently. As the tenant access is role-dependent and a contact can modify roles during a session, contact tenant access is also subject to change.

Most of the CA SDM objects when multi-tenancy is installed include a tenant attribute to specify owner of the object. Objects fall into three groups, depending on their tenant attribute and how it is used:

Untenanted

Defines objects without a tenant attribute. All data in these objects is public.

Examples: Priority and urgency.

Tenant Required

Defines objects with a tenant attribute that cannot be null (enforced by CA SDM, not the DBMS). All data in these objects is associated with individual tenants; there is no public data.

Examples: Ticket tables (Request, Issue, and Change Order).

Tenant Optional

Defines objects with a tenant attribute that can be null. Some of the data in these objects is public, and some is associated with specific tenants. Each tenant's view of the object is a merged view of the public data and their tenant-specific data.

Examples: Category and location.

When a user queries the database, CA SDM restricts the results to objects belonging to tenants. The user is authorized to access these tenants. This restriction applies in addition to any data partition restrictions that are in effect.

If a user creates an object with update access to multiple tenants, the user must specify the tenant explicitly.

> **NOTE**
> Few SREL references (such as the assignee of an incident) are permitted to reference objects that belong to tenants in their containing object's tenant hierarchy. Such references are designated as SERVICE_PROVIDER_ELIGIBLE in the CA SDM object schema (the Majic). The SERVICE_PROVIDER_ELIGIBLE flag makes a difference only if the service provider tenant is not in the tenant hierarchy; if the service provider tenant is in the hierarchy, tenant validation rules permit service provider references.

> **NOTE**
> If CA SDM limits a user from updating tenant data, an error message can announce a data partition limitation. If you receive this error message, either data partition or multi-tenancy restrictions are in effect.

The Multi-Tenancy Option

You activate multi-tenancy by installing one of the following multi-tenancy options:

> **NOTE**
> When tenancy is in setup mode, do not perform any transactions on the server as this may lead to data loss. Before performing any operations or transactions, it's recommended that you complete the task of creating tenants and set it to ON state.

- **Off** -- Multi-tenancy is not in use. Multi-tenancy features are not available, and objects do not have a tenant attribute. This option is the default setting at a new CA SDM installation.
- **Setup** -- Multi-tenancy features are in effect for administrators, so that tenant-related objects and attributes are visible and editable. However, CA SDM does not enforce tenancy restrictions, and non-administrator users see no changes. This setting allows an administrator to prepare for multi-tenancy by performing such tasks as defining tenants or assigning objects to tenants without impacting normal use of CA SDM.
- **On** -- Multi-tenancy is fully operational. All users see the UI changes appropriate to them, and CA SDM enforces tenancy restrictions.

  > **NOTE**
  > Recommendation is to create tenants in setup mode and make multi tenancy to ON state before performing operations.

Tenant Information

You create and update tenants when you install multi-tenancy (in either setup or full enforcement mode). Information that is maintained for a tenant is similar to the data maintained for Organization, except for the following two attributes:

- **Logo**
  Provides a URL to an image file with the logo of the tenant. The logo is shown both on the Tenant Detail page itself, and as a substitute for the CA logo on web forms that are displayed by a tenant user or showing an object that is associated with the tenant.
- **Service Provider**
  Indicates whether the tenant is the service provider. The service provider tenant is always the first tenant added. When the administrator adds the first tenant:

- The first tenant becomes the service provider. This designation cannot be changed.
- The privileged user (usually ServiceDesk) and all system contacts (such as System_AHD_Generated) are set to belong to the new service provider tenant.

NOTE
The system user "Administrator" is added in Windows only and is not assigned a tenant. The privileged user must assign a tenant to Administrator manually.

Tenant Access

The role of a CA SDM user governs both access authorization and the user interface. The set of roles available to users depends on their access type. Multi-tenancy lets you control the tenant or tenant group that a user can access within the role.

The Role Detail page provides Tenant Access and Tenant Write Access drop-down lists on its Authorization tab. Tenant Access is view-only, and Tenant Write Access also allows create and update.

You can assign the following associations to roles:

- **Same As Tenant Access (Tenant Write Access Only)**
  Sets Tenant Write Access to be the same as the Tenant Access setting. Default for Tenant Write Access and only valid for Tenant Write Access.
- **All Tenants**
  Removes tenant restrictions. CA SDM allows a user in a role with this access to view any object in the database (read access) or create and update (write access) any tenanted object in the database. When users with All Tenant access create an object, CA SDM requires that they select the tenant of the new object.
- **Single Tenant**
  Sets a role's tenant access to a named tenant. When this option is selected, a second field appears on the web UI that allows selection of a specific tenant. CA SDM restricts a user in a role with this access to view (read access) or create and update (write access) only those objects that are associated with the named tenant. This selection is valid for either Tenant Access or Tenant Write Access.
- **Tenant Group**
  Sets a role's tenant access to a user-defined or system-maintained tenant group. When the Tenant Group option is selected, a second field appears on the web UI that allows selection of a specific tenant group. CA SDM restricts a user with the role to view (read access) or create and update (write access) only those objects that are associated with one of the tenants in the group. When a user with tenant group access creates an object, CA SDM requires that they select the tenant for the new object. This selection is valid for either Tenant Access or Tenant Write Access.
- **Contact's Tenant**
  Sets a role's tenant access to the tenant of the contact using it. CA SDM restricts a user in a role with this access to view (read access) or create and update (write access) only those objects that are associated with their own tenant. This selection is valid for either Tenant Access or Tenant Write Access.
- **Contact's Tenant Group (Analyst Only)**
  Sets an analyst's role access to the tenant group that the analyst works with, as specified on the analyst's contact record. If the user with the role is not an analyst, this selection has the same effect as Contact's Tenant. It is valid for either Tenant Access or Tenant Write Access.
- **Contact's Subtenant Group**
  Sets a role's tenant access to the Subtenant group of the contact using it. CA SDM restricts a user in a role with this access to view (read access) or create and update (write access) only those objects that are associated with their own Subtenant group. This selection is valid for either Tenant Access or Tenant Write Access.
- **Contact's Supertenant Group**
  Sets a role's tenant access to the Supertenant group of the contact using it. CA SDM restricts a user in a role with this access to view (read access) or create and update (write access) only those objects that are associated with their own Supertenant group. This selection is valid for either Tenant Access or Tenant Write Access.
- **Contact's Related Tenant Group**

Sets a role's tenant access to the Related Tenants Group of the contact using it. CA SDM restricts a user in a role with this access to view (read access) or create and update (write access) only those objects that are associated with their own Related Tenants Group. This selection is valid for either Tenant Access or Tenant Write Access.

All users can view public data, regardless of their current role's access rights. The Update Public check box controls whether a service provider user in the role has the authorization to create or update public data. The Tenant users (users belonging to a tenant other than the service provider) cannot update public data, regardless of their role.

Edit Tenant Access for a Role

The role of a CA SDM user governs both access authorization and the user interface. The set of roles available to users depends on their access type. Multi-tenancy lets you control the tenant or tenant group that a user can access within the role.

The Role Detail page provides Tenant Access and Tenant Write Access drop-down lists on its Authorization tab. Tenant Access is view-only, and Tenant Write Access also allows create and update.

You can assign or edit tenant access for a role.

Follow these steps

1. Navigate to Security and Role Management, Role Management, Role List.
2. Click a role and click Edit.
3. Select options for Tenant Access and Tenant Write Access.
   Note: Exercise caution if you select different options for these settings.
4. Click Save.
   The updated tenant access options are saved for the role.

Tenant Terms of Usage

An end user is presented with a terms of usage statement while logging in the CA SDM application. The terms of usage statement is regarding proper usage, terms and conditions of using the product by the end user. The user must agree to the terms before logging to CA SDM. Entries are written to the standard log and in the user event log after the attempted session login.

You can perform the following terms of usage actions:

- Create, update, and delete a terms of usage statement.
- Associate a terms of usage statement with a tenant.

  NOTE
  You must enable multi-tenancy and configure one or more tenants before associating a terms of usage statement with a tenant.
- Force the end user to accept the statement every time they log in.
- Let the end user ignore the initial statement by presenting a blank terms of usage statement.

For more information about creating terms of usage statement, see Setting Up Terms of Usage.

How to Configure Terms of Usage

The terms of usage statement presents the end user with an initial page statement when they log in to CA SDM. The statement reminds the user about the proper use of the product. The user must agree to the terms before they can continue to log in to CA SDM. If the end user selects Accept, CA SDM proceeds with the login and displays the main form. If the user selects Reject, CA SDM returns to the login. Entries are written to the standard log and in the user event log after the attempted session login.

Typically, you configure the contact tenant terms of usage statement. If the contact tenant is configured with an inactive terms of usage statement, the terms of usage is not configured, or if <empty> is selected in the Terms of Usage drop-down list, CA SDM displays the terms of usage statement for the tenants parent, grandparent, and so on.

If no terms of usage statement is found at any level, CA SDM proceeds with the login. If you configure a tenant with a blank terms of usage statement, CA SDM proceeds with the login and displays the main form.

You can configure terms of usage as follows:

1. Enable multi-tenancy.
2. Configure one or more tenants.
3. Define a terms of usage statement.
4. Update a tenant to use the specific terms of usage statement.

User Interface Impact

This article contains the following topics:

Installing the multi-tenancy feature changes the user interface, depending on the authorization and tenant access that is associated with the user role. The changes affect both tenant users and service provider users.

Tenant Users

If the role of a user is restricted to a single tenant and the user is not an administrator, substitute a custom tenant logo for the default CA Technologies logo on all pages. Substitution depends on whether a logo is defined on the Tenant Detail page for the tenant.

For non-administrator tenant users, the user interface menu items or buttons allowing update or edit are suppressed for public objects. The tenant users are not authorized to update the public object.

Tenant Administrators

List pages for these objects automatically include a Public column specifying whether the list row is public data. In addition, the first element in the search filter is a Public Data drop-down list containing the following selections:

- Include (default)
- Exclude
- Only

A tenant administrator with access to multiple tenants sees a Tenant column on list pages for any tenanted object. This column takes the place of the Public column in lists of tenant optional tables.

Users with View access to a Tenant Group

A user role allows view access to multiple tenants.

- **Untenanted objects**
  Untenanted objects contain only public data. The service provider user is allowed to create or update an untenanted object only if their role has Update Public authorization. If not, the UI suppresses menu items or buttons allowing update or edit, such as the Edit button itself, or the Create New button on a list page. Tenant users cannot update public objects, and these users never see an Edit or Create New button on a list page for an untenanted object.
- **Tenant-required objects**
  Tenant-required objects contain only data associated with a particular tenant. List forms for these objects automatically include a Tenant column after the last link column. In addition, the search filter contains a tenant selector allowing the user to restrict a list to a single tenant.
- **Tenant-optional objects**

Tenant-optional objects contain both public and tenant-specific data. List forms for these objects automatically include a Tenant column (a blank tenant indicates a public object). In addition, the search filter contains both a tenant selector and a Public Data drop-down list (the same one seen by tenant administrators).

NOTE
If tenant required tables incorrectly contains untenanted data in a multi-tenancy system, a public data drop-down list appears in tenant required tables. The following message: “AHD05358 is displayed. There were untenanted active xxx objects at Service Desk startup.”

Users with Update Access to Multiple Tenants

If a user role allows access to multiple tenants, or a service provider user role has Update Public authorization (typical for an analyst that works for a service provider), detail pages change as follows:

- **Untenanted objects**
  Untenanted objects contain only public data. There are no changes to their detail pages for a service provider user with Update Public authorization. If the user is in a role without Update Public authorization, or does not belong to the service provider, read-only pages for untenanted objects are with no Edit button.
- **Existing Tenant-required objects**
  Tenant-required objects contain only data that are associated with a particular tenant. Detail pages for existing tenant-required object show the object tenant as part of the standard page header.
- **Tenant-optional objects**
  Tenant-optional objects contain both public and tenant-specific data. The detail page for these objects depends on whether the user belongs to the service provider and is in a role with Update Public authorization:
  – If a service provider user role has Update Public authorization, the detail page is the same as that for tenant-required objects.
  – If the user role does not have Update Public authorization, or the user does not belong to the service provider, detail pages for public objects do not have an Edit button. Other detail pages are the same as those for tenant-required objects.

Knowledge Management Impact

This article contains the following topics:

The impact of multi-tenancy on your knowledge environment depends on the tenant restrictions that are placed on users:

- **Tenant Users**
  Substitutes the logo of the tenant for the default, if the role is restricted to a single tenant.
- **Tenant Administrators**
  This option enables administrators to view both public and tenant-specific data. List pages for these objects automatically include a Public column specifying whether or not the list row is public.
  When searching for knowledge, the filter contains a Public Data drop-down list with selections of Include (default), Exclude, and Only.

NOTE
Objects such as approval process templates, categories, documents, files and forums are tenant optional.

Knowledge Categories and Documents

Knowledge Documents and Knowledge Categories are both tenant optional. Consider the following information for tenant optional and public objects:

- Public Knowledge Documents can only be added under public Knowledge Categories.
- Public Knowledge Categories can only be added under public categories.
- Tenant categories can be added under public categories and under tenant categories.
- Only tenant documents can be added under tenant categories.
- Public and tenanted documents can be added under public categories.

NOTE
The Cut/Copy/Paste category functionality is allowed only if the source and destination have the same tenant, or if the destination is public.

Repositories are defined as tenant optional, so the administrator can create different repositories for different tenants.

NOTE
Embedded images are allowed only when the document and image are set to the same tenant. Attachment Folders, Attachments and Attachments to document links are also defined as tenant optional.

FAQ Rating

When viewing the FAQ rating for tenant users, consider the following information about public documents:

- Public documents are viewed by a larger audience than tenant users.
  The FAQ rating of public documents is higher than a tenant-specific document.
- Each tenant has different needs, so usage patterns are different between tenants.

The Top Solutions on the CA SDM home page displays the top five public documents, as well as a tenant's top five documents.

You can configure the Top Solutions by navigating to Knowledge, Solution Survey, FAQ Settings on the Administration tab.

Knowledge Report Card

The Knowledge Report Card allows analysts and administrators to view various metrics such as document creation, publication, hits, and votes. This report is for a predetermined time period, per analyst, category, and organization.

When using the Knowledge Report Card to provide information for a role having single tenant access, the data is limited by the tenant criteria.

Create a Remote Reference

You can configure remote references to execute external programs from the CA SDM application. CA SDM lets you create a remote reference to an external program.

Follow these steps

1. Select Service Desk, Application Data, Remote References on the Administration tab.
   The Remote References List page appears.
2. Click Create New.
   The Create New Remote Reference page appears.
3. Fill in the remote reference fields as appropriate.
4. Click Save.
   The remote reference definition is saved and the Remote Reference Detail page appears.

Remote Reference Fields

You can use the fields on the Remote Reference pages to define, search, and edit Remote Reference definitions.

NOTE

All search fields that allow text entry support use of the % wildcard character.

The following fields require explanation:

- **Function Access**
  Specifies the functional area where the remote reference belongs. For example, if the reference pertains to administrative functions, select Admin. If the reference pertains to notifications, select Notify.
- **Architecture Type**
  Specifies the appropriate operating environment for the remote reference (Windows or UNIX).
- **NT Server, UNIX Server, or UNIX Client Exec Command**
  Specifies the character string that executes the remote reference on a Windows server or a UNIX operating environment.

  NOTE
  You can click the More icon to display the Additional Search Arguments field. This field is intended only for expert users who understand SQL and Majic. You can enter a SQL WHERE clause in this field to specify an additional search argument.

Audit Log List

The Audit Log List displays information regarding changes to the issue, change order, requests, and data partition tables. The audit log captures the user ID and an associated day/date/time stamp. The log also records the before and after values of the operation performed (update or insert). You can search the Audit Log List, and can use the results for report generation.

The Audit Log is automatically installed. Enable it by installing the Audit Log options with the Options Manager:

- audit_ins
- audit_upd (see Audit Log Options)

Generate Activity Log for Custom Fields

When you add a custom field, you can set up an activity log for reporting and auditing.

Follow these steps:

1. Log in to CA Service Desk Manager, as an administrator.
2. Navigate to **Administration**, **Notifications**, **Activity Associations**, and create an activity association.
   For example, create an activity association for a field: string1 in the call_request object
   - **Symbol** = string1
   - **Code** = string1
   - **Object Type** = Request/Incident/Problem
   - **Object Type Attribute** = string1
   - **Activity Type** = Field Update
   - **Log Me** = Yes
3. Open Web Screen Painter Schema Designer and perform the following tasks:
   - Navigate to the *cr* object.
   - Click the *string1* field.
   - Click **Advanced** in the **Properties** section on the right.
   - In Site-Defined UI_INFO, type: *+AUDITLOG()*
   - Click **Save**.
   - Navigate to **File**, **Save-and-publish**.

NOTE

To use the string1 field on an incident or request, use Web Screen Painter to add the field to the *detail_in* or *detail_cr* form. Now, create a ticket and save it, then edit it again and update the form field that is linked to the string1 field, and save. You can now see an activity log entry generated for the update of that field.

4. Exit the Schema Designer and Web Screen Painter.
5. Stop the CA Service Desk Manager services.
6. Execute **pdm_publish**.
7. Start the CA Service Desk Manager services back up.

Define Form Groups

Form groups define the sets of pages in the web interface that are available to a role. Each role can have one form group. Users can display only the web pages that are included in the form group that is assigned to their role.

You can create a form group to specify which forms are visible to the members of a group.

Follow these steps

1. From the Administration tab, navigate to Service Desk, Form Groups.
   The Form Group List page appears.
2. Click Create New.
   The Create New Form Group page appears.
3. Fill in the following fields:

- **Symbol** -- A unique identifier for the form group.
- **Record Status** -- Indicates whether the form group is active or inactive.
- **Description** -- Provides a detailed description of the form group. Identifies the forms that are contained in the group or the roles that would use this form group.

SOAP Web Services Policy

This section contains the following articles:

- Create a SOAP Web Services Error Type
- Create a SOAP Web Services Policy

Create a SOAP Web Services Error Type

CA SDM SOAP Web Services provide a defined set of default error types, which are created for*every* policy. The default types that are designated as *internal* error types, can be deactivated but cannot be deleted. Use the Web Services Access Policy Detail page to see the default error types when a policy is created.

CA SDM provides the following default error types:

The following information describes each internal error type:

- **ACCESS_ERROR**
  Indicates that the system failed to connect to or find a resource, such as a file or website.
- **EXCEPTION_FATAL**
  Indicates that the application is shutting down unexpectedly.
- **EXCEPTION_RUNTIME**
  Indicates that the application code encountered an exception.
- **LOGIN_ERROR**
  Indicates that the operator failed to gain access to the application.

You can create SOAP Web Services error types.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps

1. Select SOAP Web Services Policy, SOAP Error Types on the Administration tab.
   The SOAP Web Services Error Type List page appears.
2. Click Create New.
   The SOAP Create New Web Services Error Type page appears.
3. Fill in the SOAP Web Services Error Type fields as appropriate.
4. Select the Duplicate Handling tab to define the actions to take if the system detects an identical ticket that exists in the system. Select one of the following options:
   – Create the ticket and ignore duplicates. This option is the default action.
   – Do not create a ticket; add an activity log to the existing duplicate.
   – Do not create a ticket; add an entry to the CA SDM standard log.
   – Create a ticket and attach it as a child to the duplicate.

   A ticket is considered a duplicate when all following statements are true:
   – At least one ticket of the same type (request, issue, change order) and is ACTIVE.
   – The existing ticket was created using the web service.
   – The existing ticket was created with the same Policy and Error Type as the potential new ticket.
   – The create date of the existing ticket is within a specified threshold (for example, it was opened less than two days ago). This threshold is specified in the Maximum time interval for searching for duplicates field. Use the format of: HH:MM:SS.
5. Select the Return Data tab to specify data that indicates an action to take or a message displayed to the user.
6. Click Save.
   The SOAP Web Services error type is created.
   The error type definition is saved and the SOAP Web Services Error Type Detail page appears.

SOAP Web Services Error Type Fields

The following fields require explanation:

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

- **Code**
  Specifies the unique code that identifies the error type to the system. This field is required.
- **Owning Policy**
  Specifies the web services access policy that owns this error type.
- **Default**
  Indicates that this type is the default error type for the policy. Only one active default error type is allowed. If this option is selected while creating an error type, the default setting of the current default error type is removed.
- **Internal**
  Indicates whether this type is a default error type.
- **Ticket Template Type**
  Specifies the type of ticket that is created when this error is reported.
- **Ticket Template Name**
  Identifies the template to use to create a ticket when this error is reported.

NOTE
The owning policy contact is the end user of the ticket.

Create a SOAP Web Services Policy

To minimize web services ticket flooding, and to maintain the stability of the server, CA SDM SOAP Web Services uses an Access Control and Management system. Handles excessive service activities by trusted user applications resulting from programming errors or exceptions. This system also works as a barrier and controls access to CA SDM SOAP Web Services from malicious attackers.

NOTE
CA SDM provides a default access policy that has a code of DEFAULT. The default access policy contains no access restrictions and is only applied to sessions authenticated through username and password.

You can create SOAP web services policies to control access from SOAP web services applications.

NOTE
If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps

1. Select Administration tab, SOAP Web Services Policy, SOAP Policies.
2. Click Create New.
3. Fill in the fields as appropriate.
4. Select the Access Control tab to define the number of operations permitted per hour for each web services category.
5. Enter a number in each operation counter field to represent the number of operations that are permitted per hour.

   NOTE
   The default value of -1 in any operation counter indicates that no restrictions apply to the corresponding operation. A value of 0 (zero) indicates that the corresponding operation is not allowed.
6. Select the Error Types tab to review the error types in effect for the web services policy.
7. To add an error type, click Add an Error Type.
8. Click Save.
   The policy definition is saved and the SOAP Web Services Policy Detail page appears.

SOAP Web Services Policy Fields

The following fields require explanation:

- **Code**
  Specifies the unique code that identifies the access policy to the system. This field is required.
- **Proxy Contact**
  Specifies the contact to use for all web service operations and security. You can enter a contact name directly into this field, or click the search icon to select the contact name.
- **Default**
  Indicates that this policy is the default policy. Only one active default policy is allowed. If you select this option when creating an access policy, the default setting of the current default policy is removed.
- **Has Key**
  Indicates whether a public key as been associated with this policy. This field is updated when a public key is associated with a policy using the pdm_pki utility and cannot be changed.
- **Allow Impersonate**
  Allows the policyholder to invoke the impersonate() web services method and create a web services session. Additional access authentication is not performed when creating the session. However, only when the grant_level of

the new user access type is less than or equal to the access_level of the proxy user access type, can this method be successfully called.

NOTE
If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

Manage Service Type and Service Type Events

This article contains the following topics:

Service types identify the levels of support service provided to or contracted by your service desk customers.

Create a Service Type

You can also use service types to categorize customers.

NOTE
If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps

1. Select Service Desk, Service Types on the Administration tab.
   The Service Type List page appears.
2. Click Create New.
   The Create New Service Type page appears.
3. Fill in the service type fields as appropriate.
4. (Optional) Use service type tabs to add events to the service type.
5. Click Save.
   The service type definition is saved and the Service Type Detail page appears.

Service Type Fields

The following fields require explanation:

- **Symbol**
  A unique identifier for this service type. Use the symbol to identify the service type. For example, to set up a service type that requires a response within an eight hour time period, assign the symbol 08_hr_resolution.
- **Ranking**
  Service types can be associated with various objects, such as contacts, organizations, categories, and priority codes. If different service types are associated with a ticket, the one that is used depends on the value in this field. The one with the lowest number has the highest priority. For example, if an issue is opened and the Affected End-User contact has a 12-hour resolution service type, which is ranked as 2, while the priority code has a four-hour resolution service type and is ranked as 1. The service type for the issue is four hour resolution.
  Enter a value of one or greater; values of 0 or negative numbers are not allowed.
- **Workshift**
  Specifies the dates, days, and hours when the service type is in effect. Usually coincides with your service desk is operating time period (for example, 24 Hours, Regular, or Non-Business Hours).
  The following rules apply to a workshift:

- If you apply a workshift to a service type, stop and restart the service for the workshift to take effect immediately.
- If a workshift for a service type has been specified, but one has not been specified for an event, the service type workshift is in effect.
- If a workshift for an event has been specified, but one has not been specified for a service type, the service type workshift is ignored.
- If a workshift for event and for service type have been specified, the service type workshift is ignored.

NOTE

The C**reate Service Type**, **Workshift** option will work when the option **sla_workshift_override** is installed. Navigate to CA SDM **Administration**, **Options Manager**, **Request-Change-Issue**, **sla_workshift_override**. Verify if the status= Installed and you have a workshift defined for your events on a Service Type.

- **Timezone**
  The time zone for the service type. You can enter the time zone directly in this field, or click the search icon to select the time zone from a list.
  This time zone is used for triggering events in the system if the Use End-User Time Zone option is not selected.
- **Use End User's Timezone**
  Select this option if you want to use the timezone that is defined for the affected end user on a ticket for triggering events in the system.
- **Violation Cost**
  The cost that is incurred if the service type time limit is violated.

NOTE
If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

Service Type Tabs

The following tabs are available on the Create Service Type, the Service Type Detail, and the Update Service Type pages:

Requests

- **Requests**
  Allows you to add service type events to request tickets. The event object type must be Request.
- **Request Tasks**
  Allows you to add service type events to Request, Incident, and Problem workflow tasks. The event object type must be Request/ Incident/ Problem Workflow Task.
- **Request Targets**
  Allows you to link a Service Type to a Service Target Template for managing Request, Incident, and Problem tickets.

Change Orders

- **Change Orders**
  Allows you to add service type events to change order tickets.
- **Change Order Tasks**
  Allows you to add service type events to change order workflow tasks. The event object type must be Issue Workflow Task (not Workflow Task).
- **Change Order Targets**
  Allows you to link a Service Type to a Service Target Template for managing Change Orders.

Issues

- **Issues**
  Allows you to add service type events to issue tickets.
- **Issue Tasks**

Allows you to add service type events to issue workflow tasks. The event object type must be Issue Workflow Task (not Workflow Task).

- **Issue Targets**
  Allows you to link a Service Type to a Service Target Template for managing Issues.

Attach a Service Type Event

Service types determine the level of support that is given for a ticket. You can associate service type events with tickets. An event that is associated with a service type is attached to records defined with that service type.

Follow these steps:

1. On the Service Desk tab, select Incidents, Problems, Requests, Change Orders, or Issues.
2. Navigate to Assigned or Unassigned, and select the priority for the desired ticket.
3. Select the desired ticket from the list that appears.
4. Choose the Service Type tab, and click Attach Service Type Event.
   The Attach Service Type Event page appears.
5. Click Service Type Event.
   The Event Search page appears.
6. Complete one or more of the search fields, and click Search.
   The Event List displays the events that match your search criteria.
7. Select the desired event.
   The Attach Service Type Event page appears.
8. Click OK.
   The Delay Time field appears. The delay time indicates the interval of time after which the event occurs.
9. Specify the delay time as hh:mm:ss (hours, minutes, seconds).
   For example, if the event is attached to the incident at 2:20, and the Delay Time is 01:00:00 (one hour), the event occurs at 3:20.
10. Click OK.
    The ticket detail page appears with the event listed on the Service Type tab.

Create Service Type Event

You can create service type events for requests, change orders, change order tasks, issues, and issue tasks.

Follow these steps

1. From the Service Type Detail page, select the appropriate tab (Requests, Change Orders, Change Order Tasks, Issues, or Issue Tasks).
   The Event List for the ticket type appears.
2. Click Add Service Type Event.
   The Create New Service Type Event page appears.
3. Fill in the fields on this page, including:
   - **Event**
     The name of the event to which this service type event is attached. An event is a procedure that is automatically followed after a certain amount of time has elapsed. Events consist of checking for conditions and performing actions that are based on the results of those conditions. Attach events to tickets to make it easier to monitor and take measures to close these tickets.
     For example, an event may send a message if a priority 1 issue is not resolved in an hour. The event begins when an hour has elapsed. First, the system checks for the condition, which is "open issues with the priority of 1." If the condition is true, the system performs an action, which is sending the message.
   - **Delay Time**
     The interval of time after which the event occurs. Specify the delay time as hh:mm:ss, For example, if an event is attached to an issue at 2:20, and the Delay Time is 01:00:00 (one hour), the event occurs at 3:20.

4. Click Save.
   The tab definition is saved and the Tab Detail page appears.

Delay or Resume a Service Type Event

You can delay and resume service type events.

Follow these steps:

1. Select the desired ticket from the list page on the Service Desk tab.
   The ticket detail page appears.
2. Select the Service Type tab.
   A list of any service type events that have been added to the ticket appears.
3. Click Delay or Resume.
   The Reason page appears.
4. Enter the reason, and click OK.
   The ticket detail page displays the Status of the event.

Create a Service Target Template

A service target is the estimated time frame for completing and resolving a ticket. To minimize Service Level Agreement (SLA) violations, create service target templates for analyzing each stage of ticket resolution. *A* service target template *is* a set of service targets with conditions for managing and prioritizing tickets. Each template identifies a condition, deadline, and relative cost for failing to complete the service target on time.

You can apply service target templates to requests, incidents, problems, change orders, and issues. When you link the service target template to a service type for a ticket, the service target automatically attaches only to new tickets.

On new tickets with service targets, analysts can use the Remaining Time and Violation Cost fields to prioritize ticket resolution. During ticket creation, a service type assigns one or more service targets to track each stage of the ticket resolution. Each time the ticket opens or changes, the active service target evaluates the condition. If the condition is met, the ticket and activity log show the actual completion time.

> **NOTE**
> If you are also using priority calculation, fields such as Priority automatically update and can affect the way service targets attach to tickets. After you configure a service target template, test your service targets on the intended ticket type.

You can create a service target template to measure service targets for requests, incidents, problems, issues, or change orders. After you create the template, you can link it to a service type.

> **NOTE**
> If multi-tenancy is installed, specify the tenant in the Tenant field.

Follow these steps:

1. Select **Service Desk**, **Service Target Templates** on the **Administration** tab.
2. Click **Create New**.
3. Select a ticket type from the **Object Type** field, and click **Continue**.
4. Complete the following fields as appropriate:
   - **Name**
     Defines a descriptive identifier for the service target.
   - **Object Type**
     Identifies the ticket type for this service target.
   - **Target Duration**

Specifies the maximum amount of time for service target completion in the hh:mm:ss format.

- **Workshift**
  Specifies the workshift which contains a range of working hours that are used in time calculations for a service target, for example, M-F 08:00-17:00.
- **Cost**
  Defines information such as the estimated cost associated with missing the service target.
- **Record Status**
  Indicates whether this target is active or inactive.
- **Condition**
  Specifies the name of the condition macro or site-defined condition. Evaluates the ticket data and determines whether the service target is met.
- **Required Outcome**
  Specifies a True or False value that indicates whether the service target is complete.
- **Allow Set Actual**
  Specifies whether to allow users to set the date and time for a service target.
- **Allow Reset Actual**
  Specifies whether to allow users to reset the date and time for a service target.

5. Click **Save**.
   The service target template is created.

Create Log Interval Configuration

Interval Logging allows you to collect resource usage data from the CA SDM servers. The collected interval log is used to analyze and troubleshoot any performance or memory-related issues in the CA SDM servers. You can also share the collected data with CA Support Online to help identify and resolve CA SDM server problems.

> **NOTE**
> For Windows, install and add the pslist.exe tool to the windows OS %PATH% environment variable on each of the CA SDM servers.

To collect the usage data on the CA SDM servers, add the servers to the Interval Log Configuration list. Select the type of data you want to collect from each server, and change the logging options for the servers.

For example, if you select only the CPU usage option for a particular server, the utility collects the server CPU usage data.

Follow these steps:

1. Select **System**, **Interval Logging** from the **Administration** tab.
2. Click **Create New** to add an interval logging configuration.
3. Complete the fields as appropriate for the configuration.
   - **Server Name**
     Specifies the server for which you want to collect the log data. Use the Search icon to view the list of servers you can add for the interval logging.
   - **Recurrence Interval**
     Specifies the time interval during which the log data is collected for the server in hh:mm:ss format. For example, if the recurrence interval is set to 3 minutes, the log data is collected for every 3 minutes.
     - **Default: 3 minutes**
     - **Minimum: 2 minutes**
   - **Enabled**
     Indicates whether logging is enabled or disabled for the server. If enabled, log data is generated for the server.
   - **Record Status**
     Indicates whether the interval logging configuration is active or inactive.
   - **Scheduled Start Date**

Specifies the start date and time for collecting the log data. If you do not enter a start date, interval logging starts immediately till the configuration is active, enabled, or scheduled end date.

- **Scheduled End Date**
  Specifies the end date and time for collecting the log data. If you do not enter an end date, interval logging continues to run till the configuration is active and enabled.

4. Select the appropriate interval log you want to collect from the server. For example, select CPU Usage if you want to collect the log for CPU usage data.
   - **CPU Usage**
     Collect the CPU usage statistics for the server by executing the pslist - x on Windows, or “ps” on Unix.
   - **Memory Usage**
     Collects the memory usage data for the server by executing the pslist - m on Windows, or “ps” on Unix.
   - **Network Status**
     Collects the information of all active connections and network statistics by executing the netstat /b or /a.
   - **Task List**
     Collects the application and services information for all the tasks running on the server.
   - **Web Session Counts**
     Collects the CA SDM sessions and user statistics for the web engine processes by executing the pdm_webstat command.
   - **Server Status**
     Collect the information about all CA SDM daemons or processes on the server by executing the pdm_status command.
   - **DB Report**
     Collects the information of database record types by executing the db_report command.
   - **Virtual DB Info**
     Collects the information that is related to queued database requests by executing the pdm_vdbinfo command.
   - **List Server Connections**
     Collects the information on active connections for the server by executing the pdm_listconn command.
   - **List Slump Processes**
     Collects the information about slump connections and processes by executing the slstat command.

     NOTE
     Depending on the selected server type, the interval logging options are enabled or disabled for the server.
5. Click **Save**.
   The server is configured for interval logging.
6. (Optional) Repeat Steps 1-5 to create more interval log configurations.

How to Manage Contact Groups

This article contains the following topics:

A group is a collection of contacts that represent a specific area of responsibility within your service desk. Groups let you assign responsibility for resolving a ticket.
For example, you can have a Dallas Human Resources group responsible for dealing with personnel issues in the Dallas office. You can assign employee issues and request redressal from the Dallas Human Resources group.
You can associate request areas, locations, and a work shift with a group. You can also determine how contacts within a group can accept the automatic assignment of a request.

Create a Group

A group is a collection of contacts that represent a specific area of responsibility.

NOTE

: If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. Select Security and Role Management, Groups on the Administration tab.
2. Click Create New.
3. Complete the fields as appropriate for the group.
4. (Optional) Use the controls available on the tabs at the bottom of this page to configure the group.
5. Click Save.
   The group record is saved and the Group Detail page appears. The following buttons are now available for configuring the group:

- – **Update Environment**
  The Configuration Item/Asset Search window for the group to specify search criteria for assets. Click Search and on the Environment Update window add and remove assets for this group.
  – **Update Members**The Contact Search window to specify search criteria for the contacts for a group. Click Search and on th eContacts Update window to add and remove contacts for this group.

Set Up Group Notification Parameters

You can define the contact information and the method you want to use to notify a group.

Follow these steps:

1. On the Group Detail page, select the Notification tab.
2. Enter the appropriate contact information, such as Telephone Number, Fax Number, Email Address, etc.
3. Select the notification method you want to use for each message urgency level for this group (low, normal, high, and emergency).
   For example, you may want to notify this group using the Email method for normal level activities, but you may want to use the Pager_Email notification method for emergency level activities.
4. Select the workshift that is valid for each notification urgency level.
   For example, you may assign a Regular workshift (five-day week, eight-hours a day) to the normal level notification, but a 24 hour workshift to the emergency level notification.
5. Click Save to save the notification parameters.

Assign a Group to a Location

Locations precisely identify a specific physical place, such as the address of a particular company, or an office address. You can assign contacts, groups, and configuration items to a specific location.

Follow these steps:

1. On the Group Detail page, select the Address tab.
2. Enter the location where you want to assign the group, or click the search icon to select the location.
   The address information for the location is automatically filled in on the Address tab.
3. Click Save to assign the group to the selected location.

Assign a Group to an Organization

Organizations describe internal departments and divisions or external companies that can be assigned to tickets, CI classes, groups, and contacts.

Follow these steps:

1. On the Group Detail page, select the Organizational Info tab.
2. Enter the field values as follows:
   - **Functional Organization**
     The organization for which the group works.
   - **Department** The department where the group works.
   - **Administrative Organization**
     The organization to which the group reports for accounting purposes.
   - **Cost Center**
     The code to which this group's expenses are charged.
   - **Vendor** The name of the vendor associated with this group.
   - **Supervisor** The name of the person to which this group reports.

     NOTE
     You can enter the values directly in the fields or click the search icon to search for the desired field value.
3. Click Save to save the organizational information to the group detail record.

Set Up a Group Environment

A group's environment consists of the equipment, software, and services they use.

Follow these steps:

1. On the Group Detail page, select the Environment tab.
2. Click Update Environment.
3. Enter the search criteria to display the desired configuration items and click Search. For more information, see Search Configuration Items.
4. From the list on the left, choose the configuration items you want to add to this group's environment. To choose multiple items, hold down the CTRL key while clicking the left mouse button.
5. When you have selected all the configuration items you want, click the double right-arrow button.
   The selected configuration items move to the Group Environment list on the right.
6. Click OK.
   The Group Detail page displays with the selected items listed on the Environment tab.

Group Remarks

You can enter notes regarding a group on the Remarks tab. These notes can be anything related to the group's role, or any other pertinent information that needs to be saved to the group detail record.

Follow these steps:

1. On the Group Detail page, select the Remarks tab.
2. Type the remarks in the Contact Notes field.
3. Click Save to save the remarks to the group detail record.

Assign Members to a Group

A group is a collection of contacts that represent a specific area of responsibility. Adding members (contacts) to your group lets you assign responsibility for resolving a ticket to several individuals who share this responsibility.

Follow these steps:

1. On the Group Detail page, select the Members tab.
2. Click Update Members.

3. Enter the search criteria to display the desired contacts and click Search. For more information, see Search Contacts.
4. From the list on the left, choose the contacts you want to assign to this group. To choose multiple items, hold down the CTRL key while clicking the left mouse button.
5. When you have selected all the contacts you want, click double right-arrow button.
   The selected contacts move to the Members list on the right.
6. Click OK.
   The Group Detail page displays, with the selected contacts listed on the Members tab.

Group Auto Assignments

The Auto Assignment tab lets you establish a relationship between change category, issues categories, and locations. You can automatically assign tickets (requests, change orders, and issues) using the request areas or category to eligible group members.

> **WARNING**
> The system administrator must enable automatic assignment for each request area and category individually. Listing them on the Group Auto Assignment tab does not enable this function.

Follow these steps:

1. On the Group Detail page, click Edit and select the Auto Assignment tab.
2. Click one of the following buttons:
   - Update Request Areas to select request areas for auto assignment.
   - Update Change Categories to select change categories for auto assignment.
   - Update Issue Categories to select issue categories for auto assignment.
   - Update Locations to select locations for auto assignment.

   The Search page for the selected item displays.
3. Enter the search criteria to display the items you want, and click Search.
   The Update page displays, listing the items that matched the search criteria.
4. From the list on the left, choose the desired items. To choose multiple items, hold down the CTRL key while clicking the left mouse button.
5. When you have selected all the items you want, click double right-arrow button.
   The selected items move to the Assigned list on the right.
6. Click OK.
   The Group Detail page displays, with the selected items listed on the Auto Assignment tab.

Assign Roles to a Contact

Assigning roles to a contact limits the contact's access to functional areas based on the definition of the role they are using. For example, assigning the Customer Service Representative role to a contact allows the contact to access all functionality related to managing customer issues. Assigning the Knowledge Administrator role allows the contact to access all functionality related to configuring and monitoring the knowledge management process.

To assign roles to a contact

1. Select the Roles tab on the Contact Detail page.
   The Role List appears.
2. Click Update Roles.
   The Roles Search page appears.
3. Enter the search criteria to display the roles and click Search.
   The Roles Assigned Update page displays, listing the roles that matched the search criteria.
4. From the list on the left, select the roles you want to assign to this contact. To select multiple items, hold down the CTRL key while clicking the left mouse button.

5. When you have selected all the roles you want, click

   The selected roles move to the Roles Assigned list on the right.
6. Click OK.
   The Contact Detail page appears.
7. Click Save.
   The Contact Detail page displays the selected roles listed on the Roles tab.
8. Select the role you want to be the default for this contact when they log in, and then click Set Default Role.
   Note: The contact role default takes precedence over the default role defined on Access Type pages.

CA SDM Environment Promotion

CA Service Desk Manager Environment Promotion lets you promote configuration, customization, and object data changes between the source and target systems with minimal manual intervention. As an administrator, you can promote any configuration, customization, or object data changes between the development, test, pre-production, and production systems. For example, you can promote modified forms, added tables, columns, spel, majic files, object data to the production system. Based on the type of change (Configuration/Customization or Object Data), the environment promotion is initiated either through command-line interface or CA SDM User Interface.

How CA SDM Environment Promotion Works

CA SDM Environment Promotion works in the following two-phased manner:

1. Promotion of the configuration and customization of files
   For more information, see CA SDM Configuration and Customization
2. Promotion of the object data
   For more information, see Object Data Promotion

All modifications made to the files, .htmpl, .mods, .spl, .maj, and so on, are directly stored in the *NX_ROOT/site/mods* folder. These files are copied from the *NX_ROOT/site/mods* folder and creates an export package. Any modification made to add tables and columns using the Web Screen Painter are stored in the database. These modifications are captured in the form of .dat files as part of the export package.

Prerequisites

Verify the following prerequisites before performing any environment promotion processes:

- To promote changes from the source to target system, ensure that the source and target system are on the same release version.
- Close all the SDM-related files.
- Ensure that the CA SDM services are running before you perform Export or Import.
- When the environments are tenanted, both source and target systems must have the same tenants. The environment promotion does not create any tenants.
- To execute the export or import process, the user must have the operating system administrator privileges to create files and folders.
- The user must have access to Web Screen Painter to successfully execute the import process.
- The user must have read and write access to the *$NX_ROOT\site* folder.
- Do not change the name of the exported zip file. If the exported zip file name is changed, the import process fails.
- For configuration and customization, the export or import process must be executed on the same type of servers, database, and operating systems.
  For example:

  - If an export is performed on a primary server, import of the package is allowed only on the primary server.
  - If an export is performed on a background server, import of the package is allowed only on the background server.
  - If an export is performed on a secondary server or application server, the import of the package is allowed only on the secondary server or application server.
    For more information about additional scenarios, see Approaches in Typical Environment Promotion Scenarios
- Administrators are required to copy the .htmpl files to the secondary servers or application servers manually or modify the version control file to propagate the .htmpl files on the secondary servers or application servers.
- Ensure that the Apache Tomcat service is running before you perform database export.
- On Non-windows system, ensure that you enable X-Windows or an equivalent session to launch the browser for Web Screen Painter login.
- On Non-windows system, set the CA SDM bin path and CA SDM LIB path environment variables before executing the export or import process.
  For example:

```
For AIX:
export PATH=/opt/CA/ServiceDeskManager/bin:$PATH
export LIBPATH=/opt/CA/ServiceDeskManager/lib:$LIBPATH

For Solaris and Linux:
export PATH=/opt/CA/ServiceDeskManager/bin:$PATH
export LD_LIBRARY_PATH=/opt/CA/ServiceDeskManager/lib:$LD_LIBRARY_PATH
```

TIP

Performing environment promotion affects the CA SDM application performance and also restarts the services. We recommend that you perform the CA SDM Environment Promotion operations during off peak hours.

Approaches in Typical Environment Promotion Scenarios

Scenario 1: To perform environment promotion when the number of servers vary in the test and production systems.

On the source system, there is one primary server and one secondary server. Whereas, on the target system, there is one primary server and multiple secondary servers.

The environment promotion is successful when the configuration on all the servers is the same.

CA SDM Configuration and Customization

To promote your configuration and customization changes from the source to the target systems, perform the following tasks:

1. Verify the promotion prerequisites
2. Export the configuration and customization
3. Import the configuration and customization

You can use the following SDMP command to execute the export, dryrun, or import processes:

```
sdmp -a <export/dryrun/import> -p <package name> -version
```

-a denotes actions such as export, dryrun, or import.

-p denotes the package name. This parameter is mandatory for dryrun and import processes.

-version displays the version details of CA SDM, Java, OS, Tomcat, and Database type.

-h denotes Help.

The history file *SDMP_HISTORY.HIS* located at *NX_ROOT*, captures the details of the operations performed when promoting configuration and configuration changes.

This section contains the following topics:

- Export
- Import
- How to Customize the SDMP CSV File
- How CA SDM Environment Promotion Utility Works in Advanced Availability and Conventional Mode

Export

CA SDM Environment Promotion lets you export the configuration and customization changes from the source system. These configuration and customization changes are captured on the source system, which you can import on the target system using the CA SDM Environment Promotion import process. The export process can be run on the primary, secondary, background, and application servers.

WARNING

Warning: Do not initiate multiple export processes on your system.

Before initiating the export process, perform the following tasks:

- Set the **export_import_path, disk_space, timeout** property values in the SDMP Config properties file. For more information, see SDMP Config Properties File.
- On Non-Windows, define the following environment variables:
  - Solaris and Linux: define the PATH and LD_LIBRARY_PATH environment variables.
  - AIX: define the PATH and LIBPATH environment variables.
    For example:

```
For Solaris and Linux:
export PATH=/opt/CA/ServiceDeskManager/bin:$PATH
export LD_LIBRARY_PATH=/opt/CA/ServiceDeskManager/lib:$LD_LIBRARY_PATH

For AIX:
export PATH=/opt/CA/ServiceDeskManager/bin:$PATH
export LIBPATH=/opt/CA/ServiceDeskManager/lib:$LIBPATH
```

Export Configuration and Customization Changes

To export the configuration and customization changes, execute the following export command:

```
Windows: sdmp -a export
Non-windows: sdmp.sh -a export
```

After initiating the ***sdmp -a export*** command, the export process validates the CA SDM release version and captures information about Java version, Tomcat Version, host name, OS name and version, CA SDM locale, database type, and server type. Files are copied as per the entries made in the *sdmp_files.csv* file. For more information, see How to Customize the SDMP CSV File

WARNING

Warning: After export, do not rename the exported package name.

After successful completion of the export process, package and summary report files are generated. To view these files, navigate to *<export_import_path>/Promotion/Export* as specified in the SDMP Config Properties file. If the *export_import_path* property is empty, navigate to *NX_ROOT/Promotion/Export* to view the exported package and summary report. For more information, see SDMP Config Properties File.

- Package

  ```
  sdmp_export_yymmdd_hhmmss.zip
  ```

  For example: **sdmp_export_151008_122725.zip**
- Summary Report

  ```
  sdmp_export_summary_yymmdd_hhmmss
  ```

  For example: **sdmp_export_summary_151008_122725.txt**

  NOTE

  You can skip the customization and promote only the configuration changes by setting *skip.validation.wsp* as *true*. For more information, see SDMP Config Properties File.

If you encounter any errors or failures during the export process, verify the *sdmp.log* file in the *NX_Root/log* folder.

Import

For the import process, CA SDM 14.1 or CA SDM 14.1.01 versions must have similar installation configuration on both the source and target systems.

Before you initiate the import process, verify the following:

- Verify Prerequisites
- Manually copy the exported package from the source system to the target system and place this package under the *NX_ROOT/Promotion/Import* folder. However, if you have set *export_import_path* in the SDMP config properties file, create *the Promotion/Import* folder under *export_import_path* and place the exported package.

  WARNING

  Warning! Do not rename the exported zip file.
- Ensure that you have set the timeout value in the SDMP config properties file as per your environment. For more information, see SDMP Config Properties File
- On a Non-windows system, set PATH and LD_LIBRARAY_PATH environment variables.
  For example:

  ```
  For AIX:
  export PATH=/opt/CA/ServiceDeskManager/bin:$PATH
  export LIBPATH=/opt/CA/ServiceDeskManager/lib:$LIBPATH
  For Solaris and Linux:
  export PATH=/opt/CA/ServiceDeskManager/bin:$PATH
  export LD_LIBRARY_PATH=/opt/CA/ServiceDeskManager/lib:$LD_LIBRARY_PATH
  ```

- (Optional) Set the **export_import_path**, **disk_space**, **timeout** property values in the SDMP Config properties file. For more information, see SDMP Config Properties File.
- The user must have access to Web Screen Painter to successfully execute the import process.

How to Execute the Import Process

The CA SDM Environment Promotion performs *pdm_configure* as a part of the import process.

WARNING

Warning: Do not initiate multiple import processes to avoid the target system resulting in an inconsistent state.

TIP

We recommend that you execute the dryrun process before executing the import process.

Follow these steps:

1. Execute the dryrun command:

   ```
   Windows: sdmp -a dryrun -p <exported package>
   Non-windows: sdmp.sh -a dryrun -p <exported package>
   ```

   The dryrun process reports failures, if any.
   For example: **sdmp -a dryrun -p sdmp_import_151008_122725**
   The process generates a dryrun summary report that is located in the *<export_import_path>/Promotion/Dryrun* folder. The summary report contains the number of files that needs to be imported, number of records to be loaded, and failure messages related to read and write access and validation failures (Text fixes, CA SDM version, locale, Java version, server type, and database Type.)

   NOTE

   When you encounter a patch level mismatch error, to fix or ignore the error, and proceed with the import.

2. Create a backup of *NX_ROOT* folder before you proceed with the import.

   NOTE

   If the import fails, restore the application files from the location where you saved the backup.

3. Execute the Import command and follow the on-screen instructions.

   ```
   Windows: sdmp -a import -p <exported package>
   Non-windows: sdmp.sh -a import -p <exported package>
   ```

   For example: **sdmp -a import -p sdmp_import_151008_122725.zip**
   The import process performs the following tasks on the target system:
   - Validates that the CA SDM version, text fixes, locale, Java version, Tomcat version, server type, and database type of the target and source systems are similar.
   - Generates a summary report, which contains the number of files and records imported, in the *<export_import_path>/Promotion/Import* folder.

4. After completing the import process, verify that the changes are reflected on the target system.

   NOTE

   To verify errors or failures that are encountered during the import process, verify the ***sdmp.log*** file in the *NX_Root/log* folder.

5. If any failures are encountered during import, rerun the import process using the same package.
   The import process restarts from the point of failure.

How to Customize the SDMP CSV File

The import process uses the *SDMP_FILES.CSV* file as the input file, which manages the files for promotion. This article explains how to customize the *SDMP_FILES.CSV* file.

WARNING

The .exe and .dll files must not be considered for promotion.

Navigate to *NX_ROOT/site/cfg* directory, edit the *SDMP_FILES.CSV* file, and create entries in a sequential order as listed:

- **path**: Specifies the relative path from NX_ROOT.
- **file**: Specifies the file name or wildcard characters (for example: ***.***,*.xml, eiam*.config).
- **extension**: Specifies the file extension.
- **group**: Specifies the logical group of files for exporting:
  - Onetime - Files that are rarely modified and require administrator intervention to compare the differences and merge manually on the target system. These files are not replaced on the target system.
  - Copy - Copies a specific file.
  - Environment - Files specific to the environment.
  - Recursive - Applicable for folders and sub folders recursively.
  - Reports - Files related to reports.
- **parser:** Controls file processing.
  - Copy: Copies a specific file.
  - Onetime: Files that are grouped as onetime require administrator intervention to compare the differences and merge manually on the target system. These files are not replaced on the target system. These files are available at *export_import_path/Promotion/Import/<import package>/<exported package>*. Select the respective folder. For more information, see SDMP Config Properties File.
    If the export_import_path property value is empty, navigate to the NX_ROOT/Promotion/Export folder to view the exported package and summary report.
  - Recursive: Performs a recursive copy of files and folders.
- **OS**: Specifies the operating system.
  - Both: Applicable for Windows and Non-Windows.
  - Win: Applicable for Windows.
  - Non-Win: Applicable for Non-Windows.

For example:

Path	File	Extension	Group	Parser	OS
/site/mods/interp/	*.*		Recursive	recursive	Both
/pdmconf/	NX.env.tpl	tpl	Copy	copy	Win
/pdmconf/	NX.env.tpl	tpl	Copy	copy	Non-Win
/fig/cfg/	*.fmt	fmt	Reports	copy	Both
/pdmconf/	eiam*.config	properties	onetime	onetime	Both

How CA SDM Environment Promotion Utility Works in Advanced Availability and Conventional Mode

Complete the following tasks before you initiate the export process:

- Verify Prerequisites.
- Set the **export_import_path, disk_space, and timeout** property values in the SDMP Config properties file. For more information, see SDMP Config Properties File.
- For Configuration and customization promotion, the export or import process must be executed on the same type of servers.
  For example:
  - If an export is performed on a primary server, import of the package is allowed only on the primary server.
  - If an export is performed on a background server, import of the package is allowed only on the background server.
  - If an export is performed on a secondary server or application server, the import of the package is allowed only on the secondary server or application server.

NOTE
CA SDM Environment Promotion is not supported on CA SDM Standby servers.

Environment Promotion for Advanced Availability (Background Server)

CA SDM administrators must copy the .htmpl files to the application servers manually or modify the version control file to sync the .htmpl files on the application servers.

Follow these steps:

1. Export the package.
   For more information about how to export the package, see Export.
2. Log in to the CA SDM target system.
3. Copy the exported zip (*sdmp_export_<timestamp>.zip*) to the *export_import_path/Promotion/Import* folder.
   If the *export_import_path* is empty, copy the exported zip to *NX_ROOT/Promotion/Import* folder.
4. Run the **dryrun** command.
   ```
   Windows: sdmp -a dryrun -p <exported_zip>
   Non-windows: sdmp.sh -a dryrun -p <exported_zip>
   ```
5. After successful dryrun, run the import command and follow the on-screen instructions.
   ```
   Windows: sdmp -a import -p <exported_zip>
   Non-windows: sdmp.sh -a import -p <exported_zip>
   ```
   The administrator performs the failover manually when the instructions for performing failover are displayed on the Command Line interface.
6. Log in to the Standby server and run the failover command:
   ```
   pdm_server_control -b
   ```
7. After failover is complete, use the new standby server (old background server). Press **Y** to continue the import process.
8. After successful import, perform failover again on the new standby server to make it as the background server. Run the following command:
   ```
   pdm_server_control -b
   ```

NOTE

If you want to export or import from one application server to another, see Import.

Environment Promotion for Conventional Setup

CA SDM Environment Promotion does not support propagation of files (.htmpl) from the primary server to the secondary server. Either copy the files manually or use version control to propagate the files (.htmpl) to the secondary servers.

Follow these steps:

1. Export the package.
   For more information about how to export the package, see Export.
2. Log in to the CA SDM target system.
3. Copy the exported zip (*sdmp_export_<timestamp>.zip*) to the *export_import_path/Promotion/Import* folder.
   If the *export_import_path* is empty, copy the exported zip to *NX_ROOT/Promotion/Import* folder.
4. Run the **dryrun** command.
   ```
   Windows: sdmp -a dryrun -p <exported_zip>
   Non-windows: sdmp.sh -a dryrun -p <exported_zip>
   ```

5. After successful dryrun, run the import command.

```
Windows: sdmp -a import -p <exported_zip>
Non-windows: sdmp.sh -a import -p <exported_zip>
```

NOTE

To export or import from one secondary server to another, see Import.

SDMP Config Properties File

Edit the *sdmp_config.properties* file to control the behavior of the export or import process. By default, the file is installed at *NX_ROOT/site/cfg/sdmp_config.properties*.

The SDMP configuration file contains the following properties:

export_import_path

This property defines the environment promotion root folder. Set to local path, mapped drive path, or authenticated network shared path, only in the supported formats.

For example: *C:\\data\\work* or *C:/data/work*

\\\\sharedpath\\work or *//sharedpath/work*

If you do not specify a value, the *NX_ROOT* path is used.

disk_space:

This property is applicable for configuration and customization promotion. Ensure that you have enough disk space available on the system to execute the export or import process.

Default: 10 MB

Specify the value in MB.

timeout

CA SDM Environment Promotion uses the CA SDM pdm command utilities. If any command line utility does not respond in the specified time out period, the utility exits.

Default: 600 seconds

Specify the value in seconds.

db_data_disk_space

This property is applicable for object data promotion. Ensure that you have enough disk space available on the system to execute the export or import process.

Default: 10 MB

Specify the value in MB.

skip.validation.tomcat

Indicates if the validation or comparison of source and target Tomcat versions is skipped. If the value is set to True, validation is skipped.

Default: False

skip.validation.java

Indicates if the validation or comparison of source and target Java versions is skipped. If the value is set to True, validation is skipped.

Default: False

skip.validation.wsp

Indicates if the customization changes (WSP) on the source system is skipped. If the value is set to True, customization is skipped.

Default: False

Object Data Promotion

Perform the Export process from Command Line interface or navigate to **Administration**, **System**, **Data Export** page on the CA SDM User Interface to promote customized data from the source to the target system.

- Verify the prerequisites before initiating object data promotion
- Export Object Data from the CA SDM User Interface
- Export Object Data from the Command Line Interface

You can manage the export and the import processes by configuring the property files.

- Import Object Data
- Property Files for Object Data Promotion

> **NOTE**
>
> If CA SDM is tenanted, data from a source tenanted system is promoted to a target tenanted system. CA SDM Environment Promotion does not support creation of tenants.

Export Object Data from the CA SDM User Interface

This article explains the export process when you promote object data from the CA SDM user interface.

Before initiating the export process, perform the following tasks:

- Verify the Prerequisites.
- Set the **export_import_path** and **db_disk_space** values in the SDMP configuration properties file. For more information, see SDMP Config Properties File
- When promoting the *risk_level* and *prptpl* objects, update the following property files located in the *NX_ROOT\pdmconf* \ folder, on the source and target systems:
  - Edit the *dbpromote.uniqkey* file and update the following properties:
    - risk_level.uniquekey=value
    - prptpl.uniquekey=object_type, object_attrname, object_attrval, sequence
  - Edit the *DBImport.properties* file and update the following property:
    - resolve.prptpl.attributes=object_attrval

Follow these steps:

1. Log in to CA SDM user interface with privileged user access.
2. Navigate to **Administration**, **System**, **Data Export**.
3. Specify the **Start Date**, **End Date**, **Last Modified by**, **Active** fields based on your requirement.
   - **S tart Date**: Specifies the earliest modified date of the object data.
   - **End Date**: Specifies the last modified date of the object data
4. Click **Search** to view the list of objects that match the filters.

5. Select the objects that you need to promote from the **Object List** page.
6. Click **Data Export** and enter appropriate description.

   NOTE

   Special characters such as ; : { } / * \ (# + % & ' " are not allowed in the description.
7. (Optional) Click **Data Export/Import History** in the left navigation pane to verify that no other job is in progress before you submit your job.
8. Click **Submit** on the description pop-up.
9. Click **Data Export/Import History** to verify the status and location of the data export package. One of the following status messages may be displayed as per the status of the data export package:

- – **In Progress**: Indicates that the data export process is running.
  – **Complete**: Indicates that the data export process for the selected objects is complete.
  – **Abort:** Indicates that the in progress operation was cancelled using the **Abort** button.

You can view the exported package in the <*export_import_path*>*/Promotion/Export/DB* location on the CA SDM server.
If any errors or failures are encountered during the export process, check the *stdlog* file and the *dbpromote.log* file in the *NX_Root/log* folder.

Export Object Data from the Command Line

Apart from the CA SDM user interface, you can also export the object data from the command line.

Before initiating the export process, perform the following tasks:

- Verify the Prerequisites.
- Set the **export_import_path** and **db_disk_space** values in the SDMP configuration properties file. For more information, see SDMP Config Properties File
- When promoting the *risk_level* and *prptpl* objects, update the following property files located in the *NX_ROOT\pdmconf* \ folder, on the source and target systems:
  – Edit the *dbpromote.uniqkey* file and update the following properties:
    - risk_level.uniquekey=Value
    - prptpl.uniquekey=object_type, object_attrname, object_attrval, sequence
  – *Edit the DBImport.properties file and update the following property:*
    - resolve.prptpl.attributes=object_attrval

Execute any of the following commands to promote the object data changes from the source system to the target system:

```
db_promote export -d <description> -w <whereclause> -i <inputfile>
db_promote export -d <description> -w <whereclause> -f <factoryname>
```

The command line options are described as follows:

-d

(Optional) Specifies the description about the package that is exported.

-w

(Optional) Specifies the *Where* clause to filter objects for export.

WARNING

We recommended that you use the *Where* clause to optimize the search results.

-f

Specifies the factory name of the object.

-i

Specifies the input text file, which lists the object factory names in each row.

-h

(Optional) Denotes help for a command.

```
 For Example:
db_promote export -d "Export categories" -w "last_mod_dt > {unix timestamp}" -i "C:/objectlist.txt"
db_promote export -d "Export categories" -w "last_mod_dt > {unix timestamp}" -f "pcat"
```

After successful completion, the export process generates a zip file in the *<export_import_path>/Promotion/Export/DB* folder. If the *export_import_path* property is empty, navigate to the *NX_ROOT/Promotion/Export/DB* folder to view the exported package and summary report. The report file contains the number of objects that are exported. The exported package and the corresponding report file are shown in the following format:

```
sdmp_db_data_export_yymmdd_hhmmss.zip

sdmp_db_data_export_report_yymmdd_hhmmss.txt

 For Example:
sdmp_db_data_export_151009_140940.zip

sdmp_db_data_export_report_151009_140940.txt
```

Import Object Data

You can import a data object using the Command Line interface.

Follow these steps:

1. Log in to the CA SDM target system as an administrator.
2. Create a backup of the existing database.

   NOTE
   If the import fails, restore the database from the backup location.
3. Manually copy the exported zip file, *sdmp_db_data_export_yymmdd_hhmmss.zip,* from the source system to target system.
4. Execute the following command to validate the object schema (number of xml files, table names, and column names) in the target system and follow the on-screen instructions.

   ```
   db_promote import -v -p <path_to_the_zip_file>
   ```

   For example: *db_promote import -v -p c:/temp/sdmp_db_data_export_151008_102754.zip*
5. The command line options are described as follows:

   -p
   Specifies the import package name.
   -u
   (Optional) Performs the import using the specified user name.
   -h

(Optional) Denotes the help for a command.

-v

(Optional) Validates the schema.

6. Execute the following command to import the object data:

```
db_promote import -p Path_to_the_zip_file
db_promote import -p Path_to_the_zip_file -u <sdm contact userid>
```

For example: *db_promote import -p c:/temp/sdmp_db_data_export_151008_102754.zip*

7. Verify the import status by accessing Administration, System, Data Export/Import History in the CA SDM User Interface and refer the following report files generated at the *NX_ROOT/log* location:
   - **dbimport.dryrun.report.<timestamp>**: The DryRun report validates all the object schema. Schema related issues, if any, are listed here.
   - **dbpromote.log**: This file contains logs for the export/import operation.
   - **Failure_Report_<package ID>_<timestamp>**: This file contains object data failures that were encountered during import.
   - **Imported_Object_Report_<package ID>_<timestamp>**: This file contains the successfully imported object data related information.
   - **<export_import_path>/Promotion/Import/DB/Import_report_<package name>_<timestamp>**: This file contains the consolidated import summary.

     NOTE

     The **DBImport.properties** file is available *in the NX_ROOT/pdmconf/* location, which contains all configurable variables for object data import utility (db_promote). Configure this file to customize as per your environment requirements. This property file includes the following:

     - **temp.work.area**
       Specifies where import extracts the exported package. If it is empty or undefined, it uses the system temp folder provided by jvm.
       For example, temp.work.area=c:\\mywork\\temp
     - **domsrvr.default=domsrvr**
       Specifies the default dom server used for the data import process. Update the value if there are multiple dom servers.
       For example: domsrvr.default=domsrvr:02

Behavior of Abort in Import

We recommend that you do not abort the import process. The following scenarios explain the impact of Abort during the Import process.

Scenario 1: When the user closes the Command Line interface or selects the Ctrl+C key during the import process.

The status gets updated as *Aborted* in the database. In this scenario, the data import is partial, which can cause the CA SDM application to be in an inconsistent state.

Scenario 2: When the user clicks the **Abort** button on the **Administration**, **System**, **Data Export/Import History** page during the import process.

The status gets updated as *Aborted* in the database. The process keeps running on the server until the process is complete.

WARNING

Using the Abort button from the UI updates **only** the status as Aborted in the database and allows you to execute another process. Typically, you must abort an import process from the user interface when it terminates abruptly.

Property Files for Object Data Promotion

This section lists the properties files that you can manage to control the behavior of object data promotion tasks:

- Database Promote Unique Key File
- Export Configuration File
- Promotable Objects File

Database Promote Unique Key File

The **dbpromote.uniqkey** property file that contains the secondary key in the form of key value pair for each factory is available in the *NX_ROOT/pdmconf/* location. By default, *common_name* is assumed to be the secondary key. If the *common_name* of any factory has duplicate entries, specify the attribute that works as a secondary key or combination of attributes to form a unique key in order to identify the record. This file can be customized based on requirements. Ensure that the same file exists on the source system and target system after customization.

Specify the unique key in the following format:

```
<factory name1>.uniqKey=<attribute1>
<factory name2>.uniqKey=<attribute1>,<attribute2>,<attribute3>
```

Export Configuration File

The *export.cfg* configuration file is available in the *NX_ROOT/pdmconf/* folder. You can define the following values for the object data export process:

- **MAXDEPTH:** Sets the maximum depth level for export. As this is a recursive export, it limits the export till the MAXDEPTH level, and the dependent data after the MAXDEPTH level is not exported.
  For example: If the value set to 4, data is exported till 0-3 levels.
  Default: 10
- **EXCLUDE**: Contains a list of excluded objects (factory) specified in each row. Data for the excluded objects is not exported. However, when an excluded object is referred to as a dependent object, its value is exported. Importing the objects on the target system does not load the data for the excluded objects. If the excluded object exists on the target system a link is established between the objects.

Promotable Objects File

This article lists the out-of-the-box objects that CA SDM supports for object data promotion.

S. No.	Object Name	User Interface Label	User Interface Navigation Path
1	arcpur_rule	Archive and Purge Rules	Administration tab, Archive and Purge
2	grc	CI Classes	Administration tab, CA CMDB
3	nrf	CI Families	Administration tab, CA CMDB
4	mfrmod	CI Models	Administration tab, CA CMDB
5	ci_rel_type	CI Relationship Types	Administration tab, CA CMDB
6	rss	CI Service Status	Administration tab, CA CMDB
7	ci_managed_attribute	Manage Attributes	Administration tab, CA CMDB ,Configuration Control

8	ci_managed_chgstat	Manage Change Status	Administration tab, CA CMDB ,Configuration Control
9	mailbox_rule	Mailbox Rules	Administration tab, Email
10	mailbox	Mailboxes	Administration tab, Email
11	KT_ACT_CONTENT	Action Content	Administration tab, Knowledge
12	CI_WF_TEMPLATES	Approval Process Templates	Administration tab, Knowledge,Approval Process Manager
13	CI_STATUSES	Document Status	Administration tab, Knowledge,Approval Process Manager
14	QUERY_POLICY	Automated Policy	Administration tab, Knowledge, Automated Policies
15	KT_FLG_TYPE	Comment Types	Administration tab, Knowledge, Documents
16	CI_DOC_TEMPLATES	Document Templates	Administration tab, Knowledge, Documents
17	KEIT_TEMPLATES	Export/Import Templates	Administration tab, Knowledge, Documents, Export/Import
18	search_source	Search Sources	Administration tab, Knowledge, Federated Search
19	EBR_NOISE_WORDS	Noise Words	Administration tab, Knowledge, Search, KT Search Engine
20	EBR_ACRONYMS	Special Terms	Administration tab, Knowledge, Search, KT Search Engine
21	EBR_SYNONYMS	Synonyms	Administration tab, Knowledge, Search, KT Search Engine
22	act_type_assoc	Activity Association	Administration tab, Notifications
23	ntfm	Message Templates	Administration tab, Notifications
24	cmth	Notification Methods	Administration tab, Notifications
25	notification_phrase	Notification Phrases	Administration tab, Notifications
26	acctyp	Access Types	Administration tab, Security and Role Management
27	dcon	Data Partition Constraints	Administration tab, Security and Role Management, Data Partitions
28	dmn	Data Partitions	Administration tab, Security and Role Management, Data Partitions
29	func_access	Functional Access	Administration tab, Security and Role Management
30	menu_bar	Menu Bar	Administration tab, Security and Role Management, Role Management
31	menu_tree_res	Menu Tree Resource	Administration tab, Security and Role Management, Role Management

32	menu_tree_name	Menu Trees	Administration tab, Security and Role Management, Role Management
33	role	Role List	Administration tab, Security and Role Management, Role Management
34	tab	Tabs	Administration tab, Security and Role Management, Role Management
35	web_form	Web Forms	Administration tab, Security and Role Management, Role Management
36	ca_tou	Terms of Usage	Administration tab, Security and Role Management
37	attr_alias	Attribute Aliases	Administration tab, Service Desk, Application Data, Codes
38	auto_close	Auto Close Settings	Administration tab, Service Desk, Application Data, Codes
39	ctp	Contact Types	Administration tab, Service Desk, Application Data, Codes
40	cost_cntr	Cost Centers	Administration tab, Service Desk, Application Data, Codes
41	country	Countries	Administration tab, Service Desk, Application Data, Codes
42	dept	Departments	Administration tab, Service Desk, Application Data, Codes
43	perscnt	End User Roles	Administration tab, Service Desk, Application Data, Codes
44	ical_event_template	iCalendar Event Templates	Administration tab, Service Desk, Application Data, Codes
45	imp	Impacts	Administration tab, Service Desk, Application Data, Codes
46	loc	Locations	Administration tab, Service Desk, Application Data, Codes
47	outage_type	Outage Types	Administration tab, Service Desk, Application Data, Codes
48	position	Positions	Administration tab, Service Desk, Application Data, Codes
49	pri	Priorities	Administration tab, Service Desk, Application Data, Codes
50	prod	Products	Administration tab, Service Desk, Application Data, Codes
51	typecnt	Reasons	Administration tab, Service Desk, Application Data, Codes
52	rptmeth	Reporting Methods	Administration tab, Service Desk, Application Data, Codes
53	resocode	Resolution Codes	Administration tab, Service Desk, Application Data, Codes

54	resomethod	Resolution Methods	Administration tab, Service Desk, Application Data, Codes
55	rc	Root Causes	Administration tab, Service Desk, Application Data, Codes
56	sev	Severities	Administration tab, Service Desk, Application Data, Codes
57	site	Sites	Administration tab, Service Desk, Application Data, Codes
58	special_handling	Special Handling Types	Administration tab, Service Desk, Application Data, Codes
59	state	States/Provinces	Administration tab, Service Desk, Application Data, Codes
60	symptom_code	Symptom Codes	Administration tab, Service Desk, Application Data, Codes
61	ctimer	Timer Setup	Administration tab, Service Desk, Application Data, Codes
62	tspan	Timespans	Administration tab, Service Desk, Application Data, Codes
63	tz	Timezones	Administration tab, Service Desk, Application Data, Codes
64	urg	Urgencies	Administration tab, Service Desk, Application Data, Codes
65	ca_cmpny	Companies	Administration tab, Service Desk, Application Data, Configuration Items
66	vpt	Company Types	Administration tab, Service Desk, Application Data, Configuration Items
67	chgcat	Categories	Administration tab, Service Desk, Change Orders
68	chg_trans	Change Order Transition	Administration tab, Service Desk, Change Orders
69	chgtype	Change Type	Administration tab, Service Desk, Change Orders
70	chgcnf_status	Change Conflict Status	Administration tab, Service Desk, Change Orders
71	risk_svy_tpl	Risk Survey Template	Administration tab, Service Desk, Change Orders
72	chgstat	Change Order Status	Administration tab, Service Desk, Change Orders
73	tskstat	Workflow Task Status Code	Administration tab, Service Desk, Change Orders
74	tskty	Workflow Task Types	Administration tab, Service Desk, Change Orders
75	fmgrp	Form Group	Administration tab, Service Desk
76	isscat	Issue Categories	Administration tab, Service Desk, Issue
77	iss_trans	Issue Transition	Administration tab, Service Desk, Issue

78	issstat	Issue Status	Administration tab, Service Desk, Issue
79	mobile_attrs	Mobile Attribute	Administration tab, Service Desk, Mobile
80	response	Personalized Responses	Administration tab, Service Desk
81	prpval_rule	Property Validation Rules	Administration tab, Service Desk
82	pcat	Areas	Administration tab, Service Desk,Request/Incident/ Problems
83	in_trans	Incident Transitions	Administration tab, Service Desk,Request/Incident/ Problems
84	pri_cal	Priority Calculation	Administration tab, Service Desk,Request/Incident/ Problems
85	pr_trans	Problem Transitions	Administration tab, Service Desk,Request/Incident/ Problems
86	cr_trans	Request Transitions	Administration tab, Service Desk,Request/Incident/ Problems
87	transition_type	Transition Types	Administration tab, Service Desk,Request/Incident/ Problems
88	seq	Sequence Numbers	Administration tab, Service Desk
89	svc_contract	Service Contracts	Administration tab, Service Desk
90	tgt_time_tpl	Service Target Templates	Administration tab, Service Desk
91	risk_level	Risk Level	Administration tab, Service Desk, Change Order
92	mgs	Managed Survey List	Administration tab, Service Desk Surveys, Managed Surveys, Managed Survey List
93	mgsstat	Managed Survey Status	Administration tab, Service Desk, Surveys, Managed Surveys
94	svy_tpl	Survey Templates	Administration tab, Service Desk, Surveys
95	saprobtyp	Error Types	Administration tab, SOAP Web Services Policy
96	sapolicy	Policies	Administration tab, SOAP Web Services Policy
97	closure_code	Closure Codes	Administration tab, Service Desk, Change Orders
98	org	Organizations	Administration tab, Service Desk, Application Data, Codes
99	wrkshft	Workshifts	Administration tab, Service Desk, Application Data, Codes
100	crsq	Stored Queries	Administration tab, Service Desk, Application Data

101	crs	Status	Administration tab, Service Desk,Request/Incident/ Problems
102	evt	Events	dministration tab, Events and Macros
103	macro_type	Macro Types	Administration tab, Events and Macros
104	macro	Macros	Administration tab, Events and Macros
105	aty	Activity Notifications	Administration tab, Notifications
106	ntfr	Notification Rules	Administration tab, Notifications
107	ntfl	Object Contact Notification	Administration tab, Notifications
108	sdsc	Service Type	Administration tab, Service Desk

CA SDM Environment Promotion Limitations

The following limitations apply to CA SDM Environment Promotion:

1. Delete operation is not supported.
   For example, removing a role from an access type on the source system does not get promoted to the target system.
2. Selective customization promotion is not supported.
   For example, a user who has created multiple schema changes in the development system cannot promote selective changes to the target system.
3. *Objects in the exclude list that are defined in the NX_ROOT/pdmconf/export.cfg file get linked to the promoted object only if those records are available in the target system.*
   For example, OOTB cnt object is in the exclude list and you try exporting a pcat object with area *as testarea* having an assignee as *testuser*. After importing the pcat object to the target system, if the user (*testuser)* exists on the target system, the *testarea* record is linked to the *testuser* as an assignee. If the user (*testuser*) does not exist on the target system, the assignee field for this record is empty.
4. CA SDM administrators cannot promote a soft link on a Non-Windows system.
5. When you press **CTRL+C** on the console where the promotion operation is running, a confirmation message appears **(Y/N)** to terminate the job. The promotion operation is terminated irrespective of the selection.
6. When exporting or importing large amount of data, you may observe the following scenarios:
   a. If you encounter the error *Exception in thread "main" java.lang.OutOfMemoryError: Java heap space* during import, do the following tasks:
      - **Update the JVM:**
        - *Navigate to the NX_ROOT/bin/ folder and open the db_promote.bat file in edit mode.*
        - Add the memory heap size parameter to the data import manager jar file: %NX_JRE_INSTALL_DIR%/bin/ java" -cp %JAVA_CP% com.ca.ServicePlus.dataimport.DataImportManager -a %*
          For example: Add the heap size as 1024MB to the data import manager jar file: %NX_JRE_INSTALL_DIR %/bin/java" **-Xmx1024M** -cp %JAVA_CP% com.ca.ServicePlus.dataimport.DataImportManager -a %*

          NOTE

          For a 32-bit operating system, you can define the maximum heap size as 4 GB only.
      - **Reduce the export package size in the following ways:**
        - Select a smaller filter range from the User Interface.
          For example, search filter is set to 30 days, reduce the date range to 15 days and export again.
        - Decrease the value of the MAXDEPTH property in the *NX_ROOT/pdmconf/ export.cfg file.*
          The objects that are exported are restricted to the defined maximum level. Manually export the dependent objects that are beyond the maximum depth level.

b. When large amount of MDB data is selected for promotion, export or import may take several hours to complete. Before you initiate export or import, perform MDB level setting. For more information about the MDB level setting, see How to Move the CA MDB Data from the Source to the Target Systems.

Web Services for CA SDM - Best Practices and Recommendations

The usage of Web Service Calls in CA SDM is affected due to the following reasons:

- Performance Issues
- CA SDM Not responding
- CA SDM Process Memory Growth (domsrvr)

To avoid performance and other issues, ensure that you have taken into consideration the following:

Add a Dedicated domsrvr to the Web Services Calls

If you are using CA Process Automation with CA SDM - or if any other application in your environment is integrated with CA SDM, consider installing a secondary CA SDM Server and point your Web Services Application or web services to the secondary CA SDM Server. Additionally, it is recommended to add a webengine/domsrvr pair for each secondary server used for this reason.

For conventional installations, you can make the changes in the CA SDM **Administration** tab, under **Options Manager**, **Web Service**, **webservice_domsrvr** option. If the option is not installed, install it and set the new domsrvr process name to be used by web services. If you have already installed this option, edit it and adjust the domsrvr the process name. Restart CA SDM for changes to take effect.

For Advanced Availability configuration, you can consider the following:

- A dedicated domsrvr. Setting a dedicated domsrvr requires different configuration and planning in Advanced Availability. For more information, see Configure a Dedicated Domsrvr for SOAP Web Service - Advanced Availability
- A dedicated Application server for Web Services calls.
- If it is not possible to have a dedicated Application server, consider pointing the applications that are communicating with CA SDM via Web Services to the Background server, but preferentially to an Application server with less use/load.

Make use of logoff() when complete

After every set of Web Services calls, the Web Services client must perform a logoff() when complete. Web Services is intended to be an "on demand" resource, use what you need and then, close the connection when possible, as opposed to re-using the same session ID for hours at a time or leaving the session inactive. The **Web Services Session Timeout** option is a configurable option in CA SDM Options Manager. Evidences of these can be found in the **stdlog**.

For example, we can see a login and a logout session done by the Web Services call in the following:

```
  ada.cpp 2180 Web Services session ended 1814552582; user(ServiceDesk); IP Address; session count 0
ada.cpp 2180 web Services session created 779115035; user (serviceDesk); IP Address; session count 1
```

Search for **sda** in the **stdlogs** and if it shows **Web Services session created,** it means a logoff is not used.

Each Web Services Client Must Use a Different CA SDM Web Services Policy

Ensure that the web services client is using a different CA SDM web services policy.

Use a Different CA SDM Login for Web Services Client

In the web services client that calls CA SDM Web Services, there is a location to enter the WSDL and username. It is not recommended that you use the CA SDM privileged user (**Service Desk**) for integrations.

Do not allow Web Services Clients to Search with Wild Cards

In other words, if the Web Services Client does searches (that is a *doquery* or *getlist*), it must not allow a wild card search. In case of large MDB tables, CA SDM performance is negatively impacted as the Web Services query runs on top of the CA SDM object layer and all data retrieved by the query is loaded in the domsrv. This can cause performance issues or even make the domsrvr unresponsive and reach the max 2GB size limit on 32-bit applications.

Avoid requesting large data sets with Web Service Method calls like "doSelect()"

For example, setting the **maxRows** with **-1** to return all rows has an adverse impact on the CA SDM performance. Using **-1** is a not recommended. Ensure that all Web Service calls are tested prior to being placed into production to see if there is any impact on the CA SDM performance. This is especially true for Web Services calls that utilize searches or requests for lists.

> **NOTE**
>
> The maximum rows retrieved by a *doSelect* is **250**. This is by design and is non-configurable to prevent performance issues in CA SDM.

Fully Tested Web Services Client

CA Support highly recommends that all Web Services clients must be tested fully by a user or implementer in a test environment before implementing it in the production environment.

Distributing Web Services Requests for Multiple CA SDM Secondary Servers

It is very important to plan how the web services requests will be distributed across across multiple CA SDM Secondary Servers. You can address this by creating a routine which works as a pool of Web Services connections. The routine maintains permanent connections to CA SDM and distributes the calls among the available sessions that are recreated on a need basis.

Increase the Tomcat Memory (JVM) Size

Increase the Tomcat Memory (JVM) size to at least 1 GB.

Increase the Tomcat Max Threads

By default, the max threads for Tomcat is set to **75**. It is recommended that you set this value to **300** when using web services.

Cumulative Patch Implementation

If this is a new CA SDM implementation, ensure that you have installed the latest cumulative patch level. To verify the latest cumulative patch level, see CA Service Desk Manager Solutions and Patches.

CMDB GRLoader

The CMDB GRLoader utility uses Web Services. It is recommended to run the GRLoader utility against the CA SDM Web Services Interface on a dedicated CA SDM Secondary Server.

Macros and Events

You can use macros to add actions to an object, or check for a certain characteristic or condition. For example, you can use a macro to add an activity log to an issue, or to search for issues with a high priority. CA SDM comes equipped with macros that perform tasks such as searching for unassigned issues, notifying a caller, and creating an activity log.

This article contains the following topics:

Create a Macro

You can create a macro, which can be used to add actions to objects or check for certain characteristics or conditions.

If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps to create a macro:

1. Select Events and Macros, Macros on the Administration tab.
   The Macro List page opens.
2. Click Create New.
   The Create New Macro page appears.
3. Complete the fields as follows:
   SymbolEnter a descriptive identifier for the macro.
   Macro Type
   Select the type for this macro. The macro type that is selected controls the remaining data that needs to be supplied.

   NOTE

   : The Execute CA IT PAM Action selection is only available when CA IT PAM Workflow is configured with CA SDM. The Execute CA Workflow Action selection is only available when CA Workflow is configured with CA SDM.

   Object Type Select the type of object on which the macro can be run.
4. Click Continue.
   The page fills in with the remaining data needed for the macro that is based on the selected macro type.

Execute Remote Reference Macro

This type of macro launches an external program on the server.

Follow these steps:

1. Complete the following fields:
   Remote Reference
   The name of the remote reference to the external program you want the macro to launch. Type the remote reference name directly into this field, or click the search icon to select the remote reference.
   Parameters to Command
   Additional parameters required for running the command.
2. Click Save.
   The execute remote reference macro is created.

Multiple Notification Macro

This macro type lets you send a notification to one or more contacts. You can specify the message to send, the recipients of the message, and the urgency level.

Follow these steps:

1. Type the template name directly in the Message Template field, or click the search icon to select the desired template. This template is used to create the notification message.
2. Click Save.
3. Choose the appropriate contacts to notify from the following tabs:
   Note: Use the Update Contacts button that appears on each tab to search for and select more contacts to notify.
   Object Contacts
   Displays the available organizations, vendors, and configuration items for the selected Object type that receive notification about tickets. For example, you can select Affected End User or Affected End User's Org to notify.
   Contacts
   Displays the individuals who are added to the notification macro, regardless of their association with the ticket.

Contact Types
Displays the users who are defined within the notification macro with the same classification, such as analyst or customer.

4. Click Save.
   The multiple notification macro is created.

Site-Defined Condition Macro

This type of macro evaluates true and false conditional statements, and can be created by administrators to provide specific behaviors to individual sites.

Follow these steps:

1. Complete the following fields:
   If All Conditions Succeed Return Select the value you want to return if all conditions succeed. Valid values are True or False. Record Status Select whether the macro is Active or Inactive.
2. Click Save.
3. Click Add Condition on the Conditions tab.
   The Create New Atomic Condition page opens.
4. Fill in the following fields as appropriate:
   Sequence
   The order in which the condition is evaluated within the site-defined condition.
   Select an Attribute
   The attribute to be evaluated. Enter the attribute name directly in this field or click the search icon to select the desired attribute.
   Choose Operator
   Select the conditional operator to use in the evaluation. Valid values are: Equals, Does Not Equal, Greater Than, Less Than, Empty/Null, Not Empty/Null.
   Select Attribute or Data Value
   Choose either Attribute or Data Value to be evaluated.
   Attribute Value/Data Value
   The value to be evaluated. Enter the attribute or data value directly into the field, or click the search icon to select the desired value.
5. Click Save.
6. Repeat this process until all the desired conditions have been created.
   The site-defined macro is created.

Search Macros

You can view summary information for each macro on the Macro List page. To display this page, select the Administration tab, Events and Macros, Macros.

You can set up a filter to display only those macros that match certain criteria.

> **NOTE**
>
> If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

To filter the macro list follow these steps:

1. Click Show Filter on the Macro List page.
   The top portion of the page displays search filter fields.
2. Fill in the fields with the appropriate:

TypeThe symbol for the macro type as defined on the Macro Type detail page.
Object Type The symbol for the macro type as defined on the Macro Type detail page.
SymbolThe symbol for the macro.
ActiveIndicates whether the macro record is active or inactive in the database.

NOTE

You can click the More icon to display the Additional Search Arguments field. This field is intended only for expert users who understand SQL and Majic and can use it to specify search arguments that are not available in the standard search filter fields. You can enter a SQL WHERE clause in this field to specify an additional search argument.

3. Click **Search**.
   The Macro List page lists the macros that match your search criteria.

NOTE

: You can export list results to Excel for use outside of CA SDM by clicking the Export button on the List page.

Edit a Macro

You can update a macro that has already been created.

Follow these steps:

1. On the Administration tab, browse to Events and Macros, Macros.
   The Macro List appears.
2. Select the desired macro.
   The Macro Detail window opens.
3. Click Edit.
   The Update Macro window opens.
4. Edit the fields as appropriate.
5. Click one of the following buttons:
   - **Save** -- Saves the macro and closes the window.
   - **Cancel** -- Closes the window without saving any changes.
   - **Reset** -- Resets all fields to the last saved values.definitions.

Macro Types

You can view summary information for each macro type on the Macro Type List page. You cannot edit a Macro Type, but you can create a macro with a specific Macro Type, which impacts the behavior and appearance of the macro. To display this page, select Events and Macros, Macro Types on the Administration tab.

The Macro Type detail page contains the following tabs:

- **Execute Script**
  Displays the internal CA Technologies code that runs when this macro starts. For example, this macro runs when the status of a task changes from WAIT to PENDING.
- **Validate Script**Displays the internal CA code that identifies the requirements for this type of macro.

Filter the Macro Types List

You can set up filters to restrict the display of macro types to only those that match certain criteria.

Follow these steps:

1. Click the Show Filter button on the Macro Type List page.

The top portion of the page displays search filter fields.

2. Fill in the fields with the appropriate values for filtering the Macro Types list.
   - **Enum Code**
     An internal code that defines the macro type to the system.
   - **Symbol**
     An identifier for the macro type that indicates its purpose.
   - **Active**
     Indicates whether the macro type record is active or inactive in the database.
3. Click **Search**.
   The Macro Type List displays items that match your search criteria.

Event

Events are procedures that your issue management system follows after a certain amount of time has elapsed. For example, an event might send a message to a service desk analyst if a "priority 1" issue is not resolved within an hour.

This article contains the following topics:

Create a Event

An event can be created and a macro can be attached to this event. The event is executed after certain time is elapsed. If any macro is attached, an action is performed.

If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. Select Events and Macros, and Events on the Administration tab.
   The Event List page opens.
2. Click **Create New**.
   The Create New Event page opens.
3. Complete the event fields:
   Name
   The name of the event
   Object Type
   Indicates whether the event is attached to an issue, request, change order, workflow task, knowledge document, knowledge report card, assistance session, or managed survey. This field can only be edited when creating an event. This field is read-only when you want to update the event.
   Record Status
   Indicates whether the event is active or inactive. Only active events can be used.
4. Complete the configuration information:
   NameThe name of the event
   Object Type Indicates whether the event is attached to an issue, request, change order, workflow task, knowledge document, knowledge report card, assistance session, or managed survey. This field can only be edited when creating an event. This field is read-only when you want to update the event.
   Record Status
   Indicates whether the event is active or inactive. Only active events can be used.
   Delay Time
   Specifies the time after which the event is triggered.
   Repeat Delay Time Specifies the interval of time after which you want the event to be triggered again.
   Allow Time Resetting Specifies if a service desk analyst can change the Delay Time. This option must be selected in order for an event to be used as a Service Type event (an event associated with a service level, such as 4-hour resolution.).

Work Shift Specifies the dates, days, and hours when the service type is in effect.
On Done Event Flag Specifies the action to take once the event has been completed.
Repeat Event Repeats the event at the specified time interval until the issue is closed.
Save History Records the history of activities taken on the event.
No History Do not record any history of the event. The event does not appear in the Event History window.
ConditionDisplays the macro (if associated with the event) indicating the condition checked for by the event.
TextDefines the event configuration. For some action macros, this field is used for a specific purpose. For example, in the Transfer to Event Contact action macro, it contains the userid of the person to whom you want the ticket to be transferred.

5. Click **Save**.
6. To add the macro to this event, complete the action information.The new event is saved.

Search Events

You can view the summary information for each event on the Events List page. To display this page, select the Administration tab, Events and Macros, Events.

You can set up search filters to display only the events of interest. If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

To filter the event list follow these steps:

1. Click **Show Filter** on the Event List page.
   The top portion of the page displays the available filter fields.
2. Fill in the fields with the appropriate selection criteria for filtering incidents.
3. Click **Search**.
4. The Events List displays the events that match your search criteria.

   NOTE

   You can export list results to excel for use outside of CA SDM by clicking the Export button on the List page.

Event Search Fields

The following fields are available for filtering the Events List to display only the events of interest.

- **Name**
  The name of the event displayed on the Event List page.
- **Object Type**
  Indicates whether the event is attached to an issue, request, change order, workflow task, knowledge document, knowledge report card, assistance session, or managed survey.
- **Record Status**
  Indicates whether the event is active or inactive.
- **Description**
  Provides a description of the event's intended purpose.

  NOTE

  : You can click the More icon to display the Additional Search Arguments field. This field is intended only for expert users who understand SQL and Majic and can use it to specify search arguments that are not available in the standard search filter fields. You can enter a SQL WHERE clause in this field to specify an additional search argument.

Edit an Event

You can update an event that has already been created.

Follow these steps:

1. On the **Administration** tab, select **Events and Macros**, **Events**.
   The Event List appears.
2. Select the desired event.
   The Event Detail window opens.
3. Click **Edit**.
   The Update Event window opens.
4. Edit the fields as appropriate.
5. Edit the actions associated with the event as follows:
   a. Click the Action Information tab.
   b. Click Update Actions on True.
      The Macro Search page opens.
   c. Search for the macros to be performed if the event condition is true.
   d. Select the desired macros from the list on the left, and click

      The selected macros are added to the list on the right.
   e. When all desired macros are in the list on the right, click OK.
      The selected macros appear in the Actions on True Macro List
   f. Click Update Actions on False, and repeat the above procedure to select the macros to be performed if the event condition is false.
   g. If desired, select one or both of the SLA Violation for Actions on True/False Macro checkboxes. These checkboxes, if selected, cause a Service Level Agreement (SLA) violation to be logged when a true or false condition is encountered for the event.
6. Click **Save**.
   Your updates to the event are saved.

Tags Used with Maileater

The *$NX_ROOT\site\text_api.cfg* file defines the variables from the incoming mails that is interpreted by CA Service Desk Manager Desk. You can add your own tag mappings to this file, When you wish to set a value on a ticket while sending an email to CA SDM, you can use one of the tags listed below. The email must be in plain text format so that it can be processed properly, and the tag must appear in this format: "**%TAG=<input>**". Each tag must appear on its own line, and replace the **<input>** field with a value that you want to set.

Incident Tags

Maileater Tag Name	Description
%AFFECTED_RC	Sets the affected resource by name
%AFFECTED_RC_PERSID	Sets the affected resource by persid
%AFFECTED_SERVICE	Sets the affected service (takes a string)
%ASSIGNEE	sets the assignee by login id
%CALL_BACK_DATE	Sets the call back date. Proper format is required.
%CATEGORY	Sets the category
%CATEGORY	Sets the problem category (takes a string name)

%CAUSED_BY_CHG	Sets the caused by change order (takes an integer change order number)
%CAUSED_BY_CHG_PERSID	Sets the caused by change order (takes the change order persid)
%CHARGE_BACK	Takes a charge back id (string)
%CUSTOMER	Sets the customer by login ID
%CUSTOMER_EMAIL	Set the customer by email
%CUSTOMER_NAME	Sets the customer by name (last first middle)
%DESCRIPTION	Populates the description (takes a string)
%DEVICE	Sets the affected device/resource by name (takes a string)
%EVENT_TOKEN	Sets the event token (Takes a string)
%GROUP	Sets group by UUID
%IMPACT	Sets the impact (integer)
%IMPACT_PERSID	Sets the impact by persid
%INCIDENT_ID	Specifies the incident by id (takes a string)
%INCR_PREDICTED_SLA_VIOLATION	Increments the predicted SLA violation (takes an integer)
%LOG_AGENT	Sets the log agent by UUID
%PRIORITY	Sets the initial priority by ENUM integer
%PRIORITY_ENUM	Sets the priority by ENUM (takes an integer)
%PRIORITY_PERSID	Sets the priority by persid
%PROPERTY	Sets the property value
%REQUESTER	Sets the requester by login id
%REQUESTOR	Sets the requester by login id
%RESOLUTION_CODE	Sets the resolution code (integer)
%RESOLUTION_CODE_PERSID	Sets the resolution code by persid
%RESOLUTION_METHOD	Sets the resolution method (takes an integer)
%RESOLUTION_METHOD_PERSID	Sets the resolution method by persid
%SERVICE_LEVEL	Sets the service level (takes a string)
%SEVERITY	Sets the severity (takes an integer)
%SEVERITY_PERSID	Sets the severity by persid
%STATUS	Populates the status (takes a string)
%STATUS_PERSID	Sets the status by persid
%STRING1	Used to pass information into the STRING1 field
%STRING2	Used to pass information into the STRING2
%STRING3	Used to pass information into the STRING3 field
%STRING4	Used to pass information into the STRING4 field
%STRING5	Used to pass information into the STRING5 field
%STRING6	Used to pass information into the STRING6 field
%SUMMARY	Populates the summary (takes a string)
%TEMPLATE	Sets the template name (takes a string
%URGENCY	Sets the urgency (takes an integer)
%URGENCY_PERSID	Sets the urgency by persid
%UUID	Sets the affected resource by UUID

Problem Tags

Maileater Tag Name	Description
AFFECTED_RC	Sets the affected resource by name
%AFFECTED_RC_PERSID	Sets the affected resource by persid
%AFFECTED_SERVICE	Sets the affected service (takes a string)
%ASSIGNEE	sets the assignee by login id
%CALL_BACK_DATE	Sets the call back date. Proper format is required.
%CATEGORY	Sets the problem category (takes a string name)
%CAUSED_BY_CHG	Sets the caused by change order (takes an integer change order number)
%CAUSED_BY_CHG_PERSID	Sets the caused by change order (takes the change order persid)
%CHARGE_BACK	Takes a charge back id (string)
%CUSTOMER	Sets the customer by login id
%CUSTOMER_EMAIL	Set the customer by email
%CUSTOMER_NAME	Sets the customer by name (last first middle)
%DESCRIPTION	Populates the description (takes a string)
%DEVICE	Sets the affected device/resource by name (takes a string)
%EVENT_TOKEN	Sets the event token (Takes a string)
%GROUP	Sets group by UUID
%IMPACT	Sets the impact (integer)
%IMPACT_PERSID	Sets the impact by persid
%INCR_PREDICTED_SLA_VIOLATION	Increments the predicted SLA violation (takes an integer)
%LOG_AGENT	Sets the log agent by UUID
%PRIORITY	Sets the initial priority by ENUM integer
%PRIORITY_ENUM	Sets the priority by ENUM (takes an integer)
%PRIORITY_PERSID	Sets the priority by persid
%PROBLEM_ID	Specifies the problem number (takes a string)
%REQUESTER	Sets the requester by login id
%REQUESTOR	Sets the requester by login id
%ROOTCAUSE	Sets the root cause (takes an integer)
%ROOTCAUSE_PERSID	Sets the root cause by persid
%SERVICE_LEVEL	Sets the service level (takes a string)
%SEVERITY	Sets the severity (takes an integer)
%SEVERITY_PERSID	Sets the severity by persid
%STATUS – Populates the status (takes a string)	
%STATUS_PERSID	Sets the status by persid
%STRING1	Used to pass information into the STRING1 field
%STRING2	Used to pass information into the STRING2 field
%STRING3	Used to pass information into the STRING3 field
%STRING4	Used to pass information into the STRING4 field
%STRING5	Used to pass information into the STRING5 field

%STRING6	Used to pass information into the STRING6 field
%SUMMARY	Populates the summary (takes a string)
%TEMPLATE	Sets the template name (takes a string
%URGENCY	Sets the urgency (takes an integer)
%URGENCY_PERSID	Sets the urgency by persid
%UUID	Sets the affected resource by UUID

Request Tags

Maileater Tag Name	Description
%AFFECTED_RC	Sets the affected resource by name
%AFFECTED_RC_PERSID	Sets the affected resource by persid
%ASSIGNEE	sets the assignee by login id
%CALL_BACK_DATE	Sets the call back date. Proper format is required.
%CATEGORY	Sets the problem category (takes a string name)
%CAUSED_BY_CHG	Sets the caused by change order (takes an integer change order number)
%CAUSED_BY_CHG_PERSID	Sets the caused by change order (takes the change order persid)
%CHARGE_BACK	Takes a charge back id (string)
%CUSTOMER	Sets the customer by login ID.
%CUSTOMER_EMAIL	Set the customer by email
%CUSTOMER_NAME	Sets the customer by name (last first middle)
%DESCRIPTION	Populates the description (takes a string)
%DEVICE	Sets the affected device/resource by name (takes a string)
%EVENT_TOKEN	Sets the event token (Takes a string)
%GROUP	Sets group by UUID
%IMPACT	Sets the impact (integer)
%IMPACT_PERSID	Sets the impact by persid
%INCR_PREDICTED_SLA_VIOLATION	Increments the predicted SLA violation (takes an integer)
%LOG_AGENT	Sets the log agent by UUID
%PRIORITY	Sets the initial priority by ENUM integer
%PRIORITY_ENUM	Sets the priority by ENUM (takes an integer)
%PRIORITY_PERSID	Sets the priority by persid
%REQUESTER	Sets the requester by login id
%REQUESTOR	Sets the requester by login id
%RESOLUTION_CODE	Sets the resolution code (integer)
%RESOLUTION_CODE_PERSID	Sets the resolution code by persid
%RESOLUTION_METHOD	Sets the resolution method (takes an integer)
%RESOLUTION_METHOD_PERSID	Sets the resolution method by persid
%SERVICE_LEVEL	Sets the service level (takes a string)
%SEVERITY	Sets the severity (takes an integer)
%SEVERITY_PERSID	Sets the severity by persid

%STATUS	Populates the status (takes a string)
%STATUS_PERSID	Sets the status by persid
%STRING1	Used to pass information into the STRING1 field
%STRING2	Used to pass information into the STRING2 field
%STRING3	Used to pass information into the STRING3 field
%STRING4	Used to pass information into the STRING4 field
%STRING5	Used to pass information into the STRING5 field
%STRING6	Used to pass information into the STRING6 field
%SUMMARY	Populates the summary (takes a string)
%TEMPLATE	Sets the template name (takes a string
%URGENCY	Sets the urgency (takes an integer)
%URGENCY_PERSID	Sets the urgency by persid
%UUID	Sets the affected resource by UUID

Change Order Tags

Maileater Tag Name	Description
%ACTUAL_COMP_DATE	Sets the actual completion date (takes a date in the proper format)
%ACTUAL_COST	Sets the actual cost (takes an integer)
%ACTUAL_END_DATE	Sets the actual end date (takes a date in the proper format)
%ACTUAL_START_DATE	Sets the actual start date (takes a date in the proper format)
%ACTUAL_TIME	Sets the actual time (takes a time in the proper format)
%AFFECTED_USER	Sets the affected user by login id
%ASSIGNEE	sets the assignee by login id
%BACKOUT_PLAN	Populates the backout plan (takes a string)
%BUSINESS_CASE	Populates the business case (takes a string)
%CAB_APPROVAL	Populates the CAB approval…?
%CAB	Populates the CAB…?
%CATEGORY	Sets the change order category (takes a string name)
%CHANGE_ID	Matches the change id (takes a string)
%CHGTYPE	Sets the change type (takes an integer)
%CLOSURE_CODE	Sets the closure code (takes an integer)
%DESCRIPTION	Populates the description (takes a string)
%EFFORT	Populates the effort (takes a string)
%EST_COMP_DATE	Sets the estimated completion date (takes a date in the proper format)
%EST_COST	Sets the estimated cost (takes an integer)
%EST_TIME	Populates the estimated time (takes a time in the proper format)
%GROUP	Sets group by UUID
%IMPACT	Sets the impact (integer)
%INCR_PREDICTED_SLA_VIOLATION	Increments the predicted SLA violation (takes an integer)
%JUSTIFICATION	Populates the justification (takes a string)

%NEED_BY_DATE	Sets the need by date (takes a date in the proper format)
%ORGANIZATION	Sets the organization by UUID
%PRIORITY	Sets the initial priority by ENUM integer
%PRIORITY_ENUM	Sets the priority by ENUM (takes an integer)
%REQUESTER	Sets the requester by login ID
%REQUESTOR	Sets the requestor by login ID
%RISK	Sets the risk (takes an integer)
%ROOTCAUSE_PERSID	Sets the root cause by persid
%ROOTCAUSE	Sets the root cause (takes an integer)
%SCHED_DURATION	Sets the scheduled duration (takes a time in the proper format)
%SCHED_START_DATE	Sets the scheduled start date (takes a date in the proper format)
%STATUS	Populates the status (takes a string)
%STRING1	Used to pass information into the STRING1 field
%STRING2	Used to pass information into the STRING2 field
%STRING3	Used to pass information into the STRING3 field
%STRING4	Used to pass information into the STRING4 field
%STRING5	Used to pass information into the STRING5 field
%STRING6	Used to pass information into the STRING6 field
%SUMMARY	Populates the summary (takes a string)
%TEMPLATE	Sets the template name (takes a string)

Issue Tags

Maileater Tag Name	Description
%CATEGORY	Sets the change order category (takes a string name)
%DESCRIPTION	Populates the description (takes a string)
%EFFORT	Populates the effort (takes a string)
%EST_COMP_DATE	Sets the estimated completion date (takes a date in the proper format)
%EST_COST	Sets the estimated cost (takes an integer)
%EST_TIME	Populates the estimated time (takes a time in the proper format)
%FLAG1	Used to pass information into the FLAG1 field
%FLAG2	Used to pass information into the FLAG2 field
%FLAG3	Used to pass information into the FLAG3 field
%FLAG4	Used to pass information into the FLAG4 field
%FLAG5	Used to pass information into the FLAG5 field
%FLAG6	Used to pass information into the FLAG6 field
%FOR_A	Set contact type (takes an integer)
%FROM	Populates from (takes a login ID)
%GROUP	Sets group by UUID
%IMPACT	Sets the impact (integer)
%IMPACT_PERSID	Sets impact by persid

%INCR_PREDICTED_SLA_VIOLATION	Increments the predicted SLA violation (takes an integer)
%ISSUE_ID	Matches the issue number (takes a string)
%JUSTIFICATION	Populates the justification (takes a string)
%NEED_BY_DATE	Sets the need by date (takes a date in the proper format)
%NOTES	Populates the notes (takes a string)
%ORGANIZATION	Sets the organization by UUID
%PRIORITY	Sets the initial priority by ENUM integer
%PRIORITY_ENUM	Sets the priority by ENUM (takes an integer)
%PRIORITY_PERSID	Sets the priority by persid
%PRODUCT_PERSID	Sets the product by persid
%PRODUCT	Sets the product (takes an integer)
%PROMISED_DATE	Sets the need by date (takes a date in the proper format)
%REQUESTER	Sets the requester by login ID
%REQUESTOR	Sets the requester by login ID
%RESOLUTION	Set resolution (takes a string)
%RISK	Sets the risk (takes an integer)
%ROLE	Sets the role of the person creating the ticket (takes an integer)
%ROOTCAUSE	Sets the root cause (takes an integer)
%ROOTCAUSE_PERSID	Sets the root cause by persid
%ROOTCAUSE	Sets the root cause (takes an integer)
%SERVICE_DATE	Sets the service date (takes a date in the proper format)
%SERVICE_LEVEL	Sets the service level (takes a string)
%SERVICE_NUM –	Sets the service number (takes a string)
%STATUS	Populates the status (takes a string)
%STATUS_PERSID	Sets the status by persid
%STRING1	Used to pass information into the STRING1 field
%STRING2	Used to pass information into the STRING2 field
%STRING3	Used to pass information into the STRING3 field
%STRING4	Used to pass information into the STRING4 field
%STRING5	Used to pass information into the STRING5 field
%STRING6	Used to pass information into the STRING6 field
%SUMMARY	Populates the summary (takes a string)
%TEMPLATE	Sets the template name (takes a string)
%USER1	Used to pass information into the USER1 field
%USER2	Used to pass information into the USER2 field
%USER3	Used to pass information into the USER3 field

License Usage

The License Usage enable an administrator to gain insight into an organization's CA SDM license management. This allows an organization to stay compliant on licensing.

To view the license usage, log in to CA SDM as an administrator, navigate to the **Administration** tab, select **System**, and then select **License Usage**.

NOTE

: The License Usage report is always derived from the *webConcurrentTotalLicenseCt* KPI.

Tenant field specifies all the tenants (public), a single tenant, or a tenant group. If multi-tenancy is enabled, select the appropriate tenant from the drop-down list.

Licenses Purchased: An administrator manually enters a value in the Licenses Purchased field that specifies the number of licenses purchased.

Maximum Licenses Usage field specifies the number of concurrent licenses that are used from the time of installation within the organization. This value is derived from the *webConcurrentTotalLicenseCt* KPI.

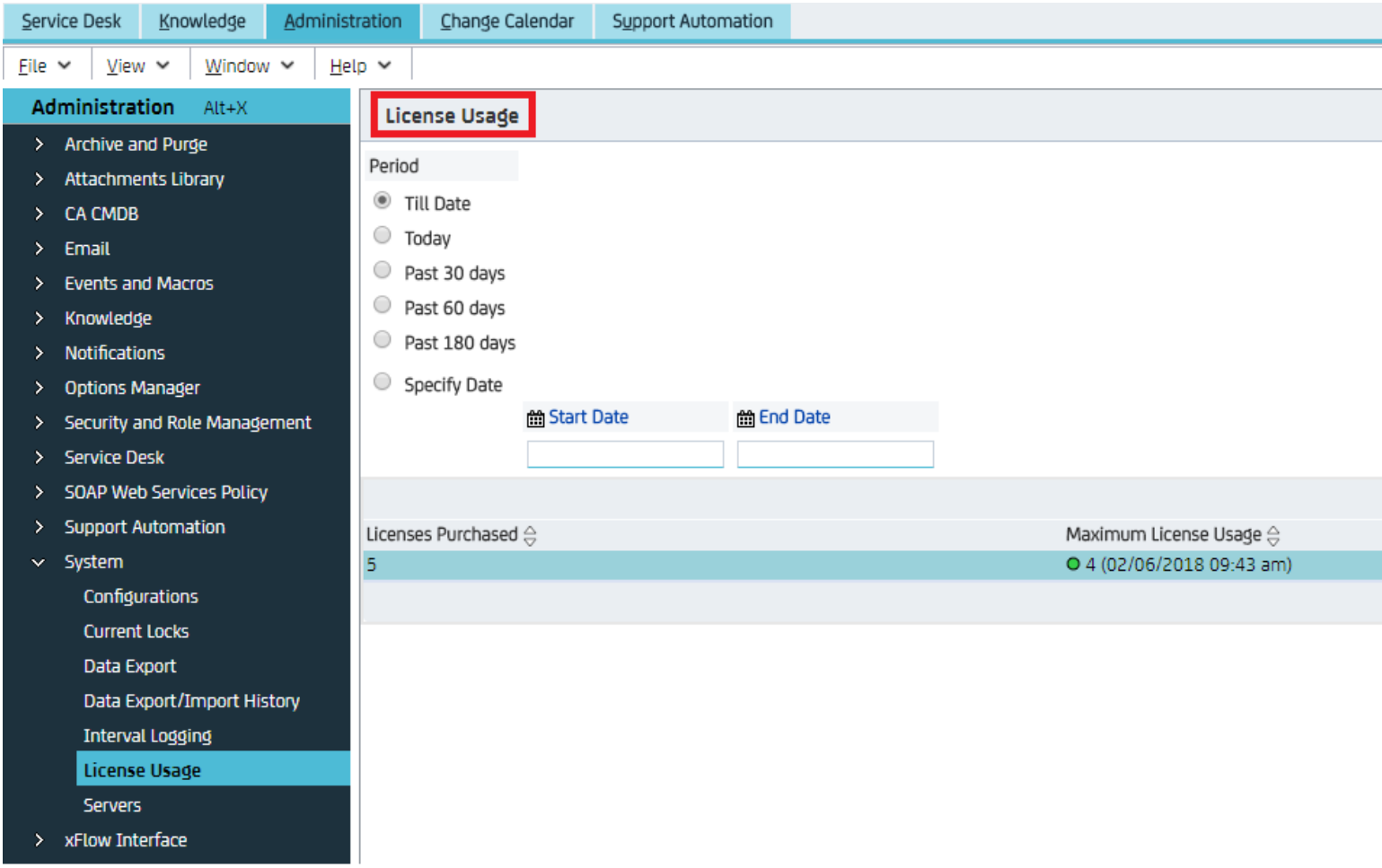
Latest Licenses Usage field specifies the number of concurrent licenses that are used today (12AM - 11:59PM) within the organization.

Color codes indicate the license usage compliance. A green dot indicates availability of license for usage, whereas, a red dot indicates number of licenses used exceeds the number of licenses purchased.

By default, the license usage results are displayed till date. The license usage can be filtered based on the following predefined filters:

- Till Date
- Today
- Last 30 days
- Last 60 days
- Last 180 days
- Date Range

The below example demonstrates license usage based on the predefined `Till Date` filter.



CA CMDB Administration

This section provides information on the Configuration Administrator role. In addition to CI management, this role includes the administration of CI classes, CI families, and relationship types.

In the following topics, the settings for each feature are explained, with references to the administrative tools that can assist you in your configuration administration tasks.

This section contains the following topics:

- Configuration Item Classes
- Configuration Item Families
- Configuration Item List
- Maintain Relationships
- Configuration Audits
- Configuration Control
- MDR-Management
- Reconciliation Management
- Transaction Work-Area
- CA CMDB Data Maintenance

Configuration Item Classes

Configuration item classes are general categories of hardware, software, and services; for example, workstation, printer, service, and so on. These configuration item classes can then be grouped into broader categories called configuration item families.

To create a new configuration item class:

1. From the Administration tab, navigate to Service Desk, Application Data, Configuration Items, Configuration Item Classes.
   The Configuration Item Class List page displays.
2. Click Create New.
   The Create New Configuration Item Class page displays.
3. Fill in the following fields:
   Class
   A unique identifier that categorizes the configuration item class you are adding to the database.
   Family
   The Configuration Item Family associated with this Configuration Item Class. Families are broader categories for configuration items than classes. The families include computers, other hardware, software, and services. This field is filled in automatically and cannot be changed.
   Record Status
   Indicates whether the configuration item class is active or inactive.
   Description
   Provides a detailed description of the configuration item class. You can use this field to further identify the configuration items assigned this class.
4. Click one of the following buttons:
   Save
   Saves the configuration item class and closes the window.
   Cancel
   Closes the window without saving any changes
   Reset
   Returns all fields to their previously saved values.

Configuration Item Families

Configuration item families are broad categories you can use to classify your network resources. You can assign configuration item classes to families. By doing this, individual configuration items are automatically assigned to families when they are assigned to classes. The configuration item family controls the kind of record displayed.

To create a new configuration item family:

1. From the Administration tab, navigate to the Configuration Item Families List.
   The Configuration Item Family List page displays.
2. Click Create New.
   The Create New Configuration Item Family page displays.
3. Fill in the following fields:
   a. **Name** - A unique identifier for the configuration item family.
   b. **Record Status** - Indicates whether the configuration item family is active or inactive.
   c. **Extension Name** - Specifies an extension value related to a physical table name in the database. When you select an extension name in the drop-down list, the Physical Table Name field appears in the read-only Physical Table Name field.
   d. **Description** - Provides a detailed description of the configuration item family. You can use this field to further identify the configuration item classes assigned this family.

4. Click one of the following buttons:
   a. **Save** - Saves the configuration item family and closes the window.
   b. **Cancel** - Closes the window without saving any changes
   c. **Reset** - Returns all fields to their previously saved values.

Configuration Item List

You can enter search criteria to filter the Configuration Item List to display only the items you want to see. You can also search for individual configuration items that you want to view or edit. To search for a configuration item

1. On the Administration tab, navigate to the Configuration Item (CI) List .
   The Configuration Item Search page appears.
2. Complete one or more of the search fields.
3. (Optional) Click More to display additional search fields.
4. Click Search.
   The Configuration Item List page opens to display the items that match your search criteria. You can select a CI to view or edit, or right-click a CI to display its menu.

   NOTE

   : You can export list results to Excel for use outside of CA SDM by clicking the Export button on the List page.

Configuration Item Detail

Configuration Item Detail pages identify a CI and determine which type of page appears so that you can create or manage the CI. CI Name and the CI Class are required. An asterisk (*) identifies a field that is required. If you use change verification, informational messages appear at the top of the detail form.

When you assign the new CI to a CI class, the CI family is automatically assigned (because a class is already associated with a family). When you click Continue on the initial Create New Configuration Item page, the appropriate CI page appears, depending on the family.

NOTE
For a list of the default CA CMDB families and the classes they contain, see the CA CMDB Technical Reference Guide.

You can display the Create New Configuration Item page by choosing New CI from the File menu on the main page. To display the Update Configuration Item page, click Edit from the CI Detail page.

You can also navigate to the CI page from the Administration tab by using the CI List item in the menu tree.

NOTE
The displayed fields can differ depending on the CI class selected. The CI Family determines what appears on the Attributes tab.

CA Business Service Insight Integration

When you integrate CA Business Service Insight and CA SDM, the BSI Service tab displays Exceeds and Violations counts. The BSI Service by Metric and BSI Contract by Metric tabs display the Metric, Violation, Compliant, and Target attributes.

NOTE
These tabs only appear for CIs with Federated Asset mappings to the CA Business Service Insight MDR.

The integration provides the following reports:

- **Service Level Compliance**
  Displays information about CA SDM Enterprise Service family CIs.
- **Modified Service Level by Metric vs. Target**
  Displays information about CA SDM Enterprise Service family CIs.
- **Modified Service Level by Contract vs. Target**
  Displays information for CA SDM Contract family CIs when you enter a valid CA Business Service Insight Contract Party and you refresh the report.

Consider the following information to help you debug error messages:

- For an incorrect CA Business Service Insight MDR userID or SharedSecret, the error message displays Invalid username/org.name.
- For an incorrect CA Business Service Insight host name, the error message displays BSI host name is unknown.
- For an invalid Contract Party in the BSI Contract by Metric tab, the error message displays there is no metric data for the contract for the Reporting Period.
- If no CA Business Service Insight reports are associated with a particular Service or Contract, the error message displays one of the following messages:
  - For Services, there are no service details report for federated service id.
  - For Contracts, there is no metric data for the contract for the Reporting Period.
- If CA Business Service Insight services are down on the CA Business Service Insight computer, the error message displays the CA Business Service Insight host name is unknown.

Maintain Relationships

You can create and manage the relationships between configuration items.

To create a new relationship:

1. From the Administration tab, navigate to the CI Relationship List.
   The Configuration Item Relationship List page appears.
2. Click Create New.
   The Create New Configuration Item Relationship page appears.
3. Fill in the following fields:
   - **Tenant (if multi-tenancy is installed)**
     The tenant level for this relationship.
   - **Provider/Peer CI**
     The configuration item that provides the relationship. Enter the configuration item name directly into this field, or click the search icon to select the configuration item from a list.
   - **Relationship Type**
     Use the Relationship Type Search and List to select relationship type.
   - **Dependent/Peer CI**
     The dependent configuration item in this relationship. Enter the configuration item name directly into this field, or click the search icon to select the configuration item from a list. This is a required field.
   - **Symbol**
     (Optional) The unique identifier for this relationship. You should assign a symbol that makes the relationship easily recognizable from the list.
   - **Description**
     (Optional) A detailed description of the relationship. You can use this field to describe the repositories assigned in the relationship, or the purpose of the relationship.
   - **Source Repository**

(Optional) A detailed description of the relationship. You can use this field to describe the repositories assigned in the relationship, or the purpose of the relationship.

– **Cost**
(Optional) The dollar value of the relationship between the configuration items.

4. Click Save.
The relationship is created and assigned to the selected CIs.

Configuration Audits

View CACF Change Specifications, Incidents, Change Orders, and Verification Log entries. The Configuration Audit node lets you perform the following actions:

- Manage Incidents that CACF created for rogue or improperly executed changes.
For example, View which changes have failed verification and which changes require additional investigation.
- Manage Changes Specifications.
For example, view all Change Specifications that completed successfully or are pending verification.
- Manage open Change Orders with active Change Specifications.
For example, view which changes orders are still open with pending change specifications.
- View the Verification Log history that shows all verification activities that have occurred in the system.
For example, view which MDR is responsible for rogue updates and the affected CIs.

Managing Change Specification

The Change Manager manages change specifications in your CACF environment. Change Analysts create Change Specifications when they create Change Orders. The Change Analyst defines the change in terms of the specific CI and CI attribute being changed.

The Change Specifications list shows all verified and outstanding change specification verifications in your environment. The ability to modify a change specification depends on the status of the Change Order. By default, you can only modify change specifications in Change Orders with the RFC status. After CACF verifies this change, it considers all subsequent changes as rogue inserts or updates

WARNING
If multiple discoveries occur during change specification verification, discovery data can invalidate a previously verified change. At the end of verification, the CI is in the state that all the change specifications desire. For example, a variance occurs for a CI with a change specification in the Verified state, but the status changes to Failed Verification.

A single CI can have many outstanding change specification verification pending from many Change Orders. CACF evaluates each verification independently, and a single MDR update can update multiple pending change verification.

NOTE
The Final field on the Change Specification List indicates whether a change has been completed. The Change Order is eligible for to move to the next default state when all change specifications are final and the Promote Change Order After Verification option is enabled for the managed change state.

The following fields require explanation:

- **Verify Status**
Specifies the status of the change specification, such as Verification Pending.
- **Change Order Status**
Specifies the status of the Change Order.
- **Incident**
Specifies the Incident that the verification policy creates optionally for an incorrectly implemented Change Order.

If the Configuration Administrator specified the policy option to create Incidents, the Incident documents the expected and discovered values. If all discoveries result in failed verification, the Incident receives log comments indicating that the incident represents multiple failures.

- **CI Name**
  Specifies the CI that you want to define the change specification.
  Note: When you make several identical changes to a set of CIs, you can leave the CI Name field blank in the Change Specification. In this case, the Managed Attribute and Planned Value apply to all CIs attached to the Change Order. Change specifications for each CI are created when the Change Order moves to a status with Change Verification Active enabled for the managed change state.
- **Original Value**
  Specifies the value of the attribute at the time a user created change specification.
  The original value field shows the current attribute value for the CI when defining the change specification value for the first time.
- **Last Discovered Value**
  Specifies the last value that was discovered for the attribute during the implementation and verification active process. This field updates each time CACF discovers a new value for the attribute.
- **Last Verification Policy**
  Specifies the verification policy that was in effect during the last change verification for this change specification.
- **Verification Message**
  Shows the message associated with the last verification action such as the reason why a verification failed.
- **TWA**
  Link to the TWA record associated with this change specification and policy.

Create a Change Specification

Create a change specification for a Change Order. By default, you can only modify change specifications in the RFC state. The Configuration or Change Administrators can change the definition of the Change Order managed change state that allows modification of change specifications.

The Verification Log tab shows the change verification activity history for this change specification. The log shows details about the policy and actions that CACF takes when change verification was in effect for the change specification. Verification log entries highlighted in red indicate that the corresponding change specification is either Failed Verification or Manual Verification Active, and requires further attention by the user.

To help you set the planned value, the Discovered Attribute History tab lists all values that an MDR recently discovered, whether they were authorized or loaded into the CMDB. From this tab, you can see the format and case sensitivity of the values so that you can determine an appropriate planned value pattern.

The Change Specification History tab shows the audit history listing all previous modifications made to the change specification options with the time and the user that made the changes.

Follow these steps:

1. From a Change Order, click the Configuration Management tab and select the Change Specifications tab.

   NOTE
   You can also create change specifications from the Administration tab from CA CMDB, Configuration Audit, Change Specifications. You can also create them from a CI if you click the Related Tickets tab then select the Change Specifications tab.

   The Change Specifications List page appears.
2. Click Create New.
   The Create New Change Specification page appears.

3. Complete the following steps:
   a. Enter the Change Order number that requests the change (only required if created from a CI or the Administration tab).
   b. Enter the name of the CI associated with the Change Order.
      You can leave the CI name blank to imply that the change specification applies to all defined CIs for the Change Order.
   c. Set the change specification as Active or Inactive.
   d. (Optional) Enter a description for the change specification.
   e. Select the Managed Attribute for this change specification from the drop-down list.
      Selecting Any Managed Attribute indicates that any managed attribute can be changed during the verification of the Change Order. In this case, multiple attributes can be updated, the planned value cannot be specified, and the planned change status is never updated.
   f. Enter the planned value of the Managed Attribute.
      Specify the expected value of the attribute after executing the change. After the Change Order moves into a verification state, and the CI updates through the web interface, GRLoader, or Web Services, CACF compares the planned value with the inbound data to determine a match.
      You can embed special characters in the planned value when the exact value is not known.
   g. Select the appropriate verification state of the Change Specification from the drop-down list:
      - Manual Verification will be required
        When CACF verifies the Change Order, manual verification will be required.
      - Set after Change Verified
        Set the CI attribute to the planned value after verification completes. Use this option for setting CI attributes when the attribute is not discoverable.

        NOTE
        The set operation only occurs when the Change Order moves from a state with change verification active to a nonactive default state. For example, the set takes place when a Change Order moves from Verification in Progress to Closed. The set does not occur if the Change Order is canceled, demoted, or moved to another state.
      - Use Discovered Value
        Copy the discovered value to the CI at verification time, used when the planned value is not known prior to verification.
      - Verification Pending
        The change has not been verified and waits for a CI update.
4. Click Save.
   The change specification is saved.

Verification Log

The Verification Log list lets you view all CACF change verification activity. CACF logs each attempted update to the CI, and if the update was allowed. This log helps you determine which policy allowed or disallowed a change to a CI.

For example, search for a log that specifies a specific CI where CACF detected a variance or unverifiable change. The results display information such as the previous and discovered value, the verify status, and associated Incident.

You access the verification log from the following locations:

- On the Administration tab, click CA CMDB, Configuration Audit, Verification Log.
- On the Scoreboard click CMDB, Configuration Audit, Rogue Inserts or Rogue Updates.

The Verification Log tab for the specific object shows detailed events only pertaining to the object in question. View the log for the following objects:

- Verification Policies
- Change Orders
- Change Specifications
- CIs
- Incidents

NOTE
Verification log entries highlighted in red indicate that the corresponding change specification is either Failed Verification or Manual Verification Active, and requires further attention by the user.

The Discovered Attribute History list displayed in the Change Specification form lists all attribute values for the currently selected managed attribute and CI that were recently discovered by an MDR, whether they were actually authorized or loaded into the CMDB.

Use this list to assist in defining the planned value for the change specification. In the list you can view the format, case sensitivity, and so on of the previously discovered values so that you can determine an appropriate planned value pattern.

Configuration Control

Under the Configuration Control node in the Administration tab, you can view and manage CACF Managed Attributes, Managed Changes States and Verification Policies which control change verification. Change verification governs inbound CI data before loading the data into the CMDB. This verification helps ensure that each requested Change Order executes correctly, and detects and manages rogue changes automatically. Change verification lets you perform the following tasks:

- **Authoritative MDR Updates**
  Specifies that MDRs can only update authoritative attributes at the attribute level.
- **Rogue Change Detection**
  Detects and manages updates to CIs when no corresponding Change Order exists.
- **Change Verification**
  Provides automated verification that Change Orders executed correctly.

Change verification policies can be implemented dynamically or scheduled in advance. They can be generic or highly specific.

Managed Attributes

Managed attributes indicate those eligible CI attributes for change verification by CACF. By default, this list contains CI name and Any Managed Attribute. You add CI attributes that you want managed as part of your change verification strategy.

CACF does not consider unmanaged attributes (attributes not listed) for change verification. These unmanaged attributes update as usual. Selecting Any Managed Attribute as the Attribute Name in a verification policy or change specification applies to all managed attributes listed.

Case sensitivity in the managed attribute definition only applies when CACF compares the change specification planned value with the inbound CI transaction data. Case sensitivity does not apply to the selection patterns in the policy, which are always case-sensitive.

WARNING
By default, CACF manages the CI Name attribute to enable rogue insert rejection. If you inactivate Name from the managed attributes list, executing a transaction that inserts a CI but does not update any managed attributes is considered an unmanaged change. Consequently, this inactivation may cause CACF verification policies to ignore rogue inserts.

Create a Managed Attribute

Add managed attributes to use change verification in your environment. For example, the Configuration Administrator wants change verification enabled for a specific CI attribute and requires defined change specifications when updating a CI in the CMDB.

The Managed Attribute History tab shows the audit history that lists all previous modifications to the managed attribute options with the time and user that made the changes.

Follow these steps:

1. On the Administration tab, click CA CMDB, Configuration Control, Managed Attributes.
   The Managed Attribute List page appears.
2. Click Create New.
   The Create New Managed Attribute page appears.
3. Enter a valid CI attribute name.

   NOTE
   For information about attribute names, see the CA CMDB Technical Reference Guide. You can also use grloader -sc xxxxx (where xxxx indicates the class name) to list attribute names for a particular class.
4. Enter an attribute label.
5. Select the default Initial Status from the drop-down list of the verification criterion. When you select a managed attribute when creating a change specification, this status is used as the initial default value.
   - **Manual Verification will be required**
     Indicates that this verification requires a Configuration Administrator to verify that the change completed successfully.
   - **Set After Change Executed**
     Indicates that the planned value from the change specification sets in the CI after CACF verifies and completes the Change Order is verified and completed. This set operation only occurs when the Change Order moves from a state with change verification active to a nonactive default state. For example, the set takes place when a Change Order moves from Verification in Progress to Closed. The set does not occur if the Change Order is canceled, demoted, or moved to another state.
   - **Use Discovered Value**
     Indicates that the Configuration Administrator has decided to use the discovered value regardless of the planned value in the change specification. The original value in the verification criterion remains unchanged, but updates the CI attribute with the new value.
   - **Verification Pending**
     Indicates a pending verification.
6. Set the managed attribute as Active or Inactive to remove the attribute from being considered by CACF.
7. Enter a description for the managed attribute.
8. Enable the Ignore Case option if you want the comparisons between the discovered and planned values to remain case insensitive.
9. Click Save.
   The managed attribute is saved.

Managed Change State

CACF uses managed change states to indicate which Change Order statuses CACF manages. CACF uses managed change states to control how or when to apply change verification for CI updates made to the system. You can customize these change states to suit the needs of your organization.

The Change Order statuses represent editable changes, and Change Order change states represent verification states. By default, you can edit change specifications for a Change Order in the RFC change state, and change verification

occurs for a Change Order in the Verification in Progress state. Optionally, change tickets can be automatically promoted or closed when all associated change specifications execute successfully.

The following list describes the Managed Change States options:

- **Change Specifications Editable**
  Specifies whether you can edit the change specifications for a Change Order. Typically, after a change request receives approval, you cannot change the request, and updates are restricted to performing a small set of override options in the Verification in Progress state.
- **Change Verification Active**
  Specifies whether changes discovered for a CI while the Change Order is in this status are considered for change verification. Change orders and their related change specifications are compared with any inbound transactions to verify that they were executed successfully.
  CACF monitors all CIs for any changes to their managed attribute values. As CACF verifies each attribute level change to the CI, CACF compares it against a list of change specifications which are in a change verification active state. After a change specification enters this state, any change specification without a specific CI undergoes expansion. This expansion occurs where new change specifications are created using the list of CIs attached to the Change Order. After a change specification exits this state, the list of change specifications with a verify status of Set After Change Verified execute. This action updates the CI with the planned values, as specified in each change specification.
- **Implementation State**
  Specifies whether the state represents a state when the changes are being executed or implemented on the CI. When a Change Order enters this transitional state, assume that the attribute values in the CI are volatile. CACF cannot consider these changes rogue changes, but it also cannot consider the changes for final verification. Typically, the change verification process should only compare the inbound attribute data after the Change Order executes completely.
- **Show Change Specification Override Buttons**
  Specifies whether the Change Analyst can control change specifications, and what level of control is given. In some implementations, the Change Analyst can edit change specifications as necessary, while in other implementations, the Change Analyst can override or cancel the change specification.
- **Promote Change Order after Verification**
  Specifies whether a Change Order promotes to the next default state automatically after CACF verifies all the change specifications.

Create a Managed Change State

Create a managed change state for your change verification environment. For example, the Change Administrator wants to create the Approval In Progress state and set the verification to active.

The Managed Change State History tab shows the audit history that lists all previous modifications made to the managed change state options with the time and user that made the changes. The tab also shows any previous values and the users that specified them.

Follow these steps:

1. On the Administration tab, click CA CMDB, Configuration Control, Managed Change States.
   The Managed Change States List page appears.
2. Click Create New.
   The Create New Managed Change State page appears.
3. Enter a Change Order status, such as Approval in Progress.
   You can also click Change Order Status to search for a status.
4. Select the option to set the change state as Active or Inactive.
5. (Optional) Select any of the following change states:
   – **Change Verification Active**

Enables change verification for this change state.

- **Implementation State**
  Indicates that the Change Order is in an implementation phase.
  Use this option to prevent unwanted Incident creation for incomplete Change Orders. For example, CACF detects a rogue change where the change matches the CI and attribute names of a change specification. The verification policy states to only update if the change matches. You can reject this change because the Change Order is verified after fully implemented and moved to a verification state.

  NOTE
  You cannot use a managed change state as both as an active change verification and an implementation state.
- **Change Specifications Editable**
  Indicates that you can edit change specifications in this state.
- **Show Change Specification Override Buttons**
  Displays the override buttons when a change verification is active, but is not allowed an edit, such as Accept Discovered Value, Accept Planned Value, Mark As Verified, and Cancel.
- **Promote Change Order After Verification**
  Promotes the Change Order to the next state automatically after change specifications for a Change Order move to a verified or final state.

  NOTE
  Promotion only occurs when the Change Order is in a state with change verification active.

6. Click **Save**.
   The managed change state is created and saved.

Verification Policies

Verification policies specify the action that CACF takes when an MDR discovers an attribute value, and attempts to update the CMDB with that data. A variance occurs when the inbound data does not match the desired state information, as described in the change specification. A policy search executes to locate a policy that matches both the change order alignment and change specifications. Policies process in an ascending, sequential number order. By default, a compatibility policy is provided to let all updates occur providing the behavior observed in previous releases of CA SDM.

The change verification policy describes how CA SDM responds to the following events:

- **Updates from Unauthorized MDRs**
  Specifies that specific MDRs are authoritative for specific attributes. CI attribute updates from unauthorized MDRs can be selectively accepted or rejected.
  For example, define policies to prevent CA Configuration Automation from updating the IP address of a CI, even if a matching Change Order exists. Allow updates to the IP address to occur only if the source MDR is Spectrum.
- **Rogue Changes**
  Detects and manages updates to CIs when no corresponding Change Order exists. Specify a policy that manages rogue changes that request inserts or updates of CI data.
  For example, define a policy whenever a CI named like server* in New York changes, but does not have a matching Change Order, do not update the CI but instead load the data to the TWA.
- **Incorrectly executed changes**
  Detects whenever a Change Order is not implemented correctly.

The policy specifies any of the following actions with the discovered data inbound to the CMDB:

- Accept the data into the CMDB even though a variance exists.
- Reject the inbound attribute data.
- Reject the entire inbound transaction.
- Write the inbound data to the TWA.
- Create an Incident.
- Write to the standard log.

> **WARNING**
> The Configuration and Change Administrators must establish a change verification strategy for your environment. The default policy specifies CACF to allow all changes to all CIs, even if the change is rogue or does not match a change ticket.

Create a Verification Policy

The Configuration Administrator creates verification policies for change verification. For example, the policy can reject specific attributes and values from a transaction, or can record the transaction to the TWA for deferred processing. For authorized source MDRs, the policy can accept the transaction unconditionally.

The Verification Policy History tab shows the audit history that displays all previous modifications made to the policy with the time and user that made the changes. The Verification Log tab shows all the CACF events associated with the policy. The Verification Log tab lists all events that have occurred for the policy. The log shows details on the affected change specifications, managed attributes, and actions when the policy was effect.

For example, the log displays the modified managed attribute and the policy history shows the category values that you specified in the policy. Verification log entries highlighted in red indicate the corresponding change specification is either Failed Verification or Manual Verification Active and require further attention by the user.

Follow these steps:

1. Click Create New.
   The Create New Verification Policy page appears.
2. Complete the Policy Description:
   - **Sequence**
     (Unique) Specifies the numbered order to process the policy.

     > **NOTE**
     > CACF processes policies in ascending order.
3. On the Policy Settings tab, select the appropriate Change Order alignment:
   - **Change Orders With Specifications**
     Active if a Change Order with selection criteria matches the inbound CI and the attribute. The inbound transaction data may not match the specified change specifications. If an exact match exists between the planned value and the inbound data, the policy determines whether to close any Incidents that change verification creates automatically.
   - **Change Orders Without Specifications**
     Active for Change Orders that do not have any change specifications defined. Indicates that CACF cannot verify the variance for the inbound CI attribute data for Change Orders that do not have change specifications.
   - **Rogue Insert**
     Indicates that the variance is found without a matching CI or change ticket.
   - **Rogue Update**
     Indicates that the variance is found with a matching CI, but without a matching change ticket that updates the changed attribute.
4. Select a Managed Attribute. Select Any Managed Attributes if the policy applies to all managed attributes.
5. Enter the appropriate patterns (Role Pattern, MDR Name Pattern, MDR Class Pattern, CI Name Pattern, Class pattern, Location Pattern), considering the following rules:

- When you specify patterns, the MDR Name Pattern, MDR Class Patterns, and role patterns refer to the source of the data update. The other filters refer to the CI that a user updated.
- Specify an exact string match for the inbound attribute value.
  For example, enter **COHESION** to return all string matching COHESION.
- Specify an exact string matched suffixed with a wildcard (*) for the inbound attribute value. CACF only allows one asterisk at the end of the string.
  For example, enter **spectrum*** to match the string spectrum, and any string that starts with spectrum.
- Specify the negative of an exact string match by prefixing the string with an exclamation point.
  For example, enter **!COHESION** to match any string that does not start with COHESION.
- The MDR Name of Web Client indicates that the web interface created the source data.

  NOTE
  Policy filters do not support embedded wildcards, such as spec*rum..

6. (Optional) Select a Priority level from the drop-down list to match the CI priority.
7. (Optional) Select a Service Type from the drop-down list to match the CI Service Type.
8. Select the appropriate Update Behavior from the drop-down list:
   - **Allow Attribute Update**
     Updates to CI attribute values are always allowed.
     Use this value when you require no attribute verification, such as when you require logging and auditing.
   - **Allow Update Only if Matches Change Specification**
     Updates the CI attribute value only if it matches the planned value of the change specification. Otherwise, CACF rejects the update, and the change specification status sets to Failed.
     Use this selection when you enable Change Order verification.
   - **Always Cancel Entire Transaction**
     Cancels this update and all other updated attributes during this transaction. If any matching policy cancels the entire transaction, the transaction is canceled, even if other policies allow the change.
     This value only triggers when an update occurs to a managed attribute.
   - **Keep Old Attribute Value**
     Value rejects the inbound data.
     This value is similar to cancelling the entire transaction, but it only rejects a single attribute.
9. Select the appropriate Action:
   - **Write Data to TWA**
     Select Always or Never. If you select Always, the incoming data is written to the TWA for later processing.
   - **Create Incident**
     Select Yes or No to create Incidents based on this verification policy. If you select Yes, CACF creates an Incident if verification fails or a rogue update occurs.
   - **Incident Template**
     (Optional) Select an existing template to use if CACF creates an Incident.
   - **Close Incident after Verification**
     Automatically closes any associated Incidents with this verification policy when the data matches the Change Order and all change specifications are verified in a final state.
   - **Log Only Mode**
     Adds a warning message to the stdlog whenever this policy would take effect.

     NOTE
     Any higher sequenced policy governs the transaction, as if this mode policy did not exist.
10. Specify Activation and Deactivation dates when this policy is in effect, and click Save.
    The verification policy is saved.

MDR-Management

The MDR Launcher is an open integration tool that lets you view data in a management data repository (MDR) from a CA CMDB web page. MDR Launcher lets you obtain additional details about a CI and to gain control over it (if the MDR supports such control).

Manage Federated CI Mappings

A single CI can be associated to multiple MDRs, with each MDR having a federated_asset_id that identifies the CI. While it is not necessary for the federated_asset_id to be unique among MDRs, a federated_asset_id must be unique within each MDR.

Each MDR is identified uniquely by its combination of MDR class and MDR name.

To view the CI in a particular MDR context, click Launch. The target MDR is launched in the context of the CI that was open.

View an MDR Location for a CI

You can display the MDR location for a CI.

- **CI Name**
  Identifies the configuration item.
- **Federated Asset ID**
  Specifies the string that the source MDR uses to identify this CI. The MDR software typically determines this identifier. CMDBf Viewer uses the UUID.
- **MDR Name**
  Identifies the MDR.
- **Active**
  Denotes whether this mapping is active or not. You can inactivate mappings but not delete them.

Manage MDRs

Before you can associate a CI with an MDR, define the MDR to CA SDM.

To define a new MDR, follow these steps:

1. On the MDR List, click Create New.
   The Create New MDR Definition page appears.
2. Enter the definition for the new MDR.
3. Click Save.
   The MDR definition is saved.

   NOTE
   When you associate a CI with an MDR, the Add MDR button does not appear until the CI is first saved and then edited.

Manage an MDR Definition

You can create, view, and edit an MDR definition. The MDR Definition page provides the following settings:

- **Tenant**
  Identifies the tenant owner of this MDR (if multi-tenancy is installed).
- **Button Name**
  Defines the MDR button label that appears on the CI Detail page. This name is unique for each MDR. Required for “launch in context” and for CMDBf Viewer.
- **MDR Name**

Specifies the data that is sent in the mdr_name field in XML. While this string can be anything that the MDR chooses to send to CA CMDB, typically it is the hostname of the source. This name together with the mdr_class create a unique name for the MDR.

NOTE
CA APM MDRs must specify an MDR name of APM and MDR class of GLOBAL. Required for “launch in context", CMDBf Viewer and GRLoader.

- **MDR Class**
  Specifies the data that is sent in the mdr_class field in the XML. While this name can be anything, it must together with the mdr_name field create a unique identifier for the MDR. Global MDRs are defined with an MDR Class of “GLOBAL”.

  NOTE
  Cohesion MDRs must specify an MDR class of COHESION, which automatically sets the Path, Parameters and "URL to be Launched" fields to the required Cohesion launch-in-context values. CA APM MDRs must specify an MDR class of "GLOBAL", which automatically sets the Path, Parameters and "URL to be Launched" fields to the required CA APM launch-in-context values. Required for CMDBf Viewer. CMDBf Endpoint sets to "cmdbf" for all federated MDRs.
- **Active**
  Denotes this MDR definition as active or inactive. Inactive MDR definitions are logically deleted, but they can be made active again by using the Search utility.
- **Owner**
  The contact responsible for this MDR.
- **Description**
  A freeform text field.
- **Hostname**
  The host name, DNS name or IP address of the host, which contains the web server which hosts the web page that is launched. Required for “launch in context”.
- **Port**
  The TCP/IP port used by the MDR web server to serve up web pages. Port 8080 is the default. Required for “launch in context”.
- **Path**
  The portion of the URL that precedes the question mark (?) character. This information can be obtained from the MDR documentation. For mdr_class of Cohesion, the value is set automatically to CAisd/html/cmdb_cohesion.html and cannot be changed. For mdr_name of UAPM and mdr_class of GLOBAL, the value is set automatically to apm/frmObject.aspx and cannot be changed.
- **Parameters**
  The portion of the URL that follows the question mark (?) character. This information can be obtained from the MDR documentation. For mdr_class of Cohesion, the value is set automatically to hostname={hostname}+port={port}+family={family}+name={name}+secret={password}+federated_asset_id={federated_asset_id and cannot be changed. For mdr_name of UAPM and mdr_class of GLOBAL, the value is set automatically to ObjectID=cmdb_asset_id}&obj=11&FUNCTION=1&WinID=OBFRASSET{cmdb_asset_id}&WinContainerID= and cannot be changed.
- **Userid**
  The user ID to logon to the MDR (if necessary). This value is substituted into the URL wherever {userid} is found.
- **Shared Secret**
  Specifies information that is shared between CA CMDB and the MDR. This value is substituted into the URL wherever {password} is found. For Cohesion MDRs, the value must match the value of the “com.cendura.security.oneclickauth.secret”. For more information about creating a shared secret, see "Integrating with CA CMDB" in the *CA Cohesion ACM Implementation Guide*. Required for CMDBf Viewer.
- **CMDBf Namespace**

Specifies the MDR namespace that is passed to the query by CMDBf Viewer. If you use CA CMDB as an MDR provider, the value is "http://cmdb.ca.com/r1". For external MDRs, consult the MDR documentation. Required if MDR Class is cmdbf.

- **CMDBf Timeout**
  (Optional) Specifies time limit for CMDBf endpoint query. Default is ten (10) seconds. Only applies when MDR Class is cmdbf.
- **URL to be Launched**
  Default value of http://{hostname}:{port}/{path}?{parameters}. For some MDRs, it can be overridden if necessary to accommodate MDR-specific requirements. Required for "launch in context". For mdr_name of UAPM and mdr_class of GLOBAL, the value is http://{hostname}:{port}/{path}?{parameters}
  For mdr_class of Cohesion the default value is:
  http://cmdb_hostname:cmdb_port/{path}?{parameters}
  where:
  cmdb_hostname is the host name, DNS name or IP address of the CA CMDB web server. Defaults to the current hostname that is currently accessing the CA CMDB web server.
  cmdb_port is the TCP/IP port of the CMDB web server. Defaults to the current port number used to access the CA CMDB web server.

  NOTE

  : If you have enabled SSL support for Cohesion, set the URL to:

  ```
  http://hostname:port/{path}?{parameters}+https=yes
  ```

  For information about enabling CA Cohesion ACM HTTPS support, see the Cohesion online help topic Creating the HTTPS Certificate and Enabling HTTPS.
- **CMDBf Endpoint**
  Specifies the MDR Query Service endpoint for CMDBf Viewer. Required if MDR Class is cmdbf.

Reconciliation Management

We recommend managing ambiguity on a CI by CI basis. In each case, the Configuration Administrator must detect when ambiguity exists and determine the optimal approach. CA SDM takes a broad approach to identify and manage ambiguous CIs that are already in the CMDB.

The *ambiguity index* is an operational measure of the potential non-uniqueness of a configuration item (CI) or a CI transaction item, based on its identifying attributes. The ambiguity index measures the probability of one or more CIs representing the same real business object, or the probability that a CI transaction has more than one possible target CI.

Uses the following process to identify and resolve ambiguous CIs:

1. Identify ambiguous CIs.
   – Calculate the ambiguity index for all CIs
   – Review the list of ambiguous CIs from the Scoreboard
2. Determine if the identifying attributes for each CI are valid.
3. Resolve the ambiguous CI with one of the following actions:
   – Modify the identifying attributes
   – Reject a CI Update by inactivating the CI
   – Supersede a CI
   – Exclude a CI from ambiguity management

Manage Ambiguous CI Transactions

An administrator should review and modify transactions that have ambiguous CI targets before loading the transaction data. You can also resolve CI transactions that do not have a target CI but have identifying attributes for one or more

existing CIs. This resolution helps ensure that the appropriate CI is updated and prevents incorrect or inconsistent data from being created.

> **NOTE**
> To view ambiguity either for CIs or CI transactions, you must calculate the ambiguity index by using the cmdb_update_ambiguity utility.

Manage Ambiguous CIs

An administrator should review and resolve the ambiguity between CIs to eliminate incorrect or inconsistent data.

> **NOTE**
> To view ambiguity either for CIs or CI transactions, you must calculate the ambiguity index by using the cmdb_update_ambiguity utility.

Manage Superseded CIs

You can supersede an ambiguous CI so that data is redirected to a specific target CI. You can display superseded CIs, which are Inactive and all web interface updates to them are ignored.

> **NOTE**
> Updates sent to superceded CIs by GRLoader or CMDBf web services are redirected to the superceding CI. However, identifying attributes are ignored.

Transaction Work-Area

The transaction work area (TWA or work area) is a data warehouse where you can store, review, and modify CI and relationship transaction information from multiple sources before you load the transactions into the CMDB. In the TWA, you can modify incoming transaction data to conform to your CA CMDB requirements. You also can use the TWA to modify data input that could create false positives (multiple CI representations of the same business object) or false negatives (multiple business objects identified as the same CI) during the reconciliation process.

> **NOTE**
> Transaction information can apply to new objects or to objects that are already in the CMDB.

Manage CI Transactions

An administrator can display and manage CI transactions in the TWA in order to correct problems before the data is loaded into the CMDB.

To manage CI transactions, an administrator can do the following:

- Search for CI transactions
- Show Filter to search based on:
  - Tenant (if multi-tenancy is enabled)
  - Transaction run status
  - Active or Inactive
  - Transaction message
  - Change Order
  - Identifying and federating attributes
  - Target CI
  - Superseded By
  - CI class or family
- Edit a transaction
- Right-click on a transaction to:

  - View transaction details
  - Edit the transaction
- Create a CI transaction
- Edit in list

Manage a CI Transaction

An administrator can create, view, or modify a CI data transaction. The CI Transaction Detail page displays all nonblank attributes in the transaction. If you clear the Hide Empty Values check box, all CI attributes are displayed. For additional attribute information, point to the attribute name.

> **WARNING**
> Data that you enter using this web interface must have the same format as if specified in native SQL or XML. For example, when you enter data for a lookup field, refer to the section on XML Input. When you want to specify an owner, to look up an owner by userid, use braces { } to specify the lookup field. For example, to find the name associated with userid "admin", enter admin{userid} in the owner column.

The TWA accepts free-form text as case sensitive by default unless you use the -I GRLoader option to enable case insensitivity. Attribute values are not validated until load-from-TWA run time, so values must be entered exactly as required. If you enter string values for nonstring MDB fields, GRLoader issues warnings when loading to the CMDB. You can also simulate loading the data to predetermine (-simci) whether a set of transactions can create new CIs (ambiguities with existing CIs) or report errors in the data.

An administrator can do the following:

- View a transaction
- Create a transaction
- Edit the transaction
- Save or Cancel transaction changes (Edit mode)
- Reset the original transaction data (Edit mode)
- Show or hide data (Attributes tab)
- Set the target CI (Reconciliation tab)

Restrictions:

- If tgt_id is defined for a transaction, it displays as a CI name in the Target CI field.
- If superseded_by is defined for a transaction, it displays as a CI name in the Superseded By field.
- TWA entries are not logged.

> **NOTE**
> For more information about data values for special attributes, see How to Load the TWA Using GRLoader.

Manage Relationship Transactions

The Relationship Transaction List displays relationship transaction data input.

Using this page, an administrator can do the following:

- Search for relationship transactions
- Show Filter to search by:

- Tenant (if multi-tenancy is enabled)
- Relationship type
- Transaction run status
- Active or Inactive
- Transaction message
- Change Order
- Provider CI identifying attributes
- Dependent CI identifying attributes

On the list, an administrator can do the following:

- Create a relationship transaction
- Edit in list (except for identifying attributes)

Manage a Relationship Transaction

An administrator can view and modify data for a single relationship transaction. On the Relationship Transaction Detail page, an administrator can do the following:

- Create a relationship transactionView the transaction
- Edit the transaction
- Save or Cancel transaction changes (Edit mode)
- Reset the original transaction data (Edit mode)

To update a relationship transaction, follow these steps:

1. Click Edit.
   The Update Relationship Transaction page appears.
2. Modify and complete fields.
3. Select Active.
4. Click Save.
   The relationship transaction is saved.

The TWA accepts free-form text as case sensitive by default unless you use the -I GRLoader option to enable case insensitivity. Attribute values are not validated until load-from-TWA run time, so values must be entered exactly as required. If you enter string values for nonstring MDB fields, GRLoader issues warnings when loading to the CMDB. You can also simulate loading the data to predetermine (-simci) whether a set of transactions can create new CIs (ambiguities with existing CIs) or report errors in the data.

CA CMDB Data Maintenance

System

This section contains the following topic:

- Configuration
- Current Locks
- Interval Logging
- Servers

Configuration

This article talks about the configuration of CA CMDB Visualizer.

The CMDB Visualizer Configuration tab allows for modifying server settings and displaying Configuration Items (CIs). Login to CA CMDB. Select the **Configuration** option from the Left Menu tree to configure the following options in a graph:

1. **Display Settings**: Provides the following options to set CI labels in a graph:
   a. **Number of Lines to Display for CI Labels**:
   Based on the specified number, the CI Label displays in the graph. Example: If 3 is specified, the label displays in three lines in a graph.
   b. **Characters per Line for CI Labels**: Specify the Characters for CI labels in a graph. For example: If 10 is specified, after 10 characters, the label will display in a new line.
   c. **Display Relationship Label**: Enable or disable this option as per your requirement. If you disable this option, CI relationships are not displayed in a graph.
   d. **Relationship Label Length**: Specify the CI relationship label length - maximum number of characters in which a graph relationship label will be displayed.
   e. **CI Relationships to (Default value)**: Used to specify the default number of levels with which the graph will be displayed.
   f. **Enable Asset Search**: If enabled, you can search for assets or CIs.

Relationship Style

Use this option to define relationship characteristics in a graph.
Select the Relationship from the list. Select the option to differentiate a relationship in a graph via **line color** and **line thickness**. Use **Preview** to visualize how the graph will be displayed in real time with the specified options.

CI Status

Use this option to define CI graphical characteristics.

The user can select CI configuration Status Image along with Image color. User can **Preview** CI configuration status with selected color and category.

Current Locks

Current Locks let you view and release locks on any object. During normal operations, the system locks objects while users work with data. If there is a system failure, you can free objects for future updates. Because the initial locks default to 5 (five) minutes, you unlock the objects when another user needs the same data.

View Current Object Locks

You can view a snapshot of currently locked objects.

Follow these steps:

1. On the Administration tab, select System, Current Locks.
   The Current Locks page lists the following information about object locks:
   Lock
   Identifies the process that requested the lock.
   Name
   Identifies the lock name. If the lock is related to a ticket, the lock name can reflect the ticket number.
   Table Name
   Identifies the location of the locked object.
   Object Name
   Identifies the Majic object.
   Lock Owner
   Identifies the table user.
   Locked Since

Shows the date and time of the lock.

Process
Identifies the process that initiated the lock.

2. Refresh the browser.
   The Current Locks page shows the most recent object locks.

Interval Logging

Interval Logging allows you to collect resource usage data from your CA SDM servers. The collected interval log is used to analyze and troubleshoot any performance or memory-related issues in the CA SDM servers. You can share the collected data with CA Support Online to help identify and resolve CA SDM server problems.

Create Log Interval Configuration

To collect the usage data on the CA SDM servers, add the servers to the Interval Log Configuration list. You can select the type of data you want to collect from each server, and also change the logging options for the servers at any point.

For example, if you select only the CPU usage option for a particular server, the utility collects CPU usage data on the server.

Follow these steps:

1. Select System, Interval Logging from the Administration tab.
   The Interval Log Configuration List page opens.
2. Click Create New to add an interval logging configuration.
   The Create New Interval Log Configuration page opens.
3. Complete the fields as appropriate for the configuration.
   See Interval Log Configuration Fields for the field definitions.
4. Select the appropriate interval log you want to collect from the server.
   For example, select CPU Usage if you want to collect the log for CPU usage data.

   CPU Usage
   Collect the CPU usage statistics for the server by executing the pslist –x on Windows, or “ps” on Unix.

   Memory Usage
   Collects the memory usage data for the server by executing the pslist –m on Windows, or “ps” on Unix.

   Network Status
   Collects the information of all active connections and network statistics by executing the netstat /b, or /a from the operating system.

   Task List
   Collects the application and services information for all the tasks running on the server.

   Web Session Counts
   Collects the CA SDM sessions and user statistics for the web engine processes by executing the pdm_webstat command.

   Server Status
   Collect the information about all CA SDM daemons or processes on the server by executing the pdm_status command.

   DB Report
   Collects the information of database record types by executing the db_report command.

   Virtual DB Info
   Collects the information that is related to queued database requests by executing the pdm_vdbinfo command.

   List Server Connections
   Collects the information on active connections for the server by executing the pdm_listconn command.

   List Slump Processes

Collects the information about slump connections and processes by executing the slstat command.

NOTE

Depending on the server type you select, the interval logging options are enabled or disabled for the server.

5. Click Save.
   The server is configured for interval logging.
6. (Optional) Repeat Steps 1-5 to create more interval log configurations.

Unlock Objects

You can review and release object locks to allow other users to access the object.

Follow these steps:

1. On the Administration tab, select System, Current Locks.
   The Current Locks page shows a list of object locks.
2. Refresh the browser.
   The Current Locks page shows the most recent object locks.
3. Right click the row.
   A Delete Lock menu appears.
4. Click Delete Lock.
   A confirmation message appears.
5. Click OK.
   The Current Locks page updates and a message notifies you that the object is free.

Servers

To configure servers for CA SDM, you add servers records using the web interface and then execute the pdm_configure utility on the servers. The server list displays information about the server deployed in your CA SDM installation. You can modify existing server information, or search for a specific configuration from the list.

To filter servers from the list, click the Show Filter and specify search criteria to filter your search. Optionally, click Export to export the server list in a Microsoft Excel sheet.

Add a Server

If you do not have any existing servers, add server records for all the servers that you want to install in your CA SDM deployment.

Follow these steps:

1. Log in to the following server, depending on your CA SDM configuration:
   - Conventional: Primary server
   - Advanced availability: Background server
2. Select System, Servers from the Administration tab.
   The Server List page opens.
3. Click Create New to add a server record for the following server, depending on CA SDM configuration:
   - Conventional: Secondary server
   - Advanced availability: Application or standby server
     The Create New Server page opens.
4. Complete the server fields, as appropriate for the server.
5. Click Save.
   You added the server detail.

Create Server Fields

The following fields appear when you create a server:

Host Name Specifies the local host name of the server. The local host name is stored in the usp_servers table in local_host column.

> **WARNING**
>
> Ensure that host name is entered as case-sensitive in the usp_servers table.

Time Zone Specifies the time zone where the server is located. The time zone is used for triggering events in the system if the Use End User Time Zone option is not selected, or if no time zone is specified for the service type.

Record Status Indicates the state of the server. Active status indicates that the server is a part of the CA SDM deployment.

> **WARNING**
>
> If you have inactivated any server, it is recommended not to start CA SDM services on that server. This action may result in unexpected behaviour.

Server Type Specifies the type of server that you want to configure. Following server types can be selected, depending on your CA SDM configuration:

- Advanced Availability: Application or standby server
- Conventional: Secondary server

ConfiguredAvailable only for advanced availability configuration. Indicates the state of the configured server. The default value of this field is No. The value is updated to Yes after you successfully run pdm_configure on that server. If you edit any of the automatically entered field values of a server record, the Configured field turns to No.

Enable SAML Authentication for CA SDM

Security Assertion Markup Language (SAML) is a standard for logging users into applications based on their sessions in another context. SAML SSO works by transferring the user's identity from one place (the identity provider) to another (the service provider). This is done through an exchange of digitally signed XML documents.

> **NOTE**
> CA SDM supports NTLMv1 and NTLMv2.

This article explain how to enable SAML authentication for CA Service Desk Manager on the following servers:

Prerequisites

1. Ensure that SSL is enabled for CA SDM. For more information see, Enable SSL Authentication for CA SM 17.4.
2. **For ADFS**: If you want to use ADFS as the identity provider, you must install ADFS on Windows Server 2012 R2 and above. After completing the installation of ADFS and CA Service Desk Manager, you must configure the ADFS with following the CA SDM related information:
   a. While configuring Add Relying Party Trust on ADFS specify the following on the **"Configure URL"** screen:
      - Select the **"Enable support for the WS-Federation Passive protocol"** check-box.
      - Enter the value in the **"Relying party WS-Federation Passive protocol URL:"** as below: ***https://<SDM host name>:<https port>/CAisd/pdmweb.exe***
      - Select the **"Enable support for the SAML 2.0 WebSSO protocol"** check-box.
      - Enter the value in the **"Relying party SAML 2.0 SSO service URL:"** as below: ***https://<sdm host>:<https port>/CAisd/pdmweb.exe***
   b. While adding claim rules, specify the following in the **"Configure Claim Rule"** step:
      - **Claim rule name**

Specify a name for the claim rule. For example, SDM Claim Rule.
- **Attribute store**
  Select the datasource from the drop-down list. For example, Active Directory.
- **LDAP Attribute**
  Select "**SAM-Account-Name**" from the drop-down list.
- **Outgoing Claim Type**
  Select "**Name**" from the drop-down list.

c. Verify that the certificate you want to use to configure the Identity Provider is SHA 256 signed.

3. **For CA Siteminder**: If you want to use CA Single Sign-On (formerly known as CA Siteminder), as your Identify Provider, do the following:
   a. Launch CA Single Sign-on as Administrator.
   b. Go to the Federation Partnership Partnership Federation and click Partnerships.
      Federation Partnership List appears.
   c. Click the Federation Partnership name which you want to use.
      Partnerships properties page opens.
   d. Expand the IP Restrictions section and verify the Assertion Attributes properties.
   e. The value of the Namespace field must be the URI for the respective Claim. If the value is specified as "Unspecified" the SAML authentication will not work.
   f. Verify that the certificate you want to use to configure the Identity Provider is SHA 256 signed.
4. **For Azure Identity Provider (IdP):** If you want to use Azure Identity Provider (IdP), as your Identify Provider, do the following:
   a. Login to Microsoft Azure Portal.
   b. Click on the "**Azure Active Directory**" link under Azure Services.
   c. Click on the "**Enterprise Applications**" link from left side window.
   d. Click "**New Application**" link and again click "**Create your own application**" link.
   e. Enter name example "**CASM tomcat**" in the "**What's the name of your App**?" text box and Click **Create** on the right-pane.
   f. Click on the "2. Set up single sign on" link and Click on the "SAML" link.
   g. Click on the Edit button on section 1 i.e "Basic SAML Configuration".
   h. Enter SDM SSL tomcat/iis URL like ***https://<sdmhost>:<sdm_ssl_port>:/CAisd/pdmweb.exe*** for all below text fields:
      a. "**Identifier (Entity ID)**"
      b. "**Reply URL (Assertion Consumer Service URL**"
      c. "**Sign on URL**"
      d. "**Relay State**"
      e. "**Logout Url**"
   i. Click **Save**.

Enable SAML Authentication for CA SDM on Tomcat

Follow these steps:

1. **Configure the web.xml file.**
   a. Navigate to the following folder and edit the file: *NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF*\web.xml
   b. Copy and paste the following configuration under the *<!-- Add filter here -->* statement:

```
<filter>
 <filter-name>FederationFilter</filter-name>
 <filter-class>com.auth10.federation.WSFederationFilter</filter-class>
 <init-param>
```

```
    <param-name>login-page-url</param-name>
    <param-value>main.jsp</param-value>
  </init-param>
  <init-param>
    <param-name>exclude-urls-regex</param-name>
    <param-value>/images/|/js/|/css/</param-value>
  </init-param>
</filter>
```

c. Copy and paste the following configuration under the *<!-- Add filter-mapping here -->* statement:

```
<filter-mapping>
 <filter-name>FederationFilter</filter-name>
 <url-pattern>/*</url-pattern>
</filter-mapping>
```

d. Save and close the file.

e. Repeat Steps b, c, and d to update web.xml.tpl file in the *NX_ROOT\samples\pdmconf* folder to avoid losing these changes, when the pdm_configure is run at a later point of time.

2. **Configure the federation.properties file.**

a. Navigate to the following folder and edit the file *NX_ROOT\bopcfg\www\CATALINA_BASE\shared\resources \federation.properties*

b. Define the following details related to the Identity Provider and CA Service Desk Manager

a. **federation.trustedissuers.issuer**
Define the Identity Provider URL.
Example:
Siteminder

NOTE

http://<siteminderURL>/affwebservices/public/wsfeddispatcher

ADFS

NOTE

```
https://<trusted_issuer_URL>/<identity_provider>/ls/idpinitiatedsignon.aspx
```

Azure Idp:
Perform the following:

a. Login to Microsoft Azure portal, navigate to **Enterprise Application**, <**CASM Application Name**>, Single-Sign-On, Block/Section 4, Login URL. Copy the URL (use the copy option adjacent to the URL).
For example: *https://login.microsoftonline.com/<tenant_ID>/saml2*

b. Replace **saml2** with **wsfed** in the URL.
For example: *https://login.microsoftonline.com/<microsoft_tenentid>/wsfed*

b. **federation.trustedissuers.thumbprint**
Define the value of the certificate thumbprint provided by the Identity Provider. You must specify the thumbprint from "**token-signing"** section of the certificate.
Example: 0214c3035d002505b9e5e672a117d9bf5c5d4d02

NOTE

The certificate you want to use to configure the Identity Provider must be SHA-256 signed. The thumbprint you use from the SHA-256 certificate must be the SHA-1 thumbprint.

Azure Idp: federation.trustedissuers.thumbprint

Perform the following:

1. Login to Microsoft Azure portal, navigate to **Enterprise Application**, <**CASM Application Name**>, Single-Sign-On, Block/Section 3, Thumbprint. Copy the value.
   For example: 7844FB57B8EB854C10F2CBD8BE78B0BF574B729D

c. **federation.trustedissuers.friendlyname**
Define a common name for the Identity Provider.
Example:
ADFS

NOTE

TestIP - <trusted_issuer_URL>

Siteminder

NOTE

TestIP

Microsoft Azure Idp:
Perform the following:

a. Login to Microsoft Azure portal, navigate to **Enterprise Application**, <**CASM Application Name**>, Single-Sign-On, Block/Section 4, Azure ID Identifier (use the copy option to copy the value).
   For example: *https://sts.windows.net/<tenant_id>/*
b. Replace **saml2** with **wsfed** in the URL.
   For example: *https://login.microsoftonline.com/<microsoft_tenentid>/wsfed*

d. **federation.audienceuris**
Define the URI of the application from where to accept the tokens.
Example: https://<SDM_HOST>:<SDM_TOM_SSL_PORT>/CAisd/pdmweb.exe|https://<SDM_HOST>:<SDM_TOM_SSL_PORT>/CAisd/pdmweb.exe

e. **federation.realm**
Define the location from where the tokens are sent.
Example: https://<SDM_HOST>:<SDM_TOM_SSL_PORT>/CAisd/pdmweb.exe

f. **federation.enableManualRedirect**
Define if you want to enable manual redirection of the token. Set the value as True to enable the manual redirect and set the value as False to disable the manual redirect.
Default: false.

g. **federation.reply**
Define the URL of the location that receives responses.
Example: https://<SDM_HOST>:<SDM_TOM_SSL_PORT>/CAisd/pdmweb.exe

c. Save and close the file.

3. Restart the CA Service Desk Manager services.
4. Launch CA SDM and enable the external user authentication from UI for the user role that you want to allow the SAML based SSO access.

Enable SAML Authentication for CA SDM on IIS

Follow these steps:

1. Ensure that the Microsoft.NET Framework 4 is installed on the server where CA SDM is installed. For more information, see Enable SSL Authentication for CA SM 17.3. For more information, see Microsoft Documentation on How to Setup SSL on IIS.
2. Take a backup of web.config file. Open the web.config file in an editor. The file is typically available under the path NX_ROOT\bopcfg\www\ wwwroot\

3. Locate the following statement in the file:
   <!-- To Enable SAML, Comment above code and Uncomment below code. -->
4. Remove all the code that appears above the statement in the file and remove the above statement as shown in Step 3.
5. Uncomment all the code that appears below the statement in the file, that means removing the tag **<!-- at the beginning and the tag -->** at the end of the code.
6. Find the following lines and replace the text "Define the URI of the application from where to accept the tokens" with http://<SDM host name>:<SSL Port>/CAisd/pdmweb.exe:

```
 <audienceUris>
<add value="Define the URI of the application from where to accept the tokens" />
</audienceUris>
```

   Example:

```
 <audienceUris>
<add value="http://CAServiceDesk:443/CAisd/pdmweb.exe" />
</audienceUris>
```

7. Find the following lines and modify the value of "add thumbprint" and "name" parameters. You must specify the thumbprint from "**token-signing"** section of the certificate.

```
 <trustedIssuers>
<add thumbprint="Define the value of the certificate thumbprint provided by the Identity Provider"
 name="Define a common name for the Identity Provider" />
</trustedIssuers>
```

   Example:

```
 <trustedIssuers>
<add thumbprint="bb89519f06dc7b1ab43871e1a3310eca893b94cb" name="sdmcomputer01.casm.local" />
</trustedIssuers>
```

8. Find the following lines and modify the value of "issuer", ""realm", "reply" and "requireHttps" parameters:

```
 <federationConfiguration>
<cookieHandler requireSsl="false" />
<wsFederation passiveRedirectEnabled="true" issuer="Define the Identity Provider URL" realm="Define the
 location from where the tokens are sent" reply="Define the URL of the location that receives responses"
 requireHttps="true|false" />
</federationConfiguration>
```

9. a. **issuer**
      Define the Identity Provider URL
      Example:
      - **ADFS:** https://hostname.casm.local/adfs/ls/idpinitiatedsignon.aspx
      - **Siteminder:** http://hostname:88/affwebservices/public/wsfeddispatcher
      - **Azure Idp**
        Perform the following:
        a. Login to Microsoft Azure portal, navigate to **Enterprise Application**, <**CASM Application Name**>, Single-Sign-On, Block/Section 4, Login URL. Copy the URL (use the copy option adjacent to the URL).
           For example: *https://login.microsoftonline.com/<tenant_ID>/saml2*
        b. Replace **saml2** with **wsfed** in the URL.
           For example: *https://login.microsoftonline.com/<microsoft_tenentid>/wsfed*
   b. **realm**

Replace the text **realm="Define the location from where the tokens are sent"** with **realm="https://<SDM host name>:<SSL Port>/CAisd/pdmweb.exe"**

Example: realm="https://CAServiceDesk:443/CAisd/pdmweb.exe"

c. **reply**

Replace the text **reply="Define the URL of the location that receives responses"** with reply="https://<**SDM host name**>:<**SSL Port**>/CAisd/pdmweb.exe"

Example: reply="https://CAServiceDesk:443/CAisd/pdmweb.exe"

d. **requireHttps**

Define whether the require https is true or false.

True

Specify the value as **true,** if the Identify Provider URL is https based.

False

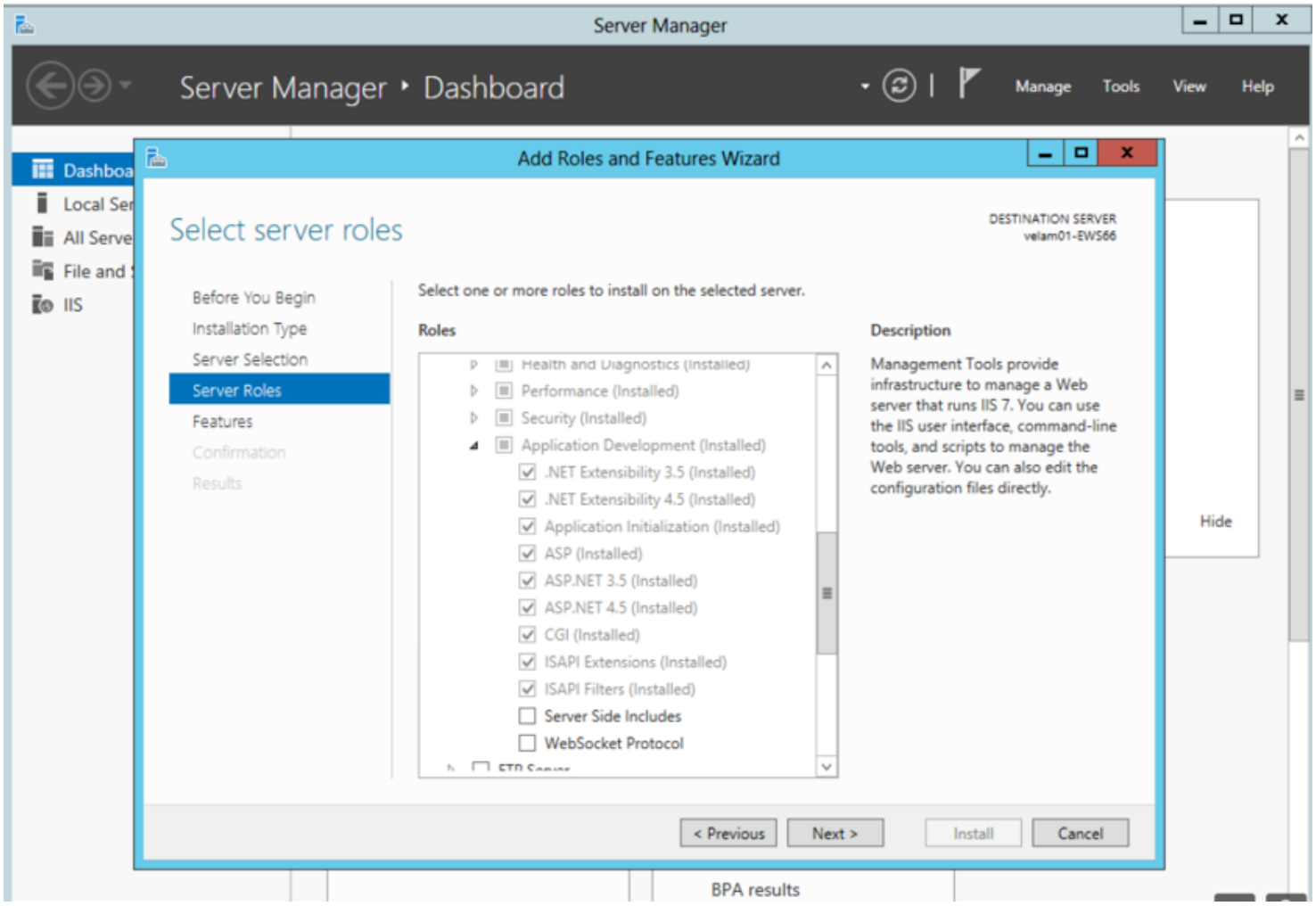
Specify the value as **false,** if the Identify Provider URL is http based.

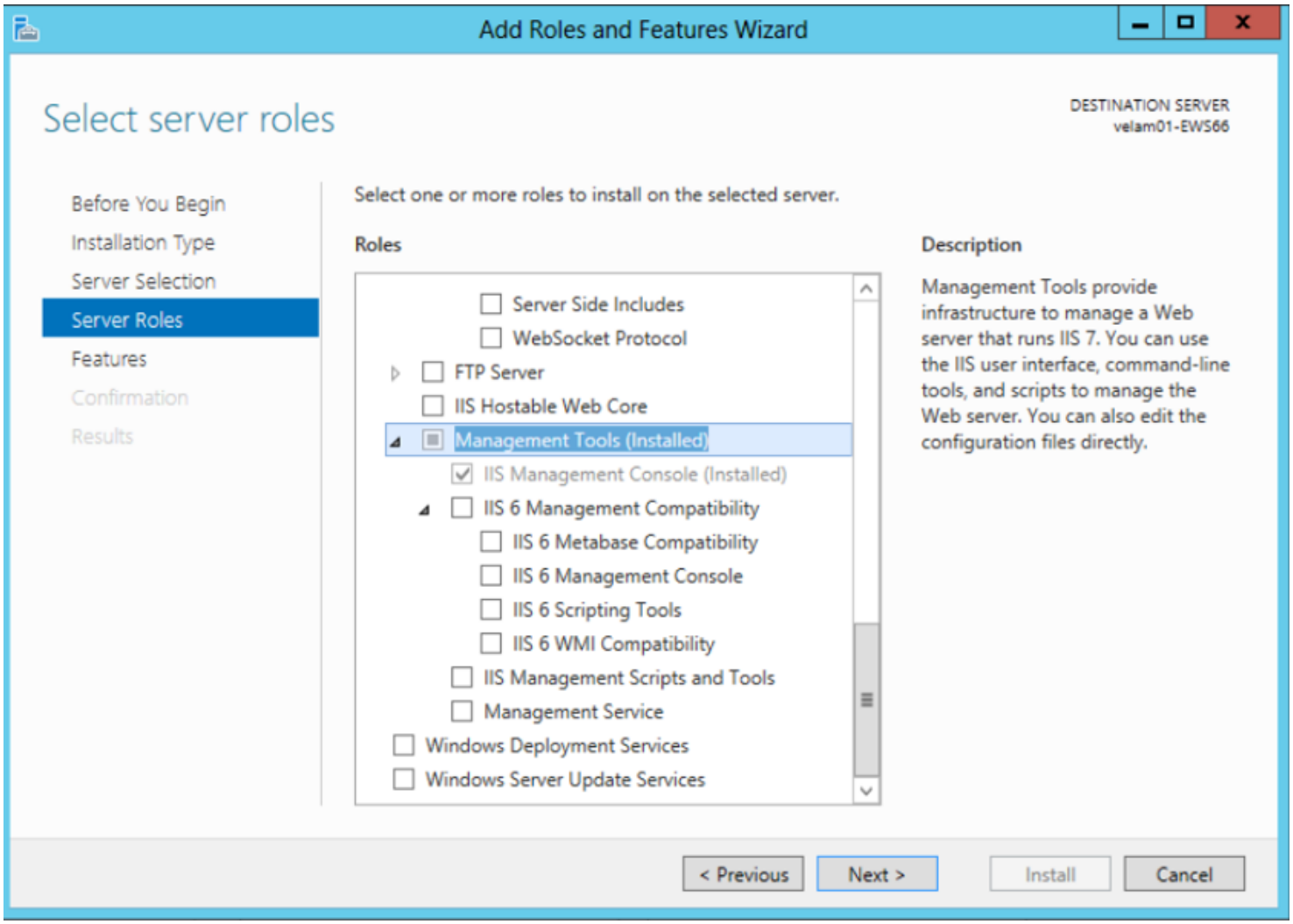
10. Save and close the file.
11. Restart the CA Service Desk Manager services.
12. Launch CA SDM and enable the external user authentication from UI for the user role that you want to allow the SAML based SSO access.

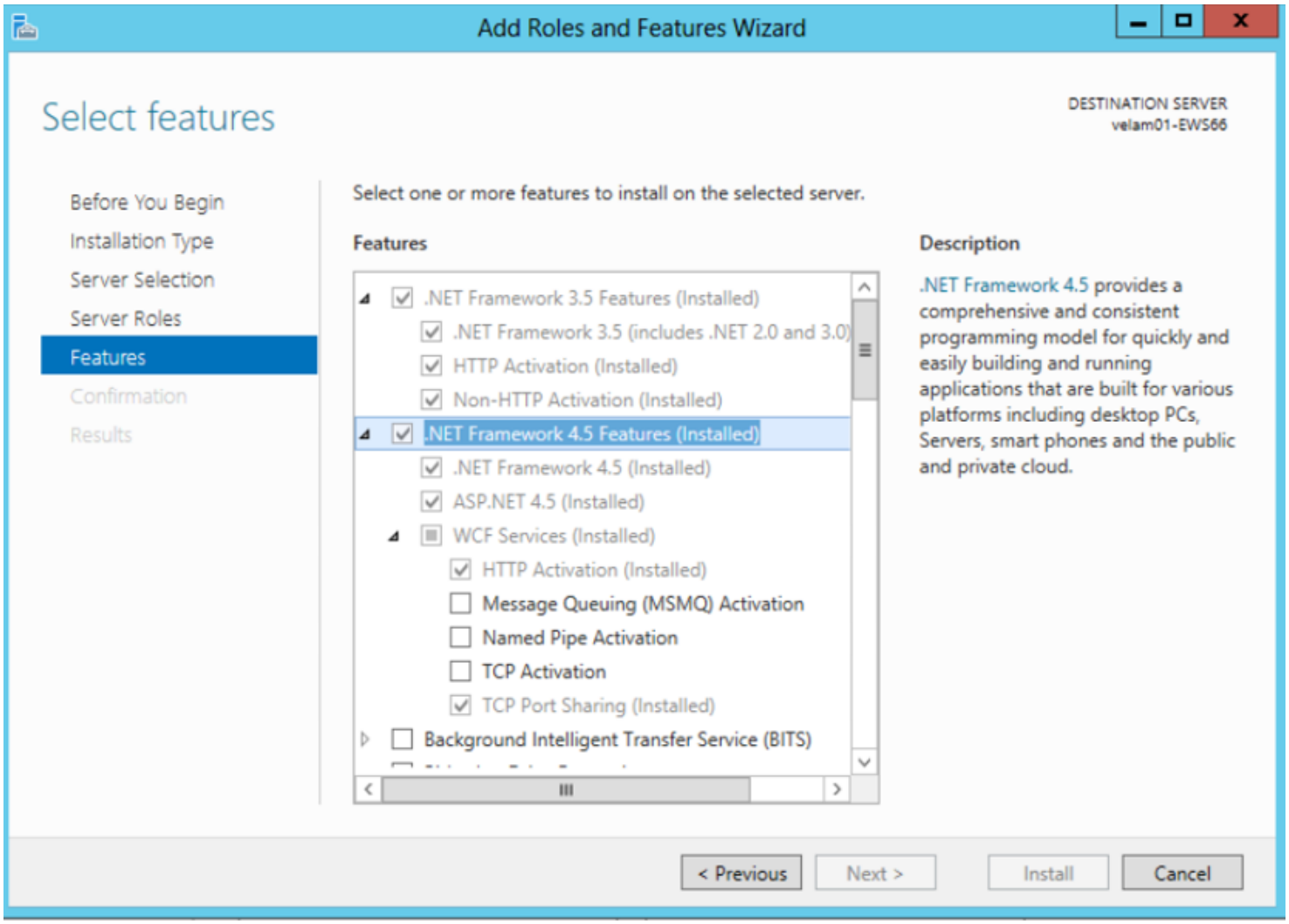
Troubleshooting - Unable to login to CA SDM application using the IIS port with any Idp (Identity Provider)

Perform the following steps:

1. Install all the Options showin the following Images:


Server Manager
Server Manager ‣ Dashboard
Manage
Tools
View
Help
Add Roles and Features Wizard
Select server roles
DESTINATION SERVER
velam01-EWS66
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
Select one or more roles to install on the selected server.
Roles
Health and Diagnostics (Installed)
Performance (Installed)
Security (Installed)
Application Development (Installed)
.NET Extensibility 3.5 (Installed)
.NET Extensibility 4.5 (Installed)
Application Initialization (Installed)
ASP (Installed)
ASP.NET 3.5 (Installed)
ASP.NET 4.5 (Installed)
CGI (Installed)
ISAPI Extensions (Installed)
ISAPI Filters (Installed)
Server Side Includes
WebSocket Protocol
Description
Management Tools provide infrastructure to manage a Web server that runs IIS 7. You can use the IIS user interface, command-line tools, and scripts to manage the Web server. You can also edit the configuration files directly.
< Previous
Next >
Install
Cancel
Hide
BPA results


Add Roles and Features Wizard
Select server roles
DESTINATION SERVER
velam01-EWS66
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
Select one or more roles to install on the selected server.
Roles
Server Side Includes
WebSocket Protocol
FTP Server
IIS Hostable Web Core
Management Tools (Installed)
IIS Management Console (Installed)
IIS 6 Management Compatibility
IIS 6 Metabase Compatibility
IIS 6 Management Console
IIS 6 Scripting Tools
IIS 6 WMI Compatibility
IIS Management Scripts and Tools
Management Service
Windows Deployment Services
Windows Server Update Services
Description
Management Tools provide infrastructure to manage a Web server that runs IIS 7. You can use the IIS user interface, command-line tools, and scripts to manage the Web server. You can also edit the configuration files directly.
< Previous
Next >
Install
Cancel



2. Open the SAML-tracer using chrome browser.
3. Try to access SDM UI functionality using normal chrome browser.
4. You can now view the first CA SDM call redirect to the Idp login page (example: login.microsoftonline.com

Enable SAML Authentication for CA SDM on Apache

Follow these steps:

1. Ensure that HTTPS is enabled for CA Service Management. For more information, see Enable SSL Authentication for CA SM 17.3
2. Download and install Shibboleth Service Provider on the Apache server where CA Service Management is installed. You must download the Shibboleth Service Provider specific to the operating system you are using. We recommend to use Shibboleth Service Provider version 2.6. Shibboleth Service Provider does not support installation on Solaris. For more information about Shibboleth installation and software download, see Shibboleth Documentation.
3. Start the Shibboleth service.
4. Launch the Identity Provider's Metadata URL in a browser and generate the **FederationMetadata.xml** file and copy to the path **\etc\shibboleth** and rename the file as **partner-metadata.xml**. For example, if you are using ADFS as identity provider, the Metadata URL is https://<ADFS server>/federationmetadata/2007-06/federationmetadata.xml
5. Modify the Shibboleth2.xml file as follows. The file is typically available under the path: \etc\shibboleth\.

a. Find the following lines in the file and modify the values of the parameters “entity ID” and “REMOTE_USER”.

```
 <
ApplicationDefaults entityID="https://<SDM host name>:<SSL Port>" REMOTE_USER="<IdP claim user detail
 property>" cipherSuites="ECDHE+AESGCM:ECDHE:!aNULL:!eNULL:!LOW:!EXPORT:!RC4:!SHA:!SSLv2"
>
```

Example:

```
 <
ApplicationDefaults entityID="https://CAServiceDesk:443"
 REMOTE_USER="smuser"
 cipherSuites="ECDHE+AESGCM:ECDHE:!aNULL:!eNULL:!LOW:!EXPORT:!RC4:!SHA:!SSLv2"
>
```

b. Find the following lines in the file and modify the values of the parameters “entity ID”.

```
 <SSO entityID="<URL of the Identity Provider> "
discoveryProtocol="SAMLDS" discoveryURL="https://ds.example.org/DS/WAYF">
SAML2 SAML1
</SSO>
```

Example:

```
 <SSO entityID="http://adfscomputer/adfs/services/trust"
discoveryProtocol="SAMLDS" discoveryURL="https://ds.example.org/DS/WAYF">
SAML2 SAML1
</SSO>
```

c. Open the shibboleth.xml file and replace the entity ID value with the entity ID value available in the partner-metadata.xml file.

d. Find the following lines and comment them, that means adding the tag <!-- at the beginning and the tag --> at the end of the code, if not already commented.

```
 <!-- Example of remotely supplied batch of signed metadata. -->
<!--
<MetadataProvider type="XML" validate="true"
uri="http://example.org/federation-metadata.xml"
backingFilePath="federation-metadata.xml" reloadInterval="7200">
<MetadataFilter type="RequireValidUntil" maxValidityInterval="2419200"/>
<MetadataFilter type="Signature" certificate="fedsigner.pem"/>
<DiscoveryFilter type="Blacklist" matcher="EntityAttributes" trimTags="true"
attributeName="http://macedir.org/entity-category"
attributeNameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:uri"
attributeValue="http://refeds.org/category/hide-from-discovery" />
</MetadataProvider>
-->
```

e. Add the following lines below the commented code as explained above.

```
 <!-- Example of locally maintained metadata. -->
<MetadataProvider type="XML" validate="false" file="partner-metadata.xml"/>
```

6. Open the file attribute-map.xml from the path: ../etc/shibboleth/ and edit the file as follows:
Add the following lines in the beginning of the file.

```
<Attribute name="<IdP claim to send to the relying party>"
nameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:unspecified" id="<IdP claim to send to the
relying party>"/>
```

Example:

```
<Attribute name="smuser" nameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:unspecified"
id="smuser"/>
```

Note: When you are creating the "Relying Party Trust" using ADFS for Shibboleth, the metadata you need to upload to ADFS can be obtained by launching the Service Provider's Metadata URL. For example, https://<Shibboleth Server Name>/Shibboleth.sso/Metadata.

7. Open the file httpd.conf and edit as follows. The file is typically available under the path: **etc/httpd/conf/**. Add the following lines at the end of the file:

```
#Shibboleth configuration
LoadModule mod_shib /usr/lib64/shibboleth/mod_shib_22.so
<Location /Shibboleth.sso>
SetHandler shib
</Location>
<Location />
AuthType shibboleth
ShibRequestSetting requireSession 1
Require valid-user
ShibUseHeaders Off
</location>
```

8. Restart Shibboleth service.
9. Restart Apache server.
10. Launch CA SDM and enable the external user authentication from UI for the user role that you want to allow the SAML based SSO access.

Configuring CA Service Catalog

This section contains the following articles:

- Manage Business Units and Tenant Administration
- Manage Users and Assign Roles
- Manage Users with CA EEM
- Enable External Authentication of Users
- Configure CA Service Catalog to Use Secure Socket Layer
- Reconfigure Service Catalog using the Setup Utility
- Configuration Files
- How to Configure the F5 Load Balancer for CA Service Catalog Server
- Enable GMSA
- Configure CA Service Catalog with CA EEM over TLS 1.2

WARNING

Any "modifications" or "adaptions" or "configurations" that are done administratively through the interface (web browser, command-line, Web Screen Painter) are "supported", meaning CA Support can assist with basic suggestions and trouble-shooting. CA Support does not do the changes for the customer, they are the customer's responsibility. For example, adding a field to a table and putting the field on a form through Web

Screen Painter is a fully supported “modification”. Another example, installing or uninstalling a feature through the Options Manager administration is a fully supported “configuration”.

Anything to do with SPEL code, Java scripting (or any language scripting), or a customer-specific change to the underlying base code-line (done by CA Services or a Partner), is not supported by CA Support. The customer can do these things, but the customer is responsible for the support, maintenance, and trouble-shooting when things go wrong. If such “customizations” affect expected out-of-the-box behavior, CA Support will ask the customer to remove the customizations and see if the behavior persists.

Manage Business Units and Tenant Administration

This article contains the following topics:

Understand Business Units and Catalog Access

Administrators create and maintain business units (tenants) as the organizational structure that controls access to data. Business units can be:

- A service provider (the highest level or *root* business unit).
- An external client company, if the service provider provides services to external customers.
- An internal corporate department or group within a department.

A business unit administrator can:

- Access the catalog that the business unit publishes.
- Subscribe accounts to services.
- Perform other administrative duties for the business unit and its child business units.

The following rules apply when you create your business unit structure:

- Business units can have child business units.
- Users can have roles in more than one business unit.
- Business units can have one or more accounts.
- Each account can have many users.
- Each user can have many accounts.
- Each user who has an account can be billed to that account.
- When you request a service, the parent business unit of your business unit defines the catalog that you see.
- When you subscribe an account to a service, the parent business unit defines the catalog that you access.

By default, users and accounts can request or subscribe to items *only* in the catalog of their parent business unit.

Example

Review the following example of a four-level hierarchy of business units, in a default configuration:

Service Provider (SP) is the top-level business unit.

- A is a child business unit of SP.
- B is a child business unit of A.
- C is a child business unit B.

In linear form, you can express the relationship as follows: SP-->A-->B-->C

- "A" Users and accounts can request or subscribe to items in the SP catalog *only*.
- "B" Users and accounts can request or subscribe to items in the A catalog *only*.
- "C" Users and accounts can request or subscribe to items in the B catalog *only*.

To change the default behavior, use the following configuration settings:

- The System configuration settings **Use Service Provider Configuration Only** and **Use Service Provider Catalog Only.**
- The Catalog configuration setting that is named **Pass Through Catalog.**

Decide Between Common And Stand-Alone Tenant Administration

Administrators decide between *common* or *stand-alone* tenant administration, as follows:

- In *common* tenant administration, you create and maintain a single tenant structure for CA Service Catalog and CA Service Desk Manager. Both parent and child tenants are created in CA Service Desk Manager *only*. You can edit common (shared) attributes of tenants in CA Service Desk Manager *only*. CA Service Catalog "inherits" the tenants, their structure, and their common attributes from the CA Service Desk Manager. These features appear as read-only in CA Service Catalog. In CA Service Catalog, you can still edit the Service Catalog specific attributes.
- In stand-alone tenant administration, you create and maintain tenants in CA Service Catalog only. The parent and child tenants apply only to CA Service Catalog. Tenant administration functions and tenant attributes are not shared between CA Service Catalog and any other product. If CA Service Desk Manager is installed, you create and maintain its tenant structures separately from CA Service Catalog. Stand-alone tenant administration is the default.
- Common tenant administration ensures efficiency and consistency in your tenant structures for applicable products providing a single point of administration. You can administer tenants once rather than multiple times in CA Service Catalog and CA Service Desk Manager. You can leverage a common data model and thus save costs for multiple service providers and other large organizations.
- Stand-alone administration provides flexibility for tenant structures across products. Thus, to use separate tenant structures for CA Service Catalog and CA Service Desk Manager, use stand-alone administration. For example,
  - You have installed CA Service Catalog *without* CA Service Desk Manager.
  - You have installed CA Service Catalog with CA Service Desk Manager. *However*, you want to administer the business units for CA Service Catalog separately from these other products.
- If you have already implemented separate tenant structures in both products, you perform synchronization activities to enable common tenant administration.
- Continue using your chosen option for the long term, preferably throughout the life cycles of the applicable products. Perform synchronization activities whenever you change from stand-alone tenant administration to common tenant administration.
- As a best practice for efficiency and consistency across applicable products, use common tenant administration.

  NOTE
  Common tenant administration has no effect on how you manage users and accounts.

  WARNING
  Do *not* move a CA Service Catalog tenant if it has a subscription or a request for an associated account. When such subscriptions exist, moving a tenant impacts its business rules and can cause problems in request management.

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{"URL":["/usm/wpf?Node=iclaunchpad.setup","/usm/wpf?
Node=iclaunchpad.account_management"],"customLabelGetStarted":"Administration","description":"IPCE_Administration","custom
[{"id":"IPCE_Manage_Business_Units","type":"manage","title":"Manage Business Units"}]}

Configure Common Tenant Administration

This article contains the following topics:

Common tenant administration (also named multi-tenant administration) lets administrators use a single administration tool to simultaneously create and maintain business units (tenants) for multiple products, including CA Service Catalog and CA Service Desk Manager. To configure CA Service Catalog to use common tenant administration, follow this process:

WARNING
This topic applies *only* if your organization has CA Service Desk Manager installed.

1. Meet the prerequisites: Verify that CA Service Catalog and CA Service Desk Manager are installed and share the same MDB.
2. Configure the tenancy settings in CA Service Desk Manager.

   WARNING
   CA Service Catalog requires a tenant of type Service Provider in CA Service Desk Manager, as a prerequisite for completing this process.
3. Create the common tenant mapping file.
4. Prepare to run the common tenant merge utility.
   This utility uses the information in the common tenant mapping file to create a shared tenant structure between CA Service Catalog and CA Service Desk Manager.
5. Run the common tenant merge utility.
6. Set the configuration variables for common tenant administration.
7. (Optional) Implement a common terms of use agreement for common tenants.
8. Note the effects of common multi-tenancy on business unit functions.

Configure the Tenancy Settings in CA Service Desk Manager

Configure the tenancy settings in CA Service Desk Manager to support the integration with CA Service Catalog.

Follow these steps:

1. Log in to CA Service Desk Manager as ServiceDesk(administrator).
2. Click Administration, Option Manager, Multi Tenancy.
3. Verify that the multi-tenancy option is on.
4. Verify that the multi-tenancy depth is 10.
5. Click Administration, Security and Role Management, Tenants.
6. Click Create New to create a tenant, if no tenant of type Service Provider exists. Enter a meaningful name for the new tenant.
7. Verify that Service Provider Checked is on and Subtenants Allowed is on.

You have configured the tenancy settings in CA Service Desk Manager to support the integration with CA Service Catalog.

Create the Common Tenant Mapping File

When you configure common tenant administration, a required task is creating the common tenant mapping file. Map tenants between CA Service Catalog and CA Service Desk Manager, so that both products use the same tenant structure.

Follow these steps:

1. Review the tenant structures in CA Service Catalog and CA Service Desk Manager. Plan the merged structure that you want to use.
2. Stop the CA Service Catalog services on the computer where MDB is installed.
3. Open the CA Service Catalog command prompt by clicking Start, Programs, CA, Service Delivery, Service Delivery Command Prompt.
4. At the CA Service Catalog command prompt, enter the following ant command:

   ```
   ant generate-merge-config
   ```

   The common tenant merge utility runs and creates the tenant mapping file, which is named merge-tenants.conf. The utility:

- Creates the initial mapping entry, which maps the Service Provider business units for each product to each other. This entry is the only *required* entry. The CA Service Catalog entry is named ca_tenant, and the CA Service Desk Manager entry is named usm_tenant_ext.
- Lists the names and database IDs of all CA Service Catalog tenants
- Lists the names and database IDs of all CA Service Desk Manager tenants

You use these names and database IDs to create the mapping entries.

5. Open the merge-tenants.conf file that is located in USM_HOME, using a text editor. After the Service Provider mapping entry, add a new line for each new mapping entry that you create. Specify each entry using the following format, using the Service Provider entry as a model:

   ```
   <usdk> business unit ID=<bcsc> business unit ID
   ```

   For example:
   0x60B4EAC8B85E41DH97E647CF84A93CFA=87958EK983987D42AB2A4PAEF808C129

   WARNING
   Mapping a parent tenant maps the parent *only*, *not* the children! Therefore, for automatic mapping, eit*her* map parents and children individually *or* omit the mapping of the parent.
6. Follow the rules for creating entries in the common tenant mapping file.
7. Review the sample common tenant mapping file and verify the contents of your mapping file.
8. Restart the CA Service Catalog services that you stopped at the beginning of this procedure.

You have created the common tenant mapping file. You are ready to run the utility.

Run the Common Tenant Merge Utility

Follow these steps:

1. Open the CA Service Catalog command prompt on the computer where the MDB is installed. Click Start, Programs, CA, Service Delivery, Service Delivery Command Prompt.
2. Enter the following command:

   ```
   ant merge-tenants
   ```

   The utility prompts you for the name (**Default:** merge-tenant.conf) and location (**Default:** USM_HOME) of the common tenant mapping. The utility also prompts you for the password for any administrator with the Service Provider role in CA Service Catalog. An example is the default user named spadmin. If running the utility creates tenants in CA Service Desk Manager, you enter this password.
3. The utility summarizes the changes to be made in the tenant table in the MDB for each product. The utility also creates any missing tenants in those tables.
4. Restart the CA Service Catalog services that you stopped at the beginning of this procedure.
5. Start CA Service Catalog and CA Service Desk Manager. Verify that the tenant structure and common attributes of tenants are the same in both products.

You have run the common tenant merge utility.

WARNING

If the same name is used for two or more tenants being created in CA Service Desk Manager the utility aborts. In such cases, check the tenant structure of each product. Rename any duplicate tenant names and ensure that they are unique. Run the utility again.

Set Configuration Variables

When you configure common tenant administration, setting the related configuration variables is a required task.

Follow these steps:

1. Log in to CA Service Catalog as a user with the Service Delivery Administrator role.

2. Click Administration, Configuration, System Information.
3. Verify that the value of the option *Common Tenant Data Synchronized* is Yes. This value is read-only. "No" indicates that a discrepancy exists between the tenant structures of CA Service Desk Manager and CA Service Catalog. Verify the accuracy of the common tenant mapping file and run the common tenant merge utility again.
4. Set the value of the option *Common Multi-tenant Administration Enabled* to Yes.
   When this setting is Yes, common multi-tenant administration is enabled for CA Service Catalog, through CA Service Desk Manager.

You have set the configuration variables for common tenant administration.

Rules for Mapping File Entries

When you configure common tenant administration, a required task is creating the common tenant mapping file. Follow these rules when creating entries in this file:

> **WARNING**
> Violations of any of these rules cause the common tenant merge utility to abort.

- Do not map business units that are at different levels in CA Service Desk Manager and CA Service Catalog.
- Do not map the same business unit more than once. If you have several business units, print the merge-tenants.conf file and mark each business unit after you map it and save the file.
- Map a child only after mapping all its parents. Map the parents from the tenant level directly above the child through the level directly under the Service Provider business unit.
- Verify that tenant names are unique across all products. For example, if you have a tenant AAA in CA Service Desk Manager and CA Service Catalog, rename at least one of them to a unique name.
- Verify that the values in CA Service Desk Manager for common attributes meet your requirements for CA Service Catalog. This requirement is for every tenant and sub-tenant that you map. Doing so is important because for common attributes, the CA Service Desk Manager values overwrite the CA Service Catalog values.
- Do *not* map that tenant or any of its sub-tenants explicitly in the merge-tenants.conf file, to add a complete tenant structure from one product to the other automatically. Instead, run the common tenant merge utility and confirm that you want to add any tenants that are not mapped in the merge-tenants.conf file. The utility then adds the complete structures of any unmapped tenants to each product automatically.

Sample Common Tenant Mapping File

Review the following sample common tenant mapping file and construct your own file:

```
...
# CA Service Catalog Tenants (id, name)
# ca.222.com   ca.222.com
# 00234A51DC4A4F70A03D3BDE5526278C   BB
# 9F6309A0CB654781B08080AD78C2280F   CC
# 9CF5655B4B8D4833A0C6E74EB56128C5   AA
#
# CA Service Desk Manager Tenants (id, name)
# 0xB3484A535A3D994B9FBCD28D3845F292   ca.111.com
# 0xBB83719AA93DFC48B909D7D72CF8B0CB   A
# 0xD9DC8FF2EB84CC43988FE71F4B489D3D   E
# 0x67958EA983987D42AB2A4BAEF808C029   D
#
# Service Provider tenants must map to each other. This mapping is mandatory.
0xB3484A535A3D994B9FBCD28D3845F292=ca.222.com
#
# Map A to AA
```

```
0xBB83719AA93DFC48B909D7D72CF8B0CB=9CF5655B4B8D4833A0C6E74EB56128C5
...
```

In this example, the following tenants exist:

- CA Service Desk Manager tenants: ca.111.com, A, D, and E.
- CA Service Catalog tenants: ca.222.com, AA, BB, and CC.

Service Provider tenants must map to each other. Hence, you merge the CA Service Desk Manager Service Provider tenant (ca.222.com) and the CA Service Catalog Service Provider tenant (ca.111.com).

Otherwise, you add a mapping entry only when you want to merge two tenants together. For example, in the sample file, you merge CA Service Desk Manager tenant A and CA Service Catalog tenant AA.

When you merge tenants, the CA Service Desk Manager tenant only overwrites common attributes in the CA Service Catalog tenant. The CA Service Catalog-specific attributes remain intact. After you run the common tenant merge utility with this sample file, the tenant ca.222.com in CA Service Catalog is renamed ca.111.com. Similarly, tenant AA in CA Service Catalog is renamed A. The tenants inherit the other common attributes of the CA Service Desk Manager tenant with which it was merged.

If you are simply adding tenants from one product to the other, do not map them in the common tenant mapping file. For this reason, the sample file does not map any other tenants. Thus, the following changes occur when you run the common tenant merge utility with this sample file:

- CA Service Catalog tenants BB and CC are added to CA Service Desk Manager.
- CA Service Desk Manager tenants E and D are added to CA Service Catalog.

(Optional) Implement Common Terms of Use Agreement

You can optionally implement a common terms of use agreement for CA Service Catalog and CA Service Desk Manager. Doing so helps enforce a consistent login policy for all common tenants in both products. CA Service Desk Manager administrators create and maintain this agreement in CA Service Desk Manager. CA Service Catalog administrators configure CA Service Catalog to adopt this agreement.

> **NOTE**
> Verify that the CA Service Desk Manager setting meets the needs of your organization. For more information about configuring terms of use agreement in CA Service Desk Manager, see the CA Service Desk Manager documentation.

1. Select Administration, Configuration, System Information in CA Service Catalog.
2. Specify Yes for the setting *Terms of Usage Prompt Enabled*. If you specify No, then users attempting to log in to CA Service Catalog are *not* prompted to accept the terms of use agreement.
3. Review the effect of the CA Service Desk Manager setting on users logging in to CA Service Catalog. The effect depends on the terms of use agreement that is configured for the common tenant in CA Service Desk Manager.

- – If the common tenant has an active, defined (not empty) terms of use agreement, prompt the user to accept it at every login.
  – If the common tenant has no terms of use agreement that is defined, or if the agreement is inactive, review the parent hierarchy of the tenant. Then perform the following steps:
    - If the nearest parent has active but empty terms of use agreement that is defined, then do not prompt the user with any terms of use agreement. Otherwise, prompt the user to accept the terms of use agreement of the nearest parent tenant that has an active terms of use agreement defined.
    - If no parent tenant has an active, specified terms of use, do not prompt the user with any terms of use.
  – If the common tenant has an active terms of use agreement defined but is configured to suppress it, then do *not* prompt the user to accept it.
  – If users attempting to log in to CA Service Catalog receive the prompt to accept the terms of use agreement but do not accept it, they cannot access CA Service Catalog.

Manage Common Tenants

When common tenant administration is enabled, all tenants exist in both CA Service Catalog and CA Service Desk Manager. Administrators can manage these tenants using CA Service Desk Manager, but not CA Service Catalog. Here, *managing* means adding or deleting tenants, or editing their common attributes. In CA Service Catalog, you can *view* tenants and all their attributes. But you can edit only the CA Service Catalog-specific attributes.

Follow these steps:

1. In CA Service Catalog, view the business units by clicking Administration, Business Unit, and:
   – Verify that any new business units that you created in CA Service Desk Manager are visible in CA Service Catalog.
   – Verify that any inactive business units in CA Service Desk Manager are not visible in CA Service Catalog.
   – Verify that you cannot add or delete business units or edit their primary attributes in CA Service Catalog.
   – Note the effects of common multi-tenancy on business unit functions.
2. In CA Service Desk Manager, perform the following actions:
   – Create one or more new tenants, as needed.
   – Make inactive any tenants that you no longer want to use, if applicable.
   – Edit the common attributes such as description, contact, location, inactive.
   – Optionally, implement a common terms of use agreement.
   – Verify the CA Service Desk Manager and CA Service Catalog tenant and business unit settings that follow. Update these settings, if necessary. These settings enable the tenant mapping between the products.

   To begin some of these tasks, select Administration, Security and Role Management, Tenants, Start. For more information about how to perform these tasks, see the CA Service Desk Manager documentation.

Tenant Setting for CA Service Desk Manager	CA Service Catalog Business Unit Setting
Service Provider selected (on)	Service Provider (SP), meaning root or highest level business unit
Subtenants Allowed selected (on)	Contains Sub Units: True, meaning Super Tenant. A super tenant is a tenant with at least one subtenant.
Subtenants Allowed not selected (off)	Contains Sub Units: False, meaning Tenant, also named leaf (end-of-tree) tenant. A subtenant is a tenant with at least one parent tenant.
Inactive Tenant	inactive, formerly deleted tenant
Parent Tenant is Empty	SP is the parent tenant

Effects of Common Tenant Administration

After you have completed configuring common tenant administration, the following results occur in CA Service Catalog:

- A new configuration item appears on the Administration, Configuration page. The name of this item is *Common Multi-Tenant Administration*. Its default setting is No. When you set this option to Yes, the following changes occur:
  – The *common attributes* of the tenant become read-only from CA Service Catalog. You can edit these common fields in CA Service Desk Manager *only*:
    - The Business Unit Name and Description in the General Information section
    - All fields in the Primary Contact Information section
    - All fields in the Location Information section
  – You can edit the *CA Service Catalog-specific attributes* in CA Service Catalog. These attributes include the following fields:

    - Federal Tax Payer ID, State Tax Payer ID, Tax Region
    - Timezone, Currency Name, Date Format, Time Format
    - Decimal Symbol, Opened Date, Email, Web Site
  - The options to add, delete, edit, cut, and paste tenants are disabled.
  - A message appears on the Administration, Business Units page explaining why these functions are disabled.
- If you implement a common terms of use agreement, the terms of use settings in CA Service Desk Manager govern the attempts of users to log in to CA Service Catalog.

Configure Stand-Alone Business Units

Stand-alone administration means that you use CA Service Catalog alone to manage business units, as follows:

1. Select **Administration**, **Business Units**.
2. Expand the business unit tree and find the business unit that you want.
3. (If applicable) Click the Add icon to add a child business unit. Complete the fields on the **Add a New Business Unit page**, and click OK.

Add a New Business Unit Page

The following fields require explanation:

- **Business Unit Login ID**
  Specifies the value that you use to log in to a business unit (except the service provider business unit). You also use this value when running IXUtil import and export commands. This value must be unique.
- **Business Unit Name**
  Specifies the name of the business unit.
  This field defaults to the same name as the Business Unit Login ID. The Business Unit Name appears throughout the user interface and in all reports and invoices.
- **Business Unit ID**
  Identifies an automatically generated, read-only value, whether you are using common or stand-alone tenant administration.
  After you create the business unit, CA Service Catalog automatically fills in this field. The only exception is the service provider business unit. You specify its business unit ID during the Catalog Component installation and cannot change this ID afterwards.
  If necessary, use the Business Unit ID field for reference. For example, you change your method of tenant administration from stand-alone to common. In that case, you use the business unit ID when you create a common tenant mapping file. You also use the business unit ID with IXUtil and web services.
- **Opened Date**
  Specifies the date when the business unit becomes part of your organizational structure, in local time.
- **Currency Name, Date Format, Time Format, Decimal Symbol**
  Specify the currency name, date and time formats, and decimal symbol that must appear on the user interface. These specifications appear, for example, when users create and manage requests. They also appear when administrators create and maintain services, users, and so forth.
- **Contains Sub Units**
  Specifies whether this business unit can have sub business units (child business units).

  NOTE
  You *cannot* change this setting after you have created the business unit.
- **Create Dashboard Library Namespace**
  Specifies whether to create a folder for the business unit in the Dashboard Library. Doing so helps organize the contents of the library.
- **Single Account Mode**
  Specifies whether the business unit can contain *only one* account.

If you select this option, specify the name of the single account.

NOTE
You *cannot* change this setting after you have created the business unit.

Create Document Namespace
Specifies whether to create a folder for the business unit in the Documents folder. Doing so helps organize the contents of the folder.

- **Primary Contact Information**
  Specifies the user ID of the primary contact for the business unit.
- **Theme**
  *S*pecifies the settings for several look-and-feel elements, including images and icons (*except for* logos), menus, and tabs. When applicable, these elements include colors, font name and point size, highlighting, and related specifications. You customize these look-and-feel elements by editing the Cascading Style Sheet (CSS) files for the theme.
  The look-and-feel of the UI matches the theme of the business unit that you are logged in to. If a business unit does not have its own theme, it uses the theme of its closest parent business unit. You can use the same theme for all business units. Alternatively, you can optionally create and use different themes for different business units. If you update a theme for a specific business unit, the change affects the users of that business unit. The change also affects any child business units that do not have their own theme specified. For more information about customizing themes, see the Modify the Branding section.
- **Logo**
  For each business unit, you can optionally specify a *business unit* logo. This logo replaces the *global* logo in the heading on product pages and request emails of the business unit. You can use a business unit logo to support the brand or other messaging uniquely for a business unit. You can update the logos for every business unit or only for specific business units. For example, you can decide to customize logos only for super tenants directly under the root business unit.

  WARNING

  If you have enabled multi-tenancy with CA Service Desk Manager, CA Service Catalog uses the logo that CA Service Desk Manager specifies. If no CA Service Desk Manager logo applies, then each business unit uses the CA Service Catalog global logo.

  If the child business unit has its own logo, users who log in to it see the child logo. Else, users who log in to it see the global logo. Users with access to multiple business units can see different header logos when they log in to each business unit.
- **Location Information**
  Specifies the location of the business unit.

You can use CA Service Catalog to add or edit business units, as follows:

NOTE
Click the Edit icon to edit the Business Unit Profile.

1. Select **Administration**, **Business Units**.
2. Expand the business unit tree and find the business unit that you want.
3. (If applicable) Click the Add icon to add a child business unit. Complete the fields on the **Add a New Business Unit page**, and click OK.

Add a New Business Unit Page

The following fields require explanation:

- **Business Unit Login ID**

Specifies the value that you use to log in to a business unit (except the service provider business unit). You also use this value when running IXUtil import and export commands. This value must be unique.

- **Business Unit Name**
  Specifies the name of the business unit.
  This field defaults to the same name as the Business Unit Login ID. The Business Unit Name appears throughout the user interface and in all reports and invoices.
- **Business Unit ID**
  Identifies an automatically generated, read-only value, whether you are using common or stand-alone tenant administration.
  After you create the business unit, CA Service Catalog automatically fills in this field. The only exception is the service provider business unit. You specify its business unit ID during the Catalog Component installation and cannot change this ID afterwards.
  If necessary, use the Business Unit ID field for reference. For example, you change your method of tenant administration from stand-alone to common. In that case, you use the business unit ID when you create a common tenant mapping file. You also use the business unit ID with IXUtil and web services.
- **Opened Date**
  Specifies the date when the business unit becomes part of your organizational structure, in local time.
- **Currency Name, Date Format, Time Format, Decimal Symbol**
  Specify the currency name, date and time formats, and decimal symbol that must appear on the user interface. These specifications appear, for example, when users create and manage requests. They also appear when administrators create and maintain services, users, and so forth.
- **Contains Sub Units**
  Specifies whether this business unit can have sub business units (child business units).

  NOTE
  You *cannot* change this setting after you have created the business unit.
- **Create Dashboard Library Namespace**
  Specifies whether to create a folder for the business unit in the Dashboard Library. Doing so helps organize the contents of the library.
- **Single Account Mode**
  Specifies whether the business unit can contain *only one* account.
  If you select this option, specify the name of the single account.

  NOTE
  You *cannot* change this setting after you have created the business unit.

  Create Document Namespace
  Specifies whether to create a folder for the business unit in the Documents folder. Doing so helps organize the contents of the folder.
- **Primary Contact Information**
  Specifies the user ID of the primary contact for the business unit.
- **Theme**
  *S*pecifies the settings for several look-and-feel elements, including images and icons (*except for* logos), menus, and tabs. When applicable, these elements include colors, font name and point size, highlighting, and related specifications. You customize these look-and-feel elements by editing the Cascading Style Sheet (CSS) files for the theme.
  The look-and-feel of the UI matches the theme of the business unit that you are logged in to. If a business unit does not have its own theme, it uses the theme of its closest parent business unit. You can use the same theme for all business units. Alternatively, you can optionally create and use different themes for different business units. If you update a theme for a specific business unit, the change affects the users of that business unit. The change also affects any child business units that do not have their own theme specified. For more information about customizing themes, see the Modify the Branding section.
- **Logo**

For each business unit, you can optionally specify a *business unit* logo. This logo replaces the *global* logo in the heading on product pages and request emails of the business unit. You can use a business unit logo to support the brand or other messaging uniquely for a business unit. You can update the logos for every business unit or only for specific business units. For example, you can decide to customize logos only for super tenants directly under the root business unit.

WARNING

If you have enabled multi-tenancy with CA Service Desk Manager, CA Service Catalog uses the logo that CA Service Desk Manager specifies. If no CA Service Desk Manager logo applies, then each business unit uses the CA Service Catalog global logo.

If the child business unit has its own logo, users who log in to it see the child logo. Else, users who log in to it see the global logo. Users with access to multiple business units can see different header logos when they log in to each business unit.

- **Location Information**
  Specifies the location of the business unit.

{"URL":["/usm/wpf?Node=iclaunchpad.setup"],"customLabelGetStarted":"Add A New Business Unit","description":"IPCE_Add_A_New_Business_Unit"}

Manage Users and Assign Roles

This article talks about managing users and roles in CA Service Catalog.

This article contains the following topics:

- Basic Information About Users
- User Groups
- Authorization Level

Basic Information About Users

A user in CA Service Catalog typically represents a person who uses the product. A person must have a user ID to log in and use CA Service Catalog. Each user requires a role and the user must belong to a business unit.

User Data in the MDB

The MDB stores user-related information for multiple CA products. For example, CA Service Catalog and other CA products share users and contacts. Therefore, use caution when managing users.

Some data in the MDB is product-specific and is not shared between products. For example, CA Service Catalog does not share role information. The MDB contains several users that CA Service Catalog does not use but that other CA products use. Examples are System_ADH_generated, System_AM_User, Systemt_Anonymous, System_Argis_User, System_MA_User, System_NSM_generated, System_SD_User, and UAPM Administrator.

The MDB stores user IDs and all other user data *except* passwords.

User Data in CA EEM

The user ID of each CA Service Catalog user maps to a matching user in CA EEM.

When a user logs in to CA Service Catalog, it passes the user ID to CA EEM for authentication. If CA EEM does not use an external directory, CA EEM authenticates the user ID. If CA EEM uses an external directory, the external directory authenticates the user ID.

If CA EEM uses an external directory, the Catalog system fetches important user details from the external directory. Examples include the first name, last name, email address, and organizational hierarchy. This fetching helps create the user quickly and efficiently.

CA EEM also controls access for each role to CA Service Catalog components.

External Directories

If you use an external directory to maintain CA Service Catalog users, perform the following tasks:

- Configure CA EEM to use an external directory.
- Synchronize the MDB users with the external directory users.

These tasks help verify that all CA EEM users have matching CA Service Catalog users.

Single Sign-On

If your organization uses a Windows domain, you can configure CA Service Catalog to use Single Sign-On. For more information, see the section Configure Single Sign-On Using Windows NTLM authentication.

User Groups

You or other administrators can organize CA Service Catalog users into the following types of user groups:

- Global and Application user groups in CA EEM

An application group that CA Service Catalog uses is the Super User.

> **NOTE**
> For information about creating these groups and assigning users to them, see your CA EEM documentation.

- User-defined groups in CA EEM

You create user-defined groups, in either your external directory (if applicable) or CA EEM. You can apply the same action to all users in the user-defined group, instead of modifying the users individually. Once you create a user-defined group in CA EEM or your external directory, it is available in CA Service Catalog.

User group memberships appear as *read-only* data in user profiles.

> **NOTE**
> For information about creating user-defined groups, see the Manage Users with CA EEM section.

Authorization Level

You can specify an authorization level for each business unit in which the user has a role. The system approval process is the only approval process that uses the authorization level. By default, the following levels are available with the text and numeric values shown:

- Level 0 (0)
- Level 10 (10)
- Level 20 (20)
- Level 30 (30)
- Level 40 (40)
- Level 50 (50)

In the system approval process, each user has an authorization level and each service has an approval level. The Catalog system automatically approves requests from users whose authorization level matches or exceeds the approval level of the service. Otherwise, the system follows a managerial hierarchy. The request proceeds to request managers until it reaches one whose authorization level matches or exceeds the approval level of the service.

Example:

If a service has an approval level of 40, then a user requesting the service requires one of the following criteria:

- The user must have an authorization level of 40.
- The user must obtain approval from a request manager who has an authorization level of 40.

  NOTE
  The Requested For user can have different roles - and therefore different authorization levels - in multiple business units. The Catalog system uses the authorization level of that user in the business unit to which the requested *service* belongs.

Administrators create users to let them access CA Service Catalog. Administrators assign a role to each user to specify the access rights of the user.

Step 1 - Manage Users

This article contains the following topics:

To define the user in CA EEM, use one of the following methods:

- If CA EEM uses an external directory:
  The administrator of the external directory must define the user in the external directory. For more information, see the documentation for the external directory. An example of an external directory is Microsoft Active Directory.
- If CA EEM uses its own repository:
  CA EEM automatically creates a new Global user in the USM/users folder when you add a new CA Service Catalog user. The Create EEM User action on the User Create event rule sets this behavior. Optionally perform one of the following tasks:
  – Disable the action that creates the CA EEM user automatically.
  – Set the format of automatically created passwords.

Administrators create and maintain users to let them access CA Service Catalog. As an administrator, you typically add a user when a new employee starts working in your organization.

Follow these steps

1. Select **Administration**, **Users** from the main menu.
2. Click **Add**.
3. Enter the data for the new user on the Add a New User page, and click OK.

You typically edit a user when an existing employee changes roles or titles or transfers to a new department. You can only edit users who have a role in the business units within the scope of your role.

WARNING
Do *not* change the user ID of a user. Do *not* reuse deleted user IDs, because all deleted users are maintained as *inactive* in the user database.

Review the Effects of Deleting a User before you delete a user.

Set the Format of Automatically Created Passwords

You can configure CA EEM to use its own repository. By default, the password is the same as the user ID. You can optionally change the format of the passwords that CA EEM creates automatically.

Follow these steps:

1. Select Administration, Tools, Events.
2. Disable the Create EEM User rule. Create a rule to use in its place, for the User Create event.
3. Disable the associated rule action that is named Create New EEM User or Modify EEM User. Create an action to use in its place.
4. Open your replacement action and locate the Java class field, which appears similar to the following text.

```
com.ca.usm.ruleEngine.action.CreateEiamUserAction,passwordTemplate=value
```

The passwordTemplate parameter specifies the text to be used as the initial password. The password is set to the user ID that is created when the passwordTemplate parameter is empty.

5. Set this parameter to a specific text value or to an available event variable.
   For example, to set the password to the user ID followed by the letter *a*, specify the following text:

```
com.ca.usm.ruleEngine.action.CreateEiamUserAction,passwordTemplate=$user_id$a
```

Add a New User Page

You complete several fields when you add a new user in your organization. The following fields require explanation:

> **NOTE**
> If CA EEM is configured to use an external directory, data is auto-populated from the external directory data, if applicable.

- **User ID**
  Specifies the ID by which CA Service Catalog identifies the user.
  The User ID value must be unique.
- **Manager**
  Specifies the manager for the user being creating.
  For system approval process, the manager must approve requests that the user submits. For other approval process, the manager can be a required approver, depending on how you configure the approval process.
- **Request Auto-Delegation: Delegate**
  Specifies the user to whom your requests pending action are delegated automatically when you auto-delegate your own requests pending action. Administrators can also auto-delegate the requests pending action of other users.
  When you clear this field in your own user profile, your requests pending action stop being auto-delegated. The requests remain in your queue. When you clear this field in the profile of another user, the requests pending action of that user stop being auto-delegated. The requests remain in the queue of that user.
  Clearing this field *does not affect* requests pending actions that were already delegated to previously assigned delegates. Therefore, after clearing this field, instruct the former delegates to handle requests pending action promptly. Alternatively, as an administrator, you can also handle them yourself or transfer them to other users.
- **Delegate Use of Catalog: Delegates**
  Specifies the users to whom you delegate the use of your catalog. These users can create and submit requests from your catalog on your behalf. In addition, administrators can delegate the catalog of one user to another user.
  This field is valid only if you or another administrator has enabled delegation of catalogs for your business unit.
  When you clear this field in your user profile, your catalog is no longer delegated. Your former delegates can no longer create and submit requests on your behalf from your catalog. When you clear this field of another user, the catalog of that user is no longer delegated. The former delegates of that user can no longer create and submit requests on behalf of that user from the catalog of that user.
- **User Location**
  Specifies the location details for the user.

  > **NOTE**
  > All CA products using the same MDB share the location. Therefore, use caution when modifying the location.
- **Select Business Unit**
  Specifies the business unit of the new user.
  The default is the business unit that you are currently logged in to.
  Only a Service Delivery Administrator or Super Business Unit administrator can change the business unit of the new user. Otherwise, you *can* create the user but *cannot* change the business unit of the user.
  Users can belong to multiple business units. However, a user can have only one role and one authorization level in each business unit.
- **Select Role**
  Specifies the role for the new user in the current business unit.
  By default, new users receive the default role for all users. Administrators can optionally assign a different role.

Select an available role and click the *Add Row* icon to add the role for the user.

- **(Optional) Select Authorization Level**
  Specifies the authorization level of the new user. This setting applies *only* if you use system approval as the approval process.

Effects of Deleting a User

Deleting a user affects its account, including its subscriptions and requests, as follows:

- **Account**
  The account remains open when you delete the user, because future transactions can exist for an account. As an administrator, you can optionally close the account or leave it open.
- **Subscriptions**
  The status of subscriptions for the deleted user change to the default cancellation status (Pending Cancellation or Cancel).
- **Requests**
  The status of requested service options for the deleted user change as follows:

Original Status	New Status
Not Submitted	None: Request is deleted
Submitted. An Approval status, Fulfillment Status, Pending Resource Assignment or Resource Assigned	Cancelled
Completed	Default Cancellation Status (Pending Cancellation or Cancel) if Accounting Component is installed Cancelled if Accounting Component is not installed
Pending Cancellation or Cancelled	Same as Original Status

Administrators create and maintain users to let them access CA Service Catalog. As an administrator, you typically add a user when a new employee starts working in your organization.

Follow these steps

1. Select **Administration**, **Users** from the main menu.
2. Click **Add**.
3. Enter the data for the new user on the Add a New User page, and click OK.

You typically edit a user when an existing employee changes roles or titles or transfers to a new department. You can only edit users who have a role in the business units within the scope of your role.

> **WARNING**
> Do *not* change the user ID of a user. Do *not* reuse deleted user IDs, because all deleted users are maintained as *inactive* in the user database.

Review the Effects of Deleting a User before you delete a user.

Add a New User Page

You complete several fields when you add a new user in your organization. The following fields require explanation:

> **NOTE**
> If CA EEM is configured to use an external directory, data is auto-populated from the external directory data, if applicable.

- **User ID**
  Specifies the ID by which CA Service Catalog identifies the user.

The User ID value must be unique.

- **Manager**
  Specifies the manager for the user being creating.
  For system approval process, the manager must approve requests that the user submits. For other approval process, the manager can be a required approver, depending on how you configure the approval process.
- **Request Auto-Delegation: Delegate**
  Specifies the user to whom your requests pending action are delegated automatically when you auto-delegate your own requests pending action. Administrators can also auto-delegate the requests pending action of other users.
  When you clear this field in your own user profile, your requests pending action stop being auto-delegated. The requests remain in your queue. When you clear this field in the profile of another user, the requests pending action of that user stop being auto-delegated. The requests remain in the queue of that user.
  Clearing this field *does not affect* requests pending actions that were already delegated to previously assigned delegates. Therefore, after clearing this field, instruct the former delegates to handle requests pending action promptly. Alternatively, as an administrator, you can also handle them yourself or transfer them to other users.
- **Delegate Use of Catalog: Delegates**
  Specifies the users to whom you delegate the use of your catalog. These users can create and submit requests from your catalog on your behalf. In addition, administrators can delegate the catalog of one user to another user.
  This field is valid only if you or another administrator has enabled delegation of catalogs for your business unit.
  When you clear this field in your user profile, your catalog is no longer delegated. Your former delegates can no longer create and submit requests on your behalf from your catalog. When you clear this field of another user, the catalog of that user is no longer delegated. The former delegates of that user can no longer create and submit requests on behalf of that user from the catalog of that user.
- **User Location**
  Specifies the location details for the user.

  NOTE
  All CA products using the same MDB share the location. Therefore, use caution when modifying the location.
- **Select Business Unit**
  Specifies the business unit of the new user.
  The default is the business unit that you are currently logged in to.
  Only a Service Delivery Administrator or Super Business Unit administrator can change the business unit of the new user. Otherwise, you *can* create the user but *cannot* change the business unit of the user.
  Users can belong to multiple business units. However, a user can have only one role and one authorization level in each business unit.
- **Select Role**
  Specifies the role for the new user in the current business unit.
  By default, new users receive the default role for all users. Administrators can optionally assign a different role.
  Select an available role and click the *Add Row* icon to add the role for the user.
- **(Optional) Select Authorization Level**
  Specifies the authorization level of the new user. This setting applies *only* if you use system approval as the approval process.

{"URL":["/usm/wpf?Node=icguinode.adduser&Args=ca.com&Args=spadmin&did=ca.com"],"customLabelGetStarted":"Add A New User","description":"IPCE_Add_A_New_User"}

Step 2 - Assign Roles

This article contains the following topics:

Relationship Between Users, Roles, and Login

Users, roles, and login have the following relationship:

- A user typically belongs to one business unit, but can optionally belong to multiple business units. A user can have only one role in a business unit.
- A user can optionally have different roles in different business units.
  Example: User A can have a catalog user role in the Finance business unit. The same user can have a catalog administrator role in the IT business unit.
- If the user does not specify a business unit at login, CA Service Catalog logs the user in to the default business unit defined for the user. The user has the role that is assigned to the user in that business unit.
- If an integrating product created the user, then the user is *not* assigned to a role or business unit. Instead, after the user logs in, the user receives the default role for all users. Examples of integrating products include CA Service Desk Manager, CA APM, and CA Business Service Insight.

Roles and Default Access Rights

Users can have one role in each business unit in which they are defined. Users can have different roles in different business units.

- Service Delivery administrators can change some default access rights for the *entire* Catalog system as follows:
  Log in to the root (highest level) business unit, select Administration, Configuration, and change the Access Control configuration settings.
- Service Delivery administrators and business unit administrators can change several default access rights for *specific business units* as follows:
  Log in to the business unit, select Catalog, Configuration, and change the Access Control configuration settings.
- All users can also delegate the use of their catalogs to other users to create requests on their behalf.
- The Catalog system creates only one user at installation time. This user, named *spadmin*, has the Service Delivery Administrator role.

Request-related functionality is available when CA Service Catalog is installed. Subscription and invoice-related functionality is available when Accounting Component is installed.

- **Catalog User**
  Is the user role for requesting services *without* subscriptions. These users can manage their own requests, such as approve, reject, fulfill, and other actions to handle requests pending action.
  Most users in the organization use this role *only*.
  This role is predefined as the default role for new users. However, administrators can optionally change the default role for new users from the catalog user to another role.
  This role is most suitable when you are *not* using subscriptions or billing in your implementation.
- **End User**
  Is the end user for all functions available through the catalog. This user includes all the same access rights as the catalog user. The end user can subscribe to services and view invoices. This role can also view and add news messages, documents, and reports.
- **Request Manager**
  Is the administrator role for managing requests, such as viewing and handling all requests in the business unit and applicable subbusiness units. Request managers handle both their own requests pending action and the requests pending action of other users. Request managers can search *all* requests in the Catalog system. But catalog users can search *only* their own requests.
- **Services Manager**
  Creates, defines, and manages services (not requests) for a specific tenant or business unit. This user also has administrative access to configure reports, dashboards, documents, and message alerts.
  This role is most suitable when you want a user to create and maintain services. This user cannot request or subscribe to services.
  This user can also handle requests pending actions, for example, by approving and rejecting requests.
- **Administrator**
  The Administrator uses Service Catalog to create a request for themselves and for other users.

This user can subscribe offering but cannot create them.
This user can also create user roles like Administrator/ Catalog User/ End User.

- **Catalog Administrator**
  Creates, defines, and manages services for a specific tenant or business unit.
  This user also has the same access rights as the request manager role.
  This user can request services but cannot subscribe to them.
- **Super Business Unit Administrator**
  Is the "root" user in a specific super tenant (super business unit). A super business unit is a business unit that contains one or more child business units. This administrator has *almost* complete access to the super business unit and all its sub business units. For example, anywhere in the super business unit, this administrator can create business units, create new users, and assign roles.
- **Service Delivery Administrator**
  Is the "root" (highest level) user in the Service Provider (highest level) business unit. This user has complete system access to all business units. For example, this user can specify default settings that apply to all users by logging in to the root business and accessing the Administration, Configuration, User Default tab. This role is available only for the Service Provider business unit, the default business unit that is created during installation.
  Only this administrator has access to data mediation, system configuration, events, rules, and actions.
  By default, at installation time, the Catalog system creates a user ID named *spadmin* with this role.
- **Default Role Specification**
  Service Delivery administrators can specify a default role for all users.

Tasks that Each Role Can Perform

The roles provide default access rights to various functions. Administrators use configuration settings to add default access rights to a role or remove default access rights from a role.

The following table lists the tasks that each role can perform. The letter **X** indicates that the role *can* perform the task. The dash (**-**) indicates that the user *cannot* perform the task.

By default, all users have all shopping functions, except as noted in Roles and Default Access Rights. However, administrators can configure the access rights of each role to create proxy requests, edit requests, and so forth.

All users can also delegate the use of their catalogs to create requests on their behalf.

Tasks	Cat Usr	Req Mgr	Cat Adm	End Usr	Adm	Svc Mgr	SBU Adm	SD Adm
Shopping								
	X	X	X	X	X	-	X	X
Managing Requests								
View, edit, delete, and cancel requests	X	X	X	X	X	X	X	X
Act on assigned requests pending action	X	X	X	X	X	X	X	X
Search for requests	X	X	X	X	X	X	X	X

View all items in a request	X	X	X	X	X	X	X	X
View request tracking and audit trail information	-	X	X	-	X	-	X	X
General								
View dashboards	X	X	X	X	X	X	X	X
Add personal dashboards	X	X	X	X	X	X	X	X
Create shared dashboards	-	-	X	-	X	-	X	X
Print dashboard reports	-	-	-	-	X	X	X	X
View subscriptions and invoices	-	-	-	X	X	-	X	X
During checkout, change the Requested For user from the current setting to another account or user. That account or user requires a role in the business unit scope of the logged in user.	-	X	X	-	X	-	X	X
View and add News Messages	-	-	-	X	X	X	X	X
View Documents (if enabled) and View Reports	-	-	-	-	-	X	X	X
Managing the Catalog								

View and alter catalog services and service option groups	-	-	X	-	-	X	X	X
View and alter CA Service Catalog configuration settings	-	-	X	-	-	-	X	X
Manage catalog entries or configuration	-	-	-	-	-	X	X	X
Manage subscriptions or invoices	-	-	-	-	X	-	X	X
Managing other elements								
Manage accounts within your business unit scope	-	-	-	-	X	-	X	X
Manage users with roles in your business unit scope	-	-	-	-	X	-	X	X
Manage the dashboard library for the business unit	-	-	-	-	X	-	X	X
Manage scheduled tasks	-	-	-	-	X	-	X	X
Manage reports	-	-	-	-	X	X	X	X
Manage Change Events and Alerts	-	-	-	-	X	-	X	X

Roles Key

Code	Role
Adm	Administrator
Cat Adm	Catalog Administrator
Cat Usr	Catalog User (none)
End Usr	end user
Req Mgr	request manager
Svc Mgr	service manager
SB Adm	Super Business Unit administrator
SD Adm	Service Delivery administrator

Tasks that Each Role Can Perform for Other Users

The following table displays the roles that can perform authorized tasks for themselves and for other accounts and users.

Can Perform Tasks for Themselves and	Cat Usr	Req Mgr	Cat Adm	End Usr	Adm	Svc Mgr	SBU Adm	SD Adm
Other accounts and users with roles in their business unit	-	X	X	-	X	X	X	X
Other accounts and users with roles in their business unit *and* any of its child business units.	-	X	X	-	-	X	X	X
Other accounts and users with roles in *all* business units, including *all* child business units	-	-	-	-	-	X	-	X

Default Role for All Users

The default role for all users applies to every user in the entire Catalog system. This default role applies to *all* users in *all* business units, including all child business units.

Only the Service Delivery administrators can set this default role. To set the role, the administrator logs in to the *root* business unit and selects Administration, Configuration, User Default Role.

The Catalog system automatically assigns this default role to every new user. However, administrators can optionally specify a different role for a user when they add or edit the user.

Manage Users with CA EEM

CA Service Catalog uses CA EEM to authenticate and authorize users. You can configure CA EEM to use an external directory, such as Microsoft Active Directory, for authentication of users.

Use the CA EEM user interface of CA Service Catalog to manage CA EEM users and groups. Start CA EEM from the Users option of the Administration Quick Start dashboard.

Integrating with embedded CA EEM and optionally with an external directory is the only required integration for CA Service Catalog. You use CA EEM and optionally an external directory to manage the CA Service Catalog user database.

After you have installed CA Service Catalog, its required components (including embedded CA EEM), and your external directory (if used), perform the following actions:

1. Import Users into the Database
2. (Optional) Create User-Defined Groups
3. (Optional) Update the host name of the CA EEM computer or the application names it uses. Perform this step periodically after CA Service Catalog has been installed and is running.
4. (Optional) Add High Availability and Failover Support as follows:
   a. Verify that you have installed CA EEM on at least two computers in your CA Service Catalog implementation.
   b. Set up high availability and failover support for CA EEM.
   c. Run the setup utility and configure CA EEM.

      NOTE
      If you have installed other CA products that use CA EEM, see the documentation for each of those products for information about setting up failover support for CA EEM.

Step 1 - Import Users into the Database

After you install CA Service Catalog, the user database contains no data. To populate the database with user information, configure and run the LDAP utility. Running the utility populates your CA Service Catalog database with the users that you specify from your LDAP server. Run the utility at regular intervals to synchronize updates in the user database from the LDAP server to the CA Service Catalog database. You can optionally use a scheduler to synchronize.

Follow these steps:

Step 1a - Perform Transition Tasks

These tasks apply if you were previously using the CA EEM synchronization utility (syncuputil) to import and synchronize users in the CA Service Catalog user database. Examples include implementations that obtained the LDAP Importer utility by upgrading CA Service Catalog or applying a patch.

These tasks help you transition efficiently from the CA EEM synchronization utility to the LDAP Importer utility.

- Stop using the CA EEM synchronization utility and its properties file (syncuputil.properties).
- Remove scheduled tasks for that utility.
- Run the LDAP Importer utility.

These tasks do *not* apply to first-time installations of CA Service Catalog.

Step 1b - Configure CA EEM to Use an External Directory

Configuring CA EEM to use an external directory is a prerequisite for using the LDAP Importer utility.

NOTE
If an external directory is enabled with SSL (LDAPS):

- You must add a CA certificate, for the self signed certificates to work.
- It is recommended that CA Service Management products should also be enabled with SSL.

Follow these steps:

1. Verify that the external directory contains at least one user ID that matches the user ID of a Service Delivery Administrator in CA Service Catalog:
   a. Log in to CA Service Catalog as the spadmin user and create at least one user ID with the Service Delivery Administrator role.
   This user ID must match a user name in the external directory.
   b. If CA EEM is configured for multiple active directories, create a user ID as Domain Name\userid in CA Service Catalog. Verify that the domain name is part of the principal name of the CA EEM user ID in the CA EEM active directory.
2. Configure the User Store in CA EEM to reference all applicable Active Directory sources.

   NOTE
   For more information, see your CA EEM documentation.
3. (For single sign-on) Verify that the domain name for Active Directory matches the domain name of the single sign-on user.
4. Select Manage Identities, Users, and validate that the users returned from a search are from the external directory. If your user base is large, limit the list of users returned.
5. Restart the CA Service Catalog service.
6. Log in to CA Service Catalog as the Service Delivery Administrator whose user name matches a user name in the external directory.

You have configured CA EEM to use an external directory.

Step 1c - (Optional) Determine the Number and Purpose of Configuration Files

If you are using multiple domains, you can use multiple configuration files. In that case, determine the number of configuration files and the purpose of each one. For example, large organizations can use several LDAP servers, one for each business unit, region, or domain. A managed service provider can use one group of LDAP servers for their internal organization and another group for their client organizations.

In such cases, as an administrator, you can use a unique configuration file for each server. You can configure the name and settings of each file to match its purpose. For example, consider a Managed Service Provider (MSP) with three internal LDAP servers and many LDAP servers for their clients. The MSP can decide to copy and customize the default file (LDAPImporter_server1.properties) as follows:

- Internal configuration files
  – LDAPImporter_Internal_Asia.properties
  – LDAPImporter_Internal_Europe.properties
  – LDAPImporter_Internal_NorthAmerica.properties
- Configuration files for clients that are retail companies in Asia.
  – LDAPImporter_Client_Azerbaijan_Retail.properties
  – LDAPImporter_Client_India_Retail.properties
  – LDAPImporter_Client_Singapore_Retail.properties

NOTE

Clustering alone does *not* require extra configuration files. If CA Service Catalog is clustered, verify that the properties file is available from the computers from which you run the utility.

Step 1d - Create the LDAP User

The LDAP Importer utility requires an LDAP user to connect to the LDAP server and import the LDAP users into the CA Service Catalog database. You specify the login credentials of the LDAP user in the configuration file properties for the utility.

NOTE
For more information about how to create LDAP users, see your LDAP documentation.

Step 1e - Create the CA Service Catalog User

To import users from the LDAP server to CA Service Catalog, the LDAP Importer requires a CA Service Catalog user. You create a CA Service Catalog user for this purpose and specify its login credentials in the configuration file properties for the utility.

NOTE
If an existing user meets the following specifications, you can use that user.

Follow these steps:

1. Log in to the *root* business unit.

   WARNING
   Creating this user in the root business unit is required for the proper role assignment and access rights.
2. Select Administration, Users from the main menu.
3. Click Add.
4. Enter the data for the new user, as follows:
   - **User Name**
     Specify the same user name, including case, as the LDAP user that you use to connect to the LDAP server.
     If your catalog works with multiple LDAP servers, include the domain name, for example, MyDomain\MyUser.
   - **Role**
     Service Delivery Administrator.

The user is added.

Step 1f - Specify the Configuration File Properties

Replace the default properties in the file with your custom values. See the comments in the file for assistance.

The default configuration file is named LDAPImporter_server1.properties. This file resides in the USM_HOME folder. Copy and modify it to create all the custom configuration files that you require for your organization.

WARNING
The configuration file includes placeholder values that you *must* customize before running the utility.

Format

Properties in the configuration file follow this format: *name*=*value*. For example, LDAP.LastSynchronizationDate=2013-12-13.

Names *must not* contain spaces. If a name contains one or more spaces, replace each space with an underscore (_).

Values *can* contain spaces and you do not need to replace spaces with underscores.

You do not need to enclose spaces or underscores in quotation marks.

LDAP Properties

- **LDAP.Base.Provider.Url=*value***
  Specifies the URL of the LDAP server. If the LDAP server is *not* using SSL, use this format:
  LDAP.Base.Provider.Url= ldap://LDAP-server-name:389
  If the LDAP server *is using* SSL, use this format: LDAP.Base.Provider.Url=ldaps://*LDAP-server-name*:636
  If the LDAP server or CA Service Catalog uses SSL, complete the SSL parameters.
- **LDAP.User.DN=*value* and LDAP.User.Domain=*domain-name***
  Specify the name of the LDAP user that you created earlier for accessing the LDAP database.
  The domain name is required if CA EEM is configured to use multiple domains.
  If you are using multiple LDAP servers, the domain name must match the domain name for the LDAP server in CA EEM.
- **Catalog.User=*domain-name**username***
  Specifies the name of the CA Service Catalog user that you created earlier for importing the users from the LDAP server into the CA Service Catalog database.
  The domain name is required if CA EEM is configured to use multiple domains.
- **LDAP.User.Password=*password* and Catalog.User.Password=*password***
  Specify the encrypted passwords for each user in the appropriate parameter.
  To generate each encrypted password, enter the following command at the CA Service Catalog command prompt:
  ```
  USM_HOME/SCRIPTS ENCRYPTER.BAT password
  ```
  Copy each encrypted value from the command line and paste it to the appropriate entry in the configuration file.
- **InsertOrUpdateUsersAfterLastSyncDateOnly=True|False**
  We recommend that you specify True. *S*pecify True if you plan to run the utility regularly with a scheduling tool. This setting helps run the synchronization process in optimized mode.
  Alternatively, specify False if you do *not* plan to run the utility regularly with a scheduling tool.
- **LDAP.LastSynchronizationDate=*yyyy-mm-dd***
  This parameter is required if you specify True for the previous parameter, InsertOrUpdateUsersAfterLastSyncDateOnly.
  We recommend that you specify this parameter when you run the file for the first time. Afterwards, this parameter is updated automatically whenever you run the utility with a scheduling tool.
  Specifying this parameter provides more efficient processing. If you use this parameter, the utility imports *only* users that were added or updated since the last time the utility ran. The utility processes deleted users according to your specifications for the next parameter, DeactivateMDBUsersUponLDAPUserDeletion.

Catalog Properties

- **DeactivateMDBUsersUponLDAPUserDeletion=True|False**
  Specifies how to process CA Service Catalog users that were deleted from the LDAP database:
  - **True**
    (Recommended) *D*eactivates these users in the CA Service Catalog database. The user IDs are no longer available in the CA Service Catalog UI.
  - **False**
    Keeps these deleted users active in the CA Service Catalog database and UI.

  In either case, these deleted users cannot log in to CA Service Catalog, because they cannot be authenticated in the LDAP database.
  This setting does not apply the *first* time that you run the utility.
- CatalogUser.Country.*xx*=*country*
  - ***xx***
    Specifies the two-letter country code from the LDAP server.
  - ***country***
    Specifies the country name from the MDB.
    To obtain country names from the MDB, query the ca_country table.

NOTE
If the LDAP server does not specify the country code, then the country name is set to NULL in the MDB.

Examples:
- CatalogUser.Country.IN=India
- CatalogUser.Country.US=United States

- **CatalogUser.*country*.*city*=*location***
  Specifies a custom location name.
  - ***country***
    Specifies the country name from the MDB, as explained in the previous entry.
  - ***city***
    Specifies the city name from the LDAP server.

    NOTE
    If the LDAP server does not specify the city, then the city is set to NULL in the MDB. If the LDAP server does specify the city, but the city is not mapped in this parameter, then the city is set to NULL in the MDB.
  - ***location***
    Specifies any custom location name relevant to your organization, for example, the name of a branch or unit in the *city*.
    If the location does not exist in the MDB, it is created.

    NOTE
    As a best practice, specify a location name that is known to other administrators in your organization. This location and all other user attributes appear when you view user attributes in the CA Service Catalog UI.

  Examples:
  - CatalogUser.United_States.Islandia=UnitedStates1
  - CatalogUser.United_States.New_York=UnitedStates2
- **CatalogUser.DefaultBusinessUnit=*business-unit***
  Specifies the CA Service Catalog business unit to which users are assigned when they are imported for the first time. Once the users are imported for the first time, their business unit assignment is the same. Only when you use the CA Service Catalog UI to change the business unit, does this value also change. This restriction applies even if you change the value of this parameter and import the users again. This restriction helps maintain the ability of users to log in to CA Service Catalog.
- **CatalogUser.DefaultRole=*role***
  Specifies the CA Service Catalog role that users are assigned when they are imported for the first time.
  Once the users are imported for the first time, their role assignment is the same. Only when you use the CA Service Catalog UI to change the role assignment, does this value also change. This restriction applies even if you change the value of this parameter and import the users again. This restriction helps maintain the ability of users to log in to CA Service Catalog.
  The following table lists the valid role names for the CatalogUser.DefaultRole parameter, according to the type of business unit.

Role	Root BU	*Node BU	**Leaf BU
spadministrator (Service Delivery Administrator)	x		
servicemanager (Service Manager)	x		
catadministrator (Catalog Administrator)	x	x	
stadministrator (Super Business Unit Administrator)		x	

administrator (Administrator)	x	x	x
requestmanager (Request Manager)	x	x	x
catalogenduser (Catalog User)	x	x	x
enduser (End User)	x	x	x

*A Node BU is a child business unit that contains one or more of its own child business units.

**A Leaf BU is a child business unit that contains no child business units of its own.

- **SSL.KeyStoreLocation=*value***
  Specifies the complete path name of the keystore file that contains the SSL Server certificates of CA Service Catalog and the LDAP server. This parameter is required if CA Service Catalog or the LDAP server use SSL. If the LDAP server uses SSL, configure the LDAP.Base.Provider.Url parameter to use SSL.
  Delimit folder names with double backslashes (\\). The standard path name follows:

```
C:\\Program_Files\\CA\\Service_Catalog\\ssl.keystore
```

  WARNING
  Verify that you have imported the complete chain of Certification Authority (CA) certificates for CA Service Catalog and the LDAP server into the Java keystore.

- **SSL.KeyStorePassword=*value***
  Specifies the encrypted password for the CA Service Catalog keystore file.
  To generate the encrypted password, enter the following command at the CA Service Catalog command prompt:

```
USM_HOME/SCRIPTS ENCRYPTER.BAT password
```

Copy the encrypted value from the command line and paste it to this entry.

Step 1g - Run the Utility

Run the LDAP Importer utility regularly to import users from the LDAP server into the CA Service Catalog database. As a CA Service Catalog administrator, determine how often you must run this utility. For example, large organizations or organizations with frequent personnel changes want to run the utility daily. However, other organizations with fewer personnel changes want to run the utility weekly or every two weeks.

The file name of the utility is LDAPImporter.bat. The file resides in the USM_HOME\scripts folder.

You can optionally use *any* of the following methods to run the utility regularly:

- Run the utility manually from the command line of any CA Service Catalog computer.
- Run the utility as a batch job at scheduled times. Use the CA Service Catalog Scheduler or any standard scheduler, such as Windows Scheduler.
  To access the CA Service Catalog Scheduler, click Administration, Tools, Scheduler. Use the following specifications when you complete the fields:
  For Action Type, select Execute Command Line.
  For Cmd Line, enter the path names of the utility and the configuration file or files. The Cmd Line field accepts a limited number of characters. If necessary, specify relative path names (from USM_HOME\view\bin) or specify folder names only, as shown in the following examples:

```
..\..\scripts\LDAPImporter.bat ..\..\scheduler.properties
..\..\scripts\LDAPImporter.bat C:\LDAPFolder\
```

  NOTE
  For more information about the CA Service Catalog Scheduler, see the Manage the Scheduler section. For more information about the Windows Scheduler, see your Windows documentation.

To run the LDAP Importer utility, enter the following command at the command prompt or in a scheduling tool:

```
LDAPImporter.bat properties-files
```

The utility processes all levels of nested groups of users. The utility imports each user and the users *above* it in organizational hierarchy, up to the top, even if the manager is not part of that group.

The utility does *not* import the users *below* the user in organizational hierarchy. For example, the utility does not import the direct reports of the user.

Logging

The log file for the utility is named LDAPImporter.log. This file resides in the USM_HOME\logs\LDAPImporter folder.

The level of logging for the utility is specified in the LDAPImporter.log4j.xml file. This file resides in the USM_HOME \scripts folder.

Examples

Examples of Running Configuration Files:

In this example, you run multiple configuration files in the USM_HOME folder:

```
LDAPImporter.bat USM_HOME\LDAPImporter_server1.properties USM_HOME\LDAPImporter_server2.properties USM_HOME
\LDAPImporter_server3.properties
```

In this example, you store all LDAP server properties in the MyFolder folder. Each LDAP server properties file name within this folder must use this format: ldapimporter_*name*.properties.

```
LDAPImporter.bat USM_HOME\MyFolder
```

Examples of Specifying Configuration File Properties

In this example, you populate the root business unit (ca.com) with users whose location is New York and who have direct reports. Assign the Service Delivery Administrator role to these users.

```
LDAP.ImportType=User
LDAP.MicrosoftAD.User.Filter = (&(objectClass=user)(!(objectClass=computer)))
LDAP.User.Filter= (&(l=New York)(directReports=*))
CatalogUser.DefaultBusinessUnit=ca.com
CatalogUser.DefaultRole= spadministrator
```

In this example, you populate the business unit that is named Europe with users who belong to the group named All Managers Europe. Assign the Super Business Unit Administrator role to these users.

```
LDAP.ImportType=Group
LDAP.Group.Name = All Managers Europe
CatalogUser.DefaultBusinessUnit=Europe
CatalogUser.DefaultRole= stadministrator
```

Step 1h - Verify the Import

Verify that the database has been populated correctly by logging in to CA Service Catalog as a user-defined in the database. A successful login indicates that the database was populated correctly.

Step 2 - (Optional) Create User-Defined Groups

Create user-defined groups in CA EEM to apply the same action to many CA Service Catalog users at once. Once you create the UDG in CA EEM, it is available in CA Service Catalog. You can then administer the UDG much like you would administer an individual user.

When you set the permissions for the UDG, those settings are applied to all individual users in the group. For example, you want to assign read-write access to a service to a group and read-only access to the service to another group. Use UDGs, to group users according to their function and purpose.

As an administrator, you can optionally assign users to one or more UDGs. The product does not require that users belong to any UDG. User who belongs to multiple UDGs have access to all resources authorized by their UDG memberships. If any UDG membership provides a user with access to a resource, then the user is granted access to that resource, regardless of whether other UDGs also grant access to the resource.

You can create and maintain user-defined groups (UDGs) in either CA EEM or your external directory (if applicable). You cannot create and maintain UDGs in CA Service Catalog directly.

Follow these steps:

> **NOTE**
> If the following steps do not match your version of CA EEM, see your CA EEM documentation for more information about assigning users to user groups in CA EEM.

1. Log in to the Service Delivery application of CA EEM. Do *not* log in to the global application of CA EEM.
2. Click Manage Identities, Groups.
3. Click Show Application Groups and click Go, in the Search Groups box on the left side of the screen.
4. Locate the Application Groups folder, in that tree; this folder contains the CA Service Catalog user groups, including the UDGs.
5. Find and click the "people" icon.
6. Enter the name and description for the new group in the fields provided.
7. Click Save to save your changes.

An important administrative task of user-defined groups (UDGs) is assigning rights to the following CA Service Catalog objects:

- Reports, including the data, data view, and layout objects
- Services, service options, and service option groups, including rights to edit, request, and subscribe to these elements.
- Documents
- Dashboards

An administrator can set permissions to a specific object (such as a service) according to the role of a user. The administrator can also set permissions to that object according to UDG.

For example, open a service in CA Service Catalog. Click Permissions, and search for a UDG or all users with a specific role. Next, you can assign rights to the service for the users and groups that are returned in the search results.

> **NOTE**
> When you search for UDGs, CA Service Catalog searches its own portions of CA EEM.

If a user has *either* a role or a group membership for which a permission has been set, the user is granted that permission. If a user has *both* a role and a group membership for which a permission has been set, the user is granted that permission.

You can optionally view the profile of an individual user who is a member of the UDG. While this step is not practical for verification on a large scale, it is helpful for spot checks.

You can assign rights to user-defined groups. By doing so, you ensure that all members of the group are assigned the same rights in a single operation.

Follow these steps:

1. Open the Set Permissions dialog for the Catalog object (such as a service) and click Groups.
2. Click Search EEM Groups.
3. Search for the group.

4. Click the group name in the search results.
   The dialog displays a new EEM Group list below the search results. The new list includes the group name that you clicked. The adjacent columns show the rights of the group to the object.
5. Select or unselect the options in the columns to grant or remove rights for the group to the object.

You have assigned right to the user-defined groups.

Enable External Authentication of Users

This article contains the following topics:

By default, CA Service Catalog uses CA EEM to authenticate users. You can configure CA Service Catalog to authenticate users with external applications such as CA SiteMinder, IBM Tivoli, and others. The process consists of the following tasks:

1. Install and implement the external authentication application, according to its documentation.
2. Review the following examples and understand how these applications typically send user authentication to CA Service Catalog. If applicable, adjust your settings to match these examples.
   - CA SiteMinder sends user identity information (authenticated user) with sm-user artifact name in the request header.
   - IBM Tivoli sends user identity information with iv_user artifact name in the request header.
   - Microsoft Internet Information Server (IIS) sends user identity information with request when configured for Windows NTLM.
   - Apache sends user identity information with request when configured for Windows NTLM.
3. Test the configuration on both CA Service Catalog and the external authentication application.
4. Verify that CA Service Catalog successfully receives and processes the authenticated users that the external authentication application passes. If necessary, adjust the parameters on both systems as needed.
5. Optionally configure Single Sign-On for the authentication method that you are using:
   - Windows NTLM authentication
   - External authentication

Configure Single Sign-on Using Windows NTLM Authentication

When you use Windows NTLM authentication, you can perform this procedure to enable Single Sign-On for CA Service Catalog. Users log in to the Windows domain, they can access CA Service Catalog without logging in to it.

Follow these steps:

1. Verify that you are *not* planning to use clustering. If you are using clustering, instead of performing this procedure, you set up NTLM authentication for each cluster.
2. Verify that your environment meets the following requirements:
   - You are using Windows domain authentication.
   - CA Service Catalog and CA EEM are installed in the same Windows domain.
   - You have configured CA EEM to use Active Directory.
     You are running a version of HTTP *higher* than 1.0.
   - If you are using Windows Server, perform one of the following tasks to use single sign-on using NTLM:
     - Use HTTP instead of HTTPS.
     - Uninstall the Internet Explorer Enhanced Security Configuration Windows Component.
   - If both of the following conditions exist, you cannot use single sign-on using NTLM with HTTPS:
     - The client computer operating system is Windows Server.
     - The Internet Explorer Enhanced Security Configuration Windows Component is installed.

3. Perform the following actions:
   a. Click **Administration**, **Configuration**, **Single Sign On Authentication**.
   b. Locate the **Single Sign On Type** and click the Modify icon.
   c. Select the option **NTLM (NT LAN Manager)** and click **Update Configuration**.
      The dialog closes, and you return to the Single Sign On Authentication page.
4. Verify that all affected users can use single sign-on to access CA Service Catalog on this computer.

You have configured NTLM Authentication.

Implement Single Sign-on for One Group of Users and Manual Login for Another Group

In this use case, you want to enable single Sign-on for one group of users. For example, internal users (Group 1) You also want to force manual login for another group of users. For example, external users such as contractors, vendors, and customers (Group 2).

Follow these steps:

1. Verify that you have two CA Service Catalog computers using the same instances of the MDB and CA EEM. This procedure calls the CA Service Catalog computers *Server 1* and *Server 2*.
2. Verify the following requirements:
   – Group 1 users log in to Server 1 *only*.
   – Group 2 users log in to Server 2 *only*.
   – If necessary, notify users in each group of this requirement.
3. On Server 2, edit the USM_HOME\webapps\usm\WEB-INF\web.xml file. Comment the following lines:

```
<!--
    <filter>
      <filter-name>NtlmAuthFilter</filter-name>
      <filter-class>com.ca.usm.httpfilter.NtlmAuthenticationFilter</filter-class>
      <init-param>
        <param-name>debug</param-name>
        <param-value>false</param-value>
      </init-param>
    </filter>
-->
 <!--
   <filter-mapping> <filter-name>NtlmAuthFilter</filter-name> <url-pattern>*.rpc</url-pattern> </filter-
mapping>
    <filter-mapping> <filter-name>NtlmAuthFilter</filter-name> <url-pattern>/wpf/*</url-pattern> </filter-
mapping>
    <filter-mapping> <filter-name>NtlmAuthFilter</filter-name> <url-pattern>/uslm/*</url-pattern> </
filter-mapping>
    <filter-mapping> <filter-name>NtlmAuthFilter</filter-name> <url-pattern>/assure/*</url-pattern> </
filter-mapping>
    <filter-mapping> <filter-name>NtlmAuthFilter</filter-name> <url-pattern>/documents/*</url-pattern> </
filter-mapping>
    <filter-mapping> <filter-name>NtlmAuthFilter</filter-name> <url-pattern>/FileStore/*</url-pattern> </
filter-mapping>
-->
```

   Commenting these lines deactivates SSO functionality from this CA Service Catalog computer.
4. Restart CA Service Catalog.

Configure Single Sign-on Using External Authentication

When you use external authentication, you can perform this procedure to enable Single Sign-On for CA Service Catalog. Users who are set up in your external authentication system can access CA Service Catalog without logging in to it.

Follow these steps:

1. Verify that CA Service Catalog and CA EEM are installed in the same Windows domain.
2. Log in to CA Service Catalog running on this computer.
3. Perform the following actions:
   a. Click **Administration**, **Configuration**, **Single Sign On Authentication**.
   b. Locate the property **Single Sign On Type** and click the Modify icon.
   c. Select the option **Artifact Based Single Sign On** and click **Update Configuration**.
   The dialog closes, and you return to the Single Sign On Authentication page.
4. Verify that all affected users can use single sign-on to access CA Service Catalog on this computer.

You have configured external authentication other than Windows NTLM.

Configure CA Service Catalog to Use Secure Socket Layer

You can configure CA Service Catalog to use Secure Socket Layer (SSL). SSL establishes an encrypted link between a server and a client. For example, a web server and a browser; or a mail server and a mail client. This link helps ensure that all data passed between the server and client remain private and integral. When you configure a product to use SSL, you change its communication method from HTTP to HTTPS.

Follow these procedures:

Create a Keystore File

A keystore file is required to enable SSL. Create a keystore file if you do not have one already for another CA product that integrates with CA Service Catalog. You can use a keystore file for a single product or for multiple products. If you must create individual keystores for each product, you can optionally merge your keystore files.

Follow these steps:

1. Open a command window on the Catalog Component server.
2. Enter the following command:

   ```
   keytool -genkey -alias alias_name -keyalg RSA -keystore "USM_HOME\.keystore" -keysize 1024
   ```

   alias_name
   Specifies the logical name for the certificate that you are using for CA Service Catalog and possibly for other products. Record this alias name for reference.
3. Enter the password at the “Enter keystore password” prompt. To make configuring Tomcat easier, you can use “changeit” as the password.
4. Record your password for reference.
5. Enter your password at the prompt.

Export CA Certificate to the Keystore

After creating the keystore file, you must export a CA certificate to the keystore. Without a CA certificate present in keystore, the self signed certificates do not work. This is required with Apache Tomcat 8.5.6 and later because Tomcat server does not accept any keystore without a CA certificate for security reasons.

Follow these steps:

1. Run the below command to export the certificate into the keystore.

```
keytool -export -alias <alias_name> -storepass <password> -file <filename.cer> -keystore
 <path_and_file_specification_for_keystore>
```

CA certificate is exported to keystore. Self signed certificates work successfully now.

Add CA Certificate to the Keystore

After creating the keystore file, you must add a CA certificate to the keystore. Without a CA certificate present in keystore, the self signed certificates do not work. This is required with Apache Tomcat 8.5.6 and later because Tomcat server does not accept any keystore without a CA certificate for security reasons.

Follow these steps:

1. Download a CA certificate using download certificate options of any web browser.
2. Run the below command to import the certificate into the **%USM_HOME%\embedded\jdk\lib\security\cacerts** file.

```
keytool - import - alias <alias_name> - file <certfile> - keystore %USM_HOME%\embedded\jdk\lib\security
\cacerts
```

CA certificate is imported to cacerts. Self signed certificates work successfully now.

Configure CA Service Catalog to Use SSL

Configure CA Service Catalog to use Secure Socket Layer (SSL).

Follow these steps:

1. Edit the server.xml file to support SSL.
   The file is updated to help support SSL for CA Service Catalog.
2. Open the USM_HOME\view\conf\viewService.conf file, using a text editor.
3. Update the following line with the path name and file name of the keystore file:

```
wrapper.java.additional.10=-Djavax.net.ssl.trustStore="USM_HOME\embedded\jdk\lib\security\cacerts"
```

4. Update the following line with the password of the keystore file:

```
wrapper.java.additional.11=-Djavax.net.ssl.trustPass=changeit
```

5. Save and close the viewService.conf file.
6. Select Administration, Configuration, Server Information on the CA Service Catalog GUI.
7. Complete the fields in this section as follows:
   For Host Name, specify the name of the host where CA Service Catalog is installed.
   For Port Number, specify the port where HTTPS is configured.
   For Enable HTTPS, specify Yes. Restart CA Service Catalog.
8. Log in to CA Service Catalog using the following URL:
   https://<hostname:port>/usm/wpf
9. You see a trusted certificate prompt, which indicates that you are using HTTPS.
10. Optionally, disable HTTP access by commenting the section for the HTTP connector, as shown in the following example:

```
<!--
<Connector port="8080" enableLookups="false" redirectPort="8443" tomcatAuthentication="false"
      maxThreads="400" minSpareThreads="25" maxSpareThreads="100" debug="0" connectionTimeout="15000"
      disableUploadTimeout="true" compression="on" compressionMinSize="2048"
      compressableMimeType="text/html,text/plain,text/xml,text/css,text/javascript,image/png,image/gif,image/
jpeg,application/json"
      useBodyEncodingForURI="false" URIEncoding="UTF-8" />
-->
```

You have configured CA Service Catalog to use SSL.

Edit the Server.xml File to Support SSL

As part of configuring CA Service Catalog to use Secure Socket Layer (SSL), edit the server.xml file to support SSL.

Follow these steps:

1. Open the USM_HOME\view\conf\server.xml file.
2. Search for the following section. Enable the commented section by removing "<!--" and "-->" from the first and last lines, as shown in the following example:

```
<!--
    <Connector port="8443" enableLookups="false" tomcatAuthentication="false" maxHttpHeaderSize="8192"
      maxThreads="400" minSpareThreads="25" maxSpareThreads="100" debug="0" connectionTimeout="15000"
      disableUploadTimeout="true" compression="on" compressionMinSize="2048"
      compressableMimeType="text/html,text/plain,text/xml,text/css,text/javascript,image/png,image/
gif,image/jpeg,application/json"
      scheme="https" secure="true" clientAuth="false" sslProtocol="TLS" SSLEnabled="true"
      keystoreFile="__USMHOME__\.keystore" keyAlias="alias_name" keystorePass="changeit"/>
    -->
```

   – ***alias name***
   Specifies the logical name for the certificate that you are using for CA Service Catalog and possibly for other products.
3. Update the default port (8444) to another secure socket layer port, if necessary.
4. Verify whether either or both of the following conditions exist:
   – You are using an existing keystore.
   – You have changed either the CA Service Catalog installation path or generated keystore name.
5. Perform the following actions, if either or both of the conditions in the previous step exist:
   – Update the keystoreFile parameter with the correct path and file name, typically USM_HOME\.keystore.
   – Update the keyAlias parameter with the *alias_name* that you specified when you created a keystore file for CA Service Catalog.
6. Save and close the server.xml file.

You have edited the server.xml file. You can continue configuring CA Service Catalog to use Secure Socket Layer (SSL).

(Optional) Adding Additional Self-Signed Certificates to the Keystore

When you use self-signed certificates for any computer that connects directly to CA Service Catalog or that CA Service Catalog connects to, add these certificates to the keystore. For example, suppose that you are using clustering with load balancing for CA Service Catalog. In that case, if you are using a self-signed certificate for the load balancing computer, add them to the keystore.

> **NOTE**
> If you are using trusted certificates for these computers, you do not need to add them to the keystore.

Follow these steps:

1. Verify the computer to be trusted, that is, the computer that has direct connection with CA Service Catalog.
   For example, suppose that you integrate CA Service Catalog with CA Service Desk Manager through a load balancing computer. In that case, CA Service Catalog connects directly to the load balancer (not CA Service Desk Manager). Therefore, the computer to be trusted is the load balancer (not the CA Service Desk Manager computer).
2. Go to a Catalog Component computer. Download the DER encoded binary X.509 file (the certificate) for the computer to be trusted.
   For example, use your web browser to visit the computer and obtain the certificate.
3. Open the CA Service Catalog command prompt and enter the following command:

```
keytool -importcert -alias aliasname -file pathname-to-certificate -keystore USM_HOME\.keystore
```

   – ***alias_name***

Specifies the logical name for the certificate that you are using for CA Service Catalog and possibly for other products.

– ***pathname-to-certificate***
Specifies the complete path name to the certificate file that you downloaded in the previous step.

You are prompted to enter a password.

4. Perform one of the following actions:
   – Enter changeit as the keystore password.
   – Enter a different keystore password.

   The password is saved.
5. Complete this step if you entered a different password than changeit in the previous step. Otherwise, skip this step.
   a. Open the viewService.conf file for editing.
   b. Find the line that contains the following phrase:

      ```
      -Djavax.net.ssl.trustPass=keystore-password
      ```

   a. Update the keystore-password to match the new password that you specified in the previous step.

   The viewService.conf file is updated with the new password.
6. Remain at this Catalog Component computer. Repeat the previous steps for every computer to be trusted which has self-signed certificates.
   The keystore file is updated with the new self-signed certificates from each applicable computer to be trusted.
7. Perform the following actions on of every other Catalog Component computer:
   a. Update the viewService.conf file to use a password other than *changeit*, if applicable.
   b. Copy the updated keystore file from this Catalog Component computer to all the remaining Catalog Component computers.

You have added self-signed certificates to the keystore.

(Optional) Merge Keystore Files

A keystore file is required to enable SSL. If you are using SSL for two or more products and have two or more keystore files, we recommend to use a single keystore for all integrated products. However, if you have multiple keystores for different products and cannot use a single keystore for them all, you can merge the contents of the individual keystore files.

Follow these steps:

1. Copy all keystore files to the USM_HOME folder. For example, any keystore files for CA Process Automation, CA CMDB, or other products that integrate with CA Service Catalog.
2. Find and record all required keystore files, keystore aliases, and keystore passwords for the products of interest. For example, for CA Process Automation, you can retrieve the c2okeystore password from KEYSTOREID property of the OasisConfig.properties file.
3. Restart Catalog Component.
4. Enter the keytool command to merge the keystore of the first product, using the following command as a model:

   ```
   keytool -importkeystore -srckeystore "product1_keystore" -destkeystore "USM_HOME\.keystore" -srcalias
    product1_alias -destalias alias_name-srckeypass "product1_password" -destkeypass "changeit"
   ```

   – **"*product1_keystore*"**
   Specifies the name of keystore file (including the complete path name) for the product you are merging.
   – ***product1_alias***
   Specifies the keystore alias for the product you are merging.
   – ***product1_password***
   Specifies the keystore password for the product you are merging.
   – ***alias_name***
   Specifies the *alias_name* that you specified when you created a keystore file for CA Service Catalog.

The following command merges the contents of the CA Process Automation keystore (c2okeystore) into the CA Service Catalog keystore:

```
keytool -importkeystore -srckeystore "%ITPAM_HOME%\server\c2o\.config\c2okeystore" -destkeystore "USM_HOME
\.keystore" -srcalias c2o-j -destalias tomcat -srckeypass "475ba811-62cd-4ec8-b757-cd7710de3fa8" -
destkeypass "changeit"
```

The following command merges the contents of the CA CMDB keystore (.cmdb_keystore) into the CA Service Catalog keystore:

```
keytool -importkeystore -srckeystore ".cmbd_keystore" -destkeystore
"USM_HOME\.keystore" -srcalias cmdb -destalias tomcat
-srckeypass "changeit" -destkeypass "changeit"
```

5. Respond No when you are prompted to overwrite the source alias.
6. Repeat the previous two steps for each product whose keystores you are merging.
7. Verify if all the certificates that you want are in one keystore, using the following command:

   ```
   keytool -list -keystore "USM_HOME\.keystore"
   ```

   This command lists the contents of the merged keystore.

You have merged keystore files.

(Optional) Configure CA Process Automation to Communicate with CA Service Catalog Using SSL

Configure CA Process Automation to communicate with CA Service Catalog using SSL.

Follow these steps:

1. On the CA Service Catalog GUI, select Administration, Configuration, CA Process Automation.
2. Complete the fields in this section as follows:
   For Host Name, specify the name of the host where CA Process Automation is installed.
   For Port Number, specify the CA Process Automation port where HTTPS is configured.
   For Enable HTTPS, specify Yes.
3. Recycle Catalog Component.
4. Click Test.
   The connection is tested, using the new values that you specified.
   If the connection fails, try using a different value.
5. Click Configure.
   The CA Process Automation configuration details are updated with the new values that you specified.
6. Perform this step if you are using content packs. Otherwise, you can skip this step.
   Edit the USM_HOME\build.xml file, and verify that the following parameters are correct. Update them, if applicable.
   - The file name and path name of the keystore
     For example, if the keystore is located in USM_HOME and the keystore file name is.mykeystore, update this line as follows:

     ```
     <sysproperty key="javax.net.ssl.trustStore" value="${env.USM_HOME}/.mykeystore" />
     ```

   - The keystore password
     For example, if the keystore password is mykeystorepw, update this line as follows:

     ```
     <sysproperty key="javax.net.ssl.trustPass" value="mykeystorepw" />
     ```

You have configured CA Process Automation to communicate with CA Service Catalog using SSL. For more information, see CA Process Automation documentation.

Enable HSTS Filter

To enable HSTS filter, you must enable Secure Socket Layer (SSL) as a prerequisite.

Follow these steps:

1. Enable SSL and then go to the location %USM_HOME%\view\webapps\usm\WEB-INF\
2. Open the web.xml file and uncomment the filter. (approx line number ~199)

```
<!--
    <filter>
        <filter-name>httpHeaderSecurity</filter-name>
        <filter-class>org.apache.catalina.filters.HttpHeaderSecurityFilter</filter-class>
        <init-param>
        .
        .
        .
        .
        .
        </init-param>
        <async-supported>true</async-supported>
    </filter>
-->
```

3. a. For 17.4.3.0 and before, after HSTS filter is uncommented, make sure to uncomment the HSTS filter mapping. (approx line number ~372)

```
<!--
<filter-mapping>
<filter-name>httpHeaderSecurity</filter-name><url-pattern>/wpf/*</url-pattern> </filter-mapping>
<filter-mapping>
<filter-name>httpHeaderSecurity</filter-name><url-pattern>*.rpc</url-pattern> </filter-mapping>
<filter-mapping>
<filter-name>httpHeaderSecurity</filter-name><url-pattern>/documents/*</url-pattern> </filter-mapping>
<filter-mapping>
 <filter-name>httpHeaderSecurity</filter-name> <url-pattern>/uslm/*</url-pattern> </filter-mapping>
<filter-mapping>
<filter-name>httpHeaderSecurity</filter-name> <url-pattern>/assure/*</url-pattern> </filter-mapping>
→
```

   b. For 17.4.4 and above, after HSTS filter is uncommented, make sure to uncomment the HSTS filter mapping. (approx line number ~372)

```
 <!--

<filter-mapping>
<filter-name>httpHeaderSecurity</filter-name><url-pattern>*.*</url-pattern> </filter-mapping>

→
```

The HSTS filter is now enabled.

Reconfigure Service Catalog using the Setup Utility

Run the CA Service Catalog setup utility to set up your database, reconfigure CA EEM, and reconfigure the product components. The product components are Catalog Component, Catalog Content, and Service Accounting Component.

> **WARNING**
> Run the CA Service Catalog setup utility on all CA Service Catalog computers in the correct sequence, as explained in this topic.

Follow these steps:

1. Run the setup utility on each CA Service Catalog computer, in the following sequence:
   a. The *first* (formerly *primary*) Service View computer
   b. All *additional* (formerly *secondary*) Service View computers
   c. All other CA Service Catalog computers, in any order

   NOTE
   To run the setup utility, perform *all* the following steps on each computer, one computer at a time. For example, the first Service View computer is Host1, the additional Service View computers are Hosts 2 and 3, and the remaining CA Service Catalog computers are Hosts 4-6. First, run the setup utility on Host1. Next, run the setup utility on Hosts 2 and 3, one computer at a time, in any order. Third, run the setup utility on Hosts 4-6, one computer at a time, in any order.
2. Start the setup utility using one of the following methods:
   – Click the option to start the setup utility *after* you have run the installation or upgrade program on the last applicable computer. The last applicable computer is the final computer on which you run the installation or upgrade program for CA Service Catalog.
   – Start the setup utility from the Windows Start menu. For example, click Start, Programs, CA, Service Catalog, Setup Utility.
3. Specify the following parameters and log in to the utility:
   – **Password**
   Specifies the password for logging in to the utility and running it using either of the methods in the previous step.

   NOTE
   Record the password for reference, because the utility requires you to specify the password each time you start it.

   If necessary, you can reset the password, as follows:
   Enter the following command:
   ```
   USM_HOME\scripts\configurator.bat -resetpwd
   ```
   The Catalog system resets the password and prompts you to perform the next action: Restart the utility and specify the new password.
   – **Remote Configuration**
   (Optional) If you have run the utility successfully on another computer (a remote computer), you can use the same configuration on this computer (the local computer). To do so, select this option and enter the URL of the remote computer. Use the following format: http://*server*:*port*/*context*. An example is http://localhost:8080/usm.
   The database setup page appears.
4. Complete this step *only* if you selected the Remote Configuration option in the previous step. Otherwise, skip to the next step.
   Supply the password for the setup utility on the remote computer.
   The local setup utility copies configuration data from the remote computer and pre-populates the corresponding local fields with this data. The local utility copies the data for the database setup, CA EEM configuration, and the name of the top-level business unit. Therefore, you do not need to enter this data manually. However, you can optionally update the copied values, if necessary. This data appears on the remaining tabs of the local setup utility and is described in the remaining steps.
   You typically use the Remote Configuration option when you plan to install multiple instances of CA Service Catalog.
5. Reconfigure the Database
6. Reconfigure CA EEM
7. Reconfigure the CA Service Catalog Components

You have completed the steps for running the setup utility on all CA Service Catalog computers.

NOTE
If you are installing CA Service Catalog for the first time on a computer, the setup utility pre-populates the fields that require input with default values. If you are re-installing or upgrading CA Service Catalog, the setup utility pre-populates these fields with the values from the current installation. The Remote Configuration option overwrites any default values or existing values with the values from the remote computer.

Reconfigure the Database

You can reconfigure database so that CA Service Catalog users and the Catalog system can function correctly.

Follow these steps:

1. Log in to the setup utility and click Database on the left menu.
   The utility displays the page for connecting to the database and configuring it.
2. Select the DBMS that you are using and provide the information to complete the fields.
3. Complete the fields for the Database Connectivity section.
   This step verifies that you can connect to the database. If the connection fails, perform these actions:
   a. Verify that the parameter values are correct. For more information about these parameter values, see Parameters for SQL Server and Parameters for Oracle.
   b. Verify that the CA Service Catalog Pre-Installation Considerations.
4. Specify the name of the CA Service Catalog database, typically MDB, in the Database Settings section.
   This step installs or upgrades the MDB, if necessary. This step also configures the CA Service Catalog database in the MDB.
5. Complete the remaining database fields.
6. Specify the details for the application user that you created earlier.
7. Save and confirm your updates.
8. (Optional) Review the log files, as follows:
   – For log information for SQL Server, see the following sources:
     • See the install_mdb.log file in your MDB target installation directory. This directory is specified in %TEMP%\MDB1.5, where %TEMP% is a Windows environment variable. If the MDB installation fails, review the log files in this location.
     • After a successful installation of the MDB, the install_mdb.log file is copied to the %ProgramFiles%\CA\SC\Mdb\Windows\logs folder. This log file is renamed to install_mdb_*mm-dd,hh_mm_ss*.log (for example, install_mdb_12-03,15_29_05.log).
     • See the view.log file. Its default location is the C:\Program Files\CA\Service Catalog\logs folder.
   – For log information for Oracle, see the following sources:
     • See the install_connectionID.log file in your MDB target installation directory. For example, if the connection ID is myhost_orcl, the install_connectionID.log file name is install_myhost_orcl.log. The MDB target installation directory is specified in %TEMP%\MDB1.5, where %TEMP% is a Windows environment variable. If the MDB installation fails, review the log files in this location.
     • After a successful installation of the MDB, the install_connectionID.log file is copied to %ProgramFiles%\CA\SC\Mdb\Windows\logs. This log file is renamed to install_ connectionID_*mm-dd,hh_mm_ss*.log, for example, install_myhost_orcl_12-03,15_29_05.log.
     • See the view.log file. Its default location is the C:\Program Files\CA\Service Catalog\logs folder.

You have reconfigured the database.

Parameters for SQL Server

Before you install CA Service Catalog, you specify values for several parameters to establish the connection to an SQL Server database. The following parameters may require explanation:

WARNING
For a migration model upgrade, when you are prompted for information about the Catalog database, use the *existing* Catalog database specifications.

For the Database Connection

- **Host Name**
  Defines the computer name of the Microsoft SQL Server (SQL Server) server.
- **Port**
  Defines the TCP Port number of the database server.
  If you are using an instance name, verify the following requirements:
  – The instance name has a port number set.
  – Your implementation is *not* using dynamic ports.
- **User Name and Password fields**
  Define the user name and password of the SQL Server database administrator (DBA). The setup utility uses these credentials to set up the CA Service Catalog database.
  You can use the login credentials System administrator (sa). This user requires dbo as its default schema.
- **Instance Name**
  Defines the SQL Server instance name for the MDB. For example, myinstance. You can specify either a primary instance or a named instance.

For the Database Configuration

- **Database Name**
  Specifies the name of the CA Service Catalog database: MDB.

  WARNING
  The name of the Catalog database must be MDB. This requirement is especially important if you integrate CA Service Catalog with another CA product, such as CA Service Desk Manager, CA APM, or CA Process Automation.

Application User Settings

- **User Name and Password fields**
  Define the user name and password that CA Service Catalog uses for accessing this database.
  This user name and password are created in the database.
  This user is the first application user. For increased flexibility, you can optionally create a second application user.

Parameters for Oracle

Before you install CA Service Catalog, you specify values for several parameters to establish the connection to an Oracle database. The following parameters may require explanation:

WARNING
For a migration model upgrade, when you are prompted for information about the Catalog database, use the *existing* Catalog database specifications.

NOTE
These parameters apply to Oracle running on both Windows and Linux.

For the Database Connection

- **Host Name**
  Defines the computer name of the Oracle server.
- **Port**
  Defines the TCP Port number of the database.
- **User Name and Password fields**

Define the user name and password of the Oracle database administrator (DBA).

- **Service Name**
  Defines the service name. Every Oracle database or service has a service name. The service name of an Oracle database is typically its global database name. Enter the service name of the database or other service that you want to access.
  A non-RAC deployment typically includes one service, one instance, and one database, each with the same name. However, in a RAC deployment, multiple instances can provide multiple services, all connecting to a single database.
- **Connection SID**
  Defines the connection ID (such as orcl) for connecting to the Oracle server, as follows:
  – For a local database, this value is typically the SID.
  – For a remote database, this value is typically the Net Service Name.

For the Database Configuration

- **Database Name**
  Specifies the name of the CA Service Catalog database: MDB.

  WARNING
  The name of the Catalog database must be MDB. This requirement is especially important if you integrate CA Service Catalog with another CA product, such as CA Service Desk Manager, CA APM, or CA Process Automation. CA Service Catalog embeds CA MDB r1.5.
- **Tablespace Path**
  Defines the complete path name of the tablespace for Oracle.
- **Data Tablespace Name**
  (Optional) Defines the tablespace name for the data.
- **Index Tablespace Name**
  (Optional) Defines the tablespace name for the indexes.
  The installer verifies whether the data and index tablespaces exist in the MDB. These tablespaces exist when either of the following conditions are met:
  – The MDB is being upgraded.
  – The CA Service Catalog database is being installed in an existing MDB, typically from an already installed CA product or group of products.
  If the data and index tablespaces exist, the installer displays their names. The installer also prompts you to specify whether to continue using the existing names or overwrite them with new names.

Application User Settings

- **User Name and Password fields**
  Define the user name and password that CA Service Catalog uses for accessing this database.
  This user name and password are created in the database.

Reconfigure CA EEM

You can reconfigure CA EEM for use with CA Service Catalog using the setup utility.

NOTE

Use the Setup Utility if you have only CA Service Catalog in your environment. Use the Service Management Administration, Common Configuration option to reconfigure CA EEM if you have integrated CA Service Catalog with at least one other CA Service Management product.

For more information, see Integrate CA Service Catalog with CA EEM Manually

CA EEM has been reconfigured for use with CA Service Catalog.

Reconfigure the CA Service Catalog Components

You can reconfigure the CA Service Catalog product components that you want on this computer.

> **NOTE**
>
> If you are installing CA Service Catalog on this computer for the first time, you can use the utility to reconfigure the product components. However, if you are upgrading or installing CA Service Catalog on this computer a second time, this value is read-only and you cannot reconfigure the product components.

Follow these steps:

1. Log in to the setup utility and click Components in the the left menu.
2. Enter the name of the service provider business unit. It is the “root” business unit above all other business units. As a best practice, specify your company domain name or a short version of your company name for the business unit ID.
3. Select each component that you want to use on this computer:
   - **Catalog Component**
     Lets you create service options and service option groups, which you use to create services that users can request from the catalog.
     This option includes the Catalog Content, which supplies the predefined services in the catalog. Examples include services for requesting hardware, software, and other IT essentials from your business unit. You can use these services as-is, or you can copy and customize them.
     This option installs a Windows service named CA Service Catalog.
   - **Service Accounting Component**
     Lets you provide billing and chargeback for the services that users request from the catalog. You can also use Service Accounting Component to allocate costs, prepare budgets, and plan IT services.
     This option installs a Windows service named CA Service Accounting.
4. Click Save.
5. Follow the prompts and complete the setup.

> **NOTE**
> You can optionally review the installation log file, view.log. Its default location is the USM_HOME\logs folder. This log file is visible after you have completed all tasks of the installation or upgrade program and have closed the program. For information about uninstallation and aborted or canceled installations (when applicable), see the CA_Service_Catalog_Uninstallation.log file in the Windows Temp folder.

CA Service Catalog components have been reconfigured.

Configuration Files

The configuration files and locations are as follows:

- Database connection configuration file: USM_HOME\DBSource.properties
- CA EEM connectivity configuration file: USM_HOME\Eiam.properties
- IXUtil utility configuration file: USM_HOME\scripts\ixutil.cfg

USM_HOME is the documentation convention that specifies the local CA Service Catalog installation directory. For 32-bit computers, the default path name is C:\Program Files\CA\Service Catalog. For 64-bit computers, the default path name is C:\Program Files (x86)\CA\Service Catalog for 32-bit installations or C:\Program Files\CA\Service Catalog for 64-bit installations.

Configuration Settings

This article contains the following topics:

- Inheritance of Configuration Settings Through the Business Unit Hierarchy
- Change the Business Unit

After the CA Service Catalog installation is complete, verify the configuration settings. Doing so helps ensure that the product functions according to your organization's needs.

Verify the settings by group, as follows:

- Let CA Service Catalog Configuration Options
- Set Accounting Configuration Options
- Set Administration Configuration Options
  These include Integration-Specific Options.
- Location of shared files (filestore)

> **NOTE**
> With one exception, CA Service Catalog sends emails in HTML format only. To receive legible emails from CA Service Catalog, recipients must configure their email software to accept emails in HTML format. Otherwise, emails from CA Service Catalog display indecipherable messages when opened. The one exception to this HTML-only rule is invoice history. You can choose to view and email invoice history in HTML, CSV, or XML format. The HTML-only rule applies to all other invoice-related areas of CA Service Catalog, including invoice generation.

Inheritance of Configuration Settings Through the Business Unit Hierarchy

The following rules govern the relationship between parent and child business units (children), including the inheritance of configuration settings:

- The child business units directly under the top-level business unit (the service provider) are named super business units. Super business units always inherit their configuration settings from the service provider business unit.
- If the configuration parameter Contains Sub Units is enabled, a super business unit can have children.
- If the *Contains Sub Units* parameter of the super business unit is enabled, you *can* edit the configuration settings of the super business unit.
- If you edit the configuration settings of a super business unit, your changes do *not* apply to the super business unit. Instead, your changes apply to its children. A child "inherits" its configuration settings from its parent.
- You can optionally create unlimited levels of children under super business units. If the*Contains Sub Units* setting of the business unit is enabled; you can edit its configuration settings. Your configuration changes do not apply to the business unit itself but instead, apply to its children. Conversely, if the *Contains Sub Units* setting of the business unit is disabled; you cannot edit its configuration settings.

Thus, the following summary applies to all business units except for the service provider:

- A business unit always inherits its configuration settings from its parent.
- You can edit the configuration settings of a business unit *only* if its Contains Sub Units setting is enabled. Your changes apply *only* to its children.

Change the Business Unit

The Catalog system default displays settings for the business unit you logged in to. If your role permits, you can change to a different business unit.

When common multi-tenant administration is enabled, you can add tenants, delete tenants, or edit the common attributes *only* through CA Service Desk Manager. When common multi-tenant administration is enabled, you can *view* tenants and all their attributes in CA Service Catalog. But you can edit only the CA Service Catalog-specific attributes.

Follow these steps:

1. Click Catalog>> Configuration.
2. Click the Change Business Unit button. The Search Business Units dialog appears.

3. Use the Expand and Collapse icons to navigate the business unit tree. Alternatively, use the selection criteria and Search button to locate the desired business unit. The list includes only the business units your role permits you to access.
4. To select a business unit, click its name in the tree. The window closes, and the configuration settings for that business unit appear.

Manage Configuration Options for CA Service Catalog

You configure CA Service Catalog to customize settings for request management, access control, request emails, and so forth. These settings specify how the Catalog system processes requests in your organization.

Follow these steps:

1. Click Catalog, Configuration.
2. Perform one of the following actions:
   – To manage configuration options for a different business unit, change the business unit before performing the next step.
   – To manage configuration options for the current business unit, go directly to the next step.
3. Select any of the following categories of options that you want to update:
   – Catalog Configuration
   – Request Management Configuration
   – System Configuration
4. View the options for the category, and click the Modify icon for the option that you want to update.
5. Update the setting as required and click Update Configuration.

You have to set the CA Service Catalog configuration options. These settings apply to the children of the business unit that you have updated, according to the inheritance of configuration settings through the business unit hierarchy.

You configure CA Service Catalog to customize settings for request management, access control, request emails, and so forth. These settings specify how the Catalog system processes requests in your organization.

Follow these steps:

1. Click Catalog, Configuration.
2. Perform one of the following actions:
   – To manage configuration options for a different business unit, change the business unit before performing the next step.
   – To manage configuration options for the current business unit, go directly to the next step.
3. Select any of the following categories of options that you want to update:
   – Catalog Configuration
   – Request Management Configuration
   – System Configuration
4. View the options for the category, and click the Modify icon for the option that you want to update.
5. Update the setting as required and click Update Configuration.

You have to set the CA Service Catalog configuration options. These settings apply to the children of the business unit that you have updated, according to the inheritance of configuration settings through the business unit hierarchy.

{"URL":["/usm/wpf?Node=iclaunchpad.catalog_config_options"],"customLabelGetStarted":"Options","description":"IPCE_Options"}

Catalog Configuration

As part of setting the CA Service Catalog configuration options, you specify the following Catalog Configuration options. These settings control the behavior of the catalog.

- **Default Effect of Service Option Element Changes**
  Specifies when to reflect changes to service option elements for accounts that have subscribed to or requested services that include them.
  Select one of the following options to use when the service option element changes:
  - **Specify when the Service Option Element Changes - Allow User to Choose**
    Enables the administrator to specify the effect of the change on existing subscribers or requestors.
  - **Beginning of Accounts' Current Billing Period - No Audit Trail**
    Implements the change retroactively to the beginning of the current billing period for existing subscribers or requestors. No audit trail applies.
  - **Beginning of Accounts' Current Billing Period**
    Implements the change retroactively to the beginning of the current billing period for existing subscribers or requestors.
  - **Beginning of Accounts' Next Billing Period**
    Implements the change at the beginning of the next billing period for existing subscribers or requestors.
  - **Immediately during Accounts' Billing Period**
    Implements the change immediately for existing subscribers or requestors.
  - **Specify a Future Effective Date**
    Enables the Administrator to specify a date on which the change takes effect for existing subscribers or requestors.

  Default: Specify when the Service Option Element Changes - Allow User to Choose.
- **Pass Through Catalog**
  Specifies whether to include the catalog of the parent business unit.
  If you select this option, the parent catalog is passed down to the child business unit. This setting is useful in a multiple-level business unit organization. This setting applies only to the settings for a child business unit.
  Default: No

Request Management Configuration

As part of setting the CA Service Catalog Configuration Options, you specify the Request Management Configuration options. These settings that control the behavior of requests for the following groups of parameters:

- Access Control Parameters
- Other Parameters
- Request Email Parameters

Access Control Parameters

As part of setting the CA Service Catalog configuration options, you specify the following Access Control parameters:

- **Access Control: Add Request**
  Specifies which user roles can add a request.
  Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, End User, Catalog User, Request Manager, Catalog Administrator
- **Access Control: Edit Request**
  Specifies which user roles can edit a request. All users can edit their own requests, if the status of the request is in the not submitted range.

Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, End User, Catalog User, Request Manager, Catalog Administrator

- **Access Control: Delete Request**
  Specifies which user roles can delete a request or a requested service. All users can delete their own requests, if the status of the request is in the not submitted range. Once a requested service is in a canceled state, only users with Delete Request permission can delete a canceled service.
  Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, Request Manager, Catalog Administrator
- **Access Control: Cancel Request**
  Specifies which user roles can cancel a request.
  This parameter works together with the Allow Cancellation Through parameter, which is described later in this list.
  Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, Request Manager, Catalog Administrator
- **Access Control: Show Amount Column**
  Specifies which user roles can view the Amount Column when viewing a request.
  Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, Request Manager, Catalog Administrator
- **Access Control: Show Period Column**
  Specifies which user roles can view the Period Column when viewing a request or shopping cart.
  Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, Request Manager, Catalog Administrator
- **Access Control: Show Related Ticket Column**
  Specifies which user roles can view the Related Ticket Column when viewing a request or shopping cart.
  Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, Request Manager, Catalog Administrator
- **Access Control: Show General Information**
  Specifies which user roles can control whether the general information form is visible when you create a request.
  Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, Request Manager, Catalog Administrator, Service Manager, End User, Catalog User
- **Access Control: Show Request Information**
  Specifies which user roles can control whether the request information form is visible when you create a request.
  Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, Request Manager, Catalog Administrator, Service Manager, End User, Catalog User
- **Access Control: Save Cart As Request**
  Specifies which user roles can view the button allowing the user to save the shopping cart as a request, clearing the cart for reuse.
  Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, Request Manager, Catalog Administrator
- **Access Control: Proxy Request**
  Specifies which user roles can view the link to change the Requested For field from your own user ID or account to another user ID or account.
  Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, Request Manager, Catalog Administrator
- **Access Control: Proxy Action**
  Specifies which user roles can see the link to approve, reject, fulfill, or transfer requests pending action that is assigned to other users.
- **Access Control: Show Fulfillment Details**
  Specifies which user roles can view statistical information regarding the time that is required to fulfill a service option in the past.

Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, Request Manager, Catalog Administrator

- **Access Control: Take/Return Request Pending Action**
  Specifies which user roles can take or return ownership of requests pending action. They can take ownership of requests pending action from the following options:
  – A group queue
  – The queue of another user in the same business unit or child business unit
  Similarly, they can return ownership of requests pending action from their own queue to their group queue.
  Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, Request Manager, Catalog Administrator
- **Access Control: Transfer Request Pending Action**
  Specifies which user roles can transfer ownership of requests pending action from the assigned user to either of the following options:
  – Another user in the same group
  – Another user in the same business unit or child business unit
  Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, Request Manager, Catalog Administrator
- **Access Control: Delegate Request Pending Action**
  Specifies which user roles can delegate ownership of requests pending action from a group queue to either of the following options:
  – A specific user in that group
  – Another user in the same business unit or child business unit
  Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, Request Manager, Catalog Administrator
- **Access Control: Push Through Requests**
  Specifies which user roles can push through ("force") a stuck request to the next level of approval or fulfillment.
  Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, Request Manager, Catalog Administrator
- **Access Control: Display Request Warning**
  Specifies which user roles can view the warning icon indicating that a request is stuck. **Default:** Service Delivery Administrator, Super Business Unit Administrator, Administrator, Request Manager, Catalog Administrator, End User, Catalog User
- **Access Control: Hold/Resume Request**
  Specifies which user roles can hold a request that is in progress and can resume a request that is in Hold status.
  Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, Request Manager, Catalog Administrator, End User, Catalog User

Other Parameters

As part of setting the CA Service Catalog configuration options, you specify the following miscellaneous parameters:

- **Administrator Contact**Catalog Administrators can configure their contact information to be displayed in the error screen, in case of a request processing failure, which will help the application users to contact Administrators.
- **Allow Attachments at Service Option Level**
  Lets users add attachments to individual service options. This setting must be Yes if you want to make attachments mandatory for any service option.
- **Allow Cancellation Through**
  Defines the status through which users can cancel requests. After the request moves to the next status, it cannot be canceled. This parameter works together with the *Access Control: Cancel Request* parameter.
  Default: Fulfilled
- **Allow Discrete Handling for Reject**

Specifies the effect of the rejection of a single service or service option upon other services and service options in the same request, as follows:

– **Yes**
Specifies that when you reject a service or service option, the remaining services or service options can advance if they are approved. If the following options are approved, they can advance in the request life cycle:
  - Other services in the same request
  - Other service options in the same service

– **No**
– Specifies that when you reject a service or service option, the entire request is rejected. All services in the same request are rejected and can no longer advance in the request life cycle, even if they were previously approved. Similarly, all service options in the same service are rejected and can no longer advance in the request life cycle, even if they were previously approved.

Default: No

- **Allow Discrete Handling of Service Options After**
Specifies the status at which request managers (approvers and fulfillers) can handle requests pending action (approval, rejection, or fulfillment) *discretely*. When the request reaches the status you specify, request managers can discretely approve, reject, or fulfill each service option in every service in the request.
Use this parameter to specify that pending actions are assigned at the service option level, rather than the service level.
The setting applies to the request *from* the starting status that you specify *through* the remainder of the request life cycle.
Valid values: Submitted, Pending Fulfillment, or Completed.
Default: Pending Fulfillment

  NOTE
  For more information about using a discrete request life cycle, see the Manage Discrete Request Life Cycle section.

- **Allow Discrete Request Life Cycle After**
Specifies the status at which individual service options in a service and individual services in a request advance independently. *Independently* means *without* requiring or waiting for any other service option in the same service or any other service in the same request to advance its status.
When the request reaches the specified status, the service options that you approve or fulfill can complete the remainder of the request life cycle. This condition exists even if other service options in the same service do not advance in status.
As soon as the request reaches the specified status, the services that you approve or fulfill can complete the remainder of the request life cycle. This condition exists even if other services in the same request are rejected or are not updated.
Valid values: Submitted, Pending Fulfillment, or Completed.
Default: Completed
The setting applies to the request *from* the starting status that you specify *through* the remainder of the request life cycle.
- **Allow Multi Service/Service Option Approval**
Lets users approve or reject multiple, selected services or service options (including all) with a single click. This ability is *multi-item approval*. Discrete approval is required to approve or reject service options using multi-item approval.
- **Allow Notes at Service Option Level**
Controls whether a user can add notes at the service and service option levels.
If this option is set to Yes, then the Add Note icon appears in the Action column for requests. Therefore, the user can add a note for a specific service or service option.
If this option is set to No, then the user can add notes for the entire request but not for a specific service or service option.
Default: No
- **Allow Only One Service Per Request**

Specifies whether a user can include more than one service in a request or cart.
If this option is set to Yes, then a user can have only one service in a cart or request. Also, the user must submit the cart or request before creating a new one.
If this option is set to No, then a user can select multiple services from the catalog. The user can add them to the cart or request before submitting the request.
Default: No

- **Allow Submission of Rejected Requests**
  Provides the ability to disable the submission of requests that are in **Not Submitted-Rejected** State. By default, this option is set to **Yes** for backwards compatibility, and **the Submit** button is enabled for the **Not Submitted-Rejected** state requests. In order to disable the **Submit** button from the requests details page and avoid requests in the **Not Submitted-Rejected** state from being submitted, set this option to **No**.
- **Browse Catalog Layout**
  Specifies the look-and-feel for the Requests page. On this page, users can create, customize, submit, and manage requests.
  Specify *one* of the following options. Click Home, Requests to verify the resulting look-and-feel on the Requests page. The Service View and Service View without Browse Section options provide the best performance. These options provide *links to* the requests, while the other options display the requests directly.
  - **Service View**
    Configures the Requests page to display the catalog, featured services, and a search-for-services field in boxes.
    A My Requests link opens a drop-down list. The list includes options to search requests, review open requests, and act on pending requests.
  - **Service View without Browse Section**
    Configures the Requests page to display the catalog alone. Featured services and the search-for-services field do *not* appear.
    A My Requests link appears, as described for the Service View option.
  - **Request View**
    Configures the Requests page to display the catalog (categories and lists of services) in a column on the left.
    Boxes to the right of the column enable users to do the following tasks:
    - Search for services.
    - View and request featured services.
    - Review their own recent requests and, if applicable, act on them.
    - Use links to review open, pending, and completed requests.
    - Search for requests.
  - **Grid View**
    Configures the Requests page to display the catalog in a grid.
    Boxes surrounding the grid enable users to perform the same tasks as listed for the Request View option.

  Default: Service View
- **Browse Catalog: Show Folder Icon**
  Specifies whether images associated with catalog folders appear in the Browse section of the Requests page.
  If this option is set to Yes, folder images appear in the Browse section. If this option is set to No, they do not appear.
  Default: No
- **Browse Catalog: Show Subfolders**
  Specifies whether the top-level subfolders under each catalog folder appear in the Browse section of the Requests page.
  If this option is set to Yes, the first level of subfolders appear under each catalog folder in the Browse section.
  If this option is set to No, only the catalog folders appear.
  Default: Yes

NOTE

For any individual catalog folder, you can override this default by changing its folder details on the Catalog, Service Offerings page.

- **Default User for Request Actions**
  Specifies the user who must be assigned a pending action when no other user is available. This setting is typically used for request approvals when the user needs manager approval and no manager exists in the system for the user. When this setting is used with the system approval process, verify that this user has the highest authorization level.
  Default: spadmin
- **Display Service Health**
  This parameter applies *only* if you are integrating CA Service Catalog with CA Business Service Insight.
  Specifies whether to enable catalog users requesting a service to view actual, current data regarding the quality or "health" of a service. The health is based on the level of compliance of the service to its associated metrics. This data includes the contractual metrics, incentive metrics, and the SLA health period. This information is specified in the contract-specific details for the service option elements in the service being requested.
  The metric data includes both the performance criteria and the actual number of violations, including the time increments measured.
  If you enable this option, catalog users can view this actual and current metric data for the service they are requesting.
- **Edit Cancelled Requests**
  Specifies that if the value is set to **Yes**, you can edit a cancelled request. If the value is set to **No**, the cancelled request cannot be edited.
- **Edit Completed Requests**
  Specifies that if the value is set to **Yes**, you can edit a completed request. If the value is set to **No**, the completed request cannot be edited.
- **Enable Delegation of Catalog**
  Specifies whether to enable the delegation of catalogs: Yes or No.
  When this option is enabled, no users are *required to* delegate use of their catalogs. But they can *optionally* do so.

  NOTE
  The delegation of a catalog does *not* affect the Catalog configuration of the CA Service Catalog, such as the *Use Service Provider Catalog Only* and *Pass Through Catalog* configuration parameters. Similarly, the delegation of a catalog is *not* directly related to the configuration options such as *Access Control: Add Request* in this Request Management Configuration section.
- **Notify users when they complete their own pending actions**
  Specifies whether request managers receive a notification email every time they address a pending action by approving, rejecting, or fulfilling a service option, service, or request.
  If you specify Yes, request managers receive such emails confirming their own actions.
  If you specify No, request managers do not receive such emails.
  This setting is especially relevant if you are using discrete approval and the *Allow Request Life Cycle to Continue After* parameter is set to Submitted. If this parameter parameter is set to Yes, request managers are likely to receive several notification emails.
  For example, if a service has five service options and the request manager approves each option, the request manager receives five confirmation emails. To prevent request managers from receiving such emails, set *Notify users when they complete their own pending actions* to No.
  The Requested For users and Requested By users always receive these notification emails, regardless of the setting of these parameters.
- **Number of Requests per Page**
  For individual users (not administrators), specifies the maximum number of requests that appear on Request List pages. When a user moves to a new Request List page, the default setting replaces their custom settings. Users can optionally customize the display for each new Request List page they view.
  Request List pages appear when you click Home, Requests, and click any of the following options:

  - Open Requests
  - My Recent Requests
  - Completed Requests
  - Pending My Action

  Advanced Search - after you search for requests and view the results.
  Administrators can set the following options for all users: The individual columns that appear on Request List pages and the default maximum number of requests per page.
- **PDA Support: Enable**
  Specifies whether to enable PDA support.
  When you specify Yes, PDA users can click the links that are provided in PDA-compliant request approval emails to access the requests. The users can approve or reject them. Support for forms are limited when you enable PDA Approval.
  When you specify No, the links for accessing the requests and approving or rejecting them do *not* appear in the request approval emails that PDA users receive. The absence of such links implicitly guides PDA users to use an office computer to view, approve, and reject requests.
- **Request Details Link for E-Mail Notifications:**
  By default, value for this parameter is not configured. You can configure this parameter in the following format if:
  - CA Service Catalog is configured with Service Point

    NOTE
    This configuration is available only if you have installed the CA Service Catalog 17.3.0.4 patch.

    http://{SP HostName}:{PortNumber}/casm.html#/bui/myTickets/$request_id$?reqType=ct
    If you configure the Service Point URL for **Request Details Link for E-Mail Notifications**, clicking on the request details hyperlinks in the email notification will take you to the Service Point login page instead of the CA Service Catalog login page. If this parameter is not configured, clicking on the hyperlinks in email notification with request details takes you to the CA Service Catalog login page.
- **Request Home Page: Include Request Information**
  Specifies which roles can view the Request Lookup and My Recent Requests sections on the Home or Requests menu option page.
  Default: Service Delivery Administrator, Super Business Unit Administrator, Administrator, End User, Catalog User, Request Manager, Catalog Administrator, Service Manager
- **Request General Information Column Configuration**
  Default: Name, ID, Requested For, Date Created, Date Required, Requested By, Priority, Last Modified, Status

Request Email Parameters

As part of setting the CA Service Catalog configuration options, you specify the following Request Email parameters:

- **Request Email: From Address**
  Specifies the email address from which the Catalog system sends emails about requests.
  - **User Defined**
    Uses the email address that the user specifies. Selecting this option with no email address specified sets this setting to EMPTY.
  - **Service Provider Email**
    Uses the email address for the service provider business unit.

  Default: CatalogSystem@serviceprovider.com, where "*serviceprovider*" is the root business unit that is specified during installation.
- **Request Email: From Name**
  Specifies the email address *name* from which emails about requests are sent.
  Default: CatalogSystem
- **Request Email: To**

Specifies the primary recipient email address to which emails about requests are sent.

- **User Defined**
  Uses the email address that the user specifies. Multiple email addresses can be specified, separated by a semi-colon. Selecting this option with no email address specified sets this setting to EMPTY.
- **Requested For User/Account Email**
  Uses the email address of the user or account to which this request applies.
- **Requested By User/Account Email**
  Uses the email address of the user or account that created this request.

Default: EMPTY, meaning this value is set when email options are used.

- **Request Email: CC**
  Specifies the “CC” recipient email address to which emails about requests are sent.
  - **User Defined**
    Uses the email address that the user specifies. Multiple email addresses can be specified, separated by a semi-colon. Selecting this option with no email address specified sets this setting to EMPTY.
  - **Requested For User/Account Email**
    Uses the email address of the user or account to which this request applies.
  - **Requested By User/Account Email**
    Uses the email address of the user or account that created this request.

  Default: EMPTY, meaning this value is set when email options are used.
- **Request Email: BCC**
  Specifies the “BCC” recipient email address to which emails about requests are sent.
  - **User Defined**
    Uses the email address that the user specifies. Multiple email addresses can be specified, separated by a semi-colon. Selecting this option with no email address specified sets this setting to EMPTY.
  - **Requested For User/Account Email**
    Uses the email address of the user or account to which this request applies.
  - **Requested By User/Account Email**
    Uses the email address of the user or account that created this request.

  Default: EMPTY, meaning this value is set when email options are used.
- **Request Email: Subject**
  Specifies the email subject that is used when emails about requests are sent.
  In addition to specific text, you can include dynamically substituted values in the subject text. The syntax for including dynamically substituted values is $Request.*variable_name*$.
  - ***variable_name***
    Specifies the name of the variable value to use.

  You can specify the following variables in the subject text:
  - **request_id**
    Specifies the internal Request ID, such as “10001”.
  - **name**
    Specifies the request name that the user specifies.
  - **status**
    Specifies the status of the request, such as “Pending Approval.”
  - **created_date**
    Specifies the date and time the request was created.
  - **modified_date**
    Specifies the date and time the request was last modified.
  - **completion_date**
    Specifies the date and time the request was completed.
  - **desired_date**

Specifies the date and time in the Date Required field of the request.
  - **comments**
    Specifies the request Description that the user specifies.
  - **priority**
    Specifies the priority of the request.
  - **req_for_account_id**
    Specifies the internal account ID for the Requested For user or account.
  - **req_by_account_id**
    Specifies the internal account ID for the Requested By user or account.
  - **req_for_user_id**
    Specifies the user ID of the Requested For user, if the request is for a user not an account.
  - **req_by_user_id**
    Specifies the user ID of the Requested By user.

  Default: Email of Request ($Request.request_id$) - $Request.name$
- **Request Email: Message**
  Specifies the email message that is used when emails about requests are sent. You can override this text.
  In addition to specific text, you can include dynamically substituted values in the subject text. The syntax for including dynamically substituted values is $Request.*variable_name*$ where *variable_name* is the name of the variable value to use.The variables available for use are the same as in the *Request Email: Subject* setting.
  Default: Request ($Request.request_id$) created on $Request.created_date$

System Configuration

As part of setting the CA Service Catalog configuration options, you specify the following System Configuration parameters. These settings control the behavior of the system, regardless of business unit. This category is displayed *only* when you are logged in to the service provider (root) business unit.

- **Use Service Provider Configuration Only**
  Specifies whether a child business unit can have its own CA Service Catalog configuration settings.
  If this setting is Yes, the Catalog system ignores all CA Service Catalog configuration settings for a business unit. So, each business unit uses the configuration settings for the service provider.
  If this setting is No, then each business unit can have its own CA Service Catalog configuration settings.
  Default: No
- **Use Service Provider Catalog Only**
  Specifies whether the service provider business unit catalog is available to all business units in the hierarchy.
  If this setting is Yes, the subscribing or requesting user of a child business unit sees only the catalog of the service provider (root) business unit. That is, the child business unit cannot have its own catalog.
  If this setting is No, the child business unit can have its own catalog.
  Default: No

Set Accounting Configuration Options

You can optionally customize several options for accounting configuration. These options cover invoicing and subscription details, to meet the requirements of your organization.

Follow these steps:

1. Log in as a user with the Service Delivery Administrator role.
2. Click Accounting, Configuration.
3. Check for the name of the current business unit, and perform one of the following actions:

- To set the accounting configuration options for a different business unit, change the business unit before performing the next step.
- To set the accounting configuration options for the current business unit, go directly to the next step.

4. Click the category of options that you want to update:
   - Accounting Profile Defaults
   - Billing Cycles
   - General
   - Invoice Engine Configuration
   - Invoice Methods
   - Payment Methods
   - Subscription Configuration
5. View the options for the category, and click the Modify icon for the option that you want to update.
6. Update the setting as required and click Update Configuration.
7. Repeat the previous steps for each accounting configuration option that you want to update.

You have set the accounting configuration options. These settings apply to the children of the business unit that you updated, according to the inheritance of configuration settings through the business unit hierarchy.

After the CA Service Catalog installation is complete, verify the configuration settings. Doing so helps ensure that the product functions according to your organization's needs.
Verify the settings for the Accounting Configuration Options.

You can optionally customize several options for accounting configuration. These options cover invoicing and subscription details, to meet the requirements of your organization.

Follow these steps:

1. Log in as a user with the Service Delivery Administrator role.
2. Click Accounting, Configuration.
3. Check for the name of the current business unit, and perform one of the following actions:
   - To set the accounting configuration options for a different business unit, change the business unit before performing the next step.
   - To set the accounting configuration options for the current business unit, go directly to the next step.
4. Click the category of options that you want to update:
   - Accounting Profile Defaults
   - Billing Cycles
   - General
   - Invoice Engine Configuration
   - Invoice Methods
   - Payment Methods
   - Subscription Configuration
5. View the options for the category, and click the Modify icon for the option that you want to update.
6. Update the setting as required and click Update Configuration.
7. Repeat the previous steps for each accounting configuration option that you want to update.

You have set the accounting configuration options. These settings apply to the children of the business unit that you updated, according to the inheritance of configuration settings through the business unit hierarchy.

{"URL":["/usm/wpf?Node=iclaunchpad.billing_configuration","/usm/wpf?Node=iclaunchpad.billing_options"],"customLabelGetStarted":"Configuration","description":"IPCE_Configuration","customCards":[{"id":"IPCE_Options","type":"manage","title":"Options"}]}

Accounting Profile Defaults

As part of setting accounting configuration options, you specify the Accounting Profile Default settings. These settings indicate default values for newly created accounts for the current business unit. The following options require explanation:

- **Account Type**
  Specifies the type of account. This setting controls how outstanding balances are handled on invoices, as follows:
  - **Open Item**
    Specifies that every invoice is unique. Once a charge appears on an invoice, it does not appear on any future invoices.
  - **Balance Forward**
    Specifies that the remaining balance of a previous invoice is added to the next invoice.
  - **Zero Balance**
    Specifies that the account is a zero balance account. A zero balance account can have charges appear on its invoice, but the total remaining balance is always zero. You cannot post payments to zero balance accounts. This option is commonly used when an aggregating account holds the charges for one or more zero balance accounts.

  Default: Open Item
- **Aggregate**
  Specifies the role of the account in charge aggregation for invoicing, as follows:
  - **No**
    Specifies that the charges for the account appear *only* in the invoice of the account. The charges do *not* appear on an invoice for a related aggregating account.
  - **Yes**
    Specifies that the charges for the account appear in the invoices for all of the following options:
    - The original account
    - Each aggregating account in the same business unit or a parent business unit, up through the hierarchy of business units
  - **Aggregating Account**
    Specifies that the invoice for the account shows charges for all accounts in the same business unit or child business units. These business units must have Aggregate set to Yes or Aggregating Account.

    NOTE
    To avoid double invoicing, do *not* specify more than one aggregating account in each business unit.

  Default: No
- **Automatic Invoicing**
  Specifies whether to create an invoice for the account, as follows:
  - **Yes**
    Specifies that the account has an invoice that is generated during a bill run.
  - **No**
    Specifies that the account does not have an invoice that is generated during a bill run.

  Default: Yes
- **Billing Cycle**
  Specifies the account invoicing cycle. Options are daily, weekly, and monthly.
  Note: This field works with the Billing Cycle Interval, which is described later in this topic.
  Default: Monthly
- **Period Start Date**
  Specifies how to set the start date for the invoice period for the account. Select from the following options:
  - **Business Unit Opened Date**
    Sets the start date as the date the business unit of the account was created (the Opened Date).
  - **Account Opened Date**

Sets the start date as the date the account was opened (the Accounting Profile Opened Date).

- **Current Date**
  Sets the start date to be the date that the accounting profile (account) was created.
- **Current Date of Next Period**
  Sets the start date as the addition of both of the following options:
  - The date the accounting profile (account) was created.
  - The interval that the Billing Cycle and the Billing Cycle Interval determine.
- **Day of Week**
  Sets the start date as the same day of the week on which the accounting profile (account) was created.
- **Day of Month**
  Sets the start date as the same day of the month on which the accounting profile (account) was created.
- **Day of Year**
  Sets the start date as the same day of the year on which the accounting profile (account) was created.
- **Day of Week Adjusted from Current Date**
  Sets the start date as the specified day of the next week after the accounting profile (account) was created.
- **Day of Month Adjusted from Current Date**
  Sets the start date as the specified day of the next month after the accounting profile (account) was created.
- **Day of Year Adjusted from Current Date**
  Sets the start date as the specified day of the next year after the accounting profile (account) was created.
- **Last Day of Current Month**
  Sets the start date as the last day of the month during which the accounting profile (account) was created.
- **First Day of Next Month**
  Sets the start date as the first day of the next month after the accounting profile (account) was created.
- **Specify Date**
  Sets the start date as the date that you specify manually.

Default: Account Opened Date

- **Billing Cycle Interval**
  Specifies the number of billing cycles between invoices.
  For example, to bill the account quarterly, select a Billing Cycle of Monthly and a Billing Cycle Interval of 3. The Billing Cycle field is described earlier in this topic.
  Default: 1
- **Default Days Due**
  Specifies the number of days after the invoice date to use as the payment due date.
  Default: 10
- **Grace Days**
  Specifies the number of days that are permitted past the invoice date without a payment. If the invoice is not paid before the grace days expire, the account incurs extra fees and penalties.
  Default: 10
- **Invoice Method**
  Specifies how to send invoices. Select one of the following options:
  - **Email**
    Sends the invoice as an email attachment to the email address of the account.
    The Catalog system:
    - Uses the format that is specified in Invoice Output Type setting.
    - Uses the delivery mechanism that is specified in the Invoice Generation Email Attachment Type setting.
    - Saves the attachment in the Invoice Email Location server folder.
  - **Fax**
    Saves the invoice using the format that is specified in Invoice Output Type setting in the Invoice Fax Location server folder.
  - **Postal**

Saves the invoice using the format that is specified in Invoice Output Type setting in the Invoice Postal Location server folder.

– **Printer**
Saves the invoice using the format that is specified in Invoice Output Type setting in the Invoice Printer Location server folder.

Default: Email

- **Period End Date**
Specifies how to set the end date for the invoice period for the account. Select one of the following options:
  – **Compute from Start Date Using Billing Cycle**
  Specifies an end date that is based on the period start date, the billing cycle, and the billing cycle interval. Each of these settings is described earlier in this topic.
  – ***other options***
  Specifies an end date equivalent to the options for period start date. That setting is described earlier in this topic.

  The dates settings presume that each new day begins at a time of 00:00:00. Specify a Period End Date that includes the last full day. For example, to bill for the entire month of July, make your period 7/1/yyyy - 8/1/yyyy to include the entire last day of the month.
  Default: Compute from Start Date Using Billing Cycle
- **Status**
Specifies the status of the account when it is created: Closed, Opened, or Suspended.
Invoicing occurs only when this setting is Opened.
Default: Opened
- **Status Reason**
Specifies the reason for the value of the Status option. You enter this text manually.
Limit: 1024 characters
Default: New account waiting approval
- **Taxable**
Specifies whether the account is taxable.
Specify Yes or No.
Default: No

Billing Cycles

As part of setting accounting configuration options, you specify the settings for the billing cycles. You specify which billing cycle choices are available on the Accounting Profile of the account.

- **Daily**
Specifies whether users can generate an invoice daily. If this option is set to Yes, Daily appears in the list of options for specifying the billing cycle.
Default: Yes
- **Monthly**
Specifies whether users can generate an invoice monthly. If this option is set to Yes, Monthly appears in the list of options for specifying the billing cycle.
Default: Yes
- **Weekly**
Specifies whether users can generate an invoice weekly. If this option is set to Yes, Weekly appears in the list of options for specifying the billing cycle.
Default: Yes

General

As part of setting accounting configuration options, you specify the General settings. This value applies to all accounts in the current business unit.

- **Default Post Payment Method**
  Specifies the default option for posting a payment: cash, check, or credit card.
  Default: Check

Invoice Engine Configuration

As part of setting accounting configuration options, you specify the Invoice Engine Configuration settings. These settings control the behavior of the invoice engine during bill runs. These values apply to transactions and invoices for all accounts in the current business unit. The following settings require explanation:

- **Invoice Style for Aggregation**
  Specifies how totals appear on invoices for aggregating accounts.
  Select from the following options:
  - **All Details**
    Displays all details.
  - **Account Totals**
    Displays only the total of the aggregating account.
  - **Drill Down Account Totals**
    Displays subtotals by account by service. Users can drill down to view details for each account.
  - **Service by Account Totals**
    Displays subtotals by account by service, without drill-down functions.
  - **Service Totals**
    Displays subtotals by service for all accounts.
  - **Type by Account Totals**
    Displays subtotals by account by service option element type.
  - **Type Totals**
    Displays subtotals by service option element type.

  Default: All Details
- **Aggregate Advanced Charges**
  Specifies how advanced charges appear on an invoice. Advanced charges span multiple periods as defined by the period start and end dates in the Accounting Profile of the account.
  - **Yes**
    Displays the total of the advanced charges on one line.
  - **No**
    Displays the advanced charges individually on multiple lines.

  For example, suppose that the cycle for a subscription is monthly and the billing cycle for an account is yearly. In this case, a Yes setting summarizes the charges on one line. Conversely, a No setting displays 12 lines instead.
  Default: Yes
- **Aggregate Current Charges**
  Specifies how to present current charges on an invoice. Current charges apply only to the current period as defined by the period start and end dates in the Accounting Profile.
  - **Yes**
    Displays the total of the current charges on one line.
  - **No**
    Displays the current charges individually on multiple lines.

  This setting is similar to Aggregate Advanced Charges, but it applies to current charges instead.

Default: Yes

- **Allow No Activity Invoices**
  Specifies whether to generate invoices for accounts with no charges.
  - **Yes**
    Generates invoices for *all* accounts, including accounts with no charges.
  - **No**
    Generates invoices *only* for accounts with charges.

  Default: No
- **Bill Run Capacity**
  Specifies the number of accounts to process for transaction and invoice generation until a commit is performed and all processing is persisted.
  The larger the value, the more server memory is required and the faster the invoices are processed. The smaller the value, the less server memory is required, and the slower the invoices are processed.
  Default: 100
- **Exclude Invoice Item**
  Specifies whether to include a chargeable item on an invoice if the quantity, unit cost, or both is 0.
  Select from the following options:
  - **Always Include**
    Invoices all items, regardless of the value for quantity and unit cost.
  - **Where Quantity is 0**
    Exclude items when the quantity is 0.
  - **Where Rate is 0**
    Exclude items when the unit cost is 0.
  - **Where Quantity and Rate are 0**
    Excludes items when *both* the quantity and unit cost are 0.
  - **Where Quantity or Rate is 00**
    Excludes items where *either* the quantity or unit cost is 0.

  Default: Always Include
- **Uninvoiced Transactions**
  Specifies whether to include a transaction on an invoice. The decision is based on a comparison of the transaction date with the period start date and end date on the Accounting Profile of the account.
  Select from the following options:
  - **Include All Uninvoiced Transactions**
    Includes all transactions that have not yet been invoiced, regardless of transaction date.
  - **Exclude Past Uninvoiced Transactions**
    Does not include transactions whose date is before the period start date, even if the transactions are not yet invoiced.
  - **Exclude Future Uninvoiced Transactions**
    Does not include transactions whose date is after the period end date, even if the transactions are not yet invoiced.
  - **Exclude Both Past and Future Uninvoiced Transactions**
    Does not include transactions whose date matches one of the following options, even if the transactions are not yet invoiced:
    - Before the period start date
    - After the period end date

    Both of these dates are outside the current period.

  Default: Include All Uninvoiced Transactions
- **Invoice From Address**
  Specifies the address for the "From" field on the invoice.
  Select from the following options: Service Provider Address and Business Unit Address.
  Business Unit Address is available for a child business unit only.

Default: Service Provider Address

- **Invoice Generation Email Attachment Type**
  Specifies the delivery mechanism for the invoice when emails about invoices are sent.
  Select from the following options:
  - **Attachment**
    Includes the invoice as an attachment in the email. The format is specified in the Invoice Output Type setting.
  - **Inline**
    Includes the invoice in HTML format in the text of the email.
  - **Link**
    Includes a link to the invoice in the text of the email.

  Default: Attachment
- **Invoice Generation Email Body Message**
  Specifies the body text for emails about invoices.
  You can specify both literal text and dynamically substituted values.
  The syntax for dynamically substituted values follows:
  - **$*object_name.variable_name*$**
    Specifies the name of the variable.
    You can specify the following values:
    - “statements” object:
      statement_label - The label of the account plus an invoice statement ID, such as “acnt1:10012”
      period_from - The Accounting Profile Period Start Date of the account
      period_to - The Accounting Profile Period End Date of the account
      due_date - The due date of the invoice
    - “billing_account” object:
      account_label - The label of the account, such as “acnt1”

  Limit: 1024 characters
  Default: Invoice Generated for $statements.statement_label$ for $statements.period_from$ - $statements.period_to$
- **Invoice Generation Email CC**
  Specifies the addresses of carbon copy (CC) recipients of emails about invoices.
  You can specify multiple email addresses. Separate each email address with a semi-colon.
  Limit: 1024 characters
  Default: EMPTY
- **Invoice Generation Email From**
  Specifies the email address that sends emails about invoices.
  Select from the following options:
  - **User Defined**
    Specifies a custom email address that you enter manually. If you select this option but enter no email address, this setting changes to the literal value of EMPTY.
  - **Service Provider Email**
    Specifies the email address of the service provider business unit.
  - **System**
    Specifies that the Catalog system determines which one of the following email addresses to use: the Services Group or the super business unit.

  Default: Service Provider Email
- **Invoice Generation Email Subject**
  Specifies the text for the subject line of emails about invoices.
  You can specify both literal text and dynamically substituted values.
  The syntax for dynamically substituted values is the same as the Invoice Generation Email Body Message setting. That setting is described earlier in this topic.
  Limits: 1024 characters

Default: Invoice: $statements.statement_label$ processed for account: $billing_account.account_label$, Due date: $statements.due_date$

- **Invoice Generation Email To**
  Specifies the primary recipient of emails about invoices.
  Select from the following options:
  - **System**
    Uses the email address of the Send Invoice To field in the Accounting Profile of the account.
  - **User Defined**
    Specifies a custom email address that you enter manually. If you select this option but enter no email address, this setting changes to the literal value of EMPTY.
    You can specify multiple email addresses. Separate each email address with a semi-colon.

  Default: System
- **Invoice History Email Attachment Type**
  Specifies the delivery mechanism for the invoice when you send emails from the Invoice History page.
  Select from the following options:
  - **Attachment**
    Includes the invoice as an attachment in the email. The format is specified in the Invoice Output Type setting.
  - **Inline**
    Includes the invoice in HTML format in the text of the email.
  - **Link**
    Includes a link to the invoice in the text of the email.

  Default: Attachment
- **Invoice History Email Body Message**
  Specifies the body text for emails about invoices, when you send these emails from the Invoice History page.
  You can specify both literal text and dynamically substituted values.
  The syntax for dynamically substituted values follows:
  - **$*object_name.variable_name*$**
    Specifies the name of the variable.

  You can specify the same variables as for the Invoice Generation Email Subject setting.
  In addition, you can specify the following variables:
  - “invoice_history” object:
    account_label - The label of the account

  Limit: 1024 characters
  Default: Invoices Generated During $invoice_history.start_date$ - $invoice_history.end_date$ total number of invoices: $invoice_history.num_invoices$ number of phases processed = $invoice_history.phases$
- **Invoice History Email CC**
  Specifies the addresses of carbon copy (CC) recipients of emails from the Invoice History page.
  You can specify multiple email addresses. Separate each email address with a semi-colon.
  Limit: 1024 characters
  Default: EMPTY
- **Invoice History Email From**
  Specifies the email address that sends emails about invoice history.
  Select from the following options:
  - **User Defined**
    Specifies a custom email address that you enter manually. If you select this option but you enter no email address, this setting changes to the literal value of EMPTY.
  - **Service Provider Email**
    Specifies the email address of the service provider business unit.
  - **System**

Specifies that the Catalog system determines which one of the following email addresses to use: the Services Group or the super business unit.
Default: System

- **Invoice History Email Subject**
  Specifies the text for the subject of emails from the Invoice History page.
  You can specify both literal text and dynamically substituted values.
  The syntax for dynamically substituted values follows:
  – **$*object_name.variable_name*$**
    Specifies the name of the variable.
    You can specify the same variables as for the Invoice Generation Email Subject setting.
    In addition, you can specify the following variables:
    - "invoice_history" object:
      account_label - The label of the account

  Limit: 1024 characters
  Default: Invoices Generated During $invoice_history.start_date$ - $invoice_history.end_date$
- **Invoice History Email To**
  Specifies the primary recipient email addresses when emails are sent from the Invoice History page. You can specify multiple email addresses, which are separated by a semi-colon.
  Limit: 1024 characters
  Default: EMPTY
- **Invoice Email Location**
  Specifies a directory path on the local file system. This path stores invoices for the account when its Accounting Profile Invoice Method is set to Email.
  Each invoice file name includes the account label and invoice statement ID.
  For example, you could specify the following setting for a Windows environment: USM_HOME\accounting\outbox\email.
  Limit: 1024 characters
  Default: EMPTY
- **Invoice Fax Location**
  Specifies a directory path on the local file system. This path stores invoices for the account when its Accounting Profile Invoice Method is set to Fax.
  Limit: 1024 characters
  Default: EMPTY
- **Invoice Postal Location**
  Specifies a directory path on the local file system. This path stores invoices for the account when its Accounting Profile Invoice Method is set to Postal.
  Limit: 1024 characters
  Default: EMPTY
- **Invoice Printer Location**
  Specifies a directory path on the local file system. This path stores invoices for the account when its Accounting Profile Invoice Method is set to Printer.
  Limit: 1024 characters
  Default: EMPTY
- **Invoice Output Type**
  Specifies the format to use when saving invoices as files.
  Select from the following options:
  – HTML
  – CSV
  – XML
  – HTML and CSV
  – HTML and XML

The file format that you select is used for the previous two settings, Invoice Postal Location and Invoice Printer Location.
Default: HTML

- **Invoice Style**
  Specifies the style to use for viewing an invoice through the user interface.
  Select from the following options:
  – Style 1 - One option for invoice format.
  – Style 2 - Another option for invoice format.
  Default: Style 1
- **Invoice URL**
  Specifies an absolute URL path to the location that stores invoices for use by the Accounting, Invoices, Batch Printing menu option.
  Limit: 1024 characters
  Default: EMPTY
- **Output Invoices**
  Specifies whether to store the invoices for the account on the server file system, as follows:
  With either setting, you can view invoices through the user interface.
  – **Yes**
    Stores the invoices on the server file system. Depending on the Invoice Method setting, the invoices are stored in the locations that are indicated in the various “location” configuration settings.
  – **No**
    Does not store invoices on the server file system.
  Default: No
- **Payment Response**
  Specifies the text to print on an invoice when payment is posted.
  Limit: 1024 characters
  Default: Payment received - Thank you
- **Pro Rate Batch**
  Specifies whether to prorate invoice charges if the service start date does not align with the Accounting Profile period of the account.
  – **No**
    Charges the invoice *only* after a full charge period has passed.
  – **Yes**
    Prorates the charges.
  For example, suppose that both of the following options are true:
  – This setting is Yes
  – The charge and the Accounting Profile period of the account are both monthly
  In this example, if you generate an invoice half way through the billing period, only half the charge appears on the invoice.
  Default: Yes
- **Pro Rate Online**
  Specifies the same options as the previous setting (Pro Rate Batch), except that this setting applies to invoices that you view through the user interface.
  Default: Yes
- **Use Time (requires restart)**
  Specifies whether to use the time portion of a transaction date and time stamp when determining the period for which to bill a transaction.
  Default: Yes

Invoice Methods

As part of setting accounting configuration options, you specify the Invoice Methods settings. These settings specify which methods of sending invoices are available on the Accounting Profile of the account.

- **Email**
  Specifies whether users can send invoices through email. If this option is set to Yes, Email appears in the list of options for specifying the delivery mechanism.
  Default: Yes
- **Fax**
  Specifies whether users can send invoices through fax. If this option is set to Yes, Fax appears in the list of options for specifying the delivery mechanism.
  Default: Yes
- **Postal**
  Specifies whether users can send invoices through the postal service. If this option is set to Yes, Postal appears in the list of options for specifying the delivery mechanism.
  Default: Yes
- **Printer**
  Specifies whether users can send an invoice by printing. If this option is set to Yes, Printer appears in the list of options for specifying the delivery mechanism\.
  Default: Yes

Payment Methods

As part of setting accounting configuration options, you specify the Payment Methods settings. These settings specify which payment methods are available for applying a payment to an invoice.

- **Cash**
  Specifies whether Cash is a payment option for an invoice.
  Default: Yes
- **Check**
  Specifies whether Check is a payment option for an invoice.
  Default: Yes
- **Coupon**
  Specifies whether Coupon is a payment option for an invoice.
  Default: No
- **Credit Card**
  Specifies whether Credit Card is a payment option for an invoice.
  Default: Yes
- **Direct**
  Specifies whether Direct is a payment option for an invoice.
  Default: No
- **Tip**
  Specifies whether Tip is a payment option for an invoice.
  Default: No

Subscription Configuration

As part of setting accounting configuration options, you specify the Subscription Configuration settings. These settings specify how the Catalog system handles subscriptions.

- **Default Subscribe State**

Specifies the status to which the selected service options are set when an account is subscribed to a service.
- Completed
- Pending

Default: Completed

- **Default Cancellation State**
  Specifies the status to which the selected service options are set when the subscription of an account to a service is cancelled.
  Select from the following options:
  - Cancel - Charges for the current billing period do not appear on the next invoice.
  - Pending Cancellation - Charges for the current billing period appear on the next invoice, at which point the status is changed to Cancelled.

  Default: Pending Cancellation
- **Default Subscription Page**
  Specifies the initial page that is displayed when the Subscriptions tab is selected.
  Select from the following options:
  - Create New Subscriptions - Show catalog of services to which the account can subscribe.
  - Existing Subscriptions - Show services to which the account is subscribed.

  Default: Create New Subscriptions
- **Allow Instance Names**
  Specifies whether a name can be associated with a subscription instance.
  If this option is set to Yes, when an account is subscribed to a service, you can add some text to the subscription to name the instance.
  Default: No
- **Enable Subscription Notes**
  Specifies whether a note can be associated with a subscription instance.
  If this option is set to Yes, when an account is subscribed to a service, you can add notes to the subscription.
  This setting is available for the service provider business unit only, and it applies to all business units.
  Default: No

Set Administration Configuration Options

To meet the requirements of your organization, you can customize several administration configuration options. The options include those for integrations, portals, request SLAs, authentication, and others.

Follow these steps:

1. Log in as a user with the Service Delivery Administrator role. Updates to these options apply to the *entire* system. You *cannot* specify different administration configuration options for different business units.
2. Click Administration, Configuration.
3. Click the link for the category of options that you want to update:
   a. Integration-Specific Options
   b. Automic Automation
   c. Event Manager
   d. File Store Information
   e. Mail Server
   f. Portal
   g. Request SLA Processor
   h. Rule Engine
   i. Server Information
   j. Single Sign On Authentication

k. System Information
l. User Default
4. View the options for the category, and click the Modify icon for the option that you want to update.
5. Update the setting as required and click Update Configuration.
6. Repeat the previous steps for each administration configuration option that you want to update.

You have to set the administration configuration options. These settings apply to the children of the business unit that you updated, according to theinheritance of configuration settings through the business unit hierarchy.

To meet the requirements of your organization, you can customize several administration configuration options. The options include those for integrations, portals, request SLAs, authentication, and others.

Follow these steps:

1. Log in as a user with the Service Delivery Administrator role. Updates to these options apply to the *entire* system. You *cannot* specify different administration configuration options for different business units.
2. Click Administration, Configuration.
3. Click the link for the category of options that you want to update.
4. View the options for the category, and click the Modify icon for the option that you want to update.
5. Update the setting as required and click Update Configuration.
6. Repeat the previous steps for each administration configuration option that you want to update.

You have to set the administration configuration options. These settings apply to the children of the business unit that you updated, according to theinheritance of configuration settings through the business unit hierarchy.

{"URL":["/usm/wpf?Node=iclaunchpad.tools_configuration"],"customLabelGetStarted":"Configuration","description":"IPCE_Configuration"}

Integration-Specific Options

If you are integrating CA Service Catalog with any of the following products, set the related configuration parameters. This task is part of setting the administration configuration options.

- CA APM
- Automic Automation
- CA Business Intelligence

  NOTE
  CA Business Intelligence packages and delivers BusinessObjects Enterprise. So, you use these CA Business Intelligence parameters to configure the integration of CA Service Catalog with BusinessObjects Enterprise.
- CMDB
- CMDB Visualizer
- CA Process Automation
- CA Business Service Insight
- CA Service Desk Manager
- Reservation Manager

Automic Automation

Introduction

As part of setting the administration configuration options, you configure the following parameters for the Automic Auotmation.

NOTE
CA Service Catalog and Automic integration must be done manually. CA Service Management Common Installer does not provide an option to auto-integrate both products.

- **Host Name**:
  Refers to Automic Auotmation host name.
- **Client ID**:
  Specifies the client id value which is required for logging into Automic.
- **Enable HTTPS**:
  Enable https while connecting to Automic Web Interface (AWI).
- **Port Number (1-65535)**:
  Provide the Port number to connect to the Automic Web Interface. The default value is 8080.
- **UC4 Port Number (1-65535)**:
  Specify the UC4 port number to connect to Automic Engine. The default value is **2217**. For more information, see **ucsrc.ini** file located at **%AutomicInstallation%\Automation.Platform\AutomationEngine\bin** to find UC4 port number.
- **User ID**:
  Specify the automic user ID.
- **User Password**:
  Specify the automic password.
- **Enable Automatic Retry**:
  Applicable when a request that is created in CA Service Catalog is not processed in Automic due to connectivity issues between CA Service Catalog and Automic. The request is placed in a queued state. Retry manually to process the request to Automic. Select **Yes** for automatic retry or **No** for manual retry.. The **ASO_QUEUE_FAILURE** automatically gets deleted from the **Home**, **Messages**, **Alerts**, **List All Failed Actions** list.
  If message disappears from the List All Failed Actions, verify that the request has changed to the appropriate status. For example, if the original status was **Submitted (Queued)**, the updated status must be Pending Approval.
- **Retry Count**:
  Specifies the number of retry attempts. This value indicates the number of times CA Service Catalog retry attempts to perform a failed CA Process Automation action. The default retry count value is **3**.If retry attempts fail, CA Service Catalog logs a queue alert as **ASO_QUEUE_FAILURE** in **Home**,**Messages**, **Alerts**. Manually retry the failed action by viewing the queue alerts in **List All Failed Actions**. Click the **Retry** button.
- **Retry Interval**:
  Specifies the number of seconds between the retry attempts described in **Retry Count**. The default retry interval is **15 seconds**.
  After providing the above details in the configuration page, click **Test** to check connectivity the from Catalog to Automic and click **Launch** to launch the Automic web interface (AWI).

Event Manager

As part of setting the administration configuration options, you configure the following parameters for the Event Manager. The Event Manager processes events. An event occurs when one or more conditions that are specified in a rule are met.

For more information about events, rules, and actions, see the Manage Events-Rules-Actions section.

- **Email From**
  Sends an email *from* this address if the Event Manager has a problem.
  Default: spadmin@*serviceprovider*, where "*serviceprovider*" is the root business unit that is specified during installation.
- **Email To**
  Sends an email *to* this address if the Event Manager has a problem.
  Default: spadmin@*serviceprovider*, where *serviceprovider* is the root business unit that is specified during installation.
- **Audit Trail Level**

Indicates the level of detail that is logged in the audit trail tables. Typically, system performance decreases as the level of log detail increases. Conversely, system performance typically increases as the level of log detail decreases. Select one of the following options:

- **No Audit Trail**
  Stores no information about the event in the database.
- **Only Object ID**
  Stores the event information, including the object ID, in the usm_system_change table. This option stores only minimal information.
- **Include Attributes**
  Stores the event information, including the object ID, in the usm_system_change table. This option stores detailed old and new values for the object attributes in the usm_system_change_detail table.
- **Include Multiple Attributes**
  Stores the event information, including the object ID, in the usm_system_change table. This option stores detailed old and new values for the object attributes in the usm_system_change_detail table. If any attributes have multiple values, the old and new values are stored in the usm_system_change_detail_ext table.
  Default: Include Attributes.

As part of setting the administration configuration options, you configure various parameters for the Event Manager. The Event Manager processes events. An event occurs when one or more conditions that are specified in a rule are met.

{"URL":["/usm/wpf?Node=iclaunchpad.tools_configuration"],"customLabelGetStarted":"Event Manager","description":"IPCE_Event_Manager"}

File Store Information

As part of setting the administration configuration options, you configure the following parameter for the File Store Information. You can set up a custom location for shared files. Having a common location improves the accuracy and efficiency of sharing files between computers.

- **File Store Location**
  Specifies the UNC path name of the shared drive.
  Default: USM_HOME\filestore

For more information about how to setup single location for shared files, see Perform CA Service Catalog Post-Installation Tasks.

As part of setting the administration configuration options, you configure the following parameter for the File Store Information. You can set up a custom location for shared files. Having a common location improves the accuracy and efficiency of sharing files between computers.

- **File Store Location**
  Specifies the UNC path name of the shared drive.
  Default: USM_HOME\filestore

{"URL":["/usm/wpf?Node=iclaunchpad.tools_configuration"],"customLabelGetStarted":"File Store Information","description":"IPCE_File_Store_Information"}

Mail Server

> **WARNING**
> Any mail-related settings in custom rule actions override these mail server parameters. For more information about events, rules, and actions, see the section Manage Events-Rules-Actions.

As part of setting the administration configuration options, you configure the following parameters for the mail server. The mail server sends automated messages from CA Service Catalog. These settings apply to the children of the business unit that you updated, according to the inheritance of configuration settings through the business unit hierarchy.

- **From Address**
  Specifies the address that emails are sent from.
- **Host Name**
  Specifies the host name of the mail server.
- **Port Number**
  Specifies the port number on the mail server that listens for incoming calls from CA Service Catalog.
- **User ID**
  Specifies the user ID for accessing the mail server.
- **User Password**
  Specifies the password for accessing the mail server.

  NOTE
  Specify these parameters and click the Test button to verify the connection between CA Service Catalog and the mail server. If the test fails, review the view.log file in the USM_HOME\logs folder.

As part of setting the administration configuration options, you configure the following parameters for the mail server. The mail server sends automated messages from CA Service Catalog. These settings apply to the children of the business unit that you updated, according to the inheritance of configuration settings through the business unit hierarchy.

- **From Address**
  Specifies the address that emails are sent from.
- **Host Name**
  Specifies the host name of the mail server.
- **Port Number**
  Specifies the port number on the mail server that listens for incoming calls from CA Service Catalog.
- **User ID**
  Specifies the user ID for accessing the mail server.
- **User Password**
  Specifies the password for accessing the mail server.

  NOTE
  Specify these parameters and click the Test button to verify the connection between CA Service Catalog and the mail server. If the test fails, review the view.log file in the USM_HOME\logs folder.

{"URL":["/usm/wpf?Node=iclaunchpad.tools_configuration"],"customLabelGetStarted":"Mail Server","description":"IPCE_Mail_Server"}

Portal

As part of setting the administration configuration options, you configure the following parameters for the Portal. The Portal settings specify how the dashboard library and document entries are made available.

- **Allow New Document Domain Namespaces**
  Specifies whether business units can have their own document name spaces. If this parameter is set to Yes, the Add A New Business Unit page contains the "Create Document Namespace" check box.
  If a business unit has its own document namespace, you can segregate documents in separate business units. This segregation allows only users in that business unit to have access to the document.
  Default: Yes
- **Allow New Library Namespaces**
  Specifies whether business units can have their own library name spaces.

If this parameter is set to Yes, the Add A New Business Unit page contains the "Create Dashboard Library Namespace" check box.
If a business unit has its own library namespace, you can segregate dashboard library content. This segregation allows only users in that business unit have access to the library content when they add a dashboard.
Default: Yes

- **Show Resource Tree**
  Specifies whether the Resource Explorer appears in the dashboard library tree.
  If this parameter is set to Yes, the Resource Explorer appears in the dashboard library tree. This option applies *only* if CA Service Catalog is integrated with CA Business Service Insight.
  Default: Yes

As part of setting the administration configuration options, you configure the following parameters for the Portal. The Portal settings specify how the dashboard library and document entries are made available.

- **Allow New Document Domain Namespaces**
  Specifies whether business units can have their own document name spaces. If this parameter is set to Yes, the Add A New Business Unit page contains the "Create Document Namespace" check box.
  If a business unit has its own document namespace, you can segregate documents in separate business units. This segregation allows only users in that business unit to have access to the document.
  Default: Yes
- **Allow New Library Namespaces**
  Specifies whether business units can have their own library name spaces.
  If this parameter is set to Yes, the Add A New Business Unit page contains the "Create Dashboard Library Namespace" check box.
  If a business unit has its own library namespace, you can segregate dashboard library content. This segregation allows only users in that business unit have access to the library content when they add a dashboard.
  Default: Yes
- **Show Resource Tree**
  Specifies whether the Resource Explorer appears in the dashboard library tree.
  If this parameter is set to Yes, the Resource Explorer appears in the dashboard library tree. This option applies *only* if CA Service Catalog is integrated with CA Business Service Insight.
  Default: Yes

{"URL":["/usm/wpf?Node=iclaunchpad.tools_configuration"],"customLabelGetStarted":"Portal","description":"IPCE_Portal"}

Request SLA Processor

As part of setting the administration configuration options, you configure the following parameter for the request SLA processor. These settings specify how the Catalog system processes request SLAs.

For information about setting up request SLA processing, see the Manage Request SLAs section .

- **Maximum Delay for Request SLA Alerts**
  Specifies how frequently the request SLA processor checks for SLA warnings or violations. This setting applies to all SLA instances managed by a Catalog Component computer.
  To minimize possible delays in SLA processing time when a Catalog Component clustered computer fails, configure this parameter. Therefore, set a smaller interval, such as one hour, to receive SLA warnings and violations quickly. Otherwise, set a larger interval, for example, one day, to receive them when the failed clustered computer is restored.

As part of setting the administration configuration options, you configure the following parameter for the request SLA processor. These settings specify how the Catalog system processes request SLAs.

For information about setting up request SLA processing, see the Manage Request SLAs section .

- **Maximum Delay for Request SLA Alerts**

Specifies how frequently the request SLA processor checks for SLA warnings or violations. This setting applies to all SLA instances managed by a Catalog Component computer.
To minimize possible delays in SLA processing time when a Catalog Component clustered computer fails, configure this parameter. Therefore, set a smaller interval, such as one hour, to receive SLA warnings and violations quickly. Otherwise, set a larger interval, for example, one day, to receive them when the failed clustered computer is restored.

{"URL":["/usm/wpf?Node=iclaunchpad.tools_configuration"],"customLabelGetStarted":"Request SLA","description":"IPCE_Request_SLA"}

Rule Engine

As part of setting the administration configuration options, you configure the following parameter for the Rule Engine. Administrators use rules to define actions to take when a specific event occurs. These actions can be running scripts or Java programs, sending emails, and so forth. The rule engine manages the execution of the actions.

For information about events, rules, and actions, see the Manage Events-Rules-Actions section.

- **Action Default Timeout (in seconds)**
  Specifies the default timeout value in seconds for rule actions.
  Default: 300

As part of setting the administration configuration options, you configure the following parameter for the Rule Engine. Administrators use rules to define actions to take when a specific event occurs. These actions can be running scripts or Java programs, sending emails, and so forth. The rule engine manages the execution of the actions.

For information about events, rules, and actions, see the Manage Events-Rules-Actions section.

- **Action Default Timeout (in seconds)**
  Specifies the default timeout value in seconds for rule actions.
  Default: 300

{"URL":["/usm/wpf?Node=iclaunchpad.tools_configuration"],"customLabelGetStarted":"Rule Engine","description":"IPCE_Rule_Engine"}

Server Information

As part of setting the administration configuration options, you configure the following parameters for the Catalog Component server. These settings apply *only* when you configure Catalog Component for use with HTTPS or for use as a load balancer in clustering. Otherwise, Catalog Component is configured automatically and the following parameters do *not* apply.

- **Enable HTTPS**
  Defines whether CA Service Catalog uses HTTPS when communicating with this Catalog Component computer.
  Default: No
- **Host Name**
  Specifies the host name accessible to catalog users, typically one of the following host names:
  - The Catalog Component computer
  - The load balancer computer (applies if you are clustering).
  - The computer that rediCrects catalog users (applies if you are using DNS aliases for redirection).
- **Port Number**
  Specifies the port number for communicating with the host named in the previous parameter.

  WARNING
  If you use clustering, replace the host name and port number in the Server Information section with the host name and port number of the load balancer. If you use DNS aliases for redirection, replace the host name and

port number in the Server Information section with the host name and port number of the computer that redirects catalog users.

As part of setting the administration configuration options, you configure the following parameters for the Catalog Component server. These settings apply *only* when you configure Catalog Component for use with HTTPS or for use as a load balancer in clustering. Otherwise, Catalog Component is configured automatically and the following parameters do *not* apply.

- **Enable HTTPS**
  Defines whether CA Service Catalog uses HTTPS when communicating with this Catalog Component computer.
  Default: No
- **Host Name**
  Specifies the host name accessible to catalog users, typically one of the following host names:
  – The Catalog Component computer
  – The load balancer computer (applies if you are clustering).
  – The computer that rediCrects catalog users (applies if you are using DNS aliases for redirection).
- **Port Number**
  Specifies the port number for communicating with the host named in the previous parameter.

{"URL":["/usm/wpf?Node=iclaunchpad.tools_configuration"],"customLabelGetStarted":"Server Information","description":"IPCE_Server_Information"}

Single Sign On Authentication

You configure the following single sign-on authentication parameters.

- **Allow Login with GET**
  Specifies whether users can log in using an HTTP GET request.
- **External Authentication Parameters**
  The external authentication parameters follow. To enable external authentication of users, configure these parameter values to match the external application that you are using.
  – **Artifact Name**
    Specifies the name of the cookie, header, or request parameter that contains the authenticated user ID. This name varies according to the external authentication system and your site-specific implementation.
    The default value is sm-user.
    For example, if you are using CA SiteMinder, select Header as the authentication type and sm-user as the artifact name. With this configuration, CA Service Catalog checks the header named sm-user whose value is the userid. When the Artifact Type is request, the artifact name is ignored.
  – **Artifact Type**
    Specifies the mechanism that the external application uses to send the authenticated user ID to CA Service Catalog, as follows:
    - Cookie: Cookie in the request
    - Header: Header in the request
    - Parameter: request parameter
    - Request: request user
  – **Bypass Nodes**
    Specifies the GUI nodes that you want the authentication check to skip. These GUI nodes typically do not require the user to log in.
    Examples include the following: icguinode.login, icguinode.logout, iclaunchpad.launch, icguinode.changepwdlockout, and icguinode.lockout.
  – **Login Page**

Specifies the page that displays to the user when either the artifact type is set incorrectly or CA Service Catalog does not find an authenticated user.
An example is wpf?Node=icguinode.login.

– **Single Sign-on Type**
Specifies *one* of the following options:
  - Disabled: Specifies that CA Service Catalog does *not* use single sign-on.
  - Artifact Based Single Sign-on: Configures CA Service Catalog to use single sign-on based on the artifacts specified on this page.
  - NTLM (NT LAN Manager): Configures CA Service Catalog to use single sign-on based on Windows NT authentication.

You configure the following single sign-on authentication parameters.

- **Allow Login with GET**
Specifies whether users can log in using an HTTP GET request.
- **External Authentication Parameters**
The external authentication parameters follow. To enable external authentication of users, configure these parameter values to match the external application that you are using.
  – **Artifact Name**
  Specifies the name of the cookie, header, or request parameter that contains the authenticated user ID. This name varies according to the external authentication system and your site-specific implementation.
  The default value is sm-user.
  For example, if you are using CA SiteMinder, select Header as the authentication type and sm-user as the artifact name. With this configuration, CA Service Catalog checks the header named sm-user whose value is the userid.
  When the Artifact Type is request, the artifact name is ignored.
  – **Artifact Type**
  Specifies the mechanism that the external application uses to send the authenticated user ID to CA Service Catalog, as follows:
    - Cookie: Cookie in the request
    - Header: Header in the request
    - Parameter: request parameter
    - Request: request user
  – **Bypass Nodes**
  Specifies the GUI nodes that you want the authentication check to skip. These GUI nodes typically do not require the user to log in.
  Examples include the following: icguinode.login, icguinode.logout, iclaunchpad.launch, icguinode.changepwdlockout, and icguinode.lockout.
  – **Login Page**
  Specifies the page that displays to the user when either the artifact type is set incorrectly or CA Service Catalog does not find an authenticated user.
  An example is wpf?Node=icguinode.login.
  – **Single Sign-on Type**
  Specifies *one* of the following options:
    - Disabled: Specifies that CA Service Catalog does *not* use single sign-on.
    - Artifact Based Single Sign-on: Configures CA Service Catalog to use single sign-on based on the artifacts specified on this page.
    - NTLM (NT LAN Manager): Configures CA Service Catalog to use single sign-on based on Windows NT authentication.

{"URL":["/usm/wpf?Node=iclaunchpad.tools_configuration"],"customLabelGetStarted":"Single Sign On Authentication","description":"IPCE_Single_Sign_On_Authentication"}

System Information

As part of setting the administration configuration options, you configure the following settings that are related to common multi-tenant administration for CA Service Catalog business units:

> **WARNING**
> These settings apply *only* if you are *both* integrating CA Service Catalog with CA Service Desk Manager *and* using common multi-tenant administration of CA Service Catalog business units. For more information about setting up common multi-tenant administration, see the section Configure Common Tenant Administration.

- **Terms Of Usage Prompt Enabled**
  Specifies that common tenants in CA Service Catalog must implement the terms of use (if any) that are created and maintained in CA Service Desk Manager. The specific effect on users attempting to log in to CA Service Catalog depends on the terms of use settings that are specified in CA Service Desk Manager.
  If this setting is Yes, the users attempting to log in to CA Service Catalog receive the terms of use prompt. If the users do not accept the terms of use, they cannot access CA Service Catalog.
  If this setting is No, the users attempting to log in to CA Service Catalog are not prompted with terms of use, regardless of the terms of use settings that are specified in CA Service Desk Manager.
- **Common Multi-Tenant Administration Enabled**
  Specifies whether you manage CA Service Catalog business units with the common multi-tenant administration framework supplied through CA Service Desk Manager.
  When this option is set to No, you manage CA Service Catalog business units directly through CA Service Catalog, and these business units are not synchronized with CA Service Desk Manager.
  When this option is set to Yes, you cannot manage CA Service Catalog business units directly through CA Service Catalog, except for CA Service Catalog-specific attributes. You use the tenant administration tool of CA Service Desk Manager to manage all business units of CA Service Catalog and CA Service Desk Manager and their common attributes.
  You cannot set this option to Yes unless the Common Multi-Tenancy Model option is set to Yes.
- **Common Tenant Data Synchronized (read-only)**
  Specifies whether the CA Service Catalog tenant (business unit) structure is synchronized with the tenant structure of CA Service Desk Manager.
  CA Service Catalog sets this option to Yes when all of the following conditions are met:
  – CA Service Desk Manager is installed.
  – The common multi-tenancy merge utility has run successfully.
  – Since the utility was last run, no business unit has been created, deleted, or had a change in a common attribute through CA Service Catalog directly.

  Conversely, CA Service Catalog sets this option to No when one or more of the following conditions exist:
  – The common multi-tenancy merge utility has not run successfully.
  – Since the utility was last run, one or more business units have been created, deleted, or had a change in a common attribute through CA Service Catalog directly.

As part of setting the administration configuration options, you configure the following settings that are related to common multi-tenant administration for CA Service Catalog business units:

> **WARNING**
> These settings apply *only* if you are *both* integrating CA Service Catalog with CA Service Desk Manager *and* using common multi-tenant administration of CA Service Catalog business units. For more information about setting up common multi-tenant administration, see the section Configure Common Tenant Administration.

- **Common Multi-Tenant Administration Enabled**
  Specifies whether you manage CA Service Catalog business units with the common multi-tenant administration framework supplied through CA Service Desk Manager.
  When this option is set to No, you manage CA Service Catalog business units directly through CA Service Catalog, and these business units are not synchronized with CA Service Desk Manager.

When this option is set to Yes, you cannot manage CA Service Catalog business units directly through CA Service Catalog, except for CA Service Catalog-specific attributes. You use the tenant administration tool of CA Service Desk Manager to manage all business units of CA Service Catalog and CA Service Desk Manager and their common attributes.
You cannot set this option to Yes unless the Common Multi-Tenancy Model option is set to Yes.

- **Common Tenant Data Synchronized (read-only)**
  Specifies whether the CA Service Catalog tenant (business unit) structure is synchronized with the tenant structure of CA Service Desk Manager.
  CA Service Catalog sets this option to Yes when all of the following conditions are met:
  – CA Service Desk Manager is installed.
  – The common multi-tenancy merge utility has run successfully.
  – Since the utility was last run, no business unit has been created, deleted, or had a change in a common attribute through CA Service Catalog directly.
  Conversely, CA Service Catalog sets this option to No when one or more of the following conditions exist:
  – The common multi-tenancy merge utility has not run successfully.
  – Since the utility was last run, one or more business units have been created, deleted, or had a change in a common attribute through CA Service Catalog directly.
- **Terms Of Usage Prompt Enabled**
  Specifies that common tenants in CA Service Catalog must implement the terms of use (if any) that are created and maintained in CA Service Desk Manager. The specific effect on users attempting to log in to CA Service Catalog depends on the terms of use settings that are specified in CA Service Desk Manager.
  If this setting is Yes, the users attempting to log in to CA Service Catalog receive the terms of use prompt. If the users do not accept the terms of use, they cannot access CA Service Catalog.
  If this setting is No, the users attempting to log in to CA Service Catalog are not prompted with terms of use, regardless of the terms of use settings that are specified in CA Service Desk Manager.

{"URL":["/usm/wpf?Node=iclaunchpad.tools_configuration"],"customLabelGetStarted":"System Information","description":"IPCE_System_Information"}

User Default

As part of setting the administration configuration options, you configure the following settings that are related to users, roles, searches, and sessions:

- **CA EEM Max Search Size**
  Defines the maximum number of users that are queried during searches in CA EEM.
- **Access Control: Allow Request Auto-Delegation via User Management**
  Defines which user roles are able to view and set auto-delegation for other users. Typically, administrator roles use this ability to set auto-delegation for users who are unable to set it for themselves. Examples of such employees:
  – Users who suddenly became unavailable due to emergency.
  – Users who left for a long absence without setting their own auto-delegation.
  – Users whose roles have no rights to set their own auto-delegation.
  The default value is the Service Delivery Administrator user role.
- **Access Control: Allow Request Auto-Delegation via User Profile**
  Defines which user roles have the Request Auto-Delegation setting appear on their User Profile window. Users with these roles can view and optionally configure this setting for themselves.
  The default value is all user roles, so that all users can change at minimum their own auto-delegation settings.
- **User Default Role**
  Defines the default business-unit role that is assigned to a user when an administrator creates or edits the user.
  If both of the following conditions are true, the Catalog system assigns the default role to the user logging in:

- The user has an MDB record that is created by another product.
- That MDB record has never been edited through CA Service Catalog.

A change of this setting requires a restart of the Catalog Component service to take effect.
The default value of this setting depends on the products installed:

- If CA Service Catalog is installed, the default value for this setting is Catalog User.
- If only Accounting Component is installed, the default value for this setting is End User.

- **User Search Scope**
  Defines how the scope of the “search user” functionality works for the Catalog User and End User roles. “Search user” functionality for these roles is available in the following areas:
  - When emailing a request, users can populate the To, CC, and BCC fields by searching from a list of users or accounts.
  - When editing the cart or a request, if the logged in user can create requests for other users or accounts (proxy requests), the user can override the Requested For value by choosing from a list of users or accounts.

    NOTE
    The scope of “search account” functionality for Catalog End Users and End Users includes *only* accounts in the same business unit as the logged in user. This scope does *not* include child business units.

  Select one of the following options:
  - Enterprise - All users who have a User ID specified without regard to the role of the user. This setting is suitable for enterprise customers where business units represent departments in one company.
    The scope of the “search user” functionality works as follows:
    - A user who can change the Requested For user for a cart or a request can select from all users.

    When a request is emailed, the list of users includes all users.
  - Business Unit - All users who have a user ID and a role in the business unit of the logged in user. This setting is suitable for customers whose business units represent separate companies or departments within those companies.
    The scope of the “search user” functionality works as follows:
    - A user who can change the Requested For user for a cart or a request can select from users who have a role in the same business unit as the logged in user.
    - When a request is emailed, the list of users includes only users who have a role in the same business unit as the logged in user.

    Default: Enterprise
- **Session Timeout**
  Defines the number of minutes of inactivity after which users are logged out. For the change to take effect, recycle the Windows service named CA Service Catalog.

As part of setting the administration configuration options, you configure the following settings that are related to users, roles, searches, and sessions:

- **Access Control: Allow Request Auto-Delegation via User Management**
  Defines which user roles are able to view and set auto-delegation for other users. Typically, administrator roles use this ability to set auto-delegation for users who are unable to set it for themselves. Examples of such employees:
  - Users who suddenly became unavailable due to emergency.
  - Users who left for a long absence without setting their own auto-delegation.
  - Users whose roles have no rights to set their own auto-delegation.

  The default value is the Service Delivery Administrator user role.
- **Access Control: Allow Request Auto-Delegation via User Profile**
  Defines which user roles have the Request Auto-Delegation setting appear on their User Profile window. Users with these roles can view and optionally configure this setting for themselves.
  The default value is all user roles, so that all users can change at minimum their own auto-delegation settings.
- **CA EEM Max Search Size**

Defines the maximum number of users that are queried during searches in CA EEM.

- **Session Timeout**
  Defines the number of minutes of inactivity after which users are logged out. For the change to take effect, recycle the Windows service named CA Service Catalog.
- **User Default Role**
  Defines the default business-unit role that is assigned to a user when an administrator creates or edits the user.
  If both of the following conditions are true, the Catalog system assigns the default role to the user logging in:
  – The user has an MDB record that is created by another product.
  – That MDB record has never been edited through CA Service Catalog.
  A change of this setting requires a restart of the Catalog Component service to take effect.
  The default value of this setting depends on the products installed:
  – If CA Service Catalog is installed, the default value for this setting is Catalog User.
  – If only Accounting Component is installed, the default value for this setting is End User.
- **User Search Scope**
  Defines how the scope of the “search user” functionality works for the Catalog User and End User roles. “Search user” functionality for these roles is available in the following areas:
  – • When emailing a request, users can populate the To, CC, and BCC fields by searching from a list of users or accounts.
    • When editing the cart or a request, if the logged in user can create requests for other users or accounts (proxy requests), the user can override the Requested For value by choosing from a list of users or accounts.

    NOTE
    The scope of “search account” functionality for Catalog End Users and End Users includes *only* accounts in the same business unit as the logged in user. This scope does *not* include child business units.

  Select one of the following options:
  – Enterprise - All users who have a User ID specified without regard to the role of the user. This setting is suitable for enterprise customers where business units represent departments in one company.
    The scope of the “search user” functionality works as follows:
    • A user who can change the Requested For user for a cart or a request can select from all users.
    When a request is emailed, the list of users includes all users.
  – Business Unit - All users who have a user ID and a role in the business unit of the logged in user. This setting is suitable for customers whose business units represent separate companies or departments within those companies.
    The scope of the “search user” functionality works as follows:
    • A user who can change the Requested For user for a cart or a request can select from users who have a role in the same business unit as the logged in user.
    • When a request is emailed, the list of users includes only users who have a role in the same business unit as the logged in user.
    Default: Enterprise

{"URL":["/usm/wpf?Node=iclaunchpad.tools_configuration"],"customLabelGetStarted":"User Default","description":"IPCE_User_Default"}

How to Configure the F5 Load Balancer for CA Service Catalog Server

F5 Load balancers help to enhance the reliability, availability, and scalability of CA Service Catalog servers and improves the overall performance. To configure an F5 load balancer in your environment, complete the following steps:

Prerequisites for configuring the F5 Load Balancer
Before you configure an F5 load balancer for CA Service Catalog, complete and keep handy the following information:

- Identify the CA Service Catalog hosts where you need the load balancing capability.
- Credentials to log in to the F5 interface.

Configure the following F5 elements so that these elements can function with the CA Service Catalog Application Servers:

1. Create a Custom Health Monitor
2. Create an F5 Pool for the CA Service Catalog Server.
3. Create an F5 Virtual Server or Node for CA Service Catalog Server
4. Verify the F5 Load Balancer Configuration.

Create a Custom Health Monitor

Create a custom monitor when the values that you have defined for a pre-configured monitor does not meet your requirements, or when a pre-configured monitor is not available for the type of monitor that you are creating. You can also provide a unique name to your monitor, specify the monitor type, and, if a monitor of that type already exists, import settings and values from the existing monitor. You can change the values of any imported settings.
To create a custom health monitor, complete the following steps:

1. Navigate to **Local Traffic,Monitors**, and Click **Create**. Complete the following field information:
   a. **Name**: Enter the name of the health monitor.
      For example: **Catalog_health**
   b. **Type**: Select HTTP from the drop-down list.
      You must base each custom monitor on a monitor type. To specify a monitor type, simply choose the one that corresponds to the service you want to check.
   c. **Send String**: POST /usm/wpf HTTP/1.1\r\nAccept: text/html, */*\r\nContent-Type: application/x-www-form-urlencoded\r\nContent-Length: 151\r\nDNT: 1\r\nConnection: Keep-Alive\r\nCache-Control: no-cache \r\n?Node=iclaunchpad.launch&previousUrl=wpf%3FNode%3Diclaunchpad.pad&username=spadmin &pass=spadmin&domain=&submit=Log+In&device=desktop\r\n/\r\n

      NOTE
      Make sure to enter valid user name and pass in the above **send string** post command.

      NOTE
      When you create a new HTTP monitor type, you must include \r\n at the end of a non-empty Send String. If you do not include \r\n at the end of the Send String, the HTTP monitor fails.

   d. **Receive String**: 200 OK
      The above string is displayed when the CA Service Catalog server is healthy.
2. Click **Finished**.
   You have successfully created a custom health monitor.
   When a user accesses the F5 load balancer URL, CA Service Catalog health check is performed, and the user is routed to a server that is functional.

Create an F5 Pool for the CA Service Catalog Server

Create an F5 pool for the CA Service Catalog server where you want to enable the F5 load balancing capability.
Perform the following steps:

1. Select **Main**, **Local Traffic**, and click **Pools**.
   The Pool List is empty if you are setting up pools for the first time. The Pool List displays the details for each pool.
2. Click **Create**.
3. Complete the Configuration section on the New Pool page:
   a. Select **Basic** from the drop-down list.
   b. Enter a name and description for the new pool.
4. Complete the Configuration section on the New Pool page:
   a. Select **Basic** from the drop-down list.
   b. Enter a name and description for the new pool.

c. Select an already pre-configured health monitor from the available health monitors list. Select HTTP and move it to the active list.
d. If a pre-configured health monitor is not available, create a custom health monitor. For more information, see Create a Custom Health Monitor.

5. Select **Round Robin** from the Load Balancing Method drop-down list.
   Default option is Round Robin.
6. Select **Disabled** from the Priority Group Activation drop-down list:
   a. Add each node or server to the new F5 pool as follows: Select **Node List** if you are adding a node that is already defined.
   b. If the nodes are not defined, select the **New Node** option to create a node.
   c. Select the CA Service Catalog Server IP address (host name) from the **Address** drop-down list that identifies the CA Service Catalog node to add to this F5 pool.
      If the applicant server is no longer valid or no longer in use, select the pool member and click **Disable**.
7. Click **Add**.
   The details that you added for this node appear in the New Members list.
8. Click **Finished**.
   The new pool is added to the F5 Pool List

Create an F5 Virtual Server or Node for CA Service Catalog Server

You can create an F5 Virtual Server for CA Service Catalog application servers to enable the F5 load balancing capability. Complete the following steps to create an F5 Virtual Server:
Perform the following steps:

1. Log in to F5. Select **Main**, **Local Traffic**, **Virtual Servers**, and **Virtual Server List**.
2. Click **Create**.
3. Complete the following General Properties on the New Virtual Server page:
   a. **Name**: Specifies the name of the virtual server, for example, cacatalog.
   b. **Destination Type**: Specifies Host for a single IP address. This IP address should be available in the network and must not be assigned to any other device. For more information, see the F5 Load Balancer documentation.
   c. **Destination Address**: Specifies the IP address of the virtual server.
   d. **Service Port**: Specifies the associated port for the virtual server.
      For example: 80 for HTTP.
   e. **State**: Select Enabled.
      Specifies whether the virtual server is available for load balancing.
4. Complete the Configuration section. Accept all defaults, except for the HTTP Profile:
   a. **HTTP Profile**: Select http.
      Specifies the HTTP profile for managing HTTP traffic.

   Complete the Resources Section:
   a. **iRules**: Specifies the iRules to enable for this virtual server.
   b. **Default Pool**: Specify the CA Service Catalog pool which was created in step 1 as the default pool. The virtual server routes traffic to this pool.
   c. **Default Persistence Profile**: Specifies the persistence profile for this virtual server. Select cookie.
   d. **Fallback Persistence Profile**: Specifies the persistence profile that this virtual server uses when the default persistence profile cannot be used.
      For example: **source_addr**
5. Click **Finished**.
   For more information about the F5 Load Balancer persistence profile, see F5 documentation.
6. Navigate to **Local Traffic**, **Profile**, **Persistence** to modify the properties of a persistence profile. Select the persistence profile that you want to modify.
   For example, click **source_addr** and change the Timeout option to **600** seconds. Click Update.

For CA Service Catalog, it is recommended that you set the Timeout option to a value of 600.

7. Click **Finished**

Verify the F5 Load Balancer Configuration

To verify the F5 load balancer configuration in your environment, complete the following steps:

1. Open any web browser, and enter the F5 Virtual Server destination IP address that you created in Create an F5 Virtual Server or Node for CA Service Catalog Server earlier in this article while creating the F5 virtual server.
   You must be able to view the CA Service Catalog login page.
2. Enter the CA Service Catalog login credentials user with any role (Ex: Service Delivery Administrator role, spadmin/spadmin) and click **Login**.
   Login must be successful.
3. Repeat step 1 and 2 on all other CA Service Catalog servers and verify that you are routed to these servers by checking the CA Service Catalog logs in the CA Service Catalog install directory.

Enable GMSA in CA Service Catalog

gMSA (Group managed service accounts)

Group managed service accounts (gMSAs) are domain accounts to help secure services. gMSAs can run on one server, or in a server farm, such as systems behind a network load balancing or Internet Information Services (IIS) server. After you configure your services to use a gMSA principal, account password management is handled by the Windows operating system (OS). Service and domain administrators do not need to schedule password changes, or manage outages as GMSA will automatically update the password with 240-byte, randomly generated passwords once in every 30 days.

> **NOTE**
> For more information, see https://learn.microsoft.com/en-us/azure/active-directory/fundamentals/service-accounts-group-managed

In 17.4, the administrator can run the service with GMSA account and can switch SQL server authentication from SQL-based authentication to Windows based authentication (using GMSA account), thereby eliminating outages completely for updating passwords regularly. Everything will be taken care of by the underlying OS.

Prerequisites:

The following are the prerequisites:

- Windows Server 2012 or later domain controller
- Windows Server 2012 or later host, which is a member of the domain

To set up a gMSA account, follow these steps:

1. Create a KDS root key to generate unique passwords for each object in your gMSA.
2. For each domain, run the following command from the Windows domain controller:
   ```
   Add-KDSRootKey -EffectiveImmediately
   ```
3. Create and configure your gMSA:
   a. Create a user group account in the following format:
      domainName\accountName$
   b. Add computer objects to the group.
   c. Use the user group you created to create the gMSA.
      Example:

```
New-ADServiceAccount -name <ServiceAccountName> -DNSHostName <fqdn> -
PrincipalsAllowedToRetrieveManagedPassword <group> -ServicePrincipalNames <SPN1,SPN2,…>
```

d. Run the Get-ADServiceAccount command to verify the service account.

4. Configure the gMSA on your hosts:
   a. Enable the Active Directory module for Windows PowerShell on the host where you want to use the gMSA account.To do this, run the following command from PowerShell:

```
PS C:\> Get-WindowsFeature AD-Domain-Services

Display Name                    Name                Install State
------------                    ----                -------------
[ ] Active Directory Domain    AD-Domain-Services    Available
Services
PS C:\> Install-WindowsFeature AD-DOMAIN-SERVICES

Success Restart Needed Exit Code      Feature Result
------- -------------- ---------      --------------
True    No             Success        {Active Directory Domain Services, Active

WARNING: Windows automatic updating is not enabled. To ensure that your newly-installed role or feature
 is automatically updated, turn on Windows Update.
```

   b. Restart your host.
   c. Install the gMSA on your host by running the following command from the PowerShell command prompt:

```
Install-AdServiceAccount <gMSA>
```

   d. Verify your gMSA account by running the following command:

```
Test-AdServiceAccount <gMSA>
```

5. Assign the administrative privileges to the configured gMSA on the host.

New Capability in Common Installer to Support Catalog Installation with gMSA Account

NOTE
If you have already enabled gmsa in 17.3 RU16 or above, the user is expected to re-enter the GMSA account related information (whereas all other configurations will be auto populated during an upgrade).

NOTE
For multi node CA Service Catalog, apply these changes across all nodes.

From 17.4, *Catalog-only* application DVDs support the installation of CA Service Catalog with a GMSA account. During the installation, Common Installer will switch to Windows-based authentication instead of SQL-based authentication to make it a password-less installation and the administrator no longer need an SQL username or password to install the application. However, there are some pre-installation steps that they need to perform before starting the installation.

Pre-Installation Steps:

1. Set up a gMSA account in the CA Service Catalog Server.
2. Create a log in for the gMSA account in the MS SQL server using the following command.

```
CREATE LOGIN [<domainName\accountName$>] FROM WINDOWS;
GO
```

3. The same domain account is used to run utility scripts in CA Service Catalog such as IXutil, Seeddata, and LDAPImporter.

To enable GMSA based installation, we should set ***allow.gmsa.installation*** property value in the *common_config.properties* file under the config folder.

Post-Installation Steps:

- Change the permission of this newly created account and grant '**usmgroup**' permission to get access to Catalog related tables under MDB database
- Change the permission of the current logged-in domain account (using which we are triggering the Common Installer) and grant '**usmgroup**' permission to get access to Catalog related tables under the MDB database.
- CA Service Management Admin installation will be skipped in gMSA-based installation
- Valid GMSA account name should be provided during the installation. Valid Format: domainName\accountName$
- During the uninstallation, DBCleanupUtility_WBA.bat should be used to cleanup DB entries using windows based authentication.

Configure and Unconfigure CA Service Catalog and CA Service Accounting to Use gMSA account and Manually Connect MS SQL Server

From 17.4, the administrator can configure gMSA or unconfigure gMSA account in CA Service Catalog using a batch script.

> **NOTE**
> For multi node CA Service Catalog, apply these changes across all nodes.

Prerequisites and Considerations:

Ensure that you have taken care of the below considerations on all nodes of CA Service Catalog where you want to configure gMSA.

1. If Accounting service is available, set the serviceNameAccounting value to ServiceAccounting and provide appropriate gMSA account name in the **gmsa.properties** file.
2. If filestore path is external to the server from where the **GMSAPostInstall.bat** is running, ensure to provide appropriate access to the server for setting the gMSA account permissions.
3. While preparing the properties file, check for any copy/paste issues, special characters, spaces and so on.

How to configure gMSA for CA Service Catalog using batch script
To configure gMSA, follow these steps:

1. Create a gMSA account for CA Service Catalog and CA Service Accounting.
2. Ensure that appropriate DB (database) permission is given to the newly created account(s).
3. Enter the below values in the **gmsa.properties** file located in **%USM_HOME%\scripts\GMSA** and save it.

   > **NOTE**
   > - All values are mandatory. Make sure that there are no leading or trialing spaces while providing these values.
   > - Here **%USM_HOME%** refers to the location in which CA Service Catalog is installed.

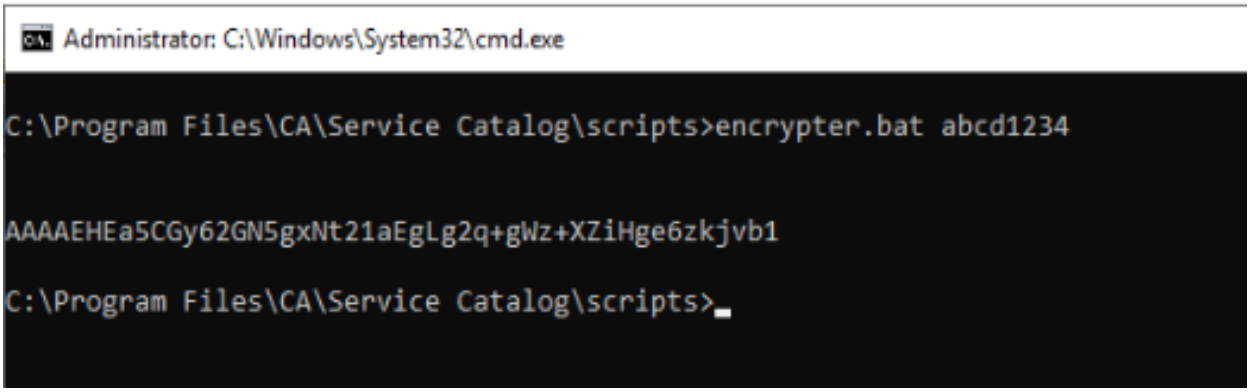
   a. **serviceNameCatalog**: Application service name to enable GMSA. For Catalog, the service name is ServiceCatalog.
   b. **serviceNameAccounting**: Application service name to enable GMSA. For Accounting, the service name is ServiceAccounting.
   c. **folderName**: Absolute path of application installation directory.
   d. **gMSAName**: Name of the GMSA service account (domain\accountName$).
   e. **filestorePath**: Shared network path (\\Server_name\filestore) or local filestore path
4. Navigate to %USM_HOME% and launch **usm.cmd**
5. Navigate to %USM_HOME%/scripts/GMSA/ folder and run the "**GMSAPostInstall.bat**" in command prompt.

   > **NOTE**
   > If the command executes without any error, "Post installation steps for GMSA executed successfully" message is displayed with exit code 0.

If there are errors, error message is shown with a non-zero exit code. Correct the error to continue and perform step **5** again.

How to unconfigure GMSA for Service Catalog Application and CA Service Accounting using batch script
To unconfigure gMSA, follow these steps:

1. Go to %USM_HOME%\scripts\GMSA\
   NOTE
   Here **%USM_HOME%** refers to the location in which CA Service Catalog is installed.
2. Launch **gMSADisableSteps.bat** in Administrator mode.
3. After the bat file is launched, it will prompt for mdb username and mdb password.
   NOTE
   Password must be provided in encrypted format.
   To encrypt the password, follow these steps:
   1. Go to the %USM_HOME%\scripts\ folder.
   2. Launch encrypter.bat file in Administrator mode along with passing the mdb password as a parameter.
      Example:



4. Wait for the script to execute.
5. If MSSQLServer service is running with the gMSA account, switch it back to the ***LocalSystem*** *account* by running the following command:
   ```
   sc config "ServiceNameofMSSQLServer" obj= LocalSystem
   ```
   Example: sc config "MSSQLSERVER" obj= LocalSystem
6. Start the MSSQLServer service.
7. Start both Service Catalog and Accounting Services.

gMSA Support in CA Service Catalog Report Data

After gMSA is configured in CA Service catalog application, we can then add additional datasource under Catalog report data, which can also acquire a DB connection using the existing gMSA account (by selecting the gMSA checkbox in the datasource page), provided sufficient permissions are given to the account to access those tables.

Configure CA Service Catalog with CA EEM over TLS 1.2

TLS 1.2 for CA Service Catalog.

To configure CA Service Catalog with CA EEM over TLS 1.2, Enable TLS 1.2 in CA EEM 12.6

Administering CA Service Catalog

This section contains the following articles:

- Migrate Data Between CA Service Catalog Systems using the Import Export Utility
- CA Service Catalog Clustering
- Archive and Purge Historical Data
- Diagnose the Health of the Product
- Deploy CA Service Catalog on a Custom Web Server
- Deploy and Undeploy Components from the Command Line
- Perform Maintenance
- Import and Export Offering in CA Service Catalog
- Enable SAML Authentication for Service Catalog
- Create Users and Services
- Install and Integrate Additional Process Automation Tools
- Upgrade JRE (AdoptOpenJDK) 11.0.18 or later
- (Recommended) Efficient Tomcat Access Log Rotatable
- (Recommended) Enhance Security
- (Recommended) Set Up Single Location for Shared Files

Migrate Data Between CA Service Catalog Systems using the Import Export Utility

Administrators can use the Import Export utility to export CA Service Catalog data between two CA Service Catalog computers. This migration is primarily intended for either:

- Moving data from test to production, when both computers are running the same release of CA Service Catalog.
- Moving data from a computer running the previous release of CA Service Catalog to a computer running a newer release of CA Service Catalog.

Follow these steps:

Step 1 - Verify the Prerequisites for Data Migration using the Import Export Utility

Before you begin to migrate data between the CA Service Catalog systems:

- Verify that CA Service Catalog is installed and running.
- Integrate CA Service Catalog with CA Service Desk Manager or CA CMDB, to export services that have CMDB CI associations.
- Understand the different object type attributes that can be exported or imported.

Step 2 - Export the Data

The Import Export utility exports the data in a proprietary XML format. The same XML format must be used in the data files when importing. You can determine the correct XML formats for each object type by examining an exported XML file.

Follow these steps:

1. Click Administration, Tools.
2. Select Import Export Utility from the left pane.
3. Select Export action in the Action and Object type.
4. In the Object Type drop-down list select the object for which, you want to export the data.
5. Specify the attributes for the selected object.
6. Click Start Export.

The exported data for the selected object type is saved as an XML file in the format:
ixutil_{object_name}_{YYYYMMDD}_{HHMMSS}.xml.
You have successfully exported the data between CA Service Catalog systems.

Step 3 - Import the Data

You can use the Import Export utility to migrate new and updated objects from one computer to another. All the object types that are exported can be imported.

Follow these steps:

1. Click Administration, Tools.
2. Select Import Export Utility from the left pane.
3. Select Import action in the Action and Object type.
4. Select the object type that you want to import in the Object Type drop-down list.
5. (Optional) Select Quick Import check box, if you do not want to specify any other attributes as inputs to import.
6. Specify the attributes for the selected object, if you do not want to perform a Quick Import.
7. Click Browse and select the xml file as input for the selected object type.

   NOTE
   For the Service Offering object type, select the exported zip file, if, the Include Translations option was selected during import.
8. Click Start Import.
   The file is imported to the business unit specified in the file.

   WARNING
   If you export report data objects that are based external data sources, verify that they are imported correctly on the target computer. To verify, review the IXUtil log file in the USM_HOME\logs\install folder on the target computer. If necessary, recreate these report data objects on the target computer.

   NOTE
   The Import Export utility does *not* import *images* of offerings. The import fails because of the variation of the image file path names on the source and target computers. To retain such images, copy and paste them manually, after the initial migration is complete.

   You have successfully imported the object type.

Step 4 - Verify the Exported or Imported Data

You can verify the imported data through the corresponding objects section in CA Service Catalog, or by querying the database.

Follow these steps:

1. Log in to CA Service Catalog on the computer where you imported the policies. If applicable, log in to the business unit to which you exported the policy or policies.
2. Click Catalog, Policies.
3. Expand the policy folders.
4. Verify that the CA Service Catalog imported the policy or policies as you want. Verify if the import was successful in Forms, Service Offerings, Events, Report Data, Business Units, and Configurations in Administration, Accounting, and Catalog.

Administrators can use the Import Export utility to export CA Service Catalog data between two CA Service Catalog computers. This migration is primarily intended for either:

- Moving data from test to production, when both computers are running the same release of CA Service Catalog.
- Moving data from a computer running the previous release of CA Service Catalog to a computer running a newer release of CA Service Catalog.

{"URL":["/usm/wpf?Node=iclaunchpad.ixutil"],"customLabelGetStarted":"Import Export Utility","description":"IPCE_Import_Export_Utility"}

Object Type Attributes

This article contains the following topics:

Depending on the selected object type to export or import, specify the information in the fields available for each attribute.

Business Unit Attribute

This section describes the business unit attribute.

- **Business Unit ID**
  Specifies the business unit login ID, the name that is used to log in to the business unit. When you use the business unit id attribute while exporting, verify that the business unit specified exists in the source. For importing a business unit, the parent business unit must exist in the target system. If it does not exist, the business unit being imported becomes a child business unit of the root business unit.
  When you import a business unit that has one or more sub-business units, first import all the business units. Then import all the accounts.

Configuration Attributes

This section describes the configuration attributes.

- **Group Name**
  Specifies the configurations that are grouped under various categories. Many groups are configured in the Administration Configurations, Accounting Configurations, Catalog Configurations, and Custom Configurations. The Configuration name to be provided as input in the Group Name text box is displayed in brackets beside the configuration name in the header section. You can import or export multiple groups at the same time. To export multiple groups, separate the names with commas.
  For example, to export a CMDB configuration group, provide the input as CMDB. All the configurations under this group are exported or imported. Leave this field blank to export all the configurations irrespective of the group.
- **Business Unit ID**
  Specifies the configurations of a group of Business Units that are exported or imported as a group. Leave this field blank to export all the configurations of the groups irrespective of the Business Unit.

Event Attributes

This section describes the event attributes.

- **Event Name**
  Specifies the name of the event as seen in the Event Type list. This option is used to export and import an event information about the specified event type.
- **Rule Name**
  Specifies the name of the rule as seen in the Rule Type list. This option is used to export and import a rule information about the specified event type.
- **Action Name**
  Specifies the name of the action as seen in the Action Type list. This option is used to export and import action information about the specified event type.

Form Attributes

The form attributes export either an individual Form Designer form or all Form Designer forms from a specified folder.

- **Form Designer Folder Name**
  Exports all the Form Designer forms from a specific folder.
- **Form Designer Form Name**
  Exports an individual Form Designer form.

  NOTE
  For Form Designer forms and folders, use only the form name or folder name; path name is not applicable.
- **Business Unit ID**
  Specifies the business unit login ID, the name that is used to log in to the business unit. Use this option to export all forms in a domain. If you omit this option, IXUtil exports all forms in all domains.
  For example, if the business unit ID is subBud, specify subBud in the text box.

Policy Attributes

This section describes the policy attributes.

- **Folder Name**
  Specifies the folder name. To export multiple folders, separate the names with commas.
- **Policy Name**
  Specifies the policy name. To export multiple policies, separate the names with commas.
- **Business Unit ID**
  Specifies the business unit login id that contains the policies being exported. If you are exporting all policies in all the business units, omit this attribute. Otherwise, this attribute is required.
  To import the policy files, optionally specify this parameter to import all the policies into a single business unit.
  If the specified business unit does not exist, the policies that are assigned to it are instead assigned to the root business unit.

Report Data Attributes

This section describes the report data attributes.

- **Folder Name**
  Specfies the folder name. To export multiple folders, separate the names with a comma.
- **ID**
  Specifies an identification of the data objects. To export multiple data objects, separate the id's with a comma.

Service Offerings Attributes

This section describes the service offering attributes.

- **Object Classifier**
  Specifies whether the folders or Service offerings are chosen specifically.
- **Business Unit ID**
  Specifies the business unit login id that contains the folders or service offerings being exported. To export all folders or service offerings in all business units, omit this attribute; otherwise, this attribute is required.
  To import folders or service offerings files, you can specify this parameter to import all the folders or service offerings into a single business unit.
  If the specified business unit does not exist, the folders or service offerings that is assigned to it is instead assigned to the root business unit.
- **Include Forms**
  Associates export of the included forms, with the export of service offerings.
- **Include CMDB CI Mapping**

Specifies that the associations between CA Service Catalog services and CMDB configuration items are exported. This option applies *only* when you have integrated CA Service Catalog with CMDB.

WARNING
Verify that the CMDB configuration items are already present on the target computer *before* you export. Also verify that the UUIDs of these configuration items are the same on both the source and target computers. For more information about how to copy the configuration items and verify the UUIDs, see your CMDB documentation.

- **Include Permissions**
  Exports or Imports the permissions for each service (offering) while exporting or importing services through service offerings.
- **Include Request SLA**
  Exports or imports the associated request SLAs for each service (offering) while exporting or importing services.
- **Include Service Hour**
  Exports or imports the associated service hours (outage calendars and business hours) for each service (offering) while exporting or importing services through service offerings.
- **Include Resource Type**
  Exports or Imports the resource type that is associated with the service (offering).
- **Include Application Metric**
  Exports or Imports the application and its metric, along with the definition of the service that is associated with the Application Rate item.
- **Include Associated Policies and Actions**
  Exports or imports the associated policies and actions for each service (offering) while exporting or importing services through service offerings.
- **Include Translations**
  Creates the output as a zip file instead of xml file during export. The zip file includes the default properties file associated with the xml file.
  During Import, if you select this option, provide the zip file as input (instead of single xml file). The zip file internally must contain the properties files corresponding to different locales with the xml file.

CA Service Catalog Clustering

A *cluster* is two or more interconnected computers that create a solution to provide higher availability, higher scalability, or both.

A CA Service Catalog administrator can optionally use clustering for CA Service Catalog to improve performance and provide failover protection. Clustering means multiple computers in a group that perform the same or similar function, essentially acting as one virtual computer. Failover protection means that if one computer malfunctions, becomes heavily loaded, or loses power, its workload is transferred to the other computers in the cluster. These computers retain and complete the active sessions. Another advantage of clustering is load-balancing. If one of the cluster components is processing a request, the load is redirected to another component in the cluster. Users of the system see no interruption of access. The loss of performance on users and business functions is minimized, even when computer availability is lost or reduced.

Clustering for CA Service Catalog is accomplished through the Catalog Component clustering. All user communications with catalog and accounting functions are accomplished through the Catalog Component. Thus, the term *Catalog Component clustering* includes both Catalog Component clustering and Service Accounting Component clustering.

In *horizontal clustering*, different physical computers comprise a cluster. As a result, the load balancer forwards requests to different computers having different IP addresses. All Catalog Component installations on all computers must use the same cluster of the MDB.

You can optionally install and cluster multiple instances of CA EEM and CA Process Automation.

WARNING
These instructions apply *only* to the Apache Tomcat web server version supplied with CA Service Catalog. If you are using any other web server instead, see your web server documentation for information about clustering.

Clustering Prerequisites

Perform the following preliminary tasks before you implement clustering:

1. Decide how many cluster nodes to create, depending on your design concerns and availability of resources.
2. Verify that the time and time zone on all clustered computers are the same.
3. Verify that the IPV6 protocol is disabled on all clustered computers.
4. Verify that no other application uses the following ports.
   By default, CA Service Catalog uses these ports for clustering, using the AJP protocol:
   - **8009** - for CA Service Catalog communicating with Apache HTTP Server
   - **8019** - for Service Fulfillment communicating with Apache HTTP server
5. Perform the following step if applicable:
   a. If another application uses one of these port numbers, eliminate the duplication. Use a different port number for either CA Service Catalog or the other application.
   b. If you use a different port number for CA Service Catalog, then also replace the old port number with the new port number. For example, if you change 8009 to a different port number, use the new port number instead of 8009.
6. Locate the multicast address for Tomcat in the server.xml file that is located in the USM_HOME\view\conf folder. This address is the value of the address attribute within the Membership tag.
   Default: 228.0.0.4
7. Locate the multicast address for Java Messaging Service (JMS) in the config.properties file that is located in the USM_HOME folder.
   Default: 239.255.2.3:6155
8. Run the following command at the DOS prompt of each clustered computer:
   ```
   netsh interface ip show joins
   ```
   The command lists all multicast addresses applicable to this computer.
9. Verify that this list includes the multicast addresses that you recorded from the server.xml and config.properties files:
   - If **Yes**, then skip the remaining steps in this topic.
   - If **No**, then use the list to determine a common multicast address applicable to all clustered computers. Complete the remaining steps.
10. Follow these steps:
    a. Update the multicast addresses in the server.xml files on all clustered computers and use this common address.
    b. In the **config.properties** files on all clustered computers, edit the JMS-related configuration section. Specify this common address, as shown in the following example.

       WARNING
       The port number *must* be the same on each computer in the cluster.

       ```
       #multicast://default = multicast://239.255.2.3:6155
       #static:(tcp://<host1>:<port>,tcp://<host2>:<port>,...)
       jms.networkConnectorUri = multicast://custom-ip-address:custom-port
       jms.discoveryUri = static: multicast://custom-ip-address:custom-port
       #jms.port = 7777
       jms.port = custom-port
       ```

    c. If you want to specify your nodes statically instead of using a multicast address, perform this step instead of **step b**.
       In the **config.properties** files on all clustered computers, edit the JMS-related configuration section. Specify the host and port for all other computers in the cluster, as shown in the following example.
       The port number can be the same value on each computer in the cluster, or it can be different.

```
#multicast://default = multicast://239.255.2.3:6155
#static:(tcp://<host1>:<port>,tcp://<host2>:<port>,...)
jms.networkConnectorUri = static:(tcp://host:custom-port)
jms.discoveryUri = static:(tcp://host:custom-port)
#jms.port = 7777
jms.port = custom-port
```

NOTE
Update the server.xml and config.properties files on all clustered computers before proceeding to the next step.

11. Restart the CA Service Catalog Windows service on all clustered computers.

How to Setup Horizontal Clustering Nodes for CA Service Catalog

This article contains the following topics:

Configure Cluster Nodes

Perform the procedures outlined in this article on *every* computer that you want to set up a horizontal cluster node for CA Service Catalog. CA Service Catalog uses Tomcat Apache for clustering. Install and configure CA Service Catalog to be able to set up cluster nodes using Tomcat. CA Service Management 17.4 uses Tomcat 9.0.x and Java 11.0.x.

NOTE
Before you perform this procedure, verify that CA Service Catalog is installed locally.

Perform the following steps:

1. As a prerequisite, perform the following steps:
   a. Verify that host name is unique where you have installed CA Service Catalog.
   b. Verify that the database is pointing to the system where you have installed CA Service Catalog.
2. Open and edit the server.xml file from ***USM_HOME\view\conf\server.xml*.**
3. Uncomment the Cluster tags, if they are commented.
4. After uncommenting the cluster tags, ensure to make the following configuration changes to set up CA Service Catalog clustering:
   a. Make the following changes to the interceptor class file from

      ```
      <Interceptor className="org.apache.catalina.tribes.group.interceptors.MessageDispatch15Interceptor"/>
      ```

      to

      ```
      <Interceptor className="org.apache.catalina.tribes.group.interceptors.MessageDispatchInterceptor "/>
      ```

   b. If you see the following lines in the cluster tag section, remove it:

      ```
      <ClusterListener className="org.apache.catalina.ha.session.JvmRouteSessionIDBinderListener"/>
      ```

   c. Make the following changes to cluster listener class name:

      ```
      <ClusterListener className="org.apache.catalina.ha.session.ClusterSessionListener"/>
      ```

5. Specify a unique port in the <*Cluster*> section for each CA Service Catalog computer:

```
<Receiver className="org.apache.catalina.tribes.transport.nio.NioReceiver"
      address="auto"
      port="4000"
      autoBind="100"
      selectorTimeout="5000"
```

```
maxThreads="6"/>
```

6. Uncomment the Apache JServ Protocol (AJP) tags shown in the following lines, if they are commented:

```
<Connector port="8009" enableLookups="false" redirectPort="8443" tomcatAuthentication="false"
      maxThreads="400" minSpareThreads="25" maxSpareThreads="100" protocol="AJP/1.3" />
```

7. (Optional) Comment the HTTP ports Connector tags to improve security as follows:

```
<Connector port="8080" enableLookups="false" redirectPort="8443" tomcatAuthentication="false"
      maxThreads="400" minSpareThreads="25" maxSpareThreads="100" debug="0" connectionTimeout="15000"
      disableUploadTimeout="true" compression="on" compressionMinSize="2048"
      compressableMimeType="text/html,text/plain,text/xml,text/css,text/javascript,image/png,image/
gif,image/jpeg,application/json"
      useBodyEncodingForURI="false" URIEncoding="UTF-8" />
```

8. Save the file.
9. Open the ***USM_HOME\view\conf\viewService.conf*** file for editing and add the following entry in the existing Java Class Path section:

```
wrapper.java.classpath.6=<catalog_install location>/embedded/tomcat/lib/*.jar
```

10. Review the ServiceCatalog.log file of each CA Service Catalog computer and verify that the replication cluster was added. For example:

```
INFO Cluster-MembershipReceiver org.apache.catalina.cluster.tcp.SimpleTcpCluster - Replication member
 added:org.apache.catalina.cluster.mcast.McastMember[tcp://141.202.143.77:4013,catalina,141.202.143.77,4013,
 alive=0]
```

11. Restart the CA Service Catalog Windows services: CA Service Accounting and CA Service Catalog.
12. If you are using NTLM authentication to enable single sign-on to CA Service Catalog,
    You have now set up clustering for CA Service Catalog.

Validate the Cluster Configuration

After catalog nodes are set up for clustering as shown before, you can do the following steps on all clustered nodes to validate it:

1. Take a backup copy of *log4.xml* file from *USM_HOME\view\conf*
2. Edit the *log4j.xml* file. Locate the following section in the log4.xml file:
3.

```
<logger name="org.apache.catalina.session" additivity="false">
<level value="INFO" />
<appender-ref ref="tomcat_view" />
</logger>
```

4. Add the following after the preceding lines:

```
<logger name="org.apache.catalina.cluster" additivity="false">
<level value="TRACE" />
<appender-ref ref="tomcat_view" />
</logger>
```

5. Save the log4j.xml file.
6. Restart the CA Service Catalog service.
7. If catalog tomcat clustering is set up properly, you see the following messages in the ServiceCatalog.log file in *USM_HOME\logs\view* folder) :

```
INFO | jvm 1 | 2018/09/13 14:25:21 | Sep 13, 2018 2:25:21 PM org.apache.catalina.ha.tcp.SimpleTcpCluster
 memberAdded
```

```
INFO | jvm 1 | 2018/09/13 14:25:21 | INFO: Replication member
 added:org.apache.catalina.tribes.membership.MemberImpl[tcp://{10, 162, 7, 4}:4000,{10, 162, 7, 4},4000,
 alive=1025, securePort=-1, UDP Port=-1, id={-120 116 59 -100 -123 -34 67 9 -68 59 -109 2 -88 -87 121
 -29 }, payload={}, command={}, domain={}, ]
INFO | jvm 1 | 2018/09/13 14:37:30 | Sep 13, 2018 2:37:29 PM org.apache.catalina.ha.tcp.SimpleTcpCluster
 memberAdded
INFO | jvm 1 | 2018/09/13 14:37:30 | INFO: Replication member
 added:org.apache.catalina.tribes.membership.MemberImpl[tcp://{10, 162, 7, 144}:4001,{10, 162, 7,
 144},4001, alive=1024, securePort=-1, UDP Port=-1, id={77 -14 -120 112 -89 26 66 -75 -119 -76 -43 85 59
 86 124 -31 }, payload={}, command={}, domain={}, ]
```

Remove a Cluster

When necessary, you can remove a horizontal CA Service Catalog cluster. Sample reasons for doing so include the following conditions:

- When the system where you have setup cluster nodes becomes obsolete.
- You want to install clustering on a different computer.
- The cluster you configured is no longer needed/valid.

Perform the following steps:

1. Open the *server.xml file* from ***USM_HOME\view\conf\server.xml.***
2. Comment the cluster tag.
3. Restart the CA Service Catalog Windows service.

You have removed the cluster node. Next, verify that the load balancer, if any, no longer references the removed cluster.

NTLM Authentication for Cluster

If you are clustering CA Service Catalog and you are using NTLM authentication to enable Single Sign-On to CA Service Catalog, set up the cluster for Single Sign-On. Perform the following steps:

1. Perform *one* of the following actions on each cluster:
   – If you are using Apache Load Balancer, perform the following steps on each cluster to set up NTLM Authentication.
      a. Download the mod_auth_sspi module from the sourceforge web site, sourceforge.net.
      b. Copy the mod_auth_sspi.so module to the <APACHE_Home>\modules directory of the Apache web server that you use for CA Service Catalog.
      c. Append the following configuration section to the <APACHE_Home>\conf\httpd.conf file.

```
LoadModule sspi_auth_module modules/mod_auth_sspi.so
<Location ~ "/usm/(wpf|documents|FileStore)">
AuthName "domain_name"
AuthType SSPI
SSPIAuth On
SSPIOfferBasic On
SSPIAuthoritative On
SSPIDomain "domain_name"
SSPIofferSSPI off
require valid-user
</Location>
```

      d. Replace *domain_name* with the name of your network domain or Windows domain in the file.
      e. Verify that the tomcatAuthentication="false" attribute is set for the Tomcat connectors that this Apache load balancer uses.

2. If you *are* using a load balancer *other than* Apache Load Balancer, configure NTLM authentication for your load balancer first. Then configure Tomcat to use NTLM authentication for CA Service Catalog for each catalog cluster node, if both of the following conditions exist:
   – Your load balancer is not configured to perform NTLM authentication.
   – Your load balancer uses the HTTP Port (default 8080) to connect to Tomcat instances of CA Service Catalog. For more information about how to set up NTLM authentication, see your load balancer documentation.
3. Log in to any CA Service Catalog computer in the cluster.
4. Click **Administration**, **Configuration**, **Single Sign On Authentication**.

   NOTE

   Perform this step and the remaining steps *once* in your implementation. You do not need to repeat these steps for individual clusters, because performing them once affects all clusters.
5. a. Locate the property that is named **Single Sign On Type** and click the Modify icon.
   b. Select the option that is named **Artifact Based Single Sign On** and click Update Configuration.
   The dialog closes, and you return to the Sign On Authentication page.
   c. Locate the property that is named **Artifact Type** and click the Modify icon.
   d. Set **Artifact Type** to the appropriate option for your load balancer, and click **Update Configuration**. For Apache Load Balancer, select **Request**.
6. Restart the CA Service Catalog Windows services to ensure that you have set up NTLM Authentication for the clusters.

Load Balancing for Cluster Nodes

To load balance the existing nodes using Apache HTTP Server, follow these instructions. Perform this process *after* you have set up all cluster nodes.

NOTE
If you are implementing horizontal clustering on multiple computers, use the computers that have the same default gateway. To obtain the default gateway, enter the ipconfig command at the DOS prompt.

1. Install Apache Web Server load balancer for Windows version 2.2.25. You can download it from the Apache web site, apache.org.
   The default installation folder is C:\Program Files\Apache Software Foundation\Apache2.2.
2. Configure Apache HTTP Server.
3. (Optional) Disable Web Server Features.
4. Create and Configure the workers.properties File.
5. Create the uriworkermap.properties file.
6. Update the httpd.conf file.
7. Verify load balancing.

Configure Apache HTTP Server

Configuring Apache HTTP Server is a required task when you set up load balancing.

Follow these steps:

1. Access the Apache website, www.apache.org.
2. Find and download the file that is named **mod_jk-1.2.30-httpd-2.2.3.so** CA Service Catalog supports **mod_jk 1.2.30**.
3. Rename the **mod_jk-1.2.30-httpd-2.2.3.so** file to ***mod_jk.so*** .
4. Copy the **mod_jk.so** file to the modules folder in the Apache Server installation.

You have configured Apache HTTP Server.

(Optional) Disable Web Server Features

Disabling the web server features of Apache HTTP Server improves performance and reduces potential security risks.

Follow these steps:

1. Delete the following folders under C:\Program Files\Apache Software Foundation\Apache2.2:
   - \cgi-bin
   - \errror
   - \htdocs
   - \icons
   - \manual

   NOTE
   The "C:\Program Files\" portion of the path names reflect the typical location that is specified at installation time. This portion of your path names can vary depending on the installation directory that is specified for the \Apache Software Foundation folder at installation time.
2. Rename httpd.conf to _httpd.conf.bak in the C:\Program Files\Apache Software Foundation\Apache2.2\confdirectory.
3. Find the httpd.minimal.conf file.
   This minimal configuration file is included in the Utilities/Apache Webserver directory of the CA Service Catalog installation media.
4. Copy the httpd.minimal.conf file to C:\Program Files\Apache Software Foundation\Apache2.2\conf.
5. Rename the httpd.minimal.conf file to httpd.conf in that directory.
6. Delete all files in that directory *except* the following files:
   - httpd.conf
   - _httpd.conf.bak
7. Change to the C:\Program Files\Apache Software Foundation\Apache2.2\modules directory.
8. Delete all files in that directory *except* the following files:
   - mod_jk.so
   - mod_log_config.so
   - mod_setenvif.so

You have disabled web server features.

Create and Configure the workers.properties File

Creating and configuring the worker.properties file is a required task when you set up load balancing.

Follow these steps:

1. Create the workers.properties file manually in the \conf directory of the Apache installation folder.
2. Verify that the file lists all the Tomcat instances to include in the clustered environment.
3. Obtain the following information for each cluster node:
   - The host name of the clustered node
   - The ajp port of the clustered node. Obtain it from the connector port in the following line in the server.xml file of each cluster node:

     ```
     <Connector port="8009"  maxThreads="150" tomcatAuthentication="false" minSpareThreads="25"
      maxSpareThreads="75" enableLookups="false" redirectPort="8443" protocol="AJP/1.3" />
     ```
   - The jvmRoute name of the node. Obtain it from the following line in the server.xml file of each cluster node:

     ```
     <Engine name="Standalone" defaultHost="localhost" debug="0" jvmRoute="hostname_USMView">
     ```

     For example, for the CA Service Catalog cluster node on the computer that is named XYZ, the jvmRoute name is XYZ_USMView.

WARNING

Verify that the jvmRoute name is unique for each CA Service Catalog cluster. This name must be unique, because the load balancer uses it to identify each cluster. This name is case-sensitive.

4. Review the following sample worker.properties file sections for the types of clustering that you can perform. In all cases, the worker.properties file defines a load balancer node named loadbalancerview. This node balances CA Service Catalog requests among the CA Service Catalog cluster nodes. For example, this node load balances the /usm context between the workers defined. The context path /usm is defined in the USM_HOME\view\conf\server.xml file pointing to the CA Service Catalog web application.
5. Review the example.
   a. Add a worker to the cluster, as follows:
      a. Create a worker section at the end of the worker.properties file, using the following lines as a model:

```
worker.jvmRouteName.port=ajpport
worker.jvmRouteName.host=HOSTNAME
worker.jvmRouteName.type=ajp13
worker.jvmRouteName.lbfactor=1
```

      b. Verify that its jvmRouteName is unique.
      c. Add the new cluster at the end of the list of balance workers for a loadbalancer node.
      For example, suppose that the jvmRoute name of the new worker is COMPUTER3_caff and its ajpport is 8019. In that case, enter the following new worker section to add the new cluster to the selected loadbalancer node:

```
worker.COMPUTER3_caff.port=8019
worker.COMPUTER3_caff.host= COMPUTER3
worker.COMPUTER3_caff.type=ajp13
worker.COMPUTER3_caff.lbfactor=1
```

      d. Remove an existing cluster from the loadbalancer setup in the worker.properties file, as follows:
         a. Delete the section for the worker.
         b. Remove any references to the deleted worker from the list of balance workers.
         The list of balance workers begins with the following line:

```
worker.<loadbalancernode>.balance_workers
```

You have created and configured the worker.properties file.

Horizontal Clustering Example for CA Service Catalog

This example illustrates how to use clustering to load balance the /usm context of CA Service Catalog between two workers. The cluster specifications are as follows:

Worker1:

- Hostname: Computer1
- JVMroutename: Computer1_USMView
- AJP port: 8009

Worker 2:

- Hostname: Computer2
- JVMroutename: Computer2_USMView
- AJP port: 8009

In these examples, Computer1 and Computer2 are the host names (computer names) of the workers.

Use the loadbalancerview node to load balance between the workers Computer1 and Computer2 as horizontal clusters for CA Service Catalog. To do so, define the loadbalancerview node sections of the worker.properties as follows.

```
worker.list=loadbalancerview
```

```
worker.status.type=status
worker.loadbalancerview.method=Busyness
worker.loadbalancerview.type=lb
worker.loadbalancerview.balance_workers=COMPUTER1_USMView,COMPUTER2_USMView
worker.loadbalancerview.sticky_session=1
worker.COMPUTER1_USMView.port=8009
worker.COMPUTER1_USMView.host=COMPUTER1
worker.COMPUTER1_USMView.type=ajp13
worker.COMPUTER1_USMView.lbfactor=10
worker.COMPUTER2_USMView.port=8009
worker.COMPUTER2_USMView.host=COMPUTER2
worker.COMPUTER2_USMView.type=ajp13
worker.COMPUTER2_USMView.lbfactor=10
```

Create the uriworkermap.properties File

Creating and configuring the uriworkermap.properties file is a required task when you set up load balancing. This file specifies which context the load balancer node selects and forwards to its cluster nodes.

NOTE
If the file exists already, update it to match the text in the following steps.

Follow these steps:

1. Manually create the uriworkermap.properties file in the \conf directory of the Apache installation folder.
2. Enter the following lines in the file:
   ```
   /usm|/*=loadbalancerview
   ```
3. Save the file.

You have created the uriworkermap.properties file.

Update the httpd.conf File

Updating the httpd.conf file is a required task when you set up load balancing.

Follow these steps:

1. Open the conf/httpd.conf file in the Apache HTTP Server installation directory and update it to include the following specifications:
   ```
   Listen 89
   LoadModule jk_module modules/mod_jk.so
   JkWorkersFile conf/workers.properties
   <VirtualHost *:89>
    ServerName load balancer host name
    JkMountFile conf/uriworkermap.properties
    JkLogFile logs/mod_jk.log
    JkLogLevel debug
   </VirtualHost>
   ```
2. Find the following lines:
   ```
   JkLogFile logs/mod_jk.log
   JkLogLevel debug
   ```
3. (Optional) Replace the lines in the previous step with the following lines:
   ```
   JkLogFile "|bin/rotatelogs.exe logs/mod_jk.log.%Y-%m-%d-%H_%M_%S 10M"
   JkLogLevel emerg
   ```

This action modifies the mod_jk.log file on your production server to record only critical errors. The action also enables automatic rollover when the log file size reaches 10 MB.

4. Add a Listen directive with the port number for every VirtualHost port in the file.
   For example, suppose that you have defined a virtual host at port 89, as follows:
   ```
   <VirtualHost *:89>
   ```
   In that case, add the Listen directive for this port in the httpd.conf file, for example:
   ```
   Listen 89
   ```
   NOTE
   The commonly used port is 89.
   This specification enables the Apache HTTP Server to listen on this port and accept incoming requests.
5. Change the Default Type from text/plain to text/html.
   NOTE
   This setting works well on most browsers. If you experience any difficulty with performance or display quality after changing this option, return it to its original value.
6. Save and close the conf/httpd.conf file.
7. Adjust related settings in other locations.
8. Disable the HTTP ports of the cluster computers.
9. Configure the server information for the cluster computers.

Adjust Related Settings

Adjusting related settings is a required task when you update the httpd.conf file.

Follow these steps:

1. Open your web browser. Set the proper values for the ServerAdmin and ServerName options.
2. Restart the Apache service, as follows:
   a. Open the Apache Service Monitor.
   b. Highlight the service to restart.
   c. Click Restart.
3. Test by accessing the link http://*hostname*:89/usm.
   Here, *hostname* specifies the load balancer computer, the computer on which the Apache HTTP Server is configured for load balancing.
4. Review the mod_jk.log file under the logs folder in the Apache directory to determine which jvmRoute node was called. An example follows:
   ```
   found worker HOST1__USMView (HOST1__USMView) for route COMPUTER1__USMView and partial sessionid
   ```

(Optional) Disable HTTP Ports

To enhance security, disable the HTTP ports of the cluster computers when you update the httpd.conf file. Doing so routes users directly to the load balancer computer rather than directly to a cluster.

Follow these steps:

1. Comment the connector tag that includes HTTP port information, in the server.xml files of each cluster computer. This port is the startup node that is provided during installation, typically 8080.
2. Recycle the CA Service Catalog Windows services of the clusters whose server.xml files were changed. The Windows services are named CA Service Accounting and CA Service Catalog.

Configure Server Information

Configuring the server information of the cluster computers is a required task when you update the httpd.conf file.

Follow these steps:

1. Click **Administration**, **Configuration**, **Server Information**, in the CA Service Catalog GUI.
2. Replace the server host name and port number with the load balancer host name and port number.
3. Recycle the CA Service Catalog Windows services of the clusters whose Server Information options were changed.

You have configured the server information of the cluster computers.

> **NOTE**
> If you do not perform this task, the following error occurs for users who request CA Service Catalog services using Internet Explorer 8.0: Forms that include a lookup field that calls a report object do not populate correctly when the user clicks the Search icon.

Verify Load Balancing

To test load balancing, verify that the load is distributed to other nodes when one of the nodes in the same cluster becomes busy. To test failover functionality, verify that the load is distributed to other nodes when one of the nodes in the same cluster shuts down.

Verify that *all* clustered nodes communicate efficiently with *all* load balancer computers using the telnet utility. If the clusters do not run efficiently, verify that this communication exists and check for any network configuration problems. For example, suppose that your load balancer host name is abc and the port is 89. Verify that you can successfully connect from all your clustered nodes using the following command from each node:

```
telnet abc 89
```

We recommend that you cluster your load balancer computers to help avoid single-point-of-failure problems with an individual load balancer computer. For more information, see the Apache website and other resources about how to cluster load balancer computers.

You can also record cluster-related logging information in the tomcat_view.log file.

Follow these steps:

1. Add the following section to the log4j.xml file for each CA Service Catalog computer:

```
<logger name="org.apache.catalina.cluster" additivity="false">
  <level value="TRACE" />
  <appender-ref ref="tomcat_view" />
</logger>
```

2. Set the log level for each log4j.xml file appropriately. For example, use a high log level while you are troubleshooting. Conversely, use a low log level, or comment out the previous lines, at other times.

CA EEM Requirements for CA Service Catalog Clustering

An installation of CA Service Catalog can include several clustered computers. Each installation of CA Service Catalog must map to an application name in CA EEM. You can set up a single instance of CA EEM and the CA Service Catalog database for multiple installations of CA Service Catalog (for example, Test and Production).

In this type of setup, the setup-related parameters (for example, the database, user, and application names) have the same values on every computer. Therefore, when you run the setup utility, you can use the remote configuration feature to copy the parameter settings from one computer to another.

If CA EEM is clustered with a load balancer, follow these steps before you run the setup utility:

1. Edit the Apache load balancer configuration file, **proxy.conf** file.
   This file is in the conf folder of the Apache installation directory.
2. Record the current setting of the **MaxKeepAliveRequests** property.
3. Set the **MaxKeepAliveRequests** property value to 0.

4. Restart the Apache load balancer services to make the configuration changes take effect.

Archive and Purge Historical Data

Administrators can archive and purge historical data concerning requests and audits from the CA Service Catalog database. Purging historical data regularly helps improve the performance of CA Service Catalog and the database.

> **WARNING**
> Once you archive the data, you cannot restore the data to production tables.

Follow these steps:

1. Step 1 - Complete the Prerequisites
2. Step 2 - Prepare the MDB for Archive and Purge for the First Time
3. Step 3 - Archive the Data
4. Step 4 - Purge the Data

Step 1 - Complete the Prerequisites

Complete these prerequisites so that you can archive and purge data successfully.

Follow these steps:

1. Understand the database-related terms that are used in this scenario, as follows:
   When you install CA Service Catalog, you install the Catalog database. The Catalog database includes the following tables:
   - The CA Management Database (MDB) tables that apply to CA Service Catalog. The names of such tables typically begin with a CA_ prefix. The MDB is the common, shared database for CA Technologies products. The MDB provides the database schema for CA Service Catalog and other CA Technologies products.
   - The CA Service Catalog-specific tables. The names of such tables typically begin with a USM_ prefix.
2. Back up the Catalog database.

   > **NOTE**
   > Log in to the Catalog database as the database application user; the user name is typically usm_user or mdbadmin.
3. Configure your DBMS to run with a minimal level of logging. This configuration helps expedite database processing for CA Service Catalog.

   > **NOTE**
   > For more information about backup procedures and how to set logging levels, see your DBMS documentation.
4. (Optional) Stop all antivirus services.

Step 2 - Prepare the MDB

To improve the performance of archive and purge activity, prepare the Catalog database. This procedure creates an index that helps the database fetch the records faster.

> **WARNING**
> Before you archive or purge data *for the first time*, perform this procedure *once*. Repeating this procedure is unnecessary. For information about how to prepare the MDB, see *Optimize the System Change Detail Tables* in Prepare for Upgrade section.

Step 3 - Archive the Data

Archiving the data moves the archival data from production tables to archival tables. After you archive the data, the data does not appear in CA Service Catalog UI.

> **WARNING**
> The data in archival tables cannot be restored to production tables. So, ensure that you no longer need the data you want to archive.

Use one of the following procedures:

- Archive the Data in Microsoft SQL Server
- Archive the Data in Oracle

Archive the Data in Microsoft SQL Server

This section describes archiving the data for Microsoft SQL Server. Perform the following steps:

> **NOTE**
> We recommend that you Use MS SQL Server Management Studio to execute the stored procedure.

1. Log in to MDB using the MS SQL Server Management Studio.
2. Click New Query and enter the following code in the new query window.

```
USE [<mdb instance name>]
GO
DECLARE    @return_value int
EXEC    @return_value = [dbo].[usm_sp_archive_data]
        @p_object_type = N'< Object Type >',
        @p_date = N'<Completion Date on or before  -  yyyy-mm-dd>',
        @p_bu = N'<Business Unit>'

SELECT    'Return Value' = @return_value
GO
```

   Replace the values inside the carets (< >) with the required data.
   - **MDB Instance Name**
     Specifies the MDB instance name that CA Service Catalog uses.
   - **Object Type**
     Specifies the type of objects to archive and purge, as follows:
     - **Request**
       Archives or purges requests that are in completed status and related data (including audit data).
     - **Audit**
       Archives or purges audit entries for all objects.
   - **Completion Date on or before - yyyy-mm-dd**
     Archives or purges only requests that are completed on or before this date.
   - **Business Unit**
     Archives or purges the records of this business unit.
3. Click **Execute**.
   The result appears on the Messages tab.

The data is archived.

Archive the Data in Oracle

This section describes archiving the data in Oracle. Perform the following steps:

> **NOTE**
> We recommend that you use Oracle SQL Developer to execute the stored procedure.

1. Log in to MDB using Oracle SQL Developer.
2. Navigate to **Connections**, **connection name**, **Procedures**.
   a. • **USM_SP_ARCHIVE_DATA**
      • **USM_SP_PURGE_DATA**
3. Right-click one of the procedures and select **Run**.
4. In the Run PL/SQL window, replace the values inside the carets (**< >**) with required data.

   NOTE

   If you are using SQLPlus on command prompt, execute the **SET SERVEROUT ON** statement before executing the stored procedure.

```
DECLARE
  P_OBJECT_TYPE VARCHAR2(200);
  P_DATE DATE;
  P_BU VARCHAR2(200);
BEGIN
  P_OBJECT_TYPE := '<Object Type>';
  P_DATE := '<Completion Date on or before  -  DD-MON-YYYY>';
  P_BU := '<Business Unit>';

  USM_SP_ARCHIVE_DATA(
    P_OBJECT_TYPE => P_OBJECT_TYPE,
    P_DATE => P_DATE,
    P_BU => P_BU
  );
END;
```

5. Click **OK**.
6. The result is displayed in the **Running - Log** tab.
   The data is archived.

Step 4 - Purge the Data

You can purge the data to remove the archived data permanently from the database. Perform the following step:

1. Follow the instructions in the Archive the Data procedure and replace the **USM_SP_ARCHIVE_DATA** stored procedure with **USM_SP_PURGE_DATA** procedure.
   When the stored procedure is successfully completed, the data is purged.

Diagnose the Health of the Product

Administrators can use the diagnostic framework to assess the health of CA Service Catalog in a clustered environment. The diagnostic framework includes web service methods, CA Remote Engineer, and the JMX client.

Follow these steps:

Step 1 - Verify the Prerequisites

Step 2 - Implement the Web Service Methods

Step 3 - Download and Run CA Remote Engineer

Step 4 - Use JMX Client to Monitor Status

Step 5 - Verify the Diagnostic Framework

Step 1 - Verify the Prerequisites

Before you use the diagnostic framework, meet the following prerequisites for each component:

- For web services, a utility to connect to the web services is required. Examples include Java Client and Simple Object Access Protocol client.
- For CA Remote Engineer, the Java Runtime Environment (JRE) version for CA Service Catalog is required. This JRE version is installed automatically when you install CA Service Catalog.
- For JVM Metrics, a Java Management Extension (JMX) client is required. An example is JConsole, which is included in the Java Development Kit (JDK).

Step 2 - Implement the Web Service Methods

Web service methods help you diagnose problems by verifying the health of the connection between CA Service Catalog and its components. The following web service methods apply during the connection time between CA Service Catalog and its components:

- Web service method for CA Embedded Entitlements Manager
- Web service method for the database
- Web service method for CA Process Automation

Web Service Method for CA Embedded Entitlements Manager

This web service method tests the connectivity of CA Service Catalog with Embedded Entitlements Manager server.

Web Service Method

```
long getEEMConnectionStatus()
```

Input Parameter

None

Return Parameter

Time that is taken in milliseconds by CA Service Catalog to connect to Embedded Entitlements Manager. If the connectivity fails, then an exception is displayed.

Pseudocode to use Embedded Entitlements Manager connectivity web service

```
URL endpoint1 = null;
AdministratorServiceSoapBindingStub adminStub = null;
endpoint1 =
new java.net.URL("http://catalog:8080/usm/services/AdministratorService");
adminStub = (AdministratorServiceSoapBindingStub) new
 AdminServiceImplServiceLocator().getAdministratorService(endpoint1);
long eemLatency;
try {
    eemLatency = adminStub.getEEMConnectionStatus();
    System.out.println("Connection successful. Time taken(milliseconds): "+ eemLatency);
} catch (Exception ex) {
    System.out.println("Connection to EEM failed. " + ex.getMessage());
}
```

Web Service Method for the Database

This web service method tests the connectivity of CA Service Catalog with the database server.

Web Service Method

```
long getDBConnectionStatus()
```

Input Parameter

None

Return Parameter

Time that is taken in milliseconds by CA Service Catalog to connect to Database. If the connectivity fails, then an exception is displayed.

Pseudocode to use database connectivity web service

```
URL endpoint1 = null;
AdministratorServiceSoapBindingStub adminStub = null;
endpoint1 =
new java.net.URL("http://catalog:8080/usm/services/AdministratorService");
adminStub = (AdministratorServiceSoapBindingStub) new
 AdminServiceImplServiceLocator().getAdministratorService(endpoint1);
long dbLatency;
try {
    dbLatency = adminStub.getDBConnectionStatus();
    System.out.println("Connection successful. Time taken(milliseconds): " + dbLatency);
} catch (Exception ex) {
    System.out.println("Connection to database failed. " + ex.getMessage());
}
```

Web Service Method for CA Process Automation

This web service method tests two-way connectivity from CA Service Catalog to CA Process Automation.

Web Service Method

```
long getITPAMConnectionStatus(String configName, String nodeUrl)
```

Input Parameter

- **configName**
  (Optional) Defines the configuration name, which is the group name of CA Process Automation as defined in Administration, Configuration, CA Process Automation. You can remove this parameter to test the default configuration.
- **nodeURL**
  (Optional) Specifies the value for connecting from CA Process Automation to CA Service Catalog. Use this parameter in a clustered setup to verify the connection to an individual node.

  NOTE
  To use this parameter, enable the HTTP connector port that is disabled while setting up a cluster. Disable the HTTP connector port again once the diagnosis is complete.

  If the nodeURL parameter is ignored, it picks CA Service Catalog URL configured with CA Process Automation.

For example:

```
http://host_name:port_no
```

For Secure Sockets Layer (SSL):

```
https://host_name:port_no
```

Return Parameter

Returns an array of the time in milliseconds taken to connect to CA Process Automation and CA Service Catalog respectively. A value of -1 is returned if there are exceptions while connecting.

Pseudocode to use CA Process Automation connectivity web service

```
URL endpoint1 = null;
AdministratorServiceSoapBindingStub adminStub = null;
endpoint1 =
new java.net.URL("http://catalog:8080/usm/services/AdministratorService");
adminStub = (AdministratorServiceSoapBindingStub) new
 AdminServiceImplServiceLocator().getAdministratorService(endpoint1);
long[] millSecs = new long[2];
try {
    millSecs = adminStub.getITPAMConnectionStatus("subton", null);
    System.out.println(String.format("Time taken for connection from Catalog to CA PAM: %d ms. and CA PAM to
 Catalog: %d ms.", millSecs[0], millSecs[1]));
} catch (Exception ex) {
    System.out.println("Connection to ITPAM failed. " + ex.getMessage());
}
```

Step 3 - Download and Run CA Remote Engineer

CA Remote Engineer collects the following product-specific critical diagnostic information and uploads it to CA Technologies:

- Log files
- Product configuration
- Hardware and software information
- A list of Microsoft installed products
- A list of CA installed products with an XML representation of the Windows registry

> **NOTE**
>
> The information in this topic is based on CA Remote Engineer Release 2.0 (the current release at publication time). If the version that you download is different, the steps can be different.

Follow these steps:

1. Download CA Remote Engineer from Broadcom Support to the CA Service Catalog computer.
2. Unzip the utility.
3. Run the re.cmd file. This file is in the RemoteEngineer folder in the location where you unzipped the utility. This file opens a command-line utility. To use the UI-based alternative instead, run the RemoteEngineer.cmd file.
4. Follow the prompts and enter the requested information.

   > **WARNING**
   > Specify *Service_Catalog* as the CA product name.

   For more information about the prompts, see the help file in the RemoteEngineer\help folder.
5. When prompted to verify the information that you entered, enter Yes or No.
   The utility proceeds as follows:
   - If you enter Yes, the utility collects the required information from your computer and zips it.
     Specify whether the utility must upload the zip file to CA Technologies through FTP. Alternatively, you can keep the zip file for reference or you can send to CA Technologies through another means. For example, email.
   - If you enter No, the utility aborts.
6. Close CA Remote Engineer.

NOTE
If you are using clustering, repeat the steps for all cluster nodes.

You have downloaded and run CA Remote Engineer.

Step 4 - JMX Client to Monitor Status

Configure the default Tomcat in CA Service Catalog to monitor the health of catalog instances with the JMX client. You can use JVM metrics to monitor the status information such as memory usage, CPU usage, and number of threads.

NOTE
This facility is not available if CA Service Catalog is deployed as web archive.

Follow these steps:

1. Enable the JMX remote monitoring without any security options. Use a free port of your choice.
2. Open the viewservice.conf wrapper configuration file from the *\view\conf* location in an editor.
3. Uncomment the following three lines in the file for the JMX remote monitor.
```
wrapper.java.additional.22=-Dcom.sun.management.jmxremote.port=9091
wrapper.java.additional.23=-Dcom.sun.management.jmxremote.ssl=false
wrapper.java.additional.24=-Dcom.sun.management.jmxremote.authenticate=false
```
4. Save and close the file.

Step 5 - Verify the Diagnostic Framework

The following results indicate a successful implementation of the diagnostic framework:

- The web service methods are called without any errors from the client.
- The web services run without errors.
- CA Remote Engineer collects product-specific data.
- The health of CA Service Catalog instances is monitored using the JMX client.

Deploy CA Service Catalog on a Custom Web Server

Deploying CA Service Catalog on a custom web server lets you use CA Service Catalog with another supported web server instead of the default Tomcat web server. Such a deployment also lets you use the same instance of the web server for multiple applications. For example, CA Service Catalog and CA RCM are installed on the same computer. You can use the same instance of the JBoss web server for CA Service Catalog and CA RCM. Such sharing helps your environment run more efficiently.

To deploy CA Service Catalog on the custom web server, follow this process:

Step 1 - Verify the Prerequisites

Verify that you have met the following prerequisites, so that you can deploy the WAR file successfully:

- Install or upgrade the following applications on the same computer:
  - CA Service Catalog
    The applications with which you want to share the instance of the custom web server as CA Service Catalog, for example, CA RCM
  - The custom web server
    You can use one of the following options as your custom web server:
    - Tomcat 6 or 7
    - JBoss 5.1

    You can use either a new or existing installation of a supported web server, one of the following options:

- A version that was supplied by installing another application. For example, CA RCM installs JBoss 5.1.
  In this case, this entire scenario applies, including the procedure to deploy the WAR file.
- A stand-alone version that you installed manually. For example, you can download a supported web server from the Apache web site or the JBoss web site.
  In this case, this entire scenario applies, *except for* the procedure to deploy the WAR file. Instead, see your web server documentation for instructions to deploy the WAR file.

- Be an experienced administrator for the web server and database that you are using with CA Service Catalog.

Step 2 - Review the Limitations

Understand the following limitations:

- You *cannot* use the same custom web server for both CA Service Catalog and CA Process Automation.
- Stop, start, or restart the web server, to stop, start, or restart CA Service Catalog. You cannot stop, start, or restart CA Service Catalog by itself. However, you can individually restart other products that use the web server.
  For example, CA Service Catalog and CA RCM use the same web server. You can restart CA RCM alone, but you cannot restart CA Service Catalog alone. To restart CA Service Catalog, you restart the web server, which restarts *both* products.
- JMX Diagnosis is not available for war deployment.

Step 3 - Create the WAR File

Create the war file from CA Service Catalog.

NOTE
The war file that is generated from a computer must be deployed on the same computer. For example, in a clustered set-up, generate the war file on each node. Now deploy the war file on the nodes.

Follow these steps:

1. Click Start, Programs, CA, CA Service Catalog Command Prompt.
2. (Optional) Run the following command if the custom web server is running on a port other than the port used for installation of CA Service Catalog (typically 8080).

   ```
   ant update-usm-host
   ```

   You are prompted to enter the new port number. Enter the port number of the custom web server.
3. Run the following command at the command prompt:

   ```
   ant create-war
   ```

4. Select one of the following options from the list:
   - **Tomcat**
     Creates the war file for Tomcat 6 or 7.
   - **JBoss**
     Creates the war file for JBoss 5.1
   - **Other**
     Do not use this option as it is reserved for future use.
     For Tomcat, the usm.war file is generated in the USM_HOME\war folder.
     For JBoss, the usm.war folder is created.
5. (Optional) Change the war file name to reflect the name of the catalog, business unit (tenant), or other meaningful entity:
   a. In Windows, manually rename the usm.war file, for example, to catalog.war or FinanceBU.war.
   b. Update the file name in the synchronization utility properties file: Specify the new name *without the .war extension* in the EIAMUser.Context=*filename* parameter in the syncuputil.properties file. This file resides in the USM_HOME folder.
   c. For example, if the new file name is FinanceBU.war, then the updated entry is EIAMUser.Context=FinanceBU.

d. Update the file name in the administration configuration settings on the UI: Log in to CA Service Catalog as a Service Delivery administrator (for example, spadmin). Then select Administration, Configuration, Server Information, Context Path, and specify the new file name.
The war file is created and available for deployment.

6. If you are migrating or upgrading CA Service Catalog, follow these steps:
   a. Apply the patch on the installed CA Service Catalog before generating the war file.
   b. Deploy the WAR file on the same web server.

Step 4 - Deploy the WAR File

Deploy the war file so that you can use the custom web server for CA Service Catalog with other applications. Using the same web server for CA Service Catalog and other applications helps your system run more efficiently. This procedure uses CA RCM for illustration. Follow this procedure as a model for deploying your WAR file with other applications.

Follow these steps:

1. Copy the usm.war folder to the *RCM_HOME*\Server\eurekify-jboss\server\eurekify\deploy folder.
   RCM_HOME is the CA RCM installation folder; the default is C:\Program Files\CA\RCM.
2. Edit the *RCM_HOME*\Server\eurekify-jboss\bin\run.bat file, as follows:
   a. Locate the *set PATH=* statement.
   b. At the end of this statement, add the path name of the database client, for example:
      - The default path name for SQL 2008 R2 is C:\Program Files\Microsoft SQL Server\100\Tools\Binn.
      - The default path name for Oracle 11g is C:\app\Administrator\product\11.2.0\dbhome_1\BIN.

Troubleshooting

After you deploy the war file with CA RCM and you cannot start it, you see the following error message:

```
Provider org.apache.xalan.xsltc.trax.TransformerFactoryImpl not found" in eurekify.log"
```

> **NOTE**
> The log files reside in this folder: C:\Program Files\CA\RCM\Server\eurekify-jboss\server\eurekify\log \eurekify.log.

If you are not able to start CA RCM, follow these steps:

1. If the xalan-x.x.x.jar and serilizer.jar files exist in the following folder, delete them.
   `RCM_HOME\Server\eurekify-jboss\server\eurekify\deploy\eurekify.war\WEB-INF\lib`
2. Copy the xalan-2.7.0.jar and serilizer.jar files from the following folder:
   `RCM_HOME\Server\eurekify-jboss\server\eurekify\deploy\usm.war\WEB-INF\lib`
   to the following folder:
   `RCM_HOME\Server\eurekify-jboss\server\eurekify\deploy\eurekify.war\WEB-INF\lib`
3. Restart the JBoss server.

Step 5 - Disable the Default Tomcat Instance

After you deploy the war file on the custom web server, stop the CA Service Catalog Windows service. Stopping the CA Service Catalog service disables the default Tomcat instance. Optionally, you can disable the service to prevent it from starting automatically.

> **NOTE**
>
> The CA Service Accounting service needs no modifications.

Step 6 - Verify the Deployment

After you deploy the custom web server, you can verify whether the deployment is successful.

Follow these steps:

1. Access the following catalog URL: http://*computer_name:port/context*
   - **Computer name**
     Defines the name of the computer on which CA Service Catalog is installed.
   - **Port**
     Defines the port number on which the web server is configured to run.
   - **Context**
     Defines the name of the war file that is created during deployment. The default is usm.war.
   - If you renamed the war file, use the new name here.
2. Verify that you can access CA Service Catalog, for example, by creating, submitting, and acting on requests.
   You have verified the deployment.

Deploy and Undeploy Components from the Command Line

This article contains the following topics:

CA Service Catalog allows you to deploy and undeploy components from the command line.

Prerequisites

Fulfill the following prerequisites before deploying:

To deploy CA EEM:

The following columns must be filled in config.properties file.

- eem.configured = true
- eiam.backend =
- eiam.application =
- eiam.admin.username = EiamAdmin
- eiam.admin.password =

To deploy the MDB:

The following columns must be filled in config.properties file.

- mdb.configured = true
  Mandatory parameters for the command line interface.
- mdb.deploy.vendor =
- mdb.deploy.host =
- mdb.deploy.port =
- mdb.deploy.user =
- mdb.deploy.password =
- mdb.deploy.dbinstance =
  Mandatory parameter specific to Oracle
- mdb.deploy.connectionid =

MDB Configuration

- mdb.deploy.dbname =
- mdb.deploy.mdbadminpwd =
- mdb.deploy.usmuser =
- mdb.deploy.usmpassword =

NOTE
For a SQL database, fill only the columns that are related to SQL.

To deploy a component:

- component.catalog = true
- component.accounting = true
- component.webserver = true
- application.service = true
- corporate.service = true
- facilities.service = true
- it.service = true
- network.service = true
- personnel.service = true
- project.service = true
- reservation.service =true

To run all the commands:

All the columns that are mentioned must be filled in config.properties. Note that the Oracle and SQL columns must not be filled at the same time. If the database is Oracle then only the Oracle columns must be filled. The same criteria holds true for SQL database.

Deploy Components

Deploy the CA Service Catalog components such as the MDB, CA EEM from the command line using the config.properties file. The config.properties file is an input file and must be located outside the USM_HOME. The file accepts passwords in plain text format only.

NOTE
You can also deploy the components from the CA Service Catalog User Interface.

Follow these steps:

1. Open the CA Service Catalog command prompt by clicking Start, Programs, CA, Service Catalog, CA Service Catalog Command Prompt.
2. Go to the Scripts path in the USM folder.
3. Enter one or more of the following commands:
   - To deploy MDB, enter the following command:
     ```
     Configurator.bat  - c mdb  - p "path to config.properties"
     ```
     For Example: C:\Program Files\ca\service catalog\scripts>Configurator.bat - c mdb - p "C:\config.properties"
   - To deploy CA EEM, enter the following command:
     ```
     Configurator.bat  - c eem  - p "path to config.properties"
     ```
     For Example: C:\Program Files\ca\service catalog\scripts>Configurator.bat - c eem - p "C:\config.properties"
   - To deploy component, enter the following command:
     ```
     Configurator.bat  - c component  - p "path to config.properties"
     ```
     For Example: C:\Program Files\ca\service catalog\scripts>Configurator.bat - c component - p "C:\config.properties"
   - To deploy all CA Service Catalog components, enter the following command:

```
Configurator.bat  - c all  - p "path to config.properties"
```

For Example: C:\Program Files\ca\service catalog\scripts>Configurator.bat - c all - p "C:\config.properties"

WARNING
Stop and start the tomcat service for the changes to take effect.

You have deployed the CA Service Catalog components using the command line.

Undeploy Components

CA Service Catalog allows you to undeploy the MDB, CA EEM, and other components individually from the command line. Undeploy these components in reverse order.

Follow these steps:

1. Open the CA Service Catalog command prompt by clicking Start, Programs, CA, Service Catalog, CA Service Catalog Command Prompt.
2. Go to the Scripts path in the USM folder.
3. Enter one or more of the following commands:
   - To undeploy all CA Service Catalog components, enter the following command:
     ```
     Configurator.bat  - u all  - p "path to config.properties"
     ```
     For Example: C:\Program Files\ca\service catalog\scripts>Configurator.bat - u all - p "C:\config.properties"
   - To undeploy component, enter the following command:
     ```
     Configurator.bat  - u component  - p "path to config.properties"
     ```
     For Example: C:\Program Files\ca\service catalog\scripts>Configurator.bat - u component - p "C:\config.properties"
   - To undeploy MDB, enter the following command:
     ```
     Configurator.bat  - u mdb  - p "path to config.properties"
     ```
     For Example: C:\Program Files\ca\service catalog\scripts>Configurator.bat - u mdb - p "C:\config.properties"
   - To undeploy CA EEM, enter the following command:
     ```
     Configurator.bat  - u eem  - p "path to config.properties"
     ```
     For Example: C:\Program Files\ca\service catalog\scripts>Configurator.bat - u eem - p "C:\config.properties"

     WARNING
     Stop and start the tomcat service for the changes to take effect.

You have undeployed the CA Service Catalog components using the command line.

Perform Maintenance

You can perform the following maintenance tasks:

- CA Service Catalog files are updated during use. Backup such files regularly. You can perform maintenance by backing up files regularly that are updated when you use CA Service Catalog. You can change the name of of the host computer running CA EEM for security compliance. You can also change the names of the applications whose access control you manage through CA EEM. To change these names, you run the setup utility on every computer that has a CA Service Catalog component.
- You can also change the database settings like host name, port number, instance name, or service name. CA Service Catalog components use the database settings for communication.
- You can also use the operating system (not the ant update-usm-host command) to change the host name. You can make use of the ant command to propagate this change for catalog systems.
- Run the setup utility to change the password of the database user, to comply with security, governance, or other requirements. CA Service Catalog components use the logon credentials (user name and password) of the database user to access the database.

Perform the following maintenance tasks:

- Files to Back Up Regularly
- Update the CA EEM Host Name and Application Names
- Update the Database Host, Password, Instance, Service Name, or Port
- Update the Host Name and Port Number Using the ant Command
- Update the Password of the Database User

Files to Back Up Regularly

Several CA Service Catalog files are updated during use. Back up such files regularly.

The following table identifies the features that use the files on the Catalog Component server and their locations:

Feature	Folder
Request Attachments	USM_HOME\filestore\documents\requests
CA Service Catalog Images for services	USM_HOME\filestore\images\offerings
Documents	USM_HOME\filestore\documents
CA Service Catalog Forms	USM_HOME\filestore\forms
Data Objects and Data Views	USM_HOME\filestore\reporting
Offline Data Objects and Data View	USM_HOME\filestore\reporting\offline
Custom files for CA Service Catalog, including custom images, data fields for business units, accounts, and users. You can optionally store these files in a filestore.	USM_HOME\filestore\custom **Example:** Suppose you customized data fields for business units, accounts, and users. Sample folder name: USM_HOME\filestore\custom\locale\icusen (for English). Suppose you customized images for service option groups. Sample folder name: USM_HOME\filestore\custom\images\rateplans
Data Mediation	USM_HOME\filestore\DataMediation
CA Service Repository Agent files (also known as Data Mediation Data Repository Agent)	USM_HOME\repagent. This folder contains these sub folders: \conf, \log, and \data

Administrators who have installed CA Service Catalog on multiple computers must keep files synchronized on all servers. Otherwise, an error can occur.

The following table identifies the features that use files on multiple computers and the folders where those files are located:

Feature	Folder
Attachments	USM_HOME\view\documents
CA Service Catalog Images	USM_HOME\FileStore\images\offerings
Documents	USM_HOME\view\documents
CA Service Catalog Forms	USM_HOME\view\forms

Update the CA EEM Host Name and Application Names

To comply with security, governance, or other requirements, you can change the name of the host computer running CA EEM. You can also change the names of the applications whose access control you manage through CA EEM. To change these names, you run the setup utility on *every* computer that has a CA Service Catalog component. This utility updates these names globally to ensure that CA Service Catalog, CA EEM, and the integrated products run efficiently.

The *application names* are the CA EEM policies for managing user access control and other resources to these products. The application name *Service Delivery* in CA EEM refers to the policies governing user access control in CA Service Catalog. CA EEM includes a *Global* application name that you can apply in part or in whole to one or more products.

Follow these steps:

1. Enter the following URL in your browser for the CA Service Catalog installation that you want to update:

   `http://localhost:portnumber/usm/config`
2. Enter the password to log in to the setup utility.
   The setup utility opens on the Database tab.
3. Click the Security tab, and specify new values for the settings that you want to change, as follows:
   – The host name of the CA EEM computer.
   If you change the host name, a “Backup and restore” option appears. To back up the CA EEM data on the current computer and restore it on the new computer, select this option.
   If you select the Backup and restore option, specify the password of the administrator on the old CA EEM computer.
   – The name of the CA EEM application instance.
   – The password of the CA EEM application instance.
4. Save your changes.
5. Restart the Windows services for Accounting Component and CA Service Catalog to make your updates take effect.
6. Make the same updates on all remaining CA Service Catalog computers in your environment.
7. Verify that CA Service Catalog is running correctly, as follows:
   a. Log in to CA Service Catalog.
   b. Perform administrative tasks. Create Services. Also, create and process requests.

The new or updated host name and application names are updated in the Windows registry.

NOTE

As a best practice, back up the Windows registry after you update the host name and application names. For more information, see your Windows documentation.

Update the Database Host, Password, Instance, Service Name, or Port

To comply with changes in your environment or for other reasons, you can change the following database settings: host name, port number, instance name, or service name. CA Service Catalog components use these settings to communicate with each other and the database.

Follow these steps:

1. Perform the following prerequisite tasks:
   a. To update the database host name, update the operating system computer name of the computer which has the database. For more information, see your operating system documentation.
   b. Update the database settings in your DBMS software (Oracle or SQL Server). For more information, see your DBMS documentation.
   c. Verify that the updated DBMS is accessible from every DBMS client on every CA Service Catalog computer. Use your DBMS client software (not CA Service Catalog) to perform this step.
2. Enter the following URL in your browser for the CA Service Catalog installation that you want to update:

   `http://localhost:portnumber/usm/config`
3. Enter the password to log in to the setup utility.
   The setup utility opens on the Database tab.
4. Enter the new password for the database user in the Application User Settings section only if *both* of the following conditions exist.

- You are using SQL Server.
- The database was backed up and restored on a new computer.

Confirm the password and save your changes.

5. (Optional) When you are prompted for your password, specify your existing password to continue using it.
6. Specify new values for the settings that you want to change, as follows:
   SQL Server
   If applicable update the instance name for the MDB.
   Default: MSSQLSERVER
   Oracle
   If applicable, update the service name. Every Oracle database or service has a service name. The service name of an Oracle database is typically its global database name. Enter the service name of the Oracle database or other service that you want to access.
7. If applicable, update the port number.
8. Save your changes.
9. Restart the Windows services for CA Service Catalog to make your updates take effect.
10. Make the same updates on all remaining CA Service Catalog computers in your environment.
11. Verify that CA Service Catalog is running correctly, as follows:
    a. Log in to CA Service Catalog.
    b. Perform administrative tasks. Create services. Also, create and process requests.

Update the Host Name and Port Number Using the ant Command

If applicable, you use the operating system (not the ant update-usm-host command) to change the host name. Afterwards, you use the ant command to propagate that host name change throughout the catalog system. Use the ant command to both update the port number directly *and* propagate that change throughout the catalog system.

You perform these steps *only* on the computer whose host name or port number has changed. You do not perform these steps on *other* computers.

Follow these steps:

1. Stop the CA Service Catalog services on the computer being updated.

   WARNING
   Stop these services on *all* CA Service Catalog computers in your environment *before* proceeding to the next step. Use the Windows Control Panel to stop the services.
2. Perform the following actions:
   a. Open the CA Service Catalog command prompt from the CA Service Catalog portion of the Windows Start menu.
   b. Enter the following command at the CA Service Catalog command prompt:

      ```
      ant update-usm-host
      ```

      NOTE
      For a list of ant commands and their descriptions, enter ant -p.

      The utility displays the current settings for the host name and port number.
   c. Update each setting that you want to change, and record all updated settings for reference.

   The command utility performs the following actions:
   - Updates the host name and port number settings in your DBMS (Oracle or SQL Server) and in configuration files
   - Records the actions that it performs in the maintenance.log file.
     This log file and the backed-up configuration files are stored in the USM_HOME\conf-backup*date-time* folder. The name of the *date-time* subfolder is based on when the ant update-usm-host command was run. Thus, this subfolder name is different on each computer in the environment. The utility writes the name of the subfolder to the screen as soon as the backup is completed.

3. Perform the following actions:
   a. View the maintenance.log file in the USM_HOME\conf-backup*date-time* folder for a record of the actions performed.
   b. Record the name of this folder for future reference.
4. Verify that you have finished updating the host name and port number on the current computer. Update the host name and port number on the next computer, if necessary.
5. Verify that you have updated the required settings on *all* CA Service Catalog computers.
6. Log in to CA Service Catalog.
7. Verify that CA Service Catalog is running correctly, as follows:
   a. Log in to CA Service Catalog.
   b. Perform administrative tasks. Create services. Create and process requests.

The new or updated host name and application names are updated in the Windows registry.

Update the Password of the Database User

You can run the setup utility to change the password of the database user, to comply with security, governance, or other requirements. CA Service Catalog components use the logon credentials (user name and password) of the database user to access the database. The setup utility updates these password references globally to help CA Service Catalog continue to run efficiently throughout your environment. You run the setup utility on *every* computer in your environment that has a CA Service Catalog component installed.

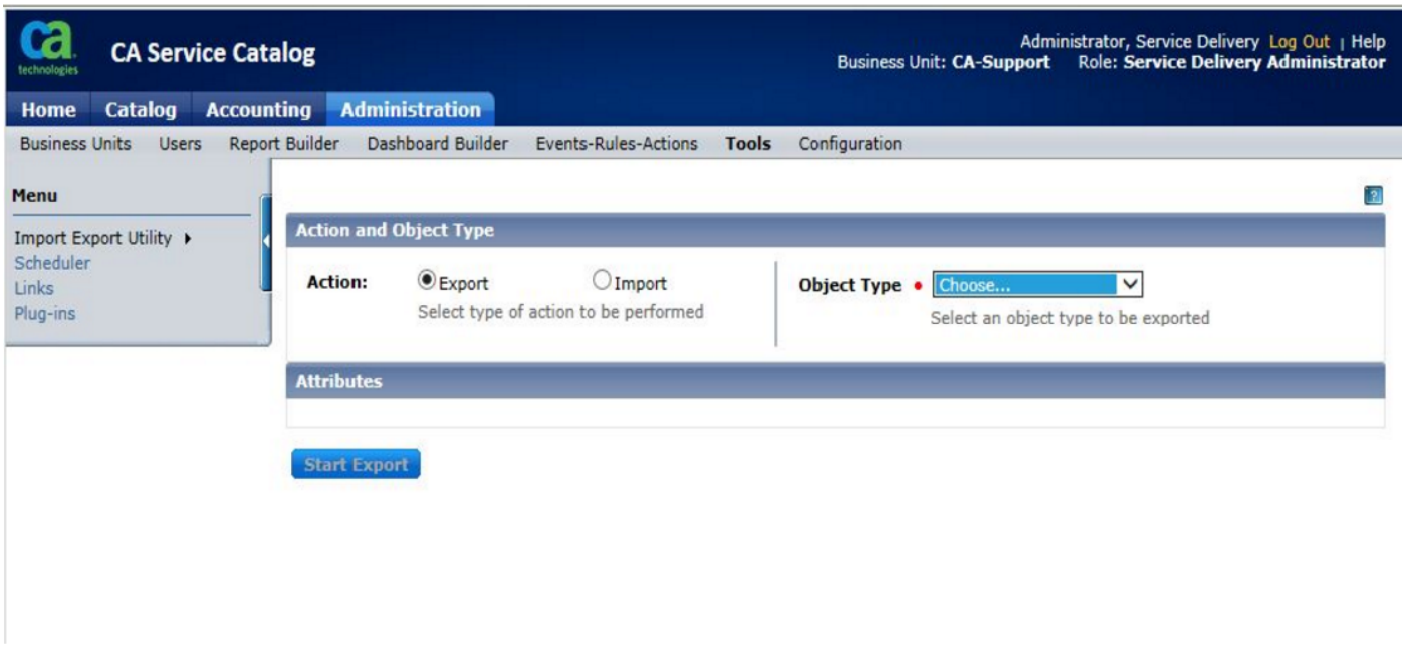
Follow these steps:

1. Start the setup utility by entering the following URL in your browser for the CA Service Catalog installation that you want to update:
   ```
   http://localhost:portnumber/usm/config
   ```
2. Enter the password to log in to the setup utility.
   The setup utility opens on the Database tab.
3. Enter the new password for the database user in the Application User Settings section. Confirm the password and save your changes.
4. Restart the Windows services for CA Service Catalog to make your updates take effect.
5. Update the password on all remaining CA Service Catalog computers in your environment.
6. Verify that CA Service Catalog is running correctly, as follows:
   a. Log in to CA Service Catalog.
   b. Create services, create and process requests, and perform administrative tasks.

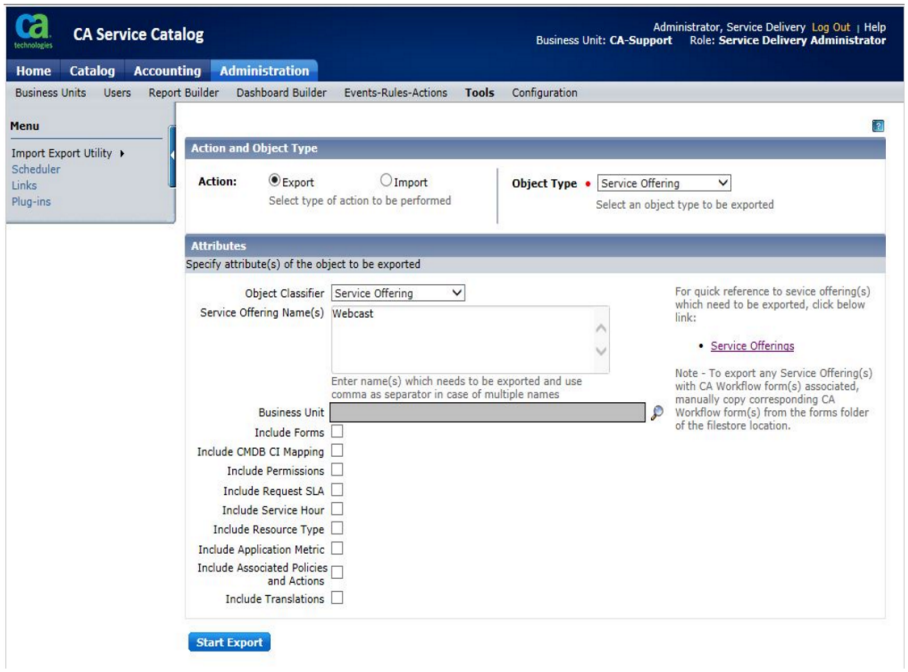
Import and Export Offering in CA Service Catalog

Importing or Exporting a CA Service Catalog Offering from one environment to another causes the offering to be placed in a non-categorized folder (Root folder) in CA Service Catalog. To move the offering and place it in a folder of your choice, perform the following steps:

1. Log in to CA **Service** Catalog as an Administrator. Navigate to **Administration, Tools, Import Export Utility**. Select Action and Object Type as **Export**.
2. Click the **Object type** to export an Event, Form, or Service offering xml separately.



3. Select the **service offering** in object type and Object Classifier. Provide the service Offering name that needs to be exported.
4. Click **Start Export** button to export the selected offering.



5. Repeat **step 1** for exporting **Event/Forms** by changing the Object type, the Form Name, and Form Folder. The xml file is exported as defined.
6. To Import Event/Form/Service XML in the same order, select the **Service Offering** for Object type and select the xml file to import the related offering links, that is, E**vent and Forms**.

WARNING

Event / Form xml's must be imported before the offerings XML

By default the offerings are linked with IDs and not by the offering name created in the root structure.

7. Select the non-categorized offering, select the cut (scissors) icon to cut the offering from the menu.
8. Select the folder and click the **paste** icon to move this offering to the selected folder.
   The selected non-categorized Offering is moved to the categories folder.

Enable SAML Authentication for Service Catalog

Security Assertion Markup Language (SAML) is a standard for logging users into applications based on their sessions in another context. SAML SSO works by transferring the user's identity from one place (the identity provider) to another (the service provider). This is done through an exchange of digitally signed XML documents.

This article explains how to enable SAML for CA Service Catalog Tomcat Servers on Active Directory Federation Services (ADFS) and CA Single Sign-On (Siteminder):

Prerequisites

Ensure that SSL is enabled for CA Service Catalog. For more information see, Enable SSL Authentication for CA SM 17.4.

Prerequisites for Enabling SAML on ADFS

If you want to use ADFS as the identity provider, you must install ADFS on Windows Server 2012 R2 and above. After completing the installation of ADFS and CA Service Catalog, you must configure the ADFS with the following information:

1. While configuring Add Relying Party Trust on ADFS specify the following on the "**Configure URL**" screen:
   a. Select the "**Enable support for the WS-Federation Passive protocol**" check-box.
   b. Enter the value in the "**Relying party WS-Federation Passive protocol URL:**" as:

      ```
      https://<catalog hostname>:<https port>/usm/wpf
      ```

   c. Select the "**Enable support for the SAML 2.0 WebSSO protocol**" check-box.
   d. Enter the value in the "**Relying party SAML 2.0 SSO service URL:**" as:

      ```
      https://<catalog hostname>:<https port>/usm/wpf
      ```

2. While adding claim rules, specify the following in the "**Configure Claim Rule**" step:
   a. **Claim rule name**: Specify a name for the claim rule. For example, **Catalog Claim Rule**.
   b. **Attribute store**: Select the datasource from the drop-down list. For example, **Active Directory**.
   c. **LDAP Attribute**: Select "**SAM-Account-Name**" from the drop-down list.
   d. **Outgoing Claim Type**: Select "**Name**" from the drop-down list.

Prerequisites for Enabling SAML on CA Single Sign-On (Siteminder)

If you want to use CA Single Sign-On (formerly known as CA Siteminder), as your Identify Provider, perform the following:

1. Launch CA Single Sign-on as Administrator.
2. Go to the Federation Partnership Partnership Federation and click Partnerships.
3. Click the Federation Partnership name from the Federation Partnership List that you want to use.
4. Expand the IP Restrictions section and verify the Assertion Attributes properties on the Partnerships properties page.
5. The value of the Namespace field must be the URI for the respective Claim.
   If the value is specified as "**Unspecified**" the SAML authentication will not work.
6. Verify that the certificate to configure the Identity Provider is **SHA 256 signed**.

Enable SAML Authentication for CA Service Catalog Microsoft Azure Idp
Perform the following steps:

1. Login to Microsoft Azure Portal.

2. Click on the "**Azure Active Directory**" link under Azure Services.
3. Click on the "**Enterprise Applications**" link from the left pane.
4. Click "**New Application**" link and again click "**Create your own application**" link.
5. Enter name example "**CASM tomcat**" in the "**What's the name of your App**?" text box and Click **Create** on the right-pane.
6. Click on the "**2. Set up single sign on**" link and Click on the "**SAML**" link.
7. Click on the Edit button on section 1 , i.e.,"Basic SAML Configuration".
8. Enter CA Service Catalog SSL tomcat URL like ***https://<slcmhost>:<slcm_ssl_port>:/usm/wpf*** for all below text fields:
   a. "**Identifier (Entity ID)**"
   b. "**Reply URL (Assertion Consumer Service UR**L"
   c. "**Sign on URL**"
   d. "**Relay State**"
   e. "**Logout Url**"
9. Click **Save**.

Enable SAML Authentication for CA Service Catalog Tomcat Web Servers

Perform the following steps on all CA Service Catalog Tomcat Web Servers:

1. Ensure that the following information is present in the **web.xml** file located where you have installed CA Service Catalog/Tomcat Web Server:
   For example: **C:\Program Files\CA\Service Catalog\view\webapps\usm\WEB-INF\web.xml**
2. Verify configuration under the **<!-- Add filter here -->** statement:

```
<filter>
 <filter-name>FederationFilter</filter-name>
 <filter-class>com.auth10.federation.WSFederationFilter</filter-class>
 <init-param>
 <param-name>login-page-url</param-name>
 <param-value>main.jsp</param-value>
 </init-param>
 <init-param>
 <param-name>exclude-urls-regex</param-name>
 <param-value>/images/|/js/|/css/</param-value>
 </init-param>
</filter>
```

3. Update the values for the parameter name "cors.allowed.origins" with SAML Identity provider host name.
4. Search the parameter name "cors.allowed.origins" and update the respective parameter value as shown here.

```
<param-name>cors.allowed.origins</param-name>
<param-value>,SAML Identity Provider Host</param-value>
```

5. Search the parameter name "cors.support.credentials" and update the respective parameter value to true as shown here.

```
<param-name>cors.support.credentials</param-name>
<param-value>true</param-value>
```

6. Close the **web.xml** file.
7. Restart CA Service Catalog Services.

Configure SAML from CA Service Catalog User Interface

Perform the following for configuring SAML on ADFS or Microsoft Azure Idp or CA SSO:

Configure SAML for ADFS or Microsoft Azure Idp

Configure the SAML Parameters in CA Service Catalog User Interface for ADFS or Microsoft Azure Idp as following:

1. Ensure that you have followed the steps shown in Enable SAML Authentication for CA Service Catalog Tomcat Web Servers.
2. Login as spadmin or CA Service Catalog Administrator.
3. Navigate to **Administrator**, **Configurations**, **Single-Sign On Authentication** and configure the following:

Parameters	Description
Single Sign-On Type	Change the Single-Sign On Type to **SAML**.
SAML-Federation Audienceuris	Define the URI of the application from where to accept the tokens. For example: `https://<catalog hostname>:<catalog-Tomcat-SSL-Port>/usm/wpf|https://<catalog hostname>:<catalog-Tomcat-SSL-Port>/usm/wpf`
SAML-Federation EnableManualRedirect	false
SAML-Federation Realm	Define the location from where the tokens are sent. For example: `https://<catalog hostname>:<catalog-Tomcat-SSL-Port>/usm/wpf`
SAML-Federation Reply	Define the URL of the location that receives responses. `https://<catalog hostname>:<catalog-Tomcat-SSL-Port>/usm/wpf`
SAML-Federation Trustedissuers Friendlyname	ADFS: Specify a common name for the Identity Provider. For example: `sun06_cert - <ADFS hostname>-I6029.sun06.local.com` **Microsoft Azure Idp:** Perform the following: 1. Login to Microsoft Azure portal, navigate to **Enterprise Application**, <**CASM Application Name**>, Single-Sign-On, Block/Section 4, Azure ID Identifier (use the copy option to copy the value). For example: *https://sts.windows.net/<tenant_id>/* 2. Replace **saml2** with **wsfed** in the URL. For example: *https://login.microsoftonline.com/<microsoft_tenentid>/wsfed*

SAML-Federation Trustedissuers Issuer	ADFS: Specify the Identity Provider URL: For example: `https://<trusted_issuer_URL>/<identity_provider>/ls/idpinitiatedsignon.aspx` Azure Idp: Perform the following: 1. Login to Microsoft Azure portal, navigate to **Enterprise Application**, <**CASM Application Name**>, Single-Sign-On, Block/Section 4, Login URL. Copy the URL (use the copy option adjacent to the URL). For example: *https://login.microsoftonline.com/<tenant_ID>/saml2* 2. Replace **saml2** with **wsfed** in the URL. For example: *https://login.microsoftonline.com/<microsoft_tenentid>/wsfed*
SAML-Federation Trustedissuers Thumbprint	ADFS: Define the value of the certificate thumbprint provided by the Identity Provider. You must specify the thumbprint from "**token-signing"** section of the certificate. For example: `375181a915d1e699ef8e8e47c20d550be9dda024` Azure Idp: federation.trustedissuers.thumbprint Perform the following: 1. Login to Microsoft Azure portal, navigate to **Enterprise Application**, <**CASM Application Name**>, Single-Sign-On, Block/Section 3, Thumbprint. Copy the value. For example: 7844FB57B8EB854C10F2CBD8BE78B0BF574B729D

Configure SAML for CA Single-Sign On

Perform the following steps:

1. Ensure that you have completed the steps shown in Enable SAML Authentication for CA Service Catalog Tomcat Web Servers
2. Login as spadmin or CA Service Catalog Administrator.
3. Navigate to **Administrator**, **Configurations**, **Single-Sign On Authentication** and configure the following:

Parameters	Description
Single-Sign on Type	Change the Single-Sign On Type to **SAML**.
SAML-Federation Audienceuris	Define the URI of the application from where to accept the tokens: `SAML-Federation Audienceuris-https://<catalog hostname>:<catalog-Tomcat-SSL-Port>/usm/wpf|https://<catalog hostname>:<catalog-Tomcat-SSL-Port>/usm/wpf`
SAML-Federation EnableManualRedirect	false

SAML-Federation Realm	Define the location from where the tokens are sent. For example: `https://<catalog hostname>:<catalog-Tomcat-SSL-Port>/usm/wpf`
SAML-Federation Reply	Define the URL of the location that receives responses. `https://<catalog hostname>:<catalog-Tomcat-SSL-Port>/usm/wpf`
SAML-Federation Trustedissuers Friendlyname	Define a common name for the Identity Provider: `ipslcm`
SAML-Federation Trustedissuers Issuer	Define the Identify Provider URL: `http://<siteminderURL>/affwebservices/public/wsfeddispatcher`
SAML-Federation Trustedissuers Thumbprint	Define the value of the certificate thumbprint provided by the Identity Provider. You must specify the thumbprint from "**token-signing"** section of the certificate. For example: `ED516ED2AE632E15A52BE0C4D1BCFDDA73B9A146`

Troubleshooting

If you provide incorrect data or values for SAML Properties in the Catalog UI as shown here, you will not be able to login to the CA Service Catalog Interface. To troubleshoot this, perform the following steps:

1. Navigate to where you have installed CA Service Catalog Tomcat Web server and locate the web.xml file.
   For example:
   C:\Program Files\CA\Service Catalog\view\webapps\usm\WEB-INF\web.xml
2. Open web.xml and locate ***<FederationFilter>*** under ***<!---SAML Auth Filter-->*** and comment the entire filter section in the following statement :

```
<filter>
  <filter-name>FederationFilter</filter-name>
  <filter-class>com.auth10.federation.WSFederationFilter</filter-class>
  <init-param>
    <param-name>login-page-url</param-name>
    <param-value>main.jsp</param-value>
  </init-param>
  <init-param>
    <param-name>exclude-urls-regex</param-name>
    <param-value>/images/|/js/|/css/</param-value>
  </init-param>
</filter>
```

3. In web.xml, locate ***<FederationFilter>*** under the ***<!—SAML Filter Mapping---->*** and comment ***<filter-mapping>*** :

```
<filter-mapping>
  <filter-name>FederationFilter</filter-name>
  <url-pattern>/*</url-pattern>
</filter-mapping>
```

4. Close the **web.xml** file.
5. Restart CA Service Catalog Services.

Create Users and Services

Create users and services after installing CA Service Catalog.

Once you have installed CA Service Catalog:

- Set up users, user groups, business units, and accounts.
- Create and customize services that users can request from the catalog.
- Configure the processes for managing, approving, and billing for requested services.

For more information about how to perform these tasks, see Service Accounting.

> **NOTE**
> Do not add users, delete users, or change user information using CA EEM. We recommend that you use CA Service Catalog for managing users. CA EEM is then updated accordingly.

Install and Integrate Additional Process Automation Tools

Install and integrate additional process automation tools for CA service catalog.

For best performance, CA Service Catalog requires a process automation tool. You can use CA Process Automation to automate processes in CA Service Catalog. Even though you can *install* CA Process Automation at any time, we recommend that you install CA Process Automation with CA Service Catalog using the CA Service Management Installer.

> **WARNING**
> If you install two or more instances of CA Service Catalog, implement clustering *before* you integrate CA Service Catalog with CA Process Automation.

Upgrade JRE (AdoptOpenJDK) 11.0.18 or later

Upgrade JRE (AdoptOpenJDK) 11.0.18 or later for CA Service Catalog

The CA Service Catalog installation program automatically installs the Java Runtime Environment (JRE). We recommend that you use the JRE version 11.0.18 that is installed by CA Service Catalog. You can configure CA Service Catalog to use or replace the JRE version, if required, as follows. **Follow these steps:**

1. Upgrade JRE version 11.0.18 or later from AdoptOpenJDK or one of its affiliated sites. Select **Windows**, **x64-bit** and download JRE zip file. Copy the zip file to the Service Catalog server and extract it.
2. Launch Service Catalog command prompt from the installed location of Service Catalog (USM_HOME\usm.cmd). Enter the following command:

   ```
   ant upgrade-jre
   ```

   > **NOTE**
   > Ignore "unable to locate**tools.jar"** warning message.
3. Stop CA Service Catalog services, when prompted.
4. Enter folder path where you have extracted JRE:
   For example: C:\OpenJDK11U-jre_x64_windows_hotspot_11.0.20
5. Enter JRE version number. For example, 11.0.18.
6. After successful upgrade, restart all CA Service Catalog services.

(Recommended) Efficient Tomcat Access Log Rotatable

Recommended Tomcat Access Log Rotatable for CA Service Catalog.

It is recommended that you change the Tomcat Rotatable Access to *True*. This is to prevent considerable or huge growth of Tomcat access log file and to prevent eventual slowdown of Tomcat Web Server.Perform the following steps:

1. Navigate to *%USM_HOME%\view\conf\ location* and *o*pen the server.xml file in Edit as an Administrator.
2. Search for:
   ```
   prefix="tomcat_view" suffix=".log" pattern="common" resolveHosts="false" rotatable="false"/>
   ```
3. Replace it to:
   ```
   prefix="tomcat_view" suffix=".log" pattern="common" resolveHosts="false" rotatable="true"/>
   ```
4. Save and close the file.
5. Restart the CA Service Catalog Services.

(Recommended) Enhance Security

Enhance security for CA Service Catalog.

To enhance security in your CA Service Catalog implementation, *consider* making the following configuration changes:

- Disable the Apache JServ Protocol (AJP) port, port 8009 while performing the initial setup, if you are *not* implementing clustering.
  To disable AJP, edit the USM_HOME\view\conf\server.xml file and verify that the AJP tags are commented out.
- Reduce the timeout of CA Service Catalog user sessions. By default, sessions time out after 60 minutes of inactivity.
  To reduce the timeout, log in to CA Service Catalog, click Administration, Configuration, User Default. Adjust the **Session Timeout** parameter.
- Configure the CA EEM password policies to be more secure, if CA EEM is *not* configured to use an external directory. Specifically, consider locking user accounts after three to five failed login attempts. To set this value, log in to CA EEM and click Configure, EEM Server, Password Policies.
- Update the list of roles that can run web services. By default, only the Certificate user and users with the service provider (SP) administrator role can run web services.
  To change this list, log in to CA EEM with the Application set to Service Catalog. Click Manage Access Policies, Policies, Access Policies, USM_Resource. Edit the policy whose permissions you want to update, and add the resource that is named usm_webservice__all to that policy.

  NOTE
  For more information about editing these policies, see your CA EEM documentation.
- Enable Secure Socket Layer (SSL) for web services so that passwords are not sent in plain text when you use the log In (String, String, String) method. If SSL is not available, consider using the logIn Token(String) method instead. This method takes a CA EEM artifact as a parameter and is encrypted.
- Install antivirus software on the filestore computer, if you are using a filestore (a single location for shared files). We recommend that you use a filestore.
- Harden CA Service Catalog computers.
  Hardening is the process of securing a computer by removing or disabling components or access points, to render the computer less vulnerable to outside attacks. Hardening can include disabling all ports on a computer initially and afterwards manually enabling individual ports as needed. Other basic hardening steps include the following: Limit the number of users permitted access to a computer, strengthen password and access control, install intrusion-detection software, and close ports.

(Recommended) Set Up Single Location for Shared Files

Set Up Single Location for Shared Files for CA Service Catalog.

If you have installed Catalog Component on multiple computers (either clustered or non-clustered), we recommend that you set up a single location for shared files. Shared files can include documents, reports, images of services, data mediation files, customizations, and forms.

By default, the location for shared files is the **USM_HOME\filestore** folder on *every* Catalog Component computer. This folder contains several subfolders. However, for optimal efficiency, you can specify *one* location on a single computer that all Catalog Component computers share. This single location is named the *central filestore* or *filestore*. The computer on which the filestore resides can have Catalog Component installed. However, Catalog Component is not required on that computer.

If you have installed Catalog Component on multiple computers and you do not set up a filestore, then verify that the individual filestores on all Catalog Component computers is synchronized.

If you have installed Catalog Component on a single computer, this entire process does not apply, so you can skip it.

Modify the Plugin Properties File

In the Service Catalog, you can add your own plugins. The plugin properties file is in the fileStore. You can access any file from the filestore by accessing an URL: *http://<localhost>:8080/usm/FileStore/custom/locale/icusen/forms/plugin.properties*

The plugin properties file contains sensitive information like database, passwords, and so on. If you want to restrict access to any such sensitive information, you can modify the **ESAPI.properties** file. To enable restriction regarding file browsing, set the **ISENABLED** value to **true** for the following. The default value is **false**.

- *filestroreValidation.ISENABLED* = **True**
- *filestroreValidation.FILEPATTREN=*
  For file pattern, you can provide values like properties, xml, or Java.
  - To block read access to properties files, configure the following:
    ```
    filestroreValidation.FILEPATTREN=properties
    ```
  - To block read access to xml files, configure the following:
    ```
    filestroreValidation.FILEPATTREN=xml
    ```

    NOTE

    You can block only one type of file. You cannot block a Java file if you have already blocked an XML file.
- *filestroreValidation.FILEPATH=*
  If you want to prevent access to any file from a particular folder, you can configure this attribute. For example, if you want to prevent access to any file in the *USMHOME/FileStore/custom/locale/icusen/forms* folder, configure the following:
  ```
  //custom//locale//icusen//forms
  ```
- *filestroreValidation.FILENAME=*
  To prevent access to any specific file with or without a parent folder, you can configure this attribute:
  ```
  filestroreValidation.FILENAME= custom_form.xml
  ```

Retain the Default Location for Shared Files

Setting up a single location for shared files helps improve the accuracy and efficiency of sharing files between computers.

Follow these steps:

1. Verify that all computers on which CA Service Catalog is installed have a trusted domain relationship. This trusted relationship enables user accounts and global groups to be used in a domain other than the domain where the accounts are defined.

2. Start the CA Service Catalog service with the login credentials of the Windows user that has read/write access to the shared location. If necessary, change the login credentials for the CA Service Catalog service to meet these requirements, as follows:
   a. Select **Administrative Tools**, **Services**.
   b. Right-click the service, click **Properties**, click the **Log On tab**, and enter the login credentials.
   c. Save the changes and restart the service.
3. Share the U**SM_HOME\filestore** folder on the first CA Service Catalog computer as the filestore for all CA Service Catalog computers.

Set Up a Custom Location for Shared Files

You can create a custom location for shared files.

Follow these steps:

1. Share the folder to use as the filestore.
2. Verify that the Windows operating system users who are updating the filestore have read/write access to this folder.
3. Use the UNC path in the format *computer-name**folder-name* to specify the location of the filestore.
4. Start the Catalog Component service with the logon credentials of the Windows user who requires access to the folder.
5. Select Administration, Configuration, Filestore Information.
6. Perform the following steps:
   a. Click the **Edit** icon for the Filestore Location variable.
   b. In the Filestore Location field, specify the UNC path name of the shared drive you defined in a previous step, for example: \\big-computer\Shared_USM\filestore or \\big-computer\filestore.
   c. Click **Update Configuration**.
   d. Click **Test** to verify the validity of the share.
   This test returns a successful connection test message if the filestore can be used to store files that are uploaded by users.

      WARNING
      Testing the filestore is mandatory.
7. Perform the action that applies:
   – If the test succeeds, copy the entire contents of the **USM_HOME\filestore** folder to the new location.
   – If the test fails, reconfigure the share. Also, verify that all the CA Service Catalog services that are accessing the share have the same, valid credentials.
8. Recycle all CA Service Catalog services on all computers.

Configuring CA Asset Portfolio Management

This section contains the following articles:

- Configurations
- Page Configuration by Asset Families and Legal Templates
- How to Configure the User Interface

WARNING

Any “modifications” or “adaptions” or “configurations” that are done administratively through the interface (web browser, command-line, Web Screen Painter) are “supported”, meaning Broadcom Support can assist with basic suggestions and trouble-shooting. Broadcom Support does not perform configuration or customization changes for the customer, they are the customer’s responsibility. For example, adding a field to a table and putting the field on a form through Web Screen Painter is a fully supported “modification”. Another example, installing or uninstalling a feature through the Options Manager administration is a fully supported “configuration”.

Anything to do with SPEL code, Java scripting (or any language scripting), or a customer-specific change to the underlying base code-line (done by Broadcom Services or a Partner), is not supported by Broadcom Support. The customer can do these things, but the customer is responsible for the support, maintenance, and troubleshooting when things go wrong. If such “customizations” affect expected OOTB (Out-of-the-Box) behavior, Broadcom Support will ask the customer to remove the customizations and see if the behavior persists.

Configurations

As an administrator, you can configure the user interface to simplify how users enter, manage, and search for data. You can also protect users from performing unauthorized tasks, ensure that you conform to your IT asset management practices. When you configure the user interface, you, and all users affected by your configuration changes, immediately see the changes. For example, you do not want anyone, except the asset manager, to see any information for sites and companies. Therefore, you hide the Site and Company tabs and specify that only users in the asset manager role can see those tabs.

When configuring the user interface, use the following types of configurations:

- **Global configuration**. Configure the product for all users, regardless of their role.
  A *global configuration* lets you modify the functionality of your product implementation. You focus on configuring the pages, objects, fields for your specific implementation. You need not worry about making configuration changes for all possible users and roles.
  For example, you do not want to use the contact management functionality. Therefore you define a global configuration on the Contact page, hide this functionality from all users, and save the global configuration. As a result, users do not see the Contact page, unless you define a local configuration to override the global configuration.
  A global configuration can apply to all asset families or legal templates or to a specific family or template. You can have only one global configuration that applies to all families or templates or that applies to a specific family or template.

  NOTE
  If you do not want to make any global user interface changes for your implementation, you do not have to define a global configuration. You can define a local configuration without defining a global configuration.
- **Local configuration**. Configure the product for specific users and roles.
  Use a *local configuration* to configure the user interface pages based on the requirements of the different users and roles.

  NOTE
  Local configuration changes override global configuration changes.

  For example, you define a global configuration to hide the contact management functionality in your implementation. However, there are users in a specific role who must be able to see and update contact information. Therefore, you define a local configuration on the Contact page, make the contact information appear, and save the local configuration. When you assign the local configuration to the users in the role, they see the contact information.
  A local configuration can apply to all asset families or legal templates or to a specific family or template. You can have multiple local configurations that apply to all families or templates or that apply to a specific family or template. You can assign to a role only one local configuration that applies to all families or templates or that applies to a specific family or template.

  WARNING
  You cannot assign a global configuration to a role. By default, a global configuration is assigned to all roles when the users log in and the security permissions for the roles are determined. You can only assign a local configuration to a role.

A global configuration is always assigned to all roles, even when you assign the role to a local configuration. Any permission from the global configuration that a local configuration does not override is applied to the role. Global configurations are used to configure the product for every user (except for the system administrator role, *uapmadmin*). To configure the user interface on a more detailed level by role, add a local configuration. In this situation, all users in all roles

will see the configured interface based on the global configuration changes, and users assigned to a local configuration will see the additional changes to the interface.

If a user is assigned to a local configuration, the local configuration is assigned to the role when the user logs in.

Page Configuration by Asset Families and Legal Templates

This article contains the following topics:

For most objects, you configure the user interface, save the configuration, and all users in the role that is assigned to the selected configuration see the page that way. No additional configuration options are available. For certain objects (assets, models, and legal documents) you can provide a more specific configuration by selecting a particular *asset family* or *legal template, or all asset families or legal templates*.

You can manage pages in the following ways:

- Configure the model or asset page by asset family.
- Configure the legal document page by legal template.

Example: Configure for a Hardware and Software Asset

- (Hardware Asset) Configure the page for assets by moving important fields (asset name, model name, quantity, serial number, operating system, purchase order number, and service status) to the top of the page and making the fields required. In addition, remove the Host Name field. When you save the configuration, select the asset family *Hardware*. As a result, when users assigned to the configuration enter a hardware asset, they see the page as you have configured it.
- (Software Asset) Configure the page for assets by moving important fields (asset name, model name, quantity, serial number, department, cost center, and purchase order number) to the top of the page and making the fields required. In addition, remove the Service Status and Service Status Date fields. When you save the configuration, select the asset family *Software*. As a result, when users assigned to the configuration enter a software asset, they see the page as you have configured it.

Example: Configure Across All Asset Families

Configure the page for assets in all asset families by making the Requisition ID and Purchase Order ID fields read only for all users except the asset fulfiller. The asset fulfiller can edit these two fields. To achieve this result, you create two configurations:

1. A global configuration that makes the Requisition ID and Purchase Order ID fields read only. Select Across all families for this configuration so that the fields are read only on the Asset page for all asset families. This global configuration applies to all users.
2. A local configuration that allows users to edit the Requisition ID and Purchase Order ID fields. Select Across all families for this configuration, also, so that the fields can be edited on the Asset page for all asset families. This local configuration is assigned to users in the asset fulfiller role.

As a result, users who are not asset fulfillers cannot edit the Requisition ID and Purchase Order ID fields on the Asset page for any asset family. However, users in the asset fulfiller role can edit the two fields on the Asset page for all asset families.

Example: Configure for a Confidentiality Agreement

Configure the page for legal documents by moving important fields (Document Identifier, Effective Date, Termination Date, and Negotiator) to the top of the page and making the fields required. In addition, remove the Status and Status Date fields. When you save the configuration, select the legal template *Confidentiality Agreement*. As a result, when users assigned to the configuration enter a confidentiality agreement, they see the page as you have configured it.

Custom Asset Families

You can extend the product by creating additional asset families to track products other than hardware and software. *Custom asset families* let you track information about almost any classification of asset in your IT environment. For example, you can create asset families for telecommunications, services. After you create a custom asset family, configure the page and save the configuration for the custom asset family. As a result, a user assigned to the configuration for the custom asset family sees the page as you configure.

Configure the Model and Asset Page by Asset Family

You can configure the Model or Asset page for a specific asset family (for example, hardware asset and software asset) or for all asset families. Users in the role assigned to the configuration see the page as you have configured it.

Follow these steps:

1. Click the Model or Asset tab.
2. On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
3. In the Configuration Information area of the page, select an asset family or select Across all families.

   NOTE
   If you are creating a global configuration and a global configuration already exists across all families, the Across all families field does not appear.
4. Specify the information for the new global or local configuration, or select an existing configuration that you want to change.

   WARNING
   Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles that are assigned to the selected configuration.
5. Complete any of the following steps:
   - Change a field label.
   - Move a field to a new location.
   - Make a field read-only, required, or optional.
   - Hide a field.
   - Make a previously hidden field appear.
   - Add a field.
6. Click Save Configuration.
   When you assign a configuration to a role, users in the role see the page as you have configured it.

Configure the Legal Document Page by Legal Template

You can configure the Legal Document page for a specific legal template or for all legal templates. Users in the role that is assigned to the configuration see the page as you have configured it.

Follow these steps:

1. Click the Legal Document page.
2. On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
3. In the Configuration Information area of the page, select a legal template or select Across all templates.

   NOTE
   If you are creating a global configuration and a global configuration already exists across all templates, the Across all templates field does not appear.
4. Specify the information for the new global or local configuration, or select an existing configuration that you want to change.

WARNING
Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles that are assigned to the selected configuration.

5. Complete any of the following steps:
   - Change a field label.
   - Move a field to a new location.
   - Make a field read-only, required, or optional.
   - Hide a field.
   - Make a previously hidden field appear.
   - Add a field.
6. Click Save Configuration.
   When you assign a configuration to a role, users in the role see the page as you have configured it.

How to Configure the User Interface

To configure the user interface, complete the following tasks:

- Protect the users from performing unauthorized tasks using object access configuration and tab and menu configuration.
- Make it easier for users to enter information for the objects they manage using field configuration.
- Validate and enforce the field format and data entry requirements using field data validation configuration.
- Protect the users from performing unauthorized tasks using hyperlink and button configuration.
- Extend the repository to store more data using extended field configuration.
- Extend the product and enhance how users enter information for the objects they manage using reference field configuration.
- Extend the product and track more information and detail about an object using hierarchy configuration.
- Make it easier for users to enter model, asset, and legal document information by family and legal template using page configuration.
- Make it easier for users to find the objects they manage in searches using search configuration.
- Make it easier for users to enter information for legal documents using legal template configuration.
- Alert users about upcoming events and verify that the appropriate tasks are performed in the correct order at the right time using event and notification configuration.
- Make it easier for users to select the correct items from lists using list management.
- Extend the product and enhance how users manage object information using custom relationships.

Custom Relationships

Custom relationships are links between two related objects. The relationship describes and provides information about the interdependencies between the objects. Through a custom relationship, you can navigate from one object to the other object. You can locate, retrieve, and modify information about the objects.

Manage Custom Relationships

You can define and update custom relationships between two objects. In a custom relationship, one of the objects is a primary object and the other object is a secondary object.

NOTE
You cannot delete a custom relationship that you define and save.

Follow these steps:

1. Define a custom relationship.
   a. Click the tab and optional subtab for the object for which you want to define a custom relationship.
   b. Select CONFIGURE: ON from the search results page.
   c. Select a global configuration across all families or templates or create and save a global configuration.

      NOTE
      You cannot define a custom relationship for a local configuration.
   d. Click Add Custom Relationship.
      The Add Custom Relationship dialog opens.
   e. Specify a name for the primary object relationship and secondary object relationship.
   f. Select the secondary object.

      NOTE
      For assets, models, and legal documents, select a family or template or select across all types.
   g. Click Save.
      The new custom relationships are displayed under the Custom Relationship menu for the primary and secondary objects.
   h. (Optional) Add simple or reference extended fields to the custom relationship.
2. Update a custom relationship.
   a. Click the tab and optional subtab for the object for which you want to update a custom relationship.
   b. Select CONFIGURE: ON from the search results page.
   c. Select a global configuration across all families or templates.
   d. Select the relationship under the Custom Relationship menu.
   e. Modify the information for the custom relationship and click Save.

Event and Notification Configuration for CA APM

This article contains the following topics:

An *event* represents an activity related to a field (default or extended) for an object. When you define an event, you specify the criteria that must be met before the event occurs. For example, you want to know when the data in a particular field changes. You can define an event that detects the data change. An event works in combination with a *notification*, which the workflow provider (for example, CA Process Automation) creates to alert your team members that an important event has occurred for a specific field or object. By using events and notifications, you alert people about upcoming events and help ensure that the appropriate tasks are performed in the correct order at the right time.

A notification is triggered when an event that you define occurs. For example, you define a date event on the Termination Date field for a legal document to notify the contract manager 15 days before a legal contract expires. The contract manager uses the 15 days to review and possibly negotiate a better contract. When the date arrives (that is, 15 days before the contract expires), the event occurs and the notification process is triggered through the workflow provider. The workflow provider constructs, issues, and manages the notification based on the configuration that you provided in the workflow provider and in CA APM.

The default notification method in CA APM supports email notifications using a workflow provider. You can send an email notification to any user or distribution list that is defined in your internal email system, even if the user is not a CA APM user. In addition, you can send an email to any external email address, if permitted by your email system.

You can also configure the notification process in the workflow provider to trigger any type of process. For example, you can set up the notification process to perform certain actions in another application when an event occurs in CA APM. For information about setting up different notification processes, see your workflow provider documentation.

You can define the following types of events to track and manage important changes to fields or objects:

- Date events. Monitor a date field for an object and have the workflow provider notify you that an important date is approaching or has passed.
- Change events. Monitor a field for an object and have the workflow provider notify you that the field value has changed.
- Watch events. Monitor a field for an object and have the workflow provider notify you about a potential obstruction to completing a task.

How to Configure Events and Notifications

Events work in combination with notifications, which the workflow provider (for example, CA Process Automation) creates, to communicate information to your team members about important events and activity. To configure events and notifications, complete the following steps:

1. Administrators grant permissions to users to manage events.
2. Users with the correct permissions open an existing configuration and define date events, change events, and watch events. For more information about defining events, see Events and Notifications for CA APM.
3. Users, when defining an event, map all workflow provider process parameters to a CA APM object attribute. For more information, see Workflow Provider Process Parameters.
4. The workflow provider initiates the notification process.
5. Users View an Audit History of Events.

Grant Permissions to Manage Events

You can grant permissions to users so they can configure the user interface and can define an event.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to configure.
2. On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
3. In the Configuration Information area of the page, select an existing global or local configuration.

   WARNING
   Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles that are assigned to the selected configuration.
4. In the Permissions area of the page, move Manage Events to the Granted Permissions list.
5. (Optional) Select the Inherit Permissions from Parent Object check box to apply any security permissions from a top-level (parent) object to the lower-level (child) object using the same configuration.
   For example, you create a local configuration for an Organization. In the configuration, you deny permissions to change field labels, move fields, make fields required, and hide fields. After you save the configuration, you open Attachments under Organization and select the Inherit Permissions from Parent Object check box. All permissions from the Organization are applied to Attachments. In this example, because permissions to change field labels, move fields, make fields required, and hide fields are applied to the Organization, the permissions are also applied to the Attachment.
6. Click Save Configuration.
   When you assign a configuration to a role, users in the role have permissions to define an event.

Extended Field Configuration

You can design your own fields (extended fields) and can add them to objects. Extended fields are additional fields to help you capture data in your repository critical to the asset management at your site. If you are not able to find an appropriate field in your repository to store key data, define an extended field for the data.

You can define extended fields for models, assets, legal documents, costs, payments, model parts and pricing, contacts, companies, organizations, locations, and sites. You can use extended fields when searching and for reports.

NOTE
You can add an extended field that you created for an asset cost to a legal document cost. Under Legal Document, click Add Existing Fields on the Costs configuration page.

WARNING
Extended fields are shared with the products integrating with CA APM (CA Service Desk Manager and CA Service Catalog). Any changes that are made to the extended field values in the integrating products are immediately reflected in CA APM. And, any changes you make to the extended field values in CA APM are reflected in the integrated products.

Define an Extended Field

You can define an extended field to help you capture all data in your repository that is critical to your asset management program. For example, when entering an asset for a blade server, there is no way to enter the chipset. Define an extended field that is named *chipset,* which adds the field to the Asset Details page. Users can enter the chipset information (for example, Intel 5520) with the other information such as the asset name, serial number, memory, processor, operating system.

WARNING

These steps work only the first-time you complete the wizard and define the extended field for the object. Before you define the extended field, verify that you have the following information for reference: table name, label, format (character, boolean, currency, date, decimal, or integer), field name, attribute name, field size, and whether an entry for the extended field is required. After you complete the wizard, you can configure the extended field like any field.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to configure.
2. On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
3. In the Configuration Information area of the page, complete the following steps:
   a. Specify the information for the new global configuration, or select an existing global configuration that you want to change.
   b. (Optional) In the Object drop-down list, select the part of the object that you want to configure. Any permission changes you make (for example, deny permissions to move a field) apply only to that part of the object.
   For example, when configuring a legal document, you select Legaldoc Status History in the Object drop-down list. You deny permissions to move fields for that part of the object (the status history). The permission changes apply only to the status history part of the object. Changes do not apply to the other parts of the object.

   WARNING
   You can only define an extended field for a global configuration. You cannot define an extended field for a local configuration.
4. Click Save Configuration to create the global configuration.
5. Click Add Extension.
   A wizard appears.
6. To enter the information for the extended field, select the Simple Field option and follow the on-screen instructions.

   NOTE
   To change the default object label for the extended field, change the label in the Object Label field. For example, change the default label *asset hardware Extension* to *Hardware Extension*.
7. Click Save Configuration.
   All users see the extended field on the page.

NOTE
After you add a field and define an extended field, and save the field to a local or global configuration, users can define an event for the field. For more information about managing events, see Event and Notification Configuration for CA APM.

Grant Permissions to Define an Extension

You can grant permissions to users so an Add Extension link appears to define an extension.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to configure.
2. On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
3. In the Configuration Information area of the page, specify the information for the new global configuration. You can select an existing global configuration that you want to change.

   WARNING
   You can only define an extended field for a global configuration. You cannot define an extended field for a local configuration.
4. In the Permissions area of the page, move Extend Object to the Granted Permissions list.
5. (Optional) Select the Inherit Permissions from Parent Object check box to apply any security permissions from a top-level (parent) object to the lower-level (child) object using the same configuration.
   For example, you create a local configuration for an Organization. In the configuration, you deny permissions to change field labels, move fields, make fields required, and hide fields. After you save the configuration, you open Attachments under Organization and select the Inherit Permissions from Parent Object check box. All permissions from the Organization are applied to Attachments. In this example, because permissions to change field labels, move fields, make fields required, and hide fields are applied to the Organization, the permissions are also applied to the Attachment.
6. Click Save Configuration.
   When you assign a configuration to a role, users in the role have permissions to define an extension.

Field Data Validation Configuration

You can create field data validation configurations to validate the data entry in fields. These field data validations ensure that users enter data in the correct format and enforce your organizational business rules.

NOTE
The data validation affects new data that is added. Existing data records are validated only when you access the data record and you save the record.

For example, you want to ensure that users enter asset names using only alphanumeric characters (no special characters). Create a data validation for the Asset Name field on the New Asset or Asset Details page. Specify that the field allows only alphanumeric entries. Users receive an error message if the characters they use are not alphanumeric.

Add a Data Validation for a Text Field

You can validate the data entry in text fields (for example, contact name, email address, or telephone number) to enforce specific format requirements. You create the data validations for text fields by defining the regular expressions that apply to the different types of text fields.

NOTE
A *regular expression* is a text string that describes a particular pattern or format. Regular expressions are used to validate text to ensure that the text matches a predefined format. For example, create a regular expression to specify the correct format for an email address, telephone number, or IP address.

WARNING
Compose and test your regular expression before creating the text field data validation. You can find resources on the Web for creating, analyzing, and testing regular expressions.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to configure.
2. On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
3. In the Configuration Information area of the page, complete the following steps:
   a. Specify the information for the new global or local configuration, or select an existing configuration that you want to change.

      NOTE
      Permissions for data validation are allowed by default. You can deny data validation permissions for the current configuration. The users that are assigned to the configuration do not then see the Data Validation icon and cannot add data validations.

   b. (Optional) In the Object drop-down list, select the part of the object that you want to configure. Any permission changes you make (for example, deny permissions to move a field) apply only to that part of the object.
      For example, when configuring a legal document, you select Legaldoc Status History in the Object drop-down list. You deny permissions to move fields for that part of the object (the status history). The permission changes apply only to the status history part of the object, and not to the other parts of the object.

      WARNING
      Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles that are assigned to the selected configuration.
4. Next to the text field that you want to validate, click the Data Validation icon.
5. Enter the regular expression that applies to the type of field (for example, telephone number, email address) and click OK.

   WARNING
   Verify that you selected the correct regular expression for the field type and that you entered the regular expression accurately.

   NOTE
   To modify or delete an existing data validation, complete one of the following steps:

   – To modify the validation, edit the regular expression in the text entry field and click OK.
   – To delete the validation, clear the regular expression in the text entry field and click OK.
6. Click Save Configuration.
   When you assign the configuration to a role, users in the role receive data validation messages if their text entries do not match the defined format.

Hierarchy Configuration

In the product, a *hierarchy* is a way to establish a logical relationship to an object field. You can define a hierarchy to extend the product and track more information and detail about an object. You can define a hierarchy for any object.

Example: Create a Hierarchy to Locate an Asset

In CA APM, when you enter an asset, use the Location Name field to enter generic asset information. Information can be the location of an asset (city and address). However, for a hardware asset family, you need a more detailed way to track the asset location. To locate an asset for maintenance and repair, find the specific office number, building number, floor number, and cubicle number. Define the following hierarchy:

```
Pittsburgh Office
```

```
Building 3
 Fourth Floor
     Cubicle 49466
```

NOTE

In the previous hierarchy, each field is related to the field above it. If you change the information for a parent field, the information for the child field is changed. However, changing a child field (Fourth Floor) does not change the parent field (Building 3).

By defining this hierarchy, you know and can track the exact location of the asset. CA APM manages the fields that you define in the hierarchy so they can be used in searches and reports.

Define a Hierarchy

You can define a hierarchy to extend the product and track more information and detail about an object. For example, define a location hierarchy for an asset to track the asset to a specific location. When asset location with an established hierarchy for the asset family is selected, a list appears inside the location section. Each location hierarchy extended field has one list. If the location selected has values for the hierarchy, the values are populated in the drop-down list.

WARNING
Before you define a hierarchy, verify that you have the following reference information: table name, label, format, field name, attribute name, field size. Also verify, if an entry for the extended field in the hierarchy is required.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to configure.
2. On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
3. In the Configuration Information area of the page, complete the following steps:
   a. Specify the information for the new global configuration, or select an existing global configuration that you want to change.
   b. (Optional) In the Object drop-down list, select the part of the object that you want to configure. Permission changes that you make apply only to that part of the object.
   For example, when configuring a legal document, you select Legaldoc Status History in the Object drop-down list. You deny permissions to move fields for that part of the object (the status history). The permission changes apply only to the status history part of the object.

   WARNING
   You can only define a hierarchy for a global configuration. You cannot define a hierarchy for a local configuration.
4. (Optional) Click Save Configuration to create the global configuration.
5. Click Add Extension.
   A wizard appears.
6. To define the hierarchy, select the Hierarchy option and follow the on-screen instructions. The following fields require explanation:
   - **Object Label**
     Specify the default object label for the hierarchy. You can change this label to meet your requirements. For example, change the default label Asset Extensions to Asset.
   - **Object Table Name**
     Specify the database table name for the hierarchy.
   - **Object Tenancy**
     If multi-tenancy is enabled, specify how multi-tenancy works for the hierarchy by selecting one of the following options. The option that you select is applied to all levels in the hierarchy.
     - **Untenanted**

Defines objects without a tenant attribute. All data in these objects is public, and any user can create and update untenanted public data.

- **Tenant Required**
  Defines objects with a tenant attribute that cannot be null (enforced by CA APM, not the DBMS). All data in these objects is associated with individual tenants; there is no public data.
- **Tenant Optional**
  Defines objects with a tenant attribute that can be null. You can either create these objects as tenanted or public. When you select a tenant in a tenant drop-down to create an object, the object becomes a tenanted object. However, when you select the Public Data option in a tenant drop-down, the object becomes a tenanted public object. A tenant drop-down does not appear for users assigned to a role that only exposes a single tenant when entering data.

– **Begin with existing field**
  Select an existing field as the basis for fields in the first hierarchy level.
– **Begin with a new field**
  Select to start the hierarchy with a new field you define. Define at least two levels for the hierarchy.

7. Click Save Configuration.
   All users see the hierarchy on the page.

Legal Template Configuration

This article contains the following topics:

A *legal template* provides the group of attributes that belong to a particular type of legal document. You typically set up and maintain legal templates, while users create legal document records based on the legal templates. When you create a legal document record, you first select a legal template on which to base the document. The legal document record inherits the attributes from the legal template.

Each legal template has *terms and conditions* that typically apply to the legal document type. For example, a legal template for an invoice includes information about the terms of payment.

You can use the product to define legal templates and change the attributes of a legal template. To change the terms and conditions assigned to a legal template, you can remove them or add new ones from the master list of terms and conditions.

A legal template cannot be deleted until all legal documents based on the template are deleted. In addition, you can search for legal documents by the legal template name, and you can use legal template names to select records to include in reports.

Define a Legal Template

You can define a legal template to group attributes that belong to a particular type of legal document. Administrators or users with appropriate privileges can define legal templates.

Follow these steps:

1. Click Directory, List Management.
2. On the left, expand Legal Document Lists and click Legal Template.
   The list of templates appears on the right.
3. Click New, enter a name for the template, and click Save.
   The new template appears in the list.
4. Click the Edit Record icon for the new template.
5. Click the View Assigned Ts & Cs hyperlink.
6. Click Select New and select the terms and conditions for the new template.
7. Click Save.

Users can select the new template when they define legal documents.

Change the Terms and Conditions for a Legal Template

You can change the terms and conditions for a legal template. Administrators or users with appropriate privileges can change the terms and conditions.

Follow these steps:

1. Click Directory, List Management.
2. On the left, expand Legal Document Lists and click Legal Template.
   The list of templates appears on the right.
3. Click the Edit Record icon for the legal template.
4. Click the View Assigned Ts & Cs hyperlink.
   All terms and conditions for the legal template appear.
5. Select any of the following options:
   - Click Select New and select the terms and conditions for the template.
   - Click the Delete icon to remove the term and condition from the template.
6. Click Save.
   The updated terms and conditions are applied to the template.

List Management

This article contains the following topics:

You can define the items that appear in lists to help make it easier for users to select the correct information from lists for the objects they manage. For example, when defining a new contact, the user can specify a contact type. You define the items that appear in the contact type list. In addition, when defining a legal document, the user must specify a legal template. You define the items that appear in the legal template list.

You can manage items in the following lists and in the reference fields you define:

- Assets
- Companies
- Contacts
- Legal Documents
- Locations
- Models
- Organizations
- Searches
- Normalization

Define List Items

You can define, update, and delete list items to help make it easier for users to select the correct information from lists when managing objects. For example, you can change the name and description of a cost center to make it easier for users to select a cost center. You can also delete a general ledger code so that users cannot select that code when they define assets.

> **WARNING**
> When you delete a list item, users cannot select the item when defining an object. Instead of deleting the list item, you can make the list item inactive. Then, if you need the list item in the future, you can make the item active again. You do not have to redefine the item.

By default, all the items in a list are displayed. However, if you have large number of list items, you can search for specific items. The search criteria depends on the properties of the list. For example, for **Cost Type** you can search the items based on **Value, Status**, and the **Description** of the cost type.

NOTE

This feature will only be available if you apply a patch for CA Service Management Release 14.1.01. Find the patch and the download details from CA Support Online.

Follow these steps:

1. Click Directory, List Management.
2. On the left, select the list that you want to manage.
3. Click New.
4. Enter the information for the list item.

   NOTE
   When multi-tenancy is enabled, select the tenant for the list item.
5. Click Save.

Define List Management Access

As an Administrator, you can define who accesses List Management. You can also determine the list items that different users can access. For example, a Contract Manager requires information related to legal document lists, company lists, and contact lists but not Asset lists. So, as an Administrator, you configure the access rights to Contract Manager role accordingly.

NOTE

This feature will only be available if you apply a patch for CA Service Management Release 14.1.01. Find the patch and the download details from CA Support Online.

Follow these steps:

1. Click Administration, User/Role Management.
2. Click the role for which you want to configure List Management access.
3. To configure which list items the role access, select the appropriate check box.
   For example, to restrict asset list access to a Contract Manager, ensure clear the Asset Lists Access check box.
4. To enable or disable List Management, select or clear List Management Access.
5. Save the configuration changes.

Define the Class and Subclass Lists for an Asset Family

You can define the class and subclass lists for an asset family to help make it easier for users to select the correct information when defining models and assets. For example, you can define the class "printer" and the subclass "laser". Then when users create or search for printer assets, this additional information helps them to define or identify the correct assets.

Follow these steps:

1. Click Directory, List Management.
2. On the left, click Asset Lists, Asset Family.
3. Complete the following steps to define the class list:
   a. Click the Edit Record icon next to the asset family for which you want to define the class.
   b. Click the Class List hyperlink.
   c. Define the class record for the asset family.
   d. Click Save.

4. Complete the following steps to define the subclass list:
   a. Click the Edit Record icon next to the class record for which you want to define the subclass.
   b. Click the Subclass List hyperlink.
   c. Define the subclass record for the asset family.
   d. Click Save.

Define Legal Document Terms and Conditions

You can define the terms and conditions that apply to legal document lists. Users can then apply the correct terms and conditions when they define legal templates or legal documents.

Follow these steps:

1. Click Directory, List Management.
2. On the left, expand Legal Document Lists and select Terms and Conditions.
3. Click New.
4. Enter the information for the new list item.

   NOTE
   To make the new item apply to date-specific terms and conditions lists only, select the Date Specific Key check box.
5. Click Save.

Exclude an Asset Family

You can exclude an asset family so that it is not available for users. If you exclude an asset family, a user who creates or modifies a model cannot select that asset family. Also, the excluded asset family is not available for managing filters, data imports, or configurations. For example, if CA APM is integrated with CA Service Desk Manager, the asset families from CA Service Desk Manager are also available to CA APM users. If you do not require the CA Service Desk Manager asset families, you can exclude them so that they are not available to users.

If a model was already associated with an asset family before you excluded the family, you can still access and edit that model. You can also still use that model to create assets. However, you cannot change the asset family for that model to another excluded asset family.

NOTE
You can change the asset family for a model only if the model does not have associated assets.

Follow these steps:

1. Click Directory, List Management.
2. On the left, select Asset Family under Asset Lists or Model Lists.
3. Click the Edit Record icon for the family that you want to exclude.
4. Clear the Is ITAM check box and click the Complete Record Edit icon.

   WARNING
   Do not select the Inactive check box. This action makes the asset family inactive for all products that are integrated with the CA MDB.
5. Click Save.

This feature will only be available if you apply a patch for CA Service Management Release 14.1.01. Find the patch and the download details from CA Support Online.

Manage Buttons

You can configure the button access to help prevent users from performing unauthorized tasks. You can configure buttons in the following ways:

- Hide a button when you do not want users to be able to see and use the New menu option and the Save, Copy, and Delete buttons. For example, you do not want a user to be able to copy or delete an asset. Therefore, you hide the Copy and Delete buttons on the Asset page.
- Make a previously hidden button appear when users must be able to see and use the New menu option and the Save, Copy, and Delete buttons. For example, you hid the Copy button for the Asset page, but the button appears. Therefore, you make the Copy button appear.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to configure. On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
   In the Configuration Information area of the page, complete the following steps:
   a. Specify the information for the new global or local configuration, or select an existing configuration that you want to change.
   b. (Optional) In the Object drop-down list, select the part of the object that you want to configure. Any permission changes you make (for example, deny permissions to move a field) apply only to that part of the object.
   For example, when configuring a legal document, you select Legaldoc Status History in the Object drop-down list. You deny permissions to move fields for that part of the object (the status history). The permission changes apply only to the status history part of the object, and not to the other parts of the object.

      WARNING
      Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.
2. In the Permissions area of the page, complete the following steps in the Granted and Denied Permissions lists:
   – To hide the New option and Save button, move Create to the Denied Permissions list.
   – To hide the Copy button, move Copy to the Denied Permissions list.
   – To hide the Delete button, move Delete to the Denied Permissions list.
   – To make the New option and Save button appear, move Create to the Granted Permissions list.
   – To make the Copy button appear, move Copy to the Granted Permissions list.
   – To make the Delete button appear, move Delete to the Granted Permissions list.
3. (Optional) Select the Inherit Permissions from Parent Object check box to apply any security permissions from a top-level (parent) object to the lower-level (child) object using the same configuration.
   For example, you create a local configuration for an Organization. In the configuration, you deny permissions to change field labels, move fields, make fields required, and hide fields. After you save the configuration, you open Attachments under Organization and select the Inherit Permissions from Parent Object check box. All permissions from the Organization are applied to Attachments. In this example, because permissions to change field labels, move fields, make fields required, and hide fields are applied to the Organization, the permissions are also applied to the Attachment.
4. Click Save Configuration.

Manage Fields

This article contains the following topics:

You can change the display and attributes of field information on the page to meet your asset management practices and help make it easier for users to enter information for the objects they manage. In the product, these fields are referred to as *configured fields*.

You can manage fields in the following ways:

- Change a field label to help make the field more familiar to users and conform to your IT asset management practices.
- Move a field to a new location to help make it easier for users to find the field on the page.
- Hide a field from display when users should not be able to view a particular field on the page.
- Make a field appear when users must be able to see a field that you previously hid. For example, you previously hid the Capacity field. Users must be able to see that field because it is required. Add the field back so users can enter a value when defining an asset.
- Make a field read-only, required, or optional.
- Add a field.
- Manage field permissions.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to configure.
   On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
   In the Configuration Information area of the page, complete the following steps:
   a. Specify the information for the new global or local configuration, or select an existing configuration that you want to change.
   b. (Optional) In the Object drop-down list, select the part of the object that you want to configure. Any permission changes you make (for example, deny permissions to move a field) apply only to that part of the object.
   For example, when configuring a legal document, you select Legaldoc Status History in the Object drop-down list. You deny permissions to move fields for that part of the object (the status history). The permission changes apply only to the status history part of the object, and not to the other parts of the object.

   WARNING
   Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.
2. To change a field label, complete the following steps:
   a. Click the field label and enter the new label.
   b. Click Save Configuration.
   When you assign a configuration to a role, users in the role see the new field label.
3. To move a filed to a new location, complete the following steps:
   a. Drag-and-drop the field to a new location in the current section.

   NOTE
   You cannot move a field from one section of the page to another. For example, you cannot move a field from the Additional Information section to the Basic Information section.
   b. Click Save Configuration.
   When you assign a configuration to a role, users in the role see the fields in the new location.
4. To hide a field, complete the following steps:
   a. Next to the field, click the Remove Field icon.
   b. Click Save Configuration.
   When you assign a configuration to a role, users in the role do not see the field. If an event is defined for the hidden field, users still receive notifications. However, any mapped attribute that is not accessible in the workflow process associated with the event is not sent as part of the notification.
5. To make a previously hidden field appear, complete the following steps:
   a. Click Expose Hidden Fields.
   b. To add the field to the page, follow the on-screen instructions.
   c. Click Save Configuration.
   When you assign a configuration to a role, users in the role see the field on the page.

NOTE
After you make a previously-hidden field appear, users can define an event for the field. You do not have to save the configuration because the field has already been added to the configuration. For more information about managing events, see Events and Notifications for CA APM.

Add a Field

You can add a field to the page when users must be able to see a field that exists in the repository but is not part of a global configuration, or any field that has been removed and you have denied access. For example, you previously removed an extended field that is named chipset from the Asset Details page. Users must be able to see and enter a value for this field, so you add the field back onto the page. In addition, if you previously added an extension but did not save the global configuration, use these steps to add the extended field to the page.

WARNING
When you add a field to an object having multiple asset families (Assets and Models) and legal templates (Legal Documents), the field is added to all families and templates for the object, regardless of the family or template to which you added the field. For example, you add a field for the Hardware asset family. The field is added to all other asset families, including Computer, Other, Projects, Service, and Software.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to configure.
2. On the left, click CONFIGURE: ON.
   The configuration to the page is enabled.
3. In the Configuration Information area of the page, complete the following steps:
   a. Specify the information for the new global configuration, or select an existing global configuration that you want to change.
   b. (Optional) In the Object drop-down list, select the part of the object that you want to configure. Any permission changes you make apply only to that part of the object.
   For example, when configuring a legal document, you select Legaldoc Status History in the Object drop-down list. You deny permissions to move fields for that part of the object (the status history). The permission changes apply only to the status history part of the object.

   WARNING
   You can only add a field for a global configuration. You cannot add a field for a local configuration.
4. Click Save Configuration to create the global configuration.
5. Click Add Existing Fields.
   A wizard appears.
6. To add to the page, select the fields.

   NOTE
   For extended fields, a link appears that matches the object label that is specified when defining an extended field. To add to the page, click the link and select the extended fields.
7. Click Save Configuration.
   All users see the field on the page.

   NOTE
   After you add a field and define an extended field, and save the field to a local or global configuration, users can define an event for the field. For more information about managing events, see Event and Notification Configuration for CA APM.

Make a Field Read-Only, Required, or Optional

A *required field* is a field that must contain a value to save the record. When you configure a field or create an extended field, you can make the field read-only, required, or optional. Making a field that is required is useful for fields that contain key pieces of data.

> **WARNING**
> When making a new required field, saved records may not have data in the field. When you save the record in the future, you must enter data into the new required field. You must also enter data when a pre-existing record is updated by an application you write using the web services. Your client application must verify that the required field contains data, or provides data for the field. If not, the record will not be updated.

We recommend that before you make a field required, you populate the field for all existing records. You can search to locate all occurrences of blank values in the field by searching for NULL or space (clear the value field).

To make a field read-only, required, or optional

1. Click the tab and optional subtab for the object that you want to configure. On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
   In the Configuration Information area of the page, complete the following steps:
   a. Specify the information for the new global or local configuration, or select an existing configuration that you want to change.
   b. (Optional) In the Object drop-down list, select the part of the object that you want to configure. Any permission changes you make (for example, deny permissions to move a field) apply only to that part of the object.
   For example, when configuring a legal document, you select Legaldoc Status History in the Object drop-down list. You deny permissions to move fields for that part of the object (the status history). The permission changes apply only to the status history part of the object, and not to the other parts of the object.

   > **WARNING**
   > Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.
2. Next to the field, click the appropriate icon to make the field read-only, required, or optional.
3. Click Save Configuration.
   When you assign a configuration to a role, users in the role see the fields as read-only, required, or optional.

Manage Field Permissions

You can grant permissions to users so they can configure the user interface to change a field label, move a field, make a field required, and hide a field.

You can also grant permissions to users so they can configure the user interface and can perform mass changes on a field. You can perform mass changes on fields that are associated with the following objects:

- Asset
- Model
- Legal document
- Organization
- Contact
- Company
- Location
- Site

Follow these steps:

1. Click the tab and optional subtab for the object that you want to configure.
2. On the left, click CONFIGURE: ON.

The configuration of the page is enabled.

3. In the Configuration Information area of the page, specify the information for the new global or local configuration, or select an existing configuration that you want to change.

   WARNING
   Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.

4. In the Permissions area of the page, do one the following:
   - To grant permissions to change a field label, move Modify Labels to the Granted Permissions list.
   - To grant permissions to move a field, move Order Fields to the Granted Permissions list.
   - To grant permissions to make a field required, move Requiredto the Granted Permissions list.
   - To grant permissions to hide a field, move Secure (Read-Only and Access)to the Granted Permissions list.
   - To grant permissions to perform mass changes on a field, move Mass Changeto the Granted Permissions list.
5. (Optional) Select the Inherit Permissions from Parent Object check box to apply any security permissions from a top-level (parent) object to the lower-level (child) object using the same configuration.

   For example, you create a local configuration for an Organization. In the configuration, you deny permissions to change field labels, move fields, make fields required, and hide fields. After you save the configuration, you open Attachments under Organization and select the Inherit Permissions from Parent Object check box. All permissions from the Organization are applied to Attachments. In this example, because permissions to change field labels, move fields, make fields required, and hide fields are applied to the Organization, the permissions are also applied to the Attachment.
6. Click Save Configuration.
   When you assign a configuration to a role, users in the role have permissions to perform the respective configuration.

View Field Information

You can view the information about any field, including extended fields, to see database-related field attributes. For any field, you can view the object label, database table name, database field name, attribute name, data type, description, and size. Use this information in the following ways:

- You want to view CA APM data outside of the product using an external reporting solution and must understand database-level information. For example, you want to know the database table name, field name, attribute name, data type, description, or field size for a particular default field or user-defined extended field.
- You have changed a field label or moved a field to a new location on the page. Use the field information to understand how the field is represented in the database. This field information can be helpful when you work with Technical Support to understand any specific configuration changes you have made to the product.

Follow these steps:

1. Click the tab and optional subtab for an object.
2. On the left, click CONFIGURE: ON.
   The configuration of the object is enabled.
3. Next to the field, click the View Details icon.
   The field information appears.

Manage Hyperlinks

You can configure hyperlink access to help prevent users from performing unauthorized tasks and only see the information appropriate for their particular job function. You can configure hyperlinks in the following ways:

- Hide a hyperlink when users should not be able to see a particular hyperlink. For example, you do not want a user to be able to see the audit history for an object. Therefore, you hide the View Audit History hyperlink.
- Make a previously hidden hyperlink appear when users must be able to see a hyperlink that you previous hid. For example, you hid the View Audit History hyperlink, but the hyperlink should now appear. Therefore, you make the View Audit History hyperlink appear.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to configure. On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
   In the Configuration Information area of the page, complete the following steps:
   a. Specify the information for the new global or local configuration, or select an existing configuration that you want to change.
   b. (Optional) In the Object drop-down list, select the part of the object that you want to configure. Any permission changes you make (for example, deny permissions to move a field) apply only to that part of the object.
   For example, when configuring a legal document, you select Legaldoc Status History in the Object drop-down list. You deny permissions to move fields for that part of the object (the status history). The permission changes apply only to the status history part of the object, and not to the other parts of the object.

   WARNING
   Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.
2. To hide a hyperlink, complete the following steps:
   a. Next to the hyperlink, click the Access Granted icon.
   The Access Denied icon appears for the hyperlink.
   b. Click Save Configuration.
   When you assign a configuration to a role, users in the role do not see the hyperlink.
3. To make a previously hidden hyperlink appear, complete the following steps:
   a. If the hyperlink is already hidden, click the Access Denied icon next to the hyperlink.
   The Access Granted icon appears for the hyperlink.
   b. Click Save Configuration.
   When you assign a configuration to a role, users in the role can see and use the hyperlink.

Manage Menu Options

You can configure the tab and menu access to:

- Help protect the integrity of the data for your objects.
- Prevent the users from performing unauthorized tasks such as adding or deleting object information.
- Support separation of duties so users only see the information appropriate for their particular job function.

NOTE
For assets, models, and legal documents, you provide a specific configuration by specifying the *asset family* and *legal template*.

You can configure the menu access in the following ways:

- Hide a menu option when users should not be able to see all, or a particular menu option.
  For example, you do not want a user to be able to view model dependencies. Therefore, you hide the Dependencies menu option.
- Make a menu option appear when users should be able to see all, or a particular menu option.
  For example, you want a user to be able to view model dependencies. Therefore, you make the Dependencies menu option appear.
- Make a menu option read-only when users should be able to see, but not use a particular menu option.

For example, you do not want a user to change any notes for a model. Therefore, you make the Notes menu option read-only.

- Make a menu option accessible when users should be able to see, and use a particular menu option.
  For example, you want a user to be able to change the generic configuration for a model. Therefore, you make the Model Configuration menu option accessible.

Combine the tab and menu configuration and field configuration to enforce more granular levels of access.

Follow these steps:

1. Click the tab and optional subtab for the object access that you want to configure. On the left, click **CONFIGURE: ON**.
   The configuration of the object access is enabled.
   In the Configuration Information area of the page, specify the information for the new global or local configuration, or select an existing configuration that you want to change.

   WARNING
   Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.
2. To hide a menu option, complete the following steps:
   a. On the menu or menu option, click the Access Granted icon.
      The Access Denied icon appears for the menu or menu option.
   b. Click Save Configuration.
      When you assign a configuration to a role, users in the role cannot see the menu option.
3. To make a menu option appear, complete the following steps:
   a. If the menu or menu option is already hidden, click the Access Denied icon on the menu or menu option.
      The Access Granted icon appears for the menu or menu option.
   b. Click Save Configuration.
      When you assign a configuration to a role, users in the role can see and use the menu option.
4. To make a menu option read-only, complete the following steps:
   a. On the menu or menu option, click the Editable icon.
      The Read Only icon appears for the menu option.
   b. Click Save Configuration.
      When you assign a configuration to a role, users in the role can see, but not use, the menu option.
5. To me a menu option accessible, complete the following steps:
   a. If the menu or menu option is already read-only, click the Read Only icon on the menu or menu option.
      The Editable icon appears for the menu option.
   b. Click Save Configuration.
      When you assign a configuration to a role, users in the role can see, and can use the menu option.

Manage Object Access

You can configure object access to help protect the integrity of the data, prevent users from performing unauthorized tasks, and provide users with only the information appropriate for their job functions. You can configure object access in the following ways:

- Hide an object to prevent users from seeing a particular area of the page for the object.
  For example, you do not want a user to be able to see pricing information for models. Therefore, you hide the Parts and Pricing area of the Models page.
- Make an object appear when users should be able to see a particular object.
  For example, you want a user to be able to see pricing information for models. Therefore, you make the Parts and Pricing area of the Models page appear.
- Make an object read-only when users should be able to see, but not change the information about a particular object.

For example, you want a user to be able to see, but not change, pricing information for models. Therefore, you make the Parts and Pricing area of the Models page read-only.

- Make an object accessible when users should be able to see and change the information about a particular object. For example, you want a user to be able to see, and edit, pricing information for models. Therefore, you make the Parts and Pricing area of the Models page accessible.
- Grant permission to users to secure objects.

Combine object access and field configuration to enforce more granular levels of access.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to configure. On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
   In the Configuration Information area of the page, complete the following steps:
   a. Specify the information for the new global or local configuration, or select an existing configuration that you want to change.
   b. (Optional) In the Object drop-down list, select the part of the object that you want to configure. Any permission changes you make (for example, deny permissions to move a field) apply only to that part of the object.
   For example, when configuring a legal document, you select Legaldoc Status History in the Object drop-down list. You deny permissions to move fields for that part of the object (the status history). The permission changes apply only to the status history part of the object, and not to the other parts of the object.

   WARNING
   Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.
2. To hide an object, complete the following steps:
   a. On the right, next to the title for the area of the page (for example, Models page, Parts and Pricing) click the Access Granted icon.
   The Access Denied icon appears for the object.
   b. Click Save Configuration.
   When you assign a configuration to a role, users in the role cannot see the object information.
3. To make an object appear, complete the following steps:
   a. If access is already denied for the object, click the Access Denied icon on the right, next to the title for the area of the page (for example, Models page, Parts and Pricing).
   The Access Granted icon appears for the object.
   b. Click Save Configuration.
   When you assign a configuration to a role, users in the role can see the object information.
4. To make an object read-only, complete the following steps:
   a. On the right, next to the title for the area of the page (for example, Models page, Parts and Pricing) click the Editable icon.
   The Read Only icon appears for the object.
   b. Click Save Configuration.
   When you assign a configuration to a role, users in the role can see, but not change the object information.
5. To make an object accessible, complete the following steps:
   a. If the object is already read-only, click the Read Only icon on the right, next to the title for the area of the page (for example, Models page, Parts and Pricing).
   The Editable icon appears for the object.
   b. Click Save Configuration.
   When you assign a configuration to a role, users in the role can access the object information.

Grant Permissions to Secure Objects

You can grant permissions to users so they can configure the user interface and hide objects, make objects appear, make objects read-only, and make objects accessible.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to configure.
2. On the left, click CONFIGURE: ON to enable the configuration of the page.
3. In the Configuration Information area of the page, specify the information for the new global or local configuration, or select an existing configuration that you want to change.

   WARNING
   Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.
4. In the Permissions area of the page, move Secure (Read-Only And Access)to the Granted Permissions list.
5. (Optional) Select the Inherit Permissions from Parent Object check box to apply any security permissions from a top-level (parent) object to the lower-level (child) object using the same configuration.
   For example, you create a local configuration for an Organization. In the configuration, you deny permissions to change field labels, move fields, make fields required, and hide fields. After you save the configuration, you open Attachments under Organization and select the Inherit Permissions from Parent Object check box. All permissions from the Organization are applied to Attachments. In this example, because permissions to change field labels, move fields, make fields required, and hide fields are applied to the Organization, the permissions are also applied to the Attachment.
6. Click Save Configuration.
   When you assign a configuration to a role, users in the role have permissions to hide objects, make objects appear, make objects read-only, and make objects accessible.

Manage Tabs

You can configure tab access to help protect the integrity of the data for your objects, prevent users from performing unauthorized tasks such as adding or deleting object information, and to support separation of duties so users only see the information appropriate for their particular job function.

NOTE
For assets, models, and legal documents, you provide a specific configuration by specifying the *asset family* (assets and models) and *legal template* (legal documents).

You can configure tab access in the following ways:

- Hide a tab when users should not be able to see a particular tab.
  For example, you do not want a user to be able to see any information for legal documents. Therefore, you hide the Legal Document tab.
- Make a tab appear when users should be able to see a particular tab.
  For example, you want a user to be able to see all information for legal documents. Therefore, you make the Legal Document tab appear.
- Make a tab read-only when users should be able to see, but not change the information on a particular tab.
  For example, you want a user to be able to see, but not change, the information for a model. Therefore, you make the Model tab read-only.
- Make a tab accessible when users should be able to view and change the information on a tab.
  For example, you want a user to be able to see, and edit, all information for a model. Therefore, you make the Model tab accessible.

Follow these steps:

1. Click the tab and optional subtab for the object access that you want to configure. On the left, click **CONFIGURE: ON**.

The configuration of the object access is enabled.

In the Configuration Information area of the page, specify the information for the new global or local configuration, or select an existing configuration that you want to change.

WARNING
Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.

2. To hide a tab, complete the following steps:
   a. On the tab, click the Access Granted icon.
      The Access Denied icon appears for the tab.
   b. Click Save Configuration.
      When you assign a configuration to a role, users in the role cannot see the tab.
3. To make a tab appear, complete the following steps:
   a. If the tab is already hidden, click the Access Denied icon on the tab.
      The Access Granted icon appears for the tab.
   b. Click Save Configuration.
      When you assign a configuration to a role, users in the role can see the tab.
4. To make a tab read-only, complete the following steps:
   a. On the tab, click the Editable icon.
      The Read Only icon appears for the tab.
   b. Click Save Configuration.
      When you assign a configuration to a role, users in the role cannot change the information on the tab.
5. To make a tab accessible, complete the following steps:
   a. If the tab is already read-only, click the Read Only icon on the tab.
      The Editable icon appears for the tab.
   b. Click Save Configuration.
      When you assign a configuration to a role, users in the role can access the information on the tab.

Reference Field Configuration

This article contains the following topics:

You can define your own *reference fields* and add them to objects to extend the product and enhance how users enter information for the objects that they manage. When you define a reference field, you can reference an existing object, or define a new object.

- Define a reference field to an *existing object* to standardize on a common set of values that the user can select when defining an object. For example, when defining an asset, you want users to select the specific terms and conditions for an asset and select an approved general ledger code. In this example, you define two reference fields to standardize the terms and conditions and general ledger codes the user can select when defining the asset.
- Define a reference field to a *new object* to establish a relationship between people, companies, assets, and so forth. For example, when defining and managing vendors, you want the user to assign a service rating to each vendor and select a quality rating (one through five stars). In this example, you define one reference field to record the quality rating when defining and managing vendors.

You can define reference fields for models, assets, legal documents, costs, payments, model parts and pricing, contacts, companies, organizations, locations, and sites.

After you define a reference field, you can configure the reference field by completing the following tasks:

- Add a field to the reference field criteria and results
- Remove a field from the reference field lookup criteria and results
- Make a previously-hidden reference field appear
- Move a reference field to a new location

Define a Reference Field

You can define your own *reference fields* and add them to objects to extend the product and enhance how users enter information for the objects they manage. When you define a reference field, you can reference an existing object, or define a new object. For example, when defining an asset, you want users to select the specific terms and conditions for an asset and select an approved general ledger code. In this example, you define a reference field to an existing object and standardize the terms and conditions and general ledger codes the user can select when defining the asset.

> **WARNING**
> These steps work only the first-time you complete the wizard and define the reference field. Before you define the reference field, verify that you have the following information for reference: table name, label, format (character, boolean, currency, date, decimal, or integer), field name, attribute name, field size, and whether an entry for the field is required. After you complete the wizard, you can configure the reference field by adding and removing fields, making a previously-hidden reference field appear, and moving a reference field to a new location.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to configure.
2. On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
3. In the Configuration Information area of the page, complete the following steps:
   a. Specify the information for the new global configuration, or select an existing global configuration that you want to change.
   b. (Optional) In the Object drop-down list, select the part of the object that you want to configure. Any permission changes you make (for example, deny permissions to move a field) apply only to that part of the object.
   For example, when configuring a legal document, you select Legaldoc Status History in the Object drop-down list. You deny permissions to move fields for that part of the object (the status history). The permission changes apply only to the status history part of the object, and not to the other parts of the object.

   > **WARNING**
   > You can only define a reference field for a global configuration. You cannot define a reference field for a local configuration.
4. (Optional) Click Save Configuration to create the global configuration.
5. Click Add Extension.
   A wizard appears.
6. Select the Reference Field option and follow the on-screen instructions to enter the information for the reference field.
   The following fields require explanation:
   – **Object Label**
   Displays the default reference field object label to appear when you add a field to the reference field criteria and results, and make a previously-hidden reference field appear. You can change this label to meet your requirements. For example, change the default label Location Extensions to Location.
   – **Label**
   Enter the label for the reference field that you want to appear in list management.
   – **Service provider eligible**
   Determines if field values from the service provider are included in the reference field. When you select this check box, public data and service provider objects are included in the reference field.
   – **Based on current object**

Select an existing object on which to base the reference field you are defining.

NOTE
When you select this option, the reference field for the object already exists and the multi-tenancy options for the object are applied.

- **Object table name**
  Specify the database table name for the reference field.
- **Object Tenancy**
  If multi-tenancy is enabled, specify how multi-tenancy works for the reference field by selecting one of the following options:
  - **Untenanted**
    Defines objects without a tenant attribute. All data in these objects is public, and any user can create and update untenanted public data.
  - **Tenant Required**
    Defines objects with a tenant attribute that cannot be null (enforced by CA APM, not the DBMS). All data in these objects is associated with individual tenants; there is no public data.
  - **Tenant Optional**
    Defines objects with a tenant attribute that can be null. You can either create these objects as tenanted or public. When you select a tenant in a tenant drop-down to create an object, the object becomes a tenanted object. However, when you select the Public Data option in a tenant drop-down, the object becomes a tenanted public object. Users assigned to a role that only exposes a single tenant will not see a tenant drop-down when entering data.

    NOTE
    When multi-tenancy is disabled, you do not see the Object Tenancy drop-down for the reference field. However, the product applies the Tenant Optional setting to the reference field. The product works this way so that if you enable multi-tenancy, the Tenant Optional setting is applied to the reference field.

7. Click Save Configuration.
   All users see the reference field on the page. When you define a reference field based on a new object, the reference field appears as a list item that can be managed using list management.

Manage Reference Fields

After you define a reference field, you can configure the field in the following ways:

- Add additional fields to extend the information that appears in your reference field criteria and results. For example, when users add an asset, they can search by model name and description to find the model describing the asset. You can configure the model reference field and add the Asset Family, Class, and Company Name fields to the reference field criteria and results to make it easier for users to find the model when defining the asset.
- Remove a field when you do not want a particular field included in the reference field criteria and results. For example, you previously configured the model reference field by adding the Asset Family, Class, Company Name, and GL Code fields. To protect users from viewing sensitive information, you remove the GL Code field from the model reference field.
- Make a reference field appear when users must be able to see a reference field that you previously hid. For example, you previously configured the model reference field by removing the Inactive field. Add the field back to the model reference field so users can find and not select inactive models when adding an asset.
- Move a reference field to a new location to help make it easier for users to find the reference field. For example, you configure the model reference field and add the Asset Family, Class, and Company Name fields to the reference field criteria and results to make it easier for users to find the model when defining an asset. You move the Company Name field to the top of the model reference field so users can find models in a particular company.

NOTE
You can complete the tasks only when your assigned role has permissions to configure the object.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to configure.
2. On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
3. In the Configuration Information area of the page, complete the following steps:
   – Specify the information for the new global or local configuration, or select an existing configuration that you want to change.
   (Optional) In the Object drop-down list, select the part of the object that you want to configure. Any permission changes you make (for example, deny permissions to move a field) apply only to that part of the object.
   For example, when configuring a legal document, you select Legaldoc Status History in the Object drop-down list. You deny permissions to move fields for that part of the object (the status history). The permission changes apply only to the status history part of the object, and not to the other parts of the object.

   WARNING

   Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.
4. Click Save Configuration.
5. Next to the field, click the Lookup Field icon.
   A list of fields in the reference field criteria and results appears.
6. To add a field to the reference field criteria and results, complete the following steps:
   a. (Global configurations only) Click Add Fields.
   b. Select the fields to add to the reference field criteria, results, or both.
   c. Click Save.
   d. Click Save Configuration.
      The field appears in the reference field criteria and results.
7. To remove a field from the reference field criteria and results, complete the following steps:
   a. Click the Mark for Deletion icon next to the field you want to remove from the reference field criteria and results.
   b. Click Save.
   c. Click Save Configuration.
      The field does not appear in the reference field criteria and results.
8. To make a previously-hidden reference field appear, complete the following steps:
   a. Click Expose Hidden Fields.
   b. Select the fields to add to the reference field criteria, results, or both.
   c. Click Save.
   d. Click Save Configuration.
      When you assign a configuration to a role, users in the role see the field in the reference field criteria and results.
9. To move a reference field to a new location, complete the following steps:
   a. Drag-and-drop the reference field to a new location in the reference field criteria and results.
   b. Click Save.
   c. Click Save Configuration.
      When you assign a configuration to a role, users in the role see the fields in the new location.

Search Configuration

This article contains the following topics:

You can configure searches to simplify how users search for information in the repository and export the results. To configure searches, complete the following tasks:

- Set a search result limit.
- Make it easier to search by assigning a default search for a role.
- Make it easier to specify search criteria by completing the following tasks:
  - Adding fields
  - Removing fields
  - Moving fields
  - Changing the field name
  - Replacing fields
- Make it easier to find information in the search results by completing the following tasks:
  - Managing Columns in Search Results.
  - Adding sort fields
  - Preventing duplicate records from appearing
  - Preventing the ability to open records
- Make it easier to search by allowing users to save searches.
- Make it easier to use search results in spreadsheets by allowing users to export search results.
- Delete a search that you do not need.

Set a Search Result Limit

When you search for an object and the results are difficult to manage because too many object records appear, you can set a limit. For example, when you search for assets, over 2,000 assets appear in the search results. The results are difficult to navigate, you cannot find the assets you want, and the performance is negatively impacted. Therefore, you set a maximum of 50 object records to return.

Follow these steps:

1. Click the tab and optional subtab for the search that you want to configure.
   On the left, click Manage Searches.
   A list of saved searches displays. Click a search in the list.
2. In the Additional Settings, Maximum Search Results Returning area, specify the total number of objects to appear.

   NOTE
   For performance reasons, we recommend that you set this value to less than 500.
3. Click Go.
   The limited search results appear and help you see the impact on the results before you save the limit. All future search results are limited to the specified number or percentage.

Assign a Default Search for a Role

You can assign a default search for a role so that all users in the role have the same default search when they click a tab or subtab. For example, all users responsible for reviewing and negotiating contracts, agreements, and services belong to the Contract and Vendor Management role. To simplify the search setup for users so they do not have to specify a default search for themselves, you configure the default legal document search. You assign the configured legal document search as the default for all users in the Contract and Vendor Management role. When users in this role click the Legal Document tab, they see the configured legal document search as their default, rather than the default legal document search provided by the product.

Consider the following information when assigning a default search for a role:

If you do not assign a default search for a role, the default search for the object appears when users in the role click the tab or subtab.

- You can assign multiple default searches for a role. However, you can only assign one default search for a particular object type (for example, model, asset, legal document, and so forth) to a role.
- When a user saves a search as their default, that search is the default for the user, even when you assign a different search as the default for the role. For example, a user in the Asset Technician role sets their default search as the asset search provided by the product. You configure the default asset search by adding a field to the search criteria and removing a field from the search results. You then assign the configured asset search as the default search for the Asset Technician role, to which the user belongs. The default asset search is the default for the user, even when you have assigned the configured asset search as the default for the role.
- A search must be available to a role before you can assign the search as the default for the role. For example, you create a legal document search. To make the search available to the Contract and Vendor Management role, assign the Contract and Vendor Management role to the search. You can then assign the legal document search as the default for the Contract and Vendor Management role.

NOTE
For more information about assigning a role to a search, see Search Security.

Follow these steps:

1. Click Administration, User/Role Management.
2. On the left, expand the Role Management menu.
3. Click Role Search.
4. Search for and select a role.
   The role details appear.
5. In the Default Searches area of the page, click Select New.
6. Select the default search for the role.
   The default search is added to the Default Searches list.
7. Click Save.
   The search is saved as the default for all users in the role.

Add a Field

CA APM lets you extend the information that appears in your search criteria and results by adding additional fields. For example, you can add the DNS Name field to the asset search. You can add fields to a new and saved search. You cannot add fields to the default searches provided by the product.

Follow these steps:

1. Click the tab and optional sub tab for the object that you want to find.
2. On the left, click Manage Searches.
   A list of saved searches displays.
3. Click a search in the list.
4. At the top of the page, click CONFIGURE SEARCH: ON.
   The configuration of the search is enabled.
5. Click Add Fields.
   The Add Fields dialog appears.
6. Select the fields to add to the search criteria, results, or both.
7. At the top of the page, click CONFIGURE SEARCH: OFF.
   The configuration of the search is complete.
8. Click Save.
   The field appears in the search criteria and results.

Remove a Field

CA APM lets you remove a field when you do not want a particular field included in the search criteria. For example, you can remove the DNS Name field from the asset search.

Follow these steps:

1. Click the tab and optional sub tab for the object that you want to find.
2. On the left, click Manage Searches.
   A list of saved searches appears.
3. Click a search in the list.
4. Complete the following steps:
   a. At the top of the page, click CONFIGURE SEARCH: ON.
   The configuration of the search is enabled.
   a. Click the appropriate icon next to the field in the search criteria.
   b. At the top of the page, click CONFIGURE SEARCH: OFF.
   The configuration of the search is complete.
5. (Optional). Complete the following steps:
   a. In the search criteria area of the page, click Advanced.
   b. At the top of the page, click CONFIGURE SEARCH: ON.
   The configuration of the search is enabled.
   a. Click the Mark for Deletion icon next to the field you want to remove from the search criteria.
   b. At the top of the page, click CONFIGURE SEARCH: OFF.
   The configuration of the search is complete.
6. Click Save.
   The field is removed from the page and does not appear in the search criteria.

Move a Field

CA APM lets you move a field in the search criteria to a new location to help make it easier for you to enter your search criteria. For example, you can move the Bar Code Number field so that the field appears before the Serial Number field.

Follow these steps:

1. Click the tab and optional sub tab for the object that you want to find.
2. On the left, click Manage Searches.
   A list of saved searches displays.
3. Click a search in the list.
4. At the top of the page, click CONFIGURE SEARCH: ON.
   The configuration of the search is enabled.
   Drag-and-drop the field to a new location in the search criteria.
5. At the top of the page, click CONFIGURE SEARCH: OFF.
   The configuration of the search is complete.
6. Click Save.
   The new location of the field is saved.

Change the Field Name

CA APM lets you change the label for a field to help make the field name more familiar in your search criteria. For example, you can change the label *Asset Quantity* to *Quantity*.

Follow these steps:

1. Click the tab and optional sub tab for the object that you want to find.

2. On the left, click Manage Searches.
   A list of saved searches appears.
3. Click a search in the list.
4. Complete the following steps:
   a. At the top of the page, click CONFIGURE SEARCH: ON.
   The configuration of the search is enabled.
   a. In the search criteria, click the field label and enter the new label.
   b. At the top of the page, click CONFIGURE SEARCH: OFF.
   The configuration of the search is complete.
5. (Optional). Complete the following steps:
   a. In the search criteria area of the page, click Advanced.
   b. At the top of the page, click CONFIGURE SEARCH: ON.
   The configuration of the search is enabled.
   a. Click the Edit Record icon next to the field for which you want to change the label.
   b. Enter the new field label.
   c. Click the Complete Record Edit icon.
   d. At the top of the page, click CONFIGURE SEARCH: OFF.
   The configuration of the search is complete.
6. Click Save.
   The new field label appears in the search criteria.

Replace a Field

CA APM lets you replace an existing field in your *advanced* search criteria with a different field. For example, when searching for companies, you can replace the field *Company ID* with *Company Name*.

> **NOTE**
> You can replace fields only in a custom search that you created. You cannot replace fields in the product default search.

Follow these steps:

1. Click the tab and optional sub tab for the object that you want to find.
2. On the left, click Manage Searches.
   A list of saved searches appears.
3. Click a search in the list.
4. Complete the following steps:
   a. In the search criteria area of the page, click Advanced.
   b. At the top of the page, click CONFIGURE SEARCH: ON.
   The configuration of the search is enabled.
   a. Click the Search icon next to the field that you want to replace with a different field.
      The Add Fields dialog appears.
   b. Select the replacement field and click OK.
   c. At the top of the page, click CONFIGURE SEARCH: OFF.
   The configuration of the search is complete.
5. Click Save.
   The existing field is replaced in the search criteria.

Manage Columns in Search Results

You can configure search results columns in the following ways:

- Add a new column to the search results to help make it easier for you to find the information you need in search result lists. For example, you have several people in your enterprise with the name John Smith. Their first and last names are the same, but their additional contact information (email address, supervisor, department, and so forth) is different. You can add columns to a new and saved search. You cannot add columns to the default searches provided by the product.
- Move a column to a new location to help make it easier for you to find the information you need in the search results. For example, you can move the Asset ID column so that the column appears before the Asset Name column.
- Change the label for a column heading to help make the label more familiar in your search results. For example, you can change the label *Asset Quantity* to *Quantity*.
- Remove a column when you do not want a particular column included in the search results. For example, you can remove the Mac Address column from the search results.

Follow these steps:

1. Click the tab and optional sub tab for the object that you want to find.
2. On the left, click Manage Searches.
   A list of saved searches displays.
3. Click a search in the list. At the top of the page, click CONFIGURE SEARCH: ON.
   The configuration of the search is enabled.
4. To add a column to the search results, complete the following steps:
   a. Click Add Fields.
   b. Select the fields to add to the search results.
   c. At the top of the page, click CONFIGURE SEARCH: OFF.
   d. Click Save.
      The column is added to the search results.
5. To move a column to a new location, complete the following steps:
   a. In the search results list, drag-and-drop the column to a new location.
   b. At the top of the page, click CONFIGURE SEARCH: OFF.
   c. Click Save.
      The new location of the column is saved.
6. To change a column heading label, complete the following steps:
   a. In the search results, select the column heading and enter the new label.
   b. At the top of the page, click CONFIGURE SEARCH: OFF.
   c. Click Save.
      The new column label appears in the search results.
7. To remove a column, complete the following steps:
   a. In the search results, click the appropriate icon next to the column.
   b. At the top of the page, click CONFIGURE SEARCH: OFF.
   c. Click Save.
      The column is removed from the page and the search results.

Add a Sorting Field

CA APM lets you add sorting fields to the search results and extend the default sort of a single column using either ascending or descending order. For example, you currently sort assets by asset name. You can add asset family to the sorting so that you can sort on both asset name and asset family.

Follow these steps:

1. Click the tab and optional sub tab for the object that you want to find.
   a. On the left, click Manage Searches.
   b. A list of saved searches displays.

c. Click a search in the list.
d. At the top of the page, click CONFIGURE SEARCH: ON.
e. The configuration of the search is enabled.
2. Click a search in the list.
3. In the Additional Settings, Search Result Sorting area, add the additional field for sorting.
4. Click Go.
The results appear with the extended sorting and help you see the impact on the results before you save the sorting. The new field is added and you can use the field to sort the search results.

Prevent Duplicate Object Records

CA APM lets you prevent duplicate object records from appearing in the search results. For example, you have several people in your enterprise with the name John Smith. Their first and last names are the same, but their additional contact information (email address, supervisor, department, and so forth) is different.

You have a saved contact search in which only the first and last name of the contact appears in the results. When you search using the saved contact search and specify *John* as the first name and *Smith* as the last name, two instances of John Smith appear in the search results. When you prevent duplicate records from appearing, only one instance of John Smith appears.

Follow these steps:

1. Click the tab and optional sub tab for the object that you want to find.
a. On the left, click Manage Searches.
b. A list of saved searches displays.
c. Click a search in the list.
d. At the top of the page, click CONFIGURE SEARCH: ON.
e. The configuration of the search is enabled.
2. Click the search for which you want to prevent duplicate records from appearing.
3. In the Additional Settings, Unique Search Characteristics area, select the Make Results Unique check box.
4. Click Go.
The results appear without the duplicate records and help you see the impact on the results before you save your settings. The DISTINCT argument is added to the SQL statement, preventing duplicate records from appearing in the search results.

Prevent Opening Records

CA APM lets you disable the ability to open individual records from the search results. For example, you do not want users to open and display contact information from the contact search results.

Follow these steps:

1. Click the tab and optional sub tab for the object that you want to find.
a. On the left, click Manage Searches.
b. A list of saved searches displays.
c. Click a search in the list.
d. At the top of the page, click CONFIGURE SEARCH: ON.
e. The configuration of the search is enabled.
2. Click a search in the list.
3. In the Additional Settings, Unique Search Characteristics area, clear the Allow Selection of Results check box.
4. Click Save.
A hyperlink does not appear in the search results to open the object.

Allow Users to Save Searches

You can grant permissions to users so a Save button appears to save searches.

Follow these steps:

1. Click the tab and optional sub tab for the object you want to configure.
2. On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
3. In the Configuration Information area of the page, specify the information for the new global or local configuration, or select an existing configuration that you want to change.

   WARNING
   Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.
4. In the Permissions area of the page, complete the following steps in the Granted and Denied Permissions lists:

   NOTE
   When you grant any combination of the following permissions, the user can save searches.

   a. To allow only the current user who is logged in to save a search, move Save Search to User to the Granted Permissions list.
   b. To allow the current user who is logged in and specific configurations to save a search, move Save Search to Configuration to the Granted Permissions list. The search is available to the current user and all users that you select for the configuration.
   c. To allow the current user who is logged in and specific roles to save a search, move Save Search to Role to the Granted Permissions list. The search is available to the current user and all users in the roles you select.
5. Click Save Configuration.
   The configuration is saved. Verify that you correctly assign a configuration to a role.

Allow Users to Export Search Results

You can grant permissions to users so they can save exported search results.

Follow these steps:

1. Click the tab and optional subtab for the object you want to configure.
2. On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
3. In the Configuration Information area of the page, specify the information for the new global or local configuration, or select an existing configuration that you want to change.

   WARNING
   Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.
4. In the Permissions area of the page, complete the following steps in the Granted and Denied Permissions lists:

   NOTE
   When you grant any combination of the following permissions, the user can save exported search results.

   a. To allow only the current user who is logged in to save the exported search results, move Export for User to the Granted Permissions list.

      NOTE
      When this is the only permission granted to a user, the user cannot select the Export to All Configurations assigned on Search and Export to All Roles assigned on Search check boxes when scheduling search requests. For information about scheduling search requests, see Search Results Export. An email is sent

to the current user only. The email includes a link to the CSV file or specifies the name of the database view, depending on the type of export.

b. To allow the current user who is logged in and specific configurations to save the exported search results, move Export for Configuration to the Granted Permissions list. The export is available to the current user and all users in the selected configurations assigned to the search used by the export.

NOTE
When this permission is granted to a user, the user can select the Export to All Configurations assigned on Search check box when scheduling search requests. For information about scheduling search requests, see Search Results Export. An email is sent to all users in the selected configurations. The email includes a link to the CSV file or specifies the name of the database view, depending on the type of export.

c. To allow the current user who is logged in and specific roles to save the exported search results, move Export for Role to the Granted Permissions list. The export is available to the current user and all users in the selected roles assigned to the search used by the export.

NOTE
When this permission is granted to a user, the user can select the Export to All Roles assigned on Search check box when scheduling search requests. For information about scheduling search requests, see Search Results Export. An email is sent to all users in the selected roles. The email includes a link to the CSV file or specifies the name of the database view, depending on the type of export.

5. (Optional) Select the Inherit Permissions from Parent Object check box to apply any security permissions from a top-level (parent) object to the lower-level (child) object using the same configuration.
   For example, you create a local configuration for an Organization. In the configuration, you deny permissions to change field labels, move fields, make fields required, and hide fields. After you save the configuration, you open Attachments under Organization and select the Inherit Permissions from Parent Object check box. All permissions from the Organization are applied to Attachments. In this example, because permissions to change field labels, move fields, make fields required, and hide fields are applied to the Organization, the permissions are also applied to the Attachment.
6. Click Save Configuration.
   The configuration is saved. Verify that you can correctly assign a configuration to a role.

Delete Saved Searches

As an Administrator, you can delete user-defined searches if they are not in use for long time.

NOTE

You cannot delete searches that are out-of-box.

Follow these steps:

1. Click the tab and optional sub tab for the object that you want to find.
2. On the left, click Manage Searches.
3. Under Search Results, select the user-defined search you want delete.
4. Click Delete.

Administering CA Asset Portfolio Management

This section contains the following articles:

- Administration
- Log In to CA APM
- Maintaining Security
- Managing Product Components
- Implementing Multi-Tenancy
- How to Secure CA APM Data with Filters
- How to Delete Unused Files from CA APM
- Purge Audit Data Administration
- Import Data
- CA APM Environment Promotion

Administration

The overall administration of CA APM involves setting up security, configuring the user interface, managing hardware reconciliation, and optionally changing configuration settings for product components. The product administration is flexible and you can complete the following administration tasks in any order that you want.

After you complete the following tasks, provide the CA APM URL and login credentials to all users so that they can log in to the product.

- **Security**
  You define security to control user access to the product and features. For example, you can provide one user with access to models and assets, and another user with access to legal documents.

  NOTE
  For more information, see Security.
- **User Interface Configuration**
  You configure the user interface to simplify how users enter, manage, and search for data. In addition, you secure and protect users from performing unauthorized tasks and ensure that you conform to your IT asset management practices. For example, you can globally hide the Contact page from all CA APM users. However, there are some users who must be able to see and update contact information. Therefore, you make the Contact page appear for those users. User interface configuration is flexible, and you can configure almost all aspects of the user interface.

  NOTE
  For more information, see How to Configure the User Interface.
- **Hardware Reconciliation**
  You use hardware reconciliation to match *discovered assets* to their corresponding *owned assets* from different logical repositories and manage your assets. You can identify unauthorized, missing, under-utilized, and over-utilized assets, which helps you to optimize your hardware asset base.

  NOTE
  For more information, see Hardware Reconciliation.
- **Product Components**
  You can change the product component configurations that were set up during the product installation. For example, you can change the listening port for Oracle. You can also add components to additional servers to maintain product performance and enable product scalability. For example, you can add a Hardware Reconciliation Engine on an additional server. The configuration is flexible, and you can change many component settings.

  NOTE
  For more information, see Add Component Servers.

Log In to CA APM

After the product is installed, your implementer verifies that all services are started and starts the web interface to verify that CA APM is ready to use. The implementer provides you with the URL and credentials to log in to the product. You can then prepare the product for users to manage assets.

By default, the login credentials (username and password) for the default System Administrator user are uapmadmin. You can change the password to meet your requirements by changing the settings for the CA EEM component.

> **NOTE**
> The default password can also be changed during the installation process.

Maintaining Security

This article contains the following topics:

Security

Before allowing user access to CA APM (ITAM), set up security to control access to the product, protect your repository from unauthorized or inaccurate changes, and make necessary data available to users. For example, you can provide one user with access to models and assets, and another user with access to legal documents.

Setting up security involves the following tasks:

1. Users. Define the users who can access the product.
2. User Roles. Define groups of users who perform similar tasks.
3. Authentication. Define how users are authenticated when they log in.
4. Searches. Define which users can use searches.
5. Configuration. Protect users from performing unauthorized tasks.

One or more system administrators perform these security tasks in CA APM. A system administrator with the user ID *uapmadmin* acts as a global system administrator, with complete control over all security aspects of the product.

You enforce security across the enterprise by using the web interface. Minimal database skills are required to perform these tasks.

User

You establish user security when you add new users to the product and assign a user ID and password. If a user does not have a valid user ID and password, they cannot log in. For each person, a user record is established, and the record is associated with a contact in the ca_contact table.

You can add users to the product in the following ways:

1. Import them.
2. Manually define them.

When manually defining users, you can immediately authorize them to use the product. However, when you import users, import them first, and then you can authorize them.

> **NOTE**
> When you define a user manually, a corresponding CA EEM user is also created. CA EEM verifies the user name and password when the user logs in to CA APM.

After you define all CA APM users, assign each user to a *user role* and assign the entire role access rights to determine what they see and can access when they log in.

Best Practices (Users and Roles)

Use the following best practices to effectively manage users and roles:

- A user must have a valid user ID and password, and be authorized, to log in.
- Remove a user from a role before assigning a new role to the user.

  NOTE
  The product does not allow you to assign a user to more than one role.
- Verify that there are no users assigned to a role before deleting the role.
- Delete users before deleting a role.

Import and Synchronize Users

WARNING

Verify that the user completing this task belongs to a role in which user management access is enabled.

You can import a list of users from an external user store such as an active directory through CA EEM, and synchronize them to be saved as contacts in CA APM. Importing users helps you to save time when defining your users, and helps ensure the accuracy of the user information. After you import and save the users, authorize them to access the product.

Follow these steps:

1. Click Administration, User/Role Management.
2. On the left, expand the User Management menu.
3. Click LDAP Data Import and Sync.
4. If multi-tenancy is enabled, select a tenant from the drop-down list.
5. Click Start LDAP Data Import and Sync.
   The import process begins and users are imported from the external store. If multi-tenancy is enabled, users are imported for the selected tenant as contacts into the ca_contact table. You can then authorize the imported users to access the product.

   WARNING
   The LDAP Data Import and Sync works for user names that begin with a letter or number. User names that begin with a special character are not imported.

Define a User

WARNING

Verify that the user completing this task belongs to a role in which user management access is enabled.

You define all users of CA APM and provide them with access to the product. After you define a user, assign a role to the user.

Follow these steps:

1. Click Administration, User/Role Management.
2. On the left, expand the User Management menu.
3. Click New User.
4. Enter the information for the new user and the contact-related information.
5. (Optional) Specify if you want to authorize the user to access to the product.
6. Click Save.
   The user is defined.

Authorize a User

WARNING

Verify that the user completing this task belongs to a role in which user management access is enabled.

You can authorize a user so they can log in and use the product. Before you can authorize a user, save the user as a contact.

Follow these steps:

1. Click Administration, User/Role Management.
2. On the left, expand the User Management menu.
3. Click Authorize Users.
4. Search to find the list of available users.
5. Select the user you want to authorize and click OK.
   The user appears in the Authorized Users list.
6. Click the Edit icon next to the user name.
7. (Optional) Select a contact to assign a user with contact details.

   NOTE
   If you do not select a contact, a new contact is created for the user.
8. (Optional) Select a role to assign to the user.
9. Click Authorize.
   The selected user is authorized to log in to the product.

Deny a User Access

WARNING

Verify that the user completing this task belongs to a role in which user management access is enabled.

You can deny a user access and prevent them from logging in to the product. For example, you hire a new asset technician and want to prevent them from using the product until they have received proper training. When you deny a user access, the contact information for the user is not deleted from the product.

Follow these steps:

1. Click Administration, User/Role Management.
2. On the left, expand the User Management menu.
3. Click Authorize Users.
4. Select the user for which you want to deny access from the Authorized Users list.
5. Click De-Authorize.
   The user is prevented from logging in to the product.

User Roles

A *user role* is the primary record that controls security and user interface navigation in the product. Each role defines a focused view of the product by exposing only the functionality necessary for users to perform the tasks that are assigned to their business roles. The default role for a user and the associated user interface configuration determine the data and functions that are available to the user. A user can belong to only a single role.

Define user roles to apply functional and field-level repository access rights. You determine and assign the level of access that is required for each role. Group the users with the same job function and assign them the corresponding role. Role assignment prevents the users from performing unauthorized tasks, such as adding or deleting data. For example, users in an Administrator role need full access to all records, while users in an Asset Technician role need limited access to fewer records.

NOTE
The product contains predefined System Administrator and user roles that you can use as the basis for user management.

You can perform several tasks to set up and manage user roles:

- Define a role.
- Assign a role to a user.
- Remove a user from a role.
- Update a role.
- Delete a role.
- Assign a configuration to a role.

Predefined Roles

The product provides a System Administrator role, which has complete control and access to all objects and tenant data. This role is associated with the System Administrator contact and cannot be deleted. A user in this role can define, update, and delete objects, in addition to defining and updating more roles to meet your business requirements. You cannot assign a configuration to the System Administrator role.

The product also provides the following predefined user roles to help you manage users:

- CA APM Asset Technician - Provides access to the data and functions that are required for working with asset information only.
- CA APM Contract Manager - Provides access to the data and functions that are required for working with legal documents and the contract management process only.
- CA APM Default User - Provides read-only access to a limited view of the product. This role can view most of the data in the product. However, this role cannot modify the product data.
- CA APM Fulfiller - Provides access to the data and functions that are required for asset fulfillment tasks only.
- CA APM Receiving - Provides access to the data and functions that are required for updating assets that are received from a fulfillment process only.

Each predefined user role has associated configurations, which provide access to the data that is required to complete the particular function. You can modify the configurations that are associated with each predefined role. The predefined roles are available only after a new installation.

Define a User Role

> **WARNING**
>
> Verify that the user completing this task belongs to a role in which role management access is enabled.

You can define customized user roles to meet your site-specific business requirements. For example, you can define one role with access to reconciliation management, and another with access to asset fulfillment.

Follow these steps:

1. Click Administration, User/Role Management.
2. On the left, expand the Role Management menu.
3. Click New Role.
4. Enter the information for the role.
   - **User Management Access**
     Select this check box so a user assigned to the role can access the user management functionality (Administration, User/Role Management, User Management). The User/Role Management subtab is available only when the role has access to the user management functionality, the role management functionality, or both.
   - **Role Management Access**
     Select this check box so a user assigned to the role can access the role management functionality (Administration, User/Role Management, Role Management). The User/Role Management subtab is available only when the role has access to the user management functionality, the role management functionality, or both.
   - **System Configuration Access**

Select this check box so a user assigned to the role can access the system configuration functionality (Administration, System Configuration).

- **Web Services Access**
  Select this check box so a user assigned to the role can access the CA APM web services documentation and WSDL (Administration, Web Services). If this check box is not selected and a user in the role attempts to access the web services from an external client application, the user receives a login error.
- **Filter Management Access**
  Select this check box so a user assigned to the role can access the filter management functionality (Administration, Filter Management).
- **Other Information Configuration Access**
  Select this check box so a user assigned to the role can access the Other Information Configuration functionality. This function allows the user to access additional related information for selected objects. The user can access this additional information by selecting menu items under Relationships on the left side of the page.
- **Data Importer User Access**
  Select this check box so a user assigned to the role can access the Data Importer functionality (Administration, Data Importer) with user permissions. Users can create imports and can modify or delete their own imports. Users can also view any import that was created by another user.
- **Data Importer Admin Access**
  Select this check box so a user assigned to the role can access the Data Importer functionality (Administration, Data Importer) with administrator permissions. Administrators can create imports and can modify or delete any import that was created by any user.
- **Reconciliation Management Access**
  Select this check box so a user assigned to the role can access the reconciliation rules management functionality (Administration, Reconciliation Management).
- **Asset Fulfillment Access**
  Select this check box so a CA Service Catalog user assigned to the role can perform asset fulfillment using CA Service Catalog.
- **Tenancy Admin Access**
  Select this check box so a user assigned to the role can access the multi-tenancy administration functionality to enable multi-tenancy, define tenants, define subtenants, and define tenant groups (Administration, Tenancy Management).
- **Normalization Access**
  Select this check box so a user assigned to the role can access the normalization rules management functionality (Directory, List Management, Normalization).
- **Mass Change Utilities Access**
  Select this check box so a user assigned to the role can access the Mass Change Utilities functionality. This function allows the user to change the asset family for a model and also to change the model for an asset.

5. (Optional) Specify the read/write permissions for tenants. Multi-tenancy expands the purpose of the role to control the tenant or tenant group that a user within the role can access. When multi-tenancy is enabled, the Tenant Information section includes Tenant Access Read and Tenant Access Write drop-down lists.

   NOTE
   The Tenant Information section is visible only when multi-tenancy is enabled. For information about how to enable multi-tenancy, see Implementing Multi-Tenancy. In addition, users associated with a tenant other than the service provider can only create or update objects associated with their own tenant. Only users associated with the service provider are permitted to create or update objects belonging to tenants other than their own.

   - **All Tenants**
     Contains no tenant restrictions. A user in a role with this access can view any object in the database (including public objects). In addition, a user associated with the service provider can update or create objects associated with

any tenant. When a service provider user with this access creates an object, the product requires the user to select the tenant of the new object.

- **Contact's Tenant**
  (Default value) Associates the role with the tenant of the contact. The product restricts a user in a role with this access to viewing, creating, and updating only those objects associated with their own tenant (and to view public objects). When a user with this access creates an object, the user cannot select a tenant. The tenant is automatically set to the tenant for the contact.
- **Contact's Tenant Group**
  Associates the role with the tenant group of the contact. The product restricts a user in a role with this access to viewing, creating, and updating only those objects associated with the tenants in their tenant group (and to view public objects). When a user with this access creates an object, the user can select any tenant belonging to the tenant group.
- **Single Tenant**
  Associates the role with a named tenant. When you select this option, select a specific tenant in either the Tenant Write or Tenant Read field. The product restricts a user in a role with this access to viewing, creating, and updating only those objects associated with the tenant you select (and to viewing public objects). When a user with this access creates an object, the user cannot select a tenant. The tenant is automatically set to the tenant you select.

  NOTE
  Only a service provider user can create or update data for a tenant other than their own. A tenant user in a role with single tenant access to another tenant is restricted to read access.
- **Tenant Group**
  Associates the role with a named tenant group. When you select this option, select a specific tenant group in either the Tenant Group Write or Tenant Group Read field. The product restricts a user in a role with this access to viewing only those objects that belong to any tenant in the tenant group. In addition, a user associated with the service provider can update or create objects associated with any tenant in the group. When a service provider user with this access creates an object, the product requires the user to select the tenant for the new object.
- **Update Public (check box)**
  Available only when you select All Tenants. Select this check box to authorize a user in the role to create or delete tenanted public data.

6. Click Save.
   The role is defined and you can assign users to the role.

Assign a Role to a User

WARNING
Verify that the user completing this task belongs to a role in which role management access is enabled. In addition, if you do not assign a role to a user, the Administration tab is hidden from the user.

You can assign a role to a user to define a focused view of the product and determine what they see when they log in. For example, assign an administrator to the system configuration role. You can assign a user to only a single role. Save a user as a contact before you assign the user to a role.

NOTE
Remove a user from their previous role before assigning a new role to the user.

Follow these steps:

1. Click Administration, User/Role Management.
2. On the left, expand the Role Management menu.
3. Click Role Search.
4. In the Role Contact area of the page, click Assign Contact.
   All users not assigned to a role appear.
5. Select the user for which you want to assign the role.
6. Click OK.

7. Click Save.
   The role is assigned to the user.

Remove a User from a Role

WARNING

Verify that the user completing this task belongs to a role in which role management access is enabled.

You can restrict the access rights for a user by removing them from a role. For example, an administrator is transferred to a different department and you remove them from the system configuration role. Remove a user from a role before assigning them to another role, or if they are no longer a part of your site or organization.

Follow these steps:

1. Click Administration, User/Role Management.
2. On the left, expand the Role Management menu.
3. Click Role Search.
4. Click the delete icon next to the user you want to remove from the role.
5. Click Save.
   The user is removed from the role.

Update a User Role

WARNING

Verify that the user completing this task belongs to a role in which role management access is enabled.

At any time, you can update a user role to change what the user sees when they log in to the product. For example, the users in a particular role no longer perform tenancy management functions. In this situation, remove the tenancy management access for the role.

WARNING
You can delete a role that is no longer active in your site or organization, or when the role functions are no longer required. You cannot delete the predefined System Administrator role.

Follow these steps:

1. Click Administration, User/Role Management.
2. On the left, expand the Role Management menu.
3. Click Role Search.
4. Search for and select a role.
5. To update the user role, change the information for the role and click Save.
   The role is updated.
6. To delete the user role, click Delete.
   The role is deleted.

Assign a Configuration to a Role

WARNING

Verify that the user completing this task belongs to a role in which role management access is enabled.

You can configure the user interface to simplify how users enter, manage, and search for data. When you assign a configuration to a role, you help ensure that any user assigned to the role sees the product as you have configured it for them.

Example: Assign a configuration to an asset manager

In this example, an asset manager must quickly view, and monitor the most important information that has been entered into the product for an asset. This information is used for reporting, cost analysis, and inventory control. The administrator

configures the search results to display the asset name, model name, quantity, serial number, operating system, purchase order number, and cost center. The administrator saves the configuration and assigns it to the asset manager role. When an asset manager logs in to the product, the configuration for the asset manager role is selected and appears.

NOTE
For more information about configurations, see How to Configure the User Interface.

Follow these steps:

1. Click Administration, User/Role Management.
2. On the left, expand the Role Management menu.
3. Click Role Search.
4. Search for and select a role.
5. Click Role Configuration.
6. Click Select New.
   The list of saved configurations appears.
7. Select the configuration you want to assign to the role.
8. Click OK.
9. Click Save.
   The configuration is assigned to the role. Any user assigned to the role sees the configuration when they log in to the product.

Authentication

Authentication is the process of obtaining identification credentials from a user such as name and password to validate their credentials to verify that the user exists. If the credentials are valid, the user is authenticated. After a user is authenticated, the authorization process determines whether the user can log in to the product.

NOTE
CA APM uses CA EEM to process user authentication.

The following types of authentication are supported:

- Form Authentication. A user is prompted for a user name and password to log in to the product.

  NOTE
  Form authentication is the default authentication type.
- Windows Integrated Authentication. A user already logged in to the Windows domain can access the product without having to provide additional login credentials.

  NOTE
  You can provide additional security by defining tab and menu configuration in the product to restrict the pages and tabs that a user can access.

Configure Form Authentication

WARNING
Verify that the user completing this task belongs to a role in which system configuration access is enabled.

You can configure form authentication so a user is prompted for a user name and password when logging in.

Follow these steps:

1. Click Administration, System Configuration.
2. On the left, click EEM.
3. Select Form from the Authentication Type drop-down list.
4. Click Save.
   Form authentication is enabled.

Configure Windows Integrated Authentication

WARNING
Verify that the user completing this task belongs to a role in which system configuration access is enabled.

You can configure Windows integrated authentication and reference the CA EEM server to the active directory used for authentication. With Windows integrated authentication enabled, a user already logged in to the Windows domain can access the product without having to provide any additional login credentials.

You can also configure Windows integrated authentication with CA EEM and CA SiteMinder. CA SiteMinder uses the active directory for authentication. For information about this configuration, see the CA EEM product documentation.

For Windows integrated authentication to work, the CA EEM server, the Active Directory, and the client computer making the authentication request must belong to the same domain.

In addition, when you create and authorize a user in the CA EEM local store with a user name that exists in the Active Directory, the corresponding Active Directory user is automatically authorized.

Follow these steps:

1. On the computer where CA EEM is installed, configure the CA EEM server to reference your Active Directory or LDAP system.

   NOTE
   For information about performing these functions, see the CA EEM product documentation.
2. In CA APM, click Administration, System Configuration.
3. On the left, click EEM.
4. Select Windows Integrated from the Authentication Type drop-down list.
5. Click Save.
   Windows integrated authentication is enabled.

Single Sign-On

Single sign-on is an authentication process where the user can enter one user ID and password and access a number of resources within the organization. Single sign-on eliminates the need to enter additional authentication credentials when switching from one solution to another.

Single sign-on lets users log in to the product automatically using Windows login information. After you add the user ID to any role, the product verifies the login credentials and displays the appropriate home page to the user.

NOTE
For single sign-on to work correctly, configure Windows user accounts as domain user accounts, and not as the local user accounts.

After enabling NTLM authentication, you may see a browser prompt in IE / Chrome /Firefox browsers for username and password to login to APM. Perform the below steps based on browser (IE, Chrome, and Firefox) that you are using:

Setting up Windows Authentication on IE and Chrome Browser

1. Open **Control Panel**, **Network and Internet**, **Internet Options**.
2. Click **Security** tab , select **Local Intranet** zone.
3. Click **Custom Level**.
4. Scroll down and select the **User Authentication** option. Set Login to **Automatic Login with current user name and password**.
5. Click Ok.
6. Click **Sites**.
7. Click **Advanced**. Add the CA APM Web URL link.

   NOTE
   Provide the correct website access protocol (http or https).

8. Click **Close**. Click **OK** to close the Internet Options.
9. Restart browser window and access the CA APM URL.

Setting up Windows Authentication on Firefox

1. Open Firefox.
2. In the Browser address bar, enter **About:Config**
3. You will receive a security warning. To continue, follow the steps in the prompt.
4. Search for the settings below by browsing through the list or searching for them individually. Locate each setting and update the value to the following:

Setting	Value **
network.negotiate-auth.delegation-uris	Enter the Fully qualified name of your IIS server.
network.automatic-ntlm-auth.trusted-uris	Enter the Fully qualified name of your IIS server.
network.automatic-ntlm-auth.allow-proxies	True
network.negotiate-auth.allow-proxies	True

5. Restart Firefox.

Search Security

Default searches let you find objects in the repository. For example, use the default searches to find assets, models, contacts, and so forth. The security for the default searches makes them available to all users and configurations. You can use these searches to create additional searches.

In contrast, you can apply security to the searches that you create to limit who can use the search. When you save a configured search, you can select specific user roles and configurations (restricted to administrators). By default, the security for the searches you create makes them available to all users and configurations. By applying unique security to your searches, you help ensure that certain users cannot view sensitive information that a search returns.

Consider the following information when applying security to searches:

- You can access all searches (default and user-defined searches) so that you can configure and troubleshoot searches for users.
- You can access all scheduled searches and exports so that you can configure and troubleshoot scheduled searches and exports for users.
- All users that are assigned to a role and configuration can access the default searches and the user-defined searches that are assigned to the role and configuration. However, the search results that users see for the default searches do not display information and fields that you hide and secure.
- When a default and user-defined search becomes invalid because of configuration changes, you may not need the search. You can delete any default and user-defined search in CA APM.

Troubleshooting Search Security

Troubleshooting tips related to search security help you when working with configured searches.

- Role Cannot Be Assigned to a Configured Search
- Configuration Cannot Be Assigned to a Configured Search

Role Cannot Be Assigned to a Configured Search

Valid on all supported operating environments.

Symptom:

When attempting to provide a role with access to a configured search, I receive an error similar to one of the following errors:

You cannot assign role <role name> to the search because the role cannot access the following field(s): <field name> on Asset Type <asset family>

You cannot assign role <role name> to the search because the role cannot access the Asset Type <asset family>

You cannot assign role <role name> to the search because the configuration cannot access the following field(s): <field name>, <field name>

Solution:

Use any of the following solutions to resolve this error:

1. Update the configuration and provide the role or user with access to the search.
2. Update the configuration and remove the hidden field from the search.
3. Do not allow the role to access the search.
4. Remove the configuration from the role.

Configuration Cannot Be Assigned to a Configured Search

Valid on all supported operating environments.

Symptom:

When attempting to provide a global or local configuration with access to a configured search, I receive an error similar to one of the following errors:

You cannot assign configuration <configuration name> to the search because the configuration cannot access the following field(s): <field name> on Asset Type <asset family>

You cannot assign configuration <configuration name> to the search because the configuration cannot access the Asset Type <asset family>

You cannot assign configuration <configuration name> to the search because the configuration cannot access the following field(s): <field name>, <field name>

Solution:

Use any of the following solutions to resolve this error:

1. Update the configuration and make the hidden field available to the search.
2. Update the configuration and remove the hidden field from the search.
3. Do not allow the configuration to access the search.

Managing Product Components

This article contains the following topics:

Product Components

After you install CA APM, you can manually configure many of the product components. In addition, you can add components to additional servers to maintain optimum performance and enable scalability. For example, you can add a Hardware Reconciliation Engine or an additional component server. The configuration is flexible, and you can change many component settings.

> **NOTE**
> For a description of each product component, see Step 9: Configure Product Components.

Configure a Product Component

WARNING
Verify that the user completing this task belongs to a role in which system configuration access is enabled.

You can change the component configurations that were set up during the product installation. For example, you can change the name of the SMTP server that is used to send email.

Follow these steps:

1. Click Administration, System Configuration.
2. On the left, select the product component.
3. Enter the new configuration settings for the component:
   - Database
     - Oracle
     - SQL Server
   - Web Server
   - Application Server
   - Hardware Reconciliation Engine
   - CA EEM
   - Export Service
   - Data Importer
   - Data Importer Engine
   - LDAP Data Import and Sync Service
   - CORA
   - Storage Manager Service
   - Event Service
   - Common Asset Viewer
   - WCF Service
4. Click Save.
   The configuration settings are saved.

Oracle Database Configuration Settings

After you install the product, if you are using Oracle as the database for the CA MDB, you can change the settings for the Oracle database server.

The following fields require explanation:

- **DBA User Name**
  The user name for connecting to the target database. The user name must be for a privileged user.
  Default: sys
- **Listen Port**
  The database connection port.
  Default: 1521
- **Oracle Service Name**
  The service name for the target database.
  Default: orcl
- **Oracle Net Service Name**
  The net service name for the target database.
  Default: orcl
- **Tablespace Path**
  The location of the Oracle tablespaces.

Default: c:\oracle\product\10.2.0\oradata\orcl

- **Data Tablespace Name**
  The name of the data tablespace.
  Default: MDB_DATA
- **Data Tablespace Size**
  The amount of disk space to be allocated for the data tablespace.
  Default: 400 MB
- **Index Tablespace Name**
  The name of the index tablespace.
  Default: MDB_INDEX
- **Index Tablespace Size**
  The amount of disk space to be allocated for the index tablespace.
  Default: 100 MB
- **mdbadmin Password**
  The password that is associated with the mdbadmin user. If a new CA MDB is being created, this is the password that is assigned to the mdbadmin user.
- **Command Timeout**
  The maximum amount of time that the application waits for a response from the database.
- **Stored Procedure Command Timeout**
  The maximum amount of time that the application waits for a response from the database stored procedures.
- **Max Connection Pool Size**
  The maximum number of requests that the database can process simultaneously.
- **Enable CORA Connection Pooling**
  Placeholder for the flag that enables CORA connection pooling. Currently, CORA does not support connection pooling.
- **CORA Connection Pool Lifetime**
  Placeholder for the lifetime of the CORA connection pool. Currently, CORA does not support connection pooling.
- **CORA Connection Pool Size**
  Placeholder for the size of the CORA connection pool. Currently, CORA does not support connection pooling.
- **Last Run Date Option**
  Determines if the imported data should update the existing CA APM data. CA APM receives discovered hardware data imports and uses the data to match ownership and discovery data. CA APM determines if the imported data is more current than the existing data by comparing their inventory dates. Then CA APM decides whether the imported data should update the existing CA APM data.
  Default: 2
  See the following table for a description of all valid options.

Option	Definition
1	Always update the existing CA APM data with the imported data.
2	(Default) Update the existing CA APM data only if the inventory date of the imported asset is more current than the inventory date of the existing asset. If the imported asset does not have an inventory date, an error occurs and the imported asset data does not update the existing CA APM data.
3	Update the existing CA APM data only if the inventory date of the imported asset is more current than the inventory date of the existing asset. If the imported asset does not have an inventory date, update the existing CA APM data with the imported data.

SQL Server Database Configuration Settings

After you install the product, if you are using SQL Server as the database for the CA MDB, you can change the settings for the SQL Server database server.

The following fields require explanation:

- **SQL Server Login**
  The user name for connecting to the target database. The user name must have sysadmin role privileges assigned in SQL Server.
  Default: sa
- **SQL Server TCP/IP Port**
  The database connection port.
  Default: 1433
- **Command Timeout**
  The maximum amount of time that the product waits for a response from the database.
- **Stored Procedure Command Timeout**
  The maximum amount of time that the product waits for a response from the database stored procedures.
- **Max Connection Pool Size**
  The maximum number of requests that the database can process simultaneously.
- **Enable CORA Connection Pooling**
  Placeholder for the flag that enables CORA connection pooling. Currently, CORA does not support connection pooling.
- **CORA Connection Pool Lifetime**
  Placeholder for the lifetime of the CORA connection pool. Currently, CORA does not support connection pooling.
- **CORA Connection Pool Size**
  Placeholder for the size of the CORA connection pool. Currently, CORA does not support connection pooling.
- **SQL Server Host Name**
  The host name of the server that is used for SQL Server.
- **Last Run Date Option**
  Determines if the imported data should update the existing CA APM data. CA APM receives discovered hardware data imports and uses the data to match ownership and discovery data. CA APM determines if the imported data is more current than the existing data by comparing their inventory dates. Then CA APM decides whether the imported data should update the existing CA APM data.
  Default: 2
  See the following table for a description of all valid options.

Option	Definition
1	Always update the existing CA APM data with the imported data.
2	(Default) Update the existing CA APM data only if the inventory date of the imported asset is more current than the inventory date of the existing asset. If the imported asset does not have an inventory date, an error occurs and the imported asset data does not update the existing CA APM data.
3	Update the existing CA APM data only if the inventory date of the imported asset is more current than the inventory date of the existing asset. If the imported asset does not have an inventory date, update the existing CA APM data with the imported data.

Web Server Configuration Settings

After you install the product, you can change the settings for the web server.

The following fields require explanation:

- **Web Server or Load Balancer IP/Host**
  The CA APM installation, by default, sets this field to the web server host name.
  - In a single web server environment, you can enter the web server host name, or the web server IP address.
  - In a multiple web server environment, you can enter either the web server host name, or the IP address of the Load Balancer.

    NOTE

    : The web server can be registered with a different name in the Domain Name System (DNS) than what is registered as the web server host name. In this situation, specify the different name in this field.
- **Authentication Timeout**
  Amount of time, in milliseconds, that users can be inactive before they are automatically logged out of the product and must log in again.
  Default: 3600000 (1 hour)
- **State Data**
  The temporary view data that is stored at the server side for each user until the user logs off. This data can be stored either in the database or on the web server file system. The value of State Data can be SERVER (FileSystem) or DATABASE.
- **Cache Timeout**
  The elapsed time-out period for cached files to be deleted from the store.
  When a user logs in, along with view data, documents are stored in system memory (Cache). If the user is not referring to these documents, the documents become stale. After some time, these documents are deleted. The time limit is the Cache Timeout.
- **Auto complete Result Count**
  The number of values to display in the Auto Complete drop-down lists. In fields with lists, the user can start to type a value and the product provides a list of matching values. The user can then select a value from this Auto Complete list.

  NOTE
  If you set a low number for this parameter, the Auto Complete list may not be useful since the user will have to type most of the value before finding it in the list. If you set a high number for this parameter, the list could perform slowly.
- **Homepage**
  The default CA APM home page that is opened after the user logs in.
- **SM Web Service Protocol**
  The protocol that is used to access the Storage Manager Service.
  The Storage Manager Service provides file storage facility to other product components. When the other components want to interact with the Storage Manager Service, the components use this protocol.
- **SM Web Service Port**
  The port on which the Storage Manager Service is running.
  This port is the HTTP port on which the Storage Manager Service is hosted. The default port is 80. If the Storage Manager Service is configured to host on a different port, that port number is shown.
- **EEM Backend**
  The name of the server where CA EEM is installed. This value is populated during installation.
- **Reporting Web Service Server**
  The CA Business Intelligence server name and port. This value is populated during installation.
- **Reporting Timeout**
  The time-out in seconds for a connection to the CA Business Intelligence (reporting) web service.
- **Reporting Name**

The name of the CA Business Intelligence (reporting) engine. This name is always "Allegheny Reporting Engine".

- **Reporting User**
  The CA Business Intelligence (reporting) user with administrative privileges.
- **Reporting Password**
  The password for the Reporting User.
- **External AuthHeader**
  This header works with the External setting of Authentication Type on the CA EEM configuration. The external authentication mechanism sets information in the HTTP header for the CA APM web pages to receive. One piece of information is the User Id. The External AuthHeader is the name of the variable external authentication that sets the User Id value. The External AuthHeader setting must match the configured setting in the external authentication for which the User Id is provided.
- **From Address**
  The From email address for the notifications that are sent from the Event Service.
- **To List**
  The list of recipients who receive the email notifications about issues from the Event Service.
- **Cc List**
  The list of recipients in the Cc list who receive the email notifications about issues from the Event Service.
- **Bcc List**
  The list of recipients in the Bcc list who receive the email notifications about issues from the Event Service.
- **Email Subject**
  The subject line in the email notification about issues that the Event Service sends to the recipients in the To List, Cc List, and Bcc List.

  NOTE
  If you change the setting on Administration, System Configuration, Web Server for the Web Server Protocol or the Web Server or Load Balancer IP/Host, you must restart the Windows service for CA Asset Portfolio Management - Export Service.

Application Server Configuration Settings

After you install the product, you can change the settings for the application server.

The following fields require explanation:

- **Application Server or Load Balancer IP/Host**
  The CA APM installation, by default, sets this field to the application server host name.
  – In a single application server environment, you can enter the application server host name, or the application server IP address.
  – In a multiple application server environment, you can enter either the application server host name, or the IP address of the Load Balancer.

  NOTE
  The application server can be registered with a different name in the Domain Name System (DNS) than what is registered as the application server host name. In this situation, specify the different name in this field.
- **Authentication Timeout**
  Amount of time, in milliseconds, that users can be inactive before they are automatically logged out of the product and must log in again.
  Default: 3600000 (1 hour)
- **EEM Backend**
  The name of the server where CA EEM is installed. This value is populated during installation.
- **USM Web Service URL**
  The URL for the USM web service. This value is used when the product is integrated with CA Service Catalog or CA Service Desk Manager.

NOTE
If you change the setting on Administration, System Configuration, Application Server for the Web Service Protocol or the Application Server or Load Balancer IP/Host, you must restart the following Windows services:

- CA Asset Portfolio Management - Export Service
- CA Asset Portfolio Management - Event Service
- CA Asset Portfolio Management - HW Reconciliation Engine
- CA Asset Portfolio Management - LDAP Import Service

Hardware Reconciliation Engine Configuration Settings

After you install the product, you can change the settings for the Hardware Reconciliation Engine.

The following fields require explanation:

- **Message Queue Retention**
  Number of days that records remain in the message queue before the Hardware Reconciliation Engine purges them.
  Default: 7
- **Refresh Lock Record Count**
  Engine performance tuning setting that works with the Lock Stale Interval setting. The number of records that the Hardware Reconciliation Engine adds or updates before refreshing the lock on the reconciliation rule record. If the lock refresh does not happen within the number of seconds specified in the Lock Stale Interval setting, the reconciliation rule record is available to another Hardware Reconciliation Engine.
  Default: 100
- **Lock Stale Interval**
  Engine performance tuning setting that works with the Refresh Lock Record Count setting. If the lock refresh does not happen on the reconciliation rule, the amount of time, in seconds, after which a reconciliation rule record is available to another Hardware Reconciliation Engine.
  Default: 600
- **Web Service Authentication Timeout**
  Amount of time, in milliseconds, that users can be inactive before they are automatically logged out of the service and must log in again.
  Default: 3600000 (1 hour)
- **Processing Mode**
  Specifies whether the Hardware Reconciliation Engine processes continuously. The product supports the following option:
  – **0**
    Process in continuous reconciliation mode.
- **Engine Snooze Time**
  Amount of time, in milliseconds, that the Hardware Reconciliation Engine waits between processing cycles.
  Default: 300000 (5 minutes)
- **Connection Retry Time**
  Amount of time, in milliseconds, that the Hardware Reconciliation Engine waits between attempts to connect to the database.
  Default: 60000 (1 minute)
- **Engine Debug Level**
  Debug level for the message queue. The levels are Fatal, Error, Warning, Info, and Debug.
  Default: Fatal
- **Web Service Batch Size**
  Number of records that are sent to the web service at one time for update processing.
  Default: 50
- **Pending Modification Retention**
  Number of days that pending modification requests remain in the queue before the Hardware Reconciliation Engine purges them. This setting affects any tenant that is processed by the Hardware Reconciliation Engine.

Default: 7

CA EEM Configuration Settings

After you install the product, you can change the settings for CA EEM.

The following fields require explanation:

- **Authentication Type**
  Type of authentication allowed:
  - **Form**. A user is prompted for a user name and password to log in to the product.
  - **Windows Integrated**. A user who is already logged in to the Windows domain can access the product without having to provide additional login credentials.
  - **External**. A user is authenticated by an external access management system, for example, CA SiteMinder.

  Default: Form
- **uapmadmin Password**
  Password for the uapmadmin user to access the web and application servers.

Export Service Configuration Settings

After you install the product, you can change the settings for the Export Service.

The following fields require explanation:

- **On Demand Request Period**
  Amount of time, in milliseconds, that the Export Service waits between On Demand Request processing cycles.
  Default: 5000 (5 seconds)
- **On Demand Maximum Threads**
  Number of On Demand Request processing threads.
  Default: 2
- **Scheduled Requests Period**
  Frequency, in milliseconds, of Scheduled Requests processing cycles.
  Default: 7200000 (2 hours)
- **SMTP Server**
  Email server name.
- **Authentication Timeout**
  Amount of time, in milliseconds, that users can be inactive before they are automatically logged out of the product and must log in again.
  Default: 360000 (6 minutes)
- **Purge Start Time**
  Time of day, in 24-hour format (Universal Time) when the purge thread starts.
  Default: 5 (5:00 a.m. Universal Time)
- **Search Batch Size**
  Maximum number of data records returned for each search in a single export request. Changing the default value may affect performance. For example, a lower value may increase the number of web service calls that are needed to retrieve all the data, and a higher value may require a longer period of time to gather all the records.
  Default: 2000
- **SM Web Service Protocol**
  The protocol that is used to access the Storage Manager Service.
- **SM Web Service Port**
  The port on which the Storage Manager Service is running.
- **Export Service Email Address**
  The email address that is used by the Export Service for notifications.

Data Importer Configuration Settings

After you install the product, you can change the settings for the Data Importer.

The following fields require explanation:

- **Authentication Timeout**
  Amount of time, in milliseconds, that users can be inactive before they are automatically logged out of the component and must log in again.
  Default: 3600000 (1 hour)
- **Maximum Batch Record Size**
  Maximum number of records in a Data Importer batch.
  Default: 50

Data Importer Engine Configuration Settings

After you install the product, you can change the settings for the Data Importer Engine.

The following fields require explanation:

- **SMTP Server Value**
  The email server name.
- **Scheduled Requests Period**
  The amount of time that the Data Importer Engine waits before checking for pending scheduled import requests. This value applies to scheduled data imports only.
  Default: 60000 (60 seconds)
- **On Demand Request Period**
  The amount of time that the Data Importer Engine waits before checking for pending on demand import requests. This value applies to on demand data imports only.
  Default: 60000 (60 seconds)
- **Max Batch Record Size**
  Maximum number of records in a Data Importer Engine batch.
  Default: 100
- **Max Job Threads**
  The maximum number of Data Importer Engine threads that can be running simultaneously for one import job.
  Default: 5
- **Max Import Threads**
  The maximum number of Data Importer Engine threads that can be running simultaneously for all import jobs.
  Default: 5

LDAP Data Import and Sync Service Configuration Settings

After you install the product, you can change the settings for the LDAP Data Import and Sync Service.

The following fields require explanation:

- **DB Check Time**
  Amount of time, in milliseconds, that the service checks the status (active or sleep mode). If the status is active, the service starts importing data.
- **EEM Backend**
  The name of the server where CA EEM is installed. This value is populated during installation.

CORA Configuration Settings

After you install the product, you can change the settings for CORA (CA APM Registration Service).

The following fields require explanation:

NOTE
Changes to the Common parameters affect the Web Server, WCF Service, Hardware Reconciliation, and Application Server components. Changes to the Registration Service parameters affect the Registration Service component.

- **(Common) Enable CORA**
  Enables Common Object Registration API features for the Web Server, WCF Service, Hardware Reconciliation, and Application Server components.
  Default: False
- **(Common) Enable CORA ID Generation**
  Allows the CORA tables to get the next CORA ID for a new record for the Web Server, WCF Service, Hardware Reconciliation, and Application Server components.
  Default: True
- **(Registration Service) Enable CORA**
  Enables the Common Object Registration API features using the Registration Service for all CA APM components.
  Default: True
- **(Registration Service) Enable CORA ID Generation**
  Allows the CORA tables to get the next CORA ID for a new record using the Registration Service for all CA APM components.
  Default: False

Storage Manager Service Configuration Settings

After you install the product, you can change the settings for the Storage Manager Service.

The following fields require explanation:

- **Purge Start Time**
  Time of day, in 24-hour format, when the Storage Manager Service starts to delete unused files.
- **Authentication Timeout**
  Amount of time, in milliseconds, that users can be inactive before they are automatically logged out of the service and must log in again.
  Default: 3600000 (1 hour)

Event Service Configuration Settings

After you install the product, you can change the settings for the Event Service.

The following fields require explanation:

- **Provider URL**
  The URL for accessing the workflow provider (for example, CA Process Automation).
  Example: The following URL is the default CA Process Automation workflow web services URL:
  ```
  http://<wf_hostname>:<wf_tomcat_port>/itpam/soap
  ```
- **Provider Authentication Type**
  The type of authentication (user or CA EEM) to be used with the Event Service.
  Default: User (CA EEM authentication is not currently supported for the Event Service.)
- **Provider User Name**
  The user ID for logging in to the workflow provider.
- **Provider Password**
  The user password for logging in to the workflow provider.
- **Provider Process Path**
  The path for accessing the start request forms for the workflow provider. These forms must be available for the CA APM integration with the workflow provider. For more information, see your workflow provider documentation.
  Default: /
- **Number of Events to be Processed**

The maximum number of events to be processed in one Web Service call.
Default: 2000

- **Time of the day for Event purge (in hours, GMT)**
  The time of day, in 24-hour format (GMT), when CA APM starts to purge event definitions that are marked for deletion.
  Default: 5 (5:00 a.m. GMT)
- **CMDB Audit Share Required**
  The indicator that enables audit sharing with the CMDB.
  The CMDB common database tables can be used by multiple applications. For example, the ca_contact table is used by CA APM, CA Service Catalog, and CA Service Desk Manager when they are integrated. An audit table maintains the changes that are made to these common tables. When there is a change to any of the CMDB objects in CA APM and this value is set to true, the change audit is posted to the CMDB audit table.
- **From Address**
  The From email address for the notifications that are sent from the Event Service.
- **To List**
  The list of recipients who receive the email notifications about issues from the Event Service.
- **Cc List**
  The list of recipients in the Cc list who receive the email notifications about issues from the Event Service.
- **Bcc List**
  The list of recipients in the Bcc list who receive the email notifications about issues from the Event Service.
- **Email Subject**
  The subject line in the email notification about issues that the Event Service sends to the recipients in the To List, Cc List, and Bcc List.
- **Interval between Event Occurrence check (in milliseconds)**
  The amount of time, in milliseconds, that CA APM waits between database checks for field changes related to defined events.
  If SAM capabilities are enabled, verify that this parameter is set to 30000. If SAM capabilities are not enabled, verify that this value matches the setting in the Event Service configuration file.
  Default (without CA SAM implementation): 3600000 (1 hour)
  Default (with CA SAM implementation): 30000 (30 seconds)
- **Interval between triggering events check (in milliseconds)**
  Amount of time, in milliseconds, that CA APM waits between database checks for triggered events that need to be sent to the workflow provider.
  If SAM capabilities are enabled, verify that this parameter is set to 60000. If SAM capabilities are not enabled, verify that this value matches the setting in the Event Service configuration file.
  Default (without CA SAM implementation): 3600000 (1 hour)
  Default (with CA SAM implementation): 60000 (60 seconds)
- **Interval between triggered events status update (in milliseconds)**
  The amount of time, in milliseconds, that CA APM waits between updates to the status of triggered events that were sent to the workflow provider.
  If SAM capabilities are enabled, verify that this parameter is set to 60000. If SAM capabilities are not enabled, verify that this value matches the setting in the Event Service configuration file.
  Default (without CA SAM implementation): 3600000 (1 hour)
  Default (with CA SAM implementation): 60000 (60 seconds)
- **Interval between asset contact update (in milliseconds)**
  The amount of time, in milliseconds, that CA APM waits between updates to asset contacts in the CA CMDB.
  Default: 43200000 (12 hours)
- **CA SAM Status Update Frequency**
  The frequency for updating the status of CA SAM import jobs in the MDB (in milliseconds).
  Default: 120000 (120 seconds)
- **On Demand Max Threads**

The maximum number of threads for processing the data synchronization between CA APM and CA SAM. The default (zero) indicates that the system creates the required number of threads, depending on the system hardware configuration. Any value other than the default value uses the same number of threads, regardless of the system configuration.
Default: 0

- **CA SAM Events Notification Email**
  The CA APM administrator email address for receiving notifications about the CA SAM data synchronization.
- **Authorization Token**
  The token that establishes communication between the CA APM Event Service and the CA SAM Import and Export Service. This value must match the CA SAM Import and Export Service configuration setting.

  NOTE
  If you change this value, you must update the value of the Authorization Token for the CA SAM Import and Export Service on the CA SAM server to match this value.

NOTE
For more information about events and notifications, see Events and Notifications for CA APM.

Common Asset Viewer

After you install the product, you can change the settings for the Common Asset Viewer.

WARNING

The Tomcat port number for CA APM defaults to 9080. If another product that is integrated with CA APM uses this port number, change the port number in CA APM so that you do not have a conflict.

The following fields require explanation:

- **Tomcat Port**
  Port that should be used for the Apache Tomcat server that is processing the Common Asset Viewer.
  Default: 9080

  NOTE
  You must first update the port in the Apache Tomcat configuration file before you change the setting in the product. For information about updating the Apache Tomcat configuration file, see Update the Apache Tomcat Configuration File.

WCF Service Configuration Settings

After you install the product, you can change the settings for the Windows Communications Foundation (WCF) Service component.

The following fields require explanation:

- **WCF Service Load Balancer IP/Host**
  The CA APM installation, by default, sets this field to the WCF Service server host name.
  – In a single WCF Service server environment, you can enter the WCF Service server host name, or the WCF Service server IP address.
  – In a multiple WCF Service server environment, you can enter either the WCF Service server host name, or the IP address of the Load Balancer.

  NOTE
  The WCF Service server can be registered with a different name in the Domain Name System (DNS) than what is registered as the WCF Service server host name. In this situation, specify the different name in this field.
- **Authentication Timeout (in milliseconds)**
  Amount of time, in milliseconds, that users can be inactive before they are automatically logged out of the service and must log in again.

Default: 3600000 (1 hour)

- **Operation Threshold**
  The maximum number of records that can be sent by or returned to the client or server. When you call the Search method from a WCF client program, this value represents the maximum number of records that can be returned to you. If you call the Create method, this value represents the maximum number of records that you can send at one time.
- **EEM Backend**
  The name of the server where CA EEM is installed. This value is populated during installation.

SAM - Import Driver Configuration Settings

After you install the product, you can change the settings for the CA SAM Import Driver.

The following fields require explanation:

- **Server**
  The name of the server where the CA SAM Import Driver component is installed.
- **Username**
  The user name that is required for adding, changing, or deleting records with the Data Importer.
- **ITAM Root Path**
  The path to the root location where the product is installed.
- **File Path**
  The path to the root location where CA SAM export files are imported.
  Example: *[ITAM Root Path]*\ITAM\Import Driver\Input
- **Data Importer Executable Path**
  The path to the Data Importer executable file (ITAM Data Importer.exe).
  Example: *[ITAM Root Path]*\ITAM\Data Importer\ITAM Data Importer.exe
- **CA SAM Server Name**
  The name of the server where CA SAM is installed.

Software Asset Management Configuration Settings

After you install the product, you can configure the settings for software asset management. After you configure these settings, restart the Apache Tomcat Common Asset Viewer service.

> **NOTE**
> Also restart the Apache Tomcat Common Asset Viewer service if you change the entries in any of the following fields at a later time:

- CA SAM Web Service WSDL URL
- CA SAM Web Service Login
- CA SAM Web Service Password

The following fields require explanation:

- **CA SAM Web Client URL**
  Specifies the URL for the CA SAM home page.

  > **NOTE**
  > You can copy the web client URL from the CA SAM home page after you log in.

  CA SAM Import Export Webservice URL
  Specifies the URL for the CA SAM web service. Use the following format:

  ```
  http://[CA SAM System Name]:[Port Number]/SAMImportExportService/Service.svc
  ```

  – Replace [CA SAM System Name] with the name of the CA SAM server.
  – Replace [Port Number] with the port number where the CA SAM Import and Export Service is hosted.

  Enable SAM Capabilities

Specifies that software asset management capabilities are enabled. If you previously had CA SCM fields on the CA APM user interface, they are removed after you select this check box.

CA SAM Web Service WSDL URL

The URL for the CA SAM Web Service Definition Language (WSDL). This URL is used to access the CA SAM web service. Use the following format:

```
http://[CA SAM System Name]:[Port Number]/prod/soap/dyn_server.php
```

– Replace [CA SAM System Name] with the name of the CA SAM server.
– Replace [Port Number] with the port number where the CA SAM Web Service is hosted.

CA SAM Web Service Login

Login name for the CA SAM web service.

> **NOTE**
> Verify that this login name and the CA SAM Web Service Password match the login name and password in the config_soap.inc file. This file is found in the following CA SAM installation folder path:

```
app\includes\prod\st\config_soap.inc
```

> **WARNING**
> The default content of the config_soap.inc file is commented. Remove the comment delimiters (/* */) and configure the login name and password.

CA SAM Web Service Password

Login password for the CA SAM web service.

CA SAM SSO Encryption Algorithm

Specifies the encryption algorithm to be used for single sign-on access to CA SAM from the CA IT Asset Manager common home page.

This entry must match the entry in CA SAM System Configuration for the security_auth_token_cipher field.

> **NOTE**
> For more information about CA SAM single sign-on, see the description of single sign-on in the *CA Software Asset Manager Administration Manual.*

CA SAM SSO Authentication Mechanism

Specifies the mechanism to be used for logging in to CA SAM.

This entry must match the entry in CA SAM System Configuration for the security_auth_method field.

> **NOTE**
> We recommend that you select auth_token_password for this mechanism. The auth_token mechanism disables the login for other CA SAM users.

CA SAM SSO Field to Authenticate User

Specifies the type of unique identifier (import ID or email address) that is used to authenticate the user.

This entry must match the entry in CA SAM System Configuration for the security_auth_token_user_identifier field.

CA SAM SSO Secret Key

Specifies the key that CA APM and CA SAM share and that is used to encrypt and decrypt the user authentication. This key ensures that CA APM users who do not have the proper authentication cannot access CA SAM.

This entry must match the entry in CA SAM System Configuration for the security_auth_token_key field.

Enable SSO from CA Asset Portfolio Management to CA Service Desk Manager

CA Service Management allows you to enable SSO from CA Asset Portfolio Management (CA APM) to CA Service Desk Manager. For more information, see Enable Single Sign-On (SSO) from CA ITAM to CA Service Desk Manager.

Add Component Servers

WARNING
Verify that the user completing this task belongs to a role in which system configuration access is enabled.

After the product is installed, you can add components to additional servers to maintain optimum performance and enable scalability as your enterprise grows. You can install the following components on one or more servers:

- Web server
- Application server
- Hardware Reconciliation Engine
- Data Importer
- WCF
- Data Importer Engine

Follow these steps:

1. Click Administration, System Configuration.
2. On the left, select the product component.
3. Specify the required server connection information in the Add Component to Server section, including the administrator username and password for the server.
4. Click Add.
   The server is added to the Component Server List.
5. Enter the configuration settings for the new component server:
   - Web Server
   - Application Server
   - Hardware Engine
   - Data Importer
6. Click Save.

NOTE
Install CA APM on the server you just added. For information about installing, see Implementing CA IT Asset Manager .

Modify the Debugging Level for Component Service Log Files

The product components have corresponding Windows services. These services create log files that allow you to verify the service status and review error details. The amount of detail information in a log file depends on the specified debugging level. The default debugging level for the service log files is Fatal. You can change the default level for the Windows services by editing the log configuration file for each component.

Follow these steps:

1. Navigate to a component folder on the application server where CA APM is installed.
   Example:
   ```
   [ITAM Root Path]\ITAM\Hardware Engine
   ```
2. Open the logging.config file in a text editor (for example, Notepad).
3. Locate the debugging level value statements.
4. Edit the statements to include your debugging levels.

NOTE
The log configuration files contain comments that explain the different debugging level values.

5. Save the log configuration file.

Implementing Multi-Tenancy

This article contains the following topics:

Multi-Tenancy

Multi-tenancy is the ability for multiple independent tenants (and their users) to share a single implementation of CA APM. Tenants only interact with each other in defined ways, as specified by their roles and tenant hierarchies. Typically, unless granted access by a role or tenant hierarchy, each tenant views the CA APM implementation as solely for its own use and cannot update or view the data for another tenant.

Multi-tenancy allows tenants to share hardware and application support resources, which reduces the cost of both, while gaining many benefits of an independent implementation.

Multi-tenancy is installed automatically during the CA APM installation. After you have installed CA APM, follow the steps in this section to implement multi-tenancy.

Service Provider

The *service provider* is the primary tenant (owner) in a CA APM multi-tenancy implementation. The first tenant added to a CA APM implementation is always the service provider tenant. The service provider tenant cannot have a parent tenant.

CA APM associates the privileged user (typically, uapmadmin) with the service provider tenant.

Only the service provider tenant can perform any of the following CA APM tasks:

- Define, edit, or delete tenants.
- Allow tenants to have subtenants.
- Update tenanted public data.

> **NOTE**
> The CA APM administrator can grant tenant users access to data other than their own. In addition, a user role can specify separate read and write access to certain tenant groups for users within that role.

How Multi-Tenancy Works

When you enable multi-tenancy, you can grant each contact access to all tenants (public), a single tenant, or a tenant group (user-defined or product-maintained). The role for a contact controls access, which specifies read and write access independently.

If multi-tenancy is enabled, most CA APM objects include a tenant attribute that specifies the tenant that owns the object. Objects are categorized into three groups, depending on their tenant attribute and how the object is used:

- **Untenanted**
  Defines objects without a tenant attribute. All data in these objects is public, and any user can create and update untenanted public data.
  Tenant Required
  Defines objects with a tenant attribute that cannot be null (enforced by CA APM, not the DBMS). All data in these objects is associated with individual tenants; there is no public data.
  Tenant Optional
  Defines objects with a tenant attribute that can be null. You can either create these objects as tenanted or public. When you select a tenant in a tenant drop-down to create an object, the object becomes a tenanted object. However, when you select the Public Data option in a tenant drop-down, the object becomes a tenanted public object. Users assigned to a role that only exposes a single tenant do not see a tenant drop-down when entering data.

When a user queries the database, the product restricts the results to objects belonging to tenants that the user is authorized to access. As a result, you never see data in tenant required tables except for the data that belongs to tenants

that you are permitted to access. If the data is tenanted public data, you can see the data in tenant optional tables because the data is also public data.

When a tenant user asks to create or update a database object, the product verifies that the object belongs to a tenant that the current role for the user can update. The product also verifies that all references from the object to other objects are to public (untenanted) objects, to objects from the same tenant, or to objects from tenants in the tenant hierarchy above the tenant for the object. That is, a tenanted object is allowed to reference objects belonging to its parent tenant, to the parent of the parent, and so on.

If a user who creates an object has update access to multiple tenants, the user must specify the tenant explicitly, either directly or indirectly.

> **NOTE**
> The referenced objects restriction has one exception. Certain references are permitted to reference objects that belong to tenants in the tenant hierarchy of their containing object. These references are designated as SERVICE_PROVIDER_ELIGIBLE in the CA APM object schema. The SERVICE_PROVIDER_ELIGIBLE setting makes a difference only if the service provider tenant is not in the tenant hierarchy above the tenant for the object; if the service provider tenant is in the hierarchy, tenant validation rules permit service provider references.

A service provider user asking to create or update an object is subject to the same restrictions as tenant users, except that service provider users can be authorized to create or update tenanted public objects. The defined role of the service provider user controls this authorization. A service provider user with authorization to multiple tenants who is creating a tenanted object must specify the tenant directly or indirectly.

User Interface Impact

Implementing multi-tenancy changes the user interface, depending on the authorization and tenant access associated with the role for the user.

Tenant Users

A tenant user who is restricted to a single tenant and who is not an administrator has the following user interface changes:

- Any user belonging to more than one tenant can select a tenant in a drop-down list when entering information and when generating a report.

  > **NOTE**
  > If you do not want a user to select a tenant when generating a report, you can remove the tenant drop-down list from the report.
- Any user having read access to more than one tenant has a Tenant Name column in search results.

How to Implement Multi-Tenancy

Multi-tenancy is the ability for multiple independent tenants (and their users) to share a single implementation of CA APM. Tenants only interact with each other in defined ways, as specified by their roles and tenant hierarchies. Typically, unless granted access by a role or hierarchy, each tenant views the CA APM implementation as solely for its own use and cannot update or view data for another tenant.

To implement multi-tenancy in CA APM, complete the following steps:

1. Verify that the CA CASM service is started.
2. Verify that the user implementing multi-tenancy is assigned to a role in which multi-tenancy administration access is enabled.
3. Enable multi-tenancy.
4. Define tenants, subtenants, and tenant groups.
5. Restart the CA APM web server and application server.

6. Log in to the product using the privileged username (typically *uapmadmin*) and complete the following steps:
   a. Define user roles with tenant access.
   b. Define contacts, or import and synchronize users.
   c. Authorize users to use the product.
   d. Assign contacts to user roles.
7. Log in to the product using the privileged username and verify that the multi-tenancy restrictions are enforced.

Enable Multi-Tenancy

Enable multi-tenancy so multiple independent tenants (and their users) can share a single implementation of CA APM. Before you enable multi-tenancy, define tenants, subtenants, tenant groups, and create user roles and assign users to roles. As soon as you enable multi-tenancy, multi-tenancy enforcement is enabled. Multi-tenancy enforcement means that when an object is tenant-required, you cannot save a record without meeting the tenant restrictions.

To enable multi-tenancy

1. Click Administration, Tenancy Management.
   The Multi-Tenancy Administration page appears.
2. Click Edit.
3. In the Status drop-down list, select one of the following options:
   – **off**
     Disables multi-tenancy.
   – **on**
     Enables multi-tenancy.
4. In the Maximum Tenant Depth field, specify the maximum depth allowed for a tenant hierarchy.
5. Click Save.
   Multi-tenancy is enabled.
6. Restart the web server and application server.

Tenant, Subtenant, and Tenant Group Administration

Define the tenants, tenant groups, and subtenants to share a single implementation of CA APM. Multi-tenancy allows tenants to share hardware and application support resources, which reduces the cost of both, while gaining many benefits of an independent implementation.

Define a Tenant

You can define as many tenants as required to manage multiple separate enterprises that provide support to clients. Define a tenant before an instance of a tenant-required object can be updated.

> **WARNING**
> The first created tenant, the service provider, is the primary tenant (owner) in a CA APM multi-tenancy implementation. The service provider tenant cannot have a parent tenant. After you define the service provider tenant, log out of the product and log in again as a member of the service provider. We recommend that you log in as the privileged user (uapmadmin), because this user automatically belongs to the service provider tenant.

To define a tenant

1. Click Administration, Tenancy Management.
   The Multi-Tenancy Administration page appears.
2. On the left, click Tenant.

   > **NOTE**
   > The Tenants page appears.
3. Click Create Tenant.
   The Create New Tenant page appears.

4. Enter the tenant information. The following fields require explanation:
   - **Tenant Number**
     (Information Only) Displays the tenant number. CA APM does not use this field.
   - **Record Status**
     Sets the tenant to active or inactive. After you define the service provider tenant, this option is read-only for the tenant.
   - **Terms of Usage**
     (Information Only) Displays the terms of usage statement for the tenant. CA APM does not use this field.
   - **Parent Tenant**
     Specifies another tenant above this tenant, making this tenant a *subtenant* in a tenant hierarchy.
   - **Subtenants Allowed**
     Allows this tenant to have subtenants. The tenant cannot modify the setting.
   - **Tenant Depth**
     (Information Only) Indicates the tenant depth of this tenant.
   - **Logo**
     (Information Only) Displays the URL for an image file that contains the logo for the tenant, which can be any web image type. CA APM does not use this field.
   - **Contact**
     Displays the Contact lookup page.
   - **Location**
     Displays the Location lookup page.
5. Click Save.
   The tenant is defined.

Update a Tenant

When necessary, you can update the information for an existing tenant.

To update a tenant

1. Click Administration, Tenancy Management.
   The Multi-Tenancy Administration page appears.
2. On the left, click Tenant.
   The Tenants page appears.
3. Search to find the tenant that you want to update.
   All tenants matching the search criteria appear in the Tenant List.
4. Click the tenant that you want to update.
   The tenant information appears.
5. Click Edit.
6. Enter the new information for the tenant.
7. Click Save.
   The tenant is updated.

Make a Tenant Active

When users must see and enter information for a particular tenant that is inactive, you can make the tenant active. For example, the service provider did not receive payment for services provided to a particular tenant. Based on the service agreement, the service provider makes the tenant inactive and stops offering services until payment is made. After the tenant provides payment for the services, the service provider makes the tenant active.

To make a tenant active

1. Click Administration, Tenancy Management.
   The Multi-Tenancy Administration page appears.
2. On the left, click Tenant.

The Tenants page appears.
3. Search to find the tenant that you want to make active.
   All tenants matching the search criteria appear in the Tenant List.
4. Click the tenant that you want to make active.
   The tenant information appears.
5. Click Edit.
6. In the Record Status drop-down list, select Active.
7. Click Save.
   The tenant is active.

How to Initialize a New Tenant

As the service provider, you can define a standard set of data for a new tenant, such as cost centers, cost types, and departments. For information about how to import data for tenants, see Import Data.

Define a Tenant Group

You can define a tenant group to classify, manage, and control access to tenants. For example, you can assign asset managers to a tenant group containing tenants belonging to a particular geographic location.

To define a tenant group

1. Click Administration, Tenancy Management.
   The Multi-Tenancy Administration page appears.
2. On the left, click Tenant Group.
   The Tenant Groups page appears.
3. Click Create Tenant Group.
   The New Tenant Group Detail page appears.
4. Enter the tenant group information.
5. Click Save.
   The tenant group is defined.
6. Click Assign Tenants.
   The Tenant Search page appears.
7. Search and select the tenant that you want to add to the group.
   The tenant is added to the group.

Update a Tenant Group

You can update a tenant group to manage the group members and detail information.

To update a tenant group

1. Click Administration, Tenancy Management.
   The Multi-Tenancy Administration page appears.
2. On the left, click Tenant Group.
   The Tenant Groups page appears.
3. Search to find the tenant group that you want to update.
   All tenant groups matching the search criteria appear in the Tenant Group List.
4. Click the tenant group in the list.
   The Tenant Group Detail page appears.
5. Click Edit.
6. Enter the new information for the tenant group.
7. (Optional) Click Assign Tenants to add a tenant to the group.

   NOTE
   Adding or removing a tenant also adds or removes the subtenants of that tenant.

8. Click Save.
   The tenant group is updated.

Tenant Hierarchies

A *tenant hierarchy* is a structured tenant group that is system-created or modified when you assign a parent tenant to a tenant. The tenant becomes a subtenant of the parent and higher tenants (if any) in that hierarchy.

> **NOTE**
> The service provider can create multiple unrelated hierarchies, or none. Even in a system with tenant hierarchies, you can define standalone tenants.

CA APM supports a tenant hierarchy of unlimited depth. However, the service provider can specify a limit on the total number of tenants and the depth of tenant hierarchies (default is four levels). The service provider also determines whether individual tenants can have subtenants.

Note: Although not required, the service provider can participate in tenant hierarchies. The service provider cannot have a parent tenant.

Define a Subtenant

Subtenancy allows you to define and modify tenant hierarchies for organizational and data-sharing purposes. To place a tenant into a tenant hierarchy, you specify a parent tenant for the tenant.

To define a subtenant

1. Click Administration, Tenancy Management.
   The Multi-Tenancy Administration page appears.
2. On the left, click Tenant.
   The Tenants page appears.
3. Click Create Tenant.
   The Create New Tenant page appears.
4. Enter the subtenant information. The following fields require explanation:
   – **Parent Tenant**
     Specifies another tenant above this tenant, making this tenant a *subtenant* in a tenant hierarchy.

     > **NOTE**
     > The Parent Tenant drop-down only displays tenants that are allowed to have subtenants.
5. Click Save.
   The tenant is a subtenant of the parent tenant.
   Note: When a tenant becomes a subtenant, the tenant belongs to the subtenant group of the parent tenant, in addition to its other subtenants (if any), and so on. The parent tenant joins the supertenant group of its new subtenant, in addition to its other supertenants (if any), and so on. Each joins the related tenants group of the other.

Update a Subtenant

When necessary, you can update the information for an existing subtenant.

To update a subtenant

1. Click Administration, Tenancy Management.
   The Multi-Tenancy Administration page appears.
2. On the left, click Tenant.
   The Tenants page appears.
3. Search to find the tenant that you want to update.
   All tenants matching the search criteria appear in the Tenant List.
4. Click the tenant in the list. The subtenant name appears in the Name column of the Tenant List.
   The tenant information appears.
5. Click Edit.

6. Enter the new information for the subtenant.
7. Click Save.
   The subtenant is updated.

Product-Maintained Tenant Groups

The product generates and maintains the following tenant groups automatically for each tenant in a tenant hierarchy (*tenant* is the tenant name):

- *tenant*_subtenants (tenant, its *child* tenants, and their lower subtenants)
- *tenant*_supertenants (tenant, parent tenant and its higher supertenants)
- *tenant*_relatedtenants (entire single hierarchy)

System-maintained groups can be used like user-defined tenant groups. However, only the name and description can be modified.

How to Secure CA APM Data with Filters

You can set up CA APM data security by creating data filters. You use the data filters to limit the data that users and user roles can view, create, or modify. Users can view, create, or modify only the data that the filter specifies. You can create filters for any of the primary objects:

- Asset
- Model
- Legal Document
- Contact
- Company
- Location
- Organization
- Site

> **NOTE**
> You cannot create a filter for a secondary object that is included in List Management. For example, you can filter assets (primary object) based on the cost center (secondary object) so that you only see assets that belong to a specific cost center. However, all cost centers are still accessible. To restrict access to cost centers (or other secondary objects), assign the user to a configuration that does not allow access to these objects.

As a system administrator, you define filters that limit data access in the following ways:

- Limit data based on the current user (contact).
  For example, define a filter that limits assets by the cost center of the contact. If you apply the filter to all users, the assets dynamically filter based on each user cost center. If the contact cost center is changed, the product filters the assets for the new cost center automatically.
- Limit data based on fixed values.
  For example, allow a user to view assets in the Development cost center. Development is a fixed value. If the cost center changes, the cost center is not automatically changed in the filter.

The data that matches the filter criteria is accessible to filter users.

Note: You can also set up CA APM security by configuring the user interface. Through this configuration, you can control user access to the different functions in CA APM.

> **WARNING**
> In this scenario, the system administrator defines the filters. However, the administrator can grant filter management permissions to any CA APM user role.

To filter CA APM data, perform these steps:

1. Review the Prerequisites.
2. Define and Apply a Filter.
3. Verify the Filter.

Example: Filter Asset Data by Cost Center

Sam, the CA APM system administrator at Document Management Company, wants to ensure that organizational data security requirements are met. Sam wants to limit user access to the company asset records so that users see only their own cost center data. Sam creates a filter for Asset (All Families) and assigns the filter to particular users. Sam verifies that the filter works to ensure the organizational data security.

Review the Prerequisites

To ensure that you can successfully filter data, verify that you have completed the following prerequisites:

1. Identify the product data that you want to limit by user or role.
2. Identify the users and roles that have limited data access.

Define and Apply a Filter

Define a filter and assign the filter to a role, a user, or a combination of roles and users. The product displays only the data that the user is authorized to access.

Follow these steps:

1. Click Administration, Filter Management.
2. On the left, click New Filter.
3. In the Filter Information area, enter the required and optional (if needed) information. The following fields require explanation:
   - **Object**
     Specifies the object data that you want to filter. The object that you select determines the fields that you can use as criteria in the Filter Criteria area.

     NOTE
     If you select Model or Asset, select a family, also. If you select Legal Document, select a template, also. The filter is applied to objects that belong to the family or template that you select. For example, a filter for the hardware asset object does not apply to any other asset family.

     WARNING
     You cannot change the object or the object family or template after you specify filter criteria or after you save the filter. Delete the filter criteria to change the object.
   - **Family or Legal Template**
     Specifies the family for the Model or Asset object or specifies the template for the Legal Document object.
4. In the Filter Security area, enter information for role or user security.

   WARNING
   If you save your filter without selecting a user or role in the Filter Security area, your filter is not applied to any users.
5. In the Filter Criteria area, click Add Fields.
6. Select a field from the dialog and click OK.
   The criterion is shown in the Filter Criteria list.
7. Click the Edit Record icon next to the criterion you added.
8. Enter information to define the filter criterion for a selected field. The following fields require explanation:
   - **Left Parenthesis**

Specifies the first criterion in a group of criteria. You can define groups of criteria to control the logic of the filter.

NOTE
If you select Left Parenthesis to define the first criterion in a group, select Right Parenthesis for the last criterion in the group.

For example, you can filter for assets with the asset name OE001 or with both the asset family Computer and the asset name Dell. In this example, the group consists of two criteria. The Left Parenthesis is selected for the first criterion, which states that the asset family is Computer. The Right Parenthesis is selected for the second criterion, which states that the asset name is Dell.

– **Field Name**
Specifies the field data that is filtered.

– **Operator**
Specifies the filter operator to use to filter object data. For all operators except Has Value and Has No Value, enter a value in the Value field.
For example, you can filter assets in which the asset name is not equal to OE001 and the asset family is equal to Computer.

NOTE
If you specify an Operator and Value, the filter limits data based on a fixed value for the Field Name. In this type of filter, the Use Contact's Value field is not applicable.

– **Use Contact's Value**
Specifies that the filter uses the Field Name value that is associated with the current user.
For example, if you select this check box and you select cost center for Field Name, the users can access product data for their cost centers only. If you apply the filter to all users in the product, the data dynamically filters based on each user cost center. If the cost center changes, the product filters the data for the new cost center automatically.

NOTE
If you select this check box, the filter limits data based on the Field Name value that is associated with the current user. In this type of filter, the Operator and Value fields are not applicable.

– **Right Parenthesis**
Specifies the last criterion in a group of criteria. You can define groups of criteria to control the logic of the filter.

NOTE
If you select Right Parenthesis to define the last criterion in a group, select Left Parenthesis for the first criterion in the group.

For example, you can filter for assets with the asset name OE001 or with both the asset family Computer and the asset name Dell. In this example, the group consists of two criteria. The Left Parenthesis is selected for the first criterion, which states that the asset family is Computer. The Right Parenthesis is selected for the second criterion, which states that the asset name is Dell.

– **Connector**
Specifies the connector between a set of two criteria:
- And - Filters the data if the current criterion and the next criterion in the list are valid.
- Or - Filters the data if either the current criterion or the next criterion in the list is valid.

For example, you can filter for assets with an asset family of Computer and an asset name of Dell.

– **Value**
Specifies the field value that you want to filter. If you specify a value, select an Operator in the Operator field. If a Search icon appears next to this field, you can also select a value by clicking the Search icon.
For example, you can filter an asset with an asset name of OE001.

NOTE
If you specify an Operator and Value, the filter limits data based on a fixed value for the Field Name. In this type of filter, the Use Contact’s Value field is not applicable.

9. Click the Complete Record Edit icon.

10. (Optional) Click Add Fields to define more filter criteria.
11. Click Save after you have completed all filter criteria.
    The filter is defined and applied to users and roles. The data that matches the filter criteria is accessible to filter users.

Example: Filter Asset Data by User Cost Center

As the system administrator, Sam defines a filter that restricts users to the asset data for their own cost centers. To define this filter, Sam makes the following selections:

1. In the Filter Information area, Sam selects Asset in the Object field and (All Families) in the Family field.
2. In the Filter Security area, Sam selects the users who are associated with the filter.
3. In the Filter Criteria area, Sam creates a criterion by performing these steps:
   a. Sam selects Cost Center in the Field Name field.
   b. Sam selects the Use Contact's Value check box.
4. Sam saves the filter.

Verify the Filter

Verify that your filter works to ensure the security of your organizational data.

Follow these steps:

1. Log in to the product as a user who was assigned to the filter.
2. View some of the data that the filter allows for the user.
3. Attempt to view some of the data that the filter does not allow for the user.
   For example, log in as another user who has access to data that is not allowed for the filter user. Copy the URL for a page that displays some data that the filter user cannot access. Paste the URL in the browser of the filter user.
   An error message appears.
4. Attempt to modify some of the data that the filter does not allow for the user.
   For example, identify a text entry field on a page. Enter data that the user cannot access (such as a specific company name that the filter does not allow) and click Save.
   An error message appears.

Example: Verify that the Data Filter Limits Asset Data

Sam created a filter for Asset (All Families) data and assigned the filter to a particular user. With the filter, the user can access only the assets that are in the user cost center. Sam then performs these steps to verify that the filter works:

1. Logs in as the assigned user and validates that assets are available on the Asset Search page.
2. Logs in as a second user with access to assets that are outside of the first user cost center.
3. Views an asset that is outside of the first user cost center and copies the URL of the asset.
4. Logs in as the first user (who was assigned the filter) and pastes the copied asset URL in the browser.
   An error message states that the asset does not exist.

How to Delete Unused Files from CA APM

The files that you use while working with the product are stored on the CA APM application server where the Storage Manager Service is installed. These files include attachment files and Data Importer source data files and legacy map files.

When you delete an attachment, you delete only the reference to the attachment in the object record. If the deleted attachment is a file, the file remains in the file system on the CA APM application server. In similar fashion, when you delete an import that uses a specific data file, you do not delete the data file from the CA APM server.

If a specific attachment file, import data file, or legacy map file is no longer needed, you can delete the file from the CA APM server. Before you delete the file, verify that the file does not have any associations.

To delete unused files, perform these steps:

1. Review the Prerequisites.
2. Query the CA MDB.
3. Locate and Delete the Unused Files.

Review the Prerequisites

To ensure that you can successfully delete the unused files, identify the names of the unused files that you want to delete.

Query the CA MDB

If a specific attachment file, import data file, or legacy map file is no longer needed, you can delete the file from the CA APM application server. Before you delete the file, verify that the file does not have any associations in CA APM (for example, a file attached to a legal document or an import data file associated with a data import).

Follow these steps:

1. Access the CA MDB that is associated with your CA APM installation.
2. Query the al_file_storage table to search for the file name and the associated tenant if applicable. The following statement is an example query:

   ```
   select COUNT(0) from al_file_storage where attachment_url = 'filename.txt'
   ```

   If no records are returned, no records are associated with the specified file name. You can delete the file.

Locate and Delete the Unused Files

After you verify that the files do not have any associations in CA APM, you can locate and delete the files.

Follow these steps:

1. On the CA APM application server where the Storage Manager Service is installed, navigate to one of the following locations, depending on whether you are using multi-tenancy:

   ```
   [ITAM Root Path]/Storage/Common Store/Attachments
   [ITAM Root Path]/Storage/Tenant_Name/Attachments
   [ITAM Root Path]/Storage/Common Store/Import
   [ITAM Root Path]/Storage/Tenant_Name/Import
   ```

2. Locate and delete the unused files.

Purge Audit Data Administration

CA Asset Portfolio Management (APM, formerly ITAM) now provides the ability to purge audit data via the user interface. Retention number of days and other such criteria (object to purge), are shown in the interface before submitting the data purge job. This new way of purging data also addresses earlier performance issues in CA APM. The Audit Purge functionality is available in the interface for the "**SYS Admin**" role and it involves the removal of various records in multiple dependent tables for purging data records.

> **NOTE**
> This feature is not supported on Oracle database.

> **NOTE**
> The purge audit data functionality as of now is only limited to Asset object (**ca_owned_resource**) table, Global ALL configuration extension table (**usp_owned_resource**), and audit tables (**aud_ca_owned_resource** and **aud_usp_owned_resource**) .

Purge Audit Data Best Practices

Consider the following best practices before you go ahead with purging the audit data:

- Take a backup of your MDB database.
- Perform Archive Purge process in a test environment first and then, proceed with your Production Environment. Both production and test environments must be synced with event definitions, extensions created in the system. This is to avoid a marked difference in the data records of both production and test environments. Once you validate the Archive Purge process in a test environment, plan for a production environment as early as possible so that your test environment and production environment data records are similar and are in sync.
- Data purge must always be done during maintenance or low network activity hours as it is a highly resource intensive task. Consult your DBA and have a database maintenance plan as per your purging requirements.
- After a failed data purge job, restore the database backup. Send the **ITAMExportService.log file** (\Program Files (x86)\CA\ITAM\Export Service) to Broadcom support for further analysis.
- When a job is in progress state, and if you have to stop or cancel the job due to some requirements, stop the **CA.ExportService.exe** service from either the task manager or navigate to Windows **services.msc** and stop **CA Asset Portfolio Management** - **Export Service**. Restore the database backup and continue.
- In case, of Multi-tenancy data, the current implementation of Archive/Purge process does not consider multi-tenancy data and purges all data records irrespective of whether the data is tenanted or not.
- Ensure that you are aware of Retention days considerations. This is to verify the number of records in the system before purging.
- Purge process depends on the data volume and hardware performance of your environment. Consider purging huge data in smaller chunks instead of purging it in a single operation. For example, consider the following use case scenarios for 4 years of data record, 365*4=1460 (4 years data record). Here 365 specifies number of days in a year multiplied by number of years for which you want to purge the data. Now, perform similar calculations for 3 years, and for 2 years, and so on.
  If a system has 5 years of audit records and you want to retain only 1 year of audit data records, then, consider purging sequentially.

  First Data Purge

  Provide the Retention days as **1460** (i.e. 365 days *4). This will retain the audit data records created in the past 4 years from the current date, and will purge the audit records created between the 4th year and the 5th year. After job completes successfully, system will have audit records created in the past 4 years.

  Second Data Purge

  Provide the Retention days as **1095** (i.e. 365 days *3), This will retain the audit data records created in the past 3 years from the current date, and will purge the audit records created between the 3rd year and 4th year. After job completes successfully, system will have audit records created in the past 3 years.

  Third Data Purge

  Provide the Retention days as **730** (i.e. 365 days *2). This will retain the audit data created in the last 2 years from the current date, and will purge the audit records created between the 2nd year and 3rd year. After job completes successfully, system will have audit records created in the last 2 years.

  Fourth Data Purge

  Provide the Retention days as **365** (i.e. 365 days *1). This will retain the audit data created in the last 1 year from the current date, and will purge the audit records created between the 1st year and 2nd year. After job completes successfully, system will have audit records created in the last 1 year.
- Run the below query to identify the number of records that needs to be removed or deleted when the Retention number of days is 365 days.

  NOTE

  We have validated in-house with the below mentioned hardware configuration. The **aud_ca_owned_resource** table was tested with14 crores of Audit data records, The Audit purge operation took around 3 hours of time to purge about 1 crore of records from this table.

- Operating System: Windows Server 2012 R2
- Database: Microsoft SQL Serve 2016
- Storage: 32 GB RAM
- Processors:16
- Hard Disk: 1TB space

Ensure that you have provided the relevant value for retention number of days with around 1 crore (or lesser number) of records by running the below query on your database. This helps in using the resultant number in the User Interface for retention number of days for every Audit Purge job.

```
DECLARE @currTime INT
SET @currTime  =  CONVERT(INTEGER, dbo.DATE2SECS(GETUTCDATE()-365));
SELECT count(*)
FROM   (SELECT CONVERT(INTEGER, dbo.Date2secs(Getutcdate() - 365)) dt_flt) dtfltr
       JOIN aud_ca_owned_resource aud
         ON dtfltr.dt_flt >= aud.last_update_date
            AND @currTime >= aud.last_update_date
       LEFT JOIN ca_owned_resource asset
              ON aud.own_resource_uuid = asset.own_resource_uuid
                 AND aud.last_update_date = asset.last_update_date
       INNER JOIN al_aud_owned_resource_potr aptr
               ON aud.own_resource_uuid = aptr.own_resource_uuid
WHERE  asset.last_update_date IS NULL
       AND aud.audit_trail_id < aptr.audit_trail_id
```

- If there are any fired/ in-progress events defined on the custom fields, the purge job will fail with below error message. Current purge functionality takes care of events defined on the OOTB fields only. If you have any such events defined on the custom fields (extensions), contact Broadcom support before you start the purge process.
  For example, consider the following error log message:

  ```
  Audit purge SP returned in error.  Error Code: 547  and Error Description: Error in delete records
   from the tables :iteration Count ends:1:Error on line 413: The DELETE statement conflicted with the
   REFERENCE constraint "usp_owned_resource_fevt_fk2". The conflict occurred in database "mdb", table
   "dbo.al_aud_usp_owned_resource_fevt", column 'audit_trail_id'.
  ```

- If an Audit Purge job was already performed in your system, and later, if you try to add any custom field (extension) from the User Interface on the Global All configuration table (**usp_owned_resource**), subsequent purge jobs fails with the below warning message in the **ITAMExportService.log** file (Program Files (x86)\CA\ITAM\Export Service):

  ```
  2020-12-14 17:40:41,179 UTC [8376] WARN  CA.ExportService.RequestsThreadAbstract - Updating Audit Purge job
   status of current job to 'SystemInterrupted'
  2020-12-14 17:40:41,179 UTC [8376] ERROR CA.ExportService.RequestsThreadAbstract - You cannot
   submit the job as system has identified inconsistencies in the ‘aud_usp_owned_resource’ and
   ‘pur_aud_usp_owned_resource' table schema. Contact Broadcom support to proceed further and resolve this
   issue.
  2020-12-30 17:40:41,180 UTC [8376] WARN  CA.ExportService.RequestsThreadAbstract -
  ```

  Contact Broadcom Support for further analysis and to proceed further.
- For better system performance, delete the archive tables (tables starting with "**pur_**") after a job completes and is verified.
- Stop all services (CA ITAM, CA SDM, CA Service Catalog, and so on) that are currently accessing your MDB database. After submitting the Audit Purge Job request, start the export service to purge the audit data.
- Event service maintains audit table records and dependent table records data at a regular interval. Resolve any Event Service issues, and then perform the Audit Purge process.
- It is recommended that you maintain the log file records of both completed and failed jobs for reference purposes.

How to Purge Audit Data?

Perform the following steps:

1. Login to CA Asset Portfolio Management.
2. Navigate to **Administration**, click **Audit Purge** to display the following options:
   a. **New Audit Purge**:
      This is to start a new purge audit data process/operation. The Audit purge page has the following options:
      a. **Retention Days**: Provide the number of days for retaining data in the system before purging data. For example: 180 days.
         Allowed number of days are **0** to **9999** only. This is a mandatory field.

         NOTE
         If you set Retention days = 30, only the last 30 days data is retained from the current date and older data prior to 30 days is purged or deleted.
      b. **Object To purge**: A drop-down list to select the required objects to be purged. For example:- Asset, contact, and so on. This is a mandatory field.

         NOTE
         As of now, only Assets are available and can be selected from the drop-down list.
      c. After providing the required data fields, click **Audit Purge**.

         WARNING
         Before proceeding further with submitting the purge data request, ensure that you have reviewed the considerations and precautions listed out in Audit Purge Precautions pop-up box. Once data purge job request is submitted, you cannot cancel it and the data selected for purging is deleted.
      d. Click **Submit** to submit the purge data request. This will delete the records as specified by you.
      e. Click **Cancel** to cancel if you are not sure about purging data and if you want to review the data records to be purged.
   b. **View Jobs**: CA APM (ITAM) Administrators can verify and monitor the status of a submitted purge job status in the View jobs screen. The View Jobs page displays the list of all jobs that are submitted for purging data.

      NOTE
      If there are any purge data jobs that are already running, that are not yet started but queued up for purging, or if the purge data process is currently ongoing, then the system does a validation in the backend and does not allow administrators to submit another purge data job. Only after completion of an existing purge data job, you can submit another audit purge job request. You can verify the purged record count and status of the job in the export service log file.
      a. To view export **ITAMExportService.log** file, navigate to Program Files (x86)\CA\ITAM\Export Service.

Import Data

This section contains the following articles:

- How to Delete Data with the Data Importer
- How to Import Data
- How to Submit a Data Import Using a Process Workflow
- How to Submit a Data Import Using the Command Line
- Managing Product-Provided Data Imports

How to Delete Data with the Data Importer

When data is no longer valid for your implementation, delete the data from CA APM using the Data Importer. In most situations, however, it is considered good IT Asset Management practice to keep the data in the repository with a status of inactive. This method allows you to access the data for historical and audit purposes. However, sometimes, the data is created by mistake. In these cases, you want to delete the data.

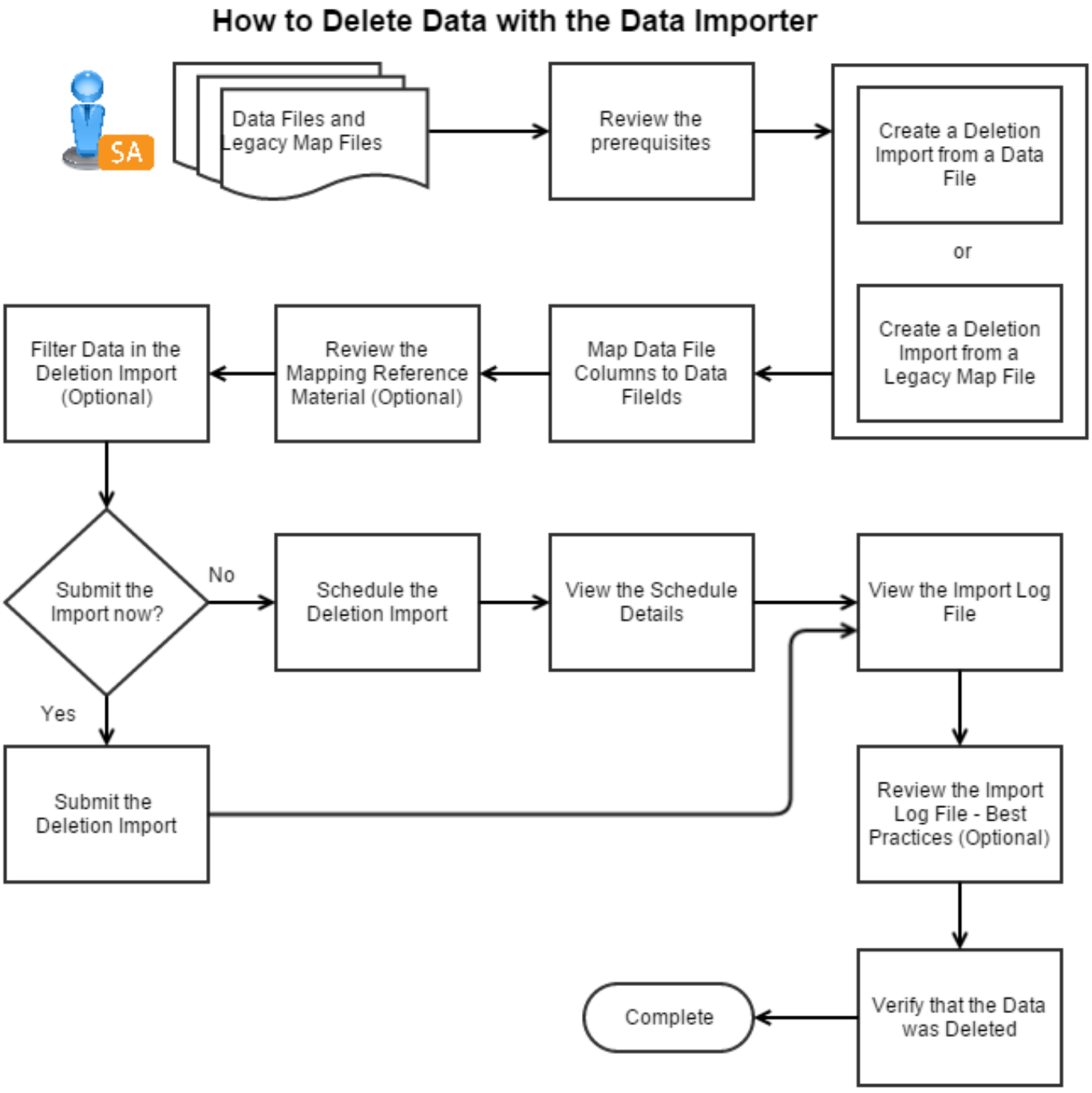
WARNING
In this scenario, the system administrator performs the data deletion. However, the administrator can grant Data Importer User Access or Data Importer Admin Access to any CA APM user role. User access allows users to create imports, modify or delete their own imports, and view any import that was created by another user. Admin access allows users to create imports and modify or delete any import that was created by any user.

You can delete primary objects and can remove the relationships to their secondary objects. For example, you delete an asset (primary object) and you remove its relationship to a legal document (secondary object).

When you delete a primary object and its relationship to a secondary object, the secondary object is not deleted. The primary object is deleted, and the relationship between the primary and secondary object is removed. For example, if you delete an asset and its associated legal document relationship, the asset is deleted but the legal document is not deleted. Only the relationship between the asset and the legal document is removed.

The following diagram illustrates how a system administrator deletes data.

Figure 28: How to Delete Data with the Data Importer



To delete CA APM data, perform these steps:

1. Review the Prerequisites.
2. Create a Deletion Import from a Data File or Create a Deletion Import from a Legacy Map File.
3. Map Data File Columns to Data Fields.
4. Review the Mapping Reference Material.

- Primary and Secondary Lookup Combinations
- Hard-coded Values
- Multiple Values for a Single Field

5. Filter Data in the Deletion Import.
6. Submit the Deletion Import or Schedule the Deletion Import.
7. View the Schedule Details.
8. View the Import Log File.
9. Review the Import Log File - Best Practices.
10. Verify that the Data was Deleted.

Example: Delete Laptops

Miriam, the CA APM system administrator at Document Management Company, wants to delete several laptops that have been retired and recycled. Miriam also wants to delete the associations with legal documents that were made for these laptops. Miriam has a data file that identifies the laptop names, the manufacturers, and the model names. Using the Data Importer and the source data file, Miriam creates a deletion import. After Miriam runs the import and views the import statistics, import log file, and user interface to verify the deletions.

Review the Prerequisites

To ensure that you can successfully delete the data, verify that you have completed the following tasks:

- Prepare a source data file in delimited text format (for example, tab or comma) containing the data that you want to delete.
  We recommend that you include the main destination object in the name of the source data file. This file naming convention helps you locate your data files when you create your import.

  NOTE
  A value of NULL in your source data file clears the corresponding destination field value. An empty field in your source data file leaves the corresponding destination field value unchanged.

  WARNING
  If a data value in your source data file contains the selected delimiter, you must use double quotation marks around the data value. For example, you select a comma as the delimiter to import companies. You want to include the data value Document Management Company, Inc. in your source data file. Specify this data value with double quotation marks: "Document Management Company, Inc".
- (Optional) Copy your source data files from your local server to one of the following locations. You can access these locations on the CA APM application server where the Storage Manager Service is installed. The location depends on whether you are using multi-tenancy.

  ```
  [ITAM Root Path]\Storage\Common Store\Import
  [ITAM Root Path]\Storage\Tenant_Name\Import
  ```

  NOTE
  If you copy the data file before you create an import, you can then specify the file name when you create the import. If you do not copy the data file first, you can upload the file from your local server when you create the import.
- (Optional) Copy your legacy map files from a previous product release (if you have these files) from your local server to one of the following locations. You can access these locations on the CA APM application server where the Storage Manager Service is installed. The location depends on whether you are using multi-tenancy.

  ```
  [ITAM Root Path]\Storage\Common Store\Import
  [ITAM Root Path]\Storage\Tenant_Name\Import
  ```

Create a Deletion Import from a Data File

You can delete data using a source data file (delimited text file) that contains the data that you want to delete. You select the file, configure the import parameters, and specify the delimiter (for example, a comma) that separates the data in the file.

You can also create a deletion import using a legacy map file from a previous product release. For more information, see creating a deletion import from a legacy map file.

Follow these steps:

1. Log in to CA APM as the administrator.

   WARNING
   In this scenario, the system administrator performs the deletion import. However, the administrator can grant Data Importer User Access or Data Importer Admin Access to any CA APM user role. User access allows users to create imports, modify or delete their own imports, and view any import that is created by another user. Admin access allows users to create imports and modify or delete any import that is created by any any user.
2. Click Administration, Data Importer.
3. Click New Import.
4. Enter the required information in the Basic Information area and supply optional information as needed. The following fields require explanation:
   - **Data File**
     Specifies the source data file.
     If this file is available on the CA APM application server, search for the data file and select the file. If this file is not available on the application server, upload the file.
   - **Upload File**
     Browse on your local server for a source data file or a legacy map file that you want to use to create mappings. This file is uploaded to the CA APM application server.

     WARNING
     The file size is limited by the product environmental settings. For more information, contact your administrator.
   - **Main Destination Object**
     Specifies the main object for the deletion import.
     Asset and Model objects are listed with their corresponding families. You can also specify All Families. Legal Document objects are listed according to legal template. You can also specify All Templates. The objects include all objects that can be imported or deleted.

     NOTE
     For assets or models that include multiple asset family types or legal documents that include multiple legal templates, use the following selections for this field. Specify the particular family or template for each record in your source data file.

     - For an asset, select Asset (All Families).
     - For a model, select Model (All Families).
     - For a legal document, select Legal Document (All Templates).

     WARNING
     Ensure that you select the correct main destination object. You cannot change the main destination object after you save or copy an import.
   - **First Row Has Column Names**
     Specifies whether the first row in the source data file contains the column names. If the first row does not contain the column names, the names display as generic names, such as Field 1 and Field 2.
   - **Tenant**

Specifies the tenant that applies to the import (if you are using multi-tenancy).
You can select a tenant only when multi-tenancy is enabled in CA APM and you are authorized to access different tenants. If you have access to public data and you have multiple tenants, you can select all tenants.
If you specify all tenants, your source data file must have a tenant name column that you map to the Tenant Name field.

NOTE
If you specify one tenant, verify that all data in your source data file belongs to your selected tenant. If you have data for more than one tenant, data for all tenants is applied to the selected tenant.

– **Data Delimiter**
Specifies the delimiter (for example, comma or tab) that you used in the source data file.

WARNING
If a data value in your source data file contains the selected delimiter, you must use double quotation marks around the data value. For example, you select a comma as the delimiter to import companies. You want to include the data value Document Management Company, Inc. in your source data file. Specify this data value with double quotation marks: "Document Management Company, Inc".

– **Data File Locale**
Specifies the locale for the source data file. This setting determines the date and time format.

5. Enter the required information in the Advanced Settings area and supply optional information as needed.
The following fields require explanation:

– **Maximum Error Threshold (in %)**
Defines the number of errors after which the import stops. The threshold is based on the percentage of records processed. We recommend a minimum threshold of 15 percent.

NOTE
The Data Importer processes the number of records that are specified on **Administration**, **System Configuration**, **Data Importer** (Maximum Batch Record Size field) before calculating, if the error threshold is reached.

– **Primary Lookup Object Processing Type**
Specifies the type of import activity. Select one of the following options:

- **Delete Primary Objects and Associated Relationships**
Select this option to delete primary objects and the relationships to their associated secondary objects. For example, you delete a company (primary object) and you remove the relationship to an associated asset allocation (secondary object).
When you select this option, verify that your mapping rules specify the primary objects only. Do not include any mapping rules for secondary objects.
Note: The secondary object that is associated with a primary object is not deleted. The relationship between the primary object and the secondary object is removed. For example, you have a primary object Company1 with an associated acquired company Company2 (secondary object). When you delete Company1, the relationship to Company2 is removed. The secondary object Company2 is not deleted.
- **Delete Relationships Only**
Select this option to remove the relationships between secondary objects and their primary objects. When you select this option, verify that your mapping rules specify the primary and secondary objects only. You include a mapping rule for a secondary object; however, do not select the Primary Lookup check box for this rule.
Note: A secondary object that is associated with a primary object is not deleted. The relationship between the primary and secondary object is removed.

– **Normalization Behavior**
Specifies whether to normalize the data or write an error message in the log file without normalizing the data.

NOTE
This field appears only if you have defined normalization rules.

- **Error on Normalization**

Writes an error message to the Data Importer log file when data that can be normalized is found in the data that you are deleting. The data that is involved is not deleted. The log file error message includes the details about the data.
For example, your data includes the company name Microsoft. The company normalization rules that you created identify Microsoft as a collected (nonauthoritative) value and specify Microsoft Corporation as the normalized (authoritative) value. If you select this option when deleting your data, the object with the company name Microsoft is not deleted and an error message is written to the log file.

- **Apply Normalization without Error**
  Uses the normalization rules to normalize the data that you are deleting. If data that can be normalized is found, the data is normalized and deleted. No error message about the data is written to the log file.
  For example, your data includes the company name Microsoft. The company normalization rules that you created identify Microsoft as a collected (nonauthoritative) value and specify Microsoft Corporation as the normalized (authoritative) value. If you select this option when deleting your data, the object with company name Microsoft is normalized. In this example, the company name is changed to Microsoft Corporation and the associated object is deleted.

6. Click Save.
   The deletion import is saved. The Mapping, Exclusion Filter, and Schedule areas of the page are now available for your input.

Example: Create a Deletion Import from a Data File

Miriam, the CA APM system administrator, performs the following actions to create the deletion import:

1. Navigate to Administration, Data Importer and clicks New Import.
2. Enter Hardware Deletions.csv in the Data File field.
   This CSV file is the source data file that contains the laptop deletions.
3. Select Asset (Hardware) for the Main Destination Object and comma for the Data Delimiter.
4. Select Delete Primary Objects and Associated Relationships in the Primary Lookup Object Processing Type field and clicks Save.

Create a Deletion Import from a Legacy Map File

You can create a deletion import using a legacy map file from a previous CA APM release. The map file defines the corresponding data file and the import parameter settings.

NOTE
We recommend that you copy your legacy map files and corresponding data files to the CA APM application server before you create the deletion imports. However, if necessary, you can use the optional steps to upload a legacy map file.

You can also create a deletion import using a data file only. For more information, see creating a deletion import from a data file.

Follow these steps:

1. Click **Administration**, **Data Importer**, **New Import**.
2. Click **Search and Load Map** to select a legacy map file that is already available on the CA APM application server.

   WARNING
   The corresponding data file must also be available on the CA APM application server.

   If the legacy map file is not available on the CA APM application server, upload the file using the Upload File field.
3. (Optional) Upload a legacy map file that is not available on the CA APM application server using the following steps:
   a. In the Upload File field, browse on your local server and select a legacy map file.
      The legacy map file is uploaded and is displayed in the Upload File field.
   b. Click Search and Load Map and select the legacy map file that you uploaded.

The legacy map file is displayed in the Legacy Map File field.
The Basic Information is loaded.

NOTE
If you receive a warning about the source data file, upload the data file using the Upload File field.

4. Specify the Advanced Settings and click Save.
   The Exclusion Filter and Mapping data mapping are loaded. The Mapping, Exclusion Filter, and Schedule areas of the page are now available for your input. The Mapping and Exclusion Filter areas display the data from the legacy map file.

   NOTE
   For information about specifying the Advanced Settings, see creating a deletion import from a data file.

Map Data File Columns to Data Fields

You can map the columns in your source data file to product fields. You perform column mapping to specify which data is deleted. You can select most objects and associated fields as destination fields during column mapping.

NOTE
If you created your deletion import from a legacy map file, the column mapping exists. You can edit the existing mapping rules if you want to change the values. You can also add new mapping rules.

When you log in, the user role that your administrator assigned to you determines the objects and fields that you can see and use. If your role specifies that you do not have permissions for an object field, the field is not available for a mapping. You can only create a mapping and import or delete data for the objects and fields for which you have permissions.

Follow these steps:

1. On the **Administration** tab, **Data Importer** page, in the Mapping area for a selected deletion import, click **New** or click **Load Source Fields**.
   - New allows you to select the source fields individually from the source data file.
   - Load Source Fields adds all source fields from the source data file.

     NOTE
     If you have existing mappings, Load Source Fields allows you to replace those mappings with the source fields in the source data file. This option also allows you to add the source fields from the source data file that you do not already have in your mappings.

   a. If you clicked Load Source Fields, click the Edit Record icon next to a field.
2. Click the Select icon next to Source Field (if this field is empty), select a column from your data source, and click **OK**.
   If this field already contains a source field (because you loaded all source fields), you can skip this step.
3. Click the Select icon next to Destination Field, select a Destination Field for the selected Source Field, and click **OK.**
   The destination fields that appear are based on your selected main destination object.

   NOTE
   The destination fields display in hierarchical order. For example, the fields that are listed under Asset Type Hierarchy are Asset Family, Class, and Subclass. The order of the fields represents the field hierarchy. Follow the field hierarchy when you specify mapping rules. For example, for Asset Type Hierarchy, specify a rule for Class before you specify Subclass.
4. Select the Primary Lookup and Secondary Lookup checkboxes as required.
   a. Select a Primary Lookup check box for each destination field that you want to use to find the primary object. Use the following guidelines when selecting this check box:
      - Select at least one Primary Lookup check box in the column mapping for an import.
      - Do not select this check box if the Destination Field is Note Text (under the Note object). The database data type for the Note Text field does not allow it to function as a lookup field.

b. Select a Secondary Lookup check box for each destination field that you want to use to find the secondary object. Use the following guidelines when selecting this check box:
   - Do not select this check box if the destination field is not one of your lookup fields for the secondary object.
   - Do not select this check box if the Destination Field is Note Text (under the Note object). The database data type for the Note Text field does not allow it to function as a lookup field.

5. Click the Complete Record Edit icon.
6. (Optional) Click New again, or click the Edit Record icon next to another source field, to specify more mapping rules.

   NOTE
   To delete a specific mapping rule from the list of mapped columns, click the Delete icon next to the mapping rule. The column mapping rule is removed from the list.

7. Click Save.
   Your column mapping is saved.

Example: Map Data File Columns to Data Fields

Miriam performs the following steps to map the columns in the source data file to the CA APM data fields:

1. Click New in the Mapping area of the Import Details page.
2. Select %Hardware Name% in the Source Field by clicking the Select icon next to Source Field and selecting this item from the dialog.
   The items that are listed in the dialog are the columns from the source data file.
3. Select Asset Name in the Destination Field by clicking the Select icon next to Destination Field and selecting this object from the dialog.
4. Select the Primary Lookup check box.
5. Click the Complete Record Edit icon and clicks Save.

Review the Mapping Reference Material

Reference the following information when setting up the column mapping for importing or deleting data.

Primary and Secondary Lookup Combinations

The fields that you select as the primary and secondary lookup in your column mapping are used to search for data in the product database.

- **Simple mapping**
  In simple mapping, you specify only the primary lookup. For example, you are importing or deleting a set of company records from a text file into the product database. You specify the Company Name as the primary lookup. If a company with a particular name does not exist in the database when you are importing data, a record is created for the company. The following table shows an example of the lookup for a simple mapping.

Source Field	Destination Field	Primary Lookup	Secondary Lookup
%Company Name%	Company.Company Name	Yes	No

- **Reference field mapping**
  In reference field mapping, you specify primary and secondary lookup values. To search for a unique object, specify more than one primary lookup. For example, to search for a company, you can specify Company Name, Parent Company, and Company Type as primary lookup values. In this example, the Data Importer searches for a company with the specified name, the specified parent company, and of the specified company type. If the object does not exist

and you are importing data, the record is created (depending on the insert or update option you selected in Advanced Settings). The following table shows an example of the lookup for reference field mapping.

Source Field	Destination Field	Primary Lookup	Secondary Lookup
%Company Name%	Company. Company Name	Yes	No
%Parent Company%	Company. Parent Company. Company Name	Yes	Yes
%Company Type%	Company.Company Type.Value	Yes	Yes

This mapping has both the Primary Lookup and the Secondary Lookup checkboxes selected for Parent Company and Company Type. The Data Importer uses the Company Name to look up the parent company and uses the Parent Company to look up the company name.

- **Secondary object mapping**
  If a mapping rule maps to a secondary object property, the primary lookup values establish a relationship between a secondary object and the reference fields. The following table shows examples of the lookup for a secondary object mapping.

Source Field	Destination Field	Primary Lookup	Secondary Lookup
%Comment%	Legal Document. Legal Party.Comment	No	Yes
%Legal Document ID%	Legal Document. Document Identifier	Yes	No
%Company Name%	Legal Document.Legal Party.Legal Party.Company Name	Yes	Yes
%Legal Template%	Legal Document.Legal Template.Template	Yes	Yes

In the first mapping rule, Legal Document is the primary object, and Legal Party is the secondary object. Comment is a property of Legal Party.

In the third mapping rule, Legal Document is the primary object, and Legal Party is the secondary object. In addition, Legal Party has a reference field in the Company table. The Secondary Lookup check box indicates that the Company Name is used to look up the Company object. The Primary Lookup check box indicates that the Company object is used to look up the Legal Party object.

Hard-coded Values

In the column mapping, the percent signs that appear before and after column names identify the names as column headers in your source data file. You can also specify a hard-coded value in the Source Field that you want to apply to all records in your source data file. You can then map the hard-coded value to a Destination Field. The hard-coded values do not display with percent signs to distinguish these values from the source data file column names.

1. Source data file column header
2. Hard-coded value

You can define a hard-coded value in the Source Field to expand your source data and to ensure that you include all required fields. Hard-coded values typically do not begin and end with a percent sign (%). If you have hard-coded values with percent signs, the values cannot match the field names in your source data file.

Example: Use hard-coded values for asset family

In this example, the assets in your source data file do not contain asset family, which is required when creating an asset. You can add a hard-coded value to your mapping. If all your assets are hardware, you can enter Hardware in the Source Field. You can map this value to the Asset Family field. If your assets belong to different families, add a column to your source data file with the corresponding asset families before importing or deleting data.

The following information illustrates the difference between values from your source data file and values that are added through hard-coded values:

- You have an Asset Family column in your source data file. The selection in the Source Field is %asset family%.
- You do not have an Asset Family column in your source data file. However, all your assets are hardware assets. You specify a hard-coded value of Hardware in the Source Field.

  NOTE
  You can also use the Main Destination Object to specify that all records in your source data file belong to a particular family or template. For example, the Asset (Hardware) selection for Main Destination Object specifies that all source records belong to the hardware asset family.

Multiple Values for a Single Field

You can add a mapping with multiple Source Field values that are mapped to a single Destination Field.

Example: Use multiple values for a single field

Your source data file has two columns with the names Manufacturer and Catalog Name. Combine these columns by selecting both in the Source Field. In this example, the Source Field selection is %Manufacturer% %Catalog Name%.

You can also enter multiple hard-coded values in the Source Field (for example, Document Management Company %model name% IT Department).

Filter Data in the Deletion Import

You can identify a subset of records in your source data file that you want to exclude from the deletion import. The Data Importer exclusion filter allows you to filter a part of your data source using exclusion filter rules.

Example: Define an exclusion filter to process returned assets

A CSV file that you receive from your hardware vendor includes assets that were ordered and returned to the vendor. You want to delete the assets that were returned to the vendor, so you want to process those records only. You define an exclusion filter to exclude records that do not have a status of Returned.

Follow these steps:

1. On the Administration tab, Data Importer page, Exclusion Filter area for a selected deletion import, select the Filter Type.
   - **And**
     Excludes a record from the source data file only if all the rules that you specify are valid for the record.
   - **Or**
     Excludes a record from the source data file if any of the rules that you specify is valid for the record.
2. Click New.
3. Click the Select icon next to Source Field, select a column from your source data file, and click OK.

   NOTE
   The percent signs before and after the column name identifies the name as a column from your source data file.
4. Select the Operator.

   NOTE
   To specify "not equal to", select the "<>" operator.
5. Enter a Filter Value for the rule.

   NOTE
   You can use special characters and wildcards in the filter value. The rules can process text, numeric, and date fields.
6. Click the **Complete Record Edit** icon.
7. (Optional) Click **New** and specify more exclusion filter rules.
8. Click Save.
   The exclusion filter rules are saved and are applied when the deletion import processes.

Submit the Deletion Import

To start a deletion import immediately, click Submit in the Schedule area of the page. The data source records from the data file for the selected deletion import are processed.

NOTE
You can specify a data file other than the default (from the Basic Information) if you want to use a different file.

You can also schedule the deletion import for a particular day and time. For more information, see schedule the deletion import.

To view the import jobs for your current selected deletion import, click Associated Jobs on the left side of the page. To view all import jobs for all imports, click Import Jobs. In the list of import jobs that appears, click Status Message to view the status of an import.

You can also view the log file for more information about the import activity. In the list of import jobs, click View Logs for the selected import.

Schedule the Deletion Import

You can schedule a deletion import for a specific time and you can specify the interval for the deletion import (for example, daily or weekly). You can schedule multiple deletion imports to process simultaneously.

Follow these steps:

1. On the Administration tab, Data Importer page, in the Schedule area for a selected deletion import, select the Is Scheduled checkbox.
2. Provide the information for the schedule. The following fields require explanation:
   - **Run Time**

Specifies the time of the day, in 24-hour format, to process the deletion import. When you schedule imports, use the local time zone on the CA APM application server.

- **Interval Day**
  Specifies the day during the Interval Type to process the deletion import. For example, if the Interval Type is Month and the Interval Day is 1, the import is processed on the first day of the month.
- **Data File**
  Specifies a data file name other than the default (from the Basic Information) if you want to use a different file. If this file is available on the application server, you can search and select the file. If this file is not available on the application server, you can locate and upload the file.
- **Upload Data File**
  Browse for the source data file. This file is uploaded to the application server.
- **First Run Date**
  Specifies the date when the first deletion import starts to process.
- **Interval Type**
  Specifies the type of interval for the deletion import (for example, Day, Month, Quarter, Week, or Year).
- **Interval**
  Specifies how often the deletion import processes. This interval is based on the specified Interval Type. For example, if the Interval Type is Weekly and the Interval is 2, the import processes every two weeks.
- **Last Day of Interval**
  Specifies that the deletion import processes on the last day of the selected Interval Type. If you select this check box, any previous value that you added to the Interval Day field is removed, and the Interval Day field is disabled.

3. Click Submit.
   The deletion import is scheduled for the specified date and time.

Examples: Using the Schedule Settings

The following examples illustrate the use of the schedule settings.

- Select Day for Interval Type and 2 for Interval. The import processes every other day.
- Select Week for Interval Type, 1 for Interval Day, and 3 for Interval. The import processes every three weeks on the first day (Sunday) of the week.
- Select Month for Interval Type, 15 for Interval Day, and 2 for Interval. The import processes every two months on the 15th day of the month.
- Select Quarter for Interval Type and select Last Day of Interval. The import processes every quarter (every three months) on the last day of the last month in the quarter.
- Select Year for Interval Type, 1 for Interval Day, and 1 for Interval. The import processes on January 1 of every year.

To view the import jobs for your current selected deletion import, click Associated Jobs on the left side of the page. To view all import jobs for all imports, click Import Jobs. In the list of import jobs that appears, click Status Message to view the status of an import.

You can also view the log file for more information about the import activity. In the list of import jobs, click View Logs for the selected import.

View the Schedule Details

You can view the schedule details for a scheduled import job that you created.

First, open the list of import jobs.

- To view the scheduled import jobs for your currently selected import, click Associated Jobs on the left side of the page, select the Scheduled check box, and click Go.
- To view all import jobs for all imports, click Import Jobs on the left side of the page, select the Scheduled check box, and click Go.

In the list of import jobs that appears, click Schedule Details for a selected import.

View the Import Log Files

You can view the Data Importer log files to see the details of all CA-provided and user-defined imports that have completed. The Data Importer creates a log file for each import that you run, including imports that were submitted immediately or scheduled for a future time. All import activities are saved in the log files.

To view the log files, first open the list of import jobs.

- To view the import jobs for your current selected import, click Associated Jobs on the left side of the page.
- To view all import jobs for all imports, click Import Jobs.

In the list of import jobs, click View Logs for a selected import. If more than one log file is available (for example, for a scheduled import that has completed a few times already), all files are listed with their corresponding creation dates.

You can view any available LDAP Import Sync log file. If you click Start LDAP Data Import and Sync on the LDAP Data Import and Sync page (Administration, User/Role Management), an import job ID is displayed. Use this job ID to locate the job in the Data Importer list of import jobs. Then click View Logs for that job.

NOTE
You can also locate and view the import log files in the following location on the CA APM application server:

```
[ITAM Root Path]\Storage\Common Store\Import\Logs
```

Review the Import Log File - Best Practices

The Data Importer log file contains information and error messages regarding the processing of import jobs. To help you understand the results of your import and to troubleshoot any errors, use the information in this log file. This section contains some recommended best practices for working with the Data Importer log file.

Match the row number in the data file with the error message in the log file.

A log file error message identifies the corresponding row number from your data file. You can also find the data file row number in the row above or below the error message in the log file.

Sometimes the error message in the log file does not show the data file row number. In this situation, the actual data file values are shown immediately after the error message in the log file.

Count the number of error messages in your log file.

1. Search for the following phrases in your log file to find the error messages in the file. These phrases are included with the error messages.
   ```
   Web Service threw exception
   Error at record
   ```
2. After you find a type of error message, search for that error in the log file and count the number of occurrences.
3. Identify and search for more error types that appear in your log file and count the number of occurrences.
4. Compare the count of all errors in your log file with the statistics that the Data Importer generated for the associated import. To view these statistics, click Status Message on the Associated Jobs list or Import Jobs list. This comparison helps you account for all relevant errors and identify error messages that are not valid and can be ignored.

Verify that the Data was Deleted

You verify that your deletion import succeeded by viewing your data in CA APM and by reviewing the Data Importer statistics.

- **Review the Data Importer Statistics.** To review the statistics for your current selected deletion import, click Associated Jobs on the left side of the page. In the list of import jobs that appears, click Status Message for your import.

You can also view the log file for more information about the import activity. In the list of import jobs, click View Logs for the selected import.

- **View the Data in CA APM.** To view the data in CA APM, navigate to the tab and subtab, if necessary, for the object that you deleted (for example, asset, company, or contact). Search for the objects that you deleted and verify that the objects are not available.

Example: Verify the Deletion of Laptops

After Miriam runs the deletion import, she performs the following steps to verify that the laptops were deleted:

1. Checks the import statistics.
   - Clicks Associated Jobs or Import Jobs on the left side of the Data Importer page.
   - Clicks Status Message for the deletion import and reviews the statistics.
2. Views the import log file and the user interface.
   - Clicks View Logs in the list of import jobs and reviews the contents of the log file.
   - Navigates to the Asset tab. Searches for the deleted laptops and verifies that the deleted laptops are not available.

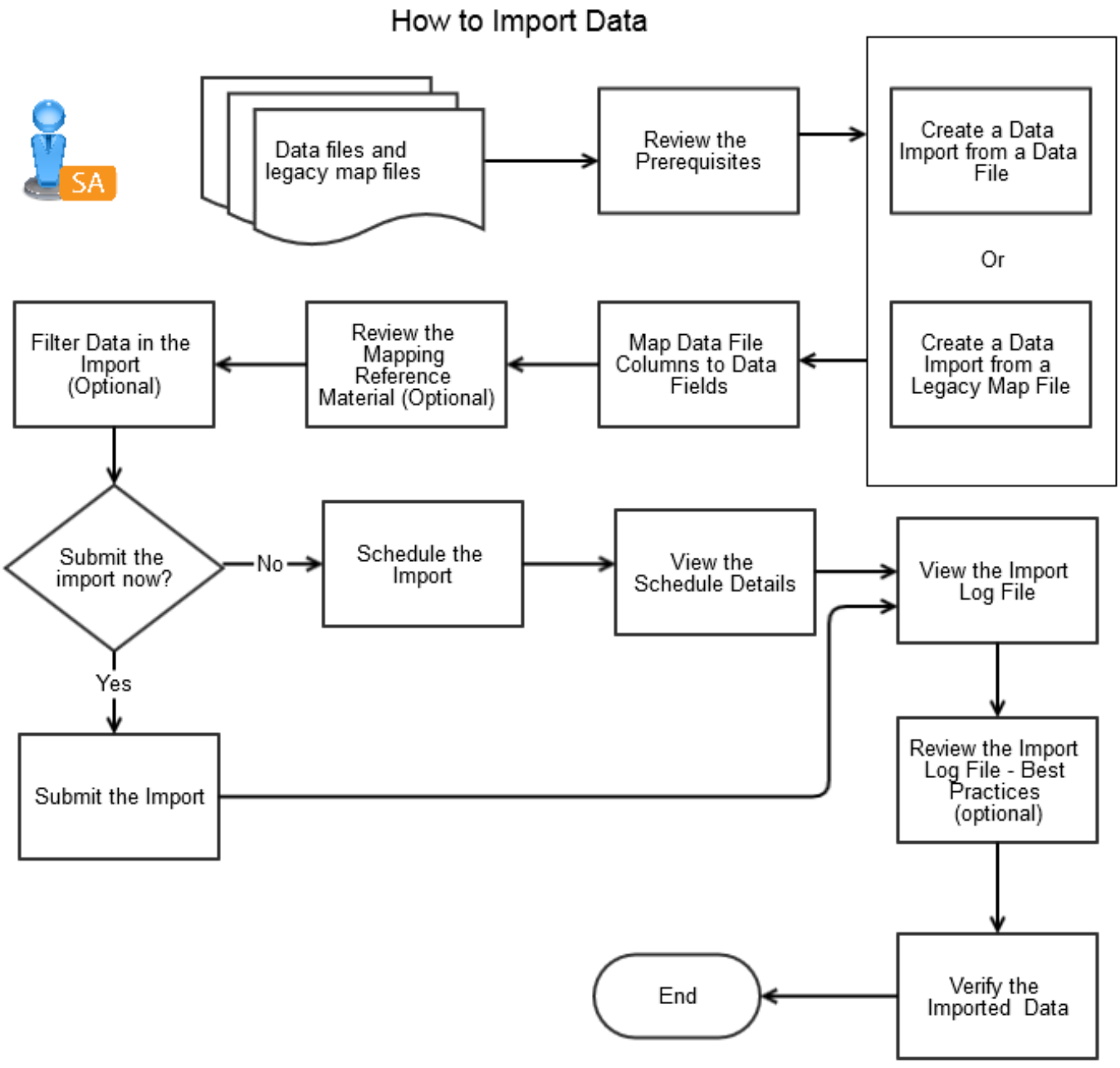
How to Import Data

When you want to add or update data, import it into CA APM using the Data Importer. The data that you import is inserted, or updates existing data, in the CA MDB.

> **WARNING**
> In this scenario, the system administrator performs the data import. However, the administrator can grant Data Importer User Access or Data Importer Admin Access to any CA APM user role. User access allows users to create imports, modify or delete their own imports, and view any import that is created by another user. Admin access allows users to create imports and modify or delete any import that is created by any user.

The following diagram illustrates how a system administrator imports data.

Figure 29: Diagram depicting how to import data



To import CA APM data, perform these steps:

1. Review the Prerequisites.
2. Create a Data Import from a Data File or Create a Data Import from a Legacy Map File.
3. Map Data File Columns to Data Fields.
4. Review the Mapping Reference Material.

- Primary and Secondary Lookup Combinations
- Hard-Coded Values
- Multiple Values for a Single Field

5. Filter Data in the Import.
6. Submit the Import.
7. Schedule the Import.
8. View the Schedule Details.
9. View the Import Log File.
10. Review the Import Log File Best Practices.
11. Verify the Imported Data.

Example: Import New Employees

Sam, the CA APM system administrator at Document Management Company, wants to import a group of new employees. The new employees use the product to manage hardware assets. Sam received a comma-separated value (CSV) source data file from Human Resources that contains the employee information. All the new asset manager employees belong to the IT department and work at the company headquarters. However, the source data file also contains data about some new employees who work at other locations and do not belong to the IT department.

Sam only wants to import the data for the IT employees at headquarters. Using the Data Importer and the source data file, Sam creates a data import to incorporate the new employees into the CA MDB. To ensure that only employees in the IT department at headquarters are imported, Sam creates an exclusion filter. After Sam runs the import, he checks the import statistics and the import log file and user interface to verify that the import was successful.

Review the Prerequisites

To ensure that you can successfully import the data, verify that you have completed the following tasks:

- Prepare a source data file in delimited text format (for example, tab or comma). This file contains the data that you want to import.

  NOTE
  We recommend that you include the main destination object in the name of the source data file. This file naming convention helps you locate your data files when you create your import. A value of NULL in your source data file clears the corresponding destination field value. An empty field in your source data file leaves the corresponding destination field value unchanged.

  WARNING
  If a data value in your source data file contains the selected delimiter, you must use double quotation marks around the data value. For example, you select a comma as the delimiter to import companies. You want to include the data value Document Management Company, Inc. in your source data file. Specify this data value with double quotation marks: "Document Management Company, Inc".
- (Optional) Copy your source data files from your local server to one of the following locations. You can access these locations on the CA APM application server where the Storage Manager Service is installed. The location depends on whether you are using multi-tenancy.

  ```
  [ITAM Root Path]\Storage\Common Store\Import
  [ITAM Root Path]\Storage\Tenant_Name\Import
  ```

  NOTE
  If you copy the data file before you create an import, you can then specify the file name when you create the import. If you do not copy the data file first, you can upload the file from your local server when you create the import.
- (Optional) Copy your legacy map files from a previous product release (if you have these files) from your local server to one of the following locations. You can access these locations on the CA APM application server where the Storage Manager Service is installed. The location depends on whether you are using multi-tenancy.

```
[ITAM Root Path]\Storage\Common Store\Import
[ITAM Root Path]\Storage\Tenant_Name\Import
```

Create a Data Import from a Data File

You create a data import using a source data file (delimited text file) that contains the data that you want to import. You select the file, configure the import parameters, and specify the delimiter (for example, a comma or a tab) that separates the data in the file.

You can also create a data import using a legacy map file from a previous product release. For more information, see creating a data import from a legacy map file.

Follow these steps:

1. Log in to CA APM as the administrator.

   WARNING
   In this scenario, the system administrator performs the data import. However, the administrator can grant Data Importer User Access or Data Importer Admin Access to any CA APM user role. User access allows users to create imports, modify or delete their own imports, and view any import that is created by another user. Admin access allows users to create imports and modify or delete any import that is created by any user.
2. Click Administration, Data Importer.
3. Click New Import.
4. Enter the required information in the Basic Information area and supply optional information as needed. The following fields require explanation:
   - **Data File**
     Specifies the source data file that you want to import.
     If this file is available on the CA APM application server, search for the data file and select the file. If this file is not available on the application server, upload the file.
   - **Upload File**
     Browse on your local server for a source data file that you want to import or a legacy map file that you want to use to create mappings. This file is uploaded to the CA APM application server.

     WARNING
     The file size is limited by the product environmental settings. For more information, contact your administrator.
   - **Main Destination Object**
     Specifies the main object for the import.
     Asset and Model objects are listed with their corresponding families. You can also specify All Families. Legal Document objects are listed according to legal template. You can also specify All Templates. The objects include all objects that can be imported.

     NOTE
     For assets or models that include multiple asset family types or legal documents that include multiple legal templates, use the following selections for this field. Specify the particular family or template for each record in your source data file.

     - For an asset, select Asset (All Families).
     - For a model, select Model (All Families).
     - For a legal document, select Legal Document (All Templates).

     WARNING
     Ensure that you select the correct main destination object for your import. You cannot change the main destination object after you save or copy an import.
   - **First Row Has Column Names**

Specifies whether the first row in the source data file contains the column names. If the first row does not contain the column names, the names display as generic names, such as Field 1 and Field 2.

- **Tenant**
  Specifies the tenant that applies to the import (if you are using multi-tenancy).
  You can select a tenant only when multi-tenancy is enabled in CA APM and you are authorized to access different tenants. If you have access to public data and you have multiple tenants, you can select all tenants.

  NOTE
  If you specify all tenants, your source data file must have a tenant name column that you map to the Tenant Name field.

  WARNING
  If you specify one tenant, verify that all data in your source data file belongs to your selected tenant. If you have data for more than one tenant, data for all tenants is imported into the selected tenant.
- **Data Delimiter**
  Specifies the delimiter (for example, comma or tab) that you used in the source data file.

  WARNING
  If a data value in your source data file contains the selected delimiter, you must use double quotation marks around the data value. For example, you select a comma as the delimiter to import companies. You want to include the data value Document Management Company, Inc. in your source data file. Specify this data value with double quotation marks:

  WARNING
  "Document Management Company, Inc"
- **Data File Locale**
  Specifies the locale for the source data file. This setting determines the date and time format.

5. Enter the required information in the Advanced Settings area and supply optional information as needed. The following fields require explanation:
   - **Maximum Error Threshold (in %)**
     Defines the number of errors after which the import stops. The threshold is based on the percentage of records processed. We recommend a minimum threshold of 15 percent.

     NOTE
     The Data Importer processes the number of records that are specified on Administration, System Configuration, Data Importer (Maximum Batch Record Size field) before calculating if the error threshold has been reached.
   - **Primary Lookup Object Processing Type**
     Specifies the type of import activity (for example, insert or update).
   - **Create Secondary Lookup Object**
     Creates new secondary lookup objects during the import process. If this option is not selected and a secondary object does not exist, an error occurs.
   - **Update Secondary Lookup Object**
     Updates the existing secondary lookup objects during the import process. If a secondary object does not exist, an error occurs.
   - **Error on Secondary Lookup Object Errors**
     Indicates that the Data Importer does not process a primary object insert or update if the secondary object process fails. If a secondary object insert or update process fails and this check box is selected, the insert or update for the primary object also fails. If this check box is not selected, the primary object is created or updated (as long as the object is not dependent on the secondary object). However, the secondary object value is not created or changed. In both situations, the secondary object error is logged in the import log file.

Example: You import a location and the location has a country. If the import fails while trying to update the country object and this check box is selected, the location record is not created. If this check box is not selected, the location record is created, and the country information is not updated.

– **Normalization Behavior**
Specifies whether to normalize the data or write an error message in the log file without normalizing the data.

NOTE
This field appears only if you have defined normalization rules.

- **Error on Normalization**
Writes an error message to the Data Importer log file when data that can be normalized is found in the data that you are importing. The data involved is not imported. The log file error message includes the details about the data.
For example, your data includes the company name Microsoft. The company normalization rules that you created identify Microsoft as a collected (nonauthoritative) value and specify Microsoft Corporation as the normalized (authoritative) value. If you select this option when importing your data, the object with the company name Microsoft is not imported and an error message is written to the log file.
- **Apply Normalization without Error**
Uses the normalization rules to normalize the data that you are importing. If data that can be normalized is found, the data is normalized and imported. No error message about the data is written to the log file.
For example, your data includes the company name Microsoft. The company normalization rules you created identify Microsoft as a collected (nonauthoritative) value and specify Microsoft Corporation as the normalized (authoritative) value. If you select this option when importing your data, the object with company name Microsoft is normalized. In this example, the company name is changed to Microsoft Corporation and the associated object is imported.

6. Click Save.
The import is saved. The Mapping, Exclusion Filter, and Schedule areas of the page are now available for your input.

Example: Create a Data Import of New Employees from a Data File

Sam, the CA APM system administrator, performs the following actions to create the data import:

1. Navigates to Administration, Data Importer and clicks New Import.
2. Enters New Employees.csv in the Data File field.
This CSV file is the source data file that Sam received from Human Resources with the new employee information.
3. Selects Contact for the Main Destination Object and comma for Data Delimiter.
4. Selects Insert or Update in the Primary Lookup Object Processing Type field and clicks Save.

Create a Data Import from a Legacy Map File

You can create a data import using a legacy map file from a previous CA APM release. The map file defines the corresponding data file and the import parameter settings.

NOTE
We recommend that you copy your legacy map files and corresponding data files to the CA APM application server before you create the data imports. However, if necessary, you can use the optional steps to upload a legacy map file.

You can also create a data import using a data file only. For more information, see creating a data import from a data file.

Follow these steps:

1. Click Administration, Data Importer, New Import.
2. Click Search and Load Map to select a legacy map file that is already available on the CA APM application server.

WARNING
The corresponding data file must also be available on the CA APM application server.

If the legacy map file is not available on the CA APM application server, upload the file using the Upload File field.

3. (Optional) Upload a legacy map file that is not available on the CA APM application server using the following steps:
   a. In the Upload File field, browse on your local server and select a legacy map file.
      The legacy map file is uploaded and is displayed in the Upload File field.
   b. Click Search and Load Map and select the legacy map file that you uploaded.
      The legacy map file is displayed in the Legacy Map File field.
      The Basic Information is loaded.

      NOTE
      If you receive a warning about the source data file, upload the data file using the Upload File field.
4. Specify the Advanced Settings and click Save.
   The Exclusion Filter and Mapping data mapping are loaded. The Mapping, Exclusion Filter, and Schedule areas of the page are now available for your input. The Mapping and Exclusion Filter areas display the data from the legacy map file.

   NOTE
   For information about specifying the Advanced Settings, see creating a data import from a data file.

Map Data File Columns to Data Fields

You can map the columns in your source data file to fields in CA APM. You perform column mapping to specify where the source data is imported. You can select most objects and associated fields as destination fields during column mapping.

NOTE
If you created your data import from a legacy map file, the column mapping exists. If you want to change the values, you can edit the existing mapping rules. You can also add or remove mapping rules and filters.

When you log in, the user role that your administrator assigned to you determines the objects and fields that you can see and use. If your role specifies that you do not have permissions for an object field, the field is not available for a mapping. You can only create a mapping and import data for the objects and fields for which you have permissions.

NOTE

We recommend that, before you map data, you review the CA APM user interface to determine the required information for a mapping. For example, review the Asset page to see that the asset name, asset family, model, and class are required. Because a model is required to create an asset, you review the Model page to see that the model name and asset family are required. By reviewing the user interface before you create a mapping, you ensure that you have all required information to create a mapping.

Follow these steps:

1. On the Administration tab, Data Importer page, in the Mapping area for a selected import, click New or click Load Source Fields.
   – New allows you to select the source fields individually from the source data file.
   – Load Source Fields adds all source fields from the source data file.

   NOTE
   If you have existing mappings, Load Source Fields allows you to replace those mappings with the source fields in the source data file. This option also allows you to add the source fields from the source data file that you do not already have in your mappings.

   a. If you clicked Load Source Fields, click the Edit Record icon next to a field.
2. Click the Select icon next to Source Field (if this field is empty), select a column from your data source, and click OK.
   If this field already contains a source field (because you loaded all source fields), you can skip this step.

NOTE
The percent signs that appear before and after the column names identify the names as column headers in your source data file. You can also specify a hard-coded value in the Source Field that you want to apply to all records in your source data file. You can then map the hard-coded value to a Destination Field. The hard-coded values do not display with percent signs so that they can be distinguished from the source data file column names. For more information, see hard-coded values.

3. Click the Select icon next to Destination Field, select a Destination Field for the selected Source Field, and click OK.
   The destination fields that appear are based on your selected main destination object.

   NOTE
   The destination fields display in hierarchical order. For example, the fields that are listed under Asset Type Hierarchy are Asset Family, Class, and Subclass. The order of the fields in the list represents the field hierarchy. Follow the field hierarchy when you specify mapping rules. For example, for Asset Type Hierarchy, specify a rule for Class before you specify Subclass.
4. Select the Primary Lookup and Secondary Lookup checkboxes as required.
   a. Select a Primary Lookup check box for each destination field that you want to use to find the primary object. Use the following guidelines when selecting this check box:
      - Select at least one Primary Lookup check box in the column mapping for an import.
      - Do not select this check box if the Destination Field is Note Text (under the Note object). The database data type for the Note Text field does not allow it to function as a lookup field.
   b. Select a Secondary Lookup check box for each destination field that you want to use to find the secondary objects. Use the following guidelines when selecting this check box:
      - Do not select this check box if the destination field is not one of your lookup fields for the secondary object.
      - Do not select this check box if the Destination Field is Note Text (under the Note object). The database data type for the Note Text field does not allow it to function as a lookup field.
5. Click the Complete Record Edit icon.
6. Click New again, or click the Edit Record icon next to another source field, to specify more mapping rules.

   NOTE
   To delete a specific mapping rule from the list of mapped columns, click the Deletion icon next to the mapping rule. The column mapping rule is removed from the list.
7. Click Save.
   Your column mapping is saved.

Example: Map Data File Columns to Data Fields

Sam performs the following steps to map the data file columns in the source data file to the CA APM data fields:

1. Clicks New in the Mapping area of the Import Details page.
2. Selects **%Login ID%** in the Source Field by clicking the Select icon next to Source Field and selecting this item from the dialog.
   The items that are listed in the dialog are the columns from the source data file.
3. Selects User ID in the Destination Field by clicking the Select icon next to Destination Field and selecting this object from the dialog.
4. Selects the Primary Lookup check box.
5. Continues to map the remaining columns in the source data file with CA APM data fields and clicks Save when finished.

Review the Mapping Reference Material

Reference the following information when setting up the column mapping for importing or deleting data.

Primary and Secondary Lookup Combinations

The fields that you select as the primary and secondary lookup in your column mapping are used to search for data in the product database.

- **Simple mapping**
  In simple mapping, you specify only the primary lookup. For example, you are importing or deleting a set of company records from a text file into the product database. You specify the Company Name as the primary lookup. If a company with a particular name does not exist in the database when you are importing data, a record is created for the company. The following table shows an example of the lookup for a simple mapping.

Source Field	Destination Field	Primary Lookup	Secondary Lookup
%Company Name%	Company.Company Name	Yes	No

- **Reference field mapping**
  In reference field mapping, you specify primary and secondary lookup values. To search for a unique object, specify more than one primary lookup. For example, to search for a company, you can specify Company Name, Parent Company, and Company Type as primary lookup values. In this example, the Data Importer searches for a company with the specified name, the specified parent company, and of the specified company type. If the object does not exist and you are importing data, the record is created (depending on the insert or update option you selected in Advanced Settings). The following table shows an example of the lookup for reference field mapping.

Source Field	Destination Field	Primary Lookup	Secondary Lookup
%Company Name%	Company.Company Name	Yes	No
%Parent Company%	Company.Parent Company.Company Name	Yes	Yes
%Company Type%	Company.Company Type.Value	Yes	Yes

This mapping has both the Primary Lookup and the Secondary Lookup checkboxes selected for Parent Company and Company Type. The Data Importer uses the Company Name to look up the parent company and uses the Parent Company to look up the company name.

- **Secondary object mapping**
  If a mapping rule maps to a secondary object property, the primary lookup values establish a relationship between a secondary object and the reference fields. The following table shows examples of the lookup for a secondary object mapping.

Source Field	Destination Field	Primary Lookup	Secondary Lookup
%Comment%	Legal Document.Legal Party.Comment	No	Yes
%Legal Document ID%	Legal Document.Document Identifier	Yes	No
%Company Name%	Legal Document.Legal Party.Legal Party.Company Name	Yes	Yes
%Legal Template%	Legal Document.Legal Template.Template	Yes	Yes

In the first mapping rule, Legal Document is the primary object, and Legal Party is the secondary object. Comment is a property of Legal Party.

In the third mapping rule, Legal Document is the primary object, and Legal Party is the secondary object. In addition, Legal Party has a reference field in the Company table. The Secondary Lookup check box indicates that the Company

Name is used to look up the Company object. The Primary Lookup check box indicates that the Company object is used to look up the Legal Party object.

Hard-Coded Values

In the column mapping, the percent signs that appear before and after column names identify the names as column headers in your source data file. You can also specify a hard-coded value in the Source Field that you want to apply to all records in your source data file. You can then map the hard-coded value to a Destination Field. The hard-coded values do not display with percent signs to distinguish these values from the source data file column names.

1. Source data file column header
2. Hard-coded value

You can define a hard-coded value in the Source Field to expand your source data and to ensure that you include all required fields. Hard-coded values typically do not begin and end with a percent sign (%). If you have hard-coded values with percent signs, the values cannot match the field names in your source data file.

Example: Use hard-coded values for asset family

In this example, the assets in your source data file do not contain asset family, which is required when creating an asset. You can add a hard-coded value to your mapping. If all of your assets are hardware, you can enter Hardware in the Source Field. You can map this value to the Asset Family field. If your assets belong to different families, add a column to your source data file with the corresponding asset families before importing or deleting data.

The following information illustrates the difference between values from your source data file and values that are added through hard-coded values:

- You have an Asset Family column in your source data file. The selection in the Source Field is %asset family%.

You do not have an Asset Family column in your source data file. However, all of your assets are hardware assets. You specify a hard-coded value of Hardware in the Source Field.

> **NOTE**
> You can also use the Main Destination Object to specify that all records in your source data file belong to a particular family or template. For example, the Asset (Hardware) selection for Main Destination Object specifies that all source records belong to the hardware asset family.

Multiple Values for a Single Field

You can add a mapping with multiple Source Field values that are mapped to a single Destination Field.

Example: Use multiple values for a single field

Your source data file has two columns with the names Manufacturer and Catalog Name. Combine these columns by selecting both in the Source Field. In this example, the Source Field selection is %Manufacturer% %Catalog Name%.

You can also enter multiple hard-coded values in the Source Field (for example, Document Management Company %model name% IT Department).

Filter Data in the Import

You can identify a subset of records in your source data file that you want to exclude from the import. The Data Importer exclusion filter allows you to filter a part of your data source using exclusion filter rules.

Example: Define an exclusion filter to process returned assets

A CSV file that you receive from your hardware vendor includes assets that were ordered and returned to the vendor. You want to process only the returned assets, so you want to import data to update those records only. You define an exclusion filter to exclude records that do not have a status of Returned.

Follow these steps:

1. On the Administration tab, Data Importer page, Exclusion Filter area for a selected import, select the Filter Type.
   - **And**
     Excludes a record from the source data file only if all the rules that you specify are valid for the record.
   - **Or**
     Excludes a record from the source data file if any of the rules that you specify is valid for the record.
2. Click New.
3. Click the Select icon next to Source Field, select a column from your source data file, and click OK.

   NOTE
   The percent signs before and after the column names identify the names as columns from your source data file.
4. Select the Operator.

   NOTE
   To specify "not equal to", select the "<>" operator.
5. Enter a Filter Value for the rule.

   NOTE
   You can use special characters and wildcards in the filter value. The rules can process text, numeric, and date fields.
6. Click the Complete Record Edit icon.
7. (Optional) Click New and specify more exclusion filter rules.
8. Click Save.
   The exclusion filter rules are saved and are applied when the import processes.

Example: Create an Exclusion Filter

Sam performs the following steps to create an exclusion filter. The filter eliminates non-IT employees and employees who do not work at the company headquarters from the data import.

1. Selects And for the Filter Type and clicks New in the Exclusion Filter area of the Import Details page.
2. Selects **%Department%** for the Source Field.
3. Selects <> for the Operator.
4. Enters IT for the Filter Value.
5. Clicks the Complete Record Edit icon and clicks New.
6. Selects %Location% for the Source Field.
7. Selects <> for the Operator.
8. Enters Headquarters for the Filter Value.
9. Clicks the Complete Record Edit icon and clicks Save.

Submit the Import

To start an import immediately, click Submit in the Schedule area of the page. The data source records from the data file for the selected import are processed.

NOTE
You can specify a data file other than the default (from the Basic Information) if you want to use a different file.

You can also schedule the import for a particular day and time. For more information, see schedule the import.

To view the import jobs for your current selected import, click Associated Jobs on the left side of the page. To view all import jobs for all imports, click Import Jobs on the left side of the page. In the list of import jobs that appears, click Status Message to view the status of an import.

You can also view the log file for more information about the import activity. In the list of import jobs, click View Logs for the selected import.

Schedule the Import

You can schedule an import for a specific time and you can specify the interval for the import (for example, daily or weekly). You can schedule multiple imports to process simultaneously.

Follow these steps:

1. On the Administration tab, Data Importer page, in the Schedule area for a selected import, select the Is Scheduled check box.
2. Provide the information for the schedule. The following fields require explanation:
   - **Run Time**
     Specifies the time of the day, in 24-hour format, to process the import. When you schedule imports, use the local time zone on the CA APM application server.
   - **Interval Day**
     Specifies the day during the Interval Type to process the import. For example, if the Interval Type is Month and the Interval Day is 1, the import is processed on the first day of the month.
   - **Data File**
     Specifies a data file name other than the default (from the Basic Information) if you want to use a different file. If this file is available on the application server, you can search and select the file. If this file is not available on the application server, you can locate and upload the file.
   - **Upload Data File**
     Browse for the source data file that you want to import. This file is uploaded to the application server.
   - **First Run Date**
     Specifies the date when the first import starts to process.
   - **Interval Type**
     Specifies the type of interval for the import (for example, Day, Month, Quarter, Week, or Year).
   - **Interval**
     Specifies how often the import processes. This interval is based on the specified Interval Type. For example, if the Interval Type is Weekly and the Interval is 2, the import processes every two weeks.
   - **Last Day of Interval**
     Specifies that the import processes on the last day of the selected Interval Type. If you select this check box, any previous value that you added to the Interval Day field is removed, and the Interval Day field is disabled.
3. Click Submit.
   The data import is scheduled for the specified date and time.

Examples: Using the Schedule Settings

The following examples illustrate the use of the schedule settings.

- Select Day for Interval Type and 2 for Interval. The import processes every other day.
- Select Week for Interval Type, 1 for Interval Day, and 3 for Interval. The import processes every three weeks on the first day of the week (Sunday).
- Select Month for Interval Type, 15 for Interval Day, and 2 for Interval. The import processes every two months on the 15th day of the month.
- Select Quarter for Interval Type and select Last Day of Interval. The import processes every quarter (every three months) on the last day of the last month in the quarter.
- Select Year for Interval Type, 1 for Interval Day, and 1 for Interval. The import processes on January 1 of every year.

To view the import jobs for your current selected import, click Associated Jobs on the left side of the page. To view all import jobs for all imports, click Import Jobs on the left side of the page. In the list of import jobs that appears, click Status Message to view the status of an import.

You can also view the log file for more information about the import activity. In the list of import jobs, click View Logs for the selected import.

View the Schedule Details

You can view the schedule details for a scheduled import job that you created.

First, open the list of import jobs.

- To view the scheduled import jobs for your currently selected import, click Associated Jobs on the left side of the page, select the Scheduled check box, and click Go.
- To view all import jobs for all imports, click Import Jobs on the left side of the page, select the Scheduled check box, and click Go.

In the list of import jobs that appears, click Schedule Details for a selected import.

View the Import Log Files

You can view the Data Importer log files to see the details of all CA-provided and user-defined imports that have completed. The Data Importer creates a log file for each import that you run, including imports that were submitted immediately or scheduled for a future time. All import activities are saved in the log files.

To view the log files, first open the list of import jobs.

- To view the import jobs for your current selected import, click Associated Jobs on the left side of the page.
- To view all import jobs for all imports, click Import Jobs.

In the list of import jobs, click View Logs for a selected import. If more than one log file is available (for example, for a scheduled import that has completed a few times already), all files are listed with their corresponding creation dates.

You can view any available LDAP Import Sync log file. If you click Start LDAP Data Import and Sync on the LDAP Data Import and Sync page (Administration, User/Role Management), an import job ID is displayed. Use this job ID to locate the job in the Data Importer list of import jobs. Then click View Logs for that job.

NOTE
You can also locate and view the import log files in the following location on the CA APM application server:

```
[ITAM Root Path]\Storage\Common Store\Import\Logs
```

Review the Import Log File - Best Practices

The Data Importer log file contains information and error messages regarding the processing of import jobs. To help you understand the results of your import and to troubleshoot any errors, use the information in this log file. This section contains some recommended best practices for working with the Data Importer log file.

Match the row number in the data file with the error message in the log file.

A log file error message identifies the corresponding row number from your data file. You can also find the data file row number in the row above or below the error message in the log file.

Sometimes the error message in the log file does not show the data file row number. In this situation, the actual data file values are shown immediately after the error message in the log file.

Count the number of error messages in your log file.

1. Search for the following phrases in your log file to find the error messages in the file. These phrases are included with the error messages.
   ```
   Web Service threw exception
   Error at record
   ```
2. After you find a type of error message, search for that error in the log file and count the number of occurrences.
3. Identify and search for more error types that appear in your log file and count the number of occurrences.
4. Compare the count of all errors in your log file with the statistics that the Data Importer generated for the associated import. To view these statistics, click Status Message on the Associated Jobs list or Import Jobs list. This comparison helps you account for all relevant errors and identify error messages that are not valid and can be ignored.

Verify the Imported Data

You verify that your data import succeeded by viewing your data in CA APM and by reviewing the Data Importer statistics.

- **Review the Data Importer Statistics.** To review the statistics for your current selected import, click Associated Jobs on the left side of the page. In the list of import jobs that appears, click Status Message for your import.
  You can also view the log file for more information about the import activity. In the list of import jobs, click View Logs for the selected import.
- **View the Imported Data in CA APM.** To view the imported data, navigate to the tab and subtab, if necessary, for the object that you imported (for example, asset, company, or contact). Search for the objects that you imported and verify that the objects are available.

Example: Verify the Data Import of New Employees

After Sam runs the import, he performs the following steps to verify the data import of new employees:

1. Checks the import statistics.
   - Clicks Associated Jobs or Import Jobs on the left side of the Data Importer page.
   - Clicks Status Message for the import and reviews the statistics.
2. Views the import log file and the user interface.
   - Clicks View Logs in the list of import jobs and reviews the contents of the log file.
   - Navigates to Directory, Contact on the CA APM user interface. Searches for the new employees. Verifies that the non-IT employees and employees who do not work at company headquarters are not available.

How to Submit a Data Import Using a Process Workflow

You can use a process workflow (for example, CA Process Automation) to submit a Data Importer data import for processing instead of using the CA APM user interface.

> **NOTE**
> You can create a data import process workflow using a company-provided sample XML file and integrating with CA Process Automation.

To submit a data import using a process workflow, perform these steps:

1. Review the Prerequisites.
2. Specify the Workflow Calls.
3. Verify the Status of the Import Job.

4. (Optional) Respond to Error Messages.

Example: Import New Hardware Devices through a Process Workflow

Sam, the CA APM system administrator at Document Management Company, has defined a business process workflow. The workflow discovers new hardware devices, adds the new devices to the company data repository, and runs reports about the new devices. Sam has already created a data import in CA APM that adds the new hardware devices to the data repository. Sam wants to execute that data import at a specific point in his overall workflow. He wants to integrate the data import with his overall business process workflow. Sam wants the data import to execute at the time that the workflow specifies without the user logging in to the product user interface. Sam updates his business process workflow to include calls to the CA APM web service operations for the Data Importer.

Review the Prerequisites

To ensure that you can successfully submit a data import using a process workflow, verify that you have completed the following prerequisites:

1. Define a data import with all mappings and settings through the CA APM user interface.
2. Verify that the data file path (if you are specifying a path) is accessible from the server where the Import Service is running. Also, the Network Service (application pool identity) user requires access to this path.
3. Define a process workflow using a workflow provider (such as CA Process Automation).

Specify the Workflow Calls

To launch the Data Importer and execute a data import from a process workflow, you provide specific workflow calls to CA APM web service operations. These operations perform the following functions:

- Login operation - Logging in to CA APM.
- Submitting a data import using one of the following ways of providing a data file:
  - SubmitImportwithfilepath Operation - The data file is available on a specified file path. This file path must be accessible from the server where the Import Service is running. The web service operation uploads the file.
  - SubmitImport Operation - The data file content has been converted to byte array binary format. The web service operation receives the byte array content from an application and submits the content to the Data Importer.

    NOTE
    To use this way of providing a data file, create an application, if one is not already available, to convert the data file content to byte array format. The application then sends the content to the web service operation.

Incorporate the calls to these operations into your business process workflow.

NOTE
For information about creating a process workflow, see the product documentation for your workflow provider.

Login Operation

This operation logs in to CA APM using the specified CA APM user ID and password. The output of this operation is the login token. The login token is used as input to other data import workflow operations.

- **Input Parameters**
  ItamUserName - CA APM user ID
  ItamUserPassword - CA APM user password
- **Output Parameters**
  loginToken - Token that is returned after the CA APM login.

SubmitImport Operation

This operation receives data file content that has been converted to byte array format and submits the content with the data import to the Data Importer. To use this operation, create an application, if one is not already available, to convert the data file content to byte array format. The application then sends the content to this web service operation.

This operation returns a data import job ID, which is used to verify the status of the import job.

- **Input Parameters**
  loginToken - Token that is returned after the CA APM login.
  ImportName - Name of the data import.
  Datafilename - Name of the data file that is associated with the data import.
  Datafilestream - Data file content in byte array format.
  Caprovided - (Optional) Indicator that specifies a product-provided data import. Set this parameter to one (1) to specify a product-provided import.
  Tenant - (Multi-tenancy only) Name of the tenant to which the import applies.
- **Output Parameters**
  Job ID - ID that is returned after a data import is submitted successfully. The GetJobStatus operation uses this ID to verify the status of an import job.

SubmitImportwithfilepath Operation

This operation uploads a data file from a specified file path and submits the data file with the data import to the Data Importer. This file path must be accessible from the server where the Import Service is running.

The operation returns a data import job ID, which is used to verify the status of the import job.

- **Input Parameters**
  loginToken - Token that is returned after the CA APM login.
  ImportName - Name of the data import.
  Datafilepath - Complete path and name of the data file that is associated with the data import. This path must be accessible from the server where the Import Service is running. Also, the Network Service (application pool identity) user requires access to this path.

  NOTE
  If the data file location is a shared path, the CA APM server and the shared computer must be in the same domain.Caprovided - (Optional) Indicator that specifies a product-provided data import. Set this parameter to one (1) to specify a product-provided import.

  Tenant - (Multi-tenancy only) Name of the tenant to which the import applies.
- **Output Parameters**
  Job ID - ID that is returned after a data import is submitted successfully. The GetJobStatus operation uses this ID to verify the status of an import job.

Verify the Status of the Import Job

The Data Importer provides a status summary of each submitted data import job. Your process workflow can include a call to the CA APM web service operation that retrieves the status of a submitted data import job. Incorporate the call to this operation into your process workflow.

GetJobStatus Operation

This operation uses the data import job ID to verify the status of an import job.

- **Input Parameters**
  loginToken - Token that is returned after the CA APM login.
  Job ID - ID that is returned after a data import is submitted successfully.
- **Output Parameters**
  Job Status - Status of the import job.

Respond to Error Messages

If errors occur during the data import workflow process, you can receive error messages. The following messages require explanation:

- **20002 - Cannot access the data import because of user permissions. Contact your administrator.**
  The user role requires Data Importer Admin access or Data Importer User access to submit a data import.
- **20005 - Cannot connect to the Import service. Contact your administrator.**
  Verify the Import Service URL in the ImportProcessor.config file, or contact your administrator.
- **21002 - The data import name is invalid.**
  The data import does not exist, or the user does not have access to the data import. If the data import is product-provided, specify a value of 1 for the Caprovided parameter.
- **21004 - The data file failed to upload.**
  This message can result from a configuration error. Review the Storage Manager Service log files.
- **21005 - No mappings are defined for the data import.**
  Define mappings and resubmit the data import.
- **22001 - The data import job ID is invalid. Verify the job ID and try to execute the import again.**
  Verify the job ID by logging in to CA APM and locating the data import job. Resubmit the data import with the valid job ID.

How to Submit a Data Import Using the Command Line

You can use a command line to submit a Data Importer data import for processing instead of using the CA APM user interface. You can execute the command line from the Import Processor folder on the application server where the product is installed. You can also copy the Import Processor folder to another computer. The users of that computer can then execute the command line, also.

The data import is submitted immediately and is executed by the Data Importer Engine with other import jobs from the CA APM user interface. You cannot schedule a data import to execute at a particular time using the command line. However, you can use a scheduler (such as the operating system scheduler) to specify dates and times to run the data import.

To submit a data import using the command line, perform these steps:

1. Review the Prerequisites.
2. (Optional) Display Version and Help.
3. Execute the Command Line.
4. Retrieve the Import Job Status.

Example: Import New Hardware Devices

Sam, the CA APM system administrator at Document Management Company, has an existing data import that adds new hardware devices to the data repository. Sam wants to execute that data import daily. Sam also wants to verify the status of the submitted import job. However, he does not want to log in to the product to perform the import because he does not always perform other product functions on a daily basis. Sam uses the command line to submit the data import and then verify the status.

Review the Prerequisites

To ensure that you can successfully submit a data import using the command line, verify that you have completed the following prerequisites:

1. Verify that Microsoft .NET Framework 4.0 is installed on the computer where you are executing the command line.
2. Define a data import with all mappings and settings through the CA APM user interface.
3. (Optional) If you change the Import Service URL, modify the ImportProcessor.exe.config file to reflect the new URL. You can locate the ImportProcessor.exe.config file in the Import Processor folder. Update the endpoint address value. Example: The following statements show an example of the endpoint address value that you modify to change the Import Service URL.

```
<endpoint address="http://localhost/ImportService/ImportService.svc"
    binding="basicHttpBinding" bindingConfiguration="BasicHttpBinding_ImportService"
```

```
contract="IImportService" name="BasicHttpBinding_ImportService" />
```

Display Version and Help

Specify the command line parameters to display the command line version and usage help.

Follow these steps:

1. Log in to the application server where you installed CA APM or to a computer that has the Import Processor folder.
2. Access the Import Processor folder.

   NOTE
   On the application server, the Import Processor folder is located in the CA APM installation path.
3. Open a command prompt window and execute the following command:

   ```
   importerprocessor -H | -V
   ```

   – **-H**
   Displays the command line version number and usage help for the command line parameters.
   – **-V**
   Displays the command line version number.

Execute the Command Line

Specify the command line parameters to submit a data import.

Follow these steps:

1. Log in to the application server where you installed CA APM or to a computer that has the Import Processor folder.
2. Access the Import Processor folder.

   NOTE
   On the application server, the Import Processor folder is located in the CA APM installation path.
3. Open a command prompt window and execute the following command:

   ```
   importerprocessor -usr "user_name" -pwd "password" -i "import_name"
    - df  "data_file_absolute_path" -t "tenant_name"  - ts -c
   ```

   – **-usr**
   Specifies the CA APM login user name.
   – **-pwd**
   Specifies the CA APM login password.
   – **-i**
   Specifies the name of the data import that was created previously through the CA APM user interface.
   – **-df**
   Specifies the absolute path of the data file that is associated with the data import. The Data Importer Engine uses this file to process the import.
   – **-t**
   (Required for multi-tenancy) Specifies the name of the tenant that is associated with the data import.
   – **-ts**
   (Optional) Specifies that the command line parameters are recorded in the Import Processor log file.

   NOTE
   The Import Processor log file is located in the Import Processor folder.

   – **-c**
   (Optional) Identifies whether the data import was provided with the product or was created by a user.
   Valid values: 1 (product provided) or 0 (created by a user)
   Default: 0

Retrieve the Import Job Status

Specify the command line parameters to verify the status of an import job.

Follow these steps:

1. Log in to the application server where you installed CA APM or to a computer that has the Import Processor folder.
2. Access the Import Processor folder.

   NOTE
   On the application server, the Import Processor folder is located in the CA APM installation path.
3. Open a command prompt window and execute the following command:

   ```
   importerprocessor -usr "user_name" -pwd "password" -j "job_id"  - ts
   ```

   - **-usr**
     Specifies the CA APM login user name.
   - **-pwd**
     Specifies the CA APM login password.
   - **-j**
     Specifies the import job ID.
   - **-ts**
     (Optional) Specifies that the command line parameters are recorded in the Import Processor log file.

     NOTE
     The Import Processor log file is located in the Import Processor folder.

Managing Product-Provided Data Imports

This article contains the following topics:

Product-Provided Data Import Types

The product provides a set of predefined data imports that already contain all mappings and settings. These imports help you get started with data management. The two types of product-provided data imports allow you to perform the following functions:

- Read-only data imports - Allow you to monitor internal system functions, such as the LDAP Sync import of contacts.
- Object imports - Allow you to perform imports of common objects, such as locations, contacts, and assets.

You cannot modify the mappings and settings in the product-provided data imports. However, you can copy the imports and modify the copies.

Monitor the Status of Product-Provided Read-Only Data Imports

The read-only data imports perform internal system functions. You can monitor the status of the read-only imports, but you cannot submit these imports. You can copy the read-only imports and modify the copies to create your own imports.

Follow these steps:

1. Navigate to Administration, Data Importer.
2. Click one of the product-provided read-only data imports (not the object data imports). The following fields require explanation:
   - **CA APM - LDAP Sync Import**
     Submits a data import with the data file that CA EEM generated. This data import creates contacts through the LDAP Sync component.
   - **CA APM - Device Delete Import**

Submits a data import with the data file that CA SAM generated. This data import deletes the information that is associated with deleted discovered assets.

- **CA APM - Device Insert or Update Import**
  Submits a data import with the data file that CA SAM generated. This data import adds or updates the information that is associated with discovered assets.

3. Click Associated Jobs on the left side of the page.
4. Click Status Message in the list of import jobs to view the status of an import.

Submit the Product-Provided Object Data Imports

The product-provided object data imports perform imports of common objects. You can submit these imports, and you can monitor the status of these imports. To submit a product-provided object data import, verify that data was added to the associated data files. You can also specify your own data file. However, the column headers in your data file must match the column headers in the product-provided data file.

You can also copy these imports and modify the copies to create your own imports.

NOTE
In a multi-tenanted environment, these imports add to or update the data in the Public Data tenant.

Follow these steps:

1. Navigate to Administration, Data Importer.
2. Click one of the product-provided object data imports (not the read-only imports). The following fields require explanation:
   - **CA APM - Company Import**
     Creates and updates companies.
   - **CA APM - Cost Center Import**
     Creates and updates cost centers.
   - **CA APM - Location Import**
     Creates and updates locations.
   - **CA APM - Contact Import**
     Creates and updates contacts.

     NOTE
     If you are using multi-tenancy, you cannot submit this import. Copy this import, add a mapping for the tenant, and submit your new contact import.
   - **CA APM - HW Model Import**
     Creates and updates hardware models.
   - **CA APM - HW Asset Import**
     Creates and updates hardware assets.
   - **CA APM - Unreconciled Discovered Assets Import**
     Creates and updates discovered assets.

     NOTE
     A CA Business Intelligence report about unreconciled discovered assets provides input for this data import. This CA Business Intelligence report does not include the Class and Status fields. Add these fields to the data file that corresponds to this import.
3. Specify your own data file in the Schedule area or use the product-provided data file.

   NOTE
   The product-provided data file does not contain data. Add the data that you want to import into the product-provided data file before you submit the import. You can find the product-provided data files at the following locations on the CA APM application server where the Storage Manager Service is installed.

```
[ITAM Root Path]\Storage\Common Store\Import
```

4. Click Submit.

CA APM Environment Promotion

The CA Asset Portfolio Management Environment Promotion lets you promote the changes across environments with minimal manual effort. As a system administrator, you can promote any change across development, test, pre-production, and production environments.

Initiate CA Asset Portfolio Management Environment Promotion either through the user interface (UI) or command-line interface (CLI).

This article contains the following topics:

Prerequisites for Environment Promotion

1. To promote content from the source to the target system, ensure that CA APM 17.4 is installed on both source and target systems.
2. During import, stop all CA APM services and resume them after the import is completed.

How Environment Promotion Works

- Environment Promotion involves the creation of an export package by promoting the configurations and content from the source system.
- To promote content to the target system, the system administrator must copy the export package and must perform a dryrun and import.

> **WARNING**
>
> Initiate Environment Promotion to promote the custom changes.
>
> Operations that are initiated through the command-line interface must be performed on the CA APM component server (First application server).

Supported Operations

> **NOTE**
>
> The supported operations must be executed after you change the directory to CA APM Promotion folder.

The following table lists the supported operations for CLI and CA APM UI:

Operation	Purpose	CLI	CA APM UI
Export	Create an export package to promote the configurations and content from the source system.	Yes*	Yes
DryRun	Validate the compatibility of the export package in the target system without actually importing it.	Yes	No
Import	Import the exported package in the target system.	Yes	No

History	Provide end-to-end execution history details such as package content, Change Initiator, and reason for change in the production environment.	Yes	No
Help	Display a list of commands applicable for Environment Promotion.	Yes	No
Version	Display the CA APM version that is installed with the patch levels.	Yes	No

* Indicates that only object-level filtering capability is supported.

WARNING

We recommend that you initiate the export operation through the CA APM UI.

Environment Promotable Objects

The following objects are eligible for Environment Promotion:

- List Management Data
- Configuration (Global and Local)
- Roles
- Search Definition
- Export Definition
- Reconciliation Rule
- Data Filter Definition
- Event Definition
- Data Import Definition

Support for Multi-Tenancy

WARNING

Before you initiate CA APM Environment Promotion, we recommend that you manually verify tenant hierarchy between the source and the target system.

- Before you promote the content, ensure that the tenant in the source and the target system is the same.
- Environment Promotion does not create, update, or delete any tenant on the target system.
- Before you promote the source content to the target system, Environment Promotion verifies if the tenant name and tenant group on the target system matches with that of the source. If the verification fails, the source content is not promoted to the target.

How to Promote Configurations and Content from the Source to the Target Systems

Syntax for Supported Operations

The following table lists the supported operation and its syntax:

Operation	Command Line Syntax	Command Line Syntax (Target Directory)
Export	>apmp -a export	apmp -a export -fp <File_Path>

Dryrun	>apmp -a dryrun -p <export_pkg_name>.zip	>apmp -a dryrun -fp <File_Path> -p <export_pkg_name>.zip
Import	>apmp -a import -p <export_pkg_name>.zip	>apmp -a import -fp <File_Path> -p <export_pkg_name>.zip
History	>apmp -a history	
Help	>apmp -help	
Version	>apmp -version / -v	

To promote custom configurations and content from the source to the target systems, perform the following tasks:

Export

Verify Prerequisites

- You must have the CA APM administrator account name and password.
- Ensure that you have sufficient disk space (at least 500 MB) on the system where you plan to export.
- Ensure that CA Asset Portfolio Management - Export Service is running.
- While exporting to a shared location, ensure that you have sufficient write permissions and sufficient disk space (at least 500 MB) for the shared path.

How to Perform Export

To promote the configurations and content from the source system, use the **Environment Promotion** option in the CA APM UI.

> **NOTE**
>
> You can initiate multiple exports. However, only one job can be processed at a time and the other jobs are scheduled in the queue.

Follow these steps:

1. Log in to CA APM as a system administrator.
2. Navigate to **Administration, Environment Promotion**.
3. (Optional) Define the following filter values:
   - From Date
   - To Date
   - Last Modified User
4. Click **Go.**
5. Select the objects from the list of **Object Type**.
6. (Optional) Click the magnifying glass icon to sub-select from the list of **Object Type**.
7. Click **Export**.
8. (Optional) Add a comment and click **Submit**.
9. Navigate to **Administration**, **Environment Promotion**, **View Jobs**, **View Log** to view the status of the submitted export package. The status can be any one of the following:

- **Not Started**: Specifies that the export package is scheduled in the queue.
- **Running**: Specifies that the export package is running.
- **Complete**: Specifies that the export package is exported successfully.
- **Canceled:** Specifies that the export is cancelled. Displayed when a user cancels any running operation using the **Cancel** button.
- **Failed:** Specifies that the export job fails.
- **Interrupted:** Specifies that the export job is interrupted.

Verify the Export

- After completion, you can view the exported package in the *Root/APM Promotion/Export* folder. Verify the **apmp_export_time stamp.zip** export package.
- To verify errors or failures that are encountered during the export process, check the *apmp.log* in the Root*/APM Promotion/logs* folder.

How to Cancel Export

You can cancel an export job only if the status is **Not Started**.

Follow these steps:

1. Log in to CA APM as a system administrator.
2. Navigate to **Administration** tab, **Environment Promotion**, **View Jobs** .
3. Select the export job that you want to cancel and click **Cancel Job.**
   The confirmation dialog appears.
4. Click **Ok**.
   The export is canceled and the job status changes to Canceled.

DryRun

Perform a DryRun to validate the compatibility of the export package in the target system without actually importing it.

For example, Visibility is a property of the foreign key search configuration attribute. The visibility of the foreign key search configuration attribute in the source is promoted to the target environment.

If the source target attributes visibility is set to **Search Criteria** and on the target system it is set to **Search Results,** after Environment Promotion the target environment attributes visibility changes to **Search Criteria**.

> **WARNING**
>
> We recommend that you perform a DryRun before initiating the import to ensure that the changes are promoted without any failure.

How to Perform DryRun

We recommend that you initiate DryRun before executing the Import process.

Follow these steps:

1. Copy the export package from the source system to target system and paste this package in the *Root/APM Promotion/Import* folder.
2. Click Start, Run to open the command prompt (cmd) window.
3. Change the directory to the CA APM Promotion folder.
4. Execute the DryRun command.
5. Enter the CA APM System Administrator User Name and press Enter.
6. Enter the CA APM System Administrator Password and press Enter.

The DryRun operation is completed and a summary report is created.

7. Press Enter to exit the console.
8. Navigate to the *Root/APM Promotion/DryRun* folder.
9. Access the apmp_dryrun_timestamp.csv file.
10. Sort the file by status, and take corrective measure for the records with status **Conflict**.

WARNING

For a successful import, we recommend that you take corrective measure for the records with status **Conflict**.

DryRun Status Definition

The following table lists the DryRun status definition:

Status	Definition
New	The source content does not exist on the target system and hence it is promoted to the target system.
Update	The source content matches the target content and hence it is promoted to the target system.
Skip	The source content is not promoted to the target system as the dependent content does not exist on the target system.
Conflict	Due to a conflict with the target content, the source content is not promoted to the target system. For example, the source and target system **global configuration** names are the same, however the object types/sub-types are different.
Conditional	The source content is promoted only when the dependent content is available on the target system while importing.

Import

Verify Prerequisites

- Back up your system database (MDB). This ensures safe data recovery if any problems are encountered.
- You must have the CA APM administrator account name and password.
- The target environment must be at the same patch level as the source environment.
- Copy the export package from the source system to target system and paste this package in the *Root/APM Promotion/Import* folder.
- Ensure that you have sufficient disk space (at least 500 MB) on the target system where you plan to import.
- The DB server (Oracle or Microsoft SQL) on the source and target system must be the same.

How to Perform Import

After a successful DryRun, initiate the Import process to promote the content to the target system.

Follow these steps:

1. Click Start, Run to open the command prompt (cmd) window.
2. Change the directory to the CA APM Promotion folder.
3. Execute the Import command.

4. Enter the CA APM System Administrator User Name and press Enter.
5. Enter the CA APM System Administrator Password and press Enter.
   After a while the import operation completes.
6. Execute the following command to stop and restart IIS:
   ```
   iisreset
   ```
7. Press Enter to exit the console.
8. Copy the Data Importer dependent files from the source system to the target system in the *Root/Storage/Common Store/Import* and *Root/Storage/Tenant Name/Import* folders.

Verify the Import

After importing the content to the target system, verify the changes are reflected on the target system.

1. A summary report is generated in the *Root/APM Promotion/Import* folder. Check the *apmp_import_timestamp.txt* file.
2. If any failures are encountered during import, rerun the import process using the same package. The import process restarts from the point of failure.

   NOTE

   To verify errors or failures that are encountered during the import process, check the *apmp.log* in the Root*/APM Promotion/logs* folder.

CA APM Environment Promotion Limitations

The following limitations apply to CA APM Environment Promotion:

- The new data importer definitions are promoted, however the job schedules are not promoted.
- The new data importer definitions are promoted, however the related data files and legacy map files are not promoted.

Object Status After Mapping from Source to Target Systems

This article describes the Environment Promotion behavior at an object level.

List Management Data

For example, when the source and target system contain the same list management data and tenant name, then the object is promoted to the target system with operation status as Skip.

List Management Data	Tenant Name	Operation
Same	Same	Skip
Same	Different	New
Different		New

Configuration

For example, when the source and target system contain the same global configuration and type/subtype, then the object is promoted to the target system with operation status as Update

Configuration Name	Global Configuration	Type /Subtype	Operation
Same	Yes	Same	Update

Same	Yes	Different	Conflict
Same	No	Same	Update
Same	No	Different	Conflict
Different	Yes	Same	Conflict
Different	Yes	Different	New
Different	No	Same	New
Different	No	Different	New

Roles

For example, when the source and target system contain the same roles, then the object is promoted to the target system with operation status as Update.

Roles	Operation
Same	Update
Different	New

Search Definition

For example, when the source and target system contain the same search definition name and type/subtype, then the object is promoted to the target system with operation status as Update.

Search Name	Type/Subtype	Operation
Same	Same	Update
Same	Different	Conflict
Different		New

Export Definition

For example, when the source and target system contain the same export definition and tenant name, then the object is promoted to the target system with operation status as Update.

Export Definition	Tenant Name	Operation
Same	Same	Update
Same	Different	Conflict
Different		New

Reconciliation Rule

Source reconciliation rules overwrite the target reconciliation rules.

Data Filter Definition

For example, when the source and target system contain the same data filter definition and type/subtype, then the object is promoted to the target system with operation status as Update.

Data Filter Definition	Type/Subtype	Operation
Same	Same	Update
Same	Different	Conflict
Different		New

Event Definition

For example, when the source and target system contain the same event definition and CA Process Automation workflow name, then the object is promoted to the target system with operation status as Update.

Event Definition	CA Process Automation Workflow Name	Operation
Same	Same	Update
Same	Different	New
Different		New

Data Import Definition

For example, when the source and target system contain the same data import definition and tenant name, then the object is promoted to the target system with operation status as Skip.

Data Import Definition	Tenant Name	Operation
Same	Same	Skip
Same	Different	New
Different		New

Extended Field

For example, when the source and target system contain the same extended field and type/subtype, then the object is promoted to the target system with operation status as Skip.

Extended Field	Type /Subtype	Operation
Same	Same	Skip
Same	Different	Skip
Different		New

Enable SSL Configuration for CA Service Management Solution

Perform the following steps to enable SSL configuration in CA Service Management.

1. Configure SSL for Service Management Solution.
2. Enable SSL Integration.

Configure SSL for Service Management Solution

You must first configure SSL authentication in each product you installed as part of the solution.

For information on how to configure SSL authentication in the products of the solution:

- SSL Configuration for CA Asset Portfolio Management
- SSL Configuration for CA Service Desk Manager
- SSL Configuration for CA Service Catalog
- SSL Configuration for xFlow Interface
- SSL Configuration for Jasper Reports Server

> **NOTE**
> CA Service Management provides the SSL Configurator Utility to enable SSL for CA Service Desk Manager and its components, xFlow Interface/Service Point, and CA Service Catalog. For more information, see SSL Configurator Utility.

Enable SSL Integration For All The Products

After you configured SSL authentication in each of the products, you must enable SSL integration through the common administration user interface.

Perform the following steps:

1. Log in to CA Service Catalog as an Administrator.
2. Access the service offerings from the CA Service Catalog user interface as an Administrator.
3. In the Home page, click **Request a Service**.
4. Under **Categories**, click **Service Management Administration**.
5. Click **Product Integrations**.
6. Under each product tab, check the **HTTPS** check box and specify the SSL port numbers.
7. Click **Submit**.

Authenticate the Users of xFlow Interface

Basic Authentication for xFlow Interface users is enabled by default. In Basic authentication, users are required to provide the user ID and password in the login page. If you want to change the authentication from **BASIC** to **NTLM**, follow these steps:

> **NOTE**
>
> NTLM Authentication is supported only on the Windows platform.

1. Login into CA Service Desk Manager as an **Administrator**.
2. Navigate to the **Administration** tab, select xFlow Interface, and select **General**.
3. Click Authentication type, under the Configuration Key column.
   Authentication type Update Configuration page opens.
4. Click Edit.
5. Modify the value to NTLM in Authentication type text box.
6. Click Save.
7. Restart the xFlow Interface service.

NOTE

The changes must be made for each instance and the services that are deployed.

SSL Configurator Utility Wizard for CA Service Management

This article contains the following topics:

CA Service Management now provides the SSL Configurator Utility Wizard to make it easier to configure SSL/TLS settings for CA Service Desk Manager, xFlow Interface, and CA Service Catalog. In addition, this utility provides the option to generate and import an SSL certificate for Apache Tomcat and IIS Web Servers.

NOTE

SSL Configurator Wizard Utility is currently supported on CA SDM (Tomcat/IIS Web Servers), CA service Catalog, and on xFlow Interface (Play Web Server).

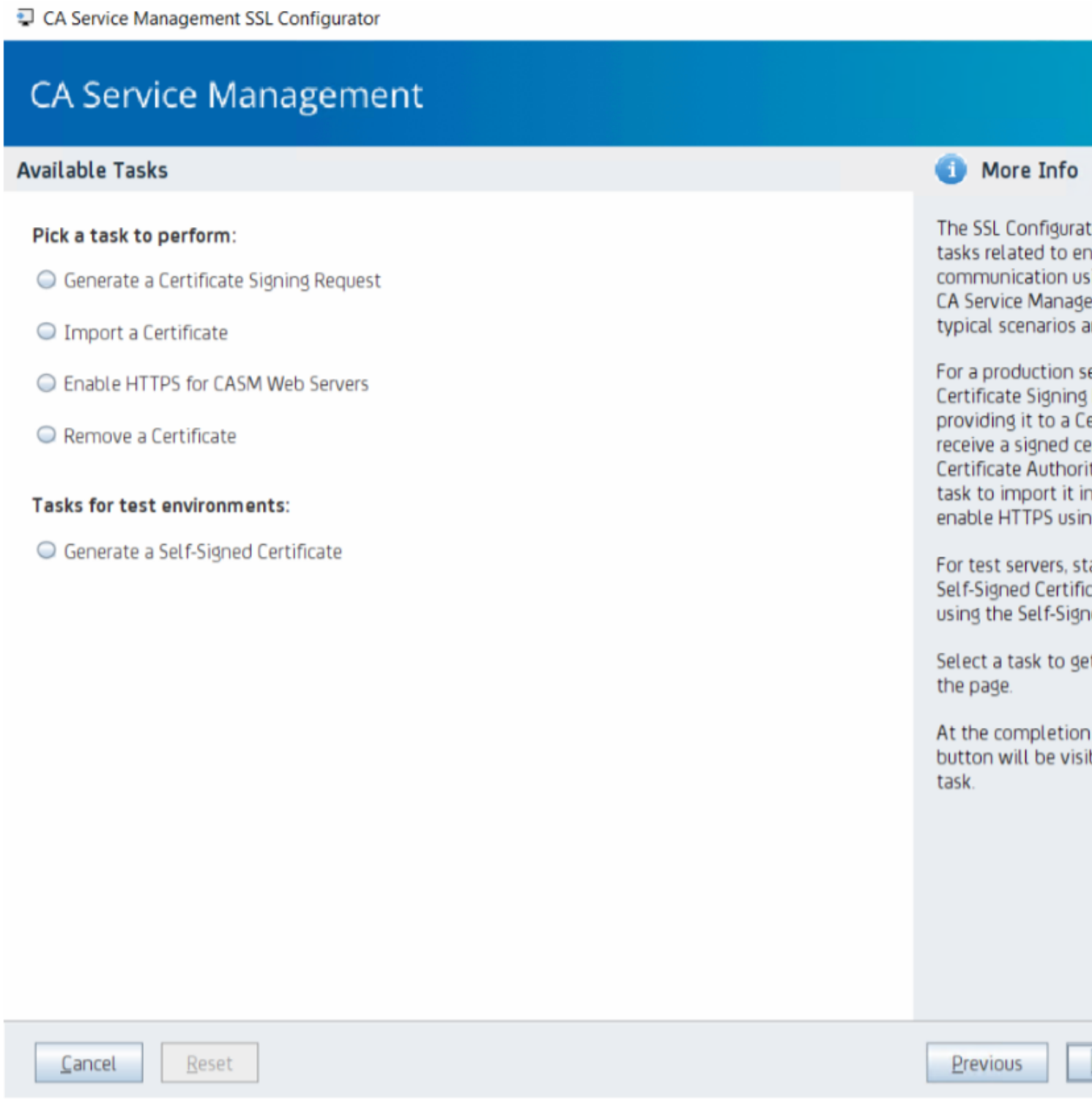
NOTE
From CASM 17.4 RU4 and above, SSL utility will retain the existing secure configurations.

The SSL Configurator Wizard utility does not provide a history of completed tasks via user interface. To review previously completed tasks, open the log file *<root install>\log\jstd.log* where each wizard task reports a successful or failed message.

Launch the SSL Configurator Utility

1. Download any one of the CA Service Management 17.4 DVD mentioned below, from Broadcom Support.
   a. CA SERVICE MANAGEMENT SDM PACKAGE 17.4 FOR WINDOWS (WITH CATALOG).iso
   b. CA SERVICE DESK MANAGER 17.4 FOR WINDOWS (WITHOUT CATALOG).iso
   c. CA SERVICE DESK MANAGER 17.4 FOR LINUX.iso
2. Extract the contents of the DVD and navigate to the Filestore location. The SSL Utility Wizard Installation files are available in: *\filestore\SSL_TLS_Utility)*
   a. The SSL Configurator tool has been enhanced as part of 17.4RU4, with automated steps to eliminate mandatory manual tasks in SSL configuration.
   Scenario 1: If the environment is not SSL-configured, it is recommended to upgrade to 17.4RU4 and run the SSL Configurator to enable SSL across ITSM applications (SDM, Catalog, xFlow/ServicePoint) for all configuration types.
   Scenario 2: If the environment is already SSL-enabled, upgrading to 17.4RU4 will preserve existing SSL configurations, and running the SSL Configurator tool is optional. However, if there's a need to change ports, running the tool will update the necessary SSL configurations across ITSM applications.
3. The SSL_TLS Utility folder has:
   a. CASM-SSL-Configurator-17.4.0.zip (Windows)
   b. CASM-SSL-Configurator-17.4.0.tar.gz (Non-Windows)
4. Create the following folder(s) in the shared components location *(\Program Files\CA\SC* or *opt/CA/SC*) for CA Service Management:
   a. CASM SSL Configurator (Windows)
   b. CasmSslConfigurator (NonWindows: Create a folder without spaces for Non-Windows.
5. Now, copy and extract CASM-SSL-Configurator.zip (Windows) or CASM-SSL-Configurator.tar.gz (Non-Windows) to the folder created in **step 4**.
6. Open the **casm_sslconfig.bat** file (Windows) or **casm_sslconfig.sh** file (Non-Windows) and update shared components JAVA home path.
   Example Window: SC_JAVA_HOME=C:\Program Files\CA\SC\JRE\11.0.18
   Example Non-Windows: SC_JAVA_HOME=/opt/CA/SC/JRE/11.0.18
7. Run as Administrator the SSL Configurator Utility batch file from **step 6** as per your platform requirements.

8. The SSL Configurator Utility Wizard is launched.
9. Select a language of your choice: The language is defaulted to the default system locale. You can change it to a different locale as per your preference.
10. Click **Next**.
11. Provide the CASM Keystore password. Perform the following steps:
    a. On the CASM Keystore page, provide the password, if you have an existing keystore already stored in your system.
    b. For users generating a keystore for the first time, provide a new password for the CASM Keystore. Click **Next**
12. Select the following options based on your requirements:



a. Pick a task to complete
   a. Generate a Certificate Signing Request
   b. Import a Certificate
   c. Enable HTTPS for CASM Web Servers
   d. Remove a Certificate

b. Tasks for Test Environment (Non-Production Environment)
   a. Generate a Self-Signed Certificate

Generate a Certificate Signing Request

Generate a Certificate Signing Request (CSR). This task generates a certificate signing request (CSR) document which needs to be submitted to a Certificate Authority. The CSR is generated using the information provided on the wizard page.

Perform the following steps:

1. On the Available Tasks page, select the **Generate a Certificate Signing Request** option to create a **Certificate Signing Request (CSR)** document.
   a. Complete the field information shown on this page as shown below:

Field Information	Description
Alias	Specifies the alias name and in most cases, it refers to the local host name.
FQDN	The fully qualified domain name (FQDN) of the local server. Must match exactly the server name used in URL accessing the web interface.
Organization	Specifies the legal name of your organization. Note that this must not be abbreviated and must include all suffixes as well.
Organization Unit	The division or unit of your organization.
City	The city where your organization is located.
State	The state/region where your organization is located. This should not be abbreviated.
Country Code	The 2-letter ISO country code where your organization is located.
Days Valid	The number of days the certificate is valid for.
Key Algorithm:	The algorithm to be used to generate the key pair. Possible values: RSA, DSA, EC. The following mapping for key size works and is a valid entry: Key Algorithm: Key Size RSA: 2048, 4096 DSA: 2048 EC: 256, 512
Key Size	The size of each key (public and private) to be generated. Default: 2048 **Note:** The Key Size is relevant to the key algorithm that is selected.

2. Click **Next** to review the Summary page.
   The .**csr** file location is mentioned in the Review page.
3. Click **Finish**. After successful completion, the task summary will display the location of the generated CSR document.

Import a Certificate

1. Launch the Wizard and select the **Import an SSL Certificate** option to import an SSL certificate or certificate chain provided by a Certificate Authority (CA) into the keystore created earlier as part of generating a Certificate Signing Request (CSR).

   NOTE

   You may need to import multiple certificates into the keystore based on your configuration requirements.

   Click **Next**.
2. Provide the following information to import the SSL certificate:
   a. Provide the alias or entry name that you used while creating the Certificate Signing Request (CSR)
   b. Browse and select the location of the SSL certificate.
3. Click **Next** to review the summary page.
4. Click **Finish** to complete importing the SSL certificate.

Enable HTTPS for Service Management (Service Desk Manager, xFlow Interface, and Service Catalog)

Perform the following steps to complete the SSL Configuration for SDM along with xFlow interface and Service Catalog:

Enable HTTPS for CA Service Desk Manager Web Servers and xFlow Interface

Launch the SSL Configurator Wizard utility and select the option to enable HTTPS for Service Desk Manager Web Servers.

NOTE

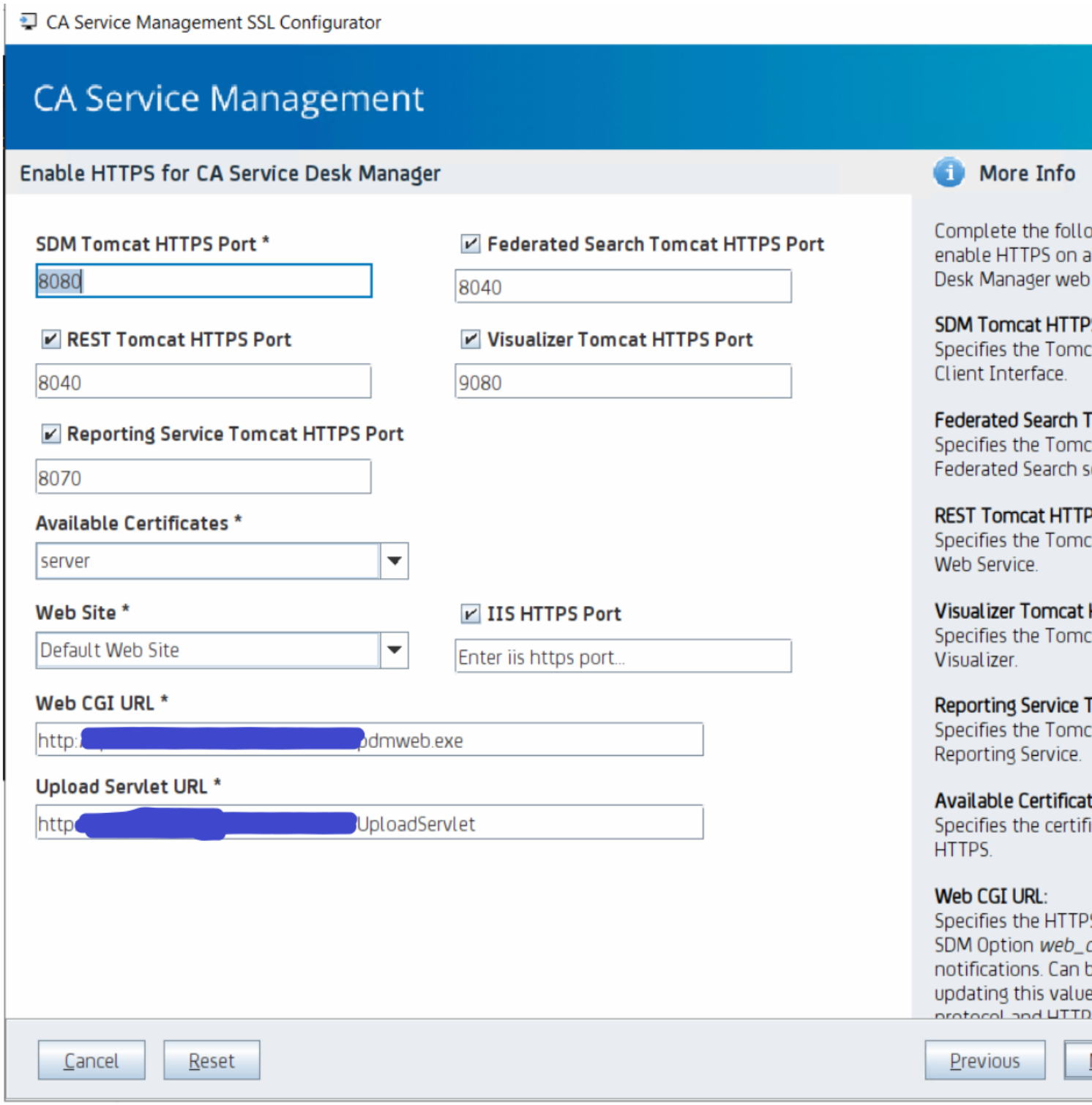
Ensure that the Service Desk Manager services are up and running in order to complete this task.

Perform the following steps:

1. After launching the SSL Configurator Wizard, select the Enable HTTPS for Service Desk Manager option from the Main page of the SSL Wizard.
   Click **Next**.
   The ports for Tomcat server instances that are not installed are greyed out.

   NOTE

   If IIS is not installed in your environment, **IIS SSL Port** fields will be hidden.



Provide the following details:

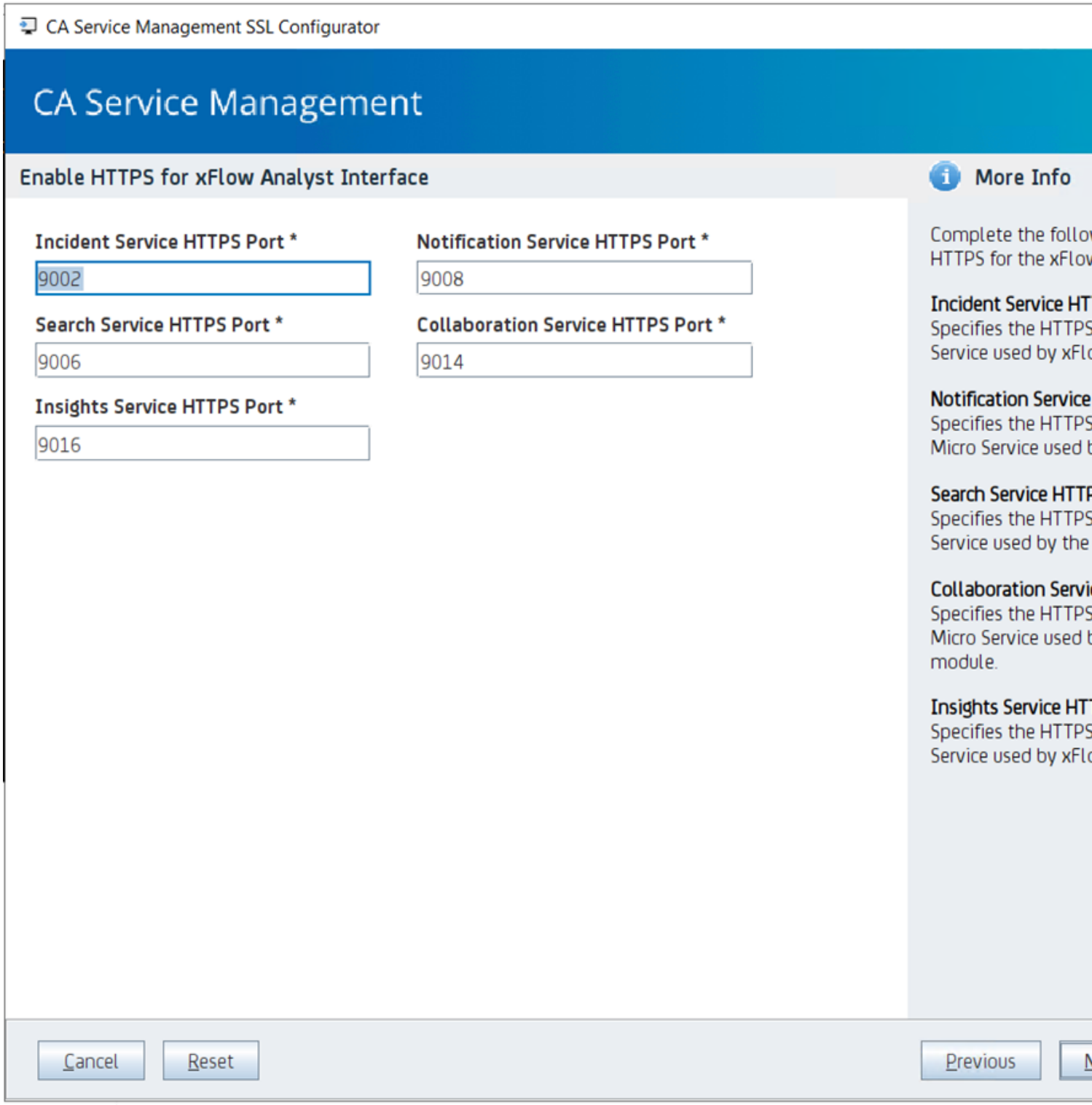
Fields	Description
SDM Tomcat HTTPS Port	Specify the Tomcat HTTPS port for the Web Client Interface.

Federated Search Tomcat HTTPS Port	Specify the Tomcat HTTPS port for the Federated Search Service.
REST Tomcat HTTPS Port	Specify the Tomcat HTTPS Port for REST Web Service.
Visualizer Tomcat HTTPS Port	Specify the Tomcat HTTPS port for CMDB Visualizer port, if you have enabled this CA SDM Component.
Available Certificates	Specify the certificate entry to be used for HTTPS.
Website	Specifies the IIS Web Server with CA SDM
IIS HTTPS Port	Specifies the HTTPS port number for IIS Web Server.
Web CGI URL	Specifies the HTTPS URL value for the global SDM Option web_cgi_url mainly used for user notifications. Can be left unchanged. When updating this value, make sure the HTTPS protocol and HTTPS port number are part of the URL.
Upload Servlet URL	Specifies the HTTPS URL value for the Attachment Servlet Path for this local server. Can be left unchanged. When updating this value, make sure the HTTPS protocol and HTTPS port number are part of the URL.

2. Click **Next** to configure the xFlow Interface for HTTPS (if you have installed xFlow Interface in your environment). Provide the HTTPS ports for the xFlow Interface as shown in the image:

   NOTE

   If the xFlow Analyst Interface is not installed in your system, a message is displayed "*The xFlow Analyst Interface is not installed on this environment*".



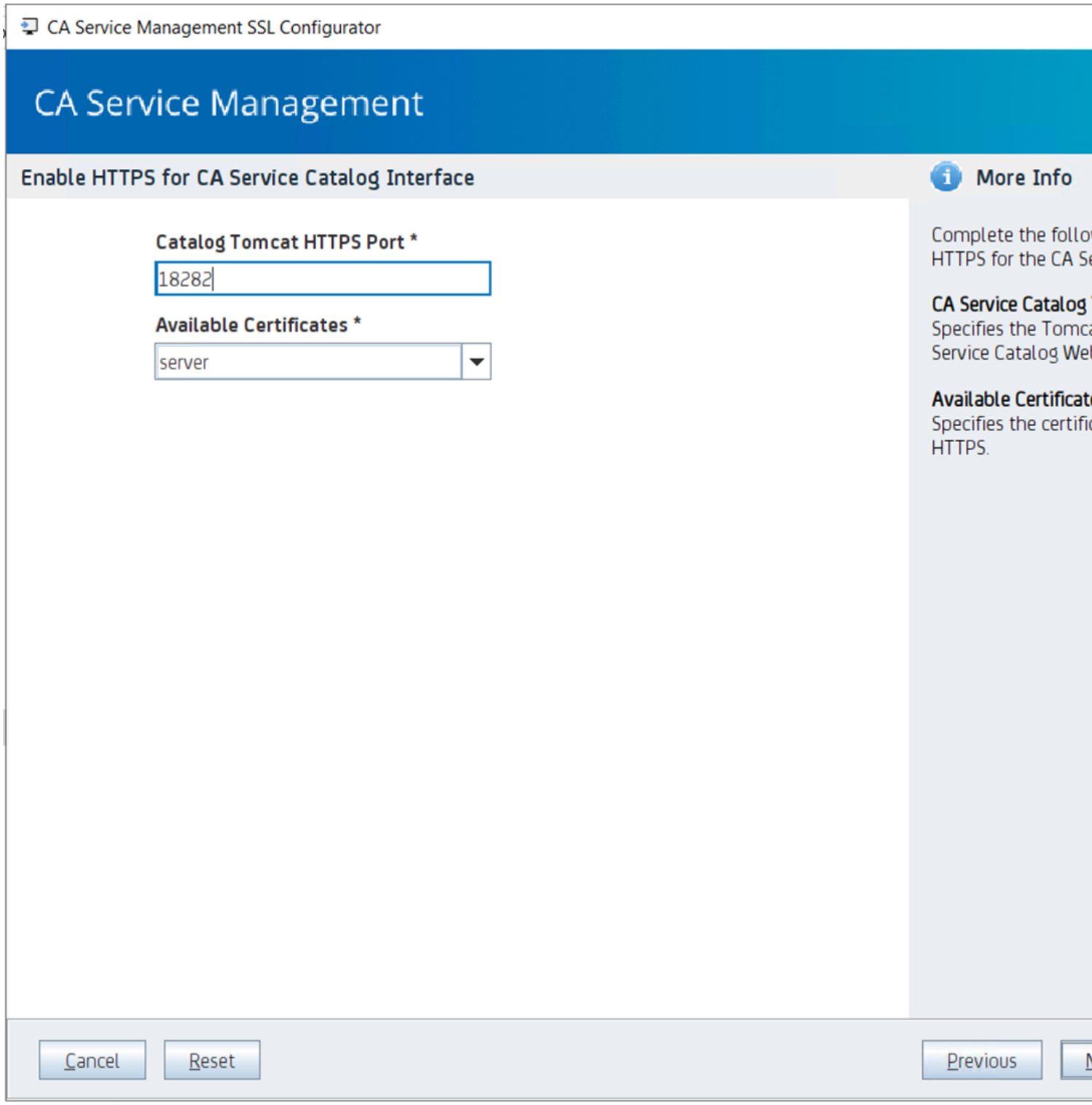
3. Click **Next**.

Enable HTTPS for CA Service Catalog

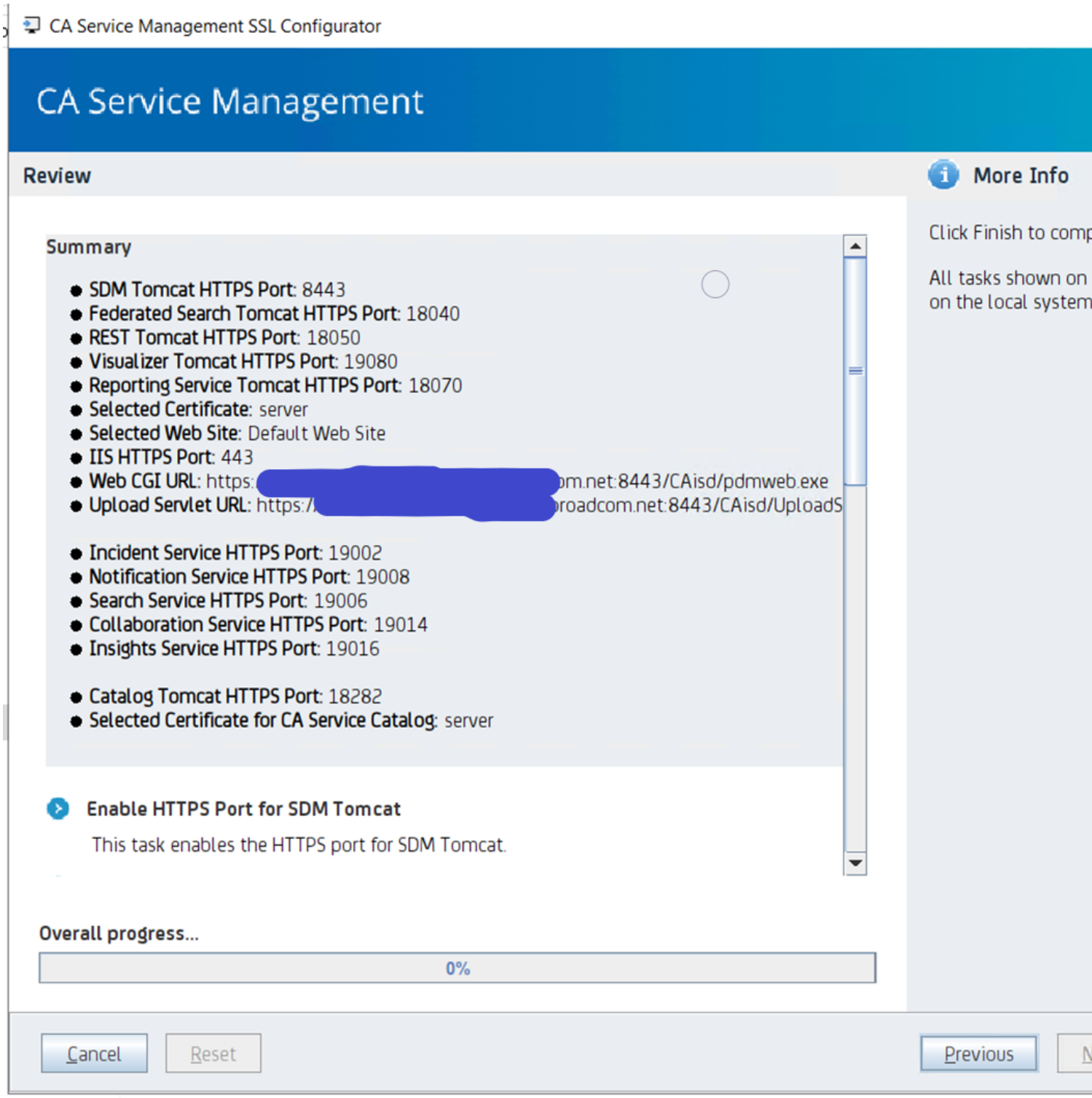
Perform the following steps:

NOTE
The SSL Configuration batch file must be updated with the JRE path used in CA Service Catalog before running the utility.

1. After launching the SSL Configurator Wizard, select the Enable HTTPS for CA Service Catalog option from the Main page of the SSL Wizard.
   NOTE
   If the catalog Analyst Interface is not installed in your system, a message is displayed " CA Service Catalog is not installed in this environment ".
2. Complete the following information:
   Catalog Tomcat HTTPS Port: Specify the Tomcat HTTPS port used by Service Catalog Web Client Interface
   Available Certificates: Specify the certificate entry to be used for HTTPS.
3. Click Next.
   CA Service Catalog tasks get added to the summary.
4. Review the Summary page.
   For any file updated by this task, a backup file is created on the same location as the source.
5. Click **Finish** to complete the process.
   Click finish will update Catalog Server and viewService.conf files (after taking the backup)in the location.



6. Summary screen:



7. To activate the SSL configuration, restart the ITSM applications based on the selected products (SDM, Catalog, xFlow). Each product must be restarted individually—if all three are in use, restart all three.

Updated Information

After completing the SSL configurator tool, the following information is updated:

1. **For CA SDM**: NX_ROOT\NX.env
   a. @NX_WEB_CGI_URL=https://<hostname>:<SDM_Tomcat_SSLPort>/CAisd/pdmweb.exe
   b. @NX_SERVLET_SERVER_URL=https://<hostname>:< SDM_Tomcat_SSLPort >
   c. @NX_LOCAL_SERVLET_SERVER_URL=https://<hostname>:< SDM_Tomcat_SSLPort >
   d. @NX_WSP_CGI_URL=https://<hostname>:< SDM_Tomcat_SSLPort >/CAisd/pdmweb.exe
   e. @NX_CMDB_VISUALIZER=https:// <hostname>:<VIZ_SSLPort> /CMDBVisualizer
   f. @NX_FS_SERVLET_URL= https://<hostname>:<FS_SSLPort>/cafedsearch/sdm/search
2. **For Visualizer**: Following are the variables in the casm.conf file of Visualizer application on the SDM server:
   a. vizserver: 'https://<hostname>:<VIZ_SSLPort> /CMDBVisualizer/viz', :::***
   b. server: 'https://<hostname>:<SDM_Tomcat_SSLPort>/CAisd/pdmweb.exe' :::***
3. For the xFlow component all the urls of the micro services also get updated.

Remove a Certificate

Launch the SSL Configurator Wizard and from the Available Tasks Page, select this option to remove a Keystore entry:

Tasks for Test Environments

From the Available Tasks Page, select this option for test environments/ Non-Production environments to generate a self-signed certificate.

1. Generate a self-signed certificate, using default values.
   Provide the **Alias and FQDN** details. All other properties are defaulted.
   Follow the steps shown in Generate a Certificate Signing Request.

Turning Debug On

To turn on debug level logging for the SSL Configurator wizard utility, open the file <root install>\cfg\log4j2.properties and change the following line:

```
rootLogger.level = info
```

to

```
rootLogger.level = DEBUG
```

Customize Tomcat Connectors

The SSL Configurator wizard utility uses a default set of properties to generate the Tomcat connector definitions. These default properties are available in <*root install*>*cfg**config.properties*. This file can be updated to add/remove or modify default properties. The file consists of variable and value pairs on each line.

Variable names that start with **base.tomcat.Connector** is applicable for all Tomcat Servers. In addition, to the base variables, single Tomcat variables can be used to override the base variables. Single Tomcat variables start with a pre-defined name:

- sdm.tomcat.Connector for SDM Tomcat
- fs.tomcat.Connector for Federated Search Tomcat
- rest.tomcat.Connector for REST Tomcat
- viz.tomcat.Connector for Visualizer Tomcat

For example, if you want all Tomcat Servers to have the property **maxThreads=250** but want the SDM Tomcat to have a value of **300**. The settings for that property would be as follows:

```
base.tomcat.Connector.maxThreads=250
sdm.tomcat.Connector.maxThreads=300
```

Backing Out Completed Task for SSL-Enabled Web Servers

If you have performed the steps shown in Enable HTTPS for CASM Web Servers , if required, you can perform the backing out steps as shown.

To backout a successfully completed task after enabling HTTPS for CASM web servers, perform the following steps:

1. Stop the Service Desk Manager Server services.
2. Navigate to Service Desk Manager install folder
3. For each Tomcat server that is updated in your environment, navigate to the corresponding Tomcat conf folder.
   For example: bopcfg\www\CATALINA_BASE\conf
4. Remove or rename the server.xml file.
5. Rename the backup file to server.xml.
   The backup file has a name pattern of *server.xml.1551384371419.bak*
6. Start Service Desk Manager Server services.
7. The Web CGI URL and the Upload Servlet URL can be updated via the Administration tab in SDM Web Interface as usual

If the xFlow Analyst Interface is installed on the system, follow the steps below.

1. Stop the xFlow Analyst Interface services.
2. Navigate to xFlow Analyst Interface Services folder.
   For example: \Program Files\CA\xFlow\APPS\Services
3. For each of the subfolders, remove or rename the text file as following:

(Current) File Name	(To) Rename or Remove File
collabmicroservice	17.0.479\COLLABMICROSERVICE_config.txt
incidentmicroservice	17.0.479\INCIDENTMICROSERVICE_config.txt
insightmicroservice	17.1.706\INSIGHTMICROSERVICE_config.txt
pushmicroservice	17.0.479\PUSHMICROSERVICE_config.txt
searchmicroservice	17.0.479\SEARCHMICROSERVICE_config.txt

4. Navigate to the web server conf folder.
   For example: *\Program Files\CA\xFlow\APPS\Services\incidentmicroservice-17.0.479\public\conf*
5. Remove or rename the **casm.conf.js** file.
6. Rename the backup file to **casm.conf.js**.
   The backup file has a name pattern of *casm.conf.js.1551384372153.bak*
7. Start the CA xFlow Analyst Interface services.

Uninstall the SSL Configurator Utility

To uninstall the SSL Configurator wizard utility, just delete the *<root install>* folder.

Enable HSTS (HTTP Strict Transport Security) for CA Service Management Solution

HTTP Strict Transport Security (HSTS) is a web security policy mechanism that forces web browsers to interact with a website only via **secure HTTPS connections**, preventing protocol downgrade attacks and cookie hijacking.

- Enable HSTS configuration for CA Service Desk Manager Tomcat Web Server for Windows and Linux
- Enable HSTS configuration for CA Service Catalog Tomcat Web Servers
- Enable HSTS configuration for CA Asset Portfolio Management
- Enable HSTS configuration for XFlow and ServicePoint

NOTE
HSTS is enabled by default for EEM and ITPAM.

Enable HSTS configuration for CA Service Desk Manager Tomcat Web Server for Windows and Linux

Perform the following steps on all CA Service Desk Manger Tomcat Web Servers:

1. **NOTE**
   1. For Windows: <NX_ROOT> = C:\Program Files (x86)\CA\Service Desk Manager for SDM Installed location.
   2. For Linux: <NX_ROOT> = /opt/CA/ServiceDeskManager for SDM Installed location.

   Ensure that SSL is enabled for CA Service Desk Manger. For more information see, Enable SSL Authentication for CA SM 17.4.
2. Ensure that the following information should be added in the **web.xml** file of **<NX_ROOT>\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF**
3. Add configuration under :
   Filter Description

```
<filter>
<filter-name>httpHeaderSecurity</filter-name>
<filter-class>org.apache.catalina.filters.HttpHeaderSecurityFilter</filter-class>
<async-supported>true</async-supported>
<init-param>
<param-name>hstsMaxAgeSeconds</param-name>
<param-value>31536000</param-value>
</init-param>
<init-param>
<param-name>antiClickJackingOption</param-name>
<param-value>SAMEORIGIN</param-value>
</init-param>
<init-param>
<param-name>blockContentTypeSniffingEnabled</param-name>
<param-value>false</param-value>
</init-param>
</filter>
```

4. Add following configuration under
   Filter Mapping

```
<filter-mapping>
<filter-name>httpHeaderSecurity</filter-name>
<url-pattern>/*</url-pattern>
<dispatcher>REQUEST</dispatcher>
</filter-mapping>
```

5. Repeat Steps 3 and 4 to update *web.xml.tpl* file in the *NX_ROOT\samples\pdmconf* folder to avoid losing these changes when pdm_configure is run at a later point of time.
6. Restart the Tomcat or CA Service Desk Manager Services.

Enable HSTS(HTTP Strict Transport Security) configuration for Visualizer

Follow these steps:

1. Open web.xml file of the Visualizer under WEB-INF (Ex: web.xml file under **<NX_ROOT>/bopcfg\www \CATALINA_BASE_VIZ\webapps\CMDBVisualizer\WEB-INF)**
2. Make sure uncomment the below section and add the init-param entry. The structure looks as follows:

```
<filter>
    <filter-name>httpHeaderSecurity</filter-name>
    <filter-class>org.apache.catalina.filters.HttpHeaderSecurityFilter</filter-class>
        <async-supported>true</async-supported>

         <init-param>
           <param-name>hstsMaxAgeSeconds</param-name>
           <param-value>31536000</param-value>
       </init-param>

   </filter>-->
```

   In the same file, uncomment the section mentioned below:

```
<filter-mapping>
       <filter-name>httpHeaderSecurity</filter-name>
       <url-pattern>/*</url-pattern>
       <dispatcher>REQUEST</dispatcher>
   </filter-mapping>
```

3. Restart the SDM services, and now you can see an HSTS header in the response when accessing the Visualizer.

Enable HSTS(HTTP Strict Transport Security) configuration for SDM with IIS

Follow these steps:

1. Navigate to **Actions** >> **Manage Website** >> **Configure** >> **HSTS**.
2. Now select the "Enable" check box.
3. Enter a value like 31536000 in the "Max-Age:" text box.
4. Select the "Redirect http to https" check box.
5. Click **OK** in the pop-up window.
   URL Rewrite Module

The SDM website is configured as virtual directories/ applications under a single website in IIS. To handle the redirection of non-SSL to SSL at the virtual directories / web application level, we need to install and configure URL Rewrite module (IIS component available separately for download/install).

Configure Inbound rules that apply to a requested URL address

In the Edit Inbound Rules page, in **Redirect URL** enter the following url: https://{SERVER_NAME} : IIS SSL PORT

Configure Headers

Open the Internet Information Services (IIS) Manager and navigate to **Start** >> **Administrative Tools** >> I**IS Manager**.

6. Click the **HTTP Response Headers.**
7. In the **Actions** panel, click **Add** .
8. Enter the following values in the Add Custom HTTP Response Headers dialog box:
   Name: **Strict-Transport-Security**
   Value: **max-age=31536000**
9. Once confirmed, close the IIS Manager.
10. Restart the IIS Services.
11. Now log in with the CA SDM IIS url.
12. In the page displayed, click F12 and in the opened developer tool, in **Networks**. In the page displayed, select any GET or POST call. In the Response Header you can see the value *max-age=31536000.*

Enable HSTS(HTTP Strict Transport Security) configuration for CA Service Desk Manager Tomcat Web Server, AXIS, CXF, and AMS

Perform the following steps on all the CA Service Desk Manger Tomcat Web Servers:

NOTE

1. For Windows: <NX_ROOT> = C:\Program Files (x86)\CA\Service Desk Manager for SDM Installed location.
2. For Linux: <NX_ROOT> = /opt/CA/ServiceDeskManager for SDM Installed location.

1. Ensure that SSL is enabled for CA Service Desk Manger. For more information see, Enable SSL Authentication for CA SM 17.4
2. Ensure that the following information should be added in the *web.xml* file in **<NX_ROOT>\bopcfg\www \CATALINA_BASE\conf.**

   cafedsearch

   Ensure that the following information should be added in the web.xml file of <NX_ROOT>\bopcfg\www \CATALINA_BASE_FS\webapps\cafedsearch\WEB-INF

   RESTFULL Web Services

   Ensure that the following information should be added in the web.xml file of <NX_ROOT>\bopcfg\www \CATALINA_BASE_REST\webapps\caisd-rest\WEB-INF

   AMS Application from SDM

   Ensure that the following information should be added in the web.xml file of <NX_ROOT>\bopcfg\www \CATALINA_BASE\webapps\AMS\WEB-INF

   AXIS Web Services

   Ensure that the following information should be added in the web.xml file of <NX_ROOT>\bopcfg\www \CATALINA_BASE\webapps\axis\WEB-INF

   CXF Web Services

   Ensure that the following information should be added in the web.xml file of <NX_ROOT>\bopcfg\www \CATALINA_BASE\webapps\cxf\WEB-INF.

   Reporting API's

   – Ensure to enable the SSL for Reporting tomcat
   – Navigate to web.xml file in the Reporting folder.

For e.g.:ex: C:\Program Files (x86)\CA\Service Desk Manager\bopcfg\www\CATALINA_BASE_RPT\webapps \CASMReports\WEB-INF

– Search for the following lines of code and uncomment them.

```
<filter>
        <filter-name>httpHeaderSecurity</filter-name>
        <filter-class>org.apache.catalina.filters.HttpHeaderSecurityFilter</filter-class>
        <async-supported>true</async-supported>
         <init-param>
         <param-name>hstsEnabled</param-name>
         <param-value>true</param-value>
         </init-param>
         <init-param>
         <param-name>hstsMaxAgeSeconds</param-name>
         <param-value>31536000</param-value>
         </init-param>
         <init-param>
         <param-name>hstsIncludeSubDomains</param-name>
         <param-value>true</param-value>
         </init-param>
    </filter>
         <filter-mapping>
       <filter-name>httpHeaderSecurity</filter-name>
       <url-pattern>/*</url-pattern>
       <dispatcher>REQUEST</dispatcher>
   </filter-mapping>
```

– Save the file and restart SDM Reporting tomcat Service, with these commands.

```
pdm_tomcat_nxd -c STOP -t RPT

pdm_tomcat_nxd -c START -t RPT
```

Telemetry

Ensure that the following information should be added in the web.xml file of <NX_ROOT>bopcfg\www \CATALINA_BASE\webapps\rest-telemetry\WEB-INF

```
<filter>
<filter-name>httpHeaderSecurity</filter-name>
<filter-class>org.apache.catalina.filters.HttpHeaderSecurityFilter</filter-class>
<async-supported>true</async-supported>
<init-param>
<param-name>hstsMaxAgeSeconds</param-name>
<param-value>31536000</param-value>
</init-param>
<init-param>
<param-name>antiClickJackingOption</param-name>
<param-value>SAMEORIGIN</param-value>
</init-param>
<init-param>
<param-name>blockContentTypeSniffingEnabled</param-name>
<param-value>false</param-value>
</init-param>
</filter>
```

```
<filter-mapping>
<filter-name>httpHeaderSecurity</filter-name>
<url-pattern>/*</url-pattern>
<dispatcher>REQUEST</dispatcher>
</filter-mapping>
```

Enable HSTS configuration for CA Service Desk Manager with Apache 2.4 Web Server on Linux environments

Perform these steps on all the CA Service Desk Manger Tomcat Web Servers:

1. Ensure that SSL is enabled for CA Service Desk Manger. For more information see, Enable SSL Authentication for CASM 17.4.
2. Update the following mentioned entry in the *httpd-ssl.conf* file in the location where apache is installed, for e.g.: (/opt/apache/httpd/conf/extra/):
   ```
   Header always set Strict-Transport-Security "max-age=31536000; includeSubDomains"
   ```
3. Restart the apache service, follow these commands:
   ```
   ./httpd -k stop
   ./httpd -k start
   ```
4. Now access the SDM url with the apache port:
   https://hostname.domainname:apacheport/CAisd/pdmweb.exe
5. You can now see the hsts response header.

Enable HSTS Configuration for CA Service Catalog Tomcat Web Servers

This page talks about HSTS configuration for CA Catalog.

Perform the following steps on all CA Service Catalog Tomcat Web Servers:

1. Ensure that SSL is enabled for CA Service Catalog. For more information see, Enable SSL Authentication for CA SM 17.4.
2. Ensure that the following information should be added in the web.xml file located where you have installed CA Service Catalog/Tomcat Web Server:
   For e.g: **C:\Program Files\CA\Service Catalog\view\webapps\usm\WEB-INF\web.xml**
3. Add configuration under thestatement:
   Filter:
   ```
   <filter>
   <filter-name>httpHeaderSecurity</filter-name>
   <filter-class>org.apache.catalina.filters.HttpHeaderSecurityFilter</filter-class>
   <init-param>
   <param-name>hstsMaxAgeSeconds</param-name>
   <param-value>31536000</param-value>
   </init-param>
   <init-param>
   <param-name>antiClickJackingEnabled</param-name>
   <param-value>true</param-value>
   </init-param>
   <init-param>
   <param-name>antiClickJackingOption</param-name>
   <param-value>SAMEORIGIN</param-value>
   </init-param>
   <init-param>
   <param-name>blockContentTypeSniffingEnabled</param-name>
   ```

```
<param-value>false</param-value>
</init-param>


<async-supported>true</async-supported>
</filter>
```

4. a. Add the following configuration under the **Add filter-mapping here >> Statement**.
   b. For 17.4.3.0 and before, after HSTS filter is uncommented, make sure to uncomment the HSTS filter mapping. (approx line number ~372)

```
<!--
<filter-mapping>
<filter-name>httpHeaderSecurity</filter-name><url-pattern>/wpf/*</url-pattern> </filter-mapping>
<filter-mapping>
<filter-name>httpHeaderSecurity</filter-name><url-pattern>*.rpc</url-pattern> </filter-mapping>
<filter-mapping>
<filter-name>httpHeaderSecurity</filter-name><url-pattern>/documents/*</url-pattern> </filter-mapping>
<filter-mapping>
 <filter-name>httpHeaderSecurity</filter-name> <url-pattern>/uslm/*</url-pattern> </filter-mapping>
<filter-mapping>
<filter-name>httpHeaderSecurity</filter-name> <url-pattern>/assure/*</url-pattern> </filter-mapping>
→
```

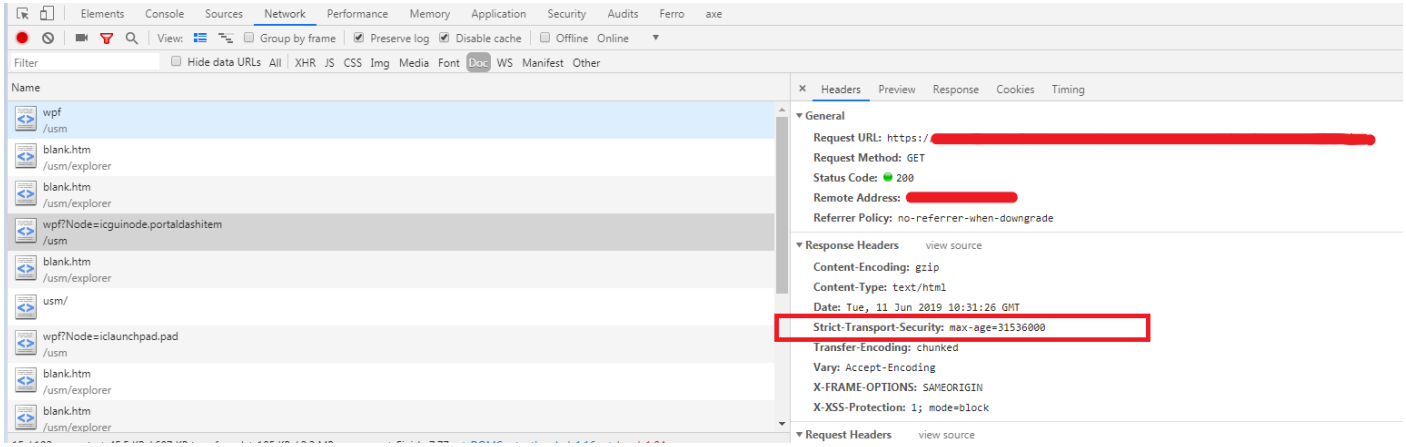
   c. For 17.4.4 and above, after HSTS filter is uncommented, make sure to uncomment the HSTS filter mapping. (approx line number ~372)

```
<!--

<filter-mapping>
<filter-name>httpHeaderSecurity</filter-name><url-pattern>*.*</url-pattern> </filter-mapping>

→
```

5. Close the *web.xml* file.
6. Restart CA Service Catalog Services.
7. Verify the HSTS header:

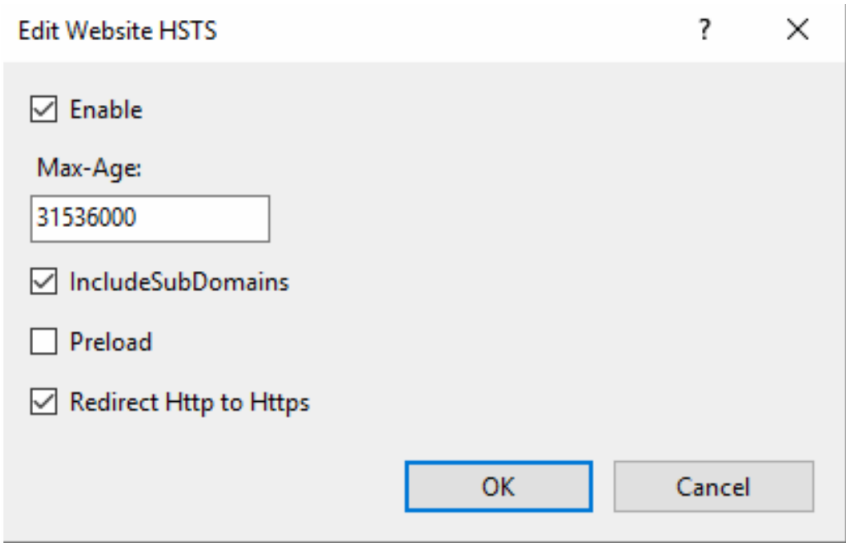


Enable HSTS configuration for CA Asset Portfolio Management

This page talks about HSTS configuration for CA APM.

Enable HSTS on CA ITAM, AMS, and CAF (Multi-Tenancy) to secure the applications.

NOTE
Ensure ITAM is configured with SSL. Follow the steps in (Optional) Configure Secure Network Communication.



Steps to enable HSTS on IIS

Follow these steps:

1. Enable HSTS and set Max-Age on IIS for ITAM website. This enables HSTS redirection at the site level.

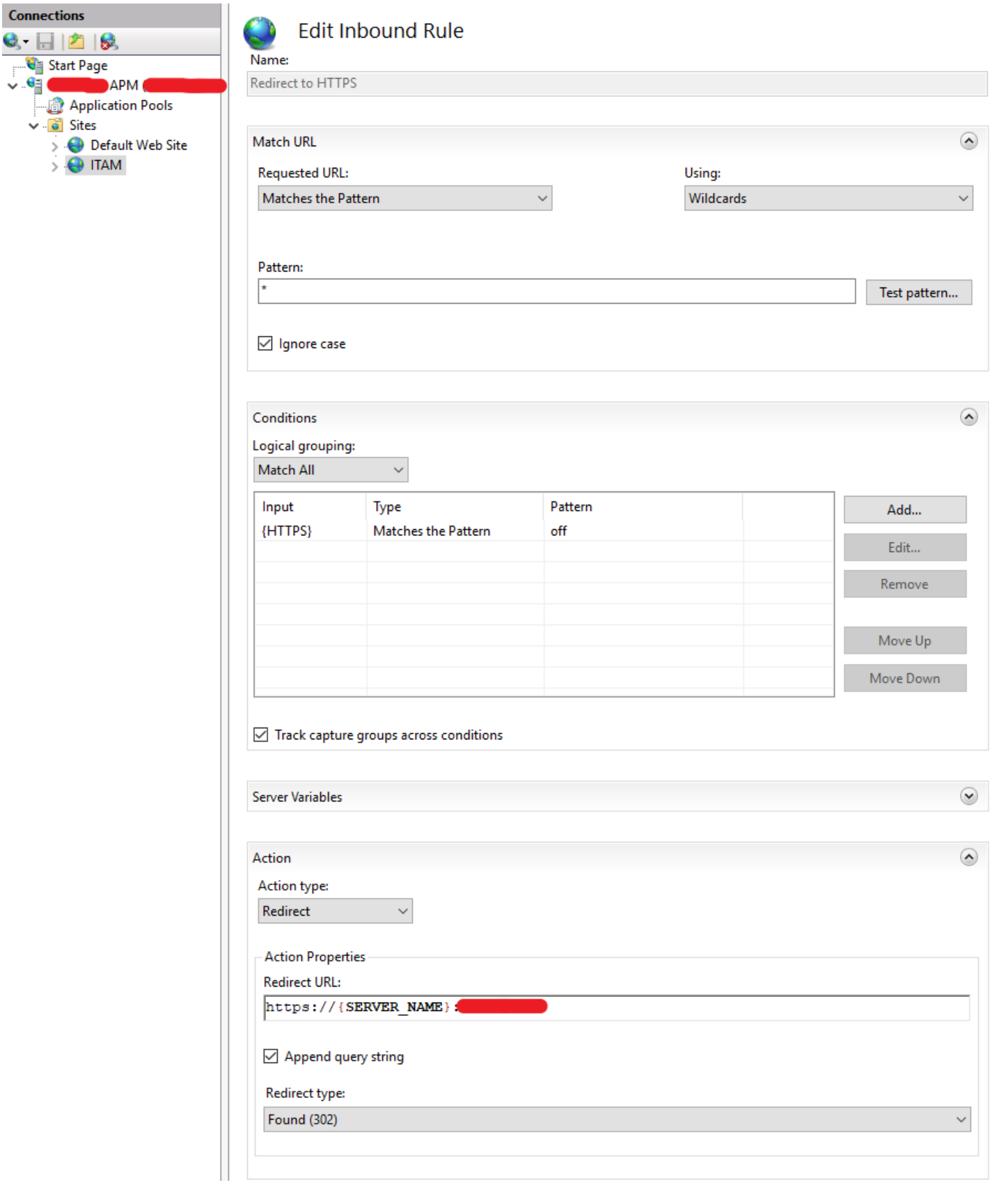
URL Rewrite Module

The ITAM website and web services are configured as virtual directories or applications under a single IIS website. To manage the redirection of non-SSL traffic to SSL at the virtual directory or application level, it is necessary to install and configure the URL Rewrite module, an IIS component available as a separate download. Download IIS from here.

NOTE
This section is not applicable if the WCF Services are SSL enabled, because WCF service works only with SSL, and there won't be any redirection.

Configure Inbound Rules

Configure Inbound Rules that apply to a requested URL address.


Connections
Start Page
[redacted]APM[redacted]
Application Pools
Sites
Default Web Site
ITAM
Edit Inbound Rule
Name:
Redirect to HTTPS
Match URL
Requested URL:
Matches the Pattern
Using:
Wildcards
Pattern:
*
Test pattern...
Ignore case
Conditions
Logical grouping:
Match All
Input
Type
Pattern
{HTTPS}
Matches the Pattern
off
Add...
Edit...
Remove
Move Up
Move Down
Track capture groups across conditions
Server Variables
Action
Action type:
Redirect
Action Properties
Redirect URL:
https://{SERVER_NAME}:[redacted]
Append query string
Redirect type:
Found (302)

Configuration File Changes

For Import Service\Web.config

```
    --- <system.serviceModel><bindings>
      <basicHttpBinding>
        <binding name="httpBinding_ImportService" maxBufferSize="2147483647"
 maxReceivedMessageSize="2147483647" messageEncoding="Text" transferMode="Streamed">
          <security mode="None" />
          <readerQuotas maxArrayLength="2147483647"/>
        </binding>
        <binding name="SecureHttpBinding">
            <security mode="Transport" />
        </binding>
      </basicHttpBinding>
</bindings>
<services>
      <service behaviorConfiguration="CA.Applications.ImportService.ImportServiceBehavior"
 name="CA.Applications.ImportService.ImportService">
        <clear/>
            <endpoint address="" binding="basicHttpBinding" bindingConfiguration="httpBinding_ImportService"
 name="BasicHttpBinding_ImportService" contract="CA.Applications.ImportService.IImportService"
 listenUriMode="Explicit"/>

            <endpoint address="" binding="basicHttpBinding" bindingConfiguration="SecureHttpBinding"
 name="BasicHttpBinding_ImportService" contract="CA.Applications.ImportService.IImportService"
 listenUriMode="Explicit"/>

            <endpoint address="mex" binding="mexHttpBinding" name="Mex" contract="IMetadataExchange"
 listenUriMode="Explicit"/>
      </service>
 </services>------</system.serviceModel>
```

For Import Processor\ImportProcessor.exe.config

```
   <?xml version="1.0"?>
<configuration>
  <startup>
    <supportedRuntime version="v4.0" sku=".NETFramework,Version=v4.5"/>
  </startup>
  <system.serviceModel>
    <bindings>
      <basicHttpsBinding>
        <binding name="BasicHttpBinding_ImportService" maxBufferSize="2147483647"
 maxReceivedMessageSize="2147483647" transferMode="Streamed">
            <readerQuotas maxArrayLength="2147483647"/>
            <security mode="Transport" />
        </binding>
      </basicHttpsBinding>
    </bindings>
    <client>
```

```
      <endpoint binding="basicHttpsBinding" bindingConfiguration="BasicHttpBinding_ImportService"
 contract="IImportService" name="BasicHttpBinding_ImportService" address="
     https://hostname:446/ImportService/ImportService.svc"/></client>
  </system.serviceModel>
</configuration>
```

Steps to enable HSTS on AMS (Tomcat web server)

1. Enable SSL for AMS with a CA server certificate. Navigate to C:\Program Files (x86)\CA\SharedComponents\AMS \TomCat\webapps\AMS\WEB-INF and edit web.xml.
2. Add the mentioned Filter and Filter Mapping: **Filter**

```
    <filter>
           <filter-name>httpHeaderSecurity</filter-name>
           <filter-class>org.apache.catalina.filters.HttpHeaderSecurityFilter</filter-class>
           <async-supported>true</async-supported>
           <init-param>
               <param-name>hstsMaxAgeSeconds</param-name>
               <param-value>31536000</param-value>
           </init-param>
           <init-param>
               <param-name>antiClickJackingOption</param-name>
               <param-value>SAMEORIGIN</param-value>
           </init-param>
           <init-param>
               <param-name>blockContentTypeSniffingEnabled</param-name>
               <param-value>false</param-value>
           </init-param>
       </filter>
```

Filter Mapping

```
  <filter-mapping>
           <filter-name>httpHeaderSecurity</filter-name>
           <url-pattern>/*</url-pattern>
           <dispatcher>REQUEST</dispatcher>
       </filter-mapping>
```

3. Now restart the AMS service from services console.

Steps to enable HSTS on CAF (Tomcat web server)

1. Enable SSL for CAF with a CA server certificate. Navigate to C:\Program Files (x86)\CA\SharedComponents\CASM \apache-tomcat\webapps\casm\WEB-INF and edit web.xml.
2. Add the following Filters and Filter Mappings :
   Filter

```
    <filter>
           <filter-name>httpHeaderSecurity</filter-name>
           <filter-class>org.apache.catalina.filters.HttpHeaderSecurityFilter</filter-class>
           <async-supported>true</async-supported>
           <init-param>
```

```
            <param-name>hstsMaxAgeSeconds</param-name>
            <param-value>31536000</param-value>
        </init-param>
        <init-param>
            <param-name>antiClickJackingEnabled</param-name>
            <param-value>true</param-value>
        </init-param>
        <init-param>
            <param-name>antiClickJackingOption</param-name>
            <param-value>ALLOW-FROM</param-value>
        </init-param>
        <init-param>
            <param-name>antiClickJackingUri</param-name>
            <param-value>https://hostname:446</param-value>
        </init-param>
        <init-param>
            <param-name>blockContentTypeSniffingEnabled</param-name>
            <param-value>false</param-value>
        </init-param>
    </filter>

    <filter>
        <filter-name>CorsFilter</filter-name>
        <filter-class>org.apache.catalina.filters.CorsFilter</filter-class>
        <init-param>
            <param-name>cors.allowed.origins</param-name>
            <param-value>https://hostname:111</param-value>
        </init-param>
        <init-param>
            <param-name>cors.allowed.methods</param-name>
            <param-value>GET, POST, PUT, DELETE, OPTIONS</param-value>
        </init-param>
        <init-param>
            <param-name>cors.allowed.headers</param-name>
            <param-value>Content-Type,Authorization</param-value>
        </init-param>
    </filter>
```

Filter Mappings

```
 <filter-mapping>
        <filter-name>httpHeaderSecurity</filter-name>
        <url-pattern>/*</url-pattern>
        <dispatcher>REQUEST</dispatcher>
    </filter-mapping>

    <filter-mapping>
        <filter-name>CorsFilter</filter-name>
        <url-pattern>/*</url-pattern>
    </filter-mapping>
```

3. Restart the CA CASM service from services console.

Enable HSTS configuration for XFlow/ ServicePoint

For Windows

To enable the HSTS configuration for XFlow/ ServicePoint, update the required HSTS filter in application.conf for *incidentmicroservice, insightmicroservice,* and *searchmicroservice*.

Add the following entry in application.conf.

```
play.filters.https.strictTransportSecurity="max-age=31536000; includeSubDomains"
```

Sample paths for the microservices for WIndows:

- incidentmicroservice: *C:\Program Files\CA\xFlow\APPS\Services\incidentmicroservice-17.0.479\conf*
- insightmicroservice: *C:\Program Files\CA\xFlow\APPS\Services\insightmicroservice-17.1.706\conf*
- searchmicroservice: *C:\Program Files\CA\xFlow\APPS\Services\searchmicroservice-17.0.479\conf*

Save and restart XFlow analyst interface services.

For Linux

To enable the HSTS configuration for XFlow/ ServicePoint, update the required HSTS filter in application.conf for *incidentmicroservice, insightmicroservice,* and *searchmicroservice*.

Add the following entry in application.conf.

```
play.filters.https.strictTransportSecurity="max-age=31536000; includeSubDomains"
```

Sample paths for the microservices for Linux:

- incidentmicroservice: */opt/CA/xFlow/APPS/Services/incidentmicroservice-17.0.479/conf*
- insightmicroservice: */opt/CA/xFlow/APPS/Services/insightmicroservice-17.1.706/conf*
- searchmicroservice: */opt/CA/xFlow/APPS/Services/searchmicroservice-17.0.479/conf*

Save and restart XFlow analyst interface services.

Axis to CXF Migration for CA Service Management

For CASM this guide serves as a resource to assist users in initiating and completing the AXIS to CXF migration process.

> **NOTE**
> From 17.4.5.0 , the Axis endpoint has been fully decommissioned from CA Service Management, and CXF migration is fully automated.

- CXF Migration for CA SDM
- CXF Migration for CA Service Catalog

> **NOTE**
> CA APM and CA ITPAM are both consumers of CA SDM and CA Service Catalog. The process of configuring CXF Endpoint references for CA APM and ITPAM are available in the links mentioned above.

CXF Migration for CA SDM

This page contains:

- Overview
- Generate Stub Classes with CXF WSDL2Java
- Java Clients
- SOAP CXF Web Services Configuration
- SOAP Error Handling
- Axis Removal
- Update CA SDM CXF process into ITPAM
- CA SDM CXF webservices for CA APM
- CA Service Catalog CXF webservices for CA APM

NOTE
To use web services for automating business processes, follow these steps mentioned here.

NOTE
The following Web Services frame works are supported:

- **For Axis:**
  `http://<servername>:<portno>/axis/services`

- **For CXF:**
  `http://<servername>:<portno>/cxf/services`

Overview

The samples directory of the CA SDM installation provides a sample Java client application for CXF Web Services. This sample assists developers with webservices client application development.

NOTE
Ensure you update the AXIS webservice URL to CXF webservice for all the endpoints.

NOTE
From **17.4 RU5**, the Axis endpoint has been fully decommissioned from the product. CXF migration is automated from RU5 and all the steps are taken care by common installer.

Many of the Web Services methods require arrays as input parameters. For example, the method createIssue() permits an empty array for propertyValues. Sometimes these arrays are optional, but the service requires an empty array for a successful pass. When you use Visual Studio .NET to access Web Services, specify an empty array with one of the following arrays:

- **C# language**
  ```
  String[] emptyArray = new string[0];
  ```
- **Visual Basic .NET**
  ```
  Dim emptyArray As String() = {}
  ```
- **Java**
  ```
  ArrayOfString attr = new ArrayOfString();
  ArrayOfString is a proprietary class.
  ```

You can then pass emptyArray to array parameters that accept empty arrays.

NOTE

The CA SDM CXF Web Services use the Apache implementation of standards that are established by the World Wide Web Consortium (W3C). Ideally, a client on any type of operating environment can access the services,

but vendor implementations vary. Many programming environments provide a tool to generate proxy classes from a Web Services Description Language (WSDL) description.

Generate Stub Classes with CXF WSDL2Java

You can generate the stub classes for the CA SDM CXF web services.

To generate stub classes with CXF WSDL2Java

1. Open a command prompt and navigate to the "<drive>:\Program Files (x86)\CA" directory.
   The directory appears.
2. Navigate to the samples\sdk\cxfwebsvc\java folder.
3. Rename the run_java_bat_txt file to run_java.bat file.
4. Set the USD_SHORT_PATH and CXFLIBPATH path (Download from apache website).

   NOTE
   Download the CXF 3.4.5 from https://cxf.apache.org/download.html or directly from https://archive.apache.org/dist/cxf/3.4.5/apache-cxf-3.4.5.zip

Java Clients

The TableOfContents.doc in $NX_ROOT/samples/sdk/cxfwebsvc lists several Java sample programs.

Each sample program contains notes on how it can be compiled and run using the script files run_java_test_bat.txt (Windows) and run_java_test_sh.txt (UNIX). These scripts demonstrate how to use org.apache.cxf.tools.wsdlto.WSDLToJava to generate the CA SDM web services client-side stub files.

Running WSDL2Java as shown will generate the stub files in the subdirectory com/ca/UnicenterServicePlus/ServiceDesk.

Import these classes with the following statement:

```
import com.ca.UnicenterServicePlus.ServiceDesk.*;
```

Many Web Service methods have parameters of the type ArrayOfString, a proprietary class. For example, the createRequest() method's attrVals, propertyValues, and attributes parameters are all ArrayOfString parameters.

To set the values in an ArrayOfString variable, instantiate the variable and then use getString().addAll() as follows:

```
ArrayOfString attrVals = new ArrayOfString();
attrVals.getString().addAll(new ArrayList<String>(Arrays.asList("customer", customerHandle,

"description", "description text")));
```

To set it to empty, instantiating the variable is sufficient.

Use a variable of type ListResult, another proprietary class, as the return value from the List methods: doQuery(), getRelatedList(), getNotificationsForContact(), getPendingChangeTaskListForContact() and getPendingIssueTaskListForContact(). A ListResult contains listHandle and listLength elements, which can be retrieved using getListHandle() and getListLength() as shown in this example:

```
ListResult doQueryResult = new ListResult();
doQueryResult = USPSD.doQuery(sid, "iss", "active = 1");
int listHandle = doQueryResult.getListHandle();
int listLength = doQueryResult.getListLength();
```

The getListValues() method uses the listHandle, retrieving the values from a subset of the list.

The Handles parameter of the freeListHandles() method is an ArrayOfInt, another proprietary class. Call freeListHandles() using the listHandle taken from a ListResult:

```
ArrayOfInt handleList = new ArrayOfInt();
handleList.getInteger().addAll(new ArrayList<Integer>(Arrays.asList(listHandle)));
USPSD.freeListHandles(sid, handleList);
```

Some methods have pass by reference parameters of type Holder<String>. For example, createRequest() has three parameters of this type, NewRequestHandle, NewRequestNumber, and cr_result.

```
Holder<String> NewRequestNumber = new StringHolder();
Holder<String> NewRequestHandle = new StringHolder();
Holder<String> cr_result = new Holder<String>();
USPSD.createRequest(sid, creatorHandle, attrVals, propertyValues, template, attributes,
NewRequestHandle, NewRequestNumber, cr_result);
```

The Request's handle and reference number (ref_num) can then be obtained from NewRequestHandle.value and NewRequestNumber.value respectively.

SOAP CXF Web Services Configuration

You can configure the CA SDM SOAP CXF Web Services with entries in special web configuration files. The following table summarizes the names and descriptions of the configuration options:

Option Name	Description
design_mode_stubs	Sets the Web Servic*e to design mode* (CA SDM only).
require_secure_logon	Requires the login() and loginService() web methods to be called with a secure protocol, such as https.
require_secure_connection	Requires that every web method be called with a secure protocol.
disable_user_logon	Disables both login() and loginService() web methods, so only loginServiceManaged() can be used to log in.

CA SDM adds protection to the integrity of the running Tomcat server by verifying the length of the attribute values that pass to Web Service methods. By default, web service calls return an WebServiceException if the length of an attribute exceeds 900,000 bytes.

You set the following parameters in the deploy.wsdd file:

- **fatal_max_string_length**
  Sets the length of the largest attribute value that a web service method accepts.
  Default: 900,000 bytes
- **validate_parameters**
  Sets whether the attribute value length checking is performed. Set the parameter to 0 to turn off the validation.
  Default: 1 (on)
- **exception_methods**
  Displays a comma-delimited list of Web Service methods that are exempt from the attribute value length validation.

Redeploy the CXF Web Services

New configuration settings take effect when you redeploy CA SDM CXF Web Services. Complete the following steps to redeploy the Web Services:

1. Recycle Tomcat by recycling the CA SDM service. You can avoid shutting down the entire CA SDM system by recycling Tomcat by simply using the following commands:
   ```
   pdm_tomcat_nxd  - c stop
   pdm_tomcat_nxd  - c start
   ```
   The Web Service is redeployed.

2. Verify that the service deployed by viewing the CXF services listing page at the following default URL:

```
http://<servername>:< port>/cxf/services
```

NOTE

The exact URL depends upon your installation settings.

SOAP Error Handling

If an error occurs with a Web Services method, a SOAP Fault is returned. The SOAP Fault is the standard means of returning exception information for Web Services.

The Fault message contains standardized <Message> and <Code> elements, but the most informative is the <Detail> element. The <Detail> element contains <ErrorCode> and <ErrorMessage> elements. The <ErrorCode> element returns an enumerated error code specific to either the CA SDM or Knowledge Management product. The <ErrorMessage> element contains an English string describing the errors. The <ErrorMessage> elements are more suitable for aiding the developer and more appropriate messages should display to users.

For example, the following illustrates a SOAP Fault when a bad parameter is supplied to the CA SDM getObjectValues() method:

```
<soap:Fault>
<faultcode>soap:Client</faultcode>
<faultstring>Error on fetch with attribute list:persistent_id,first_name,last_nameParamErrorHere<faultstring>
<detail>
<ErrorCode>1001</ErrorCode>
<ErrorMessage>Error on fetch with attribute list: persistent_id,first_name,last_nameParamErrorHere </
ErrorMessage>
</detail>
</soap:Fault>
```

In some cases, the servlet container may write errors and therefore, display in the servlet container logs. In other cases, error information may be written to CA SDM logs. These logs are located in the following subdirectories:

- In the /bopcfg/www/CATALINA_BASE/logs subdirectory of CA SDM installation
- In the /log subdirectory of the CA SDM installation and to all logs that have the prefix “jsrvr”.

NOTE
We recommend that you constantly monitor these logs, as the server may log its own errors without reporting them to the CA SDM Web Services.

Lock Errors

CA SDM objects are locked during updates. Methods that update objects (such as, updateObject() or transfer()) may return the following lock error code:

```
UDS_LOCK_ERR
```

This code indicates that another user is updating the record. Often the locking user’s handle is returned in the ErrorMessage element.

Time Outs

If the CA SDM server is heavily loaded, a method may take a long time to process. In rare cases, a method may never return because a separate process failed to reply or some other error occurred. To guard against excessive blocking, every Web Services method times out after several seconds. Web Services method time-out is a CA SDM server time-out, *not* a Web server time-out, network time-out, and so on.

NOTE

The Web Services delay a few seconds the first time accessed after the J2EE application server is recycled. This happens because the application is initializing, loading DLLs, libraries, and so on, and occurs only with the first Web Services method call. All subsequent calls return faster.

If a method times out, it returns the following error code:

```
UDS_TIMEOUT_ERR
```

The operation is not aborted! The server may have received the request and processes it successfully, although slowly. This type of problem may occur when using the *doSelect( )* method to retrieve several 1000 records.

NOTE
For information about the *doSelect* method, see the CA Service Desk Manager Reference Commands section.

Error Codes

The following table lists the possible values for the <ErrorCode> value in a SOAP Fault returned from a Web Services call:

Error Name	Value	Description
UDS_OK	0	Successful.
UDS_FAILURE	1	General failure, check system logs.
UDS_BAD_PARAM	1000	A bad parameter was passed to a method. This error occurs if a required parameter is missing, the wrong type was passed, or an invalid value was used.
UDS_INTERNAL_ERR	1001	Signals that an internal error occurred. A description is found in the return array and the system logs.
UDS_LOCK_ERR	1002	An attempt was made to update an object locked by another user or process. Usually the ID of the contact responsible for locking the object is returned in the return data.
UDS_UPDATE_ERR	1003	An error occurred updating an object. Make sure all required attributes were are set and check the system log.
UDS_CREATION_ERR	1004	An error occurred creating an object. Make sure all required attributes were are set and check the system logs.
UDS_NOT_FOUND	1005	A search method failed to find any matches or failed to find an object specified. This can happen if a bad or invalid handle is passed to any method.
UDS_TIMEOUT_ERR	1006	The current method timed out, the CA SDM server may be heavily loaded or the method itself was bad.
UDS_SERVER_GONE	1007	The CA SDM server connection is lost, UDS methods will no longer function and all list references are lost.
UDS_FETCH_ERR	1008	An error occurred while retrieving list data.

UDS_BAD_SESSION	1010	An invalid SID was used.
UDS_CNTXT_TIMEOUT	1011	The SID timed out.
UDS_SECURE_CHANNEL_REQUIRED	1012	The Web Services (or a web service method) requires a secure channel (for example: SSL) for access, but an unsecured channel is being used.
UDS_SECURITY_VIOLATION	1013	The attempted operation violates CA SDM security and was aborted.
UDS_OVER_POLICY_LIMIT	3002	The attempted request is refused because it exceeds the limit that is defined in the policy.

(Optional) Axis Removal

To remove axis references from CA SDM, run:

For Windows:

```
removeSDMAxis.bat under %NX_ROOT%/bin
```

The backup of axis specific jars are located under “<NX_ROOT>/replaced/axis.OLD”

For Linux:

```
removeSDMAxis.sh path %NX_ROOT%\bin
```

The backup of axis specific jars are located under “:::/opt/CA/ServiceDeskManager/replaced/axis.OLD”

NOTE

- **Before 17.4.3.0:** The `.bat` / `.sh` file could be executed only when ITAM (CA Asset Management) was **not** included in the solution.
- **From 17.4.3.0:** The `.bat` / `.sh` file can be executed even if ITAM is part of the solution.
- **Upgrade note**: When upgrading from 17.4.2.0 to 17.4.3.0, re-run the removeSDMAxis.bat or removeSDMAxis.sh file if it was already executed in 17.4.2.0.

NOTE
After running the *removeSDMAxis.bat/* and *removeSDMAxis.sh/*, to avoid incompatibility in existing OOTB PAM processes, ensure to update the CXF endpoint for SLCM, SDM Global datasets in PAM.
For e.g.: /cxf/services/USD_WebService

NOTE
From **CASM 17.4.5.0**, the ***removeSDMAxis.bat*** is no longer available, as it is taken care by the common installer.

Update CA SDM CXF process into ITPAM

Reload OOTB ITPAM content (which are CXF compatible) by manually importing the xml file into CA ITPAM.
This xml file is located under the “*filestore/PAM/PAMProcessFlows/SDM*” in Common Installer DVD for 17.4 RU2 and above.

All CA SDM endpoints have to be updated to CXF in PAM global datasets.

NOTE
To update ITPAM with the new CXF connector, import the latest ITPAM SDM Connector to support CXF webservices, download it from here.

NOTE
From **CASM 17.4.5.0**, the above mentioned process is automated.

CA SDM CXF webservices for CA APM

CA APM supports SDM CXF webservices

From 17.4 RU2, CA APM supports both axis and cxf web services of SDM when integrated. By default, the axis Web Service URL is configured and we can manually change it to cxf Web Service URL.
To perform this operation:

- Login to ITAM as Administrator.
- Navigate to **Administration** > **System Configuration** > under **Menu** select the CA SDM link, and change the CA SDM Web Services URL.
  Example: *http(s)://<SDMHost>:<port?/axis/services/USD_R11_WebService?wsdl* should be manually changed to *http(s)://<SDMHost>:<port?/cxf/services/USD_WebService?wsdl.*
- Perform the IISRESET on both ITAM Web and Application servers.

NOTE
From **CASM 17.4.5.0**, the above mentioned process is automated.

CA Service Catalog CXF webservices for CA APM

CA Service Catalog CXF webservices for CA APM

From 17.4 RU3, CA APM supports both axis and cxf web services of CA SLCM when integrated. By default, the axis Web Service URL is configured and we can manually change it to cxf Web Service URL.
To perform this operation:

- Login to ITAM as administrator.
- Navigate to **Administration** > **System Configuration** > under **Menu** select the Application Sever, and change the USM Web Services URL.
  Example: *http(s)://{Server}:8080/usm/services/RequestService?wsdll* should be manually changed to *http(s)://<SLCM hotname>:<port>/usm/webservices/RequestService?wsdl*
- Perform the IISRESET on both ITAM Web and Application servers.

NOTE
From **CASM 17.4.5.0**, the above mentioned process is automated.

CXF Migration for CA Service Catalog

NOTE
To use web services for automating business processes, follow these steps mentioned here.

Introduction

Starting from version 17.4 RU2, the CASM suite of products now fully supports CXF-based web services, enabling a transition away from the deprecated AXIS-based web services. Due to security vulnerabilities associated with AXIS, these services will eventually be phased out in future CASM versions. To mitigate risks, users are strongly encouraged to migrate their content from AXIS WS to CXF WS. This guide serves as a resource to assist users in initiating and completing the migration process.

This guide covers four major topics:

1. Using the new CXF Web Services.
2. Removal of AXIS references in the current product version as required.

3. Steps to migrate content from AXIS web service to CXF web services.
4. Differences in Request/ Response payload.

NOTE
From **17.4 RU5**, the Axis endpoint has been fully decommissioned from the product. CXF migration is automated from RU5 and all the steps are taken care by common installer.

Migration from AXIS WS to CXF WS in CA Service Catalog

In CA Service Catalog, CXF based webservices are available from 17.4 RU2. These services can be accessed through a new endpoint, without impacting the existing integration.

Service	Axis WSDL path	CXF WSDL path
Request Service	http://localhost:8080/usm/services/RequestService?wsdl	http://localhost:8080/usm/webservices/RequestServi
Catalog Service	http://localhost:8080/usm/services/CatalogService?wsdl	http://localhost:8080/usm/webservices/CatalogServic
User Service	http://localhost:8080/usm/services/UserService?wsdl	http://localhost:8080/usm/webservices/UserService?w
Admin Service	http://localhost:8080/usm/services/AdministratorService?wsdl	http://localhost:8080/usm/webservices/AdministratorS
Business Unit Service	http://localhost:8080/usm/services/BusinessUnitService?wsdl	http://localhost:8080/usm/webservices/BusinessunitS
Reporting Service	http://localhost:8080/usm/services/ReportingService?wsdl	http://localhost:8080/usm/webservices/ReportingServ
Account Service	http://localhost:8080/usm/services/AccountService?wsdl	http://localhost:8080/usm/webservices/AccountServic
Billing Account Service	http://localhost:8080/usm/services/BillingAccountService?wsdl	http://localhost:8080/usm/webservices/BillingAccount dl
Form Designer Service	http://localhost:8080/usm/services/FormDesignerService?wsdl	http://localhost:8080/usm/webservices/FormDesigner dl
List Services	http://localhost:8080/usm/services	http://localhost:8080/usm/webservices/list

Removal of Axis References from CA Service Catalog (optional):

To remove axis references from CA Service Catalog, run:

```
Remove_Axis.bat under USM_HOME/scripts/RemoveAxis
```

The backup of axis specific jars are located under “REPLACED/AXIS_BACKUP”.

NOTE

- Till RU2, this *.bat* file can be run only if ITAM (CA Asset Management) is not part of the solution.
- From RU3, this *.bat* file can be run even if ITAM is a part of this solution.
- While upgrading from 17.4.2.0 to 17.4.3.0, re-run the *removal_axis.bat*, if already ran it on 17.4.2.0.

NOTE

- Starting from **CASM 17.4.5.0**, this process is automated.

Steps to migrate from Axis based webservices to CXF based webservices

Follow these steps to make OOTB content compatible with CXF web services:

Import **CA Service Management Content Pack 3.0** contentpack to make all OOTB plugins use CXF based webservices, located under the %USM_HOME%\filestore\contentpacks.
Restart the Service Catalog services.

NOTE
Starting from **CASM 17.4.5.0**, this process is automated.

CA Service Catalog CXF webservices for CA SDM:

NOTE

- Ensure you update the AXIS webservice URL to CXF webservice for all the endpoints.
- For notes and attachments synchronization, ensure you update the CA Service Catalog endpoint to “http[s]://hostname:port/usm/webservices” in **Service Desk Manager** > **Options Manager** > **CA Service Catalog** > **casc_endpoint**

To update SDM and CMDB web service endpoints to CXF in database, run *Update_Axis_Endpoints.bat* under USM_HOME/scripts/RemoveAxis

NOTE
After running the *Update_Axis_Endpoints.bat*, to avoid incompatibility in existing OOTB PAM processes, ensure to update the CXF endpoint for SLCM, SDM Global datasets in PAM.
For e.g.: /cxf/services/USD_WebService.

NOTE
Starting from **CASM 17.4.5.0**, this process is automated.

CA Service Catalog CXF webservices for CA APM:

From 17.4 RU3, CA APM supports both axis and cxf web services of CA SLCM when integrated. By default, the axis Web Service URL is configured and we can manually change it to cxf Web Service URL.
To perform this operation:

- Login to ITAM as administrator.
- Navigate to **Administration** > **System Configuration** > under **Menu** select the Application Sever, and change the USM Web Services URL.
  Example: *http(s)://<<servicecatalog_hostname>>:<<port>>/usm/services/RequestService?wsdl* to *http(s)://<<servicecatalog_hostname>>:<<port>>/usm/webservices/RequestService?wsdl*
- Perform the IISRESET on both ITAM Web and Application servers.

NOTE

- Starting from **CASM 17.4.5.0**, this process is automated.
- Although the CA APM UI displays the **CA Service Catalog AXIS Web Service URL** for reference, the system internally uses **CXF Web Services**, this is configurable.

CA Service Catalog CXF webservices for CA ITPAM:

NOTE
Before importing the CXF web services: If there are any customizations to OOTB content, take a backup of existing PAM content and reapply those customizations.

NOTE
To update ITPAM with the new CXF connector, import the latest ITPAM SDM Connector to support CXF webservices, download it from here.

1. Reload OOTB ITPAM content (which are CXF compatible) by manually importing the xml files into CA ITPAM.
2. These xml files are located under the "filestore\PAM CXF\SLCM\Custom Operators\CA SLCM" in Common Installer DVD for 17.4 RU2 and above.
3. All Catalog endpoints have to be updated to CXF in PAM global datasets, as shown in the example image:

NOTE

- The above steps are automated from **CASM 17.4.5.0**.
- The Catalog password has to be updated in **SLCM_GlobalDataSet**.

Name	Value
Login Parameters CHANGE ME	
userID__	spadmin
password__	**********
businessUnit__	ca.com
certpath__	
keypath__	
certpass__	**********
APP URLs CHANGE ME	
Misc Parameters CHANGE ME	
Parameters	
SoapCallPaths	
AdministratorSvc	/usm/webservices/AdministratorService
CatalogSvc	/usm/webservices/CatalogService
RequestSvc	/usm/webservices/RequestService
UserSvc	/usm/webservices/UserService
SDM_Service_Path	/cxf/services/USD_WebService
APM_setCurrentStatusForAsset	setCurrentStatusForAsset
APM_updateObjects	updateObjects
SLCM Status Codes	
APM Parameters	
SDM Parameters	

SOAP Request/Response payload differences:

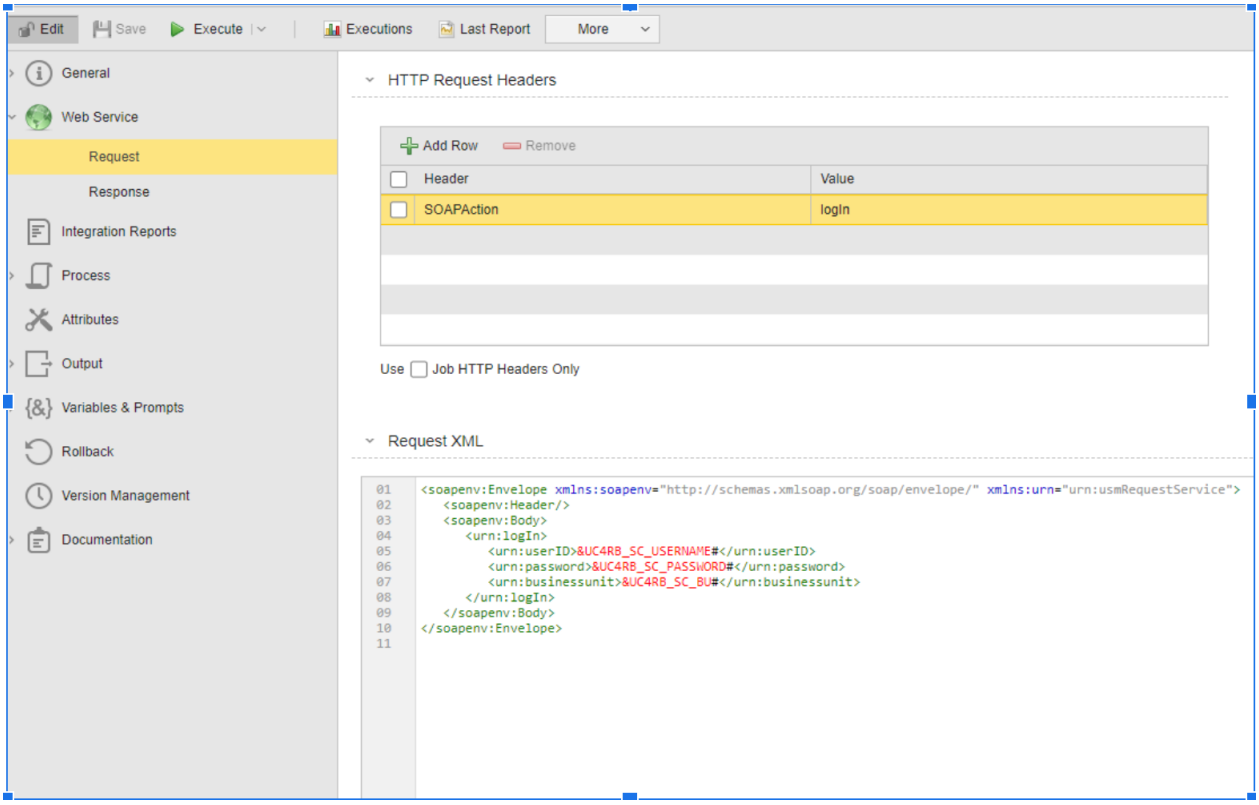
While accessing CA Service Catalog from an external applications like PAM or Automic, ensure that you update the request payload as follows:

- Unlike Axis web service, namespace is necessary for CXF based webservices.
  For example:

```
<getUser xmlns="urn:usmRequestService">
<sessionID>sessionID__</sessionID>
<userid>userid__</userid>
</getUser>
```

- The “xmlns” value should be appended to the SOAP method, which will be applied across the attribute. This value varies based on the type of service we are using. For example:

  - while accessing RequestService, then the namespace value is *urn:usmRequestService*.
  - while accessing CatalogService, then the namespace value is *urn:usmCatalogService*.
- For Automic integration, please pass “SOAPAction” header with method name under HTTP Request Headers section, when making a webservice call to CA Service Catalog, as shown in the example image:



Examples to update PAM processes

write the followng code, see the sample format for SOAP responses for sutom scripts.

Operation: *getRequestLevelNotes*

SOAP Response change in custom script:

```
if (Process[OpName].Result ==1) {


// This is Axis specific condition
if(Process[OpName].SOAPResponseData.getRequestLevelNotesReturn != undefined &amp;amp;&amp;amp;amp;
 Process[OpName].SOAPResponseData.getRequestLevelNotesReturn[0].getRequestLevelNotesReturn!=undefined)
{
Process.TotalComments =
 Process[OpName].SOAPResponseData.getRequestLevelNotesReturn[0].getRequestLevelNotesReturn.length;
Process.Successful__ = "TRUE";
}
//This is CXF specific condition
else if(Process[OpName].SOAPResponseData.getRequestLevelNotesResponse != undefined &amp;amp;&amp;amp;amp;
 Process[OpName].SOAPResponseData.getRequestLevelNotesResponse[0].getRequestLevelNotesReturn!=undefined)
{
```

```
Process.TotalComments =
 Process[OpName].SOAPResponseData.getRequestLevelNotesResponse[0].getRequestLevelNotesReturn.length;
Process.Successful__ = "TRUE";
}
}
```

Operation: *getRequestItemAttachments*

SOAP Response change in custom script:

```
if (Process[OpName].Result ==1) {
if(Process[OpName].SOAPResponseData.getRequestItemAttachmentsReturn != undefined &amp;amp;&amp;amp;
 Process[OpName].SOAPResponseData.getRequestItemAttachmentsReturn[0].getRequestItemAttachmentsReturn!
=undefined)
{
Process.TotalAttachments =
 Process[OpName].SOAPResponseData.getRequestItemAttachmentsReturn[0].getRequestItemAttachmentsReturn.length;
Process.Successful__ = "TRUE";
}
else if(Process[OpName].SOAPResponseData.getRequestItemAttachmentsResponse != undefined &amp;amp;&amp;amp;
 Process[OpName].SOAPResponseData.getRequestItemAttachmentsResponse[0].getRequestItemAttachmentsReturn!
=undefined)
{
Process.TotalAttachments =
 Process[OpName].SOAPResponseData.getRequestItemAttachmentsResponse[0].getRequestItemAttachmentsReturn.length;
Process.Successful__ = "TRUE";
}
}
```

Operation: *getTicketFromRequestItem*

SOAP Response change in custom script:

```
if(Process[OpName].SOAPResponseData.getTicketFromRequestItemReturn[0].getTicketFromRequestItemReturn !=
 undefined)
Process.TicketNumber =
 Process[OpName].SOAPResponseData.getTicketFromRequestItemReturn[0].getTicketFromRequestItemReturn[0].persid[0].text_;
//CXF
else
Process.TicketNumber = Process[OpName].SOAPResponseData.getTicketFromRequestItemReturn[0].persid[0].text_;
```

Operation: *getAssetUUIDByRequestItemID*

SOAP Response change in custom script:

```
if (Process[OpName].Result ==1) {
```

```
var bSOAPResponseDataReturn =
 Process[OpName].SOAPResponseData.getAssetUUIDByRequestItemIDReturn[0].getAssetUUIDByRequestItemIDReturn !=
 null &amp;amp;&amp;amp;
 Process[OpName].SOAPResponseData.getAssetUUIDByRequestItemIDReturn[0].getAssetUUIDByRequestItemIDReturn !==
 undefined;
//Axis
if (bSOAPResponseDataReturn) {
Asset_UUID__ =
 Process[OpName].SOAPResponseData.getAssetUUIDByRequestItemIDReturn[0].getAssetUUIDByRequestItemIDReturn[0].assetUUID[0
Process.Successful__ = "TRUE";
}
//CXF
else
{


bSOAPResponseDataReturn = Process[OpName].SOAPResponseData.getAssetUUIDByRequestItemIDReturn[0] != null
 &amp;amp;&amp;amp; Process[OpName].SOAPResponseData.getAssetUUIDByRequestItemIDReturn[0] !== undefined;
if (bSOAPResponseDataReturn) {
Asset_UUID__ = Process.getAssetUUID.SoapResponseData.getAssetUUIDByRequestItemIDReturn[0].assetUUID[0].text_;
Process.Successful__ = "TRUE";


}
}
}
```

Operation: *getRequestItemRow*

SOAP Response change in custom script:

```
var prefix = Process[OpName].SoapResponseData.getRequestItemRowReturn[0].getRequestItemRowReturn;
//Axis
if(prefix != null &amp;amp;&amp;amp; prefix != undefined){
prefix = Process[OpName].SoapResponseData.getRequestItemRowReturn[0];
}
//CXF
else{
prefix = Process[OpName].SoapResponseData;
}
for ( i = 0; i &amp;lt; prefix.getRequestItemRowReturn.length; i++ ) {
if ( prefix.getRequestItemRowReturn[i].subscriptionDetailID[0].text_ == Process.requestItemID__ ) {
   Process.RateItemID__ = prefix.getRequestItemRowReturn[i].itemID[0].text_;
   Process.RequestID__ = prefix.getRequestItemRowReturn[i].requestID[0].text_;
   Process.OfferingID__ = prefix.getRequestItemRowReturn[i].offeringID[0].text_;
   Process.BusinessUnit__ = prefix.getRequestItemRowReturn[i].domain[0].text_;
   Process.ParentBusinessUnit__ = prefix.getRequestItemRowReturn[i].parentDomain[0].text_;
   Process.GroupID__ = prefix.getRequestItemRowReturn[i].groupID[0].text_;
}
```

```
if ( prefix.getRequestItemRowReturn[i].numeric1[0].text_ != "0.0" ||
   prefix.getRequestItemRowReturn[i].numeric2[0].text_ != "0.0" ) {
   candidateCostColumns__[numCandidateCostColumns__] = prefix.getRequestItemRowReturn[i].itemID[0].text_;
   candidateQuantity__[numCandidateCostColumns__] = prefix.getRequestItemRowReturn[i].numeric1[0].text_;
   candidateUnitCost__[numCandidateCostColumns__] = prefix.getRequestItemRowReturn[i].numeric2[0].text_;
   numCandidateCostColumns__++;
}
}
```

Operation: *assignpendingaction*

Change in request payload of custom script.

SOAP request to Axis based WebService:

```
<assignPendingAction xmlns="urn:usmRequestService">
<sessionID>SESSIONID__</sessionID>
<pendingActionType>PATYPE__</pendingActionType>
<requestItemID>REQITEMID__</requestItemID>
<pendingActionStatus>REQITEMSTATUS__</pendingActionStatus>
<pendingActionUsersOrGroups>APPRLIST__</pendingActionUsersOrGroups>
<descriptions>DESCLIST__</descriptions>
<workItemID></workItemID>
<processInstanceID></processInstanceID>
</assignPendingAction>
```

SOAP request to CXF based WebService:

```
<assignPendingAction xmlns="urn:usmRequestService">
<sessionID>SESSIONID__</sessionID>
<pendingActionType>PATYPE__</pendingActionType>
<requestItemID>REQITEMID__</requestItemID>
<requestItemStatus>REQITEMSTATUS__</requestItemStatus>
<pendingActionUsersOrGroups>APPRLIST__</pendingActionUsersOrGroups>
<pendingActionDescriptions>DESCLIST__</pendingActionDescriptions>
<workItemID></workItemID>
<processInstanceID></processInstanceID>
</assignPendingAction>
```

Automated Migration for Axis to CXF Web Services

Automated Migration for Axis to CXF Web Services for CASM 17.4.5.0

Starting with version 17.4 RU5, the Axis endpoint has been fully decommissioned from the product. CXF migration is automated from RU5 and all the steps are taken care by common installer.

All out-of-the-box (OOTB) content for PAM Operators and Processes is reloaded to ensure compatibility with CXF-based SOAP responses. This update also refreshes global datasets to reference the correct CXF endpoints. However, to ensure

successful integration between CASM products, it is essential to re-enter the password in PAM global datasets after the RU installation.

The PAM service must remain running throughout the RU installation process to enable the reloading of OOTB content.

> **NOTE**
> Before this reload, a backup of existing PAM objects are created and saved under “C://Program Files/CA/PAM/Wildfly” in the format: “export_PAM_backup_<timestamp.xml>”.

Changes after installation of RU5

1. **For CA SDM**
   a. **SDM Endpoints in PAM Global Datasets:** The legacy Axis-based SDM endpoints have been deprecated. As part of the RU5 upgrade, the global datasets in PAM are now updated to point to the CXF-based SDM endpoints, ensuring all workflows and automations interact with the new web service layer seamlessly.
   b. **Loading of SDM CXF Content into PAM:** To support the updated integration, the CXF content package for SDM is automatically loaded into PAM during the upgrade. This package includes revised process definitions and operators that enable PAM to communicate reliably with SDM over CXF web services.
   c. **Change in SLCM Endpoint within SDM UI:** Within the SDM user interface, the SLCM endpoint path has been changed from:
      `/services`
      to
      `/webservices`
2. **For SLCM:**
   a. **Loading of SLCM CXF Content Pack into the SLCM:** The RU5 upgrade includes the deployment of the required CXF content pack directly into the SLCM product. This content pack contains updated definitions, processes, and operators that enable SLCM to interact through CXF-based web services instead of the older Axis endpoints.
   b. **Loading of SLCM CXF Content into PAM:** To facilitate cross-product integration, the same SLCM CXF content pack is also loaded into PAM. This ensures that workflows and automations originating from PAM can seamlessly communicate with SLCM using CXF.
   c. **Updating Endpoints in PAM Global Datasets:** As part of the transition, the endpoints for both SLCM and SDM are updated in the PAM global datasets to point to the CXF-based web services. This guarantees that all calls made from PAM reference the new ***/webservices*** path instead of the legacy ***/services*** path.
3. **For ITAM:**
   a. **Update of SDM Endpoint to CXF:** As part of the upgrade, the SDM endpoint referenced within ITAM is now updated to the CXF-based web service path. This ensures that all ITAM-to-SDM communication routes through the modern CXF framework, removing reliance on the older Axis /services endpoint.
   b. **SLCM Endpoint Behavior**: For SLCM integration, ITAM will also operate over CXF web services. However, it is important to note that in the UI or configuration screens, the SLCM endpoint may still appear in the legacy Axis format. Despite this representation, the underlying calls are executed via CXF, guaranteeing compatibility with the updated integration model.
4. **SDM PAM CXF connector:**
   a. **Automatic Update of Out-of-the-Box (OOTB) Process URLs:** All out-of-the-box process URLs associated with SDM and Catalog are automatically updated from the legacy Axis endpoints to the modern CXF endpoints. This ensures that default processes continue to function seamlessly without requiring any administrator intervention.
   b. **Manual Update Required for Customized Processes:** For any customized ITPAM processes that have been added or integrated with SDM or Catalog, administrators must manually update the service URLs from Axis to CXF. These customized workflows are not automatically migrated during the upgrade to avoid overwriting or altering user-defined configurations.

Uninstallation

1. When uninstalling SDM or SLCM version 17.4 RU5, the system is expected to return to the previous stable version that was in use before the upgrade. However, this rollback may not automatically restore all associated components, such as PAM (Process Automation content).
2. In such cases, manual steps may be required to reload the appropriate PAM content to ensure full functionality is restored. Therefore, the technical documentation must include clear instructions on how to back up PAM content before the upgrade and how to restore it after uninstallation if needed.
3. Additionally, any service endpoints that were modified as part of the 17.4 RU5 upgrade—across SDM, SLCM, ITAM, and PAM—must be manually reverted to their original (pre-upgrade) state. This ensures all components communicate correctly after the rollback.

Moving the CA MDB Data from the Source to the Target Systems

You can move the CA MDB database from the source system and configure it to work on the target system.

> **WARNING**
>
> We recommend that you test the procedure in a test environment before implementing on a production system. Currently, moving the CA MDB data is not supported for clustered database, load balancer, and application cluster.

Prerequisites

Before you move the CA MDB data, perform the following tasks:

1. Back up the CA MDB database on your target system.
2. Install the following products on the source and target systems:
   - CA Service Desk Manager 17.4 and above
   - CA Service Catalog 17.4 and above
   - CA Asset Portfolio Manager 17.4 and above
3. Ensure that the source and target systems have an identical installation configuration setup.

Update Database Login Credentials used by CA Search Server

To update the login credentials, perform the following steps:

1. Open the **config.yml** file located at, for example, in the C:\Program Files\CA\SC\CASearchServer\search folder. Locate the attribute named as “user” and “password”. The values associated these attributes are username and password respectively.
2. Change the user field value.
   The user field value can be changed only for the account which has read/write access to the database. Example: mdbadmin
3. Encrypt the password values using SDM encryption (SDME) format. To encrypt, run the following commands on CA SDM server.
   a. To change the prompt and directory to NX_ROOT\bin, run the following command.

      ```
      nxcd bin
      ```

   b. To replace your old password for mdbadmin account with the new password for the database user and to generate the encrypted string, run the following command.

      ```
      encrypt_pwd -sdme <old password>_for_mdbadmin_account
      ```

   c. Copy the encrypted string and use as password value in the **config.yml** file.

d. Save the **config.yml** file.

4. Perform a reindex of CA Search Server. See How to perform a reindex of CA Search Server.
5. Restart the xFlow Interface services.

Impacted CA Service Management Configuration Tables

The following CA Service Management Configuration tables are impacted when you move the CA MDB database on a Microsoft SQL Server database or Oracle database. Take the backup of the following tables and restore them after moving the CA MDB database.

Common Tables	CA SDM	xFlow Interface	CA Service Catalog	CA APM	CA Unified Self-Service Tables
al_cdb_comp_installstate_bkup	ci_mdr_provider,	usp_es_servers, l1admin_general_config	usm_configuration	al_meta_refresh_state This table is applicable only for CA Service Management 14.1.01 installation.	os_ExternalSource
al_cdb_componentinstallstate	options			al_meta_binary_store	os_ExternalSourceProperties
al_cdb_config_params_backup	sapolicy			al_process_account	PortalPreferences
al_cdb_configurationparameters	usp_servers			al_apmp_def al_apmp_operation	
	object_promotion promo_hist **Note:** These tables are displayed only if CA Service Management 14.1.02 is installed in your environment.			al_apmp_content_types al_apmp_progress_status al_apmp_progress al_apmp_record_filter al_apmp_run_stage These tables are displayed only if CA Service Management 14.1.02 patch is installed in your environment.	
al_cdb_files,ca_application_registration,	usp_configuration				
	usp_webeng_domsrvr				
	usp_webengine				
	usp_webeng_alias				
	usp_domsrvr				
	usp_config_other_daemons				

Moving the CA MDB Database on Microsoft SQL Server

This page also contains the following section:

If you have installed Microsoft SQL Server database in your system, perform the following steps:

1. Review CA Service Management Tables to verify the tables that are impacted when you move the CA MDB database on Microsoft SQL Server.
2. Stop the following CA Service Management services on both the source and target application servers:

   NOTE

   Some of these services might not be applicable if the associated products or components are not installed in your environment.

   a. **Common Admin Service**:
      - CA Service Management Admin

   b. **CA SDM**:

      NOTE

      If you have any shared UNC paths, ensure that you copy the paths from the source system to the target system.

      - CA Service Desk Manager Server

   c. **CA Service Catalog Services** :
      - CA Service Accounting
      - CA Service Catalog

   d. **CA Asset Portfolio Management Services**:
      - IIS Admin Service
      - CA Asset Portfolio Management - Data Importer Engine
      - CA Asset Portfolio Management - Event Service
      - CA Asset Portfolio Management - Export Service
      - CA Asset Portfolio Management - HW Reconciliation Engine
      - CA Asset Portfolio Management - LDAP Import Service
      - CA Asset Portfolio Management - Registration Service
      - CA SM Server

   e. **CA xFlow Interface Services**:
      - CA xFlow Interface

   f. **CA Search Server Services**:
      - CA Service Management Search Server

3. On the source system, back up the **mdb** database, and copy the **mdb**back up files to the target system.
4. (Optional) If you are using a named instance (non-default MDB name), ensure that you have selected the appropriate MDB instance name for the backup.
5. On the target system, perform the following steps to back up the tables:
   a. Navigate to Microsoft SQL Management Studio, right-click **MDB** (or the appropriate MDB instance name applicable for your system), and select **Tasks**, **Generate Scripts**.
   b. Click **Select Specific database objects**. Take a backup of the CA Service Management Tables.
   c. Select the **Tables** option and choose the appropriate tables.
   d. Click **Next**.
   e. Select **Advanced** and navigate to **Types of Data**.
   f. Change it from **Schema Only** to **Schema and Data**, and click **OK**.
      By default, your selection is now saved to a file **%user%\documents\script.sql**.
   g. Specify different file names for MDB.

   script file is generated.

6. On the target system, perform the following steps to detach the existing *mdb:*
   a. Navigate to Microsoft SQL Management Studio, right-click **MDB** (or appropriate MDB name), **Tasks**, **Detach**.
   b. On the Detach Database screen, select **Drop Connections**.
   c. Click **OK**.
7. Restore *mdb* backup on the target system that was copied from the source system. Select **Overwrite the existing database** (WITH REPLACE) from the **Options** tab.
   Ensure that the database name on the target system is retained.

   NOTE

   Provide read and execute privileges to the mdb.bak file that you copied from the source database to the SQLServerMSSQLUser$$MSSQLSERVER service account on the target MDB server. Failure to grant the read and execute privilege to the mdb.bak file may result in an error that the file folder cannot be located or it does not have the correct permissions.

   NOTE

   Restore must complete without errors.
8. Run the following Microsoft SQL auto_fix_user script for usermdbadmin users:
   – sp_change_users_login 'AUTO_FIX','mdbadmin'
   – sp_change_users_login 'AUTO_FIX', 'apmdba' (Applicable only if CA APM is installed and upgraded from previous release versions)
   – sp_change_users_login 'AUTO_FIX', 'uapmadmin' (Applicable only if CA APM is installed and upgraded from previous release versions)
   – sp_change_users_login 'AUTO_FIX', 'uapmbatch' (Applicable only if CA APM is installed and upgraded from previous release versions)
   – sp_change_users_login 'AUTO_FIX', 'uapmreporting' (Applicable only if CA APM is installed and upgraded from previous release versions)
   – sp_change_users_login 'AUTO_FIX', 'usmadmin' (Applicable only if CA Service Catalog is installed and upgraded from previous release versions)
   – sp_change_users_login 'AUTO_FIX', 'ServiceDesk' (Applicable only if CA SDM is installed and upgraded from previous release versions)
9. Verify that the mdbadmin users are now mapped to **mdb** and have the **bulkadmin**, **public** roles.
10. On the target system, delete data from the following tables and commit the changes:
    a. Use **mdb** and run the following queries to delete data from the required tables:
       Common Tables:
       - delete from al_cdb_comp_installstate_bkup
       - delete from al_cdb_componentinstallstate
       - delete from al_cdb_config_params_backup
       - delete from al_cdb_configurationparameters
       - delete from al_cdb_files
       - delete from ca_application_registration

       CA SDM Tables:

- delete from options
- delete from usp_servers
- delete from sapolicy
- delete from ci_mdr_provider
- delete from usp_domsrvr
- delete from usp_configuration
- delete from usp_webeng_alias
- delete from usp_webeng_domsrvr
- delete from usp_webengine
- delete from usp_config_other_daemons
- delete from object_promotion (applicable only if CA Service Management 14.1.02 is installed)
- delete from promo_hist (applicable only if CA Service Management 14.1.02 is installed)

CA Service Catalog tables:

- delete from usm_configuration

CA APM tables:

- delete from al_meta_binary_store
- delete from al_process_account

Delete the following CA APM tables, if you have CA Service Management14.1.02 installed in your system:

- delete from al_apmp_def
- delete from al_apmp_operation
- delete from al_apmp_content_types
- delete from al_apmp_progress_status
- delete from al_apmp_record_filter
- delete from al_apmp_run_stage

CA xFlow Interface tables:

- Delete from l1admin_general_config

CA Search Server tables:

- Delete from usp_es_servers

11. Execute the **script.sql** scripts that were created in **Step 5**.

    NOTE

    Ignore the warning message, There is already an object named ... in the Database.
12. In CA Asset Portfolio Management, when restoring the database from one server to another, the following updates are required to run on the destination db server:
    – Update ca_application_registration set product_specific_data = 'HTTP://ITAM-Web-server:99/ITAM', where product_code = 2002
    – Update ca_application_registration set product_specific_data = 'http://ITAM-App-Server:9060/itsm/itsm.html', where product_code = 2070
13. (Optional) To disable email notifications, modify the CA contact table query: **Update ca_contact set email_address = ''**.
14. Start the services on both the source and the target systems and on the target system, perform the following steps for each product:
    a. **Common Admin Service**:
       CA Service Management Admin
    b. **CA SDM Services**:

Perform the following steps before you start the CA SDM services:

a. If your environment has customized CA Process Automation content, export all the relevant CA Process Automation Content from the source CA Process Automation server and import to the target CA Process Automation server. Complete the required CA Process Automation content configuration.
For more information, see CA Process Automation.

b. Copy the **site/mods** directory from the source CA SDM server to the target CA SDM server.

NOTE

This is applicable only if your application environment is customized. This is not applicable for the out-of-the-box (OOTB) content.

c. Run the ***pdm_configure*** command without selecting the load default data option.

NOTE

- For Advance Availability environment run the pdm_configure -S on a background server.
- All the remaining servers run with pdm_configure.

d. Start the following CA SDM service:
   - CA Service Desk Manager Server

e. Execute the **pdm_k_reindex** command on the CA SDM target system to index knowledge documents that are created and copied from the source system.

f. Repositories.
   a. Log in to CA SDM as an administrator.
   b. Navigate to **Administration tab**> **Attachments Library**> **Repositories**.
   c. Open the repositories one by one and update the target CA SDM server name in the Servlet Server and Repository Server fields.

NOTE

Update the CA SDM server name in all the repositories(OOB and Created Repositories).

g. Attachments.
   - Copy all the attachments using upload Path of all the repositories from a source server to the same path in a target server.
     For Example: ServiceDesk repository upload path is: <NX_ROOT>/site/attachments/default/servicedesk/file on source server. Copy the attachments from a source server to the target server.

c. **CA Service Catalog Services** :

Perform the following steps before you start the CA Service Catalog services:

- If your source system has customized CA Process Automation content, export all relevant CA Process Automation content from the source CA Process Automation server and import to the target CA Process Automation server. Complete the required configurations for CA Process Automation content.
  For more information, see CA Process Automation.
- If your source system has customized CA EEM policies, export all the relevant CA EEM policies from the source CA EEM server and import to the target CA EEM server.
  For more information, see Installing Common Components.
- Copy filestore directory from the source CA Service Catalog server to the target CA Service Catalog server.

NOTE

This is applicable only if customization is carried out on the application environment. This is not applicable for the out-of-the-box content.

- Launch the CA Process Automation server and access **SLCM_Globaldataset**. and validate that the domain and the CA Service Catalog root business unit names are same.
- Start the following CA Service Catalog services:

- CA Service Accounting
- CA Service Catalog

d. **CA Asset Portfolio Management Services**:
- IIS Admin Service
- CA Asset Portfolio Management - Data Importer Engine
- CA Asset Portfolio Management - Event Service
- CA Asset Portfolio Management - Export Service
- CA Asset Portfolio Management - HW Reconciliation Engine
- CA Asset Portfolio Management - LDAP Import Service
- CA Asset Portfolio Management - Registration Service
- CA SM Server

e. **CA xFlow Interface Services**:
- CA xFlow Interface

f. **CA Search Server Services**:
- CA Service Management Search Server

Run the rebuild index on Search Server.
To Run the rebuild index, execute one of the files based on your operating system:
- $ES_HOME/search/bin/pdm_es_rebuild_index.bat (Windows)
- $ES_HOME/search/bin/pdm_es_rebuild_index.sh (Linux)

15. Verify that CA Service Management applications are accessible. All applications must be up and running.
16. Validate that all CA Service Management Integration scenarios are working seamlessly.

Move an Existing CA Service Catalog Installation to a New MDB

You may often want to restore an MDB for testing or move environments around. The following steps help you point a working CA Service Catalog to a new MDB.

1. Open a command prompt and change the directory to USM_HOME/scripts.
   The encrypter.bat file is displayed.
2. Run the following command to encrypt your MDB password (usually the MDBADMIN password):
   ```
   encrypter.bat <Your Password>
   ```
   Your MDB password is now encrypted.
3. Copy the password and update the config.properties file under USM_HOME directory for the following parameters:
   – DBSource.password = <Paste the new encrypted Password here>
   – BSource.username = your mdbadmin or equivalent database user
4. Under the mdb.deploy values set the correct values for your new MDB database.

   NOTE

   - Settings such as "mdb.deploy.usmpassword" will need the above to re-encrypt the password.
   - The "DBSource.url" may also need to be updated to reflect the new database server.
5. Open a Service Management CMD prompt from the start menu and run the following command, if your MDB has a different CA Service Catalog installation running on it:
   ```
   ant update-usm-host
   ```
6. Run the CA Service Catalog Setup utility, if your MDB does not have an existing CA Service Catalog running on it.

Moving the CA MDB Database on Oracle

Perform the following steps to copy and restore CA MDB if you have installed Oracle database in your system:

1. Review CA Service Management Tables to verify the tables that are impacted when you move the CA MDB database on Oracle.
2. Stop the following CA Service Management services on both the source and target application servers.

   NOTE

   Some of these services might not be applicable if the associated products or components are not installed in your environment.

   a. **Common Admin Service**:
      - CA Service Management Admin

   b. **CA SDM Services**:

      NOTE

      If you have any shared UNC paths, ensure that you copy the paths from the source system to the target system.

      - CA Service Desk Manager ODBC Data Access
      - CA Service Desk Manager ODBC Agent
      - CA Service Desk Manager Server

   c. **CA Service Catalog Services**:
      - CA Service Accounting
      - CA Service Catalog

   d. **CA Asset Portfolio Management Services**:
      - IIS Admin Service
      - CA Asset Portfolio Management - Data Importer Engine
      - CA Asset Portfolio Management - Event Service
      - CA Asset Portfolio Management - Export Service
      - CA Asset Portfolio Management - HW Reconciliation Engine
      - CA Asset Portfolio Management - LDAP Import Service
      - CA Asset Portfolio Management - Registration Service
      - CA SM Server

   e. **CA xFlow Interface Services**:
      - CA xFlow Interface

   f. **CA Search Server Services:**
      - CA Service Management Search Server

3. On the source system, navigate to *C:\app\Administrator\product\11.2.0\dbhome_1\BIN* path to export the database. To Create and grant permissions to the directory, execute the following commands:

```
CREATE OR REPLACE DIRECTORY test_dir AS 'C:\app\Administrator';
GRANT READ, WRITE ON DIRECTORY test_dir TOmdbadmin
;
```

NOTE

Ensure that you use the appropriate MDB instance name applicable for your system.

```
expdp mdbadmin/N0tallowed@mdb directory=test_dir dumpfile=mdb.dmp logfile=expdbmdb.log
```

4. Copy the **mdb.dmp** files to *C:\app\Administrator* on the target system.
   Ontarget system, take a backup ofmdb first and then, take a backup of the required tables. To create and grant permissions to the directory, execute the following commands for mdb:

   ```
   CREATE OR REPLACE DIRECTORY test_dir AS 'C:\app\Administrator';
   GRANT READ, WRITE ON DIRECTORY test_dir TOmdbadmin;
   ```

   NOTE

   Ensure that you use the appropriate MDB instance name.

   ```
   expdp mdbadmin/N0tallowed@mdb directory=test_dir dumpfile=mdb.dmp logfile=expdbmdb.log
   ```

5. If CA Service Management14.1.02 is installed in your system, the additional tables listed here for CA SDM and CA APM also must be included in the following query:
   - **CA SDM**: object_promotion, promo hist
   - **CA APM**: al_apmp_def, al_apmp_operation, al_apmp_content_types, al_apmp_progress_status, al_apmp_progress, al_apmp_record_filter, l_apmp_run_stage

   ```
   expdp mdbadmin/N0tallowed@mdb tables=al_cdb_comp_installstate_bkup,
   al_cdb_componentinstallstate,
   al_cdb_config_params_backup,
   al_cdb_configurationparameters,
   al_cdb_files,
   ca_application_registration,
   options,
   usp_servers,
   sapolicy,
   ci_mdr_provider,
   usp_domsrvr,
   usp_configuration,
   usp_webeng_domsrvr,
   usp_webengine,
   usp_webeng_alias,
   usm_configuration,
   al_meta_refresh_state,
   al_meta_binary_store,
   al_process_account grants=y indexes=y rows=y constraints=y triggers=y
   directory=test_dir dumpfile=expdptables.dmp logfile=expdbtables.log
   ```

6. Open a Windows terminal and enter the **SQL*Plus** command: *sqlplusw*. The **SQL*Plus** Windows GUI opens and the login dialog is displayed.
7. Log in to the **SQL*Plus** as System User on the Oracle target system and remove the **MDBADMIN** user (schema) by running the following command:

   ```
   DROP USERmdbadmincascade;
   ```

8. Run the 'exit' command on the SQL*Plus to log out.
9. Open the command prompt and run the following command to change the current directory:

   ```
   cd C:\Program Files\CA\SC\Mdb\Windows\oracle\ddl
   ```

10. To recreate the MDBADMIN user again, run the following command:

```
sqlplussystem/<password>@<Net Service Name>  @create_mdb_oracle.sql <Password of the MDBADMIN> MDB_DATA
 MDB_INDEX
```

11. To import the data that is exported on the source system, run the following command:

```
impdpsystem/<Password>@<Net Service Name> full=y directory=test_dir dumpfile=mdb.dmplogfile=impdbmdb.log
```

12. Drop the tables after import by executing the following commands. Dropping the tables removes the table definitions and rows.
    a. **Common Tables**

```
DROP TABLE al_cdb_comp_installstate_bkup;
DROP TABLE al_cdb_componentinstallstate;
DROP TABLE al_cdb_config_params_backup;
DROP TABLE al_cdb_configurationparameters;
DROP TABLE al_cdb_files;
DROP TABLE ca_application_registration;
```

    b. **CA SDM**

```
DROP TABLE ci_mdr_provider;
DROP TABLE options;
DROP TABLE usp_servers;
DROP TABLE sapolicy;
DROP TABLE usp_domsrvr;
DROP TABLE usp_configuration;
DROP TABLE usp_webeng_alias;
DROP TABLE usp_webeng_domsrvr;
DROP TABLE usp_webengine;
DROP TABLE object_promotion;
DROP TABLE promo_hist;
```

    c. **CA Service Catalog**

```
DROP TABLE usm_configuration;
```

    d. **CA APM**

```
DROP TABLE al_meta_binary_store;DROP TABLE al_process_account;
DROP TABLE al_meta_refresh_state;
DROP TABLE al_apmp_def;
DROP TABLE al_apmp_operation;
DROP TABLE al_apmp_content_types;
DROP TABLE al_apmp_progress_status;
DROP TABLE al_apmp_progress;
DROP TABLE al_apmp_record_filter;
DROP TABLE al_apmp_run_stage;
```

    e. **CA xFlow Interface**

```
DROP TABLE l1admin_general_config
```

    f. **CA Search Server**

```
DROP TABLE usp_es_servers
```

13. Run the following command as a system user:

```
impdp system/<Password>@<Net Service Name> directory=test_dir dumpfile=expdptables.dmp
 logfile=expdbtables.log
```

14. Start the services on both the source and the target systems. On the target system, perform the following steps for each product:
    a. **Common Admin Service**:
       - CA Service Management Admin
    b. **CA SDM Services**:
       Perform the following steps before you start the CA SDM services:
       a. If your environment has customized CA Process Automation content, export all the relevant CA Process Automation Content from the source CA Process Automation server and import to the target CA Process Automation server. Complete the required CA Process Automation content configuration.
          For more information, see Installing Common Components (CA EEM, CABI JasperReports, CA Process Automation)
       b. Copy the **site/mods** directory from the source CA SDM server to the target CA SDM server.

          NOTE

          This is applicable only if your application environment is customized. This is not applicable for out-of-the-box (OOTB) content.
       c. Run the ***pdm_configure*** command without selecting the load default data option.
       d. Now, start the following CA SDM services:
          - CA Service Desk Manager ODBC Data Access
          - CA Service Desk Manager ODBC Agent
          - CA Service Desk Manager Server
       e. Run the **pdm_k_reindex** command on the CA SDM target system to index knowledge documents that are created and copied from the source system.
       f. Repositories.
          a. Log in to CA SDM as an administrator.
          b. Navigate to **Administration tab**, **Attachments Library**, **Repositories**.
          c. Open the repositories one by one and update the target CA SDM server name in the Servlet Server and Repository Server fields.

             NOTE
             Update the CA SDM server name in all the repositories(OOB and Created Repositories).
       g. Attachments.
          - Copy all the attachments using upload Path of all the repositories from a source server to the same path in a target server.
            For Example: ServiceDesk repository upload path is: <NX_ROOT>/site/attachments/default/servicedesk/file on source server. Copy the attachments from a source server to the target server.
    c. **CA Service Catalog Services**:
       Perform the following steps before you start the CA Service Catalog services:
       - If your source system has customized CA Process Automation content, export all relevant CA Process Automation content from the source CA Process Automation server and import to the target CA Process Automation server. Complete the required configurations for CA Process Automation content.
         For more information, see Installing CA Common Components.
       - If your source system has customized CA EEM policies, export all the relevant CA EEM policies from the source CA EEM server and import to the target CA EEM server.
       - Copy filestore directory from the source CA Service Catalog server to the target CA Service Catalog server.

NOTE

This is applicable only if customization is carried out on the application environment. This is not applicablefor the Out-of-the-box content.

- Launch the CA Process Automation server and access **SLCM_Globaldataset**.
  Validate that the domain and the CA Service Catalog root business unit names aresame.
- Now, start the following CA Service Catalog services:
  - CA Service Accounting
  - CA Service Catalog

d. **CA Asset Portfolio Management Services**:
   - IIS Admin Service
   - CA Asset Portfolio Management - Data Importer Engine
   - CA Asset Portfolio Management - Event Service
   - CA Asset Portfolio Management - Export Service
   - CA Asset Portfolio Management - HW Reconciliation Engine
   - CA Asset Portfolio Management - LDAP Import Service
   - CA Asset Portfolio Management - Registration Service
   - CA SM Server

e. **CA xFlow Interface Services**:
   - CA xFlow Interface

f. **CA Search Server Services**:
   - CA Service Management Search Server
   - Run the rebuild index on Search Server

   Run the rebuild index on Search Server.To Run the rebuild index, execute one of the files based on your operating system:
   - $ES_HOME/search/bin/pdm_es_rebuild_index.bat (Windows)
   - $ES_HOME/search/bin/pdm_es_rebuild_index.sh (Linux)

15. Verify that CA Service Management applications are accessible. All applications must be up and running.
16. Validate that all CA Service Management Integration scenarios are working seamlessly

Building

CA Service Desk Manager

Modifying Web Interface

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More...

Building CA Service Desk Manager

CA SDM lets you fulfill various IT Service Management functions. The product provides a broad feature set, and various best-practice content to help meet your service management needs.

The default implementation of CA SDM closely matches the processes and terminology that is used in most IT organizations. You can extend CA Service Desk Manager to work with the unique aspects of your organization. The product includes the following spectrum of approaches to building the product to meet your unique needs:

- End-user personalization
- System-wide configuration
- Tool-based adaptation
- Code-level modifications

This section contains the following articles:

- Modify Notification Methods
- Query and Message Modifications
- Web Interface Modifications
- Event Log Data Storage Modification
- Print CA SDM Web Pages
- Legacy Reports Modification
- Web Services Management
- CA SDM REST API
- How to Version System Customizations Across CA SDM Servers
- Using the Web Screen Painter (WSP)

Modify Notification Methods

This article contains the following topics:

The CA SDM automatic notification methods notify personnel at key points in the service desk management process. The standard notification methods are shown as follows:

- Email
- Notification (Log)
- Pager_Email

You can define modified notification methods to specify new methods of transmission. For example, voice mail, display boards, or a specific printer. You can also access data from another application and include it in the notification message.

The Notification Process

Ticket notifications (applicable to issues, change orders, and requests) are processed when the ticket is saved:

- If the notification method is other than Notification, such as Email, the notification processor executes the notification method for each contact in the list. This method is typically an executable or shell script, which is launched in a new process. Details about the notification are stored in environment variables for easy access by the executable/script.
- For each notification requested, the notification processor sets the NX_NTF_MESSAGE and NX_NTF_SUMMARY environment variables using the Notification Message Title and Notification Message Body information that is provided on the Message Template notebook page of the Activity Notifications Detail window. If the recipient is a valid contact, other environment variables are created using information in their Contact Detail record.
- If the Write To File option is selected for the notification, a text file is created. The text file indicates that the notification method can use to obtain more detailed information.
- A list of contacts to receive the notification is built from the information that is provided on the Objects, Contacts, Types, and Survey notebook pages of the Activity Notifications Detail window. For those having a notification method matching the Notify Level and the log_all_notify Options Manager option that is installed, a notification is generated first to the notification log.

Notification Method Variables

Two sets of variables are created and made available to the notification method.

Basic Environment Variables

The first set of variables is created for every notification that is sent, independent of whether you select the Write To File option for the notification. They are written to the environment as environment variables that the notification method can access in the standard way.

The following environment variables give you basic information about the notification. They are always defined, even if the corresponding value is empty:

Environment Variable	Description
NX_NTF_MESSAGE	The completed message template text, including full expansion of all variables
NX_NTF_SUMMARY	The completed message template header, including full expansion of all variables
NX_NTF_URGENCY	The notification urgency (1 is low, 4 is emergency)

The following environment variables are created only if the recipient is a valid CA SDM contact. The variables are set using values from the Contact Detail record of the recipient as shown in the following table:

Variable	**Contact Detail Window Fields**
NX_NTF_BEEPER_PHONE	Pager Number

NX_NTF_COMBO_NAME	Last Name, First Name, Middle Name
NX_NTF_CONTACT	Contact ID
NX_NTF_EMAIL_ADDRESS	Email or Pager Email Address (depending on notification type)
NX_NTF_FAX_PHONE	Fax Number
NX_NTF_PUBLIC_PHONE	Phone Number
NX_NTF_USERID	User ID
NX_NTF_VOICE_PHONE	Alt. Phone #

NOTE
These variables are not created if the corresponding values are empty (except for NX_NTF_CONTACT, which cannot be empty).

Attribute Variables

The second set of variables is available only if you select the Write To File option when you define the notification method. The following attributes are called attribute variables and are written to the notification file only -- not to the environment. They are of the form:

NX_NTF_attribute[.secondary_attribute]=value

- ***attribute***
  The name of the attribute whose value you want to obtain. This value is the attribute name as defined for the object. For a complete list of all attribute names for any object, see CA Service Desk Manager Reference Commands. The most common objects that are associated with notifications are the ticket, which has an object name dependent of the type of ticket (for example, cr for requests), and the contact identifying the recipient, which has an object name of cnt. For example, the environment variable for the description attribute of a ticket can look as follows in the notification file:
  – NX_NTF_DESCRIPTION=<sample description>
- ***secondary_attribute***
  If the first *attribute* is an internal identifier for another object, a secondary attribute is attached to give more meaningful information using the dot notation. In database terms, attribute is a foreign key that points to a row in another table, rather than a simple data value. Using this raw key value would probably have little meaning. Many of these types of fields are resolved or de-referenced for you. The secondary_attribute is the value in the referenced table. For example, instead of writing the value for the assignee attribute, which is stored as the unique ID of the contact record for the assignee, the assignee's combined name is the combo_name attribute for the contact object, as shown in the following example:
  NX_NTF_ASSIGNEE.COMBO_NAME=Armstrong, Beth
  If an attribute does not have a value, the corresponding value is usually (NULL) or blank. For example:
  NX_NTF_CALL_BACK_DATE=(NULL)
  NX_NTF_GROUP.COMBO_NAME=

  NOTE
  An attribute variable that exists for both the ticket and the recipient is NX_NTF_ID (the id attribute), which is the unique database ID for the object.

The Notification File

If you select the Write To File option when you define a notification method, all the basic environment and attribute variables are written to a text file. This file is closed before executing the notification method script or program. This notification file is updated every time the notification method is invoked for a contact. This method is a handy mechanism for passing relevant information to the notification script that is not otherwise available in the environment.

The full path of the notification file is set in the NX_NTF_FILENAME environment variable, which is available to the notification method . The file name is also added to the end of value you enter in the Notification Method field when

defining the notification method. For example, if the Notification Method is ‘pdm_perl - w mymethod.pl’, then the actual process executes ‘pdm_perl - w mymethod.pl *unique_notification_file_name*’.

> **WARNING**
> The administrator can clean up the notification files. This clean-up is especially important for a site using a high volume of notifications. The files are located in the standard temporary directory (TEMP on Windows and TMP on UNIX). One suggestion is to delete the file at the end of the notification method script/program.

The notification file is a standard text file that is divided into sections. Each line contains either an attribute/value pair or a section marker. Each notification file has three sections. All sections begin with “-----” followed by a new line.

- **SECTION=obj**, where obj identifies the object type of the ticket
  - **Iss**
    Provides information about the issue.
  - **Chg**
    Provides information about the change order.
  - **Cr**
    Provides information about the request.
- **SECTION=cnt**
  Provides information about the recipient.
- **SECTION=notification**
  Provides the same information that is available from the basic environment variables.

> **NOTE**
> The section names for the ticket and recipient are actually the object names for the attributes in that section. For a complete list of all attribute names for any object, see the CA Service Desk Manager Reference Commands section.

Several lines of attribute/value pairs, each of which represents an attribute of the corresponding object, are contained in each section. The Attribute Variables in this section provide the detailed information about how these lines are formatted and what they mean.

Line breaks in an attribute value are reproduced as new lines in the notification file. Your notification method process can only use the attribute or value lines that begin with NX_NTF, and section markers. Generate a sample file and look at its contents before working with a notification file in your notification method process.

Using Perl Scripts

Most notification methods invoke an executable or shell script to read the environment variables and send the message. This action works well on most UNIX servers, but difficulties arise reading the environment variables on a Windows server.

You can use a Perl script to overcome environment problems on Windows. CA SDM includes a ready-to-use installation of the Perl interpreter named pdm_perl. Any Perl script that is invoked with pdm_perl as a notification method can reliably obtain the environment variables. The Perl script can read and format the environment variable values. The script can also carry on with the rest of the notification, such as invoking a pager or sending an email.

For Windows-based servers, consider using the launchit utility. You can invoke your scripts or programs in a shell environment with the proper environment variables set.

For example, suppose that you write a Perl script that is named read_env.pl to read several of the environment variables. You can invoke this script for a notification by entering the following command in the Notification Method field on the Notification Method Detail window:

pdm_perl *script_path*/read_env.pl

This notification method starts the Perl interpreter and executes the instructions in read_env.pl script.

Create a Notification Method

Follow these steps:

1. Create a script to process the message template and transmit it to the recipient. The script can be any executable, depending on the platform. Third-party or public domain interpreters can also be used. Typically, Bourne shell scripts are used on UNIX and .bat files are used on Windows. If your script requires a special template, create it.
2. Add the new notification method to your site using the web interface.

Create a Script

You can create a notification method script.

Follow these steps:

1. Determine how you want the notification to be delivered (for example, printed on a particular printer).
2. Determine the contents of the notification message.
3. Specify what information from the message template to include in the notification.
4. Set up a script to transmit the notification.
5. Place the script in an executable file in the path of the CA SDM server.

Add the Notification Method

After you create a script, define the new notification method to CA SDM. You can use *one* of the following methods to add a notification method:

- Using the web interface
- Using a UNIX shell script

Add a Notification Method Using the Web Interface

Use the web interface to add a notification method.

Add a Notification Method Using a UNIX Shell Script

The following steps create a notification method shell script that sends the notification message to the service desk printer, SDPR2. In this example, the notification message consists of the message header and the message text from the message template:

1. Set up the shell script to assemble the notification text and transmit it, as follows:

```
#!/bin/sh
echo "
TO:       $NX_NTF_USERID
SUBJECT:  $NX_NTF_SUMMARY
MESSAGE:
$NX_NTF_MESSAGE" |lp -dSDPR2
```

2. Name the executable file sd_print. Place it in any directory that is used for common scripts at your site, such as /usr/local/netbin.
3. Make the shell script an executable file using chmod.
4. Select Notification Methods from Notifications on the Administration Interface.
5. Select New from the File menu.
6. Enter data in these fields:
   - **Symbol**
     SDPR2
   - **Description**
     Send backup notification to service desk printer SDPR2
   - **Notification Method**
     /usr/local/netbin/sd_print

7. Click the Save button to save the new record. Then click Close Window to close the detail window.

Query and Message Modifications

CA SDM provides a number of features that let you narrow the focus of information so you can concentrate on requests, change orders, and issues that apply to your immediate situation. One of these functions stores queries that you can use to see relevant information on the scoreboard of the administrative or web interface. Another lets you modify the messages that notify key personnel of ticket activities.

Stored queries can provide a focus on tickets related to the logged-in user and modify the counter fields in the scoreboard area of the administrative and web interfaces. You can modify activity notification messages to include attributes from the activity log object and information on specific tickets.

Activity Notification Messages Modifications

This article contains the following topics:

Notification messages can be sent automatically when request activities occur.

Two of the fields that must be defined on the Activity Notifications Detail window are Notification Message Title and Notification Message Body. Both of these fields can contain attributes from the activity log object (alg for Requests/Incidents/Problems, chgalg for Change Orders and issalg for Issues. These three activity log objects are almost identical) and can identify the specific request related to the activity.

Formatting Attributes for Activity Notifications

Optional formatting and escaping of individual attributes can be achieved using the properties listed below. This can be useful especially if formatting HTML notification where the data in the attribute may need to be escaped to conform to HTML standards.

To include formatting, use the following syntax:

```
@{property=value property=value:attribute_name}
```

Property values pairs are separated by at least one space and are not case sensitive. A colon separates the formatting properties from the attribute name. If no properties are listed, no formatting or escaping will be done on attribute.

The following table the available formatting properties:

Property	Description
DATE_FMT	Specifies the date format for attribute. Valid values are: MM/DD/YYYY MM-DD-YYYY DD/MM/YYYY DD-MM-YYYY YYYY/MM/DD YYYY-MM-DD Valid only for Date attributes. Dates embedded in strings are not affected.

ESC_STYLE=NONE | HTML | URL	Specifies the escape type of the formatted text. Valid values are: **NONE** Default setting. Specifies that no special treatment be given to any character in the content body. **HTML** Give special treatment to the following characters, which are meaningful in HTML text: & becomes & ”_becomes " < becomes < > becomes %gt; **URL** Translate all characters other than letters, digits, and '@*-_.#' to '%xx', where xx is the hexadecimal coding of the translated character.
JUSTIFY=LEFT | CENTER | RIGHT | TRUNCATE| WRAP | LINE	Specifies the justification of the formatted text. Valid values include: **TRUNCATE** (default if formatting) Truncates text to WIDTH property value if a positive integer. If ESC_STYLE=HTML, eliminates HTML formatting by replacing '<' and '>' with < and > (see KEEPLINKS and KEEPTAGS). **LEFT|CENTER|RIGHT** Produces exactly WIDTH characters, truncated or padded with spaces as necessary, with any embedded new lines replaced by a single space. If ESC_STYLE=HTML, the output text is delimited by [set the pre variable for your book] and </pre> tags. The WIDTH argument must be specified as a positive integer. **WRAP** Same as LEFT, except that text wrapping honors word boundaries (line breaks are not placed within words). **LINE** Same as TRUNCATE, except that it also replaces all embedded line breaks with tags if ESC_STYLE=HTML. The syntax must be used in the following format: `@{justify=truncate width=30:call_req_id.summary}` You can change the width parameter accordingly.
KEEPLINKS=YES|NO	If KEEPLINKS=YES is specified, the action of JUSTIFY=LINE or JUSTIFY=TRUNCATE is modified to preserve HTML anchor tags (Action:) while converting all other '<' and '>' characters. Mutually exclusive with KEEPTAGS. Only valid if ESC_STYLE=HTML.
KEEPNL=YES|NO	The normal action of PDM_FMT is to convert all embedded new lines and any following spaces to a single space. If KEEPNL=YES is specified, embedded new lines are preserved. This argument is ignored for JUSTIFY=LINE.
KEEPTAGS=YES|NO	If KEEPTAGS=YES is specified, the action of JUSTIFY=LINE or JUSTIFY=TRUNCATE is modified to preserve all HTML tags. Mutually exclusive with KEEPLINKS. Only valid if ESC_STYLE=HTML.

PAD=YES|NO	If PAD=NO is specified, PDM_FMT does not convert empty strings to a single space. This is the normal action when WIDTH is non-zero, or JUSTIFY is TRUNCATE or WRAP.
WIDTH=*nn*	When non-zero, specifies that the text should be formatted to exactly WIDTH characters.

For example, to format the Request description for an HTML notification by escaping HTML specific characters, adding <BR> tags for line breaks and keeping any HTML Links as links, enter the following:

```
@{ESC_STYLE=HTML JUSTIFY=LINE KEEPLINKS=YES:call_req_id.description}
```

To format the open_date of a Request to European format, enter the following:

```
@{DATE_FMT=DD-MM-YYYY:call_req_id.open_date}
```

Attributes from the Activity Log Object

To include an attribute from the activity log object, include this in the Notification Message Title or Notification Message Body field:

```
@{att_name}
```

The name of the object, alg or chgalg or issalg, is the default and need not be specified. For example, to include the type of activity in the message title, enter this in the Notification Message Title field (along with the rest of what you want in the title):

```
@{type}
```

To include the description of the activity in the message body, enter this in the Notification Message Body field (along with the rest of what you want in the body):

```
@{description}
```

Information on Specific Change Orders

For messages to provide information on the specific change order that triggered the notification, the Notification Message Title or Notification Message Body fields must contain an attribute in the activity log object that references the change order object. Enter the reference in this format:

```
@{change_id.chg_att_name}
```

In this reference, the following information applies:

- **@**
  Indicates to replace this expression.
- **change_id**
  The attribute in the activity log object that links it to a specific instantiation of the change order object (chg).
- ***chg_att_name***
  Any attribute in the chg object.

For example, to include the priority of the change order in the message title, enter the following in the Notification Message Title field, along with the rest of what you want in the title:

```
@{change_id.priority.sym}
```

To identify who reported the change order (Affected End User) in the message body, enter the following in the Notification Message Body field, along with the rest of what you want in the body:

```
@{change_id.requestor.combo_name}
```

If you want to reopen a specific change order by number, and want the message to appear as follows, use the following syntax:

```
Reopen Change Order @{change_id.chg_ref_num}
```

> **NOTE**
> For messages to provide information about an issue that triggered a notification, the Notification Message Title or Notification Message Body fields must contain an attribute in the activity log object that references the issue object, iss. For more information about objects and attributes, see the CA Service Desk Manager Reference Commands section.

For example, to include the priority of the issue in the message title, enter the following in the Notification Message Title field, along with the additional information you want in the title:

```
@{issue_id.priority.sym}
```

Information on Specific Requests

For messages to provide information on the specific request that triggered the notification, the Notification Message Title or Notification Message Body fields must contain an attribute in the activity log object that references the request object. Enter this reference in this format:

```
@{call_req_id.cr_att_name}
```

- **@**
  Indicates to replace this expression.
- **call_req_id**
  The attribute in the activity log object that links it to a specific instantiation of the request object (cr).
- ***cr_att_name***
  Any attribute in the cr object.

For example, to include the impact of the request in the message title, enter this in the Notification Message Title field (along with the rest of what you want in the title):

```
@{call_req_id.impact.sym}
```

To identify the affected resource in the message body, enter this in the Notification Message Body field (along with the rest of what you want in the body):

```
@{call_req_id.affected_resource.name}
```

If you want to reopen a specific request by number, and want the message to appear as follows, use the following syntax:

```
Reopen Request @{call_req_id.ref_num}
```

There are several other mechanisms by which messages can be sent which are in the context of the request itself (or change order or issue). When the context is the request itself, you do not need (and cannot use) the "call_req_id" part of the reference. So, in these cases, you need to use:

```
"@{ref_num}" rather than "@{call_req_id.ref_num}"
```

ITIL-Specific Queries

Problems and Incidents are requests with one of two values in the *type* attribute: “I” for Incidents or “P” for Problems.

The following stored query obtains all Incidents in which the Assignee's Organization or the Group's Organization equals the logged-in Analysts Organization:

```
assignee.organization IN @cnt.organization OR group.organization IN @cnt.organization) AND active = 1 AND type
 = \'I\'
```

For Problems, the query is identical except for type = \'P\'

Scoreboard Queries

This article contains the following topics:

One of the tables in the database, Cr_Stored_Queries, defines stored queries. These stored queries, which are similar to SQL queries, can be used to customize the counter fields on nodes in the scoreboard area of the administrative and web interfaces. The counter fields tell how many records match the query. For example, they can tell how many of various types of requests have been assigned to the logged-in user.

Each user can customize the counter fields that appear on his or her scoreboard (this is explained in the online help.) However, the system administrator must first define the various types of requests that can be counted in these counter fields as stored queries.

> **NOTE**
>
> Scoreboard counts will be incorrect if database query values are equal to NULL. For example, if your Scoreboard query specifies that assignee.organization = xyz, and an assignee field is blank (NULL) for a record, then that record will not be part of the Scoreboard count.

Stored Queries for Logged in User

Two of the fields that must be defined on the Stored Query Detail window are Where Clause and Label. Both of these fields can contain expressions that are customized to the logged-in user. Stored queries refer to objects and attributes, rather than to table names and columns. A stored query that is customized to the logged-in user consists of two parts, as follows:

- **The object (such as cr for a request)**
  This is usually specified on the left of the equal (=) sign. The syntax for this part of the stored query is:
  ```
  att_name[.att_name...].SREL_att_name
  ```

A stored query always has a Type, which is an object name that the query is executed against and provides context for the query. In the syntax above, the first att_name must be an attribute name of the context object.

- **The logged-in user (the instance of the cnt object for this user)**
  This must be specified on the right of the equal (=) sign if the tickets are to be selected based on an attribute of the logged-in user. The syntax for this part of the stored query is:
  @att_name[.*att_name*...].*SREL_att_name*

  > **NOTE**
  > For more information about objects and attributes, see the CA Service Desk Manager Reference Commands section.

Syntax for cr Object

Use this syntax if the reference is to the request (cr) object:

```
att_name[.att_name...].SREL_att_name
```

This example identifies the location of the person assigned to handle a ticket. In this example, the object name is omitted, as the type of the Stored Query implies the cr object:

```
assignee.location=@cnt.location AND active=1
```

- **assignee**

The attribute in the request object that maps to the assignee field in the corresponding table. For example, the assignee attribute is defined in the cr object with SREL agt, which means it refers to the agt factory. The agt factory is part of the cnt object definition.

- **location**
  The attribute in the cnt object that maps to the c_l_id field in the Contact table. The location attribute is defined in the cnt object with SREL loc, which means it refers to the loc object.

WHERE Clause

The following example demonstrates a value you can code in a WHERE clause:

```
assignee.location=@cnt.location AND active=1
```

Given the Stored Queries type is a Request, this query selects all active requests where the assignee's location is the same as the location of the logged-in user.

Label

Attributes in the cnt object can be included in labels the same way they are included in WHERE clauses. Here is an example of the use of an attribute in the cnt object in a label:

```
@cnt.location.name Calls
```

This label will include the name of a location, for example, Phoenix, where Phoenix is substituted for @cnt.location.name when the label is displayed on a window. The label will be displayed as Phoenix Calls.

The IN Keyword

The IN keyword allows a stored query to reference two (or more) tables without creating a join. This can result in significant efficiency in executing the query. It is coded as follows:

```
SREL_att_name IN ( value1 [, value2 [,...]]  )
```

For example, a request query could be coded as:

```
category.sym IN (\'Soft%\', \'Email\')
```

This results in the following SQL WHERE clause:

```
category IN (SELECT persid FROM prob_ctg WHERE sym LIKE 'Soft%' OR sym = 'Email')
```

One use of IN is to avoid Cartesian products. For example, the following query results in a Cartesian product and is very inefficient:

```
assignee.last_name LIKE 'MIS%' OR group.last_name LIKE 'MIS%'
```

By using IN, the query does not create a Cartesian product; in fact, it creates no joins at all, as illustrated by the following example:

```
assignee.last_name IN 'MIS%' OR group.last_name IN 'MIS%'
```

> **NOTE**
> The parentheses that normally enclose the list of values on the right side of IN can be omitted if there is only one value in the list. Similarly, you should avoid joins in data partitions by converting a data partition, illustrated as follows:

```
assignee.last_name LIKE 'Smith'
to:
assignee = U'374683AA82ACE34AB999A042F3A0BA2E'
```

where:

- **U**

indicates that the value is a uuid.

- **'374683AA82ACE34AB999A042F3A0BA2E'**
  The 32 characters in single quotes indicates the string representation of an actual uuid.

This avoids the join with some loss in CA. Using IN, the same partition can be written as illustrated in the next example, with the CA of the first version and almost the same efficiency as the second version:

```
assignee.last_name IN 'Smith'
```

CA SDM supports the IN clause applied to QREL or BREL lists. For example, if you want to find all the Requests with Assets that are parents of another specific Asset (with id 374683AA82ACE34AB999A042F3A0BA2E), the appropriate where clause is as follows:

affected_resource.[parent]child_hier.child IN (U’374683AA82ACE34AB999A042F3A0BA2E’)

The first part of the clause, *affected_resource*, is an SREL (foreign key) of the cr (Request) object, pointing to the Network_Resource table. The *child_hier* portion is a list of hier objects pointing to the hierarchical relationships. The last part, *child*, forms the first part of the where clause for the IN sub query. The *374683AA82ACE34AB999A042F3A0BA2E* portion is the foreign key value to match on *child*. *[parent]*specifies thesub query return. Since the id value is a string representation of a UUID it must be indicated as such and written as U’374683AA82ACE34AB999A042F3A0BA2E’

The following is an example of the actual SQL generated, which provides all the Requests where the Asset is a parent of a specific Asset:

```
SELECT Call_Req.id FROM Call_Req WHERE Call_Req.affected_rc IN (SELECT hier_parent FROM Asset_Assignment WHERE
 hier_child = U'374683AA82ACE34AB999A042F3A0BA2E')
```

To query on multiple parents, you can provide a comma-separated list in the () portion of the SQL, as shown by the following example:

```
affected_resource.[parent]child_hier.child IN (U'374683AA82ACE34AB999A042F3A0BA2E',
 U'374683AA82ACE34AB999A042F3A0BA2E')
```

The attribute name in brackets ([]) is used to form the SELECT portion of the sub-clause. Bracket notation is not used for the group Stored Queries shipped with CA Service Desk Manager Version 6.0, as illustrated in this example:

```
(assignee = @cnt.id OR group.group_list.member IN (@cnt.id)) AND active = 1
```

> **NOTE**
> If bracket notation is not used, the SQL subsystem assumes that it is the attribute name of the first symbol in the dot-notation portion. It works in this case, more out of luck, that the group_list object has an attribute named ‘group’ in it. If it were named anything else, the where clause would fail to parse! The equivalent clause with brackets illustrated as follows:

```
(assignee = @cnt.id OR group.[group]group_list.member IN (@cnt.id)) AND active = 1
```

> **NOTE**
> You cannot extend the dot notation. For example, the following does not work:

```
affected_resource.[parent]child_hier.child.name IN ('chicago1')
```

Query Based on Priority

In the database, the Priority table has two columns named sym and enum. The value the users see are the sym values. But the application sees the sym based on the enum values. At present, the default sym values 1 to 5 are reversed in their enum value.

Example

Sym	Enum
1	5
2	4
3	3
4	2
5	1

Therefore, when writing the stored query, when you reference a value of 5, you are actually looking for priority 1 unless you use a .sym to specify which attribute to look at.

> **WARNING**
> Do not change the default enum values the product assigns. Instead, when adding new sym values, just continue from the highest enum value and so on.

Time-Based Queries

Time spans can be used to create time-based stored queries. A time span specifies a period of time, which can be relative to the current date. For example, a time span could refer to today, yesterday, last week, or last month. A time span has a name, such as TODAY or YESTERDAY. You refer to a time span in a stored query by using either of two built-in functions, as follows:

- **StartAtTime (*timespan-name*)**
  This refers to the beginning of the period described by the time span.
- **EndAtTime (timespan-name)**
  This refers to the end of the period described by the time span.

The syntax rules for stored queries require that the time span name be enclosed in single quotes, with each single quote preceded by a backslash. For example, to refer to the beginning of last week, you would specify:

```
StartAtTime(\'PAST_WEEK\')
```

The passage of time makes it necessary to periodically refresh a stored query containing a reference to a time span. For example, the interval described by “yesterday” changes at midnight. You specify the Start Time, End Time, and Trigger Time for refreshes in the Timespan Detail window.

Start Time

Start Time specifies the beginning of the time span in absolute or relative terms. The following table describes the fields within the Start Time section of the Timespan Detail window:

- **Year**
  An explicit year, such as 2000, or a relative year, such as +1 (next year) or - 1 (last year)
- **Month**
  An explicit month from 1 (January) to 12 (December), or a relative month, such as +1 (next month) or - 1 (last month)
- **Day**
  An explicit day from 1 to 31, or a relative day, such as +1 (tomorrow) or - 1 (yesterday)
- **Hour**
  An explicit hour from 0 to 24, or a relative hour, such as +1 (next hour) or - 1 (last hour)
- **Minute**
  An explicit minute from 0 to 59, or a relative minute, such as +1 or - 1

End Time

End Time specifies the end of the time span in absolute or relative terms. The End Time fields of the Timespan Detail window are the same as the Start Time fields of the Timespan Detail window.

Trigger Time

The Trigger Time field specifies when the WHERE clause of a stored query containing a reference to the time span is recreated and the stored query refreshed. Trigger Time must be relative to the current time as described in the following table:

- **Year**
  Must be a relative year from - 1 (last year) to +36 (36 years from now).
- **Month**
  Must be a relative month from - 1 (last month) to +11 (11 months from now).
- **Day**
  Must be a relative day from - 1 (yesterday) to +31 (31 days from now).
- **Hour**
  Must be a relative hour from - 1 (last hour) to +23 (23 hours from now).
- **Minute**
  Must be a relative minutes from +9 (9 minutes from now) to +59 (59 minutes from now).

Web Interface Modifications

This article contains the following topics:

The CA SDM web interface(also referred to as the browser interface) provides you with CA SDM functionality through the Internet. This functionality includes the ability to open, update, or close tickets, display and post announcements, and access supporting data tables. It enables independent browsing of the knowledge base to help reduce the number of calls to the service desk and speeding resolution times. The web interface can be fully customized and can be used with many web browsers.

If you installed and configured the web interface, you can integrate it into your existing web interface and customize it to suit your needs. For customization, be familiar with HTML and the web browser in use at your site.

> **NOTE**
> WSP Design view works for CA SDM controls (PDM_MACROs). When working on forms that do not contain CA SDM controls, you can only work on the Source tab. The Employee and Customer web forms do not contain CA SDM controls and therefore appear on the Source tab rather than the Design tab. Some Analyst forms do not contain CA SDM controls, and therefore would appear on the Source tab too.

> **WARNING**
> CA SDM no longer uses any customized .mac files for pdm_macro in the NX_ROOT\site\mods\www\macro directory. You cannot customize macros.

> **WARNING**
> Technical support cannot provide assistance with design or debugging of customizations (including documentation, such as online help systems). We provide general information for customizing the CA SDM web interface. When doing so, be aware that you are solely responsible for your own customizations. CA SDM technical support can assist you in interpreting and understanding customization.

Support for the customization techniques here extends to helping ensure that the techniques and facilities perform as documented. Be careful not to exploit undocumented features or to extend documented features beyond their documented capabilities. Such exploitation is not supported and can result in system problems or instability that may appear unrelated to the customization. For this reason, support may ask you to remove customizations to reproduce the problems. Sites should prepare for this eventuality by carefully following the guidelines on placing all modifications in the site mods directory tree and maintaining change logs. Sites that make frequent, complex, or extensive changes should

consider approaching CA SDM customization as a software engineering project with disciplined source control, testing, and controlled releases to production.

Migrating customizations between releases can present unique challenges, and we have developed the product in ways to preserve the efforts put into customization. In addition, if Level Two support supplies a patch to a system, the patch is written with these same assumptions. Patching or upgrading a system with undisciplined customizations is a risky undertaking that often results in costly system down time. Avoid it by following these guidelines and practising sound software engineering principles.

Customize the Scoreboard

The Scoreboard on this tab allows you to view the requests, change orders, issues, call backs, and tasks assigned to you or your group. You can customize your Scoreboard to display only specific folders and nodes. By default, the Scoreboard displays nodes that itemize the records for the entire application. These records can include requests, issues, change orders, assets, configuration items, callbacks, documents, and tasks.

Follow these steps:

1. Select File, Customize Scoreboard.
   The Customize Scoreboard window opens.
2. Select the Scoreboard you want to customize from the following options:
   UserDisplays the Scoreboard for the username under which you are logged in. RoleAllows you to customize the Scoreboard for all members of one type of user, such as administrators. Select a value from the drop-down list.
3. Navigate the Scoreboard tree and select an item to work with.
   To update or delete an item, select the item. To add a new node or folder, select the item after which you want to add the new item.

   NOTE

   : The Add Node and Add New Folder options are disabled when the system reaches the Scoreboard limit. To add more nodes or folders, you can remove them or ask your administrator to use Options Manager to increase the scoreboard_entry_limit.
4. Complete the entry fields for the task you want to complete. Depending on the task, you can edit the following fields:
   a. **Node Label**: Displays the name of the node that you want to add or update.
   b. **Node's Stored Query**: Shows the stored query of the node that you want to add or update. Enter a value directly or select the Node's Stored Query link to search for a stored query.
   c. **Folder Label**: Displays the name of the folder that you want to add.
5. Click the button for the task you want to complete.
   The Scoreboard tree refreshes to display your changes.
6. (Optional) Click Reset Tree to undo your changes to the Scoreboard tree.
7. Click Finished.
   The Customize Scoreboard window closes and the Scoreboard displays your changes.

Set Preferences

Setting preferences allows you to define the desired default behaviors.

Follow these steps:

1. Choose View, Preferences in CA SDM Web UI.
   The Preferences window appears.
2. Set the appropriate preferences
   For example, select the Using Screen Reader option.

NOTE

For more information about using a screen reader with CA SDM, click Help, Screen Reader Usage.

3. Click Save to save the preferences. The new settings take effect immediately.

Preferences Fields

General Settings

- **Avoid Popups**
  Opens new forms in the main browser window whenever possible, reducing the number of popups.Note: If this option is selected, the Back to List button is activated and is used to navigate from detail forms back to the previously displayed list window. The Back to List button appears in the upper right-hand side of the window.
- **Display Score Count**
  Displays the score count as left-justified.Keep Log Reader WindowKeeps the Log Reader window open when you close all popups, and when you log off. This setting has no effect when the Log Reader window is not open.
- **Preserve Popup Size**
  Causes new popup windows to open with the same dimensions as the most recently resized popup window. The popup size that is preserved is dependent on the window size used by that type of popup (Large, Medium, Small, or Xsmall). For example, if you resize a Large popup, such as a Request detail page, the new size is preserved for all Large popups. If you resize a Medium popup, such as an activity log page, the new size is preserved for all Medium popups.Note: If you maximize a popup window, subsequent popup windows may cover any other window you have open. However, new popup windows appear slightly off the screen, to the right and lower. This is because there is a 10 pixel (left and top) offset for popups to prevent them from completely overlaying the currently displayed window. We recommended that you do not maximize popup windows when using this option.
- **Mouseover Menus**
  Causes a menu to display when the mouse pointer is over the menu's link, without clicking the mouse button. You must reload any active forms for this setting to take effect on the page.
- **Using Screen Reader**
  Modifies system behavior for optimal use with a screen reader for blind and limited vision users. You must log off and log back on for this change to take effect. From the Help menu, select Screen Reader Usage for an overview of using CA SDM with a screen reader.Use Default RoleUses the default role assigned to a Contact or a Contact's Access Type as the initial role when the Contact logs in.
- **Disable Mouseover Previews**
  Select this check box if you do not want to see preview forms pop up automatically.Note: Mouse-over previews are disabled automatically when either Using Screen Reader or Mouseover Menus is enabled.

Support Automation Analyst UI Localization

English is the default language.

Knowledge Search Document Settings

- **Search Type**
  Select either Keyword Search or Natural Language Search.
- **Match Type**
  Select the default method to use for text matching during a search. Possible values are Any of the Words (OR), All of the Words (AND), and Exact Phrase.

  WARNING

  Match Type and Match preferences only set the default search criteria when you search in Knowledge Management. For example, you log in as an analyst and click the Knowledge tab of any ticket. Knowledge searches from within a ticket always default to Match Type=Any of the words (OR) and Match=Whole Words, regardless of your preference settings.
- **Match**

Select the default method by which CA SDM searches documents, either Whole Words or Words Beginning With.

- **Order By**
  Select the default property for sorting retrieved documents.
- **Search In**
  Select the document fields where you want to search for specified keywords. These options only display when Keyword Search is the selected search type.

Knowledge Document List Settings

- **Documents Per Page**
  The number of documents (10, 25, or 50) that the product that display on each page of the Knowledge Document List pane.
- **Show Document List Details**
  If selected, the Knowledge Document List pane on the Knowledge tab includes the following detailed information:
  Title
  Summary
  Document ID
  Modify Date
  If not, only the document title displays.

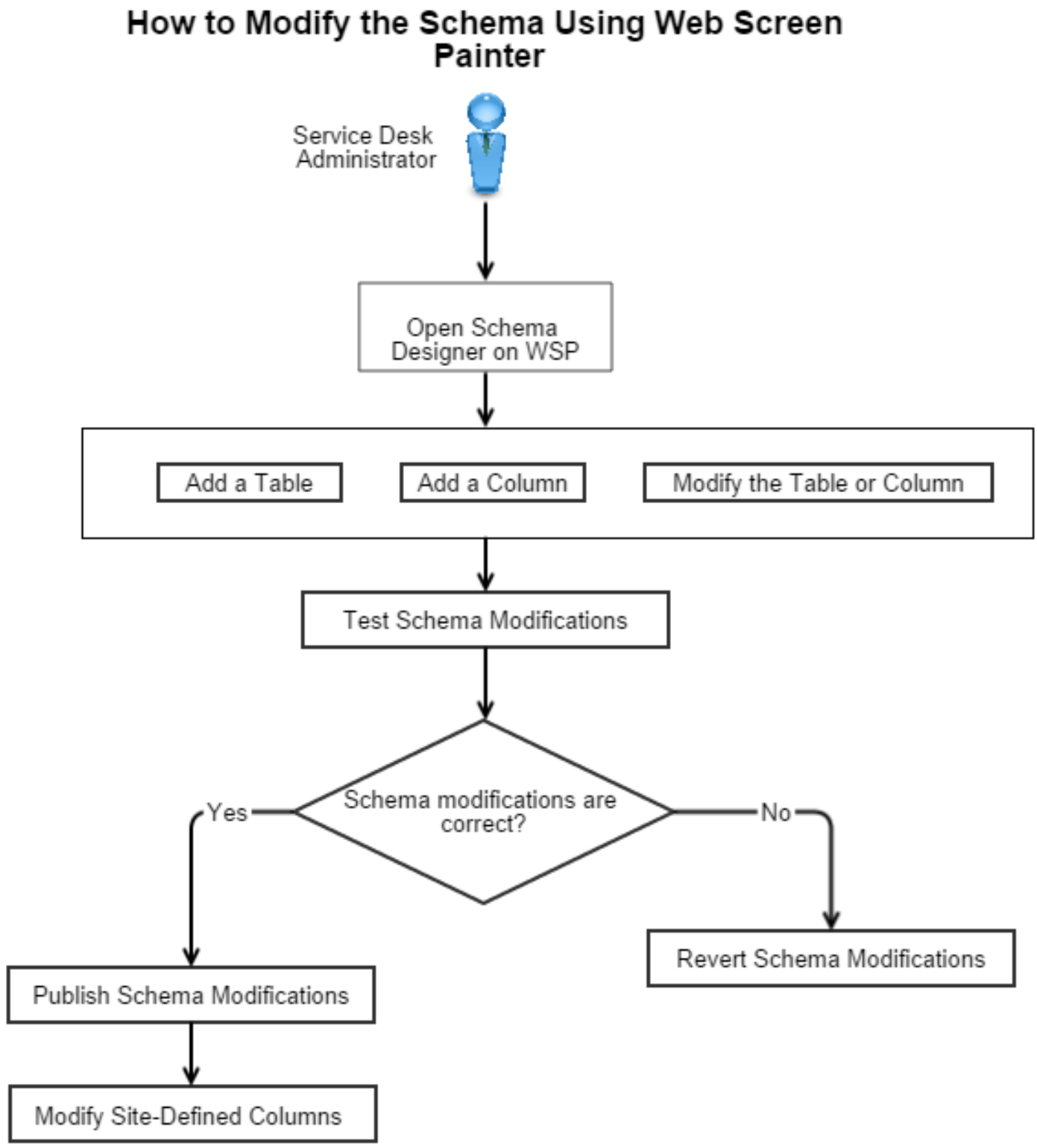
Knowledge Categories Document List Settings

- **Documents Per Page**
  The number of documents (10, 25, or 50) that the product that display on each page of the Knowledge Document List pane.
- **Attributes to be shown in list**
  Select the properties to be listed for each document in the Knowledge Document List pane. Select the desired attributes from the Available list and click the arrows to move them to the Selected list.

How to Modify Schema Using Web Screen Painter

Use the Schema Designer of Web Screen Painter to modify the database schema of CA SDM. Schema Designer provides a graphical user interface to review and modify this schema. The following diagram shows how to modify the schema using Web Screen Painter:

Figure 30: Diagram depicting how to customize schema using WSP



Follow these steps:

1. Open Schema Designer on WSP.
2. Add a Table or Add a Column or Modify the Table or Column.
3. Test Schema Modifications.
   - If the schema modifications are correct, Publish Schema Modifications.
   - If the schema modifications are incorrect, Revert Schema Modifications.
4. (If necessary) Modify Site-Defined Columns After Publishing.

Open Schema Designer on WSP

To start working on the Schema Designer, ensure that you have WSP installed on the CA SDM server. For more information about the WSP installation, see Install Web Screen Painter.

Follow these steps:

1. Log in to the following CA SDM server where WSP is installed, depending on your CA SDM configuration:
   - Conventional: Primary server
   - Advanced availability: Background server
2. Start WSP using any of the following actions, depending on the operating system that is installed on the CA SDM server:
   - (Windows) From the Start menu, select Program Files, CA, CA SDM, Web Screen Painter.
   - (UNIX) Enter the command pdm_wsp with $NX_ROOT/bin in your path.

   The Web Screen Painter login window opens.
3. Enter your login credentials.
4. Select Tools, Schema Designer.
   The Schema Designer window opens. The left side of the Schema Designer window shows the CA SDM database in a tree format. The tables and columns are displayed by their Object Name. If the Display Name differs from the Object Name of the table or column, the Display Name is displayed in parentheses along with the Object Name.

Add a Table

Use the Schema Designer to add a table in the database.

Follow these steps:

1. Select Edit, Add Table.
   The Add New Table dialog opens.
2. Enter the table name in the New Table Name field and click OK. Ensure that you begin the name of a site-defined table with the letter z to prevent conflict with possible future standard tables.
   WSP adds a z to the beginning of the table name if you do not add.
3. Complete the following fields, as appropriate:
   - **Name**
     (Read-only) Specifies the object name of the table. For example, the object name of the cr table is cr.
   - **Display Name**
     Specifies the user-friendly name of the table. For example, the Display Name of the cr table is Request. You can change the Display Name of a table by entering a new name in this field.
   - **Schema Name**
     (Read-only for standard tables) Specifies the name used to refer to the table in CA SDM utilities, such as pdm_userload. For site-defined tables, Schema Name defaults to the Object Name. You can change the Schema Name by entering a new value in this field.
   - **DBMS Name**

Specifies the name used to refer to the table in the physical DBMS. This field is read-only for all tables. For site-defined tables, it is always the same as Schema Name.

- **Default Display Field (common name)**
  Specifies the column displayed on the UI for a field that references this table. For example, the assignee field of a request is a reference to the Contact table. Because the common name of the Contact table is combo_name (last, first middle), the combo name of the referenced contact displays as assignee. You cannot change the value of common name.
- **Foreign Key Field (rel attr)**
  Specifies the column stored in the database for a field that references this table. For example, the assignee field of a request is a reference to the Contact table. Because the rel attr of the Contact table is id, the assignee column in a Request contains the id of the referenced contact. You cannot change the value of rel attr.
- **Function Group**
  Specifies the name of the group that controls the level of access that users have to records in this table. Each access type of a contact specifies whether they have read, modify, or no access to data in tables in each function group. You can change the value of rel attr by selecting a new value from the drop-down list.

  NOTE
  The Schema Designer includes an Advanced tab. Information on this tab is intended for CA Technologies Support and field representatives. You will not need to work with this tab for most uses of the Schema Designer, and it will not be discussed further in this document.

The Advanced tab of the Schema Designer column dialog shows information meaningful only to persons with internals knowledge of CA SDM. Its contents vary with the column type. We recommend that you modify the values in this tab only on direction from a CA Technologies employee. It contains the following fields if a table is selected:

- **Active Triggers**
  Enter the triggers currently active in the Object Engine for the column. A trigger is a small program that runs under the direction of the Object Engine at certain times when a column is modified. The active trigger list includes both standard and site-defined triggers (so that site-defined triggers are listed both in the active list and in the site-defined list).
- **Site-Defined Triggers**
  Enter the triggers that have been installed at your site. For example, these may be triggers that have been activated as a result of installing a CA SDM option, or triggers that have been written for your site by CA Technical Services. The active trigger list includes both standard and site-defined triggers (so that site-defined triggers are listed both in the active list and in the site-defined list).
- **LRel Specification**
  Enter the specification of a many-to-many relationship between two tables, in the form:
  table1 column1 <> table2 column2
  This shows that the LREL is a many-to-many relationship between table1 and table2, with virtual column column1 containing the relationship in table1, and virtual column column2 containing the relationship in table2.
- **Active xRel Query Information**
  Enter the active specification of the query that defines a BREL or QREL virtual column. The specification is in Object Engine Majic format.
- **Site-Defined xRel Query Information**
  Modify the query associated with a BREL or QREL. It is recommended that you enter data into this field only under the direction of a CA employee.
- **Active Derived Expression**
  Enter the active specification of the expression that the Object Engine uses to construct the value of a DERIVED virtual column.
- **Site-Defined xRel Query Information**
  Modify the expression associated with a DERIVED column. It is recommended that you enter data into this field only under the direction of a CA employee.

The Domsets tab of the Schema Designer Table dialog contains the following:

- **MLIST_DYNAMIC**
  A dynamic domset with no WHERE clause.
- **MLIST_STATIC**
  A static domset with no WHERE clause.
- **RLIST_DYNAMIC**
  A dynamic domset with a WHERE clause defined in the STANDARD_LISTS section of the FACTORY statement.
- **RLIST_STATIC**
  A static domset with a WHERE clause defined in the STANDARD_LISTS section of the FACTORY statement.

If you double-click one of the Domsets, the Properties dialog opens and contains the following fields:

- domset_name
- fetch_columns
- max_fetch
- sort columns **(**This is the only editable field)
- volatility
- where

4. Do one of the following to save the table:
   - If you are working on the test system, select File, Save.
   - If you are working on the production system, select File, Save and set to Test Mode.
     This selection saves your changes in the database, and creates a file (wsptest.mods) on the server defining your changes to the Object Engine. This file is stored in the site/mods/majic subdirectory of your CA SDM installation directory. After creating the wsptest.mods file, WSP causes its Object Engine to recycle so that it will use the new changes. This may take from a few seconds to a couple of minutes, depending on the complexity of your schema. A message is prompted. Click Yes to continue. The wsptest.mods file affects only the Object Engine designated by the wsp_domsrvr option. Other Object Engines on the same server do not process this file, and the file is not distributed to other servers. In addition, new tables and columns in Test mode are defined to the Object Engine as local objects. This means that the Object Engine knows about them and you can use them on web forms. However, they do not exist in the database, and do not affect other users. Typical CA SDM users do not use WSP Object Engine, so they are unaffected by the schema modifications you are testing. The table is added.

Add a Column

Use the Schema Designer to add a column in the database.

Follow these steps:

1. Select the table for which you want to add a column (or select any of its existing columns).
2. Select Edit, Add Column.
   The Add New Column dialog opens.
3. Enter the column name in the New Column Name field and click OK. Ensure that you begin the names of a column with the letter z to prevent conflict with possible future standard columns.
   WSP verifies that you added the prefix, but adds a z to the beginning of the column name if necessary.
4. Complete the following fields as appropriate:
   - **Name**
     (Display-only) Specifies the object name of the column. For example, the object name of the Contact alt_phone column is alt_phone.
   - **Display Name**
     Specifies the user-friendly name of the column. You can change the Display Name of a column by entering another name in this field. For example, the display name of the Contact alt_phone column is alternative phone.
   - **Schema Name**

(Read-only for standard tables) Specifies the name used to refer to the column in CA SDM utilities, such as pdm_userload. For site-defined tables, Schema Name defaults to the Object Name. You can change the Schema Name by entering another value in this field.

– **DBMS Name**
(Read-only for all tables) Specifies the name used to refer to the table in the physical DBMS. For site-defined tables, the DBMS Name equals the same as Schema Name.

– **Description**
Provides a brief description of the column.

– **Field Type**
(Read-only for all standard columns in standard tables, and saved site-defined columns) Specifies the data type of the column. You can specify or change the field type of new site-defined columns by selecting a value from the drop-down. The following list describes the available Field Types:

- **INTEGER**
  Indicates a numeric value.
- **STRING**
  Indicates a text string. The String Length field indicates the number of characters in a string.
- **DATE**
  Indicates a date and time. The integer value stored in the database contains the number of seconds since midnight on January 1, 1970.
- **DURATION**
  Indicates a period of time. The value stored in the database is an integer containing a number of seconds.
- **DOUBLE**
  Indicates a real (floating point) number.
- **SREL**
  Indicates a foreign key reference to another table. The SREL Table field specifies the referenced table. The value stored in the database is the rel attr of the referenced table, which can be either an integer or a string. The value displayed in the product is the common name of the referenced table row. For information on setting SREL attributes with foreign key values, see CA Service Desk Manager Reference Commands.
- **BREL**
  Indicates a virtual column representing the set of all objects with an SREL to this table. It exists only in the Object Engine and is not physically stored in the database. Select this field type only on direction from a CA Technologies employee.
- **QREL**
  Indicates a virtual column representing a set of objects selected by the where clause on the Advanced tab. It exists only in the Object Engine and is not physically stored in the database. Select this field type only on direction from a CA Technologies employee.
- **DERIVED**
  Indicates a virtual column constructed by the Object Engine from the values of other columns, under the direction of a formula specified on the Advanced tab. It exists only in the Object Engine and is not physically stored in the database. Select this field type only on direction from a CA Technologies employee.

– **String Length**
The length of a string column. This field is blank for non-string columns. It is read-only for all standard columns, and for site-defined columns that have been saved. You can specify or change the length of new site-defined STRING columns by entering an integer between 1 and 32767 in this field.

– **SRel Table**
The table referenced by an SREL column. This field is blank for non-SREL columns. It is read-only for all standard columns, and for site-defined columns that have been saved. You can specify the table referenced by a new site-defined SREL by selecting it from the drop-down list.

– **On New Default**
The default value assigned to this column when a new row of the table is defined. It should be a value appropriate to the field type. Some keyword values are available for particular field types:

- **NOW**
  Specifies the current date and time for a DATE column.
- **USER**
  Specifies the active user for an SREL to the Contact table.

– **On Save Set**
The value assigned to this column when a row of the table is updated. It should be a value appropriate to the field type. Some keyword values are available for particular field types:
- **NOW**
  Specifies the current date and time for a DATE column.
- **USER**
  Specifies the active user for an SREL to the Contact table.

– **Required**
When checked, this option indicates that a value must be supplied for the column before a row of the table containing it can be saved. You can set this option for both standard and site-defined columns, and you can disable an option that you have set. However, you cannot turn off the option of a standard column unless it was set by your site.

– **Updatable only for new record**
When checked, this option indicates that a value for this column can be provided only when a row of its table is initially created, and cannot thereafter be changed. You can set this option for both standard and site-defined columns, and you can disable an option that you have set. However, you cannot turn off the option of a standard column unless it was set by your site.

– **Key for pdm_userload**
When checked, this option indicates that this column is one of the columns tested by pdm_userload to determine whether or not its input is an update to an existing row. This option is available only for STRING columns. It is read only for all columns in standard tables.

– **DBMS Index Options**
These options specify characteristics of a column that is an index of the physical DBMS. They are available only for columns in site-defined tables.
- **Unique**
  Specifies that the column is unique within the table and that no two rows have the same value for the column.
- **Ascending**
  Specifies that the DBMS index is listed in ascending sequence by this column. Mutually exclusive with Descending.
- **Descending**
  Specifies that the DBMS index is listed in descending sequence by this column. Mutually exclusive with Ascending.

  WARNING
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The Advanced tab of the Schema Designer column dialog shows information meaningful only to persons with internals knowledge of CA SDM. Its contents vary with the column type. We recommend that you modify the values in this tab only on direction from a CA Technologies employee. It contains the following fields if a table is selected:

– **Active Triggers**
Enter the triggers currently active in the Object Engine for the column. A trigger is a small program that runs under the direction of the Object Engine at certain times when a column is modified. The active trigger list includes both standard and site-defined triggers (so that site-defined triggers are listed both in the active list and in the site-defined list).

– **Site-Defined Triggers**

Enter the triggers that have been installed at your site. For example, these may be triggers that have been activated as a result of installing a CA SDM option, or triggers that have been written for your site by CA Technical Services. The active trigger list includes both standard and site-defined triggers (so that site-defined triggers are listed both in the active list and in the site-defined list).

- **LRel Specification**
  Enter the specification of a many-to-many relationship between two tables, in the form:
  table1 column1 <> table2 column2
  This shows that the LREL is a many-to-many relationship between table1 and table2, with virtual column column1 containing the relationship in table1, and virtual column column2 containing the relationship in table2.
- **Active xRel Query Information**
  Enter the active specification of the query that defines a BREL or QREL virtual column. The specification is in Object Engine Majic format.
- **Site-Defined xRel Query Information**
  Modify the query associated with a BREL or QREL. It is recommended that you enter data into this field only under the direction of a CA employee.
- **Active Derived Expression**
  Enter the active specification of the expression that the Object Engine uses to construct the value of a DERIVED virtual column.
- **Site-Defined xRel Query Information**
  Modify the expression associated with a DERIVED column. It is recommended that you enter data into this field only under the direction of a CA employee.

The Domsets tab of the Schema Designer Table dialog contains the following:

- **MLIST_DYNAMIC**
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  A static domset with no WHERE clause.
- **RLIST_DYNAMIC**
  A dynamic domset with a WHERE clause defined in the STANDARD_LISTS section of the FACTORY statement.
- **RLIST_STATIC**
  A static domset with a WHERE clause defined in the STANDARD_LISTS section of the FACTORY statement.

If you double-click one of the Domsets, the Properties dialog opens and contains the following fields:

- domset_name
- fetch_columns
- max_fetch
- sort columns (This is the only editable field)
- volatility
- where

5. Do one of the following to save the column:
   - If you are working on the test system, select File, Save.
   - If you are working on the production system, select File, Save and set to Test Mode.
     This selection saves your changes in the database, and creates a file (wsptest.mods) on the server defining your changes to the Object Engine. This file is stored in the site/mods/majic subdirectory of your CA SDM installation directory. After creating the wsptest.mods file, WSP causes its Object Engine to recycle so that it will use the new changes. This may take from a few seconds to a couple of minutes, depending on the complexity of your schema. A message is prompted. Click Yes to continue. The wsptest.mods file affects only the Object Engine designated by the wsp_domsrvr option. Other Object Engines on the same server do not process this file, and the file is not distributed to other servers. In addition, new tables and columns in Test mode are defined to the Object Engine as local objects. This means that the Object Engine knows about them and you can use them on web forms. However, they do not exist in the database, and do not affect other users. Typical CA SDM users do not use WSP Object Engine, so they are unaffected by the schema modifications you are testing.

The column is added to the table.

6. To access the added attributes through Rest API, run the following command:
   - `pdm_rest_util -newdeploy`

     or
   - `pdm_rest_util -undeploy`

     and then

     `pdm_rest_util -deploy`
7. Restart the CA SDM services.

Modify the Table or Column

To modify information about a table or column, click table or column on the Schema Designer, and enter the new information in the appropriate fields. The information you can modify depends on the status of the table or column:

- **Standard Tables**
  Lets you modify the Display Name, Description, and Function Group fields.
- **Standard Columns**
  Lets you modify Display Name, Description fields, the On New Default Value, and the On Save Set value. In addition, if the check boxes for Required or Updatable only for new record are not selected, you can select them. You cannot remove these options if they are set by default. However, you can reverse your own changes.
- **Site-Defined Table**
  If the table is not published, you can modify all fields, except Name, which cannot be changed after the new table has been saved. After a site-defined table has been published, you can modify only the Display Name, Description, and Function Group fields.
- **Site-Defined Column**
  If the column is published, you can modify all fields, except Name, which cannot be changed after the new column has been saved. After a site-defined column has been published, you can modify only the Display Name and Description fields, the On New Default Value, the On Save Set value, and the check box for Required, Updateable only for new record, Key for pdm_userload, and the DBMS index options.

Test Schema Modifications

You can test your schema modifications and create, update, and view web forms using them before making any changes to the physical database. Putting schema changes in Test mode defines them to the Object Engine, but does not physically store their data in the database. Because putting schema modifications in Test mode has the potential of impacting other users, this option is available only if your installation has installed the wsp_domsrvr and wsp_webengine options to dedicate an Object Engine to WSP.

To put schema changes in Test mode, select Save and set to Test Mode from the File menu. This selection saves your changes in the database, and creates a file on the server (where you are logged in) defining your changes to the Object Engine. This file is called wsptest.mods, and is stored in the site/mods/majic subdirectory of your CA SDM installation directory.

After creating the wsptest.mods file, WSP causes the Object Engine to recycle so that it will use the new changes. This may take from a few seconds to a couple of minutes, depending on the complexity of your schema.

Once the Object Engine has completed recycling, your schema modifications are available, and can be used and test on web forms created with WSP. Data in table and columns in test mode is saved only in internal storage in the WSP Object Engine. It cannot be seen by, and does not affect, other users of the system.

Publish Schema Modifications

After you are satisfied with your schema modifications, you can make them available to all users by publishing them. WSP stores your new or updated tables and columns in the wsptbl and wspcol tables of the database, respectively.

Follow these steps:

1. Create or update files describing the modified schema to the Object Engine and to CA SDM utility programs. WSP creates the following files on the web engine designated by the wsp_webengine option (which defaults to web:local):
   - **wsp.mods**
     Describes all Web Screen Painter-maintained schema changes to the Object Engine.
   - **wsp_schema.sch**
     Describes all Web Screen Painter-maintained tables and columns.
   - **wsp_index.sch**
     Describes DBMS indexes for Web Screen Painter-maintained tables.
   - **wsp.altercol**
     Names new columns created by WSP but not yet defined to the DBMS.
   - **wsp.altertbl**
     Names new tables created by WSP but not yet defined to the DBMS. In addition, WSP distributes the wsp.mods file to all CA SDM servers with an Object Engine.
2. Select File, Save and Publish.
   The necessary files are created on the CA SDM servers, but does not recycle any of them. Thus, the new files have no immediate impact. However, after the files are created, they will be used the next time CA SDM services are recycled.
3. If you are using the *conventional configuration*, complete the following steps:
   - Shut down the CA SDM services on the primary server and run the following command:
     ```
     pdm_publish
     ```
     This command modifies the physical DBMS to contain information about the new schema.

     WARNING
     Running pdm_publish process has a significant impact on other users. Ensure that you carefully plan the publishing of the schema changes. We recommend you use CA SDM Change Orders to schedule and obtain approval for your planned schema publication.
4. If you are using the *advanced availability configuration,* complete the following steps:
   a. Execute the following command on the background server to notify all active users using Support Automation to save their work:
      ```
      sa_server_notifier [-h] | [-q seconds] | [-c]
      ```
      - **-h**
        Displays the help page.
      - **-q seconds**
        This option notifies a local server (background) to quiesce in a specified time interval. This interval is the number of seconds before the server goes offline. This option cannot be used for a standby server or application server.
      - **-c**
        This option cancels a previously sent quiesce request.

      A pop-up message is displayed to all the active users using Support Automation on the background server. This message notifies the users about the server shutdown and the scheduled time left for the shutdown. The users must save their work and logout within that scheduled time.
   b. Shut down the CA SDM services on the background server.

      WARNING
      Do not restart the CA SDM services on standby or application servers after "save and publish" is performed from WSP. This action corrupts the advanced availability configuration. If the CA SDM services on standby or application servers are stopped and you want to start the services, run the pdm_server_control -v command on the servers to suppress the version control before starting the CA SDM services.

WARNING
If the background server fails during the publishing activity, ensure that you recover the WSP changes. For more information, see the Recover WSP Changes During Background Server Failure topic.

c. Execute the following command on the standby server that you wish to promote as the new background server:

```
pdm_server_control  -b
```

- **-b**
  Notifies a local standby server to become the background server. The standby server must already be running. If the server is not running, it is started but no failover is performed; to start a failover, run the command again.

The background server shuts down automatically and the standby server is promoted as the new background server. This change does not affect the end-user sessions. The in-progress updates (if any) are stored and delayed, until the new background server comes online.

d. Run the following command on the original background server (now the standby server) to update the DBMS with the schema changes:

```
pdm_publish
```

The pdm_publish command creates a control file that causes the next CA SDM startup to suppress synchronizing the standby server with the background server. This action is necessary to preserve the schema file changes made by pdm_publish. This command optionally performs the second fail-over after successful publishing of the schema changes. The following message is prompted to the user at the end of successful publishing:

```
Do you want pdm_publish to start CA Service Desk Manager in this standby server and perform fail-
over(Y/N)?
```

- If you enter Y, pdm_publish starts the CA SDM services on the standby server and performs fail-over automatically. Skip to step g to apply the schema changes on all the application servers.
- If you enter N, go to Step e.

e. Start CA SDM services on the standby server (original background server).
The startup detects the control file that is created by pdm_publish, but does not synchronize the standby server with the background server. This lack of synchronization preserves the changes made by pdm_publish for this startup.

WARNING
Ensure that you follow these directions exactly, as the failure to failover to the original background server after a pdm_publish results in corrupted services.

f. Run the following command on the standby server (original background server) to make it the background server again:

```
pdm_server_control  -b
```

This command also deletes the control file, so that version control works normally when this server again becomes a standby server.

g. Execute the following command on the application servers:

```
pdm_server_control -q interval -s server_name
```

- **-q interval -s server_name**
  Notifies a local or remote application server to quiesce in a specified time interval. This interval is the number of seconds before the server goes offline. When using this option without a server_name, the local server is notified to quiesce. This option cannot be used for a background or a standby server.

A pop-up message is displayed to all the active users on the specified application server. This message notifies the users about the server shutdown and the scheduled time left for the shutdown. The users must save their work and logout within that scheduled time. The users log on to the updated application server to resume their work.

h. Restart all standby servers.

Recover WSP Changes During Background Server Failure

You can recover the MDB schema changes when the background server fails during the publishing activity.

WARNING
We recommend you not to perform the recovery steps directly on the production environment. Ensure that you first validate them on the test or development environment.

Follow these steps:

- If the background server has crashed before publishing, the last saved schema changes are preserved in MDB. You log in to the new background server and resume your publishing tasks.
- If the background server crashed after publishing, perform the following actions:
  a. Stop CA SDM services on the crashed background server.
  b. Choose a standby server that you want to promote as the new background server.
  c. Copy the following files from crashed background server to the same location on the standby server:
     - "NX_ROOT/site/mods/majic/wsp.mods"
     - "NX_ROOT/site/mods/wsp.altertbl"
     - "NX_ROOT/site/mods/wsp.altercol"
     - "NX_ROOT/site/mods/wsp_index.sch"
     - "NX_ROOT/site/mods/wsp_schema.sch
  d. Run the following command on the standby server to publish the schema changes and to perform the automatic fail-over:
     ```
     pdm_publish
     ```
  e. Select Y on the message prompt that appears after successful publishing of schema changes.
     The CA SDM services are started on the standby server.

     NOTE
     If automatic fail-over does not occur, run the pdm_server_control -b command from the standby server to promote it as the new background server.
  f. Quiesce and recycle each application server. Restart all standby servers. For more information, see the Publish Schema Modifications topic.

Revert Schema Modifications

If you change your mind about your schema modifications after putting them in test mode, you can revert back to the published, version of the schema. Reverting schema modifications has the potential of impacting other users. For this reason, this option is available only if your installation has installed both the wsp_domsrvr and wsp_webengine options to dedicate an Object Engine and a Web Engine to WSP.

Follow these steps:

Select File, Revert Test Mode.
WSP deletes the wsptest.mods file, causing WSP Object Engine to revert its schema back to the published version.
After deleting the wsptest.mods file, WSP causes its Object Engine to recycle so that it can rebuild its internal schema. This may take from a few seconds to a couple of minutes, depending on the complexity of your schema.
After the Object Engine has completed recycling, the active schema is back to its published version.

NOTE
Web forms modified to work with the new schema are not automatically reverted, and may not work correctly when used with the published schema.

Modify Site-Defined Columns after Publishing

After site-defined schema modifications are published, WSP treats them similarly to the standard schema and restricts further changes. You can delete a site-defined column or can change the length of a site-defined string column by manually updating the DBMS and schema external to WSP. Then you run the pdm_wspupd script to update the database wspcol table to synchronize WSP with the external changes.

Follow these steps:

1. Log in to the following server, depending upon your CA SDM configuration:
   - Conventional: Primary server
   - Advanced availability: Background server
2. Find the site/mods subdirectory in your CA SDM installation directory.

   TIP

   Create a backup of the wsp_schema.sch file, at a location other than the site/mods subdirectory, to avoid processing the duplicate files.
3. Edit the wsp_schema.sch file to delete unwanted site-defined columns or change the length of site-defined STRING columns. These updates are the only changes supported by this procedure. You can use any standard text editor to edit the wsp_schema.sch file.

   WARNING
   If any of the index options (such as, UNIQUE) were specified for deleting a column, edit the wsp_index.sch file and remove references to the column. If the column was the only indexed column in the table, remove all references to the table from wsp_index.sch.
4. Edit majic/wsp.mods file with the same changes made to wsp_schema.sch:
   - Delete unwanted site-defined columns.
   - Change the length of site-defined STRING columns.
5. Enter the following command using the command prompt:

   ```
   pdm_wspupd
   ```

   The pdm_wspupd script reads wsp_schema.sch and compares it with the wspcol table in the database, writing a line to the console for any differences. For example, see the following output:

   ```
   PDM_WSPUPD - Update wspcol table from wsp_schema.sch
   Reading wsp_schema.sch to for current DBMS information...
   Reading wspcol table for WSP schema information...
   String column zSalesOrg.description length changed from 350 to 400
   Column zSalesOrg.sym not found in wsp_schema.sch - deleting wspcol row
   pdm_wspupd found 1 WSP-maintained column(s) to update and 1 to delete. Please verify that your DBMS has
    been manually updated to correspond to wsp_schema.sch, then reply Y to update wspcol or anything else to
    cancel.
   ```
6. Verify that the changes found by pdm_wspupd correspond exactly to the changes you made to wsp_schema.sch. If they match, type **Y** to confirm the changes.
   After you confirm the update, the script uses standard CA SDM utilities to update the wspcol table. Then, Schema Designer shows your changes.
7. Stop the CA SDM servers.
8. Using the appropriate utility for your DBMS, alter the DBMS definition of the columns you changed:
   - Delete any columns from the database that you deleted from wsp_schema.sch.
   - Change the database length of any string columns you changed in wsp_schema.sch.

   Take care that the changes you make to the DBMS correspond exactly to the changes you made to wsp_schema.sch.
9. Publish Schema Modifications.
10. Start the CA SDM servers.

How to Modify the Web Interface using Web Screen Painter

Web Screen Painter (WSP) lets you modify web forms to the requirements of your site without programming.

Some of the Knowledge Management forms cannot be modified in the design view of WSP. For these forms, there are alternate approaches to providing the customization, such as:

- Document view -- the document template that is used when creating the document determines the contents of this page. These templates can be modified on the administration tab under documents, document templates.
- Knowledge categories document list -- you can modify this page by using WSP, but it is also managed by user preferences. The “preferences” screen provides personalization per user for defining which document properties display in the document list and how many documents display per page.

Follow these steps:

1. Verify the Prerequisites
2. Start WSP
3. Select a Form to Modify
   – Create a Form.
   – Open an Existing Form.
4. Modify a Form
   WSP always saves changes on the server where WSP is installed. When you save a file, it becomes accessible to other WSP users in a preview session, but it is invisible to typical CA SDM users. This is because WSP saves all files in the site/mods/wsp directory, and this directory is not used by a typical CA SDM session.
5. If you are making web form changes in the test system, Migrate from Test to Production System.
6. Publish Form Changes.
7. (If necessary) Delete Forms After Publishing.

Verify the Prerequisites

Verify the following prerequisites before you begin the customization:

- Modified the schema that you want to add in the web forms.
- (For advanced availability configuration only) Verified that the following prerequisites to publish the web forms successfully:
  – (Recommended) CA SDM services are up and running on all the CA SDM servers. Otherwise WSP publishes only on the CA SDM servers that are up and running.
  – (Required) At least one webengine instance running on every CA SDM server.

Start WSP

Start WSP to modify web forms.

Follow these steps:

1. Log in to the computer where you have WSP installed.
2. Start WSP.
   – (Windows) Click Start, Programs, CA, Service Desk, WSP.
   – (UNIX) Enter the command *pdm_wsp* with $NX_ROOT/bin in your path.

   WARNING
   When you use UNIX, ensure that you have Firefox that is installed to use WSP.

   The WSP login window opens.
3. Enter your credentials.
   WSP displays the main form.

Select the Form to Modify

You can create a form or can open an existing form to modify. Select from the following possibilities:

- Create a Form.
- Open an Existing Form.

Create a Form

Create a form in WSP.

Follow these steps:

1. Click File, New.
   The New Form dialog opens.
2. Complete the following fields, as appropriate:
   - **Interface or File Type**
     Indicates the file type of the form. For example, to create an HTMPL form select an interface (Analyst, Customer, Employee, Default, or PDA). To create a form of other file type by select the type directly (CSS style sheet, HTML, or JavaScript). When you select an interface or file type, WSP displays a list of all available templates for the selected file type in the File Name field.
   - **Form Group**
     Indicates the form group (as defined for your CA SDM installation) where you want to create your new form or file. If you have not defined any form groups, only the DEFAULT form group is listed.
   - **File Name**
     Indicates the template for the selected interface or file type. A template contains the basic requirements for a new form or file of the desired type.
   - **Select table for new list or detail form**
     Indicates the CA SDM tables for which you can create a new list or detail form. This field is populated according to your selection of an interface (Analyst, Customer, Employee, Default, or PDA).

     NOTE
     Only one detail or list form can exist for each table in a forms group, so edit an existing form (rather than create one) for tables that already have an existing form. If you want to have multiple versions of a form, create one or more form groups to hold the additional versions.
3. Click New.
4. The form is displayed for your customizations. The following two tabs are displayed:
   a. **Design**
      Available for detail forms, list forms, and menu bar forms, and shows the controls on the form that is laid out more or less as a user would see them. It is not an image of how the form looks to an end user. To see this, select **Tools**, **Preview**.
   b. **Source**
      A Notepad-style editor allowing you to review and edit the source code for a form. Some forms are editable only in the Source tab. For those forms, the edit window opens up on the Source tab, and the Design tab is disabled.

   The form is ready for customization.

   WARNING
   When you create or edit a detail or list form, use the list_ and detail_ prefixes to name the HTMPL file. For example, use list_test.htmp" and detail_test.htmpl. This prefix lets you correctly preview a form. When you save a detail template with a custom name, you must also manually edit the <PDM_WSP> tag. For example, <PDM_WSP mode=edit preview="test.htmpl+OP=CREATE_NEW" factory=cr>.

Open an Existing Form

Open an existing form in WSP to modify it.

Follow these steps:

1. Select File, Open.
   The Open Form dialog opens.
2. Select the Interface (Analyst, Customer, Employee, Default) or File Type (CSS style sheet, JavaScript, or HTML) and the forms group that contains the form you want to edit.
3. Select either the form you want from the list, or enter its name in the textbox.

When you enter a name in the text box, WSP automatically scrolls the list to the first name matching the characters entered. You can use the Files of Status drop-down list to restrict the list of files displayed:

- **Site Modified with Unpublished Changes (+)**
  Restricts the list to files that have been modified with WSP, but not yet published. These files are identified with a plus sign (+) after the file name.
- **Site Modified (*)**
  Restricts the list to forms modified at your site, both published and unpublished. The unpublished files are identified with a plus sign (+) after the file name. The published site modifications are identified with an asterisk (*) after the file name.
- **All**
  Displays the list with no restrictions. The unpublished files are identified with a plus sign (+) after the file name. The published site modifications are identified with an asterisk (*) after the file name.

4. Click Open.
   The form is displayed and ready for your customization.

Modify a Form

After you open the form you want to edit in WSP, you can use the toolbar, menu commands, and shortcuts to customize it. You can perform the following customization:

- Insert a Control
- Edit Control Properties
- Modify Menu Bars
- Modify style sheets
- Modify Mouse-Over Preview Form
- Modify Data Grid List on List Form
- Modify Notebooks on Detail Form
- Modify HTML and JavaScript Files

Insert a Control

Add a control on the form. For example, add a textbox in the form.

The controls that can be inserted on both list and detail forms are the following:

Control	Icon	Description
Insert Row		Causes the selected control to be the last in its current row (moves following controls to the next row).
Delete Row	N/A	Deletes all controls on the same row as the currently selected control.
Text box		Inserts a single or multi-line text box for editing a string or text field.
Dropdown		Inserts a drop-down list for editing a field validated against a table.
Lookup		Inserts a lookup control for editing a field validated against a table. The control consists of a text box with a hyperlink in the label that pops up a select form.
Button		Inserts a button.

Hierarchical Lookup		Similar to a Lookup control, except that it is used for a field with a hierarchical selector (such as request category).
Date		Inserts a date field. The control consists of a text box with a hyperlink in the label that pops up a date selector.

The following additional controls are available for detail forms only:

Control	Icon	Description
Checkbox		Inserts a check box.
HTML Editor		Inserts an HTML editor for a text field that contains HTML.
Read Only text box		Inserts a non-editable text field.
Read Only Lookup		Inserts a non-editable lookup field. The field is displayed as a hyperlink to pop up the detail form defining it.
Read Only		Inserts a non-editable date field.
Notebook		Inserts a notebook. There can only be one notebook on a detail form, so this control can be inserted only on forms that do not already contain a notebook.

The following additional control is available for list forms only:

Control	Icon	Description
List		Inserts a list. There can only be one list on a list form, so this control can only be inserted on new list forms.

Follow these steps:

1. Right-click the form at the location where you wish to add the control and select Insert Control.
   A drop-down list is displayed.
2. Select the insert control dialog options.
   The control is placed on the form.
3. Select Tools, Preview to check how the form is displayed to the end user. Although it resembles a standard CA SDM window, and most buttons and menus are functional, it is not a standard session, and you should not attempt to use it that way.
4. Click File, Save.
   The controls are added on a form.

Insert Control Dialog Options

Open the Insert Control dialog from the File menu. The dialog lists all the appropriate UI controls for the section of the form you select to edit.

The dialog contains the following options:

- **Insert Before**
  Inserts the selected control from the list before the currently selected control on the base form, and makes it the currently selected control.
- **Insert After**
  Inserts the selected control from the list after the currently selected control on the base form, and makes it the currently selected control.
- **Prev**
  Moves the currently selected control to the control that precedes it on the form.
- **Next**
  Moves the currently selected control to the control that follows it on the form.
- **Properties**
  Opens the Properties dialog for the currently selected control.
- **Close**
  Closes the Insert Control form.
- **Close Form After Insert**
  (Enabled) Inserts the selected control and closes the Insert Control dialog when you click Insert Before or Insert After. (Disabled) Inserts the selected control when you click Insert Before or Insert After. The Insert Control dialog remains open to insert additional controls or request properties for the currently selected control. After you clear the check box, the Close Form After Insert check box remains unchecked until you either select it again or end your WSP session.
  Default: Enabled

Edit Controls Properties

Edit the properties of a control.

Follow these steps:

1. Select the control and press F4.
   The Properties dialog opens.
2. Change properties as appropriate. For example, the Caption property specifies the header label that is displayed above a control. To specify this property, enter the desired value in the cell to the right of the Caption property.
3. Close the Properties dialog.
4. Select Tools, Preview to check how the form is displayed to the end user. Although it resembles a standard CA SDM window, and most buttons and menus are functional, it is not a standard session, and you should not attempt to use it that way.
5. Click File, Save.
   The properties of the control are modified and the form is saved.

Modify Menu Bars

Forms with names beginning *menubar_* define a menu bar. The Design view for a menu bar displays the menu at the top. You can click a menu item to lower the menu, but cannot otherwise edit the menu bar directly in Design view. To edit a menu bar, double-click the menu item to display the Menu Designer.

> **NOTE**
> Menus (and menubar forms) are used only in the analyst interface. The customer and employee interfaces use a "launch bar" containing actual links, not drop-down lists. To modify the customer or employee launch bar, edit form std_body_site.htmpl from the appropriate interface.

Follow these steps:

1. Select Tools, Menu Designer or double-click the menu shown in the Design tab of a menubar form.
   The Menu Designer dialog is displayed.
2. Complete the fields in the Menu Designer dialog to add or edit menu items. For more information about adding menu items, see the Functions Useful in Menu Items topic.
3. Click OK to close the dialog.

4. Select Tools, Preview to check how the form is displayed to the end user. Although it resembles a standard CA SDM window, and most buttons and menus are functional, it is not a standard session, and you must not attempt to use it that way.
5. Click File, Save.
   The menu bar is modified in the form.

Menu Design Dialog

The Menu Designer dialog appears when you select Menu Designer from the Tools menu, or double-click the menu shown in the Design tab of a menubar form. Use the Menu Designer dialog to add a menu bar, menus, submenus, and menu commands to the open form.

> **NOTE**
>
> : You can access the Menu Designer only while editing HTMPL forms with a name beginning with “menubar_”, such as menubar_admin.htmpl.

The Menu Editor dialog contains the following controls:

- **Menu List**
  From the menu list, select the menu item that you want to change. To change the menu label, for example, select a menu label from the list, type a new name in the Caption text box, and click Apply. You can add, insert, delete, and move the menu items using the controls in the Menu Designer.

  > **NOTE**
  >
  > Only one level of indentation is possible.
- **Caption**
  Enter the name for the selected menu item. See the property for more information.
- **Function**
  Enter the JavaScript function to execute when the user clicks the menu. See the description of the Function property for more details, including some pre-defined functions that may be useful in menus.
- **ID**
  Enter the HTML/JavaScript id to be assigned to the menu item.
- **Hot Key**
  Enter a list of characters directing CA SDM’s selection of the hotkey for this menu item. The hotkey is underlined in the menu caption when it is displayed. CA SDM typically selects the first key in the menu caption not already in use as a hotkey. You can specify one or more characters here to restrict the selection to those characters, or specify one or more characters that are preceded by an explanation point to prevent selection of those characters.
- **Image**
  Enter the location of the image that you want to appear next to the menu item.
- **Internal**
  Select this option to specify that the JavaScript invoked by the function should be executed in the context of the current window (which could be a pop-up detail window). Leave this option unchecked to specify that the function should be executed in the context of the main form.
- **Variable**
  Enter the JavaScript variable to be assigned to the menu item.
- **Tool Bar**
  Adds a toolbar icon and tooltip that corresponds to a non-top-level menu item.
- **Icon File**
  Identifies the location of the toolbar icon.
- **Tip**
  Specifies the tooltip text.

Functions Useful in Menu Items

CA SDM provides a menu bar on almost every form to control its functions. The menu bar is generated by an HTMPL form with a name of the form menubar_*xx*.htmpl. We recommend that you use WSP to modify existing menu bars and define new ones.

The following predefined functions may be useful for scripts invoked by menu items:

- **upd_frame(form)**
  Loads a new form into the main window content frame.
- **create_new(factory, use_template, width, height [,args])**
  Pops up a form to define a new record.
  Popup_window(name, form[, width, height [,features [,args]]])
  Pops up a new window.
- **showDetailWithPersid(persid)**
  Pops up a detail record.

The following terms and definitions apply to the previous functions:

- **Form**
  This is either an HTMPL file name of the form xxx.htmpl or an operation code (for example CREATE_NEW).
- **factory**
  This is the name of a database object.
- **use_template**
  This is either true or false.
- **width**
  This represents the desired form width or zero for default.
- **height**
  This represents the desired form height or zero for default.
- **features**
  This is a list of window features, in the same format used with the standard window.open function.
- **args**
  This is one or more args of the form "keyword=value" for the operation specified for form.
- **persid**
  This is a persistent ID in the form factory:ID.

Modify Style Sheets

You can use WSP to edit or create CSS (cascading style sheet) files.

> **NOTE**
>
> For performance reasons, CA SDM style sheets are delivered in two forms: Individual files (such as search_filter.css) and combination files grouping various individual files with comments and excess white space removed (such as analyst_styles.css). WSP always edits the individual files; you cannot edit a combination file directly. When you publish style sheet changes, WSP automatically builds the associated combination file if necessary.

Follow these steps:

1. Create or open CCS style sheet file.
   WSP displays the Source view of the style sheet.
2. You can edit directly in Source view, or can display the Style Designer by selecting Tools, Style Designer.
3. Complete the fields in the Style Designer dialog, as appropriate. There are a number of style attributes, such as margin and border that cannot be seen or edited in the Style Designer. These must be edited in Source view.
4. Click OK in the Style Designer.
   WSP reformats the style sheet and updates the Source view.
5. Click File, Save.

The style sheet is modified.

Style Designer Dialog

The Style Designer lets you modify or customize style sheets. By default the Style Designer dialog opens in the Font and Color tab and contains the following controls:

- **Style Classes**
  From the drop down list, select a style element you want to modify.
- **Add**
  Click to add a new style class.
- **Rename**
  Click to rename the style class selected in the Style Classes drop-down list.
- **Delete**
  Click to delete the style class selected in the Style Classes drop-down list.
- **Installed Fonts**
  This property lists the fonts that are installed on the system. Click the left and right arrows to move the selected fonts between the Installed Fonts list and the Selected Fonts list.
- **Selected Fonts**
  This property specifies a hierarchical list of preferred fonts that a browser uses to draw the style class element. A browser uses the first font in the list that is installed on the system it is running on.
  Click the up and down arrows to move the selected fonts up or down the hierarchy.
- **Font Size**
  Select the font size from the drop-down list.
- **Bold**
  Select a font style from the drop-down list.
- **Italics**
  Select a font style from the drop-down list.
- **No Effect**
  Select this option if you do not want any text decoration or special effects.
- **Underline**
  Select this option if you want the text to be underlined.
- **Strikethrough**
  Select this option if you want the text to appear with a line through it.
- **Overline**
  Select this option if you want the text to appear with a line above it.
- **Foreground Color**
  Select the foreground color for the text element by clicking the Browse button to the right of the Foreground Color property. Then select the color you want from the Color palette, and click OK.
- **Background Color**
  Select the background color for the text element by clicking the Browse button to the right of the Background Color property. Then select the color you want from the Color palette, and click OK.
- **Transparent**
  Select this option if you want the background of the style element to be transparent.

A sample of the style element is displayed in the preview area at the bottom of the dialog.

To set the position of the style element, click the Position tab. The Position tab contains the following controls:

- **Position**
  From the drop-down list, select one of the positions for the element. The position property places an element in a static, absolute or relative position. Selecting a Static value will place the element in accordance with the normal flow.

Selecting an Absolute value places the element anywhere on a page. Selecting a Relative value moves the element relative to its normal position.

- **Left**
  Enter a value for the left margin in the text box for the element. Select a size from the drop-down list. This property is disabled when the selected position is Static.
- **Top**
  Enter a value for the top margin in the text box for the element. Select a size from the drop-down list. This property is disabled when the selected position is Static.
- **Width**
  Enter a value for the width in the text box for the element. Select a size from the drop-down list.
- **Height**
  Enter a value for the height in the text box for the element. Select a size from the drop-down list.
- **Z-Index**
  Enter a value for the stack order in the text box for the element. An element with greater stack order is always in front of an element with lower stack order. Z-index only works only on elements that have Absolute position

  NOTE

  Elements can have negative stack orders.

Click the Other tab to set some of the specific properties for the element. The Other tab contains the following controls:

- **Visibility**
  Select the type of the visibility for the element from the drop-down list. The Visibility property sets how the content of an element is displayed if it overflows its area.
- **Overflow**
  Select the type of overflow for the element from the drop-down list. The Overflow property sets what happens if the content of an element overflow its set area.
- **Display**
  Select the type of the display for the element from the drop-down list. The Display sets how an element is displayed.
- **Cursor**
  Select the type of cursor for the element from the drop-down list. The cursor property specifies the type of cursor to be displayed when pointing on an element.
- **OK**
  Click to close this dialog and save your changes.
- **Cancel**
  Click to close the Style Designer dialog.

Modify Mouse-Over Preview Form

The Mouse-over previews let you view key details within the current form without the need to click a link or open a new web page. The mouse-over preview appears when you place a cursor over an object link on a list or detail form for a certain amount of time. If you move the mouse away from the link before the delay time expires, the preview does not appear. Mouse-over previews appear by default on all list and detail forms in read-only mode.

You can create mouse-over previews for forms that do not have preview_ forms by default. You can also create mouse-over previews for custom forms you created in your CA SDM environment.

The following predefined mouse-over preview forms are available:

- preview_chg.htmpl (Change Order)
- preview_cnt.htmpl (Contact)
- preview_cr.htmpl (Request)
- preview_in.htmpl (Incident)
- preview_iss.htmpl (Issue)
- preview_KD.htmpl (Knowledge Document)
- preview_nr.htmpl (Configuration Item)
- preview_pr.htmpl (Problem)

Follow these steps:

1. Open one of the following forms for the customization:
   – Open an existing form that does not have a preview_ form by default.For example, detail_loc.htmpl
   – Open an existing mouse-over preview form. For example, preview_chg.htmpl.
2. Modify the form. For example, add or remove custom controls from the form.
3. Click File, Save As, specify a file name using the preview_ prefix, and click Save. For example, enter preview_loc.htmpl.
   The mouse-over preview form is created and modified.

Modify Data Grid List on List Form

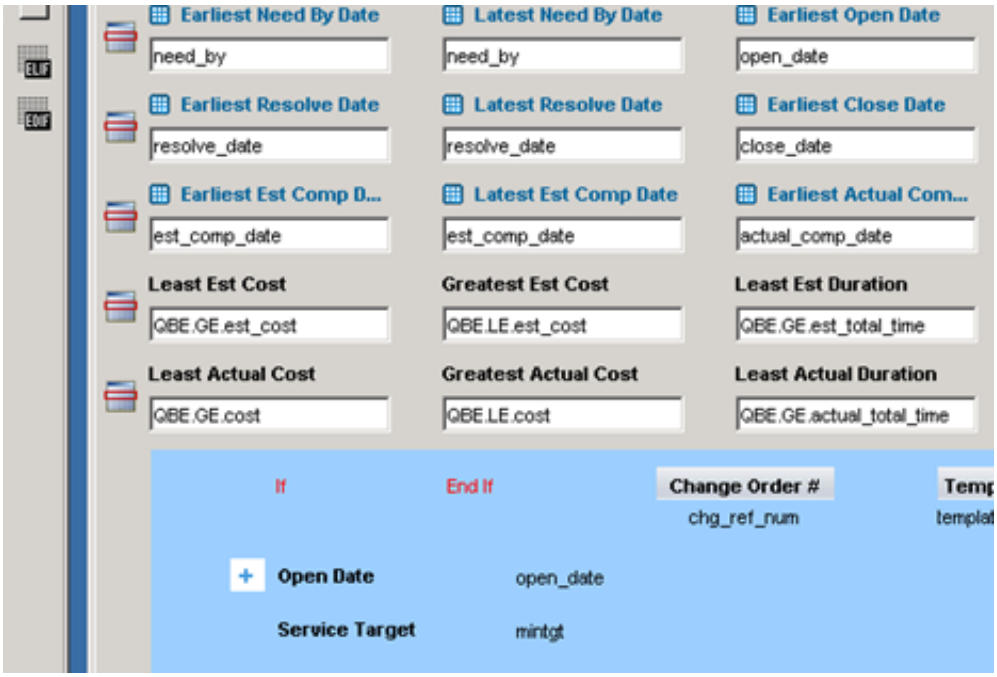
Data grid lists let you view the contents of a product page on a list form without opening a new page. This data-bound control lists items from the data source in a table so you can select items, sort items, and fetch data. For example, the expand and collapse options on the Incident List form.

Follow these steps:

1. Open a list form.
2. Locate the blue area after the labeled fields near the end of the form in the Design view. If you prefer Source view, the data grid appears between the following lines of code:

   ```
   <PDM_MACRO name=lsStart>
   <PDM_MACRO name=lsEnd>
   ```

   The following example shows this data grid area in Design view:



The plus sign in the data grid area represents the beginning of the expansion section of the row (the fields that are displayed only when a user clicks plus on the row in the list form grid). The following code generates the plus sign in the Source view:

```
<PDM_MACRO name=lsCol attr=open_date label="Open Date" sort="DESC" startrow=yes>
```

The *startrow=yes* parameter specifies to start a row, and it starts the expansion section of the row.

- Drag-and-drop columns to move them in a list. You can move columns between the main part of the row and the expansion section. You cannot move an existing control after the grid area.
- To insert a list column, right-click a control or anywhere within the blue background, and select either Insert Column or Insert Control.
  - If you select Insert Column, WSP inserts a column to the left of the currently selected control.
  - If you select Insert Control, WSP displays the Insert Control dialog that lets you add the desired control to the form.
- Select Tools, Preview to verify how the form is displayed to the end user. Although it resembles a standard CA SDM window, and most buttons and menus are functional, it is not a standard session, and you should not attempt to use it that way.
- Click File, Save.
  The data grid list is modified on the list form.

Modify Notebooks on Detail Form

The Nested tabs (notebook) control lets you expand or collapse key details within the current form. For example, use the control to customize how you organize tabs on the Incident Detail Form. From the Design View, you can use the Notebook control to add nested tabs to a detail form that does not already contain one. Double-click the Notebook control to modify it. Use drag-and-drop to add, insert, and delete notebook tabs, and to change their captions. You can also use the up and down arrow buttons to rearrange tabs by changing the position of the currently selected tab. The New Row check box specifies whether the selected tab begins a new row in the notebook header.

Follow these steps:

1. Open a detail form.
2. Locate the blue area of the form in the Design view that contains the first numbered label.
   If you prefer Source view, the notebook area appears between the following line of code:

```
<PDM_MACRO name=startNotebook hdr=cng_nb>
<PDM_MACRO name=endNotebook>
```

NOTE

In CA SDM r12.6, a notebook can contain nested tabs. In WSP, a high-level tab (a tab containing other tabs) is named a tab group. WSP displays a tab group as a dark blue solid bar that spans the entire blue notebook area, with a numbered label in its center. A low-level tab (a tab that does not contain other tabs) is named a tab. WSP displays low-level tabs as rectangles with rounded corners.

3. Click a tab to select it. Selecting a tab highlights and displays a link to the contents of the tab at the bottom of the notebook.
   You can move tabs and tab groups within a notebook using drag-and-drop. Moving a tab group moves all the tabs within the group with it.
4. To insert a tab or tab group, right-click a control or anywhere on the blue notebook background and select Insert Tab, Insert Tab Group, or Insert Control.
   If you select Insert Tab or Insert Tab Group, WSP inserts a new tab or tab group to the left of the currently selected control. If you select Insert Control, WSP displays the Insert Control dialog that lets you add the desired control to the form.
5. Select Tools, Preview to verify how the form is displayed to the end user. Although it resembles a standard CA SDM window, and most buttons and menus are functional, it is not a standard session, and you should not attempt to use it that way.
   Click File, Save.

Modify HTML and JavaScript Files

You can use WSP Source view to edit HTML and JavaScript forms.Open an HTML or JavaScript file and make the necessary changes.

Note: For performance reasons, some CA SDM JavaScript files are delivered in two forms: Individual files (such as window_manager.js) and combination files grouping a number of individual files with comments and excess white space removed (such as std_head.js). WSP always edits the individual files; you cannot edit a combination file directly. When you publish script changes, WSP automatically builds the associated combination file if necessary.

Migrate from Test to Production System

One of the design goals for WSP was to make it safe to develop and test forms modifications on a production database. Such features as a WSP-only directory tree on the server, dedicated WSP server processes, and read-only preview sessions support this goal. However, many users prefer to develop their forms modifications in an independent test system and then migrate the forms to a separate production system after they are complete as follows:

1. Copy any HTMPL forms to be migrated from the appropriate subdirectory of site/mods/www/htmpl on the test system to the same subdirectory of site/mods/wsp/project on the production system.
2. Copy any CSS, JavaScript, and HTML files to be migrated from the appropriate subdirectory of site/mods/www/wwwroot on the test system to the same subdirectory of site/mods/www/wwwroot/wsp/project on the production system.

   NOTE
   Ensure that you copy the files and forms on the following servers of the production system, depending on your CA SDM configuration:

   – **Conventional:** Primary server
   – **Advanced availability:** Background server

NOTE
You can use any file copying method supported by your operating system to perform the copying described in steps 1 and 2 above. Windows users should substitute backslash (\) for slash (/) in the directory paths shown.

Publish Form Changes

When you are satisfied with the changes, you can make them available to all CA SDM users by publishing them. Publishing updates all CA SDM servers with new or revised forms.

Follow these steps:

1. Select File, Publish.
   If you have any unsaved changes, WSP prompts you to save them, and then displays a confirmation dialog showing all pending Web Screen Painter changes (including those saved in previous sessions, or saved by other Web Screen Painter users). By default, all changes are selected for publication. You can change the selection of changes to be published by clicking them.
2. Click OK when you are satisfied with the selection.
3. (Advanced availability configuration only) If the webengine is not running on any of the CA SDM servers, an error message is displayed with the list of CA SDM servers on which WSP failed to publish. Complete the following steps:
   a. Configure the CA SDM server (the server specified in the error message) to make the corrections.
   b. Edit the file in WSP to add a white space character in the <PDM_IF 0> macro and save it.
   c. Go back to step 1 to republish the web form.

   NOTE
   If you do not want to make any configuration changes, you have to manually copy the nx_root/site/mods/www folder from the background server to all the other servers.
4. (Advanced availability configuration only) If the background server crashes while publishing, recover the HTMPL changes.
5. (For mouse-over preview forms) Execute the following command after publishing the changes:
   ```
   pdm_webcache  - H
   ```
   The web cache is refreshed. WSP makes the selected changes available to all users active on all the servers.

   NOTE
   Only site-modified forms can be deleted. Requests to delete a previously published form take effect when you publish changes. To cancel a pending delete request, select File, Undelete Form. You undo changes to a form after publication.

Recover the HTMPL Changes

You recover the HTMPL changes when the background server fails during the publishing activity.

WARNING
We recommend you not to perform the recovery steps directly on the production environment. Ensure that you first validate them on the test or development environment.

Follow these steps:

1. Go to the following directory on the crashed background server:
   ```
   $NX_ROOT$/site/mods/www/wwwroot/wsp/project/web
   ```
2. Copy all the content from the web folder.
3. Log in to the new background server and paste all the copied content in the NX_ROOT/site/mods/www/wwwroot/wsp/project/web folder.
4. Resume publishing the changes on the new background server.

HTML Templates (HTMPL Form)

This article contains the following topics:

Forms in the CA SDM web interface are delivered as HTML templates, in files with a suffix of .htmpl. These are called HTMPL forms in the remainder of this document.

An HTMPL form contains standard HTML (including JavaScript) plus language extensions that are interpreted by a CA SDM server daemon (or service) called the web engine that delivers standard HTML to the browser. These extensions are:

- References to server variables. These are indicated by a name beginning with a dollar sign. They can be the values of columns in the CA SDM database, references to web engine configuration properties, or other server information.
- Special tags directing the web engine to perform tasks on the server, such as read information from the CA SDM database. These tags have names of the form <PDM_...> or <pdm_...>.

> **NOTE**
>
> You do not need to understand the HTMPL extensions or even HTML itself to be able to customize CA SDM forms with WSP.

Template Naming Conventions

The following naming conventions are used to identify the four basic types of HTMPL files, where *xxx* is the object:

Template Type	Name
List (search filter and results)	list_*xxx*.htmpl
Combined read-only and edit detail form (analyst interface)	detail_*xxx*.htmpl
Read-only detail form	detail_*xxx*_ro.htmpl
Edit detail form	detail_*xxx*_edit.htmpl

You can find the definitions of the objects and their properties in the following locations:

- (UNIX) $NX_ROOT/bopcfg/majic/*.maj
- (Windows) *installation-directory*\bopcfg\majic*.maj

For information about the objects and attributes that define CA SDM, see CA Service Desk Manager Reference Commands.

HTMPL Directories

There are different sets of HTMPL files supplied to implement these interfaces, as shown in the following table:

Operating System	Directory Containing HTMPL Files
Windows	*Installation-directory*\bopcfg\www\htmpl\web*interface*
UNIX	$NX_ROOT/bopcfg/www/htmpl/web/*interface*

In this table, *interface* is the name of the interface (analyst, customer, or employee).

> **NOTE**
> There is no separate directory for guest interface files; by default, this interface uses the employee interface files. You can change the guest user interface by changing the access type associated with user System_Anonymous. Both the customer and employee files dynamically modify themselves depending on whether the current user is a known user or a guest, using the <PDM_IF> template command described in this document.

There are three additional interface subdirectories under the htmpl directory:

- **default:**
  Contains HTMPL files common to all interfaces. When searching for a file, the web engine looks first in the directory corresponding to the current user's interface, and then in the default directory.
- **pda/analyst: (UNIX)**
- **pda\analyst: (Windows)**
  Contains HTMPL files used by the mobile device interface. In CA Service Desk Manager r11.0, the mobile device interface is provided only for analysts.
- **web/*interface*/legacy: (UNIX)**
- **web*interface*\legacy: (Windows)**
  Contains HTMPL files from your previous release of CA SDM that are no longer used. This directory is automatically created if you upgrade from a previous release when you install CA SDM. You can delete the legacy directory when none of its files are referenced by your customized files.

We strongly recommend that you do not directly modify the supplied HTMPL files. Instead, either use WSP, or manually copy the file you want to modify to the site mods directory, and modify it there. The CA SDM web server looks for a new form in the appropriate site mods directory before checking the distribution directory. The standard site mods directories for each of the interfaces are as follows:

Operating System	Directory For Site-Modified HTMPL Files
Windows	***installation-directory*\site\mods\www\htmpl\interface*interface***
UNIX	$NX_ROOT/site/mods/www/htmpl/*interface/interface*

NOTE
If you change the form and save it into the *install directory**site**mods**www**htmpl**interface* directory, the form will be seen by everyone, regardless of the form group to which they belong. If you save it into the *install directory*\site\mods\www\htmpl\interface*interface* directory, only those Contacts that are defined as belonging to that form group will see the changed forms.

In the previous table, *interface* is the name of the interface (analyst, customer, or employee). There is no separate directory for guest interface files; this interface uses the employee interface files. The advantage of storing your modified HTMPL files in the site mods directory is that this directory is preserved when you install CA SDM maintenance or a new release. In addition, keeping your modified files in site mods while preserving the originals ensures that you always have a correct copy of the originally distributed HTMPL file.

Each web interface page has a primary function, as indicated in the following table that lists the major HTML templates. However, you can add <PDM_FORM> blocks to any template to directly access any web interface supported operation. For example, you can modify the main menu to include fields for submitting an issue without using the intermediate page, or you can add search criteria fields and a search button to a list form:

Web Page	HTML Template
Main form	menu_frames.htmpl
Display/create/update a change order	detail_chg.htmpl
Display a list of change orders	list_chg.htmpl
Display/create/update and issue	detail_iss.htmpl
Display a list of issues	list_iss.htmpl
Display/create/update a request	detail_cr.htmpl
Display a list of requests	list_cr.htmpl
Display announcement detail information	detail_cnote_html

Display a list of announcements	list_cnote.html
Login	login.htmpl

NOTE
For a complete list of templates, view the contents of the directories in the table at the beginning of this section.

Web Form Groups

You can collect customized web pages into one or more form groups. Form group directories are in the following directories:

- **Windows**
  install-directory\site\mods\www\htmpl\web*interface*
  install-directory\site\mods\www\wwwroot\subdirectory
- **UNIX**
  $NX_ROOT/site/mods/www/htmpl/web/*interface*
  $NX_ROOT/site/mods/www/wwwroot/subdirectory

Each form group is a subdirectory under these directories. You specify the customized form directory in the Customization Form Group field of the access type.

When a user requests a form, the web engine looks first in the appropriate customized form group directory, then in the standard directory for the user's web interface, and finally in the default directory. You can define more than one access type for the same web interface, each with a different customized form group. This lets you define a few specialized forms for different types of users, and still take the majority of the forms from the standard interface.

A similar process occurs when a web page requests a file from one of the subdirectories of wwwroot (css, html, img, or scripts). The webengine examines an HTMPL reference of the form CAisd/img/xxx.gif and converts it to one of the following GIF files, selecting the first one where it finds xxx.gif:

- /CAisd/sitemods/img/formgroup/xxx.gif
- /CAisd/sitemods/img/xxx.gif
- /CAisd/img/xxx.gif

How to Create a Web Form Group

You can create a web form group.

Follow these steps:

1. If you want a form group besides the predefined Analyst, Customer, or Employee form groups, create a form group by selecting Save As from the File menu in WSP and clicking the Add Form Group button on the Save Form As dialog. For example, if you want to provide two separate customized versions of the Analyst interface, you might create form groups called Analyst1 and Analyst2 to handle these. You might also define a new form group if the interface you are defining does not logically fit into one of the predefined form groups.
2. On the web interface (not a Web Screen Painter preview session), select Security, Access Types from the Administration menu. Then click an access type (or create one) and use the Customization Form Group drop-down list on the Access Type Detail window to assign a form group to an access type. CA SDM determines the access type when a contact logs in and uses customization form group to determine where to look in the site mods directory structure for customized forms. If the web engine does not find a form in the form group directory, it looks first in the standard directory for the user's access type, and then in the default directory.
3. In WSP, select Save from the File menu, or manually copy the customized HTMPL files to the following directory:
   In Windows: *installation-directory*\site\mods\www\htmpl\web*form_group_name* directory
   In UNIX: $NX_ROOT/site/mods/www/htmpl/web/*form_group_name*

After you set up a new web form group and copied any supporting files to the appropriate subdirectories, you must restart the web service before the changes take effect.

PDM_MACRO Insert Text from a Macro File

The <PDM_MACRO> tag is used to insert a macro file into an HTMPL file. Its functionality is similar to PDM_INCLUDE, with two important differences:

- A file included by PDM_MACRO has a formal argument list, with required arguments and arguments with default values.
- A file included by PDM_MACRO always comes from the directory specified for the configuration property MacroPath, regardless of the current user's access type.
- **NAME=*macroname***
  (Required) Specifies the macro to include. The web engine affixes the suffix ".mac" and searches for the file in the path specified by configuration file property MacroPath.

Other properties may be required, depending on the macro included. A macro file has the general layout:

```
comments
#args
name1 [= value1]
name2 [= value2]
...
#data
data to insert
```

The following descriptions explain the file layout, line by line:

- **comments** -- The only valid statements in a macro before the #args statement are comments. Comments are indicated by either a # sign or a // as their first non-blank character or characters.
- **#args** -- Must be coded exactly as shown, with the # sign-in column one and no other information on the line. This statement begins the args section, which can contain argument definitions and comments.
- **name [= value]** -- Defines an argument for the macro. Only arguments explicitly mentioned in the args section are valid for the macro. A value specified for an argument in the args section is that argument's default value. Arguments without a default value are required, and must be supplied by the caller on the <PDM_MACRO> statement itself.
- **#data** -- Must be coded exactly as shown, with the # sign-in column one and no other information on the line. This statement begins the data section, which is the part of the macro inserted into the file using PDM_MACRO. Everything in the data section is inserted into the calling file, including lines that would be comments before the data section.
- **data to insert** -- The data to insert into the calling file. This data can contain references to arguments in the form:
- **&{arg_name}** -- These references are replaced with the value of the argument supplied by the caller, or with the default value if the caller did not supply a value.

The web engine normally reads a macro file only once, the first time it is used, and then stores the parsed macro in its own memory. This improves performance, but can be inconvenient if you are developing a macro. Use the configuration file property SuppressMacroCache to prevent this behavior and cause the web engine to discard all macros in its memory each time it begins processing a new form.

To Comment Out PDM_MACRO Tags

To comment out <PDM_MACRO> tags, enter an exclamation point in front of the P as follows: <!PDM_MACRO>.
To prevent the browser from processing the commented out portion of the form, place <PDM_IF 0> before the <! PDM_MACRO> tag, and </PDM_IF> after the line you commented out.Example:

```
<PDM_IF 0>

<!PDM_MACRO NAME=dtlDropdown hdr="Status" attr=status lookup=no
evt="onBlur=\\\"detailSyncEditForms(this)\\\"">
<!PDM_MACRO NAME=dtlDropdown hdr="Priority" attr=priority lookup=no
```

```
evt="onBlur=\\\"detailSyncEditForms(this)\\\"">

</PDM_IF>
```

Predefined Macros Used by WSP

This article contains the following topics:

CA SDM includes a number of predefined macros. Most of these macros insert JavaScript text to create an element on a web form. Use Web Screen Painter to create and modify forms using these macros.

Detail Form Macros

The following list describes the Detail Form macros:

- **button**
  Inserts a graphic button.
- **dtlCheckbox**
  Inserts a check box on a detail form.
- **dtlDate**
  Inserts a date field on a detail form.
- **dtlDateReadonly**
  Inserts a read-only date field on a detail form.
- **dtlDropdown**
  Inserts a drop-down list on a detail form.
- **dtlEnd**
  Ends a detail form.
- **dtlEndTable**
  Ends a table within a detail form.
- **dtlForm**
  Begins a detail form.
- **dtlHTMLEditBox**
  Inserts a detail form field that is a text box containing an HTML editor.
- **dtlHier**
  Inserts a detail form field that is a text box validated against an external table with a hierarchical lookup.

  WARNING
  We recommend that you do not change the dtlHier macro to dtlLookup in the HTMPL file. Use the Options Manager suppress_web_hier_search option instead so that autocomplete works correctly.
- **dtlLookup**
  Inserts a detail form field that is a text box validated against an external table.
- **dtlLookupReadonly**
  Inserts a detail form field that is a read-only hyperlink to an external table.
- **dtlReadonly**
  Inserts a read-only text field on a detail form.
- **dtlStart**
  Begins the first table in a detail form.
- **dtlStartExpRow**
  Begins an expandable row on a detail form.
- **dtlStartRow**
  Begins a normal row on a detail form.
- **dtlTextbox**
  Inserts a text box on a detail form.

contactLookup

The contactLookup macro creates a contact lookup. This macro has the following arguments:

```
contactLookup("&{header}","&{frameName}","&{factory}","&{lookupName}");
```

- **header**
  Identifies the lookup header.
- **frameName**
  (Required) Identifies the form name.
- **factory**
  Specifies the factory.
  Default: agt
- **lookupName**
  (Required) Identifies the lookup name.

You can also enable and disable this element by using the following:

```
contactLookupDisable( Name, bDisable )
```

- **bDisable=**
- true
  Disables the element.
- false
  Enables the element.

dtlCheckboxReadonly

The dtlCheckboxReadonly macro specifies a readonly checkbox field on an HTMPL detail form. The macro has the following arguments:

```
detailCheckboxReadonly("&{hdr}","&{attr}",&{colspan},"$args.&{attr}", "&{on}", "&{off}");
```

- **hdr**
  Specifies the text of the header.
  Default: "$args.&{attr}.DISPLAY_NAME"
- **attr**
  (Required) Specifies the name of the attribute.
- **on = "X"**
  Specifies the value shown on readonly form when field is checked.
- **off = ""**
  Specifies the value shown on readonly form when field not checked.
- **colspan = 1**
  Specifies the number of columns on the form.

On both the readonly and edit forms, the field is displayed as specified in "on" and "off" arguments.

> **NOTE**
> This macro is similar to dtlCheckbox.mac, except that it is always read-only, even in Edit mode.

List Form Macros

- **lsCol**
  Specifies a column in a list form.
- **lsEnd**

Ends the list portion of a list form.
- **lsStart**
  Begins the list portion of a list form.
- **lsWrite**
  Inserts text into the repeating section of a list form.
- **sfDate**
  Inserts a date field in a search filter.
- **sfDropdown**
  Inserts a drop-down list on a search filter.
- **sfEnd**
  Ends a search filter.
- **sfHier**
  Inserts a search filter field that is a text box validated against an external table with a hierarchical lookup.
- **sfLookup**
  Inserts a search filter field that is a text box validated against an external table.
- **sfStart**
  Begins a search filter.
- **sfStartRow**
  Begins a row within a search filter.
- **sfTextbox**
  Inserts a textbox into a search filter.

Menubar Macros

The following list describes the Menubar macros:

- **endMenu**
  Ends a menu within a menu bar.
- **menuItem**
  Defines a global item on a menu.
- **endMenubar**
  Ends a menu bar.
- **menuItemLocal**
  Defines an item on a menu invoked in the context of the current window.
- **menubarItem**
  Defines a menu within a menu bar.
- **startMenu**
  Begins a menu within a menu bar.
- **startMenubar**
  Starts a menu bar.

HTMPL Tags

This article contains the following topics:

This section outlines PDM commands to add HTMPL tags.

PDM_EVAL - Insert the Value of a Pre-Processor Variable

The pdm_eval tag is used to insert the value of a pre-processor variable into the input of the webengine parser. If used inside a macro, its effect is deferred until the macro completes.

The pdm_eval tag works similarly to pdm_include or pdm_macro. It inserts the text into the parser at the point of the tag, exactly as if the value of its variable had been coded in place of the tag.

pdm_eval has the following syntax:

```
<PDM_EVAL TEXT=PRE.name>
```

- **name**
  (Required) The name of the pre-processor variable whose value is to be inserted into the webengine's input.

PDM_FORM - Start an HTML Form with a Session ID

<PDM_FORM> and </PDM_FORM> can be added to any web interface HTML template to create an HTML form including two hidden fields for the server variables SID (session ID) and FID (form ID). The optional OP operand creates an additional hidden field for one of the supported operations, as with the PDM_LINK tag. Except for the automatically-generated hidden fields, <PDM_FORM> and </PDM_FORM> are used in the same way as the standard HTML <form> and </form> tags (and generate these tags as part of their expansion).

PDM_FMT - Format Text from a Server Variable

The <PDM_FMT> and </PDM_FMT> tags are used to format blocks of text inserted by server variables ($args.xxx) as directed by its arguments.

> **NOTE**
> <PDM_FMT> is ignored for literals, including $prop.xxx variables.

The following table describes these tags:

Property	Description
ESC_STYLE=**NONE** | C | HTML | JS | JS2 | URL	Specifies the escape type of the formatted text. Valid values are: **NONE** Default setting. Specifies that no special treatment be given to any character in the content body. **C** Give special treatment to the characters ', ", \, \r, `, and \n, which are meaningful in C programs. These characters will be escaped. **HTML**Give special treatment to the following characters, which are meaningful in HTML text: & becomes & ' becomes ' " becomes " < becomes < > becomes %gt; **JS**Give special treatment to the following characters, which are meaningful in JavaScript text: ' becomes %27 " becomes %22 / becomes %2F \ becomes %5C \r becomes %0D \n becomes %0A **JS2** Same as JS, but give no special treatment to the character, /, and give special treatment to two additional characters: - % becomes %25 - Line breaks are suffixed with %0A **URL** Translate all characters other than letters, digits, and '@*-_.#' to '%xx', where xx is the hexadecimal coding of the translated character.

JUSTIFY=LEFT | CENTER | RIGHT | **TRUNCATE**| WRAP | LIN	Specifies the justification of the formatted text. Valid values include: **TRUNCATE** Default setting. Eliminates HTML formatting by replacing '<' and '>' with < and > **Note:** For more information, see the following information about KEEPLINKS and KEEPTAGS. **LEFT|CENTER|RIGHT** Produces exactly WIDTH characters, truncated or padded with spaces as necessary, with any embedded new lines replaced by a single space, and the output text delimited by [set the pre variable for your book] and </pre> tags. The WIDTH argument must be specified as a positive integer. **WRAP** Same as LEFT, except that text wrapping honors word boundaries (line breaks are not placed within words). **LINE** Same as TRUNCATE, except that it also replaces all embedded line breaks with tags.
KEEPLINKS=YES|**NO**	If KEEPLINKS=YES is specified, the action of JUSTIFY=LINE or JUSTIFY=TRUNCATE is modified to preserve HTML anchor tags (Action:) while converting all other '<' and '>' characters. Mutually exclusive with KEEPTAGS.
KEEPNL=YES|**NO**	The normal action of PDM_FMT is to convert all embedded new lines and any following spaces to a single space. If KEEPNL=YES is specified, embedded new lines are preserved. This argument is ignored for JUSTIFY=LINE.
KEEPTAGS=YES|**NO**	If KEEPTAGS=YES is specified, the action of JUSTIFY=LINE or JUSTIFY=TRUNCATE is modified to preserve all HTML tags. Mutually exclusive with KEEPLINKS.
PAD=**YES**|NO	If PAD=NO is specified, PDM_FMT does not convert empty strings to a single space. This is the normal action when WIDTH is non-zero, or JUSTIFY is TRUNCATE or WRAP.
WIDTH=*nn*	When non-zero, specifies that the text should be formatted to exactly WIDTH characters.

<PDM_FMT> without WIDTH or JUSTIFY does no formatting on the enclosed text, but surrounds the text with [set the pre variable for your book]and </pre>.

For example, to produce a multi-line description, enter the following:

```
<PDM_FMT WIDTH=50 JUSTIFY=WRAP>$args.description</PDM_FMT>
```

To produce multi-column output, enter the following:

```
<PDM_FMT><PDM_FMT WIDTH=20 JUSTIFY=LEFT>$cst.last_name</PDM_FMT>
 <PDM_FMT WIDTH=20 JUSTIFY=LEFT>$cst.first_name</PDM_FMT>
  <PDM_FMT WIDTH=20 JUSTIFY=TRUNCATE>$cst.middle_name</PDM_FMT>
</PDM_FMT>
```

PDM_IF - Conditional Processing

These tags are used to conditionally include text. <PDM_IF> blocks can be placed anywhere in an HTMPL file - in HTML, in JavaScript, and even within HTML tags. <PDM_IF> and <PDM_ELIF> (else if) both take a simple conditional clause as their properties rather than name-value pairs. If the clause is true, the text after the tag to the closing tag is included in the file; if the clause is false, the server discards the text between the tag and the closing tag. The closing tag can be <PDM_ELIF>, <PDM_ELSE>, or </PDM_IF>.

The <PDM_ELSE> and <PDM_ELIF> tags are optional. If both are specified, all <PDM_ELIF> tags must precede <PDM_ELSE>. There can be any number of <PDM_ELIF> tags between <PDM_IF> and <PDM_ELSE> (or </PDM_IF> if <PDM_ELSE> is omitted).

The syntax of the conditional in <PDM_IF> and <PDM_ELIF> is as follows:

- 0 is false; any other number is true
- "" is false; "*any-string*" is true
- "*value op value*" evaluates the left and right values against each other according to *op*. If both values consist of digits (optionally preceded by - or +), the comparisons are done numerically. Otherwise, they are done lexically (ASCII collation). Valid *op* values include:

op Value	Description
==	Equal to
!=	Not equal to
>=	Equal to or greater than (must be written as \>= or >=)
<	Less than (must be written as \< or <)
>	Greater than (must be written as \> or >)
<=	Equal to or less than (must be written as \<= or <=)
&	Performs a bit-and of the left and right values. True if any bits are set; false if none are set.
%	Returns true if the left value is an even multiple of the right value, and false otherwise (useful for building two-dimensional tables).
:	Performs a byte-oriented pattern match like the UNIX grep command. It returns true if the left value contains the regular expression defined by the right value.

Example:

```
<PDM_IF $count \>= 10> . . .
<PDM_ELIF $count &#60; 5>  . . .
<PDM_ELSE> . . .
</PDM_IF>
```

There can be more than one conditional in a PDM_IF statement. Conditionals are separated by connectors, either && (and) or || (or). There is no precedence for either connector. The web engine examines a conditional from left to right until it reaches a connector. If the initial condition is true and the connector is ||, it considers the entire condition to be true without further evaluation. If the initial condition is false and the connector is &&, it considers the entire condition to be false without further evaluation. Otherwise, it considers the condition undetermined, and evaluates the conditional from after the connector.

PDM_INCLUDE - Inserting from a Different File

The <PDM_INCLUDE> tag is used to insert text from a second file into an HTMPL file. The server replaces the <PDM_INCLUDE> tag with the contents of the second file.

Included files can contain <PDM_INCLUDE> tags. There is no limit to the depth of nesting.

The <PDM_INCLUDE> tag supports the following properties:

Property	Description
FILE=filename	(Required) Specifies the file to include. The web engine searches the directories used for HTMPL files, as defined in the current user's access type.
FIXUP=[**YES**|NO]	(Optional) Indicates whether the file should be interpreted by the web interface like a normal HTML template file, such as expanding variables beginning with dollar signs ($) and interpreting other CA SDM tags, such as PDM_LIST, and PDM_FORMAT. The value YES, indicates that the file should be treated as a regular HTML template file, and the value NO means that the included file should be treated as literal text. The default is YES. **Note:** For compatibility with previous releases, the values TRUE or 1 can be substituted for YES, and the values FALSE or 0 can be substituted for NO. These values are deprecated, and should not be used in new pages.
propname=value	Specifies that property propname should have the specified value. The property's value can be accessed within the included file by prefixing propname with $prop. For example, the following specification would allow the included file to reference $prop.menubar:<PDM_INCLUDE … menubar=no> Global properties can also be specified in the web.cfg configuration file. For information about web.cfg, see How to Configure the Web Interface. **Note:** For compatibility with previous releases, property values specified on <PDM_INCLUDE> can be referenced without the preceding 'prop.', in the form $propname. This usage is deprecated and should not be used in new pages.

PDM_JSCRIPT - Conditionally Include a JavaScript File

The <PDM_JSCRIPT> tag is used to conditionally include a JavaScript file on a form. This tag has two forms:

```
<PDM_JSCRIPT file=xxxx.js [include=yes|no]>
```

Pdm_jscript with file=xxx.js specifies that JavaScript file xxx.js is required by this form. The webengine adds the file to a list of JavaScript files required by the form. Processing of the tag occurs while the form is being parsed, and is not affected by pdm_if. That is, a pdm_jscript tag referencing a file adds that file to the list of JavaScript files if it occurs anywhere in the file or in an included file, or in a macro.

The optional argument *include=no* can be specified to instruct the webengine to ignore the tag. This argument provides conditional processing for the tag, and is primarily useful when the tag is invoked in a macro. For example, the dtlTextbox macro specifies the following:

```
<PDM_JSCRIPT file=spellcheck.js include=&{spellchk}>
```

This indicates that any form containing a dtlTextbox macro that specifies spellchk=yes requires the JavaScript file spellcheck.js.

The second form of the pdm_jscript tag is the following:

```
<PDM_JSCRIPT insert=here>
```

Pdm_jscript with insert=here requests the webengine to insert standard HTML <script> tags for all required JavaScript files. The webengine processes this form of the tag during the HTML generation phase, so that it is affected by pdm_if. A pdm_jscript tag with insert=here is part of std_head_include.htmpl, so it is present on virtually every form.

> **NOTE**
> The webengine inserts script tags only the first time it encounters pdm_jscript insert=here.

PDM_LINK - Create a Hyperlink Invoking an HTMPL Operation

<PDM_LINK> and </PDM_LINK> can be added to any web interface HTML template to create links a link that invokes an HTMPL operation. The <PDM_LINK> tag generates the standard HTML <a href=...> tag and has similar arguments, except that it allows specification of a CA SDM operation in place of a URL.

The format is as follows, where *operation* is one of the supported operations:

```
<PDM_LINK OP=operation> ... </PDM_LINK>
```

Example:

```
<PDM_LINK OP=MENU> Menu </PDM_LINK>
<PDM_LINK OP=CREATE_NEW FACTORY=iss> Submit Issue </PDM_LINK>
<PDM_LINK OP=LOGOUT> Logout </PDM_LINK>
```

PDM_LIST - Format a List of Database Rows

The <PDM_LIST> and </PDM_LIST> tags are used to delimit repeating sections of HTML for multi-record output. Everything between <PDM_LIST> and </PDM_LIST> is repeated once for each record to be output. There are two types of PDM_LISTs:

- Lists taken from an object attribute that implies a list. For example, the properties attribute of the request object is the list of properties associated with that request. This type of PDM_LIST always has a SOURCE property.
- Lists with an explicit where clause. This type of PDM_LIST always has a WHERE property.

An object attribute <PDM_LIST> takes the following properties:

Property	Description
ESC_STYLE=**NONE** | C | HTML | JS | JS2 | URL	Specifies the escape type of the formatted text. Valid values are: **NONE**Default setting. Specifies that no special treatment be given to any character in the content body. **C**Give special treatment to the characters ', “, \, \r, ', and \n, which are meaningful in C programs. These characters will be escaped. **HTML**Give special treatment to the following characters, which are meaningful in HTML text: & becomes &' becomes '" becomes "< becomes <> becomes %gt; **JS**Give special treatment to the following characters, which are meaningful in JavaScript text: ' becomes %27" becomes %22/ becomes %2F\ becomes %5C\r becomes %0D\n becomes %0A **JS2** Same as JS, but give no special treatment to the character, /, and give special treatment to two additional characters: - % becomes %25 - Line breaks are suffixed with %0A **URL** Translate all characters other than letters, digits, and '@*-_.#' to '%xx', where xx is the hexadecimal coding of the translated character.
LENGTH=*nn*	Specifies the number of rows of output (defaults to all).
PREFIX=*prefix*	Specifies the prefix on references to attributes from records in the list. These are referenced in the form *$prefix.attr_name* in the text between <PDM_LIST> and </PDM_LIST>. The PREFIX property is optional in an object variable list. If PREFIX is omitted, the value of SOURCE is also used for the prefix.
SEARCH_TYPE=DISPLAY| **GET_DOB**	Specifies the method the server should use to build the list form: DISPLAY specifies the server should issue a single query for the entire form GET_DOB specifies the server should issue separate queries for each row of the form The choice affects list performance, and depends on the complexity of the list (the number of joins required to display it) and the characteristics of your DBMS. GET_DOB has more predictable performance than DISPLAY, and is the default.
SORT=*index-name*	Specifies the index name to use for sorting. The default value of this argument is DEFAULT (which means the first sort index for the underlying factory).

SOURCE=*source*	Specifies the object variable defining this list. This field is required. Do not put a dollar sign ($) in front of *source* on the PDM_LIST statement itself. If the PREFIX property is not specified, *source* is also used as the prefix for references to attributes from records on the list, in references of the form *$source.attr_name*. When used in a reference, *source* does require a preceding dollar sign.
START=*nn*	Specifies the first output row (defaults to zero).

Example:

```
<table border>
<tr>
<th>Child Change Order Number</th>
<th>Summary</th>
</tr>
<PDM_LIST SOURCE=args.children>
<tr>
[assign the value for TD in your book]$args.children.chg_ref_num</td>
[assign the value for TD in your book]$args.children.summary</td>
</tr>
</PDM_LIST>
</table>
```

Because no prefix was specified, references to attributes of the listed records are prefixed by $args.children, the source value.

A where clause PDM_LIST takes the following properties:

Property	**Description**
FACTORY=*name*	Specifies a class of object to be searched. This property is required.
LENGTH=*nn*	Specifies the number of rows of output (defaults to all).
ORDER_BY=*attr-name*	Specify the attribute name to sort by. It can contain the DESC (descending) or ASC (ascending) modifiers.
PREFIX=*prefix*	Specifies the prefix on references to attributes from records in the list. These are referenced in the form *$prefix.attr_name* in the text between <PDM_LIST> and </PDM_LIST>. The PREFIX property is required in a where clause list.
START=*nn*	Specifies the first output row (defaults to zero).
WHERE=*where-clause*	Specify the where clause for the search. It can contain (dotted) attributes. This property is required.

For example:

```
<table>
<tr>
<th>Child Change Order Number</th>
<th>Summary</th>
</tr>
<PDM_LIST PREFIX=list FACTORY=chg WHERE="status = 'OP'">
<tr>
```

```
[assign the value for TD in your book]$list.chg_ref_num</td>
[assign the value for TD in your book]$list.summary</td>
</tr>
</PDM_LIST>
</table>
```

PDM_NOTEBOOK - Create a Notebook

Several of the forms in the CA SDM analyst interface use nested tabs (notebooks). Nested tabs display several sets of fields in the same physical area of the screen, with only one set visible at a time. The user selects the set of fields that is visible by clicking a named tab at the top of the notebook, or by pressing the access key combination Alt+*n*, where *n* is the number of the tab. An example of a form using a notebook is the Issue Detail (detail_iss.htmpl). We recommend that you use WSP to modify the contents of notebooks, or insert a notebook into a form that does not already contain one.

The following tag marks the end of a notebook:

```
<PDM_MACRO name=endNotebook>
```

PDM_PRAGMA - Specify Server Information

The <PDM_PRAGMA> tag is used to specify information used by the web engine, such as form release and version. It does not generate any HTML code, and can be placed anywhere in a form. Possible arguments are:

Argument	Description
RELEASE=value	Specifies the CA SDM release number corresponding to this form. This value is “110” on all CA Service Desk Manager r11.0 forms. It is accessible within the form in the $prop.release variable.
SITEMOD=value	Specifies a site-defined string identifying the modifications applied to this form. It is accessible within the form in the $prop.sitemod variable.
VERSION=value	Specifies a CA Technologies-defined string identifying the version number of this form. It is accessible within the form in the $prop.version variable.
OVERIDE=YES|**NO**	Specifies whether or not values in this PDM_PRAGMA statement override values in previous PDM_PRAGMA statements.

CA Technologies uses PDM_PRAGMA statements to document form versions. All CA Service Desk Manager r11.0 forms include the following PDM_PRAGMA statement:

```
<PDM_PRAGMA RELEASE=110>
```

In addition, the std_head.htmpl form includes the following JavaScript statement:

```
cfgFormRelease = "$prop.release" - 0;
```

The PDM_PRAGMA statement and the cfgFormRelease variable allows the CA SDM web interface to distinguish CA Service Desk Manager r11.0 forms from previous release forms. Releases prior to CA Service Desk Manager r6.0 did not support the PDM_PRAGMA statement.

Normally, only PDM_PRAGMA statements in the highest-level file of a form (that is, a file not brought in by PDM_INCLUDE) are used to set $prop.release, $prop.sitemod, and $prop.version. In addition, a PDM_PRAGMA statement will not override a non-empty value set by a previous PDM_PRAGMA statement. You can specify OVERRIDE=YES to specify that a PDM_PRAGMA statement can override previous PDM_PRAGMA statements, or that a PDM_PRAGMA statement in an included file can be used.

PDM_SCOREBOARD - Build a Scoreboard Tree

The <PDM_SCOREBOARD> tag is used to generate the scoreboard shown on the left side of the main form. It takes the following property:

- **TARGET=*value***
  Specifies the name of the target frame for lists requested by clicking a node on the scoreboard. Lists are loaded into the target specified, which can be any value supported for the target attribute of a link. The default value is *_self* (the window containing the PDM_SCOREBOARD tag).

Any HTMPL form including a <PDM_SCOREBOARD> tag must also include the fldrtree.js JavaScript file. This file can be included with the following statement in <HEAD> section of the form:

```
<SCRIPT LANGUAGE="JavaScript" SRC="$CAisd/CAisd/fldrtree.js"></SCRIPT>
```

In addition, it is desirable to include a link with the name scoreboard_asof_data to display the effective date of the numbers in the tree. See the distributed file scoreboard.htmpl for an example of the use of this tag.

The queries included on the scoreboard are defined by the contents of the User_Query table (object name usq) for the current user. A record in this table defines each line on the tree (folder or node).

Initially, users have no entries in their User_Query table. A user with no User_Query entries receives the default set of scoreboard queries associated with their access type. A user with administrative authority can also customize the default scoreboard for an access type.

PDM_SET - Set the Value of a Server Variable

> **NOTE**
>
> The *<PDM_SET>* tag is not supported for integer-based arithmetic operations like addition (for example: *<PDM_SET args.z_counter=$args.z_counter + 1>*).
>
> It is only supported for setting a string property and string concatenation operation.

The <PDM_SET> tag is used to assign a value to a server variable. It has the following syntax:

```
<PDM_SET arg.name[+]=value>
```

- **arg**
  (Required) Specifies the variable type, and must be arg for normal use.

  > **NOTE**
  > There is no $ character.
- **Name**
  (Required) Specifies the name of the variable.
- **+**
  (Optional) Specifies that the value should be appended to the existing value of the variable. There cannot be any spaces before or after.
- **=**
  (Required) Must be specified exactly as shown, with no spaces before or after.
- **value**
  (Required) Specifies the text to be assigned or appended to the variable.

The PDM_SET tag can also be used in the preprocessor phase to create or update a preprocessor variable.

PDM_TAB - Create a Tab within a Notebook

The <PDM_MACRO name=startNotebook hdr=cng_nb> tag is used to define a notebook tab. We recommend that you use WSP to modify the contents of notebooks, or insert a notebook into a form that does not already contain one.

PDM_WSP - Control WSP Preview

The <PDM_WSP> tag is used to control the WSP preview feature. It does not generate any HTML code, and can be placed anywhere in a form.

By default, WSP determines how to preview a form by examining the form name:

- For detail forms (names of the form detail_*factory*.htmpl), WSP displays the form in edit view, with data from the most recently created row of the appropriate table. If there is no data you are allowed to view in the table, WSP displays the form set up to create a row. WSP preview sessions are typically prohibited from updating the database. WSP displays forms in edit view to allow you to preview all features. However, CA SDM ignores a Save request from a read-only preview session. The web engine changes the text on the Save button to noSave as a visual reminder of this.
- For list forms (names of the form list_*factory*htmpl), WSP displays the form in list view, with the list displaying data from the most recently created row of the appropriate table. If there is no data you are allowed to view in the table, WSP displays the form in search view, with the filter open.
- For other forms, WSP displays the form with no database context.

You can change this default behavior by placing a PDM_WSP tag anywhere on the form. For example, you can display a notebook tab form on its associated detail form, or provide prerequisite arguments for forms normally invoked with an environment provided by another form. Possible arguments are the following:

Property	Description
FACTORY=*value*	Specifies the Object Engine factory used by this form.
PREVIEW=*name*.htmpl | *value* | no	Specifies the preview URL. This can be an HTMPL file name, in the form xxxx.htmpl; a CA SDM URL (used unaltered if it begins with “OP=”); or the keyword “no”, indicating the form cannot be previewed. A value not beginning OP= is modified by replacing a reference of the form {*factory*} or {*factory*:} with an ID or persistent ID (respectively) of the most-recently created row from the referenced factory that the current user is authorized to view.
WHERE=*value*	Specifies a where clause used to search for a representative row or rows to show on the previewed form.
MODE=*value*	Specifies the mode of the constructed URL. Can be the following: GENERAL. General format. Determine the mode by examining the preview argument: detail_xxxx.htmpl - READONLY list_xxxx.htmpl - LIST any other - GRONK READONLY. Detail file in read-only view. EDIT. Detail file in edit view. LIST. List file. GRONK. Unspecified file. In this situation, gronk the file.

Server Variables

This article contains the following topics:

CA SDM information is included in the HTML template using variables beginning with a dollar sign ($). Each page is created with some variables that are documented in the template file. These variables can be put on the page or used in conditional statements:

- Simple Variables
- Property Variables
- Environment Variables
- Business Object Variables
- List Variables

Simple Variables

Simple variables specify flags that are passed to the web page. To access a simple variable, use the variable name preceded by a dollar sign ($). This makes the value of the variable available. For example, two such variables are $CAisd and $cgi. Putting $CAisd in a template results in the substitution of the main CA SDM web server installation directory, whereas $cgi refers to the URL of the pdmweb.exe program. Simple variables are documented in the upper section of the HTMPL file that uses them.

The following shows a list of variables that can be used in all the HTMPL files:

- **$ACCESS.*group***
  The user access privilege object contains the privilege settings on the function group *group* for the current login user. For example, $ACCESS.admin holds the privilege value for the admin functional group. Valid privilege values are:
  - 0-NO ACCESS
  - 1-VIEW
  - 2-MODIFY

  This variable is not available in the login form.
- **$cgi**
  The URL of the pdmweb.exe program.
- **$cst**
  The data object of the current login user. This variable is not available in the login form. You can reference individual attributes of this object with the form $cst.*attrname*; for example, $cst.first_name.
- **$CAisd**
  The URL of main CA SDM web server installation directory.
- **$MachineName**
  The MachineName defined in the web.cfg file.
- **$ProductName**
  The product name defined in the NX.env file.
- **$SESSION**
  The session object saves all session variables including session ID ($SESSION.SID) and all variables defined in the web.cfg file.
- **$USER_STATE**
  User-defined state information.

Property Variables

Property variablesrepresent a property of the configuration file, web.cfg. You can access any entry in the web.cfg file (including user-defined entries) within an HTML template file by prefixing it with “$prop.”

For example, one of the lines in web.cfg, which specifies the number of entries displayed in a single page on a list form is as follows:

```
ListPageLength 10
```

You can refer to this variable in an HTML template with the specification:

```
$prop.ListPageLength
```

If you use the <PDM_INCLUDE> special tag to incorporate another file into a template file, you can specify additional properties as attributes of the <PDM_INCLUDE> tag. You can reference these properties in the included file in the same way as web.cfg properties. A property specified as a <PDM_INCLUDE> attribute that has the same name as a web.cfg property overrides the web.cfg property within the included file.

For example, the following <PDM_INCLUDE> tag creates a property called $prop.menubar that can be referenced within the std_body.htmpl file:

```
<PDM_INCLUDE FILE=std_body.htmpl menubar=no>
```

NOTE
You can refer to configuration file property *xxx* in two ways: $prop.*xxx* or *$SESSION.xxx*. Both return the same value. However, the $prop.*xxx* syntax is preferred because it involves less server overhead.

In addition to properties from web.cfg, there are several predefined properties that can be accessed with $prop. These are:

- **$prop.browser**
  A string identifying the browser in use. This will be “IE” for Internet Explorer.
- **$prop.combo_name**
  A string containing the current user's name, in the form “last_name, first_name middle_name.”
- **$prop.factory**
  A string containing the factory associated with the current form, such as “cr” for requests or “iss” for issues.
- **$prop.FID**
  A string containing the numeric form ID of the current form.
- **$prop.form_name**
  A string containing the name of the current HTML template, in the form *xxx*.htmpl.
- **$prop.form_name_1**
  A string containing the substring of the form name before the first underscore. For example, for the form detail_chg_edit.htmpl, form_name_1 would be “detail.”
- **$prop.form_name_2**
  A string containing the substring of the form name after the first underscore and before the last underscore (or dot). For example, for the form detail_chg_edit.htmpl, form_name_2 would be “chg.”
- **$prop.form_name_3**
  A string containing the substring of the form name after the last underscore and before the dot. For example, for the form detail_chg_edit.htmpl, form_name_3 would be “edit.” For the combination detail form, which has a file name of the form detail_*xxx*.htmpl, $prop.form_name_3 is set to the current view, either “ro” or “edit”.
- **$prop.release**
  A string containing the release level of the form. The PDM_PRAGMA statement contains more details on this property.
- **$prop.SID**
  A string containing the numeric session ID of the current session.
- **$prop.sitemod**
  A string containing the site-defined modification name of the form. The PDM_PRAGMA statement contains more details on this property.
- **$prop.user_type**
  A string containing “analyst,” “customer,” “employee,” or “guest.”
- **$prop.version**
  A string containing the version of the form. The PDM_PRAGMA statement contains more details on this property.

Environment Variables

Environment variablesrepresent an entry within the NX.env configuration file. You can reference any entry in NX.env within an HTMPL template file by prefixing it with “$env.”

For example, one of the lines in NX.env, which specifies the host name of the CA SDM server is as follows:

```
@NX_SERVER=hostname
```

You can refer to this variable within an HTMPL template file with the specification:

```
$env.NX_SERVER
```

Business Object Variables

Businessobject variablesrepresent a CA SDM object, such as an issue or a request. To access an object, you need to start with the variable name, followed by a period (.), followed by whatever attribute names you want to display. For example, on an issue where, by convention, the object is represented by the variable args, you can display the description, the open date, the assignee's phone number, the number of activities on the issue, and the description of the first activity, as shown by the following:

```
$args.description
$args.open_date
$args.assignee.phone_number
$args.act_log.length
$args.act_log.0.description
```

You can use braces to delimit the variable name if it is not surrounded by white space. For example, $foo bar and ${foo}bar are both valid. You can also use the variable args to access non-attribute values (for example, $args.KEEP.*name* as described in Supported Operations).

It is possible that a non-attribute variable may not be defined. For example, it may be possible to get to a form from two different places, only one of which provides a value for $args.KEEP.foo. You can provide a default value for a $args reference with the following syntax, where the string after the colon is substituted for the reference if *variable* is undefined:

```
${args.variable:default}
```

- **Time Zone Date Variables**
  Time zone date variables are a special case of business object variables. They provide a means to convert universal dates (UTC) represented as integers to string dates adjusted for the time zone of the user's browser. The variable for representing integer dates is:
  ```
  $args.attr_name_INT_DATE
  ```
  Example: $args.open_date_INT_DATE
- **Factory Data Variables**
  Factory data variables are a special case of business object variables. A factory data variable is replaced by information about a referenced object. There are seven such variables available:
  - **$args.*attr_name*.COMMON_NAME**
    The common name (externally readable string) of the table referenced by the attribute. For example, on the Request Detail form, the value of $args.assignee.COMMON_NAME is the assignee's combo name (“last, first, middle”).
  - **$args.attr_name.COMMON_NAME_ATTR**
    The attribute name of the common name in the table referenced by the attribute. For example, on the Request Detail form, the value of $args.assignee.COMMON_NAME_ATTR is “combo_name”.
  - **$args.FACTORY_*attr_name***
    The name of the factory associated with the specified attribute. For example, on the Request Detail form, the value of $args.FACTORY_assignee is “agt”.
  - **$args.LENGTH_*attr_name***
    The maximum length of the attribute. For example, on the Request Detail form, the value of $args.LENGTH_summary is 240.
  - **$args.*attr_name*.REL_ATTR**

The rel attr (foreign key) of the attribute. For example, on the Request Detail form, the value of $args.assignee.REL_ATTR is the value of the assignee's ID field.

- **$args.*attr_name*.REL_ATTR_ATTR**
  The attribute name of the rel_attr in the table referenced by the attribute. For example, on the Request Detail form, the value of $args.assignee.REL_ATTR_ATTR is “id”.
- **$args.REQUIRED_*attr_name***
  A string, either “0” or “1” indicating whether the referenced attribute is required.
- **$args.*attr_name*.SELECTIONS**
  A list of valid selections for *attr_name*. This value is an empty string if *attr_name* is not a reference to another table, or if the size of table referenced by *attr_name* exceeds the value of the configuration file property SelListCacheMax. Otherwise, the SELECTIONS variable is a string containing the common name and rel attr of all the entries in the referenced table. Successive values are separated by the string “@,@”, so the variable's value has the form: "cname1@,@rel_attr1@,@cname2@,@rel_attr2"
- **$args.*factory*_SEL_UNDER_LIMIT**
  A string, either “0” or “1”, indicating whether the current number of rows in the table corresponding to *factory* is less than the value of the configuration file property SelListCacheMax. This variable is deprecated in favor of the SELECTIONS variable, which should be used in all new forms.

Factory data variables containing a dotted reference (COMMON_NAME, REL_ATTR, and SELECTIONS) can be used with a dotted reference of any length. For example, on a Request Detail form $args.assignee.organization.COMMON_NAME is replaced by the external name of the assignee's organization.

List Variables

List variables are used to iterate through data. They are accessed using list tags as described in PDM_LIST: Format a List of Database Rows.

Supported Server Operations

This article contains the following topics:

The following operations are supported to let you integrate the CA SDM web pages with your web pages:

- **CREATE_NEW**
  Provides a generic interface to let the user create a row in a specified table. The object name must be specified, and by default a template named *detail_xxx*_edit.htmpl is used for object *xxx*. You can override the .htmpl file by specifying the HTMPL property.
  - **Required specifiers:**
    FACTORY=*object-name*
  - **Optional specifiers:**
    ALG_PRESET=*preset_expression*
    *ALG_PRESET_REL=preset_expression*CREATE_ALG=*activity_log_type*HTMPL=zdetailxxx_*factory*.htmplKEEP.*attr_name=expression*SET.*attr_name=value*use_template=1 | 0 (0 is the default)

    NOTE
    To use the HTMPL specifier with CREATE_NEW, the referenced form must have a name conforming to the naming convention zdetailxxx_*factory*.htmpl. The name must begin with the string zdetail, followed by any alphanumeric characters (including a null string), followed by an underscore and the factory name.
- **ENDSESSION or LOGOUT**
  Ends the current logged-in session. ENDSESSION is the preferred operation.
- **GENERIC_LIST**
  Provides a generic interface to allow the user to display a list from any table in the database. The object name must be specified, and by default a template named list_*xxx*.htmpl is used for object *xxx*. You can override the .htmpl file by specifying the HTMPL property.

  - **Required specifiers:**
    FACTORY=*object-name*KEEP.*attr_name=value*
- **DISPLAY_FORM**
  Provides a generic interface to let the user display any customized form.
  - **Required specifiers:**
    HTMPL=*htmpl_file*

    NOTE
    DISPLAY_FORM replaces JUST_GRONK_IT. Existing implementations can continue to use JUST_GRONK_IT, which functions exactly like DISPLAY_FORM. DISPLAY_FORM is the preferred operation.
- **MENU**
  Displays the main menu page, which is defined in the web.cfg file in the Menu property.
  - **Optional specifiers:**
    HTMPL=*menufile*
    menufile is the name of an alternate main menu file.
- **PAGE_EXTENSION**
  Allows the webmaster to specify additional extensions to the interface.
  - **Required specifiers:**
    NAME=*html_file*
    html_file is one of the file names listed in the configuration file UserPageExtensions directive.
  - **Optional specifiers:**
    REQUIRES_LOGIN=1
    If present, a login page appears first if the user is not currently logged in. If omitted or set to zero, the file is shown without checking if the user is currently logged in.
- **RELOG**
  Displays the login page.
- **SEARCH**
  Provides a generic interface to allow the searching of any table in the database. This operation assumes that an appropriate search_*xxx*.htmpl has been created, where *xxx* is the *object-name*, as defined in the .maj files in the majic directory in bopcfg.

    NOTE
    For more information, see CA Service Desk Manager Reference Commands. By default, the results of this search are displayed in list_*xxx*.htmpl, but this can be overridden by specifying the HTMPL property.

  - **Required specifiers:**
    FACTORY=*object-name*QBE.*op*.*attr_name=value*
  - **Optional specifiers:**
    ALG_PRESET=*preset_expression*ALG_PRESET_REL=*preset_expression*CREATE_ALG=*activity_log_type*HTMPL=*list_htm*
- **SEC_REFRESH**
  Refreshes the user access information from the security subsystem. A hyperlink for this operation is provided to users who have MODIFY privileges (for the admin functional group) on the menu screen. After updating user access privileges with the security program, this operation provides a means to refresh access information. (This operation refreshes the security information for all users.)

    NOTE
    Security refresh is an asynchronous process. When the security refresh is done, a message shows in the standard log file (stdlog).
- **SET_MENU**
  The behavior of this operation is the same as MENU when MENU is used with the HTMPL variable. The only difference is that this operation also sets the default menu form to the menu form specified with the HTMPL property.
  - **Required specifiers:**

HTMPL=*htmpl_file*

NOTE
This operation overrides the MENU set in the web.cfg until the web service is restarted.

- **SHOW_DETAIL**
  Provides a generic interface to allow the user to display a read-only detail of a row in a specified table. The persistent ID name must be specified (from which the object name is inferred). By default, a template named detail_*xxx*_ro.htmpl is used for object *xxx*. The .htmpl file can be overridden by specifying the HTMPL property.
  - **Required specifiers:**
    PERSID=*persistent-id*
  - **Optional specifiers:**
    ALG_PRESET=*preset_expression*ALG_PRESET_REL=*preset_expression*CREATE_ALG=*activity_log_type*HTMPL=*readonl*
- **UPDATE**
  Provides a generic interface to editing any table. The ID and object name must be passed in and a detail form that the user can edit is displayed to the user. By default, the user has exclusive access to the record for two minutes, and is guaranteed to get changes into the database if they are submitted in this time.
  - **Required specifiers:**
    PERSID=*persistent-id* orSET.id=*id-of-row-to-update* FACTORY=*object-name*
  - **Optional specifiers:**
    NEXT_PERSID=*persistent-id* (of record to display after successful update)KEEP.*attr_name*=*value*KEY.*attr_name*=*value*HTMPL=zdetailxxx_*factory*.htmpl

NOTE
To use the HTMPL specifier with UPDATE, the referenced form must have a name conforming to the naming convention zdetailxxx_*factory*.htmpl. The name must begin with the string “zdetail”, followed by any alphanumeric characters (including a null string), followed by an underscore and the factory name.

NOTE
For information about web.cfg, see How to Configure the Web Interface.

Operation Variables

This table lists the variables that can be set for each of the operations in the supported operations:

Variables	Description	Operations
ALG_PRESET ALG_PRESET_REL	Specifies values for one or more of the attributes of the activity log created as a result of the CREATE_ALG variable. If CREATE_ALG is not specified, ALG_PRESET and ALG_PRESET_REL are ignored.	CREATE_NEWSEARCHSHOW_DETAIL

CREATE_ALG	Specifies the activity log type of an activity log to be created as a side effect of the operation. Use the ALG_PRESET or ALG_PRESET variables to specify values for the attributes of the new activity log. The timing of creation of the activity log depends on the operation, as follows: **CREATE_NEW**The activity log is created when the new record is saved. If the new record is not saved, no activity log is created. **SEARCH**The activity log is created when a record is selected from the list form. If the record is viewed instead of selected (that is, the user explicitly selects the View command from the list form's mouse-over menu), no activity log is created. **SHOW_DETAIL**The activity log is created before the record is displayed.	CREATE_NEWSEARCHSHOW_DETAIL
FACTORY	Specifies the class of object to be searched, created, or updated. You can use any name specified as an OBJECT in the *.maj files in $NX_ROOT/bopcfg as listed in CA Service Desk Manager Reference Commands.	CREATE_NEWGENERIC_LISTSEARCHUPDATE
HTMPL	Allows the HTMPL author to override the default template naming convention and explicitly specify the HTMPL file to display, instead of the default template. **Note:** When the HTMPL specifier is used with CREATE_NEW or UPDATE, the name of the referenced form must conform to the naming convention zdetailxxx_factory.htmpl, where xxx are any characters, and *factory* is the factory name.	CREATE_NEWDISPLAY_FORMJUST_GRONK_ITME UPDATE
KEEP.*name*	Specifies the value that can be saved and passed between pages.	CREATE_NEWGENERIC_LISTSEARCHUPDATE
KEY.*attr_name*	Similar to the SET.*attr_name*, except that this specifies a lookup on *attr_name*, which must be a reference to another table or object.	UPDATE
NEXT_PERSID	Specifies the persistent ID of the record to be displayed next.	UPDATE

PERSID	Specifies the persistent ID of a record to be displayed. You can specify this in either of the following ways: Directly, with a persistent ID consisting of a factory name, a colon (:), and a unique integer database ID. For example, PERSID=chg:1234, specifies the change order with database ID 1234. Indirectly, with a persistent ID consisting of a factory name, a colon (:), an attribute name, a second colon (:), and a value. This form of PERSID specifies the record of the specified factory that has an attribute of the specified value. For example, PERSID=chg:chg_ref_num:demo:3 specifies the change order with reference number demo:3.	SHOW_DETAILUPDATE
PRESET PRESET_REL	Specifies values for one or more of the attributes of the record created as a result of the CREATE_NEW variable. If CREATE_NEW is not specified, PRESET is ignored.	CREATE_NEW
QBE.*op*.*attr_name*	Specifies the values to use when performing a search. These values are identified using a QBE keyword, where *attr_name* identifies any attribute name on a ticket that can be set and *op* indicates to search where the attribute: EQ is equal to the value NE is not equal to the value GT is greater than the value LT is less than the value GE is greater than or equal to the value LE is less than or equal to the value NU is null NN is not null IN matches the SQL LIKE expression KY contains the text entered If you do not define any QBE variables, the standard search window is displayed.	SEARCH
SET.*attr_name*	Specifies an attribute name to use when a ticket is created, where *attr_name* identifies any attribute in a ticket that can be set. The attribute names will vary depending on the underlying object. All objects and their attributes can be found in the *.maj files in the majic directory in bopcfg as listed in CA Service Desk Manager Reference Commands.	CREATE_NEWUPDATE

SET.*id*	Specifies the database ID of the row to be updated.	UPDATE
SKIPLIST	When set to 1, searches that result in 1 hit do not display the search result list. Instead, the read-only detail is displayed directly.	SEARCH
use_template	When set to 1, the SEARCH operation will return a list of templates. The returned template selected will be used in the CREATE_NEW operation to populate a new record. This variable is valid for change orders, issues, and requests.	CREATE_NEWSEARCH

Syntax of PRESET, PRESET_REL, ALG_PRESET, and ALG_PRESET_REL

The PRESET, PRESET_REL, ALG_PRESET and ALG_PRESET_REL keywords in the URL specify initial values for attributes of the ticket and its activity log, respectively. There are two possible formats:

- **[ALG_]PRESET=*attr*:*value***
  Indicates that the specified attribute of the ticket or activity log should be set to the specified value. For example, the following specification sets the description of the new ticket to “Hello:”
  ```
  PRESET=description:Hello
  ```
- **[ALG_]PRESET_REL=attr:obj.relattr:testattr:value**
  Indicates that the specified attribute of the ticket or activity log should be set to a value copied from another database table. The value is copied from the *relattr* attribute of the *obj* whose *testattr* has the specified *value*. For example, the following specification sets the analyst attribute of the new ticket to the ID of the contact with user ID xyz123:
  PRESET_REL=analyst:cnt.id:userid:xyz123
  When this format is used, the implied query must retrieve a unique record. If more than one contact has a user ID of xyz123 (or none), the example PRESET specification has no effect.

The PRESET, PRESET_REL, ALG_PRESET and ALG_PRESET_REL keywords can occur as many times as desired in a URL, allowing the setting of multiple attributes. Alternatively, a single keyword operand can specify multiple values separated by @@. If the '@@' separator is used, you cannot mix value formats for [ALG_]PRESET and [ALG_]PRESET_REL keywords. For example, the following example shows two different ways of specifying values for ticket description, summary and analyst:

```
PRESET=description:Hello+PRESET=summary:HelloThere+PRESET_REL=analyst:cnt.id:userid:xyz123
PRESET=description:Hello@@summary:HelloThere+PRESET_REL=analyst:cnt.id:userid:xyz123
```

For requests, issues, incidents, problems, and change orders, both PRESET and PRESET_REL support a keyword attribute ASSET to link an object to an asset. The ASSET attribute updates the affected_resource attribute of a request, incident, or problem, or the asset LREL of an issue or change order.

Link Examples

The following link examples do not include the path to CA SDM. All CA SDM URLs begin with coding of the following form:

http://*hostname*[:*port*]/CAisd/pdmweb.exe

In this example, *hostname* is the name of your server and *port* (optional) is the port number if you are using Tomcat. This coding is shown as an ellipsis (…) in the following URL examples:

- To create a request with an affected end user with the userid tooda01, use the following example URL:
  ```
  ...?OP=CREATE_NEW+FACTORY=cr+PRESET_REL=customer:cnt.id:userid:tooda01
  ```
- To display a list of all requests assigned to userid tooda01, use the following example URL:

```
...?OP=SEARCH+FACTORY=cr+QBE.EQ.assignee.userid=tooda01
```

- To display the detail form for request 1234, use the following example URLs:

```
...?OP=SHOW_DETAIL+FACTORY=cr+PERSID=cr:ref_num:1234 (read-only view)
...?OP=UPDATE+FACTORY=cr+PERSID=cr:ref_num:1234 (update view)
```

NOTE
You can bypass the logon challenge by using Web Services for authentication. For information about the getBopsid() method, see CA Service Desk Manager Reference Commands.

Advanced Modifications

This article contains the following topics:

You must be aware of various aspects of modifying web pages if you elect to use tools other than Web Screen Painter to modify HTMPL, or if you have unusually complex customization requirements. However, we strongly recommend that you work with WSP to modify CA SDM web pages before trying any other approach. WSP is capable of doing almost any modification you need, and it automatically handles housekeeping issues, such as placing updates in the site mods directory, and distributing published files to all servers.

The Web Engine and Its Cache

When modifying web pages, it is helpful to understand the structure of the CA SDM web server. The web interface uses either a J2EE servlet container, such as Tomcat, or a standard HTTP server, such as Apache or Microsoft Internet Information Server (IIS). When a user requests a CA SDM web page, the HTTP server invokes the supplied program pdmweb.exe.

After it starts, pdmweb.exe sets up a connection with a CA SDM daemon (or Windows service) called the web engine. The web engine interprets the user's request. Most requests require the web engine to look up a template (HTMPL) file and translate it into standard HTML. Usually, the translation process requires the web engine to communicate with a CA SDM server to read or update the database, and include database information in the generated HTML. After the HTML is complete, the web engine sends it to pdmweb.exe, which in turn sends it back to the user's browser.

To maximize performance, the web engine typically reads each HTMPL file only once. After parsing the file and determining how to translate it to HTML, the web engine stores the parsed file in its cache, significantly reducing the processing time the next time the file is requested. While the cache is beneficial in a production environment, it can be inconvenient in development, as it means that changes to HTMPL files do not take effect until either the web engine is recycled or the pdm_webcache utility is used. In a development environment, you can avoid this behavior by specifying the configuration file property SuppressHtmplCache. However, we recommend that you do not suppress the HTMPL cache in a production environment because it severely impacts overall performance of the web engine.

The web pages served up by pdmweb.exe are generated by reading HTMPL files and using them to generate HTML. HTML template files are identified by a file suffix of .htmpl. You can modify these template files, and thereby modify the CA SDM web pages.

The pdm_webcache Utility

Use the pdm_webcache utility to remove one or more HTMPL forms from the web engine cache. This forces the web engine to fetch these forms from the disk the next time they are used, allowing changes to forms to take effect.

```
pdm_webcache [-f form-name] [-g form-group] [-i interface] [-p process] [-v]
```

- **-f *form-name***
  Specifies the name of the form to be removed from the cache, such as detail_cr.htmpl. You can use '%' (or '*') as a wildcard character to select more than one form. For example, the specification:
  -f detail%
  selects all detail forms.

This argument is optional. If it is omitted, all forms in the cache are selected.

- **-g *form-group***
  Specifies the name of the form group to be removed from the cache, such as Analyst. You can use '%' (or '*') as a wildcard character to select more than one form group**.** For example, the specification:
  -g Anal%
  selects all form groups beginning with "Anal".
  This argument is optional. If it is omitted, all form groups in the cache are selected.
- **-i *interface***
  Specifies the name of the web interface to be removed from the cache, such as analyst, customer, or employee. You can use '%' (or '*') as a wildcard character. For example, the specification:
  -i a%
  selects the analyst interface.
  This argument is optional. If it is omitted, all interfaces in the cache are selected.
- **-p *process***
  Specifies the name of the web engine process whose cache is to be modified, such as web:local.
  This argument is optional. If it is omitted, all web engines are selected.
- **-v**
  Specifies verbose output. When this argument is specified, pdm_webcache lists the full name of every form removed from the cache, in the form:
  interface:*form-group*:*form-name*
  This argument is optional. If it is omitted, pdm_webcache reports only a count of forms removed from each web engine's the cache.

How to Modify HTML Templates

Typically, you can make two types of changes to the HTML templates:

- You can make modifications that will be visible to the user but will not be altered by the web interface prior to display. For example, you could add a GIF file for your company logo to the web interface pages (a “pass through”) by adding the reference to the appropriate template file or you could add JavaScript to your page to validate input. Any changes you make to the HTMPL file that are not contained within a PDM tag, as defined in the following, are passed unchanged in the HTML returned to the user.
- You can modify the replaceable sections of the templates. For example, you can add new application data to the request detail page.

Several kinds of template entries let you do the following:

- Display information from CA SDM to the user.
- Set up a query page.
- Create links to other CA SDM pages using link tags.

Files That Should Not be Modified

Certain HTMPL templates and JavaScript files contain information required by many CA SDM web forms. The information in these templates is both release dependent and critical to the successful operation of the CA SDM web interface. Therefore, these files are always replaced when a new version of CA SDM is released; changes made to them are not upgraded.

The templates affected by this restriction are as follows:

- **ahdtop.htmpl**
  Contains styles, scripts, and JavaScript variables used throughout the CA SDM web interface. This file is part of the main frameset of the web interface, and is always present during a session. All CA SDM forms have access to the JavaScript variable ahdtop that references the window containing ahdtop.htmpl.
- **menu_frames.htmpl**

Defines the HTML frameset used by the CA SDM main form.

- **msg_cat.js**
  Contains the text of all messages used in CA SDM JavaScript files.
- **reports.htmpl**
  Contains data required for web reports.
- **std_body.htmpl**
  Contains standard information used at the beginning of the BODY section of most HTMPL templates.
- **std_footer.htmpl**
  Contains standard information used at the end of the BODY section of most HTMPL templates.
- **std_head.htmpl**
  Contains standard information used at the beginning of the HEAD section of almost all HTMPL templates.
- **styles.htmpl**
  Contains CSS styles used throughout the CA SDM web interface.

Although you cannot modify these files directly, you can add additional information to them. Each restricted file xxx.htmpl (except for menu_frames.htmpl and reports.htmpl) has a corresponding xxx_site.htmpl file that you can modify. For example, you can add additional information to ahdtop.htmpl by modifying ahdtop_site.htmpl, or add new messages by modifying msg_cat_site.js.

The xxx_site.htmpl file corresponding to each restricted file is loaded after the main file so you can override or change JavaScript in the main file. Use caution when adding information, as badly designed changes to these files can cause unexpected problems throughout the CA SDM web interface.

Guidelines for New HTMPL Files

You can add your own HTMPL files to the CA SDM web interface. Follow these guidelines to help ensure your HTMPL file works well with the rest of the CA SDM interface:

1. Include the following statement somewhere in the <HEAD> section of the file. This statement should follow the <TITLE> statement (if any). It defines several JavaScript global variables required by CA SDM web interface, and also registers your page with the CA SDM window manager:
   ```
   <PDM_INCLUDE FILE=std_head.htmpl>
   ```
2. Include the following attribute as part of the <BODY> tag of the file. This attribute helps the CA SDM window manager keep track of your page:
   ```
   onUnload="deregister_window()"
   ```
3. Include the following statement at the beginning of the <BODY> section of your file. The “menubar=no” argument is optional; if specified, it suppresses the CA SDM menu bar:
   ```
   <PDM_INCLUDE FILE=std_body.htmpl [menubar=no]>
   ```
4. Include the following statement at the end of the <BODY> section of your file.
   ```
   <PDM_INCLUDE FILE=std_footer.htmpl>
   ```

How to Add User Defined State Information

Many customers want to be able to embed their own state information in the CA SDM web pages, and have CA SDM pass the state information to all subsequent pages it serves up to the user's session. This information can be interrogated with conditional statements in the HTMPL files.

State information for a user's session is accomplished by setting the special attribute USER_STATE in your links or forms. After it is submitted into the CA SDM web engine, every page that is presented to the user will have the HTMPL variable USER_STATE available and set to the value last submitted for USER_STATE.

The following examples show how you might set up an entry into CA SDM from some other part of your site, such as from pages that are oriented to your sales force:

- Using a hyperlink

```
<a href="/CAisd/pdmweb.exe?USER_STATE=Sales">Service Desk</a>
```

- Using a form with a hidden field

```
<form action="http://yourhost.com/CAisd/pdmweb.exe">
<input type=hidden name=USER_STATE value=Sales>
```

  Click the button for the Service Desk

```
<input type=submit>
</form>
```

Then you can modify your HTMPL forms based on the state information:

```
<PDM_IF "$USER_STATE" == "Sales">
```

custom information for sales audience

```
<PDM_ELIF "$USER_STATE" == "Engineering">
```

custom information for engineers

```
<PDM_ELSE>
```

information for everyone else

```
</PDM_IF>
```

How to Directly Create a Request from a Template

It is possible to create a Request directly from a Template using an URL.

Example

```
http://machinename/CAisd/pdmweb.exe?FACTORY=cr+OP=CREATE NEW+PERSID=cr:3106+use_template=1
```

where cr:3106 is the persid of the template.

Directories Used by Your HTTP Server

The default installation of CA SDM defines two virtual directories to your HTTP server:

- The CAisd virtual directory points to the following directory in your CA SDM installation:
  a. In Windows: *installation-directory*\bopcfg\www\wwwroot
  b. In UNIX: $NX_ROOT/bopcfg/www/wwwroot
- The CAisd/sitemods virtual directory points to the following directory in your CA SDM installation:
  a. In Windows: *installation-directory*\site\mods\www\wwwroot
  b. In UNIX: $NX_ROOT/site/mods/www/wwwroot

Subdirectories under these virtual directories are:

Subdirectory	Stores
css	Style sheets
help	Web interface help
html	HTML files
img	Graphic files
scripts	JavaScript
sitemods	Site-defined modifications

If you decide to create versions of any of the files in the css, html, img, or scripts directories, we strongly recommend that you do not update the file in /CAisd. Instead, store the file in the appropriate subdirectory of /CAisd/sitemods. For example, if you decide to modify a style sheet in /CAisd/css, store your modified version in /CAisd/sitemods/css. When the web engine parses an HTMPL file, it automatically modifies file names beginning with $CAisd to point to sitemods if the file exists in a subdirectory of sitemods.

Using the /CAisd/sitemods directory has these advantages:

- It allows you to keep a record of the distributed files you have changed.
- It gives you easy access to the original version in case there is a question or a problem.
- It makes the process of installing maintenance or a new release easier, since CA SDM installation never places anything in the /CAisd/sitemods directory.

  NOTE
  There is no /CAisd/sitemods/help subdirectory. Because the help data is in standard HTML files (not HTMPL templates), the web engine cannot dynamically change file references. If you need to modify help, you must make your changes in /CAisd/help.

The HTML subdirectory contains a few heavily used files that do not need to be processed by the web engine and can improve performance when cached on the browser. If you create version of any of these files, carefully check the file for references to other modified files. Because there is no web engine processing, you must manually insert a reference to sitemods where appropriate.

Download PDF Attachments

When you download and try to view a PDF attachment in CA SDM, the PDF file may not display correctly, or a blank window may appear after you upgrade to Adobe Acrobat release 7.0 or 8.0. With CA SDM, you can display the PDF file correctly by completing the following steps:

1. Set the *forceDecompressOnDownload* parameter to YES in $NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF\web.xml.

   NOTE
   On Linux, $NX_ROOT is /opt/CAisd
2. Repeat Step 1 to update *web.xml.tpl* file in the *NX_ROOT\samples\pdmconf* folder to avoid losing these changes when pdm_configure is run at a later point of time.
3. Restart the CA SDM services.

Edit in List Modification

This article contains the following topics:

Several list forms, such as the Request and Issue lists, include an Edit in List button. When this button is available and a result set is displayed, the user can click Edit in List to replace the search filter with a small edit form. The edit form allows the user to update records directly on the list form. The user can even update everything selected in the list by placing the desired new data in the edit form and clicking Change All.

Editing list data involves no communication with the server until the user clicks Save. When the user clicks Save, all the updates (marked by yellow highlights on the form) are sent to the server, which applies all the changes in a single operation, returning a status message and redisplaying the list.

You can modify this feature by controlling whether the Edit in List button is available on a particular list form, and by controlling the fields that appear in the edit form displayed when the user clicks Edit in List.

To place an Edit in List button on a list form, including the following statement somewhere in the <HEAD> section of the form:

```
<SCRIPT LANGUAGE="JavaScript" SRC=$CAisd/CAisd/list_edit.js></SCRIPT>
```

Simply adding this statement puts the button on the form. However, the button is disabled unless JavaScript statements specifying the contents of the edit form are also included in the form. These statements must be placed immediately prior to the results set specification, and have the following format:

Statements	Comments
startListEdit(_search_filter);	Specify exactly as shown
listEditStartRow();	Specify exactly as shown
listEditField("attr"[, "hdr"]);	Specify zero or more
listEditReadonly("attr[", "hdr"]);	Specify zero or more
endListEdit();	Specify exactly as shown

The endListEdit() statement must be followed by the ResultSet() statement that begins the results set. You specify the fields in the edit form and their sequence on the form by coding one or more listEditReadonly() or listEditField() statements.

startListEdit(_search_filter)

This statement begins the list edit form. It must be coded exactly as *startListEdit(_search_filter);*.

listEditStartRow()

This statement begins a new row of fields on the list edit form. It must be coded exactly as *listEditStartRow();*. You must place a listEditStartRow() statement immediately after the startListEdit() statement. You can optionally include additional listEditStartRow() statements among the listEditField() and listEditReadonly() statements that specify the fields on the form.

listEditField(attr_name ,hdr)

This statement specifies an attribute to be included on the list edit form.

- **attr_name**
  Specifies the name of the attribute to be included in the edit form (including dots, if appropriate). All attributes specified for a list edit form must also be in the results set. The *attr_name* specified must be identical to that specified in the IsWrite or IsCol macros that adds the attribute to the results set.
  The attribute appears on the edit form in the same format that it appears in the search filter. If the attribute is not in the search filter, it is edited in a 20-character text box.
  attr_name is a required argument.
- **hdr**
  Specifies the text of the header on the field in the edit form. This argument is optional; if omitted, the header text is taken from the search filter. If hdr is omitted and the attribute is not in the search filter entry for attr_name, the header text defaults to the attribute name surrounded by question marks.

listEditReadonly(attr_name ,hdr)

This statement specifies a non-editable attribute to be included on the list edit form. Its arguments have the same significance as those for listEditField().

endListEdit()

This statement ends the list edit form. It must be coded exactly as *endListEdit();*.

Integrating with Your Own Web Pages

This article contains the following topics:

You can integrate the CA SDM web interface functionality with your web pages to present a seamless interface for your users.

> **NOTE**
> The web engine, which is the executable that acts as the gateway between the web server and the CA SDM server, allows multiple simultaneous connections from a given user. More than one frame at a time can have an open connection to the CA SDM web engine process.

You can integrate the web interfaces in the following ways:

- By creating links from any of your web pages to the appropriate CA SDM web page without having to go through the web interface menu page.
- By adding HTML forms to your web pages that collect input and perform supported operations directly, without displaying any CA SDM web data entry pages.
- By creating web form groups that can be used to associate HTML web-based forms to users through their access type. Similar to the form groups used by the administrative interface, web form groups can be used to customize your HTML pages.

Linking to CA SDM Functions

You can link directly to major CA SDM functions without displaying the main page. You typically do this by accessing the pop-up window for the new window containing the CA SDM information. You can also replace your web page with the CA SDM page.

In both cases, the product displays the requested page in the same way that the user sees it in a typical session, but without the main page and scoreboard. If you are an analyst, display the main page and scoreboard by selecting File, Restore Scoreboard, which is available only on pages displayed by bypassing the main page.

To create a link that bypasses the main page, specify a URL of the following form:

```
http://hostname[:port]/CAisd/pdmweb.exe?OP=operation+var=value+...
```

In this example URL, *hostname* is the web server host computer; *port* is the port number (typically 8080) required only if you are using Tomcat as your http server; *operation* is one of the supported operations, and *var=value* is one or more of the variables allowed with the operation.

For example, a link that loads the form for creating a request can be specified as the following:

```
<A HREF=http://hostname/CAisd/pdmweb.exe?OP=CREATE_NEW+FACTORY=cr>Define Request</A>
```

Posting Forms to CA SDM

You can also access CA SDM functionality by adding HTML forms to your web pages that refer to supported operations. If the form is submitted with sufficient information to perform the operation, such as creating a request, the operation is performed without displaying a form to collect additional input.

When you add an HTML form to your web page:

- The ACTION for the form is the URL for pdmweb.exe.
- The METHOD is POST.
- Either the name of the SUBMIT button should be one of the supported operations, or you should have a hidden field named OP whose value is one of the supported operations.

For example, to create an HTML form that loads the page for creating a request, specify the following code:

```
<FORM ACTION=/CAisdCAisd/pdmweb.exe METHOD=POST>
<INPUT type=HIDDEN NAME=FACTORY VALUE=iss>
.
.
.
<INPUT type=SUBMIT NAME=CREATE_NEW VALUE=" OK ">
</FORM>
```

JavaScript Modification

This article contains the following topics:

The CA SDM web interface makes extensive use of JavaScript and includes a number of JavaScript files in the /CAisd/scripts directory. If you decide to modify any of these script files, place the modified version in /CAisd/sitemods/scripts, as described in Directories Used by Your HTTP Server.

For performance reasons, the JavaScript files delivered in the /CAisd/scripts directory are compressed, with comments and unnecessary white space removed. This compression can make them difficult to read. You can find uncompressed versions of all JavaScript files in one of the following directories:

- (UNIX) $NX_ROOT/sdk/scripts
- (Windows) $NX_ROOT/sdk/scripts

If possible, avoid creating modified versions of entire JavaScript files, because each file contains a number of functions and you may only want to modify one function. In most cases, you can override individual functions by placing a modified version in the JavaScript file sitemods.js. We strongly recommend that you take this approach when modifying JavaScript.

sitemods.js

A skeleton *sitemods.js* file is distributed with CA SDM. All distributed HTMPL files include this file at the end of their <head> section, making it the last JavaScript file loaded. Because it is the last file, any functions defined in it override functions with the same name included earlier. This lets you provide your own version of a distributed JavaScript function without directly modifying distributed code.

This approach is not effective for functions invoked at load time in the <head> section, such as those in **menubar.js** and **ahdmenus.js**.

However, you can modify most JavaScript functions by completing the following steps:

1. Place a modified version of the function in sitemods.js.
2. Store the updated copy of sitemods.js in CAisd/site/mods/www/wwwroot/scripts.

Modifying Context Menus

A number of forms within CA SDM use context menus, accessed by right-clicking an object. Using the Web Screen Painter, you can modify context menus to add, remove, or modify their items.

> **NOTE**
> For more information about adding menu items, see the *Web Screen Painter Help*.

Updating and Creating Change Orders as Employee User

By default, a user can only view change orders from the Employee web interface. You can enable creating and updating change orders by employees:

Follow these steps:

1. Sign on to the web as the Administrator, and select the Administration tab.

2. Select Access Type from the Security menu.
   The Access Type List appears.
3. Select the Employee link to display the Employee Access Type Detail window.
4. Set the Change Orders to "modify" under the Function Access tab and save.
5. Click the Back button to return to the Administration tab, and then select Data Partitions, Data Partition List.
6. Click Employee to display the Data Partition Detail window. On the Constraints List portion of the window, review the Type column for following Change_Request Tables:
   - Pre-Update
   - Create
7. For each Table that you want to edit, click the Table name to display that table's Data Partition Constraint Detail window.
8. Click the Edit button.
9. Edit the constraint as follows:
   change "id = 1" to "affected_contact = @root.id".
10. Click Save.

Now when you login to the web interface as an employee user, the *Create Change Order* link appears.

Add a "Closed Change Orders" link to the Employee Scoreboard

You can use the product to add a Closed Changes node option to the Employee web interface scoreboard.

Follow these steps:

1. Log in to the product as an Administrator.
2. Click the Service Desk tab.
3. Select File, Customize Scoreboard.
   The Customize Scoreboard dialog appears.
4. Click the Role option and select Employee in the drop-down list.
5. Under Add New Node, click the Node's Stored Query link.
   The Stored Query List dialog appears.
6. Search and select Closed Changes from the Stored Query list. This is typically displayed as code CHGUBIN7.
7. Specify a location for the new node by selecting an item in the scoreboard tree on the left.
8. Click Add New Node.
   The new node named Closed Changes is added to the scoreboard tree.
9. Click Finished.

Download Attachments

When you download an attachment in CA SDM, it automatically displays the attachment in the browser window without prompting for a response from you. This action can be dangerous if a virus is associated with the attachment.

With CA SDM, you can force a save-as dialog that prompts you to respond if you want to save the attachment on the disk or open it. Saving an attachment can be a secure method because you can save the attachment on the disk and scan it before you can actually open it. You also have the option to force the save-as dialog only on certain attachment types.

You can force the save-as dialog to appear through the web.xml servlet configuration file. The web.xml file is located at the following paths:

Windows: $NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF\web.xml

Linux: $NX_ROOT is “/opt/CAisd”

Free-Form Modification of Detail Forms

This article contains the following topics:

The topics in this section describe how to perform free-form modification of Detail Forms.

Using JavaScript on Detail Forms

Use WSP to add your own fields to a detail form, or rearrange or change edit characteristics of fields provided on the form by default. However, sometimes you want to modify a form beyond simply adding new fields to a grid. There are a number of a JavaScript functions provided with CA SDM to make it easy to merge your own modifications into a combination detail form and give it any appearance you want. These functions are summarized as follows:

- You can place any HTML whatsoever prior to the DetailForm() statement or after the endDetail() statement without affecting the operation of the detail form at all.
- You can use the detailEndTable() function to close the table that lays out detail form elements in a grid. After you have done this task, you can lay out your own HTML in any desired format. In this case, your HTML is inside the detail form, and any form fields within it are submitted to the web engine when the user clicks Save. You can use the detailNextID() function to generate ID fields for your HTML elements that allow them to participate in mouse-less navigation of the detail form. You can see several examples of this technique in the notebook tabs, such as **xx_alg_tab.htmpl**.
- You can follow your own HTML with a dtlStartRow macro to restart standard detail form formatting. This starts a second grid, whose fields will not necessarily be aligned with the first. This technique is used in every notebook tab.
- If you want to insert a custom element at the end of a row, you can use the detailWriteRow() function to write out the contents of a row without closing it. You can see an example of this technique in the code that generates the “24 Hour” button in detail_cr.htmpl and detail_iss.htmpl.
- If you want to explicitly specify the contents of an element in a row without closing out the table that lays out the grid, you can use the detailRowHdr() function to specify the header text and the detailSetRowData() function to specify the data text. You can see an example of this technique in the code that generates the timer field in detail_cr.htmpl and detail_iss.htmpl.
- If you provide a function to validate a field's value (normally in an event handler), and want its results reported during browser-side validation (so that an erroneous field is redrawn with a thick red border, and an error message appears in a yellow band at the top of the form), use the function detailReportValidation(). You can see an example of this in the validate_duration() function used to validate the duration fields in xx_candp_tab.htmpl. The validate_duration() function is in the file val_type.js.
- If you want review the HTML generated for a detail form, you can use functions docWrite() and docWriteln() in place of the standard functions document.write() and document.writeln(). Then if you invoke the function holdHTMLText() anywhere in the <HEAD> section of your form, CA SDM will pop up a debugging form containing a TEXTAREA with all of the HTML generated for the form, which you can review or copy and paste into a validation tool.

While you are composing your modifications, remember that the combination detail form is displayed in both a read-only and an edit view. If your modifications apply specifically to one view or the other, you can test the current view in one of two ways:

- In JavaScript, the expression _dtl.edit is true in the edit view and false in the read-only view.
- In either JavaScript or open HTML, the statements:

```
<PDM_IF "$prop.form_name_3" == "edit">
</PDM_IF>
```

(Used only in the edit view)

or

```
<PDM_IF "$prop.form_name_3" == "ro">
```

(Used only in the read-only view)

```
</PDM_IF>
```

Used to bracket code intended only for the edit or read-only view, respectively.

detailEndTable()

This function closes the HTML table that lays out the detail form elements in a grid. It has no arguments.

You can start a new grid with the dtlStartRow() macro. However, elements in a new grid are not necessarily aligned with elements in a previous grid.

detailNextID(colspan, lastelement)

This function returns a string of the form:

```
" ID=df_nn_nn TABINDEX=n onFocus=func onBlur=func"
```

Inserting this string into an HTML element causes the element to follow the conventions of CA SDM mouse-less navigation, including accessibility with the arrow keys and turning pale yellow when focused. The returned string begins with a space and ends with no space.

- **colspan**
  Specifies the number of columns in the grid occupied by the element. This argument is optional; it defaults to one if not provided. If omitted, the element is assumed to occupy one column of the grid. This affects arrow key behavior. The *colspan* argument can be omitted even if the *lastelement* argument is provided.
- **lastelement**
  A Boolean value specifying whether the element for which the ID being generated is the last one in its row. If omitted, the element is assumed to be followed by other elements. This affects arrow key behavior.

detailNextLinkID()

This function returns a string of the form:

```
" ID=dflnk_nn_nn TABINDEX=0 onFocus=func onBlur=func"
```

Inserting this string into an HTML element defining a link element causes the element to follow the conventions of CA SDM mouse-less navigation, including accessibility with the up arrow key from the base element and turning pale yellow when focused. The returned string begins with a space and ends with no space.

This function takes no arguments.

detailReportValidation(field, has_error, emsg)

This function reports the result of external field validation. If validation is reported to have failed, the field is redrawn with a thick red border and the error message provided is shown in a yellow band at the top of the form. The user is not permitted to save the record until a subsequent call to detailReportValidation() reports the field as error-free.

The detailReportValidation() function is functional only for fields registered for browser-side validation. All fields created with detail form macros are automatically registered for validation. You can register other fields with the detailSetValidateFunction().

- **field**
  (Required) Specifies the form element object containing the field. The easiest way to obtain this is to pass this argument to the event handler performing the validation. Another way is to use the standard JavaScript function document.getElementById().
- **has_error**
  (Required) A Boolean or integer value specifying whether the field is in error. Setting a field in error prevents the user from saving the record, causes the field to be highlighted with a thick red border, and places the error message

supplied as the third argument in a yellow band at the top of the form. Setting a field as not in error reverses these changes.

- **emsg**
  A text string specifying the message to display in the yellow band at the top of the detail form when the *has_error* flag is set. This argument is required if *has_error* is set.

detailSetValidate(hdrtext, is_required, maxsize)

This function specifies that the most recent field created with an ID supplied by detailNextID() is subject to browser-side validation. Validation for required fields and for fields with a maximum size is automatic. Other forms of validation may be provided through JavaScript functions or event handlers calling detailReportValidation().

You should call detailSetValidate() only for form fields you have defined yourself whose ID was created by detailNextID(). The detailSetValidate() function must be called immediately after creating a field that you want validated. It is unnecessary (and will caused unexpected results) to call detailSetValidate() for fields created by detail form macros.

- **hdrtext**
  (Required) Specifies a string used to identify the field in error messages.
- **is_required**
  (Required) A Boolean or integer value specifying whether the field is required. CA SDM automatically verifies that all required fields are provided whenever the user attempts to save a record.
- **maxsize**
  An integer specifying the maximum length of data allowed for the field. CA SDM automatically verifies that all fields with a *maxsize* value have a length within limits whenever the user attempts to save a record. This argument is required. To suppress *maxsize* validation, specify a value of 0.

detailRowHdr(hdrtext, colspan, is_required)

This function stores text for the header (TH) element of an item in the grid. The text is not actually written to the form until a detailWriteRow() function or dtlStartRow macro is invoked.

- **hdrtext**
  Specifies the text in the header element. This argument is required.
- **colspan**
  Specifies the number of columns in the grid occupied by the element. This argument is optional; it defaults to one if not provided. If omitted, the element is assumed to occupy one column of the grid. This affects arrow key behavior. The *colspan* argument must be provided if the *is_required* argument is provided.
- **is_required**
  Specifies whether the *hdrtext* should be displayed in the style corresponding to a required field. The argument can be a Boolean, a number, or a string. A number or a string is interpreted as false if zero and true otherwise. This argument is optional; if omitted, the *hdrtext* is styled as a non-required field.

detailSetRowData(text)

This function stores HTML text for the data (TD) element of an item in the grid. The text is not actually written to the form until a detailWriteRow() function or dtlStartRow macro is invoked. The single argument is the HTML text of the element to be stored.

detailWriteRow()

This function writes the HTML stored for the current row. This creates two HTML table rows, one for the header (TH) elements and one for the data (TD) elements. The function also writes the [assign the value for TD in your book] tag that begins a new data element. The TD tag is automatically closed by the dtlStartRow macro, so it is unnecessary (and incorrect) to provide the [assign the value for TD in your book] tags in HTML text that follows detailWriteRow(). This function has no arguments.

Create a Quick Close Ticket With Preset Options

In the Quick Profile View, you can create a Quick Close ticket, for example, a Quick Close Incident. Add a preset string to the URL when you create a Quick Close ticket to add a description, a summary, or other field information automatically.

Follow these steps:

1. Copy the ahdtop_site.htmpl file from NX_ROOT/bopcfg/www/htmpl/default to NX_ROOT/site/mods/htmpl/www/default.
2. Edit the ahdtop_site.htmpl file to add the appropriate variable (depending on the type of Quick Close ticket) with the preset string.
   - Quick Close Incident -- var quick_close_preset_in
   - Quick Close Problem -- var quick_close_preset_pr
   - Quick Close Request -- var quick_close_preset_cr
   - Quick Close Issue -- var quick_close_preset_iss

   For example, the following string sets the description to HelloIncident and summary to HelloIncidentSummary for a Quick Close Incident.

   ```
   var quick_close_preset_in = "PRESET=description:HelloIncident@@summary:HelloIncidentSummary";
   ```
3. Log in to CA SDM.
4. Select View, Quick Profile in the Service Desk tab.
   The Quick Profile Contact Search page appears.
5. Complete one or more of the search fields for the contact, and click Search.
   The Quick Profile Contact List populates with those contacts that match your search criteria.
6. Select a contact.
   The right pane displays the information for that contact.
7. Click Quick Close.
   The ticket is created with the preset information.

For more information, see Supported Server Operations.

Looking Up Information in Reference Tables

Input fields on a detail form editing a database record are namedSET.attr_name. When the record is saved, data from SET fields are copied directly to the underlying record. Thus, an input field for an attribute that references another table should contain the REL_ATTR (foreign key) of that table. This is normally the id, persistent_id, or code of the reference record.

Users do not directly provide REL_ATTR values, and the SET fields for attributes referencing another table are hidden. The visible field on the form is named KEY.*attr_name*, and it contains the common name of the referenced record. A common name must be converted into a REL_ATTR to update the record. There are several times when this might be done:

- For fields with a drop-down list, the SET value is provided directly by the drop-down.
- For fields with a lookup when the user clicks the lookup and selects an item, the SET value is copied from the selected item.
- For fields with a lookup where the user provides a partial key that uniquely identifies the record and then clicks the label, the browser requests the SET value from the server and copies both it and the full key back to the form.
- If the Autofill configuration file property is provided or defaulted, and the user both provides a partial key that uniquely identifies the record and clicks Notebook to exit the field, the browser requests the SET value from the server and copies both it and the full key back to the form.

Otherwise, when the record is saved with a KEY value and no SET value, the web engine resolves the value during the save. If any KEY values cannot be resolved to a unique SET value, the save is prevented, and the edit form is redisplayed.

If a form has been redisplayed as a result of a save that failed due to a lookup resolution failure, the following variables are available in the HTMPL for each attribute field for which a lookup was performed:

- **LIST_*attr***
  Contains all the matches found. Typically this is specified as the right-hand side of the SOURCE= field in a <PDM_SELECT> statement.
- **FLAGS_*attr***
  This is set to one of the following values:
  - **0**
    Display initial search field.
  - **1**
    More than one and fewer than MaxSelectList were found (typically a <PDM_SELECT> list would be displayed in this case).
  - **2**
    No matches were found.
  - **3**
    Too many matches were found (more than MaxSelectList).
- **SEARCH_STATUS_*attr string***
  Contains the TooManyMatches text string from the web.cfg file.

Specifying Lookups on Contacts

When specifying a contact (last name, first name, middle name) in an editable form, you can delimit the contact name with commas (,) or blank spaces, but not both. Commas are preferable because names often have embedded spaces, which cause problems.

Since a combination of commas and blank spaces is not allowed, the presence of commas implies that all parts of the name are comma-separated; if no commas are present, names are delimited by spaces.

Since the information is eventually passed to an SQL query, the percent symbol (%) serves as a wildcard character. For example, 'P%, J%' would match 'Public, John', 'Penxa, Jane', and any other names whose last name begins with P and first name begins with J. (Case-sensitivity depends on the underlying database.) Similarly, 'P% J%' would bring up the same names.

However, 'P%, Jon D' would not bring up all contacts with a first name of Jon, a middle initial of D, and a last name beginning with P, because the presence of one comma means all delimiters are commas. Therefore, the last name would be looked up as 'P%' and the first name would be looked up as 'Jon D'. To avoid this error, specify 'P%, Jon, D' instead.

Understanding List Forms

The following information provides background information about the internals of CA SDM list forms. We recommend that you use the Web Screen Painter Design View to modify these forms.

CA SDM list forms are defined with the following macros (invoked with the PDM_MACRO tag):

- **lsStart**
  Begins a list
- **lsCol**
  Defines a column in a list
- **lsWrite**
  Inserts text into the pdm_list part of a list
- **lsEnd**
  Ends a list

The general form of a list using these macros is the following:

```
<pdm_macro name=lsStart>
<pdm_macro name=lsCol hdr=hdr1 attr=attr1>
<pdm_macro name=lsCol hdr=hdr1 attr=attr1>
<pdm_macro name=lsEnd>
```

This results in text similar to the following example in the output HTML:

```
var rs = new Resultset();    From lsStart
rs.startList(); From lsStart
rs.header("hdr1"); From lsCol
rs.setData("attr1","options"); From lsCol
rs.header("hdr2"); From lsCol
rs.setData("attr2","options"); From lsCol
<PDM_LIST SOURCE=list> From lsEnd
rs.data(attr1) From lsCol/lsEnd
rs.data(attr2) From lsCol/lsEnd
</PDM_LIST> From lsEnd
```

NOTE
There are two distinct sections to the output list: the setup section before the <PDM_LIST> tag, and the actual list between the <PDM_LIST> and </PDM_LIST> tags. The lsCol macro makes use of preprocessor variables and the <PDM_SET> tag to output data to both sections of the list. The entire list section of the list is created by a <PDM_EVAL> tag generated by the lsEnd macro.

To insert your own JavaScript in the setup section of the list, simply include it where needed. Use the lsWrite macro to insert your own code into the list section of the list.

The lsWrite Macro

The lsWrite macro specifies text for the list section of a list (the portion between the <pdm_list> and the </pdm_list> tags). Text specified for the text argument of this macro is deferred, and not written to the output HTML until the lsEnd macro.

lsWrite [both=no|yes]

text=”xxx”

- **both**
  Specifies that the text operand is to be written both immediately to the output HTML and to the deferred text buffer. This can be useful to output JavaScript to conditionally bypass both the setup and the list information output by a subsequent lsCol macro. Optional; defaults to no.
- **text**
  Specifies the text generated by this macro. Text specified is deferred until the lsEnd macro.

It is often desirable to include pdm tags and references to form variables in the text output by an lsWrite macro. To prevent these from being interpreted by the web engine during parsing of the lsWrite macro itself, follow these syntax rules:

- If the lsWrite macro generates a pdm_tag, omit the surrounding “<” and “>” delimiters of the tag. For example, to insert a <pdm_else> statement into the list section of the list, code:
  ```
  <PDM_MACRO NAME=lsWrite text="pdm_else">
  ```
  The web engine automatically inserts the “<” and “>” before producing the text when it detects that the first four characters are “pdm_” (or “PDM_”).
- If the lsWrite macro generates a reference to a form variable, code an @ character in place of the $ character that designates the variable. For example, to generate a reference to the list variable $list.persistent_id, code:
  ```
  <PDM_MACRO NAME=lsWrite text="@list.persistent_id">
  ```
  The web engine automatically converts the “@” to “$” before producing the text. To produce a literal @ sign, precede it with a backslash.

Web Engine PreProcessing

This article contains the following topics:

The web engine goes through two phases when processing an HTMPL file:

- The preprocessing phase, when it reads the HTMPL file and any referenced files (including files referenced by PDM_INCLUDE and PDM_MACRO tags). The output from preprocessing is an entry in the web engine's internal cache.
- The generation phase, where it reads the form from its cache and generates HTML. The output from generation is HTML delivered to the browser.

The pre-processing phase is typically done once for each form in the lifetime of the web engine. The generation phase is done each time a form is requested.

You can use the PDM_SET and PDM_EVAL tags during the preprocessor phase to generate and store information, such as HTML text, that the web engine can use in the generation phase.

Preprocessor Variables

Preprocessor variables begin with the string “$PRE.”. They are created and updated with the PDM_SET tag. This tag has the following syntax when used with a preprocessor variable:

```
<PDM_SET PRE.name[+]=value>
```

This tag assigns or updates a preprocessor variable, creating it if necessary. It is processed when the web engine encounters it while reading a form. Only the invariant PDM_IF statements affect PDM_SET of a preprocessor variable; others are ignored.

Invariant PDM_IF Detection

When parsing a form, the web engine detects invariant PDM_IF statements. An invariant PDM_IF is one whose argument consists entirely of literals, environment variables, constant properties, and preprocessor variables. When the web engine detects an invariant PDM_IF, it evaluates its condition immediately. This has the following effects:

- PDM_SET and PDM_EVAL tags that are bypassed by an invariant PDM_IF are ignored. All other pdm_eval tags and PDM_SET tags referencing preprocessor variables are executed when processed, even if they are within a non-invariant PDM_IF.
- Form variable references bypassed by an invariant PDM_IF are ignored, and their value is not fetched when the form is used. You can use this technique to improve the performance of a form. For example, if a form contains the following, the web engine fetches the value of $args.def before it displays the form:

  ```
  <PDM_IF "$env.NX_OTB_MARKET == "itil" && "$args.a" == 1>
  <h1>This is form $args.def</h1>
  </PDM_IF>
  ```

  However, if the following segment has been written, the web engine determines that the first PDM_IF is invariant, and retrieves the value of $args.def only if $NX_OTB_MARKET is “itil”.

  ```
  <PDM_IF "$env.NX_OTB_MARKET == "itil">
  <PDM_IF "$args.a" == 1>
  <h1>This is form $args.def</h1>
  </PDM_IF>
  </PDM_IF>
  ```

PDM_EVAL Insert Text from a Preprocessor Variable

The PDM_EVAL tag inserts the value of a preprocessor variable into the input to the web engine parser. If used inside a macro, its effect is deferred until the macro completes.

The PDM_EVAL tag works similarly to PDM_INCLUDE or PDM_MACRO. It inserts the text into the parser at the point of the tag, exactly as if the value of its variable had been coded in place of the tag.

PDM_EVAL has the following syntax:

```
<PDM_EVAL text=PRE.name>
```

where PRE.name specifies the name of the preprocessor variable whose value is to be inserted into the web engine's input

Execution of the PDM_EVAL tag can be controlled by invariant PDM_IF statements.

WARNING

On UNIX, the LIBPATH needs to be set before running the utility. Use pdm_task to perform this task. For example, before running the utility, input "pdm_task pdm_eval".

Event Log Data Storage Modification

The system environment variable @*NX_EVENT_LOG_EXCLUDE*, which is set in the *NX.env* file and requires a restart of the CA SDM services, lets you control the amount of data that is stored in the event log (event_log table). This variable lets you store only the events you want to track, report on, and use as part of the Recent Activity that can be launched as a button from the Quick Profile page.

In this variable, commas separate list items (for example, @*NX_EVENT_LOG_EXCLUDE* = *FAQ,KD_OPEN*). For example, if you use the LOGIN,LOGOUT events from the following table (@NX_EVENT_LOG_EXCLUDE value of LOGIN,LOGOUT), the product does not record login and logout events.

Refer to the following information when modifying data to store in the event log using this variable:

Event	Enum	By	Sets	Comments
LOGIN	1	CA SDM		Specifies that the User logs in to the system.
LOGOUT	2	CA SDM	numdata1	Specifies that the User logs out, where numdata1=logout reason:**0** -- normal**1** -- timeout**2** -- abnormal
CR_CREATE	3	CA SDM	sd_ob_type, sd_obj_id, kd, numdata1	Indicates that the User creates a request, where numdata1=id of affected end user.
ISS_CREATE	4	CA SDM	sd_ob_type, sd_obj_id, kd, numdata1	Indicates that the User creates a change order, where numdata1=id of the affected end user.
CHG_CREATE	5	CA SDM	sd_ob_type, sd_obj_id, kd, numdata1	Indicates that the User creates an issue, where numdata1=id of the affected end user.
EMAIL	6	Knowledge Management	kd	Specifies that the Analyst emails a document.
LINK	7	Knowledge Management	kd, sd_obj_type, sd_obj_id	Indicates that the User accepts a solution, and links it to a ticket.

UNLINK	8	CA SDM	sd_id, sd_obj_type,sd_obj_id	Specifies that the User unlinks a solution from a ticket.
SEARCH	9	Knowledge Management	numdata1,	Indicates that the User searches knowledge, where numdata1=CI_ASKED_QUES id.
FAQ	10	Knowledge Management	numdata1	Indicates a FAQ search, where numdata1=O_INDEXES id (category).
DT_NAVIGATE	11	Knowledge Management	kd,numdata1,textdata1	Indicates that the User navigates a decision tree, where numdata1=ES_NODES ID textdata1=path.
KD_BOOKMARK	12	Knowledge Management	kd	Indicates that the User bookmarks a KD.
KD_COMMENT	13	Knowledge Management	kd, numdata1	Indicates that the User adds a comment to a KD, where numdata1=O_COMMENTS id.
KD_CREATE	14	Knowledge Management	sd_ob_type, sd_obj_id, kd	Specifies that a User creates a document. CA SDM IDs are used when a KD is created using submit knowledge from a request or an issue.
KD_OPEN	15	Knowledge Management	kd, numdata1	Indicates that a User opens a KD, where numdata1=BU_TRANS ID.
KD_RATE	16	Knowledge Management		Indicates that a User rates a KD, where numdata1=BU_TRANS ID.
KD_NEW	17	Knowledge Management	numdata1	Specifies that a User clicks on the New Documents folder in the Knowledge tab.

NX_ATTACH_ AUDIT_TO_NEW_TICKET	18	CA SDM		When a User opens a new ticket, all events for the current session appear by default on the Event Log tab of the ticket. **0** -- Only events relevant to the ticket appear on the Event Log tab. **1** -- All events for the current session appear on the Event Log tab of the ticket.
TICK_OPEN	19	CA SDM		Indicates that the ticket was viewed.
TICK_SEARCH	20	CA SDM		Indicates the user who searched for tickets and links the number of searches.
KD_PRNT	21	Knowledge Management	kd	Indicates the knowledge document was printed.

Print CA SDM Web Pages

CA SDM uses background graphics to format its buttons and notebook tabs. The default setting for many browsers is *not* to print these background images. Therefore, if you select File, Print on either the browser menu or the CA SDM menu, the printed page shows only the corners of the buttons or tabs.

You can print CA SDM web pages on Internet Explorer.

Follow these steps:

1. Select Tools, Internet options.
   The Internet Options dialog appears.
2. Select the Advanced tab.
3. Scroll down to the Printing heading, and select the Print Background Colors and Images option.
   The printed CA SDM web pages include background graphics.

You can print CA SDM web pages on Firefox.

Follow these steps:

1. Select File, Page Setup.
2. Select the Format & Options tab.
3. Select the Print Background (color & images) check box.
   The printed CA SDM web pages include background graphics.

Legacy Reports Modification

CA SDM lets you modify legacy reports or design your own reports. You can:

- Modify legacy Summary, Detail, and Analysis reports to contain exactly the information you need, such as additional fields.
- Produce a new report with any information available from the database in a format that is useful to you.
- Pass arguments of variable information into the report by including command line arguments. Arguments can be values or expressions, such as the current value of a field or an SQL WHERE clause expression.
- Generate reports at the command line, from a script file, or from a menu option.

Follow these steps:

1. Design the report:
   – Decide what data you want to include in the report.
   – Create a report template that contains SQL-like queries, expressions, and functions to manipulate data, and statements to format the data for the printed page.
2. Generate the report from:
   – The command line
   – A CA SDM menu option
   – A script file

> **NOTE**
> If you have a third-party database system you can use its report generating tools to create reports with data from the CA SDM database. CA SDM provides several database views that simplify the process of creating modified reports using third-party database systems. See the documentation for your database system for information about reporting on databases.

Modify Crystal Reports

Before you can display any of these reports, the following conditions apply:

- You must make the Crystal reports available to the Crystal Report Selector by copying them to the Crystal directory: $NX_ROOT/bopcfg/rpt.
- Your database client must be up and running, with connectivity established to the database server running on the same or another computer. If you are using a CA SDM Client to run your Crystal or Access reports you need to have installed a database client for the specific database and have established connectivity with the database server in order to run these reports.

After creating any custom Crystal reports, perform the following:

1. Copy custom Crystal reports to the following crystal directory:
   `$NX_ROOT/bopcfg/rpt`
2. Add the file names of custom Crystal reports to the following configuration file:
   `crystal.cfg`

You can then access the Crystal reports by clicking Start on the taskbar, and then choosing Reporting, Service Desk Reporting (Crystal Reports) from the CA SDM menu (accessible from the Programs menu). The Service Desk Reporting (Crystal) window appears.

> **WARNING**
> CA SDM clients cannot be upgraded. Therefore, if you create and use Crystal reports on the CA SDM Server and you plan to upgrade your version of CA SDM, you need to copy all custom reports to a different location so you will not lose them. Following the upgrade, copy the reports back to the $NX_ROOT/bopcfg/rpt Crystal directory and modify the crystal.config file to make them accessible from the Report Selector.

Custom Report Design

This article contains the following topics:

To design a custom report, you must have a basic understanding of the following concepts:

- Writing SQL queries.
- Programming, especially in C.
- Creating special programs or script files that you may need to execute before you execute the report template program. For example, you may want to create a program that prompts the user to enter an argument, such as the conditions for a WHERE clause.

> **NOTE**
> Before you create a custom report, be sure to check if the report you need is already provided. CA SDM provides a wide variety of Crystal and Microsoft Access reports, and runtime versions of these products to let you run the reports.

Selecting Information for the Report

To help you select data from the CA SDM database for customized reports, see CA Service Desk Manager Reference Commands. It lists database tables, fields, descriptions, and other database information.

How to Create a Report Template

A report template is a file that, when executed by a CA SDM report program, generates a report of a particular design. A report template contains variable expressions, functions, and statements that define how the data is fetched, calculated, and printed.

To create a report template, create a file containing the following types of report statements:

- **Block statements**
  Defines the CA SDM database tables from which data will be fetched and the actions that are to be performed on the fetched data.
- **Layout statements**
  Defines how the data variables and literal text display on the report output.

> **NOTE**
> Store all your .rpt files in a new directory, $NX_ROOT/site/mods/rpt (UNIX) or *installation-directory*\site\mods\rpt (Windows). This directory preserves them when you upgrade to a new release of CA SDM.

Block Statements

Block statements provide the report template with its framework. They define the data to be manipulated and control the execution of the report. Block statements begin with a name that must be unique throughout the report template. They then have the following two sections:

- **Data query section**
  Contains SQL SELECT, WHERE, and SORT clauses to define which data is fetched from the database.
- **Output program section**
  Defines the actions that are to be performed on the fetched data. It contains variable declarations, functions, and other block statements, including nested statements, which can be used to create conditional reports. It can also contain layout statements, which format and print the data as ASCII text.

A simplified version of the syntax of a block statement that shows the relationship between the two sections follows:

```
BLOCK blockname ("SELECT clause", "WHERE clause")
   SORT clause {output program statements}
```

The BLOCK in the Reference section discusses the detailed version of the syntax, along with a description of each clause and parameter.

Layout Statements in Report Templates

Layout statements define how variables and literal text will appear on the report output:

- You can use the PAGE HEADER and PAGE FOOTER statements to place information at the top and bottom of each report page.
- You can nest HEADER, HEADER2, FOOTER, and PRINT statements within the braces section of the parent BLOCK statement to create titles and summary totals for the various *reporting sections* (parts of the report output).

  NOTE
  When nesting, be careful not to confuse the braces used in layout statements with the braces that encompass the nested statements within a parent BLOCK statement.
- You can include literal text to create labels and line drawing characters to enhance the appearance of the report.

The layout statements are as follows:

- **PAGE HEADER**
  Places information at the top of each report page. It is placed outside the BLOCK statement.
- **PAGE FOOTER**
  Places information at the bottom of each report page. It is placed outside the BLOCK statement.
- **HEADER**
  Places information at the top of each reporting section. It is placed inside the BLOCK statement.
- **HEADER2**
  Places continuation header information at the top of each succeeding page of a reporting section, if that reporting section extends over multiple pages. It is placed inside the BLOCK statement.
- **FOOTER**
  Places information at the bottom of each reporting section. It is placed inside the BLOCK statement.
- **PRINT**
  Places the data in a reporting section. It is placed inside the BLOCK statement.

You can also use the following predefined variables in layout statements:

- CT prints the current time
- CD prints the current date
- PG prints the page number
- **Data Fields**
  Specifies any variable in a layout statement that results in a piece of data when you generate the report. Use the following guidelines when placing fields in your report template:
  - Enclose data fields in square brackets ([ ]).
  - The field's square brackets define its print space on each line of output. This space is the number of characters delimited by the square brackets, including the brackets. If the output of a variable is longer than the print space, the output is truncated. To ensure that the field has enough print space, you can add trailing spaces between the variable name and the closing bracket. For example, these trailing spaces allow for contacts with long names:

    ```
    [contact          ]
    ```
  - For output that is less than one line, the field can be closed with a greater than right angle bracket (>). This extends the print space to the right margin. For example, the right angle bracket used in a HEADER statement allows the current date to print without being truncated:

    ```
    [CD               >
    ```

    NOTE
    When the field is more than one line and the variable is flagged as MULTILINE, the right angle bracket (>) acts exactly the same as the right square bracket (]). If the print statement for a MULTILINE variable is closed with the right angle bracket (>), characters wrap on white space to stay within the field defined

by the left bracket ([) and the right angle bracket (>). Also, if the variable is not MULTILINE, the right angle bracket (>) causes all the data to be displayed on the current line regardless of its length.

- A field in a layout statement can refer to a previously defined variable or a column name.
- To reference a variable or column name in another block statement, use the following syntax:

```
blockname::column | variable-name
```

- **Literal Text**
  Literal text allows you to include supplementary information in your report. It will appear on the report output exactly as specified in the template. To include literal text in a layout statement, place it on any line after the opening brace ({) and before the closing brace (}). Do not enclose it in quotes or square brackets.
  In this example, “ACME Company” and “Page: ” are interpreted as literal text by the CA SDM report program:

```
PAGE HEADER {
                ACME Company           Page: [PG]
}
```

Variable Expressions in Report Templates

Every value that you want to appear on the report output can be assigned to a variable. Variable expressions let you:

- Manipulate CA SDM data
- Use functions to perform calculations on fetched values

The following example creates a variable named *desc* to reference the contents of the chg_desc field in the Change Order window. The MULTILINE flag allows the variable to print in its entirety over multiple lines:

```
desc = description MULTILINE;
```

The following example prints the description. The output will be as long as the length defined in the brackets. If you want a longer description to appear, increase the number of spaces in the brackets.

```
PRINT { [desc  ]                                  }
```

Example Report Template

The following Affected Contact Report template shows how to create a report template. It produces a report that lists open change orders with the same affected contact:

```
PAGE HEADER {
                                                        As Of: [CD>
                                                               [CT>
}
PAGE FOOTER {
                          Page: [PG>
}
BLOCK chg ("SELECT \
        chg_ref_num, description, priority, \
        status, category, assignee \
      FROM Change_Request",
    "WHERE #Change_Request.status = 'OP' \
AND #Change_Request.requestor = #ca_contact.id \
AND #ca_contact.last_name = ? \
AND #ca_contact.first_name = ? \
AND #ca_contact.middle_name = ? " , $1, $2, $3)
{
    BLOCK st ("SELECT sym FROM Change_Status",
            "WHERE code = ? ", chg::status) {}
    BLOCK (strlen(category)) cat ("SELECT sym FROM Change_Category",
```

```
                    "WHERE code = ? ", chg::category) {}
HEADER {
        OPEN CHANGE ORDERS WITH SAME REQUESTOR/FROM CONTACT
CHANGE ORDER Summary          Pri   Status    Category          Assignee
}
HEADER2 {
CHANGE ORDER Summary          Pri   Status    Category          Assignee
-------------------------------------------------------------------------
}
    num = chg_ref_num;
    desc = description MULTILINE;
    pr = deref (priority);
    stat = st::sym;
    catgry = cat::sym;
    asgn = deref (assignee);
PRINT {
[num      ] [desc            ][pr ] [stat   ] [catgry          ] [asgn ]
}
}
```

- **Page Header**
  Specifies what to print on the top of each page of the report. CD and CT are predefined variables that give the current date and time. They will appear in the header on the top of each page. Each of these fields ends with an angle bracket, which allows the field to expand towards the right margin. Because “As Of:” is outside of a field and because it is on a line after the opening brace, it will appear as literal text on the report output.

```
 PAGE HEADER {                                As Of:  [CD>
                                                      [CT>
 }
```

- **Page Footer**
  Includes the page number with “Page: ” as literal text.

```
 PAGE FOOTER {
                                  Page: [PG>
 }
```

  NOTE
  Since PAGE HEADER and PAGE FOOTER statements produce global headers and footers, they are not included in a BLOCK statement.

- **Reporting Section**
  Creates a reporting section for the main BLOCK statement, along with its nested statements. A reporting section is usually only part of the data in the report, but this report has only one reporting section. The unique name of this block is chg.
  The SELECT clause selects the columns to be included in the data for the report FROM three tables, but only where conditions specified by the WHERE clause are met.
  The last three AND expressions in the WHERE clause contain question marks, which act as argument placeholders that take the values of the $1, $2, and $3 arguments, in order. Thus $1 is for ca_contact.last_name, $2 is for ca_contact.first_name, and $3 is for ca_contact.middle_name. The $1, $2, and $3 arguments obtain the values of command line arguments.

```
 BLOCK chg ("SELECT \
 ...",
 "WHERE \
 ...\
 AND #ca_contact.last_name = ? \
 AND #ca_contact.first_name = ? \
```

```
AND #ca_contact.middle_name = ? ", $1, $2, $3)
```

- **Reporting Section Headers**
  Specifies that the opening brace starts the output program part of the BLOCK statement: its statements tell what to do with the data fetched by the SELECT and WHERE clauses. This example has nested HEADER and HEADER2 statements that will apply to this reporting section only. HEADER2 prints only if the report output is on multiple pages.

```
{
   ...
   HEADER {
               OPEN CHANGE ORDERS WITH SAME REQUESTOR/FROM CONTACT
CHANGE ORDER Summary            Pri    Status  Category  Assignee
}
   HEADER2 {
CHANGE ORDER Summary            Pri    Status  Category  Assignee
------------------------------------------------------------------
}
```

- **Variable Assignments**
  Specifies variable expressions that act on the data specified by the SELECT clauses. They assign variables to the values of columns and to the results of expressions. These variables match the fields in the PRINT statement that follows.
  The MULTILINE flag on the *desc* variable causes them to print or display on multiple lines rather than being truncated. The deref function is used to return the string expression contained in the referenced columns.

```
num = chg_ref_num;
desc = description MULTILINE;
pr = deref (priority);
stat = st::sym;
catgry = cat::sym;
asgn = deref (assignee);
```

- **Printing**
  Contains the fields to be printed. This statement could have also included literal text of lines that could enhance the appearance of the report. The final ending brace matches the opening brace of the output program section of the BLOCK statement.

```
PRINT {
[num ] [desc          ] [pr]  [stat] [catgry] [asgn         ]

}
}
```

Report Template Reference

You can use variable expressions, functions, and statements in a report template.

This section contains the following articles:

- Variable Expressions
- Report Template Functions
- Report Template BLOCK Statements
- Report Template FOOTER Statements
- HEADER
- Report Template HEADER2 Statements
- Report Template PAGE FOOTER Statements
- Report Template PAGE HEADER Statements
- Report Template PRINT Statements

Variable Expressions

This article contains the following topics:

Variable expressions define the data to be printed or displayed in a report template. They are placed in a layout or block statement.

Variable Expression Syntax

Syntax refers to the rules governing the formation of statements in a programming language. The structure of this string is as follows:

```
variable-name = expression [flags]
```

Variable Expression Flags

Flags format the result of a variable expression. Use these flags to format text fields:

- **MULTILINE**
  Displays on multiple lines rather than truncating.
- **RIGHT**
  Right justifies.

Use these flags to format numeric fields:

- **BLANKZERO**
  Functions as null-value fields, which do not print a zero.
- **BOOL**
  Converts zero to no or non-zero to yes.
- **REAL**
  Displays as floating point (default is integer).
- **ZEROFILL**
  Shows leading or trailing zeros

Use these flags to format date and time fields:

- **DATE**
  Shows only date portion of date/time.
- **DAYS**
  Displays durations with days.
- **HOURS**
  Displays durations with hours.
- **MINUTES**
  Displays durations with minutes.
- **SECONDS**
  Displays durations with seconds.
- **TIME**
  Shows only time portion of date/time.

Variable Expression Example

The following line displays a Variable Expression example:

```
desc = description MULTILINE
```

Variable Expression Remarks

Variable names must be unique within a BLOCK statement and must not duplicate any column in the SELECT clause for the block. The same variable name can be used in different BLOCK statements but it cannot be repeated within a BLOCK statement.

Follow these syntax rules when including expressions in your report template:

- Use any valid C expression.
- Do not enclose variable or column names in quotes.
- Enclose string constants in single or double quotes.
- You can refer to a nested block, but only if it contains exactly one row.
- To include a column name that is the same as a keyword, precede the column name with a backslash (\). For example, ALIAS is a keyword and \alias is a column name.
- Use the dollar sign ($) to reference environment variables, such as $name, and to reference command line arguments, such as $*n*, where *n* is the position of the argument on the command line.
- To specify the number of command line arguments, use $#. For example, the following expression means that if the number of command line arguments is greater than one, use the additional argument as an argument; otherwise, set the value of the argument to an empty string. The report template itself is considered a command line argument. Therefore, the number of arguments is at least one.

  ```
  $# > 1 ? $1 : "
  ```
- Use ## to concatenate two strings, for example:

  ```
  title = "This is the " ## "first line. "
  long_name = fn
  irst_name ## last_name
  ```
- The following casts are supported:
  - (number)
  - (string)
  - (date_time)
  - (duration)
- To reference a variable or column name in another block, precede the name with its block name and two colons. For example:

  ```
  blockname::column | variable-name
  ```

Report Template Functions

The following functions can be used in your report template:

- **is_null (*expr*)**
  This function returns true if the expression is null.

  ```
  false = 0
  true = is_null (false)
  ```
- **sqrt (*expr*)**
  This function calculates the square root of the expression.

  ```
  nine = 9
  three = sqrt (nine)
  ```
- **pow (*expr1, expr2*)**
  This function raises *expr1* to the power *expr2.*

  ```
  two = 2
  three = 3
  ```

```
eight = pow (two,three)
```

- **log (*expr, expr*)**
  This function calculates the natural log of the expression.

```
ten = 10
result = log (ten)
```

- **catname (*expr*, *expr*, *expr*)**
  This concatenates three strings representing a contact name into a string with commas, according to rules in the field format file.

```
last = "Murphy"
first = "Fred"
middle = "P"
contact_name=catname (last, first, middle)
```

- **strlen (*string*)**
  This function returns the length of the string.

```
buffer = "A thirty character long string"
thirty = strlen(buffer)
```

- **strindex *(string, pattern [, start_index]*)**
  This function returns the index of the first pattern match, or the next pattern match after the start_index, in the string. Returns -1 if there is no match.

```
buffer = "A thirty character long string"
zero = strindex(buffer, " [A-Z] ")
two = strindex(buffer, " [a-z] ")
```

- **substr (*string, pattern [, length]*)**
  This function returns the portion of the string after the first pattern match. It length is defined, it limits the length of the output string. Returns a string of zero length, if there is no match.

```
buffer = "A thirty character long string"
last_word = substr(buffer, " [a-z]*$ ")
first_capital_letter = substr(buffer, " [A-Z] ",
    1)
```

- **substr *(string, index [, length]*)**
  This function returns the portion of the string after the index. Its length is defined and limits the length of the output string. Returns a string of zero length, if there is no match.

```
buffer = "Summary: The network card displays a
    code of ... "
summary = substr(buffer, 9)
30_char_summary = strindex(buffer, 9, 30)
```

The remaining functions (pseudofunctions) perform on a block of data rather than on variable expressions. These functions are usually placed in a BLOCK statement to get information about a nested BLOCK statement's data.

- **count (*block-name*)**
  Returns the number of rows in the block specified in the BLOCK statement. The block-name must be a simple string.

```
BLOCK sample ("SELECT id  FROM Contact") {
entries = count (sample)
}
```

- **sum (*block-name, expr*)**
  Executes the expression for each row of the specified block and sums the result.

```
BLOCK sample ("SELECT actual_cost, est_cost FROM Change_Request") {
difference = sum (sample, est_cost-actual_cost)
}
```

- **average (*block-name*, *expr*)**

Executes the expression for each row of the block and returns the average of the result.

```
BLOCK sample ("SELECT actual_cost, est_cost FROM Change_Request"){
avg_difference = average (sample, est_cost-actual_cost)
}
```

- **prev (*expr*)**
  Returns the previous value of the expression. This function should be used with caution so its value does not overwrite the latest value by accident.
- **downtime *(sla_schedule, expr1, expr2 [, delay-block, expr, expr]*)**
  Invokes an SLA downtime calculation. The first argument must be a string that identifies a workshift. The other arguments are start and end times:
  expr1 is the start date/time of the event
  expr2 is the end date/time of the event
  In this example, the wrkshft BLOCK fetches the work shift schedule, the evt_dly BLOCK statement fetches the delays and the downtime function uses these records to calculate the downtime.
  BLOCK attevt ("SELECT start_time, fire_time, event_tmpl, obj_id FROM Attached_Events")

```
{
BLOCK evt ("SELECT persid, sym, work_shift FROM Events ", "WHERE persid = ?", attevt::event_tmpl) {}
BLOCK wrkshft ("SELECT sched FROM Bop_Workshift", "WHERE persid = ?", evt::work_shift) {}

BLOCK evt_dly ("SELECT start_time, stop_time FROM Event_Delay", "WHERE obj_id = ?", attevt::obj_id) {}
total_downtime = downtime(wrkshft::sched,
attevt::start_time,  attevt::fire_time,
evt_dly,
evt_dly::start_time, evt_dly::stop_time);
}
```

- **deref (*column-name*)**
  Returns the string representation of the pointer by performing an automatic lookup in the appropriate table.

```
BLOCK chg ("SELECT organization FROM Change_Request") {
org = deref (organization)
}
```

  Because this pseudofunction involves lookups, it is valid only if it is the only thing in the expression. For example, this is valid:

```
model = deref (nr_model)
This is not valid:
model = "model"  ## deref (nr_model)
```

  NOTE
  Forward references to variables or blocks are not allowed.

Report Template BLOCK Statements

This article contains the following topics:

Block statements define the database tables from which data will be fetched, and can include actions to perform on the fetched data.

Block Statement Syntax

Syntax refers to the rules governing the formation of statements in a programming language. The structure of this string for BLOCK is as follows:

```
BLOCK blockname (
       "SELECT [ALIAS,] field_name[, field_name ...]
```

```
        FROM table_name[, table_name ...] "
        [,"WHERE where_clause"][, arguments,] )
        [SORT "sort clause"]
 {
 output program statements
 }
```

Block Statement Parameters

- ***blockname***
  Identifies the block. Each *blockname* must be unique.
- **SELECT clause**
  Follows the *blockname* and is delimited by double quotes. Lists the columns to be fetched, followed by the keyword FROM, followed by the tables from which the columns are to be fetched. It is required. Here is an example with three tables specified:

  ```
  "SELECT open_date, chg_ref_num \
  last_name, first_name \
  FROM Change_Request, \
  ca_contact"
  ```

  You cannot include an SQL alias, such as:

  ```
  "SELECT open_date As OpenDate"
  ```
- **WHERE clause**
  (Optional) Follows the SELECT clause and further qualifies the information selected. It may be a string constant or an expression evaluating to a string. If the WHERE clause is an empty string, all records are returned. WHERE clauses can contain replacement arguments (which refer to variables or command line arguments) using the syntax of a question mark (?). The following WHERE clause could follow the previous SELECT clause:

  ```
  "WHERE #Change_Request.open_date >= ? \
  AND #Change_Request.active_flag = 1 \
  AND #ca_contact.last_name = ? ", $1
  ```

  NOTE
  The WHERE clause must be separate from the SELECT clause because the WHERE clause can be an expression evaluating to a string, whereas the SELECT clause is exclusively a string constant. This gives you more flexibility and data manipulation capabilities in producing your report.
- **SORT clause**
  (Optional) Follows the SELECT and WHERE clauses and sorts the fetched rows of data. The SORT clause is formatted like the SQL ORDER BY clause. Here is an example:

  ```
  SORT "open_date"
  ```
- **Output program statements**
  Controls execution of the report. Before the data query, the HEADER statement, if included, prints the heater test for the block. The data query then runs. If data is returned, each statement executes in the order written, with one exception. Block functions, like sum and average, behave as thought they were at the end of the output program. In fact, their values are not stable until execution begins on the next data record.

  WARNING
  The output program depends on the success of the data query. If no data is returned from the query, then, except for the HEADER statement, the output program will not execute.

Block Statement Example

This BLOCK statement assumes that an argument will be passed that holds an integer equal to the change order priority. The WHERE clause first checks the number of arguments passed ($#). If one is present, it is used to evaluate the expression to produce the WHERE clause; otherwise a null WHERE clause is substituted (" ").

```
BLOCK chg ("SELECT priority FROM Change_Request",
$# > 1 ? "WHERE priority =" ## $1 :  "") {}
```

Footer Statement Example

The following lines display a Footer Statement example:

```
FOOTER {
     Summary Information:
             Total Failures:  [Fail_count >
             Total Downtime:  [Downtime >
}
```

Block Statement Remarks

HEADER, HEADER2, FOOTER, PRINT and variable expressions can be placed in the braces. Any statement will be executed for each row selected.

> **NOTE**
> PAGE HEADER and PAGE FOOTER statements cannot be placed in a BLOCK statement.

Report Template FOOTER Statements

This article contains the following topics:

This layout statement places information at the bottom of a reporting section.

Footer Statement Syntax

Syntax refers to the rules governing the formation of statements in a programming language. The structure of this string for FOOTER is as follows:

```
FOOTER {parameters}
```

Footer Statement Parameters

The parameters are as follows:

- **CD**
  A predefined variable used to display the current date.
- **CT**
  A predefined variable used to display the current time.
- **PG**
  A predefined variable used to display the current page number.
- **column | variable-name**
  This field can be a variable from an earlier variable expression or a reference to a column in the SQL clause of a BLOCK statement.
- **literal-text**
  Any text that is not a predefined variable or a column or variable name is interpreted as literal text. Literal text that you include in the FOOTER statement appears in the exact horizontal location where you enter it.

Footer Statement Remarks

FOOTER statements are printed at the bottom of a reporting section. A typical use might be to present summary information or statistics. You can include a FOOTER statement in a BLOCK statement.

A field's content occupies the exact space delineated by the square brackets. Any excess characters are truncated. However, you can close a field with an angle bracket (>) to permit its content to expand in its entirety towards the right margin.

To reference a variable or column name in another BLOCK statement, use the following syntax:

```
blockname::column | variable-name
```

HEADER

This article contains the following topics:

This layout statement places information at the top of a reporting section.

Header Statement Syntax

Syntax refers to the rules governing the formation of statements in a programming language. The structure of this string for HEADER is as follows:

```
HEADER {parameters}
```

Header Statement Parameters

For a list and explanation of the valid parameters for this statement, see Report Template PAGE HEADER Statements.

Header Statement Example

The following lines display a Header Statement example:

```
HEADER {
                 Contact Summary Report
     Contact Name   Contact Alias   Organization
}
```

Header Statement Remarks

HEADERS are printed at the beginning of a reporting section and can be included in a BLOCK statement. HEADERS are typically used to present section and/or column headings.

A field's content occupies the exact space delineated by the square brackets. Any excess characters are truncated. However, you can close a field with an angle bracket (>) to permit its content to expand in its entirety towards the right margin.

> **NOTE**
> If the print statement for a MULTILINE variable is closed with the right angle bracket (>), characters wrap on white space to stay within the field defined by the left bracket ([) and the right angle bracket (>). Also, if the variable is not MULTILINE, the right angle bracket (>) causes all the data to be displayed on the current line regardless of its length.

To reference a variable or column name in another BLOCK statement, use the following syntax:

```
blockname::column | variable-name
```

Report Template HEADER2 Statements

This article contains the following topics:

This layout statement places continuation HEADER information at the top of each succeeding page of a reporting section, if that reporting section extends over multiple pages.

Header2 Statement Syntax

Syntax refers to the rules governing the formation of statements in a programming language. The structure of this string for HEADER2 is as follows:

```
HEADER2{parameters}
```

Header2 Statement Parameters

Header2 Statement Example

```
HEADER2 {
   Contact Summary Report (continued)
   Contact Name   Contact Alias   Organization
}
```

Header2 Statement Remarks

A HEADER2 statement can be included in a BLOCK statement.

A field's content occupies the exact space delineated by the square brackets. Any excess characters are truncated. However, you can close a field with an angle bracket (>) to permit its content to expand in its entirety towards the right margin.

To reference a variable or column name in another BLOCK statement, use the following syntax:

```
blockname::column | variable-name
```

Report Template PAGE FOOTER Statements

This article contains the following topics:

This layout statement places information at the bottom of each report page.

Page Footer Statement Syntax

```
PAGE FOOTER {parameters}
```

Page Footer Statement Parameters

With the exception that you cannot use column and variable names, the parameters for this statement are the same as those for FOOTER.

Parameters for a Page Footer Statement Example

```
PAGE FOOTER {
                Page Number: [PG>
}
```

Parameters for a Page Footer Statement Remarks

A field's content occupies the exact space delineated by the square brackets. Any excess characters are truncated. However, you can close a field with an angle bracket (>) to permit its content to expand in its entirety towards the right margin.

To reference a variable or column name in another BLOCK statement, use the following syntax:

```
blockname::column | variable-name
```

Report Template PAGE HEADER Statements

This article contains the following topics:

This layout statement places information at the top of each report page.

Page Header Statement Syntax

Syntax refers to the rules governing the formation of statements in a programming language. The structure of this string for PAGE HEADER is as follows:

```
PAGE HEADER {parameters}
```

Page Header Statement Parameters

With the exception that you cannot use column and variable names, the parameters for this statement are the same as those for FOOTER. For a list and explanation of the valid parameters for this statement, see Report Template PAGE FOOTER Statements.

Page Header Statement Example

```
PAGE HEADER {
   Date of Report:  [CD>
   Time of Report:  [CT>
}
```

Page Header Statement Remarks

PAGE HEADERS are printed at the top of every report page. They can be defined at any point within the report template file, but they cannot be included within a BLOCK statement.

A field's content occupies the exact space delineated by the square brackets. Any excess characters are truncated. However, you can close a field with an angle bracket (>) to permit its content to expand in its entirety towards the right margin.

To reference a variable or column name in another BLOCK statement, use the following syntax:

```
blockname::column | variable-name
```

Report Template PRINT Statements

This article contains the following topics:

This layout statement places data in a reporting section.

Print Statement Syntax

Syntax refers to the rules governing the formation of statements in a programming language. The structure of this string for PRINT is as follows:

```
PRINT {parameters}
```

Print Statement Parameters

Refer FOOTER for a list and explanation of the valid parameters for this statement.

Print Statement Example

```
PRINT {
[num ] [desc          ] [pr]  [stat] [catgry] [asgn               ]
}
```

Print Statement Remarks

Place PRINT where you want the data for a reporting section to appear in the report. You can include a PRINT statement in a BLOCK statement.

A field's content occupies the exact space delineated by the square brackets. Any excess characters are truncated. However, you can close a field with an angle bracket (>) to permit its content to expand in its entirety towards the right margin.

> **NOTE**
> If the print statement for a MULTILINE variable is closed with the right angle bracket (>), characters wrap on white space to stay within the field defined by the left bracket ([) and the right angle bracket (>). Also, if the variable is not MULTILINE, the right angle bracket (>) causes all the data to be displayed on the current line regardless of its length.

To reference a variable or column name in another BLOCK statement, use the following syntax:

```
blockname::column | variable-name
```

Web Services Management

This article contains the following topics:

Web Services are a set of data exchange standards that enable communication between products, even if they are on different operating environments. This ability is analogous to browsing the Web on a personal computer -- all remote websites are accessible regardless of whether they are hosted on Solaris, AIX, Windows, and so on. In the same manner, Web Services allow products to communicate over HTTP to various servers regardless of operating environment. For example, a Microsoft Office product can communicate with a program on a UNIX server, and a Java Server Page can access a server hosted on a Windows server. This platform-neutral communication allows for powerful integrations.

The Web Services take advantage of this technology, allowing almost any product to access CA SDM and Knowledge Management. Web Services clients can create tickets, update assets, search the knowledge base, and so forth.

The following Web Services frame work are supported:

- Axis
  `http://<servername>:<portno>/axis/services`
- CXF
  `http://<servername>:<portno>/cxf/services`

WARNING

For additional information on web services, see the CA SDM Reference.

CA SDM Components

CA SDM provides the installation files for this version of the J2EE Web Services in the following directory:

```
<NX_ROOT>/sdk/websvc/R11
```

NOTE

<NX_ROOT> specifies the root installation path for CA SDM.

Web Service Options

These options control the web service session:

- **rest_webservice_access_duration**
  Specifies the number of hours that the REST Web Service Access Key remains active before expiration. The Access Key timeout is not based on inactivity time, but it is based on duration time since the Access Key creation. After the Access Key meets the specified duration, the Access Key ends regardless of whether it is being used.
  Optionally, the REST client can also provide the Access Key duration time for the specific Access Key during the Access Key request. To provide the duration value, set it directly on the expiration_date attribute of the rest_access resource, as part of the POST request payload.
  Valid Range: 1-8760 hours
  Defaut: 168
- **rest_webservice_disable_basic_auth**
  Disables Basic Authentication in REST Web Services.
  Default: No
- **rest_webservice_list_max_length**
  Specifies the maximum number of rows that a REST Web Service query returns.
  Default: 500
- **rest_webservice_list_page_length**
  Specifies the default number of rows that a REST Web Service query returns per page.
  Valid Range: 1-500
  Default: 25
- **rest_webservice_resources_to_expose**
  Specifies the list of Majic factories (resources) that CA SDM exposes through REST Web Services. This option overrides the default behavior. By default, CA SDM exposes all factories through REST Web Services.
  If you do not enter values in this option, the default behavior exposes any Majic factory that does not have the REST_OPERATIONS property set to NONE. By default, no Majic fatory has this property set to NONE.
  Use the REST_OPERATIONS property to set the specific HTTP CRUD (CREATE, READ, UPDATE, DELETE) methods for CA SDM to expose on a given Majic factory.
  Default: rest_access
  Example: rest_access, cnt, grp, cr, crs, pri, alg, urg, imp, pcat, org
- **hmac_algorithm**
  Specifies the algorithm that you use to compute the signature for Custom/Secret Key Auhentication in REST Web Services.
  Default: HmacSHA1
- **string_to_sign_fields**
  Specifies the fields that you use to compute the signature for Custom/Secret Key Authentication in REST Web Services, in addition to the default REQUEST_METHOD, REQUEST_URI, and QUERY_STRING fields.
  Default: blank
- **webservice_domsrvr**

Specifies the name of the object engine that SOAP web services use. If not installed, SOAP web services use "domsrvr".
The value of the option must be a string beginning with the characters "domsrvr:"

- **webservice_session_timeout**
  Sets the timeout value (in minutes) for SOAP Web Service sessions. When the time between successive web method calls is greater than the value specified, the session ID is marked expired. The session is then no longer valid.
  To prevent sessions from expiring due to activity, set the value for this option to 0. Other methods, such as logoff routines, can still invalid sessions.

  NOTE
  These options require restarting the CA SDM server.

Web Services Installation

Depending on your configuration type, CA SDM installs web services for the following servers:

- Conventional: Both primary and secondary servers. For web service clients to use a URL on a secondary server, you add a web engine to the secondary server
- Advanced Availability: Application server

Web services use the default object manager that is installed on the CA SDM server. To use any other object manager, install and set the *webservice_domsrvr* option in Options Manager.

NOTE
For information about adding and configuring object managers, web directors, and web engines, see Options Manager section.

How to Activate Design-Time

The CA SDM Web Services includes a method stub configuration feature for developers in the Java version. When activated, the Web Services ignores the CA SDM server and returns simulated data for method calls so that Web Services calls can be made without running a CA SDM server.

Follow these steps:

1. Edit deploy.wsdd to uncomment the sections for “design_mode_stubs”.
2. Reverse the deployment and redeploy the server.
3. Restart the application server.
   The design-time feature is activated.

   NOTE
   The design-time feature applies to CA SDM Web Services methods only.

Web Services Security

When you deploy web services, understand the important security considerations. The default configuration when using HTTP is insecure, as it is for all information in web service calls sent between the client and the server in plain text over the network using the HTTP protocol. This includes not only application data, such as ticket descriptions and contact names, but also web service session identifiers (SID). Depending upon the web service application login methods used, it can include passwords.

We recommand that Administrators deploying web services review this information carefully, and to take additional configuration steps at the application and network levels to secure their web service environment.

WARNING
The default web service configuration used with HTTP is insecure and vulnerable to security threats, which can include password discovery, session fixation, and data spying, among others.

There are three interrelated key security considerations in deploying Web Services:

- What (application level) access authentication schemes should this deployment support?
- What additional networking level security features does this deployment require?
- How will these requirements be enforced through web service configuration options?

The following describes each security feature:

- **Web Service Application Level Authentication Schemes** -- To access Web Services, a web service client application must be authenticated with the web service application. Web Services provides two schemes of access authentication. The first is by username/password, and the other is by Public Key Infrastructure (PKI) technology. Both work with the Access Control and Management component in Web Services, using access policy. Access authentication and access management are the most important security features of Web Services.
  Authentication with username/password methods may be disabled using the following security configuration command:

  ```
  disable_user_logon
  ```

  Before enabling this option, the administrator needs to determine if each web service client for which an enterprise is requesting Web Services access, can actually provide support for the alternative authentication method, which is the PKI-based login method. The key advantage to the PKI technology is that Web Services client applications do not require *maintained* system user accounts, that is; the maintenance, storage, and transmission of their passwords.
- **Networking Level Security Configuration** -- In both authentication schemes, username/password and Public Key Infrastructure (PKI), notice that the session identifier returned from the specific login method (as well as all subsequent information), are transmitted in plain text when using HTTP. Furthermore, if the username/password authentication scheme is used, the password is sent unprotected (in plain text) from the web service client application to the Web Services. During product development, the W3C did not have recommended standards for web services security. Subsequently, WS-Security is not used by these Web Services implementations to provide a security context. Instead, point-to-point transport layer security (SSL/TLS) and other network level security mechanisms (for example, IPSec), are recommended to protect the otherwise plain text transmission of the application-level authentication exchange(s), and subsequent session identification and data.

  WARNING
  We recommend using SSL (or https) when deploying Web Services to protect the application-level authentication exchanges and subsequent transmissions of session identification and data.
- **Web Service Configuration** -- To allow administrators to enforce communications protocol-level security at the level of the Web Services application, the following two security configuration commands are supported:

  ```
  require_secure_logon
  ```

  This security feature requires you to use SSL (or https) for calling the Login() and LoginService() methods. This feature also provides a handy method for protecting the username and password, while avoiding the overhead of SSL for the rest of the web services.

  WARNING
  If you use the require_secure_logon command, the Web Services application will not confirm that communications protocol-level security is enforced for methods other than Login() and LoginService(). Unless other precautions are taken, the other Web Services methods may be invoked insecurely, causing greater vulnerability to security threats.

  ```
  require_secure_connection
  ```

  This security feature requires you to use SSL to access any part of the web service. If https is required but not used, then a SOAP Fault with code UDS_SECURE_CHANNEL_REQUIRED is returned.

  NOTE
  For information about how to configure SSL, see your J2EE Servlet Container documentation.

Use the Web Services

The information in this section provides you with the fundamentals for using the CA SDM Web Services. Example code using the Web Services exists in the following CA SDM installation directory:

```
<NX_ROOT>/samples/sdk/websvc/java
CA Service Management 17.3.0.18 and above versions, <NX_ROOT>/samples/sdk/cxfwebsvc/java
```

The sample code is written in Java using Apache Axis and Apache CXF for SOAP messaging.

Logins

Before any Web Services method can be used, a SID (session ID) must be obtained from one of these methods: login(), loginService(), and loginServiceManaged(). The first two methods require a username and password that are validated exactly the same as the CA SDM web interface; the contact's Access Type specifies the validation method. The third method requires a public/private key pair, where login request encrypted with the private key can only be decrypted through the public key, and vice versa.

How to Perform Common Tasks

The Web Services provides a flexible and powerful API into CA SDM, but it requires some knowledge of the object structure used by the product as follows:

1. Familiarize yourself with the information about objects and attributes in CA Service Desk Manager Reference Commands.
   This guide lists the attributes of each object in the system, which is essential because many of the Web Services methods require attribute names.
2. Review the Web Services methods, especially generic ones. For example, if your application must display all the activity logs for a request, first identify how the activity logs relate to the request.
   The CA Service Desk Manager Reference Commands section shows that the request object has two lists of Activity Logs: The act_log (which shows only noninternal logs), and act_log_all (which lists all activity logs).
3. Identify which Web Services methods you require. To get lists attached to an object, use getRelatedList() or getRelatedListValues().

Default Handles

Some default data provided by the product is frequently used. Instead of looking up handles for these objects, some of the commonly used ones are listed in the following tables.

> **NOTE**
> While the handles do not change the legible symbols may be edited.

Contact Type (Object name: ctp)

Handle	Note
ctp:2307	The “Analyst” type
ctp:2310	The “Customer” type
ctp:2305	The “Employee” type
ctp:2308	The “Group” type

Impact (Object name: imp)

Handle	Note
imp:1605	Impact ‘None’
imp:1600	Low impact ‘5’
imp:1601	Medium-low impact ‘4’
imp:1602	Medium impact ‘3’
imp 1603	Medium-high impact ‘2’

imp:1604	High impact ‘1’

Priority (object name: pri)

Handle	Note
pri:505	Unassigned priority ‘None’
pri:500	Low priority ‘5’
pri:501	Medium-low priority ‘4’
pri:502	Medium priority ‘3’
pri:503	Medium-high priority ‘2’
pri:504	High priority ‘1’

Severity (object name: sev)

Handle	Note
sev:800	Low severity ‘1’
sev:801	Medium-low severity ‘2’
sev:802	Medium severity ‘3’
sev:803	Medium-high severity ‘4’`
sev:804	High severity ‘5’

Call Request Type (object name: crt)

Handle	Note
crt:180	Request
crt:181	Problem
crt:182	Incident

Query for Requests, Issues, or Change Orders Assigned to a Contact

One of the most common operations is retrieving the active requests assigned to an analyst (assignee). You can use one of several methods, such as doQuery() (to get a list reference), or doSelect() (to get the values immediately). Assuming the assignee’s handle is already known, the where clause to use is as follows:

```
assignee.id = U'<assigneeID>' AND active = 1
```

In this where clause, <assigneeID> is the id portion of a contact handle, value, such as “555A043EDDB36D4F97524F2496B35E75”.

This where clause works for requests, change orders and issues because they all have the ‘assignee’ and ‘active’ attributes, and they mean the same thing for all three object types. The ‘active = 1’ portion of the where clause restricts the search to active requests.

The Active Flag

Most CA SDM objects have a field called ‘active’ or ‘delete_flag’. This is actually an SREL pointer to the Active_Boolean_Table object or Boolean_Table object. Consider adding these fields to your queries to filter objects marked as Inactive by the system administrator. For querying purposes, search for ‘delete_flag = 0’ to find active records and ‘delete_flag = 1’ for inactive records. For example, the following pseudo-code demonstrates using doSelect() to retrieve values for all active Request Status objects:

```
doSelect(SID, "crs", "delete_flag = 0", -1, new String[0]);
```

To set an object to active or inactive, you need to pass the handle of the Boolean object representing either true or false. These handles do not change, so you can safely hard-code them. These are listed as follows:

Active_Boolean_Table	Boolean_Table
actbool:4551 = ‘Active’	bool:200 = ‘False’
actbool:4552 = ‘Inactive’	bool:201 = ‘True’

Retrieve Related List Length

When requesting attribute values from an object, such as with getObjectValues(), you can get the length of a related list by requesting the following attribute:

```
"<listName>.length"
```

For example, to get the number of Activity Logs for a certain request, pass the following to getObjectValues():

```
"act_log_all.length"
```

> **NOTE**
> This is the only way you can use list names in these types of methods.

Access Control and Management

This article contains the following topics:

To minimize the potential problem of web services ticket flooding and to maintain the stability of the CA SDM server, this version of CA SDM Web Services uses an Access Control and Management system. It works primarily to handle the excessive service activities initiated by trusted user applications that can result from programming errors or exceptions. It also works as a barrier for controlling access to CA SDM Web Services from malicious attackers. An administrator of a web service application is able to create and define an access policy in CA SDM that controls access to CA SDM Web Services from a web service application.

> **NOTE**
> A default access policy with a code of DEFAULT is provided. The default access policy contains no access restrictions and is only applied to sessions authenticated through username and password.

Define an Access Policy

To create any SOAP web services access policy, an administrator defines an access policy.

Follow these steps:

1. Click the Administration tab.
2. In the tree on the left, click SOAP Web Services Policy, Policies.
   The SOAP Web Services Access Policy List page appears.
3. Click Create New.
   The Create New SOAP Web Services Access Policy dialog appears.
4. Enter the information for the new access policy:

   > **NOTE**
   > The defaultvalue of -1 in any operation counter indicates that no restrictions apply to the corresponding operation. A value of 0 (zero) indicates that the corresponding operation is not allowed.

   – **Symbol**

(Required) Identifies a symbolic name of the access policy.

- **Code**
  (Required) Indicates the unique text that identifies this access policy.
- **Status**
  (Required) Identifies the status of an access policy. An inactive policy is not used.
- **Proxy Contact**
  Identifies the contact to use for all web services operations and CA SDM security.
- **Default**
  Identifies the default policy. Set this policy as the default policy. Only one active default policy is allowed to exist. Creating a default policy automatically sets the current default policy to a nondefault status.
- **Has Key**
  (Read-Only) indicates whether a public key has been associated with this policy. This field is updated when a public key is associated with a policy through the pdm_pki utility.
- **Allow Impersonate**
  Identifies the allow impersonate permission. When you set this field, the policyholder can invoke the impersonate() web services method and create a web services session in the name of the user to be impersonated. Additional access authentication is not performed when creating the session. However, only when the access_level of the new access type for the user is less than or equal to the grant_level of the proxy access type for the user can this method be successfully called.
- **Description**
  Indicates the detailed description of this access policy.
- **Ticket Creation**
  Indicates the number of ticket (call request, change order, and issue) insertion operations allowed per hour.
- **Object Creation**
  Indicates the number of CA SDM object (other than ticket object) insertion operations allowed per hour.
- **Object Updates**
  Indicates the number of CA SDM object update operations allowed per hour.
- **Attachments**
  Indicates the number of attachment-related operations allowed per hour.
- **Data Queries**
  Indicates the number of data query operations allowed per hour.
- **Knowledge**
  Indicates the number of knowledge-related operations allowed per hour.

5. Click Save.
   The SOAP web services policy is defined.

Web Services Methods by Category

Each CA SDM Web Services method belongs to a specific category. The following lists each category and their corresponding methods:

- Ticket Creation
- Object Creation
- Object Updates
- Attachments
- Data Queries
- Knowledge

When an access policy is updated by CA SDM, Web Services dynamically updates the corresponding policy information. Active Web Services sessions controlled under this policy remain controlled with configurations in the policy. New Web Services sessions for this policy to manage and control, take the latest configurations in effect.

NOTE
For information about each method, see the CA Service Desk Manager Reference Commands section.

Define an Error Type

Error types are assigned when creating tickets and an access policy defines one set of these error types. A CA SDM Web Services user application may use low-level web methods to create a ticket (request, change order or issue), specifying one of these types to categorize the error addressed in the ticket. Error types can be used only with the high-level createTicket() method. Low-level methods, such as createRequest(), do not use error types.

Web Services Error Types

CA SDM Web Services also provides a defined set of default error types, which are created for *every* policy. These default types, designated as *internal* error types, can be deactivated, but cannot be deleted. In the product, you can use the Web Services Access Policy Detail page to see the default error types provided when a new policy is created.

The following information describes each internal error type:

- **ACCESS_ERROR**
  Indicates that the system failed to connect to or find a resource, such as a file, website, and so on.
- **EXCEPTION_FATAL**
  Indicates that the application is shutting down unexpectedly.
- **EXCEPTION_RUNTIME**
  Indicates that the application code encountered an exception.
- **LOGIN_ERROR**
  Indicates that the operator failed to gain access to the application.

Additional Error Types

The administrator of an access policy can add additional error types as described in the following information:

Error Type	Description
Ticket Template	Identifies a template of a Incident or Error, issue, or change order that you use to create a ticket when this error type is reported. **Note:** The owning policy's contact is used as the end user. The Ticket Type and Ticket Template Name define the ticket template.
Default	Indicates if this error type is the default for the policy. Only one default is allowed per policy. **Note:** A new default error type overwrites the existing default error type associated with the policy.
Active	Represents an active error type. **Note:** An inactive type does not create tickets.
Internal	Identifies the field as read-only, which indicates whether this error type is an internal, default error type.
Symbol	Indicates the symbolic name of the error type.
Code	Identifies the unique text identifier of the error type.
Description	Describes the detailed description of the error type.
Duplicate Handling	Defines the action to take when the product detects that an identical ticket already exists.

Return Data	Identifies the user-defined return message you can specify for the web method “createTicket()” to return to client applications. Return data might be used for indicating an action the application should take (Application Data Return), or a message (User Data Return) to display to the end user.

Duplicate Ticket Handling

The Web Service Access Policy can detect and handle duplicate tickets, which is helpful for preventing ticket flooding. A ticket created with the potential of being a duplicate applies if all of the following conditions are true:

- At least one ticket of the same type (cr, iss, or chg) already exists and is ACTIVE.
- The existing ticket was created by the web service.
- The existing ticket was created with the same Policy and Error Type as the ticket being created.
- The “create date” of the existing ticket is within a specified threshold (for example, it was opened less than 2 days ago).

  NOTE
  The create date field is configured using the Maximum time interval for searching duplicates.
- The duplicate ID matches the one provided by users when invoking the createTicket() method.

Users can also assist in preventing duplicates by classifying tickets as being unique or different, based on criteria known to the user. To do this, add an optional string parameter to the createTicket Web Services call. If duplicate handling is on, the string parameter is inspected after other duplicate handling criteria match to determine whether this is a unique or duplicate call to this method.

Duplicate Ticket Results

If the create ticket action results in a duplicate, the existing Error Type may be configured to do one of the following:

Reconfigured Error Type	Results
Create the New Ticket and Ignore Duplicates	A new ticket handle and number are returned (default).
Do Not Create a New Ticket; Add an Activity Log to the Existing Duplicate Instead	The ticket handle and existing ticket number are returned.
Do Not Create a New ticket; Add an Entry to the CA SDM Standard Log Instead	A ticket handle and existing ticket number are returned.
Create a New Ticket and Attach it as a Child to the Duplicate	A new ticket handle and number are returned.

Simplified Web Services Access

CA SDM Web Services provides an abbreviated set of high-level web services methods that are simplified versions of existing web services methods. The majority of users applications do not have to completely rely on a large set of web services methods before requesting service desk services through CA SDM Web Services. Working closely with user-defined access policies and using default parameters defined in the policies, this set of high-level web services methods can function with little knowledge of the CA SDM object schema. Also, the high-level methods cover a common set of CA SDM functionalities that most service-aware applications need.

The following describes the use of these high-level web services methods:

- **createTicket (SID, Description, Error_Type, Userid, Asset, DuplicationID)**
  You must specify an error type for the reported error if you use this method. The error type should contain the ticket template appropriate for the ticket you want to create. It should define the action to take in the case of a duplicate ticket, specify the data outputs, and finally, it must be associated to the access policy that is defined for the user application.

When this method is invoked, CA SDM Web Services locates the current access policy and the error type required for the ticket creation. The following shows the sequence that CA SDM Web Services uses for locating the proper error type:

- If a specific error type code is provided as input and it matches a error type that is associated to the policy, this error type is used, regardless of whether it is internal.
- If an error type is not specified or the previous step fails to locate an error type, the default error type is used if there is one defined for the policy.
- If a default error type is not defined for the policy or the previous step fails, the default error type defined for internal error types is used.

After an error type is defined, CA SDM Web Services uses it to create a ticket. The proxy user defined in the access policy is used for the ticket creation if the userid is empty, and asset information is added to the ticket (if the input is not empty). After the ticket is created, CA SDM Web Services returns both user data and application data, as specified by the error type.

- **closeTicket (SID, Description, TicketHandle)**
  Users can call this function to close an open ticket. It simply sets the status of an open ticket to ‘close’ and adds the input description to the activity log.
- **logComment (SID, TicketHandle, Comment, Internal_Flag)**
  Adds an entry with the input comment to the activity log for the open ticket.
- **getPolicyInfo (SID)**
  Lets users obtain the policy information that controls the current web services session. You can use this information as an indicator of server capacity for this user application. Users may want to adjust their web services calls to fit into the capacity.

By having this set of simplified Web Services APIs, a majority of users are spared the tremendous effort of understanding the complete set of web services API and CA SDM schema. Using them simplifies and accelerates the process of creating service-aware enabled applications for these users.

CA SDM Objects

This article contains the following topics:

CA SDM treats each entity, such as a contact or an issue, as an *object*. These high-level objects are defined in majic (.maj) and mod (.mod) files on the CA SDM server in the following directory:

```
/bopcfg/majic
```

Customized objects are defined in the following directory:

```
/site/mods/majic
```

Objects are essentially high-level wrappers around a database table.

An object’s type (sometimes referred to as *factory*) defines the object. For example, request objects belong to the ‘cr’ type. Each object’s type is defined by the “OBJECT” declaration in a majic file.

> **NOTE**
> All objects shipped with CA SDM are enumerated in the CA Service Desk Manager Reference Commands section.

An object has *attributes*, which are essentially columns in a database table (do not confuse these with XML attributes). Web Services offers many methods to retrieve values for attributes. Many methods require you to name attributes for setting or retrieving values. You must use the attribute name assigned in the majic or mod file that defines the object, which can be different from the actual database name. Client sites can add additional attributes as a customization.

> **NOTE**
> For a list of all the attributes for each object, see the CA Service Desk Manager Reference Commands section.

Web Services uniquely identifies an object by its *handle,* which is a string value of the form *objectType:ID*, where *objectType* is the object's type (factory) name, and ID is a unique value. The ID value matches that of the 'id' attribute found in every CA SDM object. Because the 'id' attribute is almost always indexed in the DBMS, using the ID portion of the object handle is especially valuable for forming efficient queries. Each object, regardless of its type, stores this value in an object attribute named "persistent_id".

NOTE
In prior releases, the ID portion of the handle was always a string of integers. In CA Service Desk r11.0 and later, the ID portion may also be the string representation of a UUID, typically 32 characters.

The following information lists the object and factory names of the entities that use UUIDs:

Object Name	Factory Name
Contact	cnt
Asset	nr
Organization	org
Location	loc
Company/Vendor	ca_cmpny
Model	mfrmod

Handles are *persistent;* a handle representing a particular object is always unique for its lifetime, even across database migrations. Clients may want to take advantage of this persistence when working with fairly static objects, for example, Status or Contact Types.

Object handles are the key to using CA SDM Web Services. Many methods, especially those that update data, require handles. Most methods that return object data also include the object's handle.

System Updates and Caching

Web Services caches information for object types. Type information is not cached until the type is referenced for the first time, and will cause a small delay.

To avoid any server or caching delays, you may want to run a primer client to activate the Web Services and cache the most popular object type information. The easiest way to cache the object type information is to perform repeated calls to GetObjectTypeInformation(). The object types to consider for this technique could be one of the following:

Object Type	Definition
cr	Request
chg	Change Order
iss	Issue
cnt	Contact
nr	Asset
prp	Property (for Change and Issue)
prptpl	Property Template (for Change and Issue)
cr_prp	Request Property
cr_prptpl	Request Property Template
cr_wf	Classic Workflow (Request/Incident/Problem)

Add any additional object types your client code references.

Categories and Properties

The request, change order and issue objects all have a category field, which is used to classify the nature of the ticket. A category may have property objects, which are attached to the ticket when the category is assigned. Some of these may be marked *required*, which means a value must be supplied before the ticket can be saved (applies to both insert and update operations).

CA SDM Web Services automatically supplies default values for any ticket created with the Web Services. The default value (currently, “-“) is obtained from the CA SDM localized message catalog.

If you need to set property values at creation time, there are three ticket creation methods: createChangeOrder, createIssue, and createRequest. Each has a parameter with which you can pass in values for any properties. To discover which properties will be attached, you must find out the properties associated with the category you intend to assign to the ticket. The easiest method to use is getPropertyInfoForCategory().

> **NOTE**
> For more information about the getPropertyInfoForCategory(), see the CA Service Desk Manager Reference Commands section.

To identify the valid values for a property, first find the property validation rule for the appropriate property template. To do this, request the validation_rule attribute when calling the getPropertyInfoForCategory method. Then, retrieve the associated validation_type for that rule. If the type is dropdown, you can then use the getRelatedList method to retrieve the values associated with the rule, using the "values" BREL attribute in the prpval_rule object.

> **NOTE**
> For more information, see the CA Service Desk Manager Reference Commands section.

To set property values after an update operation with updateObject(), you must query the property list after the update. getRelatedList() can help with this task.

Validation of property values through Web Service methods is not currently supported. For example, to assign property values to a validation rule with a validation type of drop-down option, you would have to write additional code to create property values while creating the drop-down option validation rule. Do not attach a property value to a check box validation rule.

> **NOTE**
> For more information about property validation rules, see the Setting Up Category or Area section.

XML Object Returns

Many of the Web Services methods return an XML representation of CA SDM objects. The Web Services uses a standard XML structure beginning with the following root element:

```
<UDSObject>
```

The format of the XML representation is described in the following table:

XML Element	Type	Description
<UDSObject>	N/A	Identifies the root node.
<Handle>	String	Identifies the object’s handle.
<Attributes>	Sequence	Identifies the attribute values. This holds zero or more elements for the object’s attribute values.

<*attrName0* DataType = *“typeEnum"*>	String	Identifies the *AttrName0,* which is an object attribute name as defined in the CA SDM majic (.maj) or mod (.mod) file. This name may use dot-notation depending on the web method used. The element’s value is the attribute’s value. An empty element indicates a null/empty value for this object’s attribute. The DataType attribute is an integer indicating the attribute’s data type in the CA SDM environment.

For example, a call to getObjectValues() can return information illustrated by the following:

```
<UDSObject>
    <Handle>cnt:555A043EDDB36D4F97524F2496B35E75</Handle>
    <Attributes>
        <Attribute DataType="2003">
            <AttrName>first_name</AttrName>
            <AttrValue>first name</AttrValue>
            <DisplayValue>Yaakov</DisplayValue>
        </Attribute>
        <Attribute DataType="2005">
            <AttrName>organization</AttrName>
                <AttrValue>342</AttrValue>
        <DisplayValue>Accounting Crew</DisplayValue>
    </Attribute>
</Attributes>
    <Lists>        <List name="mylist1">
            <UDSObject>...</UDSObject>
            <UDSObject>...</UDSObject>
        </List>
    </Lists>
</UDSObject>
```

Some methods, such as doSelect(), return a sequence of <UDSObject> elements contained inside a <UDSObjectList> element.

The <Lists> section holds zero or more <List> nodes. A <List> node holds zero or more <UDSObject> nodes. <List> elements are generally returned only when a specific request for list values is made.

When you want to return a list of values related to a specific object, you should use the *getRelatedListValues* method.

If a request is made just for a list with no attribute name, such as actlog, then the entire <UDSObject> is returned in the <List> section.

Specialized methods, like getDocument(), can of course be different. When a request is made for an attribute, the database value is returned. For SREL attributes, this may not be so useful. Requesting the assignee attribute of a Request returns an integer because the Contact REL_ATTR (foreign key) is its ID. For CA Service Desk r11.0, the return data for attributes includes elements for the DBMS and common name value of SREL references.

Credential Authentication

This article contains the following topics:

If you plan to use the User Name/Password type of access authentication, the user application needs to invoke one of following two web services methods to gain access to CA SDM Web Services.

> **NOTE**
> The login user that you specify in the username parameter (not the proxy contact specified in the policy) is responsible for activities initiated in a session. All function group security and data partition is enforced for this login user.

login (Username, Password)

This method is provided for backward compatibility, where access authentication is performed on the username and password supplied. A SID (session ID) is returned only if the access is authenticated. All subsequent web services calls need to include this SID. Default access policy is then applied to all subsequent web services accesses labeled with the SID.

Username and password are required fields that require plain text when you define them.

loginService (Username, Password, Policy)

This method is similar to the previous login function in that access authentication is performed on the username and password supplied. A SID is returned only if the access is authenticated. However, a specific access policy, as identified in the third parameter, is applied to control and manage all subsequent Web Services accesses. Empty content in the policy parameter automatically applies the default policy.

Username and password are required fields that require plain text when you define them. Policy is required, but can be empty, and you must use plain text. Use the policy code defined in a policy.

How a login is validated depends on the contact's assigned *access type*. The access type object is hosted by CA SDM and sets the validation type. You can use the product to view the access type record, and you can also use the getAccessTypeForContact() web method to retrieve any access type object information.

> **NOTE**
> For more information about access types, see the Security and Role Management section.

ITIL Methodology

This article contains the following topics:

By default, the Web Services fully support the ITIL methodology. CA SDM ITIL features let you take advantage of the ITIL methodology.

Incident or Problem Creation

CA SDM supports ITIL methodology so that incidents and problems can be created using the CA SDM Web Services. Both incidents and problems are held in the cr (Call_Req) object. Its *type* attribute distinguishes the record as an incident, problem, or request. To create an incident, problem, or request, call createRequest and specify the appropriate value for the type attribute.

The type attribute is a pointer (SREL) to the crt (Call_Req_Type) object, so you must pass a handle as the value.

The following code examples illustrate how to create an incident or problem by passing the correct crt object handle to the createRequest method. Setting the *type* attribute in the name-value pair that is passed as a parameter to createRequest, creates the tickets:

Example: Syntax for a Problem

```
attrVals = {"summary", "A new problem", "description", "new problem", "type", "crt:181", "catagory",
 "pcat:40002"}
```

```
USPSD.createRequest(SID, creatorHandle, attrVals, template, new String[0], new String[0])
```

NOTE

All the workflow tasks that are attached to a category are available for a problem. The workflow task is selected on the configuration of CA Process Automation or Classic workflow.

Example: Syntax for an Incident

```
attrVals = {"summary", "A new incident", "description", "new incident", "type", "crt:182", "catagory",
 "pcat:40002"}
USPSD.createRequest(SID, creatorHandle, attrVals, template, new String[0], new String[0])
```

NOTE

All the workflow tasks that are attached to a category are available for an incident. The workflow task is selected on the configuration of CA Process Automation or Classic workflow.

Query for Incidents or Problems

To retrieve incidents or problems, include the *type* attribute of the cr object in the where clause. The following example illustrates a where clause for retrieving all active Incidents. This where clause could be used with methods that perform queries for ‘cr’ objects, such as doSelect and doQuery:

```
type.id = 182 AND active = 1
```

The ‘182’ is the ID portion of the handle representing Incident types.

NOTE
For more information, see the crt (Call_Req_Type) objects table illustrated in Default Handles. For more information about forming proper queries, see WHERE Clause.

Attach an Incident to a Problem

You can associate one or more incidents to a problem. The problem attribute of an incident relates the incident to a problem.

Example: Associate an Incident to a Problem

This example demonstrates how to relate a newly created incident to an existing problem.

To associate the incident to the problem, use UpdateObject to set the problem attribute of the incident. The following example code sets the problem attribute to the handle of an existing problem ticket:

```
attributeValues = {"problem", "cr:12346"}
USPSD.UpdateObject(SID, incidentHandle, attributeValues, new String [0])
```

Attach a Problem to a Change Order

Incidents and problems can be linked to change orders with the attachChangeToRequest method. The following example code uses this method to simultaneously create a change order and attach it to a problem. In the example, “cr:12347” is the object handle of the problem -- it passes a blank handle for the fourth parameter, which causes the method to create a change:

```
UPSPSD.attachChangeToRequest(SID, creatorHandle, "cr:12347", "", new String[0], "activity description")
```

Change Order Creation

CA SDM supports ITIL methodology so that a Change Order can be created using the CA SDM Web Services. Change Order are held in the chg (Change_Request) object. To create an change order, call createRequest and specify the appropriate value for the type attribute. The following code examples illustrate how to create a change order:

Example: Syntax for a Change Order

```
attrVals = {"summary", "A new change order", "description", "new change order","catagory", "chgcat:400002"}
USPSD.createChangeOrder(SID,creatorHandle,attrVals,template,new String[0], new String[0])
```

Issue Creation

CA SDM supports ITIL methodology so that an Issue can be created using the CA SDM Web Services. Issue are held in the iss (Issue) object. To create an Issue, call createRequest and specify the appropriate value for the type attribute. The following code examples illustrate how to create an change order:

Example: Syntax for Issue

```
attrVals = {"summary", "A new Issue", "description", "new Issue", "catagory", "isscat:400002"}
USPSD. createIssue(SID,creatorHandle,attrVals,template,new String[0], new String[0])
```

Configuration Items

ITIL methodology uses the term *configuration item* (CI) to refer to hardware, software, and other IT resources. This term refers to the “nr” object stored in the CA Technologies-owned resource database table. All methods that use *asset* objects also work with CIs. This is only a difference in terminology.

Public Key Infrastructure (PKI) Authentication

This article contains the following topics:

If you plan to use the PKI authentication, realize that the content of the login request is encrypted with a private key that can only be decrypted by its matching public key. The response of the login request is returned as plain text.

Generally, each application accessing CA SDM Web Services is assigned with a policy. CA SDM Web Services stores detailed information about a policy, along with the public key of a digital certificate. An application, as the policy holder, uses the private key of the digital certificate and the policy code (as policy identifier) to assemble a login request.

loginServiceManaged (Policy, Encrypted_Policy)

CA SDM Web Services performs the user authentication by locating the policy through the plain text policy code, retrieving the policy holder’s public key associated with the policy, decrypting the encrypted policy code, matching the decrypted content with the policy code, and finally, opening a session with a back-end server. The plain text session ID (SID) is returned and can be used for subsequent method invocations. Only the policyholder holds the private key that matches the policy’s associated public key stored in CA SDM.

All subsequent web services calls must include the returned session ID (SID). The Proxy contact specified in the policy is responsible for all web services activities initiated in this session. All function group security and data partition is enforced for the proxy contact.

> **WARNING**
> The Encrypted_Policy parameter should be in the BASE64 text format. The user application must perform proper conversion from the binary format.

Policy is a required field. When you define it, use plain text policy code as defined in a policy. Encrypted_Policy (the digital signature of the policy code encrypted with the policy holder’s private key) is required. When you define Encrypted_Policy, use the algorithm SHA1 with RSA to obtain the digital signature.

Implement loginServiceManaged in Java

The following shows how to generate Certificates and then use these generated Certificates to access the CA SDM web services.

In the following example, the login process completes using the CA SDM Certificate and then performs two common web services calls. The getBopsid() web services method call allows you to obtain a token that is linked to a specific user. This token can be used to login to the CA SDM web interface as the linked user without being prompted for a password. This allows seamless integration to be enabled between different applications.

WARNING
The generated BOPSID token expires after 30 seconds, so it must be used promptly.

WARNING

There is a known issue when using the 1.4 version of the AXIS tool. For more information, see the *Release Notes*.

Follow these steps:

1. Generate the stub classes with AIXS Tool WSDL2Java. For more information, see the Generating Stub Classes with AXIS Tool WSDL2Java section from the PKI_loginServiceManaged_JAVA_steps file. Find the file in the following location:
   ```
   $NX_ROOT/samples/sdk/websvc/java/test1_pki
   ```
2. Start the CA SDM service.
3. Run pdm_pki -p DEFAULT.
   DEFAULT.p12 is created in the current directory. This policy will have the password equal to the policy name (in this case DEFAULT).

   NOTE
   This command will also add the Certificate's public key to the field pub_key field (public_key attribute) in the sapolicy table/object.
4. Log in to CA SDM.
5. Select SOAP Web Services Policy, Policies on the Administration tab.
   The SOAP Web Services Access Policy List page opens.
6. Click DEFAULT.
   The SOAP Web Services Access Policy Detail page opens.
7. Complete the Proxy Contact field (in this example, ServiceDesk) and confirm that the DEFAULT policy record Has Key field displays "Yes."
8. Copy DEFAULT.p12 (from the directory where command pdm_pki is executed), the JSP file called *pkilogin.jsp* and the HTML file called *pkilogin.htm* (from the $NX_ROOT/samples/sdk/websvc/java/test1_pki directory) to the following directory:
   ```
   $NX_ROOT/bopcfg/www/CATALINA_BASE/webapps/axis
   ```
9. Open the HTML form (from the axis directory). For example, http://localhost:8080/axis/pkilogin.htm
   Complete the appropriate fields.

   NOTE
   The Directory field identifies the location of the Certificate file. Modify the path to the correct location.
10. Click Log me in!
    The results page opens.
11. Click the BOPSID URL.

WARNING
Click this immediately! The BOPSID has a limited life token of about 30 seconds.

The format of a URL using a BOPSID is as follows:
http://<server name>:CA Portal/CAisd/pdmweb.exe?BOPSID=<BOPSID value>

NOTE
In order to use the loginServiceManaged method for a Java client program running on AIX, you may need to replace a pair of security policy files within your JAVA_HOME. Go to http://www.ibm.com and search for "developerworks java technology security information AIX". In the "developerWorks : Java technology : Security" document, follow the link to "IBM SDK Policy files". Download the unrestricted policy files, local_policy.jar and US_export_policy.jar. Use these files to replace the original files in your JAVA_HOME/lib/security directory."

Configuration for the PKI Authentication Type

To configure for PKI authentication, you must first create an access policy. The process flow is as follows:

- **Create an Access Policy**
  The administrator performs this task using the product (Web Interface only), and as part of the process, needs to assign a unique text code to each access policy.
- **Obtain a Digital Certificate with a Public/Private Key Pair and Associate it with the Access Policy**
  For PKI access authentication, a user application needs to obtain a digital certificate that contains both a public key and private key pair. An administrator can obtain the digital certificate through third-party Certificate Authority (CA) or security products that support digital certificates. CA SDM also provides a server-side utility that can generate a digital certificate. It is located in <NX_ROOT>/bin directory as follows:

  ```
  pdm_pki  - p policy_code [ - l certificate file] [ - f] [-h]
  ```

  – **-p**
    Identifies a unique policy code.
  – **-f**
    Allows the utility to replace the existing public key with a new public key.
  – **-l**
    Loads the public key stored in a X509 V3 certificate.
  – **-h**
    Displays help on the command line window.

  If you obtain a digital certificate through a third-party, CA Technologies, or security products, import it to where the CA SDM server is located, and then associate it to an access policy. The administrator of the user application should obtain a digital certificate file that includes the content of an X509 V3 certificate in DER/ASN.1 format.
  In addition, the certificate should contain only the public key of the public/private key pair. Using the - l option, the administrator should invoke the pdm_pki utility to load the certificate. The utility then loads the certificate, extracts the public key, converts the public key to BASE64 text format, and saves it with the access policy specified by the policy code.
  When a digital certificate is generated by the pdm_pki utility, the administrator invokes the command in CA SDM without the - l option. The utility then generates a public and private key pair (keys are RSA1024 bit keys). The public key is converted to BASE64 text format where it is stored along with the access policy specified by the policy code. An X509 V3 certificate is also created to hold the public key along with other information (the default pass phase is set as the policy code). Finally, the X509 V3 certificate is packaged with the private key to a standard portable certificate format of PKCS12. It is then saved in a file with a file name of *policy_code*.p12, depending on the policy code supplied. This file can then be exported to clients.

  NOTE
  If an access policy has already been associated with a public key of a certificate, users need to specify the - f option when calling the pdm_pki command in order to overwrite the existing public key with a new public key.

Login to Web Services

The following describes the process flow for logging in to Web Services configured with PKI authentication:

Process	Description
Load the Digital Certificate and Extract the Private Key	The digital certificate must be stored in secure storage on the user side, where it can be retrieved and used for logging in to Web Services. Example of secure storages include the following: Windows Certificate Store Java Certificate Store (managed by java_keytool utility) Certificate store (created by other CA Technologies security products). A user application should be able to load the digital certificate and extract the private key using appropriate APIs, depending on user environments.
Create a Digital Signature of the Plain Text Policy Code with the Private Key	After the private key is extracted from the digital certificate, it can be used to generate a digital signature of policy code. Creating a digital signature encrypts a digest of a text with a private key. The digest algorithm must be standard SHA1, and the encryption algorithm should be RSA. Also, the binary digital signature should be converted to BASE64 text format before it can be used for logging in to Web Services. Depending on user environments, appropriate API calls should be used to archive this information.
Invoke the Web Service Call	A user application should invoke the Web Services method loginServiceManaged(), along with the plain text policy code and the BASE64 text formatted digital signature of the policy code.
Obtain the Returned SID	If the access request is authenticated, a plain text SID is automatically returned.

After a SID is generated, it establishes a successful binding between a Web Service session and an access policy. The user application can invoke other web services methods with this SID, and all of its access to Web Services becomes controlled and managed by this access policy.

Public Key Infrastructure (PKI) Authentication for CXF

This article contains the following topics:

If you plan to use the PKI authentication, realize that the content of the login request is encrypted with a private key that can only be decrypted by its matching public key. The response of the login request is returned as plain text.

Generally, each application accessing CA SDM Web Services is assigned with a policy. CA SDM Web Services stores detailed information about a policy, along with the public key of a digital certificate. An application, as the policy holder, uses the private key of the digital certificate and the policy code (as policy identifier) to assemble a login request.

loginServiceManaged (Policy, Encrypted_Policy)

CA SDM Web Services performs the user authentication by locating the policy through the plain text policy code, retrieving the policy holder’s public key associated with the policy, decrypting the encrypted policy code, matching the decrypted content with the policy code, and finally, opening a session with a back-end server. The plain text session ID (SID) is returned and can be used for subsequent method invocations. Only the policyholder holds the private key that matches the policy’s associated public key stored in CA SDM.

All subsequent web services calls must include the returned session ID (SID). The Proxy contact specified in the policy is responsible for all web services activities initiated in this session. All function group security and data partition is enforced for the proxy contact.

WARNING
The Encrypted_Policy parameter should be in the BASE64 text format. The user application must perform proper conversion from the binary format.

Policy is a required field. When you define it, use plain text policy code as defined in a policy. Encrypted_Policy (the digital signature of the policy code encrypted with the policy holder's private key) is required. When you define Encrypted_Policy, use the algorithm SHA1 with RSA to obtain the digital signature.

Implement loginServiceManaged in Java

The following shows how to generate Certificates and then use these generated Certificates to access the CA SDM web services.

In the following example, the login process completes using the CA SDM Certificate and then performs two common web services calls. The getBopsid() web services method call allows you to obtain a token that is linked to a specific user. This token can be used to login to the CA SDM web interface as the linked user without being prompted for a password. This allows seamless integration to be enabled between different applications.

WARNING
The generated BOPSID token expires after 30 seconds, so it must be used promptly.

Follow these steps:

1. Generate the stub classes with CXF Tool WSDL2Java. For more information, see the Generating Stub Classes with CXF Tool WSDL2Java section from the PKI_loginServiceManaged_JAVA_steps file. Find the file in the following location:
   ```
   $NX_ROOT/samples/sdk/cxfwebsvc/java/test1_pki
   ```
2. Start the CA SDM service.
3. Run pdm_pki -p DEFAULT.
   DEFAULT.p12 is created in the current directory. This policy will have the password equal to the policy name (in this case DEFAULT).

   NOTE
   This command will also add the Certificate's public key to the field pub_key field (public_key attribute) in the sapolicy table/object.
4. Log in to CA SDM.
5. Select SOAP Web Services Policy, Policies on the Administration tab.
   The SOAP Web Services Access Policy List page opens.
6. Click DEFAULT.
   The SOAP Web Services Access Policy Detail page opens.
7. Complete the Proxy Contact field (in this example, ServiceDesk) and confirm that the DEFAULT policy record Has Key field displays "Yes."
8. Copy DEFAULT.p12 (from the directory where command pdm_pki is executed), the JSP file called *pkilogin.jsp* and the HTML file called *pkilogin.htm* (from the $NX_ROOT/samples/sdk/cxfwebsvc/java/test1_pki directory) to the following directory:
   ```
   $NX_ROOT/bopcfg/www/CATALINA_BASE/webapps/cxf
   ```
9. Open the HTML form (from the cxf directory). For example, http://localhost:8080/cxf/pkilogin.htm
   Complete the appropriate fields.

   NOTE
   The Directory field identifies the location of the Certificate file. Modify the path to the correct location.
10. Click Log me in!

The results page opens.

11. Click the BOPSID URL.

WARNING
Click this immediately! The BOPSID has a limited life token of about 30 seconds.

The format of a URL using a BOPSID is as follows:
http://<server name>:CA Portal/CAisd/pdmweb.exe?BOPSID=<BOPSID value>

Configuration for the PKI Authentication Type

To configure for PKI authentication, you must first create an access policy. The process flow is as follows:

- **Create an Access Policy**
  The administrator performs this task using the product (Web Interface only), and as part of the process, needs to assign a unique text code to each access policy.
- **Obtain a Digital Certificate with a Public/Private Key Pair and Associate it with the Access Policy**
  For PKI access authentication, a user application needs to obtain a digital certificate that contains both a public key and private key pair. An administrator can obtain the digital certificate through third-party Certificate Authority (CA) or security products that support digital certificates. CA SDM also provides a server-side utility that can generate a digital certificate. It is located in <NX_ROOT>/bin directory as follows:

  ```
  pdm_pki  - p policy_code [ - l certificate file] [ - f] [-h]
  ```

  – **-p**
    Identifies a unique policy code.
  – **-f**
    Allows the utility to replace the existing public key with a new public key.
  – **-l**
    Loads the public key stored in a X509 V3 certificate.
  – **-h**
    Displays help on the command line window.

  If you obtain a digital certificate through a third-party, CA Technologies, or security products, import it to where the CA SDM server is located, and then associate it to an access policy. The administrator of the user application should obtain a digital certificate file that includes the content of an X509 V3 certificate in DER/ASN.1 format.
  In addition, the certificate should contain only the public key of the public/private key pair. Using the - l option, the administrator should invoke the pdm_pki utility to load the certificate. The utility then loads the certificate, extracts the public key, converts the public key to BASE64 text format, and saves it with the access policy specified by the policy code.
  When a digital certificate is generated by the pdm_pki utility, the administrator invokes the command in CA SDM without the - l option. The utility then generates a public and private key pair (keys are RSA1024 bit keys). The public key is converted to BASE64 text format where it is stored along with the access policy specified by the policy code.
  An X509 V3 certificate is also created to hold the public key along with other information (the default pass phase is set as the policy code). Finally, the X509 V3 certificate is packaged with the private key to a standard portable certificate format of PKCS12. It is then saved in a file with a file name of *policy_code*.p12, depending on the policy code supplied. This file can then be exported to clients.

  NOTE
  If an access policy has already been associated with a public key of a certificate, users need to specify the - f option when calling the pdm_pki command in order to overwrite the existing public key with a new public key.

Login to Web Services

The following describes the process flow for logging in to Web Services configured with PKI authentication:

Process	Description
Load the Digital Certificate and Extract the Private Key	The digital certificate must be stored in secure storage on the user side, where it can be retrieved and used for logging in to Web Services. Example of secure storages include the following: Windows Certificate Store Java Certificate Store (managed by java_keytool utility) Certificate store (created by other CA Technologies security products). A user application should be able to load the digital certificate and extract the private key using appropriate APIs, depending on user environments.
Create a Digital Signature of the Plain Text Policy Code with the Private Key	After the private key is extracted from the digital certificate, it can be used to generate a digital signature of policy code. Creating a digital signature encrypts a digest of a text with a private key. The digest algorithm must be standard SHA1, and the encryption algorithm should be RSA. Also, the binary digital signature should be converted to BASE64 text format before it can be used for logging in to Web Services. Depending on user environments, appropriate API calls should be used to archive this information.
Invoke the Web Service Call	A user application should invoke the Web Services method loginServiceManaged(), along with the plain text policy code and the BASE64 text formatted digital signature of the policy code.
Obtain the Returned SID	If the access request is authenticated, a plain text SID is automatically returned.

After a SID is generated, it establishes a successful binding between a Web Service session and an access policy. The user application can invoke other web services methods with this SID, and all of its access to Web Services becomes controlled and managed by this access policy.

Session and Authorization

A successful validation returns a SID that is associated with the validated username, whether it is the user name supplied for login or the proxy contact specified in a policy. Because of this process, each CA SDM user is assigned security rights that you may want enforced in your web service application.

For example, a specific user may have a Data Partition restricting which Requests the user can view. When using a SID for the user to get Request information, the CA SDM system ensures the data partition is enforced.

Function Group security is also applied. For example, a user may not have access to the Call Manager function group. Invoking any web services methods, such as viewing or creating Requests, is denied because access is denied to the Call Manager function group.

When your application is finished doing work for a user, call the Logout() method to invalidate the SID.

Each SID expires after a period of inactivity. That is, a SID expires if the interval between method calls is greater than a certain timeout value. The timeout interval is set in Options Manager and is specified by the following CA SDM option:

```
'webservice_session_timeout'
```

If this value is set to zero (0), a SID never times out. If this option is missing or not set, the default is one hour. If a Web Service method is called with an expired SID, a Fault is returned with an error code of UDS_CNTXT_TIMEOUT the first time it is referenced, and UDS_BAD_SESSION each time thereafter.

To keep a SID active, call any web service method before the time out is reached. To keep the SID active without working the server, call serverStatus().

Tips for SOAP Web Services Clients

This article contains the following topics:

The samples directory of the CA SDM installation provides a sample Java client application for Web Services. This sample assists developers with web services client application development.

Many of the Web Services methods require arrays as input parameters. For example, the method createIssue() permits an empty array for propertyValues. Sometimes these arrays are optional, but the service requires an empty array for a successful pass. When you use Visual Studio .NET to access Web Services, specify an empty array with one of the following arrays:

- **C# language**

  ```
  String[] emptyArray = new string[0];
  ```

- **Visual Basic .NET**

  ```
  Dim emptyArray As String() = {}
  ```

- **Java**

  ```
  ArrayOfString attr = new ArrayOfString();
  attr.setString(new String[0]);
  ArrayOfString is a proprietary class.
  ```

You can then pass emptyArray to array parameters that accept empty arrays.

> **NOTE**
>
> The CA SDM Web Services use the Apache implementation of standards established by the World Wide Web Consortium (W3C). Ideally, a client on any type of operating environment can access the services, but vendor implementations vary. Many programming environments provide a tool to generate proxy classes from a Web Services Description Language (WSDL) description.

Java Clients

The TableOfContents.doc in $NX_ROOT/samples/sdk/websvc lists several Java sample programs.

Each sample program contains notes on how it can be compiled and run using the script files run_java_test_bat.txt (Windows) and run_java_test_sh.txt (UNIX). These scripts demonstrate how to use org.apache.axis.wsdl.WSDL2Java to generate the CA SDM web services client-side stub files.

The - w parameter is required to properly generate the stub files when using Axis 1.4. Running WSDL2Java as shown will generate the stub files in subdirectory com/ca/www/UnicenterServicePlus/ServiceDesk. The following files are generated:

- ArrayOfInt.java
- ArrayOfString.java
- ListResult.java
- USD_WebService.java
- USD_WebServiceLocator.java
- USD_WebServiceSoap.java
- USD_WebServiceSoapSoapBindingStub.java.

Import these classes with the following statement:

```
import com.ca.www.UnicenterServicePlus.ServiceDesk.*;
```

Many Web Service methods have parameters of type ArrayOfString, a proprietary class. For example, the createRequest() method's attrVals, propertyValues and attributes parameters are all ArrayOfString parameters.

To set the values in an ArrayOfString variable, instantiate the variable and then use setString() as follows:

```
ArrayOfString attrVals = new ArrayOfString();
attrVals.setString(new String[]{"customer", customerHandle, "description", "description text"});
```

To set it to empty

```
attrVals.setString(new String[0]);
```

Use a variable of type ListResult, another proprietary class, as the return value from the List methods: doQuery(), getRelatedList(), getNotificationsForContact(), getPendingChangeTaskListForContact() and getPendingIssueTaskListForContact(). A ListResult contains listHandle and listLength elements, which can be retrieved using getListHandle() and getListLength() as shown in this example:

```
ListResult doQueryResult = new ListResult();
doQueryResult = USPSD.doQuery(sid, "iss", "active = 1");
int listHandle = doQueryResult.getListHandle();
int listLength = doQueryResult.getListLength();
```

The getListValues() method uses the listHandle, retrieving the values from a subset of the list.

The Handles parameter of the freeListHandles() method is an ArrayOfInt, another proprietary class. Call freeListHandles() using the listHandle taken from a ListResult:

```
ArrayOfInt handleList = new ArrayOfInt();
handleList.setInteger(new java.lang.Integer []{ new java.lang.Integer(listHandle) });
USPSD.freeListHandles(sid, handleList);
```

Some methods have pass by reference parameters of type javax.xml.rpc.holders.StringHolder. For example, createRequest() has two parameters of this type, NewRequestHandle and NewRequestNumber.

```
StringHolder NewRequestNumber = new StringHolder();
StringHolder NewRequestHandle = new StringHolder();
String result;
result = USPSD.createRequest(sid, creatorHandle, attrVals, propertyValues, template, attributes,
 NewRequestHandle, NewRequestNumber);
```

The Request's handle and reference number (ref_num) can then be obtained from NewRequestHandle.value and NewRequestNumber.value respectively.

SOAP Web Services Configuration

You can configure the CA SDM SOAP Web Services with entries in special web configuration files. The following table summarizes the names and descriptions of the configuration options:

Option Name	Description
design_mode_stubs	Sets the Web Servic*e to design mode* (CA SDM only).
require_secure_logon	Requires the login() and loginService() web methods to be called with a secure protocol, such as https.
require_secure_connection	Requires that every web method be called with a secure protocol.
disable_user_logon	Disables both login() and loginService() web methods, so only loginServiceManaged() can be used to log in.

CA SDM adds protection to the integrity of the running Tomcat server by verifying the length of the attribute values that pass to Web Service methods. By default, web service calls return an Axis Fault if the length of an attribute exceeds 900,000 bytes.

You set the following parameters in the deploy.wsdd file:

- **fatal_max_string_length**
  Sets the length of the largest attribute value that a web service method accepts.
  Default: 900,000 bytes
- **validate_parameters**
  Sets whether the attribute value length checking is performed. Set the parameter to 0 to turn off the validation.
  Default: 1 (on)
- **exception_methods**
  Displays a comma-delimited list of Web Service methods that are exempt from the attribute value length validation.

Redeploy the Web Services

New configuration settings take effect when you redeploy CA SDM Web Services. Complete the following steps to redeploy the Web Services:

1. Open a command prompt and set the CLASSPATH environment variable to include the required Axis jar files, which are supplied in <NX_ROOT>/java/lib.
   For example, to set it on Windows, execute the following command:
   ```
   set AXISHOME=%NX_ROOT%\java\lib
   set classpath= %AXISHOME%\axis.jar;%AXISHOME%\jaxrpc.jar;%AXISHOME%\saaj.jar;%AXISHOME%\commons-
   logging.jar;%AXISHOME%\commons-discovery.jar;%AXISHOME%\wsdl4j.jar;%AXISHOME%\log4j-1.2.8.jar;%classpath%;
   ```
2. Modify the directory to <NX_ROOT>/sdk/websvc/R11 and execute the following commands:
   ```
   java org.apache.axis.client.AdminClient undeploy.wsdd
   java org.apache.axis.client.AdminClient deploy.wsdd
   ```
3. Recycle Tomcat by recycling the CA SDM service. You can avoid shutting down the entire CA SDM system by recycling Tomcat by simply using the following commands:
   ```
   pdm_tomcat_nxd  - c stop
   pdm_tomcat_nxd  - c start
   ```
   The Web Service is redeployed.
4. Verify that the service deployed by viewing the Axis services listing page at the following default URL:
   ```
   http://<servername>:< port>/axis/services
   ```

   NOTE

   The exact URL depends upon your installation settings.

SOAP Error Handling

If an error occurs with a Web Services method, a SOAP Fault is returned. The SOAP Fault is the standard means of returning exception information for Web Services.

The Fault message contains standardized <Message> and <Code> elements, but the most informative is the <Detail> element. The <Detail> element contains <ErrorCode> and <ErrorMessage> elements. The <ErrorCode> element returns an enumerated error code specific to either the CA SDM or Knowledge Management product. The <ErrorMessage> element contains an English string describing the errors. The <ErrorMessage> elements are more suitable for aiding the developer and more appropriate messages should display to users.

For example, the following illustrates a SOAP Fault when a bad parameter is supplied to the CA SDM getObjectValues() method:

```
<soap:Fault>
   <faultcode>soap:Client</faultcode>
```

```
   <faultstring>Error on fetch with attribute
 list:persistent_id,first_name,last_nameParamErrorHere<faultstring>
   <detail>
<ErrorCode>1001</ErrorCode>
<ErrorMessage>Error on fetch with attribute list: persistent_id,first_name,last_nameParamErrorHere </
ErrorMessage>
   </detail>
   </soap:Fault>
```

If you are using a client built with Microsoft .NET managed code, a failed Web Services method call raises a "SOAPException" exception. All errors cancel the operation invoked.

In some cases, the servlet container may write errors and therefore, display in the servlet container logs. In other cases, error information may be written to CA SDM logs. These logs are located in the following subdirectories:

- In the /bopcfg/www/CATALINA_BASE/logs subdirectory of CA SDM installation
- In the /log subdirectory of the CA SDM installation and to all logs that have the prefix “stdlog”.

> **NOTE**
> We recommend that you constantly monitor these logs, as the server may log its own errors without reporting them to the CA SDM Web Services.

Lock Errors

CA SDM objects are locked during updates. Methods that update objects (such as, updateObject() or transfer()) may return the following lock error code:

```
UDS_LOCK_ERR
```

This code indicates that another user is updating the record. Often the locking user’s handle is returned in the ErrorMessage element.

Time Outs

If the CA SDM server is heavily loaded, a method may take a long time to process. In rare cases, a method may never return because a separate process failed to reply or some other error occurred. To guard against excessive blocking, every Web Services method times out after a number of seconds. Web Services method time-out is a CA SDM server time-out, *not* a Web server time-out, network time-out, and so on.

> **NOTE**
>
> The Web Services delay a few seconds the first time accessed after the J2EE application server is recycled. This happens because the application is initializing, loading DLLs, libraries, and so on, and occurs only with the first Web Services method call. All subsequent calls return much faster.

If a method times out, it returns the following error code:

```
UDS_TIMEOUT_ERR
```

The operation is not aborted! The server may have received the request and processes it successfully, although slowly. This type of problem may occur when using the *doSelect( )* method to retrieve several 1000 records.

> **NOTE**
> For information about the *doSelect* method, see the CA Service Desk Manager Reference Commands section.

Error Codes

The following table lists the possible values for the <ErrorCode> value in a SOAP Fault returned from a Web Services call:

Error Name	Value	Description
UDS_OK	0	Successful.
UDS_FAILURE	1	General failure, check system logs.
UDS_BAD_PARAM	1000	A bad parameter was passed to a method. This error occurs if a required parameter is missing, the wrong type was passed, or an invalid value was used.
UDS_INTERNAL_ERR	1001	Signals that an internal error occurred. A description is found in the return array and the system logs.
UDS_LOCK_ERR	1002	An attempt was made to update an object locked by another user or process. Usually the ID of the contact responsible for locking the object is returned in the return data.
UDS_UPDATE_ERR	1003	An error occurred updating an object. Make sure all required attributes were set and check the system log.
UDS_CREATION_ERR	1004	An error occurred creating an object. Make sure all required attributes were set and check the system logs.
UDS_NOT_FOUND	1005	A search method failed to find any matches or failed to find an object specified. This can happen if a bad or invalid handle is passed to any method.
UDS_TIMEOUT_ERR	1006	The current method timed out, the CA SDM server may be heavily loaded or the method itself was bad.
UDS_SERVER_GONE	1007	The CA SDM server connection is lost, UDS methods will no longer function and all list references are lost.
UDS_FETCH_ERR	1008	An error occurred while retrieving list data.
UDS_BAD_SESSION	1010	An invalid SID was used.
UDS_CNTXT_TIMEOUT	1011	The SID timed out.
UDS_SECURE_CHANNEL_REQUIRED	1012	The Web Services (or a web service method) requires a secure channel (for example: SSL) for access, but an unsecured channel is being used.
UDS_SECURITY_VIOLATION	1013	The attempted operation violates CA SDM security and was aborted.
UDS_OVER_POLICY_LIMIT	3002	The attempted request is refused because it exceeds the limit defined in the policy.

Tips for SOAP Web Services Clients for CXF

This article contains the following topics:

The samples directory of the CA SDM installation provides a sample Java client application for CXF Web Services. This sample assists developers with web services client application development.

> **NOTE**
> Ensure you update the AXIS webservice URL to CXF webservice for all the endpoints.

Many of the Web Services methods require arrays as input parameters. For example, the method createIssue() permits an empty array for propertyValues. Sometimes these arrays are optional, but the service requires an empty array for a successful pass. When you use Visual Studio .NET to access Web Services, specify an empty array with one of the following arrays:

- **C# language**

  ```
  String[] emptyArray = new string[0];
  ```

- **Visual Basic .NET**

  ```
  Dim emptyArray As String() = {}
  ```

- **Java**

  ```
  ArrayOfString attr = new ArrayOfString();
  ArrayOfString is a proprietary class.
  ```

You can then pass emptyArray to array parameters that accept empty arrays.

> **NOTE**
>
> The CA SDM CXF Web Services use the Apache implementation of standards established by the World Wide Web Consortium (W3C). Ideally, a client on any type of operating environment can access the services, but vendor implementations vary. Many programming environments provide a tool to generate proxy classes from a Web Services Description Language (WSDL) description.

Generate Stub Classes with CXF WSDL2Java

You can generate the stub classes for the CA SDM CXF web services.

To generate stub classes with CXF WSDL2Java

1. Open a command prompt and navigate to the "<drive>:\Program Files (x86)\CA" directory.
   The directory appears.
2. Navigate to the samples\sdk\cxfwebsvc\java folder.
3. Rename the run_java_bat_txt file to run_java.bat file.
4. Set the USD_SHORT_PATH and CXFLIBPATH path (Download from apache website).

   > **NOTE**
   > Download the CXF 3.4.5 from https://cxf.apache.org/download.html or directly from https://archive.apache.org/dist/cxf/3.4.5/apache-cxf-3.4.5.zip

Java Clients

The TableOfContents.doc in $NX_ROOT/samples/sdk/cxfwebsvc lists several Java sample programs.

Each sample program contains notes on how it can be compiled and run using the script files run_java_test_bat.txt (Windows) and run_java_test_sh.txt (UNIX). These scripts demonstrate how to use org.apache.cxf.tools.wsdlto.WSDLToJava to generate the CA SDM web services client-side stub files.

Running WSDL2Java as shown will generate the stub files in subdirectory com/ca/UnicenterServicePlus/ServiceDesk.

Import these classes with the following statement:

```
import com.ca.UnicenterServicePlus.ServiceDesk.*;
```

Many Web Service methods have parameters of type ArrayOfString, a proprietary class. For example, the createRequest() method's attrVals, propertyValues, and attributes parameters are all ArrayOfString parameters.

To set the values in an ArrayOfString variable, instantiate the variable and then use getString().addAll() as follows:

```
ArrayOfString attrVals = new ArrayOfString();
attrVals.getString().addAll(new ArrayList<String>(Arrays.asList("customer", customerHandle,

"description", "description text")));
```

To set it to empty, instantiating the variable is sufficient.

Use a variable of type ListResult, another proprietary class, as the return value from the List methods: doQuery(), getRelatedList(), getNotificationsForContact(), getPendingChangeTaskListForContact() and getPendingIssueTaskListForContact(). A ListResult contains listHandle and listLength elements, which can be retrieved using getListHandle() and getListLength() as shown in this example:

```
ListResult doQueryResult = new ListResult();
doQueryResult = USPSD.doQuery(sid, "iss", "active = 1");
int listHandle = doQueryResult.getListHandle();
int listLength = doQueryResult.getListLength();
```

The getListValues() method uses the listHandle, retrieving the values from a subset of the list.

The Handles parameter of the freeListHandles() method is an ArrayOfInt, another proprietary class. Call freeListHandles() using the listHandle taken from a ListResult:

```
ArrayOfInt handleList = new ArrayOfInt();
handleList.getInteger().addAll(new ArrayList<Integer>(Arrays.asList(listHandle)));
USPSD.freeListHandles(sid, handleList);
```

Some methods have pass by reference parameters of type Holder<String>. For example, createRequest() has three parameters of this type, NewRequestHandle, NewRequestNumber, and cr_result.

```
Holder<String> NewRequestNumber = new StringHolder();
Holder<String> NewRequestHandle = new StringHolder();
Holder<String> cr_result = new Holder<String>();
USPSD.createRequest(sid, creatorHandle, attrVals, propertyValues, template, attributes,
NewRequestHandle, NewRequestNumber, cr_result);
```

The Request's handle and reference number (ref_num) can then be obtained from NewRequestHandle.value and NewRequestNumber.value respectively.

SOAP CXF Web Services Configuration

You can configure the CA SDM SOAP CXF Web Services with entries in special web configuration files. The following table summarizes the names and descriptions of the configuration options:

Option Name	Description
design_mode_stubs	Sets the Web Service *to design mode* (CA SDM only).
require_secure_logon	Requires the login() and loginService() web methods to be called with a secure protocol, such as https.
require_secure_connection	Requires that every web method be called with a secure protocol.

disable_user_logon	Disables both login() and loginService() web methods, so only loginServiceManaged() can be used to log in.

CA SDM adds protection to the integrity of the running Tomcat server by verifying the length of the attribute values that pass to Web Service methods. By default, web service calls return an WebServiceException if the length of an attribute exceeds 900,000 bytes..

You set the following parameters in the deploy.wsdd file:

- **fatal_max_string_length**
  Sets the length of the largest attribute value that a web service method accepts.
  Default: 900,000 bytes
- **validate_parameters**
  Sets whether the attribute value length checking is performed. Set the parameter to 0 to turn off the validation.
  Default: 1 (on)
- **exception_methods**
  Displays a comma-delimited list of Web Service methods that are exempt from the attribute value length validation.

Redeploy the CXF Web Services

New configuration settings take effect when you redeploy CA SDM CXF Web Services. Complete the following steps to redeploy the Web Services:

1. Recycle Tomcat by recycling the CA SDM service. You can avoid shutting down the entire CA SDM system by recycling Tomcat by simply using the following commands:
   ```
   pdm_tomcat_nxd  - c stop
   pdm_tomcat_nxd  - c start
   ```
   The Web Service is redeployed.
2. Verify that the service deployed by viewing the CXF services listing page at the following default URL:
   ```
   http://<servername>:< port>/cxf/services
   ```

   NOTE

   The exact URL depends upon your installation settings.

SOAP Error Handling

If an error occurs with a Web Services method, a SOAP Fault is returned. The SOAP Fault is the standard means of returning exception information for Web Services.

The Fault message contains standardized <Message> and <Code> elements, but the most informative is the <Detail> element. The <Detail> element contains <ErrorCode> and <ErrorMessage> elements. The <ErrorCode> element returns an enumerated error code specific to either the CA SDM or Knowledge Management product. The <ErrorMessage> element contains an English string describing the errors. The <ErrorMessage> elements are more suitable for aiding the developer and more appropriate messages should display to users.

For example, the following illustrates a SOAP Fault when a bad parameter is supplied to the CA SDM getObjectValues() method:

```
<soap:Fault>
   <faultcode>soap:Client</faultcode>
   <faultstring>Error on fetch with attribute
 list:persistent_id,first_name,last_nameParamErrorHere<faultstring>
   <detail>
<ErrorCode>1001</ErrorCode>
<ErrorMessage>Error on fetch with attribute list: persistent_id,first_name,last_nameParamErrorHere </
ErrorMessage>
```

```
</detail>
</soap:Fault>
```

In some cases, the servlet container may write errors and therefore, display in the servlet container logs. In other cases, error information may be written to CA SDM logs. These logs are located in the following subdirectories:

- In the /bopcfg/www/CATALINA_BASE/logs subdirectory of CA SDM installation
- In the /log subdirectory of the CA SDM installation and to all logs that have the prefix “jsrvr”.

NOTE
We recommend that you constantly monitor these logs, as the server may log its own errors without reporting them to the CA SDM Web Services.

Lock Errors

CA SDM objects are locked during updates. Methods that update objects (such as, updateObject() or transfer()) may return the following lock error code:

```
UDS_LOCK_ERR
```

This code indicates that another user is updating the record. Often the locking user’s handle is returned in the ErrorMessage element.

Time Outs

If the CA SDM server is heavily loaded, a method may take a long time to process. In rare cases, a method may never return because a separate process failed to reply or some other error occurred. To guard against excessive blocking, every Web Services method times out after a number of seconds. Web Services method time-out is a CA SDM server time-out, *not* a Web server time-out, network time-out, and so on.

NOTE

The Web Services delay a few seconds the first time accessed after the J2EE application server is recycled. This happens because the application is initializing, loading DLLs, libraries, and so on, and occurs only with the first Web Services method call. All subsequent calls return much faster.

If a method times out, it returns the following error code:

```
UDS_TIMEOUT_ERR
```

The operation is not aborted! The server may have received the request and processes it successfully, although slowly. This type of problem may occur when using the *doSelect( )* method to retrieve several 1000 records.

NOTE
For information about the *doSelect* method, see the CA Service Desk Manager Reference Commands section.

Error Codes

The following table lists the possible values for the <ErrorCode> value in a SOAP Fault returned from a Web Services call:

Error Name	Value	Description
UDS_OK	0	Successful.
UDS_FAILURE	1	General failure, check system logs.
UDS_BAD_PARAM	1000	A bad parameter was passed to a method. This error occurs if a required parameter is missing, the wrong type was passed, or an invalid value was used.

UDS_INTERNAL_ERR	1001	Signals that an internal error occurred. A description is found in the return array and the system logs.
UDS_LOCK_ERR	1002	An attempt was made to update an object locked by another user or process. Usually the ID of the contact responsible for locking the object is returned in the return data.
UDS_UPDATE_ERR	1003	An error occurred updating an object. Make sure all required attributes were set and check the system log.
UDS_CREATION_ERR	1004	An error occurred creating an object. Make sure all required attributes were set and check the system logs.
UDS_NOT_FOUND	1005	A search method failed to find any matches or failed to find an object specified. This can happen if a bad or invalid handle is passed to any method.
UDS_TIMEOUT_ERR	1006	The current method timed out, the CA SDM server may be heavily loaded or the method itself was bad.
UDS_SERVER_GONE	1007	The CA SDM server connection is lost, UDS methods will no longer function and all list references are lost.
UDS_FETCH_ERR	1008	An error occurred while retrieving list data.
UDS_BAD_SESSION	1010	An invalid SID was used.
UDS_CNTXT_TIMEOUT	1011	The SID timed out.
UDS_SECURE_CHANNEL_REQUIRED	1012	The Web Services (or a web service method) requires a secure channel (for example: SSL) for access, but an unsecured channel is being used.
UDS_SECURITY_VIOLATION	1013	The attempted operation violates CA SDM security and was aborted.
UDS_OVER_POLICY_LIMIT	3002	The attempted request is refused because it exceeds the limit defined in the policy.

SDM CXF webservices for CA APM

CA APM supports SDM CXF webservices

From 17.4 RU2, CA APM supports both axis and cxf web services of SDM when integrated. By default, the axis Web Service URL is configured and we can manually change it to cxf Web Service URL.
To perform this operation:

- Login to ITAM as administrator.
- Navigate to **Administration** > **System Configuration** > under **Menu** select the CA SDM link, and change the CA SDM Web Services URL.
  Example: *http://<SDMHost>:<port?/axis/services/USD_R11_WebService?wsdl* should be manually changed to *http://<SDMHost>:<port?/cxf/services/USD_WebService?wsdl.*
- Perform the IISRESET on both ITAM Web and Application servers.

CA Service Catalog CXF webservices for CA APM

CA Service Catalog CXF webservices for CA APM

From 17.4 RU3, CA APM supports both axis and cxf web services of CA SLCM when integrated. By default, the axis Web Service URL is configured and we can manually change it to cxf Web Service URL.
To perform this operation:

- Login to ITAM as administrator.
- Navigate to **Administration** > **System Configuration** > under **Menu** select the Application Sever, and change the USM Web Services URL.
  Example: *http://{Server}:8080/usm/services/RequestService?wsdl should be manually changed to http://<SLCM hotname>:<port>/usm/webservices/RequestService?wsdl*
- Perform the IISRESET on both ITAM Web and Application servers.

User Authentication and Authorization

CA SDM lets you specify user access authentication and the features available by access control and management.

User Access Authentication

CA SDM Web Services provides two access authentication schemes. They are associated with the new access control and management feature, which uses an access policy.

- **User Name/Password**
  Verifies the User Name/Password, as described in previous releases of the product.
- **Public Key Infrastructure (PKI) Technology**
  Verifies that the person requesting the access has ownership of a certain private key.

  WARNING
  If you plan to use an application that accesses this version of CA SDM Web Services, we recommend strongly that you first define a Web Service Access Policy, complete with its code value, in CA SDM. A default access policy with a policy code of DEFAULT is available when CA SDM is installed and configured.

CA SDM REST API

This article contains the following topics:

Representational State Transfer (REST) is a style of software architecture for distributed hypermedia systems such as the World Wide Web. The CA SDM REST API lets application, integration, and web developers build UIs and applications for devices such as tablet computers and smartphones. Users such as analysts, employees, and customers can then use the UI or application on these devices. For example, application developers can develop a CA SDM UI that lets analysts use devices to update tickets.

The REST API accesses resources by using a Uniform Resource Identifier (URI) -- a character string that identifies a name or resource on the Internet. In CA SDM, resources can be objects such as tickets, assets, contacts, and so on. An application using the REST API makes an HTTP request to a URI and parses the response. Such identification enables interaction with representations of the resource over a network. Each client to server request contains all the information necessary to understand the request, and does not use any stored context on the server.

Developers use the REST API directly to send HTTP requests to the server for the resource they want to manipulate. Developers only need an HTTP client library, which is available with most programming languages. Because the REST API is based on open standards, you can use any Java programming language to access it.

NOTE
For information about REST HTTP methods, see the REST HTTP Methods.

REST and SOAP

CA SDM provides REST and SOAP web services APIs. The audience for the REST API is a UI client whereas the audience for the SOAP API is a program. REST services are about resources (manipulating objects, changing object states, exchanging representations, and using nouns rather than verbs). SOAP services are about services (calling methods, using verbs, and doing actions).

REST provides the following advantages over SOAP:

- REST is lightweight, HTTP-based, and stateless (for scalability).
- REST supports client bookmarking and caching.
- REST maintains data contract loosely.
- REST is easily consumed by front-end technologies such as WEB 2.0 and AJAX.
- REST supports XML and JSON data formats.
- REST improves performance.

REST Security

Security uses multiple authentication mechanisms including a custom approach that uses shared secret keys.

The product supports the following security authentication schemes:

- REST web services secret key authentication (uses SSL and HMAC for login)
- REST basic authentication (clear text encoded username/password)
- REST BOPSID authentication (validates CA SDM BOPSIDs)
- External (CA EEM) artifact authentication (CA EEM artifact token)

CA SDM Secret Key Authentication

The CA SDM Secret Key authentication is a process that verifies the following:

- The identity of the request sender.
- That the sender is a registered user.

Secret Key authentication requires that each request includes information about the identity of the request sender. The request must also include additional information that CA SDM can use to verify the authenticity of the user. When the request passes this verification test, the request is determined to be authentic. During authentication for an Access Key request, CA SDM secret authentication does the following:

1. Assigns an access key to a client. The access key identifies the client responsible for a request and uses the CA SDM session ID as the key value. Because an access key is sent as a request parameter, it is not secret. Anyone sending a request to CA SDM can use the request parameter, therefore, a secret key is needed.
2. Assigns a secret key. A secret key is a 40-character alphanumeric sequence dynamically generated by CA SDM during login. The product encrypts this secret key before storing it in the database.
3. Uses client-provided information (a request signature using the secret key) to identify the client and verify that the request is legitimate. This additional information protects users from impersonation and demonstrates possession of a shared secret known only to CA SDM and the sender of the request.

How to Use the Secret Key Authentication with REST API?

CA Service Desk Manager's REST API supports Secret Key Authentication. For more information, see REST HTTP Methods -REST Secret Key Authentication. This article gives a high-level overview and other considerations while implementing the Secret Key Authentication in CA SDM REST API.

Prerequisites and Considerations

Consider the following before you implement the Secret Key Authentication in CA SDM REST API:

1. Ensure there is communication between the client (a third-party program of your choice) and the SDM REST server. You must first consider implementing the HTTPS between these two components. You can secure the CA SDM REST Tomcat using an SSL certificate and use that certificate + HTTPS URL when connecting from the client.
2. CA SDM must be configured to Support *HMAC_ALGORITHM*. Login to CA SDM, **Administration**, **Options Manager, Web Services, hmac_algorithm**.
   This option can be set to a preferred value. Install the option (In this case, it is set to *hmacSHA1*). Follow the required steps that you usually perform while installing a CA SDM Option.
3. The CA SDM *secret_key* is a 40-character alphanumeric sequence and is dynamically generated by CA SDM during REST access key creation. This *secret_key* is encrypted before it is stored in the CA SDM database (*usp_rest_access table*).
4. The *secret_key* needs to be used thereafter by the client to be able to properly authenticate itself as a valid client against the CA SDM REST server. This is performed by the client program which sends a Signed Header as part of its requests.
5. The signature, a Keyed-Hash based Message Authentication Code (*HMAC - Hash-based Message Authentication Code*) is calculated using the following:
   a. the *Secret_Key*
   b. the header fields (for example: **date**, **accept**) provided by *NX_STRING_TO_SIGN_FIELDS* (if the option is not installed in the same order.
   c. the cryptographic hash function provided by NX.env variable *NX_HMAC_ALGORITHM* (supported algorithms are **HmacSHA1, HmacSHA256, HmacSHA384, HmacSHA512** and **HmacMD5**).
6. Client sends the request data, the signature and the Access Key to CA SDM.
7. CA SDM uses the Access Key to look up the Secret Key from persistence store.
8. CA SDM uses the request data and the Secret Key to generate the signature using the same hash algorithm the Client used.
9. If the signature generated by CA SDM matches the signature sent by the Client, then the request is considered authentic, otherwise the request is discarded and CA SDM returns an error response.
10. The installation folder contains few samples in **$NX_ROOT/samples/sdk/rest/java/test2_auths** withREADME.txt in *$NX_ROOT/samples/sdk/rest/java'SampleSDMAuth.java', 'SampleUsingSecretKey.java' and 'HMACUtil.java*'.

Create Access Key and Secret from CA SDM REST API:

The client must obtain an access_key and secret_key from SDM REST API. This is done by doing a *POST to /caisd-rest/rest_access*. Below procedure was created by using the Postman extension of Chrome. For more instructions on how to use Postman, check out this article or search for Postman on the Google Chrome webstore.

1. First the client needs to obtain an access_key and secret_key from CA SDM REST API.
   This is done by doing a POST to */caisd-rest/rest_access.*
   a. Set your Authorization Type to: Basic Auth
   b. Populate a Username / Password with correct values
   c. Click on **Update Request** button.
2. Change the **Type now** to **No Auth** and click the **Save** button.
3. Switch to the **Headers** tab and ensure that the Authorization shows up as Basic with a base-64 encrypted string next to it.
4. Change string "**Basic**" to "**SDM**", leave the rest of the base64 string as is.

   NOTE

   There is a space character after Base, leave it as is.
5. Click **Save**.

NOTE

Basically , the Authorization header must be in the format: the string "SDM" space and the base-64 encoded username/password of a CA SDM User.

In the above example its like this:

6. Click **Send** to send the POST to the CA SDM REST server.

```
POST /caisd-rest/rest_access HTTP/1.1
Host: YourSDMHostName:8553
Authorization: SDM cnVkcmEwMjpJbmRUciFwMjAxNw==
Content-Type: application/xml
x-obj-attrs: access_key,secret_key,content-type,date
Cache-Control: no-cache

<rest_access></rest_access>
```

7. CA SDM Server gives the following response:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<rest_access id="400502" REL_ATTR="400502" COMMON_NAME="845787692">
<link href="http://localhost:8050/caisd-rest/rest_access/400502" rel="self"/>
<access_key>845787692</access_key>
<expiration_date>1503521363</expiration_date>
<secret_key>2504166E48DC19294B86773F798DEE7996D3973E</secret_key>

</rest_access>
```

8. Use the *secret_key* as well as the *access_key* to make rest of the REST operations that we need.
   For this, do the HMAC encryption of the string that you need to request. This is done by creating a Pre-Request script section of Postman. For example - try to get some attributes from the "cnt" object of SDM REST API as following:

```
var str = "GET\n/caisd-rest/cnt"
var secret = "2504166E48DC19294B86773F798DEE7996D3973E";
postman.setGlobalVariable("hmac", encodeURIComponent(CryptoJS.enc.Base64.stringify(CryptoJS.HmacSHA1(str,
secret))));
```

   Here the secret key is what we got as a response for **step 6** above.
   You can leave the rest of the information as is. This is basically going to encrypt the resource string: *GET\n/caisd-rest/cnt* and the secret key together and encode it using the *HmacSHA1* algorithm.
9. Navigate to the **Authorization** tab and change the **Authorization** to look like the following:

```
SDM 845787692:{{hmac}}
```

   Here, it is a literal string of CA SDM followed by a space, followed by the access-key from CA SDM that we obtained in **Step 6**, followed by literal string *:{{hmac}}*
10. Ensure that the *GET* section contains */caisd-rest/cnt* as the resource, because that is what is encrypted in the pre-request script.
11. Add the *X-Obj-Attrs* header key with values: *userid,last_name* (basically we are trying to get the userid, last_name field values from the resource: /caisd-rest/cnt). Your code should look like the following:

```
GET /caisd-rest/cnt HTTP/1.1
Host: SDMHostName.ca.com:58553
Authorization: SDM 1842290659:jkd32qsCPwaBcWH0NX93V8zu6sI%3D
Content-Type: application/xml
X-Obj-Attrs: userid, last_name
```

```
Cache-Control: no-cache
```

12. Save and send.
13. The following response is received:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<collection_cnt COUNT="25" START="1" TOTAL_COUNT="36">
<link href="http://localhost:8050/caisd-rest/cnt?start=26&amp;size=25" rel="next"/>
<link href="http://localhost:8050/caisd-rest/cnt?start=1&amp;size=36" rel="all"/>
..
..

..
<cnt id="U'793ED69B4E87A545BD8E911834D829FC'" REL_ATTR="U'793ED69B4E87A545BD8E911834D829FC'"
COMMON_NAME="System_AHD_generated">
<link href="http://localhost:8050/caisd-rest/cnt/U'793ED69B4E87A545BD8E911834D829FC'" rel="self"/>
<last_name>System_AHD_generated</last_name>
<userid>ahd</userid>
</cnt>
<cnt id="U'7A0E651346BF0E4491EBD37D13962417'" REL_ATTR="U'7A0E651346BF0E4491EBD37D13962417'"
COMMON_NAME="System_Argis_User">
<link href="http://localhost:8050/caisd-rest/cnt/U'7A0E651346BF0E4491EBD37D13962417'" rel="self"/>
<last_name>System_Argis_User</last_name>
<userid>System_Argis_User</userid>

</cnt>
..
..


</collection_cnt>
```

How to Version System Customizations Across CA SDM Servers

CA SDM version control helps you to manage the system customizations across all CA SDM servers (client and servers). Depending upon the CA SDM configuration, the following client and servers are used:

- Conventional configuration,
  - Client: Secondary server
  - Server: Primary server
- Advanced availability configuration,
  - Client: Application and Standby servers
  - Server: Background server

Example: As a system administrator, you added a field in the detail_usp_server.htmpl page of CA SDM. Now, you want to synchronize this change across the client and the servers. This example is used throughout the scenario to explain how the customization is synchronized.

Follow these steps:

1. Verify the Prerequisites.
2. Make changes in CA SDM. In this example, add the new field in the CA SDM HTMPL page.

3. Modify the Server Version Control File.
4. Restart CA SDM on Client.
5. Verify the Customizations on the Client.

Verify the Prerequisites

Verify the following prerequisites:

- Ensure that the ver_ctl option set to the **upgrade** value. When a version discrepancy is detected, an upgrade of the affected components is attempted on the client. If the upgrade is successful, the client connection with the server continues; otherwise, the connection terminates. For more information about the ver_ctl option, see the *Online Help*.
- Ensure that the server_secondary_custom.ver file is created at $NX_ROOT\site\mods directory of the primary server or background server (depending upon the CA SDM configuration). All customizations to a version control component must be done in this file. If the file does not exist, ensure that you create it at the same location.

Modify the Server Version Control File

After you complete the customization, (for example, added a field in the HTMPL page) add or update the version control components on the server version control file. A version control component can represent a file, directory, or the client_nx.env environment variable file.

Follow these steps:

1. Log in to the following server depending on your CA SDM configuration:
   - Conventional: Primary server
   - Advanced availability: Background server
2. Go to the following directory:
   ```
   $NX_ROOT\site\mods
   ```
3. Open the server_secondary_custom.ver file.
4. Add the following components:
   ```
   [ MY_USP_SERVERS_HTMPL ]
   directory="$NX_ROOT/site/mods/www/htmpl/web/analyst/Analyst"
   filename="detail_usp_servers.htmpl"
   version="2.0 , 20121119"
   o_mode="RW"
   g_mode="RW"
   w_mode="RW"
   file_ctl
   ```

   NOTE
   For more information about adding or updating version control components, see the Version Control Component topic.
5. Save the server_secondary_custom.ver file.
   The version control component is added in the version control file.

Version Control Components

To define a new component,

- Use the following syntax. Items in *italics* represent data that you supply. The *component-name* and version parameters are always required. Other parameters are required, depending on the value of *control-type*. All other items represent keywords that you must enter exactly as shown in the following example:
  ```
  [ component-name ]
  version = "x.x, yyyymmdd"
  control-type
  ```

```
filename = "filename"
directory = "directory"
link = "link-directory"
source = "source-directory"
effectivity = "effect-spec"
checksum
min_release = "release"
max_release = "release"
component_type = "file-type"
o_mode = "owner-mode"
g_mode = "group-mode"
w_mode = "world-mode"
```

NOTE
For more information about the parameters, see Version Control Parameters. For more information about the structure and syntax of version control files, see the .ver files that are installed in the $NX_ROOT\site directory. These files have useful comments and example settings that can help you.

To update an existing component entry,

- Change the parameter. For example, you change the version number.

To remove control from a component,

- Edit the component as follows:

```
! uncontrol component-name
```

Version Control Parameters

The following parameters apply to version control:

- **[component-name]**
  Specifies the name of an item under version control. The name must be unique and enclosed in square brackets. *component-name* is not case-sensitive. This parameter is required to begin a component definition.
- **version="x.x. yyymmd"**
  Specifies a version number (*x.x*) and a date (*yyyymmdd*) that define the version of the component. This parameter is required, and must be enclosed in double quotes. Version control verifies the version of a component by comparing the version number and date on the server with the version number and date on the client. Both version number and date must match for the component to be considered in sync between the client and server. Optionally, if the checksum property is enabled, the file is verified by checksum verification before being updated.
- **control-type**
  Specifies the type of version control for this component. The following settings are valid for control-type:

Setting	Description
dir_ctl	Specifies that the component represents a directory. You must provide the directory parameter to specify the path to the directory. You can also provide the filename parameter to specify the filename parameter to filter a set of files in the directory. Subdirectories are not upgraded on either UNIX or Windows.
file_ctl	Specifies that the component represents a file. You must provide the directory and filename parameters to specify the path to the file.

Nxenv_ctl	Specifies that this component represents the client_nx.env file, which is used to store internal CA SDM environment variables. CA SDM version control and the Options Manager automatically maintain this file. There is one nxenv_ctl component, and its component name must be CLIENT_NXENV.
ver_ctl	This is the default control type. It specifies that the component is generic; that is, not associated with any specific external object. You can use a generic component to provide version control for the client as a whole, or for a file or directory too large for an automatic upgrade. Components with a control type of ver_ctl cannot be upgraded; a version mismatch on a ver_ctl component when the server is in UPGRADE mode causes the client connection to fail.

- **filename="filename"**
  Specifies the name of a file under version control. It does not contain directory specifications. This parameter is required for file_ctl components, but is optional for directory (dir_ctl) control components. When used with directory components, the filename parameter acts as a file mask to restrict the files associated with the dir_ctl component. For example, if the filename for a dir_ctl component is *.README, then an upgrade from that directory includes only files ending with ".README.".
- **directory="directory"**
  Specifies the path to the directory for dir_ctl components, or to the directory containing the file for file_ctl components. This parameter is ignored for ver_ctl and nxenv_ctl components. The directory path must be enclosed in quotes, and can contain references to environment variables preceded with a $.

  NOTE
  Always use forward slashes (not backslashes) to separate subdirectories in the path name, even on a Windows server.
- **link="link-directory"**
  Specifies a link directory on the client in the same format described previously for directory parameter. This parameter is optional for file_ctl and dir_ctl components, and ignored for ver_ctl and nxenv_ctl components. If it is specified, an upgrade to a Linux client causes a symbolic link to be placed in the link directory, pointing to the actual file copied to the location specified by the directory parameter. An upgrade to a Windows client causes the actual file to be copied to both the link and directory locations.
- **source="source-directory"**
  (Optional) Specifies a different directory on the server where the server can retrieve files for delivery. This parameter has the same format described previously for the directory parameter. It is useful if the files that are to be delivered to the client are different from the same files in the directory location on the server. This parameter is used to tell the server to retrieve the file from *source-directory* and deliver it to the location on the client specified by the directory parameter. The directory parameter is required if you specify the source parameter.
- **effectivity="effect-spec"**
  (Optional) Specifies whether the client should get this component. It lets you exclude download to some clients. If a client is not included in the effectivity specification, it does not get the component. If this parameter is omitted, all clients receive the component. The effectivity specification uses the following symbols:
  - **+ (plus sign)**
    Indicates to add a client group.
  - **- (minus sign)**
    Indicates to exclude a client group.

The following client groups are valid:

- SUN4SOL
- AIX
- LINUX
- LINUX390
- HP
- WINDOWS_CLIENTS
- UNIX_CLIENTS

For example, the following specification indicates that only UNIX clients get the files:

```
effectivity = "+ UNIX_CLIENTS"
```

- **checksum**
  Directs the component to upgrade if the checksum of the component on the client does not match the corresponding checksum on the server. If it is applied to a directory, then checksum is applied to each file.
- **min_release="release" and max_release="release"**
  Specifies the oldest and latest client to which this component should be distributed. Statements in the server_default.ver file define releases. These parameters are in the following form, where G*axx* indicates the release, and the values following are genlevels associated with the release.

  ```
  ! Release GA50 50MVV000900 50W7T000900
  ! Release GA45 45MW000900 50WTT000900
  ```

  The order indicates that GA50 is newer than GA45.
- **component_type**
  Specifies the type of component used. Following types of components are used:

Setting	Description
file	This is the default component type. It specifies that files copied to the client be obtained directly from the location on the server indicated by the directory parameter.
exe_file	Specifies that files copied to the client be obtained from a location on the server that is dependent on the client's operating system, as shown by the following: windows (Windows) sun4Sol (Solaris) hp (HP-UX) aix -- AIX) linux (Linux) linux390 (Linux390) Locations for these subdirectories are dependent on the directory parameter setting. If this parameter is set, then subdirectories are located under the indicated *directory*. Otherwise, they are located under the bin directory of the main CA SDM installation directory.

- **o_mode="owner-mode"**
  Specifies file access permissions for the owner of the file.
- **g_mode="group-mode"**
  Specifies file access permissions for users in the file owner group (used for UNIX clients only).
- **w_mode="world-mode"**
  Specifies file access permissions for users not in the file owner group (used for UNIX clients only).

The three mode parameters allow different versions of the same executable to be maintained on the server. They specify access controls on the file when copied to the client. These parameters are used only during an upgrade operation. They consist of one to three characters, with the following significance:

Setting	Description
R	Read
W	Write
X	Execute

PC clients ignore Write and Execute permissions.

You can specify any combination of one or more file access modes. On UNIX clients, the file is given the access mode of specified. On PC clients, the file is made writable or read-only, depending on whether w_mode has been specified.

Restart CA SDM on Client

You restart CA SDM on the client servers to update the client version control files with the customizations.

NOTE

Select **Start**, **Settings**, **Control Panel**, **Administrative Tools**, **Services**. Right-click the **CA SDM Server** and choose **Start** to restart or start a server.

Follow these steps:

1. For the conventional configuration, restart the secondary server.
2. For the advanced availability configuration, complete the following steps:
   a. Restart all standby servers.
   b. Choose the less active application server.
   c. Restart the less active application server.
   d. Stop the other application server.
   e. Start the application server.
   f. Performs steps d and e for more application servers.

   The client connects to the server to send a list of its controlled components. The server compares the list to its own master list. The affected components on the client are upgraded.

Choose the Less Active Application Server

You choose an application server with the least user activity. Run the following command on each application server to choose the one with no or minimal active sessions.

```
pdm_webstat
```

NOTE

: This command does not capture the SOAP or REST Web Service sessions.

Stop the Other Application Server

You inform all the active users on an application server to move to the less active application server before you stop it. Ensure that you have restarted the less active application server before moving all the users to it.

Follow these steps:

1. (Recommended) Inform all active Support Automation analysts on the application server which you want to stop, to create a ticket in CA SDM with their session information. This process ensures that the session information is not

lost. For example, the Support Automation analyst is in a session with a customer to resolve a hardware issue. In such a case, the Support Automation analyst can create an issue in CA SDM with the session information before the application server shuts down.

2. Send a notification (for example, an email notification) to all the active users on the application server to move to the less active application server that you just restarted. This notification can include the details of the updated application server.
3. Execute the following command on the application server:

```
pdm_server_control [-h] -q interval -s server_name
```

   - **-h**
     Displays the help page.
   - **-q interval -s server_name**
     Notifies a local or remote application server to quiesce in a specified time interval. This interval is the number of seconds before the server goes offline. When using this option without a server_name, the local server is notified to quiesce. This option cannot be used for a background or a standby server.

   A pop-up message is displayed to all the active users on the application server to notify them about the server shutdown and the time left for the shutdown. The users must save their work and logout within that time. The application server stops after the specified time. The users log on to the other application server to resume their work. The Support Automation analyst can refer to the ticket and resume their work.
   The application server is stopped successfully.

Verify the Customizations on the Client

You verify the version control file on the client to check if all customizations have been synchronized.

Follow these steps:

1. Log in to the following client depending on your CA SDM configuration:
   - Conventional: Secondary server
   - Advanced availability: Standby server and Application server
2. Open the stdlog file from the following location:
   ```
   $NX_ROOT\log
   ```
3. Find out if all the customizations made on the server are applied on the client.

Using the Web Screen Painter (WSP)

Web Screen Painter (WSP) is a component of the CA SDM that provides tools for designing and customizing CA SDM forms. You can modify the existing forms to suit your needs, and you can create forms from scratch. WSP helps you reduce the time and effort it takes to modify and test HTMPL forms. With customized forms, your service desk analysts and customers can quickly find and enter the information that is important to your organization.

With WSP, it is also very easy to modify the database schema, such as adding new tables or columns to the database. It provides a user-friendly interface and automates the complex process of updating a schema. Using WSP, you can build and test forms using custom schema before making any changes to the database.

Readonly Preview Session

In order to allow you to preview how an updated form looks with real data, Web Screen Painter (WSP) previews forms with data from your CA SDM database. Forms you preview in WSP appear in this browser window. Since this is a readonly session, all update operations are ignored. As a reminder of this, the caption of all "Save" buttons is changed to "noSave". Please use WSP itself to logout from this session.

For detail forms, WSP shows the edit version of a form (you can display the readonly version by pressing the noSave button). When you preview a tab, WSP shows the entire detail form containing the tab, with the notebook opened to the previewed tab.

To prevent inadvertent damage to production data from a previewed form in edit mode, WSP normally suppresses updates from a preview session, so that you cannot modify the database from a preview session. To indicate this, WSP automatically changes the text on a "Save" button to "noSave", and displays a small red WSP palette in the upper left corner of the form. You can still press the noSave button (or take other actions that would normally modify the database), but CA SDM will discard your changes.

Optionally, your CA SDM system administrator may choose to enable updates from preview. When this is done, "Save" buttons are displayed normally, and the WSP palette is yellow. You should be cautious when updating the database from a preview session, as WSP preview sometimes bypasses the normal procedure for reaching a form, and the form's environment may not be completely set up. To ensure updates work correctly, return to the CA SDM main form and navigate to your form as you normally would in a standard (non-preview) CA SDM session. For more information about enabling updates from the preview session, see Assign Web Screen Painter Permissions to an Access Type.

Open the CA Standard Version of a Form

The CA standard version of a form is the same form with no site modifications (although it may contain CA-supplied patches). The edit window for the CA standard version of a form includes the words CA STANDARD VERSION in its title.

Follow these steps:

1. Open the form in Web Screen Painter.
2. Click File, Open CA Standard Version.

Insert Controls to a Form

The Insert menu on WSP inserts a field on the form. The insert menus follow:

- **Insert Row**
  Inserts a new row in the form on which you can place controls. If a control is selected, it causes the selected control to be the last in its row, and moves following controls to the new row.
- **Delete Row**
  Deletes a complete row. If there are controls on the row, WSP prompts you before deleting the row along with the controls.
- **Insert Textbox**
  Inserts a simple text field on the form. A text field is displayed as a box where the user can enter unedited text.
- **Insert Dropdown**
  Inserts a dropdown (select) field on the form. A dropdown box is displayed as a list of selections for a field. The user can click the dropdown arrow to choose a response from the list.
- **Insert Lookup**
  Inserts a lookup field on the form. A lookup field is displayed as a text box with a header consisting of a small magnifying glass icon followed by a link that the user can click to display a selection list.
  Lookup fields have an autofill feature. If a user clicks the hyperlink and the text specified is sufficient to identify a value in the underlying table, the pop-up search form is suppressed and CA SDM fills in the value. In addition, if the Autofill configuration property is set and the user presses the Tab key to leave a field after supplying a value, the effect is the same as clicking the hyperlink; either the value is filled in or a pop-up search form appears.
- **Insert Hierarchical Lookup**
  Inserts a hierarchical lookup field on the form. A hierarchical lookup field is displayed as a text box with a header consisting of a small plus sign icon followed by a link that the user can click to pop-up a hierarchical selection list.
  Hierarchical search fields have an autofill feature. If a user clicks the hyperlink and the text specified is sufficient to identify a value in the underlying table, the pop-up list form is suppressed and CA SDM fills in the value. In addition, if

the Autofill configuration property is set and the user presses the Tab key to leave a field after supplying a value, the effect is the same as clicking the hyperlink; either the value is filled in, or a pop-up search appears.

- **Insert Date**
  Inserts a date field on the form. A date field is displayed as a text box with a header consisting of a small calendar icon followed by a link that the user could click to pop-up a date helper to allow specification of the date.
- **Insert HTML Editor**
  Inserts an HTML editor on the form for a text field that contains HTML. An HTML editor field is displayed as a multi-line text box with a header containing icons that a user can click to insert HTML tags into the text.
- **Insert Checkbox**
  Inserts a checkbox field on the form. Use checkboxes when an option may be toggled on and off.
- **Insert Read Only Textbox**
  Inserts a read-only version of the textbox field. A read-only textbox field is displayed as simple text.
- **Insert Read Only Lookup**
  Inserts a read-only version of the of the lookup field. A read-only lookup field is displayed as a hyperlink that the user can click to display the detail form for the field's value.
- **Insert Read Only Date**
  Inserts a read-only version of the date field. A read-only date is displayed as simple text, formatted as a date.
- **Insert Button**
  Inserts a rectangular button on the form. The button invokes the JavaScript string associated with it.
- **Insert Notebook**
  Inserts a Notebook with three tabs. You can use the Notebook Designer to delete tabs or insert additional tabs. A notebook allows several sets of fields to be displayed in the same physical area of the screen, with only one set visible at a time. The user can select the set of fields that is visible by clicking a named tab at the top of the notebook.
- **Insert If/Else If/Else/End If**
  Inserts conditional logic that allows a form's layout and contents to be controlled at the time the form is displayed.
- **Insert Object**
  Inserts an Object tool to place an object control on the form. Place an object control on the form when you want to introduce specific code. You can place the object control on the form in Design mode and edit the code in the Object Designer.
  If WSP cannot recognize something on a page, it converts it into an Object control. For example, an Object tag could represent straight JavaScript or Java between PDM_MACRO tags.

Add Controls to Detail Forms

This article contains the following topics:

The Controls available in the ToolBox vary depending on the type of form that is open currently. When a Detail type form is open, the following controls are available in the ToolBox.

- Checkbox
- Command Button
- Date
- Dropdown Box
- Hierarchical Lookup Box
- Tab
- Lookup Box
- Read Only Date
- Read Only Lookup Box
- Read Only Text Box
- Text Box
- Conditional If
- Conditional Else If
- Conditional Else
- Conditional End If
- Object Button
- Row

Add a Checkbox

You can add a Checkbox control to a form.

Follow these steps:

1. Click the CheckBox button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location.
   The Checkbox control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the dropdown list.
   The Properties dialog lists the following properties for a checkbox control:

- – Attribute
  – Caption
  – Column Span
  – Event
  – On
  – Off

Add a Date

You can add a Date control to a form.

Follow these steps:

1. Click the Date button

   on the Control Palette and then, click the left mouse button and hold it down as you drag-and-drop it on the desired location.
   The Date control is placed on the form.

2. Click the control, click the right mouse button, and select Properties from the dropdown list.
   The Properties dialog lists the following properties for a date control:

- – Attribute
  – Caption
  – Column Span
  – Size
  – Time

Add a Dropdown Box

You can add a Dropdown box control to a form.

Follow these steps:

1. Click the Dropdown button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location.
   The Dropdown Box control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the dropdown list.
   The Properties dialog lists the following properties for a dropdown control:

- – Attribute
  – Autofill
  – Caption
  – Column Span
  – Default
  – Event
  – Factory
  – Initial
  – Link
  – Lookup
  – Size
  – Where Clause

Add a Hierarchical Lookup Box

You can add a Hierarchical Lookup Box control to a form.

Follow these steps:

1. Click the Hierarchical Lookup button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location.
   The Hierarchical Lookup control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the dropdown list.
   The Properties dialog lists the following properties for a hierarchical lookup control:

- – Attribute
  – Autofill
  – Caption
  – Column Span
  – Event
  – Factory
  – Size

Add a HTML Editor

You can add a HTML Editor to a form.

Follow these steps:

1. Click the HTML Editor button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location.
   The HTML Editor is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the dropdown list.
   The Properties dialog lists the following properties for a HTML editor control:

- – Attribute
  – Caption
  – Column Span
  – Event
  – Max Length
  – Option ID
  – Preview
  – Read Only
  – Rows
  – Size
  – Spell Check
  – Toolbar

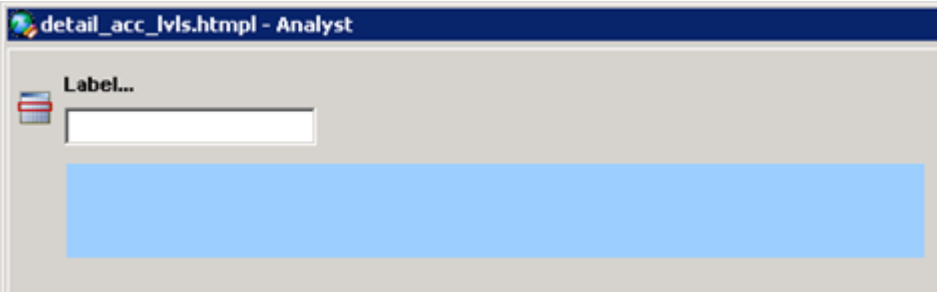
Add a Notebook

Add a notebook to modify how HTMPL forms appear.

Follow these steps:

1. Create a form in WSP, such as a form based on detail.template.
   The HTMPL file opens.
2. Click the Source view and add the following lines of code:

   ```
   <PDM_MACRO name=startNotebook hdr=cng_nb>
   <PDM_MACRO name=endNotebook>
   ```
3. Click the Design view.
   The notebook area appears after the text box, such as shown in the following example:



4. Click the Source view and add the following example code after the <PDM_MACRO name=startNotebook hdr=cng_nb> line:

```
<PDM_MACRO name=startTabGroup title="Additional Information">
```

The following example shows the updated notebook area in Design view:

5. Click the Source view and add the following example code after the previous addition:

```
<PDM_IF "$args.id" == "0">
<PDM_MACRO name=tab title="Attachments" height=300 id=attmnt src="OP=SHOW_DETAIL+HTMPL=xx_attmnt_tab.htmpl
+FACTORY=cr+PERSID=$args.persistent_id+NO_DP=yes">
<PDM_ELSE
```

The following example shows the updated notebook area in Design view:

6. Continue adding tab groups and end the notebook with the following code:

```
<PDM_MACRO name=endNotebook>
```

NOTE

Open default forms in WSP to view how notebooks and nested tab groups appear. For example, open the detail_in.htmpl form. You can also move tabs and tab groups within a notebook using drag-and-drop in Design view. Moving a tab group moves all the tabs within the group with it.

7. To insert a tab or tab group, right-click a control or within the notebook background and select Insert Tab, Insert Tab Group, or Insert Control.
   If you select Insert Tab or Insert Tab Group, WSP inserts a new tab or tab group to the left of the currently selected control. If you select Insert Control, WSP displays the Insert Control dialog that lets you add the desired control to the form.
8. Click Preview.
   Your unpublished changes appear in the web browser.
9. Save the form.
   The notebook is added.

Add a Lookup Box

You can add a Lookup Box control to a form.

Follow these steps:

1. Click the Lookup button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location.
   The Lookup Box control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the dropdown list.
   The Properties dialog lists the following properties for a lookup control:

- – Attribute
  – Autofill
  – Caption
  – Column Span
  – Event
  – Factory
  – Link
  – Size

Add a Read Only Date

You can add a Read Only Date control to a form.

Follow these steps:

1. Click the Read Only Date button

   on the Control Palette and then, click the left mouse button and hold it down as you drag-and-drop it on the desired location.
   The Read Only Date control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the dropdown list.
   The Properties dialog lists the following properties for a read-only date control:

- – Attribute
  – Caption
  – Column Span
  – Time

Add a Read Only Lookup Box

You can add a Read Only lookup box control to a form.

Follow these steps:

1. Click the Read Only Lookup button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location.
   The Read Only Lookup control is placed on the form.

2. Click the control, click the right mouse button, and select Properties from the dropdown list.
   The Properties dialog lists the following properties for a read-only lookup control:

- – Attribute
  – Caption
  – Column Span
  – Link

Add a Read Only Text Box

You can add a Read Only text box control to a form.

Follow these steps:

1. Click the Read Only Text Box button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location.
   The Read Only Text Box control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the dropdown list.
   The Properties dialog lists the following properties for a read-only text box control:

- – Attribute
  – Caption
  – Column Span
  – ID

Add a Text Box

You can add a text box control to a form.

Follow these steps:

1. Click the Text Box button

   on the toolbar, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location.
   The Text Box control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the dropdown list.
   The Properties dialog lists the following properties for a text box control:

- – Attribute
  – Caption
  – Column Span
  – Event
  – Keep Links
  – Keep Tags
  – Max Length
  – Rows
  – Size
  – Spell Check

Add Conditional If

You can add a conditional if control to a form.

Follow these steps:

1. Click the If button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location.
   The Conditional If control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the dropdown list.
   The Properties dialog lists the following properties for an if condition:
   Condition

Add Conditional Else If

You can add a conditional else if control to a form.

Follow these steps:

1. Click the Elif button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location.
   The Conditional Else If control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the dropdown list.
   The Properties dialog lists the following properties for an else if condition:
   Condition

Add Conditional Else

To add a conditional else control to a form:

1. Click the Else button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location.
   The Conditional Else control is placed on the form.

Add Conditional End If

To add a conditional end if control to a form:

1. Click the Endif button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location.
   The Conditional End If control is placed on the form.

Add Object Button

You can add an Object Button control to a form.

Follow these steps:

1. Click the Object button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location.
   The Object Button control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the dropdown list.
   The Object Code dialog appears so you can define the source associated with the Object control.

Add a Row

To add a Row control to a form:

1. Click the ROW button

   on the Control Palette and then, click the left mouse button and hold it down as you drag-and-drop it on the desired location.
   A blank Row is placed on the form. You can place the controls on the row.

Add Controls to List Forms

This article contains the following topics:

The controls available for a List form are different from the controls available for a Detail form. Some of the controls in the List form have a different set of properties compared to the controls in the Detail forms. When a List form is open, the following controls are available in the ToolBox in addition to the conditional logic controls like IF, Else, and so on, which are exactly the same the controls in the Detail form.

- Date
- Drop down Box
- Hierarchical Lookup Box
- Lookup Box
- Results List
- Text Box

Add a Command Button

To add a Command Button control to a form

1. Click the Command Button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and- drop it on the desired location.
   The Command Button control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the drop down list.
   The Properties dialog lists the following properties for a command button control:

Caption
Disabled
Function
ID
Tooltip
Width

Add a Date

To add a Date control to a form

1. Click the Date button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location of the form.
   The Date control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the drop down list.
   The Properties dialog lists the following properties for a date control:
   Attribute
   Caption
   Column Span
   QBE Condition

Add a Dropdown Box

To add a Dropdown Box control to a form

1. Click the Drop down button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location of the form.
   The Drop down Box control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the drop down list.
   The Properties dialog lists the following properties for a drop down box control:
   Attribute
   Caption
   Column Span
   Default
   OP Code

Add a Hierarchical Lookup Box

To add a Hierarchical Lookup Box control to a form

1. Click the Hierarchical Lookup button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location of the form.
   The Hierarchical Lookup Box control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the drop down list.

The Properties dialog lists the following properties for a hierarchical lookup control:
Attribute
Caption
Column Span
Factory

Add a Lookup Box

To add a Lookup Box control to a form

1. Click the Lookup button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location of the form.
   The Lookup control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the dropdown list.
   The Properties dialog appears so you can define properties for the control. The Properties dialog lists the following properties for a lookup control:
   Attribute
   Caption
   Column Span
   Factory

Add Results List

To add a Results List control to a form

1. Click the List button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location of the form.
   The Results List control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the dropdown list.
   The Properties dialog lists the following properties for a lookup control:
   Custom

Add Text Box

To add a Text Box control to a form

1. Click the Text Box button

   on the Control Palette, and then click the left mouse button and hold it down as you drag-and-drop it on the desired location of the form.
   The Text Box control is placed on the form.
2. Click the control, click the right mouse button, and select Properties from the dropdown list.
   The Properties dialog lists the following properties for a text box control:
   Attribute
   Caption
   Column Span

Disabled
QBE Condition
QBE Display
QBE Value
Size
Spell Check

Customize a Form

This article contains the following topics:

After you open the form you want to edit in WSP, you can use the toolbar, menu commands, and shortcuts to customize it. You can perform the following customization:

- Insert a Control
- Edit Control Properties
- Modify Menu Bars
- Modify Stylesheets
- Modify Mouse-Over Preview Form
- Modify Data Grid List on List Form
- Modify Notebooks on Detail Form
- Modify HTML and JavaScript Files

Add Controls

There are four ways to add a new control to the screen. To add any type of control to a form, follow one of these examples:

Example 1: Drag-and-Drop

1. Click on the button in the Control Palette for the control that you want to add by holding down the left mouse button as you drag the control to the desired location on the form.
   Release the mouse button to place the control.
   The control is placed on the form.
2. Save the form when you finish adding controls.

Example 2: Double-Click and Reposition

1. Double-click the control you want to add in the Control Palette.
   The control appears on the form.
2. Place the mouse pointer over the control, then click the left mouse button and hold it down as you reposition the control on the form.
   The control is placed in the desired location on the form.
3. Save the form when you finish adding controls.

Example 3: Right-Click and Select

1. Right-click the form at the location where you wish to add the control.
   A dropdown list appears.
2. Select the control that you want to add from the dropdown menu.
   The control is placed on the form.
3. Save the form when you finish adding controls.

Example 4: Controls Menu

1. Click the Controls menu.

Select the control you want to add from the dropdown menu..
The control appears on the form.

2. Place the mouse pointer over the control, then click the left mouse button and hold it down as you reposition the control on the form.
   The control is placed on the form.
3. Save the form when you finish adding controls.

Arrange Controls on a Form

You can reposition a control on a form by selecting and dragging it. This changes the control's positioning properties, which set the relative placement of a control on a form.

To move controls:

1. On the form, select the control you want to move by left-clicking on it and holding down the mouse button.
   Handles appear around the selection as the following example shows:

2. Drag the selected control to a new position on the form and release the mouse button.

   NOTE
   If necessary, choose Undo once from the Edit menu to reverse the most recent editing action, or choose Undo repeatedly to undo a series of actions.

Move a Control

To move a control by dragging, place the mouse pointer over the control, then click the left mouse button and hold it down as you reposition the control on the form.

To resize a control by dragging it:

1. Click to select the control.
2. Place the mouse pointer over a resizing handle so that the pointer changes to
   ↘
   ,
   ⤢
   ,
   ↕
   , or
   ↔ .
3. Click the left mouse button and hold it down as you drag the control to the appropriate size.

   NOTE
   Some controls, for example, conditionals, have a white border instead of black and cannot be resized. Also, you can change the height of only certain controls, for example, text boxes.

Control Properties

This article contains the following topics:

Attribute

The Attribute property specifies the field in a CA SDM table that is associated with the control. To set this value in the Properties dialog, click the Browse button that displayed at the right side of the selection box when you click on the cell to the right of the Attribute property. Then select the attribute you want from the Select Attribute tree, and click Select.

Autofill

The Autofill feature verifies and completes a lookup field before the form is submitted. If a user clicks the hyperlink and the text specified is sufficient to identify a value in the underlying table, the pop-up search form is suppressed and CA SDM fills in the value. Autofill is also activated when a user types a new value into a lookup field and presses Tab. This causes CA SDM to query the server for a list of all records eligible for the lookup field whose first few characters match the value specified. If it finds a match, it copies the full value into the field. If it finds more than one match, it pops-up a selection form.

The Autofill property is provided for dropdown fields because CA SDM can automatically convert a dropdown field to a lookup when the number of items exceeds a configurable threshold (see the Lookup property).

You can set the Autofill property for a dropdown box, hierarchical lookup box, or a lookup box control in the Properties dialog. To specify this property, select Yes or No from the dropdown displayed when you click in the cell to the right of the Autofill property in the Properties dialog. No means that the control will not be filled in automatically when you tab out of it. True is the default Autofill value.

Caption

The Caption property specifies the header label that is displayed above a control. To specify this property, enter the desired value in the cell to the right of the Caption property in the Properties dialog.

Notebook tabs and list columns also have captions. Specify them on the List Designer that replaces the Properties dialog for these controls.

You can set the caption for a menu item from the Menu Designer dialog. To do so, select the appropriate item from the list and enter text in the Caption text box.

Column Span

CA SDM lays out controls on a detail form or a search filter in a HTML table, with controls in rows and columns of the table. The Column Span property specifies the number of columns occupied within the table by a control. To specify this property, enter the desired numeric value in the cell to the right of the Column Span property in the Properties dialog.

You can set the column span for a control from the Properties dialog. To do so, enter an integer in the cell to the right of the Column Span property.

Condition

The Condition property sets the conditional logic for the Conditional If and Conditional Else If controls. To specify this property, enter the desired logical condition in the cell to the right of the Condition property in the Properties dialog.

The syntax of the Condition property is as follows:

- 0 is false; any other number is true
- "" is false; “any-string” is true
- “value op value” evaluates the left and right values against each other according to op. If both values consist of digits (optionally preceded by - or +), the comparisons are done numerically. Otherwise, they are done lexically (ASCII collation). Valid op values include:

op values	What it means
==	Equal to
!=	Not equal to
>=	Equal to or greater than (must be written as \>= or >=)
<	Less than (must be written as \< or <)
>	Greater than (must be written as \> or >)
<=	Equal to or less than (must be written as \<= or <=)
&	Performs a bit-and of the left and right values. True if any bits are set; false if none are set
%	Returns true if the left value is an even multiple of the right value, and false otherwise (useful for building two-dimensional tables).
:	Performs a byte-oriented pattern match like the UNIX grep command. It returns true if the left value contains the regular expression defined by the right value

For example:

```
<PDM_IF $count \>= 10> . . .
<PDM_ELIF $count &#60; 5>  . . .
<PDM_ELSE> . . .
</PDM_IF>
```

A condition can include connectors, either && (and) or || (or). There is no precedence for either connector. The web engine examines a conditional from left to right until it reaches a connector. If the initial condition is true and the connector is ||, it considers the entire condition to be true without further evaluation. If the initial condition is false and the connector is &&, it considers the entire condition to be false without further evaluation. Otherwise, it considers the condition undermined, and evaluates the conditional from after the connector.

Default

The Default property specifies a default value for a dropdown field when the associated attribute is empty. To specify this property, enter the desired value in the cell to the right of the Default property in the Properties dialog.

When this property is specified and the underlying attribute has an empty value (normally only when a new record is created), the select box has the specified value automatically selected. This property has no effect if the specified value does not correspond to one of the selections in the dropdown, or if the field is displayed as a lookup. This property is optional; if omitted, the default for an empty attribute is to leave it empty.

Disabled

The Disabled property specifies whether the control is disabled on initial display. A disabled control is displayed in gray text, and the user can neither select it nor enter text into it.

To specify this property, select Yes or No from the dropdown displayed when you click in the cell to the right of the Disabled property in the Properties dialog. To specify that a control should be disabled by default, select Yes for this property. A disabled control can be enabled only by JavaScript executed after the form is displayed.

If you specify Yes for the Disabled property in a textbox in the search filter part of a list form, you must also specify values for the QBE Condition, QBE Display, and QBE Value properties.

Event

The Event property specifies one or more event handlers to be associated with the attribute.

To specify this property, enter the desired value in the cell to the right of the Event property in the Properties dialog. Specify the value in the same format, as it would appear on an HTML statement, except that any double quotes in the handler string must be prefixed by three backslashes. For example, the Event property for the Request Area control on form detail_cr.htmpl is:

```
onChange=\\\"change_category_func('cr')\\\"
```

This specifies that the JavaScript function change_category_func(‘cr’) should be invoked whenever the value of the field changes.

You can specify multiple event handlers in the same Event property by separating them with spaces. Consult HTML or JavaScript documentation for a list of available event handlers and their usage.

Factory

The Factory property specifies the CA SDM table referenced by a field in a lookup or dropdown control. This property is normally left blank, meaning that the factory defaults to the one associated with the attribute in the CA SDM object definition. It is seldom necessary or useful to override this value.

To specify this property, select the desired value from the dropdown displayed when you click in the cell to the right of the Factory property in the Properties dialog.

Function

The Function property of a menu item specifies the JavaScript invoked when the user selects the item. To specify this property, enter the desired value in the cell to the right of the Function property in the Menu Designer dialog. Any valid JavaScript can be specified for Function, except that double quotes must be preceded by a backslash.

The following predefined JavaScript functions may be useful:

- **upd_frame(form)**
  Loads a new form into the main window content frame.
- **create_new(factory, use_template, width, height [,args])**
  Pops-up a form to define a new record.
- **popup_window(name, form[, width, height [,features [,args]]])**
  Pops-up a new window.
- **showDetailWithPersid(persid)**
  Pops-up a detail record.

In the above functions:

- **form**
  Is either an HTMPL file name of the form xxx.htmpl or an operation code (example: CREATE_NEW).
- **factory**
  Is the name of a database object.
- **use_template**

Specifies whether or not the new object should be created with a template that the user must select from a list. It can either be true or false.

- **width**
  Is the desired form width in pixels or zero for default.
- **height**
  Is the desired form height in pixels or zero for default.
- **features**
  Is a list of window features, in the same format as used with the JavaScript window.open() function.
- **args**
  Is one or more arguments of the form "keyword=value" for the operation specified for form.
- **persid**
  Is a persistent ID in the form factory:id.

ID

The ID property sets the HTML or JavaScript id of the command button, tab, or read-only text box controls for use in JavaScript that references the control. To specify this property, enter the desired value in the cell to the right of the ID property in the Properties dialog.

Initial

The Initial property specifies the initial value of a dropdown field. When this property is specified, the database value of the attribute associated with the control is ignored and the select box has the specified initial value selected.

To specify this property, enter the desired value in the cell to the right of the Initial property in the Properties dialog. The specified value should correspond to one of the selections in the dropdown; if it does not, the initial value for the dropdown is empty. This argument is optional; if not specified, the dropdown displays the database value of its associated attribute.

Keep Links

The Keep Links property specifies whether or not HTML Action: tags within a text box on the read-only view of a detail form are displayed as hyperlinks or as simple (non-clickable) text.

To specify this property, select Yes or No from the dropdown displayed when you click in the cell to the right of the Keep Links property in the Properties dialog. Specify Yes for this property to specify that Action: tags should be displayed as hyperlinks. Specify the default value of No to specify that Action: tags should be displayed as literal text. HTML tags other than Action: tags are always displayed as literal text (unless the Keep Tags property is specified).

The Keep Links property is ignored if the Keep Tags property is specified as Yes. Note that this property affects only the read-only view of a detail form. HTML is always displayed as literal text in Edit view.

Keep Tags

The Keep Tags property specifies whether or not HTML tags within a text box on the read-only view of a detail form are formatted as HTML or as simple (non-clickable) text.

To specify this property, select Yes or No from the dropdown displayed when you click in the cell to the right of the Keep Tags property in the Properties dialog. Specify Yes for this property to specify that HTML tags should have their normal effect. Specify the default value of No to specify that HTML tags should be displayed as literal text.

A value of Yes for Keep Tags overrides the Keep Links property. Note that this property affects only the read-only view of a detail form. HTML is always displayed as literal text in Edit view.

Link

The Link property specifies whether or not a Dropdown, Lookup, or Read Only Lookup field should be displayed as a clickable hyperlink in the read-only view of a detail form. A user can click on a field displayed as a hyperlink to show the detail of the record referenced by the field.

To specify this property, select Yes or No from the dropdown displayed when you click in the cell to the right of the Autofill property in the Properties dialog. Select the default value of Yes to specify that the field should be displayed as a link on the read-only view; specify No to specify that the field should be displayed as simple non-clickable text on the read-only view.

Lookup

The Lookup property specifies whether or not a Dropdown control is converted into a Lookup control if the number of entries for the Dropdown control is exceeds an installation-specified threshold (normally ten).

To specify this property, select Yes or No from the dropdown displayed when you click in the cell to the right of the Lookup property in the Properties dialog. Select the default value of Yes to specify that the Dropdown control should become a lookup control if the number of entries in the dropdown exceeds the threshold. Select No to specify that the control should always remain a dropdown, regardless of how many entries it contains.

Max Length

The Max Length property specifies the maximum number of characters that can be typed in the Textbox control. This property is normally left blank, meaning that the maximum defaults to the actual length of the database column associated with the textbox.

To specify this property, enter the desired numeric value in the cell to the right of the Max Length property in the Properties dialog.

Off

The Off property sets the value that is displayed on read-only screens when a checkbox value is false. To set this value, enter text in the cell to the right of the Off property.

On

The On property sets the value that is displayed on read-only screens when a checkbox value is true. To set this value, enter text in the cell to the right of the On property.

OP Code

The Op Code property specifies the SQL select operator used with the value of a dropdown on a search filter. The value must be a character string beginning with “QBE.” followed by an operator chosen from:

Operator	Description
EQ	equals
NE	not equals
LT	less than
GT	greater than
LE	less than or equals
GE	greater than or equals
IN	LIKE

The default value of QBE.EQ specifies that the search filter should select rows where the attribute associated with the control exactly matches the value selected from the dropdown.

To specify this property, enter the desired value in the cell to the right of the OP Code property in the Properties dialog.

Option ID

The Options ID sets the HTML or JavaScript id of the option buttons for use in JavaScript that references the control. To specify this property, enter the desired value in the cell to the right of the Options ID property in the Properties dialog.

Preview

The Preview property specifies whether or not to show the Quick View mode in the dtlHTMLEditBox. To specify this property, select Yes or No from the dropdown displayed when you click in the cell to the right of the Preview property in the Properties dialog. Select the default value of Yes to specify that the dtlHTMLEditBox control will have the quick view mode. Select No to specify that the control will have only the HTML source mode.

Preview Clause

The Preview Clause property specifies a where clause to retrieve data for the form. WSP uses this where clause to retrieve data to be displayed on the form when it is previewed. If no preview clause is specified, WSP finds the most recently added row in the table associated with the form, and displays that data on the previewed form.

Preview Mode

The Preview Mode property specifies how the form will be previewed during preview mode. To set the Preview Mode of the form, select edit or read-only from the dropdown list to the right of the Preview Mode property. Edit is the default Preview Mode value. If you are previewing the form, in read-only mode, you will not be able to edit any of the control boxes on the form. If the Preview Mode is set to edit, you can edit the control boxes.

Preview URL

The Preview URL property specifies the preview URL. This can be an HTMPL file name, in the form xxxx.htmpl; a CA SDM URL (used unaltered if it begins with “OP=”); or the keyword “no”, indicating the form cannot be previewed. A value not beginning OP= is modified by replacing a reference of the form {factory} or {factory:} with an id or persistent id (respectively) of the most-recently created row from the referenced factory that you are authorized to see.

QBE Condition

The QBE Condition property specifies the SQL select operator used with the QBE Value property of a disabled textbox field on a search filter. The value must be a character string chosen from:

Operator	Description
EQ	equals
NE	not equals
LT	less than
GT	greater than
LE	less than or equals
GE	greater than or equals
IN	LIKE

For example, a value of IN specifies that the search filter should select rows where the attribute associated with the disabled textbox control matches the QBE Value property according to the rules of an SQL LIKE comparison.

To specify this property, enter the desired value in the cell to the right of the QBE Condition property in the Properties dialog.

QBE Display

The QBE Display property specifies the value displayed to the user for a disabled textbox field on a search filter. To specify this property, enter the desired value in the cell to the right of the QBE Display property in the Properties dialog.

QBE Value

The QBE Value specifies the value used for comparison in the portion of an SQL select where clause generated from a disabled textbox field on a search filter. The generated where clause includes a portion selecting rows where the attribute associated with the control compares with the value specified for QBE Value using the operator specified for QBE Condition.

To specify this property, enter the desired value in the cell to the right of the QBE Value property in the Properties dialog.

Read Only

Like all other detail controls in detail form, when you are in read only mode you are not able to edit the control.

> **NOTE**
> This parameter is ignored when Bound is set to Yes.

Rows

The Rows property specifies the number of rows that a text box control or a HTML editor will occupy on a form. To specify this property, enter the desired numeric value in the cell to the right of the Rows property in the Properties dialog.

> **NOTE**
> This property is available only for text boxes on the detail form and not for text boxes in the list forms.

Size

The Size property specifies the width of the control in pixels. If this property is left blank, the control is displayed at whatever width is required to hold its data.

To specify this property, enter the desired numeric value in the cell to the right of the Size property in the Properties dialog.

Spell Check

The Spell Check property specifies whether or not a Textbox control should have Spelling button to the right of its caption to allow the user to spellcheck the contents of the text box. To specify this property, select Yes or No from the dropdown displayed when you click in the cell to the right of the Spell Check property in the Properties dialog. Select Yes to specify that a Spelling button should be displayed; select the default value of No to specify no button.

Time

The Time property specifies whether or not the time will be displayed along with the date for the Date and Read Only Date controls. To specify this property, select Yes or No from the dropdown displayed when you click in the cell to the right of the Time property in the Properties dialog. Select the default value of Yes to specify that the time should be displayed; select No to specify that only the date should be displayed.

Title

The Title property specifies the title of the element.

To specify this property, enter the desired value in the cell to the right of the Title property in the Properties dialog.

Toolbar

The Toolbar property specifies which toolbar set will be shown in the HTML Editor associated to the dtlHTMLEditBox control.

- **tmpl**
  Displays the toolbar set for document template editor.
- **reportcard**
  Displays the toolbar set for report cards editor.
- **default**
  Displays the default toolbar set.

Tooltip

The Tooltip property specifies the text displayed when the user places the mouse pointer over a control item.

To specify this property, enter the desired value in the cell to the right of the Tooltip property in the Properties dialog.

Where Clause

The Where Clause property specifies a where clause to select the values included in a Dropdown control on a detail form. This property is optional; if left empty, a dropdown on a detail form includes all rows from the table referenced by the attribute associated with the control.

To specify this property, enter the desired value in the cell to the right of the Where Clause property in the Properties dialog.

Wildcard

The Wildcard feature enables a wildcard search for Lookup and Hierarchical Lookup boxes on a List form. A wildcard search is equivalent to adding the "%" symbol around a text string when performing a search. For example, using the string %option% when performing a search on a text field, would retrieve all of the records that contain the word "option" in that text field. To specify this property, select Yes or No from the dropdown displayed when you click in the cell to the right of the Wildcard property in the Properties dialog. Yes means that the control will automatically have the wildcard property enabled. No is the default Wildcard value. If this option is true, the % symbols are no longer required on searches.

Width

The Width property specifies the width of a button control in pixels. If this property is left blank, the control is displayed at whatever width is required to hold its data.

To specify this property, enter the desired numeric value in the cell to the right of the Size property in the Properties dialog.

Menu Designer

Use the Menu Designer dialog in WSP to add a menu bar, menus, submenus, and menu commands to the open form.

Add a Menu Bar

You can add a menu bar to a form.

Follow these steps:

1. Select File, New.
2. On the New Form window, select menubar.template as the File Name.
3. On the menubar.htmpl window, double click to open the Menu Editor dialog. The list box at the top left area of the dialog shows the structure of the menu as you define it.
4. To add the first item, enter its name in the Caption field. To add subsequent items, click Add before entering the caption.

List Designer

This article contains the following topics:

List forms (HTML templates with names of the form **list_xxx.htmpl**) are used to search the database and display search results. They consist of a search filter at the top of the form, where the user specifies search criteria, and a results list at the bottom of the form, listing records retrieved by the search. Use the drag-and-drop method to design the results list.

Add a Results List

To add the results list control to the List form, use drag-and-drop on the List button to place it on the desired area. This action places a blank frame on the form. Double-click the frame to define the columns and headers.

Modify Results List

To modify the Results List, double-click it and modify the properties. Click **OK** after making the required changes.

How to Disable Tenant Column Sorting

By default, you can sort the Tenant column on a list form. To disable Tenant column sorting, add the following value to the appropriate list form:

```
tenantSort=no
```

> **NOTE**
> If a form includes a Tenant column, and you do not want it sorted, you can add the following line to the list form:

```
var tenantSort = "no"
```

The Tenant column is disabled on the list form.

Disable the Export Button on a List Form

You can disable the Export button that appears on the list form pages in CA SDM.

Follow these steps:

1. Open a list form, such as list_cr.htmpl (Request List).
2. Double-click the results list.
   The properties dialog appears, showing the existing list definition.
3. On the Source tab, locate the sfStart macro.
4. Change the value to export=no as follows:
   ```
   <pdm_macro name=sfStart factory=nr export=no>
   ```
   The Export button is disabled on the list form.

How to Export Fields Not Included on List Forms

You can include a new field to export but not list.

Follow these steps:

1. Open the Properties dialog and click Add.
2. Select an attribute from the Attribute list, for example, Urgency.
3. Select the Export option.
4. On the Source tab, replace NAME=lsCol with NAME=lsExport as follows:
   ```
   <PDM_MACRO NAME=lsExport hdr="Urgency" attr=urgency justify=left>:
   ```
5. Click OK.
   When a user clicks the Export button on a list form, the Urgency field is exported.

Export the Relative Attribute Value

By default, the common name value of an SREL column is exported. You can export the Relative Attribute value by modifying the lsCol macro.

Follow these steps:

1. Open a list form for editing, for example, list_cr.htmpl.
   The list form opens.
2. Click the Source tab and locate the lsCol macro as follows:
   ```
   <PDM_MACRO NAME=lsCol hdr="Priority/Parent" attr=priority sort="DESC">
   ```
3. Copy the code string to the next line, and do the following:
   - Change the lsCol value to lsExport.
   - Add the common_name= no option.
4. Click OK.
   When a user clicks the Export button on the list form, the Relative Attribute value exports.

PDM Macro Definitions

This article contains the following topics:

PDM_Macro

CA SDM builds web forms in a language named HTMPL. HTMPL extends standard HTML with references to server variables and a number of proprietary tags of the form PDM_xxx. One of these tags, PDM_MACRO, copies a named JavaScript code segment from the database into the form. Most PDM_MACRO tags invoke client-side JavaScript that builds the web forms just in time, immediately before they are presented to the user. Most web form controls are built with the PDM_MACRO tag. For example, the PDM_MACRO tag that builds the site field on the Location Detail form is coded as follows:

```
<PDM_MACRO NAME=dtlLookup hdr="Site" attr="site">
```

A PDM_MACRO tag contains one or more keyword parameters (properties). The NAME parameter is required on every PDM_MACRO to specify the name of the macro. Other parameters are required or optional depending on the macro.

> **NOTE**
> Site customization of PDM_MACRO definitions stored in the database is not supported and should not be attempted.

btnEndRow

The btnEndRow macro marks the end of a group of one or more buttons displayed in a horizontal row on a form. The button row starts with either a btnStartRow macro or with a button macro with the NewRow=yes argument. The button row ends with either a btnEndRow macro or with a button macro with the EndRow=yes property. For example:

```
<PDM_MACRO name=btnStartRow . . .>
<PDM_MACRO name=button . . .>
<PDM_MACRO name=button . . .>
. . .
<PDM_MACRO name=btnEndRow>
```

This macro has no properties.

btnStartRow

The btnStartRow macro marks the start of a group of one or more buttons displayed in a horizontal row on a form. The button row starts with either a btnStartRow macro or with a button macro with the NewRow=yes argument. The button row ends with either a btnEndRow macro or with a button macro with the EndRow=yes property. For example:

```
<PDM_MACRO name=btnStartRow . . .>
<PDM_MACRO name=button . . .>
<PDM_MACRO name=button . . .>
. . .
<PDM_MACRO name=btnEndRow>
:
```

This macro has the following properties:

- **centered=true|*false***
  Specifies whether the row of buttons is centered on the form. When you do not specify this property, the row of buttons is left-justified.
- **padding=*0*|*number***
  Specifies the number of hard (nonbreaking) spaces between buttons in the row. When you do not specify this property, buttons are placed as close together as possible.

button

The button macro defines a button that a user of a form can click to invoke an action.

This macro has the following properties:

- **btnType=negative|*positive***
  Specifies whether a button performs a positive action (such as Save) or a negative action (such as Cancel). This specification is for documentation only; specifying a button as positive or negative has no effect on the appearance or use of the button.
- **Caption=*text***
  (Required) Specifies the text for the label of the button. You can optionally end the caption with a hotkey hint enclosed in square brackets, for example, Caption=”Save[Sv]”. When you provide a hotkey hint, CA SDM selects the hotkey for the button from the characters in the hint. When you do not provide a hotkey hint, CA SDM selects the hotkey from the characters of the entire caption.
- **Disabled=Yes|*No***
  Specifies whether the button is disabled on the initial display. When you do not specify this property, the button is enabled.
- **EndRow=yes|*no***

Specifies whether this button is the last of a horizontal row of buttons. If specified as EndRow=yes, CA SDM automatically inserts a btnEndRow macro after this button. When you do not specify this property, CA SDM does not configure the button as the last one in a row.

- **Func=*string***
  (Required) Specifies the JavaScript that is invoked when the user of the form clicks the button.
- **hotkey_name=*string***
  (Required) Specifies the caption for selecting a hotkey. You must specify this property and an ASCII string. CA SDM selects the hotkey for the button from the characters in the hotkey_name string. The hotkey_name macro is ignored in locales using the Latin alphabet; these locales always use the Caption to determine a hotkey.
- **ID=*string***
  (Required) Specifies a JavaScript identifier for the button.
- **NewRow=yes|*no***
  Specifies whether this button is the first of a horizontal row of buttons. When you specify NewRow=yes, the product automatically inserts a btnStartRow macro before this button. When you do not specify this property, the product does not configure the button as the first one in a row.
- **tabIndex=n|*-1***
  Specifies the HTML tabIndex for the button. TabIndex is meaningful only in relation to other HTML elements on the form with a tabIndex. A user of the form tabs from element to element in sequence by the tabIndex values of the element. When you do not specify this property, the button receives a default tabIndex.
- **Tooltip=*string***
  Specifies the tooltip text for the button. When you do not specify this property, the property defaults to the button caption.
- **Width=*0*|*number***
  Specifies the width of the button in pixels. If you omit this property or specify it as 0, the product creates the button exactly wide enough for its caption.

cmdbMetadata

The cmdbMetadata macro provides attribute-level metadata for the versioning, TWA viewer, and CMDBf viewers for use when displaying CI and transaction information. This metadata is used for the following purposes:

- To categorize and provide help information about each attribute.
- To map each MDR provider attribute name with its corresponding CA CMDB attribute name so they are displayed together. This macro does not create a UI control and typically, use it in an HTMPL file. The HTMPL file is then included in a display form, which the versioning, TWA viewer, and CMDBf viewers use.

This macro has the following properties:

- **attr=*attributeName***
  Specifies a CA CMDB attribute name. This property is required in all cases except hiding a provider attribute from the CI displays. All subsequent properties refer to the handling of this attribute. The attribute name specified must be a valid, case sensitive, CI attribute name.
- **category=*attributeCategoryName***
  Specifies the category of this attribute to group-related attributes on the versioning and CMDBf viewer attribute displays. The category is typically the tab that contains the attribute on the CI detail display. Attributes are sorted by category on those forms.
- **common=yes|*no***
  (Deprecated) Specifies whether this attribute is a common attribute (in the nr object). Do not use this property. Do not specify an extension for the metadata for common attributes.
- **currentcivalue=*dottedNotation***
  Explicitly specifies special dotted notation for the current value of the attribute. This property is required when the default dotted notation to access the current CI value is incorrect. For example, this property is required when defining common attributes, and those attributes are not in the extension table.

NOTE
For dotted notation examples, see the cmdb_metadata_common.htmpl form.

- **dbcolumn=*UAPMcolumnName***
  Specifies a UAPM database column name so that it can be associated with the specified CA CMDB attribute name.
- **extension=*objectName***
  Specifies the extension object name that contains the attribute name. This property is required when defining attributes that are in an extension table.
- **heading=*string***
  Specifies a short label that describes this attribute. Typically, you set this property the default label that the CI detail form displays.
- **help=*string***
  Specifies a brief description of the attribute.
- **hide_provider_attr=yes|*no***
  (For use with the CMDBf viewer only) Hides the provider attribute from the CMDBf viewer display when you specify YES. Requires the specification of provider_attr and either provider_name or provider_name_regexp.
- **provider_attr=*MDRattributeName***
  (For use with the CMDBf viewer only) Specifies the MDR provider attribute name that maps to the current attribute (attr). Requires the specification of attr, and either provider_name or provider_name_regexp.
- **provider_name=*MDRname***
  (For use with the CMDBf viewer only.) Specifies the MDR provider name that is associated with the MDR attribute name (provider_attr). This property specifies an exact match with a particular MDR provider name. Requires the specification of provider_attr. This property is mutually exclusive with provider_name_regexp.
- **provider_name_regexp=*regularExpression***
  (For use with the CMDBf viewer only.) Specifies the MDR provider name regular expression that is associated with the MDR attribute name (provider_attr). This property specifies a regular expression pattern to match with the MDR provider names. Regular expressions use a valid JavaScript RegExp() pattern. Requires the specification of provider_attr. For example, \"MyMdr.*\" matches all provider names starting with \"MyMdr\" such as \"MyMdr1\", \"MyMdrAbcd\", and so on. The characters \".*\" indicate to match all provider names. This property is mutually exclusive with provider_name.
- **standardcivalue=*dottedNotation***
  Explicitly specifies special dotted notation for the corresponding value of the attribute for the standard CI. This property is required when the default dotted notation to access the standard CI value is incorrect. For example, this property is required when defining common attributes, and those attributes are not in the extension table.

dtlCheckbox

The dtlCheckbox macro specifies a check box control on an HTMPL detail form. The control appears as follows:

- A check box on the edit view of the form.
- The text specified by the on or off properties on the read-only view.

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **evt=*”eventName=’script’”***
  Specifies one or more HTML event handlers in the same way you specify them in an HTML statement, with quotes escaped as required. For example:
  ```
  evt="onclick='myfunc()' onchange=\\\"myfunc2()\\\""
  ```
- **hdr=*text***

Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.

- **off=*0*|*text***
  Specifies the text displayed on a read-only form when the check box is not selected.
- **on=*1*|*text***
  Specifies the text displayed on a read-only form when the check box is selected.
- **title=*text***
  Specifies the title for screen reader users.

dtlCheckboxReadonly

The dtlCheckboxReadonly macro specifies a read-only check box control on an HTMPL detail form. The control appears as the text specified by the on or off properties on both the edit and read-only views of the form.

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **off=*0*|*text***
  Specifies the text displayed on a read-only form when the check box is not selected.
- **on=*1*|*text***
  Specifies the text displayed on a read-only form when the check box is selected.

dtlCheckboxWithDesc

The dtlCheckboxWithDesc macro defines a check box control with a description.

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **code=*string***
  Specifies the internal value of the check box. This property is not intended for customer use.
- **desc=*text***
  Specifies the text of the description appearing after the check box.
- **evt=*”eventName=’script’”***
  Specifies one or more HTML event handlers in the same way you specify them in an HTML statement, with quotes escaped as required. For example:
  ```
  evt="onclick='myfunc()' onchange=\\\"myfunc2()\\\""
  ```
- **func=*script***
  Provides the function to select or clear the check box. This property is not intended for customer use.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **title=*text***
  Specifies the title for screen reader users.
- **padding=*0*|*number***

Specifies the number of spaces before the check box.

- **sameCol=true|*false***
  Specifies that the check box and description are in one column. By default, the check box and description are in two columns.
- **title=*text***
  Specifies the title for screen reader users.
- **value**
  Specifies the value of the attribute in the read-only view.

dtlDate

The dtlDate macro specifies a date control on an HTMPL detail form. A date control contains a date and time that can be edited with a date picker pop-up in the edit view of a detail form.

This macro has the following properties:

- **attr**
  Specifies the name of the attribute.
- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **make_required=YES|*NO***
  Makes a field required when you specify YES regardless of whether the associated attribute is required at the object level. A required field forces the user to specify a nonblank value.
- **size=*20*|*number***
  Specifies the width of the input field.
- **time=*yes*|no**
  Specifies whether the field contains both a date and time, or only a date.

dtlDateDropdown

The dtlDateDropdown macro specifies a date control on an HTMPL detail form. When empty, this field displays as a drop-down control on the edit view of the form. The drop-down control includes date selections such as "In One Day" or "In One Week". This control displays date fields with a nonempty value in the same way as the dtlDate control.

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **codename=*string***
  Specifies the name of a set of values from the ui_selection object that are displayed in the date drop-down control. The values are those values with code attributes that match the value specified for codename.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **evt=*”eventName=’script’”***
  Specifies one or more HTML event handlers in the same way you specify them in an HTML statement, with quotes escaped as required. For example:

```
evt="onclick='myfunc()' onchange=\\\"myfunc2()\\\""
```

- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **make_required=YES|*NO***
  Makes a field required when you specify YES regardless of whether the associated attribute is required at the object level. A required field forces the user to specify a nonblank value.
- **size=*20*|*number***
  Specifies the width of the input field.

dtlDateReadonly

The dtlDateReadonly macro specifies a noneditable date control on an HTMPL detail form.

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **time=*YES*|NO**
  Specifies whether the field contains both a date and time, or only a date.

dtlDropdown

The dtlDropdown macro specifies a drop-down selection control on an HTMPL detail form. The control appears as a drop-down or lookup on the edit view of the form, and as the text in the read-only view.

> **NOTE**
> The CA SDM administrator can specify the maximum number of entries a drop-down list should contain (default 10); when the size of a drop-down list exceeds this number, CA SDM automatically converts the list to a lookup. You can override this behavior with the lookup property of this macro.

This macro has the following properties:

- **attr*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **autofill=*yes*|no**
  Specifies whether the field allows autofill when it is displayed as a lookup. Autofill lets a user enter a value in the field by typing the first few characters of a value and pressing Tab. These actions cause the product to perform one of the following actions:
  – Request the full value for the field from the server.
  – Pop up a selection form when the specified value is missing or ambiguous.
- **cbwidth=*0*|*number***
  Specifies the width of the drop-down list in pixels. If omitted or specified as zero, the list automatically sizes to the width of its widest entry.
- **codename=*string***
  Specifies the name of a set of values from the ui_selection object that are displayed in the date drop-down control. The values are those values with code attributes that match the value specified for codename.

- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **default=*text***
  Specifies the default value when the attribute is null.
- **evt=*”eventName=’script’”***
  Specifies one or more HTML event handlers in the same way you specify them in an HTML statement, with quotes escaped as required. For example:
  ```
  evt="onclick='myfunc()' onchange=\\\"myfunc2()\\\""
  ```
- **extraURL=*string***
  Specifies a where clause predicate in URL format to restrict the contents of the control. For example, extraURL with "ADDITIONAL_WHERE" is displayed as:
  ```
  PDM_MACRO name=dtlLookup hdr="My own Field" attr="zField" extraURL="ADDITIONAL_WHERE=delete_flag=0"
  ```
- **factory=*name***
  Specifies the name of a Majic factory for the selection list. Defaults to the factory referenced by the associated attributes, and so is not typically required.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **title=*text***
  Specifies the title for screen reader users.
- **initial=*value***
  Specifies the initial selection in the drop-down list, overriding the value of the attribute in the database.
- **link=*yes*|no**
  Specifies whether the control on the read-only view is a link to detail for the value of the attribute.
- **list_display=*attributeName***
  Specifies the common attribute name of the table. Effective only if use_list_display=1, codename is blank, and the factory, rel_attr_name, and whereclause properties are supplied.
- **list_orderby=*attributeName***
  Specifies the order of the drop-down list. Effective only if use_list_display=1, codename is blank, and the factory, rel_attr_name, and whereclause properties are supplied.
- **lookup=*yes*|no**
  Specifies whether the field should automatically convert to a lookup if the number of entries in the drop-down list exceeds a configured value (typically 10).
- **make_required=YES|*NO***
  Makes a field required when you specify YES regardless of whether the associated attribute is required at the object level. A required field forces the user to specify a nonblank value.
- **rel_attr_name=*attributeName***
  Specifies the attribute from the referenced table that is stored. This property is required when use_list_display=1 is specified, codename is blank, and the factory, list_display, and whereclause properties are supplied.
- **size=*20*|*number***
  Specifies the width of the input field.
- **title=*text***
  Specifies the title for screen reader users.
- **use_list_display=*0*|1**
  Specifies whether the drop-down control should be built from an explicit query instead of from the referenced attribute. When use_list_display=1, the factory, list_display, rel_attr_name, and whereclause properties must also be supplied.
- **whereclause=*string***

Specifies a where clause.

dtlDropdownWithDesc

The dtlDropdownWithDesc macro specifies a drop-down list on an HTMPL detail form. A caption (header) precedes the drop-down list with a text description following it.

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **desc=*text***
  Specifies the description that appears after the drop-down list.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **sel_fac=*factoryName***
  Specifies a factory whose entire contents will display in the drop-down list. This property is effective only if sel_list is empty.
- **sel_list="text@,@value@,…,@text@,@value"**
  Specifies a comma-separated list of values for the drop-down. Values are specified in pairs, with the displayed value followed by the corresponding internal value. At symbols (@) delimit the individual values, except for the first and last value in the list.
- **title=*text***
  Specifies the title for screen reader users.
- **value=*string***
  Specifies the value of the attribute in the read-only view.

dtlEnd

The dtlEnd macro marks the end of the macros that define the contents of a detail form. This macro must be the last macro on the form. You structure all detail forms with the following code:

```
<PDM_MACRO name=dtlForm . . .>
<PDM_MACRO name=dtlStart . . .>
<PDM_MACRO name=dtlStartRow . . .>
. . .
<PDM_MACRO name=dtlEnd>
```

This macro has no properties.

dtlEndDiv

The dtlEndDiv macro marks the end of a group of detail form macros that JavaScript can show or hide. Pair this macro with a dtlStartDiv macro and use a dtlEndTable macro. For example:

```
<PDM_MACRO name=dtlStartDiv divid="my_div" . . .>
. . .
<PDM_MACRO name=dtlEndTable>
<PDM_MACRO name=dtlEndDiv>
```

This macro has no properties.

dtlEndTable

The dtlEndTable macro marks the end of an HTML table, and is not intended for customer use. No dtlStartTable macro exists; the detail form table automatically starts and ends HTML tables as required for the layout. A dtlEndTable macro is required only when a macro group must be separated into HTML blocks, such as when dtlStartDiv and dtlEndDiv create a division.

dtlEndTable has no properties.

dtlForm

The dtlForm macro marks the beginning of the macros that define the contents of a detail form. This macro must be the first macro on the form. All detail forms must be structured with the following code:

```
<PDM_MACRO name=dtlForm . . .>
<PDM_MACRO name=dtlStart . . .>
<PDM_MACRO name=dtlStartRow . . .>
. . .
<PDM_MACRO name=dtlEnd>
```

This macro has the following properties:

- **button=*true*|false**
  Specifies whether to show the default buttons on the edit view and the Edit button on the read-only view. The default buttons are Save, Cancel, and Reset.
- **factory=*name***
  Specifies the name of a Majic factory for the selection list. Defaults to the factory referenced by the associated attributes, and so is not typically required.
- **hideeditbtn=true|*false***
  Specifies whether to hide the Edit button in the read-only view.
- **onsubmit=*script***
  Overrides the default form submit handler. The default handler is sufficient for most cases.
- **saveclose=true|*false***
  Specifies whether to show a "Save and Close" button in the edit view.
- **skip_tenant_hdr=yes|*no***
  Specifies whether to suppress the tenant field at the header of the form.
- **skiphdr=yes|*no***
  Specifies whether to skip standard form headers, such as default buttons.

dtlHier

The dtlHier macro specifies a hierarchical lookup control on a detail form. In the edit view, a user can click the caption (header) of the control to pop up a hierarchical selection form.

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **autofill=*yes*|no**
  Specifies whether the field allows autofill when it is displayed as a lookup. Autofill lets a user enter a value in the field by typing the first few characters of a value and pressing Tab. These actions cause the product to perform one of the following actions:
  - Request the full value for the field from the server.
  - Pop up a selection form when the specified value is missing or ambiguous.
- **colspan=*1*|*number***

Specifies the number of columns on the form.

- **common_name=*attributeName***
  Specifies the name of the attribute from the referenced table that should be displayed on the detail form. This macro is not intended for customer use.
- **evt=*”eventName=’script’”***
  Specifies one or more HTML event handlers in the same way you specify them in an HTML statement, with quotes escaped as required. For example:
  ```
  evt="onclick='myfunc()' onchange=\\\"myfunc2()\\\""
  ```
- **factory=*name***
  Specifies the name of a Majic factory for the selection list. Defaults to the factory referenced by the associated attributes, and so is not typically required.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **make_required=YES|*NO***
  Makes a field required when you specify YES regardless of whether the associated attribute is required at the object level. A required field forces the user to specify a nonblank value.
- **size=*20*|*number***
  Specifies the width of the input field.

dtlHTMLEditbox

The dtlHTMLEditbox macro defines an HTML edit field in a detail form.

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **bound=*yes*|no**
  Specifies whether the control is bound to an attribute.
- **className=*value***
  Specifies a style sheet class.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **escape=*JS2*|C|JS|HTML**
  Specifies how data is escaped in the read-only view. This property is not intended for customer use.
- **evt=*”eventName=’script’”***
  Specifies one or more HTML event handlers in the same way you specify them in an HTML statement, with quotes escaped as required. For example:
  ```
  evt="onclick='myfunc()' onchange=\\\"myfunc2()\\\""
  ```
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **make_required=YES|*NO***
  Makes a field required when you specify YES regardless of whether the associated attribute is required at the object level. A required field forces the user to specify a nonblank value.
- **maxlength=*number***
  Specifies the maximum length of the edit field.
- **optionid=*string***

Specifies the JavaScript identification of the HTML edit field control. This property is not intended for customer use.

- **persid=*value***
  Specifies the persistent identifier of the object displayed in the edit field. This property is not intended for customer use.
- **preview=*yes*|no**
  Specifies whether the edit field supports Quick View mode.
- **readonly=yes|*no***
  Specifies whether the HTML editor is read-only. This property is ignored unless bound=no.
- **rows=1|number**
  Specifies the number of rows on the form occupied by the textbox.
- **size=*20*|*number***
  Specifies the width of the input field.
- **spellchk=yes|*no***
  Specifies whether to display a Spell button next to the field label in the edit view.
- **tenant=*value***
  Specifies the internal identification (UUID) of the tenant. This property is not intended for customer use.
- **tenantName=*value***
  Specifies the name of the tenant. This property is not intended for customer use.
- **toolbar=*default*|reportcard|tmpl**
  Specifies the set of HTML editing tools displayed in the edit field. This property has the following possible values:
  – default -- the default toolbar set
  – reportcard -- a toolbar set appropriate for the knowledge report card
  – tmpl -- a toolbar set appropriate for the document template editor

dtlLookup

The dtlLookup macro specifies a lookup control on a detail form. In the edit view, a user can click the caption (header) of the control to pop up a selection form.

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **autofill=*yes*|no**
  Specifies whether the field allows autofill when it is displayed as a lookup. Autofill lets a user enter a value in the field by typing the first few characters of a value and pressing Tab. These actions cause the product to perform one of the following actions:
  – Request the full value for the field from the server.
  – Pop up a selection form when the specified value is missing or ambiguous.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **common_name_attr=*attributeName***
  Specifies the name of the attribute from the referenced table to display on the detail form. This property is not intended for customer use.
- **evt=*"eventName='script'"***
  Specifies one or more HTML event handlers in the same way you specify them in an HTML statement, with quotes escaped as required. For example:

```
evt="onclick='myfunc()' onchange=\\\"myfunc2()\\\""
```

- **extraURL=*string***
  Specifies a where clause predicate in URL format to restrict the contents of the control.
- **factory=*name***

Specifies the name of a Majic factory for the selection list. Defaults to the factory referenced by the associated attributes, and so is not typically required.

- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **link=*yes*|no**
  Specifies whether the control on the read-only view is a link to detail for the value of the attribute.
- **make_required=YES|*NO***
  Makes a field required when you specify YES regardless of whether the associated attribute is required at the object level. A required field forces the user to specify a nonblank value.
- **size=*20*|*number***
  Specifies the width of the input field.

dtlLookupReadonly

The dtlLookupReadonly macro specifies a read-only lookup control on an HTMPL detail form.

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **link=*yes*|no**
  Specifies whether the control on the read-only view is a link to detail for the value of the attribute.

dtlLrelMultiselbox

The dtlLrelMultiselbox macro displays a multiselect field (side-by-side selection) control for a many-to-many selection form. This macro displays a set of input controls, including the following:

- A left-side multiselect field and its header
- A right-side multiselect field and its header
- Two select buttons between the multiselect fields
- Pagination text below the multiselect fields if any
- Two Clear Selection buttons at the bottom of the multiselect fields

This macro has the following properties:

- **colspan=*1*|*number***
  Specifies the number of columns on the form.

dtlRadio

The dtlRadio macro displays a group of option button controls with optional text following the control.

This macro has the following properties:

- **actcode=*string***

Specifies the action code.

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **codename=*string***
  (Required) Specifies the name of a set of values from the ui_selection object that are displayed in the date drop-down list control. These values have code attributes that match the value specified for codename.
- **dataclass=*string***
  Specifies the CSS class name used for the text of the option button.
- **evt=*"eventName='script'"***
  Specifies one or more HTML event handlers in the same way you specify them in an HTML statement, with quotes escaped as required. For example:
  ```
  evt="onclick='myfunc()' onchange=\\\"myfunc2()\\\""
  ```
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **title=*text***
  Specifies the title for screen reader users.
- **sameRow=true|*false***
  Specifies whether to display the option buttons on the same row.
- **title=*text***
  Specifies the title for screen reader users.

Example: dtlRadio

NX_ROOT\bopcfg\www\htmpl\web\analyst\detaikl_pri_cal.htmpl file has the following use of the macro:

<PDM_MACRO name=dtlRadio hdr="Duplicate Ticket Action" attr="action" actcode="$args.action" codename="dupActions">

dtlReadonly

The dtlReadonly macro specifies a read-only textbox on an HTMPL detail form.

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **fmtfunc=*name***
  Specifies the name of a JavaScript function that formats the field for display. The function is passed a single argument containing the value of the attribute, and must return a string that is displayed on the form.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **hdrclass=*text***
  Specifies the CSS class used to format the caption. This property is not intended for customer use.
- **title=*text***
  Specifies the title for screen reader users.
- **inline=yes|*no***

Specifies whether the header text and the data are on the same line, separated by a colon (:). Use this property only when attr="n/a".

- **value=*value***
  Specifies the value displayed on the form when attr="n/a".

dtlShowtext

The dtlShowtext macro displays plain text on the form without any label or header that is associated with the text. This macro lets you do the following activities:

- Associate the text to display with an attribute.
- Format and display the text when needed.
- Show a bar image below the text.

This macro has the following properties:

- **argumts=*string***
  Specifies the arguments that are passed to the fmtmsg format function.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **dataclass=pageHeader|hdr|required|alertmsg|className**
  Specifies the CSS class that is applied to displayed text.
- **fmtmsg=*string***
  Specifies the name of the function to format the text for display. The arguments for this function are specified in the argumts parameter. The function must return a text string that is the value displayed.
- **keeplinks=yes|*no***
  Specifies whether HTML links (Action: tags) are displayed as links or formatted as raw text. Preserves HTML links when the text to display contains HTML links. The default value is no.
- **keeptags=yes|*no***
  Specifies whether HTML tags are interpreted as HTML links or formatted as raw text. keeptags=yes overrides the keeplinks property.
- **showbarimg=yes|*no***
  Specifies whether to display a bar image below the text.
- **value=*text***
  (Required) Specifies the value to display. The value can be associated to an attribute.

Example: dtlShowtext

NX_ROOT\bopcfg\www\htmpl\web\analyst\detaikl_pri_cal.htmpl file uses the following macro:

<PDM_MACRO name=dtlShowtext colspan=6 dataclass=hdr value="Priority Calculation Options">

- **Colspan**
  Specifies the number of field columns we want to span.
- **Dataclass**
  Specifies the css we want to apply for the text.
- **Value**
  Specifies the displayed text value.

dtlStart

The dtlStart macro begins a detail form, and must be the second macro on the form. Structure all detail forms with the following code:

```
<PDM_MACRO name=dtlForm . . .>
```

```
<PDM_MACRO name=dtlStart . . .>
<PDM_MACRO name=dtlStartRow . . .>
. . .
<PDM_MACRO name=dtlEnd>
```

This macro has the following properties:

- **center=true|*false***
  Specifies whether the form is centered.
- **scroll=true|*false***
  Specifies whether the form always has a scroll bar.

dtlStartDiv

The dtlStartDiv macro marks the beginning of a group of detail form macros that JavaScript can show or hide. Pair this macro with a dtlEndDiv macro. For example:

```
<PDM_MACRO name=dtlStartDiv divid="my_div". . .>
. . .
<PDM_MACRO name=dtlEndTable>
<PDM_MACRO name=dtlEndDiv>
```

This macro has the following properties:

- **align=*left*|center|right**
  Specifies the alignment of the DIV.
- **class=*className***
  Specifies the CSS class of the DIV.
- **divid=*string***
  (Required) Specifies the JavaScript identification of the DIV.
- **style=*string***
  Specifies the style of the DIV.

dtlStartExpRow

The dtlStartExpRow macro specifies the start of an expandable row on a detail form. An expandable row lets the user hide the row controls by clicking the title bar of the row.

This macro has the following properties:

- **class=*className***
  Specifies the CSS class of the HTML table containing the controls in the new row.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **exp_rows="1,2,…"**
  Specifies the row numbers in the expandable section as a quoted string containing a sequence of numbers from 1 to *N*. *N* represents the number of rows desired in the expandable section. The section includes the controls following dtlStartExpRow, and the next n-1 dtlStartRow macros.
- **form_name=*name***
  Specifies the name of the containing HTMPL file name without the htmpl extension. The name value is the name of the file. When the expandable section is in a tab, this argument specifies the name of the file containing the tab.
- **label=*string***
  Specifies the label shown on the expandable section bar.

dtlStartRow

The dtlStartRow macro marks the start of a row on a detail form.

This macro has the following properties:

- **class=*className***
  Specifies the CSS class of the HTML table containing the controls in the new row.
- **hrClass=*className***
  Specifies the CSS class of an optional dividing line. This property is effective only when hrSpan specifies a nonzero value.
- **hrSpan=*0*|*number***
  Specifies the number of columns spanned by an optional dividing line above the new row. The default value of zero specifies no dividing line.

dtlSurvey

The dtlSurvey macro displays multiple choice answers for each survey question. Survey questions can accept single answers for which option buttons are displayed or can accept multiple answers for which check boxes are displayed. You can use this macro in survey forms like preview risk survey, submit risk survey, and view submitted survey.

This macro has the following properties:

- **answerselected=*number***
  Specifies the index of the selected answer.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **factory=*name***
  Specifies the factory containing the survey, usually risk_svy_atpl.
- **parentid=*number***
  Specifies the database id of the question, usually $question.id.
- **parentmultiflag=*number***
  Specifies whether to display a check box or a option button. This value is usually $question.mult_resp_flag.
- **parentsequence=*number***
  Specifies the question sequence. This value is usually $question.sequence.
- **view=*preview*|doview|viewsubmitted**
  Specifies the survey view to display. You can specify any of the following views:
  - preview -- specifies to display preview survey page
  - doview -- specifies to display survey page which is ready to be submitted
  - viewsubmitted -- specifies to display the previously submitted survey page

dtlTextbox

The dtlTextbox macro specifies a text field on a detail form. In the edit view, the field allows entry of free-form text.

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **disp_entities=yes|*no***

Specifies whether to display HTML entities (for example, & or &#26237) in read-only view. If disp_entities=no, HTML entities are displayed exactly as entered; if disp_entities=yes, HTML entities are converted to their external value.

- **evt=*"eventName='script'"***
  Specifies one or more HTML event handlers in the same way you specify them in an HTML statement, with quotes escaped as required. For example:
  ```
  evt="onclick='myfunc()' onchange=\\\"myfunc2()\\\""
  ```
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **title=*text***
  Specifies the title for screen reader users.
- **JSButton=function()**
  Specifies a JavaScript function that creates a button next to the header of the control.
- **keeplinks=yes|*no***
  Specifies whether HTML links (Action: tags) are displayed as links or formatted as raw text. Preserves HTML links when the text to display contains HTML links. The default value is no.
- **keeptags=yes|*no***
  Specifies whether HTML tags are interpreted as HTML links or formatted as raw text. keeptags=yes overrides the keeplinks property.
- **make_required=YES|*NO***
  Makes a field required when you specify YES regardless of whether the associated attribute is required at the object level. A required field forces the user to specify a nonblank value.
- **maxlength=*number***
  Specifies the maximum length of the edit field.
- **password=yes|*no***
  Specifies whether the text field contains a password field. CA SDM replaces passwords with a string of asterisks in both the edit and read-only views.
- **rows=1|number**
  Specifies the number of rows on the form occupied by the textbox.
- **size=*20*|*number***
  Specifies the width of the input field.
- **spellchk=yes|*no***
  Specifies whether to display a Spell button next to the field label in the edit view.
- **srchknow=yes|*no***
  Specifies whether to display a search knowledge button next to the field label.
- **title=*text***
  Specifies the title for screen reader users.
- **value=*text***
  Specifies the value to display when the value cannot be retrieved from the attribute.

dtlWriteproperty

The dtlWriteproperty macro specifies a custom property for a change or issue category, or a request, incident, or problem area. Properties appear on the properties tab of the respective tickets. This macro formats a single property with the label on the left, input field value in the middle, and a sample value on the right. The Input field can be a textbox, drop-down list, or a check box.

This macro has the following properties:

- **label=prop*n*.label**
  Specifies a label for the property, in the form propn.label. n is the property number specified in the propNum argument.
- **make_required=YES|*NO***
  Makes a field required when you specify YES regardless of whether the associated attribute is required at the object level. A required field forces the user to specify a nonblank value.
- **propNum**
  (Required) Specifies a number for the property. Properties are displayed in sequence by property number.
- **sample=prop*n*.sample**
  Specifies an example of a value for the property, in the form propn.sample. n is the property number specified in the propNum argument.
- **validation_rule=prop*n*.validation_rule.id**
  Specifies the validation rule for the property, in the form propn.validation_rule.id. n is the property number specified in the propNum argument.
- **validation_type=prop*n*.validation_type**
  Specifies the validation type for the property, in the form propn.validation_type. n is the property number specified in the propNum argument.
- **value=prop*n*.sample**
  Specifies the value of the property. n is the property number specified in the propNum argument.

ebr_search_filter

The ebr_search_filter macro specifies an Advanced filter for a related knowledge search. This macro is not intended for customer use.

This macro has the following properties:

- **factory=*name***
  Specifies the name of a Majic factory for the selection list. Defaults to the factory referenced by the associated attributes, and so is not typically required.
- **order_by=*true*|false**
  Specifies whether the filter should include an Order by combo box.
- **view=Architect|*Generic*|Knowledge**
  Specifies whether the filter should be displayed in a Knowledge tab, an Architect tab, or generically (any tab except Knowledge or Architect).

elsEditField

The elsEditField macro specifies an editable field for the Edit in List feature of a list form. You specify that a list form has the Edit in List feature by including the elsStartEdit and elsEndEdit macros. You define the fields on the Edit in List form with one or more elsEditField and elsEditReadOnly macros. For example:

```
<PDM_MACRO name=elsStartEdit search_filter="__search_filter">
<PDM_MACRO name=elsEditField . . .>
<PDM_MACRO name=elsEditField . . .>
<PDM_MACRO name=elsEditReadonly . . .>
. . .
<PDM_MACRO name=elseEndEdit>
```

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.

- **datatype=0**
  Specifies the data type for input validation. The only datatype supported is 0, which causes the field to be validated as an integer.
- **EndRow=yes|*no***
  Specifies whether the field is the last one in a row on the Edit in List form. Specifying EndRow=yes causes CA SDM to display the next field on a new line of the form.
- **extraEvt=function()**
  Specifies a JavaScript function that is called as an onchange event handler when a user modifies the field.
- **extraURL=*string***
  Specifies a where clause predicate in URL format to restrict the contents of the control.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **size=*20*|*number***
  Specifies the width of the input field.
- **StartRow=yes|*no***
  Specifies whether the field starts a new row on the Edit in List form.

elsEditReadonly

The elsEditReadonly macro specifies a read-only field for the Edit in List feature of a list form. You specify that a list form has the Edit in List feature by including the elsStartEdit and elsEndEdit macros. You define the fields on the Edit in List form with one or more elsEditField and elsEditReadOnly macros. For example:

```
<PDM_MACRO name=elsStartEdit search_filter="__search_filter">
<PDM_MACRO name=elsEditField . . .>
<PDM_MACRO name=elsEditField . . .>
<PDM_MACRO name=elsEditReadonly . . .>
. . .
<PDM_MACRO name=elseEndEdit>
```

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **EndRow=yes|*no***
  Specifies whether the field is the last one in a row on the Edit in List form. Specifying EndRow=yes causes CA SDM to display the next field on a new line of the form.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **size=*20*|*number***
  Specifies the width of the input field.
- **StartRow=yes|*no***
  Specifies whether the field starts a new row on the Edit in List form.

elsEndEdit

The elsEndEdit macro marks the end of the fields specified for the Edit in List feature of a list form. You specify that a list form has the Edit in List feature by including the elsStartEdit and elsEndEdit macros. You define the fields on the Edit in List form with one or more elsEditField and elsEditReadOnly macros. For example:

```
<PDM_MACRO name=elsStartEdit search_filter="__search_filter">
<PDM_MACRO name=elsEditField . . .>
<PDM_MACRO name=elsEditField . . .>
<PDM_MACRO name=elsEditReadonly . . .>
. . .
<PDM_MACRO name=elseEndEdit>
```

This macro has no properties.

elsStartEdit

The elsStartEdit macro activates the Edit in List feature of a list form, and marks the start of the list of fields. You specify that a list form has the Edit in List feature by including the elsStartEdit and elsEndEdit macros. You define the fields on the Edit in List form with one or more elsEditField and elsEditReadOnly macros. For example:

```
<PDM_MACRO name=elsStartEdit search_filter="__search_filter">
<PDM_MACRO name=elsEditField . . .>
<PDM_MACRO name=elsEditField . . .>
<PDM_MACRO name=elsEditReadonly . . .>
. . .
<PDM_MACRO name=elseEndEdit>
```

This macro has the following properties:

- **search_filter="__search_filter"**
  (Required) Specifies the JavaScript object name of the search filter. Specify this property exactly as shown.

endFrameset

The endFrameset macro marks the end of a frameset definition. You define a page of the frame with the startFrameset, endFrameset, and frame macros. For example:

```
<PDM_MACRO name=startFrameset. . .>
<PDM_MACRO name=frame. . .>
<PDM_MACRO name=frame. . .>
. . .
<PDM_MACRO name=endFrameset>
```

This macro has no properties.

endMenu

The endMenu macro marks the end of a menu within a menubar definition. You define a menubar page with the startMenubar, menubarItem, endMenubar, startMenu, menuItem, menuItemLocal, and endMenu macros. For example:

```
<PDM_MACRO name=startMenubar. . .>
<PDM_MACRO name=menubarItem. . .>
<PDM_MACRO name=menubarItem. . .>
. . .
<PDM_MACRO name=endMenubar>
<PDM_MACRO name=startMenu. . .>
```

```
<PDM_MACRO name=menuItem. . .>
<PDM_MACRO name=menuItemLocal. . .>
. . .
<PDM_MACRO name=endMenu>
<PDM_MACRO name=startMenu. . .>
. . .
<PDM_MACRO name=endMenu>
. . .
```

This macro has no properties.

endMenubar

The endMenubar macro marks the end of definition of a menubar, and is followed by the definitions of the menus on the menubar. You define a menubar page with the startMenubar, menubarItem, endMenubar, startMenu, menuItem, menuItemLocal, and endMenu macros. For example:

```
<PDM_MACRO name=startMenubar. . .>
<PDM_MACRO name=menubarItem. . .>
<PDM_MACRO name=menubarItem. . .>
. . .
<PDM_MACRO name=endMenubar>
<PDM_MACRO name=startMenu. . .>
<PDM_MACRO name=menuItem. . .>
<PDM_MACRO name=menuItemLocal. . .>
. . .
<PDM_MACRO name=endMenu>
<PDM_MACRO name=startMenu. . .>
. . .
<PDM_MACRO name=endMenu>
. . .
```

This macro has no properties.

endNotebook

The endNotebook macro marks the end of the definition of a notebook on a detail page. Only one notebook can exist on a page and it must be the last item on the page. You define a notebook with the startNotebook, startTabGroup, tab, and endNotebook macros. For example:

```
<PDM_MACRO name=startNotebook. . .>
<PDM_MACRO name=startTabGroup. . .>
<PDM_MACRO name=tab . .>
<PDM_MACRO name=tab . .>
. . .
<PDM_MACRO name=startTabGroup. . .>
<PDM_MACRO name=tab . .>
. . .
<PDM_MACRO name=endNotebook>
```

This macro has no properties.

frame

The frame macro specifies a frame on a frameset page. You define a frameset page with the startFrameset, frameab, and endFrameset macros. For example:

```
<PDM_MACRO name=startFrameset. . .>
<PDM_MACRO name=frame . .>
<PDM_MACRO name=frame . .>
. . .
<PDM_MACRO name=endFrameset >
```

This macro has the following properties:

- **extra=*string***
  Specifies a string of keywords in the same way as in an HTML <frame> statement, with quotes escaped as required.
- **frame_name=*name***
  (Required) Specifies a name for the frame (the "name" argument on an HTML <frame> statement).
- **frameborder=*yes*|no**
  Specifies whether the frame has a border (the "frameborder" argument on an HTML <frame> statement).
- **id=*name***
  (Required) Specifies the JavaScript ID of the frame (the "id" argument on an HTML <frame> statement).
- **marginheight=*number***
  Specifies the size of the top and bottom margins (the "marginheight" argument on an HTML <frame> statement).
- **marginwidth=*number***
  Specifies the size of the top and bottom margins (the "marginwidth" argument on an HTML <frame> statement).
- **noresize=true|*false***
  Specifies whether the frame is resizable (the "noresize" keyword on an HTML <frame> statement).
- **scrolling=*auto*|yes|no**
  Specifies whether the frame has a scroll bar (the "scrolling" argument on an HTML <frame> statement).
- **style=*string***
  Specifies CSS style information for the frame (the "style" argument on an HTML <frame> statement).
- **tabindex=*number***
  Specifies the tab index of the frame (the "tabindex" argument on an HTML <frame> statement).
- **title=*text***
  Specifies the title for screen reader users.
- **web_form_name=*string***
  Specifies the code name of an object in the web_form factory where CA SDM can obtain the URL for the form displayed in the frame. This property is effective only when web_form_url is not specified.
- **web_form_url=*string***
  Specifies the URL of the form that displays in the frame.

kt_Categories_Tree

The kt_Categories_Tree macro creates a knowledge category frame.

This macro has the following properties:

- **frameborder=*0*|*number***
  Specifies the size of the border.
- **height=*100px*|*value***
  Specifies the frame height.
- **iframe=*yes*|no**
  Specifies whether to generate an iframe for the tree. Specify iframe=no when you are using a predefined frame.
- **menu=*name***
  Specifies the tree menu type. Specify adm_tree, DOCUMENT_EDITOR, HTML_EDITOR, ATTACHMENTS_IMAGES, ATTACHMENTS_TAB, ATTACHMENTS_ADMIN, ARCHITECT, or FAQ.
- **ParentTenant=*value***

Specifies the parent tenant. Typically, you code this property as ParentTenant="$args.ParentTenant"

- **tabindex=*number***
  Specifies the tab index of the frame.
- **UseTenant=*0*|1**
  Specifies whether to use a tenant.
- **view=*ArchitectTree*|FAQ**
  Specifies the view: ArchitectTree or FAQ.
- **width=*100px*|*value***
  Specifies the frame width.

lsCol

The lsCol macro specifies a column on a list form. You define the contents of a list form with the lsStart, lsCol, and lsEnd macros. You can optionally include lsWrite and lsExport macros. For example:

```
<PDM_MACRO name=lsStart. . .>
<PDM_MACRO name=lsCol . .>
<PDM_MACRO name=lsCol . .>
<PDM_MACRO name=lsWrite . .>
<PDM_MACRO name=lsCol . .>
. . .
<PDM_MACRO name=lsEnd>
```

HTML for the entire list is output by the lsEnd macro. The other list macros save information used by lsEnd to generate the list.

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.

  NOTE
  This property is not used and is ignored. This property is retained for compatibility with previous releases.
- **common_name_option=*yes*|no**
  Specifies whether the column contains an SREL attribute. The common name of the referenced table when the list is exported replaces the attribute value.
- **disp_entities=yes|*no***
  Specifies whether to display HTML entities (for example, & or &#26237 in read-only view. This property has the following values:
  - **YES** -- HTML entities are converted to their external value.
  - **NO** -- HTML entities are displayed exactly as entered.
- **display_attr=*COMMON_NAME*|attrName**
  Specifies the column from the referenced table that is displayed on the list for columns containing an SREL attribute. This property is ignored for columns that are not SRELs.
- **escape=*C*|HTML|JS|JS2**
  Specifies how the value of the column is escaped. This property has the following values:
  - **C** -- Prefix quotes, backslashes, and newlines with a backslash
  - **HTML** -- Convert quote, backslash, '<', and '>' to HTML entities
  - **JS** -- Convert quote, backslash, and newline to JavaScript hex (%nn)
  - **JS2** -- Same as JS, but also convert '%' to '%25'
- **export=*yes*|no**

Specifies whether this column is exported when the user clicks the Export button.

- **export_hdr_default**
  Specifies the default header text. The list result export uses this property.
- **exportFmt=*function***
  Specifies the name of a JavaScript function (without parentheses) that returns a string format code which controls formatting the column when it is exported. This property has the following function values:
  - **"YES_NO"** -- Return either "yes" or "no"
  - **"FACTORY_LINK"** -- No formatting (reserved for future use)
  - **"LIST_LOOKUP:v1,v1a,v2,v2a,.."** -- Convert from a list. A value matching the first element in a pair is converted to the second element in the pair.
- **exportHdr=*text*|DISPLAY_NAME**
  Specifies the header for the column when it is exported. You can specify the value as explicit text, or as the keyword value DISPLAY_NAME, which sets the header to the value of export_hdr_default.
- **export_hdr_default=*text***
  Specifies the text of the export column header when exportHdr has the keyword value DISPLAY_NAME. You rarely must specify this property, which defaults to the display name of the attribute for the column.
- **fmtfunc=*function***
  Specifies the name of a JavaScript function that formats the field for display. The function is passed as a single argument containing the value of the attribute, and must return a string that is displayed on the form.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **hidden=yes|*no***
  Specifies whether the column is invisible. A hidden column can be useful for Edit in List or export.
- **img=*name***
  Specifies a JavaScript variable containing the URL of an image to display in the column. Set the JavaScript variable with a preceding lsWrite macro.
- **justify=*left*|center|right**
  Specifies the position of content within the columns.
- **keeplinks=yes|*no***
  Specifies whether HTML links (Action: tags) are displayed as links or formatted as raw text. Preserves HTML links when the text to display contains HTML links. The default value is no.
- **keeptags=yes|*no***
  Specifies whether HTML tags are interpreted as HTML links or formatted as raw text. keeptags=yes overrides the keeplinks property.
- **label=*string***
  Specifies text for a label positioned to the left of data.
- **link=*yes*|no**
  Specifies whether the control on the read-only view is a link to detail for the value of the attribute.
- **max_char=*0*|*number***
  Specifies the maximum number of characters to display in column.
- **nowrap=yes|*no***
  Specifies whether to suppress wrapping of text in a call. When you specify nowrap=yes, text for the column is not permitted to wrap to multiple lines.
- **required=yes|*no***
  Specifies whether text must appear in the column. When you specify required=yes and the attribute has no value, its value is replaced with "Not Available."
- **sort=*ASC*|DESC|no**
  Specifies the sort sequence of the list when a user clicks the column as follows:

  - ASC (ascending)
  - DESC (descending)
  - no (column is not a sort column)
- **startrow=yes|*no***
  Specifies whether the column is the first one in the expansion section. This property works as follows:
  - Columns defined by lsCol macros prior to one specifying startrow=yes are displayed on the main portion of the list.
  - Columns defined by the lsCol macro specifying startrow=yes are included in the expansion section that is hidden from view unless the user clicks the plus sign at the beginning of the list row.
  - The startrow property is effective only for the first lsCol macro specifying startcol=yes; it is ignored on subsequent lsCol macros.
- **style=*string***
  Specifies CSS formatting for the column.
- **uid=*string***
  Specifies an additional identifier to distinguish between columns with the same attribute. This property is not intended for customer use.
- **width=*0*|*number***
  Specifies the maximum width of a column.

lsEnd

The lsEnd macro marks the end of a list form specification. You define the contents of a list form with the lsStart, lsCol, and lsEnd macros. You can optionally include lsWrite and lsExport macros. For example:

```
<PDM_MACRO name=lsStart. . .>
<PDM_MACRO name=lsCol . .>
<PDM_MACRO name=lsCol . .>
<PDM_MACRO name=lsWrite . .>
<PDM_MACRO name=lsCol . .>
. . .
<PDM_MACRO name=lsEnd>
```

HTML for the entire list is output by the lsEnd macro. The other list macros save information used by lsEnd to generate the list.

This macro has the following properties:

- **alt_data_src=*name***
  Specifies the name of a JavaScript variable containing the contents of the list. This property is used to display data from a source other than the database. For example:
  ```
  var alternative_source = new Array();
  var row1 = new Object();
  row1. column0 = "column0";
  row1. column1 = 4;
  alternative_source[0] = row1;
  ......
  <PDM_MACRO name=lsEnd alt_data_src=alternative_source>
  ```
- **factory=*name***
  Specifies the name of a Majic factory for the selection list. Defaults to the factory referenced by the associated attributes, and so is not typically required.
- **isTicketFactory=*0*|1**
  Specifies a list of tickets (requests, incidents, problems, changes, or issues).
- **list=*name***

Specifies the source variable of a database list. This property is not intended for customer use.

- **sort=*attributeName***
  Specifies the initial sort sequence of the list.
- **start=*number***
  Specifies the ordinal number of the first row of the list to display. This property is not intended for customer use.

lsExport

Use the lsExport macro in place of lsCol to specify a column for a list form that is not included in the web UI list. However, this column is included in the spreadsheet generated by exporting the list. You define the contents of a list form with the lsStart, lsCol, and lsEnd macros. You can optionally include lsWrite and lsExport macros. For example:

```
<PDM_MACRO name=lsStart. . .>
<PDM_MACRO name=lsCol . .>
<PDM_MACRO name=lsCol . .>
<PDM_MACRO name=lsWrite . .>
<PDM_MACRO name=lsCol . .>
. . .
<PDM_MACRO name=lsEnd>
```

HTML for the entire list is output by the lsEnd macro. The other list macros save information used by lsEnd to generate the list.

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **common_name_option=*yes*|no**
  Specifies whether the column contains an SREL attribute. The common name of the referenced table replaces this attribute value when the list is exported.
- **export=*yes*|no**
  Specifies whether this column is exported when the user clicks the Export button.
- **export_hdr_default**
  Specifies the text of the header.
- **exportFmt=*function***
  Specifies the name of a JavaScript function (without parentheses) that returns a string format code used to control formatting the column when it is exported. The function can return the following values:
  - "YES_NO" -- Return either "yes" or "no"
  - "FACTORY_LINK" -- No formatting (reserved for future use)
  - "LIST_LOOKUP:v1,v1a,v2,v2a,.." -- Convert from a list. A value matching the first element in a pair is converted to the second element in the pair.
- **exportHdr=*text*|DISPLAY_NAME**
  Specifies the header for the column when it is exported. You can specify the value as explicit text, or as the keyword value DISPLAY_NAME, which sets the header to the value of export_hdr_default.
- **export_hdr_default=*text***
  Specifies the text of the export column header when exportHdr has the keyword value DISPLAY_NAME. You rarely must specify this property, which defaults to the display name of the attribute of the column.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **label=*string***
  Specifies text for a label positioned to the left of data.

lsStart

The lsStart macro marks the beginning of a list form specification. You define the contents of a list form with the lsStart, lsCol, and lsEnd macros, and can optionally include lsWrite and lsExport macros. For example:

```
<PDM_MACRO name=lsStart. . .>
<PDM_MACRO name=lsCol . .>
<PDM_MACRO name=lsCol . .>
<PDM_MACRO name=lsWrite . .>
<PDM_MACRO name=lsCol . .>
. . .
<PDM_MACRO name=lsEnd>
```

HTML for the entire list is output by the lsEnd macro. The other list macros save information used by lsEnd to generate the list.

This macro has the following properties:

- **search_type=*DISPLAY*|GET_DOB**
  Specifies how data for the list is retrieved. The default of DISPLAY causes all data to be retrieved directly from the database. The alternate value of GET_DOB causes data to be formatted by the object engine (domsrvr), and may be required if the list includes local attributes not stored in the database. A specification of search_type=GET_DOB reduces the performance of the list form, and only use it when necessary.

lsWrite

The lsWrite macro specifies JavaScript code executed for every row of a list. The text specified by lsWrite is inserted into the HTML form multiple times, once for each row of the list. You define the contents of a list form with the lsStart, lsCol, and lsEnd macros. You can optionally include lsWrite and lsExport macros. For example:

```
<PDM_MACRO name=lsStart. . .>
<PDM_MACRO name=lsCol . .>
<PDM_MACRO name=lsCol . .>
<PDM_MACRO name=lsWrite . .>
<PDM_MACRO name=lsCol . .>
. . .
<PDM_MACRO name=lsEnd>
```

HTML for the entire list is output by the lsEnd macro. The other list macros save information used by lsEnd to generate the list.

This macro has the following properties:

- **both=yes|*no***
  Specifies whether the JavaScript text defined by the macro is inserted before the list and for every row on the list.
- **text=*script***
  Specifies the text to insert.

menubarItem

The menubarItem macro defines an item on a menubar. You define a menubar page with the startMenubar, menubarItem, endMenubar, startMenu, menuItem, menuItemLocal, and endMenu macros. For example:

```
<PDM_MACRO name=startMenubar. . .>
<PDM_MACRO name=menubarItem. . .>
<PDM_MACRO name=menubarItem. . .>
. . .
<PDM_MACRO name=endMenubar>
```

```
<PDM_MACRO name=startMenu. . .>
<PDM_MACRO name=menuItem. . .>
<PDM_MACRO name=menuItemLocal. . .>
. . .
<PDM_MACRO name=endMenu>
<PDM_MACRO name=startMenu. . .>
. . .
<PDM_MACRO name=endMenu>
. . .
```

This macro has the following properties:

- **hotkey=*string***
  Provides a hotkey suggestion for this menubar item. CA SDM selects the hotkey for the menubar item from the characters in the string; otherwise, it selects the hotkey from the characters in the label.
- **id=*name***
  (Required) Specifies a JavaScript identifier for the menubar item.

  NOTE
  The startMenu macro references this identifier. The startMenu macro defines the contents of the menu dropped down from the menubar for this item.
- **img=*url***
  Specifies the URL of an image to display on the toolbar section of the menubar.
- **label=*text***
  (Required) Specifies the text of the menu bar item.
- **variable=*name***
  Specifies a JavaScript variable that can be used to reference the menubar item. This property is not intended for customer use.

menuItem

The menubarItem macro defines an item on a menu. The item invokes a JavaScript function in the main CA SDM form, even if the menu is in a pop-up form. Use the menuitemLocal macro to invoke a local function. You define a menubar page with the startMenubar, menubarItem, endMenubar, startMenu, menuItem, menuItemLocal, and endMenu macros. For example:

```
<PDM_MACRO name=startMenubar. . .>
<PDM_MACRO name=menubarItem. . .>
<PDM_MACRO name=menubarItem. . .>
. . .
<PDM_MACRO name=endMenubar>
<PDM_MACRO name=startMenu. . .>
<PDM_MACRO name=menuItem. . .>
<PDM_MACRO name=menuItemLocal. . .>
. . .
<PDM_MACRO name=endMenu>
<PDM_MACRO name=startMenu. . .>
. . .
<PDM_MACRO name=endMenu>
. . .
```

This macro has the following properties:

- **extended=*0*|1**

Specifies whether this item is an extended menu item supporting disabling, hiding, and changing an image.

- **function=*script***
  (Required) Specifies the JavaScript function invoked when the menu item is selected. This function must be available in the main CA SDM form, and not in the pop-up form containing the menu. Menu items invoking functions in the pop-up form should be defined with the menuitemLocal macro.
- **hotkey=*string***
  Provides a hotkey suggestion for this menubar item. CA SDM selects the hotkey for the menubar item from the characters in the string; otherwise, it selects the hotkey from the characters in the label.
- **icon_name=*string***
  Specifies the name of an image file displayed as an icon on the menu next to the menu item.
- **id=*name***
  Specifies a JavaScript ID for the menu item.
- **label=*text***
  (Required) Specifies the text of the menu item.
- **local=*0*|1**
  Specifies whether extended menus use the function in the local frame. This property is not intended for customer use.
- **tooltip=*string***
  Specifies a tooltip for the icon.

menuItemLocal

The menuItemLocal macro defines an item on a menu that invokes a JavaScript function in the form containing the menu. You define a menubar page with the startMenubar, menubarItem, endMenubar, startMenu, menuItem, menuItemLocal, and endMenu macros. For example:

```
<PDM_MACRO name=startMenubar. . .>
<PDM_MACRO name=menubarItem. . .>
<PDM_MACRO name=menubarItem. . .>
. . .
<PDM_MACRO name=endMenubar>
<PDM_MACRO name=startMenu. . .>
<PDM_MACRO name=menuItem. . .>
<PDM_MACRO name=menuItemLocal. . .>
. . .
<PDM_MACRO name=endMenu>
<PDM_MACRO name=startMenu. . .>
. . .
<PDM_MACRO name=endMenu>
. . .
```

This macro has the following properties:

- **function=*script***
  (Required) Specifies the JavaScript function invoked when the menu item is selected. This function must be available in the form containing the menu.
- **hotkey=*string***
  Provides a hotkey suggestion for this menubar item. CA SDM selects the hotkey for the menubar item from the characters in the string; otherwise, it selects the hotkey from the characters in the label.
- **icon_name=*string***
  Specifies the name of an image file displayed as an icon on the menu next to the menu item.
- **id=*name***

Specifies a JavaScript ID for the menu item.

- **label=*text***
  (Required) Specifies the text of the menu item.
- **tooltip=*string***
  Specifies a tooltip for the icon.

priMatrix

The priMatrix macro specifies a priority calculation matrix.

This macro has the following properties:

- **impact_label=*name***
  Specifies an impact label.
- **matrix_name=*name***
  (Required) Specifies the name of this matrix.
- **reset_btn_name=*name***
  Specifies a reset button name.
- **urgency_label=*name***
  Specifies an urgency label.

schedAttr

The schedAttr macro specifies an attribute included on a schedule form, either the change schedule (list_chgsched_config.htmpl) or the knowledge schedule (list_kdsched_config.htmpl). You define a schedule form with the schedConfig, schedAttr, and schedGroup macros. For example:

```
<PDM_MACRO name=schedConfig. . .>
<PDM_MACRO name=schedAttr . .>
<PDM_MACRO name=schedAttr . .>
. . .
<PDM_MACRO name=schedGroup . .>
<PDM_MACRO name=schedGroup . .>
. . .
```

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **attrRef=*.COMMON_NAME*|attributeName**
  Specifies the attribute in the table referenced by an SREL attribute that should be shown on the schedule.
- **detail=0|*1***
  Specifies whether to include the attribute in the detailed n-day view.
- **fmtfunc=*none*|function**
  Specifies the name of a JavaScript function that formats the field for display. The function is passed a single argument containing the value of the attribute, and must return a string that is displayed on the form.
- **hoverInfo=*0*|1**
  Specifies whether to include the attribute in the monthly view info pop-up.
- **ident=*0*|1**
  Specifies whether to include the group name on the n-day view.
- **label=0|string|*$args.&{attr].DISPLAY_NAME***
  Specifies a label for the attribute. A value of zero signifies no label; the default is the display name of the attribute.
- **summary=*0*|1**
  Specifies whether to include the attribute in the summary n-day view.

schedConfig

The schedConfig macro controls the appearance and contents of a schedule form, either the change schedule (list_chgsched_config.htmpl) or the knowledge schedule (list_kdsched_config.htmpl). You define a schedule form with the schedConfig, schedAttr, and schedGroup macrosle. For example:

```
<PDM_MACRO name=schedConfig. . .>
<PDM_MACRO name=schedAttr . .>
<PDM_MACRO name=schedAttr . .>
. . .
<PDM_MACRO name=schedGroup . .>
<PDM_MACRO name=schedGroup . .>
. . .
```

This macro has the following properties:

- **autoSearch=0|1**
  Specifies whether to reissue a search automatically if necessary.
- **defaultView=0|1|7|*30*|99**
  Specifies the initial view as 1 (day), 7 (week), 30 (month), 99 (*n*day), or 0 (list).
- **export=0|*codename***
  Specifies the iCalendar export template code (0=no export).
- **firstday=*0*|1|2|3|4|5|6**
  Specifies the first weekday from 0 (Sunday) to 6 (Saturday).
- **hoverMax=*35*|number**
  Specifies the maximum number of rows in a hover information pop-up.
- **legend=0|*1*|2**
  Specifies the position of a legend (1=top, 2=bottom, 0=none).
- **maxgroups=*4*|*number***
  Specifies the maximum number of groups per cell in month view.
- **ndays=0|(*3,7,14,28*)|(*n1,n2,..*)**
  Specifies the contents the n-days drop-down list as a comma-separated list of numbers. A value of 0 signifies no n-days drop-down list.
- **round=0|(*0,15*)|(*hh,mm*)**
  Specifies event times in hours and minutes enclosed in parentheses. The default of (0,15) specifies rounding to the nearest 15 minutes. A value of 0 signifies no rounding.
- **timefmt=*24hr*|(*am,pm*)**
  Specifies time format as either 12 or 24 hour. You specify the 12-hour value as two comma-separated strings, the first for am and the second for pm.
- **tzSelect=*string***
  Specifies where clause restrictions on the time zone drop-down list.

schedGroup

The schedGroup macro controls the name and styling of groups of items on a schedule form, either the change schedule (list_chgsched_config.htmpl), or the knowledge schedule (list_kdsched_config.htmpl). You define a schedule form with the schedConfig, schedAttr, and schedGroup macrosle. For example:

```
<PDM_MACRO name=schedConfig. . .>
<PDM_MACRO name=schedAttr . .>
<PDM_MACRO name=schedAttr . .>
. . .
<PDM_MACRO name=schedGroup . .>
<PDM_MACRO name=schedGroup . .>
```

. . .

This macro has the following properties:

- **bgcolor=*white*|color**
  Specifies the background color for the group as any valid web color.
- **color=*black*|color**
  Specifies the foreground (text) color for the group as any valid web color.
- **grpname=*name***
  (Required) Specifies the name of the group.
- **icon=*filename***
  Specifies an icon for the group as the name of a file in $NX_ROOT/bopcfg/www/wwwroot/img.
- **label=*string***
  Specifies a label for the group.
- **legend=0|*string***
  Specifies description of the group for the legend. A value of 0 signifies that group is not in the legend.
- **style=*normal*|bold|italic**
  Specifies the text style for the group.

sfDate

The sfDate macro specifies a date control in the search filter of a list form.

This macro has the following properties:

- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
  colspan=*1*|*number*
  Specifies the number of columns on the form.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **qbe_condition=*condition***
  Specifies one of the following conditions for comparing the value in this field to the database value:
  - EQ is equal to the value.
  - NE is not equal to the value.
  - GT is greater than the value.
  - LT is less than the value.
  - GE is greater than or equal to the value.
  - LE is less than or equal to the value.
  - NU is null
  - NN is not null.
  - IN matches the SQL LIKE expression.
  - KY contains the text entered.

sfDropdown

The sfDropdown macro specifies a drop-down selection control in the search filter of a list form.

This macro has the following properties:

- **attr=*attributeName***

(Required) Specifies the name of the attribute associated with the control.

- **codename=*string***
  Specifies the name of a set of values from the ui_selection object that are displayed in the date drop-down control. The values are those values with code attributes that match the value specified for codename.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **default=*text***
  Specifies the default value when the attribute is null.
- **factory=*name***
  Specifies the name of a Majic factory for the selection list. Defaults to the factory referenced by the associated attributes, and so is not typically required.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **lookup=*yes*|no**
  Specifies whether the field should automatically convert to a lookup when the number of entries in the drop-down list exceeds a configured value (typically 10).
- **Noempty=yes|*no***
  Specifies whether the empty value is included in the drop-down list. Specify Noempty=yes to suppress the empty value.
- **op=QBE.op**
  Specifies one of the following conditions for comparing the value in this field to the database value:
  - EQ is equal to the value.
  - NE is not equal to the value.
  - GT is greater than the value.
  - LT is less than the value.
  - GE is greater than or equal to the value.
  - LE is less than or equal to the value.
  - NU is null
  - NN is not null.
  - IN matches the SQL LIKE expression.
  - KY contains the text entered.
- **win=*windowName***
  Specifies the target window. This property is not intended for customer use.

sfEnd

The sfEnd macro marks the end of the definition of a search filter on a list page. You define a search filter with the sfStart, sfEnd, and other sfxxx macros. For example:

```
<PDM_MACRO name=ssfStart. . .>
<PDM_MACRO name=sfxxx. . .>
<PDM_MACRO name=sfxxx. . .>
. . .
<PDM_MACRO name=sfEnd>
```

The macro has no properties.

sfHier

The sfHier macro specifies a hierarchical lookup control in the search filter of a list form. A hierarchical lookup control allows a user to click the caption (header) of the control to pop up a hierarchical selection form.

This macro has the following properties:

- **addPercent=yes|no**
  Specifies whether to allow wildcard character searches. This property defaults to the value of the web_wildcard_search option.
- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **factory=*name***
  Specifies the name of a Majic factory for the selection list. Defaults to the factory referenced by the associated attributes, and so is not typically required.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.

sfLookup

The sfLookup macro specifies a lookup control in the search filter of a list form. A lookup control allows a user to click the caption (header) of the control to pop up a selection form.

This macro has the following properties:

- **addPercent=yes|no**
  Specifies whether to allow wildcard character searches. This property defaults to the value of the web_wildcard_search option.
- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **extraURL=*string***
  Specifies a where clause predicate in URL format to restrict the contents of the control.
- **factory=*name***
  Specifies the name of a Majic factory for the selection list. Defaults to the factory referenced by the associated attributes, and so is not typically required.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.

sfMultiLookup

The sfMultiLookup macro specifies a multi-value lookup control in the search filter of a list form. A multi-value lookup control allows a user to selection one or more values for the attribute associated with the control.

This macro has the following properties:

- **attr=*attributeName***

(Required) Specifies the name of the attribute associated with the control.

- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **factory=*name***
  Specifies the name of a Majic factory for the selection list. Defaults to the factory referenced by the associated attributes, and so is not typically required.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **size=*20*|*number***
  Specifies the width of the input field.

sfStart

The sfStart macro marks the beginning of the definition of a search filter on a list page. You define a search filter with the sfStart, sfEnd, and other sfxxx macros. For example:

```
<PDM_MACRO name=ssfStart. . .>
<PDM_MACRO name=sfxxx. . .>
<PDM_MACRO name=sfxxx. . .>
. . .
<PDM_MACRO name=sfEnd>
```

This macro has the following properties:

- **button=*true*|false**
  Specifies whether to show the default search filter buttons: Search, Show Filter, Clear Filter, and Create New.
- **color=*string***
  This property is not used in this release (reserved for future use).
- **create=*true*|false**
  Specifies whether to show the Create New button.
- **export=*yes*|no**
  Specifies whether to show the Export button.
- **extraCreateURL=*string***
  Specifies additional arguments for the URL used to create an object.
- **factory=*name***
  Specifies the name of a Majic factory for the selection list. Defaults to the factory referenced by the associated attributes, and so is not typically required.
- **ForceSearchWithKeywords=true|*false***
  Specifies whether to request a nonempty search text. This property is effective only when KnowledgeSearchText is true.
- **KnowledgeSearchText=true|*false***
  Specifies whether to show the knowledge search text at the top of the filter.

sfStartRow

The sfStartRow macro marks the start of a row on a search filter.

This macro has the following properties:

- **type=Set**
  Specifies that the new row begins a row set that is hidden from view until the user clicks the green filter icon at the end of the previous row.

sfTextbox

The sfTextbox macro specifies a textbox in a search filter.

This macro has the following properties:

- **addPercent=yes|no**
  Specifies whether to allow wildcard character searches. This property defaults to the value of the web_wildcard_search option.
- **attr=*attributeName***
  (Required) Specifies the name of the attribute associated with the control.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **disabled=yes|*no***
  Specifies whether the textbox is disabled for input. If you disable the textbox, you must also specify the value, display_value, and qbe_condition properties.
- **display_value=*text***
  Specifies the human-readable value of a disabled textbox. This property is effective only when disabled=yes.
- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **qbe_condition=*condition***
  Specifies one of the following conditions for comparing the value in this field to the database value:
  - EQ is equal to the value.
  - NE is not equal to the value.
  - GT is greater than the value.
  - LT is less than the value.
  - GE is greater than or equal to the value.
  - LE is less than or equal to the value.
  - NU is null
  - NN is not null.
  - IN matches the SQL LIKE expression.
  - KY contains the text entered.
- **size=*20*|*number***
  Specifies the width of the input field.
- **value=*string***
  Specifies the value of a disabled textbox used in the search. This property is effective only when disabled=yes.

startFrameset

The startFrameset macro marks the beginning of a frameset definition. You define a frame page with the startFrameset, endFrameset, and frame macros. For example:

```
<PDM_MACRO name=startFrameset. . .>
<PDM_MACRO name=frame. . .>
<PDM_MACRO name=frame. . .>
. . .
<PDM_MACRO name=endFrameset
```

This macro has the following properties:

- **border**

Specifies the space between the frames.

- **cols**
  Specifies the quantity and sizes of the columns in the frame set.
- **frameborder=*yes*|no**
  Specifies whether the frame has a border (the "frameborder" argument on an HTML <frame> statement).
- **id=*name***
  (Required) Specifies the JavaScript ID of the frame (the "id" argument on an HTML <frame> statement).
- **onload**
  Specifies the onLoad handler of the frame set.
- **onunload**
  Specifies the onUnload handler of the frame set.
- **rows**
  Specifies the quantity and sizes of the rows in the frame set.

startMenu

The startMenu macro marks the beginning of a menu within a menubar definition. You define a menubar page with the startMenubar, menubarItem, endMenubar, startMenu, menuItem, menuItemLocal, and endMenu macros. For example:

```
<PDM_MACRO name=startMenubar. . .>
<PDM_MACRO name=menubarItem. . .>
<PDM_MACRO name=menubarItem. . .>
. . .
<PDM_MACRO name=endMenubar>
<PDM_MACRO name=startMenu. . .>
<PDM_MACRO name=menuItem. . .>
<PDM_MACRO name=menuItemLocal. . .>
. . .
<PDM_MACRO name=endMenu>
<PDM_MACRO name=startMenu. . .>
. . .
<PDM_MACRO name=endMenu>
. . .
```

This macro has the following properties:

- **parentid=*name***
  (Required) Specifies the ID of the menubarItem macro specifying this menu.

startMenubar

The startMenubar macro marks the beginning of a menubar definition. You define a menubar page with the startMenubar, menubarItem, endMenubar, startMenu, menuItem, menuItemLocal, and endMenu macros. For example:

```
<PDM_MACRO name=startMenubar. . .>
<PDM_MACRO name=menubarItem. . .>
<PDM_MACRO name=menubarItem. . .>
. . .
<PDM_MACRO name=endMenubar>
<PDM_MACRO name=startMenu. . .>
<PDM_MACRO name=menuItem. . .>
<PDM_MACRO name=menuItemLocal. . .>
. . .
<PDM_MACRO name=endMenu>
```

```
<PDM_MACRO name=startMenu. . .>
. . .
<PDM_MACRO name=endMenu>
. . .
```

This macro has no properties.

startNotebook

The startNotebook macro marks the beginning of the definition of a notebook on a detail page. A page can only have one notebook and it must be the last item on the page. You define a notebook with the startNotebook, startTabGroup, tab, and endNotebook macros. For example:

```
<PDM_MACRO name=startNotebook. . .>
<PDM_MACRO name=startTabGroup. . .>
<PDM_MACRO name=tab . .>
<PDM_MACRO name=tab . .>
. . .
<PDM_MACRO name=startTabGroup. . .>
<PDM_MACRO name=tab . .>
. . .
<PDM_MACRO name=endNotebook>
```

This macro has the following properties:

- **hdr=*text***
  Specifies the text of the caption on the control; defaults to the DISPLAY_NAME of the attribute associated with the control.
- **ilayer_height=*number***
  This property is deprecated; not used in this release.
- **nb_height=*number***
  Specifies the height of the notebook in pixels.

startTabGroup

The startTabGroup macro specifies the beginning of a group of tabs in a notebook on a detail page. This macro indicates that the notebook is organized in tab groups. A notebook with tab groups shows the groups as the highest level of tab. Tabs following the group are visible only when the group is selected.

You define a notebook with the startNotebook, startTabGroup, tab, and endNotebook macros. For example:

```
<PDM_MACRO name=startNotebook. . .>
<PDM_MACRO name=startTabGroup. . .>
<PDM_MACRO name=tab . .>
<PDM_MACRO name=tab . .>
. . .
<PDM_MACRO name=startTabGroup. . .>
<PDM_MACRO name=tab . .>
. . .
<PDM_MACRO name=endNotebook>.
```

This macro has the following properties:

- **title=*string***
  Specifies the name of the tab group that is shown at the top of the notebook. If you omit this property, CA SDM creates the name of the tab group by concatenating the names of all the tabs within the group.

tab

The tab macro specifies a tab in a notebook on a detail page. You define a notebook with the startNotebook, startTabGroup, tab, and endNotebook macros. For example:

```
<PDM_MACRO name=startNotebook. . .>
<PDM_MACRO name=startTabGroup. . .>
<PDM_MACRO name=tab . .>
<PDM_MACRO name=tab . .>
. . .
<PDM_MACRO name=startTabGroup. . .>
<PDM_MACRO name=tab . .>
. . .
<PDM_MACRO name=endNotebook>
```

This macro has the following properties:

- **accordion_specific_tab=true**
  Specifies that this macro defines an CA SDM Release 12.9 notebook. If present, you must specify this property exactly as shown.
- **file=*filename*.htmpl**
  Specifies an immediate tab. The contents of an immediate tab are part of the form containing it, and are formatted at the same time the containing form is formatted. This property specifies the HTMPL file containing its contents. CA SDM looks for a <PDM_FORM> tag in the referenced file, and includes the contents of this form as the contents of the tab. This property is mutually exclusive with the src property.
- **filename=*string***
  Specifies the value of the $prop.filename server variable that is passed to the tab contents. This property has no relationship to the similarly named file property.
- **height=*number***
  Specifies the height of this tab.
- **id=*string***
  Specifies the JavaScript identification of this tab.
- **new_row=yes|*no***
  (Deprecated) Indicates that the tab starts a new row of tabs. This property is informational only and CA SDM ignores it. The startTabGroup macro handles the grouping of tabs into rows.
- **src=*url***
  Specifies that this tab is a deferred tab. The contents of a deferred tab are independent of the main form and are loaded only when the user selects the tab. This property specifies the URL invoked to load the content of the tab. This property is mutually exclusive with the file property.
- **title=*string***
  Names the tab on the notebook.
- **tooltip=*string***
  Specifies a tooltip for a tab on the notebook.

tabBanner

The tabBanner macro specifies a banner (header) within a tab. You can only use this macro within an immediate tab loaded by the file property of the tab macro. The banner can contain text and one or more buttons.

This macro has the following properties:

- **add_btns=true|*false***

Specifies whether the tab banner contains buttons. When this property is true, the containing form must include a JavaScript that defines the buttons. The btnfunc property specifies the name of the function and defaults to add_button_to_tab_banner().

- **btnfunc=*add_button_to_tab_banner*|*functionName***
  Specifies the name of the function that creates the buttons in the banner when add_btns=true.
- **btns_in_two_rows=*true*|false**
  Specifies whether to include an additional line in the banner for scrolling in the Internet Explorer browser.
- **show_bar=*true*|false**
  Specifies whether to display a horizontal bar below the tab banner.
- **title=*string***
  The text displayed in the tab banner. The string can be an empty string.

tabList

The tabList macro specifies a list included within an immediate tab loaded by the file property of the tab macro.

This macro has the following properties:

- **brBefore=yes|*no***
  Specifies whether to insert a line break before the list for spacing.
- **btnTitle=*string***
  Specifies a title for the Load button displayed when the Screen Reader preference is active.
- **colspan=*1*|*number***
  Specifies the number of columns on the form.
- **frmName=*string***
  Specifies a JavaScript identification of the HTML iframe that contains the list.
- **height=*number***
  Specifies the height of the generated list.
- **src=*url***
  (Required) Specifies the URL to invoke to generate the list.

Security and Role Management

This article contains the following topics:

Set up security and role management for user authentication, user access level, and the mode of authentication while logging in.

Access Types

Access types define contact roles, contacts authentication, and whether the contacts can modify web forms or the database schema.

Modify the predefined access types and create new ones.

Assign Access Type Using LDAP Groups

Assign AccessTypesvaluestocontactsautomatically with a Lightweight Directory Access Protocol (LDAP) server.

> **NOTE**
>
> To enable this feature, install the ldap_enable_group and ldap_group_object_class options.

Perform the following steps:

1. Select Security and Role Management, Access Types on the Administrator tab.

2. Select the Access Type you want to associate with an LDAP Group. For example, select Administration.
   If the ldap_enable_group option is installed, the LDAP Access Group field appears on the Web Authentication tab.

   NOTE

   : If an LDAP Group is already associated with the selected Access Type, a link to the LDAP Group Detail appears. Click the link for a read-only description of the LDAP Group and a listing of its members.
3. Click Edit on the Access Type Detail page to associate an Access Type with an LDAP Group.
4. Click the LDAP Access Group link.
5. (Optional) Enter filter criteria to limit the search to the LDAP groups of interest.
6. Select the LDAP Group that you want to associate with this Access Type.
7. Click Save.
   Association of the selected LDAP Group with the Access Type is complete.

Create an Access Type

Access types define how contacts are authenticated when they log in to the web interface, whether the contacts can modify web forms or the database schema using Web Screen Painter, and which roles are available for the contacts.

You can modify the predefined access types and create new ones.

The access types define all aspects of security. Several predefined access types are included, and you can modify them or can define new ones. Each access type for a user controls the following aspects of system behavior:

- How CA SDM performs the web authentication when the user logs in.
- The access level for the user.
- Whether the user can modify web forms or the database schema using Web Screen Painter.
- What are the roles available to the user.

Perform the following steps:

1. Select **Security and Role Management**, **Access Types** on the Administration tab.
   The Access Type List page is displayed.
   Default: 15
2. Click **Create New** and complete the access type fields, as appropriate, on the Create New Access Type page.
3. Use the tabs to complete the following tasks:
   - Configure Web Authentication for an Access Type
   - Assign Web Screen Painter Permissions to an Access Type
   - Assign Roles to an Access Type
4. Click **Save**.
   The access type is created.

Access Type Fields

The following fields appear on the Create Access Type, Access Type Detail, and Update Access Type pages.

- **Symbol**
  Specifies a unique identifier for the access type.
- **Default?**
  Indicates whether this access type is the default that is associated with contacts.
- **Record Status**
  Specifies whether this access type is Active or Inactive.
- **Description**
  Describes the access type. Use this field to help identify the characteristics of the access type.
- **Receive Internal Notification**

Determines whether the contacts associated with the access type receive internal notification of activities that are related to tickets.

- **Access Level**
  Determines which access types a user can grant to another user. A user can assign an access typeto the contact record of another user only if the access level of the access type they are attempting to assign is ranked the same as or lower than thegrant level for their own access type. The levels are ranked as follows:
  - Admin (highest)
  - Analyst
  - Cust/Emp
  - None (lowest)

  Licensed?
  Determines whether thiscontact is a licensed access type. Contacts assigned to unlicensed access types can only view or update their own personal data.

> **NOTE**
>
> KPIs counts the concurrent usages of the users from the system (for example, CA SDM Web UI, SOAP Web Services, REST Web Services, and so on). For example, the webConcurrentLicenseCt KPI counts the maximum number of unique users (with the "Licensed?" option selected) logged in to the CA SDM Web UI during the interval.

Configure Web Authentication for an Access Type

You can configure the web authentication and validation type to specify how roles assigned to this access type are authenticated when users attempt to access the CA products. Complete the following fields in the **Web Authentication** tab.

- **Allow External Authentication**
  Select this check box if you want to allow contacts to be authenticated externally, for example by the HTTPD server or the operating system. If you select this option, users with this access type are validated by the appropriate external method as configured during installation.Checks ensure that no external validation has taken place (for example, that the user has not attempted access through a non-secure server) and that the user is defined as a valid contact in the system using the login ID. Then, it uses the access type to determine the correct interface to use.
- **Validation Type**
  Defines how users are authenticated when an external authorization is either not permitted or fails (for example, if the user is attempting access through a non-secure server). The available options are:
  - **No Access**
    Denies access to CA products unless external authentication is allowed and is valid.
  - **Open**
    Access to the CA products are always allowed, with no additional authentication required.
  - **OS**
    Access to the CA products are allowed through operating system user name and password.
  - **PIN / PIN Number**
    Users of this type can access only if they enter the correct value for the PIN field in their contact record. If you select the PIN option, you can choose which field in the contact record stores the PIN by entering the field attribute name in the PIN Field edit box.
  - **CA EEM**
    Access to the CA products are allowed through CA EEM. This option is available only if CA SDM is integrated with CA EEM.
  - **OS-Use Operating System**
    When the Administration Access Type Validation Type drop-down is set to **OS-Use Operating System Authentication** and if you want to login using the CASMAdmin user, you must first create the CASMAdmin user in the Operating System for the login to be successful.

Assign Web Screen Painter Permissions to an Access Type

The Web Screen Painter (WSP) utility allows CA SDM users to build and publish web forms and schemas. The Web Screen Painter tab also controls the database access for Web Screen Painter preview sessions. For the details about WSP, see Using the Web Screen Painter (WSP).

Select the permissions that you want to allow for an access type in the Web Screen Painter tab.

- **Modify Forms**
  Allows the users todo the changes to existing forms withoutdoing the changes available to all users.
- **Modify Schema**
  Allows the users todo the changes to an existing schema withoutdoing the changes available to all users.
- **Publish Forms**
  Allows the users to make their modified forms available to all users.
- **Publish Schema**
  Allows the users to make their modified schema available to all users.
- **Preview Session Can Update Database**
  Allows the users to do the changes to the database during a preview session. By default, database changes are not allowed during a preview session.

Assign Roles to an Access Type

Assign the roles to an access type to limit contacts access to functional areas for assigned roles.

Perform the following steps:

1. Select the following roles for this access type:
   - **Reporting Role**
     Defines the reporting access for this type.
   - **REST Web Service API Role**
     Defines the access to the REST web services for this type.
   - **SOAP Web Service API Role**
     Defines the access to the SOAP web services for this type.
   - **Command Line Utility Role**
     Defines the access to the command-line utilities and attachments for this type.
2. Click Update Roles.
3. Enter any search criteria that you want to limit the list tot he roles of interest, and then click Search.
   The Roles Assigned - Update page opens, listing the roles that matched the search criteria.
4. Select the roles that you want to assign. To select multiple items, hold down the CTRL key while clicking the left mouse button.
5. Click the double right-directional arrows, after you have selected all the roles that you want.
   The selected roles move to the Roles Assigned list on the right.
6. Click OK.
   The Access Type Detail page opens, with the assigned roles listed on the Roles tab.
7. Click Save.
   The Access Type Detail page opens, with a confirmation message that your changes have been saved.
8. Select the role that you want to be the default for this access type upon login. Click Set Default Role.
   Your selection for the default role is saved.

Search Access Types

You can enter search criteria to filter the Access Type List to display only the types of interest.

To search for an access type

1. Select **Security and Role Management**, **Access Types** on the Administration tab.

The Access Type List page appears.
2. Click **Show Filter** and complete one or more of the search fields.
3. Click **Search**.
   The Access Type List page displays the types that match your search criteria.

Define Contact Notification Parameters

You can define the contact information and the method you want to use to notify a contact.

To set up contact notification parameters:

1. On the Contact Detail page, select the **Notification** tab.
2. Enter the appropriate contact information, such as Telephone Number, Fax Number, Email Address, etc.
3. Select the notification method you want to use for each message urgency level for this contact (low, normal, high, and emergency).
   For example, you may want to notify this contact using the Email method for normal level activities, but you may want to use the Pager_Email notification method for emergency level activities.
4. Select the workshift that is valid for each notification urgency level.
   For example, you may assign a Regular workshift (five-day week, eight-hours a day) to the normal level notification, but a 24 hour workshift to the emergency level notification.
5. Click **Save** to save the notification parameters.

Create Contacts Manually

If you do not want to use an active directory such as LDAP for your contacts information, you can create the contacts manually in CA SDM.

> **NOTE**
>
> If multi-tenancy is enabled, select the appropriate tenant from the drop-down list.

Follow these steps:

1. From the Classic CA SDM Web Interface, select **File**, **New Contact** from the menu bar on the Scoreboard.
   The Create New Contact window opens.
2. Complete the following fields as appropriate for the contact:

Tenant

Specifies the tenant that is associated with the contact (for multi-tenancy installations).

Contact ID

Specifies a unique identifier for the contact. If the default user authentication is being used, the value in this field is used as the password when the user logs in.

User ID

Specifies the user name of the contact. The contact uses this value to log in to the system.

Service Type

Specifies the level of support that is received by the contact.

Data Partition

Specifies the data partition for this contact. This value determines the records that this contact can access.

Access Type

Specifies the access type. The access type determines the system functions the contact can access.

Available

Indicates whether the contact is available for ticket assignments.

Confirm Self-Service Save

Indicates whether the contact receives a confirmation when saving a record from the self-service interface.

Analyst's Tenant Group

(Analyst Contact Type Only) Specifies the tenant group that the analyst is responsible for.
To configure the contact, use the following controls available on the tabs.

Notification

Defines the contact information and method for notifying the contact.

- – Select the notification method from the drop-down list (Email, Notification, Pager_Email, xMatters/Email, xMatters/Notification, and xMatters/Pager_Email) that you want to use for each message urgency level for this contact.

  NOTE

  CA SDM supports only one notification method at a time. If you are using Email, then you cannot use Notification at the same time. This applies to all out of the box notification methods like Email, Notification, Pager_Email, xMatters/Email, xMatters/Notification, and xMatters/Pager_Email.

  NOTE

  CA SDM Administrators must update the notification method manually in the contact details page if the xMatters and CA SDM integration is disabled. For more information, see Options Manager.

- – Select the workshift that is valid for each notification urgency level.

For example, you may assign a Regular workshift (five-day week, eight-hours a day) to the normal level notification, but a 24 hour workshift to the emergency level notification.

Address

Specifies the location of the contact.

Organizational Info

Specifies the functional or administrative organization, department, cost center, or vendor information of the contact.

Environment

Specifies the environment of the contact, such as equipment, software, and services.

Groups

Assigns a contact to a group (a collection of contacts with a common area of responsibility).

Roles

Assigns the contact to one or more roles.

Service Contracts

Displays any service contracts that have been associated with the contact.

Special Handling

Lists the special handing contacts and lets you search for and associate a contact to a special handling type, such as a visitor or security risk type.

Event Log

Lists events that are associated with the contact, such as self service and knowledge activities.

Activities

Lists the activity log for the contact.

Click **Save**.
The contact information is saved.

Role Management

Roles define the functionality that users are allowed to access. You can assign one or more roles to an individual contact record, or to an access type to define function access for all of the access type's associated contacts.

Role List

Roles are the primary records that control security and user interface navigation. Each role defines a focused view of the system by exposing only the functionality necessary for users to perform the tasks typically assigned to the role they perform within their business organization.

Predefined roles are provided that are designed to align with ITIL v3 Best Practices and thereby reduce the amount of site-specific customization required to bring your IT organization into ITIL compliance. You can use the predefined roles in their default configuration, modify them to meet your business requirements, or create new roles.

Role Tabs

> **NOTE**
>
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

The following tabs are available on the Role Detail and Update Role pages:

Authorization Allows you to define the authorization level assigned to the role.

Function Access Allows you to define the role's access to each functional area.

Web Interface Allows you to customize the web interface for the role by defining the web pages and online help content the users can access.

Knowledge Management Allows you to specify the Knowledge Management privileges for the role

KT Document Visibility Allows you to specify which document statuses the role is allowed to view (for example, draft, retired, and published).

Tabs Allows you to define the tabs that appear when a user assigned to this role is logged in.

Report Web Forms Allows you to define the report web forms that are available to this role.

Go Resources Allows you to specify which record types appear in the Go drop-down list for the role.

Define Knowledge Management Privileges

You can define the areas of Knowledge Management you want to be available for this role.

> **NOTE**
>
> Not all of the Knowledge Management privilege options are available for the Customer and Employee roles. For access to all of the options, you must select the Analyst web user interface type on the Web Interface tab and save the role definition.

Follow these steps:

1. Select the Knowledge Management tab on the Create Role or Role Update page.

The Knowledge Management fields appear.

2. Fill in the following fields as appropriate:

Open Issue/Request
Allows the role to open an issue and request.

Open Issue/Request based on Document
Allows the role to open an issue and request based on a Knowledge Document.

Bypass Approval Process
Allows the role to bypass the knowledge approval process.

Change Approval Process Template
Allows the role to change the approval process template.

Add, Edit, Copy and Paste Categories
Allows the role to manage Knowledge Categories.

Delete Multiple Categories and Documents
Allows the role to delete parents and child categories and documents.

Delete Category
Allows the role to delete a Knowledge Category.

Create Document
Allows the role to Create a Knowledge Document.

Create Document with Attachments
Allows the role to create a document with attachments.

> **NOTE**
>
> If this option is disabled, the Attach File and Attach File from Library buttons do not appear for that role.

Delete Knowledge
Allows the role to delete knowledge documents, forums and files.

View Forum
Allows the role to view forums.

Create Forum
Allows the role to create a forum.

Reply Forum
Allows the role to reply to a forum.

Edit Forum
Allows the role to edit a forum.

View File
Allows the role to view files.

Edit File
Allows the role to edit files.

View External Repository
Allows the role to view the external repository.

View Related Tickets
Allows the role to view related support tickets.

Add Bookmark
Allows the role to add bookmark to My Bookmarks.

Use Preferences
Allows the role to use preferences.

Allow Export
Allows the role to export knowledge packages.

Allow Import
Allows the role to import knowledge packages.

Edit Functional Access Areas for a Role

A *functional access area* is a group of objects that let you restrict user access. Many functional access areas are available for fine-tuning access levels for roles. You can edit access levels for roles or assign roles to new functional access areas to limit user access.

NOTE

Note: You can also use Web Screen Painter to edit functional access areas. Web Screen Painter refers to a functional access area as a function group. For help on assigning tables to function groups, see the *Web Screen Painter Help*.

Follow these steps:

1. Select the Function Access tab on the Create Role or Update Role page.
   The Functional Access to Role List appears with the following fields:
   Access Level Identifies how the role can use the functional access area.
   Comments Describes how the role can use the functional access area.
2. Click Edit in List.
3. Click a function name.
   The row highlights.
4. Select from one of the following Access Levels:
   None Denies the role access to the function object. View Grants read-only capability to the function object. Modify Grants read/write access to the function object.
   The Access Level highlights.
5. (Optional) Click Change All.
   The Access Level highlights for every functional access area.
6. Click Save.
   A message confirms the change. The role can immediately use the functional access area at the specified access level.

Assign Tabs

You can specify which tabs are available when a user assigned to this role logs in to the system.

Follow these steps:

1. Select the Tabs tab on the Create Role or Role Update page.
   The Tab List page appears.
2. Click Update Tabs.
   The Tabs Search page appears.
3. Enter the search criteria to display the tabs of interest and click Search.
   The Tabs Assigned Update page appears, listing the tabs that matched the search criteria.
4. From the list on the left, select the tabs you want to assign to this role. To select multiple items, hold down the CTRL key while clicking the left mouse button.
   Important! Include *only* tabs that contain forms that are in the form group assigned to the role you are creating or editing. For example, assigning the Customer tab or Employee tab to the Administrator role causes an error when users attempt to access that tab. For a listing of the web forms included in each form group, see the *Administration Guide*.

   NOTE

   The role's form group is specified in the Customization Form Group field on the Role Detail page, and is also displayed in the Form Group column on the Role List page.

   WARNING

   Including more tabs than your browser window can display causes some tabs to be inaccessible to the user.

5. When you have selected all the tabs you want, click

   The selected tabs move to the Tabs Assigned list on the right.
6. Click OK.
   The Role Detail page appears, with the selected tabs listed on the Tabs tab.

Assign Go Resources

Users can search for items in by a number, name, or ID. In the upper right-hand corner of the main window, there is a drop down list containing the searchable record types. These searchable record types are referred to as "go" resources. You can define the "go" resources available for each role.

To assign go resources to a role, **follow these steps:**

1. Select the Go Resources tab on the Role Detail page.
   The Go Resources List page appears.
2. Click Update Go Resources.
   The Web Form Search page appears.
3. Enter the search criteria to display the desired resources and click Search.
   The Go Resources Assigned Update page appears, listing the resources that match your search criteria.
4. From the list on the left, choose the resources you want to be available for this role. To choose multiple items, hold down the CTRL key while clicking the left mouse button.
5. When you have selected all the resources you want,
   click
   The selected resources move to the Go Resources Assigned list on the right.
6. Click OK.
   The Role Detail page displays the selected resources listed on the Go Resources tab.
7. Choose the resource you want to be the default for this role, and click Set Default.
8. Click Save.
   The available Go Resource list and default setting are saved.

To change a go resource's relationship to a menu bar, **follow these steps:**

1. Select the Go Resources tab on the Role Detail page.
   The Go Resources List page appears.
2. Click the Menu Bar link for the go resource.
   The Go Resource Relationship Update page appears.
3. Click the Menu Bar link.
   The Menu Bar Search page appears.
4. (Optional) Complete the filter fields to list only the menu bars of interest.
5. Click Search.
   The Menu Bar List page displays the menu bars that match your search criteria.
6. Select the desired menu bar.
   Your selection appears in the Menu Bar field on the Go Resource Relationship Update page.
7. Click Save.
   The Go Resources tab of the Role Detail page displays your update.

Security and Role Management - Contacts

You can define every person who uses your system on a regular basis, such as analysts and customers. These people are referred to as *contacts*.

Create a Contact

Every person who uses your system on a regular basis needs to be defined as a contact.

To create a new contact

1. Select Security and Role Management, Contacts on the Administration tab.
   The Contact Search page appears.
2. Fill in the filter fields as desired (or leave all filter fields blank to see a listing of all contacts), then click Search.
   The Contact List page appears.
3. Click Create New.
   The Create New Contact page appears.
   Note: If your installation is configured to access an LDAP server, and has the necessary options installed, the Merge LDAP button appears in the upper right corner of the Create New Contact page.
4. Complete the fields as appropriate for the contact.
5. (Optional) Use the controls available on the tabs at the bottom of this page to configure the contact.
6. Click Save.
   The contact record is saved and the Contact Detail page appears. The following buttons are now available for configuring the contact:
   - **Update Environment** -- Displays the Configuration Item/Asset Search window for the contact or organization, where you can specify search criteria for the assets you want to consider. When you click Search, the Environment Update window is displayed, where you can add and remove assets for this contact or organization.
   - **Update Groups** -- Displays the Group Search window, where you can specify search criteria for the groups you want to consider for this contact. When you click Search, the Groups Update window is displayed, where you can add and remove groups for this contact.

Enter Contact Remarks

You can enter notes regarding a contact on the Remarks tab. These notes can be anything related to the contact's role, or any other pertinent information that needs to be saved to the contact detail record.

To enter contact remarks:

1. On the Contact Detail page, select the Remarks tab.
2. Type the remarks in the Contact Notes field.
3. Click Save to save the remarks to the contact detail record.

Tab

A *tab* is a graphical display entity that links to a role to present features to the users of that role. When a user logs in to the system, the main window displays the tabs assigned to the user's default role.

You can configure tabs to include the following kinds of display features:

- A starting page (default web page) that is displayed when a user selects the tab. The starting page is a required element of all tabs. You can assign only one starting page to each tab.
- A menu bar, which presents drop-down lists of commands, such as File, View, and Search commands. The menu bar is optional. You can assign only one menu bar to each tab.
- A toolbar, which presents tool buttons for easy access to commonly used features.

Tabs define major subdivisions in the web interface main window. Each tab is configured to expose an appropriate set of user interface features to the role or roles that use it.

You can associate one or more tabs with a role. Each tab has a sequence number that controls its display order in the main window. If only one tab is associated with a role, the tab's starting page is displayed and not the tab.

Several predefined tabs are provided. You can assign the predefined tabs to a role, modify the predefined tabs, and create custom tabs.

Create a Tab

You can create your own customized tabs to display on the main page. You can then define the tabs that display when users of different roles log into the system.

To create a tab

1. Select Security and Role Management, Role Management, Tabs on the Administration tab.
   The Tab List page appears.
2. Click Create New.
   The Create New Tab page appears.
3. Fill in the fields as appropriate.
4. Click Save.
   The tab definition is saved and the Tab Detail page appears.

Tenants

You can define Tenants from the Security and Role Management node on the Administrative Interface. You can set the Service Provider tenant and add subtenants to tenant groups.

Create a Tenant

You can use the product to create a tenant.

To create a tenant, **follow these steps:**

1. Select Security and Role Management, Tenants on the Administration tab.
   The Tenant List page appears.

   NOTE

   The Security and Role Management, Tenants option is available only when multi-tenancy is installed (either on or setup).
2. Click Create New.
   The Create New Tenant page appears.
3. Complete the editable fields if necessary:
   Name
   Displays the tenant name.
   Service Provider
   Identifies whether a tenant is the service provider. The first created tenant is always the Service Provider.
   Tenant Number
   (Information Only) Displays the tenant number. This field is not used by CA SDM.
   Record Status
   Sets the tenant to Active or Inactive.
   Parent Tenant
   Specifies another tenant above this tenant, making this tenant a subtenant in a tenant hierarchy.
   Subtenants Allowed

Allows this tenant to have subtenants. The tenant cannot modify the setting.
Tenant Depth
(Information Only) Indicates the tenant depth of this tenant.
Supertenant Group
(Information Only) Identifies the system-maintained tenant group that contains this tenant and all tenants above it in the tenant hierarchy.
Subtenant Group
(Information Only) Identifies the system-maintained tenant group that contains this tenant and all tenants below it in the tenant hierarchy.
Foreign Key Group
(Information Only) Identifies the system-maintained tenant group that contains tenants that can be referenced from an SREL in data that belongs to this tenant. The foreign key group is the same as the supertenant group.
Related Tenant Group
(Information Only) Identifies the system-maintained tenant group consisting of both the supertenant and subtenant groups for this tenant.
Terms of Usage
Specifies the Terms of Usage statement for the tenant.
Logo
Specifies the URL for the tenant logo file, which can be any web image type.
Location
Displays the Location lookup page.
Contact
Displays the Contact lookup page.

NOTE

If no contact is associated with the respective tenant, the Email Address and Pager Email Address fields are inactive.

4. Click Save.
5. The tenant is created.
6. Close the window.
7. Right-click the Tenant list and click Refresh.
8. The Tenant List is updated and displays the created tenant.
9. (Optional) To assign this tenant to user-defined tenant groups, click Update Tenant Groups on the Tenant Groups tab.

Tenant Groups

You can define tenant groups for the purpose of classifying and managing tenants. You can assign new tenants to these groups.

Create a Tenant Group

You can use the product to create a tenant group.

To create a tenant group

1. On the Administration tab, select Security and Role Management.
2. Click Tenant Groups.
   The Tenant Group List appears.

   NOTE

   The Security and Role Management, Tenant Groups option is available only when multi-tenancy is installed (either on or setup).
3. Click Create New.
   The Create New Tenant Group page appears.

4. Complete the following fields:
   Tenant Group Name Displays the name of the tenant group. Record Status Sets the tenant group as active or inactive. Description Displays a description of the tenant group.
5. Click Save.
   The tenant group is created.
6. Close the window.
   The Tenant Group List appears.
7. Right-click the Tenant List and select Refresh.
   The Tenant Group List is updated.
8. Click Update Tenants on the Tenant Group Detail page to add tenant members to the group.

Web Forms

You can create your own customized web forms to display as the starting pages for your custom tabs. Web forms also can be a report to display as the starting page for a tab, a "go button" resource, or a URL to display anywhere in your system.

Create a Web Form

You can create your own customized web pages to be the starting pages for tabs, reports to display on tabs, "go button" resources, or another URL.

To create a web page

1. From the Administration tab, navigate to Security and Role Management, Role Management, Web Forms.
   The Web Forms List page appears.
2. Click Create New.
   The Create New Web Form page appears.
3. Fill in the fields as appropriate.
4. Click Save.
   The web page definition is saved and the Web Form Detail page appears.

Building CA Service Catalog

This section contains the following articles:

- Guidelines for Modifying Catalog Content
- Add Custom Fields to the User Interface
- Modify the Category, Class, and Subclass Lists
- Modify the User and Service Approval Level List
- Modify the Request Status List
- Modify the Request Priority List
- Modify the Typefaces Available for Notes in Requests
- Modify XSL, XML, JavaScript, and Image Files
- Modify the Branding
- Add a Custom Time Zone
- Customize the Online Help
- Use Web Services to Automate Business Processes
- Use API Plug-ins to Load Data into Policies and Forms

Guidelines for Modifying Catalog Content

When you modify the catalog content, follow these guidelines, mostly in sequential order, but with iteration as needed:

- Install all content onto a test system, and then browse and understand the default catalog content.
- Review the frequently asked questions.
- Understand the basic structure of catalog entries.
- Understand the main parts of a sample service specification for further details.
- Illustrate the fulfillment of the service as steps in a business process that you can follow logically.
- Create a detailed logical design to help you design the service in the CA Service Catalog.
- Determine what parts of the default catalog you want to modify in your production catalog.
- On subsequent installations, load only those sections of the content that fit into your test catalog.
- Configure all customizations in a test catalog. Test and refine them until you approve them.
- Copy only the approved modifications into your production catalog.
- Consider maintaining a separate development catalog in addition to a test catalog. Keeping the two separate ensures that testing and development can occur in parallel.

Frequently Asked Questions

The following information provides answers to frequently asked questions and includes instructions for managing your catalog content.

- **How do I modify the catalog content?**
  Modify the catalog content by using the standard CA Service Catalog component of the web browser user interface.
- **How do I back up, import, and export catalog content?**
  Use the IXUtil utility to back up the content. You can then export this content from one system, and to import it into another system.
- **How do I make bulk changes to the catalog content?**
  Make bulk changes to the catalog, as follows:
  a. Export the catalog folder or service to be changed, using the IXUTIL utility.
  b. Edit the exported XML file using a text editor.
  c. Delete and reimport the folder into the catalog.

  This process works well for simple changes in descriptions and naming conventions. For example, suppose that your IT organization is named General Information Services (GIS). In that case, consider a bulk change to replace the word "IT" with "GIS" within the catalog.
  You must be familiar with using a text editor for editing the XML files. Be especially careful to use a text editor like Notepad, not a word processor like Microsoft Word.
  Also follow these guidelines:
  – Back up frequently. If referential integrity issues occur, the import process does not always import all content.
  – Review the ixutil.log in the USM_HOME\logs\install folder after each import. Review it for duplicate name errors and other errors that incorrect syntax in the imported data can cause.
- **How do I copy catalog entries from test catalogs to production catalogs?**
  Copy catalog entries from test catalogs to production catalogs, as follows:
  a. Use a test catalog to modify the default catalog content.
  b. Back up the test catalog.
  c. Export the updated catalog content from the test catalog.
  d. Import the updated catalog content into your production catalog.

  The two entities included in your catalog are the catalog entries and the associated images and URL links.
  Use the IXUtil utility to export the catalog entries from the test catalog. Use the same utility to import the exported content into the production catalog.
  The catalog entries include folders, services, service option groups, and service option elements.

Copy the following entities to the production environment separately: category, class, and subclass definitions (category.xml), images, and status definitions.

- **How do I create bulk contacts in catalog?**
  You can create bulk contacts using the 'UserService' web service provided by catalog. Run the "createUser" method for contacts creation in Catalog. You can loop the web services method multiple times for each contact to be created.
  Below is the sample envelope for single user creation using web services:

```
You can create bulk contacts using the 'UserService' web service provided by catalog. Run the "createUser"
method for contacts creation in Catalog. You can loop the web services method multiple times for each
contact to be created.
Below is the sample envelope for single user creation using web services
<soapenv:Envelope xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://
www.w3.org/2001/XMLSchema" xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://
services.soap.usm.ca.com" xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/">
<soapenv:Header/>
<soapenv:Body>
<ser:createUser soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">
<sessionID xsi:type="xsd:string">SESSIONID</sessionID>
<userObj xsi:type="urn:User" xmlns:urn="urn:usmUserService">
<userID xsi:type="xsd:string">USERID</userID>
<firstName xsi:type="xsd:string">FIRSTNAME</firstName>
<lastName xsi:type="xsd:string">LASTNAME</lastName>
</userObj>
</ser:createUser>
</soapenv:Body>
</soapenv:Envelope>
```

Catalog Entries

As an administrator, you use the CA Service Catalog to create and maintain catalog entries. The following types of catalog entries exist:

Understand these types before you create or modify catalog entries.

Many entries in a single container can be difficult for catalog users to navigate. Therefore, as a general guideline, we recommend that you limit the number of entries in a container to ten. For example, verify that no folder contains more than ten sub-folders or services. Similarly, verify that no service contains more than ten service options.

NOTE
The out of the box content does *not* add new statuses to the predefined statuses in CA Service Catalog.

Folders

The installation process creates the folders that you select. If you select all folders, then the installation process creates the following top-level folders:

- IT Services
- Telecom Services
- Network Services
- Application Services
- Project Services
- Corporate Services
- Personnel Services
- Facilities Services
- Reservation Services

The names of the folders and services in your catalog must be unique.

If you create your own folders, you can name them any unique name. You can modify the content from the predefined folders. We recommend short descriptive names for folders.

Services

The catalog supports services that contain multiple service option groups. In this starter implementation, however, most services contain one associated service option group. As the catalog designer, you decide to use either one or multiple service option groups per service. In both cases, we recommend that you organize related service option groups in services logically and intuitively.

Focus on the user when designing services and service option groups. A common flaw in catalog design is placing service option groups in a non-intuitive service or folder. When service option groups in a service or folder are unrelated, users have difficulty finding the services that they want. For example, consider a service option group for a network design program. You could place it in a service or folder named Development Tools. However, such a service option group fits better in a service or folder named Miscellaneous Software. A network administrator is more likely to view the latter folder first for this program.

The catalog provides a search tool that can be helpful. This tool supplements thoughtful design, but does not replace it.

Service Option Groups

Most service option groups in the predefined catalog are named the same as the corresponding service, for simplicity. They have multiple service options that are contained within them.

As your catalog becomes more mature, this one-to-one relationship between services and service option groups is likely to decrease. Some examples of having multiple service option groups in a service exist in the "Procure Laptop" and "Procure Desktop" services. The hardware configuration of laptops and desktops have differences. However, the standard and optional software that is bundled with them is typically identical. Therefore, the catalog does not duplicate the service options that are related to software choices in the two services. Instead, both services include service option groups for standard and optional software bundles.

The naming conventions become more important as the number of services and service option groups increases. The naming standards are as follows:

- All folders and services have unique names.
- If a service contains only one service option group, their names are the same.
- If a service contains multiple service option groups, one service option group has the same name as the service, and the others have unique names.

Service Options and Service Option Elements

A service option is the most basic element that users can request or subscribe to in the catalog. A service option consists of one or more service option elements. The service options in the Best Practice content have the following service option elements. Your modified catalog can have more or fewer service option elements, depending on its design.

- **Short Description**
  Specifies a plain text field (not rich text) that describes the service being requested or subscribed to.
  Use this option to describe the service in cases where rich text or HTML is not processed properly. For example, the workflows use this column to title the email.
- **Long Description**
  Specifies a rich text field.
  This text describes the service in detail. This text can also include a hyperlink to an internal web page containing more information about the service.
- **Rate**
  Specifies the cost of the service option.
  Consumers must understand the cost of the service to the corporation, regardless of whether the request is charged back.
  Because the cost structure at your location is unique, all service option elements have been set to a one-time charge. Determining cost and rate structures depends on a number of factors. In some cases, the rate posted in the catalog is only advisory: The rate reminds users that although they are not charged, the service is not free. In other cases, however, the posted rate is linked to a chargeback policy. In these cases, it is designed to recover the cost of the service. Accounting Component is designed for automating the process of tracking service costs.
- **Service Level**
  Describes the level of service the user can expect. The user can click the "More Information" link to get a description of the basic levels of service that they can subscribe to. The sample information that is displayed when you click the "More Info" hyperlink is contained in "sladescription.html" located in the USM_HOME\filestore\images\offerings directory.
  You can change this location, as follows: Use the CA Service Catalog, Service Offerings, Options Group to modify each service option element in your catalog to reference a file in another location.
- **Special Instructions**
  Specifies more instructions for the user. Many services require users to specify more detail information. An example is a "backup production server" service on which you must back up files and must specify the backup interval. The instructions tell users to place this information in the notes that are associated with a request.
  You can use the Form service option element type to present custom forms to gather more information from the requester.

Images

You can optionally associate images with every folder, service, and service option. All images reside in USM_HOME \filestore\images\offerings.

Images whose size is 32x32 pixels fit best in the catalog.

Images can be in any format suitable for a web browser (For example, .jpg, .bmp).

Many predefined images exist in this directory. The Best Practice content does not use all of them. You can optionally use these predefined images and add others to meet the needs of your organization. Use images that help users searching for a service to find the service they need.

Category, Class, and Subclass

The first service option element in a service option determines the Category, Class, and Subclass for the other items in that row. The Best Practice: Foundation catalog uses several Category/Class/Subclasses which are defined in category.xml. If you are building your catalog "from scratch," you can use any Class/Subclass structure. The Category drives associated Workflow processing and other downstream activities. So, do not change the predefined Category corresponding numeric values.

The predefined service option elements reference the predefined category, class, and subclass values in the category.xml file. If you build your catalog on top of the best practices content, do not *change* the predefined settings in the file. Instead, *add* new categories, classes, or subclasses in the file, if necessary.

Service Specification

When you design your catalog, consider the following service specification factors. These attributes help your customers understand the service.

Attribute Name	Definition
Service Description	Brief description of the service which explains what the service includes.
Service Exclusions	Brief description regarding what is not included in the service.
Service Core Dependencies	Listing of the underlying services and processes that are necessary for this service.
Service Options	The various components and options available to users when ordering the service from the catalog.
Service Hours	The time period during which the service is typically available.
Service Maintenance and Planned Down Time	The time period during which the service is typically not available due to scheduled maintenance.
Service Availability	The targeted percentage (of the defined Service Hours) during which service is available for requests.
Service Owner(Performance Measure Owner)	The person responsible for the service.
Service Users	The intended group of requesters for this service.
Performance Measures/Descriptions	One or more defined performance measurements for the service fulfillment with a base line target.
Key Performance Indicators	The performance measurement which is the primary indicator of the performance of the fulfillment of this service.
Key Goal Indicators	The overall company benefit from this service.
Data Collection Method/Frequency	An overview of the intended data collection method and frequency. If tools and methods are known, specify them in this attribute.
Cost Categorization (to Business Unit)	Definition of the cost categorization of the service regarding the following aspects of costs: direct and indirect, capital and operational, fixed and variable, cost type, cost elements, cost units.
Charging Policies	The charging policy for this service. The template gives examples both for internal and external providers.

Sample Service Specification

This sample server specification illustrates the following costs that are associated with the "Personal Computer (PC)" service:

- Direct (associated costs that are allocated directly to the IT organization) for the request of the other vendor software.
- Capital for the acquisition of the new PC.
- Variable depending on the number of new PC requests.

Charges are based on the following elements:

- Internal Service Provider
- Initial setup of the service at the time of SLA signature plus Hardware
- Fixed subscription per year
- Cost of usage per request
- External Service Provider
- Market Standard

Element	Description
Service Name	Personal Computer
Service Description	New Desktop or Laptop
Service Exclusions	No support, No Equipment Returns
Service Core Dependencies	Architectural standards, including approved server hardware lists Procurement process for IT infrastructure equipment, including appropriate vendor agreements
Service Options	Options, such as Standard Laptop, Deluxe Laptop, Standard Desktop, Deluxe Laptop
Service Hours	24 hours per day, 7 days per week, 365 days per year This value can vary, depending on Service Level that the customer selects.
Service Maintenance and Planned Down Time	Last Sunday of every month from 2AM - 3AM EST This value can vary, depending on Service Level that the customer selects.
Service Availability	Availability Target: 99% of Service Hours This value can vary, depending on Service Level that the customer selects.
Service Owner (Performance Measure Owner)	John Doe - Director, IT Organization
Service Users	Finance, Marketing, Operations, and Customer Service Organizations
Performance Measures and Descriptions	Acknowledgment of order Number of hours that have elapsed between user submitting the request and user receiving acknowledgment of the order. Target: Average less than eight hours Order Fulfillment: Normal Number of days that have elapsed between user submitting the request and user receiving acknowledgment that the requested items are available. Target: Average less than seven days
Key Performance Indicators	Number of days that have elapsed between the time the user submitted the request and the time the vendor notified the user that order was shipped
Key Goal Indicators	Better capacity planning and inventory management through standardized procurement processes Compliance to SLAs by third-party vendors and providers Standardization of computers throughout the organization
Data Collection Method/Frequency	Data is collected for “time of request”, “time of acknowledgment” and “PC available” by feeds. The feeds are available from tools that are used at each event. If the feeds are not available, collect the data daily.

Cost Categorization	Direct and Indirect - Direct cost of service Capital and Operational Capital Fixed and Variable - Fixed based for subscription - variable per request Cost Type - Service + Hardware Cost Elements - Hardware + Services Defined in Service Dependencies Cost Units - Per Request + Hardware
Charging Policies	Internal Service Provider - Fixed Setup Price + Hardware External Service Provider - Market Price

Add Custom Fields to the User Interface

On the CA Service Catalog user interface, you can optionally add custom field that is related to business units, accounts, or users. You can add a new field to meet a custom requirement for your organization or one of your customers. To do so, follow this process:

Step 1 - Review the Additional Data Fields

The Business Unit, Account, and User schemas in the CA Service Catalog database tables provide extra data fields. By default, these fields do not have labels. The fields do not appear on the CA Service Catalog user interface.

The additional data fields and their data types follow.

- The additional data fields (and their types) for business units in the usm_tenant table follow:
  data1: nvarchar(32)data2: nvarchar(32)data3: nvarchar(32)data4: nvarchar(64)data5: nvarchar(64)data6: nvarchar(128)data7: nvarchar(128)
- The additional data fields (and their types) for accounts in the usm_account table follow:
  data1: nvarchar(32)data2: nvarchar(32)data3: nvarchar(32)data4: nvarchar(64)data5: nvarchar(64)data6: nvarchar(64)data7: nvarchar(128)
- The additional data fields (and their types) for users in the usm_contact_extension table follow:
  data1: nvarchar(512)data2: nvarchar(512)data3: nvarchar(512)data4: nvarchar(512)data5: nvarchar(512)data6: nvarchar(512)data7: nvarchar(512)

Step 2 - Review the Sample custom.xml File

The custom.xml file lists all additional data fields for CA Service Catalog. Optionally expose an extra data field on the CA Service Catalog user interface by adding a label to the field.

The custom.xml file can be different based on the language that is chosen for the system. This file is located in a different folder for each language. For example, for English (icusen), the custom.xml file is located in the USM_HOME\view \webapps\usm\locale\icusen folder.

Step 3 - Expose Additional Data Fields

You can expose the data fields fields to add custom fields to the user interface.

Follow these steps:

1. Edit the custom.xml file for the language of your system. For example, for English, edit the USM_HOME\view\webapps \usm\locale\icusen\custom.xml file.
2. Type the label for the field between the start and end tags for the appropriate field (data1-data7) for the appropriate object: account, business unit (tenant), or user.
   This label appears on the user interface to help users identify the purpose of the field.
3. Save the custom.xml file.

4. Verify that the label appears as specified when you view the related object on the GUI. Examples include the associated add, edit, and profile pages.

Example: Expose Additional Fields in the User Interface

This example configures the account data1 and account data 5 fields in the custom.xml file to expose Cost Center and Department data on the GUI:

```
<account>
    <data1>Cost Center</data1>
    <data2></data2>
    <data3></data3>
    <data4></data4>
    <data5>Department</data5>
    <data6></data6>
    <data7></data7>
</account>
```

NOTE
To expose a field without specifying a new label on the user interface, specify a space as the value of the label.

Modify the Category, Class, and Subclass Lists

Each service option in the catalog has a category, class, and subclass assigned to it. CA Service Catalog assigns a value to each category, class, and subclass in the category.xml file.

Each <option> section in the category.xml file represents a separate category and contains one or more <class> sections. Each of these sections contains one or more <subclass> sections.

The category.xml file can be different based on the language that is chosen for the system. The file is located in a different folder for each language. For example, for English (icusen), the category.xml file is located in the USM_HOME\view \webapps\usm\locale\icusen\billing folder.

Each service option in the catalog has a predefined category, class, and subclass assigned to it. You can update the existing values or can add new values to meet the needs of your organization or customer.

NOTE
After you add a category, class or subclass value in the category.xml file, do *not* remove it. Ensure that you do not change the meaning of the category values, because business logic in the product is based on the category values.

Follow these steps:

1. Review the inline comments in the file. Verify that synchronized sections of the file remain synchronized. Verify that you do not use, certain status values.
2. Edit the appropriate category.xml file for the language of your system. For example, for English, edit the USM_HOME \view\webapps\usm\locale\icusen\billing\category.xml file.
3. Add a line or section for the category, class, or subclass you want to add. Select an unused numeric value for the category, class, or subclass you are adding.

   NOTE
   To add a new category, add a new “option” section containing at least one class and subclass. To add a new class, add a new “class” section containing at least one subclass.

   Use a unique numeric value within the list of related objects. For example, suppose you add a new class to the Other category. In that case, you cannot use a value of 10, because the IT class already uses that value.
4. Update any existing values as needed; do *not* use the same value twice.
5. Save the category.xml file and test by using the CA Service Catalog user interface screens.

Example: Add a New Subclass

To add a new subclass named “Mouse” to the IT class of the Hardware category, modify the category.xml file as illustrated in the following example:

```
<option value="1" name="Hardware">
      <class value="10" name="IT">
          <subclass value="10" name="Desktop" />
          <subclass value="20" name="Laptop" />
          <subclass value="30" name="Monitor" />
          <subclass value="40" name="Memory" />
          <subclass value="50" name="Printer" />
          <subclass value="60" name="Server" />
          <subclass value="70" name="Storage" />
          <subclass value="71" name="Mouse" />
          <subclass value="999" name="Other" />
      </class>
    ...
  </option>
```

Modify the User and Service Approval Level List

Each user is assigned an approval level, in its user profile. Each service that requires approval is also assigned an approval level, in its service definition.

When a service uses the “System approval process,” the approver requires an approval level equal to or greater than the approval level of the service. Otherwise, the approver cannot approve or reject the service.

The approval level values are maintained in a file named approval_shared.xml. This file can be different based on the language that is chosen for the system. The file is located in a different folder for each language. For example, for English (icusen), the approval_shared.xml file is located in the USM_HOME\view\webapps\usm\locale\icusen folder.

You maintain the approval levels for users and services in a file named approval_shared.xml. You can add or change the values in this file to meet the needs of your organization.

NOTE
After you add an approval level to the file, do not remove it.

Follow these steps:

1. Edit the appropriate approval_shared.xml file for the language of your system, using an editor, such as Notepad. For example, for English, edit the approval_shared.xml file in the USM_HOME%\view\webapps\usm\locale\icusen folder.
2. Add a line for the approval level that you want to add. Select an unused numeric value for the approval level you are adding.
3. (Optional) Update an existing name or value in the file.
4. Save the approval_shared.xml file.
5. Verify that the new approval level appears as an option when you edit user profiles and service definitions on the product interface.

You have maintained the approval levels for users and services.

Example: Add a New Approval Level

This example illustrates a new approval level added to approval_shared.xml file. The name of the level is Director. The approval level value is 60.

```
<?xml version="1.0" encoding="UTF-8"?>
<shared>
```

```
  <approval_level>
   <option value='0'>Level 0</option>
   <option value='10'>Level 10</option>
   <option value='20'>Level 20</option>
   <option value='30'>Level 30</option>
   <option value='40'>Level 40</option>
   <option value='50'>Level 50</option>
   <option value='60'>Director</option>
  </approval_level>
 </shared>
```

Modify the Request Status List

This article contains the following topics:

Each service and service option in a request has a status. In addition, the request has an overall status. CA Service Catalog supplies an extensive list of status values by default. Modify status values for the approval and fulfillment phases of the request life cycle, by editing the requestshared.xml file. You can also change the spelling of existing status values.

You maintain the request status values in the requestshared.xml file. This file can be different based on the language that is chosen for the system. The file is located in a different folder for each language. For example, for English (icusen), the requestshared.xml file is located in the USM_HOME\view\webapps\usm\locale\icusen\request folder.

The request status of the *entire* request is visible on the Pending Actions page. The request statuses of individual items in the request are visible in the Item Status drop-down list on the request-related user interface pages: Request Details, Approve Request, Fulfill Request, and Push Through Request. Updates that you make to the request status values in the requestshared.xml file are reflected in the options that users see on those pages.

Follow these steps:

1. Become familiar with the requestshared.xml file.
2. Back up the original requestshared.xml file and save it for reference.
3. (Optional) Modify existing status values or add additional request statuses.
4. (Optional) Hide request statuses.
5. (Optional) Restrict the status changes available for a request item based on its status.
   By default, all options (all statuses) are available for an item always, until the entire request is completed. In other words, you can change the status of request items to any value at any time. However, your organization can optionally restrict the menu options available for a request item that is based on its status.
6. Test your changes by verifying that they are correctly reflected on the request-related user interface screens.
   In the Item Status drop-down list on the request-related user interface pages, the status is designated with an asterisk (*).

   WARNING
   After you have added and used status numeric values to the default list, do *not* remove them.

Review requestshared.xml

Become familiar with the requestshared.xml file, as follows:

- Understand the purpose of the major sections of the file: request_header, request_item, request_item_approval_action, and request_item_fulfillment_action.
- Understand that the status values in these sections must remain synchronized.
- Understand that the order of the statuses in the status drop-down lists on the GUI matches the order of the statuses in the requestshared.xml file. For example, suppose that the status 800 (Reject) is defined above the status 600

(Approved) in this file. In this case, the Reject status appears above the Approved status in the status drop-down lists on the GUI.

- Review the inline comments and note the *reserved* status values.
- Review the ranges for custom status values to see which ranges are reserved for specific types of status values.

WARNING
Do *not* modify or delete the opening and closing lines that define the default statuses in the requestshared.xml file . Even when you modify statuses, these lines must remain as shown. This requirement applies to both the <request_item_approval> and <request_item_fulfillment> sections. These lines help ensure proper status behavior when the customizations are not used or are defined incorrectly.

```
- <custom_menu current_status_value="default">
status lines
</custom_menu>
```

Major Sections

The major sections of the requestshared.xml file are as follows:

- **request_header**
  Maintains all possible status values for the entire request.
- **request_item**
  Maintains all possible status values for a specific item (such as a service option element or service option group) in a request.
- **request_item_approval_action**
  Maintains all possible status values for a specific item in a request, when the request has been submitted but has not been approved or rejected.
- **request_item_fulfillment_action**
  Maintains all possible status values for a specific item in a request, when the request has been approved but has not yet been fulfilled.

The list of possible status values must be synchronized in the request_header and request_item sections: They must have the same values with same meaning.

Every value in the request_item_approval_action and request_item_fulfillment_action sections must have a matching value in both the request_header and request_item sections. The value in the request_item_approval_action and request_item_fulfillment_action sections must be a complete set or a subset of the values in the request_header and request_item sections.

Ranges for Custom Status Values

When adding a new request status in the requestshared.xml file, define it within the range that is specified for the custom statuses:

- 300 to 399 - custom *submit* status
- 500 to 599 - custom *pending approval* status
- 900 to 990 - custom *approved* status

Add an Additional Request Status

You can modify the request status list by adding request statuses. One common purpose is for approval or rejection of a request by a specific department. For example, the Finance department.

Follow these steps:

1. Edit the requestshared.xml file for the language of your system in a text editor. For example, for English, edit the file in the USM_HOME\view\webapps\usm\locale\icusen\request folder.

2. Add a line, including the number and text for the new status to the following sections: <request_header> and <request_item>.
3. Select an unused numeric status value in the appropriate range for custom status values for the status you are adding.

   NOTE
   If possible, limit the text of the status value to 40 characters. Text longer than 40 characters can be truncated in the drop-down status menu lists and request status fields. In such cases, the entire text string is displayed to catalog users *only* in the tooltip text.
4. Copy that line to the custom section or sections where you want them to appear. Examples include <request_item_approval>, <request_item_fulfillment>, <request_item_stuck_approval_action>, and <request_item_stuck_fulfillment_action>.
5. Delete the text from the line you copied and modify the line to include the statval="*value*" expression.
6. Save the requestshared.xml file.
7. Verify that the changes are correctly reflected on the request-related user interface screens (the Request Details, Approve Request, and Fulfill Request screens).

Example: Add New Approval Statuses

To add approval statuses 500, 700, and 900, all related to financial approval, add the new lines for these statuses to the request_header, request_item, and request_item_approval_action sections of the requestshared.xml file. Examples follow, in **bold**.

Specify the numeric value and text in the request_header and request_item sections. Specify only the numeric value (without the text) in the request_item_approval_action section.

```
<?xml version="1.0" encoding="UTF-8" ?>
...
- <request_header>
  <!--  status values must be synchronized with the status list in request_item, request_item_approval_action
 and/or request_item_fulfillment_action   -->
  <st_1>Pending</st_1>
  <st_2>Completed</st_2>
...
  <!--  400 to 499 are reserved   -->
  <st_400>Pending Approval</st_400>
  <!--  500 to 599 can be used for custom pending approval status   -->
  <st_500>Pending Financial Approval</st_500>
  <!--  600 to 699 are reserved   -->
  <st_600>Rejected</st_600>
  <!--  700 to 799 can be used for custom rejected status   -->
  <st_700>Rejected by Financial Approver</st_700>
  <!--  800 to 899 are reserved   -->
  <st_800>Approved</st_800>
  <st_801>Approval Not Needed</st_801>
  <!--  900 to 990 can be used for custom approved status   -->
  <st_900>Approved by Financial Approver</st_900>
  !--  991 to 999 are reserved   -->
  <st_999>Approval Done</st_999>
...
 </request_header>
- <request_item>
  <!--  status values must be synchronized with the status list in request_header,
 request_item_approval_action and/or request_item_fulfillment_action   -->
  <st_1>Pending</st_1>
  <st_2>Completed</st_2>
```

```
...
  <!--  400 to 499 are reserved   -->
  <st_400>Pending Approval</st_400>
  <!--  500 to 599 can be used for custom pending approval status   -->
  <st_500>Pending Financial Approval</st_500>
  <!--  600 to 699 are reserved   -->
  <st_600>Rejected</st_600>
  <!--  700 to 799 can be used for custom rejected status   -->
  <st_700>Rejected by Financial Approver</st_700>
  <!--  800 to 899 are reserved   -->
  <st_800>Approved</st_800>
  <st_801>Approval Not Needed</st_801>
  <!--  900 to 990 can be used for custom approved status   -->
  <st_900>Approved by Financial Approver</st_900>
  <!--  991 to 999 are reserved   -->
  <st_999>Approval Done</st_999>
...
  </request_item>
- <request_item_approval_action>
  <!--  status values must be synchronized with the status list in request_header and request_item   -->
  <!--  A "default" value for the attribute "current_status_value" indicates these statuses will be listed by
 default in the "item status" menu if no other custom statuses are defined   -->
  <custom_menu current_status_value="default">
  <!--  400 to 499 are reserved   -->
  <!--  500 to 599 can be used for custom pending approval status   -->
    <st_500 statval="500"/>
  <!--  600 to 699 are reserved   -->
  <!--  700 to 799 can be used for custom rejected status   -->
  <st_700 statval="700"/>
  <!--  800 to 899 are reserved   -->
  <st_800 statval="800">Approve</st_800>
  <!--  900 to 999 can be used for custom approved status   -->
   <st_900 statval="900"/>
...
```

Hide Request Statuses

The default list of request statuses can include more options than you need for certain categories. You can hide some of the values in that category. So, users do not see these options on the GUI when they handle requests pending action.

Follow these steps:

1. Edit the requestshared.xml file for the language of your system using a text editor. For example, for English, edit the file in the USM_HOME\view\webapps\usm\locale\icusen\request folder.
2. Edit the line in the <request_header> and <request_item> sections for each status that you want to hide. Enter the comment characters before and after the original expression.
3. Hide the corresponding lines in all other sections that use it, such as the <request_item_approval> or <request_item_fulfillment> section.

   NOTE
   Hide the exact same lines in all relevant sections of the file. Doing so is required for the status to appear correctly in the user interface.
4. Save the requestshared.xml file.

5. Test your changes by verifying that they are correctly reflected on the following request-related pages: Request Details, Approve Request, and Fulfill Request.

Example: Hide Request Statuses

To hide certain default fulfillment-related statuses, enter the comment characters before and after the original expression. Example in **bold**. Enter the comment characters in the request_header, request_item, and request_item_fulfillment_action sections of the requestshared.xml file.

In this example, *before* the comment markers are added, the following statuses are visible on the following request-related pages: Ordered, Shipped, Received, Order Cancelled, Staged, and Configured. *After* the comment markers are added, the following statuses are visible on those pages: Ordered, Shipped, and Configured.

```
- <request_header>
...
  <st_1004>Ordered</st_1004>
  <st_1006>Shipped</st_1006>
   <!--<st_1007>Received</st_1007>-->
   <!--<st_1008>Order Cancelled</st_1008>-->
   <!--<st_1017>Staged</st_1017>-->
  <st_1019>Configured</st_1019>
...
  </request_header>
- <request_item>
...
  <st_1004>Ordered</st_1004>
  <st_1006>Shipped</st_1006>
   <!--<st_1007>Received</st_1007>-->
   <!--<st_1008>Order Cancelled</st_1008>-->
   <!--<st_1017>Staged</st_1017>-->
  <st_1019>Configured</st_1019>
...
  </request_item>
- <request_item_fulfillment_action>
...
  <custom_menu current_status_value="default">
...
  <st_1004 statval="1004"/>
  <st_1006 statval="1006"/>
   <!--<st_1007 statval="1007"/>-->
   <!--<st_1008 statval="1008"/>-->
   <!--<st_1017 statval="1017"/>-->
  <st_1019 statval="1019"/>
...
</shared>
```

Restrict the Status Changes Available for a Request Item

You can restrict the status changes available for a request item that is based on its current status. For example, suppose that when an item is approved (Approved status), you no longer want users to be able to change the status *except* to a fulfillment-related status. You can configure the Item Status drop-down list to display *only* fulfillment-related options for items whose status is approved.

Follow these steps:

1. Edit the requestshared.xml file for the language of your system using a text editor. For example, for English, edit the file in the USM_HOME\view\webapps\usm\locale\icusen\request\requestshared.xml folder.
2. Locate the section of the file in which you want to restrict the available changes for one or more status values.
   You can restrict status changes in any section *except* the <request_item> and <request_header> sections.
3. Decide which status you want to restrict and the statuses to which you permit it to be changed.
   The list of all existing status values appears under the following line:
   ```
   - <custom_menu current_status_value="default">
   ```
4. Add a request status to permit in your list, if necessary. The statuses that you want to permit must exist already.
5. Locate the custom menu section. By default, this section is indented and commented out. Delete the opening and closing comment lines of this section.
   ```
   - <!--
          <custom_menu current_status_value="400">
          <st_400 statval="400"/>
          <st_600 statval="600"/>
          </custom_menu>
     -->
          <custom_menu current_status_value="400">
              <st_400 statval="400">/>
              <st_600 statval="600"/>
          </custom_menu>
          <custom_menu current_status_value="400">
              <st_400 statval="400"/>
              <st_600 statval="600"/>
                   <st_801 statval="801"/>
            </custom_menu>
   ```
6. Verify that the value in the current_status_value="*nnn*" expression in the first line of this section matches the status value that you want to restrict.
   For example, in the previous step, 400 corresponds to the Pending Approval status. Therefore, the status values shown appear on request-related pages when the status value is Pending Approval. If you want to restrict a different status, replace the current value with your new value.
7. Verify that the value in the current_status_value="*nnn*" expression is defined in one of the "<st_*nnn* statval=..." lines, as shown for Pending Approval in the previous example.
8. Copy and paste the additional status or statuses to which you want to allow the status to be changed. You can copy and paste it from the list of status values earlier in the section.
   For example, in the <request_item_approval> section, copy and paste the new "<st_801..." line to the custom_menu section for Pending Approval:
9. Now Approval Not Needed status is added to the statuses to which items in Pending Approval status can be changed.
10. Restrict status changes for another status in the same section of requestshared.xml file, if necessary. Use these guidelines:
    a. Copy the entire custom_menu element and its children.
    b. Paste it after the element that you updated.
    c. Modify the newly copied element.
    d. Modify existing custom_menu element and children, if necessary. For example, in the <request_item_fulfillment> section, suppose you want to restrict items in Pending Fulfillment status to be changed to either Fulfillment Cancelled or Fulfilled. In that case, modify the existing custom_menu element as follows:
       ```
       <custom_menu current_status_value="1000">
           <st_1000 statval="1000"/>
           <st_1999 statval="1999"/>
           <st_2000 statval="2000"/>
       ```

```
</custom_menu>
```

11. Save the requestshared.xml file.
12. Verify that the changes are correctly reflected on the request-related user interface pages (the Request Details, Approve Request, and Fulfill Request pages).

You have restricted the status changes available for a request item that is based on its current status.

Modify the Request Priority List

This article contains the following topics:

Every request has a priority that is assigned to it. CA Service Catalog supplies predefined priority values in the requestinfoshared.xml file. You can add or change the values in this file. You can also use this file to specify which user roles can assign specific priority values to requests. For example, you can add a new priority value "Immediate" that only the Service Delivery role can assign to a request.

The requestinfoshared.xml file can be different for each language and is located in a different folder for each language. For example, for English (icusen), the requestinfoshared.xml file is located in the USM_HOME\view\webapps\usm\locale \icusen\request folder.

You can modify the predefined priority levels:

- Add a new priority level for a specific role.
- Add a new priority level for multiple roles.

Priority Levels

The <priority_levels> section of the request and priority list in the requestinfoshared.xml file defines the numeric and text values for each priority. Each <levels> section defines those priority values that are available to the role code specified in the role attribute. The “default” role specification is used when no section exists for the role of the user.

The priority list appears in the order that is specified in the <priority_levels> section, regardless of role. You can use the defaultSel attribute to specify the default value for a new request according to role. The following table lists the roles and codes:

Role	Code
Catalog User	catalogenduser
Request Manager	requestmanager
Catalog Administrator	catadministrator
End User	enduser
SMA End User	smaenduser
Administrator	administrator
Service Manager	servicemanager
Super Business Unit Administrator	stadministrator
Service Delivery Administrator	spadministrator

You can edit this file to add values or change the spelling of existing values.

Suppose that a request uses a priority value that is *not* available to the role of the user editing the request. In that case, the user sees that priority in the list of priority values. Suppose that the request is set to another priority that *is* available the role of the user editing the request. In that case, the user can see *only* the priorities available to its role.

Example: Customizations for the requestinfoshared.xml File

If the Catalog End User role does not include priority 1 (High), a user who has that role does not see "High" listed in the priority list. Moreover, that user cannot set the priority of a request to High.

Suppose that the following situation occurs:

- A Request Manager who can use all the status values later sets the priority of the request to High.
- A user with the Catalog End User role later edits the request.

In this case, the status of High does appear in the priority list.

Thus, administrators *can* configure the product to prevent a certain role from using a particular priority value. Administrators cannot prevent users in that role from viewing and editing their requests when another user has set the priority.

Add a New Priority Level for Multiple Roles

If necessary for your organization, you can add a new priority level to the predefined priority levels.

Follow these steps:

1. Edit the requestinfoshared.xml file for the language of your system, using an editor, such as Notepad. For example, for English, edit the USM_HOME\view\webapps\usm\locale\icusen\request\ requestinfoshared.xml file.

   NOTE
   requestinfoshared.xml file is applicable only for Service Catalog. To reflect the priority value in Service Point, update the following files:

   - USM_HOME\view\webapps\usm\locale\icusen\request\request_edit.xml
   - USM_HOME\view\webapps\usm\locale\icusen\request\request_create.xml
   - USM_HOME\view\webapps\usm\locale\icusen\request\request_cart.xml
2. Add a line in the priority_levels section for the priority you want to add. Specify a unique numeric value for the new priority level.
   The priority values are listed in the user interface in the order that they appear in this file section.
3. Perform *one* of the following actions:
   – Add a line in the levels section for each role that you permit to use the new priority.
   – Add a line in the levels section for the role. The line makes the new priority available to all users who do not have a role-specific priority list.
4. Save the file.
5. Log in to CA Service Catalog as a user with the role that you modified. Verify your updates on the request-related pages.

Example: Add a New Priority for All Roles

This example adds the following **bold** line to the requestinfoshared.xml file. This example adds a new priority named “Urgent” and makes it available to all users who do not have a role-specific priority list.

```
<priority_levels>
    <priority_6 propval="6">Urgent</priority_6>
    <priority_1 propval="1">High</priority_1>
    <priority_2 propval="2">Medium-High</priority_2>
    <priority_3 propval="3">Medium</priority_3>
    <priority_4 propval="4">Medium-Low</priority_4>
    <priority_5 propval="5">Low</priority_5>
    </priority_levels>
    <priority_level_roles>
<levels role="default">
    <level propval="1" />
    <level propval="2" />
```

```
        <level propval="3" defaultSel="true"/>
        <level propval="4" />
        <level propval="5" />
        <level propval="6" />
    </levels>
    </priority_level_roles>
```

Add a New Priority Level for a Specific Role

If necessary for your organization, you can add a new priority level to the predefined priority levels. You can add a new priority level for a specific role only.

Follow these steps:

1. Edit the requestinfoshared.xml file for the language of your system, using an editor such as Notepad. For example, for English, edit the USM_HOME\view\webapps\usm\locale\icusen\request\ requestinfoshared.xml file.

   NOTE
   requestinfoshared.xml file is applicable only for Service Catalog. To reflect the priority value in Service Point, update the following files:

   - USM_HOME\view\webapps\usm\locale\icusen\request\request_edit.xml
   - USM_HOME\view\webapps\usm\locale\icusen\request\request_create.xml
   - USM_HOME\view\webapps\usm\locale\icusen\request\request_cart.xml
2. Add a level section for the affected role.
3. Perform *all* of the following actions in that section:
   - Specify the role code in the role attribute.
   - Include only the lines from the priority_levels section that you want users in the role to see.
   - Specify the default priority for new requests by using the defaultSel attribute.
4. Save the file.
5. Log in to CA Service Catalog as a user with the role that you modified. Verify your updates on the request-related pages.

Example: Add a New Priority Level for a Specific Role

This example adds a new priority list for users with the Catalog User role. This example sets the default priority to Medium-Low and does *not* allow users to set the priority to High. This example achieves these goals by adding the new section that is shown in **bold**:

```
<priority_levels>
        <priority_1 propval="1">High</priority_1>
        <priority_2 propval="2">Medium-High</priority_2>
        <priority_3 propval="3">Medium</priority_3>
        <priority_4 propval="4">Medium-Low</priority_4>
        <priority_5 propval="5">Low</priority_5>
    </priority_levels>
    <priority_level_roles>
        <levels role="default">
            <level propval="1" />
            <level propval="2" />
            <level propval="3" defaultSel="true"/>
            <level propval="4" />
            <level propval="5" />
    </levels>
        <levels role="catalogenduser">
            <level propval="2" />
```

```
            <level propval="3" />
            <level propval="4" defaultSel="true"/>
            <level propval="5" />
    </levels>
</priority_level_roles>
```

Modify the Typefaces Available for Notes in Requests

Users can add notes to requests by accessing the Request Details page and clicking Add in the Notes section. The Add Notes dialog appears and presents several type faces and type sizes as options. For these notes, the default typeface is Arial, and the default type size is 8-point. In addition, users can apply several formatting and highlighting options to these notes.

By default, all supported type faces are enabled. The Add Notes dialog displays all type faces listed in the <fonts> section of requestshared.xml file. Administrators can optionally limit or expand the typefaces that appear to users in the drop-down list of the dialog.

Users can select a typeface in the drop-down that the local computer being used to display or print the request does not support. In that case, the typeface changes to the default typeface, Arial.

Follow these steps:

1. Open the requestshared.xml file and move to the <fonts> section of the file.
2. Leave the type faces that you want to be available to users for notes in request. Comment out the type faces that you do not want to appear in the dialog.
   Comment the lines whose font you want to exclude from the Add Notes dialog.
   For example, to comment out the Courier New and Bookman Old Style type faces, modify their lines as follows:
   ```
   <!--<courier_new>Courier New</courier_new>-->
   <!--<bookman_old_style>Bookman Old Style</bookman_old_style>-->
   ```
3. Add a new typeface to the list that appears in the drop-down list in the dialog: Enter a new line and specify the new typeface. Use the following convention: If the font name is X Y, use the format <x_y>X Y</x_y>.
   For example, to add a typeface, "my company font", add the following line to the <fonts> section:
   ```
   <my_company_font>my company font</my_company_font>
   ```
   The fonts that you added become available to users for notes in requests.
4. Save and close the requestshared.xml file.
5. Verify that the local computer being used to display or print the request supports the font.
6. Verify your changes by adding a note to a request and reviewing the available typefaces.

You have modified the typefaces available to users for notes in requests.

Modify XSL, XML, JavaScript, and Image Files

CA Service Catalog includes several XSL, XML, JavaScript, and image files. Together, they are used to form every page and every page element in the product. Each file represents one page or a part of a page. For example, a dialog, menu option, form field control, message, or picture.

You can optionally modify any of these files to meet your requirements. To modify XSL, XML, JavaScript, and image files, follow this process:

1. Determine the specific page or part of a page that you want to modify.
2. Locate the file whose name matches the element you want to change. For example:

- To modify the configuration information of any of the integrated products, locate the toolsconfig.xsl file.
- To modify the states of request life cycle, locate the requestshared.xml file.
- To modify the messages that appear when a user tests the connection for a new administration configuration setting, locate the toolsconfig.js and toolsconfig.xsl files.
- To modify the edit button of the Administration, Configuration page, locate the modify.gif and toolsconfig.xsl files.
- Modify the request status list by editing the requeststatus.xml file.

3. Open the file, review its contents, and verify that it controls the element or behavior that you want to change.
4. Copy the file from its original location to the custom location. Use the table as a reference.
5. Modify the file according to your requirements.
6. If you modified a JavaScript or image file, perform this step. Otherwise, skip it.
   a. Copy the modified JavaScript file to the \FileStore\custom\explorer\scripts folder.
   b. Copy the modified images to the \FileStore\custom\images folder.
   c. Locate the XSL file in the custom location in which the JavaScript or image file is used.
   d. Update this XSL file and specify the new custom path name of the JavaScript or image file. To do so, prepend "FileStore/" to the relative path of the JavaScript or image file. Use the following example as a model:

      ```
      <script src="FileStore/custom/explorer/scripts/custom_form_example.js"></script>
      ```

   This action is required because the XSL file references the filestore location for the modified script files.

   WARNING
   If you are using multiple Catalog Component computers, verify that the filestore is shared among *all* of them.
7. Clear the USM_HOME\view\translets folder on all Catalog Component computers: Delete all files in this folder, but do *not* delete the folder itself.
8. Restart all Catalog Component computers.
9. Verify that the changes are working in CA Service Catalog as you intended.

In the following table, the parent folder is USM_HOME/view/webapps/usm. The filestore folder is %USM_HOME/filestore. The folder entries, such as /explorer and /custom/explorer, are subfolders under the parent and filestore folders.

File Type	Original Location in Parent Folder	Custom Location in Filestore Folder
XSL	/explorer	/custom/explorer
XSL	/explorer/request	/custom/explorer/request
XML	/locale/icusen*	/custom/locale/icusen*
XML	/locale/icusen*/request	/custom/locale/icusen*/request
image	/images	/custom/images
image	/images/billing	/custom/images/billing
JS	/explorer/scripts	/custom/explorer/scripts

*The folder name *icusen* applies to English-language implementations only. If you are using a non-English implementation, your locale-specific folder name is different. In such cases, use your locale-specific folder name instead of *icusen*.

Increase the Number of Values for a Drop-Down Variable

An administrator can add a query runtime variable that appears as a drop-down list to users. The limit for the number of values in the list is 1000. If the report query returns more than 1000 values, the system truncates these additional values. So, the user cannot view them in the drop-down list. You can increase the number of values that appear in the drop-down list to be greater than 1000.

Follow these steps:

1. Verify that your implementation has set up a filestore, a single location for shared files.
2. Copy the reportsgenericgetvariables.xsl file from its *parent* folder (USM_HOME\View\webapps\usm\explorer\reports) to the *filestore* folder.
3. Open the reportsgenericgetvariables.xsl file in the filestore and locate the following line:
   ```
   <input type="hidden" name="Args" value="1000"/> <!---1-->
   ```
4. Increase 1000 to the value you want.
5. Save the file.
6. Restart all Catalog Component computers.
7. Run a report and test the increased limit and verify the results.

You have increased the number of values for a drop-down variable.

Modify the Branding

As an administrator, you can modify the look-and-feel of the CA Service Catalog UI. The following are the main categories of look-and-feel elements that you can modify:

- Logos are image files that uniquely identify a company, business unit, or super business unit.
  These logos include the login logo, global logo, and business unit logo. For more information, see Change the logo
- The login page enables a user to access the product.
  The same login page (including the login logo) applies to all users in all business units. For example, images and icons (*except for* logos), menus, and tabs. When applicable, these elements include colors, font name and point size, highlighting, and related specifications. You modify these look-and-feel elements by editing the Cascading Style Sheet (CSS) files for the login page.
- *A* Modify the Theme specifies the settings for several look-and-feel elements. For example, images and icons (*except for* logos), menus, and tabs. When applicable, these elements include colors, font name and point size, highlighting, and related specifications. You modify these look-and-feel elements by editing the CSS files for the theme.
  The look-and-feel of the UI matches the theme of the business unit that you are logged in to. If theme is not set for a business unit, CA Service Catalog checks the business unit hierarchy and it finds a theme. Thus, if a business unit does not have its own theme, it uses the theme of its closest parent business unit. You can use the same theme for all business units. Alternatively, you can optionally create and use different themes for different business units.
- Global page elements appear on several or all product pages. They include the product name, shopping icon, and footer. Global page elements are always the same, on every product page where they appear.
  Like the elements of the login page, global page elements apply to all users, regardless of their business unit. You *cannot* override them with business unit-specific settings. The global page elements also apply regardless of whether you have modified the themes of one or more business units.
  You modify global page elements by editing the includes_shared.xml file.

You can modify any, all, or none of the items in this list. This separation provides flexibility.

> **WARNING**
> Verify that the changes you plan to make to each one are compatible with each other. Such consistency helps provide a consistent look-and-feel to users. For example, use similar colors and elements on both the login page and the product pages.

Upgrade Considerations

Depending upon your implementation and if you have customized any of these elements in previous releases, perform *one* of the following actions:

- Verify that the modified files reside in a *custom* subfolder of the USM_HOME\FileStore folder.
- Recreate your changes by following the instructions in this section to:

- Change the Logos
- Modify the Login Page
- Modify the Theme
- Modify Global Page Elements

Change the Logos

This article contains the following topics:

Logos are image files that uniquely identify a company, business unit, or super business unit. As a service delivery manager, you can modify the logos that are used on the CA Service Catalog UI. Using custom logos helps reinforce branding or other messaging in your organization.

> **NOTE**
> We recommend that you size your custom logo to be approximately the same size as the predefined logo.

Change the Login Logo

CA Service Catalog includes a predefined login logo that you can optionally replace with a custom login logo. The login logo *always* applies to the login page, regardless of whether you have specified different logos for different business units.

Follow these steps:

1. Determine the custom login logo that you want to use.
2. Access the USM_HOME/FileStore/themes/common/images/logo folder on your filestore computer.
3. Copy and rename the predefined login logo (login_logo.png).
4. Copy the custom logo to the same folder. Rename it to the name of the original logo.
5. Access the login page for CA Service Catalog.
6. Verify that the predefined login logo no longer appears and that the custom login logo appears correctly.

You have changed the login logo.

Change the Global Logo

CA Service Catalog includes a *predefined* global logo that applies to all users. You can optionally replace it with a *custom* global logo. The global logo appears in the heading of all product pages (except the login page) and request emails.

> **NOTE**
> If a business unit has its own logo, users who log in to it see the business unit logo instead of the global logo.

Follow these steps:

1. Determine the custom global logo that you want to use.
2. Access the USM_HOME/FileStore/themes/common/images/logo folder on your filestore computer.
3. Copy and rename the predefined global logo (header_logo.png).
4. Copy your custom logo to the same folder. Rename it to the name of the original logo.
5. Log in to CA Service Catalog.
6. Verify that your custom global logo is legible on the UI and in request emails.

You have changed the global logo.

Change the Business Unit Logo

For each business unit, you can optionally specify a *business unit* logo. If you specify this logo, it replaces the *global* logo in the heading on product pages and request emails for users of the business unit. You can update the logos for every

business unit or only for specific business units. For example, you can decide to customize logos only for super tenants directly under the root business unit.

WARNING
If you have enabled multi-tenancy with CA Service Desk Manager, CA Service Catalog ignores any of its settings for business logos. Instead, CA Service Catalog uses the logo or logos that the CA Service Desk Manager setup specifies, if applicable. If no CA Service Desk Manager logo applies, then each business unit uses the CA Service Catalog global logo.

Follow these steps:

1. Determine the business unit logo that you want to use.
2. (Optional) Perform the following actions:
   - Rename the custom logo file intuitively to match its business unit. For example, for a business unit that is named Vienna_123, name the logo Vienna_123_header_logo.png.
   - Create a subfolder that is named "custom logos" in USM_HOME/FileStore/themes/common/images/logo directory.
3. Copy your custom logo to the USM_HOME/FileStore/themes/common/images/logo folder on your filestore computer.
4. Perform *one* of the following actions, whichever applies to your level of administrative access:
   - Log in to the business unit whose logo you want to change.
   - Log in to the root business unit:
     a. Select Administration, Business Units.
     b. Drill down the tree to the business unit whose logo you want to change.
5. Edit the business unit.
6. Enter the URL of the logo that you copied in the Logo field and save your changes.
7. Refresh your browser.
8. Verify that your business unit logo is legible on the UI and in request emails.
9. (If the business unit has child business units) Verify:

- – If the child business unit has its own logo, users who log in to it see the child logo, not the parent logo.
  – If the child business unit does not have its own logo, users who log in to it see the global logo.

Thus, users with access to multiple business units can see different header logos when they log in to each business unit.

You have specified a business unit logo.

Modify the Login Page

You can modify the login page, as follows:

- Modify the login logo.
- Modify the *content* or *text* of the Predefined Login Page.
- Modify the *look-and-feel* of several elements of the login page by editing the logon.css file.

You can modify the login page *without* changing the theme for one or more business units. The same login page applies to all users, regardless of whether you have changed any theme. This separation provides greater flexibility to your organization. This separation also enables you to update the login page alone more quickly and efficiently.

As an administrator, ensure that you have expertise in the following areas before modifying the login page:

- UI design, especially look-and-feel elements
- Standard specifications for CSS files
- Customization of the CSS files, using a CSS file editor for your web browser

Predefined Login Page

The following table lists the elements of the login page that you can customize, except for the login logo. When applicable, each row lists the section of the login.css file that affects the look-and-feel of the element.

GUI Element	File Location under USM_HOME*	Section in Logon.css File
Text for the browser window title and title bar	view\webapps\usm\locale\icusen\logon.xml	.loginpage
Text for product name	view\webapps\usm\locale\icusen\logon.xml	.loginproductname
Login input text	view\webapps\usm\locale\icusen\logon.xml **Note:** This text supplies the User Name, Password, and Advanced-Business Unit fields.	.logininputtext
Copyright text and date	view\webapps\usm\locale\icusen\logon.xml	.loginpagecopyright

USM_HOME is the documentation convention that specifies the local CA Service Catalog installation directory. For 32-bit computers, the default path name is C:\Program Files\CA\Service Catalog. For 64-bit computers, the default path name is C:\Program Files (x86)\CA\Service Catalog for 32-bit installations or C:\Program Files\CA\Service Catalog for 64-bit installations.

NOTE
The favorites icon appears in the browser address bar and the product title bar of every product page, including the login page. The favorites icon is a global page element that you can modify.

NOTE
For the Text elements, the File Name column displays the folder location for an English system (icusen). For other languages, the location is different.

Edit the logon.css file

Modify the look-and-feel of the login page to reinforce branding or other messaging. The same login page applies to all users, regardless of business unit that they are logging in to.

Follow these steps:

1. Open your web browser and access the login page for CA Service Catalog.
2. Back up the USM_HOME\filestore\themes\common\css\logon.css file. *A lways* back up CSS files and other configuration files before editing them.
3. Open the file, using the CSS editor for your web browser.
4. Find the lines that control the look-and-feel specification that you want to update.
   For example, to configure the look-and-feel of the product name, find the following line:

```
.loginproductname {
  text-align: left;
  font-family: CAlibri, Verdana, Arial, Helvetica, sans-serif;
  color: #ffffff;
...
```

   To change the color of the product name from the current color to blue, update the lines:

```
.loginproductname {
  text-align: left;
  font-family: CAlibri, Verdana, Arial, Helvetica, sans-serif;
  color: blue;
...
```

5. Repeat the previous step for other look-and-feel changes that you want to make.

For example, to configure the background color of the login page, find the following lines:

```
.login_page {
  background-color: #00174A;
  text-align: center;
...
```

To change the background color of the login page to yellow and align the affected text to the right, update the lines:

```
.login_page {
  background-color: yellow;
  text-align: right;
...
```

6. Refresh your browser and verify your updates on the login page.
7. Update the elements of the login page to meet your requirements. Verify each change by saving the file and refreshing the login page.

You have modified the login page.

Modify the Theme

A theme specifies the settings for the following look-and-feel elements:

- Images and icons (*except for* logos)
- Menus
- Tabs
- Toolbars
- Wizards

When applicable, these elements include the following specifications:

- Colors (especially background colors)
- Font name and point size
- Highlighting, such as bold or underline
- Position on a page or dialog

You modify these look-and-feel elements by editing the Cascading Style Sheet (CSS) files for the theme. Each theme includes the following CSS files:

- logon.css, which applies to the login page *only*
- main.css, which applies to other product pages

A theme is organized in folders, with one top-level folder for each theme. In addition to the CSS files, a theme includes several other supporting files and several folders. You do not need to edit these additional files and folders. However, you copy them as a group when you copy and modify a CSS file.

The look-and-feel of the UI matches the theme of the business unit that you are logged in to. If theme is not set for a business unit, CA Service Catalog checks the business unit hierarchy until it finds a theme. Thus, if a business unit does not have its own theme, it uses the theme of its closest parent business unit. You can use the same theme for all business units. Alternatively, you can optionally create and use different themes for different business units.

As an administrator, edit the theme for one or more business units *only* if you have expertise in:

- UI design, especially look-and-feel elements
- Standard specifications for CSS files
- Customization of CSS files, using a CSS file editor for your web browser

To customize a theme:

Step 1 - Understand Predefined and Custom Themes

A theme is organized in folders, with one top-level folder for each theme. In addition to the CSS files, a theme includes several other supporting files and several folders. You do not need to edit these additional files and folders. However, you copy them as a group when you copy and modify a CSS file.

In the USM_HOME\filestore\themes folder, CA Service Catalog includes the following top-level folders for each predefined theme:

- **CA_Technologies_R7**
  Specifies the predefined look-and-feel elements of CA Service Catalog.
- **common**
  Contains look-and-feel elements that apply to *all* predefined and custom themes.

  WARNING
  Do *not* copy and modify the *common* folder.

If you have already created custom themes, your top-level folders for those themes also appear in the USM_HOME \filestore\themes folder.

To create a custom theme, you copy either a predefined folder (such as CA_Technologies_R7) or a custom folder that you created earlier. Afterwards, you modify the CSS file of interest in the folder that you copied.

You store all custom theme folders under the filestore folder. You store them on the same folder level as the CA_Technologies_R7 folder.

Each top-level folder name becomes the name of an option for a theme. When you edit a business unit, you can select a theme for it.

If you update a theme for a specific business unit, the change affects the users who belong to that business unit. The change also affects any child business units that do not have their own theme specified. Child business units inherit the theme of their parent business unit. However, they can optionally override the inherited theme by specifying their own theme.

Step 2 - Create a Custom Theme

You create a custom theme by copying and modifying an existing theme. You can copy and modify either a predefined CA Service Catalog theme or another existing theme that you created earlier.

WARNING
As a best practice, do *not* modify a predefined CA Service Catalog theme directly. Instead, copy and modify it, so that you can efficiently return to the original version, if necessary. *Always* back up the CSS files before editing them.

Follow these steps:

1. Access the computer on which the filestore resides.
2. Find and expand the USM_HOME folders. Expand the \filestore\themes folder. Review the organization of the predefined and custom themes in that folder.

   NOTE
   The name of each top-level folder is an option that you can select when you select a theme for a business unit.
3. Copy the top-level folder of existing theme that you want to use as a starting point for your new theme.
4. Rename the new theme.
   For example, suppose that the existing theme was named Rome_Super_Tenant_A. If the new theme is for a second super tenant, you can name it Rome_Super_Tenant_B. Conversely, if the new theme is for a new child business unit of the parent super tenant, you can name it Rome_Super_Tenant_A--Child-1.

5. Add or edit the business unit for which you want to use this theme. In the Available Branding field, select the new theme that you created.
   For example:
   – To apply the theme to all business units that do not have their own theme, edit the root business unit and apply this theme.
   – To apply the theme to a specific business unit, add or edit the business unit and apply the theme.
   The theme also applies to all child business units that do not have their own theme.
6. Change the new theme by editing its main.css file.

The new theme is ready for modifications.

Step 3 - Change the New Theme

After you have created a theme, you can change it, to give it a unique look-and-feel.

Follow these steps:

1. Log in to CA Service Catalog and note the look-and-feel of the home page.
2. Back up the main.css file in the custom theme folder that you created. A sample path name is USM_HOME\filestore \themes*custom_theme*\css\main.css file.
3. Open the main.css file of your custom theme. Use a suitable CSS editor for your web browser.
4. Find the lines that control the look-and-feel specification that you want to update.
   For example, To change the background color globally on product pages, perform the following actions:
   a. Find the following default setting:
      ```
      td.pagebg{background-image:url(../images/grid/page-bg.png);background-repeat:repeat-x;background-color:#D9E2F3;}
      ```
   b. Delete the following phrase:
      ```
      background-image:url(../images/grid/page-bg.png);
      ```
      The line now appears as follows:
      ```
      td.pagebg{background-repeat:repeat-x;background-color:#D9E2F3;}
      ```
   c. Change the background-color:#D9E2F3 to the color of your choice, for example, background-color:red.
   d. Save the file.

      NOTE
      The setting in this example *does* affect the background color of the *entire* product page. However, the background colors of specific sections of the page *override* the background color of the entire page.
5. Refresh the CA Service Catalog UI. Verify the changes to the product pages.
6. Update other elements of the theme to meet your requirements. Verify each change by saving the file and refreshing the login page. For example, you can customize the look-and-feel of the top-level menu tabs (Home, Service Builder, Accounting, and Administration).

Modify Global Page Elements

Global page elements include both text and icons. The same global page elements apply to all users, regardless of business unit that they are logging in to. The global page elements also apply regardless of whether you have modified the themes of one or more business units.

Global page elements appear on several or all product pages. They include the product name, shopping icon, and footer.

Follow these steps:

1. Back up the USM_HOME\view\webapps\usm\locale\icusen\ includes_shared.xml. *Always* back up XML files and other configuration files before editing them.
2. Open the includes_shared.xml file, using an XML editor.

NOTE
The folder location is for an English system (icusen). For other languages, the location is different.

3. Find the line that controls the text that you want to update. For example,
   – Text for the product name: <product_title> value
   Default: CA Service Catalog
4. Update the lines to the values you want, and save the file.
5. Access the folder that contains the icons that you want to update. For example, USM_HOME\view\webapps\usm\images or USM_HOME\view\webapps\root folders contain icons.

   NOTE
   To view an icon, double-click its file name.
6. Perform the following steps for each icon that you want to change:
   a. Rename the original file to filename_OLD.png.
   b. Copy your new file to the folder, and rename it to the original file name.
7. Log in to CA Service Catalog and verify your updates on the product UI.
8. Copy the icons that you customized to the USM_HOME\filestore\custom\images folder.

   WARNING
   This step is required to help ensure that your updates apply to all users in your organization.

You have modified the global page elements.

Add a Custom Time Zone

You select the time zone for a business unit when you create or edit it. The time zone selection determines the date and time settings for the business unit, including the following components: the Scheduler, outage calendars, and business hours, date and time fields in forms, and the availability dates for services. You can also select a time zone for conditions in policies. CA Service Catalog supplies several predefined time zones that you can use in these settings. You can add a custom time zone.

Follow these steps:

1. Verify that you have set up the filestore.
2. Copy the appropriate sharedxml.xml file for the language of your system to the filestore. For example, for English, copy the file USM_HOME%\view\webapps\usm\locale\icusen\sharedxml.xml to the filestore.
3. Edit the sharedxml.xml file in the filestore using a text editor. Locate the <timezones> tag and the existing time zone entries under it.
4. Enter the custom time zone under the existing entries. Use the format that is shown in the following example:

```
<t53>
        <id>GMT-06:00 America/North_Dakota/Center</id>
        <text>(GMT-06:00) Central Time (US&amp;amp;Canada)</text>
</t53>
```

5. Verify that the new custom time zone is implemented, as follows:
   – Edit a business unit and verify that the available time zones include the new custom time zone.
   – Set the business unit to this time zone. Verify that the time zone is used in affected components.
   – Verify that this time zone works accurately in conditions in policies, if applicable.

   NOTE

   If You have implemented CA Service Catalog in multiple languages, repeat steps 2 through 5 for all other languages.

You have added a custom time zone.

Customize the Online Help

Service delivery administrators can optionally replace the predefined online help with custom online help, for some or all roles. Here, the predefined online help refers to the *CA Service Management Wiki*. Custom online help appears instead of the predefined online help when users with the specified roles click the **Help** button.

> **NOTE**
> If the user selects Administration, Tools, Links, CA Service Management Wiki, the predefined online help *always* appears. Customization for online help does not affect the menu selection that appears when users click the **Help** button.

Follow these steps:

1. Create your custom online help file files and copy them to the USM_HOME\filestore\custom\help\<***language***> folder.
   - ***language***
     Specifies the language of the operating system on which you installed CA Service Catalog. For example, specify en_US for U.S. English or ja_JP for Japanese.
     For example, the file location for your custom English help files must be USM_HOME\filestore\custom\help\en_US folder.

     > **NOTE**
     > The folder must include an index.html file that, when clicked, opens the custom online help.
2. Click **Administration**, **Configuration**, **Filestore**.
3. Select the option that is named **Enable Custom Help**.
4. Specify the roles that must see the custom help rather than the predefined online help.
5. Restart Catalog Component.
6. Verify that CA Service Catalog functions as expected when users with the specified roles click the **Help** button:
   - If applicable, the custom online help opens instead of the predefined online help.
   - If the index.html file does *not* exist in the USM_HOME\filestore\custom\help\<***language*** > folder, the index.html file in the USM_HOME\filestore\custom\help\default folder runs. The predefined online help opens.
   - If the custom online help is disabled, the predefined online help opens.

Use Web Services to Automate Business Processes

Administrators can use the CA Service Catalog web services to automate business processes and to reduce manual input by catalog users. You can access the web services from any client that uses standard web service protocol.

A web service is any collection of software operations or methods that is available over the Internet. A web service uses a standardized XML messaging system. The web services use XML to encode all communications: A client invokes a web service operation by sending an XML message and then waits for a corresponding XML response.

CA Service Catalog includes several predefined web services. For example, the UserService web service provides a *getUser* method and an *editUser* method that you can use to manage information about users. The BusinessUnitService web service provides similar functionality for business units. This set of web services constitutes an application programming interface (API) for CA Service Catalog.

The web services use Simple Object Access Protocol (SOAP). SOAP is a lightweight, XML-based communication protocol and encoding format for inter-application communication.

CA Service Catalog's SOAP implementation is compatible with both Axis and CXF frameworks. This allows developers to access its web services using any programming language they are comfortable in, as long as they can make SOAP requests. The services support WSDL, enabling automatic generation of client stubs and machine-readable descriptions of the deployed services.

The web services support the Web Service Description Language (WSDL). You can use WSDL to build stubs to access remote services. You can also use WSDL to export automatically machine-readable descriptions of CA Service Catalog deployed services from Axis.

> **WARNING**
> For more information about how to use the web services, log in to CA Service Catalog and click Administration, Tools, Links, Web Services API. The API documentation is automatically generated Java documentation that is based on the CA Service Catalog web service methods. Alternatively, to access this documentation, click Start, Programs, CA, Service Catalog, Documentation, Web Service API on the Catalog Component computer.

Follow these steps:

- Step 1 - Verify the Prerequisites for Clients
- Step 2 - Deploy Web Services
- Step 3 - Generate the WSDL File
- Step 4 - Generate Java Stubs
- Step 5 - Call each Web Service
- Step 6 - (Optional) Specify Special Characters
- Step 7 - Clients Invoke Login and Logout Methods
- Step 8 - Add Attachments to Requests
- View the API Documentation

Step 1 - Verify the Prerequisites for Clients

CA Service Catalog Axis based web services:

- Verify that you are using apache Axis 1 for your web service implementation.
- Axis clients require a Web Service Description Language (WSDL) file for the initialization.
- Verify that a WSDL file is generated for each web service when the Axis server is initialized. The WSDL file for a web service is updated each time when the service is dynamically deployed or undeployed successfully.
  A WSDL file is required for each web service.
  Typically, when a web service is made available using Axis, a unique URL is associated with that web service. The URL name is typically http://localhost:8080/usm/services/*webservice*. An example is http://localhost:8080/usm/services/UserService for the web service named UserService.
  - Name your WSDL files using the convention *webservice*.wsdl, where *webservice* is the name of the web service.
- Verify that each WSDL file contains the information, including method signatures, required to call currently deployed services. The WSDL files help you to differentiate the following parts of a service:
  - Abstract functionality description
  - Concrete details description, such as message format and communication protocol. Examples include SOAP, HTTP, and MIME.

  Thus, you can reuse a WSDL file for different types of clients.
- If you use a Java program to call the methods, verify that the following jar files are in the class path. These files are installed in USM_HOME/webapps/usm/WEB-INF/lib:
  - axis.jar
  - jaxrpc.jar
  - commons-logging.jar
  - commons-discovery.jar
  - wsdl4j.jar
  - mail.jar

CA Service Catalog CXF based web services:

NOTE

- Ensure you update the AXIS webservice URL to CXF webservice for all the endpoints.
- For notes and attachments synchronization, ensure you update the CA Service Catalog endpoint to “http[s]://hostname:port/usm/webservices” in **Service Desk Manager** > **Options Manager** > **CA Service Catalog** > **casc_endpoint**.

Clients must meet the following prerequisites to call CA Service Catalog CXF based web services:

- Verify that you are using Apache CXF for your web service implementation.
- CXF based clients require a Web Service Description Language (WSDL) file for the initialization.
- Verify that a WSDL file is generated for each web service when the CXF server is initialized. A WSDL file is required for each web service.
  Typically, when a web service is made available using CXF, a unique URL is associated with that web service.
  The URL name is typically http://localhost:8080/usm/webservices/webservice.
  An example is http://localhost:8080/usm/webservices/UserService for the web service named UserService.
  - Name your WSDL files using the convention webservice.wsdl, where webservice is the name of the web service.
- Verify that each WSDL file contains the information, including method signatures, required to call currently deployed services. The WSDL files help you to differentiate the following parts of a service:
  - Abstract functionality description
  - Concrete detailed description, such as message format and communication protocol. Examples include SOAP, HTTP, and MIME.
- Now you can reuse a WSDL file for different types of clients.
- If you use a Java program to call the methods, verify that the following jar files are in the class path. These files are installed in USM_HOME/webapps/usm/WEB-INF/lib:
  - cxf-rt-databinding-jaxb.jar
  - istack-commons-runtime.jar
  - jaxb-runtime.jar
  - stax-ex-1.2.jar (download this jar from Maven)
  - stax2-api.jar
  - woodstox-core.jar
  - wsdl4j.jar
  - cxf-rt-wsdl.jar
  - cxf-rt-transports-http.jar
  - cxf-rt-frontend-simple.jar
  - cxf-rt-frontend-jaxws.jar
  - xmlschema-core.jar
  - cxf-rt-bindings-soap.jar
  - cxf-core.jar
  - jaxws-api.jar
  - jaxb-api.jar
  - javax.xml.soap.jar
  - javax.annotation-api.jar
  - javax.jws-api.jar
  - activation.jar
  - commons-logging.jar
  - mail.jar

Step 2 - Deploy Web Services

CA Service Catalog Axis based web services:

To make a web service accessible to a client application, deploy the web service. After you deploy, enable the method on the server. An authenticated user with proper permissions can deploy and undeploy services dynamically when the server starts.

By default, the Catalog system deploys all web services and their methods. With the proper login credentials, a client application has access to all web services functionality. If you decide to undeploy a web service, check if any client applications use it. Therefore, undeploying a web service can affect request approval and fulfillment business processes.

When a web service is deployed or undeployed, the WSDL file for the web service is updated. Each web service has its own WSDL that contains the information about currently deployed services, including method signatures.

The URL format for Axis based web services:

```
http://hostname:port/usm/services/servicenameService?wsdl
```

For example: *http://localhost:8080/usm/services/UserService?wsdl*

> **NOTE**
> You can remotely invoke methods of the Axis server at runtime through an Axis-compatible client.

- ***hostname*:*port***
  Specifies the Catalog Component server name and port number.
- ***servicename***
  Specifies the web service name.

CA Service Catalog CXF based web services:

> **NOTE**
> Ensure you update the AXIS webservice URL to CXF webservice for all the endpoints.

The URL format for CXF based web services:

```
http://hostname:port/usm/webservices/servicenameService?wsdl
```

For example: *http://localhost:8080/usm/webservices/UserService?wsdl*

> **NOTE**
> You can remotely invoke methods of the CXF server at runtime through a CXF-compatible client.

- ***hostname*:*port***
  Specifies the Catalog Component server name and port number.
- ***servicename***
  Specifies the web service name.

The results display the following data:

- All web service methods and data structures for the User web services, in XML format.

Step 3 - Generate the WSDL File

Generating the WSDL file for each web service is required when you call web services with a Java client.

Follow these steps:

1. Access the web service URL in a browser.
   You typically see a message referencing an Axis service and SOAP access.

2. Append **?wsdl** to the URL.
   a. **Axis** generates a service description for the deployed service and returns it as XML in the browser, for example:
      `http://localhost:8080/usm/services/AccountService?wsdl`
   b. **CXF** generates a service description for the deployed service and returns it as XML in the browser, for example:
      `http://localhost:8080/usm/webservices/AccountService?wsdl`
3. Save the output as a file named *webservice.wsdl.* Record the path name for future reference.

   NOTE
   All the WSDL files generated through Axis-based endpoint follows rpc-encoded style whereas WSDL files that are generated through CXF-based endpoint follows document-literal style.

For Axis: Use the webservice.wsdl file as input to web service calls (proxy-generation) when you generate java stubs for each web service.

For CXF: If you are using CXF based web services, you can get the Java client (common.cxf.stubs.jar) from “USM_HOME\catalog\cxf-stubs" location.

Step 4 - Generate Java Stubs

Generating Java stubs for each web service is required when you call web services with a Java client.

Generate Java stubs for Axis based web services

Follow these steps for generating AXIS compatible client:

1. Verify that you are using the Axis WSDL2Java tool for generating the Java stubs.

   NOTE
   You can obtain the Axis tools from the Apache Axis website (apache.org).
2. Open a command prompt and enter the command to generate Java stubs. Use the following command as a model.
   ```
   java org.apache.axis.wsdl.WSDL2Java -o . -ptesting.soap AccountService.wsdl
   ```
   This action generates the following files in the package named testing\soap:
   – *web_service_name*Impl.java
     This new interface file contains the appropriate java.rmi.Remote usages.
   – *web_service_name*ImplService.java
     This file is the Service Interface of the web service. The Service Locator implements this interface.
   – *web_service_name*ImplServiceLocator.java
     This file is the Helper factory to retrieve a handle to the service.
   – *web_service_name*SoapBindingStub.java
     This file is the client-side stub code that encapsulates the client access.
   – Serialized beans
     These beans function as input and return types. They also catch exceptions.
3. Verify that the command generated files like the ones in the previous step.

Generate Java stubs for CXF based web services

CXF compatible Java client:

For CXF based web services, you can get the Java client jar (common.cxf.stubs.jar) from “*USM_HOME**catalog**cxf-stubs*" location.

Step 5 - Call each Web Service

This article contains the following topics:

Use a Java Program to Call Axis Web Service

The Java-related specifications and parameter-related specifications are specific to your program. For more information, see the Axis and web service related-related sections of apache.org website.

Use a JavaScript Program to Call a Axis Web Service

Developers can access web services directly through JavaScript programs. This ability enables web programmers and system administrators to invoke methods remotely through DHTML or Windows Scripting Host. The ability to call web services through client-side scripting gives web developers greater flexibility to create dynamic web sites.

For more information, see the web service-related portions of the workshop sections of the Microsoft Developer Network (MSDN).

Follow these steps:

1. Review the following sample files and use them as models to create your own programs. These files reside at: USM_HOME\ view\webapps\usm\admin
   - soapTest_index.html
   - soapTest_bottom.html
   - soapTest.html. This file contains the JavaScript code to call web service.

   The sample files provide a sample HTML that gets the list of all accounts for a business unit, using synchronous method calls.

   NOTE
   The sample files support both synchronous and asynchronous method calls. When you use asynchronous calls, the web browser does not lock during calls and responds to user input.
2. Open the soapTest_index.html file in your browser.
3. Complete the fields and run the file.
   The file runs and calls the web service. The page dynamically creates an HTML table with the list of accounts.

Example for Axis client JAVA code

```
package testing.soap;

public class axisTester {

  public static void main(String [] args) throws Exception {
    try
    {
      // constructing the endpoint for calling the Account WebService for CA Service View running on the local
 box
      java.net.URL endpoint= new java.net.URL("http://localhost:8080/usm/services/AccountService");

      //calling the Locator which implements the Account WebService interface.
      testing.soap.AccountServiceImplService service =  new testing.soap.AccountServiceImplServiceLocator();

     // Now use the service to get a Stub to the Account WebService
      testing.soap.AccountServiceSoapBindingStub fib =
 (AccountServiceSoapBindingStub)service.getAccountService(endpoint);
```

```
      // The initial call for logging into the webservice.
      // loginID  : The LoginID of the user(Eg: spadmin).
      // password : The User Password.
      // domain   : The businessunit which the user wants to logIn to (Eg: ca.com.)
      //            This is optional and if the value is null the default domain in taken.
      // sessionID: handle to the user used for calling the other methods.This is used for single sign on to
  other webservices.

      String sessionID =  fib.logIn(,,);

      // using the sessionID the stub calls the method getAccounts of BusinessUnit
      // businessunitid : the id of BU for which the user wants to extract the accounts //for.
      Account[] retAccounts = fib.getAccountsOfBusinessUnit(sessionID>,);

      for(int i =0; i< retAccounts.length;i++)
      {
           System.out.println("value of AccountID = " +retAccounts[i].getAccountID());
          System.out.println("value of AccountLabel = " +retAccounts[i].getAccountLabel());
      }

      // logging out: This erases the session of the user Logged In.
      fib.logOut(sessionID);

      }
      catch(Exception e)
      {
      System.out.println("Throwing exception" +e.getMessage());
    }
   }}
```

Use a Java Program to Call a CXF Web Service

Java client specific changes for CXF:

1. Convert Array to List as input; use Arrays.asList();
   For example:
   ```
   offerings = services.getCatalogWebService().getOfferingsByFilter(sessionID,(Arrays.asList(filterBeans));
   ```
2. Convert Array to List as an output; use toArray();
   For example:
   ```
   Offering[] offering = catalogStb.getOfferingsByFilter(sessionId, Arrays.asList(filterBeanArr)).toArray(new
    Offering[0]);
   ```
3. The method getOfferingsByFilter(String, List<FilterBean>) in the type CatalogService is not applicable for the arguments (String, FilterBean[])
   For example:
   ```
   subscriptionDetail = reqStb.getRequestService().getRequestItem(sessionId,
    Integer.parseInt(SubscriptionDetailID))
   ```
4. Type mismatch: cannot convert from org.apache.xml.xml_soap.Map to java.util.Map
   For example:
   ```
   map = (Map) services.getCatalogWebService().getConfigurationValues(sessionID, configGroup, buID);
   ```

Example for CXF client JAVA code

```
package org.cxfclientjar.testing;

import java.net.MalformedURLException;
import java.net.URL;
import java.rmi.RemoteException;
import java.util.ArrayList;
import java.util.List;

import javax.xml.namespace.QName;

import com.ca.usm.soap.axisInterfaces.ContentAccessDeniedException;
import com.ca.usm.soap.axisInterfaces.FilterBean;
import com.ca.usm.soap.axisInterfaces.FormItem;
import com.ca.usm.soap.axisInterfaces.InvalidFormatException;
import com.ca.usm.soap.axisInterfaces.ItemNotFoundException;
import com.ca.usm.soap.axisInterfaces.PropertyNotSetException;
import com.ca.usm.soap.axisInterfaces.Request;
import com.ca.usm.soap.axisInterfaces.RequestRelationship;
import com.ca.usm.soap.axisInterfaces.SubscriptionDetail;
import com.ca.usm.soap.axisInterfaces.WebServiceException;
import com.ca.usm.soap.request.RequestServiceImpl;

public class CxfRequestServiceTest {
    public static void main(String args[]) {
        try {
            String hostname = "http://localhost:8083/usm/webservices";
            //=========================Request Start=================================
            RequestServiceImpl reqServcImpl = new RequestServiceImpl(new URL(hostname+"/RequestService?wsdl"),
                    new QName("http://request.soap.usm.ca.com/","RequestServiceImplementationService"));
            String sessionId = reqServcImpl.getRequestService().logIn("spadmin", "spadmin", "");
            System.out.println("request "+sessionId);

            System.out.println(reqServcImpl.getRequestService().getFormRateItemValue(sessionId, 10005,
 "empName"));
            //reqServcImpl.getRequestService().updateRequestStatus(sessionId, 10001, 400);
            FormItem formItem = reqServcImpl.getRequestService().getFormRateItemValues(sessionId, 10005);
            System.out.println(formItem.getFormName());
            System.out.println(formItem.getRateItemID());
            System.out.println(formItem.getSubscriptionDetailID());

            List<FilterBean> filterBeanList = new ArrayList<>();
            FilterBean  filtbean = new FilterBean();
            filtbean.setField("request_id");
            filtbean.setOperator("=");
            filtbean.setValue("10001");
            List<RequestRelationship>  requestRelationShipList =
 reqServcImpl.getRequestService().getRequestRelationships(sessionId, filterBeanList);

            for(RequestRelationship reqRelationShip : requestRelationShipList) {
                System.out.println("getOfferingName "+reqRelationShip.getOfferingName());
                System.out.println("getRequestID "+reqRelationShip.getRequestID());
```

```
                System.out.println("getRequestName "+reqRelationShip.getRequestName());
                System.out.println("==");
            }

            SubscriptionDetail subscriptionDetail = reqServcImpl.getRequestService().getRequestItem(sessionId,
 10005);
            System.out.println("getAccountNo "+subscriptionDetail.getAccountNo());
            List<Request> reqList = reqServcImpl.getRequestService().getRequests(sessionId);
            for(Request req: reqList) {
                System.out.println("getName "+req.getName());

            }

            System.out.println(reqServcImpl.getRequestService().getRequestItemStatus(sessionId, 10001));

            reqServcImpl.getRequestService().addRequestItemNote(sessionId, 10001, "test note", 1);
            System.out.println(reqServcImpl.getRequestService().getPrimaryServiceDesk(sessionId));

            reqServcImpl.getRequestService().getRequestsByUserID(sessionId, "spadmin");
            List<Request> reqList1 = reqServcImpl.getRequestService().getRequests(sessionId);
            for(Request req: reqList1) {
                System.out.println("getCostCenter "+req.getCostCenter());

            }

            //============================Request End=====================================================
        }catch(MalformedURLException ex) {
            ex.printStackTrace();
        } catch (ContentAccessDeniedException e) {
            // TODO Auto-generated catch block
            e.printStackTrace();
        } catch (ItemNotFoundException e) {
            // TODO Auto-generated catch block
            e.printStackTrace();
        } catch (PropertyNotSetException e) {
            // TODO Auto-generated catch block
            e.printStackTrace();
        } catch (InvalidFormatException e) {
            // TODO Auto-generated catch block
            e.printStackTrace();
        } catch (WebServiceException e) {
            // TODO Auto-generated catch block
            e.printStackTrace();
        } catch (RemoteException e) {
            // TODO Auto-generated catch block
            e.printStackTrace();
        }
    }
}
```

Notable changes between Axis and CXF webservices:

1. CXF Return types and input param data types are changed from Arrays to List.
2. Responses do not contain null values, instead CXF web services return empty lists.

3. Object parameters are ignored in the response, if the field is an empty list. Consequently, null checks are required before accessing such list fields.
4. SOAP Request and Response payload structure changes as Axis and CXF webservices generated WSDLs follow different styles.
5. getFormDefenitions method that belongs to FormDesignerService returns response in json format compared to XML format in Axis webservices.

NOTE
The Java and parameter-related specifications are specific to your program.

Step 6 - (Optional) Specify Special Characters

If necessary, your web service calls can include special characters, as explained in the sections that follow.

Select Special Characters

WARNING
This section applies only if the special character does *not* function as a separator for the parameter in the web service call.

You can use the following XML character entities to specify special characters in web service calls:

- & (ampersand)
- ' (apostrophe)
- " (double quotation mark)
- < (less than)
- > (greater than)

For example, use the following entities to specify the business unit named Smith&Jones Hardware&Software Supplies:

```
Smith&#38;Jones Hardware&#38;Software Supplies
```

Special Characters Other Than Separator Characters

WARNING
This section applies only if the special character does *not* function as a separator for the parameter in the web service call.

To use CDATA tags to specify special characters in web service calls, use the following format:

```
<![CDATA[...]] >
```

For example, use the following expression to specify the business unit named Smith&Jones Hardware&Software Supplies:

```
<![CDATA[Smith&Jones Hardware&Software Supplies]] >
```

Separator Characters

The following special characters are typically used as separators:

- | (vertical bar)
- ! (exclamation point)

If the special character functions as a separator for the parameter in the web service call, specify the special character as a dynamic variable:

1. On the CA Service Catalog UI, enter the special character or characters in the field that you reference in the web service call. For example, in the Description field for the current business unit, enter !&.

NOTE
Dynamic variable can also handle other special characters (for example, & and ,) although they are not separators for web service methods.

2. In the web service call, replace the special character with the dynamic variable for the field of previous step. For example, $bu.description$.

The following sample web service call uses the example from the previous steps:

Axis compatible call:

```
soapenv:Envelope xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://
www.w3.org/2001/XMLSchema" xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://
services.soap.usm.ca.com">
   <soapenv:Header/>
   <soapenv:Body>
      <ser:saveRequestForm soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">
         <sessionID xsi:type="xsd:string">e2f6b05b85247d35b4d7371edc9c6fe398fba60d</sessionID>
         <subscriptionDetailID xsi:type="xsd:int">10009</subscriptionDetailID>
         <formValuesData xsi:type="xsd:string">text1:M$bu.description$</formValuesData>
      </ser:saveRequestForm>
   </soapenv:Body>
</soapenv:Envelope>
```

CXF compatible call:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:urn="urn:usmRequestService">
   <soapenv:Header/>
   <soapenv:Body>
      <urn:saveRequestForm>
         <urn:sessionID>e2f6b05b85247d35b4d7371edc9c6fe398fba60d</urn:sessionID>
         <urn:subscriptionDetailID>10009</urn:subscriptionDetailID>
         <urn:formValuesData>text1:M$bu.description$</urn:formValuesData>
      </urn:saveRequestForm>
   </soapenv:Body>
</soapenv:Envelope>
```

Step 7 - Clients Invoke Login and Logout Methods

When you call web services with a Java client, a required process is clients invoking login and logout methods for each web service. A typical process follows:

1. The client uses a method for login and authentication.
   Each web service has a set of login methods. The client applications can use several login methods for authentication. For example, the *logIn* method takes the same parameters as the Login window: User ID, Password, and Business Unit.
   To view the method parameter information, including signatures, use the following resources:
   – Web Services API documentation.
     To access the Web Services API documentation, log in to CA Service Catalog, click Administration, Tools, Links, Web Services API.
   – SOAP administration interface for deploying or undeploying web services
2. The Catalog system authenticates the user and determines its role.
   Subsequent method calls operate within the scope of the access rights of the user.
3. The client:

- States the service that it is calling.
- Provides the method name and its corresponding parameters before initiating the remote procedure call.
- Checks the WSDL file for this information, if it is unknown.
  Typically, the client already has this information.

4. The web service returns a session ID. This session ID is a required parameter that the client uses for the remaining web service calls. Because the underlying transport protocol can be either HTTP or non-HTTP, the authentication uses a common logIn web service.
   You can share the session ID across web services. For example, you can use the UserService logIn method to obtain a session ID. You can then use the session ID in a call to a Business Unit web service method.
5. This session ends when one of the following events occurs:
   - The client calls the logOut web service method.
     Once you are finished using a session ID, you call the *logOut* web service to terminate the session. The *logOut* web service also invalidates the session ID. Managing the sessions efficiently in this manner helps you obtain the best product performance.
   - The client is idle for a period longer than the session timeout value.
     The client can use the session ID repeatedly within the timeout period. If the session times out, the session ID becomes invalid.
     To change the timeout value, update the administration configuration setting named User Default: Session Timeout.

Step 8 - Add Attachments to Requests

Administrators can use the *addRequestAttachmentWithPath* web service as an automated, efficient, and reliable mechanism for adding attachments to requests.

Use the following option to use this web service to add attachments to requests:

Add Attachments Using An Absolute Path Name

You can optionally attach files to requests by using the absolute path name of the file. Perform the following steps:

1. Verify that the attachment is stored on the CA Service Catalog computer. If necessary, copy or move the attachment to meet this requirement.
2. Call the web services that use *addRequestAttachmentWithPath* and pass the values that you want to the input parameters. Pass the absolute path name of the attachment, for example: C:\\TEST\\test.txt.

View the API Documentation

You can view the CA Service Catalog web services API documentation for more information about how to use the web services. The API documentation is automatically generated Java documentation based on the CA Service Catalog web service methods.

Perform the following steps:

1. Log in to CA Service Catalog.
2. The CA Service Catalog main menu appears.
3. Click **Administration**, **Tools**, **Links**, **Web Services API (CXF)**.
   The web services API documentation appears.

Axis to CXF Migration Guide

For CASM this guide serves as a resource to assist users in initiating and completing the AXIS to CXF migration process.

Introduction

Starting from version 17.4 RU2, the CASM suite of products now fully supports CXF-based web services, enabling a transition away from the deprecated AXIS-based web services. Due to security vulnerabilities associated with AXIS, these services will eventually be phased out in future CASM versions. To mitigate risks, users are strongly encouraged to migrate their content from AXIS WS to CXF WS. This guide serves as a resource to assist users in initiating and completing the migration process.

This guide covers four major topics:

1. Using the new CXF Web Services.
2. Removal of AXIS references in the current product version as required.
3. Steps to migrate content from AXIS web service to CXF web services.
4. Differences in Request/ Response payload.

NOTE

- Ensure you update the AXIS webservice URL to CXF webservice for all the endpoints.
- For notes and attachments synchronization, ensure you update the CA Service Catalog endpoint to “http[s]://hostname:port/usm/webservices” in **Service Desk Manager** > **Options Manager** > **CA Service Catalog** > **casc_endpoint**.

Migration from AXIS WS to CXF WS in CA Service Catalog

In CA Service Catalog, CXF based webservices are available from 17.4 RU2. These services can be accessed through a new endpoint, without impacting the existing integration.

Service	Axis WSDL path	CXF WSDL path
Request Service	http://localhost:8080/usm/services/RequestService?wsdl	http://localhost:8080/usm/webservices/RequestServi
Catalog Service	http://localhost:8080/usm/services/CatalogService?wsdl	http://localhost:8080/usm/webservices/CatalogServic
User Service	http://localhost:8080/usm/services/UserService?wsdl	http://localhost:8080/usm/webservices/UserService?v
Admin Service	http://localhost:8080/usm/services/AdministratorService?wsdl	http://localhost:8080/usm/webservices/AdministratorS
Business Unit Service	http://localhost:8080/usm/services/BusinessUnitService?wsdl	http://localhost:8080/usm/webservices/BusinessunitS
Reporting Service	http://localhost:8080/usm/services/ReportingService?wsdl	http://localhost:8080/usm/webservices/ReportingServ
Account Service	http://localhost:8080/usm/services/AccountService?wsdl	http://localhost:8080/usm/webservices/AccountServic
Billing Account Service	http://localhost:8080/usm/services/BillingAccountService?wsdl	http://localhost:8080/usm/webservices/BillingAccount dl
Form Designer Service	http://localhost:8080/usm/services/FormDesignerService?wsdl	http://localhost:8080/usm/webservices/FormDesigner dl
List Services	http://localhost:8080/usm/services	http://localhost:8080/usm/webservices/list

Removal of Axis References from CA Service Catalog (optional):

To remove axis references from CA Service Catalog, run:

```
Remove_Axis.bat under USM_HOME/scripts/RemoveAxis
```

The backup of axis specific jars are located under “REPLACED/AXIS_BACKUP”.

NOTE

- Till RU2, this .bat file can be run only if ITAM (CA Asset Management) is not part of the solution.
- From RU3, this .bat file can be run even if ITAM is a part of this solution.
- While upgrading from 17.4.2.0 to 17.4.3.0, re-run the removal_axis.bat, if already ran it on 17.4.2.0.

Steps to migrate from Axis based webservices to CXF based webservices

Follow these steps to make OOTB content compatible with CXF web services:

1. Import **CA Service Management Content Pack 3.0** contentpack to make all OOTB plugins use CXF based webservices.
2. To update SLCM AXIS web service endpoints to CXF in CA APM, navigate to **Administration**> **System Configuration**> **Application server**> change the **USM Webservice URL** from *http://localhost:8080/usm/services/RequestService?wsdl* to *http://localhost:8080/usm/webservices/RequestService?wsdl*
3. To update SDM and CMDB web service endpoints to CXF in database, run *Update_Axis_Endpoints.bat* under USM_HOME/scripts/RemoveAxis

   NOTE
   After running the *Update_Axis_Endpoints.bat*, to avoid incompatibility in existing OOTB PAM processes, ensure to update the CXF endpoint for SLCM, SDM Global datasets in PAM.
   For e.g.: /cxf/services/USD_WebService.

   NOTE
   Before importing the CXF web services: If there are any customizations to OOTB content, take a backup of existing PAM content and reapply those customizations.
4. Reload OOTB ITPAM content (which are CXF compatible) by manually importing the xml files into CA ITPAM. These xml files are located under the “filestore/PAM CXF/SLCM/Custom Operators/CA SLCM” in Common Installer DVD for 17.4 RU2 and above.
5. All Catalog endpoints have to be updated to CXF in PAM global datasets, as shown in the below image:

SOAP Request/Response payload differences:

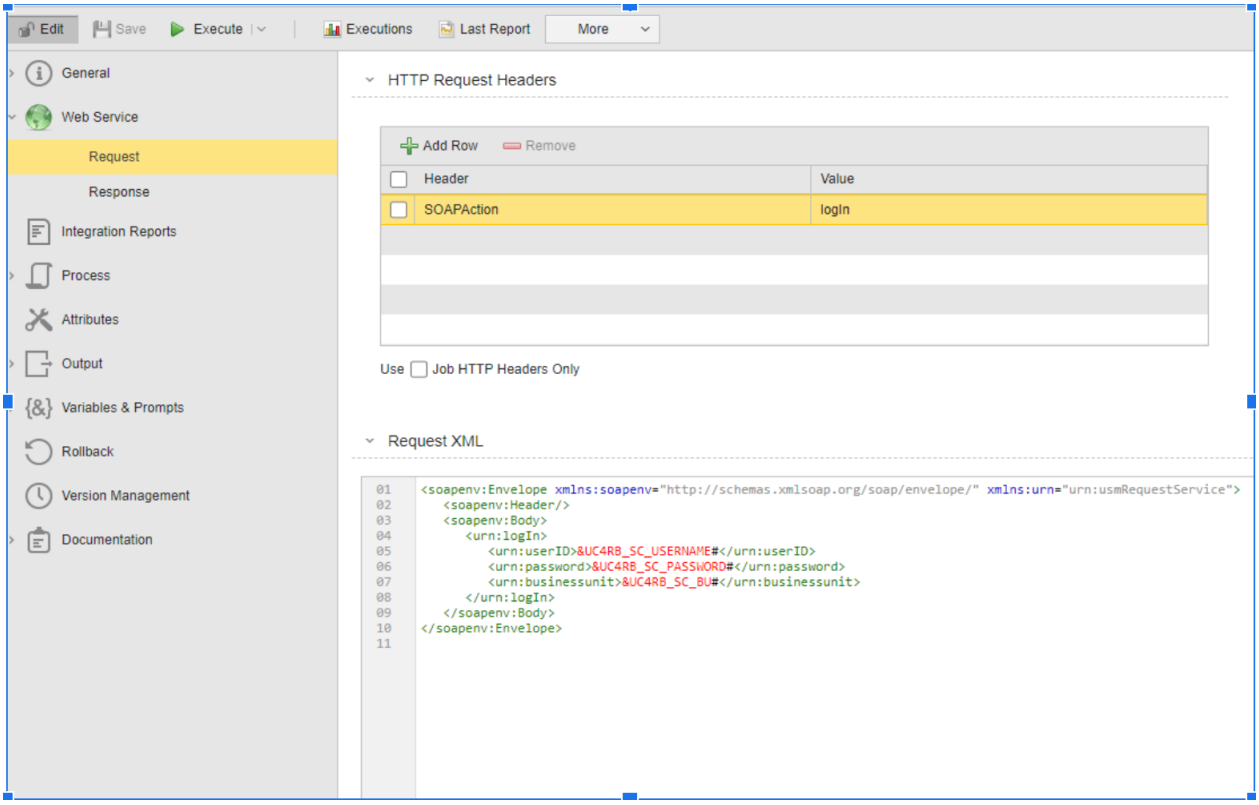
While accessing CA Service Catalog from an external applications like PAM or Automic, ensure that you update the request payload as follows:

- Unlike Axis web service, namespace is necessary for CXF based webservices.
  For example:

```
<getUser xmlns="urn:usmRequestService">
<sessionID>sessionID__</sessionID>
<userid>userid__</userid>
</getUser>
```

- The “xmlns” value should be appended to the SOAP method, which will be applied across the attribute. This value varies based on the type of service we are using. For example:

  - while accessing RequestService, then the namespace value is *urn:usmRequestService*.
  - while accessing CatalogService, then the namespace value is *urn:usmCatalogService*.
- For Automic integration, please pass “SOAPAction” header with method name under HTTP Request Headers section, when making a webservice call to CA Service Catalog, as shown in the following image:



Use API Plug-ins to Load Data into Policies and Forms

This article contains the following topics:

Administrators and service designers work together to write API plug-ins that dynamically load data into policies and forms, as follows:

- For a policy, this data specifies the assignees (approvers) for a request. You use the plug-in instead of specifying assignees manually using the Action Builder. When a user submits a request, the API plug-in dynamically specifies the assignees.
- For a form, this data specifies the value for any of the following fields: dynamic tables, dual lists, or select boxes and their options. For example, when a user completes a form to reserve a virtual computer, the report data object populates the list of available computers. You can also write other report data objects to populate related fields, for example, options for RAM and disk space.

API plug-ins query the MDB or another data source. The API plug-in returns the number of objects that meet the criteria that you specify. You can either write your own API plug-ins or copy and modify the predefined plug-ins to meet your requirements. In both cases, you meet the prerequisites and compile your plug-ins *before* you can use them.

API plug-ins are deployed as jar files in the plugins directory of the filestore. API plug-ins run in the same Java Virtual Machine instance as CA Service Catalog.

Perform the tasks that apply:

- Use API plug-ins for policies
- Use API plug-ins for forms

Use API Plug-ins for Policies

To write and use an API plug-in for policies, follow this process:

1. Define the purpose or goal of the plug-in.
   An API plug-in is useful when you query an external system for data to specify the assignees. An API plug-in is also useful when the identities vary according to the data in the request. Based on this data, the plug-in dynamically creates the assignee list and specifies the assignee levels.
2. Meet the prerequisites:
3. Be able to do the following proficiently:
   – Program in Java.
   – Create policies, including conditions, and understand the types of assignees required.
4. Review the API Plug-in documentation, as follows:
   a. Select Administration, Tools, Plug-ins.
   b. Click API Documentation.
   c. Review the com.ca.usm.plugins.apis.policies package.

   The API documentation is automatically generated Java documentation from the Java class methods for the plug-ins. You use the interfaces, classes, methods, and so on, to implement your plug-in.
5. Download and review the sample API plug-in for policies, as follows:
   a. Select Administration, Tools, Plug-ins.
   b. Click the Sample Policy Plug-in, review the details, and download the source code.
   c. Open and review the SamplePolicyPlugin.java file in the \src\java\com\ca\usm\plugins\samples\policy folder. Use this sample policy plug-in as a model.
6. Create a Java class that implements the interface, com.ca.usm.plugins.apis.policies.AssignmentPolicyPlugin. The sample policy plug-in illustrates how to implement this interface.
7. (Optional) If you use content configuration forms, retrieve values from the fields on these forms. Use the forms as needed.
8. Create a properties file for the plug-in. You can use the plugin.properties file in the sample policy plug-in as a model for your properties files.
9. Create a folder to store the properties file and any JAR files that provide the classes and supporting libraries.

   WARNING

   Store the properties file at the *top* level of the folder. Do *not* store the properties file in a subfolder.
10. Activate the plug-in:
    a. Stop the CA Service Catalog Windows service.
    b. Copy your folder (including all subfolders, if any) to the USM_HOME\filestore\plugins folder.
11. Start the CA Service Catalog Windows service.
12. Verify that the plug-in was successfully adopted:
    a. Select Administration, Tools, Plug-ins.
    b. Verify that the plug-in is listed and that its details appear properly.
13. Test this plug-in:
    a. Use it to specify assignees for a policy.
    b. Submit a request that activates the policy and verify that the policy assigns approvers dynamically as intended.

Use API Plug-ins for Forms

To write and use an API plug-in for use in a form, follow this process:

1. Define the purpose of the plug-in. For example, to populate a select field with meeting rooms that a user can reserve for a specified time period. Other examples include options for the meeting room, such as projectors, video conferencing units, and microphones.
2. Meet the prerequisites. Be able to do the following proficiently:
   - Program in Java.
   - Create forms using the Form Designer.
   - Create the following fields in Form Designer forms:
     - Single select, multi-select, and dual list fields
     - Dynamic table fields
3. Review the API Plug-in documentation, as follows:
   a. Log in to CA Service Catalog and select Administration, Tools.
   b. Select Links.
   c. Click Plug-in Documentation.

   The API documentation is automatically generated Java documentation from the Java class methods for the plug-ins. You use the interfaces, classes, methods, and so on, to implement your plug-in.
4. Create a Java class for the type of Form Designer field for which the plug-in applies, as follows:
   - For single select, multiselect, and dual list fields: Create a Java class that implements the com.ca.usm.plugins.apis.forms.FDSelectDataProvider interface. A sample implementation of this interface is provided in the Sample Select Plug-in, with id ca.catalog.samples.select-plugin.
   - For dynamic table fields: Create a Java class that implements the com.ca.usm.plugins.apis.forms.FDTableDataProvider interface. A sample implementation of this interface is provided in the Sample Table Plug-in, with id ca.catalog.samples.table-plugin.

   To access Java documentation for the interfaces, click Administration, Tools, Plug-ins, and click API Documentation. To download sample source code, click the sample plug-ins on the same page and click Download Source Code.
5. Create a properties file for the plug-in, as follows:
   - Use the sample plug-ins as models for your properties files. In that case, change the plug-in id property in the plugin.properties file.
   - Use the Java Development Kit 1.6 (or higher) and Apache Ant 1.8 (or higher) to use the included build file to compile the plug-in.
   - Use a private classloader for your plug-ins. To use a private classloader, add the following line to your plugin.properties file:

     ```
     classloader.type=private
     ```

     NOTE
     Custom plug-ins from CA Service Catalog Release 12.7 do *not* require any updates after you upgrade CA Service Catalog. These plug-ins continue to function as they originally did.
6. (Optional) If you use content configuration forms, retrieve values from the fields on these forms. Use the forms as needed.
7. Perform the actions that apply:
   - Configure Pagination Parameters for Select Fields.
   - Configure Attributes for Select Boxes Only.
   - Configure Sorting and Pagination Parameters for Dynamic Tables.
8. Create a folder to store the properties file and any JAR files that provide the classes and supporting libraries.

   WARNING

   Store the properties file at the *top* level of the folder. Do *not* store the properties file in a subfolder.

9. Activate the plug-in:
   a. Copy your folder (including all subfolders, if any) to the plugins folder of the filestore.
   b. Select Administration, Tools, Plug-ins, and click the Reload Plugins method.
10. Verify that the plug-in was successfully adopted:
   a. Log in to CA Service Catalog and select Administration, Tools.
   b. Select Plug-ins.
   c. Verify that the plug-in is listed and that its details appear properly.

You are ready to test this API for use in a form field.

Configure Pagination Parameters for Select Fields

Your plug-in can return a large amount of data in a select field on a form. In that case, you often specify the page size of the results in the select field. An example is displaying ten results per page in the select field. To accomplish this task, configure the parameters for pagination in the related Java class and object.

Follow these steps:

1. Edit the Java class that implements the interface, com.ca.usm.plugins.apis.forms.FDSelectDataProvider. Implement the following method in FDSelectDataProvider:

   ```
   List<FDOption> getOptions(int start, int numToReturn);
   ```

   – **start**
   Specifies the first row to return. This parameter is an integer.
   – **numToReturn**
   Specifies the number of rows to return. This parameter is an integer.

   This method returns a list of FDOption objects:

   – • The key value pair (id and value) in a report data object for the options of the select field.
   • Data that complement the key value pair. You can optionally display this additional data in other fields (except the select field) on the form.

   The value of the _id attribute of each field must match the one of the keys in the additional data.
2. Also in FDSelectDataProvider, implement the following method:

   ```
   int totalCount();
   ```

   This method returns the total number of rows that exist.

   NOTE
   For more information and examples, see the API plug-in documentation.

Configure Attributes for Select Boxes Only

Some HTML Attributes apply to select boxes only. For more information, see the Attributes for Select Boxes Only section.

Configure Sorting and Pagination Parameters for Dynamic Tables

Your plug-in can return a large amount of data in a dynamic table in a form. In such a scenario, you specify the page size of the results. An example is displaying ten results per page on the form. Similarly, you often want to let users sort the results on each page in ascending or descending order. To accomplish both tasks, configure the parameters for sorting and pagination in the related Java class and object.

Follow these steps:

1. Click Catalog, Forms, and open the form of interest.
2. Open the table and verify that the attribute named Sortable is set to True.
   If this attribute is enabled (True), then users can click arrows on the table column headers to sort the results.
3. Edit the Java class that implements the interface, com.ca.usm.plugins.apis.forms.FDTableDataProvider. Implement the following method in FDTableDataProvider:

   ```
   List<FDTableRow> getTableRows(int start, int numToReturn, String sortField, boolean sortAscending);
   ```

Users determine these values by interacting with the form, for example, by clicking to view the next page.

- **start**
  Specifies the first row to return. This parameter is an integer.
- **numToReturn**
  Specifies the number of rows to return. This parameter is an integer.
- **sortField**
  Specifies the rows to sort. When the value is null, no sorting occurs. This parameter is a string.
- **sortAscending**
  Specifies whether to sort the results in ascending or descending order. This parameter is Boolean.

This method returns FDTableRow objects, which are described in a later step.

4. Also in FDTableDataProvider, implement the following method:

```
int totalCount();
```

This method returns the total number of rows that exist.

5. Use the FDTableRow object and its methods to return table row data. You can use the following methods:
   - **public void setColumnValue(String columnId, String data)**
     Specifies the following to set the value for a column:
     - String columnid - Specifies “_id” attribute of a component of a table.
     - String data - Specifies a value that you can parse and place into the field on the table.
   - **public String getColumnValue(String columnId)**
     Specifies the matching GET function for the previous SET function (public void setColumnValue).
   - **public Set getColumns()**
     Returns the set of column IDs stored in this object.

     NOTE
     For more information and examples, see the API plug-in documentation.

Using

Audience: IT Analysts, Knowledge Managers, Configuration Administrators, Configuration Managers, Asset Managers, Service Designers

CA Service Desk Manager

Incident Management

Knowledge Management

Request Management

More...

Change Management

Problem Management

Configuration Management

CA Service Catalog

Service Catalog Management

Service Accounting

Request Management

CA Asset Portfolio Management

Hardware Asset Management

Software Asset Management

Request Management

More...

Contract Management

Financial Management

Vendor Management

Common Capabilities

Mobility

Self-Service

Service Point

Welcome to the Service Point Documentation Center.

This section includes information about Service Point capabilities, features, and other resources. The goal of this section is to help you quickly get started with Service Point.

NOTE

Information: To access Service Point on the go, download the CA Service Point mobile app.

What is Service Point and its Benefits?

Service Point for CA Service Management provides a self-service experience with embedded assistive intelligence for business-users. Service Point uses Natural Language Processing (NLP) that provides business users with context-based solutions and suggestions.

Business users can perform the following actions using Service Point:

- Access knowledge
- Create, view, and modify tickets
- Contextually collaborate on tickets using live chat
- View allocated resources
- Manage ticket notifications
- Approve your Service Catalog requests

With a business-centric focus, Service Point helps an organization achieve the following:

- Broad adoption of services
- Context-based solutions
- Improved productivity
- Reduced costs
- Improved collaboration and knowledge sharing
- Elevated perception of IT organization

Get to Know Service Point with a Walkthrough

Take a tour of what Service Point has to offer in a walkthrough:

TIP

To play the video in full screen, click the YouTube logo in the right of Settings at the bottom of the video.

To change the video quality, navigate to **Settings**, and set **Quality** to **720p**.

Service Point Terminology

The following terminology is used in Service Point:

Term	Definition
All Feed	All Feed is the representation of activities on a ticket displayed in a chronological order. Every activity on the ticket is captured on the All Feed with the contact information and the time stamp. Each activity type is represented by a unique icon.

Role-based Documentation

This section provides you with links to topics that are organized by professional roles.

Business User

Getting Started with Service Point

Log a Ticket From Service Point

Manage Tickets Through Service Point

Resolve Your Ticket from Service Point

Request Approval Process

Product Administrator

Define Permissions

Define the User Context

Define the Default Categories

Manage Ticket Attributes

Manage Features

Set Up My Resources

Manage CA Service Catalog Featured Offerings

Manage Smart Search

CA Service Point REST API

Enable the Guest Login

Getting Started with Service Point

New to Service Point and not sure where to begin? Don't worry, we have you covered. Follow these steps to get started:

Login to Service Point

1. Launch a supported web browser and enter the URL.
   `For example, http://<hostname>:<port number>`
2. Specify your CA Service Desk Manager (CA SDM) username and password.
3. (Optional) Select 'Remember Me' to save your credentials.
4. Click **Log In**.

View Announcements in Service Point

Announcements alert users about events, outages, or anything that is important to them. These are displayed as a banner in the announcements widget.

Announcements can differ for each user and are displayed based on the location and organization.

View Notifications in Service Point

Notifications inform users of events, ticket updates, and contextual collaboration (chat) that concern them.

The number of unread notifications are displayed on the notification icon.

The types of notifications are:

1. **Event Notifications:**You are notified when a ticket is opened, closed, or modified through its historical information.
2. **Chat Notifications:**
   You are notified when an analyst initiates a chat.

Log a Ticket from Service Point

You can report an issue and log a ticket from Service Point in the following ways:

Use the Smart Search

1. Use the Smart Search field to describe your issue. Auto-Suggest popup is displayed as you type few characters.
2. Select from the auto-suggest and press Enter.
   A relevant category or service offering is selected.
3. Optionally, you can select a different category or service offering from the list.
4. Based on the issue description, the 20 most relevant Suggested Solutions are displayed in the right vertical segment. The Suggested Solutions are derived from the list of published knowledge base articles, closed incidents and requests.
   a. Click **Like** if the solution is useful in resolving the issue.
   b. Click **Dislike** to remove the solution from the list, and proceed with ticket creation.
5. Enter the other information in the grid. These are attributes that are configured by your organization.
6. Optionally, you can add attachments to your ticket. Attachments can be added by a drag and drop or by copy-pasting the attachments in the ticket creation page.
7. Click **Submit your Ticket.**To view or manage your ticket, see Manage Tickets Through Service Point.

Use the Default Categories and Service Offerings

1. You can now browse Catalog Service Feature Offerings from a menu tree in Servce Point. Click any of the Featured Offerings to open the Create Request Form page.
   Configured featured offerings are displayed on the right pane, and offerings are displayed in the center for a selected folder.

   NOTE
   If you have not configured any offerings and have only configured featured offerings, then only featured offerings will be displayed and vice-versa.
2. From the default category and service offering list, select an option that is related to your issue.
3. Enter the information in the grid. These are attributes that are configured by your organization.
4. Optionally, you can add attachments to your ticket. Attachments can be added by a drag and drop or by copy-pasting the attachments in the ticket creation page.
5. Click **Submit your Ticket.** To view or manage your ticket, see Manage Tickets Through Service Point.

Use My Resources

The segment, **My Resources**, lists all the assets assigned to you, along with the asset information. The list displays the top four most recently updated assets. To view other assets, scroll down the list.

Facing trouble with any of your assets?
Select the asset with which you are facing an issue, click **Report an issue**, and the ticket creation form is auto-filled with the details. You can provide a detailed explanation, and click **Submit your Ticket.**

To view or manage your ticket, see Manage Tickets Through Service Point.

Ability to browse SDM Categories

This configuration lets the administrator give access to the end users of ServicePoint to browse SDM Categories(Areas).

1. ServicePoint displays SDM Categories under “More” icon by default.
2. In SDM, navigate to **Administration** > **xFlow Interface** > **ServicePoint** > **Configuration** > **Feature Configuration** > **Add Feature**. Set the name as “Browse SDM Categories” and configure the value to “Yes” or "No".

If the value is set to "No", this feature is disabled.

Based on this configuration, end users can browse the SDM categories maintaining the hierarchy of areas.

NOTE
This ServicePoint Feature configuration is available from 17.4 RU2.

Bookmark Service Folders/ SDM Service Categories

NOTE
This is applicable from CASM 17.4.5.0 and above.

You can now access Catalog Services or Folders directly through bookmarked links, enabling faster navigation to frequently used requests or offerings. While browsing catalog offerings or service requests, you can bookmark any folder by hovering over the desired item and clicking on it. The selected catalog folder or request opens in a new window, allowing you to easily save and reuse the link for quick access. This enhancement applies to both Service Desk Manager Requests and Catalog Offerings.

While browsing **catalog offerings** or **service requests**, you can:

- Hover over any service or folder and click the **bookmark icon**.
- The selected item opens in a **new browser tab or window**, allowing you to **save and reuse the link** anytime.

This feature improves navigation efficiency and helps analysts or end users quickly access **commonly used requests or catalog offerings**. The enhancement applies to both **Service Desk Manager Requests** and **Service Catalog Offerings**.

Configure SDM Ticket Category

This configuration allows the administrator to configure the type of category that should be displayed in ServicePoint.

1. ServicePoint displays both Request and Incident areas by default.
2. In CA SDM, navigate to **Administration** > **xFlow Interface** > **ServicePoint** > **Configuration** > **Feature Configuration** > **Add Feature**
3. Set name as “Ticket category Type”, and its value can be configured to “Request”, “Incident” or “Request/Incident”

Based on this configuration, the SDM category areas are displayed in ServicePoint.
If the configuration is set to “Request”, only Request Areas are displayed, and if the configuration is set to "Incident" the Incident Areas are displayed.

NOTE
This ServicePoint Feature configuration is available from 17.4 RU2.

Manage Tickets from Service Point

This section explains how you can effectively manage your tickets and perform tasks from Service Point.

Follow the navigation to view your tickets within Service Point.

- **Home > My Recent Tickets > Open** lists the three latest open tickets.
- **Home > My Recent Tickets > Resolved** lists the three latest resolved tickets.
- **My Tickets > Open** lists all your open tickets.
- **My Tickets > Resolved** list all your resolved tickets.

Manage Open Tickets

The following non-customizable predefined attributes are displayed for an open ticket: Summary, Ticket ID, Last Modified Date, and a link to the Live Chat session (if it was initiated), and the ticket Status.

The following actions can be performed on an open ticket from the ticket details section:

- Add a comment to the ticket.
- View all the various activities on a ticket that are displayed on ALL FEED.
- Filter the ALL FEED by activity type.
- Add an attachment to a ticket.
- Edit a ticket.
- **Allow Ticket Edit**: This feature Controls the business users ability to edit the ticket in Service Point. When set to '**No**', edit button in ticket detail page will not be displayed in Service Point. To enable this feature, navigate to **Administration**, **xFlow Interface**, **Service Point**, **Configurations, Feature Configuration,** select a respective configuration, **Add Feature**, **Allow Ticket Edit**.

  NOTE
  This feature is currently available only in Service Point and not in xFlow Interface.
- Resolve a ticket.

Manage Resolved Ticket

For a resolved ticket, the following non-customizable predefined attributes are displayed: Summary, Ticket ID, and Created date.

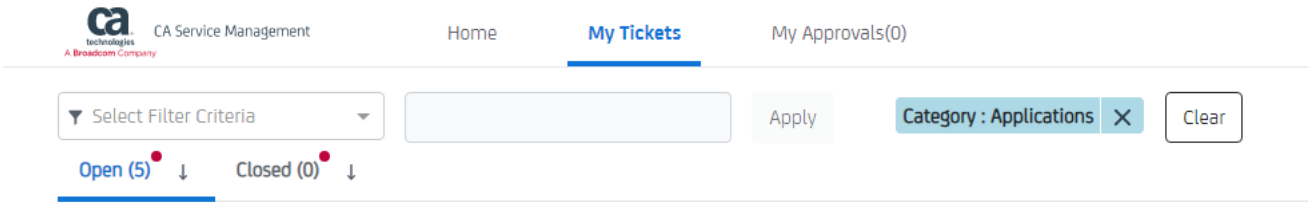
After the issue is resolved, a ticket is set to Resolved state. If you are not satisfied with the resolution, you can click **Reopen Ticket**.

Filter My Tickets

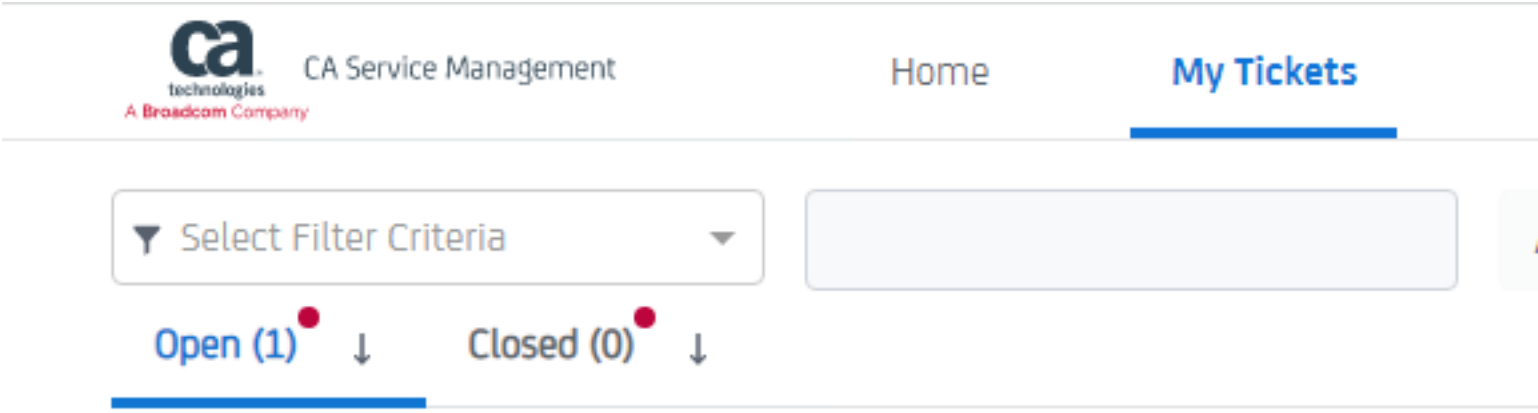
Filter "My Tickets" You can now track tickets from the My Tickets section of Service Point. With growing ticket volume, you may find it difficult to look for a particular ticket from a long list of Open and Resolved Tickets. To simplify your ticket search experience you can use filters for the following attributes:

- Ticket ID
- Summary
- Description
- Category

Filtering based on category:



Filtering based on category and description:



NOTE
In the My Tickets page, you can sort tickets by:

- *Creation Date* for Active tickets.
- *Completion Date* for Closed tickets.

Resolve Your Ticket with Live Chat

Live Chat is a real-time messaging capability that enables business users to chat with analysts, share attachments, and collaborate on their tickets.

Live Chat provides the following benefits:

- Reduces time to resolve tickets.
- Provides in-context collaboration.
- Improves end-user satisfaction.
- Improves ticket resolution rate.
- Allows uploading, and downloading attachments.
- All ticket conversations are saved. This is a convenient way to understand the ticket context when it is reopened or reassigned.

Using Live Chat

To initiate a Live Chat with an analyst, click the Chat icon from the My Recent Tickets vertical segment on the home page, or My Tickets tab, or the ticket details section. All Live Chats land in analyst queue first and then are picked manually by analysts. The live chat entry is created and the conversation history is saved in All Feed, for any future references.

When an analyst initiates a Live Chat, you receive a notification. When you click the notification, the latest message in the live chat is displayed. The live chat entry is created and the conversation history is saved in All Feed, for any future references.

Live Chat Session Timeout

After a set period of inactivity between the business user and and the analyst, the live chat session will timeout. At that point, the business user can start a new Live Chat.

Browser Desktop Notifications

Desktop notification gets you pop-up notifications on your desktop whenever you receive a new message in Live Chat. Notifications also come with a short text. This helps you to know what the live chat message received is all about at a quick glance.

These notifications are available for the following browsers: Chrome, Firefox, Safari, and Windows Edge.

xFlow Interface

The xFlow Interface provides analysts the ability to understand the IT environment, prioritize work and make decisions. Data driven intelligence improves the productivity of the analyst and enables them to understand the workload for the day and the list of activities to be completed.

This section helps an analyst get started with the xFlow Interface and includes the following topics:

NOTE

A ticket in the xFlow Interface can be an incident, request, or a problem.

Access the xFlow Interface

Access type is the primary component that controls the permission to access the xFlow Interface. For example, a contact with **Access Level** as *Customer/Employee* cannot access the xFlow Interface. Contact the administrator for access.

1. Launch a supported web browser and enter the xFlow Interface URL.
   `For example, http://<hostname>:<port number>`
2. Specify your CA Service Desk Manager (CA SDM) username and password.
3. (Optional) Select 'Remember Me' to save the username.
4. Click **Log In**.

Understand the Environment

This section helps you have a better understanding of the environment and how it works in concert with the xFlow Interface.

Weather

Weather enables an analyst to view the workload forecast for the day, based on the tickets assigned to the analyst, analyst groups, and the planned environmental changes.

Suggestions are provided to the analyst with a list of tickets to be actioned.

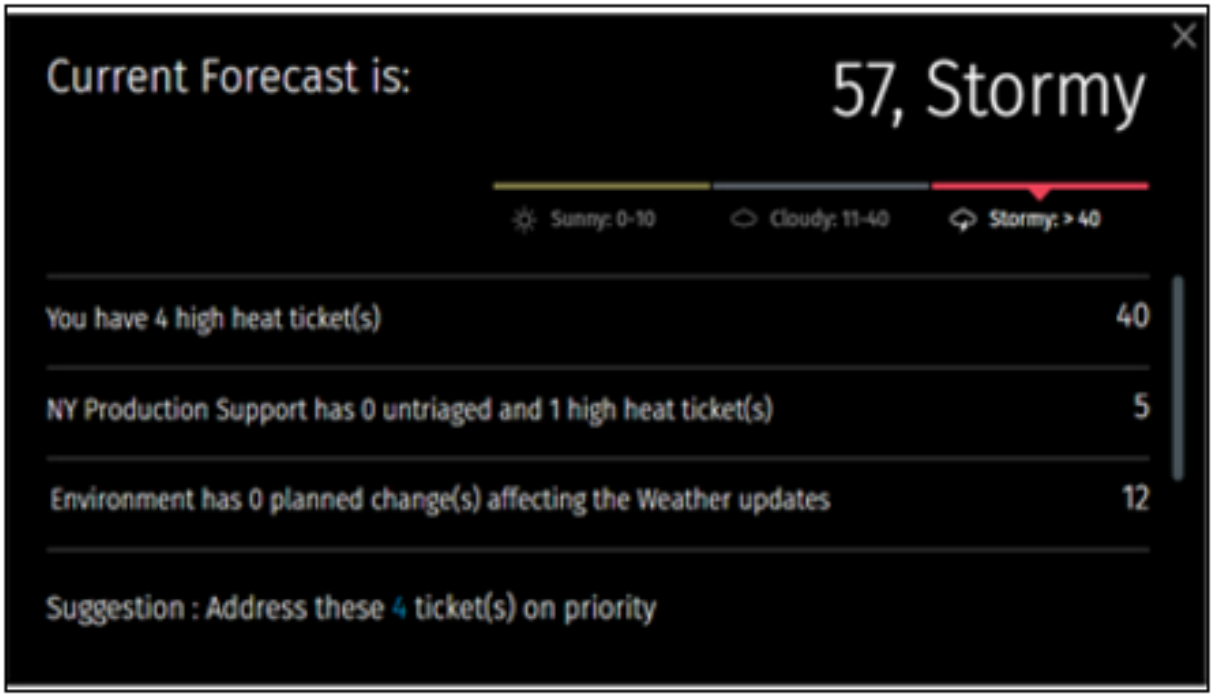
NOTE

Contact the administrator if the weather icon or Suggestions are not displayed on the xFlow Interface.

An analyst can click the weather icon to view the following details:

- Current weather.
- Weather range (Sunny, Cloudy, or Stormy)
- Criteria contributing to the weather.
- Weightage for each criteria.
- Suggestion based on the weather.

The following is an example of Stormy weather:



A red circle around the weather icon indicates that the weather is unread or has changed. The weather is refreshed based on the reindexing interval which is set by the administrator. The following is an example:

Notifications

The icon displays all the notifications for the analyst that include:

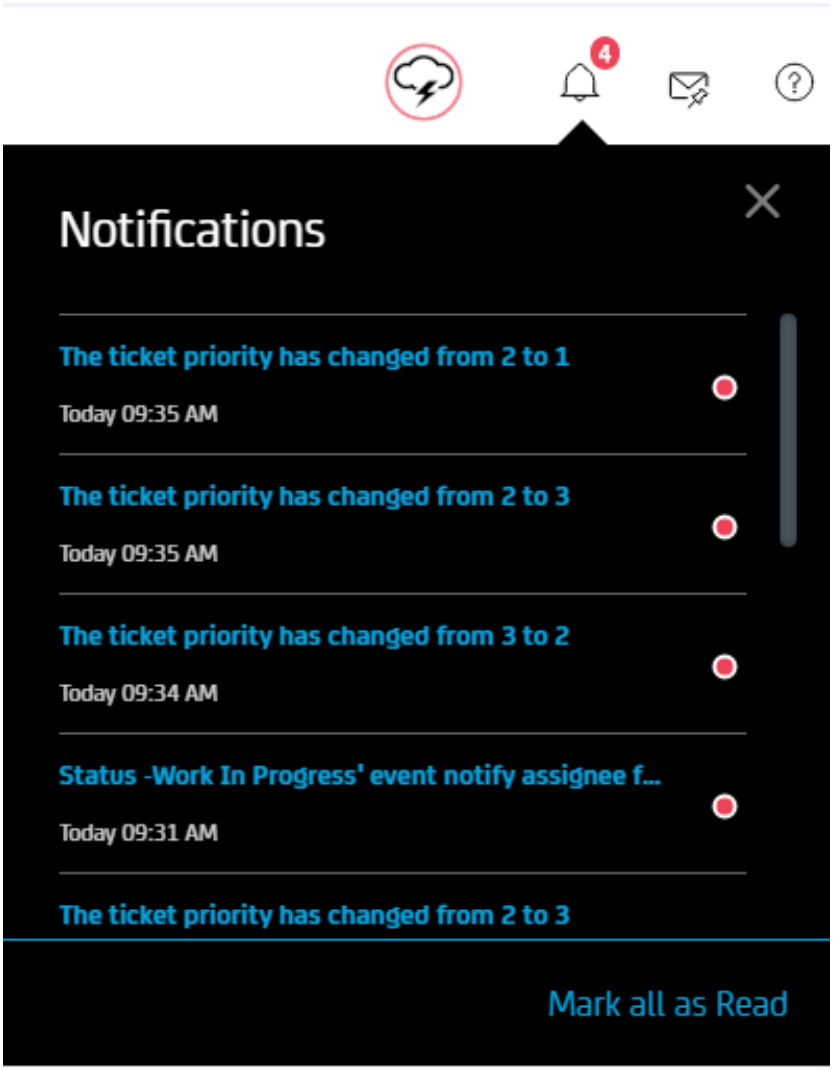
- Ticket activities, events, and system actions on a ticket in the last seven days.
- Reminders for follow up on a ticket.
- Notifications as per the configuration by the administrator.

An unread notification is indicated by a red dot.

Click the notification icon to perform the following actions:

- Mark the notification as read.
- Navigate to the ticket details.
- Click **Mark all as Read** to mark all the notifications as read and to reset the notification count to zero.

The following example shows four unread notifications:



Team Messages

Team Messages allows you to post messages to multiple groups and individual group members.

> **NOTE**
>
> Team Messages only display the groups that the analyst is a member of.

View Announcements in xFlow

> **NOTE**
> This is applicable from CASM 17.4 RU5 and above.

This enables the rendering of announcements within xFlow, extending a feature previously available only in Classic Service Desk and Service Point.

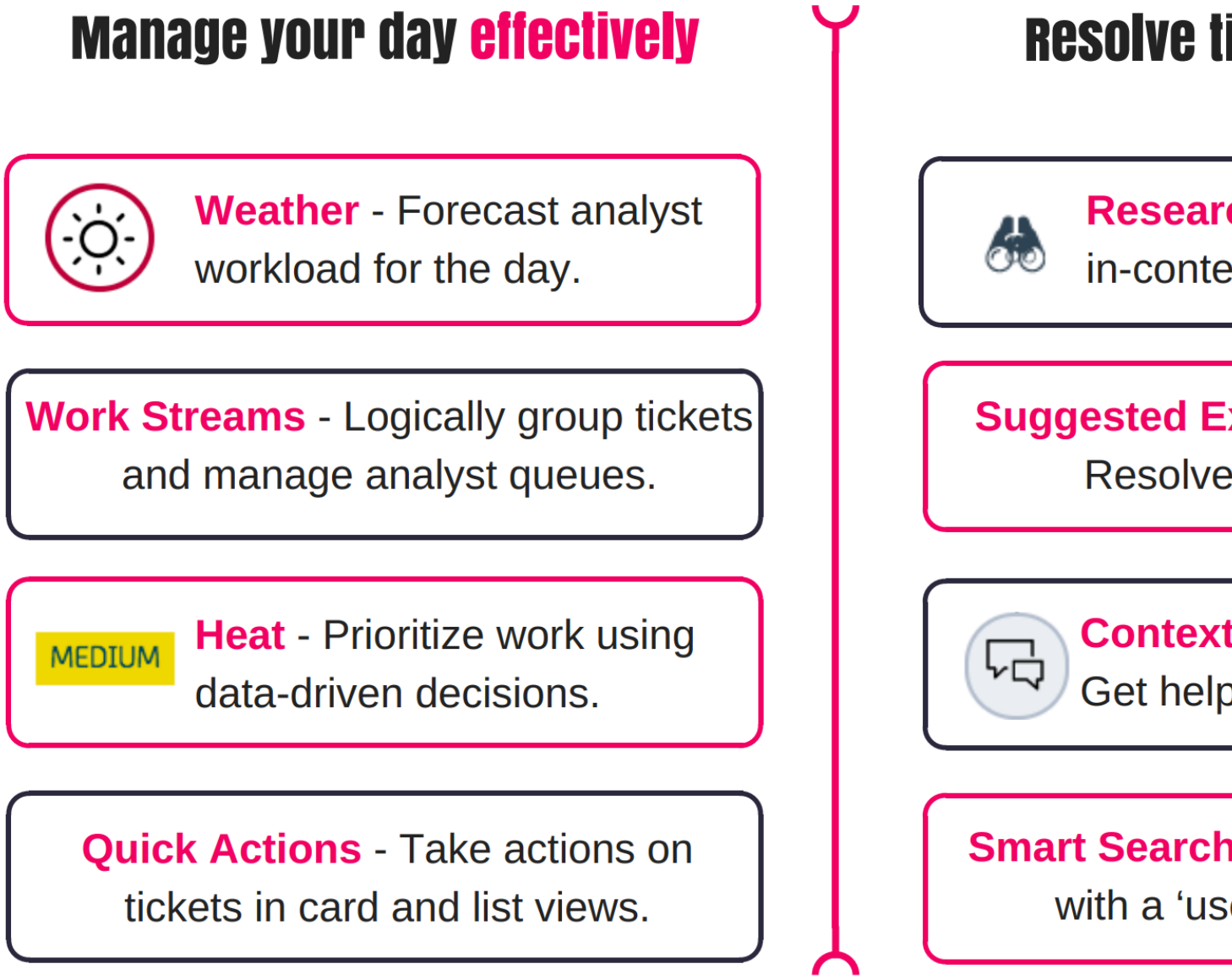
FAQs About xFlow Interface

This article contains some of the most frequently asked questions about xFlow Interface:

What are the elements of the xFlow Interface?

The xFlow Interface provides the following elements and capabilities:

- Weather
- Notifications
- Team Messages
- Work Streams
- Workflow Task
- Card and List view
- Quick Actions
- Heat
- Ticket Timeline
- Suggested Experts
- Suggested Solutions
- Contextual Collaboration
- Research Tool
- Command Bar
- Ticket Attributes



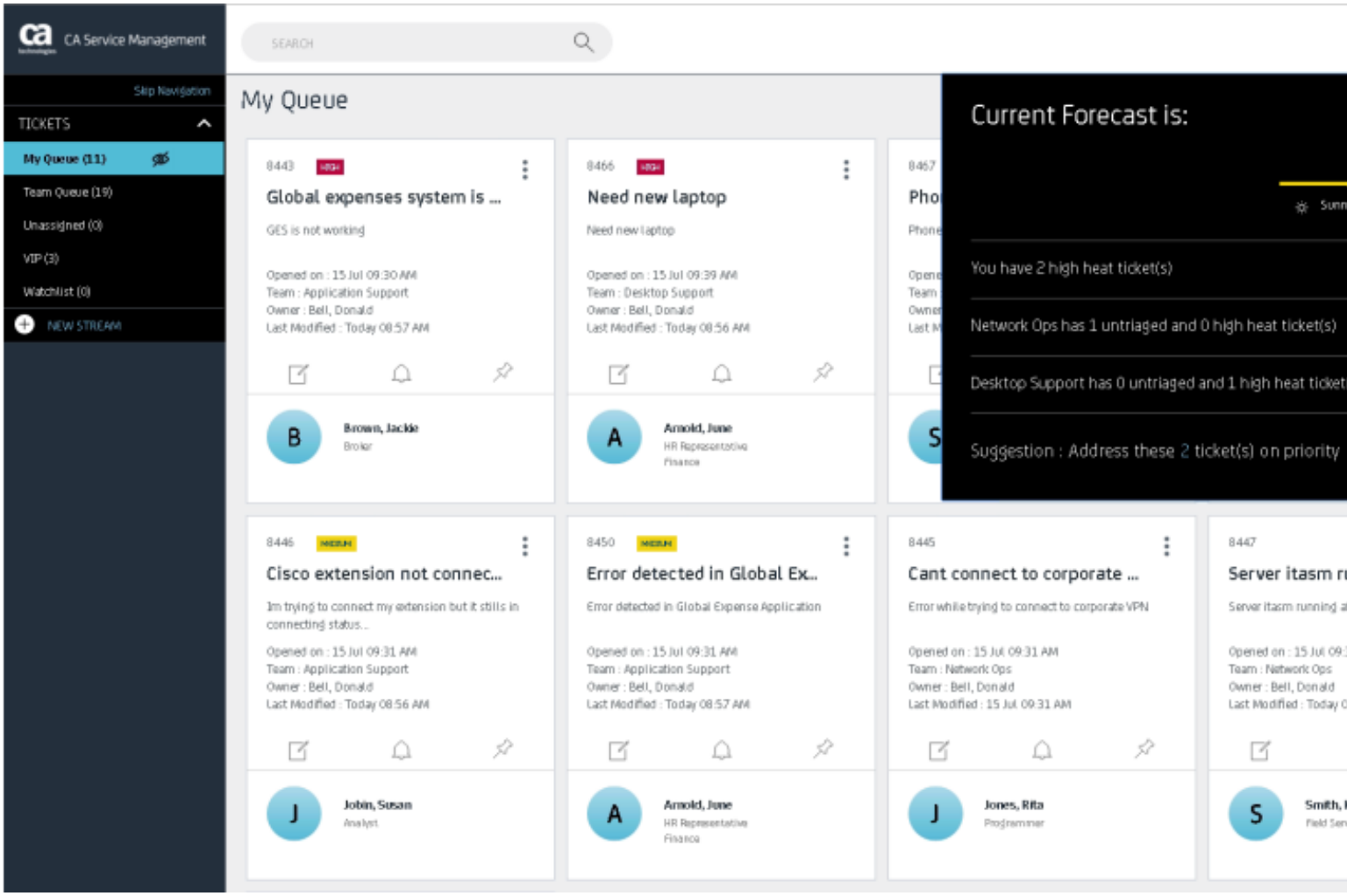
What is Weather?

Weather in the xFlow Interface forecasts analyst workload for the day and indicates the real-time state of the IT environment.

The weather icon is displayed at the top right section of the xFlow Interface and displays the metrics used to forecast the weather.

Weather is calculated based on the tickets assigned to the analyst, analyst groups (un-triaged and high heat), and the planned environmental changes. Suggestions are provided to the analyst with a list of tickets to be actioned.

This example demonstrates the **Sunny** weather (This can be Sunny, Stormy, or Cloudy).



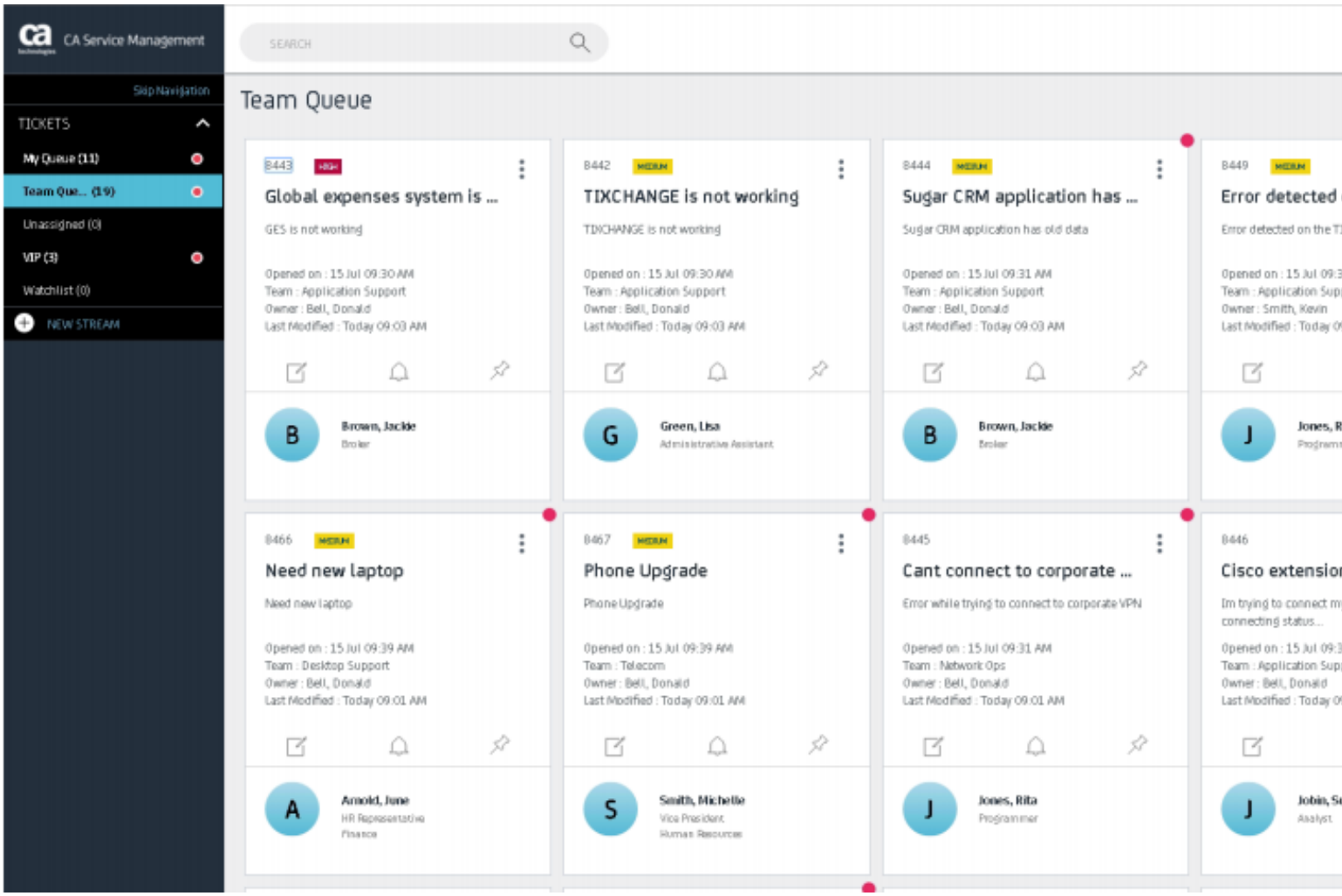
What are Work Streams?

Work streams are the independent ticket queues that analysts work on.

Work Streams provide a logical grouping of tickets based on specific search criteria.

Work streams are displayed in the left navigation pane. The following work streams are predefined: My Queue, Team Queue, Watchlist, Unassigned, and VIP.

This example displays the tickets of the **Team Queue** work stream.



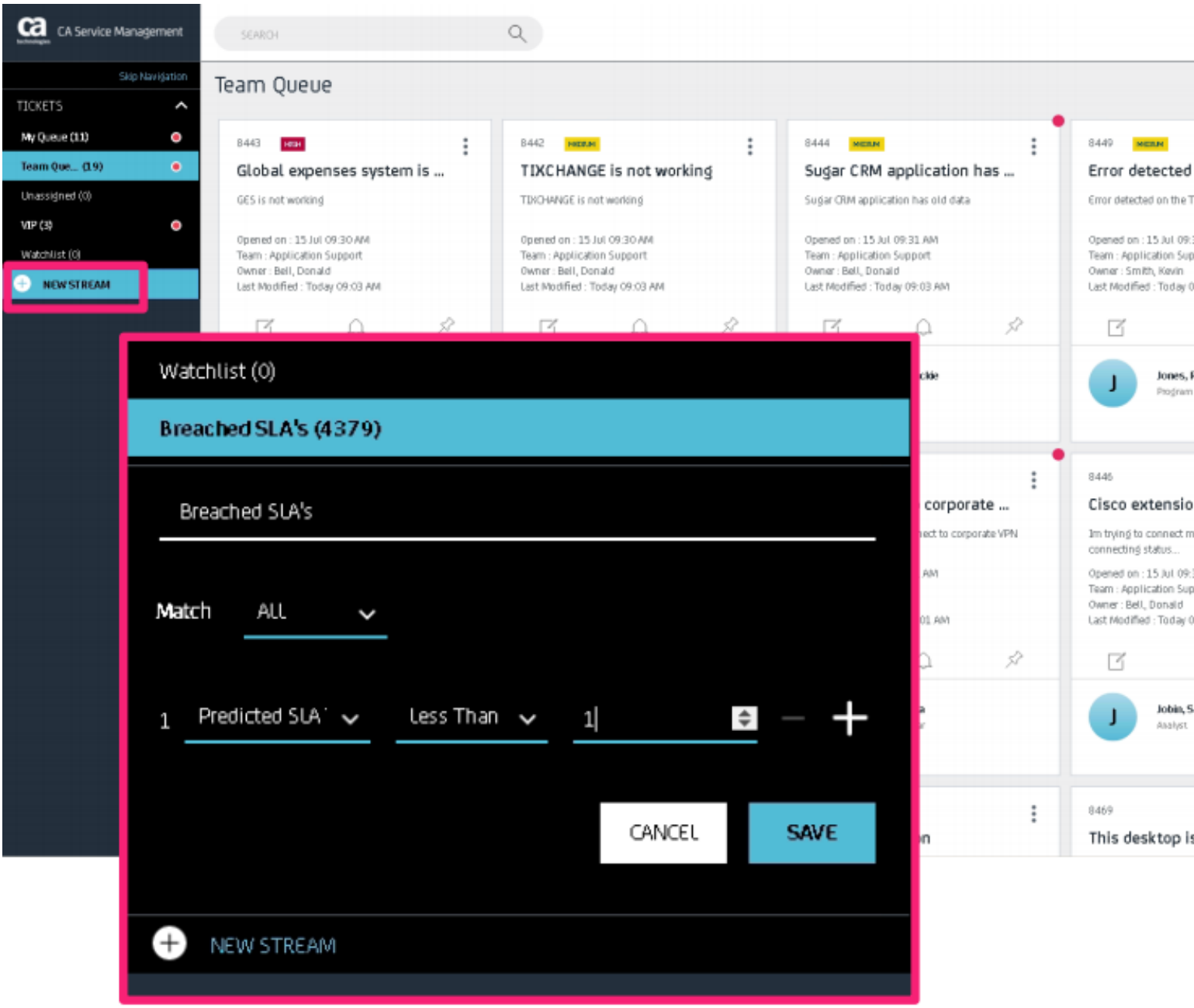
How can I create a work stream?

Work Streams provide logical grouping of tickets based on specific search criteria.

To create a work stream, select a set of attributes with their values and save it.

For example, perform the following steps to create a work stream to display the tickets with breached SLAs:

1. Click **NEW STREAM** from the left navigation pane.
2. Enter a stream name.
3. Select **Predicted SLA Violation** from the drop-down list in the left column.
4. Select **less than** from the drop-down list in the middle column and then enter **1** in the right column.
5. Enter **Breached SLAs** as its name and click the **SAVE** icon.

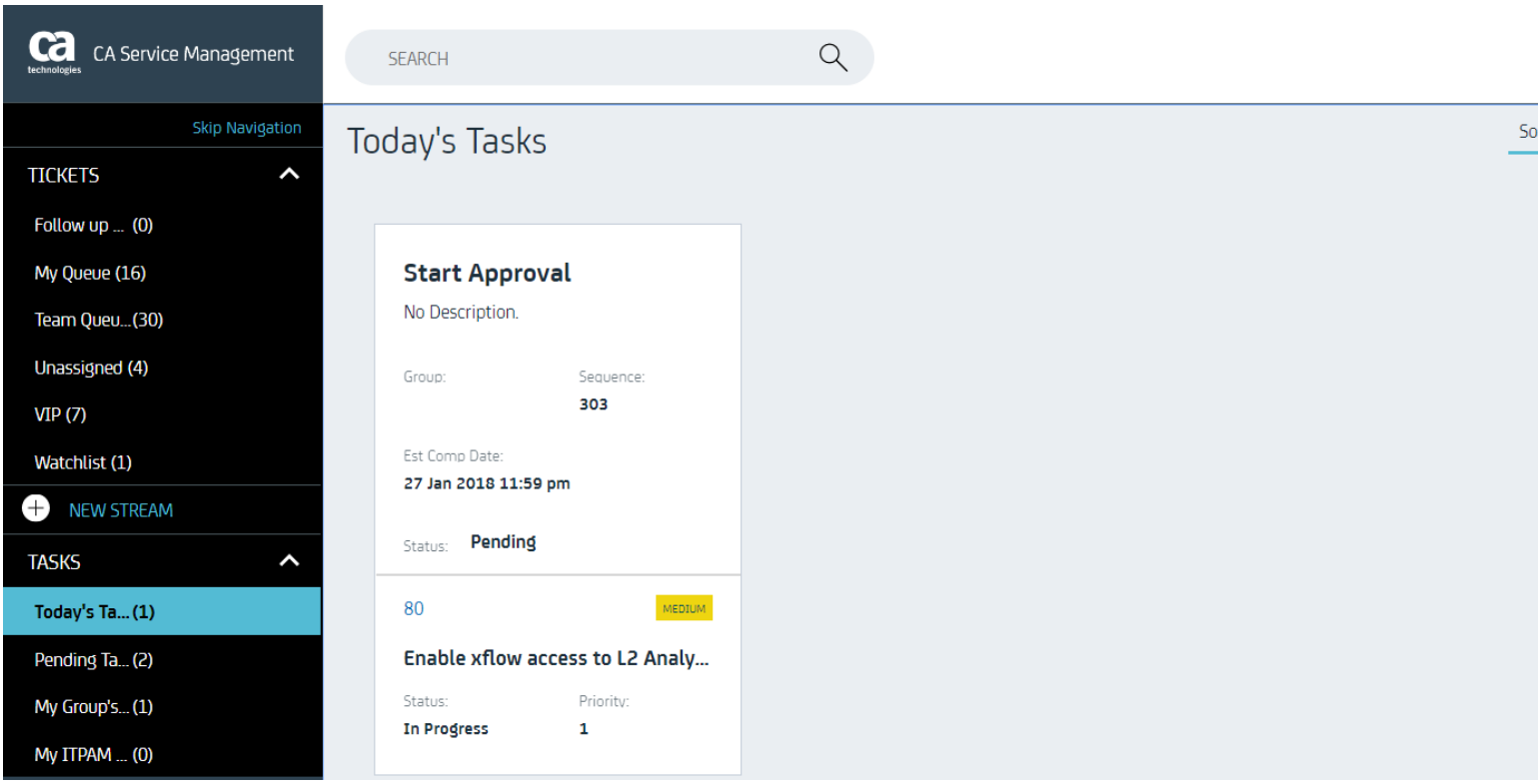


What are Workflow Tasks?

A workflow specifies the sequence of tasks to be performed to resolve a ticket. An analyst can manage workflow tasks from the xFlow Interface.

Workflow tasks are displayed below the Work Streams in the left navigation pane. The following workflow tasks are predefined: Today's Tasks, Pending Tasks, My Group's Tasks, and My ITPAM Tasks.

This example displays **Today's Tasks** that are assigned to the analyst and are due for completion the same day.



How can I view the sequence of workflow tasks for a ticket?

A visual representation of workflow tasks allows an analyst to understand the end-to-end workflow. Type

```
/workflow
```

in the Command Bar to launch the graphical workflow. For more information, see Manage Workflow Tasks.

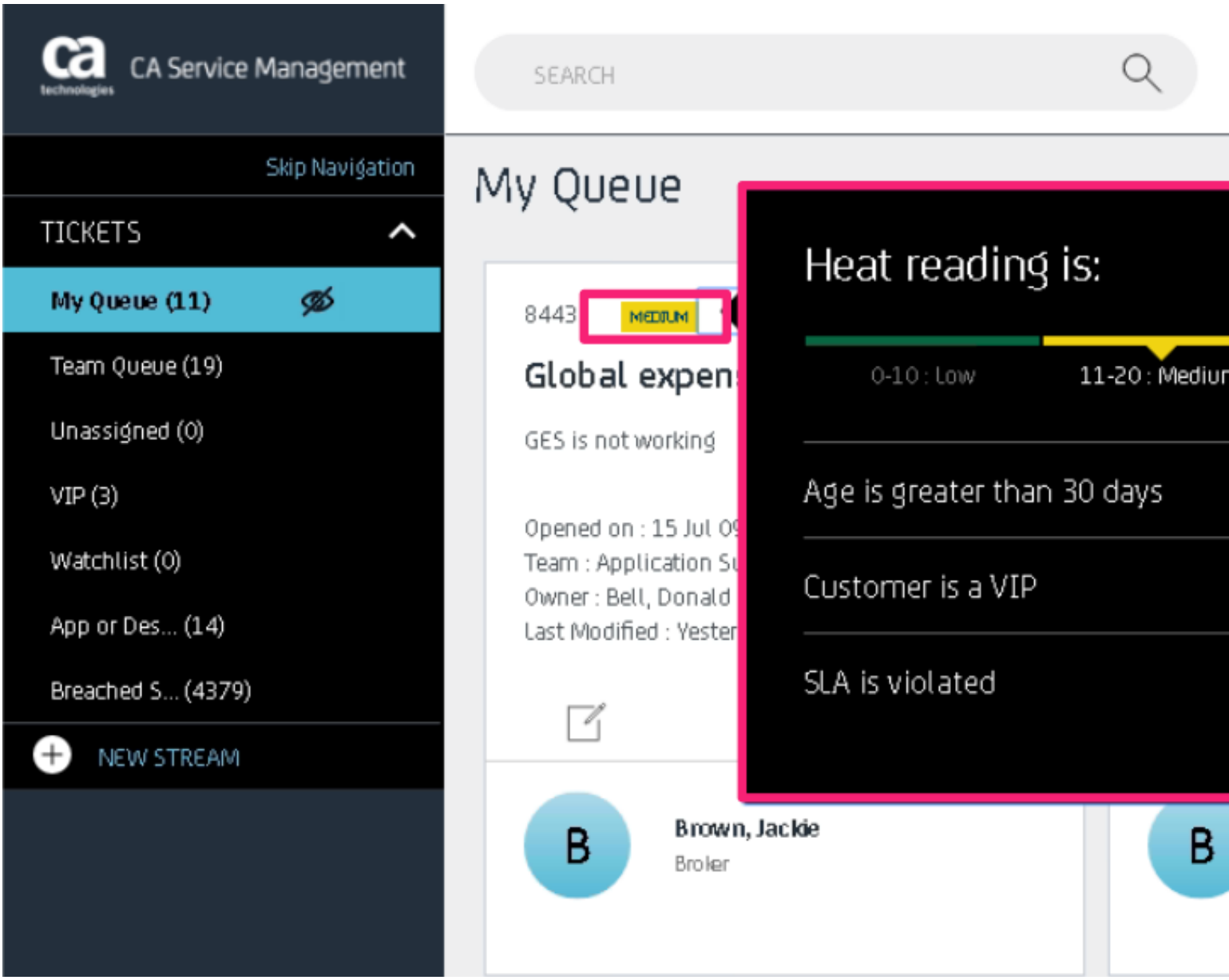
What do the red and orange indicators on a ticket mean?

The red and orange indicators above the ticket title represent the heat of a ticket.

Heat defines the criticality of a ticket and provides a dimensional scoring model to prioritize tickets. Heat is a key factor that influences Weather.

The heat score of a ticket depends upon the various attributes such as priority, impact, and age.

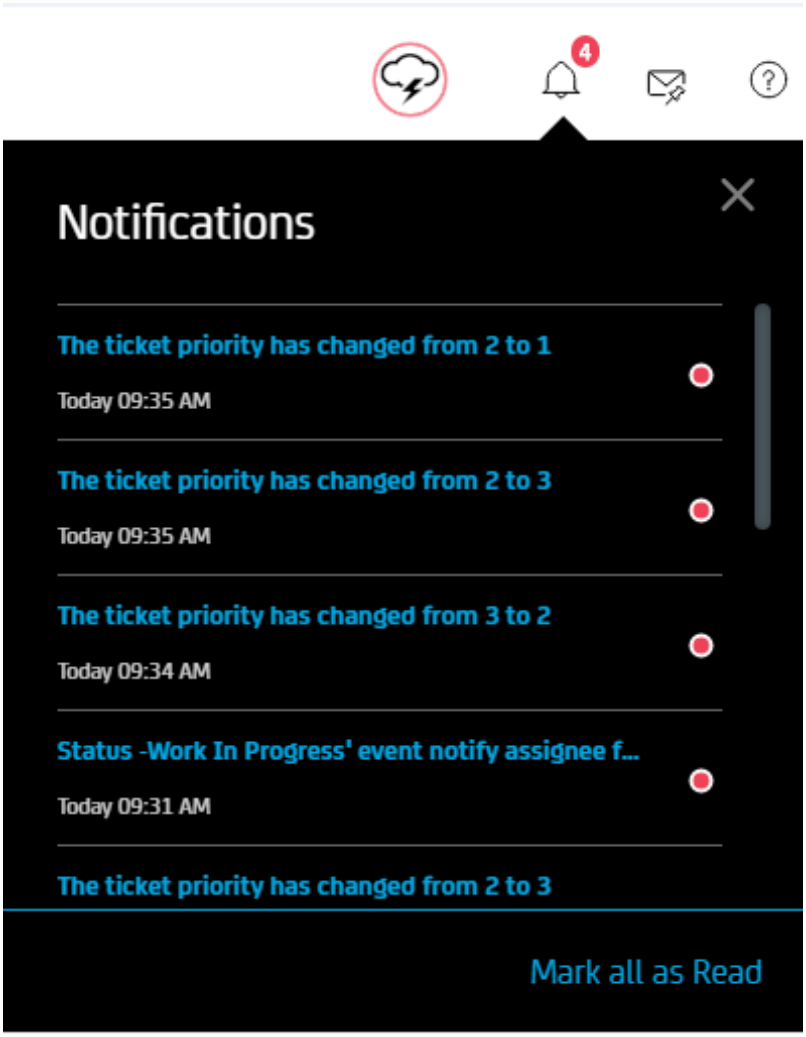
This example displays a ticket with **Medium** heat.



How can I read my notifications?

At the top right section of the xFlow Interface, the Notification icon displays the notifications for an analyst. This icon displays all the ticket notifications such as ticket activities, events, and system actions in the last seven days.

The example displays four unread notifications for an analyst.



What are the quick actions that I can perform on a ticket?

The following actions can be performed on a ticket from the card or list view without accessing the ticket details:

- Comment
- Notify
- Follow up
- Update Status
- Change Priority
- Transfer ticket

How can I comment on a ticket without accessing the ticket details?

From the card of list view, click the Comment icon

to comment on a ticket.

How can I notify the end user on a ticket?

From the card of list view, click the Notify icon

to notify the end user on a ticket.

How can I follow up on a ticket without accessing the ticket details?

From the card of list view, click the Follow up icon

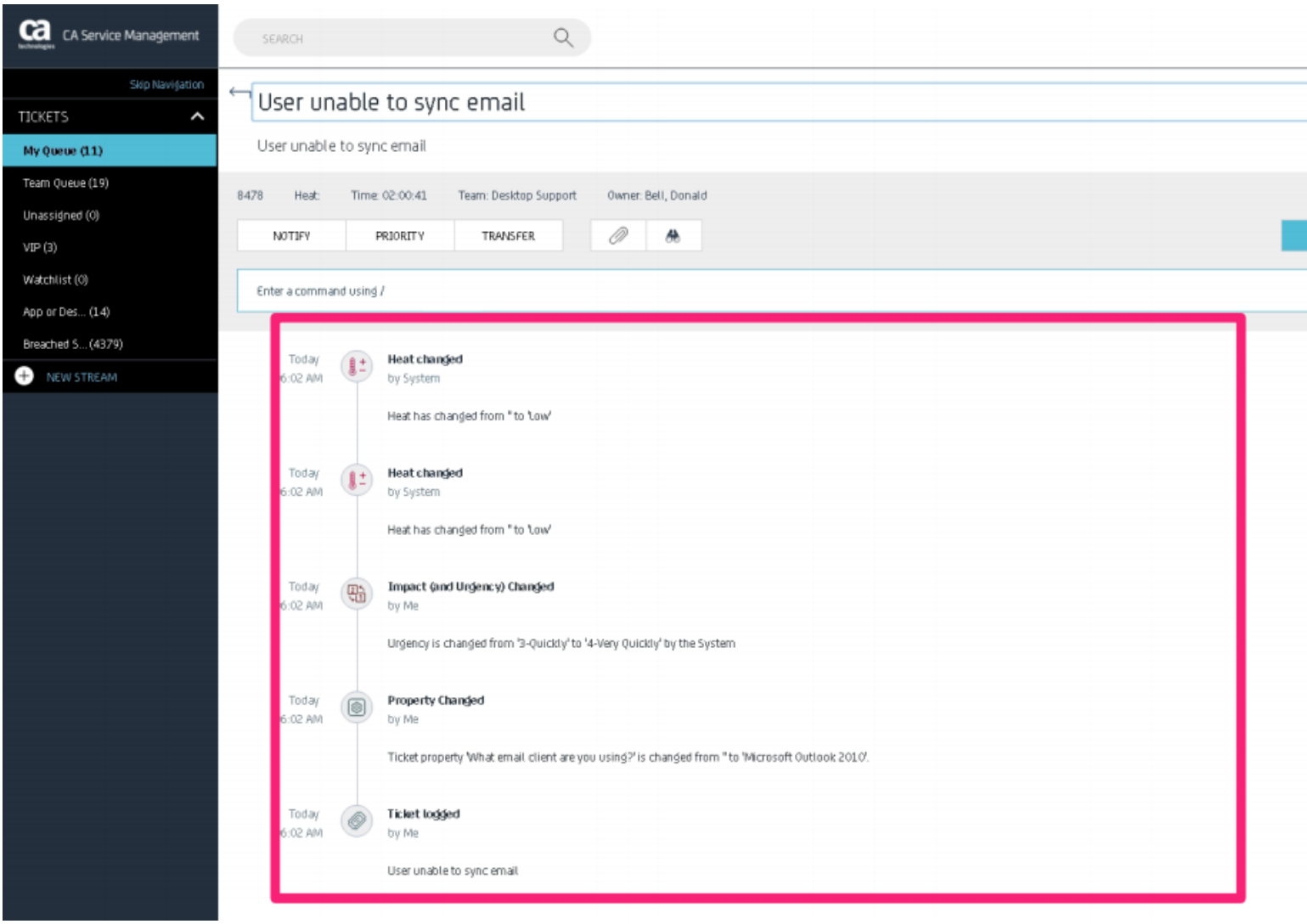
to followup on a ticket and add it to the Watchlist work stream.

How can I view the activities associated with a ticket?

A ticket timeline displays the activities associated with a ticket in chronological order.

Open the ticket to access the timeline.

The timeline makes it easier for the analyst to analyze the historical actions and collaborations associated with the ticket.

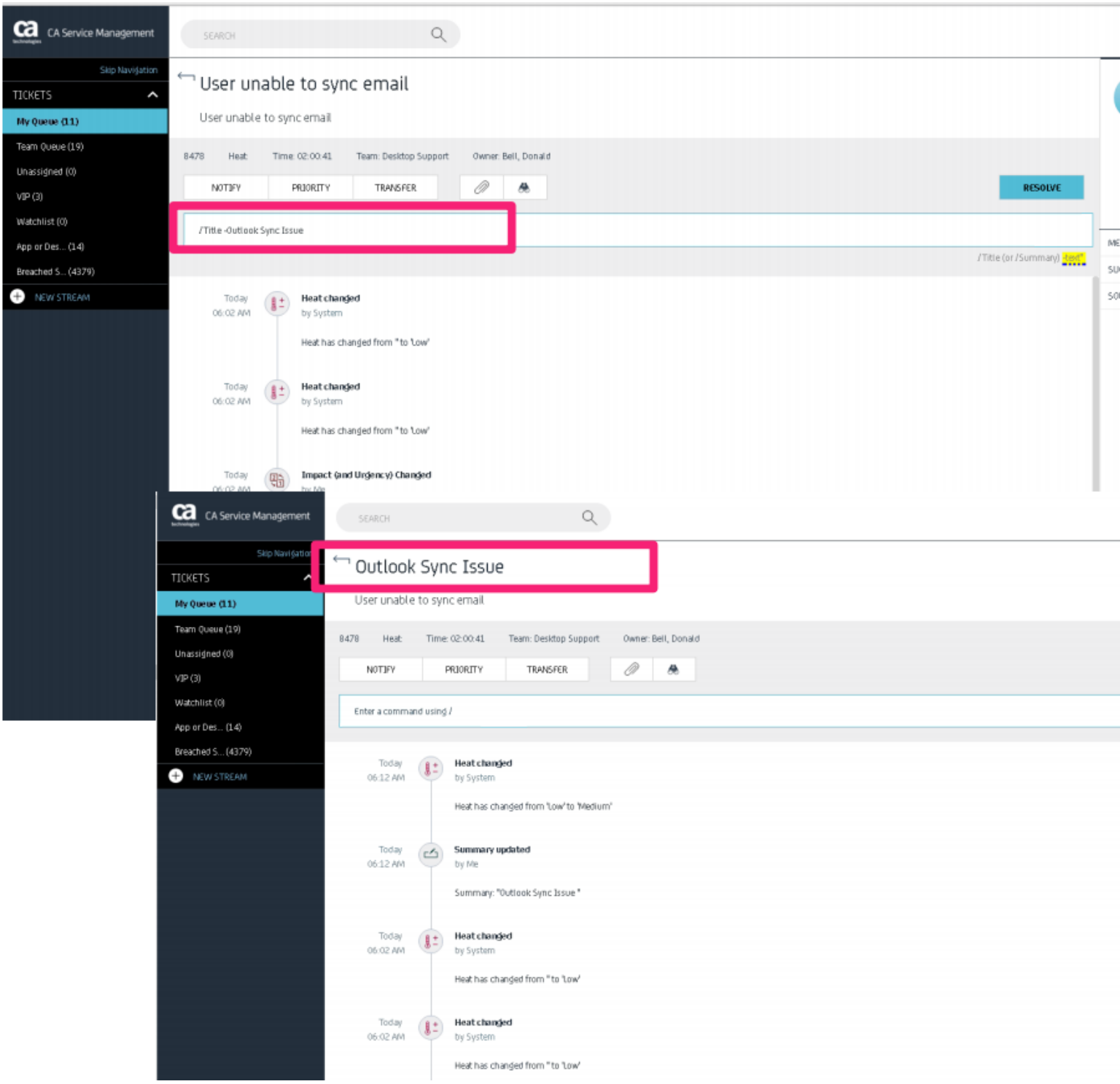


What is a command bar and how can I use it?

Command bar provides the ability to perform tasks using text commands.

To view the list of available commands, enter **/** in the command bar.

For example, to add a title to the ticket, enter **/Title –<*Outlook Sync Issue*>** and press **Enter**.



How can I search and resolve similar issues?

The **Solutions** displays knowledge base articles, tickets, and forum posts that could help resolve the issue similar to the present ticket details.

The **Solutions** are displayed on the right side of the ticket timeline.

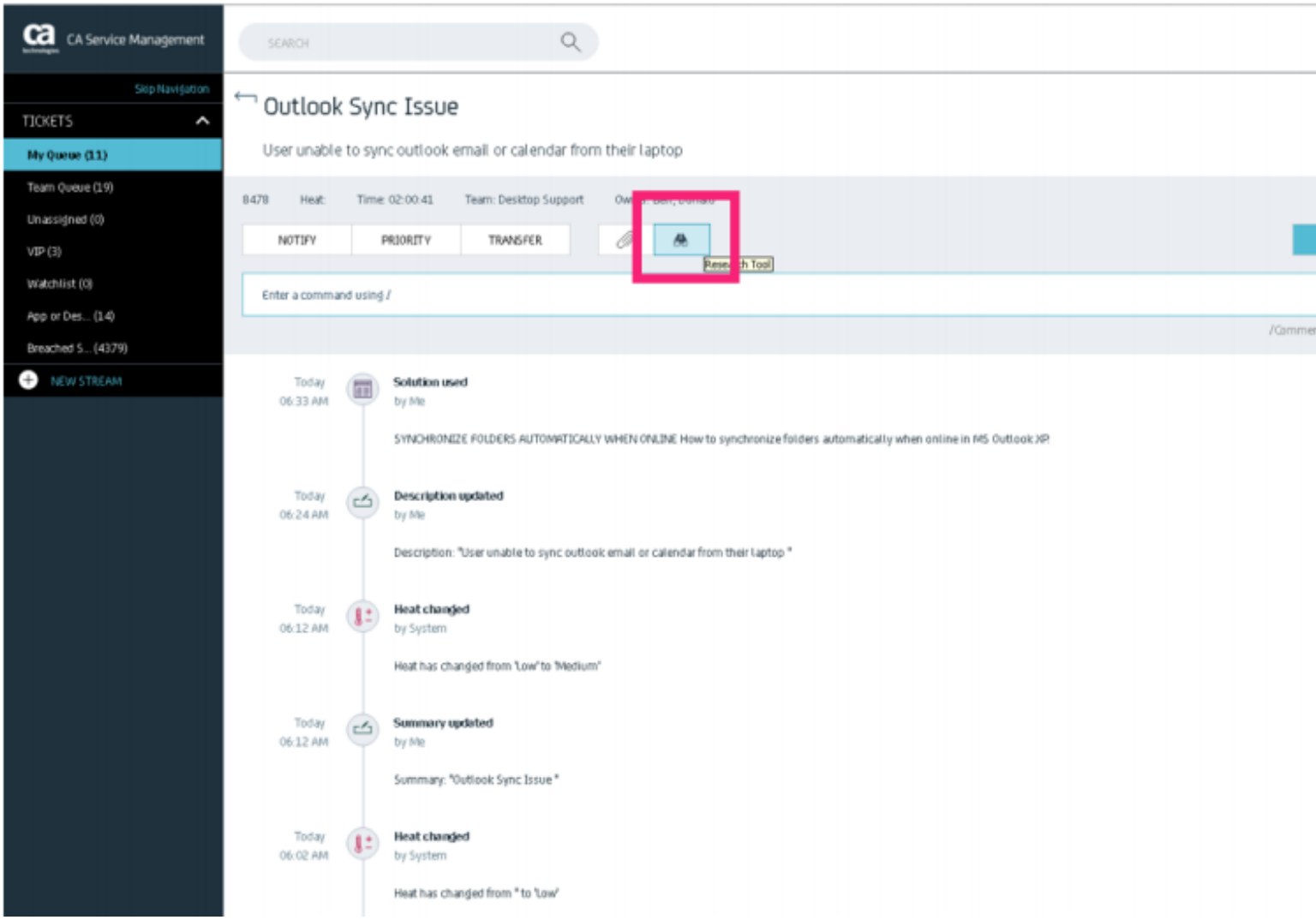
Click **Use** for a suggested solution that you find useful for resolving the ticket and it gets appended to the ticket timeline.

How can I perform an external search on a ticket?

The Research Tool allows an analyst to perform an external search to resolve a ticket.

The Research Tool icon is displayed on the ticket details page to view the Google search results based on the ticket summary.

This icon allows an analyst to attach URLs, selected text, and screenshots to a ticket timeline.



How to configure the Research Tool Extension?

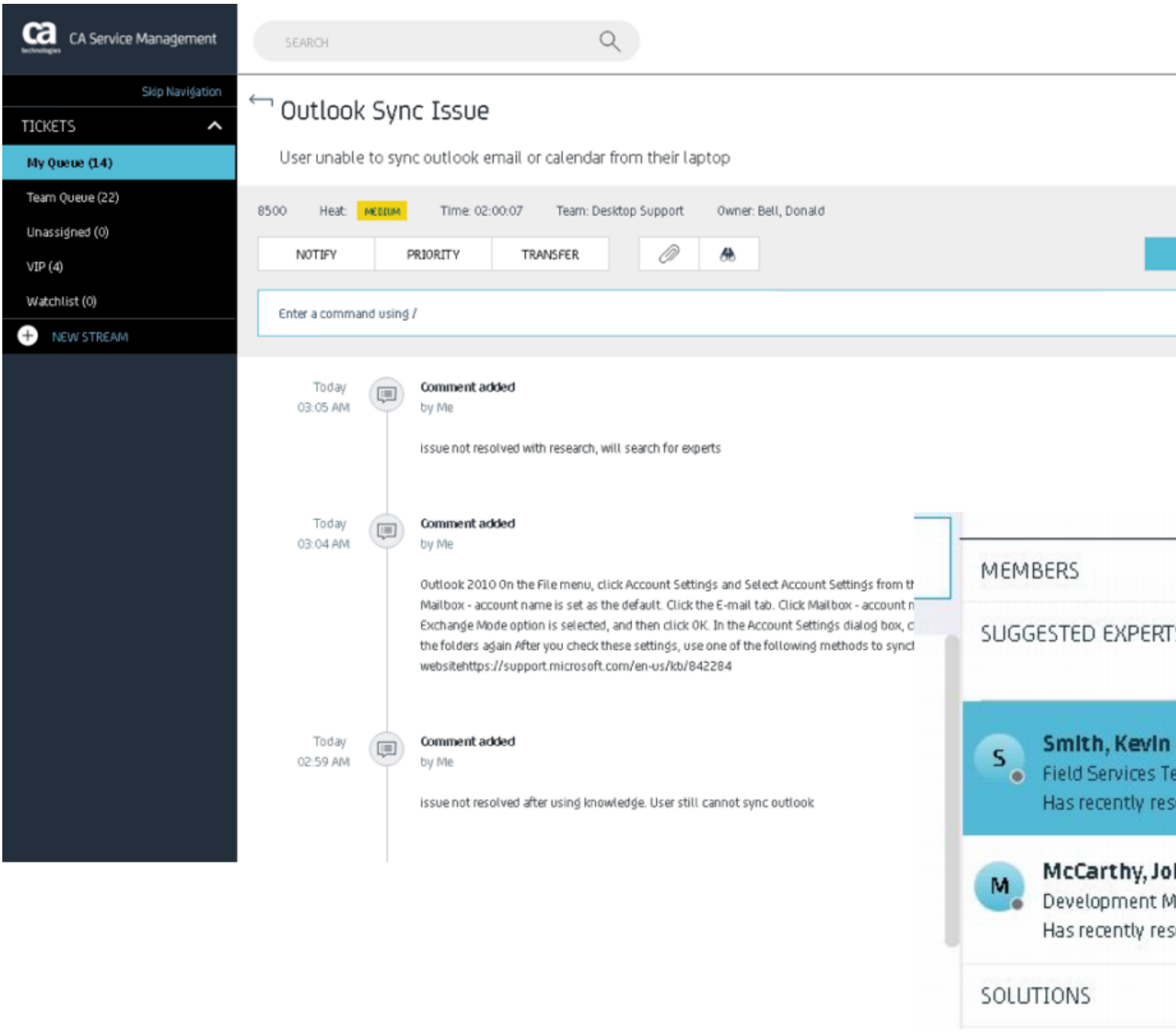
To enable the Research Tool for your browser, see Enable the Research Tool Extension.

How can I receive expert assistance on a ticket?

The **Suggested Experts** displays the list of expert analysts that can assist in resolving the ticket. The widget automatically understands the ticket context and suggests the experts that are best suited to resolve the ticket.

The **Suggested Experts** are displayed on the right side of the ticket timeline.

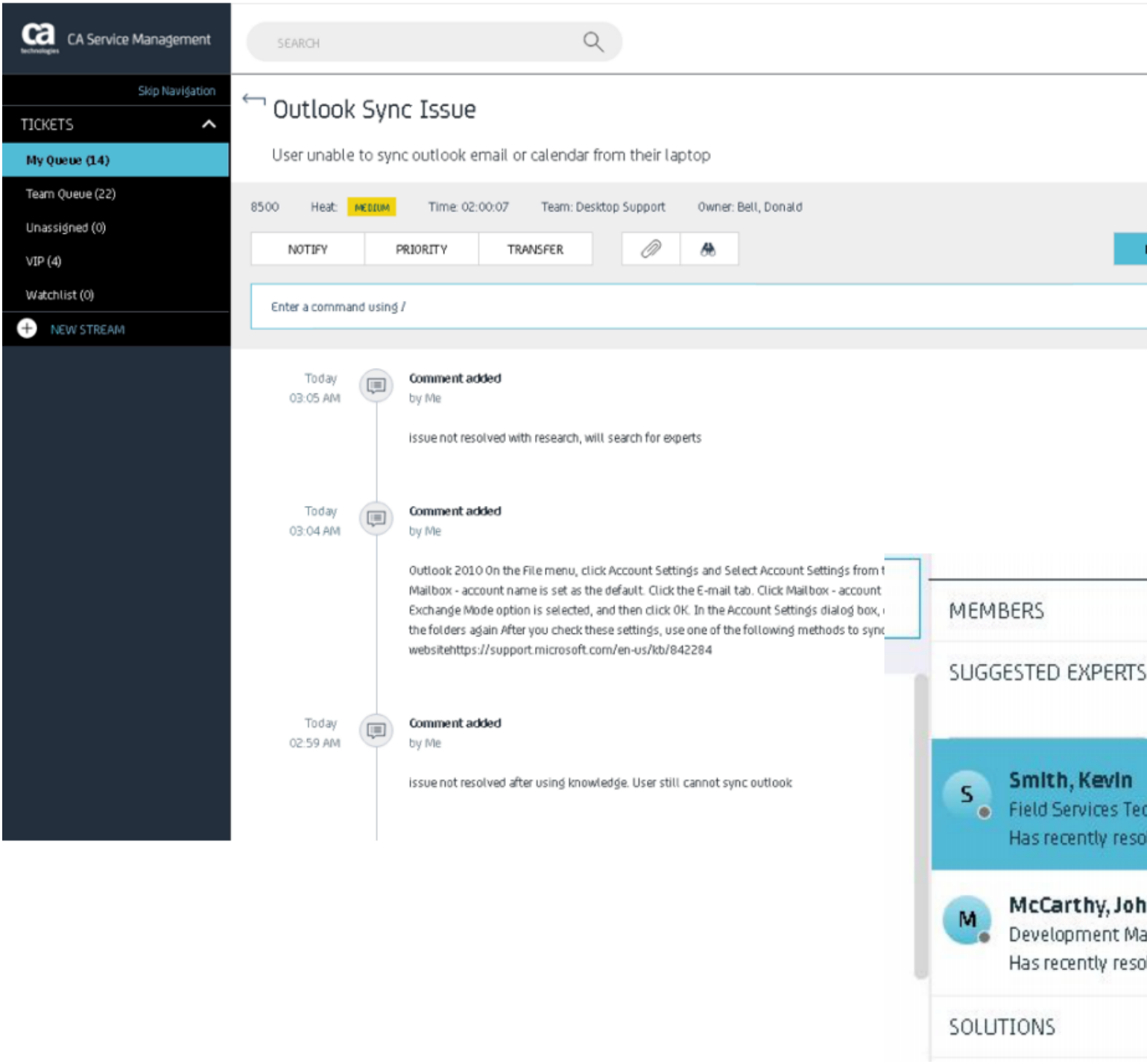
The list displays the first five contacts based on the maximum number of similar tickets they have resolved. To contact an expert, click the email or chat icon against the name.



How can I collaborate with experts on a ticket?

The Suggested Experts on the right side of the ticket timeline displays the list of expert analysts that can assist in resolving the ticket.

The list displays the first five contacts based on the maximum number of similar tickets they have resolved. To collaborate with an expert, click the chat icon against the name. The conversation history is auto-saved in the ticket timeline for future reference.



How can I change or customize the ticket attributes on the xFlow Interface?

The xFlow Interface displays eight predefined attributes. The first four attributes (Ticket Number, Heat, Summary, and Description) cannot be customized. To configure the last four attributes, see View Configuration.

How can I view all the ticket attributes at a glance?

To view all the ticket attributes, type **/ViewAttributes** in the command bar and press **Enter**.

How can I view all the apps from the xFlow Interface?

From the top right pane of the xFlow Interface, click the App Launcher icon

to view or select an app.

Access to Apps is based on the role and user permissions. For more information, see App Launcher - Apps Configuration and Role Switching.

Configure the Mobile Attributes for Tickets

CA Service Management 17.4 continues to provide the ability for Administrators to configure new mobile attributes. Analysts can view these attributes in the mobile application UI while creating or editing a ticket. Administrators can add any mandatory or custom attributes while creating a ticket and make these attributes available to analysts. Analysts can view these attribute in the mobile application while creating or editing a ticket and fill in the required information. Analysts can also search a ticket through ticket ID, summary, affected user. Based on the search criteria, matching tickets are displayed.

NOTE

Mobile attributes for xFlow Interface is supported on CA Service Point Mobile App 1.2 and onwards.

To configure mobile attributes for tickets, complete the following steps:

Follow these steps:

1. Log in to CA SDM as an Administrator.
2. Navigate to **Administration**, **xFlow Interface**, **xFlow Mobile Attributes**.
3. Click **Create New** to create a mobile attribute.
4. Select an **Object Type** from the drop-down list.
5. Select an attribute for the object type from **Select an Attribute**.
   You can also search for the object type attribute from the Object Attributes List page..
6. You can also make an attribute as mandatory on the ticket creation form. For more information, see CA Service Point Mobile App for Analysts.
   NOTE
   CA Service Management 17.4 provides Administrators with the ability to make an attribute mandatory on the ticket creation form.
7. Save the form.
   Analyst can view the configured mobile attribute while creating or editing a ticket from the mobile application.

Restricted Object Type Attributes for Mobile Applications

Currently, xFlow Interface Object type attributes are supported for Requests/Incidents only. Restricted supported object type attributes are as follows:

- ACTIVE
- Call Back Flag
- Charge Back ID
- Closed on
- Heat
- Last Modified
- Modified by
- Opened on
- Predicted SLA Violation
- Reported By
- Resolved On
- SLA Violation
- Ticket Number
- Ticket Type
- Incident Priority

Manage Work from the xFlow Interface

This section explains how an analyst can effectively manage work and contains the following topics:

- All About Work Streams
- Prioritize Work with Heat
- Manage Tickets
- Manage Workflow Tasks
- View the CA Process Automation Workflows
- Create a Ticket in the xFlow Interface
- Search the xFlow Interface
- View Knowledge Base Articles from the xFlow Interface
- Research and Resolve a Ticket
- Resolve Tickets with Contextual Collaboration
- Manage Service Type Events (SLA) - xFlow Interface
- Create Contact in the xFlow Interface

All About Work Streams

This topic provides an overview of Work Streams and includes the following:

Introduction to Work Streams

Work streams provide logical grouping of tickets based on the specific search criteria.

For example, *My Queue* work stream provides quick access to the active tickets assigned to the analyst.

The top eight streams are displayed at first. You can click the **Show More** option to view the entire list of work streams.

A red dot icon on a work stream label indicates a ticket is added or updated in the work stream.

Predefined Work Streams

A few work streams are available by default. An analyst can hide or show these work streams but cannot edit or delete them.

- **Follow up Queue:** Active tickets with a follow up scheduled.
- **My Queue:** Active tickets assigned to the analyst.
- **Team Queue:** Active tickets assigned to the analyst groups.
- **Unassigned:** Active tickets assigned to the analyst groups with no assignee.
- **VIP:** Active tickets associated with contacts (Affected End Users) that have a VIP special handling type.
- **Watchlist:** Active tickets the analyst is following to monitor the progress.

NOTE

xFlow Interface now provides the ability to reorder/rearrange the work streams by drag and drop method.

Custom Work Streams from the xFlow Interface

Analysts can create custom work streams from the xFlow Interface. You can hide, show, edit, or delete these work streams.

Create

An analyst can create a custom work stream.

Follow these steps:

1. Click **NEW STREAM** from the navigation pane.
2. Enter the work stream name.

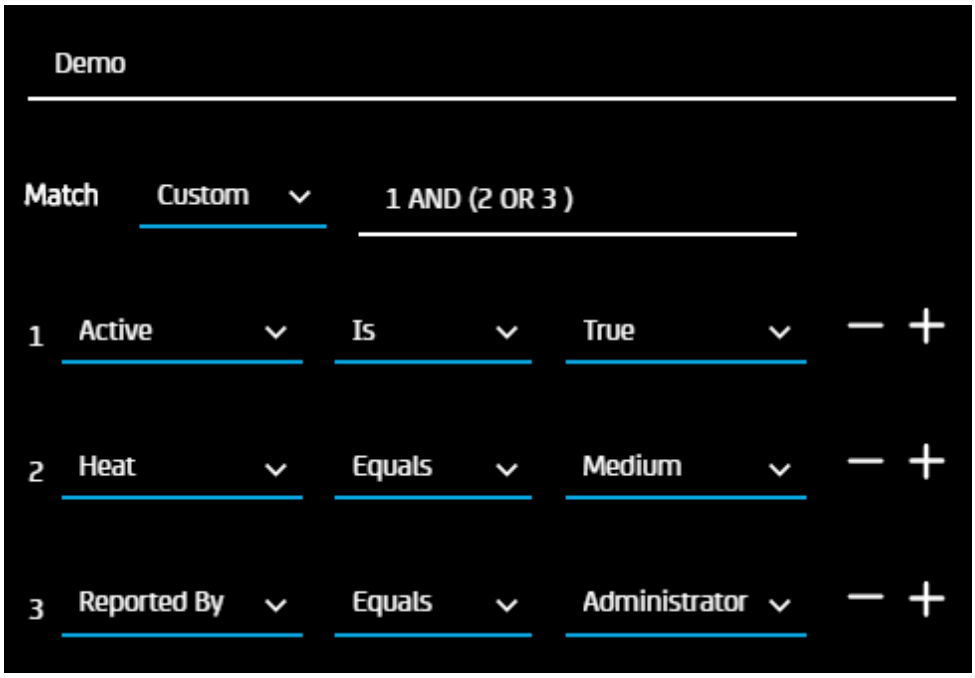
   NOTE

   Stream name can be upto 50 characters in length and can contain alphanumeric characters (A-Z, 0-9) and special characters, except for the ampersand (&), at the rate (@), hash (#), percentile (%), plus (+), quotation marks("), and backslash(\).

   : Stream name can be duplicate.
3. Select one of the following match conditions: **ANY**, **ALL**, or **CUSTOM**.
   The default match condition is **ALL**.
4. Assign a value to each field in the rule.

   NOTE

   : Add the rule: **Active Is True** to view only the active tickets.
5. For a CUSTOM condition, use the following operators: AND, OR, brackets, and the rule number.
   For example, a custom condition can be specified as:



6. Click **Save**.
   The new work stream is successfully added.

Edit

An analyst can edit a custom work stream.

Follow these steps:

1. Hover over the work stream and click the Settings icon.
   The work stream is now in edit mode.
2. Edit the work stream and click **Save**.

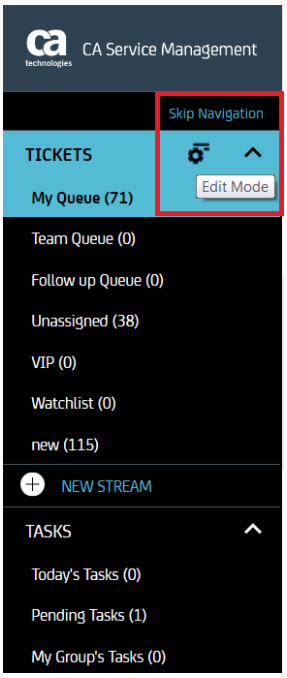
   WARNING

   When you edit a work stream with custom condition, the condition and rules are rearranged. However, the work stream remains unchanged.

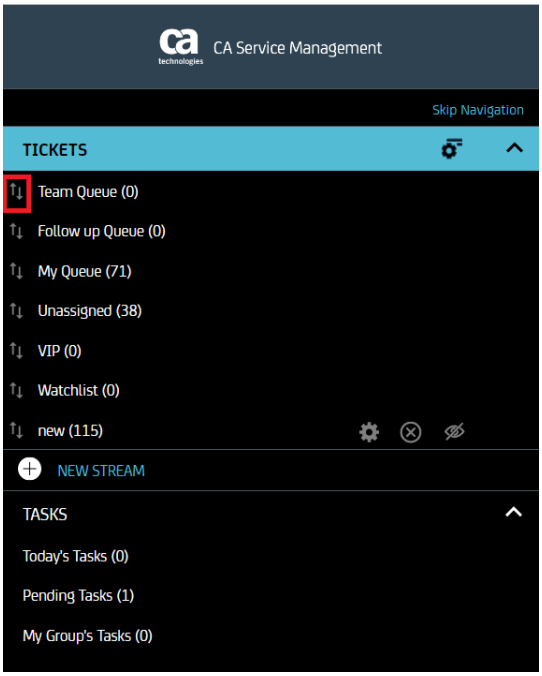
Reorder a Work Stream

xFlow Interface now provides the ability to reorder your work streams by the drag and drop method. Perform the following steps to reorder a workstream:

a. From the left pane of the xFlow Interface, click and expand the Edit Mode Icon on the Tickets section.



b. Select a workstream and click and hold the drag and drop arrows to drag and drop a work stream.



Hide or Show

An analyst can hide or show a custom work stream.

NOTE

Out-of-the-box work streams does not support the hide and show icons. The hide and show icons are supported for Custom work streams.

Follow these steps:

1. To hide a work stream:
   a. Hover over the work stream and click the Hide icon.
2. To show the hidden work stream:
   a. Hover over the TICKETS menu in the navigation pane, and click the Settings icon.
   b. Click the Show icon on the work stream label.

Delete

An analyst can delete a custom work stream. Once a work steam is deleted, it cannot be retrieved.

Follow these steps:

1. Hover over the TICKETS menu in the navigation pane, and click the Settings icon.
2. Click the Delete icon on the work stream label.
3. On the confirmation page, select **Delete**.

Display the Scoreboard Query as a Custom Work Stream

The SDM interface enables an analyst to customize the scoreboard. An analyst can enable the scoreboard query to be displayed as a work stream.

Displaying the scoreboard query as a work stream is specific to the analyst, and is not role based.

Follow these steps:

1. Log in to SDM as an analyst.
2. Select **File** and **Customize Scoreboard**.
   The Customize Scoreboard window opens.
3. Navigate to the scoreboard tree, and select an existing scoreboard node.
4. Verify the **Selected Node's Label** and **Node's Stored Query**.
5. Select the **Show in my workstreams** check box to display them as a work stream.
6. Click **Update Item** and **Finished**.
7. Logout and re-login to the xFlow Interface.

Edit

Any modification to the scoreboard query from the SDM interface will be updated on the work stream.

Follow these steps:

1. Log in to SDM as an analyst.
2. Select **File** and **Customize Scoreboard**.
   The Customize Scoreboard window opens.
3. Navigate to the scoreboard tree, and select the an existing scoreboard node you wish to edit.
4. Eedit the existing **Selected Node's Label** or **Node's Stored Query**.
5. Select the **Show in my workstreams** check box.
6. Click **Update Item** and **Finished**.
7. Logout and re-login to the xFlow Interface.

> **WARNING**
>
> The work stream is deleted from the xFlow Interface if the **Show in my workstreams** check box is not selected.

Delete

An analyst can delete the scoreboard query from the SDM interface. As a result, the corresponding work stream on the xFlow Interface is also deleted.

Follow these steps:

1. Log in to SDM as an analyst.
2. Select **File** and **Customize Scoreboard**.
   The Customize Scoreboard window opens.
3. Navigate to the scoreboard tree, and select an item to delete.
   The selected node form appears.
4. Click **Delete Selected Item** and **Finished**.
5. Logout and re-login to the xFlow Interface.

Hide or Show

An analyst can hide or show the work stream.

Follow these steps:

1. To hide a work stream:
   a. Hover over the work stream and click the Hide icon.
2. To show the hidden work stream:
   a. Hover over the TICKETS menu in the navigation pane, and click the Settings icon.
   b. Click the Show icon on the work stream label.

Prioritize Work with Heat

Analysts can prioritize work by leveraging the Heat feature.

Heat is a composite representation of various attributes and conditions that define the criticality of a ticket.

Heat provides the analysts with a dimensional scoring model to prioritize tickets and understand the amount of attention a ticket must receive.

> **WARNING**
>
> Heat is not displayed for tickets with low heat or when the heat configuration is set to Inactive status.

An analyst can click the heat icon to view the following details:

- Heat of the ticket.
- Heat range of the ticket (High, Medium, or Low).
- Attributes contributing to the heat.
- Weightage for each attribute.

Heat is refreshed based on the reindexing interval which is set by the administrator or when a ticket is updated.

Below is an example of a high heat ticket:

Manage Tickets

You can choose to view the tickets of the work stream in card or list view. You can switch between the views using the Toggle icon on the top right corner of the interface.

Perform the following actions to manage tickets from the xFlow Interface:

Actions

You can perform the following actions on a ticket from the card or list view:

- **Comment**: Allows you to add a comment to the ticket.
- **Notify**: Allows you to notify the end user.
  Optionally, you can add another contact, whose email address is defined in the CA SDM database.
  Optionally, you can attach multiple files while notifying the end user.
- **Follow Up**: Allows you to schedule a follow up for the ticket.
  Optionally, you can set a reminder for the follow up. When the reminder is due, it is displayed as a pop-up message.
  Once a follow up is scheduled, the ticket is added to the ***Follow up Queue*** Work Stream.
  You can schedule, reschedule, or delete a follow up from the card and list view or the Command Bar.

  NOTE

  : There can be only one follow up per ticket.

Click the **Ellipsis** icon to perform the following actions:

- **Status**Allows you to update the ticket status.
- **Priority**
  Allows you to modify the impact and urgency of the ticket if APC is enabled in CA SDM. Allows you to modify the ticket priority if APC is disabled in CA SDM.
- **Transfer**Allows you to transfer the ticket to a group and member of the group (assignee).

Ticket Information Panel

The Ticket Information panel, displayed at the top of ticket details section, lists the following attributes:

- **Status**Allows you to update the ticket status.
- **Priority**
  Allows you to modify the impact and urgency of the ticket if APC is enabled in CA SDM. Allows you to modify the ticket priority if APC is disabled in CA SDM.
- **Category**
  Allows you to update the ticket category.
- **Opened on**
  Displays the date and time when the ticket was created.

Quick Actions

You can perform the following actions on a ticket from the ticket details section:

- **Comment**Allows you to add a comment to the ticket.
- **Notify**Allows you to notify the end user.
  Optionally, you can add another contact, whose email address is defined in the CA SDM database.
  Optionally, you can attach multiple files while notifying the end user.
- **More**
  Allows you to perform the following actions:
  - **Transfer**Allows you to transfer the ticket to a group and member of the group (assignee).
  - **View Details**Allows you to view all the ticket attributes.
  - **Follow Up**Allows you to schedule a follow up for the ticket.

Sort By

Use the **Sort By** feature to reorder the ticket list based on a selected attribute. For example, you can sort the list by *Ticket Number*.

> **NOTE**
>
> When heat is enabled by the administrator, it is set as the default sort option.

Filter

Click the Filter icon to select multiple attributes to narrow your search results.

Export

Click the Export icon to download the ticket details listed in the work stream.

The default file name is *Export.xls*.

Command Bar

The command bar provides the ability to perform tasks through text commands.

You can use the following shortcuts to navigate to the command bar from the ticket details page.

- **Alt+R**: Chrome and Internet Explorer
- **Alt+Shift+R**: Firefox
- **Control+R**: Safari

Following are the description and syntax for the commands:

NOTE

: Enter a command using /

: Optional parameters are indicated by the **skip** factor. You can choose to hit Tab or press Enter to skip the optional parameters.

: Few of the command parameters contain text fields. If the text field has a hyphen (-) symbol, prefix it with the backslash (\) symbol. For example, refer to the following table.

S.No	Command	Description	Syntax and example
1	Chat	Initiates a conversation with another analyst to resolve a ticket.	/chat @<username> or /cht @<username> To initiate a conversation from the command bar, type /chat @ John Long
2	ChangeCategory	Assigns a category to the ticket.	/ChangeCategory -<*category*> (or) /Cat -<*category*> To select Hardware as the ticket category, type /ChangeCategory -Hardware
3	ChangeOrder	Populates the change order form to create a change order.	/ChangeOrder (or) /CO To create a change order, type /ChangeOrder
4	Close	Closes a ticket.	/Close (or) /CL To close the ticket, type /Close
5	Comment	Adds a comment to the ticket.	/Comment -<*comment*> (or) /Log -<comment>To add the comment: The test has failed, type /Comment -The test has failed
6	Description	Adds a description to the ticket.	/Description -<*description*> (or) /Dsc -<*description*> To add the description: exchange outage impacts the office users, type /Description -Exchange outage impacts the office users

7	Update Status	Changes the ticket status.	/Updatestatus -<*status*> (or) /St -<*status*> The command displays the current status of the ticket. To change the status, select the appropriate status from the list. For example, to acknowledge the ticket, type /Updatestatus -Acknowledged
8	Escalate	Assigns impact and urgency to the ticket when APC is enabled in CA SDM. Assigns priority to the ticket when APC is disabled in CA SDM.	/Escalate -<*impact*> -<*urgency*> (or) /Escl -<*impact*> -<urgency>(APC is enabled in CA SDM) To assign impact 4 and urgency 4 to the ticket, type /Escalate -Impact 4 -Urgency 4 (APC is disabled in CA SDM) To assign priority 4 to the ticket, type /Escalate -*Priority* 4
9	Filter	Allows to filter the ticket timeline by activity type. Analysts can filter the ticket timeline by multiple activity types.	/Filter -<Type> (or) /Fltr -<Type>To filter the timeline by Field Update, type/Filter - Field Update
10	Follow up	Allows to schedule a follow up for a ticket. **Note**: There can be only one follow up per ticket.	/Followup (or) /CBTo schedule a follow up, type /CB and press Enter.Select the date and time from the calendar. (**Optional**) You can add a comment and set a reminder for the follow up.Navigate to the command bar again and press Enter to save the follow up. To reschedule a follow up, type/CB and press Enter.Modify the date or time from the calendar. (**Optional**) You can modify the comment and the reminder for follow up. To remove a scheduled follow up, type/CB and press Enter.Click the Remove icon.
11	Incident	Creates an Incident.	/Incident (or) /IN To create an incident ticket, type /Incident
12	Problem	Populates the problem ticket form to create a problem ticket.	/Problem (or) /PR To create a problem ticket, type /Problem
13	Request	Creates a Request.	/Request To create a request, type /Request

14	Set	Allows to set value for the defined ticket attributes.	/Set -<*Attribute*> - <*Attribute Value*> (or) /MD -<*Attribute*> - <*Attribute Value*> To add the Major Incident attribute to the ticket, type/Set -Major Incident -Yes
15	SwitchView	Allows to view the existing ticket in classic SDM UI.	/SwitchView (or) /SV To view the existing ticket in classic SDM UI.
15	Title	Adds a title to the ticket.	/Title -<title> (or) /Summary -<*summary*>To modify the ticket title or summary: Wireless connectivity Issue, type /Title -Wireless connectivity Issue
16	Transfer	Allows transfer of a ticket.	/Transfer -<assignee> -<group> (or) /Tr -<*assignee*> -<*group*> To transfer the ticket to Dustin from NY Support Group, type /Transfer -Dustin -NY Support Group
17	UpdateProperties	Allows to modify the property value associated with the ticket.	/UpdateProperties
18	ViewAttributes	Allows to view all the ticket attributes.	/ViewAttributes (or) /VA

WARNING

For a limitation with Work Streams, see the known issues in Service Management 17.4 Release Notes.

Members

Analysts can tag contacts as members to notify them of all activities on the ticket through an email.

Follow these steps:

1. Click the Members drop-down list.
2. In the search field, enter the contact name.
3. Hover over the search result and click the **+** icon.
   The contact is added as a Member.

To delete a member, click the **x** icon on the contact.

Manual Notifications for xFlow

NOTE

This is applicable from CASM 17.4.5.0 and above.

You can send a message about an issue to an end user, vendor, analyst, or anyone else in your service desk database. Record of the manual notification is stored in the timeline for the issue.

Manual Notify Fields

To Recipients:
Lists the names of the people who receive the notification. Click **Add Contacts** to select relevant contacts or contact objects to add to the recipients list.

CC Recipients:
Added separate **CC (Carbon Copy)** recipient list alongside the primary **TO** recipient list. CC recipients receive copies of the notification without being the primary audience.

> **NOTE**
> This field appears when the option ***mail_show_to_cc_list*** is installed using Options Manager.

Contact:
Enter characters into the contact field of the ticket. Results are displayed as you type, allowing you to select a contact from the suggested list.

Urgency:
Specifies the relative importance of the notification (for example: Low, Normal, High, or Emergency). Urgency levels are predefined by the system administrator.

Preferred Method:
Specifies the notification method to use, such as Email or Notification.

> **NOTE**
>
> When you specify an email address, set the method to one that supports SMTP, otherwise, remove email addresses from the Selected Recipients list.

Internal:
Makes the notification visible to internal contacts only.

Message Title:
Specifies the title of the notification message. This is a short description similar to an email subject line. By default, the title includes the ticket type, ticket number, and the words "Manual Notify" *(for example, Incident 32 Manual Notify).*

Message Text:
Specifies the full text of the notification. The message area now includes a **rich text editor (CKEditor)** for creating professional, formatted messages. It also supports *predefined templates* configured in Service Desk Manager for quick and consistent responses.

Personalized Response:
Specifies a personalized response such as a generic or administrator signature. If any personal responses are configured in the classic UI they would appear in xFlow Interface as a Dropdown menu. The response is added at the location where the mouse cursor is pointed.

> **NOTE**
>
> The administrator must set up a personalized response before you can use this option. For more information about personalized responses, see the *Administration Guide*.

Add Stakeholders:

One-click functionality to add all ticket stakeholders to recipient lists "Add Stakeholders" button is available now for both TO and CC list recipients Firstly, Stakeholders should be added under the Ticket Members section. Later stakeholders can be added to the To/CC recipient list as per the need.

> **NOTE**
> This field appears when the option**notification_prepopulate_stakeholders** is installed using Options Manager.

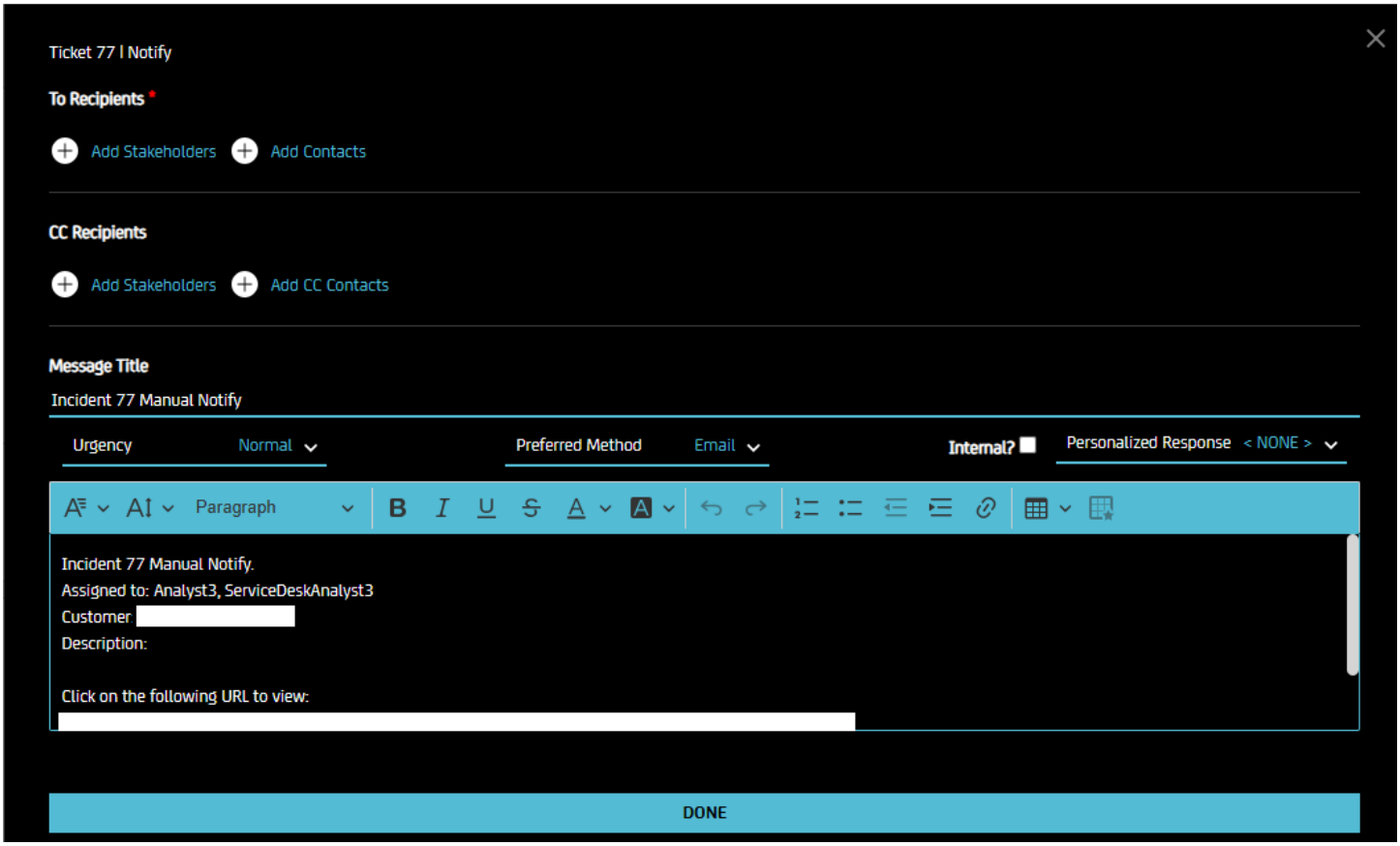
Allow Temporary Email Addresses:

(External Email Access) users have now ability to send notifications to email addresses outside the system ensuring proper email validation. You can enter an external email address in the *To recipient* list or the *CC recipient* list.

NOTE
This field appears when the option **notification_allow_temp_address**is installed using Options Manager.

Manual notify screen:

Figure 31: 17_4_ru5_manualnotify_xflow



Manage Workflow Tasks

This topic provides an overview of the classic workflow, and its tasks and how an analyst can manage their tasks from the xFlow Interface:

Workflow Tasks

A workflow specifies the sequence of tasks to be performed to resolve a ticket. A workflow may require some of the tasks to be completed before the ticket can be closed. The only exception being the ticket status is set to Cancelled.

Predefined Workflow Tasks

The following workflow tasks are available by default:

- **My Group's Tasks**
  Lists all the pending tasks that are assigned to groups that the analyst is a member of.

NOTE
Stream Name called "My Group's Tasks" lists only the classic CA Workflow tasks that are eligible. The query execution or result does not involve any integration.

- **Pending Tasks**
  Lists all the pending tasks that are assigned to the analyst.
- **Today's Tasks**
  Lists all the pending tasks assigned to the analyst that is due for completion on the current date (12AM - 11:59PM)

Filter

Click the Filter icon to select multiple attributes to narrow your search results.

Sort By

Use the **Sort By** feature to reorder the task list based on a selected attribute. For example, you can sort the task by Status.

NOTE

Ticket Number is set as the default sort option.

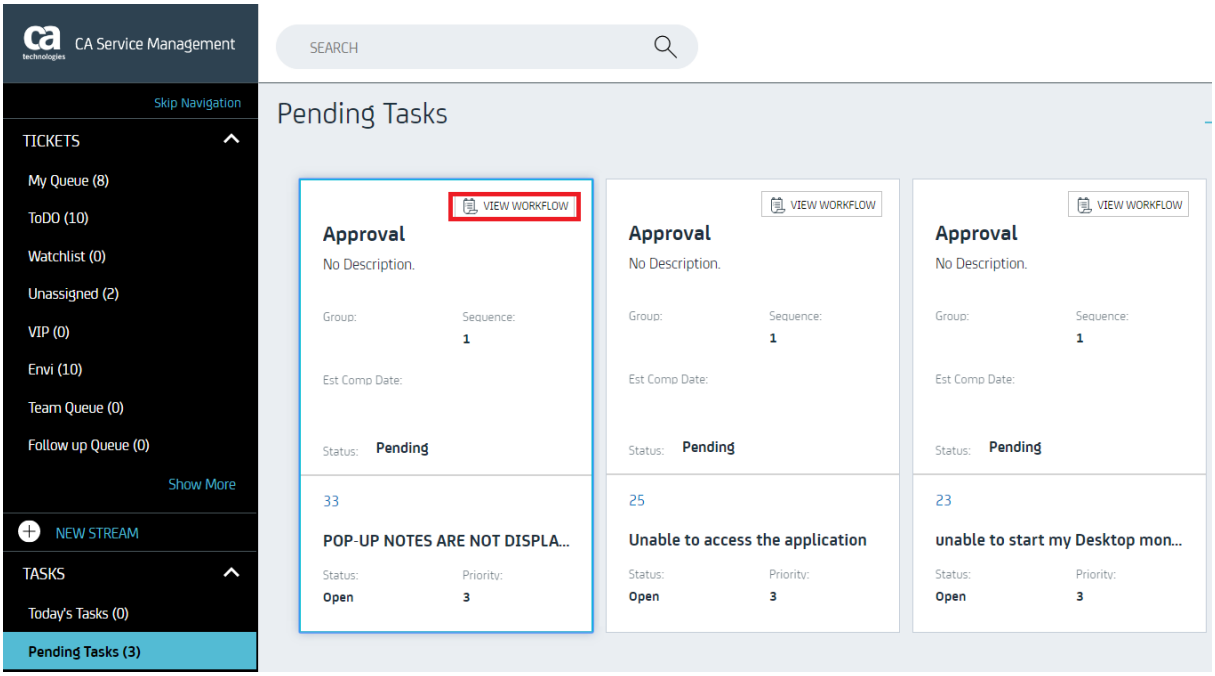
Graphical View of Workflow

A graphical viewer introduces a visual representation of workflow tasks to help analysts understand the end-to-end workflow. The updates that are made to the workflow and its tasks are displayed on the ticket timeline, and are arranged in chronological sequence.

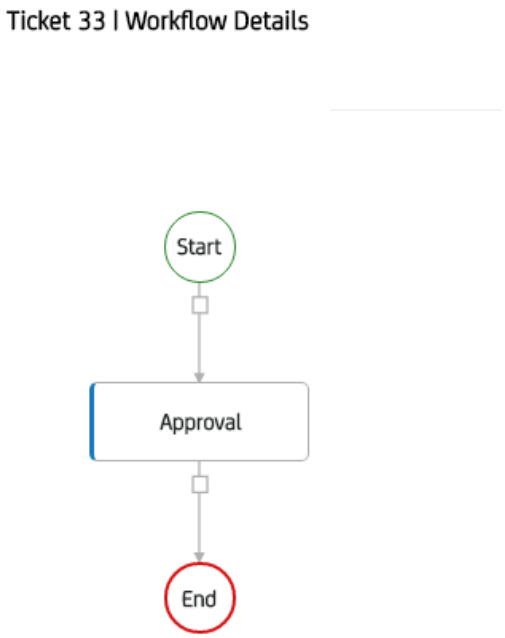
NOTE

xFlow Interface now provides the ability to view the Workflow Tasks attached to a ticket from Task Details view or Card view.

The following diagram displays the Workflow attached to a ticket in Card View:



Click **View Workflow** to see Workflow details attached to a ticket.



The following information is available to an analyst:

- Task sequence
- Current state of the task

An analyst can modify or delete a task.

A workflow can be accessed from either the ticket timeline or from the Command Bar. Type the following command in the Command Bar to launch the graphical workflow viewer.

```
/workflow
```

Legend

Green color indicates that the task is complete.

Blue color indicates that the task is pending.

Orange color indicates it is a future task.

Notifications

A red dot icon on a task indicates that the task is added or updated. Refresh the task for most recent changes.

> **WARNING**
>
> When a workflow task is added to an active ticket, the task count is not updated real time.

Edit a Task

An analyst can edit a workflow task. To edit a task, navigate to the task either from the work stream or from the graphical viewer.

Follow these steps:

1. Click the task that needs to be modified.
2. Update the task and click **Save**.

Delete a Task

An analyst can delete a workflow task. Tasks that are configured as deletable would have the delete option enabled. To delete a task, navigate to the task either from the work stream or from the graphical viewer.

Follow these steps:

1. Click the task that needs to be deleted.
2. Click **Delete**.
3. On the confirmation page, select **Delete**.

View the CA Process Automation Workflows

The xFlow Interface now offers the ability to view the CA Process Automation Workflow Process from the following:

Ensure that you have integrated CA Process Automation Workflow with CA SDM to be able to view the workflow entry from the Ticket Timeline and Tasks stream in the xFlow Interface.

> **NOTE**
>
> Only view functionality for CA Process Automation tasks is provided in the xFlow Interface. On clicking a workflow task, you will be redirected to CA Process Automation My Tasks page, where you can view and modify the workflows as required.

Ticket Timeline

Perform the following steps to view the Process Automation Workflow entry from the Ticket Timeline in the xFlow Interface:

1. Log into the xFlow Interface.
2. Select a ticket that has a Process Automation Workflow process attached to it. Click on the Workflow process hyperlink from the Ticket Timeline.
   You are redirected to the CA Process Automation Process Details page of the selected workflow.
3. Alternatively, you can also use the xFlow Interface command bar to view the CA Process Automation Workflows. Enter the following in the command bar:

   ```
   /wf
   ```

   ```
   Or
   ```

   ```
   /Workflow
   ```

4. Press **Enter** to execute the command. You will be redirected to CA Process Automation Process Details page for the selected workflow.

Tasks Stream

If you have integrated CA process Automation with CA SDM, you can view **My ITPAM Tasks** in the task stream section of the xFlow Interface (left Navigation pane). When you click **My ITPAM Tasks**, by default the tasks stream are displayed in the card view. You can change view type to **List view**.

> **NOTE**
> Stream Name called "My ITPAM Tasks" lists all the ITPAM Tasks that are eligible to appear and does not show classic CA Workflow tasks. The query execution or result works only in conjunction with integration and these results are retrieved from CA ITPAM.

In both the view types (Card and List), you can see the **Estimated End Date**, **Ticket number**, **status**, and other details related to a task.

> **NOTE**
>
> CA Process Automation Tasks stream do not have any filter options available in the xFlow Interface while the filter option is available for CA SDM Classic Workflow tasks stream.

> **NOTE**
>
> This **Estimated End Date** may or may not be available for all tasks. This date must be configured by a CA Process Automation Designer on the CA Process Automation server. If it is not configured, you cannot see the **Estimated End Date** details in the **xFlow Interface - Card View** and **List View**.

Create a Ticket in the xFlow Interface

This section explains how to create a ticket in the xFlow Interface:

Create a Ticket from the Multi-faceted Search Bar

Multi-faceted Search bar allows you to search for a contact, view the number of open tickets for that contact, and create a ticket.

The type of ticket that an analyst creates depends on the assessment of the user's requirement during the call.

Scenario 1

The following scenario explains how an analyst creates an Incident when the default ticket type is Request.

1. Quan, the manager at Example.com, encounters the Blue Screen of Death after rebooting his laptop. He contacts the Service Desk and Dustin, an IT Analyst, receives the call or email.
2. Dustin searches for Quan's name in the multi-faceted search field. The search results display Quan's details and the number of open tickets for him.
3. Dustin hovers over Quan's name from the search results and selects the **New Ticket** option.
4. As this scenario fulfils the criteria for an incident, Dustin enters the following command in the command bar:
   ```
   /Incident (or) /IN
   ```
5. As Quan describes his problem, Dustin enters a brief description in the command bar.
6. The command bar evaluates what Dustin types and if a Smart Word is detected, it is highlighted in yellow. A Smart Word provides suggestions for categories and templates.
   Optionally, Dustin can apply a category and template to the ticket, and can press Enter in the command bar.

   > **NOTE**
   >
   > Dustin can still create the ticket if a Smart Word is not detected. To add a category after creating the ticket, see Command Bar.

7. A grid requesting additional information is displayed. As and when Dustin enters the additional information, it is removed from the grid and is displayed on the Timeline.
8. Dustin must press the Enter key in the command bar, and review the details by selecting the **Show All** button on the grid.
9. Dustin can complete creating the ticket by pressing the Enter key in the command bar.

Scenario 2

The following scenario explains how an analyst (Dustin) creates a problem ticket.

1. Dustin access an open ticket, and navigates to the ticket details.
2. In the command bar, enter the following command:

   ```
   /Problem (or) /PR
   ```

   Populates the problem ticket form to create a problem ticket.
3. Dustin enters brief description and summary for the problem ticket.
4. A grid requesting additional information is displayed. As and when Dustin enters the additional information, it is removed from the grid and is displayed on the Timeline.
5. Dustin must press the Enter key in the command bar, and review the details by selecting the **Show All** button on the grid.
6. Dustin can complete creating the ticket by pressing the Enter key in the command bar.

In-Context End User Information

The In-Context End User Information viz-a-viz the Quick Profile equips the analyst with the contact and environment information of the user. This enables the analyst to deliver an enhanced customer experience and an effective ticket resolution. For detailed information, click the **More details** option.

Information Available

The following information of the user is available:

1. **Contact Information**:
   - **Name:** Displays the complete name of the contact.
   - **Tenant**: Displays the tenant associated with the contact.

     NOTE

     This field is available only when multi-tenancy is enabled.
   - **Access Type**: Specifies the access type. The access type determines the system functions the contact can access.
   - **Data Partition**: Specifies the data partition for the contact. This value determines the records that the contact can access. Data partitions are subsets of the database with restricted access to data records, based on their content.
   - **Service Type**: Specifies the level of support received by the contact. For example, some users have 24-hour support, and others receive on-site training.
   - **Time Zone**: Specifies the time zone where the contact is located.
   - **Phone Number:** Specifies primary phone number for the contact.
   - **Alternate Phone Number**: Specifies an alternate phone number for the contact.
   - **Email**: Specifies email address of the contact.
   - **Location**: Specifies the contact's physical address.
   - **Contact Notes**: Displays contact specific notes.
2. **Environment Information**:

- **Name**: Specifies the asset name associated with the contact.
- **Class**: Specifies the general category for the Configuration item (CI)
- **Family**: Specifies the CI family associated with the CI class.
- **Serial Number**: Specifies the serial number for the CI item.
- **Product Version**: Specifies the version number of the CI item.
- **Asset**: Categorizes an asset for filtering purposes and to control display in CA CMDB.

How to Access In-Context End User Information

You can access In-Context End User Information from the following:

- **Search**: Search for a contact in the multi-faceted search field and click the profile picture or contact name.
- **Card or List view**: Access a ticket in the card or list view and click the profile picture or contact name.
- **Ticket Details**: On the ticket details page, click the profile picture or contact name.

Search the xFlow Interface

Multi-faceted Search

Multi-faceted search provides you quick access to an existing record or and you can take action that is based on your search. For example, you can search for a user and can create a ticket on their behalf. Multi-faceted search provides relevant suggestions as you type and displays the first five results that match your query, for each facet. You can search for users, assets, tickets (incidents and requests), and knowledge documents.

> **NOTE**
>
> If there are no matching results for a facet in the search bar, it is removed from the drop-down list.

Create a Contact

You can also create a new contact from the search bar by entering the required name for the contact in the search bar. Click on the Create New contact option to create a new contact. For more information, see Create a Contact in CA SDM.

User Facet

In the multi-faceted search text box, type a user name to view the matching users. You can view a maximum of five users, by default. For each user, you can view details, such as first name; last name; title (for example, analyst, administrator, and so on); number of open tickets, and also create a ticket on their behalf.

- Click on the username to view the details in the CA SDM UI.
- Click on the ticket number to view the list of open tickets. When you click on any ticket the ticket details page opens.
- Click **New Ticket** or **Alt+N** to create a ticket.

For the user facet, the search is conducted only on the active users records.

Shortcut Key

@ <*username*>

For example, to search for John, type *@John* in the search field.

> **NOTE**
> The number of search results that display when you use shortcut keys are 20 for each facet.

Incidents/Requests Facet

Type the ticket number or title to view the matching incidents or requests. For each ticket, you can view details, such as incident or request number; summary; description; ticket status; affected end user. The search for tickets is done on the active and inactive tickets.

Click any ticket to view the ticket details in the xFlow Interface.

Shortcut Key

! <incident or requests>

For example, to search for the incident number 45321, type !*45321* in the search field.

> **NOTE**
> The number of search results that display when you use shortcut keys are 20 for each facet.

Search External System Ticket ID

xFlow Interface provides the ability to search for an External System Ticket ID. If any incident is created from an external system, L1 Analysts can search for the corresponding incident mapped to the external system ID from the xFlow Interface's Search Bar.

For example: If Service Desk Manager is integrated with Service Catalog, and a user reports an issue from Service Catalog, corresponding incident with an **External System Ticket ID** is created in SDM. L1 Analysts can search for the corresponding incident from the Search bar using the **External System Ticket ID**.

> **NOTE**
> In case of more than one incident in the xFlow Interface search results, L1 analysts must check for the matching **External System Ticket ID** that is reported from the displayed search result for incidents. This will help in mapping the SDM incident to the user provided **External System Ticket ID**.

Assets Facet

Type the asset details to view the matching results. The search is performed only on the active assets. You can view details that are configurable, such as asset name; asset family name; hostname (if applicable).

Click any asset in the list to view the asset details in the CA SDM UI.

Shortcut Key

$ <assets>

For example, to search for the configuration item Laptop, type *$laptop* in the search field.

> **NOTE**
> The number of search results that display when you use shortcut keys are 20 for each facet.

Knowledge Base Facet

Knowledge base facet displays the list of published knowledge articles that might be useful for the analyst in resolving a ticket. You can view details for an article, such as, document title; summary.

Click any knowledge base article to view the details in the CA SDM UI.

Shortcut Key

? <knowledge document name>

For example, to search for the knowledge document on how to resolve connectivity issue, type *?How to resolve connectivity issue* in the search field.

> **NOTE**
> The number of search results that display when you use shortcut keys are 20 for each facet.

View Knowledge Base Articles from the xFlow Interface

xFlow Interface now offers the ability for analysts to view the KB documents. Analysts do not have to switch to the Classic CA SDM every time they want to view a KB document.

NOTE

Few exceptions are listed here where a KB article can still be redirected to the Classic CA SDM UI. If a KB article has an embedded link around the following items: **Document Link**, **Create New Ticket Link, Action Content**, and **New Change Order Link**, the KB article will be redirected to the classic CA SDM UI when an analyst clicks on it. Ticket Template and Internal Links to incident search, and so on, will also be redirected to the classic CA SDM UI.

You can perform the following in the xFlow Interface for finding knowledge base solutions for a reported problem:

View Knowledge Base Documents from Search Results

Without a ticket context, you can search for a solution on the Search bar of the xFlow Interface. You can base your search criteria on the following attributes: **Title**, **Summary**, **Problem**, and **Resolution**. If there are related KB articles, search results will be displayed. Click the desired KB article to view the solution for a reported issue.

From **17.4 RU5**, Analysts can now view real-time knowledge and solution suggestions while typing the problem description on the xFlow ticket creation page. Previously, suggestions appeared only after a ticket was created, limiting the effectiveness of the knowledge base in reducing ticket inflow.

With this enhancement:

- Relevant **knowledge articles** and **solution recommendations** are suggested instantly as analysts type.
- Analysts can **review and apply solutions** before submitting the ticket, reducing duplicate or unnecessary incidents.
- This improves **first-call resolution rates** and helps **minimize ticket inflow** into the service desk.

View a Knowledge Base Document from Suggested Solutions

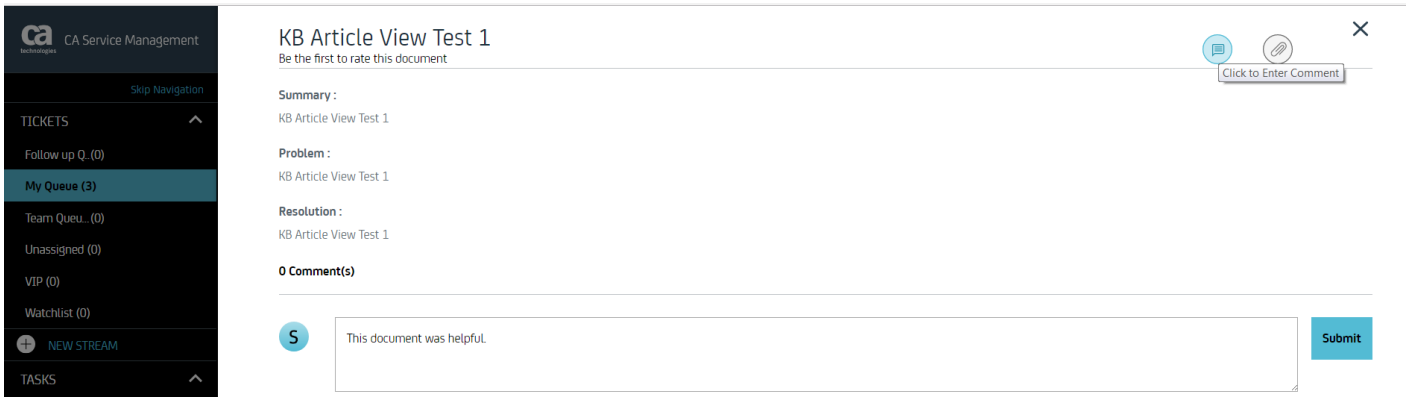
The Suggested Solutions widget displays the list of published knowledge base articles. In context of tickets, it will have suggested solutions which provides a short synopsis of a solution for a reported issue. To decide whether this is a solution or not, click the Expand Arrow icon or details icon (just below the short synopsis) to view the complete KB article. If the KB document addresses a reported issue, click **Use as a Solution** option that is available on the **Rate This Document** ribbon. The suggested solution gets appended to the Ticket Timeline as a link > **Solutions Used**. Click **Ignore** if the KB article did not satisfy your requirement and to remove the solution from the list. However, when you access the ticket again, the earlier ignored or rejected solution might reappear in the list. Click the **Expanded View** Icon (just below the short description for a KB solution) to view the complete KB article to see if it provides solution for a recorded issue.

NOTE
L1 analysts can also share a Knowledge base solution from the Suggested Solutions pane from the Chat Interface.

Provide Comments and Feedback for a KB Solution

After viewing the complete KB article, you can provide your comments in the following two ways:

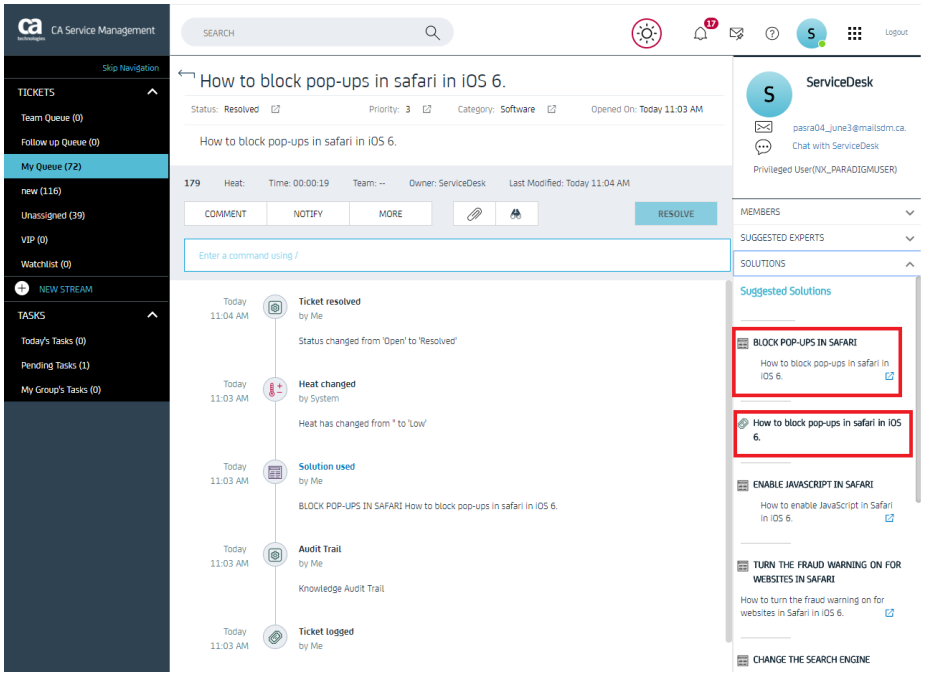
- Click the **Comments** icon on the top right corner of an opened KB Document. Enter your feedback in the comment area.
  Or
- Scroll down to the very end of the KB article and provide your feedback in the comment area. Any existing comments on a KB document is also available for you to view.
  Note that the **Comments** Icon is available when you view the KB articles from Ticket Timeline and Suggested Solution.



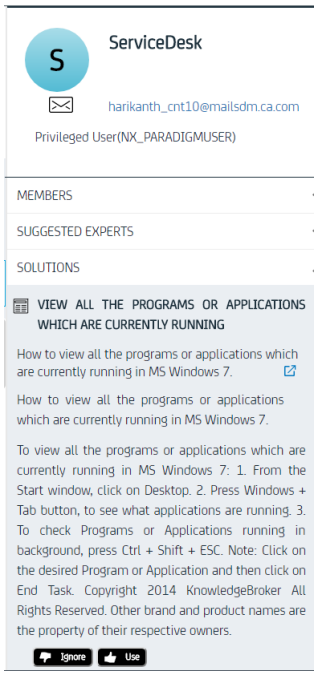
– Additionally, the **Attached Document** icon is available on the top right corner of an Opened KB Document. It shows the attached documents to a KB article. You can download the attachment and use it. The **Attached Document** icon is also available when you view a KB article from the Ticket Timeline and Suggested Solutions.

NOTE

xFlow Interface now provides the ability to differentiate between an old ticket and KB solution through different icons. Select a ticket and from the ticket timeline view, click Suggested Solutions (right pane). Select a ticket or KB solution to get a detailed view.



Click the icon meant for a KB article to have a detailed view about a KB solution.



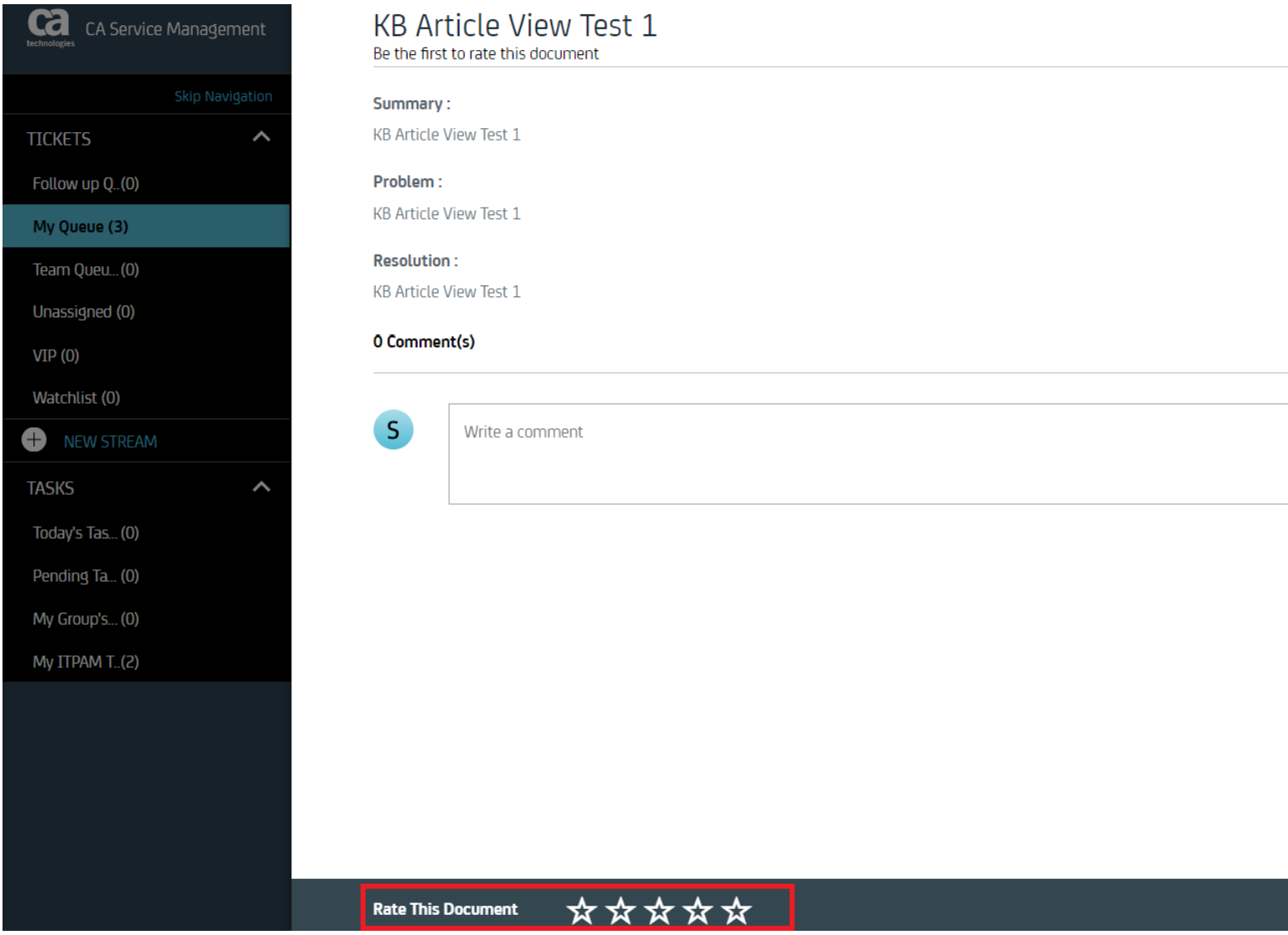
View a Knowledge Base Documents from Ticket Timeline

When an analyst accepts a solution for a ticket, the Knowledge Base document gets linked to that ticket and is logged in the ticket timeline. Analysts can always view the accepted Solution from the ticket timeline. In xFlow Interface, navigate to the ticket timeline, look for or click the **Solutions Used** Hyperlink and view the KB article that is linked to the ticket.

Rate a Knowledge Base Document

You can rate a KB article and give a rating on a scale of **1-5**. When a user rates a KB document on a scale of **3** or less, the Rate this Document bar expands and displays all the following comment types to provide feedback for a low rating:

- Broken Link
- Candidate for Publication
- Content not Clear
- Difficult to Find
- Candidate for Retirement
- Review
- Incorrect Information
- Missing Information
- Recommended New Content
- Solution does not Work



To add your own comment type reasons like the above, navigate to the CA SDM Classic Interface, **Administration**, **Knowledge, Documents, Comments Types**.

NOTE

The only special comment Type that is not listed in this section is the **General Comment**.

If a KB article is not rated, "**Be the first to rate this Document**" is displayed so that user can rate it. if it is rated, the rating is displayed on a scale of 1-5 star representation. Additionally, if a KB article is accepted as a solution for a ticket, that information is available as ***Number of times it has been accepted as a solution.***The rating on a scale of 1-5 represents the following:

- 1 Star & 1.5 star - Poor
- 2 star & 2.5 star - Fair
- 3 star & 3.5 star - Average
- 4 star & 4.5 star - Good
- 5 Star - Excellent

Research and Resolve a Ticket

To resolve a ticket you can view the ticket Timeline, use the Suggested Experts, Suggested Solutions, and the Research Tool.

Timeline

The timeline is the representation of activities on a ticket displayed in chronological order.

Every activity on the ticket is captured on the timeline along with the contact information and the time stamp. Each activity type is represented by a unique icon.

Timeline Features

Filter by Activity Type

Select the icon to filter the timeline by activity type and deselect to turn-off the filter.

Inline Edit

Inline editing is enabled only for the **Log Comment** activity type.

Add an Attachment

To add an attachment to a ticket, drag and drop files from any location or click the Attachment icon.

Select the attachment to perform the following actions:

- **Preview**: Preview the attachment.
- **Download**: Download and save the attachment in your local drive.
- **Delete**: Delete the attachment from the timeline.

Supported File Formats

You can attach the following file formats:

File Type	File Format
Word Processing Format	.doc, .docx, .txt
Spreadsheet Formats	.xls, .xlsx
Presentation Formats	.ppt, .pptx
Graphic Formats	.pdf, .jpg, .jpeg, .png

NOTE

: An empty file cannot be uploaded as an attachment.

: Attachments are stored in the predefined repository, **Service Desk** and cannot be stored in any other repository.

: To attach other file formats, contact the administrator.

For a limitation with Timeline, see the known issues in CA Service Management 17.4 Release Notes.

Suggested Solutions

Suggested Solutions widget displays the list of published knowledge base articles and closed incidents and requests. You can search and view the solution detail in the CA SDM UI. Click **Use** for a solution that you find useful for resolving the ticket and it gets appended to the ticket timeline. Click **Ignore** to remove the solution from the list. However, when you

access the ticket again, the ignored solution might appear in the list. The number of solutions that you can view at any instance is configurable from the CA SDM UI.

NOTE

The **Use** button is available only for the knowledge base articles.

Suggested Experts

Suggested Experts widget displays the list of contacts that can assist the analyst in resolving the ticket quickly. The widget automatically understands the ticket context and suggests the experts that are best suited to resolve the ticket. A suggested expert is a contact in any role; for example, the expert can be an analyst or administrator. The list displays the first five contacts that are based on the maximum number of similar tickets that they have resolved. To contact an expert, click the email icon against the name.

Research Tool

Research Tool allows an analyst to perform an external search to resolve a ticket.

Access the Research Tool

There are two ways to access the research tool:

- Click the Research Tool icon from the ticket details page to view the Google search results based on the ticket summary.
- Access the research tool without logging into **xFlow Interface** :When you click the research tool extension on a browser, a prompt appears where you can define the **xFlow Interface** URL or any other Single Sign-on enabled URL. After you login, navigate to the ticket and click the research tool icon on the ticket details page, to search and clip the information useful in resolving the ticket.

Use the Research Tool

The Research Tool allows an analyst to attach URLs, selected text, and screenshots to a ticket in the following ways:

- **URL** Clips the selected URL and adds as an attachment to the ticket.
- **Selected Text**Clips the selected text and adds it as a comment with the URL to the ticket.
- **Visible Screenshot**Clips the visible search results of the page and adds a screenshot to the ticket.
- **Full Page Screenshot**Clips the entire search results page and adds a screenshot to the ticket.

You can also link the search results to another existing ticket if the search results match the criteria for that ticket. To search for an existing ticket, enter one of the following details in the **Search Tickets** text box and click **Attach** to add the information.

- Incident or request number
- Ticket summary
- Ticket description

NOTE

: By default, the last five tickets that the analyst worked upon are displayed in the **Search Tickets** section.

Resolve Tickets with Contextual Collaboration

Why Use Contextual Collaboration?

Contextual Collaboration provides you the capability to initiate conversations with analysts, experts, and the end user who can help resolve a ticket faster. As an analyst, you can initiate the conversation, which opens the collaboration interface

in the ticket timeline. You can invite the end-user from the card or list view, one or multiple analysts from the Suggested Experts widget and the Command Bar. The conversation history is auto-saved in the ticket timeline for future reference.

The benefits of using Contextual Collaboration are as follows:

- Reduce the ticket resolution time
  Seek help of experts and analysts to resolve the ticket.
- Collaborate in-context
  Use the single interface to collaborate with experts for a ticket in-context.
- View saved conversation history
  View the saved history to understand the context of a particular ticket when it is reopened or reassigned.

Actions Available with Contextual Collaboration

Based on the privileges to the ticket available with the analyst, certain actions are restricted on the ticket and Contextual Collaboration.

Privilege	Available Actions
Read and write access to the ticket	• Contribute to the conversation • View the ticket details • Upload, download, or delete attachments • Invite other analysts to the conversation
Read access to the ticket	• Contribute to the conversation • Add other analysts to the conversation • View the ticket details • Download the attachments
No read or write access to the ticket	The analyst cannot perform any action on the ticket because of access restrictions.

Using the Contextual Collaboration Interface

Individual Conversation

You can initiate individual conversations from the following:

- **Card or List View**From the Card or List view, click the chat icon ( ) to initiate a conversation with the end user. This launches a conversation between the analyst and the end user.
- **Command Bar**To initiate a conversation with an analyst, type the following command in the Command Bar. The command launches the conversation view and creates an entry in the ticket timeline.

  ```
  /chat @<username> or /cht
  ```

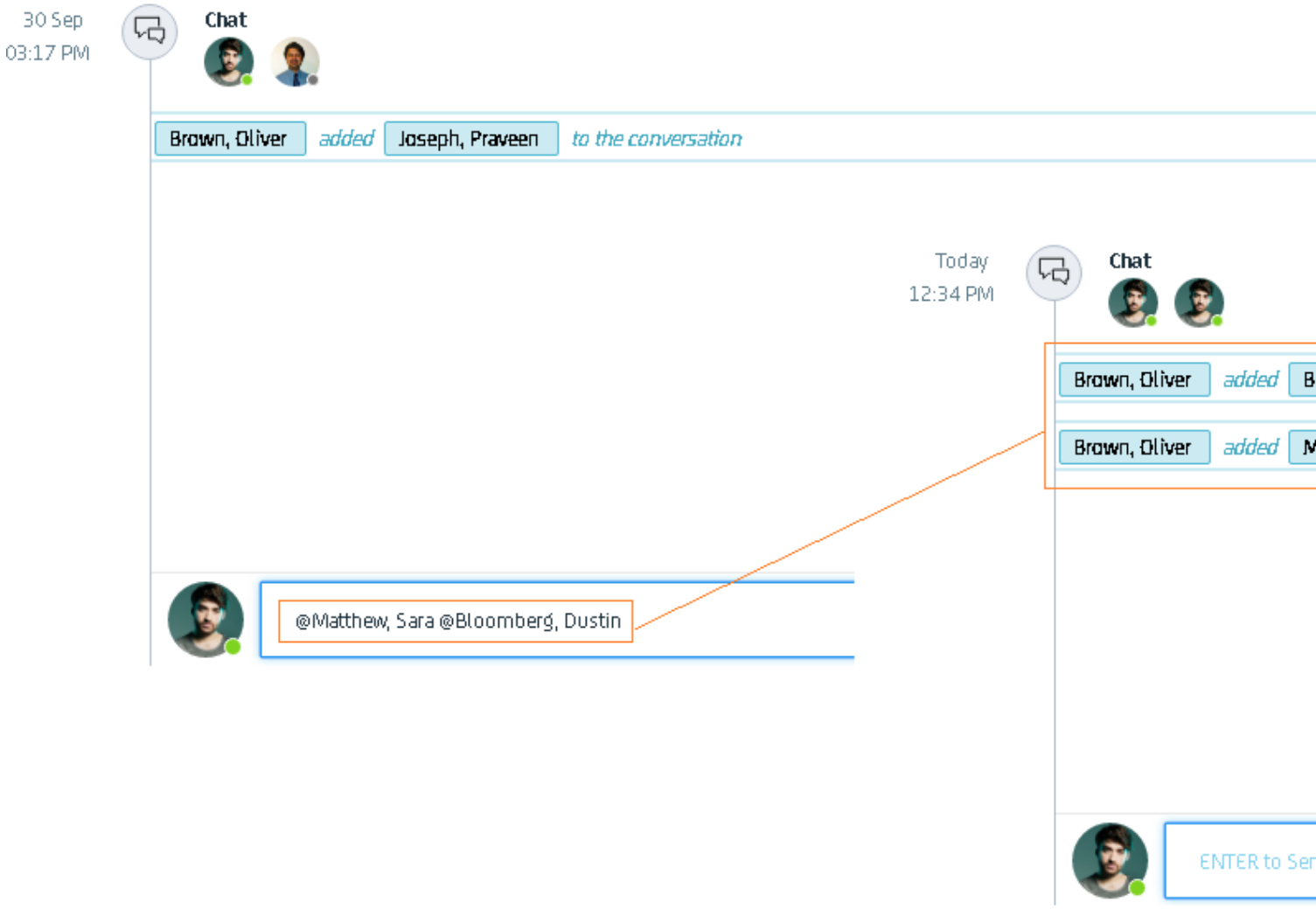
- **Suggested Experts Widget**From the Suggested Experts widget, hover your mouse over the suggested expert, and click on the chat icon
  ( )
  to initiate a conversation with the expert. This action launches a conversation between the analyst and the expert.

Group Conversation

You can add end user, other analysts, and experts to a collaboration session in the following ways:

- **Conversation Bar**

Use @ in the conversation bar to add an end user and analyst to the collaboration session:



- **Suggested Experts**
  From the Suggested Experts widget, click the chat icon
  ( ) that
  is visible when you hover your mouse to add the expert to the ongoing conversation.

Chat

Chat is one of the most important service channels. It consistently delivers on-demand, and speedy information to customers.

The Chat menu in the left navigation pane lists the following chats: **Queued** and **Active**.

Queued

- The Queued list displays chats in the queues that have been initiated by an end user and are yet to be accepted by an analyst.
- The timer indicates the number of minutes the Chat is in the queue.
- A red dot icon in the Queued list indicates a new chat request in the queue.
- To accept a chat request that is in queue, click the Handle icon. This opens the collaboration interface in the ticket timeline.
- The Chat is assigned to the analyst, and the accepted Chat is displayed in the Active list.

Active

- The active list displays the accepted chats that are assigned to analysts.
- A red dot icon indicates a new message in the active Chat.
- The entire conversation is auto-saved in the ticket timeline for future reference.
- You can perform the following actions from an active chat:
  - **Return**: Return the Chat to the queue
  - **Continue chat**: Continue the chat conversation
  - **Transfer**: Transfer the Chat to another queue that the analyst is a member of.

Notification

- Analysts are notified of an incoming chat with a desktop notification and a sound alert.
- The desktop notification includes the ticket number, the name of the chat message user, and the message preview.
- By default, the toggle is set to On. Turn the Chat toggle off to hide the desktop notifications.
- When an analyst clicks the desktop notification, the collaboration view with the latest message in the conversation is displayed.

Notification limitations

The following are limits that apply to desktop notifications.

- Focus assist must be disabled for windows.
- Exit private browsing mode, if browsing privately.
- You need a trusted certificate to access the xFlow Interface in an SSL-enabled environment while using the Firefox browser.
- No sound alert for an incoming chat while using the Firefox browser.

Manage Service Type Events (SLA) - xFlow Interface

CA Service Desk Manager defines and enforces SLAs with Service Types and Service Targets. The term, “service level agreement” or SLA is an industry and ITIL-standard term. It defines the expected level of service between a vendor and a client and is used as a metric for the vendor’s performance. It is often part of the contract between the two parties, and failure to meet the SLA could result in financial or other penalties for the vendor. For the helpdesk, a basic SLA could define how quickly a ticket of a particular severity should be acknowledged, diagnosed and fixed. For more information about how to create SLAs from the classic CA SDM Interface, see Service Level Agreements.

In xFlow Interface, you can now manage service types and service targets in the context of tickets. An Analyst can perform the following:

- Easy to access Service Types
- Attach Service Type Events
- Delay/Resume and Set/Reset Actual Service Targets
- Cancel Attached Service Type Events

This article contains the following topics:

NOTE

To execute a command on the Command bar, use the forward slash (/) first, followed by the actual command. For example: **/ViewServiceType**.

View Attached Service Type Events

To view the Service Types (SLA) that are already attached to a ticket in the xFlow Interface.

To view the Attached Service Types for a ticket, log in to the xFlow Interface and perform the following steps:

1. From the xFlow Interface, select a ticket that has a Service Type (SLA) attached to it.
   The ticket details/timeline page is displayed.
2. Select **Service Type Attached/Updated** hyperlink.
   The Service Type Event details page is displayed for **Service Types in Effect**, **Service Targets,** and **Attached Service Type Events**.
   Click **Service Type Event** to view additional attributes associated with a ticket.
3. Click **Service Targets** to view additional attributes associated with a ticket.
4. Alternatively, you can also use the command bar to view the attached Service Types (SLAs) to a ticket. Execute **/ ViewServiceType or /VST** and click **Enter** from the command bar in the xFlow Interface.

Delay or Resume a Service Type Event

You can delay and resume service type events in the xFlow Interface.

Perform the following steps to delay or resume a Service Type Event:

1. From the xFlow Interface, select a ticket that has a Service Type Event attached to it.
   The ticket details page is displayed.
2. Select **Service Type Attached/Updated**.
   Service type events details are displayed.
3. Click the **Delay** or **Resume** button. Enter the reason for delay/resume. Click **OK**.
   You can toggle between the Delay/Resume options.
   The ticket detail page displays the status of the event.
4. Alternatively, you can delay/resume a service type event from the xFlow Interface Command bar option also. Execute **/ UpdateServiceType or /UST** and click **Enter** on the command bar to Delay/Resume Service Types (SLA).
   /UpdateServiceType or /UST is single command for both Delay and Resume.

Set/Reset the Actual Time for a Service Type Event

You can set or reset the actual time for a Service Type event in the xFlow Interface. Perform the following steps:

1. From the xFlow Interface, select a ticket that has a Service Type Event attached to it.
2. Click Service Type Events Attached from the Ticket Details/Timeline page.
   Service type events details are displayed.
3. Select S**et Actual/Reset Actual** from **Service Targets**.

- – The **Set Actual** option indicates that you can set the actual time for a service target.
  A message is displayed to show that the Set Actual Date/Time of a Service Target is successful.
  – The **Reset Actual** option indicates to restart/reset the time.
  Clicking **Yes** indicates that you can reset the Actual Date/Time of a Service Target. Clicking **No** indicates that you cannot reset the Actual Date/Time.

Attach a Service Type (SLA) - Command Line Option

To attach a Service Type Event using the xFlow Interface Command Bar option, perform the following steps:

1. Log in to the xFlow Interface.
2. Select a ticket to which you want to attach a Service Type Event.
3. Navigate to the command bar option on the Tickets detail/timeline page.
4. Enter one of the following commands to attach a Service Type Event to a ticket:

   ```
   /AttachServiceTypeEvent

   or

   /AST
   ```

5. Click **Enter** to attach the Service Type Event to the selected ticket.
6. From the Service Type Event drop-down list, select the Priority category that you want to assign to this ticket.
7. Enter the delay time in the Delay box.
   Specify the delay time as *hh:mm:ss* (hours, minutes, seconds).
   The delay time indicates the interval of time after which the event occurs.
   For example, if the event is attached to the incident at 2:20, and the Delay Time is 01:00:00 (one hour), the event occurs at 3:20.
   Press **Enter** to attach the Service Type Event.
   "You are caught up for now!" is displayed. Press Enter in the command bar to execute the command.
   The Service Type Event is attached.

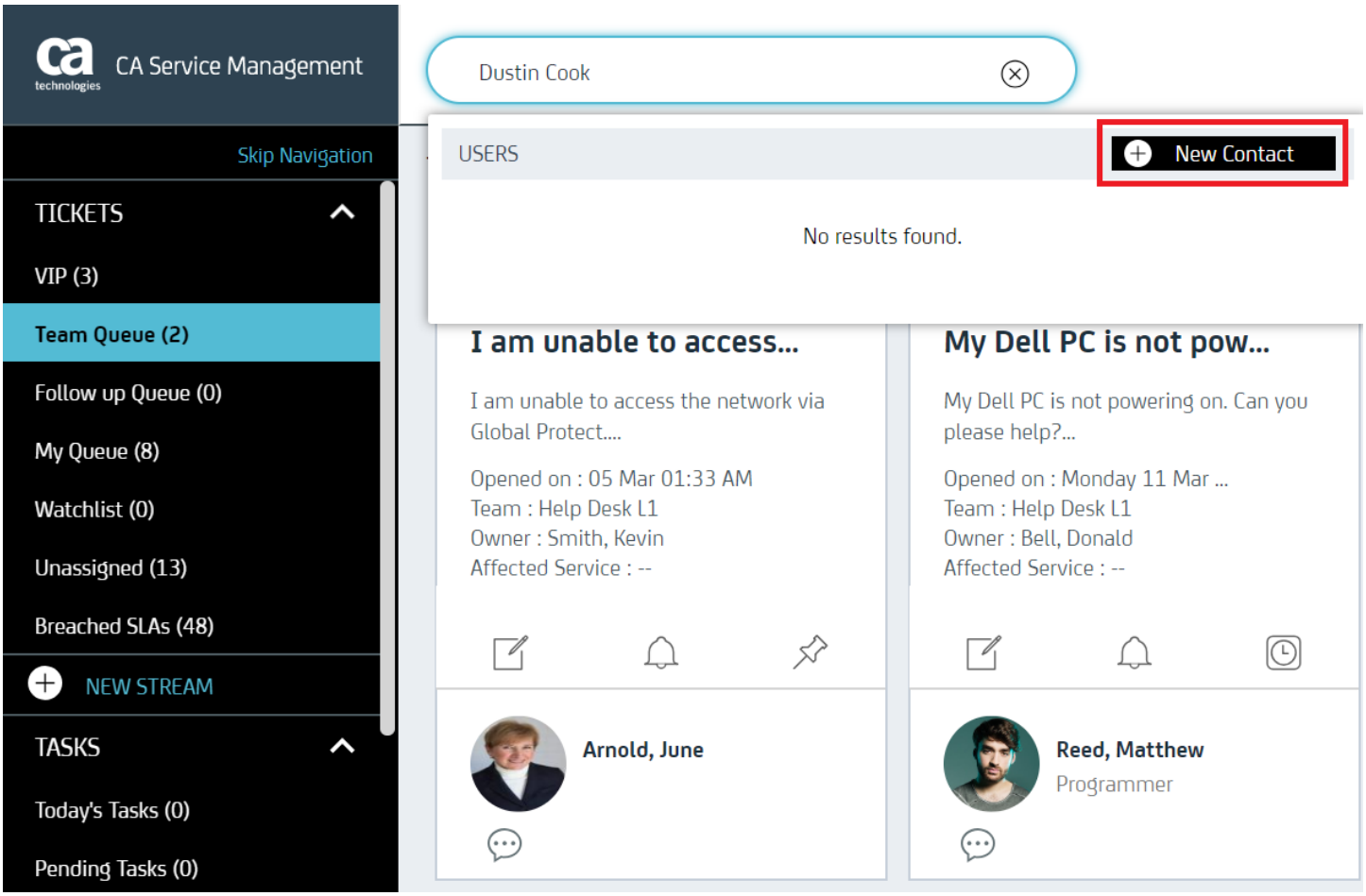
For more information on how to use the xFlow Interface Command bar option, see Command Bar.

Create Contact in the xFlow Interface

A contact is a person who uses your system regularly, such as an analyst or customer. After you have created the business structure and groups, you create contacts and map them to their respective location and organization.

xFlow Interface now provides the ability to create contacts from the search bar. Type in the name of the contact in the search bar for which you want to create a contact. Click the **New Contact** button. If you do not want to use an active directory such as LDAP for your contacts information, you can create the contacts manually in CA SDM.

> **NOTE**
>
> If multi-tenancy is enabled, select the appropriate tenant from the drop-down list.



For more information, see Create a Contact Manually.

Insights

This article contains the following topics:

Insights in CA Service Management enables decision makers with self-service analysis capability. You can save the information as a report, display it as a dashboard, and share it in different formats.

> **NOTE**
>
> For information about using Jaspersoft, access the Supportability Matrix for the Jaspersoft version that corresponds to your CA Service Management release.

Insights Overview Video

This video explains about Insights as a capability, and all it's features. For access to Insights, see the articles that follows the video.

> **TIP**
>
> To play the video in full screen, click the YouTube logo in the right of Settings at the bottom of the video.
>
> To change the video quality, navigate to **Settings**, and set **Quality** to **720p**.

Access to Insights

1. Login to xFlow Interface.

   ```
   http://<hostname>:<port number>
   ```

2. Click the **App Launcher** icon and select **Insights**.
3. The following information is displayed:

- – **Recently Viewed Items**
  Displays the recently viewed Ad Hoc Views, Reports, and Dashboards for the last seven days.
  – **Ad Hoc Views**
  Allows you to create an adhoc view to visualize your data for analysis and report creation.
  – **Reports**
  Allows you to create an interactive and analytical report using the existing Ad Hoc Views.
  – **Dashboard**
  Allows you to create a dashboard by combining Ad Hoc Views and Reports.

Using Transactional and Aggregate Reports

For more information about how to create a domain, add custom attributes to a domain, add dimensions and so on, see Using Aggregate Reports.

FAQs About Insights

This article contains some of the most frequently asked questions about Insights:

CA Business Intelligence JasperReports Server

What is CA Business Intelligence JasperReports Server?

CA Business Intelligence JasperReports Server (CABI JasperReports Server) is now the default reporting technology for CA Service Management. CABI JasperReports Server is built on industry-leading TIBCO Jaspersoft technology that offers comprehensive and easy to use business intelligence capabilities like Centralized Reporting, Dashboards, and Data Analytics.

Why has the default reporting technology changed to CABI JasperReports Server?

Effective April 1, 2016, to meet customer's demands for a self-service reporting technology, with capabilities such as ad-hoc analysis, easy administration, and drag-and-drop features, CA stopped distributing CA Business Intelligence 4.1 SAP Business Objects with all of its products globally. The default reporting technology has been changed to CABI JasperReports Server.

What is the End-of-Service date for CABI 4.1 with SAP Business Objects?

CABI 4.1 with SAP Business Objects is supported till December 31, 2018 or until the End-of-Service for CA Service Management - 14.1, whichever is earlier.

We have multiple reports in CABI 4.1 with SAP Business Objects. Can we easily migrate reports from CABI 4.1 with SAP Business Objects to CABI JasperReports Server?

No, there is no automated mechanism to migrate the report or it's content.

Does CABI JasperReports Server offer adhoc reporting? What is the equivalent of CABI 4.1 with SAP Business Objects 'Universe' in CABI JasperReports Server?

Yes, CABI JasperReports Server offers adhoc analysis. The equivalent of Universe is **Jaspersoft Domains**. The creation of domains is simple, however, it requires architectural considerations and data structure planning. As of now, CABI JasperReports Server includes a predefined domain for incidents and requests.

Where can I find education or training materials for CABI JasperReports Server?

To access education materials for CABI JasperReports Sever, see Tutorial for CABI JasperReports Server .

Install and Set Up Insights

Can I set the date to the beginning of time while loading the aggregated data?

Yes, however, during the Aggregate Configuration, it is recommended you specify the data aggregation period for not more than 2 years.

After upgrading CABI JasperReports Server, how can we integrate it with the xFlow Interface?

For more information, see JasperReports for CA Service Management.

How do I enable Insights?

Follow the post-integration steps to Integrate JasperReports for CA Service Management.

What is an ideal deployment for Insights?

The deployment for Insights varies for every organization as it depends on the number of concurrent analyst sessions. For more information on the architecture that suits your organization requirements, see Aggregate Reports Components.

A typical deployment for Insights:

Category	Concurrent Analyst Sessions	Datamart
Tiny	1 – 10	CA SDM Server
Small	1 – 200	Dedicated server
Medium	200 – 5000	Dedicated server
Large	5000 +	Dedicated server

Using Insights

What is Insights? How is it different from the other basic reports?

Insights provide pre-processed operations data from Service Management to analyze the key metrics and historical trends.

The interactive reports and dashboards allow you to view the key metrics of the IT environment in one place and perform adhoc analysis of relevant metrics, using the drag-and-drop feature.

Are the Insights reports and dashboards available real-time?

No. The reports and dashboards do not reflect the real-time data.

The data available is one day prior to the ETL run date. The data is as recent as one day prior to the ETL run date.

For example:

- ETL was run on February 13, 2018, and the user creates a report on February 14, 2018, then the data is as recent as February 12, 2018.
- ETL was run on February 10, 2018, and the user creates a report on February 13, 2018, the data is as recent as February 9, 2018.

Which objects does Insights support?

Insights currently support Incident, and Request Management objects.

Does Insights provide any predefined content (Ad hoc views, reports, and dashboards)?

Yes. The following table displays the list of all the predefined Ad hoc Views, Reports, and Dashboards.

TIP

It is recommended **not** to overwrite the predefined content

To access the predefined repository, perform the following steps:

1. From the home page, click the **View** tab and select **Repository**.
2. From the left navigation pane, select **casm_insights**, and **capability.**
3. Select among the following folders:
   a. For Ad Hoc View: **adhoc**
   b. For Reports: **report**
   c. For Dashboards: **dashboard**

The List of Predefined Content for Insights

S.No	Ad Hoc Views	Reports	Dashboard
1	Active Incidents by Analysts	Active Incidents by Analysts Report	MTTR Dashboard
2	Active Incidents by Category	Active Incidents by Category Report	Operational Effectiveness
3	Active Incidents by Group	Active Incidents by Group Report	Service Demand Incidents
4	Category Trends for Incidents Over the Last Month	Category Trends for Incidents Over the Last Month Report	
5	Category Trends for Requests Over the Last Month	Category Trends for Requests Over the Last Month Report	
6	Incident Resolution Cost by Group	Incident Resolution Cost by Group Report	
7	Incident Resolution Cost Trend By Group	Incident Resolution Cost Trend By Group Report	
8	Incident Resolution Effort by Group (In Hours)	Incident Resolution Effort by Group (In Hours) Report	
9	Incident Resolution Effort Trend By Group (In Hours)	Incident Resolution Effort Trend By Group (In Hours) Report	
10	Incident Resolution Volume by Group	Incident Resolution Volume by Group Report	
11	Incident Resolution Volume Trend By Group	Incident Resolution Volume Trend By Group Report	
12	Incidents Cost For Categories	Incidents Cost For Categories Report	

13	Incidents Cost Trend For Categories	Incidents Cost Trend For Categories Report	
14	Incidents Effort For Categories(In Hours)	Incidents Effort For Categories(In Hours) Report	
15	Incidents Effort Trend For Categories(In Hours)	Incidents Effort Trend For Categories(In Hours) Report	
16	Incidents Volume For Categories	Incidents Volume For Categories Report	
17	Incidents Volume Trend For Categories	Incidents Volume Trend For Categories Report	
18	MTTR For Categories(In Hours)	MTTR For Categories (In Hours) Report	
19	MTTR Quarterly Trend For Categories(In Hours)	MTTR Quarterly Trend For Categories (In Hours) Report	
20	MTTR Weekly Trend For Categories(In Hours)	MTTR Weekly Trend For Categories(In Hours) Report	
21	MTTR Yearly Trend For Categories(In Hours)	MTTR Yearly Trend For Categories(In Hours) Report	
22	Trending Ticket Count by Ticket Type Over the Last 3 Months	Trending Ticket Count by Ticket Type Over the Last 3 Months Report	

I can access the predefined content (Ad hoc Views, Reports, and Dashboards) in Insights. How can I create my own content?

From the home page, click the **Create** tab and select one of the following options. For more information, see Insights - Domains, Reports, Dashboards.

- Ad hoc Views
- Dashboard
- Report

Why am I unable to create my own content (Ad hoc Views, Reports, and Dashboards) in Insights?

You do not have the required permissions. Contact your administrator.

How can I access the newly created content (Ad hoc Views, Reports, and Dashboards) in Insights?

The newly created Ad hoc Views, Reports, and Dashboards in Insights are saved in the user-specific folder.

You can access the repository for Ad hoc Views, Reports, and Dashboards from the following path: **View**, **Repository**, **casm_insights**, **users**, <**User Folder**>.

For an adhoc analysis, the data is displayed in a Crosstab format. How can I change the view?

By default, the data for an adhoc analysis is displayed in a crosstab format. From the **NEW AD HOC VIEW** screen, you can change the view to **Chart** or **Table**.

How can I view the complete data for an adhoc analysis?

By default, the data option is set as **Sample Data**. To view the complete data, from the **NEW AD HOC VIEW** screen, change the option to **Full Data**.

When I create a new Ad hoc view, I see a date filter applied by default and the date is set to 2017-01-01. What is this date filter?

The date filter is meant for the ETL run date. By Default, the date is set to 2017-01-01.

For example:

- ETL was run on February 13, 2018, and the user creates a report on February 14, 2018, then the data is as recent as February 12, 2018. In this case, the date field must be set to 2018-02-12 or Day-1
- ETL was run on February 10, 2018, and the user creates a report on February 13, 2018, the data is as recent as February 9, 2018. In this case, the date field must be set to 2018-02-09 or Day-4

  TIP

  During an Ad Hoc View creation, when a date range is applied on a time-based filter (**is between,** or **on or after**), a corresponding interval (**day**, **week**, **month**, **quarter,** or **year**) must be added as a row.

How to filter data or apply filter?

You can add or edit existing filters from the filters pane. The filter logic governs how filters apply to your report.
For example, say you have a report with analyst name, date, and tickets status like Open, Closed, Pending, In-Progress. You require a snapshot for closed tickets on a `daily/weekly/monthly/quarterly/yearly` basis.

- In the left navigation pane from the Fields section, drag-and-drop **Ticket Status** to the Rows.
- To filter by status, right click and select the **Create Filter** option.
- From the Ticket Status filter, select the filter type as **Closed**.
- For `daily/weekly/monthly/quarterly/yearly` snapshot, select the corresponding filter type and click **Apply**.

Snapshot	Filter Type	Value Range
Daily	Date	Day-1
Weekly	Week	Week-1
Monthly	Month	Month-1
Quarterly	Quarter	Quarter-1
Yearly	Year	Year-1

Troubleshooting

For more information about how to troubleshoot issues related to Insights, see the Troubleshooting Insights section.

Insights - Domains, Reports, Dashboards

Insights in CA Service Management enables decision makers with self-service analysis capability. You can save the information as a report, display it as a dashboard, and share it in different formats. This article contains information about the various components for Insights.

Domains

Insights contains pre-defined domains, that provide a business view of the data warehouse. You can also create ad hoc views in the form of charts, tables, and cross tabs to analyze data from the domains.

We recommended not to modify the pre-defined domains. To create a domain that suits your business requirements, replicate a pre-defined domain and make modifications. However, any updates by CA SDM will be applicable only to the pre-defined domains. For more information about how to create a domain, see Create Domains.

Domain Terminology

The following terms are commonly used in reference to domains:

- **Topics**
  A topic is a predefined set of fields, filters, and input controls, that allows you to create reports without defining data relationships.
- **Fields**
  A field is used to display qualitative information. For example, a name or location.
- **Measures**
  A measure is used to display numeric or aggregate values.
- **Ad Hoc Viewer and Editor**
  The ad hoc viewer and editor is the interactive designer for viewing, creating, and editing an ad hoc view, to explore and analyze data.

Common Characteristics of Domains

The following characteristics are applied to all the domains:

- **Object Domains**
  Domains are provided for the most frequently used objects. For example, **Fields** pane (text fields and dates) and **Measures** pane (hours and costs).
- **Domain Security**
  Domains are accessible based on the logged-in user permissions.

Create Dashboards

You can design the Dashboard adding multiple reports to analyse multiple metrics in a single view.

NOTE

: To create a dashboard, you must first create a report.

: To create a dashboard with an Ad Hoc View, you must first create a report with the Ad Hoc View, and then add the report to the dashboard.

1. On the Insights home page, from the **Create** drop-down list, select **Dashboard**.
2. Drag and drop the Reports or Ad Hoc Views to create a dashboard.
   The Reports or Ad Hoc Views are added as Dashlets on the Dashboard.
   - Drag the edge of a dashlet to resize it.
   - Double-click to edit dashlet properties.
   - Right-click to delete a dashlet.
3. Click the **Save** icon and select **Save Dashboard**.
   Specify the **Name** and **Description**.
4. Expand the repository view, select **casm_insights**, **users**, and <**Logged-in User Folder**>.
5. Click **Save**.

Create an Ad-Hoc View

The Ad Hoc View editor is used to create and modify an Ad Hoc View to explore and analyze data from a topic or domain data source.

Follow these steps:

1. On the Insights home page, from the **Create** drop-down list, select **Ad Hoc View**.
2. Select the **Domain Topic** and click **OK**.
   The following panels are displayed:

- **Data Source Selection**
  Displays a list of available metrics and dimensions for analysis categorized as Fields and Measures.
- **Ad Hoc View**
  Displays the main view design panel.
- **Filters**
  Defines a subset of data to retrieve from the data source.

3. To design the adhoc view, drag and drop **Fields** and **Measures** to columns and rows. The following table displays the available Domain Topics, Fields, and Measures:
4. From the Date filter in the Filters pane, select the last ETL run date.
5. Click the Save icon and select Save Ad Hoc View.
6. Specify the Data View Name and Data View Description of the Ad Hoc View.
7. Expand the repository view, select casm_insights, users, and <Logged-in User Folder>.

   NOTE

   You can save the Ad Hoc View only to the folder corresponding to the logged in user, and not to any other public folder.

   While creating the user ID, any special character is converted to underscore (_) in Insights. The only exception is single quote ('), which is considered as null, when used.

   For example, if the user ID is created as John@H-Kennedy, it is displayed as John_H_Kennedy on the Insights interface and if the user ID is created as John@H'Kennedy, it is displayed as John_HKennedy.
8. Click **Save**.

Best Practices for Ad Hoc Views

Following are the best practices for Ad Hoc Views to generate the desired reports and reduce any performance issues:

- To create an Ad Hoc View, use either of the measures, **Call Req KPIs** or **Activity Log KPIs**. Incorrect data is displayed if both the measures are used.
- Do not select all the fields. Instead, only select the fields that meet your business requirement.
- Limit the size for Crosstab views, as they generally perform slower than table or chart views.
- Create Ad Hoc Views in No Data or Sample Data mode. Once the Ad Hoc view is created, you can switch to Full Data mode. This will improve the performance.
- Limit the values on X axis & Y axis for Chart views, to avoid overlapping of data.
- Specify the correct **Time** filter to get expected results. The following options are displayed in the **Time** filter:
  - **Weekly:** Allows you to measure data by week.
  - **Monthly:** Allows you to measure data by month.
  - **Quarterly:** Alows you to measure data by quarter.
  - **Yearly:** Allows you to measure data by year.

Create a Report

The Report editor allows you to create, view a report, apply filters, and export the report content to various formats.

NOTE

: To create a report, you must first create an Ad Hoc View.

1. On the Insights home page, from the **Create** drop-down list, select **Report**.
2. From the **Create Report** window, select an **Ad Hoc View** based on the report you want to create.
3. Select **Default Report Template** or **Custom Report Template**, and click **OK**.
   a. If you select **Custom Report Template**, click **Browse**, select a template and click **OK**.
4. Click the **Save** icon.

Specify the **Name** and **Description**.

5. Expand the repository view, select **casm_insights**, **users**, and <**Logged-in User Folder**>.
6. Click **Save**.

Run and Schedule a Report

You can perform the following actions after a report is created:

1. On the Insights home page, click **Library**.
2. Identify the Report you created and right click to perform the following actions:
   - **Run**
     Run and update the report with the latest data.
   - **Run in New Tab**
     Run the report in another tab.
   - **Run in Background**
     Run the report in the background, select a file format and location to save the report.
   - **Schedule**
     Select this option to run the report at a scheduled date and time.

     WARNING

     The Run in Background and Schedule functionality has restricted access. If your administrator restricted this option, you see the below user message when you click Schedule.
     "Please contact your system administrator or log in as a user with permission."

     We recommend not to use the "Schedule" functionality. The administrator can restrict the users from using this functionality.

     Following these steps:

     1. Open the applicationContext-security-web.xml file in a text editor. The file is typically available under the path, <JasperReports Server Installed Folder>\apache-tomcat\webapps\jasperserver-pro\WEB-INF\
     2. Find the following lines:
        - <security:intercept-url pattern="/scheduler/jobsummary.html" access="ROLE_USER,ROLE_ADMINISTRATOR" />
        - <security:intercept-url pattern="/scheduler/jobdetails.html" access="ROLE_USER,ROLE_ADMINISTRATOR" />
        - <security:intercept-url pattern="/scheduler/main.html" access="ROLE_USER,ROLE_ADMINISTRATOR" />
     3. Modify the above three lines as below:
        - <security:intercept-url pattern="/scheduler/jobsummary.html" access="ROLE_ADMINISTRATOR" />
        - <security:intercept-url pattern="/scheduler/jobdetails.html" access="ROLE_ADMINISTRATOR" />
        - <security:intercept-url pattern="/scheduler/main.html" access="ROLE_ADMINISTRATOR" />
     4. Save and close the file.
     5. Restart the CABI installed server.

     If you do not want to apply the restriction and experience any issues with the Run in Background and Schedule functionality, please note that CA does not provide the support.

Using Transactional and Aggregate Reports

For more information about how to use the Transactional and Aggregate Reports provided with CA Service Management 17.4, see Using Aggregate Reports.

Using Transactional Reports

This article contains the following topics:

Out-of-the-Box (OOTB) Domains

OOTB Domains	Attached OOTB Domain Reports
Incident Management	Incident Management Domain
Request Management	Request Management Domain
Problem Management	Problem Management Domain
Change Management	Change Management Domain
Survey Management	Survey Domain
Configuration Management	Configuration Management Domain

Out-of-the-Box Reports/Ad Hoc Views

Name	Description
Active Incidents by Analysts	This ad-hoc view represents the total active tickets grouped by Analysts. It displays data as of the last ETL run date -1. Total Active tickets represent all tickets that are not closed. This metric requires Ticket KPI tables to be turned ON.
Active Incidents by Category	This ad-hoc view represents the total active tickets grouped by Categories. It displays data as of the last ETL run date-1. Total Active tickets represent all tickets that are not closed. This metric requires Ticket KPI tables to be turned ON.
Active Incidents by Group	This ad-hoc view represents the total active tickets grouped by Groups. It displays data as of the last ETL run date-1. Total Active tickets represent all tickets that are not closed. This metric requires Ticket KPI tables to be turned ON.
Category Trends for Incidents Over the Last Month	This ad-hoc view represents the trend of incidents reported by Category over each day of the last Month with respect to the ETL run date.
Category Trends for Requests Over the Last Month	This ad-hoc view represents the trend of requests reported by Category over each day of the last Month with respect to the ETL run date.
Incident Resolution Cost by Group	This ad-hoc view presents a snapshot of the cost of incidents resolved by groups as of the ETL run day-1. This metric considers both resolved and closed tickets for its calculation. This metric requires Ticket KPI tables to turned ON.
Incident Resolution Cost Trend By Group	Represents quarterly trend of cost in resolving incidents by Groups. It considers incidents in resolved and closed states. It displays data from the last run ETL day-1 till the past 365 days. This metric requires Ticket KPI tables to turned ON.

Incident Resolution Effort by Group (In Hours)	This ad-hoc view presents a snapshot of the effort on incidents resolved by categories as of the ETL run day-1. This metric considers both resolved and closed tickets for its calculation. This metric requires Ticket KPI tables to turned ON.
Incident Resolution Effort Trend By Group (In Hours)	Represents quarterly trend of effort spent in resolving incidents by Groups. It considers incidents in resolved and closed states. It displays data from the last run ETL day-1 till past 365 days. This metric requires Ticket KPI tables to turned ON.
Incident Resolution Volume by Group	This ad-hoc view represents the total incidents resolved grouped by Groups. It displays data as of the last ETL run date-1. This metric requires Ticket KPI tables to be turned ON.
Incident Resolution Volume Trend By Group	Represents quarterly trend for the number of incidents resolved by groups for the last 365 days from the ETL run day-1. This report requires Ticket KPI tables to turned ON.
Incidents Cost For Top Categories	This ad-hoc view presents a snapshot of the cost of working on incidents by categories as of the ETL run date-1. This metric considers all reported incidents for its calculation. This metric requires Ticket KPI tables to turned ON.
Incidents Cost Trend For Top Categories	Represents quarterly trend for the cost of working on incidents by categories for the last 365 days from the ETL run day-1. This metric requires Ticket KPI tables to turned ON.
Incidents Effort For Top Categories(In Hours)	This ad-hoc view presents a snapshot of the effort spent on incidents by categories as of the ETL run day-1. The effort is in Hours. This metric requires Ticket KPI tables to turned ON.
Incidents Effort Trend For Top Categories(In Hours)	Represents quarterly trend for the effort spent in hours on incidents by categories for the last 365 days from the ETL run day-1. This metric requires Ticket KPI tables to turned ON.
Incidents Volume For Top Categories	This ad-hoc view represents the total incidents grouped by Categories. It displays data as of the last ETL run date-1. Total incidents represent all incidents were reported. This metric requires Ticket KPI tables to be turned ON.
Incidents Volume Trend For Top Categories	Represents quarterly trend for the number of incidents reported by categories for the last 365 days from the ETL run day-1. This metric requires Ticket KPI tables to turned ON.
MTTR For Top Categories(In Hours)	This ad-hoc view represents the Meantime to Resolution for incidents grouped by Categories. It displays data as of the last ETL run date-1. MTTR is calculated in terms of Hours. This metric requires Ticket KPI tables to be turned ON.
MTTR Quarterly Trend For Top Categories(In Hours)	Ad Hoc view compares quarter on the quarter trend of MTTR for all incidents by categories for the last 365 days from the ETL run day-1. MTTR is calculated in terms of Hours. This metric requires Ticket KPI tables to be turned ON.
MTTR Weekly Trend For Top Categories(In Hours)	Ad Hoc view compares weekly trend of MTTR for all incidents by categories for the last three months from the ETL run date-1. MTTR is calculated in terms of Hours. This metric requires Ticket KPI tables to be turned ON.

MTTR Yearly Trend For Top Categories(In Hours)	Compares the yearly trend of MTTR for all closed/resolved incidents by categories. The data set is for the last two years. MTTR is calculated in terms of Hours. This metric requires Ticket KPI tables to be turned ON.
Trending Ticket Count by Ticket Type Over the Last 3 Months	This ad-hoc view represents the trend of the number tickets in various ticket types (Incidents/Requests) over a period of the last three months. This metric requires Ticket KPI tables to be turned ON.

CA Service Desk Manager Reports

Main Report	Main Report Location	Main Report Description	Sub Reports	Sub Reports Location
Operational Effectiveness	View, Repository, root, Organizations, servicemanagement, capability, reports, Business Value Dashboards	Provides insight into metrics surrounding productivity of Groups of Teams that address the request and incident demand coming from business users.	• Categories cost by groups in USD • Categories Effort By Groups In Hours • Categories Volume By Groups • FCR Top Incident Categories Table • FCR Trend For Top Categories Table • Incident Resolution Group Trend By Cost • Incident Resolution Group Trend By Effort • Incident Resolution Group Trend By Volume • MTTR Top Incident Categories • MTTR Trend Top Incident Categories • Top Groups By Cost Incurred On Resolution • Top Groups By Time Spent On Resolution • Top Groups Resolving Incident Volume	View, Repository, root, Organizations, servicemanagement, capability, reports, Subreports

Service Demand - incidents	View, Repository, root, Organizations, servicemanagement, capability, reports, Business Value Dashboards	Provides insight into incident demand from business users to IT, and supports analysis of data through various time-periods, location, tenant or service.	• Incident Cost details for Top Categories • Incident Cost trend details for Top Categories • Incident Effort details for Top Categories • Incident Effort trend details for Top Categories • Incident Volume details for Top Categories • Incident Volume trend details for Top Categories	View, Repository, root, Organizations, servicemanagement, capability, reports, Subreports
Active Change Orders Aging	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Change order, Aging	Displays the count of all open Change Orders by the number of weeks open.		
Active Change Orders Aging by Priority for Groups	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Change order, Aging	Displays the count of open Change Orders by the number of weeks open. Adjacent bars break up counts by Priorities. User must input a Group name for a search of Change Orders in that Group.		
Active Change Orders Aging by Priority for Status	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Change order, Aging	Displays the count of open Change Orders by the number of weeks open. Adjacent bars break up counts by Priorities. User must input a Status name for a search of Change Orders in that Status.		
Active Incidents	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Incident and Problem Management	Provides information about Active Incidents based upon Analysts, Groups, Categories and Priorities.		

Active Incidents Aging	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Incident and Problem Management, Aging	Displays the count of all open Incidents by the number of weeks open.		
Active Incidents Aging for Groups	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Incident and Problem Management, Aging	Displays the count of all open, group assigned Incidents by the number of weeks open for selected groups.		
Active Incidents Aging for Incident Areas	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Incident and Problem Management, Aging	Displays the count of all open Incidents by the number of weeks open for selected Incident Areas.		
Active Incidents Aging for Status	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Incident and Problem Management, Aging	Displays the count of all open Incidents by the number of weeks open for the selected Statuses.		
Active Problems Aging	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Incident and Problem Management, Aging	Displays the count of all open Problems by the number of weeks open.		
KPI Report	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Key Performance Indicator	Renders a list and a line graph showing values collected by a selected Key Performance Indicator over a selected time frame.		
Active Requests Aging	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Requests, Aging	Displays the count of all open Requests by the number of weeks open.		

Active Requests Aging for Groups	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Requests, Aging	Displays the count of open, group assigned Requests by the number of weeks open.		
Active Requests Aging for Request Areas	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Requests, Aging	Displays the count of all open Requests by the number of weeks open for the selected Request Areas.		
Active Requests Aging for Status	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Requests, Aging	Displays the count of all open Requests by the number of weeks open for the active request Statuses selected.		
Active Request List	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Requests, Volume	Displays a list of active Requests grouped by Analysts.		
Active Requests at Weeks End	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Requests, Volume	Displays the counts of opened Requests for today, 7 days ago (1 week ago), 14 days ago (2 weeks ago), and so on for five weeks. The opened Request counts, for each Week Ending date, include the previous 6 days, up to and including the 7th day.		
Request List	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Requests, Volume	Displays a list of Requests.		
Request List by Organization	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Requests, Volume	Displays a list of Requests grouped by Customer Organization.		

Request List by Priority	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Requests, Volume	Displays a list of Requests grouped by Priority.		
Request List by Request Area	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Requests, Volume	This report shows a list of Requests grouped by Request Area.		
Total Volume of Requests	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Requests, Volume	Displays the count of both opened and closed Requests for today, 7 days ago (1 week ago), 14 days ago (2 weeks ago), and so on for about five weeks. Bar charts for opened and closed Requests are shown.		
Total Volume of Requests by Interface	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Requests, Volume	Displays the count of both opened and closed Requests by interface for today, 7 days ago (1 week ago), 14 days ago (2 weeks ago), and so on for about five weeks. Bar charts for opened and closed Requests are shown.		
Urgency Summary by Customer	View, Repository, root, Organizations, servicemanagement, capability, reports, CA Service Desk, Requests, Volume	Lists all High Urgency Requests by individual Customer as opposed to Organization.		

Using Aggregate Reports

This article contains the following topics:

Aggregate Domain Reports

Domain Topic	Fields	Measures
Incident Request Management Topic	• **Analyst** – Analyst Name – Analyst Country – Analyst Location – Analyst Cost Center – Analyst State – Analyst Department – Analyst Organization • **Category** • **Time** – Date – Week – Month – Quarter – Year • **Group** – Group Name – Group Country – Group Location – Group Cost Center – Group State – Group Department – Group Organization • **Priority** • **Status** • **Tenant** • **Ticket Type** • **Business User** – Business User Name – Business User Country – Business User Location – Business User Cost Center – Business User State – Business User Department – Business User Organization	• **Call Req KPIs** – Total Tickets – Total Open Tickets – Total Resolved Tickets – Total Closed Tickets – Total Active Tickets – Average Resolution Time(h) – Average Active Tickets Tenure(h) – Total Tickets Meeting SLA – Total Tickets Violating SLA – Total First Call Resolved Tickets – Total Major Incidents – Total Incorrectly Categorized Tickets • **Activity Log KPIs** – Total Reopened Tickets – Total Transfers – Total Escalations – Total Effort – Total Cost

Problem Management

This article contains the following topics:

Problems are difficulties encountered when following normal procedures. Records of problems are recorded, along with the steps taken to correct the problems. If you create a ticket as a copy of another ticket, the Status field displays all Status values.

WARNING

Unless otherwise noted, all the CA SDM Request features are also available for Incidents and Problems.

NOTE

Depending on your role, you may not have access to all the functionality described in this section. For example, some predefined roles can edit records but cannot create records.

Create a Problem

You can create a new problem either using a template or without using a template.

NOTE
If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. Do one of the following steps:
   - On the Service Desk tab, select File, New Problem.
   - On the Service Desk tab, select File, New Problem from Template.
2. Complete the Problem Fields as appropriate for the problem.
3. Use the controls available on the problem tabs to process the problem as appropriate.
4. Click one of the following buttons:
   - **Use Template** -- Displays a list of available problem templates. Select the template you want to use for creating this problem.
   - **Quick Profile** -- Displays the contact information for the user entered in the Affected End User field. If you are creating a new problem, a list of available users appears. Select the user that is the Affected End User.
   - **Find Similar** -- Opens the Find Similar page to search for similar problems.
   - **Create Change Order** -- Opens the Create New Change Order window so you can create a change order ticket associated with this problem.

Problem Fields

The following fields require explanation to create or update a problem:

NOTE
If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

- **Problem Reference Number**This is a unique reference number assigned by CA Service Desk Manager for all problem tickets. This is used by analysts and customers to refer to a particular problem ticket.
- **Requester**
  Specifies the name of the person who initiated the ticket. This person must be a defined contact. You can enter a value directly or click the magnifier to search for the name.
- **Affected End User**
  Specifies the contact name of the person who is affected by the record. If the contact is assigned to a special handling type, special handling indicators are displayed. You can enter a value directly or click the magnifier to search for a contact name.
- **Problem Area**
  Indicates the general area of your IT environment that is affected by the problem (for example, Applications, E-mail, Hardware, and Software). A problem area provides default values that are entered automatically on all problem tickets that are assigned to the area. In addition to the predefined problem areas, your system administrator can define custom problem areas. You can enter a value directly or click the magnifier to search for a problem area.

NOTE
Your system administrator has the option of adding custom properties to problem areas. If custom properties have been added, they are displayed on the Properties tab when you create, edit, or view a problem. Some custom properties require that you enter a value.

- **Status**Specifies the status code of the record. For example, you can list only the tickets with a status code of Fix in Progress, or can Close Requested. You can enter a value directly or click the magnifier to search for a status. The blue button (on the left side of the Status field) lets you change the current status to the next default status.
- **Priority**
  Specifies the priority ranking of the record. The ranking determines the amount of attention the ticket receives. The predefined priority levels are 1 (highest) through 5 (lowest). Your system administrator or an active priority calculation can generate the appropriate Priority values for various installations and tenants. When priority calculation is enabled, this field updates based on Impact, Urgency, Affected Service, and Affected User settings. When your administrator disables priority calculation and uninstalls the urgency_on_employee option, Self Service Users see the Priority field on the Request Detail page.

Detail Fields

- **Reported By**Specifies the name of the person reporting the record.
- **Assignee**
  Specifies the name of the person who is assigned to handle the record. You can enter a value directly or click the magnifier to search for a name. Selecting an assignee populates the groups that the assignee belongs to in the **Group** field.
- **Group**
  Specifies the group that is responsible for this record. Your system administrator defines groups of contacts that are responsible for different types of issues, requests, incidents, or problems. Any contact who is part of the group can handle the record after it is assigned to the group. You can enter a value directly or click the magnifier to search for a group. Selecting a group populates the **Assignee** field with the corresponding assignee name, which belong to the group.
- **Urgency**
  Specifies the urgency of the record. The urgency is determined by the importance of the user tasks that are affected by the record. Urgency codes indicate the importance of a ticket based on the degree to which it affects user tasks. For example, a network outage is more urgent than a printer failure. Your system administrator can modify the default urgency codes, so they can vary from one installation to another. Urgency values can update automatically based on an active priority calculation.
- **Impact**
  Specifies an impact code, such as 1 - Entire Organization, that indicates how a ticket affects the work being performed. For example, a ticket that requires a network outage for several hours would have a higher impact than a ticket that takes a printer off-line. Your system administrator can modify the default impact codes, so they can vary from one installation to another.
- **Active**Indicates whether the record is Active or Inactive. This value applies to the current record only, not the associated template.
- **Configuration Item**
  Specifies the hardware, software, or service that is affected by the record. Your system administrator creates a record that uniquely identifies each configuration item for your organization and indicates its precise location. You can enter a value directly or click the magnifier to search for an item.
- **Charge Back ID**
  Specifies the ID that is charged for the service.
- **Call Back Date/Time**

Specifies the date and time to follow up on this record. You can enter the date and time in the format mm/dd/yyyy hh:mm am | pm, or click the calendar icon to open a calendar window where you can select the date and time for follow-up.

- **Symptom**Specifies a code that describes a primary symptom of the record. For example, Slow Response.
- **Root Cause**
  Identifies the code associated with the core reason why the ticket was opened. Your service desk can use generic root cause codes, such as Hardware Failure or Software Failure, or more specific codes, such as Network.Cable, Network.Card, or Network.Response. You can enter a value directly or click the magnifier to search for a code.
- **Change**
  Specifies the number and name of the change order that is associated with this record. Enter the number or name of the change order directly in this field.
- **Caused by Change Order**
  Specifies only tickets that were opened as a result of a specific change order ticket. You can enter a value directly or click the magnifier to search for a change order ticket.
- **External System Ticket**
  Specifies an identification for a ticket that belongs to an external system that integrates with CA SDM. This field stores hyperlinks and displays functional links in read-only mode.
- **Timer**
  Tracks the incremental amount of the time spent working on various phases of ticket processing. The timer is reset to 00:00:00 each time you open the ticket record for updates. You cannot edit this field.

  NOTE
  The amount of time that is spent on each activity is shown on the Activities tab of the ticket record.
- **Action**
  Sets or resets the Actual Date/Time to the current date and time.
- **Target**
  Specifies the current Service Target.
- **Target Date/Time**
  Specifies date and time when this Service Target is due. If the ticket is in a Hold status, this value is blank.
- **Actual Date/Time**
  Specifies the time when the target condition was met. If no value appears, the target condition has not been met.
- **Time Left**
  Specifies the amount of remaining time for the service target when the ticket is on hold. If the Service Target has been met, Time Left shows the unused time. A negative value indicates the time that elapsed since the missed target date.
- **Violation Cost**
  Specifies the incurred cost when the service type time limit is violated.
- **Affected Service**
  Specifies the primary service that affects the problem or incident. The CIs of type Service have a class that is defined in the Enterprise Service Family field. The ticket stores the currently affected service information in the ticket for reporting. You can enter a value directly or click the magnifier to search for a CI.
- **Affected End User**
  Specifies the contact name of the person who is affected by the record. If the contact is assigned to a special handling type, special handling indicators are displayed. You can enter a value directly or click the magnifier to search for a contact name.
- **Priority**
  Specifies the priority ranking of the record. The ranking determines the amount of attention the ticket receives. The predefined priority levels are 1 (highest) through 5 (lowest). Your system administrator or an active priority calculation can generate the appropriate Priority values for various installations and tenants. When priority calculation is enabled, this field updates based on Impact, Urgency, Affected Service, and Affected User settings. When your administrator

disables priority calculation and uninstalls the urgency_on_employee option, Self Service Users see the Priority field on the Request Detail page.

- **Summary**Provides a brief description of the record.
- **Description**Describes the record in detail.
- **Last Modified Date/Time**
  Displays the date that this ticket was last modified.
- **Last Modified By**
  Displays the name of the last person who edited the ticket.
- **Ticket Open Date/Time**
  Displays the date and time the ticket was opened.
- **Ticket Resolved Date/Time**
  Displays the date and time the ticket was resolved.
- **Ticket Closed Date/Time**
  Displays the date and time the ticket was closed.
- **Created Via**Specifies the component reporting the record (detail_pr.htmpl).

Problem Tabs

The following tabs are available on the Create New Problem, Problem Detail, and Update Problem pages:

- **Activities:** Displays a log of the activities performed to resolve the problem. For more information, see Add an Activity from the Ticket Management topic.
- **Event Log:** Displays a record of significant actions that occur regarding the problem.
- **Attachments:** Allows you to attach a document or a link to a URL to the problem.
- **Workflow Tasks:** Lists the process instance and related audit trail messages for a CA Process Automation, or Classic Workflow that is associated with the ticket. The Workflow Tasks tab shows fields that apply to an attached workflow. The workflow may require some of the work items to complete before the ticket can close. The Workflow Tasks tab appears only if your administrator configured workflows for the ticket area or category. For more information, see Define a Category or Area topic.
- **Service Type:** Allows you to attach a service type event to indicate the level of support for the ticket.
- **Attached Incidents:** Allows you to view a list of any incident tickets attached to the problem ticket.
- **Knowledge:** Allows you to search for or submit information to the CA SDM Knowledge Base to help resolve problems. The Federated Search capability helps you to get the knowledge search results from multiple sources. For example, Google, SharePoint, CA Open Space, and so on.
- **Solutions:** Allows you to store information about the problem solution with the problem record for future reference.
- **Parent / Child:** Allows you to create a parent/child relationship between the problem and another CA SDM record.
- **Properties:** Your CA SDM administrator can add custom properties to problem areas.
- **Templates:** Allows you to create a template using the current ticket as a model.

Create a Problem from an Incident

From an incident, you can create an associated problem if necessary.

Follow these steps:

1. From the Incident Detail Page, click Create Problem.
   The Create New Problem page displays. The fields are filled with the values from the original incident, but they can be edited for the problem. See the Problem Fields as appropriate for the problem.
2. Use the controls available on the problem tabs to process the problem as appropriate.
3. Click one of the following buttons:

- **Use Template** -- Displays a list of available problem templates. Select the template you want to use for creating this problem.
- **Quick Profile** -- Displays the contact information for the user entered in the Affected End User field. If you are creating a new problem, a list of available users appears. Select the user that is the Affected End User. For more information about Quick Profile, see the Ticket Management topic.
- **Find Similar** -- Opens the Find Similar page to search for similar problems.
- **Create Change Order** -- Opens the Create New Change Order window so you can create a change order ticket associated with this problem.

For more information, see Manage Problem Status.

Attach an Incident to a Problem

There may be times when an incident entered in the system is related to a problem record. You can attach these related incidents to the problem record.

To attach an incident to a problem:

1. On the Service Desk tab, browse to Problems, Assigned or Unassigned. Select the Priority for the problem you want to edit.
   The Problem List appears.
2. Select the problem to edit.
   The Problem Detail page displays.
3. Click Edit.
   The Update Problem page displays.
4. Select the Attached Incidents tab and click Attach/Remove Incidents.
   The Incident Children Search page displays.
5. Complete one or more of the search fields and click Search, or click Search without entering search criteria to display a list of all available incidents. Refer to Incident Search Fields for field definitions.
   The Incidents Update page displays, listing all incidents that match the search criteria in the Incidents list on the left.
6. From the list on the left, choose the incidents you want to attach to this problem. To choose multiple items, hold down the CTRL key while clicking the left mouse button.
7. When you have selected all the incidents you want, click

   The selected incidents move to the list on the right.
8. Click OK.
   The Problem Detail page displays, listing the selected incidents on the Attached Incidents tab.
9. Click Save to save the problem record with the incidents attached.

Issue Management

Issues are entered by customers when they encounter questions or difficulties when following normal procedures. Records of issues are recorded, along with the steps taken to correct the issues. If you create a ticket as a copy of another ticket, the Status field displays all Status values.

Create an Issue

You can either create an issue from scratch or use an existing template.

> **WARNING**
> Depending on your role, you do not have access to all the functionality described in this section. For example, some predefined roles can edit records but cannot create new ones.

NOTE
If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. On the Service Desk tab, select File, New Issue.
2. Fill in the issue fields as appropriate for the ticket.
3. Use the controls available on the tabs to process the ticket as appropriate.
4. Click one of the following buttons:
   Auto Assign -- Triggers an auto assignment task, and updates the activity log. This button appears only when the ticket specifies a category or area that has auto-assignment enabled.
   Find Similar -- Opens the Find Similar page to search for similar problems.
   Quick Profile -- Displays the contact information for the specified user in the **Affected End User** field. You can also view their environment details and their entire ticket history.
   Use Template -- Displays a list of available templates for the selected ticket type. You can select the template that you want to use for creating this ticket.

   NOTE

   You can use the Quick Profile to identify a contact to be the affected end user of your new ticket. Quick Profile allows you to search for a contact, and view the history of the issues, requests, change orders, incidents, and problems that are assigned to that contact. When you have identified a contact, you can create the incident ticket directly from the Quick Profile.

Issue Fields

The following fields require explanation:

- **Affected End User**
  Specifies the name of the person affected by the record. You can enter a name directly or click the magnifier to search for a contact.
- **Category**
  Indicates the general category of the issue within your IT environment (for example, Hardware.pc.install and Software.pc.install). Issue categories provide default values that are entered automatically on all issue tickets assigned to the category. In addition to the predefined issue categories, your system administrator may define custom issue categories. You can enter a value directly, or click the Lookup icon to select from the defined categories. When you edit the Category and a CA Process Automation workflow is already running, the workflow cancels.
- **Status**
  Specifies the status of the record. You can enter the status directly or click the magnifier to search for a status.
- **Priority**
  Specifies the priority ranking of the record, determined by the amount of attention it should receive. Your system administrator can modify the default priority codes, so they can vary from one installation to another.
- **Product**
  Indicates the product that is associated with this issue. Select a product from the drop-down list.

Detail Fields

- **Organization**
  Specifies the name of the organization responsible for submitting the issue. You can enter a value directly or click the magnifier to search for an organization.
- **Position**
  Specifies the job title of the contact within the organization that submitted the issue, such as CEO, Director, or Manager.
- **Role**

Indicates the role of the person submitting the issue to your service desk. For example, the person might be a customer or a potential customer.

- **On**
  Displays the date and time the record was created, in the time zone of the server. This field is read-only and is automatically filled during creation. The date and time display in mm/dd/yyyy hh:mm am | pm format.
- **Reason**
  Identifies the basic reason for opening the issue. For example, the reason might be a complaint, inquiry, or suggestion from a customer. Select a value from the drop-down list.
- **Reporting Method**
  Indicates the reporting method used to submit the current issue. Select a value from the drop-down list.
- **Assignee**
  Specifies the name of the person assigned to handle the record. You can enter the name of the person directly, or click the magnifier to search for a name. Selecting an assignee populates the groups that the assignee belongs to in the **Group** field.
- **Group**
  Specifies the group that is responsible for the record. Your system administrator defines groups of contacts that are responsible for different types of issues, requests, incidents, or problems. Any individual contact assigned to the group can handle the record once it has been assigned to the group. You can enter the group name directly, or click the magnifier to search for the group. Selecting a group populates the **Assignee** field with the corresponding assignee name, which belong to the group.
- **Service #**
  Specifies an issue number for an on-site service call.
- **Service Date**
  Specifies the date of the last on-site service call. You can enter the date and time in mm/dd/yyyy hh:mm am | pm format or click the calendar to select a date.
- **Need By Date**
  Specifies the date that the issue needs to be completed by. You can enter the date and time in mm/dd/yyyy hh:mm am | pm format or click the calendar to select a date.
- **Call Back Date/Time**
  Specifies the date and time to follow up on this record. You can enter the date and time in the format mm/dd/yyyy hh:mm am | pm, or click the calendar icon to select the date and time for follow up.
- **Root Cause**
  Specifies the code associated with the core reason why the ticket was opened. Your service desk can use generic root cause codes, such as Hardware Failure or Software Failure, or more specific codes, such as Network.Cable, Network.Card or Network.Response. You can enter the root cause directly into the field, or click the Lookup icon to display the defined root causes and select one.
- **External System Ticket**
  Specifies an identification for a ticket that belongs to an external system that integrates with CA SDM. This field stores hyperlinks and displays functional links in read-only mode.

Summary Information Fields

- **Issue Summary**
  Describes the record briefly.
- **Spelling**
  Checks the spelling of the text you enter in the Issue Summary field.
- **Issue Description**
  Describes the record.
- **Spelling**
  Checks the spelling of the text you enter in the Issue Description field.
- **Open Date**

Displays the date and time when the issue was started in mm/dd/yyyy hh:mm am | pm format.

- **Resolve Date**
  Displays the date and time when the issue was resolved in mm/dd/yyyy hh:mm am | pm format.
- **Close Date**
  Displays the date and time when the issue was closed in mm/dd/yyyy hh:mm am | pm format.
- **Timer**
  Tracks the incremental amount of the time spent working on various phases of ticket processing. The timer is reset to 00:00:00 each time you open the ticket record for updates.

  NOTE
  The amount of time spent on each activity is shown on the Activities tab of the ticket record.

Issue Tabs

The following tabs are available on the Create Issue, Issue Detail, and Update Issue pages:

- **Properties:** Administrators have the option of defining custom properties for issue categories, to provide additional detail on issue tickets assigned to that category. When you select an issue category while creating or editing an issue, if any custom properties have been defined for that category, they appear on the Properties tab.
- **Workflow Tasks:** Lists Workflow tasks associated with the issue. For more information, see Define a Category or Area topic.
- **Config Items:** Links configuration items (CIs) to an issue or change order in order to give analysts information about the system that is affected by the ticket.
- **Knowledge:** Searches for or submits information to the knowledge base to help resolve issues. The Federated Search capability helps you to get the knowledge search results from multiple sources. For example, Google, SharePoint, CA Open Space, and so on.
- **Solutions:** Stores information about the issue solution with the issue record for future reference.
- **Resolution:** Displays a text description of the resolution to the issue.
- **Related Issues:** Create a parent/child relationship between the issue and another issue.
- **Activities:** Displays a log of the activities performed to resolve the issue. For more information, see Add an Activity from the Ticket Management topic.
- **Event Log:** Displays a record of significant actions that take place regarding the issue.
- **Attachments:** Attaches a document or a link to a URL to the issue.
- **Service Type:** Attaches a service type event to indicate the level of support for the ticket.
- **Time / Cost:** Calculates the estimated cost and duration of the entire issue by adding together the estimated cost and duration of each task.
- **Custom Fields:** Defines custom fields for tracking information about the issue.
- **Templates:** Create a template using the current ticket as a model.
- **Support Automation:** Displays the assistance session log and lets you invite the end user to an assistance session.

  NOTE

  If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

Accumulate Costs and Time to an Issue

The accumulate feature allows you to calculate the estimated cost and duration of the entire issue by adding together the estimated cost and duration of each task. You can also choose to accumulate the issue and all of its children. Children are additional records entered as a result of attempting to resolve the original issue.

Follow these steps:

1. Select Issues, Assigned or Unassigned, and a priority level.

The Issue List appears.
2. Select the issue of interest.
   The Issue Detail page appears.
3. Select Actions, Accumulate Issue on the menu bar.
   The Accumulate Issue page appears.
4. Click one of the following buttons:
   - **Accumulate this Issue and All Children**
     Calculates the estimated cost and duration of the issue by adding together the estimated cost and duration of tasks for this issue and all of its children.
   - **Accumulate Only this Issue**
     Calculates the estimated cost and duration of the issue by adding together the estimated cost and duration of tasks for only this issue.
   - **Cancel**
     Closes the Accumulate Issue page without accumulating the issue.

   The estimated cost and duration are accumulated and you are returned to the Issue Detail page.
5. Select the Time/Cost tab on the Issue Detail page.
   The tab displays the estimated cost and duration for the issue based on the values accumulated.

Add Comments to an Issue

You can add comments to an open issue ticket.

NOTE

After you have saved a comment, you cannot edit or delete it.

Follow these steps:

1. Click the link to your open issues in the "Look up my existing Issues" section of your home page.
   Your list of open issue tickets appears.

   NOTE

   If you know the issue number, you can enter it in the search field and click Go to access the issue ticket directly.
2. Click the Issue # link for the issue ticket you want to comment on.
   The Issue Detail page appears.
3. Click the Add Comment button in the upper right-hand corner of the window.
   The Add Comment page appears.
4. Enter your comment in the User Description field, and then click the Save button.
   Your comment appears in the History section of the Issue Detail page.

Expedite an Issue

You can use the Expedite Issue command to change the status of unnecessary tasks to Skip so the issue can be completed quickly.

NOTE
Skip must be defined as a valid status for the task.

Follow these steps:

1. Select Issues from the Scoreboard.
   The Issues folder expands to reveal nested folders for Assigned, Unassigned, and All Scheduled issues.
2. Select the appropriate folder under Assigned, Unassigned, and All Scheduled for the issue whose children you want to close.

The Issue List page appears.

3. Click the issue number.
   The Issue Detail page appears.
4. Select Expedite Issue from the Actions menu.
   The Expedite Issue page appears.
5. (Optional) Enter comments in the Remarks field to explain the action.
6. Click OK.
   The Expedite Issue page closes and the Workflow Tasks tab on the Issue Detail page shows all unnecessary tasks with status of Skip.

Define Status Transitions

You can define a subset of the full status list and determine the next status of the ticket as it continues through its lifecycle. You can also specify how strictly the system enforces status transition policies by configuring the Status Policy Violations option in Options Manager (General Options).

> **NOTE**
>
> : Because status transitions can be shared between integrations such as Events and Macros, do not inactivate predefined status transitions unless explicitly requested.

Follow these steps:

1. On the Administration tab, select Service Desk, Requests/Incidents/Problems, and specify one of the following:
   - Incident Transitions
   - Problem Transitions
     Request Transitions

   The Transition List page appears.
2. Click Create.
   The Update Status Transitions page appears.
3. Complete the fields as appropriate. The following fields require explanation:
   - **From Status**
     Defines the current status of the ticket, for example, Open.
   - **To Status**
     Specifies a valid next status of the ticket, for example, Assigned.
   - **Default Transition**
     Specifies the default status transition. CA SDM uses the default transition when a user clicks the default transition button on the ticket detail page, or when a user (including a web services user) updates the status to a <d> value. You can only configure one default transition for each status (or one for each tenant in a multi-tenanted system).
   - **Must Comment**
     Indicates that a comment for the transition must be supplied. Specifying this option indicates that an analyst must supply an activity log comment when changing the status on a request.
   - **Condition**
     Specifies the condition by which the transition is allowed. For example, when the condition "nonprty 1&2 assigned req" is associated with the Request Transition of Acknowledged to Hold, the condition verifies if the transition to move the status from Acknowledged to Hold is allowed.
   - **Transition Type**
     Links a transition type to this transition. Transition types and their corresponding statuses control when employees can close and reopen Incidents and Requests using self-service. This field only displays for Incident and Request status transitions.
   - **Condition Error Message**
     Specifies the message returned to the user if the transition is rejected.

   The transition is defined.

4. Click Save, Close Window.
   The transition appears in the Transition List when you refresh the page.

Search for Transitions

You can use the search function to filter the Request/Incident/Problem Transition List page to show only the items you want to see. You can also locate a specific transition type to view or update.

> **NOTE**
>
> If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

To search for a transition

1. On the Request/Incident/Problem Transition List page, click Show Filter.
2. Complete one or more of the following search fields:
   Status
   Defines the current ticket status, for example, Open.
   New Status
   Defines the next ticket status, for example, Closed.
   Default
   Specifies whether this transition appears as the default transition on the ticket detail page.
   Must Comment
   Indicates that a comment for the transition must be supplied. Specifying this option indicates that an analyst must supply an activity log comment when changing the status on a request.
   Active
   Specifies if the transition is currently active or inactive.
   Transition Type
   Specifies a transition type for this transition. Transition types and their corresponding statuses control when employees can close and reopen Incidents and Requests using self-service. This field only displays for Incident and Request status transitions.
   Status Description
   Provides a brief description of the record.
3. Click Search.
   The Request/Incident/Problem Transition List displays all transitions that match your search criteria.

Status Transitions and Dependent Attribute Controls

This article contains the following topics:

You can use the following configurable controls to restrict ticket status flows for change orders, issues, incidents/problems/requests, and determine which fields are shown or required for each ticket status:

- **Transitions**
  Controls how users select available statuses on the incident/problem/request, issue, or change order form. For example, a problem is in a status of Open, and the transition flow only allows the analyst to update the status to Closed. In this example, the analyst has no other status options, which reinforces the problem management process. Transitions let you define a subset of the full status list and specify the default new (or next) status of a ticket based on the current status. You can define unique status transitions for each ticket type. Consider using transitions when you want to restrict status workflows for your end users.
- **Dependent Attributes**

Controls how attributes are designated as required (must supply) or locked (cannot update) depending on ticket status. For example, the Change Manager can prevent an analyst from editing the Summary attribute after a change order is approved. Consider using attribute controls when you want to restrict certain attributes based on the status.

NOTE
You can specify how strictly the system enforces Status policies by configuring the Status Policy Violations option in Options Manager (General Options). This option only applies to automated system processes, such as integrations and macros.

Work with Status Transitions and Dependent Attribute Controls

To work with status transitions and dependent attribute controls, do the following:

1. On the Administration tab, configure the appropriate tenants, contacts, and roles for your environment.
2. On the Service Desk node, specify a ticket type (Change Order, for example), and select Status.
   The Status List page displays active status codes.
3. Edit the appropriate status code (Acknowledged, for example) and use the controls available on the tabs at the bottom of the ticket's status detail page to do the following:
   - Configure Status Transitions
   - Configure Dependent Attribute Controls

   NOTE
   You can configure unique transitions and dependent attribute controls for each ticket type.

Configure Status Transitions

You can configure a subset of the full status list and specify the default next status of a ticket based on the current status. Transitions are enforced when the status is changed on the ticket detail form.

To configure status transitions

1. Click the Transitions tab at the bottom of the ticket's status detail page.
   The Transitions List page shows all valid transitions for the status.

   NOTE
   When configured, linked transition types appear on the Incident and Request Transition list.
2. Click the Update Transitions button.
   The Update Ticket Status Transitions page appears.
3. Configure the following check boxes as appropriate:
   - **Allowed**
     Specifies a valid transition for the status. Use this option to restrict status workflows.
   - **Default**
     (Optional) Specifies the default status transition. CA SDM applies the default transition when a user clicks the default transition button on the ticket detail form, or when a user (including a web services user) updates the status to a <d> value. There is only one default transition for each status (or one for each tenant in a multi-tenanted system).
   - **Must Comment**
     (Optional) Specifies that an activity log comment for the transition is required when changing the status on a ticket.

     NOTE
     This option applies to CA SDM tickets only. It does not apply to other areas, such as web services or the edit in list functionality.
4. (Optional) Select a status code in the Name column to update its details.
5. Click Save.

The list of transitions configured for the new status appears on the Transition list. When the analyst selects the Status drop-down on the ticket form, the new status list appears.

Configure Dependent Attribute Controls

You can determine which fields are shown or required for the ticket status.

NOTE
Before you configure dependent attributes as "required" for the ticket status, be aware that the Edit in List option that appears on the ticket's list page may not display the attribute field values that are required. If the required attribute field value is not already part of the saved ticket, and if it is not presented in the Edit In List format, then the ticket is not saved. Consequently, the analyst must edit the required dependent attribute field values on the ticket's detail page instead of using the Edit in List option.

To configure a dependent attribute control

1. On the ticket's status detail page, select the Dependent Attribute Control tab at the bottom of the page.
   The Attribute Control List appears.
2. Click Create.
   The Update Status Dependent Attribute Control page appears.
3. Complete the following fields:
   - **Tenant**
     (Optional) In a system with multi-tenancy installed, specifies an optional tenant name. If a tenant is specified, the dependency applies only to that tenant and to its sub-tenants.
   - **Attribute**
     Specifies the name of the attribute you want to control, for example, Summary.
   - **Locked**
     Specifies the attribute as locked. A locked attribute associated with a status cannot be updated in a ticket with the same status. The attribute is unlocked when the status is changed.
   - **Required**
     Specifies the attribute as required. A required attribute for the status cannot use a null value in a ticket with the same status.
4. Click Save.
   The new attribute control for the status appears in the Attribute Controls List when you redisplay the page. When a user updates the ticket status, the system retrieves the list of required attributes corresponding to the new status, and updates the ticket form as appropriate. An error message appears at the top of the ticket form when a user attempts to close the ticket without filling in a required field.

Web Services Methods

You can configure the following status transition and dependent attribute control SOAP web services methods:

- **getValidTransitions**
  Lists the transitions for a ticket. This method is modeled on the existing getValidTaskTransitions method, except that the argument can be a ticket or a status.
- **getDependentAttrControls**
  Lists the locked and required attributes for an attribute of an object that persistent id specified. The Status attribute is supported at this time.

Predefined Transition Flows

For each ticket type, you can use the predefined status transitions provided with the product and modify them to accommodate your desired transition flow.

To view the list of predefined transitions, do the following:

On the Administration tab, expand the Service Desk node, and select one of the following:

- Change Order Transitions
- Issue Transitions
- Request/Incident/Problem Transitions

The Transitions List displays the predefined transitions that let you control how a ticket (incident/request/problem, change order, and issue) continues through its lifecycle.

Incident Transition Flow

The following table shows the predefined Incident transition flow:

Current Status	Default Transition	Available Next Statuses
Acknowledged	In Progress <d>	Avoided, Awaiting Vendor, Cancelled, Closed, Closed Unresolved, In Progress, Open, Pending Change, Resolved
Avoided		Acknowledged, Avoided, Awaiting End User Response, Closed, Closed Unresolved, In Progress, Reject Solution, Researching, Resolved
Awaiting End User Response	Researching <d>	Closed, Closed Unresolved, In Progress, Open, Researching, Resolved
Awaiting Vendor	Researching <d>	Acknowledged, Closed, Closed Unresolved, In Progress, Open, Pending Change, Researching, Resolved
Cancelled		Closed
Closed		Open
Closed Unresolved		Acknowledged
Closed Unresolved		Closed
Closed Unresolved		Open
Hold		Acknowledged, Closed, In Progress, Open, Pending Change, Resolved
In Progress	Researching <d>	Acknowledged, Awaiting End User Response, Awaiting Vendor, Closed, Closed Unresolved, Open, Pending Change, Researching, Resolved
Pending Change	Researching <d>	Acknowledged, Closed, In Progress, Open, Resolved
Researching	Resolved <d>	Closed, Open, Resolved
Resolved	Closed <d>	Awaiting End User Response, Closed, Open

Problem Transition Flow

The following table shows the predefined Problem transition flow:

Current Status	Default Transition	Available Next Statuses
Acknowledged	In Progress <d>	Acknowledged, Approved, Cancelled, Closed, Fix in Progress, Open, Rejected, Researching
Analysis Complete	Approved <d>	Acknowledged, Cancelled, Closed, Fix in Progress
Approved	Fix in Progress <d>	Closed, Fixed, Pending Change
Awaiting Vendor	Researching <d>	Acknowledged, Closed, Closed Unresolved, Fixed, In Progress, Open, Pending Change, Researching
Cancelled	Closed <d>	Closed, Closed Unresolved, Open
Closed Unresolved		Acknowledged, Closed, Open
Fix in Progress	Fixed <d>	Approved, Cancelled, Fixed, Fix in Progress, Researching, Rejected
Fixed	Closed <d>	Closed
Hold	Researching <d>	Acknowledged, Closed, Fixed, In Progress, Open, Pending Change, Researching
In Progress	Researching <d>	Acknowledged, Approved, Cancelled, Closed, Fix in Progress, Pending Change, Rejected, Researching
Known Error	Fix in Progress <d>	Closed, Fix in Progress, Fixed
Open	Acknowledged <d>	Acknowledged, Approved, Cancelled, Closed, Fix in Progress, In Progress, Rejected, Researching
Pending Change	Fixed <d>	Closed, Fixed, Researching
Rejected	Closed <d>	Closed, Closed Unresolved, Open
Researching	Analysis Complete <d>	Acknowledged, Analysis Complete, Approved, Cancelled, Closed, fix in Progress, Fixed, Rejected

Issue Transition Flow

The following table shows the predefined Issue transition flow:

Current Status	Default Transition	Available Next Statuses
Acknowledged	In Progress <d>	Awaiting End User Response, Awaiting Vendor, Closed, Closed Unresolved, In Progress, Open, Pending Change, Resolved
Awaiting End User Response	Researching <d>	Acknowledged, Closed, Closed Unresolved, In Progress, Open, Researching, Resolved
Awaiting Vendor	Researching <d>	Acknowledged, Closed, In Progress, Open, Pending Change, Researching, Resolved

Cancelled	Closed <d>	Closed
Closed		Acknowledged, Open
Closed Unresolved		Acknowledged, Closed, Open
Hold		Acknowledged, Closed, In Progress, Open, Pending Change, Resolved
In Progress	Researching <d>	Acknowledged, Awaiting End User Response, Awaiting Vendor, Closed, Closed Unresolved, Open, Pending Change, Researching, Resolved
Open	Acknowledged <d>	Acknowledged, Avoided by Self Service, Awaiting End User Response, Awaiting Vendor, Closed, Closed Unresolved, In Progress, Pending Change, Resolved
Pending Change	Researching <d>	Acknowledged, Closed, In Progress, Open, Researching, Resolved
Researching	Resolved <d>	Closed, Open, Resolved, Awaiting End User Response, Closed, Open

Change Order Transition Flow

The following table shows the change order transition flow:

Current Status	**Default Transition**	**Available Next Statuses**
Approval in Progress	Approved <d>	Approved, Cancelled, Closed
Approved	Scheduled <d>	Cancelled, Closed, Implementation in Progress, Scheduled
Backed Out		Approval in Progress, Closed, Open, RFC
Cancelled		Closed
Customer Hold		Cancelled, Closed, Implementation in Progress, Rejected, Scheduled, Verification in Progress
Hold		Cancelled, Closed, Implementation in Progress, Scheduled
Implementation in Progress		Backed Out, Cancelled, Closed, Customer Hold, Rejected, Scheduled, Vendor Hold, Verification in Progress
Open	RFC <d>	Approval in Progress, Cancelled, Closed, Customer Hold, Implementation in Progress, Rejected, RFC, Scheduled, Vendor Hold
Rejected	Closed <d>	Approval in Progress, Cancelled, Closed
RFC	Approval in Progress <d>	Approval in Progress, Cancelled, Closed, Customer Hold, Implementation in Progress, Open, Rejected, Scheduled, Vendor Hold

Scheduled	Implementation in Progress <d>	Cancelled, Closed, Customer Hold, Implementation in Progress, Vendor Hold, Verification in Progress, Cancelled, Closed, Implementation in Progress, Scheduled, Backed Out, Closed

Best Practice Predefined Status Transitions

The predefined status transitions delivered with the product are Active in a new installation and Inactive after the upgrade. For every status listed on the Transitions List page, there is a default status transition (or next status). The path taken by the default status transition reflects the best practice. The additional status transitions listed on the Transitions List page are provided to fulfill various ticket management workflows. However, only Active status transitions that use this best practice can ensure that the proper workflow for managing Requests, Incidents, Problems, and Change Orders occurs. This best practice helps promote the movement of tickets to resolution and closure within the IT environment.

For example, the following predefined incident transitions listed on the Incident Transition list page are set to Inactive to help promote the resolution and closure of incidents:

Status	New Status	Default	Status Description	Record Status
Acknowledged	Closed	No	Acknowledged to Closed Status Transition	Inactive
Acknowledged	Closed Unresolved	No	Acknowledged to Close Unresolved Status Transition	Inactive
Acknowledged	Open	Yes	Acknowledged to Open Status Transition	Inactive
Awaiting End User Response	Acknowledged	No	Awaiting End User Response to Acknowledged Status Transition	Inactive
Awaiting End User Response	Open	No	Awaiting End User Response to Open Status Transition	Inactive
Awaiting Vendor	Acknowledged	No	Awaiting Vendor to Acknowledged Status Transition	Inactive
Awaiting Vendor	Closed	No	Awaiting Vendor to Closed Status Transition	Inactive
Awaiting Vendor	Open	No	Awaiting Vendor to Open Status Transition	Inactive
Closed	Acknowledged	No	Closed to Acknowledged Status Transition	Inactive
Closed Unresolved	Acknowledged	No	Closed Unresolved to Acknowledged Status Transition	Inactive
Closed Unresolved	Closed	No	Closed Unresolved to Closed Status Transition	Inactive

Hold	Acknowledged	No	Hold to Acknowledged Status Transition	Inactive
Hold	Closed	No	Hold to Closed Status Transition	Inactive
Hold	Open	No	Hold to Open Status Transition	Inactive
In Progress	Acknowledged	No	In Progress to Acknowledged Status Transition	Inactive
In Progress	Closed	No	In Progress to Closed Status Transition	Inactive
In Progress	Open	No	In Progress to Open Status Transition	Inactive
Open	Closed	No	Open to Closed Status Transition	Inactive
Pending Change	Acknowledged	No	Pending Change to Acknowledged Status Transition	Inactive
Pending Change	Closed	No	Pending Change to Closed Status Transition	Inactive
Pending Change	Open	No	Pending Change to Open Status Transition	Inactive
Researching	Open	No	Researching to Open Status Transition	Inactive

Status Transitions for Self-Service

This article contains the following topics:

Status transitions let you control the movement of a ticket from one discrete state to another (for example, from Open to Closed). For employees using self-service, you can include buttons on the Incident and Request detail forms to represent any status transition.

Status transition buttons for incident and request process workflows appear in the employee interface when incident or request transitions are linked to active transition types. A transition type defines the button text and controls the behavior of the ticket detail form. When buttons are defined, the legacy Close Incident (or Request) and Reopen Incident (or Request) buttons are not displayed on the ticket detail forms. Instead, the employee can only change the status of the Incident or Request using the status transition buttons configured by the administrator.

By default, all predefined transition types delivered with the product are inactive, so status transition buttons are not in effect. As a system administrator, you can activate and modify predefined transition types or create transition types to accommodate your status transition workflows. After you create or modify a transition type, you can link them to any incident or request status transition.

How Transitions for Self-Service Work

Transition types and their corresponding statuses control when employees can close and reopen tickets as follows:

1. Active transition types are linked to incident (or request) status transitions by the administrator.
2. The employee creates an incident using self-service.

3. The analyst assigned to the incident finds a solution and moves the ticket to the Resolved status.
4. When the ticket is in a Resolved status, the employee detail form displays status transition buttons to Accept or Reject the resolution.
   - If the employee accepts the resolution, the Resolved to Closed transition occurs.
   - If the employee rejects the resolution, the Resolved to Open transition occurs.
5. After the employee clicks a button, they can add their remarks in the resolution form that appears.

How to Create or Update Transition Types for Transitions

As a system administrator, you can create new or update existing transition types for incident and request status transition workflows on the Transition Types List page.

To create a transition type for a status transition, do the following:

1. On the Administration tab, select Service Desk, Requests/Incidents/Problems, Transition Types.
2. Click Create on the list page.
3. Edit the fields as appropriate on the detail page.
   The transition type for the status transition is created.
4. Click Save.
   The new transition type appears in the Transitions Type List when you redisplay the page.

To update a transition type, do the following:

1. Open the desired transition type for editing on the Transition Types List page.
2. Edit the fields as appropriate.
3. Click Save.
   The updated transition type appears on the Transition Type list.

How to Link Transition Types to Transitions

When status transitions are linked to transition types, the employee ticket detail form displays status transition buttons to Accept or Reject the resolution. To link a transition type with a status transition, do the following:

1. On the Administration tab, select Service Desk, Requests/Incidents/Problems, Incident (or Request) Transitions.
2. Open the desired status transition for editing on the Request or Incident Transition List page.
3. Specify the desired transition type in the Transition Type field.
4. Click Save.
   The transition type is linked to the status transition.

Activate Predefined Transition Types

By default, all predefined transition types that are delivered with the product are Inactive, so status transition buttons are not in effect. You can activate and modify these transition types to accommodate your desired status transition flow.

To activate a predefined transition type

1. Select Show Filter on the Transition Type List page.
   The top portion of the page reveals additional search fields.
2. Select Inactive in the Record Status field and click Search.
   The Transition Type List displays all inactive transition types.
3. Right-click the desired transition type and select Edit from the menu.
4. Select Active in the Record Status field.
5. Click Save, Close window.
6. Click Search.
   The Transition Type List displays the active transition type.

Predefined Transition Types for Incident Status Transitions

The following table describes the predefined transition types for incident status transitions:

Symbol	Button Text	Form Header Text	Incident Status Transition
Accept incident resolution button	Accept	Accept Resolution	Resolved to Closed
Reject incident resolution button	Reject	Reject Resolution	Resolved to Open
Reject Closure incident button	Request Closure	Request Closure	To Closed Unresolved from Open Awaiting End User Response Awaiting Vendor In Progress Acknowledged

Predefined Transition Types for Request Status Transitions

The following table describes the predefined transition types for request status transitions:

NOTE
The Closed Requested status for Requests is the equivalent of the Resolved status for Incidents.

Symbol	Button Text	Form Header Text	Request Status Transition
Accept request resolution button	Accept	Accept Resolution	Close Requested to Closed
Reject request resolution button	Reject	Reject Resolution	Close Requested to Open
Request Closure request button	Request Closure	Request Closure	To Canceled from Open Awaiting End User Response Awaiting Vendor Approval In Progress Acknowledged
Close request button	Close	Close Requested	From In Progress to Closed

Define Issue Transitions

Issue status transitions control the movement of a issue from one discrete state to another (for example, from Open to Closed). With transitions, you can define a subset of the full status list and determine the next status of the ticket as it continues through its lifecycle. You can use the predefined transitions (listed on the Issue Transitions List page), modify the transitions, or create transitions.

NOTE
You can specify how strictly the system enforces Status policies by configuring the Status Policy Violations option in Options Manager (General Options). This option only applies to automated system processes, such as integrations and macros.

You can define a subset of the full status list and determine the next status of the ticket as it continues through its lifecycle.

NOTE

Because status transitions can be shared between integrations such as Events and Macros, do not inactivate predefined status transitions unless explicitly requested.

Follow these steps:

1. On the Administration tab, select Service Desk, Issues.
   The Issue Transition List page appears.
2. Click Create.
   The Update Status Transitions page appears.
3. Complete the fields as appropriate. The following fields require explanation:
   - **From Status**
     Defines the current status of the ticket, for example, Open.
   - **To Status**
     Specifies a valid next status of the ticket, for example, Assigned.
   - **Default Transition**
     Specifies the default status transition. CA SDM applies the default transition when a user clicks the default transition button on the ticket detail page, or when a user (including a web services user) updates the status to a <d> value. You can only configure one default transition for each status (or one for each tenant in a multi-tenanted system).
   - **Must Comment**
     Indicates that a comment for the transition must be supplied. Specifying this option indicates that an analyst must supply an activity log comment when changing the status on a request.
   - **Condition**
     Specifies the condition by which the transition is allowed. For example, when the condition "nonprty 1&2 assigned req" is associated with the Request Transition of Acknowledged to Hold, the condition verifies if the transition to move the status from Acknowledged to Hold is allowed.
   - **Condition Error Message**
     Specifies the message returned to the user if the transition is rejected.
   - **Description**
     Specifies the message returned to the user if the transition is rejected.

   The transition is defined.
4. Click Save, Close Window.
   The new transition appears in the Issue Transition List when you refresh the page.

Define Transition Types

This article contains the following topics:

For employees using self-service, you can modify incident and request process flows by including buttons to represent any *status transition*. Status transition buttons for incident and request process workflows are configured through *transition types*. A transition type lets you define the button text, form header text, and control the behavior of the Incident and Request detail page. You can link a transition type to any incident or request status transition.

To define a transition type

1. On the Administration tab, select Service Desk, Requests/Incidents/Problems, Transition Types.
   The Transitions Type List page appears.
2. Click Create.
   The Update Transition Types page appears.
3. Complete the fields as appropriate.
   The transition type for the status transition is created.
4. Click Save, Close Window.
   The new transition type appears in the Transitions Type List when you refresh the page.

Link Transition Types to Incident/Request Status Transitions

You can link a transition type to any incident or request status transition.

To link a transition type to an incident or request status transition

1. On the Administration tab, select Service Desk, Requests/Incidents/Problems, Request (or Incident) Transitions.
   The Transition List page appears.
2. Open the desired status transition.
   The Transition Details page appears.
3. Click Edit.
   The Update Transition page appears.
4. Specify the transition type that you want to link to in the Transition Type field. You can enter a value directly or click the magnifying glass to search for a service type.
   The transition type is associated with the status transition.
5. Click Save, Close Window.
   The transition type appears on the Status Transition list when you redisplay the list.

Activate Predefined Transition Types

By default, all predefined transition types delivered with the product are inactive, so status transition buttons are not in effect. You can activate and modify these transition types to accommodate your desired status transition flow.

To activate a predefined transition type

1. Select Show Filter on the Transition Type List page.
   The top portion of the page reveals additional search fields.
2. Select Inactive in the Record Status field and click Search.
   The Transition Type List displays all inactive transition types.
3. Right-click the title of the transition type and select Edit from the menu.
4. Select Active in the Record Status drop-down list.
5. Click Save, Close Window.
6. Click Search.
   The Transition Type List displays the active transition type.

Issue Workflows

Contents

Update a Workflow Task

You can edit an individual workflow task to change the person or group assigned to the task, the expected date of completion, the expected cost, and so on.

> **NOTE**
>
> The following procedure applies only to issues assigned to a category with an associated CA SDM classic workflow. For information about working with issues assigned to a category with an associated CA Workflow process definition, see Process CA Workflow Workitems.

To update a workflow task for an issue, **follow these steps:**

1. On the Service Desk tab, browse to Issues, Assigned or Unassigned. Select the priority for the issue you want to update.
   The Issue List appears.
2. Select the issue.

The Issue Detail page appears.
3. Choose the Workflow Tasks tab, and select the workflow task to edit.
   The Change Workflow Detail page appears.
4. Click Edit.
   The Update Change Workflow page appears.
5. Edit the fields as appropriate.
   Refer to Update Workflow Task Fields for field definitions.
6. Click Save.
   The Change Workflow Detail page displays a successful save message.
7. Click Close Window.
   The Issue Detail page appears.
8. Click Refresh Workflow Tasks on the Workflow Tasks tab.
   The updated workflow task appears on the Workflow Tasks tab on the Issue Detail page.

Add Workflow Tasks

You can add new workflow tasks to an issue, and specify the sequence in which they are performed.

> **NOTE**
>
> The following procedure applies only to issues assigned to a category with an associated CA SDM classic workflow. For information about working with issues assigned to a category with an associated CA Workflow process definition, see Process CA Workflow Workitems.

To add a workflow task to an issue, **follow these steps:**

1. Select Issues, Assigned or Unassigned on the Service Desk tab, and then select the Priority level for the issues you want to display in the list.
   The Issue List appears.
2. Select the issue to update.
   The Issue Detail page appears.
3. Select the Workflow Tasks tab, and click Insert Tasks.
   The Insert New Workflow Task page appears.
4. Complete the fields as appropriate.
   See Insert Workflow Task Fields for field definitions.
5. Click Add.
   The new workflow task appears on the Workflow Tasks tab on the Issue page.

Issue Workflows

A *workflow* is an automated or partially automated business process that specifies the sequence of tasks that must be performed to resolve a ticket.

Two workflow solutions are supported for issues:

- **CA Workflow**a high performance, scalable workflow management product that can optionally be integrated with CA SDM.
- **CA SDM "classic" workflow**internal workflow automation features provided as a standard CA SDM component.
- **CA Process Automation**workflow with task-based steps that run remotely on the CA Process Automation web service.

Your administrator can configure issue categories to use either type of workflow. When an issue ticket is assigned to a category with an associated workflow, a series of tasks appear on the ticket's Workflow Tasks tab.

Some workflow tasks may be optional, which allows you to delete them from the task list.

You must complete all mandatory workflow tasks before you can close the ticket. The only exception is that if the ticket status is set to Cancelled, the workflow process terminates automatically.

If a ticket is removed from a category with an attached workflow, or assigned to a different category, the workflow process terminates automatically. If the new category has an attached workflow, its process starts automatically.

Add Activities to an Issue

All work that is done to resolve a ticket is recorded on the Activities tab at the bottom of the ticket detail page. Activities can include actions such as research, calling a customer, and transferring responsibility to a different analyst. Some activities, such as "Initial" when a record is created or "Field Update" when a field is changed, are posted automatically. Others, such as returning a call, changing the status, or transferring the ticket are posted when you select items from the Activities menu or from the Activities page itself.

- Activities Fields
- Callback
- Escalate
- Log Comment
- Manual Notify
- Research
- Solution
- Transfer
- Update Status

Activities Fields

The following fields are present on one or more of the pages available through the Activities menu on ticket detail pages:

NumberSpecifies the number of the item for which activity is being recorded. Cannot be edited.

SummarySpecifies the summary description of the record. Cannot be edited.

Activity Type Specifies the type of activity being recorded. For example, if you select Log Comment from the Activities menu, the Activity Type is Log Comment. Cannot be edited.

Current Status Specifies the current status of the record. Cannot be edited.

New Status Specifies a new status to be assigned to the record.

Current Priority Specifies the current priority level of the record. Cannot be edited.

New Priority Specifies a new priority level to be assigned to the ticket.

Current Assignee Specifies the analyst currently assigned to the ticket. Cannot be edited.

New Assignee Specifies a new analyst to be assigned to the ticket.

Current Group Specifies the group currently assigned to the ticket. Cannot be edited.

New Group Specifies a new group to be assigned to the ticket.

AnalystSpecifies the name of analyst who performed the activity. This is the currently logged in user. Cannot be edited.

Internal?Select this check box if the activity should only be visible to internal users.

Time Stamp Specifies the date and time the activity began. This is the system date and time the activity record was opened. Cannot be edited.

Date of Activity The date and time the activity was performed. Defaults to the date and time the activity record was opened, but can be changed.

Time Spent Specifies the amount of time spent on the activity. Enter this value in hours, minutes, and seconds (hh:mm:ss).

Related Ticket Activity Type Specifies the type of activity generated for a related ticket, for example, Update Status.

User Description Specifies the description of the action taken. Click Spelling to check the spelling of the text entered. Click Submit Knowledge to submit the solution to the Knowledge Base.

System Description Specifies the system description of the action taken. Cannot be edited.

Personalized Response Specifies the personalized response you want to send for this item. See Personalized Responses for more information.

> **NOTE**
>
> The Personalized Response field is present only if an active personalized response exists.

RecipientsSpecifies the contacts who will receive manual notification pertaining to the ticket. Click Add Recipient to search the contact list, or Remove Recipient to remove the contact from the notification list.

UrgencySpecifies the urgency level associated with the manual notification. Select the appropriate urgency code from the drop-down list.

Current Urgency Specifies the current value for Area Urgency that is assigned to this ticket.

New Urgency Specifies the new Urgency that is assigned to this ticket.

Preferred Method Specifies the preferred method for delivering the manual notification message (for example, Email or Pager_Email).

Message Title Specifies the subject heading that will appear on the manual notification.

Message Text Specifies the text that will appear in the body of the manual notification message.

ChangeSelect an existing change order to which you want to attach the ticket.

RemarksEnter text explaining why the ticket is being attached to an existing change order.

ImpactIndicates the effect the record has on tasks being performed. Your system administrator can modify the default impact codes, so they can vary from one installation to another.

Current Impact Specifies the current value for Service Impact that is assigned to this ticket.

New Impact Specifies the new Service Impact that is assigned to this ticket.

Callback

When editing an issue, you can record information on the date, time, and nature of a callback. The callback record is stored in the issue history.

To record a callback for an issue, **follow these steps:**

1. Select the issue from the Issue List page on the Service Desk tab.
   The Issue Detail page appears.
2. Select Activities, Callback on the menu bar.
   The Create New Activity page appears.
3. Change the date of the activity, if appropriate, and record the time spent on the call.
   See Activities Fields for field descriptions.
4. Enter a description for the callback, such as "left message for customer."

5. Click Save.
   The activity is recorded on the Activities Tab on the Issue Detail page.

Escalate

You can escalate the priority of an issue in order to direct more attention to the issue and increase the speed with which it is resolved. You may need to escalate an issue that has been outstanding for too long or demands an immediate response.

To escalate an issue, **follow these steps:**

1. Select the issue from the Issue List page on the Service Desk tab.
   The Issue Detail page appears.
2. Select Activities, Escalate on the menu bar.
   The Escalate Issue page appears.
3. Enter the New Priority and update the other activities fields as appropriate.
   See Activities Fields for field descriptions.
4. Click Save.
   The priority of the issue is escalated, and the activity is recorded on the Activities tab on the Issue Detail page.

Log Comment

When editing an issue, you can record comments in the Activity Log to assist in resolving the issue. A comment can be any additional information about the issue that needs to be recorded.

To log a comment for an issue, **follow these steps:**

1. Select the issue from the Issue List page on the Service Desk tab.
   The Issue Detail page appears.
2. Select Activities, Log Comment on the menu bar.
   The Create New Activity page appears.
3. Enter your comment in the User Description field.
   See Activities Fields for field descriptions.
4. Click Save.
   The comment is recorded on the Activities Tab on the Issue Detail page.

Manual Notify

You can send a message about an issue to an end user, vendor, analyst, or anyone else in your service desk database. Record of the manual notification is stored in the Activities List for the issue.

To notify users of an issue manually, **follow these steps:**

1. Select the issue from the Issue List page on the Service Desk tab.
   The Issue Detail page appears.
2. Select Activities, Manual Notify on the menu bar.
   The Manual Notification page appears.
3. Complete the fields as appropriate.
   The notification is defined.
4. Click Notify.
   The message is sent and the notification activity is recorded.

Manual Notify Fields

Selected Recipients Lists the names of the people who receive the notification.

Available Recipients Lists relevant contacts and Contact Objects that you can add to the Recipients list, as specified in the Manual Notify settings in Activity Notifications options. When you select a contact object name, only the contact names that are associated with the object are added to the Recipients list. Use the CTRL or SHIFT keys plus the left mouse button to select multiple contacts.

NOTE

After you add contacts or Contact Objects to the Recipients list, the selected contacts or Contact Objects remain in the Available Recipients list.

Contact Enter characters into the contact field of the ticket. The field displays results as you type and generates a list of suggested results. You can click a suggested result in the list, use the keyboard down arrow to scroll through the suggested results, or use the look up to search for a required contact.

Email address Accepts an SMTP address for an email recipient. You can specify any valid email address including addresses that are not associated with a contact record. Separate multiple addresses with a semi-colon.

NOTE

This field appears when the option notification_allow_temp_address is installed using Options Manager.

Urgency Specifies the relative importance of the notification (by default: low, normal, high, or emergency). The system administrator predefines Urgency levels.

Preferred Method Specifies the notification method to use, such as Email or Notification.

NOTE

When you specify an email address, set the method to one that supports SMTP, otherwise, remove email addresses from the Selected Recipients list.

Internal
Makes the notification visible to internal contacts only.

Message Title Specifies a title for the notification message. This title is a short description of the message, similar to an email subject line. The default title is the ticket type and number, and the words "Manual Notify" (for example, "Incident 32 Manual Notify").

Message Text Specifies the entire text of the notification. You can click Spelling to check the spelling of the message text. Personalized Response Specifies a personalized response such as a generic or administrator signature. The response is added to the end of the message text.

NOTE

The administrator must set up a personalized response before you can use this option. For more information about personalized responses, see the *Administration Guide*.

Research

You can record information about research you performed to resolve the issue.

To record research performed for an issue, **follow these steps:**

1. Select the issue from the Issue List page on the Service Desk tab.
   The Issue Detail page appears.
2. Select Activities, Research on the menu bar.
   The Create New Activity page appears.
3. Change the date of the activity, if appropriate, and record the time spent conducting the research.
   See Activities Fields for field descriptions.

4. Enter a description of the research, such as "consulted development engineer for workaround."
5. Click Save.
   The activity is recorded on the Activities tab on the Issue Detail page.

Solution

When an issue is resolved, you can record the solution with the issue record for future reference.

To log a solution for an issue, **follow these steps:**

1. Select the issue from the Issue List page on the Service Desk tab.
   The Issue Detail page appears.
2. Select Activities, Solution on the menu bar.
   The Create New Activity page appears.
3. Change the date of the activity, if appropriate, and record the time spent resolving the issue
   See Activities Fields for field descriptions.
4. Enter a description of the solution in the User Description field. For example, if a workaround for the problem was discovered, document the workaround here.
5. Click Save.
   The solution is recorded on the Activities tab and Solutions tab on the Issue Detail page.
   OR
   Click Save and Submit Knowledge to save the solution as a Knowledge Document. For more information, see Knowledge Documents.

Transfer

You can reassign an issue that has been outstanding for too long, demands an immediate response, or should be handled by another technician. When you transfer an issue, the new analyst or group is automatically notified, as are any others who should receive notification based on the notification rules established by your system administrator.

To transfer an issue to another technician, **follow these steps:**

1. Select the issue from the Issue List page on the Service Desk tab.
   The Issue Detail page appears.
2. Select Activities, Transfer on the menu bar.
   The Transfer Issue page appears.
3. Enter a New Assignee and a New Group. You can enter the values directly or click the search icon to search for a contact or group.
4. Change any other activities fields as appropriate.
   See Activities Fields for field descriptions.
5. Click Save.
   The issue is transferred, and the activity is recorded on the Activities tab on the Issue Detail page.

Update Status

You can change the status of an issue to record progress toward resolving the issue. Status changes become part of the issue history and can be useful when you research and update issues.

To update the status of an issue

1. Select the issue from the Issue List page on the Service Desk tab.
   The Issue Detail page appears.

2. Select Activities, Update Status on the menu bar.
   The Status Change page appears.
3. Edit the status and other activities fields as appropriate.
   See Activities Fields for field descriptions.
4. Click Save.
   The activity is recorded on the Activities tab on the Issue Detail page.

Define Custom Fields for an Issue

You can add custom fields to an issue.

To define custom fields for an issue

1. On the Service Desk tab, browse to Issues, Assigned or Unassigned. Select the priority for the issue you want to edit.
   The Issue List displays.
2. Select the issue to edit.
   The Issue Detail page displays.
3. Click Edit.
   The Update Issue page displays.
4. Select the Custom Fields tab.
5. Select the Flag check boxes you want to use and enter a text string in the appropriate User text box.
6. Click Save.
   Your custom field definitions now appear on the issue's Custom Fields tab.

Incident Management

This article contains the following topics:

Incidents are events outside of normal operations. When an incident occurs, it disrupts the normal operational processes of an organization.

> **WARNING**
>
> Unless otherwise noted, all the CA SDM Request features are also available for Incidents and Problems.
>
> **NOTE**
>
> Depending on your role, you may not have access to all the functionality described in this section. For example, some predefined roles can edit records but cannot create records.

Create an Incident

You can either create an incident from scratch or use an existing template.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. On the Service Desk tab, click the File menu and select the ticket type that you want to create.
   For example: To create a request from scratch, click File, New Request. To create a request from a template, click File, New Request from Template.
2. Fill in the incident fields as appropriate. Use the controls available on the issue tabs to process the ticket as appropriate.

3. Click one of the following buttons:

   Auto Assign -- Triggers an auto assignment task, and updates the activity log. This button appears only when the ticket specifies a category or area that has auto-assignment enabled.

   Create Change Order -- Opens the Create New Change Order page. You can create a change order ticket that is associated with this incident. This button appears only when you create incidents, problems, and requests.

   Create Problem -- Opens the Create New Problem page so you can create a problem ticket associated with this incident. This button appears only when you create incidents and requests.

   Find Similar -- Opens the Find Similar page to search for similar problems.

   Quick Profile -- Displays the contact information for the specified user in the **Affected End User** field. You can also view their environment details and their entire ticket history.

   Use Template -- Displays a list of available templates for the selected ticket type. You can select the template that you want to use for creating this ticket.

   NOTE

   You can use Quick Profile to identify a contact to be the affected end user of your new ticket. Quick Profile allows you to search for a contact, and view the history of the issues, requests, change orders, incidents, and problems that are assigned to that contact. When you have identified a contact, you can create the incident ticket directly from Quick Profile.

Incident Fields

The following fields are required to create or update an incident:

- **Incident Reference Number**This is a unique reference number assigned by CA Service Desk Manager for all incident tickets. This is used by analysts and customers to refer to a particular incident ticket.
- **Requester**Specifies the name of the person who initiated the ticket. This person must be a defined contact. You can enter a value directly or click the magnifier to search for the name.
- **Affected End User**Specifies the contact name of the person who is affected by the record. If the contact is assigned to a special handling type, special handling indicators are displayed. You can enter a value directly or click on the magnifier icon to search for a contact name.
- **Incident Area**
  Indicates the general area of your IT environment that is affected by the incident. For example, Applications, E-mail, Hardware, and Software. An incident area provides default values that are automatically entered on all the incident tickets pertaining to that area. In addition to the predefined incident areas, your system administrator can define custom incident areas. You can enter a value directly, or you can expand the node to display the defined incident areas and select one.

  NOTE
  Your system administrator has the option of adding custom properties to incident areas. The custom properties are available on the Properties tab when you create, edit, or view an incident ticket. Some custom properties require that you enter a value.
- **Status**Specifies the status code of the record. For example, you can list only the tickets with a status code of Fix in Progress, or Close Requested. You can enter a value directly or click on the magnifier icon to search for a status. The blue button (on the left side of the Status field) lets you change the current status to the next default status.
- **Priority**Specifies the priority ranking of the record. The ranking determines the amount of attention the ticket receives. The predefined priority levels are 1 (highest) through 5 (lowest). Your system administrator or an active priority calculation can generate the appropriate priority values for various installations and tenants. When priority calculation is enabled, this field is updated based on Impact, Urgency, Affected Service, and Affected User settings. When your administrator disables priority calculation and uninstalls the urgency_on_employee option, Self Service Users see the Priority field on the Request Detail page.
- **Active**Indicates whether the record is Active or Inactive. This value applies to the current record only and not the associated template.

Detail Fields

- **Reported By**Specifies the name of the person reporting the record.
- **Assignee**Specifies the name of the person who is assigned to handle the record. You can enter a value directly or click on the magnifier icon to search for a name. Selecting an assignee populates the groups that the assignee belongs to in the **Group** field.
- **Group**Specifies the group that is responsible for this record. Your system administrator defines groups of contacts that are responsible for different types of issues, requests, incidents, or problems. Any contact who is part of the group can handle the record after it is assigned to the group. You can enter a value directly or click on the magnifier icon to search for a group. Selecting a group populates the **Assignee** field with the corresponding assignee name that belongs to the group.
- **Affected Service**Specifies the primary service that affects the problem or incident. The CIs of type Service have a class that is defined in the Enterprise Service Family field. The ticket stores the currently affected service information for reporting. You can enter a value directly or click on the magnifier icon to search for a CI.
- **Urgency**Specifies the urgency of the record. The urgency is determined by the importance of the user tasks that are affected by the record. Urgency codes indicate the importance of a ticket based on the degree to which it affects user tasks. For example, a network outage is more urgent than a printer failure. Your system administrator can modify the default urgency codes, so they can vary from one installation to another. Urgency values can update automatically based on an active priority calculation.
- **Impact**Specifies an impact code, such as 1 -- Entire Organization, that indicates how a ticket affects the work being performed. For example, a ticket that requires a network outage for several hours would have a higher impact than a ticket that takes a printer off-line. Your system administrator can modify the default impact codes, so they can vary from one installation to another.
- **Major Incident**Specifies that the incident is major or significant. Because of its importance, changes to this value on a ticket generate an activity log entry.

  NOTE
  When you copy an incident, the value of this field is cleared. In addition, related tickets (child incidents) do not include the Major Incident value.
- **Configuration Item**Specifies the hardware, software, or service that is affected by the record. Your system administrator creates a record that uniquely identifies each configuration item for your organization and indicates its precise location. You can enter a value directly or click on the magnifier icon to search for an item.
- **Problem**Provides the number and name of the problem that is associated with this record. Enter the number or name of the problem directly into this field or click the search icon to search for the problem.
- **Symptom**Specifies a code that describes a primary symptom of the incident. For example, Slow Response.
- **Resolution Code**Indicates the action that the analyst had taken to resolve an incident or request. Resolution codes specify the general resolution of the ticket. For example, an Applied Patch resolution code indicates that the analyst used a software patch to address an incident.
- **Resolution Method**Indicates *how* an analyst implemented the resolution. For example, a Chat Session resolution method indicates that an analyst used a chat session to address an incident.
- **Call Back Date/Time**Specifies the date and time to follow up on this record. You can enter the date and time, in the format *mm/dd/yyyy hh:mm am* | *pm*, or click the calendar icon to open a calendar window where you can select the date and time for follow-up.
- **Change**
  Specifies the number and name of the change order that is associated with this record. You can enter the number or name of the change order directly in this field. Or, click the search icon to search for the change order.
- **Caused by Change Order**
  Specifies the change order number when the Incident ticket is the result of changes that are implemented from a change order.
- **External System Ticket**Specifies an identification for a ticket that belongs to an external system that integrates with CA SDM. This field stores hyperlinks and displays functional links in read-only mode.

Summary Information Fields

- **Summary**Provides a brief description of the record.
- **Spelling**
  Checks the spelling of the text that you enter in the Summary field.
- **Search Knowledge**
  Searches the CA SDM knowledge base to help resolve tickets. If you do not find any relevant knowledge base articles, you can submit your own knowledge base article after the ticket is resolved.
- **Total Activity Time**
  Displays a running total of the overall amount of time that is spent on the ticket. This value is updated each time a change is made to the ticket record. You cannot edit this field.
- **Timer**
  Tracks the incremental amount of the time spent working on various phases of ticket processing. The timer is reset to 00:00:00 each time you open the ticket record for updates. You cannot edit this field.

  NOTE

  The amount of time that is spent on each activity is shown on the Activities tab of the ticket record.
- **Description**Describes the record in detail.
- **Created via**Specifies the component reporting the record (Form Name: detail_in.htmpl)
- **Ticket Open Date/Time**Displays the date and time the ticket was opened.
- **Last Modified Date/Time**Displays the date that this ticket was last modified.
- **Ticket Resolved Date/Time**
  Displays the date and time the ticket was resolved.
- **Ticket Closed Date/Time**
  Displays the date and time the ticket was closed.
- **Last Modified By**
  Displays the name of the last person who edited the ticket.

Incident Tabs

The following tabs are available on the Create New Incident, Incident Detail, and Update Incident pages:

- **Activities:** Displays a log of the activities performed to resolve the incident. For more information, see Ticket Management.
- **Event Log:** Displays a record of significant actions that occur regarding the incident.
- **Verification Log:** Displays the change verification activity history for this Incident. The log shows details about the policy and actions taken during the change verification operations involved in this Incident. For example, this log identifies the policy, CI, and managed attribute values that caused a change specification to create the Incident.

  NOTE

  This tab only appears if a CACF policy created the Incident.
- **Attachments:** Attaches a document or a link to a URL to the incident.
- **Service Type:** Attaches a service type event to indicate the level of support for the ticket.
- **Workflow Tasks:** Lists the process instance and related audit trail messages for CA Process Automation or Classic Workflow that is associated with the ticket. The Workflow Tasks tab shows fields that apply to an attached workflow. The workflow may require some of the work items to complete before the ticket can close. The Workflow Tasks tab appears only if your administrator configured workflows for the ticket area or category. For more information, see Define a Category or Area topic.
- **Efficiency Tracking:** You can specify criteria to track incidents. The information that you specify provides your organization with metrics about incidents for reports. For example, you can indicate that an incident was assigned incorrectly. When a large percentage of incorrectly assigned incidents appears in a report, your organization is aware that assignments must be adjusted. Efficiency tracking options appear only when the efficiency_tracking Options Manager option is installed. For more information about installing efficiency_tracking, see Incident Tracking. Complete the following fields as appropriate:

  - **Resolvable at a Lower Level**
    Flags incidents that are resolvable at a lower level.
  - **Incorrectly Assigned**
    Flags incidents that are assigned incorrectly.
  - **Remote Control was Used**
    Flags incidents in which remote control was used.
  - **Action**
    Sets or resets the Actual Date/Time to the current date and time.
  - **Target**
    Specifies the current Service Target. Service Targets measure whether the required service or repair is completed within the required time frame.
  - **Target Date/Time**
    Specifies date and time when this Service Target is due. If the ticket is in Hold status, this value is blank.
  - **Actual Date/Time**
    Specifies the time when the target condition was met. If no value appears, the target condition has not been met.
  - **Time Left**
    Specifies the amount of remaining time for the service target when the ticket is on hold. If the Service Target has been met, Time Left shows the unused time. A negative value indicates the amount of time that elapsed since the missed target date.
  - **Violation Cost**
    Specifies the incurred cost when the service type time limit is violated.
- **Parent / Child:** Create a parent/child relationship between the incident and another CA SDM record.
- **Knowledge:** Searches for or submits information to the knowledge base to help resolve incidents. The Federated Search capability helps you to get the knowledge search results from multiple sources. For example, Google, SharePoint, CA Open Space, and so on.
- **Solutions:** Stores information about the incident solution with the incident record for future reference.
- **Properties:** Adds custom properties to incident areas.
- **Outage:** Specifies information about service outages for an incident. Complete the Service Outage fields as appropriate:
  - **Start Time**
    Specifies the start date and time for an outage period related to the subject of the incident ticket. This value, together with the End Time, lets you track incident outages with the tickets that are opened relative to the outage. You can click the calendar icon to open the Date Helper window to select a date.
  - **End Time**
    Specifies the end date and time for an outage period related to the subject of the incident ticket. This value, together with the Start Time, lets you track incident outages with the tickets that are opened relative to the outage. You can click the calendar icon to open the Date Helper window so you can select a date.
  - **Type**
    Specifies the type of outage, such as a network outage. You can use the predefined outage types or create outage types.
  - **Return to Service**
    Indicates that service was restored.
  - **Percent of Service Restored**
    Specifies the percentage of service that was restored.
- **Template:** Allows you to create a Problem template using the current ticket as a model.
- **Support Automation:** Displays the assistance session log and lets you invite the end user to an assistance session.

Incident Attachments

You can attach a document or a link to a URL to an incident to provide additional explanation or justification for the incident report.

Edit Attachment Details

You can edit the details of a document or URL that has been attached to a ticket.

To edit attachment details

1. Select the desired ticket from the appropriate list page on the Scoreboard.
   The ticket's detail page appears.
2. Select the Attachments tab, and click Attach Document.
   Any documents or URLs that have been attached to the ticket are listed.
3. Right-click the attachment link and select Edit Details.
   The Edit Attachment page appears.
4. Update the editable fields as desired and then click Save.
   The View Attachment page displays your edits.
5. Click Close Window.
   Your edits appear on the Attachments tab of the ticket detail page.

Request Management

This section contains the following topics.

- Request Management Using CA IT Asset Manager
- Request Management Using CA SDM
- Request Management Using CA Service Catalog

Request Management Using CA IT Asset Manager

You can automate the request of IT assets through a repeatable service offering that is accessible through a service catalog. This function can increase customer satisfaction and can standardize your asset base to improve response times and service levels.

To automate request fulfillment, use the CA APM integration with CA Service Catalog.

Request Fulfillment

When CA APM and CA Service Catalog are integrated, you can perform request fulfillment using the two products. Use request fulfillment to associate requested items from a CA Service Catalog service request with CA APM assets. During the fulfillment process, you can perform the following tasks:

- View the assets that are assigned to a request.
- Assign the assets to a request.
- Remove the assets from a request.
- Deny the fulfillment of a request for assets.

For information about creating and managing requests in CA Service Catalog, see Integrate with CA Service Catalog Manually.

How to Fulfill Requests from Inventory

When you integrate CA Service Catalog with CA APM, you can associate assets with items requested from the catalog during request fulfillment. To fulfill requests from inventory, complete the following steps:

1. In CA APM, verify that the user fulfilling the request belongs to a role in which asset fulfillment access is enabled.

   NOTE
   For more information about creating user roles, see User Roles.
2. In CA Service Catalog, create a request for an asset.
   The request contains information about the requester and the type of hardware or software asset being requested.
3. In CA Service Catalog, open the request and click the gold brick action icon that is associated with the request.
   CA APM opens and the CA APM Fulfillment page appears.

   NOTE
   For information about creating and managing requests, see Integrate with CA Service Catalog Manually.
4. In CA APM, complete the following steps:
   a. Search for the assets from the inventory request that you want to fulfill.
   b. Complete any of the following steps:
      - Fulfill the inventory request for a hardware asset.
      - Fulfill the inventory request for a software asset.
      - (Optional) Deny an inventory request.
      - Display the list of assigned assets to verify that they match the request.
      - (Optional) Remove an assigned hardware asset from an inventory request.
      - (Optional) Remove an assigned software asset from an inventory request.
5. In CA Service Catalog, verify the status of the fulfilled inventory request.

Deny an Inventory Request

You can deny an inventory request to indicate that none of the assets that are requested are fulfilled. When you open CA APM from a CA Service Catalog request, the CA APM Fulfillment page appears with a list of all assets available in CA APM.

NOTE
Verify that the user fulfilling the request belongs to a role in which asset fulfillment access is enabled. For more information about creating user roles, see User Roles.

On the CA APM Fulfillment page, click Not Fulfilled from Inventory. The inventory request is denied, and the status of the request is updated in CA Service Catalog.

Display Assets Assigned to a Request

CA APM lets you display all assets that are currently assigned to a CA Service Catalog request. The enables you to manage the request fulfillment. When you open CA APM from a CA Service Catalog request, the CA APM Fulfillment page appears with a list of all assets available in CA APM.

NOTE
Verify that the user fulfilling the request belongs to a role in which asset fulfillment access is enabled. For more information about creating user roles, see User Roles.

To display assets that are assigned to an inventory request, click Assigned Asset in the menu on the left of the CA APM Fulfillment page. A list of all assets that are assigned to the request appears in the search results.

Fulfill an Inventory Request for a Hardware Asset

You can fulfill a CA Service Catalog inventory request so that hardware assets are correctly assigned and fulfilled. When you open CA APM from a CA Service Catalog request, the CA APM Fulfillment page appears with a list of all hardware assets available in CA APM.

> **NOTE**
> Verify that the user fulfilling the request belongs to a role in which asset fulfillment access is enabled. For more information about creating user roles, see User Roles.

Follow these steps:

1. On the CA APM Fulfillment page, complete one of the following steps to search for an asset:
   a. Scroll through the list of all available hardware assets that appear in the search results.
   b. Specify the search criteria and click Go.
      A list of matching hardware assets appears in the search results.
2. In the search results, select the assets that you want to fulfill.
3. (Optional) In the Fulfillment Changes area of the page, make field-level changes to all selected assets. For example, you can change the department, cost center, general ledger (GL) code, contact, and location for all selected assets.
4. Click Fulfill.
   The request is fulfilled, the request status is updated in CA Service Catalog. The asset information is updated in CA APM.

Fulfill an Inventory Request for a Software Asset

You can fulfill a CA Service Catalog inventory request so that a software asset is correctly assigned and fulfilled. When you open CA APM from a CA Service Catalog request, the CA APM Fulfillment page appears with a list of all software assets available in CA APM.

> **WARNING**
> If you have a CA SAM implementation, we do not recommend that you fulfill software asset requests in CA APM. We recommend that you use CA SAM to manage your software assets and licenses.

> **NOTE**
> Verify that the user fulfilling the request belongs to a role in which asset fulfillment access is enabled. For more information about creating user roles, see User Roles.

Follow these steps:

1. On the CA APM Fulfillment page, complete one of the following steps to search for a software asset:
   a. Scroll through the list of all available assets that appear in the search results.
   b. Specify the search criteria and click Go.
      A list of matching software assets appears in the search results.
2. In the search results, click the software asset that you want to fulfill.
   The Asset Details page for the selected software asset appears.
3. Define or update software internal allocations for the software asset by clicking Asset Allocation in the Relationships menu.
4. Click New and select the hardware asset that you want to associate with the software asset.
5. Click Fulfill and Save.
   The software request is fulfilled, the internal allocations are saved, the request status is updated in CA Service Catalog. The asset information is updated in CA APM.

Remove an Assigned Hardware Asset from an Inventory Request

CA APM lets you remove an assigned hardware asset from an inventory request. For example, you have a laptop that was mistakenly added to a request and you want to remove it. When you open CA APM from a CA Service Catalog request, the CA APM Fulfillment page appears with a list of all hardware assets available in CA APM.

> **NOTE**
> Verify that the user fulfilling the request belongs to a role in which asset fulfillment access is enabled. For more information about creating user roles, see User Roles

Follow these steps:

1. On the left of the CA APM Fulfillment page, click Assigned Asset.
   A list of all hardware assets that are assigned to the request appears in the search results.
2. In the search results, select the hardware assets that you want to remove from the request.
3. (Optional) In the Fulfillment Changes area of the page, make field-level changes to all selected assets. For example, you can change the general ledger (GL) code, cost center, department, contact, location.
4. Click Remove Assignment.
   The hardware asset is removed from the request, the status of the request is updated in CA Service Catalog. The asset information is updated in CA APM.

Remove an Assigned Software Asset from an Inventory Request

CA APM lets you remove an assigned software asset from an inventory request. For example, a graphics software package was mistakenly added to a request and you remove the software from the request. When you open CA APM from a CA Service Catalog request, the CA APM Fulfillment page appears with a list of all software assets available in CA APM.

> **NOTE**
> Verify that the user fulfilling the request belongs to a role in which asset fulfillment access is enabled. For more information about creating user roles, see User Roles.

Follow these steps:

1. On the left of the CA APM Fulfillment page, click Assigned Asset.
   A list of all software assets that are assigned to the request appears in the search results.
2. In the search results, click the software asset that you want to remove from the request.
   The Asset Details page for the selected software asset appears.
3. Remove or update software internal allocations for the software asset by clicking Asset Allocation in the Relationships menu.
4. Click the Delete icon next to your selected asset.
5. Click Remove Fulfillment and Save.
   The asset is removed from the request, the internal allocations are saved. The status of the request is updated in CA Service Catalog and the asset information is updated in CA APM.

Request Management Using CA SDM

This article contains the following topics:

Requests are records of reports made to the help desk and the activities performed to resolve the requests. If you create a ticket as a copy of another ticket, the Status field displays all Status values.

> **WARNING**
> Depending on your role, you do not have access to all the functionality described in this section. For example, some predefined roles can edit records but cannot create new ones.

Create a Request

You can either create a request from scratch or use an existing template.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. On the Service Desk tab, click the File menu and select the ticket type that you want to create. For example, to create a request from scratch, click File, New Request. To create a request from a template, Click File, New Request from Template.
2. Fill in the request fields as appropriate.
   Use the controls available on the tabs to process the ticket as appropriate.
3. Click one of the following buttons:
   Auto Assign -- Triggers an auto assignment task, and updates the activity log. This button appears only when the ticket specifies a category or area that has auto-assignment enabled.
   Create Change Order -- Opens the Create New Change Order page. You can create a change order ticket that is associated with this incident. This button appears only when you create incidents, problems, and requests.
   Create Problem -- Opens the Create New Problem page so you can create a problem ticket associated with this incident. This button appears only when you create incidents and requests.
   Create Incident – Opens the Create New Incident page so you can create an associated incident ticket. This button appears only when you create change orders and requests.
   Find Similar -- Opens the Find Similar page to search for similar problems.
   Quick Profile -- Displays the contact information for the specified user in the **Affected End User** field. You can also view their environment details and their entire ticket history.
   Use Template -- Displays a list of available templates for the selected ticket type. You can select the template that you want to use for creating this ticket.

   > **NOTE**
   >
   > You can use the Quick Profile to identify a contact to be the affected end user of your new ticket. Quick Profile allows you to search for a contact, and view the history of the issues, requests, change orders, incidents, and problems that are assigned to that contact. When you have identified a contact, you can create the incident ticket directly from the Quick Profile.

Request Fields

The following fields are required to create or update a request:

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

- – **Request Reference Number**This is a unique reference number assigned by CA Service Desk Manager for all request tickets. This is used by analysts and customers to refer to a particular request ticket.
  – **Requester**Specifies the name of the person who initiated the ticket. This person must be a defined contact. You can enter a value directly or click the magnifier to search for the name.
  – **Affected End User**Specifies the contact name of the person who is affected by the record. If the contact is assigned to a special handling type, special handling indicators are displayed. You can enter a value directly or click the magnifier to search for a contact name.
  – **Request Area**
  Indicates the general area of your IT environment affected by the request (for example, Applications, E-mail, Hardware, and Software). Request areas provide default values that are entered automatically on all requests

assigned to the area. In addition to the predefined request areas provided, your system administrator can define custom request areas.
You can enter the request area directly into the field, or click the Lookup icon to select from the defined request areas.

NOTE
Your system administrator has the option of adding custom properties to request areas. If custom properties have been added, they are displayed on the Properties tab when you create, edit, or view a request. Some custom properties require that you enter a value. For details, see Request Properties.

- **Status**Specifies the status code of the record. For example, you can list only the tickets with a status code of Fix in Progress, or can Close Requested. You can enter a value directly or click the magnifier to search for a status. The blue button (on the left side of the Status field) lets you change the current status to the next default status.
- **Priority**
  Specifies the priority ranking of the record. The ranking determines the amount of attention the ticket receives. The predefined priority levels are 1 (highest) through 5 (lowest). Your system administrator or an active priority calculation can generate the appropriate priority values for various installations and tenants. When priority calculation is enabled, this field is updated based on Impact, Urgency, Affected Service, and Affected User settings. When your administrator disables priority calculation and uninstalls the urgency_on_employee option, Self Service Users see the Priority field on the Request Detail page.

Detail Fields

- **Reported By**Specifies the name of the person reporting the record.
- **Assignee**Specifies the name of the person who is assigned to handle the record. You can enter a value directly or click the magnifier to search for a name. Selecting an assignee populates the groups that the assignee belongs to in the **Group** field.
- **Group**Specifies the group that is responsible for this record. Your system administrator defines groups of contacts that are responsible for different types of issues, requests, incidents, or problems. Any contact who is part of the group can handle the record after it is assigned to the group. You can enter a value directly or click the magnifier to search for a group. Selecting a group populates the **Assignee** field with the corresponding assignee name, which belong to the group.
- **Configuration Item**Specifies the hardware, software, or service that is affected by the record. Your system administrator creates a record that uniquely identifies each configuration item for your organization and indicates its precise location. You can enter a value directly or click the magnifier to search for an item.
- **Severity**Specifies the severity of the record, determined by the effect on people. Your system administrator can modify the default severity codes, so they can vary from one installation to another.
- **Urgency**Specifies the urgency of the record. The urgency is determined by the importance of the user tasks that are affected by the record. Urgency codes indicate the importance of a ticket based on the degree to which it affects user tasks. For example, a network outage is more urgent than a printer failure. Your system administrator can modify the default urgency codes, so they can vary from one installation to another. Urgency values can update automatically based on an active priority calculation.
- **Impact**Specifies an impact code, such as 1 -- Entire Organization, that indicates how a ticket affects work being performed. For example, a ticket that requires a network outage for several hours would have a higher impact than a ticket that takes a printer off-line. Your system administrator can modify the default impact codes, so they can vary from one installation to another
- **Active**Indicates whether the record is Active or Inactive. This value applies to the current record only, not the associated template.
- **Charge Back ID**
  Identifies the ID that is charged for service.
- **Call Back Date/Time**

Identifies the date and time to follow up on this record. Enter the date and time in the format mm/dd/yyyy hh:mm am | pm, or click the calendar icon to select the date and time for follow-up.

- **Resolution Code**Indicates the action that the analyst had taken to resolve an incident or request. Resolution codes specify the general resolution of the ticket. For example, an Applied Patch resolution code indicates that the analyst used a software patch to address an incident.
- **Resolution Method**Indicates *how* an analyst implemented the resolution. For example, a Chat Session resolution method indicates that an analyst used a chat session to address an incident.
- **Change**
  Specifies the number and name of the change order associated with this record. You can enter a value directly or click the magnifier to search for a change order.
- **Caused by Change Order**
  Specifies only tickets that were opened as a result of a specific change order ticket. You can enter a value directly or click the magnifier to search for a change order ticket.
- **External System Ticket**Specifies an identification for a ticket that belongs to an external system that integrates with CA SDM. This field stores hyperlinks and displays functional links in read-only mode.

Summary Information Fields

- **Summary**
  Provides a brief description of the record.
- **Spelling**
  Checks the spelling of the text you enter in the Summary field.
- **Total Activity Time**Displays a running total of the overall amount of time spent working on the ticket. This value is updated each time a change is made to the ticket record. You cannot edit this field.
- **Timer**Tracks the incremental amount of the time spent working on various phases of ticket processing. The timer is reset to 00:00:00 each time you open the ticket record for updates. You cannot edit this field.

  NOTE

  The amount of time spent on each activity is shown on the Activities tab of the ticket record.
- **Search Knowledge**
  Searches for or submits information to the CA SDM Knowledge Base to help resolve tickets.
- **Description**
  Describes the record in detail.
- **Ticket Open Date/Time**Displays the date and time the ticket was opened.
- **Last Modified Date/Time**Displays the date that this ticket was last modified.
- **Ticket Resolved Date/Time**
  Displays the date and time the ticket was resolved.
- **Ticket Closed Date/Time**
  Displays the date and time the ticket was closed.

Request Tabs

The following tabs are available on the Create Request, Request Detail, and Update Request pages:

- **Activities:** Displays a log of the activities performed to resolve the request. For more information, see Add an Activity from the Ticket Management topic.
- **Event Log:** Displays a record of significant actions that take place regarding the request.
- **Attachments:** Attaches a document or a link to a URL to the request.
- **Workflow Tasks:** Lists the process instance and related audit trail messages for CA Process Automation or Classic Workflow that is associated with the ticket. The Workflow Tasks tab shows fields that apply to an attached workflow. The workflow may require some of the work items to complete before the ticket can close. The Workflow Tasks tab

appears only if your administrator configured workflows for the ticket area or category. For more information, see Areas topic.

- **Service Type:** Allows you to attach a service type event to indicate the level of support for the ticket.
- **Parent / Child:** Allows you to create a parent/child relationship between the request and another CA SDM record.
- **Knowledge:** Searches for or submit information to the CA SDM Knowledge base to help resolve requests. The Federated Search capability helps you to get the knowledge search results from multiple sources. For example, Google, SharePoint, CA Open Space, and so on.
- **Solutions:** Stores information about the request solution with the request record for future reference.
- **Properties:** Adds custom properties to request areas.
- **Templates:** Allows you to create a template using the current ticket as a model.
- **Support Automation:** Displays the assistance session log and lets you invite the end user to an assistance session.

Request Properties

Aministrators have the option of defining custom properties for request areas, to provided additional detail on request tickets assigned to that area. When you select a request area while creating or editing a request, if any custom properties have been defined for that area, they appear on the Properties tab.

The Properties tab contains the following fields:

Name

Identifies the descriptive name assigned to the property, and indicates whether a value entry for the property is required or optional. Names of properties that require a value are shown with an asterisk after the name.

Value

Specifies the value you assign to the property. There are three kinds of property value controls:

Freeform text fields

The values you enter for these properties are not validated. Be sure to read the information in the Examples column for guidance on the kind of information this property is intended to contain.

Checkboxes

Two-state checkboxes. Checked = Yes. Cleared = No. Property values for checkboxes are validated.

Dropdown lists

A list of options. If your administrator has defined this field as required, you must select one of the values in the list. If a default value has been assigned, it is displayed when the list first appears. Property values for dropdown lists are validated.

> **NOTE**
>
> On the Request Detail page, only the specified values are displayed, with no indication of whether the value was set via freeform text field, checkbox, or dropdown list.

Examples

Provides examples of the kinds of values that are intended for this property. For example, if a freeform text field is intended to contain information about a network type, this field might include LAN, WAN, and SAN as example values.

Request Attachments

You can attach a document or link to a URL to a request to provide additional explanation or justification for the request report.

Attach a Document to a Request

You can attach a document file to a request so that it may be viewed when viewing or editing the request.

To attach a document to a request

1. On the Service Desk tab, select the desired request from the Request List page.
   The Request Detail page appears.
2. Select the Attachments tab, and click Attach Document.
   The Add File page appears.
3. Enter the file name you want to attach, or click Browse to navigate to the desired file name.
4. Select the repository where the document is stored.
5. Enter a name for the attachment.
   This name appears in the Attachments List on the Request Detail page.
6. Enter a description for the attachment.
7. Click Upload.
   The document is uploaded to the request record and is listed on the Attachments tab of the Request Detail page.

Attach a URL to a Request

You can attach a link to a URL to a request so that it may be viewed when viewing or editing the request.

To attach a URL to a request

1. On the Service Desk tab, select the desired request from the Request List page.
   The Request Detail page displays.
2. Select the Attachments tab, and click Attach URL.
   The Create New Attachment page displays.
3. Enter the URL you want to attach.
4. Enter a name for the attachment.
   This name displays in the Attachments List on the Request Detail page.
5. Enter a description for the attachment.
6. Click Save.
   A link to the URL is added to the request record and is listed on the Attachments tab of the Request Detail page.

Edit Attachment Details

You can edit the details of a document or URL that has been attached to a ticket.

To edit attachment details

1. Select the desired ticket from the appropriate list page on the Scoreboard.
   The ticket's detail page appears.
2. Select the Attachments tab, and click Attach Document.
   Any documents or URLs that have been attached to the ticket are listed.
3. Right-click the attachment link and select Edit Details.
   The Edit Attachment page appears.
4. Update the editable fields as desired and then click Save.
   The View Attachment page displays your edits.
5. Click Close Window.
   Your edits appear on the Attachments tab of the ticket detail page.

Add Activities to a Request

All work that is done to resolve a ticket is recorded on the Activities tab at the bottom of the ticket detail page. Activities can include actions such as research, calling a customer, and transferring responsibility to a different analyst. Some activities, such as "Initial" when a record is created or "Field Update" when a field is changed, are posted automatically. Others, such as returning a call, changing the status, or transferring the ticket are posted when you select items from the Activities menu or from the Activities page itself.

Solution

When a request is resolved, you can record the solution with the request record for future reference.

To log a solution for a request

1. Select the request from the Request List page on the Service Desk tab.
   The Request Detail page appears.
2. Select Activities, Solution on the menu bar.
   The Create New Activity page appears.
3. Change the date of the activity, if appropriate, and record the time spent resolving the request.
4. Enter a description of the solution in the User Description field. For example, if a workaround for the problem was discovered, document the workaround here.
   See Activities Fields for field descriptions.
5. Click Save.
   The solution is recorded on the Activities tab and on the Solutions tab on the Request Detail page.

Attach to Existing Change Order

You can attach a request to a related change order. Change orders are requested changes that vary from the original scope of a specific project or task.

To attach a request to an existing change order

1. Select the request from the Request List page on the Service Desk tab.
   The Request Detail page appears.
2. On the menu bar, select Activities, Attach to Existing Change Order.
   The Attach Change Order to Request page appears.
3. Click Change.
   The Change Order Search page opens.
4. Complete one or more of the search fields, and click Search.
   The Change Order List displays the change orders that match your search criteria.
5. Select the desired change order.
   The Attach Change Order to Request page displays the selected change order in the Change field.

   NOTE

   If you know the Change Order number, you can type it in the Change field.
6. (Optional) Enter comments in the Remarks field.
7. Click Attach.
   The request is attached to the change order, and the activity is recorded on the Activities tab on the Request Detail page.

Detach Change Order

You can end the association between a request and an attached change order.

To detach a change order from a request

1. On the Service Desk tab, select the desired request from the Request List page.
   The Request Detail page appears.
2. Select Activities, Detach Change Order on the menu bar.
   The Detach Change Order from Request page appears.
3. Click Detach.

The request is detached from the change order, and the activity is recorded on the Activities tab on the Request Detail page.

Request Management Workflows

This article contains the following topics:

Request/Incident/Problem Workflows

A workflow is an automated or partially automated business process that specifies the sequence of tasks to be performed to resolve a CA SDM ticket. The workflow solutions are supported for Request/Incident/Problem:

- **CA SDM "classic" workflow**This is an internal workflow automation features provided as a standard CA SDM component.
- **CA Process Automation**This is a workflow with task-based steps that run remotely on the CA Process Automation web service.

Your administrator can configure Request/Incident/Problem areas to use either type of workflow. When a Request/Incident/Problem ticket is assigned to an area with an associated workflow, a series of tasks appear on the ticket's Workflow Tasks tab.

Some workflow tasks may be optional, which allows you to delete them from the task list.

You must complete all mandatory workflow tasks before you can close the ticket. The only exception is that if the ticket status is set to Cancelled, the workflow process terminates automatically.

If a ticket is removed from an area with an attached workflow or assigned to a different area, the workflow process terminates automatically. If the new area has an attached workflow, its process starts automatically.

NOTE

To use the process viewer, you must associate a CA Workflow process to the Request/Incident/Problem area.

Add Workflow Tasks

You can add a new workflow task to a Request/Incident/Problem and specify the sequence in which they are performed.

Follow these steps:

1. Select **Service Desk** tab, in **Search** drop-down, select the desired among **Incidents**, **Problems**, and **Requests**.
2. Click **Search** to view the respective list.
3. Based on your selection, select any Incident or Problem or Request for which the area is already added.
4. Navigate to the **Relationships** tab, **Workflow Tasks** tab.
5. Click **Insert tasks**.
6. Select the Sequence and New workflow task type.
7. Click **Add**.
8. The new workflow task appears on the Workflow Tasks tab in the Request/Incident/Problem Workflow List.

Update a Workflow Task

You can edit existing workflow tasks.

Follow these steps:

1. Select **Service Desk** tab, in **Search** drop-down, select the desired among **Incidents**, **Problems**, and **Requests**.
2. Click **Search** to view the respective list.
3. Based on your selection, click any Incident or Problem or Request.

4. Navigate to the **Relationships** tab, **Workflow Tasks** tab.
5. Select the desired Workflow and then click **Edit**.
6. Edit the fields as appropriate and then click **Save**.

   NOTE

   The Service Type tab on the Request/Incident/Problem Workflow Detail page provides information for service types assigned to workflow tasks. Service types assigned to workflow tasks are independent of the service types assigned to a Request/Incident/Problem ticket on the ticket record's Service Types tab. For information about working with service type events, see Delay or Resume a Service Type Event.
7. The updated workflow task appears on the Workflow Tasks tab in the Request/Incident/Problem Workflow List.

Request Management Using CA Service Catalog

This section contains the following articles:

- Approval Processes and Fulfillment Processes
- Status Values
- Subscriptions and Requests
- Request Service Levels Reports
- Request Management from an Administrator Perspective
- Request Management from a User Perspective
- Approve and Reject Requests Pending Action from a PDA
- Manage Requests Pending Action
- Delegate Catalog

Approval Processes and Fulfillment Processes

Approval Processes

When a user submits a request, approvers must approve or reject each service in the request. For each service, the administrator specifies one of the following approval processes on the Service Details tab:

- **No Approval** Requires no approval. When a request containing the service is submitted, the status is changed to Approval Done.
- **System Approval Process**
  Determines the approver and the number of approval levels by using a combination of the following parameters:
  – The management hierarchy.
  – The authorization level of each user in the hierarchy.
  – The approval level that is specified for the service.

  When a request containing the service is submitted and the Requested For value is a user, not an account, the Catalog system:
  a. Finds the Requested For authorization level of the user in the user profile.
  b. Compares it with the approval level of the service.

  If the authorization level of the user matches or exceeds the approval level that the service specifies, then no further approval is required. The system moves the request to the next status, typically Pending Fulfillment, or Completed. Otherwise, then the system determines an approver, as follows:
  – The Requested For value is a user. In this case, the system determines the manager of the user and assigns the request to that manager for approval.
  – The user does not have a manager or the Requested For value is an account. The system assigns the approval task to the user-specified in the configuration option that is named Default User For Request Actions.

After the manager or other approver approves the service, the system uses similar logic to determine whether another level of approval is required. If yes, then the system routes the request to a request manager whose authorization level matches or exceeds the approval level of the service.

- **Workflow Driven Approval Process**
  Uses a CA Process Automation process to determine the approval process.
  The process includes the business logic to determine the approver and the number of approval levels. CA Service Catalog provides sample processes and process definitions, including default ones for a single level of manager approval.
  For any service, if you use this approval process with a CA Process Automation process, you can optionally use policies. In that case, the approval process proceeds the same as with Policy driven approval process, except as follows:
  - The Workflow driven approval process uses the CA Process Automation workflow engine to evaluate and implement policies.
  - Policy driven approval process uses the Catalog Policy Engine to evaluate and implement policies. This option is typically more efficient for an approval process because this engine is internal. Verify that the rules and actions are enabled for the option that you specify.
- **Policy driven approval**
  Uses a policy to determine the approval process for requests. You specify conditions in policies, which are based on the attributes of service options, services, requested items, and users. If a policy is active and a submitted request meets the condition in the policy, then the assignees in the policy receive a request pending action to approve, reject, or fulfill a service option, service, or request.

Policy driven approval and system approval use a few common terms. For example, in both methods, the *level of approval* refers to the authority of an approver in numeric terms: the higher the number, the greater the authority of the approver. However, in policy driven approval, the administrator assigns each approver and authority level uniquely, with no relation to system approval.

If a policy does not apply to a request, the Catalog system uses the approval flow as defined in the workflow driven approval process. For example, you are using the predefined workflow approval process. Also, no predefined sample policy applies to a request pending action. In that case, the Catalog system assigns request pending action to the manager of the Requested For user. If the user has no manager, then the system assigns the request pending action to the Default User for Request Actions. This user is specified in the Catalog Configuration.

Fulfillment Processes

Fulfillment is required for each service option included in the services in a request. When the status of a service option is set to Pending Fulfillment, it enters the fulfillment phase. When the status is set to Fulfilled or Fulfillment Cancelled, the service option leaves the fulfillment phase. For each service Approval Process setting, the fulfillment phase of the requested service options can be managed separately.

- No Approval and System Approval Process: After approval, the status of the service options for the service is set to either Fulfilled or Pending Fulfillment.
- Workflow Driven Approval Process: After approval, the status of the service options for the service is set to Pending Fulfillment.

When you set the requested service options status to Pending Fulfillment, a Workflow process definition manages the fulfillment process. Several process definitions manage the fulfillment process. These process definitions and associated event rules support three different approaches to fulfillment.

Generally, fulfillment options do not fit your business processes. More commonly, alter the distributed components to more accurately fit your business processes. Before altering the existing components or adding to them, it is important to understand what they do.

NOTE
Enabling all the rules that are related to all the options is not recommended, because the fulfillment options overlap.

- **Complex Fulfillment**
  This process considers the current status value of the service option and new status being assigned to determine the next step in the fulfillment process. The Complex Fulfillment process supports major fulfillment phases that are related to:
  – Checking availability of the requested service option when initially requested or when the service option status is Received.
  – (Optional) Opening a CA Service Desk Manager Change Order (for Hardware or Software) or notifying appropriate fulfillment personnel when the service option is found in the available inventory.
  – Notifying Procurement of the requested service option when it is not found in available inventory. In this case, the service option is ordered and marked as Received when it arrives.
- **Simple Fulfillment**
  This process notifies the appropriate fulfillment personnel that the item must be fulfilled with no special regard for fulfillment phases.
- **CA Service Desk Manager Request Fulfillment**
  This process:
  – Opens a CA Service Desk Manager request.
  – Notifies the appropriate fulfillment personnel that the service option must be fulfilled.
  – Lays responsibility for the fulfillment process on the CA Service Desk Manager request logic.

Change in Approval Process and Fulfillment Scenarios

After Upgrading to CA Service Catalog 17.3.0.3 and above, review the following:

- By default, all forms are in the disabled state and if you have any pending items (Pending Approval or Pending Fulfillment) in the request, a checkbox appears at the service/service option level depending on the Discrete Handling settings in the Request Management Configuration.
- Select the checkbox to edit/update form and to perform an action on the pending items.

Status Values

On a viewed request, the name of the current status displays in the Status column or Status drop-down list. To display other available statuses, click the list. Optionally, change the request to a new status, such as Approved, Cancelled, or Fulfilled. To change the request, click its status.

The following table shows the number, name, and meaning of the default statuses of Service Accounting Component and CA Service Catalog. The number of a status does not appear in the drop-down list. However, the number is used in the requeststatus.xml file. This file contains specifications for all default statuses and any customized statuses. Optionally, create new statuses and customize the statuses in the drop-down list by editing the requeststatus.xml file.

Some text in this table refers to the two request fulfillment models: simple and complex. In certain situations, the request flow differs for each model. These differences, when applicable, are noted in the following table.

Number	Name	Meaning
2	Completed	Catalog system-assigned status. When a request finishes with approval and fulfillment cycles, it reaches a Completed status. Once completed, the underlying request subscription reaches an active subscription status.

3	Pending Cancellation	This status applies when: • CA Service Catalog is installed. • Service is not subscribed. • The default Cancellation status under Accounting, Configuration is set to Pending Cancellation. On cancellation, a request first attains the Pending Cancellation status. Then, after an invoice is run, it attains Cancelled status. If a request attains the complete state and is cancelled, then the request stays in pending cancellation state until the invoice is run.
4	Cancelled	A request attains this status on cancellation. A request does not attain Cancelled status when the Service Accounting Component is installed and the default Cancellation status is Pending Cancellation. The default Cancellation status is set under Accounting, Configuration.
6	Pending Resource Assignment	This status applies only when: • CA Business Service Insight is integrated. • A request that includes a service option of type application or agreement reaches Fulfilled status. Such a service option requires provisioning; therefore, the request is moved to Pending Resource Assignment status.
7	Resource Assigned	This status applies only when a request is at Pending Resource Assignment status. Once the service option in such a request is provisioned, the request moves to Resource Assigned status.
100	Not Submitted	This status is user-initiated. When creating a request, optionally, click Save as Request to save the request in Not Submitted status. This option is an alternative to submitting the request.
101	Not Submitted - Cart	This status is Catalog system-assigned. This status is attained when a request item is added to the cart. **Note:** This status appears only in reports, not on the GUI.
103	Not Submitted - Rejected	This status is system-assigned. After a service in a request is rejected, the request eventually reaches a Not Submitted - Rejected status. The request is returned to the open queue of the requestor for review, modification, and resubmission.

104	Not Submitted - Approved	This status is catalog system-assigned. When the same request contains both approved and rejected services, the approved services eventually reach a Not Submitted - Approved status. This status helps the user to differentiate between a previously rejected request items and approved request items. In this case, the entire request is in Not Submitted - Rejected status.
200	Submitted	This status is user-initiated. A request reaches this status when a requestor submits the request. For each service in the request, the configured approval process type drives three scenarios: • For the Workflow driven approval process, a rule action is triggered to activate the approval process. This process determines the approver for the service and changes the status to Pending Approval. • For the System Approval process, the Catalog system determines the next action and status. This determination is based on: – The approval level that is assigned to the service. – The management hierarchy of the user for whom the service is requested. • For the No approval process, if the service is configured for fulfillment, then the status eventually changes to Pending Fulfillment. If the service is not configured for fulfillment, then the status eventually changes to Fulfilled. When all services in the request reach the Fulfilled status, the request reaches the Completed status.
201	Re-submitted	This status is User-initiated. When a rejected request is submitted, it reaches a Re-submitted status. The eventual flow is similar to Submitted status.

400	Pending Approval	This status is Catalog system-assigned. A request reaches this status when one or more services in a request require approval. For each service in the request, the configured approval process type drives two scenarios: • For the Workflow driven approval process, a rule action is triggered to activate the approval process. This process determines the approver for the service and changes the status to Pending Approval. By default, all Catalog Content services use the Workflow driven approval process. • For the System Approval process, the Catalog system determines the next action and status. This determination is based on: • • The approval level that is assigned to the service. • The management hierarchy of the user for whom the service is requested.
600	Rejected	This status is User-assigned. If a service in the request is configured for approval, a pending action is assigned to the approver. The approver can optionally either approve or reject each service in the request. The entire request is rejected when a service is rejected. Then, the request is returned to the queue of the requestor for review, modification, and resubmission.
800	Approved	This status is User-assigned. If a service in the request is configured for approval, a pending action is assigned to the approver. The approver can either approve or reject each service in the request. When you approve a service, the status can either: • Attain Approval Done status • The status can get reassigned to Pending Approval status, when further approval is required. When all services in the request have received all required approvals, the request reaches the Approval Done status.
801	Approval Not Needed	This status is Catalog system-assigned. If a service in the request is configured for no approval, then the service reaches this status after the request is submitted. The service eventually reaches Approval Done status.

999	Approval Done	This status is Catalog system-assigned or workflow-assigned. When all required approvers approve a service in a request, that service attains Approval Done status. When all services in a request attain Approval Done status, the entire request is approved. The request eventually reaches Approval Done status. The next status after Approval Done is either Pending Fulfillment or Fulfilled. • Using Pending Fulfillment allows for more fulfillment business processes to be followed by using CA Process Automation processes. • If the request does not require fulfillment, then it reaches a state of Fulfilled.
1000	Pending Fulfillment	This status is Catalog system-assigned or workflow-assigned. In a request containing multiple services, the services can belong to any or all of the following categories: • If a service in the request is configured for fulfillment, then a fulfillment process is triggered. This process determines the fulfiller for the service option and changes its status to Pending Fulfillment. Using Pending Fulfillment allows for more fulfillment business processes to be followed. If the request does not require fulfillment, then it reaches a state of Fulfilled. Once the service is configured for Pending Fulfillment, one of the following actions occurs: – If a service option in the request is configured for simple fulfillment, then a fulfillment process is triggered. The status is set to Pending Fulfillment. – If a service in the request is configured for complex fulfillment, then a fulfillment process is triggered. The status is set to Check Availability.

1001	Check Availability	This status is Workflow-assigned. This status applies when the Catalog system is configured for complex fulfillment. If a service option in the request is configured for fulfillment and it belongs to a hardware or software category, then an associated rule action triggers a fulfillment process definition. The process definition typically assigns a pending action to an IT Services User (that is, a fulfiller). The fulfiller can: • Choose between Filled from Inventory and Not Filled from Inventory. • Bypass fulfillment completely by selecting Fulfillment Cancelled or Fulfilled.
1002	Filled From Inventory	This status is User-assigned. This status applies when the Catalog system is configured for complex fulfillment. If an IT Services user is assigned a service option that is in Check Availability status, then the fulfiller can choose between Filled from Inventory and Not Filled from Inventory. This status is set in one of the following ways: • If CA APM is integrated with CA Service Management, then this status is set when a CA APM user assigns an asset. • If CA APM is not integrated with CA Service Catalog, then the IT Services user: – Verifies that the requested items are available. – Sets manually the status to Filled from Inventory.

1003	Not Filled From Inventory	This status is User-assigned. This status applies when the Catalog system is configured for complex fulfillment. If an IT Services user is assigned a service option that is in Check Availability status, then the fulfiller can choose between Filled from Inventory and Not Filled from Inventory. This status is set in one of the following ways: • If CA APM is integrated with CA Service Catalog, then this status is set when an asset is not available in CA APM. • If CA APM is not integrated with CA Service Catalog, then the IT Services user sets manually the status to Not Filled from Inventory when one or more requested items are not available. When the status of the service option is set to Not Filled From Inventory, then the status eventually reaches Pending Procurement. The requested hardware or software must then be ordered.
1004	Ordered	This status is User-assigned. This service option status applies when the Catalog system is configured for complex fulfillment. Once the status is set to Pending Procurement, then the IT Services user sets its status to the actual status of the ordered item, such as Ordered.
1005	Backordered	This status is User-assigned. This service option status applies when the Catalog system is configured for complex fulfillment. Once the status is set to Pending Procurement, then the IT Services user sets its status to the actual status of the ordered item, such as Backordered.
1006	Shipped	This status is User-assigned. This service option status applies when the Catalog system is configured for complex fulfillment. Once the status is set to Pending Procurement, then the IT Services user sets its status to the actual status of the ordered item, such as Shipped.

1007	Received	This status is User-assigned. This service option status applies when the Catalog system is configured for complex fulfillment. Once the status is set to Pending Procurement, then the IT Services user sets its status to the actual status of the ordered item, such as Received. If the status of the service option is set to Received, then it eventually reaches a status of Check Availability for re-assignment. This re-assignment is performed by an IT Services user or by a CA APM user (if CA APM is integrated with CA Service Catalog).
1008	Order Cancelled	This status is User-assigned. This service option status applies when the Catalog system is configured for complex fulfillment. Once the status is set to Pending Procurement, then the IT Services user sets its status to the actual status of the ordered item, such as Order Cancelled. If the status of the service option is set to Order Cancelled, then it eventually reaches a status of Check Availability for re-assignment. This re-assignment is performed by an IT Services or user or by a CA APM user (if CA APM is integrated with CA Service Catalog).
1012	Pending Procurement	This status is User-assigned. This status applies when the Catalog system is configured for complex fulfillment. If the status of the service option is set to Not Filled From Inventory, then it eventually reaches a status of Pending Procurement. The requested hardware or software must then be ordered.
1013	CA SDM Change Order Opened	This status is Workflow-assigned. This status applies when the Catalog system is configured for complex fulfillment and a rule action for creating a CA Service Desk Manager change order is enabled. A status change occurs when the following actions occur: • An IT Services user is assigned a service option in Check Availability status. • The IT Services user then sets the status of the service option to Filled from Inventory. • The CA Service Desk Manager is integrated with CA Service Catalog The status is then set to CA SDM Change Order Opened and a Change Order is created in the CA Service Desk Manager.

1015	Notified IT Services	This status is Workflow-assigned. This status applies when the Catalog system is configured for complex fulfillment. A status change occurs when the following conditions apply: • An IT Services user is assigned a service option in Check Availability status. • The IT Services user sets the status of the service option to Filled from Inventory. • The CA Service Desk Manager is not integrated with CA Service Catalog. A process definition is triggered which assigns a pending action to an IT Services User and sets the status to Notified IT Services.
1016	Being Staged	This status is User-assigned. This service option status applies when the Catalog system is configured for complex fulfillment. Once the status is set to Notified IT Services, then the IT Services user sets its status to the actual status of the requested service option, such as Being Staged.
1017	Staged	This status is User-assigned. This service option status applies when the Catalog system is configured for complex fulfillment. Once the status is set to Notified IT Services, then the IT Services user sets its status to the actual status of the requested service option, such as Staged.
1018	Being Configured	This status is User-assigned. This service option status applies when the Catalog system is configured for complex fulfillment. Once the status is set to Notified IT Services, then the IT Services user sets its status to the actual status of the requested service option, such as Being Configured.
1019	Configured	This status is User-assigned. This service option status applies when the Catalog system is configured for complex fulfillment. Once the status is set to Notified IT Services, then the IT Services user sets its status to the actual status of the requested service option, such as Configured.

1020	CA SDM Request Opened	This status is Workflow-assigned. This service option status applies when the Catalog system is configured for simple fulfillment and a rule action for creating a CA Service Desk Manager change order is enabled. A status change occurs when the following conditions apply: • An IT Services user is assigned a service option in Pending Fulfillment status. • The CA Service Desk Manager is integrated with CA Service Management. The status is then set to CA SDM Request Opened and a request is created in CA Service Desk Manager.
1999	Fulfillment Cancelled	This status is User-assigned or Catalog system-assigned. This status applies when one of the following actions is true: • The IT Services user sets the status to Fulfillment Cancelled when the service option does not require fulfillment. • The Catalog system sets the status to Fulfillment Cancelled when a user cancels a request that has been approved but has not been fulfilled. If all services are set to Fulfillment Cancelled, then the request eventually reaches Cancelled status. If the request contains services with both Fulfilled and Fulfillment Cancelled status, then the request eventually reaches Completed status.
2000	Fulfilled	This status is User-assigned. This status applies when one of the following actions is true: • The IT Services user sets the status of the service option to Fulfilled. • The CA Service Desk Manager is integrated and the process definition in CA Service Desk Manager sets the status to Fulfilled. This occurs when an associated change order is closed in the CA Service Desk Manager. When all services are set to Fulfilled, the request eventually reaches Completed status. If the request contains services with both Fulfilled and Fulfillment Cancelled status, then the request eventually reaches Completed status.

3000	Hold	This status is User-assigned. This status applies when either an administrator or the Catalog system sets the status of the service option to Held. When an item is held, no status changes can occur to it until you change its status to Resume. If a service option is held, then the monitoring of time is stopped for any request SLAs attached to the service option. This stoppage prevents related SLA warnings and SLA violations from being issued inordinately.
4000	Resume	This status is User-assigned. This status applies when an administrator sets the status of a previously held service option to Resume. When resumed, service options that were formerly held move to a temporary Resume status before automatically moving back to their previous status before being put on Hold. When you resume a previously held service or service option, the monitoring of time is resumed for any request SLAs attached to the service or service option.
5001	Request Queued	This status is assigned to the requests that are held at a specific point in the process queue. This status indicates that the request will be attempted again for further processing.

Subscriptions and Requests

You use the same service catalog for both subscriptions and requests. CA Service Catalog enables catalog users to request services. Service Accounting Component enables accounts in the catalog to have subscriptions.

Requests have a Requested For value. When you add services to your cart, the Requested For value is set to your user ID by default. As an administrator, you can change the Requested For value to be a user or account within your business unit scope. When you set the Requested For value to a user, you are also setting the value to the *user account* associated with the user.

Subscriptions apply to an account. You can optionally associate an account with a user. However, an account is not required to be associated with a user. As an administrator, you can manage subscriptions for an account within your business unit scope. Users who are not administrators can view subscriptions for their user account but cannot manage these subscriptions.

Request Service Levels Reports

Three phases of the request life cycle are monitored: Approval, Fulfillment, and Completion. Each phase represents the time that is taken for a requested service option to move from one status to another.

- Approval - The duration of moving from **Submitted** to **Approval Done**.
- Fulfillment - The duration of moving from **Approval Done** to **Completed**.
- Completion - The duration of moving from **Submitted** to **Completed**.

For each service option, service providers can establish SLA warning and violation thresholds for each phase.

Example

This example uses the Standard Desktop service option. The fulfillment phase warning time is five days and the violation time is seven days. Therefore, the service provider expects that a standard desktop be fulfilled (configured, delivered, and set up) within five days. A violation occurs if the time frame exceeds seven days.

The violation setting for a service option displays in the catalog as the *estimated time to fulfill*. In this example, the estimated time to fulfill the Standard Desktop service is seven days.

> **NOTE**
> (optional) Set up reports to monitor other request life cycles phases by altering the STATUS_RANGES value that is used in a report data object.

Several report data views and underlying data objects enable you to monitor request service levels.

- **Request Fulfillment Data View**
  This report accepts a date range and business unit. Each row represents one request. The columns represent the duration in days, hours and minutes to approve, fulfill, and complete the requests. The date range filters for requests whose initial creation is within the range specified. The business unit value filters for requests for a user or account from a business unit.
  Display the Request Item Fulfillment subreport for the service options that are contained in the request by clicking the Request ID field.
  - Yellow = Time durations that exceed the warning duration for the service option.
  - Red = Time durations that exceed the violation duration for the service option.
- **Request Item Fulfillment Averages Data View**
  This report accepts a date range and business unit. Each row represents one service option. The columns represent the average duration in days, hours, and minutes to approve, fulfill, and complete the requests. The date range filters for requests whose initial creation date is within the range. The business unit value filters for requests for a user or account from a business unit.
  - Yellow = Average time duration exceeds the warning duration for the service option.
  - Red = Average time duration exceeds the violation duration for the service option.

Click the Service Option field to display the Request Item Fulfillment report. This sub-report displays requests containing the service option.

- Yellow = Time durations that exceed the warning duration for the service option.
- Red = Time durations that exceed the violation duration for the service option.

Request Management from an Administrator Perspective

Requests have a life cycle reflected in their status. The status of a request falls into one of the following several phases:

- Not Submitted
- Submitted
- Approval
- Fulfillment
- Completed

For the request and its services and service options to move through the request life cycle, its status must change.

The requesting user controls the **Not Submitted** phase. A request is in the Not Submitted phase when one of the following conditions is true:

- The request is in the user's cart.
- The user has saved the cart as an unsubmitted request.
- The submitted request has been rejected.

NOTE

The user must submit the request or cart before the request can exit the **Not Submitted** phase.

The **Submitted** phase is brief and controlled by the system. Each service option in the request has a specified approval process. The submitted phase is used only until the approval process begins.

An approving user or the system controls the **Approval** phase. This control depends on the approval process for each service in the request. All services in the request must be approved (or not require approval) before the request enters the Fulfillment phase. If any services are rejected, the entire request is returned to the Not Submitted phase for the requesting user to manage.

A fulfilling user or the system controls the **Fulfillment** phase. This control depends on the fulfillment process for each service option in the request. Set the status of all service options in the request to either Fulfilled or Fulfillment Cancelled before the request enters the Completed phase.

An administrator or the system controls the **Completed** phase. This control depends on the type of service option elements requested.

- If CA Business Service Insight is integrated and the request includes an Agreement service option element that depends on resource metering, then:
- • The status of the service option is changed to Pending Resource Assignment.
  • When a resource is assigned, the status changes to Completed.
- For other types of service options elements, the status is set to Completed. After the status is set to Completed and if Service Accounting Component is installed, invoices then include the service option.

Enable Approval and Rejection of Multiple Items Simultaneously

Administrators enable *multi-item approval* so that request managers can simultaneously approve and reject multiple items in a request. Examples include all services in a request or all service options in a service. Multi-item approval lets request managers process requests quickly and efficiently, especially when they are already familiar with the requests:

- Request managers access their requests pending action, select the services that they want to act on. Then, they click Approve or Reject. Multi-item approval provides this ability *regardless* of whether discrete approval is implemented. Without multi-item approval, request managers must approve or reject every service *individually* by updating its status in the Status drop-down list.
- If discrete approval is implemented, request managers approve or reject multiple *service options* in a service simultaneously, as follows: Request managers access their requests pending action and open the services. In each service, they select the service options that they want to act on, and click Approve or Reject.
- If discrete approval is implemented *without* multi-item approval, request managers can approve or reject each service option *individually* by updating its status in the Status drop-down list.

The benefits of multi-item approval are more pronounced as the number of services or service options in a request increases. Multi-item approval is especially helpful for mobile users, who have limited space and typically have limited time.

Follow these steps:

1. Click Catalog, Configuration, Request Management, and verify that the following option is enabled: Allow Multi Service/Service Option Approval.
2. Verify whether you have implemented discrete approval.
3. Inform request managers of the new approval option for your implementation:

  - Multi-item approval only
  - Multi-item approval and discrete approval
4. Verify that multi-item approval functions as you intended.

Requests with Multiple Approval or Rejection Statuses

The default statuses of the request life cycle include these statuses: Approved and Rejected. When request managers approve or reject a service option or service, the status of the item changes accordingly. The changed status is displayed to request managers before they confirm their approval or rejection of services or service options. If an approval or rejection changes change the default status, the Catalog system highlights the service or option in green.

Catalog administrators can optionally create custom approval and rejection statuses, such as Approved - Manager, Approved - Finance, Rejected - No Budget. To define custom statuses, edit the requeststatus.xml file. For more information, see the Modify the Request Status List section.

If one or more custom statuses exist for this service or service option, the Catalog system performs the following actions:

- Displays an informational message that multiple statuses exist.
- Selects the first custom status that is defined in requeststatus.xml file.
- Highlights the service or option in orange.

For all status changes (custom or default), before confirming the approval or rejection, request managers can choose one of these options:

- Accept the status changes
- Update one or more statuses, for example, from a custom status to a default status

Services or options that are not selected are not highlighted and do not change status. They remain as pending items in the request until they are either approved or rejected.

Implement Discrete Request Life Cycle

This article contains the following topics:

The predefined CA Process Automation processes supplied for use with CA Service Catalog require *no* additional configuration to support the discrete request life cycle. However, verify that they work efficiently with the discrete request life cycle after you implement it.

Minimize impact on requests in progress by following the best option for introducing the use of discrete life cycle.

- Instruct the users to stop creating requests in your organization temporarily. When most or all of the requests in progress are completed, begin using a discrete life cycle by setting the related configuration parameters.
- In-progress requests may become stuck as a result of the configuration changes. If the requests become stuck, retry or override (push-through) the alerts to complete the requests.
  Retries and overrides may be required when you use the configuration setting Allow Discrete Handling of Service Options After.
- Give special attention to canceled items, rejected items, and items pending fulfillment while others are still in pending approval, when one or both of the following are true:
  - You expect some requests to reach pending fulfillment after all the items are approved.
  - You expect to reject the entire request if one item is rejected.

  In these cases, take special care before changing the configuration to support the discrete request life cycle. Verify that such requests have completed before you make the configuration changes.
- Review the following important considerations:
  - Overall status of the request

The service option with the lowest status value determines the overall status of the request. To see the status of a request, click Home, Requests, and, if applicable, use the My Requests drop-down list to display requests. Open the request, and view the status.
When the request contains services and service options having different status values, the lowest status value appears with asterisk (*) next to it, indicating that at least one service or service option in the same request has a higher status. When determining the overall status of the request, the Catalog system ignores services and service options that are rejected, completed, deleted, cancelled, pending cancellation, or on hold.

- Billing
  If you are using Service Accounting Component, the invoices you generate do include completed services and service options, even if other services and service options in the same request are not completed. The uncompleted services and service options are *not* included in the same billing run as the completed ones.
- Request SLAs
  This point is relevant only if you use request SLAs. The monitoring and recording of request SLA data is not affected when you use a discrete request life cycle. The Catalog system monitors and records the status changes of service options that have request SLAs the same way, regardless of your discrete request life cycle configuration settings.
- CA Business Service Insight-related metrics
  This point is relevant only if you are integrating CA Service Catalog with CA Business Service Insight. Your discrete request life cycle configuration settings do not affect the monitoring and recording of any CA Business Service Insight contract-related metrics that are associated with service options in the Catalog system.

Understand the Discrete Settings

The following table shows whether the request manager acts on a service, service option, or either one, based on the settings for Allow Discrete Handling of Service Options After (ADHSOA).

The table applies to all settings of Allow Discrete Request Life Cycle After.

Request Manager Action	ADHSOA=Submitted	ADHSOA=Pending Fulfillment	ADHSOA=Completed
Approve or Reject	service option	service	service
Hold or Resume	either	either	either
Transfer or Delegate	service option	**A**-service **B**-service option	service
Override	service option	**A**-service **B**-service option	service
Take or Return	service option	**A**-service **B**-service option	service
Pus-Through	service option	**A**-service **B**-service option	service
Cancel	service	service	service
Fulfillment Cancelled	service option	service option	service
Fulfill	service option	service option	service
CA Service Desk Manager Change Order	service option	service option	service

A-service, from Submitted status through Approval Done -service option, from Pending Fulfillment status through Fulfilled status.status.**B**

Set the Configuration Parameters

To enable the discrete handling of requests pending action, set the related configuration parameters. You can use these configuration parameters individually. However, they are most effective when you use them together to achieve the desired request processing.

Follow these steps:

1. Click Catalog, Configuration, Options, Request Management Configuration.
2. Set the following parameters:

- – **Allow Discrete Handling of Service Options After**
  Specifies the status at which request managers can discretely (individually) approve, reject, or fulfill each service option in every service in the request. Use this parameter to specify that pending actions are assigned at the service option level, rather than the service level.
  The setting that you specify applies to the request *from* the starting status that you specify *through* the remainder of the request life cycle.
  Valid values: Submitted, Pending Fulfillment, or Completed
  Default: Pending Fulfillment
- – **Allow Discrete Request Life Cycle After**
  Specifies the status at which individual service options in a service and individual services in a request advance to further statuses in the request life cycle independently. When the request reaches the specified status, the service options that you approve or fulfill can complete the remainder of the request life cycle. These service options complete the remainder of the request life cycle even if other service options in the same service are not acted on. When the request reaches the status that you specify, the services that you approve or fulfill can complete the remainder request life cycle, even if other services in the same request are rejected or not acted on.
  Valid values: Submitted, Pending Fulfillment, or Completed
  Default: Completed
- – **Allow Discrete Handling for Reject**
  This parameter specifies the effect of the rejection of a single service or service option upon other services and service options in the same request:
  - **Yes**: When rejecting a service or service option, the remaining services or service options can advance if they are approved. Other services in the same request can advance in the request life cycle if they are approved. Similarly, other service options in the same service can advance in the request life cycle if they are approved.
  - **No**: When you reject a service or service option, the entire request is rejected. All services in the same request are rejected and cannot advance in the request life cycle, even if they were previously approved. All service options in the same service are rejected and cannot advance in the request life cycle, even if they were previously approved.
    Default: No

Discrete Request Life Cycle Parameters

This article contains the following topics:

Administrators can implement a discrete request life cycle to enable services and service options to advance to higher request statuses *without waiting* for all other services and service options to reach the same statuses. Consequently, services and service options can reach Fulfilled and Completed statuses, and can be invoiced earlier than without a discrete life cycle, even if a request manager rejects one or more services or service options in the same request.

An administrator can optionally specify discrete settings for the request life cycle for individual business units, as follows:

- Depending on the configuration you choose, you can specify when the request managers (approvers and fulfillers) can discretely (individually) approve, reject, or fulfill each service option in every service in the request. The setting applies to the request *from* the starting status that you specify *through* the remainder of the request life cycle.
- The status of individual service options in a service and individual services in a request advance to further statuses in the request life cycle independently. Here, *independently* means *without* requiring or waiting for any other service

options in the same service or any other services in the same request to advance to any further statuses. The setting applies to the request *from* the starting status that you specify *through* the remainder of the request life cycle.

- The effect of the rejection of a single service or service option upon other services and service options in the same request:
  - If this setting is Yes, when you reject a service or service option, the remaining services or service options can advance if they are approved.
  - If this setting is No, when you reject a service or service option, the entire request is rejected. Even other services or service options that were previously approved change to Rejected status and can not advance in the request life cycle.

Sample Settings and Their Meanings

For all examples, if the request manager rejects one or more services or services options in a request that includes other services and service options, the setting of Allow Discreet Handling for Reject configuration determines effect of the rejection on the other service options and services in the request.

Examples for Allow Discrete Handling of Service Options After

The *Allow Discrete Handling of Service Options After* parameter specifies the status at which request managers (approvers and fulfillers) can handle requests pending action (approval, rejection, or fulfillment) discretely. When the request reaches the status you specify, request managers can discretely (individually) approve, reject, or fulfill each service option in every service in the request.

Example 1--Submitted

In this example, you want *both* discrete approval and discrete fulfillment. Thus, you want to approve or reject service options individually, rather than approve or reject *all* service options in the service in one action. Similarly, you want to fulfill service options individually, rather than fulfill *all* service options in the service in one action.

To achieve this goal, specify Submitted as the value for this parameter. When the user submits the request, approvers can approve or reject each service option in a service individually. Moreover, fulfillers can fulfill each approved service option individually.

To ensure that the request managers receive notification emails when they address pending actions:

- Set the *Allow Discrete Handling of Service Options After* parameter to Submitted.
- Keep the setting of Yes (predefined) for the parameter named *Notify users when they complete their own pending actions*.

For example, if a service has five service options and the request manager approves each option, the request manager receives five confirmation emails.

Example 2-- Pending Fulfillment

In this example, you do *not* want discrete approval but want discrete fulfillment. Thus, you do *not* want to approve or reject service options individually but rather want to approve or reject all service options in the service in one action. However, once the entire request is approved, you want to *fulfill* service options individually.

To achieve this goal, specify Pending Fulfillment as the value for this parameter. So, as soon as the entire request is approved, fulfillers can fulfill each approved service option in every service individually.

Example 3--Completed

In this example, you do *not* want discrete approval or discrete fulfillment. You do *not* want to approve, reject, or fulfill service options individually at any point during the request life cycle. Instead, you always want to approve, reject, or fulfill all services in the request as a single unit. To achieve this goal, specify Completed as the value for this parameter.

Essentially, this setting "turns off" discrete handling of approval and fulfillment activities. So, request managers must approve, reject, and fulfill all services in the request at once.

Examples for Allow Discrete Request Life Cycle After

The *Allow Request Life Cycle to Continue After* parameter specifies the status at which individual service options in a service and individual services in a request advance to further statuses in the request life cycle independently. Here, *independently* means *without* requiring or waiting for any other service options in the same service or any other services in the same request to advance to any further statuses.

Example 1--Submitted

For example, suppose that you want each service option in a service to advance to the remaining statuses in the request life cycle independently after being approved rather than requiring or waiting for any other service options in the same service to be approved. To achieve this goal, you specify Submitted as the value for this parameter.

As soon as the user submits a request, each service option in all services can advance independently to the remaining statuses in the request life cycle as soon as it (the service option) is approved, *without* waiting for any others to be approved or rejected.

Similarly, as soon as the user submits a request, each service in all requests can advance independently to the remaining statuses in the request life cycle as soon as it (each service) is approved *without* waiting for any others to be approved.

If an approver does not act on one or more service options, the approved service options in the same service can advance to the remaining statuses in the request life cycle. If an approver does not act on one or more services, the approved services in the same request can advance to the remaining statuses in the request life cycle.

Thus, with Submitted as the value for this parameter, the following results occur in a service that has two options: one approved and one no action, as follows:

Service Option	Approver's Action	Resulting Status
Option-1	Approve	Pending Fulfillment
Options-2	No action	Pending Approval

After all service options in the same service are approved, the service reaches Pending Fulfillment status.

After all services in the same request are approved, the request reaches Pending Fulfillment status.

Example 2--Pending Fulfillment

For example, suppose you want each service option in a service to be able to advance to the remaining statuses in the request life cycle independently after all services in the request are approved. To achieve this goal, you specify Pending Fulfillment as the value for this parameter.

So, as soon as the service is approved, it reaches Pending Fulfillment. Each service option in the service can advance independently to the remaining statuses in the request life cycle independently. Similarly, as soon as the request is approved, it reaches Pending Fulfillment, and each service in the request can advance independently to the remaining statuses in the request life cycle independently.

A setting of Pending Fulfillment as the value for this parameter differs from a setting of Submitted in these ways:

- Approved service options *cannot* advance to the remaining statuses in the request life cycle *until* all other service options in the same service are approved.
- Approved services *cannot* advance to the remaining statuses in the request life cycle *until* all other services in the same request are approved.

Thus, with Pending Fulfillment as the value for this parameter, the following results occur in a service having two options. The two are approved sequentially, as follows:

Service Option	Approver's Action	Resulting Status
Option-1	Approve	Approval Done

Option-2	Approve	Pending Fulfillment
Option-1	Fulfilled	Completed
Option-2	Fulfilled	Completed

After all service options in the same service are approved, the service reaches Pending Fulfillment status.

After all services in the same request are approved, the request reaches Pending Fulfillment status.

Example 3--Completed

In this example, you do *not* want individual service options in a service or individual services in a request to be able to change statuses independently of each other at any point in the request life cycle.

Instead, you always want complete services and complete requests to change statuses *only* as a single unit, not independently. To achieve this goal, specify Completed the value for this parameter.

Essentially, this setting "turns off" any possibility of service options in a service or services in a request changing statuses independently of each other at any point in the request life cycle.

Common Settings

The following points apply to all combinations of the *Allow Discrete Handling of Service Options After* and *Allow Discrete Request Life Cycle After:*

- If *Allow Discrete Handling for Reject* is No and you reject any service or service option, the entire request is rejected. The request life cycle stops immediately. After the rejection, the following status changes occur:
  - The status for the entire request changes to Not Submitted - Rejected.
  - Previously approved service options change to Not Submitted - Approved status.
  - All other service options move to Not Submitted - Rejected status.

  The requestor can optionally modify and resubmit the request.
  However, if this setting is Yes and you reject any service or service option, the remaining services or service options continue their request life cycle according to settings for the *Allow Discrete Handling of Service Options After* and *Allow Discrete Request Life Cycle After* parameters, as specified in the tables.
- If the request manager rejects one or more services or services options in a request that includes other services and service options, the setting of *Allow Discreet Handling for Reject* determines effect on the other service options and services in the request, regardless of whether the other services and service options were previously approved.
- The Perform Action icon or button appears on the GUI in all cases, except when *Allow Discrete Handling of Service Options After* is set to Pending Fulfillment and *Allow Discrete Request Life Cycle After* is set to Completed.
  When you click the Perform Action, a new page opens, enabling you to perform all applicable actions for each service or service option in the request, such as approve, reject, fulfill, transfer, delegate, take, return.

Use Policies to Manage Requests

This article contains the following topics:

CA Service Catalog administrators can create policies. Policies define conditions for automatically designating specific users as the assignees for a task during the request life cycle. The most common use is to define conditions for automatically assigning specific approvers for services and service options in a request pending action.

Policies provide the flexibility to assign tasks (such as approval or rejection of requests pending action) to users *other than* the managers of the requestor. However, you can optionally include these managers in your list of assignees.

The policy that you create consists of the following main components:

- The condition under which the policy applies
- The assignment to occur if the condition is met.

If the condition is met, the user or users you specify are assigned to perform a pending action.
You can specify the list of assignees according to any of several attributes, including but not limited to:
  – User name (first name and last name)
  – Membership in a user group
  – Management hierarchy, meaning the manager of the user, the manager of that manager
- An active-or-inactive setting
- Description and priority fields

You create *global* policies and global actions for general use with any service. In contrast, you create *attached* policies and actions for use with a specific service option only. You can create an attached policy or action *only* from the Policies & Actions tab of that service option.

Comparison to Events, Rules, and Actions

The conditions and actions that you specify in policies are similar to the rules and actions that CA Service Catalog supplies in events. You can view and edit these events, rules, and actions by selecting Administration, Tools.

In both the cases, you specify conditions that become part of the request life cycle workflow in CA Service Catalog. In both cases, when a request meets the specified conditions, CA Service Catalog assigns the users that are specified in the policy. For example, suppose you create a policy that applies only to requests with cost greater than $100. When a user submits a request that meets this condition, the assignees that are specified in the policy are assigned a pending action to approve, reject, or fulfill the request.

The major difference is that events, rules, and actions are system wide. But, you can optionally specify policies to be either system wide or specific to a business unit.

Step 1 - Review the Criteria to Apply Policies to Individual Requests

CA Service Catalog matches policies to individual requests according to priority and business unit (tenant) level in the following order.

1. High-priority policies at the lowest level business unit (farthest from the root)
2. High-priority policies at the next lowest level business unit
3. High-priority policies at the remaining business unit levels, from the lowest though the highest (the root)
4. Low-priority policies at the lowest level business unit
5. Low-priority policies at the next lowest level business unit
6. Low-priority policies at the remaining business unit levels, from the lowest though the highest
7. When CA Service Catalog matches a policy to a request, it considers other policies *only* if they are at the same priority and business unit level.
8. If CA Service Catalog finds two or more policies at the same priority and business unit level, it assigns requests pending action to *all* applicable assignees from *all* applicable policies.

The Catalog system requires *all* approvals from these assignees to advance the request pending action to the next status in the request life cycle. To help avoid situations in which multiple policies and approvers apply to a single request pending action, create and maintain conditions to use specific criteria.

Step 2 - Create Folders for Policies

You can create and maintain folders (as needed) to store your policies. Organize policies in folders according to the business units and child business units to which they apply, using intuitive names. Similarly, suppose that the policies apply to all business units. In that case, you can organize them according to categories under a folder that is named For All Business Units.

Follow these steps:

1. Click Catalog, Policies.

2. Add a folder, as follows:
   a. Select the folder to which you want to add the new folder and click Add, Folder.
      For example, to add a folder to the root folder (the Policies folder), select it and click Add Folder.
   b. Enter the name of the new folder and click OK.
      The folder is created and appears under the parent folder.

You can create, rename, move, copy, and delete folders.

> **NOTE**
> The names of folders are case-aware, but not case-sensitive. Thus, when you rename a folder, change more than the case. For example, you cannot rename an existing folder from spg policies to Spg Policies or SPG POLICIES.

Step 3 - Enable Rules and Actions for Policies

Policies require an engine. The rules and actions that you enable depend on the engine you use. On the Details Tab for any service that wants to associate with a policy, you specify the approval process as *one* of the following options:

- Enable policy rules and actions for CA Process Automation.
  Perform this task if any services in the catalog specify the Workflow driven approval process to apply policies. If your implementation uses CA Process Automation for approving and fulfilling requests, you can use CA Process Automation as the engine for policies.
- Enable policy rules and actions for the Catalog Policy Engine.
  Perform this task if any services in the catalog specify the Policy driven approval process to apply policies. The Catalog Policy Engine is part of the Policy Builder, and is available to all CA Service Catalog implementations. The Catalog Policy Engine does not require CA Process Automation. This approach is typically more efficient and therefore is typically preferred.

If necessary, copy and modify a predefined rule or action to specify a custom rule or action. If both options apply, then you can use either option as the policy engine for any service.

Enable Policy Rules and Actions for CA Process Automation

If any services in your catalog specify the Workflow driven approval process, enable the required rules and actions. If necessary, also enable the policy rules and actions for the Catalog Policy Engine.

Follow these steps:

1. Click Administration, Events-Rules-Actions.
2. Click the Event Type named *Request/Subscription Item Change*.
3. Select the rule *When Status is Submitted and Approval Process is driven by Workflow*.
   The Rule Details page for this rule appears. This rule is enabled by default. If it has been disabled, click the Enable button. This button appears on the Rules bar.
4. Select the action that is named Launch Policy Driven Approval SRF and click the Enable button. The Enable button appears on the Actions bar.
5. Click Done.
   The Event Details page appears.
6. To configure CA Service Catalog so that you can use policies to fulfill requests, using *simple* fulfillment, perform the following actions:
   a. Open the *When Status is Pending Fulfillment* rule. Do not exit the Event Type named Request/Subscription Item Change.
   b. Create a CA Process Automation action and select the Policy_Approval Start Request Form.
   c. Complete the parameters that appear.
   d. Specify the following parameters as indicated:
   e. **RequestStatusUponAssignment =1000**

Specifies that the requested item is set to this status (Pending Fulfillment) when the request pending action is assigned.

f. **RequestStatusUponSuccess =2000**
Specifies that the requested item is set to this status (Fulfilled) when the request pending action is completed.

The Policy_Approval Start Request Form supports *simple* fulfillment but does *not* support multilevel fulfillment and complex fulfillment. However, you can optionally create a CA Process Automation process that *does* support multilevel fulfillment and complex fulfillment. To do so, use the assignPolicyBasedPendingActions method in the web service named RequestService.

The rules and actions are enabled.

Enable Policy Rules and Actions for Catalog Policy Engine

If any services in your catalog specify the Policy driven approval process, enable the required rules and actions.

Follow these steps:

1. Click Administration, Events-Rules-Actions.
2. Click the Event Type named Request/Subscription Item Change.
3. Locate the rule *When Status is Submitted and Approval Process is driven by Policy*.
This rule is enabled by default. If the rule is disabled, select its check box. Click Enable, and click OK.
4. Click the rule name.
5. Click the Edit icon for the action *Evaluates Policy Driven Approval Process*.
The Edit Action page appears. This action is disabled by default.
6. Select Enabled in the Status drop-down list, and click OK.
This action checks the policy setting on the Policies and Actions tab of each service option that triggers the action. According to that setting, the action evaluates global policies, attached policies, or both for the service option.

The rules and actions are enabled.

Step 4 - (Optional) Export and Import Policies

The IXUTIL command-line utility is an importing and exporting command-line utility that allows the preservation and migration of CA Service Catalog data between computers. You can use IXUTIL to export and import new and updated objects, including policies, from one computer to another. You can export and import policies in the following cases:

- When you upgrade or migrate CA Service Catalog.
- When you upgrade your computer.
- When the server fails.

NOTE

You can also use the Import Export Utility on the product UI to import and export policies.

Follow these steps:

1. Open the CA Service Catalog command prompt on any Catalog Component computer.
2. Enter one or more of the following commands:
   - To export all policies in all business units, enter the following command:
     ```
     ixutil export policy -f file
     ```
   - To export all policies in a business unit, enter the following command:
     ```
     ixutil export policy -f file - businessunit businessunit id
     ```
   - To export a single policy, use the following command:
     ```
     ixutil export policy -f file -policy name -businessunit id
     ```
     If the name contains spaces, enclose the name in double quotation marks.

```
ixutil export policy -policy "Mobile Device Policy--Tier 1" ...
```

- To export multiple policies, enclose the names in double quotation marks, and separate the names with commas.

```
ixutil export policy -policy "Mobile Device Policy--Tier 1,Mobile Device Policy--Tier 2,Mobile Device
 Policy--Tier 3" ...
```

- To export a folder containing one or more policies, enter the following command:

```
ixutil export policy -f file -folder name -businessunit id
```

  To export multiple folders, enclose the names in double quotation marks, and separate the names with commas. Use the previous command for multiple polices as a model.

- To import a previously exported file containing one or more policies, enter the following command:

```
ixutil import policy -f file -businessunit id
```

  The -businessunit *businessunit id* option is optional when you import the policy file. However, consider the important factors explained in the description of this option later in this topic.

- To import a previously exported file containing one or more policies in disabled status, enter the following command:

```
ixutil import policy -f file -businessunit businessunit id -disable
```

3. Log in to CA Service Catalog on the computer where you imported the policies. If you had imported policies into a specific business unit, log in to the respective business unit.
4. Click Catalog, Policies.
5. Expand the policy folders. Verify that the Catalog system imported the policy or policies as you want.
6. Close the CA Service Catalog command prompt.

Step 5 - Create a Policy

Administrators create policies to assign approvers automatically for services and service options in a request pending action. In this way, you use policies to manage requests.

Follow these steps:

1. Click Catalog, Policies.
2. Select the folder to which you want to add the new policy and click Add, Policy.
3. Enter an intuitive name for the policy and click OK.
4. Enter meaningful descriptions in the policy fields for the policy that is created under the parent folder.
5. Select either High or Low in the Priority drop-down list.

   NOTE

   When a user submits a request, CA Service Catalog checks for and the system applies any matching *high-priority* policies. Only if no high-priority policy applies to the request, the system checks for and applies any matching *low-priority* policies.
6. Select either Active (ready to use the policy) or Inactive in the Status drop-down list.
7. Save the policy.

Step 6 - Create Conditions for a Policy

Administrators create the condition as the major decision point of the policy. If the condition is met, the Catalog system automatically assigns the pending action to the assignees. You specify the condition using the attributes of CA Service Catalog elements such as users, requests, services, business units. In addition, you can use match functions for creating conditions that are based on service options and service option elements.

Create simple conditions that are based on known attributes, such as category, external_id, code, item type, cost, status. Specify the criteria that the value of the specified attribute must meet to assign the pending action.

The condition builder is the tool in the Condition field that helps you specify valid conditions, one segment at a time. When you initially move the cursor to the field, the condition builder prompts you with valid options for the first part of the condition. These options appear in a drop-down list under the Condition field. Select the option that you want from the list. As you complete each part of the condition, the condition builder continues to prompt you with valid options for the next part. This process continues until the condition is finished, typically with a closing parenthesis.

The condition must be a valid JavaScript expression. Typically specify one condition per policy, using the following format:

$(_.*group.attribute operator 'value'*)

- ***group***
  Specifies service, request, business unit, or any other group that is illustrated in the types of conditions.
- ***attribute***
  Specifies any attribute of that group
- ***operator***
  Specifies one of the following options:
  == (equal to)
  != (not equal to)
  > (greater than)
  < (less than)
  >= (greater than or equal to)
  <= (greater than or equal to)
- ***value***
  Specifies a literal value, typically the name of a business unit, request, service, service option group, or user.
  Enter numeric values without quotation marks, for example: $(_.request.bu.status==0).
  Enclose string values in single quotation mark; for example: $(_.request.bu.taxRegion =='South').
  If a string value includes a single or double quotation mark, precede that mark with a backslash (\) as the "escape" character. For example, if the service name is Demandes d'IP Statique, then specify the condition as follows:

  ```
  $(_.service.name=='Demandes d\'IP statique')
  ```

  As you construct an expression in the Condition Builder, the data type (string or numeric) of the attribute appears on the right, letting you know whether to use quotation marks around the value.

  ```
  $(_.service.name=='Procure Server')
  ```

  This condition means that when the name of the service is Procure Server, the users you specify are assigned as actors, typically approvers, or fulfillers.
  For example: $(_.request.estimatedCost >==1000)
  This condition assigns the pending action to the specified approvers or fulfillers when the estimated cost of the total request is greater than or equal to $1,000.

Step 7 - Specify Assignees for a Policy

Administrators specify assignees for a policy to configure the Catalog system to assign automatically the request pending action that triggered the policy. If the condition of the policy is met, the Catalog system automatically assigns the request to the specified assignees. The assignees typically process the request by approving, rejecting, or fulfilling it.

To specify assignees for a policy, follow this process:

1. Review Information about Assignees for Policies
2. Decide how to Specify Assignees

The Catalog system searches for and assigns approvers and fulfillers in the following order:

1. A relevant policy
2. The management hierarchy of the requestor

3. The Service Provider administrator (spadmin)

Review Information about Assignees for Policies

Assignees are typically request managers or other administrators who act on the requests pending action. The Catalog system assigns these requests, either to the assignees personally or to an administrative group to which they belong.

For example, suppose that the condition specifies that the pending action is triggered when both of the following conditions are true:

- The name of the requested service is Procure Laptop.
- The status is Submitted (by default, 200).

In this example, the assignee can be any of the following conditions, depending on the size and scope of the organization:

- The person or group that approves hardware requests.
- The manager of the requestor or another manager.
- The IT Finance Group.
- The Purchasing department.

You can assign only one approver or several approvers. Thus, you can assign either of the following levels:

- Only one *level* of approval, such as the manager only.
- Several levels of approval, such as the manager first, followed in succession by an IT approver and a financial officer.

When you are creating a policy or editing one, you *decide* how many approvers and how many levels of approval to specify. Examples follow:

- For requests to purchase third-party software, the following conditions apply:
  - First, both the immediate manager and the next-level manager must approve the request, in succession.
  - Next, a member of the Software Purchasing department must approve the request.
- If the Requested For user or the Requested By user is a delegate, include the manager of that user in the list of assignees.
- If the Requested For user belongs to a specific business unit, specify the approvers or fulfillers for that business unit in the list of assignees.

Decide How to Specify Assignees

Use the following use cases as guidelines:

- One known user is the assignee. Use either the Action Builder or an API plug-in.
- Multiple known users at specific levels comprise the list of assignees. Use either the Action Builder or an API plug-in.
- You query an external system for data to specify the assignees. Use the API plug-in.
- The identities and number of assignees vary, depending on the data in the request. Use the API plug-in.

Use the Action Builder to Specify Assignees

The Action Builder lets you manually find and select one or more assignees for one or more levels of approval.

Follow these steps:

1. Click Catalog, Policies, and open the policy.
2. Add the first assignee by clicking the plus sign (+) in the Action Builder (the box under the Condition field).
3. Complete the first Requirement field (the one next to the Level 1 column), as follows:
   In the drop-down list, select ANY or ALL.
   - **ANY**

- The first approval or fulfillment for a requested item changes the status of the item and closes any remaining pending actions for the item. (*Item* means a service or service option.)
- If all users reject or cancel an item, then the item is rejected or canceled.

– **ALL**
- Every assignee must approve or fulfill the requested item.
- If any assignees are groups, then one member of each assigned group must act on the item.
- If any assignee cancels or rejects the item, then all pending actions for the item are rejected or canceled.

4. Complete the first Assignees field (the one next to the first Requirement field) by clicking the magnifying lens:

   NOTE
   In the Find Approvers dialog, if you select User or Group, the name of the next field remains Name Filter. If you select Manager, the name of the next field changes to Manager Level.

   The assignees you selected are recorded. The Find Approvers dialog closes. You return to the Action Builder page.
5. Click OK under the first row of approvers, and click Save (above the Policy tree).
6. Specify one or more assignees, as follows:
   If you specify multiple assignees, specify whether the pending actions are to be sequential or parallel.
   Parallel actions occur at the same time and at the same level of approval.
   Sequential actions occur in order, one after the other, and at different levels of approval.
7. To specify a complex (OR) or compound (AND) expression for the first-level approval, click the first row.
   The Operation field and the second Requirement and Assignees fields open for input.
   a. Complete the second Assignees field and second Requirement fields.
   b. Specify AND to require that *both* the first and second assignees you specify must approve (or fulfill) the pending action. Otherwise, the pending action is rejected (or is not fulfilled). Specify OR to require that *only one* of the first and second assignees you specify must approve (or fulfill) the pending action. Otherwise, the pending action is rejected (or is not fulfilled).
   c. Click OK under the first row of approvers, and click Save (above the Policy tree).

   If you are finished specifying approval levels and approvers, navigate away from this window.
8. To specify a second level of approval, go to the next step.
   Optionally complete the steps for the previous bullet (if applicable) before going to the next step.
9. If you are finished specifying approval levels and approvers, navigate away from this window.
10. (Optional) Click the plus sign (+) in the Action Builder to add a second level of approval.
    A second approval row appears, with the fields open for input.
    a. Complete the first Assignees and Requirement fields in this row.
    b. If applicable, specify a complex (either-or) or compound (both-and) expression for this row.
11. (Optional) Specify a third or more level of approval, until you are finished specifying approvers for this policy.

You have specified the assignees for this policy.

Use an API Plug-in to Specify Assignees

An API plug-in is most useful when you query an external system for data to specify the assignees. An API plug-in is also useful when the identities and number of assignees vary, depending on the data that is supplied in the request. Based on this data, the plug-in dynamically creates the assignee list and specifies the assignee levels.

Follow these steps:

1. Click Catalog, Policies, and open the policy.
2. Select Use Plug-in for Assignees. This check box appears under the Condition field.
3. Complete the fields:
   – **Plug-in Id**
     Specifies the ID of your custom plug-in for dynamically populating the list of assignees. You or another administrator must have previously written, tested, and loaded this plug-in.

To view the list of plug-ins, select Administration, Tools, Plug-ins.

– **(Optional) Variables**
Specify the list of variables for the plug-in, if necessary.
If applicable, open the plug-in that you selected to display its details, including variables. On the details page, the Inputs section lists the ID values and descriptions of the input variables for the plug-in. Copy the ID values of the variables that you want from that page and paste them into the value of the Variables attribute. Enter the variables as a JSON expression.

You have specified assignees for this policy.

Example: Use of Variables

$({'form_field_value':_.sog['sog1'].serviceoption[2].form['form1'].txt1, 'est_service_cost':_.service.estimatedCost, 'est_sog_cost':_.sog['sog1'].estimatedCost, 'req_status':_.request.status})

These variables return data to the plug-in, as follows:

- form_field_value hold the value of the field named txt1. This field is in the form whose ID is form1. This form is in the service option in row 2 of the service option group whose ID is sog1.
- If necessary, use the product UI to find the row number of a service option. (The linked topic references policy *conditions* but the row number has the same value in this context. The row number is an object property that remains constant, whether you reference it in a policy condition, a JSON expression, or another type of code.)
- est_service_cost holds the estimatedCost property for the service.
- est_sog_cost holds the estimatedCost property for the service option group.
- req_status holds the request status.

The data from the variables populates the list of assignees and the levels of approval, according to the code specified in the plug-in. For example, you can write a plug-in to specify that if the following *conditions* are true, then trigger the following *action*:

Conditions:

- If The txt field in form_field_value equals Deluxe.
- If est_service_cost is greater than 5,000.
- If est_sog_cost is greater than 500.
- If req_status equals 200 (Submitted).

Action: Create the following assignment table:

- Level 1 Assignee is the manager of the IT department.
- Level 2 Assignee is the manager of the Purchasing department.
- Level 3 Assignee is the Vice President of the business unit of the requested-for user.

Policies define conditions for automatically designating specific users as the assignees for a task during the request life cycle. The most common use is to define conditions for automatically assigning specific approvers for services and service options in a request pending action.

Policies provide the flexibility to assign tasks (such as approval or rejection of requests pending action) to users other than the managers of the requestor. However, you can optionally include these managers in your list of assignees.

Policies consist of the following components:

- The condition under which the policy applies
- The assignment is to occur if the condition is met.
  If the condition is met, the user or users you specify are assigned to perform a pending action.
  You can specify the list of assignees according to any of several attributes, including but not limited to:

  - User name (first name and last name)
  - Membership in a user group
  - Management hierarchy, meaning the manager of the user, the manager of that manager
- An active-or-inactive setting
- Description and priority fields

You create global policies and global actions for general use with any service. In contrast, you create attached policies and actions for use with a specific service option only. You can create an attached policy or action only from the Policies and Actions tab of that service option.

For more information on policies, see the following topics:

- Use Policies to Manage Requests
- Policies and Actions tab
- Policy-driven approval process

{"URL":["/usm/wpf?Node=iclaunchpad.policy_builder"],"customLabelGetStarted":"Policies","description":"IPCE_Policies"}

Types of Conditions

When a condition you specify is met, CA Service Catalog assigns the related pending actions. The pending action is typically to approve, reject, or fulfill a requested item.

Specify one of the following types of conditions:

- Conditions based on the attributes of requests
- Conditions based on the attributes of users
- Conditions based on the attributes of the business unit
- Conditions based on the attributes of services
- Conditions based on the attributes of service option groups
- Conditions based on the attributes of service options
- Conditions based on the attributes of service option elements
- Match functions for service options and service option elements
- Conditions based on the fields of Form Designer forms
- Conditions based on the attributes of the location

Conditions Based on the Attributes of Requests

You can specify conditions that are based on the following attributes of the request that the policy affects:

- **completionDate**
- **dateCreated**
- **dateRequired**
- **description**
- **estimatedCost**
  Specifies the total estimated cost of all services (including all service options) in the request. The Catalog system calculates this cost when the request is submitted.

NOTE

To find the cost of the request, click Home, Requests. If applicable, use the My Requests drop-down list to display requests. Find the request and view the details.

- **id**
- **lastModified**
- **name**
- **priority**
  Specifies the priority of the request as a number, using one of the following values:
  1 = high
  2 = medium_high
  3 = medium
  4 = medium_low
  5 = low
- **requestedBy**
- **requestedByAccountId**
- **requestedFor**
- **requestedForAccountId**
- **status**
  Specifies the numeric value of the status of the request.
  To use request status as a condition for *approval*, specify this attribute in your condition, using the *approval range*, by default, less than 800. For example, the following condition is met when the service is named "Procure Laptop" and the request status is approved:

  ```
  $(_.service.name=='Procure Laptop' && _.request.status < 800)
  ```

  To use request status as a condition for *fulfillment*, specify this attribute in your condition, using the *fulfillment range*, by default, greater than or equal to 999. For example, the following condition is met when the service is named "Procure Laptop" and the request status is fulfilled:

  ```
  $(_.service.name=='Procure Laptop' && _.request.status >= 999)
  ```

  If your organization is *not* using custom statuses, you can specify the default status values.
  If your organization *is* using custom statuses, find all *actual* status values (both default and custom). Open the requestshared.xml file. Record the values that you want to use in your conditions.
- **other attributes**
  You can view most of the other attributes when you view the request list pages. Otherwise, open a request to view its additional details.

Examples

- To assign a pending action when the estimatedCost of the request equals 100 monetary units (by default, dollars):

  ```
  $(_.request.estimatedCost == 100)
  ```
- To assign a pending action when the priority of the request equals 1, meaning high priority, use this condition:

  ```
  $(_.request.priority==1)
  ```
- To assign a pending action when a request item has not been approved (status 800) and the Requested For user has a manager:

  ```
  $(anySoWith('status', lt, 800) &&_.request.requestedForUser.manager != '')
  ```
- To assign a pending action when a request item has been approved or when the Requested For user does not have a manager:

```
$(anySoWith('status', gteq, 800) || _.request.requestedForUser.manager == '')
```

The assignee list for this example can specify the default user (spadmin) or another suitable user.

- To assign a pending action when the request status is greater than 200 (submitted) and the business unit of the assignee is ca.com:

```
$(_.request.status>200 && _.request.bu.id=='ca.com')
```

Conditions Based on the Attributes of Users

You can specify conditions that are based on the attributes of users, as follows:

- Conditions that are based on the attributes of the Requested For user
  The Requested For user is the user who receives the requested services.
- Conditions that are based on the attributes of the Requested By user
  The Requested By user is the user who created and submitted the request.
  The Requested For user and Requested By user can be the same user or different users. If you submit a request for yourself, you are both the Requested For user and the Requested By user. If you submit a request for another user, you are the Requested By user, and the other user is the Requested For user.
- Conditions that are based on the attributes of the assigner
  Typically, the assigner is a workflow process utilizing an automated user like CERT-Service Delivery. Relatively few conditions are likely to use these attributes.

You can use the following attributes in conditions based on the attributes of users:

- **alias**
- **commonName**
- **defaultDomain**
  Each user has a default domain (business unit) in the user profile. Administrators set this default business unit when they add or edit the user.
  Specifies different values for different parameters, as follows:
  - For the Requested For user parameter, specifies the default business unit of the Requested For user.
  - For the Requested By user parameter, specifies the business unit of the request.

  Enter the ID of the business unit you want from the usm_tenant_ext table.
  For example, to list the values of all business unit IDs in that table, run the following query on the MDB from your database client:

  ```
  select tenant_id from usm_tenant_ext
  ```

  To list the default business unit of a user, run the following query on the MDB from your database client:

  ```
  select domain from usm_contact_domain_role where user_id=userid and default_domain=1
  ```

  - ***userid***
    Specifies the user ID whose default business unit you want to find.
  - **default_domain=1**
    Specifies the default business unit of the user.
- **defaultRole**
  Specifies the default role of a user in a domain.
  Enter the ID of the role ID you want from the usm_role table.
  For example, to list the IDs of all roles in that table, run the following query on the MDB from your database client:

  ```
  select role_id from usm_role
  ```

  To list the default role of a user in a domain, run the following query on the MDB from your database client:

  ```
  select role_id from usm_contact_domain_role where user_id=userid and default_domain=1
  ```

  - ***userid***
    Specifies the user ID whose default domain you want to find. For example:
    ```
    select role...where user_id='john smith'...
    ```
    If the user ID includes one or more spaces, enclose it with single quotation marks, as shown in the example on the previous line.
  - **default_domain=1**
    Specifies that the role is for the default domain of the user.
- **delegate**
  Specifies the user ID of a delegate for auto-delegation of the requests pending action of the user. Such delegates appear in the user profile in the Request Auto Delegation section. Users and their administrators can specify such delegates in the profile.
  If any delegate specified in the condition matches any delegate of the Requested For user, the Catalog system assigns the pending action.
  Enter the user ID of the delegate you want from the MDB.
  For example, to list the delegates for a user ID, run the following query on the MDB from your database client:

```
select delegate_id from usm_request_Auto_delegation where delegator_id=userid and delegation_type=0
```

- ***userid***
  Specifies the user ID of the delegator.
- **delegation_type=0**
  Specifies that the type of delegation is auto-delegation.

Contains all the CA EEM groups to which the user belongs. You can, for example, create a policy condition to assign a request pending action that is based on whether a user belongs to a specific CA EEM user group.
To find the names of CA EEM user groups, query CA EEM. Log in to CA EEM and review the group names.
Use the groups property to check if the requested by user is a member of a CA EEM group. For example, you can create a policy that requires all requests that are created by developers to be assigned to architects. To do so, use the following permission to check the requested by user group membership in the developers group:

```
_.request.requestedByUser.groups.indexOf('developers') >= 0
```

In contrast, you can also create a policy for users who are not members of a CA EEM group. To do so, specify the following expression:

```
_.request.requestedForUser.groups.indexOf('developers') < 0
```

- **description**
- **email**
- **fax**
- **firstName**
- **groups (user groups)**
- **id**
- **lastName**
- **localeCountry**

Specifies the ISO 3166 two-letter country code of the logged in user.
Commonly used ISO 3166 two-letter country codes include the following country codes:

- – Brazil - BR
  – China - CN
  – France - FR
  – Germany - DE
  – Italy - IT
  – Japan - JP
  – Spain - ES
  – United Kingdom - GB
  – United States - US

  NOTE

  The *localeCountry* attribute uses different values than the *country* attribute (from the ca_country table) used in several other conditions.

- **localeLanguage**
- **manager**

Enter the value of the manager user ID you want from the ca_contact table.
For example, to list the values of all manager user IDs in that table, run the following query on the MDB from your database client:

```
select supervisor_contact_uuid from ca_contact
```

- **middleName**
- **mobile**
- **pager**
- **phone**
- **roles**
- **status**

Specifies the status of the Requested For user.
Values:

- – 0 - Active
  – 1 - inactive (deleted)

- **timezone**

Specifies the code for the time zone for the business unit.

Examples:

- Eastern USA
- Greenwich Mean Time
- Amazon Time

Enter the time_zone_code from the ca_time_zone table.
For example, run the following query on the MDB from your database client:

```
select contact_uuid from ca_contact
```

- **title**
- **uuid**
  Enter the value of the contact_uuid you want from the ca_contact table.For example, run the following query on the MDB from your database client:
  ```
  select contact_uuid from ca_contact
  ```

- **Title**
  Enter the value of the job title for the user. The ca_contact table stores the job title of each user.
  For example, list the job title of a specific user (here, Omar PE Patel) in that table. To do so, run the following query on the MDB from your database client:

  ```
  Select job_title from ca_contact where userid='Omar PE Patel'
  ```

  To list the values of all job titles available in CA Service Catalog, run the following query:

  ```
  select id from ca_job_title
  ```

Conditions Based on the Attributes of the Requested For User

To specify conditions that are based on the attributes of the Requested For user that is affected by the policy, use the following format:

```
$(_.request.requestedForUser.attribute operator 'value'')
```

Enclose string values in single quotation marks. Enter numeric values without quotation marks. Do not enter spaces.

Examples

Sample conditions follow:

- To assign a pending action when the country of the business unit of the Requested For User is Brazil, use this condition:

  ```
  $(_.request.requestedForUser.localeCountry==BR)
  ```

  NOTE

  BR is the ISO 3166 two-letter country code for Brazil.
- To assign a pending action when the role of the Requested For user is request manager, use this condition:

  ```
  $(_.request.requestedForUser.role=='request manager')
  ```
- To assign a pending action when the title of the Requested For user is Procurement Officer, use this condition:

  ```
  $(_.request.requestedForUser.title=='Procurement Officer')
  ```

Conditions Based on the Attributes of the Requested By User

To specify conditions that are based on the attributes of the requested by user the policy affects, use the following format:

```
$(_.request.requestedByUser.attribute operator 'value'')
```

Enclose string values in single quotation marks. Enter numeric values without quotation marks. Do not enter spaces.

Examples

Sample conditions follow:

- To assign a pending action when the local language of the requested by user is Arabic, use this condition:

  ```
  $(_.request.requestedByUser.localeLanguage='Arabic')
  ```
- To assign a pending action when the time zone of the requested by user is GMT-11:00 MIT, use this condition:

  ```
  $(_.request.requestedByUser.timezone=='GMT-11:00 MIT')
  ```

The time zone code for each user is specified in the user profile. To retrieve the time zone code that is associated with a specific user (Isabella Lauderos), run the following query:

```
select time_zone_code from usm_contact_extension where user_id='Isabella Lauderos'
```

Conditions Based on the Attributes of the Assigner

To specify conditions that are based on the attributes of the assigner, using the following format:

```
$(_.user.attribute operator 'value')
```

Enclose string values in single quotation marks. Enter numeric values without quotation marks. Do not enter spaces.

Examples

Sample conditions follow:

- To assign the pending action when the title of the assignee is Purchasing Officer, use this condition:

  ```
  $(_.user.title=='Purchasing Officer')
  ```

- To assign the pending action when the default role of the user in the business unit is either Service Delivery Administrator (spadmin) or the Super Business Unit Administrator (stadmin), use this condition:

  ```
  $(_.user.defaultRole=='spadmin' || _.user.defaultRole=='stadmin')
  ```

Conditions Based on the Attributes of the Business Unit

You can specify conditions that are based on the attributes of the business unit, as follows:

- Conditions that are based on the attributes of the business unit of the request; that is, on the business unit of the Requested For user
- Conditions that are based on the attributes of the business unit of the assignee
  The assignee is the logged in user attempting to complete the pending action triggered by the policy condition being met. Typically, the assignee performs an approval or fulfillment task.

In both of these types of conditions, you can use the following attributes:

- **Business unit type**
  Specifies the code for the type of the business unit.
  V**alues:**
  – TE - Tenant
  – ST - Super Tenant
  – SP -Service Provider
- **country**
  Specifies the country code for the business unit.
  Enter the code for the country you want from the ca_country table.
  An example follows: To list the code of a specific country (here, India) in that table, run the following query on the MDB from your database client:

  ```
  Select id from ca_country where country='India'
  ```

  The results show that the country code for India is 114.
  To list all country codes for CA Service Catalog, run the following query on the MDB from your database client:

  ```
  select id from ca_country
  ```

- **currency**
  Specifies the code for the currency unit for the business unit.
  Enter the value of the currency_type_code you want from the ca_currency_type table.

For example, run the following query on the MDB from your database client:

```
select currency_type_code from ca_currency_type
```

- **dateFormat**
  Specifies the date format for the business unit.
  Values:
  - M/d/yyyy
  - M-d-yyyy
  - d/M/yyyy
  - d-M-yyyy
  - yyyy/M/d
  - yyyy-M-d
  - dd.MM.yyyy
- **decimalFormat**

Specifies the decimal symbol for the business unit.

Values:

- – 1 implies that the decimal symbol is a comma (,)
  - 0 implies that the decimal symbol is a dot (.)
- **description**
- **email**
- **federalTaxId**
- **id**
- **name**
- **loginId**
- **openedDate**
- **parent**

Specifies the tenant_id of the parent business unit.
Enter the parent_tenant_id of the usm_tenant_ext table.
For example, run the following query on the MDB from your database client:

```
select parent_tenant_id from usm_tenant_ext
```

- **primaryContact**
- **Single account mode**
  Specifies whether the business unit contains a single account only.
  Values:
  - 0 - The business unit users can have multiple accounts.
  - 1- The business unit can have only a single account.
- stateTaxId
- **status**
  Specifies the status of the business unit (not the request).
  Values:
- – 0 = Inactive (Deleted)
- – 1= Active (Open)
- taxRegion
- **timeformat**
  Specifies the time format for the business unit.
  Values:

  - HH:mm:ss
  - HH.mm.ss
- **Timezone**
  Specifies the code for the time zone for the business unit.
  Enter the time_zone_code from the ca_time_zone table.

For example, run the following query on the MDB from your database client:

```
select time_zone_code from ca_time_zone
```

- **type**
- **website**
- **data1, data2, data3...**
  Specify custom data fields, if applicable, that you have created and use.

Conditions Based on the Attributes of the Business Unit of the Request

To specify conditions that are based on the attributes of the business unit of the request, using the following format:

```
$(_.request.bu.attribute operator 'value')
```

Enclose string values in single quotation marks. Enter numeric values without quotation marks. Do not enter spaces.

Examples

Sample conditions follow:

- To assign the pending action that is based on the status of the business unit, use this condition:

  ```
  $(_.request.bu.status==0)
  ```

  NOTE
  The value of 0 specifies that the business unit is inactive. You can, for example, use a policy to assign all requests from an inactive business unit to a specific user.
- To assign the pending action when the tax region of the business unit of the request is named South, use this condition:

  ```
  $(_.request.bu.taxRegion =='South')
  ```
- To assign the pending action when the currency of the business unit of the request is named Euro, use this condition:

  ```
  $(_.request.bu.currency =='Euro')
  ```

Conditions Based on the Attributes of the Business Unit of the Assignee

To specify conditions that are based on the attributes of the business unit of the assignee, use the following format:

```
$(_.user.bu.attribute operator 'value')
```

Enclose string values in single quotation marks. Enter numeric values without quotation marks. Do not enter spaces.

Examples

Sample conditions follow:

- To assign the request pending action when the name of the business unit of the assignee is Western Regional Sales, use this condition:

  ```
  $(_.user.bu.name=='Western Regional Sales')
  ```

- You can assign the request pending action when the value of the singleAccountMode attribute of the assignee's business unit is 1 (meaning Yes). To do so, use this condition:

  ```
  $(_.user.bu.singleAccountMode==1)
  ```

- To assign the pending action when the business unit of the assignee is located in India, use this condition:

  ```
  $(_.user.bu.location.country==114)
  ```

  This condition uses 114, because 114 is the ID number for India in the ca_country table.

Conditions Based on the Attributes of Services

You can specify conditions that are based on the following attributes of the service that the policy affects:

- **bu**
- **code**
- **dateAvailable**
- **dateUnavailable**
- **dateCreated**
- **dateCancelled**
- **description**
- **estimatedCost**

Specifies the total estimated cost of a service in a request. The Catalog system calculates this cost when the request is submitted.

> **NOTE**
>
> To find the cost of the service, click Home, Requests. If applicable, use the My Requests drop-down list to display requests. Find the request that contains the service and view the details.

- **id**
- **name**
- **status**

Specifies the numeric value of the status of the service (*not* the request), as follows:

- – 0 - Deleted
  – 1 - Available
  – 2 - Unavailable
  – 3 - Created
  – 4 - Canceled
- **website**
- **version**

You can view most of the attributes when you add or edit a service.

Use the following format:

```
$(_.service.attrbute operator 'value')
```

Enclose string values in single quotation marks. Enter numeric values without quotation marks. Do not enter spaces.

Examples

Sample conditions follow:

- You can assign a pending action that is based on the service name. For example, if the service name is Knowledge Management Tools, use this condition:

  ```
  $(_.service.name=='Knowledge Management Tools')
  ```

- You can assign a pending action that is based on the estimated cost of the service being greater than or equal to 100 monetary units. (The default unit is dollars). To do so, use this condition:

  ```
  $(_.service.estimatedCost<=100.0)
  ```

- You can assign a pending action for any service option (requested item) whose status is greater than 1000 when the service ID is 9999. To do so, use this condition:

  ```
  anySoWith('status', gt, 1000) || _.service.id==9999
  ```

  By default, 1000 is the value for a status of Pending Fulfillment.

Conditions Based on the Attributes of Service Option Groups

You can specify conditions that are based on the following attributes of the service option group that is affected by the policy. Use the following format:

```
$(_.sog['sogname'].serviceoption[rownumber].attribute operator 'value')
```

Enclose string values in single quotation marks. Enter numeric values without quotation marks. Do not use spaces.

The service option group has the following attributes:

- **bu**
- **code**
- **dateAvailable**
- **dateUnavailable**
- **dateCreated**
- **dateCancelled**
- **description**
- **estimatedCost**
  Specifies the total estimated cost of a service option group in a service in a request. The Catalog system calculates this cost when the request is submitted.

  NOTE

  To find the cost of the service option group, click Home, Requests. If applicable, use the My Requests drop-down list to display requests. Find the request that contains the service option group and view the details. Add the costs of all service options that belong to the service option group.

  You can verify the service options that belong to a specific service option group by clicking Catalog, Service Offerings, Option Groups. Review the details for the service option group.
- **id**
- **name**
- **status**

Specifies the numeric value of the status of the service option group. The status of the service option group is the same as the status of the service (*not* the request).

Values:

- 0 - Deleted
- 1 - Available
- 2 - Unavailable
- 3 - Created
- 4 - Canceled

Examples

Sample conditions follow:

- You can assign a pending action that is based on a service option group being named Procure Server and having an estimated cost greater than 1,000 monetary units. (The default unit is dollars). To do so, use this condition:

  ```
  sog['Procure Server'].estimatedCost>1000
  ```

- You can assign a pending action that is based on a service option group being named Create email Account and having an estimated cost greater than or equal to 200 monetary units. To do so, use this condition:

  ```
  $(_.sog['Create email Account'].estimatedCost >=200)
  ```

- You can assign a pending action that is based on a service option group being named New Hire Onboarding and being contained in a business unit named Eastern District. To do so, use this condition:

  ```
  sog['New Hire Onboarding'].bu=='Eastern District'
  ```

Conditions Based on the Attributes of Service Options

Conditions that are based on service options can apply to either global or attached policies.

Formats

Use the following format for conditions *with* a match function:

```
$(anySoWith('attribute',operator,'value'))
```

Use the following format (without spaces) for conditions *without* a match function:

- For global policies, you specify conditions that begin with the service option group to which the service option belongs. Use the following format:

  ```
  $(_.sog['sogname'].serviceoption[rownumber] operator 'value' )
  ```

  To find the row number of a service option, use the product UI. Row numbers apply to global policies *only*.
- For attached policies, you specify conditions *without* the service option group name and row number. Use the following format:

  ```
  $(_.serviceoption operator 'value' )
  ```

  NOTE

  Attached policies let you specify a policy that you can share among multiple service options efficiently, using a simple condition. When you specify a global policy for a service option, you must find its row number and must use that number in the condition.

For all formats, enclose string values in single quotation marks, and enter numeric values without quotation marks.

To specify conditions that are based on service options, use the following attributes:

category	external_id
category_class	keywords
category_subclass	status
estimatedCost	track_as_asset

1. For service options and service option elements, you can specify conditions that use match functions.

- **category, category class, and category subclass**
  Specifies the values for category, category class, and category subclass from the category.xml file.
  View this file and record the values that you want to use in your conditions. This file is located in a different folder for each localized version of CA Service Catalog. For example, for English (icusen), the category.xml file is located in the USM_HOME\view\webapps\usm\locale\icusen\billing folder.
- **estimatedCost**
  Specifies the estimated cost of a service option in a service in a request. The Catalog system calculates this cost when the request is submitted.

  NOTE

  To find the cost of the service option, click Home, Requests. If applicable, use the My Requests drop-down list to display requests. Find the request that contains the service option and view the details.
- **external_id and keywords**
  Specifies the values for the attributes that are named external_id and keywords.
  Service builders specify these values when defining the service options of a service option group. Service builders typically use these attributes to add meta information about services, especially for categorizing services.
  You can find the values of these attributes and can record them for use in this condition.
- **status**
  Specifies the request status of the service option.
- **track_as_asset**
  Specifies a numeric value indicating whether to track this service option as an asset in CA APM, as follows:
  0 - No
  1 - Yes

  NOTE

  This attribute is relevant *only* when CA Service Catalog is integrated with CA APM.

  You can verify whether a service option uses this attribute, as follows: View the value of the Track as an Asset field on the Service Option Element Options window--Options tab.
  You identify service options in conditions by their row number in the service option group. On the CA Service Catalog GUI, you can find this row number by selecting Catalog, Service Offerings, Option Groups. Click the service option group on left pane, and click the Definition tab on right pane.
  When you do so, each service option in the group appears in a table; each row contains one service option. In the condition, specify the row number of the service option of interest. For example, suppose row 2 contains a service option named Windows server. In that case, specify the following conditions to include this service option.
  For a global policy:

```
$(_.sog['sogname'].serviceoption[2]
```

  For an attached policy:

```
$(_.serviceoption
```

Examples

Consider the following examples.

- You can specify the assignee when the name of the service option group is Procure Laptop and the category of the first service option is 1. To do so, use this condition:
  For a global policy:

  ```
  $(_.sog['Procure Laptop'].serviceoption[1].category==1)
  ```

  This policy applies to the service option in row 1 of the service option group.
  For an attached policy:

  ```
  $(_.serviceoption.category==1)
  ```

  This condition specifies that the pending action is assigned to the approver or fulfiller when the service option group is named Procure Laptop and the first service option in it belongs to Category 1. By default, Category 1 signifies hardware.
- You can specify the assignee when both of the following conditions are true:
  - The name of the service option group is New Hire Onboarding.
  - The estimated cost of the third service option is 30 monetary units. A sample unit is Euros.

  To do so, use this condition:
  For a global policy:

  ```
  $(_.sog['New Hire Onboarding'].serviceoption[3].estimatedCost==30.0)
  ```

  This policy applies to the service option in row 3 of the service option group.
  For an attached policy:

  ```
  $(_.serviceoption.estimatedCost==30.0)
  ```

- You can specify the assignee when both of the following conditions are true:
  - The name of the service option group is Handheld Devices.
  - The estimated cost of the service option in the third row of the group is 300 monetary units.

  To do so, use this condition:
  For a global policy:

  ```
  $(_.sog['Handheld Devices'].serviceoption[3].estimatedCost==300)
  ```

  This policy applies to the service option in row 3 of the service option group.
  For an attached policy:

  ```
  $(_.serviceoption.estimatedCost==300)
  ```

Examples with the Match Function

Consider the following examples:

- You can specify the assignee when the service option group includes a service option (in any row) with a category class greater than 10. To do so, use this condition:

  ```
  $(anySoWith('category_subclass',gt,10))
  ```

  This condition specifies that the pending action is assigned to the approver or fulfiller when any service option belongs to a category class greater than 10. By default, a category class greater than 10 signifies that the request is not related to any IT category, such as hardware or software.
- To specify the assignee when any service option in the request includes a category subclass greater than 10, use this condition:

  ```
  anySoWith('category_subclass',gt,10)
  ```

- To specify the assignee when the external_id (a string) of any service option ends with the string "MB," use this condition:
  ```
  $(anySoWith('external_id',endsWith,'MB'))
  ```

- You can specify the assignee when the category of any service option in the request is greater than 10 but less than 30. To do so, use this condition:
  ```
  $(anySoWith('category',gt,10) && anySoWith('category',lt,30))
  ```

- You can specify the assignee when *both* of the following are true:
  - The value of the external_id attribute of any service option in the request begins with the case-sensitive word "Memory"
  - The request contains a service option group that is named Procure Server. The category attribute of its first service option (row 1) has a value of 1, signifying hardware

  To do so, use this condition:
  ```
  $(anySoWith('external_id',startsWith,'Memory') && _.sog['Procure Server'].serviceoption[1].category==1)
  ```

- To specify the assignee when any service option in the request is tracked as an asset in CA APM, use this condition:
  ```
  $(anySoWith('track_as_asset',eq,1))
  ```

  NOTE

  This attribute is relevant only when CA Service Catalog is integrated with CA APM.

Find the Row Number of a Service Option

When you specify a condition that is based on the attributes of a service option, specify the row number of the service option. This row number is the position of the service option in its service option group. Row numbers apply to global policies *only*.

Follow these steps:

1. Click Catalog, Service Offerings.
2. Click the Option Groups tab.
3. Click the service option group of interest.
4. Click the Definition tab.
5. Find the service option of interest and click its Edit icon.
   The Service Option Details page opens. The row number appears under the image, near the Description field.
6. Record the row number.

   NOTE

   Row numbers start at 1 and increase sequentially by 1 as you add service options to a service option group.

You have found and recorded the row number of the service option. You can now use the row number in a condition that is based on the attributes of this service option.

Examples

For example, a service option group contains these service options: Small, Medium, and Large. The Small service option is in row 1. The Medium service option is in row 2, and the Large service option is in row 3.

To specify a condition for the Small service option in a global policy, use the following format:

```
$(_.sog['sogname'].serviceoption[1] operator 'value' )
```

Similarly, to specify a condition for the Medium service option in a global policy, use the following format:

```
$(_.sog['sogname'].serviceoption[2] operator 'value' )
```

Find the Values of the Attributes of a Service Option

You can find the values of the attributes of a service option when required. For example, you can create conditions that are based on the attributes of service options. In that case, you can specify the values of the external_id, code, and keyword attributes as part of the condition.

Follow these steps:

1. Click Catalog, Service Offerings, Option Groups.
2. Click the service option group that you want to open.
3. To display the service options, click the Definition tab.
4. Open the service option of interest and view its details, including its service option elements.
5. Find and optionally record the values of the attributes you want, for example, the external_id, code, and keywords.

You have found and optionally recorded the values of the attributes of a service option.

Conditions Based on the Attributes of Service Option Elements

Conditions that are based on service option elements can apply to either global or attached policies.

For service options and service option elements, you can specify conditions that use match functions.

You can specify conditions that are based on the following attributes of service option elements:

- code item_type
- estimatedCost status
- item_text

The following attributes require explanation:

- **Code**
  Is a user-specified text value to represent the product code, subscription code, SKU # or any other applicable code. To find the value of this attribute for a service option element, click Catalog, Services Offerings, Option Groups. View the service option and service option element of interest. On the Service Option Element Definition dialog, click the Options tab and find the value of the Code field.
- **estimatedCost**
  Specifies the estimated cost of a service option element in a service option in a service in a request. The Catalog system includes the cost of all service option elements in the cost of the service option to which they belong. The Catalog system calculates this cost when the request is submitted.

  NOTE

  To find the cost of the service option, click Home, Requests. If applicable, use the My Requests drop-down list to display requests. Find the request that contains the service option and view the details.
- **item_type**
  Specifies a valid value for the item type, as specified in the Type label of the service option element. For example, if the type of the service option element is CA BSI contract (for CA Business Service Insight contract), the value of item_type is 5. Similarly, if the type is Form, the value of item_type is 14.
- **item_text**
  Specifies the value of the Display Text field on the service option element definition page. You can specify the condition to require either an exact match or an approximate match, as follows:

- To require an exact match, specify the condition as follows:
  For global policies:
  ```
  $(_.sog['ab'].serviceoption[1].soe[2].item_text=='abc')
  ```
  For attached policies:
  ```
  $(_.serviceoption.soe[2].item_text=='abc')
  ```
  In this format, the text must match exactly, including case and spaces. For example, suppose "Premium Laptop" is the value of the Display Text field. In that case, the value of item_text must also be 'Premium Laptop': The value *cannot* be 'premium laptop' or 'Premium laptop'; the value *cannot* be any value except an exact match.
- To require an approximate match, specify the condition as follows:
  ```
  $(anySoeWith('item_text',contains,'abc'))
  ```
  In this format, the string that is specified must be either the same as the Display Text or a substring of Display Text. The string is not required to match exactly and is case-sensitive.
  For example, if "Premium Laptop" is the value of the Display Text field, then the string that is specified can be any of the following options:
  - 'Premium Laptop'
  - 'Laptop'
  - 'Premium'

- **status**
  Specifies the request status of the service option that contains this service option element.

Formats

For all formats, enclose string values in single quotation marks, and enter numeric values without quotation marks.

Use the following format for conditions *with* a match function:

```
$(anySoeWith('attribute',operator,'value'))
```

Use the following formats (without spaces) for conditions *without* a match function.

- For global policies, conditions for service option elements use this format:
  ```
  $(_.sog[sogname].serviceoption[rownum].soe[colnum].attribute operator 'value')
  ```
  - ***sogname***
    Specifies the name of the service option group.
  - ***rownum***
    Specifies the row number of the service option.
    To find the row number of the service option in the service option group, use the product UI.
    Row numbers apply to global policies *only*.
  - ***colnum***
    Specifies the column number of the service option element in its service option.
    To find the column number of the service option element in the service option, use the product UI.
- For attached policies, conditions for service option elements use this format:
  ```
  $(_.serviceoption.soe[colnum].attribute operator 'value')
  ```
  The same *colnum* attribute as for global policies also applies to attached policies.

Examples

Consider the following examples:

- You can specify the assignee for the request pending action when all of the following conditions are true:
  - The service option group is named Reserve Virtual Machine. This condition applies to global policies only.
  - The service option is in row 2. This condition applies to global policies only.
  - The service option element is in column 3. This condition applies to both global and attached policies.
  - The item type of the service option element is 15 (for reservation). This condition applies to both global and attached policies.

  To do so, use this condition:
  For global policies:
  ```
  $(_.sog['Reserve Virtual Machine'].serviceoption[2].soe[3].item_type==15)
  ```
  For attached policies:
  ```
  $(_.serviceoption.soe[3].item_type==15)
  ```
  Thus, this condition is met when a service option for creating or extending a reservation meets the specified criteria.
- You can specify the assignee for the request pending action when all of the following conditions are true:
  - The service option group is named Increase Mailbox Size. This condition applies to global policies only.
  - The service option is in row 3 of the service option group. This condition applies to global policies only.
  - The estimated cost for the service option element is greater than 200 monetary units. The default unit is dollars. This condition applies to both global and attached policies.
  - The service option element is in column 2 of the service option. This condition applies to both global and attached policies.

  To do so, use this condition:
  For global policies:
  ```
  $(_.sog['Increase Mailbox Size'].serviceoption[3].soe[2].estimatedCost >200)
  ```
  For attached policies:
  ```
  $(_.serviceoption.soe[2].estimatedCost >200)
  ```
  Thus, this condition is especially useful to specify approvers or fulfillers for service option elements costing more than a specified amount.
- You can specify the assignee for the request pending action when all of the following conditions are true:
  - The service option group is named Application Hosting.
  - The estimated cost is greater than or equal to 2500 monetary units for the sixth service option element in the fifth service option.

  To do so, use this condition:
  For global policies:
  ```
  $(_.sog['Application Hosting'].serviceoption[5].soe[6].estimatedCost>=2500)
  ```
  For attached policies:
  ```
  $(_.serviceoption.soe[6].estimatedCost>=2500)
  ```

Examples with the Match Function

Consider the following examples:

- You can specify the assignee for the request pending action when the following condition is true: Any service option element in the request has an estimated cost greater than 30.0 monetary units (for example, pounds). To do so, use this condition:
  ```
  $(anySoeWith('estimatedCost',gt,30.0))
  ```

- You can specify the assignee for the request pending action when the following is true: The value of the item_text attribute of any service option element in the request contains the 'share' text string. To do so, use this condition:

```
$(anySoeWith('item_text',contains,'share'))
```

Find the Column Number of a Service Option Element

When you specify the row number of the service option in its service option group.

Follow these steps:

1. Click Catalog, Service Offerings.
2. Click the Option Groups tab.
3. Click the service option group of interest.
4. Click the Definition tab.
5. Edit the service option of interest.
6. Edit or add the service option element of interest, for example, a rate or text element.
   The Service Option Element Definition dialog appears.
7. Click the Options tab and view the Column field near the top of the dialog. Record the number for use in conditions.

   NOTE

   Column numbers start at 1 and increase sequentially by 1 as you add service option elements to a service option.

You have found the column number of the service option element. You can use the column number in a condition that is based on the attributes of a service option element.

Examples

For example, a service option group lists these service options, in sequential order: Gold, Silver, and Bronze. The Gold service option is in row 1. The Silver service option is in row 2, and the Bronze service option is in row 3. Each service option has a rate service option element in column 5.

The following examples apply to global policies *only*:

To specify a condition for the Gold service option, use the following format:

```
$(_.sog['sogname'].serviceoption[1].soe[5].attribute operator 'value')
```

Similarly, to specify a condition for the Silver service option, use the following format:

```
$(_.sog['sogname'].serviceoption[2].soe[5].attribute operator 'value')
```

The following example applies to attached policies *only*:

To specify a condition for any of the three options, use the following format:

```
$(_.serviceoption.soe[5].attribute operator 'value')
```

Valid values for item_type

The valid values for item_type and their meanings follow:

item_type	Meaning
0	Text
1	Header
2	Numeric range
3	Rate
4	Usage based price
5	CA Business Service Insight contract
6	Numeric
7	Boolean
8	Adjustment
9	Date
10	Date range
11	Day of billing
12	Allocation
14	Form Designer form
15	Reservation

Service designers specify the item type when they complete the Service Option Element Definition window--Definition tab. They do so as part of the process of creating or editing a service option.

Match Functions for Service Options and Service Option Elements

Use match functions when defining conditions for service options and service option elements. To specify an attribute, value, and relationship in a condition *without* specifying any other elements that are required in other conditions, use a match function.

Elements that are required in other conditions but *not* in match functions include the following elements:

- The name of the request, service, or service option group
- The row number of a service option or service option element

Match functions thus provide flexibility by enabling you to specify the attribute condition across multiple components of a request.

To specify a match function for a service option, use the following format:

```
$(anySoWith('attribute',operator,'value'))
```

To specify a match function for a service option element, use the following format:

```
$(anySoeWith('attribute',operator,'value'))
```

For both formats, enclose string values in single quotation marks, and enter numeric values without quotation marks.

- ***attribute***
  For service options, you can use all the same attributes that are used in conditions for other service option elements without a match function.
- ***operator***

For both service options and service option elements, *operator* can be any of the following values:

Operator	Meaning
eq	equal to
neq	not equal to
lt	less than
gt	greater than
lteq	less than or equal to
gteq	greater than or equal to
startsWith	starts with
contains	contains
endsWith	ends with

- ***value***
  Specifies a string or numeric value for the attribute.
  For more information, see the examples of conditions that use the match function for service option elements.

Conditions Based on the Fields of Form Designer Forms

You can specify conditions that are based on the values of fields of Form Designer forms. Use the following format to specify such conditions:

```
$(_.sog[sogname].serviceoption[rownum].form[form-name].value of _id attribute string-operator

'value of value attribute')
```

- ***rownum***
  Specifies the row number of the service option that includes the Form Designer form.
- ***form-name***
  Specifies the name of the Form Designer forms, from the Form Designer tree. Do *not* use the value of name attribute for the form.
- ***value of _id attribute string-operator 'value of value attribute'***
  Specifies the attribute operator value portion of the conditions, using the same format as other conditions. For forms, the following requirements apply:
  - The *attribute* specification must be the value of the _id attribute of the field you want.
  - The *value* specification must be based on the value of the value attribute of the same field.

  Typically, the *value* specification is an option that is based on user input or user selection.

  WARNING
  The Form Designer stores the values for all fields in string format. Even the fields that contain numeric input that users or the Catalog system specify are stored in string format. Therefore, conditions that are based on Form Designer fields can use *only* the comparison operators suitable for strings: equal (==) and not equal (!=). Moreover, always enclose these values in quotation marks, because they are always strings.

  If necessary, you can combine two or more expressions that use these comparison operators. Use operators for specifying complex or compound conditions.
  See the following examples for details about specifying conditions that are based on Form Designer fields.

Examples Using a Predefined Form for Product Information

This example is based on a fictitious service option group that is named Insider Products. The service option group contains an actual predefined form named Product Info. To see this form, select Form Designer, expand the Forms tree, and select Product Info.

Users complete this form while requesting services to obtain services from Insider Products. This form includes a Discount Program Code field with an _id attribute whose value is discount_code and a value attribute whose default value is null. However, users can change this value by entering their discount code, if applicable.

You can specify the assignee for the request pending action, which is based on which of the following actions the user performs:

- User leaves the Discount Program Code field empty.
- User enters a discount code.

To meet each specification, use the following conditions, respectively:

- For an empty field:

```
$(_.sog['Insider Products'].serviceoption[1].form['Product Info'].discount_code==’null’)
```

- For a field with user input:

```
$(_.sog['Insider Products'].serviceoption[1].form['Product Info'].discount_code!=’null’)
```

Examples Using a Predefined Form with Radio Buttons

This example is based on a fictitious service option group that is named File Shares. The service option group contains an actual predefined form named File Share Access. To see this form, select Form Designer, expand the Forms tree, and select File Share Access. This form is useful for services that users use to request access to a network file share.

The sample conditions are based on the following options for access levels to the share and the values of their _id attributes:

- None: Permits no access.
- Read: Allows viewing of contained files and directories, loading of files, and executing software.
- Change: Provides all read permissions plus creating, deleting, and changing the directories and files.
- Full Control: Provides all change permissions, including file system changes and ownership.

For all these examples, both of the following conditions apply: The service option group is named File Shares, and the third service option includes a form that is named File Share Access.

The field is a select box that forces the user to select an option (using a radio button) on the form. Therefore, the form fieldportion of the condition uses the following format:

```
id_attribute of parent field=='value attribute of child field'
```

- **id_*attribute of parent field***
  Specifies the value of the id_attribute of parent field; here, the field that defines the select box.
  In this example, on the File Share Access form, the value is access_level. The select box appears as a radio button box on the form. Users must select one of the options by clicking its adjacent radio button.
- **'*value attribute of child field*'**
  Specifies the value of the value attribute (*not* the _id attribute) for the child attribute (the radio button option) that the user selected.
  In this example, on the File Share Access form, the valid values of these fields are none, read, change, and full. These values represent the options for each level of access to the file share.

View these form attributes on the Form Designer together with the following conditions to help gain a complete understanding of the relationship between them.

The following sample conditions are based on the radio button that the user selects on the File Share Access form:

- To specify the assignee for the pending action when the user selects No Access, use this condition:

```
$(_.sog['File Shares'].serviceoption[3].form['File Share Access'].access_level =='none')
```

- To specify the assignee for the pending action when the user selects Read (for read-only access rights), use this condition:

```
$(_.sog['File Shares'].serviceoption[3].form['File Share Access'].access_level =='read')
```

- To specify the assignee for the pending action when the user selects Change (for read and update access rights), use this condition:

```
$(_.sog['File Shares'].serviceoption[3].form['File Share Access'].access_level =='change')
```

- To specify the assignee for the pending action when the user selects Full Control. Full Control provides access rights for read, update, create, delete, and so forth, use this condition.

```
$(_.sog['File Shares'].serviceoption[3].form['File Share Access'].access_level =='full')
```

Conditions Based on the Attributes of the Location

You can specify conditions that are based on the attributes of the location, as follows:

- Conditions that are based on the attributes of the location of the assignee.
  The assignee is the logged in user attempting to complete the pending action. When the policy condition is met, it triggers the pending action. Typically, the assignee performs an approval or fulfillment task.
- Conditions that are based on the attributes of the location of the business unit of the assignee.

In both of these types of conditions, you can use the following attributes:

- **address**
- **city**
  Specifies the code for the city for the business unit of the assignee. Users enter their own values, rather than run a database query. This data is user-defined, and the MDB does not store this data.
- **country**
  Specifies the country code for the business unit.
  Enter the code for the country you want from the ca_country table.
  For example, to list the code of India in that table, run the following query on the MDB from your database client:

  ```
  Select id from ca_country where name="India"
  ```

  The results show that the country code for India is 114.
  To list all country codes for CA Service Catalog, run the following query on the MDB from your database client:

  ```
  select id from ca_country
  ```

- **county**
  Specifies the *user-specified* value, such as a name or code, for a county. Examples include a county in a state of the United States or in another country that has counties.

Values for this attribute are user-specified. The CA Service Catalog and the MDB do not validate or maintain them.

- **description**
- **fax**
- **name**
- **state**
  Specifies the 50 states in the United States, and some but not all of the United States territories, such as American Samoa. Also applies to all provinces in Canada. This attribute is null for other countries and territories.
  This attribute specifies the ID for the applicable state, territory, or province for the business unit of the assigner. Examples include 7404 for California and 7434 for New York.
  Enter the ID you want from the ca_state_province table.
  To list the IDs for all states, territories, and provinces, run this query on the MDB from your database client:

  ```
  select id from ca_state_province
  ```

  Verify the two-letter symbol for the state, territory, or province from the United States Postal Service (www.usps.com). You can use that symbol to find the ID. For example, the symbol for the state of New York is NY. Therefore, to list the ID for New York, run the following query on the MDB from your database client:

  ```
  select id from ca_state_province where symbol='NY'
  ```

  This query returns a value of 7434, which you use in your condition to assign the pending action based on the user's location being the state of New York. For details, see the sample state attributes in the conditions that are based on the attributes of the location of the business unit of the assignee.
  To list the IDs, symbols, and names for all applicable states, territories, and provinces, run the following query:

  ```
  Select id,symbol,description from ca_state_province
  ```

  The results appear similar to the following output:

  ```
  7400 AK  Alaska
  7401 AL  Alabama
  ...
  ```

- **phone**
- **postalCode**
- **uuid**

Conditions Based on the Attributes of the Location of the Assignee

To specify conditions that are based on the attributes of the location of the assignee, use the following format:

```
$(_.user.location.attribute operator 'value')
```

Enclose string values in single quotation marks. Enter numeric values without quotation marks. Do not use spaces.

Examples

Sample conditions follow:

- To assign the pending action when the name of the location of the user is Home, use this condition:

  ```
  $(_.user.location.name=='Home')
  ```

- To assign the pending action when the country of the user's location is India, use this condition:

  ```
  $(_.user.location.country==114)
  ```

  NOTE

  The code for India in the ca_country table is 114.

- To assign the pending action when the city of the location of the user is Hamburg, use this condition:

```
$(_.user.location.city=='Hamburg')
```

Conditions Based on the Attributes of the Location of the Business Unit of the Assignee

To specify conditions that are based on the attributes of the location of the business unit of the assignee, use the following format:

```
$(_.bu.location.attribute operator 'value')
```

Enclose string values in single quotation marks. Enter numeric values without quotation marks. Do not use spaces.

Examples

Sample conditions follow:

- To assign the request pending action when the business unit is in the state of New York in the United States, use this condition:

  ```
  $(_. bu.location.state==7434)
  ```

  7434 is the code for New York in the ca_state_province table.
- To assign the request pending action when the business unit is in the country of Mozambique, use this condition:

  ```
  $(_. bu.location.country==169)
  ```

  169 is the code for the country of Mozambique in the ca_country table.

Manage Request SLAs

CA Service Catalog administrators can create request SLAs to monitor whether service options in a request are processed within the time period that you specify for each monitored state. Your SLAs specify time to warning and time to violation for the selected service option. A single request SLA specifies the amount of time that is permitted between specified statuses. For example, the time that is taken to move from Submitted to Approved or from Approved to Completed.

For each request SLA, you specify the starting and ending statuses to monitor. You also specify the length of time to reach warning and violation thresholds, the expected level of compliance, and related settings.

Setting SLA warning and violation thresholds helps service providers to track the progress of a request. The request SLA data is stored in the CA Service Catalog database so that it can be included in reports.

The monitoring of time for an SLA warning or violation (SLA time) is started and stopped according to the service hours that both the outage calendars and the business hours specifications that are associated to the service determine. Administrators optionally associate one outage calendar and one business hours specification to each service. Only during service hours, the SLA time is monitored.

You can optionally specify automated actions to be initiated when a request SLA reaches warning status or violation status. These actions include a predefined action for sending email alert messages. For more information, see Configure Automated Email Alerts for SLA Warnings and Violations section.

> **NOTE**
>
> Request Service Level Agreements (SLAs) are a feature of CA Service Catalog. Quality of Service (QoS) SLAs are available only if CA Service Catalog is integrated with CA Business Service Insight.

Create Request SLAs

Request SLAs are processed according to both the fixed rules specified by CA Service Catalog and the Maximum Delay for Request SLA Alerts setting that you specify.

Fixed Rules

Request SLAs are processed according to the following fixed rules specified by CA Service Catalog:

- After a request containing one or more SLAs is submitted, the SLA clock for each SLA instance starts tracking time when the request reaches the starting status that is specified for the SLA. For example, if an SLA monitors the period from Pending Approval to Approved, the SLA clock for this SLA instance starts tracking time when the request status changes to Pending Approval.
- An SLA instance completes when either of the following situations occur:
  - The request reaches the ending status of the SLA before the SLA violation time expires. In this case, the SLA instance completes successfully. The SLA violation time is the maximum length of time that is permitted for the request reach the ending status of the SLA.
  - The SLA violation time expires before the request reaches the ending status of the SLA. In this case, the SLA instance completes unsuccessfully, causing an SLA violation to occur.
- The SLA clock tracks time for each active SLA instance and alerts the Catalog system if the SLA warning or violation time expires. The SLA clock is paused and resumed *only* when a CA Service Catalog user holds or resumes the request containing the SLA instance. If a user changes a request with an active SLA instance to Hold status, then the SLA clock is paused. When a user changes such a request to Resume status, the SLA clock is resumed.

  NOTE
  If the user of another product that is integrated with CA Service Catalog puts a request on hold in that product, the CA Service Catalog request, the SLA instance remains active. To pause the SLA clock, hold the request in CA Service Catalog.
- The SLA clock is stopped immediately when a request is canceled.
- The SLA clock is never restarted (reset to zero) under any circumstances, even if the request is rejected and resubmitted.
  For example, consider an SLA named “Approved within 1 Day SLA.” This SLA monitors from “Submitted” status to “Approved” status with a one day SLA violation time. In this example, the approval process also follows this request status cycle, in the following order:
  a. Submitted
  b. Pending Approval
  c. Approved
  d. Approval Done or Submitted
  e. Pending Approval
  f. Rejected

  In this example, if an end user requests a service, a day passes, and an approver rejects the request, then the SLA “Approved within 1 Day SLA” is violated. After the rejection, if the affected end user updates the request detail and submits it again, the request approval process is repeated, but the SLA instance is not monitored again because it has already been monitored once and completed with a violation.
- If an SLA instance completes, successfully or with a violation, that same instance is never monitored again under any circumstances.
- Once a monitored request SLA instance is violated, the violation is included in SLA-related BusinessObjects Enterprise reports, even if the related request is canceled or rejected after the violation occurs. To exclude such records from a report, customize the report.

Request SLA Processor

The request SLA processor is part of Catalog Component and:

- Listens for status changes for requests that have active SLA instances.
- Sends alert messages when SLA warnings or SLA violations occur.
- Tracks time for the SLA clock.

Maximum Delay for Request SLA Alerts

The request SLA processor checks for SLA instances to be processed at these times:

- When the SLA clock for an active SLA instance alerts the Catalog system that the SLA warning or violation time has expired.
- When it receives a message of a change in the status of a request with an active SLA instance.
- When the time period specified in the Maximum Delay for Request SLA Alerts setting expires. To access this setting, select Administration, Configuration, Request SLA. This setting specifies how frequently the request SLA processor checks for SLA warnings or violations. This setting applies to all SLA instances managed by a Catalog Component computer.

In a Catalog Component-clustered environment, if a clustered computer fails, then event notifications, including SLA alert messages, can be delayed until the failed computer is restored or until other computers in the cluster begin doing the work of the failed computer. So, SLA warning and violation messages may not be issued on time.

To minimize this possible delay in SLA processing time when a Catalog Component- clustered computer fails, you can configure the Maximum Delay for Request SLA Alerts to meet your needs. The smaller the value you set, the greater the frequency with which the request SLA processor checks the SLA clock for warning and violation times. Therefore, set a less frequent time period, such as one hour, to be informed of SLA warnings and violations quickly. Otherwise, set a larger time period, such as one day, to be notified about SLA warnings and violations later, when the failed clustered computer is restored.

Setting the Maximum Delay for Request SLA Alerts helps reduce any delay of SLA processing in case the clustered computer that started the SLA clock becomes unavailable for a length of time longer than a warning or violation period. In such cases, other active clustered computers take over SLA processing from the failed computer, when one of the following events occurs:

- The Maximum Delay for Request SLA Alerts time period expires.
- An SLA warning or violation is issued.
- A status change occurs in a request with an active SLA instance.

The following criteria can help CA Service Catalog administrators determine which service options to monitor with SLAs:

- Check whether any *critical* business rules, business needs, or contractual obligations with service consumers affect any service options. Indicate that such service options require SLAs.
- Check whether it is expected that *all* service options in your implementation are monitored with SLAs.

Request SLAs monitor whether service options in a service option group are processed within the specified time period. Your SLAs specify time to warning and time to violation for the selected service option.

Follow these steps:

1. Select Catalog, Service Offerings.
2. Expand the tree and select the service to which you want to add the SLA.
3. Click the Definition tab of the service.
4. Click the expand sign on the header next to the service option group name.
5. Locate the service option to which you want to add the SLA.
6. Click the SLA icon and click Add to create the SLA.
7. Specify the values for each SLA.
8. If you create multiple SLAs, select one of them as the primary SLA. To do so, click the option button in the Primary column. If a service option contains only one SLA, then that SLA is the primary SLA.

When a service option has multiple SLAs, the status of the primary SLA determines the status of the service option as a whole, regardless of the status of the other SLAs. For example, suppose you create four SLAs for a service option. If the primary SLA has not reached a warning or violation when the request containing the service option is completed, then the SLA for that service option is satisfied, even if all of the other three SLAs *did* reach a warning or violation. Conversely, if three of the four SLAs have no warning or violation, but the primary SLA has a warning or violation, then the service option as a whole has a warning or violation.
Both warnings and violations are reflected in SLA Reports.

(Optional) Configure Automated Email Alerts for SLA Warnings and Violations

As an administrator, you can create automated actions to be initiated when a request SLA reaches warning status or violation status. Such actions can include using the predefined actions for sending email alerts and running your own custom actions, such as running command-line commands. Using automated actions can help you find the root cause of a problem sooner and correct it earlier. So, you can reduce the number of any additional related SLA warnings or violations.

To create automated email alerts for SLA warnings and violations, follow this process:

1. Select Home, Administration, Events-Rules-Actions.
2. Click the Event Type named Request SLA Alerts.
3. Decide whether to enable one or both of the predefined SLA-related rules: they are *When a Request SLA Instance is Warned* and *When a Request SLA Instance is Violated*.
4. Click Enable to enable one or both of these rules.
5. Click Enable to enable the matching action or actions for each rule that you enabled.
   The predefined (built-in) action for the SLA instance warning is named Email Alert for Warned Request SLA. Similarly, the predefined action for the SLA instance violation is named Email Alert for Violated Request SLA.
6. Click the Edit icon in the Edit column of the action. Specify the information.

   NOTE

   The type is preset to Email. The Subject and Message fields are also preset.
7. Test the action. Submit a request with at least one SLA and letting the SLA reach warn status, violation status, or both.
8. Ensure that both actions (Email Alert for Warned Request SLA and Email Alert for Violated Request SLA) are configured

(Optional) Save SLA History and Reset SLAs

One likely reason to change your SLA definition is a change in service agreement or contract that arises after the monitoring of SLAs has already started. To handle such cases, CA Service Catalog enables you to save your current SLAs as history and reset SLAs. SLA history is *not* accessible using the GUI but *is* reflected in SLA reports.

For example, suppose that you created SLAs for a service on the first day of the month but later change the SLAs. At the end of the month, the SLA report for the entire month shows the different warning and violation settings that are used for the SLAs before and after the changes.

WARNING
When you change SLA settings for a service option, the change takes effect for *future* requests that include the service in which the service option resides. Requests that are in-progress are *not* affected.

Follow these steps:

1. Select Catalog, Service Offerings.
2. Expand the tree and select the service to which you want to add the SLA.
3. Click the Definition tab of the service.
4. Click the expand sign on the header next to the service option group name.

5. Locate the service option to which you want to add the SLA.
6. Click the SLA icon.
7. Click Save as History. This button is enabled *only* if the defined SLA exists.
8. When prompted, specify a reason for saving as history and resetting the SLAs.
9. Confirm the action.
   The existing SLAs are made inactive. Inactive SLAs are saved for use in both reports and in-progress requests, However, inactive SLAs are *not* applied to any future requests.
   Inactive SLAs are removed from the service and are no longer accessible through the CA Service Catalog GUI.
10. As needed, create request SLAs for use in future requests, to replace the inactive SLAs.

NOTE

Inactive SLAs are reflected in the SLA reports for related requests.

SLA Reports

In both the Report Builder and BusinessObjects Enterprise, you can create, configure, and view reports concerning SLA-related data. This data covers both current SLAs and inactive SLAs that have been saved as history.

Review the following information when you check SLA reports for warnings and violations:

- If a key SLA is warned or violated, then the service option that is associated with that SLA is warned or violated.
- If any service option in a service is warned or violated, then the service containing that service option is warned or violated.
- If any service in a request is warned or violated, then that request is warned or violated.
- For a violation, the Whole Violated Mark appears, at both the service and request levels.

Create Advanced Searches for Requests

Administrators can define the scope of user searches by clicking **Administration**, **Configuration**, **User Default**. They can specify the User Search Scope as either the business unit of the user or the entire Catalog system.

Follow these steps to create an advanced search query:

1. Click Home, Requests.
2. Click the advanced search icon.
3. Click the Search icon in the Advanced Search box to enter a new advanced search query. The fields next to the Search icon expand so that you can enter an advanced search condition in the format property - operator - value, as follows:
   a. In the first (property) search field (a drop-down list), select the property for you want to search for, such as Action By User ID, Business unit Name, Request Priority, or Service Option Element Name.
   b. In the second (operator) search field (a drop-down list), select the operator to use for your search, such as Equals, Not Equals, Contains, Starts With, Ends With, or In. *In* is a special operator that you can use to select or specify multiple values as search criteria.
   c. In the third (value) search field, type or select your search criteria, as follows:
   – For most properties, you type custom search criteria directly into this field. Examples include actual request IDs, user names, and business units.
   For example, to search for the request whose ID is 101, specify the following in the search fields:
   property: Request ID operator: Equals value: 101
   Similarly, to search for three requests whose IDs are 101, 102, and 103, specify the following in the search fields:
   property: Request ID operator: In value: 101,102,103

NOTE

Use a comma to separate multiple values in the value field.

- If you are a Service Delivery administrator, and you plan to share the advanced search query with other users:
  - For the properties that include the user ID, specify the value as the $USERID$ variable.
  - For the properties that include the business unit ID, specify the value as the BU$ variable.

  When other users run this query, the Catalog system dynamically replaces the variable with the user ID or business unit ID of the user running the query. This dynamic replacement is useful for complex advanced search queries that other users find beneficial but lack the expertise to create.
- For certain properties, this field changes to a drop-down list, and you select the value you want from the list. For example, if you are searching for a property related to the request status, you select the request status you want from the drop-down list. If your implementation includes custom statuses, they appear in this list.
  For example, to search for requests whose status is Approved, specify the following in the search fields:
  property: Request Current Status operator: Equals value: Approved
  Similarly, to search for requests whose statuses are Approval Done or Approval Not Needed, specify the following in these fields:
  property: Request Current Status operator: In value: Approval Done and Approval Not Needed
- If you are searching for a property related to the dates, if required, you can include multiple dates in your query by using operators such as Before or After, and by using multiple search conditions.
  For example, to find request created between July 27, 2011 and July 30, 2011, inclusive, specify both of the following conditions in your query:
  property: Request Date Created operator: After or On value: 07/27/2011
  property: Request Date Created operator: Before or On value: 07/30/2011
- To find requests related to inactive users, specify one or both of the following properties: *Requested For User Status* and *Requested By User Status*. For example, to find requests created by an inactive user, specify the following
  property: Requested By User Status operator: Equals value: Inactive

  NOTE

  For queries involving inactive users, you must specify these properties first, before you specify any other search criteria.

  You have completed this search condition.

4. Click the Plus sign (+) after the third search field to specify another search condition, if required.
5. Click the Remove (-) icon after the third search field of any additional search conditions that you want to remove from the advanced search query, if required.
   You have finished specifying this advanced search query.

Only Service Delivery administrators can share queries. A shared query appears as a saved query for all users when they click the Load icon. All users in all business units can view and run shared queries, but cannot modify them.

Follow these steps to share an advanced search query:

1. Click the Save icon.
2. Enter meaningful data in the Save Query dialog.
3. (Optional) Click Share to share the query with other users.

   NOTE
   The Catalog system includes several predefined shared queries at installation time. Service Delivery administrators can optionally unshare or delete any queries that do not apply to their implementation.

You can load an advanced search.

Follow these steps to load an advanced search:

1. Click the Load icon.

2. Filter the list to display all queries, shared queries only, or personal queries.
3. Select a query and click Search.

Use Widgets for Request Management

Service Managers work with application administrators, such as portal administrators and solution designers to embed CA Service Catalog widgets and let users access request life cycle functions.

You can embed CA Service Catalog widgets that fit either a broad or special-interest context. For example, on a web page that features virtual desktop services, you can embed the Browse widget to let catalog users view the services in the catalog. You can also add the Request widget to enable the users to request the virtual desktop services they want. Similarly, on a service desk page, you can use the Status, Request List, and Edit Request widgets. These widgets work together to let request managers approve and reject their requests pending action. This setup enhances the user experience for both catalog users and request managers.

Embedding CA Service Catalog widgets provides the following benefits:

- Segregate the request life cycle functions from other software, including other portals and portlets.
- Simplify the process. Users can easily access the catalog, create and submit requests, and manage requests.
- Create different pages for a special folder, interest, or user group.

Follow these steps:

- Step 1 - Review the Prerequisites and Guidelines for Widgets Implementation
- Step 2 - Understand the CA Service Catalog Widgets
- Step 3 - Implement the Widgets in SharePoint
- Step 4 - Test the Widgets

Step 1 - Review the Prerequisites and Guidelines for Widgets Implementation

This article contains the following topics:

Prerequisites for Embedding Widgets

To use CA Service Catalog widgets in other applications, verify that you meet these prerequisites:

General Prerequisites

- Widgets are snippets of code that you can embed in an HTML page. The widget provides a UI that is typically a service. Examples of widgets that are embedded in other applications include the Facebook Like button, Google embedded maps, and Twitter feeds.
  The CA Service Catalog widgets can include CSS, HTML, and JavaScript code in a script.
- (Recommended) Configure the portal or application in which you want to use widgets to use Single Sign-On (SSO). Typically, you use either Windows NTLM domain authentication or an SSO application, for example, CA SiteMinder. Configure CA Service Catalog to implement SSO using the same method as the portal or application.
- Decide which of the following widgets to use on which pages of your portal or application, and in what context.
  - Browse Widget
  - Request Widget
  - Status Widget
  - Request List
  - Edit Request

  For example, you can use the Browse and Request widgets together to let catalog users request a service.

You must also be:

- A proficient user and administrator of CA Service Catalog.
- A proficient user and administrator of the portal or application from which you want to call the widget.
- Familiar with JavaScript, HTML, and web page design.

Guidelines for Widgets Implementation

Review the following guidelines so that you can implement widgets efficiently:

Portal Pages

- Do not use the Request and Edit Request widgets on the same portal page.

  WARNING
  Placing both widgets on the same page can result in an error that indicates that one of the widgets cannot render forms or process JavaScript correctly. So, catalog users cannot complete and submit requests and request managers cannot approve and reject requests pending actions.

  To address this limitation, embed the widgets as designed in this scenario:
  - On page 1, embed the Browse and Request widgets.
  - On page 2, embed the Status, Request List, and Edit Request widgets.
- All widgets on the same portal page *must* connect to the same CA Service Catalog host. For example, the Browse and Request widgets must specify the same CA Service Catalog host.
- Specify uniform configuration parameters for all widgets on the same page.
  For example, suppose the value of the Business Unit parameter of the Status widget is SaoPaulo12. In that case, the value of this parameter for the Request List, and Edit Request widgets must also be SaoPaulo12.
- Ensure that your request URL does not exceed 8KB. You can, however, configure this value as follows.
  a. a. Include the maxHttpHeaderSize attribute in the connector tag of the server.xml file that is at USM_HOME/view/conf path.
  b. Specify the new value for the maxHttpHeaderSize attribute. For example, specify maxHttpHeaderSize=16384 to allow request URL up to 16KB.
- Use the CA Service Catalog UI and not the widgets to transfer, delegate, take, return, hold, resume, and push through requests.
- The following functionalities are not available in the widgets:
  - Pending Override icon
  - Gold brick icon for assigning assets from catalog request
  - Display Service Health option

Accessibility

Widgets are accessible to users with visual disabilities using a screen reader such as JAWS. This access affects widgets only if you intend to utilize this access. You must ensure that the *container* is also accessible.

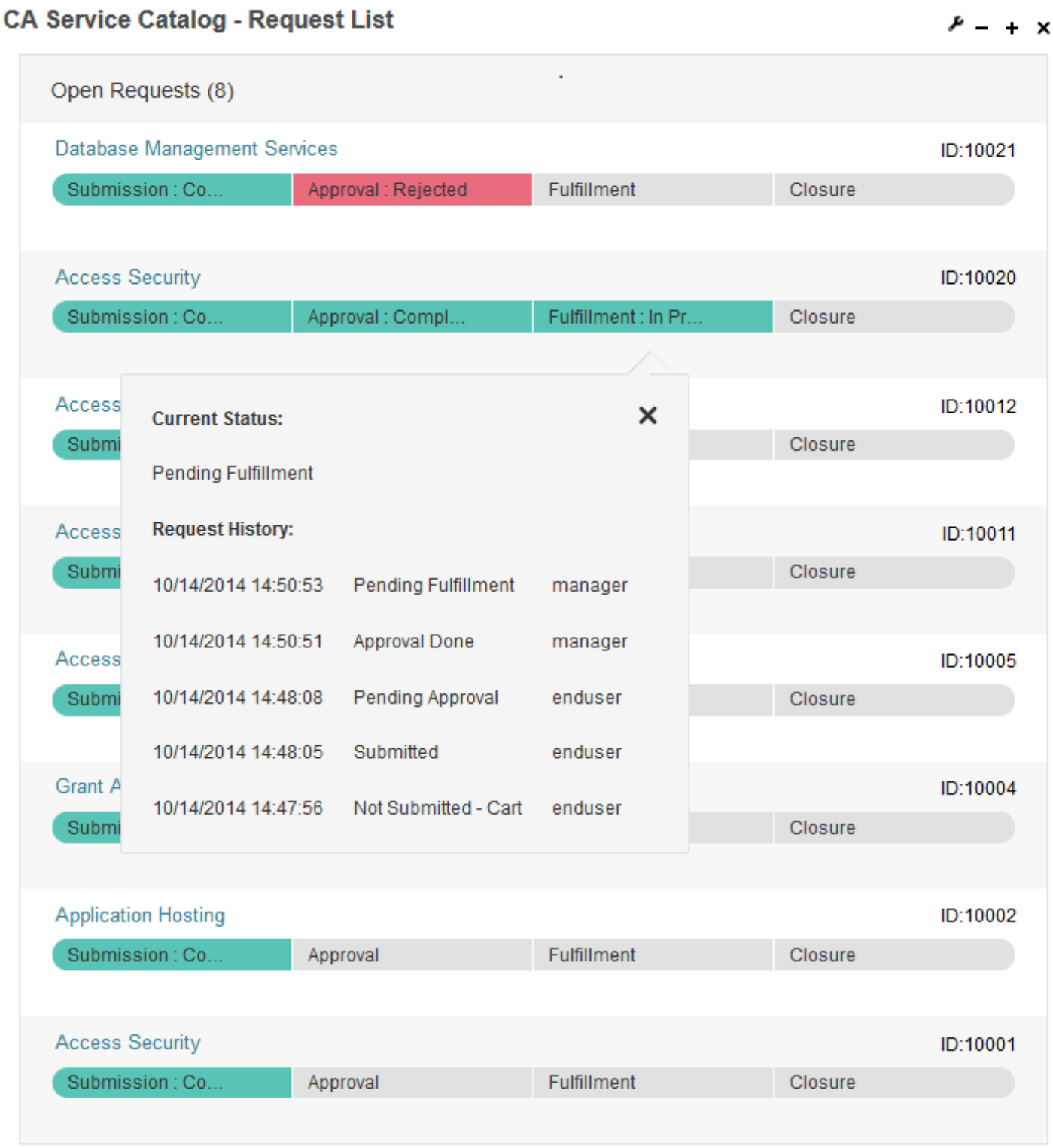
The keyboard shortcuts are as follows:

- Press Ctrl+Shift+Z to reach the first service offering from the category tree in the browse widget.
- Press letter V (small letter) to open the popup window when you click the links in the list widget.
- Press Ctrl+Shift+O to reach the first button in the button area when a request form is loaded and the focus is at any place in between the form.

Step 2 - Understand the CA Service Catalog Widgets

Before implementing widgets, understand which widget to use when. On the first portal page, you embed the Browse and Request widgets. On the second portal page, you embed the Status, Request List, and Edit Request widgets.

The Request List widget displays the state that the request is in. Click on the bar to see the current status and request history of the request as shown in the following graphic:


CA Service Catalog - Request List
Open Requests (8)
Database Management Services
ID:10021
Submission : Co...
Approval : Rejected
Fulfillment
Closure
Access Security
ID:10020
Submission : Co...
Approval : Compl...
Fulfillment : In Pr...
Closure
Access
ID:10012
Submi
Closure
Access
ID:10011
Submi
Closure
Access
ID:10005
Submi
Closure
Grant A
ID:10004
Submi
Closure
Current Status:
Pending Fulfillment
Request History:
10/14/2014 14:50:53 Pending Fulfillment manager
10/14/2014 14:50:51 Approval Done manager
10/14/2014 14:48:08 Pending Approval enduser
10/14/2014 14:48:05 Submitted enduser
10/14/2014 14:47:56 Not Submitted - Cart enduser
Application Hosting
ID:10002
Submission : Co...
Approval
Fulfillment
Closure
Access Security
ID:10001
Submission : Co...
Approval
Fulfillment
Closure

Use the Browse Widget to Display Services to Users

Use the Browse widget to offer services to users. You can use the Browse widget to offer the entire catalog or one specific folder or list of services. In this scenario, you offer the entire catalog through the Browse widget. Users can click any folder to display the services in the folder. Users can view the predefined folders and featured services to view their content.

Use the Request Widget to Let Users Request Services

Use the Request widget to let users request the services that you offer through the Browse widget. In this scenario, you add the Request widget next to the Browse widget. When the users click any service, the Request widget displays the service, including any forms. Users can complete the form and can submit the request.

For example, the user clicks the service "To host an application". The Request widget displays the service, so that the user can complete the form and can request the service. This action occurs automatically when you use the Browse and Request widgets on the same page and you select **Send local event which other widgets can listen** in the **Open In** parameter of the Browse widget.

Use the Status, Request List, and Edit Request Widgets to Let Users Manage Their Unsubmitted Requests

Use the Status and Request List widgets together, to provide the following function: When users click **Cart** on the status widget, the Request List widget opens the cart. **Cart** displays the number of items that the user has added to the cart but not submitted.

When the page includes Edit Request widget, users can view, add attachments, add notes, and change certain request details. Users *cannot* add more services, service option groups, or service options to the request.

Use the Status, Request List, and Edit Request Widgets to Let Users Manage Submitted Requests

Use the Status and Request List widgets together, to provide the following functions:

- Let users click **Open** to manage their open requests. Click **Open** to see the number of requests that the user has submitted but that have not been approved or rejected.
- Let users click **Closed** to view their completed or cancelled requests.

When the page includes Edit Request widget, the user can also act on these requests.

Use the Status, Request List, and Edit Request Widgets to Let Users Manage Their Open Requests

Use the Status and Request List widgets together, to display the *open requests* of the user when user clicks **Open**. Open requests are requests that the user has submitted but that the aprover has not yet approved or rejected. When the page includes the Edit Request widget, users can also make updates: Users can view, update (add notes or attachments), or cancel requests.

The user clicks **Open** on the Status widget, and the Request List widget opens the submitted requests.

Use the Status, Request List, and Edit Request Widgets to Let Approvers Approve or Reject Requests

Use the Status and Request List widgets together, to let users who have request approval rights open the requests pending action in their queues when they click **Pending** in the Status widget.

When the page includes the Edit Request widget, request approvers can also approve or reject these requests.

Step 3 - Implement the Widgets in SharePoint

You can implement the CA Service Catalog widgets in SharePoint using source code.

(Recommended) Configure Widgets in Liferay using Menu Options

In Liferay 6.1.2 CE, as a CA Service Catalog administrator you deploy the CA Service Catalog WAR files and then configure the widgets using the menu options.

Follow these steps:

Step a - Deploy the WAR files in Liferay 6.1.2 CE

Deploy the CA Service Catalog Web Application Archive (WAR) file in Liferay 6.1.2 CE. The WAR files let you configure the display and behavior of CA Service Catalog widgets in portlets.

Follow these steps:

1. Copy the following WAR files stored in the *usm_home**filestore**portlets* folder from the CA Service Catalog computer to the deploy folder of the Liferay computer. The deploy folder is in the Liferay install location.
   - browse.war
   - request.war
   - request-list.war
   - request-edit.war
   - status.war

   NOTE
   All widgets on a page must use the same CA Service Catalog computer. If you are using a cluster, specify the computer name of the load balancer.
2. Run the startup.bat file on the Liferay computer.

   NOTE

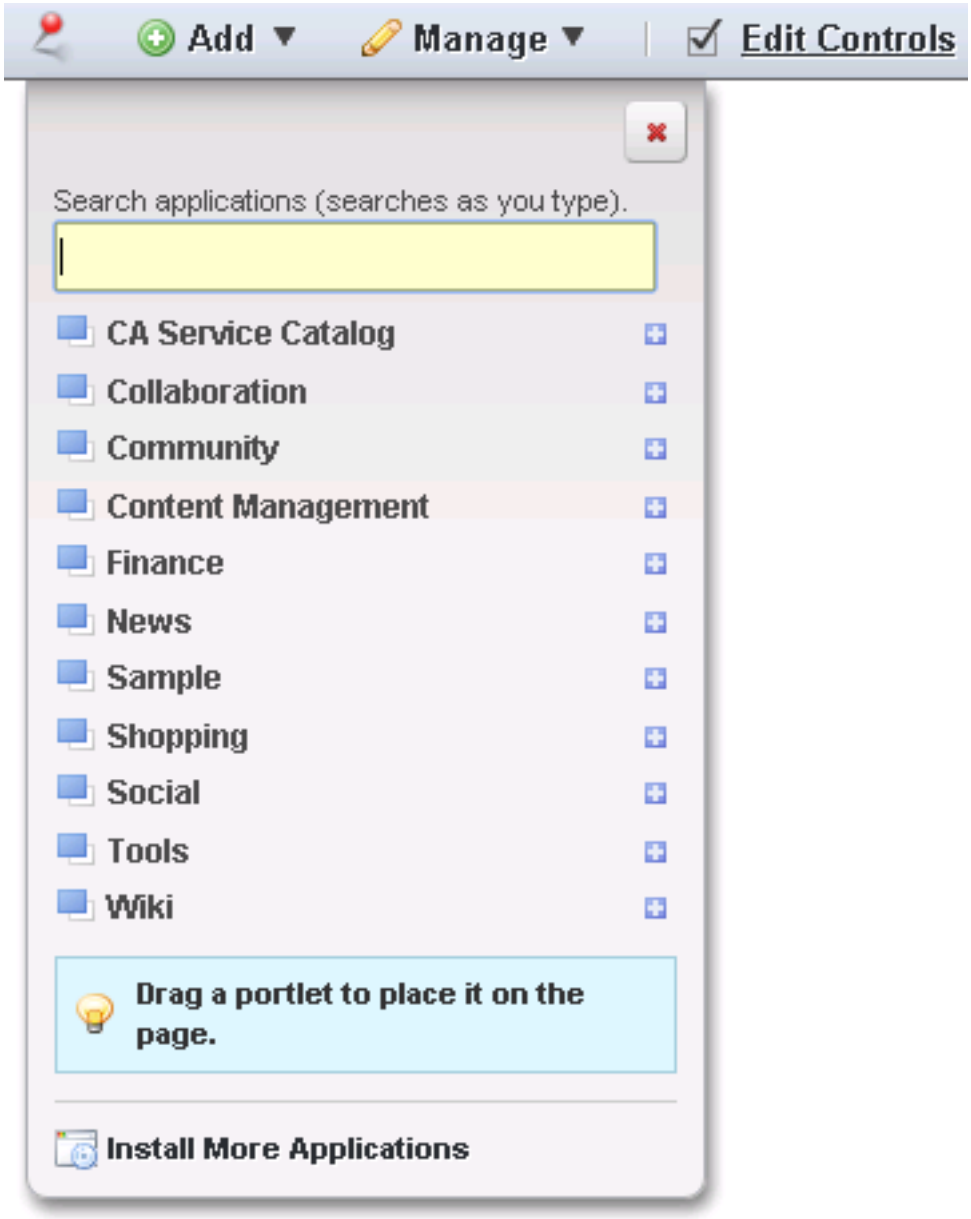
   The startup.bat file is in tomcat-7.0.40\bin folder of the Liferay install location.
3. Log into Liferay portal with the credentials provided by your Liferay administrator.
   CA Service Catalog folder is displayed when you click **Add**, **More**. The WAR files have been deployed in Liferay.

Step b - Configure the Widgets

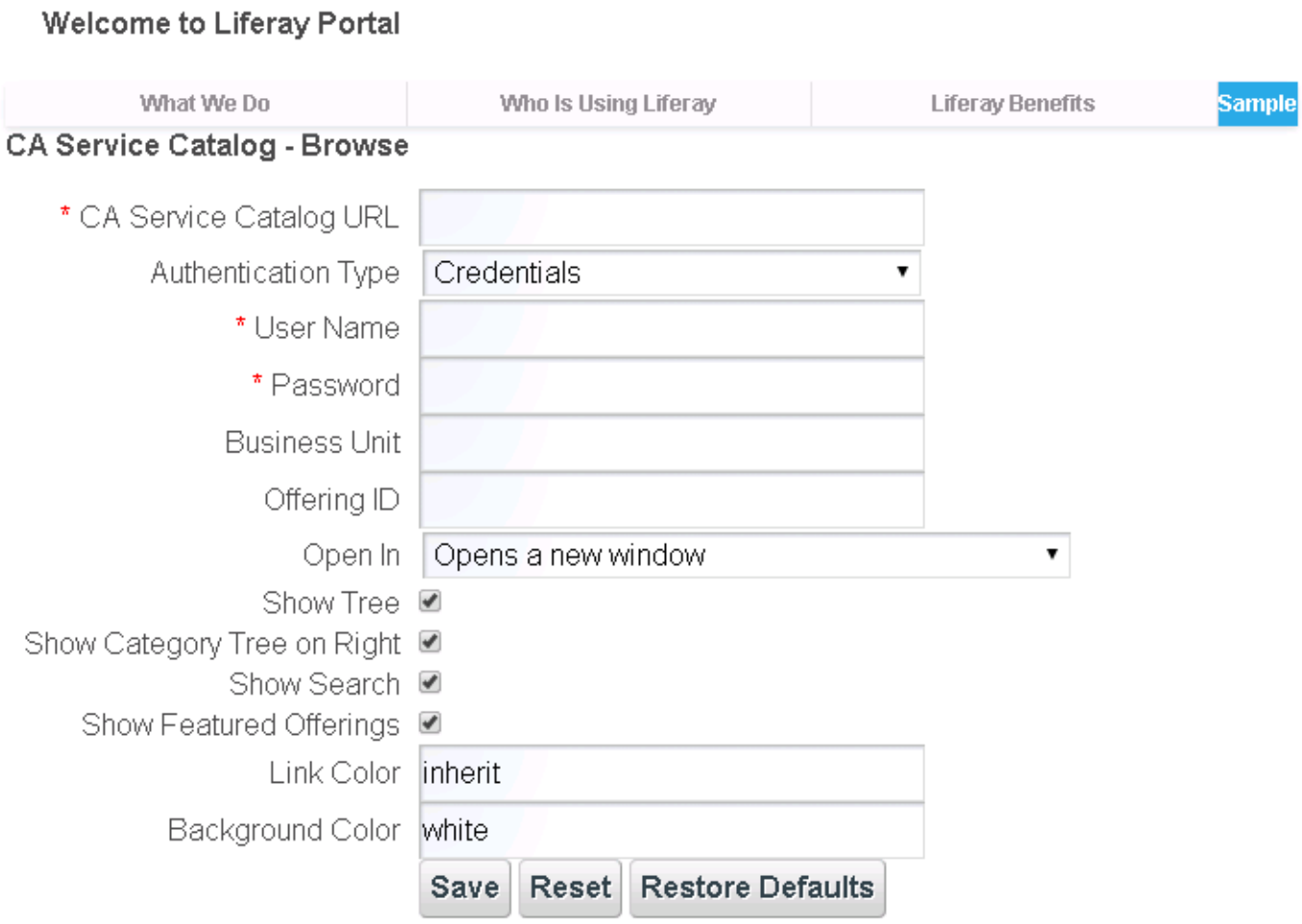
Use the CA Service Catalog WAR files to configure the display and behavior of CA Service Catalog widgets in portlets by using menu options.

Follow these steps:

1. Log into Liferay Portal with the credentials provided by your Liferay administrator.
2. Click **Add**, **Page** to create a page for your widgets.
3. Click **Add**, **More** on your portal page.
   The list of preconfigured portlets including a folder for CA Service Catalog appears as displayed in the following graphic:



4. Expand **CA Service Catalog** on the list.
5. Select the widget you want to implement and click **Add**. Alternatively, drag-and-drop the widget onto the location you want.
6. Ensure that **Edit Controls** at the top of the page is selected.
7. Mouse over the widget, click the wrench (Options) icon, and select **Preferences** from the drop-down list.
8. Review and modify the parameters for the different widgets, as shown in the following graphic:



The key parameters for the different widgets are in the following sections:

- Key Parameters for the Browse Widget
- Key Parameters for the Request Widget
- Key Parameters for the Status Widget
- Key Parameters for the Edit Request Widget
- Key Parameters for the Request List Widget

NOTE

There are other parameters available for the widgets. For example, background color or link color for the browse widget. You can modify any of these parameters depending upon your requirements.

9. Save your settings and review the portlet.
   The widget has been configured in Liferay.

NOTE

Embed the Browse and Request widgets on the first page (the Service Catalog page). These widgets work together to let catalog users browse the catalog and request services. Embed the Status, Request Edit, and Request List widgets in the second page (the Service Request page).

For each widget, ensure that:

Catalog URL=http://*host-name*:*port-number*/usm specifies the same CA Service Catalog host name and CA Service Catalog port number that you deployed the WAR Files in Liferay. Include the "/usm" in the URL for the portlet to render correctly.

Authentication Type specifies how to authenticate users. We recommend Single-Sign On for widgets.

Key Parameters for the Browse Widget

The key parameters for the Browse widget follow:

- **Business Unit**
  Specifies the business unit that catalog users can access while utilizing this Browse widget. Catalog users can browse services that they are entitled to. If you do not specify a value, the Catalog system uses the default business unit of the user accessing the widget.
- **Offering ID**
  Specifies the object ID of a single folder only or list of services that the Browse widget displays.
  You can specify either a single folder or a comma-separated list of services, using their object IDs.

  NOTE

  The folder or list of services that you specify must exist in a business unit that the user can access. Do not provide both folder IDs and service IDs in the list.
- **Open In**
  Specifies how a service opens in the Request view when the user clicks the service on the Browse widget.
  Values:
  - **Opens a new window**
    Opens the service in the catalog, on a new page. The user requests the service on that page.
  - **Raise request in top most frame**
    Performs the same function as **Opens a new window**, except that the service opens in the top-most frame of the browser. If the service is a frame, then the first associated frame in service option element is selected.
  - **Raise request in same page**
    Opens the service in the catalog, on the same page.
  - **Send local event which other widgets can listen**
    Specifies that another widget on the same page listens to events from the Browse widget and responds to them. When the user clicks a service, the Request widget responds by opening the service. Add the Request widget to this page to enable this function.
  - **URL**
    Opens the service using a custom URL. The URL can include a placeholder for the object ID of the service. For example, http://www.google.com?id={action}

Key parameters for the Request Widget

The key parameters for the Request widget follow:

- **Business Unit**
  Specifies the business unit that catalog users can access while utilizing this Request widget. Users can request services that they are entitled to. If you do not specify a value, the Catalog system uses the default business unit of the user accessing the widget.
- **Offering Id**
  Specifies the service to display when the Request widget opens initially. When the user clicks a service in the Browse widget, the Request widget displays that service.
  Allowed Values: Valid Offering ID that must exist in a business unit that the user can access. If you specify -1 or leave this parameter blank, the widget displays nothing.

Key Parameters for the Status Widget

The key parameters for the Status widget follow:

- **Business Unit**
  Specifies the business unit that catalog users can access while utilizing this Status widget. Users can view the status of the requests that they are entitled to.
  - Users who are not request managers can view the status of their own requests only.
  - Request managers can view the status of both their own requests and the requests pending action that is assigned to them.

  If you do not specify a value, the Catalog system uses the default business unit of the user accessing the widget.
- **Layout**
  Specifies how the options on the Status widget are displayed.
  Box Layout displays the options as buttons in a single row.
  Row Layout displays the options in each row in a table with a description.
- **Open In**
  Specifies how a target opens when the user clicks it on the Status widget. In this scenario, when the user clicks a service, another widget responds by opening the target. The targets are as follows:

Option	Target Function	Widget
Cart	Shopping Cart	Edit Request
Open	Open Requests	Request List
Closed	Closed Requests	Request List
Pending	Requests Pending Action	Request List

  For example, to enable these target functions to complete correctly when users click the option **Cart** in the cart, add the Edit Request widget .

Values:

- – **New window**
  Opens the target on a new page.
  – **Top frame**
  Performs the same function as **New window**, except that the target opens in the top-most frame of the browser. If the target is a frame, then the first associated frame in the service option element is selected.
  – **Same page**
  Opens the target in the catalog, on the same page.
  – **Other Widget**
  Indicates that another widget on the same page listens to events from the Status widget and responds to them.
  – **URL**
  Opens the target using a custom URL. The URL can include a placeholder for the object ID of the source context, for example, the service.
  An example follows:

```
http://www.google.com?id={action}
```

Key Parameters for the Edit Request Widget

The key parameters for the Edit Request widget follow:

- **Business Unit**
  Specifies the business unit that catalog users can access while utilizing this Edit Request widget. Users can edit requests that they are entitled to.
  If you do not specify a value, the Catalog system uses the default business unit of the user accessing the widget.
- **Request ID**
  Specifies the request to display when the Request List widget opens initially.

When the user clicks an option on the Status widget, the Request List widget displays the matching item: the cart, open requests, completed requests, or requests pending action.
Allowed Values: Valid Request ID that must exist in a business unit that the user can access. If you specify an invalid Request ID or leave this parameter blank, the widget displays nothing.

Key Parameters for the Request List Widget

The key parameters for the Request List widget follow:

- **Business Unit**
  Specifies the business unit that catalog users can access while utilizing this Request List widget. Users can list and view requests that they are entitled to.
  If you do not specify a value, the Catalog system uses the default business unit of the user accessing the widget.
- **Type**
  Specifies that the default setting for this Request List widget is to display pending requests (requests pending action). Other options include open, closed, or recent requests.

Configure Widgets in Liferay using Source Code

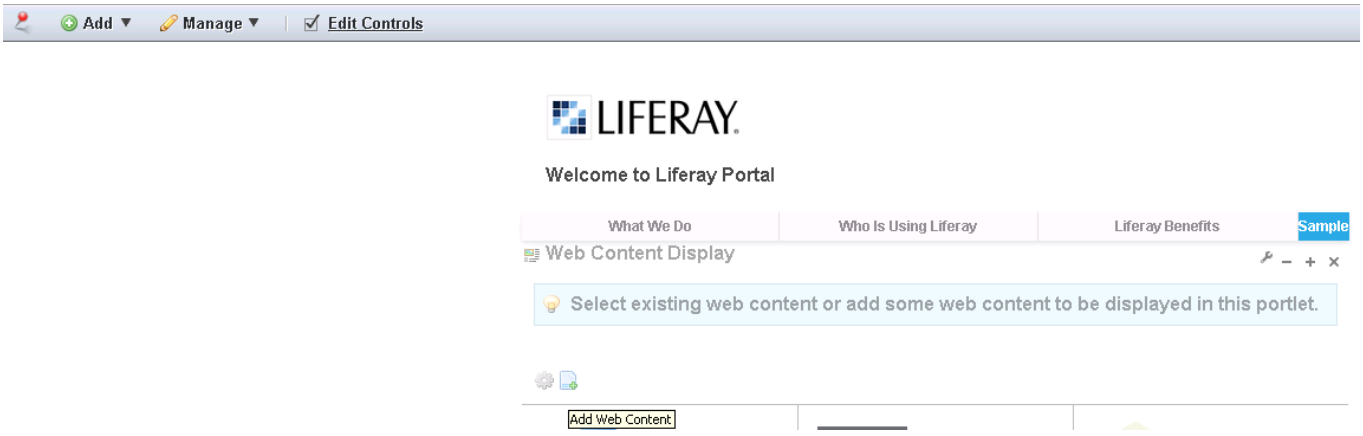
In Liferay 6.1.2 CE, as a CA Service Catalog administrator you can also call the widget by creating the portlet and specifying the source code. To configure the display and behavior of the widget in your implementation, follow the source code examples as a model.

WARNING

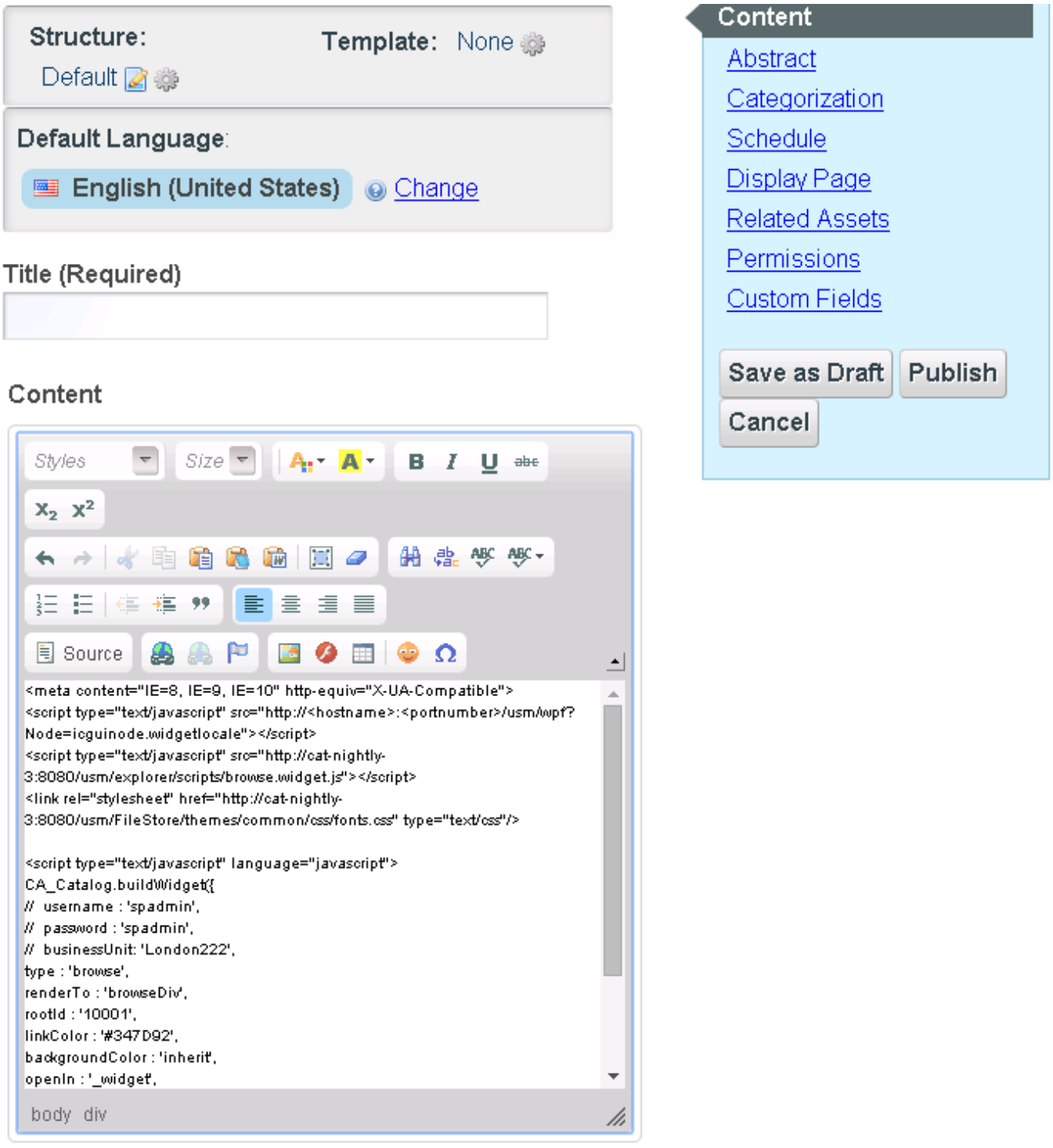
Ensure that you review the source code HTML example files for the widgets you want to implement before modifying the parameters. The source code HTML files are available at USM_HOME\view\webapps\usm path.

Follow these steps:

1. Log into Liferay Portal with the credentials provided by your Liferay administrator.
2. Click **Add**, **Page** to create a page for your widgets.
3. Perform these actions to create the portlet in the portal page:
4. 1. Click **Add**, **Web Content Display**.
   2. Click **Add Web Content** as shown in the following graphic:



3. Copy the source code for the widget that you want to embed in the **Content** section as shown in the following graphic:



The following HTML files are available for the widgets:

- browse html
  Specifies the HTML code for the Browse widget
- broswe_request html
  Specifies the combined HTML code for the Browse and Request widgets.
- status

Specifies the HTML code for the Status widget

- status_list
  Specifies the combined HTML code for the Status and Request List widgets.
- status_list_editrequest
  Specifies the combined HTML code for the Status, Request List, and Edit Request widgets.

NOTE

Embed the Browse and Request portlets on the first page (the Service Catalog page). Embed the Status, Request Edit, and Request List portlets in the second page (the Service Request page).

4. Specify the host name and the port number of the CA Service Catalog computer in each line of the source code for the different widgets as follows:

```
  <script type="text/javascript" src="http://<hostname>:<portnumber>/usm/explorer/scripts/
 browse.widget.js"> </script>
```

Update each of the following lines in the source code for the different widgets with the host name and port number information of the CA Service Catalog computer:

- browse.html file:

```
<script type="text/javascript" src="/usm/wpf?Node=icguinode.widgetlocale"></script>
<script type="text/javascript" src="/usm/explorer/scripts/browse.widget.js"></script>
<link rel="stylesheet" href="/usm/FileStore/themes/common/css/fonts.css" type="text/css"/>
```

- browse_request.html file:

```
<script type="text/javascript" src="/usm/wpf?Node=icguinode.widgetlocale"></script>
<script type="text/javascript" src="/usm/explorer/scripts/browse.widget.js"></script>
<script type="text/javascript" src="/usm/explorer/scripts/request.widget.js"></script>
<script type="text/javascript" src="/usm/gwt/fdRenderer/fdRenderer.nocache.js"></script>
<link rel="stylesheet" href="/usm/FileStore/themes/common/css/fonts.css" type="text/css"/>
```

- status.html file:

```
<script type="text/javascript" src="/usm/wpf?Node=icguinode.widgetlocale"></script>
<script type="text/javascript" src="/usm/explorer/scripts/status.widget.js"></script>
<link rel="stylesheet" href="/usm/FileStore/themes/common/css/fonts.css" type="text/css"/>
```

- status_list.html file:

```
<script type="text/javascript" src="/usm/wpf?Node=icguinode.widgetlocale"></script>
<script type="text/javascript" src="/usm/explorer/scripts/status.widget.js"></script>
<script type="text/javascript" src="/usm/gwt/requestList/requestList.nocache.js"></script>
<script type="text/javascript" src="/usm/explorer/scripts/request-list.widget.js"></script>
<link rel="stylesheet" href="/usm/FileStore/themes/common/css/fonts.css" type="text/css"/>
```

- status_list_editrequest.html file:

```
<script type="text/javascript" src="/usm/wpf?Node=icguinode.widgetlocale"></script>
<script type="text/javascript" src="/usm/gwt/fdRenderer/fdRenderer.nocache.js"></script>
<script type="text/javascript" src="/usm/explorer/scripts/status.widget.js"></script>
<script type="text/javascript" src="/usm/explorer/scripts/request-list.widget.js"></script>
<script type="text/javascript" src="/usm/explorer/scripts/edit.request.widget.js"></script>
<link rel="stylesheet" href="/usm/FileStore/themes/common/css/fonts.css" type="text/css"/>
```

5. Specify the other required fields and close the window.

NOTE

There are other parameters available for the widgets. For example, background color or link color for the browse widget. You can modify any of these parameters depending upon your requirements.

The new portlet gets added to Liferay.

5. (Optional) Perform these actions to edit the portlet in the portal page.
   a. Ensure that **Edit Controls** at the top of the page is selected.
   b. Mouse-over the portlet, and click the pencil (Edit Web Content) icon.
   c. Click **Source** in the Content window.
   d. Modify the parameters in the source file.
      The portlet is modified.
6. Close the Source container and click **Publish** to review the updated portlet.
   The widget is configured in Liferay using source code.

Configure Widgets in SharePoint using Source Code

In Microsoft SharePoint 2013, as a CA Service Catalog administrator you call the widget by creating the portlet and specifying the source code. To configure the display and behavior of the widget in your implementation, follow the source code examples as a model:

WARNING

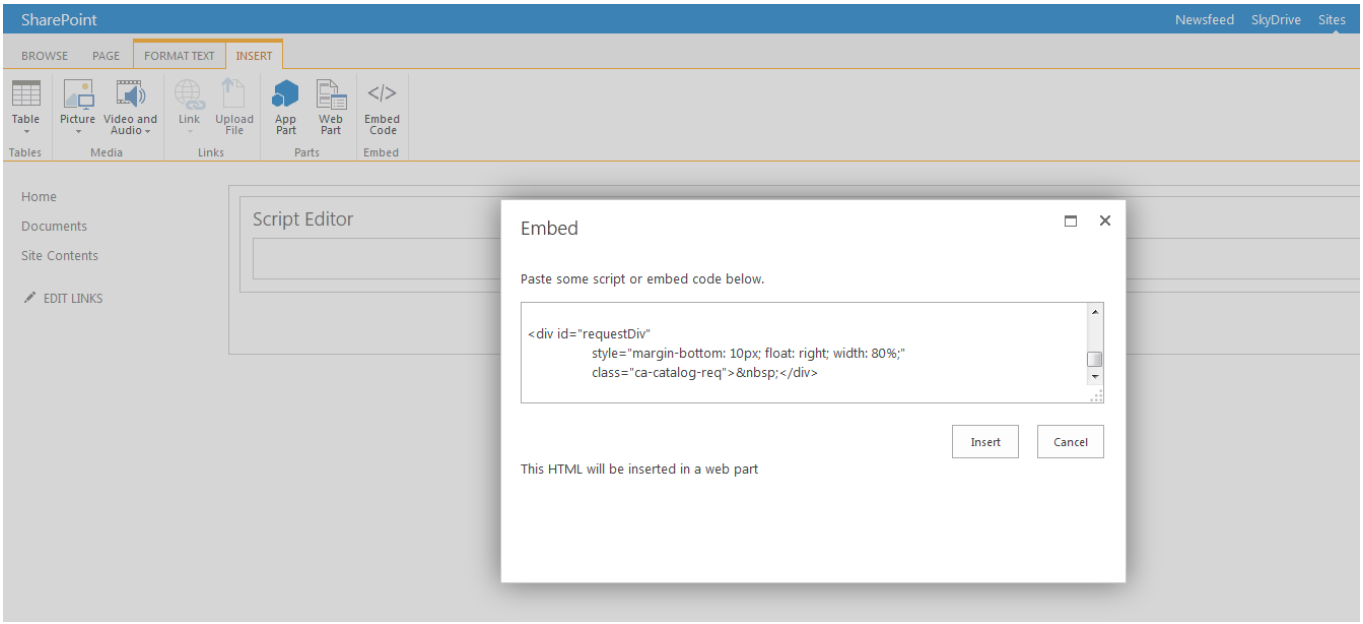
Ensure that you review the source code HTML example files for the widgets you want to implement before modifying the parameters. The source code HTML files are available at USM_HOME\view\webapps\usm path.

Follow these steps:

1. Verify with your Microsoft SharePoint administrator that a site page has been created for your organization. Ensure that you can access this site page.
2. Log into your Microsoft SharePoint site page with the credentials provided by your SharePoint administrator.
3. Perform these actions to create the portlet in the portal page:
   a. Click **EDIT** in the upper right corner of your SharePoint page.
   b. Click **INSERT**, **Embed Code** to insert the source code of the widget you want to embed in the page.
   c. Copy the source code for the widget that you want to embed. The following HTML files are available for the widgets:
      - browse html
        Specifies the HTML code for the Browse widget
      - broswe_request html
        Specifies the combined HTML code for the Browse and Request widgets.
      - status
        Specifies the HTML code for the Status widget
      - status_list
        Specifies the combined HTML code for the Status and Request List widgets.
      - status_list_editrequest
        Specifies the combined HTML code for the Status, Request List, and Edit Request widgets.

NOTE

Embed the Browse and Request portlets on the first page (the Service Catalog page). Embed the Status, Request Edit, and Request List portlets in the second page (the Service Request page).



d. Click **Insert**.

e. Specify the host name and the port number of the CA Service Catalog computer as follows:

```
<script type="text/javascript" src="http://<hostname>:<portnumber>/usm/explorer/scripts/
browse.widget.js"> </script>
```

Update each of the following lines in the source code for the different widgets with the host name and port number information of the CA Service Catalog computer:

- browse.html file:

```
<script type="text/javascript" src="/usm/wpf?Node=icguinode.widgetlocale"></script>
<script type="text/javascript" src="/usm/explorer/scripts/browse.widget.js"></script>
<link rel="stylesheet" href="/usm/FileStore/themes/common/css/fonts.css" type="text/css"/>
```

- browse_request.html file:

```
<script type="text/javascript" src="/usm/wpf?Node=icguinode.widgetlocale"></script>
<script type="text/javascript" src="/usm/explorer/scripts/browse.widget.js"></script>
<script type="text/javascript" src="/usm/explorer/scripts/request.widget.js"></script>
<script type="text/javascript" src="/usm/gwt/fdRenderer/fdRenderer.nocache.js"></script>
<link rel="stylesheet" href="/usm/FileStore/themes/common/css/fonts.css" type="text/css"/>
```

- status.html file:

```
<script type="text/javascript" src="/usm/wpf?Node=icguinode.widgetlocale"></script>
<script type="text/javascript" src="/usm/explorer/scripts/status.widget.js"></script>
<link rel="stylesheet" href="/usm/FileStore/themes/common/css/fonts.css" type="text/css"/>
```

- status_list.html file:

```
<script type="text/javascript" src="/usm/wpf?Node=icguinode.widgetlocale"></script>
<script type="text/javascript" src="/usm/explorer/scripts/status.widget.js"></script>
<script type="text/javascript" src="/usm/gwt/requestList/requestList.nocache.js"></script>
<script type="text/javascript" src="/usm/explorer/scripts/request-list.widget.js"></script>
<link rel="stylesheet" href="/usm/FileStore/themes/common/css/fonts.css" type="text/css"/>
```

- status_list_editrequest.html file:

```
<script type="text/javascript" src="/usm/wpf?Node=icguinode.widgetlocale"></script>
<script type="text/javascript" src="/usm/gwt/fdRenderer/fdRenderer.nocache.js"></script>
<script type="text/javascript" src="/usm/explorer/scripts/status.widget.js"></script>
<script type="text/javascript" src="/usm/explorer/scripts/request-list.widget.js"></script>
<script type="text/javascript" src="/usm/explorer/scripts/edit.request.widget.js"></script>
<link rel="stylesheet" href="/usm/FileStore/themes/common/css/fonts.css" type="text/css"/>
```

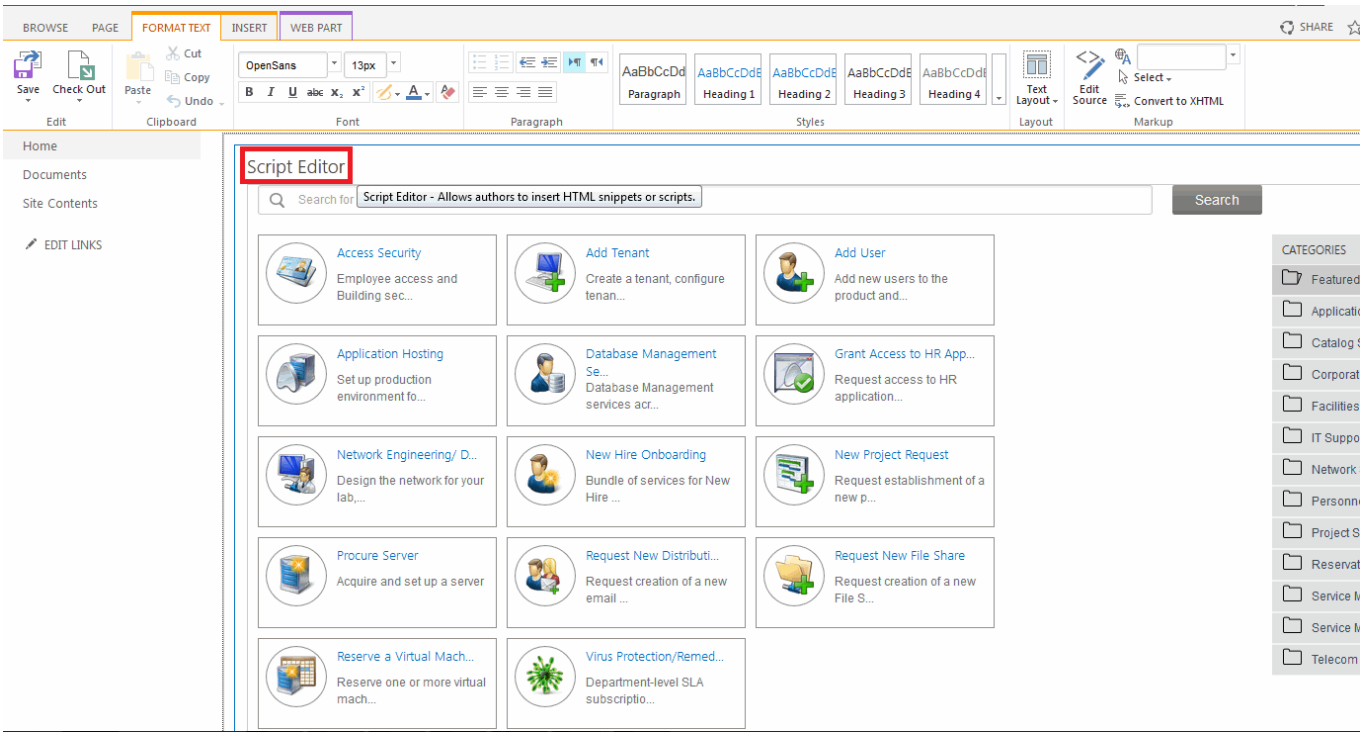
NOTE

There are other parameters available for the widgets. For example, background color or link color for the browse widget. You can modify any of these parameters depending upon your requirements.

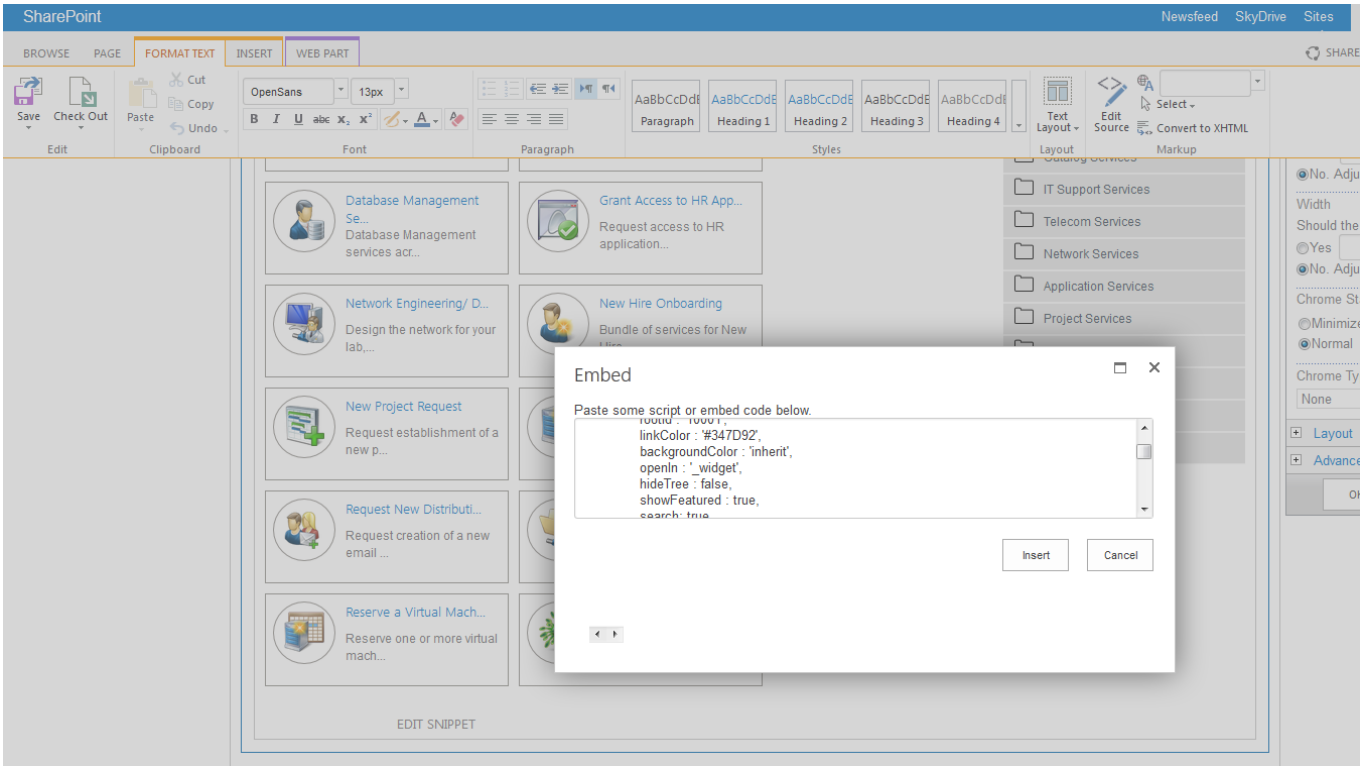
f. Click **SAVE**.

The widget is added to your Microsoft SharePoint page.

4. (Optional) Perform these actions to edit the portlet in the portal page.
   a. Click **EDIT** in the upper right corner of the SharePoint page.
   b. Click **Script Editor**.



   c. Click **WEB PART**, **Web Part Properties**.
   d. Scroll to the end of the page and Click **EDIT SNIPPET** in Script Editor.
   e. Modify the required parameters and click **Insert**.



f. Click **SAVE**.
The widget is modified.

Step 4 - Test the Widgets

Verify that the widgets appear and function as intended on portal pages or other pages on which you embedded them.

Follow these steps:

1. Access the pages that include the widgets.
2. Verify the following functions on the first page:
   - The Browse and Request widgets appear as intended.
   - The widgets work together to let catalog users view and request services.
3. Verify the following functions on the second page:
   - The Status, Edit Request, and Request List widgets appear as intended.
   - Catalog users can perform the following tasks:
     - Review the status of their submitted requests.
     - Add notes and attachments for submitted requests that have not yet been approved or rejected.
   - Request managers can manage their requests pending action by approving or rejecting them.

You have tested the widgets.

Request Management from a User Perspective

After you have submitted your cart, you have launched your new request into its life cycle. Each requested service option in the request has a status that reflects its current step in the request life cycle. The overall request also has a milestone status that reflects the aggregate values of the statuses of the requested service options. The milestone statuses are:

- Not Submitted
- Submitted
- Pending Approval
- Approved/Rejected
- Pending Fulfillment
- Fulfilled
- Completed/Cancelled

Each status range contains various detailed status values. The status values are customizable by your catalog administrator. During the approval and fulfillment phases for your request, you can receive emails informing you of the progress of your request through its life cycle. The business process for approval and fulfillment, the status values used and whether you receive emails depends on the approval and fulfillment business processes of your service provider.

For example, you request a Desktop service which includes a Standard Desktop service option and Microsoft Visio service option. Your request includes one service which contains two service options. If your service provider requires manager approval, the first step is for your manager to approve or reject your request for the Desktop service. After approval phase is complete for the two service options Standard Desktop and Microsoft Visio, the fulfillment phase begins. If the fulfillment process for these two service options includes checking available inventory, an asset manager checks the inventory.

Assuming both the Standard Desktop and a Microsoft Visio license are found in inventory, the status of each service option changes to Filled From Inventory. The next step in the fulfillment process is to notify IT Services to configure and deliver the Standard Desktop and Microsoft Visio so the status of each service option changes to Notified IT Services. After each service option is configured and delivered to you, the status of each service option changes to Fulfilled. After each service option in the request is set to Fulfilled, the status of all service options is set to Completed. The request is considered completed or closed.

For this example, for each requested service option, you can view the following status values over time (in chronological order):

- Not Submitted - Cart
- Submitted
- Pending Approval
- Approved
- Approval Done
- Pending Fulfillment
- Check Availability
- Filled From Inventory
- Notified IT Services
- Fulfilled
- Completed

Manage Requests

You can view, modify, copy, and search requests. View requests to check their status, tracking, audit trail, and other details. You can also email copies of requests. The details of a request that you can view and modify vary according to:

- Your role
- The request status
- The configuration options set by your catalog administrator

Follow these steps:

1. Log in to CA Service Catalog, as follows:
   a. Enter the URL for CA Service Catalog, in the format http://*computer-name*:*port-number*. Your catalog administrator provides the URL.
   If CA Service Catalog is configured to use single sign-on, you are automatically authenticated and the initial page appears. Otherwise the Log In page appears.
   b. If the Log In page appears, enter your assigned user name and password.

      NOTE

      If you have a user ID in multiple business units, click Advanced. Specify the business unit that you want to log in to.
2. Click Home, Requests.

   NOTE

   The Requests page can either display requests directly or provide access to them indirectly through the My Requests drop-down list. Administrators optionally configure the page to use either setup.
3. Click the My Requests drop-down if the page does not list requests and select the option of your choice.
4. If applicable, click the list of requests that contains the request that you want to copy. For example, if you want to copy a completed request, click Completed Requests.
5. Click the name of the request that you want, and proceed as follows:
   – View and optionally update the request, as applicable.
   – Copy the request.

      NOTE

      The notes and attachments for the original request are not copied to the new request. The status of the new request is Not Submitted, so that you can delete the existing services or can add new services in the request.
      Update the fields as needed, including the *Requested By* and *Requested For* users.
   – To search for requests by request ID or according to the predefined categories on the screen, use the Request Lookup section of the Requests window. You can view only the requests that your role permits.

Review the following information to understand what you can do when you view and can modify requests:

All Statuses

- View the requested services and service options, and other details of the request.
  View the status, which appears in the Status column of any request list, for example, My Recent Requests, Completed Requests. The request status also appears when you view the details of a request.
- View the request status history, by clicking the name of the status in the request details or the Status column.
  Status values appear for the entire request and for each service and service option in the request. Review the numbers, names, and meanings of default status values.
- Display and add and attachments.
- Refresh the display.
- Copy the request, as explained earlier in this topic.
- Email a copy of the request.
  To email the details of a request, click the Email button or link, specify the details, and click Send.
  Optionally click Include Request Snapshot to include details of the request in the email. Otherwise, the email contains only the text that you specify for the Subject line and the Message Body.

In the address fields, you can enter email addresses, user IDs, account names, or the *last name, first name* values of user names. Separate multiple entries with a semi-colon (;). If you enter an account name, the Catalog system uses the email address on the account profile.

- View and edit the additional information sections.
- View the subscriptions for service options, if applicable.

Not Submitted Requests

- Submit the request.
- Add or delete services.
- Delete the request.

Open Requests

An open request is one whose status is not Completed or Cancelled.

- View the status history.
- View the details of a request.
- Edit the request.
- Email the request.

Administrators can configure the display of columns on the Open Requests page to display some or all of the following columns.

The default view displays the following columns:

- ID - Assigned by the system
- Name
- Requested By
- Requested For
- Priority
- Date Created
- Date Modified
- Status
- Actions

Click the icon to configure the display of columns.

Requests Pending Action

Request managers can approve, reject, fulfill, and perform several other actions for the requests pending action that are assigned to them.

Completed Status

- Cancel a requested service.
- Cancel the request which cancels all the services in the request.

Tracking

If permitted by your role and the configuration options set by your catalog administrator, you can view and click the Tracking tab. This tab displays any CA Process Automation process instances that are related to the requested services and service options. This tab also provides the status and other details of these processes.

Audit Trail

If permitted by your role and the configuration options set by your catalog administrator, you can view and click the Audit Trail tab. This tab displays each service option element and its status changes over time. Each service option is comprised of service option elements.

You can view, modify, copy, and search requests. View requests to check their status, tracking, audit trail, and other details. You can also email copies of requests. The details of a request that you can view and modify vary according to:

- Your role
- The request status
- The configuration options set by your catalog administrator

Follow these steps:

1. Log in to CA Service Catalog, as follows:
   a. Enter the URL for CA Service Catalog, in the format http://*computer-name*:*port-number*. Your catalog administrator provides the URL.
      If CA Service Catalog is configured to use single sign-on, you are automatically authenticated and the initial page appears. Otherwise the Log In page appears.
   b. If the Log In page appears, enter your assigned user name and password.

      NOTE

      If you have a user ID in multiple business units, click Advanced. Specify the business unit that you want to log in to.
2. Click Home, Requests.

   NOTE

   The Requests page can either display requests directly or provide access to them indirectly through the My Requests drop-down list. Administrators optionally configure the page to use either setup.
3. Click the My Requests drop-down if the page does not list requests and select the option of your choice.
4. If applicable, click the list of requests that contains the request that you want to copy. For example, if you want to copy a completed request, click Completed Requests.
5. Click the name of the request that you want, and proceed as follows:
   – View and optionally update the request, as applicable.
   – Copy the request.

      NOTE

      The notes and attachments for the original request are not copied to the new request. The status of the new request is Not Submitted, so that you can delete the existing services or can add new services in the request.
      Update the fields as needed, including the *Requested By* and *Requested For* users.
   – To search for requests by request ID or according to the predefined categories on the screen, use the Request Lookup section of the Requests window. You can view only the requests that your role permits.

Review the following information to understand what you can do when you view and can modify requests:

All Statuses

- View the requested services and service options, and other details of the request.
  View the status, which appears in the Status column of any request list, for example, My Recent Requests, Completed Requests. The request status also appears when you view the details of a request.
- View the request status history, by clicking the name of the status in the request details or the Status column.

Status values appear for the entire request and for each service and service option in the request. Review the numbers, names, and meanings of default status values.

- Display and add and attachments.
- Refresh the display.
- Copy the request, as explained earlier in this topic.
- Email a copy of the request.
  To email the details of a request, click the Email button or link, specify the details, and click Send.
  Optionally click Include Request Snapshot to include details of the request in the email. Otherwise, the email contains only the text that you specify for the Subject line and the Message Body.
  In the address fields, you can enter email addresses, user IDs, account names, or the *last name, first name* values of user names. Separate multiple entries with a semi-colon (;). If you enter an account name, the Catalog system uses the email address on the account profile.
- View and edit the additional information sections.
- View the subscriptions for service options, if applicable.

Not Submitted Requests

- Submit the request.
- Add or delete services.
- Delete the request.

Open Requests

An open request is one whose status is not Completed or Cancelled.

- View the status history.
- View the details of a request.
- Edit the request.
- Email the request.

Administrators can configure the display of columns on the Open Requests page to display some or all of the following columns.

The default view displays the following columns:

- ID - Assigned by the system
- Name
- Requested By
- Requested For
- Priority
- Date Created
- Date Modified
- Status
- Actions

Click the icon to configure the display of columns.

Requests Pending Action

Request managers can approve, reject, fulfill, and perform several other actions for the requests pending action that are assigned to them.

Completed Status

- Cancel a requested service.
- Cancel the request which cancels all the services in the request.

Tracking

If permitted by your role and the configuration options set by your catalog administrator, you can view and click the Tracking tab. This tab displays any CA Process Automation process instances that are related to the requested services and service options. This tab also provides the status and other details of these processes.

Audit Trail

If permitted by your role and the configuration options set by your catalog administrator, you can view and click the Audit Trail tab. This tab displays each service option element and its status changes over time. Each service option is comprised of service option elements.

{"URL":["/usm/wpf?Node=icguinode.catalogrequestsearchpre"],"customLabelGetStarted":"Manage Requests","description":"IPCE_Manage_Requests"}

Request a Service

Users browse or search the catalog, view services, and request the services that they want. The entries in the service catalog that you view are managed by your service provider's catalog administrator. Your catalog administrator can choose to feature certain services from the catalog when you select a category or other service. If so, you can view the Featured Services section of the window appear when you select a related category or service.

The catalog is organized by a category hierarchy created by your catalog administrator. As you drill down into a category, the sub-categories contained in that category appear in the Browse section. Services can be contained in any category or in no category (shown by clicking the Not Categorized link in the initial Browse window). After you have selected a category that contains services, they appear in the Catalog Services section of the window. To further examine or select a service, click its name in the Catalog Services section of the window.

After you have selected a service by clicking its name, the Service Options page appears. This page displays all the service options available with the service. Related service options are organized into a service option group. A service can include several service option groups, each displayed in its own collapsible section of the Service Options page. Some service option groups require a choice of one or more options (presented using a radio button). Other service option groups allow you to select one or more check boxes with the option of not selecting any check boxes. The third type of service option group automatically includes the service options displayed and you cannot clear the service options.

Follow these steps:

1. Log in to CA Service Catalog, as follows:
   a. Enter the URL for CA Service Catalog, in the format http://*computer-name*:*port-number*. Your catalog administrator provides the URL.
      If CA Service Catalog is configured to use single sign-on, you are automatically authenticated and the initial page appears. Otherwise the Log In page appears.
   b. If the Log In page appears, enter your assigned user name and password.

      NOTE

      If you have a user ID in multiple business units, click Advanced. Specify the business unit that you want to log in to.
2. Select **Home**, **Requests**.
3. Browse or search the catalog to find the service you want.
   For example, if you search for "Asset Service", all the services that contain "Asset Service" in any of the following attributes are displayed in the search results:

- Service Offering Name
- Offering Description
- Option Group Name
- Option Group Description
- Option Name
- Option Description

4. Select the required service offering.
5. Select the options that you want. Complete the fields and confirm your selections as needed.
6. (Optional) Update the *Requested For* field if your role permits and if applicable. Request this service for a different user or account.
7. Click Submit if the service includes a Submit button (not a shopping cart).

NOTE

Service designers configure each service to use *either* a shopping cart or a Submit button as the method for users to request the service. If the service uses a cart, you can add more services to a cart, remove services, or save the cart.

If one or more service options or services in your request requires approval, the Catalog system sends them to the required request manager or managers.

Occasionally, you cannot submit a service. For example, because a form does not appear correctly or the Submit or Check Out button does not function. In such cases, contact your administrator. Your administrator can typically fix such problems. For example, by updating forms for the service or by updating configuration settings for your business unit or your user ID.

Add Notes or an Attachment

You can add notes or an attachment for a request or a service option, if the administrator for your business unit has enabled these features for your role. If you want to add notes for a request or service option but do not see the corresponding option on the UI, contact your administrator. The UI behavior varies slightly depending on whether the service uses a shopping cart or one-click submit. The service designer selects one of these options when configuring the details of the service.

NOTE

To enable these features, administrators select **Catalog**, **Configuration**, **Request Management Configuration**. To let users add notes for a request, use the *Access Control: Show General Information* option. To let users add notes for a service option, use the Allow Notes at Service Option Level option.

Follow these steps:

1. Click **Home**, **Requests**.
2. Perform the action that applies:
   - If the request is already submitted, click My Request, Open Requests, find the request, and open it.
   - If the request is not yet submitted and includes a service that uses a shopping cart click, Click Cart (if necessary) to display the request in the cart. Next, click Checkout.
   - If the request is not yet submitted and includes a service that uses one-click submit, go to the next step. You select such services from the catalog and submit them using the same single page. So, you add notes while completing the fields on this page.
3. Add notes or an attachment to a service option, as follows:
   a. Click Show Details in the My Selections box.
      This box appears near the top left portion of the screen.
   b. Find the service option and click the Add Note icon or the Add Attachment in the Action column.

c. Enter the text of the note or browse to select a file and attach. Click OK. For an attachment, verify the Display Name and also provide a Description.

4. Add notes or an attachment to the entire request, as follows:
   a. Verify that the service contains only one service option. If the service contains multiple service options, you cannot add notes to the entire request.
   b. Scroll down to the Notes box or the Attachments box on the right side of the page.
   The Notes box appears under the Request Information box and above the Attachments box.
   c. Click Add or browse to select a file and attach.
   d. Enter the text of the note and click OK. For an attachment, verify the Display Name and also provide a Description.

      NOTE
      Use any of the rich text options available on the Add Note dialog to enhance the appearance of your note. You cannot delete or alter a note after you click OK. To delete an attached file, locate the file in the Attachments box, scroll right, and click the Delete icon. To update the display name or description of an attached file, click the Edit icon.

   You have added notes and/or attachment.

      NOTE

      CA Service Catalog sends an email notification
      to end users and approvers when a note and/or attachment is added or edited for a request or an issue. CA Service Catalog also provides you the option of configuring the Request Details Link for E-Mail Notification. This capability is available only if you have installed the CA Service Catalog 14.1.01 patch updates from CA Support.

      For more information on Request Details Link for E-Mail Notifications parameter, refer to Request Management Configuration parameters.

Users browse or search the catalog, view services, and request the services that they want. The entries in the service catalog that you view are managed by your service provider's catalog administrator. Your catalog administrator can choose to feature certain services from the catalog when you select a category or other service. If so, you can view the Featured Services section of the window appear when you select a related category or service.

The catalog is organized by a category hierarchy created by your catalog administrator. As you drill down into a category, the sub-categories contained in that category appear in the Browse section. Services can be contained in any category or in no category (shown by clicking the Not Categorized link in the initial Browse window). After you have selected a category that contains services, they appear in the Catalog Services section of the window. To further examine or select a service, click its name in the Catalog Services section of the window.

After you have selected a service by clicking its name, the Service Options page appears. This page displays all the service options available with the service. Related service options are organized into a service option group. A service can include several service option groups, each displayed in its own collapsible section of the Service Options page. Some service option groups require a choice of one or more options (presented using a radio button). Other service option groups allow you to select one or more check boxes with the option of not selecting any check boxes. The third type of service option group automatically includes the service options displayed and you cannot clear the service options.

Follow these steps:

1. Log in to CA Service Catalog, as follows:
   a. Enter the URL for CA Service Catalog, in the format http://*computer-name*:*port-number*. Your catalog administrator provides the URL.
   If CA Service Catalog is configured to use single sign-on, you are automatically authenticated and the initial page appears. Otherwise the Log In page appears.
   b. If the Log In page appears, enter your assigned user name and password.

NOTE

If you have a user ID in multiple business units, click Advanced. Specify the business unit that you want to log in to.

2. Select **Home**, **Requests**.
3. Browse or search the catalog to find the service you want.
   For example, if you search for "Asset Service", all the services that contain "Asset Service" in any of the following attributes are displayed in the search results:
   - Service Offering Name
   - Offering Description
   - Option Group Name
   - Option Group Description
   - Option Name
   - Option Description
4. Select the required service offering.
5. Select the options that you want. Complete the fields and confirm your selections as needed.
6. (Optional) Update the *Requested For* field if your role permits and if applicable. Request this service for a different user or account.
7. Click Submit if the service includes a Submit button (not a shopping cart).

{"URL":["/usm/wpf?Node=iclaunchpad.my_requestscatalog"],"customLabelGetStarted":"Request A Service","description":"IPCE_Request_A_Service"}

Approve and Reject Requests Pending Action from a PDA

After the CA Service Catalog administrator has finished setting up CA Service Catalog for PDA approval, approvers can approve and reject requests from their PDAs. Depending on the setup specified by the administrator, you may be able to approve or reject any or all of the following in each request pending action you receive:

- Entire requests
- Individual services
- Individual service options

To approve and reject requests pending action from a PDA

1. Using your PDA, open the request approval email as soon as possible after you receive it.
2. Review the services and service options in the request and the assigned users. Decide whether to approve or reject the request, services, or service options assigned to you. Before doing so, you can optionally view further details in the request. Your organization can use policies to manage requests and thereby direct that only the users assigned by the policy (rather than any other users) approve or reject requests, services, and service options. In such cases, approve or reject only the individual services and service options explicitly assigned to you, even if you have proxy authority to approve or reject other users' requests pending action. Do not approve or reject an entire request pending action unless all services and all service options in the request are explicitly assigned to you.
3. Click the link for the action you want to perform.
   Depending on how the administrator set up the request approval email for PDA approval, you may see only a subset of these options, for example, only the options for approving or rejecting entire requests or only the options for approving or rejecting individual service options.
   Approve Request without further review
   This option approves all the requests pending action that are assigned to you.
   In addition, this option approves all other requests pending action remaining for the request, if your role permits you to do so, even if you are not the assignee. For example, if your role is Service Delivery administrator (named spadmin by default), you can approve all requests pending actions for all requests.
   Approve Service without further review

This option approves the highlighted service in the request pending action; the service assigned to you is highlighted in the request summary.

Approve Service Option without further review

This option approves the highlighted service option in the request pending action; the service option assigned to you is highlighted in the request summary.

Reject Request without further review

This option rejects all the requests pending action that are assigned to you.

In addition, this option rejects all other requests pending action remaining for the request, if your role permits you to do so, even if you are not the assignee. For example, if your role is Service Delivery administrator (named spadmin by default), you can reject all requests pending actions for all requests.

Reject Service without further review

This option rejects the highlighted service in the request pending action; the service assigned to you is highlighted in the request summary.

Reject Service Option without further review

This option rejects the highlighted service option in the request pending action; the service option assigned to you is highlighted in the request summary.

Review the details

This option displays the CA Service Catalog login page, so you can log in and review the additional information about the request to help decide whether to approve or reject it.

If you chose to log in to CA Service Catalog, complete the remaining steps.

4. Enter your user ID and password to log in to CA Service Catalog when prompted, if applicable, as follows:
   If your implementation uses direct email approval, you are not prompted or required to log in to CA Service Catalog to complete the action.
   a. If your user ID is pre-populated, enter your password.
   b. If the approval email is sent to a user group or to multiple individual users, then the user ID is not pre-populated, and you must enter both your user ID and password.
   If your implementation is using logged in approval, each option prompts and requires you to log in to CA Service Catalog. After your login credentials are validated, the action is completed, and a confirmation message displays.
5. Do the following if you require more information than you can view using the PDA, or if you cannot complete required fields on a form. These steps also apply if you are unable to approve or reject the request for any reason.
   Do not use your PDA to approve or reject any request, service, or service option that brings concerns or questions to mind, especially for services that include forms requiring your input. At publication time, browsers for some PDAs (especially older ones) display only limited information about the request.
   a. Cancel your action to use the PDA to approve or reject any service or request of concern.
   b. Log in to CA Service Catalog from a laptop or desktop computer.
   c. At that computer, review the complete request details and approve or reject the service option, service, or request of concern.
   d. If applicable, enter any input required on forms.
6. If applicable, click Logout to exit CA Service Catalog from your PDA.

PDA Approval

You can optionally use a PDA to review, approve, and reject requests.

Customize Your Notify Approver Action to Support PDA Approval

When you use CA Process Automation for approving CA Service Catalog requests, you typically use the Notify Approver action (or a customized version of it) to send request approval emails. For more information, see Customize Your Notify Approver Action to Support PDA Approval.

Manage Requests Pending Action

NOTE

After you have submitted a request, a message is displayed *Request Queued Successfully*. The request is queued for further processing. Additionally, on the Request Details page, click **Refresh** to verify the status of your submitted requests. Failed requests are shown as *Request Failed*.

In case of a failed request, you will receive a notification if you have configured the mail server option. It is recommended that you configure the mail server option to receive notifications. For more information, see Manage Rules.

WARNING

Warning! All active MQ messages are logged in: *USM_HOME\jms-data*

Do not delete the folder ending with **RequestQueue** as it contains submitted request data.

When a user submits a request, it enters the approval and subsequent fulfillment phase. Your service provider determines the business processes for approval and fulfillment of the services, and service options in the catalog. During approval and fulfillment, usually a person is assigned an approval or fulfillment task.

You can receive an email notification if a request approval or fulfillment task is assigned to you or to a group of which you are a member. You can view the requests which have an approval or fulfillment task that you are assigned. To do so, view your Pending Action list. Click the **Pending My Action** link in the **Request Lookup** section of the Requests window.

The Requests Pending Action list displays the list of requests which have an approval or fulfillment task that you are assigned. You can optionally filter this list of assignments by user name, group name, or all.

Request administrators can *search* for all requests, including requests that are assigned to other users. Request administrators have one of the following roles: Service Delivery Administrator, Super Business Unit Administrator, Administrator, Catalog Administrator, and Request Manager. If the requests are already assigned to a user for approval or fulfillment, administrators can either complete these requests or they can transfer them to other users.

Rules for Requests Pending Action

Here, the term *request pending action* refers to one or more services or service options in a request that are pending approval or pending fulfillment. In both single-service and multi-service requests, all services are part of the request, but the term request pending action applies only to the services or service options that are pending approval or pending fulfillment. Specifically, pending *approval* status applies to *services* only, while pending *fulfillment* status applies to *service options* only.

- Both CA Service Catalog administrators and non-administrator users can transfer or delegate requests pending action in their own queues to other users. When you either transfer or delegate a request pending action, the user who receives the request becomes the new assignee. The major difference between transferring and delegating follows: When you delegate a request, it stays in both your queue and the queue of the new assignee. When you transfer a request, it leaves your queue and moves to the queue of the new assignee.
- Thus, as an administrator, you can decide to delegate a request pending action if you want to monitor it within your own queue. Conversely, you can decide to transfer a request pending action if you do not want to monitor it. After you delegate a request, the new assignee (User B) or an administrator can transfer the request to someone else (User C). When User B or another transfers the request, the request moves from both your queue and User B's queue to User C's queue only.
- Only requests pending action that is assigned to a user can be delegated. When a request pending action is assigned to a user and a group, only the one that is assigned to a user can be delegated. The one assigned to a group cannot be delegated.
- A request pending action can be delegated only once. This limitation includes auto-delegation; a request that has already been delegated manually cannot be auto-delegated. Therefore, before you delegate a task to a user, ensure

that the user does not have auto-delegation enabled. However, the request can be transferred multiple times. You can delegate requests to an individual user only, not to a group.

- A request pending action is transferred or completed by the primary assignee of a pending action, by administrators, and by a user with the required access control settings. A request pending action can be transferred to only one user or one group.
- Both administrators and non-administrative users can also take ownership of a request pending action that is assigned to a group queue to which they belong. Users can return to the group queue a request pending action that they have previously taken but have not completed.
- A request pending action can only be returned by the primary owner of a taken pending action. The administrator (or any other user) cannot return a request pending action on another user's behalf.

To manage requests pending action, follow this process:

1. Act on requests pending action that can be assigned to you or other request managers, as follows:
   - Approve or reject
   - Fulfill
   - Transfer
   - Delegate or auto-delegate
2. Act on requests pending action that can be assigned to you, other request managers, or a group queue:
   - Take or return
   - Ignore, retry, or override alerts
   - Hold or resume
   - Cancel

Typically, the Status drop-down list includes several options. The names of these options and the order in which they appear are specified in the requestshared.xml file. For more information about how to update the values and the order of the options, see the Modify the Request Status List section.

NOTE

Warning messages can appear if a user with the Services Manager role handles a requests pending action, for example, by approving and rejecting requests. In such cases, the approvals and rejections proceed successfully even though the warning messages appear.

To prevent such warning messages from appearing, follow these steps:

1. Log in to CA Service Catalog as a Service Delivery administrator or business unit administrator.
2. Change the default access rights of the Services Manager role for a specific business unit, as follows:
   a. Log in to the business unit.
   b. Select **Catalog**, **Configuration**, **Request Management Configuration**.
   c. Add the **Access Control: Add Request** setting to this role.
   d. Save your changes.

Approve or Reject Requests Pending Action

Request managers approve and reject the requests pending action that is assigned to themselves or other request managers. Approving or rejecting requests pending action is an essential step in the process of managing requests from creation through fulfillment.

All CA Service Catalog users can approve or reject requests pending action that is assigned to them.

Follow these steps:

1. Click Home, Requests. If applicable, click the My Requests drop-down list on the Requests page to display requests. The Requests page can either display requests directly or provide access to them indirectly through the My Requests drop-down list. Administrators optionally configure the page to use either setup.
2. Click Pending My Action.
3. Locate the request and click the Approve/Reject icon in the Actions column.
4. View the details of requested services, including their statuses.
5. Focus on the services for which you are assigned an approval task.
   Decide which items in the request to approve or reject.
6. If you have *not* implemented multi-item approval, perform this step only.
7. In the Item Status drop-down list, select the appropriate status to approve or reject each service, and click OK. The statuses are typically named Approved and Rejected, respectively.
   – If you have *not* implemented discrete approval, the following rules apply:
     • Rejecting a service rejects the entire request. The request is returned to the user who submitted it. That user decides whether to update the request and resubmit it or accept the rejection as final.
     • Once all services in the request are approved, the request moves the request to the next phase of the request life cycle. For example, the second level of approval (if applicable) or the fulfillment stage. After you have approved a service, the approval processes of your service provider can require a second level of approval. In such a scenario, the second-level approver is assigned an approval task.
   – You can optionally implement discrete approval so that you can approve or reject the following *individually*:
     • Services in a request
     • Service options in a service

   If the approval task was assigned to more than one user or a group, once any assigned user completes that pending action, the request is removed from the Pending Action list for all assigned users. If the request contains multiple services, it is not removed until either all services are approved or one service is rejected.
   You are returned to the Request Details window. If you return to the Pending Action list, you can see that the request is removed from your list.
8. If you have implemented multi-item approval perform this step only.
9. Perform one or both of the following actions:

- – Select all or multiple services in the request and click Approve or Reject.
  This option appears when multi-item approval is implemented but discrete approval is *not* implemented.
  – Select all or multiple service options in each service and click Approve or Reject.
  This option appears when both multi-item approval and discrete approval are implemented.

Approve or Reject the Requests Pending Action of Other Users

Only CA Service Catalog administrators and other users with the required access control setting can approve or reject requests pending action that is assigned to other users. The access control setting *Access Control: Proxy Action* under Catalog, Configuration, Options, Request Management Configuration determines the such roles.

NOTE
For more information, see the section Set CA Service Catalog Configuration Options.

Follow these steps:

1. Click Home, Requests. If applicable, click the My Requests drop-down list on the Requests page. The Requests page can either display requests and the search box directly or provide access to them indirectly through the My Requests drop-down list. Administrators optionally configure the page to use either setup.
2. Search for requests, using the Advanced Search.
3. In the Query drop-down list, perform the following actions:
   a. Ensure that Action Request is selected.

b. (Optional) Enter the user ID of interest, in the Userid field.
c. Click Search.

4. Click the Approve/Reject icon next to the name of the service of interest, in the Actions column.

   NOTE
   The Item Status drop-down list is disabled. You enable it and use it in the next steps.

5. Click the Override (Push-Through) icon in the Actions column.
6. Confirm that you want to override a pending action that is not assigned to you.
   The Item Status drop-down list is enabled, meaning that you can now approve or reject each service in the request.
7. (If the request contains multiple services) Click the alert status icon. Enable the Item Status drop-down list for each service that you want to approve or reject.
   The status of each service option matches the status of the service during the approval phase of the request life cycle.
8. Decide which items in the request to approve or reject.
9. If you have *not* implemented multi-item approval, perform this step only.
   In the Item Status drop-down list, select the appropriate status to approve or reject each service, and click OK. The statuses are typically named Approved and Rejected, respectively.
   – If you have *not* implemented discrete approval, the following rules apply:
     • Rejecting a service rejects the entire request. The request is returned to the user who submitted it. That user decides whether to update the request and resubmit it or accept the rejection as final.
     • Once all services in the request are approved, the request moves the request to the next phase of the request life cycle. For example, the second level of approval (if applicable) or the fulfillment stage. After you have approved a service, the approval processes of your service provider can require a second level of approval. In such a scenario, the second-level approver is assigned an approval task.
   – You can optionally implement discrete approval so that you can approve or reject the following *individually*:
     • Services in a request
     • Service options in a service

   If the approval task was assigned to more than one user or a group, once any assigned user completes that pending action, the request is removed from the Pending Action list for all assigned users. If the request contains multiple services, it is not removed until either all services are approved or one service is rejected.
10. If you have implemented multi-item approval perform this step only.
    Perform one or both of the following actions:
    – Select all or multiple services in the request and click Approve or Reject.
      This option appears when multi-item approval is implemented but discrete approval is *not* implemented.
    – Select all or multiple service options in each service and click Approve or Reject.
      This option appears when both multi-item approval and discrete approval are implemented.

Fulfill Requests Pending Action

Request managers fulfill the approved requests pending actions that are assigned to themselves or other request managers. Fulfilling requests pending action is the final step in the process of managing requests from creation through fulfillment.

All CA Service Catalog users can fulfill requests pending action that is assigned to them.

Follow these steps:

1. Click Home, Requests. If applicable, click the My Requests drop-down list on the Requests page. The Requests page can either display requests directly or provide access to them indirectly through the My Requests drop-down list. Administrators optionally configure the page to use either setup.
2. Click Pending My Action.
3. Locate the request and click the Fulfill icon in the Actions column.
4. View the details of requested services. Focus on the services for which you are assigned a fulfillment task.

The status of each service option matches the status of the service during the fulfillment phase.

5. Decide and select the next status for each service for which you are assigned a fulfillment task. The exact statuses available can vary, based on the business processes of your service provider. Two significant statuses are Fulfilled and Fulfillment Cancelled, both of which end the fulfillment process for that service option.
   - If you are using CA APM and the service is eligible for asset assignment, select the available inventory of assets. Assign assets or indicate that no assets are available for assignment. Using the assign assets functionality changes the status of the related service option.
   - If you are *not* using CA APM or the service is *not* eligible for asset assignment, in the Item Status drop-down list, select the appropriate status for each service.
6. Optionally add notes or attachments as supporting documents.
7. When you are finished, click OK.
   You are returned to the Request Details window. If all pending action assigned to you have been completed, the request is removed from your list. If the fulfillment task was assigned to more than one user or a group, once any assigned user completes that pending action, the request is removed from the Pending Action list for all assigned users. If the request contains multiple services, it is not removed until either all services are either fulfilled or have their fulfillment that is cancelled.

   NOTE
   The fulfillment processes of your service provider can require more fulfillment steps. In such cases, the next fulfiller is assigned a fulfillment task.

Fulfill the Requests Pending Action of Other Users

Only CA Service Catalog administrators and other users with the required access control setting can fulfill requests pending action that is assigned to other users. The access control setting *Access Control: Proxy Action* determines such roles. This setting is under Catalog, Configuration, Options, Request Management Configuration.

Follow these steps:

1. Click Home, Requests. If applicable, click the My Requests drop-down list on the Requests page. The Requests page can either display requests and the search box directly or provide access to them indirectly through the My Requests drop-down list. Administrators optionally configure the page to use either setup.
2. Search for requests, using the Advanced Search.
3. In the Query drop-down list, perform the following actions:
   a. Ensure that Action Request is selected.
   b. (Optional) Enter the user ID of interest in the Userid field
   c. Click Search.
4. Click the name of the request of interest to open it.
5. View the details of requested services. Focus on the services whose fulfillment status you want to update.
   The status of each service option matches the status of the service during the fulfillment phase.
6. Click the alert status icon in the Actions column.
7. Confirm that you want to override a pending action that is not assigned to you, when prompted.
   The Item Status drop-down list is enabled, meaning that you can now update the fulfillment status of the service.
8. Click the alert status icon. Enable the Item Status drop-down list for each service whose fulfillment status you want to update, if the request contains multiple service. The status of each service option matches the status of the service during the fulfillment phase.
9. Decide and select the next status for each service whose fulfillment status you want to update. The exact statuses available can vary, based on the business processes of your service provider. Two significant statuses are Fulfilled and Fulfillment Cancelled, both of which end the fulfillment process for that service option.

- If you are using CA APM and the service is eligible for asset assignment, select the available inventory of assets. Assign assets or indicate that no assets are available for assignment. Using the assign assets functionality changes the status of the related service option.
- If you are *not* using CA APM or the service is *not* eligible for asset assignment, in the Item Status drop-down list, select the appropriate status for each relevant service.

10. Optionally add notes or attachments as supporting documents, and click OK.
    You are returned to the Request Details window. If you return to the Pending Action list after all pending action assigned to you have been completed, the request is removed from your list. If the fulfillment task was assigned to more than one user or a group, once any assigned user completes that pending action, the request is removed from the Pending Action list for all assigned users. If the request contains multiple services, it is not removed until either all services are either fulfilled or have their fulfillment cancelled.

    NOTE
    The fulfillment processes of your service provider can require more fulfillment steps. In such cases, the next fulfiller is assigned a fulfillment task.

Transfer Requests Pending Action

Request managers can transfer the requests pending actions that are assigned to themselves or other request managers. You can transfer a request pending action that is assigned to you or another user. For example, when you or other user become sick or otherwise are unavailable to perform or approve an assigned task.

All users can transfer their own requests pending action.

Follow these steps:

1. Click Home, Requests.
2. Click Pending My Action.
   To open the request of interest, click the request name.
   The Requests Details window appears, displaying the details of the request you just opened.
   The details for the selected request appear, with a Requested Services section at the top of the screen. That section lists all services in the request. If the request contains multiple services, you can transfer them either individually or in groups of two or more.

   NOTE
   A service can be transferred if you can select it. If you cannot select the service, it cannot be transferred. For example, a multiservice request can contain one or more services that are not pending approval or fulfillment and therefore cannot be transferred.
3. Select the service or services and click Transfer.
   The Search User window appears.
4. Review the list of users and select *one* user to whom the selected services are transferred.
   You return to the Requested Services section. In the Item Status column, the selected request or requests are marked for Transfer to *username*. *username* is the user to whom you transferred those requests.
5. Verify that the correct user is designated for the transfer and click OK.

Transfer the Request Pending Action of Other Users

Only CA Service Catalog administrators and other users with the required access control setting can transfer requests pending action that is assigned to other users. The access control setting *Access Control: Proxy Action* determines such roles. This setting is under Catalog, Configuration, Options, Request Management Configuration The *Access Control: Transfer Request Pending Action* setting determines the roles that have access to the Transfer button. This button is hidden from other roles.

Follow these steps:

1. Click Home, Requests. If applicable, click the My Requests drop-down list on the Requests page. The Requests page can either display requests and the search box directly or provide access to them indirectly through the My Requests drop-down list. Administrators optionally configure the page to use either setup.
   Search for requests, using the Advanced Search.
2. In the Query drop-down list, perform the following actions:
   a. Ensure that Action Request is selected.
   b. (Optional) Enter the user ID of interest, in the Userid field.
   c. Click Search.
3. To open the request, click the request name.
   The Requests Details window appears. The Requested Services section lists all services in the request. If the request contains multiple services, you can transfer them either individually or in groups of two or more.

   NOTE
   A service can be transferred if you can select it. If you cannot select the service, it cannot be transferred. For example, a multiservice request can contain one or more services that are not pending approval or fulfillment and therefore cannot be transferred.
4. Click the Transfer button at the top of the window.

   NOTE

   The Item Status drop-down list is disabled. You enable it and use it in the next steps.
5. Click the alert status icon in the Actions column.
6. Confirm that you want to override a pending action not assigned to you, when prompted.
7. If the request contains multiple services, click the alert status icon and enable the Item Status drop-down list for each service that you want to approve or reject.
   The status of each service option matches the status of the service during the approval phase.
8. Select the service and click Transfer.
   The Search User window appears.
9. Review the list of users and select *one* user to whom the selected services are transferred.
   You return to the Requested Services section. In the Item Status column, the selected request or requests are marked for Transfer to *username. username* is the user to whom you transferred those requests.
10. Verify that the correct user is designated for the transfer and click OK.

Delegate or Auto-Delegate Request Pending Action

Request managers can *delegate* their requests pending action. Request managers can *auto-delegate* requests pending actions that are assigned to themselves or other request managers. For example, when you or others are unavailable to perform or approve an assigned task. Auto-delegation is useful for managers who monitor requests but who do not address requests directly. Thus, auto-delegation helps ensure that requests are addressed in a timely manner. A request pending action can be delegated only once.

Only the owner of the request pending action can delegate, if the owner has the required access control setting. The access control setting named **Access Control: Delegate Request Pending Action**. This setting is under Catalog, Configuration, Options, Request Management Configuration.

Follow these steps:

1. Click Home, Requests. If applicable, click the My Requests drop-down list on the Requests page to display requests. The Requests page can either display requests directly or provide access to them indirectly through the My Requests drop-down list. Administrators optionally configure the page to use either setup.
2. Perform *one* of the following actions:
   – Click **Pending** in **My Request(s)**.

Click the **Delegate** button for the request pending approval or fulfillment.

– Click **Find Requests** in **My Request(s)**.
  a. Search in the Request Lookup section.
  b. Verify that **Action By User ID** is selected.
  c. Enter your user ID in the search field and click **Search**.
     The Search Results window appears, displaying all requests that are assigned to you.
  d. Click the **Delegate** button for a request pending approval or fulfillment.

The details for the selected request appear, with a Requested Services section at the top of the screen. That section lists all services and service options in the request. If the request contains multiple services and service options, they can be delegated either individually or in groups.

NOTE

A service can be delegated if you can select it. If you cannot select the service, it cannot be delegated. For example, a multi-service request can contain one or more services or service options that are not pending approval or fulfillment. Such a request cannot be delegated.

3. Select the service or service options and click **Delegate**.
   The Search User Accounts window appears.

   NOTE
   The Search User scope is based on the administration configuration of either all users or only users in the same business unit.
4. Review the list of users and select *one* user to whom you delegate the selected services or service options.
   You return to the Requested Services section. In the Item Status column, the selected service or service options are marked for delegation to *username. username* is the user whom you selected.
5. Verify that the correct user is named for the delegation and click OK.

Use auto-delegation and specify all of your newly assigned or transferred requests pending action are automatically delegated to another user.

NOTE
The access control setting **Access Control: Allow Request Auto-Delegation via User Profile** determines whether users can view and change auto-delegation settings, according to their user roles.

Follow these steps:

1. Move the cursor to the top right portion on the screen. Click the Profile button next to the Help button.
   The User Profile window appears; the fields are read-only.
2. Click the Edit button.
3. Click the down-arrow icon next to the Request Auto-Delegation heading.
   The Request Auto-Delegation setting for your user ID opens, so that you can view and optionally change it.
4. To remove an existing auto-delegation (if applicable), click the red "minus sign" icon.
5. To add or change auto-delegation, click the magnifying glass icon.
   The Search and Select User Search dialog appears.
6. Click the name of the user to designate for auto-delegation.
   The dialog closes, and you return to the Edit User Profile window. The user that you selected is named in the Delegate field.
7. Verify that the correct user is designated for auto-delegation and click OK. Click Done.

Auto-Delegate the Requests Pending Action of Other Users

Use auto-delegation and specify that any *new* requests assigned to a specific user are automatically delegated to a different user. Here, *new* request means any newly assigned or transferred pending action.

Follow these steps:

1. Click Administration, Users.
2. Enter search criteria in the User ID field to find the user whose requests you want to auto-delegate to another user.
3. Click the name of the user whose requests you want to auto-delegate, and click Edit.
4. Click the drop-down icon next to the Request Auto-Delegation heading.
   The user in the Delegate field is the one to whom requests are currently auto-delegated.
5. To remove an existing auto-delegation (if applicable), click the red "minus sign" icon.
6. To add or change auto-delegation, click the magnifying glass icon, and review the list of users.
7. Click the name of the user to designate for auto-delegation.
8. Verify that the correct user is designated for auto-delegation and click OK. Click Done.

Take or Return a Request Pending Action

From the Take/Return window, you can take or return ownership of requests pending action. You can take ownership from a group queue or the queue of another user in the same business unit or sub-business unit. Similarly, you can return ownership of requests pending action from your queue to the queue of your group. You can take or return ownership of requests pending action that is assigned to your group.

The Take and Return functions work *only* if you are logged in as a user of the group to which the request pending action is assigned. Only CA Service Catalog administrators and other users with the required access control setting can do so. The access control option *Access Control: Take/Return Request Pending Action* under Catalog, Configuration, Options, Request Management Configuration determines such roles. For more information about setting this option, see the Request Management Configuration section.

Follow these steps:

1. Click Home, Requests. If applicable, click the My Requests drop-down list on the Requests page to display requests.
   The Requests page can either display requests directly or provide access to them indirectly through the My Requests drop-down list. Administrators optionally configure the page to use either setup.
2. Click Pending My Action.
3. Click the Take/Return button for a request pending approval or fulfillment.
   The Take/Return window shows the details for the selected request, including a Requested Services section. That section lists all services and service options in the request. If the request contains multiple services or service options, they can be taken or returned individually or in groups.

   NOTE
   A service or service option can be taken or returned if you can select it. If you cannot select the service or service option, it cannot be taken or returned. For example, a multi-service request can contain one or more services or service options that are not pending approval or fulfillment. Therefore, the service or service option cannot be taken or returned.
4. (To *take* one or more services from a request pending action) Select the services or service options. Click Take.
   The request is marked and your user ID can take it.
   If a request is already taken, the status indicating “taken by *username*” appears. Therefore, you cannot take it. Other users in the same group can still approve or fulfill a taken service, if necessary. Either the user who has taken the service must return it or an authorized user must transfer it to the new assigned user. If a request is not taken but the Take option is not available, then the action is not assigned to your group.
5. (To *return* one or more service or service options from a request pending action) Select the services or service options. Click Return.
   The only services that you can return are services already marked as taken by your user ID.
6. Click OK to perform the take or return action.
   Your action is confirmed: Any requests that you have taken are marked as taken by your user ID. Similarly, any requests that you have returned are no longer marked as taken.

Ignore, Retry, or Override Alerts

During the submitted state or the approval process, requests can occasionally become marked with an alert status indicating a potential problem. For example, because of a system error or a user error. The alert is indicated in the Status column of several request windows. If the role of a user does not permit access to this option, then that user cannot see the status, even if the user's request is stuck. We recommend that you ignore, retry, or override (push through) alerts as soon as you become aware of them. Only CA Service Catalog administrators and other users with the required access control setting can ignore, retry, or override alerts.

The access control option *Access Control: Push Through Request* determines such roles. This setting is under Catalog, Configuration, Options, Request Management Configuration. The access control option named *Access Control: Display Request Warning* determines which roles are able to see the warning icon indicating that a request is stuck.

Follow these steps:

1. Click Home, Requests. If applicable, click the My Requests drop-down list on the Requests page to display requests. The Requests page can either display requests directly or provide access to them indirectly through the My Requests drop-down list. Administrators optionally configure the page to use either setup.
2. Find the "alert" request on any of the following windows: Open Requests, Completed Requests, Pending My Action, Request Search, or Recent Requests Lists.
   The request can be marked with an alert status, because of one or more information-only alerts that are logged during the request flow. In this case, ignore the alerts, as follows:
   a. Open the window that includes the alert request.
   b. In the Action column, click the Actions drop-down and select Push Through from the list.
      The Push Through Request window appears, and the Push Through Request tab is selected by default.

      NOTE
      The details for the selected request appear, with a Requested Services section at the top of the screen. That section lists all services in the request. If the request contains multiple services, they can be pushed through. You can push them either individually or in groups of two or more at a time.
   c. In the Override Status column, select the status for the request to acquire from the drop-down list.
   d. Click Save.
3. For each stuck service, click the Item Status drop-down list. Select the Approval or Fulfillment status that best meets your needs.
   The options for any service vary, according to its status. To view all available options, use the drop-down list.
4. Click OK.
   Each service whose status you changed has its alerts that are removed (overridden). The service is "pushed through" to the selected status.

   NOTE
   If a request is in queued status before being pushed through, it carries its (queued) prefix state along with new status of the request. Push Through can be performed for both queued and nonqueued requests.

   The alerts are made inactive, and the status is removed.
   When the status of a requested service is either Pending or Completed, you can either ignore or push through the alert.

Hold and Resume Requests, Services, or Service Options

Holding requests, services, or service options ensures to anyone monitoring a request that the processing of the request, service, or service option has been *intentionally* suspended. Only CA Service Catalog administrators and other users with the required access control setting can hold and resume requests.

The access control setting *Access Control: Hold/Resume Request* determines such roles. This setting is under Catalog, Configuration, Options, Request Management Configuration. For more information about setting this option, see the Request Management Configuration section.

How Hold and Resume Actions Work

Reasons for holding a request include:

- The temporary unavailability of the requestor or approver.
- The temporary unavailability of requested items, such as the items being in back-ordered status.

When a service option is held, the monitoring of time stops for any request SLAs attached to the service option. The request SLAs are stopped to prevent related SLA warnings and SLA violations from being issued.

When an item is held, no status changes can occur for it until you *resume* it. That is, you change its status to Resume. When you resume an item, it returns to its last previous status before it was held.

If you hold only selected services within a request or only selected service options within a service, the services or services options that you did not hold continue the normal request life cycle, except that they cannot reach Completed status until the held items are resumed and are also completed.

When resumed, service options that were formerly held, move to a temporary Resume status. The service options then automatically move back to the status they were at before they put on Hold. Checking Resume status is useful for history purposes. Doing so helps you set up optional rules and actions, automated processing using CA Process Automation, or automatic email notifications.

Automatic Email Notifications

You can use the events, rules, and actions for sending automatic email notifications when a request, service, or service option is held or resumed.

The events *Request Change* and *Request/Subscription Item Change* include two rules that are related to hold-resume functions:

- When Status is Hold
- When Status is Resumed

For the *Request Change* event, the *When Status is Hold* rule includes an action that sends an automatic email notification when a request is held. Similarly, the *When Status is Resumed* rule includes an action that sends an automatic email notification when a request is resumed.

For the *Request/Subscription Item Change* event, the *When Status is Hold* rule includes an action that sends an automatic email notification when a service option is held. Similarly, the *When* Status is Resumed rule sends includes an action that sends an automatic email notification when a service option is resumed.

Hold and Resume Requests, Services, or Service Options

Follow these steps:

1. Click Home, Requests. If applicable, click the My Requests drop-down list on the Requests page to display requests.
   The Requests page can either display requests directly or provide access to them indirectly through the My Requests drop-down list. Administrators optionally configure the page to use either setup.
2. Click Pending My Action or Open Requests. Alternatively, click Advanced Search for requests.
   The Requests Pending Action window appears, or the Open Requests window appears.
   Click the Hold/Resume icon for the request you want.
   The Hold/Resume window appears including a Requested Services section. That section lists all services and service options in the request. If the request contains multiple services or service options, you can hold or resume them either individually or in groups.
3. To *hold* one or more services or service options, select the services or service options and click Hold.
   Each service option that you selected is marked *to be held* by your user ID.

If a request is already held, the status indicating “held by *username*” appears next to the Item Status field of the request item. Therefore, you cannot hold it again. The *username* identifies the user who has held the request.

4. To *hold* an entire request, select the All check box for the request and click Hold.
   All service options in the request are marked *to be held* by your user ID.
5. Add a note to the request for the services or service options that you held explaining why you held them. Provide any information that helps other administrators determine when to resume a service or service option.
6. Verify that all reasons have been addressed before resuming the request.
7. To *resume* one or more services or service options, select the services or service options and click Resume.

   NOTE
   When you resume a previously held service or service option, the monitoring of time is resumed for any request SLA attached to the service or service option.

   The only services that you can resume are those services already marked as held.
8. Click Save to perform the hold or resume action.

Cancel a Request

Administrators can cancel a request for several possible reasons. For example, the wrong items were requested or the requested items are no longer wanted or needed.

Only CA Service Catalog administrators and other users with the required access control setting can see the Cancel button. The access control setting *Access Control: Cancel Request* determines such roles. The setting is under Catalog, Configuration, Options, Request Management Configuration. For more information, see the Set CA Service Catalog Configuration Options section.

You can cancel a request *only* if the status of the request has not *exceeded or is equal to* the status that is specified by *Allow Cancellation Through* configuration option. You can set this configuration option in the CA Service Catalog User Interface, navigate to **Catalog**, **Configuration**, **Options**, **Request Management Configuration**. This setting defines the status through which a request can be canceled. After the request moves to the next status, it cannot be canceled.

NOTE
By default, for the event *Request/Subscription Item Change*, for the rule *When Status is Canceled*, an action is enabled. The action aborts CA Process Automation instances when a request is canceled. For best performance, keep this default setting. For more information about setting this option, see the Set CA Service Catalog Configuration Options section.

Follow these steps:

1. Click Home, Requests. If applicable, click the My Requests drop-down list on the Requests page to display requests. The Requests page can either display requests directly or provide access to them indirectly through the My Requests drop-down list. Administrators optionally configure the page to use either setup.
2. Click the list of requests that contains the request that you want to cancel. For example, if you want to cancel a request pending action, click Pending My Action.
3. Click the request that you want, and click Cancel.

   NOTE
   If the requirements for cancellation are not met, the Cancel button is disabled. The request cannot be canceled. Moreover, items in the same request can be at different statuses. Some items may have exceeded the *Allow Cancellation Through* status while others have not yet done so. In such cases, you can cancel *only* the items that have *not* exceeded the *Allow Cancellation Through* status.
4. Confirm the cancellation and click OK.

Delegate Catalog

This article contains the following topics:

Delegating the use of your catalog to another user (a delegate) lets the delegate browse your catalog. The delegate can also create and submit requests on behalf of you (the delegator). Administrators assign this right to individual roles by setting the configuration option named *Access Control: Add Request* in the Request Management Configuration section of the Administration Configuration page.

The catalog of a delegator typically contains services that are not available in the catalog of the delegate.

Delegators are typically executives or managers but can be any user whose role permits creating and submitting requests. Delegates are typically employees who report to the delegator. But the delegates can also be any user whose role permits creating and submitting requests.

> **NOTE**
> Typically, a delegate creates and submits requests for the delegator from the catalog of the delegator. In such a scenario, the *Requested For field* shows the name of the delegator. The field cannot be changed.

History, auditing, and logging records for a request show that the delegate created and submitted the request on behalf of the delegator.

Comparison to Related Features

Delegation of catalogs is related to, but not the same as, the following features:

- Requesting a service for another user, using your own catalog. Specify the name of the other user in the *Requested For* field when you view and change more information in the request before submitting it.
- Delegating a request. You manually delegate a request pending action, from either yourself or another request manager, to a different request manager. The delegated request manager can then approve, reject, or transfer the request.
- Auto-delegating your own requests pending action or auto-delegating requests pending action of other users. The auto-delegated request manager can then approve, reject, or transfer the request.

In contrast to these features, delegating the use of your catalog lets another user browse your catalog. The other user can create, edit, and submit requests on your behalf.

Limitations of Catalog Delegation

The following limitations apply when you delegate the use of your catalog to another user:

- Delegates must create and submit requests themselves from the catalog of the delegator catalog.
- Delegates *cannot* change the Requested For field to request services for any user other than the delegator.
- Users who have a role that cannot create a request (such as Service Manager) *can* delegate their catalogs. However, delegates of such users *cannot* browse or create requests from the catalogs of those delegators. Therefore, as a best practice, users whose roles prohibit them from creating requests do *not* delegate their catalogs.
- All users can delegate the use of *their own catalog* to another user. If you have the Super Business Unit Administrator or Service Delivery Administrator role, you can delegate the use of either *your own catalog* or *the catalog of any other user ID* to another user. If an administrator delegates the use of your catalog to another user, the Catalog system does not automatically notify you or the other user. However, in such cases, the delegate does appear in the list of delegates on your User Profile page. You can manually remove that delegate. As a best practice, administrators who perform such actions notify affected users personally.

  > **WARNING**
  > The delegate must have a role in the business unit of the delegator. Otherwise, the catalog of the delegator is not made available to the delegate.

Considerations for Business Units

You can delegate your catalog to a user in your own business unit or any other business unit. For example, the top-level (service provider) business unit.

If you delegate your catalog to a user in the same business unit, the delegate can use your catalog immediately. When the delegate logs in to CA Service Catalog, the delegate can see and select your name in the *Use Catalog Of* field of the Requests home page. (This page appears when you select Home, Requests). Then, the delegate can browse and use your catalog.

When you delegate your catalog to a user in another business unit, you must have a role in that business unit. Otherwise, the delegate cannot see and select your name in the *Use Catalog Of* field of the Requests home page and cannot use your catalog. If necessary, you or an administrator can edit your user profile to add a role for your user ID in that business unit.

For example, suppose that your organization is grouped into business units according to department. If you work in the Finance department, and you delegate use of your catalog to a user in the Human Resources department (a different business unit). Verify that you have a role in Human Resources.

Sample Scenarios

The following scenarios illustrate possible instances of delegating the use of a catalog--either your own catalog or another catalog.

- **Scenario 1**
  A CIO delegates her catalog to her executive assistant, with instructions to create and submit requests for ten items on her behalf.
  The items are not available in the assistant's catalog, so the CIO delegates use of her catalog to the assistant. The assistant uses the CIO's catalog as a delegate and requests the ten items for the CIO.
  History, auditing, and logging records for the requests show that the assistant created and submitted the requests on behalf of the CIO.
- **Scenario 2**
  An administrator (User A) delegates the use of the catalog of another user (User B) to a different user (User C). Doing so requires the Super Business Unit Administrator or Service Delivery Administrator role.
  In this scenario, the CIO requires the ten items that are requested not for herself but for a vice president who reports to her. That vice president is on leave without access to CA Service Catalog but needs the items when returning to work next week.
  In this scenario, the CIO delegates the use of the vice president's catalog (not her own catalog) to her assistant. Before the vice president returns to work, the assistant uses the vice president's catalog as a delegate and requests the ten items for the vice president.
  Scenario 3
  The CIO begins to create a request for herself but must stop and save the request to address an emergency issue. When she realizes that the issue requires long, uninterrupted effort, she stops the effort. She cancels the request, and instructs her assistant to create and submit the request as a delegate.
  History, auditing, and logging records for the request show that the delegator created the request, but the assistant edited and submitted the request on behalf of the delegator.

Administrators configure CA Service Catalog to let users (*delegators*) delegate their catalog to other users (*delegates*). Delegates work on behalf of delegators who are unable or unavailable to create and submit their own requests. The catalog of a delegator typically contains services that are not available in the catalog of a delegate.

1. Administrators configure CA Service Catalog to enable the delegation of catalogs.
2. Delegators edit their profile to delegate the use of their catalogs to one or more delegates.
   The delegates use the catalog of a delegator to create, edit, and submit requests.
   If necessary, delegates, delegators, and administrators work together to manage unsubmitted requests that delegates create.

Step 1 - Enable or Disable Delegation of Catalogs

Users with the Super Business Unit Administrator or Service Delivery Administrator role can enable or disable delegation of catalogs within the business unit--and any subbusiness units--to which their role applies. Users with other roles can optionally delegate the use of their catalogs to any other user. Delegation of catalogs is enabled by default, but enable it explicitly if it was disabled earlier.

Follow these steps:

1. Click Catalog, Configuration, Request Management Configuration.
2. Locate the option *Enable Delegation of Catalog*. Set this option to Yes to enable the feature or No to disable it.
   - If this option is set to No, enable it. To enable it, click the Edit icon, select the check box, and click Update Configuration.
   - If this option is set to Yes, disable it. To disable it, click the Edit icon, clear the check box, and click Update Configuration.

When this option is enabled, no users are *required to* delegate use of their catalogs. When this option is enabled, users can *optionally* delegate use of their catalogs.

> **NOTE**
>
> Delegating a catalog does *not* affect the Catalog configuration of the CA Service Catalog, such as the *Use Service Provider Catalog Only* and *Pass Through Catalog* parameters. Similarly, the delegation of a catalog is *not* directly related to the configuration options named *Access Control: Add Request*, *Access Control: Edit Request that also exists* in the same Request Management Configuration section as Enable Delegation of Catalog.

Step 2 - Delegate the Use of a Catalog

When you delegate the use of your catalog to other users, they can browse your catalog. They can also create and submit requests on your behalf. Typically, you delegate the use of your catalog to other users when you are unavailable to perform this work yourself but the work must continue with minimal supervision. As a prerequisite, review the limitations of delegation of catalogs.

Follow these steps:

1. Verify with your Super Business Unit Administrator or Service Delivery Administrator that this feature is enabled.
2. Click Home, Administration, Users.
3. Click the Search button on the Users search dialog.
   The user IDs that you can update appear.
4. Click the Edit icon for the user ID you want to edit.
5. Click the Open icon for the Delegate Use of Catalog box.
6. Click the Search icon next to the Delegates field.
7. Click the user ID to which you want to delegate use of your catalog.
8. Click the Add icon.
   The user ID you selected is moved from the Delegates field to the list of delegates underneath it. The user ID is added to your list of delegates.

   > **WARNING**
   > The delegate *must* have a role in the business unit of the delegator. Otherwise, the catalog of the delegator is *not* made available to the delegate.

   Also:
   - Delegates *cannot* edit and submit existing requests that the delegator creates.
   - A single user can be a delegate for multiple delegators.
   - Delegates can create and submit requests using the catalog of the delegator.

9. If applicable, add another delegate by clicking the Search icon and repeating the previous two steps.
10. When you are finished adding delegates, click OK.

If you delegated the use of your own catalog, all delegates you selected can browse your catalog. They also can create, edit, and submit requests on your behalf. If you are a Super Business Unit Administrator or Service Delivery Administrator who delegated the use of the catalog of another user, the delegates you selected can browse the catalog. They also can create, edit, and submit requests on behalf of that user.

Step 3 - Use the Catalog of a Delegator

When another user specifies you as a delegate, you can use the catalog of the delegator instead of your own. The catalog of the delegator likely includes more services and service options than your own. You can access the catalog and can browse it. You can use the catalog to create, edit, and submit requests on behalf of the delegator.

Follow these steps:

1. Click Home, Requests.
2. Click the Use Catalog Of drop-down list.
   The list displays your delegators, the users for whom you are a delegate in your currently logged in business unit or one of its subbusiness units.
   If no users appear:
   - Contact users for whom you are a delegate. Verify that they have delegated the use of their catalogs to you.
   - Verify the business unit in which the delegator has specified you as a delegate. Verify that you have a role in the business unit.
   - Verify that the delegation of catalogs option is enabled.
3. Select the delegator.

You can now browse the catalog of the delegator and create, edit, and submit requests on behalf of the delegator. History, auditing, and logging records show that the delegate created, edited, and submitted the request on behalf of the delegator.

Remove a Delegate

When you remove a user from your list of delegates, the user can no longer browse your catalog. The user also cannot create and submit requests on your behalf. Remove a user from your list of delegates when appropriate for business reasons. For example: the delegate leaves the organization, or you transfer to another department, or you no longer need the user as a delegate. Typically, you remove delegates from using your catalog. If you are a Super Business Unit Administrator or Service Delivery Administrator, you can remove delegates from using the catalog of another user.

Follow these steps:

1. Click Home, Administration, Users.
2. Click the Search button on the Users search dialog.
3. Click the Edit icon for the user ID you want to edit.
   To reveal the list of delegates for a user, select the Open icon for the Delegate Use of Catalog check box.
4. Select the user or users whom you want to remove as delegates.
5. Click the Delete icon.
6. Click OK.

Step 4 - Manage Unsubmitted Requests Created by Delegates

A delegate can create and save requests using a catalog of a delegator. The delegate can also submit the requests later, on behalf of the delegator. However, before the delegate submits the request, the following events can occur:

- The delegator (or an administrator) removes a delegate from using the catalog of the delegator.
- An administrator disables the delegation of catalogs in one or more business units or throughout the Catalog system.

In both cases, delegates are *not* notified automatically, and delegates are no longer able to submit the requests. The only exception is that delegates whose roles permit them to override alerts can optionally "push through" the request to the

next status and so, submit the request. Similarly, delegates can optionally delete such requests, because delegates own the requests.

Delegators can view requests that are made on their behalf (whether submitted or not) in the Open Requests queue. However, delegators *cannot* delete or submit such requests. The only exception is that delegators whose roles permit them to override alerts can optionally "push through" the request to the next status and so, submit the request.

Delegators and delegates decide together how to manage saved but requests not submitted, so that such requests do not remain open. Options are as follows:

- Delete the request. Only the delegate can delete the request.
- Push through the request. The delegate, delegator, or an administrator with the required role (upon request).
- Temporarily make the user as a delegate.
- Ask an administrator to re-enable the delegation of catalogs in your business unit long enough for the delegate to submit the request.

Change Management

Change Management for CA SDM is a set of features for Change Managers and Change Advisory Board (CAB) members to coordinate the review and approval of change requests for CI components and services. For example, Change Managers can review and approve all changes to CI components and services to ensure that no new security vulnerabilities are introduced into the production environment. The Change Manager leads the CAB and is responsible for the ultimate approval of change requests.

Change Management includes the following components:

- **Change Calendar:** Displays a calendar view. The calendar also displays black-out dates, which are freeze periods. Users can create a change order from the context menu on the daily, weekly, and monthly calendar views. The Change Manager, Level 2 Analysts, and CAB members use this feature.
- **Change Scheduler:** Displays scheduled time periods during which CI changes can or cannot occur in the Change Calendar. The Change Manager uses this feature to view and create change schedules and CI associations during the time period.
- **Conflict Analysis and Collision Detection:** Analyze change orders to help identify potential implementation conflicts that could increase risk of failure. The Change Manager uses this feature.
- **CAB Console and Reporting:** Displays a dashboard that facilitates quick online approvals of change orders that require CAB approval. In CA SDM, the Change Manager can generate a report with details about the proposed requests for changes ready for CAB review and electronically distribute the reports to all CAB members.

  NOTE
  Users with appropriate privileges can display Compliance, Forecast, Trend, and Volume predefined reports for change management in Business Intelligence Launch Pad with CA Business Intelligence.
- **Risk Assessment:** Provides the ability to attach risk assessments for each change request submitted. The Risk Survey Option lets you create surveys to evaluate risks, and associate them with change categories. The risk survey lists a series of single or multiple choice questions.
- **Impact Explorer:** Displays CI-to-CI relationships for CIs that are attached to a change order. Impact Explorer is launched from the Change Order Detail page to facilitate on-demand impact analysis. The Change Manager uses this feature.
- **Change Verification:** Verifies that changes execute accurately and no unauthorized changes occur, so that CMDB contains an accurate and current representation of all managed CIs.

Configuration Audit and Control Facility

This article contains the following topics:

Configuration Audit and Control Facility (CACF) combines Change Management, Configuration Management (CMDB), and Discovery Management to provide change verification. Change verification helps ensure that the CMDB reflects any changes accurately, and the Discovery Management tools verify the changes.

Select the Configuration Management tab to define change specifications, CIs, and view the Verification log.

You administer the CACF components in the Configuration Control node in the CMDB section of the Administration tab and define and view CACF change specifications from the Change Order, CI and Incident forms.

The CMDB Administrator defines the CIs and attributes to manage and defines the policy for updating those CIs and attributes.

The Administrator must consider allowing standard changes, defining authoritative and trusted sources of data, and defining which CIs and CI attributes are under CACF management. If an incorrectly executed change or rogue change occurs (also referred to as a variance), CA SDM can accept the new value, create an Incident, or copy the data to the TWA for later processing. CA SDM can also use any combination of these actions.

> **WARNING**
> The Configuration and Change Administrators *must* define a change verification strategy for your environment. For more information, see Planning and Implementing Change Verification topic.

How to Define Policies for Change Verification

Complete the following recommended steps to begin using CACF in your CA SDM environment:

1. Review the Managed Change States in the CA SDM environment. Update the default Managed Change States or add states as necessary to suit your change management strategy.
   Change verification initiates when the status of a Change Order changes to a Change State that CACF manages, such as Verification in Progress.
2. Identify the available CI Managed Attributes that you want CACF to manage for change verification.
   - For example, you want to manage changes to Memory Installed (phys_mem) so that the attribute *must* have a matching Change Order for the test* server. You also consider changes to IP Address (alarm_id) as normal and always allowed.
   - For example, you want to manage changes to Memory Installed (phys_mem), MAC Address (mac_address), and IP Address (alarm_id) so that the attribute must have a matching Change Order for prod*. You do not want MDR1 to update IP Address and prefer MDR2 to complete this update. When you have a problem, create an Incident. Closing the Change Order also closes the Incidents automatically.
3. Determine what CIs or subset of the CIs in your system that you want to monitor.
   For example, you want to control all servers name test* and prod*. In addition, you control what CIs based on CI Class and CI Location.
4. Determine what MDRs and sources of data that you want to monitor.
   For example, you want to restrict data from MDR1 for the IP Address attribute from updating any CIs.
5. Determine the appropriate verification policy to manage the following types of change:
   - The CI changes that match change orders.
   - The CI changes that match the attributes in a Change Order, but do not match the value.
   - The CI changes that you consider as rogue updates.
   - Then CI changes that you consider as rogue inserts.
6. Determine the appropriate update behavior that CACF takes when it detects a variance for a policy:

- Always allow the attribute update request.
- Allow the attribute update only when a corresponding change specification is specified for the update.
- Always cancel the entire transaction.
- Keep the existing attribute value.
- Write data to the TWA.
- Create an Incident when CACF detects a variance.
- Automatically close any Incidents after Change Order verification.

NOTE

For more information about change verification planning and implementation, see Planning and Implementing Change Verification.

Example Change Specification in the Pending Verification Status

In this example, the Configuration Administrator established a verification policy for the Memory Installed (phys_mem) attribute to Allow Update Only if Matches Change Specification. CACF monitors a change specification for this attribute with the status Verification Pending, and sets the Change Specification to the Verified or Failed Verification status.

The following steps describe the lifecycle of this change specification:

1. A Change Order requests an update to the Memory Installed (phys_mem) attribute of a CI.
2. You create a change specification for the Memory Installed (phys_mem) Managed Attribute with a Planned Value of 4gb with the initial status as Verification Pending.
3. The Change Order moves to a status with change verification active such as Verification in Progress.
4. The change specification waits for a discovery tool to discover CI details and export the values to the CMDB.
5. One of the following actions occurs after the discovery tool exports the value to the CMDB:
   - The value for Memory Installed matches the 4gb value that you specified in the change specification. CACF sets the status from Verification Pending to Verified. CACF closes the Change Order.
   - The value for Memory Installed does not match the value that you specified. CACF sets the status from Verification Pending to Failed Verification.

Example Change Specification in the Verified Status

In this example, CACF verified a change specification and set the status to Verified. This change specification contained an initial status such as Verification Pending.

The following steps describe the lifecycle of this change specification:

1. A Change Order requests an update to the Serial Number (serial_number) attribute of a CI.
2. Create a change specification for the Serial Number managed attribute and enter **12345** as the planned value.
3. The Change Order moves to a managed state so that Change Verification is active.
   For example, Verification in Progress.
4. The change specification waits for a discovery tool to discover CI details and imports the updated Serial Number values to the CMDB.
5. The discovery tool exports the value to the CMDB and matches the value that you specified in the change specifications.
6. CACF sets the status to Verified.
   Note: If the CI update to the attribute does not match the planned value, CACF considers the update as a variance and sets the status to Failed Verification.

Example Change Specification in the Set After Change Verified Status

In this example, CACF updates the CMDB with the value that you want to use after the Change Order exits a managed change state with change verification active. For example, your environment uses Verification in Progress as the managed change state with change verification active enabled.

The following steps describe the lifecycle of this change specification:

1. A Change Order requests an update to the Serial Number (serial_number) attribute of a CI.
2. You create a change specification, specify Serial Number, and enter a planned value for an attribute.
3. You set the Change Order state to from RFC to Verification in Progress to Closed.
4. The CI attribute value is updated with the planned value after the Change Order exits a state with Change Verification Active (Verification In Progress) to a state that does not have change verification active (Closed).
5. The change specification sets to Was Set to Planned Value to indicate that the set was completed successfully.

Example Accept the Planned Value

In this example, the Change Analyst researches the correct attribute value for a CI and accepts the planned value.

Follow these steps:

1. Open a Change Specification that an end user created in your environment.
2. View the detail page to see the planned value for the CI attribute that the Change Analyst modified.
3. Research the attribute value for the CI.
   For example, your research indicates that the Change Analyst entered the correct value.
4. Click Accept Planned Value to update the CI with the corresponding planned value.
   CACF sets the state of the Change Specification to Accepted Planned Value.

Example Accept the Discovered Value

In this example, the Change Analyst determines that the CI has been updated to a correct attribute value for a CI and accepts the discovered value.

Follow these steps:

1. Open a Change Specification that a Change Analyst created in your environment.
2. View the detail page to see the planned value for the CI attribute that the Change Analyst specified.
3. Research the attribute value for the CI.
   For example, discovery indicates that the Change Analyst entered the incorrect value.
4. Click Accept Discovered Value.
   The CI updates with the discovered value and CACF sets the status of the change specification from Verification Pending to Accepted Discovered Value.
5. If you configured the managed state to enable the Promote Change Order After Verification option, CACF closes the Change Order after all change specifications are in a final state.

Example Manage Changes from an Unauthorized MDR

In this example, the Configuration Administrator established a verification policy to manage changes from an unauthorized MDR named MDR1. The policy rejects all updates to CIs from MDR1.

The following steps describe the lifecycle of this verification policy:

1. You create a Verification Policy with an MDR Name pattern of MDR1 and an Update Behavior of Always Cancel Entire Transaction.
2. Enable all Change Order alignment options.
3. Specify Any Managed Attribute and an asterisk (*) for all filters.
4. When the discovery tool (MDR1) runs and exports the data to the CMDB, the verification policy rejects all the updates.

How to Archive and Purge Audit Data

We recommend that you archive and purge obsolete Verification Log entries, CACF audit history, and CI audit history as part of your periodic database maintenance.

WARNING
The CACF CA Business Intelligence reports typically present monthly reports with a yearly summary. The default archive purge time value for the archive purge rules for the log, CACF audit history, and CI audit history tables is 30 days. Use the same value in the archive and purge rules to help ensure data consistency between verification log entries, Incidents, and Change Orders. You can probably change the 30 day default to something more in keeping with your reporting requirements.

Complete the following actions to archive and purge audit data:

1. Use the current Incident rule to archive and purge inactive Incidents older than *nn* days.
   CA SDM archives and purges verification log entries only after archiving and purging associated Incidents. If you associate a Change order with an Incident, CA SDM does not check if the Change Order is active.
2. Use the current Change Order rule to archive and purge inactive Change Orders older than *nn* days.
   CA SDM archives and purges verification log entries and change specifications only after archiving and purging associated Change Orders. If you associated an Incident with a Change Order, CA SDM does not archive and purge the Change Order.
3. Use the Rogue Change Verification Log rule for rogue changes that did not create an Incident.
   Use this rule to archive and purge verification log entries older than *nn* days. By definition, rogue changes are not associated with Change Orders.
4. Use the CMDB Audit rule to archive information that is shown in the Change Specification History, Verification Policy History, Managed Change State History, and Managed Attribute History tabs show in the respective Change Specifications, Verification Policy, Managed Change State, and Managed Attributes detail forms. CA SDM stores this information in the ci_audit table.

Change Manager Responsibilities

This article contains the following topics:

The Change Manager is responsible for the overall enterprise change management process and for the ultimate approval of change orders. They also generate the change management metrics analysis reports and they do the following tasks:

- Review change requests and add appropriate stakeholders and approvers as needed.
- Facilitate resolution of issues, such as detecting collisions and scheduling conflicts in the calendar.
- Review installation, back-out, and fallback plans for accessibility and soundness.
- Understand the risk of each change and ensure that the appropriate risk level is assigned to the change.
- Monitor changes for their respective areas to ensure that they comply with Technology Change Management requirements.
- Represent their respective areas and communicate the impact of high-level risk changes at weekly CAB meeting.
- Facilitate in reviews after the installation is complete for problem installations and failed changes.
- Serve as the escalation point for change requesters, stakeholders, approvers, implementers, and support groups.

How the Change Manager Role Works

Change Managers are responsible for monitoring change orders to ensure that the operation team members comply with business policies and processes. The Change Calendar can facilitate resolution of issues such as change order conflicts by scheduling black-out dates and freeze periods during which a CI or a set of CIs can be changed. Change Managers also do the following tasks:

1. Oversee the CAB Console from which online and quick approvals of change orders and requests for changes are managed.
2. Organize CABs with members appropriate for the change orders under consideration and conduct regularly scheduled CAB meetings to review incoming change orders.
3. Create reports with details about the proposed requests for changes ready for CAB review and electronically distribute the reports to all CAB members.

4. Perform a real-time review of each request for change and update the record with the CAB decision during the CAB meeting.
5. Use BusinessObjects Business Intelligence Launch Pad to manage Compliance, Forecast, Trend, and Volume reports and create on-demand reports.
6. Represent their respective areas and communicate the impact of high-level risk changes at CAB meetings.
7. Evaluate the risk of each change and ensure that the appropriate risk level is assigned to the change.

Define Tasks for the Change Manager Role

You can define tasks for the Change Manager role.

Follow these steps:

1. On the Administration tab, select Security and Role Management, Role Management, Role List.
   The Role List page appears.
2. Select the Change Manager role.
   The Change Manager Role Detail page appears.
3. Click Edit.
   The Change Manager Update Role page appears.
4. Use the following tabs and fields to configure tasks and access permissions for the Change Manager role:
   - Authorization
   - Function Access
   - Web Interface
   - Knowledge Management
   - KT Document Visibility
   - Tabs
   - Report Web Forms
   - Go Resources
   - Tenant Read Access
   - Tenant Write Access
5. Click Save, Close Window.
   The Change Manager role record is updated.

Configure Change Manager Options

You can configure options for the Change Manager role.

Follow these steps:

1. Select Options Manager on the Administration tab.
2. Expand the Change Order Mgr node.
   The Option List appears.
3. Right-click the option that you want and select Edit from the context menu.
   The Update Option page appears.
4. Edit the option as appropriate.
5. Click Save, Close Window.
   The updated option appears on the Options list.

Change Categories, Status, and Risk Levels

You can define how change orders operate within your service environment. You can edit the default values that are installed with CA SDM, or can define your own.

Follow these steps:

1. Select Service Desk, Change Orders on the Administration tab.
2. Expand the Change Order node and select *one* of the following items:
   - Categories
   - Change Types
   - Closure Codes
   - Conflict Status
   - Risk Level
   - Risk Survey
   - Status
   - Workflow Task Status Code
   - Workflow Task Types

   The List page for the selected item appears.
3. Select the item to edit.
   The Update Details page appears.
4. Use the controls available on the tabs at the bottom of the page to define how change orders operate within your environment.
5. Click Save, Close Window.
   The updated item appears in the list.

View the Change Order Scoreboard

The Change Order scoreboard shows the change orders, conflicts, and scheduled tasks that are assigned to Level 2 Analysts, Change Managers, Change Coordinators, or CAB members. Users can view their assigned and unassigned records by priority.

Follow these steps:

1. Navigate to Change Orders on the CA SDM scoreboard.
2. Expand the folders to reveal nested folders for the following items:
   - Assigned or unassigned Open and Closed items
   - Resolved or unresolved conflicts
   - Scheduled tasks that start today or next week.
3. Select the folder for the items you want to see.
   The List page appears.
4. (Optional) Click Show Filter and complete one or more of the fields to specify search criteria that restrict the list to comments of interest.
5. Click Search.
   The List page displays summaries of the items that match your search criteria.
6. (Optional) Click the Edit in List button to display some additional fields that can be associated with an item.

Define a Change Order Stored Query

Defining the stored queries that are available to users on the Change Order scoreboard is an administrative task. You can modify the predefined stored queries that are installed with CA SDM, or can define your own.

Follow these steps:

1. Select Service Desk, Application Data, Stored Queries on the Administration tab.
   The Stored Query List appears.
2. Select the stored query that you want to edit.
   The Stored Query Detail page appears.

3. Click Edit.
   The Update Stored Query page appears.
4. Edit the field values as appropriate.
5. Click Save, Close Window.
   The updated stored query appears in the Stored Query List.

Example: Define a Stored Query to List Change Orders Assigned to a CAB to Which the Logged In User Belongs

This example demonstrates how you can create a stored query that lists only change orders that are assigned to a CAB to which the logged in user belongs.

Follow these steps:

1. Navigate the Scoreboard to the Update Stored Query page.
2. Edit the field values as follows:
   a. Select Scoreboard Usage.
   b. Set Type to Change Order.
   c. Enter the Where clause:
   ```
   cab.[group]group_list.member IN (@cnt.id) AND active = 1
   ```
3. Click Save, Close Window.
   The stored query appears in the Stored Query List.

CAB Responsibilities and CAB Groups

CAB Responsibilities

The Change Advisory Board (CAB) coordinates the review and approval or rejection of change requests for CI components and services. The CAB members are responsible for the following tasks:

- Reviewing all major changes to production systems.
- Attending all relevant CAB meetings as required by the Change Manager.
- Reviewing all submitted requests for changes to determine their impact, resources that are required to implement them and any ongoing costs.
- Participating in scheduling and coordination of the Change Calendar.
- Helping to ensure that all changes are adequately assessed and prioritized.
- Participating in reviews after the installation is complete.

CAB Approvals

If the CAB Approval field is set to YES, the CAB must approve a change order before implementing the change order. During the approval process, the change manager clicks Approve or Reject. If you want to use different status values, you can update the Approve or Reject button code using Web Screen Painter.

How the CAB Process Works

The CAB is responsible for reviewing all major changes to production systems. The CAB members are notified that a change request requires their approval, and they take the following actions:

1. List change requests that the CAB must review.
2. Open the CAB Console and view the information that is contained in the request.
3. View the business justification, implementation plan, configuration items, and supporting documentation that is associated with the request and decide to approve it or reject it.
4. Approve or reject the change order, and the next change order in the list appears automatically.
   The person requesting that the change is notified automatically that the CAB status is updated for the change request.

Manage CAB Groups

You can create and manage the CAB groups with members appropriate for the change orders under consideration. The CAB can include members from the application team, development manager, component owner, QA, support, and any additional parties deemed necessary.

NOTE
Before you implement a CAB group, configure the appropriate contacts for your business structure.

Create a CAB Group

1. On the Administration tab, select Groups.
   The Group Search page appears.
2. Click Create New.
   The Create New Group page appears.
3. Complete the fields as appropriate.
4. Click Save.
   The CAB Group appears on the Group List page.

Assign Members to the CAB Group

1. On the Group Detail page, select the Members tab.
2. Click Update Members.
   The Contact Search page displays.
3. Enter the search criteria to display the desired contacts and click Search.
   The Members Update page displays, listing the contacts that matched the search criteria.
4. From the list on the left, select the contacts that you want to assign to this group. To select multiple items, hold down the Ctrl key while clicking the left mouse button.
5. When you have selected all the contacts that you want, click the selection button (>>).
   The selected contacts move to the Members list on the right.
6. Click OK.
   The Group Detail page displays, with the selected contacts listed on the Members tab.

Add the CAB Group to a Change Category

NOTE
Ensure that the Category_Defaults option is installed, so that the CAB field on change orders defaults to a CAB group.

1. On the Administration tab, select CA SDM, Change Orders, Categories.
   The Change Category List appears.
2. Select a category from the list.
   The Update Change Category page appears.
3. Select the appropriate CAB group from the CAB field.
4. Complete other fields as appropriate.
5. Click Save.
   The Change Category Detail page displays a successful save message.
6. Click Close Window.
   The CAB group is associated with the change category and the change order.

CAB Console and Reporting

Contents

The CAB Console is a dashboard that facilitates online and quick approvals of change orders that require CAB approval. The Change Manager and other CAB members use the console to view details about a change order (and its associated Workflow tasks and configuration items) and approve or reject the change order. The CAB Console lets team members review, approve or reject a change order, and continue to the next change order quickly. For change order requests, the CAB can deal with the request by satisfying it directly or by escalating/referring the request to an appropriate group.

The Change Manager can use CA SDM built-in summary and detail reporting options to accomplish the following tasks:

- Report approved and rejected change orders.
- Report change orders awaiting approval.

To print or view summary and detail reports, first select the records that you want to include in the report. You can select specific records for a report using the search feature of the list pages.

For example, from the Change Order list you can search for a list of change orders that are awaiting CAB approval that you can then use to generate a report.

Work with the CAB Console

The CAB Console provides Change Managers and CAB members with a visual rendering of the approval process for a selected change request. The CAB deals with the request in the most appropriate manner, either by satisfying the request directly or by escalating/referring the request to an appropriate group.

CAB Status Update Methods

Change Managers can update the CAB status for change orders (and associated CA Workflow tasks) CAB status, without opening each change and updating the details. Change Managers can launch the CAB Console to display details of change orders, CA Workflow tasks, or workitems, depending on the type of approval the group follows, and approve or reject them. The Change Manager can enter comments while the approvals or rejections are in-progress.

> **WARNING**
>
> Change and issue categories can use either CA SDM internal (classic) workflow system or the CA Workflow application. The CAB approval process uses the classic workflow system. The CAB Console lets you list and view CA Workflow workitems; however, updating workitems for Approvals is dependent on the definition of CA Workflow.

Approve or Reject Change Orders

Change Managers can quickly approve or reject change orders that are awaiting CAB approval, without needing to open each change order.

Follow these steps:

1. On the Service Desk tab, search for change orders that include values to indicate that CAB Approval is required and that the approval is not completed. For example, you can use the following values:
   - YES for the CAB Approval drop-down list.
   - Not Approved for the Status field.

   The Change Order List page shows change orders that are awaiting CAB approval.
2. Do one of the following steps:
   - Click CAB Console.
     The CAB Console appears and shows the change order details in read-only form for the first change order in the list.
   - Right-click any change order and select CAB Console from the shortcut menu.
     The CAB Console appears and shows the change order details in read-only form for the change order you selected.
3. Review the change order details and decide whether to approve or reject the change order.
4. (Optional) Enter the comments about the approval or rejection in the Comments field.

5. Click Approve Change or Reject Change.
   The first change order status is changed, any comments are saved, and the details for the next change order in the list appear automatically.

   NOTE
   You can use the Previous and Next buttons to navigate to next or previous change orders without updating the change.
6. Continue reviewing change orders, and approving or rejecting them until you reach the final change order in the list.
   The change orders are approved or rejected and their statuses are changed to Approved or Rejected, respectively.
7. Close the CAB Console.

Change CAB Console Properties

Using Web Screen Painter, you can change the CAB Console properties that appear on web forms in the CAB Console. For example, you can do the following actions:

NOTE

For more information, see Using the Web Screen Painter.

- Rename the Approve and Reject buttons. This button is the property on the order_approval_console.htmpl (change orders) web forms.
- Modify the status values of the Approve and Reject buttons by changing the 'REJ' or 'APR' values.
  - The ‘Reject Task’ button calls a function approve_reject(‘REJ’).
  - The ‘Approve Task’ button calls a function approve_reject(‘APR’).

WARNING
You can only associate an active transition with a button. Do not deactivate a predefined status transition that is associated with a button. Otherwise, the predefined status transition no longer functions.

Follow these steps:

1. In Web Screen Painter, open the CAB Console form that you want to change.
   Web Screen Painter opens and displays the form.
2. On the Design tab, right-click the control that you want to change, and select Properties.
   The Properties - *control* page appears.
3. Change the properties that you want by entering a new value for each one.
   Changes take effect as soon as you click outside the field, and when you close the Properties page.
   The Web Screen Painter displays a brief summary of the significance of a property in a note that appears at the bottom of the Properties form when you select the property.

Example: Modify the CAB Console for Change Workflow Tasks

This example shows how to modify the task status using Web Screen Painter.

1. In Web Screen Painter, open the orderwf_approval_console.htmpl web form.
   Web Screen Painter opens and displays the form.
2. On the Design tab, right-click the Reject Task button, and select Properties.
   The Properties - Button page opens.
3. Find the function approve_reject(‘REJ’).
   ‘REJ’ is the status code for the Task status Reject.
4. Enter a new value for 'REJ'.
   Changes take effect as soon as you click outside the field, and when you close the Properties page.
   The Web Screen Painter displays a brief summary of the significance of a property in a note that appears at the bottom of the Properties form when you select the property.

Change Management Reporting

The Change Manager with appropriate privileges can use Business Intelligence Launch Pad to accomplish the following tasks:

- Report change volume by operating system, change category, group, implementer, risk, status, implementation date, affected configuration items (CIs), and changes originating from incident or problem tickets.
- Report successfully implemented changes grouped by change category, urgency, priority, impact, % successful vs. total for the specified period, and for the group of the change requestor
- Report failed changes grouped by category, urgency, priority, impact, reason for back out, % failed vs. total for the specified period, and the change requestor's group.
- Report the total number of change requests grouped by change category Change Coordinator, Change Manager, risk level, priority, and affected CIs for a specific time period.

Conflict Analysis and Collision Detection

Conflict analysis detects and shows collisions that occur when two or more changes to the same configuration item are scheduled for implementation at the same time. The change management team handles scheduling collisions in the following ways:

- Uses the change calendar to see RFC-related CI collisions and makes adjustments to them to reduce potential impact.
- Searches for and detects collisions in a log.

You use the conflict status to help manage and identify conflicts. CA SDM provides the following default conflict statuses:

- **Researching**

Indicates the conflict is being researched by a user.

- **Resolved**

Indicates the conflict was researched and resolved by a user.

- **Unresolved**

Indicates the conflicts was not researched and resolved.

Resolve Scheduling Collisions

Conflict analysis detects and shows collisions that occur when two or more changes to the same configuration item are scheduled for implementation at the same time. You handle scheduling collisions as follows:

1. Navigate to the Change Order Detail page, and click the Conflicts tab.
   The Conflict List page appears.
2. Click Conflict Analysis.
   CA SDM detects scheduling collisions and the Conflict List page lists Schedule Collisions.

   NOTE
   The Conflict List and Change Calendar report any collisions to change orders. Each entry in the list or calendar represents a single collision that puts that change order at risk of failure. Collisions are detected only *after* you click Conflict Analysis for a change order.
3. Click Edit, and then click the Scheduler button.
   The Schedule for Change Order page appears and lists CIs that have scheduling collisions.
4. Investigate and resolve the collisions by updating the Schedule Start Date and Schedule End Date of a change order.

   NOTE
   The implementation schedule is the period between the Schedule Start Date and Schedule End Date of a change order.

5. Record your investigation of the collision.

Detect and Investigate Conflicts

You can detect and investigate change order conflicts to resolve scheduling problems.

Follow these steps:

1. Click the Conflicts tab on a change order.
   The Conflict List page appears.
2. Click Conflict Analysis.

   NOTE
   You can use the Search facility to search the Conflict List, however, searching does not run Conflict Analysis. Click Conflict Analysis to create or remove Conflict List entries.

   The conflicts that are detected for the change order are listed. The following listed information is not self-explanatory:
   - **Type Conflicting Change**
     Displays the type of change that is causing a conflict, for example, Scheduling Collision.
   - **Configuration Item Conflicting Change Summary**
     Identifies the configuration item that is causing the conflict and displays the configuration item summary.
3. Click a change or configuration item link.
   The details about the change or configuration item appear.
4. Investigate the cause of the conflict.
5. (Optional) Adjust the change schedule.

   NOTE
   The change schedule is the period between the Schedule Start Date and Schedule End Date. You have detected and investigated conflicts.

Define Conflict Logging

You can configure the Activity Log to generate entries when conflicts are created or updated so that you can track them. The Activity Log is predefined to generate entries when a conflict is updated. The Options Manager controls the Activity Log.

Follow these steps:

1. On the Administration tab, navigate to Options Manager, Change Order Mgr.
   The Option List page appears.
2. Install the following options:
   - **conflict added**
     Generates the Activity Log entries when conflicts are added.
   - **conflict updated**
     Generates the Activity Log entries when conflicts are updated.
3. Save and restart the CA SDM server.
   Conflict logging is defined.

Report Change Order Conflicts

You can report details about change orders including information about change order conflicts, such as the conflict type, the change order that is affected by a conflict, and other conflict details.

Follow these steps:

1. Select the Service Desk tab.
2. Search for change orders.

3. Click Reports, Details.
   The Change Order Detail report appears.
4. (Optional) Scroll toward the end of the report to see conflict details.

Implement the Risk Survey

This article contains the following topics:

Risk assessments let you identify, evaluate, and quantify the risks of change orders that belong to change categories, before modifying a system or service in your environment. You create risk surveys to evaluate risks, and associate the surveys with change categories. When a user creates a change order and specifies a change category, the survey that is associated with that category is available for completion and submission.

The risk survey lists a series of single or multiple choice questions. Each answer has a weightage value. When creating a change order, the user selects the appropriate answers and submits the survey. The evaluated risk level is based on the weightages of answers that are selected by the user.

Implement the risk survey as follows:

1. Establish risk levels for your organization.
2. Create a risk survey or select from the default list. Create or modify risk survey questions and answers.
3. Modify risk ranges for the risk survey.
4. Associate the risk survey with a change category.
5. After you associate a risk survey with a change category, the Risk Survey button appears for the requester when they save a change order using the specified change category.
6. View the evaluated risk based on the risk survey results.
7. (Optional) Override the evaluated risk value from the Activities menu.

Create a Risk Survey

You can create a risk survey to assess risks for change orders that belong to a change category.

Follow these steps:

1. On the Administration tab, click Service Desk, Change Orders, Risk Survey.
   The Risk Survey Template List appears.
2. Click Create New.
   The Create New Risk Survey page appears.
3. Complete the following fields:
   - **Risk Survey Name**
     Specifies the name of the risk survey.
   - **Include Comments**
     Specifies to include comments in the survey.
   - **Comment Label**
     Specifies the caption to display for comments in the survey.
   - **Active?**
     Specifies if the survey is active or inactive.
   - **Risk Survey Description**
     Specifies a description of the risk survey.
4. Click Save.
   The Risk Survey Detail page appears. You can add questions and answers to the survey.

Add Risk Survey Question

You can add questions to a survey to help assess risks for change orders that belong to a change category.

Follow these steps:

1. Open a risk survey, click the Questions tab, and click Add Question.
   The Create New Risk Survey Question Template page appears.
2. Complete the following fields:
   - **Sequence**
     Defines the order in which the question appears on the risk survey.
   - **Include Comments**
     Adds a text box to the end of the survey form for customer free-form comments.
   - **Comment Label**
     Adds a label to the comments text box.
   - **Active?**
     Specifies if the question is active or inactive.
   - **Multiple Response Question?**
     Indicates that this question has multiple answers.
   - **Question Text**
     Specifies the question text that appears in the survey.
3. Click Save.
   The question is added to the risk survey.
4. Close the page or create answers.

Add a Risk Survey Answer

After you save a risk survey question, you can add answers to the question to help assess risks for change orders that belong to a change category.

Follow these steps:

1. Open a risk survey, click the Answers tab, and click Add Answer.
   The Create New Risk Survey Answer Template appears.
2. Complete the following fields:
   - **Sequence**
     Defines the order in which the answer appears on the risk survey question.
   - **Active?**
     Specifies if the answer is active or inactive.
   - **Weightage**
     Specifies a numerical value that calculates risk.
   - **Answer Text**
     Specifies the answer text that appears in the survey.
3. Click Save.
   The Risk Survey Question Template detail page appears and the Answers tab updates with your answer.
4. Close the page or create more answers.

View a Default Risk Survey

CA SDM provides default risk surveys that you can associate with change categories.

Follow these steps:

1. Navigate to Service Desk, Change Orders, Risk Survey.
2. Select General from the Risk Survey Name column.
   The General Risk Survey Detail page appears..
3. Click View Survey.

The risk survey appears, listing questions, answers, and weightage values for each answer. This survey applies to general changes within your organization.
4. Close the survey.
5. The General Risk Survey Detail page appears; close it.

Modify Risk Ranges

You can update risk range values for a survey. The sum of all selected answer weightages are compared with each risk range and the risk level are calculated.

Follow these steps:

1. Click Edit on the Risk Survey Template Detail page.
2. Click Edit in List to edit list items of risk range.
   The Minimum and Maximum Value fields appear.
3. Click on each row to modify minimum and maximum values. The values cannot overlap or you are unable to save the risk survey. Specify the range of values for each category of change that is based on the questions and their weight as follows:
   - High Risk Level -- 21-25
   - Medium-High Risk Level -- 16-20
   - Medium Risk Level -- 8-15
   - Low Risk Level -- 0-7
4. On the Risk Range tab, click Save(Y).
   The Risk Range List page updates.
5. Set the risk survey to Active.
6. Save and close the window.

Associate Risk Survey with a Change Category

You can associate a risk survey to help assess risk for change orders that belong to a change category.

Follow these steps:

1. Navigate to Service Desk, Change Orders, Categories.
   The Change Category List appears.
2. Click a change category in the Symbol column.
   The Change Category Detail page appears.
3. Click Edit.
   The Update Change Category page appears.
4. Click Risk Survey.
   The Risk Survey Template Search appears.
5. Search for the appropriate survey.
   Select the survey from the Risk Survey Name column.
   The Update Change Category page appears.
6. Click Accept.
   The change category updates and is associated with the risk survey.

Example: Deploy a Risk Survey

This example demonstrates how to deploy a default risk survey on your system using a default change category. **Follow these steps:**

1. Navigate to Service Desk, Change Orders, Categories.
   The Change Category List appears.

2. Select Add.IT.Other from the Symbol column.
   The Add.IT.Other Category detail page appears.
3. Click Edit.
   The Add.IT.Other Update Change Category page appears.
4. Click Risk Survey.
   The Risk Survey Template Search appears.
5. Search for the risk survey. In this example, search for General and select the risk survey to add it to the detail form.
   The Risk Survey field populates with General.
6. Save and close the window.

If a user creates a change order using the Add.IT.Other category, the Risk Survey button when they save the change order.

View a Risk Survey

You can view risk surveys that are associated with change orders. If a user submitted the risk survey for that change order, you can view the survey answers to assess the change order risk.

To view a risk survey, **follow these steps:**

1. Open the change order.
   The Change Order Detail page appears.
2. Click View Survey.
   The survey appears.

Update Risk Range

You can update risk range values for a survey. The sum of all selected answer weightages are compared with each risk range and the risk level are calculated.

To update a risk range

1. Click Edit on the Risk Survey Template Detail page.
2. Click Edit in List to edit list items of risk range.
   The Minimum and Maximum Value fields appear.
3. Click on each row to modify minimum and maximum values.

   NOTE

   The values cannot overlap or you are unable to save the risk survey.
4. On the Risk Range tab, click Save(Y).
   The Risk Range List page updates.
5. Set the risk survey to Active.
6. Save and close the window.

Impact Explorer

This article contains the following topics:

Impact Explorer is an advanced tool for managing and controlling change within an organization. Impact Explorer allows the Change Manager to explore CIs that are attached to a change order and also interact directly with an attached CI or its child CIs.

Impact Explorer provides these advantages:

- Displays all CIs that are attached to a change order.
- Displays all child and peer-to-peer relationships for attached CIs.
- Provides the ability to attach any related CI to the change order.
- Displays a Descendants List of successively related CIs for any attached CI.
- Provides the ability to launch CMDB Visualizer for a CI.

Launch Impact Explorer

You can access Impact Explorer from the Config. Items tab of any change order.

Follow these steps:

1. Open a Change Order Detail page.
2. Select the Config. Items tab.
3. Click Impact Explorer.
   The Impact Explorer page appears.

   NOTE
   If you close the Change Order Detail page, the Impact Explorer page also closes. If Impact Explorer is launched from multiple change orders, multiple Impact Explorer pages are displayed.

Explore Attached CIs

The Impact Explorer tree contains a node for each CI that is attached to a change order. A plus (+) symbol indicates that a CI has at least one child CI.

NOTE
If more than 100 CIs are attached to the change order, only the first 100 are displayed. To view the next 100 CIs, click More...

To display the relationships for an attached CI, click the plus (+) symbol for the CI. The related CIs appear in the tree. In addition to the related CI names, the tree also displays relationship types in brackets.

View a CI in Impact Explorer

The Impact Explorer tree displays a node for each CI that is linked to a change order. A plus (+) symbol indicates that a CI has at least one child CI.

Follow these steps:

1. Click a CI node in the left pane.
   The Configuration Item Detail page appears in the right pane.

   NOTE
   You can also display the CI Detail page in the right pane by right-clicking on the CI and selecting View from its context menu.
2. Click the change order node at the top of the left pane.
   The Change Order Detail page appears.

   NOTE
   If the change order has more than 100 attached CIs, only the first 100 are displayed. To view the next 100 CIs, click More...

Add a Related CI to a Change Order

To maintain CI relationships, you may want to attach a related CI to the change order.

Follow these steps:

1. Click the Config. Items tab for a change order.
   The Configuration Items List page appears.
2. Click Impact Explorer.
   The Impact Explorer tree displays attached CIs.
3. Click any plus (+) for an attached CI node.
   CIs related to the node appear.
4. Right-click it a related CI and select Add to Change Order from the context menu.
   The new attached CI appears in the Config. Items tab for the change order.

Display the CI Descendants List

For any CI, you can display lines of related CIs (named *descendants*). In addition to basic information about each CI, the CI Descendants List displays relationship levels, with the originating CI as level 1. CI children begin at level 2, and their children at 3, and so on. If a CI is encountered more than once among the relationships, only its closest relationship level is displayed.

Example: Multiple Descendant Paths for One CI

Due to multiple relationships, a descendant search finds the same related CI at level 2 and also at level 4. The descendant CI is displayed only at level 2.

To list descendants for a CI

1. From a Change Order Detail page, select the Config. Items tab.
2. Click Impact Explorer to display attached CIs.
3. Expand a CI node if needed.
4. Right-click a CI and select List Descendants.
   The CI Descendants List is displayed.

Launch CMDB Visualizer from Impact Explorer

Impact Explorer provides the capability to launch CMDB Visualizer from an attached or related CI.

Follow these steps:

1. From a Change Order Detail page, click Impact Explorer.
   The Impact Explorer page displays.
2. Right-click any CI and select Launch Visualizer from its context menu.
   CMDB Visualizer is launched with the CI as the focus.

Configuring Impact Explorer

An Administrator can configure the Impact Explorer as follows:

- Specify the number of child CIs displayed for an attached CI.
- Specify the number of relationship levels that are displayed in the CI Descendants List,
- Suppress the display of child CIs,

The number of child CIs displayed is configurable by adding a NX_IMPACT_EXPLORER_MAX_CHILD_NODES setting to the NX.env configuration file. The default is 100.

Example: Configure the Display of Child CIs

Using the following setting, Impact Explorer only displays ten children at a time for an attached CI:

@NX_IMPACT_EXPLORER_MAX_CHILD_NODES=10

The default depth of the CI Descendants List is nine (9) levels. The depth is configurable by adding a NX_IMPACT_EXPLORER_MAX_LEVELS setting to the NX.env configuration file.

Example: Configure the Number of Descendant Levels

Using the following setting, the CI Descendants List displays the originating CI and only its relationally adjacent child CIs:

@NX_IMPACT_EXPLORER_MAX_LEVELS=2

Use the following Options Manager option to suppress the display of a certain type of child CI:

IMPACT_EXPLORER_EXCLUDE_HIER=boolean

If the IMPACT_EXPLORER_EXCLUDE_HIER option is installed and set to Yes, child CIs that are related through the CMDB Relationships tab of a CI Detail page are not displayed in the Impact Explorer tree or in the CI Descendants List.

> **NOTE**
> Child CIs that are related through CA CMDB are still displayed.

Define a Change Order Stored Query

Defining the stored queries that are available to users on the Change Order scoreboard is an administrative task. You can modify the predefined stored queries that are installed with CA SDM, or can define your own.

Follow these steps:

1. Select Service Desk, Application Data, Stored Queries on the Administration tab.
   The Stored Query List appears.
2. Select the stored query that you want to edit.
   The Stored Query Detail page appears.
3. Click Edit.
   The Update Stored Query page appears.
4. Edit the field values as appropriate.
5. Click Save, Close Window.
   The updated stored query appears in the Stored Query List.

Example: Define a Stored Query to List Change Orders Assigned to a CAB to Which the Logged In User Belongs

This example demonstrates how you can create a stored query that lists only change orders that are assigned to a CAB to which the logged in user belongs.

To create the stored query:

1. Navigate the Scoreboard to the Update Stored Query page.
2. Edit the field values as follows:
   a. Select Scoreboard Usage.
   b. Set Type to Change Order.
   c. Enter the Where clause:
   ```
   cab.[group]group_list.member IN (@cnt.id) AND active = 1
   ```
3. Click Save, Close Window.
   The stored query appears in the Stored Query List.

Change Orders

Change orders are requested changes that vary from the original scope of a specific project or task.

Create a Change Order

You can either create a change order from scratch or use an existing template.

WARNING
Depending on your role, you do not have access to all the functionality described in this section. For example, some predefined roles can edit records but cannot create new ones. Other roles, such as Configuration Viewer, can view Change Order information, but cannot create or edit change orders.

NOTE
If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. On the Service Desk tab, click the File menu and select the ticket type that you want to create. For example, to create a request from scratch, click File, New Request. To create a request from a template, Click File, New Request from Template.
2. Complete the Change Order Fields as appropriate for the change order.
3. Use the controls available on the Change Order Tabs as appropriate.
4. Click one of the following buttons:
   Auto Assign Triggers an auto assignment task, and updates the activity log. This button appears only when the ticket specifies a category or area that has auto-assignment enabled.
   Create Incident Opens the Create New Incident page so you can create an associated incident ticket. This button appears only when you create change orders and requests.
   Find Similar Opens the Find Similar page to search for similar problems.
   Quick Profile Displays the contact information for the specified user in the **Affected End User** field. You can also view their environment details and their entire ticket history.
   Use Template Displays a list of available templates for the selected ticket type. You can select the template that you want to use for creating this ticket.

   NOTE

   You can use the Quick Profile to identify a contact to be the affected end user of your new ticket. Quick Profile allows you to search for a contact, and view the history of the issues, requests, change orders, incidents, and problems that are assigned to that contact. When you have identified a contact, you can create the incident ticket directly from the Quick Profile.

Add Comments to a Change Order

Depending on the access level your administrator has granted, you may be able to add comments to an open change order ticket.

NOTE

After you have saved a comment, you cannot edit or delete it.

NOTE

If the Add Comment button does not appear on the Order Detail page, you have not been granted access to update change order tickets.

To add a comment to a change order

1. Click the link to your open change orders in the "Look up my existing tickets" section of your home page.
   Your list of open change order tickets appears.

NOTE

If you know the change order number, you can enter it in the search field and click Go to access the change orders directly.

2. Click the Change Order # link for the ticket you want to comment on.
   The Change Order Detail page appears.
3. Click the Add Comment button in the upper right-hand corner of the window.
   The Add Comment page appears.
4. Enter your comment in the User Description field, and then click the Save button.
   Your comment appears in the History section of the Change Order Detail page.

Change Order Properties

Administrators have the option of defining custom properties for change categories, to provide additional detail on change order tickets assigned to that category. When you select a change order category while creating or editing a change order, if any custom properties have been defined for that category, they appear on the Properties tab.

The Properties tab contains the following fields:

NameIdentifies the descriptive name assigned to the property, and indicates whether a value entry for the property is required or optional. Names of properties that require a value are shown with an asterisk after the name.

ValueSpecifies the value you assign to the property. There are three kinds of property value controls:

- Freeform text fields
  The values you enter for these properties are not validated. Be sure to read the information in the Examples column for guidance on the kind of information this property is intended to contain.
- Checkboxes
  Two-state checkboxes. Checked = Yes. Cleared = No. Property values for checkboxes are validated.
- Dropdown lists
  A list of options. If your administrator has defined this field as required, you must select one of the values in the list. If a default value has been assigned, it is displayed when the list first appears. Property values for dropdown lists are validated.

  NOTE

  On the Change Detail page, only the specified values are displayed, with no indication of whether the value was set via freeform text field, checkbox, or dropdown list.

Create a Change Order from an Incident, Problem, or Request

While creating or editing an incident, problem, or request, you can define a change order to be associated with the incident, problem, or request.

Follow these steps:

1. On the Create New or Update page for your incident, problem, or request, click Create Change Order.
   The Create New Change Order page displays.

   NOTE
   The Create New Change Order page opens only if all the required fields are defined for your incident, problem, or request.
2. Complete the fields as appropriate for the change order.
   See Change Order Fields for field definitions.
3. Use the controls available on the tabs at the bottom of this page to process the change order as appropriate.
   See Change Order Tabs for more information.
4. Save and close the window.

Create a Change Order from the Calendar

You can create a Change Order from the context menu on the daily, weekly and monthly calendar views. The context menu also allows you to create a Change Order from a template.

> **NOTE**
> You cannot create a change order by right clicking on a range of dates in the weekly view.

Follow these steps:

1. Right click a date on the calendar.
   The context menu appears.
2. Select one of the following:
   - **Create Change Order**
     The Create Change Order detail page appears and pre-populates the following fields:
     - Requester and Affected End User (always set to the login user)
     - Schedule Start Date (set to midnight of the selected day)
     - Scheduled Duration (always set to 00:00:00)
   - **Create Change Order from Template**
     The Change Order Template list view appears.
     Select a template and the Change Order detail page appears with pre-populated fields.
3. Complete the fields as appropriate for the change order.
   See Change Order Fields for field definitions.
4. Use the controls available on the tabs at the bottom of this page to process the change order as appropriate.
   See Change Order Tabs for more information.
5. Click an execution option. Following are options that are not self-explanatory:
   Create Incident
   Opens the Create New Incident page so you can create a incident ticket associated with this change order.
   Quick Profile
   Displays the contact information for the user entered in the Affected End User field.
   If you are creating a change order, a list of available users appears. You can select the user that is in the Affected End User field.
   Use Template
   Displays a list of available change order templates. You can select the template you want to use for creating this change order.

Change Order Fields

The following fields are required to create or update a change order:

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

> **NOTE**
>
> When the Change Order is in Edit mode, the Update CI button is disabled to prevent the possibility of duplicate CIs being added. To add CIs to a Change Order, save the Change Order, and the Update CI button is available for your use.

- **Change Reference Number**This is a unique reference number assigned by CA Service Desk Manager for all change order tickets. This is used by analysts and customers to refer to a particular change order ticket.
- **Requester**
  Specifies the name of the person who initiated the ticket. This person must be a defined contact. You can enter a value directly or click the magnifier to search for the name.
- **Affected End User**

Specifies the contact name of the person who is affected by the record. If the contact is assigned to a special handling type, special handling indicators are displayed. You can enter a value directly or click the magnifier to search for a contact name.

- **Category**
  Indicates the general category of the change within your IT environment (for example, Change.IT.Server.Configuration or Move.IT.Workstation). Change categories provide default values that are entered automatically on all change orders assigned to the category. In addition to the predefined change categories, your system administrator may define custom change categories. You can enter a value directly or click the magnifier to search for a category. When you edit the Category and a CA Process Automation workflow is already running, the workflow cancels.

  WARNING
  If a risk is associated with the specified change category, a Risk Survey button appears when you save the change order. This option opens a risk survey questionnaire for that change category.

  NOTE
  Your system administrator has the option of adding custom properties to change order categories. If custom properties have been added, they are displayed on the Properties tab when you create, edit, or view a change order. Some custom properties require that you enter a value.
- **Status**Specifies the status code of the record. For example, you can list only the tickets with a status code of Fix in Progress, or can Close Requested. You can enter a value directly or click the magnifier to search for a status. The blue button (on the left side of the Status field) lets you change the current status to the next default status.
- **Priority**Specifies the priority ranking of the record. The ranking determines the amount of attention the ticket receives. The predefined priority levels are 1 (highest) through 5 (lowest). Your system administrator or an active priority calculation can generate the appropriate Priority values for various installations and tenants. When priority calculation is enabled, this field updates based on Impact, Urgency, Affected Service, and Affected User settings. When your administrator disables priority calculation and uninstalls the urgency_on_employee option, Self Service Users see the Priority field on the Request Detail page.
- **Type**
  Specifies the ITIL change type as Standard, Normal, or Emergency. A default value may be defined the change category.
- **Risk**
  Identifies the risk level of the change order. The risk level is determined by evaluating the risk survey associated with the change order.

Detail Fields

- **Created By**
  Specifies the name of the person who created or reported the record. This field is filled automatically with the current user's login information when the record is created.
- **Assignee**
  Specifies the name of the person who is assigned to handle the record. You can enter a value directly or click the magnifier to search for a name. Selecting an assignee populates the groups that the assignee belongs to in the **Group** field.
- **Group**
  Specifies the group that is responsible for this record. Your system administrator defines groups of contacts that are responsible for different types of issues, requests, incidents, or problems. Any contact who is part of the group can handle the record after it is assigned to the group. You can enter a value directly or click the magnifier to search for a group. Selecting a group populates the **Assignee** field with the corresponding assignee name, which belong to the group.
- **CAB**
  Specifies the group that is responsible for reviewing Requests for Changes (RFCs). The CAB provides multiple perspectives necessary to ensure proper decision making about implementing changes. The CAB can include

members from the application team, development manager, component owner, QA, support, and any additional parties deemed necessary. You can enter a value directly or click the magnifier to search for a group.

- **Impact**Specifies an impact code, such as 1 -- Entire Organization, that indicates how a ticket affects work being performed. For example, a ticket that requires a network outage for several hours would have a higher impact than a ticket that takes a printer off-line. Your system administrator can modify the default impact codes, so they can vary from one installation to another
- **Active**Indicates whether the record is Active or Inactive. This value applies to the current record only, not the associated template.
- **Need By Date**
  Displays the date that the change order needs to be completed by. You can enter the date in mm/dd/yyyy hh:mm am | pm format, or click the calendar icon to select a date.
- **Call Back Date/Time**
  Indicates the date on which a follow-up call for this change order should be made. On the date you specify, the change order displays on your scoreboard under the Today's Issue Callbacks category. You can enter the date in mm/dd/yyyy hh:mm am | pm format, or click the calendar icon to select a date.
- **Root Cause**Identifies the code associated with the core reason why the ticket was opened. Your service desk can use generic root cause codes, such as Hardware Failure or Software Failure, or more specific codes, such as Network.Cable, Network.Card, or Network.Response. You can enter a value directly or click the magnifier to search for a code.
- **Organization**
  Specifies the company, division, or department that is associated with the change order. You can select a value from the drop-down list.
- **Project**
  Identifies the project. You can associate a change order with a project in order to establish a connection to an external project in another product such as CA CA PPM or CA SCM. The Project you enter in this field contains the external project information, such as the project name or ID, to make the connection and integration between CA SDM and the external project. You can enter a value directly or click the search icon to search for a project.
- **Closure Code**
  Indicates the final outcome of a completed change as Successful, Unsuccessful, or Successful with Errors. You can also create custom closure codes.
- **Business Classification**Classifies the change order into Major, Minor, or Significant. The classification helps to assess the change order magnitude.
- **External System Ticket**Specifies an identification for a ticket that belongs to an external system that integrates with CA SDM. This field stores hyperlinks and displays functional links in read-only mode.

Summary Information Fields

- **Order Summary**
  Gives an abbreviated description of the change order.
- **Spelling**
  Checks the spelling of the text you enter in the Order Summary field.
- **Order Description**
  Gives a detailed description of the change order.
- **Spelling**
  Checks the spelling of the text you enter in the Order Description field.
- **Schedule Start Date**
  Specifies the start date and time a change order appears on the Change Calendar pages. This field is optional, but must contain a date value if the Schedule Duration field contains a date value. Changes without an implementation date do not appear on the Change Calendar pages.
- **Schedule Duration**
  Indicates the amount of time required to implement the change in hours and minutes.
- **Schedule End Date**

Indicates the end date and time a change order appears on the Change Calendar pages. This field is read-only. Its value is calculated from Schedule Start Date and Schedule Duration values.

- **CAB Approval**
  Indicates (Y/N) whether the change requires approval from a CAB. You can specify which changes are to be considered for CAB approval, by doing any of the following:
  – Set this option at change creation time or any subsequent time throughout the approval process.
  – Add an Action macro to set this field to Yes or No for use within classic workflow.
- **Open Date**
  Indicates the date and time at which the record was first created, in the time zone of the server. This field is filled in automatically when the change order is created. The date and time appear in mm/dd/yyyy hh:mm am | pm format.
- **Actual Start Date**
  Indicates the date and time at which the word Pending appears in the Status field for the change order. The date and time appear in mm/dd/yyyy hh:mm am | pm format.
- **Last Modified Date/Time**Displays the date that this change order was last modified.
- **Last Modified By**
  Displays the name of the last person who edited the change order.
- **Resolve Date**
  Indicates the date and time at which the change order is resolved. The date and time appear in mm/dd/yyyy hh:mm am | pm format.
- **Close Date**
  Indicates the date and time at which the change order is closed. The date and time appear in mm/dd/yyyy hh:mm am | pm format.

Change Order Tabs

The following tabs are available on the Create Change Order, Change Order Detail, and Update Change Order pages:

Related Tickets

- **Related Orders:** Allows you to create a parent/child relationship between the change order and another CA SDM record.
- **Incidents / Problems:** Attaches related incidents and problems to a change order for analysts to reference.
- **Caused Requests:** Attaches requests that the change order causes for analysts to reference.

Configuration Management

- **Configuration Items:** Links configuration items (CIs) to a change order to provide information to analysts about the system that the ticket affects. Configuration Items are automatically added to this list when a Change Specification is created.
- **Change Specifications:** Lists the change specifications that are associated with the CI and the verification status. Change specifications are highlighted in red if they have failed verification or require manual verification.
- **Verification Log:** Displays logged information about change verification activity for this Change Order. For example, this log identifies a CI where CACF detected a variance or rogue change. Verification log entries highlighted in red indicate the corresponding change specification is either Failed Verification or Manual Verification Active and may need further attention from the user.

Additional Information

- **Properties:** Defines custom properties for change order categories.
- **Workflow Tasks:** Associates workflow tasks with a change order. For more information, see Define a Category or Area topic.
- **Service Type:** Allows you to attach a service type event to indicate the level of support for the ticket.
- **Attachments:** Attaches a document or a link to a URL to the change order.
- **Conflicts:** Identifies change orders that conflict with the one being viewed. For more information, see Conflict Analysis and Collision Detection topic.
- **Templates:** Allows you to create a template using the current ticket as a model.
- **Costs / Plans:** Calculates estimated cost and duration of the entire change order by adding the estimated cost and duration of each task.

Logs

- **Activities:** Displays a log of the activities performed to resolve the change order. For more information, see Add an Activity from the Ticket Management topic.
- **Event Log:** Displays a record of significant actions that occur regarding the change order.
- **Support Automation:** Displays the assistance session log and lets you invite the end user to an assistance session.

> **NOTE**
> If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

View Global Change Order Queue List

This window only appears in a multi-site installation and allows you to search and list change orders across the regions defined to your multi-site setup.

The Global Queue list forms can be used by analysts that work tickets across multiple regions. Each queue is a read-only list page that relates to tickets on remote (or local) regions. Once an item is selected from the queue, the analyst is re-directed to the appropriate regions to continue to process the ticket. Only enough information is replicated to allow analysts to make decisions on which ticket to process.

Only tickets that are assigned to groups belonging to a Global Group are selected for replication. Once selected for replication, it is replicated from that point forward even if no longer assigned to a Global Group. A Global Queue ticket is marked Inactive if no longer assigned to a Global Group. Marking the Global Queue ticket inactive does not affect the status of the original ticket that the Global Queue ticket references.

- **Change #**
  Specifies the number assigned to the change order.
- **Priority**
  Specifies a priority ranking for the change order.
- **Queue**
  Specifies is the name of the multi-site Region. This value is for display purposes only.
- **Status**
  Displays a specific status code for the change order. For example, you may want to list only the tickets with a status code of Fix in Progress, or Close Requested. Click the search icon to search for the status of interest.
- **Open Date**
  Displays the date the change order was opened. This field also displays the region of the ticket.
- **Contacts**
  Displays contacts associated with the change order, such as the assignee and end user.
- **Category**
  Displays the category of the change order. Categories designate an area of responsibility, such as adding or moving a workstation, and auto-populate certain fields on the tickets assigned to that category.

Attach Incidents, Problems, or Requests to a Change Order

You can attach related incidents, problems, or requests to a change order so that they can be referenced when editing or reviewing the change order.

Follow these steps:

1. On the Service Desk tab, browse to Change Orders, Assigned or Unassigned. Select the priority for the change order you want to edit.
   The Change Order List displays.
2. Select the change order.
   The Change Detail page displays.
3. Choose the Incidents/Problems tab and click Attach Incidents.
4. Complete the appropriate search fields, and click Search.
5. Select the incidents/problems/requests you want to attach to the change order, and click

   The selected items are added to the list on the right.

   NOTE
   Use the CTRL or SHIFT keys plus the left mouse button to select multiple incidents.
6. Click OK.
   The selected items appear in the Related Requests List on the Incidents/Problems tab of the Change Detail page.

Expedite a Change Order

You can use the Expedite Change Order command to change the status of unnecessary tasks to Skip so the change order can be completed quickly.

NOTE

Expedite Change Order is supported only for Classic Workflows.

NOTE
Skip must be defined as a valid status for the task.

Follow these steps:

1. Select Change Orders from the Scoreboard.
   The Change Orders folder expands to reveal nested folders for Assigned, Unassigned, and All Scheduled change orders.
2. Select the appropriate folder under Assigned, Unassigned, and All Scheduled for the change order whose children you want to expedite.
   The Change Order List page appears.
3. Click the change order number.
   The Change Order Detail page appears.
4. Select Expedite Change Order from the Actions menu.
   The Expedite Change Order page appears.
5. (Optional) Enter comments in the Remarks field to explain the action.
6. Click OK.
   The Expedite Change Order page closes and the Workflow Tasks tab on the Change Order Detail page shows all unnecessary tasks with status of Skip.

How to Schedule Change Orders

Contents

You can view the schedule of all configuration items associated with a change order. Consider scheduling information when you create, edit, or view a change order, or when you update the Change Calendar. Viewing the schedule can help you avoid scheduling collisions.

Follow these steps:

1. Create or open a change order.
2. Associate CIs with the change order.

   NOTE
   If you do not associate a CI with the change order, an error appears when you click the Scheduler button.
3. Click View Scheduler.
   The Change Scheduler appears and displays the following views:
   - **Daily**
     (Default) Displays the time period of the change order for the selected date and time. If no date is associated with the change order, the view defaults on the current day.
   - **Weekly**
     Displays the time period of the change order in the current week and includes the implementation start date, which can be configured to the Change Calendar start date.

   Note: The schedule duration determines the length of shading on the schedule. For example, if you create a change order with a two hour duration, the light-yellow shading on the scheduler displays for the two hour defined duration.
4. (Optional) Click a CI to view its details. Hover over the CI name to view its full name.
5. (Optional) Modify the Implementation Start Date or Duration.
   a. Click View Schedule to preview the updated schedule.
   b. Click Update Schedule to update the schedule.

   Note: You can only modify the schedule in edit mode.
6. Save the change order.
7. Refresh the Change Calendar to view global and CI-associated change windows.

Use the Change Scheduler Example

The following example demonstrates how to use the change scheduler when creating a change order.

Follow these steps:

1. On the Service Desk tab, select File, New Change Order.
   The Create New Change Order page appears.
2. Select Update CIs on the Config. Items tab.
   The Configuration Item Search page appears.
3. Create or search for CIs.
4. Using the Affected Configuration Items Update page, add CIs to the change order.
5. Click OK.
   The Create New Change Order page updates.
6. Click the Scheduler button.
   The Schedule for Change Order page appears.
7. Select a view and do *any* of the following:
   - Click within the table cells to modify the Schedule Start Date.
   - Click Schedule Start Date to use the Date Helper.
8. Modify the Duration and click View Schedule.
   The schedule previews your changes.

9. Click Update Schedule.
   The schedule updates with your changes.
10. Save the change order.

Set the Maximum Number of CIs

You can set the maximum number of CIs that display in the change scheduler.

Follow these steps:

1. Open *NX.env*.
   You can locate this file in the following directory:
   ```
   $NX_ROOT\
   ```
2. Modify the value of *@NX_CHANGE_SCHEDULER_MAX_CI_CNT* to the maximum number you want.

   NOTE
   By default, the variable is set to 100. If you set the maximum number of CIs greater than 100, only the first 100 display and you get a warning.
3. Save *NX.env* and cycle CA SDM.
   The maximum number of CIs is set.

Example: Set the Maximum Number of CIs to 25

With the following setting, only the first 25 CIs will display on the change scheduler.

```
@NX_CHANGE_SCHEDULER_MAX_CI_CNT=25
```

Modify Change Order Duration Background Color

You can change the background color of the change order duration. Modify the following lines in *schedule.css* to change the highlighting of the green change order duration bar.

Follow these steps:

1. Open *schedule.css*.
   You can locate this file in the following directory:
   ```
   $NX_ROOT\bopcfg\www\wwwroot\css
   ```

   WARNING
   $NX_ROOT\sdk also contains a file called schedule.css that contains helpful comments about css controls.
2. Modify the following lines with the color code you want:
   ```
   td.noBorderBackColor{border-left:none;border-right:none;background-color:#F6E3CE;}
   td.withBorderBackColor{border-right:none;border-left:1px solid;background-color:#F6E3CE;}
   ```
3. Save *schedule.css*.

In the following example, the change duration background color is #FF0000.

Example: Modify the Change Duration Background Color to #FF0000

```
td.noBorderBackColor{border-left:none;border-right:none;background-color:#FF0000;}
td.withBorderBackColor{border-right:none;border-left:1px solid;background-color:#FF0000;}
```

Modify Change Window Colors in the Change Scheduler

You can change colors that indicate change windows by modifying lines in *schedule.css.*

Note: If you modify *schedule.css* to change the formatting of windows on the Change Scheduler, you must also modify the schedGroup macros used by the Change Calendar if you want its formatting to be consistent with the Scheduler.

Follow these steps:

1. Open *schedule.css* in a text editor.

You can locate this file in the following directory:

```
$NX_ROOT\bopcfg\www\wwwroot\css
```

2. Modify the following lines with the color code you want:

```
.schedConflict { background-color: #ff0000; font-size: 4px }
.schedBusy { background-color: #0176ff; font-size: 4px}
.schedCurrent { background-color: #008000; font-size: 4px; }
.schedMW { background-color: #40ff40; font-size: 4px; }
.schedBW { background-color: black; font-size: 4px; }
.schedNone {font-size: 4px; }
.schedDialog { width: 100%; height: 4px; margin-left: 0px; margin-right: 0px; margin-top: 4px; margin-
bottom: 4px; }
```

3. Save *schedule.css*.
   The background color is modified.

View or Update a Change Schedule

You can view and update a change schedule to help avoid scheduling collisions, or to streamline and balance the work effort.

Follow these steps:

1. Create, edit, or view a change order.

   NOTE
   You can only modify the schedule in edit mode.

   The Change Order Detail page appears.
2. On the Config. Items tab, click Update CIs to add CIs to the change order.
3. Click the Scheduler button.
   The Schedule for Change Order page appears.
4. Modify the Schedule Start Date and Duration as appropriate, for example, to resolve scheduling collisions.

   NOTE
   You can edit the start date by clicking within the table cells. Clicking the table cells on the daily view shifts the schedule in 10-minute increments. Clicking the table cells on the weekly view shifts the schedule in one hour increments.

   Details of the change order display when you hover over a line in the table cells.
   The change scheduler indicates the following types of schedules:
   - Blue -- Regular change order
   - Green -- Global maintenance window or CI-associated window
   - Dark Green -- Schedule of the current change order
   - Black -- Blackout window
   - Red -- If the current change order conflicts with other scheduled change orders.

     NOTE
     This indicator only displays conflicts with the current change order, not change conflicts in general. The red conflict only appears after the change order is saved. Use the highlighted change scheduler durations to determine conflicts.

   For information about modifying colors in the change scheduler, see Modify Change Window Colors in the Change Scheduler
5. Click Update Schedule and save the change order.
6. View the updated Change Calendar.

Update the Change Scheduler

You can update the Change Scheduler when you create or edit a change order.

Follow these steps:

1. Select the change order from the Change Order List page on the Scoreboard.
   The Change Order Detail page appears.
2. Select Activities, Update Schedule on the menu bar.
   The Update Schedule page appears.
3. Enter a new start date, duration, and date of activity.

   NOTE
   You can enter the values directly or click the icon to open the Date Helper.
4. Change any other activities fields as appropriate.
5. Click Save.
   The Change Scheduler is updated, and the activity is recorded on the Activities tab on the Change Order Detail page.

Create Change Windows

Create change windows to avoid collisions in the Change Calendar.

Follow these steps:

1. On the Administration tab, click Service Desk, Change Orders, Change Windows.
   The Change Windows List appears.
2. Click Create New.
   The Create New Change Window page appears.
3. Complete the appropriate fields.
4. Save and close the change window.
   The change window appears in the Change Windows List.

Associate a CI with a Maintenance Window

To control when a CI undergoes maintenance, you can associate that CI to a maintenance window.

Follow these steps:

1. Navigate to the Change Window Detail page.
2. Click the Associated CIs tab.
   The Associated CIs list appears.
3. Click Update Configuration Items.
   The CI Search page appears.
4. Specify information for the required CIs.
5. Click Search
   The Available CIs page appears.
6. Move any required CIs in the Available CIs list to the Associated CIs list.
7. Click OK.
   The Change Window Detail page appears.
8. Click Save.
   The CI is associated with a maintenance window.

The change window, the CIs and the selected window associations are saved.

Accumulate Costs and Time to a Change Order

The accumulate feature allows you to calculate the estimated cost and duration of the entire change order by adding together the estimated cost and duration of each task. You can also choose to accumulate the change order and all of its children. Children are additional records entered as a result of attempting to resolve the original change order.

Follow these steps:

1. Select Change Orders, Assigned or Unassigned, and priority level on the Service Desk tab.
   The Change Order List appears.
2. Select the change order of interest.
   The Change Detail page appears.
3. Select Actions, Accumulate Change Order on the menu bar.
   The Accumulate Change Order page appears.
4. Click one of the following buttons:
   - **Accumulate this Change Order and All Children**
     Calculates the estimated cost and duration of the change order by adding together the estimated cost and duration of tasks for this change order and all of its children.
   - **Accumulate Only this Change Order**
     Calculates the estimated cost and duration of the change order by adding together the estimated cost and duration of tasks for only this change order.
   - **Cancel**
     Closes the Accumulate Change Order page without accumulating the change order.

   The estimated cost and duration are accumulated and you are returned to the Change Detail page.
5. Select the Costs/Plans tab on the Change Detail page.
   The tab displays the estimated cost and duration for the change order based on the values accumulated.

Change Order Configuration Items

Configuration items include all the hardware, software, and services associated with your system. Configuration item records uniquely identify each item and its exact location. These records are part of the inventory components that make up your database.

You can link configuration items to an issue or change order in order to give analysts information about the system that is affected by the ticket. Consider using change verification in your CMDB environment to verify that changes execute correctly.

Follow these steps:

1. On the Scoreboard, browse to Change Orders, Assigned or Unassigned. Select the priority for the change order you want to edit.
   The Change Order List displays.
2. Select the change order.
   The Change Detail page displays.
3. (Optional) Select the Change Specifications tab to view or create change specification to associate with the ticket.
4. Select the Configuration Items List tab, and click Update CIs.
   The Configuration Item Search page displays.
5. Complete one or more of the search fields, and click Search.
   A list of available configuration items displays.
6. Select the configuration items you want to link to the change order, and click

   The selected configuration items are added to the list on the right.

NOTE
Use the CTRL or SHIFT keys plus the left mouse button to select multiple configuration items.

7. Click OK.
   The selected configuration items appear on the Config. Items tab on the Change Detail page.
8. Click Close Window.
   The updated change order appears in the Change Order List when you redisplay the list.

Change Order Workflows

This article contains the following topics:

Process CA Workflow Workitems

You can process workitems on the ticket's Workflow Tab to indicate progress as you work through the tasks to resolve the ticket.

NOTE

The following procedure applies only to change orders assigned to a category with an associated CA Workflow process definition. For information about working with change orders assigned to a category with an associated CA SDM classic workflow, see Add Workflow Tasks or Update a Workflow Task sections in this topic.

To process workitems on a change order

1. Select Change Orders, Assigned or Unassigned on the Service Desk tab, and then select the Priority level for the change orders you want to display in the list.
   The Change Order List appears.
2. Select the change order to update.
   The Change Detail page appears.
3. Select the Workflow Tasks tab.
   The following read-only fields appear at the top of the tab:
   Definition Name
   Indicates the CA Workflow process definition your administrator has assigned to the ticket's change category.
   Description
   Provides a general summary of the objectives of this workflow.
   Start Date
   Indicates the date and time the change order was assigned to the category associated with the workflow.
   Status
   Indicates the overall status the of the entire workitem list.
4. Locate the workitem you want to process on Workflow Tab, and then click the workitem's Activity Name.
   The CA Workflow Worklist GUI appears.

   NOTE

   If your administrator has not configured automatic login, you are prompted for your user name and password to access the CA Workflow Worklist.
5. Process the tasks on the worklist as appropriate for this change order.

   NOTE

   For information about processing tasks, see the CA Workflow Worklist online help.
6. Select Refresh from the View menu to see updates to workitems on the Workflow Tasks tab.

Add Workflow Tasks

You can add new workflow tasks to a change order, and specify the sequence in which they are performed.

NOTE

The following procedure applies only to change orders assigned to a category with an associated CA SDM classic workflow.

To add a workflow task to a change order, follow these steps:

1. Select Change Orders, Assigned or Unassigned on the Service Desk tab, and then select the Priority level for the change orders you want to to display in the list.
   The Change Order List appears.
2. Select the change order to update.
   The Change Detail page appears.
3. Select the Workflow Tasks tab, and click Insert Tasks.
   The Insert New Workflow Task page appears.
4. Complete the fields as appropriate.
5. Click Add.
   The new workflow task appears on the Workflow Tasks tab on the Change Detail page.

Update a Workflow Task

You can edit existing workflow tasks.

NOTE

The following procedure applies only to change orders assigned to a category with an associated CA SDM classic workflow.

To update a workflow task for a change order, follow these steps:

1. Select Change Orders, Assigned or Unassigned on the Service Desk tab, and then select the priority level for the change orders you want to display in the list.
   The Change Order List appears.
2. Select the change order.
   The Change Detail page appears.
3. Choose the Workflow Tasks tab, and select the workflow task to edit.
   The Change Workflow Detail page appears.
4. Click Edit.
   The Update Change Workflow page appears.
5. Edit the fields as appropriate.
   See Update Workflow Task Fields for field definitions.

   NOTE

   The Service Type tab on the Update Change Workflow page provides information for service types assigned to workflow tasks. Service types assigned to workflow tasks are independent of the service types assigned to a change order ticket on the ticket record's Service Types tab.
6. Click Save.
   The Change Workflow Detail page displays a successful save message.
7. Click Close Window.
   The Change Detail page appears.
8. Click Refresh Workflow Tasks on the Workflow Tasks tab.
   The updated workflow task appears on the Workflow Tasks tab on the Change Detail page.

Change Order Workflows

A *workflow* is an automated or partially automated business process that specifies the sequence of tasks to be performed to resolve a CA SDM ticket.

Two workflow solutions are supported for change orders:

- **CA Workflow**This is a high performance, scalable workflow management product that can optionally be integrated with CA SDM.
- **CA SDM "classic" workflow**This is an internal workflow automation features provided as a standard CA SDM component.
- **CA Process Automation**This is a workflow with task-based steps that run remotely on the CA Process Automation web service.

Your administrator can configure change order categories to use either type of workflow. When a change order ticket is assigned to a category with an associated workflow, a series of tasks appear on the ticket's Workflow Tasks tab.

Some workflow tasks may be optional, which allows you to delete them from the task list.

You must complete all mandatory workflow tasks before you can close the ticket. The only exception is that if the ticket status is set to Cancelled, the workflow process terminates automatically.

If a ticket is removed from a category with an attached workflow, or assigned to a different category, the workflow process terminates automatically. If the new category has an attached workflow, its process starts automatically.

NOTE

To use the process viewer, you must associate a CA Workflow process to the change category.

View Change Order Events

Contents

View the Event Delay History

You can view the event delay history for a change order to see descriptions of the service type event delays. Service type event delays let you suspend a service type event to prevent a service level violation.

To view the event delay history for a change order

1. On the Scoreboard, browse to Change Orders, Assigned or Unassigned. Select the Priority for the change order.
   The Change Orders List appears.
2. Select the change order.
   The Change Order Detail window opens.
3. From the menu select View, Event Delay History.
   The Event Delay History window opens.
   Start Time
   Indicates the date and time the delay began.
   Stop Time
   Indicates the date and time the delay ended.
   Actual Delay
   Indicates the actual duration of delay.
   Effective Delay
   Indicates the effective duration of delay.
   Service Type Defines the support level assigned to the ticket (for example, 4-hour service requirement).

View the Event History for a Change Order

You can view the event history for a change order to see what actions have been performed on the change order.

To view the event history for a change order

1. On the Scoreboard, browse to Change Orders, Assigned or Unassigned. Select the Priority for the change order.
   The Change Order List appears.
2. Select the change order.
   The Change Order Detail window opens.
3. From the menu select View, Event History.
   The Event History window opens.
   Condition
   Indicates the condition checked for by the event. CA SDM provides macros that check for standard conditions such as priorities and object status (for example, open or closed).
   Event
   Indicates the name of event associated with this row in the Event History list.
   Status
   Indicates whether the event is active or inactive.
   Check Time
   Indicates the time the event was checked, based on time parameters specified in the event configuration.

Add Activities to a Change Order

All work that is done to resolve a ticket is recorded on the Activities tab at the bottom of the ticket detail page. Activities can include actions such as research, calling a customer, and transferring responsibility to a different analyst. Some activities, such as "Initial" when a record is created or "Field Update" when a field is changed, are posted automatically. Others, such as returning a call, changing the status, or transferring the ticket are posted when you select items from the Activities menu or from the Activities page itself.

Update Status

You can change the status of a change order to record progress toward resolving the change order. Status changes become part of the change order history and can be useful when you research and update change orders.

To update the status of a change order

1. Select the change order from the Change Order List page on the Scoreboard.
   The Change Detail page appears.
2. Select Activities, Update Status on the menu bar.
   The Status Change page appears.
3. Edit the status and other activities fields as appropriate.
   See Activities Fields section in this topic for field descriptions.
4. Click Save.
   The status is updated and the activity is recorded on the Activities tab on the Change Detail page.

Update Risk

You can change the risk level of a change order to record progress toward resolving the change order. Risk changes become part of the change order history and can be useful when you research and update change orders.

To update the risk level of a change order, **follow these steps:**

1. Select the change order from the Change Order List page on the Scoreboard.
   The Change Detail page appears.

2. Select Activities, Update Risk on the menu bar.
   The Override Risk of Change Order page appears.
3. Edit the risk value other activities fields as appropriate.
   See Activities Fields for field descriptions.
4. Click Save.
   The risk level is updated and the activity is recorded on the Activities tab on the Change Detail page.

Callback

When editing a change order, you can record information on the date, time, and nature of a callback. The callback record is stored in the change order history.

To record a callback for a change order

1. Select the change order from the Change Order List page on the Scoreboard.
   The Change Detail page appears.
2. Select Activities, Callback on the menu bar.
   The Create New Activity page appears.
3. Change the date of the activity, if appropriate, and record the time spent on the call.
4. Enter a description for the callback, such as "left message for customer."
5. Click Save.
   The activity is recorded on the Activities Tab on the Change Detail page.

Research

You can record information about research you performed to resolve the change order.

To record research performed for a change order

1. Select the change order from the Change Order List page on the Scoreboard.
   The Change Detail page appears.
2. On the menu bar, select Activities, Research.
   The Create New Activity page appears.
3. Change the date of the activity, if appropriate, and record the time spent conducting the research.
   See Activities Fields section in this topic for field descriptions.
4. Enter a description of the research, such as "consulted development engineer for workaround."
5. Click Save.
   The activity is recorded on the Activities tab on the Change Detail page.

Log Comment

When editing a change order, you can record comments in the Activity Log. A comment can be any additional information about the change order that needs to be recorded.

To log a comment for a change order

1. Select the change order from the Change Order List page on the Scoreboard.
   The Change Detail page appears.
2. Select Activities, Log Comment on the menu bar.
   The Create New Activity page appears.
3. Enter your comment in the User Description field.
4. Click Save.
   The comment is recorded on the Activities Tab on the Change Detail page.

Transfer

You can reassign a change order that has been outstanding for too long, demands an immediate response, or should be handled by another technician. When you transfer a change order, the new analyst or group is automatically notified, as are any others who should receive notification based on the notification rules established by your system administrator.

To transfer a change order to another technician

1. Select the change order from the Change Order List page on the Scoreboard.
   The Request Detail page appears.
2. Select Activities, Transfer on the menu bar.
   The Transfer Change page appears.
3. Enter a New Assignee and a New Group. You can enter the values directly or click the search icon to search for a contact or group.
4. Change any other activities fields as appropriate.
   See Activities Fields section in this topic for field descriptions.
5. Click Save.
   The change order is transferred, and the activity is recorded on the Activities tab on the Change Detail page.

Escalate

You can escalate the priority of a change order in order to direct more attention to the change order and increase the speed with which it is resolved. You may need to escalate a change order that has been outstanding for too long or demands an immediate response.

To escalate a change order

1. Select the change order from the Change Order List page on the Scoreboard.
   The Change Detail page appears.
2. Select Activities, Escalate on the menu bar.
   The Escalate Change page appears.
3. Enter the New Priority and update the other activities fields as appropriate.
   See Activities Fields section in this topic for field descriptions.
4. Click Save.
   The priority of the change order is escalated, and the activity is recorded on the Activities tab on the Change Detail page.

Manual Notify

You can send a message about a change order to an end user, vendor, analyst, or anyone else in your service desk database. Record of the manual notification is stored in the Activities List for the change order

To notify users of a change order manually

1. Select the change order from the Change Order List page on the Scoreboard.
   The Change Detail page appears.
2. Select Activities, Manual Notify on the menu bar.
   The Manual Notification page appears.
3. Complete the fields as appropriate as specified in the Manual Notify Fields section in this topic.
   The notification is defined.
4. Click Notify.
   The message is sent and the notification activity is recorded.

Manual Notify Fields

Selected Recipients

Lists the names of the people who receive the notification.

Available Recipients

Lists relevant contacts and Contact Objects that you can add to the Recipients list, as specified in the Manual Notify settings in Activity Notifications options. When you select a contact object name, only the contact names that are associated with the object are added to the Recipients list. Use the CTRL or SHIFT keys plus the left mouse button to select multiple contacts.

> **NOTE**
>
> After you add contacts or Contact Objects to the Recipients list, the selected contacts or Contact Objects remain in the Available Recipients list.

Contact

Enter characters into the contact field of the ticket. The field displays results as you type and generates a list of suggested results.

You can click a suggested result in the list, use the keyboard down arrow to scroll through the suggested results, or use the look up to search for a required contact.

Email address

Accepts an SMTP address for an email recipient. You can specify any valid email address including addresses that are not associated with a contact record. Separate multiple addresses with a semi-colon.

> **NOTE**
>
> This field appears when the option notification_allow_temp_address is installed using Options Manager.

Urgency

Specifies the relative importance of the notification (by default: low, normal, high, or emergency). The system administrator predefines Urgency levels.

Preferred Method

Specifies the notification method to use, such as Email or Notification.

> **NOTE**
>
> When you specify an email address, set the method to one that supports SMTP, otherwise, remove email addresses from the Selected Recipients list.

Internal

Makes the notification visible to internal contacts only.

Message Title

Specifies a title for the notification message. This title is a short description of the message, similar to an email subject line. The default title is the ticket type and number, and the words "Manual Notify" (for example, "Incident 32 Manual Notify").

Message Text

Specifies the entire text of the notification. You can click Spelling to check the spelling of the message text.

Personalized Response

Specifies a personalized response such as a generic or administrator signature. The response is added to the end of the message text.

> **NOTE**
>
> The administrator must set up a personalized response before you can use this option. For more information about personalized responses, see the Administration Guide.

Update the Change Scheduler

You can update the Change Scheduler when you create or edit a change order.

To update the Change Scheduler

1. Select the change order from the Change Order List page on the Scoreboard.
   The Change Order Detail page appears.
2. Select Activities, Update Schedule on the menu bar.
   The Update Schedule page appears.
3. Enter a new start date, duration, and date of activity.

   NOTE

   You can enter the values directly or click the icon to open the Date Helper.
4. Change any other activities fields as appropriate.
5. Click Save.
   The Change Scheduler is updated, and the activity is recorded on the Activities tab on the Change Order Detail page.

Add Schedule Information to a Change Order

You can add schedule information when you create or edit a change order.

To add schedule information

1. Do one of the following actions:
   – Click File, New Change Order.
   – Open a change order and click Edit.
2. Complete the following schedule fields:
   Type
   Specifies the ITIL change type as Standard, Normal, or Emergency. A default value may be defined for a change category that CA SDM uses to initialize new change orders of the category.
   Schedule Start Date
   Specifies the start date of the change order.
   Duration
   Specifies the duration of the change order, in 00:00:00 format.
   The schedule contains only those change orders with a non-empty schedule start date and duration. Type is useful to group change orders on the schedule, but does not directly affect scheduling.
3. Continue creating the change order.
4. Click Save when you finish.
5. The schedule information is added to the change order.

Activities Fields

The following fields can be present on Activities pages:

- **Number**The number of the item for which activity is being recorded. Cannot be edited.
- **Summary**
  The summary description of the record. Cannot be edited.
- **Activity Type**
  The type of activity being recorded. For example, if you select Log Comment from the Activities menu, the Activity Type is Log Comment. Cannot be edited.
- **Current Status**
  The current status of the record. Cannot be edited.

- **New Status**
  Specifies a new status to be assigned to the record.
- **New Closure Code**
  Specifies a new closure code to be assigned to the change order.
- **Current Priority**
  The current priority level of the record. Cannot be edited.
- **New Priority**
  Specifies a new priority level to be assigned to the change order.
- **Current Assignee**
  The analyst currently assigned to the ticket. Cannot be edited.
- **New Assignee**
  Specifies a new analyst to be assigned to the ticket.
- **Current Group**
  The group currently assigned to the change order. Cannot be edited.
- **New Group**
  Specifies a new group to be assigned to the change order.
- **Analyst**
  Name of help desk analyst or technician who performed the activity. This is the currently logged in user. Cannot be edited.
- **Internal**
  Select this check box if the activity should only be visible to internal users.
- **Time Stamp**
  The date and time the activity began. This is the system date and time the activity record was opened. Cannot be edited.
- **Date of Activity**
  The date and time the activity was performed. Defaults to the date and time the activity record was opened, but can be changed.
- **Time Spent**
  The amount of time spent on the activity. Enter this value in hours, minutes, and seconds (hh:mm:ss).
- **User Description**
  Description of the action taken. Click Spelling to check the spelling of the text entered. Click Submit Knowledge to submit the solution to the Knowledge Base.
- **System Description**
  The system description of the action taken. Cannot be edited.
- **Personalized Response**
  Choose the personalized response you want to send for this item. See Personalized Responses for more information.

  NOTE
  The Personalized Response field is present only if an active personalized response exists.
- **Recipients**
  Specifies the contacts who receives manual notification pertaining to the change order. Click Add Recipient to search the contact list, or Remove Recipient to remove the contact from the notification list.
- **Urgency**
  Specifies the urgency level associated with the manual notification. Select the appropriate urgency code from the drop-down list.

- **Preferred Method**
  Specifies the preferred method for delivering the manual notification message (for example, Email or Pager_Email).
- **Message Title**
  Specifies the subject heading that appears on the manual notification.
- **Message Text**
  Specifies the text that appears in the body of the manual notification message.
- **Change**
  Select an existing change order to which you want to attach the change order.
- **Remarks**
  Enter text explaining why the change order is being attached to an existing change order.

Change Calendar

This article contains the following topics:

The *Change Calendar* provides a graphical view of change events with implementation start and end times. This calendar view of the change schedule provides analysts and managers a quick way to identify when events occur and how they affect the environment, organization, and resources.

The calendar lets you create change orders from the daily, weekly, and monthly views and also lets you view global change windows for scheduled change orders. If multiple change windows exist in a month, they are grouped, such as in a single Blackout Window group. Drill down to a weekly or daily view, or view hover information to see the individual window details.

NOTE

You can only export Change Order schedules, not Change Windows.

View the Change Calendar

You can configure the view of the Change Calendar page to display a calendar of the change orders of interest.

You can create a Change Order from the File menu, the Create New button on the Change Calendar page, or the context menu on the daily and monthly calendar views. The context menu also lets you create a Change Order from a template.

Follow these steps:

1. Click the Change Calendar tab.
   The Change Calendar filter fields appear.
2. Display a calendar view showing the schedule for the change orders of interest by completing the Change Calendar search fields. Click Search.
   The Change Calendar provides the following filter fields to restrict the change orders displayed on the schedule:
   - **Active**
     Filters by whether the change orders are active (open) or inactive (complete).
   - **Priority**
     Filters by the priority ranking of the change orders.
   - **Status**
     Filter by the status of the change order. You can enter the status directly or click the search icon to search for the desired status.
   - **Category**
     Filters by the category of change order. You can enter a value directly or click the browse icon to select from defined categories.
   - **Group**

Filters by the group responsible for the change order. Your system administrator defines groups of contacts that are responsible for different types of issues, requests, incidents, change orders, and so on. Any individual contact assigned to the group can handle the record once it has been assigned to the group. You can enter the value directly or click the search icon to search for the group.

– **Parent**
Filters by the parent change order.
– **Change Type**
Filters by the type of change order as Emergency, Normal, or Standard.
– **CI Name**
Filters by the name of the Configuration Item. You can enter a value directly or click the browse icon to search for the desired Configuration Item.
– **CI Class**
Filters by the class of the Configuration Item. You can enter a value directly or click the browse icon to select from defined categories.
– **CI Family**
Filters by one of the following Configuration Item families:
  • Computer
  • Hardware
  • Other
  • Projects
  • Service
  • Software
– **Conflict Status**
Filters by the conflict status of a change order. You can enter a value directly or click the browse icon to select from defined categories.
– **Closure Code**
Filters by the closure code of a change orders.
– **CAB**
Filters by CAB member. You can enter a value directly or click the browse icon to select from defined groups.
– **CAB Approval**
Filters by change orders that do not require CAB approval (N) or that do require CAB approval (Y).
– **Risk**
Filters by the risk level of a change order.
– **Schedule Start Date**
Specifies the beginning of an implementation date range for the change orders you want displayed on the schedule. If you do not enter a value in this field, the filter selects all change orders with a non-null target implementation start date. Regardless of whether or not you enter a value in this field, the initial view always includes the first change order actually selected for the schedule.
– **Schedule End Date**
Specifies the end of an implementation date range for the change orders you want displayed on the calendar. If you do not enter a value in this field, the filter selects all change orders with a non-null target implementation end date.
– **Schedule Timezone**
Specifies a time zone to view your search results.

NOTE
If no time zone is selected, the events are displayed in your current time zone.

– **Initial View**
Select the view of the Change Order Calendar you want to see:
  • **Month**

Displays a schedule for the full month that includes the first change order selected by the filter. The schedule shows change orders grouped by change type and schedule start and end date.

- **Week**
  Displays a schedule for a the week that includes the first change order selected by the filter. The schedule seven consecutive days in a single column, beginning with the weekday configured as the starting weekday at your installation. The schedule shows individual change orders, and includes summary information about each one
- **Day**
  Displays a view similar to the week view, except that it shows only a single day's change orders.
- **n Days**
  Displays a view similar to the week view, except that it can begin on any day of the week, and continues for the specified number of days.
- **List**
  Displays a standard list page.

iCalendar Event Templates

iCalendar event templates control the information that is exported to iCalendar format. The following predefined templates are installed with CA SDM:

- Change Schedule
- KnowledgeScheduleCreation
- KnowledgeScheduleExpired
- KnowledgeScheduleReview
- KnowledgeScheduleStart

NOTE
You can edit the predefined iCalendar event template codes, but you cannot delete them or create new ones.

WARNING
The SchedExpMaximum variable in web.cfg controls the maximum events allowed for an export. Increasing the default (1000) could cause system instability. If you attempt to export more than the value specified in SchedExpMaximum, a message appears refusing your exporting request.

iCalendar Event Template Fields

The following fields require explanation:

- **Symbol**
  Defines the display name by which you want to identify the event template within your system.
- **Code**
  Defines a localized and independent code for the event template.
- **Record Status**
  Specifies whether the record is active or inactive.
- **Object Type**
  Displays the object type such as Change Order or Knowledge Document.
- **Alarm**
  Allows you to set a display alarm (such as 1 Day before or 1 Hour before) for each event that creates a calendar.
- **Category**
  Displays the field property for the event template. This is similar to how message templates are processed.
- **URL**
  Displays the field property for the event template. This is similar to how message templates are processed.
- **Extra Entries**
  Allows you to add other properties (not managed by CA SDM) to the event.

Export Schedules to iCalendar

CA SDM lets you export Change Orders in the standard iCalendar format. This data exchange lets you import Change Order schedules into many widely used calendaring applications, including Microsoft Outlook and Lotus Notes.

NOTE
When exporting schedules on some calendaring programs, choosing the Open option instead of Save causes the file to import incorrectly. To avoid this issue on Knowledge Management and Change Order schedules, select the Save option instead of Open. After you save the exported file, import it through the interface of the calendar program interface. You cannot export Change (blackout and maintenance) Windows, only Change Orders.

WARNING
The data you export is based on the current view. If you want to export a custom range of dates, such as 32 days, the export should be done from the list view. Otherwise, the view is truncated to a month or week, and it only exports that amount.

Follow these steps:

1. Click the Change Calendar tab.
   The Change Calendar filter fields appear.
2. Complete the search fields to display a calendar or list view that includes the change orders of interest.
3. Click Export.
   The Schedule Export page appears.

   WARNING
   The SchedExpMaximum variable in web.cfg controls the maximum events allowed for an export. Increasing the default (1000) could cause system instability. If you attempt to export more than the value specified in SchedExpMaximum, a message appears refusing your exporting request.
4. Enter the location where you want to save an iCalendar file.
   An iCalendar file containing all events in the current view is saved at the specified location.

Use the Change Calendar Tab

NOTE
This tab is displayed, only if you have access to the Change Calendar.

The Change Calendar tab provides access to the schedule to view change order impacts on your system. From the Change Calendar tab, you can:

- Search for change orders
- Filter and sort change orders and configuration items
- Create change orders
- View CIs associated with change orders
- View change windows
  - Global maintenance window
  - Non-Global maintenance window
  - Global blackout window
  - Non-Global blackout window
- View CIs associated with change windows

You can use the schedule to view and manage change orders. Create change orders from the schedule and search for change orders and configuration items.

To search for items within the schedule, use the search filter. Complete one or more of the search entry fields and click Search. The list pane populates with all items that match your search criteria.

View and Configure Scheduling Views

This article contains the following topics:

Schedule Views

CA SDM provides the following scheduling views:

- **List**
  Displays a list page sorted by schedule start and end date.
- **Month**
  Displays a calendar for a full month.
  The view shows change orders in groups, with each entry collecting one or more change orders. You can see detailed information about the change orders in a group by hovering over the group with the mouse; by pressing Alt+Right Arrow when the focus is on the group; or by clicking on the group to display its contents in an n-day view.
- **Week**
  Displays a full week in a single column, starting with the day configured as the first weekday.
  The view shows changes individually and includes detailed information about each change order scheduled during the week.
- **Day**
  Displays change orders for a single day.
  The view shows changes individually and includes detailed information about each change order scheduled during the day.
- ***n* Days**
  Displays change orders for the number of days specified by the dropdown selector.
  The view shows changes individually and includes detailed information about each change order scheduled during the days selected.

Navigating the Schedule Views

You can use the arrow and tab keys to navigate the scheduling views. The following keyboard shortcuts are available:

- **Tab**
  Navigates to a later date. Use this from the last cell in the schedule to navigate to the Search button.
- **Shift+Tab**
  Navigates to a previous date. Use this from the first cell in the schedule to navigate to the Next Month button.
- **Shift+Arrow**
  Navigates around the calendar from date to date in the direction of the arrow.
- **Right Arrow**
  Displays the context menu for the currently focused date or event. If there is no context menu, it navigates to the next higher date (similar to Shift+Right Arrow).
- **Alt+Right Arrow**
  Displays a hover information popup for the currently focused date or event. If there is no hover information, it navigates to the next higher date (similar to Shift+Right Arrow).
- **Down Arrow**
  Navigates to the next event in the current cell. If there are no events in the current cell, or if the focus is already on the last event, it navigates to the date in the next cell down (similar to Shift+Down Arrow).
- **Up Arrow**
  Navigates to the previous event in the current cell. If focus is not on an event, it navigates to the date in the next cell up (similar to Shift+Up Arrow).

Calender View Hotkeys

Each schedule view supports hotkey access to its buttons. The following are the supported hotkeys:

- **Alt+0**

Switch to list view.
- **Alt+1**
  Switch to daily view.
- **Alt+7**
  Switch to weekly view.
- **Alt+3**
  Switch to monthly view.
- **Alt+9**
  Switch to *n*-day view.
- **Alt+<**
  Move to previous time period in current view.
- **Alt+>**
  Move to next time period in current view.

Scheduling View Configuration

You configure the monthly and weekly scheduling views by specifying pdm_macro statements in the <head> section of the HTMPL forms defining the schedule. We recommend using the Source View of Web Screen Painter to edit these forms.

Any form that displays a schedule must contain the following:

- A schedConfig macro
- At least one schedAttr macro
- At least one schedGroup macro

The configuration macros are in a separate source file referenced by a pdm_include statement in the main source file. This file lets you configure your schedule without modifying the main source file.

For example, the configuration macros for the Change Calendar form list_chgsched.htmpl are in a file named list_chgsched_config.htmpl. For the Knowledge Lifecycle Schedule, you can modify the list_kdsched_config.htmpl using the same macros.

You can find list_chgsched_config.htmpl and list_kdsched_config.htmpl in the following directory:

$NX_ROOT\bopcfg\www\htmpl\web\analyst\

schedConfig Macro -- Configure Schedule

The schedConfig macro specifies that a form contains a schedule and provides basic configuration information. The following values are valid macro arguments:

- **autosearch=1|0**
  Specifies whether the schedule form reloads data from the server when the user selects a view outside the currently selected date range. Setting the value to 1 (default) causes the form to search automatically when the user selects a view with one or more days outside the date selection range of the search filter. Setting the value to 0 requires the user to press the Search button to initiate a search.
- **defaultView=0|1|7|30|99**
  Specifies the default view for the search filter as 0 (list), 1 (day), 7 (week), 30 (month), or 99 (n-day).
  The specification for defaultView affects only the initial display of the search filter. After the schedule displays, CA SDM automatically keeps the filter view selection aligned with the current view.
  Default: 30
- **firstday=0|1|2|3|4|5|6|7**
  Specifies the first weekday on the monthly view as a number between 0 (Sunday) and six (Saturday).
  Default: 0
- **export=xxx|0**

Specifies the code name of the template used for exporting in iCalendar format. Setting the value to 0 indicates the export feature and button are disabled.
Default: ChangeSchedule

- **legend=1|2|0**
  Specifies the location of the schedule legend showing the name and formatting of the groups on the schedule. You can set the value to 1 to position the legend above the schedule, or 2 to position the legend below the schedule. Set the value to 0 to disable the legend.
  Default: 2
- **maxGroups=0/*n***
  Specifies the maximum number of groups to be displayed in a single cell of the calendar month view.
  If there are more than maxGroups scheduled for a single day, CA SDM displays only the first maxGroups-1, and replaces the last with a "..nn more changes" hyperlink that the user can mouseover or click to see the full list. Set the value to 0 to disable this feature and allow an unlimited number of events in a calendar cell.
  Default: 4
- **nday=(n,n,...)**
  Specifies selections for the drop-down list for the n-day view.
  The specification is a list of day counts that are to be included in the drop-down list, or 0 to indicate that the n-day drop-down list is omitted from the schedule. The first value specified is the default for the drop-down list.
  Default: (3,7,14,28)
- **round=(*hr,min*)|0**
  Specifies whether schedule start and end dates are rounded when collecting change orders or knowledge documents into groups. Specify round=0 to disable rounding.
  By default, schedule start and end date groups objects. All CA SDM dates include a time, and without rounding, objects scheduled as little as a minute apart would be in separate groups. Rounding determines the group after adjusting the start date to an earlier hour or minute and the end date to a later hour or minute.
  The value of round specifies either an hour or a minute (but not both). Times are rounded to the nearest multiple of the value specified, for example:
  round=(0,15) rounds to the nearest quarter hour
  round=(0,30) rounds to the nearest half hour
  round=1 rounds to the nearest hour
  round=12 rounds to the nearest half day (12:00 AM or PM)
  round=24 rounds to the nearest day
  Default: (0,15)
- **timefmt=24hr|([am],[pm])**
  Specifies the format of times on the calendar views of the schedule.
  The default value of 24hr specifies that times are displays in 24 hour format (0:01 - 23:59). The alternative value of (am,pm) specifies a suffix for either morning or afternoon times, or both.

  NOTE
  All schedConfig arguments are optional.

schedAttr Macro -- Specify a Stored Attribute

The schedAttr macro specifies an attribute stored for each item selected for the list. Stored attributes are available for hover information on the monthly view, for the detailed or summary information in views other than the monthly view, and in the setSchedEvents() JavaScript function. The following values are valid macro arguments:

- **attr=xxxx**
  (Required) Specifies an attribute from the object on the schedule, such as a change order or Knowledge Document. Dotted attributes are permitted. The keyword attribute name CI*nn* can be used on the Change Calendar to specify that first *nn* CIs associated with the change order are included with the information stored.

  NOTE
  This argument is the only required argument for the schedAttr macro.
- **attrRef=.COMMON_NAME|xxxx**

Stores the attribute of the referenced table stored for an SREL attribute (ignored for non-SREL attributes). The attribute name specified must be prefixed with a dot.
Default: .COMMON_NAME

- **label=**
  Displays a label for the attribute on the n-day view.
  Default: the Majic DISPLAY_NAME of the attribute
- **ident=1|0**
  Specifies whether the attribute is an identifier for the object (such as a reference number of a change order). Identifier attributes are displayed without a label in front of the group name in hover information and the n-day view.
  Default: 0
- **detail=1|0**
  Specifies whether the attribute is included in the detail information shown on views other than the monthly view. Detail information is the information shown when the Summary Only check box on the view is not selected.
  Default: 1
- **hoverInfo=1|0**
  Specifies whether the attribute is included in the hover information pop-up displayed on the monthly view when the mouse pointer hovers over a group, or the user presses Alt+Right Arrow when focus is on the group.
  Default: 0
- **summary=1|0**
  Specifies whether the attribute is included in the detail information shown on views other than the monthly view. Detail information is the information shown when the Summary Only check box on the view is not selected.
  Default: 0

  NOTE
  CA SDM displays attributes in summary, detail, or hover information in the same order as their schedAttr macros.

schedGroup Macro -- Specify an Event Group

The schedGroup macro specifies the name and color coding of a group of items. The monthly view aggregates all items in a group into a single event. Views other than the monthly show individual items in the format for the group to which they belong. The following optional values are valid macro arguments:

- **grpname=xxx**
  (Required) Specifes the name of the group. The macro automatically assigns a number to the group and assigns the number to a JavaScript variable with a name of the form schedGroup_xxx, where xxx is the name of the group. This variable can be used in the setSchedEvents() JavaScript function to create an event belonging to the group.

  NOTE
  This argument is the only required argument for the schedGroup macro.
- **label=xxx**
  Specifies a label for the group. If specified, the label is displayed in all views.
- **legend=xxx|0**
  Displays a description of the group for the legend that appears at the bottom of the schedule. Groups appear in the legend if at least one example of the group exists in the current view. Specifying 0 causes the group always to be excluded from the legend.
  Default: 0
- **color=black|*color***
  Specifies the color of text in items of this group. You can specify color in CSS format, either a valid web color or a hexadecimal value preceded by a pound sign.
  Example: Enter either "#FF0000" or "red" for red.
  Default: black
- **bgcolor=white|*color***

Specifies the background color of items of this group. You can specify bgcolor in CSS format, either a valid web color or a hexadecimal value preceded by a pound sign.
Example: Enter either "#FF000" or "red" for red.
Default: white.

- **style=normal|bold|italic**
  Specifies the style of text of this group in the normal, bold, or italic style.
  Default: normal

Configure Blackout and Maintenance Windows on the Change Calendar

You edit PDM_MACRO statements in list_chgsched_config.htmpl to modify colors, labels, legends, and icons for change windows.

> **NOTE**
> If you use schedGroup macros to change the formatting of windows on the Change Calendar, you must also update *schedule.css* if you want the formatting to be consistent with the Change Scheduler.

To configure change windows

1. Open the *list_chgsched_config.htmpl* form in a text editor or WSP.
   You can locate this file in the following directory:
   ```
   $NX_ROOT\bopcfg\www\htmpl\web\analyst\
   ```
2. On the schedGroup macro, modify the following PDM_MACRO statements:
   For maintenance windows:
   ```
   <PDM_MACRO NAME=schedGroup grpname=maintWindow
       bgcolor=lightgreen
       label="Maintenance"
       legend="Maintenance Window"
       icon= "confirmation_12.png">
   ```
   For blackout windows:
   ```
   <PDM_MACRO NAME=schedGroup grpname=blackoutWindow style=italic color=white
       bgcolor=black
       label="Blackout"
       legend="Blackout Window"
       icon= "warning_12.png">
   ```
   - **bgcolor**
     Specifies the background color of the window.
   - **label**
     Specifies the window label as Blackout or Maintenance.
   - **legend**
     Specifies the text of the legend as it appears on the Change Calendar.
   - **icon**
     Specifies an optional URL to a 12x12 pixel web graphic.
     This icon displays with a change order or group on the Change Calendar if the change order lies completely within a maintenance window or overlaps a blackout window.
3. Save the form.
   The change windows are configured.

The setSchedEvents() JavaScript Function

The setSchedEvents() JavaScript function creates events in the schedule. Modify this function when you want to view any new group objects. The predefined group objects appear by default.

CA SDM calls setSchedEvents() once for each object (change order or knowledge document) selected by the schedule search filter. The function creates events for the object by calling a second function, schedEvent(), and passing the group ID, start date, and end date of the event.

The function can create any number of events (including zero) for an object. The default setSchedEvents() function for the Change Calendar (list_chgsched.htmpl) creates one event for each change order and groups change orders by change type. This function is coded as follows:

```
1.    function setSchedEvents( chg )
2.    {
3.    var grpnum;
4.    switch( chg["chgtype"] - 0 ) {
5.        case 100: grpnum = schedGroup_std;  break;
6.        case 300: grpnum = schedGroup_emer; break;
7.        default:  grpnum = schedGroup_norm; break;
8.    }
9.    chg.schedEvent( grpnum, chg["sched_start_date"], chg["sched_end_date"] );
10.    }
```

The case parameter specifies the change type ID.

The function has a single argument of a JavaScript object containing the attributes specified by schedAttr macros. The switch statement in lines 4-8 examines the chgtype attribute of the change order, and assigns the appropriate group number from one of the schedGroup_xxxx variables defined by previous schedGroup macros. On line 9, it calls the schedEvent() function to create an event in the schedule, passing the group number previously assigned and the schedule start and end dates. The dates are available in the argument object because they were specified in earlier schedAttr macros.

Change Window Management

Change Window management in CASM.

This page contains:

- What is a Change Window
- Define Change Windows
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What is a Change Window

A Change Window is a scheduled period when a Configuration Item (CI) can be modified or is off-limits for changes. You can view these windows in the Change Calendar tab.

There are two types of windows:

1. **Maintenance Windows**:
   A Change Window is a scheduled period when a CI, a set of CIs, or all CIs can be changed. These windows help minimize disruptions to critical business processes, especially since changes may require downtime. Typically, scheduled changes should occur within a maintenance window.
   There are two types of Maintenance Windows:
   a. **Global**: A Global Maintenance window is applicable for all the CIs.
   b. **Non-Global**: A Non-global Maintenance window can be linked to specific CIs, indicating that the changes apply only to those associated CIs.
2. **Blackout Window:**

A Blackout Window is a scheduled period when a CI, a set of CIs, or all CIs cannot be changed. This may be due to reduced support during holidays, corporate events, or critical times like fiscal year-end. Scheduled changes should typically occur only outside of blackout windows.

a. **Global**: A Global Blackout Window applies to all CIs, meaning that no changes can be made to any CI during this period.
b. **Non-Global**: A Non-global Blackout Window is associated with specific CIs, meaning that changes cannot be made to those particular CIs during this period.

For example: To create a Global Blackout Window named Holiday, set it to start in November and end in January in the Eastern Time Zone (American). It's recommended not to schedule any changes during this period. While this blackout window will alert users of any scheduled changes, it does not prevent those changes from occurring.

Define Change Windows

You can define change windows to avoid collisions on the organization. In the Change Calendar tab look at the existing schedules and define the duration required, to create the window.
Understand the recurrence pattern, like the window should be one time/daily/weekly/monthly/yearly with an end date and it should be applicable for all the CIs in the organization or applicable for a set of CIs in the organization.
Define how often the change window occurs. It can be a one-time event or happen daily, weekly, monthly, or yearly. Set an end date if needed, and decide if it applies to all Configuration Items (CIs) or just a specific group.

The following recurring patterns are applicable to all types of Change Windows:

- **None**:
  Specifies no recurrence pattern.
- **Daily**:
  Specify either the number of days between recurrences, or that the window should recur every day.
- **Weekly**:
  Specifies the number of weeks between recurrences, and the weekdays when the window recurs. Windows longer than 24 hours can only recur on a single weekday.
- **Monthly**:
  Specifies the number of months between occurrences, and the day of the recurrence. The day can be specified either absolutely (for example, the 7th of the month) or positionally (for example, the second Monday of the month).
- **Yearly**:
  Specifies the number of years between occurrences, and the day of the recurrence. The day can be specified either absolutely (for example, March 7th) or positionally (for example, the second Monday of March).
- **End by**:
  Specifies the end of the recurrence.

Set up the appropriate change windows for your organization.

Create a Change Window

Creating Global Maintenance Windows:

1. Navigate to **Administration** > **Service Desk** > **Change Orders** > **Change Windows**. The **Change Windows List** page appears.
2. Click **Create New**, the **Create New Change Window** page appears.
3. Complete the required fields as mentioned below:
   a. Name: Weekend_Maintenance.
   b. Select the type as Maintenance.
   c. Select the Active status.
   d. Check the Global checkbox.
   e. Select the upcoming Friday and 6:00 PM from the Date Helper as the Start Date.

f. Select the upcoming Sunday and 6:00 PM from the Date Helper as the End Date.
g. Select your time zone.
h. (Optional) Enter a description for the blackout window.
i. On the Recurrence Pattern tab update the recurrence information along with the end date.
j. Click on Save to save the maintenance window.
k. Refresh the change calendar tab to view the Global Maintenance windows created.

Creating Non- Global Maintenance Windows:

1. Navigate to **Administration** > **Service Desk** > **Change Orders** > **Change Windows**. The **Change Windows List** page appears.
2. Click **Create New**, the **Create New Change Window** page appears.
3. Complete the required fields as mentioned below:
   a. Name: Weekend_Maintenance_For_Set_Of_CIs.
   b. Select the type as Maintenance
   c. Select the Active status.
   d. Uncheck the Global checkbox.
   e. Select the upcoming Friday and 6:00 PM from the Date Helper as the Start Date.
   f. Select the upcoming Sunday and 6:00 PM from the Date Helper as the End Date.
   g. Select your time zone.
   h. (Optional) Enter a description for the blackout window.
   i. On the Recurrence Pattern tab update the recurrence information along with the end date.
   j. Click on Save to save the maintenance window.
   k. Refresh the change calendar tab to view the Non Global Maintenance windows created.

Creating Global Blackout Windows:

1. Navigate to **Administration** > **Service Desk** > **Change Orders** > **Change Windows**. The **Change Windows List** page appears
2. Click **Create New**, the **Create New Change Window** page appears.
3. Complete the required fields as mentioned below:
   a. Name: Weekend_Blackout.
   b. Select the type as Blackout.
   c. Select the Active status.
   d. Check the Global checkbox.
   e. Select the upcoming Friday and 6:00 PM from the Date Helper as the Start Date.
   f. Select the upcoming Sunday and 6:00 PM from the Date Helper as the End Date.
   g. Select your time zone.
   h. (Optional) Enter a description for the blackout window.
   i. On the Recurrence Pattern tab update the recurrence information along with the end date.
   j. Click on **Save** to save the Blackout window.
   k. Refresh the change calendar tab to view the Global Blackout windows created.

Creating Non- Global Blackout Windows:

1. Navigate to **Administration** > **Service Desk** > **Change Orders** > **Change Windows**. The **Change Windows List** page appears
2. Click **Create New**, the **Create New Change Window** page appears.
3. Complete the required fields as mentioned below:
   a. Name: Weekend_Blackout_For_Set_Of_CIs.
   b. Select the type **Blackout**.

c. Select the **Active** status.
d. Uncheck the **Global** checkbox.
e. Select the upcoming Friday and 6:00 PM from the Date Helper as the Start Date.
f. Select the upcoming Sunday and 6:00 PM from the Date Helper as the End Date.
g. Select your time zone.
h. (Optional) Enter a description for the blackout window.
i. On the Recurrence Pattern tab update the recurrence information along with the end date.
j. Click on Save to save the blackout window.
k. Refresh the change calendar tab to view the Non Global Blackout windows created.

Associate a CI with Change Window

CIs can only be associated to Non-Global Maintenance Windows and Non-Global Blackout Windows.

Steps to associate CIs to a Non Global Maintenance/Blackout window :

1. Navigate to the **Change Window Detail** page.
2. Click **Associated CIs**, and the **Associated CIs** list page appears.
3. Click **Update Configuration Items**, and the **CI Search** page appears.
4. Specify information for the required CIs.
5. Click **Search**, the **Available CIs** page appears.
6. Move any required CIs in the Available CIs list to the Associated CIs list.
7. Click **OK**, and the **Change Window Details** page is displayed.
8. Click **Save**.
9. The CI is associated with a maintenance/blackout window.

The change window, the CIs, and the selected window associations are saved.

Ticket Management

This article contains the following topics:

A ticket can be an incident, problem, change order, issue or request

Use the Quick Profile

You can view detailed information about the contacts and organizations that make up your enterprise. Depending on your role, you can access the Quick Profile in the following ways:

- For the roles that use the Quick Profile frequently, click the Quick Profiles tab.
- For the roles that use the Quick Profile less often, select the View, Quick Profile command on the menu bar of a Scoreboard tab.
- For the roles that create new tickets using the Quick Profile, click the Quick Profile button on ticket detail pages.

You can use the Quick Profile as follows:

- Shows contact and organization information, environment, and history in the left pane.
- Display the windows for different types of contact history depending on your latest selection in the right pane.
- View the Scratchpad, which allows you to enter description information to the record you are creating in the bottom pane.

Quick Profile Scratchpad

You can use the Scratchpad Area at the bottom of the Quick Profile screen to merge information about the current user into new incidents, requests, issues, change orders, and activity logs. You can merge the text and contact information into a new record. You can also use existing templates to create records.

NOTE
The type of tickets that you can create from the Scratchpad Area depends on the Preferred Document setting for the Access type of the selected contact. Your administrator determines this setting. This setting also determines the type of tickets that the "Quick" button labels use in the Scratchpad Area.

The Quick Profile Scratchpad lets you perform the following tasks:

- Check the spelling of the text in the scratchpad by clicking Spelling.
- Search the knowledge base using the keywords that are entered in the scratchpad by clicking Search Knowledge.
- Clear all text from the scratchpad by clicking Clear Scratchpad.
- Create a ticket using the data from the scratchpad and immediate close the ticket by clicking Quick Close.
- Create a ticket quickly using the selected template by clicking Quick. No detail page displays, but a message indicating the results of the quick create process is displayed.
- Create a ticket by opening the detail page for the ticket type that is selected in the Type field. The text that is entered in the Scratchpad Area displays in the Description field of the new ticket, and the current contact is the Affected End User of the new ticket.

Quick Profile Contact Information

The left pane of the Quick Profile displays the following contact and organization information, environment, and history details for the selected contact:

- **Information**
  Displays general information about the contact, such as name, status, access type, and system login information. When a contact requires special handling, a banner indicates the handling classification.
- **Environment**
  Displays a list of the configuration items associated with the selected contact, and the number of incidents, problems, requests, change orders, and issues for each configuration item. Click Update Environment to add or remove configuration items for this contact.
- **Issue History**
  Displays a list of the issues for which the selected contact is the Affected End User. You can search for a specific issue, filter the list of issues displayed, or create an issue from this list.
- **Request History**
  Displays a list of the requests for which the selected contact is the Affected End User. You can search for a specific request, filter the list of requests displayed, or create a request from this list.
- **Incident History**
  Displays a list of the incidents for which the selected contact is the Affected End User. You can search for a specific incident, filter the list of incidents displayed, or create an incident from this list.
- **Problem History**
  Displays a list of the problems for which the selected contact is the Affected End User. You can search for a specific problem, filter the list of problems displayed, or create a problem from this list.
- **Change Order History**
  Displays a list of the change orders for which the selected contact is the Affected End User. You can search for a specific change order, filter the list of change orders displayed, or create a change order from this list.

Add Activities to a Ticket

You can record all the actions that you are taking to resolve the ticket on the Activities tab. Activities can include actions such as research, calling a customer, and transferring responsibility to a different analyst. Some activities, such as "Initial" or "Field Update" are posted automatically when you create the ticket or update a field. Others, such as returning a call,

changing the status, or transferring the ticket are posted when you select items from the Activities menu or from the Activities page itself.

Follow these steps:

1. On the Service Desk tab, browse to the ticket type such as, Incidents, Problems, Requests, Change Orders, or Issues.
2. Navigate to Assigned or Unassigned, and select the priority for the desired ticket.
3. Open the desired ticket from the list that appears.
4. On the menu bar, click Activities, and then select one of the following options:
   - **Update Status** to update the status of the ticket.
   - **Callback** to record the time that is spent on the call.
   - **Research** to record the time spent conducting the research.
   - **Log Comment** to record your comments.
   - **Solution** to enter a description of the solution.
   - **Transfer** to reassign a problem to another technician.
   - **Escalate** to escalate the priority of a problem.
   - **Manual Notify** to send a manual notification to the user.
5. Edit the fields as appropriate. The following fields require explanation:
   - **Internal?**
     Select this check box if the activity must only be visible to internal users.
   - **Time Stamp**
     Specifies the date and time the activity began. This field shows the system date and time the activity record was opened. This field cannot be edited.
   - **Date of Activity**
     The date and time the activity was performed. Defaults to the date and time the activity record was opened, but can be changed.
   - **Time Spent**
     Specifies the amount of time that is spent on the activity. Enter this value in hours, minutes, and seconds (hh:mm:ss).
   - **Related Ticket Activity Type**
     Specifies the type of activity that is generated for a related ticket, for example, Update Status
   - **Change**
     Select an existing change order to which you want to attach the ticket.
   - **Impact**
     Indicates the effect that the record has on tasks being performed. Your system administrator can modify the default impact codes, so they can vary from one installation to another.
6. Click Save.
   The activity is recorded on the Activities tab on the ticket details page.

Provide a Reason for Escalating the Ticket

If your priority calculation settings let you change Urgency and Impact values, you provide a reason for escalating the ticket. If priority calculation is not enabled, you can edit the priority field.

Follow these steps:

1. Open a ticket, such as an Incident or a Problem.
   The Ticket Detail page appears.
2. Modify the Urgency or Impact of the ticket.
3. Click Save.
   The Escalate Detail page appears and displays the current, read-only values for Priority, Urgency, and Impact.
4. Enter a reason for why you want to change the Urgency or Impact values.

5. Click Accept.
   The ticket is modified and the activity log updates with your changes.

Send a Manual Notification to a Temporary Email Address

You can send a manual notification to a temporary email address. For example, one that is not associated with a contact record. This operation is available only if the Administrator has installed the notification_allow_temp_address Update Options (Options Manager, Notifications).

Follow these steps:

1. On the Service Desk tab, browse to the ticket type such as, Incidents, Problems, Requests, Change Orders, or Issues.
2. Navigate to Assigned or Unassigned, and select the priority for the desired ticket.
3. Open the desired ticket from the list that appears.
4. On the menu bar, click Activities, Manual Notify.
   The Manual Notification page appears.
5. Complete the following field, and click Add Email.
   - **Email address**
     Specifies an SMTP address for an email recipient. You can specify temporary email addresses (for example, addresses not associated with a contact record). Separate multiple addresses with a semi-colon. This field appears only if the Administrator installed the notification_allow_temp_address Update Options (Options Manager, Notifications).

   The address is added to the Recipients list.
6. Complete the remaining fields as appropriate.

   NOTE

   You can also add a personalized response such as a generic or administrator signature to the notification. The response is added at the end of the message text. For more information about personalized responses, see Personalized Responses.
7. Click Notify.
   The manual notification is sent and the activity is recorded on the Activities tab of the ticket Detail page.

Attach to Existing Change Order

You can attach an incident, problem, or request, to a related change order. Change orders are requested changes that vary from the original scope of a specific project or task.

Follow these steps:

1. On the Service Desk tab, select the incident, problem, or request to which you want to attach the change order.
   The ticket detail page appears.
2. Select Activities, Attach to Existing Change Order on the menu bar.
   The Attach Change Order to Incident page appears.
3. Click Change.
   The Change Order Search page opens.
4. Enter the search criteria that you want to use and click Search.
   A list of change orders matching the search criteria appears.
5. Select the desired change order from the list.
   The Attach Change Order to Incident page displays the selected change order number in the Change field.

   NOTE
   If you know the Change Order number, you can type it in the Change field.
6. (Optional) Enter the comments in the Remarks field.
7. Click Attach.

The ticket is attached to the change order, and the activity is recorded on the Activities tab on the ticket detail page.

Detach Change Order

You can end the association between an incident and an attached change order.

Follow these steps:

1. On the Service Desk tab, select the desired incident, problem, or request from which you want to detach the change order.
   The ticket detail page appears.
2. Select Activities, Detach Change Order on the menu bar.
   The Detach Change Order from Incident page appears.
3. Click Detach.
   The ticket is detached from the change order, and the activity is recorded on the Activities tab on the ticket detail page.

Solution

When a problem is resolved, the solution can be recorded with the problem record for future reference.

To log a solution for a problem

1. Select the problem from the Problem List page on the Service Desk tab.
   The Problem Detail page appears.
2. Select Activities, Solution on the menu bar.
   The Create New Activity page appears.
3. Change the date of the activity, if appropriate, and record the time spent resolving the problem.
4. Enter the description of the solution in the User Description field. For example, if a workaround for the problem was discovered, document the workaround here.
   See Activities Fields for field descriptions.
5. Click Save.
   The solution is recorded on the Activities tab and on the Solutions tab on the Problem Detail page.
   OR
   Click Save and Submit Knowledge to save the solution as a Knowledge Document.

Activities Fields

The following fields are present on one or more of the pages available through the Activities menu on ticket detail pages:

Number

Specifies the number of the item for which activity is being recorded. Cannot be edited.

Summary

Specifies the summary description of the record. Cannot be edited.

Activity Type

Specifies the type of activity being recorded. For example, if you select Log Comment from the Activities menu, the Activity Type is Log Comment. Cannot be edited.

Current Status

Specifies the current status of the record. Cannot be edited.

New Status

Specifies a new status to be assigned to the record.

Current Priority

Specifies the current priority level of the record. Cannot be edited.

New Priority

Specifies a new priority level to be assigned to the ticket.

Current Assignee

Specifies the analyst currently assigned to the ticket. Cannot be edited.

New Assignee

Specifies a new analyst to be assigned to the ticket.

Current Group

Specifies the group currently assigned to the ticket. Cannot be edited.

New Group

Specifies a new group to be assigned to the ticket.

Analyst

Specifies the name of analyst who performed the activity. This is the currently logged in user. Cannot be edited.

Internal?

Select this check box if the activity should only be visible to internal users.

Time Stamp

Specifies the date and time the activity began. This is the system date and time the activity record was opened. Cannot be edited.

Date of Activity

The date and time the activity was performed. Defaults to the date and time the activity record was opened, but can be changed.

Time Spent

Specifies the amount of time spent on the activity. Enter this value in hours, minutes, and seconds (hh:mm:ss).

Related Ticket Activity Type

Specifies the type of activity generated for a related ticket, for example, Update Status.

User Description

Specifies the description of the action taken. Click Spelling to check the spelling of the text entered. Click Submit Knowledge to submit the solution to the Knowledge Base.

System Description

Specifies the system description of the action taken. Cannot be edited.

Personalized Response

Specifies the personalized response you want to send for this item. See Personalized Responses for more information.

> **NOTE**
>
> The Personalized Response field is present only if an active personalized response exists.

Recipients

Specifies the contacts who will receive manual notification pertaining to the ticket. Click Add Recipient to search the contact list, or Remove Recipient to remove the contact from the notification list.

Urgency

Specifies the urgency level associated with the manual notification. Select the appropriate urgency code from the drop-down list.

Current Urgency

Specifies the current value for Area Urgency that is assigned to this ticket.

New Urgency

Specifies the new Urgency that is assigned to this ticket.

Preferred Method

Specifies the preferred method for delivering the manual notification message (for example, Email or Pager_Email).
Message Title

Specifies the subject heading that will appear on the manual notification.
Message Text

Specifies the text that will appear in the body of the manual notification message.
Change

Select an existing change order to which you want to attach the ticket.
Remarks

Enter text explaining why the ticket is being attached to an existing change order.
Impact

Indicates the effect the record has on tasks being performed. Your system administrator can modify the default impact codes, so they can vary from one installation to another.
Current Impact

Specifies the current value for Service Impact that is assigned to this ticket.
New Impact

Specifies the new Service Impact that is assigned to this ticket.
Impact

Indicates the effect the record has on tasks being performed. Your system administrator can modify the default impact codes, so they can vary from one installation to another.
Current Urgency

The current value for Area Urgency that is assigned to this ticket.
Current Impact

The current value for Service Impact that is assigned to this ticket.
New Urgency

Specifies the new Urgency that is assigned to this ticket.
New Impact

Specifies the new Service Impact that is assigned to this ticket.

Create a Quick Close Ticket

You can create a Quick Close ticket, for example, a Quick Close Issue. The Quick Close option of the Quick Profile feature lets you open and close a new ticket in one step, for example, if you resolve the issue in the same session.

To create a quick close ticket

1. Select View, Quick Profile on the Service Desk tab.
   The Quick Profile Contact Search page opens.
2. Complete one or more of the search fields for the contact, and click Search.
   The Quick Profile Contact List populates with those contacts that match your search criteria.
3. Select a contact.
   The right pane displays the information for that contact.
4. Click Quick Close.
   The ticket is created.

Quick Profile Search Fields

> **NOTE**
>
> If you are using multi-tenancy, a tenant drop-down list appears in the search filter. If you select <empty> in this drop-down list, the search is public.

Last Name/First Name/Middle Name

The last, first, and middle names of the contact. Only the Last Name field is required.

Contact Type

The contact type from the drop-down list.

Status

Indicates whether the contact is Active or Inactive.

Contact ID

A unique identifier for the contact. If the default user authentication is being used, the value in this field is used as the password when the user logs in.

User ID

The user name of the contact. This value is used to log the contact into the system.

Access Type

The access type from the drop-down list. The access type determines the functions the contact can access.

Location

The contact's location. Enter the location directly, or click the search icon to search for the desired location.

Organization

The contact's organization. Enter the organization directly, or click the search icon to search for the desired organization.

Department

The contact's department. Enter the department directly, or click the search icon to search for the desired department.

Phone Number

The contact's telephone number.

> **NOTE**
>
> You can click the More icon to display the Additional Search Arguments field. This field is intended only for expert users who understand SQL and Majic and can use it to specify search arguments that are not available in the standard search filter fields. You can enter a SQL WHERE clause in this field to specify an additional search argument.

Attach a Service Type Event

Attach a Service Type Event

Service types determine the level of support that is given for a ticket. You can associate service type events with tickets. An event that is associated with a service type is attached to records defined with that service type.

Follow these steps:

1. On the Service Desk tab, select Incidents, Problems, Requests, Change Orders, or Issues.
2. Navigate to Assigned or Unassigned, and select the priority for the desired ticket.
3. Select the desired ticket from the list that appears.
4. Choose the Service Type tab, and click Attach Service Type Event.
   OR
   Choose Actions, Attach Service Type Event.
   The Attach Service Type Event page appears.
5. Click Service Type Event.
   The Event Search page appears.
6. Complete one or more of the search fields, and click Search.
   The Event List displays the events that match your search criteria.

7. Select the desired event.
   The Attach Service Type Event page appears.
8. Click OK.
   The Delay Time field appears. The delay time indicates the interval of time after which the event occurs.
9. Specify the delay time as hh:mm:ss (hours, minutes, seconds).
   For example, if the event is attached to the incident at 2:20, and the Delay Time is 01:00:00 (one hour), the event occurs at 3:20.
10. Click OK.
    The ticket detail page appears with the event listed on the Service Type tab.

Delay or Resume a Service Type Event

You can delay and resume service type events.

Follow these steps:

1. Select the desired ticket from the list page on the Service Desk tab.
   The ticket detail page appears.
2. Select the Service Type tab.
   A list of any service type events that have been added to the ticket appears.
3. Click Delay or Resume.
   The Reason page appears.
4. Enter the reason, and click OK.
   The ticket detail page displays the Status of the event.

Edit Service Targets

Service targets determine whether Service Level Agreements (SLA) have been met within the required time frame. If a service type has a set of service targets, you can view the status and deadlines for completing each target on the ticket. If necessary, you can update or override service target values such as Workshift.

Follow these steps:

1. On the Service Desk tab, select Incidents, Problems, Requests, Change Orders, or Issues.
   The ticket detail page appears.
2. Click the Service Type tab.
   Additional Service Type information appears at the bottom of the ticket.

   NOTE
   Service targets appear on tickets that meet the conditions that the Administrator sets up. Priority calculation can be a factor in how target information calculates and displays.
3. In the Target column, click the target.
   The Service Target Detail page appears.
4. Click Edit.
   The Update Service Target page appears.
5. Complete the following fields as appropriate:
   - **Name**
     Identifies the service target.
   - **Target Duration**
     Specifies the amount of allotted time to perform the service target. You can only override this value by editing the ticket.
   - **Workshift**
     Displays the schedule to use for service target time calculations.
   - **Condition**

Specifies the name of the condition or site-defined condition. The condition evaluates the ticket data to determine whether the service target is met.

- **Required Outcome**
  Displays the required result of the condition or site-defined condition Macro.
- **Cost**
  Specifies the penalty that incurs for missing the target. This information also displays on the ticket.
- **Target Date/Time**
  Specifies the deadline for completing the target. If the ticket is in a Hold status or the target has been met, this field is blank.
- **Actual Date/Time**
  Specifies the date and time that the condition was satisfied or the user clicked Set Actual.
- **Time Left**
  Specifies the amount of remaining time for the service target. A negative value indicates the amount of time that exceeded the target time frame.
- **Allow Set Actual**
  Lets the users set the date and time of a service target.
- **Allow Reset Actual**
  Lets the users reset the service target.
- **Last Modified Date/Time**
  Displays the date that this ticket was last modified.
- **Service Type**
  Displays the Service Type that is attached to this service target.
- **Service Target Template**
  Specifies the name of the service target template to link to the service type.
- **Last Modified By**
  Displays the name of the last person who edited the ticket.
- **Lock Target Date/Time From Hold Recalculations**
  Prevents automatic target date and time updates when the ticket goes on hold or when the ticket is delayed.

6. Click Save.
   The system saves and updates the service target.

View Ticket Counters and Timers for Service Targets

If an analyst assigned a set of service targets to a ticket, you can view the status and deadlines for completing each target.

Follow these steps:

1. On the Service Desk tab, display a list of Incidents, Problems, Requests, Change Orders, or Issues.
   The ticket detail page appears.
2. Click the Service Type tab.
   Additional Service Type information appears at the bottom of the ticket.
3. In the Target column, click the Service Target for additional information.
   The Ticket Counters and Timers section appears near the bottom of the Assigned Service Target Detail page. The Assigned Service Target Detail page displays the following fields:
   - **Name**
     Displays the name of the service target.
   - **Target Duration**
     Displays the amount of allotted time to perform the service target. You can only override this value by editing the ticket.
   - **Workshift**

Displays the schedule used for time calculations for the service target.

- **Condition**
  Displays the condition or site-defined condition macro that evaluates the ticket data to determine whether the work can complete within the target time frame.
- **Required Outcome**
  Displays the required result of the condition or site-defined condition Macro.
- **Cost**
  Displays the penalty that incurs for missing the target. This information also displays on the ticket.
- **Target Date/Time**
  Displays the deadline for completing the target. If the ticket is in a Hold status or the service target has been met, this field is blank.
- **Actual Date/Time**
  Specifies the date and time that the condition was satisfied or the user clicked Set Actual.
- **Time Left**
  Displays the amount of remaining time for the service target. A negative value indicates the amount of time that exceeded the target time frame.
- **Allow Set Actual**
  Displays whether you can set the actual time. Yes indicates that you can set the Actual Date/Time of a Service Target. No indicates that you cannot override the Actual Date/Time.
- **Allow Reset Actual**
  Displays whether you can restart the time. Yes indicates that you can reset the Actual Date/Time of a Service Target. No indicates that you cannot reset the Actual Date/Time.
- **Last Modified Date/Time**
  Displays the date that this ticket was last modified.
- **Last Modified By**
  Displays the name of the last person who edited the ticket.
- **Service Type**
  Displays the name of the service type that attached this service target.
- **Service Target Template**
  Displays the name of the service target template that was linked to the service type that was used to create this Service Target.
- **Lock Target Date/Time From Hold Recalculations**
  Locks the Target Date/Time from being automatically updated when the ticket goes on hold or is delayed.
- **Last Start Date/Time**
  Displays the last time the service target timer was started.
- **Ticket Status**
  Displays the value of the Status field of the ticket.
- **Hold Status**
  Displays whether the ticket status has placed the ticket on hold.
- **Last Hold Date/Time**
  Displays the last time that the ticket was placed on hold.
- **Hold Count**
  Displays the number of times the ticket was placed on hold.
- **Last Resolved Date/Time**
  Displays the last time that the ticket transitioned to a resolved status.
- **Resolved Count**
  Displays the number of times that the ticket changed to resolved status.
- **Last Closed Date/Time**
  Displays the last time the ticket was changed to a closed status.
- **Closed Count**

Displays the number of times the ticket was changed to a closed status.

- **Ticket Open Date/Time**
  Displays the date and time the ticket was opened.
- **Ticket Resolved Date/Time**
  Displays the date and time the ticket was resolved.
- **Ticket Closed Date/Time**
  Displays the date and time the ticket was closed.

View Service Target Status

On an open ticket, you can view the status for each service target. Status information such as Time Left and Violation Cost help you prioritize your work.

Follow these steps:

1. On the Service Desk tab, display a list of Incidents, Problems, Requests, Change Orders, or Issues.
   The respective ticket list displays with the following Service Target information:
   - **Service Target**
     Displays the time that the next service target is due.
   - **Projected Violation**
     Displays the incurred cost when the service type time limit is violated.
2. Select the ticket that you want from the list page.
   The ticket detail page appears.
3. Select the Service Type tab.

   NOTE
   Service targets appear on tickets that meet the conditions that the administrator sets up. Priority calculation can be a factor in how target information calculates and displays.

   If the ticket meets predefined target conditions, the Service Targets List the following information about service targets:
   - **Action**
     Sets or resets the Actual Date/Time to the current date and time.
   - **Target**
     Specifies the current service target for the ticket.
   - **Target Date/Time**
     Specifies date and time when this service target is due. If the ticket is in a Hold status, this value is blank.
   - **Actual Date/Time**
     Specifies the time when the target condition was met. If no value appears, the target condition has not been met.
   - **Time Left**
     Specifies the amount of remaining time for the service target when the ticket is on hold. If the service target has been met, the Time Left field shows the unused time. A negative value indicates the amount of time that elapsed since the missed target date.
   - **Violation Cost**
     Displays the penalty that incurs for missing the target. This information also displays on the ticket.

Personalized Response

Personalized Responses are frequently used when you respond to customers. You can create and attach these predefined responses to all ticket types when updating activities for the record. For example, you can append personalized response information to an updated status or a logged comment.

Create a Personalized Response

Create a personalized response so that you can add them in your response to customers.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. From the Scoreboard, choose File, New Personalized Response.
   The Create New Personalized Response page displays.
2. Fill in the fields as follows:
   - **Name** -- Unique identifier and key for the response. This field can be up to 50 alphanumeric characters long.
   - **Response Owner** -- Name of the contact that owns the personalized response. If this field is left blank, the response is available to all analysts. Enter the contact name directly into this field, or click the search icon to search for the desired contact name.
   - **Record Status** -- Indicates whether the response is Active or Inactive.
   - **Response** -- The personalized response text that is delivered to the customers. This field can be up to 1000 characters long.
     You can use variables in this field, for example:
     - Ticket ref_num: @{call_req_id.ref_num}
     - Assignee: @{call_req_id.assignee.combo_name}
     - Customer: @{call_req_id.customer.combo_name}
     - Description: @{call_req_id.description}
   - **Display the Response For** -- Select the type of records for which you would like this response available. By default, Requests, Change Orders, and Issues are selected. For example, if you select the Requests option, you can append this response to a Request record.

   The personalized response is created.

Add Personalized Response to a Ticket

Add a personalized response when you are sending a communication to a customer about a ticket.

Follow these steps:

1. On the Service Desk tab, select Incidents, Problems, Requests, Change Orders, or Issues.
   The ticket detail page appears.
2. Open the ticket for editing.
3. From the Activities menu, select the activity for which you want to add a personalized response.
   The Create New Activity form opens. You see the Personalized Response drop-down at the end of the form.
4. Select the response that you want to add to the ticket and save the form.
   The description is automatically updated with the personalized response. You also see this response in the Log, Activities tab of the ticket.

Search Personalized Responses

You can enter search criteria to filter the Personalized Response List to display only the responses you want to see. You can also search for individual responses that you want to view or edit.

Follow these steps:

1. On the menu bar, select Search, Personalized Responses.
2. Complete one or more of the search fields, as follows, and click Search.
   a. **Active** -- Indicates whether the response is Active or Inactive.

b. **Name** -- The unique identifier for the response. Enter the first few letters of a name to display all responses that begin with those letters.
c. **Response Owner** -- The name of the contact that owns the personalized response. Enter the contact name directly in the field, or click the search icon to search for the desired contact.
d. **Response** -- The text that is delivered to all those who receive this response. Enter the first few letters of the response text to display all responses that begin with those letters.

3. The Personalized Response List page opens to display the responses that match your search criteria. You can select a response to view or edit.

Create a Ticket Template

You can define a ticket, such as an incident, problem, issue, change order, or request, to serve as a model for future tickets. You can then save it as a template. Tickets that are created with that template have certain fields auto-populated with your specified values.

Create a Template from a New Ticket

You can use a new ticket to create a template to be used when creating future tickets.

Follow these steps:

1. On the Service Desk tab, click the File menu and select the ticket type for the template. For example, to create an incident template, click File, New Incident.
2. Complete the fields as appropriate for the template in the page that opens.
3. Select the Template tab at the bottom of the page.
4. Complete the following fields:
   Template Name This name appears in the list of available templates.
   Template Class(Optional) You can assign a class to the template.
   Quick Template Type You can use quick templates to define new requests on the Quick Profile Scratchpad. Select one of the following options:
   – None
   – Quick
   – Review Before Save
   Template Active Specifies whether the template is active or inactive in the database.
   DescriptionDescribe the template settings and intended usage. This field can be used as a filter when searching for a template. Wildcard searches are supported.
5. Click Save.

Create a Template from an Existing Ticket

You can use an existing ticket to create a template.

Follow these steps:

1. On the Service Desk tab, browse to the ticket type such as, Incidents, Problems, Requests, Change Orders, or Issues.
2. Navigate to Assigned or Unassigned, and select the priority for the desired ticket.
3. Select the desired ticket from the list that appears.

   NOTE

   Some roles can only access certain ticket types through the Search menu on the Service Desk menu bar.

4. Click Edit.
5. (Optional) Edit the fields as appropriate.

6. Select the Template tab at the bottom of the page.
7. Complete the following fields:
   Template Name This name appears in the list of available templates.
   Template Class(Optional) You can assign a class to the template.
   Quick Template Type You can use quick templates to define new requests on the Quick Profile Scratchpad. Select one of the following options:
   – None
   – Quick
   – Review Before Save
   Template Active Specifies whether the template is active or inactive in the database.
   DescriptionDescribe the template settings and intended usage. This field can be used as a filter when searching for a template. Wildcard searches are supported.
8. Click Save.

Save Search Filters

This article contains the following topics:

Basic Search

You can set filter values on the search page of an object to list the corresponding details, or leave the fields blank to display all the details.

> **NOTE**
>
> If you are using multi-tenancy, a tenant drop-down list appears in the search filter. If you select <empty> from the drop-down list, the search is public.

Follow these steps:

1. Navigate to **Service Desk**, **Search**, and select any object.
   The search page related to the object is displayed.
2. (Optional) Define the required fields for your search criteria. Click **Save As** to save the filters as your personalized search.
3. Click **Search** to display the search results.
4. Click any link to view the details of that specific result in the list.
   The object details are displayed.

Search Using Personalized Filters

As an Analyst, you can save your search conditions. These saved searches are displayed on the analysts' scoreboard. You need not define the search fields for such objects need not be defined every time a search is performed. Saving the search filters enables users to update the search criteria values and save it for future use. Other analysts can also customize their scoreboards and add these saved searches.

> **NOTE**
>
> : You can save a filter only for those roles for which the scoreboard appears.

By default, you can save a filter for the following objects:

- Incident
- Problems
- Requests
- Change Orders
- Issues
- Announcements
- Configuration Items
- Contacts
- Change Workflow Tasks
- Issue Workflow Tasks

By default, personalized search options are enabled. For disabling the personalized search options, see the *Enable or Disable Personalized Search Options* section below.

Follow these steps:

1. Click a saved search from **My Saved Search** in the scoreboard.
   The search results appear based on the saved search criteria.
2. (Optional) Click **Show Filter** and update the required fields for your search criteria.
3. Do any *one* of the following tasks:
   a. Click **Save** to save the updated filters.
   b. Click **Save As** to save the search with another name.
   c. Click **Search**.
      The page displays search results based on the updated search criteria.
4. Click any link to view the details of that specific result in the list.
   The object details are displayed.

Enable or Disable Personalized Search Options

As an administrator, execute the following command to disable the search options:

```
pdm_options_mgr -c -s DISABLE_SAVED_SEARCHES -v 1 -a pdm_option.inst
```

To ensure that the search options update takes effect while running the pdm_configure command, execute the following command with the **-t** option.

```
pdm_options_mgr -c -s DISABLE_SAVED_SEARCHES -v 1 -a pdm_option.inst -t
```

Restart the **CA Service Desk Manager Server** service.

Create an Issue Template

You can define a ticket, such as an incident, problem, issue, change order, or request, to serve as a model for future tickets. You can then save it as a template. Tickets that are created with that template have certain fields auto-populated with your specified values.

Follow these steps:

- You can use a new ticket to create a template to be used when creating future tickets. Complete the following fields:

1. a. On the Service Desk tab, click the File menu and select the ticket type for the template. For example, to create an incident template, click File, New Incident.
   b. Complete the ticket template fields as appropriate for the template in the page that opens.
   c. Select the Template tab at the bottom of the page.

d. Click Save.

- You can use an existing ticket to create a template.Complete the following fields:

1. a. On the Service Desk tab, browse to the ticket type such as, Incidents, Problems, Requests, Change Orders, or Issues.
   b. Navigate to Assigned or Unassigned, and select the priority for the desired ticket.
   c. Select the desired ticket from the list that appears.

      NOTE

      Some roles can only access certain ticket types through the Search menu on the Service Desk menu bar.
   d. Click Edit.
   e. (Optional) Edit the ticket template fields as appropriate.
   f. Select the Template tab at the bottom of the page.
   g. Click Save

Ticket Template Fields

Complete the following fields:

Template Name This name appears in the list of available templates.

Template Class(Optional) You can assign a class to the template.

Quick Template Type You can use quick templates to define new requests on the Quick Profile Scratchpad. Select one of the following options:

- None
- Quick
- Review Before Save

Template Active Specifies whether the template is active or inactive in the database.

DescriptionDescribe the template settings and intended usage. This field can be used as a filter when searching for a template. Wildcard searches are supported.

Create a Parent-Child Relationship

Records that are entered in the CA SDM system can be related to other records that are entered into the system. For example, a problem could result in two more problems when you attempt to resolve the original problem. The original record is referred to as the *parent*, and the additional records are referred to as *children*.

You can record such relationships in the CA SDM system for future references. The Parent/Child tab on the ticket detail page lists the parent and children tickets of a ticket.

Follow these steps:

1. On the Service Desk tab, browse to the ticket type such as, Incidents, Problems, Requests, Change Orders, or Issues.
2. Navigate to Assigned or Unassigned, and select the priority for the desired ticket.
3. Open the desired ticket from the list that appears.
4. Select the Parent/Child tab and click Update Children.
   The search page opens.
5. Complete one or more of the search fields for related ticket records, and click Search.
   A list of tickets matching the search criteria are displayed.
6. Select the related tickets from the list on the left, and click **>>**.
   The selected tickets are added to the list on the right.

7. When all related tickets are in the list on the right, click OK.
   The detail page displays with the selected tickets listed on the Parent/Child tab.

Close All Children

You can close all the children tickets before closing the parent ticket.

Follow these steps:

1. Select the appropriate ticket type from the Scoreboard. For example, select Requests.
2. Open the parent ticket.
3. Click the Action menu and select Close All Children.

Create Tickets

You can create a ticket such as an incident, problem, request, issue, or change order, using various methods. This article explains these methods.

> **WARNING**
> Depending on your role, you do not have access to all the functionality described in this section. For example, some predefined roles can edit records but cannot create new ones.

Create a Ticket from the File Menu

You can either create a ticket from scratch or use an existing template.

> **NOTE**
>
> : If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. On the Service Desk tab, click the File menu and select the ticket type that you want to create. For example, to create a request from scratch, click File, New Request. To create a request from a template, Click File, New Request from Template.
2. Fill in the fields as appropriate for the ticket. See the field definitions at the end of this page.
   Use the controls available on the tabs at the bottom of this page to process the ticket as appropriate.
3. Click one of the following buttons:
   Save
   Saves the incident and closes the page.
   Auto Assign
   Triggers an auto assignment task, and updates the activity log.

   > **NOTE**
   >
   > This button appears only when the ticket specifies a category or area that has auto-assignment enabled.

   Create Change Order
   Opens the Create New Change Order page. You can create a change order ticket that is associated with this incident.

   > **NOTE**
   >
   > This button appears only when you create incidents, problems, and requests.

   Create Problem
   Opens the Create New Problem page so you can create a problem ticket associated with this incident.

NOTE

This button appears only when you create incidents and requests.

Create Incident
Opens the Create New Incident page so you can create an associated incident ticket.

NOTE

This button appears only when you create change orders and requests.

Reset
Resets all fields to the last saved values.

Find Similar
Opens the Find Similar page to search for similar problems.

Quick Profile
Displays the contact information for the specified user in the **Affected End User** field. You can also view their environment details and their entire ticket history.

Use Template
Displays a list of available templates for the selected ticket type. You can select the template that you want to use for creating this ticket.

Create a Ticket Using Quick Profile

You can use the Quick Profile to identify a contact to be the affected end user of your new ticket. Quick Profile allows you to search for a contact, and view the history of the issues, requests, change orders, incidents, and problems that are assigned to that contact. When you have identified a contact, you can create the incident ticket directly from the Quick Profile.

Follow these steps:

1. Select View, Quick Profile on the menu bar of the Scoreboard.
   The Quick Profile Contact Search window opens.
2. (Optional) Complete one or more of the filter fields to list only the contacts of interest.
3. Click Search.

   NOTE
   All search fields that allow text entry support use of the % wildcard character.

   The Quick Profile Contact List page lists the contacts that match your search criteria.
4. Select the user who is the affected end user of the new ticket.
   The contact pane displays contact information and the ticket history for the contact.
5. In the Scratchpad pane, do the following actions:
   a. Enter a description for the new record in the Scratchpad text field.
   b. Select the ticket type from the Type drop-down list.
   c. (Optional) Enter a Template name for the new record. You can enter a value directly or click the search icon to search for available templates.
6. Click New.
   A new window opens and displays the information that you entered in the Scratchpad.
7. Complete the empty fields as appropriate.
   See the field definitions at the end of this page.
8. Use the controls available on the tabs at the bottom of this page to process the ticket as appropriate.

Add Attachments to Tickets

You can attach a document or a link to a URL to a ticket. The document or URL provides more explanation or justification for the report.

Follow these steps:

1. On the Service Desk tab, browse to the ticket type such as, Incidents, Problems, Requests, Change Orders, or Issues.
2. Navigate to Assigned or Unassigned, and select the priority for the desired ticket.
3. Open the desired ticket from the list that appears.
4. Select the Attachments tab, and then click one of the following options:
   – Attach Document
   – Attach URL
5. Follow the on-screen instructions to attach a file or URL to the incident.
   The URL or the document is added to the record and is listed on the Attachments tab of the Incident Detail page.

View Tickets

You can view summary information for tickets after you create them.

> **NOTE**
>
> : If you are using multi-tenancy, a tenant drop-down list appears in the search filter. If you select <empty> in this drop-down list, the search is public.

Follow these steps:

1. On the Service Desk tab, browse to the ticket type such as, Incidents, Problems, Requests, Change Orders, or Issues.
2. Navigate to Assigned or Unassigned, and select the priority for the desired ticket.
3. Select the folder for the tickets that you want to see.
4. (Optional) Click Show Filter and complete one or more of the fields to specify search criteria that restricts the list to the tickets of interest.
5. Click Search.
   The search result displays the tickets that match the criteria.
6. Click the ticket # link to view the incident of interest.
   The detail page appears.

Use Knowledge to Resolve a Ticket

You can search the CA SDM Knowledge Base to help resolve a ticket, such as an incident, problem, request, issue, or change order.

Follow these steps:

1. On the Service Desk tab, browse to the ticket type such as, Incidents, Problems, Requests, Change Orders, or Issues.
2. Navigate to Assigned or Unassigned, and select the priority for the desired ticket.
3. Open the desired ticket from the list that appears.
4. Select the **Knowledge Management** tab.
5. Enter text in the search field to describe the problem, or copy text from the **Summary** or **Description** fields of the ticket.
6. From the drop-down list, select where you want to search.
7. Click **Search**.
   A list of documents matching the search criteria displays.
8. Look for the document that provides the solution to your ticket.
9. Right-click the document and select **Accept as Solution**.

The knowledge document information is copied into the **Summary** and **Description** fields.

Submit a Knowledge Document

When you log a solution activity for a ticket, you can submit the solution to the knowledge base administrator as a candidate for publication.

Follow these steps:

1. Open a ticket, such as an Incident.
   The ticket detail page appears.
2. On the Knowledge tab, click Submit Knowledge.
   The Create New Document page displays.

For more information on creating knowledge documents, refer to Create Knowledge Documents.

After you submit a solution to the knowledge base, it becomes a candidate for publication in the knowledge base. A Knowledge Management administrator reviews the information and publishes it if it is deemed appropriate and accurate. After it is published, the information becomes available to other users.

View Events

The Event List contains a record of the events that are associated with the ticket. Your administrator maintains events, or procedures the system perform after a certain amount of time has elapsed. For example, an event might send a message to an analyst if a Priority 1 ticket is not resolved within an hour.

Follow these steps:

1. On the Service Desk tab, select Incidents, Problems, Requests, Change Orders, or Issues.
2. Navigate to Assigned or Unassigned, and select the priority for the desired ticket.
3. Select the desired ticket from the list that appears.
4. Click the Event Log tab at the bottom of the details page.
   A list of events for this incident appears.

View the Event History

You can view the event history for a ticket to see what actions have been performed to resolve the ticket.

Follow these steps:

1. On the Service Desk tab, select Incidents, Problems, Requests, Change Orders, or Issues.
2. Navigate to Assigned or Unassigned, and select the priority level for the desired ticket.
3. Select the desired ticket from the list that appears.
4. Select View, Event History on the detail page menu bar.
   The Event History window displays the following fields:
   - **Condition**
     Indicates the condition checked for by the event. CA SDM provides macros that check for standard conditions such as priorities and object status (open or closed).
   - **Event**
     Indicates the name of event that is associated with this row in the Event History list.
   - **Status**
     Indicates whether the event is active or inactive.
   - **Check Time**
     Indicates the time that the event was checked, based on time parameters that are specified in the event configuration.
   - **Time Loaded**

Indicates the time that the event was loaded, based on time parameters that are specified in the event configuration.

View the Event Delay History

You can view the event delay history for a ticket to see descriptions of the service type event delays. Service type event delays let you suspend a service type event to prevent a service level violation.

Follow these steps:

1. On the Service Desk tab, select Incidents, Problems, Requests, Change Orders, or Issues.
2. Navigate to Assigned or Unassigned, and select the priority level for the desired ticket.
3. Select the desired ticket from the list that appears.
4. Select View, Event Delay History on the detail page menu bar.
   The Event Delay History window displays the following fields:
   - **Start Time**
     Indicates the date and time the delay began.
   - **Stop Time**
     Indicates the date and time the delay ended.
   - **Actual Delay**
     Indicates the actual duration of delay.
   - **Effective Delay**
     Indicates the effective duration of delay.
   - **Service Type**
     Defines the support level that is assigned to the ticket (for example, four-hour service requirement).

Event Logs

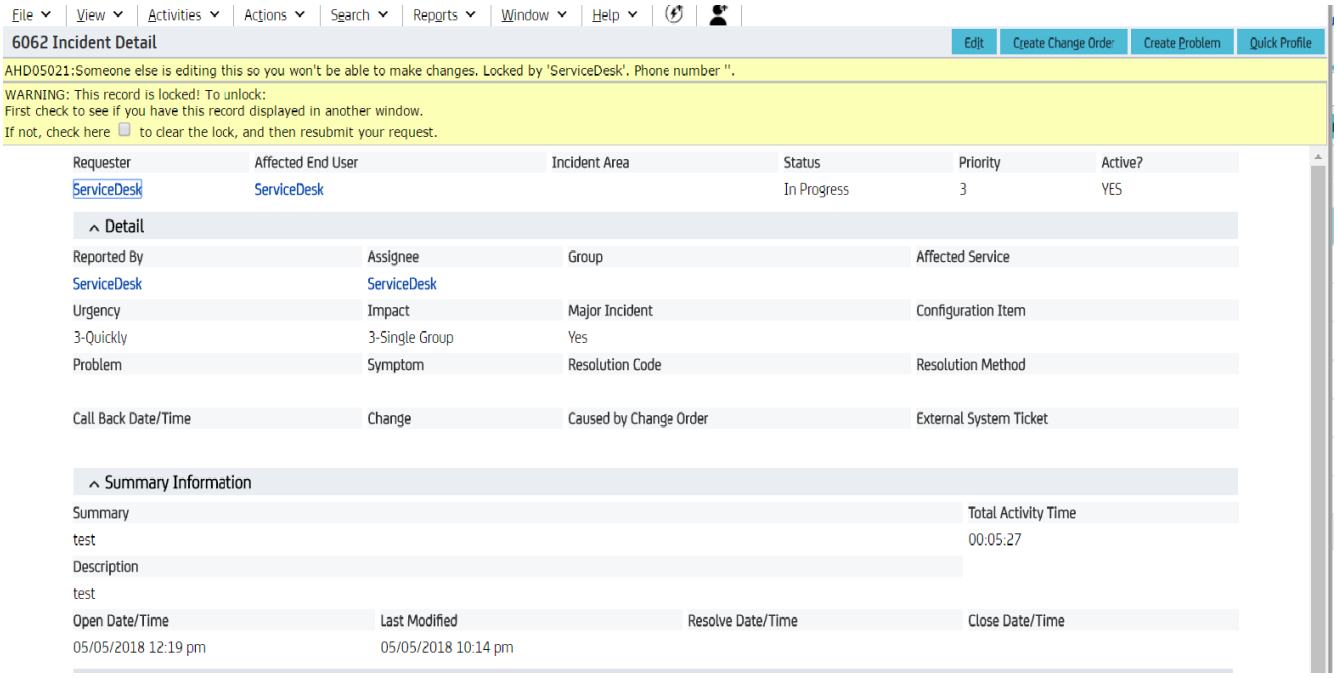
The Event Log displays significant actions that are performed by the user, such as on tickets. For example, you submit a Knowledge Document and link it to an Incident. The Incident logs a failed status transition or detected Change Conflict. In this example, the Contact detail can display the following types of information:

- The log displays the time that the user logged in to CA SDM and when they closed the session.
- The user accepted the Terms of Usage.
- The user created a Knowledge Document.
- The user created a ticket, such as an Incident, Problem, Request, or Change Order.
- The user used a Knowledge Document to avoid creating a ticket.

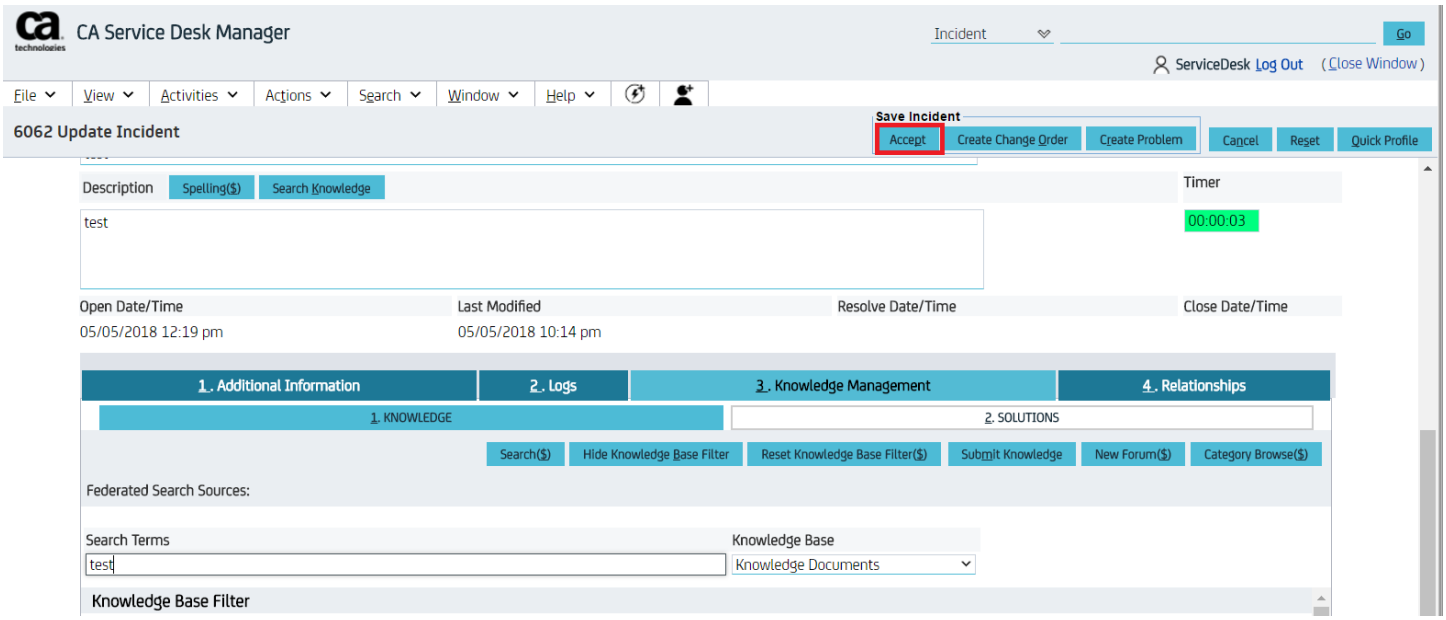
How does Service Desk handle multiple updates in the same session to the same ticket?

CA Service Desk users are unable at times to save updates to a ticket as the same ticket may already have pending updates. This article explains how CA Service Desk Manager handles such a situation.

"**Pending updates**" occur when a user performs multiple updates on the same ticket. For example, when you are editing a ticket, you can also **Log a Comment** or perform other activities prior to saving the ticket. Pending updates are allowed only for the same user in the same session. If you open a new browser window and login as the same user, and try to update the same ticket, you will get a warning message saying the record is locked.



The situation where there are pending updates when you open an activity window (eg. the Log comment activity), you will see the **Accept** button instead of the **Save** button on the activity screen.



The '**Accept**' button shows that there is already a pending update on this ticket, that is, you already have a lock on the ticket. If you have administrative privileges in Service Desk, you can see the lock on the ticket in the **Administration** tab, **System, Current Locks**:

The rule is that, if the initial pending update is cancelled, or terminated abnormally, all of the pending updates are abandoned, even if the message " **Save Successful - Activity Log created**" is displayed after clicking the **Accept** button. The update is, in fact, not saved at that point , as it is still pending the initial update. The examples of initial updates:

1. Create a new ticket, (it will be in edit mode immediately)

2. Edit a ticket
3. Open an Activities window, such as Update Status, Escalate, etc.
4. Other activities to update a ticket.
5. If you then open another Activities window, you will see the Accept button.

NOTE

: If you open multiple update windows on the same ticket, you must save the initial update window to save all the updates.

Where the performance of the system is slow, and the user tries to update a ticket (for example, open an Activities window) and then closes the window before the window details are displayed and the hour-glass is still spinning. At this point no updates on this ticket can be saved and the user will always see the **Accept** button on the Activities screen. This may leave the ticket in a locked state and unable to be edited by other users.

Under this scenario, the solution is to login as an Administrator, navigate to **Administration** tab, **System, Current Locks**, find the locked ticket, right click on it, and select '**Delete Lock**' to release the lock. A user without admin rights should be able to release the lock by logging out of Service Desk and logging back in.

Asset Management

This section contains the following articles:

- Audit History
- Contract Management
- Financial Management
- Hardware Asset Management
- Searching
- Software Asset Management
- Vendor Management

Audit History

An *audit history* is a chronological list of changes that are made to an object record over time. Auditing records all changes that are made to each field in an object record. CA APM lets you use an audit history to view a list of changes that are made to a record.

You can use an audit history to see the changes that are made to the value of a particular field. You can also see who changed the value. If a field contains an incorrect entry or value, use the audit history to verify the last correct value and update the object record.

The following objects enable auditing by default and without any additional configuration:

- Models (including Model Pricing)
- Assets
- Companies
- Contacts
- Organizations
- Locations
- Sites
- Legal Documents (including Terms and Conditions)
- Legal Assets (including Terms and Conditions)
- Costs
- Payments
- Relationships
- Asset Configurations (Models and Assets)
- Software Internal Allocations (Locations, Companies, Contacts, and Assets)

View an Audit History of Object Changes

You can view an audit history for an object record to see the changes that are made to the record. The audit history maintains and displays the historical information of the changes that are made to an object.

Example: Review the Audit History to Change the Cost Center

In this example, a line manager discovers that the incorrect department is being charged for a laptop. The line manager investigates the situation and determines that the laptop is assigned to the incorrect cost center. To understand how this error occurred, the line manager configures and filters the audit history search criteria and results for the laptop to include the cost center. The line manager reviews the audit history for a specific time period to determine when the cost center value changed and who changed the value. The line manager contacts the asset manager to have the cost center changed to the correct value.

Follow these steps:

1. Click the tab and optional subtab for the object for which you want to view an audit history.
2. Search to find the list of available objects.
3. Click the object in the search results for which you want to view an audit history.
4. Click View Audit History.
   All changes that are made to the object record appear.
5. (Optional) To view the audit history for a relationship, complete the following steps:
   a. Select the relationship (for example, Image Partitions).
   b. Click the Edit Record icon, and click View Audit History.
6. (Optional) Complete any of the following steps in the Search Criteria to filter the audit history search results:
   a. Select a specific field in the Target Field drop-down list and click Go.
      The Search Results are limited to the field that you select. In addition, the standard auditing fields identify the user who changed the field and the type and date of change.
   b. Select the Highlight Changes check box and click Go.
      The Search Results highlight changes in adjacent rows. You can use this check box, in combination with a specific field in the Target Field drop-down list, to identify field-level changes that need correction.
7. Export the audit history to a CSV file or to return to the object details, click the hyperlink.

View an Audit History of Events

CA APM lets you view an audit history of all events that are created for an object. For example, you define a date event on the Terminate Date field for a legal document and a notification for the event is sent to the appropriate person on 3/1/2010. The audit history maintains and displays the event history for an object.

Follow these steps:

1. Click the tab and optional sub tab for the object for which you want to view an audit history.
2. Search to find the list of available objects.
3. Click the object in the search results for which you want to view an audit history.
4. Click View Audit History.
   All changes that are made to the object record appear.
5. In the audit history results, click the icon in the Event History column.
   All events and their associated status that is created for the audit record appear.

Event Status

When you view an audit history of events, you can see the associated status for each event.

Event Status	Description
Unprocessed	CA APM has created the event. However, the event has not been processed to map the workflow process attributes for notifications in CA Process Automation.
Started	The mapping for the workflow process attributes in CA Process Automation is complete. The associated notification process for the event has started.
In Progress	The CA Process Automation notification process for the event is processing.
Completed	The CA Process Automation notification process for the event is completed.
Failed	The CA Process Automation notification process for the event has failed.
Aborted	The CA Process Automation notification process for the event has stopped.

Contract Management

After you negotiate contracts with your vendors, the supporting documentation is frequently archived and the terms are forgotten. Making contract information easily accessible lets you properly administer the terms of an agreement. CA APM lets you manage legal documents and standardize on terms and conditions for reporting and analysis. You can see the relationships between agreements and costs associated with vendor contracts to understand the financial impact. Finally, CA APM lets you attach electronic files or URL pages containing supporting documentation to objects, such as contract profiles, to quickly access the original document.

Contract management in CA APM involves working with the following objects:

- Legal documents
- Terms and conditions
- Attachments

Legal Documents

This article contains the following topics:

A *legal document* describes a legal relationship or agreement between two or more parties. For example, contracts, notification letters, master agreements, lease agreements, volume purchase agreements, additions to agreements, letters of intent to purchase, and so forth, are all considered legal documents.

Legal document records are based on legal templates, which your CA APM administrator defines. When you define a legal document, you start by selecting the appropriate template. Templates provide fields that apply to specific types of legal documents. Regardless of the legal template you use, you can use legal documents to track the following information:

- Record information about parties to the legal document, both primary and other parties.
- Create relationships to associate related records (for example, to associate an amendment to its original agreement).
- Store attachments with your legal document record (for example, a scanned image of a document).
- Record associated cost information.

You can retrieve information from the repository about any object by searching. You can then select, view, and manage individual object records from the search results.

Manage Legal Documents

You can define, update, or delete a legal document. For example, you can define a legal document for a contract or negotiation letter, or you can change the termination date for an equipment lease. You can delete a contract or lease agreement that has expired. You cannot delete a legal document that is associated with an asset.

> **WARNING**
> When you delete an object, you can no longer view the audit history for the object. We recommend that instead of deleting the object, you make the object inactive. Then, you can still view the audit history for the object.

Follow these steps:

1. Click Legal Document.
2. Perform one of the following actions.
3. Define a legal document.
   a. Click New Legal Document.
   b. Enter the legal document information.
   c. Click Save.

   > **NOTE**
   > You can also define a legal document by copying an existing legal document, supplying a new name, changing the information, and saving the new legal document.
4. Update a legal document.
   a. Search to find the list of available legal documents.
   b. Click the legal document that you want to update.
   c. Enter the new information for the legal document.
   d. Click Save.

   > **NOTE**
   > You can also view the details for an object that is related to your legal document, if the related object has a Browse icon. When you click the Browse icon, you leave the legal document page and you navigate to the related object page. To keep the legal document page in view and preserve the legal document information, right-click the Browse icon and select Open Link in New Window. Close the new window when you are finished viewing the related object details.

5. Delete a legal document.
   a. Search to find the list of available legal documents.
   b. Click the legal document that you want to delete.
   c. Click Delete and confirm that you want to delete the legal document.

Associate a Governing Legal Document with a Legal Document

A governing legal document is the document on which a legal document is based. CA APM lets you associate a governing legal document with the legal documents on which it is based. This association is useful when tracking the source of the legal terms and conditions of a legal document.

NOTE
You can view an audit history for this relationship.

Follow these steps:

1. Click Legal Document.
2. Search to find the list of available legal documents.
3. Click the legal document to which you want to associate a governing legal document.
4. On the left, expand Relationships and click Governing Legal Document.
5. Click Select New and select a different legal document, other than the legal document previously selected.
6. Click Save.
   The governing document is associated with the legal document.

Track Amendments to a Legal Document

CA APM lets you create and track amendments that have been made to a legal document. Save the amendments as a separate legal document and associate the amendments with the parent legal document.

NOTE
You can view an audit history for this relationship.

Follow these steps:

1. Click Legal Document.
2. Search to find the list of available legal documents.
3. Click the legal document for which you want to enter amendment details.
4. On the left, expand Relationships and click Legal Amendment.
5. Click Select New and select a different legal document, other than the legal document previously selected.
6. Click Save.
   The amendment details are saved.

Associate an Asset with a Legal Document

CA APM lets you associate assets and legal documents to identify the assets that a legal document covers. Initiate this association from either the legal document or asset. You can associate multiple assets to a single legal document and multiple legal documents to a single asset.

NOTE
You can view an audit history for this relationship.

Follow these steps:

1. Click Legal Document.
2. Search to find the list of available legal documents.
3. Click the legal document that you want to associate with an asset.

4. On the left, expand Relationships and click Legal Asset.
5. Click Select New in the Legal Asset section, search for and select an asset.
   The asset name appears.
6. Click Save.
   The asset is associated with the legal document.

Add and Remove Asset Legal Document Terms and Conditions

Terms and conditions are specific areas of agreement that are defined in legal documents. For example, legal documents can have terms and conditions for a multi-product discount, a new pricing model, copyright protection, and so forth. After you associate an asset with a legal document, CA APM lets you add or remove terms and conditions for the asset legal document from the Asset or the Legal Document page.

NOTE
You can view an audit history for this relationship.

Follow these steps:

1. Click Legal Document.
2. Search to find the list of available legal documents.
3. Click the legal document for which you want to add or remove terms and conditions.
4. On the left, expand Relationships and click Legal Asset.
5. Click the Edit Record icon for the asset for which you want to add or remove terms and conditions.
6. Click View Assigned T's & C's.
7. Select one of the following options:
   - Click Select New for the date-specific or non-date-specific T's and C's to add to the asset legal document.
   - Click the Mark for Deletion icon for the terms and conditions that you want to remove from the asset legal document.
8. Click Save.
   The terms and conditions are added or removed.
   For more information about defining date-specific and non-date-specific terms and conditions for legal documents, see Define Legal Document Terms and Conditions.

Associate a Legal Party with a Legal Document

CA APM lets you associate the people and entities involved in creating a legal document to the document record. For example, you can associate the lawyers and the law firm as legal parties for the legal document.

NOTE
You can view an audit history for this relationship.

Follow these steps:

1. Click Legal Document.
2. Search to find the list of legal documents.
3. Click the legal document for which you want to associate a legal party.
4. On the left, expand Relationships and click Legal Party.
5. Click Select New to search for and select a company.
6. Click Save.
   The company is defined as the legal party.

Make an Obsolete Legal Document Inactive

As a best practice, change the status of an obsolete legal document to inactive rather than deleting the legal document. We recommend this approach because when you delete a legal document record, the historical information for the legal document is permanently removed from the repository. In this way, you retain the legal document information for future reporting and reference.

Follow these steps:

1. Click Legal Document.
2. Search to find the list of available legal documents.
3. Click the legal document that you want to make inactive.
4. Select the Inactive check box.
5. Click Save.
   The legal document status is changed to inactive.

Assign and Track the Status of a Legal Document

CA APM lets you assign and track the status of a legal document. For example, if the document has been signed, assign the status as executed. You can track the changes in the status of a legal document over time. The status identifies the stage of completion or implementation of the legal document.

Follow these steps:

1. Click Legal Document.
2. Search to find the list of available legal documents.
3. Click the legal document for which you want to track the status.
   The legal document details appear. The status details are available in the Status section.
4. Click New in the Status section.
5. Specify the new status for the legal document.
6. Select the Current check box to indicate that the selected status is the current status of the legal document.
7. Click Save.
   The status is updated and the previous status is added to a chronological status list for the legal document. As a best practice, change the current status each time the status of a legal document changes. Only one status can be entered as the current status of a legal document. All dates are automatically updated.

Terms and Conditions

This article contains the following topics:

Terms and conditions are areas of agreement specified in legal documents. CA APM lets you track terms and conditions for the following reasons:

- To enforce order or comply with an existing legal document.
- To negotiate future legal documents.

Administrators and users with appropriate privileges create and maintain a master list of terms and conditions. When you work with legal documents, you use terms and conditions from this list. You can assign terms and conditions when you define legal document records or asset records.

When you define a legal document record, you start by selecting a legal template. The legal template contains the terms and conditions that typically apply to the legal document type. You can update the terms and conditions provided by the legal template to help ensure that the legal document record contains only the terms and conditions that apply.

A master terms and conditions list must exist in your repository before you can assign terms and conditions to your legal templates. You can assign a term or condition to one or more legal templates and legal document records.

Terms and conditions can be date-specific or non-date-specific. Date-specific terms and conditions include information about the start and end dates. For example, if one of your terms and conditions is Installation Date, you can have start and end date information included about the Installation Date. You determine which terms and conditions are date-specific when you create new terms and conditions in Directory, List Management.

For more information about defining date-specific and non-date-specific terms and conditions for legal documents, see Define Legal Document Terms and Conditions.

We recommend that you do not include the following terms and conditions in your master list:

- Effective and termination dates, as they are recorded directly on legal document records.
- Cost-related terms and conditions, as they are recorded directly on cost records.
- Software licensing terms, as they are recorded on license records.

> **NOTE**
> When you define an asset, you can associate the asset to the legal document records that govern the asset. The terms and conditions that apply to the legal documents also apply to the asset. If a particular asset requires terms and conditions that the legal documents do not provide, you can change the terms as you associate the asset and legal document. The legal documents covering a particular asset must contain the terms and conditions that apply only to the asset.

Add and Remove Legal Document Terms and Conditions

Terms and conditions are specific areas of agreement that are defined in legal documents. For example, legal documents can have terms and conditions for a multi-product discount, a new pricing model, or a copyright protection. CA APM lets you add the terms and conditions that are required in a legal document and remove any that are inherited from the legal template but are not required in the legal document.

> **NOTE**
> After you associate an asset with a legal document, you can add or remove terms and conditions for the legal document from the Asset or Legal Document page.

Follow these steps:

1. Click Legal Document.
2. Search to find the list of available legal documents.
3. Click the legal document for which you want to add or remove terms and conditions.
4. On the left, click T's and C's.
5. Select one of the following options:
   - Click Select New for the date-specific or non-date-specific T's and C's to add to the legal document.
   - Click the Mark for Deletion icon for the terms and conditions that you want to remove from the legal document.
6. Click Save.
   The terms and conditions are added or removed for the legal document.

For more information about defining date-specific and non-date-specific terms and conditions for legal documents, see Define Legal Document Terms and Conditions.

Manage an Attachment

Attachments are electronic files or URL pages containing supporting documentation that you associate with an object. For example, you can add an attachment of a scanned contract to the legal document record that represents the contract.

You can use the following types of attachments:

- **Web URL**. Provides direct access to the page specified in the URL. When you add this type of attachment, include the prefix http:// for the link to work correctly.
- **File**. Provides direct access to a file. The file opens using the default program for the file type. At the time that you create this attachment type, the file is copied from your file system to the file system on a CA APM server.

You can add attachments to the following objects:

- Model
- Asset
- Legal Document
- Contact
- Company
- Organization
- Location

You can add, update, and delete an attachment for an object. The following examples illustrate how you can manage an attachment:

- You can attach a scanned copy of an invoice to record all assets that are listed on the invoice.
- You can edit the URL to a scanned copy of an invoice if the location has changed.

When you delete an attachment, you delete only the reference to the attachment in the object record. If the deleted attachment is a file, the file remains in the file system on the CA APM server.

Follow these steps:

1. Click the tab and optional subtab for the object for which you want to manage an attachment.
2. Search to find the list of available objects.
3. Click the object for which you want to manage an attachment.
4. On the left, click Attachments.
5. **Add an Attachment.** Follow these steps:
   a. Click New.
   b. Complete the required information.
   c. In the File Path field, select a file from your local server, or enter a URL for the new attachment. For a URL, include the following prefix:
      ```
      http://
      ```
      If you are adding a file attachment, your selected file is copied to a CA APM server. This copy activity can take a few moments to complete. Wait for the file to finish copying before you click Save.

      WARNING
      The file size is limited by the product environmental settings. For more information, contact your administrator or CA Support.

      NOTE
      If you select a file that exists on the server, you are prompted to overwrite the existing file. Click Yes to overwrite the file, click No to use the existing file on the server, or click Cancel to clear the file selection.
6. **Update an Attachment.** Follow these steps:
   a. Click the Edit Record icon for the attachment.
   b. Change the information for the attachment.

      NOTE
      For a file attachment, you can update the name and description for the attachment record. However, to update the file for a file attachment, you delete the existing attachment record and add a new attachment with a new file.
7. **Delete an Attachment.** Click the Mark for Deletion icon next to the attachment.

8. Click Save.
   The attachment is added, updated, or deleted.

Financial Management

Understanding the financial impact of your asset base lets you decide what costs are relevant to your business. CA APM helps you track, manage, and categorize all asset-related expenditures, including historical, current, and projected costs, so that you can analyze their financial impact.

Financial management in CA APM involves working with the following objects:

- Models
- Assets
- Asset Configurations
- Costs and Payments
- Events and Notifications
- Notes

Models

This article contains the following topics:

The repository contains two types of records that describe the basic characteristics of IT products:

- A *model* describes a product type that you have purchased or might purchase.
- An *asset*, which is based on a model, describes a product that you own or plan to acquire. You can create many assets that are based on a single model, and the new asset inherits the model attributes. For example, if you own 100 laptops of a particular model, you would have one model record describing the laptop and 100 asset records describing the individual laptops.

A model can be any of the following examples:

- A *computer model* such as Dell Optiplex GX270 and Dell Precision 410.
- A *hardware component* such as 104+ keyboard, 256 MB-RAM, Monitors, and Intel Pentium processor.

You manage *hardware models* to describe the following models:

- Configurations that the manufacturer offers.
- Substitutions or additions that you typically make to the configurations when you purchase them.

Each model configuration is composed of links between the models and their component model records (such as a monitor and a keyboard).

You can have some products that you want to group together to manage as models, such as the following examples:

- Standard, relatively inexpensive computer components in a configuration. For example, you can record a particular type of network card as a model. This model lets you determine which systems are using that type of network card without tracking details about individual network cards of that type. Other similar components include SCSI hard drives, memory, graphics cards, and CD or DVD devices.
- Products that your company purchases from different vendors at different times. This additional information lets you track pricing information for those products and select the best price.

Define models first in the repository for each asset that you want to manage. After you define a model, the model acts like a template for the physical assets that are created. The assets inherit the properties or attributes from the model, such as the name, asset family, and manufacturer. By using this template, a single model can represent many assets and the model information is common to many assets. However, each asset record contains information specific to that asset.

Manage Models

Managing models includes defining, updating, and deleting models for an asset. For example, you define Dell Precision Workstation 410 as a model to describe the workstation.

Follow these steps:

1. Click Model.
2. Perform one of the following actions.
3. **Define a model.**
   a. Click New Model.
   b. Enter the model information.

      NOTE
      You can also define a model by copying an existing model, supplying a new name, changing the model information, and saving the new model.
4. **Update a model.**
   a. Search for the list of available models.
   b. Click the model that you want to update.
   c. Enter the new information for the model.

      NOTE
      You can also view the details for an object that is related to your model, if the related object has a Browse icon. When you click the Browse icon, you leave the model page and you navigate to the related object page. To keep the model page in view and preserve the model information, right-click the Browse icon and select Open Link in New Window. Close the new window when you are finished viewing the related object details.
5. **Delete a model.**
   a. Search for the list of available models.
   b. Click the model that you want to delete.
   c. Click Delete and confirm that you want to delete the model.
6. Click Save.

Define an Asset from a Model

You define an asset, which is based on a model, to describe a product that you own or plan to acquire. For example, you have defined Dell Precision Workstation 410 as a model. Your enterprise has purchased 100 individual computers, and you define 100 asset records that are based on the model to describe the individual computers.

NOTE
You cannot define an asset from a model that does not have an asset family.

Follow these steps:

1. Click Model, New Model.
2. Enter the model information with a valid asset family.
3. Click Save.
   The model is defined.
4. Click the Create Asset hyperlink.
5. Enter the asset-specific information.
6. Click Save.

Make an Obsolete Model Inactive

As a best practice, change the status of an obsolete model to inactive rather than deleting the model. For example, after replacing all Dell Precision Workstation 410 computers with laptops, you can change the status of the associated model to inactive.

We recommend this approach because deleting a model record permanently removes the historical information from the repository. Changing the status to inactive lets you retain the model information for future reporting and reference.

> **NOTE**
> Changing the status to inactive has no impact on the assets that are based on the model. If you want the assets to be inactive, search for the assets that are based on the model and manually indicate that they are inactive.

Follow these steps:

1. Click Model and search for the list of available models.
2. Click the model that you want to make inactive and select the Inactive check box.
3. Click Save.

Change a Model Asset Family

You can change the asset family of selected models. For example, you realize that a set of related computer models were imported into the product using the hardware asset family. You search for these models and select them. Then you can change the asset family for all selected models simultaneously.

> **NOTE**
> When you change the asset family for a model, the asset family is also changed for the associated assets.

Follow these steps:

1. Click Model.
2. Click Change Model Asset Family under Mass Change Utilities on the left.
3. Select the original asset family and the new asset family.

   > **NOTE**
   > If the original asset family has extended fields that the new asset family does not have, a warning message and a check box appear. To proceed with changing the model asset family, select the check box Check to change model to new asset family. The extended fields are lost. To avoid the data loss, define the extended fields for the new asset family before you change the model asset family.
4. Click Go.
   All models that have the original asset family are listed in the Search results.
5. Select either the individual models that you want to change or all models.
6. Select a class for the models using one of the following actions:
   - Select the Assign a Single Class to All Models check box and select a class if you want to apply the same class to all selected models.
   - Select the class for each individual selected model.
7. Click Change Model Asset Family.
   Your request is submitted.
8. Click View Mass Change Utilities Progress to see the status of your job and verify completion.

Add Components in a Model Configuration

You associate a hardware model to its component models to define a model configuration. A generic model is a model that is not acquired as an asset, for example, keyboards and network cards. Generic models comprise a model configuration. This configuration is available to the inherited assets from the model.

Follow these steps:

1. Click Model and search for the list of available models.
2. Click the model for which you want to add components.
3. Expand Configurations on the left and click Model Configuration.
4. Click Select New, search for and select a model.
5. Click the Edit Record icon next to the model name to define the association between the hardware model and model components.
6. Click Save.

Assets

This article contains the following topics:

An *asset* is an IT product that you own or plan to acquire. Assets represent physical products with unique identifiers such as a serial number, a configuration, or a contact. You define assets that are based on a model and the information specific to the asset is added to the asset record. Define an asset record when you want to track cost, legal, and other ownership purposes of an asset.

You can retrieve information from the repository about any object by searching. You can then select, view, and manage individual object records from the search results.

> **NOTE**
> The search results only include assets that CA APM manages. If your repository contains assets that the product does not manage, those assets do not appear in the asset search results.

You can assign one Lifecycle status to an asset to indicate its availability. If your administrator configured the asset search to include the Lifecycle Status field, you can retrieve information about assets based on the status.

Best Practices for Tracking Assets

When you manage assets, we recommend the following best practices:

- Enter useful information for cost, productivity, and decision making.
- If you cannot think of an immediate need, reconsider tracking the information.
- Keep the information accurate and up-to-date. If the information becomes outdated and obsolete, you do not know what information to trust. As a result, you can make incorrect purchasing and business decisions.
- Track expensive products and key components (such as system units, mainframes, large devices, printers, and monitors) as assets. Do not maintain information about IT assets that are not important when making purchasing and business decisions.
- Track a product as an asset only if you want to know the detailed information (such as the cost, serial number, bar code number, location) about each instance of that product. For example, if you want to know the location of each monitor of a particular model being used, or the cost of each monitor of a model.
- If you are using a discovery product to track the usage of hardware or software products, consider the following information:
  - The discovery product, instead of CA APM, tracks the product types.
  - The product types that the discovery product and CA APM must track support a comparison of usage and ownership information.

Asset Families

You can classify assets and models that are based on an asset family. An *asset family* is a way to organize and classify assets to track specialized information about products, services, or equipment that you use. The asset family determines the information that you see on the page when you enter an asset.

The product provides a set of predefined asset families, such as computer, hardware, other, projects, service, and software. Your administrator can create user-defined asset families based on the internal structure of your organization. When you define an asset, select either a predefined asset family or a user-defined asset family.

For each asset family, you can create a hierarchy to track an asset to a precise location. Assets inherit the asset family from the model records on which they are based.

Asset Classification

Asset classification helps reduce the number of records that are returned in searches and reports. When you define a model and asset, you can classify the objects by asset family. You can further classify models and assets in the following ways:

- **Class**. Broadly categorizes the product, such as a printer.
- **Subclass**. Provides additional refinement, such as inkjet or laser printer.

Assets inherit class and subclass assignments from the models on which they are based. Consider the following information when classifying assets:

- If you change the class and subclass for a model, the change is not automatically made to saved assets based on the model. The changes are not retroactive.
- You can change the class or subclass that an asset inherits from the model.

Manage Assets

You can add information about each asset to the repository so that you can manage assets throughout their lifecycle. Assets are defined based on models, and inherit data from the selected model. For example, you have defined Dell Precision Workstation 410 as a model. Your enterprise has purchased 100 individual computers, and you define 100 asset records that are based on the model to describe the individual computers.

Review the following capabilities to manage your assets more effectively:

- Update an Asset (not assigned to a user) from ITAM UI
- Ability to Set the Lifecycle Status as Active

Follow these steps:

1. Click Asset and perform one of the following actions.
   - **To define an asset, complete the following steps:**
     a. Click New Asset.
     b. Enter the asset information.
     c. Associate the asset with other objects such as contacts, organizations, and companies.
     Associating assets with other objects enables data access to roles or users when you implement multi-tenancy.

     NOTE
     You can also define an asset by copying an existing asset, supplying a new name, changing the asset information, and saving the new asset.
   - **To copy an asset, complete the following steps:**
     a. Search for the list of available assets.
     b. Click the asset that you want to copy.
     c. Click Copy.
     The Asset Copy wizard appears.
     d. Follow the on-screen instructions to define unique assets and to specify the data that you want to copy to the new assets.
     e. Click Finish when you have completed the wizard.

f. Click Close and then click Save.

– **To update an asset, complete the following steps:**
  a. Search for the list of available assets.
  b. Click the asset that you want to update.
  c. Enter the new information for the asset.

NOTE
You can also view and edit an object that is related to your asset, if the related object has a Browse icon. When you click the Browse icon, you leave the asset page and you navigate to the related object page. To keep the asset page in view and preserve the asset information, right-click the Browse icon and select Open Link in New Window. Close the new window when you are finished.

– **To delete an asset, complete the following steps:**
  a. Search for the list of available assets.
  b. Click the asset that you want to delete.
  c. Click Delete and confirm that you want to delete the asset.

NOTE
If the CI check box is selected (on the Asset Details page) for the asset you want to delete, you cannot delete the asset. The selected CI check box indicates that CA Service Desk Manager manages the asset. Delete the asset from CA Service Desk Manager.

WARNING
We do not recommend that you delete an asset because the audit history for the asset is permanently removed. For example, legal, cost, relationship, and other information that is related to the asset is also deleted. Instead, update the asset status to indicate that the asset has been disposed of, or is *obsolete*. Using this approach, you retain the information about the asset in your repository for future reporting and reference purposes.

2. Click Save.

Update an Asset (not assigned to a user) from ITAM UI

CA Asset Portfolio Management 17.3.0.7 provides users the ability to move an asset (not assigned to a user) using the ITAM interface. This capability is not supported on Data Importer/WCF. Administrators can now save a record field (like Cost Center) by updating a value from the ITAM User Interface. This value is updated in the configuration file in the ITAM web server file. For example, earlier it was not possible for users to save or delete an asset if the user does not have adequate permission to modify the data. With this new capability, Administrators can use the new configuration option in the web server file (\Program Files (x86)\CA\ITAM\Web Server) and update the following default value:

```
<add key="Disable_FiltersValidation" value="false" />
```

to

```
<add key="Disable_FiltersValidation" value="true" />
```

.
After updating the above value, users can modify the record for not just the cost center but also other attributes like GL Code, Cost Center, and so on. After updating or moving an asset, below message is shown from the Asset Details page: `"The saved record will not be shown because you have selected an item for which you do not have access to."`

NOTE
Only Administrators can view the record. Due to the filter constraints, the asset details page of the updated asset will not be valid and users cannot use any links on the left navigation pane except the Asset search link.

NOTE
Non-admin users cannot perform ‘Delete’ or ‘Create’ operations on assets with limited access.

Ability to Set the Lifecycle Status as Active

CA Asset Portfolio Management 17.3.0.6 now provides the ability to set the default status of Lifecycle Status field of a new Asset to **Active** from the User Interface.

NOTE
This capability is currently supported from the user interface only when the asset is created from the Create New Asset link.

IMPORTANT
Do not delete an active status Asset from the List Management. When Default LifeCycle Status parameter is set to True, then, Lifecycle status value 'Active' must not be deleted from List Management, Asset Lists.

- When the **DefaultLifeCycleStatus** set to **False** (default value), the newly created asset life cycle status field value is set as **null**.
- When **DefaultLifeCycleStatus** is set to **True**, the newly created asset life cycle status field value is set as **Active**. This parameter is available in the **web.config** file of application server (…\Program Files (x86)\CA\ITAM\Application Server).

Consider the following while creating assets:

- **Data Importer**: For creating assets via Data Importer, set the Lifecycle Status field to **Active** in mappings explicitly.

Add an Asset that CMDB Manages

You can add assets to your repository that CMDB manages. CMDB is integrated with CA Service Desk Manager. After you add an asset that CMDB manages, you can then open CA Service Desk Manager from CA APM and view and manage the asset.

Note: You can also create a configuration item (CI) in CMDB and then view it as an asset in CA APM. For more information about creating a configuration item, see Configuration Items.

WARNING
You cannot delete an asset in CA APM that CMDB manages. You can only delete assets in CA APM that CA APM manages.

Follow these steps:

1. Click Asset, New Asset.
2. Enter the asset information.
3. Select the CI check box to have CMDB and CA Service Desk Manager manage the asset.

   WARNING
   After you save an asset with the CI check box selected, you cannot update the asset to clear the CI check box.
4. Associate the asset with other objects such as contacts, organizations, and companies. This approach enables specific data access to roles or users when you implement multi-tenancy.
5. Click Save.

6. (Optional) View the new asset in CA Service Desk Manager by clicking the CMDB hyperlink at the top of the page.
7. Log in to CA Service Desk Manager.
   A CA Service Desk Manager window opens and the details of the selected asset appear.

Associate a Legal Document with an Asset

You associate assets and legal documents to identify the assets that a legal document covers. For example, you associate a volume purchase agreement with 100 laptops. Initiate this association from either the legal document or asset. You can associate multiple assets to a single legal document and multiple legal documents to a single asset.

NOTE
You can view an audit history for this relationship.

Follow these steps:

1. Click Asset.
2. Search to find the list of available assets.
3. Click the asset that you want to associate with a legal document.
4. Expand Relationships (on the left) and click Legal Documents.
5. Click Select New in the Legal Document section, search for and select a legal document.
6. Click Save.

Add and Remove Asset Legal Document Terms and Conditions

Terms and conditions are specific areas of agreement that are defined in legal documents. For example, legal documents can have terms and conditions for a multiproduct discount, a new pricing model, or copyright protection. After you associate a legal document with an asset, add or remove terms and conditions for the asset legal document from the Asset or Legal Document page.

NOTE
You can view an audit history for this relationship.

Follow these steps:

1. Click Asset.
2. Search for the list of available assets.
3. Click the asset for which you want to add or remove legal document terms and conditions.
4. Expand Relationships (on the left) and click Legal Documents.
5. Click the Edit Record icon for the legal document for which you want to add or remove terms and conditions.
6. Click View Assigned T's & C's.
7. Select one of the following options:
   - Click Select New for the date-specific or non-date-specific terms and conditions to add to the asset legal document.
   - Click the Mark for Deletion icon for the terms and conditions that you want to remove from the asset legal document.
8. Click Save.

Change an Asset Model

You can change the model for selected assets. For example, you identified 15 assets that were entered into the product with an incorrect server model. You search for the assets and select them. You then can change the server model for all selected assets simultaneously.

Follow these steps:

1. Click Asset.
2. Click Change Asset Model under Mass Change Utilities on the left.

3. Select the original model and the new model and click Go.
   All assets that have the original model are listed in the Search results. If you do not select an original model, all assets that do not have models are listed.

   NOTE
   If the original model asset family has extended fields that the new model asset family does not have, a warning message appears. If you proceed with changing the asset model, the extended fields are lost. To avoid the data loss, define the extended fields for the new model asset family before you change the asset model.
4. (Optional) Select Assign Model Class to All Assets if you want all the listed assets to use the new model class.

   NOTE
   This check box is available only if the new model has a class.
5. Select the individual assets that you want to change or select all assets.
6. Click Change Model.
   Your request is submitted.
7. Click View Mass Change Utilities Progress to see the status of your job.

View an Asset in CA Service Desk Manager

When you integrate the product with CA Service Desk Manager, you can select an asset in CA APM and view the asset information in CA Service Desk Manager.

Follow these steps:

1. Click Asset.
2. Search for the list of available assets.
3. Click the asset that you want to view in CA Service Desk Manager.
4. Click the CMDB hyperlink at the top of the page.

   NOTE
   You can see the CMDB hyperlink only if the product is integrated with CA Service Desk Manager and if your user role includes asset configuration privileges.
5. Log in to CA Service Desk Manager.
   A CA Service Desk Manager window opens and the details of the selected asset appear.

   NOTE
   Asset information that you enter in the product is not immediately available in CA Service Desk Manager. By default, a 10-minute delay exists from the time an asset is updated and the time you can view the asset in CA Service Desk Manager.

View Discovered and Owned Information for an Asset

You can view the discovered information for the owned assets that are linked with discovered assets during hardware reconciliation. Discovered information for an asset includes system configuration, operating system, system devices, file systems, and other information. You can also view the owned information for the assets that you define in the product. Owned information for an asset includes asset properties (for example, operating system and product version), legal information, installed software, components, and other information.

Follow these steps:

1. Click Asset.
2. Search for the list of available assets.
3. Click the asset for which you want to view the information and perform one of the following steps:
   – Click Discovered Information.

NOTE
The Discovered Information hyperlink appears only for assets that are reconciled.

– Click Owned Information.

NOTE
If the product is integrated with CA Service Desk Manager, CA Client Automation, and CA SAM, you can see the asset information from those products.

An asset viewer window opens and displays the information. The links on the window let you access more details.

4. Click each link to view the specific information.
5. Close the window when you are finished viewing the information.

Hardware Assets

CA APM lets you track the hardware asset information that your company is entitled to use. For each hardware asset, you can track the following information:

- Maintain a record of cost and legal information.
- Maintain a record of components in asset configuration and the configurations where the asset is a component.
- Maintain the availability status of the asset.
- Track the software details that are allocated to the asset.

Add Components in an Asset Configuration

An asset can inherit the components from a model. Unicenter APM lets you add components to an asset configuration, for example, an external hard drive or a DVD writer.

Follow these steps:

1. Click Asset.
2. Search for the list of available assets.
3. Click the asset for which you want to add components.
4. Expand Configurations on the left and click Asset Configuration.
5. Click Select New in the Asset Configuration section, search for and select an asset.
6. Click Save.

Update the Asset Status

CA APM lets you change the lifecycle status of an asset. For example, you can update the status when the asset changes from received to available.

To update the asset status

1. Click Asset.
2. Search to find the list of available assets.
3. Click the asset for which you want to update the status.
4. Change the Lifecycle Status and Lifecycle Status Date fields to the appropriate status and date for the asset.
5. Click Save.
   The status for the asset is updated.

Track Image Partitions

CA APM lets you enter image partition details for an existing asset and track them. For example, you can specify that the hard disk on a computer has two partitions and enter the size in GB for each partition. Create an image partition as a separate asset and associate it with the parent asset.

NOTE
You can view an audit history for this relationship.

To track image partition details

1. Click Asset.
2. Search to find the list of available assets.
3. Click the asset for which you want to enter image partition details.
4. On the left, expand Relationships and click Image Partitions.
5. Click Select New and select a different asset, other than the asset previously selected.
6. Click Save.
   The image partition details are saved and can be tracked.

Asset Groups

An *asset group* is a collection of assets that you acquire together and that are based on the same model. You define an asset group in one asset record. The value in the Quantity field on the Asset Details page indicates the number of assets in the group. The product tracks the asset information for the entire group, rather than for the individual assets. All members of an asset group share asset information.

Asset groups are useful in the following situations:

- The assets are covered under the same purchase or license agreement. You do not need to track cost, payment, licensing, and legal information separately for each asset.
- The assets share some common information, but some of the individual assets in the group have different information. In this situation, you first define the assets as a group and enter the common information one time. Then, you divide the asset group into individual assets (subgroup the asset group) and enter any information that is unique to the individual assets.

You do not have to use asset groups when you are billed for assets on a single invoice. You can record payments for assets that are tracked separately, regardless of whether they were billed on the same invoice.

Asset Subgrouping

CA APM lets you divide (subgroup) an asset group into individual assets using the product. When you subgroup assets, the data for the asset group is copied to the new individual assets, *except* for the following information:

- Serial Number
- Alt Asset ID
- Host Name
- DNS Name
- MAC Address

You can change any existing data or add unique information to the new individual assets. You track the new assets separately from that point forward.

Example: Subgroup Software Licenses

In this example, you purchase 100 licenses of Microsoft Visual Studio under a single purchasing agreement. You record the licenses as an asset group with a quantity of 100. Later, you allocate ten licenses to members of your development staff. You then subgroup the ten allocated licenses into individual assets and enter their allocation information. The 90 licenses that are not allocated remain in the asset group.

Define an Asset Group

CA APM lets you add information to the repository about a group of assets that share common information. You can then manage these assets as a group throughout their lifecycle. You can divide (subgroup) the asset group into individual assets later so that you can specify and track unique information for the individual assets.

Follow these steps:

1. Click Asset, New Asset.

2. Enter the asset information.
3. In the Quantity field, enter the total number of assets in the group. You enter a value greater than one for an asset group.
4. Associate the asset group with other objects such as contacts, organizations, and companies. This approach enables specific data access to roles or users when you implement multi-tenancy.
5. Click Save.

Subgroup an Asset Group

CA APM lets you divide an asset group into individual assets when you want to manage assets in the group separately. When you subgroup an asset group, the product removes assets from the group (decreases the group quantity) and creates either one new asset or multiple new assets (depending on your criteria). The product copies the information for the group to the new assets, except Serial Number, Alt Asset ID, Host Name, DNS Name, and MAC Address.

Follow these steps:

1. Click Asset.
2. Search to find the list of available assets.
3. Click the asset that you want to subgroup.
   The Asset Details page for the selected asset appears.
4. Click Subgroup.

   NOTE
   The Subgroup button is enabled only for an asset with a quantity greater than one.

   The Subgroup Asset wizard appears.
5. Select a subgrouping option and click Next. The following fields require explanation:
   - **Create a new asset**
     Divides the original asset group into two assets. You specify the quantity for the new asset. The remaining quantity is left with the original asset.
     For example, you have an asset group with a quantity of 50. You want to divide the group into two assets with a quantity of 35 for the new asset and 15 for the original asset. You use this subgrouping option and specify 35 for the new asset quantity.
   - **Create individual assets each with a quantity of one**
     Divides the entire quantity of the original asset group into multiple assets. Each asset has a quantity of one.
     For example, you have an asset group with a quantity of 25. You want to divide the group into 25 individual assets. You use this option.
6. Follow the wizard to define unique asset names and to specify the data that you want to copy to the new assets.

   NOTE
   If you do not define unique asset names, the new assets retain the name of the original asset group.
7. Click Finish when you have completed the wizard.
8. Click Save.

Asset Registration

Use the *asset registration process* to report on a single asset and identify the instances of the asset within all asset functional areas through asset registration fields. These fields help ensure that assets within an asset functional area are not duplicated.

The following fields on the Asset Details page represent the asset registration fields:

- Asset Name
- Serial Number
- Asset Tag (represented on the page as Alt Asset ID)
- Host Name
- DNS Name
- MAC Address

Asset Configurations

This article contains the following topics:

An *asset configuration* is a description of an asset (for example, a desktop computer) and its individual components (for example, personal productivity software, monitor, modem, and so forth). CA APM lets you track the configuration information for both an asset and model. In addition, you can associate an asset or model record to their component asset and model records to initiate a relationship between them.

This information helps you maintain historical information about the changes to a configuration for an asset over the course of its life. You can define configuration records for any model or asset in the repository.

Asset configuration records can be of the following types:

Model configuration. Describes the configurations that manufacturers currently offer and any substitutions or additions that your company typically makes to those configurations when you purchase them. Use this type of configuration record to describe configurations for hardware models.

- **Asset configuration**. Describes the configurations of existing hardware assets. Use this type of configuration record when the asset configurations are different from the standard model configurations to indicate the changes between the model and asset configurations. You can also use this configuration type to describe changes to existing configurations.

You can change these configuration types at any time and manage the changes.

Add Components in a Model Configuration

You associate a hardware model to its component models to define a model configuration. A generic model is a model that is not acquired as an asset, for example, keyboards and network cards. Generic models comprise a model configuration. This configuration is available to the inherited assets from the model.

Follow these steps:

1. Click Model and search for the list of available models.
2. Click the model for which you want to add components.
3. Expand Configurations on the left and click Model Configuration.
4. Click Select New, search for and select a model.
5. Click the Edit Record icon next to the model name to define the association between the hardware model and model components.
6. Click Save.

Define Generic Model Configuration Details

CA APM lets you define configuration details for a generic model on which an asset can be based.

Follow these steps:

1. Click Model.
2. Search for the list of available models.

3. Click the model for which you want to define a generic configuration.
4. Expand Configurations on the left and click Model Configuration.
5. Click Select New to search for and select a model.
6. Click Save.

Define Generic Asset Configuration Details

CA APM lets you define configuration details for a generic asset.

Follow these steps:

1. Click Asset.
2. Search for the list of available assets.
3. Click the asset for which you want to define a generic configuration.
4. Expand Configurations on the left and click Model Configuration.
5. Click Select New to search for and select a model.
6. Click Save.

Define Specific Asset Configuration Details

CA APM lets you define a specific asset configuration.

Follow these steps:

1. Click Asset.
2. Search for the list of available assets.
3. Click the asset for which you want to define a specific configuration.
4. Expand Configurations on the left and click Asset Configuration.
5. Click Select New to search for and select an asset.
6. Click Save.

Add Components in an Asset Configuration

An asset can inherit the components from a model. Unicenter APM lets you add components to an asset configuration, for example, an external hard drive or a DVD writer.

Follow these steps:

1. Click Asset.
2. Search for the list of available assets.
3. Click the asset for which you want to add components.
4. Expand Configurations on the left and click Asset Configuration.
5. Click Select New in the Asset Configuration section, search for and select an asset.
6. Click Save.

Update or Delete a Configuration Record

CA APM lets you update or delete a configuration record.

> **WARNING**
> When you delete an object, you can no longer view the audit history for the object. We recommend that instead of deleting the object, you make the object inactive. Then, you can still view the audit history for the object.

Follow these steps:

1. Click Model or Asset.

2. Search for the model or asset for which you want to update or delete the configuration record.
3. Click the model or asset in the search results list.
4. Expand Configurations on the left and click the appropriate configuration option. For example, click Asset Configuration or Model Configuration.
5. Perform the steps for one of the following actions.
6. Update the configuration record.
   a. Click the Edit Record icon next to the configuration record.
   b. Select the new information for the configuration record.
   c. Click Save.
7. Delete the configuration record.
   a. Click the Mark for Deletion icon next to the configuration record.
   b. Click Save.

View the Configuration When an Asset is a Component

CA APM lets you view the configuration when an asset is a component.

Follow these steps:

1. Click Asset.
2. Search for the asset for which you want to view the configuration.
3. Click the asset in the search results list.
4. Expand Configurations on the left and click Asset Configuration.
   The list of assets in the specific configuration appears.

Costs and Payments

This article contains the following topics:

CA APM lets you track financial information that is associated with assets and legal documents. Tracking costs and payments helps reduce the risk of overpaying or underpaying vendors and suppliers. This information is also useful when deciding about future equipment purchases and deployment.

In addition to tracking cost-related information for assets and legal documents, use cost records with payments to create *payment schedules.* The schedules can help you make timely payments and reduce the risk of overpayment. The information that you provide on the Cost page is used to calculate the payment schedules. If you define the cost as a recurring cost, the schedule includes multiple payment records. For example, if you enter a cost that recurs monthly for one year, the product automatically creates 12 payment records. If the cost is a one-time cost (not a recurring cost), the schedule includes only a single payment record.

Cost records store information about costs and billing, payees, recurring charges, and one time charges. Payment records maintain information about individual payments that are related to a specific cost. A payment record also calculates and displays the total scheduled payment amount, total paid amount, and the balance due for a given cost. A payment record can represent a single payment of the full cost or multiple partial payments.

Define the Parts and Pricing for a Model

You can define the parts and pricing details for a model to identify the individual component costs that comprise the total cost for a model. For example, you have a model named Dell Precision Workstation 410. You can define the parts and pricing for the internal drives, video connectors, USB connectors, SCSI port connectors, and power supply.

> **NOTE**
> You can view an audit history for this relationship.

Follow these steps:

1. Click Model.
2. Search to find the list of available models.
3. Click the model for which you want to define the parts and pricing.
4. In the Parts and Pricing area of the page, click New and enter the parts and pricing details.
5. Click Save.

Define the Cost for an Asset or Legal Document

Defining asset costs helps you track and manage the financial information that is related to a specific asset or legal document. You can add multiple cost records for an asset or legal document. You can also add cost information to multiple assets or legal documents and update multiple payment records.

Follow these steps:

1. Click Asset or Legal Document.
2. Search to find the list of available items.
3. Click the asset or legal document that you want to update.
4. On the left, click Costs.
5. Click New and enter the cost information.
   The following fields require explanation:
   - **Unit Amount**
     Specifies the cost per unit.
   - **Total Amount**
     Displays the total cost for the total quantity of the selected asset. This amount is calculated automatically by multiplying the Unit Amount on the Costs page by the Quantity on the Asset Details page.
   - **Recurring Period**
     Specifies the number of Recurring Period Units (days, months, or years) after which the cost recurs. For example, if you enter 1 for the Recurring Period and select Month for the Recurring Period Units, the cost recurs every month.

     NOTE
     If you specify a value other than zero, select the Recurring Period Units and Termination Date.
   - **Recurring Period Units**
     Specifies the unit (days, months, or years) for the Recurring Period.
   - **Termination Date**
     Specifies the last date for the recurring schedule.
6. Click Save.

   NOTE
   If you specify recurring period information for an asset cost, a payment schedule with multiple payments is automatically created. If you do not specify recurring period information, the payment schedule includes only one payment.
7. (Optional) View the payment schedule for the cost that you defined using the following steps:
   a. Click the Edit Record icon for the cost record.
   b. Click Show Payments on the right.

   NOTE
   You can record payments that you made for this cost.

Add Cost Records to Multiple Assets

CA APM lets you add a cost record to multiple assets at a time. For example, you procured 20 new laptops and want to enter cost records for these laptops.

You can choose to update individual asset cost record or make changes to all the records at a time. For example, all the laptops that you procured have the same Unit Amount. You enter the Unit Amount and apply the value to all the asset cost records.

Follow these steps:

1. Click the Asset tab.
2. Click Add Asset Cost under Mass Change utilities on the left.
3. Search for the list of the available assets.

   NOTE

   : You can only add new cost records to the assets. To update existing cost records, you must open the respective asset cost record and must update it.
4. To make cost record changes to multiple assets at the same time, complete the following steps:
   a. Under Search Results, select the assets that have common cost record fields.
   b. Click Fill All Values in a Column.
   c. Click New.
   d. In the Select Column drop-down list, select the cost record field name that you want to fill for the assets.
   e. In the Value text box, enter the data that you want to update.
   f. Click the Check button.
      The selected field column is updated with the value you entered.
5. Under Search Results, enter the appropriate values in the cost record fields.
6. Click Submit.
   The cost records are added to the assets.

Delete a Cost Record

You can delete the cost record of an asset or legal document. The payment records that are associated with the deleted cost record are also deleted.

WARNING
When you delete a cost record, you can no longer view the audit history for the record.

Follow these steps:

1. Click Asset or Legal Document.
2. Search for and select the asset or legal document for which you want to delete the cost record.
3. On the left, click Costs.
4. Click the Mark for Deletion icon for the records you want to delete.

   NOTE
   To reverse the Mark for Deletion selection, click the Undo Record Deletion icon.
5. Click Save.

Payment Schedules and Recalculation

The Payment Schedule is a list of the payments due, based on the cost information you provide on a cost record. CA APM automatically generates the payment schedule in the form of a table when you define and save a cost record. The schedule generated depends on whether the cost is a one-time cost or a recurring cost.

For a one-time cost, the schedule assumes that you will pay the entire amount on the date you incurred the cost. One payment record is scheduled.

For a recurring charge, the schedule assumes that you will make payments of equal amounts for each recurring period. One payment record for each recurring period is scheduled.

Each row in the payment schedule represents a payment that you expect to make. If a payment is done, you can manually enter the date and amount paid details in the respective columns.

By default, every schedule is system-maintained as CA APM creates schedules when you create and save cost records. However, when you manually change information in a payment schedule, it becomes user-maintained.

System-Maintained Payment Schedules

A system-maintained payment schedule is a schedule that CA APM creates and automatically calculates the payments and dynamically updates whenever the cost record is modified. CA APM recalculates payment schedule based on the criteria you provide.

For example, you expect costs to escalate 3% a year and specify an escalation percentage. CA APM creates a payment schedule on this data. Later, if you change the escalation percentage to 3.5%, CA APM automatically recalculates the schedule. This helps you adjust cost information affecting the payment schedule to meet your requirements.

Payments are calculated automatically when you change the following values of the cost record:

- Unit Amount
- Recurring Period
- Recurring Period Units
- Escalation Percent
- Begin Date
- Termination Date
- Asset Quantity

User-Maintained Payment Schedule

The system-maintained payment schedule becomes user-maintained when you manually change the information on the payment schedule.

For example, to change the escalation percentage, you perform one of the following tasks:

- Recalculate each payment manually and update the amount due in each row.
- Specify the cost information and Save. CA APM automatically recalculates and updates the payments.

Events and Notifications for CA APM

This article contains the following topics:

An *event* represents an activity related to a field (default or extended) for an object. When you define an event, you specify the criteria that must be met before the event occurs. For example, you want to know when the data in a particular field changes. You can define an event that detects the data change. An event works in combination with a *notification*, which the workflow provider (for example, CA Process Automation) creates to alert your team members that an important event has occurred for a specific field or object. By using events and notifications, you alert people about upcoming events and help ensure that the appropriate tasks are performed in the correct order at the right time.

A notification is triggered when an event that you define occurs. For example, you define a date event on the Termination Date field for a legal document to notify the contract manager 15 days before a legal contract expires. The contract manager uses the 15 days to review and possibly negotiate a better contract. When the date arrives (that is, 15 days before the contract expires), the event occurs and the notification process is triggered through the workflow provider. The workflow provider constructs, issues, and manages the notification based on the configuration that you provided in the workflow provider and in CA APM.

The default notification method in CA APM supports email notifications using a workflow provider. You can send an email notification to any user or distribution list that is defined in your internal email system, even if the user is not a CA APM user. In addition, you can send an email to any external email address, if permitted by your email system.

You can also configure the notification process in the workflow provider to trigger any type of process. For example, you can set up the notification process to perform certain actions in another application when an event occurs in CA APM. For information about setting up different notification processes, see your workflow provider documentation.

You can define the following types of events to track and manage important changes to fields or objects:

- Date events. Monitor a date field for an object and have the workflow provider notify you that an important date is approaching or has passed.
- Change events. Monitor a field for an object and have the workflow provider notify you that the field value has changed.
- Watch events. Monitor a field for an object and have the workflow provider notify you about a potential obstruction to completing a task.

Escalation of Notifications

When an event occurs, the workflow provider sends an email notification to the recipients that you specified when you defined the event. CA APM lets you send email notifications to different levels of recipients.

- *Initial recipients* are the primary recipients of the notification. They are the first users to receive the notification and to respond and acknowledge the notification. The notification contains information about the event that you specified when you defined the event. The recipient receives a reminder email before the acknowledgment due date arrives. If the recipient still does not acknowledge the notification by the due date, the notification is escalated if you selected a workflow process with escalation.
- *Escalation recipients* are secondary recipients of the notification. If the initial recipients do not acknowledge the notification within a specified time frame, the product escalates the notification to the escalation recipients if you selected a workflow process with escalation. Escalations help ensure that someone is notified about an important date or event when the initial recipient is not available to acknowledge the notification. The product includes the following workflow processes and escalations:
  - Notification and One Escalation - Sends a notification to the initial recipient and sends a reminder email. If the recipient acknowledges the notification, the process marks the event as completed. If the recipient does not acknowledge in the specified time frame, the process escalates the notification to the escalation recipient. If the escalation recipient responds, the process marks the event as completed. If the escalation recipient does not respond, the process marks the event as failed.
  - Notification without ACK - Sends a notification to the initial recipient and marks the event as completed. The recipient does not need to respond and the process does not escalate the notification.
  - Notification without Escalation - Sends a notification to the initial recipient and sends a reminder email. If the recipient does not respond, the process marks the event as failed. If the recipient responds in the specified time frame, the process marks the event as completed.

The notification levels let you notify one or more users about an event and provide separate instructions to each user. You define the recipients and the notification escalation levels when you specify the workflow provider process parameters for an event.

Acknowledgements

An email notification is acknowledged when the recipient opens the email, clicks the link to CA Process Automation, logs in to CA Process Automation, and acknowledges receipt of the notification. You acknowledge an email notification in the workflow provider.

The email that is sent to the initial recipient contains the message that you specified when you defined the event. If a user does not acknowledge receipt of the email, the notification is escalated to the next responsible recipient if the selected workflow process includes escalations. When a notification is acknowledged, the product does not perform any future escalations for the same event notification.

Email Notification Process Selection

When a date, change, or watch event occurs, the email notification process is started in the workflow provider (for example, CA Process Automation). You define and set up the email notification process in the workflow provider and in CA APM (for example, you define the email recipients, levels of escalation, and notification text). CA APM lets you define different types of notification processes in the workflow provider. For example, you can have an email notification process (provided with the product) and another user-defined process that initiates actions in an external application, such as an asset management dashboard. You select the notification process that you want to use with a specific event when you define the event.

NOTE
For more information about defining notification processes in the workflow provider, see your workflow provider documentation.

The type of email notification process that is started after an event occurs depends on the process that you selected when you defined the event. The following email notification processes are provided with the product and apply to the CA Process Automation workflow provider:

- Notification and One Escalation - Notifies the initial recipient and sends a reminder email. If the recipient acknowledges the notification, the process marks the event as completed. If the recipient does not acknowledge in the specified time frame, the process escalates the notification to the escalation recipient. If the escalation recipient responds, the process marks the event as completed. If the escalation recipient does not respond, the process marks the event as failed.
- Notification without ACK - Notifies the initial recipient and marks the event as completed. The recipient does not need to acknowledge the notification.
- Notification without Escalation - Notifies the initial recipient and sends a reminder email. If the recipient does not acknowledge the notification within the specified time frame, the process marks the event as failed. If the recipient acknowledges within the specified time frame, the process marks the event as completed.

You can define additional notification processes in the workflow provider that perform other actions (in addition to email notifications) when events occur. You can then select one of your own defined processes when you define an event.

How to Manage Events and Notifications

Events work in combination with notifications, which the workflow provider (for example, CA Process Automation) creates, to communicate information to your team members about important events and activity. To manage events and notifications, complete the following steps:

1. Administrators grant permissions to users to manage events.
   For more information about the permissions to manage events, see Grant Permissions to Manage Events.
2. Open an existing local or global configuration and define any of the following events:
   – Date event
   – Change event
   – Watch event
3. When defining an event, map all required workflow provider notification parameters to a CA APM object attribute.
4. The workflow provider initiates the email notification process.
5. View an audit history of events.
6. (Optional) The notification recipient acknowledges the notification.

Date Events

Use a *date event* to monitor a date field for an object and have the workflow provider (for example, CA Process Automation) notify you that an important date is approaching or has passed. Date events are based on the value that is stored in a specific date field (default and extended fields), and the notification from the workflow provider provides an advanced warning to alert someone to complete a follow-up task. The date on which a user is notified about the upcoming

or passed event is based on the field value for the object and the information that you specify when you define the event, including the Days After value.

CA APM lets you define one or more date events for a single field.

> **NOTE**
> As a rule, events are not triggered (and notifications are not sent) for field changes that happened before the event was defined. An exception to this rule occurs for date events. If the notification date occurs *after* the event was defined, an event is triggered (and a notification is sent) even if the field change happened *before* the event was defined.

Example: Define a Date Event to Terminate a Legal Document

In this example, a contract negotiator must review contracts thirty days before they expire. You define a date event on the Terminate Date field for a legal document. For example, when the contract manager adds a legal document for which the termination date is 3/31/2010, a notification is sent to the appropriate person on 3/1/2010.

Define a Date Event

CA APM lets you define a date event to monitor a date field and have the workflow provider (for example, CA Process Automation) notify you that an important date is approaching or has passed. For example, you can define a date event on the Terminate Date field for a legal document. You can define one or more date events for a single field.

> **NOTE**
> If your administrator has granted you the correct permissions, you can complete this task.

Follow these steps:

1. Click the tab and optional subtab for the event definition that you want to configure. On the left, click CONFIGURE: ON.
   The configuration of the event is enabled. In the Configuration Information area of the page, select an existing global or local configuration.

   > **WARNING**
   > Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.

   Next to the field, click the Event Configuration icon.
   The Events page for the selected field appears.
2. Click New.
3. Specify the information for the date event.
   The following fields require explanation:
   - **Event Type**
     Select the type of event as a date event. After you select the event type and save the event, you cannot change the event type. If you select the incorrect event type when defining an event, delete the event and define it again using the correct event type.
   - **Days After**
     Specify a positive or negative number to indicate how many days before or after a field change occurs to create a date event.
     - A positive number indicates how many days after the original value has passed to create a date event.
     - A negative number indicates how many days before the original value approaches to create a date event.
   - **Inactive**
     Select this check box to indicate that the date event is inactive. When you make a date event inactive, no new notifications are created for the event. However, pending notifications are processed.
   - **Event Provider**

Select the workflow provider to notify users that the date event has occurred (for example, CA IT Process Automation Manager). When you select a provider, all available workflow processes for the selected provider appear in the Workflow Process field.

– **Workflow Process**
Identifies the workflow process for the workflow provider. When you select a workflow process, all available process parameters for the workflow provider appear.
  - Notification and One Escalation - Notifies the initial recipient and sends a reminder email. If the recipient acknowledges the notification, the process marks the event as completed. If the recipient does not acknowledge in the specified time frame, the process escalates the notification to the escalation recipient. If the escalation recipient responds, the process marks the event as completed. If the escalation recipient does not respond, the process marks the event as failed.
  - Notification without ACK - Notifies the initial recipient and marks the event as completed. The recipient does not need to acknowledge the notification.
  - Notification without Escalation - Notifies the initial recipient and sends a reminder email. If the recipient does not acknowledge the notification within the specified time period, the process marks the event as failed. If the recipient acknowledges within the specified time period, the process marks the event as completed.
  - (Optional) Additional Process Types - Uses processes you defined in the workflow provider.

– **Notification Parameters**
Specify each process parameter for the workflow provider by doing one of the following in each field:
  - Enter an actual (hard-coded) value.
  - Click Map Fields to map the parameter to a CA APM object attribute.
  - Enter an actual (hard-coded) value and, in the same field, click Map Fields to map the parameter.

NOTE
Refer to the field tooltips for specific information about the format and content of each parameter field.

4. Click Save.
5. Click CONFIGURE: OFF.
   The configuration of the date event is complete.

Update a Date Event

CA APM lets you update the information for an existing date event. For example, you can change the event name and description, and you can make the event inactive.

NOTE
If your administrator has granted you the correct permissions, you can complete this task.

Follow these steps:

1. Click the tab and optional subtab for the event definition that you want to configure. On the left, click CONFIGURE: ON.
   The configuration of the event is enabled.
   In the Configuration Information area of the page, select an existing global or local configuration.

   WARNING
   Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.

   Next to the field, click the Event Configuration icon.
   The Events page for the selected field appears.
2. Click the Edit Record icon next to the date event that you want to update.
3. Enter the new information for the date event.

NOTE
After you define and save an event, you cannot change the Event Type, Event Cause, Value Changed From, and Value Changed To. If you enter the incorrect information, delete the event and define it again using the correct information.

4. Click the Complete Record Edit icon.
5. Click Save.
6. Click CONFIGURE: OFF.
   The configuration of the date event is complete.

Delete a Date Event

CA APM lets you delete a date event that you do not need. For example, when you do not want to be notified about a change to the Terminate Date field for a legal document, you can delete the associated date event. If your administrator has granted you the correct permissions, you can complete this task.

NOTE

Any pending notifications from the workflow provider (for example, CA Process Automation) about the event are sent before the event is deleted. When you delete an event, all historical information about the event is deleted. We recommend that instead of deleting the event, you make the event inactive. That way, if you need the event in the future, you do not have to redefine it.

Follow these steps:

1. Click the tab and optional subtab for the event definition that you want to configure.
   On the left, click CONFIGURE: ON. The configuration of the event is enabled.
   In the Configuration Information area of the page, select an existing global or local configuration.

   WARNING
   Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.

   Next to the field, click the Event Configuration icon. The Events page for the selected field appears.
2. Click the Mark for Deletion icon next to the date event that you want to delete.
3. Click Save.
4. Click CONFIGURE: OFF.
   The configuration of the date event is complete.

Change Events

Use a *change event* to monitor a field for an object and have the workflow provider (for example, CA Process Automation) notify you that the field value has changed. Change events are based on the value that is stored in a specific field (default and extended fields), and the notification from the workflow provider provides a warning to alert someone when the value of a field is set or changes.

CA APM lets you define one or more change events for a single field.

Example: Define a Change Event to Find Equipment for a New Employee

In this example, asset technicians provide the appropriate equipment when an employee is hired or transferred to a different department. You define a change event that issues a notification to a member of the asset technician team when the Department field value for the contact is set or changes. This notification alerts the technician to find equipment for the new or transferred employee.

Define a Change Event

CA APM lets you define a change event to monitor a field and have the workflow provider (for example, CA Process Automation) notify you that the field value has changed. For example, you can define a change event on the Department field for a contact. You can define one or more change events for a single field.

NOTE
If your administrator has granted you the correct permissions, you can complete this task.

Follow these steps:

1. Click the tab and optional subtab for the event definition that you want to configure.
   On the left, click CONFIGURE: ON. The configuration of the event is enabled.
   In the Configuration Information area of the page, select an existing global or local configuration.

   WARNING
   Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.

   Next to the field, click the Event Configuration icon.
   The Events page for the selected field appears.
2. Click New.
3. Specify the information for the change event.
   The following fields require explanation:
   - **Event Type**
     Select the type of event as a change event. After you select the event type and save the event, you cannot change the event type. If you select the incorrect event type when defining an event, delete the event and define it again using the correct event type.
   - **Event Cause**
     Select the type of action that must happen to the field for the change event to occur. Supported event causes include when a field is changed, a record is added, and a record is deleted.
   - **Value Changed From**
     Select the initial state of the field value for the change event to occur. Supported field value changes include the following options:
     - **Any value**. Any field value sets the initial state.
     - **Blank**. A blank field value sets the initial state.
     - **Old value**. A specific value sets the initial state.
   - **Value**
     Available when you select *Old value* in the Value Changed From field. Enter a specific value to set the initial state.
   - **Value Changed To**
     Select the final state of the field value for the change event to occur. Supported field value changes include the following options:
     - **Any value**. Any field value, except a blank value, sets the final state.
     - **Blank**. A blank field value sets the final state.
     - **New value**. A specific value sets the final state.
   - **Value**
     Available when you select *New value* in the Value Changed To field. Enter a specific value to set the final state.
   - **Inactive**
     Select this check box to indicate that the change event is inactive. When you make a change event inactive, no new notifications are created for the event. However, pending notifications are processed.
   - **Event Provider**
     Select the workflow provider to notify users that the change event has occurred (for example, CA IT Process Automation Manager). When you select a provider, all available workflow processes for the selected provider appear in the Workflow Process field.
   - **Workflow Process**

Identifies the workflow process for the workflow provider. When you select a workflow process, all available process parameters for the workflow provider appear.

- Notification and One Escalation - Notifies the initial recipient and sends a reminder email. If the recipient acknowledges the notification, the process marks the event as completed. If the recipient does not acknowledge in the specified time frame, the process escalates the notification to the escalation recipient. If the escalation recipient responds, the process marks the event as completed. If the escalation recipient does not respond, the process marks the event as failed.
- Notification without ACK - Notifies the initial recipient and marks the event as completed. The recipient does not need to acknowledge the notification.
- Notification without Escalation - Notifies the initial recipient and sends a reminder email. If the recipient does not acknowledge the notification within the specified time period, the process marks the event as failed. If the recipient acknowledges within the specified time period, the process marks the event as completed.
- (Optional) Additional Process Types - Uses processes you defined in the workflow provider.

– **Notification Parameters**
Specify each process parameter for the workflow provider by doing one of the following in each field:

- Enter an actual (hard-coded) value.
- Click Map Fields to map the parameter to a CA APM object attribute.
- Enter an actual (hard-coded) value and, in the same field, click Map Fields to map the parameter.

NOTE
Refer to the field tooltips for specific information about the format and content of each parameter field.

4. Click Save.
5. Click CONFIGURE: OFF.
   The configuration of the event is complete.

Update a Change Event

CA APM lets you update the information for an existing change event. For example, you can change the event name and description, and you can make the event inactive.

NOTE
If your administrator has granted you the correct permissions, you can complete this task.

Follow these steps:

1. Click the tab and optional subtab for the event definition that you want to configure. On the left, click CONFIGURE: ON.
   The configuration of the event is enabled.
2. In the Configuration Information area of the page, select an existing global or local configuration. Next to the field, click the Event Configuration icon.
   The Events page for the selected field appears.

   NOTE

   Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration
3. Click the Edit Record icon next to the change event that you want to update.
4. Enter the new information for the change event.

   NOTE

   After you define and save an event, you cannot change the Event Type, Event Cause, Value Changed From, and Value Changed To. If you enter the incorrect information, delete the event and define it again using the correct information.
5. Click the Complete Record Edit icon.
6. Click Save.

7. Click CONFIGURE: OFF.
   The configuration of the change event is complete.

Delete a Change Event

CA APM lets you delete a change event that you do not need. For example, when you do not want to be notified about a change to the Department field for a contact, you can delete the associated change event. If your administrator has granted you the correct permissions, you can complete this task.

> **NOTE**
>
> Any pending notifications from the workflow provider (for example, CA Process Automation) about the event are sent before the event is deleted. When you delete an event, all historical information about the event is deleted. We recommend that instead of deleting the event, you make the event inactive. That way, if you need the event in the future, you do not have to redefine it.

Follow these steps:

1. Click the tab and optional subtab for the event definition that you want to configure. On the left, click CONFIGURE: ON.
   The configuration of the event is enabled. In the Configuration Information area of the page, select an existing global or local configuration.

   > **WARNING**
   > Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.

   Next to the field, click the Event Configuration icon.
   The Events page for the selected field appears.
2. Click the Mark for Deletion icon next to the change event that you want to delete.
3. Click Save.
4. Click CONFIGURE: OFF.
   The configuration of the change event is complete.

Watch Events

Use a *watch event* to monitor a field for an object and have the workflow provider (for example, CA Process Automation) notify you about a potential obstruction to completing a task. Watch events are based on inactivity on a particular field (default and extended fields) within a specified time period, and the notification from the workflow provider provides an advanced warning to alert someone about a potential obstruction. If the field value changes within the time period, the workflow provider does not send a notification.

CA APM lets you define one or more watch events for a single field.

Example: Define a Watch Event to Configure and Deploy New Laptops

In this example, you require that asset technicians configure and deploy all new laptops to employees within five days of receiving the laptop. To meet this requirement, you define a watch event that creates an event when an asset is assigned a status (Lifecycle Status field) of received. If the status remains received for more than five days, a notification is sent to an asset technician.

Define a Watch Event

CA APM lets you define a watch event to monitor a field and have the workflow provider (for example, CA Process Automation) notify you about inactivity on a particular field. For example, you can define a watch event on the Lifecycle Status field for an asset. You can define one or more watch events for a single field.

> **NOTE**
> If your administrator has granted you the correct permissions, you can complete this task.

Follow these steps:

1. Click the tab and optional subtab for the event definition that you want to configure. On the left, click CONFIGURE: ON.

   The configuration of the event is enabled. In the Configuration Information area of the page, select an existing global or local configuration.

   WARNING
   Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.

   Next to the field, click the Event Configuration icon.

   The Events page for the selected field appears.
2. Click New.
3. Specify the information for the watch event.

   WARNING
   The Value Changed From and Value fields work together to start the timer for the watch event. The timer continues for the duration that you specify in the Days After field. The Value Changed To and Value fields work together to indicate the value that you want to achieve and stop the timer. If the value that you want to achieve does not occur in the specified time period after the timer starts, the watch event occurs and indicates that the defined workflow did not happen.

   The following fields require explanation:

   – **Event Type**
   Select the type of event as a watch event. After you select the event type and save the event, you cannot change the event type. If you select the incorrect event type when defining an event, delete the event and define it again using the correct event type.

   – **Event Cause**
   Select the type of action that must happen to the field for the watch event to occur. A supported event cause is when a field is changed.

   – **Days After**
   Specify a positive number to indicate how many days to wait after the Value Changed From field value to start the watch timer.

   NOTE
   If the Value Change To field value does not occur, the watch timer expires after the number of days that you specify. After the watch timer expires, the watch event is created. However, the watch timer will stop when the Value Change To field value changes to the specified value and no watch event is created.

   – **Value Changed From**
   Select the initial field value to start the timer to create the watch event. Supported field value changes include the following options:
   - **Any value**. Any field value starts the timer for the watch event.
   - **Blank**. A blank field value starts the timer for the watch event.
   - **Old value**. A specific initial value starts the timer for the watch event.

   – **Value**
   Available when you select *Old value* in the Value Changed From field. Enter a specific value to start the timer for the watch event.

   – **Value Changed To**
   Select the final state of the field value that stops the timer for the watch event and prevents the watch event from occurring. Supported field value changes include the following options:

- **Any value**. Any field value stops the timer for the watch event and prevents the watch event from occurring.
- **Blank**. A blank field value stops the timer for the watch event and prevents the watch event from occurring.
- **New value**. A specific field value stops the timer for the watch event and prevents the watch event from occurring.

– **Value**
Available when you select *New value* in the Value Changed To field. Enter a specific value to stop the timer for the watch event and prevent the watch event from occurring.

– **Inactive**
Select this check box to indicate that the watch event is inactive. When you make a watch event inactive, no new notifications are created for the event. However, pending notifications are processed.

– **Event Provider**
Select the workflow provider to notify users that the watch event has occurred (for example, CA IT Process Automation Manager). When you select a provider, all available workflow processes for the selected provider appear in the Workflow Process field.

– **Workflow Process**
Identifies the workflow process for the workflow provider. When you select a workflow process, all available process parameters for the workflow provider appear.

- Notification and One Escalation - Notifies the initial recipient and sends a reminder email. If the recipient acknowledges the notification, the process marks the event as completed. If the recipient does not acknowledge in the specified time frame, the process escalates the notification to the escalation recipient. If the escalation recipient responds, the process marks the event as completed. If the escalation recipient does not respond, the process marks the event as failed.
- Notification without ACK - Notifies the initial recipient and marks the event as completed. The recipient does not need to acknowledge the notification.
- Notification without Escalation - Notifies the initial recipient and sends a reminder email. If the recipient does not acknowledge the notification within the specified time period, the process marks the event as failed. If the recipient acknowledges within the specified time period, the process marks the event as completed.
- (Optional) Additional Process Types - Uses processes you defined in the workflow provider.

– **Notification Parameters**
Specify each process parameter for the workflow provider by doing one of the following in each field:

- Enter an actual (hard-coded) value.
- Click Map Fields to map the parameter to a CA APM object attribute.
- Enter an actual (hard-coded) value and, in the same field, click Map Fields to map the parameter.

NOTE
Refer to the field tooltips for specific information about the format and content of each parameter field.

4. Click Save.
5. Click CONFIGURE: OFF.
   The configuration of the watch event is complete.

Update a Watch Event

CA APM lets you update the information for an existing watch event. For example, you can change the event name and description, and you can make the event inactive.

Note: If your administrator has granted you the correct permissions, you can complete this task.

Follow these steps:

1. Click the tab and optional subtab for the event definition that you want to configure. On the left, click CONFIGURE: ON.
   The configuration of the event is enabled. In the Configuration Information area of the page, select an existing global or local configuration.

WARNING
Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.

Next to the field, click the Event Configuration icon.
The Events page for the selected field appears.

2. Click the Edit Record icon next to the watch event that you want to update.
3. Enter the new information for the watch event.

NOTE
After you define and save an event, you cannot change the Event Type, Event Cause, Value Changed From, and Value Changed To. If you enter the incorrect information, delete the event and define it again using the correct information.

4. Click the Complete Record Edit icon.
5. Click Save.
6. Click CONFIGURE: OFF.
   The configuration of the watch event is complete.

Delete a Watch Event

CA APM lets you delete a watch event that you do not need. For example, when you do not want to be notified about a change to the Lifecycle Status field for an asset, you can delete the associated watch event. If your administrator has granted you the correct permissions, you can complete this task.

NOTE

Any pending notifications from the workflow provider (for example, CA Process Automation) about the event are sent before the event is deleted. When you delete an event, all historical information about the event is deleted. We recommend that instead of deleting the event, you make the event inactive. That way, if you need the event in the future, you do not have to redefine

Follow these steps:

1. Click the tab and optional subtab for the event definition that you want to configure. On the left, click CONFIGURE: ON.
   The configuration of the event is enabled. In the Configuration Information area of the page, select an existing global or local configuration.

WARNING
Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.

Next to the field, click the Event Configuration icon.
The Events page for the selected field appears.

2. Click the Mark for Deletion icon next to the watch event that you want to delete.
3. Click Save.
4. Click CONFIGURE: OFF.
   The configuration of the watch event is complete.

Workflow Provider Process Parameters

You perform some of the setup and configuration of the email notification process in the workflow provider. However, you also specify the process parameters for the workflow provider when you define an event in CA APM. The process parameters include items such as user IDs, email addresses, email subject, content of email, and other items. The workflow provider uses this information to construct, issue, and manage the email notification.

NOTE
For information about workflow provider process parameters that you must specify in CA Process Automation, see Integrate with CA Process Automation Integration for a Notification Process Manually . For information about setting up a notification process, see your workflow provider documentation.

You can provide actual (hard-coded) values when you specify the process parameters. If you use an actual value for an email address (or another parameter), you must verify that the address (or other data) is valid.

You can also map the process parameters to CA APM object attributes. If you map a process parameter to a CA APM object attribute, the workflow provider accesses CA APM to find the current value of the mapped attribute and uses that value to construct and manage the notification. For example, you map the process parameter Initial Email Addresses to the CA APM attribute Contact Email Address when you define an event. When that event occurs, the workflow provider determines the current value of the CA APM Contact Email Address and uses that value for the Email ID.

WARNING
To map CA Process Automation process parameters to CA APM objects successfully, you must understand the CA APM data objects and the CA Process Automation parameters. You need to determine which CA APM object is an appropriate match for each CA Process Automation parameter.

Notification and One Escalation Process Parameters

The process parameters that appear when you define an event depend on the workflow provider process type that you select. The Notification and One Escalation process is an email notification process that is included with the product. This process sends an email notification to the initial recipient when an event occurs and, if the initial recipient does not respond, escalates the email to the first escalation level recipients. When you select this process as the workflow process for an event, a list of parameters appears.

WARNING
CA APM and CA Process Automation do not validate the information that you enter for the parameters. You must verify that your input is valid and that you entered the data in the correct format.

The following fields require explanation:

- **Initial User IDs**
  CA Process Automation user ID (CA APM or non-CA APM user) for acknowledging the initial notification. You can specify more than one user ID, separated with colons. Do not enter spaces between entries.
  Example: This example contains a user ID (John) that is a text entry value and the mapped field {legaldoc.owner.userid}. The two items are separated with a colon. The mapped field {legaldoc.owner.userid} represents the user ID of the user in the Owner field of the Legal Document. To obtain the mapped field {legaldoc.owner.userid}, click the Map Fields button, select the Owner link in the Add Fields dialog, and select User ID from the list of attributes.
  ```
  John:{legaldoc.owner.userid}
  ```
- **Initial Groups**
  CA Process Automation group name for acknowledging the initial notification. You can specify more than one group name, separated with colons. Do not enter spaces between entries.
  Example: This example contains a CA Process Automation group name (ITAM) that is a text entry value.
  ```
  ITAM
  ```
- **Initial Email Addresses**
  Email address (or distribution list) of the initial recipient of the email. You can specify more than one address, separated with semicolons. You can enter spaces between entries.
  Example: This example contains an email address (john.doe@company.com) that is a text entry value and the mapped field {legaldoc.owner.emailid}. The two items are separated with a semicolon. The mapped field {legaldoc.owner.emailid} represents the email address of the user in the Owner field of the Legal Document. To obtain the mapped field {legaldoc.owner.emailid}, click the Map Fields button, select the Owner link in the Add Fields dialog, and select Email Address from the list of attributes.

```
john.doe@company.com;{legaldoc.owner.emailid}
```

- **Initial Email Copy Addresses**
  Email address (or distribution list) of the initial copy recipient of the email. You can specify more than one address, separated with semicolons. You can enter spaces between entries.
  Example: This example contains an email address (jane.doe@company.com) that is a text entry value and the mapped field {legaldoc.requestor.emailid}. The two items are separated with a semicolon. The mapped field {legaldoc.requestor.emailid} represents the email address of the user in the Requestor field of the Legal Document. To obtain the mapped field {legaldoc.requestor.emailid}, click the Map Fields button, select the Requestor link in the Add Fields dialog, and select Email Address from the list of attributes.
  ```
  jane.doe@company.com;{legaldoc.requestor.emailid}
  ```
- **Initial Email Subject**
  Subject of the email message for the initial recipient. Enter text or combine text entry with mapped fields. You do not need to enter colons or semicolons to separate entries.
  Example: This example contains text entry content (Acknowledgement required for) and the mapped field {legaldoc.documentidentifier}.
  ```
  Acknowledgment required for {legaldoc.documentidentifier}
  ```
- **Initial Email Message**
  Message of the email for the initial recipient. Enter text or combine text entry with mapped fields. You do not need to enter colons or semicolons to separate entries.
  Example: This example contains text entry content (The Legal Document Document Identifier), the mapped field {legaldoc.documentidentifier}, and more text entry content.
  ```
  The Legal Document Document Identifier {legaldoc.documentidentifier} requires your acknowledgment using the
   link in the Subject of this email.
  ```
- **Acknowledgment Title**
  Title that appears on the acknowledgment task that the user accesses in CA Process Automation to acknowledge receipt of the notification. You do not need to enter colons or semicolons to separate entries.
  Example: This example contains text entry content (The Legal Document Document Identifier) only.
  ```
  The Legal Document Document Identifier
  ```
- **Acknowledgment Description**
  Description that appears on the acknowledgment task that the user accesses in CA Process Automation to acknowledge receipt of the notification. You do not need to enter colons or semicolons to separate entries.
  Example: This example contains text entry content (The Legal Document Document Identifier) and the mapped field {legaldoc.documentidentifier}.
  ```
  The Legal Document Document Identifier {legaldoc.documentidentifier}
  ```
- **Escalation User IDs**
  CA Process Automation user ID (CA APM or non-CA APM user) for acknowledging the escalation notification. You can specify more than one user ID, separated with colons. Do not enter spaces between entries.
  Example: This example contains a user ID (Mary) that is a text entry value and the mapped field {legaldoc.owner.supervisor.userid}. The two items are separated with a colon. The mapped field {legaldoc.owner.supervisor.userid} represents the user ID of the supervisor of the user in the Owner field of the Legal Document. To obtain the mapped field {legaldoc.owner.supervisor.userid}, click the Map Fields button, select the Owner link in the Add Fields dialog, select the Supervisor link, and select User ID from the list of attributes.
  ```
  Mary:{legaldoc.owner.supervisor.userid}
  ```
- **Escalation Groups**
  CA Process Automation group name for acknowledging the escalation notification. You can specify more than one group name, separated with colons. Do not enter spaces between entries.
  Example: This example contains a CA Process Automation group name (ITAM) that is a text entry value.
  ```
  ITAM
  ```
- **Escalation Email Addresses**

Email address (or distribution list) of the recipient of the escalation email. You can specify more than one address, separated with semicolons. You can enter spaces between entries.
Example: This example contains an email address (mary.doe@company.com) that is a text entry value and the mapped field {legaldoc.owner.supervisor.emailid}. The two items are separated with a semicolon. To obtain the mapped field {legaldoc.owner.supervisor.emailid}, click the Map Fields button, select the Owner link in the Add Fields dialog, select the Supervisor link, and select Email Address from the list of attributes.

```
mary.doe@company.com;{legaldoc.owner.supervisor.emailid}
```

- **Escalation Email Copy Addresses**
  Email address (or distribution list) of the copy recipients of the escalation email. You can specify more than one address, separated with semicolons. You can enter spaces between entries.
  Example: This example contains an email address (jane.doe@company.com) that is a text entry value and the mapped field {legaldoc.requestor.emailid}. The two items are separated with a semicolon. The mapped field {legaldoc.requestor.emailid} represents the email address of the user in the Requestor field of the Legal Document. To obtain the mapped field {legaldoc.requestor.emailid}, click the Map Fields button, select the Requestor link in the Add Fields dialog, and select Email Address from the list of attributes.

  ```
  jane.doe@company.com;{legaldoc.requestor.emailid}
  ```

- **Escalation Email Subject**
  Subject of the escalation email. Enter text or combine text entry with mapped fields. You do not need to enter colons or semicolons to separate entries.
  Example: This example contains text entry content (Escalation for), the mapped field {legaldoc.documentidentifier}, and additional text entry content.

  ```
  Escalation for {legaldoc.documentidentifier}. Acknowledgment required
  ```

- **Escalation Email Message**
  Message of the escalation email. Enter text or combine text entry with mapped fields. You do not need to enter colons or semicolons to separate entries.
  Example: This example contains text entry content (The Legal Document Document Identifier), the mapped field {legaldoc.documentidentifier}, and more text entry content.

  ```
  The Legal Document Document Identifier {legaldoc.documentidentifier} requires your acknowledgment using the
   link in the Subject of this email.
  ```

- **Reminder Email Subject**
  Subject of the reminder email message for initial and escalation recipients. The product sends a reminder when half of the Acknowledgment Time-out period has passed. Enter text or combine text entry with mapped fields. You do not need to enter colons or semicolons to separate entries.
  Example: This example contains text entry content (Reminder for), the mapped field {legaldoc.documentidentifier}, and additional text entry content.

  ```
  Reminder for {legaldoc.documentidentifier}. Acknowledgment required.
  ```

- **Reminder Email Message**
  Message of the reminder email for initial and escalation recipients. The product sends a reminder when half of the Acknowledgment Time-out period has passed. Enter text or combine text entry with mapped fields. You do not need to enter colons or semicolons to separate entries.
  Example: This example contains text entry content (Reminder: The Legal Document Document Identifier), the mapped field {legaldoc.documentidentifier}, and more text entry content.

  ```
  Reminder: The Legal Document Document Identifier {legaldoc.documentidentifier} requires your acknowledgment
   using the link in the Subject of this email.
  ```

- **Acknowledgment Time-out (Days:Hours:Minutes)**
  The amount of time that is allowed for acknowledgment after the email notification is sent before the escalation process begins. This time-out applies to initial and escalation notifications. Days, hours, and minutes can be any numeric value. The format must be *days*:*hours*:*minutes* (separated with colons).
  Example: This example specifies a time-out period of exactly four days.

  ```
  4:00:00
  ```

Notification without ACK Process Parameters

The process parameters that appear when you define an event depend on the workflow provider process type that you select. The Notification without ACK process is an email notification process that is included with the product. This process sends an email notification to the specified recipient when an event occurs. The recipient does not need to acknowledge the notification, and the product does not escalate the notification. When you select this process as the workflow process for an event, a list of parameters appears.

> **WARNING**
> CA APM and CA Process Automation do not validate the information that you enter for the parameters. You must verify that your input is valid and that you entered the data in the correct format.

The following fields require explanation:

- **Email Address**
  Email address (or distribution list) of the recipient of the email. You can specify more than one address, separated with semicolons. You can enter spaces between entries.
  Example: This example contains an email address (john.doe@company.com) that is a text entry value and the mapped field {legaldoc.owner.emailid}. The two items are separated with a semicolon. The mapped field {legaldoc.owner.emailid} represents the email address of the user in the Owner field of the Legal Document. To obtain the mapped field {legaldoc.owner.emailid}, click the Map Fields button, select the Owner link in the Add Fields dialog, and select Email Address from the list of attributes.
  ```
  john.doe@company.com;{legaldoc.owner.emailid}
  ```
- **Email Copy Address**
  Email address (or distribution list) of the copy recipient of the email. You can specify more than one address, separated with semicolons. You can enter spaces between entries.
  Example: This example contains an email address (jane.doe@company.com) that is a text entry value and the mapped field {legaldoc.requestor.emailid}. The two items are separated with a semicolon. The mapped field {legaldoc.requestor.emailid} represents the email address of the user in the Requestor field of the Legal Document. To obtain the mapped field {legaldoc.requestor.emailid}, click the Map Fields button, select the Requestor link in the Add Fields dialog, and select Email Address from the list of attributes.
  ```
  jane.doe@company.com;{legaldoc.requestor.emailid}
  ```
- **Email Subject**
  Subject of the email message for the recipient. Enter text or combine text entry with mapped fields. You do not need to enter colons or semicolons to separate entries.
  Example: This example contains text entry content (Acknowledgment required for) and the mapped field {legaldoc.documentidentifier}.
  ```
  Acknowledgment required for {legaldoc.documentidentifier}
  ```
- **Email Message**
  Message of the email for the recipient. Enter text or combine text entry with mapped fields. You do not need to enter colons or semicolons to separate entries.
  Example: This example contains text entry content (The Legal Document Document Identifier), the mapped field {legaldoc.documentidentifier}, and more text entry content.
  ```
  The Legal Document Document Identifier {legaldoc.documentidentifier} requires your acknowledgment using the
   link in the Subject of this email.
  ```

Notification without Escalation Process Parameters

The process parameters that appear when you define an event depend on the workflow provider process type that you select. The Notification without Escalation process is an email notification process that is included with the product. This process sends an email notification to the specified recipient when an event occurs. If the recipient does not respond in the specified time period, the process does not escalate the notification. However, the process marks the associated event as failed. When you select this process as the workflow process for an event, a list of parameters appears.

WARNING
CA APM and CA Process Automation do not validate the information that you enter for the parameters. You must verify that your input is valid and that you entered the data in the correct format.

The following fields require explanation:

- **User IDs**
  CA Process Automation user ID (CA APM or non-CA APM user) for acknowledging the notification. You can specify more than one user ID, separated with colons. Do not enter spaces between entries.
  Example: This example contains a user ID (John) that is a text entry value and the mapped field {legaldoc.owner.userid}. The two items are separated with a colon. The mapped field {legaldoc.owner.userid} represents the user ID of the user in the Owner field of the Legal Document. To obtain the mapped field {legaldoc.owner.userid}, click the Map Fields button, select the Owner link in the Add Fields dialog, and select User ID from the list of attributes.
  ```
  John:{legaldoc.owner.userid}
  ```
- **Groups**
  CA Process Automation group name for acknowledging the notification. You can specify more than one group name, separated with colons. Do not enter spaces between entries.
  Example: This example contains a CA Process Automation group name (ITAM) that is a text entry value.
  ```
  ITAM
  ```
- **Email Addresses**
  Email address (or distribution list) of the recipient of the email. You can specify more than one address, separated with semicolons. You can enter spaces between entries.
  Example: This example contains an email address (john.doe@company.com) that is a text entry value and the mapped field {legaldoc.owner.emailid}. The two items are separated with a semicolon. The mapped field {legaldoc.owner.emailid} represents the email address of the user in the Owner field of the Legal Document. To obtain the mapped field {legaldoc.owner.emailid}, click the Map Fields button, select the Owner link in the Add Fields dialog, and select Email Address from the list of attributes.
  ```
  john.doe@company.com;{legaldoc.owner.emailid}
  ```
- **Email Copy Addresses**
  Email address (or distribution list) of the copy recipient of the email. You can specify more than one address, separated with semicolons. You can enter spaces between entries.
  Example: This example contains an email address (jane.doe@company.com) that is a text entry value and the mapped field {legaldoc.requestor.emailid}. The two items are separated with a semicolon. The mapped field {legaldoc.requestor.emailid} represents the email address of the user in the Requestor field of the Legal Document. To obtain the mapped field {legaldoc.requestor.emailid}, click the Map Fields button, select the Requestor link in the Add Fields dialog, and select Email Address from the list of attributes.
  ```
  jane.doe@company.com;{legaldoc.requestor.emailid}
  ```
- **Email Subject**
  Subject of the email message for the recipient. Enter text or combine text entry with mapped fields. You do not need to enter colons or semicolons to separate entries.
  Example: This example contains text entry content (Acknowledgment required for) and the mapped field {legaldoc.documentidentifier}.
  ```
  Acknowledgment required for {legaldoc.documentidentifier}
  ```
- **Email Message**
  Message of the email for the recipient. Enter text or combine text entry with mapped fields. You do not need to enter colons or semicolons to separate entries.
  Example: This example contains text entry content (The Legal Document Document Identifier), the mapped field {legaldoc.documentidentifier}, and more text entry content.
  ```
  The Legal Document Document Identifier {legaldoc.documentidentifier} requires your acknowledgment using the
   link in the Subject of this email.
  ```
- **Acknowledgment Title**

Title that appears on the acknowledgment task that the user accesses in CA Process Automation to acknowledge receipt of the notification. You do not need to enter colons or semicolons to separate entries.
Example: This example contains text entry content (The Legal Document Document Identifier) only.

```
The Legal Document Document Identifier
```

- **Acknowledgment Description**
  Description that appears on the acknowledgment task that the user accesses in CA Process Automation to acknowledge receipt of the notification. You do not need to enter colons or semicolons to separate entries.
  Example: This example contains text entry content (The Legal Document Document Identifier) and the mapped field {legaldoc.documentidentifier}.

  ```
  The Legal Document Document Identifier {legaldoc.documentidentifier}
  ```

- **Acknowledgment Time-out (Days:Hours:Minutes)**
  The amount of time that is allowed for acknowledgment after the email notification is sent. Days, hours, and minutes can be any numeric value. The format must be *days*:*hours*:*minutes* (separated with colons).
  Example: This example specifies a time-out period of exactly four days.

  ```
  4:00:00
  ```

- **Reminder Email Subject**
  Subject of the reminder email message. The product sends a reminder when half of the Acknowledgment Time-out period has passed. Enter text or combine text entry with mapped fields. You do not need to enter colons or semicolons to separate entries.
  Example: This example contains text entry content (Reminder for), the mapped field {legaldoc.documentidentifier}, and additional text entry content.

  ```
  Reminder for {legaldoc.documentidentifier}. Acknowledgment required.
  ```

- **Reminder Email Message**
  Message of the reminder email. The product sends a reminder when half of the Acknowledgment Time-out period has passed. Enter text or combine text entry with mapped fields. You do not need to enter colons or semicolons to separate entries.
  Example: This example contains text entry content (Reminder: The Legal Document Document Identifier), the mapped field {legaldoc.documentidentifier}, and more text entry content.

  ```
  Reminder: The Legal Document Document Identifier {legaldoc.documentidentifier} requires your acknowledgment
   using the link in the Subject of this email.
  ```

Make an Event Inactive

CA APM lets you make an event inactive so that the workflow provider (for example, CA Process Automation) does not send future notifications for the event. The history about important dates and events is retained.

> **NOTE**
> If your administrator has granted you the correct permissions, you can complete this task.

To make an event inactive

1. Click the tab and optional subtab for the event definition that you want to configure. On the left, click CONFIGURE: ON.
   The configuration of the event is enabled. In the Configuration Information area of the page, select an existing global or local configuration.

   > **WARNING**
   > Global configuration changes affect all users, regardless of their role. Local configuration changes only affect users in the roles assigned to the selected configuration.

   Next to the field, click the Event Configuration icon.
   The Events page for the selected field appears.
2. Click the Edit Record icon next to the event that you want to make inactive.
3. Select the Inactive check box.

4. Click the Complete Record Edit icon.
5. Click Save.
6. Click CONFIGURE: OFF.
   The configuration of the event is complete.

Notes

This article contains the following topics:

Notes are free-form explanatory text that you associate with any object, and supplement the information for an object. Notes are categorized by a *type* that you specify when you attach a note to an object. Use this information to search for objects that have a particular type of note assigned to them. For example, if another company acquires one of your primary suppliers, you can attach a company acquisition note to the company record for that supplier.

Default note types are provided for the following objects. Your administrator can define additional types.

- Models
- Assets
- Legal Documents
- Contacts
- Companies
- Organizations
- Locations
- Sites

Attach a Note

CA APM lets you attach a note to supplement the information for an object. For example, if another company acquires one of your primary suppliers, you can attach a company acquisition note to the company record for that supplier.

Follow these steps:

1. Click the object for which you want to attach a note. For example, click Model, Asset, Legal Document, Contact, Company, Organization, or Location.
2. Search for the list of available objects.
3. Click the object record for which you want to attach a note.
4. Click Notes on the left.
5. Click New and enter the note.
6. Click Save.

Update or Delete a Note

CA APM lets you update or delete a note that is attached to an object record.

> **WARNING**
> When you delete an object, you can no longer view the audit history for the object. We recommend that instead of deleting the object, you make the object inactive. Then, you can still view the audit history for the object.

Follow these steps:

1. Click the object for which you want to update a note. For example, click Model, Asset, Legal Document, Contact, Company, Organization, or Location.
2. Search for the list of available objects.
3. Click the object record for which you want to update or delete a note.
4. Click Notes on the left.

5. Perform one of the following actions.
6. Update a note.
   a. Click the Edit Record icon for the note you want to update.
   b. Update the note information.
   c. Click Save.
7. Delete a note.
   a. Click the Mark for Deletion icon next to the note you want to delete.
   b. Click Save.

Hardware Asset Management

This section contains the following articles:

- Hardware Reconciliation
- How to Reconcile
- Data Normalization
- Define a Reconciliation Rule
- Define Reconciliation Update Options
- Asset Matching Criteria
- Exclude an Ownership Asset from the Reconciliation Process
- Exclude an Asset Family from the Reconciliation Process
- Exclude an Asset Family Class or Subclass from the Reconciliation Process
- View the Reconciliation Results
- Add Assets from Unreconciled Discovered Records
- Manage Reconciliation Rules
- Export the Reconciliation Results

Hardware Reconciliation

This article contains the following topics:

The hardware reconciliation process matches *discovered assets* to corresponding *owned assets* from different logical repositories so that you can manage the assets based on your business practices. Use this process to identify the discrepancies between your owned and discovered assets. Hardware reconciliation identifies unauthorized, missing, under-utilized, and over-utilized assets, which helps you to optimize your hardware asset base.

- Owned assets provide IT asset information from a financial and ownership perspective. The product supports the entire lifecycle of an owned asset from procurement, acquisition, allocation, use, and disposal, including legal and cost information. Owned assets have purchase records, including a purchase order number, invoice number, associated costs (lease, payment schedule, maintenance), associated contracts (terms and conditions), and vendor information. The owned-asset data is entered and imported using the CA APM user interface, Administration tab, Data Importer.
- Discovered assets provide information about the assets that an enterprise is using or has deployed. External discovery products and the discovery component of CA Client Automation scan the assets on a network and store them as discovered assets in the CA MDB. Discovered assets contain evidence about the following information:
  – The asset is deployed and can be found on your network.
  – The asset is being used and has metrics.
  – The asset contains up-to-date configuration information for inventory.

The CA APM hardware reconciliation process matches the discovered asset data and the owned asset data that are stored in the CA MDB. The hardware reconciliation process may detect assets that cannot be reconciled with any of your

owned assets. You can decide to add the unreconciled assets to your repository so that you can track and manage all assets in your network.

Hardware reconciliation automates the synchronization of ownership and discovered data. Hardware reconciliation supports the discovery component of CA Client Automation and third-party discovery products. These components and products are supported through the combination of the CA Asset Converter and the asset collector component of CA Client Automation. When CA SAM is installed, discovery connectors load the discovery data into the CA SAM repository. The discovery data is then synchronized with CA APM.

Hardware Reconciliation Engine

The *Hardware Reconciliation Engine* is a continuously processing Windows service that is responsible for the following tasks during the reconciliation process:

- Synchronizes discovered assets with ownership assets using reconciliation rules.
- Reconciles owned and discovered assets based on asset matching criteria.
- Maps discovery data to ownership data based on normalization rules.
- Updates selected asset fields in the product based on changes to the corresponding discovered assets.
- Completes the actions that are defined in the reconciliation rule that is being executed.

How Reconciliation Engines Process Reconciliation Rules

The date and time at which a reconciliation rule was last executed determines when the rule will be processed again. Hardware Reconciliation Engines process reconciliation rules in the following sequence, which allows for multiple tenant support as well as multiple Hardware Engines:

1. Each Hardware Reconciliation Engine searches for reconciliation rules that are not being processed by another engine and selects the rule that has the oldest processing date and time.
2. A Hardware Reconciliation Engine locks the reconciliation rule so it cannot be accessed by another engine, executes the rule, updates the rule date-and-time value, and then unlocks the rule. The Hardware Reconciliation Engine searches for the next available reconciliation rule with the oldest date-and-time value and repeats the process.
3. The process continues with all engines continuously operating and searching for and executing the next available reconciliation rule with the oldest date-and-time value.

How to Reconcile

The reconciliation process compares data from discovery products with CA APM ownership data in the CA MDB. This process reconciles discovered and owned assets, tracks changes to critical fields, and tracks discrepancies as the result of missing or deleted assets. To reconcile, complete the following steps:

1. Establish data normalization rules to map data values between discovery repositories and the product.
2. Define a reconciliation rule to specify how to limit the data being processed and how to process the records found.
3. (Optional) Define reconciliation update options to specify the owned-asset fields that you want the Hardware Reconciliation Engine to update automatically with changes found in the corresponding discovered assets.
4. Define asset matching criteria to match owned and discovered assets for a reconciliation rule.
5. (Optional) Exclude an ownership asset from the reconciliation process.
6. (Optional) Exclude an asset family from the reconciliation process.
7. View the reconciliation results in the message queue.
8. (Optional) Add unreconciled assets to your repository so that you can track and manage all assets in your network.

NOTE
You can generate reports to view information about your reconciliation results and environment.

Data Normalization

This article contains the following topics:

Data normalization is a step in the reconciliation process where you establish a list of rules to standardize, organize, and consolidate data between the product and discovery repositories. Normalization reduces, eliminates, and consolidates redundant data that is imported into the product from multiple sources, such as purchase order products, human resource products, procurement products, the Data Importer, and so forth.

When you normalize data, you reduce the time and effort necessary to manage the data. You also reduce the possibility of the user selecting the incorrect information when defining assets, models, and other objects, and when generating reports. The product guides you through the normalization process, enabling you to perform it much more efficiently and accurately. The consolidated discovery data is then reconciled with your owned assets during the reconciliation process and can be reported on using the reconciliation reports.

You normalize three of the fields that can be used as asset matching criteria: company, operating system, and system model. These fields are normalized because they often have multiple values that represent one normalized value. For example, the discovery tool may find many variations for the name of one company. You normalize all the variations to one value for the company name and include the normalized company name in asset matching criteria. These fields are normalized so that you can include them in asset matching criteria.

The Hardware Reconciliation Engine normalizes the data that is imported into the product by referencing the following normalization rules, which you define:

- Company normalization rules
- Operating System normalization rules
- System Model normalization rules

Example: Normalize Company Data

In this example, when you import company data into the product using CA Client Automation, multiple variations of Document Management Company are discovered in various formats, including the following values:

- Document Management Company
- Document Management Co
- Doc Management Company
- Doc Management Co

To help users select the correct company when defining an asset and model and when generating reconciliation reports, you define company normalization rules to map all variations to Document Management Company. During the reconciliation process, the normalization rules map the values as follows, before an asset is updated in the CA MDB:

> **NOTE**
> A collected company that is mapped to an authoritative company affects Hardware Reconciliation only if the collected company is a discovered company.

Collected (Nonauthoritative Value)	Normalized (Authoritative Value)
Document Management Co	Document Management Company
Doc Management Company	Document Management Company
Doc Management Co	Document Management Company

Company Normalization Rules

Company normalization rules are intended for key organizations with which you have a business relationship, for example, Microsoft, Adobe, Lenovo, and so forth.

- **Collected Company**
  A *collected company* is either a discovered company or a *product-defined company*. Product-defined companies are collected from user input and other products that share the CA MDB. Hardware reconciliation reconciles only *discovered* companies. Collected companies have a *nonauthoritative* status. You map nonauthoritative collected companies to normalized *authoritative* companies.

  WARNING
  Only *discovered* collected companies that are mapped to normalized companies are reconciled during Hardware Reconciliation.

- **Normalized Company**
  A normalized company is either a *CA-content company* or a product-defined company. CA-content companies are provided with CA APM and have an *authoritative status*. Authoritative status allows a company to have a normalization rule. You define a normalization rule for an authoritative normalized company when you map one or more collected companies to the authoritative company.
  Product-defined companies initially have a nonauthoritative status. You can change the status of a product-defined company from nonauthoritative to authoritative. You can then map a collected company to the authoritative product-defined company to define a normalization rule. Only company normalization rules for discovered collected companies affect Hardware Reconciliation.
- **Subordinate Company**
  When you map a collected nonauthoritative company to a normalized authoritative company, the nonauthoritative company becomes *subordinate* to the authoritative company in the normalization rule.

Define Company Normalization Rules

WARNING

Normalization rules apply to all tenants and public data that are associated with a service provider. Verify that the user completing this task belongs to a role in which reconciliation management access is enabled.

You can define company normalization rules that are applied during the matching process whenever the reconciliation rule contains asset matching criteria. Any asset matching criterion that contains company information automatically applies the company normalization rules.

When you map a collected nonauthoritative company to a normalized authoritative company, the nonauthoritative company becomes *subordinate* to the authoritative company in the normalization rule. The subordinate company no longer appears in the Collected Company list or the Normalized Company list. If you delete the normalization rule that contains the subordinate company, the subordinate company returns to nonauthoritative status and appears in the Collected Company list and, if it was a product-defined company, also in the Normalized Company list.

Follow these steps:

1. Click Directory, List Management.
2. On the left, expand Normalization and select Company Normalization.
   The discovered collected company values and the normalized company values appear.
3. If the term you want to use as the normalized value does not appear in the Normalized Company list or the Collected Company list, click New Company to add the company, and then repeat the previous steps.
4. (Optional) Change a nonauthoritative company to an authoritative normalized company.
5. Map the Collected Company list discovered values to a Normalized Company value.
   The list of company normalization rules is defined and is referenced during the reconciliation process.

Update Company Normalization Rules

WARNING

Normalization rules apply to all tenants and public data that are associated with a service provider. Verify that the user completing this task belongs to a role in which reconciliation management access is enabled.

You update a company normalization rule when you want to change how a discovered company is normalized. This update may affect the reconciliation process. To update a normalization rule for a company, you delete the rule and define a new rule.

The product monitors updates to normalization rules. If you change a normalization rule, the Hardware Reconciliation Engine processes the asset matching so that the assets are matched using the new rule. Any assets that are matched as a result of a previous Hardware Reconciliation Engine process are evaluated again to determine if their matching should change based on the new normalization rule.

When you delete a company normalization rule, the subordinate company returns to nonauthoritative status and appears in the Collected Company list. If the subordinate company was a product-defined company, it also appears in the Normalized Company list on the Normalization Rule page.

Follow these steps:

1. Click Directory, List Management, Company Rules.
2. Delete the normalization rule that you want to change.
3. Define the new company normalization rule.

Change a Nonauthoritative Company to an Authoritative Normalized Company

> **WARNING**
>
> Normalization rules apply to all tenants and public data that are associated with a service provider. Verify that the user completing this task belongs to a role in which reconciliation management access is enabled.

You can map collected companies to only authoritative companies in the Normalized Company list on the Company Normalization page. Product-defined companies in the Normalized Company list initially have a nonauthoritative status. You can change a product-defined company that is in the Normalized Company list to an authoritative normalized company, to which you can map collected companies.

> **NOTE**
> Only company normalization rules for discovered collected companies affect Hardware Reconciliation.

Follow these steps:

1. Click Directory, List Management.
2. On the left, expand Normalization and select the normalization type.
   The discovered collected values and the normalized values appear.
3. Clear the Show only Authoritative records check box in the Normalized Company section.
4. Click Go in the Normalized Company section.
   The collected product-defined companies that are nonauthoritative and not yet mapped to an authoritative company appear in the normalized list with an Override icon next to the company name.
5. Click the Override icon to the left of the nonauthoritative company that you want to change to an authoritative company, in the *normalized* list.
   The nonauthoritative company changes to an authoritative normalized company.
6. Map a collected company to the new authoritative normalized company before performing any other action in the Normalized Company section.

   > **NOTE**
   > The company is not saved as an authoritative company until at least one collected company is mapped to the new authoritative company.

   The new authoritative normalized company and its normalization rule are saved.

Operating System Normalization Rules

Operating system normalization rules are intended for operating systems managing your computers, for example, Windows XP Professional, Windows Server 2008 Enterprise Edition, Windows Vista Enterprise Edition, and so forth.

- **Collected Operating System**
  A collected operating system is always a discovered operating system. Collected operating systems have a nonauthoritative status. You map nonauthoritative collected operating systems to normalized authoritative operating systems.
- **Normalized Operating System**
  A normalized operating system is always a *product-defined operating system*. Product-defined operating systems are collected from user input and other products that share the CA MDB. Product-defined operating systems always have an authoritative status. Authoritative status allows an operating system to have a normalization rule. You define a normalization rule for an authoritative normalized operating system when you map one or more collected operating systems to the authoritative operating system.
- **Subordinate Operating System**
  When you map a collected nonauthoritative operating system to a normalized authoritative operating system, the nonauthoritative operating system becomes subordinate to the authoritative operating system in the normalization rule.

Define Operating System Normalization Rules

> **WARNING**
>
> Normalization rules apply to all tenants and public data that are associated with a service provider. Verify that the user completing this task belongs to a role in which reconciliation management access is enabled.

You can define operating system normalization rules that are applied during the matching process whenever the reconciliation rule contains asset matching criteria. Any asset matching criterion that contains operating system information automatically applies the operating system normalization rules.

When you map a collected nonauthoritative operating system to a normalized authoritative operating system, the nonauthoritative operating system becomes subordinate to the authoritative operating system in the normalization rule. The subordinate operating system no longer appears in the Collected Operating System list. If you delete the normalization rule that contains the subordinate operating system, the subordinate operating system returns to nonauthoritative status and appears in the Collected Operating System list.

Follow these steps:

1. Click Directory, List Management.
2. On the left, expand Normalization and select Operating System Normalization.
   The discovered Collected Operating System values and the Normalized Operating System values appear.
3. If the term you want to use as the normalized value does not appear in the Normalized Operating System list or the Collected Operating System list, click New Operating System to add the operating system, and then repeat the previous steps.
4. Map the Collected Operating System list discovered values to a Normalized Operating System value.
   The list of operating system normalization rules is defined and is referenced during the reconciliation process.

Update Operating System Normalization Rules

> **WARNING**
>
> Normalization rules apply to all tenants and public data that are associated with a service provider. Verify that the user completing this task belongs to a role in which reconciliation management access is enabled.

You update an operating system normalization rule when you want to change how a discovered operating system is normalized. This update may affect the value of the operating system in a matching asset. To update a normalization rule for an operating system, you delete the rule and define a new rule.

The product monitors updates to normalization rules. If you change an operating system normalization rule and the reconciliation rule update option for operating system is enabled, the Hardware Reconciliation Engine changes the value of the operating system to the new normalized name in matching assets.

WARNING
If you change a normalization rule for an operating system that applies to public tenant data, the change may affect discovered assets for all tenants, depending on whether they use the specific operating system.

When you delete an operating system normalization rule, the subordinate operating system returns to nonauthoritative status and appears in the Collected Operating System list on the Normalization Rule page.

Follow these steps:

1. Click Directory, List Management, Operating System Rules.
2. Delete the normalization rule that you want to change.
3. Define the new operating system normalization rule.

System Model Normalization Rules

System model normalization rules are intended for hardware devices such as a computer, for example, Lenovo ThinkPad T400, Lenovo ThinkCentre M58, and so forth.

- **Collected System Model**
  A collected system model is always a discovered system model. Collected system models have a nonauthoritative status. You map nonauthoritative collected system models to normalized authoritative system models.
- **Normalized System Model**
  A normalized system model is always a *product-defined system model*. Product-defined system models are collected from user input and other products that share the CA MDB. Product-defined system models always have an authoritative status. Authoritative status allows a system model to have a normalization rule. You define a normalization rule for an authoritative normalized system model when you map one or more collected system models to the authoritative system model.
- **Subordinate System Model**
  When you map a collected nonauthoritative system model to a normalized authoritative system model, the nonauthoritative system model becomes subordinate to the authoritative system model in the normalization rule.

Define System Model Normalization Rules

WARNING

Normalization rules apply to all tenants and public data that are associated with a service provider. Verify that the user completing this task belongs to a role in which reconciliation management access is enabled.

You can define system model normalization rules that are applied during the matching process whenever the reconciliation rule contains asset matching criteria. Any asset matching criterion that contains system model information automatically applies the system model normalization rules.

When you map a collected nonauthoritative system model to a normalized authoritative system model, the nonauthoritative system model becomes subordinate to the authoritative system model in the normalization rule. The subordinate system model no longer appears in the Collected System Model list. If you delete the normalization rule that contains the subordinate system model, the subordinate system model returns to nonauthoritative status and appears in the Collected System Model list.

Follow these steps:

1. Click Directory, List Management.
2. On the left, expand Normalization and select System Model Normalization.
   The discovered collected system model name and manufacturer name values and the normalized system model name and manufacturer name values appear.
3. If the term you want to use as the normalized value does not appear in the Normalized System Model list or the Collected System Model list, click New System Model to add the system model, and then repeat the previous steps.
4. Map the Collected System Model list discovered values to a Normalized System Model value.
   The list of system model normalization rules is defined and is referenced during the reconciliation process.

Update System Model Normalization Rules

WARNING

Normalization rules apply to all tenants and public data that are associated with a service provider. Verify that the user completing this task belongs to a role in which reconciliation management access is enabled.

You update a system model normalization rule when you want to change how a discovered system model is normalized. This update may affect the reconciliation process. To update a normalization rule for a system model, you delete the rule and define a new rule.

The product monitors updates to normalization rules. If you change a normalization rule, the Hardware Reconciliation Engine processes assets so that the assets are matched using the new rule. Any assets that are matched as a result of a previous operation of the Hardware Reconciliation Engine are evaluated again to determine if their matching should change based on the new normalization rule.

When you delete a system model normalization rule, the subordinate system model returns to nonauthoritative status and appears in the Collected System Model list on the Normalization Rule page.

Follow these steps:

1. Click Directory, List Management, System Model Rules.
2. Delete the normalization rule that you want to change.
3. Define the new system model normalization rule.

View Normalization Rules

WARNING

Normalization rules apply to all tenants and public data that are associated with a service provider. Verify that the user completing this task belongs to a role in which reconciliation management access is enabled.

You can search for and view the normalization rules to see the mapping between collected values and the normalized values. This information is used during the hardware reconciliation process.

Follow these steps:

1. Click Directory, List Management.
2. On the left, expand Normalization and select the normalization rule type that you want to view.
3. Search for the normalized value or the collected value for which you want to view the normalization mapping rules. For each normalization rule, the collected value and its corresponding normalized value appear in the Search Results section.

Delete a Normalization Rule

WARNING

Normalization rules apply to all tenants and public data that are associated with a service provider. Verify that the user completing this task belongs to a role in which reconciliation management access is enabled.

You delete a normalization rule when you no longer want asset matching criteria to apply the normalization rule during the reconciliation process or when you want to change a normalization rule. To change a normalization rule for a company, operating system, or system model, you delete the rule and define a new rule.

When you delete a company normalization rule, the subordinate company returns to nonauthoritative status and appears in the Collected Company list. If the subordinate company was a product-defined company, it also appears in the Normalized Company list on the normalization rule page. When you delete an operating system normalization rule, the subordinate operating system returns to nonauthoritative status and appears in the Collected Operating System list on the normalization rule page. When you delete a system model normalization rule, the subordinate system model returns to nonauthoritative status and appears in the Collected System Model list on the normalization rule page.

Follow these steps:

1. Click Directory, List Management.
2. On the left, expand Normalization and select the normalization rule type for the rule that you want to delete.
3. Search for the normalized value or the collected value for which you want to delete a normalization rule.
   For each normalization rule, the collected value and its corresponding normalized value appear in the Search Results section.
4. Select the rule that you want to delete.
5. Click Delete Rule.
   The normalization rule is deleted. Asset matching criteria do not apply the rule during the reconciliation process.

Updates to Normalization Rules

You update a system model or company normalization rule when you want to change how a discovered system model or company is normalized. This update may affect the reconciliation process. You update an operating system normalization rule when you want to change how a discovered operating system is normalized. This update may affect the value of the operating system in a matching asset. To update a normalization rule, you delete the rule and define a new rule.

The product monitors updates to normalization rules. If you change a system model or company normalization rule, the Hardware Reconciliation Engine performs the asset matching process so that the assets are matched using the new rule. Any assets that are matched as a result of a previous operation are evaluated again to determine if their matching should change based on the new normalization rule. If you change an operating system normalization rule, the Hardware Reconciliation Engine changes the value of the operating system to the new normalized name in matching assets if the reconciliation rule update option for operating system is enabled.

When you delete a company normalization rule, the subordinate company returns to nonauthoritative status and appears in the Collected Company list. If the subordinate company was a product-defined company, it also appears in the Normalized Company list on the Normalization Rule page. When you delete an operating system normalization rule, the subordinate operating system returns to nonauthoritative status and appears in the Collected Operating System list on the Normalization Rule page. When you delete a system model normalization rule, the subordinate system model returns to nonauthoritative status and appears in the Collected System Model list on the Normalization Rule page.

Define a Reconciliation Rule

NOTE

Verify that the user completing this task belongs to a role in which reconciliation management access is enabled.

Use a reconciliation rule to define the processing options and actions that the Hardware Reconciliation Engine performs. You can define one reconciliation rule for each tenant.

NOTE
Each tenant can have only one reconciliation rule. Therefore, if you make a rule inactive, the inactive rule is the only reconciliation rule that is associated with the tenant. If you want to change the reconciliation rule for a tenant, you can update the current reconciliation rule or delete the current reconciliation rule and define a new rule. If you delete a rule, all asset matching links between discovered and owned assets that are associated with the rule are also deleted.

Follow these steps:

1. Click Administration, Reconciliation Management.
2. On the left, click New Reconciliation Rule.
3. If applicable, select the tenant that is associated with the reconciliation rule that you are defining.
4. Enter the reconciliation rule information and click Save.

NOTE
Select the Inactive check box if you want to suspend reconciliation processing for an individual tenant. If you select the Inactive check box, the Hardware Reconciliation Engine does not process the rule (no asset matching or data updates occur for the rule). For example, you can make a rule inactive temporarily while you define normalization rules or troubleshoot asset matching errors. If you make an existing rule inactive and the discovered and owned assets were already matched, the matching links are saved.

5. (Optional) Select the Monitor Asset Updates check box to define the reconciliation update options that you want the Hardware Reconciliation Engine to update automatically.
6. (Optional) Select the Match Assets check box to define the asset matching criteria that you want the Hardware Reconciliation Engine to apply.

   NOTE
   If you do not define matching criteria, the Hardware Reconciliation Engine uses the default asset matching criterion, which matches the owned serial number to the discovered serial number.
7. Click Save.
   The new reconciliation rule is defined.

Define Reconciliation Update Options

NOTE

Verify that the user completing this task belongs to a role in which reconciliation management access is enabled.

You can apply changes to selected, critical owned-asset fields when the Hardware Reconciliation Engine detects new values in the corresponding discovered-asset fields. The Hardware Reconciliation Engine monitors the critical fields that you select and updates the owned-asset fields when changes are detected in the corresponding discovered assets. You specify the owned-asset fields that you want to be automatically updated with changes that are found in the corresponding discovered assets.

Follow these steps:

1. Click Administration, Reconciliation Management.
2. On the left, click Reconciliation Rule Search.
3. Search to find the list of available reconciliation rules.
4. Click the reconciliation rule for which you want to define reconciliation update options.
5. Select the Monitor Asset Updates check box to have the Hardware Reconciliation Engine apply automatic updates.
6. Select the check boxes for the fields that you want to be automatically updated.

   NOTE
   Select the Host Name check box to enable and select the Copy Host Name to Asset Name update option.
7. Click Save.
   The new reconciliation update options are defined.

Asset Matching Criteria

This article contains the following topics:

The Hardware Reconciliation Engine matches the owned and discovered assets based on the *asset matching criteria* that you define for a reconciliation rule. The criteria define the matching factors for owned and discovered assets based on a list of field values from both CA APM and the discovery products. The Hardware Reconciliation Engine identifies all assets being managed and provides the required data during the reconciliation process.

If you modify the asset matching criteria associated with a reconciliation rule, the Hardware Reconciliation Engine reprocesses reconciled assets using the new criteria the next time the engine processes the rule.

You can match the following asset field values:

Owned Asset Field	Discovered Asset Field
Alternate Host Name	Host Name
Alternate Host Name	Registry Asset Name
Alternate ID	BIOS Asset Tag
Asset Alias	BIOS Asset Tag
Asset Alias	Host Name
Asset Name	Host Name
Class	System Type
Host Name	Host Name
Host Name	Registry Asset Name
MAC Address	MAC Address
MAC Address and Host Name	MAC Address and Host Name
Manufacturer	System Vendor
Model Name	System Model
Previous Asset Tag	BIOS Asset Tag
Serial Number	Serial Number
Subclass	System Type

The product monitors the asset matching fields of owned and discovered assets. If you modify the value of an owned asset field that can be used for asset matching, the Hardware Reconciliation Engine reprocesses the modified owned asset using the new value, during the next reconciliation process.

Similarly, if the discovery component of CA Client Automation or a third-party discovery product modifies the value of a discovered asset field in the CA MDB that can be used for asset matching, the Hardware Reconciliation Engine reprocesses the modified discovered asset using the new value, during the next reconciliation process.

The product also monitors changes to normalization rules for companies and system models. These rules affect the asset matching of owned and discovered assets. If you change one of the normalization rules, the Hardware Reconciliation Engine runs the asset matching process so that assets are matched using the new rules. Any assets that are matched as a result of a previous run of the Hardware Reconciliation Engine are evaluated again to determine if their matching should change based on the new normalization rules.

Inactive Status Effect on Asset Matching

Hardware Reconciliation processes all active owned assets created by CA APM that are not excluded from reconciliation. An inactive asset, model, asset family, or company affects reconciliation links between owned and discovered assets as follows:

- Inactive owned asset
  - If an owned asset is inactive before the Hardware Reconciliation Engine matches the asset to a discovered asset, the owned asset is not matched.
  - If an owned asset is made inactive after asset matching, the Hardware Reconciliation Engine clears the matching link the next time the engine processes the reconciliation rule.
- Inactive Model

  - When a model is made inactive before the Hardware Reconciliation Engine matches a discovered asset to an owned asset that is based on that model, the owned asset is not matched.
  - When a model is made inactive after an owned asset that is based on that model is matched, the Hardware Reconciliation Engine clears the matching link for that asset and all others that are based on the inactive model the next time the engine processes the reconciliation rule.
- Inactive Asset Family
  - When an asset family, class, or subclass is made inactive before the Hardware Reconciliation Engine matches a discovered asset to an owned asset with that asset family, class, or subclass, the owned asset is not matched.
  - When an asset family, class, or subclass is made inactive after an owned asset with the asset family, class, or subclass is matched, the Hardware Reconciliation Engine clears the matching link for that asset and all others that belong to the inactive asset family, class, or subclass the next time the engine processes the reconciliation rule.
- Inactive Company
  - When a company is made inactive before the Hardware Reconciliation Engine matches a discovered asset to an owned asset with that company as the manufacturer, the owned asset is not matched.
  - When a company is made inactive after an owned asset with that company as the manufacturer is matched, the Hardware Reconciliation Engine clears the matching link for that asset and all others that are associated with the inactive company the next time the engine processes the reconciliation rule.

Define Asset Matching Criteria

NOTE

Verify that the user completing this task belongs to a role in which reconciliation management access is enabled.

Based on the matching criteria that you define for a reconciliation rule, the product attempts to match the owned and discovered assets. The Hardware Reconciliation Engine performs the asset reconciliation when the ownership field value matches the discovered field value.

Follow these steps:

1. Click Administration, Reconciliation Management.
2. On the left, click Reconciliation Rule Search.
3. Search to find the list of available reconciliation rules.
4. Click the reconciliation rule for which you want to define an asset matching criterion.
5. Select the Match Assets check box to have the Hardware Reconciliation Engine apply asset matching criteria.
6. Select the owned and discovered fields that you want to match and click Add Criteria.
   A new asset matching criteria record is added to the Matching Criteria section and a new trimming record is added to the Trimming section.
7. Click the Edit Record icon next to the new criterion in the Matching Criteria section.
8. Select the matching criterion options.
9. (Optional) In the Trimming section, click the Edit Record icon and select trimming options for the asset matching criterion. For example, discovered computer names at one site have a three-character location code as a prefix, which is not in the owned asset computer name. You create a trimming record for the asset matching criterion that trims three characters from the left side of the discovered computer names.
10. (Optional) Continue to add matching criteria to the reconciliation rule.
11. Click Save.
    The new asset matching criteria are defined and the reconciliation rule is saved.

NOTE

After a reconciliation rule is saved, if you want to change the matched fields in an asset matching criterion to different fields, delete the criterion and create a new one.

Exclude an Ownership Asset from the Reconciliation Process

You can exclude individual owned assets from the reconciliation process. Some reasons for excluding assets from reconciliation include the following examples:

- An owned asset has no matching discovered asset and continues to be included in the list of unreconciled assets. For example, assets that are never attached to the network, like some laptops, or assets that are retired, but their ownership data is stored in CA APM.
- A company does not want to include a particular class of asset, for example, laptops, in reconciliation.

If an owned asset is excluded from the reconciliation process before the Hardware Reconciliation Engine matches the asset to a discovered asset, the owned asset is not available for matching. If an owned asset is excluded from the reconciliation process after asset matching, the Hardware Reconciliation Engine clears the matching link the next time the engine processes the reconciliation rule. The owned asset is not available for matching.

> **NOTE**
> If the asset family of an excluded asset is set to be included in the hardware reconciliation process, the asset continues to be excluded from the process. If an asset family is excluded from the hardware reconciliation process, all assets that belong to the excluded asset family are excluded.

Follow these steps:

1. Click Asset, Asset Search.
2. Search to find the list of available assets.
3. Click the asset that you want to exclude from the reconciliation process.
4. In the Basic Information section, select the Exclude Reconciliation check box.
5. Click Save.
   The ownership asset is not included in future reconciliations.

Exclude an Asset Family from the Reconciliation Process

You can exclude all owned assets in an asset family from the hardware reconciliation process. Some reasons for excluding asset families from reconciliation include the following examples:

- Assets in an asset family have no matching discovered assets and continue to be included in the list of unreconciled assets. For example, assets that are in the service asset family are not attached to the network, but their ownership data is stored in CA APM.
- The asset family is software. The product does not reconcile software.
- A company wants to make an asset family inactive. Hardware Reconciliation processes active owned assets created by CA APM.

If an asset family is excluded from the reconciliation process before the Hardware Reconciliation Engine matches owned assets in that asset family to discovered assets, the owned assets are not available for matching. If an asset family is excluded from the reconciliation process after owned assets in that asset family are matched, the Hardware Reconciliation Engine clears the matching links the next time the engine processes the reconciliation rule. The owned assets are not available for matching.

Follow these steps:

1. Click Directory, List Management.
2. On the left, expand Asset Lists and click Asset Family.
3. Click the Edit Record icon for the asset family that you want to exclude from the reconciliation process.
4. Complete one of the following options:
   - Clear the Reconcile Hardware check box.
   - Select the Is Software check box.
   - Select the Inactive check box.

5. Select the Complete Record Edit icon for the asset family object.
6. Click Save.
   Assets in the excluded asset family are not included in future reconciliations.

Exclude an Asset Family Class or Subclass from the Reconciliation Process

You can exclude all owned assets in an asset family class or subclass from the hardware reconciliation process. For example, you can exclude assets because your company wants to make an asset family class or subclass inactive. Hardware Reconciliation processes active owned assets created by CA APM.

If an asset family class or subclass is excluded from the reconciliation process before the Hardware Reconciliation Engine matches discovered assets to owned assets in that asset family class or subclass, the owned assets are not available for matching. If an asset family class or subclass is excluded from the reconciliation process after owned assets in that asset family class or subclass are matched, the Hardware Reconciliation Engine clears the matching links the next time the engine processes the reconciliation rule. The owned assets are not available for matching.

Follow these steps:

1. Click Directory, List Management.
2. On the left, expand Asset Lists and click Asset Family.
3. Click the Edit Record icon for the asset family with a class or subclass you want to exclude from the reconciliation process.
4. Click Class List.
5. Click the Edit Record icon for the class you want to exclude from the reconciliation process (or the class with a subclass you want to exclude).
6. Select the Inactive check box and click the Complete Record Edit icon for the asset family class if you want to exclude the entire class.

   NOTE
   Skip this step if you want to exclude a subclass, but not the entire class, from the reconciliation process.
7. Click Subclass List if you want to exclude a subclass.
8. Click the Edit Record icon for the subclass you want to exclude from the reconciliation process.
9. Select the Inactive check box and click the Complete Record Edit icon for the asset family subclass.
10. Click Save.
    Assets in the excluded asset family class or subclass are not included in future reconciliations.

View the Reconciliation Results

NOTE

Verify that the user completing this task belongs to a role in which reconciliation management access is enabled.

When the Hardware Reconciliation Engine processes actions for a reconciliation rule, the engine writes records to the message queue in the database. You can search the message queue for reconciliation log messages. The message queue retains log messages for a configurable number of days.

NOTE
You can control the level of detail written to the message queue by changing the Hardware Reconciliation Engine logging level. You can also control the number of days that messages are retained in the message queue. For more information about logging level and message queue retention settings, see Hardware Reconciliation Engine Configuration Settings.

Follow these steps:

1. Click Administration, Reconciliation Management.

2. On the left, click Reconciliation Message Search.
   The message queue displays reconciliation log messages in the Search Results section.
3. (Optional) Search to find a message in the message queue.

   NOTE
   You can export the message queue to a comma-separated values (CSV) file for use in a spreadsheet application. You can also generate reports to view information about your environment relative to reconciliation.

Add Assets from Unreconciled Discovered Records

The hardware reconciliation process can detect assets that cannot be reconciled with any of your owned assets. You can decide to add the unreconciled assets to your repository so that you can track and manage all assets in your network.

You can add the assets to your repository in one of the following ways:

- Configure CA APM to automatically add unreconciled discovered assets to your ownership repository.
- Add the unreconciled assets by generating and exporting the results of a report and then importing the report results through the Data Importer.

  WARNING
  Before you import data into CA APM, review the data to ensure accuracy and uniqueness.

Follow these steps:

- **To add assets from unreconciled discovered assets to your ownership repository automatically, complete the following steps:**
  a. Click Administration, Reconciliation Management.
  b. Click New Reconciliation Rule or search and select a reconciliation rule you created.
  c. The Reconciliation Rule Details page opens.
  d. Check the Add Assets to Ownership Repository check box.
  e. Under Add Assets to Ownership Repository, select the contact update details and the asset status.
  f. Select Add Models to add model details to the repository.

     NOTE

     If you select Add Models, and the unreconciled asset has no model information, CA APM does not add the asset details to the repository.
  g. Click Save.
- **To add assets from unreconciled discovered records assets to your ownership repository manually, complete the following steps:**
  a. Log in to Business Intelligence Launch Pad.
     The Reports pane opens.
  b. Click Document List.
  c. Expand Public Folders, CA Reports.
  d. Click CA ITAM.
  e. Double-click the icon to the left of the report that identifies the discovered assets that are not matched to any owned assets.
  f. Enter the search criteria for the report.

     NOTE
     Select one tenant only when you generate the report. You can import data into only one tenant at a time.
  g. Click Run Query.
  h. Click the link for the flat file format that you can export.
     The report is converted to a document format that you can view and export.
  i. Save the document as a CSV file.

j. Log in to CA APM as the administrator.
k. Navigate to Administration, Data Importer, New Import.
l. Specify the CSV file name in the Data File field.
m. Select the main destination object and the delimiter.

WARNING
Select the same tenant that you selected when you generated the report.

n. In the Advanced Settings area, verify that the following options are selected:
Insert or Update
Create Secondary Lookup Object
Update Secondary Lookup objects
Error on Secondary Lookup Object Errors
o. Click Save.
p. Specify the column mapping.
q. Click Submit in the Schedule area to start the import process.
The unreconciled assets are added to your data repository.

Manage Reconciliation Rules

NOTE

Verify that the user completing this task belongs to a role in which reconciliation management access is enabled.

You can manage reconciliation rules in the following ways:

- Update the information for a reconciliation rule.
If you modify the asset matching criteria associated with a reconciliation rule, the Hardware Reconciliation Engine reprocesses reconciled assets using the new criteria the next time the engine processes the rule.
- Delete a defined reconciliation rule.
If you delete a rule, all asset matching links between discovered and owned assets that are associated with the rule are also deleted. Each tenant can have only one reconciliation rule. If you want to change the reconciliation rule for a tenant, update the current reconciliation rule or delete the current rule and define a new reconciliation rule.

NOTE
You can also make a rule inactive if you want to suspend reconciliation processing temporarily for an individual tenant. To make a rule inactive, update the rule and select the Inactive check box. If the discovered and owned assets were already matched for the rule, the matching links are saved.

Follow these steps:

1. Click Administration, Reconciliation Management.
2. On the left, click Reconciliation Rule Search.
3. Search to find the list of available reconciliation rules.
4. To update a reconciliation rule, complete the following steps
   a. Click the reconciliation rule that you want to update.
   b. Enter the new information for the reconciliation rule.

NOTE
Select the Inactive check box if you want to suspend reconciliation processing for an individual tenant. If you select the Inactive check box, the Hardware Reconciliation Engine does not process the rule (no asset matching or data updates occur for the rule). For example, you can make a rule inactive temporarily while you define normalization rules or troubleshoot asset matching errors. If you make an existing rule inactive and the discovered and owned assets were already matched, the matching links are saved.

c. Click Save.

NOTE
After a reconciliation rule is saved, if you want to change the matched fields in an asset matching criterion to different fields, delete the criterion and create a new one.

5. To delete a reconciliation rule, complete the following steps:
   a. Click the rule that you want to delete.
   b. Click Delete and confirm that you want to delete the reconciliation rule.

Export the Reconciliation Results

NOTE

Verify that the user completing this task belongs to a role in which reconciliation management access is enabled.

After you view the message queue, you can export the queue to a comma-separated value (CSV) file for use in a spreadsheet application.

Follow these steps:

1. Click Administration, Reconciliation Management.
2. On the left, click Reconciliation Message Search.
   The message queue displays reconciliation log messages in the Search Results section.
3. Search to find the reconciliation log messages that you want to export.
4. Click Export to CSV.
   The message queue search results are exported to a CSV file and a link to the CSV file appears.

Searching

This section describes the following articles:

- Object Searching
- Search Results Export
- Search Results Mass Change
- How to Configure Searches

Object Searching

An *object* represents something that you record and track in your repository. The primary objects in CA APM are models, assets, legal documents, contacts, companies, organizations, locations, and sites. At any time, you can search to find objects in the repository to manage. You can also search the audit history to see all changes made to an object record over time.

For example, you can search for objects for the following reasons:

- Search for a model so that you can define an asset from the model.
- Search for an asset so that you can define a support contact for the asset.
- Search for an asset from an inventory request that you want to fulfill.
- Search for a legal contract so that you can define terms and conditions for the contract.
- Search the audit history so that you can see when the cost center for a laptop changed, and who changed the cost center.

Based on the search criteria that you specify, a list of matching objects appear in the search results.

Most of the time, the objects that you want to view or update appear with the default search provided for each object. This type of search is intended for when you want to locate a single record for update from a simple list of objects. For example, you can enter the serial number when searching for a laptop to find the laptop matching the serial number.

When you search, you can use search operators and search connectors to make your searches more precise and get more useful results.

Search Tips

Use the following information and techniques to help make your searches more effective:

- **Keep it simple**. For example, if you are looking for a specific asset or model, and you have the name available, enter the name. If you have additional information to identify the object, such as the serial number for an asset, enter that number. This additional information increases the chance that your search returns the object you want.
- **Expand your search with a wildcard**. Use the asterisk (*) or percent sign (%) wildcard character as a substitute for any number of characters in your search string to return search results. Use as many wildcard characters in your search string as you want. For example, enter *S** or *S%* when searching for contacts by last name to find contacts having the last name Sanders, Shelley, Smith, Spencer, Solomon, and so forth. Enter *Dell *" Monitor* when searching for assets by name to find any size Dell monitor; Dell 19" Monitor, Dell 21" Monitor, Dell 30" Monitor, and so forth.
- **Searches are not case-sensitive**. You can ignore capitalization in your searches. For example, a company search for *Document Management Company* returns the same results for *document management company.*
- **Search titles are unique in a tenant**.
  When you save a search and specify a title, the title must be unique within a tenant. You cannot save a search with the same title in a single tenant.
- **If you receive too many, or too few, search results** and cannot find an object, try the following suggestions:
  - Use different search criteria. For example, instead of searching for a contact by last name, search for their first name, department, user ID, or location.
  - Use operators and connectors to increase or decrease the objects returned in the results.
  - Verify that you have spelled the name of the object correctly, and have entered any additional search criteria correctly.
- **Expand your search with check box criteria**. Include check box fields such as Inactive in your searches. For example, search for all inactive assets and generate a report for management analysis. You can also search for all inactive assets to make them active again.
- **Expand your search within multiple asset families and legal template types**. Search within one or more asset families and legal template types. For example, search for a specific model under both hardware and software, or search for a legal document under a service contract and software addition.
- **Invalid Searches**. You can use the Invalid drop-down list when you manage searches to display or hide invalid searches. A search becomes invalid when your CA APM administrator configures the user interface and restricts access to any field that a search uses, or deletes an extended field that a search uses. If you have a search that you cannot use because of security restrictions, contact your CA APM administrator for assistance.
- Sort the search results to help make it easier to find information.
- Save a search you frequently use so that you do not have to enter the search criteria each time you use the search.
- Copy a search and use the search as a template for creating another search that is similar.
- Export search results for use in spreadsheet applications or reports.
- Configure your searches.

Search Security

Default searches let you find objects in the repository. For example, use the default searches to find assets, models, contacts, and so forth. The security for the default searches makes them available to all users and configurations. You can use these searches to create additional *user-defined searches*.

Consider the following security information for user-defined searches:

- All users that are assigned to a role and configuration can access the default searches and the user-defined searches that are assigned to the role and configuration. However, the search results that you see for the default searches do not display information and fields that your CA APM administrator hides and secures.
- When you configure the user interface and restrict access to any field that a search uses, or delete an extended field that a search uses, the search is invalid. As a result, the search is not available to any user that is associated with the role and configuration.
- Invalid and active searches appear together when you manage searches. When you try to use an invalid search, you receive a message. You should contact your CA APM administrator to troubleshoot the invalid search and make the search valid again.
- When a search becomes invalid, you may have no default search. When attempting to search, you receive a message. You should contact your CA APM administrator to have a default search assigned, or you can save a new search and set the search as your default search.

Example: Limit Asset Searches by Cost Center and General Ledger Code

In this example, an asset in your organization is assigned to a particular cost center and general ledger code. This information is used to identify the department responsible for the expense, and to allocate costs on order releases, shipments, and payment invoices. This information is sensitive and should not be available to all users who search for assets in your repository.

You can copy the default asset search and configure the search by adding the cost center and general ledger code to both the search criteria and search results. When you save the new configured search, you assign the search to only those individuals in your finance and procurement department (that is, users assigned to the finance user role). Users in the finance user role can now search and find assets based on the cost center and general ledger code. Users not assigned to this role cannot search for an asset using that information.

Search for Objects

At any time, you can search to find objects in the repository to manage. For example, you can search for a model and define an asset from the model. Based on the search criteria that you specify, a list of matching models appear in the search results.

To search for objects

1. Click the tab and optional subtab for the object that you want to find.
   In the Search Criteria area, specify the search criteria.
2. (Optional) In the Search Criteria area, click Advanced.
3. Click the Edit icon to specify the search criteria:
   - **Left Parenthesis**
     Determines if left parentheses are used to group search criteria and control the logic of the search. For example, you can select this check box to search for assets in which the asset name is OE001 or both the asset family is Computer and the asset name is Dell.
   - **Operator**
     Determines the standard search operators to use to find objects. For example, you can search for assets in which the asset name is greater than OE001 and the asset family is Computer.
   - **Value**
     Determines the specific field value that you want to find. For example, you can search for an asset in which the asset name is OE001.

     NOTE
     When you enter a search value in combination with the Like or Not Like operators, you can use all supported wildcard characters to expand your search.
   - **Right Parenthesis**

Determines if right parentheses are used to group search criteria and control the logic of the search. For example, you can search for assets in which the asset name is OE001 or both the asset family is Computer and the asset name is Dell.

– **Connector**
Determines the standard connectors to use to find objects. For example, you can search for assets in which the asset name is OE001 or both the asset family is Computer and the asset name is Dell.

4. Click the Complete icon to accept your search criteria changes.
5. Click Go.
A list of matching objects appears in the search results.

Search Operators

When you search for objects, CA APM lets you use the following standard search operators to find objects in the repository:

Operator	Description
Equal	Search for objects having the exact value specified.
Not Equal	Search for objects not matching the value specified.
Greater Than	Search for objects greater than the value specified.
Greater Than or Equal	Search for objects greater than or equal to the value specified.
Less Than	Search for objects less than the value specified.
Less Than or Equal	Search for objects less than or equal to the value specified.
Like	(Similar) Search for objects that match the value specified. When you use the Like search operator, the % wildcard is added after the characters you enter and searches for all objects that start with the characters. You can also add the % wildcard before the characters to search for all objects that end with the specified characters.
Not Like	(Not Similar) Search for objects that do not match the value specified. When you use the Not Like search operator, the % wildcard is inserted after the characters you enter and searches for all objects that do not start with the characters. You can also insert the % wildcard before the characters to search for all objects that do not end with the specified characters.

Search Connectors

When you search for objects, CA APM lets you use the following standard AND/OR connectors to connect search strings and find objects in the repository.

- **AND**
The search must satisfy the criteria for both the current search field and the following search field.
- **OR**
The search must satisfy the criteria for either the current search field or the following search field.

Sort the Search Results

CA APM lets you sort the search results to help make it easier for you to find the information in your search results. For example, you can sort the results in ascending order by asset family to find assets in the hardware asset family first, followed by assets in the software family.

To sort the search results

1. Search for objects.
   The search results appear.
2. In the Search Results area, click the appropriate icon next to a column heading.
   The sort results appear in either ascending or descending order.

   NOTE
   You can extend the default sort of a single column by adding sort fields.

Save a Search

CA APM lets you save a search that you frequently use so that you do not have to enter the search criteria each time you use the search. For example, you can save a search to find assets by asset name, asset family, model, cost center, and creation date.

To save a search

1. Click the tab and optional subtab for the object that you want to find.
2. On the left, click New Search.
   The Add Fields dialog appears.
3. Specify the fields to appear in the search criteria and results.

   NOTE
   When you search within multiple asset families and legal template types, only the assets, models, and legal documents for the selected families and template types appear in the search results.
4. At the top of the page, click CONFIGURE SEARCH: OFF.
   The configuration of the search is complete.
5. In the Search Details area of the page, specify the information to uniquely identify the search.

   NOTE

   When you save a search and specify a title, the title must be unique within a tenant. You cannot save a search with the same title in a single tenant.
6. In the Search Security area of the page, select the user roles for which the search is available. Roles are helpful so you can make the search available to all users having the roles you select. Administrators can also select specific configurations for the search.

   NOTE

   If you do not select either a role or configuration, the search is available to the current user.
7. Click Save.
   The search is saved and is available for future searches.

Update a Search

CA APM lets you change the search criteria in a saved search. For example, you can add the contact ID to a saved asset search.

To update a saved search

1. Click the tab and optional subtab for the object that you want to find.
   On the left, click Manage Searches.
   A list of saved searches displays.
2. Click the search that you want to update.
3. Update the search criteria.

4. (Optional) In the Search Security area of the page, select the user roles for which the search is available. Roles are helpful so you can make the search available to all users having the roles you select. Administrators can also select specific configurations for the search.

   NOTE

   If you do not select either a role or configuration, the search is available to the current user.
5. Click Save.
   The updates to the search are saved and available for future searches.

Search for Objects Using a Saved Search

CA APM lets you search for objects using a saved search. For example, you can use a saved weekly asset search to find all assets that have been added during the past week.

To search for objects using a saved search

1. Click the tab and optional subtab for the object that you want to find.
   On the left, click Manage Searches.
   A list of saved searches displays.
2. Click the search that you want to use.
3. Click Go.
   A list of matching objects appears in the search results.

Copy a Search

CA APM lets you copy a saved search and use the search as a template for creating another search that is similar. For example, you can copy the default asset search and add the asset creation date and user ID of the person who created the asset.

To copy a search

1. Click the tab and optional subtab for the object that you want to find.
   On the left, click Manage Searches.
   A list of saved searches displays.
2. Click the search that you want to copy.
3. Click Copy.
   A new search is created based on the copied search.
4. Change the information for the new, copied search.
5. (Optional) In the Search Security area of the page, select the user roles for which the search is available. Roles are helpful so you can make the search available to all users having the roles you select. Administrators can also select specific configurations for the search.

   NOTE

   If you do not select either a role or configuration, the search is available to the current user.
6. Click Save.
   The search is saved and is available for future searches.

Delete a Search

CA APM lets you delete a saved search that you do not need. You cannot delete the default searches provided by the product.

To delete a saved search

1. Click the tab and optional subtab for the object that you want to find.

On the left, click Manage Searches.
A list of saved searches displays.
2. Click the search that you want to delete.
3. Click Delete and confirm that you want to delete the search.
The search is deleted.

Search Results Export

CA APM lets you export the results of an object, asset fulfillment, and audit history search so that you can report on and analyze the search results. For example, the asset manager, purchasing manager, and facilities manager require a weekly report that lists all of the assets that have been added during the past week. Based on this requirement, the administrator configures the search to return the list of assets and schedules the search and export to process at 10 p.m. every Friday. After the search and export processes, the managers receive an email notification that includes a link to the weekly report.

You can export the results of a search to the following formats:

- CSV file
- Database view

You can schedule a search and export for a particular time and automatically notify contacts by email so that the contact can access the latest information.

CSV File Export

When you export search results to a CSV file, the export includes the current data found by the search. CA APM lets you use the following methods to export to a CSV file:

- On demand export from a real-time search. This type of search includes the current data.
- Scheduled export from a saved search. This type of search includes the current data found each time you use the search according to the schedule.

Example: Export a Search for New Assets to a CSV File

In this example, a company adds new assets to its repository every week. The asset manager, purchasing manager, and facilities manager require a weekly report that lists all of the assets that have been added during the past week. Based on this requirement, the administrator configures the search to return the list of assets and schedules the search and export to process at 10 p.m. every Friday. After the search and export processes, the managers receive an email notification that includes a link to the weekly report.

Database View Export

When you export search results to a database view, the export contains the SQL statement that defines the columns and data in the search. The view does not collect data until you (or an external application) access and use the view in the database.

You can schedule the export from a saved search to export to a database view.

Example: Export a Search for Expiring Assets to a Database View

In this example, a company develops an external dashboard application that monitors asset allocation. The asset manager views the dashboard daily to verify that expiring assets are allocated properly - reallocated, retired, or returned to vendor. The company wants to configure the dashboard application to display CA APM asset data.

Based on these requirements, the administrator configures the search to display the list of expiring assets and their status and exports the search results to a database view. The administrator schedules the search and export to process weekly. The external dashboard application accesses the database view to gather the asset data, and the asset manager views the data in the external dashboard.

How Exporting Search Results Works

When you export search results to a *CSV file*, the process uses the following general steps:

1. The user defines a new search and exports the results; or the user accesses an existing saved search and schedules the export.
2. The search processes and the results include the current data.

> **NOTE**
>
> The results for a scheduled search and export only include the results found by the current search and export. If you change the search or any part of the export criteria, you will not see the new search results until the next time the search and export process runs. For example, if you change the Export Format from CSV File with Column Headers to Database View or you change the Frequency of the Export Schedule, you do not see those changes in the current search and export.

1. The search results data is saved to a CSV file. The column heading labels in the CSV file (if requested with the export) match the column heading labels from the search results. If the user configured the default column heading labels in the search results, the configured labels are included in the CSV file.
2. An email notification is sent to the users assigned to the search. The email includes a link to the CSV file.

When you export search results to a *database view*, the process uses the following general steps:

1. The user accesses an existing saved search and schedules the export.
2. The search processes and the database view is exported. The column heading labels in the database view match the column heading labels from the search results. If the user configured the default column heading labels in the search results, the configured labels are included in the database view.

> **NOTE**
>
> The results for a scheduled search and export only include the results found by the current search and export. If you change the search or any part of the export criteria, you will not see the new search results until the next time the search and export process runs. For example, if you change the Export Format from CSV File with Column Headers to Database View or you change the Frequency of the Export Schedule, you do not see those changes in the current search and export.

3. An email notification is sent to the users assigned to the search. The email specifies the name of the database view.

Export Search Results to a CSV File (On Demand)

CA APM lets you export the results of a search to a CSV file by searching for objects and exporting the results on demand or by scheduling a saved search and exporting the results.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to find.
2. Search for objects by entering search criteria or by selecting a saved search.
   A list of matching objects appears in the search results.
3. Click Export to CSV.
   The search results are exported. The column heading labels in the CSV file match the column heading labels from the search results. An email notification with a link to the CSV file is sent to all contacts associated with the export request.

Export Search Results to a Database View

CA APM lets you export the results of a search to a database view by scheduling a saved search and exporting the results.

Follow these steps:

1. Click the tab and optional subtab for the object for which you want to schedule a search and export.

2. On the left, click Manage Searches.
   A list of saved searches appears. If there are no saved searches, define and save a search that you want to schedule and export.
3. Click the search that you want to schedule and export.
4. On the left, click New Export.
5. Enter the basic information, schedule information, and security information as described in the steps to schedule the search and export.
   The search is processed and the results are exported according to the schedule. The column heading labels in the database view match the column heading labels in the search results. An email notification specifying the database view name is sent to all contacts associated with the export request.

   NOTE

   The results for a scheduled search and export only include the results found by the current search and export. If you change the search or any part of the export criteria, you will not see the new search results until the next time the search and export process runs. For example, if you change the Export Format from CSV File with Column Headers to Database View or you change the Frequency of the Export Schedule, you do not see those changes in the current search and export.

Schedule Searches and Exports

CA APM lets you schedule searches to process periodically and export the search results to a CSV file or a database view. For example, you can schedule a search to process at the end of the week and export all updated assets during that week to a CSV file. You can schedule the following types of searches:

- **Predefined**. Specify a recurring time for the search. For example, you can process the search at 3:00 p.m. on the 21st of every month.
- **Calculated**. Specify the start time and frequency for the search. For example, you can process the search at 10:00 p.m. on the 21st of September and use the search every five days thereafter.

Follow these steps:

1. Click the tab and optional subtab for the object for which you want to schedule a search.
2. On the left, click Manage Searches.
   A list of saved searches appears. If there are no saved searches, define and save a search that you want to schedule and export.
3. Click the search that you want to schedule.
4. On the left, click New Export.
5. Enter the basic export information.
   The following fields require explanation:
   - **Export Name**
     Specify the export name.
   - **Export Format**
     Select the format for the exported search results.
   - **View Name**
     Specify the database view name.

     NOTE
     The View Name is required if you select Database View for the Export Format. The name must be a valid database view name. See your database product documentation for information about database view name requirements.
   - **Description**

Specify a description for the exported search results.

- **Retention Days**
  Specify the number of days that the exported search results are retained before the results are purged.
- **Folder Name**
  Specify the folder for the exported CSV file search results.
- **Never Expires**
  Select this check box to specify that the number of days of the selected period (Period Type) that the CSV file or database view is stored before being deleted never expires (the CSV file or database view is never purged). When you select this check box, any previous value that you added to the Retention Days field is removed, and the Retention Days field is disabled.

6. Schedule the search.
   The following fields require explanation:
   - **Run Time**
     Select the time of the day, in 24-hour format, to process the search. When you schedule searches, use the local time zone on the CA APM application server.
   - **Interval Type**
     Select the type of interval for the search. For example, you can select Day, Month, Quarter, Week, or Year.
   - **Interval Day**
     Specify the day during the Interval Type to process the search. For example, if the Interval Type is Month and the Interval Day is 1, the search is processed on the first day of the month.
   - **First Run Date**
     Select the date when the first search starts to process.
   - **Interval**
     Specify how often the search processes, based on the specified Interval Type. For example, if the Interval Type is Weekly and the Interval is 2, the search processes every two weeks.
   - **Last Day of Interval**
     Select this check box to specify that the search processes on the last day of the selected Interval Type. When you select this check box, any previous value that you added to the Interval Day field is removed, and the Interval Day field is disabled.
7. Specify whether all roles and configurations assigned to the search receive the exported search results.
8. Click Save.
   The search is saved. The search processes at the scheduled time and the search results are exported.

NOTE

The results for a scheduled search and export only include the results found by the current search and export. If you change the search or any part of the export criteria, you will not see the new search results until the next time the search and export process runs. For example, if you change the Export Format from CSV File with Column Headers to Database View or you change the Frequency of the Export Schedule, you do not see those changes in the current search and export.

How Exported Search Results are Retained and Purged

All exported CSV file and database view search results are retained and purged based on configuration settings you specify when you schedule searches and exports. The exported search results that are no longer needed are purged to release disk space.

The following information describes how exported search results are retained and purged:

- You specify the retention days when you schedule searches and exports. After the retention days elapse, the exported search results (CSV files and database views) are considered expired and are purged.
- The Export Service processes the purge once a day (by default, 5:00 a.m. Universal Time) and purges exported search results based on the specified retention days. Your administrator can configure the purge start time.

NOTE
For information about the Export Service configuration settings, see Export Service Configuration Settings.

- The retention period depends on the processing time of the export and the time at which the Export Service purges the search results.

Example: Purge Search Results

In this example, you schedule an asset search named Weekly Asset Search to process every Friday and export all updated assets during the week. You set the retention days to 1 and the export completes at 5:00 p.m. Eastern time. The Export Service is scheduled to purge search results at midnight, Eastern time. Because the search results are less than one day old, the Export Service does not purge them until the next scheduled time (in this example, shortly after midnight the following day).

Search Results Mass Change

This article describes the following:

You can perform the following mass changes on a search results list:

- Change the value of a specific field for all objects or for selected objects in the list.
- Add a value to a specific field for all objects or for selected objects in the list that have blank values in that field.

NOTE
The following restrictions can affect the mass change functions that you can perform:

Configuration restrictions for your user role. For example, if your configuration does not allow you to modify a particular field, you cannot perform mass changes on that field.

- Multi-tenancy. If you have a multi-tenanted environment, the tenant name can be a part of the search results list. However, you cannot perform a mass change on the tenant name.

Example: Perform a mass change on the cost center

A data center administrator wants to assign all assets in the London data center to cost center 3218. The administrator configures the search to return a list of all assets in the London data center. The administrator then performs a mass change to change the cost center assignment for the assets to 3218.

How the Search Results Mass Change Works

When you perform a mass change on a search results list, the process uses the following general steps:

1. The user defines a new search, selects an existing search, or uses the default search.
2. The search processes and the results appear.
3. The user defines the mass change settings for all fields or selected fields and executes the mass change.

NOTE
The user must be assigned to a configuration that permits mass changes.

4. The mass change job is created and displayed in the Mass Change Jobs list.
5. The user verifies the status of the mass change job and performs the original search again when the job is completed.
6. The search results list data is updated to show the new field values.

Perform a Mass Change on a Search Results List

You can change specific field values for all or selected objects in a search results list.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to find.

2. Search for objects by entering search criteria or by selecting a saved search.
   A list of matching objects appears in the search results.
3. Leave all objects selected, or clear the Select All check box and select specific object check boxes.
4. Click Mass Change Settings.
5. Click New.
6. Enter the information to define the mass change. The following fields require explanation:
   - **Mass Change Field**
     Specifies the field in the search results that you want to change.

     NOTE
     The fields that are included in the search results list are available for selection.
   - **Value**
     Specifies the new value for the selected field.

     NOTE
     Leave the Value field blank to specify a null value for a field.
   - **Update Only Blank Values**
     Specifies that only blank values in the selected field are updated with the new value for the selected objects.
7. Click the Complete Record Edit icon.
8. (Optional) Click New to define more mass change settings.
9. Click the Mass Change button above the search results list when you have defined all mass change settings.
   A mass change job is created.
10. Click Mass Change Jobs on the left to see the Mass Change Jobs list.

    NOTE
    You can click Go to update the jobs list.
11. Click Status Message for your mass change job to verify when the processing is completed.
12. Click View Log for your selected job in the Mass Change Jobs list after the job is completed.
    The log file provides more information about the mass change job activity.
13. Perform your original search again.
    The search results display the new field values.

How to Configure Searches

This article contains the following topics:

CA APM lets you configure object, asset fulfillment, and audit history searches to simplify how you search for information in the repository. To configure searches, complete the following steps:

- Set a search result limit.
- Make it easier to search by specifying a default search.
- Make it easier to specify search criteria by completing the following tasks:
  - Adding fields
  - Removing fields
  - Moving fields
  - Changing the field name
  - Replacing fields
- Make it easier to find information in the search results by completing the following tasks:

- Adding columns
- Moving columns
- Changing the column label
- Removing columns
- Adding sort fields
- Preventing duplicate records from appearing
- Preventing the ability to open records

Set a Search Result Limit

When you search for an object and the results are difficult to manage because too many object records appear, you can set a limit. For example, when you search for assets, over 2,000 assets appear in the search results. The results are difficult to navigate, you cannot find the assets you want, and the performance is negatively impacted. Therefore, you set a maximum of 50 object records to return.

Follow these steps:

1. Click the tab and optional subtab for the search that you want to configure.
2. On the left, click Manage Searches.
   A list of saved searches displays.
3. Click a search in the list.
4. In the Additional Settings, Maximum Search Results Returning area, specify the total number of objects to appear.

   NOTE
   For performance reasons, we recommend that you set this value to less than 500.
5. Click Go.
   The limited search results appear and help you see the impact on the results before you save the limit. All future search results are limited to the specified number or percentage.

Specify a Default Search

CA APM lets you specify a search that you frequently use as the default each time you click a tab or subtab. For example, to find the contact search you click Directory, Contact. You copy the contact search, rename it, and then set it as your default. The next time you access the Contact Search page, the default search appears instead of the previous search.

NOTE
You cannot specify a search as the default when the search does not have a selectable column in the results.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to find.
2. On the left, click Manage Searches.
   A list of saved searches appears.
3. Click a search in the list.
4. Click Set As Default.
5. Click Save.
   The search is saved as the default.

Manage Fields

CA APM lets extend or modify your new and saved searches by letting you manage fields in search criteria. Management of fields includes adding, removing, moving, and changing a field in your search criteria. As a result, you can extend or modify the information that appears in your search criteria and results.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to find.
2. On the left, click Manage Searches.
   A list of saved searches appears.
3. Click a search in the list.
4. At the top of the page, click CONFIGURE SEARCH: ON.
5. Complete the following steps to add a field.
   a. Click Add Fields.
   b. Select the fields to add to the search criteria, results, or both.

      NOTE
      You can add fields to a new and saved search. You cannot add fields to the default searches provided by the product.
6. (Optional) To move a field complete the following step:
   a. Drag-and-drop the field to a new location in the search criteria.
      The new location of the field is saved.
7. (Optional) To remove, replace, and change a field, complete the following steps:
   a. In the search criteria area of the page, click Advanced.
   b. Click the Mark for the Deletion icon next to the field you want to remove from the search criteria.
      The field is removed.
   c. Click the Search icon next to the field that you want to replace with a different field.
   d. Select the replacement field and click OK.
      The field is replaced.
   e. Click the Edit Record icon next to the field for which you want to change the label.
      Enter the new field label.
   f. Click the Complete Record Edit icon.
      The field name is changed.
8. At the top of the page, click CONFIGURE SEARCH: OFF.
   The configuration of the search is complete.
9. Click Save.
   The addition, movement, removal, replacement, and name change of a field is completed.

Manage Columns in Search Results

CA APM makes it easier for you to find the information you need in search results by letting you manage columns in search results. Management of columns includes adding, removing, moving, and changing a column in your search result. As a result, you can easily find the information in your search results.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to find.
2. On the left, click Manage Searches.
   A list of saved searches appears.
3. Click a search in the list.
4. At the top of the page, click CONFIGURE SEARCH: ON.
5. Complete the following steps to add a column.
   a. Click Add Fields.
   b. Select the fields to add to the search results.

      NOTE
      You can add columns to a new and saved search. You cannot add columns to the default searches provided by the product.

6. (Optional) To move a column complete the following step:
   a. In the search results list, drag-and-drop the column to a new location.
      The new location of the column is saved.
7. (Optional) To remove a column and change a column label, complete the following steps:
   a. In the search results, click the appropriate icon for the Deletion next to the column.
      The column is removed.
   b. In the search results, select the column heading and enter the new label.
8. At the top of the page, click CONFIGURE SEARCH: OFF.
   The configuration of the search is complete.
9. Click Save.
   The addition, movement, removal, and change of a column is completed.

Add a Sorting Field

CA APM lets you add sorting fields to the search results and extend the default sort of a single column using either ascending or descending order. For example, you currently sort assets by asset name. You can add asset family to the sorting so that you can sort on both asset name and asset family.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to find.
2. On the left, click Manage Searches.
   A list of saved searches displays.
3. Click a search in the list.
4. In the Additional Settings, Search Results Sorting area, add the additional field for sorting.
5. Click Go.
   The results appear with the extended sorting and help you see the impact on the results before you save the sorting.
   The new field is added and you can use the field to sort the search results.

Prevent Duplicate Object Records

CA APM lets you prevent duplicate object records from appearing in the search results. For example, you have several people in your enterprise with the name John Smith. Their first and last names are the same, but their additional contact information (email address, supervisor, department, and so forth) is different.

You have a saved contact search in which only the first and last name of the contact appears in the results. When you search using the saved contact search and specify *John* as the first name and *Smith* as the last name, two instances of John Smith appear in the search results. When you prevent duplicate records from appearing, only one instance of John Smith appears.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to find.
2. On the left, click Manage Searches.
   A list of saved searches displays.
3. Click the search for which you want to prevent duplicate records from appearing.
4. In the Additional Settings, Unique Search Characteristics area, select the Make Results Unique check box.
5. Click Go.
   The results appear without the duplicate records and help you see the impact on the results before you save your settings. The DISTINCT argument is added to the SQL statement, preventing duplicate records from appearing in the search results.

Prevent Opening Records

CA APM lets you disable the ability to open individual records from the search results. For example, you do not want users to open and display contact information from the contact search results.

Follow these steps:

1. Click the tab and optional subtab for the object that you want to find.
2. On the left, click Manage Searches.
   A list of saved searches displays.
3. Click a search in the list.
4. In the Additional Settings, Unique Search Characteristics area, clear the Allow Selection of Results check box.
5. Click Save.
   A hyperlink does not appear in the search results to open the object.

Software Asset Management

This section describes the following:

- Software License Management
- Software Internal Allocations
- Software Assets

Software License Management

CA APM provides visibility into the software environment in your organization by tracking detailed information about software licenses. This information includes internal allocations (locations, companies, contacts, and assets), payment history, purchasing information, and the location of hard copy license agreements or attachments of the relevant license and payment documents required by an audit.

Software license management in CA APM involves working with the following objects:

- Software internal allocations
- Software assets

Software Internal Allocations

This article describes the following topics:

A *software internal allocation* describes how your organization is internally approved to use a software asset, as specified in your software license agreement. For example, a license can stipulate that the software can be used only on a particular computer, or that a limited number of users can use the software at one time.

You track and maintain allocations in CA APM using an allocation record. Define an allocation record for any software asset in your repository.

CA APM lets you record an allocation in the following ways:

- For a software asset, you can add, edit, or delete a relationship to a hardware asset on which the software is internally approved for use.
- For a hardware asset, you can edit or delete information about an allocation that is specific to that hardware asset.

> **NOTE**
> Internal allocations are not legal restrictions. Legal information, including software usage constraints, can be maintained in legal document records and in CA SAM.

Add a Location Allocation

CA APM lets you add a location allocation to list the places where your organization is internally approved to use a software asset, as specified in your software license agreement. For example, you are licensed to use 100 copies of version 4.0 of a software product in your North American office.

NOTE

You can view an audit history for this relationship.

Follow these steps:

1. Click Asset.
2. Search to find the list of available assets.
3. Click the software asset for which you want to add a location allocation.
4. On the left, expand Relationships and click Location Allocation.
5. Click Select New to search for and select a location.
6. Click the Edit Record icon and enter the location allocation details.
7. Click Save.
   The location where your organization is internally approved to use the software asset is added.

Add a Company Allocation

CA APM lets you add a company allocation to list the subsidiary companies where your organization is internally approved to use a software asset, as specified in your software license agreement. For example, you are licensed to use 25 copies of version 4.0 of a software product in a sales office.

NOTE

You can view an audit history for this relationship.

Follow these steps:

1. Click Asset.
2. Search to find the list of available assets.
3. Click the software asset for which you want to add a company allocation.
4. On the left, expand Relationships and click Company Allocation.
5. Click Select New to search for and select a company.
6. Click the Edit Record icon and enter the company allocation details.
7. Click Save.
   The company where your organization is internally approved to use the software asset is added.

Add a Contact Allocation

CA APM lets you add a contact allocation to list the people who are internally approved to use a software asset, as specified in your software license agreement. For example, the members of your IT department are licensed to use ten copies of version 4.0 of a software product in a development office.

NOTE

: You can view an audit history for this relationship.

Follow these steps:

1. Click Asset.
2. Search to find the list of available assets.
3. Click the software asset for which you want to add a contact allocation.
4. On the left, expand Relationships and click Contact Allocation.

5. Click Select New to search for and select a contact.
6. Click the Edit Record icon and enter the contact allocation details.
7. Click Save.
   The contact who is internally approved to use the software asset is added.

Add an Asset Allocation

CA APM lets you add an asset allocation to list the hardware assets on which your organization is internally approved to use a software asset, as specified in your software license agreement. For example, the members of your IT department are licensed to use 10 copies of version 4.0 of a software product on their Dell Precision Workstation 410 computers.

NOTE

: You can view an audit history for this relationship.

Follow these steps:

1. Click Asset.
2. Search to find the list of available assets.
3. Click the software asset for which you want to add an asset allocation.
4. On the left, expand Relationships and click Asset Allocation.
5. Click Select New and select a different hardware asset, other than the asset previously selected.
6. Click the Edit Record icon and enter the asset allocation details.
7. Click Save.
   The hardware asset on which your organization is internally approved to use the software asset is added.

Delete an Allocation

CA APM lets you delete the details of an allocation record. For example, your organization purchases Adobe Acrobat Professional and internally allocates the licenses to 100 users. One user does not need the license. Therefore, you remove the software from the computer and delete the allocation. By completing these steps, you make the license available to another user in a new allocation record.

Follow these steps:

1. Click Asset.
2. Search to find the list of available assets.
3. Click the software asset for which you want to delete the allocation.
4. On the left, expand Relationships and click the appropriate allocation type.
   The allocation list appears.
5. Click the Mark for Deletion icon next to the allocation that you want to delete.
6. Click Save.
   The allocation record is deleted.

Software Assets

CA APM lets you track and manage licensing information for *software assets* that your company is entitled to use. The following fields on the Asset Details page represent the licensing information that you can manage for software assets:

- License Class
- License Count
- License Key
- License Duration
- License Duration Units

You can define and maintain this license information for software assets only.

Vendor Management

Take control of your vendor relationships by understanding their interdependencies between your organization and among one another. CA APM lets you track and manage detailed information about the vendors with whom you do business, including contact information and their relationships to other companies. You can gather complete information about the total amount of money spent and what you will spend with a vendor, allowing you to negotiate product prices and purchases with your vendors.

Vendor management in CA APM involves working with the following objects:

- Directories
- Companies
- Contacts
- Organizations
- Locations
- Sites

Directories

Directory information is maintained in the repository so that you can locate the contact, company, location, and organization information you need for your IT assets. Having a directory offers you consistency for all of your assets to help make analysis easier. In addition, the directory serves as a contact repository when you must contact someone associated with an asset.

Companies

This article contains the following topics:

A *company* buys, sells, services, manages, or uses your IT assets in CA APM. You define company records for key organizCAations with which you have a business relationship, such as the following examples:

- Your own company, its parent company, or subsidiaries.
- IT manufacturers, vendors, escrow agents, maintenance providers, and service providers.

Before you define a company record, you must define records for the parent company, if any, and the default location, such as the headquarters. This additional information makes it easier to enter the information when defining the company record.

You can have multiple associations between locations and companies. These associations are useful to track companies with worldwide offices. For example, to track the contact details of a large vendor with worldwide offices, define location records for each office and associate them with the company record of the vendor.

You can specify one of the locations as the default location. The default location can be the headquarters of the company or the location that you contact most frequently.

> **NOTE**
> Although associating locations is not mandatory, it is considered good practice. Location records must exist in your repository before you can select locations for any objects.

You can retrieve information from the repository about any object by searching. You can then select, view, and manage individual object records from the search results.

Manage Companies

You can define, update, and delete company records for key organizations with which you have a business relationship. For example, you can define a company as an IT manufacturer, vendor, escrow agent, maintenance provider, or service provider.

> **WARNING**
> When you delete an object, you can no longer view the audit history for the object. We recommend that instead of deleting the object, you make the object inactive. Then, you can still view the audit history for the object.

Follow these steps:

1. Click Directory, Company.
2. Perform one of the following actions.
3. Define a company.
   a. Click New Company.
   b. Enter the new company information and click Save.

   > **NOTE**
   > You can also define a company by copying an existing company, supplying a new name, changing the information, and saving the new company.
4. Update a company.
   a. Search for the list of available companies.
   b. Click the company that you want to update.
   c. Enter the new information for the company and click Save.

   Note: You can also view the details for an object that is related to your company, if the related object has a Browse icon. When you click the Browse icon, you leave the company page and you navigate to the related object page. To keep the company page in view and preserve the company information, right-click the Browse icon and select Open Link in New Window. Close the new window when you are finished viewing the related object details.
5. Delete a company.
   a. Search for the list of available companies.
   b. Click the company that you want to delete.
   c. Click Delete and confirm that you want to delete the company.

Associate Locations to a Company

Associating multiple locations to a company is a recommended best practice to track companies with worldwide offices. For example, you can associate your company with the North America office, Latin America office, Asia Pacific office, and European office. Location records must exist before you can associate the location with a company.

Follow these steps:

1. Click Directory, Company.
2. Search for the list of available companies.
3. Click the company that you want to associate with multiple locations.
4. Click Locations on the left.
5. Click Select New to display the list of all available locations.
6. Select the company locations.
7. Click Save.

Add an Acquired Company

CA APM lets you maintain the details of the companies that you acquire and track acquisitions made by external companies. This information is useful when tracking the association between parent and subsidiary companies. For example, to understand the relationship between two companies, you can review the list of acquired companies.

NOTE
You can view an audit history for this relationship.

Follow these steps:

1. Click Directory, Company.
2. Search for the list of available companies.
3. Click the company to which you want to add an acquired company.
4. Expand Relationships on the left and click Company Acquisition.
5. Click Select New and select a different company, other than the company previously selected.
6. Click Save.
   The acquired company is added to the list.

Add a Company Allocation

You can add a company allocation to list the software assets that your organization is internally approved to use, as specified in your software license agreement. For example, you are licensed to use 25 copies of version 4.0 of a software product in a sales office.

NOTE
You can view an audit history for this relationship.

Follow these steps:

1. Click Directory, Company.
2. Search for the list of available companies.
3. Click the company for which you want to add a company allocation.
4. Expand Relationships on the left and click Software Allocation.
5. Click Select New to search for and select an asset.
6. Click the Edit Record icon and enter the company allocation details.
7. Click Save.
   The software asset that your company is internally approved to use is added.

Contacts

This article contains the following topics:

'

A *contact* is a person or department who buys, sells, services, manages, or uses your IT assets in CA APM. Define contact records for key individuals and departments in which you have a business relationship, such as the following examples:

- Users, asset management staff, contract administrators, and IT staff.
- Representatives of other companies, such as manufacturers, vendors, escrow agents, maintenance, and service providers.

Before you define a contact record, define a company record to associate with the contact. Having the company record available makes it easier to add the company information when you are defining the contact record.

You can retrieve information from the repository about any object by searching. You can then select, view, and manage individual object records from the search results.

Manage Contacts

You can define, update, and delete contact records for key people or departments in which you have a business relationship. For example, you can define contacts as asset management staff, contract administrators, IT staff, manufacturers, vendors, and service providers.

WARNING

When you delete an object, you can no longer view the audit history for the object. We recommend that instead of deleting the object, you make the object inactive. Then, you can still view the audit history for the object.

Follow these steps:

1. Click Directory, Contact.
2. Perform one of the following actions.
3. Define a contact.
   a. Click New Contact.
   b. Enter the new contact information.
   c. Click Save.

   NOTE
   You can also define a contact by copying an existing contact, supplying a new name, changing the information, and saving the new contact.
4. Update a contact.
   a. Search for the list of available contacts.
   b. Click the contact that you want to update.
   c. Enter the new information for the contact.
   d. Click Save.

   NOTE
   You can also view the details for an object that is related to your contact, if the related object has a Browse icon. When you click the Browse icon, you leave the contact page and you navigate to the related object page. To keep the contact page in view and preserve the contact information, right-click the Browse icon and select Open Link in New Window. Close the new window when you are finished viewing the related object details.
5. Delete a contact.
   a. Search for the list of available contacts.
   b. Click the contact that you want to delete.
   c. Click Delete and confirm that you want to delete the contact.

Add a Contact Allocation

You can add a contact allocation to list the software assets that people in your organization are internally approved to use, as specified in your software license agreement. For example, the members of your IT department are licensed to use ten copies of version 4.0 of a software product in a development office.

NOTE

You can view an audit history for this relationship.

Follow these steps:

1. Click Directory, Contact.

2. Search for the list of available contacts.
3. Click the contact for which you want to add a contact allocation.
4. Expand Relationships on the left and click Software Allocation.
5. Click Select New to search for and select an asset.
6. Click the Edit Record icon and enter the contact allocation details.
7. Click Save.

Enter Contact Remarks

You can enter notes regarding a contact on the Remarks tab. These notes can be anything related to the contact's role, or any other pertinent information that needs to be saved to the contact detail record.

To enter contact remarks:

1. On the Contact Detail page, select the Remarks tab.
2. Type the remarks in the Contact Notes field.
3. Click Save to save the remarks to the contact detail record.

Organizations

This article contains the following topics:

An *organization* is an internal department. CA APM lets you assign organizations to assets, locations, and contacts. For example, use an organization to identify the department for which an employee works.

> **NOTE**
> Administrators or users with administrative privileges can manage organizations. In addition, if you are using CA Service Desk Manager, you do not have to create the organization. You can use the information from the service desk.

Manage Organizations

You can define, update, and delete an organization for an internal department, division, or external company. For example, you can define an organization for research and development, corporate finance, worldwide law, or global human resources.

> **WARNING**
>
> When you delete an object, you can no longer view the audit history for the object. We recommend that instead of deleting the object, you make the object inactive. Then, you can still view the audit history for the object.

Follow these steps:

1. Click Directory, Organization.
2. Perform one of the following actions.
3. Define an organization.
   a. Click New Organization.
   b. Enter the new organization information.
   c. Click Save.

   > **NOTE**
   > You can also define an organization by copying an existing organization, supplying a new name, changing the information, and saving the new organization.
4. Update an organization.
   a. Search for the list of available organizations.
   b. Click the organization that you want to update.

c. Enter the new information for the organization.
d. Click Save.

NOTE
You can also view the details for an object that is related to your contact, if the related object has a Browse icon. When you click the Browse icon, you leave the contact page and you navigate to the related object page. To keep the contact page in view and preserve the contact information, right-click the Browse icon and select Open Link in New Window. Close the new window when you are finished viewing the related object details.

5. Delete an organization.
   a. Search for the list of available organizations.
   b. Click the organization that you want to delete.
   c. Click Delete and confirm that you want to delete the organization.

Locations

This article contains the following topics:

A *location* is a place where assets, companies, contacts, and legal documents are placed or situated. CA APM lets you associate locations with assets, companies, and contacts. You can define locations for the following objects:

- Assets
- Company offices and other locations where you track IT assets.
- Manufacturing companies, vendors, escrow agents, maintenance, and service providers.
- Contacts within and outside the company.

You can define multiple associations between locations and companies. The associations are useful when you want to track a large vendor with worldwide offices. For example, to track the contact details of a large vendor with worldwide offices, you can define location records for each office and associate them with the company record of the vendor. You can define location records for each office and associate them with the vendor company record.

You can specify one of the locations as the default location. The default location can be the headquarters of the company, or the location that you contact most frequently.

NOTE
Although associating locations is not mandatory, it is considered a best practice. Location records must exist in your repository before you can select locations for any objects.

You can retrieve information from the repository about any object by searching. You can then select, view, and manage individual object records from the search results.

Manage Locations

You can define, update, or delete a location to manage the addresses of assets, contacts, and companies. For example, you can define the address for your North American office, Latin America office, and Asia Pacific office.

NOTE
You can view an audit history for this relationship.

Follow these steps:

1. Click Directory, Location.
2. Perform one of the following actions.
3. Define a location.
   a. Click New Location.
   b. Enter the new location information.

c. Click Save.

NOTE
You can also define a location by copying an existing location, supplying a new name, changing the information, and saving the new location.

4. Update a location.
   a. Search for the list of available locations.
   b. Click the location that you want to update.
   c. Enter the new information for the location.
   d. Click Save.

NOTE
You can also view the details for an object that is related to your location, if the related object has a Browse icon. When you click the Browse icon, you leave the location page and you navigate to the related object page. To keep the location page in view and preserve the location information, right-click the Browse icon and select Open Link in New Window. Close the new window when you are finished viewing the related object details.

5. Delete a location.
   a. Search for the list of available locations.
   b. Click the location that you want to delete.
   c. Click Delete and confirm that you want to delete the location.

Associate Companies to Locations

Associating companies to locations is a best practice to track companies with worldwide offices. For example, you can associate your company with the North America office, Latin America office, Asia Pacific office, and European office.

Follow these steps:

1. Click Directory, Location.
2. Search for the list of available locations.
3. Click the location that you want to associate with companies.
4. Click Companies.
5. Associate the new location with the company.
6. Click Save.

Add a Location Allocation

You can add a location allocation to list the software assets that the locations in your organization are internally approved to use, as specified in your software license agreement. For example, you are licensed to use 100 copies of version 4.0 of a software product in your North American office.

NOTE
You can view an audit history for this relationship.

Follow these steps:

1. Click Directory, Location.
2. Search for the list of available locations.
3. Click the location for which you want to add a location allocation.
4. Expand Relationships on the left and click Software Allocation.
5. Click Select New to search for and select an asset.
6. Click the Edit Record icon and enter the location allocation details.
7. Click Save.

The software asset that the location is internally approved to use is added.

Sites

This article contains the following topics:

A *site* is a group of locations, which lets you use the new site in the location. For example, a site can be a city in which your enterprise has one or more physical locations, or a region in which you have customers that you support.

For more information about defining date-specific and non-date-specific terms and conditions for legal documents, see Define Legal Document Terms and Conditions.

You can retrieve information from the repository about any object by searching. You can then select, view, and manage individual object records from the search results.

Manage Sites

You can define, update, and delete sites. Sites specify groups of locations, such as a city in which your enterprise has one or more locations, or a region in which you have customers.

> **WARNING**
> When you delete an object, you can no longer view the audit history for the object. We recommend that instead of deleting the object, you make the object inactive. Then, you can still view the audit history for the object.

Follow these steps:

1. Click Directory, Site.
2. Perform one of the following actions.
3. Define a site.
   a. Click New Site.
   b. Enter the new site information.
   c. Click Save.

   > **NOTE**
   > You can also define a site by copying an existing site, supplying a new name, changing the information, and saving the new site.
4. Update a site.
   a. Search to find the list of available sites.
   b. Click the site that you want to update.
   c. Enter the new information for the site.
   d. Click Save.

   > **NOTE**
   > You can also view the details for an object that is related to your site, if the related object has a Browse icon. When you click the Browse icon, you leave the site page and you navigate to the related object page. To keep the site page in view and preserve the site information, right-click the Browse icon and select Open Link in New Window. Close the new window when you are finished viewing the related object details.
5. Delete a site.
   a. Search to find the list of available sites.
   b. Click the site that you want to delete.
   c. Click Delete and confirm that you want to delete the site.

Configuration Management

This section contains the following articles:

- CORA
- Configuration Items
- CI Relationships
- Versioning
- Stage CI Transactions Before Loading into CMDB
- Define the Business Infrastructure
- About MDR
- CMDB Management
- Using Configuration Audit
- Configuration Audit and Control Facility (CACF)
- Planning and Implementing Change Verification
- Verify a CI Attribute Value Update Manually
- MDR Management
- Using the Configuration Control

CORA

The Common Object Registration API (CORA) provides an interface to register assets into the CA-MDB, reconciling asset information coming from several data sources. Creating an asset starts with passing the basic registration properties to check for uniqueness Name, Host Name, Serial Number, MAC Address, Alt Asset ID and DNS Name.

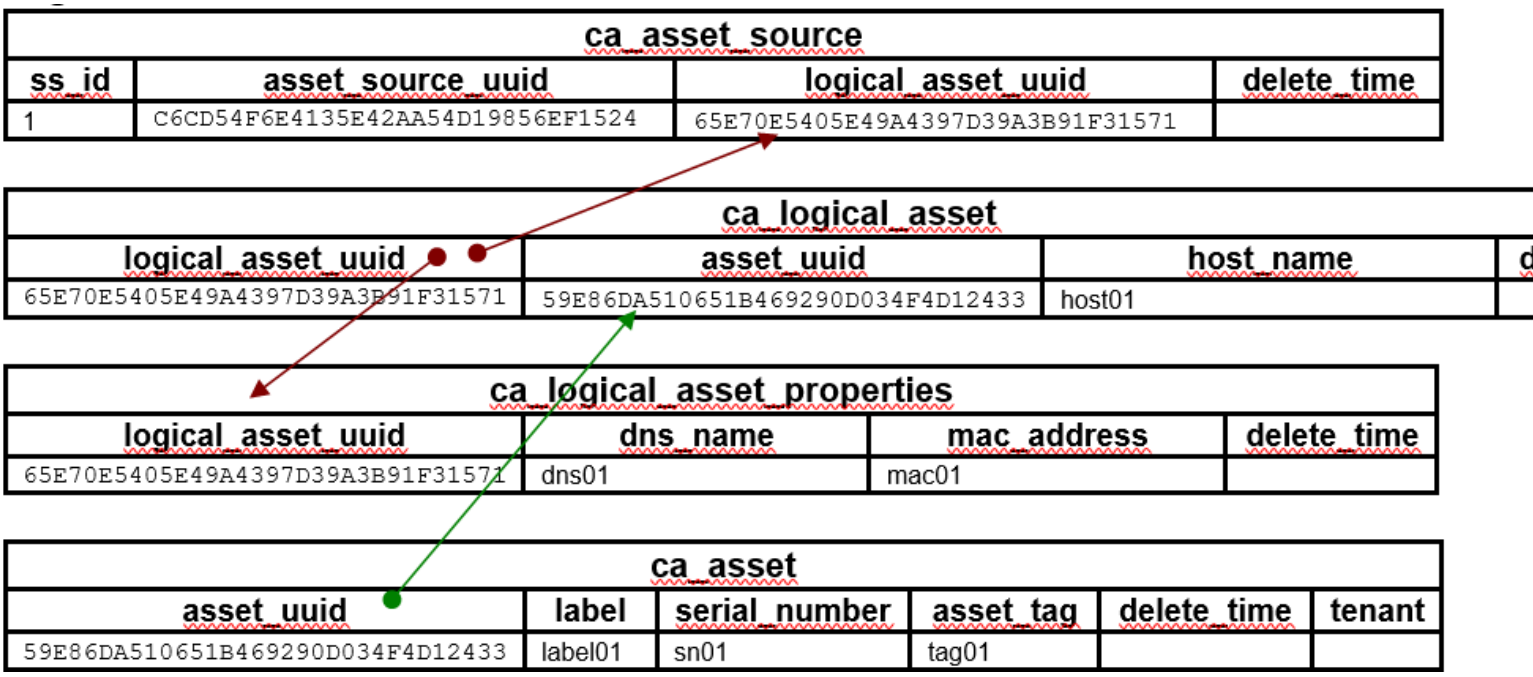
Underlying Data

At a database level CORA saves the asset registration properties to other **CA_MDB** tables: **ca_asset**, **ca_logical_asset**, **ca_logical_asset_property**, and **ca_asset_source.**

Service Desk UI field	ca_owned_resource	ca_asset	ca_logical_asset	ca_logical_asset_property
Name	resource_name	label		
Host Name	host_name		host_name	
Serial Number	serial_number	serial_number		
MAC Address	mac_address			mac_address
Alt Asset ID	resource_tag	asset_tag		
DNS Name	dns_name			dns_name

The c**a_asset_source** table contains the **asset_source_uuid**, which is generated for each registered asset. The **asset_source_uuid** is also stored in the **ca_owned_resource** table by CA Service Desk Manage or CA ITAM

Registration table structure



ca_asset_source

ss_id	asset_source_uuid	logical_asset_uuid	delete_time
1	C6CD54F6E4135E42AA54D19856EF1524	65E70E5405E49A4397D39A3B91F31571	

ca_logical_asset

logical_asset_uuid	asset_uuid	host_name
65E70E5405E49A4397D39A3B91F31571	59E86DA510651B469290D034F4D12433	host01

ca_logical_asset_properties

logical_asset_uuid	dns_name	mac_address	delete_time
65E70E5405E49A4397D39A3B91F31571	dns01	mac01	

ca_asset

asset_uuid	label	serial_number	asset_tag	delete_time	tenant
59E86DA510651B469290D034F4D12433	label01	sn01	tag01		

CORA Registration Queries

When creating or updating an asset, Service Desk/ITAM first uses issues a ca_reg_query call to check if matching asset is already registered. ca_reg_query can be called with any or all 6 of the registration properties: label, host_name, serial_number, asset_tag, mac_address and dns_name.

If the queried values match a registered asset, that asset's asset_source_uuid is returned. With certain exceptions a match occurs when all of the provided properties are identical to those of an existing asset. Not all of an existing asset's registered properties must be included to find a match. ca_reg_query can also find matches when given more properties than actually exist in the asset.

CORA Asset Matching Logic

The following table shows how CORA determines uniqueness of an Asset. The intent here is not to show every single combination but to show enough of the behavior so that one could determine what would occur based on the properties they choose to include when registering an asset.

Service Desk UI field	ca_owned_resource	ca_asset	ca_logical_asset	ca_logical_asset_property	ca_logical_asset	ca_logical_asset_property
Unique	**Unique**	**Unique**	**Unique**	**Unique**	**Unique, Duplicate, or Null**	**New Asset**
Unique	**Null**	**Null**	**Null**	**Null**	**Unique, Duplicate, or Null**	**New Asset**
Null	**Unique**	**Null**	**Null**	**Null**	**Unique, Duplicate, or Null**	**New Asset**
Null	**Null**	**Unique**	**Null**	**Null**	**Unique, Duplicate, or Null**	**New Asset**

Null	**Null**	**Null**	**Unique**	**Null**	**Unique, Duplicate, or Null**	**New Asset**
Null	**Null**	**Null**	**Null**	**Unique**	**Unique, Duplicate, or Null**	**New Asset**
Null	Null	**Null**	**Null**	**Null**	**Unique**	**New Asset**
Null	**Null**	**Null**	**Null**	**Null**	**Duplicate**	**Duplicate**
Null	**Null**	**Null**	**Unique**	**Duplicate**	**Unique, Duplicate, or Null**	**Duplicate**
Null	**Null**	**Null**	**Duplicate**	**Unique**	**Unique, Duplicate, or Null**	**Duplicate**
Null	**Null**	**Null**	**Unique**	**Unique**	**Unique, Duplicate, or Null**	**New Asset**
Unique	**Duplicate**	**Duplicate**	**Duplicate**	**Duplicate**	**Unique, Duplicate, or Null**	**New Asset**
Duplicate	**Unique**	**Duplicate**	**Duplicate**	**Duplicate**	**Unique, Duplicate, or Null**	**New Asset**
Duplicate	**Duplicate**	**Unique**	**Duplicate**	**Duplicate**	**Unique, Duplicate, or Null**	**New Asset**
Duplicate	**Duplicate**	**Unique**	**Duplicate**	**Duplicaet**	**Unique, Duplicate or Null**	**New Asset**
Duplicate	**Duplicate**	**Duplicate**	**Unique**	**Duplicate**	**Unique, Duplicate or Null**	**Duplicate**
Duplicate	**Duplicate**	**Duplicate**	**Duplicate**	**Unique**	**Unique, Duplicate or Null**	**Duplicate**
Duplicate	**Duplicate**	**Duplicate**	**Unique**	**Unique**	**Unique, Duplicate or Null**	**Duplicate**
Duplicate	**Duplicate**	**Duplicate**	**Duplicate**	**Duplicate**	**Unique, Duplicate or Null**	**Duplicate**

Another behavior of CORA which is represented below shows other ways duplicates can occur.

Serial Number	**Asset Tag**	**Host Name**	**DNS Name**	**MAC Address**	**Asset Label**	**Results**
Null	ABC	Null	Null	Null	Unique or Null	New Asset

123	ABC	Null	Null	Null	Unique, Duplicate or Null	Duplicate of ABC
789	XYZ	Null	Null	Null	Unique or Null	New Asset
Null	XYZ	Null	Null	Null	Unique, Duplicate or Null	Duplicate of XYZ

Hierarchy

The database table structure and CORA support an asset property hierarchy. At the top are physical asset properties in the **ca_asset** table: **label**, **serial_number** and **asset_tag**.

A serial_number / asset_tag combination can be used by assets with different host_names, so ca_asset has a one-to-many relationship with ca_logical_asset. In this case, two logical assets have their own ca_asset_source records, though they share a ca_asset record.

		ca_asset_source
ss_id	**asset_source_uuid**	**logical_asset_uuid**
1	C6CD54F6E4135E42AA54D19856EF1524	65E70E5405E49A4397D39A3B91F3
2	8C1DD57B3BAD2C438CF56AAFB538C345	D8DF0ACF5EDB69419211CF85EC9

		ca_logical_asset
ss_id	**logical_asset_uuid**	**asset_uuid**
1	65E70E5405E49A4397D39A3B91F31571	59E86DA510651B469290D034
2	D8DF0ACF5EDB69419211CF85EC943AC4	59E86DA510651B469290D034

		ca_logical_asset_proper
ss_id	**logical_asset_uuid**	**dns_name**
1	65E70E5405E49A4397D39A3B91F31571	dns A
2	D8DF0ACF5EDB69419211CF85EC943AC4	dns B

		ca_asset	
ss_id	**asset_uuid**	**label**	**serial_number**
1	C6CD54F6E4135E42AA54D19856EF1524	label01	sn01

As a host_name can have multiple dns_name / mac_address combinations, the ca_logical_asset has a one-to-many relationship with ca_logical_asset_properties. This relationship is possible with CORA, but Service Desk cannot represent it.

		ca_asset_source
ss_id	**asset_source_uuid**	**logical_asset_uuid**
1	C6CD54F6E4135E42AA54D19856EF1524	65E70E5405E49A4397D39A3B91
2	8C1DD57B3BAD2C438CF56AAFB538C345	D8DF0ACF5EDB69419211CF85E

		ca_logical_asset
ss_id	**logical_asset_uuid**	**asset_uuid**

1	C6CD54F6E4135E42AA54D19856EF1524	65E70E5405E49A4397D39A3B91F3

		ca_logical_asset_proper
ss_id	**logical_asset_uuid**	**dns_name**
1	65E70E5405E49A4397D39A3B91F31571	dns_A
2	65E70E5405E49A4397D39A3B91F31571	dns_B

		ca_asset
ss_id	**asset_uuid**	**label**
1	59E86DA510651B469290D034F4D12433	label01

The additional layer of ca_asset_source is used when one asset is represented in multiple subschemas. For Service Desk this comes into play when an asset is imported from an outside source, like NSM, resulting in multiple ca_asset_source records for the asset. Each of these rows has the same logical_asset_uuid, but different asset_source_uuids and of course the different subschema_ids. Service Desk and ITAM assets are in subschema 1.

		ca_asset_source
ss_id	**asset_source_uuid**	**logical_asset_uuid**
1	C6CD54F6E4135E42AA54D19856EF1524	65E70E5405E49A4397D39A3B91F31
2	D91ED85FD5463C4B9F0881F060A20C6C	65E70E5405E49A4397D39A3B91F31

		ca_logical_asset
logical_asset_uuid	**asset_uuid**	**host_name**
1	65E70E5405E49A4397D39A3B91F31571	59E86DA510651B469290D

	ca_logical_asset_properties	
logical_asset_uuid	**dns_name**	**mac_address**
65E70E5405E49A4397D39A3B91F31571	dns01	mac01

ca_asset			
asset_uuid	**label**	**serial_number**	**delete_tim**
59E86DA510651B469290D034F4D12433	label01	sn01	tag01

Label

The label (or Name) field is handled a bit differently than other fields. First, Service Desk requires each asset to have a label, but they do not have to be unique. All other fields are optional. From the CORA perspective, if **ca_reg_query** is provided with properties in addition to label, the label will be only used to find label-only registered assets. Consider a registered asset with properties:

- label = "**asset1**"
- host_name = "**host1**"
- All other properties are empty.

A **ca_reg_query** that is provided with label = "**newasset**" and host_name = "**host1**" will find the existing asset, despite the label value difference.
Alternatively, consider a registered asset that is label-only:

- label = “**asset2**”
- All other properties are empty.

In this case a **ca_reg_query** that uses label = “**asset2**” and **asset_tag** = “**tag54321**” will find the existing label-only asset. Because of this, multiple assets can have the same label. However, there can only be one asset with a given label that is label-only.

Mac_address and dns_name

The mac_address and dns_name properties are also treated somewhat differently. These fields are considered properties of a host. A host (such as a router) can have multiple dns_name / mac_address pairs. Consider a registered asset that has properties for **host_name**, **dns_name** and **mac_address**. A **ca_req_query** using the **host_name** but a different **mac_address** and/or **dns_name** will still find the existing asset.
However, if an asset only has a label and a **mac_address** and/or **dns_name**, **ca_reg_query** will not find the existing asset if the **mac_address** or **dns_name** does not match.

> **NOTE**
> Service Desk cannot create a single asset that has multiple **dns_name** / **mac_address** pairs.

Using the CORA Cleanup Utility

This article contains the following topics:

Introduction

The CORA Cleanup utility allows the administrator to determine the action required on the unregistered assets.

Set the environment variable as follows:

- CA Asset Portfolio Manager (stand alone)
  Set the JAVA_HOME env. variable pointing to 32-bit Java
- CA Service Desk Manager and CA Asset Portfolio Manager Application Server (installed on the same host)
  Set the JAVA_HOME env. variable pointing to 64-bit Java

In certain circumstances, the data within the CORA cache gets corrupt or inconsistent. This results in the inability to edit the Configuration Items (CI) that exist in the CMDB or create new CIs. In such scenarios, it is recommended you contact the Broadcom Support.

> **WARNING**
>
> It is recommended to use the CORA cleanup utility only under the guidance of Broadcom Support.

How the CORA Cleanup Utility Works

CORA Cleanup utility enforces the CORA rules to rebuild its internal tables: ca_asset_source; ca_logical_asset; ca_logical_asset_property; ca_asset. CORA Cleanup utility does not delete or update any Configuration Item or Asset data, it only uses the data available in ca_owned_resource, ca_discovered_hardware tables and uses it to rebuild the internal cache. The utility creates a log file that can be reviewed to identify the assets that are not registered due to various reasons. In most scenarios, CORA considers the Configuration Item and Asset as a duplicate of another already registered item.

The utility provides the following real-time statistics:

- Number of total objects to be processed
- Number of objects processes successfully
- Number of objects that failed to process
- Time Estimate to complete objects processing

Based on the log file and the real-time progress report, the Administrator can take further actions to manually clean up the data for the product on which it is executed.

TIP

Stop the CA SDM or CA APM services before executing the CORA Cleanup utility. Stopping the services improves the utility performance and prevents any erroneous creation or update of Configuration Items or Assets while the CORA cache is being rebuilt.

Prerequisite

- Set JAVA_HOME environment variable
  Set the environment variable as follows:
  - CA Asset Portfolio Manager (stand alone)
    Set the JAVA_HOME env. variable pointing to 32-bit Java
  - CA Service Desk Manager and CA Asset Portfolio Manager Application Server (installed on the same host)
    Set the JAVA_HOME env. variable pointing to 64-bit Java
- Ensure that the required product version from where you want to execute the utility is installed: CA SDM r17.1 and later and / or CA ITAM r17.1 and later.

Perform Cleanup

Follow these steps:

1. Navigate to the location that contains the CORA Cleanup utility.
   a. *For CA SDM: NX_ROOT\bin*
   b. *For CA APM: <ITAM installation folder>\SC\CORA*
2. Execute the utility:
   – Windows: *coracleanup.cmd*
   – Linux/Solaris: *coracleanup.sh*
3. Define the following parameters:
   – *Database server name*
   – *Database type*: MSSQL or Oracle
   – *Username*: database username
   – *Password*: database password
   – *Scope*: Define the product scope against which the Cora cleanup runs. For example, select *USD* to execute the cleanup on CA Service Desk Manager.
     - *USD*: CA Service Desk Manager
     - *ALL*: Select the option to execute the cleanup utility on all supported products.
   – *Method*: Define the action that the utility takes on the results:
     - *REINIT*: Select this option to refresh the CORA data.
     - *DELETE*: Select this option to delete the data from the following tables: ca_asset_source; ca_logical_asset; ca_logical_asset_property; ca_asset
       The utility does not delete any data from the product(s) defined in the scope.
     - *APITEST: Select this option to test the API threads.*
     - *PRODTEST: Select this option to test the production setup.*
     - *UPDTEST*: Select this option to execute the various database queries to select, update, insert, and so on. To perform this action, CORA utility establishes a JDBC connection with the database.

NOTE

Note: Use the *APITEST*, *PRODTEST*, *UPDTEST* options only upon request from Support, typically for debugging the connectivity issues.

- *REPAIR*: Select this option to restart the CORA cleanup from where the REINIT failed. We recommend that you use the option after a REINIT and if it fails for any reason.

– *MSSQL database name*: If you are using MSSQL database, provide the corresponding database name.
– *Oracle server instance*: If you are using Oracle database, provide the corresponding database name.
– *Database owner*: Define the database owner if it is different from the default database owner.
– *Port number*: Define the port number if it is different from the default JDBC port number.
– Following is the Oracle syntax for CoraCleanup utility

```
CORACLEANUP /SERVER <servername>  /TYPE 3 /USERNAME mdbadmin /PASSWORD <password> /SCOPE <scope> /METHOD
 reinit /INSTANCE <instance>
```

– Following is the SQL syntax for CoraCleanup utility

```
CORACLEANUP /SERVER <servername>  /TYPE 2 /USERNAME <dbuser> /PASSWORD <password> /SCOPE <scope> /METHOD
 reinit /DATABASE mdb
```

4. Navigate to the location that contains the output log file.
   a. *For CA SDM: NX_ROOT/log*
   b. *For CA APM: <ITAM installation folder>\SC\CORA\corautil.log*

Additional Information

For additional information about CORA Cleanup and performance, see Troubleshooting and CORA Cleanup Performance .

Configuration Items

Configuration items are the devices, software, and services that make up your business infrastructure. The information that is associated with a configuration item uniquely identifies the configuration item and indicates its precise location. Configuration items can be associated with contacts (private configuration items) and organizations (shared configuration items). Configuration items let you:

- Identify configuration items by name, class, and family.
- Specify inventory information.
- Specify additional properties to define the configuration item.
- Log and view comments that are associated with the configuration item.
- Specify location information for the configuration item.
- Specify service information, such as a service type, for the configuration item.
- View and define contacts and organizations that are assigned to the configuration item.
- Identify hierarchical and peer-to-peer relationships between configuration items.
- View tickets that are associated with the configuration item.

You are not required to define all this information for configuration items. Defining an optimal amount of configuration item information gives you a clearer picture when you perform impact analysis for the organization.

WARNING
You can attach configuration items only if they are Active. You cannot search for an Inactive CI when working on a ticket.

WARNING
Depending on your role, you do not have access to all the functionality described in this section. For example, some predefined roles can edit records but cannot create new ones.

Depending on your role, you can perform user and administration tasks on configuration items (CIs) by using the following features:

- Scoreboards allow you to perform user tasks and manage CIs and CI relationships.
- The Administration tab lets you perform administration tasks with CIs and relationships.
- CMDB Visualizer lets you perform user and administration tasks for visualizing CIs and relationships.

Create a Configuration Item

You can create a configuration item to be linked to tickets. Configuration items give analysts information about the entities that tickets affect.

Follow these steps:

1. On the Scoreboard, select File, New Configuration Item on the menu bar.
2. Enter a name and class for the configuration item.
   The class is a unique identifier that categorizes the CI class you are adding to the database. You can enter a class directly into the field or click the search icon to search for a defined class.
3. Click Continue and complete the Configuration Item fields as appropriate.
4. Click Save.
   The configuration item definition is saved and the Configuration Item Detail page appears. After you create a CI, the CI Detail page does not display some identifying attributes in the following cases:
   – The attributes have no associated value.
   – The attributes do not apply to the CI family.
   However, identifying attributes with non-blank values always display. Configuration Item Detail pages identify a CI and determine which type of page appears so that you can create or manage the CI. CI Name and the CI Class are required. An asterisk (*) identifies a field that is required. If you use change verification, informational messages appear at the top of the detail form.
   When you assign the new CI to a CI class, the CI family is automatically assigned (because a class is already associated with a family). When you click Continue on the initial Create New Configuration Item page, the appropriate CI page appears, depending on the family.

   NOTE

   A notification appears for successful and unsuccessful changes. A warning message is displayed for an unsuccessful change indicating the policy, attribute, and the corresponding action that occurred.

   The following buttons are available for viewing configuration item information:
   Asset Viewer: Opens the Common Asset Viewer. The viewer displays a detailed view of the configuration item that includes all properties that are managed or discovered.
   Affected Tenants: Displays the tenants that are affected by this CI, based on its associated contacts and organizations.

   NOTE

   This button only appears when multi-tenancy is installed.

   CMDBf Viewer: Displays the Federated View, a side by side view of CI attributes across registered MDRs.

Visualizer: Opens the Visualizer, which displays a graphical representation of the configuration item showing its relationship to other resources.

Configuration Item Fields

The following fields are required to create or update or search for a CI:

- **Family**
  Categorizes the classes of CIs.
- **Standard CI**
  Configuration item that is used for comparison.
- **Host Name**
  Identifies the IP host name of the CI.
- **MAC Address**
  Identifies the Media Access Code for the device. This field is automatically filled when the system creates a record for a device it detects.
- **Alt CI ID**
  Specifies the alternate identifier for a CI; for example, a barcode.
- **DNS Name**
  Identifies the Domain Name System name of the CI.
- **Serial Number**
  Identifies the serial number for the CI. The serial number applies to devices or software.
- **Active?**
  Specifies whether the record is active in the database. The inactive CI records do not appear on display lists.
- **Asset?**
  Categorizes an asset for filtering purposes and to control display in CA CMDB or other products such as CA APM. You cannot change the Asset flag to NO when an asset is managed by CA APM.
- **CI?**
  Categorizes a CI for filtering purposes and to control display in CA CMDB or other products such as CA APM. By default, a CI created by CA CMDB is flagged as a CI but not as an Asset.
- **Superceded By**
  Indicates the CI that has rendered this CI inactive due to its ambiguity.
- **Action**Sets or resets the Actual Date/Time to the current date and time.
- **Target**Specifies the current Service Target.
- **Target Date/Time**Specifies date and time when this Service Target is due. If the ticket is in a Hold status, this value is blank.
- **Actual Date/Time**Specifies the time when the target condition was met. If no value appears, the target condition has not been met.\
- **Time Left**Specifies the amount of remaining time for the service target when the ticket is on hold. If the Service Target has been met, Time Left shows the unused time. A negative value indicates the time that elapsed since the missed target date.
- **Violation Cost**Specifies the incurred cost when the service type time limit is violated.

  NOTE
  The CI Detail page does not display identifying blank attributes and attributes that do not apply to the CI family. However, identifying attributes with non-blank values always display.

Configuration Item Tabs

The following tabs can be available on the Create New Configuration Item, Configuration Item Detail, and the Update Configuration Item pages:

CMDB Attributes

- **Attributes**

Itemizes and manages CI attributes associated with the class that is assigned. The CI Family determines what appears on the Attributes tab.

- **CMDB Relationships**
  Displays the relationships for this CI. All links display the CI Relationship Detail page.
- **Versioning**
  Displays and manages the history of the CI, associated snapshots, milestones, and other views. Versioning also displays pending Change Specifications that are associated with Change Orders. For example, view details about the future state of a CI.
- **Reconciliation**
  Lists all active CIs that are ambiguous with this CI. You can select Exclude Ambiguity to remove this CI from ambiguity calculations and ambiguity management. For more information, see the Reconcile CI Ambiguities Using MDR.
- **Inventory**
  Stores detailed inventory information about the CI.
- **Service**
  Stores detailed service information that is related to servicing a CI. The following sub-tabs are required for a Service:
  a. **Service Type**: The level of support service received for this configuration item. For example, some items may receive problem resolution within four hours, while problems with others may be resolved within 72 hours. Enter the service type directly, or click the search icon to select the desired type.
  b. **Cost Center**: The code to which expenses related to this configuration item are charged. Enter the cost center directly, or click the search icon to select the desired cost center.
  c. **Responsible Organization**: The organization that is responsible for this configuration item.
  d. **Maintenance Organization**: The organization responsible for the maintenance of this configuration item.
  e. **Priority**: The level of attention given to problems with this configuration item.
  f. **Supply Vendor**: The vendor responsible for maintaining supplies for this configuration item.
  g. **Responsible Vendor**: The vendor responsible for maintaining the service provided by this configuration item.
  h. **Maintenance Vendor**: The vendor responsible for the maintenance of this configuration item.
  i. **Service Impact**: The level of importance of this service to the business.

Contacts, Location, Organizations

- **Contacts**
  Stores detailed contact information about the contacts responsible for the CI. The following sub-tabs are required for Contacts:
  a. **Primary Contact**: The primary person or group in charge of the CI.
  b. **Phone Number**: The primary telephone number of the person or group in charge of the CI
  c. **Email Address**: The email address of the person or group in charge of the CI
  d. **Billing**: The address that is listed by room number and building
  e. **Support 1**: The first contact for support in charge of the CI
  f. **Support 2**: The second contact for support in charge of the CI
  g. **Support 3**: The third contact for support in charge of the CI
  h. **Disaster Recovery**: The contact for disaster recovery in charge of the CI
  i. **Backup Services**: The contact for backup services in charge of the CI
  j. **Network Operations**: The contact for network operations in charge of the CI
- **Location**
  Stores detailed information about the geographical location of the CI.
- **Organizations**
  Stores detailed organization information about the organizations responsible for the CI.

Related Tickets

Information about related CA SDM tickets is displayed through Incidents, problems, requests, change orders, change specifications, and verification log.

Additional Information

- **Maintenance Windows**
  Displays global maintenance windows that are associated with the CI.
- **Service Contracts**
  Displays information about service contracts that are associated with the CI, to identify the level of support the item receives.
- **Activities**
  Displays the activity log for the CI. You can display the analyst or activity details.
- **Attachments**

  NOTE

  This feature will only be available if you apply a patch for CA Service Management Release 14.1.01. Find the patch and the download details from CA Support Online.

  Enables you to attach documents or URLs with this CI. Depending on the requirement, click Attach Documents or Attach URL and follow the on-screen instructions. The URL or the document is added to the record and is listed on the Attachments tab of the Configuration Item Detail page. A notification is sent to the related contacts of the CI when the attachment is attached successfully. Similarly when an attachment is removed from the CI, a notification is sent.

  NOTE

  CA SDM is integrated with CA IT Asset Management and this CI is also an asset in CA IT Asset Management. If you have added an attachment to this CI from CA SDM, it will not be visible when this CI is viewed in CA IT Asset Management.

Knowledge Management

- **Knowledge**
  Lists the Knowledge documents that are related to the CI.

Inactivate a Configuration Item

If a configuration item is no longer used, you can edit the configuration item details to make it inactive. An inactive status removes the configuration item from the Scoreboard Configuration Item List. You cannot delete a configuration item.

Edit a CI and select Inactive from the Active drop-down list. The configuration item continues to appear in any existing relationships until the relationships are inactivated.

NOTE

Inactivating a Family or Class does not affect existing CIs in that Family and Class. Inactivating only prevents new CIs from being created in that Family and Class.

Deactivate CIs with Relationships

When inactivating a Configuration Item (CI) in Service Desk Manager, the application (system) validates if there are any dependencies (connected CIs). In case of dependencies, it notifies the user to remove the dependencies before inactivating the CI.

Reactivate a Configuration Item

To use a configuration item that is inactive, you can reactivate it. Edit a CI and select Active from the Active drop-down list.

View CI Attributes in Other CA Products

You can use the Common Asset Viewer page to view configuration item attributes in other CA products.

Follow these steps:

1. Locate and open the configuration item.
2. Click Asset Viewer.
   The Common Asset Viewer page appears.
3. Click the links on the Owned Resources tab to locate attribute information from other CA products.
4. Click Close Window when you are done.
   You have viewed configuration item attributes in other CA products.

Add a Discovered Asset

You can change non-owned assets in the Management Database (MDB) into owned configuration items.

Follow these steps:

1. Click the Discovered Assets button on the Configuration Item List page.
   The Discovered Asset Search page appears.
2. Click Search to display the Discovered Asset List.
3. Select the asset from the list that you want to add as a configuration item.
4. Right-click the asset and select Create New Configuration Item from the pop-up menu that appears.
   The Create New Configuration Item page appears with information about the item populating some of the fields.
5. Complete any remaining fields that apply to the new configuration item and click Continue.
6. Enter data as required in the appropriate fields on the Attributes tab.
   The family of the class that you selected for the configuration item determines the attributes that appear on the tab. Your business processes and the information you want to store and view for a configuration item determines the information that you enter here.
7. Click Save.
   The discovered asset is added.

Asset and CI Flags

To categorize the type of CI/asset for filtering purposes and to control what entities are visible in CA SDM or other products such as CA APM, use the following flags:

- **CI? (YES/NO)**
  Identifies configuration items.
- **Asset? (YES/NO)**
  Identifies the assets.

By default, a CI created by CA CMDB is flagged as a CI and not as an Asset. You can override this behavior if necessary. A CI can also be an Asset. CA CMDB does not allow the Asset flag to be changed once it has been set to Yes. The Asset flag is typically set to Yes when CA APM creates an asset.

These flags appear on all CI detail forms and in the CI search facility. The values can be updated in the CI detail form, GRLoader, and CA CMDB web services. The Edit in List feature also supports updating the new flags in multiple records.

The object attribute name for the CI flag is **is_ci**. The object attribute name for the Asset flag is **is_asset**. These names are SREL attributes that reference bool. The attribute names can be used as GRLoader XML tags. For example, to change the default value of the Asset flag when creating a CI with GRLoader, add the following XML to define the new CI:

```
<is_asset>YES</is_asset>
```

Create a CI Company

The companies supply the hardware, software, and services associated with your system.

Follow these steps:

1. From the Administration tab, navigate to Service Desk, Application Data, Configuration Items, Companies.
2. Click Create New and fill in the field information as appropriate.
   - **Company Type**
     The company type assigned to this vendor. For example, a company type of Lessor can be assigned to suppliers leasing out products. A company type of Provider can be assigned to companies offering services. A company type of Vendor can be assigned to companies selling assets.
   - **Primary Contact**
     The name of the person primarily responsible for issues related to this company. This person must be a defined contact in your system.
   - **Alias**
     An alternate name for this company. For example, if the full company name is John Smith Company, Incorporated, you might enter an alias of Smith Company.
   - **Parent Company**
     If this company is part of or a subsidiary of a larger organization, enter the name of that larger organization. The parent company must be a defined company in your system.
     You can enter the company name directly into this field, or click the search icon to select the company.

   A new company is created for CI use.

Create a Company Type

Company types are supplier classification for the hardware, software, and services that are associated with your system. Company Types classification helps to organize and select the vendor on the basis of services provided. From the Administration tab, navigate to Service Desk, Application Data, Configuration Items, Company Types.

Families and Classes

Families classify configuration items by type and assign meaningful attributes for each one. *Classes* identify general categories of configuration items that your enterprise supports. Families are broad categories of configuration items such as hardware, software, and services. Classes are more specific categories within the broader family category. For example, the family hardware could contain classes such as, modem, router, repeater, and bridge.

Organizing your configuration items into families and classes makes it easier to manage your configuration items. For example, you can generate a list of configuration items that belong to a particular family or class.

Create a Configuration Item Class

Configuration item classes are general categories of hardware, software, and services. For example, workstation, printer services. The configuration item classes are grouped into broader categories known as configuration item families.

When you assign the new CI to a CI class, the CI family is automatically assigned (because a class is already associated with a family). When you click Continue on the initial Create New Configuration Item page, the appropriate CI page appears, depending on the family.

Follow these steps:

1. From the Administration tab, navigate to Service Desk, Application Data, Configuration Items, Configuration Item Classes.
2. Click Create New.
3. Fill in the following fields:
4. • **Family** -- The Configuration Item Family associated with this Configuration Item Class. Families are broader categories for configuration items than classes. The families include computers, other hardware, software, and services. This field is updated automatically and cannot be modified.
   • **Record Status** -- Status indicates whether the configuration item class is active or inactive.

5. Click Save.

Create a Configuration Item Families

Configuration item families are broad categories you can use to classify your network resources. You can assign configuration item classes to families. By doing this, individual configuration items are automatically assigned to families when they are assigned to classes. The configuration item family controls the kind of record displayed.

Follow these steps:

1. From the Administration tab, navigate to the Configuration Item Families List.
   The Configuration Item Family List page displays.
2. Click Create New.
   The Create New Configuration Item Family page displays.
3. Fill in the following fields:
   - **Record Status** -- Indicates whether the configuration item family is active or inactive.
   - **Extension Name** -- Specifies an extension value related to a physical table name in the database. When you select an extension name in the drop-down list, the Physical Table Name field appears in the read-only Physical Table Name field.
4. Click Save.

Create a Service Status

A service status indicates the disposition of a configuration item (for example, In Service, Discontinued, In Repair, and so on).

Follow these steps:

1. Select Service Desk, Application Data, Configuration Items, Service Status on the Administration tab.
2. Click Create New.
3. Fill in the following fields:
   - **Name**
     A unique identifier for the service status.
   - **Record Status**
     Status whether the service is active or inactive.
   - **Description**
     A detailed description of the service status.
4. Click Save.
   The service status definition is saved and the Service Status Detail page appears.

Inventory Information

You can store detailed inventory information on each configuration item.

To define inventory information for the configuration item

1. On the Scoreboard, browse to Configuration Items, Active or Inactive. Select the Priority, Warranty, and Ownership for the configuration item you want to edit.
   The Configuration Item List appears.
2. Select the Configuration Item to edit.
   The Configuration Item Detail page appears.
3. Click Edit.
   The Update Configuration Item page appears.
4. Fill in the inventory fields as appropriate.

5. Click Save.
   The configuration item definition is saved and the Configuration Item Detail page appears.

The following buttons are available for viewing configuration item details:

Asset Viewer

Opens the Common Asset Viewer, a detailed view of the configuration item that includes all properties that are managed or discovered.

Affected Tenants

Displays the tenants affected by this CI, based on its associated contacts and organizations. Note: This button only appears when multi-tenancy is installed.

CMDBf Viewer

Displays the Federated View, a side by side view of CI attributes across registered MDRs.

Visualizer

Opens the Visualizer, which displays a graphical representation of the configuration item showing its relationship to other resources.

Inventory Fields

The following fields require explanation:

IP Address

The identifier for the configuration item on the network.

Model

The manufacturer's model identifier for the configuration item.

Manufacturer

The manufacturer of the item. This field is automatically populated, based on the value of the Model field, and cannot be changed.

License Number

The license number of the service provider (if applicable).

Service Status

The service status of the item. Available values are based on the Configuration Item Service Status List as defined by the administrator.

Acquire Date

The date and time the item was acquired, in the format mm/dd/yyyy hh:mm am | pm. Enter the date and time directly or click the calendar icon to select the date and time.

Installation Date

The date and time the item was installed, in the format mm/dd/yyyy hh:mm am | pm. Enter the date and time directly or click the calendar icon to select the date and time.

Expiration Date

The date and time the item expires, in the format mm/dd/yyyy hh:mm am | pm. Enter the date and time directly or click the calendar icon to select the date and time.

Warranty Start/End Date

The date and time the item's warranty begins and ends, in the format mm/dd/yyyy hh:mm am | pm. Enter the date and time directly or click the calendar icon to select the date and time.

Product Version

The version number of the item, typically used for software and/or hardware.

Financial Reference

Indicates the association of the configuration item with your financial software.
Quantity

The number of the item currently on hand in inventory.

Configuration Item Events and Logs

View Configuration Item Event History

You can view a record of significant actions that occur regarding a CI. The event list contains a record of the events that are associated with the CI. You maintain events, or procedures the system follows after a certain amount of time has elapsed. For example, an event can send a notification to an analyst if a CI has been updated.

You can view historical events and the status of a configuration item.

Follow these steps:

1. On the Administration tab, select Service Desk, Application Data, Configuration Items, Configuration Item List.
   The Configuration Item Search page appears.
2. Search for a Configuration Item and select it.
3. Click the Event History button.
   The Event History page displays the events and their status. The following fields require explanation:
   - **Condition**
     Indicates the condition that is checked for by the event. CA SDM provides macros that verify standard conditions such as priorities and object status (open or closed).
   - **Check Time**
     Indicates the time that the event was checked, based on the time parameters that the event configuration specifies.
   - **Time Loaded**
     Indicates the time that the event was loaded, based on the time parameters that the event configuration specifies.

Add Configuration Item Log

You can add log entries to record activities for the configuration item.

Follow these steps:

1. On the Scoreboard, browse to Configuration Items, Active, or Inactive. Select the Priority, Warranty, and Ownership for the configuration item you want to edit.
   The Configuration Item List appears.
2. Select the Configuration Item.
   The Configuration Item Detail page appears.
3. Click Edit.
   The Update Configuration Item page appears.
4. Select the Log tab.
   A list of log entries for this configuration item appears.
5. Click Add Log.
   The Create New Configuration Item Log page appears.
6. Enter the information that you want to record in the log and click Save.
   The Configuration Item Detail page displays the new log entry that is listed on the Log tab.

View Log History

You can view the log entries to see a history of activity for the configuration item.

Follow these steps:

1. On the Scoreboard, browse to Configuration Items, Active, or Inactive. Select the Priority, Warranty, and Ownership for the configuration item you want to edit.
   The Configuration Item List appears.
2. Select the Configuration Item.
   The Configuration Item Detail page appears.
3. Click Edit.
   The Update Configuration Item page displays.
4. Select the Log tab.
   A list of log entries for this configuration item displays.

CI Naming Conventions and Restrictions

CIs have the following naming conventions and restrictions:

- CI Name
  The CI Name is the common or display name that is used in all CI lists. The total length of the name must not exceed 255 characters. The CI name does not need to be unique, but we recommend that it is globally unique. In addition, when MDR determines the name, we recommend that MDR administrators emphasize the human readability factor when populating this field.
- Software CIs
  For third-party software, follow the same naming convention as the names that you manually create using the Administration tab. Matching naming conventions lets Cohesion ACM-discovered CIs reconcile to the manually created ones. If you do not follow this convention, you can create multiple CIs with only one software instance.
  Use of systemname Attribute
  The systemname attribute uniquely associates a *single* CI with a particular host name.
  If multiple CIs are imported and specify the same systemname as an existing CI, reconciliation results only one CI.

Example

The following output lines show the creation of a Server CI (provider), a Software CI (dependent), and a *runs* relationship between them:

```
CI:  Name: Server1     Class: Server   Systemname: Server1
CI:  Name: Apache1   ON Server1      Class: Software
Relationship:  Server1 runs Apache1   ON Server1
```

The resulting relationship represents a server that runs Apache software.

Add a Discovered Asset

CA SDM does not operate on the entire set of resources that are stored in any database. Only configuration items (CIs) registered by CA Service Desk Manager are immediately available. From a business process perspective, a help desk is only concerned about supporting *owned* resources that are part of the organization through a purchasing process. For example, if a visiting consultant from an outside company plugs his laptop computer into the network, should this device automatically be entered into the help desk's inventory of supported configuration items? Probably not. The collection of configuration items that are used by CA Service Desk Manager are represented in the owned resource table: (ca_owned_resource).

You may need to track issues with configuration items that are registered in the database, but do not yet exist as owned resources. The Discovered Asset Selector allows the users to browse non-owned configuration item resource records in the database and transform them into owned resource records that can be used by CA Service Desk Manager.

The information from the nonowned asset is merged with the CA Service Desk Manager information into the same master record. If you search for Discovered Assets again, you can see that the original record has been updated to include the CA Service Desk Manager information you entered.

Follow these steps:

1. From the Configuration Item List page, click the Discovered Assets button.
   The Discovered Asset Search page appears.
2. To display the Discovered Asset List, click Search.
3. Select the asset from the list that you want to add as a configuration item. Right-click the asset and select Create New Configuration Item.
   The Create New Configuration Item page appears with information about the item populating some of the fields.
4. Complete any remaining fields that apply to the new configuration item and click Continue.

   NOTE
   Only name and class are required values for creating a configuration item.
5. Enter data as required in the appropriate fields on the Attributes tab.
   The family of the class that you selected for the configuration item determines the attributes that appear on the tab. The information you enter here is determined by your business processes and the information you want to store and view for a configuration item.
6. Click Save.
   The discovered asset is added.

Configuration Item Service Information

Service Information

You can store detailed information related to servicing a configuration item.

To define service information for the configuration item:

1. On the Service Desk tab, browse to Configuration Items, Active or Inactive. Select the Priority, Warranty, and Ownership for the configuration item you want to edit.
   The Configuration Item List appears.
2. Select the Configuration Item to edit.
   The Configuration Item Detail page displays.
3. Click Edit.
   The Update Configuration Item page displays.
4. Select the Service tab and fill in the fields as appropriate.
5. Click Save.
   The configuration item definition is saved and the Configuration Item Detail page appears.

The following buttons are available for viewing configuration item details:

Asset Viewer
Opens the Common Asset Viewer, a detailed view of the configuration item that includes all properties that are managed or discovered.

Affected Tenants
Displays the tenants affected by this CI, based on its associated contacts and organizations.

NOTE

This button only appears when multi-tenancy is installed.

CMDBf Viewer

Displays the Federated View, a side by side view of CI attributes across registered MDRs.

Visualizer
Opens the Visualizer, which displays a graphical representation of the configuration item showing its relationship to other resources.

Service Fields

- **Service Type**
  The level of support service received for this configuration item. For example, some items may receive problem resolution within four hours, while problems with others may be resolved within 72 hours. Enter the service type directly, or click the search icon to select the desired type.
- **Cost Center**
  The code to which expenses related to this configuration item are charged. Enter the cost center directly, or click the search icon to select the desired cost center.
- **Responsible Organization**The organization that is responsible for this configuration item.
- **Maintenance Organization**The organization responsible for the maintenance of this configuration item.
- **Priority**
  The level of attention given to problems with this configuration item.
- **Supply Vendor**
  The vendor responsible for maintaining supplies for this configuration item.
- **Responsible Vendor**
  The vendor responsible for maintaining the service provided by this configuration item.
- **Maintenance Vendor**The vendor responsible for the maintenance of this configuration item.
- **Service Impact**The level of importance of this service to the business.

CI Relationships

CA SDM provides a list of predefined relationship types that help you describe a relationship, or an association between configuration items. How the relationship is expressed depends on which configuration item is the focus. For example, a provider supports a dependent configuration item, but the dependent is supported by the provider. These are different expressions of the same relationship. The relationship type label that you see depends on whether you are viewing a provider-to-dependent or a dependent-to-provider relationship.

CIs in the CMDB can be related to other CIs. For example, a computer as a configuration item can result in additional configuration items, such as a printer and a monitor. The CI Relationship List displays sets of related CIs. For information about the relationship types that CMDB provides, see CMDB Technical Reference.

The CI relationships have the following characteristics:

- The *relationship type* defines how two configuration items are related.
- Hierarchical relationship types are paired as *provider/dependent*. For example, A (provider) *hosts* B; B (dependent) *is hosted by* A.
- Nonhierarchical relationship types are defined as *peer-to-peer*. For example, *is connected to*, which is the same in either direction.

You can do the following relationship functions by using the CMDB Relationships tab:

- View Active or Inactive relationships by searching on the Active? field.
- From the CI Relationship List, click any link for a relationship to launch its CI Relationship Detail page.

Create a Relationship Type

You can create a relationship type, and can make it available to create or update relationships.

Follow these steps:

1. On the Administration tab, expand the CMDB folder from the left pane, click CI Relationship Types.
   The CI Relationship Type List appears in the right pane.
2. Click Create New.
   The Create New CI Relationship Type page appears.
3. Complete the following fields:
   - **Provider to Dependent Label**
     Describes the relationship from the provider to the dependent. This is the name that appears in the list of relationship types. For example, "services" or "manages" is a Provider to Dependent relationship.
   - **Dependent to Provider Label**
     Describes the relationship from the dependent to the provider. This is the name that appears in the list of relationship types. For example, "is serviced by" or "is managed by" is a Dependent to Provider relationship.
   - **Peer-to-Peer?**
     Specifies that the relationship does not have a provider CI and dependent CI but instead consists of two equal CIs. For example, "connects to" and "fails over" are Peer-to-Peer relationships.
   - **Active?**
     Makes the relationship type available for selection in configuration item relationships and lists.
4. Click Save.
   The relationship type is created and you can use it to create or update relationships.

Create a CI Relationship

You can add a CI relationship by starting from a focal configuration item. To launch a CI Relationship Detail page from the list, click any link in a relationship row. The Create New button allows you to create a new relationship from the CI detail.

The Impact Explorer button lets you view related providers and dependents using a search. The default search retrieves immediate providers one level deep. The related CIs are listed with the Relationship type with these additional fields:

- Family
- Contact

Follow these steps:

1. Select Search, Configuration Items on the menu bar of the Scoreboard.
   The Configuration Item Search page appears.
2. (Optional) Complete the filter fields to limit the search to the configuration items of interest and click Search.
   The Configuration Item List page displays the items that match your search criteria.
3. Select the Configuration Item to edit.
   The Configuration Item Detail page appears.
4. Select the CMDB Relationships tab.
   The tab lists any existing relationships for this CI.
5. Click Create New.
   The Create New Relationship page appears.
6. Enter the required data for this relationship:
   - **Tenant (if multi-tenancy is installed)**
     The tenant level for this relationship.
   - **Provider/Peer CI**
     The configuration item that provides the relationship. Enter the configuration item name directly into this field. To select the configuration item from a list, click the search icon.
   - **Relationship Type**
     Use the Relationship Type Search and List to select relationship type.
   - **Dependent/Peer CI**

The dependent configuration item in this relationship. Enter the configuration item name directly into this field. To select the configuration item from a list, click the search icon. This field is required.

- **Symbol**
  (Optional) The unique identifier for this relationship. Assign a symbol that makes the relationship easily recognizable from the list.
- **Description**
  (Optional) A detailed description of the relationship. You can also use this field to describe the repositories that are assigned in the relationship.
- **Source Repository**
  (Optional) A detailed description of the relationship. You can also use this field to describe the repositories that are assigned in the relationship.
- **Cost**
  (Optional) The dollar value of the relationship between the configuration items.

7. Click Save.
   The CI relationship is saved and the CMDB Relationships tab displays the new relationship.

Manage a CI Relationship

The Configuration Item Relationship page provides all necessary settings in one location to simplify the CI relationship process.

You can use the page to do the following functions:

- Toggle the provider/dependent relationship between two CIs.
  Click Reverse to switch the Provider CI and Dependent CI for the relationship. The focal CI changes. If the reversed relationship is not valid, clicking Reverse clears the Relationship field.
- Set the Optional fields such as Symbol, Description, and Cost.
  Click Optional to set optional fields.
  If a Symbol is not supplied, a unique Symbol is automatically generated with prefix **cmdb bmhier:** and a unique number when the relationship is created.

Inactivate a CI Relationship

There are two ways to inactive a relationship. Consider a process depending on your organization needs.

Process 1: By setting the CI relationship status to Inactive.

Follow these steps:

1. Navigate to the CI Detail page, click the CMDB Relationships tab, and click Edit.
   The CMDB Relationships tab displays in Edit mode.
2. Choose a relationship and click any link on its row.
   The CI Relationship Detail page displays.
3. Click Edit.
4. Set the Active? attribute to Inactive. Click Save.
   The relationship is made Inactive. On the CI Relationship Detail page, the “Active?” attribute displays the Inactive attribute value.

Process 2: Using GRLoader to submit XML

Follow these steps:

1. Write XML that uses a Relation tag to identify the following parameters:
   - Provider CI
   - Dependent CI
   - Relationship type

2. Set the delete_flag to true (or yes or 1).
3. Submit the XML.
   The CI Relationship is deleted.

For more information about GRLoader, see the General Resource Loader topic.

Example: Delete a CI Relationship Using GRLoader

In the following XML example, the "connects to" relationship between ci_1 and ci_2 is deleted.

```
<GRLoader>
    <relation>
        <dependent>
            <name>ci_2</name>
        </dependent>
    <type>connects to</type>
        <provider>
            <name>ci_1</name>
        </provider>
    <delete_flag>true</delete_flag>
    </relation>
</GRLoader>
```

Reactivate a CI Relationship

There are two ways to reactivate a CI relationship. Consider a process depending on your organization needs.

Process 1: Reactivate a deleted CI relationship

Follow these steps:

1. Navigate to the CI Detail page.
   The CI Detail page displays in Read-Only mode.
2. Click Edit.
   The CI Detail page displays in Edit mode.
3. Click the CMDB Relationships tab.
4. To open a CI Relationship Detail page, click any link on a row.
5. Click Edit.
   The Update CI Relationship page appears.
6. Change the Active? attribute to “Active”. Click Save.
   The relationship is reactivated.

Process 2: Reactivate using GRLoader to submit XML

Follow these steps:

1. Write XML that uses a relation tag to identify the following parameters:
   - Relationship type
   - Dependent CI
   - Provider CI
2. Set the delete_flag to false (or no or 0).
3. Submit the XML.
   The CI Relationship is reactivated.

For more information about GRLoader, see the General Resource Loader topic.

Example: Reactivate a CI Relationship

In the following XML example, the "connects to" relationship between ci_1 and ci_2 is reactivated.

```
<GRLoader>
        <relation>
                <dependent>
                        <name>ci_2</name>
                </dependent>
        <type>connects to</type>
                <provider>
                        <name>ci_1</name>
                </provider>
        <delete_flag>false</delete_flag>
        </relation>
</GRLoader>
```

CI Relationship History and Comparison

The Versioning tab displays CI relationships and lets you do the following actions:

- View the relationship history for any CI in the CMDB. Changes that are made to a relationship are logged automatically when the relationship is created, updated, or deleted. You can view all current and historical relationships for a CI.

  NOTE
  Relationships that are created with previous versions of CMDB do not have audit history information.
- Compare the relationships at any snapshot or milestone.

Configuration Item Information

Contents

Define Contact Information

You can define detailed information about the contacts that are responsible for a configuration item.

To define contact information for a CI, **follow these steps:**

1. On the Scoreboard, select Configuration Items, Active or Inactive.
2. Select the Priority, Warranty, and Ownership for the configuration item you want to edit.
   The Configuration Item List appears.
3. Select the Configuration Item that you want to edit.
   The Configuration Item Detail page appears.
4. Click Edit.
   The Update Configuration Item page appears.
5. Select the Contacts tab.
6. Complete the tab fields as appropriate.
7. (Optional) Click Update Contacts.
   The Contact Search page displays.
8. Complete one or more of the search fields, and click Search.
   The Contact List page displays the contacts that match your search criteria.
9. From the list on the left, select the contacts that you want to assign to this configuration item. To select multiple contacts, hold down the CTRL key while clicking the left mouse button.
10. When you have selected all the contacts you want, click the move button (>>).
    The selected contacts move to the Assigned to CI list on the right.
11. Click OK.

The Configuration Item Detail page displays the selected contacts listed on the Contacts tab.

Location Information

You can store detailed information about the location of a configuration item.

To define location information for a configuration item:

1. On the Scoreboard, browse to Configuration Items, Active or Inactive. Select the Priority, Warranty, and Ownership for the configuration item you want to edit.
   The Configuration Item List appears.
2. Select the Configuration Item to edit.
   The Configuration Item Detail page displays.
3. Click Edit.
   The Update Configuration Item page displays.
4. Select the Location tab, and fill in the fields as appropriate for the configuration item.

   NOTE

   The Address fields are filled in automatically when you select a location from the drop down list.
5. Click Save.
   The location information is stored in the configuration item record.

Define Organization Information

You can define detailed information about the organizations that are responsible for a configuration item.

To define organization information for a CI

1. On the Scoreboard, select Configuration Items, Active or Inactive. Select the Priority, Warranty, and Ownership for the configuration item you want to edit.
   The Configuration Item List appears.
2. Select the Configuration Item to edit.
   The Configuration Item Detail page appears.
3. Click Edit.
   The Update Configuration Item page appears.
4. Select the Organizations tab and click Update Organizations.
   The Organization Search page displays.
5. Complete one or more of the search fields, and click Search.
   The Organization List page displays the organizations that match your search criteria.
6. From the list on the left, click an organization that you want to assign to this configuration item. To highlight multiple organizations, hold down the CTRL key while selecting.
7. When you have selected all the organizations that you want, click Move (>>).
   The selected organizations move to the Assigned to CI list on the right.
8. Click OK.
   The Configuration Item Detail page displays the selected organizations listed on the Organizations tab.

View Related Tickets

Tickets, such as incidents, problems, requests, change orders, and issues can be associated with configuration items.

To see the tickets associated with a configuration item, select the appropriate tab on the Configuration Item Detail page.

NOTE

Configuration items are assigned to tickets when the ticket is created or edited. They cannot be assigned from this tab on the CI Detail page.

View Service Contracts

Service contracts can be associated with configuration items to identify the level of support the item receives.

To see the service contracts associated with a configuration item, select the Service Contracts tab on the Configuration Item Detail page.

NOTE

Configuration items are assigned to Service Contracts when the contract is created or edited. They cannot be assigned from this tab in the Detail page.

View CI Activities

You can view all activities for a configuration item (CI), such as when a CI is inserted or inactivated. All activities for a CI are recorded on the Activities tab on the CI Detail page.

Versioning

This article contains the following topics:

Versioning promotes control over the IT infrastructure. Use versions for tracking and managing the life cycles of the CIs that constitute the authorized state of the CMDB. Versioning also applies to auditing the history of an object. Versioning provides the following functions to control the IT infrastructure:

- **Snapshots**
  Recorded automatically for changes to the value of any update to an object, such as when you update a CI. Versioning creates a snapshot with the complete state of the hardware CI after the memory size of a computer is modified. Versioning can display the snapshot and can indicate which CI attributes have changed. Versioning can also compare it against all other historical snapshots of that CI. You require this critical information to understand the impact of changes on the availability and performance of a CI.
- **Milestones**
  Recorded as special-purpose snapshots with a user-defined label, such as First Day of Production or January Baseline. These labels can help find specific snapshots more quickly.

  NOTE

  Milestones do *not* apply to change specifications, verification policies, managed change states, and managed attributes.
- **Comparisons**
  Compare a CI to another CI that acts as a standard. You can use any CI as a standard for comparison, but we recommend that you dedicate specific CIs as *Standard CIs*. A Standard CI lets you define a standard configuration to which other operational CIs in the same family can be compared. For example, define Server Large as a Standard CI to define the attribute values of this type of server. The Standard CI for a family can be made an attribute of an operational CI in that family. This relationship enables you to compare the current state or any historical state of a CI to its associated standard configuration. Like other comparisons, you can print this information or can export it as a comma-delimited file. Milestones can act as unshared, unchangeable baselines, but Standard CIs provide shared, dynamic baselines for your CIs.

WARNING
A Standard CI must never contain values for any attribute that is used for CI reconciliation.

- **Change Specifications**
  View pending scheduled or unscheduled change specifications that a user specified for Change Orders for the CI. You can compare the change specification with the current state of the CI. For example, compare a planned value to the current state of a CI. This comparison can show conflicts with changes, identify overlapping changes in the schedule, and so on.

Versioning helps you complete the following tasks:

- Automatic capture of all CI modifications
- Snapshots of a CI at any time that are based on date or attribute changes
- Various levels of display including detailed Log, Basic, and Advanced views
- Multiple attribute comparisons with other snapshots, user-defined milestones for CIs, or a Standard CI
- Automatic snapshots whenever CA SDM change tickets change states

NOTE

The automatic snapshot does not apply to change specifications, verification policies, managed change states, and managed attributes.

- Advanced filtering and comparison with Print and CSV Export support

Uses of Versioning

Versioning includes the following uses:

- - **Problem Determination**
    To correct a problem with a CI, identify what changed in the environment of the CI. Versioning shows its current defective state, and its state at any point in the past. To identify potential problems, compare the two states.
  - **Performance and Capacity Planning**
    By reviewing the history of a CI, a performance or capacity planner can determine the causes of performance issues. The history also helps in planning for future system growth.
  - **Compliance**You can compare the state of a CI to its family Standard CI. To verify CI compliance and to identify attributes that need corrections, compare the state at any point in its change management lifecycle.
  - **Change Verification**You can view the audit history of the object. For example, see who modified a verification policy, the date of the request, and the attributes and values that were modified. You can compare and contrast the changes between specific dates.

CI Versioning and Future State

View change Specifications in the CI Detail form Versioning tab. Provide more information, identify Change Order and corresponding change specifications. This tab also shows the future state of the CI as if the changes were applied. View snapshots of the CI, as of the scheduled date of the Change Order. If the Change Order has not been scheduled yet, it shows as an unscheduled change.

Versioning enables the following functionality:

- Quickly identify overlapping or conflicting change specification.
- Launch directly to change specification and view the corresponding change details by clicking Change Order in the detail view.
- Snapshots showing an *Unscheduled Change* indicated a change that is scheduled for some unspecified time in the future.
- The Change Order number is displayed with each change specification shown on the right side of snapshot label.
- Informational text displays in the bottom pane providing the following information about the change specification as hover text:

- Change Order number.
- Date of the scheduled change, if the Change Order specifies a scheduled date.
- Date need by, only shown if Change Order specifies a need by date.

Shared Asset and CI Audit Trail Records

The Versioning feature includes audit trail changes made to CA APM. Versioning also supports launch in context from CMDB to CA APM directly. The launch can be from an attribute log entry that is associated with each CA APM change. The CA APM audit trail is only available for CI/Assets from the CMDB families with CA APM audit logging enabled.

The Versioning tab that includes the CA APM audit trail information supports all families for which logging is enabled.

By default, CA APM does not log Asset attribute changes. To use CMDB Versioning with a CA APM-managed Asset/CI, enable Store Audit Trail Data on each asset attribute that requires logging. For more information about how to enable auditing, see the CA APM documentation.

When asset attributes are modified in CA APM, the Versioning tab data on the CI Detail page can be updated before the Attributes tab data due to caching activity. To resynchronize the values, click Edit.

Sources of Versioning Data

Versioning data generates whenever you create or modify an object by any component that supports the Audit Log facility. Snapshots that are generated from multiple sources are indistinguishable from one another. In a properly configure environment, snapshots appear automatically whenever CIs and their relationships are changed.

All CMDB families are enabled for versioning. CIs in CA Service Desk base families must be converted to CMDB families to take advantage of versioning.

Versioning data includes the following sources:

- **CMDB CI updates** -- Updates to common CI attributes and family-specific CI attributes by using the user interface.

  NOTE
  For more information, see CMDB Technical Reference .
- **CMDB relationship updates** -- Updates to relationships on the Relationships tab and the CI Relationship List by using the Edit and Edit In List feature.
- **GRLoader** -- CI insert and attribute updates, relationship updates, Standard CI assignments, and Milestones imported by using GRLoader. When the updates from an MDR are sent using GRLoader, the MDR source is also recorded. Recording the source helps you track attribute changes directly to their source.
- **CA SDM Change History** -- A snapshot is automatically created for all CIs associated with a change ticket. Up to four snapshots are taken when the CI is opened, closed, active, or resolved.
- **CA Asset Portfolio Management (CA APM) CI changes** -- Changes that CA APM makes to CIs that have logging enabled.
- **Change Verification** -- Updates to change specifications, verification policies, managed change states, and managed attributes.

  NOTE
  CIs created in CA SDM or previous CMDB releases only contain modification changes. The initial attribute values such as name, family, and class were not logged and do not appear in snapshots. In addition, Relationships created with CA Service Desk or previous versions of CMDB do not have audit history information.

CA SDM Change Management Integration

Versioning data is generated for each CI that is associated with a CA SDM change ticket. As the change ticket moves from *open* to *active* to *resolved* to *closed*, a snapshot is taken for each of the associated CIs. Thus, snapshots are created for opened, closed, active, or resolved statuses. If no CIs are associated with a change request, no snapshots are taken.

NOTE
The versioning feature requires no special configuration and is supported automatically in integrated installations.

The versioning feature handles the change ticket as follows:

1. Compares the state of the CI at the time the ticket was opened with its current state.
2. Verifies that all the required changes have been properly executed and that no additional changes were introduced.
3. Closes the ticket after the changes have been validated.

For auditing purposes, you can readily compare the state of the CI before and after each change was made. The comparison information helps answer questions including the state of a CI before and after a change request, for example:

- Did the appropriate change occur as requested?
- What other changes occurred to the CI not related to the change ticket?
- What time did the changes occur?
- How does this CI compare against the standard configuration, both before and after the change?

CA APM Integration

Versioning displays changes that CA APM makes to its CIs. Versioning also supports launch in context from CMDB to CA APM directly. The launch can be from any attribute log entry that is associated with a CA APM change. CA APM audit information is only available for CIs from CMDB families when CA APM auditing logging is enabled.

NOTE
When the CI attributes are modified in CA APM, the Versioning tab on the CI Detail page can be updated before the Attributes tab. This is due to caching activity. To resynchronize the values, click Edit.

By default, CA APM does not log CI attribute changes. To use Versioning with a CA APM-managed CI, enable Store Audit Trail Data on each asset attribute that requires logging.

By default, a 10-minute delay exists from the time a CI is updated in CA APM and the time it becomes available to versioning in the CMDB. You can modify this delay by changing the @NX_DBMONITOR_TIMER_MINUTES variable in an integrated installation.

NOTE
For information about how to enable attribute logging for assets, see the CA APM documentation.

CA SDM provides the following integration capabilities with CA APM:

- Shared records and audit log that differentiate between CA APM CIs and assets
- CMDB and CA APM launch in context to CI/asset information
- Shared extension table fields
- A CA SDM contact updates when CA APM changes a primary contact
- CA APM asset type modification to use CI families

CI Versioning Management

You can display and manage the history of the CI, associated snapshots, milestones, unscheduled changes, and other views. You use the left pane to navigate and the right pane to display CI details and the audit log.

NOTE

Versioning works similarly for managed attribute history and managed change state history. For example, the Managed Attribute History tab displays versioning information about a managed attribute.

To identify change specifications and the associated Change Orders, use versioning information. Versioning identifies unscheduled changes in the change specifications in the snapshot view. These changes correspond to Change Orders with no scheduled start date. You can view the snapshot and log of a change specification in CACF.

Example: Unscheduled Change

In this example, a user tries to change the value of a managed attribute. You view the Versioning tab on the CI detail page that displays Unscheduled Change in the snapshot. This snapshot displays information about the change, such as the date, time, username, and attribute value.

Select a snapshot or milestone in the left pane. The right pane displays the attribute values of that CI state, event, or comparison. The left pane displays CI snapshots or milestones by date and time (Basic mode) or by CI characteristics (Advanced mode). The left pane also includes the following links to help you manage a CI:

- **Create Milestone**
  Labels the state of a CI.
- **Show Log**
  Displays unfiltered CI history.
- **Advanced/Basic**
  Toggles between snapshot views.
- **Hide Empty Values**
  Permits filtering of blank data fields. When not selected, all CI attributes appear.
- **Print and Export**
  Print or cut-and-paste to save the versioning data on display.

Configuration Item Log

The log lets you view configuration item history. The Filter option permits filtering of log rows. Print and Export allow you to print or cut-and-paste to save the versioning data on display.

The Versioning information pane displays the following fields:

- Category
- Date
- Log
- Attribute
- New Value
- Old Value
- Changed By
- MDR

> **NOTE**
> The log only displays attribute *updates* for CIs that were created in previous releases of CA SDM or CA CMDB. The log does not display their initial CI attribute values. Relationships that were created in previous releases of CA SDM or CA CMDB do not include audit history information and do not appear in the log.

Log Filtering

The log can be filtered by entering a simple string in the Filter field. The Filter is case insensitive. If the filter string appears anyplace in a log row, the row is displayed. No special characters or wildcards are used in the filter. Log rows matching the filter criteria are displayed hiding rows that do not match.

You do not need to press Enter or refresh to update the display. The log view is filtered as each keystroke is entered and applies to all fields, Date, Log, Attribute, Old value, New Value, MDR, and Name.

CSV Export Support

You can export the CA CMDB log content to a Comma Separated Value (CSV) format file. Third-party reporting tools or applications can import the CSV file. The export is based on what is currently displayed, which can be filtered or unfiltered.

Integrated CA SDM and CA APM Logs

In addition to changes that were made using CA CMDB facilities, the CI log also includes CA SDM change ticket update activities and CA APM updates.

MDR Launch-in-Context and Source Identification

CA CMDB provides a way to launch in context, directly from the log to the MDR data provider for a particular CI log entry. You can also launch in context to the change specification. This launching feature does the following tasks:

- Tracks the attribute changes back to the source MDR, when the MDR used the <mdr_name> <mdr_class> and <federated_asset_id> tags in the GRLoader input.
- Identifies when multiple MDRs update a CI attribute. This situation occurs when multiple MDRs contribute data independently to a CI definition.
- Identifies which MDR is acting as the authoritative source.

NOTE
Log entries that are created in CA SDM or previous CA CMDB releases do not contain MDR launch in context information. In addition, CA Configuration Automation provides MDR launch information for most but not all attributes or relationships.

Attribute Names

Attribute names that are displayed in the log match the attribute name that is shown in the user interface. The attribute name is not same as the internal object name. For example, CA CMDB displays “Maintenance Type” instead of “mtce_type”.

Logging Custom Families and Attributes

To enable logging for modified families and attributes, specify new MDR attributes and audit triggers. If you create new families or attributes, you also must create metadata files to specify human readable attribute names to display in the log; otherwise, the internal object name appears.

For more information, see the Extending CA CMDB topic.

Create a Milestone

You can create a milestone from the user interface or by using GRLoader.

Follow these steps:

1. To create a milestone from the user interface, complete the following steps:
   a. Select a CI and click its Versioning tab.
   b. Click Create Milestone.
   c. Enter descriptive text for the milestone. Click Save.
      The Create Milestone page closes.
   d. Select View, Refresh.
      The Basic view displays a new snapshot with the current date and time that represents the milestone that you created. Switching to the Milestone view, displays only those snapshots with assigned names. Milestones are displayed in descending date/time sequence, with the most recent at the top.
2. To create a milestone using GRLoader, complete the following steps:
   a. Add the <milestone> tag to the XML for that CI.
   b. Apply the same reconciliation rules apply as when associating a milestone with a CI.

NOTE
The value of the milestone tag is the label that is associated with that milestone.

c. Save and close the XML for the CI.
The milestone is created.

Example: Use GRLoader to Create a Milestone

The following GRLoader sample creates the milestone **Fiscal year end 2008**.

```
<ci>
<name>server1 </name>
<milestone>Fiscal year end 2008</milestone>
</ci>
```

Create a Standard CI

You create a Standard CI like you create any other CI. Any CI can act as a Standard CI, but the following cautions apply:

- A Standard CI must follow a naming convention so that it is not confused with regular CIs.
- A Standard CI must only act as a baseline for CIs in the same family as the Standard CI.
- A Standard CI must *not* be assigned values for any attributes that are used in reconciliation.
  For example, MAC Address, Serial Number, Alt CI ID, DNS, and so on.
- Because standard CIs are indistinguishable from regular CIs, they appear in your Scoreboard and count in the total number of CIs.
- As a best practice, Standard CI must not represent any actual instance of a business object.

Follow these steps:

1. Select Create New Configuration Item from the File menu.
2. Select the family that you want to use for the new Standard CI.
3. Define the attributes, as with any regular CI.
   The Standard CI is created.

Assign a Standard CI to a CI

You can assign a Standard configuration item to a configuration item by using the CI Detail page. You can assign a Standard CI to a list of CIs or using Edit in List.

Follow these steps:

1. To assign a Standard CI to a CI using the CI Detail page, complete the following steps:
   a. Select a CI and click Edit.
   The Update Configuration Item page appears.
   b. Click the icon next to the Standard CI field.
   The CI List displays.
   c. Select a CI to serve as the Standard CI. Click Save.
   The Standard CI is assigned.
2. To assign a Standard CI to a list of CIs, using edit in list, complete the following steps:
   a. From the Scoreboard or Administration tab, specify the search filter and click Search.
   b. Click the Edit In List button at the top right of the list results.
   The CI Edit in List controls is displayed.
   c. Select a row containing CI to assign the Standard CI too.
   The row is shown in highlighted color.
   d. Click the icon next to the Standard CI field.
   The CI List displays.
   e. Select a CI to serve as the Standard CI. Click Save to update the single CI or Change All to update all CIs in the list.

The Standard CI is assigned to the single CI (Save) or all CIs listed (Change All)

Note: Any CI can act as a Standard CI, but the following cautions apply:

- – A Standard CI must follow a naming convention so that it is not confused with regular CIs.
  – A Standard CI must only act as a baseline for CIs in the same family as the Standard CI.
  – A Standard CI must *not* be assigned values for any attributes that are used in reconciliation. For example, MAC Address, Serial Number, Alt CI ID, DNS, and so on.
  – Because standard CIs are indistinguishable from regular CIs, they appear in your Scoreboard and count in the total number of CIs.
  – As a best practice, Standard CI must not represent any actual instance of a business object.

Assign a Standard CI to a CI Using GRLoader

To assign a Standard CI to a CI, include the <standard_ci> tag in the description of a CI.

Example: Assign a Standard CI to a CI

If you have a standard workstation configuration that is stored in a CI named **standard workstation configuration.** You can assign the Standard CI to Joe's Workstation by using GRLoader to load the following XML:

```
<ci>
        <name>Joe's Workstation</name>
        <class>Server</class>
        <standard_ci>standard workstation configuration</standard_ci>
</ci>
```

Use the Basic View

You can select snapshots in either Basic or Advanced view. The same kinds of versioning data can be displayed in either mode. The Basic view lets you easily view the state of a CI.

Follow these steps:

1. Open the CI in the user interface and click the Versioning tab.
   The existing snapshots are listed on the left side of the page.
2. Select a snapshot type from the Snapshot type drop-down list:

   NOTE
   If you select a Standard CI for the focal CI, it displays at the top of the snapshot or milestone list. If you click a Standard CI, the attributes for the Standard CI appear and not the focal CI.

   – **Snapshot**
   Lists all CI snapshots identified based on date/time and the Standard CI. The Snapshot view is the default option.
   – **Milestone**
   Lists all user-defined Milestones and the Standard CI.

   NOTE
   You can select two or more snapshots, milestones, or Standard CIs to compare them.

Use the Advanced View

You can select snapshots in either Basic or Advanced view. The same versioning data can be displayed in either mode. The Advanced view lets you view snapshots that are based on the attribute type, value, and time stamp. The Advanced view also lets you compare any with attributes, milestones, or a Standard CI with each other. A tree hierarchy shows a folder for each attribute of the CI, and each attribute folder contains a history of the values of the attribute. The hierarchy is organized as follows:

```
Root
     Attribute name
          Attribute value1
          Attribute value2
```

This hierarchy lets you view history of unique values for any particular attribute at a glance.

Follow these steps:

1. Open the CI in the user interface and click the Versioning tab.
   The existing snapshots are listed on the left side of the page.
2. Click Advanced.
   Advanced Selection shows a folder hierarchy.
3. Navigate the folder hierarchy, and click the folder that includes the information you want to view:
   - **Date**
     Lists Snapshots that are based on date/time, which is identical to the Basic view. If there are 30 or more snapshots for the CI, the snapshots can be divided further based on year/month. If a Standard CI is assigned to the focal CI, it is also listed in this folder.
   - **Milestone**
     This folder lists all user-defined Milestones. This view is identical to the Basic view. If a standard CI is assigned to the focal CI, it also appears in this folder.
     Standard CI attributes are displayed as special tree leaf nodes with a "Standard CI". Standard CI values only appear for attributes for which there is a standard CI value. If the value for an attribute is not set, it does not appear.
   - **Relationship**
     Contains all the relationships (past and present) for the CI. The folder hierarchy conveys the following information:

     ```
     Relationship
          Relationship Type
               Partner CI
                    Status and Date
     ```

     Relationship Type specifies the kind of relationship, such as “is contained by”, “hosts” or “communicates with”.
     Partner CI is name of the CI associated with the relationship.
     Status and Date specify the Status of the relationship at the specified Date and Time. Status values include the following ones:
     - Relationship Created -- The state of the CI when the Relationship was created.
     - Relationship Terminated -- This CI is no longer involved in the relationship. The relationship still exists at the partner end of the relationship, but the focal CI is not involved.
     - Relationship Deleted -- The relationship was marked as deleted.
     - Relationship Changed -- The relationship was reactivated from deleted state.
     - New Partner and Type -- The partner end of the relationship was assigned to a new CI, and the relationship type was changed simultaneously.
     - New Relationship Type -- The relationship type between two CIs has changed.
     - Partner CI Assigned -- The partner end of the relationship has changed.
4. Click an attribute value.
   The complete state of the CI at the time that the attribute value was set displays.

Example: Use Advanced View to Show Disk Space Attributes

In the following example, Advanced Selection shows that disk space increased from 10 to 20 to 100 GB.

```
Root
     Disk space
          10 GB
          20 GB
          100 GB
```

Click any value to see the following information:

- When the change occurred
- The state of the other attributes when the change occurred
- The change request number that was open when the disk space was changed

View Snapshot Details

You can view snapshot details in the Basic or Advanced view. For example, the basic view displays a snapshot of an unscheduled change to a CI.

Follow these steps:

1. Open the object in the user interface and click the Versioning tab.
   A list of existing snapshots appears on the left side of the page.
2. Click a snapshot.
   Details are displayed in the right pane of the Basic and Advanced views. When you select only a single item, the right pane shows information about the selected snapshot, milestone (only for CIs), or standard.
   The displayed data includes the following details and indicators:
   - **Hide Empty Values**
     Permits filtering of blank data fields. When unchecked, all CI attributes are displayed.
   - **Bolded Value field text**
     Indicates that an attribute or relationship has changed since the last snapshot was taken. When viewing details for a Standard CI, all values are bold.
     Value
     Shows the previous value of the attribute and the time of the last change.
   - **"(blank)" Value**
     Indicates whether a previous value was cleared.
   - **Relationship Category**
     Shows information about the relationship, including the type and partner CI information.
   - **MDR Launch in Context and Source Identification**
     Provides launch-in-context to a provider MDR from the CI detail entry.

     NOTE
     CIs created in CA SDM or previous versions of CA CMDB can lack MDR and Changed-By information. In addition, CA Cohesion ACM provides MDR launch-in-context for most but not all attributes or relationships.

     Tracks the attribute changes back to the source MDR.
     Detects when multiple MDRs update a CI attribute. This situation occurs when multiple MDRs contribute data independently to a CI definition.
     Identifies which MDR acts as the authoritative source.

Launch the MDR That Set an Attribute

If the history for a CI references an MDR, you can launch an MDR to view CI details.

Follow these steps:

1. Select a CI and navigate to its CI Detail page.
2. Click the Versioning tab.
3. Click Show Log, or select the CI snapshot or milestone you want to view.
4. Select the attribute row that you want to investigate.
5. If the CI history references an MDR, click the MDR link.
   The provider MDR shows more CI details.

Export Data

The Versioning Export to CSV page lets you export the snapshot and log information in a Comma Separated Value (CSV) format. Data displays in an Export page. The formatted data corresponds to the view you obtained. The filtering and data comparison you select on the Versioning tab is what is formatted for export.

Follow these steps:

1. Select an object and click the Versioning tab.
   Click the view that you want of the CI.
   For example, click Show Log, a snapshot comparison, milestone (CIs only).
2. Click Export.
   The Export Log to CSV for CI page appears.
3. Use the Select All action (from either context menu or keyboard shortcut).
4. Use cut and paste to transfer the data from the formatted page to a CSV file or third-party application. For example, you can Export to an Excel spreadsheet.
   The data is exported.

CI Family Changes and Snapshots

Family changes can affect a snapshot of a CI.

A snapshot includes the following different kinds of attributes:

- Common attributes
- Family-specific attributes

When you change the family of a CI, the following snapshot changes occur:

- The Family-specific attributes that are associated with the CI change.
- The Snapshot details reflect the family at the time of the snapshot.

The family of a CI determines the attributes of the CI. If you first change a *family-specific* attribute and then the family of the CI, the resulting CI no longer possesses the *family-specific* attribute you changed. Changing the family of a CI has no impact on its *common* attributes.

Example: Change the Family Associated with a CI

This example shows how CI family changes affect a snapshot of a CI.

Time	Status/Changes
0	The CI is in Family1.
1	One Family1-specific attribute value is changed.
2	Several common attribute values are changed.
3	The family of the CI is changed from Family1 to Family2. When this occurs, all Family1-specific attributes become unavailable to this CI.
4	One Family2-specific attribute value is changed.
5	Several common attributes are changed.
6	The CI family is changed back to Family1. When this occurs, all Family2-specific attribute values become unavailable to this CI, but previous Family1-specific attribute values are restored.
7	Several common attributes are changed.
8	One Family1-specific attribute is changed.

In the example, a snapshot at Time=7 does not contain information from the changes that are made at Times 3 or 4. Those changes represent changes that are made to Family2-specific attributes.

Compare Snapshots (Basic View)

You can compare two or more snapshots, milestones, or Standard CIs.

Follow these steps:

1. Select a CI and navigate to its CI Detail page.
2. Click the Versioning tab.
3. Select a Milestone or Snapshot from the Snapshot type drop-down list.
   Milestones or snapshots are listed.
4. Select two or more snapshots, milestones, or Standard CIs.
   The comparison view displays the following results:
   - A column is displays how the snapshots or milestones differ.
   - Each row displays the attribute values for the selected snapshots.
   - Only the attributes with differences are displayed. Date is always displayed unless snapshots are identical.
   - Standard CI comparisons are not time-specific, so their Date fields are set to say **Standard CI**.
   - Comparison results can be printed or exported to a CSV file.

Compare Snapshots (Advanced View)

You can compare two or more snapshots, milestones, or Standard CIs.

Follow these steps:

1. Select a CI and navigate to its CI Detail page.
2. Click the Versioning tab, and click Advanced.
   Advanced Selection displays the CI.
3. Navigate the folder hierarchy to the value you want to use for the snapshot comparison.
4. Select a value for each snapshot comparison.
   The comparison for the CI displays; for example, snapshot comparison that is based on the folder values selected.
   The comparison view displays the following results:
   - A column is displays how the snapshots or milestones differ.
   - Each row displays the attribute values for the selected snapshots.
   - Only the attributes with differences are displayed. Date is always displayed unless snapshots are identical.
   - Standard CI comparisons are not time-specific, so their Date fields are set to say **Standard CI**.
   - The comparison results can be printed or exported to a CSV file.

     NOTE
     In Advanced view, two CIs can have a relationship with no relationship type assigned (for example, CIs created by CA NSM). Such relationship nodes are identified as (blank).

CA SDM Change Management

Versioning data is automatically generated for each CI that is associated with a CA Service Desk change ticket. As the change ticket moves from *open* to *active,* and from *resolved* to *closed*, a snapshot is taken for each of the associated CIs. Versioning lets you do the following tasks:

- Compare the state of the CI to its Standard CI at any point in a change management lifecycle.
- Help ensure that the CI is in compliance and help identify those attributes which need remediation.
- Perform comparisons between states of the CI at any point in lifecycle of the change order, and any date or time in-between.
- View progress at each stage of the change, including any milestones.

The integration between CA Service Desk and CA CMDB is illustrated by demonstrating how you can audit changes that are performed during the service management lifecycle of a CI. In this example, a change request is used to upgrade components of a hardware server.

Follow these steps:

1. To create a change order and associate it with a CI, complete the following steps:
   a. Create a CA Service Desk change ticket by clicking File, New Change Order.
   b. Enter the change request information and order summary, for example: Upgrade hard drive to 500 GB.
   c. Click the Configuration Items tab, and then click Update CIs.
   d. Specify CI search criteria (for example, the name of the hardware server) and click Search.
      In the Affected Configuration Items Update form select the CIs associated with the change ticket (for example, the hardware server from Step 4). Add them to the Affected Configuration Items list and click OK.
   e. Save the change request.
2. To perform the changes, complete the following steps:
   a. Make the changes for the CI. For example, install the hard drive on the physical computer, and then update the Disk Capacity attribute for the hardware server CI.
   b. Close the change request.
   c. Ensure that the change was completed: View the CI, click the Versioning tab, and compare snapshots of the CI before and after the change.
      Suppose that you compare the state of the CI between the times when the change order was opened and closed. You notice that the hardware server drive size was 100 GB before the change and 500 GB after the change was completed. You also see the date and times that the changes occurred. Any other attributes that were modified between the open and close times also display.

CMDB Versioning Terminology

This article contains the following topics:

Versioning is the practice of representing and tracking *service transitions*. Versioning typically uses a naming convention to identify the dates, sequences, and meanings of such transitions. These records can be used to identify and compare specific states of a configuration item (CI).

Example: Version Comparison

For a Payroll Application CI, a comparison of Version 3 against Version 2 indicates the enhanced features and other differences.

States

In the context of versioning, the *state* of a CI represents all the attribute values of that CI at a single point in time. The state of a CI can be the result of attribute changes from multiple MDRs.

Versions

A *version* is an identified instance of an object such as a CI within a product breakdown or configuration structure. Use versions for purposes of tracking and auditing change history.

Snapshots

A *snapshot* is a representation of the complete state of an object at a single point in time. For example, a typical CI has many snapshots that are associated with it. A snapshot consolidates all modification events that occur to an object during a one-minute time interval. For example, if you update a CI (edited and saved) several times within one minute, CA SDM creates a single snapshot and includes all the updates during that minute.

Snapshot is a general term that refers to time and date-based snapshots, and milestones or Standard CIs. To help you to locate meaningful points in time, CA SDM lets you label snapshots meaningfully. These named snapshots are named milestones.

Every time an object is changed, CA SDM creates a snapshot automatically. Versioning captures all changes to the object. For example, snapshots originating at an MDR and came to the CMDB through GRLoader or CMDB Web Services.

Categories

A *category* identifies a class of attributes. The category is typically the name of the tab that displays the attributes.

> **NOTE**
>
> Items such as name, serial number, active flag are assigned to a category that is not a tab name.

Examples: Tabs and Categories

The following examples show how categories correspond to tabs:

- Because the *disk space* appears on the Attributes tab, its category is *Attributes*.
- Because *IP address* appears on the Inventory tab, its category is *Inventory*.
- *Milestone* appears in the *General* category.
- A *Standard CI* appears in the Classification category.
- *Relationships* appear in the Relationship category.

Milestones

A *milestone* is an on-demand labeled snapshot of a CI. It is created to mark an event, a logical breakpoint, or an accumulation of changes. The milestone contains the actual state of the CI at the time that the snapshot was created. Milestones let you quickly identify and navigate to significant points in the history of a CI. Creating a milestone, creates an equivalent “date” snapshot.

Milestones are CI-specific, not shared. The subcomponents of a milestone for a high-level object such as a service do not automatically create milestones. To take a milestone for an object which is composed of several subcomponents, you can generate several independent milestones.

Milestones are static and can never be changed or be deleted. When you create a milestone, the snapshot is of the current state of the CI. Later, when displayed or used in a comparison, milestones always reference a point in the past. A good naming convention for your milestones helps you easily identify critical points in the life of a CI.

> **WARNING**
> Milestones do *not* apply to change specifications, verification policies, managed change states, and managed attributes.

Standard CIs

A *Standard CI* is an abstracted configuration for a CI family. It can be used for baseline comparisons to "real" CI instances in the same family. A Standard CI can be a real CI in the sense that it represents a physical object or it can exist only for comparison. Standard CIs can have the following associations:

- A Standard CI can be shared among several CIs.
- A CI can only have a single Standard CI associated with it at any given time. When the Standard CI is changed, the change is shown in any comparison between the associated CIs and the Standard CI.
- A family can have multiple Standard CIs. For example, one family might have standard CIs named “Standard Test Server,” “Standard Production Server,” “Standard Acceptance Server.” We recommend that you name Standard CIs in such a way that they can be easily searched for and identified.

Because a Standard CI is itself a CI, it can be managed. It has an audit trail, security, a history of changes, and so on, like any other CI. After you define a Standard CI for a particular CI, you can use it to verify compliance with corporate standards.

The concept of “date” or “time” does not apply when comparing a Standard CI with a snapshot or milestone. Only the Standard CI attribute values are used for comparison, snapshots and milestones specific to the Standard CI do not apply.

Example: Standard CI Use

A company defines an **Employee Workstation** configuration as a Standard CI to compare with its actual desktop computers in the Hardware.Workstation family. A comparison reveals that a specific computer has only 1 GB of RAM, instead of the Standard CI memory value of 2 GB.

Stage CI Transactions Before Loading into CMDB

CMDB provides a facility where you can store data before you load it into the CMDB. This "staged" data is stored as transactions in the transaction work area (TWA). A staged transaction is a unit of work that creates or updates a CI or Relationship. The TWA can contain many transactions for a given CI or relationship.

You can capture data being loaded into the CMDB before that data is committed so that you can:

- Clean up and standardize nonstandard data.
- Supplement incomplete data. For example, CI names starting with "NY" can have their location set to "New York".
- Modify data that does not match existing lookup tables (SRELs).
- Schedule transactions for implementing later.
- Reconcile transactions to existing CIs in the CMDB before you load the data.
- Validate the CI and relationship transactions. Validating prevents the creation of invalid data or creation of new CIs when a single or existing CI can be updated. View each transaction and the potential CIs that it can update. This helps you reconcile the transaction manually to the target CI.

You can use the TWA to help you proactively manage the reconciliation process. You can configure GRLoader to load the data into the TWA. When the data is loaded, CMDB lets you modify it to handle transactions that can potentially update or reference the wrong CI.

For more information, see Review and Modify Inbound Data Using Transaction Work Area (TWA).

How To Load Transactions into the CMDB

This article contains the following topics:

Instead of creating CIs and relationships through XML, specify the following options to select the TWA as its input source:

- lftwa (load from TWA)

- lftwai (load from TWA and inactivate)

> **NOTE**
> Unlike other GRLoader modes that use an err.xml file, loading from the TWA returns error messages to the transaction Message field.

As the transactions are processed, GRLoader sets the transaction status to indicate success or failure of each TWA transaction. If GRLoader -lftwa is run multiple times continuously, set the status of the transaction to Ready between runs of GRLoader. GRLoader does not attempt to load data that is already loaded.

The transaction is executed from the TWA as if GRLoader were running using XML input. Transaction security is based on the user that loads from the TWA, not the user that loaded the TWA transaction originally.

Auditing occurs when the transactions are processed successfully.

> **NOTE**
> Before you load the transactions, you can predetermine whether a set of transactions will create new CIs or relationships. Use the simulation options (-simci and -simrel). See How to Simulate TWA Operations for more information.

How To Prevent Data Regression

While transaction data for a CI is in the TWA, some other action can update that CI in the CMDB. The update can then cause the TWA transaction data to become invalid. To prevent unwanted data regression, GRLoader compares the TWA tran_dt (Transaction Date) field with the CI last_update_dt (Last Modified Date) or relationship last_mod_dt (Last Modified Date). If the last_update_dt is more recent, GRLoader rejects the TWA transaction and posts an error message to the tran_message column.

You can resolve this situation by completing one of the following actions:

- Use the web interface to edit the CI or relationship tran_dt (Transaction Date) field to a more appropriate value.
- Set the GRLoader option grloader.workarea.ignore_transaction_dates="yes" to ignore the transaction dates and insert the data.

> **NOTE**
> If GRLoader rejects a transaction as older than the target CI, you must verify that the transaction is running. Se t the transaction date to a current or recent date and rerun GRLoader.
>
> **WARNING**
>
> The ignore transaction dates option can cause data back-leveling or regression. When a single CI or relationship is updated multiple times in the same GRLoader -lftwa run, the last_update date is set to the current time during the first update. To allow for multiple updates to the same CI or relationship, another verification is made to see if the CI or relationship was updated after GRLoader started. If it was, then the update is allowed. If it was, then the update is allowed.

Filter By Change Order Number

The GRLoader -chg option can be used to select only transactions that are associated with a change ticket. To filter by change number, set the Change Order attribute (tran_chg_ref_num) to the appropriate change request number.

Note: The Change Order string is not validated when loaded into the CMDB.

Example: Filter By Change Ticket

The following option loads only transactions that are related to Change Order 23434.

```
grloader -lftwa -chg 23434
```

How to Use the Web Interface to Update Data in the TWA

Content

You can stage CI and relationship transactions before execution, by copying data into TWA staging area. Once in the staging area, you can manipulate CIs and relationships by using the web interfaces.

You also can validate the CI and relationship transactions. Validating prevents the creation of invalid data or creation of new CIs when a single or existing CIs must be updated. In this approach, you view each transaction and the potential CIs that it can update. It helps you reconcile the transaction manually to the target CI.

To manage the transaction work area, select the Administration tab and navigate to the CA CMDB, Transaction Work Area node. From this node, you can manage the Transaction Work Area pages.

Manage CI Transactions

The CI Transaction Detail page displays all nonblank attributes in the transaction. If you clear the Hide Empty Values check box, all CI attributes are displayed. For more attribute information, point to the attribute name.

You can do the following actions:

- View a transaction
- Create a transaction
- Edit the transaction
- Save or Cancel transaction changes (Edit mode)
- Reset the original transaction data (Edit mode)
- Show or hide data (Attributes tab)
- Set the target CI (Reconciliation tab)

The TWA accepts free-form text as case sensitive by default unless you enable case insensitivity. Use the -I GRLoader option to enable case insensitivity. Attribute values are not validated until load-from-TWA run time, so values must be entered exactly as required. If you enter string values for nonstring MDB fields, GRLoader issues warnings when loading to the CMDB. You can also simulate loading the data to predetermine (-simci) whether a set of transactions can create new CIs or report errors in the data.

Restrictions:

- If tgt_id is defined for a transaction, it displays as a CI name in the Target CI field.
- If superseded_by is defined for a transaction, it displays as a CI name in the Superseded By field.
- TWA entries are not logged.

> **WARNING**
>
> ! Data that you enter using this web interface must have the same format as native SQL or XML. For example, when you enter data for a lookup field, refer to the section on XML Input. When you want to specify an owner to look up by userid, use braces { } to specify the lookup field. For example, to find the name that is associated with the userid "admin", enter admin{userid} in the owner column.

After you save a single transaction, click the Reconciliation tab for that transaction. Verify whether any ambiguity exists in its intended target CI. If there is any ambiguity, you can specify a target CI. When all data entry is complete, you can simulate loading the data to validate the data and verify reconciliation.

> **WARNING**
> Data that you enter into the TWA using this interface must follow certain rules. For more information about the rules, see the "How to load data into the TWA Using GRLoader" section.

The web UI does not validate case-sensitive data in the TWA. Verify that the values you enter exactly match the lookup values. Alternately, when you run GRLoader, you can specify the - I option to enable case insensitive lookups.

> **NOTE**
> Although tgt_id or superseded_by are stored as UUIDs in the transaction, they are displayed as CI names in the web interface.

Manage Ambiguous CI Transactions

Review and modify transactions that have ambiguous CI targets before loading the transaction data. Resolve CI transactions that do not have a target CI, but have identifying attributes for one or more existing CIs. This resolution ensures that the appropriate CI is updated and prevents incorrect or inconsistent data from being created.

After you save a single transaction, click the Reconciliation tab for that transaction. Verify whether any ambiguity exists in its intended target CI. If there is any ambiguity, you can specify a target CI. When all data entry is complete, you can simulate loading the data to validate the data and verify reconciliation.

For more information about reconciling the data in the TWA, see Review and Modify Inbound Data Using Transaction Work Area (TWA).

Manual Reconciliation

To reconcile a CI transaction manually, determine the target CI that matches the identifying attributes of the transaction. Set the tgt_id of that transaction to the UUID of the target CI.

When GRLoader processes a transaction with a tgt id, it updates the target CI with the transaction information. GRLoader registers that CI again when its identifying attributes have changed.

You can specify the target CI explicitly in the transaction. You also can use the Reconciliation tab to select a target CI.

For more information, see Review and Modify Inbound Data Using Transaction Work Area (TWA).

Specify a Target CI for an Ambiguous CI Transaction

You can specify a CI for an ambiguous transaction so that its data is directed to a specific target CI regardless of the values of the identifying attributes specified in the transaction.

Follow these steps:

1. Select a CI transaction from the Ambiguous Transaction List page.
   The Configuration Item Transaction Detail page appears.
2. Click Edit.
   The Update Configuration Item Transaction page appears.
3. On the Reconciliation tab, select a CI from the list to designate the CI as the Target CI for the transaction. Click Set Target and then click Save.
   The selected CI becomes the target CI for the transaction.

Populating the TWA

This article contains the following topics:

Input data can come from a number of sources, including the following sources:

- CA products such as CA Configuration Automation, CA Wily CEM, CA Introscope, and CA Spectrum that import data by using GRLoader
- Any other MDR that imports data by using GRLoader
- Another CMDB
- Microsoft Excel spreadsheets
- Database tables
- An ETL tool that a vendor chooses

See Database Limitations for more information.

How to Load Data into the TWA Using GRLoader

The following GRLoader options support the uses of the TWA:

- **-lttwa**
  Loads XML data to TWA in initial state.
- **-lttwar**
  Loads XML data to TWA in ready state.
- **-lftwa**
  Loads transactions to the CMDB from TWA.
- **-lftwai**
  Loads transactions to the CMDB from TWA and inactivates successful transactions.

GRLoader input XML documents can be loaded into the CMDB or the TWA without modification.

Loading data into the transaction work area can be accomplished by using GRLoader with the - lttwa (load to TWA) option. In the GRLoader configuration file:

```
grloader.loadtotwa=yes
```

In TWA mode, instead of creating CIs and relationships directly, GRLoader inserts the information into the transaction work area tables. Data that has been loaded into the TWA can be edited, changed, and verified. After the data modification process is complete, individual transactions can be loaded into the CMDB by using - lftwa or -lftwai.

When loading CI data into the CMDB, CIs undergo automatic reconciliation. Successful reconciliation results in data from a single business object updating a single CI in the CMDB. When loading *CI transactions* into the TWA, no automatic reconciliation occurs. Multiple transactions with identifying attributes targeting a single CI can appear in the TWA. The existing rows in the TWA are not updated when new rows are added in load to TWA mode, despite an identical match to an existing CI. Even identical CI definitions can appear many times in the TWA.

GRLoader validates data values when running in load from TWA mode (-lftwa or -lftwai) to create objects in the CMDB. Or, when running using the simulation mode (-simci or - simrel). See How to Simulate TWA Operations for more details. The data values are not validated when GRLoader is run in load to TWA (-lttwa or -lttwar) mode.

The -lttwai parameter (or configuration option grloader.loadtotwa.inactivate) inactivates successfully loaded TWA transactions. After a successfully loaded transaction has been marked inactive, it can be permanently purged from the TWA by using Archive/Purge.

Example: Load to CMDB

The following command loads data directly into the CMDB:

```
grloader ... -i mydata.xml  - a  - n
```

The XML file mydata.xml contains:

```
<GRLoader>
  <ci>
     <name>server1</name>
     <class>Server</name>
   </ci>
</GRLoader>
```

Example: Load to TWA

The following command loads the data into the TWA, so that it can be manipulated before loading into the CMDB:

```
grloader ... -i mydata.xml -lttwa
```

NOTE

- *description* field for ci_twa_relation is not supported
- custom fields of bmhier are not supported inTWA

XML Input

GRLoader uses XML input data. When loading to TWA, conflicts may occur when the column names overlap with the standard column names in the database. To prevent the conflict, the following XML columns are mapped:

FROM XML Name	TO Database Column
tenant	tgt_tenant
id	tgt_id
delete_flag	tgt_delete_flag

For relationships, the following mapping applies:

FROM XML Name	TO Database Column
name	provider_name or dependent_name
dns_name	provider_dns_name or dependent_dns_name
delete_flag	tgt_delete_flag

Mappings are reversed when loading from the TWA database tables.

lookup

When you specify data for lookup (SREL) attributes in the TWA, maintain the same format as specified in native XML for GRLoader. Lookup attributes accept only a specific set of values that must be defined in related tables in CA SDM. These attributes also can have more restrictions and exceptions that must be met for the assignment to occur. The specified values are the same whether specified using XML, SQL, or the TWA web interface.

To determine whether an attribute is an SREL, refer to the following *CMDB Technical Reference* sections:

- Families and Classes (SREL attributes are identified)
- Contact and Other Lookup Fields
- Fields Validated Against Data in Existing Tables (SREL)

Example: lookup XML

In the following example, the alternate SREL column is set by specifying the lookup parameter:

```
<ci>
<name>server1</name>
<owner lookup="userid">mckpe99</owner>
</ci>
```

In the TWA, the same thing can be accomplished by suffixing the alternate SREL column to the data value, which is included between delimiters. For more information, see the grloader.workarea.delimiters configuration given later. The equivalent transaction is represented in the transaction work area as:

ID	Name	Owner
100	server1	mckpe99 {userid}

You also can set the delimiter character that is used previously in the GRLoader configuration file:

```
grloader.workarea.delimiters=xy
```

x and y are different characters that typically do not appear in the work area.

Date Format

GRLoader supports the following date format options in XML:

- datefmt=UTC
- datefmt=localtime (the default)

Examples: datefmt XML

```
<ci>
<name>server1</name>
<purchase_date datefmt="UTC"> 1241197235</purchase_date>
</ci>
<ci>
<name>server3</name>
<purchase_date> 2009/05/01</purchase_date>
</ci>
```

The equivalent transaction is represented in the transaction work area as:

ID	Name	purchase_date
101	server1	1241197235
102	server2	2009/05/01

NOTE
If a date contains a special character such as a slash (/), then the format is assumed to be "localtime". If a date contains no special characters, the date is assumed to be UTC.

EMPTY

GRLoader supports the update_if_null option in the XML, which clears a field in the CMDB. The following example clears the owner field for server1. Without that attribute, the owner field is not affected. When using the TWA, you can use the keyword EMPTY instead.

Example: update_if_null XML

```
<ci>
<name>server1</name>
<owner update_if_null="yes"></owner>
</ci>
```

In the TWA, the database value is cleared by specifying the keyword EMPTY as the string value. The equivalent transaction in the work area is:

ID	Name	Owner
102	server1	EMPTY

The keyword value can be set by using the grloader.emptyvalue configuration option:

```
grloader.emptyvalue=xxxx
```

xxxx represents any string that typically does not appear in the work area data.

How To Simulate TWA Operations

You can predetermine whether a set of transactions can create new CIs or relationships and therefore create new ambiguities for other CIs. Use the following options:

- **- simci**

Simulates processing CI transactions only. The option can be used to determine whether transactions create or update CIs. When the -simci option is used, GRLoader performs data validation.

- **- simrel**
  Simulates processing relationship transactions only. The option can be used to determine whether relationship transactions create or update relationships. The - simrel option checks relationships for the existence of the provider and dependent CIs, validates relationship types, and so on.

The output from simulation mode is directed to the TWA or to the _err.xml file. In normal load mode, the _err.xml file contains the CI input and a comment indicating whether the CI was inserted or updated. When a simulation is used, the GRLoader message on the CI Transaction List indicates whether the CI or relationship was inserted or updated. The transaction state remains unchanged.

Simulation can also be enabled in a configuration file by using the **grloader.simulateloadci** and **grloader.simulateloadrelation** options.

> **NOTE**
> If the GRLoader input creates CIs and relationships simultaneously, the - simrel option can only process actual CIs. The option cannot process CIs that are scheduled to be created. Because of this limitation, -simci and - simrel are mutually exclusive.

How to Use SQL to Update Data into the TWA

The TWA is stored in the MDB and you can update it directly with SQL queries.

TWA Tables

Transaction Work Area (TWA) tables include:

- **ci_twa_ci**
  A single table that contains all attributes across all CI families. The table stores data in a de-normalized form to enable customers and services to understand and manipulate the content.
- **ci_twa_relation**
  Complements the ci_twa_ci table. This table contains CI relationship information.

TWA column names correspond to the attributes that are used with GRLoader, with some exceptions.

TWA SQL Column Names

Column names in the ci_twa_ci and ci_twa_relation tables match the CI and Relationship attribute names with some exceptions.

TWA fields common to ci_twa_ci and ci_twa_relation

- **apply_after_dt (Apply After Date)**
  GRLoader executes the transaction only if the current date is after the Apply After Date. Measured in number of seconds since the epoch. If the value is zero (0), this field is ignored.
- **tran_chg_ref_num (Change Order)**
  Specifies the change order identifier that is associated with this transaction. The GRLoader - chg option uses this attribute to select only transactions with the specific change order number.
  Note: For more information about using the - chg option option, see Filter By Change Order Number.
- **tran_status (Transaction Status)**

Specifies the status of the transaction. The tran_status must be one of the following values. Only the transactions with tran_status=1 are executed.

Value	Short Description	Long Description
0	Initial	Default value. Requires intervention to move to "Ready" for GRLoader to act upon this transaction.
1	Ready	The transaction is ready to be loaded into either the appropriate CI or relationship tables.
2	Successful	Transaction that is processed successfully.
3	Warning	Warning that is detected during load into ca_owned_resource/bmhier.
4	Error	Error that is detected during load into ca_owned_resource/bmhier.
40000+	For customer use	

- **tran_message (Message)**
  GRLoader message showing results of the load or simulation.
- **tran_dt (Transaction Date)**
  Prevents accidental overlaying of current information by old transaction data. The transaction date is compared with the last modification date of the CI. GRLoader rejects the transaction if the CI is more recent.
  If GRLoader rejects a transaction as older than the target CI, verify that the transaction is still applicable. Set the transaction date manually to be more current. Run GRLoader again.
- **tgt_delete_flag (Active?)**
  Inactivates the target CI or relationship. Set this value to 1 (one). This setting is not the same as setting delete_flag to 1, which indicates that the transaction must be deleted.

ci_twa_ci columns

In addition to the following column names, you can specify CI attribute names as column names in your SQL statements:

- **tgt_id (Target CI)**
  Specifies the UUID of the target CI. Used when performing manual reconciliation.
- **superseded_by (Superseded By)**
  Specifies the UUID of the superseded CI.
- **tgt_tenant (Tenant)**
  Specifies the tenant name that is assigned to the target CI. Tenant can only be assigned at the time of creating CI if the GRLoader user has proper authorization. This setting applies only when multi-tenancy is enabled.

ci_twa_relation columns

In addition to the following column names, you can specify relationship attribute names as column names in your SQL statements:

When using SQL to define relationship transactions, use the following database column names:

provider_name (Provider Name)

provider_mac_address (Provider MAC Address)

provider_serial_number (Provider Serial Number)

provider_system_name (Provider Host Name)

provider_asset_num (Provider Asset Number)

provider_dns_name (Provider DNS Name)

provider_tenant (Provider Tenant)

AND

dependent_name (Dependent Name)

dependent _mac_address (Dependent MAC Address)

dependent _serial_number (Dependent Serial Number)

dependent _system_name (Dependent Host Name)

dependent _asset_num (Dependent Asset Number)

dependent _dns_name (Dependent DNS Name)

dependent_tenant (Dependent Tenant)

If provider or dependent CI UUIDs are known, you can use the following fields:

provider_id specifies the UUID of the provider CI

dependent_id specifies the UUID of the dependent CI

> **NOTE**
> provider_tenant and dependent_tenant only apply when multi-tenancy is enabled.

Insert New Records in the TWA

You can insert a new record into the ci_twa_ci and ci_twa_relation tables from outside of CA SDM.

To insert records into the ci_twa_ci and ci_twa_relation tables, specify 0 in the ID column as:

```
insert into ci_twa_ci values(id, name) values(0,'test-ci');
insert into ci_twa_relation values(id, type) values(0,'test-relation-type');
```

> **NOTE**
> If multi-tenancy is enabled, provide the tenant, tgt_tenant, provider_tenant or dependent_tenant attributes as appropriate. If you create a TWA transaction without specifying a tenant, the Public tenant is assumed.

How to Use the ODBC Driver

You can use SQL instead of GRLoader to load the TWA. We recommend that you use the ODBC driver that is provided with CA SDM, instead of native SQL services. Using the ODBC driver provides the following benefits:

- Performance
- Security
- Database independence
- Data integrity when the database updates come from the CA SDM server and the bulk editing tool is running simultaneously

Cautions

- Only the CA SDM ODBC drivers honor security features such as multi-tenancy, data partitioning, and role access. SQL that is executed using the native DBMS drivers does not enforce security.
- Make frequent data backups when performing updates using native SQL.

CI Identifiers in the TWA

In the ci_twa_ci table, three identifiers are associated with a CI transaction: ID, tgt_id, and superseded_by, These attributes are:

- ID is the transaction sequence number. **Do not modify this field.** IDs 1-2,000,000,000 are reserved for use by systems that modify the TWA outside the CA SDM server. IDs from 2,000,000,001 to approximately 4 billion are reserved for use by CA SDM. When the system runs out of IDs, reinitialize the table.
- tgt_id, provider_id, or dependent_id are the UUID of the target CI. Set this field whenever you perform manual reconciliation.
- superseded_by is the UUID of the superseding CI. Set this field whenever you manually designate the superseding CI for the transaction.

How to Set Transaction Status

You can set the transaction status (tran_status) of each transaction in the TWA. For GRLoader to process a transaction, it must be set to Ready (tran_status=1).

Value	Short Description	Long Description
0	Initial	Default value. Requires intervention to move to "Ready" for GRLoader to act upon this transaction.
1	Ready	The transaction is ready to be loaded into either the appropriate CI or relationship tables.
2	Successful	Transaction that is processed successfully.
3	Warning	Warning was detected during load into ca_owned_resource/bmhier.
4	Error	Error was detected during load into ca_owned_resource/bmhier.
40000+	For customer use	

If GRLoader - lftwa is run multiple times, the status is updated after the first run and all subsequent runs do not find any transactions. Reset the status of the ready transactions before every GRLoader - lftwa execution.

For information about how to represent XML data in the TWA tables, see the XML examples in previous sections.

How to Share Tables with CA SDM

SQL Server Only

When using SQL to update the ci_twa_ci and ci_twa_relation tables, set the last_mod_dt column manually, as follows:

```
SET last_mod_dt = DATEDIFF(s,'19700101', GETUTCDATE())
```

Example: Set last_mod_dt

The following example sets the last_mod_dt:

```
UPDATE ci_twa_ci
    SET tran_status=1,
        last_mod_dt = DATEDIFF(s,'19700101', GETUTCDATE())
    WHERE tran_status=1
```

NOTE
TWA table updates can take several minutes to appear in the web interface or GRLoader. For more information about update timing, see the NX_DBMONITOR_TIMER_MINUTES option.

Transaction Work Area

Use the Transaction Work Area (TWA or work area) to inspect and modify CI and relationship data before loading the data into the CMDB.

The transaction work area is a data warehouse to store, review, and modify CI and relationship transaction information from multiple sources. In the TWA, you can modify incoming transaction data to conform to your CMDB requirements. You also can use the TWA to modify data input that could create false positives or false negatives during the reconciliation process. False positives are multiple CI representations of the same business object. False negatives are multiple business objects that are identified as the same CI.

Transaction information can apply to new objects or to objects that are already in the CMDB.

The TWA process works as follows:

1. Load the data into the TWA. The content can include CI transactions or relationship transactions. The input process does not attempt to reconcile records. The process simply populates the work area with CI and relationship transaction records in one of the following ways:
   - GRLoader - Reads XML data and stores it in the TWA using the -lttwa or -lttwar options.
   - Native SQL - Places data into the TWA using standard SQL processing.
   - Create a CI transaction or relationship transaction using the web interface.
2. (Optional) Manually reconcile transactions to existing CIs in the CMDB before you load the data. You can also Review and Modify Inbound Data Using Transaction Work Area (TWA).
3. Modify the data. You can modify the TWA from the following sources:
   - **CA SDM user interface**
     The web-based user interface lets you view and modify transactions in the work area.
   - **Native SQL**
     When performing many changes to multiple transactions, native SQL can modify the data in the TWA.

     NOTE
     Changes that are made using native SQL can bypass all CA SDM security.

   GRLoader loads the data from the TWA to the CMDB using the -lftwa or -lftwai options. Each TWA row is treated as a separate transaction.
   a. If there is an error after a GRLoader run, the error code is populated into the transaction to indicate that it is incomplete (to facilitate future retries).
   b. All other records are identified as completed transactions.
4. Review the transaction results and correct any errors using the web interface. Consider the following points:
   - If you want the columns of custom families or attributes to appear in the work area, modify the work area to include the modified columns. For more information, see Extend the TWA Object.
   - To take advantage of the TWA, use the latest version of GRLoader. If you are using a product that includes a previous release of GRLoader, upgrade GRLoader to the latest version.
   - Changes made to transactions in the TWA are not audited.
5. Repeat steps 3 as needed.

TWA Administration

This article contains the following topics:

The TWA may require the following administration:

- Extend the TWA - required when you extend any OOB families or when you define your own custom families.
- Archive and purge of the TWA
- How to use the TWA with CA Configuration Automation
- Database limitations

Extend the TWA Object

Any modifications to the CMDB family schema require parallel modifications to the TWA schema. If no custom families or attribute have been defined, the TWA requires no modifications.

To extend the TWA object:

1. Modify the user-defined families and attributes *using the procedures in* Extending CMDB
2. Add the new family attributes to the ci_twa_ci schema using Web Screen Painter Schema Designer.
   To add a new attribute to ci_twa_ci schema:
   a. Using Web Screen Painter Schema Designer, open the schema for the ci_twa_ci table.
   b. Add the desired family attribute to the extension table.
   c. Publish the modified ci_twa_ci schema.

   NOTE
   The new attributes must be type STRING and must hold the longest possible text value.
3. Add the metadata for the custom family to the TWA as follows:
   a. Using Web Screen Painter, open the cmdb_metadata_site_families.htmpl file.
   b. Add an PDM_INCLUDE for the appropriate cmdb_metadata_*extension*.htmpl file that was created for the custom family.

Archive and Purge of the TWA

To maintain the TWA, CA SDM provides the following archive/purge rules:

- **CI Inactive Transactions**
  Archives and purges the CI transactions that are marked for deletion.
- **CI Successful Transactions**
  Archives and purges the CI transactions that have completed successfully.
- **Relationship Inactive Transactions**
  Archives and purges relationship transactions that are marked for deletion.
- **Relationship Successful Transactions**
  Archives and purges relationship transactions that have completed successfully.

You can select how often the rules run or whether they are enabled.

You can modify and enable the rules for enabling archive and purge of the TWA. If you are a heavy user of the TWA, be aware of the limitations on the number of records in the TWA imposed by the DBMS.

If you have to reinitialize the TWA to clear out all data in the TWA, complete the following tasks:

- Set all transactions to inactive
- Run archive purge

  WARNING
  Do not use SQL to delete all records in the TWA because it deletes required header records.

TWA GRLoader Command Options

CA SDM provides the following command options for use in GRLoader TWA processing:

- **-lttwa**
  Loads XML into the initial state.
- **-lttwar**
  Loads XML into the ready state.
- **-lftwa**

Loads transactions from the TWA.

- **-lftwai**
  Loads transactions from the TWA and inactivates successful transactions.
- **-chg *nnnn***
  Used with - lftwa and -lftwar. Loads only those transactions that are associated with change order *nnnn*.
  Note: The Change Order string is not validated when loaded into the CMDB.
- **-l**
  Ignore case of lookup attributes. By default free-form text for lookup attributes is processed as case sensitive unless you use this option.
- **-simci**
  Determines whether the CI transactions can create or update CIs. When this option is used, GRLoader perform error checking against CIs only.
- **-simrel**
  Determines whether relationship transactions can create or update relationships. This option verifies relationships for the existence of the provider and dependent CIs, validates relationship types, and so on.

TWA GRLoader Configuration File Options

CMDB provides the following TWA options for the GRLoader configuration file:

grloader.emptyvalue=EMPTY

grloader.loadfromtwa=yes

grloader.loadfromtwa.inactivatesuccessful=yes/no

grloader.loadtotwa=yes

grloader.loadtotwa.ready=yes/no

grloader.workarea.delimiters={ }

grloader.workarea.ignore_transaction_dates=yes

grloader.simulateloadci=yes/no

grloader.simulateloadrelation=yes/no

How to Use the TWA with CA Configuration Automation

To use the TWA with CA Configuration Automation processing, apply patch RO08739 to CA Configuration Automation r5. This patch allows you to specify the - lttwa option in the Other Options field on the Export Options tab of the CMDB Export Report definition. Upgrade the GRLoader component. Contact CA support for questions.

Database Limitations

The TWA is subject to the data limitations of the underlying database, as follows:

- When using TWA and Microsoft SQL Server, the total length of one data transaction must not exceed 8060 bytes.
- When using the dbload and pdm_load utilities, a load operation is restricted to 512 columns.
- Oracle has a 1000-column limitation. If you modify CMDB, verify that the total number of attributes does not exceed 1000 across all families (both CA-supplied and user-defined).
- SQL Server 2005 limitations as described in the following table:

Limitation Type	Limit (32-bit and 64-bit)
Columns per wide table	30000

Columns per base table	1024
Columns per SELECT statement	4096
Bytes per row	8060
Rows per table	Limited by available storage

Define the Business Infrastructure

Implement your service desk using CA SDM by defining your business infrastructure with the following objects:

- Configuration item families and classes
- Manufacturers and models
- Service statuses
- Vendors and vendor types

The following information provides a general description for each object and explains how the object is used in the product.

Object Definition Order

You start at the bottom level of the object hierarchy when you define objects. Therefore, when you define objects at higher levels, you can select from existing objects at lower levels in the hierarchy. For example, because a class has (references) a family, you define families first, followed by classes. Define configuration objects last, after you define all the supporting objects, because they are the top of the hierarchy. Therefore, you define data objects in the following order:

Define First	Define Second	Define Third
Families	Classes	Configuration items
Manufacturers	Models	Configuration items
Service statuses	Configuration items	
Vendor types	Vendors	Configuration items

Manufacturer and Models

Manufacturers identify the manufacturers of the various configuration items of concern to your enterprise. *Models* contain specific information about the products that a particular manufacturer provides to your enterprise. For example, you might define as a manufacturer a particular software company. Then, you would define as models each one of the applications that the company provides for your enterprise.

Defining manufacturers and models makes it easier to manage your configuration items. For example, you can generate a list of models that are provided by a particular manufacturer. You can also generate a list of configuration items of a particular model.

You can enter information about the manufacturers and models of the hardware, software, and services that are associated with your system. You can also reference this information in configuration item records. To create a model from the Administration tab, navigate to Service Desk, Application Data, Configuration Items, Models. Click Create New and enter the fields as appropriate.

Service Status

Service status identifies the readiness condition of configuration items. Examples of service status include: *in service*, *in repair*, or *discontinued*. Defining service status lets you track the availability and use of configuration items in your enterprise. For example, you can generate a list of configuration items that are currently in repair.

Vendor Types and Vendors

Vendor types are classifications for vendors that identify the type of company providing configuration items. For example, you could classify vendors that you lease configuration items from as leasers. You can classify vendors that provide service to you as providers.

Vendors identify the companies that supply your enterprise, including the type of company and a primary contact. You can also reference a vendor in a user's contact record.

Defining vendor types and vendors gives you a convenient way to organize your configuration items. For example, you can generate a list of vendors that fall under a particular vendor type. You can also generate a list of configuration items from a particular vendor.

External Asset Management Tools

You can integrate CA SDM with other asset management tools such as CA Client Automation, and CA APM. The asset management features of these tools from CA SDM include the following features:

- CA Client Automation provides a complete set of hardware and software inventory functions. When viewing a CI in the CA SDM client, click Asset Viewer to display more information about the asset.
  Integration between CA SDM and CA Asset Management is activated when the products are installed on the same MDB.
- CA Asset Portfolio Management provides a complete set of asset lifecycle functions. When viewing a CI in the CA SDM client, click Asset Viewer to display more financial information about the asset.
  Integration between CA SDM and CA Asset Portfolio Management is activated when the products are installed on the same MDB.

About MDR

This article contains the following topics:

The Configuration Management Database Federation (CMDBf) is a working group that is composed of representatives from CA, IBM, HP, Microsoft, and other companies. The CMDBf defines a Management Data Repository (MDR) as anything that collects information about configuration items (CIs).

To create the relationship between an MDR and its CIs when implementing MDR Launcher, complete the following tasks:

1. Define the MDR.
2. Define the CIs that reference that MDR.
   You cannot have a CI that references a non-existent MDR. You can define a CI without defining an MDR association. You can add the MDR information during an update or can edit to take advantage of the MDR Launcher capability.

Multiple MDRs can discover the same CI. After the CI is discovered, each MDR attempts to manage that CI, and an MDR can do the following actions:

- Discover detailed attributes about the CI.
- Attempt to modify the CI state.

Example: A CI Discovered by Multiple MDRs

Both Network Management software and an Asset Management software package discover a CI.

- The network management software maintains information about network configuration and network topology.
- The asset management software contains information regarding cost, depreciation, licensing, and maintenance contracts for that particular CI.

MDR Classes and MDR Names

An IT enterprise can include many MDRs. Each MDR has an identifier that is known as an *MDR name (mdr_name*). It is common for an MDR to use the host server name as the mdr_name. The server name is used to allow multiple MDRs to reside on the same host server. Therefore, an MDR class (mdr_class) is appended to the mdr_name to identify each MDR uniquely.

CA Configuration Automation is an enterprise discovery tool that integrates tightly with CMDB Each CA Configuration Automation MDR defined to the CMDB must have an *mdr_class* of **Cohesion**. For more information about CA Configuration Automation, see the CA Configuration Automation online help. For CA Configuration Automation-CMDB integration, see the *CA Configuration Automation Implementation documentation*.

How does an MDR Complement CA SDM?

An MDR typically contains more detailed CI information than CMDB. However, a single MDR is not typically aware of the existence of other MDRs. It does not focus on the relationships that a particular CI can have with other CIs, especially if they are contained in other MDRs. CMDB manages this type of environment well, because it focuses on managing CIs regardless of their MDR source.

CMDB is not intended to store all attributes for all CIs. CMDB is used to consolidate the most important attributes that must be managed centrally. Attributes that are under the control of change management are excellent candidates for inclusion in the CMDB. Attributes that CMDB does not manage can be accessed by using the MDR Launch capability. In addition, CMDB provides the CMDBf Viewer. CMDBf Viewer allows side by side comparison of CI attributes across multiple CMDBs and MDRs.

MDR Launcher

The MDR Launcher lets you view data from almost any MDR by using a web page, without any coding. MDR Launcher lets anyone viewing a CI to obtain more details about the CI, and to gain control over it (if the MDR supports such control).

Some uses of the MDR Launcher include the following ones:

- From the hardware.server detail page, launch CA Configuration Automation to verify a change.
- From an Air Conditioning CI detail, launch to a vendor web page for diagnostic information and incident reporting.
- From a Contract CI, launch a contracts management system to learn contract details.
- From an SLA CI, launch CA Service to review Service Level Agreements before modifying.
- From a Server CI, launch CA Remote Control to take over the server to diagnose and correct a problem.

Define a URL to Launch an MDR

CMDB uses a URL to start a web page session with the source MDR to operate the MDR Launcher. You define the URL that CMDB uses.

Follow these steps:

1. Click the Administration tab.
2. In the left pane, open the CMDB, MDR Management tree.
3. Click MDR List.
   The Management Data Repository (MDR) List page appears.
4. Click an existing MDR (or create and save a new one).
   The MDR Provider Definition page appears.
5. Click Edit.
   The Update MDR Definition page appears.
6. Complete the following parameters:
   – **Hostname**

Specifies the Network address or DNS name of the web server that serves up web pages for the MDR.

- **Port**
  Specifies the port number that the Hostname web server uses.
- **Path**
  Specifies the path to the web page, including the page itself.
  Parameters
  Specifies any parameters that are used to identify the desired CI to the MDR. CMDB posts this information to the MDR.
- **Userid**
  Specifies the userid, if a common userid is allowed access to the MDR.
- **Shared Secret**
  Specifies information that is shared between CMDB and the MDR. For CA Cohesion MDRs, this value must match the value that is specified in the CA Cohesion properties file, for the com.cendura.security.oneclickauth.secret value.
- **CMDBf Namespace** Specifies the federated_asset_id that is passed to the query as a local ID. For CA CMDB, the value is http://cmdb.ca.com/r1.
- **CMDBf Timeout** (Optional) Specifies time limit for CMDBf endpoint query.
  Default: 10 seconds
- **CMDBf Endpoint** Specifies the Query Service endpoint for the MDR. Required for CMDBf Viewer and retrieving updated MDR data. If you use CA CMDB as an MDR provider, the value is as follows.

  ```
  http://<cmdb_hostname>:<cmdb_port>/axis/services/QueryPort.
  CA Service Management 17.3.0.18 and above versions, http://<cmdb_hostname>:<cmdb_port>/cxf/services/
  QueryPort.
  ```

7. Save the definition.

The URL is defined. In addition, the URL can contain the substitution variables to further qualify the CI to the MDR. For more information, see the Using the MDR Launcher topic.

Set Up a CA APM MDR Provider

You can set up an MDR to be the CA APM provider.

Follow these steps:

1. On the Administration tab, click CMDB, MDR Management, MDR List.
2. Click Create New to specify the CA APM MDR.
   The MDR Provider definition appears.
3. Enter the following required MDR provider information:
   - Button Name -- Specify APM or any other valid button name. We recommend using a Button Name of APM.
   - MDR Name -- Specify APM for CA Asset Portfolio Management r11.3.4 or ITAM for CA APM r12.6
   - MDR Class -- Specify GLOBAL.
   - Hostname -- Specifies the CA APM server name by using the network address or the DNS name of the CA APM web server.
   - URL for Launch in Context -- Specifies http://{hostname}:{port}/{path}?{parameters} and must not be changed.

   The MDR provider form automatically populates the path and parameter values with the required CA APM launch in context information.
4. Click Save.
   The CA APM MDR provider is set up. For more information about the MDR launcher, see the MDR Launcher topic.

Launch in Context from CMDB to CA APM

The CMDB MDR Launcher facility supports launch in context to CA APM when the tools share the same MDB. The CMDB UI provides a launch in context button on the Attributes tab in the CI detail form. The button appears when the user creates a special CA APM MDR provider definition.

The CA APM MDR definition has all the capabilities of a traditional MDR. The CMDB Versioning feature also supports launch in context directly from an attribute log entry that is associated with each CA APM change.

> **WARNING**
> Unlike other MDRs, the CA APM MDR is automatically associated with each CI or Asset. The MDR class of GLOBAL and MDR name of APM are used to identify the CA Asset Portfolio Management r11.3.4 MDR. The MDR class of GLOBAL and MDR name of ITAM are used to identify the CA APM r12.6 MDR. Use of the CA APM MDR is fully compatible with other MDRs, even for the same CI.

Using the MDR Launcher

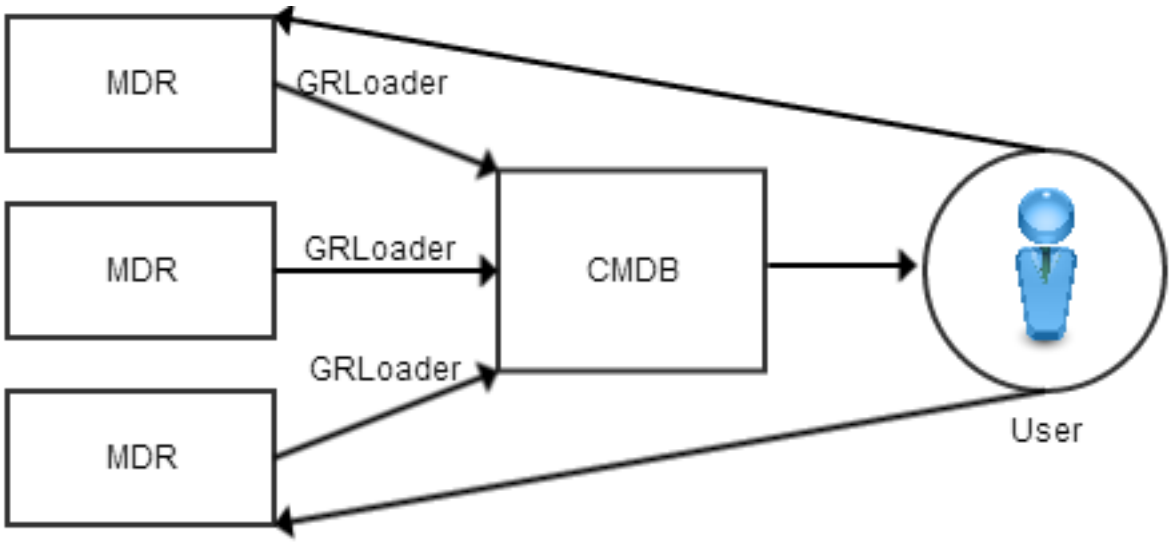
This article contains the following topics:

The MDR Launcher

One of the main purposes for implementing CMDB is to aggregate data from multiple data sources (known as MDRs). However, a CI must always include a reference back to its MDR origin.

CMDB provides facilities for importing and loading CIs and also for associating the CIs with their origins. To view a CI in the CMDB, use the MDR Launcher. When you use the MDR Launcher for this purpose, you can return seamlessly to the CI-originating system. See the following diagram.

Figure 32: BCMDB--r12.1--IG_MDR_1_Diagram



Using the MDR Launcher, it is possible to implement a “closed loop” change management process such as the following one:

1. Create a change record.
2. Implement the change.
3. Verify the change by checking the MDR source.
4. To indicate that the change has been made, update the CMDB.

From a Problem Management process perspective, you can use the MDR Launcher in the following way:

1. Detect a problem.
2. Determine the severity and pervasiveness of the problem. To determine what dependent CIs are affected, use the CI relationship data.
3. Determine possible causes of the problem by researching provider CIs.
4. Perform an in-depth analysis if necessary using the detailed information available in the MDR. To take corrective action, use the MDR.

MDR Terminology

The following terms are used in CMDB-MDR integration:

A management data repository (MDR) represents software or data that contains source information about a CI. An MDR generally contains more unrefined CI information than the CMDB, which contains a managed subset of that data.

An *MDR class* (MDR_CLASS) is used to group MDRs that CMDB processes similarly. Three special MDR classes include: **COHESION**, **GLOBAL**, and **cmdbf**.

An *MDR name* (MDR_NAME) is the name that an MDR uses to reference itself. Verify that the mdr_name and mdr_class value combination must be unique within your enterprise.

A Federated asset ID (FEDERATED_ASSET_ID) is a unique MDR identifier for a CI.

The different CI families typically use different respective MDRs as data providers. However, a single CI can have multiple MDR data providers. For example:

CI Family	MDR_CLASS
Contact	human resources system telephone directory single sign-on authentication system
Document	document management system
Air Conditioning	document management system contract management system air conditioning control system
Mainframe	tape management system DASD management system performance management system job scheduler
Storage	storage management system asset management system
Location	asset management system education calendar office directory
Network	network management systems problem management system

There can be multiple MDRs in each MDR class, and each MDR can contribute data to multiple CIs. A given CI can receive data from zero or more MDRs. A CI also can have data contributed to it independently. Consider the following example: One mainframe CI has data that Disk Management System 1 donated. Another mainframe CI has data

that Disk Management System 2 and Job Scheduler 2 donated. CMDB manages the relationships among CIs and all their related MDRs.

MDR Mapping

Every MDR has a unique way of identifying the CIs that it manages. Those identifiers are seldom synchronized across MDRs. For example, when referencing a specific Contact CI, different MDRs can use different identifiers. A national identity number, telephone number, license number, or Employee ID can identify the same person. The process of associating these disparate identifiers with the same unique identifier (UUID) maintained in the MDB is named *mapping*. Mapping occurs automatically when data is imported using GRLoader when the CI contains the <mdr_name> <mdr_class> and <federated_asset_id> tags. Mapping can also be accomplished manually through the Administration functions in the user interface. A CI that has no mappings that are associated with it is named *unfederated*. Every CI is automatically mapped to global MDRs using the UUID as the federated_asset_id.

MDR Launching

When you view a CI with the CMDB user interface, click buttons to launch an MDR user interface directly. One button per MDR mapping exists for the focus CI. To verify that a change request has completed successfully, use this launching. To obtain more information about a CI than the CMDB has collected, use this launching.

CMDBf Viewer

CA SDM provides the CMDBf Viewer with the results of CI federation across MDRs. From a CI Detail page, click CMDBf Viewer to see CI attributes of federated CMDBs and MDRs in parallel. On the Federated View page, you can click Retrieve to update the information from any of the federated MDRs. For better readability, CMDB metadata files can reconcile MDR attribute names and CMDB attribute names.

> **NOTE**
> This feature requires MDRs that support Query. You configure the MDR CMDBf Endpoints to display their results on Federated View.

Define an MDR to CMDB

Before you use the CMDBf Viewer, define the MDR to CMDB.

Follow these steps:

1. On the Administration tab, navigate to CMDB, MDR Management, MDR List.
2. Click Create New.
   The Create New MDR Definition page appears.
3. Complete the following fields:
   - **Tenant**
     Identifies the tenant owner of this MDR (if multi-tenancy is installed).
   - **Button Name**
     Specifies the button label to appear on the CI Detail page. This name must be unique for each MDR. Required for “launch in context” and CMDBf Viewer.
   - **MDR Name**
     Specifies the string to match the XML data that is sent in the mdr_name field. While the MDR can use any string, the host name is used frequently. This name together with the mdr_class form a unique name for the MDR. Required for “launch in context” and CMDBf Viewer.
   - **MDR Class**
     Specifies the class that must match the data that is sent in the mdr_class field in the XML. While this name can be anything, it must together with the mdr_name field form a unique identifier for the MDR. Global MDRs are defined with an MDR Class of GLOBAL.

- CA Configuration Automation MDRs must specify an MDR class of COHESION, which automatically sets the Path, Parameters and URL to be Launched fields to the required CA Configuration Automation launch-in-context values.
- CA Asset Portfolio Management r11.3.4 MDRs must specify an MDR name of APM and MDR class of GLOBAL, which sets the Path, Parameters, and URL to be Launched fields to the required CA Asset Portfolio Management r11.3.4 launch-in-context values.
- CA APM r12.5 MDRs must specify an MDR name of ITAM and MDR class of GLOBAL, which sets the Path, Parameters, and URL to be Launched fields to the required CA APM 12.5 launch-in-context values.
- For CMDBf Viewer, MDR Class must be cmdbf.

– **Active**
Denotes this MDR definition as active or inactive. The Inactive MDR definitions are logically deleted, but they can be made active again by using the Search utility.

– **Owner**
Specifies the contact responsible for this MDR.

– **Description**
Specifies a description in free-form text.

– **Hostname**
Specifies the host name, DNS name, or IP address of the host, which contains the web server. The web server is the one that hosts the web page to be launched. Required for “launch in context”.

– **Port**
Specifies the TCP/IP port that is used by the MDR web server to serve up web pages. Port 80 is the default. Required for “launch in context”.

– **Path**
Specifies the portion of the URL that precedes the question mark (?) character. This information can be obtained from your MDR documentation.

- For mdr_class of Cohesion, the value is set automatically to "CAisd/html/cmdb_cohesion.html" and cannot be changed.
- For mdr_name of APM and mdr_class of GLOBAL, the value is set automatically to apm/frmObject.aspx and cannot be changed.
- For mdr_name of ITAM and mdr_class of GLOBAL, the value is set automatically to ITAM/Pages/Asset.aspx and cannot be changed.

– **Parameters**
Specifies the portion of the URL that follows the question mark (?) character. This information can be obtained from the MDR documentation.

- For mdr_class of Cohesion, the value is set automatically to "hostname={hostname}+port={port}+family={family}+name={name}+secret={password}+federated_asset_id={federated_a The value cannot be changed.
- For mdr_name of APM and mdr_class of GLOBAL, the value is set automatically to ObjectID={cmdb_asset_id}&obj=11&FUNCTION=1&WinID=OBFRASSET{cmdb_asset_id}&WinContainerID=. The value cannot be changed.
- For mdr_name of ITAM and mdr_class of GLOBAL, the value is set automatically to ParentClass=Asset&assetid={cmdb_asset_id}&TicketID={itam_ticketid}. The value cannot be changed.

– **Userid**
Specifies the MDR user logon, if necessary. This value is substituted into the URL wherever {userid} is found. If blank, userid defaults to whomever is signed on.
For CA Configuration Automation, "Shared Secret" is the secret that is used to access CA Configuration Automation, if necessary. This value is substituted into the URL wherever {password} is found. For more information about defining the MDR, see the *CA Configuration Automation Implementation documentation*.

– **Shared Secret**

Specifies information that is shared between CMDB and the MDR. This value is substituted into the URL wherever {password} is found. For CA Configuration Automation MDRs, the value must match the value of the “com.cendura.security.oneclickauth.secret”. For more information about creating a shared secret, see "Integrating with CA CMDB" in the *CA Configuration Automation Implementation documentation*. Required for CMDBf Viewer.

– **CMDBf Namespace**
Specifies the federated_asset_id that is passed to the query as a local ID. For CMDB, the value is http://cmdb.ca.com/r1.

– **CMDBf Timeout**
(Optional) Specifies time limit for CMDBf endpoint query. Default is ten (10) seconds.

– **URL to be Launched**
Default value of http://{hostname}:{port}/{path}?{parameters}. For some MDRs, it can be overridden if necessary to accommodate MDR-specific requirements. Required for "launch in context".
For mdr_name of APM and mdr_class of GLOBAL, the value is http://{hostname}:{port}/{path}?{parameters}
For mdr_name of ITAM and mdr_class of GLOBAL, the value is http://{hostname}:{port}/{path}?{parameters}
For mdr_class of Cohesion the default value is http://*cmdb_hostname*:*cmdb_port*/{path}?{parameters}
where:
cmdb_hostname is the host name, DNS name, or IP address of the CMDB web server. Defaults to the current hostname that is accessing the CMDB web server.
cmdb_port is the TCP/IP port of the CMDB web server. Defaults to the current port number used to access the CMDB web server.

NOTE
If you have enabled SSL support for CA Configuration Automation, set the URL to: http://hostname:port/{path}?{parameters}+https=yes

For information about enabling CA Configuration Automation HTTPS support, see the CA Configuration Automation online help topic Creating the HTTPS Certificate and Enabling HTTPS.

– **CMDBf Endpoint**
Specifies the Query Service endpoint for the MDR. Required for CMDBf Viewer and retrieving updated MDR data. If you use CA CMDB as an MDR provider, the value is as follows.

```
http://cmdb_hostname:cmdb_port/axis/services/QueryPort.
CA Service Management 17.3.0.18 and above versions, http://cmdb_hostname:cmdb_port/cxf/services/QueryPort.
```

4. Click Save.
The MDR is defined.

MDR URL Definitions

The URL to be launched has the default value of http://{hostname}:{port}/{path}?{parameters}. If necessary, you can modify this expression to accommodate any MDR-specific requirements. The URL is required for "launch in context".

For mdr_name of APM or ITAM and mdr_class of GLOBAL, the default value is:

```
http://{hostname}:{port}/{path}?{parameters}
```

For mdr_class of Cohesion, the default value is:

```
http://cmdb_hostname:cmdb_port/{path}?{parameters}
```

- ***cmdb_hostname***
The cmdb_hostname variable specifies the hostname, dnsname, or IP address of the CMDB web server. The variable defaults to the current hostname currently accessing the CMDB web server.
- ***cmdb_port***
The cmdb_port variable specifies the TCP/IP port of the CMDB web server. The variable defaults to the current port number that is used to access the CMDB web server.

Include http=yes when you set the URL if you have enabled SSL support for CA Configuration Automation. See the following example:

```
http://hostname:port/{path}?{parameters}+https=yes
```

For information about enabling CA Configuration Automation HTTPS support, see the CA Configuration Automation online help.

MDR Launch URL

The MDR launch URL has the following default value:

```
http://{hostname}:{port}/{path}?{parameters}
```

You can modify this expression to accommodate any MDR-specific requirements. The URL is required for "launch in context."

- For mdr_name of APM or ITAM and mdr_class of GLOBAL, the default value is:
  ```
  http://{hostname}:{port}/{path}?{parameters}
  ```
- For mdr_class of Cohesion, the default value is:
  ```
  http://cmdb_hostname:cmdb_port/{path}?{parameters}
  ```
  – ***cmdb_hostname***
  The cmdb_hostname variable specifies the host name, DNS name, or IP address of the CMDB web server. The variable defaults to the host name currently accessing the CMDB web server.
  – ***cmdb_port***
  The cmdb_port variable specifies the TCP/IP port of the CMDB web server. The variable defaults to the current port number that is used to access the CMDB web server.

  Set the URL to include https=yes in the query string, if you have enabled SSL support for CA Configuration Automation. See the following example:
  ```
  http://hostname:port/{path}?{parameters}+https=yes
  ```
  For information about enabling CA Configuration Automation HTTPS support, see the Cohesion online help.

Defining Launch Parameters for URL Substitution

When defining an MDR, the following parameters can be used to construct its URL for display. These parameters are substituted with their appropriate values at run time. These variables must be specified in the fields that were described previously.

hostname is the MDR host name from the MDR definition.

alarm_id is the IP address of the selected CI.

federated_asset_ID is the unique identifier of the selected CI in the MDR.

cmdb_asset_id is the asset ID for the CI.

port is the MDR port number from the MDR definition.

userid is the user ID from the MDR definition. If blank, userid defaults to whomever is currently signed on.

password is the shared secret from the MDR definition.

mdr_name is the mdr_name from the MDR definition.

mdr_class is the mdr_class from the MDR definition.

class is the class of the selected CI.

family is the family of the selected CI.

path is the path as described in the MDR definition.

name is the name of the selected CI.

model is the model of the selected CI.

manufacturer is the manufacturer of the selected CI.

itam_ticketid is the ticket id to log in to CA APM.

Example: Launching an MDR

A CMDB user is looking at a Server CI named server1. Server1 has a map to an internally developed application that is named Comet. Comet uniquely identifies server1 as server:server1.

Comet is defined as an MDR with the following properties:

- Hostname: CometServer
- Port: 80
- Path: index.php
- Parameters: item={federated_asset_id}
- Launch_url: http://{hostname}:{port}/{path}?{parameters}

In CMDB, when the user clicks Comet on the Attributes tab of the server1 CI, a web browser opens the following URL:

http://CometServer:80/index.php?item=server:server1

Parameters for URL Substitution

When defining an MDR, you can use the following parameters to construct its URL for display. These parameters are substituted with their appropriate values at runtime. These variables must be specified in the MDR field definitions.

- **{hostname}**
  Specifies the MDR host name from the MDR definition.
- **{alarm_id}**
  Specifies the IP address of the selected CI.
- **{federated_asset_ID}**
  Specifies the unique identifier of the selected CI in the MDR.
- **{cmdb_asset_id}**
  Specifies the asset ID for the CI.
- **{port}**
  Specifies the MDR port number from the MDR definition.
- **{userid}**
  Specifies the user ID from the MDR definition. If blank, userid defaults to whomever is currently signed on.
- **{password}**
  Specifies the shared secret from the MDR definition.
- **{mdr_name}**
  Specifies the mdr_name from the MDR definition.
- **{mdr_class}**
  Specifies the mdr_class from the MDR definition.
- **{class}**
  Specifies the class of the selected CI.
- **{family}**
  Specifies the family of the selected CI.
- **{path}**
  Specifies the path as described in the MDR definition.
- **{name}**
  Specifies the name of the selected CI.
- **{model}**

Specifies the model of the selected CI

- **{manufacturer}**
  Specifies the manufacturer of the selected CI

Example: Use Parameters for URL Substitution

A CMDB user is looking at a Server CI named server1. The Server CI has a map to an internally developed application that is known as Comet. Comet uniquely identifies server1 as server:server1.

Comet is defined as an MDR with the following properties:

- Hostname: CometServer
- Port: 80
- Path: index.php
- Parameters: item={federated_asset_id}
- Launch_url: http://{hostname}:{port}/{path}?{parameters}

In CMDB, when the user clicks Attributes, Comet in the server1 CI, a web browser opens the following URL:

```
http://CometServer:80/index.php?item=server:server1
```

Federation Using GRLoader

When using GRLoader, the following XML tags must be populated in every CI in the XML document. These tags apply to every MDR family.

- <mdr_name>
- <mdr_class>
- <federated_asset_id>

> **NOTE**
>
> CA Configuration Automation automatically provides mdr_name, mdr_class and federated_asset_id.

if the XML tags are missing, “launch in context” is not possible. This is because, the origin of the CI cannot be determined.

To identify the source of a CI, you can modify the XML before it is input into GRLoader. You can use a text editor to modify the XML and make a global change. You can also program this task.

For more information about MDR identification and GRLoader, see the GRLoader XML topic.

Federate a CI

If CIs are loaded into CMDB before their corresponding MDR is defined, they are unfederated. Unfederated means that the CIs are not yet connected to an MDR and they do not yet support “launch in context”.

Follow these steps:

1. Define the required MDR.
2. Do one of the following tasks:
   – Manually map the CI.
   – Rerun the CA Configuration Automation report which created the CI, specifying Allow update of existing CIs on the report. For more information about CA Configuration Automation Reports, see the *CA Configuration Automation Product documentation*.
3. Rerun GRLoader, specifying the same input file that was used to create the CIs.
   The CMDB reconciliation engine merges the MDR information into the existing CIs.
4. Create an XML document that describes the CI and its MDR and run GRLoader in Update mode.
   The CMDB reconciliation engine merges the new information into the existing CI. The existing CI is federated.

Example: Specify CI Location

To locate the CI you want to update, verify that the reconciliation engine is provided with enough information. In the following example, end tags are removed and spaces added for readability.

```
<ci>
   <name>         server3
   <mac_address>
   <serial_number>
   <asset_num>
   <dns_name>
   <mdr_name>     mdr_one
   <mdr_class>    Cohesion
</ci>
```

Define Multiple MDRs to a CI Using GRLoader

You can define multiple MDRs to a CI by using GRLoader.

To define multiple MDRs to a single CI, the XML document can repeat the <ci> node. With each duplicated <ci> mode, specify a different mdr_name and mdr_class. In other words, each MDR can contribute its attributes independently of any other MDR that contributes data to the CI.

Example: Define Multiple MDRs to a CI

If MDR1 and MDR2 both contribute data to the server2 CI, the XML document looks something like the following example. In the example, end tags are removed and spaces added for readability.

```
<ci>
   <name>      server2
   <mdr_name>  mdr1
   <mdr_class> Cohesion
   <diskspace> 500 gb
   <disktype>  SCSI-3
</ci>
<ci>
   <name>     server2
   <mdr_name> mdr2
   <mdr_class>Service Assure
   <sla>
</ci>
```

CMDB reconciles the two previous CIs to the same CI and associates each MDR to that single CI.

> **NOTE**
> The CIs can be imported in one or two runs of GRloader.

Map Between MDR CIs and CMDB CIs

After you define a CI manually using the File, New Configuration Item... option, manually define the mapping between this CI and the CI in the federated MDR. Associate a CI with an MDR in the following ways:

- Editing the CI.
- Using the Federated CI Mapping node on the CMDB Administration tab.

Follow these steps:

1. To create a mapping by editing the CI, complete the following steps:
   a. Display the CI Detail page of the CI that you want to associate with an MDR.

b. Click Edit.
c. Display the Attributes tab.
d. Click Add MDR.
The CI is associated with the MDR.

2. To create a mapping using the Federated CI Mapping page, complete the following steps:
a. Click the CMDB Administration tab.
b. Open the Management Data Repository node.
c. Select the Federated CI Mapping node.
The Federated CI Mapping List appears.
d. Click Create New.
The Create New Federated CI Mapping For page appears.
e. Associate the CI with the MDR by completing the Federated CI Mapping fields:
- **CI Name**
Specifies the name to use to identify the configuration item.
- **Federated Asset ID**
Specifies the string identifier that is used by the source MDR to identify this CI. The MDR software determines the identifier.
- **MDR Name**
Specifies the name that identifies the MDR (and its MDR button).
- **Active**
Denotes whether this mapping is active or not. Mappings cannot be deleted, only inactivated.
f. Click Save.
The mapping between this CI and the CI in the federated MDR is defined.

How To Configure MDRs for CMDBf Viewer

To use the CMDBf Viewer, set up your federated MDR providers to point to the CMDBf query service, as follows:

- External MDRs must provide a query service that can handle InstanceIdConstraint query.
- Button Name, MDR Name, and MDR Class are required to show the CMDBf Viewer button on the CI Detail page.
- MDR Class must be defined as **cmdbf**
- For CMDB, CMDBf Namespace must be set to **http://cmdb.ca.com/r1**. For other CMDBs and MDRs, see the appropriate documentation.
- Timeout is optional. Default is ten (10) seconds.
- To display a working Retrieve button on the Federated View, define CMDBf Endpoint, Userid, and Shared Secret.

NOTE

An existing CMDB system can be set up as a CMDBf provider by specifying a CMDBf Endpoint of "http://servername:port/axis/services/QueryPort" **OR** "http://servername:port/cxf/services/QueryPort" where:

- hostname is the computer where CMDB is installed (localhost or computer name).
- port is the port where CMDB is configured.

Launching the MDR Web Browser Interface

After a mapping between a CI and an MDR is created, a button is placed automatically on the Attributes tab. If multiple MDRs are associated with this CI, multiple buttons appear.

When you click an MDR button, a new page opens. The fully substituted MDR URL that is defined in the MDR definition appears on the page.

Example: Launch CA Cohesion

When a CI has been correctly associated with its MDR, a Cohesion button appears on the Attributes tab. If a button does not appear on the Attributes tab, review the mapping for the displayed CI. Verify that there is a mapping for this CI. Also verify that the target MDR has a URL that can be launched. Clicking the button to launch the MDR causes a new window to open. The window opens to the target URL to launch CA Cohesion.

CA Cohesion Integration

Consider the following items when integrating CA Configuration Automation with CMDB:

- Integrating Cohesion with CMDB

  NOTE
  For information about Cohesion-CMDB integration, see the *CA Configuration Automation Implementation documentation*.
- Importing CIs from a Cohesion MDR
  For information about how to import CIs from a Cohesion MDR, see the help in the CA Configuration Automation Report Templates tab.
- Launch-in-Context for Cohesion MDRs
  For launch-in-context integration to work best with CA Configuration Automation, we recommend that you define the Cohesion MDR *before* running the Cohesion CMDB Report. You can define Cohesion MDR in the CMDB Administration tab.

  NOTE
  CA Configuration Automation does not support a unique Federated asset ID for NIC or File System CIs. Therefore, Cohesion does not support MDR Launcher for NIC or File System CIs. As a result, a Cohesion-based NIC or a File System CI does not display an MDR launch button even when it was imported successfully.

Reconcile CI Ambiguities Using MDR

Reconciliation ensures that updates from multiple data sources that refer to the same business object only update a single CI. A single CI is updated even if the data sources possess different identifying information.

Ambiguity represents the possibility that a CI is not unique. CIs are *ambiguous* when they represent the same real business object. The CI transactions are ambiguous when they have more than one possible target CI.

Ambiguous CIs can lead to incorrect data in the CMDB, which negates the value of the CMDB. A negated value can lead to incorrect business actions.

Automatic CI reconciliation is a key CMDB advantage that saves significant time, when compared with manual data maintenance. The process of reconciling CIs uses several specific identifying attributes. However, automatic reconciliation can result in the following matches:

- False positive matches
  Existing CIs are updated instead of creating a CI.
- False negative matches
  New CIs are created instead of updating an existing CI. The set of CIs with similar identifying attributes are ambiguous, because they resemble the same real business object with similar identifying attributes.

CA Service Desk Manager supports the following reconciliation approaches:

- **MDR-Based Reconciliation (Passive)**
  Allow the CMDB to reconcile any ambiguous data based on the MDR-Based Reconciliation process.
- **Identify and Resolve ambiguous CIs (Reactive)**
  Identify and resolve ambiguous CIs through identification and management of existing CIs in the CMDB.
- **Review and modify inbound data using a transaction work area (Proactive)**

Review and modify inbound data before loading into the CMDB using a transaction work area (TWA).

How to Identify and Resolve Ambiguous CIs

Contents

We recommend managing the ambiguity for a CI by reviewing each CI. In each case, the Configuration Administrator must detect when the ambiguity exists and must determine the optimal approach. CA SDM takes a broad approach to identify and manage ambiguous CIs that are already in the CMDB.

The ambiguity index is an operational measure of the potential non-uniqueness of a configuration item (CI) or a CI transaction item, which is based on its identifying attributes. The ambiguity index measures the probability of one or more CIs representing the same real business object. Or, it measures the probability that a CI transaction has more than one possible target CI.

> **WARNING**
> If you manually delete CIs with database queries, errors can occur for CIs ambiguity indexes. To avoid these errors, execute the cmdb_update_ambiguity utility and set the -full parameter to 1. This parameter ensures that you receive the accurate ambiguity indexes when you execute cmdb_update_ambiguity.bat or the shell script.

CI Ambiguity Example

Data from four different data sources is loaded into the CMDB. Each data source uses its own subset of identifying characteristics. Because of this inconsistency, more CIs exist in the CMDB than are desired.

Example: Ambiguous CIs

The following four CIs reside in the CMDB:

- Name(Server1) DNS Name(dns1) Serial Number(serial1)
- Name(Server1) DNS Name(dns1)
- Name(Server2) DNS Name(dns1) Serial Number(serial1)
- Name(Server3) MAC Address(mac1)

Due to shared identifying characteristics, the two instances of Server1 and Server2 are ambiguous with each other. Server3 is not ambiguous.

Every CI has an ambiguity index that is associated with it. The ambiguity index is approximately the number of existing CIs that match on any of the identifying attributes. The greater the index, the more CIs that match on the identifiers Therefore, the greater the probability that CI data was entered inconsistently and that more CIs are incorrectly created. CIs with an ambiguity index of zero are unique across all identifiers and are therefore unambiguous.

The ambiguity index of each of the previous CIs is

- First Instance for Server1: Count of other CIs matching name + dnsname + serial number = 1+2+ 1=4
- Second Instance of Server1: Count of other CIs matching name + dnsname = 1+2 = 3
- Server2: Count of other CIs matching name + dnsname + serial number = 0 +2+1 = 3
- Server3; Count of other CIs matching name + mac address = 0+0 = 0

Follow these steps:

1. Identify ambiguous CIs.
   - Calculate the ambiguity index for all CIs.
   - Review the list of ambiguous CIs from the Scoreboard.
2. Determine if the identifying attributes for each CI are valid.
3. Resolve the ambiguous CI with one of the following actions:

- Modify the identifying attributes.
- Exclude a CI from ambiguity management.
- Reject a CI Update by inactivating the CI.
- Supersede a CI.

Identify Ambiguous CIs

Investigate any CI with a nonzero ambiguity index to determine if it is unique or it was inadvertently created. Ambiguous CIs can be created because of incorrect reconciliation. CIs are not ambiguous by themselves, they are ambiguous with other CIs. A system can have many sets of ambiguous CIs, each set containing CIs with common values for the identifying attributes.

When you research the ambiguity of a CI, you also research the ambiguity of other CIs in the same set. When you resolve the ambiguity of a single CI, you reduce the ambiguity of other CIs in that set.

CA SDM has reconciliation management tools, which enable you to find ambiguous CIs. The tools also help you find the set of CIs of which the ambiguous CI is a member. You can research the underlying cause of the ambiguity and can resolve it.

When managing ambiguity, follow these steps:

1. Calculate the ambiguity index for all CIs.
2. To view the list of ambiguous CIs, use the scoreboard. The scoreboard lists all CIs that are ambiguous, in descending order of ambiguity.
3. Starting with the CI that has the highest level of ambiguity in the scoreboard, inspect all CIs in its ambiguity set. The reconciliation tab on CI detail form of a single CI lists all other CIs in the ambiguity set.
   a. Determine if all CIs in the set are legitimate or if an error was made in reconciliation. Review the identifying attributes. Determine if the CIs in the set are created correctly or are created due to a false negative reconciliation match.
   b. Determine which CIs in the set, if any, are false negatives.
   When you determine that a false negative reconciliation match has occurred, and more CIs are created, determine the valid CIs and the incorrectly-created CIs. Consider factors such as identifying attributes and attributes such as last update date, related problems, and issues.

Calculate the Ambiguity Index

Before you can begin managing ambiguity, update the ambiguity index for the existing CIs and CI Transactions. You update the ambiguity index for CIs and CI transactions by running the cmdb_update_ambiguity command.

> **WARNING**
> Run cmdb_update_ambiguity at least once to measure CI or CI transaction ambiguity. If you do not run the command, all ambiguity indexes are 0 (zero). An ambiguity index of zero implies no ambiguity.

> **WARNING**
> You can run the utility before and during ambiguity management. Schedule the utility to run periodically, so the ambiguity indexes reflect the current state of your system.

Follow these steps:

1. Determine the necessary parameters to run the utility.
2. Run the utility.
3. Start the CA SDM Web Client and navigate to the Ambiguous CI or Ambiguous CI Transaction Lists.

cmdb_update_ambiguity Command

You can calculate CI and CI transaction ambiguity indexes by entering a command syntax similar to the following command:

```
cmdb_update_ambiguity [parameters]  - m { ci | twa | all }
```

For command parameters, see cmdb_update_ambiguity Parameters.

By default, the CI scan is done from the last scan date. If "-full 1" is specified, a complete scan is performed.

Example: Calculate CI ambiguity on a Microsoft SQL Server database

The following command calculates the ambiguity index for all existing CIs and Transactions in the TWA:

```
cmdb_update_ambiguity  - m all  - d MSSQL  - u servicedesk  - p dbpassword -s dbserver1
```

Example: Calculate CI ambiguity on an Oracle database

The following command calculates the ambiguity index for CIs and specifies database information.

```
cmdb_update_ambiguity  - m ci  - d Oracle  - u mdbadmin  - p dbpassword -s server1 -port 1521  - SID orcl
```

Use a Configuration File

You can specify many cmdb_update_ambiguity parameters in a configuration file. You can use the configuration file to secure parameter settings in an encrypted form using operating system tools. The valid keywords and the corresponding command-line options are listed in the parameter table.

> **NOTE**
> If you specify a parameter both on the command line and in a configuration file, the command-line value overrides the configuration file value.

Example: Use a configuration file to specify the parameters for a Microsoft SQL Server database

The following command runs the configuration file ambiguity_mssql.cfg.

```
cmdb_update_ambiguity  - m all -c ambiguity_mssql.cfg
```

Configuration File Format

Configuration file options are specified as *keyword=value*. On Windows, the directory separator can be a double backslash (\\) or a single forward slash (/). The path name must not be enclosed in double quotation marks ("").

The valid keywords and the corresponding command-line options are listed in the parameter table.

> **NOTE**
> A hash mark (#) in column 1 starts a comment line.

Example: Microsoft SQL Server configuration settings

```
#Sample configuration file for Microsoft SQL Server
DBType=MSSQL
DBUser=servicedesk
DBPassword=dbpassword
DBHost=dbserver1
LogLocation=C:\\Program Files\\CA\\Service Desk Manager\\log
LogLevel=ERROR
SchemaName=dbo
```

Example: Oracle configuration settings

```
#Sample configuration file for Oracle
DBType=Oracle
DBUser=mdbadmin
DBPassword=dbpassword
DBHost=dbserver1
LogLocation=/tmp/ambiguity/log
LogLevel=INFO
DBPort=1521
DBSID=orcl
SchemaName=mdbadmin
```

cmdb_update_ambiguity Parameters

You can specify parameters on the command line or in a configuration file (some parameters are command-line only). On the command line, use quotation marks ("") to enclose any path name with spaces. In the configuration file, do not use quotation marks. If any parameter is specified on the command line and in the configuration file, the command-line value overrides the configuration file value.

The cmdb_update_ambiguity command takes the following parameters:

Option	Configuration File Keyword	Values	Notes
-m	(none)	twa, ci, all	(Required) Command line only twa = compute ambiguity on TWA only. ci = compute ambiguity for CIs only. all = compute ambiguity for CIs and TWA.
-d	DBType	MSSQL or Oracle	(Required. Windows only.) Database type. On Linux/UNIX, only Oracle is supported and this option is not needed.
-u	DBUser	<db user name>	(Required) Database user name. A user name with spaces must be enclosed in double quotation marks (for example: -u "sys as sysdba"). The quotation marks are not required in the configuration file.
-p	DBPassword	<db password>	(Required) Password for database user. A password with spaces must be enclosed in double quotation marks (for example: -p "secret code"). The quotation marks are not required in the configuration file.

-c	(none)	<configuration file>	(Optional) Command line only The full pathname of the configuration file. The pathname must be enclosed in double quotes if there is a space in the pathname.
-log	LogLocation	<directory to place log file>	(Optional) Log file directory. The Default is the NX_ROOT\log directory.
-level	LogLevel	INFO, ERROR, DEBUG	(Optional) Level of detail to write to the log file. Default value is ERROR.
-s	DBHost	<server name>	(Required) Database server host name. To use a Microsoft SQL Server named instance, specify *server**instance* on the command line, or *server**instance* in the configuration file.
-CI	CI	<CI uuid>	(Optional) Compute ambiguity for the specified CI and all CIs that are ambiguous with it.
-full	(none)	0,1	(Optional) Command line only Optimizes the performance of the scan by only considering those CIs updated since the last time the utility was run. If set to 1, forces a full scan of all CIs in the computation of the ambiguity index. The default is 0. This parameter does not apply to the calculation of transaction ambiguity. The utility always evaluates all transactions in the TWA.
-port	DBPort	<port number>	(Required. Oracle only) Oracle port number
-SID	DBSID	<SID name>	(Required. Oracle only) Oracle SID name
-h	(none)		(Optional) Prints help usage message.
-schema	SchemaName	<db schema name>	(Optional) Default is mdbadmin for Oracle; or dbo for Microsoft SQL Server.

Resolve Ambiguous CIs

After you identify ambiguous CIs and determine if their identifying attributes are valid, resolve the ambiguity among the CIs in the ambiguity set. Use one or more of the following ways:

- **Modify the identifying attributes**

If the identifying attributes are not complete or invalid, set the CI identifying attributes so that the CI is unique. Use either the Web interface, GRLoader, or CMDBf.

NOTE
When the MDR updates the CI, the MDR can undo the manual reconciliation changes.

- **Exclude a CI from ambiguity management**
  If the CIs identifying characteristics are correct and represent a known, valid ambiguity, remove the CI from the ambiguity management lists and the ambiguity calculation of other CIs. You can mark the CI as not ambiguous (exclude ambiguity).
- **Reject a CI Update by inactivating the CI**
  When you determine that the identifying attributes of a CI are incorrect and that updates using those attributes cause data corruption, you can inactivate the CI and prevent further updates. The user or MDR generating the information receives an error and the entire transaction is rejected.
- **Supersede a CI**
  Sometimes the updates to the CMDB are beyond the control of the administrator. Invalid identifying data must be in the system with valid nonidentifying attribute data. The incoming transaction attribute data can be transparently redirected to a superseding CI.

  NOTE
  The transparent redirection of attribute data from one CI to another can cause confusion. This is because transaction data may not be stored in the same CI as the transactions identifying attributes would lead you to believe. Use this method with discretion, when the previous methods cannot be used.

Exclude CIs from Ambiguity Management

Sometimes you recognize that although a particular CI receives an ambiguity index greater than zero, it must be left as is. Any CI with its Exclude Ambiguity check box selected is not considered part of the ambiguity index or ambiguity management features.

Exclude Ambiguity for a CI

You can update the exclude ambiguity option for a CI from the web interface.

Follow these steps:

1. Select the CI that you want to remove from ambiguity management from the Ambiguous CI List page.
   The Configuration Item Detail page appears.
2. Click Edit.
   The Update Configuration Item page appears.
3. Select the Exclude Ambiguity check box on the Reconciliation tab, and click Save.
   The CI is excluded from the ambiguity index calculation and other ambiguity management features.

Exclude Ambiguity Using GRLoader

You can update the exclude ambiguity option for a CI by using GRLoader to set the not_ambiguous flag.

not_ambiguous values are as follows:

- **YES (1)**
  Removes the CI from ambiguity management. The CI is identified as unique regardless of the identifying attributes of other CIs. The ambiguity index of the CI remains zero.
- **NO (0) (default)**
  This CI is eligible for ambiguity management. The uniqueness of the CIs identifying attributes determines the ambiguity index of this CI. The identifying attributes of this CI are considered when evaluating the ambiguity index of other CIs.

NOTE
For more information about GR Loader, see theCMDB Technical Reference.

Reject Updates

When a CI is marked as Inactive, no updates can be performed on it. Set the CI to Inactive if you want CA CMDB to reject data for a given CI and you want the MDR to know of the rejection.

In this way, you can reject problematic data immediately and can reflect it back to the source. The MDR can correct the rejected input at the source.

To set a CI to Inactive in the web interface, edit the CI, set it to Inactive, and click Save. You also can set a CI to Inactive by using GRLoader or CMDBf web services. Inactivated CIs can also be reactivated.

Supersede Ambiguous CIs

You can supersede an ambiguous CI to redirect data to a specific target CI. You can display superseded CIs, which are Inactive and all web interface updates to them are ignored.

NOTE
Updates that are sent to superseded CIs by GRLoader or CMDBf web services are redirected to the superseding CI. However, updates to identifying attributes are ignored.

Follow these steps:

1. Select the CI that you want to be the focal (superseding) CI from the Ambiguous CI List page.
   The Configuration Item Detail page appears.
2. Click Edit.
   The Update Configuration Item page appears.
3. Click the Reconciliation tab.
   All CIs that are ambiguous with the focal CI display.
   Determine which CIs you want to supersede by the focal CI. Inspect each CI in the list by clicking each CI to launch its Configuration Item Detail page.
4. Select one or more ambiguous CIs that you want to supersede with the focal CI and click Supersede.
   The focal CI supersedes the selected CIs.

MDR-Based Reconciliation

This article contains the following topics:

MDR-based reconciliation is performed at the management data repository. This helps reduce the occurrence of multiple CIs that refer to the same object in the physical world.

MDR-based reconciliation treats the MDR as a trusted source that always uses the same federated asset ID when it communicates information about a single CI. All updates from a given MDR to a given federated asset ID always update the same CI, even when identifying attributes are changed.

MDR-based reconciliation, reconciliation management, and the Transaction Work Area (TWA) described in these sections help you take control of the reconciliation process. However, to use reconciliation management and the TWA successfully, first understand how CA SDM uses reconciliation attributes.

WARNING
If you reinstall or reinitialize any external data provider, inactivate and reactivate its MDR definition in the CMDB. If the MDR reuses its federated asset IDs, inadvertent CI data overlay can occur.

How MDR Reconciliation Matches CIs

MDR-based reconciliation uses the following process to identify the appropriate matching CI:

1. If the transaction specifies an ID, the CI is identified and reconciliation is complete.
2. If the transaction does not specify an ID, CA SDM checks whether federated identification attributes are specified and match a CI. If there is a match, the transaction reconciles to the matching CI.
3. If the transaction does not specify an ID or federated identifying attributes, the CI transaction uses the identification attributes.

How MDR Reconciliation Identifies CIs

MDR-based reconciliation uses the following precedence to identify a CI:

1. ID (if a transaction specifies an ID, reconciliation is successful)
2. Federated identification attributes
   - Federated asset ID
   - MDR name
   - MDR class
3. CI identifying attributes
   - Tenant (if multi-tenancy is installed)
   - Name
   - Serial number
   - MAC address
   - System name
   - Alternate asset ID
   - DNS name

Review and Modify Inbound Data Using Transaction Work Area (TWA)

You can stage CI and relationship transactions before execution, by copying data into the TWA staging area. Once in the staging area, you can manipulate CIs and relationships by using the web interface or native SQL.

You can also validate the CI transactions to prevent the creation of new CIs when you update existing CIs. In this approach, you view each transaction and the potential CIs that it can update. This helps you reconcile the transaction manually to the target CI. Likewise, relationship transactions can be validated to reference the correct CIs.

For more information, see How to Identify and Resolve Ambiguous CIs.

CI Transaction Ambiguity Example

Transaction data from different data sources is loaded into CMDB. Each data source uses its own subset of identifying characteristics and may not fully identify the target CIs for the transactions. Because of this inconsistency, more CI Transactions may exist in CMDB than are valid.

Example: Ambiguous CI Transactions

The following CI transaction resides in the TWA:

- Name(Server1) DNS Name(dns1), Serial Number(serial1)

The following CIs reside in CMDB:

- Name(Server2) DNS Name(dns1)
- Name(Server3) DNS Name(dns1), Serial Number(serial1)
- Name(Server4) MAC Address(mac1), Serial Number(serial1)

Due to shared identifying characteristics, the Server1 transaction is ambiguous with Server2, Server3, and Server4 in CMDB.

Every CI Transaction has an ambiguity index that is associated with it. The ambiguity index is approximately the number of existing CIs that match on any of the identifying attributes, minus one, specified in the CI transaction. The greater the index, the greater the number of other CIs that match on the transaction identifiers. Therefore the greater the probability that CI data was entered inconsistently and the possibility that more CIs are incorrectly created. CI Transactions with an ambiguity index of zero have identifying attributes that are unique across all CIs. Or, they have a target CI specified and are therefore unambiguous.

Example: Calculate the Ambiguity Index

The following CI transactions reside in the TWA:

- Name(Server1) DNS Name(dns1), Serial Number(serial1)
- Name(Server2) DNS Name(dns1), Serial Number(serial1)

The following CIs reside in CMDB:

- Name(Server1) DNS Name(dns1), Serial Number(serial1)
- Name(Server2) DNS Name(dns1), Serial Number(serial1), Mac Address(mac1)

The first transaction (Server1) ambiguity is 0 because there is an exact match with the Server1 CIs identifying attributes. The only possible target CI to this transaction is Server1 CI.

The second transaction (Server2) is ambiguous with Server1 CI and Server2 CI.

The ambiguity index for Server2 transaction consists of the following components:

- Number of matching CIs with Name (Server2) = 1
- Number of matching CIs with DNS Name(dns1) = 2
- Number of matching CIs with Serial Number (serial1) = 2

Based on shared CI identifying characteristics, the ambiguity index for server2 transaction is (1-1) + (2-1) + (2-1) = 2.

CI Properties that Support MDR Federation

This article contains the following topics:

Configuration item properties (attributes) identify assets for MDR federation purposes.

Federated Asset ID

People are known to different organizations by different identifiers. You may have the following identifiers:

- A unique nickname to your close friends
- Driver license (unique ID associated with you)
- A national service ID (for example, a selective service card)
- A health insurance number
- A national tax identification number

Each of these unique identifiers refers to you. However, the identifier is only valid when used to describe you to the appropriate repository.

Similarly, a CI can have multiple identifiers to associate it with its source MDRs. Each CI is known to an MDR by only a single identifier. We call this identifier the *federated asset ID*. The process of associating a CI with one or more MDRs is *mapping the CI*.

Mapping occurs when CIs are loaded into the MDB in one of two ways:

- Defining the CI mapping using the Administration user interface
- Loading CIs using the GRLoader utility

MDR Name and Class

The MDR name identifies the MDR to CMDB when exporting data using XML and GRLoader. The MDR typically has its own naming convention for how it identifies itself: a combination of host server name plus an identifying instance name or number. Because only one MDR exists on a particular host, the MDR name is often set to the host server name. **Required for CMDBf Viewer.**

> **NOTE**
>
> The MDR Name for CA Asset Portfolio Management release 11.3.4 is APM and the MDR Name for CA APM r12.6 is ITAM. Both products are supported. However, we recommend that you verify product availability at CA Support Onlinebefore implementing the products

The customer defines the MDR class to group MDRs.

> **NOTE**
> An MDR Class of CMDBf is required for CMDBf viewing.
>
> **WARNING**
> The MDR Name plus the MDR Class must be unique within your enterprise.

MDR Definition with CA Configuration Automation Installation

CA Configuration Automation MDRs have the following requirements:

- The mdr_name specified in the MDR definition on the CMDB server must match exactly the value of the attribute com.cendura.installation.name in the cendura.properties file on the target CA Configuration Automation server.
- CA Configuration Automation MDRs must have an MDR class of Cohesion.
- The MDR must specify the hostname and port number of the CA Configuration Automation server.

To run MDR Launcher, edit the following portion of the cendura.properties file:

```
# -- Configure One-Click Authentication --
com.cendura.security.oneclickauth.secret=shared_secret
com.cendura.security.oneclickauth.scheme=
com.cendura.security.oneclickauth.user=userid
```

> **WARNING**
> The secret that is specified in the cendura.properties file must match the *shared secret* in the MDR definition.

The MDR Launcher logs in as the *userid* specified in the properties file and inherits its security attributes. To use functionality such as Refresh for CI attributes, verify that this user has sufficient privileges. For more information about creating a user, setting security options, and modifying the properties file, see the *CA Configuration Automation Implementation documentation.*

CA Configuration Automation MDRs

This article contains the following topics:

CA Configuration Automation MDRs must be defined before importing data from a CA Configuration Automation server. A CA Configuration Automation MDR must specify an MDR Class of Cohesion.

Example: Cohesion1 MDR Definition

In the following Cohesion1 MDR definition, the XML specifies an MDR Name of cohesion_server and an MDR Class of Cohesion. These values are required for the CIs to be imported.

- **Button Name**
  Cohesion1
- **MDR Name**
  cohesion_server
- **MDR Class**
  Cohesion
- **Active?**
  Active
- **Owner**
  CMDBAdmin
- **Description**
  CA Configuration Automation server in Chicago
- **Hostname**
  cohesion_server
- **Port**
  8090
- **Path**
  CAisd/html/cmdb_cohesion.html
- **Parameters**
  hostname={hostname}+port={port}+family={family}+name={name}+secret={password}+federated_asset_id={federated_asset_i
- **userid**
  cohesion_userid
- **Shared Secret**
  Chicago01
- **URL to launch in Context**
  http://cmdb_hostname:8080/{path}?{parameters}

In addition, because this is a Cendura server, the values that are listed previously must match the values in the cendura.properties file on that server. See the following example:

```
com.cendura.security.oneclickauth.secret=Chicago01
com.cendura.installation.name=cohesion_server
```

You can modify the URL syntax to handle special requirements.

How to Associate an MDR to a CI Manually

You can associate MDRs with a CI manually. Use the Federated CI Mapping facility in the CMDB Administration tab, in the MDR Management branch of the tree.

Before you associate a CI to an MDR, complete the following tasks:

1. Create the MDR definition (if it does not exist).
2. Create the CI definition (if it does not exist).
3. Identify the unique Federated Asset ID of the CI that you want to connect to the MDR. This ID is MDR-specific, so it is beyond the scope of this document.

   NOTE
   To identify Federated Asset ID, consult your MDR documentation.

CA Cohesion Automatic Import

You can import CIs directly from CA Configuration Automation. Run a CA Configuration Automation report specifying the host name, port, user ID, and password to a CMDB server. If the MDR is defined in CMDB, CA Configuration Automation automatically generates the XML. The XML includes CIs and relationships, with the information necessary to perform the MDR Launch on a CI. The report automatically imports the CIs into CMDB. For more information about exporting CIs from CA Configuration Automation, see the online help from the Reports, Report Templates tab.

CI to MDR Mapping

Because each MDR uses a federated_asset_id to identify a CI, one CI can be related to multiple MDRs. A federated_asset_id does not have to be unique across MDRs, but a federated_asset_id must be unique within an MDR. Each MDR must have a unique MDR class and MDR name.

> **WARNING**
> Whenever a CI or an MDR provider is made Inactive, all Federated CI Mappings that are associated with the CI or the MDR provider are also made Inactive.

After you create an MDR Data Provider definition, complete the following tasks:

1. Create a CI in the CMDB that references an MDR.
2. Verify that the MDR definition works.
   Because you can only launch from within the context of a CI, it is not possible to test directly from the MDR definition, which has no CI context.

You can view the Federated CI Mapping List in the Administration tab, in the Federated CI Mapping node.

To view the CI in a particular MDR context, click an MDR Launch button.

The target MDR is launched in the context of the CI that was opened.

Example: CI Mapping

1. Click the Administration tab.
2. Navigate to MDR Management, Federated CI Mapping.
   The Federated CI Mapping List appears.
3. Enter server1 in the CI Name field.
   The following columns show the values:
   - **Federated Asset ID**
     1000234
     1000235
   - **CI Name**
     server1
     server1
   - **MDR Provider**
     Cohesion1
     Cohesion2
   - **Active**
     Active
     Active

   The Cohesion1 MDR Provider and the Cohesion2 MDR Provider know the existence of server1. The example shows that they each have independently assigned the server a unique ID.
4. Click the CI Name server1.
   The server1 Configuration Item Detail page appears that includes launch buttons would be named Cohesion1 and Cohesion2.
5. Click either MDR Provider launch button.

More details about the CI appear.

MDR Definition Administration

Administering the MDR definitions is a flexible process. You can modify the parameters in an MDR definition even when CIs reference it. For example, you can modify the button name, host name, user ID, and shared secret, after the MDR is defined and the CIs are loaded.

CA Configuration Automation Report

CA Configuration Automation provides a facility for scheduling recurring reports. Use this facility to simplify the process of keeping the CMDB synchronized with the data in the CA Configuration Automation MDR. Common errors such as an invalid password (due to password change or expiration) can prevent successful importing of data. You can choose to receive notification of any errors that occur during background data execution of the data import. Enable the Notification option in the CMDB Export report in CA Configuration Automation. You will receive an email when an import error occurs. If you run the Cohesion report in the background as a scheduled task, we recommend that you enable the Notification option. For information about scheduling execution of the CMDB Export report regularly, see the *CA Configuration Automation Product Documentation*.

How to Deploy CMDBf Web Services for Apache Axis

After you install CA SDM, you can deploy CMDBf web services for Apache Axis.

Follow these steps:

1. Verify that the web server is up and running.
2. Navigate to the CMDBHOME\sdk\websvc\CMDBF directory.
3. Run deploy_cmdbws.bat.
   The CMDBf web services are deployed and started. For more information about CMDBf web services deployment, see the Web Services Deployment topic.

Using GRLoader

When using GRLoader to load a CI, populate the following fields in the XML for the MDR Launch to operate:

CA

- <mdr_class>
- <mdr_name>
- <federated_asset_id>

The values that you provide for <mdr_name> and <mdr_class> in the XML must match the values in the MDR definition exactly.

> **WARNING**
> The MDR name and class must be defined using the Administration interface before importing the CIs that reference the MDR. If the MDR specified in the XML is not defined, the CI is not imported.

The GRLoader supports importing CIs and CI relationships from spreadsheets in the XLS and XLSX formats. To load CI data into CA SDM, format the source data into XML or Microsoft Excel spreadsheets. For more information about using GRLoader to load spreadsheet data, see the How to Prepare for Loading Spreadsheet Data topic.

system_name Naming Convention

We recommend the following naming standards for software CIs and all MDRs. These naming standards facilitate the integration of MDRs with CA Configuration Automation and other MDRs. The standards are also helpful for reconciling CIs properly. CA Configuration Automation follows the same standards.

- **System_name**
  Identifies software CIs uniquely. When you define a relationship that involves a software CI, specify the same system_name as the one in that CI definition. If multiple instances of the same version of a software are installed on the same hostname, modify the system_name. Modifying the system_name enforces uniqueness. The total length of the system_name must not exceed 255 characters. If you ignore this restriction, data corruption can occur. System_name must be a unique identifier for this instance of the software on a single host.

Use the following syntax:

```
hostname | softwarename | version | business-application
```

- **pipe (|) character**
  Separates the various fields in the concatenation of the syntax to let the user use the search facility.
- ***hostname***
  Specifies the hostname that contains the software.
- ***softwarename***
  Specifies a common name for the software.
- ***version***
  Specifies the version number of the software if available.
- ***business-application*** Specifies a unique identifier for this instance of the software on *hostname.* If the instance is associated with a business application or service, the name of that service is the qualifier. When you cannot determine *business-application*, you can use the installation directory to identify the software. If the total length of this field exceeds 255 characters, use ellipses (…) to shorten the total length of this field.

Examples: Use the UI Search Facility

You can use the UI search facility to search software CIs, as in the following examples:

Use case	Name	System_name
Find all software CIs on host	xxx	Xxx%
Find all instances of software	yyy	yyy%
Find all instances of software	yyy version 123.0	yyy% %|%|123.0%

In the results list that the search returns, the user sees only the Name field.

```
mac_address     Null.  - Inappropriate for software
asset_num       Null   - Inappropriate for software
serial_number   Null   - Inappropriate for software
dns_name        Null   - Inappropriate for software
```

How to Update Metadata Files for CMDBf Mapping

This article contains the following topics:

Use CA CMDB metadata files to translate between MDR attribute names and CMDB attribute names. Using CMDB metadata files improves readability of attribute comparisons. For any MDR attribute without CMDB mapping, the Federated View displays the attribute name. The attribute name is sent by the MDR. Metadata can be defined for CMDBf data providers to do the following tasks:

- Display MDR attribute values using CMDB attribute names.
- Prevent MDR provider attributes from being displayed in Federated View.
- Define MDR provider attributes that do not have CMDB equivalents.

You define metadata using the cmdb_metadata_federation_viewer_site_attr.htmpl file. This file contains instructions on how to update the file. Metadata can apply to all CMDB families (common attributes) or family-specific attributes.

To map external MDR attribute names to CMDB labels, you update the respective cmdb_metadata_*extensiontable*.htmpl form. Use the following fields in the cmdbmetadata macro:

- mdr_attr - the name of the MDR attribute to be translated.
- mdr_name - the name of the MDR being translated. Regular expressions are supported.

Example: Attribute mapping

The code in this example defines metadata that equates the CMDB "phys_mem" attribute with the provider attribute "mdr_memory" for all providers named "myMdr" or starting with "MDR". In addition, "physical_memory" is equated to with "phys_mem" for all other providers.

```
<macro name=cmdbMetadata attr="phys_mem"  provider_attr="mdr_memory" provider_name="myMdr">
<macro name=cmdbMetadata attr="phys_mem"  provider_attr="mdr_memory" provider_name_regexp="MDR.*">
<macro name=cmdbMetadata attr="phys_mem" provider_attr="physical_memory" provider_name_regexp=".*">
```

Example: Attribute hiding

The following statement hides the MDR provider "widget_cost" attribute for all providers named "myMdr".

<macro name=cmdbMetadata hide_provider_attr="YES” provider_attr="widget_cost" provider_name="myMdr">

Example: Setting an attribute label

The following statement defines an attribute name “ext_mem_capacity” using the label “External Memory Capacity” under the attributes category in the CMDBf Viewer.

```
<macro name=cmdbMetadata  attr="ext_mem_capacity" category="Attributes" heading="External Memory Capacity"
 help="Total external memory">
```

ow To Display MDR Attribute Values With CA CMDB Attribute Names

Metadata can create an association between MDR provider attributes and CMDB attributes. The association helps to display the attributes together to see differences and share labels. By default, MDR attributes that do not have a mapping are displayed as **Not in CMDB** in the viewer. cmdbMetadata macro arguments to equate a CMDB attribute with an MDR provider attribute include:

- attr - CMDB attribute name
- provider_attr - MDR provider attribute name
- provider_name - MDR provider name
- provider_name_regexp - MDR provider name regular expression

provider_name or provider_name_regexp are required.

Example: Associate MDR Attributes with CMDB Attribute Names

The following three metadata statements do these respective actions:

- Equate the CMDB "phys_mem" attribute with the provider attribute "mdr_memory" for all providers named "myMdr".
- Equate the CMDB "phys_mem" attribute with the provider attribute "mdr_memory" for all provider names starting with "MDR".
- Equate "physical_memory" with "phys_mem" for all other providers.

```
<macro name=cmdbMetadata attr="phys_mem"  provider_attr="mdr_memory" provider_name="myMdr">
```

```
<macro name=cmdbMetadata attr="phys_mem"  provider_attr="mdr_memory" provider_name_regexp="MDR.*">
<macro name=cmdbMetadata attr="phys_mem"  provider_attr="physical_memory" provider_name_regexp=".*">
```

Hide MDR Provider Attributes

Some MDR provider attributes do not need to appear in the Federated View. You can hide metadata for a particular MDR provider. This option only applies for MDR provider attributes and does not apply for CMDB attributes.

cmdbMetadata macro arguments to hide a provider attribute include:

- hide_provider_attr - "YES" - hides the MDR provider attribute
- provider_attr - MDR provider attribute name
- provider_name - MDR provider name
- provider_name_regexp - MDR provider name regular expression

provider_name or provider_name_regexp are required.

Example: Hide an MDR-Only Attribute

To hide the MDR provider *widget_cost* attribute for all providers that are named *myMdr,* use the following metadata statement:

```
<macro name=cmdbMetadata  hide_provider_attr="YES" provider_attr="widget_cost" provider_name="myMdr">
```

Define MDR Attributes Without CA CMDB Equivalents

You can define labels and help text for MDR provider attributes that do not correspond with any CMDB attributes. Attributes are labeled Not in Family in the Federated View.

cmdbMetadata macro arguments to define an MDR provider attribute include:

- attr - CMDB attribute name
- category - Category name where attribute is displayed
- heading - Heading label for attribute
- help - Brief description of the attribute

Example: Define an MDR-Only Attribute

The following statement defines an attribute name "ext_mem_capacity" using the label "External Memory Capacity" under the Attributes category in the Federated View .

```
<macro name=cmdbMetadata  attr="ext_mem_capacity" category="Attributes" heading="External Memory Capacity"
 help="Total external memory">
```

Define CMDBf Data Provider Metadata

You can control how data is displayed in the Federated View.

To define metadata for CMDBf Viewer

1. To open the cmdb_metadata_federation_viewer_site_attr.htmpl file, use Web Screen Painter.
2. Determine which metadata changes are needed.

   NOTE
   Sample templates are provided in the file.
3. Copy the appropriate template and substitute the required arguments according to the instructions in the file.
4. Save and publish the changes.

CMDB Management

This article describes the following:

Depending on your role, CMDB lets you manage configuration items and their relationships:

- The Configuration Administrator (CMDB Administrator) manages the tools in your CMDB, such as change verification. In addition to CI management, Configuration Administrator role includes administration of CI classes, CI families, and relationship types.
- The Configuration Manager plans and audits the type of information to store in the CMDB.
- The Configuration Analyst enters data into the CMDB by using approved methods and checks the accuracy of the data.
- The Change Manager helps ensure that users complete the following tasks:
  - Document the requests for change correctly.
  - Review the requests for change.
  - Execute the changes according to environment rules. The environment rules include priority, completeness, compliance, and signoffs.

    NOTE
    The Change Manager is not responsible for the CMDB, but servers as a major customer of the CMDB.

Create a CI from a Base Object

A CA CMDB installation can define CIs for *base objects*: Contacts, Locations, and Organizations. These CIs can be managed so, and they can establish relationships with other CIs as desired. For some customers, this approach supports the configuration management process better than employing base objects only as CI attributes.

NOTE

You cannot select a contact that is already represented by another CI in the Contact family.

The Scoreboard lets you create a configuration item from a base object.

Follow these steps:

1. Click File, New Configuration Item.
   The Create New Configuration Item page appears.
2. Complete the CI fields. Name and Class are required. Click Continue.

   NOTE
   The common CI attributes Host Name, Serial Number, MAC Address, and DNS Name are not relevant to a CI for a base object.

   The Create New Configuration Item page appears.
3. Click the base object link to define the object that this CI represents.
4. Select the object. You can search for an existing object or click Create New to create an object. If you want to create an object, click Save to continue.
   The selected object appears at the top of the Configuration Item Detail page.
5. Click Save.
   The main attributes of the selected object appear on the Attributes tab of the Configuration Item Detail page.

Create a Base Object CI Using GRLoader

You can use GRLoader to create a base object CI from an existing base object.

Follow these steps:

1. Write XML that identifies the following parameters:

- CI name
- Family
- Class

2. Submit the XML.
   The base object is created and the GRLoader displays that one CI was read and inserted.

Example: Create a CI From an Existing Contact

The following example creates a CI from the existing contact **Gibbs, Keith**.

```
<GRLoader>
<ci>
    <name>Gibbs, Keith</name>
    <family>Contact</family>
    <class>Technical</class>
    <base_contact>Gibbs, Keith</base_contact>
</ci>
</GRLoader>
```

How to View a Federated CI

The Federated View page displays parallel CI attribute values from CMDB and CMDBf-registered MDRs. If no external MDRs are configured for CMDBf, only the CMDB attributes are displayed.

To update the fields from an MDR, click Retrieve.

Legend:

bold - Attribute value differences between CMDB and MDRs.

Not in Family - MDR-only attribute customized to display in a CMDB category.

Not in CMDB - Category for MDR-only attributes.

> **NOTE**
> To customize the way MDR attributes are displayed, see MDR documentation.

> **NOTE**
> To display federated CI attributes, the MDR class must be **cmdbf**.

Populating CMDB

This article contains the following topics:

Database Population

Populating the CMDB with the CIs and relationships in your IT infrastructure is a part of using the application efficiently. You can populate the CMDB with data manually by:

- Using the built-in Configuration Item Editor
- Using GRLoader
- Importing items from other asset management tools.

How GRLoader Populates the Database

T populate the database by loading CIs and relationships, complete the following tasks:

1. Convert input data containing information about CIs and their relationships to XML or in a spreadsheet.
2. The CA CMDB GRLoader program uses the XML data as input.

3. GRLoader loads the data into the database.For information about GRLoader parameters, see the GRLoader Command topic.

Use GRLoader to Import the Data

Import data by using the GRLoader program that is provided with CA SDM. The program creates CIs based on the data in an XML file or a Microsoft Excel spreadsheet.

Follow these steps:

1. From the Start menu, select Run.
2. Enter **cmd**.
   A DOS command window appears.
   Enter the following command:
   ```
   Grloader -u <username> -p <password> -s http://<ca_sdm_servername>:8080
   -i <xml_document or spreadsheet>
   ```
   GRLoader creates CIs using the data in the XML file. If errors are found during this process, an error file is created. The error file lists the CIs that could not be imported and the reason.
   GRLoader import completes.
3. Start CA SDM and verify that the CI data has been correctly populated.
4. To verify that relationship data has been populated correctly, start the CMDB Visualizer.
   The data is imported and verified. For more information about GRLoader, see the General Resource Loader topic.

Family and Class Assignments

Apply a classification scheme to each CI; this scheme involves assigning each CI the following attributes:

- Family -- A collection of configuration items having similar attributes
- Class -- A subset of configuration items within a family

You can create assignments in the following ways:

- Populate manufacturer data identifying family and class for each hardware asset.
- Include nonblank values for family and class in the respective columns in the input file.

GRLoader does not import an CI that cannot resolve family and class to an existing family and class.

How to Load CA APM Data

The primary input to the CA APM Loader program is a table or view that contains an extract of the CA APM import data. This data is contained in the CMDB_Export_Asset_Data database table or view. In some cases, the import data into the MDB is located in the same database as the target. However, in many circumstances, the import data resides in a different database, for example, when importing data across subsidiaries. In either case, create a view on the same database as the source data.

The CA APM data view does not contain the class and family attributes, and can come from a system with a different classification scheme. For example, if the source of the asset data is a different company, a different classification system can be in effect.

The CA_MODEL_DEF table contains a list of models, which are matched against the data in CMDB_Export_Asset_Data. If there is a match, the family and class from the model are assigned to the asset being imported. If there is no match, define a new model for the asset. Consider either copying entries from the source CA_MODEL_DEF table to the target MDB, or updating the CA_MODEL_DEF table with entries for all new hardware makes and model numbers.

CMDB Visualizer Overview

CA SDM lets you align your IT components (*configuration items*, or CIs) with your business services. CMDB defines *relationships* among CIs, as when a group of CIs work to provide a business service. CMDB Visualizer lets you see the entire picture of your CI relationships, and provides functions to manage the relationships. Working from a *focus CI*, you can use the Visualizer to display up to nine levels of related CIs.

CA uses a provider/dependent model to define relationships among CIs. All CIs that contribute to a business service are *providers* to that business service (the *dependent*). In much the same way, providers can also be *dependents* that rely on other CIs. You can use Visualizer to perform the following provider/dependent analyses:

- **Browse**
  Displays unfiltered view of all CIs.
- **Impact Analysis**
  Starts with a focal CI (provider) and searches for its dependents.
- **Root Cause**
  Starts with a business service (dependent) and view all the CIs that are providers to that service.
- **Cause and Effect CIs**
  Combines impact analysis and root cause in one search.
- **Trace Relation**
  Displays all possible relationships that are based on levels. If you select only one CI, this filter displays the Browse view.
- **Shortest Path**
  Displays the shortest chain of relationships that are based on levels.

CMDB Visualizer lets you do the following actions:

- Visualize multiple levels of CIs from a configurable graphical view
- Monitor or cancel rendering progress
- Search using flexible criteria
- Filter based on CI families, relationship types, and other attributes
- Display CI relationships
- Trace a relationship between two CIs
- Visualize a dependency chain
- Invoke CMDB directly from Visualizer
- Display CI attributes and properties
- Save graph metadata
- Print the graph layout
- Find a specific CI on a displayed graph
- Create a CI (depending on role)
- Create new CI relationships (depending on role)
- Use the Scratchpad to store key CIs
- Display CI status
- Hide or reveal a CI in the Visualizer layout
- Role-based data security
- Launch external MDR

How To Use the Visualizer Interface

This article contains the following topics:

CA CMDB Visualizer provides a graphical interface with toolbars and a graphical canvas. You can click the button on the left side of the canvas to display the retractable Search pane.

Graphical Display Tasks

With the Visualizer graphical display, you can display the CIs and relationships that you select for analysis. You can also do the following tasks:

NOTE
Visualizer uses the space bar to select or confirm an operation. For example, if the focus is on a button, then instead of pressing Enter, press the space bar.

- Create, manage, and apply analysis filters.
- Change the graph layout and number of CI levels displayed.
- Select CIs for graphical viewing and right-click to provide various functions, including the ability to view and modify CI properties.
- Use the Toolbar icons to provide various functions.
- Zoom in to view areas of interest, or zoom out to see the larger context.
- Use Full Screen Mode display a larger work area.

NOTE
When you enable Full Screen mode, you cannot use the keyboard functionality.

Search for CIs by Attributes

You can search for CI (and asset) attributes to locate and manage CIs.

NOTE
If you only search for CIs, you can disable the Asset search. To disable the Asset search, clear the CA APM check box on the Visualizer administrative interface.

Follow these steps:

1. Open the retractable Search pane on the left side of the Visualizer graph.
2. Click the Search tab.
3. Expand the sets of search criteria that you want.
4. Complete one or more of the attribute fields on the search criteria areas.
   The search criteria is specified.

   NOTE
   Visualizer searches for both CIs and Assets by default. To restrict the search, change the CI or Asset general options.
5. Click Search.
   The Search Results tab displays all CIs/Assets that match your search criteria.

Primary

Primary fields focus your search that is based on the following basic CI attributes:

- CI Family
- CI Class
- CI Status
- Location
- Organization
- Department
- Asset
- CI

Network

Network fields search on the following common CI attributes for network addressing:

- IP Address
- DNS Name
- Host Name
- MAC Address

Asset

Asset fields search on the following basic CI/asset attributes:

- Manufacturer
- Model
- Serial Number
- Cost Center
- License
- Product Version

Maintenance

Maintenance fields provide searching on CI attributes for the following maintenance tasks:

- Service type
- Responsible Vendor
- Responsible Organization
- Maintenance Vendor
- Maintenance Organization
- Acquire Date
- Installation Date
- Expiration Date
- Warranty Start Date
- Warranty End Date

Filters and Layouts in Visualizer

This article describes the following topics:

Visualizer Filters

CA CMDB Visualizer provides built-in analysis filters, and you also can create filters.

Built-in filters include the following filters:

- **Browse**
  Displays the default unfiltered display.
- **Impact Analysis**
  With the focal CI as provider, shows all the CIs affected when any change occurs to the focal CI.
- **Root Cause**
  With the focal CI as dependent, shows all the CIs that could be the root cause for the focal CI to go down. Or, shows the CIs that can affect the focal CIs behavior.
- **Cause and Effect CIs**
  Combination of Root Cause and Impact Analysis. From the focal CI, shows all the CIs that could be the root cause. Or, shows the CIs that could be affected when a change occurs to the focal CI.
- **Trace Relation**

Displays all possible relationships among focal CIs, based on the Levels setting. If the graph has only one focal CI, applying this filter displays only the focal CI.

- **Shortest Path**
  Displays the shortest chain of relationships among focal CIs, based on the Levels setting. If the graph has only one focal CI, applying this filter displays only the focal CI.

In addition to the built-in filters, any available administrator-defined and user-defined filters display.

Layout

The Layout drop-down list lets you switch your CI display between two different layout options, depending on your analysis needs. The following types of layouts are available:

- Hierarchical
- Circular

Hierarchical Layout

Use a hierarchical layout when you want to view related CIs that can be logically distant from the focal CI.

Circular Layout

You use a circular layout when you want to emphasize all relationships closest to a focal CI.

Change Levels

You can change the number of levels of CIs that are displayed relative to the focal CI. The maximum number of levels that are allowed is nine. The default value is two levels, which you can modify by using the Administration interface.

Follow these steps:

1. Click the Levels drop-down list.
   The numbered levels appear.
2. Select the number of levels that you want.
   The graph adjusts to reflect the new setting.

Working with CIs in Visualizer

This article contains the following topics:

Create CI

Use the Create CI button to launch CMDB to provide CI definition capability directly from Visualizer.

> **NOTE**
> This function is available to roles that have MODIFY permission for Configuration Items and VIEW permission or higher for Relationships.

Follow these steps:

1. Click Create New CI on the Visualizer toolbar.
   The CMDB Create New Configuration Item page launches in context.
2. Define the new CI and click Save.

The CI is created.

View CI Properties

The Properties button provides a CMDB launch in context capability to view and edit selected CI details. Other methods include the following options:

- Select CI on Search Results, right-click, and select Properties.
- Select CI on the graph, right-click, and select Properties.

Create Relationship

Use the Create Relationship button to launch CMDB to provide relationship definition capability directly from Visualizer. For more information about how to create relationship between a CI or two CIs dynamically, see Visualizer Graphs.

> **NOTE**
> This function is available to roles that have MODIFY permission for Relationships and VIEW permission or higher for Configuration Items.

Follow these steps:

1. From the graph, select up to two CIs.
2. On the Visualizer graph, click Create Relationship.
   The Create New Configuration Item Relationship page launches in context. If two CIs are selected, one is shown as Provider and the other one as Dependent.
3. Select the relationship type from the drop-down list.

   > **NOTE**
   > Specify the Relationship Type to save the graph with the new relationship.
4. Click Save.
   The new relationship is saved and the Create Relationship page closes.

Set a Focal CI

You can set a focal CI to display relationships from its point of view.

Follow these steps:

1. Select a CI to serve as the focus of the analysis from Search or graph.
2. Make the focal CI in one of the following ways:
   - Select CI on Search Results and hit Enter.
   - Select CI on Search Results and click its Make Focal CI icon.
   - Select CI on Search Results, right-click, and select Make Focal CI.
   - Select CI on the graph, and click Make Focal CI on the toolbar.
   - Select CI on the graph, right-click, and select Make Focal CI.

   The graph is reorganized around the focal CI.

Scratchpad

You use the Visualizer Scratchpad like the Favorites or Bookmark feature of internet browsers. You can store some CIs in the Scratchpad for convenient use later. You can select one or more CIs from the Scratchpad to add them to the Visualizer graph. You also can use the Scratchpad to make a CI the focal CI.

> **NOTE**
> The Scratchpad is saved locally. If you launch the Visualizer from a different server later, a saved Scratchpad is not available.

The Scratchpad provides two different CI views: Tiles and Details.

NOTE
If you change a CI name or family in CMDB, refresh the Scratchpad to view the changes.

Hidden CI(s)

Use the Hidden CI(s) button to display the CIs that were hidden by using the Hide/Unhide commands. On the Hidden CI(s) list, you can select one or more CIs and can redisplay them. Perform one of the following actions:

Follow these steps:

- Click Unhide.
- Drag and drop the CIs on the graph canvas.
- Using the CI right-click menu to select Unhide.

Hidden CIs can display either the Tiles or Details view.

Show Configured Status

You use the Show Configured Status button to switch the CI status display. The status bar displays the current setting.

When Show Configured Status is on, each CI displays an icon on the graph to indicate its configured status. To define icons for each Configured Status, use the CI Status page in the Administrator interface.

Working with Graphs in Visualizer

This article contains the following topics:

Open Graph

Follow these steps:

1. Click Open Graph on the toolbar.
   The Open Graph list displays graphs with saved metadata.
2. From the list, complete one of the following actions:
   – Double-click a graph item.
   – Select a graph and click Open.
   The graph is rebuilt based on its saved metadata and its current CMDB data.

Follow these steps:

1. Click Open Graph on the toolbar.
   The Open Graph list displays graphs with saved metadata.
2. Select the graph from the Open Graph list.
3. Click Delete.
   The graph is deleted.

Invert Graph

By default, Visualizer displays Provider CIs at the top and Dependent CIs at the bottom, relative to the Focal CI.

Follow these steps:

1. Click the Invert Graph button to reverse this arrangement to a vertical axis.
   You can view the data from a different perspective.

Find CIs on the Graph

You can find specific CIs to display as nodes on the graphical display. Use the Find CIs button to open a new window over the graph to prompt for what string to match. The Find string is matched using graphical node attributes such as the CI name and family. Find shows the count of CIs which matched the users input criteria. The count is displayed in the bottom left corner.

Follow these steps:

1. Select the nodes to use from the Find CI results.
2. Click Select On Graph.
   The selected CIs display.

Show Graph for a Single CI

Follow these steps:

1. On the Visualizer graph, select a CI to serve as the focus of the analysis.
2. Right-click the CIs menu and select Make Focal CI.
   The selected CI becomes the *Focal CI* in the Visualizer. Its associated relations display, including the configured number of levels.

Add Multiple CIs to Graph

Follow these steps:

Make Focal CI, Add to Graph, or drag-and-drop CIs from the following sources:

- Search Results
- Scratchpad
- Saved graphs
- Hidden CI list

> **NOTE**
> Users can add maximum of 10 focal CIs or 10 graphs.

Visualizer Procedures

This article contains the following topics:

The topic describes the Visualizer features that are available when the CA SDM server with Visualizer is running.

Note: (For Advanced availability configuration only) If the visualizer makes a server request during the application server quiesce period, the following message is displayed:

“This CMDB Visualizer server is scheduled to shut down for maintenance in xx:xx. Please save your work and logout.”.

You can hide the message box but the message is displayed on the top panel throughout the quiescing period. If the server quiescing is canceled, you receive a message about the cancellation.

Launch Visualizer in CI Context

Follow these steps:

1. In the CA SDM web interface, navigate to a CI List, or Search for the CIs to analyze.
   The CI List displays.
2. Click a CI of interest.

The CI Detail page is displayed and the Visualizer button is available. If CA CMDB Visualizer was not installed, the Visualizer button is inactive.

3. Click Visualizer.
   Visualizer starts and the relationships for the focal CI display.
   Default: Cause and Effect CIs

Launch Visualizer with Applied Filter

Follow these steps:

1. Navigate to the CI List, or search for the CIs you want to analyze.
   The CI List displays.
2. Click a CI of interest.
   The Configuration Item Detail page displays.
3. On the Filters list, select a filter.
4. Click Visualizer.
   Visualizer starts and the graphical relationships for the focal CI display according to the selected filter. If CA CMDB Visualizer was not installed, the Visualizer button is inactive.

Trace Relation with Focal CIs

You can trace all relationships between the selected CI and any focal CIs on the graph.

Follow these steps:

1. Select a CI on the graph.
2. Right-click and select Trace Relations with Focal CIs.
   Any relationships are displayed.

   NOTE
   If the selected CI is also the only focal CI, only that CI is displayed.

Launch MDRs

Follow these steps:

1. Right-click a CI on the Visualizer graphical display.
   A pop-up menu displays.
2. Select the Launch MDR option on the pop-up menu.
   All external data providers that are associated with the CI appear.
3. Select an application name and click Launch to start the associated application.
   Visualizer launches the external MDR application as it is defined in the CMDB.

   NOTE
   Visualizer does not validate the external MDR applications credentials and URLs.

Hide/Display CIs

To promote clearer viewing, you can hide and then redisplay graphed CIs. Hidden CIs are available only for the current graph. When a new graph is generated, or when a Focal CI is set, the Hidden CIs List is cleared.

Follow these steps:

1. To select a CI, click it.
2. Right-click the CI and select Hide from the drop-down list.
   The graph updates so that the CI is hidden.

Follow these steps:

1. Click Hidden CIs on the toolbar or status bar.
   The Hidden CIs list displays.
2. Select one or more hidden CIs from the list and perform one of the following actions:
   – Click Unhide.
   – Drag to the graph canvas.
   – Right-click and select Unhide.
   The graph updates so that the CIs display.

View Change History for a CI

Follow these steps:

1. Select the CI you want to view.
2. Click Properties.
   CA CMDB is launched in context, displaying the CI Detail page.
3. Select the Versioning tab and click Show Log.
   The tab displays the details of the changes that were made to the CI.

Apply a Filter

Follow these steps:

1. Build the desired graphical display.
2. Click the Filter drop-down list.
   The Filter list is displayed.
3. Select a filter from the list.
   The filter is applied to the graph, which reorients to display the new perspective.

Use the Scratchpad

To view the Scratchpad, click the Scratchpad button on the Visualizer toolbar.

The Scratchpad displays.

Follow these steps:

1. Select one or more CIs on the Scratchpad.
2. Complete one of the following actions:
   – Click the Make Focal CI button on the Scratchpad toolbar.
   – Drag and drop one or more CIs from the Scratchpad to the graph.
   – Right-click on a CI and select Make Focal CI.

Follow these steps:

1. Select one or more CIs on the Scratchpad.
2. Complete one of the following actions:
   – Click the Add Focal CI button on the Scratchpad toolbar.
   – Right-click on the CI and select Add To Graph.

Follow these steps:

1. Select one or more CIs to delete.
2. Click the Delete button on the Scratchpad toolbar.
   The CI is deleted from Scratchpad.

To synchronize the Scratchpad to the Database, click Refresh. The Scratchpad removes all the CIs that were deleted from the database.

To save a CI on the Scratchpad, right-click the CI and select Add to Scratchpad.

The CI is added to the Scratchpad.

Visualizer Administration

Use the CMDB Visualizer administration interface to perform the following functions:

- Install and configure the Visualizer server.
- Modify the display of CI status and relationships.
- Create and manage administrative filters.

> **NOTE**
> These functions are only available to roles with administrator privileges.

How to Configure CMDB Visualizer

You can configure a CMDB Visualizer server as a primary or secondary server. The primary Visualizer server must reside on the same computer as the primary CA SDM server. All other Visualizer servers are treated as secondary servers. You cannot change a secondary Visualizer server to primary unless a CA SDM server on the same computer is changed to primary. The primary server address is required to reconfigure a secondary server.

Managing CI Family Icons

CA CMDB Visualizer can represent CI families by using either SVG or JPEG graphics icons. By default, CA SDM provides SVG icons for CA-supplied CI families at installation. You can use either SVG or JPEG images for user-defined CI families.

You manage SVG and JPEG graphics using properties files in the following directory:

```
visualizer-install-dir\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\assets
```

SVG graphics are defined as key=value pairs in the serverjpegfamilyimagemap.properties file.

- ***key***
  Specifies the family ID.
- ***value***
  Specifies the file name of the SVG graphic.

Example

400007=HARDWARE_MAINFRAME

> **NOTE**
> SVG images cannot be previewed; they can only be viewed in the application.

You also maintain JPEG graphics as key=value pairs in the serverjpegfamilyimagemap.properties file.

- ***key***
  Specifies the family ID.
- ***value***
  Specifies the file name of the JPEG graphic.

Example

400039=conx_32.jpg

NOTE
JPEG images reside in the following directory:

```
$NX_ROOT\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\assets\images\viz_icons
```

Important: When you create and maintain graphics in the .properties files, verify that no duplicate Family ID (key) occurs.

Visualizer displays an SVG or JPEG graphic for a CI Family that is based on the following priority:

- Family ID in the serverjpegfamilyimagemap.properties file
- Family ID in the serverjpegfamilyimagemap_OOTB.properties file
- JPEG image with key DEFAULT in the JPEG image map file

Example

DEFAULT=default_32.jpg

Built-In SVG Graphics

CA SDM supplies SVG graphics for the following families:

- CLUSTER
- CLUSTER_RESOURCE
- CLUSTER_RESOURCE_GROUP
- COMPUTER
- CONTACT
- DOCUMENT
- FACILITIES_AIR_CONDITIONING
- FACILITIES_FIRE_CONTROL
- FACILITIES_OTHER
- FACILITIES_UNINTERRUPTIBLE_POWER_SUPPLY
- HARDWARE
- HARDWARE_LOGICAL_PARTITION
- HARDWARE_MAINFRAME
- HARDWARE_MONITOR
- HARDWARE_OTHER
- HARDWARE_PRINTER
- HARDWARE_SERVER
- HARDWARE_STORAGE
- HARDWARE_VIRTUAL_MACHINE
- HARDWARE_WORKSTATION
- INVESTMENT_IDEA
- INVESTMENT_OTHER
- INVESTMENT_PROGRAM
- INVESTMENT_PROJECT
- LOCATION
- NETWORK_BRIDGE
- NETWORK_CONTROLLER
- NETWORK_HUB
- NETWORK_NETWORK_INTERFACE_CARD
- NETWORK_OTHER
- NETWORK_PORT
- NETWORK_ROUTER
- NETWORK_SWITCH
- ORGANIZATION
- OTHER
- PROJECT
- PROJECTS
- SAN_INTERFACE
- SAN_SWITCH
- SECURITY
- SERVICE
- SERVICE_LEVEL_AGREEMENT
- SOFTWARE
- SOFTWARE_APPLICATION
- SOFTWARE_BESPOKE
- SOFTWARE_COTS
- SOFTWARE_DATABASE
- SOFTWARE_IN_HOUSE
- SOFTWARE_OPERATING_SYSTEM
- TELECOM_CIRCUIT
- TELECOM_OTHER
- TELECOM_RADIO
- TELECOM_VOICE
- TELECOM_WIRELESS

How to Change the Icon of CI on the Visualizer

You can change the icon of a CI on the Visualizer. Complete the following:

Follow these steps:

1. Navigate to *$NX_ROOT\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\assets*.
2. Open the**serverjpegfamilyimagemap.properties** file and add the family class and image name (300002=new_hardware_server.jpg). .
3. Navigate to *$NX_ROOT\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\viz-ux\assets\images \viz_icons*and place the image file mentioned in step 2.
4. Restart the CA Service Desk Manager service.
   You have successfully changed the icon of a CI on the Visualizer.

CMDB Data Maintenance

This article contains the following topics:

The tasks that you perform to maintain CMDB data require administrator privileges. The tasks include setting up the families, classes, and relationship types that are used to manage configuration items and relationship information.

CMDB Family/Class Structure

CMDB provides default configuration item families and the CI classes they contain.

The following Service Desk and CA APM base families do not have their own CA CMDB extension tables:

- Computer
- Contact (as base object)
- Hardware
- Location (as base object)
- Organization (as base object)
- Other
- Project
- Software

In CA CMDB, CIs in these base families receive CI Detail pages with some extraneous fields and without Attributes tab. Use the Change Family and Class capability of CA CMDB to convert these CIs to CA CMDB families. The advanced features include the ability to track family-specific attributes, versioning, snapshots, and baselines.

Change Family/Class of a Single CI

You can change the Family and Class for a single CI.

Follow these steps:

1. Select the CI that you want to change.
2. Click Edit.
   The Update Configuration Item page displays.
3. Change the Class value for the CI. Selecting the class also determines the family for the CI. You can enter a value directly or click the magnifying glass to select from a list of classes.
4. Click Save.
   The Family and Class of the single CI is changed.

Change CI Family/Class Using GRLoader

You can change the Family and Class for a CI by using GRLoader.

Follow these steps:

1. Create XML code to change the CI attribute.
2. Using GRLoader, open the CI you want to modify.
3. To change the CI attribute, run the XML through GRLoader.
   The Family and Class for a CI are changed.

Example: Set the Class of an Open CI to Document

The following example shows an XML fragment that sets the class of an open CI to *Document*.

```
<ci>
    <name>document number 1
    <class>Document
</ci>
```

> **WARNING**
> Do not attempt to update both the Family and Class attributes of a CI simultaneously. To change both, use two separate CI updates in two separate invocations of GRLoader.

Extending CMDB

This article contains the following topics:

CMDB is a highly flexible system that can be extended to include extra families, classes, and attributes according to your business needs. New attributes can be family-specific or *common* (applicable across all families). While CMDB provides predefined families with many classes and attributes based on industry standards, some business cases require one or more of the following activities:

- Extend one or more of the CI families by adding new attributes. For example, to add a GPS coordinate for every device on your office campus, you can define a gps_coordinate attribute to add to any family. If you only want to extend one family, use Web Screen Painter Schema Designer to define the new attributes in the existing extension table. In addition, whenever you add an attribute, you also must modify the Detail page, Attribute tab, and metadata forms that use the attribute. For more information, see Add Family Attributes.
- Extend all CI families by adding a common attribute. For more information, see Add Common Attributes.
- Add new classes to an existing family to support more classification detail in your system. For example, instead of the generic Server class, you can create a separate class for every model of server device. For more information, see Define a New CI Class.
- Add a new family by using an existing extension table and its attributes. A new family provides an alternative way of organizing or classifying CIs. For more information, see Define a New CI Family.
- If the existing class or family structure does not match your requirements, you can start over with a minimum set of attributes. To add a new family using a new extension table, define the new extension table and its attributes using Web Screen Painter Schema Designer. Also, create the forms that are required for display and update. For more information, see Constructing a New Attribute Framework.

> **WARNING**
> Extending CMDB requires specialized knowledge of CA SDM data structures and tables, and familiarity with Web Screen Painter (WSP). We recommend that you contact CA Services to help with this activity and also read and understand thoroughly the following sections before attempting to extend CMDB families and attributes.

Adding New CMDB Attributes

Adding a new CMDB attribute requires careful planning. CMDB already provides many attributes in its predefined families and classes, so determine whether they are sufficient for your needs before considering customization.

If you determine that a new attribute is necessary, we recommend proceeding conservatively by asking whether you must:

- add one or more new attributes to an existing family?
- add the new attribute(s) to more than one family?
- add a new common attribute, which applies across all families?

You can extend an existing family by adding new attributes. For example, if some devices on your office campus are assigned a GPS coordinate, you can add a gps_coordinate attribute to any applicable CI family. For more information, continue to Add Family Attributes.

After your attribute requirements are identified, if you determine that they require the use of new families and classes, see Adding a New CA CMDB Family or Class.

Add Family Attributes

You can add a new attribute to one or more existing families by using the Web Screen Painter Schema Designer, which updates the database to include the new attribute and also updates the associated CA SDM tables and files. This method is preferred over updating the tables and file changes manually.

To add a new attribute to a family

1. Using Web Screen Painter Schema Designer, open the schema that corresponds to the family extension table.
2. Add the desired attribute to the extension table.
3. Publish the modified extension table.

> **NOTE**
> GRLoader and Versioning automatically pick up new attributes without further action. However, we also recommended that you enable logging. Logging is required for auditing purposes and Versioning enabled to record all snapshots.

To enable logging, verify that UI_INFO for the attribute is set to **AUDITLOG**.

After a new family attribute is created, it must be added to each display form so that users can see and update that attribute and to the metadata form specific to that extension table. For more information, go to Add Attributes to Forms.

In addition, Versioning requires metadata, including information about column headings and the related Standard CI attributes. You define new metadata for all new attributes that you create; for instructions, continue to Create Metadata.

Add Common Attributes

Common attributes are stored in the **ca_owned_resource** table in the **nr** object. This table provides the following customizable fields:

- smag_1
- smag_2
- smag_3
- smag_4
- smag_5
- smag_6

If you require less than seven modified attributes among all CI families, these fields provide a convenient solution.

As supplied, these custom fields do not display on any form. To allow CMDB users to view or update a smag_*n* field, use the Web Screen Painter (WSP) to add it to any display form. All logging and GRLoader functions are already enabled.

NOTE

For Web Screen Painter Schema Designer procedures, see How to Implement CA SDM and the Web Screen Painter online help.

Adding a New CMDB Family or Class

Adding a new CMDB family requires careful planning. CMDB provides many predefined families and classes, so determine whether these are sufficient for your needs before considering customization. If not, some new requirements can be met by one of the following actions:

- Defining a new class in an existing family
- Defining a new family that uses an existing extension table

If you determine that existing extension tables are insufficient for your needs, skip this section and continue to the section Constructing a New Attribute Framework to create an extension table and the forms that it requires.

Define a New CI Class

You can add new classes to support higher levels of classification granularity. For example, instead of using any of the CMDB-provided classes in the Hardware.Server family, you can define more classes for different server devices.

Before you create a new configuration item class, determine whether a suitable class already exists in the Configuration Item Class List.

NOTE
If you also require a new family to govern a new set of classes, skip this topic and proceed to Define a New CI Family. Then you can return and populate the new family with classes.

To define a CI class using the Administration interface

1. Navigate to the Configuration Item Class List.
2. Click Create New.
   The Create New Configuration Item Class page appears.
3. Enter a unique name for the new class.
4. Enter the appropriate family name in the Family field, or click the icon over the field to search for a family.
5. Verify that the Record Status field is set to Active.
6. Click Save.
   The new CI class is defined and ready for you to create new CIs.

Define a New CI Family

Before you create a configuration item family, determine whether a suitable family already exists in the Configuration Item Family List. If not, you can create new families as an alternative method of organizing or classifying CIs using existing attributes.

If the new family that you need cannot use an existing extension table (or the default table), complete all the additional preparations that are outlined in Constructing a New Attribute Framework. The new extension table can be used to define new families.

To define a CI family using the Administration interface

1. Navigate to the Configuration Item Family List.
2. Click Create New.
   The Create New Configuration Item Family page appears.
3. Enter a unique name for the new family.
4. Verify that the Record Status field is set to Active.

5. In the Extension Name field, select the extension table that identifies the type of family that you want to create. This may be a new extension table that you created recently.
   For example, if you are adding a family for an unspecified type of hardware, select ci_hardware_other. This type ensures that when you create configuration items whose classes use the new family, the appropriate attributes display on the Attributes tab. If you do not select a table name in the Extension Name field, the default table is used and only default attributes appear when you create a configuration item within the new family.
6. Type a description in the Description field.
   The description that you enter displays in the Configuration Item Family List for informational purposes.
7. Click Save.
   The configuration item family is defined.

Now that your new family is defined, you can add classes to it as described in Define a New CI Class.

Constructing a New Attribute Framework

If none of the existing class or family structures match your requirements, you can start over with a minimum set of new attributes. This requires construction of a new extension table and other supporting structures.

Before using the CMDB administration interface to define a new CI family that is based on a new extension table, use the Web Screen Painter Schema Designer to create the new extension table.

To use the new extension table, you must also create new HTMPL forms for:

- New CI Detail page
- New Attributes tab with its associated attributes
- New metadata form and metadata

CA CMDB supplies templates that you can use to build these HTMPL forms. The following sections provide more detailed information about what is required.

> **NOTE**
>
> For Web Screen Painter Schema Designer procedures, see How to Implement CA SDM and the Web Screen Painter online help.

To use the new framework in the CMDB user interface, define:

- One or more CI families that use the new extension table
- CI classes to classify your CIs by type

Create a New Extension Table

Before you can define a family that is based on a new extension table, update the database with the new table and also update the CMDB schema with information about that table.

To create an extension table

1. Using the Web Screen Painter Schema Designer, define the new extension table and extension name.
2. Save and publish the new extension table.

Note: WSP Schema Designer automatically creates the logging trigger in CMDB.

To create the CI Detail page, continue to the next section.

Create a CI Detail Page

A CI Detail page is required to support attribute display for CIs that are associated with a new extension table.

To create the CI detail page

1. Using the Web Screen Painter Schema Designer, click File, New, and create a form that is based on detail_extension.template.

2. Save this new form as **detail_** *extension* **.htmpl**, where *extension* is the name of the extension table.
3. Follow the instructions that are listed in the file, replacing the ***EXTENSION*** string with the name of the new extension table (defined previously).
4. Save the file with all changes.

The CI Detail page includes two attribute sections:

- Common attribute section named cmdb_detail.htmpl
- Family-specific section (the Attributes tab) named nr_cmdb_extension_tab.htmpl, where *extension* is the name of the new extension table.

Continue to the next section to Create the CI Attributes Tab.

Create a CI Attributes Tab

The Attributes tab displays the family-specific attributes for a CI.

To create the Attributes tab

1. Using the Web Screen Painter Visual Editor, click File, New, and create a form that is based on nr_cmdb_extension_tab.template.
2. Save this file as **nr_cmdb_** *extension* **_tab.htmpl**, where *extension* is the name of the new extension table.
3. Follow the instructions that are listed in the file, replacing the ***EXTENSION*** string with the name of the new extension table (defined previously).
4. Save and publish the file with all changes.

Continue to the next section to populate the Attributes tab.

Add Attributes to Forms

After Web Screen Painter Schema Designed has been used to create a new attribute in an extension table, that attribute must be added to any form that is used for display or update. For new family-specific attributes, the only form that must be changed is the Attributes tab that is named **nr_cmdb_** *extension* **_tab.htmpl**, where *extension* is the name of the extension table. This form must be edited to include any new attributes.

To edit an attribute form

1. Using Web Screen Painter Visual Editor, click File, Open to access the appropriate form.
2. Drag and drop the new attribute or attributes on the form.

   NOTE
   CMDB-provided forms do not lend itself to editing using the Web Screen Painter Visual Editor, so use the Web Screen Painter text editor on the Source tab.
3. Save and publish the form.

If you have not yet created a metadata form, continue to the section Create a Metadata Form. To define metadata for a new attribute on the form, continue to the section Create Metadata.

Create a Metadata Form

A new extension table requires its own metadata form to define column headings and Standard CI information for Versioning.

To create the metadata form

1. Using the Web Screen Painter Visual Editor, click File, Open to access cmdb_metadata_extension.template.
2. Save the file as **cmdb_metadata_** *extension* **.htmpl**, where *extension* is the name of the new extension table.
3. Follow the instructions that are listed in the file, replacing the ***EXTENSION*** string with the name of the new extension table (defined previously).
4. Save and publish the form with all changes.

To populate the metadata form, continue to the next section.

Create Metadata

Metadata includes information about attribute column headings and Standard CI information that the Versioning feature requires.

> **WARNING**
> Metadata requires careful planning to ensure correct data in Snapshots, correct titles in Versioning, and successful Standard CI comparisons.

To create metadata

1. Using the Web Screen Painter Visual Editor, click File, Open to access **cmdb_metadata_** *extension* **.htmpl**, where *extension* is the name of the extension table.
2. Following the instructions that are listed in the form, copy and modify the indicated row for each attribute in the new extension table.

   NOTE

   The following attributes, although required, do not need metadata:

   - ID
   - Last_modified_by
   - Etc (to be provided)

If you are adding metadata to an existing CMDB family, audit changes are displayed correctly on the Versioning tab. However, if you are defining metadata for a new extension table, you must have a new family and class for your attributes; for more information, see Adding a New CA CMDB Family or Class.

The structures for your new extension table are now in place. To define a new family for your attributes, continue to Define a New CI Family.

Example

In this sample scenario, you create an **Automobile** family and a **Sedan** class in it for tracking inventory for manufacturing, shipping, rental transportation, or other purposes. You could also create many other automobile classes; this example is for demonstration purposes only. ***Step 1 Create the New Extension Table***

1. In Web Screen Painter Schema Designer, click Add Extension Table.
2. Enter a name for the new extension table. In this example, **vehicle**.
   The **zvehicle** extension table is created and its properties displayed.

   > **NOTE**
   > “z” is appended to the beginning of all new table names to distinguish them from application-provided tables.
3. In the Table Info tab, set the Function Group field to **inventory**. Other fields are populated with default values.
4. Add new columns and attribute information as desired.
5. Save and publish the new extension table.

Step 2 Create a New Family **To create a family**

1. Navigate to the CI Family List.
2. Click Create New.
   The Create New Configuration Item Family page displays.
3. Enter a unique name for the new family. In this example, **Automobile**.
4. Verify that Record Status is set to Active.

5. Select the extension table name. In this example, **zvehicle**.
6. Click Save.
   The new CI family is created.

Step 3 Create a New Class

1. Navigate to the CI Class List.
2. Click Create New.
   The Create New Configuration Item Class page displays.
3. Enter a unique name for the new class. In this example, **Sedan**
4. Verify that Record Status is set to Active.
5. Select the Family. In this example, **Automobile**.
6. Click Save.
   The new CI class is created.

Step 4 Create the New CI Detail Form **To create the CI Detail form**

1. In the Web Screen Painter Visual Editor, open the form detail_extension.template.
2. Save the template with the name **detail_zvehicle.htmpl**.
   The new form is saved under NX_ROOT\site\mods\wsp\project\web\analyst
3. In the new form, replace all instances of ***EXTENSION*** with **zvehicle**
4. Save and publish the new CI Detail form.

Step 5 Create the Attributes Tab **To create the Attributes tab form**

1. In the Web Screen Painter Visual Editor, open the form nr_cmdb_extension_tab.template
2. Save the template with the name **nr_cmdb_zvehicle_tab.htmpl**
   The new form is saved under NX_ROOT\site\mods\wsp\project\web\analyst
3. In the new form, replace all instances of ***EXTENSION*** with **zvehicle**
4. Add attributes to the form as desired.
5. Save and publish the new Attributes tab form.

Step 6 Create the Metadata Form **To create the metadata form**

1. In the Web Screen Painter Visual Editor, open the form cmdb_metadata_extension.template.
2. Save the template with the name **cmdb_metadata_zvehicle.htmpl**
   The new form is saved under NX_ROOT\site\mods\wsp\project\web\analyst
3. Replace all occurrences of ***EXTENSION*** with **zvehicle**
4. Using the heading and attribute placeholders, add metadata for all family-specific attributes.
5. Save and publish the new metadata form.

CMDB Visualizer Interface

This article describes the CA CMDB Visualizer Interface optimizations from an end-user perspective.

CA CMDB Visualizer provides a graphical interface with toolbars and a graphical canvas. You can click the button on the left side of the canvas to display the retractable Search pane.

> **NOTE**
> Visualizer uses the space bar to select or confirm an operation. For example, if the focus is on a button, then instead of pressing Enter, press the space bar.

Graphical Interface

CDMB Visualizer Graphical interface provides the following capabilities:

1. A way to create, manage and apply analysis filters.
2. Change the graphical layout and number of CI levels to be displayed.
3. Select CIs for graphical viewing and right-click to provide various functions, including the ability to view and modify CI properties.
4. Use the Toolbar icons to provide various functions.
5. Zoom in to view areas of interest, or zoom out to see the larger context.
6. Use the Full Screen Mode to display a larger work area.

   NOTE
   When you enable Full-Screen mode, you cannot use the keyboard functionality. Search for CIs by Attributes.

For more information, see the following topics:

- Visualizer Graphs
- Change Icons of CIs on Visualizer

CA Business Service Insight Integration

When you integrate CA Business Service Insight and CA Service Desk Manager, the BSI Service tab displays Exceeds and Violations counts. The BSI Service by Metric and BSI Contract by Metric tabs display the Metric, Violation, Compliant, and Target attributes.

NOTE

These tabs only appear for CIs with Federated Asset mappings to the CA Business Service Insight MDR.

The integration provides the following reports:

- **Service Level Compliance**
  Displays information about CA Service Desk Manager Enterprise Service family CIs.
- **Modified Service Level by Metric vs. Target**
  Displays information about CA Service Desk Manager Enterprise Service family CIs.
- **Modified Service Level by Contract vs. Target**
  Displays information for CA Service Desk Manager Contract family CIs. Enter a valid CA Business Service Insight Contract Party and refresh the report.

Consider the following information to help you debug error messages:

- For an incorrect CA Business Service Insight MDR userID or SharedSecret, the error message displays Invalid username/org.name.
- For an incorrect CA Business Service Insight hostname, the error message displays BSI hostname is unknown.
- For an invalid Contract Party in the BSI Contract by Metric tab, the error message displays there is no metric data for the contract for the Reporting Period.
- If no CA Business Service Insight reports are associated with a particular Service or Contract, one of the following error messages is displayed:
  - For Services, there are no service details reports for federated service id.
  - For Contracts, there is no metric data for the contract for the Reporting Period.
- If CA Business Service Insight services are not available on the CA Business Service Insight computer, the error message displays that the CA Business Service Insight hostname is unknown.

Database Views

This article contains the following topics:

The CA SDM database is a repository for data entered and used to operate your service desk. CA SDM provides several database views and enables you to create customized reports of the database.

Basic Views

The basic views are based on CA SDM's tables. These views show data as long as the implementation makes use of Requests, Incidents, Problems, Change Orders, Issues, Assets, or Contacts / Groups.

Using the basic views, you can design and produce many reports, including:

- Detailed and summarized lists of open priority 1 requests sorted by request area.
- Detailed and summarized lists of change orders assigned to the level 1 group that have been open for more than 24 hours, sorted by priority.
- Counts of issues open for more than a specified number of days, sorted by priority, assigned group, or priority and assigned group.

The following basic views are provided:

View Name	Description
View_Contact_Full	All contacts, including their contact type, location, organizations, and service types
View_Contact_to_Environment	All contacts and their environment (assets)
View_Group	All the groups defined in the database
View_Group_To_Contact	All contacts (including managers) within their group assignments
View_Request	All requests, including their service types, severities, urgencies, categories, assets, impact numbers, assignees by name, customers by name, groups, statuses, and priorities
View_Act_Log	The request activity log information, including activity type and analyst name
View_Request_to_Act_Log	All requests with their activity logs (this view joins View_Request and View_Act_Log)
View_Request_to_Properties	Requests and their properties (this may not include all requests, especially if no properties are assigned to them)
View_Change	All change orders, including their statuses, priorities, categories, organizations, affected end users by name, requesters by name, assignees by name, groups, service types, and impact numbers
View_Change_Act_Log	The change order activity log information
View_Change_to_Change_Act_Log	All change orders with their activity logs (this view joins View_Change and View_Change_Act_Log)
View_Change_to_Properties	Change orders and their properties (this may not include all change orders, especially if no properties are assigned to them)
View_Change_to_Change_WF	Change orders and their workflow tasks (this may not include all change orders, especially if no workflow tasks are assigned to them)

View_Change_to_Assets	Change orders and their assets (this may not include all change orders, especially if they have no assets)
View_Change_to_Requests	Change orders with basic information about attached requests (this view joins View_Change and the Call_Request table, which may not include all change orders -- especially if no requests are attached to them)
View_Issue	All issues, including their statuses, priorities, categories, organizations, affected end users by name, assignees by name, groups, service types, and so on
View_Issue_Act_Log	The issue activity log information
View_Issue_to_Issue_Act_Log	All issues with their activity logs (this view joins View_Issue and View_Issue_Act_Log)
View_Issue_to_Properties	Issues and their properties (this may not include all issues, especially if no properties are assigned to them)
View_Issue_to_Issue_WF	Issues and their workflow tasks (this may not include all issues, especially if no workflow tasks are assigned to them)
View_Issue_to_Assets	Issues and their assets (this may not include all issues, especially if they have no assets)

Advanced Views

The advanced views are based on CA SDM's audit_log table. Install the audit_ins and /or the audit_upd options in the Options Manager and restart the system before using these views.

Using the advanced views, you can report on the duration of a ticket (that is, request, change order, or issue) in a particular state. For example, you can report on the following:

- Show the length of time, between opening and assignment to L2 for requests opened since January 1, 2002
- Show the length of time, issues remained priority 3 before escalating to priority 2 for issues opened since January 1, 2002.

The following advanced views are provided, which are views of the audit_log table in the CA SDM database that specifically query for the status, priority, group, or assignee attributes:

View Name	Description
View_Audit_Status	All tickets sorted by the time in each status (does not include tickets that have no status)
View_Audit_Priority	All tickets sorted by the time in each priority
View_Audit_Group	All tickets sorted by the time in each group assignment (does not include tickets that have no group)
View_Audit_Assignee	All tickets sorted by the time in each individual assignment (does not include tickets that have no assignee)

WARNING
You must install audit logging options using the Options Manager, to view data in the advanced views. The descriptions for the Audit Options audit_ins and audit_upd in the Online Help provide more information.

Using Configuration Audit

The Configuration Audit node lets you perform the following actions:

- Manage Incidents that CACF created for rogue or improperly executed changes.
  For example, View which changes have failed verification and which changes require more investigation.
- Manage Changes Specifications.
  For example, view all Change Specifications that completed successfully or are pending verification.
- Manage open Change Orders with active Change Specifications.
  For example, view the changes orders that are still open with pending change specifications.
- View the Verification Log history that shows all verification activities that have occurred in the system.
  For example, view which MDR is responsible for rogue updates and the affected CIs.

Managing CACF Incidents

The Change Manager manages CACF incidents generated due to a rogue change or improperly executed Change Orders that require further attention.

CACF policies provide conditions under which Incidents are created, and can specify that Incidents close automatically after failed verification remediation.

Managing Change Orders

The Change Manager manages Change Orders with active specifications from this menu.

Verification Log

The Verification Log list lets you view all CACF change verification activity. CACF logs each attempted update to the CI, and if the update was allowed. This log helps you determine which policy allowed or disallowed a change to a CI.

For example, search for a log that specifies a specific CI where CACF detected a variance or unverifiable change. The results display information such as the previous and discovered value, the verify status, and associated Incident.

You access the verification log from the following locations:

- On the Administration tab, click CA CMDB, Configuration Audit, Verification Log.
- On the Scoreboard click CMDB, Configuration Audit, Rogue Inserts or Rogue Updates.

The Verification Log tab for the specific object shows detailed events only pertaining to the object in question. View the log for the following objects:

- Verification Policies
- Change Orders
- Change Specifications
- CIs
- Incidents

> **NOTE**
>
> Verification log entries highlighted in red indicate that the corresponding change specification is either Failed Verification or Manual Verification Active, and requires further attention by the user.

The Discovered Attribute History list displayed in the Change Specification form lists all attribute values for the currently selected managed attribute and CI that were recently discovered by an MDR, whether they were actually authorized or loaded into the CMDB.

Use this list to assist in defining the planned value for the change specification. In the list you can view the format, case sensitivity, and so on of the previously discovered values so that you can determine an appropriate planned value pattern.

Managing Change Specifications

This article contains the following topics:

The Change Manager manages change specifications in your CACF environment. Change Analysts create Change Specifications when they create Change Orders. The Change Analyst defines the change in terms of the specific CI and CI attribute being changed.

The Change Specifications list shows all verified and outstanding change specification verifications in your environment. The ability to modify a change specification depends on the status of the Change Order. By default, you can only modify change specifications in Change Orders with the RFC status. After CACF verifies this change, it considers all subsequent changes as rogue inserts or updates

> **WARNING**
> If multiple discoveries occur during change specification verification, discovery data can invalidate a previously verified change. At the end of verification, the CI is in the state that all the change specifications desire. For example, a variance occurs for a CI with a change specification in the Verified state, but the status changes to Failed Verification.

A single CI can have many outstanding change specification verifications pending from many Change Orders. CACF evaluates each verification independently, and a single MDR update can update multiple pending change verifications.

The Final field on the Change Specification List indicates whether a change has been completed. The Change Order is eligible for to move to the next default state when all change specifications are final and the Promote Change Order After Verification option is enabled for the managed change state.

The following fields require explanation:

- **Verify Status**
  Specifies the status of the change specification, such as Verification Pending. For more information about change specification statuses, see Change Specification Statuses.
- **Change Order Status**
  Specifies the status of the Change Order.
- **Incident**
  Specifies the Incident that the verification policy creates optionally for an incorrectly implemented Change Order.
  If the Configuration Administrator specified the policy option to create Incidents, the Incident documents the expected and discovered values. If all discoveries result in failed verifications, the Incident receives log comments indicating that the incident represents multiple failures.
- **CI Name**
  Specifies the CI that you want to define the change specification.

  > **NOTE**
  > When you make several identical changes to a set of CIs, you can leave the CI Name field blank in the Change Specification. In this case, the Managed Attribute and Planned Value apply to all CIs attached to the Change Order. Change specifications for each CI are created when the Change Order moves to a status with Change Verification Active enabled for the managed change state.
- **Original Value**
  Specifies the value of the attribute at the time a user created change specification.
  The original value field shows the current attribute value for the CI when defining the change specification value for the first time.
- **Last Discovered Value**

Specifies the last value that was discovered for the attribute during the implementation and verification active process. This field updates each time CACF discovers a new value for the attribute.

- **Last Verification Policy**
  Specifies the verification policy that was in effect during the last change verification for this change specification.
- **Verification Message**
  Shows the message associated with the last verification action such as the reason why a verification failed.
- **TWA**
  Link to the TWA record associated with this change specification and policy.

Create a Change Specification

Create a change specification for a Change Order. By default, you can only modify change specifications in the RFC state. The Configuration or Change Administrators can change the definition of the Change Order managed change state that allows modification of change specifications.

The Verification Log tab shows the change verification activity history for this change specification. The log shows details about the policy and actions that CACF takes when change verification was in effect for the change specification. Verification log entries highlighted in red indicate that the corresponding change specification is either Failed Verification or Manual Verification Active, and requires further attention by the user.

To help you set the planned value, the Discovered Attribute History tab lists all values that an MDR recently discovered, whether they were authorized or loaded into the CMDB. From this tab, you can see the format and case sensitivity of the values so that you can determine an appropriate planned value pattern.

The Change Specification History tab shows the audit history listing all previous modifications made to the change specification options with the time and the user that made the changes.

Follow these steps:

1. From a Change Order, click the Configuration Management tab and select the Change Specifications tab.

   NOTE
   You can also create change specifications from the Administration tab from CA CMDB, Configuration Audit, Change Specifications. You can also create them from a CI if you click the Related Tickets tab then select the Change Specifications tab.

   The Change Specifications List page appears.
2. Click Create New.
   The Create New Change Specification page appears.
3. Complete the following steps:
   a. Enter the Change Order number that requests the change (only required if created from a CI or the Administration tab).
   b. Enter the name of the CI associated with the Change Order.
      You can leave the CI name blank to imply that the change specification applies to all defined CIs for the Change Order.
   c. Set the change specification as Active or Inactive.
   d. (Optional) Enter a description for the change specification.
   e. Select the Managed Attribute for this change specification from the drop-down list.
      Selecting Any Managed Attribute indicates that any managed attribute can be changed during the verification of the Change Order. In this case, multiple attributes can be updated, the planned value cannot be specified, and the planned change status is never updated.
   f. Enter the planned value of the Managed Attribute.
      Specify the expected value of the attribute after executing the change. After the Change Order moves into a verification state, and the CI updates through the web interface, GRLoader, or Web Services, CACF compares the planned value with the inbound data to determine a match.

You can embed special characters in the planned value when the exact value is not known.

g. Select the appropriate verification state of the Change Specification from the drop-down list:
   - Manual Verification will be required
     When CACF verifies the Change Order, manual verification will be required.
   - Set after Change Verified
     Set the CI attribute to the planned value after verification completes. Use this option for setting CI attributes when the attribute is not discoverable.

     NOTE

     The set operation only occurs when the Change Order moves from a state with change verification active to a nonactive default state. For example, the set takes place when a Change Order moves from Verification in Progress to Closed. The set does not occur if the Change Order is canceled, demoted, or moved to another state.
   - Use Discovered Value
     Copy the discovered value to the CI at verification time, used when the planned value is not known prior to verification.
   - Verification Pending
     The change has not been verified and waits for a CI update.

4. Click Save.
   The change specification is saved.

Change Specification Considerations

Consider the following information about change specifications:

- The CI name can be left blank which implies the change specification applies to all CIs defined for the change order. This option is useful when all of the CIs for Change Order are using the same planned value. The change specification applies to all CIs that are attached to the Change Order when the managed change state moves to a state with change verification active.
- You can create a Change Specification that requires manual verification. This action lets you verify managed attributes manually.
- Change Specification planned values for Lookup field managed attributes must be unique.
  For example, if you want to create a planned value for the Primary Contact (primary_contact), the contact specified in the planned value must be unique.
- You set case sensitivity only for comparisons made for change specification managed attribute planned values using the *Case Sensitive* option in the managed attribute. Specifying case-sensitive managed attributes do *not* affect the policy selection patterns.
- When the exact value is not known for the planned value, you can use wildcard (*), range (> or <) or negation (!) values.

Special Characters

You can embed special characters when you do not know the exact value. Use the asterisk wildcard character to match any number of characters. The pattern *must* match the discovered data in the same sequence, and spaces are significant.

For example, enter *10.*.*.** as the Planned Value to match any IP addresses that begins with *10.* and has two periods that follow any value between the periods.

For example, the inbound server_type value contains Windows 2003 (WIN32) 5.2.Service Pack 2 (Build 3790) Intel x86. To verify this value, specify a planned value of * *Service Pack 2* * in the change specification.

A word at the beginning of the planned value indicates that the discovered value must start with that value. Similarly, an asterisk at the beginning of the planned value indicates that the discovered value can begin with any value and end with the value specified following the asterisk.

The following table provides examples about how to use the asterisk:

Planned Value	Discovered Value	Match or No Match
a	b	No Match
a	a	Match
a	aba	Match
a	bab	Match
a*	a	Match
a*	ab	Match
a*	ba	No Match
*a	a	Match
*a	ab	No Match
*a	ba	Match

A pattern that begins with an exclamation point results in the negation of the value. You can only use the exclamation point as the first character in the pattern. For example, you cannot use a pattern of 10.!*.*.*.

To compare numeric values contained within string values, use greater than (>) or less than (<) as the first character in the planned value. If there is a leading exclamation point, it must be the second character.

WARNING
CACF ignores leading or trailing nonnumeric values in the discovered value and planned value patterns.

The following table provides examples about how to use the exclamation point, and the greater than and less than symbols:

Planned Value	Discovered Value	Match or No Match
>200	aaa 201 bbb	Match
>200GB	aaa 200 bbb	No Match
>200GB	300 GB	Match
!<200GB	200 Bytes	Match
!<200	200 Bits	Match

If you do not know the planned value in advance, but the value may change, you can set the status of the change specification to Use Discovered Value. This behavior requires discovery to update the CI before CACF considers the change specification as validated. You require this behavior for numeric fields and SRELs where an asterisk cannot be accepted as a planned value.

To help you with setting the planned value, the Discovered Attribute History tab lists all values that were recently discovered by an MDR, and whether were actually authorized, or loaded into the CMDB. This tab displays the format, case sensitivity, and other information about values so that you can determine an appropriate planned value pattern.

Change Specification Statuses

Change specifications use status to indicate the type of verification to perform and the current state of verification. The status values are also used in the verification log when recording change verification operations during the verification process.

- **Initial Statuses**

Used when creating a Change Specification to specify the type of verification for CACF to perform. These statuses are non-final and the Change Order is not considered verified when change specifications are in either of these statuses.
Verification Pending -- The change specification has not been verified, or the change is waiting for the CI update.
Manual Verification will be required -- When the changes for a Change Order are verified, manual verification will be required.
Use Discovered Value -- Copy the discovered value to the CI at verification time, and used when the planned value is not known prior to verification.
Set After Change Executed -- Sets a CI attribute to the planned value after verification has completed. Use this status for setting CI attributes when the attribute is not discoverable. For example, set the CI Primary Contact to User1 after a change completes.

- **Final Statuses**
  Indicates a completed change specification and considered as final. When all change specifications enter either of these final states, the Change Orders are eligible for promotion to the next default state.
  Verified -- The change specification has been successfully verified.
  Was Manually Verified -- The change specification has been manually verified.
  Used Discovered Value -- The CI has been updated with the discovered value.
  Was Set To Planned Value -- The CI has been updated with the planned value after Change Order verification.
  Accepted Planned Value -- The CI has been updated with the planned value and overwritten during verification.
  Accepted Discovered Value -- The CI has been updated with the last discovered value and overwritten during verification.
  No Change -- The planned value matched the CI at verification time and no verification was required.
  Cancel -- The change specification was canceled by the change manager.
- **Intervention Statuses**
  Indicates a change specification requires manual intervention for verification. CA SDM highlights these statuses in a red color on list forms. These statuses are not final and the Change Order is not considered verified when change specifications are in either of these statuses.
  Failed Verification -- Discovery found the value to be other than what was specified in the Change Order. The Change Analyst must determine if the change was correctly executed and the Change Order was incorrectly specified, or if the Change Order was correct and the change requires verification again. Depending on the definition of the Managed Change State and the current change order status, the Change Analyst can override, change, or cancel the failed change specification.
  Manual Verification Active -- Manual verification is required before the change specification can be marked final.
- **Action Override Statuses**
  Indicates an action to initiate during change verification. These statuses are not final and the Change Order is not considered verified when change specifications are in either of these statuses.
  Accept Planned Value -- Request to override the CI attribute with the planned value.
  Accept Discovered Value -- Request to override the CI attribute with the last discovered value.
- **Reporting Status**
  Indicates the result of policy operations that the verification log displays for logging purposes and not used by change specifications.
  Update Was Allowed -- A policy allowed a request to update a CI and did not match any change specifications.

Managing Failed Verifications

When a verification fails, the CMDB, Configuration Audit, Failed Verifications node in the scoreboard lists the number of failed verifications and change specifications that require manual intervention in a red color. The Change Specifications can also be viewed under the Change Specifications tab in the Change Order, CI or Administration tab with the verify status of "Failed Verification". When a verification fails, the Change Analyst can intervene or wait until the next discovery occurs.

If the managed change state allows the Show Change Specification Override Buttons action, the Change Analyst can open the change specification, review the data, and click one of the following options to close the verification by moving it to a final state:

- **Accept Discovered Value**
  The Analyst determines that the change specification is incorrect and that the discovery tool has discovered the correct value.
- **Accept Planned Value**
  The Analyst determines that the discovery tool is wrong or has not performed discovery, and to accept planned value, as if it was discovered correctly.
- **Cancel**
  This portion of the Change Order was not executed and this specification was canceled.

Managing Undiscoverable Attributes

When a Change Specification requires manual verification, the Configuration Analyst must manually verify the change. The CMDB, Configuration Audit, Manual Intervention node in the scoreboard shows the number of change specifications that require manual intervention in a red color. The Change Specifications can also be viewed under the Change Specifications tab in the Change Order, CI or Administration tab with the verify status of "Manual Verification Active."

If the managed change state allows the Show Change Specification Override Buttons action, the Change Analyst can open the change specification, review the data, and click one of the following options to close the verification by moving it to a final state:

- **Mark as Verified**
  Used when the attribute is not discoverable and manually verified.
- **Cancel**
  This portion of the change order was not executed and this specification was canceled.

Defining Bulk Changes

Creating identical changes to a large number of CIs, for example, when changing the location for a collection of CIs, with the following steps:

1. Create a Change Order.
2. Define a single change specification which describes the change and leave the CI name blank.
3. Attach all the CIs to the Change Order.

If you do not know the exact details of a change in advance of the implementation, for example, you acquire a new Server and you only know the CI Name, complete the following steps:

1. Create a Change Order.
2. Create a change specification that specifies Any Managed Attribute as the attribute name.
3. Leave the planned value empty, since it is unknown.
4. Move the ticket through the change management process.
5. When the Change Order is in a verification active state, and discovery runs, the inbound CI data loads into the CI (assuming that the policy allows it).
6. When all discoveries for the CI complete, the Change Manager must mark the change specification as verified manually.

Follow these steps when there are a large number of dissimilar changes to a large number of CIs. For example, when moving a collection of servers from one location to another, and also assigning each one a unique IP address.

1. Create a Change Order
2. Create a spreadsheet and list on each row the Change Order number, CI and, the new attribute values. Each row results in a distinct change specification.
3. Load the spreadsheet with GRLoader to create the required change specifications.
4. Promote the Change Order and its change specifications as usual.

NOTE
For more information about using GRLoader, see CMDB Technical Reference.

Configuration Audit and Control Facility (CACF)

Configuration Audit and Control Facility (CACF) unifies three disciplines: Change Management, Configuration Management (CMDB), and Discovery Management. CACF verifies that changes are executed accurately and no unauthorized changes occur.

Change verification ensures that the CMDB reflects any changes accurately, and the Discovery Management tools verify the changes.

Change verification provides the following benefits:

- **Change Management**
  - Assurance that authorized changes execute correctly.
  - Detection of incorrectly executed changes.
  - Detection of unauthorized changes.
  - Management reporting.
  - Full auditing of all changes down the attribute level.
  - Detection of overlapping or conflicting changes.
- **Configuration Management**
  - CMDB contains an accurate and current representation of all managed CIs.
  - Ability to view the future state of the CI with the proposed change specifications.
  - Full auditing of CIs.

CACF has two major sections: the CMDB administrative interface and the change management interface.

CACF Administration and Policy Definition

This article contains the following topics:

The CMDB Administrator defines CIs and attributes that are managed. The CMDB Administrator also defines the policy for updating those CIs and attributes. You administer the CACF components in the Configuration Control node in the CA CMDB section of the Administration tab. Define CACF change specifications from the Change Order, CI, Incident forms, or the Configuration Audit node on the Administration tab.

The administrator must consider allowing standard changes and whether to define the following items:

- Authoritative and trusted sources of data.
- CIs and the CI attributes that are under CACF management.

An incorrectly executed change or rogue change can occur (also referred to as a variance). In this case, CA SDM can respond in any of the following ways:

- accept the new value
- create an Incident
- copy the data to the TWA for later processing.

CA SDM can use any combination of these actions.

WARNING
The Configuration Administrator *must* establish a change verification strategy for your environment. The default policy for CACF allows all changes to all CIs. This policy applies even if the change is rogue, or if the change does not match a change ticket.

You can implement change verification policies dynamically or scheduled in advance. You can define these policies as generic or highly specific. The change verification policy describes how CA SDM responds to the following events:

- **Updates from Unauthorized MDRs**
  Indicates that specific MDRs are authoritative for specific attributes. CI attribute updates from unauthorized MDRs can be selectively accepted or rejected.
  For example, define policies to prevent CA Application Configuration Manager from updating the IP address of a CI, even if a matching Change Order exists. Let updates to IP address only if the source MDR is Spectrum.
- **Rogue Changes**

Detects and manages updates to CIs when no corresponding Change Order exists. Specify a policy that manages rogue changes requesting inserts or updates of CI data. For example, you can define a policy that loads data into TWA and does not update the CI for the following conditions:

- A CI named similar to server* in New York changes
- The change does not have a matching Change Order

- **Incorrectly executed changes**
  Detects whenever a Change Order is not implemented correctly.

Auditing facilities provide the Configuration Manager with capabilities to view changes to policies and CACF object definitions. CACF logs each attempted update to the CI, whether the update was allowed. This log helps you determine which policy allowed or disallowed a change to a CI.

The Configuration Administrator defines which Change Order statuses represent changes that can be edited, and which Change Order change states represent verification states. By default, you can edit change specifications for Change Orders in the *RFC* change state. In addition, you can edit when change verification occurs for a Change Order in the *Verification in Progress* state. Optionally, change tickets can be automatically promoted or closed when all associated change specifications execute successfully.

> **WARNING**
> Performance may be degraded if the Configuration Administrator enables the Create Incident option with more than one active verification policy.

Note: By default, CACF considers updates to attribute values to Change Orders with change verification active as non-rogue changes. The changes are effectively in the Verification in Progress status. The Configuration Administrator can modify this behavior.

How Change Analysts Define Change Specifications

The Change Management interface portion of change verification integrates into the change ticket order form. Change Analysts typically create change specifications when they create a Change Order. The Change Analyst defines the change in terms of the specific CI and CI attribute that they want to change.

The Change Analyst can describe the change specification using any of the following templates:

- Provide the exact CI, CI attribute, and value.
  For example, after the change has been implemented, the IP Address for server1 sets to 10.10.10.10.
- Provide the CI attribute and value, but omit the CI name.
  For example, after the change, all CIs attached to the Change Order upgrade to 8 GB of memory.
- Provide an attribute value to set when the attribute is not discoverable.
  For example: after the change, the CMDB reflects that all CIs are located in NY.
- Use the discovered value.
  For example, you do not know the new IP address in advance, but you know that the IP address will definitely change. Set the IP Address of the CI to whatever value the authoritative MDR discovers.

When the change order is approved, the change specifications are approved as part of the same approval process. The subsequent modifications to the specification can be prohibited and are fully audited.

Change Verification

Change verification ensures that Change Orders execute according to change specifications. A Change Order is verified when its state is Verification in Progress and when the CI is updated. Now, when data from any MDR or web client user imports into the CMDB, CACF compares the incoming attribute data with active change specifications. If the CI attribute data matches, CACF verifies the change. If the inbound data does not match the Change Order specifications, a corrective action can occur based on the policy.

The executed policies can perform any of the following actions:

- Reject a single attribute update from the transaction.
- Reject the entire transaction, including all attribute and value pairs.
- Load the entire transaction to the TWA for deferred processing after approval and verification of the transaction.
- Create an Incident that describes each variance.

  NOTE
  This action can trigger existing notification and automation. To prevent multiple incident creation, rogue change detection creates a single incident for all rogue changes to a single CI.
- Accept the transaction unconditionally when accepting data from authorized source MDRs.

For example, the CMDB Administrator can define a policy that creates an Incident when a change to a production server executes incorrectly. When imported data from an authorized MDR does not match a pending change specification, CACF creates an Incident. The policy can allow or reject a CI update by the nonmatching import data. In addition, the Change Manager may accept the newly discovered value, even if the value does not match the planned change specification exactly.

NOTE
If multiple discoveries occur during change specification verification, discovery data can invalidate a previously verified change. At the end of verification, the CI is in the state that all the change specifications want.

How Change Managers Use Change Specifications

The Change Manager can perform the following actions:

- View and modify the verification status of each change specification.
  For example, the Change Manager wants to know which change specifications are still pending for Change Order 12345. The change order is in *Pending Verification* status.
- Manage Incidents for rogue or improperly executed changes.
  For example, the Change Manager wants to see which changes have failed verification and which changes require more investigation.
- Manage the changes that require manual intervention.
  For example, a Change Order specifies that manual verification is required for a nondiscoverable change. The Change Analyst must complete the task.
  For example, the Change Manager investigates and determines that Change Order 12345 specifies ten different change specifications. Due to a typo, change specification number 9 was incorrect and can be canceled.

Auditing facilities provide the change capabilities to view the status of each update to the change specification and any overrides to the change specification. The scoreboard shows the status of the running system and any exceptional conditions that require intervention or investigation. CA Business Intelligence reports provide a longer term view of the efficiency of Configuration Manager policies to monitor the health of your environment.

How Configuration Audit and Control Facility Works

This article contains the following topics:

CA

CACF governs inbound CI data before loading the data into the CMDB. This verification ensures that each requested Change Order executes correctly, and detects and manages rogue changes automatically. The Configuration Administrator (CMDB Administrator) defines verification policies. The policies determine how CA SDM responds when any deviation or variance from the requested change occurs.

The following diagram shows how CACF works:

1. Inbound CI data loads from Web Services, GRLoader, MDR, or the Web Interface to request an attribute update for a CI.
2. Based on the verification policies that your Configuration Administrator establishes, CACF completes one or more of the following actions:
   - CACF accepts the update and modifies the CI data in the CMDB.
   - CACF defers the update to the TWA for further processing.
     For example, your Change Manager handles transactions in the TWA.
   - CACF creates an Incident and lets you manage the ticket.
     For example, your Incident Management process requires managing the Incident for reporting purposes.
   - CACF rejects the update and the CI data is not changed in the database.

Verification Policy

First the Configuration Manager and the Change Manager agree on a change verification policy. Next, the decisions transcribe into a CA SDM change verification policy. A system can have many verification policies. When a CI update occurs, CACF selects a single verification policy for each updated attribute. The verification policy can update the CI, create a TWA entry, or create an Incident if a matching change specification exists.

NOTE
To monitor updates through the web client, specify the MDR Class Pattern as **Web Client** (case sensitive) in the policy. This pattern usage applies to English and localized versions of CA SDM when you want to monitor updates through the web client.

WARNING
Determining what policies you need in your environment requires significant thought into the goals of the organization and how you want to introduce those policies.

Verification Policy Selection

When an update occurs to a CI, CACF selects a policy for each updated attribute. This selection depends on the source of the update, the target CI, and whether matching change specification exists. When multiple policies match, CACF selects the single policy with the lowest sequence number to manage the attribute update.

NOTE
CACF can select multiple policies to manage a single transaction that updates multiple attributes. For example, updating a CI with a new description and new IP Address can trigger two different policies.

Each policy has four sections that CACF uses in policy selection:

- **Identification and Priority**
  Identifies a policy and specifies the order to evaluate the policy.
- **Change Order Alignment**
  Selects transactions that are based on how closely the attribute update matches an existing change specification.
- **Transaction Filter**
  Specifies the data sources to which this policy applies.
- **CI Filter**
  Specifies the CIs to which this policy applies.

A policy matches an attribute update when it passes each of these filters. If a policy does not match a transaction, CACF bypasses the policy. Next CACF evaluates other policies in the system. If CACF bypasses all policies, the default action lets the CI attribute update occur.

Verification Policy Filter Syntax

To help with establishing generic policies, the Change Order Alignment, Transaction Filter, and CI Filter accept an asterisk as a final pattern character as a wildcard. To indicate negation, use an exclamation point at the beginning of these filters.

> **WARNING**
> Policy pattern rules differ from change specification planned value rules. For policy filters, you cannot use embedded asterisks (*) or the greater than (>) and less than (<) special characters.

Policy filters are case-sensitive for verification policy selection patterns, such as CI Name or CI Class. The filters are case-sensitive even if you define those attributes as case insensitive.

> **NOTE**
> A blank "" (no value is specified in the field) policy filter matches only the blank value. A policy filter that uses an asterisk matches all values.

Consider the following information with the following example verification policies:

- Policy1 selects CIs name(prod*) attribute(IP Address) where a matching change specification exists.
- Policy2 selects CIs name(prod*) attribute(IP Address) where there is a rogue update.
- Policy3 selects CIs name(test*) attribute(Memory Installed) from mdr(Cohesion).
- Policy4 selects CIs name(“”) attribute(Memory Installed).

The following list describes the possible outcomes of these policies:

- When an update to CI(test2) attribute(Memory Installed) occurs, Policy3 manages the update. CACF ignores Policy1 and Policy2.
- When an update to CI(prod3) attribute(Memory Installed) occurs, and a matching change specification exists, Policy1 manages the update.
- When an update to CI(development4) attribute(IP Address) occurs, CACF updates the CI with the new IP address as no matching policy exists.
- When an update to CI(prod1) attribute(Disk Capacity) occurs, no policies for attribute(Disk Capacity) exist. CACF updates the CI with the new value. In this example, CACF does *not* manage the Disk Capacity attribute.
- Policy4 only matches CIs with blank names. Because blank CI names are invalid, the change is never executed.

Verification Policy Prioritization by Sequence Number

When multiple policies match an attribute update, CACF chooses the policy with the lowest sequence number. Consider the following policies and their sequence numbers.

- Policy name(Policy1) sequence(1000) CI(prod*)
- Policy name(Policy2) sequence(2000) CI(prod-NY*)

When an update to CI(prod1) occurs, Policy1 manages the transaction. When an update to CI(prod-NY-1) occurs, Policy1 still manages the transaction as it has a lower sequence number. In this example, Policy2 never takes effect.

We recommend that you change the sequence numbers so that the most specific policy has the lower sequence number. Refer to the following sample policies:

- Policy name(Policy1) sequence(2000) CI(prod*)
- Policy name(Policy2) sequence(1000) CI(prod-NY*)

When an update to CI(prod1) occurs, Policy1 manages the transaction. When an update to CI(prod-NY-1) occurs, Policy2 manages the transaction as it has a lower sequence number.

Change Order Alignment

Policies filter on how closely a CI attribute update matches a change specification. After a CI update, CACF searches for a change specification that matches both the CI and the attribute name. This search can verify the change specification. This Change Order Alignment can occur as follows:

- Change specifications exist for this attribute and CI.
- Change Orders exist for this CI, and those Change Orders do *not* contain change specifications.
- A rogue insert transaction occurs which indicates and new CI. A Change Order does *not* exist with this CI attached to it.
- A rogue update transaction occurs which indicates an existing CI. A Change Order does not exist with this CI attached to it.

Only Change Orders in a state where *change verification is active* are considered when looking for a matching Change Order. Closed, unapproved, unscheduled, or otherwise nonexecuted Change Orders are not considered when searching for a matching Change Order.

After CACF determines the Change Order alignment of a transaction, it performs a search for a corresponding policy. A policy can manage one or more Change Order alignment types:

- **Change Order with Specifications**
  Indicates that a Change Order with a change specification which specifies the updating CI and attribute name exists.
- **Change Orders Without Specifications**
  Indicates Change Orders that specify this updating CI, and those Change Orders do not have any change specifications.
- **Rogue Insert**
  Indicates an inserted CI, by definition, the CI does not have any matching Change Order in a verification state.

  WARNING
  Policies which prevent rogue inserts must specify a managed attribute of *Name* or *All Managed Attribute. Name* specifies an active managed attribute.

  CACF always allows rogue inserts and updates when a transaction only has unmanaged attributes.
- **Rogue Update**
  Indicates an updated CI that has no matching Change Orders in a verification state.

  Example Change Order Alignment Policies

  The following example policies specify Change Order Alignments:

  - Policy name(legacy) filters CI(*) attribute(Any Managed Attribute) alignment(Change Orders Without Specifications) action(Allow Attribute Update)
  - Policy name(new) filters CI(*) attribute(Any Managed Attribute) alignment(Change Orders with Specifications) action(Allow Update Only if Matches Change Specification)
  - Policy name(no-rogue) filters CI(*) attribute(Any Managed Attribute) alignment(rogue insert or rogue update) action(Always Cancel Entire Transaction, create Incident)

  These examples provide the following functionality:

  - Policy(legacy) allows CA SDM r12.6 change orders without change specifications to update the CI as they did before a CACF implementation.
  - Policy(new) enforces change verification for all new change orders which have matching change specifications.
  - Policy(no-rogue) prevents updates to CIs when no Change Orders exist in a Verification Active state.

Transaction Filter

Policies filter transactions that are based on the source of the transaction. The source includes the attribute name, MDR name, MDR class, and role of the user that performs the update.

Note: If you want to filter users logged on through the web interface only, specify the keyword **Web Client** for the MDR class. The userid of the contact specifies the MDR name. This method of identifying users applies only to verification policies. The method does not appear in any other part of the product.

Example: Transaction Filters

The following example policies specify Transaction Filters:

- Policy name(root_access) filters ci(*) attribute(All Managed Attributes) role(Administrator) action(Allow Attribute Update)
- Policy name (cohesion_not_authorized) filters ci(*) attribute(IP Address) MDRclass(Cohesion) action(Keep Old Attribute Value)
- Policy name(john) filters ci(user1*) attribute(All Managed Attributes) MDRName(user1) MDRClass(Web Client) action(Allow Attribute Update)

These examples provide the following functionality:

- Policy(root_access) lets any user with Administrator access update any value. We recommend this type of policy to have a low sequence number.
- Policy(cohesion_not_authorized) prevents any MDR of MDR class(Cohesion) from updating the IP Address of the filtered CI. This example shows how to prevent an unauthorized MDR from updating data.
- Policy(user1) lets user1 update the CIs that the contact owns, but only when using the web client. This example shows how to provide specific users full control over their data.

CI Filter

Policy filters transactions on the characteristics of the updated CI. This selection criteria includes CI Name, class, priority, service type, and location.

> **WARNING**
> The CI filter is based on the attribute value in the CI before the update occurs. The filter is not based on the value in the inbound transaction data.

Example: CI Filter Policies

The following example policies specify CI Filters:

- Policy name(priority1) filters Service Type(priority1 resolution) action(Allow Update Only if Matches Change Specification, Create Incident)
- Policy name(prod-NY) filters ci(prod*) location(NY) attribute(All Managed Attributes) action(Allow Update Only if Matches Change Specification)
- Policy name(prod-not-NY) filters ci(prod*) location(!NY) attribute(All Managed Attributes) action(Allow Attribute Update)

These examples provide the following functionality:

- Policy(priority1) requires CIs with a service type of *priority1 resolution* to have a matching Change Order. This policy is a best practice because it helps control the most important CIs in the CMDB. This policy also requires verification for all changes under change management to have change specifications. It also requires the changes to be verified and completed.
- Policy(prod-NY) requires CIs in the NY location to have matching Change Orders. Using location to filter policies can help gradually implement change verification on a site by site basis.
- Policy(prod-not-NY) illustrates using the exclamation point in the location pattern to indicate CIs not located in NY.

Policy Actions

After CACF selects a policy, the action section of the policy determines the outcome of the attribute update. This section identifies the most important part of the verification policy because it affects the integrity of the CMDB. This section also affects the change management workflow, related notifications, and created Incidents.

A policy can have one of the following Update Behaviors:

- **Allow Update Only if Matches Change Specification**
  If the CI matches a change specification, conditionally applies the inbound attribute update to the CI. This update occurs when the Change Order is in a Verification Active state. Selecting this option enables change verification.
- **Allow Attribute Update**
  Unconditionally applies the inbound attribute update to the CI. This option effectively disables all change verification. Use this behavior for standard changes, when you have a trusted data source, authoritative, and you do not require a Change Order.
- **Always Cancel Entire Transaction**
  Cancels the update and any other attribute update in this transaction, even if a matching change specification exists. To prevent MDRs from updating CIs where they are not authorized to update, use this behavior. If any policy cancels a transaction, the entire transaction is canceled. The transaction is canceled even if other policies would have allowed the change to occur.
  For example, specify this behavior to stop an unauthorized MDR from inserting a CI. In addition, specify a common attribute name such as *Name*.
- **Keep Old Attribute Value**
  Cancels the single attribute update, but allows other attributes in the transaction to update if their policy allows it. Use this behavior when an MDR is unauthoritative at the attribute level.

A verification policy can specify that Incidents close automatically when failed verifications are remediated. The policy can specify that the inbound transaction data copies to the TWA. This type of policy is useful when data from an unauthorized MDR requires review before updating the CMDB.

To let the Configuration Manager identify that CACF policies that created the TWA record, the Change Order field in TWA is set to the name of the policy that created it. Writing data to TWA is independent of updating the CI in the CMDB, so a policy can update the CI, write to the TWA, or both.

Policy Scheduling

You can specify activation and deactivation dates on verification policies, which let the Configuration Administrator schedule unattended policy changes.

Multiple Policies

As you introduce more verification policies into your environment and more attributes are managed, organizing the policies becomes increasingly complex. More overlaps exist between the policies. For example, you have one policy for Server CIs and another policy for CIs in NY. Consider the appropriate policy for Server CIs in NY. Use the policy sequence number to force one policy to take precedence when multiple policies can potentially control a single attribute. As you manage more attributes, the possibility that more policies manage a single transaction can increase.

Consider the following information when you have multiple active policies for a single transaction:

- If any selected policy requests writing data to the TWA, data also writes to the TWA.
- If any selected policy requests to cancel the transaction, CACF cancels the entire transaction.

  WARNING
  Facilities that use the createAsset web service (including GRLoader and CMDBf) break the transaction into three separate transactions. The transactions include two inserts, and an update. Updates to ca_owned_resource can be allowed, while updates to the CI extension table are canceled. Updates to CI family-specific attributes in the extension table may not be detected as inserts, but rather as updates.

Example Multiple Policies

The following examples specify multiple policies:

- Policy name(IP Address) attribute(IP Address) action(Allow Attribute Update)
- Policy name(Memory Installed) attribute(Memory Installed) action(cancel transaction)
- Policy name(Disk Capacity) attribute(Disk Capacity) action(Keep Old Attribute Value)

These examples provide the following functionality:

- If a transaction only updates IP Address, the update is performed.
- If a transaction only updates Memory Installed, the update is not performed.
- If a transaction updates both IP Address and Memory Installed, the transaction is canceled in the web interface.
- If a transaction updates both IP Address and Disk Capacity in the web interface, IP Address is updated, but Disk Capacity is not updated.
- If a transaction updates both IP Address and Memory Installed in the web interface, neither attribute is updated.

Incident Consolidation

When a verification policy requests to create an Incident, CACF can reduce the number of open Incidents as follows:

- CACF creates only one open Incident for a single CI for rogue changes. More rogue changes update the single open Incident for rogue changes.
- CACF creates only one open Incident for each change specification verification failure. More verification failures for the change specification update the open Incident.

How to Define Policies for Change Verification

To begin using CACF in your CA SDM environment, complete the following recommended steps:

1. Review the Managed Change States in the CA SDM environment. Update the default Managed Change States or add states as necessary to suit your change management strategy.
   Change verification initiates when the status of a Change Order changes to a Change State that CACF manages. For example, Verification in Progress.
2. Identify the available CI Managed Attributes that you want CACF to manage for change verification.
   - For example, you want to manage changes to Memory Installed (phys_mem) so that the attribute *must* have a matching Change Order for the test* server. You also consider changes to IP Address (alarm_id) as normal and always allowed.
   - For example, you want to manage changes to Memory Installed (phys_mem), MAC Address (mac_address), and IP Address (alarm_id) so that the attribute must have a matching Change Order for prod*. You do not want MDR1 to update IP Address and prefer MDR2 to complete this update. When you have a problem, create an Incident. Closing the Change Order also closes the Incidents automatically.
3. Determine what CIs or subset of the CIs in your system that you want to monitor.
   For example, you want to control all servers name test* and prod*. In addition, you control what CIs based on CI Class and CI Location.
4. Determine what MDRs and sources of data that you want to monitor.
   For example, you want to restrict data from MDR1 for the IP Address attribute from updating any CIs.
5. Determine the appropriate verification policy to manage the following types of change:
   - CI changes that match change orders.
   - CI changes that match the attributes in a Change Order, but do not match the value.
   - CI changes that you consider as rogue updates.
   - CI changes that you consider as rogue inserts.

6. Determine the appropriate update behavior that CACF takes when it detects a variance for a policy:
   - Always allow the attribute update request.
   - Allow the attribute update only when a corresponding change specification is specified for the update.
   - Always cancel the entire transaction.
   - Keep the existing attribute value.
   - Write data to the TWA.
   - Create an Incident when CACF detects a variance.
   - Automatically close any Incidents after Change Order verification.

For more information about change verification planning and implementation, see Planning and Implementing Change Verification.

Change Verification Best Practices

Consider the following best practices when implementing change verification:

- Define a small number of policies.
- Avoid the use of negative logic (exclamation point in a policy pattern).
- A single CI update save must match a small number of policies.
- Organize the policies in a numerically tiered structure.
- Use Log Only Mode policies before you implement a change verification policy.
- Minimize overlapping policies, where multiple policies manage a single attribute being updated.
- Limit the number of Incidents that CACF creates.
- Implement archive and purge of CACF data.

Organizing Policies

You can use several strategies to track policies. We recommend that you use a multi-tier approach. To allow for future inserts, use increments of 100 between policy numbers.

Consider the following example tiers for this strategy:

- **Fundamental Policies**
  100,000-199,999
  - Allow Change Administrators to perform inserts.
  - Disallow the inserts by MDR Class(xxxx).
  - Exclude all test CIs from incident creation.
- **Temporary or Transient Policies**
  200,000-299,999: Bulk loads this week
- Application-Specific Policies
  - 301,000-301,999: Exceptions to the following general application-specific policies:
    For example, server1 in NY policies
  - 311,000-311,999: Policies that relate to Service Type priority1 resolution.
  - 320,000-320,999: Policies that relate to NY.
  - 331,000-331,999: Policies that relate to Lisle.
- Technology-Specific Policies
  - 410,000-410,999: Policies that are related to Servers.
  - 411,000-411,999: Policies that are related to Network.
- **Default Policies**
  You can use these policies if the previous policies do not apply to your environment.
  900,000-999,999: Default policies, such as all updates to managed attributes (Any Managed Attribute) must have a change specification.

Multi-Tenancy Considerations

Consider the following information when using CACF in a multi-tenancy environment:

- Managed attributes, verification policies, Change Orders, and change specifications are tenanted.
- Tenants up their hierarchy can view all CACF objects.
  For example, if a tenant creates a managed attribute, this action may block a tenant in the hierarchy from creating a duplicate managed attribute. The hierarchy requires unique policy sequence numbers.
- To determine the policy that manages a change to an attribute, consider only policies from the tenant hierarchy of the object (CI).
- A subtenant can create a policy that overrides the supertenant by assigning a lower sequence number to their local policy.
- CACF only considers policies at the tenant level and its parents in the hierarchy.
- CACF does not consider the tenant of that contact that performs the change.

NOTE
Verification policies are specific to your system and do not vary by user or role.

CACF Roles and Functional Access

The following table describes the default roles that CACF uses:

Role/Functional Access	Administration	CI	Incident/Problem/ Request	Change Order
Administrator	Modify	Modify	Modify	Modify
Configuration Administrator	Modify	Modify	Modify	Modify
Configuration Analyst	View	Modify	Modify	Modify
Configuration Viewer	None	View	Modify	View
Change Manager	View	Modify	Modify	Modify
Service Desk Administrator	Modify	Modify	Modify	Modify
Service Desk Manager	View	Modify	Modify	Modify
System Administrator	Modify	Modify	View	View
Level 1 Analyst	View	View	Modify	View
Level 2 Analyst	View	Modify	Modify	Modify
Incident Manager	View	View	Modify	Modify
Problem Manager	View	View	Modify	Modify

- **Administration**
  Indicates creating, updating, and viewing CACF administration, policies, and attribute management.
- **CI**
  Indicates creating, updating, and viewing CI and relationship attribute data.
- **Incident/Problem/Request**
  Indicates modifying and viewing outstanding CACF issues, including rogue changes and incorrect change execution variances.
- **Change Orders**
  Indicates creating, updating, and viewing change specifications.

For example, the Change Manager role can view CACF policies and can manage attributes, but cannot update them.

WARNING
Update access to the Change Order and its status determine who can edit change specifications. For example, the Change Administrator provides this access to the Change Manager.

Example Change Specification in the Pending Verification Status

In this example, the Configuration Administrator established a verification policy for the Memory Installed (phys_mem) attribute to Allow Update Only if Matches Change Specification. CACF monitors a change specification for this attribute with the Verification Pending status. It sets the Change Specification to the Verified or Failed Verification status. The following steps describe the lifecycle of this change specification:

1. A Change Order requests an update to the Memory Installed (phys_mem) attribute of a CI.
2. You create a change specification for the Memory Installed (phys_mem) Managed Attribute with a Planned Value of 4 GB with the initial status as Verification Pending.
3. The Change Order moves to a status with change verification active such as Verification in Progress.
4. The change specification waits for a discovery tool to discover CI details and export the values to the CMDB.
5. One of the following actions occurs after the discovery tool exports the value to the CMDB:
   – The value for Memory Installed matches the 4gb value that you specified in the change specification. CACF sets the status from Verification Pending to Verified. CACF closes the Change Order.
   – The value for Memory Installed does not match the value that you specified. CACF sets the status from Verification Pending to Failed Verification.

Example Change Specification in the Set After Change Verified Status

In this example, CACF updates the CMDB with the value that you want to use after the Change Order exits a managed change state with change verification active. For example, your environment uses Verification in Progress as the managed change state with change verification active enabled. The following steps describe the lifecycle of this change specification:

1. A Change Order requests an update to the Serial Number (serial_number) attribute of a CI.
2. You create a change specification, specify Serial Number, and enter a planned value for an attribute.
3. You set the Change Order state to from RFC to Verification in Progress to Closed.
4. The CI attribute value is updated with the planned value after the Change Order exits a state with Change Verification Active (Verification In Progress). The value is updated to a state that does not have change verification active (Closed).
5. The change specification sets to Was Set to Planned Value to indicate that the set was completed successfully.

Accept the Planned Value

In this example, the Change Analyst researches the correct attribute value for a CI and accepts the planned value.

Follow these steps:

1. Open a Change Specification that an end user created in your environment.
2. To see the planned value for the CI attribute that the Change Analyst modified, view the detail page.
3. Research the attribute value for the CI.
   For example, your research indicates that the Change Analyst entered the correct value.
4. To update the CI with the corresponding planned value, click Accept Planned Value.
   CACF sets the state of the Change Specification to Accepted Planned Value.

Accept the Discovered Value

In this example, the Change Analyst determines that the CI is updated to a correct attribute value for a CI and accepts the discovered value.

Follow these steps:

1. Open a Change Specification that a Change Analyst created in your environment.
2. To see the planned value for the CI attribute that the Change Analyst specified, view the detail page.
3. Research the attribute value for the CI.
   For example, discovery indicates that the Change Analyst entered the incorrect value.
4. Click Accept Discovered Value.
   The CI updates with the discovered value. CACF sets the status of the change specification from Verification Pending to Accepted Discovered Value.
5. If you configured the managed state to enable the Promote Change Order After Verification option, CACF closes the Change Order. The closure happens after all change specifications are in a final state.

Manage Changes from an Unauthorized MDR

In this example, the Configuration Administrator established a verification policy to manage changes from an unauthorized MDR named MDR1. The policy rejects all updates to CIs from MDR1.

The following steps describe the lifecycle of this verification policy:

1. You create a Verification Policy with an MDR Name pattern of MDR1 and an Update Behavior of Always Cancel Entire Transaction.
2. Enable all Change Order alignment options.
3. Specify Any Managed Attribute and an asterisk (*) for all filters.
4. When the discovery tool (MDR1) runs and exports the data to the CMDB, the verification policy rejects all the updates.

Managed Attributes

Managed attributes indicate those eligible CI attributes for change verification by CACF. By default, this list contains *CI Name* and *Any Managed Attribute*. You add the CI attributes that you want managed as part of your change verification strategy. Define managed attributes as part of your change verification strategy in the Configuration Control, **Managed Attributes** node in the CA CMDB section of the Administration tab.

CACF does not consider unmanaged attributes (attributes not listed) for change verification. These unmanaged attributes update as usual.

> **WARNING**
> Change verification ignores unmanaged attributes and lets them update the CI, unless a verification policy specifies the Always Cancel Entire Transaction behavior.

> **NOTE**
> Case sensitivity in the managed attribute definition only applies when CACF compares the change specification planned value with the inbound CI transaction data. Case sensitivity does not apply to the selection patterns in the policy, which are always case-sensitive.

For a list of CI attribute names, see CMDB Technical Reference .

Managed Change States

This article contains the following topics:

CACF uses managed change states to indicate which Change Order statuses CACF manages. CACF uses managed change states to control how or when to apply change verification for CI updates made to the system. You can customize these change states to suit the needs of your organization.

You configure managed change states on the Administration tab in the CA CMDB, Configuration Control, Managed Change States node.

The following list describes the Managed Change States options:

- **Change Specifications Editable**
  Specifies whether you can edit the change specifications for a Change Order. Typically, after a change request receives approval, you cannot change the request, and updates are restricted to performing a small set of override options in the *Verification in Progress* state.
- **Change Verification Active**
  Specifies whether changes discovered for a CI while the Change Order is in this status are considered for change verification. Change orders and their related change specifications are compared with any inbound transactions to verify that they were executed successfully.
  CACF monitors all CIs for any changes to their managed attribute values. As CACF verifies each attribute level change to the CI, CACF compares it against a list of change specifications that are in a *change verification active* state.
  After a change specification enters this state, any change specification without a specific CI undergoes expansion. This expansion occurs where new change specifications are created using the list of CIs attached to the Change Order.
  After a change specification exits this state, the list of change specifications with a verify status of *Set After Change Executed* complete. This action updates the CI with the planned values, as specified in each change specification.
- **Implementation State**
  Specifies whether the state represents a state when the changes are being executed or implemented on the CI. When a Change Order enters this transitional state, it is understood that the attribute values in the CI are volatile and may be updated as requested by the pending change specifications. CACF cannot consider these changes rogue changes, but it also cannot consider the changes for final verification. Typically, the change verification process should only compare the inbound attribute data after the Change Order executes completely.
- **Show Change Specification Override Buttons**
  Specifies whether the Change Analyst can control change specifications, and what level of control is given. In some implementations, the Change Analyst can edit change specifications as necessary, while in other implementations, the Change Analyst can override or cancel the change specification.
- **Promote Change Order after Verification**
  Specifies whether a Change Order promotes to the next default state automatically after CACF verifies all the change specifications.

Default Managed Change States

You can define Change Order states during which change specifications are created, overwritten, verified, or ignored.

The following list describes the default managed change state definitions that are provided by CA SDM as an example:

- **RFC**
  Define the Change Order specifications in this state, but are not considered during change verification. Changes to the CI detected while the Change Order is in this state are typically considered as rogue.
- **Approval in Progress**
  Change specifications in this state have the same characteristics as RFC and are editable and not considered for verification.
  After Change Order approval, the change specifications are approved as part of the same approval process. CACF prohibits more modifications to the specification and provides full auditing.
- **Implementation in Progress**
  Changes to CIs can occur, but are not considered rogue or used for verification. You may want to change the definition of this change state to let verification occur for Change Orders in this state.
- **Verification in Progress**
  Change specifications cannot be edited, but an analyst can override them. Change verification is active in this state.

Change Specifications

A Change Order contains change specifications that define the specific CI changes that are requested for a CI. CACF uses these change specifications when the Change Order enters a verification state to validate and confirm that the

actual changes to the CI completed correctly. Create change specifications from a Change Order, CI, or the Configuration Control, Change Specifications node the CA CMDB section of the Administration tab.

The change specification contains the following main sections:

- **Change Order Number**
  Specifies the Change Order that requests the change.
- **CI Name**
  Contains the name of the CI that you want to update.
  You can leave the CI Name field blank to imply that the change specification applies to all CIs defined for the Change Order. Use this option when all the CIs for a Change Order use the same managed attribute and planned value. The change specification applies to all CIs that are attached to the Change Order when the change order status moves to a managed change state with change verification active, referred to as expansion.
- **Attribute Name**
  Specifies the name of the managed attribute that you want to update.
  Selecting Any Managed Attribute indicates that any managed attribute can change during the verification of the Change Order. Because you can update multiple attributes in this case, you cannot specify the planned value.
- **Planned Value**
  Indicates the expected value of the attribute after executing the change. After the Change Order moves into a verification state, and the CI updates through the web interface, GRLoader, or Web Services, CACF compares the planned value with the inbound data to determine a match.
  Special characters can be embedded in the planned value when the exact value is not known.
- **Status**
  Specifies the status of the change specification, such as Verification Pending.

If you do not know the planned value in advance, but the value may change, you can set the status of the change specification to Use Discovered Value. This behavior requires discovery to update the CI before CACF considers the change specification as validated. You require this behavior for numeric fields and SRELs where an asterisk cannot be accepted as a planned value.

To help you with setting the planned value, the Discovered Attribute History tab lists all values that were recently discovered by an MDR, and whether were actually authorized, or loaded into the CMDB. This tab displays the format, case sensitivity, and other information about values so that you can determine an appropriate planned value pattern.

Special Characters

You can embed special characters when you do not know the exact value. Use the asterisk wildcard character to match any number of characters. The pattern *must* match the discovered data in the same sequence, and spaces are significant.

For example, enter *10.*.*.** as the Planned Value to match any IP addresses that begins with *10.* and has two periods that follow any value between the periods.

For example, the inbound server_type value contains Windows 2003 (WIN32) 5.2.Service Pack 2 (Build 3790) Intel x86. To verify this value, specify a planned value of * *Service Pack 2* * in the change specification.

A word at the beginning of the planned value indicates that the discovered value must start with that value. Similarly, an asterisk at the beginning of the planned value indicates that the discovered value can begin with any value and end with the value specified following the asterisk.

The following table provides examples about how to use the asterisk:

Planned Value	Discovered Value	Match or No Match
a	b	No Match
a	a	Match
a	aba	Match

a	bab	Match
a*	a	Match
a*	ab	Match
a*	ba	No Match
*a	a	Match
*a	ab	No Match
*a	ba	Match

A pattern that begins with an exclamation point results in the negation of the value. You can only use the exclamation point as the first character in the pattern. For example, you cannot use a pattern of 10.!*.*.*.

To compare numeric values that are contained within string values, use greater than (>) or less than (<) as the first character in the planned value. If there is a leading exclamation point, it must be the second character.

WARNING
CACF ignores leading or trailing nonnumeric values in the discovered value and planned value patterns.

The following table provides examples about how to use the exclamation point, and the greater than and less than symbols:

Planned Value	Discovered Value	Match or No Match
>200	aaa 201 bbb	Match
>200GB	aaa 200 bbb	No Match
>200GB	300 GB	Match
!<200GB	200 Bytes	Match
!<200	200 Bits	Match

If you do not know the planned value in advance, but the value may change, you can set the status of the change specification to Use Discovered Value. This behavior requires discovery to update the CI before CACF considers the change specification as validated. You require this behavior for numeric fields and SRELs where an asterisk cannot be accepted as a planned value.

To help you with setting the planned value, the Discovered Attribute History tab lists all values that an MDR recently discovered, and whether they were authorized, or loaded into the CMDB. This tab displays the format, case sensitivity, and other information about values so that you can determine an appropriate planned value pattern.

Defining Bulk Load

Managing new CIs without first creating a Change Order that requires defining those CIs presents an issue within CMDB environments. Consider the following information when you define a large number of CIs

- Define a special bulk loading policy that allows rogue insert to complete successfully, restricted by userid, MDR, or role.
- Define a special bulk loading policy that reroutes all new CIs to the TWA for verification and later processing. This action also requires the previous special bulk loading policy.

Defining Bulk Changes

Creating identical changes to a large number of CIs, for example, when changing the location for a collection of CIs, with the following steps:

1. Create a Change Order.
2. Define a single change specification which describes the change and leave the CI name blank.

3. Attach all the CIs to the Change Order.

If you do not know the exact details of a change in advance of the implementation, for example, you acquire a new Server and you only know the CI Name, complete the following steps:

1. Create a Change Order.
2. Create a change specification that specifies Any Managed Attribute as the attribute name.
3. Leave the planned value empty, since it is unknown.
4. Move the ticket through the change management process.
5. When the Change Order is in a verification active state, and discovery runs, the inbound CI data loads into the CI (assuming that the policy allows it).
6. When all discoveries for the CI complete, the Change Manager must mark the change specification as verified manually.

Follow these steps when there are many dissimilar changes to many CIs. For example, when moving a collection of servers from one location to another, and also assigning each one a unique IP address.

1. Create a Change Order
2. Create a spreadsheet and list on each row the Change Order number, CI and, the new attribute values. Each row results in a distinct change specification.
3. To create the required change specifications, load the spreadsheet with GRLoader.
4. Promote the Change Order and its change specifications as usual.
   For more information about using GRLoader, see the General Resource Loader topic.

How Change Verification Occurs

When a user requests a save to a CI, CA SDM searches for applicable change specifications as it searches for all matches. The following information describes an applicable change specification:

- The CI in the change specification is the same as the one that is being saved.
- The Change Order is in a managed change state that is defined with Change Verification Active.
- The attribute in the change specification is the same as the attribute being updated.
- The change specification is active.
- If there are no managed attributes being updated or policy in effect, the save proceeds uninhibited.

If the inbound data values match a change specification exactly, CACF considers the change specification as validated. Depending on the policy, verification closes any Incident that change verification created.

Verification takes place when information from web clients, web services, or GRLoader update the CI. At this time, the Change Order and all underlying change specifications are undergoing verification. After all verification for a Change Order complete, the Change Order can optionally promote to the next Change Order state automatically.

If the inbound data values do not match the values in the change specification, it is considered an incorrectly implemented or failed change. Incidents are created or appended to, depending on the policy.

For example, CA SDM does not find applicable change specifications, but Change Orders exist in a *change verification active* state. These Change Orders specify the target CI and have no change specifications. CACF manages this change by policies which handle *Change Orders without Specifications*.

- If there are no applicable change specifications or Change Orders without specifications, the save operation is considered rogue.
- After all attributes have been processed, any policy that requested the inbound data to write to the TWA triggers.

Managing Change Specifications

After a change is executed or implemented, the Change Order enters a change state that is managed by CACF. The change state is set to *change verification active*. When the Change Order is in this state, CACF reviews all related CI activity. CACF reports the status of each change specification to the scoreboard.

The Change Manager uses the scoreboard to verify that the required changes are correctly executed and to perform any manual verification that is required.

After all verification for a Change Order are complete, the Change Order may optionally be promoted to the next Change Order state (typically closed). In this state all change specifications are automatically marked inactive.

Change Specification Statuses

Change specifications use status to indicate the type of verification to perform and the current state of verification. The status values are also used in the verification log when recording change verification operations during the verification process.

- **Initial Statuses**
  Used when creating a Change Specification to specify the type of verification for CACF to perform. These statuses are not final. The Change Order is not considered verified when change specifications are in either of these statuses.
  Verification Pending -- The change specification has not been verified, or the change is waiting for the CI update.
  Manual Verification will be required -- When the changes for a Change Order are verified, manual verification is required.
  Use Discovered Value -- Copy the discovered value to the CI at verification time, and used when the planned value is not known before verification.
  Set After Change Executed -- Sets a CI attribute to the planned value after verification has completed. Use this status for setting CI attributes when the attribute is not discoverable. For example, set the CI Primary Contact to User1 after a change completes.
- **Final Statuses**
  Indicates a completed change specification and considered as final. When all change specifications enter either of these final states, they are eligible for promotion to the next default state.
  Verified -- The change specification has been successfully verified.
  Was Manually Verified -- The change specification has been manually verified.
  Used Discovered Value -- The CI has been updated with the discovered value.
  Was Set To Planned Value -- The CI has been updated with the planned value after Change Order verification.
  Accepted Planned Value -- The CI has been updated with the planned value and overwritten during verification.
  Accepted Discovered Value -- The CI has been updated with the last discovered value and overwritten during verification.
  No Change -- The planned value matched the CI at verification time and no verification was required.
  Cancel -- The change manager canceled the change specification.
- **Intervention Statuses**
  Indicates that a change specification requires manual intervention for verification. CA SDM highlights these statuses in a red color on list forms. These statuses are not final. The Change Order is not considered verified when change specifications are in either of these statuses.
  Failed Verification -- Discovery found the value to be other than what was specified in the Change Order. The Change Analyst must determine if the change was correctly executed and the Change Order was incorrectly specified. Or, if the Change Order was correct and the change requires verification again. Depending on the definition of the Managed Change State and the current change order status, the Change Analyst can override, change, or cancel the failed change specification.
  Manual Verification Active -- Manual verification is required before the change specification can be marked final.
- **Action Override Statuses**
  Indicates an action to initiate during change verification. These statuses are not final. The Change Order is not considered verified when change specifications are in either of these statuses.
  Accept Planned Value -- Request to override the CI attribute with the planned value.
  Accept Discovered Value -- Request to override the CI attribute with the last discovered value.
- **Reporting Status**

Indicates the result of policy operations that the verification log displays for logging purposes and not used by change specifications.
Update Was Allowed -- A policy allowed a request to update a CI and did not match any change specifications.

Managing Failed Verifications

When a verification fails, the CMDB, Configuration Audit, Failed Verifications node in the scoreboard lists the number of failed verifications. The scoreboard also lists the number of change specifications that require manual intervention in red. The Change Specifications can also be viewed under the Change Specifications tab in the Change Order, CI, or Administration tab. Failed verifications have the verify status of "Failed Verification". When a verification fails, the Change Analyst can intervene or wait until the next discovery occurs.

If the managed change state allows the Show Change Specification Override Buttons action, the Change Analyst can open the change specification and review the data. The Change Analyst can click one of the following options to close the verification by moving it to a final state:

- **Accept Discovered Value**
  The Analyst determines that the change specification is incorrect and that the discovery tool has discovered the correct value.
- **Accept Planned Value**
  The Analyst determines that the discovery tool is wrong or has not performed discovery. The Analyst decides to accept planned value, as if it was discovered correctly.
- **Cancel**
  This portion of the Change Order was not executed and this specification was canceled.

Managing Undiscoverable Attributes

When a Change Specification requires manual verification, the Configuration Analyst must manually verify the change. The CMDB, Configuration Audit, Manual Intervention node in the scoreboard shows the number of change specifications that require manual intervention in red. The Change Specifications can also be viewed under the Change Specifications tab in the Change Order, CI, or Administration tab with the verify status of "Manual Verification Active."

If the managed change state allows the Show Change Specification Override Buttons action, the Change Analyst can open the change specification and review the data. The Change Analyst can click one of the following options to close the verification by moving it to a final state:

- **Mark as Verified**
  Used when the attribute is cannot be discovered and manually verified.
- **Cancel**
  This portion of the change order was not executed and this specification was canceled.

How to Archive and Purge Audit Data

We recommend that you archive and purge obsolete Verification Log entries, CACF audit history, and CI audit history as part of database maintenance.

> **WARNING**
> The CACF reports typically present monthly reports with a yearly summary. The default archive purge time for the archive purge rules for the log, CACF audit history, and CI audit history tables is 30 days. Use the same value in the archive and purge rules for verification log entries, Incidents, and Change Orders. Using the same value ensures data consistency. You can probably change the 30-day default to something more in keeping with your reporting requirements.

To archive and purge audit data, complete the following actions:

1. To archive and purge inactive Incidents older than *nn* days, use the current incident rule.
   CA SDM archives and purges verification log entries only after archiving and purging associated Incidents. If you associate a Change order with an Incident, CA SDM does not verify whether the Change Order is active.

2. To archive and purge inactive Change Orders older than *nn* days, use the current Change Order rule.
   CA SDM archives and purges verification log entries and change specifications only after archiving and purging associated Change Orders. If you associated an Incident with a Change Order, CA SDM does not archive and purge the Change Order.
3. Use the Rogue Change Verification Log rule for rogue changes that did not create an Incident.
   To archive and purge verification log entries older than *nn* days, use this rule. By definition, rogue changes are not associated with Change Orders.
4. Use the CMDB Audit rule to archive information that is shown in the Change Specification History, Verification Policy History, Managed Change State History, and Managed Attribute History tabs. These tabs are shown in the respective Change Specifications, Verification Policy, Managed Change State, and Managed Attributes detail forms. CA SDM stores this information in the ci_audit table.

Implement a Change Verification Strategy

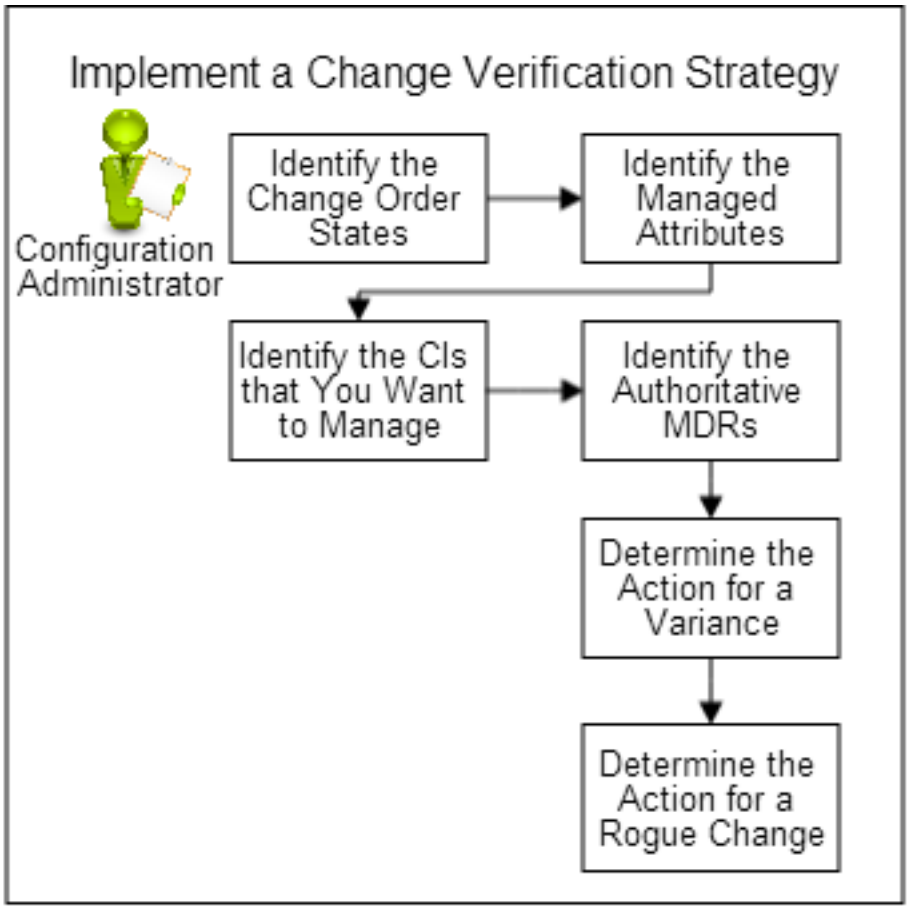
This article contains the following topics:

The Configuration Administrator determines how aggressively to implement the change verification strategy for your CMDB environment. You identify areas of your Change Management process that require a change verification strategy. These areas include attributes under change control, the Change Order states that indicate that change verification are active, when you can modify change specification, and authoritative MDRs.

For example, you only want to allow updates to the IP Address attribute from MDR1. You also determine the appropriate actions when CACF detects a variance and rogue change.

The following diagram explains how a Configuration Administrator implements a change verification strategy:

Figure 33: BSVC--Implement a Change Verification Strategy--OTH



1. Identify the Change Order States.
2. Identify the Managed Attributes.
3. Identify the CIs that You Want to Manage.
4. Identify the Authoritative MDRs.
5. Determine the Action for a Variance.
6. Determine the Action for a Rogue Change.

Identify the Change Order States

The Configuration Administrator identifies the Change Order states for when change verification is in effect. The verification happens after a change is executed. Change states help you determine specific conditions, such as if you can edit change specifications in a particular state. For example, you want to review the RFC change state for Change Orders that request updates to CIs.

Follow these steps:

1. On the Administration tab, click CA CMDB, Configuration Control, Managed Change States.
2. To view the change state details or create a Change Order state that has not been already defined, click RFC.

NOTE
By default, the RFC Change Order state lets you only edit change specifications. The Implementation in progress state does not let you edit change specifications. The Verification in progress state enables change verification and displays override buttons for the Change Manager or other authorized user.

3. Specify the CACF options and behavior for when a Change Order enters this state.
4. Click Save.

Identify the Managed Attributes

Identify which CI attributes you want to manage for the change verification strategy. For example, you want to manage the IP Address (alarm_id) attribute with change verification.

Follow these steps:

1. On the Administration tab, click CA CMDB, Configuration Control, Managed Attributes.
2. Click Create New.
3. Complete the following steps:
   a. Enter **alarm_id** as the Attribute Name.
   b. Enter **IP Address** as the Attribute Label.
   c. Select an Initial Verify Status from the drop-down list.
   Default: Verification Pending
   d. (Optional) If you want to enforce case sensitivity for change specification planned value comparisons, select the Case Sensitive option.
   Default: Disabled
   e. Click Save.

Identify the CIs that You Want to Manage

Identify the CIs that you want to manage with a verification policy. For example, you want to manage the IP Address (alarm_id) for all high priority CIs with names beginning with NY_Server.

Follow these steps:

1. On the Administration tab, click CA CMDB, Configuration Control, Verification Policies.
2. Click Create New.
3. Complete the following steps:
   a. Enter a sequence, such as **100**.
   b. Enter **NY Server IP Addresses** as the Policy Name.
   c. Select the appropriate Change Order Alignment options.
   For example, you want the verification policy for all options except Change Orders Without Specifications.
4. To specify the transaction and CI filters, complete the following steps:
   a. Select IP Address from the Managed Attribute drop-down list.
   b. Enter a Role Pattern or leave the asterisk to apply to all roles.
   c. Enter **NY** as the Location Pattern.
   d. Enter **NY_Server*** as the CI Name Pattern.
   For example, this filter verifies CIs named NY_Server1, NY_Server2.
5. Select *Allow Update Only if Matches Change Specification* from the Update Behavior drop-down list.
6. (Optional) Use Log Only Mode if you want to experiment with the policy. Also use this mode to view the results only in the standard log, and not affect the active CMDB environment.
7. Click Save.

Identify the Authoritative MDRs

Identify the authoritative MDRs in your CMDB environment. For example, you want to allow updates to CIs from CA Configuration Automation that you identified as MDR1. You consider updates from Web Services that you identified as MDR2 as unauthoritative. You want to send those requests to the TWA.

Follow these steps:

1. Open the NY Server IP Addresses policy, click Edit.
2. Enter **MDR1** as the MDR Name Pattern.
   Note: If you want to exclude an MDR named MDR2, but you want to allow MDR1, MDR3, and so on, enter **!MDR2** as the pattern.
3. Create a separate verification policy, such as MDR2 NY Server.
4. Enter the same information as in the previous policy except the following fields:
   a. Enter **MDR2** as the MDR Name Pattern.
   b. Select *Always Cancel the Entire Transaction* as the Update Behavior.
   c. Select Always from the Write Data to the TWA drop-down list.
5. Click Save.

Determine the Action for a Variance

Determine the action for a variance. For example, a change to a CI does not match the values that are specified in the Change Order. You want the verification policy to create an Incident for the variance.

Follow these steps:

1. Open the MDR2 NY Server policy that you created and click Edit.
2. Select Yes from the Create Incident drop-down list and assign a template.
3. Click Save.
4. An end user creates a Change Order with a change specification to update a CI from MDR2.
5. CACF creates an Incident if MDR2 requests a CI update with a value that does not match the planned value.

Determine the Action for a Rogue Change

Determine the action for a rogue change. For example, CACF detects a change to a CI that does not have any change specifications from any active Change Orders. The Configuration Administrator wants to reject these types of requests.

Follow these steps:

1. Create a verification policy and enter **Rogue NY Server** as the name.
2. Complete the appropriate pattern fields.
3. Verify that only Rogue Insert and Rogue Update are selected as the Change Order Alignment options.
4. Select *Allow Update Only If Matches Change Specification* from the Update Behavior drop-down list.
5. Select Yes from the Create Incident drop-down list and assign a template.
6. Click Save.

Change Verification Notification Messages

The CI detail form displays a notification message when a change is completed. The message indicates whether a change performed in the web interface was successful or unsuccessful. If a change does not complete successfully, a warning message displays with the policy, attribute, and corresponding action. For example, if a user attempts to update a CI but the change was not saved. The reason could be the policy rejected the change when the user attempted to update the CI.

The Verification Log tab also shows details corresponding to the message for future reference.

The notification message indicates the following information:

- Which policy was in effect and corresponding policy action
- Whether a rogue update was detected and if the changes do not match any change orders pending verification
- Whether an Incident was created, updated, or closed, and the corresponding reference number

Examples for Implementing Change Verification

This article contains the following topics:

For Implementing Change verification, consider the following examples:

Example Upgrade Laptops in Your Organization

In this example, an Asset Manager wants to upgrade all laptops in your organization. The Configuration Administrator creates a verification policy for the Product Version attribute. The verification policy is to filter all laptops with the Windows XP operating system. The Configuration Administrator creates a Managed Attribute definition with the *Use Discovered Value* status, reviews Managed States, and creates a verification policy. CA SDM receives updates to the CMDB from CA Configuration Automation from laptops that require an operating system update. The Asset Management team completes the upgrades. **Follow these steps:**

1. Create the following managed attribute:
   - Enter **product_version** as the Attribute Name.
   - Enter **Product Version** as the Attribute Label.
   - Select Use Discovered Value from the Initial Status drop-down list.
   - Enter a label and description for details about the managed attribute.
2. Create the following verification policy:
   - Enter a sequence that you base on other policies in your organization.
     For example, you enter **101**.
   - Enter **Windows* XP*** as the CI Name Pattern.
   - Enter **Workstation** as the Class Pattern.
   - (Optional) Enter a Location Pattern if you want to associate the policy with a specific office in your organization.
   - Select Allow Update Only if Matches Change Specification from the Update Behavior drop-down list.
   - Select Yes from the Create Incident drop-down list.
3. Create a Change Order and change specifications that specify Windows 7 for the Product Version managed attribute for each of the laptop CIs.
4. Move the Change Order to Implementation in Progress status and wait for your Asset Management team to start the upgrades.
5. When the implementations complete, move the Change Order to Verification in Progress.
   CA Configuration Automation discovers laptop information and imports the data in the CMDB.
6. View the open CACF Incidents list for any variances that CACF detected.
7. Select one of the Incidents and review the details about the laptop that CA Configuration Automation discovered.
8. Click Create Change Order and associate the CI with the ticket. Specify change specifications for Product Version for the outstanding changes.
   Wait until discovery verifies that all remaining changes completed.
9. Repeat Steps 6-8 as necessary for any new Incidents that CACF creates.

Example Lock Down Nonverified Change Orders

In this example, the Configuration Administrator wants to only allow a CI update for a matching Change Order. Only CIs in the Server class that is located in NY update for verified Change Orders. Any variance creates an Incident. **Follow these steps:**

1. Create a verification policy.
2. Complete the following steps:
   a. Enter a sequence number.
   b. Select Any Managed Attribute as the Managed Attribute.
   c. Enter **NY** as the Location Pattern.
   d. Enter **Server** as the Class Pattern.
   e. Select Allow Update Only if Matches Change Specification as the Update Behavior.
   f. Select Yes from the Create Incident drop-down list.
3. Save the policy.

Example Allow a CI Update If No Matching Change Order Exists

In this example, the Configuration Administrator allows updates for all CIs named test*, even if no matching Change Order exists. This policy accepts updates from all users in an administrative role. **Follow these steps:**

1. Create a verification policy.
2. Complete the following steps:
   a. Enter a sequence number.
   b. Select Any Managed Attribute as the Managed Attribute.
   c. Enter **test*** as the CI Name Pattern.
   d. Enter **Administrator*** as the Role Pattern.
   e. Select Allow Attribute Update as the Update Behavior.
3. Save the policy.
4. A user with an administrative role creates a change specification for a CI named test5.
   The CI updates successfully.

Example Defer All Updates from CA Configuration Automation to the TWA

In this example, the Configuration Administrator wants to defer all CI updates from CA Configuration Automation to the TWA. This policy does not update the CI in CMDB. It writes the data for all rogue changes to the TWA for further evaluation instead. **Follow these steps:**

1. Create a verification policy.
2. Complete the following steps:
   a. Enter a sequence number.
   b. Select Any Managed Attribute as the Managed Attribute.
   c. Select Rogue Insert and Rogue Update as the Change Order Alignment.
   d. Enter **CCA** as the MDR Class Pattern.
   e. Select Always Cancel Entire Transaction as the Update Behavior.
   f. Select Always for the Write Data to TWA option.
3. Save the policy.

Example Only Log the Policy Results as a Test

In this example, the Configuration Administrator wants to test a new policy before implementing it in your CMDB environment. The Log Only option lets CACF write the potential CI impacts of the policy to the standard log file. **Follow these steps:**

1. Create a verification policy.

2. Complete the information for the alignment, filters, and action.
3. Select the Log Only Mode option.
4. Save the policy.
5. To simulate executing the policy, view the standard log file after you perform CI updates that match the policy specifications and filter criteria.

Example Reject a CI Update

In this example, the Configuration Administrator rejects updates from an MDR named Cohesion for the IP Address (alarm_id) attribute only. This policy does not update the IP address of the CI in CMDB, while it may update other attributes. CACF writes all attributes to the TWA for further evaluation. **Follow these steps:**

1. Add IP Address (alarm_id) to the managed attributes list.
2. Create a verification policy.
3. Complete the following steps:
   a. Enter a sequence number.
   b. Select IP Address as the Managed Attribute.
   c. Select Rogue Insert and Rogue Update as the Change Order Alignment.
   d. Enter **Cohesion** as the MDR Pattern.
   e. Select Keep Old Attribute Value as the Update Behavior.
   f. Select Always for the Write Data to TWA option.
4. Save the policy.

Example Allow Change Orders Created Without Specifications

In this example, the Configuration Administrator wants to trust Change Orders that users created without specifications. This policy assumes that the CI text describes all changes accurately and lets the CI update. **Follow these steps:**

1. Create a verification policy.
2. Complete the following steps:
   a. Enter a sequence number.
   b. Select Any Managed Attribute as the Managed Attribute.
   c. Select Change Orders Without Specifications as the alignment.
   d. Select Allow Attribute Update as the Update Behavior.
3. Save the policy.

Example Do Not Allow Change Orders Created Without Specifications

In this example, the Configuration Administrator does *not* want to trust Change Orders that users created without specifications. This policy ignores Change Orders without specifications and creates Incidents. **Follow these steps:**

1. Create a verification policy.
2. Complete the following steps:
   a. Enter a sequence number.
   b. Select Any Managed Attribute as the Managed Attribute.
   c. Select Change Orders Without Specifications as the alignment.
   d. Select Always Cancel Entire Transaction as the Update Behavior.
   e. Select Yes from the Create Incident drop-down list and select an Incident template.
3. Save the policy.

Example Allow Rogue Inserts from Selected Sources

In this example, the Configuration Administrator wants to allow new CIs from selected sources. The z/OS MDRs can create CIs without requiring a Change Order. **Follow these steps:**

1. Create a verification policy.
2. Complete the following steps:
   a. Enter a sequence number.
   b. Select Any Managed Attribute as the Managed Attribute.
   c. Select Rogue Insert as the alignment.
   d. Enter **z/OS** as the MDR Class Pattern.
   e. Select Allow Attribute Update as the Update Behavior.
3. Save the policy.

Example Allow a Rogue Update for a Nonproduction CI

In this example, the Configuration Administrator allows updates to the IP Address attribute from the Spectrum MDR, but the name cannot begin with *PROD*. **Follow these steps:**

1. Create a verification policy.
2. Complete the following steps:
   a. Enter a sequence number.
   b. Select IP Address (alarm_id) as the Managed Attribute.
   c. Select Rogue Update as the alignment.
   d. Enter **!PROD*** as the CI Name Pattern.
   e. Enter **Spectrum** as the MDR Class Pattern.
   f. Select Allow Attribute Update as the Update Behavior.
3. Save the policy.

Example Allow Rogue Updates Only From a Specific Location

In this example, server CIs in your New York office require repair. The vendor that repairs your servers also resides in New York. The Asset Manager requires that your organization ships all defective servers to New York. The Configuration Administrator wants to allow rogue updates to the CIs when the hardware arrives in New York. The Configuration Administrator creates a verification policy for the Maintenance Vendor attribute. This policy lets a Service Desk Analyst in New York verify that the vendor receives the servers. **Follow these steps:**

1. Create the following managed attribute:
   – Enter **vendor_repair** as the Attribute Name.
   – Select verification pending from the Initial Status drop-down list.
   – Enter **Maintenance** as the label and description for details about the managed attribute.
2. Create the following managed change status:
   – Enter **Vendor Hold** as the Change Order Status.
   – Enable the Change Verification Active option.
3. Create the following verification policy:
   – Select Rogue Insert and Rogue Update as the change order alignments.
   – Enter a sequence number for the policy.
   For example, you enter **201**.
   – Enter **server*** as the CI Name Pattern.
   – Enter **Server** as the Class Pattern.
   – Enter **New York** as the Location Pattern.
   – Select Allow Attribute Update from the Update Behavior drop-down list.

4. The Service Desk Analyst creates a Change Order and change specifications for the server CIs and specifies New York for the location.
5. The Service Desk Analyst sets the Change Order Status to Vendor-Hold.
6. The Service Desk Analyst updates the location for the server CIs as New York.
   For example, defective servers from the Chicago office ship to New York. The Service Desk Analyst verifies that the servers arrived in New York.
7. The Service Desk Analyst enters the vendor information in the CI.
   The Change Order closes after all the CI locations are set to New York and all the change specifications are verified.

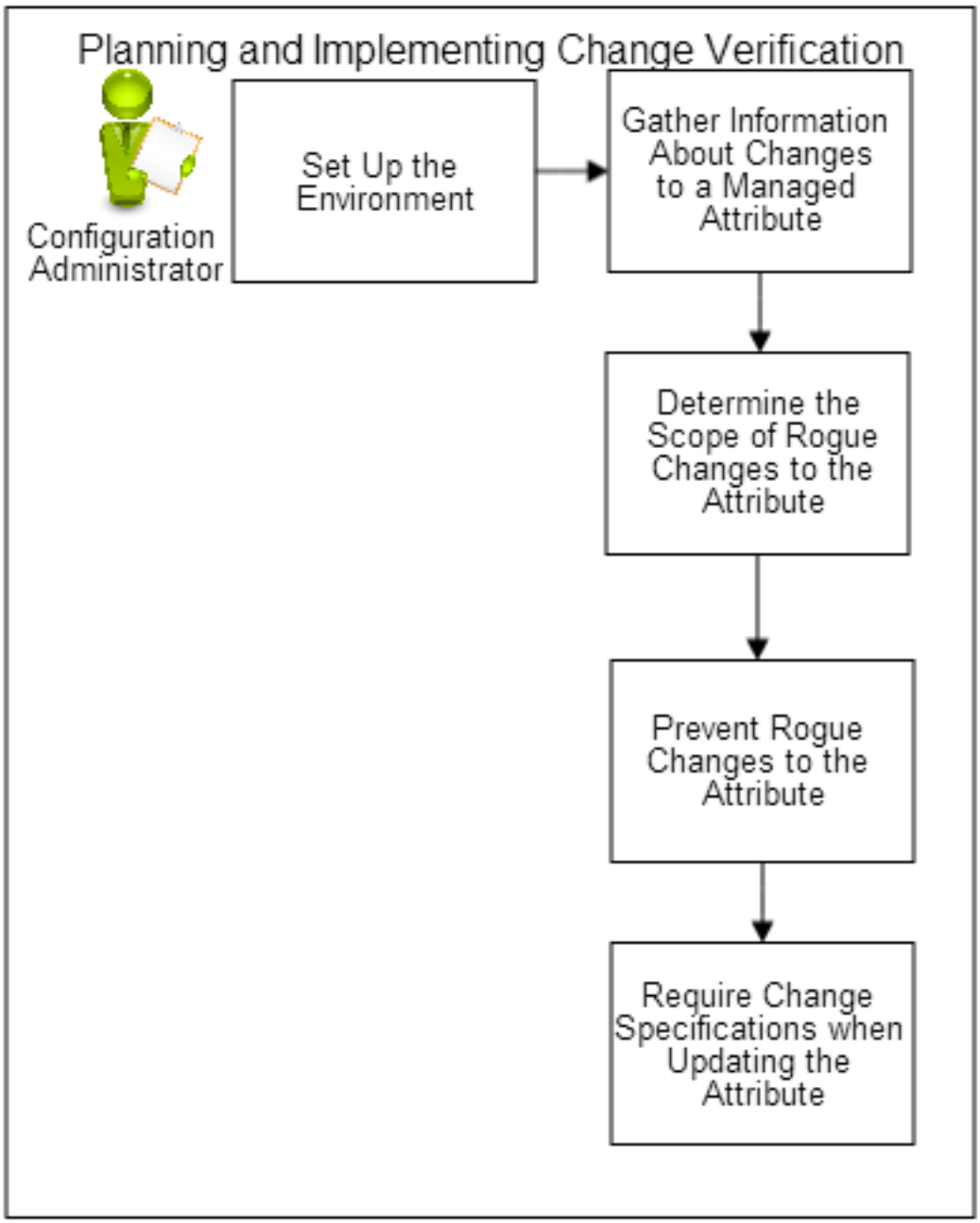
Planning and Implementing Change Verification

The Configuration Manager wants to implement change verification so that the CMDB contains correct data. The Change Manager wants to ensure that changes execute correctly. The Change Manager wants to track the number of rogue changes that occur and their data sources. Both managers agree implementing change verification provides significant value to the organization. The Change Manager wants to ensure that changes do not harm the production environment.

The Configuration Manager and the Change Manager agree to a conservative-phased implementation. The Configuration Administrator implements policies for specific locations and specific CIs in those locations. This scenario describes example phases about completing the verification policy implementation to a broader audience for your organization.

The following diagram shows how a Configuration Administrator completes the example phases to implement change verification:

Figure 34: BSVC--Planning and Implementing Change Verification--OTH



Follow these steps:

1. Set Up the Environment.
2. Gather Information About Changes to a Managed Attribute.
3. Determine the Scope of Rogue Changes to the Attribute.
4. Prevent Rogue Changes to the Attribute.
5. Require Change Specifications when Updating the Attribute.

Set Up the Environment

You set up the environment so that Configuration Administrators can always update CIs. The Configuration Administrator can create and update CIs in the CMDB as needed, without specifying a change order. The Configuration Administrator is trusted to provide correct CI data.

Follow these steps:

1. Create a verification policy that is named **Policy0.1** with the following information:
   - Enter **1** as the sequence.
   - Enter **Allow Administrator to Always Update CIs** as the description.
   - Select all Change Order Alignment options.
   - Enter **Configuration Administrator** as the Role Pattern.
   - Enter **Web Client** as the MDR Class Pattern.
   - Enter an asterisk (*****) as the CI Name and Class, and all other patterns.
   - Select Any Managed Attribute for the Managed Attribute.
2. Select Allow Attribute Update as the Update Behavior action.
3. Click Save.
   The policy is enabled to only allow Configuration Administrators to update CIs always.

Gather Information about Changes to a Managed Attribute

You want to gather information about changes to a specific CI-managed attribute. The change information helps you identify the CIs that are being updated, the source of the changes, and the specified values. This information helps when defining a verification strategy for the attribute and corresponding CIs. For example, you want to understand the scope of all changes to IP Address (alarm_id) and log this to the CA SDM standard log (stdlog). Gathering this information can take several weeks, depending on the number of changes to attributes that occur in your organization.

1. Define IP Address (alarm_id) as a Managed Attribute.
2. Define Implementation in Progress and Verification in Progress as Managed Change States in your environment. Enable Change Verification Active in both states.
3. Create a verification policy that is named Policy1.1with the following information:
   - Enter **3000** as the Sequence.
   - Enter **Log All Changes to IP Address** as the description.
   - Select all Change Order Alignment options.
   - Enter an asterisk as the CI Name and Class and all other patterns.
   - Enable Log Only Mode.
4. Click Save.
5. After a few weeks, review the stdlogs to view the sources of the updates.
   For example, the log displays rogue CI updates and updates with matching Change Orders.
6. You determine that users located in NY have been updating the IP Address of CIs without creating Change Orders.
7. After you complete your analysis, inactivate Policy1.1 by editing the policy and setting Active? to Inactive.

Determine the Scope of Rogue Changes to the Attribute

You want to determine the scope of rogue changes to the IP Address attribute. Knowing the scope of rogue changes helps you understand the impact of rogue changes on your organization. For example, based on your previous analysis using the Log Only option, you want to create Incidents whenever a change occurs to IP Address to CIs that begin with "test" located in NY without a Change Order. To ignore new Incidents that this process creates, communicate with Change Managers in your organization.

Follow these steps:

1. Complete the following initial implementation actions:

- Define Implementation in Progress as a managed change state and enable the Implementation State option.
- Define Verification in Progress as a managed changes state and disable the Promote Change Order After Verification option.
- Define IP Address (alarm_id) as a Managed Attribute if not already defined.

2. Create a verification policy that is named Policy2.1 with the following information:
   - Enter 3001 as the Sequence.
   - Enter a description about this policy. For example, you enter **Rogue inserts and rogue updates cause Incidents to be created. All other changes are not impeded**.
   - Select Rogue Inserts and Rogue Updates as the alignments.
   - Select IP Address from the Managed Attribute drop-down list.
   - Enter **test*** as the CI Name and **NY** as the Location and an asterisk for all other Patterns.
   - Select Allow Attribute Update, Yes to Create Incident, and an Incident Template from the actions.
3. Click Save.
   After you complete this phase, CACF creates Incidents for all computers in NY. You discover the following information after reviewing the Incidents:
   - Several MDRs report different IP addresses for the same CI.
   - Some MDRs are authoritative, other MDRs are not authoritative.
   - Some CIs cannot be managed based on Location, Family, Name, Service Type, and so on.
4. The Change Manager meets with other IT Managers to review the CMDB, CI detail forms, verification logs, and filtering by attribute name to see the rogue data.
5. Your organization decides to update the CI filter in Policy 2.1 to manage Priority 1 Servers only.

Prevent Rogue Changes to the Attribute

You want to prevent rogue changes to the IP Address attribute to ensure CMDB data integrity. You want to require a Change Order for the update. The Change Order must specify the CI. However, a change specification is not required for the IP Address for the change to occur. Change verification occurs at the CI/Change Order level but not at the attribute level. For example, prevent any rogue updates to and use Incident creation to track the users that request these changes.

Follow these steps:

1. Set Policy1.1 and Policy2.1 to Inactive to disable these policies so that they are not in effect.
2. Create Policy 3.1 with the following information:
   - Enter **3100** as the Sequence.
   - Enter **Prevent rogue changes without any Change Order** as the description.
   - Select Rogue Update as the alignment.
   - Enter **test*** as the CI Name and **NY** as the Location and Any Managed Attribute. Enter an asterisk for all other patterns.
   - Select the Always Cancel Entire Transaction action.
3. Click Save.
4. Create Policy3.2 with the following information:
   - Enter **3200** as the Sequence.
   - Enter **Require a Change Order for IP address changes** as the description.
   - Select Change Orders Without Specifications as the alignment.
   - Enter **test*** as the CI Name and **NY** as the Location and an asterisk for all other patterns.
   - Select IP Address from the Managed Attribute drop-down list.
   - Select Allow Attribute Update as the action.
5. Click Save.
   This policy eliminates rogue changes, because Change Orders must accompany the change.

With this policy, the Configuration Manager prevents data that is not defined in the Change Order being updated. For example, requesting a change to increase Memory Installed while changing the IP Address simultaneously, would not be considered a rogue change because there is a Change Order. They also would like the Change Orders to be automatically verified and promoted and determine they want verification at the attribute level.

Require Change Specifications when Updating the Attribute

Your Configuration Manager may be concerned that the CMDB is updated multiple times although changes are not specified in the Change Order. For example, a user changes the value of Memory Installed (phys_mem) and IP Address (ip_address) simultaneously. In the previous phase, CACF does not consider this request as a rogue change because a Change Order exists.

The Configuration Manager wants to enforce change verification at the attribute level. The Configuration Administrator creates a policy with the following requirements:

- Use change specifications when updating IP Address.
- If no rogue update is attempted, create an Incident.

Follow these steps:

1. The Configuration Manager contacts Change Analysts that changes to IP Address must contain a change specification.
2. Set Policy3.1 and Policy3.2 to Inactive to disable these policies so they are not in effect.
3. Create Policy4.1 with the following information:
   - Enter **4000** as the Sequence.
   - Enter **Require Change Specifications for IP Address changes** as the description.
   - Select Rogue Insert, Rogue Update, and Change Orders Without Specifications as the alignment.
   - Enter **test*** as the CI name and **NY** as the Location, and an asterisk for all other patterns.
   - Select IP Address from the Managed Attribute drop-down list.
   - Select Always Cancel Entire Transaction as the action.
4. Click Save.
5. Create Policy4.2 with the following information:
   - Enter **4100** as the Sequence.
   - Enter **Require Change Specifications for IP Address changes** as the description.
   - Select Change Orders With Specifications as the alignment.
   - Enter **test*** as the CI Name and **NY** as the Location pattern.
   - Select IP Address from the Managed Attribute drop-down list.
   - Select Allow Update only if Matches Change Specification as the action.
6. Click Save.

You have successfully completed the appropriate example phases to implement change verification in your environment. You can expand the change verification strategy to include more attributes, CIs, MDRs, and tenants.

Verify a CI Attribute Value Update Manually

The Configuration Administrator determines that your CA SDM environment requires a verification policy for the Memory Installed (phys_mem) attribute. The Configuration Administrator creates a Managed Attribute definition with the *Manual Verification will be required* status. This status is appropriate because CA SDM does not have an MDR to discover the value. The Configuration Administrator reviews the managed states and creates a verification policy. The Change Manager views a Change Order that requests an update to the Memory Installed value of a CI. This change request requires manual verification.

Follow these steps:

1. Create a Managed Attribute definition.
2. Review the Managed Change states in your environment.
3. Create a Verification Policy for the Managed Attribute.
4. Review the Change Specifications list.
5. Accept the Planned Value.

Create a Managed Attribute Definition

The Configuration Administrator creates a Managed Attribute definition for the Memory Installed (phys_mem) attribute.

Follow these steps:

1. On the Administration tab, click CA CMDB, Configuration Control, Managed Attributes.
2. Click Create New.
3. Enter **phys_mem** as the attribute name.
4. Enter **Memory Installed** as the Attribute Label.
5. Select *Manual Verification will be required* from the Initial Verify Status drop-down list.
6. (Optional) Select the Case Sensitive option to enforce case sensitivity for change specification planned value comparisons.
   Default: Disabled
7. Click Save.
   The Managed Attribute is saved.

Review the Managed Change States in Your Environment

The Configuration Administrator reviews the Managed Change States in the CA SDM environment. Change verification initiates when the status of a Change Order changes to a Change State that CACF manages. For example, Verification in Progress.

> **NOTE**
>
> The Configuration Administrator can modify which state in the Change Order lifecycle initiates change verification. The administrator can also modify which states allow you to modify change specifications when implementation starts.

Follow these steps:

1. On the Administration tab, click CA CMDB, Configuration Control, Managed Change States.
2. Click Implementation in Progress to open the managed status detail page.
3. Review the details about the managed status.
   For example, you make the Implementation in Progress status allow editing change specifications, and also enable Change Verification Active. This example indicates an active change verification process. It also indicates that you can edit values when the Change Order sets to the Implementation in Progress status.

Create a Verification Policy for the Managed Attribute

The Verification Policy specifies the CACF action when an MDR discovers an attribute value. The MDR attempts to update the CMDB with that data.

Follow these steps:

1. On the Administration tab, click CA CMDB, Configuration Control, Verification Policies.
2. Click Create New to open the detail page.
3. Enter **1000** as the policy Sequence.

NOTE

You can enter a higher or lower number, which is based on other verification policy priorities in your environment.

4. Enter a Policy Name, such as **RAM Management** and a brief description of the policy.
5. Select Rogue Insert and Rogue Updates under Change Order Alignment.
6. You can configure the policy to prevent a specific MDR from updating the Memory Installed (phys_mem) attribute. Complete the following tasks:
   - Select Memory Installed as the Managed Attribute.
   - Enter a Role Pattern or use an asterisk to apply to all roles.
   - Enter a CI Name Pattern or use an asterisk to apply to all CIs.
   - Enter a Class Pattern or use an asterisk to apply to all classes.
   - Select Keep Old Attribute Value as the Update Behavior.
   - **NOTE**

     To allow updates to this attribute from a web interface user, enter Web Client as the MDR Name Pattern.
7. Click Save.

Review the Change Specifications List

A Change Order with an RFC status wants to change the value of the Memory Installed (phys_mem) attribute in a CI named server1. The Change Manager sets the verification status of the Change Specification to *Manual Verification will be required*. The Change Manager can view the Change Specification in CA SDM.

Follow these steps:

1. The Change Manager moves the Change Order from RFC to Verification in Progress.
   All the change specifications with Manual Verification become Manual Verification Active.
2. From the Change Order, select the Change Specifications tab from the Configuration Management tab.
   The tab highlights all change specifications in red to indicate that manual verification is required.

   NOTE

   You can also view change specifications from the Administration tab when you click CA CMDB, Configuration Audit, Change Specifications. Additionally, view change specifications from a CI after you select the Change Specifications tab from the Related Tickets tab.
3. Search for change specifications with *requires manual verification* as the Verify Status.
   For example, change specifications of Change Order 21 contain this status.
4. Research the appropriate value for the Memory Installed attribute for the server1 CI.

   NOTE

   Change specifications that failed verification also display in red to help show change specifications that require further attention.

Accept the Planned Value

The Change Manager researches the correct attribute value for the CI and makes the appropriate decision on the change request.

Follow these steps:

1. Open the Change Specification to view the detail page.
2. You research the specific Memory Installed (phys_mem) attribute for CI and confirm the planned value.
3. Verify that the Change Order of the Change Specification is in Verification in Progress State.
4. Click Accept Planned Value.

The Verify Status changes to Accepted Planned Value.

NOTE

After you click Accept Planned Value, the CI updates with the Planned Value automatically.

MDR Management

This article contains the following topics:

The MDR Launcher is an open integration tool that lets you view data in a management data repository (MDR) from a CMDB web page. MDR Launcher lets you obtain additional details about a CI and to gain control over it (if the MDR supports such control).

Manage Federated CI Mappings

A single CI can be associated to multiple MDRs, with each MDR having a federated_asset_id that identifies the CI. While it is not necessary for the federated_asset_id to be unique among MDRs, a federated_asset_id must be unique within each MDR.

Each MDR is identified uniquely by its combination of MDR class and MDR name.

To view the CI in a particular MDR context, click Launch. The target MDR is launched in the context of the CI that was open.

View an MDR Location for a CI

You can display the MDR location for a CI.

- **CI Name**
  Identifies the configuration item.
- **Federated Asset ID**
  Specifies the string that the source MDR uses to identify this CI. The MDR software typically determines this identifier. CMDBf Viewer uses the UUID.
- **MDR Name**
  Identifies the MDR.
- **Active**
  Denotes whether this mapping is active or not. You can inactivate mappings but not delete them.

Create MDRs

Before you can associate a CI with an MDR, define the MDR to CA SDM.

Follow these steps:

1. Click Create New on the MDR List.
2. Enter the definition for the new MDR.
3. Click Save.

NOTE

When you associate a CI with an MDR, the Add MDR button does not appear until the CI is first saved and then edited.

MDR Definition

You can create, view, and edit an MDR definition. The MDR Definition page provides the following settings:

- **Tenant**

Identifies the tenant owner of this MDR (if multi-tenancy is installed).

- **Button Name**
  Defines the MDR button label that appears on the CI Detail page. This name is unique for each MDR. Required for “launch in context” and for CMDBf Viewer.
- **MDR Name**
  Specifies the data that is sent in the mdr_name field in XML. While this string can be anything that the MDR chooses to send to CMDB, typically it is the hostname of the source. This name together with the mdr_class create a unique name for the MDR. Note: CA APM MDRs must specify an MDR name of APM and MDR class of GLOBAL. Required for “launch in context", CMDBf Viewer and GRLoader.
- **MDR Class**
  Specifies the data that is sent in the mdr_class field in the XML. While this name can be anything, it must together with the mdr_name field create a unique identifier for the MDR. Global MDRs are defined with an MDR Class of “GLOBAL”. Note: Cohesion MDRs must specify an MDR class of COHESION, which automatically sets the Path, Parameters and "URL to be Launched" fields to the required Cohesion launch-in-context values. CA APM MDRs must specify an MDR class of "GLOBAL", which automatically sets the Path, Parameters and "URL to be Launched" fields to the required CA APM launch-in-context values. Required for CMDBf Viewer. CMDBf Endpoint sets to "cmdbf" for all federated MDRs.
- **Active**
  Denotes this MDR definition as active or inactive. Inactive MDR definitions are logically deleted, but they can be made active again by using the Search utility.
- **Owner**
  Denotes the contact responsible for this MDR.
- **Description**
  Specifies text field.
- **Hostname**
  Specifies the host name, DNS name or IP address of the host, which contains the web server which hosts the web page that is launched. Required for “launch in context”.
- **Port**
  Specifies the TCP/IP port used by the MDR web server to serve up web pages. Port 8080 is the default. Required for “launch in context”.
- **Path**
  Specifies the portion of the URL that precedes the question mark (?) character. This information can be obtained from the MDR documentation. For mdr_class of Cohesion, the value is set automatically to CAisd/html/cmdb_cohesion.html and cannot be changed. For mdr_name of UAPM and mdr_class of GLOBAL, the value is set automatically to apm/frmObject.aspx and cannot be changed.
- **Parameters**
  Specifies the portion of the URL that follows the question mark (?) character. This information can be obtained from the MDR documentation. For mdr_class of Cohesion, the value is set automatically to hostname={hostname}+port={port}+family={family}+name={name}+secret={password}+federated_asset_id={federated_asset_i and cannot be changed. For mdr_name of UAPM and mdr_class of GLOBAL, the value is set automatically to ObjectID={cmdb_asset_id}&obj=11&FUNCTION=1&WinID=OBFRASSET{cmdb_asset_id}&WinContainerID= and cannot be changed.
- **Userid**
  Specifies the user ID to logon to the MDR (if necessary). This value is substituted into the URL wherever {userid} is found.
- **Shared Secret**
  Specifies information that is shared between CMDB and the MDR. This value is substituted into the URL wherever {password} is found. For Cohesion MDRs, the value must match the value of the “com.cendura.security.oneclickauth.secret”. Required for CMDBf Viewer.
- **CMDBf Namespace**

Specifies the MDR namespace that is passed to the query by CMDBf Viewer. If you use CMDB as an MDR provider, the value is "http://cmdb.ca.com/r1". For external MDRs, consult the MDR documentation. Required if MDR Class is cmdbf.

- **CMDBf Timeout**
  (Optional) Specifies time limit for CMDBf endpoint query. Default is ten (10) seconds. Only applies when MDR Class is cmdbf.
- **URL to be Launched**
  Default value of *http://{hostname}:{port}/{path}?{parameters}*. For some MDRs, it can be overridden if necessary to accommodate MDR-specific requirements. Required for "launch in context".
  For mdr_name of UAPM and mdr_class of GLOBAL, the value is *http://{hostname}:{port}/{path}?{parameters}*
  For mdr_class of Cohesion the default value is: *http://cmdb_hostname:cmdb_port/{path}?{parameters}*
  where
  cmdb_hostname is the host name, DNS name or IP address of the CMDB web server. Defaults to the current hostname that is currently accessing the CMDB web server.cmdb_port is the TCP/IP port of the CMDB web server. Defaults to the current port number used to access the CMDB web server.If you have enabled SSL support for Cohesion, set the URL to:*http://hostname:port/{path}?{parameters}+https=yes*
- **CMDBf Endpoint**
  Specifies the MDR Query Service endpoint for CMDBf Viewer. Required if MDR Class is cmdbf.
  `http://<host name:<port number>/axis/services/QueryPort`
  `CA Service Management 17.3.0.18 and above versions, http://<host name:<port number>/cxf/services/QueryPort`

Using the Configuration Control

Under the Configuration Control node in the Administration tab, view and manage CACF Managed Attributes, Managed Changes States, and Verification Policies. Change verification governs inbound CI data before loading the data into the CMDB. This verification ensures that each requested Change Order executes correctly, and detects and manages rogue changes automatically. Change verification lets you perform the following tasks:

- **Authoritative MDR Updates**
  Specifies that MDRs can only update authoritative attributes at the attribute level.
- **Rogue Change Detection**
  Detects and manages updates to CIs when no corresponding Change Order exists.
- **Change Verification**
  Provides automated verification that Change Orders executed correctly.

Change verification policies can be implemented dynamically or scheduled in advance. They can be generic or highly specific.

Managed Attributes

The managed attributes indicate those eligible CI attributes for change verification by CACF. By default, this list contains *CI name* and *Any Managed Attribute*. You add the CI attributes that you want managed as part of your change verification strategy. For more information about establishing a change verification strategy, see the Implement a Change Verification Strategy topic.

CACF does not consider unmanaged attributes (attributes not listed) for change verification. These unmanaged attributes update as usual. Selecting Any Managed Attribute as the Attribute Name in a verification policy or change specification applies to all managed attributes.

Case sensitivity in the managed attribute definition only applies when CACF compares the change specification planned value with the inbound CI transaction data. Case sensitivity does not apply to the selection patterns in the policy, which are always case-sensitive.

WARNING
By default, CACF manages the CI *Name* attribute to enable rogue insert rejection. If you inactivate *Name* from the managed attributes list, executing a transaction that inserts a CI but does not update any managed attributes is considered an unmanaged change. So, inactivating may cause CACF verification policies to ignore rogue inserts.

For a list of CI attribute names, see the *CA CMDB Technical Reference Guide*.

Create a Managed Attribute

Add managed attributes to use change verification in your environment. For example, the Configuration Administrator wants to enable change verification for a specific CI attribute and requires to define change specifications when updating a CI.

The Managed Attribute History tab shows the audit history. The audit history lists all previous modifications to the managed attribute options with the time and user that changed.

Follow these steps:

1. On the Administration tab, click [set the cmdb value at the book level], Configuration Control, Managed Attributes.
   The Managed Attribute List page appears.
2. Click Create New.
   The Create New Managed Attribute page appears.
3. Enter a valid CI attribute name. For information about attribute names, see the *CA CMDB Technical Reference Guide*. You can also use grloader -sc xxxxx (where xxxx indicates the class name) to list attribute names for a particular class.
4. Enter an attribute label.
5. Select the default Initial Status from the drop-down list of the verification criterion. When you select a managed attribute when creating a change specification, this status is used as the initial default value.
   - **Manual Verification will be required**
     Indicates that this verification requires a Configuration Administrator to verify that the change completed successfully.
   - **Set After Change Executed**
     Indicates that the planned value from the change specification sets in the CI after CACF verifies and completes the Change Order. This set operation only occurs when the Change Order moves from a state with change verification active to nonactive. For example, the set takes place when a Change Order moves from Verification in Progress to Closed. The set does not occur if the Change Order is canceled, demoted, or moved to another state.
   - **Use Discovered Value**
     Indicates that the Configuration Administrator uses the discovered value regardless of the planned value in the change specification. The original value in the verification criterion remains unchanged, but updates the CI attribute with the new value.
   - **Verification Pending**
     Indicates a pending verification.
6. Set the managed attribute as Active or Inactive to remove the attribute from being considered by CACF.
7. Enter a description for the managed attribute.
8. Enable the Ignore Case option if you want the comparisons between the discovered and planned values to remain case insensitive.
9. Click Save.
   The managed attribute is saved.

Managed Change States

CACF uses managed change states to indicate which Change Order statuses CACF manages. CACF uses managed change states to control how or when to apply change verification for CI updates. You can customize these change states to suit the needs of your organization.

The Change Order statuses represent changes that can be edited. Similarly, Change Order change states represent verification states. By default, you can edit change specifications for a Change Order in the RFC change state. The change verification occurs for a Change Order in the Verification in Progress state. Optionally, change tickets can be automatically promoted or closed when all associated change specifications execute successfully.

The following list describes the Managed Change States options:

- **Change Specifications Editable**
  Specifies whether you can edit the change specifications for a Change Order. Typically, after a change request receives approval, you cannot change the request. The updates are restricted to performing a small set of override options in the *Verification in Progress* state.
- **Change Verification Active**
  Specifies whether changes discovered for a CI while the Change Order is in this status are considered for change verification. Change orders and their related change specifications are compared with any inbound transactions to verify that they were executed successfully.
  CACF monitors all CIs for any changes to their managed attribute values. As CACF verifies each attribute level change to the CI, CACF compares it against a list of change specifications which are in a *change verification active* state. After a change specification enters this state, any change specification without a specific CI undergoes expansion. This expansion occurs where new change specifications are created using the list of CIs attached to the Change Order. After a change specification exits this state, the change specifications with a verify status of *Set After Change Verified* execute. This action updates the CI with the planned values, as specified in each change specification.
- **Implementation State**
  Specifies whether the state represents a state when the changes are being executed or implemented on the CI. When a Change Order enters this transitional state, assume that the attribute values in the CI are volatile. CACF cannot consider these changes rogue changes, but it also cannot consider the changes for final verification. Typically, the change verification process only compares the inbound attribute data after the Change Order executes completely.
- **Show Change Specification Override Buttons**
  Specifies whether the Change Analyst can control change specifications, and what level of control is given. In some implementations, the Change Analyst can edit change specifications as necessary. In other implementations, the Change Analyst can override or cancel the change specification.
- **Promote Change Order after Verification**
  Specifies whether a Change Order promotes to the next default state automatically after CACF verifies all the change specifications.

Create a Managed Change State

Create a managed change state for your change verification environment. For example, the Change Administrator wants to create the Approval In Progress state and set the verification to active.

The Managed Change State History tab shows the audit history that lists all previous modifications to the managed change state options. The tab also shows any previous values and the users that specified them.

Follow these steps:

1. On the Administration tab, click CA CMDB, Configuration Control, Managed Change States.
   The Managed Change States List page appears.
2. Click Create New.
   The Create New Managed Change State page appears.
3. Enter a Change Order status, such as **Approval in Progress**.
   You can also click Change Order Status to search for a status.
4. Select the option to set the change state as Active or Inactive.
5. (Optional) Select any of the following change states:
   - **Change Verification Active**
     Enables change verification for this change state.
   - **Implementation State**

Indicates that the Change Order is in an implementation phase.
Use this option to prevent unwanted Incident creation for incomplete Change Orders. For example, CACF detects a rogue change where the change matches the CI and attribute names of a change specification. The verification policy states to only update if the change matches. You can reject this change because the Change Order is verified after fully implemented and moved to a verification state.

NOTE

You cannot use a managed change state as both as an active change verification and an implementation state.

- **Change Specifications Editable**
  Indicates that you can edit change specifications in this state.
- **Show Change Specification Override Buttons**
  Displays the override buttons when a change verification is active. However, editing such as Accept Discovered Value, Accept Planned Value, Mark As Verified, and Cancel is not allowed.
- **Promote Change Order After Verification**
  Promotes the Change Order to the next state automatically after change specifications move to a verified or final state.

NOTE

Promotion only occurs when the Change Order is in a state with change verification active.

6. Click Save.
   The managed change state is created and saved.

Verification Policies

Verification policies specify the action that CACF takes when an MDR discovers an attribute value, and the MDR attempts to update the CMDB with that data. A variance occurs when the inbound data does not match the desired state information, as described in the change specification. A policy search executes to locate a policy that matches both the change order alignment and change specifications. Policies process in an ascending, sequential number order. By default, a compatibility policy lets all updates occur providing the behavior observed in previous releases of CA Service Desk Manager.

The change verification policy describes how CA Service Desk Manager responds to the following events:

- **Updates from Unauthorized MDRs**
  Specifies that specific MDRs are authoritative for specific attributes. CI attribute updates from unauthorized MDRs can be selectively accepted or rejected.
  For example, define policies to prevent CA Configuration Automation from updating the IP address of a CI, even if a matching Change Order exists. Allow updates to the IP address to occur only if the source MDR is Spectrum.
- **Rogue Changes**

Detects and manages updates to CIs when no corresponding Change Order exists. Specify a policy that manages rogue changes that request inserts or updates of CI data. For example, you can define a policy that loads data into TWA and does not update the CI for the following conditions:

- – A CI named similar to server* in New York changes
  – The change does not have a matching Change Order
- **Incorrectly executed changes**
  Detects whenever a Change Order is not implemented correctly.

The policy specifies any of the following actions with the discovered data inbound to the CMDB:

1. Accept the data into the CMDB even though a variance exists.
2. Reject the inbound attribute data.
3. Reject the entire inbound transaction.

4. Write the inbound data to the TWA.
5. Create an Incident.
6. Write to the standard log.

WARNING
The Configuration and Change Administrators *must* establish a change verification strategy for your environment. The default policy allows all changes to all CIs, even if the change is rogue, or the change does not match a change ticket. For more information about implementing a strategy see the Implement a Change Verification Strategy topic.

Create a Verification Policy

The Configuration Administrator creates verification policies for change verification. For example, the policy can reject specific attributes and values from a transaction. Or, it can record the transaction to the TWA for deferred processing. For authorized source MDRs, the policy can accept the transaction unconditionally.

The Verification Policy History tab shows the audit history that displays all previous modifications that are made to the policy. The Verification Log tab shows all the CACF events that are associated with the policy. The Verification Log tab lists all events that have occurred for the policy. The log shows details on the affected change specifications, managed attributes, and actions when the policy was effect.

For example, the log displays the modified managed attribute and the policy history shows the category values that you specified in the policy. Verification log entries in red indicate that the change specification is either Failed Verification or Manual Verification Active and require further attention by the user.

Follow these steps:

1. Click Create New.
   The Create New Verification Policy page appears.
2. Complete the Policy Description:
   a. **Sequence**
      (Unique) Specifies the numbered order to process the policy.
      Note: CACF processes policies in ascending order.
3. On the Policy Settings tab, select the appropriate Change Order alignment:
   a. **Change Orders With Specifications**
      Active if a Change Order with selection criteria matches the inbound CI and the attribute. The inbound transaction data may not match the specified change specifications. If an exact match exists between the planned value and the inbound data, the policy determines whether to close any Incidents that change verification creates automatically.
   b. **Change Orders Without Specifications**
      Active for Change Orders that do not have any change specifications defined. Indicates that CACF cannot verify the variance for the inbound CI attribute data for Change Orders that do not have change specifications.
   c. **Rogue Insert**
      Indicates that the variance is found without a matching CI or change ticket.
   d. **Rogue Update**
      Indicates that the variance is found with a matching CI, but without a matching change ticket that updates the changed attribute.
4. Select a Managed Attribute.
5. Select Any Managed Attributes if the policy applies to all managed attributes.
6. Enter the appropriate patterns such as Role Pattern, MDR Name Pattern, MDR Class Pattern, CI Name Pattern, Class pattern, Location Pattern. Consider the following rules:
   – When you specify patterns, the MDR Name Pattern, MDR Class Patterns, and role patterns refer to the source of the data update. The other filters refer to the CI that a user updated.
   – Specify an exact string match for the inbound attribute value.

For example, enter **COHESION** to return all string matching *COHESION*.

- Specify an exact string matched suffixed with a wildcard (*) for the inbound attribute value. CACF only allows one asterisk at the end of the string.
  For example, enter **spectrum*** to match the string *spectrum*, and any string that starts with *spectrum*.
- Specify the negative of an exact string match by prefixing the string with an exclamation point.
  For example, enter **!COHESION** to match any string that does not start with COHESION.
- The MDR Name of Web Client indicates that the web interface created the source data.

NOTE

Policy filters do not support embedded wildcards, such as spec*rum.

7. (Optional) Select a Priority level from the drop-down list to match the CI priority.
8. (Optional) Select a Service Type from the drop-down list to match the CI Service Type.
9. Select the appropriate Update Behavior from the drop-down list:
   a. **Allow Attribute Update**
      Updates to CI attribute values are always allowed.
      Use this value when you require no attribute verification, such as when you require logging and auditing.
   b. **Allow Update Only if Matches Change Specification**
      Updates the CI attribute value only if it matches the planned value of the change specification. Otherwise, CACF rejects the update, and the change specification status sets to Failed.
      Use this selection when you enable Change Order verification.
   c. **Always Cancel Entire Transaction**
      Cancels this update and all other updated attributes during this transaction. If any matching policy cancels the entire transaction, the transaction is canceled, even if other policies allow the change.
      This value only triggers when an update occurs to a managed attribute.
   d. **Keep Old Attribute Value**
      Value rejects the inbound data.
      This value is similar to canceling the entire transaction, but it only rejects a single attribute.
10. Select the appropriate Action:
    a. **Write Data to TWA**
       Select Always or Never. If you select Always, the incoming data is written to the TWA for later processing.
    b. **Create Incident**
       Select Yes or No to create Incidents that is based on this verification policy. If you select Yes, CACF creates an Incident for failed verification or a rogue update.
    c. **Incident Template**
       (Optional) Select an existing template to use if CACF creates an Incident.
    d. **Close Incident after Verification**
       Automatically closes any associated Incidents with this verification policy. The criteria for closing are that the data matches the Change Order and all change specifications are in a final state.
    e. **Log Only Mode**
       Adds a warning message to the stdlog whenever this policy would take effect.
       Note: Any higher sequenced policy governs the transaction, as if this mode policy did not exist.
11. Specify the Activation and Deactivation dates when this policy is in effect, and click Save.
    The verification policy is saved.

Knowledge Management

Knowledge Management

This article contains the following topics:

Knowledge management refers to the concept of finding, organizing, and publishing knowledge. Knowledge management captures information quickly and efficiently and then delivers this information to a user or group. The information that is captured and made available for retrieval is referred to as a knowledge base.

Users access a knowledge base by using a search engine. Knowledge Management lets you create and manage content that resides in a knowledge base. You set category and document permissions to use groups or roles. Knowledge Management helps you provide customers with solutions to complex issues. Effective knowledge management quickly delivers solutions to customers through a process that is user-friendly and easy to navigate.

To manage knowledge effectively, you take the following actions:

- Create a meaningful hierarchy of content.
- Identify the gaps in existing knowledge.
- Perform the updates and maintenance to help ensure relevance of content.
- Measure the value of available content.

Key Features

Some of the key features of Knowledge Management are as follows:

- **Document Preferences**
  Users can define preferences that help them work with documents. For example, users can set preferences for the contents that display on the Knowledge Document List page. To view the Preferences Settings page:
  – Select Preferences from the View menu.
  – Click Edit on the Preferences Detail window.
- **Natural Language Search**
  Natural Language Search (NLS) lets the analyst specify a natural language string, such as "How do I install a network printer?" to query the knowledge base for solutions. NLS instantly pinpoints candidate solutions, ranking them in order of relevance. As NLS learns from every solution that is captured, it continuously and automatically refines the knowledge base.
- **Knowledge Categories**
  Knowledge categories allow the analyst to manage the content in the knowledge base. They provide a mechanism for building and editing knowledge underlying the retrieval tools. Knowledge categories also include the ability to assign ownership of a particular knowledge solution to an expert. Hence, they ensure that the solution for a given problem is kept current and accurate.
- **Knowledge Tree Designer**
  The Knowledge Tree Designer tool helps the analyst develop and deploy business policies and intelligence by mapping any reasoning process into a knowledge tree structure.
- **HTML Editor**
  The HTML Editor lets the analyst define the layout and static content of a knowledge document template. The template defines the layout and content of the Resolution section of a knowledge document. In a knowledge tree document, it defines the layout and content of a node.
- **Knowledge Report Card**
  The Knowledge Report Card provides feedback to analysts, knowledge engineers, knowledge managers, supervisors, category owners, and system administrators about which knowledge documents are most effective. You can use the report card to improve the processes of creating knowledge documents and providing the best support to customers.
- **Multi-Tenancy**
  Multi-Tenancy allows analysts to create and modify knowledge documents and knowledge categories publically, or for specific tenants. A tenant dropdown appears in the search filter, and the search can include or exclude Public data.

Types of Knowledge Documents

Knowledge documents can be of two types:

- Knowledge document
- Knowledge tree document

Knowledge Document

Knowledge documents are placed into categories that some organizations assign to owners. Under the Top category in the Knowledge Tree, many sub categories can exist. These sub categories can in turn have many sub categories.

A knowledge document contains the following components:

- **Template**
  Specifies the content and format of documents that are displayed in the defined user view. The following templates are available by default:
  – Knowledge Documents
  – Knowledge Tree Documents
  – Quick Editing
- **Document Fields**
  Provides a consistent structure to the content in a knowledge document. For example, the Title and Summary make the content easy to scan when viewed in a list of items. The Resolution field stores the body of the solution, in rich text, tables, graphical images, and much more. These fields and their associated attributes define properties such as categorization, ownership, permission, modification date, and a range of other metadata that can help with management and retrieval.

Links to Other Forms

Allows knowledge documents to link to other forms of knowledge, which is stored either within or outside of Knowledge Management, including unstructured content such as a text file. You can also add action content (a live URL) so that the user viewing the document can perform certain actions. For example, you can insert a link into the Resolution field, which creates a ticket or performs some other action.

Knowledge Tree Documents

A knowledge document and knowledge tree document share much of the same descriptive data (such as Title, Summary, Modify Date). However, a knowledge tree document prompts the user with a series of questions with a list of multiple choice answers. By responding to the questions, they are directed to the correct answer or to the information they require. This interaction is governed by a decision tree, which is designed by the creator of the Knowledge Tree document.

Knowledge Tree Documents is well-suited to the following situations:

- Common incidents that need decision-based diagnosis/process guidance.
- Highly structured content.
- Relatively unchanging content.

Knowledge Management Roles and Functions

Knowledge Management is designed for a wide variety of users, from administrators and knowledge managers, who manage the product. It is also for customers, and employees, who use the system to find solutions to their problems. Although one person can fill multiple roles, the following roles are the basic user roles in Knowledge Management:

- **Customer --** An external end user who performs basic self-service tasks.
- **Employee --** An internal end user who performs basic self-service tasks.
- **Knowledge Analyst --** A user that is responsible for one or more steps within the knowledge management process. This user interacts with service desk analysts to create and maintain a quality solution base.
- **Knowledge Manager --** A supervisor for the Knowledge Analyst. This role handles knowledge document reassignments and escalations, and manages day-to-day administrative aspects of the solution. For example, creating

the category structure, defining the approval process, managing noise words, special terms, synonyms, and other settings and options that are more dynamic in nature than what the Knowledge Management Administrator controls.

- **Administrator --** The administrator who has access to all the functionality in CA SDM and Knowledge Management. This role is typically used when implementing CA SDM to help ensure that all users and roles are set up properly. This role is also applicable for a CA SDM environment that has a single person performing all administration tasks.
- **Knowledge Management Administrator --** An administrator who is responsible for configuring and monitoring the knowledge management process. This role includes creating the category structure, defining the approval process, and configuring default search and security settings.

Each role has a different level of access in the CA SDM environment. These levels help define the tasks that each role performs.

Knowledge Documents Lifecycle

Knowledge documents provide you with information about knowledge that is stored in the knowledge base. Creating quality knowledge requires input from several individuals. Each individual is responsible for performing specific tasks throughout various stages in the lifecycle of a knowledge document.

Knowledge documents reside in the knowledge base and are managed as part of the following ongoing process:

1. Identify content to include in the knowledge base.
2. Create a knowledge document.
   Knowledge documents are placed into categories that some organizations assign to owners. If the Incident/Request Area in CA SDM matches the knowledge categories in Knowledge Management, the category is automatically selected for the knowledge submission.

   NOTE
   When a document is created or updated, it is placed in an owner Inbox. Until the items are published, the items in the Inbox do not appear as resolutions and are not added to the knowledge base.
3. Revise the document.
   After a document arrives in the inbox, users can modify the documents according to their assigned roles.
   All users with full (read/write) permission to the document can modify the document. The current owner has full permissions to the document, but not necessarily have explicit write permissions. Users can create versions or can roll back to a previous version when a problem with the document is found.
4. Submit the document.
   In addition to submission from the customer or employee self-service interface, knowledge can also be submitted from CA SDM. This option lets the analyst submit a new resolution from an existing ticket. This option also provides a link between a problem and its resolution, and can help other users with similar problems to find a resolution.
5. Publish the document.
   After a document has passed through the complete approval process cycle, it can be published. A document that has been published becomes part of the viewable knowledge base on the start date, which is the current date by default. The document is only viewable by groups that have been granted access rights to read it. A user with full permissions can edit a published document.
6. Evaluate and decide whether to perform the following tasks:

1. • **Unpublish the document --** When a knowledge document is published, the user is not permitted to modify the document unless it is unpublished first. During this time, the knowledge document is offline and unavailable to users. The owner of the document, a knowledge manager, or a system administrator can unpublish the document

using the Rework button and the Unpublish check box. Unpublishing a document returns it to draft status. An administrative user can then select the next step in the workflow process.

- **Create a Rework Version --** Users with full editing permissions can create a rework-draft version of a published document while it remains online and available for viewing and searches. A rework version create a copy of the document. This document replaces the original document in the knowledge base after it is verified and republished.
- **Retire the document --** The owner of the document, a knowledge manager, or a system administrator can retire the document from the knowledge base.

You can define the tasks and the roles that perform each of these tasks to meet the approval process structure that exists in your organization.

Note: You can track knowledge activities from the Event Log tab on contact detail pages. For example, the event log records the actions that an user performed on the document and provides a link to the document.

Knowledge Management User Interfaces

The following user interfaces help you manage knowledge:

- **Self-Service --** In the self-service interface, customers and employees using CA SDM can access knowledge documents and can submit knowledge for further consideration. Customers can search, browse, or use bookmarks to locate and view knowledge documents.
- **Knowledge Documents --** The knowledge documents interface is accessed from the Knowledge Documents node on the CA SDM Scoreboard. All users of the system can view their Inbox and follow-up comments using this interface. The *administrator* manages their unassigned/unindexed documents, automated document lifecycle policies, and user forums.
- **Knowledge Management --** The knowledge management interface is accessed from the Knowledge tab in CA SDM. The *knowledge analyst* or *knowledge manager* can find, organize, and publish knowledge using this interface.. They can also filter the documents that are displayed using advanced options, and sort the results by relevance, modified date and much more.
- **Knowledge Administration --** The knowledge administration interface is accessed from the Knowledge node on the CA SDM Administration tab. The *administrator, knowledge manager, or knowledge management administrator* can set system options using this interface. Settings can help conform the functionality and use of Knowledge Management.

Knowledge Management Configuration and Management Functions

You can perform the following configuration and management functions in Knowledge Management:

- Create "action content" (a live action URL) that you can insert into the Resolution field of a document.
- Set up the approval process and define the knowledge document lifecycle process.
- Set up automated policies that automate certain tasks in the knowledge document approval process.
- Set up document options that are related to comments, submitting knowledge, templates, and document settings.
- Create the templates that control how a document displays information.
- Manage the default Knowledge Management search engine and configure noise words, special terms, and synonyms that are used to perform keyword and natural language searches.
- Create "recommended documents" that display in the self-service interface when users search for knowledge solutions.
- Manage the knowledge category structure to make document access easier.
- Set up the Knowledge Report Card and general system settings.
- Define the surveys that collect and analyze the customer feedback about knowledge document performance.
- Manage integration of Knowledge Management into CA SDM, including the field mapping and request and issue search configuration.

Web Services

Knowledge can be accessed through SOAP web services. Various methods are available, permitting the search, retrieval, creation, and updating of documents, and a range of other operations.

> **NOTE**
> For more information about SOAP web services, see Web Services Management.

Knowledge Base Monitoring

You can monitor the efficiency of the knowledge base using the following reporting tools. These tools let you view statistics on the usefulness of your documents and their effectiveness in solving problems.

- **Knowledge Report Card**
  Lists statistics for documents you have created. Each user has an individual Knowledge Report Card.
- **Web-Based Reports**
  Displays metrics that describe how knowledge is meeting user needs. Some of the most commonly used features include:
  - Listing the most frequently accessed documents.
  - Displaying user searches that did not return any results.

My Recent Documents

You can view a knowledge document after closing it without searching again. My Recent Documents displays your most recently accessed documents based on the following:

Title

Identifies the title of the Knowledge Document.

Hit Date

Identifies the date and time the document was accessed.

Vote

Identifies the vote rating of the document.

Default: Somewhat Helpful (if nothing is selected)

Click the title to open the corresponding Knowledge Document.

> **NOTE**
>
> If multi-tenancy is installed, the list page displays a tenant column and a tenant drop-down list in the search filter. Specifying <empty> in the search filter searches for public objects. On detail pages, select the appropriate tenant from the drop-down list. If you select <empty>, the object is public.

Getting Started with Knowledge Management

Effective knowledge management consists of more than the development of a large knowledge base. Knowledge management also involves the implementation of processes and procedures for maintaining relevant and accurate content.

To set up a knowledge base, do the following tasks:

1. Analyze how you plan to use the knowledge base and the scope of it.
2. Consider who the customers are and what information they are likely to need.
3. Develop a strategic plan that identifies the potential issues that can result in the need for customer support. The problems that you identify can guide you in determining what content must reside in your knowledge base.
4. Execute the development of your knowledge base. Create, organize, maintain, and secure this data. You can perform all these functions by using Knowledge Management.

Import Sample Knowledge Data

Sample Knowledge data from Knowledge Broker and Knowledge Accelerators is provided for your use. You can import the sample data to view example knowledge articles on How-to, knowledge broker, and so on.

Note: The sample files are not localized. Importing sample data does not work on localized environments.

For the advanced availability configuration, run ImportSampleData.bat or ImportSampleData.sh from the background server.

Windows

1. Go to $NX_ROOT\samples\data and unzip SampleData.zip into the same directory.
2. From the command window, go to $NX_ROOT/bin and run ImportSampleData.bat.

UNIX

1. Run the command tar -xvf SampleData.tar from $NX_ROOT/samples/data.
2. From the command window, go to $NX_ROOT/bin and run ImportSampleData.sh.

Setting Up the Knowledge Management System

This section includes the following articles:

- Configure Knowledge Management Settings
- Create a Knowledge Category
- Define a Document Approval Process
- Define Comment Types
- Define Document Templates
- Define FAQ and Solution Survey Settings
- How to Set Up the KT Search Engine
- Work with Forums

Configure Knowledge Management Settings

You can configure certain role-specific and document-specific settings when you are setting up your knowledge management system. You can also change these settings later to suit your requirements.

How to Manage Role Privileges and Document Visibility

You can set up Knowledge Management security permissions by managing role privileges for users in your environment. Use these permissions to define the information that users can access in a knowledge document, when they view or create knowledge. You can also define how users are authenticated when they log in to the system.

Follow these steps:

1. From the Administration tab, select Security and Role Management, Role Management, Role List.
   The Role List page appears.
2. Select the role that you want. For example, Knowledge Manager.
   The Role Detail and Update Role pages display the following tabs:
   - **Knowledge Management**
     Specifies the Knowledge Management privileges for the role.
   - **KT Document Visibility**
     Specifies which document statuses the role is allowed to view, such as draft, retired, and published.
3. Complete the tab pages and save changes.

Security and role privileges are defined.

NOTE
For more information about setting up security and defining roles, see Managing Roles.

Configure General Settings

You can set the default information to display on the Knowledge tab at logon. You can specify the format in which categories display in the Knowledge Categories pane on the Administration tab, and the number of documents to list in the Top Solutions list on the Knowledge Management home page.

Follow these steps:

1. Select **Knowledge, System, General Settings** on the **Administration** tab.
   The **General Settings** page opens.
2. Complete the following fields as appropriate:
   - **Search Tool Opening Screen**
     Specifies the information that is displayed by default on the **Knowledge** tab. You can select one of the following options:
     - **Open with FAQ / Search**
       Displays the **Category, Knowledge Search**, and **Knowledge Document List** panes.
     - **Open with Knowledge Tree Document ID**
     - Displays the knowledge tree with the document ID specified in the field provided. You can return to the knowledge tree document itself, and then to the **Category**, **Knowledge Search**, and **Knowledge Document List** panes. **Default:** Open with FAQ/Search
   - **Category Viewing**
     Specifies the format in which document categories display in the **Knowledge Categories** pane on the **Administration** tab. You can select one of the following options:
     - **Display categories in tree view**
       Presents the categories in a hierarchical tree structure in the **Knowledge Categories** pane. Categories expand to reveal associated subcategories. In this manner, you can view all the categories in the tree simultaneously.
     - **Display categories in list view**
       Presents the categories in a list format in the **Knowledge Categories** pane. When you select a category, its subcategories display in a list. You can view only the current level of categories or subcategories at one time. Use the **Up One Level** link to return to the previous category level.

     Note: If you have more than 250 categories under the top category, or under any category, use the **Display Categories in List View** option and not the tree view.
   - **Top Solutions**
     Specifies the number of documents to list in the **Top Solutions** list on the CA SDM home page.
     Default: 10

     Path for EBR Index Files

     Defines the location for storing EBR index files. CA SDM creates EBR index files when you save and publish any knowledge document. Depending on your configuration type, consider the following points while defining the EBR index file path:
     - Conventional: If you are upgrading from the CA SDM Release 12.9 from r11.2 or r12.X, you may choose not to use UNC shared path. If you have not created a UNC path, CA SDM uses the default path to store EBR index files. If you are using a UNC shared drive, manually copy the ebr/ebr_ADM folders from the default location ($NX_ROOT/site/) to the UNC shared path.
     - Advanced availability: If you are upgrading to the advanced availability configuration from CA SDM r11.2 or r12.X, you must create the UNC shared path and use it to store EBR index files. The UNC credentials are not required for the default path. After you create the UNC path, manually copy the ebr/ebr_ADM folders from the default location ($NX_ROOT/site/) to the UNC shared path.

WARNING
EBR Index Files path & **KEIT Files path** must refer to the same UNC credentials and the path must reside on a same server to support this.

Default: $NX_ROOT/site/ebr

- **Path For Knowledge Import/Export Files**

Defines the location for storing KEIT import/export packages during an import/export operation. Depending on your configuration type, consider the following points while defining KEIT file path:

- Conventional: If you are upgrading to CA SDM Release 12.9 from r11.2 or r12.X, you may chose not to use UNC shared path. If you have not created a UNC path, CA SDM uses the default path to store the KEIT files. If you are using a UNC shared drive, manually copy the Import/Export package folders from the default location ($NX_ROOT/site/keit) to the UNC shared path.
- Advanced availability: If you are upgrading to the advanced availability configuration from CA SDM r11.2 or r12.X, you must create the UNC shared path and use it to store the KEIT files. The UNC credentials are not required for the default path. After you create the UNC path, manually copy the Import/Export package folders from the default location ($NX_ROOT/site/keit) to the UNC shared path.

WARNING
EBR Index Files path & **KEIT Files path** must refer to the same UNC credentials and the path must reside on a same server to support this.

Default: $NX_ROOT/site/keit

- **UNC Credentials**

You can use this option to create UNC credentials to access the network shared drive to access EBR indexing files and import/export packages. Use the **UNC Credentials** link to create the UNC credentials.

NOTE
UNC paths and UNC credentials are required in case of the advanced availability configuration. Restart the CA SDM service when you change any of the UNC details (UNC paths or UNC credentials).

– **Document Indexing Notifications**
Sets a user to receive email notifications about status or when errors occur with document indexing. The user must have an email address in the ca_contacts table to receive these email notifications. Use the Notification Page for the assignee contact record for setting notification methods.

WARNING
You must set an **Assignee** to receive document indexing notifications in the **Document Indexing Notifications** section. A valid email address must be defined in the **Notification** tab of this contact to enable email notifications.

3. Click Save.
General settings are configured.

NOTE
When migrating from an older CA SDM release to the advanced availability configuration, manually copy the import/export packages to UNC shared location (specified in **Path For Knowledge Import/Export Files** field). For EBR index files, you can manually move the EBR/ebr_ADM folders to the UNC shared location or execute the pdm_k_reindex command path.

Specify Document Settings

As a system administrator, you can specify settings that are related to comments, submitting knowledge, and viewing knowledge tree documents.

Follow these steps:

1. On the Administration tab, select Knowledge, Documents, Document Settings.
The Document Settings page appears.

2. Complete the following fields:
   - **Knowledge Tree Document Viewing**
     Specifies the viewing mode in which knowledge tree documents open. Select Open in Tree Mode (default) to open the knowledge tree directly or Open in Document Mode to open the document in document view. If you open in Document Mode, click Display to show the knowledge tree.
   - **Comments**
     Specifies if users can submit comments for documents and view document comments. Select one of the following options:
     - • **Allow comment submission and comment viewing (default)**
       Displays a Comment field at the bottom of an open document so users can submit comments for the document. Users can view comments that are already associated with the open document.
     - **Allow comment submission but not comment viewing**
       Displays a Comment field on the right of an open document so users can submit comments for the document. Users cannot view comments that are already associated with the open document.
     - **Allow neither comment submission nor comment viewing**
       Disallows the users from submitting or viewing comments. The Comment field does not display in an open document.
   - **Submit Knowledge**
     Defines the repository for user-submitted documents. The product populates the list with the names of repositories that are defined on the Attachment Library pane.
     Consider the following information:
     - When an analyst creates a document in a category that has an owner and assigns the document to the category owner:
       Category owner becomes the assignee and owner of that document.
       Category owner must receive a document assigned notification.
     - When an analyst creates a document in a category that has no owner and assigns the document to the category owner:
       No one becomes the assignee or owner of that document.
       Submit KD notification must send as defined in the administration.
     - When an analyst creates a document and does assign the document to the category owner:
       That user becomes the assignee and owner.
       A notification does not send.
     - When an employee or customer creates new documents, CA SDM takes the action as stated in the first two points.
   - **Maximum Resolution Size**
     Defines the maximum size (in characters) that the Resolution field in a document can contain.
     Limits: The maximum characters allowed is 256000.
     Default: 32768.
   - **Duplicate Document Avoidance**
     Forces a search for similar documents when the user creates a knowledge document.
   - **Notification before Expiration**
     Defines the number of days before the document expires and a notification is sent.
     Default: 7

     NOTE
     This value only applies to documents that have been updated or created in CA SDM. If you migrate documents from CA SDM r11.2, and the expiration dates are set before the migration, this option applies only when you update the document after the migration.
3. Click Save.
   The changes apply when you open a knowledge document or a knowledge tree document.

Setting Up Self-Service Knowledge Options

Customers and employees using CA SDM have access to knowledge documents through the self-service interface. Customers can search, browse, or use bookmarks to locate and view knowledge documents. Giving the customers access to knowledge documents and services permits customers to find answers themselves and reduces the pressure on resources.

In the self-service user interface, employees, and customers can view the following knowledge solution options:

- **Search for Knowledge Solutions**
  Employees and customers can enter keywords and phrases in a search field that retrieves a list of knowledge solutions. You can configure this option in the following location: Administration, Knowledge, Search, Search Settings.
- **View Top Solutions**
  Employees and customers can display a list of top solutions in the self-service interface. The following Administrator setting: Administration, Knowledge, System, General Settings, Top Solutions **--** Number of Documents to Display, determines the number of documents to display.
- **Prompt for Knowledge Survey**
  The users are prompted for a knowledge survey after they have read the document. The survey has a series of questions. One of these questions lets the customer indicate through a prompt whether they feel that their problem has been solved. You can configure this option in the following location: Administration, Knowledge, Solution Survey, Survey Settings.
- **Suggest Knowledge**
  Employees and customers can, where permitted, view a list of knowledge suggestions when they create a ticket in the self-service interface. You can configure this option in the following location: Administration, Knowledge, Service Desk Integration, Suggest Knowledge.

Define Document Preferences

You can define preferences that help you work with information in Knowledge Management. The Preferences Settings window opens when you do either of the following actions:

- Select Preferences from the View menu
- Click Edit on the Preferences Detail window

In General Settings, the following fields require explanation:

- **Avoid Popups**
  Specifies whether to reduce the number of new browser windows opened by displaying new forms in the main browser window whenever possible.
  – Select the check box to minimize the number of new browser windows opened.
  – Clear the check box to open new browser windows according to the properties set in the form definition.
  Default: Cleared.

  NOTE
  If this option is selected, the Back to List button can be used to navigate from detail forms back to the previously displayed list window. The Back to List button appears in the upper right-hand side of the window.
- **Keep Log Reader Window**
  Specifies whether to keep the Log Reader window open when you select the Close All Popups command from the Window menu and after you log off of Knowledge Management. This setting has no effect when the Log Reader window is not open.
  – Select the check box to keep the Log Reader window open.
  – Clear the check box to close the Log Reader window when you select the Close All Popups command or log off of Knowledge Management.
  Default: Selected.
- **Preserve Popup Size**

Specifies whether to open new popup windows with the same dimensions as the most recently resized popup window.
– Select the check box to use the new dimensions for new popup windows.
– Clear the check box to open new popup windows with default dimensions.
Default: Selected.

NOTE
If you select Preserve Popup Size, maximizing a popup window will cause subsequent windows to cover any other open window. However, new popup windows appear slightly off the screen, to the right and down. As there is a 10 pixel (left and top) offset for popups, they do not completely overlay the currently displayed window. We recommended that you do not maximize popup windows when using the Preserve Popup Size option.

- **Using Screen Reader**
  Selecting this preference modifies behavior for optimal use with a screen reader for blind and limited vision users. Log off and log back in again after changing this preference. From the Help menu, select Screen Reader Usage for an overview of using CA SDM with a screen reader.

In Knowledge Search Document Settings, the following fields require explanation:

- **Attributes to be shown in list**
  Displays two lists with which you can define the properties that display for each document in the Knowledge Document List pane on the Knowledge Categories pane of the Administration tab.

NOTE

Choosing the attributes different from the Preference default can impact performance. If you believe that extra display columns are needed, contact the system administration and ask them to use the Web Screen Painter utility to modify the default page columns for the Document List page.

Create a Knowledge Category

Knowledge documents are arranged into *knowledge categories*. Knowledge engineers, knowledge managers, and administrators can manage the categories. Each of these individuals uses Knowledge Management to create, copy, and modify categories. However, only knowledge managers and administrators can delete categories. The category structure in Knowledge Categories helps you create a hierarchical structure that service desk employees, customers, and analysts use to navigate to relevant documents.

Assign each document to a primary category. For example, any knowledge that is related to Microsoft Word must be added to the Microsoft Word category. In addition, Knowledge Management lets you associate a document with multiple secondary categories and other documents. This feature allows a document to be classified under many different applicable categories and results in better search results.

The category structure performs the following functions:

- Organizes knowledge solutions into manageable groups.
- Makes it easier to assign access rights.
- Makes it possible to search for solutions using FAQ/Browse.
  To use category browse functionality from an incident, the incident area and the knowledge category must match. When you create an incident that is based on a knowledge document, the document category sets the incident area; therefore, the category and the area always match.
- Creates a document link -- A *see also* link appears when viewing either of the linked documents. The *see also* link lets you go directly from one linked document to the other.

For each category, you can define properties that identify attributes or qualities to be associated with the ticket. You can also create a workflow that identifies all the individual tasks that are required to fulfill the ticket.

You can use categories to specify default values for certain fields in tickets, or automatically associate a level of service to tickets by assigning a default service type to categories. Whenever an analyst assigns a category to a ticket, all the information you associate with the category is automatically associated with the ticket.

NOTE
If you are using multi-tenancy, a tenant drop-down list appears in the Knowledge Document search filter. If you select <empty> in this drop-down list, the search is public. A tenant column also appears on the list page.

Follow these steps:

1. On the Administration Tab, browse to Knowledge, Knowledge Categories.
   The Knowledge Categories page appears.
2. Right-click the category under which you want to create the category. Select New Category from the shortcut menu.
   The Create New Category page opens to the Content tab.
3. Complete the fields as appropriate.
4. Click Permissions.
   The Permissions tab appears.
5. Select one of the following permissions options for the category:
   - **Inherit from Parent**
     Specifies that the new category has the same permission settings as its parent category.
     Note: The Inherit from Parent option is not available if you select the TOP category before opening the Create Category page.
   - **Control by Group**
     Specifies category permissions for groups to have read or write access to the category.
   - **Control by Role**
     Specifies category permissions for roles to have read or write access to the category.

     NOTE

     If you change controls, such as changing the category permission from group to role, a warning appears that previous permissions are deleted for that category.
   - **Grant Write Permission to Everyone**
     Specifies that all users have write access to the category. Write access indicates that you can edit or delete this category.

     NOTE
     The Grant Read Permission to Everyone check box is automatically selected if you select the Grant Write Permission to Everyone check box.
   - **Grant Read Permission to Everyone**
     Specifies that all users have read access to the category. Read permission indicates that you can view the category, but you cannot edit or delete it. Users with administrative rights can edit a folder even if their associated permission group cannot. If a user belongs to multiple permission groups with varying levels of access to the category, the user gets the highest available access level (for example, if one group has read-only access and the other write access, the user gets write access).

     NOTE
     The Grant Read Permission to Everyone check box is automatically selected if you select the Grant Write Permission to Everyone check box.

     WARNING
     When you grant permissions for Everyone, the access by role or group is the same. If you selected Everyone and Control by Role, after you save the permissions, the Control by Group becomes selected.
6. (Optional) Specify the read-write permissions to specific groups or roles from the Available and Selected lists.
   You use this option to manage which groups or roles have read or write access to the category. You can select one or more permission groups or roles from the Available Groups/Roles list, and then use the Add and Remove buttons to

move the selected groups or roles to the Groups/Roles with Write Permission and Groups/Roles with Read Permission lists.

7. Click Save.
   The Category Detail page appears.
8. Click Close Window.
   The Knowledge Categories pane refreshes to include the new category.

Category Fields

- **Title**
  Names the category.
- **Description**
  Describes the category.
- **Category Owner**
  Indicates the person responsible for the category. When a contact is defined as the owner of a category, the contact has a link on the Knowledge Report Card named "My Categories," from which they can view statistics for that category and the documents it contains. This person is also the default owner for new documents in the category when the user who creates the documents is not an analyst, or an analyst creates the documents with 'Assign to Category Owner' selected.
- **Documents Template**
  Defines the document template to use for all documents that are associated with this category. The <empty> option means that none have been defined, but by default, the predefined template is used.
- **Approval Process Template**
  Defines the default template to use for the approval process for all documents that are associated with this category. The approval process template defines the workflow steps a document must go through before it is published. The default is <empty>, which indicates the application default template is used.
- **Allow forums to be created in this category**
  Specifies whether analysts can create forums within this category.
- **Request/Incident/Problem Area**
  Designates a Request/Incident/Problem area that your administrator defines to designate an area of responsibility. You can click the search icon to select from the available areas.
- **Issue Category**
  Designates an Issue Category that your administrator defines to designate an area of responsibility. You can click the search icon to select from the available areas.

Manage Category Permissions

You can manage category permissions in your Knowledge Management environment by granting specific read/write permissions based on group or role. You can specify who can view, edit, delete, or add subcategories to a category.

> **NOTE**
>
> You can use empty write permissions for documents and categories. Empty write permissions are set when you do not select write permissions. Only privileged users can modify categories and documents with empty write permissions.

To manage category permissions

1. On the Administration tab, browse to Knowledge, Knowledge Categories.
   The Knowledge Categories page appears.
2. Right-click the category under which you want to create the category, and select New Category from the shortcut menu to manage the permissions of a new category.
   The Create New Category page displays the Content tab.
3. Click Permissions.
   The Permissions tab appears.

4. Select one of the following permissions options for the category:
   Inherit from Parent
   Specifies that the new category has the same permission settings as its parent category.

   NOTE

   If you select the Top category before opening the Create Category page, the Inherit from Parent option is not available.

   Control by Group
   Specifies category permissions for groups to have read or write access to the category.
   Control by Role
   Specifies category permissions for roles to have read or write access to the category.

   NOTE

   If you change controls, such as setting the permission from group to role, a warning appears that says existing permissions are deleted.

   Grant Write Permission to Everyone
   Specifies that all users have write access to the category. Write access indicates that you can edit or delete this category.

   NOTE

   The Grant Read Permission to Everyone check box is automatically selected if you select the Grant Write Permission to Everyone check box.

   Grant Read Permission to Everyone
   Specifies that all users have read access to the category. Read permission indicates that you can view the category, but you cannot edit or delete it. Users with administrative rights can edit a folder even if their associated permission group cannot. If a user belongs to multiple permission groups with varying levels of access to the category, the user gets the highest available access level (for example, if one group has read-only access and the other write access, the user gets write access).

   NOTE

   The Grant Read Permission to Everyone check box is automatically selected if you select the Grant Write Permission to Everyone check box.

   NOTE

   When you grant permissions for Everyone, the access by role or group is the same. If you selected Everyone and Control by Role, after you save the permissions, the Control by Group becomes selected.
5. Grant the appropriate read/write permissions to specific groups or roles from the Available and Selected lists.
   You can select one or more permission groups or roles from the Available Groups/Roles list, and then use the Add and Remove buttons to move the selected groups or roles to the Groups/Roles with Write Permission and Groups/Roles with Read Permission lists.

   NOTE

   All groups or roles added to the permission list are automatically added to the read permission list. When you select the Grant Read Permission to Everyone check box, you do not need to add groups or roles to the read permission list.
6. Click Save.
   The Category Detail page appears.
7. Click Close Window.
   The modified category appears in the Knowledge Categories pane.

Copy a Category with Document Links

You can put a copy of a category, its subcategories, and all associated document links into another category without removing the selection from its original location.

To copy and paste a category with document links

1. On the Administration Tab, browse to Knowledge, Knowledge Categories.
   The Knowledge Categories page appears.
2. Right-click the category to copy, and select Copy Category with Document Links from the shortcut menu.
   The product puts the selected category, its subcategories, and all associated document links in memory.
3. Right-click the category into which to paste the copied information, and select Paste Category from the shortcut menu.
   The copied information appears beneath the selected category. The Knowledge Categories pane refreshes to display the new category structure.

Copy a Category without Document Links

You can place a copy of a category and its subcategories into another category without removing the selection from its original location or copying associated document links.

To copy and paste a category without document links

1. On the Administration Tab, browse to Knowledge, Knowledge Categories.
   The Knowledge Categories page appears.
2. Right-click the category to copy, and select Copy Category from the shortcut menu.
   The product puts the selected category and its subcategories in memory.
3. Right-click the category into which to paste the copied information, and select Paste Category from the shortcut menu.
   The copied information appears beneath the selected category. The Knowledge Categories pane refreshes to display the new category structure.

Define a Document Approval Process

Contents

For the administrators who want to control the management of their knowledge base, the ability to modify the document editing and approval process is essential. You can design Approval Process templates that specify how, when, and by which an employee document can be modified and published to the public. Approval Process templates can designate different approval processes best suited to your business environment. The approval process that you implement can be modified over time to become simpler or more complex.

The Approval Process Manager lets you define Approval Process templates. By default, the Built-in Approval Process template is used. However, you can create a template or can edit an existing template. When creating an Approval Process template, you define statuses and you add tasks to the template. The approval process involves a series of tasks that are performed on a knowledge document. The owner that is assigned a task in the Approval Process template performs each task.

The following statuses are the various states that the document is associated with during the stages of the approval process:

- **Draft**
  Specifies a new document.
- **Published**
  Specifies a document that has passed through the complete approval cycle and becomes part of the viewable knowledge base.
- **Rework-Draft Version**
  Specifies a rework version of a copy of the document that is replaced in the knowledge base after it is verified and republished.
- **Retired**

Specifies a document that has reached its expiration date. You can also create your own statuses, which you later associate with tasks.

NOTE
You can also create new document statuses. For more information, see Create a Document Status.

Approval Process Manager

Knowledge Administrators can use the Approval Process Manager to perform these actions:

- Determine which groups can read a knowledge document and which groups can write (or edit) the knowledge document.
- Identify the tasks in an approval process template that determine the lifecycle of documents that are created with the template.
- Define the various statuses that the document can be associated with during the approval process.

WARNING
When creating a knowledge document, verify that document permissions include users that can later be assigned to the document through the approval process. Users from that group do not necessarily have the permission to view the document. If the document is assigned to a specific user, default data partition constraints let the user view the document.

Define an Approval Process for Document Editing

Knowledge administrators can specify who can edit documents before the approval process and after publishing. Users with full (read/write) permissions can edit published documents.

Follow these steps:

1. On the Administration tab, browse to Knowledge, Approval Process Manager, Approval Process Settings.
   The Approval Process Settings page appears.
2. Specify who can edit documents *before* the documents are published. Select one of the following options:
   - **Documents may be edited by a task assignee, an owner or users with the appropriate Access Type views**
     Specifies that the following contacts can edit documents:
     - A contact that is assigned to the current task
     - A contact that is specified as an owner of the document for the current task
     - A knowledge manager
     - A system administrator
   - **Documents may be edited by users with full permissions**
     Specifies that any user with write permissions to the document can edit it.
3. Specify who can edit documents *after* the documents are published. Select one of the following options:
   - **User with full permissions may edit documents after they have been published**
     Specifies that a user with full permissions can edit published documents.
   - **User with full permissions can change published document's attributes**
     Specifies that any user with write permissions to the document can change only attributes of published documents such as configuration items or products.
   - **Document must be unpublished before editing is allowed**
     Specifies that the user must unpublish a document before editing it.

   Click Save.
   The approval process is defined.

Create an Approval Process Template

The tasks in an approval process template define the lifecycle of documents that are created with the template.

NOTE
If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. From the Administration tab, select Knowledge, Approval Process Manager, Approval Process Templates.
2. Click Create New.
3. Enter a name for the template and a description.
4. Click Save.
   The Approval Process Template Detail page displays more fields.
5. Select the task that you want to perform when you create a rework version of the document using this template.
6. Select the task that you want to perform when you unretire a document that was created using this template.
7. Click Insert Task to create a task to add to the template.
   The Create New Task page displays.
8. Complete the fields.
9. Click Save.
   The approval process template is created.

Add Alternate Assignees to a Task

You can add alternate assignees when you edit a task.

Follow these steps:

1. Select the task you want to edit from the Approval Process Template list.
   The Task Detail page appears.
2. Edit the fields as appropriate.
3. Click the Add button on the Assignee List.
   The Create New Assignee page appears.
4. In the Assignee field, enter the name of the person that you want to assign to the task, or click the search icon to select the name.
5. Repeat Step 4 as necessary to create a list of alternate assignees.
6. Click Save.
   The alternate assignees are added to the task.

Search for an Approval Process Template

You can enter search criteria to filter the Approval Process Template List to display only the items you want to see. You can also search for individual items to view or edit.

NOTE

If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

Follow these steps:

1. From the Administration tab, select Knowledge, Approval Process, Approval Process Templates.
   The Approval Process Template List page appears.
2. Click Show Filter.
3. Type in the first few characters of the template name you want.
4. (Optional) Click More.
   Additional fields are displayed, letting you to restrict the items that display in the Approval Process Template List.
5. Click Search.

The Approval Process Template List page displays the items that match your search criteria. You can select an item to view or edit it.

Document Status Definitions

You can add and delete user-defined document statuses, and modify the names and descriptions of predefined document statuses.

Create a Document Status

You can add and delete user-defined document statuses, and modify the names and descriptions of predefined document statuses.

Follow these steps:

1. From the Administration tab, select Knowledge, Approval Process Manager, Document Statuses.
   The Document Status List appears.
2. Click Create New.
3. Enter a name for the status and a description.
4. Click Save.
   The document status is created.

Define Comment Types

This article contains the following topics:

When a user opens a knowledge document, they can add a comment that flags the document for correction, promotion, or retirement, and much more. The Comment Type List page lets you manage the details of the comment types.
To display this page, select the Administration tab, Knowledge, Documents, Comment Types.

> **NOTE**
>
> If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

If the analyst notices a typo or problem with the content in a document, they can add a comment. The comment flags the document for correction and then assign the problem to another analyst for followup. Administrators can define the comment types that appear in various list views within the end-user interface.

Create a Comment Type

You can define the comment types that appear in various list views within the end-user interface.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. On the Administration tab, select Knowledge, Documents, Comment Types.
   The Comment Type List page appears.
2. Click Create New.
   The Create New Comment Type page appears.
3. Complete the fields as appropriate. The following fields require explanation:
   - **Time to Complete (Days)**

Defines the number of days by which the user must follow up on this type of comment.

– **Show in User View**
  Displays the comment in various list views within the user interface.
– **Follow-up Required**
  Specifies if the user is required to respond to this type of comment.

Click Save.
The new comment type appears on the Comment Types List page.

Set Up a Follow-Up Comment Notification

Several default activity notifications that are listed on the Activity Notification List page let you set up notifications to users when a follow-up comment is assigned to them.

The Follow-Up Comments queue on the scoreboard is the repository for assigned and unassigned follow-up comments.

Follow these steps:

1. On the Administration tab, browse to Notifications, Activity Notifications.
   The Activity Notification List page appears.
2. Select *one* of the following activity notifications:
   – Follow-Up Comment Assigned
   – Follow-Up Comment Closed

   The Activity Notification Detail page appears.
3. Click Edit.
4. Change the fields as appropriate.
5. Click Save.

Define Document Templates

You can use document templates to control the format and default content of knowledge documents. Every knowledge document uses a document template to define its properties and appearance when opened. By default, a built-in template is associated with new knowledge documents.

The Template Editor lets you do the following tasks:

- Design a document template that can later be associated with a document.
- Modify the built-in template and other templates.
  By editing templates, you can create templates to associate with documents. You can select the Properties options and can edit the Header and Body sections using the HTML Editor.
- Update document templates from a previous release to support knowledge relationships, such as parent-child links.

A document template specifies the content and appearance of documents in the knowledge base. You can apply the default templates:

- Built In -- Knowledge Document
- Built In -- Knowledge Tree
- Built In -- Quick Editing

The knowledge documents and knowledge tree documents use the default templates unless you create document templates and you associate them with your documents.

> **NOTE**
> When you are creating a knowledge document, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. On the Administration tab, browse to Knowledge, Documents, Document Templates.
   The Document Templates List page appears.
2. Click Create New.
   The Create New Document Template page appears.
3. Complete the following fields:
   - **Template**
     (Required) Defines a unique name for the template.
   - **Detail**
     Displays the static content that appears in documents that are created using the current template. If you select the HTML Source option, you can edit HTML code for the body directly in the Body. Select the Quick View option to view the body content as it appears at run time.
4. (Optional) Click Set Default Values.
   The Default Values Template page appears.
   You can set default values when creating a document. If you change the template on an existing document, there is no impact.
5. (Optional) Hide the Title, Summary, Problem, or Resolution from appearing in your template.
   You can hide these fields when you want a simple document, such as a question and answer template.
6. Click Edit Detail.
   The HTML Editor page appears and you can specify the static content and layout of documents that use the template. You can edit the code using the toolbar to insert placeholder tags.

   WARNING
   You can remove knowledge relationships, such as parent, child, and related links, by deleting the {TAG_PARENT} and {TAG_RELATED} tags from the template.
7. Click OK.
   The Detail field shows the updated content.
8. (Optional) Click Quick View.
   A quick view of the content as it appears in documents that are based on the template is displayed.
9. (Optional) Click HTML View.

   NOTE

   You can update knowledge document templates from a previous release of CA SDM to display knowledge relationships. You modify the template by adding the {TAG_PARENT} and {TAG_RELATED} tags. The tags allows documents that are using the template to display document relationships, such as parent-child links and tenant. To do so, select *Select Template Placeholder* on the Edit Detail dropdown to add tags to the document template.
10. Click Save.
11. The Document Template Detail page appears.
12. Click Close Window.
    The new documents that use the template show the new content and layout.
    The name of the template appears in the Template List when you display the list again.

    NOTE
    Changes to the document using the template display after starting a new session by logging in to the system.

Define FAQ and Solution Survey Settings

Solution Surveys let you collect and analyze customer feedback about Knowledge Document performance. You can modify the survey settings by selecting Knowledge, Solution Survey from the Administrative Interface. The survey

appears on a published Knowledge Document. The customers, guests, and employees can take the survey and rate the effectiveness of the Knowledge Document.

This feature contains the following components:

- FAQ Settings
- Survey Settings

Define FAQ Settings

You can set the parameters by which the product calculates the FAQ rating that is assigned to each document. The product bases the FAQ rating on the following criteria:

- How frequently the document was accessed in the past?
- How helpful the document was to users?
- How the effectiveness of the document has decreased over time?

By default, the document list page displays documents that are sorted in the order of FAQ rating (in the order of usefulness). The most useful documents "bubble up" to the top of the document list. Over time, documents tend to move downward in the document list as users learn solutions to the problems.

Follow these steps:

1. Select Knowledge, Solution Survey, FAQ Settings from the Administration tab.
   The FAQ Settings page appears.
2. Complete the following fields as appropriate:
   - **Last Updated**
     Specifies whether to run the FAQ Rating Service and displays the date on which FAQ ratings were last updated.
     - Select the Run the FAQ Rating Service check box to run the FAQ calculation service using the settings on this window.
     - Clear the Run the FAQ Rating Service check box to turn off the FAQ calculation service.
   - **Schedule**
     Defines the frequency at which the product updates FAQ ratings. This field contains the following components:
   - **Perform the FAQ Calculation Every...**
     Specifies the time that elapses before the product updates the FAQ rating for documents.
     Default: 1 day
   - **From...**
     Specifies the time of day that the product must begin recalculating FAQ ratings.
     Default: 00:00 (12:00 A.M.)
   - **To...**
     Specifies the time of day that the product must stop recalculating FAQ ratings, regardless of whether the calculation is finished. This setting initially takes effect the day after product installation. For example, if you install the product on April 19, 2008, the FAQ server runs for the first time on April 20, 2008.
     Default: 07:00 (7:00 A.M.)
   - **Aging**
     Defines the number of times a document FAQ rating is recalculated before it reaches 0. Based on the specified value, the document FAQ rating decreases and eventually becomes 0. At this point it appears at the bottom of the document list (when the list is sorted by FAQ Rating).
     Default: 180
     For example, if the Aging value is 180 for a document with a rating of 4 (very helpful), the FAQ rating of the document is 0 (zero) when the product has recalculated the FAQ rating 180 times.

NOTE
By default, the FAQ bubble-up calculation requires bu_trans data for the last 180 days, where 180 is the aging factor. So, if you change the aging factor for FAQ to more than 365 days, you must extend the archive rules for the bu_trans table accordingly.

- **Days New**
  Specifies the number of days that a newly created or imported document displays in the New Documents folder on the Knowledge tab.
  Default: 5 days
- **Rating**
  Specifies the default rating (Not Helpful at All, Somewhat Helpful, or Very Helpful) for documents that users have opened but not rated.
  Default: Somewhat Helpful

Click Save.
The FAQ settings are defined.

Define Solution Survey Settings

You can configure the survey options that you want to show to the users when they access the document.

Follow these steps:

1. Select Knowledge, Solution Survey, Survey Settings from the Administration tab.
   The Survey Settings page appears.
2. Select the survey options that you want. Select the Display 'Voting Summary' option to display the number of votes that the document received and the average rating. The summary is displayed on the Employee interface.
   The solution survey settings are defined.

How to Set Up the KT Search Engine

The Search feature enables administrators to perform the following tasks:

- Manage the Knowledge Management search engine.
- Define the settings that are used to manage noise words, special terms, and synonyms that are excluded or included in searches.
- Define the settings that are used to parse documents.
- Define default search settings.
- Create "recommended documents" that display in the search results. Dynamic FAQ listing is used to push (bubble up) the recommended documents to users.

NOTE
A document can have different permissions than the attachments linked to the document.

KT Search Engine

Content:

After you install CA SDM, the KT search engine is configured as the default. Searches of the knowledge base are limited to knowledge documents. The search engine is on the Administration tab, Knowledge, Search, KT Search Engine node.

NOTE
You can define accessibility and defaults to all knowledge sources based on a user role. By default, knowledge documents are searchable for all user roles.

To configure the search engine, go to Options Manager, Search Engine from the Administration tab and edit the ebr_version as KT Search Engine.

Noise Words, Synonyms, and Special Terms

You can define words (synonyms, noise words, and special terms) that affect natural language and keyword searches that are performed in CA SDM. Adding or deleting the following terms has a significant effect on search results that are returned to the user:

- Noise Words
- Special Terms
- Synonyms

Create Noise Words

Specifies the words that do not generally contribute to the search process and can, therefore, be ignored. For example, prepositions such as a, an, the, or, and to are often identified as noise words. The search engine ignores the noise words in documents and queries without affecting search results.

Follow these steps:

1. On the Administration tab, select Knowledge, Search, KT Search Engine, Noise Words.
   The Noise Words List page appears.
2. Click Create New.
   The Create New Noise Word page appears.
3. Enter the word that you want to define as a noise word in the Noise Word FIELD.

   NOTE
   You cannot define a noise word that exists as a synonym or keyword.
4. Click Save
   The new noise word appears on the Noise Words List page. You can click the Edit button to update the new noise word.
   Note: You cannot define a noise word that exists as a synonym or keyword.

   NOTE

   Run the knowledge re-index utility to reindex the knowledge base when you create, modify, or delete Parse Settings or KT Search Engine settings (Noise Words, Special Terms, and Synonyms).

Create Special Term

Specifies a term that you want identified as a single word during the search process, although it can consist of several words or contain special characters. For example, words that have a nonalphanumeric character, such as the forward slash (/) in TCP/IP, the hyphen (-) in dial-up, or the underscore (_) in LOCAL_SERVER can be added as special terms. In your evaluation of which words to define, consider valid words that can be divided during the search process because they have a nonalphanumeric character.

Follow these steps:

1. On the Administration tab, browse to Knowledge, Search, KT Search Engine, and Special Terms.
   The Special Terms List page appears.
2. Click Create New.
   The Create New Special Term page appears.
3. Enter the word or phrase you want to define as a special term in the Special Term field.
4. Click Save.
   The Special Term Detail page opens so you can review the word or phrase you added. You can use the Edit button to update the new term. The new special term appears on the Special Terms List page.

Create a Synonym

Specifies a word that has the same meaning as another word. When a user searches for a particular word and a corresponding synonym for that word exists in your knowledge base, the information can be found. You can define several synonyms for the same word. The system automatically creates reverse synonyms from the keywords you define. For example, if you define a computer as a synonym for the word PC, PC automatically becomes a synonym for the word computer. Use the Synonyms List page to specify keyword/synonym pairs that the product uses interchangeably when parsing documents and queries. These keyword/synonym pairs can improve search results.

NOTE
Run the knowledge re-index utility to reindex the knowledge base when you create, modify, or delete Parse Settings or KT Search Engine settings (Noise Words, Special Terms, and Synonyms).

If you define a new complex synonym (that is, a synonym containing multiple words that are separated by spaces or other delimiters), also create an identical special term so that the product can treat the synonym as a single entity. For example, if you define "Computer Associates" as a synonym for "CA", also define "Computer Associates" as a special term.

You cannot define a synonym or keyword that exists as a noise word.

Follow these steps:

1. On the Administration tab, browse to Knowledge, Search, KT Search Engine, Synonyms.
   The Synonyms List page appears.
2. Click Create New.
   The Create New Synonyms page appears.
3. Enter the word or phrase you want to define as a synonym in the Synonyms field.
4. Click Save.
   The Synonyms Detail page opens so you can review the word or phrase you added. The new synonym appears on the Synonyms List page.
5. Right-click the synonym and select Edit if you want to edit the synonym,

Define Parse Settings

When you publish a document to the knowledge base, the product parses the information in the Title, Summary, Problem, and Resolution fields of the document into keywords. When a user searches the knowledge base, the product compares keywords from the user query with the keywords parsed from the knowledge base to produce a result list.

Run the knowledge re-index utility to reindex the knowledge base in the following scenarios. Re-indexing the knowledge base ensures that the keyword search results are current and most accurate.

1. When you create, modify, or delete Parse Settings or KT Search Engine settings (Noise Words, Special Terms, and Synonyms).
2. When you create or modify a knowledge document using the KT Search Engine settings (Noise Words, Special Terms, and Synonyms).

NOTE

Based on the size of the knowledge base, reindexing can take significantly longer.

To re-index the knowledge base, see instructions on the pdm_k_reindex utility from the CA SDM PDM Database Commands page.

Follow these steps:

1. Browse the Administration tab in Knowledge, Search, Parse Settings.
   The Parse Settings appears.
2. Use the following fields to define settings:

- – **Maximum Search Keywords**
  Defines the maximum number of keywords to extract when the product parses the search text.
  Default: 20

NOTE
The valid range is 1-100, so that a CA SDM Knowledge administrator can change the value within this range, which is based on search needs and parameters of a specific Knowledge database. Use a lower number of search keywords for faster performance.

– **Language**
Specifies the language type to use for parse processing. Select one of the following settings:
- **English**
  Performs certain types of processing specific to the English language (for example, de-pluralizing search terms) during a search, if applicable.
- **Other European**
  Performs only European specific processing during the search.
- **Korean**
  Performs only Korean specific processing during the search.
- **Other Far East**
  Performs processing for other far east languages during the search.

NOTE

: When you are in a Chinese, Japanese, or Korean operating environment, verify that you understand the available parsing approaches, and limitations of Multi-Byte Character Set (MBCS) languages, before implementing your Knowledge Management system to help ensure that user expectations are set appropriately.

– **Valid Character Range**
Defines the range of alphanumeric characters to consider valid when parsing the Title, Summary, Problem, and Resolution fields in a document. The product treats any other characters as separators.

NOTE
When you select Yes from the Recognize Special Terms list, the product does not parse words and phrases defined as special terms.

Default: a-z, which indicate that the alphabetic characters a through z are valid characters for parsing.
The Valid Character Range field contains the appropriate letters that parsing uses. The letters that are not presented in the Valid Character Range are removed.
The recommended values for different languages are as follows:

Language	Valid Character Range
German	a-zäöüß
Spanish	a-záÉíóúüñ
French	a-zàäâçÉèêëîïôùû
Portuguese Brazil	a-zàáãçÉêíóúü
Italian	a-zàèÉìîïù
Simplified Chinese	a-z
Japanese	a-z
Traditional Chinese	a-z
Korean	a-z

NOTE
Japanese contains the "a-z" range plus a list of Katakana valid characters, excluding punctuation marks.

– **Remove Similar Words**
Specifies whether the product removes structurally similar keywords from the groups that are used in a search. You can select one of the following settings:
- **Yes**
  Removes structurally similar keywords from the search criteria.

NOTE
When you select Yes, the product also removes similar words when you save or publish the document. This setting can impact whether a document is searchable if the Remove Similar Words field was set to Yes. The similar word may have not been indexed and used in the later search and retrieval of the document.

- **No**
  Leaves structurally similar keywords in the search criteria.

Default: No

– **Remove Noise Words**
Specifies whether the product removes noise words when parsing the Title, Summary, Problem, and Resolution fields in a document. You can select one of the following settings:
- **Yes**
  Removes noise words from the search criteria.
- **No**
  Leaves noise words in the search criteria.

Default: Yes

– **Recognize Special Terms**
Specifies whether the product considers special terms as single entities or as multiple words when parsing the Title, Summary, Problem, and Resolution fields in a document. You can select one of the following settings:
- **Yes**
  Processes special terms as single entities in the search criteria.
- **No**
  Processes the words that comprise special terms as separate entities in the search criteria.

Default: Yes

Multi-Byte Character Set Search Limitations

Make sure that you understand the available parsing approaches, and limitations of MBCS languages, before implementing your Knowledge Management system to help ensure that user expectations are set appropriately. This limitation of the product impacts search features using Japanese, Chinese, or Korean language text within the system. The word parsing mechanism used by the search mechanism is controlled on the Parse Settings page.

For the English, Other European, and Korean settings, the product assumes that punctuation, “white space,” or both characters separates words. This assumption allows the document text to be broken into specific words, and allows noise words to be ignored and application of known synonyms and special terms to search terms.

Alternatively, when the Far East language setting is selected, the parsing routine uses a character-by-character parsing approach to accommodate some Far East language text approaches of not using white-space delimiters between words. This setting tells the parser to assume that each character is treated as a full word. The setting applies to all text to be searched. Because the language setting changes the way that the search parsing works, the entire search index must be recreated if the language setting is changed to or from Far East.

Create Recommended Documents

The CA SDM users can specify the criteria about an item of interest. The search engine then finds the matching knowledge documents and displays them on the search results page as a set of "recommended document" links. The search query can be expressed as a keyword or set of words (phrase) that identify the desired concept that one or more documents can contain.

The list of documents that meet the search criteria is sorted, and ranked (from highest to lowest) to place the most relevant documents first in the search results. Using recommended documents helps users reduce the time that is required to find the desired information.

To provide a set of matching documents that are sorted according to some criteria quickly, the search engine collects data through the condition type (phrase, keywords, or category), which the administrator configures on the Create Recommended Documents page.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. On the Administration tab, browse to Knowledge, Search, Recommended Documents.
   The Recommended Documents List appears.
2. Click Create New.
   The Create New Recommended Documents page appears.
3. Complete the following fields as appropriate.
   - **Knowledge Document**
     Specifies a knowledge document or click the search icon to open the Knowledge Document Search page.
   - **Condition Type**
     Specifies a condition type by which the search engine sorts and matches the document.
     - If the condition type is Full Match, Exact Phrase, or Keywords, a text field appears. You can enter a phrase or keyword that identifies the concept you want for the document to contain.
     - If the condition type is Knowledge Category, the Knowledge Category link appears. You can specify a knowledge category to associate with this document.
   - **Status**
     Defines the status of this record as active or inactive.

   Click Save.
   The new recommended document is saved to the knowledge base and appears on the Recommended Documents List page.

Condition Type Field

The search engine locates documents by the following condition types:

- **Full Match**
  Searches for documents by the search phrase entered in the search text. A match occurs only when the search engine finds all the same words in a phrase.
- **Exact Phrase**
  Searches for documents by the exact phrase entered in the search text. A match occurs only when the search engine finds the exact set or sequence of words in a phrase.
- **Keywords**
  Searches for documents by the keywords entered in the search text. A match occurs only when the search engine finds all of the keywords.
- **Knowledge Category**
  Searches for documents by knowledge category. A match occurs only when a user navigates to a category configured for recommended documents.

Set Up Default Search Settings

You can set up default search options that appear when users search for knowledge using the search field.

> **NOTE**
>
> These search options are overwritten by any personal search settings users define in the Preferences window, or any additional search options that are specified in the following places:

- In the Knowledge Search pane on the Knowledge tab.
- In the Knowledge Categories pane on the Administration tab.

Follow these steps:

1. From the Administration tab, navigate to Knowledge, Search, Search Settings.
   The Search Options page displays.
2. Select the following options as appropriate:
   – **Recommended Results**
   Specifies the number of documents to display in the search results list.
   – **Default Search Fields**
   Specifies which document fields to include by default in keyword searches. The following document fields are available for searching:
   - Title
   - Summary
   - Problem
   - Resolution
   - Attachments

   – **Search Settings for All Sources**
   Specifies whether searches can include all knowledge sources. For example, knowledge categories and request areas.
   – **Search Settings in a Ticket Context**
   Specifies whether searches can include all fields that are defined in a service desk ticket (incident, problem, issue, change order, or request).
   For these options, select *one* of the following match types:
   - **Any of the words (OR)**
     Includes a document in the result set when it contains any of the words in the Search field. This option is the default selection.
   - **All of the words (AND)**
     Includes a document in the result set only when it contains all the words in the Search field.
3. Click Save.
   The default search settings are set up.

How to Set Up Knowledge Management with Ticket Categories

This article contains the following topics:

Define the configuration options available for CA SDM Integration.

Define Field Mapping

Administrators can specify which fields to populate with Knowledge Management information, and whether to overwrite existing information.

Follow these steps:

1. On the Administration tab, browse to Knowledge, Service Desk Integration, Field Mapping.
   The Field Mapping page appears.
2. Complete the following fields as appropriate:
   – **Populate Service Desk Values from Knowledge Management**
   Specifies whether to use information from Knowledge Management to populate fields in service desk issues or requests.

- Select the check box to make fields in the Knowledge Management, Populate Empty Service Desk Values, and Overwrite Service Desk Values columns available so you can specify which Knowledge Management information to use to populate fields in service desk issues or requests.
- Clear the check box to make fields in the Knowledge Management, Populate Empty Service Desk Values, and Overwrite Service Desk Values unavailable. In this case, users must manually populate service desk issues or requests that are created from Knowledge Management.

Default: This check box is selected.

– **Service Desk**
Identifies the fields in issues or requests that correspond to fields listed in the Knowledge Management column. For each check box selected in the Populate Empty Service Desk Values column, information from the corresponding field in the Knowledge Management column populates the issue or request.

– **Knowledge Management**
Identifies the Knowledge Management fields that correspond to the service desk fields listed in the Service Desk column.
For each check box selected in the Populate Empty Service Desk Values column, information from the corresponding field in the Knowledge Management column populates the issue or request.
The Knowledge Management column contains two drop-down lists:

- The first drop-down list corresponds to the Summary field in the Service Desk column. It also specifies the Knowledge Management field (Title, Summary, or Problem) with which to populate the Summary field in an issue or request.

Default: Summary.

- The second drop-down list corresponds to the Description field in the Service Desk column. It also specifies the Knowledge Management field (Title, Summary, or Problem) with which to populate the Description field in an issue or request.
- **Default**: Problem.

– **Populate Empty Service Desk Values**
Specifies which empty fields in a service desk issue or request to populate with information from Knowledge Management.

- Select a check box to map information from the Knowledge Management field to the corresponding service desk field if that field currently contains no information.
- Clear a check box if you do not want to map information from the Knowledge Management field to the corresponding service desk field.

Default: The check boxes corresponding to the
Summary, Description, Product, Asset, and Request Area fields in service desk are selected.

– **Overwrite Service Desk Values**
Specifies which fields in a service desk issue or request to overwrite with information from Knowledge Management.

- Select a check box to replace information in the service desk field with information from the corresponding Knowledge Management field.
- Clear a check box if you do not want to replace information in the service desk field with information from the corresponding Knowledge Management field.

These check boxes are only available when the corresponding check boxes in the Populate Empty Service Desk Values column are selected.
Default: All Overwrite Service Desk Values check boxes are cleared.

3. Click Save.
Field mapping is defined.

Define Issue Search Configuration

You can define the fields within an Issue to search in when you click the Search Knowledge button on a ticket. The fields that you select are copied to the corresponding fields in the Search Filter on the Knowledge tab of the Issue Detail

window. The population of the Search Filter fields from the ticket occurs when the Knowledge tab is selected or the Reset Filter button (on the Knowledge tab) is clicked.

Follow these steps:

1. From the Administration tab, select Knowledge, CA SDM Integration, Issue Search Configuration.
   The CA SDM Integration page displays.
2. Select the fields that you want to be available for Knowledge Base searches.
   - Summary
   - Description
   - Configuration Item
   - Priority
   - Category
   - Root Cause
   - Product

   Note: You cannot select both the Summary and Description fields.
3. Select the "Automatically run search when the Knowledge tab of an issue is selected" option if you want to search the knowledge base automatically when the Knowledge tab on the detail page is selected.
4. Click Save.
   Issue search is configured.

Define Request/Incident/Problem Search Configuration

You can define the fields within a request, incident, or problem to search in when you click the Search Knowledge button on a ticket. The fields that you select are copied to the corresponding fields in the Search Filter on the Knowledge tab of the ticket detail page. The population of the Search Filter fields from the ticket occurs when the Knowledge tab is selected or the Reset Filter button (on the Knowledge tab) is clicked.

Follow these steps:

1. From the Administration tab, select Knowledge, CA SDM Integration, Request/Incident/Problem Search Configuration.
   The CA SDM Integration page displays.
2. Select the fields that you want to be available for Knowledge Base searches.
   - Summary
   - Description
   - Configuration Item
   - Severity
   - Impact
   - Urgency
   - Priority
   - Request Area
   - Root Cause

   NOTE

   You cannot select both the Summary and Description fields.
3. Select the "Automatically run search when the Knowledge tab of a request is selected" option if you want to search the knowledge base automatically when the Knowledge tab on the detail page is selected.
4. Click Save.
   The fields in a request, incident, or problem to use for Knowledge Base searches search is configured.

Knowledge Suggestions

Employees and customers can, where permitted, view a list of knowledge suggestions when they create a ticket in the self-service interface.

If a solution is found, and the ticket is not saved, the documents that were suggested can be credited through a self-service rating system in the document form. This rating system differs depending on the self-service policy settings that are defined on the Search Settings page.

The data that is retrieved can be used for reports, dashboards and also while searching the knowledge base, where users can filter the documents that have successfully resolved their tickets.

The benefits of self-service are in the form of fewer support calls and redundant tickets that are created, which translates into reduced operational costs.

The administrator must enable this feature before use and must configure the appropriate issue categories and request areas for which knowledge is suggested in the self-service interface.

Define Issue Categories

You can define issue categories for which knowledge is suggested to customers and employees during ticket creation.

You can also mark the Suggest Knowledge feature as active or inactive. When you mark this feature as inactive, it is no longer available to customers and employees, but it remains available in the database for future use. If you decide to use this feature in the future, you can go back and can mark it as active.

Follow these steps:

1. On the Administration tab, browse to Knowledge, Service Desk Integration, Suggest Knowledge, Issue Categories.
   The Suggest Knowledge for Issue Categories page appears.
2. Select the **Do not suggest knowledge** option to mark the Suggest Knowledge feature as active.
   Default: Inactive
   If you mark this feature active, the following options appear:
   - **By default knowledge will be:**
   - **Suggested**
     Indicates that you *do* want knowledge to be suggested for all issue categories except the definitions in the issue categories list.
   - **Not Suggested**
     Indicates that you *do not* want knowledge to be suggested for all issue categories except the definitions in the issue categories list.
   - **For all Issue Categories except the following:**
     Displays the list of request areas where knowledge is either suggested or not suggested to employees and customers in the self-service interface. The self-service user is not allowed to edit the request areas, they are read-only.
3. Click *one* of the following buttons:
   - **Add**
     Adds the selected request area to the list.
   - **Remove**
     Removes the selected request area from the list.
   - **Remove All**
     Removes all request areas from the list.
   - **Save**
     Saves the request area information to the knowledge base.

   The Issue categories are defined.

Define Request/Incident/Problem Areas

You can define request, problem, and incident areas for which knowledge is suggested to customers and employees during ticket creation.

You can also mark the Suggest Knowledge feature as active or inactive. When you mark this feature as inactive, it is no longer available to customers and employees, but it remains available in the database for future use. If you decide to use this feature in the future, you can go back and can mark it as active.

Follow these steps:

1. On the Administration tab, browse to Knowledge, CA SDM Integration, Suggest Knowledge, Request/Incident/Problem Areas.
   The Suggest Knowledge for Request/Incident/Problem Areas page appears.
2. Select the **Do not suggest knowledge** option to mark the Suggest Knowledge feature as active.
   Default: Inactive
   If you mark this feature as active, some additional options appear:
   - **By default knowledge will be:**
   - **Suggested**
     Specifies this option if you want knowledge be suggested for all request/incident/problem areas except those that are defined in the Request Area list.
   - **Not Suggested**
     Specifies this option if you *do not* want knowledge be suggested for all request/incident/problem areas except those that are defined in the request area list.
   - **For all Request/Incident/Problem Areas except the following:**
     Displays the list of request areas where knowledge is either suggested or not suggested to employees and customers in the self-service interface. The self-service user is not allowed to edit the request areas, they are read-only.
3. Click Save.
   The suggest knowledge request/incident/problem areas are defined.

Configure Self-Service Policies

You can configure self-service policies that credit documents based on a set of user scenarios.

Follow these steps:

1. On the Administration tab, browse to Knowledge, CA SDM Integration, Suggest Knowledge, Self-Service Configuration.
   The Search Settings page appears.
2. Specify the appropriate policy settings that credits documents and save avoided tickets that are based on the following user scenarios:
   - User did not open any suggested document.
   - User opened a suggested document.
   - User accepted a suggested document as a solution to their problem.
   - User searched for knowledge, opened the document, and exited.

   Click Save.
   Self-service policy settings are saved.

Work with Forums

Contents

Forums allow you to communicate about existing knowledge documents. Using forums allows documents to be globally shared or shared among predefined groups that work together in knowledge-sharing and brainstorming over existing challenges.

Forums broaden the scope of knowledge contributions by allowing discussion on general questions, usability tips, and so on.

Create a New Forum

You can create a new forum from the Knowledge tab.

> **NOTE**
>
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants

To create a new forum, **follow these steps:**

1. Click the Knowledge tab.
   The Knowledge Search appears.
2. Click File, New Forum.
   The Create New Forum page appears.
3. Complete the appropriate fields:
   Category Opens the Knowledge Category List page. Use this to specify a category for your new forum. Submitter Displays the submitter of the new forum. Priority Sets the priority of the new forum. Title Defines the title of the new forum. Question Defines the question for the new forum. Click Spell Check All Fields. (Optional) The Spell Check Results page appears if there are spelling errors.
4. Click Save.
   The Update Forum page appears.
5. Click the following tabs and add appropriate information:
   - Content
   - Attributes
   - Attachments
   - Notifications
   - History
6. Do any of the following:
   - Click Save and continue editing.
   - Click Save and Close.
   - Click User View.

Content Tab

Use the Content tab to verify or modify the following:

StatusSet to Open by default (read-only). You can change this field when editing a forum.

PrioritySets the priority of the new forum.

TitleDefines the title of the new forum.

QuestionDefines the question for the new forum.

(Optional) Click Spell Check All Fields. The Spell Check Results page appears if there are spelling errors.

Click Save when you finish.

Attributes Tab

Use the Attributes tab to verify or modify the following:

InitiatorDisplays the name of the contact who created the forum. Click the name to open the Detail window for the contact. This field is read-only.

AssigneeDisplays the Contact Search page. Use this to assign a contact to the forum for the duration of the current approval process task.

Creation Date Displays the date and time at which the forum was created. This field is read-only.

Disregard Life-Cycle Policies Ignores your organization's life-cycle policies.

FAQ Rating Displays the forum's current FAQ rating, which the product calculates from the frequency at which a forum is accessed, how helpful it has been to users, and its age. This field is read-only.

HitsDisplays the number of times the forum has been accessed. This field is read-only.

Self-Service Count Specifies the number of times that the forum was used to solve a problem. It is a count of the number of times the customer responded with Yes to the question "Did this document solve your problem? " in the document Solution Survey.

Average Rating Specifies an average rating value. Each rating response (very helpful, somewhat helpful, and not helpful at all) has an assigned numeric value. The average rating is the total of the forum response ratings divided by the total number of votes.

User Votes Specifies the total number of votes for the forum. Submission of the forum's Solution Survey is considered a vote.

PrioritySpecifies the priority to associate with the forum, indicating the urgency of problems the document is meant to resolve. Valid values are 1, 2, 3, 4, 5, and None.

SeveritySpecifies the level of effect that the problem the forum is meant to resolve may have on users. Valid values are 1, 2, 3, 4, and 5.

ImpactSpecifies the level of impact that you expect the forum to have on work being performed. Valid values are 1, 2, 3, 4, 5, or None.

UrgencySpecifies the importance level of the user tasks associated with the forum. Valid values are 1, 2, 3, 4, and 5.

ProductDefines a product to associate with the forum by opening the Product Search dialog so you can locate and select a product.

Configuration Item Opens the Configuration Item Search page. Use this to define a configuration item (hardware, software, or service) to associate with the forum.

Root Cause Opens the Root Cause Selection page. Use this to define a root cause or core reason to associate with the forum.

Solution Count Displays the number of tickets resolved by the forum. This field is read-only.

Parent Problem/Incident/Request Defines an associated parent problem, incident, or request record.

Parent Issue Defines an associated parent issue record.

ProblemDefines a problem linked to this forum.

Click Save when you are finished.

Attachments Tab

You can add both file and link attachments to a forum using the Attachments tab. This tab contains the following:

Attach File Opens the Add File page. Use this to attach a local file to the forum.

Attach File from Library Opens the Attachments Library. Use this to attach a file already upload to your environment.

Add URL Link Adds a URL link to the forum.

AttachmentsDisplays attachments added to the forum. This field is read-only.

Notifications Tab

Use this tab to add notifications to a forum. This tab displays the following information:

Alternate Email Displays Email address of the user who added the notification.

User ID Displays the system login of the user who added the notification.

NameDisplays the name of the user who added the notification.

Click Add Notification to open the Create New Notification for Document page.

NOTE

If you are using multi-tenancy, a tenant column appears on the list page.

For more information, see How to Set Up Notification for an Activity.

History Tab

The History tab lets you search for event details on a forum, such as when it was created, edited, and so on. This tab lets you search the history as well as displays the following read-only fields:

EventDisplays the event such as, Document was created.

NameDisplays the user that caused the event.

Time Stamp Displays the time of the event.

Version Number Displays the version number of the event.

Event Detail Displays a link to details about the event.

Edit a Forum

You can edit a forum thread after you create or search for it.

To edit a forum, **follow these steps:**

1. Open a forum.
2. Click Edit.
   The Update Forum page appears.
3. Complete the appropriate fields:
   Category
   Opens the Knowledge Category List page. Use this to specify a category for your new forum.
   Submitter
   Displays the submitter of the new forum.
   Priority Sets the priority of the new forum.
   Title Defines the title of the new forum.
   Question Defines the question for the new forum.
   Click Spell Check All Fields.
   (Optional) The Spell Check Results page appears if there are spelling errors.
4. Click the following tabs and add appropriate information:
   - Content
   - Attributes
   - Notifications
   - History
5. Do any of the following:
   - Click Save and continue editing.
   - Click Save and Close.
   - Click User View.

Reply to a Forum

You can reply to a forum from the Knowledge tab.

To reply to a forum

1. Click the Knowledge tab.
   Search for a forum.
2. Click the forum name.
   The Forum page opens.
3. Click Reply in the Page Options menu.
   The Create New Reply page opens.
   Complete the appropriate fields:
   Reply Type
   Choose one of the following
   - Answer
     Allows you to answer the question.
   - Clarification
     Allows you to ask for a clarification of the question.

   Quick View
   - Displays the answer in Quick View mode.
   - **HTML Source**
   - Allows you to answer using HTML.
   - **Attach Document**
   - Allows you to attach a local file to the reply.
4. Click Save.
   The reply is saved.
   Close the window.

Create Knowledge Documents

This article contains the following topics:

Create the Knowledge Document

You can create knowledge documents to capture and organize information and expand your knowledge base. Knowledge documents provide users in your environment with solutions to complex issues.

Follow these steps:

1. Click the Administration tab and navigate to Knowledge, Knowledge Categories.
2. Select the desired knowledge category.
3. Click File, New Knowledge Document.
   The Create New Document page appears.
4. (Optional) Use the Find Similar tab to search for related documents.
5. (Required) Choose a document template from the drop-down list. The template specifies the content and format of documents that are displayed in the defined user view.
   You can also use templates to populate default values.
6. Depending on the template definition, complete the following fields display:

   NOTE
   If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

- **Doc ID** A numeric unique ID is assigned to a knowledge document. The Doc ID is used by analysts and customers to refer to a particular knowledge document.
- **Title**
  Defines a short, descriptive name for the document. For knowledge tree documents, Title defines the name of the primary node in the knowledge tree.
- **Summary**
  Defines a summary that briefly describes the problem that is associated with the document.
- **Problem**
  Defines the full description of the problem that is associated with the document.
  NLS search type is based on this field.
- **Parent Document**
  Specifies the parent document of the knowledge document you are viewing. The tenant of the parent document also appears, depending on your permissions.
  Note: Modifying a parent document alerts you that the change can affect child documents.
- **Related Documents**
  Specifies the related documents of the knowledge document you are viewing. The tenant of the related documents also appears, depending on your permissions.
- **Resolution**
  Defines the description of how to resolve the problem.
- **Quick View**
  Specifies to display content in the associated field as it appears at runtime. This field only displays when you are creating or updating a knowledge document.
- **HTML Source**
  Specifies to display content in the associated field as HTML source code you can edit. Select this option to make simple changes directly to the HTML code. This field only displays when you are creating or updating a knowledge document.
- **Notes**
  Defines additional information about the document. The contents of this field are available only to analysts.

  NOTE

  Modifying a parent document alerts you that the change can affect child documents.

7. Click Save and Close.
   The document appears on the Document List page and in the appropriate inboxes.
8. Edit the knowledge document and add permissions to the users to whom you have assigned the document.

Copy Knowledge Documents

You can create a new document by copying an existing document.

To create a copy of a knowledge document

1. From the Document list, right-click the title of the document to copy, then select Create Copy of this Document from the short-cut menu.
   The Copy Knowledge Document page appears.
2. Select the desired tabs to display when the document is opened.
3. Click Save.
   The new document opens.

   NOTE

   : If you copy a knowledge document with one or more attachments, only files uploaded and attached from the Library, Repositories list are copied with the document. File attachments uploaded from other locations are not copied with the document.

Edit the Knowledge Document

You can edit a knowledge document that has not yet been published. All users with full write permissions can edit documents by default, but the administrator specifies your knowledge document permissions. For example, your administrator configures Knowledge Management so that only an assignee, the document owner, or a knowledge manager can edit documents.

Follow these steps:

1. On the Knowledge tab, select the appropriate category.
   The Knowledge Document List page displays documents in the selected category.
2. Right-click the title of the document, and select Edit from the shortcut menu. If the document has been published, create a rework version to modify the content.
   The Update Knowledge Document page appears.
3. (Optional) The properties of a knowledge document can be changed on the Attributes tab, but you must have write access to the document to modify the properties.
4. (Optional) Complete or change the fields on the various tabs as appropriate.
   - Click the Attributes tab to define the document attributes such as the Assignee, Approval Process Template.

     NOTE
     You must have write access to the document to modify the properties.
   - Click the Permissions tab to define the permissions for the contacts who will be working on the document.
   - Click the Related Knowledge tab to add the related knowledge documents.
   - Click the Attachments tab to add a file or URL to the knowledge document.
   - Click the Comments tab to add a comment for follow-up and assign the comment to a contact.
   - Click the History tab to view the events that occurred on the knowledge document.
   - Click the Versions tab to view the document versions and revisions.
   - Click the Find Similar tab search for similar documents that may contain knowledge solutions. Locating similar documents can help you avoid creating duplicate information. You can also create knowledge relationships from the context menu.
5. (Optional) Do *one* of the following, as appropriate:
   - **Knowledge Document**
   - Click Edit Resolution on the Content tab to open the HTML Editor.
   - Modify the body (resolution) of the document.
   - Click OK to save your changes and close the HTML Editor.
   - **Knowledge Tree Document**
   - Click Design Tree on the Content tab to open the Knowledge Tree Designer.
   - Edit the knowledge tree.
   - Click Save and Close.
6. (Optional) Click Spell Check to review the spelling in the content you entered.
7. (Optional) Click User View to open the Preview window, which presents the document as it appears when a user displays it.
   You can close the Preview window when you finish reviewing the document.
8. Click Save when you finish editing the document.
   The product saves your changes.
9. Do *one* of the following actions:
   - Click Save Version to save a new version of the document.
   - Click Publish to put the document in the knowledge base and make it available to users with the appropriate permissions.
   - Click Cancel to close the Update Knowledge Document window and open the Knowledge Document Detail window.

Update Document Attributes

You can use the Attributes tab to set properties of a knowledge document that define ownership, scheduling, approval process, layout, associations, and so on.

NOTE
You must have write access to a document to set its attributes.

NOTE
If the Priority Calculation is set up and a user opens a ticket on the Knowledge Document, the Priority, Urgency, and Impact fields on the Attributes tab automatically appear on the ticket.

The tab contains the following fields:

- **Initiator**
  (Read-only) Displays the name of the contact who created the document. You can click the name to open the Detail window for the contact.
- **Assignee**
  Defines the name (in "last name, first name" format) of the contact to whom the document is assigned during the current approval process task. You can enter or search the name of the contact in "last name, first name" format.
- **Author**
  Defines the name (in "last name, first name" format) of the contact with responsibility for the document content.
- **Owner**
  Defines the name (in "last name, first name" format) of the contact assigned to maintain the document. A document can have the same or a different owner in each stage of the approval process. All analysts with write access to a document can modify it, even if they do not own it.

  NOTE
  If a guest user or customer submits a document using the Submit Knowledge feature, the document has no owner.
- **Subject Expert**
  Defines the name (in "last name, first name" format) of a contact with expertise in the subject matter of the document. Enter or search the name of the contact in "last name, first name" format.
- **Creation Date**
  (Read-only) Displays the date and time at which the document was created.
- **Modify Date**
  (Read-only) Displays the date and time at which changes to the document were last saved.
- **Published Date**
  (Read-only) Displays the date and time at which the document was published. If the document has not been published, the field is blank.
- **Last Accepted Date**
  (Read-only) Displays the date and time at which the document was last accepted as the solution to a ticket. If the document has not been accepted, the field is blank.
- **Start Date**Defines the date (in mm/dd/yyyy format) on which to make the document available in Search Tools. You can click the calendar icon to open the Date Helper window so you can select a date.

  NOTE
  A document can only become available on the Search Tools window when it is published and reaches its start date. For example, if you publish a document on March 1 with a start date of March 8, the document does not become available on the Search Tools window until March 8. If you publish a document with no start date specified, it immediately becomes available on the Search Tools window. The date format in Knowledge Management must match the date format that you have set up on your Windows operating system.
- **Expiration Date**Defines the date (in mm/dd/yyyy format) on which the product removes the document from the knowledge base and the Search Tools window. After the specified date, the document is available from the Administration tab to users with the appropriate permissions. You can click the calendar icon to open the Date Helper window so you can select a date.

NOTE
The date format in Knowledge Management must match the date format that you have set up on your Windows operating system.

- **Review Date**Defines the date on which the document must be next reviewed. Click the calendar icon to open the Date Helper window so you can select a date.

  NOTE
  The date format in Knowledge Management should match the date format that you have set up on your Windows operating system.
- **Disregard Life-Cycle Policies**
  Ignores the lifecycle policies in your organization.
- **Approval Process Template**
  (Required) Specifies an approval process template to associate with the document. The approval process template specifies the tasks in the document lifecycle. You can choose the default template provided with the product or a user-defined template. If left empty, it is set to the default template used in your knowledge environment.
  Note: For knowledge managers, knowledge engineers, and system administrators: The access types that can select an approval process template during the first task (typically Create Knowledge Document) are selected on the Knowledge tab Access Type window. In subsequent stages of the approval process, the Approval Process Template box is read-only and displays the name of the template selected for the document.
- **Approval Process Priority**
  Specifies the approval process priority to associate with the document. Valid values include <empty>, Low, Normal, High, and Emergency.
  Default: Normal.
- **User Defined ID**
  Defines an ID for the document to use as an alternative to the product-defined ID.
- **FAQ Rating**
  (Read-only) Displays the current document FAQ rating, which the product calculates from the frequency at which a document is accessed, how helpful it has been to users, and its age.
- **Hits**
  (Read-only) Displays the number of times the document has been accessed.
- **Self-Service Count**
  (Read-only) Specifies the number of times that the knowledge document was used to solve a problem. It is a count of the number of times the customer responded with Yes to the question “Did this document solve your problem? “ in the document Solution Survey.
- **Average Rating**
  (Read-only) Specifies an average rating value. Each rating response (very helpful, somewhat helpful, and not helpful at all) has an assigned numeric value. The average rating is the total of the document response ratings divided by the total number of votes.
- **User Votes**
  (Read-only) Specifies the total number of votes for the document. Submitting a Solution Survey is considered a vote.
- **Priority**
  Specifies the priority to associate with the document, indicating the urgency of problems the document is meant to resolve.
- **Severity**
  Specifies the level of effect that the problem the document is meant to resolve may have on users.
- **Impact**
  Specifies the level of impact that you expect the document to have on work being performed.
- **Urgency**
  Specifies the importance level of the user tasks that are associated with the document.
- **Product**

Defines a product to associate with the document. Enter the name of the product or click the magnifier to open the Product Search dialog so you can locate and select a product.

- **Configuration Item**
  Defines the name of a configuration item (hardware, software, or service) to associate with the document. Enter the name of the configuration item or click the magnifier to open the Configuration Item Search dialog so you can locate and select an asset.
- **Root Cause**
  Defines a root cause or core reason to associate with the document. Enter a root cause or click to open the Root Cause Selection window so you can locate and select a root cause.
- **Solution Count**
  (Read-only) Displays the number of tickets that are resolved by the document.
- **Parent Request**
  Defines an associated parent request record.
- **Parent Issue**
  Defines an associated parent issue record.
- **Problem**
  Defines a problem that is linked to this document.

Grant Permissions to Document Assignees

You can determine which groups or roles can access a knowledge document and which groups or roles can modify the knowledge document by modifying permissions for Knowledge Documents and Knowledge Categories.

> **WARNING**
>
> When creating a knowledge document, make sure that document permissions include users that can be assigned later on the document through the approval process. When a group is assigned to a document, users from that group cannot have the permission to view the document. If the document is assigned to a specific user, default data partition constraints allow the user to view the document. If you do not specify group permissions, the document is only viewable by privileged users.

You can view and grant permissions as follows:

- **From the Knowledge Category page**
  a. Right-click a category and select Edit Category from the shortcut menu.
  b. Click the Permissions tab.
- **From a Knowledge Document**
  If it is a published Knowledge Document:
  a. Click Rework.
  b. Create a Rework version, click Save.
  c. Select the Permissions tab and make the appropriate changes.

  > **NOTE**
  > By default, a Knowledge Document inherits its permissions from Category Permissions. You can use empty write permissions for documents and categories. Empty write permissions are set when you do not select write permissions. Only privileged users can modify categories and documents with empty write permissions.

Use the HTML Editor

Use the HTML Editor for the following tasks:

- To define the layout and static content of a knowledge document template.
- To define the layout and content of the Resolution section of a knowledge document.
- To define the layout and content of a node in a knowledge tree document.

The HTML Editor opens when you do any of the following actions:

- Click Edit Detail on either of the following windows:
  - Create New Document Template
  - Update Document Template
- Click Edit Resolution on either of the following pages:
  - Create New Document
  - Update Knowledge Document
- Click Edit (HTML) in any of the following DT Builder panes:
  - Conclusion Text (HTML)
  - Query Text (HTML)
  - Tree Description (HTML)

Click the Insert menu to perform the following tasks:

- Insert a Horizontal Rule
- Insert a Link to a Document
- Insert a New Line
- Insert an Image
- Insert a Link to a New Ticket
- Insert a Link to Create a Change Order
- Insert Action Content

Insert a Link to a New Ticket

In the HTML Editor you can add a link to a form that creates a new issue or request.

To insert a link to a new ticket form, **follow these steps:**

1. Select Insert, New Ticket Link.
   The Create a Ticket Link dialog appears.
2. In Link Text, specify the text to appear as a hyperlink to the new ticket form.
3. Complete the fields to specify how to create the issue or request for the new ticket.
4. Click Save.
   The Insert Document closes and the HTML Editor shows a hyperlink to the new ticket form.

Insert a Link to a URL

In the HTML Editor, you can add a link to a URL from the Insert/Modify Link dialog to create a ticket.

To insert a link to URL, **follow these steps:**

1. In the HTML Editor select the text for the link.
2. Select Insert, URL.
   The Insert/Modify Link dialog appears.
3. Specify the URL for the link.
4. In Title (tooltip), specify the text to display when the mouse hovers over the hyperlink.
5. From the Target drop-down list, select how to display the URL. You have the following options:
   New window (_blank) Opens the link in a new window. Top Frame (_top) Opens the link in the top frame. Other Specifies another location for the link to open. Example: "_parent"
6. Click OK.
   The Insert/Modify Link dialog closes and the HTML Editor shows the hyperlink.

Insert a Link to Create a Change Order

In the HTML Editor, you can add a link to a form from the Create a Change Order Link dialog to create a new ticket by selecting Insert, New Ticket Link.

To insert a link to create a change order, **follow these steps:**

1. Complete the following fields:
   Link Text
   Defines the text of the change order link.
2. When you click the link at runtime, the Create New Change Order page appears.
3. When the user clicks on this link, create:
   – Specifies whether to create a blank change order or use a specific template to create it. Select one of the following options:
   – An empty change order. Specifies to create a blank change order.
   – Change order using the Template. Specifies to create a change order based on values set in the specified template.
4. Click Template to open the Change Order Template Search window so you can locate and select the appropriate template.
5. Click Save.
6. The Create a Change Order Link dialog closes and the HTML Editor shows a hyperlink to the new Create a Change Order form.

Insert an Image

You can insert an image into your document or template in the HTML Editor.

To insert an image, **follow these steps:**

1. Select Insert, Image.
   The Insert Image dialog appears.
2. Specify the location of the image by doing *one* of the following:
   – Click Browse to locate the file on your computer.
   – In Image URL specify the Uniform Resource Locator of the image.
   Example: \\c:\images\trees.png
   You can use .bmp, .gif, .jpg, or .png images.
3. Complete the fields to customize your image settings.
4. Click OK.
   The image appears in the HTML editor.

Email Documents

You can email a knowledge document from the Page Options pane.

To email a document

1. Open a knowledge document from the Document List page.
   The detail page opens.
2. Select Email from Page Options.
   The Email Document page appears.
3. Complete the following fields:
   Your Name
   Your Email
   Send To—Click this link to open a user search.
   Subject
   Personal Message
   Send Document—Select In HTML or Link Only.
4. Click OK.
   The email is sent.

Add the Search Source

You can add a source from where you want to search for information.

Follow these steps:

1. Click Create New to add a Federated Search Source.
   The Create New Search Source page opens.
2. Enter the following field values:
   Search Source Name
   Specifies a unique name for the search source.
   Code
   Specifies a unique identifier for the search source.
   Record Status
   Specifies the status of the search source. If the search source is inactive, you cannot find information from this source and this source option will not display in the Knowledge Search Source page.
3. (If multi-tenancy is enabled) Click Update Tenants and add the tenants.
4. Click Save.
   You added the search source.

Working with Knowledge Documents

This article contains the following topics:

Work with Knowledge Categories

You can view the Knowledge Categories and folders that contain knowledge documents and knowledge tree documents from the Knowledge Tree. Each document resides in a primary category and may also be linked to other related categories.

To view the Knowledge Categories

1. Click the Knowledge Tab.
   The Knowledge Tree content appears.
2. Click Top.
   The Knowledge Categories tree displays the folders that contain documents.

You can perform the following tasks in Knowledge Categories:

- Find categories and documents with the Advanced Search/Filter
- Create a knowledge document within a category
- Create a knowledge tree document within a category
- Create a knowledge file within a category
- Add or modify a category, and delete one or more categories (available in category view)
- Cut, copy, or paste a category (available in category view)
- Refresh the Document List
- Edit document properties in the document list
- Use preferences to select document attributes that appear in the document list

You can perform these tasks using the commands on the File and View menus, or the shortcut menu that displays when you right-click a category.

View Unassigned Documents

Documents that are not yet assigned to users are stored in the Unassigned queue on the scoreboard.

NOTE
If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

Follow these steps:

1. On the Service Desk tab, browse to Knowledge Documents, Unassigned.
   The Document List page displayed the documents that are unassigned.
2. (Optional) Right-click the title to edit the document.

View Inbox or Group Inbox

When a document is created or updated, it is placed in the Inbox of the owner. Items that appear in an Inbox require the attention of the user as part of the publishing process. Until they are published, items in the Inbox will not appear as resolutions and are not added to the knowledge base. You must regularly monitor your Inbox to check for new documents.

You can view summary information for the documents that are contained in your Inbox or Group Inbox on the Document List page.

NOTE
If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

Follow these steps:

1. On the Service Desk tab, browse to Knowledge Documents, Inbox, or Group Inbox.
   The Document List page displays the documents in the inbox.
2. (Optional) Right-click the title to edit a document.
3. All users with full permission to the document can edit the document. The current owner has full permissions to the document but may not have explicit write permissions. Only the owner can change the owner of the document (on the Attributes tab).

View Follow-Up Comments

You can view summary information for your follow-up comments on the Comment List page.

NOTE
If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

Follow these steps:

1. On the Service Desk tab, browse to Knowledge Documents, Follow-Up Comments.
2. Do *one* of the following actions:
   - Expand the My Follow-Up Comments folder to reveal nested folders for your assigned Open and Closed items.
   - Expand the All Follow-Up Comments folder to reveal the nested folders.
3. Select the folder for the comments you want to see.
   The Comment List page appears. This page contains the following columns:
   - **Comment Type**
     Identifies the type of comment that is used to flag a document for correction, promotion, or retirement.
   - **Date Logged**
     Identifies the date on which the comment was logged in the system.
   - **Status**

Indicates whether the comment is active or inactive.
- **Target Date**
  Identifies the date by which the comment must be followed-up on.
- **Contacts**
  The contact who received the follow-up comment.

4. (Optional) Click Show Filter and complete one or more of the fields to specify search criteria that restrict the list to the comments of interest.
5. Click Search.

The Comment List page displays summaries of the comments that match your search criteria.

1. (Optional) Click the Edit in List button to display some additional fields that can be associated with a comment.
   - **Assignee**
     The name of the person that is assigned to handle the comment. Enter the name of the person directly, or click the search icon to search for the name.
   - **Target Date**
     Identifies the date by which the comment must be followed-up on. Enter the target date directly, or click the calendar icon to specify a target date.
   - **Status**
     Indicates whether the comment is active or inactive.
   - **Reply**
     Allows you to enter more information.
2. (Optional) Click the More link to display the Additional Search Arguments field.

Publish Knowledge Documents

When you are ready to add a document to the knowledge base, you can publish it from the Update Knowledge Document page. Analysts can publish documents when they are at the last stage of the approval process. Administrators can publish documents at any stage of the approval process.

Follow these steps:

1. Open the document for editing.
   The Update Document page appears.
2. Click Publish.
   The Publish Document page appears.
3. Complete the following fields as appropriate:
   - **Owner**

   (Required) Defines the name of the contact that is assigned to maintain the document. Enter the name of the contact in "last name, first name" format or click the magnifier to open the Contact Search dialog so you can search for a contact.
   - **Comment**

   (Optional) Enter a brief comment about the document.
4. Click OK.
   The published document is added to the knowledge base.

Create Rework Versions

If you have full (read/write) permissions to edit documents, you can create a *Rework-Draft* version of a published knowledge document while the document is online and available to users.

A rework version starts as a copy of the document that is replaced in the knowledge base after it is verified and republished.

Follow these steps:

1. From the Document List page, right-click the title of the document to edit, then select Edit from the shortcut menu.
   The Update Knowledge Document page appears.
2. (Optional) Complete the various tabs as appropriate.
3. Click Rework.
   A page opens displaying the list of approved document tasks from which you can select tasks to start with (if you are permitted to do so). By default, a built-in approval process template allows you to create a document.
4. Select the appropriate task from the list.
   The Create Rework Version page appears.
5. Complete the appropriate fields:
   - **Document Title**
     Displays the published document name.
   - **Document Status**
     Displays the status of the record (published).
   - **Keep Published Version Available**
     Specify whether or not to keep the original published version available to all users.
     Default: Available
   - **KD Approval Task**
     Displays the workflow tasks that you are able to perform if you have permission to do so.

   Note: You can cancel the rework version and can add comments about why you want to cancel the rework version.
6. (Optional) Select the Assignee link to assign this rework version to another person or group for review or follow-up.
   Once saved, the rework version appears in their Inbox.

   NOTE
   If *Keep Published Version Available* is checked in the *Create Rework Version* form, it does not copy the assignee value to the rework document. If *Keep Published Version Available* is unchecked, it copies the assignee value to the rework document.
7. (Optional) Enter information about this document in the Comment field.
8. Click Save.
   The Update Document page appears.
9. Click Save.
   The *rework-draft* version appears in your Knowledge Documents Inbox on scoreboard. To display this page, select the Service Desk tab, then Knowledge Documents, Inbox.

Save Versions

You can create a draft version of a document for backup. The Save Version option is available for draft documents and rework versions that have not yet been published.

Follow these steps:

1. On the Document List page, open a draft document to edit, and select Edit from the shortcut menu.
   The Update Document page appears.
2. Complete or change the fields on the various tabs as appropriate.
3. Click Save Version.
   The Create New Version page appears.
4. (Required) Add a comment in the Comment field.
5. Click Save.
   The draft version displays on the Versions tab.

Rollback to a Previous Version

If there is a problem with the current version, you can roll back to a previous version of a published document.

Before you begin, consider the following factors:

- When performing a rollback of a published document, the earlier version appears in the document list as a new *rework-draft* document, which must be re-published in order to replace the currently published document.
- When performing a rollback of a draft document, the content of the current document is replaced with the content of the earlier version. The document remains in draft status until it is published.

Follow these steps:

1. From the Documents List page, right-click the published document to edit, then select Edit from the short-cut menu.
   The Update Knowledge Document page appears.
2. Click the Versions tab.
   The Versions List appears.
3. In the Version Number column, click the version number link.
   The Save Version page appears.
4. Click Rollback.
   The document rolls back to the previous version. If the document is published, a rework version of the document is started.

Retire a Knowledge Document

A knowledge document can be retired after it reaches an expiration date, or if the product removes it from the knowledge base. A privileged user can retire or rework a knowledge document manually.

Follow these steps:

1. Open the document for editing.
2. Click Retire.
   The Retire Knowledge Document page appears.
3. Enter a comment about why you want to retire the document.
   For example, the knowledge document described a workaround task that no longer applies to your organization.
4. Click Save.
   The document is retired.

Unretire Knowledge Documents

When a published document reaches its expiration date, the product typically retires it. A retired document is removed from the knowledge base and the approval process). A system administrator or a user with full permissions can unretire or republish the document to return it to the approval process and the knowledge base.

Follow these steps:

1. Open the document for editing.
2. Click Unretire.
   The Unretire Document page appears.
3. Do *one* of the following, as appropriate:
   - If you are a system administrator, you can republish the document. To do so, click Publish this Document. A confirmation message appears.
   - If you are a system administrator and there are multiple approval process tasks to which the document might revert, you can choose the task to which to assign the document, choose an assignee, and (optionally) enter a comment:

- Click the name of a previous task to which to assign the unretired document. The Unretire Document page refreshes.
- Select the name of a contact to assign as the owner for the document from the Assignee list.
- (Optional) Enter a brief comment in the Comment box.
- Click OK. A confirmation message displays.

– If you are a knowledge administrator or analyst, you can choose an assignee for the task and (optionally) enter a comment:
- Select the name of a contact to assign as the owner for the document from the Assignee list.
- (Optional) Enter a brief comment in the Comment box.
- Click OK. A confirmation message displays.

4. Click OK.
The product assigns the document to the specified task and owner and returns the document to the knowledge base and the workflow.

Document Search Fields

NOTE
If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

- **Keywords for Advanced Search**
Specifies the keywords (separated with comma or space) to use for the search in the Search field. The search result displays the keyword with the document frequency. This means the search result displays all the occurrences of the document names with that keyword. For example, in the following screenshot, we searched for a knowledge document associated with a CI and used the keyword as "updated". The search result shows updated(29), but does not display any document associated with this CI. This search result is correct. It means that 29 documents in the knowledge base contains the word "updated", but none are associated with the CI.

- **Search In**
Specifies in which fields to search for specified keywords. When you click Search, the product returns only items that contain the specified keywords in the fields that are specified by the selected *Search In* check boxes.
The check boxes only display when Keyword Search is the selected search type. If you select Natural Language Search from the Search Type list, the Search In check boxes do not display and the product only searches the Problem fields of documents.
- **Reset to my defaults**
Resets all filters to their default values. This option is only available for knowledge documents.
- **Set as my defaults**
Saves all filters and applies them as the default values. This option is only available for knowledge documents.
- **Search Type**
Specifies the type of search to perform. Select one of the following options:
  – **Keyword Search**

Searches for the specified keywords in the Title, Summary, Problem, and Resolution fields of documents. Use the Search In check boxes to exclude one or more of these fields.

- **Natural Language Search**Searches for the specified text only in the Problem fields. Natural Language Search (NLS) lets you perform a search using natural language, which compares the pattern of words that are contained in the query with the patterns of words that are contained in the Problem field of knowledge base documents.
  Default: Keyword Search

  NOTE
  Depending on how your administrator configured search capabilities, you may not have access to Keyword Search and Natural Language Search.

- **Match Type**
  Specifies the method for text matching during the search. Select one of the following options:
  - **Any of the Words (OR)**Retrieves the matches for *any* of the specified words in a document.
  - **All of the Words (AND)**Retrieves the matches only when the product finds *all* the specified words in a document.
  - **Exact Match**Retrieves the matches for *exactly* the specified words in a document.
    Default: Any of the Words (OR)

    WARNING
    Match Type and Match preferences only set the default search criteria when you search in Knowledge Management. For example, you log in as an analyst and click the Knowledge tab of any ticket. Knowledge searches from within a ticket always default to Match Type=Any of the words (OR) and Match=Whole Words, regardless of your preference settings.

- **Match**
  Specifies the method by which the product searches documents. Select one of the following options:
  - **Whole words**Retrieves only documents that contain the entire words entered.
  - **Words beginning with...**
    Retrieves the documents that contain the entire words entered or words that begin with the words entered. For example, a search for the word "print" also returns documents containing the words "printer" or "printing."
    The Match list setting overrides the default set on the Preferences window and is only available when you specify keywords or phrases for which to search.
    Default: Whole words

- **Order by**
  Specifies a criterion by which to sort search results. Select one of the following options:
  - **Relevance**Sorts documents by their relevance to the specified search criteria (expressed as EXCELLENT, GOOD, and so on). Documents with the highest relevance (EXCELLENT) are listed first.
  - **FAQ Rating**
    Sorts documents that are based on their FAQ rating, which considers how often the document is accessed, how helpful it has been to users, and its age. Documents with the highest FAQ ratings are listed first.
  - **Hits**Sorts documents by the number of times users have accessed them, with the most frequently accessed documents listed first. Hits may not be available for all record types. This field only displays for advanced searches on the Knowledge tab.
  - **Solution Count**
    Sorts documents by the number of issues or requests they have resolved, as reported by users. Documents with the highest solution counts are listed first. This field only displays for advanced searches on the Knowledge tab.
  - **Modify Date**Sorts the documents by the date on which they were last modified, with the most recently modified documents listed first.

- **Return only documents that solved tickets**
  Returns the documents that have successfully resolved tickets.

- **User Defined ID/Created By/Submitter**
  Define a user-defined ID or enter a name by which to filter documents retrieved. This field only displays for advanced searches of knowledge documents and free text content.

- **Product**

Defines a product by which to filter documents retrieved. Enter the name of the product in the box, or click to open the Product Search window so you can search for and can select one. When you click Search, the product returns only documents that are associated with the specified product.

- **Configuration Item**
  Defines the name of an asset (hardware, software, or service) by which to filter documents retrieved. Enter the name of the asset in the box, or click to open the Configuration Item Search window so you can search for and can select one. When you click Search, the product returns only documents that are associated with the specified asset.
- **Category**
  Define a category by which to filter documents retrieved. Enter the name of the category in the box or click to open the Category Search window. When you click Search, the product returns only documents that are associated with the specified category. Use the Remove and Clear Categories options as needed.

(Optional) Click the More link to display the following additional fields:

- **Owner**
  Defines the name (in "last name, first name" format) of the contact that is assigned to maintain the document. Enter the name of the contact in "last name, first name" format or click to open the Contact Search dialog so you can locate and select a contact.
- **Author**
  Defines the name (in "last name, first name" format) of the contact with responsibility for the document content. Enter the name of the contact in "last name, first name" format or click to open the Contact Search dialog so you can locate and select a contact.
- **Subject Matter Expert**
  Defines the name (in "last name, first name" format) of a contact with subject-matter expertise for the document. Enter the name of the contact in "last name, first name" format or click to open the Contact Search dialog so you can locate and select a contact.
- **Document Type**
  Specifies the type of document (knowledge document or knowledge tree document) to retrieve.
- **Approval Process Priority**
  Specifies the level (emergency, high, low, normal) by which to filter approvals retrieved.
- **Root Cause**
  Defines a root cause (that is, the core reason for opening the ticket) by which to filter documents retrieved. Possible root causes include Hardware Failure, Software Failure, and Network Cable. Enter the root cause in the box, or click the icon to open the Root Cause Selection window so you can search for and can select one. When you click Search, the product returns only documents that are associated with the specified root cause.

Priority/Severity/Impact/Urgency
Specifies the level (1, 2, 3, 4, 5, or None) by which to filter documents retrieved. When you leave <empty> as the field value, the product does not consider a level when filtering documents. When you click Search, the product returns only documents with the specified level.

NOTE
You can click the More icon to display the Additional Search Arguments field. This field is intended only for expert users who understand SQL and Majic. You can enter a SQL WHERE clause in this field to specify an additional search argument.

Knowledge Advanced Search Fields

NOTE

Depending on how your administrator configured the system for your role, you may not have access to all of the functionality described in this topic.

Knowledge Base Specifies the type of knowledge to retrieve. When you click Search, the search engine returns only documents of the specified type.

Select one of the following options:
Knowledge Documents

Returns only knowledge documents.
Forums

Returns only user forum information.
Knowledge Documents and Forums

Returns all knowledge sources.
External Repositories

Returns unstructured knowledge from an external repository related to a problem or resolution.

Knowledge Search

Defines a space-delimited list of words or a phrase by which to filter documents retrieved. When you click Search, the product returns only documents that contain the specified keywords. Wildcard search text should contain at least two letters before a percent sign, for example, 'xy%'; leading percent as in '%xy' is prohibited. Words beginning with search text should contain more than one letter. One digit is allowed.

NOTE

The match count for the wildcard based keyword search is always set to 0.

Search Only In category

Searches for documents only in the selected category and its subcategories. If you do not want to limit your search to the selected category and its subcategories, select the Search All Categories option instead. This field only displays on the Knowledge tab when you select a category other than TOP from the Category pane.
Search All Categories

Searches for documents in all categories and subcategories. If you want to limit your search to the category selected in the Category pane and its subcategories, select the Search Only In category option instead. This field only displays on the Knowledge tab when you select a category other than TOP from the Category pane.
Search In

Specifies the fields in which to search for specified keywords. When you click Search, the product returns only items that contain the specified keywords in the fields specified by the selected Search In check boxes.

The check boxes only display when Keyword Search is the selected search type. If you select Natural Language Search from the Search Type list, the Search In check boxes do not display and the product only searches the Problem fields of documents.
Reset to my defaults

Returns all filters to their default values. This option is only available for knowledge documents.
Set as my defaults

Saves all filters and applies them as the default values. This option is only available for knowledge documents.
Search Type

Specifies the type of search to perform. Select one of the following options:
Keyword Search

Searches for the specified keywords in the Title, Summary, Problem, and Resolution fields of documents. Use the Search In check boxes to exclude one or more of these fields.
Natural Language Search

Searches for the specified text only in the Problem fields. Natural Language Search (NLS) allows you to perform a search using natural language, which compares the pattern of words contained in the query with the patterns of words contained in the Problem field of knowledge base documents.

Default: Keyword Search

Match Type

Specifies the method for text matching during the search. Select one of the following options:
Any of the Words (OR)

A match occurs when the product finds any of the specified words in a document.
All of the Words (AND)

A match occurs only when the product finds all of the specified words in a document.
Exact Match

A match occurs only when the product finds exactly the specified words in a document.

Default: Any of the Words (OR)

> **NOTE**
>
> Match Type and Match preferences only set the default search criteria when you search in Knowledge Management. For example, you log in as an analyst and click the Knowledge tab of any ticket. Knowledge searches from within a ticket always default to Match Type=Any of the words (OR) and Match=Whole Words, regardless of your preference settings.

Match

Specifies the method by which the product searches documents. Select one of the following options:
Whole words

Retrieves only documents that contain the entire words entered.
Words beginning with...

Retrieves documents that contain the entire words entered or words that begin with the words entered. For example, a search for the word "print" also returns documents containing the words "printer" or "printing".

The Match list setting overrides the default set on the Preferences window and is only available when you specify keywords or phrases for which to search.

Default: Whole words

Order By

Specifies a criterion by which to sort search results. Select one of the following options:
Relevance

Sorts documents by their relevance to the specified search criteria (expressed as EXCELLENT, GOOD, and so on). Documents with the highest relevance (EXCELLENT) are listed first.
FAQ Rating

Sorts documents based on their FAQ rating, which considers how often the document is accessed, how helpful it has been to users, and its age. Documents with the highest FAQ ratings are listed first.

Hits

Sorts documents by the number of times users have accessed them, with the most frequently accessed documents listed first. Hits may not be available for all record types. This field only displays for advanced searches on the Knowledge tab.

Solution Count

Sorts documents by the number of issues or requests they have resolved, as reported by users. Documents with the highest solution counts are listed first. This field only displays for advanced searches on the Knowledge tab.

Modify Date

Sorts documents by the date on which they were last modified, with the most recently modified documents listed first.

Search for documents in these statuses

The following check boxes define the stage by which users can search for documents during their lifecycle. For example, when the Draft check box is selected, draft documents that are not yet published are displayed in the search results.

- Published
- Draft
- Retired
- Not Started
  Default: Published documents appear in the search results.

  NOTE

  Custom status fields may appear in this list (depending on how your administrator configured the system).

Return only documents that solved tickets

Returns documents that have successfully resolved tickets.

User Defined ID/Created By/Submitter

Define a user-defined ID or enter a name by which to filter documents retrieved. This field only displays for advanced searches of knowledge documents and free text content.

Product

Defines a product by which to filter documents retrieved. Enter the name of the product in the field, or click the magnifier to open the Product Search window so you can search for and select one. When you click Search, the product returns only documents associated with the specified product.

Configuration Item

Defines the name of an asset (hardware, software, or service) by which to filter documents retrieved. Enter the name of the asset in the field, or click the magnifier to open the Configuration Item Search window so you can search for and select one. When you click Search, the product returns only documents associated with the specified asset.

Category

Defines a category by which to filter documents retrieved. Enter the name of the category in the field or click the magnifier to open the Category Search window. When you click Search, the product returns only documents associated with the specified category.

(Optional) Click the Spigot icon to display the Additional Search Arguments field. More link to display the following additional fields:

Owner

Defines the name (in "last name, first name" format) of the contact assigned to maintain the document. Enter the name of the contact in "last name, first name" format or click the magnifier to open the Contact Search dialog so you can locate and select a contact.

Author

Defines the name (in "last name, first name" format) of the contact with responsibility for the document content. Enter the name of the contact in "last name, first name" format or click the magnifier to open the Contact Search dialog so you can locate and select a contact.

Subject Matter Expert

Defines the name (in "last name, first name" format) of a contact with expertise in the subject matter of the document. Enter the name of the contact in "last name, first name" format or click the magnifier to open the Contact Search dialog so you can locate and select a contact.

Document Type

Specifies the type of document (document or tree document) to retrieve.

Approval Process Priority

Specifies the level (emergency, high, low, normal) by which to filter approvals retrieved.

Root Cause

Defines a root cause (that is, the core reason for opening the ticket) by which to filter documents retrieved. Possible root causes might include Hardware Failure, Software Failure, and Network Cable. Enter the root cause in the field, or click the magnifier to open the Root Cause Selection window so you can search for and select one. When you click Search, the product returns only documents associated with the specified root cause.

Priority/Severity/Impact/Urgency

Specifies the level (1, 2, 3, 4, 5, or None) by which to filter documents retrieved. When you leave <empty> as the field value, the product does not consider a level when filtering documents. When you click Search, the product returns only documents with the specified level.

Additional Search Arguments

Defines additional criteria by which to search. This field only displays when you click the More link in the Knowledge Search page.

Manage Document Versions

Using the document versioning capabilities of Knowledge Management, an analyst with editing privileges can create a *Rework-Draft* version of a document. A rework version starts as a copy of the document that is replaced in the knowledge base after it is verified and republished. The need to unpublish the document first is avoided.

Users with editing privileges can also perform the following versioning tasks:

- Save a draft version of a document.
- Roll back to a previous version if a problem with the current version occurs (Draft or Published). The Versions tab on the Update Document page is where users can select different versions for rollback to replace the current version.
- Track the number of document versions that are saved, deleted, and archived in the knowledge base.

The Knowledge Documents Inbox on the CA SDM Scoreboard is the repository for documents in all statuses including saved and assigned draft and rework-draft documents.

Administrators can set up and can manage document versions by performing the following steps:

1. Identify who can edit published documents and can create the Rework-Draft versions. The role being used for a particular contact record controls editing privileges.
2. Define an approval process template that groups tasks or steps to complete during the document lifecycle. By default, a built-in approval process template allows users to create documents.
3. Determine whether to use the document approval process. Analysts who are permitted to bypass the approval process can identify which tasks they want to start with when they create a Rework-Draft version.
4. Create archive and purge rules for document version maintenance.

Create Knowledge Document Links

You can maintain your Knowledge Management environment by creating document links. Knowledge relationships let you create document hierarchies and manage changes to your knowledge documents.

Follow these steps:

1. Define knowledge document templates to display or hide the parent-child relationships in view mode.
2. Create or modify a knowledge document. You can only add links to an unpublished knowledge document by default. If the document is already published, open the document in edit or rework mode to create document links. You can modify the permissions for documents before the approval process and after publishing.
3. Create *any* of the following document links from the Related Knowledge tab, as appropriate to your Knowledge Management environment:
   - Link the document as a non-hierarchal See Also to link knowledge documents to other existing documents. Right-click the document in the Documents pane and select the option *Link this Document as See Also*.
   - Link the document as a parent or child. Right-click the document in the Documents pane and select the *Link this as Parent* or *Link this as Child* option to create the link between the documents. If you modify a parent document, an alert appears saying that child documents can be affected.
   - Link the global document to a tenanted document.
     You can link knowledge documents to a single tenant, or to multiple tenants. For example, a child document can have a different tenant than the parent document.

   The History tab updates when you create document links to help track changes.
4. Save the document and verify that the links appear when opening the document in User View.
   You can verify that the document links appear according to the permissions you established.

View Related Knowledge and Document Links

You can manage category and document links within a document. Related knowledge and document links let users find a document in multiple places, such as in a parent or child document.

You can use the default templates (Built-In Knowledge Document and Built-In Knowledge Tree) to create document relationships, such as See Also links, parent-child links, and tenant.

> **NOTE**
>
> You can only add links to an unpublished knowledge document. If the document is already published, open the document in edit or rework mode to create document links.

To view Related Knowledge and Document Links

1. Open a Knowledge Document from the Knowledge Categories page.
   The Knowledge Document opens.
2. Click the Related Knowledge tab.
   The Categories page appears.
   The following information is listed:

Primary Category

Displays the location of the knowledge document when navigating categories.This field is read-only.

Example: TOP>Email

Categories

Lists the category and subcategory hierarchy currently defined in the knowledge base. When you select a category from the list, its associated documents display in the Documents list.

To link a category to a document, right-click the row and select Link this Category from the shortcut menu.

Documents

Displays the ID and title of each document in the category selected from the Categories list. You can right-click a row to link documents from the shortcut menu.

NOTE

You can create and link documents in different languages.

Category Links

Displays the category name and path for each category link defined for the document. You can right-click a row to delete category links or set the primary category from the shortcut menu.

Document Links

Displays the IDs and titles of documents to which the current document is linked. You can right-click the row, select the Remove this Link option to remove the link, and you can view the document that is linked by clicking on the document ID link.

Create Action Content

This article contains the following topics:

You can create "action content" (a live URL), which can be inserted into the Resolution field of a knowledge document. When the end user click the action content, it creates an incident, or performs some other action. Using action content, a substantial degree of definition and classification can be achieved without the user even realizing it.

The steps to insert a live "action content" link into a document are simple and no coding is required. The Knowledge Management HTML editor handles the generation of the HTML code.

Action content is primarily used for interaction with external applications.

Create Action Content (Action URL)

You can create Action Content for a knowledge documents. These action URLs can launch a live website that is accessible to all users of your system. You can also link action URLs to automated tasks on your server (any server where the KT daemon is running), and you can embed these scripts into knowledge documents, tree documents, document templates, and knowledge forums.

Follow these steps:

1. On the **Administration** tab, browse to **Knowledge**, **Action Content**.
   The Action Content list page appears.
2. Click **Create New**.
   The Create New Action Content page displays.
3. Complete the fields. The following fields require explanation:
   - **Code**
     Specifies a unique identifier for this action content item.
   - **Use Internal HTMPL**
     Creates an internal link in the application that dynamically passes information, such as the user name and session ID, from knowledge documents into third-party applications. Do not select this option.
   - **Action URL**
     Specifies a URL that links to a web page, template, or automated script, for example: http://www.ca.com.
4. Click **Save**.
   The action content is created.

Create Action Content (Internal HTMPL)

These action URLs can launch self-service scripts and also third-party applications. This link is generated dynamically and automatically transfers attributes about the user, such as the user name, to the target application (http://www.ca.com?USERNAME=BBB, for example). User attributes are specified in an HTMPL file.

Follow these steps:

1. Create an internal HTMPL file that passes data to the target application.

   NOTE
   The act_content_sample.htmpl file is available in the following location: NX_ROOT\bopcfg\www\htmpl\default.
2. Save the HTMPL file in the following location: NX_ROOT\site\mods\www\htmpl\default directory.
3. From the **Administration** tab, navigate to **Knowledge**, **Action Content**.
   The Action Content list page appears.
4. Click **Create New**.
   The Create New Action Content page appears.
5. Specify a unique identifier for this action content item in the **Code** field.
6. Select the **Use Internal HTMPL** check box.
7. Specify the appropriate HTMPL file in the Action URL field, for example: act_content_sample.htmpl

   NOTE

   You can use Support Automation scripts by using the format *SA_SCRIPT=[Self-Service Script ID]* in the URL.
8. Click **Save**.
   The action content is created. When the user clicks this Action Content link within a knowledge document, attributes about the user, such as the user name, are dynamically passed on to the target application.

Create Knowledge Tree Documents

This article contains the following topic:

The Knowledge Tree Designer is a visual tool that is used to create knowledge trees quickly and easily. A knowledge tree is a representation of expert knowledge in a particular area. The Knowledge Tree Designer lets analysts and knowledge engineers build detailed trees that guide the user through a series of questions and potential answers until they reach a resolution. This tool eliminates the need for specialized scripting or programming skills and only involves an analyst working with a subject expert to create a tree design.

Knowledge trees can be complex in design. Therefore, it is recommended that you construct a diagram to map the design before creating a knowledge tree. The diagram must contain a series of question, possible responses, and associated resolutions. The hierarchy of questions and responses in the diagram you construct is the foundation on which the knowledge tree is built.

After you have mapped out a diagram of your knowledge tree, you build the tree using the Knowledge Tree Designer.

You can create a Knowledge Tree Document from the Knowledge tab.

NOTE
If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. Select the desired knowledge category.
2. Click File, New Knowledge Tree Document.

NOTE
You cannot create a document directly under the TOP category. If necessary, add a new category in which to create the document.

3. Complete the required fields.
4. Choose a Document Template from the drop-down.
5. Click Save.
6. Click Design Tree on the Content tab to edit the knowledge tree in the Tree Designer.

Use the Tree Designer

The Knowledge Tree Designer tool employs rule-based modeling functionality that helps you develop and deploy business policies and intelligence by mapping any reasoning process into a knowledge tree structure. The knowledge tree is perfectly suited to guiding users through a logical process such as changing a toner cartridge in a printer or changing a hard drive in a server.

You can view the Tree Designer when you open a Knowledge Tree Document for editing.

Follow these steps:

1. Create or Edit a Knowledge Tree Document.
   The document opens.

   NOTE

   If the document has been published, you may need to unpublish it before you can view it.
2. Click Design Tree on the Content tab.
   The Tree Designer opens.

   NOTE
   You cannot perform any add or edit actions on the primary node. For example, you cannot add a query or conclusion to a primary node.
3. Click Node Actions, and select one of the following nodes:
   - **How to Use the Tree Designer**
     Allows you to write a question.
   - **How to Use the Tree Designer**
     Allows you to write an answer.
   - **How to Use the Tree Designer**
     Links the currently selected jump node to the next node you select. You cannot link a jump node to the primary node or to another jump node.
   - **Add Jump**
     Allows you to link to another document.
   - **Delete**
     Deletes the currently selected node and its subnodes from the knowledge tree without putting them on the clipboard.

     NOTE
     When you delete a query node, the query and its responses are removed from the knowledge tree. You cannot delete the primary node.

   You can use the Quick View or HTML Source to modify the appearance of the nodes.
4. Click the Edit menu to perform copy and paste operations.

   NOTE
   Use the Copy option to copy of the currently selected node (and its subnodes). Use the Copy Single Node option to copy the currently selected node (but not its subnodes or responses).

5. Click the Options menu to personalize your interaction with the Tree Designer. This menu contains the following commands:
   - **Copy Response**
     Specifies whether to use response text as the name of an associated node when you create it.
     - Select this command to automatically use the text of a node as its name when you create it. When the command is active, a check mark displays to the left of the command name on the menu.
     - Clear this command to use the default name (Default Node Name) when you create a node. You can use the Rename command on the Node Actions menu to change the name of the node.

     This command is also available on the shortcut menu that opens when you right-click a node in the Tree pane.
   - **Prompt on Node Delete**
     Specifies whether the Tree Designer prompts you before deleting nodes.
     - Select this command to display a confirmation prompt before deleting nodes. When the command is active, a check mark displays to the left of the command name on the menu.
     - Clear this command if you do not want the Tree Designer to prompt you to confirm node deletion.

     This command is also available on the shortcut menu that opens when you right-click a node in the Tree pane.
6. Click the Quick View option to view a preview of the node design.
7. Click Save.
   The Knowledge Tree Document is saved as a draft.

Access Knowledge Documents from the Self-Service Interface

This article contains the following topics:

Knowledge Submission from Self-Service

By default, any user that is logged in to Knowledge Management can submit knowledge for consideration . This page lets a customer submit knowledge without contacting their local service desk representative. After you submit knowledge, it passes through a publishing process. During this process, the knowledge is reviewed and edited before it is added to the knowledge base.

Specify information into every field that appears in the Submit Knowledge page. Pay special attention when completing the Summary field. Typically, this field contains a succinct overview of the document that you are submitting.

You can submit knowledge from the Search for a Solution pane on the CA SDM home page by performing the following steps:

1. Click Submit Knowledge.
   The CA SDM page is displayed.
2. Click **Category**, select the category matching your knowledge.
   The category name appears in the Category field.
3. Complete the following:
   - **Title**: Displays the title of the document.
   - **Summary:** Displays a summary of the document.
   - **Problem**: Displays the problem of the document.
   - **Resolution**: Displays how the problem was resolved.
   - **Comment**: Displays additional comments about the document.
4. Click Save.
   The Knowledge Document is displayed.
5. Click **Home**.
   The Self-Service page is displayed.

Knowledge Search

Knowledge Management provides users with the following options for retrieving knowledge:

- **Category browsing**
  Finds the solutions that are based on categories. In each category, additional subcategories can narrow the search results to a set of solutions that are likely to be most relevant to an issue.

  NOTE
  When you search for knowledge documents using the category, the search result displays the keyword count and the list of documents. The count within the parentheses is the total number of documents in the knowledge base that contains at least one occurrence of the keyword. For example, when searching for the string, "live assistance", if 3 documents each contain that keyword at least once, then "live assistance(3)" is displayed.

  The count is not filtered according to the specified search criteria. So, it may not match the number of documents that are listed. Additionally, the count only includes documents that have been processed by pdm_k_reindex. So, documents which contain the specified string but which have been published after the most recent pmd_k_reindex are not included in the count.

  To turn off the parenthesized count, add the following code to the $NX_ROOT/NX.env file:

  ```
  @EBR_SHOW_WORDS_DF=No
  ```

- **Knowledge tree**
  Asks a series of questions to guide the user to possible solutions. The responses lead the individual to a solution that appears to be most relevant.
- **Bookmark**
  Provides access to frequently viewed documents that are included in a bookmark list.

Advanced Search

In Advanced Search Settings, customers and employees can use the following options to refine a search for the solution to a problem:

- **Knowledge Management Search**
  Searches for certain keywords, which serve as preliminary matches.
- **Natural Language Search**
  Searches for words and accounts for word proximity, word order, and relevance.

The search results are listed by relevance. The relevance is determined by the specified search criteria (expressed as EXCELLENT, GOOD, and so on). Documents with the highest relevance (EXCELLENT) are listed first.

Each result can include a title that appears as a link, a summary of the document, and additional statistics relevant to the document, such as Relevance Rating, Document ID, Modify Date, FAQ Rating, and Hits Received.

Depending on how the system administrator configures Knowledge Management, users can open an incident, rate a document, and can provide comments when a knowledge document is open.

Document Browsing and Bookmarking

You can categorize knowledge documents to let customers browse for information that is based on frequently asked questions (FAQs). By selecting a knowledge category, the related subcategories and knowledge documents appear. You can select the document that you would like to view or select a subcategory to narrow your search further.

To enhance self-service capabilities, a dynamic list of the most frequently used documents appear.

NOTE
Users can specify criteria about an item of interest and the search engine finds the matching knowledge documents and displays them on the search results page as a set of "recommended document" links. The

search query can be expressed as a keyword or set of words (phrase) that identify the desired concept that one or more documents possibly contain. For more information, see Create New Recommended Documents .

You can bookmark the document for easy access. The My Bookmarks folder stores links to the most frequently referenced documents. When you click My Bookmarks in the Category pane on the Knowledge tab, the list of bookmarked documents displays in the Knowledge Document List pane.

Incidents and Problems

Customers sometimes encounter problems that cannot be solved simply by searching for knowledge. Not all problems have a direct solution in the knowledge base. When a customer has a problem that cannot be solved, they can create an incident that is sent to an analyst for further processing. The incident describes the problem, and it can also be based on an existing knowledge document. The more information that is added to an incident, the easier it is for the analyst to solve.

Many ITIL-defined activities are supported in Knowledge Management, including the following activities:

- Incident Management
  - Knowledge searches can help find known errors during further incident investigation and diagnosis
  - Incident categorization
- Problem Management
  - Accessing information about known errors, and helping with problem matching to obtain the resolution when the problem has occurred before
  - Maintaining and providing access to information about workarounds
  - Recording information about procedures, work instructions, diagnostic scripts, and known errors (while supporting rich content, editing tools, measurement, and a definable approval process for the development of resolutions)
  - Problem analysis (through the linkage and analysis of incidents)

How to Use the Tree Designer

Contents

The following topics provide procedures for using the Tree Designer to create and edit knowledge trees.

Tree Pane

The Tree pane displays the nodes that comprise the knowledge tree. Each node represents a step in the process that is presented by the knowledge tree. A well-constructed knowledge tree guides the user along a systematic path to answering a question, resolving a problem, or completing a process. Click ⊞ or appropriate to open or close nodes in the Tree pane. When you click a node in the Tree pane, the associated information displays in the right pane.

The following information displays in the right pane of the Tree Designer window when you select a node in the Tree pane:

Icon	Node Type	Right Pane Display
	Primary	Tree Description (HTML) pane
	Query	Query Text (HTML) pane
	Conclusion	Conclusion Text (HTML) pane
	Jump	N/A

You can right-click a node in the Tree pane to open a shortcut menu that contains a subset of the menu commands.

NOTE
The right pane is empty when you select a Jump node.

Add a Conclusion Node

Use a conclusion node to describe the resolution for a particular problem, the final step in a procedure, or a possible termination point in a query. The Conclusion nodes do not have subnodes. You would not typically nest a conclusion node immediately beneath the root node of a knowledge tree. You can add up to seven subnodes to a query node.

Follow these steps:

1. Select the query node under which to add the conclusion node in the Tree pane.
   The Query Text (HTML) pane opens.
2. Click Add Conclusion.
   The Tree Designer adds a conclusion node below the selected node in the Tree pane and adds a response field in the Query Text (HTML) pane.
3. Enter a response to the question or statement that is posed in the parent node in the new field in the Query Text (HTML) pane.
4. (Optional) To rename the new node, right-click it in the Tree pane, then select Rename from the shortcut menu. The Prompt window opens. Enter a new name for the node and click OK to close the Prompt window. The Tree pane refreshes to show the new node name.
5. Select the new conclusion node in the Tree pane.
   The Conclusion Text (HTML) pane opens.
6. Do one of the following actions in the Conclusion Text (HTML) pane:
   – Enter the displayed text (that is, the information that is associated with the node) in the Conclusion Text (HTML) box. You can use plain text or can enter HTML codes as appropriate. To preview the text as it appears at run time, select the Quick View option.
   – Click Edit Conclusion Text (HTML) to open the HTML Editor so you can design the information that is associated with the node in WYSIWYG (what you see is what you get) format. When you finish editing the information, click OK to save your work and close the HTML Editor.
7. Click Save.
   The Tree Designer saves your changes.

Add a Jump Node

Use a jump node to link to previously defined nodes in the knowledge tree. For example, if the answer to the query posed in a parent node requires repetition of previous steps in a procedure. Jump nodes do not have subnodes. You would not typically nest a jump node immediately beneath the root node of a knowledge tree.

Follow these steps:

1. Select the query node under which to add the jump node in the Tree pane.
   The Query Text (HTML) pane opens.
2. Click Add Jump.
   The Tree Designer adds a jump node below the selected node in the Tree pane and adds a response field in the Query Text (HTML) pane.
3. Enter a response to the question or statement that is posed in the parent node in the new field in the Query Text (HTML) pane.
4. Right-click the new jump node in the Tree pane, then select Link Jump Node from the shortcut menu.
5. Select the query or conclusion node to which to link the jump node in the Tree pane.

The jump node's name in the Tree pane changes to include the name of the linked query or conclusion node. For example, if you linked a node that is called "Wednesday", the jump node name changes to "NODE: Wednesday." When a user selects the jump node in the knowledge tree at runtime, the linked query or conclusion node displays.

6. Click Save.
   The Tree Designer saves your changes.

Add a Query Node

Use a query node to supply a possible answer to the question posed in its parent node and to serve as a parent node for further subnodes in the tree. The query text typically contains a question (the answer to which determines the next node in the tree) or a step in a procedure. The query nodes can have conclusion nodes, jump nodes, and other query nodes as subnodes.

NOTE
You can add up to seven subnodes to a query node.

Follow these steps:

1. Select the query node under which to add the new query node in the Tree pane.
   The Query Text (HTML) pane opens.
2. Click Add Query.
   The Tree Designer adds a query node below the selected node in the Tree pane and adds a response field in the Query Text (HTML) pane.
3. Enter a response to the question or statement that is posed in the parent node in the new field in the Query Text (HTML) pane.
4. (Optional) To rename the new node, right-click it in the Tree pane, then select Rename from the shortcut menu. The Prompt window opens. Enter a new name for the node and click OK to close the Prompt window. The Tree pane refreshes to show the new node name.
5. Select the new query node in the Tree pane.
   The Query Text (HTML) pane opens.
6. Do one of the following actions in the Query Text (HTML) pane:
   - Enter the displayed text (that is, the information that is associated with the node) in the Query Text (HTML) box. You can use plain text or can enter HTML codes as appropriate. To preview the text as it appears at run time, select the Quick View option.
   - Click Edit Query Text (HTML) to open the HTML Editor so you can design the information that is associated with the node in WYSIWYG (what you see is what you get) format. When you finish editing the information, click OK to save your work and close the HTML Editor.
7. Click Save.
   The Tree Designer saves your changes.

Add the Tree Description

The tree description displays at runtime when a user clicks the Tree Description link while viewing a knowledge tree document in tree view. For example, if you design a knowledge tree to help a customer choose a credit card, you can define a tree description such as "Use this document to find the best credit card for you."

Follow these steps:

1. Select the knowledge tree's primary node in the Tree pane.
   The Tree Description (HTML) pane opens.
2. Do one of the following action in the Tree Description (HTML) pane:

- Enter the displayed text (that is, the information for the tree description) in the Tree Description (HTML) box. You can use plain text or can enter HTML codes as appropriate. To preview the text as it appears at run time, select the Quick View option.
- Click Edit Tree Description (HTML) to open the HTML Editor so you can design the tree description in WYSIWYG (what you see is what you get) format. When you finish editing the information, click OK to save your work and close the HTML Editor.

3. Click Save.
   The Tree Designer saves your changes.

Add the Displayed Text for a Node

Use the Tree Description (HTML), Query Text (HTML), and Conclusion Text (HTML) panes to design the information that is displayed for a particular node. The displayed text presents the description of the knowledge tree, poses a question, or presents information to help the user navigate the knowledge tree.

Follow these steps:

1. Select the primary node, query node, or conclusion node for which to add displayed text in the Tree pane.
   Depending upon the node you selected, one of the following panes opens:
   If you selected the primary node, the Tree Description (HTML) pane opens.
   - If you selected a query node, the Query Text (HTML) pane opens.
   - If you selected a conclusion node, the Conclusion Text (HTML) pane opens.
2. Do one of the following actions in the Tree Description (HTML), Query Text (HTML), or Conclusion Text (HTML) pane:
   - Enter the displayed text (that is, the information that is associated with the node) in the field provided. You can use plain text or can enter HTML codes as appropriate. To preview the text as it appears at runtime, select the Quick View option.
   - Click the Edit *xxx* (HTML) button to open the HTML Editor so you can design the information that is associated with the node in WYSIWYG (what you see is what you get) format. When you finish editing the information, click OK to save your work and close the HTML Editor.
3. Click Save.
   The Tree Designer saves your changes.

Add the Response Text for a Node

When you add a node to an existing query node in the Tree pane, the Tree Designer adds a response box for the added node in the right pane. Use the box to specify a response to the question or statement posed in the Query Text (HTML) pane.

Follow these steps:

1. Select the query node that contains the node for which to add response text in the Tree pane.
   The Query Text (HTML) pane opens.
2. Locate the response box for which to specify text in the Query Text (HTML) pane, then enter a response to the question or statement that is posed by the parent node.

   NOTE
   By default, the response text you enter replaces the node name in the Tree pane (that is, the Tree Designer synchronizes the response text and the node name). To disable synchronization, select the Copy Response command from the Options menu to clear the check mark from it.
3. Click Save.
   The Tree Designer saves your changes.

Check a Knowledge Tree for Errors

Follow these steps:

1. Select Check Errors from the Tree menu.
   The Tree Designer scans the knowledge tree for missing displayed text, responses, and links, then displays the results in the Errors pane at the bottom of the window.
2. Select an error in the Errors pane to display its associated node.
   The Tree Designer selects the node in the Tree pane and displays its definition in the right pane.
3. Correct the indicated error.
4. Click Save.
   The Tree Designer saves your changes.

Save a Knowledge Tree

To save changes to the open knowledge tree

1. Click Save.
   The Tree Designer saves all changes to the knowledge tree since you last saved. TheTree Designer window remains open.
2. Click Save and Close.
   The Tree Designer saves all changes to the knowledge tree since you last saved, then closes the Tree Designer window.

Set Tree Designer Options

Use the commands on the Options menu to control certain Tree Designer behaviors, as follows:

- To set whether the Tree Designer uses response text as the name of an associated node, select Copy Response from the Options menu.
  - If a check mark displays next to the command on the menu, the option is active and the response text you specify for a node replaces its name in the Tree pane. This is the default setting.
  - If no check mark displays, the option is inactive and you manually rename the node.
- To set whether the Tree Designer prompts you before deleting a node, Select Prompt on Node Delete from the Options menu.
  - If a check mark displays next to the command on the menu, the option is active and the Tree Designer displays a confirmation prompt when you attempt to delete a node. Click OK to close the prompt and delete the node and its subnodes. This is the default setting.
  - If no check mark displays, the option is inactive and the Tree Designer deletes the selected node and its subnodes without displaying a confirmation prompt.

The Tree Designer Interface

This article contains the following topics:

The Tree Designer includes a number of windows and dialogs to help you create and edit knowledge trees.

CA

You can design and build knowledge tree documents for a knowledge base. A knowledge tree represents expert knowledge in a particular area. Each tree is constructed to guide the user through the process of resolving a problem, performing a procedure, or achieving a specified goal.

You can use the Tree Designer to do the following:

- Add, delete, rename, copy, cut, and paste decision nodes in a hierarchical tree structure.
- Design query nodes that pose questions, supply possible answers to questions, or serve as parent nodes for further subnodes in the tree.
- Design conclusion nodes to supply the final answer to a query or to end a procedure.
- Link jump nodes to previously defined nodes in a procedure or query.
- Check and correct errors in the decision tree logic.
- View node layout that displays at run time.

The menu bar at the top of the right pane of the Tree Designer window provides access to features with which you can build and edit knowledge trees. The Tree pane shortcut menu in the Tree Designer opens when you right-click a branch in the tree.

The Tree Menu

The Tree menu lets you check for errors, save changes to a knowledge tree, and close the Tree Designer window. This menu contains the following commands:

Save and Close

Saves changes made to the knowledge tree since you last saved it and closes the Tree Designer window. This command is also available on the toolbar.

Save

Saves changes made to the knowledge tree since you last saved it, but does not close the Tree Designer window. This command is also available on the toolbar.

Check Errors

Checks the knowledge tree for errors, and (if errors exist) displays them in the Results of Error Check pane. This command is also available on the shortcut menu that opens when you right-click a node in the Tree pane.

Exit

Prompts you to save your changes, then closes the window.

The Node Actions Menu

The Node Actions menu lets you add, rename, and delete nodes in the knowledge tree. This menu contains the following commands:

Add Query

Adds a query node to the query node currently selected in the Tree pane. You cannot add a query node to the primary node, a conclusion node, or a jump node. This command is also available on the toolbar and on the shortcut menu that opens when you right-click a node in the Tree pane.

Add Conclusion

Adds a conclusion node to the query node currently selected in the Tree pane. You cannot add a conclusion node to the primary node, a conclusion node, or a jump node. This command is also available on the toolbar and on the shortcut menu that opens when you right-click a node in the Tree pane.

Add Jump

Adds a jump node to the query node currently selected in the Tree pane. You cannot add a jump node to the primary node, a conclusion node, or a jump node. This command is also available on the toolbar and on the shortcut menu that opens when you right-click a node in the Tree pane.

Rename

Opens the Prompt window so you can change the name of the node currently selected in the Tree pane. You cannot rename the primary node. This command is also available on the shortcut menu that opens when you right-click a node in the Tree pane.

Link Jump Node

Links the currently selected jump node to the next node you select. You cannot link a jump node to the primary node or to another jump node. This command is also available on the shortcut menu that opens when you right-click a node in the Tree pane.

Delete

Deletes the currently selected node and its subnodes from the knowledge tree without putting them on the clipboard.

> **NOTE**
>
> When you delete a query node, the query and its responses are removed from the knowledge tree. You cannot delete the primary node. This command is also available on the toolbar and on the shortcut menu that opens when you right-click a node in the Tree pane.

> **NOTE**
>
> By default, the Tree Designer prompts you to confirm node deletion. Click OK to close the prompt and delete the node and its subnodes.

The Options Menu

The Options menu lets you personalize your interaction with the Tree Designer. This menu contains the following commands:

Copy Response

Specifies whether to use response text as the name of an associated node when you create it.

- Select this command to automatically use the text of a node as its name when you create it. When the command is active, a check mark displays to the left of the command name on the menu.
- Clear this command to use the default name (Default Node Name) when you create a node. You can use the Rename command on the Node Actions menu to change the node's name.

This command is also available on the shortcut menu that opens when you right-click a node in the Tree pane.

Default: Selected

Prompt on Node Delete

Specifies whether the Tree Designer prompts you before deleting nodes.

- Select this command to display a confirmation prompt before deleting nodes. When the command is active, a check mark displays to the left of the command name on the menu.
- Clear this command if you do not want the Tree Designer to prompt you to confirm node deletion.

This command is also available on the shortcut menu that opens when you right-click a node in the Tree pane.

Default: Selected

The Help Menu

This menu contains the following commands:

Help

Opens the Online Help.

Help on This Window

Opens a window that provides online help information about the currently active pane or window.

Power User Tips

Opens a window that provides help information such as keyboard shortcuts for expert users.

Screen Reader Usage

This Help menu item opens a window explaining how to use a Screen Reader.

Support

Opens the CA Support web site.

Operations Info

Typically, this command displays a window showing information about your product installation, such as hours of operation and contact information.

The information displayed on this window is contained in the file:

$NX_ROOT/bopcfg/www/htmpl/web/analyst/about.htmpl.

If you want to customize this file for your installation, please copy it to:

$NX_ROOT/site/mods/www/htmpl/web/analyst/about.htmpl and then make the required changes.

About

Opens a window displaying information about this version of the product, third party notices, technical support information, and system information. Click Done to close the About window.

The Tree Pane Shortcut Menu

The Tree pane shortcut menu opens when you right-click a node in the Tree pane on the Tree Designer window. This menu contains the following commands:

Copy

Puts a copy of the currently selected node (and its subnodes) on the Clipboard. You cannot copy the primary node to the Clipboard. When you copy a query node, its responses are also copied.

Copy Single Node

Puts a copy of the currently selected node (but not its subnodes or responses) on the Clipboard. You cannot copy the primary node to the Clipboard.

Cut

Removes the currently selected node (and its subnodes) from the Tree pane and puts it on the Clipboard. When you cut a query node, the query and its responses are removed from the knowledge tree. Use the Cut command to move a node and its subnodes from one branch of the knowledge tree to another. You cannot cut the primary node or the root node to the Clipboard.

Paste

Pastes a previously copied or cut node (and its subnodes) under the selected query node. You can only paste a node under a query node.

Add Query

Adds a query node to the query node currently selected in the Tree pane. You cannot add a query node to the primary node, a conclusion node, or a jump node.

Add Conclusion

Adds a conclusion node to the query node currently selected in the Tree pane. You cannot add a conclusion node to the primary node, a conclusion node, or a jump node.

Add Jump

Adds a jump node to the query node currently selected in the Tree pane. You cannot add a jump node to the primary node, a conclusion node, or a jump node.

Rename

Opens the Prompt window so you can change the name of the node currently selected in the Tree pane. You cannot rename the primary node.

Link Jump Node

Links the currently selected jump node to the next node you select. You cannot link a jump node to the primary node or to another jump node.

Delete

Deletes the currently selected node (and its subnodes) from the knowledge tree without putting them on the clipboard. When you delete a query node, the query and its responses are removed from the knowledge tree. You cannot delete the primary node.

Check Errors

Checks the knowledge tree for errors, and (if errors exist) displays them in the Results of Error Check pane.

Copy Response

Specifies whether to use response text as the name of an associated node when you create it.

- Select this command to automatically use the text of a node as its name when you create it. When the command is active, a check mark displays to the left of the command name on the menu.
- Clear this command to use the default name (Default Node Name) when you create a node. You can use the Rename command on the Node Actions menu to change the node's name.

This command is also available on the shortcut menu that opens when you right-click a node in the Tree pane.

Default: Selected

Prompt on Node Delete

Specifies whether the Tree Designer prompts you before deleting nodes.

- Select this command to display a confirmation prompt before deleting nodes. When the command is active, a check mark displays to the left of the command name on the menu.
- Clear this command if you do not want the Tree Designer to prompt you to confirm node deletion.

This command is also available on the shortcut menu that opens when you right-click a node in the Tree pane.

Default: Selected

Tree Designer Components

The Tree Designer opens when you do either of the following:

- Click the Design Tree button on the Content tab of the Create New Tree Document window
- Click the Design Tree button on the Content tab of the Update Tree Document window

This window has the following main components:

Tree Pane

Displays the nodes that comprise the knowledge tree. Each node represents a step in the process presented by the knowledge tree. A well-constructed knowledge tree guides the user along a systematic path to answering a question, resolving a problem, or completing a process. Click or as appropriate to open or close nodes in the Tree pane. When you click a node in the Tree pane, the associated information displays in the right pane.

NOTE

The right pane is empty when you select a Jump node.

Menu Bar

Provides access to features with which you can build and edit knowledge trees.

Toolbar

Contains shortcuts for some of the features available from the menu bar.

Tree Description (HTML)

Contains the description that displays when a user clicks the Tree Description link while viewing the document in tree view. You can edit the description directly in the Tree Description (HTML) pane, or you can click Edit Tree Description (HTML) to open the HTML Editor so you can more precisely format the text.

Query Text (HTML) Pane

Contains the text defined in the knowledge tree for the selected node. The query text typically contains a question (the answer to which determines the next node in the tree) or a step in a procedure. The Query Text (HTML) pane also displays a field for each possible response (query, jump, or conclusion) to the question or instruction displayed in the Query Text (HTML) box. Each response corresponds to a subnode of the selected query node in the Tree pane.You can edit text directly in the Query Text (HTML) pane, or you can click Edit Query Text (HTML) to open the HTML Editor so you can more precisely format the text.

Conclusion Text (HTML) Pane

Contains the text defined in the knowledge tree for the selected node. The conclusion text typically describes the resolution for a particular problem, the final step in a procedure, or a possible termination point in a query. You can edit text directly in the Conclusion Text (HTML) pane, or you can click Edit Conclusion Text (HTML) to open the HTML Editor so you can more precisely format the text.

Results of Error Check Pane

Displays detected errors in the knowledge tree. The error messages displayed help you locate and fix errors in the knowledge tree. When the Tree Designer finds no errors in the tree, the pane does not open. Use the commands on the View menu, which are only available when the Tree Designer finds errors, to open and close the Results of Error Check pane.

Tree Description (HTML)

When you select the primary node in the Tree pane, the right pane contains the description that displays when a user clicks the Tree Description link while viewing the document in tree view. You can edit the description directly in the Tree Description (HTML) pane, or you can click Edit Tree Description (HTML) to open the HTML Editor so you can more precisely format the text.

You can use the following keyboard shortcuts when editing text in the pane:

Ctrl+C

Puts a copy of the currently selected text on the Clipboard.

Ctrl+X

Removes the currently selected text and puts it on the Clipboard.

Ctrl+V

Pastes previously copied or cut text at the current cursor position.

Ctrl+Z

Reverses the previous action.

Ctrl+Y

Reverses the previous Undo action.

For more robust editing, click Edit Tree Description (HTML) to use the HTML Editor to define the layout and content of the text.

This pane contains the following fields:

Tree Description HTML

Defines the tree description.

Quick View

Specifies to display content in the associated field as it appears at run time.

HTML Source

Specifies to display content in the associated field as editable HTML source code. Select this option to make simple changes directly to the HTML code.

This pane contains the following button:

Edit Tree Description (HTML)

Opens the HTML Editor so you can use plain text or HTML elements to design a tree description for the knowledge tree document.

Query Text (HTML) Pane

When you select a query node in the Tree pane, the right pane contains the text defined in the knowledge tree for the selected node. The query text typically contains a question (the answer to which determines the next node in the tree) or a step in a procedure. The Query Text (HTML) pane also displays a field for each possible response (query, jump, or conclusion) to the question or instruction displayed in the Query Text (HTML) box.

Each response corresponds to a subnode of the selected query node in the Tree pane. You can edit text directly in the Query Text (HTML) pane, or you can click Edit Query Text (HTML) to open the HTML Editor so you can more precisely format the text.

You can use the following keyboard shortcuts when editing text in the pane:

Ctrl+C

Puts a copy of the currently selected text on the Clipboard.

Ctrl+X

Removes the currently selected text and puts it on the Clipboard.

Ctrl+V

Pastes previously copied or cut text at the current cursor position.

Ctrl+Z

Reverses the previous action.

Ctrl+Y

Reverses the previous Undo action.

For more robust editing, click Edit Query Text (HTML) to use the HTML Editor to define the layout and content of the text.

This pane contains the following fields:

Query Text (HTML)

Defines the text of the question or step.

Quick View

Specifies to display content in the associated field as it appears at run time.

HTML Source

Specifies to display content in the associated field as editable HTML source code. Select this option to make simple changes directly to the HTML code.

Responses

Defines a field for each possible response (query, jump, or conclusion) to the question or instruction displayed in the Query Text (HTML) box. Each response corresponds to a subnode of the query node selected in the Tree pane.

- To edit the name of a response, enter your changes in the box provided.
- To edit the text of a response, select the response node in the Tree pane and edit it in the Query Text (HTML) pane or the Conclusion Text (HTML) pane, as appropriate.

This pane contains the following button:

Edit Query Text (HTML)

Opens the HTML Editor so you can use plain text or HTML elements to design a query for the knowledge tree document.

Conclusion Text (HTML) Pane

When you select a conclusion node in the Tree pane, the right pane contains the text defined in the knowledge tree for the selected node. The conclusion text typically describes the resolution for a particular problem, the final step in a procedure, or a possible termination point in a query. You can edit text directly in the Conclusion Text (HTML) pane, or you can click Edit Conclusion Text (HTML) to open the HTML Editor so you can more precisely format the text.

You can use the following keyboard shortcuts when editing text in the pane:

Ctrl+C

Puts a copy of the currently selected text on the Clipboard.

Ctrl+X

Removes the currently selected text and puts it on the Clipboard.

Ctrl+V

Pastes previously copied or cut text at the current cursor position.

Ctrl+Z

Reverses the previous action.

Ctrl+Y

Reverses the previous Undo action.

For more robust editing, click Edit Conclusion Text (HTML) to use the HTML Editor to define the layout and content of the text.

This pane contains the following fields:

Conclusion Text (HTML)

Defines the text of the resolution for a particular problem, the final step in a procedure, or a possible termination point in a query. Quick View Specifies to display content in the associated field as it appears at runtime. HTML Source Specifies to display content in the associated field as editable HTML source code. Select this option to make simple changes directly to the HTML code.

This pane contains the following button:

Edit Conclusion Text (HTML)

Opens the HTML Editor so you can use plain text or HTML elements to design a conclusion for a query in the knowledge tree document.

The Prompt Window

The Prompt window opens when you do either of the following:

- Select a node from the Tree pane and select Rename from the Node Actions menu
- Right-click a node in the Tree pane and select Rename from the shortcut menu

Use the window to enter a new name for the selected node.

This window contains the following field:

Please Enter the New Node Name

Defines the new name for the selected node.

Administering Knowledge Management

This article contains the following topics:

Document Permissions

You can set, view, and edit permissions for a document. These permissions can be assigned to different groups of individuals. When setting permissions, you can decide to inherit permissions from the primary category of a document or can specify new permissions. By default, documents inherit their permissions from their primary category. This default handles access permissions at the category level rather than the document level.

> **WARNING**
> When creating a knowledge document, verify that document permissions include users that can be later assigned on the document through the approval process. When a group is assigned to a document, users from that group may not have the permission to view the document. If the document is assigned to a specific user, default data partition constraints allow the user to view the document.

Version Documents

Using the document versioning capabilities of Knowledge Management, an analyst with editing privileges can create a *Rework-Draft* version of a document. A rework version starts as a copy of the document that is replaced in the knowledge base after it is verified and republished. The need to unpublish the document first is avoided.

Users with editing privileges can also perform the following versioning tasks:

- Save a draft version of a document.
- Roll back to a previous version if a problem with the current version occurs (Draft or Published). The Versions tab on the Update Document page is where users can select different versions for rollback to replace the current version.
- Track the number of document versions that are saved, deleted, and archived in the knowledge base.

The Knowledge Documents Inbox on the CA SDM Scoreboard is the repository for documents in all statuses including saved and assigned draft and rework-draft documents.

How to Manage Document Versions

Administrators can set up and can manage document versions by performing the following steps:

1. Identify who can edit published documents and can create the Rework-Draft versions. The role being used for a particular contact record controls editing privileges.
2. Define an approval process template that groups tasks or steps to complete during the document lifecycle. By default, a built-in approval process template allows users to create documents.
3. Determine whether to use the document approval process. Analysts who are permitted to bypass the approval process can identify which tasks they want to start with when they create a Rework-Draft version.
4. Create archive and purge rules for document version maintenance.

Document Expiration

When a published document reaches its expiration date, the product typically retires it (removes the document from the knowledge base and the approval process).

Document Archive and Purge

You can manage the size of your knowledge pool by using Archive and Purge, which removes old and unused documents automatically. Archive and Purge improve the efficiency of knowledge searches by returning only current documents.

Archive and Purge run as a background process and automatically removes inactive records in the CA SDM database according to rules that you configure. These rules act on CA SDM objects at specific time intervals.

Manage Export/ Import of Knowledge Documents

Contents

Knowledge Base Re-Index

The following changes to search settings require that you re-index the knowledge base using the Knowledge Re-index utility:

- After importing knowledge
- After changing parse settings
- After mass deletions
- When search failures occur

Re-indexing is necessary because existing documents do not reflect any changes that are made to the synonyms, noise words, special terms, and other research parameters until they have been re-indexed. All new synonyms, noise words, and special terms must be re-indexed to help ensure that keyword searches of the knowledge base are current and accurate.

You run Knowledge Re-index from the command line. The pdm_k_reindex.exe command is the executable file for Knowledge re-index.

> **NOTE**
> Re-indexing the documents in the knowledge base can be a time-consuming operation, depending on the size of your database. We recommend that you run the Knowledge Re-index utility after all changes have been added.

Export or Import Log Files

You can sort transactions by package, template, and status. All KEIT transactions are recorded in the following log files:

- **stdlog**
  Logs CA SDM information.
- **keitstat.log**
  Provides statistical information on the KEIT operation.
- **keitinfo.log**
  Provides detailed information on the KEIT operation.

  > **NOTE**
  >
  > : If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.
- **Package**

Displays the export package name.

- **Run By**
  Displays the user who ran the export.
- **Start Time**
  Displays the start time of the transaction.
- **End Time**
  Displays the end time of the transaction.
- **Document Count**
  Displays the number of documents exported.
- **Failure Count**
  Displays the number of failed exports.
- **Progress (%)**
  Displays the transaction's progress.
- **Status**
  Displays the transaction's status.

How to Export/Import Knowledge Documents

The Knowledge Management Export/Import tool (KEIT) migrates and synchronizes data between different Knowledge Management systems and enables export/import of Knowledge Documents. Use this tool to define export/import transactions.

> **NOTE**
> If multi-tenancy is installed, all tenanted and public categories that were imported from a previous system display as public in the new multi-tenancy system, but imported documents can be tenant-specific.

Follow these steps:

Allow Users to Export or Import Transactions

You can allow analysts to grant export and import permissions by managing your role list on the Administration tab.

Follow these steps:

1. Navigate to Security and Role Management, Role Management, Role List.
   The Role List appears.
2. Select a role with the Analyst Interface Type.
   The Role Detail page appears.
3. Click Edit.
   The Update Role page appears.
4. Click the Knowledge tab and select the options depending on the permissions that you want to grant.
5. Click Save.
   The Role Detail page reappears.
   Review your changes in the Knowledge tab and close the Role Detail page. You can also restrict users from exporting/importing by clearing the checkboxes on the Knowledge tab of the Role Detail page.

Define the Export Import Directories

Follow these steps:

1. Select the **General Setting** page under **Administrator** tab, update the **Path For Knowledge Import/Export Files** option to define package directories.
   – (Conventional configuration) By default it uses “$NX_ROOT/site/keit”. Restart the CA SDM servers after changing this path. Packages are stored in the following default directories:

- Export Packages: $NX_ROOT/site/keit/export
- Import Packages: $NX_ROOT/site/keit/import

– (Advance Availability Configuration) This path must be UNC shared drive. Restart the CA SDM servers after changing this path. Packages are stored in the following directories:
  - Export Packages: *UNC Shared Location/*export
  - Import Packages: *UNC Shared Location/*import

2. To add attributes for the export/import availability, add the following attribute names to the NX.env file:
   – • **@NX_KEIT_AVAILABLE_FIELDS**
     Adds the attribute name.
   • **@NX_KET_ADDL_FIELDS**
     Adds the attribute name, and if the new attribute is a SREL to another object, adds the attribute name that is used to synchronize the source and target systems.
     Example: Add Attributes
     @NX_KEIT_ADDL_FIELDS = STATUS_ID.STATUS,DOC_TEMPLATE_ID.TEMPLATE_NAME

   The export and import directories contain the following files:
   – • **keit_config.xml**
     Contains the configuration file for an export or import package.
   • **data.xml**
     Contains the data file containing values of knowledge documents.
   • **rep.xml**
     Contains the repositories that are referenced in the data.xml file.
   • **images**
     Specifies the image files that are embedded in knowledge documents.

View Export/Import Templates

The Knowledge Export/Import Tool (KEIT) templates are used to define export/import transactions.

> **NOTE**
>
> The default export/import template is named All-Docs and does not contain the STATUS_ID field. This template creates the data.xml file and imports documents with a draft status. If you manually add the STATUS_ID field to the All-Docs template before the export operation, the original document status is preserved upon importation.

Follow these steps:

1. Click the Administration tab.
   The Administration console appears.
2. Click Knowledge, Documents.
   Click Export/Import, Export/Import Templates.
   The Knowledge Export/Import Templates List appears.
3. Click a template in the Name column.
   The KEIT template opens.

Create an Export/Import Template

You can create and modify a template using the Knowledge Export/Import Template Settings page in the Administration tab.

Follow these steps:

1. On the Administration tab, browse to Knowledge, Documents, Export/Import, Export/Import Templates.
   The Knowledge Export/Import Templates List appears.
2. Click Create New.
   The Create New Export/Import Template page appears.

3. Complete the following fields:
   - **Template Name**
     Identifies the name of the template.
   - **Description**
     Provides a brief description of the template.
4. Complete the appropriate fields on the following tabs:
   - Export Fields
   - Export Filter
   - Import Settings
5. Click Save.
   The template is created.

Export Fields

The Export Fields tab appears and contains the following fields:

- **Available**
  Displays the Knowledge Document fields that are available for export.

  NOTE

  If you want to preserve status of the documents, such as Draft, for the export and import process, add STATUS_ID to the Exported column.
- **Exported**
  Specifies document fields to export.
- **Export Attachments**
  Exports the Knowledge Document file attachments.

You can use the arrows in the Select Document Attributes section to add document attributes export.

NOTE
If you edit the export/import template using Firefox, the list size does not change after you use the arrows in the first edit. Close and reopen the template in edit mode, and using the arrows again causes changes to the list size. This behavior does not occur when you use Internet Explorer.

WARNING
The exported fields in the KEIT have a higher priority than when the fields are selected as defaults in the Import Settings tab. If you select a default export field in the Import Settings tab, it is not processed as the default field.

Export Filter

The Export Filter tab appears and contains the following fields:

- **Category**
  Opens the Knowledge Category page. Use this option to add categories to the list on the Export Filter tab.
- **Remove Category**
  Removes categories from the list on the Export Filter tab.
- **Clear Category**
  Clears the category list on the Export Filter tab.
- **Include child categories**
  Exports the documents from child categories of exported categories.
- **Include secondary categories**
  Exports secondary categories of exported documents.
- **Include all documents linked to selected categories**
  Exports all documents that are linked to selected categories.

NOTE
Enabled only if you select Include secondary categories.

- **Additional Filter**
  Provides an additional WHERE clause.

Import Settings

The Import Settings tab appears and contains the following fields:

- **Error Threshold (%)**
  Stops the import process (sets the status to Failed) if the percent of errors exceeds the specified number.
- **Override published documents**
  Allows imported documents to override published documents of the CA SDM server.
- **Override documents in all non-published statuses**
  Allows imported documents to override non-published documents of the CA SDM server.
- **Use default values when overriding documents**
  Uses default values on overridden documents for defined fields.
- **Index documents immediately**
  Indexes the document after the import process.
- **Status**
  Select Draft, Published or Retired.
- **Priority**
  Sets the priority level of the import settings.
- **Template**
  Selects one of the following default templates:
  - Built in - Knowledge Document
  - Built in - Knowledge Tree
  - Built in - Quick Editing
- **Owner**
  Allows you to set the owner by opening the Contact Search page.
- **Author**
  Allows you to set the author by opening the Contact Search page.
- **Subject Expert**
  Allows you to set the subject expert by opening the Contact Search page.
- **Assignee**
  Allows you to set the assignee by opening the Contact Search page.
- **Expiration date**
  Opens the date helper.
- **Review Date**
  Opens the date helper.
- **Product ID**
  Allows you to set the product ID by opening the Product Search page.
- **Asset**
  Opens the Configuration Item Search page.
- **Root Cause**
  Opens the Root Cause Search page.
- **Service Desk Priority**
  Sets the Service Desk priority of the import settings.
- **Severity**
  Sets the severity level of the import settings.
- **Impact**

Sets the impact level of the import settings.

- **Urgency**
  Sets the urgency level of the import settings.

Import Packages

You can import packages from the Knowledge Import Packages List page in Knowledge Administration or by running the pdm_kit utility from the command line. For advanced availability configuration, run ImportSampleData.bat or ImportSampleData.sh from background server only.

Note: The sample files are not localized. Importing sample data does not work on localized environments. In advance availability configuration, run ImportSampleData.bat or ImportSampleData.sh from the background server only.

Follow these steps:

1. Depending on your configuration type, perform the following steps:
   - Conventional: From the default $NX_ROOT/site/keit/export directory, copy the desired knowledge package to your default $NX_ROOT/site/keit/import installation directory.
   - Advanced Availability: Copy the knowledge content package from *UNC Shared Location*/export directory to the *UNC Shared Location*/import directory.
2. From the Administration tab, select Knowledge, Documents, Export/Import, Import.
   The Knowledge Import Packages List page appears.
3. Right-click the desired package and select Import from the short-cut menu.
   Note: For advanced availability configuration, you cannot export or import from application server using CA SDM Web UI. To export or import using application server, execute the pdm_kit or pdm_ket utility if the package resides in a UNC shared drive.
4. Click OK in the message that appears.
   Note: The message may not appear if the package is larger. The import progress can be seen under the Import Transactions page.
   The package is imported to the target system and appears on the Knowledge Import Transaction List page.

Export and Import Transactions

Follow these steps:

1. Export the data to another system by clicking **Export** on the **Export/Import Template Detail** page or by running the pdm_ket utility.

   NOTE

   In the advanced availability configuration, you cannot export packages from the application server using web interface. Use the pdm_ket utility to export packages from any server.
2. Import knowledge data from the package by performing any of the following actions, depending on your CA SDM configuration:
   - In the conventional configuration, copy the desired knowledge package from the default $NX_ROOT/site/keit/export directory and place it in the $NX_ROOT/site/keit/import directory. On the **Knowledge Import Packages List** page, right-click a knowledge package, and click **Import** on the shortcut menu.
   - In the advanced availability configuration, copy the desired knowledge package from the *UNC Shared Location*/export directory and place it in the *UNC Shared Location*/import directory. You cannot import packages from the application server using web interface. Use the pdm_kit utility to import packages from any server.

     NOTE
     The knowledge package name must start with "package_" prefix for it to be listed in the Knowledge Import page.

WARNING
The r11.0 import utility has been renamed pdm_kit_txt.exe to allow importing r11.0 text files. This utility does not support any of the r12.0 import enhancements.

3. Documents that are not indexed after import are stored in the Unindexed queue on the scoreboard. Users cannot search for unindexed documents, therefore, they do not appear on the Document list. To display this queue, select the **Service Desk** tab, **Knowledge Documents**, **Unindexed**. Before you initiate the import, enable the Index Document option to ensure that you can import the knowledge document. If the documents are not searchable, you can reindex by executing the pdm_k_reindex utility.

   NOTE
   If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

Knowledge Export/Import CLI

pdm_ket Utility -- Knowledge Management Export Tool

The pdm_ket utility exports knowledge from a source computer to a knowledge package, using a template that is created from the web interface.

Attachments and their links are exported to data.xml during the export. Before you import the attachments, manually move them to following directory on the background server:

$NX_ROOT/site/attachments/default

NOTE

For advanced availability configuration, you can execute the pdm_ket utility using attachments UNC path.

Use this utility to do the following tasks:

- Create configuration file based on the related knowledge export template.
- Export data with UUID of document, and content
  such as title, summary, problem, resolution and other document attributes like owner, status, and so on.
- Export all unique images that are used by the exported documents (always exported).
- Export all unique attachments that are used by the exported documents (EXP_ATTMNT field).
  The files are copied from the remote repository to the local package folder.

The utility is invoked as follows:

```
pdm_ket -n <template name> [-h] ]-v]
```

- **-n <template name>**
  Defines the name of the template that is used for export (case sensitive).
- **-h**
  (Optional) Displays help on the utility.
- **-v**
  (Optional) Enables extensive logging (bop_logging) of program events. This option is commonly used for internal problem solving.

Example: Using pdm_ket to Export Knowledge Using the my_template Template.

```
pdm_ket -n my_template
```

The pdm_ket utility can be scheduled using a third-party scheduler for exporting Knowledge Documents.

pdm_kit Utility -- Knowledge Import Tool

The pdm_kit utility imports data to the destination server according to the settings in the configuration file of a package.

> **WARNING**
> Before you execute the pdm_kit utility from the application server, ensure that the path (local or UNC) of the package is accessible by the background server. If the background server is not able to access this path, the package is not imported from the application server. The r11.0 import utility has been renamed to pdm_kit_txt.exe to allow importing r11.0 text files. This utility does not support any of the Release 12.9 import enhancements.

The referenced data that is previously referenced by the pdm_ket utility gets the real UUID or ID value of the destination server. When running pdm_kit utility a new userid parameter is applied. The pdm_kit utility works as follows:

1. Imports the documents by replacing the userid value (for contacts) or referenced name (for fields like asset) with appropriate UUID of the destination server.
2. Imports the images.
3. Imports the attachments.
4. Uploads the files from the local package folder to the remote repositories.

> **NOTE**
> If editing published documents is disabled, then the imported document is created as a rework version.

> **NOTE**
> The utility is invoked as follows:

```
pdm_kit [-h] -f -u [-v]
```

- **-h**
  (Optional) Displays help for the utility on the interface.
- **-f**
  Specifies the path to the package.
- **-u**
  Specifies the default user.
- **-v**
  (Optional) Enables extensive logging (bop_logging) of program events. This option is commonly used for internal problem solving.

> **NOTE**
> For advanced availability configuration, you can execute the pdm_kit utility on background server. For Windows, you can also execute the pdm_kit utility on application server if the package resides in a UNC shared drive. For non-windows platform, you cannot execute pdm_kit utility on application server.

Example: Using pdm_kit to Import a Package

```
pdm_kit -f c:\package_path -u ServiceDesk
```

Managing Automated Policies

An automated policy describes the condition by which documents are flagged for correction and are promoted to publication or retirement throughout the various stages of the document lifecycle process. For example, you can specify the "fix broken links" default policy that matches documents that are found in the knowledge base with broken links. The task of fixing the problem can be assigned to an analyst.

Administrators can automate certain tasks in the knowledge document approval process that is based on document lifecycle policies and actions they define. By automating tasks, end users who are searching for solutions can solve problems faster, and they can do so without contacting other individuals, which provides a benefit to the organization.

Automated policies work with events and macros. Each policy is associated with the following components:

- **Stored Query:** Contains a set of action macros that execute when the policies identify and match a document during processing. After processing, the stored query condition event displays a role-based Knowledge Management lifecycle policy report on the CA SDM Scoreboard. The administrator is responsible for monitoring the reports and for providing feedback and recommendations to the appropriate document editors.
- **Action Macro:** Contains code that lets users set a flag, increase the priority, or perform some other action. You can modify the macros that appear on the Macro list, or can define your own.

How to Set Up Automated Policies

Administrators can set up the Automated Policies feature by performing the following steps:

1. A batch process must be defined in the Automated Policies Scheduler that executes on the background server to present the data that is required to view the Knowledge Management Lifecycle Policy reports. This action step also applies to the Knowledge Report Card.
2. For security and role management, define the stage by which users can view and search on documents during their lifecycle on the Knowledge Document Visibility tab in Role Management.
3. On the Automated Policy List page, you can edit the default policies, or can define your own.

Create an Automated Policy

You can create an automated policy that activates when an action occurs, such as when a document is published or retired from the knowledge base.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.
>
> **WARNING**
>
> For new policies, the administrator must include the "Disregard Life Cycle Policies" field in the stored query; otherwise, it does not appear on the Attributes tab.

Follow these steps:

1. Select **Administration** tab, **Knowledge**, **Automated Policies**, **Policies**.
   The Create New Automated Policy page appears.
2. Enter a name and description for the policy in the appropriate fields.
3. Enter a stored query name or select one using the search icon.
4. Click **Add Action**.
   The Macro List page appears.
5. On the Macro List page, select one of the predefined action macros, or define your own (click **Create New**).
   The action macro appears in the Action Information list on the Create New Automated Policy page.

   > **NOTE**
   > You can delete an action macro: right-click the name and select Delete from the shortcut menu.
6. Click **Save**.
   The new policy appears in the Automated Policy list.

Run the Automated Policies

The Automated Policies List page contains the details of the policies you can manage. To display this page, select the Administration tab, Knowledge, Automated Policies.

Each policy contains a stored query that executes when documents are matched during processing. After processing completes, a Lifecycle Policy report appears on the CA SDM Scoreboard. To view a report, select Knowledge Documents, Automated Policies. From the scoreboard, the analyst can manage their own documents, and by default, the administrator can manage all documents for each role.

To implement the reports, run a batch process with the Automated Policies Scheduler. The scheduler runs on the following CA SDM server, depending upon the configuration and displays the data that is required to view the reports.

- Conventional:Primary or secondary
- Advanced availability: Background

When you are finished, run the Knowledge Report Card.

Follow these steps:

1. Select the **Administration** tab, **Knowledge**, **Automated Policies**, **Scheduling**.
   The Automated Policies Scheduler appears.
2. Complete the following fields:
   - **Last Updated** -- Select the Run Calculation check box.
   - **Schedule** -- Specify a date and time from which CA SDM performs the calculation and runs the policies.
3. Click Save.

View Document LifeCycle Policy Reports

Administrators can view the Document Lifecycle Policy Reports that are generated by the automated policies on the Document List page. Each report is role-specific and provides administrative information about the documents that are flagged for correction and promoted to publication or retirement throughout the various stages of the document lifecycle process.

> **NOTE**
> If multi-tenancy is installed, the list page displays Tenant and Public Data settings in the search filter. Public Data can be Excluded or Included with Tenant data; Only searches for public objects exclusively. On detail pages, select the appropriate tenant from the list. If you select <empty>, the object is public.

Follow these steps:

1. On the **Service Desk** tab, **Knowledge Documents**, **Automated Policies**.
2. Do *one* of the following:
   - **Expand the My Documents** folder to reveal nested folders for your assigned items.
   - **Expand the All Documents** folder to reveal nested folders for all items.
3. Select the folder that you want to view.
   The Document List page displays the following columns:
   - **Title**
     Displays the title of the document that is either flagged for correction or promoted for publication or retirement.
   - **Policy/Actions**
     Displays the policy name and action content defined for the policy.
   - **Attributes**
     Displays the properties of the document that are affected by the policy it is assigned to.

Knowledge Documents Schedule

This article contains the following topics:

The Knowledge Documents Schedule provides Change Managers, Knowledge Managers, and Level 2 Analysts a high-level view of key events in the knowledge management lifecycle. The following events can be scheduled for a Knowledge Document:

- Submission
- Publication
- Review
- Expiration

The Knowledge Management Schedule tab displays a calendar format similar to the Change Calendar, but instead of showing start times and end times, it only shows specific dates.

Configuring scheduling views involves setting macros statements.

The Knowledge Document Schedule can also be exported to iCalendar format.

> **NOTE**
> When exporting schedules on some calendar programs, selecting the Open option instead of Save causes the file to import incorrectly. To avoid this issue on Knowledge Management and Change Order schedules, select the Save option instead of Open. After you save the exported file, import it through the calendar program interface.

Knowledge Schedule Filter

The Knowledge Schedule Filter displays the following fields:

- **Tenant**
  Filters the search by tenant. This field displays for the privileged user in a multi-tenancy installation.
- **Event Type**
  Filters the search by the following event types:
  - Submission
  - Review
  - Publication
  - Retirement
- **Schedule Start Date**
  Specifies the date for the beginning of a range for filtering the history to show only entries for a specified time frame.
- **Schedule End Date**
  Specifies the date to specify the end of a range for filtering the history to show only entries for a specified time frame.
- **Schedule Timezone**
  Specifies a timezone to view your search results.
  Note: If no timezone is selected, the events are displayed in your current timezone.
- **Owner**
  Defines the name of the contact that is assigned to maintain the document. Enter or search the name of the contact in "last name, first name" format.
- **Assignee**
  Defines the name of the contact that is assigned to handle the record. Enter or search the name of the contact in "last name, first name" format.
- **Subject Expert**

Defines the name of a contact with expertise in the subject matter of the document . Enter or search the name of the contact in "last name, first name" format.

- **Status**
  Select one of the following document statuses to perform your search:
  - Draft
  - Published
  - Retired
- **Category**
  Defines a category by which to filter documents retrieved. Enter the name of the category in the box or open the Category Search window. When you click Search, the product returns only documents that are associated with the specified category.
- **Initial View**
  Select the view of the Knowledge Management Calendar you want to see:
  - **Month**
    Displays a calendar for the full month that includes the earliest implementation start date that is specified in the Start Date field. The calendar shows abbreviated information about each change within the range (change number, start and end time, and first affected CI).
  - **Week**
    Displays seven consecutive days in a single column, beginning with the earliest implementation start date specified in the Start Date field. The calendar includes summary information about each change within the specified range (change number, start and end time, summary description, assignee, group, and the first ten affected CIs).
  - **Day**
    Displays a view similar to the week view, except that it shows only the day that is specified in the Start Date field.
  - ***n* Days**
    Displays a view similar to the week view, except that it continues for the specified number of days.
  - **List**
    Displays a standard CA SDM list page.

    NOTE
    You can click the More icon to display the Additional Search Arguments field. This field is intended only for expert users who understand SQL and Majic. You can enter a SQL WHERE clause in this field to specify an additional search argument.

Knowledge Schedule Views

The Knowledge Documents Schedule has the following views:

- **Month**
  Displays a calendar for the full month that includes the earliest implementation start date that is specified in the Schedule Start Date field. The calendar shows abbreviated information about each change within the range (change number, start and end time, and first affected CI).
  The Knowledge Documents Schedule has similar functionality to the Change Orders Month view with the following exceptions:
  - Knowledge events only have a date (no time) and event type groups them, similar to change orders that are grouped for each time period and change type.
  - Events types have the following default predefined colors, using the schedGroup macros:
    - Submission -- Black
    - Review -- Green
    - Publication -- Blue
    - Retirement -- Red

NOTE
If you see a review date referring to the past, it indicates that the review was probably not done.

- **Week**
  Displays seven consecutive days in a single column, beginning with the earliest implementation start date specified in the Schedule Start Date field. The calendar includes summary information about each change within the specified range (change number, start and end time, summary description, assignee, group, and the first ten affected CIs).
- **Day**
  Displays a view similar to the week view, except that it shows only the day that is specified in the Schedule Start Date field.
- ***n* Days**
  Displays a view similar to the week view, except that it continues for the specified number of days.
- **List**
  Displays a standard CA SDM list page.

Scheduling View Configuration

You configure the monthly and weekly scheduling views by specifying pdm_macro statements in the <head> section of the HTMPL forms defining the schedule. We recommend using the Source View of Web Screen Painter to edit these forms.

Any form that displays a schedule must contain the following:

- A schedConfig macro
- At least one schedAttr macro
- At least one schedGroup macro

The configuration macros are in a separate source file that is referenced by a pdm_include statement in the main source file. This file lets you configure your schedule without modifying the main source file.

For example, the configuration macros for the Change Calendar form list_chgsched.htmpl are in a file named list_chgsched_config.htmpl. For the Knowledge Lifecycle Schedule, you can modify the list_kdsched_config.htmpl using the same macros.

You can find list_chgsched_config.htmpl and list_kdsched_config.htmpl in the following directory:

$NX_ROOT\bopcfg\www\htmpl\web\analyst\

schedConfig Macro -- Configure Schedule

The schedConfig macro specifies that a form contains a schedule and provides basic configuration information. The following values are valid macro arguments:

- **autosearch=1|0**
  Specifies whether the schedule form reloads data from the server when the user selects a view outside the currently selected date range. Setting the value to 1 (default) causes the form to search automatically when the user selects a view with one or more days outside the date selection range of the search filter. Setting the value to 0 requires the user to press the Search button to initiate a search.
- **defaultView=0|1|7|30|99**
  Specifies the default view for the search filter as 0 (list), 1 (day), 7 (week), 30 (month), or 99 (n-day).
  The specification for defaultView affects only the initial display of the search filter. After the schedule displays, CA SDM automatically keeps the filter view selection aligned with the current view.
  Default: 30
- **firstday=0|1|2|3|4|5|6|7**
  Specifies the first weekday on the monthly view as a number between 0 (Sunday) and six (Saturday).
  Default: 0

export=xxx|0

Specifies

- • the code name of the template that is used for exporting in iCalendar format. Setting the value to 0 indicates the export and disables the buttons.

Default:

ChangeSchedule

- **legend=1|2|0**
  Specifies the location of the schedule legend showing the name and formatting of the groups on the schedule. You can set the value to 1 to position the legend above the schedule, or 2 to position the legend below the schedule. Set the value to 0 to disable the legend.
  Default: 2
- **maxGroups=0/*n***
  Specifies the maximum number of groups to be displayed in a single cell of the calendar month view.
  If there are more than maxGroups scheduled for a single day, CA SDM displays only the first maxGroups-1, and replaces the last with a "..nn more changes" hyperlink that the user can mouseover or click to see the full list. Set the value to 0 to disable this feature and allow an unlimited number of events in a calendar cell.
  Default: 4
- **nday=(n,n,...)**
  Specifies selections for the drop-down list for the n-day view.
  The specification is a list of day counts that are to be included in the drop-down list, or 0 to indicate that the n-day drop-down list is omitted from the schedule. The first value specified is the default for the drop-down list.
  Default: (3,7,14,28)
- **round=(*hr,min*)|0**
  Specifies whether schedule start and end dates are rounded when collecting change orders or knowledge documents into groups. Specify round=0 to disable rounding.
  By default, schedule start and end date groups objects. All CA SDM dates include a time, and without rounding, objects scheduled as little as a minute apart would be in separate groups. Rounding determines the group after adjusting the start date to an earlier hour or minute and the end date to a later hour or minute.
  The value of round specifies either an hour or a minute (but not both). Times are rounded to the nearest multiple of the value specified, for example:
  round=(0,15) rounds to the nearest quarter hour
  round=(0,30) rounds to the nearest half hour
  round=1 rounds to the nearest hour
  round=12 rounds to the nearest half day (12:00 AM or PM)
  round=24 rounds to the nearest day
  Default: (0,15)
- **timefmt=24hr|([am],[pm])**
  Specifies the format of times on the calendar views of the schedule.
  The default value of 24hr specifies that times are displays in 24 hour format (0:01 - 23:59). The alternative value of (am,pm) specifies a suffix for either morning or afternoon times, or both.

  NOTE
  All schedConfig arguments are optional.

schedAttr Macro -- Specify a Stored Attribute

The schedAttr macro specifies an attribute stored for each item selected for the list. Stored attributes are available for hover information on the monthly view, for the detailed or summary information in views other than the monthly view, and in the setSchedEvents() JavaScript function. The following values are valid macro arguments:

- • **attr=xxxx**

(Required) Specifies an attribute from the object on the schedule, such as a change order or Knowledge Document. Dotted attributes are permitted. The keyword attribute name CI*nn* can be used on the Change Calendar to specify that first *nn* CIs associated with the change order are included with the information stored.

NOTE
This argument is the only required argument for the schedAttr macro.

attrRef=.COMMON_NAME|xxxx

- • Stores the attribute of the referenced

table stored

- • for an SREL attribute (ignored for non-SREL attributes). The attribute name must be prefixed with a dot.

Default:

- .COMMON_NAME
- **label=**
  Displays a label for the attribute on the n-day view.
  Default: the Majic DISPLAY_NAME of the attribute
- **ident=1|0**
  Specifies whether the attribute is an identifier for the object (such as a reference number of a change order). The identifier attributes are displayed without a label in front of the group name in hover information and the n-day view.
  Default: 0
- **detail=1|0**
  Specifies whether the attribute is included in the detail information that is shown on views other than the monthly view. Detail information is the information shown when the Summary Only check box on the view is not selected.
  Default: 1
- **hoverInfo=1|0**
  Specifies whether the attribute is included in the hover information pop-up that is displayed on the monthly view. The hover information appears when the mouse pointer hovers over a group, or the user presses Alt+Right Arrow when focus is on the group.
  Default: 0
- **summary=1|0**
  Specifies whether the attribute is included in the detail information that is shown on views other than the monthly view. Detail information is the information shown when the Summary Only check box on the view is not selected.
  Default: 0

NOTE
CA SDM displays attributes in summary, detail, or hover information in the same order as their schedAttr macros.

schedGroup Macro -- Specify an Event Group

The schedGroup macro specifies the name and color coding of a group of items. The monthly view aggregates all items in a group into a single event. Views other than the monthly show individual items in the format for the group to which they belong. The following optional values are valid macro arguments:

- **grpname=xxx**
  (Required) Specifes the name of the group. The macro automatically assigns a number to the group and assigns the number to a JavaScript variable with a name of the form schedGroup_xxx, where xxx is the name of the group. This variable can be used in the setSchedEvents() JavaScript function to create an event belonging to the group.

  NOTE
  This argument is the only required argument for the schedGroup macro.
- **label=xxx**

Specifies a label for the group. If specified, the label is displayed in all views.

- **legend=xxx|0**
  Displays a description of the group for the legend that appears at the bottom of the schedule. Groups appear in the legend if at least one example of the group exists in the current view. Specifying 0 causes the group always to be excluded from the legend.
  Default: 0
- **color=black|*color***
  Specifies the color of text in items of this group. You can specify color in CSS format, either a valid web color or a hexadecimal value that is preceded by a pound sign.
  Example: Enter either "#FF0000" or "red" for red.
  Default: black
- **bgcolor=white|*color***
  Specifies the background color of items of this group. You can specify bgcolor in CSS format, either a valid web color or a hexadecimal value that is preceded by a pound sign.
  Example: Enter either "#FF000" or "red" for red.
  Default: white.
- **style=normal|bold|italic**
  Specifies the style of text of this group in the normal, bold, or italic style.
  Default: normal

The setSchedEvents() JavaScript Function

The setSchedEvents() JavaScript function creates events in the schedule. Modify this function when you want to view any new group objects. The predefined group objects appear by default.

CA SDM calls setSchedEvents() once for each object (change order or knowledge document) selected by the schedule search filter. The function creates events for the object by calling a second function, schedEvent(), and passing the group ID, start date, and end date of the event.

The function can create any number of events (including zero) for an object. The default setSchedEvents() function for the Change Calendar (list_chgsched.htmpl) creates one event for each change order and groups change orders by change type. This function is coded as follows:

```
1.    function setSchedEvents( chg )
2.    {
3.    var grpnum;
4.    switch( chg["chgtype"] - 0 ) {
5.        case 100: grpnum = schedGroup_std;  break;
6.        case 300: grpnum = schedGroup_emer; break;
7.        default:  grpnum = schedGroup_norm; break;
8.    }
9.    chg.schedEvent( grpnum, chg["sched_start_date"], chg["sched_end_date"] );
10.    }
```

The case parameter specifies the change type ID. To list the case IDs, see Create a Change Type.

The function has a single argument of a JavaScript object containing the attributes that are specified by schedAttr macros. The switch statement in lines 4-8 examines the chgtype attribute of the change order, and assigns the appropriate group number from one of the schedGroup_xxxx variables that are defined by previous schedGroup macros. On line 9, it calls the schedEvent() function to create an event in the schedule, passing the group number that is previously assigned and the schedule start and end dates. The dates are available in the argument object because they were specified in earlier schedAttr macros.

Integrating Multiple Search Engines Using Federated Search

The Federated Search feature extends the capabilities of the CA SDM built-in knowledge base.

- When a user performs a knowledge search, the results from the internal knowledge database are augmented with results from the external search engines. We provide out-of-the box search adapter configuration for CA Open Space, Microsoft SharePoint, and Google.
- The Federated Search architecture is flexible. Support for other search engines can be added by developing a custom search adapter using the CA Federated Search SDK. The SDK interface provides information and source code samples for extending the Federated Search functionality.
- Information that is contained in CA SDM tickets and knowledge articles can also be searched with an external search engine. The new Crawler Surface component allows an external search engine crawler to discover the CA SDM information easily. The crawler stores this information in its repository. This indexed information can be searched using Federated Search. Attachments for tickets and knowledge articles is also supported.

The main components of Federated Search are as follows:

- Configure Federated Search
- Configure Crawlers
- SDK Custom Search Adapter

How to Configure Federated Search

The Federated Search feature consists of the following components:

- **UI components**
  Enables the CA SDM Analyst users to enter search arguments and pass the search request to the Federated Search server.
- **CAFedSearch servlet**
  Main component of the federated Search feature and uses a REST interface. The Federated Search servlet runs on a dedicated Tomcat instance within the CA SDM application.
- **Plug-in search adapter**
  The plug-in search adapter interfaces the CA SDM application to an external search engine. The adapter translates the generic search requests to a search engine proprietary format and calls the search engine. The Federated Search servlet returns the configured search engine results to the CA SDM UI component.

  NOTE

  When you launch the Knowledge Document search form via Quick Profile browser, the configured Federated Search checkboxes are not displayed. As a result, users are unable to perform federated searches on Knowledge Search forms displayed via the Quick Profile browser.

Follow these steps:

Complete the Prerequisites

Complete the following prerequisites:

- Ensure to select the Federated Search option on the Configure Federated Search wizard while installing the CA SDM application.
- Know and decide the search engines that you want to use for Federated Search.
- Ensure to have an active Google Account for configuring the Google Plug-in Search Adapter. A Google API key and Google Custom Search Engine ID is also required. For more information, see Google Custom Search Engine https://www.google.com/cse/all
- Verify that IIS on the SharePoint server is configured with basic authentication. The Microsoft SharePoint adapter requires the basic authentication. By default, Basic Authentication is turned off in SharePoint. For more information about how to enable basic authentication in SharePoint, see the Microsoft SharePoint Documentation.

  NOTE
  Enable Basic Authentication for _vti_bin in IIS.
- For more information about how to integrate with Open Space version 2.0 SP1, see the Open Space documentation.

Enable Federated Search

A dedicated Tomcat instance is installed to host the CAFedSearch and FSCrawl servlets. The Federated Search Tomcat is under the control of the Daemon Manager. Tomcat Starts and stops automatically when CA SDM runs or shuts down respectively.

Configure the Federated Search (FS) Tomcat with the following installation options:

- **Configure Federated Search**
  Ensure to select this option when installing the CA SDM application. The Tomcat options are available only after selecting this option.
- **Tomcat Port**
  Specifies the Federated Search Tomcat Port.
  Default: 8040
- **Tomcat Shutdown Port**
  Specifies the Federated Search Tomcat Shutdown Port.
  Default: 8045

How to enable NTLM Authentication

CA Service Management 17.3 Federated Search Sharepoint Adapter supports NTLMv2 authentication along with the default Basic Authentication. NTLM authentication requires the following prerequisites.

- A domain user
- SharePoint server and CA SDM Federated Search server on the same domain

CA SDM Administrators can enable the NTLM authentication by performing the following steps:

1. On the SharePoint server, enable **Windows Authentication** and disable **Basic Authentication** for the Search service using the IIS Manager.
2. On the SharePoint server, add the domain user to the Web Application policy through SharePoint Central Administration.
3. On the Service Desk Federated Search server, add NTLM as the only option in the configuration properties file. For more information, see Invoke the Utility File to Configure the Search Adapters.

Generate Configuration XML Files for Federated Search

To configure Federated Search and to decrease editing errors, a utility file fs_adapters_cli is provided. A batch file for Windows and a Shell script for Linux and Unix is provided. The utility file is a Java application that is contained in two Jar files.

Three template files are supplied for CA Open Space, Google, and SharePoint adapters. The adapters-config.xml file is used for both input as well as output to the utility. The adapters.properties file is used as input to the utility. This file holds all the necessary configuration parameters for the adapters. The jfedsearch.log file contains the log file information.

The fs_adapters_cli utility is used to configure the adapters-config.xml file. The utility file installs and uninstalls the adapters by adding or removing entries from the adapters-config.xml. Individual adapter XML configuration files are generated. These files are then included in the adapters-config.xml with the <import> bean directive. Ensure to maintain a clean copy of the adapters-config.xml.

- To install an adapter, two entries are added for registering and importing the adapter in the adapters-config.xml file.
- To uninstall an adapter, two entries are removed for registering and importing the adapter from the adapters-config.xml file.

Federated Search Utility Files

The fs_adapters_ cli utility file is located in the CA SDM $NX_ROOT\samples\cafedsearch directory. The following utility file components are available in this location:

- The following Script files are located in the NX_ROOT\bin folder:
  - fs_adapters_cli .bat (for Windows)
  - fs_adapters_cli.sh (for Non-Windows)
- The following JAR files are available in the NX_ROOT\java\lib folder:
  - fs_adapters_config_cli.jar
  - fs_adapters_config_schema.jar
- The following templates and properties files are located in the $NX_ROOT\samples\cafedsearch directory:
  - openspace-tmpl.xml
  - google-tmpl.xml
  - sharepoint-tmpl.xml
  - adapters.properties

To install the federated search utility file, run the following command:

```
fs_adapters_cli -i -k <key> -b <bean> -t <filename> -p <filename>  -o <filename> [-c <filename>]
```

Use the fs_adapters_cli - h option to get Help on the Utility file. The following attribute options are available:

- **-b (Optional)**
  Attribute value to map adapters. Refers to an actual ID of an adapter bean. If not specified, then the value from - k is taken as the value for - b.
- **-c (Optional)**
  The main XML file that contains all the configured adapters.
- **-h (Optional)**
  Provides Help for the Federated Search Utility.
- **-i**
  Specifies the option that is used for installing the adapter.
- **-k**
  Specifies the adapter key attribute.
- **-o**
  Specifies the Output XML filename to be generated. Name of the XML file to create and update the merged content of the XML template and SharePoint properties file.
- **-p (Optional)**
  Indicates the properties file name to merge with the Adapter Definition XML template.
- **-t**
  Specifies the name of the XML template file for defining an adapter.
- **-u**

Specifies the name of the adapter XML file to uninstall the adapter.

Invoke the Utility File to Configure the Search Adapters

A Search engine requires specially coded plug-in adapters. A plug-in search adapter translates generic search requests to a search engine proprietary format, calls the search engine, and returns the search requests.

NOTE
If CA SDM is configured for multi-tenancy, your Tenant is passed along to the search engine. Federated Search has built-in support for multi-tenancy. You can use federated search to isolate data by a tenant in a single SharePoint implementation.

WARNING
Do not use ampersands or spaces in the adapters.properties file values.

Follow these steps:

1. Encrypt the passwords for search adapters using the encrypt password utility. To encrypt passwords, navigate to the following CA SDM directory:
   ```
   $NXROOT\bin
   ```
2. Run the following command for generating encrypted passwords:
   ```
   encrypt_pwd [-e] <search engine password>
   ```
   The default option is -e.
3. Open the adapters.properties file from the following CA SDM directory:
   ```
   $NX_ROOT\samples\cafedsearch
   ```
4. Edit the adapters.properties file. Specify the appropriate parameters for the adapters you want to configure.
5. For SharePoint, update the following values in the adapters.properties file:
   - **sharepoint_username=**
     Enter the user name for SharePoint access.
   - **sharepoint_password=**
     Enter the encrypted password for SharePoint access as shown in Step1.
   - **sharepoint_hostname=**
     Enter the hostname.
   - **sharepoint_domainName=**
     Enter the domain name.
   - **sharepoint_protocol=**
     Enter the communication protocol (http or https).
   - **sharepoint_portNumber=**
     Enter the port number. Default is 80.
   - **Search Service URI path=**Default: _vti_bin/search.asmx
     sharepoint_searchServicePath=_vti_bin/search.asmx
   - **Authentication schemes=**One or more comma-separated values {Basic, NTLM}.
     Default: Basic
     For NTLM authentication, provide the following:
     sharepoint_authSchemes=**NTLM**
6. For Google, update the following values in the adapters.properties file:
   - **google_googleCx=**
     Enter the unique key value that Google uses to decide which Google Custom Search Account to use.
   - **google_googleApiKey=**
     Enter the unique key value that helps Google determine the identity of an application. To retrieve the key in the APIs Console, activate JSON/Atom Custom search API. This API provides a new API key for Simple API Access.
7. For CA Open Space, update the following values in the adapters.properties file:
   - **openspace_protocol=**

Enter the communication protocol (http or https).

- **openspace_portNumber=**
  Enter the port number for CA Open Space. Default is 8686.
- **openspace_default_tenant_userName=**
  If CA SDM is not configured with multi-tenancy, enter a username to perform search in CA Open Space.
- **openspace_default_tenant_password=**
  Enter the CA Open Space encrypted password. For more information, see Step1.
- **openspace_default_tenant_companyHost=**
  Enter the tenant company host details.

a. If CA SDM uses multi-tenancy, add an entry for each tenant in the openspace-tmpl.xml:
For example, if CA SDM has a tenant that is named Tenant 1, provide the following values in the openspace-tmpl.xml file:

```
<entry key="Tenant1">
 <bean class="com.ca.ServicePlus.cafedsearch.adapters.openspace.OpenSpaceCompanyDetail">
  <property name="userName" value="$(openspace_tenant1_userName)"/>
   <property name="password" value="$(openspace_tenant1_password)"/>
  <property name="companyHost" value="$(openspace_tenant1_companyHost)"/>
  </bean>
</entry>
```

b. Add the following entries for Tenant 1 in the adapters.properties file:

```
openspace_tenant1_userName=
openspace_tenant1_password=
openspace_tenant1_companyHost=
```

Repeat steps a and b for all required tenants in the openspace-tmpl.xml file.

8. Invoke the fs_adapters_cli once for each adapter you want to configure.
   - If you want to configure a CA Open Space adapter, use the bean value as OpenSpaceSearchAdapter and the template as openspace-tmpl.xml as shown in the following example:

     ```
     fs_adapters_cli -i -k OpenSpace -b OpenSpaceSearchAdapter -t "openspace-tmpl.xml" -o "openspace.xml"
     ```

   - To configure a Google adapter, provide the the bean value as GoogleSearchAdapter and the template as google-tmpl.xml as shown in the following example:

     ```
     fs_adapters_cli -i -k Google -b GoogleSearchAdapter -t "google-tmpl.xml" -o "google.xml"
     ```

   - If you want to configure a SharePoint adapter, use the bean value as SharePointSearchAdapter and the template as sharepoint-tmpl.xml as shown in the following example:

     ```
     fs_adapters_cli -i -k SharePoint -b SharePointSearchAdapter -t "sharepoint-tmpl.xml" -o "sharepoint.xml"
     ```

   Modify the -k and -o attribute values with a name of your choice. For more information about the federated search utility file and attributes,.see Federated Search Utility Files.
9. After the installation, if there are any errors, check the log file that is located in the CA SDM directory:

   ```
   $NXROOT\log\jfedsearch.log
   ```

10. Optionally, you can also create your own XML file for registration. All adapter entries are registered in the adapters-config.xml located in the following directory:

    ```
    $NX_ROOT\samples\cafedsearch
    ```

11. To create your own XML file for registration, you can also make a copy of the existing adapters-config.xml file name. Optionally, change the name of the modified adapters-config.xml file.
    Use the -c option with the modified XML file (xyz.xml) to register adapters.
12. Change the resource value in bean.xml <import resource="adapters-config.xml"/>.
13. Copy the adapters-config.xml or the modified xyz.xml (step 11) and any associate adapter-specific XML files for Google (google.xml), CA Open Space (openspace.xml), SharePoint (sharepoint.xml) in the following CA SDM directory:

    ```
    $NX_ROOT\bopcfg\www\CATALINA_BASE_FS\webapps\cafedsearch\WEB-INF
    ```

14. Restart the Federated Search Tomcat instance:
```
pdm_tomcat_nxd  - c STOP  - t FS
pdm_tomcat_nxd  - c START  - t FS
```
The federated search adapters are configured.
15. Verify and check the error log files in the CA SDM directory:
```
$NX_ROOT\log\jfedsearch.log
```

Create the Federated Search Sources in CA SDM

The CA SDM web UI is loosely coupled with the CAFedSearch Servlet. Configure Search Sources in CA SDM to create a new record.

Follow these steps:

1. Log in to the CA SDM application as Service Desk Administrator.
2. Select **Administration**, **Knowledge**, **Federated Search**, **Search Source** to create a new record that represents a federated search engine.
The Search Source contains the Name of the Federated Search Source that is displayed to users.

   NOTE

   Code value must match the key value attribute of an entry in the adapters-config.xml.
3. Click Create New to add a Federated Search Source.
4. Enter the following field values:
Search Source Name
Specifies a unique name for the search source.
Code
Specifies a unique identifier for the search source.
Record Status
Specifies the status of the search source. If the search source is inactive, you cannot find information from this source and this source option is not displayed in the Knowledge Search Source page.
5. (If multi-tenancy is enabled) Click Update Tenants and add the tenants.
6. Click Save.
You added the search source.

Search the Knowledge Solution using the New Custom Search Adapter

Search for knowledge documents and articles in the CA SDM application using the new custom search adapter.

Follow these steps:

1. Log in as a CA SDM Service Desk Administrator.
2. Navigate to **Administration**, **Knowledge**, **Federated Search**, **Search Sources**.
3. Add search source **Code** Key value.
4. Click **Save**.
5. Navigate to **Knowledge Management** using a new or existing ticket.
6. Search for the knowledge source that you have created. Click **Show Filter** to enter the search criteria and click **Search**.
The search information is displayed.

Modifying the Cross-Origin Resource Sharing (CORS) Filter

The default CORS filter setting allows any server to send requests to Federated Search. We recommend that you restrict this setting to allow incoming requests only from the web interface servers.

Follow these steps:

1. Navigate to the following CA SDM directory on the server where you have installed federated search:
   ```
   $NX_ROOT\bopcfg\www\CATALINA_BASE_FS\webapps\cafedsearch\WEB-INF\web.xml
   ```
2. Open the web.xml file and check for:
   ```
   <web-app><filter><init-param><param-name>cors.allowOrigin</param-name>
   ```
3. Update the <param-value> tag with a space-separated list of domains. For example, the base URL address of the servers from which you want to allow incoming requests:
   ```
   <param-value>http://web01:8080 http://web02:8080 http://web03:8080</param-value>
   ```
4. Save the web.xml file. Tomcat is restarted automatically.
5. Test and verify the Federated Search configuration on the web client interface. Ensure that it is functional.
6. For the web interface updates to persist after running the CA SDM Configuration, repeat the same updates to the web.xml.tpl file.

Uninstall the Search Adapter

To uninstall the search adapters, use the fs_search_cli utility command with the - u option. Use the following utility file command:

```
fs_adapters_cli -u -k <key> -b <bean> [-c <filename>]-f <filename>]
```

- **-u**
  Specify the -u option to uninstall the search adapters.
- **-k**
  Specify the key name value that is provided at the time of installation.
- **-b**
  Specify the bean value that is provided at the time of installation.
- **-c**
  Specify the name of the new search adapter configuration xml, if you created any while configuring the search adapters with Utility.
- **-f**
  Gives the Generated output file name at the installation time.

For example, if you want to uninstall the Google search adapter, perform the following steps:

Follow these steps:

1. Run the utility file command with the -u option for uninstalling the Google search adapter:
   ```
   fs_adapters_cli -u -k Google-b GoogleAdapter-f Google.xml
   ```
2. Check and verify that the adapters-config.xml removes registration for Google.
3. Google.xml file is deleted from the following CA SDM directory:
   ```
   $NX_ROOT\samples\cafedsearch
   ```
4. Copy the adapters-config.xml to the following directory:
   ```
   $NX_ROOT\samples\cafedsearch\WEB-INF
   ```
   Delete the google.xml file from this location.
5. Restart the FS Tomcat:
   ```
   pdm_tomcat_nxd  - c STOP  - t FS
   pdm_tomcat_nxd  - c START  - t FS
   ```
6. In the CA SDM UI, verify the Activate Search source entry for this Adapter (Google, in this case).
7. To uninstall other external search adapters, repeat steps 1 through 6.

How to Configure SDK Custom Search Adapters

This article contains the following topics:

> **WARNING**
> CA Federated SDK sample source code is provided to users for creating and deploying alternate custom adapters. However, Broadcom Support does not provide support to write the actual custom code. We do not actively support the creation and deployment of additional adapters beyond the SDK source code samples shipped with CA SDM.

The CA Federated Search SDK architecture is designed for extensibility. The SDK contains all the necessary JAR files to develop a custom adapter. Sample Eclipse projects can be imported, and Ant build targets are also provided to build custom search adapters. The SDK is located in the CA SDM directory:

```
$NX_ROOT\samples\sdk\fedsearch\adapters-source.tar.gz
```

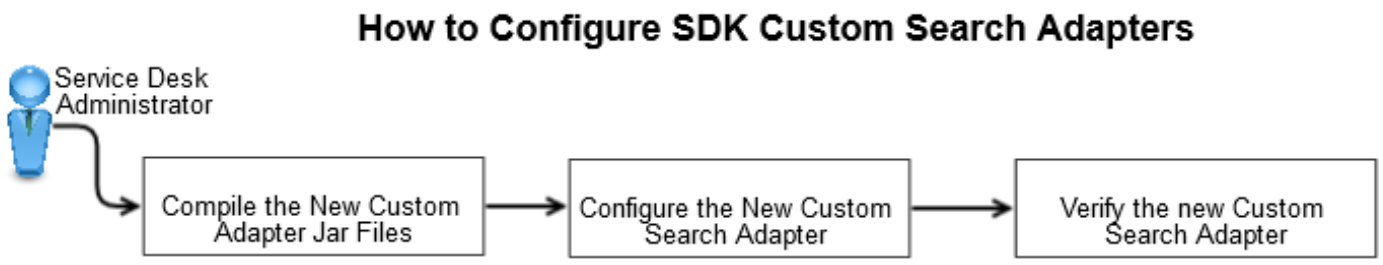
The source code of the following adapters can be used to write more custom adapters:

- Google
- CA Open Space
- Microsoft Share Point

> **NOTE**
> Source code for the CAFedSearch servlet, its underlying framework, and the security filter are not provided. No source code is provided for any of the Crawler Surface components.

The SDK component is written in java and is shared as a jar file (cafedsearch-adapter-sdk-1.0.0.jar). The SDK provides a simple interface to enable customers to develop and deploy their own search adapters.

Figure 35: How to Configure SDK Custom Adapters Diagram



Compile the New Custom Adapter Jar Files

Compile the custom adapter jar files successfully.

Follow these steps:

1. Compile the new custom search adapters. Ensure that you have the following jar files in your Java Classpath:
   - **jsr311-api-1.0.jar**
     This jar file is available in the CA SDM directory:
     ```
     %NX_ROOT%\java\lib\CXF\
     ```
   - **cafedsearch-core.jar**
     This file is available in the directory:
     ```
     %NX_ROOT%\bopcfg\www\CATALINA_BASE_FS\webapps\cafedsearch\WEB-INF\lib
     ```
   - **cafedsearch-adapter-sdk-1.0.0.jar**

This file is available in the directory:

```
%NX_ROOT%\bopcfg\www\CATALINA_BASE_FS\webapps\cafedsearch\WEB-INF\lib
```

- **log4j-1.2.15.jar (optional)**
  This file is available in the directory:

  ```
  %NX_ROOT%\java\lib
  ```

2. Write a new Java Class which extends the SearchAdapter Class and provides an implementation for the abstract method.

   ```
   search
   ```

   The CAFedSearch component invokes and passes search method parameters. These parameters are embedded inside the SearchOptions parameter for each search request from the client.

   NOTE
   Ensure that your implementation is thread safe as the CAFedSearch component maintains only one instance of the Java Class. For each search operation, search method is invoked on the same instance.
3. The SearchOptions parameter for the search method has the following methods:
   - **getSearchTerms()**
     Specifies the method for retrieving the search string.
   - **getStartIndex()**
     Specifies the start index method, the number from which the client wants to search items. Index starts from 1.
   - **getItemsPerPage()**
     Specifies the maximum number of search results that the client expects.

     NOTE
     You can also use other Java Class methods. For example: getUserId()
4. Send the collected information to the external search engine API for retrieving the search results.

   NOTE
   For information about Java Class Methods, see Java Documentation.
5. The search method returns an instance of ResultCollection class. Create an instance of ResultCollection and populate values using the following methods:
   - **setSources(String name)**
     Specifies the name of the search adapter. Names are case-sensitive and must match exactly with the name that is provided in the utility configuration file. For convenience, SearchAdapter provides a method getName() which should be used.
     For example:

     ```
     results.setSources(getName());
     ```
   - **setTotalResults (int total)**
     Specifies the total search result count.
   - **setStartIndex(int startIndex)**
     Specifies a start index of results. This value is as per results from your search engine.

     ```
     results.setStartIndex(startIndex);
     ```
6. A collection of ResultItems must be passed to the ResultCollection object by calling the setSearchResultItems method. To add an instance of ResultItem at a time, use the addSearchResultItem() method.

   NOTE
   For more information about ResultCollection Java class, see the Java documentation.
7. The ResultItem class has the following important methods, which must be filled for each search result item (row).
   - **setContentText(String txt)**
     Specifies a method for setting the search result actual content.
   - **setContentHTML(String txt)**
     Specifies a method for setting the HTML (can contain the HTML tags) content. If the search engine gives HTML highlighted, then set the highlighted text using this method.

NOTE
If your search engine does not have this feature, you can write a simple Java class method to highlight the text. The CA Open Space adapter has a simple method to bold terms in the search results.

If the search adapter requires more jar files, modify the build.xml to compile and prepare an adapter jar file. Ant binaries are required to use the build.xml. Use Ant to run the targets in build.xml to compile and make the JAR files Keep the build.xml along with your source (src) folder. The build.properties file is optional. For more information about Ant binaries, see Ant Help.

The jar file is successfully compiled.

Configure the New Custom Search Adapter with the CAFedSearch Component

Configure the new search custom adapter jar file with the CAFedSearch Component for creating the CA SDM search sources.

Follow these steps:

1. Copy the jar file from the location dist\lib created in Compile the New Custom Adapter Jar Files to the following CA SDM directory:
   ```
   %NX_ROOT%\bopcfg\www\CATALINA_BASE_FS\webapps\cafedsearch\WEB-INF\lib
   ```
2. Navigate to the following CA SDM directory:
   ```
   $NX_ROOT\sample\cafedsearch\
   ```
3. Create template xml file for the custom search adapter. For example, the XYZ-tmpl.xml file.
4. The template file contains the following contents:
   ```
   <?xml version="1.0" encoding="UTF-8"?>
   <beans default-autowire="byName" xsi:schemaLocation="http://www.springframework.org/schema/beans
   http://www.springframework.org/schema/beans/spring-beans-2.5.xsd
   http://cxf.apache.org/core
   http://cxf.apache.org/schemas/core.xsd
   http://cxf.apache.org/jaxrs
   http://cxf.apache.org/schemas/jaxrs.xsd" xmlns:cxf="http://cxf.apache.org/core"
   xmlns:jaxrs="http://cxf.apache.org/jaxrs"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
   xmlns="http://www.springframework.org/schema/beans">

       <bean autowire="autodetect"
           class=" com.abc.xyz.XYZAdapter "
               id="XYZAdapterConfiguration" scope="singleton">
   <property name="username" value="$(xyz_username)"/>
       <property name="password" value="$(xyz_password)"/>
   </bean>
   <bean autowire="autodetect" class="com.abc.xyz.XYZAdapter" id="XYZSearchAdapter" init-method="init"
    destroy-method="destroy" depends-on="XYZAdapterConfiguration">
               <property name="config" ref="OpenSpaceAdapterConfiguration" />
       </bean>
   </beans>
   ```
5. You can create property values that are based on the input values provided for the search adapters in step 4. Add xyz_username and xyz_password in the adapters.properties file and provide values to the property.
6. Run the utility file to generate configuration XML for custom search adapters:
   ```
   fs_adapters_cli -i -k XYZ - b XYZSearchAdapter -t "XYZ-tmpl.xml " -o "xyz.xml
   ```
7. The xyz.xml file is created and registered in the adapters-config.xml file.

8. Copy the xyz.xml file and adapters-config.xml to the following CA SDM directory:
   ```
   $NX_ROOT\samples\cafedsearch\WEB-INF
   ```
9. Run the following command to restart Federated Tomcat Services:
   ```
   pdm_tomcat_nxd  - c STOP  - t FS
   pdm_tomcat_nxd  - c START  - t FS
   ```
10. Create the federated search sources for the XYZ adapter in the CA SDM UI.
    For more information about federated search sources, see Create the Federated Search Sources in CA SDM.
11. In CA SDM, navigate to the **Knowledge Management** Tab. Select the XYZ check box and perform search operations. The new custom search adapter is successfully configured with the CAFedSearch component.

Verify the new Custom Search Adapter

Verify the functionality of your new custom search adapter by using the REST client.

Follow these steps:

1. Download the REST client from the Internet:
   https://code.google.com/p/rest-client/
2. Launch it using Java (JRE is required).
3. Enter the following URL in the REST client UI:
   ```
   http://sdmhostname:<FS_Tomcat_Port>/cafedsearch/sdm/search?q=search&userid=<sdmuserid>&source=<Adapter
    Name>
   ```
4. Turn off the CA SDM security interceptor. Open the following file for editing and comment out the “<jaxrs:inInterceptors> “ section:
   ```
   $NX_ROOT\bopcfg\www\CATALINA_BASE_FS\webapps\cafedsearch\WEB-INF\beans.xml
   ```

   NOTE
   Turning off the security filter is not recommended in a production environment. Turn off the Security filter in a test environment for verifying the new custom search adapter functionality.
5. Verify that the new custom search adapter is functioning properly. If search results are not displayed, try reviewing the logs, or increase the log level to Debug.

Call the CAFedSearch Servlet Using REST

The CAFedSearch servlet exposes a RESTful interface where custom search clients, programs, and User Interfaces (UI) can send search requests.

This RESTful interface only accepts the HTTP Collection GET requests following the OpenSearch specification. It provides support for JSON and XML responses. Each request must contain a Service Desk BOPSID token which can be obtained from CA SDM RESTful or SOAP Web Services.

Follow these steps:

1. Obtain a BOPSID token using a CA SDM RESTful Web Service.
2. Get a REST Access Key by sending an HTTP POST request on the rest_access resource along with login credentials. You can also obtain a BOPSID token using the REST Access Key by sending an HTTP POST request on the bopsid resource.
3. For the details on sending requests to CA SDM RESTful Web Services, see the sample files in the following CA SDM directory:
   ```
   NX_ROOT\samples\sdk\rest\java
   ```
4. To use the Federated Search API for searching, send an HTTP GET request on the search resource and pass search criteria and BOPSID token through the CA SDMURL.
   ```
   GET
   http://<sdmhostname>:<FS_TOMCAT_PORT>/cafedsearch/sdm/search?
   q=<searchTerms>&source=<adapterName>&BOPSID=<bopsidToken>&userid=<userId>
   ```

- **searchTerms**
  Specifies a space delimited list of keywords. Must be URL encoded.
- **adapterName**
  Specifies search engine name as specified in the Search Source record Code field using the adapters configuration utility.

Other supported arguments are as follows:

- **index**
  Specifies the first index that is desired in the search results, must be integer > = 1.
- **page**
  Specifies Indicates the first page that is desired in the search results, must be integer >= 1.
- **size**
  Specifies the number of results per page that is desired by the search client.
- **type**
  Valid values include JSON or XML.

How to Configure the Crawler Surface for SharePoint

This article contains the following topics:

The administrators can configure CA SDM to allow Microsoft SharePoint 2010 and SharePoint 2013 Servers to crawl the Crawler Surface. The Crawler Surface contains a FSCrawl Servlet component that lets you search the CA SDM knowledge solutions and articles.

A crawler is a search engine component that browses the Internet and indexes search terms. The Crawler Surface is a read-only web-based interface to the CA Service Desk Manager application. With this interface, external search engine crawlers can discover information using the Java Server Page (JSP) technology. The Crawler Surface uses the JSP to provide the information in plain text and individual hyperlinks to tickets and knowledge documents.

> **NOTE**
> The CA Service Desk Manager content that is exposed to a crawler can be modified using an XML configuration file. For more information, see Crawler Surface XML Configuration File.

Complete the Prerequisites

Complete the following prerequisites before you start configuring the Crawler Surface for a Microsoft SharePoint Server:

- To enable Federated Search, select the Configure Federated Search option while installing CA Service Desk Manager.
- Identify a dedicated User ID for accessing the search result information. The User ID controls the information that is present in the Crawler Surface. For multi-tenancy configuration, you can create several user IDs to allow the segregation of tenant information.

Create the CA Service Desk Manager User Crawler Surface

In your CA Service Desk Manager configuration, create user identities to segregate information by tenant.

The process of creating a user ID for the Crawler Surface is the same as creating the CA Service Desk Manager user identities for regular users. Create a contact with Access Type and Role as "Crawler" in the CA Service Desk Manager Application. The Crawler Access Type and Role provide the user with read-only access to the CA Service Desk Manager data. A contact is a person who uses the system regularly, such as an analyst or customer. After you have created the business structure and groups, you can create contacts and can map them to their respective location and organization. You can create contacts using the following ways:

- Create Contacts Manually
- Create Contacts Using the LDAP Database

Configure the Tomcat Remote IP Address Filter

Change the Tomcat Remote IP Address Filter setting to point to the remote system that hosts the SharePoint Server. The IPV4 and IPV6 addresses are supported.

The crawler surface uses the Tomcat Remote IP Address Filter mechanism to access the CA Service Desk Manager information. The Tomcat filter mechanism uses an IP Address pattern (maintained by the CA Service Desk Manager administrator) to match authorized IP addresses. By default, the Remote IP Address filter is configured with the loopback adapter IP address 127.0.0.1. For a secure communication in a production environment, consider using Secure Sockets Layer (SSL) encryption between the crawler and the crawler surface.

Follow these steps:

1. Log in to the following server, depending on your CA Service Desk Manager configuration:
   - Conventional: Primary or Secondary Server
   - Advanced Availability: Application Server
2. Open the web.xml file in the following CA Service Desk Manager directory:
   ```
   $NX_ROOT\bopcfg\www\CATALINA_BASE_FS\webapps\fscrawl\WEB-INF
   ```
3. Save a backup of the web.xml file.
4. Find the <filter-name>Remote Address Filter</filter-name> section. Change the pattern in <param-value> in the <filter>.
   This parameter allows you to specify a range of IP Address patterns and provides access from the remote system hosting the SharePoint server. For more information about the Tomcat Remote IP Address Filter, see the Apache Tomcat documentation.
5. Save the XML file.
6. Restart the Federated Search Tomcat server.
   The Tomcat Remote IP Address Filter value is changed and the Crawler Surface can now be accessed from the remote system.

   NOTE
   Do not access the crawler surface from an unconfigured IP address. You are redirected to the CA Service Desk Manager UI.
7. The Crawler Surface is accessed through a URL like other web application. To validate SharePoint logging, enter the following URL in a browser:
   ```
   http://<sdmhostname>:<FS_TOMCAT_PORT>/fscrawl/index.jsp?farm=<FarmName>
   ```

   NOTE
   All elements of the URL are case-sensitive.

Configure the Crawler Surface User ID

The CA Service Desk Manager information content that is exposed to a crawler surface can be modified using a configuration xml file. External search engine discovers information using the Java Server Page (JSP) technology. The crawler surface user ID controls the information that is present in the Crawler Surface.

To modify the crawler surface, you can configure the crawler surface user ID in the crawler_surface_config.xml file.

WARNING
Verify the language settings of your browser as Microsoft SharePoint is sensitive to the language used in the search request.

Follow these steps:

1. Log in to the following CA Service Desk Manager server that hosts the Crawler Surface depending on your install configuration:

- Conventional: Primary or secondary Server
- Advanced Availability: Application Server

2. Open the crawler_surface_config.xml from the following CA Service Desk Manager directory:
   `$NX_ROOT\bopcfg\www\CATALINA_BASE_FS\webapps\fscrawl\WEB-INF`
3. Save a backup of the crawler_surface_config.xml file.

   WARNING
   Make all XML file modifications in a test environment before porting the final changes to a production server.
4. In the original file, locate *<sdm_user>CHANGE_THIS</sdm_user>* under each *<farm>* section and replace *CHANGE_THIS* with the crawler user ID that you created earlier.
5. You can further modify and restrict the XML object attributes depending on your requirements.

   NOTE
   For more information, see the Crawler Surface XML Configuration File.
6. Save the XML file.
7. Restart the Federated Search Tomcat server to reload the file.
   The Crawler Surface XML file is modified.

Crawler Surface XML Configuration File

The crawler_surface_config.xml file contains the following XML sections.

- **<objects>**
  Specifies the information about the objects and attributes that the Crawler Surface exposes for an object. The objects section describes the layout of a detail page for each object type that is exposed to a crawler. This section does not control the selection of individual records. The <objects> section is a collection of <object> sections.
  Each object is defined in an <object> section. The default specifications for these objects are provided as:
  - **KD**
    Specifies Knowledge Documents.
  - **chg**
    Specifies Change Orders.
  - **iss**
    Specifies Issues.
  - **in**
    Specifies Incidents.
  - **pr**
    Specifies Problems.
  - **cr**
    Specifies Requests.

The XML file contains the following sections that create the <head> section of a detail page in CA Service Desk Manager:

- **<name>**
  Specify the Majic object name of the exposed object.
- **<note>**
  Specify a location for a short description of the object. This element is only for documentation purposes, and the Crawler Surface ignores this element.
- **<last_mod_dt>**
  Specify the attribute name that stores the last modified date and time. The time and date is exposed to the search engine crawler to allow the search engine to determine whether the record was updated. The search engine crawler skips the crawl when the record is not updated.
- **<title>**
  Specify the attribute that is used for the title of the detail page. The search engines use this element as the title of the document that is returned in search results. This element entry generates an HTML <title> tag in the <head> of the

detail page. For a knowledge document, the title defaults to the title of the knowledge document. The summary is used for the title for incidents, problems, requests, change orders, and issues.

- **<meta_data>**
  Specify one or more properties that is exposed as a metadata. A Metadata allows a search engine to store extra characteristics of the document in its index. Metadata is not searched directly but instead is used to filter search results. This section generates HTML <meta> tags in the <head> of the detail page.
  Each entry in the <meta_data> section contains one or more <property> entries. Each <property> element consists of a <name> element and a <content> element.
  - **<name>**
    Specify the name of the metadata property.
  - **<content>**
    Specify the attribute of the object that is used as the value for the metadata.
    Together, each <name> and <content> element pair of a <property> generate an HTML <meta> tag. The search engine crawlers use the following two metadata properties by default:
    - **Description**
      Specify the metadata property of a search engine that stores a short summary of the document.
    - **Author**
      Specify the author of the document.

The CASDMTENANT metadata property is also configured by default for each object. This property is a CA Service Desk Manager specific-metadata property. The Crawler Surface uses this property to expose the Tenant name of the object to the crawler of the search engine. During a Federated Search, the results that are obtained from the search engine are filtered based on this metadata property.

The Crawler Surface XML Configuration file contains the following attributes that are used to create the <body> section of a detail page in CA Service Desk Manager:

- **<additional_attributes_to_index>**
  Indicates a list of attributes from the object that the Crawler Surface exposes. Separate multiple entries with a comma and a space. For example, PROBLEM, RESOLUTION, SD_ASSET_ID.name.
- **<activity_logs>**
  The Crawler Surface displays information from Activity Logs for objects that have activity logs. The <activity_logs> section contains the <object>, <select_criteria>, <rel_attr>, and <attributes> elements.
  - **<object>**
    Specifies the object name that contains the activity log entries for the object. For example:
    - *alg* is the activity log object entry for Incidents, problems, and requests.
    - *chgalg* is the activity log object entry for change orders.
    - *issalg* is the activity log object entry for issues.
    - O_COMMENTS is the activity log object entry for knowledge documents.
- **<select_criteria>**
  Allows you to filter the activity log objects that are exposed. This element is important to increase the relevancy of your search results by decreasing frequently occurring words. For example, the <select_criteria> for chgalg contains the following Magic Where clause:

  ```
  "type IN ('ST', 'UPD_RISK', 'CB', 'RS', 'LOG', 'TR', 'ESC' ,'NF', 'UPD_SCHED')"
  ```

  Includes only activity log entries that allow a user to enter comments and eliminates activity log entries with fixed text like Initial or Attached document.
- **<rel_attr>**
  Specifies how an activity log entry relates to its parent object. The <rel_attr> subsection contains <parent_obj_attr> and <join_attr> elements.
  - **<parent_obj_attr>**

Indicates an attribute of an activity log that contains an SREL (or foreign key pointer) to the parent object. For example, the change_id is the attribute of activity log object *chgalg*.

– **<join_attr>**
Indicates the relational attribute (Rel Attr) of the parent object that is stored in <parent_obj_attr>. You can verify these values by using the following command:

```
bop_sinfo -df chgalg
```

You can verify both of these values by using the bop_sinfo -df chgalg command. The output must show that the value for change_id is *SREL -> chg.id* and ISS is *SREL -> iss.persistent_id.*

- **<attachments>**
The attachments subsection allows you to expose attachments to the crawler so that content can be indexed. The <attachments> section is only allowed for objects that have attachments.

The attachments are handled in a special manner by the Crawler Surface. The Crawler Surface exposes a hyperlink that the crawler follows to download the attachment from CA Service Desk Manager. If an attachment is included in the search results, you can click on the hyperlink to navigate to the parent object.

The <attachments> section contains <object>, <rel_attr>, <attmnt_id>, and <is_parent_updated> elements.

- **<object>**
This element specifies the Majic object that links the Attachment to its parent object.
- **<rel_attr>**
This subsection works the same as it does in activity logs. Specifies how the parent object relates to this object which links the parent object to the attachment.
- **<attmnt_id>**
This element specifies the attribute of this linking object that points to the attachment.
- **<is_parent_updated>**
Specifies the Crawler Surface on how to expose the last-modified date for the object. For some objects like Knowledge Documents (KDs) when an attachment is added, the last modified date of the Knowledge document is not updated. The last-modified date is important when the search engine is doing an incremental crawl.
- **<configuration_items>**
Used for objects that contain a list of configuration Items. This section contains the <object>, <rel_attr>, and <attributes> elements.
  – **<object>**
  Works the same as in activity logs and attachments.
  – **<rel_attr>**
  Work the same as they do in activity logs and attachments.
  – **<attributes>**
  This element works the same as in attachments.
- **<multi-farm_datasets>**
The <multi-farm_datasets> specifies how records are selected. The <multi-farm_datasets> section is a collection of <farm> sections.
- **<farm>**
Each <farm> section controls the CA Service Desk Manager information that is exposed to a crawler. When a crawler is configured, the <farm> section is specified in the URL. Only the information that is specified in the <farm> section is exposed to the crawler. Each <farm> section contains <name>, <data_sets>, and <sdm_user> elements.<name>.

  NOTE
  This value is case-sensitive.

Configure the SharePoint Crawler

Configure Crawlers to crawl and search for content in SharePoint. The Crawler is a multithreaded application capable of high throughput and can sometimes have a negative impact on the CA Service Desk Manager performance. To improve performance, ensure that you have considered the following factors:

- Limit the number concurrent SharePoint crawle rs accessing the Crawler Surface at any one time.
- Use SharePoint Crawler Impact Rules to throttle the crawlers
- Schedule crawls at off-peak times of the day
- Dedicate an Object Manager <sdm_domsrvr_name> to the Crawler Surface in crawler_surface_config.xml. For more information, see Crawler Surface XML Configuration File.
- For CA Service Desk Manager Advanced Availability, dedicate an entire Application Server to the Crawler Surface.

Follow these steps:

1. Create the Content Source in SharePoint
2. Create Crawler Rules
3. Run the Crawler in SharePoint
4. Configure the Metadata in SharePoint
5. Verify the Crawler Data in SharePoint

Create the Content Source in SharePoint

Create Content Sources for identifying the type of content that the SharePoint crawler processes.

> **NOTE**
> The names of SharePoint specific settings can vary depending on the SharePoint version you are using. For more information about creating content sources in SharePoint, see the Microsoft SharePoint Documentation.

Follow these steps:

1. Log in to the MS SharePoint Central Administration console.
2. Click **Manage Services Application**, **Search Service Application**.
3. Click **Content Source** for creating content Sources:
4. Enter data name for the Content Source in **Name** as CA Service Desk Manager.
5. Set the Content Source Type to Web Sites.
6. Enter the following URL in **Start Address**:

   ```
   http://<sdmhostname>:<FS_TOMCAT_PORT>/fscrawl/listObject.jsp?farm=<Farm Name>
   ```
7. Click **Save**.

Create Crawl Rules

The Crawl rules define how the SharePoint Web Crawler Surface URL is crawled. Define the following Crawl Rules:

- A crawl rule that lets SharePoint crawl the Crawler Surface.
- A crawl rule that allows SharePoint to access attachments.

Follow these steps:

1. Log in to the Microsoft SharePoint Central Administration console.
2. Click **Manage Services Application**, **Search Service Application**.
3. Click **Crawler Rule**. Create new Crawler Rule.
4. Enter the following URL in the browser:

   ```
   http:// <sdmhostname>:<FS_Tomcat_Port>/fscrawl/*farm=<farm-name>*
   ```

   For example: http:// <sdmhostname>:8040/fscrawl/*farm=kd*

   > **WARNING**
   > The Crawler Surface URL is case-sensitive. SharePoint changes uppercase hostnames to lowercase. For SharePoint 2010, ensure to select the **Match Case** check box.
5. Select '**Include all items in this path**' to configure the crawler.
6. Select '**Crawl complex URLs (URLs that contain a question mark - ?)**'
7. Select '**Specify a different content access account**'.

8. If you have Disabled the Basic Authentication in CA Service Desk Manager Crawler Surface;
   a. Select one of the following options:
      - (SharePoint 2013), Select the '**Specify a different content access account**' option.
      - (SharePoint 2010), Select the default content access account (NT AUTHORITY\NETWORK SERVICE).
   b. Save this Crawl rule and skip to step 10 for defining second crawl rule.
9. Enter the CA Service Desk Manager user account name and password for the Crawler Surface.
10. Repeat steps 3 - 6 and create a second crawler rule for the CA Service Desk Manager attachments using the following URL in step 4:

    ```
    http://<sdmhostname>:<WEB_TOMCAT_PORT>/CAisd/*
    ```

    For example: http://<sdmhostname>:8080/CAisd/*

    NOTE
    <sdmhostname> is the configured host name that is used for downloading the CA Service Desk Manager attachments.
11. Select '**Specify a different content access account**'.
12. If you have Disabled the Basic Authentication in CA Service Desk Manager Crawler Surface;
    a. Select one of the following options:
       - (SharePoint 2013), Select the '**Anonymous Access**' option.
       - (SharePoint 2010), Select the default content access account (NT AUTHORITY\NETWORK SERVICE).

       NOTE
       The Crawler Surface uses Basic Authentication. The CA Service Desk Manager Repository Daemon uses proprietary BOPSID security which is not directly supported by Microsoft SharePoint.
    b. Save this Crawl rule

    The Microsoft SharePoint Crawl rule is created.

Start a Crawl in SharePoint

Start a full or incremental crawl of the content sources in SharePoint to index the search content.

Follow these steps:

1. Navigate to the Microsoft SharePoint Central Administrator page.
2. Click **Manage Services Application**, **Search Service Application**.
3. Click **Content Sources**.
4. Select **Start Full Crawl** or **Start Incremental Crawl**.
   A full crawl crawls the entire content under a content source. Full crawls take more time and resource to complete than Incremental crawls.
   In an incremental crawl, the index remains intact, and the crawler crawls only the content that is added or modified since the last successful crawl. For more information, see the Microsoft SharePoint Documentation.

Configure the Metadata in SharePoint

Note: This topic is applicable to CA Service Desk Manager multi-tenancy environments.

When the crawler encounters a CA Service Desk Manager metadata, it stores the metadata in SharePoint as Crawled properties. These properties are discovered during the initial full crawl of the CA Service Desk Manager Crawler Surface. The SharePoint crawler discovers the metadata and creates the Crawled properties.

The metadata is used to pass extra information to a crawler in the detail pages using the <meta> tag in the <head> section. This information is available for searching and filtering. When CA Service Desk Manager is configured for multi-tenancy, the Crawler Surface exposes only the tenant metadata information.

When you perform a Federated Search, the tenant name is passed to the search engine to filter the results appropriately.

Follow these steps:

1. Log in to the Microsoft SharePoint Central Administration console.
2. Click **Manage Services Application**, **Search Service Application**, **Metadata** (SharePoint 2010), or **Search Schema** (SharePoint 2013).
3. Ensure that the SharePoint crawling is successful on the CA Service Desk Manager data.
4. Click **Managed Properties** and create the following properties:
   a. *CASDMTENANT*
   b. *CASDMCRAWLERFILE*
5. Create the *CASDMTENANT* property:
   a. Click **New Managed Property**.
   b. Enter **CASDMTENANT** as Property Name.
   CASDMTENANT Indicates the tenant name for the CA Service Desk Manager object. The sub tenant information is not displayed.
   c. Select **Text** as **Property Type**.
   d. Scroll down. Click **Add a Mapping**.
   e. Search for the *CASDMTENANT* Crawl Property and select it.
   f. Click **OK** to save the new Managed Property.
6. Repeat step 5 to create the *CASDMCRAWLERFILE* property.
   The metadata in SharePoint is configured.

Verify the Crawler Data in SharePoint

Verify the Crawler data in SharePoint to display search results.

Follow these steps:

1. Log in as CA Service Desk Manager Service Desk Administrator.
2. Click the **Knowledge Management** tab for a new or existing ticket.
3. Select **SharePoint Search Source**.
4. Enter the search key. Click **Search**.
   The Crawler displays the search results.

(Optional) Disable Basic Authentication

The Basic authentication authenticates and authorizes the users accessing CA Service Desk Manager content from an HTTP client and provides additional security to the Crawler Surface. However, basic authentication is an overhead for every HTTP request to the Crawler surface and eventually impacts product performance. If you have configured the remote IP address filter on Tomcat to limit the computers that can access CA Service Desk Manager content, you can disable basic authentication.

Follow these steps:

1. Take a backup of web.xml at $NX_ROOT\bopcfg\www\CATALINA_BASE_FS\webapps\fscrawl\WEB-INF.
2. Open the file using a text editor.
3. Comment or remove the tags <security-constraint>, <login-config>, <security-role>.
4. Save the file.
5. Restart FS Tomcat.

The basic authentication for Crawler Surface is disabled.

Troubleshooting

The Crawler Surface has the usual array of log files:

1. If you want to enable the debug mode for Federated Search, navigate to the following CA Service Desk Manager directory:

```
$NX_ROOT\bopcfg\www\CATALINA_BASE_FS\webapps\cafedsearch\WEB-INF
```

2. Open the log4j.properties file and modify info to debug mode.
3. To enable debug mode for fscrawl, navigate to the following CA Service Desk Manager directory:
   ```
   $NX_ROOT\bopcfg\www\CATALINA_BASE_FS\webapps\fscrawl\WEB-INF
   ```
4. Open the log4j.properties file and modify info to debug mode.
5. To correct the syntax errors that are encountered while configuring the SharePoint crawler surface, open the jfscrawl log file from the CA Service Desk Manager directory:
   ```
   $NX_ROOT\logs directory
   ```
6. If you locate any syntax errors, correct the XML file and restart the Federated Search Tomcat. The log is located in the CA Service Desk Manager directory:
   ```
   $NX_ROOT\logs\jfscrawl.log
   ```
   For example, if a <meta_data> tag is accidentally corrupted, then the log indicates the following error:
   ```
   08/06 15:43:52.624 [pool-2-thread-1] ERROR FSCrawlApplicationListener 302 XmlException::Problem
    loading config_file::C:\PROGRA~2\CA\SERVIC~1\bopcfg\www\CATALINA_BASE_FS\webapps\fscrawl\WEB-INF
   \crawler_surface_config.xml:274:8: error: </meta_dataxxxxx> does not close tag <meta_data>
   08/06 15:43:52.625 [pool-2-thread-1] ERROR FSCrawlApplicationListener 144 crawler_surface_config.xml could
    not be loaded, cannot read.
   ```
7. If there are no syntax errors, the following message is displayed:
   ```
   08/06 15:46:27.924 [pool-2-thread-1] INFO FSCrawlApplicationListener 58 fscrawl context had been loaded
    successfully.
   ```
8. Correct any other errors that do not show up until you try to access CA Service Desk Manager.
   For example, an unknown attribute xxxxx is requested to be exposed for Incidents in the <additional_attributes_to_index> element of crawler_surface_config.xml. The Crawler Surface application does not detect the error. But, when the Crawler Surface sends the request to the Object Manager, the error is detected and reported in the stdlog.x file as follows:
   ```
   08/06 15:51:23.92 SDMSERVER domsrvr 10860 ERROR domset.c 8049 Unknown attribute "xxxxx" requested from
    domset MLIST_STATIC of factory
   ```
9. Use the bop_sinfo -d command to resolve the error.
10. Modify the crawler_surface_config.xml file.
11. Restart the Federated Search Tomcat.
    The Crawler Surface objects are configured without any errors.

Knowledge Management Reports and Metrics

This article contains the following topics:

You can monitor the efficiency of the knowledge base using the reporting tools that are described in this article. These tools let you view statistics on the usefulness of your documents and their effectiveness in solving problems.

Knowledge Report Card

The Knowledge Report Card displays information about the knowledge contribution from each end user and provides feedback about which knowledge documents are most effective. You can use the information to improve the processes of creating knowledge documents and providing the best support to end users in your environment.

Define Knowledge Report Card Statistics

As an Administrator, you define the schedule at which the product calculates and sends notifications about the Knowledge Report Card. You also define content of Knowledge Report Card notification emails.

NOTE
Rework versions and retired documents are not presented when the Knowledge Report Card calculation is run.

Follow these steps:

1. Select the **Administration** tab, **Knowledge**, **Knowledge Report Card**.
   The Report Card page opens.
2. Complete the following fields as appropriate.
   - **Last Updated**
     Runs the Report Card calculation.
     Default: Deactivated
     Note: When the calculation is not run, and the statistics to present the data are not collected. The following message appears when the Knowledge Report Card command is specified from the View menu on the Knowledge tab: "Please Run Report Card Calculation."
   - **Schedule**
     Schedules the Report Card.
     - **Report Card Calculation will next run on xxx and will run every xxx**
       Specifies the frequency at which to recalculate the Report Card statistics.
     - **Report Card Email Notifications will be sent on xxx and will be sent every xxx**
       Specifies the frequency at which to send the Report Card notifications.
       Default: Never
     - **Report Card Email Should Display Statistics for the Past xxx**
       Specifies the amount of time for which the Report Card notification contains information. This field is only available when you select a value other than Never from the Report Card Email Notifications will be Sent Every xxx list.
       Default: 365 days
3. Click **Save**.
   The Knowledge Report Card statistics are defined.

Run the Knowledge Report Card

The Knowledge Report Card displays information about the knowledge contribution from each end user and provides feedback about which knowledge documents are most effective. You can use the information to improve the processes of creating knowledge documents and providing the best support to end users in your environment.

Note: After you run the knowledge report card, the analyst will be able to view their knowledge report card statistics.

Follow these steps:

1. Select the **Administration** tab, **Knowledge**, **Knowledge Report Card**.
   The Knowledge Report Card appears.
2. Complete the following fields and click Save.
   - **Last Updated** -- Select the Run Calculation check box.
   - **Schedule** -- Specify a date and time from when CA SDM performs the calculation and runs the Report Card.

     NOTE
     For information about using Automated Policies and the Knowledge Report Card, see View Document LifeCycle Policy Reports

Web-Based Reports for Knowledge Management

CA Business Intelligence installs a set of predefined Knowledge Management reports. These reports are automatically deployed to the CA Business Intelligence after the CA SDM installation.The reports are developed with either BusinessObjects Web Intelligence or Crystal Reports. The authorized users can display reports on the CA SDM Reports Tab.

Role-Based Report Web Forms

If you are an authorized manager or analyst, perform the following:

Follow these steps:

1. Click the Report List icon on the Reports tab to define the report web forms.
2. Additionally, you can click the **Administration**, **Security and Role Management**, **Role Management** to see how Report web forms are managed through the Role List.

Define Knowledge Report Card Statistics

The Knowledge Report Card provides feedback to analysts and administrators about which knowledge documents are most effective. You can use the provided information to improve the processes of creating knowledge documents and providing the best support to customers.

> **NOTE**
>
> These statistics are collected only when the Knowledge Report Card calculation service is run, as specified by your Knowledge Management administrator.

As an Analyst, you use the Knowledge Report Card to view statistics about knowledge you submitted and reviewed. These statistics are collected only when the calculation runs, as specified by your Knowledge Management administrator.

> **NOTE**
> The rework versions are not presented when you run the Knowledge Report Card calculation.

Follow these steps:

1. On the Knowledge tab, select View, Knowledge Report Card.

   > **NOTE**
   > Access to the Knowledge Report Card tab depends on your role.

   The Knowledge Report Card appears.
2. Complete the following fields as appropriate.
   - **My Documents:** Specifies the number of days on which to recalculate the Report Card statistics.
   - Specifies the section on which to recalculate the Report Card statistics.
   - **My Statistics:** Displays the totals of any documents the analyst has authored as follows:
     - **Submitted --** Specifies the document totals, regardless of status (Published, Draft, Rework-Draft, Retired).
     - **Published --** Specifies the number of published documents.
     - **Hit --** Specifies the total number of page visits to the knowledge documents.
     - **Voted --** Displays the number of votes an individual document has received from its Solution Survey form. You can click the Vote column header to display all documents in descending order, by document votes.

How to find the CA SDM Knowledge Documents Assigned to Other Users?

Perform the following steps to find out the CA SDM knowledge Documents assigned to other users:

1. Login to CA Service Desk Manager (CA SDM) as an Administrator.
2. Navigate to the **Administration** tab and click **Service Desk.**
3. Under Service Desk, select **Application Data**, **Stored Queries**.
4. Select **SHOW FILTER** on the top right. Under the **Type** drop down box, select **Knowledge Document** and click **Search**.
5. Right-click inbox and select copy. Change the label name as per your requirement.
6. Change the *Where* Clause to the following:

```
ACTIVE_STATE=1 and ASSIGNEE_ID!=@cnt.id
```

7. Change CA SDM Role to Knowledge Analyst and follow these steps:
   a. On the **Service Desk** tab, click **File**, **Customize Scoreboard.**
   b. Under **Add New Node**, add a node label, click **Node's Stored Query** and search for the label you created while you were logged into CA SDM Administrator by selecting Knowledge Document from the **Type** drop-down menu and click **Search**.
   c. Select the Stored Query you created. Click **Add New Node** and click **Finish**.
   You can now view the CA SDM Knowledge Documents that are not assigned to you from the scoreboard created

Edit Knowledge Documents

Contents

You can edit a knowledge document that has not yet been published. All users with full write permissions can edit documents by default, but the administrator specifies your knowledge document permissions. For example, your administration configures Knowledge Management so that only an assignee, the document owner, or a knowledge manager can edit documents.

To edit a document from the Knowledge tab

1. On the Knowledge tab, select the appropriate category.
   The Knowledge Document List page displays documents in the selected category.
2. Right-click the title of the document, and select Edit from the shortcut menu.
   The Update Knowledge Document page appears.

   NOTE

   If the document has been published, create a rework version to modify the content.
3. (Optional) The properties of a knowledge document can be changed on the Attributes tab, but you must have write access to the document to modify the properties.
4. (Optional) Complete or change the fields on the various tabs as appropriate.
5. (Optional) Do *one* of the following, as appropriate:
   Knowledge Document
   a. Click Edit Resolution on the Content tab to open the HTML Editor.
   b. Modify the body (resolution) of the document.
   c. Click OK to save your changes and close the HTML Editor.
   Knowledge Tree Document
   a. Click Design Tree on the Content tab to open the Knowledge Tree Designer.
   b. Edit the knowledge tree.
   c. Click Save and Close.
6. (Optional) Click Spell Check to review the spelling in the content you entered.
7. (Optional) Click User View to open the Preview window, which presents the document as it appears when a user displays it.
   You can close the Preview window when you finish reviewing the document.
8. Click Save when you finish editing the document.
   The product saves your changes.
9. Do *one* of the following:

- Click Save Version to save a new version of the document.
- Click Publish to put the document in the knowledge base and make it available to users with the appropriate permissions.
- Click Cancel to close the Update Knowledge Document window and open the Knowledge Document Detail window.

Attach a File

You can manage file and URL attachments in the current document. In documents created using the default Built-In Knowledge Document or Built-In Knowledge Tree template, links to attachments display under the "Attachments" heading.

> **NOTE**
>
> : The Attachments tab displays attachments, as well as a list of URLs and files currently attached to the document.

You can optionally attach a file or URL so that it may be viewed when viewing or editing the document.

To add a file attachment to a document, follow these steps:

1. Open the document for editing.
2. Click the Attachments tab.
3. Do *any* the following:
   - Enter the URL you want to attach.
   - Click Attach File to add a file from your system.
   - Click Attach File From Library to select the file name you want to attach from the Repositories list.

   The name displays on the Attachments tab.

Add Comments

You can add comments to a document that flag the document for correction, promotion, retirement, and so on. In addition, if you notice a problem with the content in a document, you can assign the problem to another analyst for follow-up. Follow-up comments appear on the Comment List page after they are saved. To display this page, select Knowledge Documents, Follow-Up Comments on the Scoreboard.

To add comments, follow these steps:

1. Open the document for editing.
2. Click the Comments tab.
   The Comment List appears.
3. Click Create New.
   The Create New Comment For Document page appears.
4. Select *one* of the following comments from the Comment Type drop down.
   - Broken Link
   - Candidate For Publication
   - Candidate For Retirement
   - Content Not Clear
   - Difficult to Find
   - General Comment
   - Incorrect Information
   - Missing Information
   - Recommend New Content
   - Review
   - Solution Does Not Work
5. Complete the comment field.

6. (Optional) Complete the following fields:
   Assignee
   You can assign the comment to another person for follow-up. Enter the name of the person assigned to handle the record, or click the search icon to search for the name.
   Target Date
   Identifies the date by which a comment must be followed-up on.
   Reply
   Enter a brief reply.
7. Click Save.
8. Click Close window.
   The comment appears on the Comment List page.

Add Notifications

You can add a notification to a document once it is open for editing.

To add a notification

1. Open the document for editing.
2. Select the Notifications Tab.
   The Document Notification List appears.
3. Click Add Notification.
   Complete the appropriate fields and click Save.
   The Save Successful - Notification created message appears.
4. Right-click the page and select Refresh.
   The notification is added to the list.
5. Click Save and Close.

Maintain Attributes

In the Document Editor, you can use the Attributes tab to set properties of a knowledge document that define ownership, scheduling, approval process, layout, associations, and so on.

> **NOTE**
>
> You must have write access to a document to set its attributes.
>
> If the Priority Calculation is set up and a user opens a ticket on the Knowledge Document, the Priority, Urgency, and Impact fields on the Attributes tab automatically appear on the ticket.

The tab contains the following fields:

Initiator(Read-only) Displays the name of the contact who created the document. You can click the name to open the Detail window for the contact.

AssigneeDefines the name (in "last name, first name" format) of the contact to whom the document is assigned for the duration of the current approval process task. You can enter the name of the contact in "last name, first name" format or click the magnifier to open the Contact Search dialog so you can locate and select a contact.

AuthorDefines the name (in "last name, first name" format) of the contact with responsibility for the document content. You can enter the name of the contact in "last name, first name" format or click the magnifier to open the Contact Search dialog so you can locate and select a contact.

OwnerDefines the name (in "last name, first name" format) of the contact assigned to maintain the document. Enter the name of the contact in "last name, first name" format or click the magnifier to open the Contact Search dialog so you can locate and select a contact. A document can have the same or a different owner in each stage of the approval process. All analysts with write access to a document can modify it, even if they do not own it.

NOTE

If a guest user or customer submits a document using the Submit Knowledge feature, the document has no owner.

Subject Expert Defines the name (in "last name, first name" format) of a contact with expertise in the subject matter of the document. Enter the name of the contact in "last name, first name" format or click the magnifier to open the Contact Search dialog so you can locate and select a contact.

Creation Date
(Read-only) Displays the date and time at which the document was created.

Modify Date(Read-only) Displays the date and time at which changes to the document were last saved.

Published Date(Read-only) Displays the date and time at which the document was published. If the document has not been published, the field is blank.

Last Accepted Date(Read-only) Displays the date and time at which the document was last accepted as the solution to a ticket. If the document has not been accepted, the field is blank.

Start Date Defines the date (in mm/dd/yyyy format) on which to make the document available in Search Tools. You can click the calendar icon to open the Date Helper window so you can select a date.

NOTE

A document can only become available on the Search Tools window when it is published and reaches its start date. For example, if you publish a document on March 1 with a start date of March 8, the document does not become available on the Search Tools window until March 8. If you publish a document with no start date specified, it immediately becomes available on the Search Tools window. The date format in Knowledge Management should match the date format you have set up on your Windows operating system.

Expiration Date

Defines the date (in mm/dd/yyyy format) on which the product removes the document from the knowledge base and the Search Tools window. After the specified date, the document is available from the Administration tab to users with the appropriate permissions. You can click the calendar icon to open the Date Helper window so you can select a date.

NOTE

The date format in Knowledge Management should match the date format you have set up on your Windows operating system.

Review Date Defines the date on which the document should next be reviewed. Click the calendar icon to open the Date Helper window so you can select a date.

NOTE

The date format in Knowledge Management should match the date format you have set up on your Windows operating system.

Disregard Life-Cycle Policies Ignores the lifecycle policies in your organization.

Approval Process Template(Required) Specifies an approval process template to associate with the document. The approval process template specifies the tasks in the document lifecycle. You can choose the default template provided with the product or a user-defined template. If left empty, it is set to the default template used in your knowledge environment.

NOTE

: For knowledge managers, knowledge engineers, and system administrators: The access types that can select an approval process template during the first task (typically Create Knowledge Document) are selected on the Knowledge tab Access Type window. In subsequent stages of the approval process, the Approval Process Template box is read-only and displays the name of the template selected for the document.

Approval Process Priority Specifies the approval process priority to associate with the document. Valid values include <empty>, Low, Normal, High, and Emergency.

Default: Normal.

User Defined ID Defines an ID for the document to use as an alternative to the product-defined ID.

FAQ Rating(Read-only) Displays the current document FAQ rating, which the product calculates from the frequency at which a document is accessed, how helpful it has been to users, and its age.

Hits(Read-only) Displays the number of times the document has been accessed.

Self-Service Count(Read-only) Specifies the number of times that the knowledge document was used to solve a problem. It is a count of the number of times the customer responded with Yes to the question “Did this document solve your problem? “ in the document Solution Survey.

Average Rating(Read-only) Specifies an average rating value. Each rating response (very helpful, somewhat helpful, and not helpful at all) has an assigned numeric value. The average rating is the total of the document response ratings divided by the total number of votes.

User Votes(Read-only) Specifies the total number of votes for the document. Submitting a Solution Survey is considered a vote.

PrioritySpecifies the priority to associate with the document, indicating the urgency of problems the document is meant to resolve.

SeveritySpecifies the level of effect that the problem the document is meant to resolve may have on users.

ImpactSpecifies the level of impact that you expect the document to have on work being performed.

UrgencySpecifies the importance level of the user tasks associated with the document.

ProductDefines a product to associate with the document. Enter the name of the product or click the magnifier to open the Product Search dialog so you can locate and select a product.

Configuration Item Defines the name of a configuration item (hardware, software, or service) to associate with the document. Enter the name of the configuration item or click the magnifier to open the Configuration Item Search dialog so you can locate and select an asset.

Root Cause Defines a root cause or core reason to associate with the document. Enter a root cause or click to open the Root Cause Selection window so you can locate and select a root cause.

Solution Count(Read-only) Displays the number of tickets resolved by the document.

Parent Request Defines an associated parent request record.

Parent Issue Defines an associated parent issue record.

ProblemDefines a problem linked to this document.

Maintain Permissions

You can determine which groups or roles can access a knowledge document and which groups or roles can modify the knowledge document by modifying permissions for Knowledge Documents and Knowledge Categories.

> **WARNING**
>
> When creating a knowledge document, make sure that document permissions include users that can be assigned later on the document through the approval process. When a group is assigned to a document, users from that group cannot have the permission to view the document. If the document is assigned to a specific user, default data partition constraints allow the user to view the document. If you do not specify group permissions, the document is only viewable by privileged users.

You can view and grant permissions as follows:

From the Knowledge Category page

1. Right-click a category and select Edit Category from the shortcut menu.
2. Click the Permissions tab.

From a Knowledge Document

If it is a published Knowledge Document:

1. Click Rework.
2. Create a Rework version, click Save.
3. Select the Permissions tab and make the appropriate changes.

View the Knowledge History

You can view the history of a Knowledge Document. You view events, such as when the document was created, published, or when the document was linked in a parent-child or tenant relationship.

To view the history, follow these steps:

1. Open the Knowledge Document.
   The Knowledge Document opens.
2. Click the History tab.
   The Event list appears.
   The following information is displayed:
   EventIndicates the name of event type associated with this row in the Event History list.
   Event Detail Displays a link to details about the event.
   NameDisplays the user that caused the event.
   Time Stamp Displays the time of the event.
   Version Number Displays the version number of the document.
3. Close the window.

View Document Versions

You can view the version summary information for each document version to see what actions have been performed.

To view the version information for a document

1. From the Documents List page, right-click the document to edit, then select Edit from the short-cut menu.
   The Update Knowledge Document page appears
2. Click the Versions tab.
   The following fields appear:
   Version Number
   Shows the number of versions retained in the system.
   Start Date
   Displays the date and time the version was saved.
   End Date
   Displays the date and time the version was last updated.
   Document Status
   Shows the status of the version.
   Started By
   Defines the level of urgency for the version (low, normal, high, or emergency).
   Published By
   The full message text for the notification.

Service Catalog Management

CA Service Catalog provides system forms for users to complete as part of their requests. A service consists of hardware, software, or other resources that users request from the catalog. Administrators can manage services to meet the needs of their users and organization. You can use events, rules, and actions to automate the processing of requests, users, and accounts in the CA Service Catalog system. All rules are initially disabled. A rule must be enabled to be used. You can add a dashboard, configure content elements, manage outages, create business hours, news, and content packs as shown in the following:

- Manage Forms
- Manage Services
- Manage Events-Rules-Actions
- Manage Dashboards
- Manage Outage Calendars
- Manage News, Change Events, and Alerts
- Manage the Scheduler
- Manage Content Packs
- Using the Service Catalog Content Packs
- Best Practices for Service Catalog Management
- How CA Service Catalog Determines the Role of a User?
- High Security Settings

Open CA Service Catlog link and enter your credentials to log in.
The Home tab opens with the Dashboards sub-tab selected by default.

You can use any of the tabs and sub-tabs and also view the relevant documentation by clicking Help.

Administrators and other users use the Dashboard Builder to manage dashboards and their content elements (dash items). Administrators can create shared dashboards and manage the dashboard library. Other users can create personal dashboards and can access the shared dashboards to which they have access.

{"URL":["http://10.252.12.131:8080/usm/wpf","/usm/wpf?Node=iclaunchpad.my_workplace"],"customLabelGetStarted":"CA Service Catalog","description":"IPCE_CA_Service_Catalog","customCards":
[{"id":"IPCE_Home","type":"manage","title":"Home"}]}

Manage Forms

CA Service Catalog provides the following system forms for users to complete as part of their requests:

- General Information Form
- Request Information Form

These forms request user-related information that is typically required to complete a request. For example, basic personal data and shipping address.

General Information Form

The General Information form asks the user to provide the required details for completing any checkout process. This form is not modifiable and therefore does not appear in the Form Designer. This form appears on the checkout page and other pages of the request.

Request Information Form

By default, the Request Information form asks the user to provide any custom descriptive information about the request. This form appears in any checkout process for services. The default version of the form appears in the Forms, System folder in the Form Designer.

You can optionally customize the Request Information form. Click the Customize button in the Component Tree when the form is selected. After confirming that you want to customize the form, you can edit it.

Use caution when modifying this form: Changes apply to all checkout-process for services in your business unit and all sub-business units. Even past (fulfilled requests, no archiving), present, and future are affected.

A tenant can have *one* active custom copy of this form.

Search the Forms

You can search for a specific form using either the Form Display name or Folder name option. You can use the Search Form UI option to modify or view an existing form. The search mechanism provides auto suggestion as the user types search terms in the form search text box. Search terms are case insensitive. The search results are displayed in a hierarchical manner.

Delete the Forms

After you customize this form, you can optionally delete the customized version to revert to the parent form.

The Delete function is disabled for the Service Provider (SP) business unit, because deleting the parent form is prohibited.

Hide the Forms

You can hide the Request Information form. Doing so ensure that the form does not appear to users or administrators during the request life cycle. By default, the request information form appears.

To hide it, open Catalog, Configuration, Request Management Configuration, and change the setting *Access Control: Show Request Information* to include no roles. Hence, no users or administrators can view or edit the Request Information form during request processing.

Restrictions

Keep in mind the following restrictions:

- Copying and pasting the Request Information form creates a copy that does *not* become the system form.
- You cannot move the Request Information form out of the Forms, System folder.
- You cannot alter the Forms, System folder, other than customizing the forms that can be edited.

Pre-Defined JavaScript Functions for System Forms

Customize the pre-defined JavaScript functions to update fields in the request information form when the general information form is changed. This optional customization helps integrate the request information form with the checkout and request edit pages. The pre-defined JavaScript functions are as follows:

- **custom_onPriorityChange(newPriority)**
  This function is invoked when the priority of the request is changed in the general information form. You can add code to update the customized request information form to reflect the priority change. By default, this function does nothing.
  newPriority is a string that corresponds to the new priority.
- **custom_onDateRequiredChange(newDateRequired)**
  This function is invoked when the date required is changed in the general information form. You can add code to this function to update the request information form. By default, this function does nothing.
  newDateRequired is a JavaScript date object set to the new required date.
- **custom_onRequestedForChange(type, id, name, shippingAddress)**
  This function is invoked when the requested for user or account for a request is changed. You can add code here to update values in your request information form. Example: Changing the shipping address fields in the request information form to have the address of the requested for user or account.
  By default, this function updates the shipping address of the request information form with the shipping address of the new user or account.

WARNING

If you change the shipping address section of the request information form, update this code to reflect those changes.

type is the user when a user is selected and account when an account is selected.

id is the CA Service Catalog ID of the selected user or account.

name is the display name of the selection.

shippingAddress is a JavaScript map with the following keys: address_1, address_2, city, state, zip code, and country. If the user or account has no address, shippingAddress can be null or empty.

For more information on how to manage CA Service Catalog Forms, see the following

CA Service Catalog provides the following system forms for users to add details about their requests:

- General Information Form
- Request Information Form

These forms request user-related information that is typically required to complete a request. For example, basic personal data and shipping address.

You can perform the following tasks related to a form:

- Create the forms that you want to include in a request.
  For example, a form typically lists the options and records the options that you select.
- Customize the Request Information forms.
- Edit the forms by adding and configuring the elements, such as text fields, text areas with checkboxes, and radio buttons.
- Delete the customized versions to revert to the parent form.
- Localize the element names and selected attributes in forms.
- Use the Script dialog on each form to create and maintain the custom JavaScript functions for the form.
  You can add JavaScript functions to perform automated tasks in the form fields.

For more information on how to manage CA Service Catalog Forms, see the following topics:

- Manage Forms
- Elements of a Form
- Form Attributes
- HTML Attributes for Elements
- JavaScript Attributes for Elements
- Customize a Predefined Form
- Create and Customize a Form
- Perform Automated Tasks in Form Fields

Users with administrator privileges can create, manage and maintain the forms and folders. Administrators can perform automated tasks in the form fields.

For more information on how to administer the CA Service Catalog Forms, see the Administer Forms.

{"URL":["/usm/wpf?
Node=iclaunchpad.form_designer"],"description":"IPCE_Description","new":"","new_video":"","admin":"IPCE_Admin","troubleshooti
{"masterkb":"","text":"the following KB articles will help you","URL": ["https://knowledge.broadcom.com/external/article?
articleId=189174"]}, "pendo":"","video":""}

Elements of a Form

For each form element, the _id value is auto generated. This value is a combination of text and numbers and depends upon the element you are creating. The first element of the form always begins with 1. For example, in a form the _id for the first text field element is txtf_1 and the _id for the first text area element is txta_1. In the same form, the _id for the second text field element is txtf_2 and the _id for the second text area element is txta_2.

You can modify this auto generated _id value of the form elements, if required.

> **WARNING**
>
> CA Service Catalog 17.3 does not support drag and drop of a form element to a Form Preview Panel.
>
> **WARNING**
>
> We recommend that you do not specify the same _id value for the different form elements in the same form. Having multiple form elements with the same _id value can cause validation errors.

To reuse the elements that you have customized, copy and paste the elements from one form to another.

The Form Designer provides the following elements for you to create and modify forms:

Check Box

Enables the user to select or decline a single option.
You can optionally use multiple check boxes to group related choices, of which users can select two or more. For example, you can add one check box each for the peripheral devices that are associated with a laptop computer. These devices include a mouse, docking station, carrying case, and external backup drive.

Column Layout

Enables the user to organize the components on the form for maximum efficiency. The column layout includes two vertical columns, Column 1 and Column 2, aligned next to each other. You can place form components in one or both of these columns. You cannot insert one column inside another, and you are not allowed to add more than two columns per Column Layout place holder. However, you can place multiple column layouts in a form.

Date

Sets the format of the date and time. The date and time that appear on the Form Designer match the date and time of the browser that you are using to access CA Service Catalog. The date and time that appear to other users accessing the form match the date and time of their browsers. The date and time of form settings -- even on the same forms -- can differ between users in different time zones. This principle applies whether users access the form through the Form Designer or while requesting a service.

- **Date**
  Use the date portion of this field whenever date information is required. **Example:** Start Date, Date Required, Estimated Date of Arrival. Users can select a date from the calendar.
  By default, the date format for the Date Time field matches the date format of the current business unit.

  > **NOTE**
  > As an administrator, you can optionally change the default format for the business unit to a different format: Edit the profile of the business unit on the Administration, Business Units page and select a new date format. Examples include M/d/yyyy, d-M-yyyy, or yyyy/M/d. While giving inputs for Date, ensure to follow the global standards for Date formats.

Specify date formats in Date Time fields on your forms according to the following rules:

Letter	Meaning	Format	Example
y	year	Number	2009
M	month in a year	Text or Number	July or 07
d	day in a month	Number	10

- **Time**
  Use the time portion of the Date Time field whenever time information is required. **Example:** Start Time, End Time, and Estimated Time of Arrival. When you use the time portion of this field, a drop-down list of time values appears next to the calendar. Users can select a time from this list.
  Default: The time format for the Date Time field matches the time format of the business unit of the user.

  NOTE
  As an administrator, you can optionally change the default format for the business unit to a different format: Edit the profile of the business unit on the Administration, Business Units page and select a new time format.

  Specify time formats in Date Time fields on your forms according to the following rules:

Format	Separator
HH:mm	colon
HH.mm	period

  If the Hide Time attribute is set to *true*, then the time portion of the field is hidden. This time format is ignored. The drop-down list for selecting the time does not appear on the form. Otherwise, the field accepts valid values that are separated by a space. CA Service Catalog supports the time format attributes supplied with the Google Web Toolkit (GWT) 1.6. For more information about these attributes, see the Google web site, www.google.com. You can add literal values that are enclosed in quotation marks (such as "hours" or "minutes") before or after the hour and minute values. The HTML attributes named *hidetime* and *increment* control whether and how incremented time values appear in each Date Time field.

Dual List

NOTE

- The admin must ensure that the dual list data do not contain pipe “|” symbol due to technical reasons.
- Selected items of a dual list are missing in CA Service Catalog 17.1 and later versions. The Dual list does not work with duplicate value fields. To resolve this, for each dual list option, ensure that the value field is unique for the dual list to work properly.

Displays two columns in a box: The left column lists available options, and the right column lists the selected options.
A toolbar of arrows appears between the columns. For example, to select an option, highlight it in the left column. Then, you click the arrow to move the option to the right column.
Users can highlight multiple options in one column and click the arrow to move them as a group to the other column.
Users can click the double arrows to move all options from one column to another. This action selects or deselects *all* options.
While you are designing the form, you can perform the same actions as the user to select and clear options.
By default, the dual list field contains *no* available options. To populate the dual list with available options, use *one* of the following methods:

- Specify a report data object for the dual list: Enter the HTML attributes named Report/Plug-in Id and Report/Plug-in Variables.
- Create a static list of individual options for the dual list: Add options to the dual list. You can add, move, and delete options to the dual list. The default dual list element in the Component Tree includes one option. You can copy and modify this option to create more options.

To enable users to specify the order of the selected options, specify *true* for the HTML attribute named ordered. When you do so, the toolbar is updated to include up and down arrows. Users can highlight selected options and click these arrows to move the options up or down in the list. When you are designing the form, you can order the selected options in the right column. The available options in the left column always appear in the order that you arrange them on the Component Tree.

Field set

Enables the user to group multiple elements as a set. Examples include the following groups:

- Accessories for a laptop such as battery charger, carrying case, and docking station.
- Telephone numbers for a user such as home, work, cell, and fax.

Elements in a field set have a box around them to show that they are related. For maximum accessibility, do *not* place one of the following elements immediately after a field set: check box, Date Time field, radio group, and radio buttons.

Image

Enables the user to provide a picture representing an item that can be included in an associated service or service option. The image that you specify must reside in the filestore, the central location for CA Service Catalog files. Use the following format:

FileStore/*path*/*filename.ext*

> **WARNING**
>
> The folder name *FileStore* is case-sensitive. Use the correct case in path names and all other references.

Label

Identifies a form or an area of the form. **Example:** A title in a form, such as Medical History Form. Within the form, you can have other labels, such as Family History, Eating Habits, and Illnesses. If you use a multiple-column format in your form, it can be helpful to use a label as a heading for each column.

Use the HTML attribute that is named Label Text to configure the label that users see on the form. You can optionally perform the following tasks for labels:

- Use the HTML attribute that is named Hidden to hide the label when the conditions you specify are met.
- Use the JavaScript attribute named onClick to run a JavaScript function when the user clicks the label.

Lookup Field

Works with the JavaScript function ca_fdDoFieldLookup(fieldId, reportId) to populate fields based on user input to a report data object.
You configure the lookup field to prompt the user to enter the data for the query of the data object. **Example:** user ID, asset ID, or city. The data object queries the data source, which is based on the user input. The results are used to populate the fields you specify.
The user clicks the search icon for the lookup field and enters the requested data. The query runs and returns the search results in rows. Each matching value appears in a row. Users review the rows and select one. When the user selects a row, the results from that row populate the matching fields on the form.

For example, the lookup field can prompt the user to enter a user ID. The query then searches the database for the first name and last name of the user ID. The user can select one row to populate the corresponding First Name and Last Name fields on the form.

Radio Group

Presents a list of options to the user. A radio group contains radio buttons *only*. A radio group cannot contain any other element. The user must select only one. **Example:** A radio group that is named Size with three buttons: one each for Small, Medium, and Large. In contrast to options in select boxes, radio group buttons *always* appear on the form. To save space, you can use a radio group only for fields having four or fewer options.

Select

NOTE

The admin must ensure that the select data do not contain pipe “|” symbol due to technical reasons.

Presents a list of options, of which the user select either one (the default) or multiple options.

A select box contains select options *only*. A select box cannot contain any other element. **Example:** You can create a select box that is named Width with two options: Narrow and Standard.
The options for select boxes appear on the form only when the user clicks the drop-down list box. Therefore, you can use a select box rather than an option group to save space on a form. Doing so is especially relevant when the number of options is four or more.
Optionally change the default setting of a select box to **True** or **False**. When you set the select box to **False** (the default). the select box appears as a combo box on the form and users can select one item from the list. You cannot enter a custom value for this attribute. When you set the attribute to **True**, the select box appears as a sequential list or list box on the form. The user sees multiple options in the select box and can select multiple options. Users can type inside a select box. As they type, the drop-down list is populated with the options that begin with the typed text. Users can then select an option from this "auto-complete" list.

To set specifications for the auto-complete list, use the Minimum Search Characters attribute. For more information, see HTML attributes for select boxes only.

NOTE

To delete an already selected value from the drop-down list, select the component and hit Delete/ Backspace.

Slider

Enables the user *slide* a control forwards or backwards to increase or decrease the selected value. Each slide updates the selected value according to the increment that you set. Specify the unit of measure for the slider in a message that appears each time that the user increases or decreases the selected value. Sample messages include *0 CPUs, 1 CPU, 2 CPUs, and 3 CPUs*.

To configure the contents and behavior of the slider, use the following attributes:

- Attributes for most or all elements
- Attributes for sliders only

Spinner Field

Enables a user to select a numerical value from a range of incremented values. **Example:** 100, 200, 300, and 400. The user clicks the up or down arrows to increase or decrement the selected value. Use the HTML attributes for spinner fields only to configure the contents and behavior of the spinner field.

Table

Enables the user to create tables. You can create either a Static Table or a Dynamic Table.

Static Table

A static table is a type of container, like a field set, that can contain certain elements of a form. You can use the columns in the table to organize the data from each type of element. In contrast to a dynamic table, a static table consists of fixed data that you specify manually.

You can create a static table to enter structured data into a form.

Follow these steps:

1. Edit or create the form in which you want to add the table. Expand the form.
2. Add the table element to the form:
   a. (Optional) Create a field set to contain the table.
   b. Expand the System folder, drag the Table element, and drop it on the form. If applicable, drop the table onto the field set that you created in the previous step.
   c. Specify an _id value for the table and save the form.
3. Add an element to the table:
   a. Expand the table and display the Row field.
   b. Drag one of the following elements from the System folder and drop it on the Row field:
      - Date Time field
      - Label
      - Select field whose Multi-Select attribute is set to false. This setting allows only a *single* selection.
      - Spinner
      - Text

      The name of the element that you drag-and-drop becomes the name of the first column. For example, if you drag-and-drop a Date element, the name of the first column becomes Date.
      Similarly, the data that you enter in the column must match its element. For example, in a Date column, you can enter only dates.
   c. Specify an _id value for the column and save the form.
      After you save the form, you can optionally rename the element that you dragged and dropped. If you rename the element, the name of the column changes accordingly. For example, if you rename the element to Start Date, the name of the column also changes to Start Date.
   d. Configure each element that you add to a table. The steps are the same as adding the element to the form without a table.
      For date fields, the value returned must be a long or appropriately formatted string. For label columns, the value is converted to a string. For spinner columns, the value must be an integer or double. For text columns, the value is converted to a string.
4. To add rows to the table, perform the following steps:
   a. Select the Row field on the Table element on the System folder.
   b. Drag-and-drop it on the Table element on the form.
   c. Repeat these steps until you have added all the rows that you want. You cannot move, copy, cut, or paste rows.
5. Define the values for each row in the table, as follows:
   a. In the first row, specify the static values of each cell, using their value attribute.
   b. In the remaining rows, specify values using the column attribute.
      The Form Designer does *not* validate any data or the data format that you enter in the table rows. When users display the form in a request, the Catalog system validates the data and displays it *only* if you use the correct format. Thus, any invalid values that you specify do *not* appear when users view the form in a request. **Example:** If you specify a string value for a date column, the corresponding table cell appears empty to the user.
   c. Perform this step if applicable; otherwise, skip it.

If you are using a Select field whose Multi-Select attribute is set to false, then the Select field does *not* contain the value attribute. In that case, perform the following actions:

- For the first row: Enter the value of the Selected Index attribute of the Select field. For example, to specify the first option, enter 0. To specify the second option, specify 1, and so on.
- For the remaining rows: Copy the value of the *value* attribute from the select option of the Select field. Paste this value into the column attribute of the row.

6. Specify any of the following attributes:
   – HTML attributes
   – Attributes for tables only
   – JavaScript functions for tables only

Dynamic Table

A dynamic table is also a type of container that can contain certain elements of a form. You can use the columns in the table to organize the data from each type of element.

You can create a dynamic table to enter structured data from a report data object into a form.

Follow these steps:

1. Create or edit the report data object or API plug-in that you plan to use to populate the dynamic table.
   The variables in the report data object or API plug-in must return data in the format that the table columns require. Otherwise, users do not see this data when they open the form in a request.
2. Edit or create the form in which you want to add the table. Expand the form.
3. Add the Table element to the form, as follows:
   a. (Optional) Create a field set to contain the table.
   b. Expand the System folder, drag the Table element, and drop it on the form. If applicable, drop the table onto the field set.
   c. Specify an _id value for the table and save the form.
4. Add the report data object or API plug-in to the table, as follows:
   a. Select the Table element.
   b. If you are using an API plug-in, specify the values of the following attributes:
      - Report/Plug-in Id: Enter the ID of the API plug-in that you want to use. You can find values for these attributes on the Administration, Tools, Plug-ins page. Report/Plug-in Variables: Open the API plug-in that you selected to display its details, including variables. On the details page, the Inputs section lists the ID values and descriptions of the input variables for the plug-in.
   c. If you are using a report data object, specify the values of the following attributes:
      - Report/Plug-in Id: Enter the ID of the report data object that you want to use. You can find values for these attributes on the Administration, Report Builder, Data Objects page.
      - Report/Plug-in Variables: Click the Edit icon for the report data object that you selected to display its properties, including variables. On the properties page, the input variables for the report data object appear as follows:
        For a Query: The input variables appear as %*expression*% statements.
        For Plug-in: The input variables appear in the Arguments field.
        For CSV: The input variables do not apply.

      For both attributes, enter variables as a JSON expression, for example:

```
$({'<variable name>' : '<variable value>', ...})
$({'userid':_.user.id,'rm_orgunit':ca_fdGetSelectedOptionValues(ca_fd.formId,'orgunit_id')})
```

WARNING
Specify the variables carefully. If you specify *no* variables, unpredictable results can occur.

5. Save the form.
   When users complete this form, the report data object or API plug-in runs and returns the data that you specified.

Next, you configure the dynamic table.

Configure the Dynamic Table

After you create a dynamic table, you configure it to contain the data and use the format that you require.

Follow these steps:

1. Review the following requirements for adding elements to the table. Each element that you add becomes a column in the table.
   - Add one element for each variable that you specified in the Report/Plug-in Variables attribute of the table element.
   - (API plug-ins) The value of the _id attribute of each element (column) in the table must match a value of an Output Id of the plug-in.
   - (Report data objects) The value of the _id attribute of each element (column) in the table must match a value of a Field of the data object.
2. Add an element to the table, according to the requirements in the previous step, as follows:
   a. Expand the table to display the Row field.
   b. Drag the element from the System folder and drop it on the Row field. You can drag-and-drop the following elements:
      - Date Time field
      - Label
      - Spinner
      - Text
      - Select the field whose Multi-Select attribute is set to False. This setting allows only a *single* selection.
   c. Perform this step if applicable; otherwise, skip it.
      If you are using a Select field whose Multi-Select attribute is set to false, then the Select field does *not* contain the value attribute. In that case, you can populate the Select field using *either* a static list *or* a report data object.
      To populate the Select field *using a static list,* perform the following actions:
      - For the first row: Enter the value of the Selected Index attribute of the Select field. For example, to specify the first option, enter 0. To specify the second option, specify 1, and so on.
      - For the remaining rows: Copy the value of the *value* attribute from the select option of the Select field. Paste this value into the column attribute of the row.

      The name of the element that you drag-and-drop becomes the name of the first column. For example, if you drag-and-drop a Date element, the name of the first column becomes Date.
      You cannot move, copy, cut, or paste rows.
   d. Specify an _id value for the column and save the form.

      WARNING
      The _id value for the column must meet the requirements noted in the previous step. Also, the data type and data format must also be the same. Otherwise, the column is not populated with data.

      Example: You are using an API plug-in, and you want a table column to specify the start date of an event. You drag-and-drop a Date Time element onto the table and specify an _id value of start_date. This element becomes a column in the table. Therefore, the Output Id of a variable of the API plug-in must also be start_date. The sequence does not matter. This variable must also return the required date and time data in the format that matches the Date Time element.
3. Configure each element that you add to a table. The procedure is the same as if you were adding the element to the form without a table.
   Each one is a basic element, except the Date Time field and the elect field. The data that you enter in the column must match its element. For example, in a Date column, you can enter only dates.
4. (Optional) Specify a custom value for pagination for the table, as follows:
   a. Specify the value of the Page Size attribute of the Table element.
   b. For API plug-ins, configure sorting and pagination parameters. For report data objects, further action is necessary.

5. (API plug-ins only) Enable users to sort the results in the table, as follows:
   a. Specify a value of True for the Sortable attribute of the Table element.
   b. Configure sorting and pagination parameters. Sorting does not apply to report data objects.
6. Specify any of the following attributes:
   – HTML attributes
   – Attributes for tables only
   – JavaScript functions for tables only

Text Area

Enables the user to enter more than one line of input. **Example:** A free-form description of the offering.

Text Field

Enables the user to enter free-form text. **Example:** Fields for the user to enter a name, address, telephone number, or multiple-digit numeric entries. Use text fields when only one line of input is required.

Page Layout

Enables the user to place the form components within a page layout for maximum efficiency. You cannot place one page layout inside another and you can have only one page layout in a form.

You can have any number of pages within a page layout.

Page layout element can be one of the following:

- Card layout - Displays the element as pages with next and back button if there are multiple pages. This is the default page layout.
- Tab layout - Displays the element as tabs on top of the page layout.

Default page height: 400 pixels

Form Attributes

Form attributes apply to the *entire* form rather than to one field only.

- **name**
  Specifies the name of the form.
  Specify this attribute for each form, and verify that this attribute is unique for each form.
- **_id**
  Specifies the identifier of the form.
- **class**
  Specifies the CSS class name to use for the form.
  Define the new CSS class in the formdesigner.css file in the folder named USM_HOME\view\webapps\usm\gwt\fdBase\css.
  For example, to apply a class that changes the form to a 12-point, blue, boldface font, Follow these steps:
  a. Back up the formdesigner.css file before you edit it.
  b. Append the following class statement to formdesigner.css, as follows:

```
/* custom class by author-name, date for purpose */
.blue12-class {
color: blue;
font-size:12px;  /* Make sure to mention font size in pixels(px) */
font-weight:bold;
}
```

NOTE

Text fields have a preset height limit and therefore ignore the font-size attribute.

c. Enter blue12-class as the value for CSS class attribute.

d. Verify your updates by using the form in a request.

- **Form Type**
  Specifies whether the form is a configuration form type, as follows:
  – Request - for a request of a service from the catalog
  – Configuration - for a content configuration form
  Default: Request
  This attribute *cannot* be written using JavaScript expression.
- **Label Align**
  Applies to labels *only*. This attribute specifies the alignment of the label in relation to the fields on the form. Select top, right, or left (default).
- **Label Width**
  Specifies the width in pixels of the field labels in the form.
  You can also specify Label Width as an HTML attribute for the column layouts and field sets on the form.
- **onLoad**
  Specifies the JavaScript function to run when the user opens the form in a request.
  Multiple forms having an onLoad attribute can appear on a single page. In that case, each JavaScript function runs when the user opens the form that includes the function.
  An alert appears if an onLoad function fails to run correctly. For example, if the function cannot be found.
  If you receive an alert, verify that function is correctly defined and referenced to ensure that it runs without errors.

  NOTE
  Use the Script dialog on each form to create and maintain the custom JavaScript functions for the form.
- **onSubmit**
  Specifies the JavaScript function to run when the user submits the request that contains the form.
  This function must return a Boolean value. If the function returns any value except *true*, the form is not submitted.
  Multiple forms having an onSubmit attribute can appear on a single page. In that case, each JavaScript function runs when the user submits the form that includes the function.
  The information about alerts for the onLoad function also applies to the onSubmit function.

HTML Attributes for Elements

You can use HTML attributes to configure the appearance and behavior of the elements of a form. For more information on how to use the JavaScript Functions, see JavaScript Function APIs.

Keep in mind the following principles:

- Not all HTML attributes apply to all elements (fields). Click an element to see which attributes apply to it.
- Required HTML attributes appear in **bold** in the Item Inspector. When you add an element to a form, complete the required attributes and save the form immediately after adding the element. Afterwards, you can add optional HTML attributes and JavaScript attributes.
- The HTML attributes are primarily internal, in that they regulate the appearance and behavior of the form. They *typically* do not supply literal text that appears to the administrator in the Design Area or to the user in a request.
- You view and modify the names and values of the HTML attributes in the Item Inspector as needed. However, you generally do not see the changes as literal text in the Design Area. The exceptions are:

  - The Checked attribute for radio buttons and check boxes
  - The Value attribute for labels, text fields, and text areas
- For an HTML attribute, you can optionally enter a JavaScript expression, but you cannot call a JavaScript function.
- For most elements, you specify the literal text that appears to end users by adding and configuring elements on the form. Specifically, you rename the elements on the form in the Component Tree, and you verify their appearance in the Preview Area.

For more information about the different HTML Attributes, see the following sections.

- Attributes for Most or All Elements
- Attributes for Date Time Field Only
- Attributes for Dual List and Select Boxes Only
- Attributes for Field Sets Only
- Attributes for Labels Only
- Attributes for Radio Groups Only
- Attributes for Select Boxes and Tables Only
- Attributes for Select Boxes Only
- Attributes for Sliders Only
- Attributes for Spinner Fields Only
- Attributes for Tables Only
- Attributes for Text Fields and Text Areas Only

Attributes for Most or All Elements

Some HTML attributes apply to *all or most* elements (fields) of a form. The following attributes of that group may require explanation:

- **_id**
  Specifies an auto-generated unique identifier for the form field.
  If the _id is not unique, a warning message appears.

  WARNING

  The Form Designer *warns* you if the _id attribute values are duplicated for multiple fields; however, it does *not* prevent you from saving them. Therefore, change any duplicate _id attributes a soon as you become aware of them. If the form contains duplicate _id attributes, JavaScript functions can run on the wrong form field and cause errors. If you change the value of the _id attribute, verify that you also change any references to it, especially in JavaScript functions.
- **CSS Class**
  Specifies a space-separated list of CSS classes. This class is associated with style information.
- **Empty Text**
  Provides extra description of a field to the user.
  The value that you specify for this attribute appears inside the field when the user opens the form. This text disappears when the user begins entering a value in the field. A sample value follows, for a password field: You can optionally use the Empty Text, Hint, and Tool Tip attributes together to assist the user.
- **Name**
  Specifies the name of the element.
- **Value**
  Specifies the value of the input component. This attribute is often useful for entering sample data or instructions.
  For example, for the value attribute of a text field named Address, you can specify something like "123 my street" or

"Enter your address here." In both examples, the value suggests that users replace the default text with their actual addresses.

- **Disabled**
  Specifies a value of true or false.
- **Hidden**
  Specifies whether to hide a field, including the label. Specify true or false. When the value is set to true, the field is *not* visible.
  You can use JavaScript expressions to specify this value as a condition, as follows: If the condition is met, hide the field. Otherwise, display the field.
  For example, you can hide a fulfillment-related field before the request reaches the Pending Fulfillment status. Similarly, you can hide a salary-related field from any user other than the manager of the user and the Human Resources user group.
  You can specify all the same conditions that apply for the Policies. Similarly, you can use the JavaScript expressions to specify the value as a condition for the **Include in Email Notification** attribute too.
- **Include In Email Notification**
  Specifies whether to display a field or component value, including the label. Specify the value as true or false. When the value is set to true, the field or component value is visible in the email. To set or edit this attribute, you must have admin rights. Navigate to **Catalog**, **Forms**, select the Form Component and click on the **Include in Email Notification** attribute.
  When you set the **Hidden** and **Include In Email Notification** element attributes to true/false in the CA Service Catalog UI, the following is reflected in UI and email notifications:

Hidden Attribute	Include In Email Notification Attribute	Result
true	true	Form component is not displayed anywhere.
true	false	Form component is not displayed anywhere.
false	true	Form component and value is displayed in the Form Designer, the Request Form while raising a request, and in the email notification.
false	false	Form component appears in the Request Form UI, the Form Designer UI, but the component value is not displayed in Email Notifications.

- **Hide Label**
  Specifies whether to hide *only* the name of a field. The field remains visible regardless of whether you name the label. This attribute provides flexibility when space is a concern.
  Specify true or false. When the value is set to true, the name for this field is *not* visible.
  You can use JavaScript expressions to specify this value as a condition, as follows: If the condition is met, hide the name. Otherwise, display the name. You can specify all the conditions that apply for the Policies.
  For example, you can hide the names of obvious fields, such as Name, Address, and Telephone Number, after the user submits the request. Doing so can be helpful to save space on the form when the additional fields become visible for approval and fulfillment statuses only.

  NOTE

  The Hide Label attribute does *not* apply to the Label element. Instead, the Hide Label attribute applies to the *name* of certain other fields that you enter in the Component Tree of the Form Designer. After you enter the name of a field in the Component Tree, the name appears in the Preview Pane.
- **Hint**

Specifies help text for a field. This text *always* appears below the field, regardless of the position of the mouse.
For example, for a password field, you can specify the following hint: "Passwords must be six to eight characters and must include both letters and numbers."
You can optionally use the Empty Text, Hint, and Too tip attributes together to assist the user.

- **Required**
  Specifies whether this field is required. Enter a value of true or false.
  When a user submits the form, the system verifies that all the required fields are not empty. A required field is a field whose required attribute is set to true. If any required fields are empty, the system prompts the user to complete them.
- **Style**
  Specifies associated style information.
- **Tab Index**
  Specifies the position in the tabbing order. Enter numeric value only.
- **Tool Tip**
  Specifies helpful text appears *only* when the user hovers the mouse over the field.
  You can optionally use the Empty Text, Hint, and Tool Tip attributes together to assist the user.
- **Text Direction**
  Specifies the supported values for left-to-right and right-to-left for the direction of the text.

Attributes for Date Time Field Only

The following HTML attributes apply to the Date Time field *only*.

- **Hide Time**
  Applies only to the time portion of the Date Time field; this field is an optional element of a form.
  Specifies whether to display the drop-down list for users to select the time. Specify *true* to hide this list or *false* to display it.
- **Time Increment**
  Applies only to the time portion of the Date Time field; this field is an optional element of a form.
  Specifies the increment (in minutes) of the times that appear in the drop-down list for users to select the time. Specify any positive integer greater than zero.
  For example, if you specify 15, the following values appear in the list: 9:00, 9:15, 9:30, 9:45, 10:00, and so forth.
  Similarly, if you specify 30, the following values appear in this list: 9:00, 9:30, 10:00, 10:30, 11:00, and so forth.
- **Minimum Value & Maximum Value**
  The Minimum and Maximum Value attributes define the minimum and maximum date time values for the date time range respectively. These fields accept both numerical value (UNIX Epoch time value) and an expression which evaluates to a numerical value (UNIX Epoch time value). To set a fixed date range, use the Epoch value and to set a dynamic date range based on the current date time, use an expression.
  For example:
  - To set a fixed date time range of Jan 10, 2018, to Jan 20, 2018, set the values as:
    Minimum value: 1515522600000
    Maximum value: 1516386600000
  - To set a dynamic date time range of three days past to three days future of the current date time, set the values as:
    Minimum value: $(new Date().getTime() -(3*86400000))
    Maximum value: $(new Date().getTime() +(3*86400000))

    NOTE

    86400000 is the number of milliseconds in a day.

Attributes for Dual List and Select Boxes Only

The following HTML attributes apply to dual lists and select boxes only.

Additional attributes apply to select boxes only.

- **Height**
  (Dual List) Controls the height of the dual list. Using this attribute displays the entire list of options *without* a scroll bar.
  (Select Box) Controls the maximum height of the drop-down list that appears when the user clicks the arrow to display the options.
  For select boxes that allow users to select multiple options, using this attribute displays the entire list of options *without* a scroll bar.

See also:

- attributes for select boxes only

Attributes for Field Sets Only

The following HTML attributes apply to field sets *only*. Fields sets are basic elements of a form.

- **Collapsed**
  Applies *only* to the field set element.
  Specifies whether the label of the field set is initially collapsed (true) or expanded (false).
- **Label Width**
  Specifies the width in pixels of the column layout and field set.
  You can also specify Label Width as a form attribute.

Attributes for Labels Only

The following HTML attributes apply to labels *only*. Labels are basic elements of a form.

- **Label Text**
  Specifies the text for a label.
  A label identifies a form or an area of the form. For example, you can use a label at the top of the form to provide the title, such as Medical History Form. Within the form, you may have other labels, such as Family History, Eating Habits, and Illnesses. If you use a multiple-column format in your form, it may be helpful to use a label as a heading for each column.
  The Label Text attribute of a label contains the text that users see on the form, for example, as a title or heading. This Label Text attribute can contain HTML tags, which are especially helpful for highlighting purposes. For example, to specify the Family History label in bold, enter the following data in the Label Text field:

```
<b>Family History Form</b>
```

Attributes for Radio Groups Only

The following HTML attributes apply to radio groups *only*. Radio groups are basic elements of a form.

- **Orientation**
  Lets the user control the orientation of the options for the group.
  Specify horizontal or vertical (default).

Attributes for Select Boxes and Tables Only

The following HTML attributes apply only to select boxes and tables (as noted).

- **Report/Plug-in Id**
  Specifies the ID of the report data object (data object). This data object is used to pre-populate a combo box or to pre-populate rows in a dynamic table.
  The data object returns two fields:
  – The first field returns the list of values in the combo box. Users can select one or more of these values, depending on the setting of the multiple attribute.
  – The second field returns the label for the combo box.
  The data object represents a stand-alone MDB query that does not require a JavaScript function to run.
- **Report/Plug-in Variables**
  Specifies the attribute to use user input to pre-populate a select box or to pre-populate rows in a dynamic table.
  For example, you can specify this attribute to use the selection in one select box to determine the values in a second select box.
- **Page Size**
  Specifies the number of options that appear on one page in a drop-down list. The drop-down list appears when a catalog user performs one of the following actions:
  – Clicks the arrow of a drop-down list
  – Types text in a field and activates the auto-complete feature
  The Page size attribute applies *only* to the following elements:
  – Select boxes whose Multi-Select attribute is set to False.
  – Static table
  – Dynamic table
  The Page size attribute is optional and accepts *only* positive integers.
  This attribute is useful when report object returns too many options to list on a single page. In such cases, the drop-down list contains a scroll bar and page numbers. To view the values in the list, users can either scroll or click a page number.
  A blank (empty) value effectively disables pagination.
  The default is 50.

See also:

- attributes for select boxes only
- attributes for tables only

Attributes for Select Boxes Only

> **NOTE**
> It is recommended to avoid using page size attribute for the Select component in CA Service Catalog Form designer, when the same form must be rendered in Service Point.

The following HTML attributes apply to select boxes only.

- **Eager**
  Applies *only* when both of the following settings are used:
  – The value of the Multi-Select attribute of the select box is False.
  – The Report/Plug-in Id attribute is set.
  Specifies a value of true or false.
  – **True**
  Loads the data from the Report/Plug-in Id attribute into the select box when the user requesting the service displays the form.
  – **False**

Loads the data from the Report/Plug-in Id attribute into the select box when the user requesting the service clicks the select box.

- **List Width**
  Specifies the minimum width of the drop-down list for a select field.

  NOTE

  To delete already selected text for a selected component from the drop-down list, select the component and hit delete/ backspace

  The Form Designer uses the *larger* of the following values:
  - The width of the select field
  - The value of the List Width attribute
- **Multi-Select**
  Specifies a value of true or false.
  - **True**
    If set to **True**, you can select multiple options from the list. You can also remove the options in the selected list by clicking the cross-symbol or X mark.

    NOTE

    When the select box is added to the table, the multi-select attribute of the box is not functional and it is displayed as a single select.
  - **False**
    If set to False, you can only select a single attribute from the list.
- **Minimum Search Characters**
  Applies for single and multi-select attribute. Catalog users cannot enter a *custom* value in a select box. However, users can type inside a select box. As they type, the drop-down list is populated with the options that begin with the typed text. Users can select an option from this "auto-complete" list. The auto-complete list starts to populate after the user types the *n*th character, where *n* is the value you set for Minimum Search Characters attribute. For example, if you specify a value of 7, the auto-complete options appear after the user types the seventh character.
- **Selected Index**
  Specifies the default selection.
  For example, the select box contains 10 values. If you specify 0 for the Selected Index attribute, the first value in the list becomes the default. Similarly, if you specify 5 for the Selected Index attribute, the fourth value in the list becomes the default.
  You can use a select box in a static table.

See also:

- attributes for select boxes and tables only
- attributes for dual list and select boxes only

Attributes for Sliders Only

The following HTML attributes apply to sliders only.

- **Increment**
  Specifies the increment by which the selected value increases or decreases with each forward or backward slide, for example, 1, 5, 10, 25, or 100.
- **Value**
  Specifies the initial value of the slider, for example, 10.
- **Maximum Value**
  Specifies the maximum value of the slider, for example, 100.
- **Minimum Value**

Specifies the minimum value of the slider, for example, 1.

- **Message**
  Specifies the message that appears as the user slides the control. Use the following format:
  - **{0} *text***
    {0}displays the selected value. As the user slides the control, this value is automatically replaced with the selected value.
    text specifies the unit of measure, for example, CPU, RAM, or another meaningful unit.

For best results, specify values for these parameters and test the slider until you obtain the results that you want.

Note: You can optionally use JavaScript functions to set the values of these attributes. For details about predefined JavaScript functions, including slider-specific functions, select Administration, Tools, Links, Form Designer JavaScript API. The values are not saved if both of the following conditions exist: You use JavaScript functions to set the minimum and maximum values of these attributes, and the value is dynamic. Therefore, use the JavaScript functions to apply the values individually on each page.

Attributes for Spinner Fields Only

The following HTML attributes apply to spinner fields *only.* Spinner fields are basic elements of a form.

This field enables users to select one of many values by clicking the up or down arrow. Each time the user clicks an arrow, the selected value increases or decreases by the amount specified by the Increment attribute.

Typically, the values for a spinner field appear in meaningful, equal increments. An example is 50, 100, 150, and so forth.

- **Allow Decimals**
  Specifies whether to use decimal values, for example, in the rate or cost.
- **Allow Negative Numbers**
  Specifies whether to use negative values.
- **Increment**
  Specifies the amount by which the selected value increases or decreases each time that the user clicks the up or down arrow. For example, suppose you want to use the following values: 100, 200, 300, and 400. In this example, the increment is 100. If the selected value is 200 and the user clicks the up arrow, the selected value increases to 300.
- **Minimum Value *and* Maximum Value**
  Specify the minimum and maximum values, respectively.
  In the example for the previous attribute (Increment), the minimum value is 100 and the maximum value is 400.
- **Number Format**
  Specifies the format of the number that appears to the user in the following formats.

  NOTE

  You must not use comma (**,**) and dot(**.**) together and multiple commas (,,) while defining a number format.

  - **Number format = #,#0**
    Must start with a number (#) separated by a comma (,) followed by sequential increase of two digit numbers separated by a comma. For example, 1, 23, 56, 78 and so forth. Here, **#** or **0** can be replaced by any numerical value. If negative value is set to true, you can increase or decrease the number with a comma or dot separated values.
  - **Number format = #.1234 or 0.1234**
    Here, **#** or **0** can be any digit, any number after precision or decimal value must contain four digits. If you click the up and down arrow to increase or decrease the numerical value, the number will be increased or decreased but the decimal value remains the same. If negative value is set to true, you can increase or decrease the number with a comma or dot separated values. For example: 1.234

Attributes for Tables Only

The HTML attributes in this topic apply to tables *only*.

Table elements have the following types of attributes:

- Table attributes apply to the entire table, that is, to *all rows* in the table.
- Row attributes apply only to the selected row.

Table Attributes

The following table attributes require explanation:

- **Report/Plug-in ID and Report/Plug-in Variables**
  Apply to dynamic tables.
- **Allow Edits**
  Specifies whether catalog users can manually update the data that is dynamically inserted into a row by a JavaScript function. The use of a JavaScript function for this purpose is described in the Note for the next attribute, Allow Inserts.
  If the value of Allow Edits is true, users *can* manually update this data in a form as they request a service.
  If this value is false, users *cannot* manually update this data in a form as they request a service.
  The Allow Edits attribute takes effect *only* if the Allow Inserts attribute is set to false.
  If the Allow Inserts attribute is set to true, users can *always* update data in a row, regardless of the following settings:
  - Whether you populate the row with default text, a JavaScript function, or not at all
  - Whether you set Allow Edits to true or false
- **Allow Inserts**
  Specifies whether catalog users can manually add rows to tables in a form when they are requesting a service.
  When this value is false, users *cannot* add rows.
  When this value is true, users *can* add rows by clicking an Add icon in the table. For example, you can design a service that contains a table with several predefined options. You can also enable the user to add custom values to the table. The advantage of using such a table over a text box is that the table stores the data in *structured* format.
  If Allow Inserts is true, the following conditions exist when users update the form as they request a service:
  - Users can click the Add button in the table to add rows. The data that users enter must meet the validation criteria for each cell, such as minimum text length, and data format. For each cell, the type of element that you specify determines this validation criteria.
  - Users *cannot* add, delete, move, or edit rows that you specify when you create the form. All rows that a user inserts are marked with red dots in their cells.
  - When users add or update rows, JavaScript functions run *only* on the row being added or updated.
  - User-created rows always appear at the *top* of the table, *before* all rows populated through a report data object, an API plug-in, or static data. Moreover, user-created rows always appear in the reverse order that they were added: The first rows in the table are the last rows added.
  - Sorting does not apply to user-created rows. However, sorting does apply to the rows populated through an API plug-in or static data.

    NOTE

    As a form designer, you can use JavaScript functions to add rows to tables in a form while users are requesting a service. For example, use a JavaScript function to populate a row when a user clicks an option button in a radio group. This ability exists regardless of the setting of Allow Inserts. You can use either a custom JavaScript function or the predefined JavaScript function named ca_fdAddTableRow. For more information about predefined JavaScript functions, select Administration, Tools, Links, and view the *JavaScript API Reference*.
- **Auto Number**
  Specifies whether the first column of each row contains the row number.
  If the value is true, then the first column of each row displays the row number.

The default is true.

- **Height**
  Specifies the height of the table, in pixels.
  The default is 125.
- **Maximum Selected Rows**
  Specifies the maximum of rows that users can select in the table while completing the form in a request.
  If the number of selections exceeds this value, an error occurs.
  If this value is not specified, then the number of rows that users can select is unlimited.
- **Minimum Selected Rows**
  Specifies the minimum of rows that users can select in the table while completing the form in a request.
  If this value is not specified, then the user is not required to select any rows.
- **Mode**
  Specifies whether the table functions in *selection mode* or *data mode* during the request life cycle, as follows:
  – In selection mode, the table automatically includes a check box in the first column, so that users can select one or more rows from the table. The user-selected rows appear in the request throughout the request lifecycle.
  – In data mode, the table does *not* include the check box. Therefore, users cannot select rows from the table.

  In *both* modes, if the value of the Allow Inserts attribute is True users *can* perform the following action: Configure the request by adding rows to the table and entering custom values. The catalog system automatically saves any rows that the user adds and displays them throughout the request life cycle. Users can remove each row that they added by clicking the row and selecting Remove.
  Selection mode is useful when you want users to select a set of related options for the request by selecting a row on the table. For example, you can specify the following table:

Check Box	Disk Space, GB	Memory, GB	Processor Speed, GHz	Operating System
_	700	4	2	Windows XP
_	900	6	2.4	Windows 7
_	1000	8	3	Windows 2008 Server
_	1400	8	4	Windows 7 Server

In this example, each row contains one set of configuration options for a computer reservation. Users cannot select each option individually, but they can select a set by selecting a row.

Row Attributes

The following table attributes apply to the selected row only, not the entire table:

- **Column Widths**
  Specifies a list of comma-separated values for the *widths* of each column in the table, from left to right.
  For example, if the table contains three columns, then the following points apply:
  – The first value applies to the first (left) column
  – The second value applies to the second (middle) column
  – The third value applies to the third (right) column

  NOTE

  This attribute *applies* to the entire table, but *appears* only in the first row.

See also attributes for select boxes and tables only.

Attributes for Text Fields and Text Areas Only

The following HTML attributes apply to text fields and text areas *only*. Text fields and text areas are basic elements of a form.

- **password**
  Specifies an attribute for text fields. Can have a value of true or false. When the value is set to true, this component becomes a password component.
- **Height**
  Specifies the number of rows in the text area. For example, if you specify 3, the text area contains three rows.
- **Maximum Length**
  Specifies the maximum number of characters that the user can enter in the field.
- **Pattern**
  In the value for this attribute, you specify regular expressions to validate numeric and address data. Such data includes credit card numbers, social security numbers, email addresses, IP addresses, and telephone numbers.
- **Pattern Message**
  Applies only when *pattern* (the previous attribute) is specified.
  You can specify an error message to appear when the user input violates the pattern attribute. You can optionally localize this error message.

JavaScript Attributes for Elements

You can use JavaScript attributes to invoke JavaScript functions while the user is completing the form in a request. JavaScript functions include predefined functions for validating user input into fields and any custom functions that you have written. All JavaScript attributes are named with the prefix *on*, such as onClick.

You can use the following JavaScript attributes in elements on forms. Not all attributes apply to all elements, however. Click an element to see which attributes apply to it.

> **WARNING**
> JavaScript attributes *must* have JavaScript functions but not JavaScript expressions as values. Conversely, HTML attributes can have JavaScript expressions but *not* JavaScript functions as values. Thus, JavaScript expressions apply to HTML attributes *only*, and JavaScript functions apply to JavaScript attributes only.

JavaScript functions in JavaScript attributes are validated when the user performs the action that the attribute specifies. Examples are clicking (for onClick) and mousing over (for onMouseOver). If the field does not validate, it is highlighted in red and an error message appears.

> **NOTE**
> Verify that the error messages (if any) returned by each JavaScript function are localized for the users of the form.

- **onFocus**
  Specifies the JavaScript function to run when the element comes into focus.
- **onBlur**
  Specifies the JavaScript function to run when the element loses the focus.
- **onChange**
  Specifies the JavaScript function to run when the element value is changed.
  > **NOTE**
  > For the Table components, onChange triggers when a user adds a new row using the Insert table (+) icon or table APIs (or) when edits/removes a table row.
- **onSelect**
  Applies to the Table components only.

Specifies the JavaScript function to run when selection/deselection of checkbox on a table row.

- **onClick**
  Specifies the JavaScript function to run when the component is clicked with the left mouse button.
- **onLoad**
  Applies to dual lists and select boxes only.
  Specifies the JavaScript function to run when the data associated with the select is loaded. This attribute is ignored if the dual list or select box specifies static options.
  An alert appears if an onLoad function fails to run correctly, for example, if the function cannot be found. If you receive an alert, verify that function is correctly defined and referenced, and that the function runs without errors.
  Use the Script dialog on each form to create and maintain the custom JavaScript functions for the form.
- **onMouseDown**
  Specifies the JavaScript function to run when a mouse button is pressed.
- **onMouseUp**
  Specifies the JavaScript function to run when a mouse button is released.
- **onMouseOver**
  Specifies the JavaScript function to run when the component is moused over.
- **onMouseMove**
  Specifies the JavaScript function to run when the mouse passes over the component.
- **onMouseOut**
  Specifies the JavaScript function to run when the mouse was moved away from the component.
- **onKeyPress**
  Specifies the JavaScript function to run when a key is pressed and released.
- **onKeyDown**
  Specifies the JavaScript function to run when a key is pressed.
- **onKeyUp**
  Specifies the JavaScript function to run when a key is released.
- **onValidate**
  Specifies the JavaScript function to run when the field is validated. The field is validated whenever the user navigates away from a field or when the user submits the form.
  If you specify a custom JavaScript function for the onValidate attribute, code that function as follows:
  - Specify an attribute that is named _val; this attribute is required to pass the current value of the field.
  - If no validation error occurs, return null.
  - If a validation error occurs, return a string with a helpful error message explaining the problem and solution. Localize error messages for each browser locale applicable to your users.
- **onLookup**
  Applies to lookup fields *only*. When the user clicks the magnifying glass for the lookup field, the JavaScript function for this attribute runs. You can use a lookup field to populate fields based on user input to a report data object.

Customize a Predefined Form

This scenario illustrates how Service Managers create a form by copying a predefined form and customizing the copy. You add fields and specify some fields to be auto-populated. You use spinner fields and radio buttons so that users can specify only valid options. You also show or hide fields according to the options selected by the user. Creating form in this manner helps reduce user errors and helps increase efficiency.

This scenario focuses on the onboarding of a new employee into the Lab Services group in your organization. You can use this scenario *as a model* to create a form using similar techniques.

For other use cases, search the **Catalog**, **Forms** folder for the form that most closely matches the form that you want to create. **Example:** To create a form for ordering hardware, review the Procure New Hardware form in the IT Support Services subfolder.

NOTE
Service Managers typically have one or more of the following roles in CA Service Catalog: Service Delivery Administrator, Services Manager, Super Business Unit Administrator, and Catalog Administrator.

Follow these steps:

WARNING

We recommend that you do not create multiple forms with the same _id value for each form. If you already have multiple forms with the same _id value, do not associate such forms to a single Service Option. Having multiple forms with the same _id value in a single Service Option can cause validation errors.

Step 1 - Review and Copy the Form

All services include at least one predefined or custom form to record essential information from the user requesting the service. In this scenario, you review and copy the predefined form included with the standard New Hire Onboarding service. Afterwards, you modify the copied form for use by the New Hire Onboarding for Lab Services group.

Follow these steps:

1. Click **Catalog**, **Forms**.
2. Expand **Forms**, **CA Catalog Content** in the Component Tree.
3. Select **New Hire Onboarding**.
4. Click Copy.
5. Move the top level of the Forms folder and complete these actions:
   a. Click **Add** and create a subfolder and call it Custom.
   b. Select the Custom folder and click Paste.
   The form is copied, and its new name is "copy of *original name.*" The copied elements under the copied form are *not* renamed.

      NOTE
      If you copy the form from one business unit to another, the pasted form, including all elements under it, belong to the business unit of the destination folder.
   c. Select the form and click Rename.
   d. Enter the new name as New Hire Onboarding for Lab Services. Click OK.

      NOTE
      The name must be unique within its business unit.
6. Specify new_hire_onboard_labservices in the _id attribute for the form, press Enter, and click Save.

      NOTE
      After you specify a new value for any attribute, press Enter to enable the Save button. This value of the _id attribute of the form is the form ID. Specify a unique ID for each form in a business unit.

Step 2 - Add Fields for Contact Information

Modify the form that you copied in the previous procedure by adding more contact information fields. These new fields increase the options for other employees to communicate with the Lab Services employees.

Follow these steps:

1. Expand the New Hire Onboarding for Field Lab Services form.
   The list of components in the form appear in the tree under the form name. This list matches the same named components in the middle pane.
2. Copy the Last Name field, to create another field, as follows:
   a. Select the Last Name field in the tree, and click the Copy icon.
   b. Select the name of the New Hire Onboarding for Lab Services form in the tree, and click Paste.

   c. In the _id attribute in the right pane, specify a unique ID. For example, employee_id. Press Enter and click Save.
3. Rename the new field, as follows:
   a. Select the field and click Rename.
      The Name dialog appears.
   b. Enter the new name as Employee ID. Click OK.
4. Move the Employee ID field, as follows:
   a. Select the name of the field in the tree.
   b. Drag it upwards until the horizontal line for the field appears above the Job Title field, and drop it.
   c. Verify that the First Name, Last Name, and Employee ID fields now appear sequentially in the tree.
5. Using steps 2-4 as a model, add the following new fields and move them under the Employee ID field:
   – Email
   – Phone number
   – Address

You have added more fields for contact information.

Step 3 - Configure Fields to be Populated Automatically

Use JavaScript expressions to retrieve the data from the user database and populate the fields automatically to reduce errors and increase efficiency.

Follow these steps:

1. Expand the Forms tree to display the fields of the New Hire Onboarding for Lab Services Only form.
2. Specify the following JavaScript expressions in the Value attributes of the following fields. Press Enter and click Save after you specify each value.
   – First name: $(_.user.firstName)
   – Last name: $(_.user.lastName)
   – Employee ID: $(_.user.id)
   – Email: $(_.user.email)
   – Phone: $(_.user.phone[0])- Primary phone
      $(_.user.phone[1])- Secondary phone
   – Address: $(_.user.address)

When a catalog user opens this form while requesting a service, these fields are automatically populated based on the user ID specified at login.

Step 4 - Copy and Modify Radio Buttons

You can add radio buttons to the New Hire Onboarding for Lab Services Only form to allow users to select the type of server. Copying and modifying the radio buttons is more efficient than creating them. Therefore, you copy the radio buttons from the Procure Sun Server form and modify them for this form.

Follow these steps:

1. Expand **CA Catalog Content** in the Forms tree and perform these actions:
   a. Expand **Procure Server** and **Procure Sun Server**.
   b. Select the Choose Server Model field and click the Copy icon.
   c. Navigate the Forms tree to the New Hire Onboarding for Lab Services Only form. Select this form and click the Paste icon.
2. Move the Choose Server Model field, as follows:
   a. Select the name of the field in the tree.
   b. Drag it upwards until the horizontal line for the field appears above the Additional Info field, and drop it.

c. Verify that the Hiring Manager, Choose Server Model, and Additional Info fields now appear sequentially in the tree, followed by the other fields.
d. Verify that the Choose Server Model field includes these options Starter, Mid Size, and High End.

NOTE

The values for _id and other attributes for these fields are already set, because you copy them rather than create them. We use the same attribute values in this scenario because the values remain unique within the same form.

Step 5 - Add Text Areas

Add a text area to provide specifications for each type of server.

Follow these steps:

1. Verify that the New Hire Onboarding for Lab Services Only form is displayed in the Preview Pane.
2. Click **Components**, **System** in the Form tree.
3. Drag and drop the the Text Area element to your form directly above the Additional Info field.
4. Verify that the text area appears above the Additional Info field in the Preview Pane.
5. Select the Text Area in the tree and click the **Rename** button. Specify the new name as Starter.
6. Specify these values for the following attributes. After you specify each value, press Enter and click Save.
   - _id: starter_description
   - Value: 10K (ULTRA 10000, Starfire) UltraSPARC® processor
   - Disabled: true
     This setting means that this field is read-only, so that users *cannot* change the text that you specified.
   - Hidden: false
7. Repeat the previous steps for the next two text areas: Mid Size and High End. Use the following specifications for the attributes:
   Mid Size Text Area:
   - Name: Mid Size
   - _id: midsize_description
   - Value: 20K UltraSPARC IV & UltraSPARC III Processors
   - Disabled: true
   - Hidden: true

   High End Text Area:
   - Name: High End
   - _id: highend_description
   - Value: E25K UltraSPARC IV & UltraSPARC III Processors
   - Disabled: true
   - Hidden: true
8. Verify that the three Text Area fields appear on the form before the Additional Info field.

Step 6 - Show and Hide Text Areas

Add JavaScript functions to the form so that *only* the description of the selected server type (Starter, Mid Size, or High End) appears. If the user selects a new server type, the previous description is replaced with the description for the new selection. This enhancement helps save time for the user by eliminating the need to scroll through multiple static descriptions. This reduced scrolling is especially helpful for mobile users.

Follow these steps:

1. Select the New Hire Onboarding for Lab Services Only form in the Preview Pane.

2. Click the Script icon at the top of the form and proceed as follows:
   a. Delete the existing sample code.
   b. Copy and paste the JavaScript code from the section that is named JavaScript for New Hire Onboarding for Lab Services into the Script dialog.
   c. Save the script, and close the dialog.
3. Expand the form to show the fields for the Choose Server Model radio buttons.
4. Enter the following JavaScript function call in the onClick attribute of the Starter, Mid Size, and High End fields. After you update the attribute for each field, press Enter and click Save.

   ```
   ca_fd.js.onFormLoad();
   ```

   This call loads the JavaScript code that you copied earlier to the Script dialog for the form.

JavaScript for New Hire Onboarding for Lab Services

```
{
  onFormLoad : function() {
    ca_fd.js.clickStarter();
    ca_fd.js.clickMidSize();
    ca_fd.js.clickHighEnd();
  },

// ShowFields/HideFields either shows or hides the description field for Starter servers when Starter radio is
 changed
// ResetFields will reset MidSize and HighEnd when selected and clicked away from Starter
// This function is called in to Starter radio's onChange
  clickStarter : function() {
    if (ca_fdIsSelectRadio('new_hire_onboard_labservices','class','starter')) {
ca_fdShowField('new_hire_onboard_labservices','starter_description'); }

    else {
ca_fdHideField('new_hire_onboard_labservices','starter_description');

        ca_fdResetFields('new_hire_onboard_labservices',['starter_description']); }
  },

// ShowFields/HideFields will either show or hide the description field for Mid Size servers when MidSize
 radio is changed
// ResetFields will reset the description field for Mid Size servers when selected and clicked away from
 MidSize
// This function is called in to MidSize radio's onChange
  clickMidSize : function() {
    if (ca_fdIsSelectRadio('new_hire_onboard_labservices','class','midsize')) {
            ca_fdShowField('new_hire_onboard_labservices','midsize_description'); }

    else {
ca_fdHideField('new_hire_onboard_labservices','midsize_description');

        ca_fdResetFields('new_hire_onboard_labservices',['midsize_description']); }
  },

// ShowFields/HideFields will either show or hide the description field for High End servers when HighEnd
 radio is changed
// ResetFields will reset hide the description field for High End servers when selected and clicked away from
 HighEnd
```

```
// This function is called in to HighEnd radio's onChange
  clickHighEnd : function() {
    if (ca_fdIsSelectRadio('new_hire_onboard_labservices','class','highend')) {
        ca_fdShowField('new_hire_onboard_labservices','highend_description'); }

    else {
ca_fdHideField('new_hire_onboard_labservices','highend_description');

        ca_fdResetFields('new_hire_onboard_labservices',['highend_description']); }

  },
}
```

Step 7 - Add Spinner Fields

As a best practice, configure each form so that users can specify only a valid number of items. This practice helps reduce errors. To implement this practice, the form uses a spinner to field to let users request 0-10 servers of each type.

Follow these steps:

1. Verify that the New Hire Onboarding for Lab Services Only form is displayed in the Preview Pane.
2. Click **Components**, **System** in the Form tree.
3. Drag and drop the Spinner Field element to your form directly above the Additional Info field.
4. Verify that the Spinner Field appears above the Additional Info field in the Preview Pane.
5. Select the Spinner Field in the tree and click the **Rename** button. Specify the new name as Number of Starter Servers.
6. Specify these values for the following attributes. After you specify each value, press Enter and click Save.
   - Maximum Value: 10
   - Minimum Value: 0
   - Increment: 1
7. Repeat the previous steps for the next two fields: Mid Size and High End.
   Letting users specify 0-10 of each type of server provides flexibility for users. To prevent users from specifying 0 for all three types, specify a value of 1 or higher as the default value of one field.
8. Verify that the three spinner fields appear on the form before the Additional Info field.

Step 8 - Test the Form in a Service

Test the form before you make the form available in a service.

Follow these steps:

1. Select the service to which you want to add the form.
2. Add the form to a service option group in that service.
3. Verify that the service is available.
4. View the service in the catalog, complete the form, and submit the request.
5. Verify the specifications if you detect any errors.
   For example, if the text areas do not show or hide as planned, verify that the related JavaScript code and attributes are correct.

Create and Customize a Form

This article contains the following topics:

Administrators add and configure elements on forms, including text fields, check boxes, radio buttons, and more. Each element of the form captures information for approving and completing a request for a service. All services include at least one

> **WARNING**
>
> We recommend that you do not create multiple forms with the same _id value for each form. If you already have multiple forms with the same _id value, do not associate such forms to a single Service Option. Having multiple forms with the same _id value in a single Service Option can cause validation errors.

Follow these steps:

1. Verify that the elements do *not* duplicate fields on other forms for the same service, including system forms.
2. Decide which elements you want on the form and in which order. Verify that each part of the draft form has a matching element on the Form Designer.
3. Using the Form Designer, create a form.
   Alternatively, consider these options:
   - Use an existing Form Designer form.
   - Use an existing Form Designer form as a starting point, and modify it to include the changes you want.
     For both possibilities, see the forms in the predefined services and any custom services that you have. See also the forms in the content packs.
   - Use the Customize a Predefined Form scenario and create a form by copying and modifying the predefined form.
4. For each element, specify the HTML attributes and JavaScript attributes.
5. Attach the form to a service option group and test it in a service.

Create a Form

You can create a form to include in a request. You can also capture user input that is required to complete the request. For example, a form typically lists options and records the ones that the user selects.

> **NOTE**
>
> The form level associations are showed only for the selected Business Unit level.

Follow these steps:

1. Click **Catalog**, **Forms**.
2. Click the **Forms** folder to select it.
3. Click the **Add** button above the **Components** folder and select **Form** from the drop-down list.
4. Enter the name of the new form and click OK.
   The new form appears in the Forms folder. The form attributes appear in the Item Inspector (the right pane).
5. Specify the required and optional form attributes. These attributes are for *the form itself*. These attributes are *not* the same as the HTML and JavaScript attributes of the *elements* on the form.

Add, Move, and Delete Elements

You can add and configure the elements, such as text fields, text areas with check boxes and radio buttons.

Follow these steps:

1. Click **Components** and expand **System**. Verify that the element you want to add appears in the tree view under the System folder.
   For example, to add a text field, verify that the System folder appears and is expanded to show the Text Field option. Similarly, to add a select box and its options, verify that the System, Radio Group folder appears and is expanded to show the Radio option.
2. Verify that the form to which you want to add the element already exists.
3. Expand the **Forms** folder. Scroll to the form to which you want to add the element.

4. Verify that both the element you want to add and the intended form are visible in the tree view simultaneously.
5. Click the element (for example, Text Field), drag and drop it on the intended form in the Component Tree.
   For example, you have a custom form named Model Options. Click Text Field under the System folder. Drag Text Field to Model Options, and drop it there. All these actions can be on the Component Tree or the Preview Pane.
   When you click and drag an element, both a checked circle and the message "1 object selected" appear next to the cursor.
   As you drag the element over a form or existing form element on which it can be dropped, the checked circle turns green. Also, the cursor changes to appear as a hand. Conversely, the checked circle turns red when the dragged element is not over a form or form element on which it can be dropped.

   NOTE

   The element is not added if the attempted action violates a rule. Examples include attempting to drop a select option onto a radio group or to drop a select group onto another select group.
6. Specify the *required* HTML attributes for the new element, and click Save.
   The required HTML attributes for each element appear in bold.

   NOTE
   Attempts to perform a different action *before* completing this step fail. For example, attempts to drag-and-drop a new element *before* saving this element fail.
7. Check the form in the Preview Area (the middle pane) to verify that the element was added to the form as intended. If it was not, you can either rearrange the elements or delete the new element and try adding it again.
8. Specify the optional attributes (if any) that you want for the new element, and click Save.
9. By default, components are organized in one column. You can optionally configure the form to use two columns, as follows:
   a. Drag Column Layout from the System folder and drop it onto a desired form.
   b. When you drop this layout, the tree updates with Column Layout and its two children, Column 1 and Column 2 automatically.
   c. Click the icon next to Column Layout to display the two columns, which are named Column1 and Column 2.
   d. Drag-and-drop other elements, such as text fields, radio groups, and select boxes, onto Column 1 and Column 2.

   NOTE
   When you highlight the Column Layout or a Column, the corresponding sections in the Preview Area of form may be rendered gray and disabled, and the elements within the column may be highlighted with red border.
10. Specify the JavaScript attributes (if applicable) for the new element, and click Save.
11. Specify a meaningful name for the new element, as follows:
    a. Select the new element in the Component Tree and click the **Rename** button at top of the Component Tree.

       NOTE
       Verify that you select the new element on *your* form in the Component Tree. Do *not* select the same-named element on the *System* folder in the Component Tree.
    b. Enter the new name and click OK.

    For example, perform these actions in a custom External Storage Drive Order Form:
    – Drag-and-drop a radio group and change its name from the default Radio Group to Capacity in GB.
    – Drag-and-drop three radio buttons onto that radio group. Change their names from the default radio button to Small (50), Medium (100), and Large (150), respectively.

       NOTE
       When you create a new element, a colon (:) is automatically included at the end of its name. The colon does not appear in the Name dialog when you rename the element. However, it does appear in the Preview Area after you click OK to close the dialog. Verify that you do not enter another colon in the Name dialog. Otherwise, the element has *two* colons after its name in the Preview Area.
12. Verify that the new element appears and functions as you intended. If necessary, refine any attributes to obtain the appearance and function you want.

13. (Optional) Move elements on the form by dragging and dropping them onto a different location in the form.
14. (Optional) Delete the elements that you no longer want on the form by selecting them and clicking **Remove**.

Specify Attributes for an Element

After administrators add an element to a form, they specify its required HTML attributes and any optional HTML attributes or JavaScript attributes. Specifying these attributes helps you modify the element to meet your requirements.

Follow these steps:

1. Click **Catalog**, **Forms**.
2. Open your form. For example, expand the Forms folder and select the form whose elements you want to configure.
3. Click the icon next to the form name to expand the folder for the form and display its elements.
4. Select the element whose attributes you want to set.
   The Item Inspector (the right pane) displays the names and values of all HTML attributes and JavaScript functions available for the selected element.
5. In the Item Inspector, search the Name column and locate the attribute, one of the following types:
   - HTML Attributes
   - JavaScript Attributes
6. Move your cursor to the adjacent field in the Value column and click.
7. Enter the value for this attribute and click Save.
   For example, you can set the value for Hidden attribute of a Label element.
8. Verify that all required attributes are set correctly.

Attach a Form to a Service Option Group

Administrators attach a form to one or more applicable service option groups. Users who select one of these service options in a request are prompted to complete the form. The form lets the user enter data to complete the request.

Follow these steps:

1. Click **Catalog**, **Service Offerings**,**Option Groups**.
2. Click the service option group name to which you want to add the form.
3. Click the **Edit** icon for the service option to which you want to add the form.
4. Click the **Add** button for the form.
5. Enter meaningful text that describes the purpose of this form, in the Display Text Field.
6. Click the magnifying glass icon in the Form Name field to select a form.
7. Navigate through the form tree and select the form that you want to attach.
8. Click the Select Form button.

   NOTE
   You can add a form to a service option only once.
9. If you do *not* want to hide or disable the entire form at any point, leave the Hidden and Disabled fields blank. In that case, skip the remainder of this step.
   If you want to hide or disable the entire form according to a criteria, enter the corresponding JavaScript expression in the Hidden or Disabled field. Use the following format: $(_.*object.property*). The expression must return a value of true or false.
   Example:
   - To hide or disable the form when the request status is Pending Approval, enter: $(_.request.status == 400)
   - To hide or disable the form for end-user roles *only*, enter: $(_.user.role == ‘enduser’)

   You can also specify other JavaScript expressions in fields.

NOTE
When a form is disabled, it is disabled but visible during all stages of the request life cycle *except* check-out. During check-out, a disabled form is both disabled and hidden.

10. Click the Options tab and complete the remaining fields on the Service Option Element Options Window--Options Tab.
11. Click Update and click Save.

Perform Automated Tasks in Form Fields

Administrators can use JavaScript expressions and JavaScript functions to perform automated tasks in form fields. At runtime, when the user fills out the form while completing a request, the JavaScript expression or function runs and performs the specified task.

WARNING
To use JavaScript expressions and functions, you must have a working knowledge of HTML, CA Service Catalog administration, and JavaScript.

Follow this process to specify automated tasks in fields, using one or more of the following options: JavaScript expressions, JavaScript functions, and report data objects.

1. Verify that you can represent the field or fields as one or more of the elements available for creating forms. The elements can be text fields, check boxes, select boxes, or radio groups.
2. Determine whether to achieve the goal you intend for the field by one or both of the following methods:
   – Specifying a JavaScript *expression* for the value of an *HTML* attribute of the element
   – Specifying a custom or pre-defined JavaScript *function* for the value of a *JavaScript* attribute of the element
3. Consider specifying a JavaScript expression for the value of an HTML attribute of the element. The JavaScript expression runs only once, as soon as the form appears.
   Examples:
   – Pre-populating user-related data in one or more fields on a form.
   – Disabling a field based on the request status, on the business unit or role of the user completing the form.
   – Hiding a field based on the request status, on the business unit or role of the user completing the form.
   JavaScript expressions retrieve the values of object properties from the CA Service Catalog database at runtime. These properties are grouped into the following categories: user, business unit, service, service option groups, and request.

   NOTE
   To hide an *entire* form, you do not specify JavaScript expressions for any fields but for the entire form when you attach the form to a service option group.
4. Consider specifying a JavaScript function for the value of a JavaScript attribute of the element. (For more information about *predefined* JavaScript functions, select Administration, Tools, Links, Form Designer JavaScript API.)
   This method is best suited for dynamic events that occur when the user performs an action while completing the form. The JavaScript function can run multiple times when a user checks a box, enters a value, or performs some other action. Thus, JavaScript functions are most useful when you want to create dynamic relationships between fields.
   – Enabling or displaying a list of options or fields under a check box when a user enables the check box.
   – Disabling or hiding a list of options or fields under a check box when a user clears the check box.
5. Test the JavaScript expressions and JavaScript functions to verify that they accomplish your goals for the fields on the form.

Use JavaScript Expressions in Fields

This article contains the following topics:

Administrators follow this process to use JavaScript expressions to perform automated tasks in fields:

1. Decide the fields on the form for which you want to specify a JavaScript expression. Also, decide the task that you want the JavaScript expression to perform. For example, you can pre-populate fields. You can hide or disable forms or fields under certain specific conditions. Another example is to pre-populate the value attribute of a field with the runtime value of one of many objects and properties from the CA Service Catalog database. To pre-populate a field for a user while the user is completing a form, specify a simple JavaScript expression for the value attribute of the field, using the following format: $(_.*object*.*property*), such as $(_.user.lastName)
2. Verify that you can represent the field as one of the elements available for creating forms.
3. Verify that the purpose or goal you intend for the field is represented in one of the following HTML attributes. You can specify JavaScript expression for the values of the following HTML attributes *only*.
   - Value - Value is the only attribute for which you can use JavaScript expressions to pre-populate values in fields. The value attribute is most frequently used for this purpose, especially for pre-populating user-related data such as address, phone numbers, business unit, and other personnel-related data
   - Disabled - You can disable a field, according to the user's role or business unit, the request status, or other criteria that you specify.
   - Hidden - You can hide a field, according to the user's role or business unit, the request status, or other criteria that you specify.
   - Checked - Check boxes and radio buttons support the checked attribute.
     You can select a check box or a radio button, according to the user's role or business unit, the request status, or other criteria that you specify.
   - Required - You can set a field as required, according to the user's role or business unit, the request status, or other criteria that you specify.
4. Keep in mind these rules when coding JavaScript expressions for use in the HTML attributes:
   - For the disabled, hidden, checked, and required attributes, the JavaScript expression must return a value of true or false.
   - If the returned value is any value except true, CA Service Catalog automatically uses a value of false. The value attribute is the only attribute that is used for pre-populating fields.
5. Review the objects and properties that you can specify in JavaScript expressions and determine which ones you need. These objects and properties are retrieved from the CA Service Catalog database at runtime, as noted earlier, and are grouped according the following categories: user, business unit, service, service option group, and request.
6. If the objects and properties that you need cannot be retrieved using a JavaScript expression, consider using the following pre-defined JavaScript function to retrieve them: ca_reportQuery(reportId, variables, onSuccess, onFailure).
7. Review and follow the instructions for performing any of the following tasks or use them as a model for a similar task, if applicable:
   - Pre-populate fields
   - Hide or disable a field based on the request status, the business unit or role of the user completing the form, or on other criteria
   - Select options for fields by default based on the request status, the business unit or role of the user completing the form, or on other criteria; you may select a check box or an option in a radio group by default, based on such criteria
   - Hide, enable, or disable an entire form under specified conditions.
8. Using all applicable information from the previous steps, construct and test the JavaScript expression to accomplish your goals for the fields on the form.

Objects and Properties that You Can Specify in JavaScript Expressions

You specify objects and properties in JavaScript expressions. You can use these expressions for the runtime values of the *value* attribute for the elements that you use as fields on forms. Typically, you specify these expressions in the following format: $(_.*object*.*property*).

For example, to capture the runtime value of the logged in user completing the form in a request, specify $(_.user.firstName). You can also concatenate strings, as explained later in this topic.

Forms

The form object contains the following property that you can use in JavaScript expressions or as a first parameter in JavaScript functions: ca_fd.formId.

formId refers to the *active* form. The form is in active when a field in the form triggers an event. The following are a few sample actions that activate forms:

- The user opens the form while working on a request, which activates the JavaScript attribute named onLoad.
- The user clicks a label or image, which activates the JavaScript attribute named onClick.
- The user submits the form, which activates the JavaScript attribute named onSubmit.

User

The user object contains an array of user properties that you can access using _.user.

Each user object has the following properties, listed in related groups:

- id, uuid, status (0 = active, 1 = inactive)
- firstName, lastName, middleName, commonName, alias, title
- groups
  The _.user.groups property specifies the name of a user group in CA EEM. You can use this property to manipulate a field based on whether the user belongs to a specific CA EEM group. For more details, see the Example section.
  The _.user.groups property contains an array with all the CA EEM groups to which the user belongs.
- manager, delegate, description
- phone (an array: phone[0] = primary and phone[1] = secondary)
- mobile, fax, pager, email
- timezone, localeLanguage, localeCountry, defaultRole, defaultDomain, location.uuid, location.name, location.city, location.state, location.country, location.postalCode, location.phone, location.fax, location.description, location.address[0-5]
- roles.<domain>

Examples

You can use the _.user.groups property to hide a field from members of a CA EEM group, for example, the group named developers. To do so, set the value of the HTML attribute named Hidden to the following:

```
_.user.groups.indexOf("developers") >= 0
```

In contrast, you can also make a field visible to users who are *not* members of a CA EEM group. To do so, set the value of the Hidden attribute to the following:

```
_.user.groups.indexOf("developers") < 0
```

Business Unit

The business unit object contains an array of business unit properties that you can access using _.bu.

Each business unit object has the following properties, listed in related groups:

- id, name
- type(business unit type where SP=service provider, ST=can have child business units, TE=cannot have child business units)
- singleAccountMode (true or false)
- status (0=inactive, 1=active)
- openedDate, description, timezone, federalTaxId, stateTaxId, taxRegion, currency, dateFormat, parent (parent bu id)
- email, website, primaryContact (contact userid), location.uuid, location.name, location.city, location.state, location.country, location.postalCode, location.phone, location.fax, location.description, location.address[0-5]
- data1, data2, data3, data4, and data5

Request

The request object contains an array of request properties that you can access using _.request.

Each request object has the following properties, listed in related groups:

- id, name, requestedFor, requstedForAccountId, requestedBy, requestedByAccountId
- description, priority, status
- dateCreated, completionDate, dateRequired, and lastModified
- newDateRequired, newName, and newPriority
  These properties reflect the new value of an object as soon as the user updates the object during the request lifecycle. (The user is typically a request manager or the requestor.)
  The user can update the object, for example, while approving, rejecting, or pushing through the pending action for the item associated with the property. The *new* value takes effect *immediately* when the user updates the property; saving is *not* required.
  When creating or editing forms, you can use these properties at your discretion. For example, you can use these properties in a custom JavaScript function for validation that you use in the JavaScript attribute named OnSubmit. For example, if a user enters an invalid date such as a holiday, your custom JavaScript function can display text explaining the valid dates.

Service

The service object contains an array of services that you can access using _.service.

Each service object has the following properties, listed in related groups:

- id, bu, name, description
- status(0=deleted, 1=available, 2=unavailable, 3=created, 4=cancelled, 5=total)
- website, code, version, dateAvailable, dateUnavailable, dateCreated, dateCancelled

Service Option Groups

The service option groups object contains an array of service options that you can be access using _.sog.name.

Each service option group has the following properties, listed in related groups:

- id, bu, name, description
- status(0=deleted, 1=available, 2=unavailable, 3=created, 4=cancelled, 5=total)
- code
- dateAvailable, dateUnavailable, dateCreated, dateCancelled

Service Options

The service option object applies *only* to the predefined JavaScript function named _.serviceoption.status( ).

The _.serviceoption.status( ) function contains a single property, which is named newStatus.

NOTE

This function does *not* take any parameters.

- **newStatus**
  Specifies the *new* value for the status when a user action updates the status of the service option. This property takes the new value for the status as soon as the user does one of the following:
  – Selects a new status value from the drop-down list in the Action column while managing requests pending action . A request manager typically performs this action. The value of newStatus changes dynamically as soon as the user selects the new status of the service option.
  – Opens a page or dialog that changes the status to a fixed value when the user clicks OK. For example, when a user creates a request, the value of newStatus changes dynamically as soon as the user creates the request. In this example, the value of newStatus immediately becomes Submitted.

  In both cases, the value of _.serviceoption.newStatus() changes immediately. In contrast, the value of _.serviceoption.status( ) remains unchanged until the page is submitted and refreshed.

Examples

You can use the newStatus property to help your implementation process reservations efficiently. For example, suppose a user submits a main request for a computer and additional requests for related options. These options include extra memory, an upgraded keyboard, and so forth. You can create a custom JavaScript function so that if the main request is rejected, the onSubmit function runs for all optional requests. Examples of optional requests include accessories, extra memory, and so forth. For example, the following expression evaluates to True when the original status is 400 and the newStatus is changed to 800:

```
if (_.serviceoption.status() == 400 && _.serviceoption.newStatus == 800)...
```

Operators

The operators most commonly used for form design are described here. For complete details about standard operators, see the JavaScript standards reference that your organization uses, for example, www.developers.sun.com, or www.javascript.com.

You can specify all standard operators in your JavaScript expression *except* assignment operators.

The assignment operators are as follows: =, +=, -=, *=, /= .

For example, the folowing expression is invalid because it uses the = assignment operator:

```
$(var x = 1+2)
```

Required Return Values

Specify JavaScript expressions for the disabled, checked, and hidden attributes that return one of the following values:

- true or “true”
- false or “false”

If the expression returns any other value, CA Service Catalog replaces it with a value of false. Therefore, if you specify $(_.user.firstName) for the disabled attribute of a text field, the field is not disabled unless "true" is the first name of the user.

Concatenation Operators

You can optionally concatenate two strings together, using the + operator,

For example, $(‘Hello ‘ + _.user.firstName + ‘ ‘ + _.user.lastName) returns the following text: Hello *first-name last-name.* "Hello John Doe" and "Hello Jane Smith" are samples.

Comparison Operators

You can optionally use the following comparison operators:

- == is equal to
- === is exactly equal to (value and type)
- != is not equal
- > is greater than
- < is less than
- >= is greater than or equal to
- <= is less than or equal to

Logical Operators

You can optionally use the following logical operators:

- and (&&)
- or (||)
- not (!)

Pre-Populate Fields Based on JavaScript Expressions

When you use an element that includes the *value* attribute as a field on a form, you can use a JavaScript expression to pre-populate the element's value attribute with the runtime value of one of many objects and properties from the CA Service Catalog database.

For example, in the text field for First Name on your form, you can specify the JavaScript expression $(_.user.firstName) to pre-populate the field with the first name of the logged in user ID who is creating the request and completing the form. Similarly, in the text field for Last Name, you can specify the JavaScript expression $(_.user.lastName) to pre-populate the field with the user's last name.

Follow these steps:

1. Design and create the form.
2. Review the guidelines for using JavaScript expressions in fields and verify that you want to pre-populate the value attribute of an element in the field.
3. Verify that the data that you want to pre-populate is one of the objects and properties that you can specify in JavaScript expressions. These objects and properties are related to the logged in user's personnel data, to one or more business units, or to service, service options, status, or other data related to the request that contains the form.
4. Specify the JavaScript expression in the value attribute of the element for the field. When specifying the expression, follow all syntax rules in objects and properties that you can specify in JavaScript expressions, especially the rules related to using operators to link properties. The following examples are some valid expressions:
   - User's first name: $(_.user.firstName)
   - User's last name: $(_.user.lastName)
   - User's first name and last name, concatenated in a single field: $(_.user.firstName + ' ' + _.user.lastName)
   - User's physical address data, concatenated in a single field: $(_.user.location.address[0] + ‘ ‘ + _.user.location.address[1] + ' ' + _.user.location.city + ' ' + _.user.location.state)
     This example returns the street address, city, and state of the user, for an address in the United States.
   - User's role: $(_.user.roles[domainId])
   - User's business unit, with parent: $(_.bu.id)
   - User's business unit, without parent: $(_.bu.id.parent)
5. Test the JavaScript expression to verify that it accomplishes your goals for pre-populating the field on the form.

Hide or Disable a Field Based on Request Status, Business Unit, Role or Other Criteria

Decide to use JavaScript expressions to hide or disable a field based on the request status, on the business unit or role of the user completing the form, or on other criteria.

As an administrator, you typically disable fields that you want users to view but *not* update. Examples include options for a service that the user's manager selected or that the request manager selected based on available inventory.

In contrast, as an administrator, you can hide fields when you either do not want users to be aware of them at all, for some reason, or when the fields add distracting information that the user does not need. Examples include cost data, inventory options available to certain roles only or certain business units only, or data that does affect the end user, such as the estimated administrative cost of fulfilling the service.

To hide a field, specify the JavaScript expression that specifies your criteria in the hidden attribute of the field element. Similarly, to disable a field, specify the JavaScript expression that specifies your criteria in the disabled attribute of the field element.

> **WARNING**
> Hiding or disabling a *field* is not completely secure. While the field and its value are not displayed on the CA Service Catalog page, the data does still exist on the form's HTML page and can be accessed by viewing the source text of the browser. Therefore, to restrict sensitive data, create two versions of the form and hide the form that contains the sensitive data from users who you do not want to see it. For more information, see How to Hide, Enable, or Disable an Entire Form Under Specified Conditions.

Follow these steps:

1. Design and create the form.
2. View the field that you want to enable or disable and verify that it includes the HTML attribute named disabled.
3. Determine the exact criteria that you want to use to disable or enable the field. Examples of the criteria are the request status, the business unit or role of the user completing the form.
   If the disabled attribute is set to true, users can view the field but cannot edit it. Conversely, if the disabled attribute is set to false, users can both view and edit the field.
4. Verify that the data that you want to use to determine whether to enable or disable the field is one of the objects and properties that you can specify in JavaScript expressions.
5. Review the guidelines for using JavaScript expressions in fields.
6. Specify the JavaScript expression in the value attribute of the element for the field. The following examples are some valid expressions:

1. - To hide or disable a field if the user has the end-user role, set the hidden or disabled attribute of that field to the following JavaScript expression: $(_.user.roles[_bu.id]==’enduser’).
     This expression returns a value of true and therefore hides or disables the field *only* if the user filling out the form has an enduser role in the current business unit.
     You can use this setting or a similar one to restrict users from either viewing or editing the text field that is named Memory in a form for a request for a new laptop computer. Conversely, this example displays or enables the field for all other roles in the business unit. Thus, when used with the disabled attribute, this example permits the request manager to edit the field, based on the available inventory.
     To display or enable a field for only request managers, hide or disable it for all other roles, by setting the hidden or disabled attribute of that field to $(_.user.roles[_.bu.id] != ‘requestmanager’).
   - To hide or disable a field in a form that is shared by all business units *only* if the current business unit is PBJ222, set the hidden or disabled attribute of that field to $(_bu.id==’PBJ222’).

Select Options for Fields by Default Based on Request Status, Role, Business Unit, or Other Criteria

As an administrator, you can use JavaScript expressions to select a check box by default or select an option for a radio group by default,.Conversely, you can clear certain check boxes or radio group options by default according to such criteria. For example, select the check boxes related to publicly known managerial duties by default for managers only, while clearing them for other users.

Selecting or clearing the check box or radio group option by default does not prevent the user from changing the default that you specified before submitting the request that contains the form. Therefore, before using this feature, verify that

permitting users to change the default is appropriate. Consider redesigning the form to either hide or disable the field the options that you want to prohibit users from changing, according to the criteria specified in your JavaScript expression.

Follow these steps:

1. Design and create the form.
2. On the Form Designer, view the check box or radio group option that you want to select by default and verify that it includes the HTML attribute named checked.
3. Determine the exact criteria that you want to use to select the check box or radio group option by default. The criteria can be based on the request status, on the business unit or role of the user completing the form, or some other criteria.
4. Verify that the data that you want to use to make the selection is one of the objects and properties that you can specify in JavaScript expressions. These objects and properties are related to the logged in user's personnel data, to one or more business units, or to service, service options, status, or other data related to the request that contains the form.
5. Review the guidelines for using JavaScript expressions in fields.
6. To select a check box or select an option for a radio group by default, specify the JavaScript expression that specifies your criteria in the checked attribute of the field element. When the user opens the form, the expression runs as follows:
   - If the criteria *are* satisfied, then value that is returned is true. Therefore the check box or radio group option is selected by default.
   - If the criteria are *not* satisfied, then value that is returned is false. Therefore the check box or radio group option is *not* selected by default.

   When specifying the expression, follow all syntax rules in objects and properties that you can specify in JavaScript expressions, especially those related to using operators to link properties. The following examples are some valid expressions: In these examples, the term *option* refers to both a check box and a radio group option.
   - To select an option by default if the user has the enduser role, set the checked attribute of that field to the following JavaScript expression: $(_.user.roles[_bu.id]==’enduser’).
     This expression returns a value of true and therefore selects the option by default *only* if the user filling out the form has an enduser role in the current business unit.
   - To select an option by default for request managers only, set the checked attribute of that option to $(_.user.roles[_.bu.id] != ‘requestmanager’).
   - To select an option by default in a form that is shared by all business units *only* if the current business unit is PBJ222, set the checked attribute of that option to $(_bu.id==’PBJ222’).
   - To select an option by default in a form that is shared by all business units *only* if the parent of the current business unit is PBJ222, set the checked attribute of that option to $(_bu.id.parent==’PBJ222’).

Hide, Enable, or Disable an Entire Form Under Specified Conditions

To hide, enable, or disable an entire form under certain conditions, you attach the form to a service option group and enter the JavaScript expression you want in the Disabled and Hidden fields of the Service Option Group Definition dialog. You can hide, enable, or disable an entire form according to the request status, the role or business unit of the logged in user, or according to other criteria that you specify. For more information, see Attach a Form to a Service Option Group.

Use JavaScript Functions in Fields

This article contains the following topics:

Administrators follow this process to use JavaScript functions to perform automated tasks in fields:

1. Consider using the pre-defined JavaScript functions.
2. If none of the pre-defined functions meets your goals, then write your own custom function:

- Follow the guidelines for custom JavaScript functions.
- As a best practice, use the Script dialog on each form to create and maintain the custom JavaScript functions for the form.

Consider these examples:

- Retrieving the values of one or more object properties from the CA Service Catalog database, the MDB, or other data source. This option especially useful when the value *cannot* be retrieved with a JavaScript expression. Consider all options to populate and pre-populate fields automatically.
- Validating user input, specifically validating the format of numeric or address data entered by the users, including social security numbers, credit card numbers, telephone numbers, email addresses, and IP addresses.

3. As needed, specify JavaScript functions to link fields in two forms for simultaneous updates.
4. Using all applicable from the previous steps, construct and test the JavaScript function to accomplish your goals for the fields on the form.
5. As a best practice, ensure that you test the function in a form used with a service in a test environment before you use the form and the service in a production environment.

Pre-Defined JavaScript Functions

CA Service Catalog provides several options for performing automated tasks for fields in forms, including report data objects, JavaScript expressions, and JavaScript functions.

WARNING
Information and documentation pertaining to Predefined JavaScript API functions, descriptions, and examples are available on the CA Service Catalog console. Select **Administration**, **Tools**, **Links**, **Form Designer JavaScript API, Components.** Click **Global, Checkbox, Date, Lookup, Radio, Select, Slider, Table,** and/or **Text options** to view the Javascript APIs, functions, and description**.**

Guidelines for Custom JavaScript Functions

Use the following guidelines for creating custom JavaScript functions in forms:

- Follow the instructions for using predefined JavaScript functions, as applicable.
- Follow the instructions for using JavaScript attributes.
- You can use either the Script dialog or the custom_form_lib.js file to store your custom JavaScript functions, as follows:
  - As a best practice, use the Script dialog on each form to create and maintain the custom JavaScript functions for the form. If only one form uses the custom JavaScript function, the Script dialog is preferred over the custom_form_lib.js file for the following reasons:
    - The Catalog system loads the form most efficiently when the custom JavaScript function is in the Script dialog.
    - Using the Script dialog enables you to more efficiently move the form, custom functions, and related services or service option groups together. This consideration is important if you move these objects because of an upgrade, migration, or other event.

      NOTE
      To move these objects, use the Import Export utility. For more information about the utility, see the Migrate Data between Catalog Systems section.
  - Alternatively, store custom JavaScript functions in the custom_form_lib.js file in the filestore. This method is preferred *only* when two or more forms use the same custom JavaScript function.

    WARNING
    If you use the custom_form_lib.js file, use a third-party JavaScript minification tool to *minify* this file on production sites. Minifying the file removes unnecessary characters from the file and reduce its size. As a

result, the Catalog system loads the form more efficiently. Several third-party JavaScript minification tools are available on the Internet.

- Do not write excessively long custom JavaScript functions.
- Design your forms with usability as a high priority.
- Take into account the browser locale, to support other language users and to localize forms as needed.

Use the Script Dialog to Maintain Custom JavaScript Functions

The Script dialog for each form lets you create and maintain JavaScript functions that apply to a form. The functions that you save in the dialog apply *only* to the form that you are editing. To use the Script dialog for a form, you require access rights to edit the form.

Follow these steps:

1. Click the name of the form in the Form Designer tree.
   The form attributes appear, and the Script button activates. The Script button appears on the toolbar with the Localize, Save, Restore, and other buttons.
2. Click the Script button.
   The Script dialog appears and displays the custom JavaScript functions already specified for the form, if any. You can enter new JavaScript functions before or after any existing ones.

   NOTE
   To view a template for a function, click Show Help on the Script dialog. The template shows the product-specific implementation of a standard JavaScript function.
3. Write and save the custom function. Use the format shown in the sample functions that appear when you click Show Help on the Script dialog. This format is known as *JavaScript object literal notation*.
4. Save the form.
5. Specify the name of the function in the JavaScript attribute from which you want to call it. Use the following format:

   ```
   ca_fd.js.function-name
   ```

   An example follows:

   ```
   ca_fd.js.calculateCostQty()
   ```

   Typically, you specify a custom JavaScript function in one of the following:
   - The onSubmit or on Load attribute of the form attributes
   - The onChange or onValidate attributes of applicable fields

You have used the Script dialog to maintain JavaScript functions that apply to a form.

Options to Populate and Pre-populate Fields Automatically

Administrators have several options for using JavaScript functions to populate and pre-populate fields automatically. *Populating* fields means that the fields are filled with the results of a database query *based on* user input. The affected fields are empty when the user opens the form. When the user enters the requested input, the database query is run and the results are used to populate the fields.

Conversely, *pre-populating* fields means that the fields are filled with the results of a database query *without* any user input. As soon as the user opens the form, the data source query is run and the results are used to populate the fields. Thus, the affected fields are populated when the user opens the form.

Review the options and choose the one that best suits your needs. Use the pre-populating options when no user input is required for the requested information to be retrieved by the database query. Use the populating options when the user's input is critical for the requested information to be retrieved by the database query.

The options for using JavaScript functions to *populate* fields automatically follow:

- Use the pre-defined JavaScript function named ca_fdDoFieldLookup with a lookup field to populate fields based on user input to a report data object.
- Use ca_fdDoFieldLookup with a lookup field, and use other pre-defined JavaScript functions like ca_fdSetTextFieldValue to set the values of elements in your form with a custom JavaScript function.

The options for using JavaScript functions to *pre-populate* fields automatically follow:

- Pre-populate a combo box based on a report data object.
- Use user input to pre-populate a select box.
- Use user input for personal data to pre-populate a select box.
- Pre-populate fields based on a report data object and JavaScript functions.

Populate Fields Based on User Input to a Report Data Object

This topic explains how to populate fields based on user input, by using a lookup field, a report data object (data object), and the pre-defined JavaScript function named ca_fdDoFieldLookup(fieldId, reportId). You use them together to run a data object and return the results in the matching field or fields on the form.

When the user clicks the magnifying glass icon for the lookup field on the form, CA Service Catalog prompts the user to enter the variable that you specified in your data object. An example is the user ID.

When the user answers the prompt or prompts in the lookup field, CA Service Catalog runs the data object and searches the data source for the data requested in fields of the data object; for example, the first name and last name of the user ID entered in the lookup field. In this example, you create matching First Name and Last Name fields on the form, and CA Service Catalog returns the results to the user. The user selects which result (if any) to use to populate the form fields.

To populate fields based on user input, by using a lookup field, a data object, and the pre-defined JavaScript function named ca_fdDoFieldLookup(fieldId, reportId), follow these steps:

1. In the Report Builder, create a data object that queries the data source (such as the MDB) for the data you want (the data copied to the matching fields on the form). The query is based on the user input to the prompt or prompts that you specify, such as user ID.
   Record the name of the data object for later reference.
   The following sample query for the MDB matches the example started earlier in this topic:
   ```
   SELECT userid,first_name,last_name FROM ca_contact WHERE userid = '%userid%'
   ```
   The ca_fdDoFieldLookup function processes this query as follows:
   – "WHERE userid = '%userid%" specifies that the user is prompted to enter this value, and this value is used in the SELECT clause.
   – "SELECT userid,first_name,last_name" specifies which values to return.
   – "FROM ca_contact" specifies that the data source is the ca_contact table in the MDB.
2. On the form, add the lookup field; it is one of the elements of a form.
   a. For the _id attribute, specify the first field returned by the query, and save the form.
      Thus, in the continuing example, you specify *userid* as the value of the _id attribute.
   b. For the value of the JavaScript attribute named onLookup, specify the pre-defined JavaScript function named ca_fdDoFieldLookup(fieldId, reportId) and save the form.
      This function runs the data object, associates it to the lookup field, and copies the data returned by the data object to the matching fields on the form.
      For fieldId, specify the value of the _id attribute of the lookup field.
      For reportId, specify the value of the _id attribute of the data object that you created earlier.
   c. For the tool tip attribute, optionally specify instructional text, such as "Click the magnifying glass and enter the requested data." Save the form.
      In the continuing example, for the tool tip attribute, you may optionally specify text such as "Click the magnifying glass and enter user ID."
   d. Optionally rename the default display text from the lookup field from "Lookup Field" to a more meaningful name, by selecting the element in the Component Tree and clicking the Rename icon at the top of the tree.

In the continuing example, you may rename the display text to "User ID" or something similar.

e. Leave value attribute empty. This value is populated with the first result returned by the query. The value appears on the form to the user as the value of the lookup field.

3. Also on the form, typically under the lookup field, add the fields that hold the results returned by the query. For each field, specify the value of the HTML Attribute named _id to match exactly the ID attribute of the corresponding object in the database.

   NOTE
   To optionally disable a field so that users cannot change the result returned by the query, set the disabled attribute of the field to true. You can also set the disabled attribute to true only under certain conditions that you specify.

   In the continuing example, create matching First Name and Last Name fields on the form, and specify first_name and last_name, respectively, in the _id attributes of these fields.
   Thus, all matching values of userid are returned in the lookup field; each result appears as a row in the search results. Users can review the search results and select a row. When the user selects a row, the lookup field closes, and the form is populated with the results from that row. The user sees something like the following on the form:
   User ID: johsmi515
   First Name: John
   Last Name: Smith
4. Test the form to verify that it works as you intended.

   WARNING

   The lookup field displays *only* the first 25 matches in the search results. If your query returns more than 25 matches, refine your query to return fewer matches.

Pre-Populate a Combo Box Based on a Report Data Object

This topic explains how to pre-populate a combo box using a data object. On a form, a combo box appears as a single line item; the user clicks its arrow to open the entire list. You configure the combo box to specify whether the user can select only one option or multiple options.

On the Form Designer, you add and configure the select group HTML Attribute named reportobjid. When the user opens the form, the data object runs and return the results in the combo box. The user then selects one or more of the options in the combo box while completing the form.

This option is useful when you want to provide the user with multiple valid choices from your data source but do not want to enable the user to specify a custom selection. This option thus helps enforce the standardization and validity of users' selections while still typically providing users with multiple options from which to choose.

Follow these steps:

1. In the Report Builder, create a data object that queries the data source (such as the MDB) for the data you want. The query must specify *two* fields from the data source, as follows:
   – The first field returns the list of all possible values and populates the combo box with this list.
   – The second field returns the label for the combo box.

   When the form appears, the report data object runs and populates the combo box with the resulting data.
   Record the name of the data object for later reference.
   Consider the following sample query for a data object for the MDB. This example returns the list of available service option groups (rate plans).

   ```
   SELECT rate_plan_id,rate_plan_name FROM usm_rate_plan
   ```

   This query does the following:
   – "SELECT rate_plan_id,rate_plan_name" specifies which values to return:

- rate_plan_id returns the list of all service option groups. Each plan becomes an option in the combo box; for example, Budget, Standard, Deluxe, and so forth.
- rate_plan_name returns the label for the combo box; for example, Service Option Groups.

– "FROM usm_rate_plan" specifies that the data source is the usm_rate_plan table in the MDB.

2. On the form, add a select box; it is one of the elements of a form.
   a. For the _id attribute, specify a meaningful name, and save the form.
   b. For the value of the HTML Attribute named reportobjid, specify the id of the data object, and save the form.

      NOTE
      When you use a data object to populate a combo box, do not add any options to the select box, because they are ignored (not used) when the user opens the form. Any options for the select box are ignored, only the data object "matters."

Use User Input to Pre-Populate a Select Box

You can use input provided by the user in one or more fields in the form to determine the values in a select box. This technique is useful when you want to provide the user with two sets of multiple valid choices from your data source but do not want to enable the user to specify a custom selection. This option thus helps enforce the standardization and validity of users' selections while still typically providing users with multiple options from which to choose.

A common application of this technique occurs on a form with select boxes for both country and state. Using the reportobjid and reportobjvars attributes, you can configure the state select box to display only the states for the country selected by the user.

To use this technique of pre-populating a select box based on a user's selection in an earlier select box, follow this process. This process uses the country-state scenario as a running example.

1. In the Report Builder, create a data object that queries the data source (such as the MDB) for the data you want in the first select box. The query must return the list of all possible values and populate the combo box with this list. This step is part of the process of pre-populating a combo box based on a report data object.
   In our running example, create a report data object to retrieve the list of countries from the database.
   When the form appears, the report data object runs and populates the combo box with the resulting data.
   Record the ID of the data object for later reference.
   Consider the following sample query for a data object for the MDB. This example returns the list of available countries for the service or service option to which the form is attached

   ```
   SELECT country_id,country_name from my_country_table
   ```

   This query does the following:
   – "SELECT " specifies which values to return. In this example, it returns the ID and the name of the country in the database.
   – "FROM my_country_table" specifies the name of the database table.
2. On the form, add the first select box; it is one of the elements of a form.
   a. For the _id attribute, specify a meaningful value, and save the form. Record the value for later reference when you specify the reportobjvars attribute.
   b. For the value of the HTML Attribute named reportobjid, specify the id of the data object, and save the form.

      NOTE
      When you use a data object to populate a combo box, do not add any options to the select box, because they are ignored (not used) when the user opens the form. Any options for the select box are ignored, only the data object "matters."
3. Create a second report data object to retrieve a list of values from the database, based on the user's selection in the first select box. In this report data object, specify some report variables in the query that are to be filled using the input provided by user in other fields in the form
   In our running example, create a second report data object to retrieve the states from the database, based on the country that the user selected. Follow the guidelines in Step 1 to create the second report data object.

4. On the form, add the second select box. This box is to be pre-populated, based on the user's selection for the first select box. Follow the guidelines in Step 2 to create the second select box, but account for the following considerations:
   – The second query contains a report variable and is similar to the following:
   ```
   select state_id,state_name from my_state_table where country_id=%selected_country%
   ```
   – The running example in this topic uses only one report variable (%selected_country%); however, your query may contain multiple report variables.
   – Record the names of all report variables for use when you specify the reportobjvars attribute for the second select box.
5. In the second select box, specify the reportobjvars attribute using the following format: $({'reportvar':*value*}).
   A valid *value* is one of the following:
   – A constant, such as 12, as in $({'reportvar':12})
   – A string, such as abc, as in $({'reportvar':'abc'})

     NOTE

     Enclose strings in single quotation marks, as shown. Within a quoted string, if needed, use a backslash (\) as an escape character to specify a literal single quote or apostrophe. For example: $({'what\'s the status?'})
   – A JavaScript expression, such as the user's last name, as in $({'reportvar':_.user.lastName})
   – A JavaScript function of the following form: $({'reportvar':foo()})
   – Combinations of the previous values, delimited with commas, using the format shown in the following example:
   ```
   $({'reportvar':_.user.lastName,'reportvar1':'abc','reportvar2':12,'reportvar3':foo()})
   ```
   In our running example, create a second select box to be populated with the states from the database, based on the country that the user selected. For the reportobjvars attribute of the state select box, specify the pre-defined JavaScript function named ca_fdGetSelectedOptionValues, as follows:
   ```
   $({'selected_country':ca_fdGetSelectedOptionValues ('<form _id> ','country' )[0]})
   ```
   – **form _id**
     Specifies the value of the _id attribute of the form containing the first select box. You reference this form in Step 2.
   – **country**
     Specifies the value of the _id attribute of the first select box. You create and record this value in Step 2.
6. In the first select box, set the onchange attribute to retrieve the data for the second select box as soon as the user makes a selection in the first select box. For the onchange attribute, specify the pre-defined JavaScript function named ca_fdFetchSelectData, using the following format:
   ```
   ca_fdFetchSelectData('<form _id>','<state field _id>');
   ```
   – **form _id**
     Specifies the value of the _id attribute of the form containing the second select box. You reference this form in Step 4.
   – **field_id**
   Specifies the value of _id attribute of the form containing the second select box. You create and record this value in Step 4.
   In our running example, set the onchange attribute for the country select box to the following:
   ```
   ca_fdFetchSelectData('<form _id>','<state field _id>');
   ```
7. Test the form to verify that it works as you intended.

Use User Input for Personal Data to Pre-Populate a Select Box

You can use the first few characters provided by the user in a select box in a form to trigger an auto-complete function. A common application of this technique occurs on a form with a select box (an element of a form) for the last name. For example, if the user types the letters *smit*, the field displays all valid options, such as Smittapopolous, Smitderski, Smitapunangala, and so forth. The user can select a valid value but cannot enter a custom value. This technique helps users find and specify valid data efficiently when the database is large.

Follow these steps:

1. In the Report Builder, create a data object that queries the data source (such as the MDB) for the data you want. The query must return the list of all possible values and populate the select box with this list. This step is part of the process of pre-populating a combo box based on a report data object. For this example, do the following:
   - Create a report data object to retrieve the list of user IDs from the database. When the form appears, the report data object runs and populates the combo box with the resulting data.
   - Record the ID of the data object for later reference.
   - Verify that the variable you want to use exists and is of type *string*. If necessary, create or update the variable. For illustration, we use the last_name_prefix variable in the following example.

   For example, consider the following sample query for a data object for the MDB. This example returns the list of available last names that match the user input in the field.

   ```
   Select userid as id, last_name from ca_contact where last_name like '%last_name_prefix%%%'
   ```

   - **Select userid as id, last_name**
     Specifies which values to return. It returns the ID and the last name of all users that match the first few characters entered by the user.
   - **From ca_contact**
     Specifies the name of the database table.
   - **where last_name like '%last_name_prefix%%%'**
     Specifies the filter clause.
     The last_name_prefix variable filters on the last_name column (the display column of the query).
     The 'like '%last_name_prefix%%%' syntax filters the results so that the last_name starts with last_name_prefix.
2. Add the select box to the form.

   NOTE

   When you use a data object to populate a select box, do not add any options to it, because they are ignored (not used). When the user opens the form, any options for the select box are ignored, only the data object is used.

   a. For the _id attribute, specify a meaningful value, and save the form.
   b. (Optional) Rename the default display text from the select group from "Select" to a more meaningful name, as follows: Select the element in the Component Tree and click the Rename icon at the top of the tree.
3. Specify values for the following attributes of the select box, and save the form.
   - Multi-Select: Specify False so that users can select only one option in the combo box.
   - (Optional) Title (tooltip text): Specify instructional text, such as "Click the arrow and scroll to select a last name.

     NOTE
     You can optionally localize the tooltip text or the name of the select box, if applicable.
4. Specify values for the following attributes of the select box, and save the form.
   - Report/Plug-in Id: Specify the name of the report data object that you created in Step 1.
   - Report/Plug-in Variable: Specify a JSON expression for the object that maps a key value pair, as follows:
     - The key is the variable for the report data object.
     - The value is a JavaScript expression that matches the data you want.

     For this example, specify the following:

     ```
     $({'last_name_prefix':_val})
     ```

     - last_name_prefix is a string constant, as noted in Step 1.
     - _val is a predefined variable that CA Service Catalog provides.

NOTE

: When a _val variable is used in the Report/Plug-in variable field, then the select search is disabled and we recommend to use the same _val variable to filter the data at the report object level.

- Minimum Search Characters: Specify a custom value, for example, 4 or 5. This attribute specifies the number of characters that users must type to trigger the auto-complete feature. The defaults is 4.

5. Test the form to verify that it works as you intended.

Pre-Populate Fields Based on a Report Data Object and JavaScript Functions

To pre-populate fields with data that is not accessible through JavaScript expressions, you can use a report data object (data object), custom JavaScript functions, and pre-defined JavaScript functions, especially ca_reportQuery(reportId, variables, onSuccess, onFailure). A representative case involves querying your organization's Human Resources database (not the CA MDB) for sensitive data, such as the banking account number and related information for a user.

For example, you may create a form for changing the bank account to which a user's pay check is automatically deposited. In this form, you may pre-fill fields with name of the current bank name and account number. Use later fields on the form for the user to enter the name of the new bank name and account number.

First, ca_reportQuery(reportId, variables, onSuccess, onFailure) retrieves from the data object the variables that match the fields queried by the data object. Next, the OnSuccess function is called, at which point you can use other pre-defined JavaScript functions to pre-populate multiple fields in the form with the results of the query. For example, you may write a custom function that calls ca_fdSetTextFieldValue(formId, _id, text) multiple times, once for each field you want to pre-populate. This scenario is typically most suitable for pre-populating text fields, but can also be used to populate other types of fields.

Follow these steps:

1. Design and create the form, if you have not already done so.
2. In the Report Builder, create a data object that queries the data source for the data you want (the data to be copied to the matching fields on the form).
   Record the ID of the data object for later reference.
   The following sample query for the MDB matches the example started earlier in this topic:
   ```
   SELECT bank_name,account_number FROM my_hr_db WHERE userid ='%userid%'
   ```
   CA Service Catalog processes this query as follows:
   a. "WHERE userid = '%userid% ' specifies that a user id must be provided as input to this data object.
   b. "SELECT bank_name,account_number" specifies which values to return.
3. "FROM my_hr_db" specifies that the database is my_hr_db, a fictitious database for illustration.
4. On the form, add text fields (or other suitable form elements instead) to receive the retrieved data, as follows:
   - Add one text field for each variable queried by the data object.
     For the continuing example, add two text fields, one for bank name and one for account number.
     For the continuing example, add two text fields, specify bank_name and account_number, respectively.
   - Rename each text field to have a meaningful name.
     For the continuing example, rename the fields to Bank Name and Account Number, respectively.
5. Select the form name to display the form attributes.
6. In the form attributes, in the JavaScript attribute named onload, enter the name of your function that calls ca_reportQuery. For illustration, these steps use getAcctData() as the name for this function. Substitute the name of your function if it is different.
7. Verify that getAcctData() is declared in a JavaScript file accessible by CA Service Catalog.
   For best results, use this location in the filestore: FileStore/scripts/custom_form_lib.js.

   WARNING
   The folder name *FileStore* is case sensitive. Therefore, use the correct case in path names and all other programmatic references.

8. In the body of getAcctData, specify the call to ca_reportQuery function, to run the data object and return the results.
9. Use the results as input values to JavaScript functions such as ca_fdSetTextFieldValue to update the fields in the form. In this example, use the results to make two calls: one to update the bank name field and another to update the account number field.
10. Customize ca_reportQuery(reportId, variables, onSuccess, onFailure) to meet your requirements. This step uses the continuing example for illustration.
    - For reportId, specify the data object that you created earlier.
    - For variables, specify the JavaScript map with the variable names and values.
      For the continuing example, specify: {“userid”:_.user.id}.
      These variables *must* match the ones that you query in the data object, but they do not need to be ordered in the same sequence.
    - For onSuccess, specify the custom JavaScript function to run when the query returns successfully. The onSuccess function must take one operand. This operand is set with an array of maps, each representing a row, returned by the query. Within your custom onSuccess function, use ca_fdSetTextFieldValue(formId, _id, text) to return the results of each field, one at a time.
      For the continuing example, specify the following:

      ```
      function updateFields(result) {
         if (result.length == 1) {
         ca_fdSetTextFieldValue(formId, 'bank_name', result[0].['bank_name']);
         ca_fdSetTextFieldValue(formId,'account_number',result[0].['account_number']);
         } else {
            alert('could not find your bank account information')
         }
      }
      ```

    - For onFailure, specify the JavaScript function to run if the query fails.
      For the continuing example, specify the following:

      ```
      function onGetAcctDataFail() { alert("can't find your account");}
      ```

11. Finish creating the form to meet your requirements.
12. Test the form to verify that it works as you intended.

Validate User Input

You can configure Form Designer forms to verify that users entered the following types of data in the correct format: credit card numbers, social security numbers, email addresses, phone numbers, postal (zip) codes, and decimal numbers with two fractional digits.

This process does *not* guarantee that the data entered by the user is authentic but does assist in the validation process by verifying that the user entered the data in the correct format.

You can thus use the Form Designer to verify that the data you want and is recorded in the format that you want. For example, you can verify that phone number fields accept only numeric input, that date fields accept only your company's standard date format, and that credit card fields accept only numbers that are entered in the correct format for the credit card type specified.

Additionally, you can specify maximum length, minimum length, and other attributes used to validate user input.

Choose the task you want to perform:

- HTML attribute named pattern to validate the format of credit card numbers, social security numbers, email addresses, phone numbers, IP addresses, postal codes, dates, and decimal numbers with two fractional digits.
- Use JavaScript functions to validate the format of credit card numbers. For example, you use the JavaScript attribute such as onValidate to verify the credit card formats. Alternatively, you can create your own custom JavaScript function.

Use Regular Expressions to Validate Numeric and Address Data

This topic explains how to use regular expressions to configure Form Designer forms to verify that users entered the following types of data in the correct format: credit card numbers, social security numbers, email addresses, IP addresses, phone numbers, postal (zip) codes, and decimal numbers with two fractional digits. .

To verify the types of user input mentioned earlier, follow these steps. These steps use a continuing example for illustration.

1. Design and create the form, if you have not already done so.
2. For each type of user input, add a text field.
3. Rename each text field from Text Field to a meaningful name, such as Social Security Number, Credit Card Number, and so forth.
4. For the value of the HTML Attribute named pattern, specify the regular expression for the type of data being validated, and save the form. Valid values follow:
   - Social security number: ^([0-6]\d{2}|7[0-6]\d|77[0-2])([\\d{2})\2(\d{4})$
   - Decimal number with two fractional digits: ^[1-9]\d*(\.\d{2})$
   - E-mail address: ^([0-9a-zA-Z]+([_.-]?[0-9a-zA-Z]+)*@[0-9a-zA-Z]+[0-9,a-z,A-Z,.,-]*(.){1}[a-zA-Z]{2,4})+$
   - IP Address: ^((25[0-5]|2[0-4][0-9]|1[0-9]{2}|[0-9]{1,2})\.){3}(25[0-5]|2[0-4][0-9]|1[0-9]{2}|[0-9]{1,2})$
   - Telephone number: ^(([0-9]{1})*[- .(]*([0-9a-zA-Z]{3})*[- .)]*[0-9a-zA-Z]{3}[- .]*[0-9a-zA-Z]{4})+$
   - URL: ^(http[s]?://|ftp://)?(www\.)?[a-zA-Z0-9-\.]+\.(com|org|net|mil|edu|ca|co.uk|com.au|gov)$
   - Postal Code (CAD): ^([A-Z][0-9]){3}$
   - For additional values, or to create your own expression, see your reference for regular expressions, such as www.regular-expressions.info or www.regexlib.com.
5. For the value of the HTML Attribute named patternMessage, specify an error message to appear when the user input violates the pattern attribute.

   NOTE
   You can optionally localize this error message.
6. Finish creating the form to meet your requirements.
7. Test the form to verify that it works as you intended.

Use a JavaScript Function to Validate the Format of Credit Card Numbers

This topic explains how to configure Form Designer forms to use a JavaScript function to verify that users entered a credit card number in the correct format for the credit card type that they specified.

This topic explains how to present two credit card types: Master Card and Visa. For each type, you create a radio group option and a text field for the account number. Each account number field is disabled by design but is enabled if the user selects the matching credit card type.

You use the pre-defined JavaScript function named ca_fdValidateCC(*number*, '*type*') to validate the format of the credit card number entered by the user, based on the credit card type that the user selected. This JavaScript function verifies the format according to the specifications set by the company issuing the credit card. Each company sets its own standard format.

To verify the credit card numbers entered by users, follow these steps. These steps use a continuing example for illustration.

1. Design and create the form, if you have not already done so.
2. Verify that the value of the form's HTML Attribute named _id specifies a meaningful name, such as ccValdtnForm1.
3. Add a radio group; it is one of the elements of a form.
   a. For the value of the _id attribute of the radio group, specify a meaningful name, such as rgCCVal, and save the form.
   b. Rename "Radio Group" to a meaningful name or phrase, such as Select a Credit Card or Credit Card Type.

4. Add the first radio group option; these options are elements of a form that apply only to radio groups.
   a. For the _id attribute, specify a meaningful name, such as mcard (signifying Master Card radio group option), and save the form.
   b. Rename "Radio" to a meaningful name, such as Master Card.
5. Add the second radio group option.
   a. For the _id attribute, specify a meaningful name, such as visa (signifying Visa radio group option), and save the form.
   b. Rename "Radio" to a meaningful name, such as Visa.
6. Add a text field onto the form
   a. For the _id attribute, specify a meaningful name, such as ccf.
   b. For the value of the JavaScript attribute named onvalidate, specify cc_val(_val).
7. Use the Script dialog to add the following function definition:

```
function cc_val(value) {
                var ccname = '';
                if (ca_fdIsSelectRadio('ccValdtnForm1', 'rgCCVal', 'mcard'))
                                ccname = 'master';
                else
                                ccname = 'visa';
                ca_fdValidateCC(value, ccname);
}
```

   As a best practice, use the Script dialog on each form to create and maintain the custom JavaScript functions for the form.
8. Finish creating the form to meet your requirements.
9. Test the form to verify that it works as you intended.

Link Fields in Two Forms for Simultaneous Updates

You can optionally specify JavaScript functions to link two fields in two Form Designer forms. When a user updates a field in one form, the linked field is updated in the second form simultaneously. This technique essentially creates an automatic cross-reference between the fields in both forms.

Follow these steps:

1. Verify that the forms are in the same service option.

   WARNING
   The forms *must* be in same service option.
2. Decide which fields in which forms you want to link. For example, you may want to ensure that a name, date, item, or phone number field has the same value in both forms in each request.
3. Specify the same exact function in both forms *except* that you switch the form IDs in each JavaScript function, as follows:
   – In the JavaScript function in Form 1, you specify the ID for Form 2.
   – In the JavaScript function in Form 2, you specify the ID for Form 1.
   – For ease of maintenance, use the same field name and same JavaScript attribute in each form.
   For example, you may decide to name the field Address in both forms and specify the JavaScript function in the onclick attribute in each form.

   WARNING
   Do *not* re-use either form in a different service. If one of the forms is re-used in a different service, the fields become un-linked. Consequently, both fields function correctly individually but are no longer cross-referenced. Therefore, they are no longer configured for simultaneous updates.

Example

This example illustrates how to use JavaScript functions to link two fields in two Form Designer forms for simultaneous updates.

In Form AA, whose form ID is form_aa, specify the following for the linked field:

- Name: Item Requested
- JavaScript function in the onclick attribute: ca_fdSetTextFieldValue('form_bb','item_requested','New Laptop')

In Form BB, whose form ID is form_bb, specify the following for the linked field:

- Name: Item Requested
- JavaScript function in the onclick attribute: ca_fdSetTextFieldValue('form_aa','item_requested','New Laptop')

Administer Forms

This article contains the following topics:

Understand Access Level Rules

Authorized users can access forms. Authorized users are users with the following roles: Super Business Unit Administrator, Catalog Administrator, Service Manager, and Service Delivery Administrator. Here, *access forms* means that the users can view, create, edit, copy, and delete forms, unless noted otherwise.

To configure access rights to forms, configure these Use Service Provider Catalog Only and Pass Through Catalog settings. Accept or change the default settings, as follows:

- By default, the *Use Service Provider Catalog Only* setting is No. Authorized users can access the forms in the current business unit and its child business units *only*. Authorized users can add forms to services and can delete forms from services in the current business unit and its child business units *only*.
  If the *Use Service Provider Catalog Only* is Yes, then authorized users can view and copy forms in the Service Provider business unit. They cannot perform any other actions on these forms. Similarly, authorized users cannot add forms to services and cannot delete forms from services in the Service Provider business unit.
- By default, the *Pass Through Catalog* setting is No. Authorized users can access forms in the current business unit and its child business units *only*. Authorized users can add forms to services and can delete forms from services in the current business unit and its child business units *only*.
  If the *Pass Through Catalog* is Yes, then authorized users can view and copy forms in the parent business unit. They cannot perform any other actions on these forms. Similarly, authorized users cannot add forms to services and cannot delete forms from services in the Service Provider business unit.
- If both the *Use Service Provider Catalog Only* and *Pass Through Catalog* settings are Yes, then the former setting takes precedence. Authorized users can view and copy forms in the Service Provider business unit, but not in any parent business unit. Similarly, authorized users cannot add forms to services and cannot delete forms from services in the Service Provider business unit.

To configure access rights to forms, proceed as follows:

- To configure the *Use Service Provider Catalog Only* setting, log in to the root (highest level) business unit as a Service Delivery administrator. Select Catalog, Configuration, System Configuration.
- To configure the *Pass Through Catalog* setting, log in to the business unit of interest as an authorized user. Select Catalog, Configuration, Catalog Configuration.

  NOTE
  Users cannot access forms in sibling business units under any circumstances.

Create and Maintain Forms

You can create forms in the Form Designer.

Follow these steps:

1. Verify that you have administrator rights to each folder that you want to access, in each affected business unit. To copy or move a form from one business unit to another, you must be an administrator in both business units. If you are an administrator in a parent business unit, then you are automatically an administrator in all its children business units.

   NOTE
   You cannot create, copy, move, or delete any forms or elements within the System folder.
2. Use caution when modifying an existing form. If the form is included in an *active* request, you receive a warning message.
   Do *not* update the form (except trivial changes) if you receive this warning message. When you no longer receive this message, proceed with your change. To make important changes to the form, create a version with the changes and use the new form.
3. Use extreme caution when deleting an existing form or moving an existing form from one business unit to another.
   If the form is included in an *active* request, you receive a warning message. Do *not* complete the move or delete operation if you receive this warning message. When you no longer receive this message, proceed with your change.

   WARNING
   When you delete a form, all elements under it are also deleted. If you want to save the elements, store them in another form before deleting this form.
4. Click Catalog, Forms.
5. Click the icon next to the Forms folder in the Component Tree.
6. Create the form, if applicable.
7. Click the Associations tab
   to display the list of service offerings and service option groups that are associated with the selected form. It also provides the ability to view where a particular form has been used in the current Business Unit and also the capability to assess the impact of modifying or deleting a form. This functionality is also available in the Form Chooser UI when attaching a form to a Service Option.
8. Edit a form, as follows:
   a. Open the folder that contains the form you want to edit.
   b. Open the form to display its existing elements.
   c. Add, modify, and configure elements on the form.
      You can rename, move, copy, or delete a form.

      NOTE

      The copied form, including all its elements, now belong to the business unit of the destination folder. The name of the copied form must be unique within the business unit. The form is copied, and its new name is "copy of *original name.*" The copied elements under the copied form are *not* renamed.

(Optional) Create and Maintain Folders

Create and maintain folders to organize your forms according to groups, categories, or business units. For example, you create and name folders that correspond to each business unit, department, owner, or service.

- If you delete a folder, all sub folders, forms, and elements under it are also deleted.
- If you move a folder from one business unit to another, the moved folder now belongs to the business unit of the destination folder. The names of each folder, form, and element must be unique within their new business unit.
- If you copy a folder from one business unit to another, the pasted folder now belongs to the business unit of the destination folder. However, the sub folders, forms, and elements under it are *not* renamed.
- If you want to delete a folder, first delete any sub folders or forms under it.

Import and Export of Forms

You can use the Import Export utility to perform the following tasks:

- Import and export Form Designer forms by themselves, without related services or service option elements.
- Import and export Form Designer forms associated to service option elements.

Localize Forms

You can use the Localization Editor to localize the element names and selected attributes in forms. You can do so for the supported languages.

When you edit the form in the Form Designer, you see localized names and values only if you are using the Localization Editor. However, when a localized form appears in a request, it displays all localized values. If no localized values exist, the form displays the default values for the form name, element names, and selected attributes.

For more information about how to set the Localization Attributes for a Form, see Localize a Single Service section.

Manage Services

A service consists of hardware, software, or other resources that users request from the catalog. Administrators *manage* services to meet the needs of their users and organization. Here, *manage* means to create or update services, or to perform other tasks to manage services, as explained in the following steps.

Administrators (typically service designers) group one or more service option groups together as a single service. You can create services for several purposes, including requesting hardware or software, on boarding new employees, and reserving physical or virtual resources. You can optionally use Service Accounting Component to specify charges for each service.

You create and maintain services in folders in a service catalog. You make the catalog available either to all business units or only to a specific business unit. You can have multiple catalogs for multiple business units.

1. Verify that you are logged in to the correct business unit. If your role permits, you can change to another business unit to access its catalog instead of the catalog of your current business unit as follows:
   a. Click **Catalog**, **Service Offerings**.
      The Services page appears and displays the folders containing services, organized hierarchically according to category.
   b. Click **Change Business Unit**.
      The **Search Business Units** window appears.
   c. Use the Expand and Collapse icons to navigate the business unit tree or search to locate the required business unit.

      NOTE

      The list includes only the business units that the role of the user permits.
   d. Select the business unit name in the tree.
      The window closes and the catalog for that business unit is displayed. You have changed business units.
2. Click Catalog, Service Offerings.
3. Expand the tree and select the folder in which you want to manage services.
4. (Optional) Create a Simple Service.
5. (Optional) Add or Update a Service.
   This task optionally includes specifying inheritance, setting featured and related services, and defining dependencies between services.
6. Perform Other Tasks to Manage Services:
   – Setting thresholds for quality-of-service (QoS) SLAs for services
   – Managing the folders that store services

As part of managing services, you Localize Multiple Services.

A service consists of hardware, software, or other resources that users request from the catalog. You can create services for several purposes, including requesting hardware or software, on boarding new employees, and reserving physical or virtual resources.

For more information on adding or updating a service, see Add or Update a Service.

You can create a simple service by copying predefined service option groups and forms and customizing the copied objects. Creating a service in this manner is more efficient than creating and configuring the objects that comprise the service. For more information, see Create a Simple Service.

Administrators (Service Designers) manage services to meet the needs of their users and organization. For more information on how to manage services, see Manage Services.

Administrators can perform the following activities related to the services:

- Group one or more service option groups together as a single service.
- Copy, cut and paste, delete, and cancel services.
- Manage the folders that store services
- Define dependencies
- Set thresholds for quality-of-service (QoS) SLAs

For more information on how to perform other tasks to manage the services, see Perform Other Tasks to Manage Services.

{"URL":["/usm/wpf?Node=iclaunchpad.catalog_builder#id=*;t=*;d=ca.com","/usm/wpf?Node=iclaunchpad.service_designer#id=*;t=*;d=ca.com"],"description":"IPCE_Description","new":"","new_video":"","admin":"IPCE_{"masterkb":"","text":"the following KB articles will help you","URL": ["https://knowledge.broadcom.com/external/article?articleId=214779"]}, "pendo":"","video":""}

Create a Simple Service

This scenario illustrates how service design managers create a simple service by copying predefined service option groups and forms and customizing the copied objects. Creating a service in this manner is more efficient than creating and configuring the objects that comprise the service. You can use this scenario *as a model* to create a simple service.

This scenario focuses on the onboarding of a new employee into the Field Services group in your organization. For a use case other than new hire onboarding, search the catalog for the service that most closely matches the service that you want to create. For example, to create a service for reserving a virtual machine using Reservation Manager, review services in the Reservation Services folder.

The predefined service in this scenario is a simple, on-premise enterprise service named New Hire Onboarding. You use it to create a similar service for the Field Services group. In this scenario, you optionally configure default options so that users can request the service with little to no effort or input. Creating simple services in this manner reduces user errors and increases efficiency, especially for users submitting requests from mobile devices.

> **WARNING**
>
> If you already have multiple forms with the same _id value, do not associate such forms to a single Service Option. Having multiple forms with the same _id value in a single Service Option can cause validation errors.

Follow these steps:

Step 1 - Create the Service Option Group for the Form

As a best practice, copy predefined objects and customize the copies, rather than changing predefined objects.

Follow these steps:

1. Click **Catalog**, **Service Offerings**.

2. Click the **Option Group** tab.
3. Select the group **New Hire Onboarding** and perform the following actions:
   a. Click the Copy icon.
   b. Click the Paste icon (*not* the Paste as Inherited icon).
   c. Specify New Hire Onboarding for Field Services Only in the prompt for the name of the new option group.
4. Select the copy of the service option group and perform the following actions:
   a. Click the Definition tab.
   b. Click the Delete icon to delete this form from the service option group.
   This action deletes the form from this service option group *only*. The form still exists in any other service option groups that include it, and in the Form Designer folder.

Step 2 - Review and Copy the Form

All services include at least one predefined or custom form to record and process essential information from the user requesting the service. In this scenario, you review and copy the predefined form that is included with the standard New Hire Onboarding service. Afterwards, you modify the copied form for use in the New Hire Onboarding for Field Services Only service.

Follow these steps:

1. Copy the New Hire Onboarding form, as follows:
   a. Click **Catalog**, **Forms**.
   b. Expand the **Forms**, **CA Catalog Content** folder in the Component Tree.
   c. Select the **New Hire Onboarding** form.
   d. Click the Copy icon.
2. Select the top level of the Forms folder and complete these actions:
   a. Click Add and create a subfolder that is named Custom.
   b. Select the Custom folder and click Paste.
   The new form appears; its name includes the "copy of" prefix. The copied elements under the copied form are *not* renamed.

      NOTE
      When you copy the form from one business unit to another, the pasted form, including all elements under it, belong to the business unit of the destination folder.
   c. Select the form and click Rename.
   d. Enter the new name as New Hire Onboarding for Field Services Only. Click OK. The name must be unique within its business unit.

Step 3 - Modify the Form

Modify the form that you copied by adding a unique field for Field Services.

Follow these steps:

1. Expand the New Hire Onboarding for Field Services Only form.
2. Copy the Additional Info field, as follows:
   a. Select the Additional Info field in the tree, and click the Copy icon. The Copy icon appears near the top left of the screen, in the row of form tools under the main menu and tab names.
   b. Select the name of the New Hire Onboarding for Field Services Only form in the tree. Click Paste.
   c. Enter a unique ID (for example, field_services_group_id) in the _id field in the right pane.
   d. Click Save.
3. Rename the new field, as follows:
   a. Select the field and click Rename.

b. Enter the new name as Field Services Group ID. Click OK.

The new field is renamed. The Field Services managers can use this field to assign new employees to a group in the organization.

Step 4 - Add the Form to the Service Option Group

Add the new, modified form to the New Hire Onboarding for Field Services Only service option group. Later, you add this service option group to the New Hire Onboarding for Field Services service.

Follow these steps:

1. Click **Catalog**, **Service Offerings**.
2. Click the **Options Group** tab.
3. Select the group named New Hire Onboarding for Field Services Only. Perform the following actions:
   a. Click the Definition tab.
   b. Click **Add Option**.
      The Service Option Details tab appears and lists the service option elements in the group.
   c. Enter a meaningful name for the service option. Example: New Hire Onboarding Form for Field Services Only.
4. Scroll to the Form field and click Add.
5. Complete the fields in the Service Option Element Definition, as follows:
   a. Specify meaningful text In the Display Text field. Describe the purpose of this form.
      This text appears to the user as a description for the form when administrators view this service option group in the catalog.
   b. Click the Search icon in the Form Name field.
   c. Navigate through this form tree and select the New Hire Onboarding for Field Services Only form.
   d. Click the Select Form button.
6. Click Update.
   The Service Option Element Definition dialog closes. You return to the Service Option Details tab.
7. Scroll to the bottom of the tab and click Save.
8. Scroll to the top of the tab and click Return to Service Option Group.

Step 5 - Create the Service and Customize the Details

You create a service to contain one or more service option groups. Specify the details of how and when catalog users can request the service. In this scenario, you create and configure a service to contain the service option group that you created previously (New Hire Onboarding for Field Services Only).

Follow these steps:

1. Click **Catalog**, **Service Offerings**.
2. Expand the **Personnel Services** folder in the tree and select the **New Hire Onboarding** service.
3. Click the Definition tab to view the service option groups. In this scenario, you use the same service option groups in your new service.
4. Select the **Personnel Services** folder, and click **Add**, **Offering**.
5. Enter the name of the new service as New Hire Onboarding for Field Services Only.
6. Complete the following fields on the Details tab of the new service, and click Save:
   – For Description, specify Onboards new employees in the Field Services group.
   – Complete the fields in the Availability section, and click Save.
      The service becomes visible and users can request it only on the date that you specify in the Available On field.
7. Complete the following fields as shown, and click Save:
   – For User Request Method, verify that One-Click Submit is selected.

This setting lets users request the service quickly, without using a cart. This setting is ideal for simple services.
- Select the option named Available from mobile device.
- For the Approval Process drop-down list, either select No Approval or verify your selection with the catalog administrator. Catalog administrators typically configure either the Workflow driven approval process or the Policy driven approval process.
- Accept the default values for the other fields.

8. Click the **Definition** tab, and perform the following actions:
   a. Click the Edit icon for the Editor (default) Overview field.
   b. Specify the following (or similar) text for the Overview, and click the Save icon:
   For onboarding new employees in the Field Services group only

TIP

If deleting the overview text is required, we recommend deleting in the **Source** tab and then click Save.

TIP

When trying to access a resource from CA Service Catalog Server, provide the absolute URL path in **Overview Text** of CA Service Catalog Offerings so that it works in Service Point as well. Do not provide a relative path in Overview Text.

Step 6 - Add and Customize the Service Option Groups

You can add several service option groups for onboarding Field staff. As you add each group, if applicable, you customize it for the Field staff by including only the deluxe option.

NOTE

The customizations in these steps affect this service *only*. You do not modify the service option groups or their service options. Instead, you customize this service to include only the service options that you select from each service option group.

Follow these steps:

1. Click the Edit Offering Selection button on the Definition tab for the New Hire Onboarding for Field Only service.
2. Scroll to the service option group named Business Cards. Perform the following actions:
   a. Click the Include checkbox.
   b. Select Order Business Cards - Embossed, and select the options for Include and Default.
   Leave both options *unchecked* for the service option named Order Business Cards - Standard.
   This action adds embossed business cards to the service and includes them by default.
3. Scroll to the service option group named Mobile Phone, and perform these actions:
   a. Click the Include checkbox.
   b. Select Mobile Phone- Deluxe, and select the option for Default.
   This action adds a deluxe mobile phone to the service and includes it by default.
4. Scroll to the service option group named Mobile Phone Accessories, and perform these actions:
   a. Click the Include checkbox.
   b. Select all the options for Include and Default.
   This action adds all mobile phone accessories to the service and includes them by default.
5. Scroll to the service option group named New Hire Onboarding for Field Services Only, and perform these actions:
   a. Click the Include checkbox.
   b. Select the options for Include and Default.
   This action adds the form that you created earlier to the service and includes it by default.
6. Scroll to the service option group named Procure Laptop, and perform these actions:
   a. Click the Include checkbox.

b. Select Laptop - Deluxe, and select the options for Include and Default.
This action adds a deluxe laptop to the service and includes it by default.

7. Scroll to the bottom of the dialog and click Save.
The Catalog system adds the service option groups to the service, with your customizations.

Step 7 - Specify Automatically Chosen Selections

You can specify the service option groups that you added to the service as automatically selected (default). This technique helps to verify that Field Services personnel can see and request all required equipment efficiently, especially from mobile devices.

Follow these steps:

1. Perform the following actions on the Definition tab for the New Hire Onboarding for Field Only service:
   a. Click the Edit icon for the first service option group, Business Cards.

      NOTE

      When you mouse-over this Edit icon, the help text is "Edit Service Option Group." Other Edit icons for the header text and for the service option also appear near this Edit icon, but each one has different help text when you mouse over it.

      The Details tab for the service option group appears.
   b. Specify Automatically Chosen as the Selection Type. Click Save.
      This service option group is automatically included when the service is requested or subscribed to. This service option group contains one service option, embossed business cards. The embossed business cards option is automatically selected for Field Services staff, and is the only option for them.
   c. Click the Back button on the browser to return to the Details tab for the service.
   d. Click the Definition tab for the service.
      You return to the Definition tab for the New Hire Onboarding for Field Only service. The list of service option groups in the service appears.
2. Perform the following actions:
   a. Click the Edit icon for the next service option group, Mobile Phone.
   b. Specify Automatically Chosen as the Selection Type. Click Save.
      This service option group is automatically included when the service is requested or subscribed to. This service option group contains one service option, the deluxe mobile phone. The deluxe mobile phone option is automatically selected for Field Services staff, and is the only option for them.
   c. Click the Back button on the browser to return to the Details tab for the service.
   d. Click the Definition tab for the service.
      You return to the list of service option groups on the Definition tab.
3. Use the previous steps as a model to edit the definitions for the following service option groups and to specify Automatically Chosen as the Selection Type:
   – Mobile Phone Accessories
   – New Hire Onboarding for Field Services Only
   – Procure Laptop

   These service option groups also contain a single, deluxe service option that is automatically included for Field Services staff.

Step 8 - Set the Permissions

You can optionally set the permissions for the service so that only specified roles or groups can request it. If a role or group has permission to access the service, the associated users can perform the following actions.

- View the service in the catalog.
- View, request, and subscribe to the service
- Access the service through searches and web service calls

Conversely, if a role or group does not have permission to access the service, the associated users do not have these rights.

Follow these steps:

1. Click the **Permissions** tab of the service named New Hire Onboarding for Field Only.
2. Clear the check box named **Grant All Access to All Roles in All Business Units**.
3. Select the Request/Subscribe box for Catalog User, and click Save.
   This selection lets all Field staff request this service. In this scenario, all Field staff have the catalog user role. In your organization, another role may be more suitable.

   NOTE

   You can optionally define a user group for Field staff in CA EEM and use the Groups tab to restrict the permissions for the service to that group.

Add or Update Services

This article contains the following topics:

Service designers create or update services. They do so by grouping service option groups together as a single service that users can request from the catalog.

Follow these steps:

1. Click **Catalog**, **Service Offerings**.
2. Expand the tree and select the folder in which you want to manage services.

   NOTE
   To add a folder at the top level (root folder), click Add and name the new folder.
3. Perform one of the following actions in the selected folder:
   – Create a service by clicking Add and naming the new service.
   – Edit a service by selecting it and updating its fields.
4. Complete the fields on the Details tab for the service.

   NOTE

   This step and the remaining steps apply *only* if you are adding or updating a service.
5. Define the service.
6. Click the Permissions tab, and set permissions for the service.
   Each role and group (if applicable) receives the access rights that you specify to the new service.
   Only if a role or group has permission to access the folder, the associated users can perform the following actions:
   – View the service in the catalog.
   – View, request, and subscribe to the service.
   – Access this service through searches and web service calls.
7. (Optional) Click the Related Offerings tab, and perform the following tasks:
   – Review Service and folder inheritance. Specify other services to inherit (include) with this service.
   – Define dependencies for this service, if applicable.
8. (Optional) Set Featured and Related Services.

Service Details Tab

When you add or edit a service, you complete the fields on the Details tab for the service.

The following fields on the Details tab for a service require explanation.

- **Image**
  Assigns an image to a service.
  To select the new image from the USM_HOME\FileStore\images\offerings folder, click the existing image.

  WARNING
  The folder name *FileStore* is case-sensitive. Therefore, use the correct case in path names and all other programmatic references.

  The recommended size for an image is no larger than 48 x 48 pixels. However, the size of an image for a featured service is fixed at 32 x 32 pixels. So, the image is reduced or enlarged to 32 x 32 pixels, regardless of its original size. If you make this service a featured service, verify that the image is legible when viewed at 32 x 32 pixels. If you add images to several services in the same folder, verify that the sizes of the images are compatible. Verify that the images blend together to form a balanced layout that aligns with the Featured Item header.
- **Service ID**
  (read-only) Specifies the object ID of the service.

  NOTE

  As part of importing or exporting objects using IXUtil or content packs, you specify the Service ID explicitly.
- **Name *and* Description fields**
  Specifies the name and description of the service for catalog users. You can optionally localize these fields for use in a multilingual catalog.
- **Code**
  Specifies the text value to represent the product code, subscription code, SKU number, or any other applicable value.
- **URL Info**
  Specifies an external URL for more information about the service. For example, a service for a new laptop can include the URL to the manufacturer specifications page for the laptop.
- **Date fields**
  The dates and times that you view and specify are based on the time zone on the CA Service Catalog server. These values can be different from your local time zone.
  The dates that you enter in any of the following fields affect the availability of the service to catalog users:
  – Available On
  – Unavailable After
  – Canceled After fields
  The changes that you make to these fields take effect immediately.
  The following date fields require further explanation:
  – **Unavailable After**
    Specifies the date when catalog users can *no longer* request or subscribe to the service.
  – **Canceled After**
    Specifies the cancellation date for all requests for this service and all accounts with subscriptions to this service. On this date, these subscriptions and requests are canceled. Requests and subscriptions that are in progress are canceled immediately.
- **Business Hour**
  Specifies the business hours for the service. The business hours are the regularly scheduled days and times of service. For example, Monday-Friday, 9:00 am to 5:00 pm. Business hours apply only when you use request SLAs to help measure the availability of a service. For more information, see Manage Outage Calendars.

NOTE

During the time periods that the business hours define, time is monitored for the request SLA attached to a service.

- **Outage Calendar**
  Specifies the outage calendar for the service. The outage calendar specifies days, dates, and times when SLA monitoring is disabled for the service. For example, weekends, holidays, and one-time outages. Outage calendars apply only when you use request SLAs to help measure the availability of a service. For more information, see Manage Outage Calendars.

  NOTE

  During the outages that the outage calendar of the service specifies, the time is not monitored for the request SLA.
- **User Request Method**
  Specifies whether users request the service by using the one-click submit method or by using a shopping cart.
  - **One-Click Submit**
    Allows users to request the service, using one click, without a shopping cart. This method submits a stand-alone request for the service.
    The one-click submit method is appropriate for the internal business and personnel services that are *not* related to shopping. Examples: Services for onboarding new employees and for setting up or accessing virtual computers.
    When you use this method for a service, users cannot include it in the same request as other services.
  - **Shopping Cart**
    Allows users to request this service by adding it to a shopping cart. The cart can include multiple services. The user finishes shopping, verifies the cart, and submits it. This method is appropriate for services that are related to internet shopping, typically procurement requests. Example: Requests for new hardware or software.
    A catalog user who has added services to a shopping cart without submitting it can request a one-click submit service. The request for the one-click submit service is submitted immediately as a separate stand-alone request. The shopping cart is not affected.
  - **Available from mobile devices**
    Enable the checkbox to raise a service request from the mobile application.
  - Display SOG Name

    NOTE

    This capability is available only if you have installed the CA Service Catalog 14.1.01 patch updates available on CA Support.

    Check or uncheck the checkbox to hide or display the Service Option Group name that has single or multiple options associated in an offering. This option is enabled only when you have a single Service Option Group in an offering. In case of multiple service option groups in an offering, this option is disabled or greyed out by default. Checking or unchecking this option will not alter the offering Preview.
- **Approval Process**
  Specifies the approval process to use for the service when a user requests it. This Approval Process setting does *not* apply to subscriptions.
  Select *one* of the following options:
  - **System approval process**
    To determine whether the request for a service requires further approval, the System approval process uses the following options:
    - The authorization level of the user requesting the service
    - The approval level of the service

    If the authorization level of the user is less than the approval level of the service, the request requires further approval. In that case, the Catalog system performs the following tasks:

- Assigns the manager of the user to approve the request
- Places the request in the Requests Pending Action queue of that manager

This process repeats until the authorization level of the approver is at least equal to the approval level for the service.

– **No approval process**
Approves the service automatically.

– **Workflow Driven Approval Process**
Uses a CA Process Automation process to determine the approval process.
The process includes the business logic to determine the approver and the number of approval levels. CA Service Catalog provides sample processes for a single level of manager approval.
For any service, if you use this approval process with a CA Process Automation process, you can optionally use policies. In that case, the approval process proceeds the same as with Policy driven approval process, except as follows:

- The Workflow driven approval process uses the CA Process Automation workflow engine to evaluate and implement policies.
- Policy driven approval process uses the Catalog Policy Engine to evaluate and implement policies. This option is typically more efficient for an approval process that uses policies, because this engine is internal.

Verify that the rules and actions are enabled for the option that you specify.

– **Policy-driven approval process**
Uses policies to determine the approval process for requests. In policies, you specify conditions that are based on the attributes of service options, services, requested items, and users. If a policy is active and a submitted request meets the condition in the policy, then the following event occurs: The assignees in the policy receive a request pending action to approve, reject, or fulfill a service option, service, or request. Policy driven approval and system approval use a few common terms. Example: The *level of approval* refers to the authority of an approver in numeric terms. The higher the number, the greater the authority of the approver. However, in policy driven approval, the administrator assigns each approver and authority level uniquely, with no relation to system approval. If a policy does not apply to a request, the Catalog system uses the approval flow that is defined in the workflow driven approval process. Example: You are using the predefined workflow approval process and no predefined sample policy applies to a request pending action. In that case, the Catalog system assigns the request pending action to the manager of the Requested For user. If the user has no manager, then the system assigns the request pending action to the Default User for Request Actions. This user is specified in the Request Management Configuration.

- **Approval Level**
Specifies the approver Authorization Level that is required for the service. Specify these values logically and consistently to verify proper approval operations in your organization. This setting applies *only* if you specify System approval process or Workflow driven approval process in the Approval Process field.
- **Status on Approval**
Specifies the status of the request items in the service once the service is approved. Select *one* of the following options:
  – Fulfilled
  – Pending Fulfillment.
  Using Pending Fulfillment enables you to specify workflow processes that assign any additional tasks to fulfill the request, if necessary.

  This setting applies *only* if you specify System approval process or No approval process in the Approval Process field.
- **Sort By**
Arranges the service option groups to appear to the user according to the category you select.
Categories include Name, Selection Type, Code, Date Created, and None. You can also select Custom - use Sort Number. This option uses the Sort Number field in the service option group details.
- **Sort Number**
Sorts the service according to the value you specify.
This field applies *only* when the parent folder uses a value of Custom - use Sort Number in the Sort By field.

Define a Service

After you create a service, define it. *Defining* a service primarily means adding one or more service option groups and specifying which service options to include.

Follow these steps:

1. Click Catalog, Service Offerings.
2. Expand the tree and display the service of interest and click the Definition tab.
3. Click Edit Offering Selection.
   - If no service option groups are already associated with this service, all service option groups appear.
     To add service options from a group to a service, select the group, expand it. Select the service options that you want to include.
   - If one or more service option groups are already associated with this service, the service options from those groups appear.
     - Expand the associated service option groups and select or remove service options for this service.
     - Click Show All to display all service option groups. Select the group that you want, and include the service options that you want in the service.
   - Verify that you have completed the following actions, as applicable:
     - Select the service options that are included in the service *by default*. To do so, select the check box for the service option in the Default column.
     - Select the service options that are *required* in the service. To do so, select the check box for the service option in the Include column.
     - Exclude any groups or service options that you do *not* want in the service.
4. Save your changes and close the Service Offering Selection dialog.

You can set thresholds for QoS SLAs for request fulfillment reports. You can also set permissions as part of managing the service.

Service and Folder Inheritance

You can configure the inheritance folders and services, as follows:

- Inherit folders and services from the catalog of one business unit to the catalog of another business unit.
- Inherit folders and services from one part of the catalog hierarchy to another.
- Copy a service or folder and paste it as inherited.

After you create the inherited folder or service, you can update both the parent and child (inherited) objects. The following tables list:

- The changes to the parent that the child inherits
- The changes to either the parent or child that break inheritance

Parent Object	Change	Change Inherited?	Breaks Inheritance?
Folder or Service	Update settings on the Details tab	Y*	N
Folder	Add subfolders or services	N	N

Folder or Service	Delete services or subfolders--Not permitted directly. To delete the service or subfolder, first break the inheritance by deleting the inherited service or subfolder. You can also break inheritance for a service by editing or deleting an inherited service option group or service option.	N	N
Service Option Group	Update settings on the Details tab	N	N
Service Option Group	Add a service option	N	N
Service Option Group	Delete a service option--Not permitted directly. To delete the service option, first break the inheritance by editing or deleting the inherited service option.	N	N
Offering or Service Option Group or Service Option***	Update permissions for Offering or Service Option Group or Service Option	N	N
Service Option	Update a service option element**	Y	N
Service Option	Add a service option element	N	N
Service Option	Delete a service option element	N	N

*Your updates in the parent folder override any existing values (including any custom values) in the child folder. The only exceptions are as follows:

- For child folders, the folder name is not changed.
- For child services, the service name, dates, and status are not changed.

** A service option element is a part of a service option. Examples include forms, reservations, and rate elements. Service option elements appear as fields on the Service Option Details tab. When you update these fields, the Service Option Element dialog opens.

*** indicates updating permissions for the child offering, child service option, child service option group are not automatically propagated to the parent offering, parent service option, and parent service option group.

Child Object	**Change**	**Breaks Inheritance?**
Folder or Service	Update settings on the Details tab	Y or N*
Folder	Add subfolders or services	N
Folder	Delete subfolders or services	N
Service Option Group	Update settings on the Details tab	N
Service Option Group	Add a service option	N
Service Option Group	Delete a service option	N
Offering or Service Option Group or Service Option***	Update permissions for Offering or Service Option Group or Service Option	N

Service Option	Update a service option element	Y**
Service Option	Add a service option element	Y**
Service Option	Delete a service option element	Y**

* For child folders, you can change the name without breaking inheritance. For child services, you can change the name, dates, and status without breaking inheritance. Any other change breaks inheritance of the folder or service details *only*. Inheritance of service options and service option elements is not broken.

** The inheritance of the service option is broken. Inheritance of the folder and service details remains.

*** indicates updating permissions for the child offering, child service option, child service option group are not automatically propagated to the parent offering, parent service option, and parent service option group.

NOTE
Inheritance applies only to the changes listed in the tables.

Set Featured and Related Services

For folders, you can specify *featured* services. Similarly, for services, you can specify *related* services. These specifications are helpful when the user viewing a service or folder considers requesting a featured or related service. **Example:** When you create an "onboarding" service for new employees, you can specify services for laptops and cell phones as related services. When new employees access the onboarding service in the catalog, they see summaries and links for these related services.

Follow these steps:

1. Click Catalog, Service Offerings.
2. Expand the tree. Perform one of the following actions for the service or folder of interest:
   - Select the folder and click the Featured Offerings tab.
   - Select the service and click the Related Offerings tab.
3. Expand the Catalog Tree and specify one or more services, as follows:
   a. On the left side of the tree, select the box next to the service name.
   b. Next to the Selected Offerings box, click the right arrow to add the selected service to the box.
   c. (Optional) Include the children (inherited services) with the featured service. You can specify this setting individually for each featured service.
4. Verify and save your updates.
5. Verify that the catalog is configured to display related services, as follows:
   a. Select Catalog, Configuration, Request Management Configuration.
   b. For the Browse Catalog Layout option, specify Request View.
   c. For the *Access Control: Show General Information* option, specify the roles that you want to see featured services.
   d. For the *Access Control: Show General Information* and *Selections in Catalog Item Details* option, specify the same roles that you specified in the previous step.

Perform Other Tasks to Manage Services

This article contains the following topics:

Administrators (typically service designers) can copy, cut and paste, delete, and cancel services. They can use a service as a template. Administrators can also manage the folders that store services, define dependencies, and set thresholds for quality-of-service (QoS) SLAs.

Perform or skip any of the following tasks, as needed.

1. Use a service as a template.

2. Copy a service *with or without* inheritance, as follows:
   a. Select the service and click Copy.
   b. Specify which *one* of the following options you want to include in the copy:
      - All associated service option groups (preferred)
      - Only the service, *without* the service option groups (primarily for backward compatibility with earlier releases of the product)
   c. Expand the tree and select the folder to which you want to paste the copied service.
   d. Click the Paste icon or the Paste as Inherited icon.
      The folder (and its child folders and services) or service is pasted to the new location.
      The inherited services are named "Inherited from *parent service*." When the parent service is updated, the same updates occur in the child service automatically.
3. Cut and paste a service, as follows: Select the service and click Cut. Expand the tree to the new location and click Paste.
4. Delete a service by selecting it and clicking the Delete icon. Before you confirm the deletion, review the effects of deleting a service.

   WARNING
   Deletion is permanent.
5. Cancel a service by selecting it and setting the Canceled After field on the Details tab.
   Before you confirm the cancellation, review the effects of deleting a service. Canceling a service has the same effects as deleting a service on the related subscriptions and requests.
6. Review thresholds for QoS SLAs and set them.
7. Manage the folders that store services in the catalog.

Define Service Dependencies

A dependency in the catalog allows a service state to change depending on the state of another service. You can also set the availability of a service to a user based on the business unit of that user.

The ability to define dependencies of services allows you to define how one service behaves based on how you select or clear another service. For example, if you are subscribing to Service A, the following events can occur for Service B:

- Service B can become disabled. The selection check box is inactive and cannot be selected or cleared.
- Service B can become subscribed to. The check box displays selection whether it is enabled or disabled.
- Service B can become unsubscribed to. The check box is cleared whether it is enabled or disabled.

Follow these steps:

1. Click Catalog, Service Offerings.
2. Click Change Business Unit, if necessary, and select the business unit whose service dependencies you want to define.
3. Click Define Dependencies.
4. Select one of the following options, and click Save Dependency.
   - **Business Unit to Service Assignment**
     Specifies which services appear in the catalog of the child business unit.
   - **Service to Business Unit Assignment**
     Specifies which services appear in the catalog of the child business unit.
   - **Subscribe Services**
     Specifies which services are included automatically in a subscription to another service. When a user subscribes to the selected service, the Catalog system subscribes the user to the services that you specify.
   - **Un-subscribe Services**

Specifies which subscribed services are canceled automatically with the cancelation of another service. When a user cancels the selected service, the Catalog system cancels the subscription (of this user) to the services that you specify.

– **Disable Services**
Specifies which services are made *unavailable* (disabled) automatically as a result of a subscription to another service. When a user subscribes to the selected service, the Catalog system disables (for this user) the services that you specify.

5. Click the Close icon.

Use a Service as a Template

You can create a service to use as a template for creating all services in your catalog. Creating a template service helps ensure that all services in your catalog have a consistent look-and-feel. To create a template service, follow this process:

1. Add a service, using the following guidelines:
   – Include the word "template" in the name.
   – Mark the service as unavailable by setting the Unavailable After date to the current date.
   – Select the service option groups that you want to be in the template service.
2. Inform other administrators about the template service. Tell them to follow the remaining steps to use it to add a new service:
   a. Copy and paste the template service to an appropriate "holding" folder.
   b. Rename the service each time that you copy it to a different folder.
   c. Add new service option groups to the service, if necessary.
   d. Remove existing service option groups from the service, if necessary.
3. Perform one of the following steps to modify a service option group in a service that is created from your template service:
   – To apply your changes to all services that use the service option group, edit the service option group definition directly.
   – To apply your changes only to the new service that you are creating from the template service:
      a. Copy and rename the service option group.
      b. Modify the new service option group as required.
      c. Add the new service option group to your new service.
      d. Remove the original service option group from the service.

      NOTE

      When you copy a service, specify whether the new service includes copies of the original service option groups or links to them. If you specify links, any changes to a linked service option group apply to all services that include it. That is, the original service, the new service that you copied, and any other service in the Catalog system that includes the link.
4. Define a service option element in a service option group, or update it, if necessary.

Effects of Deleting a Service

Deleting folders or services affects the subscriptions and requests for the deleted services, as follows:

- **Subscriptions**
  The account that is subscribed to the service being deleted no longer contains or lists that service.
- **Requests**

The status of requested service options for the account change that is based on their original status, as follows:

Original Status	New Status
Not Submitted	The service is deleted from the request.
Submitted. An approval status, a fulfillment status, Pending Resource Assignment, or Resource Assignment	The service and its service options are set to Cancelled.
Completed	The default cancellation status (Pending Cancelation or Cancel) if Service Accounting Component is installed. Cancelled if Service Accounting Component is not installed.
Pending Cancelation or Canceled	Same as original status.

Thresholds for QoS SLAs

Request Service Level Agreements (SLAs) are a feature of CA Service Catalog. Quality of Service (QoS) SLAs are available only if CA Service Catalog is integrated with CA Business Service Insight.

You can establish thresholds for QoS SLAs for use with Request Fulfillment reports. An SLA threshold specifies the following data:

- Starting status
- Ending status
- Duration (days, hours, and minutes) for warning status
- Duration for a violation status

If the request for the service option takes longer than the SLA duration, then the request reaches a warning or violation status.

The Request Fulfillment reports use the SLA threshold specification. By default, the report includes the following phases:

- The approval phase
- The fulfillment phase
- The combination of approval and fulfillment phases

The following table shows summary data for these items:

Starting Status	Numeric Value	Ending Status	Numeric Value
Submitted	200	Approval Done	999
Approval Done	999	Completed	2
Submitted	200	Completed	2

You can also configure the reports to display the following data:

- The time for the status of a service option to move from the starting to the ending phase.
- The average time in each phase.
- The violation time for the SLA phases.
- The color code of warnings and violations for the service options for which SLA thresholds have been set.

If you report on different SLA thresholds, alter the STATUS_RANGES values for the Request Fulfillment reports.

The SLA violation duration for the Submitted to Completed phase is used for the “estimated time to fulfill” value. This value appears in the catalog in the “fulfillment details” for a service option.

Set Thresholds for QoS SLAs

You can establish thresholds for QoS SLAs for use with Request Fulfillment reports. Doing so is an optional task when you define a service.

Follow these steps:

1. Select Catalog, Services Offerings.
2. Expand the tree, locate the service of interest, and click its Definition tab.
3. Select the service option of interest and click its SLA icon.
4. Click Add, specify the following parameters for the new SLA, and click OK:
   - The starting status
   - The ending status
   - The warning and violation thresholds

Manage Folders

Services exist in a folder structure. You can *manage* the predefined folder structure to meet your needs. *Manage* here means create, copy, delete, and move.

Follow these steps:

1. Click Catalog, Service Offerings.
2. Expand the tree and select the folder of interest.

   NOTE
   To add a folder at the top level (root folder), click Add and name the new folder.
3. Perform one of the following steps in the selected folder:
   - Copy a folder *with or without* inheritance, as follows:
     Select the folder and click Copy.
     Specify which *one* of the following you want to include in the copy:
     - All associated service option groups
     - Links to these service option groups

     Expand the tree and select the folder to which you want to paste the copied folder.
     Click the Paste icon or the Paste as Inherited icon.
     The folder (and its child folders and services) or service is pasted to the new location.
     Inherited folders are named "Inherited from *parent folder*." When the parent folders or services are updated, the same updates occur in the child folders or services automatically.
   - Delete a folder by selecting it and clicking the Delete icon.

     WARNING
     Deletion is permanent. Deleted folders *cannot* be recovered.

     Deleting the folder deletes all services in it. Before you confirm the deletion, review the effects of deleting a service.
     Do one of the following actions *if* the folder that you want to delete contains inherited services.
     - Delete the child, inherited service.
     - Break the inheritance by editing the child, inherited service.
4. Complete the fields on the Details tab for the folder.

   NOTE

   This step and the remaining steps apply only if you are creating or updating a folder.
5. Click the Permissions tab, and set permissions for the folder.
   Each role and group (if applicable) receives the access rights that you specify to the new folder.
   Only if a role or group has permission to access the folder, the associated users can perform the following actions:

- View the folders and its subfolders in the catalog.
- View, request, and subscribe to the services in the folder and its subfolders.
- Access this folder, its subfolders, and associated services through searches and web service calls

6. (Optional) Review featured services. Specify any services in this folder to feature in the catalog.
   (Optional) Review service and folder inheritance. Specify other services to inherit with this folder.

Details Tab for a Folder

When you add or edit a folder, you complete the fields on the Folder Details tab.

The following fields on the Folder Details tab require explanation.

- **URL**
  Specifies a URL that supplies more information about the services in the folder. For example, a URL to a manufacturer web site.
- **Display Subfolders**
  Specifies whether the top-level subfolders under this catalog folder appear in the Browse section of the Requests page. Select *one* of the following options:
  - **Use System Setting**
    Use the system or "global" setting to show or hide the subfolders that are specified in the Browse Catalog: Show Subfolders parameter in the Request Management Configuration section of the Configuration page of the Catalog tab.
    If the value of the Browse Catalog: Show Subfolders parameter changes, all catalog folders that use the system setting are automatically updated accordingly.
  - **Show Subfolder**
    Specifies that this catalog folder *always* displays its subfolders in the Browse section of the Requests page, regardless of the value of the system setting.
  - **Hide Subfolder**
    Specifies that this catalog folder *never* displays its subfolders in the Browse section of the Requests page, regardless of the value of the system setting.
    Hiding these subfolders is often helpful when the number of them is so high that the following condition exists: Users must scroll through the Browse Catalog several times to view them all or to see the remaining folders and subfolders

  Default: Use System Setting

A dependency in the catalog allows a service state to change depending on the state of another service. You can also set the availability of a service to a user based on the business unit of that user.

You can define how one service behaves based on how you select or clear another service. For more information, see Perform Other Tasks to Manage Services.

{"URL":["/usm/wpf?Node=icguinode.billdepend&Args=ca.com"],"description":"IPCE_Description","new":"","new_video":"","admin":"","troubleshooting":"","pendo":"","video":""}

Manage Service Option Groups

This article contains the following topics:

Administrators (typically service designers) create, update, and otherwise manage service option groups, which are the main components of services. Here, *manage* means create, update, copy, edit, delete, and so forth. A service option group consists of hardware, software, or other resources that you can include in a service. Service option groups are bundled groups of service options. Each service option consists of individual service option elements. You can use the same service option group in multiple services. Each business unit can have its own set of service option groups.

Follow this process:

1. Click Catalog, Service Offerings.
2. Click the Option Groups tab and perform one of the following actions:
   – View the details of a service option group by selecting it.
   – Edit a service option group by selecting it and updating its fields. Go to the next step.
   – Create a service option group by clicking the + sign below the Option groups tab and name it. The details page for the new service option group appears. Go to the next step.
   – Copy a service option group *with or without* inheritance, as follows:
      a. Select the service option group in the tree and click Copy.
      b. Specify a unique name for the new service option group.
      c. Click the Paste icon or the Paste as Inherited icon.
         On the service option group tree, the inherited options appear in green.
   – Delete a service option group by selecting it and clicking the Delete icon. Before you confirm the deletion, review the effects of deleting a service option group.

      WARNING
      Deletion is permanent. Deleted service option groups *cannot* be recovered.

      Perform one of the following actions *if* the service option group that you want to delete contains inherited service option groups. Otherwise, skip to the next step.
      - Delete the child, inherited service option group.
      - Break the inheritance by editing the child, inherited service option group. You cannot delete a folder or service that contains inherited service option groups.
3. Complete the fields on the Details Tab for the service option group.

   NOTE

   This step and the remaining steps apply only if you are adding or updating a service option group.
4. You can optionally use a tiered service option group.
5. Define the service option group by clicking the Definition tab and performing one of the following actions:
   – Create service options and add them to the group.
   – Edit a service option in the group.
   – Delete a service option from the group.
   – Copy a service option in the group, with or without inheritance.
   – Organize the service options in the group.

   After you define the service option group, you can add it to services that users can request from the catalog.
6. Publish a report layout for a service option group.
7. Associate a CA APM model with a service option, if you have integrated with CA APM.

Service Option Group Details Tab

The following fields on the Details tab for a service option group require explanation:

- **Id**
  (Read-only) Specifies the object ID of the service option group. At times, you specify object IDs explicitly. For example, as part of importing or exporting objects using IXUtil or content packs.
- **Date fields**
  The dates and times that you view and specify are based on the time zone of the CA Service Catalog server, which can be different from your local time zone.
  The following date fields require further explanation:
  – **Available On**

Specifies the date when this service option group becomes available for catalog administrators to include in a service when they define the service. The change that you make to this field take effect immediately.

  - **Unavailable After**
    Specifies the date when catalog users can *no longer* request or subscribe to the service option group. Catalog administrators can no longer include this service option group in a service when they define the service. The change that you make to this field take effect immediately.

- **Name *and* Description fields**
  Specify the name and description of the service option group for catalog users. You can optionally localize these fields for use in a multilingual catalog.
- **Type**
  Specifies the type of service option group. This value determines how you can include this service option group within other service option groups as a characteristic of a service option element. Select one of the following options:
  - Fixed - A fixed service option group presumes a fixed cost or a variable cost that is based on usage.
  - Tiered - A tiered service option group presumes a variable cost value.

  Default: Fixed
- **Selection Type**
  Specifies how catalog users select the options in the service option group while requesting a service that includes the service option group.
- **Code**
  Specifies a text value for a code of your choice. Examples include the product code, subscription code, and SKU number.
- **Sort Number**
  Specifies how to sort this service option group when a service that includes it uses a Sort By setting of Custom - use Sort Number.
- **Offering Dependencies**
  Specifies the names and statuses (for example, Available) of the services that include this service option group.
- **Account Dependencies**
  Specifies the following data:
  - The names of the accounts that are subscribed to services that include this service option group. The accounts subscribe to the service option group by subscribing to the service.
  - The names and statuses of the services to which these accounts are subscribed.

Tiered Service Option Group

A tier is a single row in a tiered service option group. You can use a tiered service option group to produce a variable cost according to a lookup value. The lookup value from the referencing service option group is matched with the tier values in the tiered service option group.

An association into a tiered service option group starts at the first row and works its way down: The first tier has the lowest tier values. When an applicable tier is identified and the Quantity Specification is set to System Specified, this event occurs: The lookup value is multiplied by the rates in other service option elements in the tier row to determine the invoice amount. If the Quantity Specification is not set to System Specified, other rates apply.

When a service option element is associated with a tiered service option group, the tier type value determines how the tiered service option group is used. The following tier types are available, depending on the referencing service option element:

- **Lookup**
  Use the first tier that matches the value being passed to the tiered service option group.
- **Lookup Multiple**
  Use each tier that matches the value being passed to the tiered service option group.
- **Variable Lookup**

Use each tier up to and including the tier that exactly matches the value being passed to the tiered service option group.

- **Fixed**
  Use the first tier that matches the value being passed to the tiered service option group. Once in that tier the tier is fixed and cannot change.
- **Fixed Incremental**
  Use the first tier that matches the value being passed to the tiered service option group. Increment to the next tier only if the value passed in exceeds the previously used tier. That is, a certain tier level is used; a lower tier level can never be used.
- **Variable Fixed**
  Is a merger between variable lookup and fixed tier types. Use each tier up to and including the tier that exactly matches the value being passed to the tiered service option group. Once in the tier that exactly matches, that tier is fixed and cannot change.
- **Variable Fixed Incremental**
  Is a merger between variable lookup and fixed incremental. Use each tier up to and including the tier that exactly matches the value being passed to the tiered service option group. Increment to the next tier only if the value passed in exceeds the previously used tier.

Inheritance of Service Option Groups

You can copy a service option group and paste it as an inherited service option group. The inherited service option group has the same characteristics as the parent service option group from which it was created. The Catalog system shows the inheritance in the inherited service option group, as follows:

- On the Details tab, a note indicates that the service options in this group are inherited.
- On the Definition tab, the inherited service options appear in green.
- If you click the Edit icon of an inherited service option, the Service Option Details tab includes the following information:
  - Text that the option is inherited.
  - A link to the parent service option.
- If you click the Edit icon of an inherited service option element on the Service Option Details tab, the resulting dialog includes:
  - Text that the service option element is inherited.
  - The ID of the parent service option element.

The following rules apply to the inheritance relationship:

- If you update a service option element in the parent service option group, the Catalog system automatically makes the same updates to the inherited service option element in the child service option group. The updated values of the parent replace the existing values in the child.
- If you make either of the following updates to the parent service option, the updates do not apply to the child service option:
  - Adding a service option element
  - Deleting a service option element
- If you make either of the following updates to the child service option, you break the inheritance relationship. If this relationship breaks, then changes to the parent no longer propagate automatically to the child.
  - Adding a service option element
  - Deleting a service option element
  - Editing a service option element

When you manually update an inherited service option element, the green color disappears. The color disappearance indicates that the element was changed after it was inherited.

Effects of Deleting a Service Option Group

Deleting a service option group affects the subscriptions and requests for the deleted service option groups, as follows:

- **Subscriptions**
  The account that is subscribed to the service option group being deleted no longer contains or lists that service option group.
- **Requests**
  The statuses of requested service options in the service option group being deleted change, as follows:

Original Status	New Status
Not Submitted	The service option group and its service options are deleted from the request.
Submitted. An approval status, a fulfillment status, Pending Resource Assignment, or Resource Assignment	The service option group and its service options are set to Canceled.
Completed	The default cancellation status (Pending Cancellation or Cancel) if Service Accounting Component is installed. Cancelled if Service Accounting Component is not installed.
Pending Cancellation or Canceled	Same as original status.

Publish a Report Layout

You can publish a report layout that appears as a service option in a special service option group. This service option group is created the first time that a report is published. This service option group is named *Published Reports*. You can manage it as you manage other service option groups.

Follow these steps:

1. Select Administration, Report Builder, Layouts.
2. Expand the folders and select the report of interest.
3. Click the Publish Layout to the Catalog icon in the Actions column.
   A new service option for the report layout is added to the Published Reports service option group. The new service option has a default cost of 0.
   The service option representing the published report contains the following data:
   – Two Text type service option elements that display the report name and comment values.
   – A Rate Type service option element that holds the report cost.
   You can edit this service option and can include it in services.

   NOTE
   You can remove a published report from the catalog, as follows: De-select the service option from a service that includes it, or remove the service option from the service option group.

A service option group consists of hardware, software, or other resources you can include in a service. Service option groups are bundled groups of service options. Each service option consists of individual service option elements.

You can create, update, and manage service option groups, which are the main components of services.

Also you can define the service option group by clicking the Definition tab and perform one of the following actions:

- Create service options and add them to the group.
- Edit a service option in the group.
- Delete a service option from the group.
- Copy a service option in the group, with or without inheritance.
- Organize the service options in the group.

For more information on how to manage service option groups, see Manage Service Option Groups.

{"URL":["/usm/wpf?Node=iclaunchpad.catalog_builder#id=*;t=SERVICE_OPTION_GROUP;d=ca.com"],"description":"IPCE_Description","new":"","new
{"masterkb":"","text":"the following KB articles will help you","URL": ["https://knowledge.broadcom.com/external/article?articleId=137077"]}, "pendo":"","video":""}

Manage Service Options and Service Option Elements

This article contains the following topics:

Administrators (typically service designers) create, update, and otherwise manage service options, which are the building blocks of a service option group. A service option group must contain at least one service option. Similarly, each service option contains one or more service option elements.

Each service option element adds value to the service. Examples include an item that the user can request or a function to complete the request lifecycle. Examples include these elements:

- A form element provides a form that the user completes to customize the request and to obtain fulfillment.
- A rate element provides billing specifications for a service option.
- A reservation element helps provide required data in a service for reserving physical or virtual computers.

Service option elements can have the following characteristics:

- **Static**
  Provides fixed information about the service option.
- **Actionable**
  Specifies that a request, service, or service option follows a specific approval process, fulfillment process, or both.
- **Financial**
  Specifies fixed, tiered, or usage-based rates. You can optionally link these rates to budget and planning functions.

You create, edit, and delete service options as part of managing service option groups.

When you create or edit a service option, you complete the fields on the following tabs, as applicable:

- Policies & Actions
- Options
- Details

After you finish creating a service option, click the Preview tab to see how it appears to users in the catalog. If necessary, update the service option after you preview it to achieve the results you want.

Policies and Actions Tab

You create *global* policies and global actions for general use with any service. In contrast, you create *attached* policies and actions for use with one or more specific service options only. You attach these policies and actions to each service option individually. You can create an attached policy or action *only* from the Policies & Actions tab of that service option. After you create an attached policy or action for that service option, you can optionally attach that policy or action to another service option. However, you cannot change an attached policy or action into a global policy or action. Similarly, you cannot change a global policy or action into an attached policy or action.

> **NOTE**
> On the Policies and Actions tab, you can use the link to access CA Process Automation.

By default, CA Service Catalog evaluates all global policies and all events, rules, and actions (global actions only) for every requested *service*. However, for individual service options, you can replace this default behavior. From the Policies & Actions tab of a service option, you can specify that the Catalog system apply the following specifications to that service option:

- For policies, one of the following options:
  - Global policies only
  - Attached policies only
  - Both global and attached policies
- For actions, one of the following options:
  - Global actions only
  - Attached actions only
  - Both global and attached actions

WARNING

Applying policies or actions at both the service and service option levels could produce conflicting, unintended, or unpredictable results. Use attached actions and policies *only* if you are using a discrete request life cycle that includes discrete approval of service options. This precaution helps avoid the unintended application of policies and actions at *both* the service and service option levels.

Service Option Element Options Window--Options Tab

Several fields appear on the Options tab of the Service Option Element Definition dialog. These fields apply to all service option elements.

- **Change to Take Effect**
  Applies *only* if the Catalog Configuration item named **Default Effect of Service Option Element Changes** is set to *Allow User to Choose*. For more information about Catalog Configuration settings, see the Catalog Configuration section.
  If applicable, this option appears when you create or update service options in a service option group that is already defined in a service. The setting for this field applies to *all* updates that you make to other fields on this tab.
  Select an option from the following list and specify when the changes take effect for existing subscribers or requesters:
  - **Beginning of Accounts' Current Billing Period - No Audit Trail**
    The change takes effect retroactively to the beginning of the current billing period for existing subscribers or requesters.
  - **Beginning of Accounts' Current Billing Period**
    The change takes effect retroactively to the beginning of the current billing period for existing subscribers or requesters.
  - **Beginning of Accounts' Next Billing Period**
    The change takes effect at to the beginning of the next billing period for existing subscribers or requesters.
  - **Immediately during Accounts' Billing Period**
    The change takes effect immediately for existing subscribers or requesters.
  - **Specify a Future Effective Date**
    The change takes effect for existing subscribers or requesters on the date specified.
- **Code**
  Specifies user-specified text value to represent the product code, subscription code, SKU # or any other applicable code.
- **URL Info**
  Displays a clickable URL with the service option element to the requesting or subscribing user.
- **Display Type**
  Specifies how the service option element appears in the request, subscription, and invoice (if Service Accounting Component is installed). This setting allows you to hide the service option element from view in special cases. Select an option from the following list:

  - Include in Request/Subscription and Invoice
  - Include in Request/Subscription, Exclude from Invoice
  - Exclude from Request/Subscription and Invoice
- **Include in services (offerings)**
  Applies only when you create a service option element in a service option that belongs to a service option group that is already defined in a service. In other words, one or more services already include this service option group.
  Select from the following list:
  - **Do not include**
    Specifies that the new service option element is not included in any existing (defined) services.
    However, newly defined services that include this service option do include the new service option element.
  - **That have this service option**
    Updates all existing services that include this service option to include the new service option element.
    The updated service definition includes the new service option element.
  - **That have this Service Option Group** Updates all existing services that include this service option group to include the new service option element, as follows:
    Did the original service definition include the service option that you updated by adding the new service option element?
    If yes, the updated service definition includes the new service option element in that service option.
    If no, the updated service definition includes a new service option that contains only the new service option element.

  If you select the second or third option, the following subscription prompt appears:
  Select from the following list:
  - **Do not subscribe**
    The new service option element is not included in any existing subscriptions.
    However, new subscriptions for this service that include this service option do include the new service option element.
  - **That have this service option**
    Updates all existing subscriptions for services that include this service option to include the new service option element.
  - **That have this Service Option Group**
    Updates all existing subscriptions for services that include this service option group to include the new service option element, as follows:
    Did the original service definition include the service option that you updated by adding the new service option element?
    If yes, the updated subscription to the service includes the new service option element in that service option.
    If no, the updated subscription to the service includes a new service option that contains only the new service option element.

Service Option Details Tab

Use this tab to add the service option elements that you want to this service option.

To add an element, click the Add icon and supply the following data. To update an element, click the Edit icon and update the following data.

- Fields that apply to all service option elements
- Fields that apply to the element only.

Basic Information

- **Name *and* Description fields**
  Specify the name and description of the service option for catalog users. You can optionally localize these fields for use in a multilingual catalog.
- **Attachment Mandatory**

Requires the user requesting the service to add an attachment for this service option. If the user does not add the attachment, the user cannot submit the request. Use this option to collect documents (such as certifications or proof of identity) or data that you cannot collect in a form.
If selected, this option is activated *only* when the Allow Attachments at Service Option Level parameter is set to Yes. To set this parameter, select Catalog, Configuration, and click Request Management Configuration.

Reservation

- **Reservation Service Option**
  Associates a Reservation Manager reservation of a physical or virtual resource.
  This option and the related fields apply *only* if Reservation Manager or an external reservation system is installed and integrated with CA Service Catalog. The related fields are as follows: Reservation Display Text, Operation, Reservation Ready Status, Reservation Failure Status, and Reservation System

Form

- **Form**
  Associates the form or forms that you select with this service option. In the catalog, the form appears under the service option. Users complete the form when they request this service option. This form is a service option element within this service option.
  Complete the fields for Form Designer form elements.
  You can create, customize, and use forms with the Form Designer (the preferred method).

Contract

- **CA Business Service Insight Contract**
  Associates a CA Business Service Insight service level agreement (SLA).
  This option applies only if CA Business Service Insight is installed and integrated with CA Service Catalog.

Categorization

- **Category**
  Specifies the major category of the service option element from a list of categories.
  The Category value helps determine which rule actions execute approval and fulfillment business processes for a request for a service including this service option element.
  In addition, this value is used to determine the asset type for assigning an asset.
  If CA APM is installed and you select Track as an Asset, the following option applies: You can use the CA APM Assign Asset dialog to associate a request item with a software asset or other type of asset.
- **Category Class**
  Specifies a value from the list to categorize the class of this element within the selected category.
- **Category Subclass**
  Specifies a value from the list to categorize the subclass of this element within the selected class.
- **Keywords**
  Specifies a list of comma-separated list of keywords that are referenced during a catalog search.
- **External ID**
  A user-specified text value to represent the product code, subscription code, SKU # or any other applicable code.
- **Track as an Asset**
  Identifies whether the service option element is eligible to be associated with an asset.
  If CA APM is installed and you select Track as an Asset, the following option applies: You can use the CA APM Assign Asset dialog to associate a request item with a software asset or other type of asset.
- **Information Service Option Only**
  Indicates that this service option is for information only and cannot be subscribed to or requested.

Cost & Pricing

- **Rate**

Associates a rate (price or cost) service option element with this service option. This price or cost is fixed; it is *not* based on application usage.
Complete the fields for rate elements.

- **Adjustment**
  Adjusts the charge of an associated service option element. It can be a fixed value or a specified multiplier of the value of the associated service option element.
  Complete the fields for adjustment elements.
- **Usage Based Price**
  Specifies the charges for a service option or other item according to usage.
  Complete the fields for usage based costs elements.

Additional Elements

- **Text**
  Specifies the text and optional image file that appear to users when they select the associated service option.
  An example text string follows: Estimate: one week to complete the request. A sample image file is a detailed photograph of the item being ordered.
  You can optionally upload an image file and specify a text value for the image.
  - **Associate Service Option Group**
    Indicates that a service option group is associated with this service option element.
    Select this field to display a list of tiered service option groups. Associate a service option group and tier type.
  - **Rich Display Text**
    Converts the Display Text field to a rich text field, so that you can add images and special formatting to the Display Text field.
    The recommended size for a Service Option Element image is no larger than 48 x 48 pixels.
  - **Information Service Option Only**
    (Read-only) This field is automatically assigned the same value as the Information Service Option Only field on the Service Option Details tab.
- **Numerics**
  Specifies either a fixed numeric value or a numeric value that is entered by the user requesting the service that contains this service option element.
  Complete the fields for numeric elements.
- **Booleans**
  Specifies a true or false value.
  Note that some fields apply only if Service Accounting Component is installed.
  - **Display Text**
    Specifies the additional text that appears for this service option element.
  - **Boolean Value**
    Determines the value of the service option element. Select from: False or True.
  - **Associate Service Option Group**
    Indicates that a service option group is associated with this service option element.
    Select this field to display a list of tiered service option groups. Associate a service option group and tier type.
- **Date**
  Specifies the date values used with a service option.
  - **Date Type**
    Specifies the type of date. Select from the following list:
    - Specify Value - The Administrator specifies the date value that appears in the catalog. This setting displays the following field:
    - Date Value: The date value for the catalog.
    - Subscription Date - The system sets this value to the date of the request or subscription.
    - Invoice Date - The system sets this value to the date of the invoice if Service Accounting Component is installed.
  - **Associate Service Option Group**

Specifies that a service option group is associated with this service option element.
Select this field to display a list of tiered service option groups. Associate a service option group and tier type.

- **Day of Billing**
  Similar to a rate element but also allows a day of the week for weekly charges or a day of the month for monthly charges.
  A Day of Billing element specifies one of the following options:
  – For weekly charges: the day of the week on which to bill the item
  – For monthly charges: the day of the month on which to bill the item
  This element applies only if Service Accounting Component is installed.
  Most of the fields for a Day of Billing element have the same meaning as the fields for rate elements.
- **Numeric Range**
  Specifies a numeric range entered by the user requesting the service that contains this service option element.
  This element applies to tiered service option groups *only*.
  Complete the fields for numeric range elements.
- **Date Range**
  Specifies a date entered by the user requesting the service that contains this service option element.
  This element applies to tiered service option groups *only*.
  – **Lower Bound**
    Specifies the numeric value of the lower bound for matches when this service option group is referenced.
  – **Upper Bound**
    Specifies the numeric value of the upper bound for matches when this service option group is referenced.
  – **Infinite Upper Bound**
    Specifies that the Catalog system uses this service option element for matches when the lookup value exceeds the value of the lower bound.

Fields for Form Elements

- **Display Text**
  Specifies the descriptive text to display for this form to administrators browsing the list of forms.
- **Form Name**
  Specifies the name of the Form Designer form that you want to define as a service option element for this service. Use the Search icon to view the list of forms and select one.
- **Hidden and Disabled fields**
  If you do not want to hide or disable the entire form under any conditions, leave the Hidden and Disabled fields empty.
  If you do want to hide or disable the entire form, enter the corresponding JavaScript expression in the Hidden or Disabled field. You can hide or disable the form according to the request status, the user role or business unit, or other criteria. Use the following format: $(_.object.property). The expression must return a value of true or false.
  You can specify JavaScript expressions in the Hidden field, the Disabled field, or both. Examples follow:
  – To hide or disable the form when the request status is Pending Approval, enter the following JavaScript expression in the Hidden or Disabled field: $(_.request.status == 400).
  – To hide or disable the form for the end user roles only, enter $(_.user.role == ‘enduser’).
  – To hide or disable the form from all business units except ca.com, enter $(_.bu.id != ‘ca.com’).
  – To disable the form when the request status is Fulfilled, enter $(_.request.status == 2000).
  When an entire form is disabled, it is disabled but visible during all stages of the request lifecycle. The exception is checkout, when the form is both disabled and hidden.
- Display Form Name

  NOTE

  This capability is available only if you have installed the CA Service Catalog 14.1.01 patch updates available on CA Support.

Check or uncheck to show or hide the Form name. The **Display Form Name** option is also dependent on the System Configuration settings. If the administrator has disabled the form name display option at the System Configuration level, the form name is not displayed, even if the the **Form Name Display** option is checked.

Fields for Rate Elements

- **Cost Type**
  Specifies the type of cost for the Service Option Element. Select from the following list:
  - **Specify Value**
    The Administrator specifies the cost value that appears in the catalog and the requesting or subscribing user cannot change the value.
    This setting displays the following field:
    - Unit Cost: The cost value that is to appear in the catalog.
  - **User Specified**
    The Administrator specifies the default cost value that appears in the catalog. The requesting or subscribing user can change the value.
    This setting displays the following fields:
    - Default Unit Cost: The default cost value that is to appear in the catalog.
  - **Allocate Cost**
    The cost is determined by using values that are associated to a Set as established in Accounting, Budgeting, and Planning worksheets. This setting applies only if you have Service Accounting Component installed.
    This setting displays the following fields:
    - Default Unit Cost: This value in this field is set to 0, as the cost is determined from the worksheet value for the associated set with the selected Allocation Method.
    - Set: List of Accounting Budgeting and Planning sets available for this cost type.
    - Allocation Method: The list of possible methods of allocating the cost that is tied to the service option element.
    - Assign: Use the value in the set for the total cost of this service option element for every subscription or request.
    - Distribute by Subscribed Account: Use the value in the set for this service option element that is divided by the number of accounts that are subscribed to this service option element.
    - Distribute by Subscription: Use the value in the set for this service option element that is divided by the number of subscriptions to this Service Option Element.
    - Weighted Distribution: Use the value in the set for this service option element that is allocated according to actual usage by the account.
  - **Standard Cost**
    The unit cost is determined by pulling values that are associated to a Set value as established in Budgeting and Planning worksheets. This setting applies only if Service Accounting Component is installed.
    This setting displays the following fields:
    - Default Unit Cost: This value in this field is set to 0, as the cost is determined from the worksheet value for the associated set with the selected Allocation Method.
    - Set: List of Accounting Budgeting and Planning sets available for this cost type.
    - Allocation Method: The list of possible methods of allocating the unit cost that is tied to the service option element.
    - Assign: Use the value in the set for the unit cost of this service option element.
- **Display Unit Type**
  Specifies a text value that appears with the cost value.
- **Charge Type**
  Indicates whether cost value must appear as a Charge or a Credit on an Accounting invoice.
- **Budget**
  Indicates whether the service option element appears in the Budget and Planning worksheet.

If Service Accounting Component is not installed, this field serves as additional categorization for the service option element.

- **Billing Cycle**
  Indicates how the cost value is applied to an invoice if Service Accounting Component is installed. Select from the following list:
  - One-Time - The charge is applied one time.
  - Installments - Cost is applied on an installment plan.
    This setting displays the following fields:
    - Periodic Type: The type of interval to be used when applying the cost: Daily, Weekly, Monthly, or N/A.
    - Periodic Type Interval: The frequency of the interval in Periodic Type field for determining the billing interval of the cost.
    - Number of Installments: The number of times the cost must be applied before no longer applying the cost.
  - **Periodic**
    This setting displays the following fields:
    - Periodic Type: The type of interval to be used when applying the cost: Daily, Weekly, Monthly, or N/A.
    - Periodic Type Interval: The frequency of the interval in Periodic Type field for determining the billing interval of the cost.
- **Quantity Specification**
  Indicates how a quantity value must be applied. Select from the following list:
  - **Flat Rate**
    The cost is applied as a flat rate without alteration.
    This setting displays the following fields:
    - Default Quantity Value: The multiplier for the cost value.
  - **Specify Quantity**
    The cost is multiplied by the value in the Quantity field.
    This setting displays the following fields:
    - Quantity: The multiplier for the cost value.
    - Show Quantity: Indicates whether the Quantity value must be shown in the catalog.
  - **Lookup Admin Specified Quantity**
    The cost is applied according to an associated service option group.
    This setting displays the following fields:
    - Service Option Group: Displays the associated service option group from the list of tiered Service Option Groups in the list or None.
    - Service Option Element: List of service option elements in the tiered service option group which must be associated with this service option element.
    - Show Quantity: Indicates whether the Quantity value must be shown in the catalog.
  - **Lookup User Specified Quantity**
    The cost is applied according to a service option element in a service.
    This setting displays the following fields:
    - Service: List of services or None.
    - Service Option Group: List of service option groups for the selected service in the list or None.
    - Service Option Element: List of service option elements in the selected service option group which must be associated with this service option element.
    - Show Quantity: Indicates whether the Quantity value must be shown in the catalog.
  - **User Specified**
    The cost is multiplied by the value in the quantity field.
    This setting displays the following fields:

    - Default Quantity Value: The multiplier for the cost value. The user can set this value.
    - Show Quantity: Indicates whether the Quantity value must be shown in the catalog.
  - **System Specified**
    The Catalog system applies the cost based on a usage quantity.
  - **Form Specified**
    The Catalog system applies the cost based on a form field.
    If you select this option, complete the additional related fields that appear for the related form.
    The Charge Effective From Form Field provides the date field on the form that specifies when billing starts.

Fields for Adjustment Elements

Note that some fields apply only if Service Accounting Component is installed. Adjustments appear on Accounting invoices.

- **Adjustment Value**
  Displays the numeric value of the adjustment to the Accounting invoice.
- **Charge Type**
  Indicates whether cost value must appear as a Charge or a Credit on an Accounting invoice.
- **Adjustment Type**
  Indicates how the Adjustment Value is applied to an invoice if Service Accounting Component is installed. Select from the following list:
  - Applied Amount - The actual amount of the associated service option element is applied.
  - Multiplier - A multiplier of the associated service option element is applied.
- **Service**
  Specifies the service to which the adjustment applies.
- **Service Option Group**
  Specifies the service option group (of the selected service) to which the adjustment applies.
- **Service Option Element**
  Specifies the service option element (of the selected service option group) to which the adjustment applies.

Usage Based Costs Element Fields

Note that some fields apply only if Service Accounting Component is installed.

- **Pricing Structure**
  Specifies the method of charging for a selected application. Select from the following list:
  - **Subscription Based**
    The cost is based on a pre-defined fixed rate. This setting structure displays the following fields:
    - Cost Type: The cost type that must be used to apply the cost of this service option element. This field has the same effect as the Cost Type field on the rate element: Depending on the Cost Type selected, additional fields appear.
    - Display Unit Type: Text value that appears with the cost value
    - Charge Type: Indicates whether cost value must appear as a Charge or a Credit on an Accounting invoice.
  - **Tier Based**
    The cost is derived from an associated tiered service option group according to a lookup value. This setting displays the following fields:
    - Service Option Group: The list of tiered service option groups, allowing the Administrator to select an associated service option group.
    - Tier Type: The list of available tier types for the selected service option group.

    The options that appear in tier type drop-down have the same meaning as the options for tiered service option groups.
  - **Usage Based**
    The cost is based on the usage information from data mediation.

NOTE
For Service Accounting Component to assign costs correctly, the fiscal period of the Budgeting and Planning Set that is used and the account Billing Cycle must be aligned. For example, if monthly fiscal periods are defined in the set, the billing cycles of associated accounts must also be set to monthly. The Period Start and Period End dates for the Accounting Profile must be aligned. The period start date and period end dates for the account period must have an end date one day later than the fiscal period end date.

This setting displays the following fields:

- Cost Type: The cost type that must be used to apply the cost of this service option element. This field has the same effect as the Cost Type field on the rate element: Depending on the Cost Type selected, additional fields appear.
- Display Unit Type: Text value that appears with the cost value
- Charge Type: Indicates whether cost value must appear as a Charge or a Credit on an Accounting invoice.

- Show Metric Result
  Shows the metric results on an Accounting invoice and provides a link to a report representing this data.
- Application
  The list of available applications appears in the list.
- Metric
  The list of metrics available for the selected Application is shown.

Fields for Numeric Elements

- **Numeric Specification**
  Specifies whether the subscriber or requester can change the numeric value of the service option element. Select from the following list:
  – Specify Value - The Administrator sets the value of the service option element. This setting displays the following field:
    Numeric Value: The numeric value of the service option element.
  – User Specified - The Administrator sets the default value of the service option element which the subscribing or requesting user can change. This setting displays the following field:
    Default Value: The default numeric value of the service option element.
- **Show Numeric Value**
  Determines whether the value of the service option element is shown to the user.
- **Associate Service Option Group**
  Indicates that a service option group is associated with this service option element.
  Select this field to display a list of tiered service option groups. Associate a service option group and tier type.

Fields for Numeric Range Elements

- **Lower Bound**
  Specifies the numeric value of the lower bound for matches when this service option group is referenced.
- **Upper Bound**
  Specifies the numeric value of the upper bound for matches when this service option group is referenced.
- **Infinite Upper Bound**
  Specifies that the Catalog system uses this service option element for matches when the lookup value exceeds the value of the lower bound.
- **Specify Median**
  Exposes the following field:
  – **Median Value**
    Specifies that when determining the tier to be used, the value to compare is the lookup value from the usage data. The value of the multiplier for any rate elements for the tier is the absolute value of the difference between the lookup value and the median value.

Localize a Single Service

Service Managers can specify localization attributes to make a single service available in multiple languages. The service appears in the catalog with localized names, descriptions, and forms. This feature lets a multilingual organization maintain one service in a single catalog for *all* supported languages. Multiple versions of a service or multiple catalogs are *not* required.

For efficiency, localize your *custom* services *only.* CA Service Catalog automatically localizes the predefined services when you install a language pack for a localized language. For example, after you install the French language pack, the predefined services appear in the catalog in French if you set your browser language to French.

Follow these steps:

Step 1 - Review the Best Practices and Limitations

The following best practices apply when you localize a single service:

- Before you localize services and the following related objects, verify that they are completed in the original language:
  - Service option groups
  - Service options
  - Service option elements
  - Forms (if applicable)
  - Folders
- If you update these objects in the original language after you localize them, update the localization attributes accordingly. Otherwise, your catalog could contain localized services and related objects that do not completely match the original-language versions.
- You can copy or inherit a service that includes localization attributes. But, the copied or inherited service retains the localization attributes of the original service.
- To make a service available in the catalog, define the service as you define a service without localization attributes.

The following limitations apply when you localize a single service:

- Once all the fields are localized the localized values are seen in end user view only, in Request page. In Service Designer page the definition remains in original language itself.
- Only the fields that you localize appear in the local language in the catalog.
- You can set *one* currency unit (for example, dollars) that applies to *all* local languages.
- You can localize attributes *only* for the languages in which CA Service Catalog is localized for this release.

> **NOTE**
>
> You can optionally replace the default icon at *USM_HOME*\view\webapps\usm\images\localization_16.png with another icon.

Step 2 - Set the Localization Attributes for the Folder

After you set the Folder attributes, they appear in the local language when users view the folder in the catalog.

Follow these steps:

1. Click Catalog, Service Offerings. Expand the folders and open the folder that you want to localize.
2. Click the Localization icon for the Name field and specify the text.
3. Save your changes and close the Localization Values dialog.
4. Click the Localization icon for the Description field and specify the text
5. Save your changes to the folder.

You have set the localization attributes for the folder.

Step 3 - Set the Localization Attributes for the Service

After you set the Service attributes, they appear in the local language when users view the service in the catalog.

Follow these steps:

1. Click Catalog, Service Offerings. Expand the folders and open the service that you want to localize.
2. Click the Localization icon for the Name field and specify the text.
3. Save your changes and close the Localization Values dialog.
4. Click the Localization icon for the Description field and specify the text.
5. Save your changes and close the Localization Values dialog.
6. (Optional) Click the Definition tab to see the Overview text. Click the Localization icon, specify localized overview text in the Localization Values dialog, and save your changes.

   TIP

   If deleting the overview text is required, we recommend deleting in the **Source** tab and then click Save.
7. Save your changes on the Overview dialog.

You have set the localization attributes for the service.

Step 4 - Set the Localization Attributes for the Service Option Group

To set the Service Option Group attributes, click the Localization icon for the Name field and Description fields of the Service Option Group you want to localize. Provide the information and save your changes.

Next, you set the localization elements for its service options and its service option elements, including forms.

Step 5 - Set the Localization Attributes for the Service Option

To set the Service Option attributes, click the Localization icon for the Name field and Description fields of the Service Option you want to localize. The Description field includes several options for rich text, including pictures, links, colors, formatting, and highlighting. You can optionally use these options for any language, including all languages on the Localization Values dialog.

You have set the localization attributes for the service options.

Step 6 - Set the Localization Attributes for the Form

You use the Form Designer to create and localize custom forms: You can localize the element names and selected attributes in forms, using the Localization Editor.

While editing the form in the Form Designer, you see localized names and values only when you are using the Localization Editor. However, when a localized form is used in a request, it displays all localized values. If no localized values exist, the form displays the default values for the selected attributes.

Localized system forms are applied *automatically* when you install the CA Service Catalog language pack for your language.

Follow these steps:

1. Select the form name in the Component Tree and click the Localize button.
2. Select a language from the Language drop-down list. Specify local values for the following attributes, for each element in the Localization Editor. Some attributes do not apply to all elements. When an attribute does not apply to an element, the attribute is omitted. *Only* the following attributes can be localized.
   - **Empty Text**
     Provides more description of a field to the user.

The value that you specify for this attribute appears inside the field when the user opens the form. This text disappears when the user begins entering a value in the field. A sample value follows, for a password field: AAbb1234

- **Hint**
  Specifies text to help users complete the field correctly. This text *always* appears below the field, regardless of the position of the mouse.
  For example, for a password field, you can specify the following hint: "Passwords must be six to eight characters and must include both letters and numbers."
- **Label**
  Specifies the name of the element. This text is used as the name of the field when the user displays the form in a request. For example, you can specify Name, Address, City, or State.
- **Pattern Message**
  Applies to text fields and text areas *only*, and only when the HTML attribute named pattern is also used.
  Specifies the error message to appear when the user input violates the *pattern* attribute.
  In the value for the pattern attribute, you specify regular expressions to validate numeric and address data.
  Examples include credit card numbers, social security numbers, email addresses, IP addresses, and telephone numbers.
- **Tool Tip**
  Specifies the Tool Tip text, if applicable. Only certain elements use this attribute.
- **Value**
  Specifies the value of the element; this text is the default value for the field when the user displays the form in a request.
  For example, you can leave the value blank for an element whose name (label) is Name or Address, because each user must specify a custom value.

3. Click Save after you enter a localized value.
4. Continue specifying local values for attributes.

You have set the localization attributes for forms.

Step 7 - Test the Localized Services

To verify that the localized services appear as intended in the catalog**,** set your Web browser to the localized language and access the localized services in the catalog. Ensure that the localized elements and attributes appear correctly in folders and services, including its forms.

Localize Multiple Services

Service Managers can localize multiple service offerings (services), including forms, to be available in the catalog in multiple languages. You use IXUtil commands, a localization properties file, and a localization agency. This process lets a multilingual organization maintain one service in a single catalog for *all* supported languages. Multiple versions of a service or multiple catalogs are *not* required.

For efficiency, localize your *custom* services *only.* CA Service Catalog automatically localizes the predefined services when you install a language pack for a localized language. For example, consider the predefined services that appear in English when you install CA Service Catalog. After you install the French language pack, these predefined services appear in the French catalog if your browser language is set to French.

Follow these steps:

Step 1 - Review the Best Practices and Considerations

The following best practices apply when you localize multiple services:

- Before you localize services and the following related objects, verify that they are completed in the original language:

  - Service option groups
  - Service options
  - Service option elements
  - Forms (if applicable)
  - Folders
- After you export these objects to an XML file, make only critical updates to the objects. Otherwise, your catalog can contain localized services and related objects that do not completely match the original-language versions. When you merge the completed localization properties files, any existing objects with the same IDs are overwritten. Thus, any changes that you make to these objects using the UI are also overwritten.

  WARNING
  If you update these objects, apply the updates again after you merge the completed file.
- You can copy or inherit a service that includes localization attributes. But the copied or inherited service retains the localization attributes of the original service.
- To make a service available in the catalog, define the service as you define for a service without localization attributes.

The following considerations apply when you localize multiple services:

- Fields appear with localized attributes in the catalog only. In all other components of the product, for example, in Catalog Component, fields appear in the original language.
- You can set *one* currency unit (for example, dollars) that applies to *all* local languages.
- You can localize attributes *only* for the languages in which CA Service Catalog is localized for this release.

Step 2 - Export the Services and Forms to an XML File

The export creates both the XML file and a default localization properties file. The default localization properties file includes the attributes of the exported objects, for the default (original) language *only*.

Follow these steps:

1. Open a CA Service Catalog command prompt.
2. Navigate to the *USM_HOME*\scripts folder.
3. (If you have access to the Catalog system and you are working individually)
   Enter the following command to export services and related data to an XML file. The XML file includes all services that you specified and the objects that comprise the services. This command also extracts the default translatable strings from the XML file into a default localization properties file.
   To export services *with* forms, use the following command:
   ```
   ixutil export service  -f filename.xml -include_forms -include_translation object-specific parameters
   ```
   To export services *only*, use the following command:
   ```
   ixutil export service  -f filename.xml -include_translation object-specific parameters
   ```
   ***filename*.xml**
   Specifies an intuitive name of your choice for the exported objects. Enclose the *filename*.xml in quotation marks if it contains one or more spaces.
   -include_translation
   Creates a default localization properties file. This file contains default localization parameters. You send this file to the localization agency, which supplies the language-specific attributes and completes the file. The file name is *xml-filename*_default.properties. For example, if you specify PersonnelServices.xml as the XML file name, then this file name is PersonnelServices_default.properties.
   object-specific parameters
   Specifies the parameters for a specific object. For example, a business unit or folder, as shown in the examples that follow this procedure.

4. (If you do not have access to the Catalog system or if you are working in a team)
   a. The first person or team exports the services to one or more XML files. Use the following command:
      ```
      ixutil export service  -f filename.xml -include_forms object-specific parameters
      ```
   b. The second person or team receives the XML file or files and then extracts the default translatable strings from each XML file into a default localization properties file. Use the following command:
      ```
      ixutil export service  -f filename.xml -extract_translation
      ```

Examples

These sample commands show how to export services, with or without forms, using a single command. All single-command operations must specify the -include_translation option to create the localization properties file. If you are using two teams and two commands, use the following examples to help construct your first export command.

```
ixutil export service  -f AllServicesAndForms.xml -include_forms -include_translation
```

This command creates the AllServicesAndForms.xml file and the AllServicesAndForms_default.properties file.

The following command exports all services in the business unit that is named SaoPaulo25, and does not include forms:

```
ixutil export service  -f SaoPaulo25.xml  -domain "SaoPaulo25" -include_translation
```

This command creates the SaoPaulo25.xml file and the SaoPaulo25_default.properties file.

The following command exports the services and forms in the folder Folder 1 in the business unit named CA:

```
ixutil export service  -f CA-Folder1.xml  -folder "Folder 1"  -domain "CA" -include_forms -include_translation
```

This command creates the CA-Folder1.xml file and the CA-Folder1_default.properties file.

Step 3 - Send the Default Localization Properties File to the Localization Agency

The agency completes the default localization properties file by adding localization attributes for the languages that you require.

Follow these steps:

1. Send the default localization properties file to the localization agency.
2. Instruct the agency to provide one new properties file for each required language. An example for the file naming convention is *filename*_pt_BR.properties for Brazilian Portuguese. Or *filename*_fr_FR.properties for French.

   WARNING
   Instruct the agency to save each file with the UTF-8 encoding. This encoding is required for the file to function properly.

   You have sent the default localization properties file to a localization agency.

Step 4 - Merge the Completed Localization Properties Files

When the localization agency returns the completed localization properties file for each language, you merge these files into the catalog. After the merge, the services are available to catalog users in the localized languages.

Follow these steps:

1. Review each completed localization properties file that the localization agency returned to you. Verify that the localization attributes appear for services and forms (if applicable) in the required languages.
2. Copy the completed localization properties files to the same folder as the XML file that you exported originally.

   WARNING
   This XML file and the properties files *must* reside in the same folder. Otherwise, the merge does not occur.
3. Open a CA Service Catalog command prompt and navigate to the *USM_HOME*\scripts folder.

4. (If you have access to the Catalog system and you are working individually)
   Enter the following command to import the completed localization properties files into an XML file. This command also imports the updated XML file into the Catalog system. Merge the completed localization properties files into the Catalog system, using the IXUtil import command.
   To import services *with* forms, use the following command:
   ```
   ixutil import service  -f filename.xml -include_translation -include_forms object-specific parameters
   ```
   To import services only, use the following command:
   ```
   ixutil import service  -f filename.xml -include_translation object-specific parameters
   ```
   ***filename*.xml**
   Specifies the name of the XML file that you exported originally. Enclose the *filename*.xml in quotation marks if it contains one or more spaces. Specify the complete path name. Enclose the complete path name in quotation marks if it contains one or more spaces.
   -include_translation
   Merges the localization attributes from the completed localization properties files into the exported XML file. Also inserts the objects from this XML file into the Catalog system.
   object-specific parameters
   Specifies the parameters for a specific object. For example, a business unit or form.
5. (If you do not have access to the Catalog system or you are working in a team)
   a. The first person or team imports the completed localization properties files into one or more XML files. Use the following command:
      ```
      ixutil import service  -f filename.xml -merge_translation
      ```
   b. The second person or team imports each updated XML file into the Catalog system after receiving it from the first person or team. This command merges the completed localization attributes from the localization properties files into the corresponding objects in the XML file. Next, the command inserts these objects into the Catalog system. The updated objects from the XML file overwrite any existing objects with the same ID in the system. Use the following command:
      ```
      ixutil import  -f filename.xml -include_forms object-specific parameters
      ```

Examples

These examples show how to import services, with or without forms, using a single command. If you are using two teams and two commands, use these examples to help construct your first import command.

The following command imports all services and forms from the AllServicesAndForms.xml file into the catalog:

```
ixutil import service  -f AllServicesAndForms.xml -include_translation -include_forms
```

The following command imports all services (without forms) from the SaoPaulo25.xml file into the business unit that is named Sao_Paulo25:

```
ixutil import service  -f LocalizationSaoPaulo25.xml -include_translation  -domain "Sao_Paulo25"
```

Step 5 - Test the Localized Services

Verify that the localized services appear as intended in the catalog.

Follow these steps:

1. Set your web browser to the localized language and access the localized services in the catalog.
2. Verify that the localized elements and attributes appear correctly in folders and services, including its forms.
3. If necessary, correct any errors. Also, if necessary, synchronize changes between the original language and the localization attributes.

You have tested the localized services to verify that they appear as intended in the catalog.

Manage Events-Rules-Actions

As a CA Service Catalog administrator, you can use events, rules, and actions for automating the requests, users, and accounts process in the system as following:

- Events
- Rules
- Actions

Events

Events are installed automatically when you install or upgrade CA Service Catalog.

- Events represent changes that occur within CA Service Catalog. Several standard events typically occur in various components. For example, the User Create event occurs when an administrator adds new user, by using either the user interface or the createUser web service method. You can also add custom events.
- Events can have rules that are associated with them. Rules can have a set of filter conditions that define when the rule applies. When the filter conditions are satisfied and the rule is enabled, the rule actions are launched.
- A rule can have one of more actions that are associated with it. When an event occurs and a rule filter is satisfied, the associated rule actions are executed, if the rule is enabled.

Perform one of the following actions when a custom event occurs:

- Use one of the postEvent web service methods for the event type
- Use a URL to post the event

To manage rules, select Administration, Events-Rules-Actions. The page displays the event types and provides access to the rules and actions that are associated with each event type. You can perform the following actions:

- Use Event Parameters
- Add a Custom Event Type
- Manage Rules
- Manage Actions
- Post an Event

Rules

All rules are initially disabled. A rule must be enabled to be used. In addition, for some of the rules, there are two mutually exclusive actions, one for use with CA Service Desk Manager and one for use without CA Service Desk Manager. By default, the actions which interface with CA Service Desk Manager are disabled.

The rule conditions that are shipped with CA Service Catalog may not completely match your business processes. They are generic and must be examined for applicability to your environment. In addition, some rules overlap in functionality. When activating rules, be careful to understand the full implications of your changes.

A rule can have one of more actions associated. When an event occurs and a rule filter is satisfied, the associated rule actions are executed. A rule action causes some task to be executed. A rule action can perform one of several action types such as run a command line command, send an email, start a CA Process Automation process, or run a Java plug-in.

The When Status is Submitted and Approval Process is Driven by Workflow rule handles the approval process. The rest of the rules handle the fulfillment process.

Actions

Actions are invoked when a condition in a rule is satisfied. Actions include running a command line command, starting a Java process, sending an email, and starting a CA Process Automation process. For more information about actions, see the descriptions of the rules.

Add a Custom Event Type

Administrators can create *custom* events by adding a new event type. You typically do so to address a need in your organization that the predefined event types do not meet.

Follow these steps:

1. Select Administration, Events-Rules-Actions.
2. Click Add.
3. Complete the fields on the page, and click OK.
   The following fields require explanation:

- – **Event Source**
  Specifies that the event source is *physical.* A physical event is based on an update to a database table.
  – **Audit Trail Level**
  Specifies the level of detail to log in an event log table.
  – **Transaction Name**
  Specifies the name of the database table.
  – **Transaction Type**
  Specifies the transaction type: Added, Modified, or Deleted.
  This type represents the general behavior of the custom event. The Transaction Type together with the Event Type Name and Event Source make a unique combination.

  NOTE

  The event type name is not required to be unique. However, we recommend that you specify a unique name.

Event Parameters

Events represent changes that occur in components. Each standard event occurs when it is caused by an action. For example, the User Create event occurs when a new user is added using the *Add User* user interface or the *createUser* web service method. You can optionally add custom events.

Each event type may have parameters that are associated with it. When an event occurs, the parameter values reflect the context of the event. For example, when the User Create event occurs, the associated parameter named **$user_id$** contains the User ID value for the new user created. Event parameter values can be used in rule filters and rule actions.

Administrators can use event parameters as conditions to determine whether rules or actions are triggered. Each event type has parameters that reflect the context of the event. **Example:** The **$user_id$** parameter of the User Create event contains the User ID value for the new user.

Most parameters for event types are self-explanatory and intuitive. However, the Event Type parameters in the following sections require explanation:

Event-Based Load (EBL) data for CA Service Catalog Offerings and Requests

To enable CA Service Catalog Offerings and Requests in Service Point, CA Service Catalog sends request and offering data to the Elastic Search Server. For more information, see Perform Initial Load and Refresh Index and Data on the Search Server.

Three new Event Rule Actions namely **Featured Offering Added**, **Featured Offering Deleted**, **Option Item Added or Modified** are added with a new rule for each event. Click **Save** to update the event type information after you have added or modified an event type.

NOTE

CA Service Catalog Offerings are displayed in the Service Point if CA Service Desk Manager, CA Service Catalog, and xFlow Interface are integrated. If CA Service Catalog is installed as a standalone product, CA Service Catalog Requests and Offerings are not displayed in Service Point and the EBL Event Rules must be disabled in such cases. For more information, see Manage CA Service Catalog Featured Offerings.

NOTE

Note: If CA Service Management SSL enabled, ensure that CA Search Server SSL certificate is imported into CA Service Catalog keystore. For more information, see Enable SSL Authentication for CA Service Management Solution.

NOTE

CA Service Catalog offerings are displayed in Service Point and will be considered for Event-Based Load (EBL) if at least one CA Service Catalog Option Group is associated with the offering. Also the Availability Date for an Offering must be set so that it can be picked up by Service Point via Elastic Search server.

NOTE

Two existing Event Rule Actions - **Service Change, Service Option Group Change**, and **Request Change** are also updated with new rules to implement Service Catalog Offerings and Requests in Service Point.

- **Service Change**: EBL Service Change Rule
- **Service Option Group Change**: EBL Featured Offering Rule
- **Request Change**: Sync Elastic search for request change rule

Featured Offering Add EBL Rule

This Event Type occurs when an offering is added as a Featured Offering.

Event Parameter	Meaning	Example
$feature_type$	Featured Offering type	
$featured_in_offering_id$	Represents the Offering ID with associated Featured Offerings	
$featured_offering_id$	Represents Offering ID which is Featured in another Offering	
all	Name=value pairs of all available event parameters including additional data not available as event parameters. Also, includes variables containing old data before saving the data, causing the event to occur.	

Featured Offering Deleted

This Event Type occurs when an offering is modified as a Featured Offering.

Event Parameter	Meaning	Example
$feature_type$	Featured Offering type	

$featured_in_offering_id$	Represents Offering id which has Featured Offerings associated	
$featured_offering_id$	Represents Offering id which is Featured in another Offering	
all	Name=value pairs of all available event parameters including additional data not available as event parameters. Also includes variables containing old data before the save causing the event to occur.	

Option Item Added or Modified

This Event Type occurs when an Option Group or an Option associated with an Offering is added or modified.

Event Parameter	Meaning	Example
$approval_level$	Approval level assigned to the Offering	
$approval_process$	Approval process assigned to the Offering	
$base_id$	Represents Unique internal ID of Base Offering	
$child_id$	Represents Unique internal ID of an Option Included in the offering definition	
id	Unique internal ID for this item	
$parent_id$	Represents Unique internal ID of Offering	
$selection$	Indicates Boolean value whether this Option item is selected in the Offering definition or not. If selected value is 1 else 0	
all	Name=value pairs of all available event parameters including additional data not available as event parameters. Also includes variables containing old data before the save causing the event to occur.	

Request Create Event and Change Event

These event types occur when the request header information is modified.

NOTE
You must have CA Service Catalog installed to use this event type.

Event Parameter	**Meaning**	**Example**
$completion_date$	Date the request is completed.	
$created_date$	Date the request is created.	
$desired_date$	Date required	
$domain$	Business unit name for the requested for user or account.	ABC Corp Sales
$modified_date$	Date the request was last modified.	
$name$	Request name	My new laptop

$priority$	Numeric value for request priority	3
$req_by_account_id$	Internal ID for Requested By user	10018
$req_by_user_id$	User ID for Requested By user	ABCUser
$req_for_account_id$	Internal ID for an account that is associated with Requested For user or account.	10054
$req_for_user_id$	User ID for Requested For user	ABCUser - for a request for a user NIL - for a request for an account
$request_id$	Internal ID for request	10023
$status$	Numeric value of status of the request, for example: 400=Pending Approval800=Approved	400
all	Name=value pairs of all available event parameters including additional data not available as event parameters. Also includes variables containing old data before the save causing the event to occur.	comments='My laptop for travel' comments_eventdatatype='String' comments_old='NIL'

Request Item Form Element Create Event and Change Event

These event types occur when a field in a form that is associated with a request item is changed.

Event Parameter	Meaning	Example
$form_elem_name$	Label for the field as it appears on the displayed form.	Employee Title
$form_elem_value$	Value for field on the form **Note:** This value is the HTML “value” field which is different from the choice list “label” field displayed.	Vice President (for an input type field) 1 (for a select list type field)
$subscription_detail_id$	Unique internal ID for this request item	10012
all	Name=value pairs of all available event parameters including additional data not available as event parameters. Also includes variables containing old data before the save causing the event to occur.	form_elem_label='Employee Title' form_elem_label_eventdatatype='String' form_elem_label_old='NIL' form_elem_name='emp_title' form_elem_name_eventdatatype='String' etc

Request Pending Action Change

The following parameters apply to the Request Pending Action Change event type and any new event types that you create based on it.

Parameter	Value
rpa_action_type	0=Default1=System2 =CA Workflow3=CA Process Automation4=Policy
rpa_id	The request_pending_action_id entry in the usm_request_pending_action table in the MDB
rpa_object_type	1=Service Offering (service)2 =Request Item

$rpa_reassigned_id$	The reassigned value of the request_pending_action_id entry in the usm_request_pending_action table in the MDB
rpa_status	0=deleted1=active2=completed by assigned user3=completed by other user4=transferred5=delegated6=terminated
$rpa_users_or_groups$	The user_id or group_id entry in the usm_request_pending_action table (mdb)

Request/Subscription Item Create Event and Change Event

These event types occur when a request or subscription item is modified. The modification is generally a change to the status of the item.

NOTE
Requests or subscriptions are for service options. Service options are comprised of service option elements. When the status of a service option that is part of a request or subscription changes, this event occurs for each service option element in the service option.

Event Parameter	Meaning	Example
$account_label$	The account name for the Requested For user	ABC Corp:ABCUser(for a request for a user) Sales (for a request for an account)
$account_no$	Internal ID for the account in $account_label$	10002
$approval_level$	Numeric value for the approval level for service	10
$approval_process$	Numeric value for the approval process, as follows: 0=No approval1=System approval process2=Workflow driven approval process	2
$category$	Numeric value for requested item category, for example: 0=Software1=Hardware	0
$category_class$	Numeric value for requested item class within category, for example: For category 0 (Software), class 10 may be “Office”.	10
$category_subclass$	Numeric valued for requested item subclass within class within category, for example: For category 0 (Software), class 10 (Office), subclass 10 may be “Microsoft”.	10
$charge$	Numeric value indicating whether associated amounts are a charge or credit for use by Service Accounting Component where: 0=credit 1=charge	1

$charge_date$	Date the item is first charged for by Service Accounting Component.	
$code$	Code for the service option element	My Element Code
$domain$	Business Unit Name for the Requested For user or account	ABC Corp Sales
$enum_1$ through $enum_5$	Vary depending on service option element type.	
$external_id$	Service option element External ID	My Element External ID
$group_id$	Numeric value indicating the occurrence of the service in a request (in case a request includes more than one copy of a service)	1 NIL (for a subscription)
$form_data_sd$ - for the Request/ Subscription Item Change Event only	If this service option includes a form, then this parameter lists the details of the form in JSON structure. To apply rules to subscription items whose type is form, append the following condition to the event filter: “AND form_data_sd <> 'NIL'”	See Sample Data for $form_data_sd$ (after this table)
$form_data_sd_row$ for the Request/ Subscription Item Change Event only	If multiple forms exist in a service option, this parameter lists the details of all these forms in JSON structure.	See Sample Data for $form_data_sd_row$ (after this table)
id	Unique internal ID for this request item	10012
$installments$	Numeric value indicating the number of installments for associated amounts for use by Service Accounting Component	0
$instance_name$	Optional value that is assigned when subscribing an account to a service.	My instance
$item_id$	Internal ID for requested service option element	9981
$last_charge_date$	Date the item was last charged for by Service Accounting Component.	
$numeric_1$ and $numeric_2$	Vary depending on service option element type.	
$offering_id$	Internal ID for requested service	10002
$offering_name$	Name of the service offering (service)	Standard Laptop Bundle
$rate_item_col$	Numeric value (zero-based) indicating the associated service option element's column position in the service option row	0
$rate_item_name$	Service option element Display Text	Standard Laptop
$rate_item_row$	Numeric value (one-based) indicating the associated service option element's row position in the service option group	1

$rate_item_text_1$	Additional text that is associated with the service option element. For example, for a Text type service option element, this is Text Value and for a Rate type service option element, this is Display Unit Type.	/month
$rate_item_type$	Numeric value for the associated service option element where: 0=Text1=Header2=Numeric Range 3=Rate4=Application5=Agreement6=Numeric7=Boolean 8=Adjustment9=Date10=Date Range11=Day12=Allocation13=Form	3
$rate_plan_id$	Internal ID for associated service option group	10035
$rate_plan_name$	Associated service option group name	Standard Laptop Accessories
$req_by_user_id$	User ID for Requested By user	ABCUser NIL (for a subscription)
$req_for_user_id$	User ID for Requested For user	ABCUser(for a request for a user) NIL (for a request for an account) NIL (for a subscription)
$request_id$	Internal ID for request	10023 (for a request) NIL (for a subscription)
$request_name$	Request Name	My new laptop NIL (for a subscription)
$request_priority$	Numeric value for request Priority	3 NIL (for a subscription)
sd_row	Numeric value (one-based) for the service option containing this service option element relative to all the service options selected with the service.	1
$status$	Numeric value of status of the requested item, for example: 400=Pending Approval800=Approved	400
$subscribed_date$	Date the item was included in a request or subscription.	
$subscription_type$	Numeric value indicating the type of item where: 0=Other1=Application2=Agreement3=Adjustment4=Form	0
$text_1$ - $text_7$	Vary depending on service option element type.	
$tiered_item_id$	Internal ID of the tiered service option group of the associated service option element where: -1=Not associated with a tiered service option group	-1

$tiered_last_date$	Date the item was last charged for by Service Accounting Component. Applicable only if the related service option element is part of a tiered service option group.	
$track_as_asset$	Numeric value indicating whether this request item must be tracked as an asset where: 0=No1=Yes	1
$unsubscribed_date$	Date the item is cancelled or unsubscribed.	NIL
all	Name=value pairs of all available event parameters including additional data not available as event parameters. Also includes variables containing old data before the save causing the event to occur.	account_label='Sales'account_label_eventdatatype='S etc

Sample Data for $form_data_sd$

The following sample data lists the details of one form:

```
{"name" : "dept",
"value" : "001|Human Resources",
"type" : "9"},
{"name" : "empid",
"value" : "spadmin",
"type" : "5"},
{"name" : "empName",
"value" : "Administrator, Service Delivery",
"type" : "5"},
{"name" : "reason",
"value" : "0",
"type" : "12"}
```

NOTE

: When CA Service Catalog data is used by CA Process Automation, then in the dual list and select box fields, a comma (,) is replaced by two tildes (~~). Before using the data, the user must replace two tildes (~~) with a comma (,).

Sample Data for $form_data_sd_row$

The following sample data lists the details of two forms (10246 and 10245) in the same service option. Form 10245 includes the form type subscription ids.

```
{
"10246" : [
{"name" : "dept",
"value" : "001|Human Resources",
"type" : "9"},
{"name" : "empid",
"value" : "spadmin",
"type" : "5"},
{"name" : "empName",
"value" : "Administrator, Service Delivery",
```

```
 "type" : "5"},
{"name" : "reason",
"value" : "0",
"type" : "12"}],
"10245" : [
{"name" : "accesstype",
"value" : "0",
"type" : "12"},
{"name" : "buildingcode",
"value" : "1|Building1",
"type" : "9"},
{"name" : "date",
"value" : "",
"type" : "14"},
{"name" : "date_fdms$$",
"value" : "",
"type" : null},
{"name" : "device_code",
"value" : "device access code value",
"type" : "5"},
{"name" : "empid",
"value" : "spadmin",
"type" : "5"},
{"name" : "empname",
"value" : "Administrator, Service Delivery",
"type" : "5"},
{"name" : "purpose",
"value" : "purpose value",
"type" : "8"}]
}
```


Notes Create and Notes Change

The Notes Create and Notes Change event types are described as follows:

Notes Create

This event type occurs when a note is created for a CA Service Catalog request. Rules associated with this event type are as follows:

- **When a note is added to a CA Service Catalog request**: When note is added to a CA Service Catalog request, email notifications are sent.
- **When notes are added to a CA Service Catalog request**: When a note is added to a CA Service Catalog request, it is also synchronized with the corresponding CA Service Desk Manager ticket.
- **When a CA Service Desk Manager note is synchronized to a CA Service Catalog request**: When CA Service Desk Manager (CA SDM) note is synchronized to CA Service Catalog request, email notifications are sent.

Notes Change

This event type occur when a Note is changed or modified for a request. The following rules are associated with this event type:

- **When note with attachment(s) is edited in a CA Service Catalog request**: When a note with attachment(s) is edited in CA Service Catalog request, E-Mail notifications are sent.
- **When note without attachment(s) is edited in a CA Service Catalog request**: When a note without attachment(s) is edited in CA Service Catalog request, E-Mail notifications are sent.

The following event parameters associated with these event types require explanation:

Event Parameter	Meaning	Example
$source_id$	The request ID of the corresponding Note.	10510
$note_text$	Description of the note.	
$ignore_notification$	Value will be true in the following cases: • on submission of request that has note(s). • When a note event is triggered for a comment with notes and attachments posted in Widgets.	

Attachment Create and Attachment Change

The Attachment Create and Attachment Change event types are described with associated rules:

Attachment Create

This event type occurs when an attachment is created for a request. Rules associated with this event type are as follows:

- **When attachment is added to CA Service Catalog request**: When an attachment is added to a CA Service Catalog request, it is added as a link to the corresponding CA Service Desk Manager ticket.
- **When attachment is added to a CA Service Catalog request:** When an attachment is added to a CA Service Catalog request, email notification is sent.
- **When note and attachment(s) are added to a CA Service Catalog request**: When note (s) and attachment(s) are added to a CA Service Catalog request from Widgets, email notifications are sent.
- **When a CA Service Desk Manager attachment is synchronized to a CA Service Catalog request**: When a CA Service Desk Manager attachment is synchronized to CA Service Catalog request, email notifications are sent.

Attachment Change

This event type occurs when an attachment is changed for a CA Service Catalog request. Following rule is associated with this event type:

- **When attachment associated with a CA Service Catalog request is edited**: When an attachment associated with a CA Service Catalog request is edited, email notifications are sent.

The following event parameters associated with these event types require explanation:

Event Parameter	Meaning	Example
$object_id$	Request ID of the corresponding attachment.	10510
$ignore_notification$	Value will be true in the following cases: • on submission of request that has attachment(s). • for the first *n-1* attachments when a comment with *n* attachments is posted in Widgets.	For example, if a comment in Widgets has 1 note and 5 attachments, so for the first four attachments (n-1=4), the value is true.
$attachment_url$	Specifies the URL(s) of the uploaded attachment(s).	

Manage Actions

Administrators can configure actions in several ways, to trigger the response you want for specific rules.

Example: You can create and configure a rule to trigger an automatic email notification when a fulfiller fulfills a request.

Perform the following steps:

1. Select **Administration**, **Events-Rules-Actions**.
2. Click the event that contains the rule and action of interest.
3. Click the name of the rule and modify the actions associated with the rule.
4. (Optional) Retry failed action, if required.

You create *global* policies and global actions for general use with any service. In contrast, you create *attached* actions and attached policies for use with a specific service option only. You can create an attached action or policy *only* from the Policies & Actions tab of that service option.

> **NOTE**
>
> You can perform significant edits on custom actions *only*. The predefined actions permit only limited editing. Similarly, you can delete custom actions *only*. You *cannot* delete predefined (built-in) actions.

If the predefined actions and existing custom actions do not meet your needs, you can add a new action to a rule. You can *add* new actions (create custom actions) in both custom rules and predefined (built-in) rules. When required, you can edit a custom action to meet a need in your organization more closely. For example, you can edit a custom action to refine the request email that the action sends.

> **NOTE**
>
> After you create, save, and close a custom action, you can edit its parameters. Note that you *cannot* change the **Type** field. To specify a different type for the action, copy, and rename it. If you do so, update the related rule to trigger the renamed action.

Parameters

You can specify the following parameters for a *custom* action when you add or edit it.

- **Type**
  Specifies the type of action that must be used. To disable a rule action, select **DISABLED**. Based on the type you select, more fields are exposed. The **Add** icon can be used to add an event parameter to the field used for each action type.
  - **Command line**
    Specifies a command that must be run.
  - **HTTP Post**
    Specifies the URL that must be posted.
  - **Java**
    Specifies the Java Class and parameters that must be passed.
  - **Email**
    Notifies users (typically administrators) about system activities that are not related to requests. Examples include the addition of new users or a new business unit, or a data mediation load.
  - **Request Email**
    Notifies stakeholders about a request and optionally includes request details.
    The following parameters require explanation:

- **Include Request Details**: Specifies whether to add the details of the request to the text of the message body in the request email. Click "**Yes**" to add these details or "**No**" to omit them.
- **Type**: Applicable *only* when you use the **$pending_action_users_or_groups$** variable. Specifies the Type as Request, Service, or Service Option. It retrieves the list of pending action users or groups applicable to the selected Type. Also assigns the retrieved list to the**$pending_action_users_or_groups$** variable.
- **Request ID**: Specifies the ID (not the name) of the request, which must be a numeric value or variable.
- **Request Item ID**: Applicable only when the selected **Type** is **Service** or **Service Option**. Specifies the ID of the subscription detail object, which must be a non-numeric value or variable. This value is used to retrieve the list of pending action users or groups applicable to the selected Type. The selected type can be Service or Service Option.
- **From Email**: Specifies the email address from which to send request email

  NOTE
  If you do not specify a value for **From Email** or **From Name**, Service Catalog assigns the value that is specified in **Catalog**, **Request Management**.

  - **To** - Specifies one or more primary recipients for the email.
    You can use specific email addresses, user IDs, and requested for user-related variables.
    To send an email to specific users or groups automatically, specify the matching variables, as follows. Click the **List** icon adjacent to a field, and select these variables from the list that appears.
    - For **requested for user**, specify **$req_for_user_id$**
    - For **requested by user**, specify **$req_by_user_id$**
    - For **pending action user**, specify **$pending_action_users_or_groups$**

    To specify more than one value, separate the entries with a semi-colon (;).
- **From Name**: Specifies the name of the email address from which to send request emails.

  NOTE
  If you do not specify a value for **From Email** or **From Name**, the Catalog system assigns the value that is specified in the Catalog, Request Management configuration options of the business unit of the request.

– **CA Process Automation**: Specifies the start request form (**SRF**) for CA Process Automation to invoke. Use the **Find Start Request Form** icon for CA Process Automation and specify the parameter values for the **SRF**.
– **Automic Automation**: Specifies the workflow (**JOBP**) (for Automic Engine) to invoke. There could be more than one promptset associated to a workflow. Specify variables for each promptset required to trigger a workflow. The promptset variable name associated with a workflow must be properly declared under the event rule actions input field while assigning any catalog request data. For Example: UC4RB_SC_REQUEST_ITEM_ID#=id Here, "UC4RB_SC_REQUEST_ITEM_ID#" is the prompset variable name for request ID and "id" is the CA Service Catalog Request ID.
– **CA Automation Suites Reservation Manager**: Specifies a parameter that must be passed to Reservation Manager.
– **Execution Mode**: Specifies whether the action runs asynchronously or synchronously.

  NOTE
  Administrators *cannot* set the order in which actions run.
– **Timeout**: Specifies the number of seconds that an action must run, before a user can cancel it. A value of 0 means no timeout.

Parameters That Accept Multiple Values

For some events, certain parameters can have multiple values. For such parameters, use a delimiter to separate multiple values. For each parameter with have multiple values, specify a separator when that event occurs. If **Multiple Values** check box is selected, the rule action is executed for each value of a parameter indicated in the Parameter field.

NOTE

The only standard event with multiple values parameters is the **Catalog Subscription Change** event. This event provides multiple values for the**$new_offerings$**, **$new_subscriptions$**, **$old_offerings$,** and **$old_subscriptions$** parameters. You can use multiple value parameters with custom event. However, the event posting logic must provide the multiple parameter values.

Retry Failed Actions

When a rule action fails to launch its associated CA Process Automation process and/or Automic Workflow, a system alert is created. Failures may occur due to the following reasons:

- When services responsible for executing workflow is unable to accept request from Service Catalog.
- Connectivity issues between Service Catalog and Process Automation,
- Connectivity between CA Service Catalog and Automic Workflow is not established. Service Delivery administrators can retry failed actions of these types.

Follow these steps:

1. Select Home, Messages, Alerts from the Messages menu.
   The Alert Messages lists all the alert types appears.
2. Click **List All Failed Actions**.
3. Use **Filter Alerts** to display the list of alerts that include the failed actions you want to retry.
4. Click **Change Business Unit**, if the failed action is in another business unit,
5. Select the alerts in the that you want to retry, and click **Retry Failed Actions**.
   For example:
   – (**CA Process Automation**) If you have retried an alert of type **ITPAM_QUEUE_FAILURE**, then it appears as a separate alert message for that request on the Alerts page. You have retried the failed actions.
   – (**Automic**) If you have retried an alert of type **ASO_QUEUE_FAILURE**, then it appears as a separate alert message for that request on the Alerts page. You have retried the failed actions.

Manage Rules

Administrators can configure rules in several ways, to trigger the actions you want under the conditions you want. For example, you can create and configure a rule to trigger an action when a user, service, business unit, or account is created or updated.

You can perform the following actions to manage rules:

- Add a Rule
- Add an Event Filter

Add a Rule

NOTE

A new event rule *Request Submission Failed* is added for failed requests. By default, this rule is enabled. It triggers and sends a mail notification in case of failed requests if a mail server is configured. It is recommended that you configure the mail server option to receive notifications. For more information, see Manage Requests Pending Action.

Administrators can add a new rule for several reasons. **Example:** Create a rule to trigger an action, such as an email notification, when an approver approves or rejects a request pending action.

Follow these steps:

1. Select Administration, Events-Rules-Actions.

2. Click the event to which you want to add the rule and click Add.
3. Complete the fields, following these guidelines:
   – Specify a unique rule name.
   – Click the Add Filter icon and create or edit an event filter.
4. Click OK.
   The rule is saved. The Event Type Details page appears, and it includes the new rule.

Add an Event Filter

Specify the conditions for invoking the rule and triggering its actions by adding an event filter. Otherwise, the rule and actions are invoked whenever the event occurs.

The properties available to a rule condition depend on the event type of the rule. You can specify both *old* and *new* (or *before* and *after*) values for each event property in the condition.

Follow these steps:

1. Select Administration, Tools, Events-Rules-Actions.
2. Click the event that contains the rule for which you want to add/edit a filter.
3. Click the Add icon next to the Event Filter field.

   NOTE

   You can add or update event filters in custom rules *only*, not predefined (built-in) rules. The predefined rules permit limited editing. You can copy a predefined rule and can save it with a new name as a custom rule.
4. Specify the values for the property, operator, and constant of your new condition in the Condition Builder section.
5. Click the Add Condition arrow and add your new condition to the Current Conditions section.
   The Condition appears in the Current Conditions field. The condition is translated into a conditional SQL-style expression that the rule engine can process. You can add as many conditions as required.
6. Click OK.
   The Event Filter appears in the Rule Information section of the Edit Rule page for the selected rule.
7. Click OK.
   The system saves the rule, including the event filter.

You have added the event filter.

Post an Event

Administrators can post an event as an automated way to mimic an action by a user. **Example:** A catalog user submitting a request or a catalog administrator updating the details of a user or business unit. In both cases (posted event or user action), when the event occurs, its rules are evaluated. If the conditions for a rule are met, the rule actions are executed.

Posting an event is useful for performing such tasks using automation and with specific values and actions. **Example:** You can post events to specify custom values for the integration between CA Service Catalog and another CA product. This topic explains how to post a CA Service Catalog event through HTTP URL. You can also use this URL in rule actions whose type is HTTP Post.

WARNING

If you post an event, test it thoroughly before moving it to a production environment. Verify that the event does *not* run cyclically if the action is triggered.

Follow these steps:

1. Gather the data from the event and construct the HTTP URL.
2. Specify the HTTP URL. This topic uses a URL example to illustrate how to post an event.

NOTE

You can also post events using web services. Both techniques produce the same results.

3. Verify that the event is posted.

Gather the Data from the Event to Construct the HTTP URL

Gather the following values from the event definition for the URL.

This example uses the Data Mediation Aggregation event on the Tools, Events-Rules-Actions page. Click the event name. You see the following details of the event. Use these details to help specify the HTTP URL, as illustrated in the following text:

- Event Type: Data Mediation Aggregation
- Event Type Name: Data Mediation Aggregation
- Event Source: Logical
- Audit Trail Level: System Default
- Transaction Name: DATA_MEDIATION_AGGREGATION
- Transaction Type: Modified
- Event Type Parameters: end_date, $start_date$, $status$, $status_date$
- Description: When Data Mediation Aggregation Status is changed

Specify the HTTP URL

To specify the HTTP URL to post an event to the system, use the following syntax:

NOTE
In the following line and in the example that follows, the line breaks are for readability *only*. In the product UI, enter this code as a single continuous line.

```
http://hostname:port/usm/wpf?Node=icguinode.postevent
&username=userid&pass=password&domain=businessunit
&Args=eventsource&Args=nsppath&Args=transactionname&Args=eventtypename
&Args=transactiontype&Args=eventdescription&Args=associatedobjectid
&Args=false&Args=param1|oldvalue1!param#|oldvalue#!
&Args=param1|newvalue1!param#|newvalue#!
```

An example URL follows:

```
http://hostname:port/usm/wpf?Node=icguinode.postevent
&username=spadmin&pass=spadmin&domain=ca.com&Args=LOGICAL
&Args=DATA_MEDIATION_AGGREGATION:MODIFIED
&Args=DATA_MEDIATION_AGGREGATION&Args=MODIFIED&
Args=LOGICAL&Args=Modified&Args=$id$&Args=false
&Args=end_date|abc!start_date|abc!status|123!status_date|abc!
&Args=end_date|abd!start_date|abd!status|124!status_date|abd!
```

NOTE
When you post an event using a URL from Java or some other tools, encode the URL by replacing unsupported characters with codes. **Example:** Replace the ampersand (&) with %26 or replace a single blank space with %20.

Note: The Catalog system uses both old and new values when it evaluates rule filters for event rules.

The following parameters require explanation:

- ***userid* and *password***

Specifies login credentials for authentication.

- ***businessunit***
  Specifies the business unit for the role of the user ID.
- ***eventsource***
  Specifies the Event Source from the event details: LOGICAL, PHYSICAL, CommonDB.
  In this example, the value is LOGICAL.
- ***nsppath***
  Specifies the namespace path, a placeholder value only. The Catalog system does not use the actual value but requires a placeholder value.
  Use the following format:
  <Transaction Name>-<Transaction Type>
  <Transaction Type> - MODIFIED, ADDED, or DELETED.
  <Transaction Name> - As displayed in event details
  In this example, the value is DATA_MEDIATION_AGGREGATION:MODIFIED.
- ***transactionname***
  Specifies the transaction name for the event.
  As shown in the event details, in this example, the value is DATA_MEDIATION_AGGREGATION.
- ***eventtypename***
  Specifies the name of the event type: MODIFIED, ADDED, or DELETED.
  In this example, the value is MODIFIED.
- ***transactiontype***
  Specifies the transaction type for the event.
  This value is the same as the value of the eventsource parameter.
  In this example, the value is LOGICAL.
- ***eventdescription***
  (Optional for web service) Specifies a description for the event.
  Specify the Transaction Type as displayed in event details.
  In this example, the value is Modified.
- ***associatedobjectid***
  Specifies the ID of an object to associate with this event.
  You can optionally specify one of the event parameters for this value. This value is used to associate the alert that was logged in Change Events.
  This example uses a dummy value: id.
- **ispartial**
  Specifies whether this event is partial; This value is always false.
- ***param#|oldvalue#***
  Specifies the parameter name and the *old* value. Delimit the name and value with a vertical bar. Separate each name and value pair with an exclamation point.
  In this example, the value is as follows:

  ```
  end_date|abc!start_date|abc!status|123!status_date|abc!
  ```

  NOTE
  This example uses dummy values for event attributes. Do *not* specify a value for the all attribute, because it reads *all* values.
- ***param#|newvalue#***
  Specifies the parameter name and *new* value. Delimit the name and value with a vertical bar. Separate each name and value pair with an exclamation point.
  In this example, the value is as follows:

  ```
  end_date|abd!start_date|abd!status|124!status_date|abd!
  ```

  The note for the previous parameter also applies to this parameter.

Verify that the Event is Posted

To verify that an event is posted, follow these steps:

1. Disable all the rules for the event that you are posting.
2. Enable only one rule with no filter and one command-line action, as follows:
   ```
   cmd /c echo Posted Event: $all$ >> C:\PostEventCheck.txt
   ```
3. Verify that the PostEventCheck.txt file is created on the C:\ drive of the application server for CA Service Catalog.

NOTE
You can also post an event using one of the postEvent Administration web service methods.

Manage Dashboards

Administrators and other users use the Dashboard Builder to manage dashboards and their content elements (dash items). Administrators can create shared dashboards and manage the dashboard library. Other users can create personal dashboards and can access the shared dashboards to which they have access in the following ways:

Add a Dashboard

You add personal or shared dashboards to provide access to information and to frequently used features of CA Service Catalog.

Follow these steps:

1. Click Home, Dashboards.
2. Click the << icon at the top right part of the page, and click Add Dashboard.
3. Name the dashboard, configure the options, and click Add.
   The following fields require explanation. Click the Help (question mark) icon for further assistance.
   - **Shared Dashboard**
     Creates a shared dashboard.
     Administrators use shared dashboards to publish information to users. If this option does not appear or if you do not select it, then this dashboard is available to you *only*.
     You can create a personal dashboard and can share it later.
     When you select Shared Dashboard, several other fields appear. These fields are mutually exclusive. Select one of the following options:
     - Accessible by Sub Business Units - Shares this dashboard with users in your business unit and its child business units.
     - Accessible by Role - Shares this dashboard with users who have the roles you specify. Only those roles can access the dashboard. If you do not specify your own role, you cannot access the dashboard after you create it.
   - **Default Dashboard**
     Sets the dashboard as the default dashboard.
   - **Full Screen**
     Configures the dashboard to open in full-screen mode when users select it.
   - **Open in New Window**
     Configures the dashboard to open in a new window when users select it.
   - **Disable Session Timeout**
     Disables the session timeout feature. Users are not logged out if they are inactive longer than the session timeout value.
   - **Auto Arrange**
     Arranges the dashboard items automatically.
   - **Make Item Snappable**
     Creates an invisible grid on the Dashboard, which lets you position and resize the Dashboard items.

You set the Grid Height and Grid Width properties of each cell in the grid. Each Dashboard item "snaps" to this grid when it is positioned or resized.
This option applies only if Auto Arrange is *not* selected.

- **Lock Down Items**
  Fixes the location of the dash items so that other users cannot move them.

The new dashboard appears in the dashboard menu and is selected. The rest of the window is blank, because a new dashboard has no dash items.

4. Add dash items as follows:
   a. Verify that the new dashboard is selected. Click the << icon at the top right part of the page, and click Show Library.
   b. Navigate the Library tree and locate the elements that you want to use on the dashboard.
   c. Drag the content elements to the place where you want them on the dashboard.
      The elements become the dash items.
   d. Adjust the size of the dash items as needed.
5. Set the properties of the dash items by clicking the Edit icon on the dash item heading.
6. Click Save Layout.

Configure Content Elements

You configure content elements in dashboards to customize them to meet the needs of your organization.

Follow these steps:

1. Click Administration, Dashboard Builder.
2. Expand the folders and subfolders and select a content element.
   The details of the element appear in the Content Preview and Content Properties panes.
3. Configure the fields on the Content Properties pane, and click Save.
   The following fields require explanation:
   - **ACL Settings**
     Specifies the access control list (ACL) settings.
     Use these settings to specify the level of access for each role to the content element.
   - **Content Type**
     Specifies the type of content element, as explained in the next section.

Type of Content Element

- **Folder**
  Configures the content element as a folder.
- **External Web Content**
  Configures the content element as a web page URL.
- **External XML Source**
  Configures the content element as an external web reference in XML format. If access to the XML content requires authentication, the web publishing framework automates the authentication login to gain access.
  The web publishing framework can use Web Services to transform the information to meet the needs of the user. You can view the XML directly. Alternatively, you can supply a custom XSL instead.
  If you use XSL, you can embed it as part of the published data definition or can obtain it through a URL.
- **Embedded HTML**
  Configures the content element to include the HTML to display.
  You supply the information in HTML format. The information is stored with the metadata in the Dashboard Library. Embedding HTML enables you to integrate with application data accessible through web controls and Java applets, such as Microsoft Outlook.
- **Embedded XML**
  Configures the content element to include the XML to display.

You supply the information in XML format. The information is stored with the metadata in the Dashboard Library. Embedding XML enables you to view it directly. Alternatively, you can optionally supply a custom XSL instead of XML.

- **GUINode**
  Configures the content element to reference a CA Service Catalog page.
  Specify the page as a Graphical User Interface (GUI) node.
- **GUINode XML**
  Configures the content element to reference internal CA Service Catalog data that you obtain through a GUI Node. This setting enables you to define custom views of information without modifying back-end methods and customizing XSL style sheets.
  Alternatively, you can optionally supply a custom XSL instead of XML.
  Access to the published data requires access to the GUI Node.
- **Managed Document**
  Configures the content element as a document maintained in CA Service Catalog.
  Specify the path in the Filename field. This document must be in the same location on all Catalog Component servers if you are using multiple Catalog Component servers.

Administer Dashboards

You administer dashboards to meet the needs of your organization.

Follow these steps:

1. Click Administration, Dashboard Builder.
   The Dashboard Library folders appear and display the dashboards that you have permission to access.
2. Expand the library tree and display the category for which you want to administer dashboards.
3. Install ActiveX components, if you are prompted to do so.
4. Select an option from the Action drop-down list and click Go. The options vary, according to the category that you select in the library tree.
5. Repeat these steps as needed for more dashboards.

Administrators and other users use the Dashboard Builder to manage dashboards and their content elements (dash items). Administrators can create shared dashboards and manage the dashboard library. Other users can create personal dashboards and can access the shared dashboards to which they have access.

You add personal or shared dashboards to provide access to information and to frequently used features of CA Service Catalog.

Follow these steps:

1. Click Home, Dashboards.
2. Click the << icon at the top right part of the page, and click Add Dashboard.
3. Name the dashboard, configure the options, and click Add.
   The following fields require explanation. Click the Help (question mark) icon for further assistance.
   - **Shared Dashboard**
     Creates a shared dashboard.
     Administrators use shared dashboards to publish information to users. If this option does not appear or if you do not select it, then this dashboard is available to you *only*.
     You can create a personal dashboard and can share it later.
     When you select Shared Dashboard, several other fields appear. These fields are mutually exclusive. Select one of the following options:

- Accessible by Sub Business Units - Shares this dashboard with users in your business unit and its child business units.
- Accessible by Role - Shares this dashboard with users who have the roles you specify. Only those roles can access the dashboard. If you do not specify your own role, you cannot access the dashboard after you create it.

– **Default Dashboard**
Sets the dashboard as the default dashboard.
– **Full Screen**
Configures the dashboard to open in full-screen mode when users select it.
– **Open in New Window**
Configures the dashboard to open in a new window when users select it.
– **Disable Session Timeout**
Disables the session timeout feature. Users are not logged out if they are inactive longer than the session timeout value.
– **Auto Arrange**
Arranges the dashboard items automatically.
– **Make Item Snappable**
Creates an invisible grid on the Dashboard, which lets you position and resize the Dashboard items.
You set the Grid Height and Grid Width properties of each cell in the grid. Each Dashboard item "snaps" to this grid when it is positioned or resized.
This option applies only if Auto Arrange is *not* selected.
– **Lock Down Items**
Fixes the location of the dash items so that other users cannot move them.

The new dashboard appears in the dashboard menu and is selected. The rest of the window is blank, because a new dashboard has no dash items.

4. Add dash items as follows:
   a. Verify that the new dashboard is selected. Click the << icon at the top right part of the page, and click Show Library.
   b. Navigate the Library tree and locate the elements that you want to use on the dashboard.
   c. Drag the content elements to the place where you want them on the dashboard.
      The elements become the dash items.
   d. Adjust the size of the dash items as needed.
5. Set the properties of the dash items by clicking the Edit icon on the dash item heading.
6. Click Save Layout.

Administrators and other users use the Dashboard Builder to manage dashboards and their content elements (dash items). Administrators can create shared dashboards and manage the dashboard library.

A Dashboard Library is a hierarchical tree structure of name spaces facilitating the sharing of information. The Dashboard Library does not hold actual data. Instead, it stores information about how the data can be accessed.

You can Create Dashboard Library Namespace and specify whether to create a folder for the business unit in the Dashboard Library. Doing so helps organize the contents of the library.

{"URL":["/usm/wpf?Node=iclaunchpad.my_dashboards","/usm/wpf?Node=icguinode.portalmyoptionsmodal","/usm/wpf?Node=iclaunchpad.portal_management"],"customLabelGetStarted":"Manage Dashboards","description":"IPCE_Manage_Dashboards","customCards":[{"id":"IPCE_Add_A_Dashboard","type":"manage","title":"Add A Dashboard"},{"id":"IPCE_Dashboard_Library","type":"manage","title":"Dashboard Library"}]}

Manage Outage Calendars

You can create and Maintain Outages, outage groups, outage calendars, create business hours, and associate outage calendars and business hours to a service in the following ways:

Before you perform these tasks, verify that you have completed the tasks to manage request SLAs.

Create and Maintain Outages

You create a scheduled outage to specify a single occurrence when a service is not available. Reasons for such outages include holidays, maintenance periods, disaster recovery activities, one-time events such as major hardware or software changes. Both outages and outage groups are independent entities that can be reused in multiple outage calendars. During the outage periods that the outage calendars define, the monitoring of time for a request SLA is stopped.

Follow these steps:

1. Click Catalog, Service Hours.
2. Check the business unit identification message under the main menu. Verify that you want to create this outage for the current business unit.
3. Click Add to create a new outage.
4. Complete all required fields and add or edit an SLA.
5. Click Save.

This outage can be used in an outage group or outage calendar.

Create and Maintain Outage Groups

Outage groups are logical sets of related outage events, such as holidays or annual maintenance days. These outage groups typically recur annually. As an administrator, you combine individual outages into outage groups. Outages and outage groups are the major ingredients of an outage calendar. Both outages and outage groups are independent entities that can be reused in several different outage calendars.

Follow these steps:

1. Click Catalog, Service Hours.
2. Check the business unit identification message under the main menu. Verify that you want to create this outage for the current business unit.
3. Click Add to create a new outage group.
4. Complete all required fields and add or edit an SLA. For existing outage groups, any outages that are already included in the group appear in the Associated Outages box.
5. Click Save.
6. Click Associate Outages to add an outage to the current outage group.

This outage group is ready to be used in an outage calendar.

Create and Maintain Outage Calendars

As an administrator, you use outages and outage groups to create *outage calendars*. For each business unit, you can optionally apply *one* outage calendar to a service. You cannot apply outages or outage groups to a service directly; they must be associated to a calendar. The criteria that you use for determining the outages, outage groups, and calendars for each service in a business unit are based on different groupings. Examples of such groupings are geographical areas, departments, time zones, and other customer-related considerations. Both outages and outage groups are independent entities that can be reused in several different outage calendars.

Follow these steps:

1. Click Catalog, Service Hours.
2. Check the business unit identification message under the main menu. Verify that you want to create this outage for the current business unit.
3. Click Add to create a new outage calendar.

4. Complete all required fields and add or edit an SLA. For an existing outage calendar, outages and outage groups that are already in the calendar appear in the Associated Outages box. For outage groups, the Group Name column is completed; for outages, the Group Name column is empty.
5. Click Save.
6. Click Associate Outages to add one or more outages to the current outage calendar.
7. Click Associate Outage Groups to add one or more outage groups to the current outage calendar.

After selecting the outages and outage groups for your outage calendar, you can associate the calendar to a service.

Create Business Hours

Business hours specify the regularly scheduled days and times of service. The service provider sets the business hours. But for best customer relations, business hours are geared towards the typical business schedule of all requestors.

As an administrator, you can optionally apply *one* outage calendar and *one* business hours specification to a service. The criteria that you use for determining the business hours for each service in a business unit can be based on different grouping.

During the time periods that the business hours define, time is monitored for the request SLA attached to a service. During the outages that the outage calendar of the service specifies, the time in not monitored.

Follow these steps:

1. Click Catalog, Service Hours.
2. In that Menu, click Business Hours.
3. Check the business unit identification message under the main menu. Verify that you want to create this outage for the current business unit.
4. Click Add to create a new Business Hours specification.
5. Complete all required fields and add or edit the Business Hours.
6. Click Save.

After you select and save your business hours specification, you can associate it to a service.

Associate Outage Calendars and Business Hours to a Service

The outage calendar and business hours that are associated to a service work together to specify the time periods during which the service is expected to be available. During these periods, the time is monitored for the Request SLAs attached to the service.

You are not required to associate an outage calendar and business hours object to a service at the same time. However, we recommend that you do so. When you associate an outage calendar and business hours object to a service at the same time, it ensures the expected up time and down time govern the Request SLAs attached to the service. **Example:** Suppose that you attach a Request SLA to a service without specifying either an outage calendar or business hours. The clients then expect that the service is to be available 24 hours per day and 365 days per year.

Follow these steps:

1. Click Catalog, Service Offerings.
2. Expand the tree and access the service you want.
3. Follow these steps to associate an outage calendar to a service.
   a. Click the Not Specified link for the business hours.
   b. Click Search in the Associate Business Hours dialog. The currently associated object (if applicable) is selected.
   c. Select the objects that you want and click Associate.
   The list of available objects disappears, and the object that you selected appears in the Associated Business Hours field.

4. Follow these steps to associate business hours to a service.
   a. Click the Not Specified link for the business hours.
   b. Click Search in the Associate Business Hours dialog. The currently associated object (if applicable) is selected.
   c. Select the objects that you want and click Associate.
      The list of available objects disappears, and the object that you selected appears in the Associated Business Hours field.

(Optional) Specify Default Outage Calendar and Business Hours

Business unit administrators can optionally specify a default outage calendar and default business hours for all services in their business units. Doing so helps verify consistency in the days and hours that services are available throughout the business unit.

If a service is associated to a different outage calendar, you can assign the default outage calendar by clicking the Use Default button of the calendar. If a service is associated to a non-default business hours, you can assign the default business hours by clicking the Use Default button of the business hours.

If the business unit administrator assigns no default outage calendar and if the administrator for a service does not associate a custom outage calendar to the service, then time is tracked for SLAs associated to the service *every day of the year*. Similarly, if the business unit administrator assigns no default business hours and if the administrator for a service does not associate custom business hours to the service, then time is tracked for SLAs associated to the service *every hour of the day*.

Therefore, as a best practice, business unit administrators and administrators of individual services work together. Together, they verify that the default outage calendar and default business hours for the business unit are reasonable and achievable. Otherwise, avoidable SLA warnings and violations are issued for such services.

Follow these steps:

1. Review the outage calendars and business hours that you have created already. Identify the ones that you want to use as defaults. Verify that your selections meet your requirements for all services in the business unit. If no suitable options exist yet, create suitable outage calendars and create suitable business hours.
2. Click Catalog, Service Offerings.
3. Check the business unit identification message under the main menu. Verify that you want to create this outage for the current business unit.
4. Click the root folder, which is also the name of the business unit that you selected in the previous step.
5. Click Not Specified in the Default Business Hours field and select the business hours.
6. Click Not Specified in the Default Outage Calendar field and select the outage calendar.
7. (Optional) Specify the default outage calendar and default business hours for each child business unit.

   NOTE
   You set the default outage calendar and default business hours for each business unit *individually*. Child business units do *not* inherit these default settings from their parent business unit.

Review SLA reports for the affected services to verify that the default settings meet your requirements for the business unit.

Service hours is the time period during which the service is typically available.
The monitoring of time for an SLA warning or violation (SLA time) is started and stopped according to the service hours that both the outage calendars and the business hours specifications that are associated to the service determine. Administrators optionally associate one outage calendar and one business hours specification to each service. Only during service hours, the SLA time is monitored.

You create a scheduled outage to specify a single occurrence when a service is not available. Reasons for such outages include holidays, maintenance periods, disaster recovery activities, one-time events such as major hardware or software

changes. Both outages and outage groups are independent entities that can be reused in multiple outage calendars. During the outage periods that the outage calendars define, the monitoring of time for a request SLA is stopped.

Follow these steps:

1. Click Catalog, Service Hours.
2. Check the business unit identification message under the main menu. Verify that you want to create this outage for the current business unit.
3. Click Add to create a new outage.
4. Complete all required fields and add or edit an SLA.
5. Click Save.

This outage can be used in an outage group or outage calendar.

Outage groups are logical sets of related outage events, such as holidays or annual maintenance days. These outage groups typically recur annually. As an administrator, you combine individual outages into outage groups. Outages and outage groups are the major ingredients of an outage calendar. Both outages and outage groups are independent entities that can be reused in several different outage calendars.

Follow these steps:

1. Click Catalog, Service Hours.
2. Check the business unit identification message under the main menu. Verify that you want to create this outage for the current business unit.
3. Click Add to create a new outage group.
4. Complete all required fields and add or edit an SLA. For existing outage groups, any outages that are already included in the group appear in the Associated Outages box.
5. Click Save.
6. Click Associate Outages to add an outage to the current outage group.

This outage group is ready to be used in an outage calendar.

As an administrator, you use outages and outage groups to create *outage calendars*. For each business unit, you can optionally apply *one* outage calendar to a service. You cannot apply outages or outage groups to a service directly; they must be associated to a calendar. The criteria that you use for determining the outages, outage groups, and calendars for each service in a business unit are based on different groupings. Examples of such groupings are geographical areas, departments, time zones, and other customer-related considerations. Both outages and outage groups are independent entities that can be reused in several different outage calendars.

Follow these steps:

1. Click Catalog, Service Hours.
2. Check the business unit identification message under the main menu. Verify that you want to create this outage for the current business unit.
3. Click Add to create a new outage calendar.
4. Complete all required fields and add or edit an SLA. For an existing outage calendar, outages and outage groups that are already in the calendar appear in the Associated Outages box. For outage groups, the Group Name column is completed; for outages, the Group Name column is empty.
5. Click Save.
6. Click Associate Outages to add one or more outages to the current outage calendar.
7. Click Associate Outage Groups to add one or more outage groups to the current outage calendar.

After selecting the outages and outage groups for your outage calendar, you can associate the calendar to a service.

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As an administrator, you can optionally apply *one* outage calendar and *one* business hours specification to a service. The criteria that you use for determining the business hours for each service in a business unit can be based on different grouping.

During the time periods that the business hours define, time is monitored for the request SLA attached to a service. During the outages that the outage calendar of the service specifies, the time in not monitored.

Follow these steps:

1. Click Catalog, Service Hours.
2. In that Menu, click Business Hours.
3. Check the business unit identification message under the main menu. Verify that you want to create this outage for the current business unit.
4. Click Add to create a new Business Hours specification.
5. Complete all required fields and add or edit the Business Hours.
6. Click Save.

After you select and save your business hours specification, you can associate it to a service.

{"URL":["/usm/wpf?Node=iclaunchpad.cat_calendar_businesshour","/usm/wpf?Node=iclaunchpad.cat_calendar_event","/usm/wpf?Node=iclaunchpad.cat_event_group","/usm/wpf?Node=iclaunchpad.cat_calendar","/usm/wpf?Node=iclaunchpad.cat_business_hour"],"customLabelGetStarted":"Service Hours","description":"IPCE_Service_Hours","customCards":[{"id":"IPCE_SC_Outage","type":"manage","title":"Outages"},{"id":"IPCE_SC_Outage_Groups","type":"manage","title":"Outage Groups"},{"id":"IPCE_SC_Outage_Calendar","type":"manage","title":"Outage Calendars"},{"id":"IPCE_SC_Outage_Businesshours","type":"manage","title":"Business Hours"}]}

Manage News, Change Events, and Alerts

Administrators can manage news, change events, and alerts from the Home, Messages tab.

News is any announcement for the user community. Administrators can perform the following tasks to manage news messages:

- Add a news message for other users or roles in the same business unit.
- Delete news messages from themselves and from other users.

Change Events are automatically generated when either the system or users (especially administrators) perform significant actions. For example, if you add a user, your action appears as a change event message on the Change Events page. Administrators can view and delete change events.

Alerts represent failed change events. Typically, CA Service Catalog components issue alert messages when a system action fails or a user action fails. For example, failed actions and alerts can result from an administrator specifying an incorrect email address for a user. In such cases, actions that use the email address can fail. An attempt by a request manager to perform email-based approval of a request from that user can fail, resulting in the "REQEMAILACTION_FAILED" alert. Administrators can view and delete system alerts and also retry failed actions.

Administrators can manage news, change events, and alerts from the Home, Messages tab.

News is any announcement for the user community. Administrators can perform the following tasks to manage news messages:

- Add a news message for other users or roles in the same business unit.
- Delete news messages from themselves and from other users.

Change Events are automatically generated when either the system or users (especially administrators) perform significant actions. For example, if you add a user, your action appears as a change event message on the Change Events page. Administrators can view and delete change events.

You can specify the conditions for invoking the rule and triggering its actions by adding an event filter. Otherwise, the rule and actions are invoked whenever the event occurs.

The properties available to a rule condition depend on the event type of the rule. You can specify both *old* and *new* (or *before* and *after*) values for each event property in the condition.

Follow these steps:

1. Select Administration, Tools, Events-Rules-Actions.
2. Click the event that contains the rule for which you want to add/edit a filter.
3. Click the Add icon next to the Event Filter field.

   NOTE

   You can add or update event filters in custom rules *only*, not predefined (built-in) rules. The predefined rules permit limited editing. You can copy a predefined rule and can save it with a new name as a custom rule.
4. Specify the values for the property, operator, and constant of your new condition in the Condition Builder section.
5. Click the Add Condition arrow and add your new condition to the Current Conditions section.
   The Condition appears in the Current Conditions field. The condition is translated into a conditional SQL-style expression that the rule engine can process. You can add as many conditions as required.
6. Click OK.
   The Event Filter appears in the Rule Information section of the Edit Rule page for the selected rule.
7. Click OK.
   The system saves the rule, including the event filter.

You have added the event filter.

Alerts represent failed change events. Typically, CA Service Catalog components issue alert messages when a system action fails or a user action fails. For example, failed actions and alerts can result from an administrator specifying an incorrect email address for a user. In such cases, actions that use the email address can fail. An attempt by a request manager to perform email-based approval of a request from that user can fail, resulting in the "REQEMAILACTION_FAILED" alert. Administrators can view and delete system alerts and also retry failed actions.

When a rule action fails to launch its associated CA Process Automation process and/or Automic Workflow, a system alert is created. Failures may occur due to the following reasons:

- When services responsible for executing workflow is unable to accept request from Service Catalog.
- Connectivity issues between Service Catalog and Process Automation,
- Connectivity between CA Service Catalog and Automic Workflow is not established. Service Delivery administrators can retry failed actions of these types.

Follow these steps:

1. Select Home, Messages, Alerts from the Messages menu.
   The Alert Messages lists all the alert types appears.
2. Click **List All Failed Actions**.
3. Use **Filter Alerts** to display the list of alerts that include the failed actions.
4. Click **Change Business Unit**, if the failed action is in another business unit.
   The Failed Actions are listed.

{"URL":["/usm/wpf?Node=iclaunchpad.my_messages","/usm/wpf?Node=iclaunchpad.message_news","/usm/wpf?Node=iclaunchpad.message_changeevents&&Args=50&&Args=1","/usm/wpf?Node=icguinode.createquery&Args=mdb&Args=usm_system_change","/usm/wpf?Node=iclaunchpad.message_alerts&&Args=50&&Args=1"],"customLabelGetStarted":"Messages","description":"IPCE_Messages","
[{"id":"IPCE_News","type":"manage","title":"News"},{"id":"IPCE_Change_Events","type":"manage","title":"Change Events"},
{"id":"IPCE_Filter_Events","type":"manage","title":"Filter Events"},{"id":"IPCE_Alerts","type":"manage","title":"Alerts"},
{"id":"IPCE_List_All_Failed_Actions","type":"manage","title":"List All Failed Actions"}]}

Manage the Scheduler

Administrators can use the Scheduler to schedule tasks that you want to run one or multiple times. Scheduling is especially useful for recurring tasks, so that you do not have to repeat them manually. To manage scheduled tasks, Click Administration, Tools, Scheduler. You can:

- View the list of scheduled tasks.
- Add or edit a scheduled task.
- Delete a scheduled task.
- View Scheduler-related system alerts.

> **NOTE**
> For optimum scheduling capabilities, use CA Process Automation to schedule tasks. For more information, see the CA Process Automation documentation.

Follow these steps:

1. Click Administration, Tools, Scheduler.
2. Click the Add button and create a scheduled task.
3. Complete the fields and click OK.
   The following options for the *Category* field require explanation:

- – **Valid Until**
  Specifies the date when the scheduled task stops running. If this field left blank, the scheduled task continues to run indefinitely.
  – **Action Type**
  Depending on the Action Type, other fields appear to help further define the action.
  - **Execute Command Line**
    Specifies a command that runs on the server.
  - **Execute Scheduler Plugin**
    Specifies an option for system use *only.*If you have scheduled data mediation tasks, you can view this option.
  - **Missed Actions**
    Specifies the action to perform when a scheduled task cannot run, as follows:
    - *Ignore* skips all missed scheduled tasks.
    - *System Alert* posts a system alert when a scheduled task is missed.
    - *Execute All* runs all missed scheduled tasks as soon as possible.

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- View the list of scheduled tasks.
- Add or edit a scheduled task.
- Delete a scheduled task.
- View Scheduler-related system alerts.

You can add and create a scheduled task.

Follow these steps:

1. Click Administration, Tools, Scheduler.
2. Click the Add button and create a scheduled task.
3. Complete the fields and click OK.
   The following options for the *Category* field require explanation:

- – **Valid Until**
  Specifies the date when the scheduled task stops running. If this field left blank, the scheduled task continues to run indefinitely.
  – **Action Type**
  Depending on the Action Type, other fields appear to help further define the action.
    - **Execute Command Line**
      Specifies a command that runs on the server.
    - **Execute Scheduler Plugin**
      Specifies an option for system use *only.*If you have scheduled data mediation tasks, you can view this option.
    - **Missed Actions**
      Specifies the action to perform when a scheduled task cannot run, as follows:
      - *Ignore* skips all missed scheduled tasks.
      - *System Alert* posts a system alert when a scheduled task is missed.
      - *Execute All* runs all missed scheduled tasks as soon as possible.

Click any of the links to view the corresponding documentation.

Administrators and service designers work together to write API plug-ins that dynamically load data into policies and forms, as follows:

- For a policy, this data specifies the assignees (approvers) for a request. You use the plug-in instead of specifying assignees manually using the Action Builder. When a user submits a request, the API plug-in dynamically specifies the assignees.
- For a form, this data specifies the value for any of the following fields: dynamic tables, dual lists, or select boxes and their options. For example, when a user completes a form to reserve a virtual computer, the report data object populates the list of available computers. You can also write other report data objects to populate related fields, for example, options for RAM and disk space.

API plug-ins query the MDB or another data source. The API plug-in returns the number of objects that meet the criteria that you specify. You can either write your own API plug-ins or copy and modify the predefined plug-ins to meet your requirements. In both cases, you meet the prerequisites and compile your plug-ins *before* you can use them.

API plug-ins are deployed as jar files in the plugins directory of the filestore. API plug-ins run in the same Java Virtual Machine instance as CA Service Catalog.

{"URL":["/usm/wpf?Node=iclaunchpad.scheduler","/usm/wpf?Node=icguinode.schedulertaskcreate&lastCategory=","/usm/wpf?Node=iclaunchpad.linksmanagement","/usm/wpf?Node=iclaunchpad.plugins"],"customLabelGetStarted":"Manage the Scheduler","description":"IPCE_Manage_the_Scheduler","customCards":
[{"id":"IPCE_Create_Scheduled_Task","type":"manage","title":"Create Scheduled Task"},
{"id":"IPCE_Links","type":"manage","title":"Links"},{"id":"IPCE_Plug-ins","type":"manage","title":"Plug-ins"}]}

Manage Content Packs

A content pack is a collection of CA Service Catalog objects, such as services, forms, policies, events, report data objects, and CA Process Automation processes. This article contains the following topics:

Who Creates, Exports, or Imports Content Packs

Both customers and CA Technologies can create content packs, as follows:

- CA Technologies typically creates content packs that include new objects or updated versions of existing objects, including sample objects and fixes.
- Customers typically create content packs that include objects that are customized to meet specific organizational requirements.
- Customers can optionally copy content packs from CA Technologies. They can then customize the content packs before they apply them across their implementations.
- You can export and import multiple content packs for a single business unit. Similarly, you can export and import multiple content packs for all business units. If a conflict occurs between the existing content pack and the one you are activating, the new content pack automatically overrides the old one.
- As a producer or designer, create and export content packs to package customized versions of the CA Service Catalog objects.
- As a consumer or adopter, import content packs and use the customized objects without having to perform the same customization processes.

Typically, the customizations in a content pack are focused to configure CA Service Catalog system for the optimal use of a specific feature, service, or environment. Content packs enable you to repeat the customizations efficiently and accurately from one system to another, multiple times.

Service Management Content Pack

Demo Content

The Demo Content (Demo) content pack contains several model services for common requests throughout an enterprise. Some services require CA Process Automation processes and integration with underlying systems to function as intended.

An example service is the **Smartphone Selector service .** This service lets catalog users select a smartphone and accessories by guiding them through several nested levels of choices. This service uses JavaScript programs in forms for this purpose. To copy and customize this service, you must have advanced knowledge of the Form Designer and a working knowledge of JavaScript.

For the complete list of services in the Demo content pack, import it and view the Import Report. Then view the services in the CA Service Catalog Demo Content folder in the catalog. You can import the Demo content pack file from this location: USM_HOME\Filestore\contentpacks.

A content pack is a collection of CA Service Catalog objects, such as services, forms, policies, events, report data objects, and CA Process Automation processes.

This content pack contains a folder named Service Management with a subfolder named Asset Services for the business unit in which you import the content pack.

The Asset Services folder (available in Catalog, Service Offerings, IT Support Services, Service Management) stores the services for *Request a Hardware Asset* and *Request a Hardware Asset (for mobile users)*. The other services are available in Catalog, Service Offerings, IT Support Services, Service Management folder.

{"URL":["/usm/wpf?Node=iclaunchpad.cat_config_content_packs"],"customLabelGetStarted":"Content Packs","description":"IPCE_Content_Packs"}

Implement the Content Packs

Implementing the content packs enables you to do *both* of the following tasks:

- As a producer or designer of content packs, package a library of objects including your updates and customizations in a single location.
- As a consumer or adopter of content packs, import the customized objects programmatically in a single operation, as many times as necessary.

You do not have to repeat individual operations for each object type. Content packs provide an efficient method of packaging and applying such updates and customizations, especially when you move from one implementation to another. For example:

- Test-to-production migrations and other same-release migrations
- Replacement of a decommissioned computer
- Restoration of CA Service Catalog customizations after upgrades

To manage the content packs, perform the following tasks:

Step 1 - Complete the Prerequisites

- If you are using CA Process Automation as your process automation tool, verify that you have:
  - Installed and configured CA Process Automation.
  - Integrated CA Service Catalog with CA Process Automation.
- If you use content packs to import CA Process Automation objects and if *both* CA Process Automation and CA Service Catalog are using secure socket layer (SSL), verify that you have completed all applicable tasks for configuring CA Service Catalog to use SSL. As part of that process, you configure CA Process Automation to communicate with CA Service Catalog using SSL.

Step 2 - Create and Export the Content Pack

You create content packs to record (export) customizations so that you can reuse them in another implementation. Using content packs provides greater efficiency and accuracy than repeating multiple customization processes manually.

Follow these steps:

1. Decide and record the objects that you want to include in the content pack.
   You can include any or all of the following categories:
   - API plug-ins
   - CA Process Automation processes
   - Events, including rules, and actions
   - Form Designer forms
   - Policies
   - Report data objects
   - Services, including service hours and request SLAs

   For each category that you select, decide and record which objects to include, as follows:
   - All objects in your implementation; that is, all objects in all business units (domains).
   - All objects from one or more specific business units only.
   - Only the objects that you specify by object names, for example, in a comma-separated list.
   - Only the objects that you specify by object-specific criteria.
     For example, for services, you can specify the last modified date.

   Events, rules, actions, and report data objects (including API plug-ins) are *not* specific to any business units. That is, they always apply to *all* business units. The exception is *attached* actions. The attached actions apply *only* to the individual service options that you specify explicitly. Attached actions are service option elements of the service option for which they were created.
2. Select **Start**, **Programs**, **CA**, **CA Service Catalog**, **Service Catalog Command Prompt** on the source computer.
3. Run the following command at this command prompt:

   ```
   ant create-contentpack
   ```

   The Catalog system creates the folder structure for the content pack.
   The Catalog system also prompts you to specify the following information:

- Simple identification data for the content pack, such as a name, author, and description.
- The message with which to prompt the user during an import.

We recommend that you name the *folder* to include the name, version, and locale (language) of the content pack. The ant command creates the content pack folder named *USM_HOME*\FileStore\contentpacks*folder-name*. This folder includes the contentpack.properties file. This file stores the identification data for the content pack. This folder also contains several subfolders, including Forms, Policies, Reports, and Services.

4. Answer the prompt about whether to export the objects into this content pack now or later, as follows:
   - If you specify Yes, skip to the next step.
   - If you specify No, run the following command on the source computer:

     ```
     ant export-to-contentpack
     ```
5. Answer the prompts about which objects must be exported, and their attributes.
   The ant command performs the following functions:
   - An XML file for each object is created, using the attributes that you specified.
   - In most cases, copies the XML file to the appropriate subfolder.

   For example, if you exported services, the ant command performs the following functions:
   - A services.xml file is created using the attributes that you specified.
   - Copies the services.xml file to the Services subfolder of the content pack folder

   The ant command copies some (but not all) categories of objects to their subfolders. Therefore, you copy the remaining categories of objects to their folders manually, as explained in the next step.
6. When prompted, copy the objects that you want to include (if any) to the following subfolders of the content pack folder:
   - **Processes**
     Stores CA Process Automation processes.
   - **Images\Offerings**
     Stores images for the services that you have included the Services subfolder.
   - **Images\RatePlans**
     Stores images for the service option groups that you have included in the Services subfolder.
   - **Prescripts**
     Stores custom scripts to run *before* you import the content pack. Examples include scripts, required to unzip files that are needed for the import or scripts to display the critical information.
   - **Postscripts**
     Stores custom scripts to run *after* you import the content pack. Examples include scripts that load data into the Catalog system or that prompt the user for configuration specifications.
   - **Plug-ins**
     Stores custom API plug-ins.

You have created and exported the content pack on the source computer. You are now ready to import it on the target computer.

Step 3 - Import Content Packs

You import content packs so that you can reuse customizations that you (or another administrator) previously exported from another implementation.

> **NOTE**
> As a best practice, use scheduled down time. Verify that no users are active on CA Service Catalog before you import, enable, or disable content packs.

You can import the content packs using one of the following methods:

- Import Content Packs using the Ant Scripts
- Import Content Packs from the GUI

Import Content Packs Using the Ant Scripts

This section describes importing the content packs using the ant scripts.

Follow these steps:

1. Copy the content pack folder (*USM_HOME\FileStore\contentpacks\folder-name* folder) from the source computer to a location on the target computer. Record the location for reference.
2. Select **Start**, **Programs**, **CA**, **CA Service Catalog**, Service Catalog Command Prompt on the target computer.
3. Run the following command at this command prompt:
   ```
   ant import-contentpack
   ```
4. Enter the complete path name of the folder that stores the content pack to import.
5. Perform the following steps:
   a. Confirm that you want to continue the import.
   b. Enter the business unit ID for the content pack. You can specify any business unit, including the root business unit.
   c. Answer the object-specific prompts. When applicable, consider carefully whether to import objects as disabled. For example, you import a new rule action that affects the emails that the Catalogs system sends. Before you enable the new rule action, you likely want to update the configuration of your mail server.

      NOTE
      If you import objects as disabled, manually enable them before you can use them.
6. Restart the Windows service named CA Service Catalog if the content pack includes events, rules, or actions. Restart this service on all Catalog Component computers in your environment.

   WARNING
   If you do not restart the CA Service Catalog service as directed, unpredictable results can occur.

You have imported the content pack using the ant scripts. All imported objects are either read-only or permit only limited editing.

Import Content Packs from the GUI

This section describes how you can import the content packs from the GUI.

Follow these steps:

1. Click **Catalog**, **Configuration** in CA Service Catalog UI.
2. Click **Content Packs** in the left menu.
3. Click **Import** tab.
4. Specify the folder where the content pack is stored. Select the content pack zip file and click Open.
5. Clear the **Continue import with default values** check box if you want to import the Content Pack without the default values.
6. Click **Start Import**.
   If the Content Pack includes any CA Process Automation Objects in it, the following three options appear:
   – Set imported version of CA Process Automation objects as the current version:
     • If the option is selected, the imported version of CA Process Automation objects and the current version is same.
     • If the option is not selected, the imported version of CA Process Automation objects and current version is different.
   – Make imported custom operators or sensors available:
     • If the option is selected the operators or sensors that are imported are in Available status in CA Process Automation.
     • If the option is not selected, the operators or sensors are imported as it is present in the content pack.
   – Enter the name of CA Process Automation configuration and import the process definitions to specific CA Process Automation instance.

7. If the content pack includes events, rules, or actions, the Import rules, actions in a disabled state option appears. If the option is selected, the events, rules, or actions in the content pack are imported in disabled state irrespective of the content pack status.
8. Restart the Windows service that is named CA Service Catalog on *all* Catalog Component computers in your environment.

   WARNING
   If you do not restart the CA Service Catalog Windows service as directed, unpredictable results can occur.

   For example, you import a new rule action that affects the emails that the Catalog system sends. Before you enable the new rule action, update the configuration of your mail server.

   NOTE
   Enable the objects manually if you import the objects as disabled.
9. If the content pack includes policies, the Import policies in a disabled state option appear. If the option is selected, the policies in the content pack are imported in disabled state irrespective of the content pack status.
10. Click Continue Import.
    The content pack import is successful.

    NOTE
    The import of the content packs is blocked with the availability of any .bat or .cmd files in the content packs. Ant scripts are used import the content packs containing .bat or .cmd files.

You have imported the content pack in CA Service Catalog UI. All imported objects are either read-only or permit only limited editing.

When the import of the content pack is complete, the images of the services are copied to the *USM_HOME\Filestore\Images* folder. The Plug-ins are copied to the *USM_HOME\Filestore\Plugins* folder.

Step 4 - Verify the Content Packs

After you import the content packs in CA Service Catalog, you can verify if the content packs were imported successfully in CA Service Catalog.

Follow these steps:

1. Click **Catalog**, **Configuration**, **Content Configuration**.
2. Select the content pack that you imported.
3. Verify the following criteria:
   - The Content Pack Details section lists the details that you specified when you created the content pack, for example, the name, version, and status.
   - The Content section lists the object that you specified, according to the criteria that you specified.
4. Verify that the user interface menus include the imported objects, for example:
   - Select **CA Service Catalog**, **Service Offerings**, **Offerings**, **Services**. Verify that the list of services includes any services that you imported.
   - Select **Administration**, **Events**. Verify that the list of events includes any services that you imported.

Step 5 - Enable or Disable the Content Packs

You can enable and disable either an entire content pack or individual objects (if applicable) in the content pack. You enable objects in a content pack so that the Catalog system can use the objects. Enabling the imported objects and setting permissions on them work together to let users view and use the imported objects. After you enable objects in a content pack you can disable them, for example, if a problem occurs.

NOTE
As a best practice, use scheduled down time, verify that no users are active on CA Service Catalog to import, enable, or disable content packs.

Follow these steps:

1. Log in to the business unit that contains the content pack that you want to enable or disable.
2. Click **Catalog**, **Configuration**, **Content Packs**.
3. Click the content pack that you want and enable.
4. Enable or disable objects in the list, as follows:
   – Activate *all objects* in the content pack by clicking the Enable button for the entire content pack. This button appears on the Content Pack Details bar.

     WARNING
     Use this option with caution. The content pack contains rules, action, or policies that perform redundant or conflicting tasks, causing unpredictable results. Therefore, if you are not certain regarding the purpose or goal of the content pack, then enable each object individually.
   – Conversely, deactivate *all objects* in the content pack by clicking the Disable button for the entire content pack.
   – Enable or disable individual objects within a category by clicking the Enable or Disable button for the object.
   You can enable or disable objects in any or all of the following categories:
   - **Services**
     Enabling a service activates its Date Available setting: The service uses its Date Available setting to determine whether and when it is available to users.
   - **Service Option Groups**
     Enabling a service option group activates its Date Available setting: The service option group uses its Date Available setting to determine whether and when it is available to users.
     Enabling a service or service option group sets its status to System Object--Available (6). Similarly, disabling a service or service option group sets its status to System Object--Unavailable (7).
   - **Policies**
     You enable a policy to set its status to Active, and disable a policy to set its status to Inactive.
     You can make only limited updates to imported polices that you have enabled. To make more updates to such policies, copy and modify them.
   - **Rules**
     You enable or disable rules individually, without affecting the status of any other rules in the same event. Enabling or disabling a rule does not automatically enable or disable the actions in the rule. The actions remain in the original status.
   - **Actions**
     You enable or disable actions individually, without affecting the status of any other actions in the same rule. Similarly, enabling or disabling an action does not affect the status of the rule that contains the action.

   Enabling and disabling does not apply to the following objects: Events, forms, reports, images, and CA Process Automation processes. You specified during the import process whether to make CA Process Automation processes active or inactive.
5. Click **Done**.
6. (Optional) Verify that the object is enabled (active) or disabled (inactive) by viewing its status. For example, select Services, *folder-name*, and open a service that you enabled. Verify that its status is System Object--Available. Also verify that its Date Available meets your requirements.

You have enabled or disabled objects in the content pack. All objects that you enabled are available in the Catalog system.

NOTE

If you copy and customize an object from a content pack, the customized object is *not* affected by enabling or disabling the content pack.

Step 6 - Modify the Imported Objects

The actions that you can perform on the objects that are imported from a content pack vary by object. When applicable, you perform certain actions on certain imported objects so that users can view and use the objects. The following table shows which actions apply to which objects.

Object	Enable or Disable	Set Permissions	Limited Editing
Services	Y	Y	Y
Service Option Groups	Y	Y	Y
Policies	Y	Y	Y
Events	N	Y	N
Rules	Y	Y	N
Actions	Y	Y	N
Forms	N	Y	N
Report Data Objects	N	Y	N
Images	N	N	N

The actions are as follows:

- **Enable or disable**
  You enable objects so that users and the Catalog system can use them. For example, suppose that your content pack includes policies. You enable these policies so that the Catalog system can use them and manages requests. Similarly, your content pack includes services and you enable these services for the users to view and request the services. For any reason, such as a problem occurring, you can disable any objects that you have enabled.
- **Set permissions**
  To set permissions for each Catalog role on an imported object, use the portion of the UI that stores and maintains the object. For example, to set permissions for services, select Catalog, Service Offerings, Services, Permission, and edit the service details.
  Edit limited attributes
  Typically, imported objects are read-only. You can perform only limited editing on certain imported objects, as follows:
  – For services and service options, you can change the date available and date unavailable.
  – For policies, you can add or remove approvers, change the priority, and set the status as active or inactive.
  Otherwise, to customize an imported object, copy, and modify it. For example, to customize an imported report data object, copy and modify it.

Step 7 - Customize the Imported Objects

To customize an imported object, copy and modify it. For example, to add a field to an imported form, copy the form, rename it, and add the field.

You can customize a service, service option group, or a form. The common customizations are as follows:

- Adding, deleting, and modifying the service option groups or the image in the service.
- Setting the dates that the service is available or unavailable.

NOTE

For more information about performing these tasks, see the Manage Forms sections.

WARNING
If you copy and customize an object that you imported in a content pack, copy and customize all *parent* objects that include the original object. This requirement exists because forms, services, and service option groups allow only limited editing.

Follow this process to copy a form, customize the copy, and include the form in a service option group and service that are based on the original service option group and service in the content pack:

1. Copy Form A and modify the copy, creating Form B.
   Service Option Group A (SOG A) contains Form A. You cannot modify SOG A to replace Form A with Form B, because SOG A allows only limited editing. So, you perform the next steps.
2. Copy SOG A to create SOG B.
3. Update SOG B by deleting Form A and adding Form B.
   Service A contains SOG A. You cannot modify Service A to replace SOG A with SOG B, because Service A allows only limited editing. Therefore, you perform the next steps.
4. Copy Service A to create Service B.
5. Update Service B by deleting SOG A and adding SOG B.
6. Copy Folder A to create Folder B.
7. Update Folder B by deleting Service A and adding Service B.

The following table lists the parent objects that you must copy and customize if you copy and customize a child object.

Child	Parent
Form	Service Option Group
Service Option Group	Service
Service	Folder
Folder	Parent Folder

A content pack is a collection of CA Service Catalog objects, such as services, forms, policies, events, report data objects, and CA Process Automation processes.

The content pack helps avoid redundant operations for each object type. Content packs enable you to repeat the customizations efficiently and accurately, especially when you move from one implementation to another. For example:

- Test-to-production migrations and other same-release migrations
- Replacement of a decommissioned computer
- Restoration of CA Service Catalog customizations after upgrades

Customers and CA Technologies can create content packs as follows:

- CA Technologies typically creates content packs that include new objects or updated versions of existing objects, including sample objects and fixes.
- Customers typically create content packs with customized objects to meet specific organizational requirements.
- Customers can optionally copy content packs from CA Technologies. They can customize the content packs before applying them across their implementations.
- You can export and import multiple content packs for a single business unit. Similarly, you can export and import numerous content packs for all business units. If a conflict occurs between the existing content pack and the one you are activating, the new content pack automatically overrides the old one.
- As a producer or designer, package a library of objects in a single location, including your updates and customizations.
- As a consumer or adopter, import the customized objects programmatically in a single operation as often as necessary.

For more information on implementing and using the content pack, see How to implement content packs and How to use content packs.

{"URL":["usm/wpf?Node=iclaunchpad.cat_config_content_packs"],"customLabelGetStarted":"Implement and Use Content Packs","description":"IPCE_sc_contentpack"}

Overview of Content Configuration Form

You can create a content configuration form to specify any custom configuration information and then use your content pack. These forms are typically not required but can be helpful, especially under the following circumstances.

- The imported objects require configuration before you can use them.
- The administrator who imports the content pack did not export it.
- You require custom values for variables in API plug-ins or CA Process Automation processes. Instead of hard-coding specific values, you can retrieve values from fields on content configuration forms.
  Example: The values can change and cause the API plug-ins or CA Process Automation processes to fail, leading to system downtime.
  Example: A server URL that can change during a migration from a low-security to a high-security environment.

Content configuration forms can be helpful when your imported content requires a custom configuration for:

- CA Process Automation processes and plug-ins that require configuration data.
- A plug-in that needs access to an external data source (not the MDB), such as a web service or database.
  An example is an Active Directory query that supplies CA EEM. The configuration form can list a *server name=field-name* field. The Active Directory query references this field in the form, rather than a hard-coded server name.
- Organization-specific requirements, such as user names and passwords that change both at regular intervals and intermittently as needed.

Content configuration forms are specific to the business unit for which you create them. Parent business units have access to the forms of their child business units.

Administrators of each business unit can define their own configuration forms in the same way as they define request forms. Catalog, Configuration, Content Configuration page includes the Change Business Unit button. This button opens a dialog that lets you select another business unit that you are authorized to access. If you change business units, the list of configuration forms on the left pane of the page updates to display the forms of the current business unit.

This article contains the following topics:

Create the Content Configuration Form

To create a content configuration form, follow this process:

> **NOTE**
> For more information about creating forms using the Form Designer, see the section Manage Forms.

1. Decide the purpose of the configuration form. Examples include specifying custom *parameter=value* expressions for API plug-ins or CA Process Automation processes.

   > **NOTE**
   >
   > These custom expressions are *not* the same as the parameters on the Administration Configuration page, the Accounting Configuration page, or the CA Service Catalog Configuration page. Moreover, these expressions are *not* the same as any other parameters on the CA Service Catalog GUI. Custom parameters can *complement* the GUI parameters, but they are not required to do so.
2. Determine the fields that are required on the form. Create the fields that require user input and determine configuration data that the Catalog system saves and uses.
   Examples include the following fields, which supply database parameters:

- Server name or URL
- User name
- Password
- Port number

NOTE
This form applies to your business unit *only*. Queries to the fields on the form from API plug-ins or CA Process Automation processes must specify the business unit.

3. Create the form in the Form Designer. Follow these guidelines:
   - For the Form attributes, specify a value of configuration for the Form Type attribute.
   - Create *unique* configuration forms, especially if you use multiple configuration packs. A unique configuration form has unique value for the _id attribute in the Form attributes. Verify that no other configuration form in your business unit has the same value for that attribute.
4. Open the form on the Catalog, Configuration, Content Configuration page and specify the values that you want in each field. Follow these guidelines:
   - Specify default values that can help run the content pack successfully without the user changing any data. Doing so is helpful if the user creating or exporting the content pack is not familiar with it.
   - Optionally specify _.bu and _.user JSON objects as values. You *cannot* specify any other JSON objects as values.
5. Save the form.

NOTE
If you import configuration forms using content packs, the values for the form fields are available only when you save them on this page. This page is the Catalog, Configuration, Content Configuration page that you opened previously.

Retrieve Values from Fields on Content Configuration Forms

You can have custom values for variables in API plug-ins or CA Process Automation processes.

WARNING

The values on a content configuration form can have a maximum length of 4,000 characters for single-byte languages. **Example:** English.

The values can have a maximum length of 2,000 characters for double-byte languages. **Example:** Chinese or Japanese. For example, a select box with many options can exceed this limit and cause errors.

Follow these steps:

1. (API plug-in) Locate the com.ca.usm.plugins.apis.PluginContext object. In that object, use one of the following methods to retrieve values from fields on a content configuration form:
   - **Object getCatalogConfigValue(String configGroup, String tenantId, String configName);**
     Queries *one* field on the form. This method returns the value of the configuration parameter.
     Specify the values for the following parameters:
     configName specifies the value of the _id attribute of the field on the form.
     configGroup specifies the literal value "ca_cc_" followed by the value of the _id attribute of the form. An example is ca_cc_form1.
     tenantId specifies the business unit ID, for example, ca.com.
   - **Map<String, Object> getCatalogConfigValues(String configGroup, String tenantId);**
     Queries *all* fields on the form. This method returns the map of key-value pairs. The keys are the values of the _id attributes of the form fields.
     For configGroup and tenantId, the same values apply as for the previous method, Object getCatalogConfigValue.
2. (CA Process Automation) Link the process to one of the following methods in the CA Service Catalog web services. You link the process and the method so that you can retrieve values from fields on a content configuration form:

- **public String getConfigurationValue(String sessionId, String configGroup, String tenantId, String configName)**
- **public Map<String, Object> getConfigurationValues(String sessionId, String configGroup, String tenantId)**
  The parameter explanations are the same as for the methods in the previous step, except session ID.
  session ID uniquely identifies the session. This parameter is required and the client uses the value for the remaining web service calls.

  NOTE
  For more information about web services and examples, see the Plug-in Documentation. To access the Plug-in Documentation, log in to CA Service Catalog and click Administration, Tools, Links, Plug-in Documentation.

You have retrieved values from the form fields.

{"URL":["usm/wpf?Node=iclaunchpad.cat_content_configuration"],"customLabelGetStarted":"Manage Content Configuration","description":"concept.dita_f0b857f9035ea1a656ddd777cd62b0a35eb0615d"}

Using the Service Catalog Content Packs

This content pack contains a folder named Service Management with a subfolder named Asset Services for the business unit in which you import the content pack.

The Asset Services folder (available in Catalog, Service Offerings, IT Support Services, Service Management) stores the services for *Request a Hardware Asset* and *Request a Hardware Asset (for mobile users)*. The other services are available in Catalog, Service Offerings, IT Support Services, Service Management folder.

CA Service Catalog Default Content

The content pack provides the following service offerings:

- **My Resources**
  Lets business users view all the devices, software, and applications that are assigned to them. For example, the resources can be computers, mobile phones, or software that is installed on the devices. Business users can access this service and the items that it includes through a mobile device, laptop, desktop, or a server.
  The My Resources service does not require any approval process. This service offering is of type informational service.
- **Problem with My Resource**
  Lets business users report an issue with the resources that are assigned to them and are listed under My Resources. The Problem with My Resource service is used only with the My Resources service. When business users use this service, the application displays the information corresponding to the selected resource.
  The Problem with My Resource service does not require any approval process.
- **Reset Application Password**
  Lets business users reset the password of any application that they have subscribed to. Typically, an application is a software program that is not installed on your computer. For example, a web-based defect tracking tool is an application. You may not have installed the tool on your computer, but you access it using a web browser.
  The Reset Application Password service is a self-service feature. It helps business users reset their passwords without having to raise Service Desk tickets. The service and the items that it includes can be accessed through a mobile device, laptop, desktop, or a server.
  The Reset Application service does not require any approval process. This service offering is only an informational service.
- **Reset User Password**
  Lets business users reset their domain passwords. The service is an example of automating the procedure to reset a user password in Active Directory.

This service offering does not require any approval process.

- **Report an Issue**
  Lets business users report an issue (incident) in CA Service Desk Manager. For example, a CA Service Catalog user can use this service to report a broken laptop in CA Service Desk Manager and arrange for a repair or replacement. Business users can access this service and the items that it includes through a mobile device, laptop, desktop, or a server.
  The Report an Issue service has no approval process, because the service is only for reporting an issue. When the request reaches Pending Fulfillment status, the incident is created in CA Service Desk Manager. After this incident is created, the request reaches Request Opened status. The incident IDs are attached to the request, and notes and attachments are synchronized between CA Service Catalog and CA Service Desk Manager.
- **Request a Hardware Asset**
  Lets catalog users select and request a hardware asset.
- **Request a Hardware Asset (for mobile users)**
  Lets catalog users request a hardware asset from a mobile device.

The following table provides the details of the Option Groups available for the services:

Service	Option Group	Option	Items in the Option Group
My Resources	My Resources	My Resources	My Resources form
Reset Application Password	Reset Application Password	Reset Application Password	Reset Application Password form
Reset User Password	Reset User Password	Reset User Password	Reset User Password form
Report an Issue	Report an Issue	Report an Issue	Report an Issue form
Request a Hardware Asset	Request a Hardware Asset	Request a Hardware Asset	Request a Hardware Asset
Request a Hardware Asset (for mobile users)	Request a Hardware Asset (for mobile users)	Request a Hardware Asset (for mobile users)	Request a Hardware Asset (for mobile users)

NOTE

All the Service Option Groups are available in Catalog, Service Offerings, Option Groups.

Forms

The content pack provides the following forms:

- **My Resources Configuration**
  Lets administrators configure the options that display the resources assigned to a user.
- **SAM Configuration**
  Lets administrators configure the CA Software Asset Manager server details. This enables retrieving the software and displaying it in My Resource for business users.
- **Reset Password Configuration**
  Lets administrators map applications with their knowledge documents. This helps end-users know the procedure to reset their passwords for applications.
- **Reset User Password**
  Lets catalog users configure the fields that are displayed for end-users to reset their domain passwords.
- **Reset Application Password**
  Lets administrators configure the fields that are displayed for end-users. These fields help end-users reset the password for the applications they have access to.
- **Report an Issue service**

This service includes a service option group named Report an Issue, which includes a form named Issue Details. This form lets catalog users provide details for an issue to the service desk and request help. Users enter a description and assign the issue to a CA Service Desk Manager category. This form is the only form that catalog users see.

- **SDM Incident Configuration**
  Lets administrators define settings or attributes that CA Service Catalog creates in CA Service Desk Manager. This form is also used to define settings or attributes that CA Service Desk Manager creates in CA Service Catalog.
- **Issue Details**
  Lets catalog users use this form to provide details of an issue to the service desk and request help. Users enter a description and assign the issue to a CA Service Desk Manager category.
- **Request a Hardware Asset**
  Lets catalog users select and request a hardware asset.
- **Request a Hardware Asset (for mobile users)**
  Lets catalog users request a hardware asset from a mobile device.

The *Request a Hardware Asset* and *Request a Hardware Asset (for mobile users)* are available in Catalog, Forms, CA Catalog Content, Service Management Forms, Asset Service. All the other forms are available in Catalog, Forms, Forms folder, CA Catalog Content, Service Management Forms.

Plug-ins and Default Locations

The plug-ins that are available in the content pack also include the source code. You can copy the code and can customize it to suit your requirements. To customize the code, copy the Java files into any folder of your choice and modify as required. Ensure that the plugin_class attribute in the plugin.properties file includes the modified folder path.

To view the plug-ins, go to Administration, Tools, Plug-ins. The following plug-ins are available in the content pack:

Imported Object	Plug-in
My Resources	USM_HOME\filestore\plugins\ ca.catalog.content.resource-plugin
Reset Application Password	USM_HOME\filestore\plugins\ ca.catalog.fetchallservicedeskapplicationareas.plugin USM_HOME\filestore\plugins\ ca.catalog.fetchkbapplications.select-plugin USM_HOME\filestore\plugins\ ca.catalog.fetchkbdocument.select-plugin USM_HOME\filestore\plugins\ ca.catalog.fetchservicedeskapplicationareas.plugin
Reset User Password	USM_HOME\filestore\plugins\ ca.catalog.updateactivedirectoryuserpassword.select-plugin
Common	USM_HOME\filestore\plugins\ ca.catalog.samples.configurationreader-plugin
Plug-in for fetching incident areas from CA Service Desk Manager	ca.catalog.servicedesk-select-plugin

Events, Rules, and Life Cycle

The following table lists the action that apply to the Request/Subscription Item Change event:

Event	Status	Action
Request/Item Subscription Change	Not Applicable	When a Note or an Attachment is added to CA Service Catalog, this event triggers CA Process Automation process to synchronize with CA Service Desk Manager.

The following table lists the action that applies to the different rules:

Rule	Status	Action
When Category is Service Management Content	Canceled	Cancel Incident
When Category is Service Management Content	Completed	Close Incident
When Category is Service Management Content	Pending Fulfillment	Create Incident

Typically, you do not need to customize these rules and actions.

Verify the General Prerequisites

To verify that you can use the content packs as intended, complete the prerequisites. For more information about the prerequisites for individual offerings and services, see the corresponding section.

Software Prerequisites:

This content pack requires integrations with other applications. Ensure that you install the following components:

- CA Service Desk Manager 12.9 or higher
- CA Asset Portfolio Management 12.9 or higher
- CA Software Asset Manager 12.9 or higher
- Microsoft Active Directory 2008 R2

General Prerequisites:

The following prerequisites apply to all content packs:

- Restart the CA Service Catalog Service.
- Have a working knowledge of content packs, including how to import and verify the import, and how to enable and disable the objects in the content pack. For more information about importing a content pack, watch this video.
- Understand that you can edit only limited attributes of the objects that you import from the content pack. To make further changes, copy the objects and customize the copies. You must have a working knowledge of how to copy and modify any object that you want to modify, and all its parent objects. For more information about how to customize the content, watch this video.
- Follow these steps to update the Problem with My Resource offering ID, Reset User Password offering Id, and Report an Issue offering Id in the Content Configuration page.

  NOTE

  Ensure that you update these IDs before using the My Resources and Reset Password service offerings.

  a. Log into CA Service Catalog as an administrator.
  b. Go to **Catalog**, **Service Offerings**, **IT Support Services**, **Service Management**.
  c. Select the ID from the required service offering.
  d. Use this ID value to update the existing value in the Content Configuration page (**Catalog**, **Configuration**, **Content Configuration**).
  e. Click **Save** after you update the configuration details.
  The offering ID has been updated.

Use My Resources Service

The *My Resources* offering lets business users view all the devices, software, and applications that are assigned to them. For example, the resources can be computers, mobile phones, or software that is installed on the devices.

Prerequisites:

Before going through this topic, ensure that you verify the general prerequisites.

The following prerequisites apply for using the *My Resources* offering:

- CA Service Catalog is integrated with CA Service Desk Manager and CA Asset Portfolio Management.
- CA Software Asset Manager configuration and My Resources configuration details are updated under **Catalog**, **Configuration**, **Content Configuration**.

The following video demonstrates how to use the My Resources offering:

Use Reset Password Service

The Reset Password offering consists of two parts:

- Reset Application Password
- Reset User Password

Prerequisites:
Before going through this topic, ensure that you verify the general prerequisites.
The following prerequisites apply for using the Reset Password offering:

- The Active Directory server is installed and configured.
- The alternate email address is added to Active Directory.
- Ensure that the Active Directory Certificate Services are installed.
- The reset password offering ID is mentioned in the Reset Password Configuration form.

Add the alternate email address to Active Directory

To add an alternate email address to Active Directory, you must first add the custom attribute Alternate Email to the Active Directory User class. To add a custom attribute, you modify the Active Directory schema. Adding custom attribute requires that the modifying user be a member of the Schema Administrators and Enterprise Administrators groups.

Follow these steps to add a custom attribute to the user object:

1. Register the schema snap-in.
2. Create a custom attribute.
3. Add the custom attribute to the user class.
4. Restart the Active Directory Domain Services.

Follow these steps to register the schema snap-in:

1. Press Windows + R on your keyboard.
2. Type RegSvr32 SchmMgmt.dll and click OK.
   When the schema snap-in is registered, a confirmation message appears.

Follow these steps to create a custom attribute:

1. Press Windows + R on your keyboard.
2. Type mmc.exe and click OK.
3. Click File, Add/Remove snap-in, or simply press Ctrl + M.
4. Select the snap-in Active Directory Schema, click Add >, and then click OK.

5. Expand the Active Directory Schema root node, right-click the node Attributes, click Create Attribute.
6. If the Schema Object Creation warning message appears, click Continue.
7. Create the custom attribute Alternateemail. Complete the following fields:
   - Common Name: Enter Alternate Email
   - LDAP Display Name: Enter alternateemail
   - Unique X500 Object ID: Enter the object ID that is applicable for your domain.
   - Syntax: Select Unicode String
8. Click OK.
   The new custom attribute Alternateemail is created and displayed in child node of Attributes.

Follow these steps to add the custom attribute to the user class:

1. Navigate to the node Active Directory Schema, Classes, and select the class User.
2. Right-click the User class and click Properties. Go to the Attributes tab and click Add.
3. Select the schema object alternateemail and click OK.
   This adds alternateemail as an optional attribute for the User class.
4. Click Apply.
   You have successfully added the alternate email attribute to the User class.

Restart the Active Directory Domain Services

After you have created the custom attribute and added it to the User class, you must restart the Active Directory domain service to apply the schema change.

Follow these steps:

1. Press Windows + R on your keyboard.
2. Type services.msc and click OK.
3. Right-click Active Directory Domain Services and click Restart, and then click Yes.
   All the related services are restarted.

Follow these steps to create the Active Directory Certificate and use it:

1. Create an Active Directory Certificate File on the computer where Active Directory is hosted.
2. Register the certificate in CA Service Catalog.

Follow these steps to create the Active Directory Certificate:

1. Log in to a computer where Active Directory is installed.
2. Open a Command Prompt window using the Run as Administrator option.
3. Type MMC and press Enter.
   The Microsoft Management Console opens.
4. Click File, Add/Remove Snap-in.
5. Select Certificate in the Available snap-ins section, click Add to move it to the Selected snap-ins section, and then click OK.
6. Select Computer account and click Next.
7. Select Local computer: (the computer this console is running on), and click Finish.
   The Add or Remove Snap ins page opens. The Certifications (Local Computer) is added to Selected snap-ins.
8. Click OK.
   The Add or Remove Snap-ins wizard closes.
9. Expand Certificates (Local Computer) in the Console and then expand Personal.
10. Right-click Certificates, select All Tasks, and then select Request New Certificate.
    The Certificate Enrollment wizard opens.
11. Read the instructions and click Next.
    The Select Certificate Enrollment Policy page opens.

12. Verify that Active Directory Enrollment Policy is selected, and click Next.
    The Request Certificates page opens.
13. Select Domain Controller, Domain Controller Authentication, and click Enroll.
    The Certificate Installation Results page opens.
14. Verify that the status is Succeeded for both policies, and click Finish.
15. Verify that there are three certificates under Certificates, in the Console.
16. Right-click the certificate with the Intended Purpose of <All>.
17. Select All Tasks and Export.
    The Certificate Export Wizard opens.
18. Click Next.
    Select the option No, do not export the private key, and click Next.
19. Select DER encoded binary X.509 (.CER), and click Next.
20. Browse and create a folder called Certificate (\Certificate).
21. Name the certificate file to identify it (for example, ITSM-AD.cer) and click Next.
    The Completing the Certificate Export page opens.
22. Verify the Certification Export information and click Finish. Click OK.
23. Close the Microsoft Management Console.
24. Click Yes and Save to save the Console.
    A Certification File (for example, ITSM-AD.cer) is created in the Certification folder on the Domain Controller.
25. Copy the Certification File (ITSM-AD.cer) to the computer where CA Service Catalog is installed.

Follow these steps to register the certificate in CA Service Catalog:

1. Log in to a computer where CA Service Catalog is installed.
2. Go to [USM_Home]\view\conf and open the viewService.conf file.
3. Find the Djavax.net.ssl.trustStore parameter under Java Additional Parameters, and get the value of the parameter. Identify the keystore.
   For example, if the key store value is "C:/Program Files (x86)/Java/jdk1.7.0_40/jre/lib/security/cacerts", navigate to the directory C:\Program Files (x86)\Java\jdk1.7.0_40\jre\bin.
4. Run the following command where ITSM-AD.cer is the name of the file from your Active Directory server:
   keytool -import -keystore ..\lib\security\cacerts -alias ADCertificateRegistration -file C:\ITSM-AD.cer
5. Enter the keytool password, which is available in the 'Djavax.net.ssl.trustPass' parameter value in the viewService.conf file.
   The message Trust this certificate? [no]: appears.
6. Enter yes and confirm the key import.

Change the Reset Password Offering ID

If you want to modify the Reset User Password form and include other details, change the Reset Password Offering ID.
Follow these steps:

1. Log in into CA Service Catalog as Service Delivery Administrator or Catalog Administrator.
2. Specify the new Reset Password Offering ID in the Reset Password Configuration form. Click Save.

   NOTE

   The new offering ID is generated when you copy the default Reset Password offering and create a custom form.
3. Log in into EEM using the CA Service Catalog instance.
4. Click Manage Access Policies and click the offering.
5. Click ACL_Guest_Offering.
6. Remove all resources except “_preventWildcardMatch__ACL_Guest_Offering”.
7. Add the resource “offering__<new reset password offering id>”.

NOTE

This offering ID is generated when you copy the default Reset Password offering and create a custom form.

8. (Optional) To find the offering ID, log in to CA Service Catalog and click the new Reset Password Service offering. The Id is displayed in the Details tab.
   For example suppose that the offering is under ca.com (Id = 10001), IT Support Services (Id = 10013), Service Management (Id = 10048). When you click a service offering or a folder, the offering ID is displayed in the Details tab. Per the example, add the following resources to the ACL_Guest_Offering policy:
   - offering__10001
   - offering__10013
   - offering__10048
9. Click Save. Click Configure, Session, Synchronize Cache.
10. Restart CA Service Catalog.

The following video demonstrates how to use the Reset Password offering:

Use Report an Issue Service

The *Report an Issue* service creates an incident in a predefined (default) implementation of CA Service Desk Manager. If you have customized CA Service Desk Manager to match your business needs, copy and customize the service to match your implementation.

Before going through this topic, ensure that you verify the general prerequisites.

Follow these steps:

1. Understand how the Catalog system creates and populates a CA Service Desk Manager incident.
2. Include custom attributes in the CA Service Desk Manager incident.
3. Customize status codes, attribute prefixes, and other details for the CA Service Desk Manager incident.
4. Review the guidelines and limitations for the Report an Issue service and your custom versions of it.
5. Follow the guidelines for using and customizing rules and actions.

The following video demonstrates how to use the Report an Issue service:

The following video demonstrates how to customize the services, forms as per your requirement:

Retry Failed Attempts to Create CA Service Desk Manager Incidents

At times, a catalog user submits a request for the Report an Issue service. But the attempt to create the incident in CA Service Desk Manager fails. In such cases, the Catalog system reports the failed attempts as alert messages (alerts). The system issues one alert per incident. For example, if a service contains two service options that create an incident, the system issues two alerts. A common reason for failed attempts is a temporary network problem that makes CA Service Catalog inaccessible to CA Service Desk Manager. After the problem is resolved, retry the failed attempts.

Follow these steps:

1. Log in to CA Service Catalog as Service Delivery Administrator.
2. Click **Home**, **Messages**, **Alerts**.
3. Click **List All Failed Actions**, and proceed as follows:
   a. If necessary, click Change Business Unit to display the list of alerts that include the failed actions you want to retry.
   b. Retry all Incident Creation Failures of the following type: INCIDENT_CREATION_FAILED.
4. Select five or fewer rows and click **Retry Failed Actions**.

If the retry fails, the Catalog system marks the alert as Retried, and issues a new failure alert that you can retry again later.
If the retry succeeds also, the Catalog system marks the alert as Retried. You can view the request and its corresponding ticket in CA Service Desk Manager.

Synchronize Attachments and Notes between CA Service Catalog and CA Service Desk Manager

You can synchronize notes and attachments between CA Service Catalog and CA Service Desk Manager.

Follow these steps:

1. Log in to CA Service Catalog as Service Delivery Administrator.
2. Click **Administration**, **Event-Rules-Actions**.
   a. Open the **Document Create** event type.
   b. Select the **When attachment is added to Service Catalog request** rule and click **Enable**.
   c. Enable the Launch PAM SRF to sync Attachments action.
   This action synchronizes the attachments between service options and tickets after the ticket is created. That is, if a business user adds an attachment to a service option after submitting the request, this action passes the attachment to the related ticket.
3. Follow these steps to synchronize notes:
   a. Open the **Notes Create** event type.
   b. Select the **When note is added to Service Catalog request** rule and click Enable.
   c. Enable the Launch PAM SRF to sync notes action.
   This action synchronizes notes between service options and tickets after the ticket is created. That is, if business user adds a note to a service option after submitting the request, this action passes the note to the related ticket.

Use Asset Services

To use the *Asset Services* offering, follow these processes.

Before going through this topic, ensure that you verify the general prerequisites.

Request a Hardware Asset Service

Follow this process to create a custom service for requesting a hardware asset from CA Asset Portfolio Management. In this process, you copy the Request a Hardware Asset service and customize the copied objects.

1. Select **Administration**, **Events-Rules-Actions**, and perform these actions:
   a. Open the event named **Request/Subscription Item Change**.
   b. Enable and open the rule named **When Category is Hardware and Status is Pending Fulfillment**.
   c. Enable the action named **Auto assign the selected asset to the user**.
   After all approvals are supplied and the request reaches Pending Fulfillment status, this action automatically assigns the CA Asset Portfolio Management asset to the Requested For user. If this action is not enabled, an administrator must fulfill the request manually, using gold brick icon.

   NOTE

   You perform this step only once per business unit. Unlike the remaining steps, you do not perform this step each time that you create a service.
2. Create a folder and copy the Request a Hardware Asset service from the content pack to the new folder.

   NOTE

   Copy all objects in the service.
3. Rename the folder, service, service option group, service option, and form to intuitive names. An example is Request a Hardware Asset--custom-V1.

4. Meet the following requirements:
   - Assign models to a service option.
   - Verify that this service option is included in a service option group in the service.
   - Understand that the Request a Hardware Asset form uses a hidden field named Selected Assets' Id to store the UUID of the IT Asset Manager asset that the user selects.

     WARNING

     Do not modify or replace this field. If you decide to modify or replace this field for any reason, the new or modified field must use the same value for the _id attribute as the original field: ca_itam_attr_asset_id. Otherwise, the selected asset is not assigned properly.
   - When the request for the selected asset is approved, the Assign Asset action from the content pack (or your custom version of that action) assigns that UUID to the Requested For user specified in the request.
5. Verify the following requirements for using the report data object (data object) from the content pack:
   - The catalog user and the assets must have the same location.
   - The assets must not have been assigned to any user.
   - The assets must be active in status.
6. (Optional) If the CA Service Catalog database runs on SQL Server, you can optionally create and use a custom query that removes one or more of the restrictions in the predefined data object for requesting a hardware asset.
   This topic illustrates the process to remove the restriction that users and assets have the same location. You can use this topic as a model to make similar customization to your queries. Follow this process:
   a. **Create a data object with your custom query:**
      a. Click **Administration**, **Report Builder**, **Data Objects**.
      b. Expand **Catalog**.
      c. Open the data object named **Fetch Hardware Assets to Assign - Table Provider**.
      d. Copy the fields of this report data object to a Notepad file. The fields include Database, Table, Fields, and Query.
      e. Create a data object. Specify an intuitive name, for example, Fetch Hardware Assets--Custom-version1.
      f. Copy and paste the text from the Notepad file to each field of the new data object.
      g. Update the query in the new data object to remove the location requirement.
         The updated query appears as follows:

```
 SELECT cc.resource_name AS ca_itam_attr_asset_name,cc.host_name AS ca_itam_attr_host_name,cs.name
 AS ca_itam_attr_resource_family,
cv.name AS ca_itam_attr_resource_class,lc.name AS ca_itam_attr_lifecycle_status,sc.name AS
 ca_itam_attr_cost_center,bb.location_name AS ca_itam_attr_location_name,cc.resource_name,
CONVERT(CHAR(100),own_resource_uuid,1) AS ca_itam_attr_asset_id_temp
FROM ca_owned_resource cc
JOIN ca_model_def cl ON cc.model_uuid = cl.model_uuid
JOIN usm_link_rateitem_model lrm ON cl.model_uuid=lrm.model_uuid
JOIN usm_rate_definition r ON lrm.rate_item_id=r.item_id
JOIN usm_offering_ratedef_inclusion ori ON r.item_id=ori.child_id
JOIN usm_offering o ON ori.parent_id=o.offering_id
JOIN ca_resource_family cs ON cc.resource_family = cs.id
JOIN ca_resource_class cv ON cc.resource_class = cv.id
LEFT OUTER JOIN ca_asset_lifecycle_status lc ON cc.lifecycle_status = lc.id
LEFT OUTER JOIN ca_resource_cost_center sc ON cc.cost_center = sc.id
LEFT OUTER JOIN ca_location bb ON cc.location_uuid = bb.location_uuid
WHERE cc.resource_contact_uuid IS NULL AND cc.inactive='0' AND cl.name=(SELECT cl.name WHERE
 cl.inactive='0' AND cl.family_id='%family_id%' AND o.offering_id='%offeringid%')
```

      h. Copy the ID of this new report data object to a Notepad file. You use this information in the next procedure

b. **Copy and customize the form to accommodate the new object**
   a. Select **Catalog**, **Forms**.
   b. Expand **Forms**, **CA Catalog Content**, **Service Management Forms**, **Asset Services**, **Request a Hardware Asset** and review the value of the Report/Plug-in Variables attribute in the original form
   c. Select the **Select An Asset** field and review the value of the Report/Plug-in Variables attribute.
   The original value of the attribute follows:
   $({'offeringid':_.service.id,'family_id':'600','location':_.user.location.uuid})
   d. Copy the form. Specify an intuitive name, for example, Request a Hardware Asset--Custom-version1.
   e. Update the following attributes of the Select An Asset field:
   f. Report/Plug-in Id: Replace the existing value with the ID of the new report data object that you created in the previous procedure.
   g. Report/Plug-in Variables: Remove the location requirement.
   The updated value appears as follows:
   $({'offeringid':_.service.id,'family_id':'600'})

7. (Oracle Users Only) Replace the SQL Server query in the content.
   The predefined queries in the data objects in the content pack are for SQL Server only. Therefore, if the CA Service Catalog database runs on Oracle, create a custom query for Oracle that performs the same functions as the predefined query.
   – Create an Oracle query for requesting hardware assets.
   – Create an Oracle query for requesting hardware assets (for mobile users).
8. You can use the predefined or custom approval process of your choice.

Create an Oracle Query for Requesting Hardware Assets

Create an Oracle query for requesting hardware assets, for use in a service that users submit from a laptop or desktop computer (not a mobile device).

Follow this process:

Step 1 - Create a data object with your custom query

1. Select **Administration**, **Report Builder**, **Data Objects**.
2. Create a data object. Specify an intuitive name, for example, Fetch Hardware Assets–RequestedFromLaptop.
3. Configure the new data object to perform the same functions for Oracle as the original data object for SQL Server. Use this query as a model to create your query. This query includes the location requirement. Remove it if you do not want it.

```
 Database: mdb
Fields:
 ca_itam_attr_asset_name,ca_itam_attr_host_name,ca_itam_attr_resource_family,ca_itam_attr_resource_class,ca_itam_att
 ca_itam_attr_location_name,ca_itam_attr_cost_center,resource_name,item_id,ca_itam_attr_model_name,ca_itam_attr_loca
Query: SELECT cc.resource_name AS ca_itam_attr_asset_name,cc.host_name AS ca_itam_attr_host_name,cs.name
 AS ca_itam_attr_resource_family, cv.name AS ca_itam_attr_resource_class,lc.name AS
 ca_itam_attr_lifecycle_status,sc.name AS ca_itam_attr_cost_center,bb.location_name AS
 ca_itam_attr_location_name,cc.resource_name, CAST(own_resource_uuid AS CHAR(100)) AS
 ca_itam_attr_asset_id_temp FROM ca_owned_resource cc
JOIN ca_model_def cl ON cc.model_uuid = cl.model_uuid
JOIN usm_link_rateitem_model lrm ON cl.model_uuid=lrm.model_uuid
JOIN usm_rate_definition r ON  lrm.rate_item_id=r.item_id
JOIN usm_offering_ratedef_inclusion ori ON r.item_id=ori.child_id
JOIN usm_offering o ON ori.parent_id=o.offering_id
```

```
JOIN ca_resource_family cs ON cc.resource_family = cs.id
JOIN ca_resource_class cv ON cc.resource_class = cv.id
LEFT OUTER JOIN ca_asset_lifecycle_status lc ON cc.lifecycle_status = lc.id
LEFT OUTER JOIN ca_resource_cost_center sc ON cc.cost_center = sc.id
LEFT OUTER JOIN ca_location bb ON cc.location_uuid = bb.location_uuid
WHERE cc.resource_contact_uuid IS NULL AND cc.inactive='0' AND CAST(cc.location_uuid AS CHAR(100))='0x'
 + '%location%' AND cl.name=(SELECT cl.name FROM ca_model_def cl WHERE cl.inactive='0' AND
 cl.family_id='%family_id%' AND o.offering_id='%offeringid%')
```

4. Copy the ID of this new report data object to a Notepad file. You use this information in the next procedure.

Step 2 - Copy and customize the form

1. Review the value of the Report/Plug-in Variables attribute in the original form in the content pack, as follows:
   a. Select **Catalog**, **Forms**.
   b. Expand **Forms**.Select **CA Catalog Content**, **Service Management Forms**, **Asset Services**, **Request a Hardware Asset**.
   c. Click the **Select An Asset** field and review the value of the Report/Plug-in Variables attribute.
   The original value of the attribute follows:
   $({'offeringid':_.service.id,'family_id':'600','location':_.user.location.uuid})
2. Copy the form. Specify an intuitive name, for example, Request a Hardware Asset--From a Laptop.
3. Update the Report/Plug-in Id attribute of the Select An Asset field: Replace the existing value with the ID of the new report data object that you created in the previous procedure.

Create an Oracle Query for Requesting Hardware Assets (for Mobile Users)

Create an Oracle query for requesting hardware assets, for use in a service that users submit from a mobile device.

Follow this process:

Step 1 - Create a data object with your custom query

1. Select **Administration**, **Report Builder**, **Data Objects**.
2. Create a data object. Specify an intuitive name, for example, Fetch Hardware Assets–RequestedFromMobile.
3. Configure the new data object to perform the same functions for Oracle as the original data object for SQL Server. Use this query as a model to create your query. Test your query thoroughly before making the service that includes it available to catalog users. This query includes the location requirement. Remove it if you do not want it.

```
Database: mdb
Fields:
 ca_itam_attr_asset_id,ca_itam_attr_asset_name,ca_itam_attr_host_name,ca_itam_attr_resource_family,ca_itam_attr_reso
ca_itam_attr_lifecycle_status,ca_itam_attr_location_name,ca_itam_attr_cost_center,resource_name,item_id,ca_itam_attr
Query: SELECT CAST(own_resource_uuid AS CHAR(100)) AS ca_itam_attr_asset_id,cc.resource_name
 + ', ' + cl.name AS ca_itam_attr_asset_name,cc.host_name AS ca_itam_attr_host_name,cs.name AS
 ca_itam_attr_resource_family,
cv.name AS ca_itam_attr_resource_class,lc.name AS ca_itam_attr_lifecycle_status,sc.name AS
 ca_itam_attr_cost_center,bb.location_name AS ca_itam_attr_location_name,cc.resource_name
FROM ca_owned_resource cc
JOIN ca_model_def cl ON cc.model_uuid = cl.model_uuid
JOIN usm_link_rateitem_model lrm ON cl.model_uuid=lrm.model_uuid
JOIN usm_rate_definition r ON  lrm.rate_item_id=r.item_id
JOIN usm_offering_ratedef_inclusion ori ON r.item_id=ori.child_id
JOIN usm_offering o ON ori.parent_id=o.offering_id
```

```
JOIN ca_resource_family cs ON cc.resource_family = cs.id
JOIN ca_resource_class cv ON cc.resource_class = cv.id
LEFT OUTER JOIN ca_asset_lifecycle_status lc ON cc.lifecycle_status = lc.id
LEFT OUTER JOIN ca_resource_cost_center sc ON cc.cost_center = sc.id
LEFT OUTER JOIN ca_location bb ON cc.location_uuid = bb.location_uuid
WHERE cc.resource_contact_uuid IS NULL AND cc.inactive='0' AND CAST(cc.location_uuid AS CHAR(100))='0x'
 + '%location%' AND cl.name=(SELECT cl.name FROM ca_model_def cl WHERE cl.inactive='0' AND
 cl.family_id='%family_id%' AND o.offering_id='%offeringid%')
```

4. Copy the ID of this new report data object to a Notepad file. You use this information in the next procedure.

Step 2 - Copy and customize the form

1. Review the value of the Report/Plug-in Variables attribute in the original form in the content pack, as follows:
   a. Select **Catalog**, **Forms**.
   b. Expand **Forms**. Select **CA Catalog Content**, **Service Management Forms**, **Asset Services**, **Request a Hardware Asset (for Mobile Users)**.
   c. Click the **Select An Asset** field and review the value of the Report/Plug-in Variables attribute.
   The original value of the attribute follows:
   $({'offeringid':_.service.id,'family_id':'600','location':_.user.location.uuid})
2. Copy the form. Specify an intuitive name, for example, Request a Hardware Asset--From Mobile Devices.
3. Update the Report/Plug-in Id attribute of the Select An Asset field: Replace the existing value with the ID of the new report data object that you created in the previous procedure.

CA Service Catalog Offerings for CA Service Management

CA Service Catalog provides several out-of-the-box service offerings. This content may be included in a Service Catalog instance by optionally selecting to install the Best Practice Foundation Content while running the Setup Utility. These out-of-the-box services are offered by CA based on extensive research on common and in-demand services implemented by various organizations. This content serves as a reference and includes a comprehensive set of simple and complex examples of service offerings that can be used as templates when organizations build and design their own services using Catalog.

- Application Services
- Corporate Services
- Facilities Services
- IT Support Services
- Network Services
- Personnel Services
- Project Services
- Telecom Services

CA Service Catalog provides several out-of-the-box service offerings. This content may be included in a Service Catalog instance by optionally selecting to install the Best Practice Foundation Content while running the Setup Utility. These out-of-the-box services are offered by CA based on extensive research on common and in-demand services implemented by various organizations. This content serves as a reference and includes a comprehensive set of simple and complex examples of service offerings that can be used as templates when organizations build and design their own services using Catalog.

{"URL":["/usm/wpf?Node=iclaunchpad.catalog_builder#id=10001;t=OFFERING_FOLDER;d=ca.com"],"customLabelGetStarted":"Catalog","description

Application Services

Following Application Services are available on the main folder level:

- Application Hosting
- Database Management Services
- Application Environment Setup

Application Access

Following Application Access services are available:

- Grant Access to HR Applications
- Grant Access to Finance Applications
- Grant Access to Sales / Marketing Applications

Application Reports

Following Application Reports are available:

- Subscribe to Finance / Accounting Reports
- Subscribe to Human Resources Reports
- Subscibe to IT Service Performance Reports
- Subscribe to Marketing / Sales Reports
- Subscribe to Procurement / Inventory Reports

The following Application Services are available on the main folder level:

- Application Hosting
- Database Management Services
- Application Environment Setup

The following Application Access services are available:

- Grant Access to HR Applications
- Grant Access to Finance Applications
- Grant Access to Sales / Marketing Applications

The following Application Reports are available:

- Subscribe to Finance / Accounting Reports
- Subscribe to Human Resources Reports
- Subscibe to IT Service Performance Reports
- Subscribe to Marketing / Sales Reports
- Subscribe to Procurement / Inventory Reports

{"URL":["/usm/wpf?Node=icguinode.cataloggetchildren&Args=10201&ObjectID=10201&NspPath=","/usm/wpf?Node=icguinode.cataloggetchildren&Args=10204&ObjectID=10204&NspPath=","/usm/wpf?Node=icguinode.cataloggetchildren&Args=10208&ObjectID=10208&NspPath="],"customLabelGetStarted":"Application Services","description":"IPCE_Application_Services","customCards":[{"id":"IPCE_Application_Access","type":"manage","title":"Application Access"},{"id":"IPCE_Application_Reports","type":"manage","title":"Application Reports"}]}

Corporate Services

Following Corporate Service offerings are available:

- Business Cards
- Corporate Charge Card
- Employee Recognition

The following Corporate Service offerings are available:

- Business Cards
- Corporate Charge Card
- Employee Recognition

{"URL":["/usm/wpf?Node=icguinode.cataloggetchildren&Args=10301&ObjectID=10301&NspPath="],"customLabelGetStarted":"Corporate Services","description":"IPCE_Corporate_Services"}

Facilities Services

Following Facilities Services offering is available:

- Access Security

The following Facilities Services offering is available:

- Access Security

{"URL":["/usm/wpf?Node=icguinode.cataloggetchildren&Args=10401&ObjectID=10401&NspPath="],"customLabelGetStarted":"Facilities Services","description":"IPCE_Facilities_Services"}

Reservation Services

Following Service offerings are available for Reservation Services:

- Reserve a Virtual Machine
- Reserve a Physical Machine
- Reserve Using Reservation Template
- Extend Reservation
- Return Reservation

The following Service offerings are available for Reservation Services:

- Reserve a Virtual Machine
- Reserve a Physical Machine
- Reserve Using Reservation Template
- Extend Reservation
- Return Reservation

{"URL":["/usm/wpf?Node=icguinode.cataloggetchildren&Args=11001&ObjectID=11001&NspPath="],"customLabelGetStarted":"Reservation Services","description":"IPCE_Reservation_Services"}

IT Support Services

Following Main folder level IT Support Services are available:

- Request New File Share
- Virus Protection / Remediation Plan SLA
- Install / Upgrade / Remove
- VPN Access
- Proxy Access
- Pre-Production Security Scans
- PC Loaner
- Backup Production Server Data
- Backup Local PC / Laptop Data
- Restore Procution Server Data
- Restore Local PC / Laptop Data
- Request Access to Existing File Share

Service Management

Following Service Management Services are available:

> **NOTE**
>
> Service Management content is not available through the Setup Utility. This is available only when the Service Management Content pack is imported.

Following Service Management Service offerings are available:

- My Resources
- Problem with My Resource
- Report an Issue
- Reset Application Password
- Reset User Password

Asset Services

Following Asset Services are available:

- Request a Hardware Asset
- Request a Hardware Asset (for mobile users)
- View My Assets
- View My Assets (for mobile users)

Procure New Hardware

Following procure new hardware service offerings are available:

- Procure Server
- Procure Laptop
- Procure Desktop
- Procure Local Printer
- Procure Network Printer

Change/Upgrade Existing PC

Following Change/Upgrade Existing PC offerings are available:

- Add Storage
- Add Memory
- Add Accessories
- Replace Monitor (PC)

Software Procurement

Following Software Procurement Services are available:

- Procure HR in-house Apps
- Procure Finance In-house Apps
- Procure Procurement In-house Apps
- Procure Marketing/Sales In-house Apps
- Procure Operations In-house Apps
- Procure Miscellaneous In-house Apps

PC Software

Following PC Software Services are available:

- Procure Desktop Operating Systems
- Procure Desktop Database Software
- Procure Desktop Office Suites
- Procure Project Management Tools
- Procure Desktop Reporting Tools
- Procure Desktop Development Tools
- Procure Desktop Management Software
- Procure Miscellaneous Desktop Software

Server Software

Following Server Software Services are available:

- Procure Server Operating System Software
- Procure Server Database Software
- Procure Miscellaneous Server Software

Vendor Software - Corporate Apps

Following Vendor Software-Corporate Apps are available:

- Procure HR Apps
- Procure Finance Apps
- Procure Procurement Apps
- Procure Marketing/Sales Apps
- Procure Operations Apps
- Procure Miscellaneous Vendor Apps

Email

Following Email Services are available:

- Create email Account
- Increase Mailbox Size
- Recover Mailbox

Email Distribution Lists

Following Email Distribution Lists services are available:

- Request New Distribution List
- Subscribe to a Distribution List
- Unsubscribe form a Distribution List
- Reove Distribution List

Email Forms

Following Email Forms services are available:

- Request Access to Form
- Remove Form

Public Folder

Following Public Folder Services are available:

- Public Folder Creation
- Email Enabled Public Folder Creation

Knowledge/Self-Service

Following Knowledge/Self-Service is available:

- Knowledge Engineering

Incident Response Service Labels-Dept Contract

Following Response Service Labels-Dept Contract Service is available:

- Department Contract - SLA

The following main folder level IT Support Services are available:

- Request New File Share
- Virus Protection / Remediation Plan SLA
- Install / Upgrade / Remove
- VPN Access
- Proxy Access
- Pre-Production Security Scans
- PC Loaner
- Backup Production Server Data
- Backup Local PC / Laptop Data
- Restore Procution Server Data
- Restore Local PC / Laptop Data
- Request Access to Existing File Share

{"URL":["/usm/wpf?Node=icguinode.cataloggetchildren&Args=10501&ObjectID=10501&NspPath="],"customLabelGetStarted":"IT Support Services","description":"IPCE_IT_Support_Services"}

Network Services

Following Network Services are available:

- Network Engineering / Design
- Jack Activations
- Static IP Requests
- Network Performance Testing / Tuning
- Network Education
- Network Vulnerability Analysis
- LAN Admin SLA
- WAN Admin SLA
- Wireless Network Admin SLA
- File Share

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{"URL":["/usm/wpf?Node=icguinode.cataloggetchildren&Args=10601&ObjectID=10601&NspPath="],"customLabelGetStarted":"Network Services","description":"IPCE_Network_Services"}

Personnel Services

Following Personnel Services Offering is available:

- New Hire Boarding

The following Personnel Services Offering is available:

- New Hire Boarding

{"URL":["/usm/wpf?Node=icguinode.cataloggetchildren&Args=10701&ObjectID=10701&NspPath="],"customLabelGetStarted":"Personnel Services","description":"IPCE_Personnel_Services"}

Project Services

Following project service offerings are available:

- New Project Request
- Vendor Product Evaluation Request
- Vendor Management Request
- Contract Negotiations Request
- Project Managemet Request
- Project Change Request
- Business Continuity Planning

The following Project Service offerings are available:

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- Vendor Product Evaluation Request
- Vendor Management Request
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{"URL":["/usm/wpf?Node=icguinode.cataloggetchildren&Args=10801&ObjectID=10801&NspPath="],"customLabelGetStarted":"Project Services","description":"IPCE_Project_Services"}

Telecom Services

Following Telecom Services are available:

- Video Conference
- Webcast
- Desk Phone
- Desk Phone Accessories
- Voicemail
- Mobile Phone
- Mobile Phone Accessories
- Handheld Devices
- Pager
- Fax
- Corporate Calling Card
- Conference Bridge
- Analog/Fax Line
- ACD

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- Webcast
- Desk Phone
- Desk Phone Accessories
- Voicemail
- Mobile Phone
- Mobile Phone Accessories
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{"URL":["/usm/wpf?Node=icguinode.cataloggetchildren&Args=10901&ObjectID=10901&NspPath="],"customLabelGetStarted":"Telecom Services","description":"IPCE_Telecom_Services"}

Best Practices for Service Catalog Management

IT organizations have focused on technology without clear accountability for adding business value, while their customers have focused on costs. Increased competition in the marketplace and increased dependence on high technology demand that IT organizations:

- Build a stronger understanding of the customers they serve.
- Work in a close partnership with them to deliver true value.

These business alignment initiatives require:

- A clear definition of the services that the IT organization provides.
- An understanding of the components and resources that constitute the services.
- An identification of the costs of the associated services.

With centralization of IT services and movement toward a utility model, implementing a service catalog becomes an increasingly important necessity. Using a service catalog leads to better IT service alignment with business goals and improved internal customer satisfaction. It also leads to standardized processes to achieve greater operational efficiency.

The ITIL™ (IT Infrastructure Library) framework stipulates that IT organizations start their process enhancement initiatives by developing a service catalog to:

- List all the services being provided
- Summarize the details of the services, customers, service designers, and service administrators.

Many IT groups produce a service catalog as part of their ITIL Service Level Management deployment. Others view the opportunity to leverage the service catalog as the focal point for communication between IT and the business.

Benefits

CA Service Catalog enables you to load catalog content and to enable related business process automation through CA Process Automation. The goal of this initiative is not to create a complete and comprehensive list of services. Instead, the goal is to help you get started in building your own catalog.

As the foundation for aligning IT with the business, service catalog represents a crucial element of success in ITIL initiatives. By using a service catalog, IT organizations can:

- Create clear service definitions.
- Define service levels and costs.
- Deliver usage and performance data in business terms.
- Track user subscriptions and service requests.
- Provide a fast, consistent, and more effective access to IT services.
- Publish or review service reports.

Developing a service catalog can provide the following improvements to drive down costs while still achieving high levels of service:

- Improved efficiencies in handling incoming requests.
- Eliminate service redundancies.
- Increased IT staff productivity.
- Better allocation of resources.
- Reduced calls to the help desk.

Guidelines for Collecting Data

The following guidelines are for using staff interviews to collect data regarding the services and service levels currently existing in your organization. These guidelines are developed based on ITIL Methodology and CobiT™ Principles. These guidelines consist of the following activities for collecting and analyzing data, in sequential order:

- Review of the purpose of the staff interviews.
- Use of the suggested questions and any related questions of your own to conduct the staff interviews.
- Analysis of service-related documents.
- Review and benchmark activities for existing services.
- Collation the results of staff interviews.
- Organization of the newly determined information into service specifications that include detailed descriptions of each service.

Purpose of the Staff Interviews

The purpose of the interviewing the staff is to obtain and understand organization-wide policies and procedures relating to provider-to-user relationships. Examples follow:

- IT policies and procedures:
  - Service level agreements
  - Operational reporting content, timing, and distribution
  - Performance tracking methods
  - Corrective action activities
- IT documentation:
  - Service level performance reports
  - Service budget and costing procedures
  - Charge-back algorithms and methodology for calculating charges
  - Service improvement programs
  - Recourse resulting from nonperformance
  - Service level agreements with internal and external users and providers of services

Questions for Staff Interviews

The staff interviews can include the following questions:

- Who delivers which services where, to which organization or organizational groups, and at what cost?
- Who requests which services from which IT group and at what volume?
- What level of service is in place within the organization?
- Who are the stakeholders?
- How can we use the template catalog?

Staff Interviews

The staff whom you Interview include the following personnel:

- Chief Information Officer
- IT senior management
- IT Contract Administrator
- Service Level Administrator
- IT Operations management
- User management

Documents to Analyze

You review and analyze the following documents:

- Service level agreement process
- Definitions of responsibilities of users and providers
- Management reports on the achievement of the specified service performance criteria and all problems encountered

The purpose of analyzing the documents is to verify whether:

- Users understand service level agreement processes and procedures.
- The level of user satisfaction with the service level process and service level agreements is sufficient.
- Service fulfillment data is available to track performance.
- A performance improvement program exists.
- The accuracy of actual charges matches agreement content.
- A comparison of historical performance and prior service improvement commitments is tracked.
- Managers appropriately use reports on service performance.

Review and Benchmark Activities

The review and benchmark activities are as follows:

- Benchmarking of service level agreements against similar organizations or appropriate international standards and recognized industry best practices.
- Review of service level agreements to determine qualitative and quantitative provisions confirming obligations are defined and being met.
- Review of selected service level agreements to confirm compliance and resolution procedures for problems, specifically nonperformance.

Results of Staff Interviews

The results of the staff interviews help the organization identify the:

- Adequacy of the provisions describing, coordinating, and communicating the relationship between the providers and the users of information services.
- Incorrect calculations for selected categories of information.
- Ongoing review and corrective action by management of service level reporting.
- Adequacy of proposed service improvements in comparison with cost-benefit analysis.
- Adequacy of the ability of providers to meet commitments for improvements.

Best Practices Foundation

This article contains the following topics:

Request and Fulfillment Automation with CA Process Automation

A key characteristic of the predefined foundation content is the automation of approvals and notifications for actions that complete the service fulfillment process. This automation makes the services "actionable." Service designers accomplish this goal through the workflow capabilities included with CA Process Automation.

All services in the predefined foundation content invoke the workflow-driven approval process. This process sends an email to the manager of the requestor, indicating the requiring approval for the request. The email includes a link to the approval page in CA Service Catalog to streamline the approval process. You can use a different approval process for individual services. To do so, select a different approval mechanism for those services through the Service Builder.

After a service request has been approved, the requested items in the service option group then follow a fulfillment workflow. The category that is specified for the requested component determines the workflow. The predefined service option groups contain components that belong to one of these categories: Hardware, Software, or Service.

The event rules for each category control the notifications that are generated as the components of the request move through the stages of fulfillment. The foundation content, as delivered, follows the predefined rules for each category. You can customize these rules to address your specific requirements.

The predefined foundation content provides a category-class-subclass structure for your use.

Category-Class-Subclass Structure

The predefined foundation content provides a category-class-subclass structure. You can use the class and subclass designations within each category to customize the workflow process. The default structure contains the following top-level categories:

- None
- Hardware
- Reservation Manager
- Service
- Service Offering Management
- Software
- Other

Multiple classes exist within each category, and in most cases multiple subclasses within each class.

To view and optionally update the complete detailed structure of category classes and subclasses, edit the category.xml file.

> **NOTE**
> You can add categories to meet your specific requirements. However, you cannot change the internal numeric values of the default categories. This field is used for logic in other portions of the product.

Service Catalog Logical Structure

You can use a table or spreadsheet to design the full logical structure of the catalog before its actual implementation. This approach facilitates communication between stakeholders and end users regarding the service catalog structure and content required.

We recommend that you agree on the general service catalog content and structure first and the details next. As the catalog and its use mature within an organization, some restructuring and fine-tuning are likely to occur.

IT Support Services						
Folder	**Service**	**Service Option**	**Category**	**Class**	**Subclass**	**Description**
Data Security						
	Virus Protection / Remediation Plan SLA					
		Bronze	Service	IT	SLA	Bronze SLA for Virus Protection/ Remediation
		Silver	Service	IT	SLA	Silver SLA for Virus Protection/ Remediation
		Gold	Service	IT	SLA	Gold SLA for Virus Protection/ Remediation
	VPN Access					
		Request VPN Access	Service	IT	Security	Request VPN access for remote user
	Proxy Access					
		Subscribe to Proxy	Service	IT	Security	Request proxy access (security request)
	Pre-Production Security Scans					
		Security Scan	Service	IT	Security	Request Security Scan before production status
Data Management						
	Backup Data					
		Backup Production Server Data	Service	IT	Data	Backup Services for Production Server Data.
		Backup Local PC / Laptop Data	Service	IT	Data	Backup Services for Local PC / Laptop Data.

	Restore Data					Services to restore production or local data.
		Restore Production Server Data	Service	IT	Data	Restore data on a production server from backup files.
		Restore Local PC / Laptop Data	Service	IT	Data	Restore data on a local PC from backup files.

We recommend limiting the number of levels that users navigate to reach specific services in your catalog. As an example, consider an IT Support Services category such as Hardware and Software Procurement. For ease of use, divide it into two separate categories. Examples include Hardware Procurement and Software Procurement. This approach is more efficient than using subfolders, for example, under Hardware and Software Procurement.

You can customize the structure of the provided content to meet the needs of your organization. For example, you can divide a single-Server service into the following services: Blade Server, Standard Server, and Mainframe Server. In this case, create a folder that is named Server to contain the other thee services. Each service contains descriptions and service option group items to assist users in their selection.

Performance Tuning Parameters

Following are the recommendation to fine tune the CA Service Catalog performance:

PARAMETER	SETTING	UNITS
Tomcat Threads	800	Number
Tomcat connection timeout	1000000	Milliseconds
Java initial Heap Size	512	MB
Java maximum Heap Size	2048	MB
Connection Pool Size	300	

The typical location of the source files to modify these parameters are as follows:

- **Java Heap**
  Viewservice.conf located in USM_HOME\view\conf
- **Tomcat**
  Server.xml located in USM_HOME\view\conf
- **Connection pool**
  Config.properties located in USM_HOME

NOTE

The Maximum heap size for Java is 1/4 of the physical memory on the server assuming that the Service Catalog is the only application installed on the server.

Customer-Focused Documentation

Prepare customer-focused documentation for the service catalog and make it available to all potential users. Sample contents follow:

- Forward from the IT Director
- Table of Contents
- IT Service Provider profile
- Version number, date created, date amended
- Changes from last published Catalog
- Service times and accessibility of the IT Service Provider
- Overview of Services and Products
- Customer-focused Service and Product descriptions
- Specifications of Services
- Deliverables
- Service Times
- Maintenance Times
- Support Times
- Delivery Times
- Quality Target (availability, reliability, usability, priority)
- Requirements
- Request and Change procedures
- Contingency policy
- Pricing and Charging
- Index and definitions

How CA Service Catalog Determines the Role of a User?

There is no direct way to extract the list of users and their corresponding roles in CA Service Catalog . CA Service Catalog authenticates a user and determines the role in the following:

1. CA Service Catalog first checks if a user exists in **ca_contact** table in MDB.
   If it cannot find a user in **ca_contact** table , it does not proceed with verification and will stop.
2. If it finds the user in **ca_contact** table, it will send the user's credential to CA EEM for authentication. If the user's credential is successfully authenticated by CA EEM, it will verify the user in CA Service Catalog User Interface by checking the following to determine the type of role that must be assigned to the user.
3. CA Service Catalog checks *usm_contact_domain_role* table to see if the said user has explicit role recorded in this table.If a record exists in usm_contact_domain_role for this user, the user is able to login to CA Service Catalog application with the role specified in that table.
4. If the user does not exist in usm_contact_domain_role table, catalog will let the user login to CA Service Catalog with the default role assigned by Catalog. The default role is specified in Catalog user interface, **Administration** tab, **Configuration, User Default, User Default Role**.

   NOTE

   To explicitly specify a catalog role for a user, you can login into catalog GUI with service delivery administrator role. Navigate to **Administration, Users,** search and locate the user and edit its catalog profile in the "User Setting" section. Note that when you explicitly grant a role to a user and save it, it generates/ updates a record in *usm_contact_domain_role* table .

The user role information is stored in usm_contact_domain_role table in CA Service Catalog. If there is no record specified in this table for a user, the user is assgined the default role.

High Security Settings

CA Service Catalog provides system forms for users to complete as part of their requests. A service consists of hardware, software, or other resources that users request from the catalog. Administrators can manage services to meet the needs of their users and organization. You can use events, rules, and actions to automate the processing of requests, users, and accounts in the CA Service Catalog system. All rules are initially disabled. A rule must be enabled to be used.

CA Service Catalog by default does not allow any third-party scripts, CSS, or images to be downloaded directly from an external website.

For example, while adding a dashboard, you need to configure any other third-party scripts, CSS, or images as a trusted source, configure the external web origins in CA Service Catalog.

Follow these steps:

1. Navigate to `<Catalog_home>\view\webapps\usm\WEB-INF` and open **web.xml** file in a text editor.
2. Search for the header, **Content-Security-Policy**.
3. Search for **<param-name>contentSecurityPolicy.**
4. Update the **<param-value>** as shown below in the following examples:
   a. To add https://ajax.googleapis.com as a trusted origin for content such as script, css, and image:

      ```
      <param-value>default-src'self' 'unsafe-inline' 'unsafe-eval' https://ajax.googleapis.com/</param-value>
      ```

   b. To add https://ajax.googleapis.com as a trusted origin for content type as script.

      ```
      <param-value>default-src 'self'; script-src 'unsafe-inline' 'unsafe-eval' https://ajax.googleapis.com;</param-value>
      ```

5. Restart the CA Service Catalog server.

NOTE

Information! For more information, see W3C's Content Security Policy documentation.

Service Accounting

Service Accounting lets you manage your invoicing and financial reports, accounts, subscriptions, exchange rates, budgets and plans, and so on. You can search and view all invoices for an account for a certain period. It offers you a consolidated view of all invoices for a selected group. Financial reports provide insight and visibility about account details, adjustment details, invoice details, and payment details. An Administrator can manage invoices for an account using the available options. Accounting configuration enables you to prorate the online as well as batch processes. You can generate an invoice for charges as per the usage rate. Administrators can schedule the on-demand invoice to produce the instant current invoices for an account.

This section contains the following articles:

- Manage Invoicing and Financial Reporting
- Manage Accounts
- Manage Subscriptions
- Manage Exchange Rates
- Manage Budgets and Plans
- Manage General Adjustments
- Manage Data Mediation
- Manage Chargeback and Pricing Models

The Search tab opens with the Accounts tab by default. To search for all Accounts, click Search without specifying a search string.

The scope of “search” functionality for users includes only accounts in the same business unit as the logged in user. This scope does not include child business units.

You can search and view all accounts. Click Search to display all available accounts. You can also specify search criteria and view the required accounts.

You can search and view all invoices for an account for a certain period. It offers you a consolidated view of all invoices for a selected group. Ensure that you have access to the accounts for which you want to view the invoices. Click Search to display all available data or specify search criteria to view the required data.

You can search and view all payments for an account for a certain period. It offers you a consolidated view of all payments for a selected group. Ensure that you have access to the accounts for which you want to view the payments. Click Search to display all available data or specify search criteria to view the required data.

{"URL":["/usm/wpf?Node=iclaunchpad.search","/usm/wpf?Node=iclaunchpad.sl-searchaccountpre","/usm/wpf?Node=iclaunchpad.sl-searchinvoicepre","/usm/wpf?Node=iclaunchpad.sl-searchpaymentpre"],"customLabelGetStarted":"Search","description":"IPCE_Search","customCards":[{"id":"IPCE_Accounts","type":"manage","title":"Accounts"},{"id":"IPCE_Invoices","type":"manage","title":"Invoices"},{"id":"IPCE_Payments","type":"manage","title":"Payments"}]}

Manage Invoicing and Financial Reporting

This article contains the following topics:

Invoice History

The Invoice history feature enables you to search for and view all invoices for a certain period for any account. Ensure that you have access to the accounts for which you want to view the invoices. Invoice History Records offer a consolidated view of all invoices for a given group. Information that is displayed includes the Invoice group and the status (successful or unsuccessful). Invoices can be viewed in HTML, CSV, or XML format, and provides the facility to email these invoices.

Invoicing and Financial Reporting

Once services are tied to accounts through a subscription, you can schedule these accounts for batch invoicing. You can process any rates, adjustments (general or SLA violation), and payments posted for the current billing period. An invoice group with global settings can be established to bill an associated list of accounts together in one instance at a scheduled future date. This list can be assembled at the time the group is created or determined dynamically during invoice execution. At the scheduled time, the invoice engine kicks off. The group is then placed into a queue for the accounting service to pick up for processing. Once selected, the engine queries the accounts that are associated with the invoice group. The engine then proceeds to generate an invoice for each of accounts.

The invoice states and justifies charges to users and business owners. These charges can be rolled up by a Business Unit organization. The stakeholders, at each level, can be given access to their IT cost information. Once processed, organizations can provide each Business Unit Web access to real time and historical chargeback invoice information. Business Units can also see detailed IT usage reports. IT owners can view actual spending versus budget and can determine the actual cost by business function. Providing such visibility into IT costs enables providing an organization with the necessary data for setting budgets and business planning.

Financial Reports

Several reports are available to help provide insight and visibility into IT. These reports include:

- Accounting Details - shows the accounting profile details for all accounts given the filter specification.
- Adjustment Details - displays details pertaining to any defined adjustments.
- Invoice Details - shows information that is related to invoicing such as dates, balance, and number of violations.
- Payment Details - provides a listing of payments posted against account balances.

Invoice Criteria for Accounts

For the invoice engine to process the account, the account must be:

- Included in the Account List associated to an active invoice group.
- Have the Billing Date field (in the Accounting profile) specified earlier than the date of the bill run.
- Have the status field (in the Accounting profile) specified.
- Have the Automatic Invoicing field (in the Accounting profile) set to Yes.
- Subscribed to a service under a valid state if the configuration is set for *Allow No Activity Invoices*.

Invoice Criteria for Subscriptions

A subscription must meet the following criteria before showing on an invoice:

- Chargeable items such as one time, daily, weekly, monthly, and installment appear when those periods for those charges are recognized.
- An element in the service option groups of a service is not specified to be Excluded From Invoice.
- Usage and transaction based charges appear when there have been metrics that are collected to represent use or transactions.
- Agreement and contract charges as defined through CA Business Service Insight appear on an invoice. The subscription is being charged as a one time, periodic or installment charge.

Invoice Groups

Administrators can create invoice groups. An Invoice Group provides the capability for grouping and scheduling accounts for future invoicing. When the invoicing process initiates, each group enters a queue to provide the accounting service the facility to retrieve and process the list of accounts in the group. This account list can be assembled and maintained manually or dynamically depending on the group type. The invoice groups generate invoices on a schedule or on demand and facilitate financial reporting.

Add a Static Invoice Group

Static Invoice Groups contain an account list that is comprised of accounts that have been manually assigned to it. Accounts assigned to a static invoice group cannot be assigned to any other group. Any accounts that are assigned to an invoice group remain unless explicitly removed or deleted from the system. You can further expand the account list for active groups by directly selecting and adding accounts to it.

Follow these steps:

1. Click Accounting, Invoices, Groups.
2. Click the Add button.
3. Complete the Invoice Group information:

   NOTE
   Do not select Dynamic for this procedure because groups are static by default.
4. Complete the Accounting Profile information
   - Billing Cycle
     Determines the cycle when an account is invoiced.
   - Billing Cycle Interval

Indicates the number of billing cycles that must elapse before an account is invoiced. This parameter is used with Billing Cycle.
For example, to specify a quarterly billed account, select a monthly billing cycle, and specify 3 as the billing cycle interval.

- Period Start Date
  Indicates when the accounting profile billing period begins. Choices include:
  - The date when department was created.
  - The date when the account was opened.
  - The current date that the accounting profile is created.
- Period End Date
  Indicates when the accounting profile billing period ends.
- Days Due Default
  Calculates the number of days past the date of the invoice that payment is due.

Review the information on the Group Confirmation window. Click Next to confirm the information.

5. Click Add Accounts and use the shuttle box to move any accounts you wish to include in the account list. Verify that all selected accounts appear in the Accounts to Save column, and click OK.
6. Click Next and apply the appropriate Scheduler settings giving the billing date and the time zone.
7. Set the status to Active and click Finish.
   The Invoice Groups window appears, reflecting your newly created group.

Add a Dynamic Invoice Group

With Dynamic Invoice Groups, account lists are generated dynamically given a specified criteria that is stored as a Data Object. As new accounts are added and fall within that criteria, they are included in the account lists in that group. In essence, Invoice Groups can now be set to maintain and manage their own account lists.

Follow these steps:

1. Click Accounting, Invoices, Groups.
2. Click the Add button.
3. Complete the Invoice Group information and select Dynamic.
4. Review the Accounting Profile information and change the fields as necessary.
5. Review the information on the Group Confirmation window.
6. The Account List window appears empty since there are no associated criteria to assemble this list. Click Define Criteria and use the Account Criteria Builder to create a query to generate a list.
7. Click the Create Criteria button.
   - Save Last Executed Search
     Saves the last search that the Account Criteria search account mechanism executes.
   - Advanced Custom Search
     Enables the user to specify directly the where clause in SQL.
8. Once a criteria type is selected, two more fields are present:
   - Name
     Specifies the name to be associated to this query. (This value is stored as a Report Data Object.)
   - Comments
     Provides a field for more information.
9. Click Save when complete.
10. Click Next.
11. Apply the appropriate Scheduler settings giving the billing date and the time zone.
12. Set the status to Active.
13. Click Finish.
    The Dynamic Invoice Group has been created and is reflected in the Invoice Groups window. Click View Accounting Tasks to view the scheduled task.

Specify the Output Location

The default locations where invoices are saved to the hard drive are as follows:

- Accounts that receive invoices through email have a copy of their invoices that are stored in %ACC_HOME%\outbox\email.
- Accounts that receive invoices through regular postal mail have a copy of their invoices that are stored in %ACC_HOME%\outbox\postal.
- Accounts that receive invoices through fax have a copy of their invoices that are stored in %ACC_HOME%\outbox\fax.
- Accounts that the administrator wants to print internally have a copy of account invoices that are stored in %ACC_HOME%\outbox\printer.

NOTE
Specify \\ as a folder separator on Windows.

Instead of emailing a business unit an invoice directly, you can also to print or fax it. You can store the files in a directory on your hard drive. To do so, set the **Invoice Method** in the accounting profiles of all your customer accounts to *Postal* by default.

Manage Invoices

An administrator has several options to manage invoices that have already been generated for an account.

Follow these steps:

1. Click Accounting, Account Management.
2. Search for and select an account.
3. Click the Invoices tab.
4. Manage invoices, as follows:
   - View the invoice history
   - Edit an invoice
   - Perform an invoice on-demand
   - View an invoice online
   - Rollback an invoice

An invoice can be edited and line items can be added, modified, or removed. These line items include text, charges, adjustments, and payments.

Proration

Prorating is the process of dividing the cost of a subscription over the billing cycle period of an account. For example, Suppose that a monthly charge is $3000.00, and an account is on a Monthly billing cycle. An invoice is generated one day into the billing cycle with 30 days that are left in that month. The charge on the invoice is $100.00 and a description that specifies that the charge is for one day.

The accounting configuration enables you to specify that both online and batch process invoices are prorated:

- Only online invoices are prorated.
- Only invoices that are generated during the batch process are prorated.
- Or neither are prorated.

By default, both online invoices and batch invoices are configured to prorate charges.

Generate Invoice

An invoice can be generated to show the charges that are computed against the usage. If User Usage-Based billing was employed, the charges come from all users who are associated to the invoiced account.

Invoice On-Demand

On-demand invoicing enables an administrator to bypass the scheduled invoicing and instantaneously produce the current standing invoice for an account.

Performing an invoice on-demand produces the same results as if it had been processed as a scheduled bill run. A new invoice is generated for the current billing period and the accounting profile is updated.

Invoice Online

Online invoicing enables you to view your charges in real time with a web browser. An online invoice displays charges that an account has incurred since its last billing period, up to and including the current date. All the charges are computed in real time. Because these values can change, the invoice does not get persisted until the end of the billing cycle when the invoicing process runs.

Similar to on-demand invoicing, online invoices do not have to go through the same account query and verification: an invoice is requested to be viewed for that account regardless of the account status, billing cycle, or automatic invoicing. The criteria still apply for choosing or selecting which subscriptions show up on the invoice as charges, as with the invoice process.

You can enable or disable prorating charges when viewing your invoice online. Set the **Pro Rate Online** to Yes in the **Invoice Engine Configuration** of Accounting Configuration.

View Invoice Status

As the Invoice Engine processes the invoices, it logs alerts to the CA Service Catalog database. These alerts can be viewed by clicking Messages from the main menu.

Your Alert messages are listed in a table displaying the following parameters:

- Date/Time
- Type
- Source
- Computer
- Message

If a failure occurs during any one of the phases, the Invoice Engine logs a critical severity alert. The Invoice Engine then displays a message.

Rollback Invoice and Rollback Invoices Utility

Rollback invoicing enables you to invoice accounts from a previously billed period. This sequential rollback of invoices can reprocess an invoice to reflect any changes that are made to charges. To roll back to a previous invoice period, select the invoices to roll back from the Invoice History window of the account. Click Rollback. The period date, bill date, and account balance for the accounting profile is updated up to the period, invoices were rolled back. These Invoices and their associated transactions are permanently deleted from the system once the rollback is complete. Invoice On-demand can be used to regenerate any or all the invoices whenever necessary.

> **NOTE**
> The rollback works *only* for invoices that a successful invoice bill creates.

The syntax for using the rollback.bat (Windows) command follows.

rollback -g *groupname* -b *businessunitid*

- **-g *groupname***
  Specifies the name of the invoice group that defines the accounts whose most recent invoices you want to roll back. This parameter is required.
- **-b *businessunitid***
  Specifies the business unit ID of the accounts--in the named invoice group--whose most recent invoices are being rolled back. For Windows, the rollback.bat file is located in the USM_HOME\accounting\scripts folder.

Examples

The following command rolls back the most recent invoices for all accounts in the InternationalAccounts invoice group, in *all* business units:

```
rollback -g InternationalAccounts
```

The following command rolls back the most recent invoices for only the accounts in the InternationalAccounts invoice group that belong to the ForwardInc business unit:

```
rollback -g InternationalAccounts -b ForwardInc
```

Batch Printing

Service Accounting Component offers a Batch Printing option. You can create print jobs and print invoices individually or as a whole. A print job is a collection of invoices that have been grouped according to the specified credentials. After it has been created, the collection remains unchanged and its invoices can be printed as a whole or individually.

Follow these steps:

1. Click Accounting, Invoices, Batch Printing.
2. Click Add Print Job.
3. Enter the Name and Description.
4. Select the appropriate Invoice Group.
   An Invoice group is a set of accounts that you can run together in one instance. With the Batch Printing feature, invoice groups serve as an invoice filter. Only those invoices that are associated with the accounts in the selected invoice groups are included in this print job.
5. Select Services. Select all Services (the default) or further refine your number of invoices by filtering by Service name.
6. Select Billing Cycles.
   By including specific types of billing cycles, the number of invoices in the collection is once again decreased. This billing cycle is for the account, not at the Invoice Group.
7. Click Add.
   Once a print job has been added, the Batch Printing user interface can be used to manage them. Clicking a print job changes its details on the right. You can delete the selected print job or can reuse it.

   NOTE
   You can return the invoices in reverse order or filter the invoices by selecting a specific Item Status.

By default, 50 invoices are returned for each page. To adjust this number, specify the desired amount and click Refresh. Clicking an invoice launches it into a new window serving as a print preview tool. This tool can also check all invoices by item status. By clicking the Reverse Order check box, the checked invoices on the current page is printed in ascending order.

After each print item is completed, the status for each item is updated. The print job status appears at the top.

You can generate invoices for accounts. Once services are tied to accounts through a subscription, you can schedule these accounts for invoicing. You can process any rates, adjustments (general or SLA violation), and payments posted for the current billing period. An invoice group with global settings can be established to bill an associated list of accounts together in one instance at a scheduled future date. This list can be assembled at the time the group is created or determined dynamically during invoice execution. At the scheduled time, the invoice engine kicks off. The group is then placed into a queue for the accounting service to pick up for processing. Once selected, the engine queries the accounts that are associated with the invoice group. The engine then proceeds to generate an invoice for each of accounts.

The Invoice history feature enables you to search for and view all invoices for a certain period for any account.
Ensure that you have access to the accounts for which you want to view the invoices. Invoice History Records offer a consolidated view of all invoices for a given group. Information that is displayed includes the Invoice group and the status

(successful or unsuccessful). Invoices can be viewed in HTML, CSV, or XML format, and provides the facility to email these invoices.

Administrators can create invoice groups. An Invoice Group provides the capability for grouping and scheduling accounts for future invoicing. When the invoicing process initiates, each group enters a queue to provide the accounting service the facility to retrieve and process the list of accounts in the group. This account list can be assembled and maintained manually or dynamically depending on the group type. The invoice groups generate invoices on a schedule or on demand and facilitate financial reporting.

Add a Static Invoice Group

Static Invoice Groups contain an account list that is comprised of accounts that have been manually assigned to it. Accounts assigned to a static invoice group cannot be assigned to any other group. Any accounts that are assigned to an invoice group remain unless explicitly removed or deleted from the system. You can further expand the account list for active groups by directly selecting and adding accounts to it.

Follow these steps:

1. Click Accounting, Invoices, Groups.
2. Click the Add button.
3. Complete the Invoice Group information:

   NOTE
   Do not select Dynamic for this procedure because groups are static by default.
4. Complete the Accounting Profile information
   - Billing Cycle
     Determines the cycle when an account is invoiced.
   - Billing Cycle Interval
     Indicates the number of billing cycles that must elapse before an account is invoiced. This parameter is used with Billing Cycle.
     For example, to specify a quarterly billed account, select a monthly billing cycle, and specify 3 as the billing cycle interval.
   - Period Start Date
     Indicates when the accounting profile billing period begins. Choices include:
     - The date when department was created.
     - The date when the account was opened.
     - The current date that the accounting profile is created.
   - Period End Date
     Indicates when the accounting profile billing period ends.
   - Days Due Default
     Calculates the number of days past the date of the invoice that payment is due.

   Review the information on the Group Confirmation window. Click Next to confirm the information.
5. Click Add Accounts and use the shuttle box to move any accounts you wish to include in the account list. Verify that all selected accounts appear in the Accounts to Save column, and click OK.
6. Click Next and apply the appropriate Scheduler settings giving the billing date and the time zone.
7. Set the status to Active and click Finish.
   The Invoice Groups window appears, reflecting your newly created group.

Add a Dynamic Invoice Group

With Dynamic Invoice Groups, account lists are generated dynamically given a specified criteria that is stored as a Data Object. As new accounts are added and fall within that criteria, they are included in the account lists in that group. In essence, Invoice Groups can now be set to maintain and manage their own account lists.

Follow these steps:

1. Click Accounting, Invoices, Groups.
2. Click the Add button.
3. Complete the Invoice Group information and select Dynamic.
4. Review the Accounting Profile information and change the fields as necessary.
5. Review the information on the Group Confirmation window.
6. The Account List window appears empty since there are no associated criteria to assemble this list. Click Define Criteria and use the Account Criteria Builder to create a query to generate a list.
7. Click the Create Criteria button.
   - Save Last Executed Search
     Saves the last search that the Account Criteria search account mechanism executes.
   - Advanced Custom Search
     Enables the user to specify directly the where clause in SQL.
8. Once a criteria type is selected, two more fields are present:
   - Name
     Specifies the name to be associated to this query. (This value is stored as a Report Data Object.)
   - Comments
     Provides a field for more information.
9. Click Save when complete.
10. Click Next.
11. Apply the appropriate Scheduler settings giving the billing date and the time zone.
12. Set the status to Active.
13. Click Finish.
    The Dynamic Invoice Group has been created and is reflected in the Invoice Groups window. Click View Accounting Tasks to view the scheduled task.

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{"URL":["/usm/wpf?Node=iclaunchpad.invoices","/usm/wpf?Node=iclaunchpad.invoices_history","/usm/wpf?Node=iclaunchpad.billing_groups&Args=ca.com","/usm/wpf?Node=icguinode.billgroupadd&Args=ca.com&Args=-1","/usm/wpf?Node=icguinode.schedulertasklist&Args=Billing&Args=next_alarm_time&Args=0&Args=&Args=ca.com","/usm/wpf?Node=iclaunchpad.batch_printing","/usm/wpf?Node=icguinode.printjobadd&Args=icaccount&Args=billinggroup&Args="],"customLabelGetStarted":"Budget and Planning","description":"IPCE_Budget_and_Planning","customCards":[{"id":"IPCE_Invoices","type":"manage","title":"Invoices"},{"id":"IPCE_Invoice_History","type":"manage","title":"Invoice History"},{"id":"IPCE_Groups","type":"manage","title":"Groups"},
{"id":"IPCE_Add_New_Invoice_Group","type":"manage","title":"Add"},
{"id":"IPCE_View_Accounting_Tasks","type":"manage","title":"View Accounting Tasks"},
{"id":"IPCE_Batch_Printing","type":"manage","title":"Batch Printing"},
{"id":"IPCE_Add_Print_Job","type":"manage","title":"Add Print Job"}]}

Manage Accounts

This article contains the following topics:

Administrators manage the accounts for a business unit. An *account* represents a corporate unit, a regional office, an individual user, or a logical group of them. CA Service Catalog applies charges to accounts.

If a user requests a service from the catalog, the product automatically creates an account for the user. The product also subscribes the account to the requested service.

> **NOTE**
>
> In Accounting Component, two digits appear after the decimal point for all supported currency units, including the Japanese yen. If your organization uses the yen, decide how to process the two digits after the decimal point because Yen is processed in whole numbers only. For example, you can decide to ignore the two digits after the decimal point or round them up or down to the nearest whole number.

Add or Edit Accounts

You can add an account so that you can bill it for services that it has requested or subscribed to. Typically, administrators edit accounts periodically as part of standard business unit operations.

Follow these steps:

1. Select Administration, Business Units from the main menu.
2. Expand the business unit tree and find the business unit that you want.
3. Click:
   - Add Account.
   - Account name of interest and edit it.
4. Complete or update the fields.

When you add an account, the Add a New Account page appears. You complete the fields on this page to configure the new account to meet the needs of your organization.

- **Opened Date**
  Specifies the date when the account becomes part of your organizational structure, in local time.
- **Status**

Specifies whether the account is open or it is closed.

- **Primary Contact Information**
  Specifies the user ID of the primary contact for the account.
- **Location Information**
  Specifies information about the location of the account.

  NOTE
  All CA products using the same MDB share location data. Therefore, be careful when you edit location information.
- **Account Settings**
  Specifies the list of users who are associated with this account. Usage-based billing references these users.

  NOTE
  When a user creates a request for the first time, the Catalog system creates a request-related account for the user.

Close and Delete Accounts

Closing an account is typically a required task as you create and maintain accounts in your business unit over time. If the status of an account is Closed, you cannot:

- Enter a subscription for the account.
- Generate the invoices.
- Edit the Accounting Profile.

If you close an account, the associated accounting profile handles any residual accounting functions before the status is Closed. The status field moves from “Close Requested” to “Closed” once the outstanding charges and credits are processed for that period.

Follow these steps:

1. Review the effects of closing or deleting an account and verify that you want to proceed.
2. Run the final invoicing for the account, if necessary.
3. Select Administration, Business Units, and expand the tree to display the business unit that contains the account that you want.
4. Click the account name.
5. Edit the account profile and change the status to Closed.
6. Delete the account. You can optionally close an account without deleting it.

Effects of Deleting or Closing an Account

Deleting or closing an account affects the subscriptions and requests for the account, as follows:

- **Subscriptions**
  The status of subscriptions for the account change to the default cancellation status (Pending Cancellation or Cancel).
- **Requests**
  The status of requested service options for the account change that is based on their original status as follows:

Original Status	New Status
Not Submitted	None. Request is deleted.
Submitted. An approval status, a fulfillment status, Pending Resource Assignment, or Resource Assigned	Cancelled

Completed	Default cancellation status (Pending Cancellation or Cancel) if Service Accounting Component is installed. Cancelled if Service Accounting Component is not installed.
Pending Cancellation or Cancelled	Same as Original Status.

Aggregation of Accounts

A business unit can designate an account to aggregate the charges and account balances for all other accounts within its scope. To do so, configure the accounting profiles for a set of accounts appropriately. Set the Aggregate option in the Accounting Profile to Aggregating Account to select the accounts for this consolidation. The aggregating account includes all accounts in the business unit hierarchy. Even sibling accounts whose Aggregate option is set to Yes are included in the aggregating account.

When the aggregating account is used to maintain the account balances of all accounts included in its aggregation, the aggregated accounts are typically zero balancing accounts. Set the Account Type option to Zero Balance in its Accounting Profile.

Post a Payment to an Account

Payments can be posted for an account to settle any outstanding balances.

Follow these steps:

1. Click Accounting, Account Management.
2. Click the Make Payment icon for the account for which you are posting the payment.
3. The Payments List and Payment Details page appears. From this window, you can perform the following actions:
   - Add payments to the Payments List by completing the payment details and clicking the Add to List button.
   - Modify Payments in the Payments List by selecting the payment, making your modifications and clicking the Update Payment button.
   - Delete Payments in the Payments List by selecting the payment and clicking the Delete Payment button.
4. The Business Unit, Account Name, and Account Balance are populated in the Payment Details window.
5. Complete the mandatory Payment Details and any optional fields.
6. Click Submit Payment.
   The payment is processed.

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You can add an account so that you can bill it for services that it has requested or subscribed to. Typically, administrators edit accounts periodically as part of standard business unit operations.

Follow these steps:

1. Click Accounting, Account Management.
2. Click:
   - Add Account.
   - Change Business Unit and edit it.
3. Complete or update the fields and click OK.

When you add an account, the Add a New Account page appears. You complete the fields on this page to configure the new account to meet the needs of your organization.

- **Opened Date**
  Specifies the date when the account becomes part of your organizational structure, in local time.
- **Status**
  Specifies whether the account is open or it is closed.
- **Primary Contact Information**
  Specifies the user ID of the primary contact for the account.
- **Location Information**
  Specifies information about the location of the account.

  NOTE
  All CA products using the same MDB share location data. Therefore, be careful when you edit location information.
- **Account Settings**
  Specifies the list of users who are associated with this account. Usage-based billing references these users.

  NOTE
  When a user creates a request for the first time, the Catalog system creates a request-related account for the user.

The Catalog system default displays settings for the business unit you logged in to. If your role permits, you can change to a different business unit.

When common multi-tenant administration is enabled, you can add tenants, delete tenants, or edit the common attributes *only* through CA Service Desk Manager. When common multi-tenant administration is enabled, you can *view* tenants and all their attributes in CA Service Catalog. But you can edit only the CA Service Catalog-specific attributes.

Follow these steps:

1. Click Catalog, Configuration.
2. Click the Change Business Unit button.
   The Search Business Units dialog appears.
3. Use the Expand and Collapse icons to navigate the business unit tree. Alternatively, use the selection criteria and Search button to locate the desired business unit.
   The list includes only the business units your role permits you to access.
4. To select a business unit, click its name in the tree.
   The window closes, and the configuration settings for that business unit appear.

{"URL":["/usm/wpf?Node=iclaunchpad.billing_management","/usm/wpf?Node=iclaunchpad.subscriptions","/usm/wpf?Node=icguinode.addregaccount&Args=ca.com","/usm/wpf?Node=icguinode.searchtenantpopup"],"customLabelGetStarted":"Accounting","description":"IPCE_Accounting","customCards":[{"id":"IPCE_Manage_Accounts","type":"manage","title":"Manage Accounts"},
{"id":"IPCE_Add_Account","type":"manage","title":"Add Account"},
{"id":"IPCE_Change_Business_Unit","type":"manage","title":"Change Business Unit"}]}

Manage Subscriptions

This article contains the following topics:

A *subscription* is a Service Accounting Component to identify the assignment of a service by an administrator to an account. Once the service is activated for the account and the subscription is active, the charges are calculated for each invoice period. If a subscription is cancelled, the service is deactivated as soon as all outstanding charges are processed. In addition, subscriptions can be suspended for any amount of time, during which no charges are incurred.

Subscription Types

Subscriptions are possible *only* when Service Accounting Component is installed.

Accounts can have *physical* subscriptions, *data mediation logical* subscriptions, or both types of subscriptions.

- A physical subscription includes one or more service options in a service. A physical subscription results from one of the following conditions:
  - When a service that a user explicitly requests is completed or a request that is only submitted but not completed. When a service is completed, it results in *request-based* subscription.
  - The business unit administrator assigns subscriptions to accounts, which are based on certain criteria, such as business needs. Such subscriptions are *account-based* subscriptions.
- A data mediation logical subscription is virtual; it includes no physical subscription records. Rather, it is a logical mapping and an aggregation mapping. This type results from an administrator using the Data Mediation component to aggregate usage data. In the Invoice component, the usage cost for this subscription is shown when data is aggregated.
  Considering data mediation logical subscriptions is important when you implement usage-based billing.

A single account can have one or more subscriptions of either type or both types. If an account has both a physical subscription and a data mediation subscription, the physical subscription *always* overrides the data mediation subscription.

On the Existing Subscriptions window, each applicable subscription type appears in the Currently Subscribed column. However, data mediation logical subscriptions appear *only* if no physical subscription exists.

Subscription types are cancelled when:

- A physical subscription (S) is canceled when you cancel the subscription for an account.
- A request subscription (R) is canceled when you cancel the request for an account.
- A Data Mediation logical subscription (D) is not shown when an administrator unloads or aggregates usage data again.

Administrators manage subscriptions by selecting Accounting, Account Management, clicking an account name, and clicking the Subscriptions tab. The subscription process is primarily an administrator task. But end users are given read access to active subscriptions for their accounts.

The Administration, Configuration, Subscription Configuration menu option enables Administrators to change the way future subscriptions of an account are handled.

Create a Physical Subscription

You can create a physical subscription.

Follow these steps:

1. Select Accounting, Account Management.
2. Click the account name for which you want to add a subscription.
3. Click the Subscriptions tab.
4. Select the service to which you want to subscribe.
5. Click the Subscribe link.
6. Click Yes.
   The physical subscription is added.

Enable Notes in Physical Subscriptions

To add notes to a physical subscription, you first enable the related notes feature.

Follow these steps:

1. Click Accounting, Configuration.
2. Click Subscription Configuration.
3. Click the Edit icon.
4. Check "Enable Subscription Notes."
5. Click Update Configuration.

The feature has been enabled and you can add notes to a physical subscription.

Add Notes to a Physical Subscription

After you have enabled notes in physical subscriptions, you can add notes to physical subscriptions.

Follow these steps:

1. Click Accounting, Account Management.
2. Click the account name for which you want to add notes.
3. Click the Subscriptions tab.
4. Click the Existing Subscriptions button.
5. Locate and expand the subscription for which you want to add notes.
6. Click the Notes link for the item of interest.
7. Click Save Notes.

The notes are added to the selected physical subscription for the selected account.

Suspend a Physical Subscription

You can *suspend* a physical subscription only for rate items that have a periodic billing cycle. You can cancel a physical subscription for all other rate items.

Follow these steps:

1. Click Accounting, Account Management.
2. Click the account name for which you want to suspend the subscription.
3. Click the Subscriptions tab.
4. Click the Existing Subscriptions button.
5. Locate and expand the subscription that you want to suspend.
6. Click Suspensions.
7. Enter the start and end dates for the suspension or select "Indefinite Suspension."
8. Click Add New.

9. Click Close.

The physical subscription is suspended for the selected account for the period indicated.

Cancel a Physical Subscription

You can *cancel* a physical subscription for all rate items except ones that have a periodic billing cycle. For such rate items, you can suspend a physical subscription.

Follow these steps:

1. Click Accounting, Account Management.
2. Click the account name for which you want to cancel a subscription.
3. Click the Subscriptions tab.
4. Click Existing Subscriptions.
5. Remove the subscription for the service.
6. Click Save.
   A message appears asking you to confirm the subscription changes.
7. Confirm or cancel the changes.
8. Verify that the physical subscriptions were successfully updated.

You can set subscription options that apply to services, service option groups, and service option elements. By default, these settings apply to all accounts. In addition, however, you can specify custom settings for one or more selected accounts.

Follow these steps:

1. Click Accounting, Configuration.
   The Accounting Configuration page appears. On the left, in the Menu under the main menu, Options is selected by default. On the right, links appear for each category (subset) of the accounting configuration options.
2. Review the name of the current business unit. To manage subscription options for a different business unit, change the business unit.
3. Click Subscription Management in the Menu on the left under the main menu.
   The Subscription Decision Tree appears on the right, replacing the links for the accounting configuration categories.
4. Decide whether to update subscription options for all accounts in the business unit or only one more selected account. In either case, decide whether to update subscription options for a specific service, service option group, or service option element.
   To help you decide, review the following order of precedence:
   – A setting for a service option element overrides the setting for the service option group to which it belongs.
   – A setting for a service option group overrides the setting for the service to which it belongs.
   – A setting for a specific account overrides the corresponding setting under “All Accounts.”
5. Perform *one* of the following actions:
   – Click All Accounts to apply your updates to every account in the business unit.
     The Services and Service Option Groups nodes appear, under all accounts.
   – Click Apply to Individual Account to apply the updates to only one or more selected accounts in the business unit.
     The Select Accounts dialog appears. Select the account or accounts you want and click OK. The Services and Service Option Groups nodes appear, under the selected accounts.
6. Click the Service or Services node (whichever you want to update), under either All Accounts or a selected account.
7. Drill down the Services and Service Option Group node to reach the subscription management options of interest. You can update the options for services, service option groups, and service option elements.

NOTE
Optionally specify *no* value for Bill In Advance field or the Effective Dates And Bill Dates field when you create an account. In that case, the system applies default settings. So, the account does *not* appear in the Subscription Decision tree. For information about creating accounts, see the Manage Accounts section.

8. Update these options as needed and save your changes.

You have managed subscriptions. These settings apply to any sub business units of the business unit that you are updating. If a business unit has sub business units, it has its own configuration settings.

Subscription Management Options

To manage subscriptions, you specify the subscription management options, using the Subscription Decision Tree. Specify separate settings that apply to all either *all* accounts in the business unit or to selected accounts *only*. In both cases, use the tree to set the following options for individual services, service option groups, or service option elements:

- **Effective Dates and Bill Dates**
  Specifies when billing begins after an account subscribes to a service option element. Select *one* of the following options:
  - **Bill Date begins on the Effective Date**
    Begins billing the account on the date the subscription is made.
  - **Specify a Bill Date separate from the Effective Date**
    Begins billing the account on a specific date, regardless of the subscription date.
  - **Specify a Bill Date relative to the Effective Date**
    Begins billing the account that is based on the *complex* option you specify, relative to the subscription date. An example is “the first day of the month of the subscription.”
  - **Specify a Bill Date offset from the Effective Date**
    Begins billing the account that is based on the "before and after" option you specify, relative to the subscription date. An example is “two months after the subscription date.”
- **Bill in Advance**
  Specifies whether a subscribing account is charged in advance of a service being rendered. Select *one* of the following options:
  - **Do not bill in advance**
    Bills the account when the service is rendered.
  - **Specify how far in advance the charge should be billed**
    Bills the account the specified number of periods in advance.

  The billing period is the billing period of the service option element being subscribe*d t*o. For example, suppose that a service option element specifies a Unit Cost of $10 with a Billing Cycle of Periodic, a Periodic Type of Monthly, and a Periodic Type Interval of 1. If this setting is 3, then a subscribing account is billed $30 on the first invoice and each interval invoice, rather than $10 per month.

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- A request subscription (R) is canceled when you cancel the request for an account.
- A Data Mediation logical subscription (D) is not shown when an administrator unloads or aggregates usage data again.

Administrators manage subscriptions by selecting Accounting, Account Management, clicking an account name, and clicking the Subscriptions tab. The subscription process is primarily an administrator task. But end users are given read access to active subscriptions for their accounts.

The Administration, Configuration, Subscription Configuration menu option enables Administrators to change the way future subscriptions of an account are handled.

{"URL":["/usm/wpf?Node=iclaunchpad.subscription_management"],"customLabelGetStarted":"Manage Subscriptions","description":"IPCE_Manage_Subscriptions"}

Manage Exchange Rates

Service Accounting Component supplies the Accounting, Configuration, Exchange Rates table. By default, this table displays exchange rates for the current time period. You can also view any previously defined exchange rates for earlier or future periods.

Using this table, you manage exchange rates for the currencies that are used in your implementation. The exchange rates are relative to the US dollar. The Catalog system uses these exchange rates to produce invoices for each account. The invoices are produced in the currency of the business unit to which the account belongs.

All exchange rates are relative to the US dollar. When you convert from one currency type to another, the system first converts the catalog amount to US dollars. Next, the system converts the amount to the currency type of the account.

By default, the exchange rates for all currency types are set to 1.0. This setting means that the exchange rate for each currency type relative to the US dollar is 1.0.

Updates to exchange rates apply to the *entire* system. You *cannot* specify different exchange rates for different business units.

You can update exchange rates in Service Accounting Component when the official exchange rates change. Updates to exchange rates apply to the *entire* system. You *cannot* specify different exchange rates for different business units.

Follow these steps:

1. Click Accounting, Configuration.
2. Click Exchange Rates in the Menu on the left under the main menu.

The Exchange Rates table for the current date range appears on the right. The table lists the currency name, symbol, date, exchange rate, and so forth. By default, The table for the current time period (the previous month) appears.

3. View the exchange rates for the currency that is defined in your system.
4. (Optional) Edit exchange rates for the current time period to provide a regularly scheduled update.
5. (Optional) Delete an exchange rate for a date range.

   WARNING

   Deleting *all* exchange rate entries for a currency name deletes that currency name from the Exchange Rates table completely! You *cannot* use the GUI to restore a deleted currency name. Therefore, use caution when deleting exchange rates from the table.
6. (Optional) Delete a currency name from the exchange rate table if your implementation does not need the currency.

   NOTE

   Delete currency names with caution, because you *cannot* use the GUI to add a missing or deleted currency name.
7. (Optional) Add an exchange rate for a date range to provide a periodic update or to prepare for the start of the next billing period.

   NOTE

   You *cannot* add a new exchange rate that overlaps the same date range as any existing exchange rate.

Service Accounting Component supplies the Accounting, Configuration, Exchange Rates table. By default, this table displays exchange rates for the current time period. You can also view any previously defined exchange rates for earlier or future periods.

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You can update exchange rates in Service Accounting Component when the official exchange rates change. Updates to exchange rates apply to the *entire* system. You *cannot* specify different exchange rates for different business units.

Fiscal Periods can be user-defined for any time period to represent an accounting period. Service Accounting Component allows for the definition of monthly, quarterly, or yearly fiscal periods. Once defined, these periods are used in Budget and Planning worksheets when defining budgets or setting costs for services. These periods are also used for data mediation when selecting a period for aggregation.

By default, you create monthly fiscal periods for the current year, which is based on your installation date. You also use fiscal periods to view Data Mediation aggregations and to manage budgeting and planning.

You cannot create different fiscal periods for different business units. Only one set of fiscal periods applies to all business units.

You can add and define monthly, quarterly, or yearly fiscal periods.

Follow these steps:

1. Click Add.
   The Add New Fiscal Window opens.

2. Enter the fields and click OK.
   The new fiscal period is defined.

{"URL":["/usm/wpf?Node=iclaunchpad.exchange_rates","/usm/wpf?Node=iclaunchpad.fiscal_periods","/usm/wpf?Node=icguinode.billbudgetdefmodal"],"customLabelGetStarted":"Manage Exchange Rates","description":"IPCE_Manage_Exchange_Rates","customCards":[{"id":"IPCE_Manage_Fiscal_Periods","type":"manage","title":"Manage Fiscal Periods"},{"id":"IPCE_Add","type":"manage","title":"Add New Fiscal Period"}]}

Manage Budgets and Plans

This article contains the following topics:

Implementing a successful budgeting process is a primary key to achieve organizational business objectives. Budgets help IT managers evaluate the performance of a business unit though comparison analysis of actual expenditures against budgeted amounts. These findings can aid in justifying IT costs or identifying why expectations have not been met. In addition, this information can be used to establish new goals and discovering the means to reach them through planning.

Through the Budget and Planning module, a financial manager can set up period budgets for business units and their services. At the end of each fiscal period, a variance report evaluates the differences between actual and budgeted costs. This information can be then used to help identify the root cause for areas of inefficiency.

Manage Fiscal Periods

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By default, you create monthly fiscal periods for the current year, which is based on your installation date. You also use fiscal periods to view Data Mediation aggregations and to manage budgeting and planning.

You cannot create different fiscal periods for different business units. *Only one* set of fiscal periods applies to all business units.

Use Worksheets

Worksheets are used to capture budget and cost totals for services. A Service Worksheet can be used to set budgets, unit cost, or total cost for services across fiscal periods. Costs can also be pooled. Then, use the Service Worksheet to set service costs by tracing pooled activities to the services that consume them. A Pool Worksheet is used to establish these pooled costs. A cost pool is comprised of contributing cost elements whose sum makes up the total cost of the pooled activity. A percentage of this total can then be allocated among the services in the Service Worksheet.

Follow these steps:

1. Click Accounting, Budget and Planning, Worksheets.
2. Click Change Business Unit and select a different business unit, if necessary.
3. Complete the Worksheet Options:
4. Click View Worksheet.

The services that are presented in the worksheet contain any chargeable service option group elements whose Budget option is selected. For the elements that employ a Cost Type of either *Allocated Cost* or *Standard Cost*, the values that are entered in the worksheet determine the cost of the service element.

For the *Allocated Cost* setting, the value represents the total cost of the service element to be allocated back to Business Unit accounts. The associated Allocation Method specification determines the distribution of that cost. For example,

suppose that the method is set to Distribute by Subscription. Then, the total cost of the chargeable element is divided by the number of subscriptions to it.

The *Standard Cost* option, is used to define a predetermined unit cost for the service option group element. This cost can be used to assign a rate to a service that can vary from period to period, or as the expected costs defined in a budget. For example, suppose that you have to pre-define a set of unit costs for a particular period and then change. Worksheets are tools that enable these service rates to change over time. Worksheets allow the definition of budgets for service during specified fiscal periods. The values that are defined in this instance are not used in the cost calculation of a service option group element. But these values are used instead in reporting variance between budgeted and actual amounts.

Create a Set

Sets enable you to define related financial and quantitative values. For example, you can use sets to represent various types of budgets, especially when you are initially creating these budgets. You can also link costs to services. Sets can be defined to represent a unit cost or total cost of a service. These types of sets are typically associated with a service during service definition when you employ a Cost Type of *Allocated Cost* or *Standard Cost.*

Follow these steps:

1. Click Accounting, Budget and Planning and click Sets.
2. Click the Add Set button.
3. Complete the following information:
   - **Name**
     Specifies the set name.
   - **Description**
     Indicates the optional text area for a description.
   - **Status**
     Specifies whether the associated worksheet values can be modified. Choices include Locked or Unlocked. Once a set is locked, values are read-only in the worksheets until the Status is changed to Unlocked.
   - **Source**
     Provides for further classification of the associated values. Choices include:
     - Allocated Cost - used when defining the total cost of a chargeable service. Once this type of set is bound to a service, an associated Allocation Method can be selected to employ a means of distributing the cost.
     - Standard Cost - used when defining the total unit cost of a chargeable service. Once this type of set is bound to a service, a predetermined service cost can be set for each period.
     - Actual Units - used to attribute unit quantities to services.
4. Click Add New Set.

Create a Cost Element

A Cost Element is a resource that is used to subdivide costs corresponding to the consumption of a particular service. The amount is paid for a resource that an activity consumes and is included in a cost pool.

Follow these steps:

1. Click Accounting, Budget and Planning, Cost Elements.
2. Click the Add Cost Element button.
3. Complete the fields for this cost element.
4. Click the Update button.

You have created a cost element.

Create a Cost Pool

A Cost Pool is the grouping of all cost elements that are associated with an activity carried out by an entity. The cost pool can be used to identify the cost of any main activities in providing a service.

Follow these steps:

1. Click Accounting, Budget and Planning and click Cost Pools.
2. Click the Add Cost Pool Button.
3. Complete the fields on the dialog:
   - **Name**
     Specifies the cost pool name.
   - **Type**
     Specifies how Cost Pool totals apply to service costs.
4. Click Add New Pool button.

Assign Cost Elements to Cost Pools

Once the main cost pools have been defined, a total cost of each cost pool can be calculated. First, the activities that are related to each cost pool are identified and assigned to the appropriate cost pool. To trace the expenses to each cost pool, identify the cost drivers for each cost element.

Follow these steps:

1. Go to Accounting, Budget and Planning, Cost Elements.
2. Select the Cost Elements to assign to a cost pool.
3. Click the Assign to Cost Pools button.
4. Assign the pools with cost elements by selecting the corresponding check-box.
5. Click the Save Assignments button.

Assign Activity Based Costing to Services

Activity Based Costing (ABC) is a process of assigning costs to services and is used as a tool for planning. The method aims initially to identify and categorize the main operating activities within an organization. Once defined, the costs attributable to these activities are assigned to cost pools. The cost pools are then allocated back to services based on how the service consumes the activity. The steps that follow outline the ABC process:

- Typically, an organization can be broken down into a set of succinct activities that accomplish business processes. The cost pools are created to assemble all cost elements that are related to each of these activities.
- Once the activities have been identified, the principle causes of the costs for each activity are identified. These expenses are categorized by establishing cost elements.
- Add the cost elements. Then, ensure that the cost elements are related to their corresponding cost pools through cost pool assignment.
- The Pool Worksheet applies values to the cost pools based on the expense attributed to each associated cost element. Go to Accounting, Budget and Planning, Worksheets and use the Pool Worksheet option. Then apply these values.
- The Service Worksheet is used as activities are traced to services. The associated pooled costs are allocated to services based on their level of consumption of activities. The Service Worksheet is used to capture these amounts, which can be fixed or a percentage of the pool total.
- Only services that contain chargeable service option group elements are eligible to appear on the Service Worksheet. Check the Budget properly in the service option group element definition dialog when defining a service. In addition to having the worksheet values applied in invoicing calculations, a cost type of *Allocated Cost* or *Standard Cost* must be selected for the service element.
- Invoices are generated for accounts that are subscribed to these services. These invoices show an expandable breakdown of services by cost pools and cost elements.

Implementing a successful budgeting process is a primary key to achieve organizational business objectives. Budgets help IT managers evaluate the performance of a business unit though comparison analysis of actual expenditures against budgeted amounts. These findings can aid in justifying IT costs or identifying why expectations have not been met. In addition, this information can be used to establish new goals and discovering the means to reach them through planning.

Through the Budget and Planning module, a financial manager can set up period budgets for business units and their services. At the end of each fiscal period, a variance report evaluates the differences between actual and budgeted costs. This information can be then used to help identify the root cause for areas of inefficiency.

Worksheets are used to capture budget and cost totals for services. A Service Worksheet can be used to set budgets, unit cost, or total cost for services across fiscal periods. Costs can also be pooled. Then, use the Service Worksheet to set service costs by tracing pooled activities to the services that consume them. A Pool Worksheet is used to establish these pooled costs. A cost pool is comprised of contributing cost elements whose sum makes up the total cost of the pooled activity. A percentage of this total can then be allocated among the services in the Service Worksheet.

You can view worksheets and the services present in the worksheet.

Follow these steps:

1. Click Accounting, Budget and Planning, Worksheets.
2. Click Change Business Unit and select a different business unit, if necessary.
3. Complete the Worksheet Options:
4. Click View Worksheet.

The services that are presented in the worksheet contain any chargeable service option group elements whose Budget option is selected. For the elements that employ a Cost Type of either *Allocated Cost* or *Standard Cost*, the values that are entered in the worksheet determine the cost of the service element.

For the *Allocated Cost* setting, the value represents the total cost of the service element to be allocated back to Business Unit accounts. The associated Allocation Method specification determines the distribution of that cost. For example, suppose that the method is set to Distribute by Subscription. Then, the total cost of the chargeable element is divided by the number of subscriptions to it.

The *Standard Cost* option, is used to define a predetermined unit cost for the service option group element. This cost can be used to assign a rate to a service that can vary from period to period, or as the expected costs defined in a budget. For example, suppose that you have to pre-define a set of unit costs for a particular period and then change. Worksheets are tools that enable these service rates to change over time. Worksheets allow the definition of budgets for service during specified fiscal periods. The values that are defined in this instance are not used in the cost calculation of a service option group element. But these values are used instead in reporting variance between budgeted and actual amounts.

Sets enable you to define related financial and quantitative values. For example, you can use sets to represent various types of budgets, especially when you are initially creating these budgets. You can also link costs to services. Sets can be defined to represent a unit cost or total cost of a service. These types of sets are typically associated with a service during service definition when you employ a Cost Type of *Allocated Cost* or *Standard Cost.*

You can add define a set to represent a budget.

Follow these steps:

1. Click Accounting, Budget and Planning and click Sets.
2. Click the Add Set button.
3. Complete the following information:
   - **Name**
     Specifies the set name.
   - **Description**
     Indicates the optional text area for a description.
   - **Status**

Specifies whether the associated worksheet values can be modified. Choices include Locked or Unlocked. Once a set is locked, values are read-only in the worksheets until the Status is changed to Unlocked.

- **Source**
  Provides for further classification of the associated values. Choices include:
  - Allocated Cost - used when defining the total cost of a chargeable service. Once this type of set is bound to a service, an associated Allocation Method can be selected to employ a means of distributing the cost.
  - Standard Cost - used when defining the total unit cost of a chargeable service. Once this type of set is bound to a service, a predetermined service cost can be set for each period.
  - Actual Units - used to attribute unit quantities to services.

4. Click Add New Set.

A Cost Element is a resource that is used to subdivide costs corresponding to the consumption of a particular service. The amount is paid for a resource that an activity consumes and is included in a cost pool.

Once the main cost pools have been defined, a total cost of each cost pool can be calculated. First, the activities that are related to each cost pool are identified and assigned to the appropriate cost pool. To trace the expenses to each cost pool, identify the cost drivers for each cost element.

Follow these steps:

1. Go to Accounting, Budget and Planning, Cost Elements.
2. Select the Cost Elements to assign to a cost pool.
3. Click the Assign to Cost Pools button.
4. Assign the pools with cost elements by selecting the corresponding check-box.
5. Click the Save Assignments button.

You can add a cost element.

Follow these steps:

1. Click Accounting, Budget and Planning, Cost Elements.
2. Click the Add Cost Element button.
3. Complete the fields for this cost element.
4. Click the Update button.

You have created a cost element.

A Cost Pool is the grouping of all cost elements that are associated with an activity carried out by an entity. The cost pool can be used to identify the cost of any main activities in providing a service.

You can add a cost pool.

Follow these steps:

1. Click Accounting, Budget and Planning and click Cost Pools.
2. Click the Add Cost Pool Button.
3. Complete the fields on the dialog:
   - **Name**
     Specifies the cost pool name.
   - **Type**
     Specifies how Cost Pool totals apply to service costs.
4. Click Add New Pool button.

{"URL":["/usm/wpf?Node=iclaunchpad.budgets","/usm/wpf?Node=iclaunchpad.planning_worksheets","/usm/wpf?Node=iclaunchpad.planning_sets","/usm/wpf?Node=icguinode.billbudgetsetsmodal","/usm/wpf?Node=iclaunchpad.planning_elements","/usm/wpf?Node=icguinode.billbudgetpoolsassign&Args=ca.com&Args=&Args=9981!9984","/

usm/wpf?Node=icguinode.billrpbuilddefmodal&ceview=true&title=Cost%20Element%20Definition%20Dialog&tid=ca.com","/usm/wpf?Node=iclaunchpad.planning_pools","/usm/wpf?Node=icguinode.billbudgetpoolsmodal"],"customLabelGetStarted":"Budget and Planning","description":"IPCE_Budget_and_Planning","customCards":[{"id":"IPCE_Worksheets","type":"manage","title":"Worksheets"},{"id":"IPCE_View_Worksheet","type":"manage","title":"View Worksheet"},{"id":"IPCE_Sets","type":"manage","title":"Sets"},{"id":"IPCE_Add_Set","type":"manage","title":"Add Set"},{"id":"IPCE_Cost_Elements","type":"manage","title":"Cost Elements"},{"id":"IPCE_Assign_to_Cost_Pools","type":"manage","title":"Assign to Cost Pools"},{"id":"IPCE_Add_Cost_Element","type":"manage","title":"Add Cost Element"},{"id":"IPCE_Cost_Pools","type":"manage","title":"Cost Pools"},{"id":"IPCE_Add_Cost_Pool","type":"manage","title":"Add Cost Pool"}]}

Manage General Adjustments

Adjustments are credits or debits applied to services, individual charges of a service option group or service option element, and SLA violations. These adjustments are a fixed dollar amount or a percentage. Applying an adjustment to a business unit applies to all accounts within the selected business unit.

The General Adjustment properties appear on the Adjustment Information window.

- **Adjustment For**
  Specifies the account number of the account being adjusted.
- **Adjustment Type**
  Specifies whether the adjustment is a fixed amount or a percentage. More fields appear depending on the option you select in **Adjustment Type**.

Administrators can apply general adjustments either globally to all accounts or individually to a specific account. Applying general adjustments globally are required when a general adjustment applies to all accounts. For example, global credit to all accounts to cover an accidental overcharge of a specific fee in an earlier bill.

Follow these steps:

1. Select Accounting, Adjustments.
2. Select General from the side menu.
3. Click the Add General Adjustment button.
4. Complete the fields.
5. Click OK.

If only one account requires general adjustments, use the following procedure to apply the adjustments to the account.

Follow these steps:

1. Select Accounting, Account Management.
   Click the Add Adjustments icon (+/-) associated with the account.
2. Complete the fields.
3. Click OK.

Adjustments are credits or debits applied to services, individual charges of a service option group or service option element, and SLA violations. These adjustments are a fixed dollar amount or a percentage. Applying an adjustment to a business unit applies to all accounts within the selected business unit.

The General Adjustment properties appear on the Adjustment Information window.

- **Adjustment For**
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Specifies whether the adjustment is a fixed amount or a percentage. More fields appear depending on the option you select in **Adjustment Type**.

Administrators can apply general adjustments either globally to all accounts or individually to a specific account. Applying general adjustments globally are required when a general adjustment applies to all accounts. For example, global credit to all accounts to cover an accidental overcharge of a specific fee in an earlier bill.

You can add a general adjustment to a specific account and apply as required.

Follow these steps:

1. Select Accounting, Adjustments.
2. Select General from the side menu.
3. Click the Add General Adjustment button.
4. Complete the fields.
5. Click OK.

If only one account requires general adjustments, use the following procedure to apply the adjustments to the account.

Follow these steps:

1. Select Accounting, Account Management.
   Click the Add Adjustments icon (+/-) associated with the account.
2. Complete the fields.
3. Click OK.

{"URL":["/usm/wpf?Node=iclaunchpad.adjustments","/usm/wpf?Node=iclaunchpad.general_adjustments","/usm/wpf?Node=icguinode.billingaddadjustment&ObjectID=ca.com"],"customLabelGetStarted":"Adjustments","description":"IPCE_Adjustmen
[{"id":"IPCE_General","type":"manage","title":"General"},
{"id":"IPCE_Add_General_Adjustment","type":"manage","title":"Add General Adjustment"}]}

Manage Data Mediation

This article contains the following topics:

Data Mediation lets you import usage event data from various external sources. Data Mediation uses the process of ETL (Extraction, Transform, and Load). You can use this process to transform operational data into event data to support billing, and reporting. After the data has been imported, other CA Service Catalog components can use this data. Data Mediation is useful when batch operational data from a source other than CA Service Catalog is available. Also, the requirements allow for historical rather than real-time data.

Data Mediation supports the following types of data feeds:

- Delimiter Separated File
- Fixed-Length File
- Database Import includes Database Table Import and Advanced Database Import.

When the external usage data is imported through Data Mediation as File Type data feeds, it is transformed into a flat file. The file is uploaded to the \datamediation subdirectory of the filestore location. By default, the filestore location is the USM_HOME\FileStore\ directory of the Catalog Component computer that you are using. To view the location of the imported flat files, select Administration, Configuration, File Store Information. Use the filestore location to find and view the flat files in the datamediation subdirectory. The value can be a local directory, or it a directory on another Catalog Component computer.

> **WARNING**
> The folder name *FileStore* is case-sensitive. Therefore, use the correct case in path names and all other programmatic references.

Depending on the completeness of the usage data, the data is loaded into the database into a *reference table*, a *temporary data table, or* a Data Mediation *event table*. Define the mapping between external data feed fields to database field types before importing data. The field definitions also contain the Driver Validation Rules which are enforced upon data import and filter irrelevant data.

Data Summary

During Data Mediation design, determine where to summarize data. For example, if your business objective is to measure average CPU usage over a month for reporting. If the data source system has running information and can determine the average over the period, then use a single, summarized data point rather than a data series of five-second samplings over the same time period. This data point serves to use computer and human resources (CPU, bandwidth, maintenance) efficiently. The best practice is to summarize information early in the process.

Data Aggregation

Data Aggregation is the gathering and expressing of data in a summary form. Data from the reference tables or temporary data tables are normalized into a set of Data Mediation event tables.

Aggregation processes one event at a time. Normalization or initial aggregation is customized using a SQL query and is specified in the Data Mediation profile. Data that is stored in Data Mediation event tables represents complete usage event data. All data from Data Mediation event tables is aggregated. The results are loaded into the set of metric-based result tables. These metric requests are processed for the Service Accounting Component rating engine. Budgeting information and billing transactions are created based on the resulting data that is contained within the metric result tables.

Imported usage data by defined profiles is processed for Service Accounting Component through Data Aggregation. The usage data is imported using the Metric Data Profile. Before usage data is aggregated, the data must be normalized into a common format. The common format consists of fields that are known as Server Mandatory fields. Five Default Server Mandatory Fields are required in every record to qualify as complete event usage data.

Service Accounting Component creates billing transactions that are based on the event results. The data that is imported through Data Mediation is aggregated based on the fiscal period. The data can be aggregated for a defined fiscal period or for all fiscal periods. Aggregation produces results by server mandatory fields, for example, by account, service offering, metric, and fiscal period.

When an aggregation is run for a defined fiscal period, the transactions of the previous aggregation are deleted. Then, new transactions are created to reflect the data being imported. This result is due to the transactions being rolled back on re-aggregation. The new transactions are created as part of the aggregating again. Therefore, regenerate the invoices to capture these new transactions, and show the associated charges.

> **NOTE**
> You can use Invoice groups and then generate your invoices in batch mode. You can also bill for two different data sets that are imported at two different times. You can do an aggregation after the first one is imported and then aggregate again after the second data imported. Generate the invoices *after* you verify that all the data for that fiscal period is valid.

The Data Mediation component aggregates data as follows:

- Generates usm_mr_itemp_XXXXXX data from the external files and databases.
- Generates usm_mr_ievent_XXXXXX data from the usm_mr_itemp_XXXXXX data based on the SQL aggregation specified in the Data Mediation profile. The XXXXXX values are sequential numbers that the system generates.
- Generates usm_mr_iresult_YYYY data from the usm_mr_ievent_XXXXXX table that is based on the built-in aggregation logic. The YYYY is the internal metric ID for the event that is specified by the aggregation.
- Generates transactions for the accounts that are based on the usm_mr_iresult_YYYY data that is created.

You can run the aggregation process any number of times for a fiscal period to reflect the latest imported data.

Custom Fields

You can use custom fields to facilitate aggregation logic. For example, business rules can indicate that the processing of rows in a data set varies based on external value.

You can also use custom fields to maintain legacy information as part of a user interface customization. For example, a charge can have a value in an external system that is used to perform cross referencing.

Aggregation Status

The possible data aggregation status values are as follows:

- Load files only -- New data has been imported since the last aggregation. Aggregation is not currently occurring.
  Numeric code: 0
- Importing -- Data is being imported and initial aggregation logic in profiles is being executed. In this phase, any data in reference tables or temporary data tables is being placed into Data Mediation event tables.
  Numeric code: 1
- Aggregating -- Aggregation is being performed. DC Importer is performing aggregation. In this phase, the data in Data Mediation event tables is being stored into the metric results tables.
  Numeric code: 6
- Profile Error -- An error occurred while executing initial aggregation logic contained in profiles. Aggregation is stopped. Go to Profile Management to locate profiles that are in error. Profiles in error are denoted with Error in the Import Status column.
  Numeric code: 8
- Profiles Pending -- A profile has not been defined. Aggregation is stopped. Profiles containing usage event data must be defined.
  Numeric code: 7
- Waiting for CA Service Accounting -- The Service Accounting Component rating engine is creating billing transactions.
  Numeric code: 5
- Aggregation Error -- A severe error occurred during aggregation.
  Numeric code: 4
- Aggregation Complete -- Aggregation completed successfully.
  Numeric code: 3

Data Management

Data Management lets administrators manage data sets imported into the CA Service Catalog by using the following functions:

- **Unload Checked**
  Unloads (removes) the data from the data mediation tables. The associated charges are removed the next time aggregation occurs.

  NOTE
  You do not need to roll back the invoices that used the data that this data set provides before unloading and aggregating again,
- **Re-import Checked**
  Reloads imported data sets and effectively resets the aggregation flag of the imported data set to "not aggregated." The associated charges are recalculated the next time aggregation occurs.
- **Archive Checked**
  Archives the selected data set. The status of the data set is changed to Archived.

  NOTE
  Archived data sets cannot be unloaded or re-imported after archiving. Use archiving only if required.

NOTE
Aggregation results for a period appear on the Aggregation Status page, although data for that period has been unloaded and aggregated again.

Reference Data

Reference data is often required to facilitate the transformation process. An example is the need to translate account codes from a legacy system to match accounts in CA Service Catalog. In such a case, you can match accounts in the following ways:

- You can customize the user interface to add a field for the legacy account code in CA Service Catalog. This option is best to use when the mapping information is static and can be maintained through data entry.
- You can use a reference table. This option is required if the mapping data is dynamic and it is maintained outside of CA Service Catalog.

Data Import

After creating a data mediation profile and defining the data feed and structure of the external data, import the data. You can import database profiles as needed and can schedule them to occur later.

You can import data from a file or from an Ingres, SQL Server, or Oracle database table. Data is loaded into database tables when imported.

The database tables that the Data Mediation creates include the following types:

- **Reference tables**
  Contains reference data to be used as lookup tables (usm_mr_iref).
- **Temporary tables**
  Contains partial usage event data, meaning that all server mandatory fields are not present, implying that normalization is necessary (usm_mr_itemp).
- **Event tables**
  Contains complete usage event data (usm_mr_ievent).
- **Metric result tables**
  Contains aggregated event data (usm_mr_iresult).

Data Profiles

Every data profile has one data source. The data profile contains a definition of the data feed of the external data. The profile also contains the initial aggregation logic in the form of a SQL statement. This initial aggregation logic is specified during profile definition. A data source can consist of an ASCII file (fixed-length, or delimited) or a database query. For database queries, multiple physical database resources (tables, databases) can be combined into a single result set. The resulting set is considered a single data source.

One set of fields exist per profile. Consider the size of the returned dataset. Determine how to deal with portions of the total data so that you can perform loads, unloads, and aggregations in smaller sets.

Unique initial aggregation logic (normalizing and aggregating raw data) indicates a unique profile. However, given the same data source and fields you can copy an existing profile to create a profile. You do not have to redefine the source information. For example, consider the situation when given a single data source, aggregation logic varies based on some field value.

The types of profiles for Data Mediation are as follows:

- Metric Data -- Usage event data for the cost allocation process of Service Accounting Component. This type of profile loads data into temporary or event tables that are based on the completeness of the usage event data.
- Reference Lookup Data -- Data contains reference data for lookup purposes. Data is loaded into reference tables.

If necessary, usage event data can be normalized or aggregated, preparing it for the aggregation process and cost allocation process. To do so, use an SQL query or procedure that is defined in the profile.

Repository Agent

The CA Service Repository Agent automates the process of importing usage data. The usage data is stored in Delimiter Separated File or Fixed-Length File format. The CA Service Repository Agent is also known as Data Mediation Data Repository Agent. The repository agent is installed as a service named CA Service Repository Agent during the installation of Catalog Component.

The Repository Agent imports usage data files into the database in the following ways:

- The agent can be configured to read files from an FTP server. The files are automatically copied from the FTP server locally and processed based on associated Data Mediation profiles.
- An external system can locally copy the usage data files. The usage data files are then processed based on associated Data Mediation profiles.
- A Data Mediation profile can be generated externally as an XML file. The XML file can then automatically be loaded and used to process a corresponding data file where both the profile definition XML file and the data file are copied locally by an external system.

  NOTE
  The integration with MICS uses this technique.

Data Import Frequency

Decide the frequency of data imports. To determine the frequency, consider your reporting requirements.

For example, consider the following situations:

- If an invoice, an SLA report, a metric report, or a custom report is run daily, then the Data Mediation feed frequency must also be daily.
- If the data is required for a monthly bill run, then a monthly feed can be appropriate.
- If millions of records are loaded monthly, import subsets daily to spread the processing. You can also minimize the end-of-month window. Another benefit of this approach is to avoid having to unload and reload larger data sets when changes are necessary. The size of the data set does not affect the commit size in the database tables. The best practice is to perform commits in small batches, rather than after the entire data set. This approach minimizes lock contention and log file growth.

Data Mediation lets you import usage event data from various external sources. Data Mediation uses the process of ETL (Extraction, Transform, and Load). You can use this process to transform operational data into event data to support billing, and reporting. After the data has been imported, other CA Service Catalog components can use this data. Data Mediation is useful when batch operational data from a source other than CA Service Catalog is available. Also, the requirements allow for historical rather than real-time data.

Data Mediation supports the following types of data feeds:

- Delimiter Separated File
- Fixed-Length File
- Database Import includes Database Table Import and Advanced Database Import.

When the external usage data is imported through Data Mediation as File Type data feeds, it is transformed into a flat file. The file is uploaded to the \datamediation subdirectory of the filestore location. By default, the filestore location is the USM_HOME\FileStore\ directory of the Catalog Component computer that you are using. To view the location of the imported flat files, select Administration, Configuration, File Store Information. Use the filestore location to find and view the flat files in the datamediation subdirectory. The value can be a local directory, or a directory on another Catalog Component computer.

Depending on the completeness of the usage data, the data is loaded into the database into a *reference table*, a *temporary data table, or* a Data Mediation *event table*. Define the mapping between external data feed fields to database field types before importing data. The field definitions also contain the Driver Validation Rules which are enforced upon data import and filter irrelevant data.

Data Management lets administrators manage data sets imported into the CA Service Catalog by using the following functions:

- **Unload Checked**
  Unloads (removes) the data from the data mediation tables. The associated charges are removed the next time aggregation occurs.

  NOTE
  You do not need to roll back the invoices that used the data that this data set provides before unloading and aggregating again,
- **Re-import Checked**
  Reloads imported data sets and effectively resets the aggregation flag of the imported data set to "not aggregated." The associated charges are recalculated the next time aggregation occurs.
- **Archive Checked**
  Archives the selected data set. The status of the data set is changed to Archived.

  NOTE
  Archived data sets cannot be unloaded or re-imported after archiving. Use archiving only if necessary.

NOTE
Aggregation results for a period appear on the Aggregation Status page, although data for that period has been unloaded and aggregated again.

{"URL":["/usm/wpf?Node=iclaunchpad.data_mediation"],"customLabelGetStarted":"Data Mediation","description":"IPCE_Data_Mediation","customCards":[{"id":"IPCE_Data_Management","type":"manage","title":"Data Management"}]}

Implement Data Mediation

This article contains the following topics:

Administrators use data mediation functions to import, aggregate, and summarize data from external sources. Administrators use the results to help calculate costs for usage-based billing and to generate reports.

WARNING
As a prerequisite, verify that your fiscal period has been defined to fit your business model.

Step 1 - Define Data Fields

Define the Extraneous fields from an external data feed to map the external data feed field to a database field type. You can also define field definitions when creating a profile. When creating a profile, you use field definitions to define the structure of the external data.

When usage data is imported, it is uploaded into the database. During data import, data validation checks are performed. The invalid records are not uploaded into the database. Before data is imported into the database, you can apply rules to filter out erroneous or irrelevant data. The field definitions hold the definition of these validation rules, specifying a validation rule on each column of data imported.

Follow these steps:

1. Select Accounting, Data Mediation.
2. Click Add to define a new field.
3. Specify the information in the Define a Field window. The following fields need explanation:
   – **Mandatory Types**

- None - indicates that both the server and the client are not mandatory.
- Server Mandatory - indicates that the field is required for data mediation aggregation to occur. The absence of these fields results in a profile error.
- Client Mandatory - includes the field in profile definitions.
- Both - indicates that both the server and client are mandatory.
- **Default:** Server Mandatory

– **Check type**
Specifies the check type, if applicable.
- **Default:** No check

4. (Optional) Specify a driver validation rule:
   – **Not Empty**
   Specifies that the field value must not be empty.
   – **Range**
   Specifies that the field value falls within the specified range.
   – **Lookup**
   Specifies that a field value exists in a database table field.
   – **Lookup Replace**
   Specifies a replacement value, if the value in the field exists in the specified database table field.

   These validation rules are enforced when usage data is imported. If the driver validation rules are violated upon import, a kickoff report is generated. The report is available through the Data Management user interface.
5. Click Submit.
   The data fields are defined.

Default Server Mandatory Fields

Normalize the usage data into a common format that data mediation can understand before it is aggregated. The aggregation results in an error if the following server mandatory fields are not included:

- **CA Service Account Number**
  Associates a data record to a Service Accounting Component account ID.
- **Event ID**
  Associates a data record to an event (Event ID).
- **Event Time**
  Associates a data record to a fiscal period (Time Stamp).
- **Metric Value**
  Associates a data record with a metric value.
- **Service Code**
  Associates a data record with a service offering (Service Offering ID).

  NOTE
  Formatting data to include all server mandatory fields before data import is unnecessary. Data Mediation features complex data mapping mechanisms, such as validation and normalization, which help the administrator include server mandatory fields.

Step 2 - Create a Profile

Profile management is a definition of the data feed of the external data. The profile also contains the initial aggregation logic in the form of an SQL statement.

If necessary, usage event data can be normalized or aggregated, preparing it for the Data Mediation's aggregation and cost allocation processes through a SQL query or procedure defined in the profile. This profile feature helps normalize the data to a common format understood by Data Mediation.

You can create a profile to define the data feed of the external data.

Follow these steps:

1. Click Accounting, Data Mediation, Profile Management.
2. Click Add New Profile.
3. Select the Data Type:
   - **Metric Data**
     Specifies the usage event data for the cost allocation process. This type of profile loads data into temporary or event tables based on the completeness of the usage event data.
   - **Reference Lookup Data**
     Data contains reference data for lookup purposes. Data is loaded into the reference tables for data mediation.
4. Select the Import Format.
5. Click the Define Fields button.
   A profile template window appears for field definitions.
6. Enter a unique name for the profile you are creating. Define the fields for the profile.
7. Click Save.
   The profile is created.

   NOTE

   You can optionally edit or delete profiles. But you cannot edit profiles after they have been aggregated. If the Import Status is “Processed,” you cannot edit the profile. You cannot delete data profiles after they have been aggregated.

Step 3 - Create an Application Metric

The Data Mediation component uses metric packages to measure usage of an application.

Follow these steps:

1. Add, edit, or delete event information, as follows:
   a. Select Accounting, Data Mediation, Application Metric, Event.
      The Event List window appears.
   b. Click Add to create an event or click Edit to modify an existing event.
      The Event Detail window appears.
   c. Specify all details about the event and click Save.
2. Add, edit, or delete metric information, as follows:
   a. Select Accounting, Data Mediation, Application Metric, Metric.
      The Metric List window appears.
   b. Click Add to create a metric or click Edit to modify an existing metric.
      The Metric Detail window appears.
   c. Specify all details about the metric and click Save.
   d. Link or unlink related events, as described in the next step.
3. Link or unlink related events for a metric, as follows:
   a. Ensure that at least one event is associated with each metric.
   b. Click Associate Event.
      The Metric Name: *metric name* window appears, displaying all events that are not already associated with the metric.
   c. Select the events that you want to associate with the metric.
   d. Click Associate Event. The selected events are linked and therefore appear in the Associated Events list.
   e. Select the events that you want to disassociate and click Disassociate Event.
4. Add, edit, or delete application information, as follows:
   a. Select Accounting, Data Mediation, Application Metric, Application.

The Application Detail window appears.

b. Click Add to create an application or click Edit to modify an existing application.
The Application Detail window appears.

c. Specify all details about the application and click Save.

d. Link or unlink related metrics, as described in the next step.

5. Link or unlink related metrics for an application, as follows:
   a. Ensure that at least one metric is associated with each application.
   b. Click Associate Metric.
   The Application Name: *application name* window appears, displaying all metrics that are not already associated with the application.
   c. Select the metrics that you want to associate with the application.
   d. Click Associate Metric. The selected metrics are linked and therefore appear in the Associated Metrics list.
   e. Select the metrics that you want to disassociate and click Disassociate Metric. The selected metrics are unlinked and therefore no longer appear in the Associated Metrics list.
6. Click Done.

The application metric is created.

Step 4 - Import the Data

You can import data by using Data Mediation.

Follow these steps:

1. Click Accounting, Data Mediation, Data Management.
2. Click Import Data.
3. Select the appropriate Data Type of the profile.
4. Select the import format.
5. Select the appropriate import profile for the data to be imported.

   NOTE

   You can specify a filter and description.
6. Click Import Now.
7. (Optional) Click Schedule Import to automate the process of importing usage data.
   The scheduling information is already populated. The administrator only schedules the task by entering an optional Comment and the required fields. Do not modify fields other than Comment and the required fields.

   NOTE

   To edit a scheduled database import, go through the Scheduler user interface and click View Scheduled Tasks.

Configure the CA Service Repository Agent

You can configure the CA Service Repository Agent by modifying the settings in the configuration file USM_HOME \repagent\config\repagent.cfg. Use the CA Service Repository Agent to automate the import of usage data that is stored in Delimiter Separated File or Fixed-Length File format

The repagent.cfg file uses the following parameters:

- **usm.url**
  Specifies the Catalog Component URL.
- **init.pause.ms**
  Specifies the pause time (for initialization) in milliseconds when the repository agent starts up.
- **repeat.interval.ms**

Specifies the polling interval in milliseconds. The Repository Agent polls the FTP server for any new files that are based on this value. This value also determines the polling interval for processing local files.

- **upload.list.file**
  Specifies the profile configuration file name.
- **history.file**
  Specifies the repository agent upload history log file name. The repository agent keeps a log of all files that are imported to the database.
- **ftp.host**
  Specifies the FTP host name.
- **ftp.user**
  Specifies the FTP user name.
- **ftp.password**
  Specifies the FTP password name

Automatic Profile Load with Data Load

If the profile to be used is not already in the database, the repository agent can automatically load the profile and can load the associated data. This approach requires a pair of files in the USM_HOME\repagent\data folder with the same file name except for the extension. The file with the extension of .xml contains the profile definition. The file with the extension of .txt contains the usage data. For example, suppose a profile definition that is named abc.xml exists. The repository agent loads that profile into the database and uses it to process an associated usage data file abc.txt.

The profile definition XML file must follow a certain format. If integrated with CA NeuMICS, CA NeuMICS automatically generates this profile definition. For loading other usage data from other systems, generate the profile XML file according to the supported format.

Profile XML File Format

To generate a profile from an external source, use the format in the sample profile XML file. This file resides in the USM_HOME\repagent\data\samples folder.

The profile XML file that the repository agent uses to create a Data Mediation profile contains two sections:

Profile section
Contains information about the data mediation profile. Only one profile section can exist for each XML file.

- profile_name - the profile name (mandatory).
- profile_type - the profile type where 0=reference, 1=metric (default is 0).
- import_format - the format of the data source where 0=delimiter-separated file, 1=fixed-length file (default is 0).
- field_separator - the delimiter between fields in the usage file. The valid values are either the character itself or the ASCII numeric value for the following characters: ampersand (&), asterisk (*), at sign (@), comma (,), dollar sign ($), exclamation mark (!), percent (%), period (.), pipe (|), or space ( ). The ASCII numeric value for a tab can be used.

Field section
Contains information about each column of the data file. You can use this section to create the fields in the Data Mediation profile. Each XML file can contain many field sections.

- field_name - database table column name (mandatory)
- display_name - display name of the field
- mandatory - mandatory status
  - 0 - both server and client are not mandatory.
  - 1 - server mandatory
  - 2 - client mandatory
  - 3 - both client and server mandatory
- data_type - data type

  - 0 - string
  - 1 - integer
  - 2 - float
  - 3 - date
  - 4 - decimal
  - 5 - double
    A data type of “double” is converted to “float” during import.
  - 6 - binary
- data_length - the length of this field
- data_format - the format of date data (only if data_type =3).
  A dash (-) can replace a slash (/)in the following formats: The separator between the date and time portion can be a slash (/), dash (-) or space. Upper or lower case letters can be used. For example, YYYY-MM-DD hh24:mi:ss is a valid format.
  - MM/DD/YY
  - MM/DD/YYYY
  - MM/DD/YYYY HH:MI:SS
  - MM/DD/YYYY HH24:MI:SS
  - MM/DD/YYYY HH:MI:SS.MSS
  - MM/DD/YYYY HH24:MI:SS.MSS
  - DD/MM/YY
  - DD/MM/YYYY
  - DD/MM/YYYY HH:MI:SS
  - DD/MM/YYYY HH24:MI:SS
  - DD/MM/YYYY HH:MI:SS.MSS
  - DD/MM/YYYY HH24:MI:SS.MSS
  - YY/MM/DD
  - YYYY/MM/DD
  - YYYY/MM/DD HH:MI:SS
  - YYYY/MM/DD HH24:MI:SS
  - YYYY/MM/DD HH:MI:SS.MSS
  - YYYY/MM/DD HH24:MI:SS.MSS
- default_value - the value for this field. The Catalog system uses this value, not the input record.
- start_position - if import_format=1 (fixed-length file), start_position is the starting position of the field in each record, starting at 1. If import_format=0 (delimiter-separated file), it is the field position starting with 1.
  For example:
  - For a fixed-length file, a record contains “abc001” and this field is the numeric portion. Therefore, the start_position is 3, and the end_position is 6.
  - For a delimiter-separated file, a record contains “abc,001” and this field is the numeric portion. Therefore, the start_position is 2, because this field is the second field. Also, the end_position is blank.
- end_position - required only when import_format=1. It is the ending position of a field in each record.
- status - status of the field (default is 1)
  - 0 - system (cannot delete)
  - 1 - active
  - 2 - inactive

Profile List

When a profile is created and the repository agent must process the usage data, the agent must be made aware of the new profile. The profile information is specified in the Profile List file that is named in the upload.list.file setting. This file is located in the USM_HOME\repagent\config folder.

The format of each record in the Profile List file is as follows:

```
profile_table_id, usage_file_name
```

- ***profile_table_id***
  Specifies the numeric ID value from the Data Mediation profile Source Table column, without the leading zeroes. For example, if the Source Table column contains entry usm_mr_itemp_001012, the *profile_table_id* entry in the Profile List file for this profile is 1012.
- ***usage_file_name***
  Specifies the full path and file name for the usage file corresponding to the profile referenced in the first parameter, for example, C:\Program Files\CA\Service Delivery\repagent\data\my_usage_data.txt.

Step 5 - Aggregate the Data

You can aggregate the data.

Follow these steps:

1. Select Accounting, Data Mediation.
2. Select Aggregation Status.
3. Complete the window options:
   - **Use selected period for aggregation**
     Aggregates data that is contained within a specific fiscal period (suggested). Only data falling within the specified date range is aggregated. Not selecting this option aggregates data for all fiscal periods.
   - **Advanced Option**
     Specifies multiple fiscal periods.
4. Click Start Aggregation.
5. (Optional) Click Refresh to check the Aggregation Status during data aggregation.
   The data is aggregated. To view the details, including history, for an invoice, navigate the Aggregation Summary pages for the account of interest. Alternatively, click Accounting, Account Management and click the action icon that is named View Invoice History for account name.

Data Mediation Aggregation Utility

You can start a data mediation aggregation using a command-line utility. For Windows, the startDMAggregation.bat file is located in the USM_HOME\scripts folder.

The syntax for using the startDMAggregation.bat (Windows) command is as follows:

```
startDMAggregation.bat [startdate] [enddate]
```

Specify the start date and end date in MM/DD/YYYY format.

Define Aggregation Logic

You can define the initial aggregation logic in the form of an SQL statement.

Follow these steps:

1. Click Accounting, Data Mediation, Profile Management.
   If the target table column is empty and the configuration status is “pending,” the initial aggregation logic for the profile is not yet defined. The presence of a target table entry and a configuration status of “Defined” indicate that the profile has been defined.

NOTE
The Target Table column is visible when the Show more columns double arrow icon is clicked.

2. Click the Define Aggregation Logic icon.
3. Drag-and-drop each field from the data source into the target table field to define the logic to be used during aggregation.
4. Click Generate Query to generate the SQL Expression.

NOTE
You can define the initial aggregation logic by using a SQL statement or procedure. Specify the initial aggregation logic to prepare the imported event usage data to include all of the server mandatory fields for the aggregation and cost allocation process. For example, suppose that you have usage data for each host in temporary tables for data mediation. You want to map the hosts to accounts using the data mediation reference tables. In that case, perform a join of these tables in the SQL procedure to prepare the event table.

5. Click Save.
The aggregation logic is defined.

Define Multiple Aggregation

Data that is imported from an external source can be aggregated in multiple ways. To fulfill this requirement, you can define multiple profiles that are based on the same data set. This definition lets you use one set of external source data. You can then aggregate the external data in different ways to meet your business requirements.

Follow these steps:

1. Go to Accounting, Data Mediation, Profile Management.
2. Click Copy for the profile for which you want to define a new profile and different aggregation logic.
A copy of the profile using the same name is created using the same source. The SQL Expression is not copied.
3. Click the new profile and define the alternate aggregation logic for this data source. Click Save.
A different target table entry is created.

When data aggregation is called, the SQL query or procedure runs, resulting in two separate event tables. If the data in both tables is normalized properly, both data sets are aggregated successfully.

NOTE
The initial aggregation logic for a profile is run without regard to sequential order.

Step 6 - Generate Invoices

You can use data aggregation to generate invoices in Service Accounting Component. If you generate an invoice after an aggregation and again aggregate the data for the same fiscal period, the previously generated invoice has zero charges. These zero charges are due to the transactions being rolled back on aggregating again. You regenerate invoices to capture the new transactions. You can use invoice groups and can create invoices in batch mode. If two different data sets are involved at two different times, you can aggregate when the first data set is imported and again aggregate after the second data set is imported. You then generate invoices after all the data for that fiscal period has been received and verified.

Follow these steps:

1. Create profiles for each data set to mediate and then aggregate.
2. Import the data (files and external database) for the associated profile.
3. Define fiscal periods depending on the business model.
4. Aggregate data so that the metric result tables and the Service Accounting Component transactions are created.
Repeat the aggregation process if any new data is imported or changed for that fiscal period.
5. After all the data is received and is aggregated, generate the invoices that reflect the aggregated data.

Data Mediation Reports

Several Data Mediation report data views are available:

Data Mediation - Profile List
Displays a list of Data Mediation profiles, including the following associated parameters:

- Index ID
- Profile Name
- Status
- Temp Table
- Event Table
- Data Import Type
- Profile Owner
- Description
- ID

Data Mediation - Rating Engine Queue Items
Displays rating engine queue items by the specified runtime variables. Enter the start and end date and times. The report displays the Data Mediation Rating Engine Queue items, along with the following associated fields:

- Index
- Group ID
- Queue Item ID
- Created Time
- Status
- Start Time
- Finish Time
- Metric Result ID
- Metric Result Table Name

Define the Extraneous fields from an external data feed to map the external data feed field to a database field type. You can also define field definitions when creating a profile. When creating a profile, you use field definitions to define the structure of the external data.

When usage data is imported, it is uploaded into the database. During data import, data validation checks are performed. The invalid records are not uploaded into the database. Before data is imported into the database, you can apply rules to filter out erroneous or irrelevant data. The field definitions hold the definition of these validation rules, specifying a validation rule on each column of data imported.

Follow these steps:

1. Select Accounting, Data Mediation.
2. Click Add to define a new field.
3. Specify the information in the Define a Field window. The following fields need explanation:
   - **Mandatory Types**
     - None - indicates that both the server and the client are not mandatory.
     - Server Mandatory - indicates that the field is required for data mediation aggregation to occur. The absence of these fields results in a profile error.
     - Client Mandatory - includes the field in profile definitions.
     - Both - indicates that both the server and client are mandatory.
     - **Default:** Server Mandatory
   - **Check type**
     Specifies the check type, if applicable.
     - **Default:** No check

4. (Optional) Specify a driver validation rule:
   - **Not Empty**
     Specifies that the field value must not be empty.
   - **Range**
     Specifies that the field value falls within the specified range.
   - **Lookup**
     Specifies that a field value exists in a database table field.
   - **Lookup Replace**
     Specifies a replacement value, if the value in the field exists in the specified database table field.

   These validation rules are enforced when usage data is imported. If the driver validation rules are violated upon import, a kickoff report is generated. The report is available through the Data Management user interface.
5. Click Submit.
   The data fields are defined.

Profile management is a definition of the data feed of the external data. The profile also contains the initial aggregation logic in the form of an SQL statement.

If necessary, usage event data can be normalized or aggregated, preparing it for the Data Mediation's aggregation and cost allocation processes through a SQL query or procedure defined in the profile. This profile feature helps normalize the data to a common format understood by Data Mediation.

You can create a profile to define the data feed of the external data.

Follow these steps:

1. Click Accounting, Data Mediation, Profile Management.
2. Click Add New Profile.
3. Select the Data Type:
   - **Metric Data**
     Specifies the usage event data for the cost allocation process. This type of profile loads data into temporary or event tables based on the completeness of the usage event data.
   - **Reference Lookup Data**
     Data contains reference data for lookup purposes. Data is loaded into the reference tables for data mediation.
4. Select the Import Format.
5. Click the Define Fields button.
   A profile template window appears for field definitions.
6. Enter a unique name for the profile you are creating. Define the fields for the profile.
7. Click Save.
   The profile is created.

   NOTE

   You can optionally edit or delete profiles. But you cannot edit profiles after they have been aggregated. If the Import Status is “Processed,” you cannot edit the profile. You cannot delete data profiles after they have been aggregated.

Data Aggregation is the gathering and expressing of data in a summary form. Data from the reference tables or temporary data tables are normalized into a set of Data Mediation event tables.

Aggregation processes one event at a time. Normalization or initial aggregation is customized using a SQL query and is specified in the Data Mediation profile. Data that is stored in Data Mediation event tables represents complete usage event data. All data from Data Mediation event tables is aggregated. The results are loaded into the set of metric-based result tables. These metric requests are processed for the Service Accounting Component rating engine. Budgeting information and billing transactions are created based on the resulting data that is contained within the metric result tables.

You can aggregate the data.

Follow these steps:

1. Select Accounting, Data Mediation.
2. Select Aggregation Status.
3. Complete the window options:
   - **Use selected period for aggregation**
     Aggregates data that is contained within a specific fiscal period (suggested). Only data falling within the specified date range is aggregated. Not selecting this option aggregates data for all fiscal periods.
   - **Advanced Option**
     Specifies multiple fiscal periods.
4. Click Start Aggregation.
5. (Optional) Click Refresh to check the Aggregation Status during data aggregation.
   The data is aggregated. To view the details, including history, for an invoice, navigate the Aggregation Summary pages for the account of interest. Alternatively, click Accounting, Account Management and click the action icon that is named View Invoice History for account name.

The Data Mediation component uses metric packages to measure usage of an application.

Follow these steps:

1. Add, edit, or delete event information, as follows:
   a. Select Accounting, Data Mediation, Application Metric, Event.
      The Event List window appears.
   b. Click Add to create an event or click Edit to modify an existing event.
      The Event Detail window appears.
   c. Specify all details about the event and click Save.
2. Add, edit, or delete metric information, as follows:
   a. Select Accounting, Data Mediation, Application Metric, Metric.
      The Metric List window appears.
   b. Click Add to create a metric or click Edit to modify an existing metric.
      The Metric Detail window appears.
   c. Specify all details about the metric and click Save.
   d. Link or unlink related events, as described in the next step.
3. Link or unlink related events for a metric, as follows:
   a. Ensure that at least one event is associated with each metric.
   b. Click Associate Event.
      The Metric Name: *metric name* window appears, displaying all events that are not already associated with the metric.
   c. Select the events that you want to associate with the metric.
   d. Click Associate Event. The selected events are linked and therefore appear in the Associated Events list.
   e. Select the events that you want to disassociate and click Disassociate Event.
4. Add, edit, or delete application information, as follows:
   a. Select Accounting, Data Mediation, Application Metric, Application.
      The Application Detail window appears.
   b. Click Add to create an application or click Edit to modify an existing application.
      The Application Detail window appears.
   c. Specify all details about the application and click Save.
   d. Link or unlink related metrics, as described in the next step.
5. Link or unlink related metrics for an application, as follows:
   a. Ensure that at least one metric is associated with each application.

b. Click Associate Metric.
The Application Name: *application name* window appears, displaying all metrics that are not already associated with the application.
c. Select the metrics that you want to associate with the application.
d. Click Associate Metric. The selected metrics are linked and therefore appear in the Associated Metrics list.
e. Select the metrics that you want to disassociate and click Disassociate Metric. The selected metrics are unlinked and therefore no longer appear in the Associated Metrics list.

6. Click Done.

The application metric is created.

{"URL":["/usm/wpf?Node=iclaunchpad.data_mediation"],"customLabelGetStarted":"Field Definition","description":"IPCE_Field_Definition","customCards":[{"id":"IPCE_Profile_Management","type":"manage","title":"Profile Management"},{"id":"IPCE_Aggregation_Status","type":"manage","title":"Aggregation Status"},{"id":"IPCE_Application_Metric","type":"manage","title":"Application Metric"}]}

Manage Chargeback and Pricing Models

This article contains the following topics:

Chargeback

Enterprises can define their services to allocate costs internally for budgeting and chargeback or to charge customers for them. Service Accounting Component supports various chargeback methodologies. Charges can be per transaction, or per session, and can vary by resource. The rates that are used can vary by service package and activity level. When CA Business Service Insight is integrated, the invoices that are based on real-time usage data can be generated. Adjustments can then be applied automatically to reflect any SLA violations.

Under most circumstances, the cost that is charged to a business unit is not disputed. The cost can be directly attributed to IT services that have a role in supporting its business operations. The primary focus is on whether or not the provided services and the corresponding costs are at the levels anticipated. In essence, business units want to be certain that they are getting everything that they are paying for. Business Units want to know how to distribute the burden of cost and how to lower it. Conversely, an organization must verify that all costs are recovered regardless of whether or not resources are fully utilized. The process of distributing the cost must be easily managed. The system that is put into place influences business unit behavior and demands.

Administrators can configure Service Accounting Component to specify chargeback and to use one or more of the following pricing models:

- Subscription-based Pricing
- Usage-based Pricing
- Tiered-based Pricing
- Combined Approach

Subscription-Based Pricing

Chargeback can be done regardless of actual consumption, but on a fixed basis using a specific unit of measure and unit cost. This method is known as subscription-based pricing. This strategy can be implemented by using any chargeable type service option element when defining services for a Business Unit. This element type allows for the specification of a unit cost, billing cycle, and quantity. Examples include subscription charges, such as $500 per month or $1000 per laptop.

This method is used when the true cost driver is unknown or it is too difficult to ascertain. Because the subscription is based on a controllable unit of specification, this approach provides the Business Unit with some control over cost expectation and a clear picture of their IT-related costs. But this method fails to meet any objectives pertaining to fairness.

Disparate resource use is not accounted for, as the treatment of all subscribers is equal. This decoupling of service cost and usage translates into a system that tends to fail to motivate Business Unit behavior. So, subscription-based pricing does not guarantee the recovery of cost or that profit is generated.

Usage-Based Pricing

Usage-based pricing enables IT to charge back to Business Units cost based proportionately to their consumption of resources. This strategy can be implemented by using the Application type service option element and choosing the Usage Based selection for its Pricing Structure when defining services for a Business Unit. This element type allows for the specification of a unit cost and a metric type.

Import the usage-based data into the system using data mediation. Use this metric result to calculate the total cost of that service for an account. In addition to global account charges, these usage-based amounts can be processed on a per user basis.

Implementing usage-based billing involves knowledge and decision at multiple levels within an organization such as:

- Financial decisions determine how to present and price these services.
- IT technical decisions on where and how to collect and process useful usage data and how to handle exceptional cases like adjustments and promotions.

This method assumes that the associated resources are shared among several Business Units and their unit cost is fairly static. Consumption is directly linked to the cost incurred. For example, when services are set to employ weighted distribution as the means of distributing charges, lowering usage levels of these services does not always guarantee lower cost. One Business Unit can decide to cut usage, thus expecting a relative change in the cost that it is allocated. However, if all other Business Units that contend for the same resources also decide to cut back on usage, but at a higher rate, the original Business Unit is forced to pay more than it did previously. Although control can be a problem, this model affects business unit behavior.

Tiered-Based Pricing

Tiered-based pricing enables IT to charge back to business units based on levels of service. A tiered structure can be constructed associating various rates to level ranges. This strategy can be implemented by constructing a tiered type service group to contain these ranges and corresponding rates, and defining services to draw their costing information from them based on a passed value. You can use this value to look up and select the appropriate tiers.

Consider a tiered service group that contains two columns. In the first column, various ranges can be defined, such as numeric values 1-100, 101-1000. In the second column, associated rates can be defined. You can configure several service group elements to use this tiered service group to determine their cost.

For example, an application service group element can be created to use this tiered service group. To do so, select Tiered-Based as its Pricing Structure. During the cost calculation for this element, the metric value can be used to select the matching tier and use the corresponding rates. The unit specification at that rate can be derived in a number of ways, such a predefined fixed quantity or the metric value itself. Agreement service group elements can employ tiered service groups to attribute a cost that is based on the number of violations.

Combined Approach

The combined approach creates a chargeback procedure that is tailored to the needs of a specific organization. For instance, a particular organization can implement a system that influences service objectives by using a tiered-based pricing with assurance adjustments on top of a fixed recurring subscription charge. Here the aim is to strengthen the control a Business Unit has over the quality and equitability of a service, while providing for a great deal of stability and predictability of cost.

A differentiation can exist between services that are bound to the infrastructure of that organization and those that are not. These infrastructure services can be positioned to do chargeback that is based on a specified allocation method as in a ‘per subscription’ basis. Chargeback for the other services can be based on actual consumption using a fixed rate.

Implement Usage-Based Billing

This article contains the following topics:

Step 1 - Verify the Requirements for Services

Administrators can set up usage-based billing to charge for services that are based on consumption. Administrators use data mediation to import usage-based data. Then this data is used to calculate the cost of a service for a business unit, user, or account.

Verify the requirements for the services for which you are implementing usage-based billing. The service option groups being billed must include a service option that is configured as follows:

- Type = Application
- Pricing Structure = Usage Based
- Cost Type = Specify Value
  Do not select User Specified for the Cost Type if you are using usage-based billing through data mediation logical mapping. If you select User Specified for the Cost Type, then users can overwrite the default unit cost. The default unit cost is defined in the original service definition. However, data mediation logical subscription is designed to use the default unit cost that is defined in the service definition.
- Unit Cost = any numeric value
- Application = any available application (for example, Operating System Platform)
- Metric = any available metric (for example, Disk Quota Used (KB) Per User).

Step 2 - Create the Users and Account

To get started implementing usage-based billing, create one or more users and associate them to an account.

> **NOTE**
> If the service activation is to be achieved through request management, then the user is automatically associated to the account.

Step 3 - Create an Application Metric

To create an application metric, see the section "Create an Application Metric" in Implement Data Mediation.

Step 4 - Create Data Mediation Profile

To create a data mediation profile, see the section "Create a profile" in Implement Data Mediation.

Step 5 - Define Aggregation Logic

To define the aggregation logic, see the section "Define Aggregation Logic" in Implement Data Mediation.

Step 6 - Import the Data

You can import database profiles as needed and can schedule them to occur later.

You can import data from a file or from an Ingres, SQL Server, or Oracle database table. Data is loaded into database tables when imported.

The database tables include the following types:

- **Reference tables**
  Contains reference data to be used as lookup tables (usm_mr_iref).
- **Temporary tables**

Contains partial usage event data, meaning that all server mandatory fields are not present, implying that normalization can be necessary (usm_mr_itemp).

- **Event tables**
  Contains complete usage event data (usm_mr_ievent).
- **Metric result tables**
  Contains aggregated event data (usm_mr_iresult).

Step 7 - Aggregate the Data

To aggregate the data, see the section "Aggregate the Data" in Implement Data Mediation.

Mobility

This section contains the following articles:

- PDA Interface
- CA Service Point Mobile App

PDA Interface

CA SDM supports ITIL terminology for the Personal Digital Assistant (PDA) interface. This interface lets end users create Requests, Incidents, and Problems, and lets them search for the following ticket types:

- Incidents
- Problems
- Requests
- Change Orders
- Issues

Analysts can use the PDA interface. This interface honors the role present in the *PDA Role* field of the Access Type of the contact.

The PDA Interface is displayed for on any unsupported device or browser.

Look up service for searching a contact works only on PDA browsers that support multiple tabs and multiple browser windows. Multi-tenancy works similar to the browser interface, but the drop-down list to select multi-tenancy is not provided. When you create the ticket, tenancy is based on the requestor, the affected enduser, and the analyst.

Automatic Priority Calculator (APC) helps in deciding priority while creating Incident/Problem, based on the attributes, urgency, impact, request area, and affected end user.

NOTE

The PDA Interface does not support attachments when you create tickets.

CA Service Point Mobile App

CA Service Point Mobile App mobile app provides the ability for business users and analysts to manage your tickets on the go to resolve issues faster and stay updated with ticket specific discussions. Analysts can create, edit and view list of CA SDM incidents and requests using the Service Point Mobile App.

CA Service Point Mobile App is available for download on both App Store and Google Play.

1. **Requirements**
   a. iOS device requirements: iOS 16 and above.
   b. Android device requirements: Android 11 and above.
   c. CA Service Management 17.3 or later.

2. **Access**
   a. Contact your administrator to verify that you have the required access rights in Service Point.
3. **Download and install**
   a. On your mobile device, go to the App store or Google Play store.
   b. Download the **CA Service Point** mobile app.
   c. Open the mobile app to configure a connection to your Service Point system.
4. **Sign In**
   a. Verify that you are connected to a wireless network or mobile data plan.
   b. Open the mobile app and perform the following:
      a. Enter the server URL. For example, `<hostname:9002>`
      b. Enter your CA SDM credentials.
   c. Securely Authenticate with SSO authentication (NTLM/ SAML/ SiteMinder).
   d. Post successful authentication, review and accept the License Agreement Information.
   e. To supplement your own authentication, enable fingerprint login with biometric authentication.

NOTE
After restarting xFlow Analyst services, you may get an error " Unable to display list of tickets. Please contact administrator." In this case, it is recommended to clear app cache, and relaunch the mobile app again.

Release Notes for Mobile App

Version 1.8

The latest version of CA Service Point mobile app now supports login via SiteMinder SSO. This update also includes defect fixes.

CV Defects

Case id: 36163634

Description: After authentication on the Service Point mobile app using SAML, the following error appeared: ‘Error while processing request. See error logs for details.’ However, no related entry was found in the *incidentMS_debug.log*.

NOTE
This defect has been fixed and is available from 17.4 RU4 and above.

Case id: 35884578

Description: The existing mobile app supports Single Sign-On (SSO) via NTLM or SAML only; support for SiteMinder SSO is not available.

Supportability Matrix for Mobile App

Device Compatibility

The capability and experience of the application differs based on the device that you use.

Mobile Platform	CA Service Point
Apple iOS 16 and above	Yes
Android 11 and above	Yes

Supported Authentication Types for Mobile App

Supported Authentication Types	Links to configure
xFlow Basic Authentication	xFlow Interface Configuration
SAML with ADFS Configuration	Enable SAML Authentication for xFlow Interface
NTLM	Authenticate the Users of xFlow Interface
SiteMinder (SSO) with Agent based Configuration	Integrating CA SiteMinder
SiteMinder (SSO) with SAML Configuration (WS-Fed)	Enable SAML Authentication for xFlow Interface
Azure SAML	Enable SAML Authentication for xFlow Interface
Okta with SAML Configuration	NA

CA Service Point Mobile App for Analysts

CA Service Point Mobile App now provides capabilities for Analysts to effectively create and manage tickets while on the move. Analysts can perform the following actions on a ticket via the mobile app:

IMPORTANT

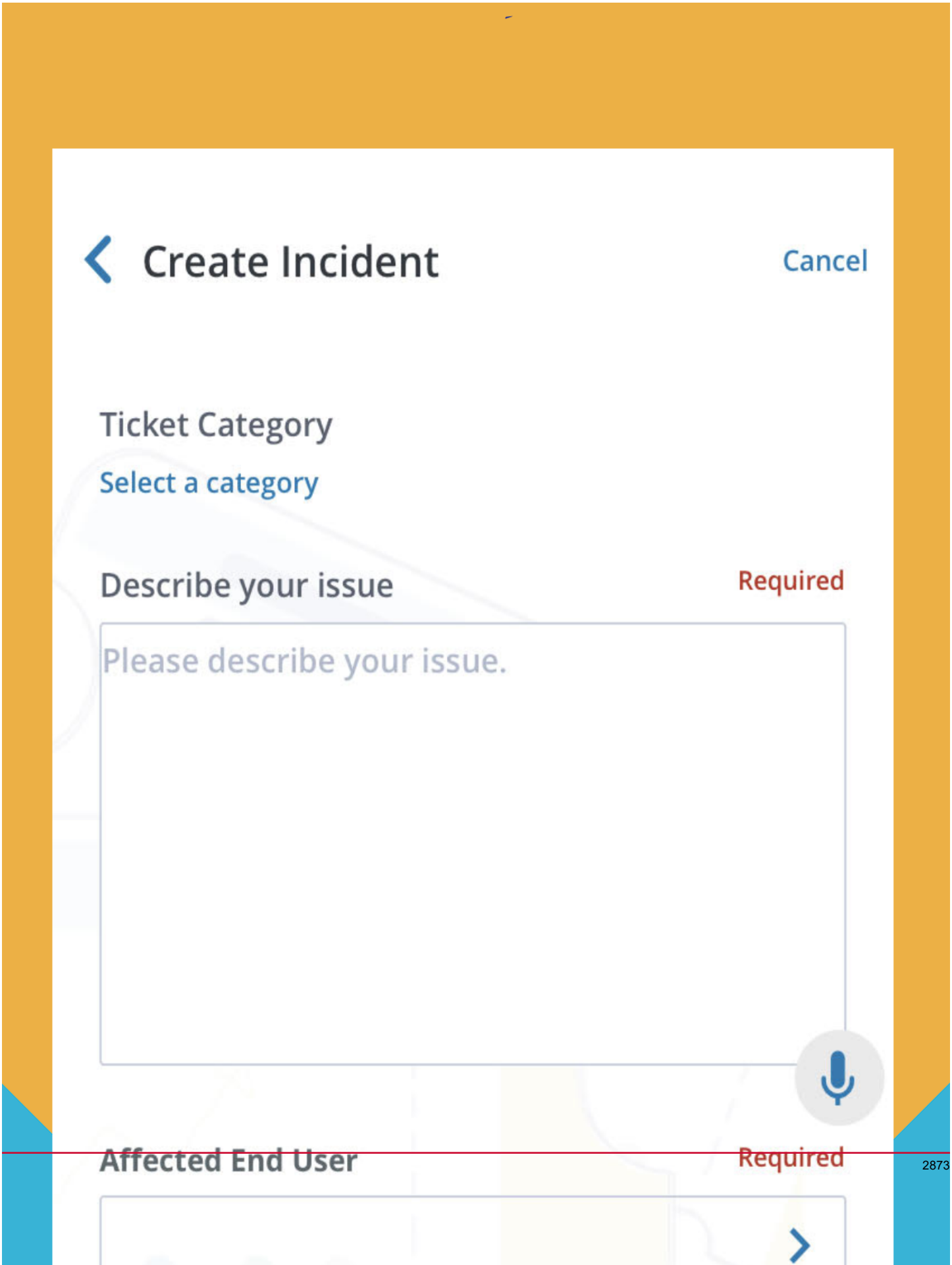
Support for Analyst Capabilities in Mobile App are continued for CA Service Management 17.4.

Create a Ticket

Analysts can create incidents and requests using the mobile app. The ticket creation form contains the following Out-of-the-box attributes:

1. **Category**
2. **Description**
3. **Affected End User**

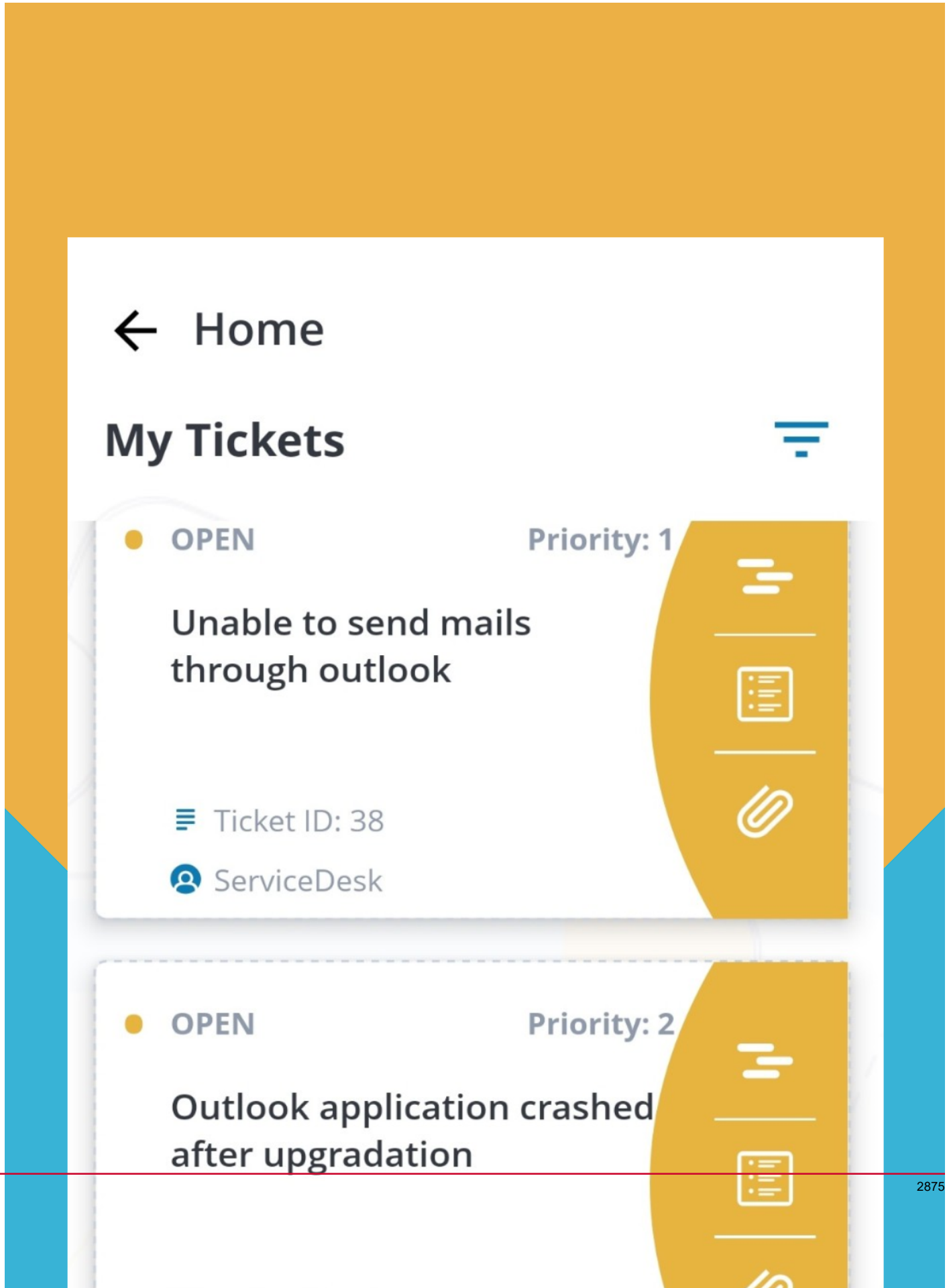
Additional attributes (Out-of-the-box or custom) can be added to the ticket creation form. These attributes can be configured through SDM Admin Configuration. For more information, see Configure the Mobile Attributes for Tickets.


Create Incident
Cancel
Ticket Category
Select a category
Describe your issue
Required
Please describe your issue.
Affected End User
Required

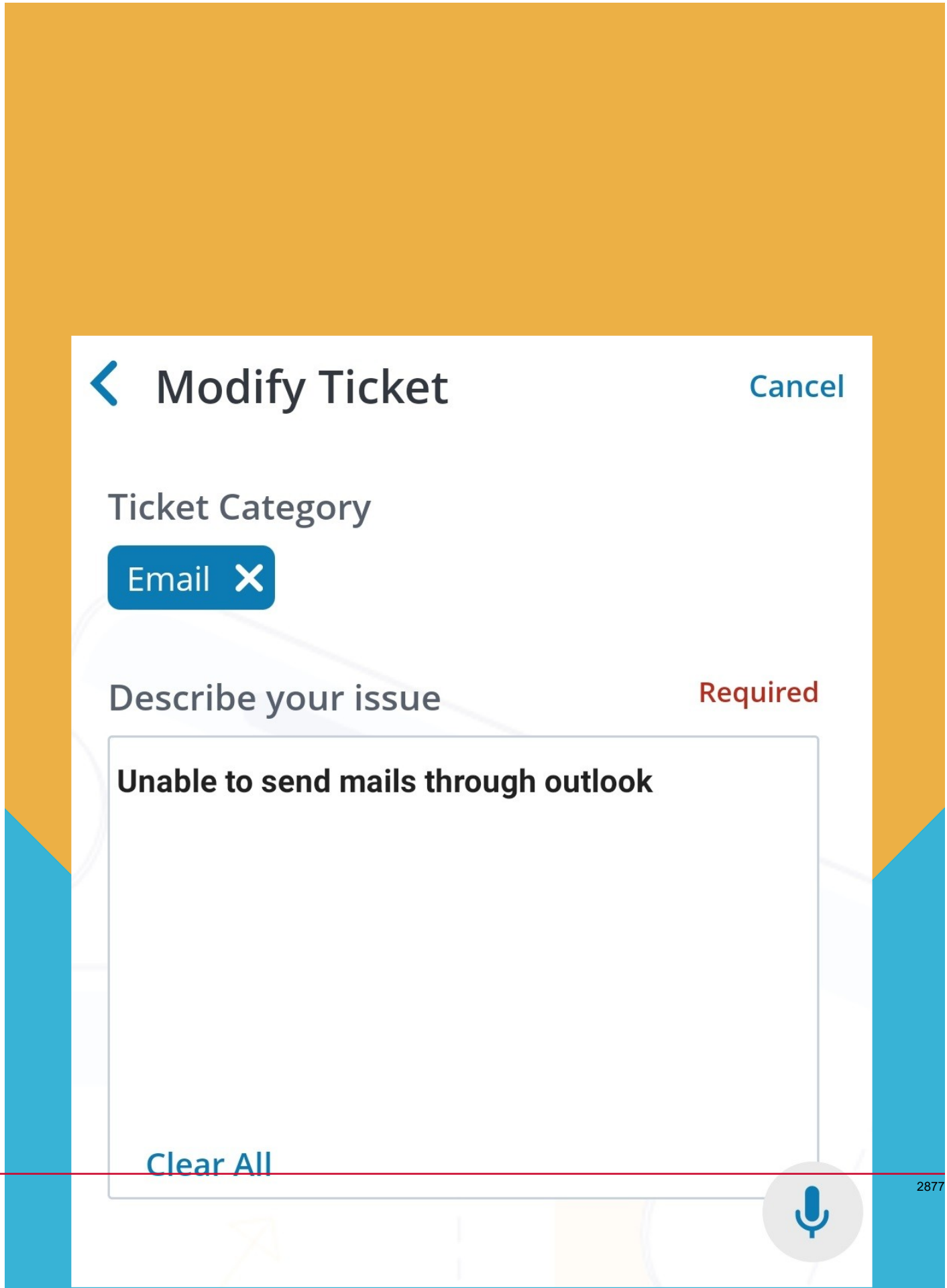
Ticket Actions (Activities on a Ticket)

The following activities can be performed on a ticket:

- **View**:
  Analysts can view tickets assigned to them or can view the the tickets of the groups they are part of. Similarly, Analysts can filter the tickets assigned to them or filter the tickets for group(s) they are part of.


Home
My Tickets
OPEN
Priority: 1
Unable to send mails through outlook
Ticket ID: 38
ServiceDesk
OPEN
Priority: 2
Outlook application crashed after upgradation

Analysts can modify any existing ticket attributes using the **Edit** action. All ticket attributes captured during ticket creation are available to Analysts for editing.


Modify Ticket
Cancel
Ticket Category
Email
Describe your issue
Required
Unable to send mails through outlook
Clear All

Ability to attach files directly to a ticket from the Service Point Mobile App. Attachments can be added to a ticket either by selecting from the mobile phone gallery or by capturing an image via the mobile camera.

- **View the feed on the ticket**:
  Analysts can view activities recorded for a ticket in the ticket details page. The feed can also be filtered based on the feed type along with the ability to add any additional comments to a ticket.

- **Ability to transfer the ticket**:
  Analysts can now transfer a ticket from the Service Point mobile app. Tickets can be assigned to a particular analyst or to a group.

Search for a Ticket

Analysts can search for tickets based on ticket ID or ticket summary.

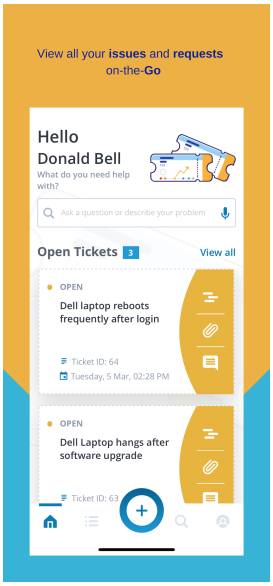
Switch between Analyst and End User Interfaces

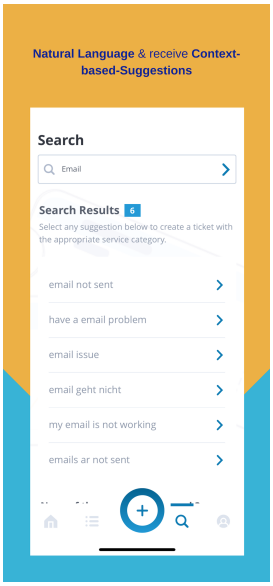
Ability to switch between analyst and end-user roles if a user has access to both Analyst and End user Interfaces. The default interface after logging into the app is based on the default app configuration for a user in CA SDM (based on Access Type). For more information, see Set the Default App for an Access Type.
After login, the interface shows **Switch to** with the Role name. You can switch to the displayed role.

> **NOTE**
> Ability to switch is not available for contacts with end-user access type only.

CA Service Point Mobile App for Business Users

Business users have all their cases right at their fingertips and they are equipped to get immediate and effective solution to their queries.





CA Service Point mobile app allows a business users to:

- **Request a service or report an issue. Anywhere, anytime!**
  With the mobile app, you can seamlessly request for a service or report an issue from your mobile device. Based on the issue description, and user context (location, organization, and department), the mobile app displays suggested categories.
- **Manage tickets**
  Manage tickets by performing actions such as update, edit, and add comments.
- **Track ongoing activity**
  Stay up-to-date on all the activities from a central location, the **All Feed**.
- **Use attachments**
  You can view, download, and add attachments to a ticket. The mobile app allow you to attach images and documents from the camera and gallery.
- **Use the Geolocation feature**
  While requesting a service, or reporting an issue, you can provide your exact location, which is captured and displayed in the **All Feed**.
- **Call Service Desk**
  Reaching the service desk can be like finding your way through a maze. Therefore, we have simplified the process! You can now call the service desk from within the mobile app. Your administrator must configure a phone number to reach the service desk using the CA SDM Options Manager.

Enable Microsoft Intune for Mobile Device Management

This article describes how to enable Microsoft Intune in CA Service Desk Manager (SDM).

SDM Mobile app supports Intune for Mobile Device Management. Microsoft Intune is a cloud-based enterprise mobility management tool. Intune helps organizations manage access to corporate apps, data, and resources. It combines Mobile Device Management (MDM) capabilities with Mobile Application Management (MAM). For more information, see Microsoft Intune App SDK.

Perform the following steps to enable Intune in SDM:

1. Login to CA CA Service Desk Manager as an Administrator.

2. Navigate to **Administration**, **xFlow Interface**, **General**, **General Configuration List** page, **Intune Enabled**. Set the value of Intune Enabled to **true**

Intune Company Portal App

After enabling Intune in SDM as shown above, perform the following steps to install the Intune Company Portal App:

1. **Android**: Install the Intune App from Google Play Store.
2. **iOS**: Install the Company Portal App from App Store.
3. Enroll your device for device management and to access corporate resources.

For more information, see Intune User Help.

Reporting

This article describes the following topics:

- Reporting Using CA Service Desk Manager
- Reporting Using CA Service Catalog

Reporting Using CA Service Desk Manager

CA SDM provides a variety of built-in reporting capabilities and options including the ability to do the following:

- Generate Analysis Reports
- Summary and Detail Reports
- Key Performance Indicators

Report Methods Setup

Report methods let you specify where report results are to be sent when you select a report. Examples are the summary and detail reports available on the Reports menu and the reports available from the Analysis menu. Several predefined report methods are provided, and you can modify them.

Report Formatting

Using the dataent.fmt file found in $NX_ROOT/fig/cfg (UNIX) or installation_directory\fig\cfg (Windows) you can customize various data formats on the reports you print. You can view and modify this file using any text editor (Windows users should use WordPad to edit the file).

The following lines in the file control some of the date and time formats:

```
default_date = "long_date"
short_date = "M/D/YY// Enter a date as MM/DD/YY"
long_date = "MM/DD/YYYY// Enter a date as MM/DD/YY"
default_tod = "hour_12"
hour_12 = "h:mm:ss a(am,pm)// Enter a time of day as hh:mm:ss am/pm"
hour_24 = "HH:mm:ss// Enter a time of day as 00:00:00 - 23:59:59"
hms_12 = "h:mm:ss a(am,pm)// Enter a time of day as hh:mm:ss am/pm"
hms_24 = "HH:mm:ss// Enter a time of day as 00:00:00 - 23:59:59"
default_date_time = "date_time12"
date_time12 = "M/DD/YYYY h:mm:ss a(am,pm)// Enter a date and time as MM/DD/YY hh:mm:ss am/pm"
date_time24 = "M/DD/YYYY HH:mm:ss// Enter a date and time as MM/DD/YY 00:00:00-23:59:59"
```

The following code is an example of how you can change these lines to support European dates and times:

```
default_date = "long_date"
short_date = "D/M/YY// Enter a date as DD/MM/YY"
long_date = "DD/MM/YYYY// Enter a date as DD/MM/YYYY"
default_tod = "hour_24"
hour_12 = "h:mm:ss a(am,pm)// Enter a time of day as hh:mm:ss am/pm"
hour_24 = "HH:mm:ss// Enter a time of day as 00:00:00 - 23:59:59"
hms_12 = "h:mm:ss a(am,pm)// Enter a time of day as hh:mm:ss am/pm"
hms_24 = "HH:mm:ss// Enter a time of day as 00:00:00 - 23:59:59"
default_date_time = "date_time24"
date_time12 = "M/DD/YYYY h:mm:ss a(am,pm)// Enter a date and time as MM/DD/YY hh:mm:ss am/pm"
date_time24 = "M/DD/YYYY HH:mm:ss// Enter a date and time as MM/DD/YY 00:00:00-23:59:59"
```

NOTE
If you are using pdm_extract to export data from the CA SDM database to another system, and want to extract data to use the date format specified in the dataent.fmt file, use the -d flag when you invoke pdm_extract.

Data Analysis Setup

An interactive viewer provides an extensive set of tools for adjusting how the report data is viewed. You can drill to various levels of detail, such as from the group, to the assignee, or to the actual requests.

The interactive viewers let you do the following:

- Change the appearance of the report by presenting the data in a different "block" format, such as a pie chart. For example, by right-clicking in the report and selecting "Turn table to," report presentation features appear.
- Perform different tasks, such as sorting, creating report breaks, calculating, filtering, and ranking the report.
- Share information with other users and groups.

Publish and Distribute Reports

Business Intelligence Launch Pad users can save report data in Excel, PDF or CSV format, and then distribute it to a file location, an inbox, email address, or FTP site.

When a report is approved for use within CA SDM, it is moved in to the public section making it available to authorized users. Security can be assigned to folders and documents to specify whether they can be accessed globally, by specific roles or by individuals. Security is administered in the BusinessObjects Central Management Console.

Generate Analysis Reports

The Administration Tab provides built-in analysis reports for a global and detailed perspective on the service desk process

- Request Area or Issue Category reports provide you with the number of requests or issues opened in the specified period for each request area or issue category. This report is sorted alphabetically by request area or issue category.
- Request or Issue reports provide you with statistics, such as number of requests or issues opened, number of requests or issues closed, average time open, and average time until closed, for a specified period. This report is sorted by date, oldest to newest.
- Request Area Priority or Issue Category Priority reports provide you with the number of requests or issues opened by priority in the specified period for each request area or issue category. This report is sorted by priority (highest to lowest) and then alphabetically by request area or issue category.

You can display the analysis report for today, for the past thirty days, year-to-date, and much more.

Follow these steps:

1. Choose the Administration tab.
2. Expand the Service Desk and Analysis nodes.

3. Choose Request or Issue.
4. Choose the appropriate report based on the time frame you want.

Summary and Detail Reports

CA Service Desk Manager has built in summary and detail reporting options. To print or view summary and detail reports, you must first select the records you want to include in the report. You can select specific records for a report using the search feature of the list windows. To generate the report, select Summary or Detail from the Reports drop-down menu.

For example, from the Request List page you can enter search criteria to create a list of requests that you can then use to generate a report.

You can also print a single page detail report for each record by choosing Print Form from the File menu within any detail window. To print a report for a newly created record, you must first save the record.

> **NOTE**
> If the ServiceDesk Manager is configured on IIS WebServer, Read/Write permissions must be given on NX_ROOT\log folder for the IUSR user.

Key Performance Indicators

Key Performance Indicators (KPIs) are metrics you can use to identify areas of your service management environment that can require administrative attention or configuration tuning.

Use the CA SDM web interface to configure your KPI definitions. The data they produce is stored in the CA SDM database.

> **NOTE**
> In addition to defining KPI queries, you can configure the KPI daemon to retrieve CA SDM ticket data whenever a ticket is opened, closed, or certain fields are modified.

By defining and monitoring a planned set of KPIs, you can measure progress toward your organization's performance-related goals, and gather valuable data for driving strategic decisions about your IT environment.

> **NOTE**
> For information about the KPI database tables, see Technical Reference.

KPI Types

CA SDM supports three KPI types:

- **System KPIs** are installed with the product. You can customize them to meet your needs, but you cannot create new System KPIs.
- **Stored Query KPIs** call a stored query to retrieve metrics from the database. You can create custom Stored Query KPIs, or modify the predefined examples installed with the product.
- **SQL KPIs** allow you to incorporate a complete SQL query directly into the KPI. You can create custom SQL KPIs, or modify the predefined examples installed with the product.

> **NOTE**
>
> KPIs, whether predefined or user-created, cannot be deleted. When a KPI is no longer needed, you can deactivate it by setting the Record Status to inactive.

Predefined KPIs

Several KPIs of each type are installed with CA SDM. You can use the predefined Stored Query and SQL KPIs with their original definitions, or as models for your custom definitions.

You can use the predefined System KPIs with their original definitions, or modify some of their fields to meet your needs.

All predefined KPIs are installed as Inactive. For a KPI to begin functioning in your system, it must be set to Active. Navigate to Service Desk, KPIs on the Administration tab and search for the inactive KPI. Open the KPI and click Activate

The Options Manager **concurrent_lic_refresh_interval** KPI option can be used to change the Refresh Time interval for the 8 KPI's that have names beginning with '**webConcurrent'** and have a Type of '**System**'. The setting affects the 8 KPIs whenever the status of the KPI is Active . The default refresh time interval is 1 hour (3600 seconds).

> **WARNING**
> Multiple versions of a KPI with the same name cannot be active at the same time.

Create a KPI

You can create custom Stored Query KPIs to produce Count metrics, and SQL KPIs to produce Count, Sum, Max, or Duration metrics.

> **NOTE**
> You cannot create new System KPIs; however, you can edit the predefined System KPIs to meet your needs.

> **NOTE**
> If multi-tenancy is installed, select the appropriate tenant from the drop-down list. The public (shared) option creates the object for all tenants.

Follow these steps:

1. Click CA SDM, KPIs on the Administration tab.
   The KPI List page appears.
2. Click Create New.
   The Create New KPI page appears.
3. Enter a name for the KPI and select either Stored Query or SQL from the dropdown list.
   Click Continue.
   The Create New KPI page displays the KPI fields. The following table describes the KPI fields. The Sys. S.Q. and SQL columns indicate the KPI types each field belongs to (System, Stored Query, or SQL).

Field	Sys.	S.Q.	SQL	Description
KPI Name	X	X	X	Identifies the display name for the KPI. Not editable.
Type	X	X	X	Defines the record as a System, Stored Query, or SQL KPI. Not editable.
Keep Existing Version	X	X	X	Specifies that the current version of the KPI record will be retained if the record is updated. Editable when the @NX_ALWAYS_KEEP_KPI_VERS setting in the NX.env is set to No.

Version	X	X	X	Identifies the version number of the KPI record. The version number is incremented automatically each time the record is updated. Not editable.
Record Status	X	X	X	Specifies whether the KPI is active (gathering data) or inactive (not gathering data). Editable only by using the "Edit in List" feature on the KPI List page.
Metric Type	X	X	X	Specifies the type of calculation the KPI will perform: Stored Query KPIs are always Count. System KPIs can be Count or Duration. SQL KPIs can be Count, Sum, Max, or Duration. Editable for SQL type KPI only. The calculated duration is based on a change to values in a ticket in real time, not in business hours.
Refresh Time	X	X	X	Specifies the time interval for retrieving KPI metrics from the database. The value is editable. Refresh time is specified in HH:MM:SS format.
Process Type	X			System KPI metrics can be derived from the following CA SDM processes: **domsrvr** -- Object Manager **bpvirtdb** -- BOP Virtual Database Server **db_agents** -- Database agents **webengine** -- The CA SDM web client

System Name	X			The internal name of the System KPI. Not editable.
User Context		X	X	(Optional) Specifies the userid of a CA SDM contact. The contact's role and tenant assignments are used to determine data partitioning for the metrics produced by the KPI. The value is editable.
Stored Query		X		The name of the stored query called by this KPI. The value is editable. The stored query definition must be active and have the KPI Usage option enabled.
SQL Query			X	The SQL code for the query. The value is editable. When you save a SQL KPI record, the syntax of your SQL query is validated. An error message appears when the code is invalid. If your SQL code does not include an aggregate, the KPI daemon logs error messages and does not return data to the kpi_data table.
Description	X	X	X	(Optional) Provides a detailed description of the KPI. The value is editable.

4. Click Save.

NOTE
When you save the initial definition for a SQL KPI, the syntax of your SQL query is validated. An error message is displayed if the code is invalid.

KPI Daemon

The KPI daemon manages the retrieval, organization, and storage of KPI metric data. The daemon runs continuously, and collects data at specified intervals from various system resources.

When the refresh time of a KPI is reached, the KPI daemon interacts with other system components as follows:

- **System KPI** -- Sends a request to a target daemon (webengine, domsrvr, bpvirtdb, or db_agents) to retrieve count or duration data.
- **Stored Query KPI** -- Sends a request to the domsrvr to collect count data.
- **SQL KPI** -- Sends a request to the domsrvr to collect count, sum, max, or duration data.

When the KPI daemon receives the requested data, it stores the resulting metrics in the database.

> **NOTE**
> The calculated duration is based on a change to values in a ticket in real time, not in business hours.

System KPIs

System KPIs enable you to collect metric data related to the operation of CA SDM processes.

The following process types are supported:

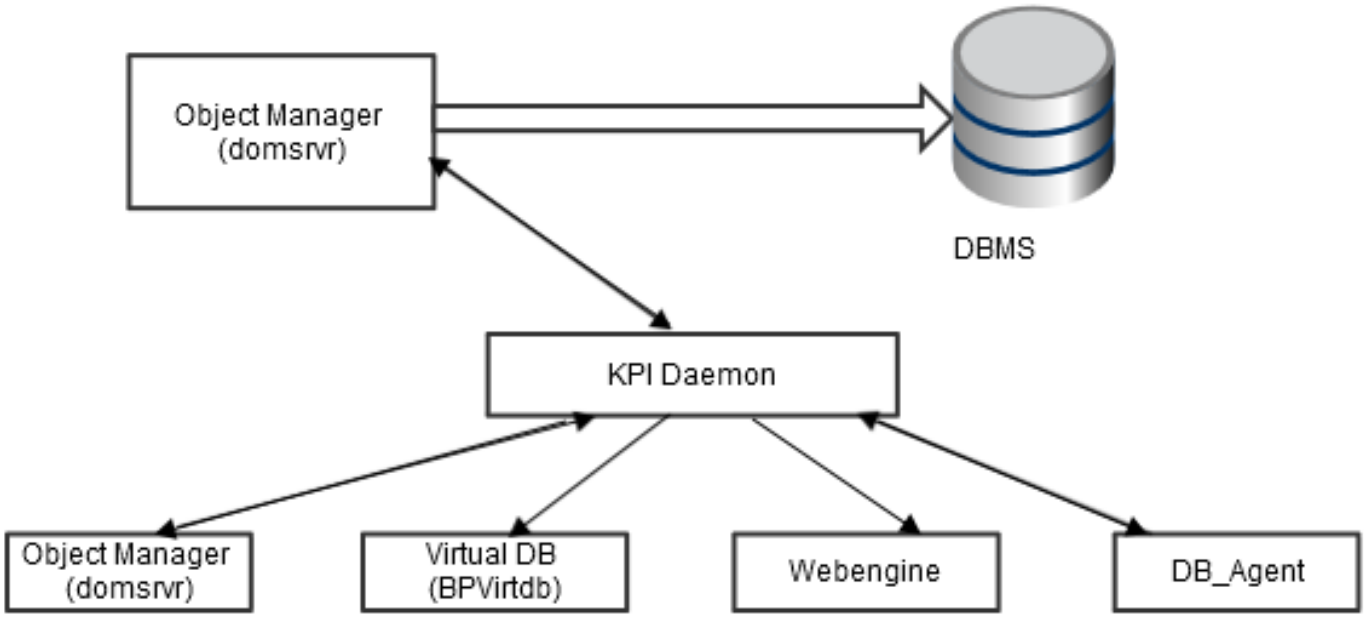
- **domsrvr** -- The CA SDM Object Manager (server process). The Object Manager also caches various records and tables for clients.
- **bpvirtdb** -- The BOP Virtual Database server enables operation of multiple Object Managers in the CA SDM environment. The Object Managers connect to the virtual database which arbitrates their access to database agents. For example, when retrieving a new range of ticket reference numbers, the virtual database ensures that only one Object Manager at a time accesses the table containing the reference numbers.
  Depending on your CA SDM configuration, the Object Managers connect to the virtual database of the following servers:
- Conventional: All Object Managers running on primary or secondary server connect to the virtual database of the primary server.
  - Advanced availability: All Objects Managers connect to the virtual database of the same server that it is running on. For example, object managers running on the application server connect to the virtual database of the application server only.

  The virtual database also performs caching of database information for Object Managers.
- **db_agents** -- Database agents perform SQL queries, and handle other interactions between CA SDM and the database management system (DBMS). Database Agents adhere to the CA SDM database schema, and translate the SQL code to the form required by the particular DBMS (for example, Oracle).
- **webengine** -- The CA SDM component that connects web browsers to an Object Manager through a pdmweb cgi running on a Microsoft IIS or Apache Tomcat web server.
  Depending on your CA SDM configuration, there must be a web engine for WSP on the following servers for WSP Schema Designer to write schema files:
- Conventional: Primary server
  - Advanced availability: Background server

  Web engines are the true client of an Object Manager for user client web browsers. Web engines cache .htmpl web forms for connected users. You can manipulate the cache using the pdm_webcache utility and see client connection statistics using the pdm_webstat utility.

The following diagram illustrates data flow for System KPIs.

Figure 36: CA_SDM_SystemKPIs



Each system KPI produce one of the following metric types:

- Count
- Duration

System KPI Examples

- This example produces a count of the update requests sent to BOP Virtual Database:
  `updateCt`
- This example reports how long requests for updates, inserts, and deletes to BOP Virtual Database took to complete:
  `virtdbUpdateResponseDt`

Stored Query KPIs

With Stored Query KPIs you can generate reports based on count metrics retrieved from the CA SDM database.

CA SDM provides a set of predefined stored queries. Many of them are useful just as they are. You can also customize them to meet your needs, or use them as models for writing your own queries.

All Stored Query KPIs have a Count metric type.

> **NOTE**
> You can use stored queries to produce counter fields for your web interface Scoreboard, or KPI metrics, or both. To use a stored query in a KPI definition, the query must be enabled for KPI usage. For more information, see Stored Query Setup.

Stored Query KPI Examples

- This example produces a count of the open change orders at the assignee's location:
  `@cnt.location.name Changes`
- This example produces a count of the workflow tasks that will violate an SLA before midnight of the current day:
  `Issue Workflow Tasks that will Violate an SLA today`

SQL KPIs

SQL KPIs provide more flexibility than Stored Query KPIs. With SQL KPIs you can write your own queries to the CA SDM database to produce the following types of metrics:

- Sum
- Count
- Max
- Duration

> **NOTE**
> Your SQL code must comply with SQL92 syntax.

The following considerations apply to SQL KPIs:

- To see the SQL code for these example queries, open the query definitions from the KPI List page.

  > **NOTE**
  > For instructions, see the *Online Help*.
- The @ symbol is not supported in SQL KPIs. Instead, use syntax as shown in the following example and an attribute alias:

  ```
  SELECT * FROM cr WHERE assignee_last_name = "Smith"
  ```

  For this example, you could use a predefined attribute alias. The Attribute Alias Detail page appears as follows:
  " Object Name = cr
  " Alias Name = assignee_last_name
  " Status = Active
  " Alias Value = assignee.last_name

Examples: SQL Query KPI

- This predefined example produces the sum of the estimated costs of all pending change order workflow tasks:

  ```
  Est Cost Sum of Pending Change Workflow Tasks
  ```
- This predefined example produces a count of the active change orders that have been closed and reopened:

  ```
  Count of Reopened Change Orders
  ```

Retrieve Ticket Data

To allow reporting on how long tickets remain in the various processing states, you can configure the KPI daemon to retrieve CA SDM ticket data whenever a ticket is opened or closed, and whenever any of the following fields values are changed:

- Active
- Assignee
- Area or Category
- Group
- Impact
- Organization
- Priority
- Root Cause
- Status
- Service Type

To enable ticket retrieval, install the KPI Ticket Data Table option available in Options Manager KPI folder.

The collection of data from new and updated tickets is enabled. The ticket data is written into the usp_kpi_ticket_data database table, and is available for generating web-based reports.

NOTE
For instructions, see the *Online Help*.

The following considerations apply to ticket retrieval:

- The KPI daemon may take up to 30 minutes to populate the ticket information to the usp_kpi_ticket_data table.
- The support_lev field allows tracking of Service Type processing when the classic_sla_processing option is installed. The classic_sla_processing option enables the Service Type processing from CA SDM version 6.0 and earlier.
- Enabling this feature may result in degraded CA SDM performance.

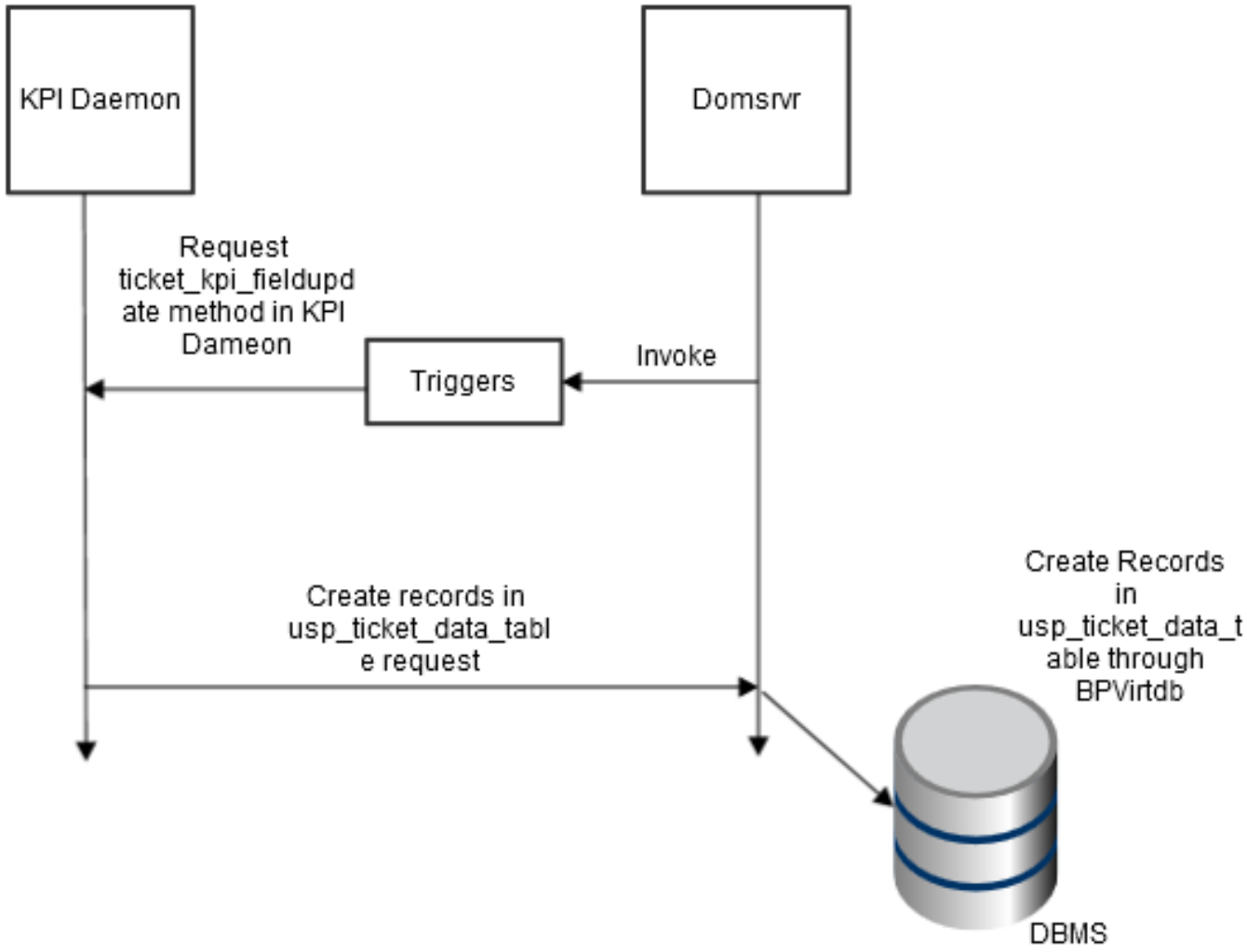
KPI Ticket Data Flow

The KPI Ticket Data Table feature works on a trigger mechanism. When the kpi_ticket_data_table option is installed, the KPI daemon begins monitoring CA SDM ticket objects to track the following events:

- Open new ticket
- Check in changes to one or more monitored fields
- Close ticket

When one of these events occurs, a POST_CI trigger fires. The trigger sends a BPMessage with a trigger attribute list to KPI daemon, and adds the returned data to the usp_kpi_ticket_data table, as illustrated in the following data flow diagram.

Figure 37: CA_SDM_KPI Ticket Data Flow1



KPI Ticket Data Table Record Types

There are five operation types in the usp_kpi_ticket_data table:

- INIT (0)
- NO_INIT_REC (1)
- REOPEN (2)
- UPDATE (3)
- CLOSE (4)

The type value (integer value shown in parentheses) is stored in the operation field. The end_time of the previous record is stored in the prev_time field. The ktd_duration field stores the duration, which is calculated as end_time minus prev_time. The previous record is defined in the following rules:

1. The INIT record indicates a ticket has been created.

2. An UPDATE record is created for each update to a ticket. It gets prev_time from a previous UPDATE record to the same ticket field if a previous UPDATE record exists. Otherwise, it gets prev_time from the INIT, NO_INIT_REC or REOPEN record, depending on which record has an end_time value closest in time to the UPDATE record.
3. The CLOSE record is the same as UPDATE.
4. If a ticket is reopened, a single REOPEN record is created to show the Active field value. The REOPEN record gets prev_time from a previous CLOSE record for the same Active field.
5. If a ticket was created before turning on the Options Manager option, it may have records for updates but without an INIT record. The program in this case creates a NO_INIT_REC record and gets prev_time from the first non-INIT record it encounters.

NOTE
For complete documentation of the table fields, see the Technical Reference .

Troubleshooting

You can diagnose problems you might experience with KPI usage.

Verify that the KPI Daemon is Running

To verify that the KPI daemon is running, run the pdm_status command-line utility as follows:

```
pdm_status
```

Examine the output for for the following:

```
KPI Daemon myserver  Running  myserver 3568  Wed Feb 06 07:23:53
```

NX.env File

Review the $NX_ROOT/NX.env file to verify that the basic KPI configuration is correctly set.

If you have installed the KPI Ticket Data Table option, the NX.env file includes the following:

```
######################################################################
# NX_KPI_TICKET_DATA_TABLE
# Enable the collection of changed fields from ticket objects
# (like Requests, Issues and Change Orders) to write record entries
# into the usp_kpi_ticket_data table.
######################################################################
@NX_KPI_TICKET_DATA_TABLE=Yes

######################################################################
# NX_KPI_FUZZ_TIME
# Specifies a tolerable delay for each KPI within which kpi_daemon
# sends a request to retreive KPI data when this kPI's refresh time
# is timeout
######################################################################
@NX_KPI_FUZZ_TIME=20
```

NOTE
The default value of @NX_KPI_FUZZ_TIME is 20 seconds. You can modify this value within the range of 0 to 40 seconds. If you set this variable to a value higher than 40, 40 seconds will be used.

```
######################################################################
# NX_ALWAYS_KEEP_KPI_VERSIONS
# The keep_version attribute in Kpi table is displayed on the KPI
# Detail Edit form as a checkbox. The NX_ALWAYS_KEEP_KPI_VERSIONS
# option specifies that checkbox is read-only and is always checked.
```

```
#####################################################################
@NX_ALWAYS_KEEP_KPI_VERSIONS=Yes
```

Turn on KPI Daemon Tracing

Use the pdm_logstat utility to turn on trace logging to monitor KPI daemon activity.

To turn on KPI daemon tracing, run the following command:

```
pdm_logstat -f cache_table_mgr.c VERBOSE
```

To turn off KPI daemon tracing, run the following command:

```
pdm_logstat -f cache_table_mgr.c
```

Examine the stdlog.0 file for entries pertaining to KPI daemon activity.

The following example shows normal KPI daemon connections to the host computer and the domsrvr:

```
02/06 05:42:14.58 garbo-2k3-bb kpi_daemon 2432 SIGNIFICANT kpi_daemon.c  117 KPIDaemon ready for action!
02/06 05:42:14.80 garbo-2k3-bb domsrvr 792 SIGNIFICANT connmgr.c 2314 Connecting client kpi_daemon:garbo-2k3-
bb
02/06 05:42:14.81 garbo-2k3-bb kpi_daemon 2432 SIGNIFICANT main.c 220 KPI daemon connected with domsrvr
```

The following example indicates an error in a KPI named webSessionCT:

```
01/16 13:24:46.74 jed web:local 2152 ERROR sys_kpi.c 96 Invalid KPI metric type: 1382180215 (KPI
 name:webSessionCT)
```

The following examples show KPI daemon activity:

```
02/06 07:23:50.47 garbo-2k3-bb  kpi_daemon 3568 TRACE cache_table_mgr.c 427 Cache Table manager created
 Kpi_Obj (KPI dob record_id:9003) (KPI type:2)
02/06 07:23:50.53 garbo-2k3-bb kpi_daemon 3568 TRACE cache_table_mgr.c  427 Cache Table manager created
 Kpi_Obj (KPI dob record_id:9103) (KPI type:1)
```

> **NOTE**
> For information about KPI types, see KPI Ticket Data Table Record Types.

Reporting Using CA Service Catalog

This article contains the following topics.

- Manage Data Objects
- Manage Layouts
- Manage Data Views
- View the Business Value Dashboards

Report Builder helps you create reports in CA Service Catalog with the following options:

- Data Objects
- Data Views
- Layouts

{"URL":["/usm/wpf?Node=iclaunchpad.reports"],"customLabelGetStarted":"Report Builder","description":"IPCE_Report_Builder"}

Manage Data Objects

This article contains the following topics:

Administrators can use data objects to define the data to use for a chart, table, or other facility, including a field in a form. A data object can have the following data sources:

- An ODBC-connected or JDBC-connected database management system (DBMS). Examples include SQL Server or Oracle database.
- A delimiter-separated value file.
- Any other data source that a Java report plug-in can access.

> **NOTE**
>
> When you create reports, ODBC-type data objects do not retrieve nvarchar-type fields. To retrieve nvarchar fields in the objects in your reports, use JDBC as the database connection type.

You perform the following tasks to manage data objects:

- View the data objects and organize them in folders.
- Become familiar with the runtime variables in data objects.
- Add a query runtime variable for use in a data object.
- Add a data object or edit one.
- Use predefined data objects.
- Delete a data object.

Runtime Variables

Data objects can use *runtime variables* to alter behavior and selection criteria dynamically.

For Query data objects, you can use runtime variables in the SQL query. As a prerequisite, define these runtime variables in the Variables list. That list includes default Query variables, and you can add your own.

For example, consider a report that is based on a SQL query that displays a list of users. The data object can take a Last_Name value as a runtime variable to indicate the starting characters of the last name. Users of the data object are prompted for the Last_Name value. You can use the runtime variable of type String that is named %Last_Name% in the SQL statement. Use this variable to restrict the results to user records that start with the value entered by the user. The following SQL statement provides a sample query:

```
SELECT first_name,middle_name,last_name FROM ca_contact WHERE (ca_contact.last_name like '%Last_Name%%')
```

> **NOTE**
> Users are prompted for only the runtime variables in the SQL query.

For plug-in data objects, you can pass runtime variables to the Java class as name-and-value pairs. As a prerequisite, verify that the Java report plug-in class takes the name-and-value pairs for the plug-in data object.

For example, consider the com.ca.usm.reporting.Plugins.RequestFulfillmentReport plug-in class. This plug-in class takes a parameter of Date type named START_DATE. Therefore, START_DATE is required in the data object that uses this plug-in class. In this case, perform either of the following actions:

- Hard-code START_DATE as a constant.
- Prompt the user for the date START_DATE value and pass that value to the plug-in.

You can use contextual *system variables* with runtime variables:

- As default values when you prompt the user.
- As constant values to pass to the plug-in.
- As values to use in a query.

The contextual system variables are as follows:

Name	Variable
Current Day	%TODAY%
Previous Day	%TODAY%-Days(1)
Next Day	%TODAY%+Days(1)
First Day Of Month	%START_OF_CURRENT_MONTH%
Last Day of Month	%END_OF_CURRENT_MONTH%
First Day Of Year	%START_OF_CURRENT_YEAR%
Last Day Of Year	%END_OF_CURRENT_YEAR%
User Domain (Business Unit)	%USER_DOMAIN%
User ID	%USER_ID%

Add a Query Runtime Variable

You can add a custom runtime variable to use with a Query data object. An example is adding a query runtime variable that is a drop-down variable.

Follow these steps:

1. Click **Administration**, **Report Builder**.
2. Perform *one* of the following actions:
   - Add a data object
   - Edit an existing data object
3. Click **Create Data Object**.
   The Create Runtime Variable or Edit Runtime Variable dialog appears.
4. Specify the **Name**, **Datatype**, and other data in the fields provided.
5. Click **Create Variable**.
   The Catalog system saves your variable definition.

You can use the variable in an SQL query for a Query data object. Users are prompted to enter a value when they run the data object.

> **NOTE**
> When you add a query runtime variable that is a drop-down variable, the number of values in the resulting drop-down list is limited to 1000. If the report query returns more than 1000 values, the system truncates these additional values. So, the user cannot view them in the drop-down list. For more information about how to increase this value, see the Modify XSL, XML, JavaScript, and Image Files section.

Add a Data Object

You can add and edit data objects to define the source and content to retrieve for a report.

Follow these steps:

1. Click **Administration**, **Report Builder**.
2. Click **Create Data Object**.
   The Data Object Properties page appears.
3. Perform the following actions:
   a. Select the Type of data object to create.
   b. Complete the fields, specifying the detailed options for the data object. The exact fields that appear vary according to your selections for Type and Show Advanced.

4. Set the permissions for the data object, as follows:
   a. Click **Permissions**.
   b. Set the access levels according to role, business unit, and CA EEM user groups, as applicable.
   c. Click **OK**.
5. Click *one* of the following options:
   – **Save**
   Saves the new data object without displaying any data.
   – **Save & Test**
   Saves the new data object and displays the first 25 rows of the resulting data.

The Catalog system saves the new data object.

NOTE
You can also click Create Data View to create a data view from this data object.

Detailed Options for Data Objects

When you add or edit a data object, you complete the fields, specifying the detailed options for the data object. The exact fields that appear vary according to your selections for **Type** and **Show Advanced** on the page for adding or editing a data object. The following fields require explanation. These fields appear according to your selection for Type: Query, CSV File, or Plugin.

- **Query**
  Specifies the data to retrieve from a database query, as follows:
  – **Database**
  Specifies the name of the database connection. The default is mdb.
  Using the Query Builder, you can create a database connection definition that can use JDBC or an existing ODBC data source on the Catalog Component server. For ODBC, administrators must define this data source identically on each Catalog Component computer in your implementation. This requirement does not apply to JDBC. Therefore, we recommend JDBC.

  NOTE
  When gMSA is enabled for CA Service Catalog, select 'Use GMSA' checkbox to connect to the new database using your gMSA account . This checkbox is shown only if the database type is **SQL** and driver type is **JDBC**. Username and password is not required if this checkbox is selected. Ensure that you have sufficient database rights/permissions for the gMSA account, to access tables in the database.

  – **Table**
  Specifies a comma-separated list of tables for use in the SQL query. Use English characters *only*.
  – **Fields**
  Specifies the names of the database table fields to produce as data object columns.
  Use the Search icon to populate these names that are automatically based on the query in the Query field. As a prerequisite, save the data object before you do so.
  If you use an alias for a field in the query, you can use either name for the value of Fields. Use English characters *only*.
  – **Query**
  Specifies the SQL query to use.
  Optionally click Query Builder for help defining the query.

  NOTE
  You can use the Query Builder to define new ODBC or JDBC data sources.

  – **Pivot, DB Locking, *and* the Create Variables and Manage Variables buttons**
  See Advanced Fields at the bottom of this topic.
- **CSV File**
  Specifies the data to retrieve from a file in delimiter-separated value format, as follows:
  – **Fields**

Specifies the names of the CSV file fields to produce as data object columns.

- **CSV file**
  Specifies the path name of the CSV file. The path can be relative to %RPT_HOME% folder on the Catalog Component server.

  NOTE
  If you use multiple Catalog Component computers, this file must exist in the same folder on all of them.
- **Delimiter**
  Specifies the delimiter that separates the values in the file. Select a value from the drop-down list.
- **Pivot**
  See Advanced Fields at the bottom of this page.

- **Plugin**
  Specifies the data to retrieve from the output of a Java plug-in, as follows:

  NOTE
  For more information about plug-ins, select Administration, Tools, Links, and see the Plug-In Documentation.

  Enter the values for the fields that are shown for this type:
- **Fields**
  Specifies the names of the plug-in output fields to produce as data object columns.
- **Class Name**
  Specifies the full class name of the plug-in.
- **Arguments**
  Specifies the arguments for the plug-in.
- **Pivot *and* DB Locking**
  See Advanced Fields at the bottom of this page.
- Advanced Fields
  Selecting the Advanced check box exposes the following fields:
- **Pivot**
  Presents subtotals and counts of selected data.
  Select Pivot to display the Select Pivot Fields dialog and configure the parameters.
  This field applies to *all* selections for Type.
- **DB Locking**
  Specifies the type of database locking to use when reading from the database.
  Select a value from the drop-down list.
  For assistance, click the Help (question mark) icon next to this field.
  This field applies when the Type is Query or Plugin.
- **Manage Variables and Create Variable buttons**
  Manage existing variables and add new ones. You can use these variables in SQL queries.
  This field applies *only* when the Type is Query.

Pre-Defined Data Objects

CA Service Catalog has many predefined report data objects (data objects) that you can use in your reports. F

NOTE
Data objects in this list that mention CA CMDB, CA Service Desk Manager, or CA APM are applicable *only* if the named product is integrated with CA Service Catalog. Otherwise, the data object returns irrelevant data. For information about the integrations with these products, see the Integrating CA Service Catalog section.

- **Requests associated to assets**
  Returns the requests that are associated to assets in the business unit that you specify.
- **Requests associated to change orders and configuration items, by business unit**

Returns CA Service Desk Manager change orders and CA CMDB configuration items that are associated to CA Service Catalog requests in the business unit you specify.

- **Requests associated to change orders and configuration items, by user**
  Returns CA Service Desk Manager change orders and CA CMDB configuration items that are associated to CA Service Catalog requests for the user ID that you specify.
- **Requests by business unit**
  Returns all requests for the business unit and date range that you specify.
- **Requests by status**
  Returns all requests for the status and date range that you specify.
- **Requests by year and month**
  Returns the total number of requests for each month in the year that you specify.
- **Request fulfillment**
  Returns the amount of time that is taken to approve, fulfill, and complete all requests within the date range that you specify.
- **Request item fulfillment**
  Returns the following values:
  - The time that is used to approve, fulfill, and complete the request for the specified service option element.
  - Request SLA violation data for each request.
- **Request item fulfillment averages**
  Returns the following values:
  - The total number of requests that include the specified service option element.
  - The total time and the average time that is used to approve, fulfill, and complete the requested service option elements.
  - Request SLA violation data for each request item average.
- **Request item fulfillment by request ID**
  Returns the following values:
  - The amount of time that is taken to approve, fulfill, and complete each service option of the specified request.
  - Request SLA violation data for each service option
- **Request SLA Instances**
  Returns all request SLA instances, with their SLA warning and violation thresholds.
- **Services associated to asset models**
  Returns the CA APM asset models associated to services and service options in the business unit that you specify.
- **Services associated to configuration items**
  Returns the CA CMDB configuration items that are associated to service in the business unit that you specify.
- **Total requests by business unit**
  Returns the total number of requests that are created per business unit for the date range that you specify.
- **Total requests by month**
  Returns the total number of requests that are grouped by month for the year that you specify.
- **Total requests by status**
  Returns the total number of requests that are grouped by status for the date range that you specify.

  NOTE
  CA Service Catalog also provides predefined data objects for accounting, data mediation, financial reports, and metric events. For more information, see the comments that are supplied with the related data objects.

Administrators can use data objects to define the data to use for a chart, table, or other facility, including a field in a form. A data object can have the following data sources:

- An ODBC-connected or JDBC-connected database management system (DBMS). Examples include SQL Server or Oracle database.
- A delimiter-separated value file.
- Any other data source that a Java report plug-in can access.

NOTE

When you create reports, ODBC-type data objects do not retrieve nvarchar-type fields. To retrieve nvarchar fields in the objects in your reports, use JDBC as the database connection type.

You perform the following tasks to manage data objects:

- View the data objects and organize them in folders.
- Become familiar with the runtime variables in data objects.
- Add a query runtime variable for use in a data object.
- Add a data object or edit one.
- Use predefined data objects.
- Delete a data object.

You can add and edit data objects to define the source and content to retrieve for a report.

Follow these steps:

1. Click **Administration**, **Report Builder**.
2. Click **Create Data Object**.
   The Data Object Properties page appears.
3. Perform the following actions:
   a. Select the Type of data object to create.
   b. Complete the fields, specifying the detailed options for the data object. The exact fields that appear vary according to your selections for Type and Show Advanced.
4. Set the permissions for the data object, as follows:
   a. Click **Permissions**.
   b. Set the access levels according to role, business unit, and CA EEM user groups, as applicable.
   c. Click **OK**.
5. Click *one* of the following options:
   – **Save**
   Saves the new data object without displaying any data.
   – **Save & Test**
   Saves the new data object and displays the first 25 rows of the resulting data.

The Catalog system saves the new data object.

NOTE
You can also click Create Data View to create a data view from this data object.

Detailed Options for Data Objects

When you add or edit a data object, you complete the fields, specifying the detailed options for the data object. The exact fields that appear vary according to your selections for **Type** and **Show Advanced** on the page for adding or editing a data object. The following fields require explanation. These fields appear according to your selection for Type: Query, CSV File, or Plugin.

- **Query**
  Specifies the data to retrieve from a database query, as follows:
  – **Database**
  Specifies the name of the database connection. The default is mdb.

Using the Query Builder, you can create a database connection definition that can use JDBC or an existing ODBC data source on the Catalog Component server. For ODBC, administrators must define this data source identically on each Catalog Component computer in your implementation. This requirement does not apply to JDBC. Therefore, we recommend JDBC.

NOTE
When gMSA is enabled for CA Service Catalog, select 'Use GMSA' checkbox to connect to the new database using your gMSA account . This checkbox is shown only if the database type is **SQL** and driver type is **JDBC**. Username and password is not required if this checkbox is selected. Ensure that you have sufficient database rights/permissions for the gMSA account, to access tables in the database.

- **Table**
  Specifies a comma-separated list of tables for use in the SQL query. Use English characters *only*.
- **Fields**
  Specifies the names of the database table fields to produce as data object columns.
  Use the Search icon to populate these names that are automatically based on the query in the Query field. As a prerequisite, save the data object before you do so.
  If you use an alias for a field in the query, you can use either name for the value of Fields. Use English characters *only*.
- **Query**
  Specifies the SQL query to use.
  Optionally click Query Builder for help defining the query.

  NOTE
  You can use the Query Builder to define new ODBC or JDBC data sources.
- **Pivot, DB Locking, *and* the Create Variables and Manage Variables buttons**
  See Advanced Fields at the bottom of this topic.

- **CSV File**
  Specifies the data to retrieve from a file in delimiter-separated value format, as follows:
  - **Fields**
    Specifies the names of the CSV file fields to produce as data object columns.
  - **CSV file**
    Specifies the path name of the CSV file. The path can be relative to %RPT_HOME% folder on the Catalog Component server.

    NOTE
    If you use multiple Catalog Component computers, this file must exist in the same folder on all of them.
  - **Delimiter**
    Specifies the delimiter that separates the values in the file. Select a value from the drop-down list.
  - **Pivot**
    See Advanced Fields at the bottom of this page.
- **Plugin**
  Specifies the data to retrieve from the output of a Java plug-in, as follows:

  NOTE
  For more information about plug-ins, select Administration, Tools, Links, and see the Plug-In Documentation.

  Enter the values for the fields that are shown for this type:
- **Fields**
  Specifies the names of the plug-in output fields to produce as data object columns.
- **Class Name**
  Specifies the full class name of the plug-in.
- **Arguments**
  Specifies the arguments for the plug-in.
- **Pivot *and* DB Locking**

See Advanced Fields at the bottom of this page.

- Advanced Fields
  Selecting the Advanced check box exposes the following fields:
- **Pivot**
  Presents subtotals and counts of selected data.
  Select Pivot to display the Select Pivot Fields dialog and configure the parameters.
  This field applies to *all* selections for Type.
- **DB Locking**
  Specifies the type of database locking to use when reading from the database.
  Select a value from the drop-down list.
  For assistance, click the Help (question mark) icon next to this field.
  This field applies when the Type is Query or Plugin.
- **Manage Variables and Create Variable buttons**
  Manage existing variables and add new ones. You can use these variables in SQL queries.
  This field applies *only* when the Type is Query.

You can add a custom runtime variable to use with a Query data object. An example is adding a query runtime variable that is a drop-down variable.

Follow these steps:

1. Click **Administration**, **Report Builder**.
2. Perform *one* of the following actions:
   - Add a data object
   - Edit an existing data object
3. Click **Create Data Object**.
   The Create Runtime Variable or Edit Runtime Variable dialog appears.
4. Specify the **Name**, **Datatype**, and other data in the fields provided.
5. Click **Create Variable**.
   The Catalog system saves your variable definition.

You can use the variable in an SQL query for a Query data object. Users are prompted to enter a value when they run the data object.

> **NOTE**
> When you add a query runtime variable that is a drop-down variable, the number of values in the resulting drop-down list is limited to 1000. If the report query returns more than 1000 values, the system truncates these additional values. So, the user cannot view them in the drop-down list. For more information about how to increase this value, see the Modify XSL, XML, JavaScript, and Image Files section.

1. Click Set Folders.
   The Change Folder Location window opens.
2. Enter a folder name or click the folder icon to select an existing folder.
3. Click OK and the folder is set.

Set the permissions for the Data Object, as follows:

1. Click Set Permissions.
2. Set the access levels according to role, business unit, and CA EEM user groups, as applicable.
3. Click OK.

1. Select a Data Object and click Delete Items.
   A pop up opens.
2. Click OK to delete the Data Object.

{"URL":["/usm/wpf?Node=iclaunchpad.report_data_objects","/usm/wpf?Node=icguinode.reportsgenericdatacreate&Args=","/usm/wpf?Node=icguinode.reportsgenericvariablesavemodal","/usm/wpf?Node=icguinode.reportsgenericfoldersmultisavemodal","/usm/wpf?Node=icguinode.acl&sh=5%2Fc9%2FTT1MJdq25nu%2FnKwcgQnx2s%3D","/usm/wpf?Node=icguinode.reportsgenericdatalist&initialView=Tree"],"customLabelGetStarted":"Manage Data Objects","description":"IPCE_Manage_Data_Objects","customCards":[{"id":"IPCE_Add_A_Data_Object","type":"manage","title":"Add A Data Object"},{"id":"IPCE_Create_Variable","type":"manage","title":"Create Variable"},{"id":"IPCE_Set_Folders","type":"manage","title":"Set Folders"},{"id":"IPCE_Set_Permissions","type":"manage","title":"Set Permissions"},{"id":"IPCE_Delete_Items","type":"manage","title":"Delete Items"}]}

Manage Layouts

Administrators use layouts to present multiple report elements as one report. You can design custom layouts using data views, text, images, URLs, and other objects. You can specify positions, sizes, colors, borders, styles, and other settings.

- View layouts and organize them in folders.
- Manage off-line layouts that are based on a layout.
- Add or edit a layout. When you add or edit a layout, you can set its status. You can also set the permissions to specify the level of access of each role to the layout.

A layout defines the presentation of multiple report elements. Create and configure layouts to obtain the format you want for reports. You can use one of several sample layouts as a starting point. You can set permissions for the layout and optionally propagate those permissions to the underlying data views and data objects.

Follow these steps:

1. Click Administration, Report Builder, and select Layouts in the left menu.
2. Create or edit a layout.
3. Click one of the options to add the new element you want to the layout. For example, New Text or New Data View.
4. Drag the object to the location you want on the layout.
5. Configure the properties of the object by clicking the properties (i) icon and completing the fields.
6. Click Permissions and set the access levels according to role, business unit, and CA EEM user groups, as applicable, and click OK.
   The Set Permissions dialog closes, and you return to the Edit Custom Report Layout page.
7. Click Save and specify the status from one of the following options:
   - **Created**
     Specifies that the layout is visible *only* to the user who created it (the creator). No other user is permitted to view the layout, regardless of the permissions set for it. This option enables the creator to complete it before anyone else can access it.
   - **Available**
     Specifies that the layout is visible to the owner and to any user who is granted permissions to it.
8. Continue editing the layout as needed; save it periodically and when you are finished. Exit the Edit Custom Report Layout page by clicking Cancel.

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You can add or edit a Layout.

Follow these steps:

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8. Continue editing the layout as needed; save it periodically and when you are finished. Exit the Edit Custom Report Layout page by clicking Cancel.

{"URL":["/usm/wpf?Node=iclaunchpad.report_layouts","/usm/wpf?Node=icguinode.reportscustomlayoutedit&Args="],"customLabelGetStarted":"Manage Layout","description":"IPCE_Manage_Layouts","customCards":[{"id":"IPCE_Add_A_Layout","type":"manage","title":"Add A Layout"}]}

Manage Data Views

Administrators use data views to format the data from a data object and present it in tables and charts.

- Add, edit, delete, or view data views and assign them to folders.
- Export a data view in PDF or CSV (delimited) format.
- Manage off-line data views based on a data view.
- Set permissions to specify the level of access to a data view for each role.

You can add or edit a Data View.

Follow these steps:

1. Click Administration, Report Builder.
2. Click Data Views in the left menu.
3. Create a data view or edit an existing one.
4. Complete the fields and click Save. Use the Choose Data Object icon to select a data object to use for this data view.
5. (Optional) Configure how each field in the data object appears in a report: Click Column Rules to configure the presentation of a data view that appears as columns in a report. Select the column that you want in the Settings for Column field. Complete the fields and configure the settings on each tab.
   - Select Show Help to display help text for the tab.
   - Click Advanced to specify advanced settings.

Linking Tab

- **Link Items in Column *column-name* To Other Pages**
  Specifies whether to link the values in the column to another web page, either within or outside CA Service Catalog. Select this field to activate it and to open the fields under it for editing.
  - **Special**
    Specifies a link within CA Service Catalog to another data view, a report layout, or a GUI node.
  - **Link Address**
    Specifies one of the following options:
    - Another data view, a report layout, or a GUI node, if you used the Special link to add one.
    - A URL to a website or file share, if you did *not* use the Special field. Enter the URL manually.

    Also, the Link Address field optionally specifies a variable from the data object on which the data view is based. Click the Insert Variables icon to add a variable to this field.

Formulas Tab

- **Apply Formula to Column *column-name***
  Specifies whether to apply a JavaScript formula to the column value.
  Select this field to activate it and to open the fields under it for editing.
  - **Special**
    Specifies an image formula to apply on the data for the column. Use the formula to format a cell or row that is based on a value. Use the following format:

    ```
    IMG:image_file_path
    ```

    image_file_path specifies the folder and file name *under* the USM_HOME\view\webapps\usm folder.
    For example, the following line displays the add.gif image in the cell for the column:

    ```
    IMG:images/add.gif
    ```

    To include the cell text also, enter the column name variable within the single quotes. An example follows:

    ```
    IMG:images/add.gif]%Col1%
    ```

    If you use multiple Catalog Component servers, verify that this file resides in the same folder location on all servers.
  - **Column Formula**
    Specifies variable from the data object to use in the column formula. To use a variable, click the Insert Variables icon.
    Examples of formulas using JavaScript follow:

    ```
    100*%name%
    Math.max(%Col1%,%Col2%)
    (10*%Col1%)+(20/%Col2%))+' Units'
    %name%'.toUpperCase()
    ```

Translations Tab

- **Apply Translations to Column *column-name***
  Specifies whether to apply a translation formula to the column value.
  Select this field to activate it and to open the fields under it for editing.
  - **Translations**
    Substitutes a translated value for each data object value. You can optionally apply the translation before applying a formula to the column value.
    For example, a column can return data as integer value *1* which means opened, and *2* which means closed. You can apply translations that:
    - Display *Opened* in place of all values of type *1* in a column.
    - Display *Closed* in place of all values of type *2.*

Formatting Tab

- **Apply Formatting Settings to Column *column-name***
  Applies formatting to a column. Examples include font, justification, highlighting, and color.

Select this field to activate it and to open the fields under it for editing.

Summary Tab

- **Summary**
  Adds summary information for the column.
  Select this field to activate it and to open the fields under it so that you can select them. The fields that you select appear in the summary of the report.

Administrators use data views to format the data from a data object and present it in tables and charts.

- Add, edit, delete, or view data views and assign them to folders.
- Export a data view in PDF or CSV (delimited) format.
- Manage off-line data views based on a data view.
- Set permissions to specify the level of access to a data view for each role.

You can add or edit a Data View.

Follow these steps:

1. Click Administration, Report Builder.
2. Click Data Views in the left menu.
3. Create a data view or edit an existing one.
4. Complete the fields and click Save. Use the Choose Data Object icon to select a data object to use for this data view.
5. (Optional) Configure how each field in the data object appears in a report: Click Column Rules to configure the presentation of a data view that appears as columns in a report. Select the column that you want in the Settings for Column field. Complete the fields and configure the settings on each tab.
   - Select Show Help to display help text for the tab.
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  Specifies whether to link the values in the column to another web page, either within or outside CA Service Catalog.
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  - **Link Address**
    Specifies one of the following options:
    - Another data view, a report layout, or a GUI node, if you used the Special link to add one.
    - A URL to a website or file share, if you did *not* use the Special field. Enter the URL manually.

    Also, the Link Address field optionally specifies a variable from the data object on which the data view is based. Click the Insert Variables icon to add a variable to this field.

Formulas Tab

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  Specifies whether to apply a JavaScript formula to the column value.
  Select this field to activate it and to open the fields under it for editing.
  - **Special**
    Specifies an image formula to apply on the data for the column. Use the formula to format a cell or row that is based on a value. Use the following format:

    ```
    IMG:image_file_path
    ```

    image_file_path specifies the folder and file name *under* the USM_HOME\view\webapps\usm folder.
    For example, the following line displays the add.gif image in the cell for the column:

```
IMG:images/add.gif
```

To include the cell text also, enter the column name variable within the single quotes. An example follows:

```
IMG:images/add.gif]%Col1%
```

If you use multiple Catalog Component servers, verify that this file resides in the same folder location on all servers.

– **Column Formula**
Specifies variable from the data object to use in the column formula. To use a variable, click the Insert Variables icon.
Examples of formulas using JavaScript follow:

```
100*%name%
Math.max(%Col1%,%Col2%)
(10*%Col1%)+(20/%Col2%))+' Units'
%name%'.toUpperCase()
```

Translations Tab

- **Apply Translations to Column *column-name***
Specifies whether to apply a translation formula to the column value.
Select this field to activate it and to open the fields under it for editing.
  – **Translations**
  Substitutes a translated value for each data object value. You can optionally apply the translation before applying a formula to the column value.
  For example, a column can return data as integer value *1* which means opened, and *2* which means closed. You can apply translations that:
    - Display *Opened* in place of all values of type *1* in a column.
    - Display *Closed* in place of all values of type *2.*

Formatting Tab

- **Apply Formatting Settings to Column *column-name***
Applies formatting to a column. Examples include font, justification, highlighting, and color.
Select this field to activate it and to open the fields under it for editing.

Summary Tab

- **Summary**
Adds summary information for the column.
Select this field to activate it and to open the fields under it so that you can select them. The fields that you select appear in the summary of the report.

{"URL":["/usm/wpf?Node=iclaunchpad.report_data_views","/usm/wpf?Node=icguinode.reportsgenericdataviewedit&Args="],"customLabelGetStarted":"Manage Data Views","description":"IPCE_Manage_Data_Views","customCards":[{"id":"IPCE_Add_A_Data_View","type":"manage","title":"Add A Data View"}]}

View the Business Value Dashboards

Business value dashboards aggregate smaller chunks of data into higher-level actionable information that enables IT Managers to make informed decisions. The dashboards correlate information across incident and request management that is required for executives users/mangers whose responsibilities span multiple functions.

CA Service Management provides two dashboards namely, Service Demand Dashboard and Operational Effectiveness Dashboard. Information that these business value dashboards provide is as follows:

- Service support metrics categorized by parameters such as time period, location, and organization unit. This information helps IT managers understand the prevailing pattern and plan capacity for upcoming months/quarters.

For example, as an IT executive, you may want to understand the volume pattern of incidents or requests over the past three months to help you manage resources better. Also, you may want to identify the category of services which involved the maximum cost and cost pattern over the last six months.
You can access the Service Demand dashboard to view this information.

- Demand for services categorized by parameters such as time period, location, and organization unit that provide insight into consumption of services in a specific category.
  For example, as an IT manager, you may want to know the categories (software, hardware, network and so on) which had the maximum number of incidents in the last one month and the category's incident/request volume of a team in a particular location.
  You can access the Service Demand dashboard to view this information.
- Functional effectiveness information that helps process managers analyse the performance of a support group the impact on cost.
  For example, as a process manager, you may want to analyse the performance of an L1 function (number of services resolved) in different locations and identify the reason for varying performances of the teams. Also, in a support group, you may want to understand the number of incidents that were transferred from L1 to L2. If the number of escalations is high, the time and effort spent increases and thus results in a significant impact on cost and customer satisfaction. The information in the dashboard helps you identify any process changes/resource requirement/ knowledge gaps required to enhance the performance of the various support groups and the functions.
  You can access the Operational Effectiveness dashboard to view this information.

To understand how cost, effort, and volume of incident/request are calculated, see Important Metrics. Each dashboard is a service offering that is exposed through CA Service Catalog.

Follow these steps:

1. Access the service offerings from the CA Service Catalog user interface as a user with permissions to access the dashboards. For example, Executive User.
2. In the Home page, click Request a Service.
   The Request Page Appears.
3. Under Categories, click Service Management Dashboards.
4. Click an appropriate service offering to see the related information.
   Note: The Service Demand dashboard displays only information based on products you installed. For example, if CA Service Catalog is installed the dashboard displays information related to requests only. Contact your administrator for more information.

REST API

This article describes how to use the REpresentational State Transfer Application Programming Interface (REST API) available for CA Service Management, and provides usage examples.

In CA Service Management, any application using the CA Service Management REST API makes an HTTP request to a URI and parses the response. Such identification enables interaction with representations of the resource over a network. Also, each client to server request contains all the information necessary to understand the request, and does not use any stored context on the server. The REST API accesses the resources using a Uniform Resource Identifier (URI).

The developers can use the REST API directly to send HTTP requests to the server for the resource they want to manipulate. The developers only need an HTTP client library, which is available with most programming languages. The REST API is based on open standards and you can use any programming language to access.

CA Service Management REST calls support the following REST API methods:

- **POST (Create)**
  Creates a resource. The web service can respond with data or status indicating success or failure.
- **GET (Read)**

Performs a query on a resource and retrieves data. The data that is returned from the web service is a representation of the requested resource.

- **PUT (Update)**
  Updates an existing resource.
- **DELETE (Delete)**
  Removes an existing resource.

These resources can be manipulated by a set of pre-defined CRUD operations: Create, Read, Update, and Delete. In the case of HTTP, these operations are the Request methods GET, PUT, POST, and DELETE.

For testing the REST APIs, follow the steps:

1. Download the Swagger-ui **dist folder** .
2. Host the dist folder in any web server container (ex: Tomcat container) or paste it inside incidentMicroService Public folder.
3. Download the **JSON** file from CA Service Point REST API page and paste it inside the dist folder.
4. Navigate to the dist folder and open the **index.html** file. Edit the "URL" value to point it to the path of the JSON file.
5. Open the **JSON** file and change the host and port value to point it to the CA Service Point URL.
6. Navigate to the hosted URL and access the **index.html** file.
   For example: http://localhost:9002/dist/index.html (if Service Point is hosted in localhost at port number 9002 and the dist folder is inside incidentMicroService public folder).

REST API - Service Point

Perform the following steps:

1. Download the Swagger-ui dist folder.
2. Host the dist folder in any web server container (ex: Tomcat container) or paste it inside incidentMicroService Public folder.
3. Download the JSON File (attached in this topic). Save it locally.
4. Navigate to the dist folder and open the index.html file. Edit the "URL" value to point it to the path of the JSON file.
5. Open the JSON file and change the host and port value to point it to the CA Service Point URL.
6. Navigate to the hosted URL and access the index.html file.
   For example:*http://localhost:9002/dist/index.html* (if Service Point is hosted in localhost at port number 9002 and the dist folder is inside incidentMicroService public folder).

none

```
{
  "swagger": "2.0",
  "info": {
    "version": "Leh-17.2 GA",
    "title": "Service Point REST End Points",
    "contact": {
      "name": ""
    },
    "license": {
      "name": "",
      "url": ""
    }
  },
  "host": "<host_name>:<port_number>",
  "basePath": "/",
  "tags": [
    {
```

```
    "name": "User Authentication",
    "description": "user authentication API"
  },
  {
    "name": "Service Point Ticket",
    "description": "all about ServicePoint users tickets."
  },
  {
    "name": "Service Point Ticket Attributes",
    "description": "all about ServicePoint ticket attributes and their configurations."
  },
  {
    "name": "Service Point Ticket Activities",
    "description": "all about ServicePoint ticket activities."
  },
  {
    "name": "Service Point All Feed Activities",
    "description": "all about ServicePoint ALL FEED operations."
  },
  {
    "name": "Service Point Attachment",
    "description": "all about ServicePoint file uploads and attachments."
  },
  {
    "name": "Service Point Notification",
    "description": "all ServicePoint notifications for activities performed."
  },
  {
    "name": "Service Point Configuration Item",
    "description": "all about ServicePoint Configuration items / Assets."
  },
  {
    "name": "Service Point Chat",
    "description": "all about ServicePoint users chat sessions."
  },
  {
    "name": "Service Point Knowledge",
    "description": "all about ServicePoint Knowledge Documents."
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  {
    "name": "Service Point Knowledge Document Activities",
    "description": "all ServicePoint knowledge documents operations."
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  },
  {
    "name": "Service Point Generic Utilities",
    "description": "all ServicePoint factory agnostic utilities."
  },
  {
    "name": "Service Point My Ticket List",
```

```
      "description": "all ServicePoint tickets for logged in user. Please point to search server to test these
APIs."
    },
    {
      "name": "Service Point Categories and Service Catalog Offering Search",
      "description": "search relevant categories from Service Desk and offerings form Service Catalog if
integrated. Please point to search server to test these APIs."
    },
    {
      "name": "Service Point Solutions",
      "description": "all about ServicePoint Solutions. Please point to search server to test these APIs."
    }
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    "https"
  ],
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      "post": {
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        ],
        "summary": "Returns the Authentication Token 'X-AccessToken', as part of response headers, if the
provided user name and password is correct.",
        "description": " This is only for BASIC AUTHENTICATION sceanrio, for other authentication types please
refer SDM Documentation. This also provides the user data such as App access, role etc. \n N.B.: PLEASE
AUTHORIZE THIS BEFORE TRYING OUT.",
        "operationId": "getAccess",
        "security":[{"basicAuth":[]}],
        "produces": [
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        ],
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            "description": "Successful Operation. For example: {\"pri_cal\":0,\"teammsgsenabled\":\"true
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\":10002,\"sym\":\"Administration\",\"commonNameField\":\"Administration\",\"access_level\":{\"entity
\":\"acc_lvls\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.acc_lvls\",\"enum1\":30,\"relAttrField
\":30,\"sym\":\"Admin\",\"commonNameField\":\"Admin\"}},\"tenant\":{\"entity\":\"tenant\",\"entityClass\":
```

\"com.ca.casm.playApi.sdm.genpojo.tenant\",\"id\":\"8B062326375F14448FCD33AED3604A23\",\"relAttrField\":
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                    "type": "string"
                }
            }
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/getBUIAllConfig": {
      "get": {
        "tags": [
          "Service Point Ticket"
        ],
        "summary": "Returns all the configuration properties for Service Point App.",
        "operationId": "getBUIAllConfig",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "tenantUuid",
            "description": "This is an optional parameter. By default logged-in user's tenant is used to fetch
 the configuration. But can be passed to fetch other tenant's configuration details.",
            "in": "query",
            "required": false,
            "type": "string",
            "default": "null",
            "x-example": "8B062326375F14448FCD33AED3604A23"
          },
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            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
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        "responses": {
          "200": {
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```
            "description": "Successful Operation. For example: \n{\"properties\":{\"configName\":\"Default
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\"stateInProgress\":\"In-Progress\",\"inheritPublicAttr\":\"0\",\"searchSdmMinWt\":\"1\",\"stateStart\":
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\"global_phone_number\":\"\",\"showCatalog\":\"1\",\"showTrending\":\"0\"}}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/getAllowedStatus": {
      "get": {
        "tags": [
          "Service Point Ticket"
        ],
        "summary": "Returns the list of available Statuses based on allowed Status transition.",
        "description": "",
        "operationId": "getAllowedStatus",
        "produces": [
          "application/json"
        ],
        "parameters": [
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            "required": false,
            "type": "string",
            "default": "in",
            "x-example": "in"
          },
          {
            "name": "$currStatus",
            "in": "query",
            "description": "Current status (e.g. 'CL'). Default is set to empty. If no value specified it
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            "required": false,
            "type": "string",
            "x-example": "'OP'"
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            "description": "Access Token of the authenticated user",
            "required": true,
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```
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/getMyTicketDetails('{persid}')": {
      "get": {
        "tags": [
          "Service Point Ticket"
        ],
        "summary": "Returns ticket details along with extra attributes provided in the $attrList parameter.",
        "operationId": "getMyTicketDetails",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "persid",
            "description": "Persistent Id of the ticket (e.g. 'cr:400001')",
            "in": "path",
            "required": true,
            "type": "string",
            "x-example": "cr:400001"
          },
          {
            "name": "$attrList",
            "description": "Comma separated list of ticket attributes to fetch (e.g.
 'status,call_back_date,category')",
            "in": "query",
            "required": false,
            "type": "string",
            "x-example": "status,call_back_date,category"
          },
          {
            "name": "$expandList",
            "description": "This field is used to fetch any extra information of an attribute mentioned in the
 $attrList (e.g. 'category($select=ss_sym)')",
            "in": "query",
            "required": false,
            "type": "string",
            "x-example": "category($select=ss_sym)"
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```
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: \n{\"results\":{\"entity\":\"cr\",
\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cr\",\"description\":\"kjhkhkhkh Server1\",\"id\":400276,
\"open_date\":1551119200,\"persistent_id\":\"cr:400276\",\"relAttrField\":\"cr:400276\",\"ref_num\":
\"233\",\"commonNameField\":\"233\",\"resolve_date\":1551172235,\"summary\":\"kjhkhkhkh Server1\",
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\"relAttrField\":0},\"assignee\":{\"entity\":\"agt\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.agt
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\",\"id\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\",\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},
\"category\":{\"entity\":\"pcat\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.pcat\",\"persistent_id
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          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/getTicketsCustomData('{persid}')": {
      "get": {
        "tags": [
          "Service Point Ticket"
        ],
        "summary": "Returns ticket's summary attributes configured at SDM.",
        "operationId": "getConfigTicketData",
        "produces": [
          "application/json"
        ],
        "parameters": [
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```
          {
            "name": "persid",
            "description": "Persistent id of the ticket (e.g. 'cr:400001')",
            "in": "path",
            "required": true,
            "type": "string",
            "x-example": "cr:400001"
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
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          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: \n{\"attributes\":{\"1.0\":{\"attrName\":
\"Urgency\",\"attrValue\":\"urgency\",\"refObj\":\"urg\",\"reference\":1,\"attr_type\":\"0\",\"data\":\"1-
When Possible\"},\"2.0\":{\"attrName\":\"Category\",\"attrValue\":\"category\",\"refObj\":\"pcat\",\"reference
\":1,\"attr_type\":\"2\",\"data\":\"Applications\"},\"3.0\":{\"attrName\":\"Configuration Item\",\"attrValue
\":\"affected_resource\",\"refObj\":\"nr\",\"reference\":1,\"attr_type\":\"9\"},\"4.0\":{\"attrName\":\"Call
 Back On\",\"attrValue\":\"call_back_date\",\"refObj\":null,\"reference\":0,\"attr_type\":\"7\"},\"5.0\":
{\"attrName\":\"Last Modified\",\"attrValue\":\"last_mod_dt\",\"refObj\":null,\"reference\":0,\"attr_type\":
\"7\",\"data\":\"1551172235\"},\"6.0\":{\"attrName\":\"Resolved On\",\"attrValue\":\"resolve_date\",\"refObj
\":null,\"reference\":0,\"attr_type\":\"7\",\"data\":\"1551172235\"},\"10.0\":{\"attrName\":\"Root Cause
\",\"attrValue\":\"rootcause\",\"refObj\":\"rc\",\"reference\":1,\"attr_type\":\"0\",\"data\":\"rc1\"}},
\"properties\":{\"1\":{\"attrName\":\"Select if license expired\",\"attr_type\":\"2\",\"attrProp\":\"true
\",\"data\":\"Yes\",\"validationTypeId\":100002},\"2\":{\"attrName\":\"Dropdown 1\",\"attr_type\":\"2\",
\"attrProp\":\"true\",\"data\":\"102\",\"validationTypeId\":100003},\"3\":{\"attrName\":\"Provide Application
 Name\",\"attr_type\":\"2\",\"attrProp\":\"true\",\"data\":\"app\",\"validationTypeId\":-1}}}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/getMyActiveChatSession": {
      "get": {
        "tags": [
          "Service Point Chat"
        ],
        "summary": "Gets the active chat session details for the logged-in Service Point user.",
        "description": "",
        "operationId": "getMyActiveChatSession",
        "produces": [
```

```
          "application/json"
        ],
        "parameters": [
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: \n{\"results\":{\"status\":2,\"ticket_ref_num
\":\"148\",\"id\":400051,\"chatroom_type\":\"cr\",\"chatroom_id\":\"400169\",\"thread_id\":\"alg:400982\"}}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/checkActiveSessions": {
      "get": {
        "tags": [
          "Service Point Chat"
        ],
        "summary": "Checks if the logged-in Service Point user has an active chat session.",
        "description": "",
        "operationId": "checkActiveChatSessions",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: {isActiveChatSessionPresent : true}"
          },
```

```
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/createChatThread": {
      "post": {
        "tags": [
          "Service Point Chat"
        ],
        "summary": "Creates an activity log entry for initiating a chat session",
        "description": "Returns the activity log persistent id to be used for creating a chat session.",
        "operationId": "createChatThread",
        "produces": [
          "application/json"
        ],
        "consumes": [
          "multipart/form-data"
        ],
        "parameters": [
          {
            "name": "chatroomId",
            "in": "query",
            "description": "Ticket's persistent id for which user has requested for a chat session (e.g.
 cr:400001)",
            "required": true,
            "type": "string",
            "x-example": "cr:400001"
          },
          {
            "name": "parentDetails",
            "in": "formData",
            "description": "Ticket activity log details (e.g. '{\"description\":\"Initiated contextual chat\",
\"type\":{\"code\":\"CHATLOG\"},\"call_req_id\":{\"persistent_id\":\"cr:400001\"}}')",
            "required": true,
            "type": "string",
            "x-example": "{\"description\":\"Initiated contextual chat\",\"type\":{\"code\":\"CHATLOG\"},
\"call_req_id\":{\"persistent_id\":\"cr:400001\"}}"
          },
          {
            "name": "childFactory",
            "in": "formData",
            "description": "Casm Activity log details (e.g. 'casmalg')",
            "required": false,
            "type": "string",
            "x-example": "casmalg"
          },
          {
            "name": "parentChildLinkAttrName",
```

```
            "in": "formData",
            "description": "The linking field between activity log object and casm activity log object (e.g.
 'activity_id')",
            "required": false,
            "type": "string",
            "x-example": "activity_id"
          },
          {
            "name": "childDetails",
            "in": "formData",
            "description": "Casm activity log object details (e.g. '{\"modified_attr_name\":\"chat\",
\"old_value\":\"\",\"current_value\":\"U'E2FF59085FE43E4E8BAB9D23FB7EB419'\"}'",
            "required": false,
            "type": "string",
            "x-example": "{\"modified_attr_name\":\"chat\",\"old_value\":\"\",\"current_value\":
\"U'E2FF59085FE43E4E8BAB9D23FB7EB419'\"}"
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: {\"entity\":\"alg\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.alg\",\"action_desc\":\"Chat Log\",\"description\":\"Initiated contextual
 chat\",\"commonNameField\":\"Initiated contextual chat\",\"id\":401017,\"internal\":0,\"k_tool\":
\"PDM\",\"last_mod_dt\":1550836496,\"persistent_id\":\"alg:401017\",\"relAttrField\":\"alg:401017\",
\"producer_id\":\"alg\",\"system_time\":1550836496,\"time_stamp\":1550836496,\"analyst\":{\"entity\":\"cnt
\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cnt\",\"id\":\"6E68E6D3F1F6064399DFCC621BE8FB76\",
\"relAttrField\":\"6E68E6D3F1F6064399DFCC621BE8FB76\"},\"call_req_id\":{\"entity\":\"cr\",\"entityClass
\":\"com.ca.casm.playApi.sdm.genpojo.cr\",\"persistent_id\":\"cr:400169\",\"relAttrField\":\"cr:400169\"},
\"tenant\":{\"entity\":\"tenant\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.tenant\",\"id\":
\"8B062326375F14448FCD33AED3604A23\",\"relAttrField\":\"8B062326375F14448FCD33AED3604A23\"},\"type\":{\"entity
\":\"aty\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.aty\",\"code\":\"CHATLOG\",\"relAttrField\":
\"CHATLOG\"}}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/getChatHistory/{roomName}": {
      "get": {
```

```
        "tags": [
          "Service Point Chat"
        ],
        "summary": "Gets the chat history for the logged-in ServicePoint user for the passed-in ticket id
(here roomName is the ticket persistent id).",
        "description": "",
        "operationId": "createObject",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "roomName",
            "description": "Persistent id of the ticket for which chat history is requested for (e.g.
'cr:400176')",
            "in": "path",
            "required": true,
            "type": "string",
            "x-example": "cr:400176"
          },{
            "name": "$skip",
            "description": "Number of records to skip (e.g. '40')",
            "in": "query",
            "required": false,
            "type": "integer",
            "default": 0,
            "format": "int32",
            "x-example": 10
          },
          {
            "name": "$top",
            "description": "Number of records to fetch (e.g. '20')",
            "in": "query",
            "required": false,
            "type": "integer",
            "default": 10,
            "format": "int32",
            "x-example": 20
          },
          {
            "name": "$orderBy",
            "description": "Order in which chat history to be fetched (e.g. 'time_stamp DESC, id DESC')",
            "in": "query",
            "required": false,
            "type": "string",
            "default": null,
            "x-example": "time_stamp DESC, id DESC"
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
```

```
            "type": "string",
            "x-example":
 "891db3b702df20b77ab3a934026fe617b65851301dff5ccaa9efc0f801a687ecd740c4eb78ed3a89cf83c526a51e11ab1588971b62edda3ae35c3
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: \n {\"total-count\":6,\"results\":[{\"entity\":
\"chat_log\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.chat_log\",\"chat_room_name\":\"cr:400176\",
\"id\":400108,\"relAttrField\":400108,\"msg_html\":\"\\\"Chat session ended\\\"\",\"persistent_id\":
\"chat_log:400108\",\"thread_id\":\"alg:401517\",\"time_stamp\":1551261705,\"type\":\"chat_session_ended
\",\"sender\":{\"entity\":\"cnt\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cnt\",\"combo_name
\":\"emp\",\"commonNameField\":\"emp\",\"id\":\"6E68E6D3F1F6064399DFCC621BE8FB76\",\"relAttrField\":
\"6E68E6D3F1F6064399DFCC621BE8FB76\",\"userid\":\"emp\"}},{\"entity\":\"chat_log\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.chat_log\",\"chat_room_name\":\"cr:400176\",\"id\":400107,\"relAttrField
\":400107,\"msg_html\":\"\\\"Chat session initiated\\\"\",\"persistent_id\":\"chat_log:400107\",\"thread_id
\":\"alg:401517\",\"time_stamp\":1551261630,\"type\":\"chat_session_initiated\",\"sender\":{\"entity\":
\"cnt\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cnt\",\"combo_name\":\"emp\",\"commonNameField\":
\"emp\",\"id\":\"6E68E6D3F1F6064399DFCC621BE8FB76\",\"relAttrField\":\"6E68E6D3F1F6064399DFCC621BE8FB76\",
\"userid\":\"emp\"}},{\"entity\":\"chat_log\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.chat_log\",
\"chat_room_name\":\"cr:400176\",\"id\":400106,\"relAttrField\":400106,\"msg_html\":\"hello\",\"persistent_id
\":\"chat_log:400106\",\"thread_id\":\"alg:401517\",\"time_stamp\":1551261630,\"type\":\"group_message
\",\"sender\":{\"entity\":\"cnt\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cnt\",\"combo_name
\":\"emp\",\"commonNameField\":\"emp\",\"id\":\"6E68E6D3F1F6064399DFCC621BE8FB76\",\"relAttrField\":
\"6E68E6D3F1F6064399DFCC621BE8FB76\",\"userid\":\"emp\"}},{\"entity\":\"chat_log\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.chat_log\",\"chat_room_name\":\"cr:400176\",\"id\":400105,\"relAttrField
\":400105,\"msg_html\":\"hello\",\"persistent_id\":\"chat_log:400105\",\"thread_id\":\"alg:401516\",
\"time_stamp\":1551261611,\"type\":\"group_message\",\"sender\":{\"entity\":\"cnt\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.cnt\",\"combo_name\":\"ServiceDesk\",\"commonNameField\":\"ServiceDesk\",
\"id\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\",\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\",\"userid
\":\"ServiceDesk\"}},{\"entity\":\"chat_log\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.chat_log\",
\"chat_room_name\":\"cr:400176\",\"id\":400104,\"relAttrField\":400104,\"msg_html\":\"hi\",\"persistent_id
\":\"chat_log:400104\",\"thread_id\":\"alg:401516\",\"time_stamp\":1551261610,\"type\":\"group_message
\",\"sender\":{\"entity\":\"cnt\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cnt\",\"combo_name
\":\"ServiceDesk\",\"commonNameField\":\"ServiceDesk\",\"id\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\",
\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\",\"userid\":\"ServiceDesk\"}},{\"entity\":\"chat_log\",
\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.chat_log\",\"chat_room_name\":\"cr:400176\",\"id\":400103,
\"relAttrField\":400103,\"msg_html\":\"\\\"Chat session initiated\\\"\",\"persistent_id\":\"chat_log:400103\",
\"thread_id\":\"alg:401516\",\"time_stamp\":1551261608,\"type\":\"chat_session_initiated\",\"sender\":
{\"entity\":\"cnt\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cnt\",\"combo_name\":\"ServiceDesk
\",\"commonNameField\":\"ServiceDesk\",\"id\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\",\"relAttrField\":
\"E2FF59085FE43E4E8BAB9D23FB7EB419\",\"userid\":\"ServiceDesk\"}}]}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/getDefaultCategories": {
```

```
      "get": {
        "tags": [
          "Service Point Ticket"
        ],
        "summary": "Returns the default list of categories for 'cr' factory defined in Service Desk which is
 configurable.",
        "description": "",
        "operationId": "getDefaultCategories",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: {\"total-count\":5,\"results\":[{\"entity
\":\"pcat\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.pcat\",\"cawf_cache\":1551163261,\"cr_flag
\":1,\"description\":\"Applications\",\"id\":5103,\"in_flag\":1,\"persistent_id\":\"pcat:5103\",
\"relAttrField\":\"pcat:5103\",\"pr_flag\":1,\"producer_id\":\"pcat\",\"ss_sym\":\"Applications\",
\"sym\":\"Applications\",\"commonNameField\":\"Applications\",\"delete_flag\":{\"entity\":\"actbool\",
\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.actbool\",\"enum1\":0,\"relAttrField\":0},\"ss_include
\":{\"entity\":\"bool\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.bool\",\"enum1\":1,\"relAttrField
\":1}},{\"entity\":\"pcat\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.pcat\",\"cawf_cache
\":1551163261,\"cr_flag\":1,\"description\":\"Email\",\"id\":5109,\"in_flag\":1,\"persistent_id\":
\"pcat:5109\",\"relAttrField\":\"pcat:5109\",\"pr_flag\":1,\"producer_id\":\"pcat\",\"ss_sym\":\"Email
\",\"sym\":\"Email\",\"commonNameField\":\"Email\",\"delete_flag\":{\"entity\":\"actbool\",\"entityClass
\":\"com.ca.casm.playApi.sdm.genpojo.actbool\",\"enum1\":0,\"relAttrField\":0},\"ss_include\":{\"entity
\":\"bool\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.bool\",\"enum1\":1,\"relAttrField\":1}},
{\"entity\":\"pcat\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.pcat\",\"cawf_cache\":1551163261,
\"cr_flag\":1,\"description\":\"Hardware\",\"id\":5101,\"in_flag\":1,\"persistent_id\":\"pcat:5101\",
\"relAttrField\":\"pcat:5101\",\"pr_flag\":1,\"producer_id\":\"pcat\",\"ss_sym\":\"Hardware\",\"sym\":
\"Hardware\",\"commonNameField\":\"Hardware\",\"delete_flag\":{\"entity\":\"actbool\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.actbool\",\"enum1\":0,\"relAttrField\":0},\"ss_include\":{\"entity
\":\"bool\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.bool\",\"enum1\":1,\"relAttrField\":1}},
{\"entity\":\"pcat\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.pcat\",\"cawf_cache\":1551163261,
\"cr_flag\":1,\"description\":\"Networks\",\"id\":5102,\"in_flag\":1,\"persistent_id\":\"pcat:5102\",
\"relAttrField\":\"pcat:5102\",\"pr_flag\":1,\"producer_id\":\"pcat\",\"ss_sym\":\"Networks\",\"sym\":
\"Networks\",\"commonNameField\":\"Networks\",\"delete_flag\":{\"entity\":\"actbool\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.actbool\",\"enum1\":0,\"relAttrField\":0},\"ss_include\":{\"entity
\":\"bool\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.bool\",\"enum1\":1,\"relAttrField\":1}},
{\"entity\":\"pcat\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.pcat\",\"cawf_cache\":1551163261,
\"cr_flag\":1,\"description\":\"Software\",\"id\":5100,\"in_flag\":1,\"persistent_id\":\"pcat:5100\",
\"relAttrField\":\"pcat:5100\",\"pr_flag\":1,\"producer_id\":\"pcat\",\"ss_sym\":\"Software\",\"sym\":
```

\"Software\",\"commonNameField\":\"Software\",\"delete_flag\":{\"entity\":\"actbool\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.actbool\",\"enum1\":0,\"relAttrField\":0},\"ss_include\":{\"entity\":\"bool
\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.bool\",\"enum1\":1,\"relAttrField\":1}}]}"

```
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/helpAbout": {
      "get": {
        "tags": [
          "Service Point Utility"
        ],
        "summary": "Returns the help about text. First priority is tenanted if not found(text is empty or
 config itself is not defined), public about text is returned",
        "description": "",
        "operationId": "helpAbout",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: {\"phone\":\"\",\"about\":\"\"}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/attrInfo": {
      "get": {
        "tags": [
          "Service Point Ticket Attributes"
```

```
        ],
        "summary": "Returns the 'required attributes' and 'custom attributes' for 'cr' factory defined in
 service point configuration at SDM admnistrative page.",
        "description": "",
        "operationId": "attrInfo",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: {\"factoryName\":\"cr\",\"listAttrInfo\":
[{\"attributeName\":\"ref_num\",\"displayName\":\"Request #\",\"attrType\":2,\"locked\":false,\"srel\":false,
\"required\":true},{\"attributeName\":\"status\",\"displayName\":\"Status\",\"attrType\":2,\"refObjName
\":\"crs\",\"refAttrName\":\"code\",\"locked\":false,\"srel\":true,\"required\":true},{\"attributeName
\":\"active\",\"displayName\":\"Active\",\"attrType\":0,\"refObjName\":\"bool\",\"refAttrName\":\"enum\",
\"defaultValue\":\"1\",\"locked\":false,\"srel\":true,\"required\":true},{\"attributeName\":\"log_agent\",
\"displayName\":\"Reported By\",\"attrType\":9,\"refObjName\":\"cnt\",\"refAttrName\":\"id\",\"defaultValue
\":\"D97BAA7DC58FAE4A95BC127FF76514A7\",\"defaultCommonName\":\"ServiceDesk\",\"locked\":false,\"srel\":true,
\"required\":true},{\"attributeName\":\"customer\",\"displayName\":\"Affected End User\",\"attrType\":9,
\"refObjName\":\"cnt\",\"refAttrName\":\"id\",\"locked\":false,\"srel\":true,\"serviceProvEligible\":true,
\"required\":true},{\"attributeName\":\"category\",\"displayName\":\"Category\",\"attrType\":2,\"refObjName\":
\"pcat\",\"refAttrName\":\"persistent_id\",\"locked\":false,\"srel\":true,\"required\":false},{\"attributeName
\":\"priority\",\"displayName\":\"Priority\",\"attrType\":0,\"refObjName\":\"pri\",\"refAttrName\":\"enum
\",\"defaultValue\":\"0\",\"locked\":false,\"srel\":true,\"required\":true},{\"attributeName\":\"urgency\",
\"displayName\":\"Urgency\",\"attrType\":0,\"refObjName\":\"urg\",\"refAttrName\":\"enum\",\"defaultValue\":
\"0\",\"locked\":false,\"srel\":true,\"required\":false},{\"attributeName\":\"call_back_date\",\"displayName
\":\"Call Back On\",\"attrType\":7,\"locked\":false,\"srel\":false,\"required\":false}]}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/addLogComment": {
      "post": {
        "tags": [
          "Service Point Ticket Activities"
        ],
```

```
        "summary": "Add log comment to the ticket",
        "operationId": "addLogComment",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
"9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          },
          {
            "name": "persid",
            "in": "query",
            "required": true,
            "type": "string",
            "description": "Current Ticket's Persistent ID (e.g. 'cr:400060')",
            "x-example": "cr:400060"
          },
          {
            "name": "body",
            "in": "body",
            "required": true,
            "description": "Log comment to be added (e.g. 'This is a comment')",
            "schema": {
              "$ref": "#/definitions/LogComment"
            }
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: {\"RETCODE\":\"alg:401366\"}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/attmnt": {
      "post": {
        "tags": [
          "Service Point Attachment"
        ],
        "summary": "Upload an attachment to SDM and also links the attachment to the specified factory object
if parentFactory and parentId parameter values are provided, else simply uploads the file.",
```

```
        "description": "",
        "operationId": "uploadFile",
        "produces": [
          "application/json"
        ],
        "consumes": [
          "multipart/form-data"
        ],
        "parameters": [
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          },
          {
            "name": "attachmentFile",
            "in": "formData",
            "description": "Uploaded file info (file name along with file binary data) as form-data",
            "required": true,
            "type": "file"
          },
          {
            "name": "parentFactory",
            "in": "formData",
            "description": "Factory name where attached file will be linked. (e.g. For ticket's attachment,
 use factory 'cr')",
            "required": true,
            "type": "string",
            "x-example": "in"
          },
          {
            "name": "parentId",
            "in": "formData",
            "description": "Persistent id of the parent factory, if attachment is for tickets it should be
 tickets persistent id (e.g. 'cr:400001')",
            "required": true,
            "type": "string",
            "x-example": "cr:400001"
          },
          {
            "name": "attachmentDetails",
            "in": "formData",
            "description": "Attachment details: \n(e.g. {\"orig_file_name\":$File_Name$ ,\n\"repository\":
{\"persistent_id\":$persistent_id_of_the_repository$},\n\"attmnt_name\": $Attachment_Name$,\n\"description\":
$Description_for_the_Image$,\n\"link_only\":{\"enum1\":0}})",
            "required": true,
            "type": "string",
```

```
            "x-example": "{\"orig_file_name\":\"pexels-photo-236047.jpeg\",\"repository\":{\"persistent_id\":
\"doc_rep:1002\"},\"attmnt_name\":\"pexels-photo-236047.jpeg\",\"description\":\"pexels-photo-236047.jpeg\",
\"link_only\":{\"enum1\":0}}"
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: attmnt:400015"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/getBopSID": {
      "get": {
        "tags": [
          "Service Point Attachment"
        ],
        "summary": "Returns BOP session ID, which can be used to download file.",
        "description": "",
        "operationId": "getBopSID",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: 1533683469"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
```

```
   "/bui/getEEMAuth": {
     "get": {
       "tags": [
         "Service Point Generic Utilities"
       ],
       "summary": "Returns EEM Artifact to be used for EEM integrated solutions requests (e.g. ITPAM tasks
and Catalog requests)",
       "description": "",
       "operationId": "getEEMAuth",
       "produces": [
         "application/json"
       ],
       "parameters": [
         {
           "name": "X-AccessToken",
           "in": "header",
           "description": "Access Token of the authenticated user",
           "required": true,
           "type": "string",
           "x-example":
"9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
         }
       ],
       "responses": {
         "200": {
           "description": "Successful Operation. For example:
UserSession;Version-1.0;186c8316a0b736d9d68729679e1f5917-5c73714e-b925c10-be"
         },
         "401": {
           "description": "Invalid Authentication Token"
         },
         "500": {
           "description": "Internal Server Error"
         }
       }
     }
   },
   "/bui/attmnt('{persid}')/$value": {
     "get": {
       "tags": [
         "Service Point Attachment"
       ],
       "operationId": "attmnt",
       "summary": "Returns an attachment as binary data.",
       "produces": [
         "application/octet-stream"
       ],
       "parameters": [
         {
           "name": "persid",
           "in": "path",
           "description": "Attachment persistent id (e.g. 'attmnt:400406')",
           "required": true,
```

```
            "type": "string",
            "x-example": "attmnt:400406"
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
"9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. (return attachment binary data)"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/updateStatusForNotification": {
      "put": {
        "tags": [
          "Service Point Notification"
        ],
        "operationId": "updateStatusForNotification",
        "summary": "Marks the notification status as READ for the given notification persistent id",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "persid",
            "in": "query",
            "description": "Notification persistent id (e.g. 'lr:401016')",
            "required": true,
            "type": "string",
            "x-example": "lr:401016"
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
"9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
```

```
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: \"{all_lr=1}\""
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/updateStatusForAllNotification": {
      "put": {
        "tags": [
          "Service Point Notification"
        ],
        "operationId": "updateStatusForAllNotification",
        "summary": "Marks all the notification's based on the $filter value provided as READ.",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "$filter",
            "in": "query",
            "description": "This field can be used to filter which notifications we need to mark as READ (e.g.
 start_date > StartAtTime('PAST_WEEK') and status eq 2 and contact eq U'E2FF59085FE43E4E8BAB9D23FB7EB419')",
            "required": true,
            "type": "string",
            "x-example": "start_date > StartAtTime('PAST_WEEK') and status eq 2 and contact eq
 U'E2FF59085FE43E4E8BAB9D23FB7EB419'"
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: [{\"entity\":\"all_lr\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.all_lr\",\"persistent_id\":\"lr:401064\",\"relAttrField\":\"lr:401064\"},
{entity\":\"all_lr\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.all_lr\",\"persistent_id\":
\"lr:401063\",\"relAttrField\":\"lr:401063\"}]"
          },
```

```
      "401": {
        "description": "Invalid Authentication Token"
      },
      "500": {
        "description": "Internal Server Error"
      }
    }
  }
},
"/bui/getMyResources": {
  "get": {
    "tags": [
      "Service Point Configuration Item"
    ],
    "operationId": "getMyResources",
    "summary": "Returns all the resources(Assets) the user is having access to",
    "produces": [
      "application/json"
    ],
    "parameters": [
      {
        "name": "$skip",
        "description": "Number of records to skip (e.g. '40')",
        "in": "query",
        "required": false,
        "type": "integer",
        "default": 0,
        "format": "int32",
        "x-example": 10
      },
      {
        "name": "$top",
        "description": "Number of records to fetch (e.g. '20')",
        "in": "query",
        "required": false,
        "type": "integer",
        "default": 10,
        "format": "int32",
        "x-example": 20
      },
      {
        "name": "$spe",
        "description": "Service Provider Elligible flag (e.g. 'true')",
        "in": "query",
        "required": false,
        "type": "boolean",
        "default": false,
        "x-example": true
      },
      {
        "name": "$useSyncFetch",
        "description": "Whether to use Sync Fetch (e.g. 'true')",
        "in": "query",
```

```
            "required": false,
            "type": "boolean",
            "default": true,
            "x-example": true
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: {\"results\":[{\"resourceName\":\"Ser\",
\"resourceId\":\"A3FDA2994DE7D6458764A992E4B106D5\",\"fields\":[]}],\"total-count\":\"1\"}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/getDocument({id})": {
      "get": {
        "tags": [
          "Service Point Knowledge"
        ],
        "operationId": "getDocument",
        "summary": "Returns a Knowledge Document and updates the document's hits count if $updateHits
 parameter is true",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "id",
            "in": "path",
            "description": "Knowledge Document id (e.g. '400100')",
            "required": true,
            "type": "integer",
            "x-example": "400100",
            "format": "int32"
          },
          {
            "name": "$updateHits",
```

```
            "description": "This flag reperesents whether we should update the hit count for the document or
 not. If it's true we will get a bubble up transaction id as part of response, which could be later used in
 'rateDocument' API.",
            "in": "query",
            "required": false,
            "type": "boolean",
            "default": false,
            "x-example": true
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. e.g: {\"id\":400100,\"persistent_id\":\"KD:400100\",\"title
\":\"VIEW ALL THE PROGRAMS OR APPLICATIONS WHICH ARE CURRENTLY RUNNING\",\"summary\":\"How to view all the
 programs or applications which are currently running in MS Windows 7.\",\"problem\":\"How to view all the
 programs or applications which are currently running in MS Windows 7.\",\"resolution\":\"
           <B>To view all the programs or  applications which are currently running in MS Windows 7:</B><BR>
      <BR>      1.     From  the <B>Start </B>window, click on <B>Desktop</B>.<BR>
    <BR>      2.     Press  <B>Windows + Tab</B> button, to see what applications are
 running.<BR>       <BR>      3.     To  check <B>Programs </B>or<B> Applications</
B> running in background, press <B>Ctrl  + Shift + ESC.</B><BR>      <B> </B><BR>      <img width=403
  height=453 src=\\\"\\\" casm-
embedded-img=\\\"RepId=doc_rep:1000&AttmntId=400227&Bpsid=0\\\">  <BR>      <B> </B><BR>      <B>Note</
B>: Click on the  desired <B>Program </B>or<B> Application</B> and then click on <B>End Task</B>.<BR>
  <BR>       <BR>       <BR>      Copyright 2014 KnowledgeBroker<BR>      All Rights Reserved.
 Other brand and  product names are the property of their respective owners.<BR>   \",\"versionCount\":1,
\"userSolvedCount\":0,\"ticketSolvedCount\":0,\"buid\":400051,\"type\":1,\"averageRating\":\"1\",\"voteCount
\":1}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/addKDComment({id})": {
      "post": {
        "tags": [
          "Service Point Knowledge Document Activities"
        ],
        "summary": "Add a comment to an Existing Knowledge Document",
```

```
        "operationId": "addKDComment",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          },
          {
            "name": "id",
            "description": "Id of the knowledge document for which the comment is to be added (e.g. 400615)",
            "in": "path",
            "required": true,
            "type": "integer",
            "x-example": 400615
          },
          {
            "name": "body",
            "in": "body",
            "required": true,
            "description": "Log comment to be added to Knowledge Document along with the comment type.
\n This values represents if the document requires any further action to be performed on the it like
 correction, promotion, or retiremnt etc.\n This value is configured by the admin. For employee user,
 as he/she does not have any adminitsrative privileges , this value sould always be 1 else it will
 throw error. \n (e.g. '{\n\"comment\": \"This is a comment\",  \n\"commenttype\": 1,  \n\"useruuid\":
 \"6E68E6D3F1F6064399DFCC621BE8FB76\"\n}')",
            "schema": {
              "$ref": "#/definitions/KDComment"
            }
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: \n{\"id\":400051,\"persistent_id\":
\"O_COMMENTS:400051\",\"comment\":\"hey fine\",\"timestamp\":1551313265,\"username\":\"ServiceDesk\",
\"comboname\":\"ServiceDesk\"}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/rateDocument({id})": {
```

```
      "post": {
        "tags": [
          "Service Point Knowledge Document Activities"
        ],
        "summary": "Rate an Existing Knowledge Document",
        "operationId": "rateDocument",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
"9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          },
          {
            "name": "id",
            "description": "Id of the knowledge document which is to be rated (e.g. 400615)",
            "in": "path",
            "required": true,
            "type": "integer",
            "x-example": 400615
          },
          {
            "name": "body",
            "in": "body",
            "required": true,
            "description": "Rating for the provided document along with the bubble up transaction Id (if any).
\n (e.g. '{\n  \"buid\": 400207,\n  \"rating\": 3\n}')",
            "schema": {
              "$ref": "#/definitions/RateDocument"
            }
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation."
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/kdLinkInfo({id})": {
      "get": {
```

```
        "tags": [
          "Service Point Knowledge Document Activities"
        ],
        "summary": "It tells whether the KD article attached to the ticket is an accepted solution for the
problem mentioned in the ticket.",
        "description" : "Response '0' -> not an accepted solution, Response '1' -> accepted as a solution.",
        "operationId": "kdLinkInfo",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
"9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          },
          {
            "name": "id",
            "description": "id of the knowledge document which is to be linked (e.g. 400615)",
            "in": "path",
            "required": true,
            "type": "integer",
            "x-example": 400257
          },
          {
            "name": "$crPersId",
            "in": "query",
            "description": "Persistent id of the ticket to which the provided knowledge document ID is a
linked (e.g. 'cr:401016')",
            "required": true,
            "type": "string",
            "x-example": "cr:401016"
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For Example:{\"kdLinkCount\":1}. \nResponse '0' -> not an
accepted solution, Response '1' -> accepted as a solution."
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/unLinkSolution({id})": {
```

```
      "post": {
        "tags": [
          "Service Point Knowledge Document Activities"
        ],
        "summary": "Marks the knowledge document as not an accepted solution for the problem mentioned in the
ticket",
        "operationId": "unLinkSolution",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
"9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          },
          {
            "name": "id",
            "description": "Id of the knowledge document which is to be unlinked (e.g. 400602)",
            "in": "path",
            "required": true,
            "type": "integer",
            "x-example": 400602
          },
          {
            "name": "body",
            "in": "body",
            "description": "Persistent id of the ticket to which the above knowledge document is linked (e.g.
'{\"crPersId\": \"cr:400602\"}')",
            "required": true,
            "schema": {
              "$ref": "#/definitions/UnlinkSolution"
            }
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation."
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/getKDAttachments({id})": {
```

```
"get": {
  "tags": [
    "Service Point Knowledge"
  ],
  "operationId": "getKDAttachments",
  "summary": "Returns all Knowledge Document attachments for the provided document id.",
  "produces": [
    "application/json"
  ],
  "parameters": [
    {
      "name": "id",
      "in": "path",
      "description": "Knowledge Document id (e.g. '400100')",
      "required": true,
      "type": "integer",
      "x-example": "400100",
      "format": "int32"
    },
    {
      "name": "$skip",
      "description": "Number of records to skip (e.g. '40')",
      "in": "query",
      "required": false,
      "type": "integer",
      "default": 0,
      "format": "int32",
      "x-example": 10
    },
    {
      "name": "$top",
      "description": "Number of records to fetch (e.g: '20')",
      "in": "query",
      "required": false,
      "type": "integer",
      "default": 10,
      "format": "int32",
      "x-example": 20
    },
    {
      "name": "$orderBy",
      "description": "Order in which Attachment Details to be fetched (e.g. 'TIMESTAMP ASC')",
      "in": "query",
      "required": false,
      "type": "string",
      "default": null,
      "x-example": "TIMESTAMP ASC"
    },
    {
      "name": "$useSyncFetch",
      "description": "Whether to use Sync Fetch (e.g. 'true')",
      "in": "query",
      "required": false,
```

```
            "type": "boolean",
            "default": true,
            "x-example": true
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: {\"total-count\":1,\"results\":[{\"id\":400408,
\"persistent_id\":\"attmnt:400408\",\"timestamp\":1551265365,\"name\":\"pexels-photo-236047.jpeg\",\"status\":
\"INSTALLED\"}]}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/getKDComments({id})": {
      "get": {
        "tags": [
          "Service Point Knowledge"
        ],
        "operationId": "getKDComments",
        "summary": "Returns Knowledge Document comments for the provided document id.",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "id",
            "in": "path",
            "description": "Knowledge Document id (e.g. '400100')",
            "required": true,
            "type": "integer",
            "x-example": "400100",
            "format": "int32"
          },
          {
            "name": "$skip",
            "description": "Number of records to skip (e.g. '40')",
            "in": "query",
```

```
            "required": false,
            "type": "integer",
            "default": 0,
            "format": "int32",
            "x-example": 10
          },
          {
            "name": "$top",
            "description": "Number of records to fetch (e.g. '20')",
            "in": "query",
            "required": false,
            "type": "integer",
            "default": 10,
            "format": "int32",
            "x-example": 20
          },
          {
            "name": "$orderBy",
            "description": "Order in which knowledge document comments to be fetched (e.g. 'TIMESTAMP ASC')",
            "in": "query",
            "required": false,
            "type": "string",
            "default": null,
            "x-example": "TIMESTAMP ASC"
          },
          {
            "name": "$useSyncFetch",
            "description": "Whether to use Sync Fetch (e.g. 'true')",
            "in": "query",
            "required": false,
            "type": "boolean",
            "default": true,
            "x-example": true
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: {\"total-count\":2,\"results\":[{\"id\":400009,
\"persistent_id\":\"O_COMMENTS:400009\",\"comment\":\"comments\",\"timestamp\":1550749248,\"username\":\"emp
\",\"comboname\":\"emp\"},{\"id\":400008,\"persistent_id\":\"O_COMMENTS:400008\",\"comment\":\"ffeddback\",
\"timestamp\":1550749243,\"username\":\"emp\",\"comboname\":\"emp\"}]}"
          },
          "401": {
            "description": "Invalid Authentication Token"
```

```
        },
        "500": {
          "description": "Internal Server Error"
        }
      }
    }
  },
  "/bui/changeStatus/{factory}('{persid}')": {
    "post": {
      "tags": [
        "Service Point Ticket Activities"
      ],
      "operationId": "changeStatus",
      "summary": "Returns updated status value of the provided ticket persistent id.",
      "produces": [
        "application/json"
      ],
      "consumes": [
        "multipart/form-data"
      ],
      "parameters": [
        {
          "name": "factory",
          "in": "path",
          "description": "Name of the factory. (e.g. For tickets it would be 'cr')",
          "required": true,
          "type": "string",
          "x-example": "cr"
        },
        {
          "name": "persid",
          "in": "path",
          "description": "Persistent id of the ticket (e.g. 'cr:400406')",
          "required": true,
          "type": "string",
          "x-example": "cr:400406"
        },
        {
          "name" : "status",
          "in" : "formData",
          "description" : "The new status value of the ticket (e.g. 'CL')",
          "required" : true,
          "type" : "string",
          "x-example" : "CL"
        },
        {
          "name" : "description",
          "in" : "formData",
          "description" : "Description for the status update (e.g. 'Issue is not fixed, hence opening it
again.')",
          "required" : false,
          "type" : "string",
          "x-example" : "Issue is not fixed, hence opening it again."
```

```
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
"9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation."
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/attrCtrl": {
      "get": {
        "tags": [
          "Service Point Ticket Attributes"
        ],
        "operationId": "attrCtrl",
        "summary": "Returns all dependent attributes for the provided ticket attribute ($attrname parameter)",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "$objname",
            "description": " Ticket Type (e.g. 'in')",
            "in": "query",
            "required": true,
            "type": "string",
            "x-example": "in"
          },
          {
            "name": "$attrname",
            "description": " Ticket attribute for which dependent attribute needs to be fetched (e.g.
'status')",
            "in": "query",
            "required": false,
            "type": "string",
            "default": "status",
            "x-example": "status"
          },
```

```
          {
            "name": "$value",
            "description": " Value of the above ticket attribute (e.g. Ticket status 'OP')",
            "in": "query",
            "required": false,
            "type": "string",
            "default": "OP",
            "x-example": "OP"
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: {\"factoryName\":\"in\",\"listAttrInfo
\":[{\"attributeName\":\"priority\",\"displayName\":\"Priority\",\"attrType\":0,\"refObjName\":\"pri
\",\"refAttrName\":\"enum\",\"defaultValue\":\"0\",\"locked\":false,\"required\":true,\"srel\":true},
{\"attributeName\":\"call_back_flag\",\"displayName\":\"Call Back Flag\",\"attrType\":0,\"defaultValue\":
\"0\",\"locked\":false,\"required\":false,\"srel\":false},{\"attributeName\":\"call_back_text\",\"displayName
\":\"call_back_text\",\"attrType\":2,\"locked\":false,\"required\":true,\"srel\":false},{\"attributeName\":
\"major_incident\",\"displayName\":\"Major Incident\",\"attrType\":0,\"refObjName\":\"bool\",\"refAttrName\":
\"enum\",\"defaultValue\":\"0\",\"locked\":false,\"required\":false,\"srel\":true}]}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/allFeeds('{persid}')": {
      "get": {
        "tags": [
          "Service Point All Feed Activities"
        ],
        "operationId": "allFeeds",
        "summary": "Returns Activity Log Information for the provided ticket",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "persid",
            "in": "path",
```

```
            "description": "Persistent ID of the ticket (e.g. 'cr:400276')",
            "required": true,
            "type": "string",
            "x-example": "cr:400276"
          },
          {
            "name": "$skip",
            "description": "Number of records to skip (e.g. '40')",
            "in": "query",
            "required": false,
            "type": "integer",
            "default": 0,
            "format": "int32",
            "x-example": 10
          },
          {
            "name": "$top",
            "description": "Number of records to fetch (e.g. '20')",
            "in": "query",
            "required": false,
            "type": "integer",
            "default": 10,
            "format": "int32",
            "x-example": 20
          },
          {
            "name": "$orderBy",
            "description": "Order in which Activity Log Details to be fetched (e.g. 'time_stamp ASC')",
            "in": "query",
            "required": false,
            "type": "string",
            "default": null,
            "x-example": "time_stamp ASC"
          },
          {
            "name": "$filter",
            "description": "Which Activity log to display (e.g. type in ('FLD'))",
            "in": "query",
            "required": false,
            "type": "string",
            "default": null,
            "x-example": "type in ('FLD')"
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
"9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
```

```
        "responses": {
          "200": {
            "description": "Successful Operation. For example: {\"total-count\":6,\"ActivityLogEntry\":
[{\"entity\":\"alg\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.alg\",\"action_desc\":\"FIELD='Summary'
 OLD='kjhkhkhkh' NEW='kjhkhkhkh Server1'\nFIELD='Description' OLD='kjhkhkhkh' NEW='kjhkhkhkh Server1'\n
\",\"description\":\"FIELD='Summary' OLD='kjhkhkhkh' NEW='kjhkhkhkh Server1'\nFIELD='Description'
 OLD='kjhkhkhkh' NEW='kjhkhkhkh Server1'\n\",\"commonNameField\":\"FIELD='Summary' OLD='kjhkhkhkh'
 NEW='kjhkhkhkh Server1'\nFIELD='Description' OLD='kjhkhkhkh' NEW='kjhkhkhkh Server1'\n\",\"id\":401322,
\"persistent_id\":\"alg:401322\",\"relAttrField\":\"alg:401322\",\"time_spent\":16,\"time_stamp\":1551169184,
\"analyst\":{\"entity\":\"cnt\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cnt\",\"combo_name
\":\"ServiceDesk\",\"commonNameField\":\"ServiceDesk\",\"combo_name2\":\"ServiceDesk,,\",\"id\":
\"E2FF59085FE43E4E8BAB9D23FB7EB419\",\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},\"type\":{\"entity
\":\"aty\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.aty\",\"code\":\"FLD\",\"relAttrField\":\"FLD
\"},\"ModifiedFields\":[{\"modified_attr_name\":\"Summary\",\"old_value\":\"kjhkhkhkh\",\"current_value\":
\"kjhkhkhkh Server1\"},{\"modified_attr_name\":\"Description\",\"old_value\":\"kjhkhkhkh\",\"current_value
\":\"kjhkhkhkh Server1\"}]},{\"entity\":\"alg\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.alg
\",\"action_desc\":\"FIELD='Root Cause' OLD='' NEW='rc1'\n\",\"description\":\"FIELD='Root Cause'
 OLD='' NEW='rc1'\n\",\"commonNameField\":\"FIELD='Root Cause' OLD='' NEW='rc1'\n\",\"id\":401248,
\"persistent_id\":\"alg:401248\",\"relAttrField\":\"alg:401248\",\"time_spent\":0,\"time_stamp\":1551119745,
\"analyst\":{\"entity\":\"cnt\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cnt\",\"combo_name
\":\"ServiceDesk\",\"commonNameField\":\"ServiceDesk\",\"combo_name2\":\"ServiceDesk,,\",\"id\":
\"E2FF59085FE43E4E8BAB9D23FB7EB419\",\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},\"type
\":{\"entity\":\"aty\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.aty\",\"code\":\"FLD\",
\"relAttrField\":\"FLD\"},\"ModifiedFields\":[{\"modified_attr_name\":\"Root Cause\",\"old_value\":null,
\"current_value\":\"rc1\"}]},{\"entity\":\"alg\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.alg
\",\"action_desc\":\"FIELD='Root Cause' OLD='rc2' NEW=''\n\",\"description\":\"FIELD='Root Cause'
 OLD='rc2' NEW=''\n\",\"commonNameField\":\"FIELD='Root Cause' OLD='rc2' NEW=''\n\",\"id\":401247,
\"persistent_id\":\"alg:401247\",\"relAttrField\":\"alg:401247\",\"time_spent\":1,\"time_stamp\":1551119731,
\"analyst\":{\"entity\":\"cnt\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cnt\",\"combo_name
\":\"ServiceDesk\",\"commonNameField\":\"ServiceDesk\",\"combo_name2\":\"ServiceDesk,,\",\"id\":
\"E2FF59085FE43E4E8BAB9D23FB7EB419\",\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},\"type
\":{\"entity\":\"aty\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.aty\",\"code\":\"FLD\",
\"relAttrField\":\"FLD\"},\"ModifiedFields\":[{\"modified_attr_name\":\"Root Cause\",\"old_value\":
\"rc2\",\"current_value\":null}]},{\"entity\":\"alg\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.alg
\",\"action_desc\":\"FIELD='Root Cause' OLD='' NEW='rc2'\n\",\"description\":\"FIELD='Root Cause'
 OLD='' NEW='rc2'\n\",\"commonNameField\":\"FIELD='Root Cause' OLD='' NEW='rc2'\n\",\"id\":401245,
\"persistent_id\":\"alg:401245\",\"relAttrField\":\"alg:401245\",\"time_spent\":0,\"time_stamp\":1551119687,
\"analyst\":{\"entity\":\"cnt\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cnt\",\"combo_name
\":\"ServiceDesk\",\"commonNameField\":\"ServiceDesk\",\"combo_name2\":\"ServiceDesk,,\",\"id\":
\"E2FF59085FE43E4E8BAB9D23FB7EB419\",\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},\"type\":
{\"entity\":\"aty\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.aty\",\"code\":\"FLD\",\"relAttrField
\":\"FLD\"},\"ModifiedFields\":[{\"modified_attr_name\":\"Root Cause\",\"old_value\":null,\"current_value
\":\"rc2\"}]},{\"entity\":\"alg\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.alg\",\"action_desc
\":\"FIELD='Root Cause' OLD='rc2' NEW=''\n\",\"description\":\"FIELD='Root Cause' OLD='rc2' NEW=''\n\",
\"commonNameField\":\"FIELD='Root Cause' OLD='rc2' NEW=''\n\",\"id\":401243,\"persistent_id\":\"alg:401243\",
\"relAttrField\":\"alg:401243\",\"time_spent\":0,\"time_stamp\":1551119522,\"analyst\":{\"entity\":\"cnt
\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cnt\",\"combo_name\":\"ServiceDesk\",\"commonNameField
\":\"ServiceDesk\",\"combo_name2\":\"ServiceDesk,,\",\"id\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\",
\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},\"type\":{\"entity\":\"aty\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.aty\",\"code\":\"FLD\",\"relAttrField\":\"FLD\"},\"ModifiedFields\":
[{\"modified_attr_name\":\"Root Cause\",\"old_value\":\"rc2\",\"current_value\":null}]},{\"entity\":
\"alg\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.alg\",\"action_desc\":\"FIELD='Root Cause'
 OLD='ROOT CAUSE1' NEW='rc2'\nFIELD='Major Incident' OLD='YES' NEW='NO'\n\",\"description\":\"FIELD='Root
```

Cause' OLD='ROOT CAUSE1' NEW='rc2'\nFIELD='Major Incident' OLD='YES' NEW='NO'\n\",\"commonNameField
\":\"FIELD='Root Cause' OLD='ROOT CAUSE1' NEW='rc2'\nFIELD='Major Incident' OLD='YES' NEW='NO'\n\",\"id
\":401240,\"persistent_id\":\"alg:401240\",\"relAttrField\":\"alg:401240\",\"time_spent\":0,\"time_stamp
\":1551119289,\"analyst\":{\"entity\":\"cnt\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cnt\",
\"combo_name\":\"ServiceDesk\",\"commonNameField\":\"ServiceDesk\",\"combo_name2\":\"ServiceDesk,,\",\"id\":
\"E2FF59085FE43E4E8BAB9D23FB7EB419\",\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},\"type\":{\"entity
\":\"aty\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.aty\",\"code\":\"FLD\",\"relAttrField\":\"FLD\"},
\"ModifiedFields\":[{\"modified_attr_name\":\"Root Cause\",\"old_value\":\"ROOT CAUSE1\",\"current_value\":
\"rc2\"},{\"modified_attr_name\":\"Major Incident\",\"old_value\":\"YES\",\"current_value\":\"NO\"}]}]}"

```
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/{factory}('{persid}')": {
      "get": {
        "tags": [
          "Service Point Generic Utilities"
        ],
        "summary": "Returns the object details specified by the persistent id of the mentioned factory",
        "description": "",
        "operationId": "getObject",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "factory",
            "in": "path",
            "description": "Name of the factory. (e.g. For tickets it would be 'cr')",
            "required": true,
            "type": "string",
            "x-example": "cr"
          },
          {
            "name": "persid",
            "description": "Persistent Id of the object. (e.g. For tickets it would be something like
 'cr:400001')",
            "in": "path",
            "required": true,
            "type": "string",
            "x-example": "cr:400001"
          },
          {
            "name": "$expand",
            "in": "query",
```

```
            "description": "This field is used to fetch any extra information, if any, for the
 fields mentioned in the below $select parameter (e.g. For tickets it would be something like
 'category($select=ss_sym:description),status($select=resolved:description)')\n If extra attributes
 are belongs to master data, add the attributes into masterdata.cache.factory.{factory} key in
 general configuration.\n For the above example 'masterdata.cache.factory.crs' key should contain
 'description,resolved'",
            "required": false,
            "type": "string",
            "default": "null",
            "x-example": "category($select=ss_sym:description),status($select=resolved:description)"
          },
          {
            "name": "$select",
            "in": "query",
            "description": "Comma separated list of fields to fetch (e.g. For ticket it would be something
 like 'status,call_back_date,category'). If not passed, will fetch all fields.",
            "required": false,
            "type": "string",
            "default": "null",
            "x-example": "status,call_back_date,category"
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example for tickets : \n{\"results\":{\"entity\":
\"cr\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cr\",\"description\":\"kjhkhkhkh Server1\",\"id
\":400276,\"open_date\":1551119200,\"persistent_id\":\"cr:400276\",\"relAttrField\":\"cr:400276\",\"ref_num
\":\"233\",\"commonNameField\":\"233\",\"resolve_date\":1551172235,\"summary\":\"kjhkhkhkh Server1\",
\"active\":{\"entity\":\"bool\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.bool\",\"enum1\":0,
\"relAttrField\":0},\"assignee\":{\"entity\":\"agt\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.agt
\",\"combo_name\":\"ServiceDesk\",\"commonNameField\":\"ServiceDesk\",\"combo_name2\":\"ServiceDesk,,
\",\"id\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\",\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},
\"category\":{\"entity\":\"pcat\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.pcat\",\"persistent_id
\":\"pcat:5103\",\"relAttrField\":\"pcat:5103\"},\"priority\":{\"entity\":\"pri\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.pri\",\"enum1\":0,\"relAttrField\":0,\"persistent_id\":\"pri:505\",
\"sym\":\"None\",\"commonNameField\":\"None\"},\"status\":{\"entity\":\"crs\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.crs\",\"code\":\"CL\",\"relAttrField\":\"CL\",\"hold\":0,\"persistent_id
\":\"crs:5201\",\"sym\":\"Closed\",\"commonNameField\":\"Closed\"},\"tenant\":{\"entity\":\"tenant\",
\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.tenant\",\"id\":\"8B062326375F14448FCD33AED3604A23\",
\"relAttrField\":\"8B062326375F14448FCD33AED3604A23\",\"name\":\"TenantSP\",\"commonNameField\":\"TenantSP
\"},\"type\":{\"entity\":\"crt\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.crt\",\"code\":\"I\",
\"relAttrField\":\"I\"},\"state\":[{\"label\":\"Submitted\",\"active\":0},{\"label\":\"In-Progress\",\"active
\":0},{\"label\":\"Closed\",\"active\":1}],\"call_help_desk\":\"\"}}"
          },
```

```
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      },
      "put": {
        "tags": [
          "Service Point Generic Utilities"
        ],
        "summary": "Updates an object with passed in details as part of form data specified by the 'persid'
 parameter for the provided factory.",
        "description": "Here parentDetails parameter is the only mandatory field. But if we want to update
 any child objects also then we have to provide childFactory, parentChildLinkAttrName and at least one
 childDetails parameter.",
        "operationId": "updateObject",
        "produces": [
          "application/json"
        ],
        "consumes": [
          "multipart/form-data"
        ],
        "parameters": [
          {
            "name": "factory",
            "in": "path",
            "description": "Name of the factory. (e.g. For tickets it would be 'cr')",
            "required": true,
            "type": "string",
            "x-example": "cr"
          },
          {
            "name": "persid",
            "description": "Persistent Id of the object. (e.g. For tickets it would be something like
 'cr:400001')",
            "in": "path",
            "required": true,
            "type": "string",
            "x-example": "cr:400001"
          },
          {
            "name": "parentDetails",
            "in": "formData",
            "description": "Parent object details (e.g. For tickets it would be something like \n
 '{\"time_spent_sum\":\"\",\"rootcause\":{\"relAttrField\":400003},\"urgency\":{\"relAttrField\":0},
\"call_back_date\":\"\",\"call_back_flag\":\"\",\"call_back_text\":\"wee\",\"major_incident\":{\"relAttrField
\":0},\"customer\":{\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},\"description\":\"test\",
\"log_agent\":{\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},\"tenant\":{\"relAttrField\":
\"8B062326375F14448FCD33AED3604A23\"},\"category\":{\"relAttrField\":\"pcat:5103\"}}')",
            "required": true,
            "type": "string",
```

```
            "x-example": "{\"time_spent_sum\":\"\",\"rootcause\":{\"relAttrField\":400003},\"urgency
\":{\"relAttrField\":0},\"call_back_date\":\"\",\"call_back_flag\":\"\",\"call_back_text\":\"wee\",
\"major_incident\":{\"relAttrField\":0},\"customer\":{\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},
\"description\":\"test\",\"log_agent\":{\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},\"tenant\":
{\"relAttrField\":\"8B062326375F14448FCD33AED3604A23\"},\"category\":{\"relAttrField\":\"pcat:5103\"}}"
          },
          {
            "name": "childFactory",
            "in": "formData",
            "description": "Child object name (e.g. For tickets properties it would be something like
 'cr_prp')",
            "required": false,
            "type": "string",
            "x-example": "cr_prp"
          },
          {
            "name": "parentChildLinkAttrName",
            "in": "formData",
            "description": "The linking field between parent object and child object (e.g. For tickets
 properties it would be something like 'owning_cr')",
            "required": false,
            "type": "string",
            "x-example": "owning_cr"
          },
          {
            "name": "childDetails",
            "in": "formData",
            "description": "Child object details (e.g. For tickets, to update ticket's category properties,
 it would be something like: \n'{\"label\":\"Select if license expired\",\"required\":0,\"sequence\":1,
\"value\":\"Yes\",\"validation_rule\":{\"entity\":\"prpval_rule\",\"id\":400001,\"relAttrField\":400001},
\"validation_type\":{\"entity\":\"prpval_type\",\"id\":100002,\"relAttrField\":100002}})'",
            "required": false,
            "type": "string",
            "x-example": "{\"label\":\"Select if license expired\",\"required\":0,\"sequence\":1,\"value
\":\"Yes\",\"validation_rule\":{\"entity\":\"prpval_rule\",\"id\":400001,\"relAttrField\":400001},
\"validation_type\":{\"entity\":\"prpval_type\",\"id\":100002,\"relAttrField\":100002}}"
          },
          {
            "name": "childDetails1",
            "in": "formData",
            "description": "Child object details (e.g. For tickets, to update ticket's category properties,
 it would be something like: \n'{\"label\":\"Dropdown 1\",\"required\":1,\"sequence\":2,\"value\":\"101\",
\"validation_rule\":{\"entity\":\"prpval_rule\",\"id\":400002,\"relAttrField\":400002},\"validation_type\":
{\"entity\":\"prpval_type\",\"id\":100003,\"relAttrField\":100003}}'",
            "required": false,
            "type": "string",
            "x-example": "{\"label\":\"Dropdown 1\",\"required\":1,\"sequence\":2,\"value\":\"101\",
\"validation_rule\":{\"entity\":\"prpval_rule\",\"id\":400002,\"relAttrField\":400002},\"validation_type\":
{\"entity\":\"prpval_type\",\"id\":100003,\"relAttrField\":100003}}"
          },
          {
            "name": "childDetails2",
            "in": "formData",
```

```
            "description": "Child object details (e.g. For tickets, to update ticket's category properties, it
 would be something like: \n'{\"label\":\"Provide Application Name\",\"required\":1,\"sequence\":3,\"value\":
\"ew\"})'",
            "required": false,
            "type": "string",
            "x-example": "{\"label\":\"Provide Application Name\",\"required\":1,\"sequence\":3,\"value\":\"ew
\"}"
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": false,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: For tickets \"{in=9, cr_prp=4}\""
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/bui/{factory}": {
      "get": {
        "tags": [
          "Service Point Generic Utilities"
        ],
        "summary": "Returns the objects based on the parameters for the mentioned factory",
        "description": "",
        "operationId": "getObjects",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "factory",
            "in": "path",
            "description": "Name of the factory. (e.g. For tickets it would be 'cr')",
            "required": true,
            "type": "string",
            "x-example": "cr"
          },
          {
            "name": "$filter",
```

```
            "in": "query",
            "description": "This field can be used to filter which set of objects to fetch for (e.g. For
tickets 'active eq 1'). This filter is dom server object filter.",
            "required": true,
            "type": "string",
            "x-example": "active eq 1"
          },
          {
            "name": "$expand",
            "in": "query",
            "description": "This field is used to fetch any extra information, if any, for the
fields mentioned in the below $select parameter (e.g. For tickets it would be something like
'category($select=ss_sym:sym),status($select=code)')",
            "required": false,
            "type": "string",
            "default": "null",
            "x-example": "category($select=ss_sym:sym),status($select=code)"
          },
          {
            "name": "$select",
            "in": "query",
            "description": "Comma separated list of fields to fetch (e.g. For ticket it would be something
like 'status,call_back_date,category'). If not passed, will fetch all fields.",
            "required": false,
            "type": "string",
            "default": "null",
            "x-example": "status,call_back_date,category"
          },
          {
            "name": "$orderBy",
            "in": "query",
            "description": "Order in which objects needs to be fetched (e.g. For tickets 'last_mod_dt DESC')",
            "required": false,
            "type": "string",
            "default": "null",
            "x-example": "last_mod_dt DESC"
          },
          {
            "name": "$skip",
            "description": "Number of records to skip (e.g. '40'). Both $skip and $top parameters are used for
pagination. Based on this value, from result set it skips the number of records and fetches number of records
specified by $top parameter from the result set.",
            "in": "query",
            "required": false,
            "type": "integer",
            "default": 0,
            "format": "int32",
            "x-example": 10
          },
          {
            "name": "$top",
            "description": "Number of records to fetch (e.g. '20')",
            "in": "query",
```

```
            "required": false,
            "type": "integer",
            "default": 10,
            "format": "int32",
            "x-example": 20
          },
          {
            "name": "$useSyncFetch",
            "description": "Set this flag to false, where it will use select query mechanism which reduces the
 load on the Object Engine (domsrvr) but it may reduce the performance as it does not use domsrvr cache. (e.g.
 'true')",
            "in": "query",
            "required": false,
            "type": "boolean",
            "default": true,
            "x-example": true
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          },
          {
            "name": "X-RoleID",
            "in": "header",
            "description": "RoleID of the default role configured for the Service point application",
            "required": false,
            "type": "string",
            "x-example": ""
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example for tickets: \n{\"total-count\":12,
\"next-page\":\"/cr?$filter=active eq 1&$select=ref_num&$skip=3&$top=3\",\"results\":[{\"entity
\":\"cr\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cr\",\"persistent_id\":\"cr:400079\",
\"relAttrField\":\"cr:400079\",\"ref_num\":\"100\",\"commonNameField\":\"100\"},{\"entity\":\"cr\",
\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cr\",\"persistent_id\":\"cr:400080\",\"relAttrField
\":\"cr:400080\",\"ref_num\":\"101\",\"commonNameField\":\"101\"},{\"entity\":\"cr\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.cr\",\"persistent_id\":\"cr:400081\",\"relAttrField\":\"cr:400081\",
\"ref_num\":\"102\",\"commonNameField\":\"102\"}]}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
```

```
      },
      "post": {
        "tags": [
          "Service Point Generic Utilities"
        ],
        "summary": "Creates a new object for the provided factory with passed in details as part of form data.
 (We can provide multiple child details.)",
        "description": "Here parentDetails parameter is the only mandatory field. But if we want to update
 any child objects also then we have to provide childFactory, parentChildLinkAttrName and at least one
 childDetails parameter.",
        "operationId": "createGenericObject",
        "produces": [
          "application/json"
        ],
        "consumes": [
          "multipart/form-data"
        ],
        "parameters": [
          {
            "name": "factory",
            "in": "path",
            "description": "Name of the factory. (e.g. For tickets it would be 'cr')",
            "required": true,
            "type": "string",
            "x-example": "cr"
          },
          {
            "name": "parentDetails",
            "in": "formData",
            "description": "Parent object details (e.g. For tickets it would be something like \n
 '{\"time_spent_sum\":\"\",\"rootcause\":{\"relAttrField\":400003},\"urgency\":{\"relAttrField\":0},
\"call_back_date\":\"\",\"call_back_flag\":\"\",\"call_back_text\":\"wee\",\"major_incident\":{\"relAttrField
\":0},\"customer\":{\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},\"description\":\"test\",
\"log_agent\":{\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},\"tenant\":{\"relAttrField\":
\"8B062326375F14448FCD33AED3604A23\"},\"category\":{\"relAttrField\":\"pcat:5103\"}}')",
            "required": true,
            "type": "string",
            "x-example": "{\"time_spent_sum\":\"\",\"rootcause\":{\"relAttrField\":400003},\"urgency
\":{\"relAttrField\":0},\"call_back_date\":\"\",\"call_back_flag\":\"\",\"call_back_text\":\"wee\",
\"major_incident\":{\"relAttrField\":0},\"customer\":{\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},
\"description\":\"test\",\"log_agent\":{\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},\"tenant\":
{\"relAttrField\":\"8B062326375F14448FCD33AED3604A23\"},\"category\":{\"relAttrField\":\"pcat:5103\"}}"
          },
          {
            "name": "childFactory",
            "in": "formData",
            "description": "Child object name (e.g. For tickets properties it would be something like
 'cr_prp')",
            "required": false,
            "type": "string",
            "x-example": "cr_prp"
          },
          {
```

```
            "name": "parentChildLinkAttrName",
            "in": "formData",
            "description": "The linking field between parent object and child object (e.g. For tickets
 properties it would be something like 'owning_cr')",
            "required": false,
            "type": "string",
            "x-example": "owning_cr"
          },
          {
            "name": "childDetails",
            "in": "formData",
            "description": "Child object details (e.g. For tickets, to update ticket's category properties,
 it would be something like: \n'{\"label\":\"Select if license expired\",\"required\":0,\"sequence\":1,
\"value\":\"Yes\",\"validation_rule\":{\"entity\":\"prpval_rule\",\"id\":400001,\"relAttrField\":400001},
\"validation_type\":{\"entity\":\"prpval_type\",\"id\":100002,\"relAttrField\":100002}})'",
            "required": false,
            "type": "string",
            "x-example": "{\"label\":\"Select if license expired\",\"required\":0,\"sequence\":1,\"value
\":\"Yes\",\"validation_rule\":{\"entity\":\"prpval_rule\",\"id\":400001,\"relAttrField\":400001},
\"validation_type\":{\"entity\":\"prpval_type\",\"id\":100002,\"relAttrField\":100002}}"
          },
          {
            "name": "childDetails1",
            "in": "formData",
            "description": "Child object details (e.g. For tickets, to update ticket's category properties,
 it would be something like: \n'{\"label\":\"Dropdown 1\",\"required\":1,\"sequence\":2,\"value\":\"101\",
\"validation_rule\":{\"entity\":\"prpval_rule\",\"id\":400002,\"relAttrField\":400002},\"validation_type\":
{\"entity\":\"prpval_type\",\"id\":100003,\"relAttrField\":100003}}'",
            "required": false,
            "type": "string",
            "x-example": "{\"label\":\"Dropdown 1\",\"required\":1,\"sequence\":2,\"value\":\"101\",
\"validation_rule\":{\"entity\":\"prpval_rule\",\"id\":400002,\"relAttrField\":400002},\"validation_type\":
{\"entity\":\"prpval_type\",\"id\":100003,\"relAttrField\":100003}}"
          },
          {
            "name": "childDetails2",
            "in": "formData",
            "description": "Child object details (e.g. For tickets, to update ticket's category properties, it
 would be something like: \n'{\"label\":\"Provide Application Name\",\"required\":1,\"sequence\":3,\"value\":
\"ew\"})'",
            "required": false,
            "type": "string",
            "x-example": "{\"label\":\"Provide Application Name\",\"required\":1,\"sequence\":3,\"value\":\"ew
\"}"
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
```

```
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example for tickets: \n{\"entity\":\"in\",\"entityClass
\":\"com.ca.casm.playApi.sdm.genpojo.in\",\"audit_userid_f\":1,\"auto_assign\":0,\"auto_imp\":-1,
\"auto_urg\":-1,\"call_back_flag\":0,\"call_back_text\":\"wee\",\"dderun_f\":0,\"description\":\"test\",
\"edit_from_play_api\":1,\"edit_from_web\":0,\"eval_service\":1,\"fldchange_log_f\":1,\"from_edit_in_list\":0,
\"group_flag\":0,\"heat_overlay\":\"{}\",\"id\":400353,\"incident_priority\":0,\"init_imp\":3,\"init_log_id_f
\":401813,\"init_urg\":2,\"last_mod_dt\":1551434350,\"macro_predicted_violation\":0,\"man_imp\":0,\"man_urg
\":0,\"needs_beep\":0,\"open_date\":1551434350,\"persistent_id\":\"cr:400353\",\"relAttrField\":\"cr:400353\",
\"predicted_sla_violation\":0,\"producer_id\":\"cr\",\"ref_num\":\"262\",\"commonNameField\":\"262\",
\"saved_f\":1,\"sla_violation\":0,\"summary\":\"test\",\"target_closed_count\":0,\"target_hold_count\":0,
\"target_resolved_count\":0,\"target_start_last\":1551434350,\"time_spent_sum\":0,\"web_url\":\"http://
localhost:8080/CAisd/pdmweb.exe?OP=SEARCH+FACTORY=cr+SKIPLIST=1+QBE.EQ.id=400353\",\"wf_attr_changed\":1,
\"active\":{\"entity\":\"bool\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.bool\",\"enum1\":1,
\"relAttrField\":1},\"assignee\":{\"entity\":\"agt\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.agt
\",\"id\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\",\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},
\"audit_userid\":{\"entity\":\"cnt\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cnt\",\"id\":
\"E2FF59085FE43E4E8BAB9D23FB7EB419\",\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},\"category
\":{\"entity\":\"pcat\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.pcat\",\"persistent_id\":
\"pcat:5103\",\"relAttrField\":\"pcat:5103\"},\"created_via\":{\"entity\":\"intfc\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.intfc\",\"id\":3576,\"relAttrField\":3576},\"customer\":{\"entity\":\"cnt
\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cnt\",\"id\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\",
\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},\"impact\":{\"entity\":\"imp\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.imp\",\"enum1\":0,\"relAttrField\":0,\"persistent_id\":\"imp:1605\",\"sym
\":\"None\",\"commonNameField\":\"None\"},\"incorrectly_assigned\":{\"entity\":\"bool\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.bool\",\"enum1\":0,\"relAttrField\":0},\"last_mod_by\":{\"entity\":\"cnt
\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.cnt\",\"id\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\",
\"relAttrField\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},\"log_agent\":{\"entity\":\"cnt\",\"entityClass
\":\"com.ca.casm.playApi.sdm.genpojo.cnt\",\"id\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\",\"relAttrField
\":\"E2FF59085FE43E4E8BAB9D23FB7EB419\"},\"major_incident\":{\"entity\":\"bool\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.bool\",\"enum1\":0,\"relAttrField\":0},\"priority\":{\"entity\":\"pri
\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.pri\",\"enum1\":0,\"relAttrField\":0,\"persistent_id
\":\"pri:505\",\"sym\":\"None\",\"commonNameField\":\"None\"},\"remote_control_used\":{\"entity\":
\"bool\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.bool\",\"enum1\":0,\"relAttrField\":0},
\"resolvable_at_lower\":{\"entity\":\"bool\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.bool
\",\"enum1\":0,\"relAttrField\":0},\"return_to_service\":{\"entity\":\"bool\",\"entityClass\":
\"com.ca.casm.playApi.sdm.genpojo.bool\",\"enum1\":0,\"relAttrField\":0},\"rootcause\":{\"entity\":
\"rc\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.rc\",\"id\":400003,\"relAttrField\":400003},
\"status\":{\"entity\":\"crs\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.crs\",\"code\":\"OP\",
\"relAttrField\":\"OP\",\"hold\":0,\"persistent_id\":\"crs:5200\",\"sym\":\"Open\",\"commonNameField\":
\"Open\"},\"tenant\":{\"entity\":\"tenant\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.tenant\",\"id
\":\"8B062326375F14448FCD33AED3604A23\",\"relAttrField\":\"8B062326375F14448FCD33AED3604A23\"},\"type\":
{\"entity\":\"crt\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.crt\",\"code\":\"I\",\"relAttrField\":
\"I\"},\"urgency\":{\"entity\":\"urg\",\"entityClass\":\"com.ca.casm.playApi.sdm.genpojo.urg\",\"enum1\":0,
\"relAttrField\":0,\"persistent_id\":\"urg:1100\",\"sym\":\"1-When Possible\",\"commonNameField\":\"1-When
 Possible\"}}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
```

```
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/gs/mytkt_active": {
      "get": {
        "tags": [
          "Service Point My Ticket List"
        ],
        "operationId": "mytkt_active",
        "summary": "Returns active tickets of the logged-in user. It tries to fetch maximum 100 tickets with
 authorization using elastic search. In case you want to fetch more, use generic factory api.",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: \n{\"type\":\"mytkt_active\",\"fields
\":[{\"tenant.name\":\"TenantSP\",\"summary\":\"test\",\"external_system_ticket\":\"\",\"last_mod_dt
\":\"1551852890\",\"description\":\"Update ticket\",\"tenant.phone_number\":\"\",\"status.desc\":\"Open
\",\"priority\":\"0\",\"status.code\":\"OP\",\"tenant.alt_phone\":\"\",\"open_dt\":\"1551434350\",
\"ref_num\":\"262\",\"assignee_f_n\":\"\",\"inc_id\":\"400353\",\"customer.last_name\":\"ServiceDesk
\",\"assignee_l_n\":\"ServiceDesk\",\"customer.first_name\":\"\",\"id\":\"cr:400353\"},{\"tenant.name
\":\"TenantSP\",\"summary\":\"Emaol\",\"external_system_ticket\":\"\",\"last_mod_dt\":\"1551852900\",
\"description\":\"Emaol\",\"tenant.phone_number\":\"\",\"status.desc\":\"Open\",\"priority\":\"0\",
\"status.code\":\"OP\",\"tenant.alt_phone\":\"\",\"open_dt\":\"1551428107\",\"ref_num\":\"261\",\"assignee_f_n
\":\"\",\"inc_id\":\"400352\",\"customer.last_name\":\"ServiceDesk\",\"assignee_l_n\":\"ServiceDesk\",
\"customer.first_name\":\"\",\"id\":\"cr:400352\"},{\"tenant.name\":\"TenantSP\",\"summary\":\"qwert\",
\"external_system_ticket\":\"\",\"last_mod_dt\":\"1551414168\",\"description\":\"qwert\",\"tenant.phone_number
\":\"\",\"status.desc\":\"Open\",\"priority\":\"0\",\"status.code\":\"OP\",\"tenant.alt_phone\":\"\",\"open_dt
\":\"1551346940\",\"ref_num\":\"259\",\"assignee_f_n\":\"\",\"inc_id\":\"400302\",\"customer.last_name
\":\"ServiceDesk\",\"assignee_l_n\":\"ServiceDesk\",\"customer.first_name\":\"\",\"id\":\"cr:400302\"},
{\"tenant.name\":\"TenantSP\",\"summary\":\"Hshshsb\",\"external_system_ticket\":\"\",\"last_mod_dt
\":\"1551345930\",\"description\":\"Hshshsb\",\"tenant.phone_number\":\"\",\"status.desc\":\"Open\",
\"priority\":\"0\",\"status.code\":\"OP\",\"tenant.alt_phone\":\"\",\"open_dt\":\"1551345930\",\"ref_num\":
\"258\",\"assignee_f_n\":\"\",\"inc_id\":\"400301\",\"customer.last_name\":\"ServiceDesk\",\"assignee_l_n
\":\"ServiceDesk\",\"customer.first_name\":\"\",\"id\":\"cr:400301\"},{\"tenant.name\":\"TenantSP\",
\"summary\":\"Dry\",\"external_system_ticket\":\"\",\"last_mod_dt\":\"1551345601\",\"description\":
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\"400300\",\"customer.last_name\":\"ServiceDesk\",\"assignee_l_n\":\"ServiceDesk\",\"customer.first_name
```

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\"cdscsdc\",\"tenant.phone_number\":\"\",\"status.desc\":\"Open\",\"priority\":\"0\",\"status.code\":\"OP\",
\"tenant.alt_phone\":\"\",\"open_dt\":\"1551243718\",\"ref_num\":\"240\",\"assignee_f_n\":\"\",\"inc_id\":
\"400283\",\"customer.last_name\":\"ServiceDesk\",\"assignee_l_n\":\"ServiceDesk\",\"customer.first_name\":
\"\",\"id\":\"cr:400283\"},{\"tenant.name\":\"TenantSP\",\"summary\":\"He of these his a trucks the kicked
\",\"external_system_ticket\":\"\",\"last_mod_dt\":\"1551181023\",\"description\":\"He of these his a trucks
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\"400282\",\"customer.last_name\":\"ServiceDesk\",\"assignee_l_n\":\"ServiceDesk\",\"customer.first_name\":
\"\",\"id\":\"cr:400282\"}]}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/gs/mytkt_inactive": {
      "get": {
        "tags": [
          "Service Point My Ticket List"
        ],
        "operationId": "mytkt_inactive",
        "summary": "Returns inactive tickets of the logged-in user. It tries to fetch maximum 100 tickets with
 authorization using elastic search. In case you want to fetch more, use generic factory api.",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",

```
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: \n{\"type\":\"mytkt_inactive\",\"fields\":
[{\"tenant.name\":\"TenantSP\",\"summary\":\"tkt from bui\",\"external_system_ticket\":\"\",\"description
\":\"tkt from bui\",\"tenant.phone_number\":\"\",\"status.desc\":\"Closed\",\"resolve_dt\":\"1551345967\",
\"priority\":\"0\",\"status.code\":\"CL\",\"tenant.alt_phone\":\"\",\"ref_num\":\"255\",\"assignee_f_n
\":\"\",\"inc_id\":\"400298\",\"customer.last_name\":\"ServiceDesk\",\"assignee_l_n\":\"ServiceDesk\",
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\",\"assignee_l_n\":\"ServiceDesk\",\"customer.first_name\":\"\",\"id\":\"cr:400297\"},{\"tenant.name\":
\"TenantSP\",\"summary\":\"rtt\",\"external_system_ticket\":\"\",\"description\":\"rtt\",\"tenant.phone_number
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\"tenant.alt_phone\":\"\",\"ref_num\":\"248\",\"assignee_f_n\":\"\",\"inc_id\":\"400291\",\"customer.last_name
\":\"ServiceDesk\",\"assignee_l_n\":\"ServiceDesk\",\"customer.first_name\":\"\",\"id\":\"cr:400291\"},
{\"tenant.name\":\"TenantSP\",\"summary\":\"knkndwjks\",\"external_system_ticket\":\"\",\"description\":
\"knkndwjks\",\"tenant.phone_number\":\"\",\"status.desc\":\"Closed\",\"resolve_dt\":\"1551267047\",\"priority
\":\"0\",\"status.code\":\"CL\",\"tenant.alt_phone\":\"\",\"ref_num\":\"247\",\"assignee_f_n\":\"\",\"inc_id
\":\"400290\",\"customer.last_name\":\"ServiceDesk\",\"assignee_l_n\":\"ServiceDesk\",\"customer.first_name
\":\"\",\"id\":\"cr:400290\"},{\"tenant.name\":\"TenantSP\",\"summary\":\"rfgfb\",\"external_system_ticket
\":\"\",\"description\":\"rfgfb\",\"tenant.phone_number\":\"\",\"status.desc\":\"Closed\",\"resolve_dt
\":\"1551266960\",\"priority\":\"0\",\"status.code\":\"CL\",\"tenant.alt_phone\":\"\",\"ref_num\":
\"246\",\"assignee_f_n\":\"\",\"inc_id\":\"400289\",\"customer.last_name\":\"ServiceDesk\",\"assignee_l_n
\":\"ServiceDesk\",\"customer.first_name\":\"\",\"id\":\"cr:400289\"},{\"tenant.name\":\"TenantSP\",
\"summary\":\"wwewewe\",\"external_system_ticket\":\"\",\"description\":\"wwewewe\",\"tenant.phone_number
\":\"\",\"status.desc\":\"Closed\",\"resolve_dt\":\"1551266662\",\"priority\":\"0\",\"status.code
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\"customer.last_name\":\"ServiceDesk\",\"assignee_l_n\":\"ServiceDesk\",\"customer.first_name\":\"\",\"id
\":\"cr:400288\"},{\"tenant.name\":\"TenantSP\",\"summary\":\"kjhkhkhkh Server1\",\"external_system_ticket
\":\"\",\"description\":\"kjhkhkhkh Server1\",\"tenant.phone_number\":\"\",\"status.desc\":\"Closed\",
\"resolve_dt\":\"1551172235\",\"priority\":\"0\",\"status.code\":\"CL\",\"tenant.alt_phone\":\"\",\"ref_num
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\":\"ServiceDesk\",\"customer.first_name\":\"\",\"id\":\"cr:400276\"},{\"tenant.name\":\"TenantSP\",
\"summary\":\"fasadfsdf Server1\",\"external_system_ticket\":\"\",\"description\":\"fasadfsdf Server1\",
\"tenant.phone_number\":\"\",\"status.desc\":\"Closed\",\"resolve_dt\":\"1551172212\",\"priority\":\"0\",
\"status.code\":\"CL\",\"tenant.alt_phone\":\"\",\"ref_num\":\"232\",\"assignee_f_n\":\"\",\"inc_id\":
\"400275\",\"customer.last_name\":\"ServiceDesk\",\"assignee_l_n\":\"ServiceDesk\",\"customer.first_name
\":\"\",\"id\":\"cr:400275\"},{\"tenant.name\":\"TenantSP\",\"summary\":\"hello \",\"external_system_ticket
\":\"\",\"description\":\"hello \",\"tenant.phone_number\":\"\",\"status.desc\":\"Closed\",\"resolve_dt
\":\"1551345989\",\"priority\":\"0\",\"status.code\":\"CL\",\"tenant.alt_phone\":\"\",\"ref_num\":\"229\",
\"assignee_f_n\":\"\",\"inc_id\":\"400272\",\"customer.last_name\":\"ServiceDesk\",\"assignee_l_n\":
\"ServiceDesk\",\"customer.first_name\":\"\",\"id\":\"cr:400272\"},{\"tenant.name\":\"TenantSP\",\"summary\":
\"Issue\",\"external_system_ticket\":\"\",\"description\":\"Issue\",\"tenant.phone_number\":\"\",\"status.desc
\":\"Closed\",\"resolve_dt\":\"1551090745\",\"priority\":\"0\",\"status.code\":\"CL\",\"tenant.alt_phone\":
\"\",\"ref_num\":\"188\",\"assignee_f_n\":\"\",\"inc_id\":\"400231\",\"customer.last_name\":\"ServiceDesk\",
\"assignee_l_n\":\"ServiceDesk\",\"customer.first_name\":\"\",\"id\":\"cr:400231\"}]}"
          },
          "401": {
```

```
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/suggestedSolutions": {
      "get": {
        "tags": [
          "Service Point Solutions"
        ],
        "operationId": "suggestedSolutions",
        "summary": "Returns all probable solutions for the provided search string.",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "$searchString",
            "description": "The problem string to search for",
            "in": "query",
            "required": false,
            "type": "string",
            "default": null,
            "x-example": "email"
          },
          {
            "name": "$tenantId",
            "description": "Tenant uuid, if passed, it retrieves supertenant group tenants as well as SP
tenant as well, else just uses the logged-in user's tenant.",
            "in": "query",
            "required": false,
            "type": "string",
            "default": null,
            "x-example": "8B062326375F14448FCD33AED3604A23"
          },
          {
            "name": "$skip",
            "description": "Number of records to skip (e.g. '10'). Both $skip and $top parameters are used for
pagination. Based on this value, from result set it skips the number of records and fetches number of records
specified by $top parameter from the result set.",
            "in": "query",
            "required": false,
            "type": "integer",
            "default": -1,
            "format": "int32",
            "x-example": 10
          },
          {
            "name": "$top",
            "description": "Number of records to fetch (e.g. '20')",
```

```
            "in": "query",
            "required": false,
            "type": "integer",
            "default": -1,
            "format": "int32",
            "x-example": 100
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
"9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation."
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/pcatSearch": {
      "get": {
        "tags": [
          "Service Point Categories and Service Catalog Offering Search"
        ],
        "operationId": "pcatSearch",
        "summary": "Returns relevant categories(and offerings if Service Catalog is integrated) based on
configured parameters in Service Point configuration.",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "$searchString",
            "description": "The string to search for, like 'problem with email client'",
            "in": "query",
            "required": true,
            "type": "string",
            "default": null,
            "x-example": "email"
          },
          {
            "name": "$top",
```

```
            "description": "Number of records to fetch (e.g. '20'), and it is restricted to configuration
 parameter esearch.max.numberOf.search.results.allowed (defaulted is 100).",
            "in": "query",
            "required": false,
            "type": "integer",
            "default": -1,
            "format": "int32",
            "x-example": 100
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
"9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: \n {\"type\":\"pcat\",\"fields\":[{\"id\":
\"pcat:5109\",\"ss_sym\":\"Email\",\"description\":\"Email\"},{\"id\":\"pcat:5103\",\"ss_sym\":\"Applications
\",\"description\":\"Applications\"}]}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/getOfferings": {
      "get": {
        "tags": [
          "Service Point Categories and Service Catalog Offering Search"
        ],
        "operationId": "getOfferings",
        "summary": "Returns featured offerings for the loggedin user's domain, if Service Catalog is
 integrated.",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "$skip",
            "description": "Number of records to skip (e.g. '10'). Both $skip and $top parameters are used for
 pagination. Based on this value, from result set it skips the number of records and fetches number of records
 specified by $top parameter from the result set. If because of authorization the count was less in last call,
 still fetch skip=skip+top in next call to get next page data, and do next page call only if hasMore is true
 in last call response.",
```

```
            "in": "query",
            "required": false,
            "type": "integer",
            "default": 0,
            "format": "int32",
            "x-example": 10
          },
          {
            "name": "$top",
            "description": "Number of records to fetch (e.g. '20'). It does not guarantee the results count
 because of authorization.",
            "in": "query",
            "required": false,
            "type": "integer",
            "default": -1,
            "format": "int32",
            "x-example": 100
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          },
          {
            "name": "X-Lang",
            "in": "header",
            "description": "Language of the locale",
            "required": false,
            "type": "string",
            "x-example": ""
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: {\"type\":\"service_offering\",\"count
\":\"13\",\"hasMore\":\"false\",\"fields\":[{\"id\":\"10021\",\"options.option_name\":[\"ID Badges\",
\"Building Keys/Access Cards\",\"Desk Keys/File Cabinet Keys\",\"Alarm Codes\"],\"offering_description\":
\"Employee access and Building security offerings\",\"offering_name\":\"Access Security\",\"ctg_source\":
\"catalog\",\"offering_status\":\"1\",\"domain\":\"ca.com\",\"path\":\"10001/10020/10021/\",\"image_file\":
\"OotbOfferings/access%20security.svg\",\"mobile_enabled\":\"0\",\"options.option_id\":[\"10243\",\"10248\",
\"10253\",\"12585\"]},{\"id\":\"10003\",\"options.option_name\":[\"Virtual Environment\",\"Dev/Test/Staging
 Environments\",\"Production Environment\"],\"offering_description\":\"Set up an application environment
 for test or production purposes\",\"offering_name\":\"Application Environment Setup\",\"ctg_source\":
\"catalog\",\"offering_status\":\"1\",\"domain\":\"ca.com\",\"path\":\"10001/10002/10003/\",\"image_file
\":\"OotbOfferings/application_env_setup.svg\",\"mobile_enabled\":\"0\",\"options.option_id\":[\"9986\",
\"9991\",\"9996\"]},{\"id\":\"10004\",\"options.option_name\":[\"Production Application Hosting\"],
\"offering_description\":\"Set up production environment for application hosting with Bronze/Silver/Gold SLA
\",\"offering_name\":\"Application Hosting\",\"ctg_source\":\"catalog\",\"offering_status\":\"1\",\"domain
\":\"ca.com\",\"path\":\"10001/10002/10004/\",\"image_file\":\"OotbOfferings/host.svg\",\"mobile_enabled
```

\":\"0\",\"options.option_id\":[10007]},{\"id\":\"10015\",\"options.option_name\":[\"Basic Information
\",\"Database Design\",\"Database Installation\",\"Backup and Recovery - Review and implementation\",
\"Security Audits\",\"Remote DBA Support\"],\"offering_description\":\"Database Management services across
 all the phases of Database Lifecycle from Planning, Design, Implementation and ongoing maintenance\",
\"offering_name\":\"Database Management Services\",\"ctg_source\":\"catalog\",\"offering_status\":\"1\",
\"domain\":\"ca.com\",\"path\":\"10001/10002/10015/\",\"image_file\":\"OotbOfferings/DatabaseManagement.svg
\",\"mobile_enabled\":\"0\",\"options.option_id\":[10177,10178,10183,10188,10193,10198]},{\"id\":\"10006\",
\"options.option_name\":[\"Request access to HR applications\"],\"offering_description\":\"Request access
 to HR applications like Recruitment Management,Induction management etc.,\",\"offering_name\":\"Grant
 Access to HR Applications\",\"ctg_source\":\"catalog\",\"offering_status\":\"1\",\"domain\":\"ca.com\",
\"path\":\"10001/10002/10005/10006/\",\"image_file\":\"OotbOfferings/grantaccess_HR.svg\",\"mobile_enabled
\":\"0\",\"options.option_id\":[10017]},{\"id\":\"10095\",\"options.option_name\":[\"Network Engineering/
Design Request\"],\"offering_description\":\"Design the network for your lab, office, etc. Also enhances
 with new network capabilities.\",\"offering_name\":\"Network Engineering/ Design\",\"ctg_source\":
\"catalog\",\"offering_status\":\"1\",\"domain\":\"ca.com\",\"path\":\"10001/10092/10095/\",\"image_file
\":\"OotbOfferings/network.svg\",\"mobile_enabled\":\"0\",\"options.option_id\":[11519]},{\"id\":\"10104\",
\"options.option_name\":[\"Laptop - Standard\",\"Laptop - Deluxe\",\"New Hire\",\"Mobile Phone - Standard
\",\"Mobile Phone - Deluxe\"],\"offering_description\":\"Bundle of services for New Hire necessities
 (PC, office space, telephone, email, security, etc.)\",\"offering_name\":\"New Hire Onboarding\",
\"ctg_source\":\"catalog\",\"offering_status\":\"1\",\"domain\":\"ca.com\",\"path\":\"10001/10103/10104/
\",\"image_file\":\"OotbOfferings/new_hire_onboard.svg\",\"mobile_enabled\":\"0\",\"options.option_id
\":[11414,11420,11634,11669,11674]},{\"id\":\"10109\",\"options.option_name\":[\"New Project Request\"],
\"offering_description\":\"Request establishment of a new project\",\"offering_name\":\"New Project Request
\",\"ctg_source\":\"catalog\",\"offering_status\":\"1\",\"domain\":\"ca.com\",\"path\":\"10001/10105/10109/
\",\"image_file\":\"OotbOfferings/new_project_request.svg\",\"mobile_enabled\":\"0\",\"options.option_id\":
[11714]},{\"id\":\"10089\",\"options.option_name\":[\"Windows Server\"],\"offering_description\":\"Acquire and
 set up a server\",\"offering_name\":\"Procure Server\",\"ctg_source\":\"catalog\",\"offering_status\":\"1\",
\"domain\":\"ca.com\",\"path\":\"10001/10022/10086/10089/\",\"image_file\":\"OotbOfferings/procure_server.svg
\",\"mobile_enabled\":\"0\",\"options.option_id\":[11298]},{\"id\":\"10030\",\"options.option_name\":[\"Create
 New List\"],\"offering_description\":\"Request creation of a new email Distribution List\",\"offering_name
\":\"Request New Distribution List\",\"ctg_source\":\"catalog\",\"offering_status\":\"1\",\"domain\":\"ca.com
\",\"path\":\"10001/10022/10023/10027/10030/\",\"image_file\":\"OotbOfferings/distribution_list_new.svg\",
\"mobile_enabled\":\"0\",\"options.option_id\":[10368]},{\"id\":\"10047\",\"options.option_name\":[\"Create
 File Share\",\"File Share Access Report\"],\"offering_description\":\"Request creation of a new File Share
 or Usage Report of the exising file share\",\"offering_name\":\"Request New File Share\",\"ctg_source\":
\"catalog\",\"offering_status\":\"1\",\"domain\":\"ca.com\",\"path\":\"10001/10022/10047/\",\"image_file
\":\"OotbOfferings/newfile_share.svg\",\"mobile_enabled\":\"0\",\"options.option_id\":[10637,10643]},{\"id
\":\"10115\",\"options.option_name\":[\"Reserve a virtual machine\"],\"offering_description\":\"Reserve
 one or more virtual machines. Specify a virtual machine template to use for deploying the new virtual
 machine. Specify software to be installed.\",\"offering_name\":\"Reserve a Virtual Machine\",\"ctg_source
\":\"catalog\",\"offering_status\":\"1\",\"domain\":\"ca.com\",\"path\":\"10001/10113/10115/\",\"image_file
\":\"OotbOfferings/reserve_virtual_machine.svg\",\"mobile_enabled\":\"0\",\"options.option_id\":[11759]},{\"id
\":\"10038\",\"options.option_name\":[\"Description\",\"Bronze\",\"Silver\",\"Gold\"],\"offering_description
\":\"Department-level SLA subscription for Virus Protection/Remediation Plans\",\"offering_name\":\"Virus
 Protection/Remediation Plan SLA\",\"ctg_source\":\"catalog\",\"offering_status\":\"1\",\"domain\":\"ca.com
\",\"path\":\"10001/10022/10038/\",\"image_file\":\"OotbOfferings/virus_protection.svg\",\"mobile_enabled\":
\"0\",\"options.option_id\":[10463,10467,10472,10477]}]}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"

```
          }
        }
      }
    },
    "/getBrowseOfferings": {
      "get": {
        "tags": [
          "Service Point Categories and Service Catalog Offering Search"
        ],
        "operationId": "getBrowseOfferings",
        "summary": "Returns all offerings for the loggedin user's domain, if Service Catalog is integrated.",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "$skip",
            "description": "Number of records to skip (e.g. '10'). Both $skip and $top parameters are used for
pagination. Based on this value, from result set it skips the number of records and fetches number of records
specified by $top parameter from the result set. If because of authorization the count was less in last call,
still fetch skip=skip+top in next call to get next page data, and do next page call only if hasMore is true
in last call response.",
            "in": "query",
            "required": false,
            "type": "integer",
            "default": 0,
            "format": "int32",
            "x-example": 10
          },
          {
            "name": "$top",
            "description": "Number of records to fetch (e.g. '20'). It does not guarantee the results count
because of authorization.",
            "in": "query",
            "required": false,
            "type": "integer",
            "default": -1,
            "format": "int32",
            "x-example": 100
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
"9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          },
           {
            "name": "X-Lang",
            "in": "header",
            "description": "Language of the locale",
```

```
            "required": false,
            "type": "string",
            "x-example": ""
          }
        ],
        "responses": {
          "200": {
            "description": "Successful Operation. For example: {\"type\":\"service_offering\",\"count
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\"Building Keys/Access Cards\",\"Desk Keys/File Cabinet Keys\",\"Alarm Codes\"],\"offering_description\":
\"Employee access and Building security offerings\",\"offering_name\":\"Access Security\",\"ctg_source\":
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\"10253\",\"12585\"]},{\"id\":\"10003\",\"options.option_name\":[\"Virtual Environment\",\"Dev/Test/Staging
 Environments\",\"Production Environment\"],\"offering_description\":\"Set up an application environment
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\",\"offering_name\":\"Application Hosting\",\"ctg_source\":\"catalog\",\"offering_status\":\"1\",\"domain
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\",\"Database Design\",\"Database Installation\",\"Backup and Recovery - Review and implementation\",
\"Security Audits\",\"Remote DBA Support\"],\"offering_description\":\"Database Management services across
 all the phases of Database Lifecycle from Planning, Design, Implementation and ongoing maintenance\",
\"offering_name\":\"Database Management Services\",\"ctg_source\":\"catalog\",\"offering_status\":\"1\",
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Design Request\"],\"offering_description\":\"Design the network for your lab, office, etc.  Also enhances
 with new  network capabilities.\",\"offering_name\":\"Network Engineering/ Design\",\"ctg_source\":
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\",\"mobile_enabled\":\"0\",\"options.option_id\":[11298]},{\"id\":\"10030\",\"options.option_name\":[\"Create
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```

\",\"path\":\"10001/10022/10023/10027/10030/\",\"image_file\":\"OotbOfferings/distribution_list_new.svg\",
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File Share\",\"File Share Access Report\"],\"offering_description\":\"Request creation of a new File Share
or Usage Report of the exising file share\",\"offering_name\":\"Request New File Share\",\"ctg_source\":
\"catalog\",\"offering_status\":\"1\",\"domain\":\"ca.com\",\"path\":\"10001/10022/10047/\",\"image_file
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machine. Specify software to be installed.\",\"offering_name\":\"Reserve a Virtual Machine\",\"ctg_source
\":\"catalog\",\"offering_status\":\"1\",\"domain\":\"ca.com\",\"path\":\"10001/10113/10115/\",\"image_file
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\":\"10038\",\"options.option_name\":[\"Description\",\"Bronze\",\"Silver\",\"Gold\"],\"offering_description
\":\"Department-level SLA subscription for Virus Protection/Remediation Plans\",\"offering_name\":\"Virus
Protection/Remediation Plan SLA\",\"ctg_source\":\"catalog\",\"offering_status\":\"1\",\"domain\":\"ca.com
\",\"path\":\"10001/10022/10038/\",\"image_file\":\"OotbOfferings/virus_protection.svg\",\"mobile_enabled\":
\"0\",\"options.option_id\":[10463,10467,10472,10477]}]}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    },
    "/getServiceRequest": {
      "get": {
        "tags": [
          "Service Point My Ticket List"
        ],
        "operationId": "getRequest",
        "summary": "Returns service requests of logged in user, if Service Catalog is integrated.",
        "produces": [
          "application/json"
        ],
        "parameters": [
          {
            "name": "$active",
            "description": "If true retrieves active requests, else inactive requests",
            "in": "query",
            "required": true,
            "type": "boolean",
            "x-example": true
          },
          {
            "name": "$skip",
            "description": "Number of records to skip (e.g. '10'). Both $skip and $top parameters are used for
pagination. Based on this value, from result set it skips the number of records and fetches number of records
specified by $top parameter from the result set.",
            "in": "query",
            "required": false,
            "type": "integer",

```
            "default": 0,
            "format": "int32",
            "x-example": 10
          },
          {
            "name": "$top",
            "description": "Number of records to fetch (e.g. '20').",
            "in": "query",
            "required": false,
            "type": "integer",
            "default": -1,
            "format": "int32",
            "x-example": 100
          },
          {
            "name": "X-AccessToken",
            "in": "header",
            "description": "Access Token of the authenticated user",
            "required": true,
            "type": "string",
            "x-example":
 "9814520f0ff3bf99baf8c3953f59284d5335f8069d77d12845caf36173df8a35fd55c16c6df59663da649fe89f4ca8e5ace365ec7f3e948be0a74
          }
        ],
        "responses": {
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            "description": "Successful Operation. For example: \n {\"type\":\"service_request\",\"fields
\":[{\"request_id\":\"10495\",\"request_name\":\"Grant Access to HR Applications\",\"requested_by\":
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\"1551260480\",\"status.code\":\"103\",\"tkt_source\":\"catalog\"},{\"request_id\":\"10494\",\"request_name
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\":\"1551514475\",\"open_dt\":\"1551255076\",\"status.code\":\"103\",\"tkt_source\":\"catalog\"},
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\":\"5722800BB13A41769E1F42285FD8950A\",\"priority\":\"3\",\"last_mod_dt\":\"1550739832\",\"open_dt\":
\"1550480393\",\"status.code\":\"103\",\"tkt_source\":\"catalog\"},{\"request_id\":\"10483\",\"request_name
\":\"Access Security\",\"requested_by\":\"5722800BB13A41769E1F42285FD8950A\",\"priority\":\"3\",\"last_mod_dt
\":\"1550233054\",\"open_dt\":\"1550233054\",\"status.code\":\"200\",\"tkt_source\":\"catalog\"},{\"request_id
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\",\"priority\":\"3\",\"last_mod_dt\":\"1550233013\",\"open_dt\":\"1550233012\",\"status.code\":\"200\",
\"tkt_source\":\"catalog\"},{\"request_id\":\"10481\",\"request_name\":\"Procure Server\",\"requested_by
\":\"5722800BB13A41769E1F42285FD8950A\",\"priority\":\"3\",\"last_mod_dt\":\"1550486565\",\"open_dt\":
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\"1550047432\",\"status.code\":\"103\",\"tkt_source\":\"catalog\"},{\"request_id\":\"10473\",\"request_name
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```

\":\"1550047107\",\"open_dt\":\"1550047106\",\"status.code\":\"200\",\"tkt_source\":\"catalog\"},{\"request_id\":\"10472\",\"request_name\":\"\",\"requested_by\":\"5722800BB13A41769E1F42285FD8950A\",\"last_mod_dt\":\"1550047459\",\"open_dt\":\"1550047064\",\"status.code\":\"101\",\"tkt_source\":\"catalog\"},{\"request_id\":\"10471\",\"request_name\":\"Procure Laptop\",\"requested_by\":\"5722800BB13A41769E1F42285FD8950A\",\"priority\":\"3\",\"last_mod_dt\":\"1550305852\",\"open_dt\":\"1550046515\",\"status.code\":\"103\",\"tkt_source\":\"catalog\"},{\"request_id\":\"10470\",\"request_name\":\"New Hire Onboarding\",\"requested_by\":\"5722800BB13A41769E1F42285FD8950A\",\"priority\":\"3\",\"last_mod_dt\":\"1550305884\",\"open_dt\":\"1550046491\",\"status.code\":\"103\",\"tkt_source\":\"catalog\"},{\"request_id\":\"10467\",\"request_name\":\"Grant Access to HR Applications\",\"requested_by\":\"5722800BB13A41769E1F42285FD8950A\",\"priority\":\"3\",\"last_mod_dt\":\"1550300544\",\"open_dt\":\"1550041155\",\"status.code\":\"103\",\"tkt_source\":\"catalog\"}]}"
          },
          "401": {
            "description": "Invalid Authentication Token"
          },
          "500": {
            "description": "Internal Server Error"
          }
        }
      }
    }
  },
  "securityDefinitions":{
      "basicAuth":{
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      }
  },
  "definitions": {
    "LogComment": {
      "type": "object",
      "properties": {
        "logComment": {
          "type": "string"
        }
      },
      "xml": {
        "name": "LogComment"
      }
    },
    "KDComment": {
      "type": "object",
      "properties": {
        "comment": {
          "type": "string"
        },
        "commenttype": {
          "type": "integer"
        },
        "useruuid": {
          "type": "string"
        }
      },
      "xml": {

```
        "name": "KDComment"
      }
    },
    "RateDocument": {
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      "properties": {
        "buid": {
          "type": "integer"
        },
        "rating": {
          "type": "integer"
        }
      },
      "xml": {
        "name": "RateDocument"
      }
    },
    "UnlinkSolution": {
      "type": "object",
      "properties": {
        "crPersId": {
          "type": "string"
        }
      },
      "xml": {
        "name": "UnlinkSolution"
      }
    }
  }
}
```

Integrating

This section contains the following articles:

- Integrating CA Service Desk Manager
- Integrating CA Service Catalog
- Integrating CA Asset Portfolio Management
- Integrating Third Party Applications

How to Enable TLS 1.2 with CA EEM 12.6

This article describes how to enable TLS 1.2 with CA Embedded Entitlements Manager (CA EEM) 12.6 Server.

Enable CA SDM with CA EEM 12.6

Enable TLS 1.2 in CA EEM

Perform the following steps:

1. Log in to CA EEM Server
2. In a text editor, open the `igateway.conf` file that is at `C:\Program Files\CA\SC\iTechnology`
3. In the <`secureProtocol` > section, add the following tag:

```
<secureProtocol>TLSV1_2</secureProtocol>
```

4. Restart CA EEM services.

Enable TLS 1.2 in CA SDM

Perform the following steps:

1. Log in to CA SDM Server
2. In a text editor, open the `eiam.config` file that is at `NX_ROOT\pdmconf`
3. In the <`secureProtocol` > section, add the following tag:

```
<secureProtocol>TLSV1_2</secureProtocol>
```

4. Restart CA SDM services.
   Verify that CA SDM and CA EEM are successfully integrated.

Enable CA APM with CA EEM 12.6

Enable TLS 1.2 in CA EEM

Perform the following steps:

1. Log in to CA EEM Server.
2. In a text editor, open the `igateway.conf` file that is at `C:\Program Files\CA\SC\iTechnology`
3. In the <`secureProtocol` > section, add the following tag:

```
<secureProtocol>TLSV1_2</secureProtocol>
```

4. Restart CA EEM services.

Enable TLS 1.2 in CA APM

Perform the following steps:

1. Open registry on the server by running the `regedit command` and navigate to `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols`
2. Change DWORD Values under Server and Client under TLS 1.2 key:

```
DisabledByDefault [Value = 0]
Enabled [Value = 1]
```

3. Restart CA APM services.
4. Verify that CA APM and CA EEM are successfully integrated.

Enable CA Service Catalog with CA EEM 12.6

Perform the following steps:

1. Apply the corresponding SQL Server patch that provides TLS 1.2 support for the SQL Server deployed in your environment.
2. Stop the following services:
   a. CA Catalog Service
   b. Accounting
3. In a text editor, open the `viewService.conf` file that is at `C:\Program Files\CA\Service Catalog\View\Conf`
4. In the <# Java Additional Parameters> section, add the following tags at the end:

```
wrapper.java.additional.23=-Djdk.tls.client.protocols=TLSv1.2

wrapper.java.additional.24=-noverify
```

5. In the <# Java Additional Parameters> section, comment the following tag:

```
wrapper.java.additional.20=-XX:-UseSplitVerifier
```

6. Restart the following services:
   a. CA Catalog Service
   b. Accounting

Integrating CA Service Desk Manager

You can integrate CA SDM with other CA products as seen in the following topics:

- Integrate CMDB with CA Configuration Automation
- Integrate with CA Configuration Automation

Integrate CMDB with CA Configuration Automation

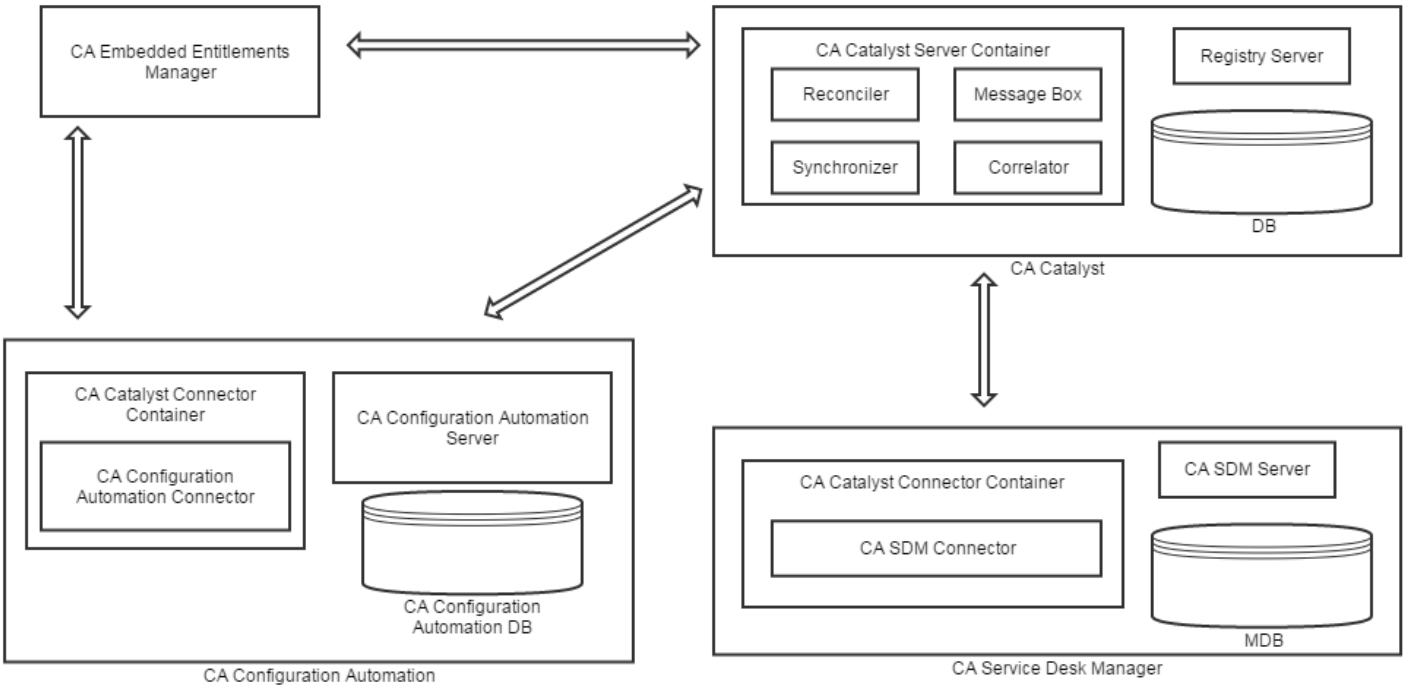
This topic contains information intended for administrators who want to configure CA Catalyst to leverage key CA Configuration Automation data in the CMDB module of CA SDM. Ensure that you are familiar with the following processes:

- Performing administrative tasks in CA Catalyst, CA Configuration Automation, and CA SDM.
- Accessing product updates from CA Support Online
- Performing system administration for Windows and Non-Windows operating environments
- Managing CIs and CI Relationship data in GRLoader and CA SDM.

Integration Architecture

CA Catalyst integrates the CMDB module of CA SDM with CA Configuration Automation. The following diagram provides a high-level view of the integration architecture:

Figure 38: Integration Architecture

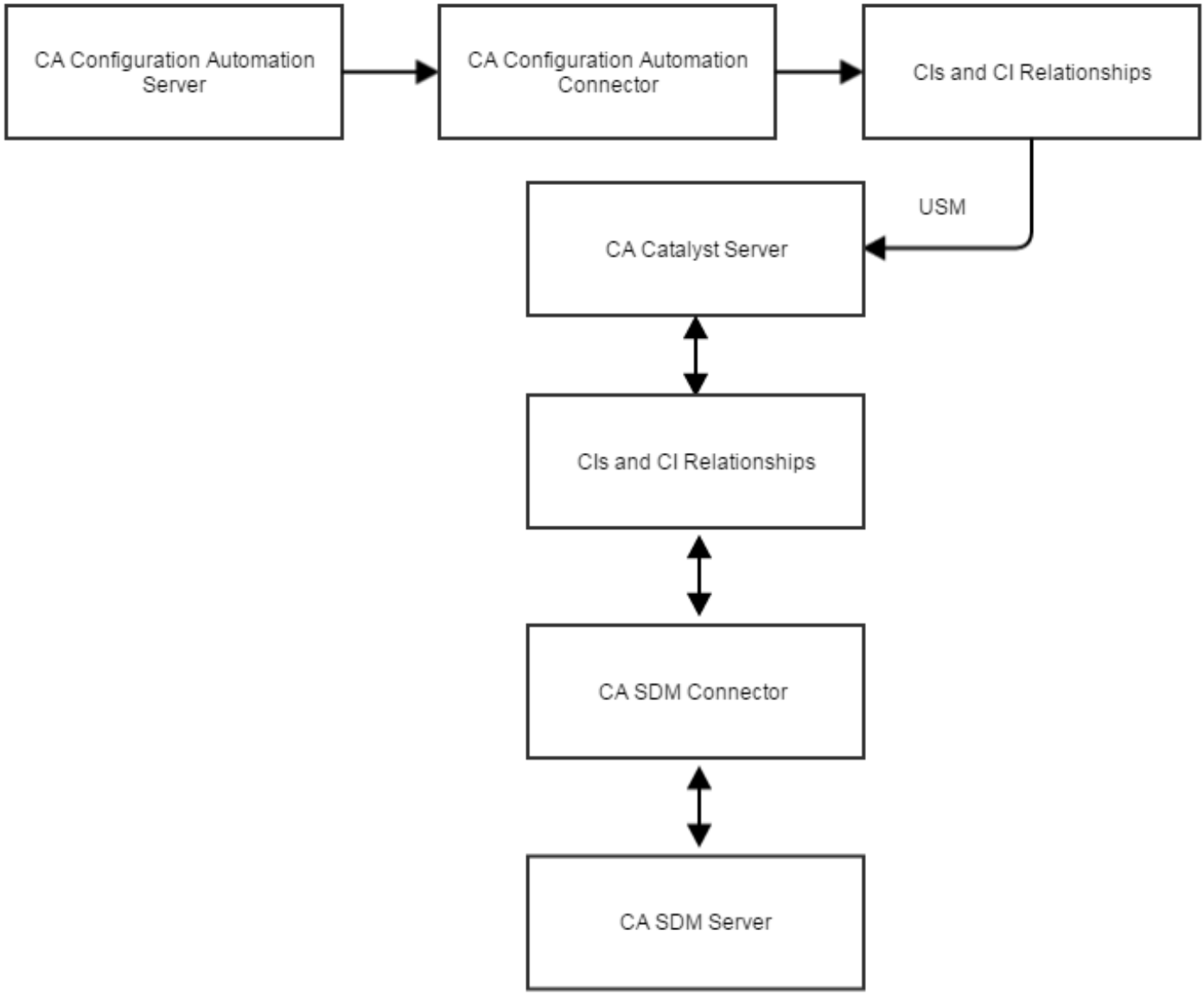


1. CA Configuration Automation server, database, and connector are installed on a set of servers.
2. CA Catalyst server and database are installed on a separate set of servers.
3. CA SDM server, MDB, and connector are installed on a separate set of servers.

   NOTE
   The CA SDM Connector must reside with one of the CA SDM servers.
4. CA Embedded Entitlements Manager is installed on another server.
5. CA Configuration Automation server and CA Catalyst server communicate with CA Embedded Entitlements Manager for authentication and authorization service.
6. CA Configuration Automation uses the CA Configuration Automation connector to communicate with the CA Catalyst server and push the discovered data.
7. CA SDM uses the CA SDM Connector to communicate bidirectionally with the CA Catalyst server and push and receive data.
8. Together all the components are integrated to form a solution and represent the CA Configuration Automation discovered data in the CMDB module of CA SDM.

The following information and diagram provide a summary of how the integration works:

Figure 39: CMDB-CCA Integration Works



1. The CA Configuration Automation Connector discovers CIs and CI relationship data in CA Configuration Automation.
2. The CA Configuration Automation Connector transforms the data to USM values and publishes the data to CA Catalyst.
3. The CA SDM Connector transforms the USM data into CA SDM values and publishes the CIs and CI data to the Transaction Work Area (TWA) or into CA SDM, directly.

NOTE

When you enable the TWA flag the CI relationships are loaded to the TWA, only after the CIs are loaded from the TWA to CA SDM.

Install and Configure the Integration

This integration requires installing and configuring the CA Catalyst Server, the CA Configuration Automation Connector, and the CA SDM Connector.

Follow these steps:

1. Verify the Prerequisites.
2. Inactivate the MDR.
3. Create an MDR Definition in CA SDM.
4. Install the CA Configuration Automation Connector.
5. If you plan to install CA SDM Connector and CA Catalyst Server on different computers, install the CA Catalyst Container on the same computer where you plan to install the CA SDM Connector (For more information about installing the CA Catalyst Container, see the *CA Catalyst documentation*).
6. Install the CA SDM Connector.
7. (Optional) Configure the CA SDM Connector
   The integration is installed and the connectors are configured.
8. Restart the CA Catalyst Container service on the CA Catalyst server.

Verify the Prerequisites

Verify the following prerequisites before starting the integration:

Hardware and Software Requirements

CA Catalyst, CA SDM, and CA Configuration Automation support a number of hardware, software, operating systems, and databases. For complete information about CA Catalyst and CA Configuration Automation, see the respective documentation. For CA SDM, see the Supportability Matrix.

Supported Version of the Integrated Products

This integration supports the following product versions:

- CA SDM r12.9 CUM1, r14.1.01
- CA Configuration Automation r12.8 SP1, r12.8 SP2
- CA Catalyst Server r3.2
- CA EEM r12.5, r12.5.1
- CA Catalyst Container r3.2, r3.4.1

Integration Considerations

Consider the following information before you begin the integration:

- If you have previously used the CA Cohesion product, note that CA Cohesion ACM r5.0 and CA Configuration Automation are different products with different architectures. The data that these two products discover may also differ.
- If you upgraded to the supported version of CA SDM CA Configuration Automation, CA Catalyst replaces CA Business Intelligence as the integration point.
- You upgraded to CA SDM 14.1.01. Ensure that you first apply RO80318 (Windows Master) patch and then the following patches:

Language	Patch
German	T52Y463
Japanese	T52Y464
French	T52Y465

French Canadian	T52Y466
Spanish	T52Y467
Italian	T52Y468
Brazilian Portuguese	T52Y469

- You installed and configured the supported version of CA Configuration Automation. For more information, see the *CA Configuration Automation Implementation Guide.*
- Ensure that the services and servers discovered in CA Configuration Automation can be exported to CMDB module of CA SDM. For more information about CA Configuration Automation discovery, see the following sections in the *CA Configuration Automation Product Guide:*
  - Service Management
  - Server Management
  - Network Management
  - Blueprint Management
- You applied the patch to enable the reactivation of the federated CI mapping. When a CI is deactivated in the CMDB module of CA SDM, the ID mappings also get deactivated by default. When this CI is activated again, the ID mappings remain deactivated because of the post trigger issue in CA SDM. Use the following text fixes to fix this issue:
  - RO42342 (for CA SDM r12.6)
  - RO42343 (for CA SDM r12.5)

    NOTE
    When you apply the patch, the CA SDM service stops automatically. Start the CA SDM server after applying the patch.
- Verify that the CA Configuration Automation Server, the CA Catalyst Server (Registry, Container, and Administrator), and CA SDM server are up and running. Verify that all these servers have network connectivity with each other.
- We recommend that you install CA Catalyst, CA Configuration Automation, and CA SDM on different computers for this integration to work properly.
- You verified the CA SDM Connector Installation Considerations.
- Reviewed the Frequently Asked Questions for the Integration.

Inactivate the MDR

If you want to import CIs and relationships from CA Configuration Automation r12.6 to the CA CMDB, you can first inactivate any MDRs defined in CA CMDB for CA Cohesion ACM r5.0, CA Application Configuration r12.0, and CA Configuration Automation r12.5. This action deactivates all the Federated CI Mappings for CA Cohesion ACM r5.0. View the Federated CI Mappings list in the Federated CI Mapping node on the Administration tab.

WARNING
If you want to retain both CA Cohesion ACM r5.0 and CA Configuration Automation as active MDRs, do *not* inactivate the r5.0 MDR. CIs imported from CA Configuration Automation reconcile to existing CIs imported from CA Cohesion ACM r5.0. For those CIs that are imported from both CA Configuration Automation and CA Cohesion ACM r5.0, both of these MDRs appear as MDR buttons on the CI detail page.

Follow these steps:

1. Launch CA SDM by entering the following URL in the web browser:
   `http://<SDM Server Name>:<Port>/CAisd/pdmweb.exe`
2. From the Administration tab, navigate to CA CMDB, MDR Management, MDR List.
   The MDR List appears.
3. Open the MDR that you want to deactivate and click Edit.
   The Update MDR Definition page appears.
4. From the Active drop-down list, select Inactive and click Save.

The MDR is inactivated.

Create an MDR Definition in CA SDM

Create the MDR definition to specify information about the MDR contributing to the CIs. The MDR definition completes the Launch-in-Context URL configuration. You can verify the configuration from a CI detail page within CA SDM

Example: Create an MDR Definition to specify the CA Configuration Automation server information.
Follow these steps:

1. Log in to the CA SDM as a user with Administrator access such as ServiceDesk.
2. On the Administration tab, click CA CMDB, MDR Management, MDR List.
   The Management Data Repository (MDR) List page appears.
3. Click Create New.
   The Create New MDR Definition page appears.
4. Complete the following required information:
   - **Button Name**: Defines the MDR button label that appears on the CI Detail page. Enter CCA.
   - **MDR Name**: Specifies the data that is sent in the mdr_name field in XML. Enter CCA.
   - **MDR Class**: Specifies the data that is sent in the mdr_class field in the XML. Enter CCA.
   - **Active**: Denotes the MDR definition as active or inactive. Inactive MDR definitions are logically deleted, but they can be made active again by using the Search utility. Select Active.
   - **Owner**: Specifies the contact responsible for this MDR. Enter ServiceDesk.
   - **Description**: Specifies the description of the MDR. Enter CCA LIC URL.
   - **Hostname**: Specifies the host name, DNS name, or IP address of the host, which contains the web server that hosts the launched web page. **Example:** CA Configuration Automation server hostname.
   - **Port**: Specifies the TCP/IP port used by the MDR web server to serve up web pages. **Example**: CA Configuration Automation Port.
   - **Path**: Specifies the portion of the URL that precedes the question mark (?) character. Enter CCAUI.jsp.
   - **Parameters**: Specifies the portion of the URL that follows the question mark (?) character. Keep the default selection.
   - **CMDBf Timeout**: Specifies time limit for CMDBf endpoint query. Enter 10.
   - **URL to Launch in Context**: Required for launch in context. Keep the default URL.
5. Click Save.
   The MDR definition is created.
6. Complete the following steps to modify the ServiceDeskManager_ConnectorSB.xml file:
   a. Open the Catalyst Server Registry UI using the following URL:
      ```
      https://<registryserver:port>/registry/carbon/admin/login.jsp
      ```
   b. Log in to Catalyst Registry UI and browse to the following directory:
      ```
      \topology\physical\<CA_Catalyst_Server>\ServiceDeskManager_ConnectorSB.xml
      ```
   c. Click Edit as Text.
   d. Search for the following lines:
      ```
      <Field output="temp_mdr_class" type="map" input="MdrProductLastUpdated">
      </Field>
      ```
   e. Enter the following code just before the </Field> tag:
      ```
      <mapentry mapin="CA:00033" mapout="CCA" />
      ```
   f. Search for the following lines:

```
<Field output="temp_mdr_name" type="map" input="MdrProductLastUpdated,MdrProdInstanceLastUpdated">
</Field>
```

g. Enter the following code just before the <Field> tag:

```
<mapentry mapin="CA:00033,.*" mapout="CCA" />
```

h. Click Save Content.

Install the CA Configuration Automation Connector

You can install the CA Configuration Automation Connector locally on the CA Configuration Automation Server (recommended), on the CA Catalyst Server, or on a Windows system that is in the same domain as the CA Configuration Automation Server.

NOTE

If you install the CA Configuration Automation Connector on the same server hosting the CA Catalyst Server, then Choose Installation Folder, CA EEM, and Peer Node Configuration screens are not displayed when you run the installation program. The information in these screens is collected when you install the CA Catalyst Server.

WARNING
We recommend that you do *not* install the CA Configuration Automation Server on the CA Catalyst Server host because performance problems can occur. If you install CA Catalyst on the CA Configuration Automation server host, you run into port conflicts. To avoid conflicts change the default ports. Ensure that the CA Catalyst Registry Server is up and running before you start the CA Configuration Automation Connector installation.

Follow these steps:

1. Open CA Configuration Automation installer and complete the following steps:
   a. Click Run the CA Catalyst CA Configuration Automation Connector Installation Wizard.
   The installer introduction page appears.
   b. Click Next, accept the terms of the License Agreement, and click Next.
   The Choose Install Folder page opens.
   Note: This page is *not* displayed if the CA Catalyst components already reside on the system.
   c. Specify the installation folder and click Next.
   Note: The maximum installation path length is 150 characters. The installation program blocks paths with more than 150 characters.
   The Administration Configuration screen appears and displays the default CA Catalyst user in the Username field.
   d. Enter the CA Catalyst administrator user password and click Next.
   The CA Embedded Entitlements Manager Server Configuration screen appears.
2. Enter the following CA Embedded Entitlements Manager information and click Next.
   – **Server**
   Specifies the CA Embedded Entitlements Manager host server.
   – **User**
   Specifies the name of the CA Embedded Entitlements Manager administrator user.
   – **Password**
   Specifies the password for the CA Embedded Entitlements Manager administrator.
   – **Application Name**
   Identifies the CA Catalyst and CA Embedded Entitlements Manager integration using the following format: CATALYST-<*CA_Catalyst_host_name*>.

   NOTE
   Use the same EEM Application name that you used during the CA Catalyst server installation.
   – **Default Catalyst Access for all users**

Select READ-WRITE as the default access for all users.

The Remote Registry Server Configuration screen appears.

NOTE
You can use the same CA Embedded Entitlements Manager Server for CA Catalyst and CA Configuration Automation.

3. Enter the following information and click Next.
   - **Registry Host**
     Specifies the CA Catalyst Registry host server. This repository contains the USM schema and policies that control the behavior of other components in the CA Catalyst container.
     Note: For detailed information about CA Catalyst architecture and components, see the *CA Catalyst Installation Guide*.
   - **HTTP Port**
     Specifies the server port number where you installed the CA Catalyst Registry.
   - **Secure Server Port**
     Specifies the secure server port number where you installed the CA Catalyst Registry.

   The Catalyst Container Server Configuration screen appears.
4. Enter the following information and click Next.
   - **Node Name**
     Specifies the name of the host where the CA Configuration Automation Connector is installed.
   - **Bus Port**
     Specifies the bus port number for the CA Catalyst server.
   - **HTTP Port**
     Specifies the HTTP listening port of the CA Catalyst server.
   - **HTTP Service Port**
     Specifies the SOAP-based web service HTTP network port on the CA Catalyst server. Other systems use this port to make API remote procedure calls to the CA Catalyst server.
   - **HTTPS Service Port**
     Specifies the SOAP-based web service HTTPS network port on the CA Catalyst server. Other systems use this port to make secure API remote procedure calls to the CA Catalyst server.

   The Peer Node Configuration screen appears.
5. Enter the following information and click Next.
   - **Peer Host**
     Specifies the name of the CA Catalyst server.
   - **Peer Port**
     Specifies the WS Endpoint port number of the CA Catalyst server.
   - **Message Bus Host**
     Specifies the server where the CA Catalyst message bus is deployed; typically the same computer that hosts the CA Catalyst server.
   - **Message Bus Port**
     Specifies the bus port number where you installed the Message Bus.

   The CA Configuration Automation Connector Configuration screen appears.
6. Enter the following information and click Next.
   - **CA Configuration Automation Server Hostname**
     Specifies the name of the CA Configuration Automation server that the CA Configuration Automation Connector monitors for alarms and updates.
   - **CA Configuration Automation Server Port**
     Specifies the port the CA Configuration Automation Connector uses to communicate with the CA Configuration Automation server.
   - **CA Configuration Automation Server User**

Specifies an administrator user who can access the CA Configuration Automation server.

- **CA Configuration Automation Server Password**
  Specifies the password for the specified user.
- **Verify CA Configuration Automation Server Password**
  Ensures that the password was entered correctly by matching the passwords.
- **CA Configuration Automation Notification Listener Port**
  Specifies the port that receives events from CA Configuration Automation.
- **HTTPS**
  Enables https on the target CA Configuration Automation Server.
- **X.509 Certificate Authentication**
  Enables client authentication on the target CA Configuration Automation Server.
- **Certificate Path**
  Specifies the path to the certificate for the CA Configuration Automation Server User that you configured.
- **Certificate Password**
  Specifies the password for the certificate file.

The Database Server screen appears.

7. Enter the following information and click Next.
   - **Database Type**
     Specifies the type of database that the CA Configuration Automation server uses.
   - **Server Name**
     Specifies the name of the CA Configuration Automation Database host.
   - **Port Number**
     Specifies the listening port that the CA Configuration Automation database host uses.
   - **Instance Name (Optional)**
     Specifies the name of the CA Configuration Automation Database instance.

   The Database Configuration screen appears.
8. Enter the following information and click Next.
   - **Database Name**
     Specifies the name of the CA Configuration Automation Database.
   - **Database User**
     Specifies the name of the CA Configuration Automation Database administrator user.
   - **Database User Password**
     Specifies the password of the CA Configuration Automation Database administrator user.
   - **Retype Password**
     Ensures that the password is entered correctly by matching the passwords.

   The Change Detection Alert Metric and Threshold Levels page appears.
9. Enter the following information and click Next.
   - **Alert Metric**
     Specifics one of the following metrics that are used to determine the severity level of the alert when combined with the threshold values:
     - **CountChange**
       Specifies the number of changes from the source to the target server that the Change Detection operation discovers. This number of changes determines the severity level of an alert.
     - **CountSource**
       Specifies the number of changes to the source server that the Change Detection operation discovers. This number of changes determines the severity level of an alert.
     - **CountTarget**
       Specifies the total number of changes to the target server that the Change Detection operation discovers. This total number of changes determines the severity level of an alert.
     - **CountTotal**

Specifies the total number of changes that the Change Detection operation discovers. This total number of changes determines the severity level of an alert.

- **Information Threshold**
  Specifies the minimum number of changes for the specified Change Detection metric that is required to assign an Information severity level to an alert.

  NOTE
  The value in the threshold fields must increase with each severity level. The Information value must be the lowest, the Minor value must be next lowest, and so on, until the Fatal value, which is the highest.
- **Minor Threshold**
  Specifies the minimum number of changes for the specified Change Detection metric that is required to assign a Minor severity level to an alert.
- **Major Threshold**
  Specifies the minimum number of changes for the specified Change Detection metric that is required to assign a Major level to an alert.
- **Critical Threshold**
  Specifies the minimum number of changes for the specified Change Detection metric that is required to assign a Critical severity level to an alert.
- **Fatal Threshold**
  Specifies the minimum number of changes for the specified Change Detection metric that is required to assign a Fatal severity level to an alert.

The Installation Summary page appears.

10. Review your selections and click Install.
    The CA Configuration Automation Connector installs on the system and integrates with the appropriate CA Configuration Automation and CA Catalyst instances. The Install Complete page appears when the installation finishes.
    If the Installation Summary page displays installation errors, view the CATALYST_HOME \CA_Configuration_Automation_Connector_InstallLog.log file to troubleshoot the installation. This file is created when you click Done after the installation finishes. CATALYST_HOME specifies the folder where you installed CA Configuration Automation Connector.

How to Export CI Data

This integration lets you manage CI data in CA Catalyst, CA Configuration Automation, and CA SDM. You can export CI data from CA Configuration Automation to CA SDM.

NOTE
Management Profiles can also be used to export data instead of Catalyst Jobs.

Complete the following steps to create a Catalyst Job in CA Configuration Automation and execute a process to export the CI data:

1. Complete the following steps to create and execute the Catalyst Job:
   a. Log in to the CA Configuration Automation server UI from the following URL:
      `http://<CA_Configuration_Automation_Server>:<port number>`

      NOTE
      The default port number is 8080.
   b. Click Administration.
   c. On the Catalyst Integration tab, click Jobs.
   d. From the Catalyst Job pane, click Table Actions, Create Catalyst Job.
   e. Enter the following information:
      - **Name**

Specifies the name of the Catalyst Job.

- **Attribute Profile**
  Attaches an attribute profile with the necessary filter criteria. If you do not want to use the filter criteria, select Use Default.
- **Services/ Server Groups/ Servers**
  Specifies the Services, Server Groups, and Servers that you want to export, in that order, from the consecutive Wizard pages.
- **Blueprint Groups/ Blueprints**
  Specifies the Blueprint Groups and Blueprints that you want to export, in that order, from the consecutive Wizard pages.
- **Schedule**
  Schedules the job, if needed for synchronizing the updates on the CIs.

f. Click Finish to save the Job.
g. Select the check box next to the Job that you created.
h. Select Run Catalyst Job under Select Actions to execute the Job.

NOTE
For more information about creating and executing the Catalyst Job, see the CA Configuration Automation and CA Configuration Automation Connector Product documentation.

WARNING
Create a Catalyst profile or use the default catalyst profile before creating and executing the Catalyst Job.

2. You can view all the available CIs in [set value for the rose variable at the book level], CA Configuration Automation, and CA SDM in the CA Catalyst USM Web View. Complete the following steps:
   a. Open the following URL to access CA Catalyst:
      `http://<CA_Catalyst_Server:port>/ca-rest/home`
   b. Log in to CA Catalyst and click Browse by CI Type.
   c. Select the CA Catalyst (CMDB-view) data repository from the Now viewing data-source drop-down list.
      The list of the available CI types appears.
   d. Click a CI Type, such as Computer System.
      The available CIs with the type set as Computer System appear.
   e. Click a CI from the results.
      The CI details appear.

      NOTE
      Similarly, you can view the CA Configuration Automation and ServiceDesk-CMDB Data Repositories.

   CA Catalyst lets you view data repositories for CA Catalyst, CA Configuration Automation, and CA SDM.
3. After data exports from CA Configuration Automation to CA Catalyst, view the data that is imported to CA SDM from CA Catalyst. Complete the following steps:
   a. Log in to CA SDM.
   b. Complete *one* of the following actions:

      NOTE
      Use TWA as a staging area for unapproved changes and for checking duplicates.

   c. Enter the search criteria and click Search to find a CI.

      NOTE
      If you click Search without entering the search criteria, the list of all CIs is displayed.

   d. Open the exported CI from the CI List.
      The Configuration Item Detail page appears.

NOTE
You can also use the CMDB Visualizer to display a visual representation of the relationship of a CI to other resources. Visualizer is available only if it was configured during the CA SDM installation.

CI details appear in CA SDM. Visualizer provides a graphical representation of the configuration item showing its relationship to other resources.

Frequently Asked Questions for the Integration

The following categories lists the commonly asked questions for CA Configuration Automation integration:

System Requirements

I am currently using CA Catalyst Container r2.5 to integrate CA SDM with CA Spectrum Service Assurance (CA SSA) or CA Service Operations Insight (CA SOI). Can I also integrate my CA SDM environment with CA Configuration Automation?

Use Catalyst connector 3.2 to integrate CA SDM with both CA SOI and CA Configuration Automation. For more information on the supported products, see the Compatibility Matrix.

Installation and Uninstallation

During installation of the CA Configuration Automation and SDM Connectors, I have to enter a Node Name on the Catalyst Container Server Configuration panel. What do I enter as the Node Name?

The Node Name is the host name of the computer on which you installing the connector. For example, if you are installing the SDM Connector on server A, enter Server A as the Node Name.

After installing the integration, there are three different application instances in the Embedded Entitlement Manager (EEM) for the integration (for example, CATALYST-<CACatalystServer>, CATALYST-<CCAConnectorServer>, CATALYST-<CASDMConnectorServer>). Is this correct?

No, there must only be one application instance for the integration with EEM. When installing the connector, enter the same application name that was used during the CA Catalyst installation (CATALYST-<CatalystServer>).
If, after installing your connector, you have multiple application instances for the integration in EEM, uninstall the connector. Next, you unregister the extraneous application instances from EEM using the following steps:

1. Log in to the EEM UI using the <Global> application instance, https://<EEMServer>:5250/spin/eiam
2. Select the Configure, Applications submenu.
3. From the list of applications in the left pane, click on the name of the application instance added by the CA Configuration Automation or CA SDM connector.
4. On the right-hand pane, click Unregister.
5. Repeat for any other incorrect application instances.
6. Log out of EEM.

Once the steps for EEM are executed, reinstall the connector, ensuring to use the same application instance name used during the CA Catalyst installation.

When installing the CA SDM Connector, I noticed there was an option to 'Enable TWA'. What does this option do? Why would I use it?

When the 'Enable TWA' option is selected, CIs are imported into a staging location within CA SDM called the Transaction Work Area (TWA) instead of directly into the CMDB. This allows you to vet the data (remove or modify) before the CIs are loaded into the CMDB. After reviewing the CIs, you would be responsible for transferring the CIs to the CMDB using the GRLoader import utility. This approach gives you much more control over which CIs are stored in the CMDB, but it is also a more manual and time-consuming method.

Exporting

How do I export CIs from CA Configuration Automation?

There are two ways to export CIs from CA Configuration Automation. The first way is to execute a CA Catalyst Job. To configure a job, you must log into the CCA UI and navigate to Jobs sub-menu option on Catalyst Integration tab within the Administration section. From the Catalyst Jobs table, click on Table Actions and select Create Catalyst Job. The second way is to run an integration-enabled Management Profile. To enable the integration within a Management Profile, you must create a Management Profile or edit an existing profile. On the Profile tab, in the Catalyst Integration, select the Enable Integration option and select an Attribute Profile. Save the Management Profile and run it.

What is the best method of pushing the data from CA Configuration Automation into CA Catalyst (and ultimately CA CMDB) – using Management Profiles or Catalyst Jobs?

The Catalyst jobs in CA Configuration Automation give you more control over the data entering the CMDB. When you define a Catalyst Job, you select exactly which services/servers and blueprints to export. When using Management Profiles, any service/server assigned to the integration-enabled management profile will be exported and any blueprints configured within the profile will also be exported.

When exporting CIs from CA Configuration Automation, do all chosen CIs get sent to CA Catalyst regardless of whether they were previously exported?

CA Configuration Automation maintains a check sum table in its database called acm_catlst_ci_cksum. When a CI is exported, CA Configuration Automation queries the table to see if a check sum for the CI exists. If the CI is not present in the table, a check sum of the CI's attributes is stored within the table and the CI is sent to Catalyst. If the CI is present in the table but the check sum is different, then the new checksum is entered into the table and the CI is sent to Catalyst for an update. If the CI is present and the checksum is the same, the CI is dropped and not sent to Catalyst as there were no changes to the CI.

After exporting my selected CIs from CA Configuration Automation, I see server CIs within SDM that are not listed in the Servers table within CA Configuration Automation and were not part of my Catalyst Job or Management Profile. Where did these CIs come from?

These server CIs most likely are a result of selecting to export communication or configuration relationships or both from CA Configuration Automation. The 'unknown' server CIs are typically the source/target servers discovered as part of those relationships. From the CA Configuration Automation UI, click on a server from the Servers table to bring up the server details, select the Relationship tab and then select the Communication or Configuration sub-menu option. View the list of relationships, noting the servers listed in the source and target columns. The 'unknown' servers should be among those in the list.

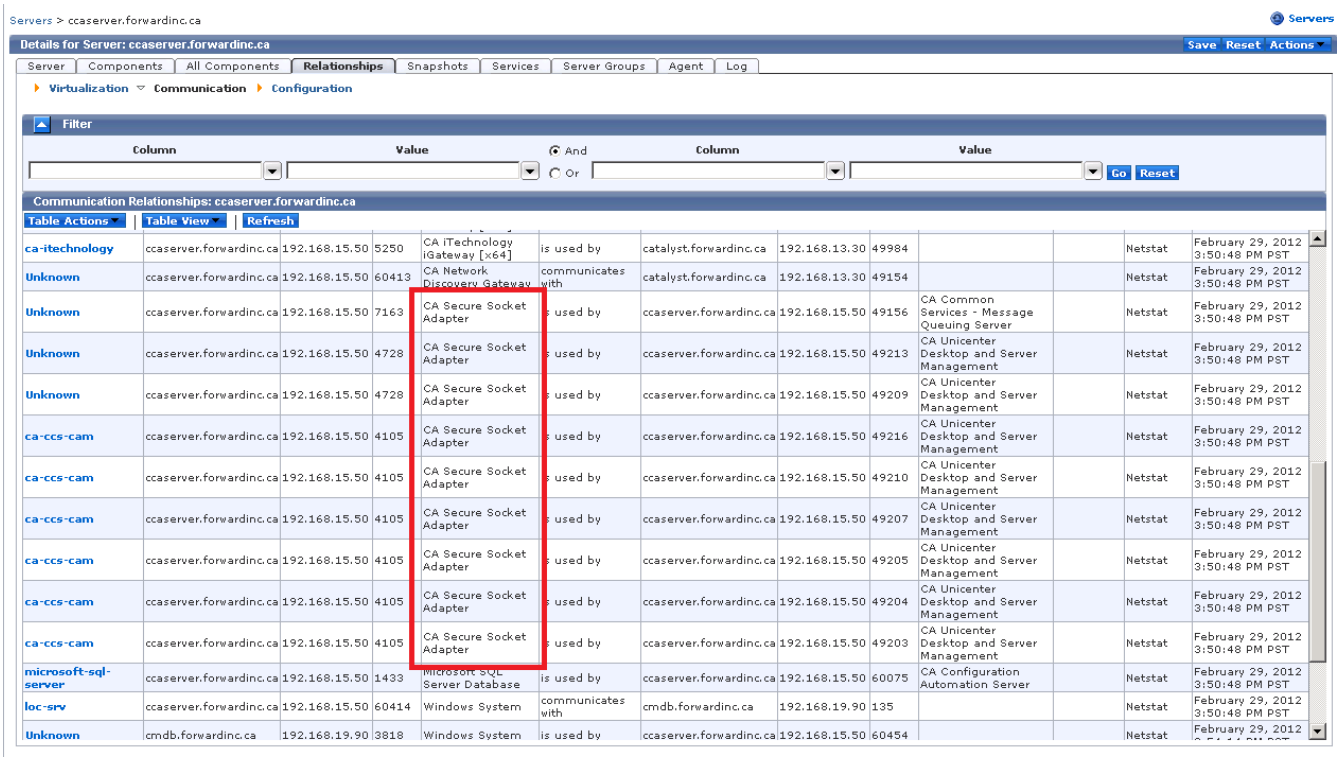
When I view my CIs within CA SDM, why do I see multiple entries for the same application?

When exporting Component CIs (for example, ProvisionedSoftware CIs) from CA Configuration Automation, the CA Configuration Automation Connector uses the Blueprint Name as the name of the CI. Although the blueprint name is not unique, the CORA attributes, specifically the system name, make the CI unique.

I ran a CA Catalyst Job or integration-enabled Management Profile in CA Configuration Automation. Then I used the CA Catalyst USM Web View to verify the CI counts in the three data repositories: CA Configuration Automation, CA Catalyst (CMDB-View) and ServiceDesk-CMDB. Why are the CI counts between the CA Configuration Automation and CA Catalyst (CMDB-View) or ServiceDesk-CMDB are different?

When CA Configuration Automation exports the CI or CI relationships, it is possible that multiple instances of the same server or same relationship exist. When the information is processed for export, the duplicate instances are reconciled and only one instance of the CI is sent to Catalyst.

For example, see the following screenshot:



This screenshot shows the communication (Netstat) relationships for the server, ccaserver.forwardinc.ca, discovered using the Softagent technology during Network Discovery. The highlighted portion shows that there are 9 instances where the CA Secure Socket Adapter application on ccaserver.forwardinc.ca is communicating with target servers.

From the CCA relationships, a BackgroundProcess CI is created in the Unified Service Model (USM) to represent the process attributes on the relationship. For the CA Secure Socket Adapter application, 9 processes would have been derived based upon what is seen in the screenshot. However, these 9 processes essentially represented the same process when each of the CCA relationships is prepared for export (the target servers are all in fact the same although communication is on different ports). Consequently, only one BackgroundProcess CI (out of the 9) is stored in Catalyst.

Also, as part of the CI processing, an 'IsHostFor' relationship is established from the ComputerSystem/ VirtualSystem to the BackgroundProcess causing the relationships to be reconciled . This is the expected behavior as Catalyst helps to reconcile all duplicate CIs.

Why do the number of relationships in CA CMDB and ServiceDesk-CMDB Projection Sheet of the CA Catalyst differ?

The Unified Service Model (USM) enforces a constraint that relationships are correlated using source, target, and scope. Scope is an attribute which represents the service object the relationship affects. When CA Catalyst sends relationships to CMDB for creation, the CA SDM connector creates the CI and traverses the CMDB network of CIs to find its scope. If the scope is not found, the CA SDM connector publishes the unscoped version of the relationship to CA Catalyst. Later, if the CA SDM connector finds the scope due to some update activity on the relationship, it publishes the scoped relationship to CA Catalyst. The scoped and the unscoped relationships do not correlate which results in extra relationships in the CA Catalyst database. There is no side-effect of having both scoped and unscoped version of relationships in the CA Catalyst database. However, the relationships tend to persist in CA Catalyst until they are removed from CMDB. When the relationship is removed from CMDB, it deletes the scoped, and the unscoped relationships from CA Catalyst.

When I delete servers or components from CA Configuration Automation, are those deletions reflected in CA Catalyst and CA SDM?

Yes. The CA Configuration Automation Connector has a timer thread and, at an interval specified by the delete_thread_interval configuration parameter defined in CA Configuration Automation (default: 15 minutes). The thread identifies the CIs marked for deletion in the checksum table and publishes a delete event to CA Catalyst.

I have multi-tenancy enabled in CA Configuration Automation but not in CA SDM (or vice versa). Will this cause any problem?

Yes. Catalyst does reference the tenant of a CI when performing a reconciliation. Tenancy that is enabled within CA Configuration Automation, but not within CA SDM may create a problem with reconciliation of information within Catalyst. We recommend that if you are using multi-tenancy, enable it in both products and ensure that the names of the tenants exist in both. If you are not using tenancy in CA Configuration Automation, but have it enabled in CA SDM you should not run into a problem if the CI is tenanted to “Public”. The problem will only occur if it were assigned to a specific tenant. Possible problems that you may see includes updates to the CIs within CA Configuration Automation not propagating to CA SDM, or relationships not being imported into CA SDM from CA Configuration Automation.

Integrate with CA Configuration Automation

Integrating CA SDM with CA Configuration Automation includes the following steps:

Step 1: Verify the Prerequisites

Step 2: Create CA Configuration Automation MDR in CA SDM

Step 3: Define the CA SDM Configuration Properties in CA Configuration Automation

Step 4: Create CA SDM Job to Export CIs to CA CMDB

Step 1: Verify the Prerequisites

Verify the following prerequisites before starting the integration:

Hardware and Software Requirements

CA SDM and CA Configuration Automation support several hardware, software, operating system, and database. For more information about CA Configuration Automation, see Supportability Matrix.

Supported Version of the Integrated Products

This integration supports the following product versions:

- CA Configuration Automation
- CA EEM

Integration Considerations

Consider the following information before you begin the integration:

- Install CA SDM.
- Install CA Configuration Automation.
- Ensure that the services and servers that are discovered in CA Configuration Automation can be exported to CMDB module of CA SDM. For more information about CA Configuration Automation discovery, see the following sections:

  - Service Management
  - Server Management
  - Network Management
  - Blueprint Management
- Verify that the CA Configuration Automation and CA SDM servers are up and running. Verify that all these servers have network connectivity with each other.
- We recommend that you install CA Configuration Automation, and CA SDM on different computers for this integration to work properly.
- Ensure that the latest version of GRLoader.jar file is available in CCA:
  a. Backup the existing GRLoader.jar file.
  b. Navigate to the `$NX_ROOT\java\lib` folder.
     a. Copy the `GRLoader.jar` file.
  c. Navigate to `<CCA Installed Directory>\tomcat\webapps\ROOT\WEB-INF\lib` folder
     a. Paste the `GRLoader.jar` file.
  d. Restart the CA SDM and CCA services.

Step 2: Create CA Configuration Automation MDR in CA SDM

For the CA SDM server to identify the details of the CA Configuration Automation server, create an MDR in the CA SDM server. For more information, see CA Configuration Automation documentation.

Step 3: Define the CA SDM Configuration Properties in CA Configuration Automation

Define the CA SDM configuration properties, before you export the CIs from CA Configuration Automation to CA SDM. For more information about defining the CA SDM configuration properties, see For more information, see CA Configuration Automation documentation.

Step 4: Create CA SDM Job to Export CIs to CA CMDB

To export the CIs to the CA CMDB, create an SDM job in CA Configuration Automation. For more information, see CA Configuration Automation documentation.

Integrating CA Service Catalog

This section contains the following articles:

- Integrate CA Single Sign-On with CA Service Catalog
- Integrate CMDB with CA Service Catalog
- Enable PKI-based Integration

Integrate CA Single Sign-On with CA Service Catalog

This article contains the following topics:

By default, CA Service Catalog uses CA EEM to authenticate the requests. You can optionally use CA Single Single Sign On to provide Web based single sign-on (SSO) and enhanced authentication to CA Service Catalog users.

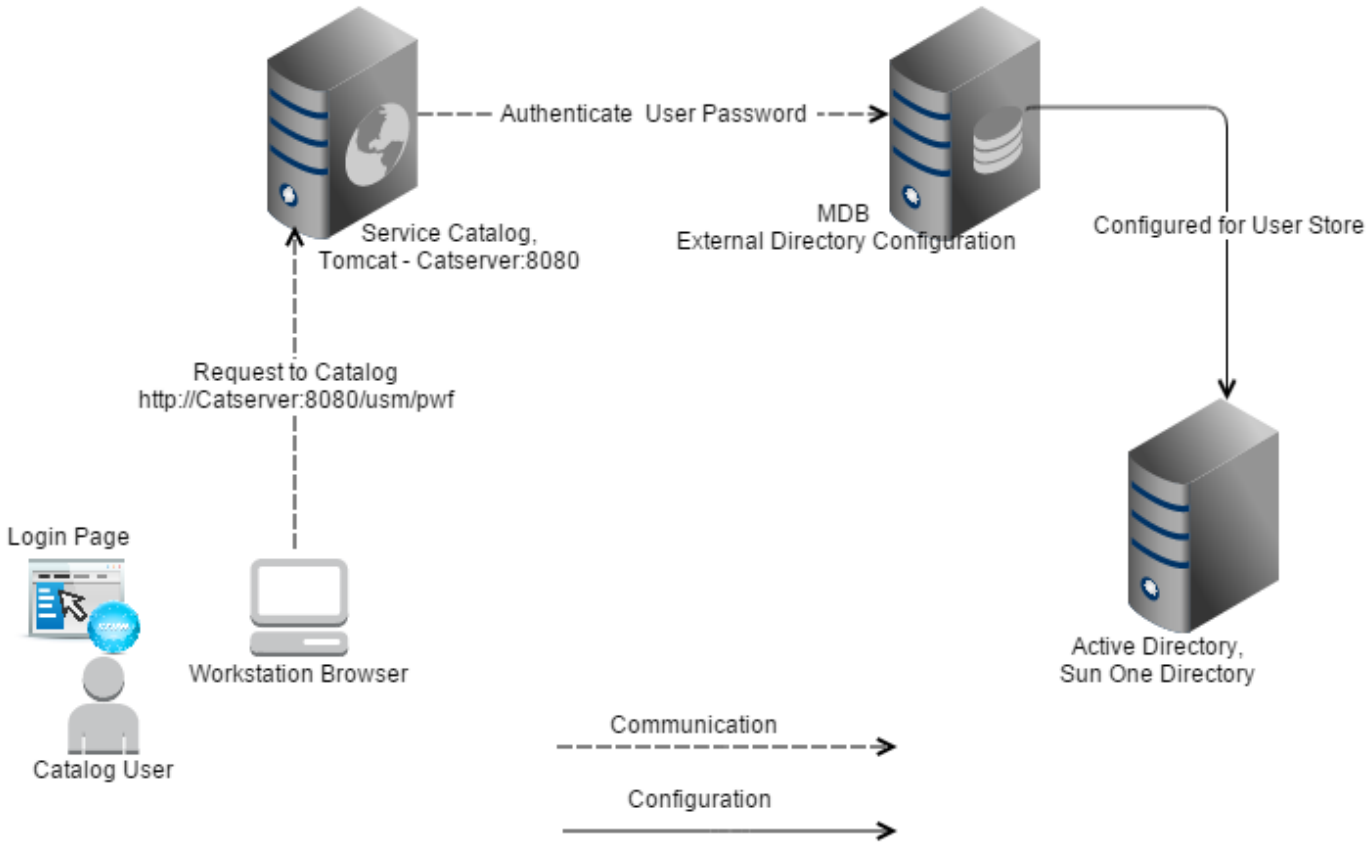
Options to Authenticate Users

This section explains the two possible authentication flows for authenticating CA Service Catalog users. The first authentication flow uses CA EEM alone. The second authentication flow uses CA EEM with CA SiteMinder, enabling single sign-on and providing enhanced security.

Authentication Flow Using CA EEM

By default, CA Service Catalog uses CA EEM to authenticate users. In this example of the basic authentication flow, requests from CA Service Catalog users first pass through Tomcat. The requests then pass through the MDB, and end at your authentication server, for example, Active Directory. The following diagram illustrates the basic authentication flow with CA EEM and Active Directory:

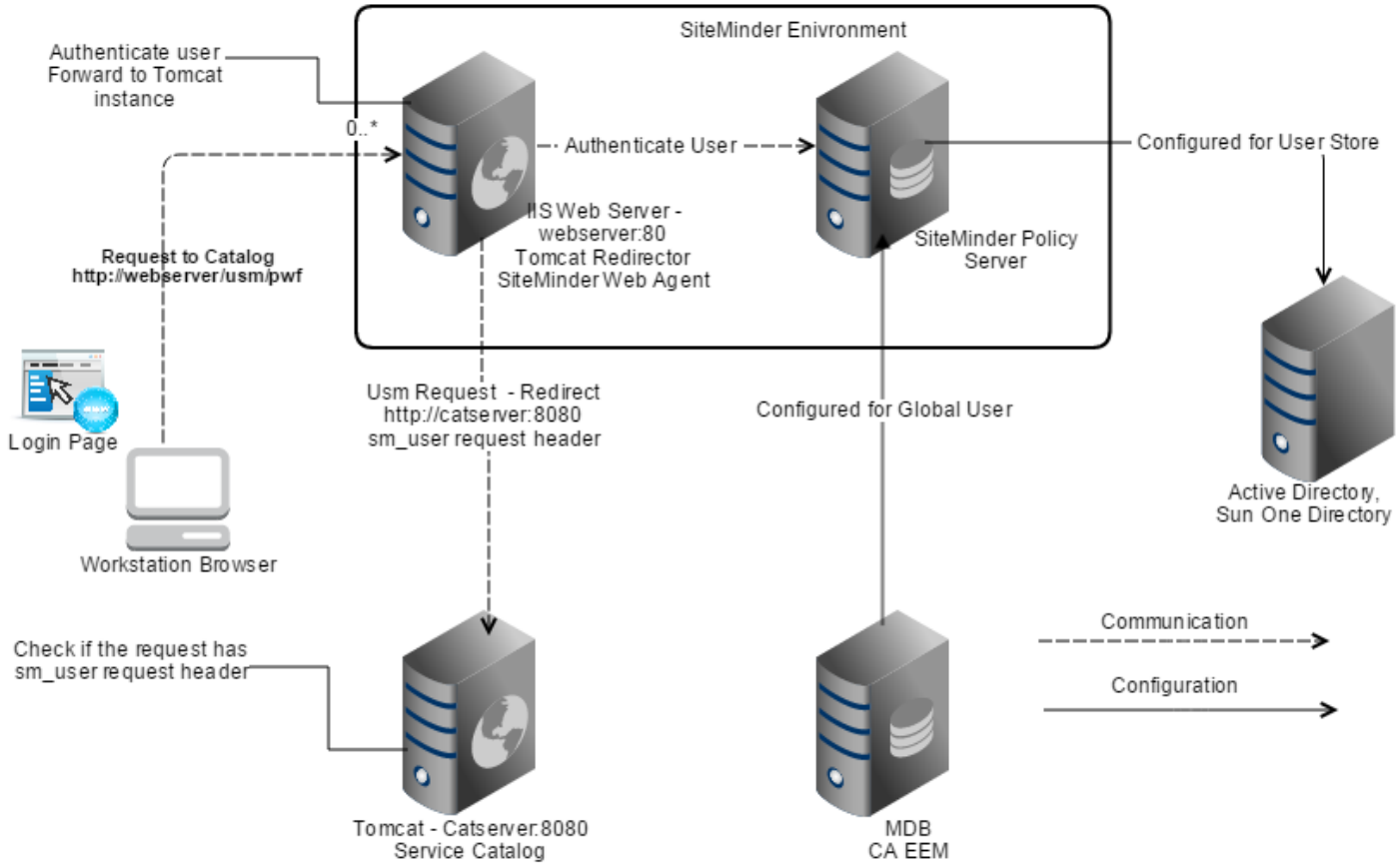
Figure 40: CA Service Catalog Authentication Flow Using CA EEM



Authentication Flow Using CA SiteMinder

In this authentication flow with CA Single Sign-On, requests from CA Service Catalog users go to CA SiteMinder for authentication. If the request is authorized, then the requests are forwarded to the Tomcat instance of Catalog Component by the web server hosting CA SiteMinder. Thus, in this flow, CA SiteMinder validates the authentication requests from CA Service Catalog users. The following diagram illustrates the enhanced authentication flow with CA SiteMinder.

Figure 41: CA Service Catalog Authentication Flow Using CA SiteMinder



Set Up Web Single Sign-on

To implement web-based (SSO) and enhanced authentication to CA Service Catalog users, complete the following tasks:

1. Install and configure CA SiteMinder, including its Policy Server and CA SiteMinder Web Agent.
   For more information, see your CA SiteMinder documentation.
2. Redirect the authentication requests from your web server to Tomcat. Here, your web server (such as Apache or Microsoft Internet Information Server [IIS]) has the CA SiteMinder web agent installed.
   For more information, see your web server documentation.
3. In the SiteMinder Administration GUI, configure CA SiteMinder to protect CA Service Catalog resources by performing the following tasks. For more infromation, see your CA SiteMinder documentation.
   a. Open the policy server UI.
   b. Create an agent object for CA Service Catalog; do *not* check support 4.x Agents".
   c. Create an agent configuration object for the agent you just created.
   d. Create a host configuration object.
   e. Optionally, create an authentication scheme.
   f. Create a realm and a rule with the resource filter as usm/*.
   g. Create a CA Service Catalog domain and add the user directories, administrator, and realm to the domain.
   h. Create a policy and add the rule that you just created to the policy.
4. In the Administration, Configuration section of the CA Service Catalog GUI, configure the single sign-on authentication parameters to match CA SiteMinder.

Integrate CMDB with CA Service Catalog

This article contains the following topics:

CMDB Overview

A configuration management database (CMDB) is a repository of asset information defining the critical configuration items in your environment and their relationships. A service catalog fits into that framework as a tool to help with financial governance, workflow, management of service requests, and service definition. The closer the relationship between the infrastructure assets that are represented by the configuration items and the logical services that the users consume, the more useful the integration between the CMDB and the service catalog is to your organization.

CMDB is a comprehensive, integrated solution for managing the IT components and services and their relationships. You can store reliable, up-to-date details on assets, which are known as configuration items, and their relationships with each other. These relationships form the basis for impact analysis, an important tool for controlling change within an organization.

When the two are integrated, the CA Service Catalog services are associated with CMDB configuration items. Using the CMDB Visualizer, you can trace the dependencies of the services on other configuration items. You can therefore determine the impact of configuration items on services. An important application is when a configuration item used by one or more services either suffers loss of performance or stops working. You can visually pinpoint the impact of this event to the service.

When a specific change affects a configuration item, use the CMDB impact analysis tools to figure out how the change affects the services. Then you can use CA Service Catalog to determine the organizations and users that have subscribed to the impacted services. This information helps you plan the change that is required. Also, you can use the service level agreements between the subscribing organization and the providers to help plan the change. For example, if a heavily used server slows down or crashes, determine the impact on CA Service Catalog services by using the integration. You can answer the following questions:

- Which specific services use that server?
- Do those services use any other server?
- Are other servers available to replace the affected server?
- Which accounts and subscriptions are affected?
- What are the possible financial risks? For example: do the affected services or configuration items have associated service-level agreements (SLAs)? Do the impacted services help generate revenue?

Your analysis can include impact analysis, efficiency analysis, financial analysis and profitability, and ITIL considerations. Both CMDB and CA Service Catalog support ITIL principles.

Key Terms

Configuration Management Database (CMDB) is a functional data repository that unifies and simplifies the management of configuration information. CMDB consolidates and reconciles disparate sources of IT-related data in the context of business priorities. CMDB provides visibility into configuration item information such as resource attributes, relationships and dependencies.

Configuration items form the basis of configuration management solutions. Typically, a configuration item is a collection of objects that are related to the specific functionality of a larger system. Examples of these objects are requirements, code, documentation, models, and other files.

Visualizer provides a unified graphical view of relationships between configuration items and inter-dependencies of business processes. It helps you determine which CA Service Catalog services are connected to which CMDB configuration items.

Types of Association Between Services and Configuration Items

You can view, modify, and create associations between any CA Service Catalog services and any CMDB configuration items. Typically, the most suitable configuration items belong to the Business Service class in the Enterprise Services family. Several configuration items in this family and class are provided by default. You can create any new ones that you require.

The following types of association between CA Service Catalog services and CMDB configuration items are possible:

- One service to one configuration item
- One service to many configuration items
- Many services to one configuration item
- Many services to many configuration items

The many-to-many relationship between CA Service Catalog services and CMDB configuration items is the most commonly used.

You can also:

- Associate existing services to existing configuration items.
- Use an existing service to create one or more new configuration items.
- Create a service for one or more existing configuration items.

Prerequisites for CMDB - CA Service Catalog Integration

- Be familiar with basic functions and administration of CMDB, such as families, classes, configuration items, and relationships between configuration items. Understand how to use the analysis tools, such as root cause analysis and impact analysis.
- You must be familiar with the basic functions and administration of CA Service Catalog.
- Ensure that both CA Service Catalog and CMDB are installed, configured, and running.
- If you are using Service Accounting Component, ensure that it is installed, configured, and running.
- Use the same MDB installation for CMDB and CA Service Catalog.

After you have installed and configured CA Service Catalog and CMDB, integrate them by performing the setup tasks first.

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When the two are integrated, the CA Service Catalog services are associated with CMDB configuration items. Using the CMDB Visualizer, you can trace the dependencies of the services on other configuration items. You can therefore determine the impact of configuration items on services. An important application is when a configuration item used by one or more services either suffers a loss of performance or stops working. You can visually pinpoint the impact of this event on the service. When a specific change affects a configuration item, use the CMDB impact analysis tools to figure out how the change affects the services. Then you can use CA Service Catalog to determine the organizations and users that have subscribed to the impacted services. This information helps you plan the change that is required. Also, you can use the service level agreements between the subscribing organization and the providers to help plan the change.

To set up integration between CMBD and CA Service Catalog, see the steps to integrate CMDB and CA Service Catalog.

{"URL":["usm/wpf?Node=iclaunchpad.cmdb_config_options"],"customLabelGetStarted":"Manage CMDB and Configuration Item

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Step 1 - Perform Common Setup Tasks

This article contains the following topics:

The CMDB and CA Service Catalog administrators can work together to perform the following common setup tasks:

Step 1a - Verify Access Rights

CA Service Catalog administrators must ensure that CMDB administrators, analysts, and other users have the required access rights in CA Service Catalog to components (such as requests, subscriptions, business units, or accounts) that are associated to configuration items.

Assign CA Service Catalog administrator roles to these CMDB users. By doing so, you ensure that they have required access rights.

CMDB administrators must ensure that CA Service Catalog administrators have the required access rights to CMDB configuration items that are associated to CA Service Catalog components.

Step 1b - Configure for Single Sign-on

Bypass user login by configuring CA Service Catalog and the CA products with which it integrates to use single sign-on.

To configure CMDB and CMDB Visualizer to use single sign-on, configure them to use either CA SiteMinder or Windows NTLM authentication.

To configure CA Service Catalog to use single sign-on, configure it to use either CA SiteMinder or Windows NTLM Authentication. For more information about how to configure, see in Enable External Authentication of Users section.

Step 2 - Perform Setup Tasks for CMDB

As a CMDB administrator, on the CMDB computer, log on to CMDB. Access the CMDB Administration tab, and perform the following tasks.

1. Using the Web Services Policy option, create the Web Services Access policy.
   a. Assign a name. The suggested name and code are USM_CMDB_Policy, but you can use any name that meets your needs.
   b. Assign a proxy user to this policy.

      WARNING
      This proxy user *must* have administrative access to CMDB.

   c. Verify that the Allow Impersonate property of CMDB is set to Yes.

   CMDB checks the access type of the CA Service Catalog user to be impersonated against the access type of the policy proxy user. If the access level of the CA Service Catalog user is less than or equal to the grant level of the proxy user's access type, CMDB replaces the CA Service Catalog user with the proxy user. For more information, see your CMDB documentation.
2. Generate the policy key file for the Web Services Access policy that you created. To generate the key file, use the pdm_pki tool.
   The suggested name is USM_CMDB_Policy.p12, but you can use any name that meets your needs.
   Verify that the HasKey property is set to Yes in the newly created Web Services access policy. The system typically configures this setting automatically when you generate the policy key file.
   Copy the policy key file to the CA Service Catalog server and save it in the USM_HOME directory.

WARNING
If you are using application clustering, copy the policy key file to the USM_HOME directory of *every* Catalog Component cluster node.

3. Verify that the Business Service class of the Enterprise Service family exists. This class is required for CA Service Catalog users to create a configuration item in CMDB. If this class does not exist, create it manually or run the CMDB installation again to load default data.

Step 3 - Perform Setup Tasks for CA Service Catalog

This article contains the following topics:

As a CA Service Catalog administrator, on the CA Service Catalog computer, log in to CA Service Catalog.

Update the Configuration Settings

Specify the configuration settings for CA CMDB.

Follow these steps:

1. Click Configuration, CMDB, on the Administration tab of CA Service Catalog.
2. Complete the Configuration settings, as follows:
   - **Enable HTTPS**
     Specifies the option (Yes or No) that matches your implementation.
   - **Host Name**
     Specifies the host name of CMDB computer.

     NOTE
     This recommendation applies after you have configured the integration and have started to use it: Do *not* change the host name of the CMDB computer or the CA Service Catalog computer, *unless* the two host computers share MDB.
   - **Port Number**
     Specifies the port number on the CMDB computer that listens for incoming calls from CA Service Catalog.
   - **Key File Name**
     Specifies the name of the Web Services policy key file that you created earlier.
     The default value is USM_CMDB_Policy.p12.
   - **Policy Code**
     Specifies the name of the Web Services policy code that you created earlier.
     The default value is USM_CMDB_Policy.
   - **Default New CI State Inactive**
     Specifies whether new configuration items that you create through this integration have a default state of *inactive* or *active*.
     Specify Yes for inactive or No for active.
3. Verify the CMDB Configuration by clicking Test Connection.
   If the test fails, perform the following actions:
   a. Verify that your settings are completed correctly, especially the host name and port number of the CMDB computer.
   b. Verify that the CMDB service is up and running.

Update the Visualizer Settings

Specify the configuration settings for the CMDB Visualizer.

Follow these steps:

1. Click Configuration, CMDB Visualizer on the Administration tab.

2. Specify the following Visualizer Configuration settings:
   - **Enable HTTPS**
     Select the option (Yes or No) that matches your implementation.
   - **Host Name**
     Specifies the host name of the CMDB Visualizer computer, typically the CMDB computer.
   - **Port Number**
     Specifies the port number of the CMDB Visualizer.
3. Verify the CMDB Visualizer by clicking Test Connection.
   If the test fails, perform the following actions:
   a. Verify that your settings are completed correctly, especially the host name and port number of the CMDB Visualizer computer.
   b. Verify that the CMDB Visualizer service is up and running.

Update CA CMDB in CA Service Catalog Configuration.

Follow these steps:

1. Click Configuration, CMDB, on the Administration tab of CA Service Catalog.
2. Complete the Configuration settings, as follows:
   - **Enable HTTPS**
     Specifies the option (Yes or No) that matches your implementation.
   - **Host Name**
     Specifies the host name of CMDB computer.

     NOTE
     This recommendation applies after you have configured the integration and have started to use it: Do *not* change the host name of the CMDB computer or the CA Service Catalog computer, *unless* the two host computers share MDB.
   - **Port Number**
     Specifies the port number on the CMDB computer that listens for incoming calls from CA Service Catalog.
   - **Key File Name**
     Specifies the name of the Web Services policy key file that you created earlier.
     The default value is USM_CMDB_Policy.p12.
   - **Policy Code**
     Specifies the name of the Web Services policy code that you created earlier.
     The default value is USM_CMDB_Policy.
   - **Default New CI State Inactive**
     Specifies whether new configuration items that you create through this integration have a default state of *inactive* or *active*.
     Specify Yes for inactive or No for active.
3. Verify the CMDB Configuration by clicking Test Connection.
   If the test fails, perform the following actions:
   a. Verify that your settings are completed correctly, especially the host name and port number of the CMDB computer.
   b. Verify that the CMDB service is up and running.

Update CA CMDB Visualizer in CA Service Catalog Configuration.

Follow these steps:

1. Click Configuration, CMDB Visualizer on the Administration tab.
2. Specify the following Visualizer Configuration settings:
   - **Enable HTTPS**

Select the option (Yes or No) that matches your implementation.

- **Host Name**
  Specifies the host name of the CMDB Visualizer computer, typically the CMDB computer.
- **Port Number**
  Specifies the port number of the CMDB Visualizer.

3. Verify the CMDB Visualizer by clicking Test Connection.
   If the test fails, perform the following actions:
   a. Verify that your settings are completed correctly, especially the host name and port number of the CMDB Visualizer computer.
   b. Verify that the CMDB Visualizer service is up and running.

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Step 4 - Set Up the CMDB - CA Service Catalog Integration

This article contains the following topics:

Follow these steps to enable the integration between CMDB and CA Service Catalog:

1. Ensure that CA Service Catalog, Service Accounting Component (if used), and CMDB are installed, configured, and running.
2. Ensure that you verify the MDB version compatibility for all CA products that you plan to integrate, including CMDB.
   - Review the level of maturity of each product in your environment.
   - Review standard process and procedures in your organization, especially external standards that you must follow. Examples of standards are ITIL, ISO, COBIT, and CORA.
     ITIL is especially applicable because both CMDB and CA Service Catalog support ITIL concepts.
   - In CMDB, review the configuration items in the Business Service class of the Enterprise Services family. These configuration items are considered to be the most likely candidates for integration with CA Service Catalog services. Remember, however, that you can associate *any* configuration item or items with one or more CA Service Catalog services.
   - In CA Service Catalog, review the existing services.
   - In both, review the existing associations if any.
3. Review the key terms for the integration.
4. Review the types of association between services and configuration items that you can create.
5. Review the associations that already exist, if any; for more information, see View Associations Between Services and Configuration Items.
6. Determine any *new* Catalog-CMDB associations that make sense for your business. Ask these questions:
   - Which existing configuration items must be associated with one or more services? Do those services already exist, or do you have to create new ones?
   - Which existing services must be associated with one or more configuration items? Do those configuration items already exist, or do have to create new ones?
   - Do you want to populate the CMDB service by service?
   - Have you started or finished populating the CMDB, using the guidelines for doing so in the CMDB documentation? Or do you want to start by seeding CA CMDB with a few mission critical applications and business services?
   - Have you checked the business continuity plan of your organization? Review this plan and determine which business services and respective components in your infrastructure are critical.

As you answer the questions, consider the following factors:

- Keep in mind that the goal of all associations is to use IT to support your business needs.
- If applicable, begin thinking about associations starting from the perspective of the product that is most mature in your environment. For example, if you have an established CMDB implementation but a newer CA Service Catalog implementation, then CMDB is the more mature product. Start by viewing your existing configuration items in the Business Service class of the Enterprise Services family. Determine which ones could use CA Service Catalog services, to serve your business needs. In this model, it is likely that you first create the needed services and next create the associations.
  Conversely, if you have mature services but your CMDB implementation is just starting, create configuration items corresponding to the services. For example, for critical services connected to service level agreements, create configuration items that are based on the hardware or other infrastructure that is required to support the CA Service Catalog services.
  For example, define your email application and server as configuration items and associate them to an email service in CA Service Catalog. Similarly, you can also associate that email application and server with a service for requesting Blackberry units and subscriptions.
- If you are using Service Accounting Component, consider creating and associating configuration items with the services related to your most important subscriptions and most important financial reporting items.

7. Create any configuration items that you need for the integration.
8. Create any services that you need for the integration.

   NOTE
   When a service is copied or inherited, its associations to CIs are *not* copied.
9. Consider service level agreements (SLAs).
   Whenever the availability of services is affected, the SLA can have a significant effect on impact analysis. For example, if a server used by a service that your company provides is down, the ability to meet the SLA obligations to your customers is affected. Therefore, you want to set up your service-to-configuration item associations to include services and computers that are involved in meeting SLA obligations.
10. Associate the services to configuration items. These associations are *from* CA Service Catalog *to* CMDB. You associate a CA Service Catalog service ID with a CMDB UUID (not a logical ID).
    For example, consider the New Hire Onboarding service, which is one of many services that are supplied with CA Service Catalog. This service can be used for requesting office equipment that is required by a newly hired employee. Create a configuration item for each of these assets and associate them to the service. In this case, the association is many to one (many configuration items to one service).
11. Using the CMDB Visualizer, leverage the analysis tools of CMDB, such as impact analysis and root cause analysis. For more information, see How to Leverage Information from Analysis Tools.

Create a Configuration Item for a Service and Associate Them

Many of the associations that you create involve CA Service Catalog services and existing CMDB configuration items. You can *create* a configuration item specifically for a service, if no suitable configuration item exists already. You do not have to create and associate a configuration item for every service. Instead, create and associate such configuration items *only* when they support your business needs and goals.

Before associating a service to a configuration item, ensure that the service is *defined* in CA Service Catalog. The service option groups and service options for that service have been specified. For more information about defining services, see the Manage Services section.

NOTE
If you associate a CA Service Catalog service with a CMDB configuration item that does *not* display the Attributes tab, you *cannot* view the service-related details of the configuration item from CA Service Catalog. For more information, see the CMDB documentation.

For more information to associate services with *existing* configuration items, see Associate Services with Configuration Items.

Follow these steps:

1. From the CA Service Catalog home page, select Catalog, CMDB CI Association.
2. Under the line "The current business unit is: *business-unit-name*," click the Catalog tab.
   The list of existing folders, sub folders, and services appears.
3. Expand the folders and sub folders as needed, until the services you want appear.
4. Check the box next to the service or services for which you want to create and associate a configuration item.
5. Click the right arrow next to the Selected Services box.
6. Click Next.
   The screen refreshes, and the CI List for Association appears. The
   names of any configuration items that are already associated with the selected service appears.
7. View the list of existing associations under the CI Name heading.
8. Click Create CI to *create* a configuration item for each selected service.
   For each selected service, a new configuration item is created in the Business Service class of the Enterprise Services family. Also, a one-to-one association is created between each service and its new, same-named configuration item. By default, the system also displays the names of several CMDB configuration item attributes. While the *names* of these attributes are listed by default, the *values* of these attributes are not populated by default. You can optionally populate these values. In addition, you can add and populate other attributes of your choosing. For more information, consult your implementation team, including your CMDB administrator and CA Service Catalog administrator.
9. Click Save to associate the new configuration item and all selected service.

Associate Services with Configuration Items

Create only the associations that support your business goals and needs.

Before associating a service to a configuration item, ensure that the service is *defined* in CA Service Catalog. The service option groups and service options for that service have been specified.

> **NOTE**
> If you associate a CA Service Catalog service with a CMDB configuration item that does *not* display the Attributes tab, you *cannot* view the service-related details of the configuration item from CA Service Catalog.

Follow these steps:

1. From the CA Service Catalog home page, select Catalog, CMDB CI Association.
2. Under the line "The current business unit is: *business-unit-name*," click the Catalog tab.

   > **NOTE**
   >
   > If you know the name of the service or folder that you want, you can optionally use the Search tab to display services.
3. Expand the folders and sub folders as needed, until the services you want appear.
4. Check the box next to the service or services that you want to associate with one or more configuration items.
5. Click the right arrow next to the Selected Services box.
   This action selects the services. The names of the selected services appear in the Selected Services box.
6. Click Next.
   The screen refreshes, and the CI List for Association appears, displaying the names of the service that you selected. The services are listed in the service column. If any service is already associated to one or more configuration items, the item is listed next to the service in the CI Name column.
   You can associate each of these services (either individually or as a group) to one or more configuration items.
7. Check the service or services that you want to associate to one or more configuration items in the same action. Leave any other services unchecked; you can optionally associate such services to configuration items in a separate action.

NOTE
To *create* a configuration item for each selected service and associate them, click Create CI, as explained in Create a Configuration Item for a Service and Associate Them. If you click Create CI, new configuration item is created for each selected service. Also, a one-to-one association is created between each service and its new, same-named configuration item.

To associate a single service with one or more existing configuration items, perform the following actions:

1. Check the service and click Associate CIs.
2. Click Search or use a search string to display the available configuration items.
3. Verify that all configuration items that you want to associate with the service that you checked, and click OK.
   If you selected only one configuration item, a one-to-one association is created between the checked service and that item.
   If you selected two or more configuration items, a one-to-many association is created between the checked service and those items.

To associate two or more services with one or more existing configuration items, perform the following actions:

1. Check the services and click Associate CIs.
2. Click Search or use a search string to display the available configuration items.
3. Verify that all configuration items that you want to associate with the services that you checked, and click OK.
   If you selected only one configuration item, a many-to-one association is created between the checked services and that item.
   If you selected two or more configuration items, a many-to-many association is created between the checked services and those items.
4. Click Save after associating the items.

Your associations are saved. The associations are used for impact analysis, root cause analysis, and other analysis functions for your integration between CMDB and CA Service Catalog.

View Associations Between Services and Configuration Items

You can quickly view existing associations from CA Service Catalog services to CMDB configuration items. It is helpful to see such quick views for many reasons, including saving time, avoiding duplicate associations, and gaining quick access to the CMDB Visualizer for the service and any associated configuration items.

NOTE
If your implementation requires you to log in to CMDB Visualizer, see your administrator to obtain a user name and password for accessing CMDB Visualizer.

Follow these steps:

1. Click Catalog, Service Offerings.
2. Search the folders and services, expanding them if necessary, until you locate the service you want.
3. Under the Actions icons, click the CMDB Association icon. Its tooltip text is "CMDB CI Associations."
   The Service Offerings dialog appears, showing the configuration items that are currently associated with the service.
4. Optionally, perform one of the following actions:
   a. Click the Visualizer icon to view the relationship of those configuration items currently associated with the service. Its tooltip text is "CMDB Visualizer."
      Log in to CMDB Visualizer and use the Visualizer features to find the most critical relationships; you use this information for impact analysis, root cause analysis, and other analysis.
   b. Delete the association by selecting the configuration item or items and clicking Remove Association.

View the Service-Related Details of a Configuration Item

The service-related details of a CMDB configuration item are the CA Service Catalog service or services associated with the item, the number of requests for each associated service, and any other related properties that your organization deems important. This information can be a critical part of your analysis activities for the associated services and configurations items. For example, a high number of associated requests, subscriptions, or accounts may indicate that a configuration item is very critical for business continuance and therefore merits special attention.

If your implementation requires you to log in to CMDB Visualizer, see your administrator to obtain a user name and password for accessing CMDB Visualizer.

> **NOTE**
> If you associate a CA Service Catalog service with a CMDB configuration item that does *not* display the Attributes tab, you *cannot* view the service-related details of the configuration item from CA Service Catalog.

Follow these steps:

To begin in CA Service Catalog, perform the following actions.

1. Click Catalog, Service Offerings.
2. Search the folders and services, expanding them if necessary, until you locate the service you want.
3. Under the Actions icons, click the CMDB Association icon. The tooltip text for this icon is "CMDB CI Associations."
   The Service Offerings dialog appears, showing the configuration item or items that are associated with the service.
4. Click Visualizer.
5. (Optional) Log in to CA CMDB Visualizer.
   The associated items appear in CMDB Visualizer. If these items have relationships with other configuration items, the related items and the relationships also appear.

To begin in CMDB Visualizer, perform the following actions:

1. Log in to CMDB Visualizer.
2. Display the configuration items that are associated with the service of interest.
   If these items have relationships with other configuration items, the related items and the relationships also appear.
3. Right-click the configuration item and select View Properties.
4. Click the Attributes tab and click the Service Catalog button, which is the managed data repositories (MDR) button for CA Service Catalog.
   The CA Service Catalog GUI opens.

   > **NOTE**
   >
   > Optionally right-click the configuration item and select View, MDR. From the list of MDRs, select Service Catalog.
5. If the CA Service Catalog GUI does not open, perform the following actions:
6. a. Verify that the CA Service Catalog host name property specified in the MDR button is correct.
   b. Verify that the CA Service Catalog services are started.
7. If necessary, log in to CA Service Catalog.
   The CI Associated Offering Details dialog appears.
8. View the count for the requests, subscriptions, accounts, and business units that are related to the configuration item and its associated service or services.
   If multiple services are associated to the configuration items whose properties are viewing, the counts for all associated services are listed.
   For requests, accounts, subscriptions, and business units, optionally click the links for further details about each count:

Use this information along with any information you obtain from the CMDB Visualizer analysis tools. These tools are summarized in Leverage Information from Analysis Tools.

Leverage Information from Analysis

After you have established the relationships between configuration items, you can use the CMDB Visualizer for the following tasks:

- Provide a 360-degree view of the relationships between business-critical configuration items within your organization.
- Perform root cause analysis for Incidents and Problems.
- Perform impact analysis on changes to configuration items.
- Create filters in the CMDB Visualizer to customize your views of configuration items.

Performing these tasks helps you detect and strengthen the connections between IT and business: specifically, between your IT components (configuration items) and business services. Each analysis also helps you understand the connections between configuration items and helps you maximize their value.

A service can both *be* a configuration item and can also involve *changing* the IT infrastructure. While the distinction sometimes seems obvious, distinguishing between these two types of services explicitly can be helpful.

Consider a managed application service example: A department requests that the IT department supply and maintain a server, install an application, and administer the application. The interaction with the CMDB is through a workflow. The workflow includes change orders for configuration items. The workflow also sets up relationships between the server, application, application service, and other configuration items.

Enable PKI-based Integration

This article describes how to enable PKI-based integration between CA Service Catalog and CA Service Desk Manager. Certain features, such as Service Management Content packs requires enabling of PKI between CA SDM and CA Service Catalog.

Follow these steps:

1. **Create a policy in CA Service Desk Manager with the same policy name as in CA Service Catalog**
   a. In CA Service Catalog, navigate to **Administration**, **Configuration**, **CA Service Desk** and identify the policy name from the Policy Code property.
   By default, the property name us USM_SD_Policy.
   b. In CA Service Desk Manager, navigate to **Administration, SOAP Web Service Policy**, **Policies**
   c. Click Create New and define the following values:
      a. Define *Symbol* as *USM_SD_Policy*
      b. Define *Code* as *USM_SD_Policy*
      c. Set *Status* as *Active*
      d. Set *Proxy Contact* as *Administrator, Service Delivery*
      You can also set another proxy contact, as required.
      e. Set *Allow Impersonate* as *Yes*
      f. Save and exit.
2. **Create a Certificate File for the Policy in CA Service Desk Manager**
   a. On CA Service Desk Manager server, navigate to the NX_Root folder and execute the following command:

      ```
      pdm_pki -p USM_SD_Policy -f
      ```

      Replace the *USM_SD_Policy* policy name with the name that you defined in in step 1 (c), if any.
   b. Restart CA Service Desk Manager services.
      This creates a file named *USM_SD_Policy.p12* in the NX_Root folder.
3. **Enable PKI in CA Service Catalog**
   a. Copy the file created in step 2 (a) to <USM_Home> folder.
   For example, copy *USM_SD_Policy.p12* to the <*USM_Home*> folder.

b. In CA Service Catalog, navigate to **Administration**, **Configuration**, **CA Service Desk**.
c. Set Enable PKI to Yes and click the Test button.
The Test action uses the certificate file to connect to CA Service Desk Manager and report a success. This indicates that the PKI integration is working and complete.

Integrating CA Asset Portfolio Management

This article describes the following post-installation tasks for CA Asset Portfolio Management (CA APM):

- Integrate with CMDB
- Integrate CA Asset Portfolio Management with CA Service Catalog Manually
- Integrate CA Asset Portfolio Management with Common Components Manually

Integrate with CMDB

The article contains the following articles:

This section explains how to integrate CA APM with CMDB that is included in CA Service Desk Manager 17.4.

CMDB is a comprehensive, integrated solution for managing the IT components and services in an enterprise and their relationships, in heterogeneous computing environments. CMDB lets you provide and store reliable, up-to-date information about assets, known as configuration items (CI), and their relationships with each other. These relationships form the basis for impact analysis, an important tool for controlling change within an organization.

CMDB integrates with CA APM in several areas, including the following areas:

The CA APM audit history records can include all of the changes that have been made to asset/CI records by CMDB, and CA APM.

- When CA Service Desk Manager is installed, the asset/CI audit history records include any CA APM audit history records on the CMDB Versioning tab.
- When you define an asset in CA APM, you can categorize and control the asset and CI records by selecting or clearing the Asset and CI check boxes. This flexibility is provided because CIs that CMDB creates may not be relevant to CA APM. Conversely, assets that CA APM creates may not be relevant to CMDB.
- CA APM can extend the fields on an asset/CI within the context of *asset families*. The extended fields can be shared in CA APM. For example, a CA APM administrator can configure the Asset page and define an asset extended field to let users view and update a CI that is created in CMDB.
- You can define an event on a field that is shared with CMDB in CA APM and trigger the event in either CA APM or CMDB. For more information about managing events and notifications, see Events and Notifications for CA APM.
- A CA Service Desk Manager user can define a Management Data Repository (MDR) and allow the CMDB CI to launch the corresponding asset in CA APM.

How to Integrate CA APM and CMDB

When you integrate CA APM and CMDB, you integrate and delineate the assets that CA APM manages from the configuration items (CIs) that CMDB manages in a simple and concise manner. The CA APM users can move to a shared classification model for the assets and CIs. To integrate CA APM and CMDB, complete the following steps:

1. Share asset and configuration item audit history records.
2. Categorize the asset and configuration item records.
3. Define an asset extended field.
4. Define an event on a shared field.
5. Define a Management Data Repository (MDR) from CMDB of CA Service Desk Manager.

Share Asset and Configuration Item Audit History Records

To integrate CA APM and CMDB, the CA APM audit history records can include all of the changes that were made to asset/CI records by CMDB, and CA APM. When CA Service Desk Manager is installed, the asset/CI audit history records in CMDB (Versioning tab) includes any CA APM audit history records.

CMDB includes audit history records from CA APM. The audit history records are updated in both CMDB and CA APM when the CA Asset Portfolio Management - Event Service service is started. For more information, see Start the Services.

Categorize the Asset and Configuration Item Records

In this step to integrate CA APM and CMDB, you can categorize and control the asset and CI records when defining an asset in CA APM by selecting or clearing the Asset and CI check boxes. This flexibility is provided because CIs that CMDB creates may not be relevant to CA APM and conversely, assets that CA APM creates may not be relevant to CMDB.

Consider the following information when using these check boxes:

- **Default Values**
  - All new asset records that CA APM creates are initially set both as an Asset only and Managed by CA APM. On the New Asset page in CA APM, the Asset check box is selected, the Managed by CA APM check box is selected, and the CI check box is not selected.
  - All the CMDB created asset records are initially set to CI only. On the CI pages in CMDB, the CI? column heading is set to Yes and the Asset? column heading is set to No.
  - Both CA APM and CMDB have the Asset and CI fields available on the New Asset and CI pages. However, the Managed by CA APM check box is only viewable in CA APM. The existing audit and security features for each product applies to these check boxes.
- **Appearance**
  - The Asset and CI fields appear in CA APM and CMDB even when other Broadcom products are installed.
  - The CA APM administrator can configure the user interface and move the Asset and CI fields to a new location, make the fields read-only, required, or optional, and hide the fields.

    NOTE
    For more information about configuring the user interface, see How to Configure the User Interface.
- **Viewing and Updating**
  - **CA CMDB**
    - By default, the CMDB analyst and administrator can update the Asset and CI field values.
    - CMDB, by default, does not allow the Asset? value to be changed when the Asset? value is set to Yes.
  - **CA APM**
    - By default, CA APM sees asset and CI records.
    - The CA APM administrator can configure the user interface and move the Asset and CI check boxes to a new location, make the check boxes read-only, required, or optional, and hide the check boxes. After you select the CI check box and save the asset, the CI check box is not available and you cannot change the setting.

      WARNING
      We strongly recommend that you configure the CI check box in CA APM as read-only and restrict changes to the check box to only the CMDB analyst and administrator.
    - An asset in CA APM in which the Managed by CA APM check box is selected is always an asset. You cannot save an asset in CA APM in which the Managed by CA APM check box is selected without also selecting the Asset check box.
- **Searching**
  - **CMDB**
    - The CMDB search initially displays, by default, all records. However, an option is provided to filter records.
  - **CA APM**

- The default asset search includes a drop-down list for Managed by CA APM, CI, and Asset. This flexibility is provided so that you can differentiate between assets and CIs.

- **Hardware Reconciliation**
  Hardware reconciliation analyzes all asset and CI records. Searches provide a way to view any CIs that are related to discovered assets as the result of running hardware reconciliation. A CA APM user can view the exceptions and determine whether they want to select the Asset check box. As a result of selecting the Asset check box, the asset records are available in a CA APM asset search.

Define an Asset Extended Field

In this step to integrate CA APM and CMDB, CA APM can extend the fields on an asset within the context of *asset families*. The extended fields can be shared in CA APM. For example, a CA APM administrator can configure the Asset page and define an asset extended field to let users view and update a CI that is created in CMDB.

> **WARNING**
> These steps work only the first-time you complete the wizard and define the asset extended field. Before you define the extended field, verify that you have the following information from the usp_owned_resource table in CA CMDB for reference: table name, format (character, boolean, currency, date, decimal, or integer), field name, attribute name, and field size. After you complete the wizard, you can configure the extended field like any field in CA APM.

Example: Define an Asset Extended Field for Warranty Start Date

In this example, you define an asset extended field for Warranty Start Date. In CA CMDB on the Inventory tab, you view the label in the CI as Warranty Start Date. Next, you view the information for the associated nr_wrty_st_dt column from the usp_owned_resource table in CMDB. In this example, the nr_wrty_st_dt column format is integer, the field name is nr_wrty_st_dt, the attribute name is nr_wrty_st_dt, and the field size is 4. Record and enter this information exactly as it appears in the appropriate Format, Field Name, Attribute Name, and Field Size fields in the wizard. We also recommend that to avoid confusion, you use the same label for the CI (Warranty Start Date) on the Label field in the wizard.

To define an asset extended field

1. Determine the CMDB extension table name and database field name by reviewing the CMDB schema files.
2. Log in to CA APM using login credentials in which you have permissions to define an extension.
3. Click Asset, New Asset.
4. On the left, click CONFIGURE: ON.
   The configuration of the page is enabled.
5. In the Configuration Information area of the page, define and save a global configuration.
6. Click Add Extension.
   A wizard appears.
7. Follow the on-screen instructions to enter the information for the extended field.
8. In the Type page of the wizard, complete the following steps:
   a. Select the Simple Field option.
   b. Select the part of the page on which the new field appears.
   c. Select the Across all extended types check box.
   d. Click Next.
9. In the Fields page of the wizard, complete the following steps:

   > **WARNING**
   > Enter the column information from the usp_owned_resource table in CMDB. We also recommend that to avoid confusion, you use the same label for the CI on the Label field.

   a. Click Add Field.
   b. Enter the field label to appear on the page.

c. Select the data format.
d. Enter the database field name.
e. Enter the attribute name.
f. Enter the field size.
g. (Optional) Enter a description for the field.
h. Specify whether an entry for the field is required.
i. Click the checkmark icon to save the field.
The product displays the field information you enter.
j. Click Next.

10. In the Summary page of the wizard, review the field information and click Save and Exit.
11. Verify that the field appears on the Asset page.
12. Click Save Configuration.
All users see the extended field on the page. You can define an event in CA APM and trigger the event in either CA APM or CMDB. For more information about managing events, see Events and Notifications for CA APM.

Define an Event on a Shared Field

You can define an event in CA APM on any field that is shared between CA APM and CMDB. When the criteria for the event occurs by a change in CMDB or CA APM, the event will complete and the notification will be sent. For example, you can define an event on the Asset page for the Contact field. If the event is a change event, the event can be completed when you change the Contact field in either the asset or the related configuration item (CI). Once the event has completed, a notification will be sent.

NOTE
For more information about managing events and notifications, see Events and Notifications for CA APM.

Define a Management Data Repository (MDR) from CMDB

In this step to integrate CA APM and CMDB, a CA Service Desk Manager user
can define a Management Data Repository (MDR) and allow the CMDB CI to launch in context the corresponding asset in CA APM.

To define a MDR from CMDB of CA Service Desk Manager

1. In the CA Service Desk Manager web interface, log in as an administrator.
2. Select the Administration tab. From the Administration browser, select CA CMDB, MDR Management, MDR List.
3. Click Create New.
The MDR Provider definition appears.
4. Enter the following required MDR provider information:
   - **Button Name**
   Specify *ITAM* as the button name.
   - **MDR Name**
   Specify *ITAM* as the MDR name.
   - **MDR Class**
   Specify *GLOBAL* as the MDR class.
   - **Hostname**
   Specify the CA APM server name by using the network address or the DNS name of the CA APM web server.

   WARNING
   The MDR provider form automatically populates the URL for Launch in Context field based on the information that you provide, so you do *not* enter a value for this field.
5. Click Save.
The CA APM MDR provider is defined.

6. In CMDB, define a CI.
7. Click the Attributes tab in the CI detail form.
8. Click the ITAM button that you previously defined.
   The corresponding asset in CA APM appears.

Integrate CA Asset Portfolio Management with CA Service Catalog Manually

WARNING

Review the following section only if you want to manually integrate CA Asset Portfolio Management with CA Service Catalog when the automatic integration between the two has failed.

CA Service Catalog integrates with CA Asset Portfolio Management in several areas, including these areas:

- Requested items can be associated with CA Asset Portfolio Management Assets. Asset managers can associate CA Asset Portfolio Management assets with items requested from the catalog during request fulfillment.
- Catalog entries can be associated with CA Asset Portfolio Management Models. CA Service Catalog catalog administrators can associate one or more CA Asset Portfolio Management models with a service option. This association can assist with automated asset creation and association during request fulfillment using a workflow process and the CA Service Catalog web services.

  WARNING
  CA Service Catalog and CA Asset Portfolio Management must share the same MDB and the same CA EEM for this integration to work properly.

Requirements for the integration:

- You must have a role defined in CA Asset Portfolio Management.
- Verify that CA Asset Portfolio Management supports your internet browser.
- Check whether the CA Asset Portfolio Management application and web servers reside on the same computer or different computers. If they reside on different computers, complete the CA Asset Portfolio Management web services section in the administration configuration settings on the CA Service Catalog GUI.
- Authorized CA Service Catalog users use the CA Asset Portfolio Management Model Assignments window to assign eligible assets during the fulfillment of a request.

You can assign the assets through CA Service Catalog services if the following requirements are met:

- The assets must be active.
- The assets must not be assigned to any user.

You can associate one or more CA APM models with a service option, if the following requirements are met.

CA Service Catalog and CA Asset Portfolio Management administrators work together to use one of the following methods to authorize CA Service Catalog users (typically request managers) to assign assets:

1. Assign to the user the Fulfiller role in CA Asset Portfolio Management.
2. Assign to the user another role in CA Asset Portfolio Management that can update the Asset Fulfillment object. If you use another role, it must have the same Asset Fulfillment security settings as the CA Asset Portfolio Management Fulfiller role.
3. Specify a proxy user for CA Asset Portfolio Management in CA Service Catalog. To specify this proxy user, select **Administration**, **CA APM Web Services**, and complete the fields. Specify this CA Asset Portfolio Management user in the **User ID** field.
   In this case, CA Service Catalog users inherit the authority of the proxy user to assign assets.

   NOTE

   For information about fulfilling requests from inventory, see How to Fulfill Requests from Inventory.

Follow these steps:

- Step 1 - Create Users in CA Service Catalog
- Step 2 - Create or Update Administrator in EEM
- Step 3- Update CA APM Web Services in CA Service Catalog Configuration
- Step 4- (Optional) Create New Rules and Actions
- Step 5- Create a Service and Request
- Step 6- Configure for Single Sign-On
- Verify CA Asset Portfolio Management - CA Service Catalog Integration

Step 1 - Create Users in CA Service Catalog

To enable the integration between CA APM and CA Service Catalog, create customized users in CA Service Catalog specifically for use with this integration.

Follow these steps:

1. Determine a CA APM user ID that CA Service Catalog can use as a proxy user.
   You can supplement the proxy user with the following CA Service Catalog users:
   – The Service Delivery administrator, service designers, and catalog administrators can associate service options to models.
   – Request managers can use the gold brick icon to fulfill the requests pending action that are assigned to them.

   NOTE
   If you specify a proxy user, the Catalog system uses the proxy user, *not* authorized request managers, to fulfill the requests.
2. As a catalog administrator, work with a CA APM administrator to meet these criteria:
   – This user must be an authorized user in CA APM.
   – In the user details in CA APM, under Permissions, for Role Details, Asset Fulfilment Access is required.
3. Log in to CA Service Catalog as the spadmin user or another user with the Service Delivery Administrator role.
4. Select Administration, Configuration, CA APM Web Services in CA Service Catalog. Specify the CA APM user ID from the previous step in the User ID field.
5. Set up the following users:
   – Catalog User
   – Request Manager (Approver for the Catalog User)
   – Service Delivery Administrator, typically spadmin (Fulfiller)
6. Enable all rules for Event Type: Request/Subscription Item Change except When Status is Pending Fulfillment.
7. Log out of CA Service Catalog.

Step 2 - Create or Update Administrator in EEM

To enable the integration between CA APM and CA Service Catalog, create customized users in CA EEM specifically for use with this integration.

Follow these steps:

1. Log in to CA EEM as EiamAdmin. For Application, select Global.
2. Select Manage Identities, Users and click Go, without specifying any search criteria.
3. If the Administrator user already exists under the Users node, skip this step.
   Otherwise, create the Administrator user under Users node, and specify Administrator for the Name and the Display Name.
   Set the password for this Administrator user and click Save.
   For more information about creating global users in CA EEM, see the CA EEM documentation.

4. Log out of CA EEM.

Step 3- Update CA APM Web Services in CA Service Catalog Configuration

Update the CA APM Web Services in CA Service Catalog Configuration.

Follow these steps:

1. Log in to CA Service Catalog as a user with the Service Delivery Administrator role.
2. Click **Administration**, **Configuration**, **CA APM Web Services**.
3. Click on Modify icon for **Host Name** property and update CA APM Application Server Host Name.
4. Click on Modify icon for **Port Number** property and update CA APM Web Services Port Number.
5. Save these configurations.
6. Restart CA Service Catalog Service.

Update the CA APM Web Services in CA Service Catalog Configuration.

Follow these steps:

1. Log in to CA Service Catalog as a user with the Service Delivery Administrator role.
2. Click **Administration**, **Configuration**, **CA APM Web Services**.
3. Click on Modify icon for **Host Name** property and update CA APM Application Server Host Name.
4. Click on Modify icon for **Port Number** property and update CA APM Web Services Port Number.
5. Save these configurations.
6. Restart CA Service Catalog Service.

{"URL":["/usm/wpf?Node=iclaunchpad.tools_configuration"],"customLabelGetStarted":"CA APM Web Services","description":"IPCE_CA_APM_Web_Services"}

Step 4- (Optional) Create New Rules and Actions

You may need to create new rule actions in CA Service Catalog to enable the integration with CA Asset Portfolio Management. As supplied, CA Service Catalog rules and actions can have only limited changes made by users. However, you can create new rules and actions, if required. For maximum efficiency, create new rules and actions by copying and modifying existing rules and actions. These steps illustrate how to copy and modify a predefined rule and its actions. You can optionally use these steps as a model to copy and modify other predefined rules and their actions.

Follow these steps:

1. Log in to CA Service Catalog as a user with the Service Delivery Administrator role.
2. Click **Administration**, **Events-Rules-Actions**.
3. Click the **Request/Subscription Item Change** event.
4. Click the Copy icon for the rule that you want to copy and modify, for example, the rule named **When Category is Hardware and Status is Pending Fulfillment**. The Copy icon apears in the icon columns next to the Description field. A dialog appears prompting you to enter the name of the new rule.
5. Enter the new name and click **OK**.
   You return to the Event Type Details page. The new rule is added to the list of rules for the event.
6. Click the Edit icon to edit the details of the rule, for example, its name, description, status, or event type.
   The Edit Rule page appears.
   a. Select Enabled in the Status drop-down list.
      The new rule is enabled. You must enable the new rule to activate it, because new rules are disabled by default.
   b. Update the remaining fields, as needed, to meet your requirements.
   c. Click OK.

You return to the Event Type Details page.

7. Click the new rule to open it.
   The Rule Details and Actions page opens.
   The rule details that you modified earlier appear in read-only format.
   The actions for the rule also appear, so that you can view and modify them as needed: Actions that were enabled in the old rule are enabled in the new rule. Similarly, actions that were disabled in the old rule are disabled in the new rule.
8. (Optional) Add, edit, or copy the action.
9. When you are finished modifying actions, click Done to close the rule.
   You return to the Event Type Details page.
10. Locate the original rule that you copied and click the Disable button; this button appears on the Rules bar.
    The rule is disabled. Disabling the original rule is important to help prevent duplicate rules and actions from being triggered.

Step 5- Create a Service and Request

A required task of integrating CA Asset Portfolio Management and CA Service Catalog is creating at least one service and request that use the features enabled by the integration.

Follow these steps:

1. Define a Service using a service option group
   As part of defining the service option group, do the following from the Options tab of the service option element:
   a. Ensure that Track As An Asset is checked.
   b. Specify the same category for the service option element as you specified in the rule condition associated with the service.
      For example, if you created a new rule and action using the Hardware category, then you *must* create your service option group elements as Hardware Category. For more information about how to create service option groups, see the section Service Option Groups.
2. Log out of CA Service Catalog.
3. Log on to CA Service Catalog as the Catalog User created earlier.
4. Submit a request using the service that you defined earlier. The request will proceed through its life cycle.
5. Log out of CA Service Catalog.
6. Log on to CA Service Catalog as the Request Manager (Approver for the End User) that you created earlier.
7. Approve the request in the Pending Actions queue.
8. Log out of CA Service Catalog.
9. Log on to CA Service Catalog as the Service Delivery Administrator (Fulfiller) that you created in an earlier step.
10. To fulfill the request that you submitted in an earlier step, do the following:
    – Become familiar with the Assign Assets window.
    – Assign an Available Asset to a Requested Service Option.

NOTE
For more information about fulfilling requests, see the section Request Management from an Administrator Perspective.

Step 6- Configure for Single Sign-On

You can optionally bypass user login by configuring CA Service Catalog and the CA products with which it integrates to use Single Sign-On.

To configure CA Asset Portfolio Management to use Single Sign-On, see Maintaining Security.

Similarly, to configure CA Service Catalog to use Single Sign-On, configure it to use either CA SiteMinder or NTLM authentication on Windows.

Verify CA Asset Portfolio Management - CA Service Catalog Integration

This article contains the following topics:

When you are viewing the Fulfill Request or Edit Request window (and the request is past the approval phase of its life cycle), you may see the Assigned Assets gold brick icon in the Actions column for a requested service option. The gold brick icon indicates this service option is eligible to have one or more assets assigned to it from available inventory.

To search available inventory, click the Assigned Assets gold brick icon. The CA APM Assign Assets window appears. By default, the Assigned Assets action is selected so the list of assets already assigned to the requested service option appears.

Using the Assign Assets window, you can do the following:

- View the list of assets assigned to the requested service option by selecting Assigned Assets in the Actions list
- Search for and assign assets to the requested service option
- Indicate that an available asset could not be found in inventory
- Display assigned assets and remove the assignment

Associate a CA APM Model with a Service Option

This information is available during workflow-driven fulfillment and can be useful if CA APM assets are automatically created or assigned to the request.

Follow these steps:

1. Click Catalog, Service Offerings.
2. Click the Option Groups tab.
3. Select any service option group and click the Definition tab.
4. Click the CA APM Model icon.
5. Click Assign Models to display the UAPM Model Assignments window and perform the following actions:
   - **View Assigned Models**
     Select Assigned Models from the Action list.
   - **Assign Models**
     Select Assign from the Action list. You can search for both active and inactive CA APM models that match the criteria you enter and check the ones you want to assign to the service option. Click Assign Models to save the assignment.

     NOTE

     To search for both active and inactive CA APM models, before running the search, do not select the option to include only active records.
   - **Remove Assigned Models**
     Select View Assigned Models from the Action list. Check the models you want to remove from the Assigned Models list and click Remove Assignment.
6. Click Go.
   The Assign Models window is closed.

Assign an Available Asset to a Service Option for Hardware

Use the Search for Assets section of the Assign Assets window to search for available assets and assign one or more of them to the requested service option for hardware. Doing so is required to fulfill a service option for hardware.

Follow these steps:

1. Select an asset in the Search Results List to assign to the requested service option.
   The Asset window appears for the selected asset.
2. Click the Assign button.
   The Asset page refreshes and displays a confirmation message.
3. Click the Refresh button on the CA Service Catalog Fulfill Request window.
   The service option status is changed.

   NOTE
   Do not change the status manually. After the Assign Assets window changes the status of the requested service option, you may no longer be assigned a fulfillment task for that service option. Therefore, you may no longer permitted to change its status.

Mark a Requested Service Option as Not Filled From Inventory

If you have used the Assign Assets window to search the available asset inventory and have been unable to find an eligible asset, you can indicate that the requested service option could not be filled from inventory.

Follow these steps:

1. Click the Not Assigned From Inventory button.
   The status of the requested service option is automatically changed to Not Filled From Inventory.

   NOTE
   You must have conducted an asset search for the Not Assigned From Inventory button to appear.
2. Click the Refresh button on the CA Service Catalog Fulfill Request window.
   The service option status is changed.

   NOTE
   Do not change the status yourself since once the Assign Assets window changes the status of the requested service option you may no longer be assigned a fulfillment task for that service option and therefore no longer permitted to change its status.

Remove Asset Assignment

If an asset has been incorrectly assigned to a requested service option, you can remove the incorrect assignment. Click Remove Assignment from the Actions drop down list on the Assign Assets window. The list of assigned assets is displayed.

To remove the assignment for one or more non-Software assets assigned to the requested service option:

1. Select the check box for each asset in the Search Results List for which assignment is to be removed for the requested service option.
2. Complete the Asset Information section of the Assign Assets window.
   The assets for which assignment is to be removed are updated with the specified information.

   NOTE
   If a field is left blank, the associated field on the assigned assets is set to blank. For more information on the meaning of the asset-related fields, see the CA APM documentation.
3. Click the Remove Assignment button.
   The Remove Assignment list of assigned assets is displayed. The status of the requested service option is not affected by a remove assignment.

To remove the assignment for a Software asset assigned to the requested service option

1. Select an asset in the Search Results List for which assignment is to be removed for the requested service option.
   The Asset window is displayed for the selected asset.

2. Navigate to the Entitlements for the Software asset and select an entitlement for which to remove the assignment.
3. Click the Remove Assignment and Save button.
   The asset is removed from the list of assigned assets. The status of the requested service option is not affected by a remove assignment.

How Assets Are Filtered During Fulfillment

By default, when a request reaches fulfillment, CA APM filters assets *only* according to models associated with service options.

You can optionally replace the default filtering with advanced filtering by issuing the equivalent custom query at the database console.

Example

For example, consider the following query:

```
update usm_configuration
set value = 'model=$model_ids$&loc=$loc_id$&org=$org_id$&dept=$dept_id$&ccenter=$costcenter$&tenant=$tenant_id
$'
WHERE name = 'asset_fulfillment_url_params'
```

This query filter assets according to the following criteria:

1. Models: Associated to service option
2. Location: Location of the user
3. Organization: Organization of the user
4. Department: Department of the user
5. Cost Center: Cost Center of the user
6. Tenant: BU of the user

Integrate CA Asset Portfolio Management with Common Components Manually

This article contains the following topics:

- Integrate CA Asset Portfolio Management with CA EEM Manually
- Integrate with Process Automation Data Importer Process Manually
- Integrate with Process Automation Notification Process Manually

Integrate CA Asset Portfolio Management with CA EEM Manually

> **WARNING**
>
> Review the following section only if you want to manually integrate CA Asset Portfolio Management with CA EEM when the automatic integration between the two has failed.

Execute the following commands in the computer where CA EEM is installed.

1. Log in to the product and navigate to Administration, System Configuration, EEM.
2. Enter in the CA EEM Backend field, the name of the server where CA EEM is installed. For example, EEMServer1, EEMServer2 if CA EEM servers are configured for Failover.
   To prevent or troubleshoot errors, see Knowledge Document.
3. On the server where CA EEM is installed, register CA APM with CA EEM using the following steps:
   a. From **Start** menu, select **Run** and open a command prompt (cmd) window.
   b. In the command prompt window, change the directory to \filestore\ITAM\Setups\EEM fetched from the 17.4 DVD.

For exmaple:
ITAM17.4DVD\\filestore\ITAM\Setups\EEM

c. Execute the following commands , with your hostname and password parameters:
For MS SQL Server

```
AppLauncher.exe" -ProductName=EEM -EEMParameter='-EEMBackend=<eem_hostname>,
-EEMAdminPass=<*****>,-EEMUapmPassword=<*****>,-EEMCasmPassword=<*****>,-EEMTelePassword=<*****>,
-DatabaseType=MSSQL,-DBServer=<itam_dbhostname>,-UserID=mdbadmin,-DBName=<itam_dbname>,-
Port=<itam_dbport>,-DBOwner=dbo,
-InstanceName=,-DBPassword=<*****>,-MdbAdminPassword=<*****>,-BundleVersion=17.0'
```

For Oracle server

```
AppLauncher.exe -ProductName=EEM -EEMParameter='-EEMBackend=<eem_hostname>,
-EEMAdminPass=<*****>,-EEMUapmPassword=<*****>,-EEMCasmPassword=<*****>,-EEMTelePassword=<*****>,
-DatabaseType=Oracle,-DBServer=<itam_dbhostname>,-UserID=mdbadmin,
-DBName=<oracle_dbservicename>,-Port=<itam_dbport>,-DBOwner=dbo,
-InstanceName=,-DBPassword=<*****>,-MdbAdminPassword=<*****>,-BundleVersion=17.0'
```

NOTE

Replace ITAM and EEM login password and server details as required.

4. Perform an IIS after the installations.

Integrate with Process Automation Data Importer Process Manually

The article contains the following topics:

CA APM and CA Process Automation integrate to let you set up and configure a Data Importer process. This integration uses a sample data import XML file that you import into CA Process Automation and integrate with an overall process workflow. The Data Importer process launches the Data Importer and executes a data import.

NOTE
This integration uses CA Process Automation and a sample data import XML file that is company-provided. You can also use any other workflow provider to create your own overall workflow and Data Importer process.

How to Set Up the CA Process Automation Data Importer Process

Use the following steps to set up the Data Importer process:

1. Install CA APM and CA Process Automation.
2. Log in to the application server where you installed CA APM.
3. Access the following folder on the CA APM application server where the Storage Manager Service is installed.
   `[ITAM Root Path]\Storage\Common Store\Import`
4. Locate the Import_Automation_Workflow.xml file.
5. In CA Process Automation, import the Import_Automation_Workflow.xml file.
6. In CA Process Automation, integrate the Import_Automation_Workflow.xml into your overall process workflow.
7. In CA Process Automation, modify the settings for the Data Importer process parameters.
   a. Change the default Import Service URL to match your required setting.
   b. Change the default CA APM user ID and password to your own settings.
   c. Change the default data import name to match your data import.
   d. Specify the data file name that corresponds to your data import.

NOTE
For information about using CA Process Automation, see the CA Process Automation documentation.

Modify CA Process Automation Workflow Process Parameters

After you install CA APM and CA Process Automation and import the Import_Automation_Workflow.xml file into CA Process Automation, the default workflow process parameters are defined in CA Process Automation. You can modify the default process parameters to include your required settings. You provide actual (hard-coded) values for the process parameters. You must verify that the values you enter are valid.

You can modify the process parameters in the main data set or in the individual process start request forms. The parameters that you specify for an individual process override the parameters in the main data set for that process.

Follow these steps:

> **WARNING**
> CA APM and CA Process Automation do not validate the information that you enter for the parameters. You must verify that your input is valid and that you entered the data in the correct format.

1. Log in to CA Process Automation and navigate to the CA Process Automation client.
2. Enter the information for the Data Importer parameters. The following fields require explanation:
   - **ITAMImportServiceURL**
     Specifies the complete URL path where the Import Service is running.
     Example:
     ```
     http://server/ImportService/ImportService.svc
     ```
   - **username**
     Specifies the CA APM user ID.
   - **password**
     Specifies the CA APM user password.
   - **Importname**
     Specifies the name of the data import that you want to execute.
   - **FilePath**
     Specifies the complete path and name of the data file that is associated with your data import.
     Example:
     ```
     C:\\CAITAMCostcenter.csv
     ```
3. Save the changes in CA Process Automation.

> **NOTE**
> For information about setting up a process in CA Process Automation, see your CA Process Automation documentation.

Integrate with Process Automation Notification Process Manually

This article contains the following topics:

CA APM and CA Process Automation integrate to let you set up and configure a notification process that delivers notifications to specific recipients after a defined event occurs. CA APM provides email notification processes with the product. These processes are delivered in files that are included on the product installation media. You import the files into CA Process Automation and specify process parameters in CA Process Automation and CA APM.

How to Set Up the CA Process Automation Notification Process

Use the following steps to set up the email notification processes that are provided with CA APM.

1. Install CA APM and CA Process Automation.
2. In CA Process Automation, import the workflow provider notification process files.
3. In CA Process Automation, configure the mail server.

4. In CA Process Automation, modify the settings for the workflow process parameters.
   a. Change the default email address for the administrator (Admin_Email_To parameter) to specify your required setting.
   b. Change the default CA APM URL (ITAM_URL parameter) to specify your required setting.
   c. Change the default CA Process Automation URL (ITPAM_URL parameter) to specify your required setting.
   d. (Optional) Change any of the other parameters for which you want to specify your required settings.
5. In CA APM and CA EEM, permit CA APM users to use CA Process Automation.
6. In CA EEM, create CA Process Automation user accounts for any non-CA APM users.
7. In CA APM, specify the workflow process parameters when you define an event.

NOTE
For information about defining an event in CA APM, see Events and Notifications for CA APM. For information about using CA Process Automation and CA EEM, see the CA Process Automation and CA EEM documentation.

Import the Workflow Provider Notification Process Files

CA APM provides default email notification process files. You import these files into CA Process Automation before you can set up and can configure email notifications in the products.

NOTE
For more information about importing and working with the files, see the CA Process Automation documentation.

Follow these steps:

1. Log in to CA Process Automation as the administrator.
2. Navigate to the CA Process Automation client.
3. Locate the ITAM.xml file on the CA APM installation media. For example:
   `DVDITAM174\filestore\PAM\PAMProcessFlows\ITAM`
4. Import the ITAM.xml file into the **/** node.

NOTE
In CA Process Automation Release 3.1, you import the XML file from the client. In Release 4.0 SP1, you import the XML file from the Library tab.

Select the import options to set the imported versions as current and to make the imported custom operators and sensors available.
The notification process files are imported into the default /ITAM folder.

Configure the CA Process Automation Mail Server

To implement email notifications between CA Process Automation and CA APM, configure the mail server for CA Process Automation.

1. Log in to CA Process Automation as the administrator.
2. Navigate to the CA Process Automation client.
3. Navigate to the library browser.
4. Locate and lock the default environment.
5. Locate the alert module and clear the inherit check box.
6. Specify the SMTP (mail) server.
   Example: mail.company.com
7. Specify the From address.
   Example: admin@company2.com
8. Save the changes.

9. Unlock the default environment.
   The changes require a few minutes to take effect.

   NOTE
   You can send an email notification to an external email address if your SMTP (mail) server settings permit email delivery to the external address. Check your mail server settings to verify if you can send email to external addresses.

Modify CA Process Automation Workflow Process Parameters

After you install CA APM and CA Process Automation and import the notification process files into CA Process Automation, the default workflow process parameters are defined in CA Process Automation. You can modify the default process parameters to include your required settings. You provide actual (hard-coded) values for the process parameters. You must verify that the values you enter are valid.

You can modify the notification process parameters in the data set that applies to all notification processes or in the individual process start request forms. The parameters that you specify for an individual process override the parameters in the main data set for that process.

NOTE
You specify some of the notification process parameters for the workflow provider when you define an event in CA APM. For more information about specifying process parameters in CA APM, see Workflow Provider Process Parameters.

To modify CA Process Automation workflow process parameters

WARNING
CA APM and CA Process Automation do not validate the information that you enter for the parameters. You must verify that your input is valid and that you entered the data in the correct format.

1. Log in to CA Process Automation and navigate to the CA Process Automation client.
2. In the ITAM folder, locate the data set that is named Dataset.
3. Enter the information for the parameters.
   The following fields require explanation:
   - **Ack_Interaction_Form_Full_Path**
     Full path to the file that contains the acknowledgment interaction form in CA Process Automation. The email notification recipient uses this form to acknowledge receipt of the notification. Each workflow process must have a unique user interaction form and a unique path to the form. You can find the acknowledgment interaction form files that are provided with the product in the folder that contains the processes and main data set (/ITAM or the folder where you imported the processes).
   - **Admin_Email_CC**
     Email address of the copy recipients for the email that is sent to the administrator when a notification error occurs.
   - **Admin_Email_To**
     Email address of the administrator for the email that is sent when a notification error occurs. Change the default value to your required setting.
   - **Log_Folder_Path**
     Full path of the error log file that is created when an error occurs with the notification process. If you do not specify a path, the log file uses the default CA Process Automation log file path.
   - **ITAM_Username**
     User name to log in to CA APM. CA Process Automation requires access to CA APM for information about notification recipients and escalation.
   - **ITAM_Password**
     User password to log in to CA APM. CA Process Automation requires access to CA APM for information about notification recipients and escalation.
   - **Admin_Email_Subject**

Subject for the email that is sent to the administrator when a notification error occurs. This parameter can be set in the main data set or in the individual process start request form.

- **Admin_Email_Header**
  Header or introduction for the email that is sent to the administrator when a notification error occurs (for example, "Hello"). This parameter can be set in the main data set or in the individual process start request form.
- **Admin_Email_Footer**
  Signature for the email that is sent to the administrator when a notification error occurs (for example, "Thank You"). This parameter can be set in the main data set or in the individual process start request form.
- **Log_File_Name**
  Name of the error log file that is created when an error occurs with the notification process. The email that is sent to the administrator when a notification error occurs contains the log file name. If you do not specify a name, the log file uses the following default CA Process Automation log file name:
  `process name_process instance number.log`
- **ITAM_URL**
  CA APM URL that CA Process Automation uses to access CA APM for information about notification recipients and escalation. Change the default value to your required setting.
  Example:
  `http://ITAMAPPSERVER:99/ITAMService/Service.asmx`
- **ITPAM_URL**
  CA Process Automation URL that is included in the email notification message. Change the default value to your required setting.
  Example:
  `http://PAMSERVER:8080/itpam`

4. Save the changes in CA Process Automation.

NOTE
For information about setting up a notification process, see your workflow provider documentation.

Permit CA APM Users to Use CA Process Automation

The CA APM users who receive notifications must access CA Process Automation to acknowledge the notifications. These users must have permission to use CA Process Automation. You permit users to use CA Process Automation by performing steps in CA APM first and then in CA EEM. In CA APM you define and authorize users to log in to and use CA APM. In CA EEM, you permit the authorized CA APM users to use CA Process Automation.

To permit CA APM users to use CA Process Automation

1. Log in to CA APM.
2. Verify that both new users and existing users are authorized to log in to and use CA APM.

NOTE
For information about defining and authorizing new users and authorizing existing users in CA APM, see Maintaining Security.

The product defines and authorizes the CA APM users. CA EEM now includes the CA APM users in the list of available users.

3. Log in to CA EEM, selecting CA Process Automation from the application drop-down list.

WARNING
You must select the CA Process Automation application when you log in to CA EEM to permit CA APM users to use CA Process Automation.

4. Select a CA APM user from the list of all users and click the application user details for the user.
5. Select a CA Process Automation user group for the user and save the selection.

NOTE
For information about using CA EEM to add applications to user details, see this topic in the CA EEM documentation.

The CA APM user can now access and use CA Process Automation.

Required Indicators and Multiple Line Text Fields for Parameters

The default notification processes that are delivered with the product contain the parameters that appear on the product Event Definition user interface and the parameters that you specify in the workflow provider. The default processes also contain XML formatting that lets you display a required indicator and a multiple line text field on the product user interface. These items are not readily available from the workflow provider, and so they are specified in the process. In the CA Process Automation start request form for each default process, the following XML statement appears before the Description of each user interface parameter:

```
<FieldDescriptor><IsRequired>true_or_false</IsRequired><Length>number</Length></FieldDescriptor>
```

- **IsRequired**
  Specifies if the parameter is required (true) or not required (false). If the parameter is required, the product displays the standard required indicator on the user interface.
  Example: <FieldDescriptor><IsRequired>true</IsRequired></FieldDescriptor>
- **Length**
  Specifies the length of the parameter text entry field. To define a multiple line text field, specify a value greater than 255.
  Example: <FieldDescriptor><Length>275</Length></FieldDescriptor>

You can change the default notification processes that are delivered with the product, and you can also create your own new notification process. To include information about the required indicator and field length in your changed or new process, you must insert the XML statement before the Description of each user interface parameter in your process.

NOTE
If you are creating a notification process, you must have a corresponding start request form for the process. For information about changing or creating notification processes, see your workflow provider documentation.

Integrating Third Party Applications

CA Service Management can integrate third party applications to work seamlessly.

- CA SDM and Jira Agile Management Integration
- CA Virtual Analyst Integration with MS Teams

CA SDM and Jira Agile Management Integration

CA Service Management now supports bi-directional integration between Service Desk Manager & Jira Agile Management (Cloud) enabling organizations to monitor & control the entire lifecycle of service operations and eliminating the information gaps.

Overview

CA Service Management now supports bi-directional integration between Service Desk Manager and Jira Agile Management (Cloud) enabling organizations to monitor and control the entire lifecycle of service operations and eliminating the information gaps.

Prerequisites

Review the following prerequisites before proceeding with the integration of CA SDM and Jira Agile Management (cloud):

1. CA Service Desk Manager 17.4 with REST enabled
2. Jira Agile Management (cloud)
3. Windows Server 2016 (referred as the CIS server) and above, must be an internet facing server as we are integrating with Jira Cloud. Install the following softwares:
   a. JRE (AdoptOpenJDK) 11.0.18
   b. apache-tomcat-9.0.71-windows-x64
   c. apache-activemq-5.15.16-bin, download it from here.

> **NOTE**
> After successfully installing ActiveMQ, verify that the installation was successful and confirm that the ActiveMQ URL is accessible and running. Example URL: http://localhost:xxxx/index.html

Installation and Integration – CA SDM and Jira

Perform the following steps to complete the integration:

1. Install the WebHook component in CIS server
2. Configure a New Webhook Listener in Jira
3. Create a WebHook Listener in CA SDM
4. Configure the CIS Adapters Component

Install the WebHook Component in CIS server

Perform these steps:

1. Log in to the CIS server, and download the latest available 17.4 DVD for CA Service Management.
2. After extracting the DVD file, navigate to \filestore\CloudIntegration.
3. You find two zip files, **SDM-CIS-Adapters.zip** and **SDM-CIS-Webhook.war**
4. Rename the **SDM-CIS-Webhook.war** as **webhook.war**
5. Copy the webhook.war to the Tomcat webapps folder.
   Example: C:\apache-tomcat-9.0.71\apache-tomcat-9.0.71\webapps
6. Restart Tomcat Services.
7. After restarting the Tomcat services, you can see the META-INF and WEB-INF folders in the webhook directory.
8. Navigate to the WEB-INF/Classes folder and update the following values in the msgQueueConfig.properties file.
   a. **For Example**: C:\apache-tomcat-9.0.71\apache-tomcat-9.0.71\webapps\webhook\WEBINF\classes:
      a. activemq.broker.url=tcp:// <activeMQserver>:<TCP-Port>
      b. activemq.queue.names=sdm-jira,jira-sdm,sdm-msteams
      c. webhook.allowed.remotheosts= JIRA_SERVER_IP Address

> **NOTE**
> JIRA_SERVER_IP will accept requests from specific IP Addresses. If you do not intend to allowlist the IPs, you can ignore this parameter and allow requests from any IP addrss.

9. Restart the Tomcat server services from apache-tomcat-9.0.71\apache-tomcat-9.0.71\bin\.
10. Launch the TOMCAT URL.
    Example: http://TOMCAT_Host_NAME:TOMCAT_PORT_NUMBER/webhook/webhook
11. Launch the above url in any browser, and you see the following message: **This is the home page for Service Desk Manager CIS Webhook Application.**
12. Refer these logs in the tomcat logs folder: **cis_webhook.log** and **localhost_access_log**

NOTE
To enable additional logging, turn on the Root level as DEBUG in APACHE_TOMCATROOT\webapps \webhook\WEB-INF\classes\log4j2.xml

13. Enbale https for webhook URL.

Configure a New WebHook Listener in Jira

Configuring webhooks in Jira:

1. Log in to Jira Agile Management (cloud).
2. Navigate to **Settings**> **System**> **WebHook**s, and click **Create a Webhook**.
   Create a WebHook listener for issues that are created and updated.
3. Enter the following mandatory field information:
   a. **Name**: Specify the webhook name.
   b. **Status**: Enabled/Disabled
   c. **URL**: Provide the webhook url: https://TOMCAT_Host_NAME:TOMCAT_PORT_NUMBER/webhook/webhook/jira-sdm/updateTicket
   d. **Issue**: select **created** and **updated**.
   e. **Comment**: select **created**.
   f. **Attachment**: select **created**
   g. Fill the remaining field information as required and click **Save**.

Create a new WebHook Listener in CA SDM

1. Log in to CA SDM Classic Interface.
2. Navigate to **Administration**> **System**> **Webhooks**> **Endpoints** to create a new Webhook. Complete the field information:

Field	Description
Symbol	Mandatory field information
Code	Mandatory field information
Endpoint URL	Provide the webhook URL details. For Example: http://TOMCAT_SERVER:TOMCAT_PORT/ webhook/webhook/sdm-jira/updateTicket

3. Complete the remaining fields and click **Save**.

Create a new notification To listen to changes in Change Orders, create a notification under webhook. Follow these steps to create a notification:

1. Log in to CA SDM Classic Interface.
2. Navigate to **Administration**> **System**> **Webhook**> **Notifications**> **Create New** to create a notification.
3. Complete the field information:

Field	Value
Symbol	Mandatory field information
Code	Mandatory field information
Object Type	Select Request/Incidents/Problems or Change Orders
Payload Body	Enter * to send the entire ticket information from CA SDM to webhook server

4. Complete the remaining fields and click **Save**.

5. In **Webhook Endpoints**, click **Update Webhook Endpoints** to map a **Notification** with an existing Webhook Endpoint.
6. In the **Endpoint Search** page displayed, provide the required endpoint details and click **Save**.
7. Select the Endpoint to which the new notification is to be mapped and move it to the adjacent box.
8. Click **OK**, and then click **Save**.
9. This Notification is successfully updated and mapped to the selected Endpoint.
10. To specify Notification Rules, select **Triggers** from the Notification Detail page.
11. Select **Update Notification Rules** and the **Notification Rules Search** page is displayed.
12. In the page displayed, follow these steps:
    a. Provide the required details about the Notification Rule that must be mapped to a notification.
    b. Click **Search**.
    c. The **Notification Rules – Update** page displays the Out-of the-box Notifications.
    For Example select:
        a. Default Attach Doc Notification Rule for request/incidents/problem
        b. Default Field Update Notification Rule for request/incidents/problem
        c. Default Initial Notification Rule for request/incidents/problem
        d. Default Escalate Notification Rule for request/incident/problem
        e. Default Log Comment Notification Rule for request/incident/problem
13. Click **OK** and the Notifications are updated for a selected **Webhook Endpoint** when a new ticket is created.

Configure the CIS Adapters Component
This component enables bi-directional integration between CA SDM and Jira by mapping the corresponding/respective attributes.
Follow these steps to install the SDM-CIS-Adapters.zip:

1. Unzip SDM-CIS-Adapters.zip to extract the Adapters folder in a location of your choice.
2. Update the configuration files.
   For example: C:\adapters\config
   a. application.properties: Provide the activeMQ Server details in this file:
   activemq.broker.url=tcp://ACTIVEMQ_SERVER:ActiveMQ_port
   activemq.queue.names=sdm-jira,jira-sdm,sdm-msteams
   Here sdm-jira refers to the flow from CA Service Desk Manager to Jira Agile Management (cloud), and jira-sdm refers to the flow from Jira Agile Management (cloud) to CA SDM.
   b. Take a backup of integrationmapping.yml, and update the object and attribute values relevant to your project in Jira and CA SDM.
   integrationmapping.yml: Contains information about the location of the CA SDM server and user credentials.
   - CA SDM Server Details

Field	Description
useConfiguration	Set this value to “**true**”
sdmRestUrl	Enter the SDM Web Interface URL: https:// SDM_SERVER:REST_PORT /caisd-rest Enable REST web services in your CA SDM server. **Note:** If there are multiple REST servers, choose one server in **Administration**> **Options Manager**> **Web Service**, provide the **Host Name** and install the sdm_rest_for_cis option.
sdmUserName	Provide the SDM User name. For example: CASM_Integration_User

Field	Description
sdmpassword	Provide an encrypted password. For example: ”SDM_USER_PASSWORD” For example: For the encrypted passwords, run the “encrypt_pwd.bat” in the C:\Program Files (x86)\CA\Service Desk Manager\bin Use the command: encrypt_pwd.bat -sdme <password>
sdmDefaultuuid	Provide the SDM Default Contact UUID. If no contacts are found, CA SDM creates a ticket with this default contact. Tickets are created in CA SDM based on the email id of Jira user.
sdmattmntRepositoryId	Provide the SDM Default Attachment Repository ID into which the JIRA attachments will be uploaded.

SDM attribute values can be identified from the SDM data base queries, for example:

- select * from ca_contact
- select * from chgcat (to get the available categories for change order)
- select * from prob_ctg (to get the available areas for incidents)
- select * from cr_stat(To Get the available status values for Incidents/Requests)
- select * from chgstat(To Get the available status values for Change)
- select * from pri(To Get the available prioirty values)

- **Global Mappings:** Update the values in this file as per your configurations. If the user does not provide mappings, integration services use the Global Mappings to create the tickets from Jira to CA SDM and from CA SDM to Jira.
- Jira Details:

Field	Value
jiraProducts	code: Jira product code name: Jira product name
jiraRestUrl	Enter the Jira cloud RestAPI URL. For example: https:// JIRA_HOST:REST_PORT/rest/api/2
jiraUserName	Provide the Jira REST API User name
jiraPassword	Provide your encrypted Jira key. For example: https://id.atlassian.com/manage-profile/security/api-tokens
jiraDefaultName	Jira default username
JIRA Projects	Update the Jira project key. For example: - key: <project_key>

For instance, use the following APIs to get Jira Object & Attributes:

- https://Account_Name/rest/api/2/project/
- https://Account_Name//rest/api/2/issuetype/
- https://Account_Name/rest/api/2/project/PROJECT_NAME/components

NOTE

To get the custom fields and its values for a given issue type, you can use an example as shown below:

https://Account_Name/rest/api/2/issue/createmeta?projectKeys=PROJECT_NAME&issuetypeNames=ISSUE_TYPE&expand=projects.issuetypes.fields

- Based on the mapping of this configuration, the attributes are identified and the tickets are created.
  mappings: Mappings for the Jira and CA SDM attributes.
  Example mapping in yml file

```
# JIRA Bug <=> SDM Incident
 ########################################
 - jiraObjectName : issue
 jiraObjectTypeName : Bug
 jiraObjectTypeId : 10004
sdmObjectName : in
 attributes:
 - name: "reporter:customer"
 jira-type: "CLU"
 sdm-type : "CLU"
 - name: "creator:requested_by"
 jira-type: "CLU"
 sdm-type : "CLU"
 flow: "JIRA-SDM"
 - name: "components.id:category.@COMMON_NAME"
 jira-type: "ARRAY_OBJS"
 values: ["10003:Email", "10002:Hardware", "10004:Printer",
"10001:Software" ]
 defaultvalue: "10005:Applications"
 - name: "status.id:status.@REL_ATTR"
 values: ["1:OP", "3:WIP", "5:RE"]
 defaultvalue: "1:OP"

 - name: "priority.id:priority.@REL_ATTR"
 values: ["5:1", "4:2","2:4", "1:5"]
 defaultvalue: "3:3"


 - name: "customfield_10108.value:zviptkt.@REL_ATTR"
 values: ["Yes:1", "No:0"]
 defaultvalue: "No:0"
```

NOTE
If no value matches the given entity, then **a ticket is created using the defaultvalue.**

3. Navigate to adapters and run the following commands to start the batch file:

```
startup.bat
```

   – This starts the CA SDM- Jira adapter and it starts listening to the ActiveMQ queue server.
4. Refer to these logs in adapters/logs: **cis_adapters_debug.log** and **cis_adapters.log**

NOTE
To enable additional logging, turn on the Root level as DEBUG in C:\adapters\config\log4j2.xml

Supported Use cases

After the successful installation and configuration of CA SDM and Jira, validate the integration with a few use cases. Make the following changes and check if the bidirectional integration between Jira and CA SDM works:

- Create a story/bug/epic in JIRA and check for corresponding incident/request/change order on the Service Desk Manager's side and vice-versa.
- Modify the status of the incident/request/change order on the SDM or corresponding story/bug/epic in JIRA.
- Modify the Summary/Description/Category/Priority on Service Desk Manager or JIRA.
- Add a Comment/Attachment on SDM or JIRA.

Limitations

1. Tickets in Jira are created by the user provided in the integrationmapping.yml.
2. Multiple categories for the same ticket are not supported in the integration. The category assigned is selected in alphabetical order.
3. Turn off **Automatic Priority Calculation** in CA Service Desk Manager.

CA Virtual Analyst Integration with MS Teams

CA Service Management now supports integration of "ARIA" (Automated Response with Intelligent Automation) with MS Teams for a simplified and connected user experience.

Overview

This article details the steps to integrate and enable CA Virtual Analyst (ARIA) in MS Teams to support the following use-cases:

- Get Knowledge Base article recommendations and view the KB article.
- Create an SDM ticket via the MS Teams Bot.
- Perform following updates in an SDM ticket via the MS Teams Bot:
  - Log comments
  - Update status
  - Close ticket
- Get notifications of any ticket updates from the MS Teams Bot.
- Get transferred to a live agent for further assistance.

Prerequisites

Review the following prerequisites before proceeding with the installation and integration of CA Virtual Analyst with MS Teams:

1. Install Open JDK v.11.0.19 from https://adoptium.net/temurin/archive/?version=11
2. Install MongoDB v.6.0.6 community server from https://www.mongodb.com/try/download/community
3. Windows Server 2016 and above with the following software installed:
   a. apache-tomcat-9.0.71-windows-x64
   b. apache-activemq-5.15.16-bin
4. Users from MS Teams must sync with CA SDM, before using MS Teams - VA Integration.

Creating a Bot for MS Teams

Follow these steps to configure MS Teams bot:

1. Log in to https://portal.azure.com with your tenant admin credentials.
2. Once logged in, go to **Create Resource** and select **Azure bot**.
3. Fill in the required details, and click **Create** to create a new bot.
4. After deploying the created bot, click **Go to resource.**
5. In **Settings**, click **Bot Profile** and give a **Display name.**

6. Click **Apply.**
7. Go to **Configuration** and in **Messaging endpoint** provide a valid internet accessible https URL that points to the bot server. For example:
   ```
   https:<host_name>/api/messages
   ```
8. In **Microsoft App id**, click **Manage password** >**New client secret**
9. Once a password is created, copy the **Value** which is the BOT_PASSWORD.

Installation and Configuration of CA Virtual Analyst with MS Teams

Deployment of MS Teams Bot Server

Follow these steps for the deployment of a bot server:

1. Copy the zip file **MSTeams-CIS-Server.zip** from 17.4GA <DVD>\filestore\TeamsIntegration, and extract them to your machine.
2. In the extracted files, navigate to **.env.sample** file and rename it to **.env**
3. In the **.env** file, make the required changes.
   For example:

```
# IMPORTANT
# values in the key-value pair should be enclosed in single quote


# App Id and App Password for the Bot

#Enter the BOT_ID of MS Team
BOT_ID=

#Enter the SECRET OF above of BOT
BOT_PASSWORD=

# Enter the BOT name for the bot
BOT_NAME=Aria

# Collab xFlow Server name - ws://xxxx:9014
COLLAB_XFlowServer='ws://xFlowServerName:9014'

# API xFlow Server name - http(s)://xxxx:9002
# This is the URL that the bot is using to communicate to xFlow
API_XFlowServer='http://xFlowServerName:9002'

# Public xFlow Server name - http(s)://xxxx:9002
# This is the URL that will be used to build the URLs sent in the chat communications
# if this is not set, then API_XFlowServer will be used
# WEB_XFlowServer='http://xFlowServerName:9002'


## NOT RECOMMENDED Disable SSL Certificate verification - when xFlow server uses a self signed certificate
## Use valid certificate signed by a valid Certificate Authority
# NODE_TLS_REJECT_UNAUTHORIZED=0

# ServiceDesk admin user
userId='ServiceDeskUser'
# ServiceDesk admin password
password='password'
```

```
# MS Team -xFlow integration works based on the userId attribute on SDM Contact table
# during syncing/importing if you updated another field (eg email, domain/user) to userId field in SDM
 contacts
# specify that format here - to make MS Team Integration to work

# possible values are
# '{username}@{domain}'
# '{username}'
# '{domain}/{username}'

# if userIdFormat is null/empty/not specified in this .env file
# it will default to '{username}'

userIdFormat = '{username}'


# Logger information
logName='va-connector'

# Log messages to the console as well, not only in the log file. Valid values true or false, default false
# logEnableConsole=true

# The detail log level. Valid log levels: "crit", "error", "warning", "notice", "info", "debug", "data",
 "silly", default "info"
# logLevel='debug'

# Timeouts
# Web socket ping interval - to keep the Websocket open (seconds) - default 55 seconds
# WS_PING_INTERVAL=55

# Timeout value for X-Access Token (minutes) - default 50 minutes
# X_ACCESS_TOKEN_TIMEOUT=50

# Bot Session Timeout (minutes) - default 30 minutes
# BOT_SESSION_TIMEOUT=30

# Bot Message Timeout - how long to wait the bot to respond to sent messages (seconds) - default 5 seconds
# VA_SERVER_TIMEOUT=5

# The connection string to mongodb
CONNECTION_STRING='mongodb://mongodb-server:27017/'
# The Default DATABASE
DB_NAME='db_name'
# The Default Collection
DB_COLLECTION='db_collection'

# Notification settings
ACTIVEMQ_STOMP_URL = 'ws://localhost:61614/stomp'
ACTIVEMQ_QUEUE = '/queue/sdm-msteams'
BOT_TENANT_ID=
NOTIFICATION_SERVICE_URL='https://smba.trafficmanager.net/amer/'
```

```
# The default Time-To-Live for records in the collection (in minutes)
# This will be applied ONLY to new or renewed documents. Documents stored already in the DB will not be
 changed.
TTL_AGE=1440

# Renew element in the registry if the TTL is less than specified when it's accessed (in minutes)
# This will take effect after a process restart
TTL_RENEW=60

# How long to wait(in minutes) before another server takes ownership of a message after the owner goes
 down.
CLUSTER_RECOVERY_TIMEOUT=1

# Interval (in seconds) at which a server sends a signal to indicate it's still operational
BOT_HEARTBEAT_INTERVAL=28
```

NOTE

1. In the .env file mentioned above for information about **BOT_ID**, **BOT_PASSWORD** and **BOT_NAME** refer to Creating a Bot for MS Teams.
2. BOT_TENANT_ID should be taken from your Azure portal.
3. For **BOT_PASSWORD** and **ServiceDesk admin** password generation, in the MSTeams-CIS-Server.zip file open command prompt and give the command:
   ```
   node encode.js botpassword/xflowpassword
   ```
4. Based on your geographical location, in Notification Settings, the NOTIFICATION_SERVICE_URL would change. For more information, refer here.
5. By default, the bot server runs on port 9239. You can configure the variable PORT in .env to run it on a different port.
6. By default, we assume that the contacts from MS Teams are imported into SDM based on the user id from the MS Teams Principal Contact Name. For example, if the PCN of a user is **janedoe@onmicrosoft.com**, we assume that the SDM contact's user id for this user would be **janedoe**. You can change this format by configuring userIdFormat variable.
7. User specified for 'userId' variable in .env should have Administration access type. This Admin is used to impersonate other users.

4. Click **start.bat** from the bot server location, for the bot to be up and running.

Uploading an application to MS Teams

1. Copy the zip file **MSTeams-AppPackage.zip** from 17.4GA <DVD>\filestore\TeamsIntegration, and extract them to your machine.
2. In the extracted files, you can find two files: **manifest.json** and **resources**.
3. Update the following in manifest.json:
   a. Copy the **BOT_ID** from .env file and paste them in **id** and **botId**.
   b. Copy the **BOT_NAME** from .env file. Search for "name" in the .json file, and replace "Aria" with the created bot name.
   For example:
   ```
   "name": {"short": "Aria", "full": "This is a bot which is running in a local server. It talks to CA
    Service Desk."}
   ```
   c. Search for **validDomains** and enter the domain for https machine. (Refer step 7 in **Creating a Bot in Microsoft Teams**)
4. Zip and rename the updated **manifest.json** and **resources** files.
5. Now, upload the renamed zip file to teams.microsoft.com. After uploading, follow these steps:
   a. In **Apps** click on **Manage your apps.**

b. Click on **Upload an app** and select **Upload an app to your org's app catalog.**
c. After uploading, navigate to **Built for your org** and click on the app that you added.
d. Click **Add**.

6. You can now start using your bot.

Configuring WebHooks with CA Service Desk Manager

Follow these steps for configuring WebHooks:

1. Copy **SDM-MSTeams-Webhook.war** file from 17.4GA <DVD>\filestore\TeamsIntegration and deploy SDM-MSTeams-Webhook.war in tomcat webapps folder
   For example: C:\apache-tomcat-9.0.71\webapps
2. Navigate to the bin folder in which **Active MQ** is installed and run the command:
   ```
   activemq start
   ```
3. In activemq console, locate the tcp and copy the tcp url. For example: tcp://<bot_server_host_name>:61616
   Navigate to **tomcat_installed_location> webapps> SDM-MSTeams-Webhook> WEB-INF> classes> msgQueueConfig.properties.**
   In msgQueueConfig.properties, look for activemq.broker.url and paste the tcp url.
4. Start the Tomcat server.

Configure CA Service Desk Manager

Follow these steps after the successful deployment of CA Service Desk Manager (CA SDM):

1. Log in to CA SDM Classic Interface.
2. In CA Service Desk Manager, go to **Administration** > **xFlow Interface** >**General**. A new option **allowApiImpersonation** has been added to SDM. The value should be set to **true** (case sensitive). This option enables the integration between Bot server and Virtual Analyst chat.
3. Now restart the xFlow Analyst Interface services.

   NOTE
   VA server should be integrated with xFlow Server. Refer Configure Virtual Analyst in Service Desk Manager.

Create a new Webhook endpoint in CA SDM

Perform the following steps after the successful deployment of CA Service Desk Manager:

1. Log in to CA SDM.
2. Navigate to **Administration**> **System**> **Webhooks**> **Endpoints** to create a new Webhook. Complete the field information:

Field	Description
Symbol	Mandatory field information
Code	Mandatory field information
Endpoint URL	Provide the webhook URL details. For Example: http://TOMCAT_SERVER:TOMCAT_PORT/SDM-MSTeams-Webhook/webhook/sdm-msteams
Content Type	JSON

3. Optionally fill in the remaining fields, and click **Save**.

Create a Notification

To listen to changes in Requests/ Incidents/ Problems, create a notification under webhook.

Follow these steps to create a notification:

1. Log in to CA Service Desk Manager.
2. Click **Administration**> **System**> **Webhook**> **Notifications**> **Create New** to create a notification.
3. Complete the field information:

Field	Value
Symbol	Mandatory field information
Code	Mandatory field information
Object Type	Select **Request/ Incidents/ Problems**
IsTemplate?	Check the box

Payload Body	Enter the values to be sent to MS Teams as part for notification. For example:
	
	`Create Ticket:`
	`{"message": "New Ticket is`
	`created: @{call_req_id.ref_num}",`
	`"userId":"@{call_req_id.customer.userid}",`
	`"userName":"@{call_req_id.customer.combo_name}",`
	`"persId":"@{call_req_id.persistent_id}"}`
	
	`Log Comment:`
	`{"message": "Comment added`
	`for : @{call_req_id.ref_num}",`
	`"userId":"@{call_req_id.customer.userid}",`
	`"userName":"@{call_req_id.customer.combo_name}",`
	`"persId":"@{call_req_id.persistent_id}"}`
	
	`Status Update:`
	`{"message": "Status updated for`
	`ticket: @{call_req_id.ref_num}`
	`to @{call_req_id.status.sym}",`
	`"userId":"@{call_req_id.customer.userid}",`
	`"userName":"@{call_req_id.customer.combo_name}",`
	`"persId":"@{call_req_id.persistent_id}"}`
	
	• The values of userId, userName, persId are mandatory and the value should configured as shown above.
	• The content for message can be configured for each notification as per the notification template syntax.
	• For information on Service Desk Manager notification template syntax, refer to Create Message Templates.

4. Optionally fill in the remaining fields and click **Save**.
5. Select **Webhook Endpoints**, **Update Webhook Endpoints** to map a Notification with an existing Webhook Endpoint.
6. In the **Endpoint Search** page displayed, provide the required endpoint details and click **Save**.
7. Select the Endpoint to which the new notification is to be mapped and move it to the adjacent box. Click **OK**, and then click **Save**. This Notification is successfully updated and mapped to the selected Endpoint.
8. To specify Notification Rules, select Triggers from the Notification Detail page.

9. Select **Update Notification Rules** and the **Notification Rules Search** page is displayed.
10. In the page displayed, follow these steps:
    a. Provide the required details about the Notification Rule that must be mapped to a notification.
    b. Click **Search**.
    c. The **Notification Rules – Update** page displays the Out-of the-box Notifications.
    For Example:
    Select **Default Initial Notification Rule** for Request/Incidents/Problems.
    Select **Default Log Comment Notification Rule** for Request/Incident/Problem.
11. Click **OK** and the Notifications are updated for a selected **Webhook Endpoint**.

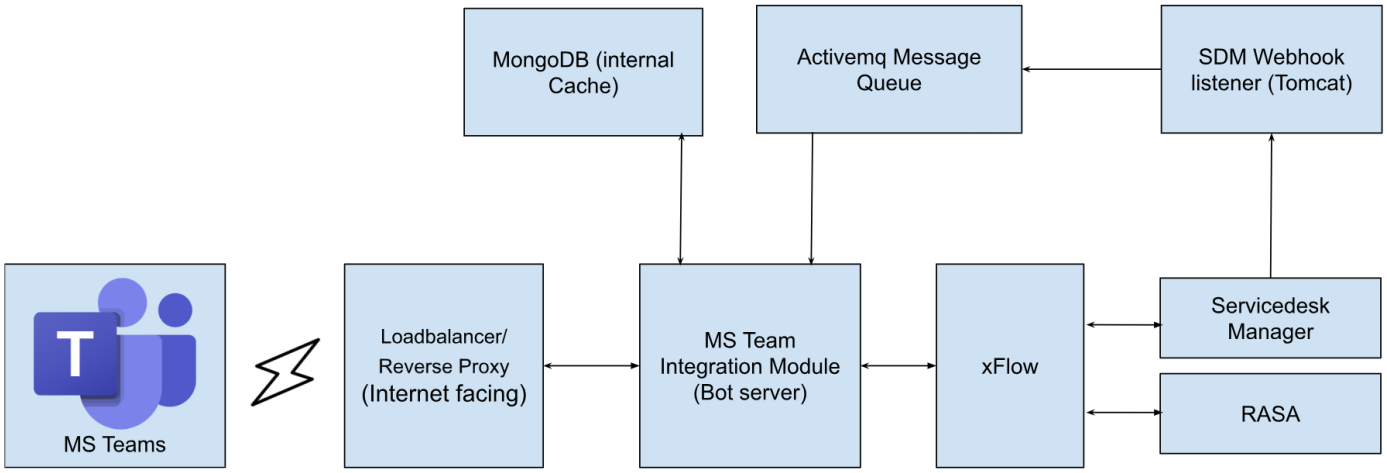
Scaling MS Teams integration using clustering

To handle increased load and improve performance, the MS Team Integration module can be scaled using clustering. There are three ways to do that:

- Simple Deployment
- Clustering Deployment

Simple Deployment

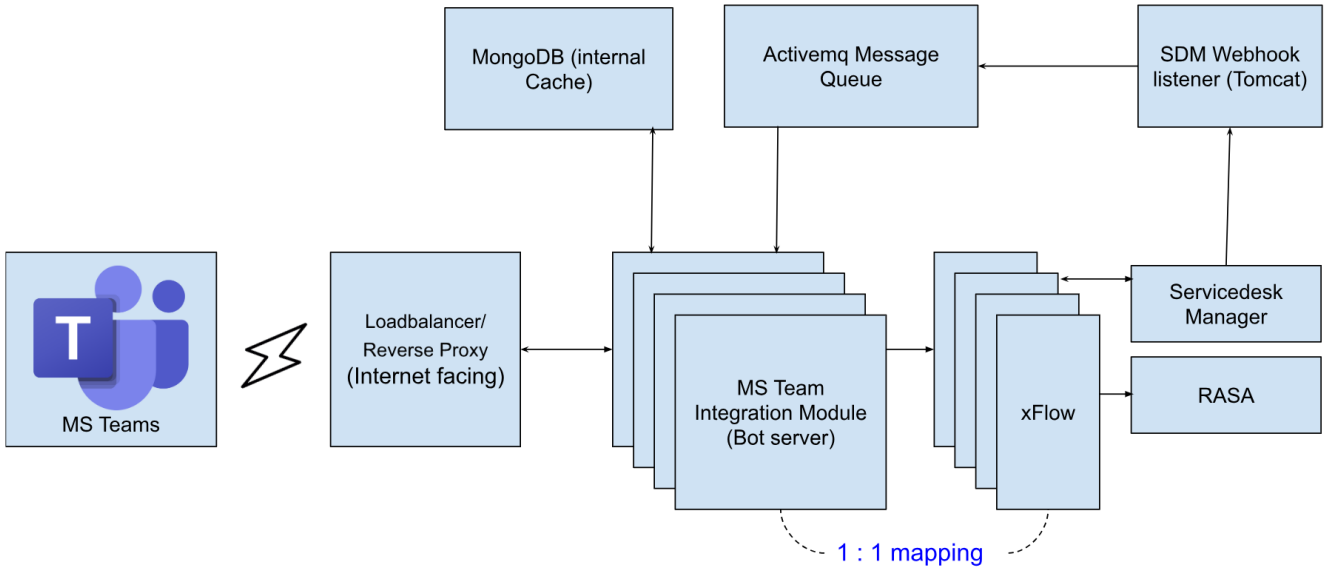
MS Teams integration with CA Virtual Analyst has multiple modules and each module is independently scalable. In a simple deployment, each module has one server instance each, and can handle sixty concurrent users. The following image shows how a simple deployment functions:



Cluster Deployment

Cluster deployment is used to handle a higher load of concurrent users. Multiple MS Teams Bot servers can be scaled either in the same server or across multiple servers. To configure multiple bot servers, follow these steps:

1. Refer Deployment of MS Teams Bot Server.
2. If multiple servers are being configured in the same server, update the PORT variable in .env file for each server.
3. Ensure all the servers are configured behind a load balancer.
4. By default load balancer's session stickiness ensures that given users' conversation goes to a single instance of MS Team Bot server. Optionally you can set the session stickiness based on the specific load balancer configuration.



In addition to scaling MS Team Bot servers, other modules like xFlow, ActiveMQ, MongoDB can be scaled as per their respective configuration.

> **NOTE**
> Refer to the pricing and licensing terms for the installation and usage of Docker in your organization https://www.docker.com/pricing/faq/.

To scale RASA servers, see RASA Clustering using Docker.

Integrating CA SiteMinder

CA product contacts are authenticated (SDM) and authorized by CA EEM while CA EEM is configured to point to CA SiteMinder for directory information. CA SiteMinder integrates with various directory services and user stores, eliminating redundant administration of user information. This integration simplifies administration and provides unique and comprehensive security capabilities. CA SiteMinder fully leverages existing user directories, from leading LDAP directories and relational databases to mainframe security directories. CA EEM is a central repository of user information known as identities. CA EEM defines user authentication and access to other applications. CA EEM server can be configured to reference global users and global groups from CA SiteMinder.

This section describes how to integrate CA SiteMinder and CA EEM.

The following high-level steps are involved in the configuration:

1. Configure CA SiteMinder by creating the following items:
   a. CA SiteMinder Web Agent
   b. CA SiteMinder Agent Configuration
   c. CA SiteMinder User Directory
   d. Agent config object and host config object
   e. CA SiteMinder Domain
   f. CA SiteMinder Realm and Rule
   g. CA SiteMinder Policy
2. Configure CA EEM to use the CA SiteMinder user store.

Configure CA SiteMinder

The following sections document in detail how to create the Agent, User Directory, Domain, Realm, Rule, and Policy in CA SiteMinder. The steps in these sections are just one of a few different ways to create these objects and are included as examples only.

Create a CA SiteMinder Agent

From the CA SiteMinder Administrative user interface, create a CA SiteMinder Web Agent for communication between CA EEM and CA SiteMinder policy server.

Follow these steps:

1. From the CA SiteMinder Administrative user interface, select the Agents(Under Agent) link on the Infrastructure tab.
2. Click on “Create Agent” button
3. Select “Create a new object of type Agent” and click Ok.
4. Provide the following information in the General section:
   a. Enter a name for the agent (for example, eemagent).
   b. Select the agent type as SiteMinder.
   c. Select the Agent Type as Web Agent.
   d. Select the Supports 4.x agents option.
   e. Enter the IP Address of the CA EEM server.
   f. Enter a Shared Secret. This entry must match what will be entered in CA EEM when configuring the CA EEM Server Global Users/Global Groups.
5. Click Submit.

Create a User Directory

Create a CA SiteMinder User Directory for authorization, which is used by CA EEM to retrieve LDAP attributes. Ensure that the UniversalID field uniquely identifies a user in the directory on the User attributes tab.

1. Go to Infrastructure-> Directory-> User Directory
2. Click on Create User Directory Button
3. Enter the Directory Name
4. Under Directory Setup, provide the server name and port. Ex: ITGAD:389
5. Select “Require Credentials” checkbox
6. Provide username and password.
7. Under LDAP Settings, provide Root as showed in the screenshot.
8. 8. Under LDAP User DN Lookup, provide the start and end fields.
9. Click Submit.

Create a Domain

Create a domain for CA EEM and associate the user directory that you created for CA EEM. For more information about creating a domain.

Create the Realm and Rule

Follow these steps:

1. From the CA SiteMinder Administrative user interface, select Domain-> Realms link on the Policies tab.
2. Click on the Create Realm button create the realm.
   a. Select the Domain created in the previous procedure and click Next.
   b. In the General section, enter a realm name (for example, eemrealm).
   c. Under Resource, select the Agent that was previously created (for example, Eemagent).
   d. Enter a backslash (\) for the Resource filter.
   e. Select Protected in the Default Resource Protection field.
   f. Select an Authentication Scheme (for example, Basic).

3. Create a Rule for the realm by clicking Create and providing the following information:
   a. Enter a Name (for example, eemRule).
   b. Under Attributes, enter iamt.html as the Resource.
   c. Select the Allow Access and Enabled options
   d. Under Action, select Web Agent actions and select Get and Post in the Action box

The realm adn ruke are created.

Create a Policy

Follow these steps:

1. From the CA SiteMinder Administrative user interface, select the link Domain->Domain Policies link on the Policies tab.
2. Click the Create Policy button.
3. Select the Domain that you previously created, EEMDomain for example, and click Next.
4. Under the General section, enter the policy name (for example, EEMPolicy) and click Next
5. Create the User Directories by selecting the Members to include in each User Directory. For example, click the Add All button under the user directory that you just created.
6. Click Next. Add a Rule for the Policy by selecting the Add Rule button.
7. Select the Rule in the list you just created and click OK.
8. Click Next and click Finish.

Configure the EEM Server to Reference CA SiteMinder

Follow these steps:

1. Log in to the CA EEM user interface.
2. On the Configure tab, select User Store.
3. On the left pane, select User Store.
4. On the right pane, select “Reference from CA SiteMinder” and enter the following values:
   – Host: IP Address of the CA SiteMinder server
   – Admin Name: CA SiteMinder Administrator name (for example, SiteMinder)
   – Admin Password: CA SiteMinder Administrator password (itgadmin@123)
   – Agent Name: SiteMinder Web Agent created (for example, eemagent)
   – Agent Secret: Secret entered during Web Agent creation (for example: password)
5. Click Save.
6. The status must confirm a successful bind, as shown in the following example.
7. Navigate to the Users node on the Manage Identities tab and click the Go button.
8. Validate that the users are listed.

Reference

This section contains the following articles:

- CA Service Desk Manager Reference Commands
- CA Service Catalog Glossary
- CA Service Management Common Data Object Field Level Mapping Details
- USM Data Mapping for CA Service Desk Manager Connector
- Default Port Numbers and Connectivity - SOI Integration with SDM
- Relationship and Service Mapping

CA Service Desk Manager Reference Commands

This section contains CA Service Desk Manager reference commands:

- Data Element Dictionary
- Comments
- Technical Reference
- CMDB Technical Reference
- API Documentation for RESTful Services
- Objects and Attributes

> **NOTE**
> The interactive ODBC SQL (pdm_isql) tool is no longer supported for CA Service Management 17.3 and above versions.

Data Element Dictionary

This article contains the following topics:

Explains the definitions for the tables in the CA SDM database. The tables are found in schema (.sch) files. See the ddict.sch file in the $NX_ROOT/site directory (Non-Windows) or *installation-directory*\site directory (Windows) for the most current list of all database tables for your specific installation.

> **WARNING**
> Although several of the tables in this section are obsolete, for backward compatibility, or reserved for future use, it is important that you not delete these -- or any other table -- from the database schema. You can add new tables and add new fields/columns to the existing tables, but never delete any of the tables. Many applications access the mdb database, so deleting or modifying existing fields or tables could cause some applications to not function properly. Ensure that you follow supported schema modification best practices by using the Web Screen Painter.

admin_tree Table

Program control table used by CA SDM applications.

- **SQL Name** -- admin_tree
- **Object** -- ADMIN_TREE

Field	Data Type	Reference	Remarks
caption	STRING 50		

description	STRING 255		Specifies the textual description of this tree entry.
has_children	INTEGER		
id	INTEGER KEY		Unique (to the table) Numeric ID
kt_admin	INTEGER		
kt_ks_caption	STRING 50		
kt_ks_flag	INTEGER		
kt_manager	INTEGER		
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
parent_id	SREL		
resource	STRING 255		
sd_admin	INTEGER		

Animator Table

Program control table used by CA SDM applications

- **SQL Name** -- anima
- **Object** -- ANI

Field	Data Type	Reference	Remarks
a_act	INTEGER		
a_delta	INTEGER		
a_lock	STRING 200		
a_name	STRING 30 S_KEY		
a_org	LOCAL_TIME		
a_string	STRING 30		
a_time	LOCAL_TIME		
id	INTEGER KEY UNIQUE NOT_NULL		Unique (to the table) Numeric ID
t_method	STRING 30 S_KEY		
t_persid	STRING 30 S_KEY		
t_type	INTEGER		

Atomic_Condition Table

These define a single condition.

- **SQL Name** -- atomic_cond
- **Object** -- atomic_cond

Field	Data Type	Reference	Remarks
cond_code	STRING 500		
connecter	INTEGER NOT_NULL		
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
description	STRING 240		Specifies the textual description of this condition.
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
l_paran	INTEGER NOT_NULL		
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified
lval	STRING 30 NOT_NULL	Act_Type_Assoc ::code	
operator	INTEGER NOT_NULL		
owning_macro	STRING 30	Spell_Macro::persid	
persid	STRING 30		Persistent ID (SystemObjectName:id)
r_paran	INTEGER NOT_NULL		
rval	STRING 50		
rval_assoc	STRING 30	Act_Type_Assoc ::code	
sequence	INTEGER NOT_NULL		Ordering

Attribute_Name Table

Provides corresponding user name for an object attribute. Default population for the table should set at_name and at_dflt to the same thing. The user sees and is able to modify only the at_name. at_dflt is used to restore the default name. at_desc could be user changeable or not. It is of use for identifying the attribute for an as yet unplanned user maintenance application. at_sys should never be seen by the user, nor should at_obj.

- **SQL Name** -- atn

Field	Data Type	Reference	Remarks
at_desc	STRING 240		Specifies the description of the attribute.
at_dflt	STRING 30		Specifies the default external name.

at_name	STRING 30		Specifies the user name for attribute.
at_obj	STRING 30 S_KEY		Specifies the system object name.
at_sys	STRING 30 S_KEY		Specifies the system name.
del	INTEGER NOT_NULL		Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID

Audit_Log Table

Contains all audit log entries.

- **SQL Name** -- audit_log
- **Object** -- audlog

Field	Data Type	Reference	Remarks
analyst	Byte 16	ca_contact::uuid	Specifies the user whose update generated this audit record.**Note:** This is a foreign key.
attr_after_val	nvarchar(160)		Specifies the new value of the object's attribute that has changed.
attr_before_val	nvarchar(160)		Indicates the previous value of the object's attribute that has changed.
attr_name	nvarchar(80)		This is the object's attribute that has changed.
aud_opr	INTEGER		Indicates the type of operation that generated this entry, such as, update, insert, and delete.
audobj_name	nvarchar(10)		Used for tracking the object that has changed.
audobj_persid	nvarchar(30)		Used for tracking the object that has changed.
audobj_trkid	nvarchar(40)		Used for tracking the object that has changed.
audobj_uniqueid	nvarchar(30)		Used for tracking the object that has changed.
change_date	INTEGER		The change date value for this Audit_Log.

id	INTEGER		Specifies the unique (to the table) numeric ID.**Note:** This is a primary key.
int1_rsrved	INTEGER		Reserved
int2_rsrved	INTEGER		Reserved
persid	nvarchar(30)		This is the Persistent ID (SystemObjectName:id).
str1_rsrved	nvarchar(25)		Reserved for future use by CA.

Behavior_Template Table

Each object includes a list of things to execute based on the state of the object. If no behavior is associated with the state transition then it simply controls whether the object can be transitioned to the state.

- **SQL Name** -- bhvtpl
- **Object** -- bhvtpl

Field	Data Type	Reference	Remarks
context_attrname	STRING 30		Specifies the attribute name, such as "state".
context_attrval	INTEGER		Specifies the attribute value.
context_type	STRING 30 NOT_NULL		Specifies information which determines whether the transition is valid for the consumercontext object. **Note:** For general use, the context_attrnamecontext_attrval pair was added. However, can't get the list of valid values very easily from the GUI so we've added an SREL to the object that context_attrnamecontext_attrval really represents. short name of object (eg. wf)
del	INTEGER NOT_NULL	Active_ Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
description	STRING 500		Specifies the textual description of this behavior template
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact:: uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.

macro_condition	STRING 30	Spell_Macro ::persid	Specifies whether the macro executes.
object_id	INTEGER NOT_NULL		Specifies the id of creator object.
object_type	STRING 30 NOT_NULL		Specifies the short name of creator object.
persid	STRING 30		Persistent ID (SystemObjectName:id)
transition_errmsg	STRING 240		Specifies the text to display on failure.
transition_test	STRING 30	Spell_Macro ::persid	Specifies the condition.

Bop_Workshift Table

Workshift definition.

- **SQL Name** -- bpwshft
- **Object** -- wrkshft

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_Table::enum	This is the Deleted flag:**0** -- Active**1** -- Inactive/marked as deleted
description	nvarchar(80)		Specifies the textual description of this workshift.
id	INTEGER		Unique (to the table) Numeric ID.
last_mod_by	Byte(16)	ca_contact:: uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).**Note:** This is a primary key.
sched	nvarchar(1000)		Describes the schedule of this workshift.
sym	nvarchar(30)		Represents the symbolic value for this workshift.

BU_TRANS Table

Program control table used by Knowledge Management.

- **SQL Name** -- BU_TRANS
- **Object** -- BU_TRANS

Field	Data Type	Reference	Remarks
BU_DATE	LOCAL_TIME		
BU_PROCESSED	INTEGER		
BU_RATING	REAL		
DOC_ID	INTEGER	SKELETONS::id	
HIT_NO_VOTE	INTEGER		
HIT_ORIGIN	INTEGER		
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
INDEX_ID	INTEGER	O_INDEXES::id	
User_slv	INTEGER		
Ticket_slv	INTEGER		
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
USER_ID	UUID	ca_contact::uuid	Specifies the UUID of the user id.
kd_duration	INTEGER		Specifies the duration of a specific event, such as how long a knowledge document was open.

Business_Management_Repository Table

Table that keeps track of CA NSM repositories used by Service Analyzer.

- **SQL Name** -- busrep
- **Object** -- bmrep

Field	Data Type	Reference	Remarks
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
hostname	STRING 64 UNIQUE NOT_NULL		
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
nx_desc	STRING 40		

password	STRING 200		
persid	STRING 30		Persistent ID (SystemObjectName:id)
sym	STRING 60 UNIQUE NOT_NULL S_KEY		
userid	STRING 40		

Column_Name Table

Column Name list used by Web Screen Painter.

- **SQL Name** -- cn

Field	Data Type	Reference	Remarks
cn_desc	STRING 240		description of column
cn_dflt	STRING 30		default external name
cn_name	STRING 30		user name for column
cn_sys	STRING 30 S_KEY		system name
cn_table	STRING 30 S_KEY		system table name
del	INTEGER NOT_NULL		Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID

Contact_Method Table

Defines contact method types. The cm_template field is a command string that gets executed as a script (with environment variables set) by the notify subsystem.

- **SQL Name** -- ct_mth
- **Object** -- cmth

Field	Data Type	Reference	Remarks
cm_template	nvarchar(240)		Specifies the method template.
del	INTEGER	Active_Boolean_Table::enum	Identifies the Delete flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted)
id	INTEGER		Specifies the primary key of this table, this is also a unique, numeric ID.

last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp of when this record was last modified.
nx_desc	nvarchar(40)		Specifies the Contact method description.
persid	nvarchar(30)		Specifies the Persistent ID (SystemObjectName:id).
supports_smtp	INTEGER		Determines if the method supports SMTP email addresses.
sym	nvarchar(60)		Identifies the Contact method symbolic name.
write_file	INTEGER		Flag that indicates the following:**1** -- write output to the file

D_PAINTER Table

Tables to be used for Form Server and Screen Painter For the new GUI.

- **SQL Name** -- D_PAINTER

Field	Data Type	Reference	Remarks
CNTLID	INTEGER		id of the control
CNTLTYPE	INTEGER		type of control
DDID	INTEGER		data dictionary id
ENTITYID	INTEGER		entity type
EXTRA_L1	INTEGER		user-definable
EXTRA_L2	INTEGER		user-definable
EXTRA_L3	INTEGER		user-definable
EXTRA_S1	STRING 50		user-definable
EXTRA_S2	STRING 50		user-definable
EXTRA_S3	STRING 50		user-definable
FORMGROUP	STRING 50		group in which the form is contained
FORMID	INTEGER		id number of the form
FORMNAME	STRING 50		name of the form
FORMTYPE	INTEGER		type of form
ID	INTEGER UNIQUE NOT_NULL KEY		key ID
MAPBACK	STRING 30		data dictionary owner

PARENTID	INTEGER		control id of parent control
PREDEFINED	INTEGER		**0** -- Normal screen **2** -- Default screen
PROPLIST	STRING 1000		properties for the control
SECLEVEL	INTEGER		security level
TSTAMP	REAL		time stamp

Delegation_Server Table

List of servers that can be delegated from this one along with xport methods.

- **SQL Name** -- dlgsrvr
- **Object** -- dlgsrvr

Field	Data Type	Reference	Remarks
anon_userid	STRING 8		Specifies the anonymous userid.
appl_addr	STRING 48		Specifies the name or address of application.
default_assignee	UUID	ca_contact::uuid	Specifies the assignee for incoming tickets.
default_userid	STRING 8		Specifies the default userid.
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
metafile	STRING 256		Specifies the path to conversion metafile def.
nx_desc	STRING 40		Specifies the description.
password	STRING 16		Specifies the server password.
server	STRING 128		Specifies the server name or ip address.
sym	STRING 64 UNIQUE NOT_NULL S_KEY		Specifies the system name.
transport	INTEGER		

Controlled_Table Table

Program control table used by CA SDM applications.

- **SQL Name** -- ctab
- **Object** -- ctab

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_ Table::enum	Indicates the Deleted flag, as follows:**0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER		Primary key to this table, it is a unique numeric ID.
nx_desc	nvarchar(40)		Specifies the Table description.
obj_name	nvarchar(30)		Specifies the Majic object name that corresponds to this table.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
sym	nvarchar(30)		Represents the symbolic name of this controlled table.

EBR_SUFFIXES Table

Program control table used by Knowledge Management.

- **SQL Name** -- EBR_SUFFIXES
- **Object** -- EBR_SUFFIXES

Field	Data Type	Reference	Remarks
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
SUFFIX	STRING 50		

Priority Table

List of Priority entries. The priority reflects the time-frame in which a ticket must be resolved. For the ticket, it represents the highest priority of any problem attached to the ticket. Problem priorities are derived from the scope (impact) and severity of the problem.

- **SQL Name** -- pri
- **Object** -- pri

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_Table::enum	Deleted flag that indicates the following:**0** -- Active **1** -- Inactive/marked as deleted
enum	INTEGER		Primary key of this table.
id	INTEGER		Specifies the unique (to the table) numeric ID.

last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
nx_desc	nvarchar(40)		Describes the priority.
service_type	nvarchar(30)	Service_Desc:: code	Classic Service Type. Foreign key to the code field of the srv_desc table.
sym	nvarchar(12)		Indicates the symbolic name for this priority.

Queued_Notify Table

Notifications that are queued due to workshifts are saved here.

- **SQL Name** -- not_que
- **Object** -- notque

Field	Data Type	Reference	Remarks
cmth_override	INTEGER		method over ride
context_instance	INTEGER 30		Contains the persistent ID of the associated Activity Log for the notifications
context_persid	STRING 30		Contains the persistent ID of object for notification
del	INTEGER NOT_NULL	Active_Boolean_Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
internal	INTEGER		internal notification
msg_ack	STRING 40		message acknowledgment
msg_body	STRING 1000		message text
msg_body_html	STRING 32768		message text
msg_title	STRING 40		Msg header text
notify_level	INTEGER		notification level
persid	STRING 30		Persistent ID (SystemObjectName:id)
transition_pt	INTEGER		transition point

Quick_Template_Types Table

Quick_Template_Types - Reference table for quick template types.

- **SQL Name** -- quick_tpl_types
- **Object** -- quick_tpl_types

Field	Data Type	Reference	Remarks
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
enum	INTEGER UNIQUE NOT_NULL		Enumerated value for this entry - specifies ordering in lists and relative values
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
nx_desc	STRING 40		Descriptive Info
persid	STRING 30		Persistent ID (SystemObjectName:id)
sym	STRING 30		Symbolic name of level

Remote_Ref Table

Remote References. Used for smart hooks. Determines what command to execute. Different command for UNIX and pc's using the same smart hook. Can apply security to smart hook.

- **SQL Name** -- rem_ref
- **Object** -- rrf

Field	Data Type	Reference	Remarks
description	STRING 500		Specifies the description of the command.
arch_type	STRING 12		Specifies the architecture to exec this on UNIX or PC. If empty, then all.
code	STRING 12 UNIQUE NOT_NULL S_KEY		Specifies the noneditable key for command.
del	INTEGER NOT_NULL	Active_Boolean_Table::enum	Deleted flag **0** -- Active **1 --** Inactive/marked as deleted
exec_str	STRING 500		Specifies the string to execute on UNIX.
function_group	STRING 30		Specifies the function group for security.
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record

last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
pcexec_str	STRING 500		Specifies the string to execute on pc.
sym	STRING 30 NOT_NULL		Specifies the name of command.

Response Table

Personalized response text used to simplify data entry when using the CA SDM applications.

- **SQL Name** -- response
- **Object** -- response

Field	Data Type	Reference	Remarks
chg_flag	INTEGER S_KEY		
cr_flag	INTEGER S_KEY		
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
in_flag	INTEGER S_KEY		
iss_flag	INTEGER S_KEY		
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
persid	STRING 30		Persistent ID (SystemObjectName:id)
pr_flag	INTEGER S_KEY		
response	STRING 1000		response text
response_owner	UUID S_KEY	ca_contact::uuid	response owner
sym	STRING 50 NOT_NULL S_KEY		symbol

Rootcause Table

Reference table to denote the rootcause type used when resolving or closing a request, change order, or issue.

- **SQL Name** -- rootcause
- **Object** -- rc

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag as follows:**0** -- Active**1** -- Inactive/ marked as deleted
description	nvarchar(240)		Provides a textual description of this root cause.
id	INTEGER		Primary key of this table.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
sym	nvarchar(60)		Identifies the symbolic value for this Rootcause.

Rpt_Meth Table

Reporting methods used to display information within the CA SDM applications.

- **SQL Name** -- rptmth
- **Object** -- rptm

Field	Data Type	Reference	Remarks
description	STRING 80		Specifies the textual description of this reporting method.
def_pg_len	STRING 80		Specifies the page length default.
default_out	STRING 80		Specifies the output default.
del	INTEGER NOT_NULL	Active_Boolean_Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
is_default	INTEGER		If set, this specifies the default reporting method.
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.

script	STRING 1000		
sym	STRING 30 NOT_NULL		

Reporting_Method Table

Reference table to denote how the contact with the customer occurred. Example: email, phone.

- **SQL Name** -- repmeth
- **Object** -- rptmeth

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag as follows:**0** -- Active**1** -- Inactive/ marked as deleted
id	INTEGER		Primary key of this table.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
sym	nvarchar(60)		Identifies the symbolic value for this Reporting Method.

Note_Board Table

Message board (announcements) on the main menu.

- **SQL Name** -- cnote
- **Object** -- cnote

Field	Data Type	Reference	Remarks
active_flag	INTEGER NOT_NULL	Boolean_Table:: enum	**0** -- Inactive **1** -- Active
close_date	LOCAL_TIME		Indicates when closed
cnote_type	INTEGER		Indicates the announcement type
control_group	UUID		Specifies the group to display to
del	INTEGER		Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
internal	INTEGER		Internal Flag

loc_id	UUID	ca_location:: location_uuid	Indicates a pointer to location
organization	UUID	ca_organization ::organization_uuid	Indicates a pointer to Organization
persid	STRING 30		Persistent ID (SystemObjectName:id)
posted_by	UUID	ca_contact::uuid	Specifies who posted the announcement
posted_date	LOCAL_TIME		Indicates the last modify time
text	STRING 4000		Indicates the message text

Prob_Category Table

Call Request call areas. Category of the issue that the customer is calling about. Can be hierarchical.

- **SQL Name** -- prob_ctg
- **Object** -- pcat

Field	Data Type	Reference	Remarks
id	INTEGER	UNIQUE NOT_NULL KEY	Unique (to the table) numeric ID.
persid	STRING (30)		Persistent ID (SystemObjectName:id)
del	INTEGER	NOT_NULL	Delete flag that indicates the following: **0** -- Active **1** -- Inactive/marked as deleted
sym	STRING (1000)	NOT_NULL S_KEY	Specifies the symbolic name of the request area.
last_mod_dt	LOCAL_TIME		Identifies the timestamp of when this record was last modified.
last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
description	STRING (500)		Identifies the textual description of the call area.
organization	UUID	ca_organization	Foreign key to the id field of the ca_organization table, this is the Organization.
assignee	UUID	ca_contact	Foreign key to the contact_uuid field of the ca_contact table, this is the Assignee.
group_id	UUID	ca_contact	Foreign key to the contact_uuid field of the ca_contact table, this is the Group.
tcode	INTEGER		Deprecated

service_type	STRING (30)	Service_Desc	Foreign key to the code field of the srv_desc table, this is the Classic Service Type.
survey	INTEGER	Survey_Template	Foreign key of the id field of the survey_tpl table, this is the Survey.
schedule	INTEGER	Bop_Workshift	Deprecated
auto_assign	INTEGER		Represents the flag that enables auto assignment.
owning_contract	INTEGER	Service_Contract	Foreign key to the id field of the svc_contract table.This is the Service Contract.
cr_flag	INTEGER		Represents the cr_flag status. When set to 1, this status is valid for requests.
in_flag	INTEGER		Specifies the Incident flag. When set to 1, the status is valid for Incidents.
pr_flag	INTEGER		Specifies the Problem flag. When set to 1, the status is valid for Problems.
suggest_knowledge	INTEGER		Specifies whether or not to suggest knowledge to users. **1** -- Suggest knowledge. **2** -- Do not suggest knowledge.
assignable_ci_attr	STRING (60)		Identifies the name of the attribute on a CI object that contains the grp contact that should be used to perform Category/CI auto group assignment.
flow_flag	INTEGER		Specifies the type of workflow:**0** -- Classic Workflow or none (default) **2** -- CA IT PAM
caextwf_start_id	INTEGER	caextwf_start_forms	Identifies the CA IT PAM process definition information to use when the user selects this category on a change order, issue, request.
tenant	UUID	ca_contact	
ss_include	INTEGER		Required. On new default: 0
ss_sym	STRING (128)		
category_urgency	INTEGER	Urgency	
sap_prop	INTEGER		

Product Table

Reference table to denote the product that the complaint relates to.

- **SQL Name** -- product
- **Object** -- prod

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_ Table::enum	Delete flag that indicates the following:**0** -- Active**1** -- Inactive/ marked as deleted
id	INTEGER		Primary key of this table.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
sym	nvarchar(60)		Specifies the symbolic value for this Product.

sa_agent_availability Table

Program control table used by Support Automation.

- **SQL Name** -- sa_agent_availability
- **Object** -- sa_agent_availability

Field	Data Type	Reference	Remarks
id	INTEGER		KEY
agentID	UUID	ca_contact	NOT_NULL
queueID	INTEGER	sa_queue	NOT_NULL
status	INTEGER		
availEpoch	LOCAL_TIME		
clientSessionID	INTEGER	sa_login_session	
matchEpoch	LOCAL_TIME		
groupID	INTEGER	sa_group	
incidentCount	INTEGER		
last_mod_by	UUID	cnt	Specifies the UUID of the contact who last modified this record.
last_mod_dt	DATE		Specifies the timestamp of when this record was last modified.

tenant	UUID	ca_tenant	Specifies the UUID of the tenant.

Table_Name Table

Table name list used by CA SDM applications.

- **SQL Name** -- tn

Field	Data Type	Reference	Remarks
del	INTEGER NOT_NULL		Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
tn_desc	STRING 240		description or table
tn_dflt	STRING 30		default external name
tn_name	STRING 30		user name for table
tn_sys	STRING 30 S_KEY		system name

usp_special_handling Table

The usp_special_handling table defines the characteristics of each special handling classification.

Attribute	Data Type	SREL References	Flags
alert_icon_url	STRING 1000		
alert_text	STRING 60		
autodisplay_notes	INTEGER		
del	INTEGER		NOT_NULL
description	STRING 4000		
escalate_urgency	INTEGER		
id	INTEGER		UNIQUE NOT_NULL KEY
last_mod_by	UUID	ca_contact	
last_mod_dt	LOCAL_TIME		
sym	STRING 60		NOT_NULL
tenant	UUID	ca_tenant	

usp_symptom_code Table

The usp_symptom_code table associates codes with a special handling classification.

Attribute	Data Type	SREL References	Flags
del	INTEGER		NOT_NULL
description	STRING 4000		
id	INTEGER		UNIQUE NOT_NULL KEY

last_mod_by	UUID	ca_contact	
last_mod_dt	LOCAL_TIME		
sym	STRING 60		NOT NULL
tenant	UUID	ca_tenant	

usp_tab Table

Information on Tabs used in the Role Based UI.

- **SQL Name** -- usp_tab
- **Object** -- tab

Field	Data Type	Reference	Remarks
id	INTEGER		Specifies the unique (to the table) numeric ID.
name	STRING (80)		Tab Name
display_name	STRING (80)		Text displayed on tab
code	STRING 30		Code
del	INTEGER	Active_ Boolean_ Table::enum	
description	STRING (255)		Description
menu_bar_obj	INTEGER	usp_menu_bar::id	Foreign key to the menu bar id field of the usp_menu_bar table.
web_form_obj	INTEGER	web_form::id	Foreign key to the web form id field of the usp_web_form table.

usp_ticket_type Table

The usp_ticket_type table lists the ticket types referenced by Action Object drop-down list (action_object) in the usp_mailbox_rule table.

Field	Data Type	Description
id	INTEGER UNIQUE KEY	Specifies the primary key of this table.
allow_mailbox	INTEGER	
code	STRING 30 UNIQUE S_KEY	Specifies the REL_ATTR value of the table, and the distinct keyword for each ticket type that is recognized by some components
del	INTEGER	
sym	STRING 60	Specifies the symbolic value for the ticket type.
ticket_object	STRING 30	Specifies the Majic object for the ticket (cr, chg, or iss).
ticket_object_type	STRING 1	Specifies the Call_Request_Type field (R, I or P) for the cr object.

The following values are the defaults for the usp_ticket_type table:

id	code	sym	ticket_obj	ticket_obj_type
100	REQUEST	Request	cr	R
200	INCIDENT	Incident	cr	I
300	PROBLEM	Problem	cr	P
400	CHANGE	Change	chg	
500	ISSUE	Issue	iss	

usp_web_form Table

Information on Web Forms used in the Role Based UI.

- **SQL Name** -- usp_web_form
- **Object** -- web_form

Field	Data Type	Reference	Remarks
id	INTEGER		Specifies the unique (to the table) numeric ID.
name	STRING (80)		Tab Name
code	STRING 30		Code
del	INTEGER	Active_Boolean_ Table::enum	
description	STRING (255)		Description
wf_resource	STRING (1024)		The URL to display the HTMPL form, Report, Go button resource or third party form.
wf_type	INTEGER		The type of the web form: **0** -- HTMPL **1** -- Report **2** -- Go Resource **3** -- Other
dflt_for_obj	STRING 30		For Go Resource type web forms only. Indicates that this Go Resource will override the role's default when displaying objects of this type (such as "Request", "Issue", "Change Order"). Only one Go Resource can be assigned a dflt_for_obj for each type.

usp_attr_control Table

A dependent_control object specifies the attribute name and locked or required value that activates dependent control.

- **SQL Name** -- usp_attr_control
- **Object** -- att_control

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE REQUIRED KEY
control	control	INTEGER REF	usp_dependent_control	
attrname	attrname	STRING 64		Identifies attribute name.
locked	locked	INTEGER		Identifies locked attribute.
required	required	INTEGER		Identifies required attribute.
last_mod_dt	last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
last_mod_by	last_mod_by	UUID REF	ca_contact	Specifies the UUID of the contact who last modified this record.
del		INTEGER	nn	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
tenant		UUID REF	ca_tenant	

usp_auto_close Table

The auto close object controls the automatic closure of a ticket (request, incident, problem, change order, issue). The usp_auto_close table lists the automatic closure settings. For each ticket type, you can define the number of business hours before the closure takes place. By definition, zero hours means that automatic closure is not implemented for the ticket type.

In an untenanted system only one active row is allowed in this table.

In a tenanted system, each tenant can have its own row and only one row is permitted. In addition, one active Public row is permitted for tenants that do not have a tenanted row. If a tenant does not have its own row of auto-close settings, and there is not an active Public row, the automatic closure feature is disabled for that tenant.

- **SQL Name** -- usp_auto_close
- **Object** -- auto_close

Label	Field	Description
id	INTEGER	Unique key
sym	STRING 80	
cr_ach	INTEGER	
in_ach	INTEGER	
pr_ach	INTEGER	
chg_ach	INTEGER	
iss_ach	INTEGER	

description	STRING 255	
last_mod_dt	LOCAL_TIME	
last_mod_by	UUID REF	ca_tenant
del	INTEGER	nn
tenant	UUID REF	ca_tenant

usp_ci_window Table

The usp_ci_window table stores the associations between windows and configuration items.

- **SQL Name** -- usp_ci_window
- **Object** -- ci_window

Attribute	Data Type	SREL References
id		
nr_uuid	UUID	nr (object)
Window_id	ID	window (object)
producer_id	STRING	
persistent_id	STRING	
last_mod_by	SREL	cnt.id
last_mod_dt	DATE	

usp_email_type Table

The usp_email_type table lists the email types that are available for the Email Type drop-down list (email_type) in the usp_mailbox table.

Field	Data Type	Description
id	INTEGER UNIQUE KEY	Specifies the primary key of this table.
code	STRING 30 UNIQUE S_KEY	Specifies the code value for the email type.
del	INTEGER	
incoming	INTEGER	Specifies whether the email type is incoming or outgoing.
sym	STRING 60	Specifies the symbolic value for the email type.

The following values are the defaults for the usp_email_type table:

id	code	incoming	sym
100	NONE	1	NONE
200	IMAP	1	IMAP
300	POP3	1	POP3
400	SMTP	0	SMTP

usp_export_list_format Table

The usp_export_list_format table lists the file format that is used to export list results outside of CA Service Desk Manager.

- **SQL Name** -- usp_export_list_format
- **Object** -- usp_exlist_format

Label	Field	Description
id	INTEGER	Unique (to the table) Numeric ID.
file_extention	STRING	Specifies the active file format to export search list results. Excel (.xls) is currently supported.
mime_type	STRING	Specifies the internet media type.
xslt_name	STRING	Specifies the XSL file name on the server.
sym	STRING	Specifies the export file format.
delete_flag	INTEGER	Delete indicator.

usp_ical_alarm Table

- **SQL Name** -- usp_ical_alarm
- **Object** -- ical_alarm

Field	Data Type	Reference	Remarks
id	INTEGER		Unique key
del	INTEGER		
alarm_value	STRING 120		
sym	STRING 60		
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID REF	ca_contact	

usp_ical_event_template Table

- **SQL Name** -- usp_ical_event_template
- **Object** -- ical_event_template

Field	Data Type	Reference	
id	INTEGER		Unique key
del	INTEGER		NOT_NULL
obj_type	STRING 30		NOT_NULL
code	STRING 60		NOT_NULL
sym	STRING 60		NOT_NULL
start_date	STRING 30		NOT_NULL

end_date	STRING 30		
alarm	INTEGER	usp_ical_alarm	
categories	STRING 128		
summary	STRING 240		
description	STRING 4000		
url	STRING 4000		
extra_entries	STRING 4000		
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	

usp_owned_resource Table

Holds information about the owned resource.

- **SQL Name** -- usp_owned_resource
- **Object** -- usp_owned_resource

Attribute	DB Field	Data Type	SREL References	Flags
argis_id	nr_argis_id	STRING		
assoc_projex	assoc_projex	BREL	projex.id	
audit_userid	audit_userid	LOCAL SREL	cnt	
assoc_in	assoc_in	QREL	cr	
assoc_pr	assoc_pr	QREL	cr	
bm_rep	nr_bm_rep	INTEGER	busrep id	
bm_label	nr_bmlabel	STRING		
bm_macro_smag	bm_macro_smag	LOCAL STRING 0		
bm_status	nr_bms	INTEGER	busstat status_no	
expiration_date	nr_exp_dt	LOCAL_TIME		
financial_num	nr_financial_id	STRING		
nsm_id	nsm_id	STRING 40		
contact_1	nr_nx_ref_1	UUID	ca_contact uuid	
contact_2	nr_nx_ref_2	UUID	ca_contact uuid	
contact_3	nr_nx_ref_3	UUID	ca_contact uuid	
linked_id_usp_ owned_resource	linked_id_usp_ owned_resource	UUID		
smag_1	nr_nx_string1	STRING		
smag_2	nr_nx_string2	STRING		
smag_3	nr_nx_string3	STRING		
smag_4	nr_nx_string4	STRING		
smag_5	nr_nx_string5	STRING		
smag_6	nr_nx_string6	STRING		

priority	nr_pr_id	INTEGER	pri enum	
name_type	nr_prim_skt_id	INTEGER		
service_type	nr_service_type	STRING	srv_desc code	
sla	nr_sla_id	INTEGER		
warranty_end	nr_wrty_end_dt	LOCAL_TIME		
warranty_start	nr_wrty_st_dt	LOCAL_TIME		
id	owned_resource_uuid	UUID		

USP_Preferences Table

Holds information about the CA SDM and Knowledge Management application preferences.

- **SQL Name** -- usp_preferences
- **Object** -- usp_preferences

Field	Data Type	Reference	Remarks
ANALYST_ID	UUID	ca_contact ::uuid	
ARC_DOCS_TO_ DISPLAY	INTEGER		
ASSIGNEE	INTEGER		
AUTHOR	INTEGER		
BU_RESULT	INTEGER		
CLASSIC_ RESULTSET_ CONTEXT	INTEGER		Specifies the Classic resultset context menu activation.
CREATED_VIA	INTEGER		
CREATION_DATE	INTEGER		Indicates the timestamp indicating when this record was created.
CURRENT_ACTION	INTEGER		
CUSTOM1	INTEGER		
CUSTOM2	INTEGER		
CUSTOM3	INTEGER		
CUSTOM4	INTEGER		
CUSTOM5	INTEGER		
CUSTOM_NUM1	INTEGER		
CUSTOM_NUM2	INTEGER		
DOC_ID	INTEGER		
DOC_TEMPLATE	INTEGER		
DOC_TYPE	INTEGER		
DOC_VERSION	INTEGER		
EXPIRATION_DATE	INTEGER		

GLOBALSD_ ACTIVE_ZONE	INTEGER		Specifies the status of the Global Service Desk Active Zone log reader:**0x01** -- Reduce popups**0x02** -- Close log reader
HITS	INTEGER		
ID	INTEGER NOT_NULL KEY		Specifies the numeric ID that is unique to this table.
INBOX_COUNTER	INTEGER		
INITIATOR	INTEGER		
ITEM	INTEGER		
KT_REPORT_ CARD_PAST_ DAYS	INTEGER		Indicates the Knowledge report card past_days status. This is a user-defined preference.
KT_REPORT_ CARD_SCREEN_ DEFAULT	INTEGER		Indicates the Knowledge report card screen default. this is a user-defined preference.
LAST_ACCEPTED_DATE	INTEGER		
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp for when this record was last modified.
MODIFY_DATE	INTEGER		
ONE_B_DOC_ VIEW_MODE	INTEGER		
ONE_B_DOCS_TO_DISPLAY	INTEGER		
ONE_B_HIDE_ DETAILS	INTEGER		
ONE_B_MATCH_ TYPE	INTEGER		
ONE_B_SEARCH_FIELDS	INTEGER		
ONE_B_SEARCH_ORDER	STRING 255		
ONE_B_SEARCH_TYPE	INTEGER		
ONE_B_WORD_ PARTS	INTEGER		
OWNER	INTEGER		
PRIMARY_INDEX	INTEGER		
PRIORITY	INTEGER		
PRODUCT	INTEGER		
PUBLISHED_DATE	INTEGER		
REVIEW_DATE	INTEGER		
SD_ACCEPTED_ HITS	INTEGER		
SD_IMPACT	INTEGER		
SD_PRIORITY	INTEGER		
SD_ROOTCAUSE	INTEGER		
SD_SEARCH_ FIELDS_CR	INTEGER		Specifies the CA SDM and Knowledge Management search fields for requests.

SD_SEARCH_ FIELDS_ISS	INTEGER		Specifies the CA SDM and Knowledge Management search fields for issues.
SD_SEVERITY	INTEGER		
SD_URGENCY	INTEGER		
START_DATE	INTEGER		
STATUS	INTEGER		
SUBJECT_EXPERT	INTEGER		
USER_DEF_ID	INTEGER		
WEB_LAST_LOGIN	LOCAL_TIME		Indicates the time of the last web login.
WEB_POPUP1_ HEIGHT	INTEGER		Specifies the maximum height for pop-up1.
WEB_POPUP1_ WIDTH	INTEGER		Specifies the maximum width for the Web pop-up1.
WEB_POPUP2_ HEIGHT	INTEGER		Specifies the medium Web pop-up2 height.
WEB_POPUP2_ WIDTH	INTEGER		Specifies the Medium Web pop-up2 width.
WEB_POPUP3_ HEIGHT	INTEGER		Specifies the Small Web pop-up3 height.
WEB_POPUP3_ WIDTH	INTEGER		Specifies the Small Web pop-up3 width.
WEB_POPUP4_ HEIGHT	INTEGER		Specifies the extra Small Web pop-up height.
WEB_POPUP4_ WIDTH	INTEGER		Specifies the extra Small Web pop-up4 width.
WEB_ PREFERENCES	INTEGER		Indicates Web Preferences Flags.
WEB_SUPPRESS_TOUR	INTEGER		Specifies the Web suppressor tour flag: **1** -- Do not Suppress**2** -- Suppress tour page
WEB_TOOLBAR_ TAB	INTEGER		Indicates the initial toolbar tab.
WF_TEMPLATE	INTEGER		

usp_pri_cal Table

Stores priority calculation data.

- **SQL Name** -- usp_pri_cal
- **Object** -- pri_cal

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE NOT_NULL KEY
name	STRING 80		NOT_NULL UNIQUE NOT_NULL S_KEY
description	STRING 1024		
del	INTEGER		NOT_NULL
in_flag	INTEGER	Boolean_Table	
pr_flag	INTEGER	Boolean_Table	
imp_def	INTEGER	Impact	
urg_def	INTEGER	Urgency	
ci_imp	INTEGER	Boolean_Table	
cat_urg	INTEGER	Boolean_Table	
bk_window	INTEGER		
cnt_vip	INTEGER		
pri_5_4	INTEGER	Priority	
pri_5_3	INTEGER	Priority	
pri_5_2	INTEGER	Priority	
pri_5_1	INTEGER	Priority	
pri_5_0	INTEGER	Priority	
pri_4_4	INTEGER	Priority	
pri_4_3	INTEGER	Priority	
pri_4_2	INTEGER	Priority	
pri_4_1	INTEGER	Priority	
pri_4_0	INTEGER	Priority	
pri_3_4	INTEGER	Priority	
pri_3_3	INTEGER	Priority	
pri_3_2	INTEGER	Priority	
pri_3_1	INTEGER	Priority	
pri_3_0	INTEGER	Priority	
pri_2_4	INTEGER	Priority	
pri_2_3	INTEGER	Priority	
pri_2_2	INTEGER	Priority	
pri_2_1	INTEGER	Priority	
pri_2_0	INTEGER	Priority	
pri_1_4	INTEGER	Priority	
pri_1_3	INTEGER	Priority	

pri_1_2	INTEGER	Priority	
pri_1_1	INTEGER	Priority	
pri_1_0	INTEGER	Priority	
pri_0_4	INTEGER	Priority	
pri_0_3	INTEGER	Priority	
pri_0_2	INTEGER	Priority	
pri_0_1	INTEGER	Priority	
pri_0_0	INTEGER	Priority	
cap_reason	INTEGER	Boolean_Table	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_tenant	

usp_properties Table

Provides a list of property/value pairs for CA SDM and Knowledge Management applications.

- **SQL Name** -- usp_properties
- **Object** -- usp_properties

The maximum number of characters (HTML or pure text) allowed in the document's resolution field is 32768 bytes by default. The system Administrator can set this limit based on the type of data being stored. The limit can be set from the Administration tab, Knowledge, Documents, Document Settings.

There is also a built-in limit of 32768 bytes for the document's pure text that will be indexed. If the resolution of a document is larger than the set limit, only the first 32768 bytes of the document will be indexed and the rest will be ignored.

Field	Data Type	Reference	Remarks
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID.
auto_policies_last	Date		Displays the *last* date and time when automated policies were run on the server; stored in seconds, in UNIX time format.
auto_policies_next_ interval	Number		Identifies the number of days between automated calculation batches.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
notify_days_bef_doc_ exp	Number		Defines the number of days before the document expires and a notification is sent.
property_default	STRING 32768		Identifies the property default.
property_description	STRING 255		Identifies the property description.

property_name	STRING 100 S_KEY		Identifies the property name.
property_type	STRING 100		Identifies the property type.
property_value	STRING 32768		Identifies the property value.
suggest_knowledge_ for _issue_cats	INTEGER		Flag that determines whether or not to suggest knowledge to customers. **1** -- Suggest knowledge by default for all issue categories. **0** -- Do not suggest knowledge.

usp_notification_phrase Table

The usp_notification_phrase table lists common phrases that notification message templates can use.

- **S QL Name** -- usp_notification_phrase
- **Object** -- notification_phrase

Field	Data Type	References	Description
id	INTEGER UNIQUE KEY		Specifies the primary key of this table.
del	INTEGER REF		
last_mod_dt	LOCAL_TIME		Specifies the time stamp of when this record was last modified.
last_mod_by	UUID REF	ca_contact	Specifies the UUID of the contact who last modified this record.
code	STRING UNIQUE S_KEY		Specifies the code name for the notification phrase that identifies the specific phrase when referenced in other text.
sym	STRING		Specifies the display name for the notification phrase.
description	STRING		Describes the notification phrase.
phrase	STRING		Specifies the phrase text to use for notification.

usp_organization Table

This table provides extensions to the ca_organization table that are used only by CA SDM products.

- **SQL Name** -- usp_organization
- **Object** -- org

Field	Data Type	Reference	Remarks
iorg_assigned_svr	INTEGER		Deprecated.

iorg_service_type	nvarchar(30)	Service_Desc::code	Foreign key to the code field of the srv_desc table, this is the Classic Service Type.
last_mod	INTEGER		Indicates the timestamp of when this record was last modified.
organization_uuid	byte(16)		Primary key. Unique identifier.
owning_contract	INTEGER	Service_Contract::id	Identifies the unique (to the table) numeric ID.

Form_Group Table

Listing of defined form groups.

- **SQL Name** -- fmgrp
- **Object** -- fmgrp

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
description	nvarchar(100)		Specifies the textual description of this form group.
id	INTEGER		Primary key of this table, it is a unique, numeric ID.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
sym	nvarchar(30)		Specifies the symbolic value for this Form_Group.

True_False_Table Table

Contains localized True or False strings that display on the UI.

- **SQL Name** -- True_False_Table
- **Object** -- true_false

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE NOT_NULL KEY Unique to the table Numeric ID.
del	INTEGER		NOT_NULL Deleted flag: **0** -- Active **1** -- Inactive/marked as deleted

enum	INTEGER		NOT_NULL Enumerated value for this entry **0** -- False **1** -- True
sym	STRING 60		UNIQUE NOT_NULL S_KEY Identifies the symbolic value for this target
desc	STRING 40		Describes the enum

Access Level and Type

This article contains the following information:

Access_Levels Table

Access level definitions for CA SDM applications.

- **SQL Name** -- acc_lvls
- **Object** -- acc_lvls

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_Table::enum	Deleted flag:**0** -- Active **1 --** Inactive/marked as deleted)
enum	INTEGER		Indicates the enumerated value for this entry, it specifies ordering in lists and relative values **Note:** This is a primary key.
id	INTEGER		Unique (to the table) Numeric ID.
nx_desc	nvarchar(40)		Specifies the description or access level.
persid	nvarchar(30)		Specifies the Persistent ID (SystemObjectName:id).
sym	nvarchar(12)		Specifies the symbolic name of the level.

Access_Type_v2 Table

Access type information for the CA SDM applications.

- **SQL Name** -- acctyp_v2
- **Object** -- acctyp

Field	Data Type	Reference	Remarks
access_level	INTEGER	Access_Levels::enum	Enumerated value for this entry, it specifies ordering in lists and relative values. **Note:** This is a foreign key.
config	INTEGER		Indicates that it is authorized for the configuration interface.
default_flag	INTEGER		Specifies the default flag value for this Access Type.
del	INTEGER	Active_Boolean_ Table::enum	Indicates the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted)
description	nvarchar(100)		Describes the textual description of this Access Type
external_auth	INTEGER		Allows external authorization for this Access Type.
grant_level	INTEGER	Access_Levels::enum	Enumerated value for this entry, this specifies ordering in lists and relative values. **Note:** This is a foreign key.
id	INTEGER		Specifies the primary key of this table.
last_mod_by	Byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
ldap_access_ group	nvarchar(512)		Identifies the ldap access group value for this Access Type.
pin_field	nvarchar(50)		Specifies the field that contains the user's pin for pin type authentication.
sym	nvarchar(30)		Indicates the symbolic value for this Access Type.
user_auth	INTEGER		Identifies the user authentication type for this Access Type.
view_internal	INTEGER		Controls access to internal logs.
wsp	INTEGER		Indicates the Web Screen Painter authorization level.
cmdline_role	INTEGER	role::id	Specifies the id of the role that is used for command line utilities.

reporting_role	INTEGER	role::id	Specifies the id of the role that is used for reporting.
web_service_ role	INTEGER	role::id	Specifies the id of the role that is used for web services.
rest_web_service_role	INTEGER	usp_role	

Activities

This article contains the following topics:

Act_Log Table

Act_Log is a history of activities associated with a call request. The types of activities are listed in the Act_Type table.

- **SQL Name** -- act_log
- **Object** -- alg

Field	Data Type	Reference	Remarks
action_desc	nvarchar(4000)		Provides the textual description of the activity log entry.
analyst	byte(16)	ca_contact::uuid	Foreign key to the Contact that created this activity log entry.
call_req_id	nvarchar(30)	Call_Req::persid	This is the Persistent ID (SystemObjectName:id).
description	nvarchar (4000)		Indicates the textual description of this Activity Log.
id	INTEGER		Identifies the unique (to the table) numeric ID.
internal	INTEGER		Marks this as Internal log.
knowledge_ session	nvarchar(80)		Indicates the reference to a knowledge tool session.
knowledge_ tool	nvarchar(12)		Specifies the Knowledge tool used for this activity.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
persid	STRING 30		Persistent ID (SystemObjectName:id).
system_time	INTEGER		Indicates the Date and Time of record creation.
time_spent	INTEGER		Specifies the time spent on this activity by the user.
time_stamp	INTEGER		Specifies the Date and Time of the user activity.

type	nvarchar(12)	Act_Type::code	(Not Used) Specifies the acknowledgement, which is a noneditable string enum.**Note:** This is a foreign key.

Act_Type Table

Identifies the activities which may be used to create a history of a ticket. Controls the creation of the activity automatically, generates a notification to those contacts specified on the call request, and controls the notification level and message.

- **SQL Name** -- act_type
- **Object** -- aty

> **NOTE**
> Adding an activity type requires customization to allow the activity to be part of a ticket history.

Field	Data Type	Reference	Remarks
id	INTEGER		Indicates the numeric ID that is unique to this table.
persid	nvarchar(30)		Specifies the Persistent ID: (SystemObjectName:id).
code	nvarchar(12)		This is the primary key of this table.
del	INTEGER	Active_Boolean_Table::enum	Specifies the status of the Deleted flag:**0** -- Active**1** -- Inactive/marked as deleted
sym	nvarchar(60)		Describes the name of the activity.
description	nvarchar(500)		Specifies the textual description of this Activity Type.
flag1	INTEGER	Boolean_Table::enum	Specifies the flag1 value: **1** -- Used by call requests.
flag2	INTEGER	Boolean_Table::enum	Specifies the flag2 value: **1** -- Used by change requests.
flag3	INTEGER	Boolean_Table::enum	Specifies the flag3 value: **1** -- Used by issues
flag4	INTEGER	Boolean_Table::enum	Specifies the flag4 value: **1** -- Used by messages
flag5	INTEGER	Boolean_Table::enum	Specifies the flag5 value: **1** -- Used by Knowledge Management
flag6	INTEGER	Boolean_Table ::enum	Specifies the flag6 value: **1** -- Used by Support Automation

krc_flag	INTEGER	Boolean_Table ::enum	Specifies the flag7 value: **1** -- Used by the Knowledge Report Card
sa_notif_flag	INTEGER	Boolean_Table ::enum	Specifies the flag8 value: **1** -- Used by Support Automation assistance sessions
cr_notifiy_info	nvarchar(30)		Specifies the request information link to macros for notify.
chg_notify_info	nvarchar(30)		Specifies the change order information link to macros for notify.
iss_notify_info	nvarchar(30)		Specifies the issue information link to macros for notify.
msg_notify_info	nvarchar(30)	Spell_Macro::persid	Specifies the message information link to macros for notify.
kd_notify_info	nvarchar(30)		Specifies the knowledge document information link to macros for notify.
kd_comment_ notify_info	nvarchar(30)		Specifies the knowledge document comment information link to macros for notify.
krc_notify_info	nvarchar(30)		Specifies the knowledge report card information link to macros for notify.
sa_notif_info	nvarchar(30)		Specifies the support automation information link to macros for notify.
cr_send_survey	INTEGER		Specifies the request information link to macros for notify.
cr_survey_ msgbody	Long Varchar		Specifies the body text of the survey.
cr_default_ survey	INTEGER	Survey_ Template	Specifies the default template used for the survey.
cr_survey_ msgtitle	nvarchar(80)		Specifies the title of the survey.
cr_survey_ method	INTEGER	Contact_ Method	Specifies the contact method used in the survey.
chg_send_ survey	INTEGER		
chg_survey_ msgbody	Long Varchar		Specifies the body text of the survey.
chg_default_survey	INTEGER	Survey_ Template	Specifies the default template used for the survey.
chg_survey_msgtitl	nvarchar(80)		Specifies the title of the survey.

chg_survey_method	INTEGER	Contact_ Method	Specifies the contact method used in the survey.
iss_send_survey	INTEGER		
iss_survey_msgbody	Long Varchar		Specifies the body text of the survey.
iss_default_survey	INTEGER	Survey_ Template	Specifies the default template used for the survey.
iss_survey_msgtitl	nvarchar(80)		Specifies the title of the survey.
iss_survey_method	INTEGER	Contact_ Method	Specifies the contact method used in the survey.
krc_send_survey	INTEGER		
krc_survey_ msgbody	Long Varchar		Specifies the body text of the survey.
krc_default_survey	SREL	svy_tpl	Specifies the default template used for the survey.
krc_survey_msgtitle	nvarchar(80)		Specifies the title of the survey.
krc_survey_method	SREL	cmth	Specifies the contact method used in the survey.
sa_send_survey	INTEGER		
sa_survey_msgbody	Long Varchar		Specifies the body text of the survey.
sa_default_survey	SREL	svy_tpl	Specifies the default template used for the survey.
sa_survey_msgtitle	nvarchar(80)		Specifies the title of the survey.
sa_survey_method	SREL	cmth	Specifies the contact method used in the survey.
internal	INTEGER		Specifies the activity type:**0** -- User activity**1** -- Internal
last_mod_dt	DATE		Indicates the timestamp of when this record was last modified.
notify	INTEGER		Specifies whether to generate notifications:**0** -- No**1** -- Yes
notify_level	INTEGER	Notification_ Urgency	Indicates the Notification level.
notify_ack	nvarchar(240)		Indicates the Notification message.
notify_body	Long nvarchar		Indicates the Notification message body.
notify_body_html	Long nvarchar		Specifies the Notification message body html.
notify_title	nvarchar(80)		Specifies the Notification message title.
ci_flag	INTEGER		Specifies the CI flag value.
cnt_flag	INTEGER		Specifies the contact flag value.

ci_notify_info	nvarchar(30)		
cnt_notify_info	nvarchar(30)		
rel_ticket	INTEGER		Specifies the related ticket of the activity.
single_tenant	UUID	ca_tenant	Specifies the UUID of the single tenant.
tenant_group	UUID	ca_tenant	Specifies the UUID of the tenant group.
tenant	UUID	ca_tenant	Specifies the UUID of the tenant.

Act_Type_Assoc Table

Maps a field of an object we want to do activity logging for to an Activity Type (Assignee field on a request is mapped to 'Transfer' Act_Type. All _rsrved fields are reserved for CA and NOT FOR CLIENT USE)

- **SQL Name** -- atyp_asc
- **Object** -- act_type_assoc

Field	Data Type	Reference	Remarks
act_type	STRING 30	Act_Type::code	Associated activity type
code	STRING 12		UNIQUE NOT_NULL S_KEY
description	STRING 80		Textual description of this Act Type Association
id	INTEGER		Unique (to the table) Numeric ID UNIQUE NOT_NULL KEY
int1_rsrved	INTEGER		Flex field
int2_rsrved	INTEGER		Flex field
int3_rsrved	INTEGER		Flex field
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified
log_me_f	INTEGER		Logging on/off flag
ob_type	STRING 30		Associated object type
ob_type_attr	STRING 50		Attribute name within associated object
persid	STRING 30		Persistent ID (SystemObjectName:id)
str1_rsrved	STRING 30		Flex fields - reserved for CA
str2_rsrved	STRING 30		Flex field
str3_rsrved	STRING 30		Flex field
sym	STRING 30 NOT_NULL		

Boolean

This article contains the following topics:

Active_Boolean_Table Table

Table to translate 0 and 1 for the del field on records.

- **SQL Name** -- actbool
- **Object** -- actbool

Field	Data Type	Reference	Remarks
del	INTEGER NOT_NULL		**0** -- Active **1** -- Inactive/marked as deleted
description	STRING 240		Textual description of this boolean value
enum	INTEGER NOT_NULL		Enumerated value for this entry
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified
sym	STRING 12 S_KEY		

Active_Reverse_Boolean_Table Table

Table to translate 0 and 1 for the del field on records.

- **SQL Name** -- acrtbool
- **Object** -- actrbool

Field	Data Type	Reference	Remarks
del	INTEGER NOT_NULL		**0** -- Active **1** -- Inactive/marked as deleted
description	STRING 240		Specifies the textual description of this boolean value
enum	INTEGER NOT_NULL		Enumerated value for this entry
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified
sym	STRING 12 S_KEY		

Boolean_Table Table

Table of boolean values that allows the customer to define the text associated with true and false.

- **SQL Name** -- bool_tab
- **Object** -- bool

Field	Data Type	Reference	Remarks
del	INTEGER NOT_NULL		Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
enum	INTEGER NOT_NULL		Enumerated value for this entry - specifies ordering in lists and relative values
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
nx_desc	STRING 40		
sym	STRING 12 UNIQUE NOT_NULL S_KEY		

Reverse_Boolean_Table Table

Reverse boolean lookup table.

- **SQL Name** -- rbooltab
- **Object** -- rev_bool

Field	Data Type	Reference	Remarks
del	INTEGER NOT_NULL		Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
enum	INTEGER NOT_NULL		Enumerated value for this entry - specifies ordering in lists and relative values
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
nx_desc	STRING 40		
sym	STRING 12 UNIQUE NOT_NULL S_KEY		

Asset

This article contains the following topics:

Am_Asset_Map Table

Maps a CA Asset Management asset to an internal object am_domain_id and am_unit_id to form a unique AM asset identifier. NOT FOR CLIENT USE.

- **SQL Name** -- am_map
- **Object** -- am_asset_map

Field	Data Type	Reference	Remarks
am_dmuuid	STRING 64		AM DMUUID of asset
am_domain_id	INTEGER		AM asset domain identifier which created asset
am_server	STRING 64		AM name of UAM domain server
am_type	INTEGER		AM asset type **1** -- Computer **2** -- User
am_unit_domain_ id	INTEGER		AM asset domain identifier of asset
am_unit_id	INTEGER		AM asset unit id
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
int1_rsrved	INTEGER		Reserved for CA
int2_rsrved	INTEGER		Reserved for CA
ob_persid	STRING 30		
ob_type	STRING 30		
persid	STRING 30		Persistent ID: SystemObjectName:id
str1_rsrved	STRING 80		Reserved for CA
str2_rsrved	STRING 80		Reserved for CA
version	INTEGER NOT_NULL		Internal version

Asset_Assignment Table

Describes a relationship between two assets. Each instance of this table is one parent-child arrangement.

- **SQL Name** -- hier
- **Object** -- hier

Field	Data Type	Reference	Remarks
hier_child	UUID NOT_NULL S_KEY	ca_owned_ resource:: uuid	Specifies the child.
hier_license_ num	STRING 40		Specifies the license serial number.
hier_log_date	LOCAL_TIME NOT_NULL		Specifies the time of assignment.
hier_parent	UUID NOT_NULL S_KEY	ca_owned_ resource:: uuid	Specifies the parent.

id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
persid	STRING 30		Persistent ID (SystemObjectName:id)

Archive and Purge

This article contains the following topics:

Archive_Purge_History Table

Historic information about archive/purge activities completed.

- **SQL Name** -- arcpur_hist
- **Object** -- arcpur_hist

Field	Data Type	Reference	Remarks
chd_obj_ deleted	INTEGER		
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
end_time	LOCAL_TIME		
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified
obj_deleted	INTEGER		
persid	STRING 30		Persistent ID (SystemObjectName:id)
rule_name	STRING 30 NOT_NULL	Archive_Purge_ Rule::persid	
start_time	LOCAL_TIME		

Archive_Purge_Rule Table

Rule definitions for the archive/purge engine.

- **SQL Name** -- arcpur_rule
- **Object** -- arcpur_rule

Field	Data Type	Reference	Remarks
add_query	STRING 240		
arc_file_name	STRING 240		
conf_obj_name	STRING 64		
days_inactive	INTEGER NOT_NULL		
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
name	STRING 30 NOT_NULL UNIQUE NOT_NULL S_KEY		Specifies the text name of this item.
oper_type	INTEGER NOT_NULL		
persid	STRING 30		Persistent ID (SystemObjectName:id)
reoccur_interv	DURATION		
sched	STRING 30	Bop_Workshift ::persid	

Attached_Events Table

This article contains the following topics:

This table lists the actual attached events and their information.

- **SQL Name** -- att_evt
- **Object** -- atev

Field	Data Type	Reference	Remarks
cancel_time	LOCAL_TIME		Time canceled
event_tmpl	STRING 30 NOT_NULL S_KEY	Events:: persid	Specifies the persid for the attached event.
fire_time	LOCAL_TIME		Datetime event will fire.
first_fire_time	LOCAL_TIME		Datetime event first fired.
group_name	STRING 30		Group smag field.
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.

num_fire	INTEGER		Number of times the event fired.
obj_id	STRING 30 NOT_NULL		The persid of the object.
owning_ast	INTEGER	Attached_ SLA::id	Owning ast object (CA Service Desk Manager Version 11.0).
persid	STRING 30		Persistent ID (SystemObjectName:id).
start_time	LOCAL_TIME		Datetime when the event was attached.
status_flag	INTEGER		
user_smag	STRING 200		User smag field
wait_time	DURATION		Time to wait before firing

Attached_SLA Table

Service Level Agreements associated to a ticket.

- **SQL Name** -- attached_sla
- **Object** -- attached_sla

Field	Data Type	Reference	Remarks
_mapped_chg	INTEGER	chg id	
_mapped_cr	STRING 30	Call_Req::persid	These are mostly required for advanced runtime queries.
_mapped_iss	STRING 30	issue persistent_id	
_mapped_iss_wf	INTEGER	isswf::id	
_mapped_wf	INTEGER	wf::id	
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
last_ttv_upd	LOCAL_TIME		
map_sdsc	STRING 30 NOT_NULL	Service_Desc:: code	
persid	STRING 30		Persistent ID (SystemObjectName:id)
sla_viol_status	INTEGER		
ticket_id	INTEGER NOT_NULL S_KEY		
ticket_type	STRING 30 NOT_NULL		
time_to_ violation	LOCAL_TIME		

ttv_event	STRING 30	Attached_Events::persid	

Attachment Table

Object Attachments. (5.0 version)

- **SQL Name** -- attmnt
- **Object** -- attmnt

Field	Data Type	Reference	Remarks
attmnt_name	STRING 240		Specifies the attachment name.
created_by	UUID	ca_contact::uuid	Specifies who created this attachment.
created_dt	LOCAL_TIME		Specifies when the attachments was created.
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
description	STRING 500		Specifies the textual description of this attachment.
exec_cmd	STRING 12	Remote_Ref::code	Specifies the unix exec string (not currently used).
file_date	LOCAL_TIME		Specifies the date of the file.
file_name	STRING 240		Specifies the server attachment filename.
file_size	INTEGER		Specifies the size of the file.
file_type	STRING 12		Specifies the file extension (not currently used).
folder_id	INTEGER	attmnt_folder::id	Specifies the folder id.
folder_path_ids	STRING 255		Specifies the folder path ids.
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
inherit_ permission_id	INTEGER		Specifies the folder ID where pgroup permissions come from.
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
link_only	INTEGER	Boolean_Table ::enum	
link_type	String		If embedded image = EMBD_IMG, otherwise NULL
orig_file_name	STRING 240		original file name, URL or UNC

persid	STRING 30		Persistent ID (SystemObjectName:id)
read_pgroup	INTEGER	P_GROUPS::id	Specifies the group of groups eligible to read the document.
rel_file_path	STRING 512		Specifies the relative path to the file.
repository	SREL	Document_ Repository::persid	Specifies the repository.
status	STRING 12		Specifies the attachment status (INSTALLED, INSTALL_FAILED, LINK_ONLY).
write_pgroup	INTEGER	P_GROUPS::id	Specifies the group of groups eligible to update the document.
zip_flag	INTEGER		Specifies if the file is zipped.

attmnt_folder Table

List of attachment repository locations.

- **SQL Name** -- attmnt_folder
- **Object** -- attmnt_folder

Field	Data Type	Reference	Remarks
description	STRING 500		Textual description of this attachment folder
folder_name	STRING 255		Specifies the folder name
folder_path_ ids	STRING 255		Specifies the folder path ids
folder_type	INTEGER		Specifies the folder type
has_children	INTEGER		Specifies if the folder has children
id	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
inherit_ permission_ id	INTEGER	attmnt_folder::id	Specifies the id of folder to inherit from
last_mod_ date	LOCAL_TIME		Specifies the last modify date
parent_id	SREL	attmnt_folder::id	Specifies the parent folder id
read_pgroup	INTEGER	P_GROUPS::id	Read p Group permissions of the attachment
repository	SREL	Document_ Repository:: persid	Specifies the repository pers id
write_pgroup	INTEGER	P_GROUPS::id	Write p Group permissions of the attachment
tenant	UUID	ca_tenant	Specifies the UUID of the tenant.

Attachment

This article contains the following topics:

Attachment Table

Object Attachments. (5.0 version)

- **SQL Name** -- attmnt
- **Object** -- attmnt

Field	Data Type	Reference	Remarks
attmnt_name	STRING 240		Specifies the attachment name.
created_by	UUID	ca_contact::uuid	Specifies who created this attachment.
created_dt	LOCAL_TIME		Specifies when the attachment was created.
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
description	STRING 500		Specifies the textual description of this attachment.
exec_cmd	STRING 12	Remote_Ref::code	Specifies the unix exec string (not currently used).
file_date	LOCAL_TIME		Specifies the date of the file.
file_name	STRING 240		Specifies the server attachment filename.
file_size	INTEGER		Specifies the size of the file.
file_type	STRING 12		Specifies the file extension (not currently used).
folder_id	INTEGER	attmnt_folder::id	Specifies the folder id.
folder_path_ids	STRING 255		Specifies the folder path ids.
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
inherit_ permission_id	INTEGER		Specifies the folder ID where pgroup permissions come from.
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
link_only	INTEGER	Boolean_Table ::enum	
link_type	String		If embedded image = EMBD_IMG, otherwise NULL
orig_file_name	STRING 240		Original file name, URL or UNC

persid	STRING 30		Persistent ID (SystemObjectName:id)
read_pgroup	INTEGER	P_GROUPS::id	Specifies the group of groups eligible to read the document.
rel_file_path	STRING 512		Specifies the relative path to the file.
repository	SREL	Document_ Repository::persid	Specifies the repository.
status	STRING 12		Specifies the attachment status (INSTALLED, INSTALL_FAILED, LINK_ONLY).
write_pgroup	INTEGER	P_GROUPS::id	Specifies the group of groups eligible to update the document.
zip_flag	INTEGER		Specifies if the file is zipped.

attmnt_folder Table

List of attachment repository locations.

- **SQL Name** -- attmnt_folder
- **Object** -- attmnt_folder

Field	Data Type	Reference	Remarks
description	STRING 500		Textual description of this attachment folder
folder_name	STRING 255		Specifies the folder name
folder_path_ ids	STRING 255		Specifies the folder path ids
folder_type	INTEGER		Specifies the folder type
has_children	INTEGER		Specifies if the folder has children
id	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
inherit_ permission_ id	INTEGER	attmnt_folder::id	Specifies the id of folder to inherit from
last_mod_ date	LOCAL_TIME		Specifies the last modify date
parent_id	SREL	attmnt_folder::id	Specifies the parent folder id
read_pgroup	INTEGER	P_GROUPS::id	Read p Group permissions of the attachment
repository	SREL	Document_ Repository:: persid	Specifies the repository pers id
write_pgroup	INTEGER	P_GROUPS::id	Write p Group permissions of the attachment
tenant	UUID	ca_tenant	Specifies the UUID of the tenant.

usp_lrel_attachments_changes Table

Relates attachments to change orders.

- **SQL Name** -- usp_lrel_attachments_changes
- **Object** -- lrel_attachments_changes

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
chg	INTEGER	Change_Request	
attmnt	INTEGER	Attachment	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_attachments_issues Table

Relates attachments to issues.

- **SQL Name** -- usp_lrel_attachments_issues
- **Object** -- lrel_attachments_issues

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
iss	STRING 30	Issue	
attmnt	INTEGER	Attachment	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_attachments_requests Table

Relates attachments to requests, problems, or incidents.

- **SQL Name** -- usp_lrel_attachments_requests
- **Object** -- lrel_attachments_requests

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
cr	STRING 30	Call_Req	
attmnt	INTEGER	Attachment	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

Document_Repository Table

Contains Information on document repositories, which are used to store attachments.

- **SQL Name** -- doc_rep
- **Object** -- doc_rep

Field	Data Type	Reference	Remarks
description	STRING 500		Specifies the description.
archive_path	STRING 255		
archive_type	INTEGER		
cgi_path	STRING 255		Specifies the location and name of CGI.
default_rep	INTEGER		
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
file_limit_size	INTEGER		Specifies the file limit size.
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
persid	STRING 30		Persistent ID (SystemObjectName:id)
prohibited_ext	STRING 500		Specifies the prohibited file extensions.
protocol	STRING 12		HTTP or SHARE
repository_type	INTEGER		Specifies the type of repository (attachments, knowledge).
retrieve_path	STRING 255		Specifies how to get back to upload_path via protocol.
server	STRING 30		Specifies the name of Doc Server.
servlet_path	STRING 255		Specifies the servlet URL.
sym	STRING 30 NOT_NULL S_KEY		Specifies the name of document repository.
upload_path	STRING 255		Specifies the server location of doc repository.

Company (CA MDB)

This article contains the following topics:

ca_company Table

Company information (CA-MDB).

- **SQL Name** -- ca_company
- **Object** -- ca_cmpny

Field	Data Type	Reference	Remarks
alias	nvarchar(30)		Identifies the Company alias or "also known as" name. For example, CA.
authentication_ password	nvarchar(20)		Specifies the authentication password.
authentication_ user_name	nvarchar(64)		Specifies the authentication user name.
bbs	nvarchar(30)		Identifies bulletin board system information.
company_name	nvarchar(100)		Identifies the company name.
company_type	INTEGER	ca_company_ type::id	Identifies the Company Type or Vendor Provider. **Note:** This is the Foreign key to the id field of the ca_company_type table.
company_uuid	Byte(16)		The Primary key of this table.
creation_date	INTEGER		Indicates the date this record was created.
creation_user	nvarchar(64)		Identifies the user or process that created the record.
delete_time	INTEGER		Specifies the delete time.
description	nvarchar(400)		Shows the textual description of this entry.
exclude_registration	INTEGER		Indicates to exclude registration.
inactive	INTEGER	Active_Boolean_Table::enum	Flag representing whether this record is active or inactive.**0** -- False (Active)**1** -- True (Inactive)
last_update_date	INTEGER		Indicates the date that this record was last updated.
last_update_user	nvarchar(64)		Identifies the user or process that last updated the record.
location_uuid	Byte(16)	ca_location::location_uuid	Identifies the location.**Note:** This is the Foreign key to the ca_location table entry.
month_fiscal_year_ends	INTEGER		Specifies the Integer value (1-12) representing the month the company's fiscal year ends.
parent_company_ uuid	Byte(16)	ca_company::company_uuid	This is the Foreign key to the company_uuid field of the ca_company table for the company's parent company.

primary_contact_ uuid	Byte(16)	ca_contact::uuid	Specifies the Primary contact. **Note:** This is the Primary foreign key to the ca_contact table.
source_type_id	INTEGER		Represents the Source type id.**Note:** This is the Foreign key to the ca_source_type table.
version_number	INTEGER		Specifies the version number for transaction integrity.
web_address	nvarchar(50)		Identifies the company web address.

ca_company_type Table

Company type information (CA-MDB).

- **SQL Name** -- ca_company_type
- **Object** -- vpt

Field	Data Type	Reference	Remarks
creation_date	INTEGER		Specifies the date this record was created.
creation_user	nvarchar(64)		Identifies the user or process that created the record.
description	nvarchar(255)		Provides the textual company type description of this entry.
id	INTEGER		Specifies the Primary key, this also identifies the unique company type numeric ID.
inactive	INTEGER	Active_Boolean_ Table::enum	Specifies Active flag, as follows:**0** -- False (Active)**1** -- True (Inactive)
last_update_date	INTEGER		Specifies the date when this record was last updated.
last_update_user	nvarchar(64)		Specifies the user or process that updated the record.
name	nvarchar(100)		Identifies the unique, company type name.
version_ number	INTEGER		Specifies the version number for transaction integrity

Contact (CA MDB)

This article contains the following topics:

ca_contact Table

Table of persons that interact with the system in some manner (CA-MDB).

- **SQL Name** -- ca_contact
- **Object** -- cnt

Field	Data Type	Reference	Remarks
admin_organization_uuid	Byte(16)	ca_organization ::organization_uuid	Identifies the Administrative Organization.**Note:** This is the Foreign key to the ca_organization table.
alias	nvarchar(30)		Identifies the Contact alias or "known as" name.
alt_phone_number	nvarchar(40)		Specifies the alternate phone number.
alternate_identifier	nvarchar(30)		Identifies the alternate contact (for example, the social security number).
comment	nvarchar(255)		Shows the text of the comment.
company_uuid	Byte(16)	ca_company::company_uuid	Identifies the Company.**Note:** This is the Foreign key to the ca_company table.
contact_type	INTEGER	ca_contact_type::id	Specifies the Contact type, such as: Customer, Analyst, and so on.**Note:** This is the Foreign key to the id field of the ca_contact_type table.
contact_uuid	Byte(16)		This is the Primary key, and is a unique identifier.
cost_center	INTEGER	ca_resource_cost_ center::id	Identifies the cost center.**Note:** This is the Foreign key to the id field of the ca_resource_cost_center table.
creation_date	INTEGER		Identifies the date that the record was created.
creation_user	nvarchar(64)		Specifies the user or process that created this record.
delete_time	INTEGER		Shows the time of deletion.
department	INTEGER	ca_resource_ department::id	Identifies the department.**Note:** This is the Foreign key to the id field of the ca_resource_department table.
email_address	nvarchar(120)		Identifies the email address.
exclude_registration	INTEGER		Indicates to exclude registration.
fax_number	nvarchar(40)		Identifies the fax number.
first_name	nvarchar(100)		Identifies the first name.

floor_location	nvarchar(30)		Identifies the floor location. (for example, the employee works on the first floor).
inactive	INTEGER	Active_Boolean_ Table::enum	Specifies the Active flag, as follows:**0** -- Active**1** -- Inactive
inrdid	INTEGER		Identifies the role ID.
job_function	INTEGER		Specifies the job function.**Note:** This is the Foreign key to ca_job_function table.
job_title	INTEGER	ca_job_title::id	Specifies the job title.**Note:** This is a Foreign key.
last_name	nvarchar(100)		(Required) Identifies the last name.
last_update_date	INTEGER		Shows the date this record was last updated.
last_update_user	nvarchar(64)		Specifies the user or process that last updated the record.
location_uuid	Byte(16)	ca_location:: location_uuid	Specifies the location.**Note:** This is the Foreign key to the ca_location table entry.
mail_stop	nvarchar(30)		Identifies the mail stop.
middle_name	nvarchar(100)		Identifies the middle name.
mobile_phone_number	nvarchar(40)		Specifies the mobile phone number.
organization_uuid	Byte(16)	ca_organization::organization_uuid	Specifies the Organization.**Note:** This is a Foreign key to the id field of the ca_organization table.
pager_email_address	nvarchar(120)		Identifies the pager email address.
pager_number	nvarchar(40)		Identifies the pager number.
pri_phone_number	nvarchar(40)		Identifies the primary phone number.
room_location	nvarchar(30)		Identifies the Room location (for example, the employee works in Cube 123).
supervisor_contact_uuid	Byte(16)	ca_contact::uuid	Identifies the Supervisor.**Note:** This is a Foreign key to the ca_contact table.
userid	nvarchar(100)		Specifies the User account id (for example, a company employee ID). This ID is unique for active records in order to prevent a user from retrieving the security settings of another user.

version_number	INTEGER		Version number for transaction integrity.

ca_contact_type Table

Definitions of type/classifications of personal information stored in the ca_contact table (CA-MDB).

- **SQL Name** -- ca_contact_type
- **Object** -- ctp

Field	Data Type	Reference	Remarks
creation_date	INTEGER		Specifies the date when this record was created.
creation_user	nvarchar(64)		Identifies the user or process that created this record.
delete_time	INTEGER		Specifies the delete time.
description	nvarchar(255)		Specifies the textual, contact type description.
exclude_ registration	INTEGER		Indicates to exclude registration.
hourly_cost	Money		Specifies the hourly cost.
id	INTEGER		Unique (to the table), numeric contact type ID.**Note:** This is a Primary key.
inactive	INTEGER	Active_Boolean_Table::enum	Identifies the Active flag, as follows:**0** -- False (Active)**1 --** True (Inactive)
last_update_date	INTEGER		Specifies the date of when this record was last updated.
last_update_user	nvarchar(64)		Identifies the user or process that last updated the record.
name	nvarchar(100)		Identifies the unique,contact type name.
user_uuid	UUID	ca_contact:: uuid	This is a unique identifier.**Note:** This is also the Primary key.
version_number	INTEGER		Version number for transaction integrity.
view_internal	INTEGER		Flag to represent whether this contact type is allowed to view internal data.

Job (CA MDB)

This article contains the following topics:

ca_job_function Table

Job function descriptions (CA-MDB).

- **SQL Name** -- ca_job_function
- **Object** -- job_func

Field	Data Type	Reference	Remarks
creation_date	INTEGER		Specifies the date when this record was created.
creation_user	nvarchar(64)		Specifies the user or process that created the record.
delete_time	INTEGER		Identifies the delete time.
description	nvarchar(255)		Specifies the Job function description.
exclude_registration	INTEGER		Indicates to exclude registration.
id	INTEGER		Identifies the unique (to the table) numeric job function ID.**Note:** This is a Primary key.
inactive	INTEGER	Active_Boolean_ Table::enum	Sets the Active flag, as follows:**0** -- False (Active)**1** -- True (Inactive)
last_update_date	INTEGER		Specifies the date of when this record was last updated.
last_update_user	varchar(64)		Identifies the user or process that last updated the record.
name	varchar(100)		Identifies the job function name.
version_number	INTEGER		Specifies the version number for transaction integrity.

ca_job_title Table

Job title descriptions (CA-MDB).

- **SQL Name** -- ca_job_title
- **Object** -- position

Field	Data Type	Reference	Remarks
creation_date	INTEGER		Specifies the date of when this record was created.
creation_user	nvarchar(64)		Identifies the user or process that created the record.
delete_time	INTEGER		Indicates the time of deletion.
exclude_registration	INTEGER		Indicates to exclude registration.

id	INTEGER		Identifies the unique (to the table) numeric job title ID.**Note:** This is a Primary key.
inactive	INTEGER	Active_Boolean_ Table::enum	Specifies the Active flag, as follows:**0** -- False (Active)**1** -- True (Inactive)
last_update_date	INTEGER		Indicates the date of when this record was last updated.
last_update_user	varchar(64)		Specifies the user or process that last updated the record.
name	varchar(100)		Specifies the job title name.
version_number	INTEGER		Specifies the version number for transaction integrity.

Organization

ca_organization Table

Used to describe a business, or associate businesses with smaller business units, or contacts to businesses.

- **SQL Name** -- ca_organization
- **Object** -- org

Field	Data Type	Reference	Remarks
abbreviation	nvarchar(30)		Identifies the abbreviation of the organization.
alt_phone_cc	INTEGER		Identifies the alternate phone number country code.
alt_phone_number	nvarchar(32)		Specifies the alternate phone number.
comment	nvarchar(255)		Shows the comment.
company_uuid	byte(16)	ca_company::company_uuid	Foreign key to the company_uuid field of the ca_company table representing the organization's company.
contact_uuid	byte(16)	ca_contact::uuid	Foreign key to the contact_uuid field of the ca_contact table representing the organization's primary contact.
cost_center	INTEGER	ca_resource_cost_center::id	Foreign key to the id field of ca_resource_cost_center table which represent this cost center.
creation_date	INTEGER		Identifies the date of when this record was created.
creation_user	nvarchar(64)		Identifies the user or process that created the record.

delete_time	INTEGER		Specifies the time of deletion.
description	nvarchar(255)		Indicates the description of the organization.
email_address	nvarchar(120)		Specifies the email address.
exclude_registration	INTEGER		Indicates to exclude registration.
fax_cc	INTEGER		Specifies the fax number's country code.
fax_number	nvarchar(32)		Identifies the fax number.
inactive	INTEGER	Active_Boolean_Table::enum	Specifies the Active flag, as follows:**0** -- False (Active)**1** -- True (Inactive)
last_update_ date	INTEGER		Specifies the date of when this record was last updated.
last_update_ user	nvarchar(64)		Specifies the user or process that last updated this record.
location_uuid	byte(16)	ca_location:: location_uuid	Foreign key to the location_uuid field of the ca_location table for the organization's parent location.
org_name	nvarchar(100)		Identifies the name of the organization.
organization_ uuid	byte(16)		This is the Primary key for Organization.
pager_email_ address	nvarchar(120)		Identifies the pager email address.
parent_org_ uuid	byte(16)		Foreign key to the organization_uuid field of the ca_organization table for the organization's parent organization.
pri_phone_cc	INTEGER		Identifies the country code for the primary phone number.
pri_phone_number	nvarchar(32)		Identifies the primary phone number.
version_number	INTEGER		Specifies the version number for transaction integrity.

Model Definitions

ca_model_def Table

Model definitions for specific manufacturer/model combinations (CA-MDB).

- **SQL Name** -- ca_model_def
- **Object** -- mfrmod

Field	Data Type	Reference	Remarks
abbreviation	nvarchar(30)		Specifies the model abbreviation.
capacity	float		Defines the capacity.
capacity_unit	INTEGER		Defines the capacity unit.**Note:** This is the Foreign key to the id field of the ca_capacity_unit table.
class_id	INTEGER	ca_resource_ class::id	Foreign key to the id field of the ca_resource_class table for the class to which this model belongs.
creation_date	INTEGER		Specifies the date of when this record was created.
creation_user	nvarchar(64)		Identifies the user or process that created the record.
current_as_of_ date	INTEGER		Specifies the date which represents the point at which the model information is considered current.
delete_time	INTEGER		Indicates the time of deletion.
description	nvarchar(255)		Identifies the model description.
exclude_registration	INTEGER		Indicates to exclude registration.
family_id	INTEGER		Foreign key to the id field of the ca_resource_family table for the family to which this model belongs.
gl_code	INTEGER		Specifies the GL code.**Note:** This is the Foreign key to the id field of the ca_resource_gl_code table.
inactive	INTEGER	Active_Boolean_Table::enum	Specifies the Active flag, as follows:**0** -- False (Active)**1 --** True (Inactive)
last_update_date	INTEGER		Specifies the date of when this record was last updated.
last_update_user	nvarchar(64)		Specifies the user or process that last updated this record.
manufacturer_uuid	byte(16)	ca_company::company_uuid	Foreign key to the company_uuid field of the ca_company table for the record that represents the manufacturer.

model_uuid	byte(16)		Specifies the model ID.**Note:** This is a Primary key.
name	nvarchar(100)		Identifies the unique Model name.
operating_system	INTEGER		Operating system.**Note:** It is a Foreign key to the id field of the ca_resource_operating_system table.
preferred_seller_ uuid	byte(16)		Foreign key to the company_uuid field of the ca_company table for the record that represents the preferred seller company of this model.
subclass_id	INTEGER		Foreign key to the id field of the ca_resource_class table for the subclass to which this model belongs.
version_number	INTEGER		Indicates the version number for transaction integrity.

Resource

This article contains the following topics:

ca_resource_class Table

Definitions of the classifications that may be applied to an asset/resource.

- **SQL Name** -- ca_resource_class
- **Object** -- grc

Field	Data Type	Reference	Remarks
creation_date	INTEGER		Specifies the date the of when this record was created.
creation_user	nvarchar(64)		Identifies the user or process that created the record.
delete_time	INTEGER		Indicates the date of deletion.
description	nvarchar(255)		Specifies the class description.
exclude_registration	INTEGER		Indicates to exclude registration.
family_id	INTEGER	ca_resource_ family::id	Primary key that is also the resource family id.
id	INTEGER		Primary key that is also the class id.
inactive	INTEGER	Active_Boolean_Table::enum	Sets the Active flag, as follows:**0** -- False (Active)**1** -- True (Inactive)

last_update_date	INTEGER		Indicates the date the record was last updated.
last_update_user	nvarchar(64)		Specifies the user or process that last updated the record.
name	nvarchar(100)		Identifies the unique class name within family.
parent_id	SREL		Foreign key back to the id field of the ca_resource_class table to allow for hierarchical class groupings (for example, subclass).
usp_nsm_class	INTEGER	buscls id	Represents the CA NSM class.**Note:** This is the Foreign key to the id field of the buscls table.
version_number	INTEGER		Specifies the version number for transaction integrity.

ca_resource_cost_center Table

An asset is associated with.

- **SQL Name** -- ca_resource_cost_center
- **Object** -- cost_cntr

Field	Data Type	Reference	Remarks
creation_date	INTEGER		Identifies the date of when this record was created.
creation_user	nvarchar(64)		User or process that created the record.
delete_time	INTEGER		Specifies the time of deletion.
description	nvarchar(255)		Indicates the cost center description.
exclude_registration	INTEGER		Specifies to exclude registration.
id	INTEGER		Primary key that is also the cost center id.
inactive	INTEGER	Active_Boolean_ Table::enum	Sets the Active flag, as follows:**0** -- False (Active)**1** -- True (Inactive)
last_update_date	INTEGER		Specifies the date of when this record was last updated.
last_update_user	nvarchar(64)		Specifies the user or process that last updated this record.
name	nvarchar(100)		Identifies the unique cost center name.

version_number	INTEGER		Specifies the version number for transaction integrity.

ca_resource_department Table

Department that a resource is assigned to and associated with.

- **SQL Name** -- ca_resource_department
- **Object** -- dept

Field	Data Type	Reference	Remarks
creation_date	INTEGER		Specifies the date of when this record was created.
creation_user	nvarchar(64)		Indicates the user or process that created this record.
delete_time	INTEGER		Indicates the time of deletion.
description	nvarchar(255)		Specifies the department description.
exclude_registration	INTEGER		Specifies to exclude registration.
id	INTEGER		Primary key that is also the resource department id.
inactive	INTEGER	Active_Boolean_Table::enum	Sets the Active flag, as follows:**0** -- False (Active)**1** -- True (Inactive)
last_update_date	INTEGER		Specifies the date of when this record was last updated.
last_update_user	nvarchar(64)		Specifies the user or process that last updated the record.
name	nvarchar(100)		Identifies the unique department name.
version_number	INTEGER		Specifies the version number for transaction integrity.

ca_resource_family Table

High level classification of a resource items, such as Computer, furniture, phone, software and so on.

- **SQL Name** -- ca_resource_family
- **Object** -- nrf

Field	Data Type	Reference	Remarks
creation_date	INTEGER		Specifies the date of when this record was created.
creation_user	nvarchar(64)		Indicates the user or process that created this record.
delete_time	INTEGER		Indicates the time of deletion.

description	nvarchar(255)		Specifies the family description.
exclude_ registration	INTEGER		Specifies to exclude registration.
id	INTEGER		Primary key that is also the resource family id.
inactive	INTEGER	Active_Boolean_ Table::enum	Specifies the Active flag, as follows:**0** -- False (Active)**1 --** True (Inactive)
include_reconciliation	INTEGER		Specifies the flag that indicates whether to include this family in reconciliation.
last_update_date	INTEGER		Specifies the date of when this record was last updated.
last_update_user	nvarchar(64)		Specifies the user or process that last updated the record.
name	nvarchar(100)		Identifies the unique family name.
table_extension_ name	nvarchar(30)		Identifies the name of the associated table to hold extended data for this family.
version_number	INTEGER		Indicates the version number for transaction integrity.

ca_resource_gl_code Table

General ledger entry a resource is assigned to/associated with.

- **SQL Name** -- ca_resource_gl_code
- **Object** -- gl_code

Field	Data Type	Reference	Remarks
creation_date	INTEGER		Specifies the date of when this record was created.
creation_user	nvarchar(64)		Indicates the user or process that created this record.
delete_time	INTEGER		Indicates the time of deletion.
description	nvarchar(255)		Specifies the GL code description.
exclude_registration	INTEGER		Specifies to exclude registration.
id	INTEGER		Primary key that is also the resource GL code id.
inactive	INTEGER	Active_Boolean_Table::enum	Specifies the Active flag, as follows:**0** -- False (Active)**1 --** True (Inactive)
last_update_date	INTEGER		Specifies the date of when this record was last updated.

last_update_user	nvarchar(64)		Specifies the user or process that last updated the record.
name	nvarchar(100)		Identifies the unique GL code name.
version_number	INTEGER		Indicates the version number for transaction integrity.

ca_resource_operating_system Table

Operating system a resource is assigned to/associated with.

- **SQL Name** -- ca_resource_operating_system
- **Object** -- opsys

Field	Data Type	Reference	Remarks
creation_date	INTEGER		Specifies the date of when this record was created.
creation_user	nvarchar(64)		Indicates the user or process that created this record.
delete_time	INTEGER		Indicates the time of deletion.
description	nvarchar(255)		Specifies the operating system description.
exclude_registration	INTEGER		Indicates to exclude registration.
id	INTEGER		Primary key that is also the resource operating system id.
inactive	INTEGER	Active_Boolean_Table::enum	Specifies the Active flag, as follows:**0 --** False (Active)**1 --** True (Inactive)
last_update_date	INTEGER		Specifies the date of when this record was last updated.
last_update_user	nvarchar(64)		Specifies the user or process that last updated the record.
name	nvarchar(100)		Identifies the unique operating system name.
version_number	INTEGER		Specifies the version number for transaction integrity.

ca_resource_status Table

Status of a resource/asset.

- **SQL Name** -- ca_resource_status
- **Object** -- rss

Field	Data Type	Reference	Remarks
creation_date	INTEGER		Specifies the date of when this record was created.

creation_user	nvarchar(64)		Indicates the user or process that created this record.
delete_time	INTEGER		Indicates the time of deletion.
description	nvarchar(255)		Specifies the operating system description.
exclude_ registration	INTEGER		Specifies to exclude registration.
id	INTEGER		Primary key that is also the resource status id.
inactive	INTEGER	Active_Boolean_ Table::enum	Specifies the Active flag, as follows:**0** -- False (Active)**1 --** True (Inactive)
last_update_date	INTEGER		Specifies the date of when this record was last updated.
last_update_user	nvarchar(64)		Specifies the user or process that last updated the record.
name	nvarchar(100)		Identifies the unique status name.
version_number	INTEGER		Indicates the version number for transaction integrity.

Tenant

This article contains the following topics:

ca_tenant Table

This table lists information about a tenant.

- **SQL Name** -- ca_tenant
- **Object** -- tenant

Column	Type	Remarks
id	UUID	ID
name	STRING 255	Specifies the tenant name.
tenant_number	STRING 30	Specifies the alternate tenant identification (customer comment; no OTB use)
service_provider	INTEGER	**1** -- Service Provider tenant
contact	SREL cnt	Specifies the primary contact for the tenant .
logo	STRING 255	Specifies the URL of the tenant logo.
description	STRING 1024	
phone_number	STRING 255	
fax_phone	STRING 255	
alt_phone	STRING 255	
location	SREL loc	Specifies the location of the tenant .

inactive	INTEGER	**1** -- tenant deleted
version number	INTEGER	Specifies the update version of this tenant (incremented whenever tenant is saved).
tenant	UUID (SREL tenant)	Always specifies the same value as the ID
ldap_tenant_group	STRING 512	Specifies the name of the LDAP group that corresponds to this tenant.
subtenants_ok	INTEGER	**1** -- tenant is allowed to have subtenants
fkey_group	UUID (SREL tenant_group)	Specifies the tenant group to which objects that belong to this tenant can reference. Same as the supertenant group. System-maintained (cannot be edited).
supertenant_group	UUID (SREL tenant_group)	Specifies the tenant group that includes the tenant and higher tenants in this hierarchy. Can include the service provider (if in the same hierarchy). System-maintained (cannot be edited).
subtenant_group	UUID (SREL tenant_group)	Specifies the tenant group that includes the tenant and lower tenants in this hierarchy. System-maintained (cannot be edited).
relatedtenant_group	UUID (SREL tenant_group)	Specifies the tenant group that includes all tenants in this hierarchy (includes the subtenant and supertenant groups). Can include the service provider (if in the same hierarchy). System-maintained (cannot be edited).
tenant_depth	INTEGER	Specifies the number of levels above the tenant in this hierarchy. System-maintained (cannot be edited).
parent	UUID	Specifies the tenant immediately above this tenant in the hierarchy.
creation_user	STRING 64	Specifies the user ID of contact who created this tenant.
creation_date	INTEGER	Specifies the date of tenant creation (number of seconds after 0000 hours on 01/01/1970).
last_update_user	STRING 64	Specifies the user ID of contact who last updated this tenant.
last_update_date	INTEGER	Specifies the date of last update (number of seconds after 0000 hours on 01/01/1970).
terms_of_usage	UUID REF ca_tou	Specifies the terms of usage associated with the tenant.

ca_tenant_group Table

This table lists information about a tenant group.

- **SQL Name** -- ca_tenant_group
- **Object** -- tenant_group

Column	Type
id	UUID
name	STRING 255
description	STRING 1024

ca_tenant_group_member Table

This table lists information about a tenant group member.

- **SQL Name** -- ca_tenant_group_member
- **Object** -- tenant_group_member

Column	Type
id	UUID
tenant	SREL tenant
tenant_group	SREL tenant_group

g_tenant Table

Column	Type	Remarks
id	integer	
remote_sys_id	SREL Global_Servers	Required (NOT NULL)
remote_id	UUID	Regional tenant id
tenant_name	STRING 255	

Call Request

This article contains the following topics:

Call_Req Table

Integration with call manager.

- **SQL Name** -- call_req
- **Object** -- cr

Field	Data Type	Reference	Remarks
active_flag	INTEGER	Boolean_Table ::enum	Sets the Active flag, as follows:**0** -- Inactive**1** -- Active
affected_rc	byte(16)	ca_owned_ resource::uuid	Foreign key to the id field of the ca_owned_resource table. It identifies the asset.

assignee	byte(16)		Foreign key to the contact_uuid field of the ca_contact table. It identifies the assignee.
call_back_date	INTEGER		Specifies the call back timestamp for this request.
call_back_flag	INTEGER		Specifies the call back flag value for this request.
category	nvarchar(30)	Prob_Category ::persid	Foreign key to the persistent_id field of the prob_ctg table. This identifies the category.
change	INTEGER	chg ID	Foreign key to the ID field of the chg table. This is the associated change order.
caused_by_chg	INTEGER	Change_Request	Indicates the change request was caused by another change order.
charge_back_id	nvarchar(12)		Indicates the user-defined string field.
close_date	INTEGER		Represents the timestamp of when this request was closed.
cr_tticket	INTEGER		Not used.
created_via	INTEGER	Interface::id	Foreign key to the id field of the interface table. Based on site-defined conditions, this reflects which interface created the request.
customer	byte(16)	ca_contact::uuid	Foreign key to the contact_uuid field of the ca_contact table. This identifies the Affected End User.
description	nvarchar(4000)		This is the textual description of this call request.
event_token	nvarchar(30)		Used by TNGeh_writer for message matching.
external_system_ ticket	STRING 4000		
extern_ref	nvarchar(30)		(Deprecated) Specifies the trouble ticket associated with the call request.
group_id	byte(16)		Foreign key to the contact_uuid field of the ca_contact table, this represents the Assigned to Group ID.
id	INTEGER		Specifies the unique (to the table) numeric ID.

impact	INTEGER	Impact::enum	Foreign key to the enum field of the impact table, this identifies the impact of the request.
incident_priority	INTEGER		Specifies the computed incident priority if this is an ITIL incident.
incorrectly_ assigned	INTEGER		Indicates that the incident is assigned incorrectly.
last_act_id	nvarchar(12)		Identifies the persid of the last activity.
last_mod_by	byte(16)		Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp of when this record was last modified.
log_agent	byte(16)	ca_contact::uuid	Foreign key to the contact_uuid field of the ca_contact table. This identifies who the request was reported by.
macro_predict_ viol	INTEGER		Indicates that it is likely to violate its sla (boolean) for action macros to predict sla violations.
major_incident	INTEGER		Identifies the incident as a major incident.
open_date	INTEGER		Identifies the timestamp of when the request was created.
orig_user_admin_org	UUID	ca_organization	
orig_user_cost_ center	INTEGER	ca_resource_ cost_center	
orig_user_dept	INTEGER	ca_resource_ department	
orig_user_ organization	UUID	ca_organization	
outage_detail_ what	STRING 4000		Describes the outage details.
outage_detail_ who	STRING 4000		
outage_detail_ why	STRING 4000		Describes why the outage occurred.
outage_reason	INTEGER	Outage_Reason	Identifies the reason for the outage.
outage_type	INTEGER	Outage_Type	Identifies the type of outage.
parent	nvarchar(30)	Call_Req::persid	Foreign key to the persistent_id field of the call_req table to allow for hierarchical request groupings (for example, "parent/ child").
pct_service_ restored	INTEGER		Represents the percentage of service restored.

persid	nvarchar(30)		Identifies the Persistent ID (SystemObjectName:id).
predicted_sla_viol	INTEGER		Indicates that an sla violation has been predicted by Neugents: **1** -- Request
priority	INTEGER	Priority::enum	Foreign key to the enum field of the pri table, this indicates the priority of the call request.
problem	nvarchar(30)		Foreign key to the persistent_id field of the call_req table to allow for linking this incident to a problem.
ref_num	nvarchar(30)		Shows a visible reference number to the user.
requested_by	UUID	ca_contact	Identifies who requested the ticket.
remote_control_ used	INTEGER		Indicates that remote control was used.
resolvable_at_lower	INTEGER		Indicates that the ticket was resolved at a lower level.
resolve_date	INTEGER		Indicates the timestamp of when this request was resolved.
return_to_service	INTEGER		Indicates whether service is fully restored.
rootcause	INTEGER	Rootcause::id	Foreign key to the id field of the rootcause table. Specifies the root cause of the request.
sched_token	nvarchar(128)		Specifies the job scheduling token.
severity	INTEGER	Severity::enum	Foreign key to the enum field of the Severity table, this identifies the severity of the Request.
sla_violation	INTEGER		If defined, this flags the request as follows:Template **1** -- Request has violated its sla
solution	nvarchar(30)	Call_Req::persid	(Decrecated) Foreign key to the persistent_id field of the crsol table for old request solutions.
status	nvarchar(12)	Cr_Status::code	Foreign key to the code field of the cr_stat table, this is the status of the problem.
string1	nvarchar(40)		Identifies the user defined string field.
string2	nvarchar(40)		Identifies the user defined string field.

string3	nvarchar(40)		Identifies the user defined string field.
string4	nvarchar(40)		Identifies the user defined string field.
string5	nvarchar(40)		Identifies the user defined string field.
string6	nvarchar(40)		Identifies the user defined string field.
summary	nvarchar(240)		Identifies the summary text.
support_lev	nvarchar(30)	Service_Desc ::code	Foreign key to the code field of the srv_desc table, this defines the Classic Service Type.
symptom_code	INTEGER	Symptom_Code	
template_name	nvarchar(30)	Cr_Template::template_name	Foreign key to the template_name field of the cr_tpl table, this specifies the name of the request template.
caextwf_instance_id	INTEGER	caextwf_instances	Indicates the CA Process Automation process instance id and process definition name and reference path launched by this Service Desk object.
tenant	UUID	ca_tenant	
outage_start_ time	LOCAL_TIME		
outage_end_time	LOCAL_TIME		
ticket_ avoided	INTEGER		**0** -- Do not search **10** -- Perform search **20** -- Open knowledge document **30** -- Ticket avoided by self-service
time_spent_sum	INTEGER		Specifies the sum of activity time spent.
type	nvarchar(10)	crt code	Foreign key to the crt table, this is the Request type.
urgency	INTEGER	Urgency::enum	Foreign key to the enum field of the urgency table, this indicates the call request urgency.
target_start_last	LOCAL_TIME		Time when target timer started or restarted.
target_hold_last	LOCAL_TIME		Time of most recent ticket hold
target_hold_count	INTEGER		Number of times ticket went into a Hold status
target_resolved_last	LOCAL_TIME		Time of most recent ticket resolution

target_resolved_ count	INTEGER		Number of times ticket went into a Resolved status
target_closed_last	LOCAL_TIME		Time ticket was last closed
target_closed_count	INTEGER		Number of times that ticket was closed
affected_service	UUID	ca_owned_ resource	UNIQUE NOT_NULL KEY

Call_Req_Type Table

Stores codes used in Call_Req.type and the detail form names that should be displayed. This is used by the ITIL vertical market customizations to display alternative forms for cr_detail based on the value of Call_Req.type

- **SQL Name** -- crt
- **Object** -- crt

Field	Data Type	Reference	Remarks
code	nvarchar(10)		Primary key of this table.
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
display_name	nvarchar(30)		The display name value for this Request type.
id	INTEGER		Specifies the unique (to the table) numeric ID.
nx_desc	nvarchar(30)		The description value for this Request type.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
sym	nvarchar(30)		The symbolic value for this Request type.

Call_Solution Table

This table exists in the schema for backward compatibility only. Although there is an interface to it, you should not use this table at all; however, it is important that you not delete it from the schema.

- **SQL Name** -- crsol
- **Object** -- crsol

Field	Data Type	Reference	Remarks
description	STRING 500		Specifies the problem description.
cr_count	INTEGER		Specifies the call request usage count:**0** -- Yes**1** -- No call request usage count
cr_flag	INTEGER		Indicates the call request manager flag used by ITIL code.

del	INTEGER NOT_NULL	Active_Boolean_Table::enum	Specifies the status of the Deleted flag:**0** -- Active**1** -- Inactive/marked as deleted.
id	INTEGER UNIQUE NOT_NULL KEY		Indicates the numeric ID that is unique to the table.
in_flag	INTEGER		Specifies the in_flag used by ITIL code.
last_mod_dt	LOCAL_TIME		Indicates the timestamp for when this record was last modified.
nx_desc	STRING 40		Specifies the name for prob_mgr cbox table.
persid	STRING 30		Indicates the Persistent ID:SystemObjectName:id
pr_flag	INTEGER		Specifies the pr_flag used by ITIL code.
sapproved	INTEGER	Boolean_Table::enum	Indicates the status of solution approved.
sname	STRING 40		Specifies the solution name.
solution	STRING 1000		Specifies the solution description.
sym	STRING 60 NOT_NULL S_KEY		Specifies the symbol for the solution.
tcode	INTEGER		This field is no longer used.

Cr_Call_Timers Table

Call Request call timers. A stop watch with various thresholds that gives the help desk analyst a visual and audio indication of elapsed time.

- **SQL Name** -- crctmr
- **Object** -- ctimer

Field	Data Type	Reference	Remarks
beep	INTEGER		A beep indicator for then the threshold is reached:**0** -- No beep**1** -- Beep
color	STRING 12		Indicates the color of the timer at the start time.
del	INTEGER NOT_NULL	Active_Boolean_Table::enum	Specifies the status of the Delete flag: **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		The ID unique (to the table) Numeric ID.

last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp for when this record was last modified.
persid	STRING 30		Identifies the Persistent ID (SystemObjectName:id).
text	STRING 240		Identifies the threshold text to display when the timer indicates elapsed time.
threshold	DURATION NOT_NULL		Identifies the threshold elapsed time.

Cr_Status Table

Call Request Status. Lists the states of the call request. May be added to at will. Allows the user to control whether the call request is active or inactive when it is changed to this status.

- **SQL Name** -- cr_stats
- **Object** -- crs

Field	Data Type	Reference	Remarks
active	INTEGER		Sets the Active flag, as follows:**0** -- Inactive**1** -- Active
code	nvarchar(12)		Primary key of this table.
cr_flag	INTEGER		When this flag is set to 1, this status is valid for Requests.
del	INTEGER	Active_Boolean_Table::enum	The Delete flag that indicates the following:**0** -- Active**1** -- Inactive/marked as deleted
description	nvarchar(500)		Identifies the textual description of the status.
hold	INTEGER		Sets the Hold flag, as follows:**0** -- Start events**1** -- Stop events
id	INTEGER		Unique (to the table) numeric ID.
in_flag	INTEGER		When this flag is set to 1, the status is valid for Incidents.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp of when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
pr_flag	INTEGER		When this flag set to 1, the status is valid for Problems.

resolved	INTEGER		Flag that indicates the following:**0** -- Not yet resolved**1** -- Resolved
sym	nvarchar(30)		Identifies the symbol of the Request status name.

Cr_Stored_Queries Table

Custom bin stored queries. System administrators may add to this table at will. Determines which queries may be used by help desk analysts to customize their scoreboard.

- **SQL Name** -- crsq
- **Object** -- crsq

Field	Data Type	Reference	Remarks
description	STRING 240		Specifies the textual description of this stored query.
code	STRING 12 UNIQUE NOT_NULL S_KEY		Specifies the non-editable handle for the query.
count_url	STRING 240		Specifies the URL for counts, if obj_type=url.
criteria	STRING 240		Specifies the where clause for querying.
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	The deleted flag **0** -- active **1** -- inactive/marked as deleted)
id	INTEGER UNIQUE NOT_NULL KEY		Specifies the key (to the table) Numeric ID.
label	STRING 80		Specifies the label to display on the scoreboard.
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp for when this record was last modified.
obj_type	STRING 30		Indicates the scoreboard, with the capability of having enough space to allow for expansions to accommodate the cr, tt, ir, and chg types.
persid	STRING 30		The Persistent ID (SystemObjectName:id).

Cr_Template Table

Request Template Table maps a template name to a Call_Req that will be used as a template.

- **SQL Name** -- cr_template
- **Object** -- cr_tpl

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_ Table::enum	The Delete flag that indicates the following:**0** -- Active**1** -- Inactive/marked as deleted
description	nvarchar(1000)		Shows the description of the template.
id	INTEGER		Identifies the nique (to the table) numeric ID.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
quick_tmpl_type	INTEGER	Quick_Template_ Types::enum	Flag that indicates the following:**1** -- Quick tmpl**2** -- Quick tmpl+review
template	nvarchar(30)	Call_Req::persid	Persistent ID (SystemObjectName:id).
template_class	nvarchar(12)		Indicates to allow subclassing templates.
template_name	nvarchar(30)		Identifies the unique name of the template.

cr_trans Table

A transition object controls the current and next ticket status. The cr_trans table lists the status, new status, and actions that need to occur for a default transition.

- **SQL Name** -- cr_trans
- **Object** -- cr_trans

Label	Field	Description
id	INTEGER	Unique key.
status	SYMBOL	Specifies the current ticket status.
new_status	SYMBOL	Specifies the new ticket status
must_comment	INTEGER	Comment required when using a transition. On new default: 0
is_default	INTEGER	Default transition that appears when the Status field is empty. On new default: 0
condition	BOP_REF_STR_REF Macro	Site condition macro to approve transition.
condition error	STRING 255	Error message for site condition.

description	STRING 255	Description of this transition.
last_mod_dt	LOCAL_TIME	Timestamp of last update to this record.
last_mod_by	UUID REF ca_contact	User who last updated this.
del	INTEGER nn	Logical database delete status.
tenant	UUID REF ca_tenant	Reference to Tenant information.

Knowledge Management Tables

This article contains the following topics:

CI_ACTIONS Table

Program control table used by Knowledge Management.

- **SQL Name** -- CI_ACTIONS
- **Object** -- CI_ACTIONS

Field	Data Type	Reference	Remarks
ACTION_ORDER	INTEGER		
ACTION_TITLE	STRING 100		
ANALYST_ID	UUID	ca_contact::uuid	
GROUP_ID	UUID	ca_contact::uuid	
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
PREDEFINED	INTEGER		
STATUS_CURRENT_ID	INTEGER	CI_STATUSES::id	
UNPUBLISH	INTEGER		
UNRETIRE	INTEGER		
WF_TEMPLATE_ID	INTEGER	CI_WF_TEMPLATES ::id	

CI_ACTIONS_ALTERNATE Table

Program control table used by Knowledge Management.

- **SQL Name** -- CI_ACTIONS_ALTERNATE
- **Object** -- CI_ACTIONS_ALTERNATE

Field	Data Type	Reference	Remarks
ACTION_ID	INTEGER	CI_ACTIONS::id	
CONTACT_ID	UUID	ca_contact::uuid	
CONTACT_TYPE	INTEGER		
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.

CI_BOOKMARKS Table

Program control table used by Knowledge Management.

- **SQL Name** -- CI_BOOKMARKS
- **Object** -- CI_BOOKMARKS

Field	Data Type	Reference	Remarks
BOOKMARK_ TITLE	STRING 100		
DOCUMENT_ID	INTEGER	SKELETONS::id	
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
USER_ID	UUID	ca_contact::uuid	

CI_DOC_LINKS Table

Program control table used by Knowledge Management.

- **SQL Name** -- CI_DOC_LINKS
- **Object** -- CI_DOC_LINKS

Field	Data Type	Reference	Remarks
DOC_ID1	INTEGER	SKELETONS::id	Specifies the parent ID.
DOC_ID2	INTEGER	SKELETONS::id	Specifies the child ID.
id	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
parent_child	INTEGER		Indicates the document has one of the following relationships with another document: **0** -- See also link **1** -- Parent Child Link
tenant	UUID	ca_tenant	Specifies the UUID of the tenant.

CI_DOC_TEMPLATES Table

Program control table used by Knowledge Management.

- **SQL Name** -- CI_DOC_TEMPLATES
- **Object** -- CI_DOC_TEMPLATES

Field	Data Type	Remarks
ID	INTEGER NOT_NULL KEY	Unique (to the table) Numeric ID

IS_DEFAULT	INTEGER	
IS_PREDEFINED	INTEGER	
KD ID	Long	Knowledge Document id that is used as a template for default values.
LAST_MOD_DT	LOCAL_TIME	Indicates the timestamp of when this record was last modified.
PAGE_HTML	STRING 32768	
TEMPLATE_NAME	STRING 255	

CI_DOC_TYPES Table

Program control table used by Knowledge Management.

- **SQL Name** -- CI_DOC_TYPES
- **Object** -- CI_DOC_TYPES

Field	Data Type	Reference	Remarks
DOC_TYPE_TXT	STRING 50		
ID	INTEGER KEY		Unique (to the table) Numeric ID
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.

CI_PRIORITIES Table

Program control table used by Knowledge Management.

- **SQL Name** -- CI_PRIORITIES
- **Object** -- CI_PRIORITIES

Field	Data Type	Reference	Remarks
ID	INTEGER KEY		Unique (to the table) Numeric ID
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
PRIORITY	STRING 50		

CI_STATUSES Table

Program control table used by Knowledge Management.

- **SQL Name** -- CI_STATUSES
- **Object** -- CI_STATUSES

Field	Data Type	Reference	Remarks
ID	INTEGER KEY		Unique (to the table) Numeric ID
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
PREDEFINED	INTEGER		
STATUS	STRING 50		

STATUS_ DESCRIPTION	STRING 255		
STATUS_ORDER	INTEGER		

CI_WF_TEMPLATES Table

Program control table used by Knowledge Management.

- **SQL Name** -- CI_WF_TEMPLATES
- **Object** -- CI_WF_TEMPLATES

Field	Data Type	Reference	Remarks
ID	INTEGER KEY		Unique (to the table) Numeric ID
IS_DEFAULT	INTEGER		
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
WF_DESCRIPTION	STRING 255		
WF_NAME	STRING 255		

Doc_Versions Table

Program control table used by CA SDM Knowledge Management.

- **SQL Name** -- DOC_VERSIONS
- **Object** -- DOC_VERSIONS

Field	Data Type	Reference	Remarks
ID	INTEGER		Specifies the primary key.
doc_ID	INTEGER	skeletons::id	Specifies the SREL to the knowledge document object.
ver_count	INTEGER		Specifies the internal version ID that indicates the number of versions created (1, 2, 3, 4...)
ver_comment	STRING (2000)		Specifies the comment text.
start_date	DATE		Specifies the date when this version started.
end_date	DATE		Specifies the date when this version ended.
started_by	UUID		Specifies the user that created this version.
published_by	UUID		Specifies the user that published this version.
status_ID	INTEGER		Specifies the document version status.

ver_status	INTEGER		Specifies the previous version's status: **0** -- Existed **1** -- Archived **2** -- Deleted (this only relates to the resolution field)
title	STRING		Specifies the document version title.
summary	STRING		Specifies the document summary text.
problem	STRING		Specifies the problem description.
notes	STRING		Specifies the document notes.
doc_type_ID	INTEGER		Specifies the document type.
ext_doc_ID	INTEGER		Specifies the Tree ID when Decision Tree is used.
last_mod_by	Byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.

EBR_ACRONYMS Table

Program control table used by Knowledge Management.

- **SQL Name** -- EBR_ACRONYMS
- **Object** -- EBR_ACRONYMS

Field	Data Type	Reference	Remarks
ACRONYM	STRING 50		
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.

EBR_KEYWORDS Table

Program control table used by Knowledge Management.

- **SQL Name** -- EBR_KEYWORDS
- **Object** -- EBR_KEYWORDS

Field	Data Type	Reference	Remarks
ENTITY_ID	INTEGER		
EXT_TABLE_ID	INTEGER		
FULL_WORD	STRING 50		
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID

EBR_LOG Table

Program control table used by Knowledge Management.

- **SQL Name** -- EBR_LOG
- **Object** -- EBR_LOG

Field	Data Type	Reference	Remarks
ASKED_DATE	LOCAL_TIME		
BEST_IDS	STRING 110		
EXTERNAL_ID	STRING 50		
FILTER_DATA	STRING 32768		
FUZZINESS	INTEGER		
ID	INTEGER KEY		Unique (to the table) Numeric ID
KEYWORDS	STRING 32768		
MATCH_TYPE	INTEGER		
METHOD_ PERFORMANCE	INTEGER		
METHOD_TYPE	INTEGER		
NUM_MATCHES	INTEGER		
ORDER_DIRECTION	INTEGER		
PRIMARY_ORDER	STRING 50		
ROWS_FOUND	INTEGER		
ROWS_TO_FETCH	INTEGER		
SEARCH_IN	INTEGER		
SEARCH_QUALITY	INTEGER		
SEARCH_TEXT	STRING 255		
SEARCH_TYPE	INTEGER		
SECONDARY_ORDER	INTEGER		
SESSION_ID	INTEGER		
SQL_TEXT	STRING 32768		
TOP_MATCH_ID	INTEGER		
UNIQUE_WORD_COUNT	INTEGER		
USER_ID	STRING 100		
WORD_COUNT	INTEGER		

EBR_METRICS Table

Program control table used by Knowledge Management.

- **SQL Name** -- EBR_METRICS
- **Object** -- EBR_METRICS

Field	Data Type	Reference	Remarks
COMMENTS	STRING 255		
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
METRIC	STRING 50		
WEIGHT	REAL		

EBR_NOISE_WORDS Table

Program control table used by Knowledge Management.

- **SQL Name** -- EBR_NOISE_WORDS
- **Object** -- EBR_NOISE_WORDS

Field	Data Type	Reference	Remarks
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
NOISE_WORD	STRING 50		

EBR_PATTERNS Table

Program control table used by Knowledge Management.

- **SQL Name** -- EBR_PATTERNS
- **Object** -- EBR_PATTERNS

Field	Data Type	Reference	Remarks
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
PATTERN_DEFAULT	STRING 255		
PATTERN_NAME	STRING 50		
PATTERN_VALUE	STRING 255		
PATTERN_VALUE_ADM	STRING 255		

EBR_PREFIXES Table

Program control table used by Knowledge Management.

- **SQL Name** -- EBR_PREFIXES
- **Object** -- EBR_PREFIXES

Field	Data Type	Reference	Remarks
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
PREFIX	STRING 50		

EBR_PROPERTIES Table

Program control table used by Knowledge Management.

- **SQL Name** -- EBR_PROPERTIES
- **Object** -- EBR_PROPERTIES

Field	Data Type	Reference	Remarks
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
PROPERTY_ADMIN	INTEGER		
PROPERTY_DEFAULT	STRING 50		
PROPERTY_NAME	STRING 50 S_KEY		
PROPERTY_TYPE	STRING 50		
PROPERTY_VALUE	STRING 32768		
PROPERTY_VALUE_ADM	STRING 32768		

EBR_SUBSTITS Table

Program control table used by Knowledge Management.

- **SQL Name** -- EBR_SUBSTITS
- **Object** -- EBR_SUBSTITS

Field	Data Type	Reference	Remarks
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
SYMBOL1	STRING 50		
SYMBOL2	STRING 50		

ES_CONSTANTS Table

Program control table used by Knowledge Management.

- **SQL Name** -- ES_CONSTANTS
- **Object** -- ES_CONSTANTS

Field	Data Type	Reference	Remarks
COMMENTS	STRING 255		
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
NAME	STRING 50		Specifies the text name of this item.
PROPERTYID	INTEGER		
PROPVALUE	INTEGER		

ES_NODES Table

Program control table used by CA SDM Knowledge Management.

- **SQL Name** -- ES_NODES
- **Object** -- ES_NODES

Field	Data Type	Reference	Remarks
DISPLAYED_ TEXT	STRING 32768		
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
LAST_MOD_ DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
LINK_ID	INTEGER	ES_NODES::id	
NODE_ID	INTEGER		
NODE_SHORT_DESC	STRING 150		
NODE_TYPE	INTEGER		
PARENT_ NODE_ID	INTEGER		
QUERY_RESP_NUMBER	INTEGER		
QUERY_RESP_TYPE	STRING 50		
RESPLINKID1	INTEGER		
RESPLINKID2	INTEGER		
RESPLINKID3	INTEGER		
RESPLINKID4	INTEGER		
RESPLINKID5	INTEGER		
RESPLINKID6	INTEGER		
RESPLINKID7	INTEGER		
RESPONSE1	STRING 100		
RESPONSE2	STRING 100		
RESPONSE3	STRING 100		
RESPONSE4	STRING 100		
RESPONSE5	STRING 100		
RESPONSE6	STRING 100		
RESPONSE7	STRING 100		
ROOT_ID	INTEGER	ES_NODES::id	
TREE_ID	INTEGER	SKELETONS::id	

ES_RESPONSES Table

Program control table used by Knowledge Management.

- **SQL Name** -- ES_RESPONSES
- **Object** -- ES_RESPONSES

Field	Data Type	Reference	Remarks
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID

LAST_MOD_ DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
PARENT_ NODE_ID	INTEGER	ES_NODES::id	
RESPONSE_ LINK_ID	INTEGER	ES_NODES::id	
RESPONSE_ LINK_ORDER	INTEGER		
RESPONSE_ LINK_TEXT	STRING 100		

ES_SESSIONS Table

Program control table used by Knowledge Management.

- **SQL Name** -- ES_SESSIONS
- **Object** -- ES_SESSIONS

Field	Data Type	Reference	Remarks
COMMENT_TEXT	STRING 50		
EVALUATION	INTEGER		
EXTERNAL_ID	STRING 50		
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
PATH_IDS	STRING 50		
PATH_QAS	STRING 32768		
SESSION_ID	INTEGER		
TREE_ID	INTEGER	ES_NODES::id	

Index_Doc_Links Table

Program control table used by Knowledge Management.

- **SQL Name** -- index_doc_links
- **Object** -- index_doc_links

Field	Data Type	Reference	Remarks
DOC_ID	INTEGER	SKELETONS::id	
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
INDEX_ID	INTEGER	O_INDEXES::id	
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
RELATIONAL_ID	STRING 255		

KD_Atmnt Table

Program control table used by Knowledge Management.

- **SQL Name** -- kd_attmnt
- **Object** -- kd_attmn

Field	Data Type	Reference	Remarks
ATTMNT_ID	INTEGER	Attachment::id	
DOC_ID	INTEGER	SKELETONS::id	
ID	INTEGER KEY		Unique (to the table) Numeric ID
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
RES_ID	Long		If there is an embedded image, the RES_ID points to the resource file id in the attmnt table.

Knowledge_Keywords Table

Knowledge Base key words for associating trouble codes and call areas to solutions.

- **SQL Name** -- km_kword
- **Object** -- kwrd

Field	Data Type	Reference	Remarks
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
persid	STRING 30		Persistent ID (SystemObjectName:id)
sym	STRING 30 NOT_NULL S_KEY		keyword

kdlinks Table

Each record indicates a link between a KD and request or issues link_type field represented by the following enum value:

1 = KD is a solution to the ticket

2 = Ticket created is based on the document last_mod_by the person that links the document and the ticket

- **SQL Name** -- kdlinks
- **Object** -- kdlinks

Field	Data Type	Reference	Remarks
cr	STRING 30	Call_Req::persid	
ID	INTEGER KEY		Specifies the Numeric ID that is unique to this table.
iss	STRING 30	Issue	
kd	INTEGER	SKELETONS::id	

last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp for when this record was last modified.
link_type	INTEGER		
sd_obj_id	INTEGER		
sd_obj_type	STRING 5		

KEIT_TEMPLATES Table

Field Name	Type	Index	Comment
ID	Int	Primary Key	
TEMPLATE_NAME	Text (50)		
EXP_SUB_CAT	int		Indicates a flag to export sub categories: **0** -- No **1** -- Yes
EXP_SEC_CAT	int		Indicates a flag to export secondary categories **0** -- No **1** -- Yes
EXP_ALL_DOCS	int		Indicates a flag to export all documents linked to the selected category **0** -- No **1** -- Yes
EXP_FILTER	Text (2048)		Additional Filter for export
EXP_ATTMNT	int		Indicates a flag to export attachments **0** -- No **1** -- Yes
EXP_INDEX_DOC	int		Indicates a flag to Index documents after import **0** -- No **1** -- Yes
OVERRIDE_PUB	int		Indicates a flag to override published documents **0** -- No **1** -- Yes

OVERRIDE_UNPUB			Indicates a flag to override unpublished documents **0** -- No **1** -- Yes
OVERRIDE_DEFAULTS			Use the default values when overriding documents
ERR_THRESHOLD	int		Error threshold for stopping the process
FIELD_LIST	Text		List of fields to export
CAT_LIST	Text		List of categories to export
DEFAULT_LIST	Text		List of fields, their types and values used as defaults on import
STATUS_ID and following fields			Used internally to fulfill the DEFAULT_LIST field
last_mod_by	UUID		
last_mod_dt	Long		

KT_act_content Table

The kt_act_content table is used for action content links.

Field	Data Type	Remarks
Id	Long	id
Name	String	Name of the action content.
Code	String (10)	Action content code - Rel_attr
Description	String	Description of the action content
URL	String	URL of the action content
HTMPL	String	Parameters for the HTMPL URL

KT_Blc Table

Program control table used by Knowledge Management to store the best link conditions data for the Recommended Documents feature.

- **SQL Name** -- kt_blc
- **Object** -- kt_blc

Field	Data Type	Reference	Remarks
ID	INTEGER		Primary key.
KD	INTEGER		SREL to the knowledge document object.
Kcat	INTEGER		SREL to knowledge category for recommended document condition type of 4.

BLC_Type	INTEGER		SREL to recommended document condition type.
BLC_Text	STRING 255		The recommended document condition term.
Author_ID	UUID		SREL to cnt
Del	INTEGER	Active_Boolean _Table::enum	Specifies the status of the Deleted flag: **0** -- Active **1** -- Inactive/marked as deleted
Last_mod_dt	DATE		Shows the date of when this record was last updated.

KT_Blc_Type Table

Program control table that is used by Knowledge Management to store the Best Link Condition types for the Recommended Documents feature.

- **SQL Name** -- kt_blc_type
- **Object** -- kt_blc_type

Field	Data Type	Reference	Remarks
ID	INTEGER		Primary key.
SYM	STRING 50		**1** -- Full Match **2** -- Exact Match **3** -- Keyword List **4** -- Knowledge Category
BLC_Condition	STRING 255		Category path if BLC type is category; otherwise, BLC text is used.
Last_mod_dt	DATE		Shows the date of when this record was last updated.

KT_Flg_Status Table

Identifies the different flag statuses used by the Automated Policies feature.

- **SQL Name** -- kt_flg_status
- **Object** -- kt_flg_status

Field	Data Type	Reference	Remarks
ID	INTEGER		Identifies the unique (to the table) numeric ID.
sym	STRING (50)		Flag type name.

KT_Flg_Type Table

Identifies the different flags types that are used by the Automated Policies feature to flag documents for correction.

- **SQL Name** -- kt_flg_type
- **Object** -- kt_flg_type

Field	Data Type	Reference	Remarks
ID	INTEGER		Identifies the unique (to the table) numeric ID.
sym	STRING (50)		Flag type name.
description	STRING (2000)		Flag type description.
init_by_users	INTEGER (1/0)		Specifies the flag type users select in the end user interface.
show_as_comment	INTEGER (1/0)		Displays the flag type as a comment when document is open for viewing.
follow_up	INTEGER (1/0)		Raises follow-up mechanism for the flag type.
time_to_complete	INTEGER		Specifies the time from which the argument is completed.
dev_assignee	UUID		For future use.
del	INTEGER		Specifies if the flag type is active or inactive.
last_mod_by	UUID	SREL to the Contacts table.	Displays the last date of when a user modified the record.
last_mod_dt	Long (date)		Displays the last modified date.

KT_Life_Cycle_Rep Table

Identifies the knowledge document lifecycle data for the Automated Policies feature in Knowledge Management.

- **SQL Name** -- KT_Life_Cycle_Rep
- **Object** -- KT_Life_Cycle_Rep

Field	Data Type	Reference	Remarks
ID	Long	Primary index.	Identifies the unique (to the table) numeric ID of the primary index.
policy	Long	Required field. SREL to query_policy.	Identifies the Policy ID.
kd	Long	Required field. SREL to the knowledge document (Skeletons table).	Identifies the Document ID.
last_mod_dt	Long (date)		Displays the last modified date.

KT_REPORT_CARD Table

Program control table used by Knowledge Management.

- **SQL Name** -- kt_report_card
- **Object** -- kt_report_card

Field	Data Type	Reference	Remarks
ID	INTEGER		KEY Unique (to the table) Numeric ID
SUBJECT_ID	nvarchar(64)		
PAST_DAYS	INTEGER		
ORG_STATISTICS	INTEGER		
DOCUMENTS_ SUBMITTED	INTEGER		
DOCUMENTS_ PUBLISHED	INTEGER		
TOTAL_HITS	INTEGER		
AVERAGE_ EFFECTIVENESS_ RATING	INTEGER		
TOTAL_SOLUTION_ COUNT	INTEGER		
USER_SLV_CNT	INTEGER		
TOTAL_VOTES	INTEGER		
AVG_RATING	REAL		
LINKED_ KNOWLEDGE_BY_ OTHERS	INTEGER		
LINKED_ KNOWLEDGE_BY_ME	INTEGER		
CLOSED_TICKETS	INTEGER		
TICKETS_WITH KNOWLEDGE	INTEGER		
TICKETS_HAD_ SEARCH_ ACTIVITES	INTEGER		
KNOWLEDGE_ SUBMIT_FROM_ TICKET	INTEGER		
TIME_TO_PUBLISH	INTEGER		
MY_COMMENTS	INTEGER		
DOCUMENTS_ RETIRED	INTEGER		
TIME_TO_FIX	INTEGER		
FLAGS_FIXED	INTEGER		
COMMENTS_ON_ MY_DOCUMENTS	INTEGER		
FIRST_CALL_RES_ WITH_KNOW	INTEGER		
FIRST_CALL_RES_ WITHOUT_KNOW	INTEGER		

creation_user	nvarchar(64)		Specifies the name of the person who created this record. Should be in form: LastName, FirstName
creation_date	INTEGER		Indicates the timestamp indicating when this record was created
last_update_user	nvarchar(64)		Specifies the contact who last modified this record. Should be in form: LastName, FirstName
last_update_date	INTEGER		Indicates the timestamp of when this record was last modified
CONTRIBUTOR_ UUID	SREL	cnt	TENANCY_ UNRESTRICTED
IS_SUPERVISOR	INTEGER		
SUPERVISOR_ UUID	SREL	cnt	TENANCY_ UNRESTRICTED
MY_ORG_REF_ID	SREL	KT_REPORT_ CARD	
ALL_ORG_REF_ID	SREL	KT_REPORT_ CARD	
PAST_DAYS_TEXT	nvarchar(20)		Populates from the resource file.
LAST_MOD_DT_ TEXT	nvarchar(50)		Specifies the KRC calculation date.
DOCUMENTS_ PUBLISHED_ PERCENT	nvarchar(4)		
AVG_HITS	nvarchar(20)		
AVG_RATING_ TEXT	nvarchar(100)		
last_mod_by	SREL	cnt	
last_mod_dt	DATE		

O_INDEXES Table

Program control table used by Knowledge Management.

- **SQL Name** -- O_INDEXES
- **Object** -- KCAT

Field	Data Type	Reference	Remarks
DESCRIPTION	STRING 255		Textual description
AUTHOR_ID	UUID	ca_contact::uuid	
CAPTION	STRING 50		
CR_CAT	STRING 30	Prob_Category	
DOC_TEMPLATE	INTEGER	CI_DOC_ TEMPLATES::id	
HAS_CHILDREN	INTEGER		
HAS_DOCS	INTEGER		
ID	INTEGER KEY		Unique (to the table) Numeric ID
iSS_CAT	STRING 30	Issue_Category	

KEYWORDS	STRING 255		
LAST_MOD_BY	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified
OWNER_ID	UUID	ca_contact::uuid	
PARENT_ID	SREL	O_INDEXES::id	
PERMISSION_INDEX_ ID	INTEGER	O_INDEXES::id	
READ_PGROUP	INTEGER	P_GROUPS::id	
RELATIONAL_ID	STRING 255		
SUBJECT_EXPERT_ ID	UUID	ca_contact::uuid	
WF_TEMPLATE	INTEGER	CI_WF_ TEMPLATES::id	
WRITE_PGROUP	INTEGER	P_GROUPS::id	

SKELETONS Table

Program control table used by CA SDM Knowledge Management.

- **SQL Name** -- SKELETONS
- **Object** -- KD

Field	Data Type	Reference	Remarks
Active_state	INTEGER		
Active_state_date	LOCAL_TIME		
assignee_ID	UUID	ca_contact::uuid	
author_ID	UUID	ca_contact::uuid	
bu_result	REAL		
created_via	INTEGER		
creation_date	LOCAL_TIME		Indicates the timestamp when this record was created.
current_action_ID	INTEGER	ci_actions::id	
custom1	STRING 50		
custom2	STRING 50		
custom3	STRING 50		
custom4	STRING 255		
custom5	STRING 255		
custom_num1	REAL		
custom_num2	REAL		
doc_template_ID	INTEGER	ci_doc_templates ::id	
doc_type_ID	INTEGER	ci_doc_types::id	
doc_version	STRING 50		

expiration_date	LOCAL_TIME		
expire_notification_sent	INTEGER		
ext_doc_ID	INTEGER		
flg_cnt	INTEGER		Number of open flags in the KD.
fullwords	STRING 32768		
hits	INTEGER		
ID	INTEGER KEY		Identifies the unique (to the table) numeric ID.
indexed	INTEGER		
inherit_permission	INTEGER		
initiator	STRING 100		
initiator_ID	UUID	ca_contact::uuid	
KD_permission_index_ID	INTEGER	o_indexes::id	
last_accepted_date	LOCAL_TIME		
last_hit_date	INTEGER (DATE)		Specifies the last hit date created.
last_mod_dt	LOCAL_TIME		Indicates the timestamp when this record was last modified.
last_vote_date	INTEGER (DATE)		Specifies the last vote date presented in the Knowledge Report Card interface.
locked_by_ID	UUID	ca_contact::uuid	
modify_date	LOCAL_TIME		
notes	STRING 32768		
owner_ID	UUID	ca_contact::uuid	
parent_cr	STRING 30	call_req::persid	
parent_iss	STRING 30	issue persistent_id	
primary_index	INTEGER	o_indexes::id	
priority_ID	INTEGER	ci_properties::id	
PROBLEM	STRING 32768		
published_date	LOCAL_TIME		
read_pgroup	INTEGER	p_groups::id	
resolution	STRING 32768		
resolution_length	INTEGER		
resolution_short	STRING 255		
review_date	LOCAL_TIME		
rework_ver	INTEGER (1/0)		**0** -- Specifies all available knowledge document version records. **1** -- Specifies a rework version record.

sd_accepted_hits	INTEGER		
sd_asset_ID	UUID	ca_owned_ resource::uuid	
sd_impact_ID	INTEGER	impact::enum	
sd_priority_ID	INTEGER	priority::enum	
sd_problem	STRING		Problem persid
sd_product_ID	INTEGER	Product::id	
sd_rootcause_ID	INTEGER	rootcause::id	
sd_severity_ID	INTEGER	severity::enum	
sd_urgency_ID	INTEGER	urgency::enum	
shortwords	STRING 32768		
skip_auto_policies	INTEGER (4) (1/0)		Boolean (1/0) field to indicate if the automated policies should skip a knowledge document.
status_ID	INTEGER	ci_statuses::id	
subject_expert_ID	UUID	ca_contact::uuid	
summary	STRING 255		
ticket_avoided	INTEGER		Specifies the counter field to update in the Customer/ Employee interface when the user accepts a document as a solution to a request/incident/ problem during the self-service process.
title	STRING 255		
user_def_ID	STRING 40		
user_slv_cnt	INTEGER		
vote_count	INTEGER		
avg_rating	FLOAT		
faq_sym	INTEGER		
ver_comment	STRING 1000		Comment logged when the rework (new) version was created
ver_count	INTEGER		Specifies the version ID for the number of versions created (1, 2, 3, 4...).
ver_cross_ref_ID	INTEGER		Specifies a cross-reference between the published document record and the new version record.
wf_template	INTEGER	ci_wf_templates ::id	
word_count_total	INTEGER		
wordcount	INTEGER		

wordcounts	STRING 32768		
wordorders	STRING 32768		
wordplace	STRING 32768		
wordspans	STRING 32768		
write_pgroup	INTEGER	p_groups::id	

LONG_TEXTS

Program control table used by CA SDM Knowledge Management.

- **SQL Name** -- long_texts
- **Object** -- long_texts

Field	Data Type	Reference	Remarks
ACTUAL_TEXT	STRING 32768		
CNT_ORDER	INTEGER		
ID	INTEGER KEY		Unique (to the table) Numeric ID
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
REF_PERSID	STRING 30		

P_Groups Table

Program control table used by Knowledge Management.

- **SQL Name** -- P_GROUPS
- **Object** -- P_GROUPS

Field	Data Type	Reference	Remarks
GRP_LIST	STRING 4096		
GRP_LIST_KEY	STRING 255		
TYPE	INTEGER		Indicates if the P Groups is based on Roles or Groups: **1** -- Groups (Default) **2** -- Roles
ID	INTEGER KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
tenant	UUID	ca_tenant	Specifies the UUID of the tenant.

Doc_Versions

Program control table used by CA SDM Knowledge Management.

- **SQL Name** -- DOC_VERSIONS
- **Object** -- DOC_VERSIONS

Field	Data Type	Reference	Remarks
ID	INTEGER		Specifies the primary key.
doc_ID	INTEGER	skeletons::id	Specifies the SREL to the knowledge document object.
ver_count	INTEGER		Specifies the internal version ID that indicates the number of versions created (1, 2, 3, 4...)
ver_comment	STRING (2000)		Specifies the comment text.
start_date	DATE		Specifies the date when this version started.
end_date	DATE		Specifies the date when this version ended.
started_by	UUID		Specifies the user that created this version.
published_by	UUID		Specifies the user that published this version.
status_ID	INTEGER		Specifies the document version status.
ver_status	INTEGER		Specifies the previous version's status: **0** -- Existed **1** -- Archived **2** -- Deleted (this only relates to the resolution field)
title	STRING		Specifies the document version title.
summary	STRING		Specifies the document summary text.
problem	STRING		Specifies the problem description.
notes	STRING		Specifies the document notes.
doc_type_ID	INTEGER		Specifies the document type.
ext_doc_ID	INTEGER		Specifies the Tree ID when Decision Tree is used.
last_mod_by	Byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.

KD_Atmnt Table

Program control table used by Knowledge Management.

- **SQL Name** -- kd_attmnt
- **Object** -- kd_attmn

Field	Data Type	Reference	Remarks
ATTMNT_ID	INTEGER	Attachment::id	
DOC_ID	INTEGER	SKELETONS::id	
ID	INTEGER KEY		Unique (to the table) Numeric ID
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
RES_ID	Long		If there is an embedded image, the RES_ID points to the resource file id in the attmnt table.

Change Request

This article contains the following topics:

Change_Act_Log Table

Change manager tables Change_Act_Log is a history of activities associated with a change request. The types of activities are listed in the Act_Type table.

- **SQL Name** -- chgalg
- **Object** -- chgalg

Field	Data Type	Reference	Remarks
action_desc	nvarchar(4000)		Shows the text description of the activity log entry.
analyst	byte(16)	ca_contact::uuid	Foreign key to the contact_uuid field of the ca_contact table. This is the Analyst who created this activity log.
change_id	INTEGER	chg id	Foreign key to the id field of the chg table to which the activity log belongs. This is a Change Order.
description	nvarchar(4000)		Textual description of this activity log
id	INTEGER		Unique numeric ID that is the Primary key of this table.
internal	INTEGER		Marks this as 'Internal' log.
knowledge_session	nvarchar(80)		The knowledge session value for this Change_Act_Log.

knowledge_tool	nvarchar(12)		The knowledge management tool value for this Change_Act_Log.
last_mod_dt	INTEGER		Specifies the date of when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
system_time	INTEGER		Specifies the date and time of record creation.
time_spent	INTEGER		Specifies the time spent by the user on the activity.
time_stamp	INTEGER		Specifies the date and time time of activity by the user.
type	nvarchar(12)	Act_Type::code	Identifies the Activity log type.Note: This is the Foreign key to the code field of the act_type table.

Change_Category Table

Change Request categories can be hierarchical.

- **SQL Name** -- chgcat
- **Object** -- chgcat

Field	Data Type	Reference	Remarks
id	INTEGER		Specifies the unique (to the table) numeric ID.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id)
del	INTEGER	Active_Boolean_Table::enum	Delete flag that indicates the following:**0** -- Active**1** -- Inactive/ marked as deleted
sym	STRING (1000)	S_KEY	REQUIRED Change Category symbolic description.
code	STRING (12)	S_KEY	Primary key of this table.
last_mod_dt	INTEGER		Identifies the timestamp of when this record was last modified.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
description	nvarchar(500)		Identifies the textual description of this change category.
organization	byte(16)	ca_organization ::organization_ uuid	Foreign key to the id field of the ca_organization table, this is the Organization.

assignee	byte(16)	ca_contact	Foreign key to the contact_uuid field of the ca_contact table, this is the Assignee.
group_id	byte(16)	ca_contact	Foreign key to the contact_uuid field of the ca_contact table. This is the Group.
children_ok	INTEGER		Specifies the handling of the change category:**0** -- Children not allowed**1** -- Children are allowed
service_type	STRING (30)	Service_Desc	
survey	INTEGER	Survey_Template	
schedule	INTEGER	Bop_Workshift	Deprecated.
auto_assign	INTEGER		Flag that enables auto assignment.
owning_contract	INTEGER	Service_Contract	Foreign key to the id field of the svc_contract table.This is the Service Contract.
flow_flag	INTEGER		Specifies the type of workflow: **0** -- Classic Workflow or none (default) **2** -- CA Process Automation
caextwf_start_id	INTEGER	caextwf_start_forms	Identifies the CA Process Automation process definition information to use when the user selects this category on a change order, issue, request.
tenant	UUID	ca_tenant	
chgtype	INTEGER	usp_change_type	
risk_survey	INTEGER	Risk_Survey_Template	
cab	UUID	ca_contact	Reference to Contact information.
ss_include	INTEGER		REQUIRED On new default: 0
ss_sym	STRING (128)		

Change_Request Table

Change requests.

- **SQL Name** -- chg
- **Object** -- chg

Field	Data Type	Reference	Remarks
id	INTEGER UNIQUE NOT_NULL KEY		Primary key of this table, this is a unique numeric ID.
persid	STRING 30		Specifies the Persistent ID (SystemObjectName:id).
chg_ref_num	STRING 30 UNIQUE NOT_NULL S_KEY;		Shows a User visible reference number.
summary	STRING 240		Identifies the Change Order summary text.
description	STRING 4000		Provides a textual description of this Change Order.
status	STRING 12	Change_Status	Foreign key to the code field of the chgstat table, this identifies the Status.
active_flag	INTEGER NOT_NULL	Boolean_Table::enum	Flag representing whether this record is active or inactive: **0** -- Inactive **1** -- Active
start_date	LOCAL_TIME		Indicates the timestamp of when processing started.
open_date	LOCAL_TIME		Shows the timestamp of when this Change Order was created.
last_mod_dt	LOCAL_TIME		Identifies the timestamp of when this record was last modified.
last_mod_by	UUID REF	ca_contact	Specifies the UUID of the contact who last modified this record.
close_date	LOCAL_TIME		Shows the timestamp of when the Change Order was closed.
resolve_date	LOCAL_TIME		Indicates the timestamp of when the Change Order was resolved.
rootcause	INTEGER	Rootcause	Foreign key to the id field of the rootcause table, this identifies the Root Cause.
est_total_time	DURATION		Identifies the sum of estimated task time.
actual_total_time	DURATION		Specifies the sum of actual task time.
log_agent	UUID NOT_NULL	ca_contact	Foreign key to the contact_uuid field of the ca_contact table, this identifies who the change request was reported by.

assignee	UUID	ca_contact	Foreign key to the contact_uuid field of the ca_contact table, this identifies the Assignee.
organization	UUID	ca_organization	Foreign key to the id field of the ca_organization table, this identifies the Organization.
group_id	UUID	ca_contact	Foreign key to the contact_uuid field of the ca_contact table, this is the Group ID.
affected_contact	UUID NOT_NULL	ca_contact	Foreign key to the contact_uuid field of the ca_contact table, this specifies the Affected End User.
requestor	UUID NOT_NULL	ca_contact	Foreign key to the contact_uuid field of the ca_contact table, this identifies the Requestor.
category	STRING 12	Change_Category	Foreign key to the code field of the chgctg table, this identifies the Category.
priority	INTEGER NOT_NULL	Priority	Foreign key to the enum field of the pri table, this identifies the Priority.
need_by	LOCAL_TIME		Shows the Need By Date timestamp.
est_comp_date	LOCAL_TIME		Shows the timestamp for the estimated completion date.
actual_comp_date	LOCAL_TIME		Specifies the actual completion date timestamp.
est_cost	INTEGER		Specifies the estimated cost of this Change Order.
actual_cost	INTEGER		Identifies the actual cost of this Change Order.
justification	STRING 4000		Identifies the justification value for this Change_Request.
backout_plan	STRING 4000		Identifies the backout plan value for this Change_Request.
risk	INTEGER	Risk_Level	
business_case	STRING 4000		
cab	UUID	ca_contact	
closure_code	INTEGER	Closure_Code	
impact	INTEGER	Impact	Foreign key to the enum field of the impact table, this defines the impact.

parent	INTEGER	Change_Request	Foreign key to the id field of the chg table to allow for hierarchical Change Order groupings (for example, "parent/child").
effort	STRING 2000		Identifies the effort value for this Change_Request.
support_lev	STRING 30	Service_Desc	Foreign key to the code field of the srv_desc table, this identifies the Classic Service Type.
template_name	STRING 30		Foreign key to the template_name field of the chg_tpl table, this identifies the Template name.
sla_violation	INTEGER		Flag indicating the following: **1** -- change order has violated its sla.
predicted_sla_violation	INTEGER		Flag that indicates the following: **1** -- change order has been predicted by neugents.
macro_ predicted_violation	INTEGER		Identifies that it is likely to violate its sla (boolean) for action macros to predict sla violations.
created_via	INTEGER	Interface	Foreign key to the id field of the interface table, this identifies which interface created the Change request.
call_back_date	LOCAL_TIME		Identifies the call back timestamp for this Change Order.
call_back_flag	INTEGER		Specifies the call back flag value for this Change Order.
string1	STRING 40		This is a user-defined string field.
string2	STRING 40		This is a user-defined string field.
string3	STRING 40		This is a user-defined string field.
string4	STRING 40		This is a user-defined string field.
string5	STRING 40		This is a user-defined string field.
string6	STRING 40		This is a user-defined string field.

service_date	LOCAL_TIME		Specifies the service date value for this Change_Request.
service_num	STRING 30		Specifies the service num value for this Change_Request.
product	INTEGER	Product	Foreign key to the id field of the product table, this identifies the Product.
actions	STRING 750		
type_of_contact	INTEGER	Type_Of_Contact	
reporting_method	INTEGER	Reporting_Method	Foreign key to the id field of the repmeth table, this identifies how this Change Order was reported.
person_contacting	INTEGER	Person_ Contacting	Foreign key to the id field of the person table, this identifies the person who made the contact.
flag1	INTEGER		This is a user-defined integer flag.
flag2	INTEGER		This is a user-defined integer flag.
flag3	INTEGER		This is a user-defined integer flag.
flag4	INTEGER		This is a user-defined integer flag.
flag5	INTEGER		This is a user-defined integer flag.
flag6	INTEGER		This is a user-defined integer flag.
user1	STRING 100		This is a user-defined string field.
user2	STRING 100		This is a user-defined string field.
user3	STRING 100		This is a user-defined string field.
caextwf_instance_id	INTEGER	caextwf_instances	Indicates the CA Process Automation process instance id and process definition name and reference path launched by this Service Desk object.
project	UUID	ca_owned_ resource	
tenant	UUID	ca_tenant	
sched_start_ date	LOCAL_TIME		
sched_end_ date	LOCAL_TIME		
sched_duration	DURATION		

actual_start_date	LOCAL_TIME		
actual_end_date	LOCAL_TIME		
chgtype	INTEGER	usp_change_ type	
cab_approval	INTEGER	Boolean_Table	
requested_by	UUID REF	ca_contact	
external_ system_ticket	STRING 4000		
orig_user_dept	INTEGER	ca_resource_ department	
orig_user_organization	UUID	ca_organization	
orig_user_ admin_org	UUID	ca_organization	
orig_user_cost_ center	INTEGER	ca_resource_ cost_center	
target_start_last	LOCAL_TIME		
target_hold_last	LOCAL_TIME		
target_hold_ count	INTEGER		
target_resolved_last	LOCAL_TIME		
target_resolved_count	INTEGER		
target_closed_last	LOCAL_TIME		
target_closed_count	INTEGER		
target_start_last	LOCAL_TIME		Time when target timer started or restarted.
target_hold_last	LOCAL_TIME		Time of most recent ticket hold
target_hold_count	INTEGER		Number of times ticket went into a Hold status
target_resolved_last	LOCAL_TIME		Time of most recent ticket resolution
target_resolved_count	INTEGER		Number of times ticket was resolved
target_closed_last	LOCAL_TIME		Time ticket was last closed
target_closed_count	INTEGER		Number of times that ticket was closed

Change_Status Table

Lists the states of the change request, which you can add to at will. This table allows you to control whether the change request is active or inactive when it is changed to this status. Possible statuses include: Open, approval in process, implementation in progress, verification in progress, cancelled, suspended, and closed.

- **SQL Name** -- chgstat
- **Object** -- chgstat

Field	Data Type	Reference	Remarks
active	INTEGER		Flag that indicates the following:**0** -- Inactive**1** -- Active

code	nvarchar(12)		Primary key of this table.
del	INTEGER	Active_Boolean_Table::enum	Delete flag that indicates the following:**0** -- Active**1** -- Inactive/ marked as deleted
description	nvarchar(500)		Identifies the textual description of this status.
hold	INTEGER		Flag that specifies the following:**0** -- Start events**1** -- Stop events
id	INTEGER		Unique (to the table) numeric ID.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp of when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
resolved	INTEGER		Flag that indicates the following:**0** -- Not yet resolved**1** -- Resolved
sym	nvarchar(30)		Identifies the Change Request status name.

Chg_Template Table

Maps a template name to a Change_Request that will be used as a template.

- **SQL Name** -- chg_template
- **Object** -- chg_tpl

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_Table::enum	Delete flag that represents the following:**0** -- Active**1** -- Inactive/ marked as deleted
description	nvarchar (1000)		Describes the template.
id	INTEGER		Unique (to the table) numeric ID.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp of when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id)
quick_tmpl_type	INTEGER	Quick_Template_Types::enum	Flag that indicates the following:**1** -- Quick tmpl**2** -- Quick tmpl+review

template	INTEGER	chg id	Unique (to the table) numeric ID.
template_class	nvarchar(12)		Allows subclassing templates.
template_name	nvarchar(30)		Unique name of the template.

chg_trans Table

A transition object controls the current and next ticket status. The chg_trans table lists the status, new status, and actions that need to occur for a default transition.

- **SQL Name** -- chg_trans
- **Object** -- chg_trans

Label	Field	Description
id	INTEGER	Unique key.
status	SYMBOL	Specifies the current ticket status.
new_status	SYMBOL	Specifies the new ticket status
must_comment	INTEGER	Comment required when using a transition. On new default: 0
is_default	INTEGER	Default transition that appears when the Status field is empty. On new default: 0
condition	BOP_REF_STR_REF Macro	Site condition macro to approve transition.
condition error	STRING 255	Error message for site condition.
description	STRING 255	Description of this transition.
last_mod_dt	LOCAL_TIME	Timestamp of last update to this record.
last_mod_by	UUID REF ca_contact	User who last updated this.
del	INTEGER nn	Logical database delete status.
tenant	UUID REF ca_tenant	Reference to Tenant information.

CI Attributes

ci_managed_attribute Table

The following table lists CI attributes that CACF manages for change verification:

- **SQL Name** -- ci_managed_attribute
- **Object** -- ci_managed_attribute

Field	Data Type	Reference	Remarks
id	INTEGER		Primary key to this table, it is a unique numeric ID.
last_mod_dt	LOCAL_TIME		Specifies the timestamp of when this record was last modified.

last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
tenant	UUID	ca_tenant	Specifies the UUID of the tenant.
del	INTEGER	Active_Boolean_ Table::enum	Indicates the Deleted flag, as follows:**0** -- Active **1** -- Inactive/marked as deleted
attribute_name	STRING 128		Specifies the name of the CI attribute.
attribute_label	STRING 50		Specifies the label for the attribute.
description	STRING 255		Specifies a description for the attribute.
initial_status	STRING 32	ci_planned_change_status	Specifies the initial status of the attribute.
case_sensitive	INTEGER		Specifies if the managed attribute is case sensitive.
attribute_type	INTEGER		Specifies the data type.
attribute_length	INTEGER		Specifies the maximum length of the attribute.
srel_factory	STRING 26		Specifies the factory associated with a SREL attribute.
srel_rel_attr	STRING 26		Specifies the factory attribute used as a foreign key.
srel_common_name_attr	STRING 26		Specifies the factory attribute that stores the human-readable value.
srel_show_dropdown	INTEGER	Boolean_Table	Specifies the factory to show in the drop-down list.

Change States

ci_managed_chgstat Table

The following table lists details about managed change states:

- **SQL Name** -- ci_managed_chgstat
- **Object** -- ci_managed_chgstat

Field	Data Type	Reference	Remarks
id	INTEGER		Primary key to this table, it is a unique numeric ID.
last_mod_dt	LOCAL_TIME		Specifies the timestamp of when this record was last modified.

last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
del	INTEGER	Boolean_Table	Indicates the Deleted flag, as follows:**0** -- Active **1** -- Inactive/marked as deleted
code	STRING 12	chgstat	Specifies the Change Order status managed by change verification.
tenant	UUID	ca_tenant	Specifies the UUID of the tenant.
can_edit_criteria	INTEGER	Boolean_Table	Specifies if the change status lets you edit criteria.
verification_active	INTEGER	Boolean_Table	Specifies if change verification is active in this state.
autopromote_chg	INTEGER	Boolean_Table	Specifies if autopromote is active for the state.
show_override_buttons	INTEGER	Boolean_Table	Specifies if the override buttons appear for the state.
is_implementation	INTEGER	Boolean_Table	Specifies if the state is an implementation state.

Change Verification

ci_verification_policy Table

The following table lists details about verification policies:

- **SQL Name** -- ci_verification_policy
- **Object** -- ci_verification_policy

Field	Data Type	Reference	Remarks
id	INTEGER		Primary key to this table, it is a unique numeric ID.
last_mod_dt	LOCAL_TIME		Specifies the timestamp of when this record was last modified.
last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
tenant	UUID	ca_tenant	Specifies the UUID of the tenant.
del	INTEGER	Active_Boolean_ Table::enum	Indicates the Deleted flag, as follows:**0** -- Active **1** -- Inactive/marked as deleted
attribute_name	STRING 128	ci_managed_attribute	Specifies the name of the attribute.

ci_name_pattern	STRING 255		Specifies the name of the CI pattern.
class_pattern	STRING 255		Specifies the name of the CI class pattern.
description	STRING 255		Specifies the name of the description of the policy.
mdr_name_pattern	STRING 255		Specifies the name of the MDR pattern.
mdr_class_pattern	STRING 255		Specifies the name of the MDR class pattern.
rolename_pattern	STRING 255		Specifies the name of the role pattern.
name	STRING 100		Specifies the name of the policy.
sequence	INTEGER		Specifies the order that CACF processes the policy.
location_pattern	STRING 255		Specifies the name of the location pattern.
isvariance	INTEGER	Boolean_Table	Specifies if the policy activates for a variance.
isnotverifiable	INTEGER	Boolean_Table	Specifies if the policy is not verifiable.
isrogue_insert	INTEGER	Boolean_Table	Specifies if the policy activates for rogue inserts.
isrogue_update	INTEGER	Boolean_Table	Specifies if the policy activates for rogue updates.
action	STRING 32	ci_verification_policy_act	Specifies the policy action.
write_twa	INTEGER	Boolean_Table	Specifies if you want the policy to write CI and relationship data to the TWA.
write_incident	INTEGER	Boolean_Table	Specifies if you want the policy to create an Incident.
incident_template	STRING 30	cr_template	Specifies the Incident template.
autoclose_incident	INTEGER	Boolean_Table	Specifies if you want the policy to close the Incident automatically.
log_only_mode	INTEGER	Boolean_Table	Specifies if you want the policy to write to the log instead of modifying a Change Order, CI, or relationship.
start_date	LOCAL_TIME		Specifies the date to start the policy.
end_date	LOCAL_TIME		Specifies the date to end the policy.
service_type	STRING 30	Service_Desc	Specifies the service type of the policy.

priority	INTEGER	Priority	Specifies the priority of the policy.

ci_verification_policy_act Table

The following table lists information about verification policy actions:

- **SQL Name** -- ci_verification_policy_act
- **Object** -- ci_verification_policy_action

Field	Data Type	Reference	Remarks
id	INTEGER		Primary key to this table, it is a unique numeric ID.
last_mod_dt	LOCAL_TIME		Specifies the timestamp of when this record was last modified.
last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
tenant	UUID	ca_tenant	Specifies the UUID of the tenant.
description	STRING 255		Specifies the long description of the policy action.
name	STRING 100		Specifies the short name of the policy action.
sym	STRING 30		Specifies the constant symbol for this action.

ci_verification_policy_log Table

The following table lists CI attributes that CACF manages for change verification:

- **SQL Name** -- ci_verification_policy_log

Field	Data Type	Reference	Remarks
id	INTEGER		Primary key to this table, it is a unique numeric ID.
last_mod_dt	LOCAL_TIME		Specifies the timestamp of when this record was last modified.
last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
tenant	UUID	ca_tenant	Specifies the UUID of the tenant.
del	INTEGER	Active_Boolean_ Table::enum	Indicates the Deleted flag, as follows:**0** -- Active **1** -- Inactive/marked as deleted
ci_planned_change	INTEGER	ci_planned_change	Specifies the change specification.

nr	UUID	ca_owned_resource	
ci_verification_policy	INTEGER	ci_verification_policy	Specifies the verification policy.
ci_managed_attribute	INTEGER	ci_managed_attribute	Specifies the managed attribute.
oldvalue	STRING 2000		Specifies the old attribute value.
discoveredvalue	STRING 2000		Specifies the discovered attribute value.
newvalue	STRING 2000		Specifies the new attribute value.
status	STRING 32	ci_planned_change_status	Specifies the change verification state.
ci_twa_ci	INTEGER	ci_twa_ci	Specifies the CI in the TWA.
ci_twa_relation	INTEGER	ci_twa_relation	Specifies the relationship in the TWA.
ci_mdr_provider	INTEGER	ci_mdr_provider	Specifies the provider of the MDR.
is_unverifiable	INTEGER	Boolean_Table	Specifies if the change is verifiable.
is_rogueinsert	INTEGER	Boolean_Table	Specifies if the change is a rogue insert.
is_variance	INTEGER	Boolean_Table	Specifies if the change is a variance.
is_rogueupdate	INTEGER	Boolean_Table	Specifies if the change is a rogue update.
ci_verification_policy_act	STRING 32	ci_verification_policy_act	Specifies the verification policy action.
incident	STRING 30	Call_Req	Specifies the incident associated with the verification policy.
role	INTEGER	usp_role	Specifies the role associated with the verification policy.
error_message	STRING 2000		Specifies the error message.

ci_verification_twa_act Table

The following table lists CI attributes that CACF manages for change verification:

- **SQL Name** -- ci_verification_twa_act
- **Object** -- ci_verification_twa_act

Field	Data Type	Reference	Remarks
id	INTEGER		Primary key to this table, it is a unique numeric ID.
last_mod_dt	LOCAL_TIME		Specifies the timestamp of when this record was last modified.

last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
del	INTEGER	Active_Boolean_ Table::enum	Indicates the Deleted flag, as follows:**0** -- Active **1** -- Inactive/marked as deleted
description	STRING 255		Specifies a description for the attribute.
name	STRING 50		Specifies the name of the TWA action.
sym	STRING 32		Specifies the constant symbol for this action.

Change Specifications

This article contains the following topics:

ci_planned_change Table

The following table lists details about change specifications:

- **SQL Name** -- ci_planned_change
- **Object** -- ci_planned_change

Field	Data Type	Reference	Remarks
id	INTEGER		Primary key to this table, it is a unique numeric ID.
last_mod_dt	LOCAL_TIME		Specifies the timestamp of when this record was last modified.
last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
tenant	UUID	ca_tenant	Specifies the UUID of the tenant.
del	INTEGER	Active_Boolean_ Table::enum	Indicates the Deleted flag, as follows:**0** -- Active **1** -- Inactive/marked as deleted
attribute_name	STRING 128		Specifies the name of the CI or relationship attribute.
attribute_value_original	STRING 2000		Specifies the original attribute value.
attribute_value_planned	STRING 2000		Specifies the planned attribute value.
attribute_value_discovered	STRING 2000		Specifies the discovered attribute value.

chg	INTEGER	Change_Request	Specifies the Change Order associated with the planned change.
ci	UUID	ca_owned_resource	Specifies the CI associated with the change specification.
description	STRING 255		Specifies a description of the change specification.
status	STRING 32	ci_planned_change_status	Specifies the status of the change.
incident	STRING 30	Call_Req	Specifies the Incident that the change specification created.
attribute_value_disc_internal	STRING 255		Specifies the internal discovered attribute value.
attribute_value_orig_internal	STRING 255		Specifies the internal original attribute value.
attribute_value_internal	STRING 255		Specifies the internal attribute value.
last_verification_policy	INTEGER	ci_verification_policy	Specifies the last verification policy associated with the change specification.
ci_twa_ci	INTEGER	ci_twa_ci	Specifies the CI in the TWA.
ci_twa_relation	INTEGER	ci_twa_relation	Specifies the CI relationship in the TWA.
verification_msg	STRING 255		Specifies the verification message.

ci_planned_change_status Table

The following table lists details about change specification states:

- **SQL Name** -- ci_planned_change_status
- **Object** -- ci_planned_change_status

Field	Data Type	Reference	Remarks
id	INTEGER		Primary key to this table, it is a unique numeric ID.
last_mod_dt	LOCAL_TIME		Specifies the timestamp of when this record was last modified.
last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
tenant	UUID	ca_tenant	Specifies the UUID of the tenant.
del	INTEGER	Boolean_Table	Indicates the Deleted flag, as follows:**0** -- Active **1** -- Inactive/marked as deleted

description	STRING 255		Specifies the long description of the policy action.
name	STRING 128		Specifies the short name of the policy action.
sym	STRING 10		Specifies the constant symbol for this action.
isinitial	INTEGER	Boolean_Table	Specifies the initial state of the state.
isfinal	INTEGER	Boolean_Table	Specifies the final state of the state.
isselectable	INTEGER	Boolean_Table	Specifies the selectable state.

Events

This article contains the following topics:

Event_Delay Table

This table lists the times that events were delayed.

- **SQL Name** -- evt_dly
- **Object** -- evtdly

Field	Data Type	Reference	Remarks
description	STRING 80		Specifies the user description of delay,
act_delay	DURATION		Specifies the actual duration of delay.
cancel_time	LOCAL_TIME		Specifies when the time delay was cancelled.
create_time	LOCAL_TIME		Specifies when the time delay was created.
delay_type	INTEGER	Event_Delay_Type:: enum	enum Event_Delay_Type
eff_delay	DURATION		Specifies the effective duration of delay.
group_name	STRING 30		Specifies the group Name of attevt (currently only SLA).
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
obj_id	STRING 30 NOT_NULL S_KEY		Specifies the persid of the object.
persid	STRING 30		Persistent ID (SystemObjectName:id)

start_time	LOCAL_TIME		Specifies when the time delay was started.
start_userid	UUID	ca_contact::uuid	Specifies the user that created and started delay.
status_flag	INTEGER		Specifies a flag for indicating an event delay status.
stop_time	LOCAL_TIME		Specifies when the time delay was stopped
stop_userid	UUID	ca_contact::uuid	Specifies the user that stopped and cancelled delay.
support_lev	STRING 30	Service_Desc::code	Specifies the service type in effect.

Event_Delay_Type Table

User-defined code for the type of event delay.

- **SQL Name** -- evtdlytp
- **Object** -- evtdlytp

Field	Data Type	Reference	Remarks
del	INTEGER NOT_NULL	Active_Boolean_Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
enum	INTEGER NOT_NULL		Enumerated value for this entry - specifies ordering in lists and relative values
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
nx_desc	STRING 40		
sym	STRING 12 UNIQUE NOT_NULL S_KEY		

event_log Table

USP event log.

- **SQL Name** -- event_log
- **Object** -- event_log

Field	Data Type	Reference	Remarks
event	INTEGER	event_type::id	Event type
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
kd_id	INTEGER	SKELETONS::id	Assoc knowledge doc
log_time	LOCAL_TIME		Server log date
millitime	INTEGER		Log date millisec

numdata1	INTEGER		Smag number
numdata2	INTEGER		Smag number
sd_obj_id	INTEGER		Assoc SD id
sd_obj_type	STRING 30		Assoc SD object
session	INTEGER	session_log::id	Session with event
textdata1	STRING 500		Smag string
textdata2	STRING 500		Smag string

event_type Table

Event type.

- **SQL Name** -- event_type
- **Object** -- event_type

Field	Data Type	Reference	Remarks
description	STRING 500		Specifies the description.
code	STRING 12 UNIQUE NOT_NULL S_KEY		Specifies the noneditable string enum.
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
persid	STRING 30		Persistent ID (SystemObjectName:id)

Events Table

Tables for Event stuff. Events attached to objects.

- **SQL Name** -- evt
- **Object** -- evt

Field	Data Type	Reference	Remarks
description	STRING 80		Specifies the event description.
condition	STRING 30	Spell_Macro:: persid	Specifies the macro function for condition.
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
delay_time	DURATION NOT_NULL		Specifies the time until condition check.

id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
modulo_time	DURATION NOT_NULL		Specifies the time increment to adjust.
obj_type	STRING 30		Specifies the object type for this event.
on_done_ flag	INTEGER NOT_NULL		Sets the *fire_time* time directly on the attached event to indicate when the event is done.
persid	STRING 30		Persistent ID (SystemObjectName:id)
sym	STRING 30 NOT_NULL unique S_KEY		Specifies the event name.
urgency	INTEGER		Specifies the urgency of the event.
user_settime	INTEGER NOT_NULL		The user_settime allows the user (or system) to override the fire time (delay time) that is defined for an event. For example, when users add an event to a Service Type, they can redefine the fire time for the event only if the user_settime flag is set. Otherwise, the fire time defined in the event is used.
user_smag	STRING 200		Specifies the user smag field.
violate_on_ false	INTEGER		
violate_on_ true	INTEGER		
work_shift	STRING 30	Bop_Workshift:: persid	

O_EVENTS Table

Program control table used by CA SDM Knowledge Management.

- **SQL Name** -- O_EVENTS
- **Object** -- o_events

Field	Data Type	Reference	Remarks
ACTION	STRING 32768		
ENTITY_ID	INTEGER	SKELETONS::id	
EVENT_NAME	STRING 50		
EVENT_TIMESTAMP	LOCAL_TIME		
ID	INTEGER KEY		Unique (to the table) Numeric ID

LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
VER_COUNT	INTEGER		Version ID indicates the number of versions created (1, 2, 3, 4...)
WF_ACTION_ID	INTEGER		
WF_USER_ID	UUID	ca_contact::uuid	

Data Partition Constraints and Controlled Table

This article contains the following topics:

Domain Table

Lists the names and descriptions of the data partitions themselves.

- **SQL Name** -- dmn
- **Object** -- dmn

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
id	INTEGER		Primary key of this table, it is a unique numeric ID.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
nx_desc	nvarchar(40)		Identifies the data partition description.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
sym	nvarchar(30)		Identifies the data partition name.

Domain_Constraint Table

Lists the constraints associated with a particular data partition and controlled table.

- **SQL Name** -- dcon
- **Object** -- dcon

Field	Data Type	Reference	Remarks
alias	INTEGER		Identifies the alias value for this Domain_Constraint.
constraint_majic	nvarchar(4000)		Specifies Constraint (Majic).
constraint_ SQL	nvarchar(4000)		Specifies Constraint (SQL).

del	INTEGER	Active_Boolean_Table::enum	Indicates the Delete flag, as follows:**0** -- Active**1** -- inactive/ marked as deleted)
dom_id	INTEGER	Domain::id	Identifies the unique (to the table) numeric ID.
error_msg	nvarchar(150)		Specifies the message on violation.
id	INTEGER		Primary key to this table, it is a unique numeric ID.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
tbl_id	INTEGER	ctab::id	Indicates the unique (to the table) numeric ID.
type	INTEGER	Domain_ Constraint_Type ::enum	Enumerated value for this entry, this specifies ordering in lists and relative values.

Domain_Constraint_Type Table

Lists the types of constraints that can be associated with a particular data partition and controlled table.

- **SQL Name** -- dcon_typ
- **Object** -- dcon_typ

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
enum	INTEGER		Enumerated value for this entry, this specifies ordering in lists and relative values.
id	INTEGER		Indicates the unique (to the table) numeric ID.
nx_desc	nvarchar(40)		Specifies the description of the domain contraint type.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
sym	nvarchar(12)		Identifies the symbolic name of this constraint type.

External Entity

External_Entity_Map Table

NOT FOR CLIENT USE. Maps an external entity to an internal object where the external entity is uniquely defined in its own namespace by the xentity_id. The namspace is uniquely defined for our use by the xschema_code and xschema_ver. The semantics of the xentity_id, and parameters (*_rsrved fields) depends upon the namespace.

- **SQL Name** -- xent_map
- **Object** -- ext_entity_map

Field	Data Type	Reference	Remarks
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
int1_rsrved	INTEGER		reserved for CA - do not use
int2_rsrved	INTEGER		reserved for CA - do not use
int3_rsrved	INTEGER		reserved for CA - do not use
int4_rsrved	INTEGER		reserved for CA - do not use
int5_rsrved	INTEGER		reserved for CA - do not use
int6_rsrved	INTEGER		reserved for CA - do not use
lstr1_rsrved	STRING 255		reserved for CA - do not use
lstr2_rsrved	STRING 255		reserved for CA - do not use
ob_persid	STRING 30		the "mapped to" object
ob_type	STRING 30		the "mapped to" object type
persid	STRING 30		Persistent ID (SystemObjectName:id)
str1_rsrved	STRING 80		reserved for CA - do not use
str2_rsrved	STRING 80		reserved for CA - do not use
xentity_id	STRING 180 NOT_NULL		Specifies the uniq. external entity reference.
xschema_code	STRING 12 NOT_NULL		Specifies the internal code for the namespace of the entity.
xschema_ver	INTEGER NOT_NULL		Specifies the internal ver for the namespace of the entity.

Form Group

Form_Group Table

Listing of defined form groups.

- **SQL Name** -- fmgrp
- **Object** -- fmgrp

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
description	nvarchar(100)		Specifies the textual description of this form group.
id	INTEGER		Primary key of this table, it is a unique, numeric ID.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
sym	nvarchar(30)		Specifies the symbolic value for this Form_Group.

Interface Definitions

intfc Object

Interface definitions used for creating requests and change orders.

- **SQL Name** -- interface
- **Object** -- intfc

Field	Data Type	Reference	Remarks
code	nvarchar(10)		Specifies the code value for this interface.
del	INTEGER	Active_Boolean_ Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
id	INTEGER		Primary key of this table, it is a unique numeric ID.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp of when this record was last modified.
nx_desc	nvarchar(240)		The desc value for this interface.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
sym	nvarchar(30)		Identifies the symbolic value for this interface.

Document Repository

Document_Repository Table

Contains Information on document repositories, which are used to store attachments.

- **SQL Name** -- doc_rep
- **Object** -- doc_rep

Field	Data Type	Reference	Remarks
description	STRING 500		Specifies the description.
archive_path	STRING 255		
archive_type	INTEGER		
cgi_path	STRING 255		Specifies the location and name of CGI.
default_rep	INTEGER		
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
file_limit_size	INTEGER		Specifies the file limit size.
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
persid	STRING 30		Persistent ID (SystemObjectName:id)
prohibited_ext	STRING 500		Specifies the prohibited file extensions.
protocol	STRING 12		HTTP or SHARE
repository_type	INTEGER		Specifies the type of repository (attachments, knowledge).
retrieve_path	STRING 255		Specifies how to get back to upload_path via protocol.
server	STRING 30		Specifies the name of Doc Server.
servlet_path	STRING 255		Specifies the servlet URL.
sym	STRING 30 NOT_NULL S_KEY		Specifies the name of document repository.
upload_path	STRING 255		Specifies the server location of doc repository.

External Application

ext_appl Table

External application.

- **SQL Name** -- ext_appl

Field	Data Type	Reference	Remarks
description	STRING 500		Appl description
code	STRING 12 UNIQUE NOT_NULL S_KEY		Noneditable string enum
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID		Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
persid	STRING 30		Persistent ID (SystemObjectName:id)
sym	STRING 30 NOT_NULL S_KEY		Application name

Group Member

Group Members.

- **SQL Name** -- grpmem
- **Object** -- grpmem

Field	Data Type	Reference	Remarks
group_id	byte(16)		Foreign key to the contact_uuid field of the ca_contact table, this is the group.
id	INTEGER		Specifies the unique (to the table) numeric ID.
manager_flag	INTEGER		Specifies the Manager flag, as follows:**0** -- No**1** -- Group manager.
member	byte(16)	ca_contact::uuid	Foreign key to the contact_uuid field of the ca_contact table, this is the Member.
notify_flag	INTEGER		Specifies the Notify flag, as follows:**0** -- No notification, **1** -- Notify.

Global Tables

This article contains the following topics:

Global_Change_Extension Table

Local copy of data that will be pushed to master's Global_Change_Queue table.

- **SQL Name** -- g_chg_ext
- **Object** -- g_chg_ext

Field	Data Type	Reference	Remarks
active_flag	INTEGER NOT_NULL	Active_Reverse_ Boolean::enum	**1** -- Active **0** -- Inactive
affected_contact	UUID NOT_NULL	ca_contact::uuid	Specifies the Effected End User of CO.
assignee	UUID	ca_contact::uuid	Specifies the assignee of CO.
category	STRING 30		Specifies the category symbol of CO.
chg_ref_num	STRING 30 NOT_NULL		Specifies the change ref num.
close_date	LOCAL_TIME		Specifies the close Date of CO.
global_queue_id	INTEGER	Global_Queue_ Names::id	Specifies the pointer to global queue.
group_id	UUID		Specifies the group assigned to CO.
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
impact	INTEGER	Impact::enum	Specifies the impact of change order.
last_mod_dt	LOCAL_TIME NOT_NULL		Indicates the timestamp of when this record was last modified.
open_date	LOCAL_TIME NOT_NULL		Specifies the open date of CO.
priority	INTEGER NOT_NULL	Priority::enum	Specifies the priority of CO.
remote_id	INTEGER NOT_NULL S_KEY		Specifies the regional request id.
requestor	UUID NOT_NULL	ca_contact::uuid	Specifies the requester of CO.
status	STRING 30 NOT_NULL	Change_Status:: code	Specifies the status symbol of CO.
summary	STRING 240		Specifies the summary of CO.

Global_Change_Queue Table

This table is an accumulation of all of the regional Global_Change_Extension tables.

- **SQL Name** -- g_chg_queue
- **Object** -- g_chg_queue

Field	Data Type	Reference	Remarks
active_flag	INTEGER NOT_NULL	Active_Reverse_ Boolean::enum	**1** -- Active **0** -- Inactive
affected_ contact	UUID NOT_NULL		Specifies the Affected End User of CO.
assignee	UUID		Specifies the assignee of CO.

category	STRING 30		Specifies the category symbol of CO.
chg_ref_num	STRING 30 NOT_NULL		Specifies the change ref num.
close_date	LOCAL_TIME		Specifies the close Date of CO.
global_queue_ id	INTEGER	Global_Queue_ Names::id	Specifies the pointer to global queue.
group_id	UUID		Specifies the group assigned to CO.
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
impact	INTEGER	Impact::enum	Specifies the impact of change order.
last_mod_dt	LOCAL_TIME NOT_NULL		Indicates the timestamp of when this record was last modified.
open_date	LOCAL_TIME NOT_NULL		Specifies the open date of CO.
priority	INTEGER NOT_NULL	Priority::enum	Specifies the priority of CO.
remote_id	INTEGER NOT_NULL S_KEY		Specifies the regional change id.
remote_sys_id	INTEGER NOT_NULL S_KEY	Global_Servers:: remote_sys_id	Specifies the regional system id.
requestor	UUID NOT_NULL		Specifies the requestor of CO.
status	STRING 30 NOT_NULL	Change_Status:: code	Specifies the status sybmol of CO.
summary	STRING 240		Specifies the summary of CO.

Global_Contact Table

Contain all contacts across all systems participating in the Global Service Desk.

- **SQL Name** -- g_contact
- **Object** -- g_cnt

Field	Data Type	Reference	Remarks
contact_num	STRING 30		Specifies the contact number.
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
email_address	STRING 120		Specifies the email address of the contact.
first_name	STRING 100		Specifies the first name of the contact.
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.

last_name	STRING 100		Specifies the last name of the contact.
loc_id	UUID		Specifies the location id.
middle_name	STRING 100		Specifies the middle name of the contact.
org_id	UUID		Specifies the organization id.
pri_phone_ number	STRING 32		Specifies the phone number of the contact.
remote_id	UUID NOT_NULL S_KEY		Specifies the regional contact id.
remote_sys_id	INTEGER NOT_NULL S_KEY	Global_Servers ::remote_sys_id	Specifies the regional system id.
userid	STRING 100		Specifies the userid of the contact.

Global_Issue_Extension Table

Local copy of data that will be pushed to master's Global_Issue_Queue table.

- **SQL Name** -- g_iss_ext
- **Object** -- g_iss_ext

Field	Data Type	Reference	Remarks
active_flag	INTEGER NOT_NULL	Active_Reverse_ Boolean::enum	**1** -- Active **0** -- Inactive
assignee	UUID	ca_contact::uuid	Assignee of Issue
category	STRING 30		Category symbol of Issue
close_date	LOCAL_TIME		Close Date of Issue
global_queue_id	INTEGER	Global_Queue_Names::id	Pointer to global queue
group_id	UUID		Group assigned to Issue
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
impact	INTEGER	Impact::enum	Impact of Issue
last_mod_dt	LOCAL_TIME NOT_NULL		Indicates the timestamp of when this record was last modified.
open_date	LOCAL_TIME NOT_NULL		Open Date of Issue
priority	INTEGER NOT_NULL	Priority::enum	Priority of Issue
product	INTEGER	Product::id	Product of Issue
ref_num	STRING 30 NOT_NULL		Issue ref num
remote_id	INTEGER NOT_NULL S_KEY		Regional request id
requestor	UUID NOT_NULL	ca_contact::uuid	Affected End User of CO
status	STRING 30 NOT_NULL	Issue_Status::code	Status symbol of Issue
summary	STRING 240		Summary of Issue

Global_Issue_Queue Table

This table is an accumulation of all of the regional Global_Issue_Extension tables.

- **SQL Name** -- g_iss_queue
- **Object** -- g_iss_queue

Field	Data Type	Reference	Remarks
active_flag	INTEGER NOT_NULL	Active_Reverse_ Boolean::enum	**1** -- Active **0** -- Inactive
assignee	UUID		Assignee of Issue
category	STRING 30		Category symbol of Issue
close_date	LOCAL_TIME		Close Date of Issue
global_queue_ id	INTEGER	Global_Queue_ Names::id	Pointer to global queue
group_id	UUID		Group assigned to Issue
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
impact	INTEGER	Impact::enum	Impact of Issue
last_mod_dt	LOCAL_TIME NOT_NULL		Indicates the timestamp of when this record was last modified.
open_date	LOCAL_TIME NOT_NULL		Open Date of Issue
priority	INTEGER NOT_NULL	Priority::enum	Priority of Issue
product	INTEGER		Product of Issue
ref_num	STRING 30 NOT_NULL		Issue ref num
remote_id	INTEGER NOT_NULL S_KEY		Regional Issue id
remote_sys_id	INTEGER NOT_NULL S_KEY	Global_Servers:: remote_sys_id	Regional system id
requestor	UUID NOT_NULL		Affected User of Issue
status	STRING 30 NOT_NULL	Issue_Status::code	Status symbol of Issue
summary	STRING 240		Summary of Issue

Global_Location Table

Contains all Location names for all regions.

- **SQL Name** -- g_loc
- **Object** -- g_loc

Field	Data Type	Reference	Remarks
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.

loc_name	STRING 100		Location Name
remote_id	UUID NOT_NULL S_KEY		Regional location id
remote_sys_id	INTEGER NOT_NULL S_KEY	Global_Servers:: remote_sys_id	Regional system id

Global_Organization Table

Contains all Organizations names for all regions.

- **SQL Name** -- g_org
- **Object** -- g_org

Field	Data Type	Reference	Remarks
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
org_name	STRING 100		Organization Name
remote_id	UUID NOT_NULL S_KEY		Regional organization id
remote_sys_id	INTEGER NOT_NULL S_KEY	Global_Servers:: remote_sys_id	Regional system id

Global_Product Table

Contains all Product names for all regions. Used by Global_Issue_Queue.

- **SQL Name** -- g_product
- **Object** -- g_prod

Field	Data Type	Reference	Remarks
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
remote_id	INTEGER NOT_NULL S_KEY		Regional product id
remote_sys_ id	INTEGER NOT_NULL S_KEY	Global_Servers:: remote_sys_id	Regional system id
sym	STRING 60		Symbol name of product

Global_Queue_Names Table

List of queues that are considered global.

- **SQL Name** -- g_queue_names
- **Object** -- g_qname

Field	Data Type	Reference	Remarks
description	STRING 100		comments field
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
sym	STRING 30 UNIQUE NOT_NULL		Descriptive name

Global_Request_Extension Table

Local copy of data that will be pushed to master's Global_Request_Queue table.

- **SQL Name** -- g_req_ext
- **Object** -- g_cr_ext

Field	Data Type	Reference	Remarks
active_flag	INTEGER NOT_NULL	Active_Reverse_ Boolean::enum	**1** -- Active **0** -- Inactive
assignee	UUID	ca_contact::uuid	Assignee of Request
category	STRING 30		Category symbol of Request
close_date	LOCAL_TIME		Close Date of Request
customer	UUID NOT_NULL	ca_contact::uuid	Affected User of Request
global_queue_id	INTEGER	Global_Queue_ Names::id	Pointer to global queue
group_id	UUID		Group assigned to Request
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
impact	INTEGER	Impact::enum	Impact of Request
last_mod_dt	LOCAL_TIME NOT_NULL		Indicates the timestamp of when this record was last modified.
open_date	LOCAL_TIME NOT_NULL		Open Date of Request
priority	INTEGER NOT_NULL	Priority::enum	Priority of Request
ref_num	STRING 30 NOT_NULL		Request ref num
remote_id	INTEGER NOT_NULL S_KEY		Regional request id
status	STRING 30 NOT_NULL	Cr_Status::code	Status symbol of Request

summary	STRING 240		Summary of Request
type	STRING 10	crt code	ITIL record type

Global_Request_Queue Table

This table is an accumulation of all of the regional Global_Request_Extension tables.

- **SQL Name** -- g_req_queue
- **Object** -- g_cr_queue

Field	Data Type	Reference	Remarks
active_flag	INTEGER NOT_NULL	Active_Reverse_ Boolean::enum	**1** -- Active **0 --** Inactive
assignee	UUID		Assignee of Request
category	STRING 30		Category symbol of Request
close_date	LOCAL_TIME		Close Date of Request
customer	UUID NOT_NULL		Affected User of Request
global_queue_id	INTEGER	Global_Queue_ Names::id	Pointer to global queue
group_id	UUID		Group assigned to Request
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
impact	INTEGER	Impact::enum	Impact of Request
last_mod_dt	LOCAL_TIME NOT_NULL		Indicates the timestamp of when this record was last modified.
open_date	LOCAL_TIME NOT_NULL		Open Date of Request
priority	INTEGER NOT_NULL	Priority::enum	Priority of Request
ref_num	STRING 30 NOT_NULL		Request ref num
remote_id	INTEGER NOT_NULL S_KEY		Regional request id
remote_sys_ id	INTEGER NOT_NULL S_KEY	Global_Servers:: remote_sys_id	Regional system id
status	STRING 30 NOT_NULL	Cr_Status::code	Status symbol of Request
summary	STRING 240		Summary of Request
type	STRING 10	crt code	ITIL record type

Global_Servers Table

Maintains a list of CA SDM installations that comprise the Global Service Desk.

- **SQL Name** -- g_srvr
- **Object** -- g_srvrs

Field	Data Type	Reference	Remarks
description	STRING 100		Specifies the comments field.
chg_prefix	STRING 5		Specifies the prefix for changes for a region.

cr_prefix	STRING 5		Specifies the prefix for requests for a region.
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
global_name	STRING 30 UNIQUE NOT_NULL		Value of NX_GLOBAL_NAME on region
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
is_master	INTEGER	Boolean_Table ::enum	Indicates if this server defined as master.
iss_prefix	STRING 5		Specifies the prefix for issues for a region.
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
remote_sys_id	INTEGER UNIQUE NOT_NULL		Regional system id
slump_addr	STRING 30		Specifies the CA SDM primary server hostname or IP address. A slump_addr is defined for each global region definition.
sym	STRING 30 UNIQUE NOT_NULL		Specifies the descriptive name.
web_protocol	STRING 10		Specifies the master region web access protocol (http https).
web_server	STRING 30		Specifies the web server name.
web_server_port	STRING 10		Specifies the web server port (can be blank).
web_url	STRING 100		Specifies the rest of URL to pdmweb.exe of remote system (or webdirector).

Usp_Servers Table

The usp_servers table lists all the servers in the CA SDM deployment.

Field	Data Type	Reference	Description
id	INTEGER UNIQUE		Unique key.

local_host	STRING 128 UNIQUE NOT_NULL S_KEY		Host name/Server name
timezone	STRING 30	Timezone	Timezone of server.
desc	STRING 1024		Textual description of the this host/server.
del	INTEGER NOT_NULL		Deleted flag 0 -- active 1 -- inactive/marked as deleted
server_id	INTEGER		Server ID or Slump ID
slump_port	INTEGER		Slump port number for communication.
server_type	INTEGER	server_types	Type of Server 0 -- primary 1 -- Secondary 2 -- Background 3 -- Stand-By 4 -- Application Server
database_type	STRING 30		Currently unused. 0 -- SQL 1 -- Oracle
external_dns_name	STRING 30		Currently unused.
platform	STRING 30		Currently unused. 0 -- Unix/Linux 1 -- Windows
linked	INTEGER NOT_NULL		Specify whether the server is configured.
last_mod_dt	LOCAL_TIME		Timestamp of when the record was last modified.
last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified the record.

Global_Table_Map Table

Contains list of table maps (local to master) used in global data transport.

- **SQL Name** -- g_tbl_map
- **Object** -- g_tblmap

Field	Data Type	Reference	Remarks
description	STRING 100		Comments field
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted

id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
map_definition	STRING 64 NOT_NULL		Map Definition name in xml file
sym	STRING 30 UNIQUE NOT_NULL		Name of this table map

Global_Table_Rule Table

Contains list of extract rules for global data transport.

- **SQL Name** -- g_tbl_rule
- **Object** -- g_tblrule

Field	Data Type	Reference	Remarks
description	STRING 100		Comments field
addl_query	STRING 240		Additional query
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
last_sync_dt	LOCAL_TIME		What was the last sync date
reoccur_interv	DURATION		How often to re-query
sched	STRING 30	Bop_Workshift:: persid	Valid run schedule
sym	STRING 30 UNIQUE NOT_NULL S_KEY		Name for rule
table_map	INTEGER	Global_Table_ Map::id	Table map

Transition Object Control

This article contains the following information:

in_trans Table

A transition object controls the current and next ticket status. The in_trans table lists the status, new status, and actions that need to occur for a default transition.

- **SQL Name** -- in_trans
- **Object** -- in_trans

Label	Field	Description
id	INTEGER	Unique key.
status	SYMBOL	Specifies the current ticket status.
new_status	SYMBOL	Specifies the new ticket status
must_comment	INTEGER	Comment required when using a transition. On new default: 0
is_default	INTEGER	Default transition that appears when the Status field is empty. On new default: 0
condition	BOP_REF_STR_REF Macro	Site condition macro to approve transition.
condition error	STRING 255	Error message for site condition.
description	STRING 255	Description of this transition.
last_mod_dt	LOCAL_TIME	Timestamp of last update to this record.
last_mod_by	UUID REF ca_contact	User who last updated this.
del	INTEGER nn	Logical database delete status.
tenant	UUID REF ca_tenant	Reference to Tenant information.

iss_trans Table

A transition object controls the current and next ticket status. The iss_trans table lists the status, new status, and actions that need to occur for a default transition.

- **SQL Name** -- iss_trans
- **Object** -- iss_trans

Label	Field	Description
id	INTEGER	Unique key.
status	SYMBOL	Specifies the current ticket status.
new_status	SYMBOL	Specifies the new ticket status
must_comment	INTEGER	Comment required when using a transition. On new default: 0
is_default	INTEGER	Default transition that appears when the Status field is empty. On new default: 0
condition	BOP_REF_STR_REF Macro	Site condition macro to approve transition.
condition error	STRING 255	Error message for site condition.
description	STRING 255	Description of this transition.
last_mod_dt	LOCAL_TIME	Timestamp of last update to this record.
last_mod_by	UUID REF ca_contact	User who last updated this.
del	INTEGER nn	Logical database delete status.

tenant	UUID REF ca_tenant	Reference to Tenant information.

Issue Template

iss_Template Table

Table to manage Issue templates. There is one entry for each Issue that is a template.

- **SQL Name** -- iss_template
- **Object** -- iss_tpl

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
description	nvarchar(1000)		Specifies the textual description of the template.
id	INTEGER		Specifies the unique (to the table) numeric ID.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp of when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
quick_tmpl_type	INTEGER	Quick_Template_Types::enum	Specifies the quick template type, as follows:**1** -- Quick tmpl**2** -- Quick tmpl+review
template	nvarchar(30)	issue persistent_id	Persistent ID (SystemObjectName:id).
template_class	nvarchar(12)		This allow subclassing of the templates.
template_name	nvarchar(30)		Identifies the unique name of the template

Issue Table

This article contains the following topics:

Request management's analyst recording of an external user's ticket.

- **SQL Name** -- issue
- **Object** -- iss

Field	Data Type	Reference	Remarks
id	INTEGER UNIQUE NOT_NULL KEY		Specifies the unique (to the table) numeric ID.

persid	STRING 30		Specifies the Persistent ID (SystemObjectName:id).
ref_num	STRING 30 UNIQUE NOT_NULL S_KEY		Identifies the reference number.
summary	STRING 240		Identifies the Issue summary text.
description	STRING 4000		Provides a textual description of this Issue.
status	STRING 12	Issue_Status	
active_flag	INTEGER NOT_NULL		Flag representing whether this record is active or inactive: **0** -- Inactive **1** -- Active
start_date	LOCAL_TIME		Indicates the timestamp of when processing started.
open_date	LOCAL_TIME		Shows the timestamp of when this Issue was opened.
last_mod_dt	LOCAL_TIME		Identifies the timestamp of when this record was last modified.
last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
close_date	LOCAL_TIME		Shows the timestamp of when the Issue was closed.
resolve_date	LOCAL_TIME		Indicates the timestamp of when the Change Order was resolved.
rootcause	INTEGER	Rootcause	Foreign key to the id field of the rootcause table, this identifies the Root Cause.
est_total_time	DURATION		Identifies the sum of estimated task time.
actual_total_time	DURATION		Specifies the sum of actual task time.
log_agent	UUID NOT_NULL	ca_contact	Foreign key to the contact_uuid field of the ca_contact table, this identifies who the change request was reported by.
assignee	UUID	ca_contact	Foreign key to the contact_uuid field of the ca_contact table, this identifies the Assignee.
organization	UUID	ca_organization	Foreign key to the id field of the ca_organization table, this identifies the Organization.
group_id	UUID	ca_contact	Foreign key to the contact_uuid field of the ca_contact table, this is the Group ID.
affected_contact	UUID	ca_contact	Foreign key to the contact_uuid field of the ca_contact table.

requestor	UUID NOT_NULL	ca_contact	Foreign key to the contact_uuid field of the ca_contact table, this specifies the Affected End User.
category	STRING 12	Issue_Category	Foreign key to the code field of the isscat table, this identifies the Category.
priority	INTEGER NOT_NULL	Priority	Foreign key to the enum field of the pri table, this identifies the Priority.
need_by	LOCAL_TIME		Shows the Need By Date timestamp for this Issue.
est_comp_date	LOCAL_TIME		Shows the estimated completion date for this Issue.
actual_comp_date	LOCAL_TIME		Specifies the actual completion date timestamp.
est_cost	INTEGER		Specifies the estimated cost value for this Issue.
actual_cost	INTEGER		Identifies the actual cost of this Issue.
justification	STRING 240		Identifies the justification value for this Issue.
backout_plan	STRING 240		Identifies the backout plan value for this Issue.
impact	INTEGER	Impact	Foreign key to the enum field of the impact table, this defines the impact.
parent	STRING 30	Issue	Foreign key to the persistent_id field of the iss table, this identifies the Parent.
effort	STRING 240		Identifies the effort value for this Issue.
support_lev	STRING 30	Service_Desc	Foreign key to the code field of the srv_desc table, this identifies the Classic Service Type.
template_name	STRING 30		Foreign key to the template_name field of the iss_tpl table, this identifies the Template name.
sla_violation	INTEGER		Flag indicating the following: **1** -- Issue has violated its sla.
predicted_sla_ violation	INTEGER		Flag that indicates the following: **1** -- Has been predicted by neugents.
macro_predicted_ violation	INTEGER		Identifies that it is likely to violate its sla (boolean) for action macros to predict sla violations.
created_via	INTEGER	Interface	Specifies the unique (to the table) numeric ID.

call_back_date	LOCAL_TIME		Identifies the call back timestamp for this Issue.
call_back_flag	INTEGER		Specifies the call back flag value for this Issue.
string1	STRING 40		This is a user-defined string field.
string2	STRING 40		This is a user-defined string field.
string3	STRING 40		This is a user-defined string field.
string4	STRING 40		This is a user-defined string field.
string5	STRING 40		This is a user-defined string field.
string6	STRING 40		This is a user-defined string field.
service_date	LOCAL_TIME		Specifies the service date value for this Issue.
service_num	STRING 30		Specifies the service num value for this Issue.
product	INTEGER	Product	Foreign key to the id field of the product table, this identifies the Product.
actions	STRING 750		Identifies the actions value for this Issue.
type_of_contact	INTEGER	Type_Of_Contact	Foreign key to the id field of the toc table, this identifies the Type of Contact.
reporting_method	INTEGER	Reporting_ Method	Foreign key to the id field of the repmeth table, this identifies how this Issue was reported.
person_contacting	INTEGER	Person_ Contacting	Foreign key to the id field of the person table, this identifies the person who made the contact.
flag1	INTEGER		This is the flag1 value for this Issue.
flag2	INTEGER		This is the flag2 value for this Issue.
flag3	INTEGER		This is the flag3 value for this Issue.
flag4	INTEGER		This is the flag4 value for this Issue.
flag5	INTEGER		This is the flag5 value for this Issue.
flag6	INTEGER		This is the flag6 value for this Issue.
user1	STRING 100		This is a user-defined string field.

user2	STRING 100		This is a user-defined string field.
user3	STRING 100		This is a user-defined string field.
caextwf_instance_id	INTEGER	caextwf_instances	Indicates the CA Process Automation process instance id and process definition name and reference path launched by this Service Desk object.
ticket_avoided	INTEGER		
tenant	UUID	ca_tenant	
requested_by	UUID	ca_contact	
external_system_ ticket	STRING 4000		
orig_user_dept	INTEGER	ca_resource_ department	
orig_user_ organization	UUID	ca_organization	
orig_user_admin_ org	UUID	ca_organization	
orig_user_cost_ center	INTEGER	ca_resource_ cost_ center	
target_start_last	LOCAL_TIME		
target_hold_last	LOCAL_TIME		
target_hold_count	INTEGER		
target_resolved_ last	LOCAL_TIME		
target_resolved_ count	INTEGER		
target_closed_last	LOCAL_TIME		
target_closed_ count	INTEGER		

Issue_Act_Log Table

History of activities associated with an issue. Types of activities are listed in the Act_Type table.

- **SQL Name** -- issalg
- **Object** -- issalg

Field	Data Type	Reference	Remarks
action_desc	nvarchar(4000)		Specifies the text description of the activity log entry.
analyst	byte(16)	ca_contact::uuid	Foreign key to the contact_uuid field of the ca_contact table, this represents the Analyst who created this activity log.
description	nvarchar(4000)		Specifies the text description of the activity log.
id	INTEGER		Primary key of this table, this is a unique numeric ID.
internal	INTEGER		Designates the log as an Internal log.

issue_id	nvarchar(30)	issue persistent_id	Foreign key to the persistent_id field of the iss table, this is the Issue for this activity.
knowledge_session	nvarchar(80)		Specifies the reference to the Knowledge Management session.
knowledge_tool	nvarchar(12)		Identifies the Knowledge Management Tool used for this activity.
last_mod_dt	INTEGER		Identifies the timestamp of when this record was last modified.
persid	nvarchar(30)		Specifies the Persistent ID (SystemObjectName:id).
system_time	INTEGER		Represents the date and time of the record creation.
time_spent	INTEGER		Identifies the time spent on the activity by the user.
time_stamp	INTEGER		Specifies the time spent on the activity by the user.
type	nvarchar(12)	Act_Type::code	(Not Used) This is an acknowledgement that is also a non-editable string enum.

Issue_Category Table

Issue categories. Can be hierarchical.

- **SQL Name** -- isscat
- **Object** -- isscat

Field	Data Type	Reference	Remarks
id	INTEGER		Unique (to the table) numeric ID.
persid	STRING(30)		Persistent ID (SystemObjectName:id)
del	INTEGER	Active_Boolean_ Table::enum	Delete flag that indicates the following:**0** -- Active**1** -- Inactive/ marked as deleted
sym	STRING(1000)		Identifies the Issue Category symbolic description.
code	STRING (12)		Primary key of this table.
last_mod_dt	LOCAL_TIME		Identifies the timestamp of when this record was last modified.
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
description	STRING(500)		Identifies the textual description of this issue category.

organization	UUID	ca_organization:: organization_uuid	Foreign key to the id field of the ca_organization table, this is the Organization.
assignee	UUID		Foreign key to the contact_uuid field of the ca_contact table, this is also the Assignee.
group_id	UUID		Foreign key to the contact_uuid field of the ca_contact table. This is the Group.
children_ok	INTEGER		Specifies the handling of the issue category:**0** -- Children not allowed**1** -- Children are allowed
service_type	STRING (30)	Service_Desc	
survey	INTEGER	Survey_Template::id	Foreign key of the id field of the survey_tpl table, this is the Survey.
schedule	INTEGER	Bop_Workshift	Deprecated.
auto_assign	INTEGER		Flag that enables auto assignment.
owning_contract	INTEGER	Service_Contract::id	Foreign key to the id field of the svc_contract table.This is the Service Contract.
flow_flag	INTEGER		Specifies the type of workflow:**0** -- Classic Workflow or none (default) **2** -- CA IT PAM
caextwf_start_id	INTEGER	caextwf_start_forms	Identifies the CA IT PAM process definition information to use when the user selects this category on a change order, issue, request.
tenant	UUID	ca_tenant	
chgtype	INTEGER	usp_change_type	
risk_survey	INTEGER	Risk_Survey_Template	
cab	UUID	ca_contact	
ss_include	INTEGER	bool	REQUIRED On new default: 0
ss_sym	STRING (128)		

Issue_Property Table

Property value pairs for an object.

- **SQL Name** -- issprp
- **Object** -- iss_prp

Field	Data Type	Reference	Remarks
description	nvarchar(240)		Specifies the textual description of this property.
error_msg	STRING 240		Specifies the error message produced if validation fails.
id	INTEGER		Primary key of this table, it is also a unique, numeric ID.
label	nvarchar(80)		Specifies the label value for this Issue property.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
owning_iss	nvarchar(30)	Issue	Foreign key to the persistent_id field of the iss table, this is also the Issue.
owning_macro	BOP_REF_STR NOT_NULL	Spell_Macro	Specifies the Spell_Macro for validation.
required	INTEGER	Boolean_Table::enum	Identifies the Required flag, as follows:**0** -- Not required**1** -- Required
sample	nvarchar(240)		The sample value for this property.
sequence	INTEGER		Used to order the properties for an Issue.
value_ description	STRING 240		Text description of the value.
validation_rule	BOP_REF_STR NOT_NULL	Property_Validation_Rule	Identifies the valid value rule.
validation_type	BOP_REF_STR NOT_NULL	Property_Validation_Type	Identifies the type of validation rule.
value	nvarchar(240)		Identifies the value of the property.

Issue_Status Table

Lists the states of the Issue, which you can also add. This table allows you to control whether the issue is active or inactive when it is changed to this status. Possible status include: open, approval in process, implementation in progress, verification in progress, cancelled, suspended, and closed.

- **SQL Name** -- issstat
- **Object** -- issstat

Field	Data Type	Reference	Remarks
active	INTEGER		Flag that indicates the following:**0** -- Inactive**1** -- Active
code	nvarchar(12)		Primary key of this table.
del	INTEGER		Delete flag that indicates the following:**0** -- Active**1** -- Inactive/ marked as deleted
description	nvarchar(500)		Identifies the textual description of this status.
hold	INTEGER		Flag that specifies the following:**0** -- Start events**1** -- Stop events
id	INTEGER		Unique (to the table) numeric ID.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp of when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
resolved	INTEGER		Flag that indicates the following:**0** -- Not yet resolved**1** -- Resolved
sym	nvarchar(30)		Identifies the symbolic name of the Issue status.

Key Control

Table of sequence numbers for requests, change orders, tickets, and foreign keys.

- **SQL Name** -- kc

Field	Data Type	Reference	Remarks
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
key_name	STRING 20		
key_value	INTEGER		Indicates the next key available for handout.

Message Status

This article contains the following topics:

Mgs_Act_Log Table

History of activities associated with a managed survey. The types of activities are listed in the Act_Type table.

- **SQL Name** -- mgsalg
- **Object** -- mgsalg

Field	Data Type	Reference	Remarks
action_desc	nvarchar(700)		Specifies the text description of the activity log entry.
analyst	byte(16)	ca_contact::uuid	Foreign key to the contact_uuid field of the ca_contact table, this represents the Analyst who created this activity log.
description	nvarchar(1000)		Specifies the text description of the activity log.
id	INTEGER		Primary key of this table, this is a unique numeric ID.
internal	INTEGER		Designates the log as an Internal log.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
mgs_id	INTEGER	Managed_Survey::id	Specifies the unique (to the table) numeric ID.
persid	nvarchar(30)		Specifies the Persistent ID (SystemObjectName:id).
system_time	INTEGER		Represents the date and time of the record creation.
time_spent	INTEGER		Identifies the time spent on the activity by the user.
time_stamp	INTEGER		Specifies the date time spent on the activity by the user.
type	nvarchar(12)	Act_Type::code	Foreign key to the code field of the act_type table, this is the Type.

Mgs_Status Table

List of valid message status definitions.

- **SQL Name** -- mgsstat
- **Object** -- mgsstat

Field	Data Type	Reference	Remarks
active	INTEGER		Flag that indicates the following:**0** -- Inactive**1** -- Active
code	nvarchar(12)		Primary key of this table.

del	INTEGER	Active_Boolean_Table::enum	Delete flag that indicates the following:**0 --** Active**1 --** Inactive/ marked as deleted
description	nvarchar(500)		Identifies the textual description of this status.
hold	INTEGER		Flag that specifies the following:**0** -- Start events**1** -- Stop events
id	INTEGER		Unique (to the table) numeric ID.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp of when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
sym	nvarchar(30)		Identifies the Managed Survey Status name.

Notification Table

This article contains the following topics:

Notification Table

Program control table used by Knowledge Management.

- **SQL Name** -- Notification
- **Object** -- Notification

Field	Data Type	Reference	Remarks
ALT_EMAIL	STRING 100		
ANALYST_ID	UUID	ca_contact::uuid	
DOC_ID	INTEGER		
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
NTF_LEVEL	INTEGER		**0** -- External and Internal **1** -- External only **2** -- Internal only

Notification_Urgency Table

Maps internal enums to strings for representing notification urgency for contacts.

- **SQL Name** -- noturg
- **Object** -- noturg

Field	Data Type	Index	Reference	Remarks
del	INTEGER NOT_NULL		Active_Boolean_Table::enum	**0** -- Present **1** -- Gone
desc	STRING 40			Specifies the description of notification urgency.
enum	INTEGER NOT_NULL			Indicates the enumeration value.
id	INTEGER UNIQUE NOT_NULL KEY	storage hash		Indicates the key ID.
sym	STRING 60 UNIQUE NOT_NULL S_KEY	sort dsc		Indicates the notify urgency symbol.

Notify_Log_Header Table

This table keeps track of each notify, and what happened to it.

- **SQL Name** -- not_log
- **Object** -- lr

Field	Data Type	Reference	Remarks
cmth_used	INTEGER	Contact_Method ::id	
cntxt_obj	STRING 30		Specifies the termination of sequence Context for notification.
context_instance	STRING 30		Contains the persistent ID of the associated Activity Log for the notifications.
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID.
last_mod	LOCAL_TIME		Specifies the last modification time, for purging.
nlh_ack_by	LOCAL_TIME		Specifies the time deadline for ack.
nlh_ack_time	DURATION		Specifies how long for ack.
nlh_c_addressee	UUID	ca_contact::uuid	Specifies the alias contact.
nlh_c_alias	UUID		
nlh_cm_method	INTEGER	Notification_ Urgency::enum	

nlh_email	STRING 50		Specifies the resolved email address (the email notification sent was not associated with the proper contact).
nlh_end	LOCAL_TIME		Specifies the time of ACK or FYI final.
nlh_hdr	STRING 40		Specifies the msg header text.
nlh_msg	STRING 4000		Specifies the message text.
nlh_msg_html	STRING 32768		Specifies the html version of notification if sent through mail.
nlh_pri	INTEGER		Generating notification enum priority of transition event
nlh_start	LOCAL_TIME NOT_NULL		Specifies the notification start date.
nlh_status	INTEGER		Pending send, sent FYI.
nlh_transition	INTEGER		Specifies the notify method used - redefined in majic, points to 'noturg' object Transition point.
nlh_type	INTEGER		FYI or ack
nlh_user_ack	STRING 40		Sent ack, acked, nacked, cleared. Who acknowledged or cleared it.

Notify_Msg_Tpl Table

List of object attribute message template definitions for notification rules.

- **SQL Name** -- ntfm
- **Object** -- ntfm

Field	Data Type	Reference	Remarks
ID	INTEGER		Primary key
persid	nvarchar(30)		Object key
del	SREL integer -- > actbool		Delete indicator
sym	nvarchar(128)		Symbolic description; required field
notify_flag	INTEGER		Auto notification
notify_level	SREL integer --> noturg		Notification level
notify_msg_title	nvarchar(255)		Notification message title (Required)
notify_msg_body	nvarchar(4000)		Notification message body
notify_msg_body_html	nvarchar(32768)		Notification HTML message
obj_type	SREL string --> ntfm_prod_list		Object type: cr, chg, iss; required field
last_mod_dt	DATE		Audit trail date

last_mod_by	SREL uuid --> cnt		Audit trail user

Notify_Object_Attr Table

List of object attribute contacts that may receive notifications.

- **SQL Name** -- ntfl
- **Object** -- ntfl

Field	Data Type	Reference	Remarks
description	STRING 240		Specifies the textual description.
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
object_attr	STRING 250		Specifies the attr name in object.
object_type	STRING 30		Specifies the object name.
persid	STRING 30		Persistent ID (SystemObjectName:id)
sym	STRING 30 NOT_NULL		

Notify_Rule Table

List of object attribute contacts and definitions for notification rules.

- **SQL Name** -- ntfr
- **Object** -- ntfr

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE NOT_NULL KEY
persid	nvarchar(30)		Specifies the object key
del	INTEGER		Delete indicator
sym	nvarchar(128)		Symbolic description REQUIRED
description	nvarchar(500)		Long description
condition	nvarchar(30)	Spell_Macro	Specifies the foreign key to the condition macro.

obj_type	nvarchar(30)		Specifies the object type as cr, chg, or iss REQUIRED
last_mod_dt	LOCAL_TIME		Audit trail date
last_mod_by	UUID	ca_contact	Audit trail user
default_rule	INTEGER		**1** -- Default rule; do not allow condition
cr_notify_info	nvarchar(30)	ntfm	Specifies the foreign key for 'cr' obj_type to condition macro.
chg_notify_info	nvarchar(30)	ntfm	Specifies the foreign key for 'chg' obj_type to condition macro.
iss_notify_info	nvarchar(30)	ntfm	Specifies the foreign key for 'iss' obj_type to condition macro.
mgs_notify_info	nvarchar(30)	ntfm	Specifies the foreign key for 'mgs' obj_type to condition macro.
kd_notify_info	nvarchar(30)	ntfm	Specifies the foreign key for 'kd' obj_type to condition macro.
kd_comment_ notify_info	nvarchar(30)	ntfm	Specifies the foreign key for 'kd_comment' obj_type to condition macro.
krc_notify_info	nvarchar(30)	ntfm	Specifies the foreign key for 'krc' obj_type to condition macro.
sa_notify_info	nvarchar(30)	ntfm	Specifies the foreigh key for 'sa' obj_type to condition macro.
cnt_notify_info	nvarchar(30)	ntfm	Specifies the foreigh key for 'cnt' obj_type to condition macro.
ci_notify_info	nvarchar(30)	ntfm	Specifies the foreigh key for 'ci' obj_type to condition macro.
tenant	UUID	ca_tenant	Specifies the UUID of the tenant.
ntfr_ntflist	LREL	ntfl	Associated Object Contacts
ntfr_ctplist	LREL	ctp	Associated Contact Types
ntfr_macrolist	LREL	macro	Macros (aty's) using this rule

Promotion

This table contains the following topics:

Object_Promotion Table

Object Promotion definition for CA SDM application.

- SQL Name: object_promotion
- Object: object_promotion

Field	Data Type	Reference	Remarks
id	INTEGER		Unique numeric ID.
last_mod_dt	INTEGER		Indicates when the record was last modified.
last_mod_by	Byte(16)	ca_contact:: uuid	Specifies the UUID of the contact who last modified the record.
delete_flag	INTEGER		Specifies the default flag value for this access type.
object_name	nvarchar(60)		Provides the name of the promotable object.
description	nvarchar(512)		Provides the description of the promotable object.
start_date	INTEGER		Indicates the start date of promotable object.
end_date	INTEGER		Indicates the end date of promotable object.

Promo_Hist Table

Promotion History definition for CA SDM application.

- SQL Name: promo_hist
- Object: promo_hist

Field	Data Type	Reference	Remarks
id	INTEGER		Unique numeric ID.
last_mod_dt	INTEGER		Indicates when the record was last modified.
last_mod_by	Byte(16)	ca_contact:: uuid	Specifies the UUID of the contact who last modified the record.
created_user	nvarchar(255)		Name of the user who performed the operation.
description	nvarchar(512)		Provides the description of the executed operation.
start_date	INTEGER		Indicates the start date of the executed operation.
end_date	INTEGER		Indicates the end date of the executed operation.
package_name	nvarchar(255)		Package name of the executed operation.

promo_status	nvarchar(255)		Status of the executed operation.
promotion_cmd	nvarchar(255)		Promotion command of the executed operation.
report_file	nvarchar(255)		Report file name of the executed operation.
sdm_filter	nvarchar(512)		Filter of the executed operation.
status	INTEGER		Status code of the executed operation.
updated_user	nvarchar(255)		Name of the user who performed the operation.

Property

This table contains the following topics:

Property Table

Property value pairs for an object.

- **SQL Name** -- prp
- **Object** -- prp

Field	Data Type	Reference	Remarks
description	nvarchar(240)		Identifies the textual description of this property.
error_msg	STRING 240		Specifies the error message produced if validation fails.
id	INTEGER		Primary key of this table.
label	nvarchar(80)		Identifies the label value for this property.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp of when this record was last modified.
object_id	INTEGER	chg id	Identifies the unique (to the table) numeric ID.
object_type	nvarchar(30)		Specifies the short name of the object to which this property belongs.
owning_macro	BOP_REF_STR NOT_NULL	Spell_Macro	Specifies the Spell_Macro for validation.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).

property	INTEGER	Property_Template ::id	Foreign key to the id field of the prptpl table, this is the Template.
required	INTEGER	Boolean_Table::enum	Specifies the Required field as follows:**0** -- Not required**1** -- Required
sample	nvarchar(240)		Indicates the sample value for this property.
sequence	INTEGER		Specifies the sequence value for this property.
value_description	STRING 240		Text description of the value.
validation_rule	BOP_REF_STR NOT_NULL	Property_Validation_Rule	Identifies the valid value rule.
validation_type	BOP_REF_STR NOT_NULL	Property_Validation_Type	Identifies the type of validation rule.
value	nvarchar(240)		Specifies the value of the property.

Property_Template Table

Additional properties for objects.

- **SQL Name** -- prptpl
- **Object** -- prptpl

Field	Data Type	Reference	Remarks
code	nvarchar(12)		Specifies the non-editable handle for the query.
del	INTEGER	Active_Boolean_Table::enum	Deleted flag that indicates the following:**0** -- Active **1** -- Inactive/marked as deleted
description	nvarchar(240)		Specifies the textual description for the property template.
label	nvarchar(80)		Display the text for the property.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp of when this record was last modified.
object_ attrname	nvarchar(30)		Specifies that this template belongs to the attribute name in the object.
object_attrval	INTEGER		Identifies the object attribute value, which drives the template attribute value.
object_type	nvarchar(30)		Specifies the short name of the object.

persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
prptbl_id			Indicates the unique (to the table) numeric ID.
required	INTEGER NOT_NULL		Identifies the Required flag as follows:**0** -- Not Required**1** -- Required
sample	nvarchar(240)		Provides a sample, or Help text.
validation_rule	BOP_REF_STR NOT_NULL	Property_Validation_Rule	Identifies the valid value rule.
sequence	INTEGER		Specifies the display order.

Property_Validation_Rule Table

Property validation rule.

- **SQL Name** -- prpval_rule
- **Object** -- prpval_rule

Field	Data Type	Reference	Remarks
id	INTEGER UNIQUE NOT_NULL KEY		Specifies the primary key of this table.
persid	BOP_REF_STR		
del	INTEGER NOT_NULL		**0** -- Present **1** -- Not present
sym	STRING 30 UNIQUE NOT_NULL S_KEY		Specifies the name of the validation rule.
description	STRING 240		
validation_type	BOP_REF_STR	Property_Validation_ Type	Specifies the type of validation rule.
error_msg	STRING 240		Specifies the error message produced if validation fails.
owning_macro	BOP_REF_STR	Spell_Macro	Specifies the spell_Macro for validation.
label	LABEL_SYM NOT_NULL		Specifies the helper to pre-populate property template.
last_mod_dt	LOCAL_TIME		Indicates the timestamp indicating when this record was last modified.
last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.

Property_Validation_Type Table

Property validation type.

- **SQL Name** -- prpval_type
- **Object** -- prpval_type

Field	Data Type	Reference	Remarks
id	INTEGER UNIQUE NOT_NULL KEY		Specifies the primary key of this table.
persid	BOP_REF_STR		Persistent ID
del	INTEGER NOT_NULL		**0** -- Present **1** -- Not present
sym	STRING 30 UNIQUE NOT_NULL S_KEY		Specifies the name of the validation rule.
description	STRING 240		
last_mod_dt	LOCAL_TIME		Indicates the timestamp indicating when this record was last modified.
last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.

Property_Value Table

Property values for an object.

- **SQL Name** -- prpval
- **Object** -- prpval

Field	Data Type	Reference	Remarks
id	INTEGER NOT_NULL		Specifies the primary key of this table.
persid	BOP_REF_STR		Specifies the persistent ID.
del	INTEGER NOT_NULL		**0** -- Present **1** -- Not present
sym	STRING 30 NOT_NULL S_KEY		Specifies the unique name of property value.
description	STRING 240		
owning rule	BOP_REF_STR NOT_NULL	Property_Validation_ Rule	Specifies the property validation rule this value applies to.
value	LONG_SYM NOT_NULL		Specifies the valid value.
is_default	INTEGER		**1** -- Default value for the rul/Pe
last_mod_dt	LOCAL_TIME		Specifies the last modified time.

last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified the record.

Query Policy

This article contains the following topics:

Query_Policy Table

Identifies the stored query event data for the Automated Policies feature in Knowledge Management.

- **SQL Name** -- query_policy
- **Object** -- query_policy

Field	Data Type	Reference	Remarks
ID	Long	Primary index.	Identifies the unique (to the table) numeric ID of the primary index.
persid	nvarchar(30)	Required field.	This is the Persistent ID.
Del	INTEGER (0/1)	Required field.	Deleted flag. **0** -- Active **1** -- Inactive/marked as deleted
last_mod_dt	Long (date)		Displays the last modified date.
last_mod_by	UUID	SREL to the ca_contacts table.	Displays the last date of when a user modified the record.
obj_type	nvarchar(30)	Required field. SREL to event_prod_list.	Kind of object for this event (knowledge document, and so forth).
query	nvarchar(30)	Required field. SREL to the cr_stored_queries.	Identifies the store query name.
sym	nvarchar(30)	Required field.	Identifies the policy name.
description	nvarchar(80)		Query policy description.

Query_Policy_Actions Table

Identifies the actions attached to the stored query event data for the Automated Policies feature in Knowledge Management. Each record represents a link between the macro and a query_policy record in the database.

- **SQL Name** -- query_policy_actions
- **Object** -- query_policy_actions

Field	Data Type	Reference	Remarks
ID	Long	Primary index.	Identifies the unique (to the table) numeric ID of the primary index.
policy	Long	Required field. SREL to query_policy.	Identifies the Policy ID.

macro	STRING	Required field. SREL to splmac.	Identifies the action macro Persistent ID.
last_mod_dt	Long (date)		Displays the last modified date.
last_mod_by	UUID	SREL to the ca_contacts.	Displays the last date of when a user modified the record.

Request Property

This article contains the following topic:

Req_Property Table

Property value pairs for a Request.

- **SQL Name** -- cr_prp
- **Object** -- cr_prp

Field	Data Type	Reference	Remarks
description	nvarchar(240)		Specifies the textual description of this property.
error_msg	STRING 240		Specifies the error message produced if validation fails.
id	INTEGER		Primary key of this table, it is a unique, numeric ID.
label	nvarchar(80)		Specifies the label value for this Request property.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
owning_cr	nvarchar(30)	Call_Req::persid	Persistent ID (SystemObjectName:id)
owning_macro	BOP_REF_STR NOT_NULL	Spell_Macro	Specifies the Spell_Macro for validation.
required	INTEGER		Identifies the Required flag, as follows:**0** -- Not required**1** -- Required
sample	nvarchar(240)		The sample value for this Request_Property.
sequence	INTEGER		The sequence value for this Request property.
value_description	STRING 240		Text description of the value.
validation_rule	BOP_REF_STR NOT_NULL	Property_Validation_ Rule	Identifies the valid value rule.
validation_type	BOP_REF_STR NOT_NULL	Property_Validation_ Type	Identifies the type of validation rule.

value	nvarchar(240)		Identifies the value of the property.

Req_Property_Template Table

Templates used to specify request properties.

- **SQL Name** -- cr_prptpl
- **Object** -- cr_prptpl

Field	Data Type	Reference	Remarks
description	STRING 240		Specifies the textual description of this template.
code	STRING 12 UNIQUE NOT_NULL S_KEY		Specifies the noneditable handle for query.
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0 --** Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
label	STRING 80 NOT_NULL		Specifies the display text for property.
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
owning_area	STRING 30 NOT_NULL	Prob_Category::persid	Specifies the parent category.
persid	STRING 30		Persistent ID (SystemObjectName:id)
required	INTEGER NOT_NULL		**0** -- Not required
sample	STRING 240		Specifies the help text.
sequence	INTEGER NOT_NULL		Specifies the display order.
validation_rule	BOP_REF_STR NOT_NULL	Property_Validation_ Rule	Identifies the valid value rule.

Survey

This article contains the following topics:

sa_survey Table

Program control table that is used by Support Automation.

- **SQL Name** -- sa_survey
- **Object** -- sa_survey

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE NOT_NULL KEY
last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Specifies the timestamp of when this record was last modified.
tenant	UUID	ca_tenant	
surveyName	STRING 32		NOT_NULL
question	STRING 512		NOT_NULL
responseType	INTEGER		NOT_NULL
isDeleted	INTEGER		

sa_survey_localized Table

Program control table that is used by Support Automation.

- **SQL Name** -- sa_survey_localizedg
- **Object** -- sa_survey_localizedg

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE NOT_NULL KEY
last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Specifies the timestamp of when this record was last modified.
surveyID	INTEGER	sa_survey	NOT_NULL
localizationID	INTEGER	sa_localization	NOT_NULL
question	STRING 512		NOT_NULL
tenant	UUID	ca_tenant	

sa_survey_result Table

Program control table that is used by Support Automation.

- **SQL Name** -- sa_sa_survey_result
- **Object** -- sa_sa_survey_result

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE NOT_NULL KEY

last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Specifies the timestamp of when this record was last modified.
surveyID	INTEGER	sa_survey	NOT_NULL
selfServeSessionID	INTEGER	sa_self_serve_session	NOT_NULL
scriptID	INTEGER	a_script	NOT_NULL
response	INTEGER		
completion	INTEGER		NOT_NULL
SurveyComment	STRING 512		
surveyEpoch	LOCAL_TIME		
tenant	UUID	ca_tenant	

Survey Table

Customer Survey.

- **SQL Name** -- survey
- **Object** -- survey

Field	Data Type	Reference	Remarks
comment_label	nvarchar(80)		Specifies the comment label value for this Survey.
conclude_text	nvarchar(400)		Indicates the text to display after the survey is completed.
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
description	nvarchar(400)		Specifies the textual description for this survey.
id	INTEGER		Specifies the primary key of this table.
include_ comment	INTEGER		Specifies the Include Comment flag, as follows: **0** -- No comment allowed
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
nx_comment	nvarchar(200)		Identifies the comment value for this Survey.
object_id	INTEGER		Identifies the ID of the object for this survey.

object_type	nvarchar(10)		Identifies the object type for this survey, for eample, CR, CHG, and so on.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
sym	nvarchar(12)		Identifies the symbolic name of this survey.

Survey_Answer Table

Customer Survey Answer.

- **SQL Name** -- survey_answer
- **Object** -- svy_ans

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_ Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
id	INTEGER		Primary key of this table.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
own_srvy_ question	INTEGER	Survey_Question ::id	Unique (to the table) numeric ID.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
selected	INTEGER		Identifies the selected value for this Survey_Answer.
sequence	INTEGER		Specifies the display order of the survey answers.
txt	nvarchar(400)		Specifies the question text.

Survey_Answer_Template Table

Customer Survey Answer Template.

- **SQL Name** -- survey_atpl
- **Object** -- svy_atpl

Field	Data Type	Reference	Remarks
atbl_id	INTEGER		Identifies the unique (to the table) numeric ID.
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted

last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
own_srvy_ question	INTEGER	Survey_Question_Template::id	Unique (to the table) numeric ID.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
sequence	INTEGER		Specifies the display order of the survey answers.
txt	nvarchar(400)		Identifies the answer text.

Survey_Question Table

Customer Survey Question.

- **SQL Name** -- survey_question
- **Object** -- svy_ques

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_ Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
id	INTEGER		Primary key of this table.
include_qcomment	INTEGER		Specifies the Include Question Comment flag, as follows:**1** -- Include comment box for this question
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
mult_resp_flag	INTEGER		Specifies the Multiple Response flag, as follows:**0** -- Choose 1 response (radio)**1** -- Choose "N" (check boxes)
owning_survey	INTEGER	Survey_id	Specifies the unique (to the table) numeric ID.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
qcomment_ label	nvarchar(80)		Specifies the label for the comment.
resp_required	INTEGER		Specifies the Required Response flag, as follows:**1** -- Respondents must answer the question

response	INTEGER		Specifies the sequence number of the response.
sequence	INTEGER		Specifies the display order.
txt	nvarchar (400)		Identifies the question text.

Survey_Question_Template Table

Customer Survey Question Template.

- **SQL Name** -- survey_qtpl
- **Object** -- svy_qtpl

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_ Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
id	INTEGER		Primary key of this table.
include_qcomment	INTEGER		Specifies the Include Question Comment flag, as follows:**1** -- Include comment box for this question
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
mult_resp_flag	INTEGER		Specifies the Multiple Response flag, as follows:**0** -- Choose 1 response (radio)**1** -- Choose "N" (check boxes)
owning_survey	INTEGER	Survey_Template::id	Specifies the unique (to the table) numeric ID.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
qcomment_ label	nvarchar(80)		Specifies the label for the comment.
resp_required	INTEGER		Specifies the Required Response flag, as follows:**1 --** Respondents must answer the question
sequence	INTEGER		Specifies the display order of the survey questions.
txt	nvarchar(400)		Identifies the question text.

Survey_Stats Table

This table contains customer survey statistics.

- **SQL Name** -- survey_statistics
- **Object** -- svystat

Field	Data Type	Reference	Remarks
cyc_counter	INTEGER		Identifies the cycle counter to be compared against the cycle (described in the next field).
cycle	INTEGER		Identifies the submission cycle.
del	INTEGER	Active_Boolean_Table::enum	Specifies the status of the Deleted flag, as follows:**0** -- Active**1 --** Inactive/marked deleted)
eval_counter	INTEGER		Identifies the number of times called to evaluate for notification submission.
id	INTEGER		Primary key of this table.
persid	nvarchar(30)		Specifies the Persistent ID: (SystemObjectName:id).
sub_counter	INTEGER		Identifies the number of times notification submission was approved.
tplid	INTEGER	Survey_Template::id	Identifies the unique (to the table) numeric ID.

Survey_Template Table

Customer Survey Template.

- **SQL Name** -- survey_tpl
- **Object** -- svy_tpl

Field	Data Type	Reference	Remarks
comment_label	nvarchar(80)		Identifies the comment label value for this Survey_Template.
conclude_text	nvarchar(400)		Specifies the text to display after the survey has been completed.
cycle_counter	INTEGER		Indicates to keep a running count for Submit Cycle.
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/marked as deleted
description	nvarchar(400)		Indicates the textual description for this template.
id	INTEGER		Primary key of this table.
include_comment	INTEGER		Specifies the Include Question Comment flag, as follows:**0** -- No comment allowed

last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
submit_cycle	INTEGER		Indicates to send the survey every 'nth' time.
sym	nvarchar(12)		Identifies the symbolic name of the survey template.
tracking_flag	INTEGER		Specifies the flag to track responses.

Survey_Tracking Table

Table used to tack status of managed surveys.

- **SQL Name** -- survey_tracking
- **Object** -- svytrk

Field	Data Type	Reference	Remarks
cntid	byte(16)	ca_contact::uuid	Primary key of this table.
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
id	INTEGER		Primary key of this table, this is a unique numeric ID.
notif_dt	INTEGER		Specifies the time of the last notification.
object_id	INTEGER		Identifies the ID of the object for this survey.
object_type	nvarchar(10)		Indicates the object type for this survey, such as, CR, CHG, and so on.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
recv_dt	INTEGER		Identifies the time the completed survey was received.
status	INTEGER		Identifies the Status flag, as follows:**1** -- Survey submitted
tplid	INTEGER	Survey_Template ::id	Unique (to the table) numeric ID.

Managed_Survey Table

Stores the definition of the managed surveys.

- **SQL Name** -- managed_survey
- **Object** -- mgs

Field	Data Type	Reference	Remarks
create_date	INTEGER		Identifies when the managed survey was created.
del	INTEGER	Boolean_Table::enum	Specifies the Deleted flag, as follows:**0** -- Active **1** -- Inactive/marked as deleted
description	nvarchar(400)		Identifies the textual description of this managed survey.
end_date	INTEGER		Specifies the end of survey period.
group_id	byte(16)		Foreign key to the contact_uuid field of the ca_contact table, this is the Group.
id	INTEGER		Primary key of this table.
initial_method	INTEGER	Contact_Method::id	Foreign key to the id field of the ct_mth table, this specifies the Contact Method.
initial_msgbody	nvarchar(1000)		Identifies the message body of the initial notification message.
initial_msgtitle	nvarchar(80)		Identifies the message title of the initial notification message.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
owner	byte(16)	ca_contact::uuid	Foreign key to the contact_uuid field of the ca_contact table, this specifies the Owner.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id)
reminder_method	INTEGER	Contact_Method::id	Foreign key to the id field of the ct_mth table, this is the Reminder Contact Method.
reminder_msgbody	nvarchar(1000)		Identifies the reminder message body of the initial notification message.
reminder_msgtitle	nvarchar(80)		Identifies the message title of the initial notification message.
start_date	INTEGER		Specifies the start of the survey period.

status	nvarchar(12)	Mgs_Status::code	Foreign key to the code field of the mgsstat table, this is the Status.
sym	nvarchar(12)		Identifies the symbolic name for this Managed Survey.
tplid	INTEGER	Survey_Template::id	Foreign key to the id field of the survey_tpl table, this is the Survey Template.

Sequence Control

Sequence_Control Table

Used to determine what to use for a prefix and suffix when generating call request numbers. Users may not create new records or delete records in this table.

- **SQL Name** -- seqctl
- **Object** -- seq

Field	Data Type	Reference	Remarks
description	STRING 240		Textual description
code	STRING 12 UNIQUE NOT_NULL S_KEY		noneditable key for record
del	INTEGER NOT_NULL	Active_Boolean_Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
prefix	STRING 5		Indicates the prefix to sequence number.
suffix	STRING 5		Indicates the suffix to sequence number.
sym	STRING 30 NOT_NULL		

Server

This topic contains the following information:

Server_Aliases Table

Contain server alias names that are valid for various server-client zones.

- **SQL Name** -- srvr_aliases
- **Object** -- srvr_aliases

Field	Data Type	Reference	Remarks
alias_name	STRING 30 NOT_NULL		Alias name
del	INTEGER NOT_NULL	Active_Boolean_Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
host_addr	STRING 30		Translated Host Address
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
zone_id	INTEGER NOT_NULL	Server_Zones::id	Zone name

Server_Zones Table

Contain regional server zone names.

- **SQL Name** -- srvr_zones
- **Object** -- srvr_zones

Field	Data Type	Reference	Remarks
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
is_default	INTEGER	Boolean_Table:: enum	1 default zone
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
zone_name	STRING 30 NOT_NULL		Zone name

Service

This topic contains the following information:

Service_Contract Table

Used to track relationships between orgs, ticket attr's and svc types. Used for SLA management.

- **SQL Name** -- svc_contract
- **Object** -- svc_contract

Field	Data Type	Reference	Remarks
active	INTEGER	Active_Boolean_ Table::enum	Specifies the status of the contract, as follows:**0** -- Inactive**1** -- Active
contract_num	nvarchar(50)		Identifies the Contract ID.
del	INTEGER	Active_Boolean_ Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
dflt_chgcat_st	nvarchar(30)	Service_Desc::code	Defines the default service type for the change category.
dflt_cnt_st	nvarchar(30)	Service_Desc::code	Defines the default service type for the end user.
dflt_isscat_st	nvarchar(30)	Service_Desc::code	Defines the default service type for the issue category.
dflt_nr_st	nvarchar(30)	Service_Desc::code	Defines the default service type for the asset.
dflt_pcat_st	nvarchar(30)	Service_Desc::code	Defines the default service type for the request area.
dflt_pri_st	nvarchar(30)	Service_Desc::code	Defines the default service type for the priority.
expiration	INTEGER		Specifies the Contract expiration date.
id	INTEGER		Primary key of this table.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp of when this record was last modified.
nx_desc	nvarchar(240)		Specifies descriptive information.
org_svc_type	nvarchar(30)	Service_Desc::code	Defines the service type for the organization.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
svc_advocate	byte(16)	ca_contact::uuid	Identifies the customer advocate for the organizations assigned to the contract.
svc_owner	byte(16)	ca_contact::uuid	Foreign key to the contact_uuid field of the ca_contact table, this is the service desk person responsible for contract.

sym	nvarchar(80)		Specifies the symbolic name of the contract level.

Service_Desc Table

This table contains the Call Request Service Types that are not related to service level agreements in Problem Manager. These may be used to associate the types of service call requests to receive, and are can be defined by the user. Examples include: Platinum, Gold, Silver, or Bronze.

- **SQL Name** -- srv_desc
- **Object** -- sdsc

Field	Data Type	Reference	Remarks
code	nvarchar(30)		Specifies the primary key of this table.
del	INTEGER	Active_Boolean_ Table::enum	Specifies the status of the Deleted flag as follows:**0** -- Active**1** -- Inactive/marked as deleted
description	nvarchar(500)		Provides the text description of service.
id	INTEGER		Specifies the primary key of this table.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp for when this record was last modified.
owning_contract	INTEGER	Service_Contract ::id	Specifies the unique (to the table) numeric ID.
persid	nvarchar(30)		Identifies the Persistent ID: (SystemObjectName:id)
rank	INTEGER		Identifies the ranking status of the service type to determine level of priority.
schedule	nvarchar(30)	Bop_Workshift::persid	Foreign key to the peristent_id field of the bpwshft table, this is the Schedule.
sym	nvarchar(30)		Identifies the name of the service type.
violation_cost	INTEGER		Specifies the monetary cost for violation.

Session

This topic contains the following information:

session_log Table

Session log.

- **SQL Name** -- session_log
- **Object** -- session_log

Field	Data Type	Reference	Remarks
contact	byte(16)	ca_contact::uuid	Foreign key to the contact_uuid field of the ca_contact table, this is the User.
id	INTEGER		Primary key of this table
login_time	INTEGER		Indicates the time of when the session began.
logout_time	INTEGER		Indicates the time of when the session ended.
policy	INTEGER	SA_Policy::id	Specifies the session policy.
session_id	INTEGER		Displays the ID if the status is okay.
session_type	INTEGER	session_type::id	Specifies the unique (to the table) numeric ID.
status	INTEGER		Identifies the Login status: **0** -- Okay

session_type Table

Session type (Web client, Java client, and so on).

- **SQL Name** -- session_type
- **Object** -- session_type

Field	Data Type	Reference	Remarks
description	nvarchar(500)		Identifies the description of the session type.
id	INTEGER		Primary key of this table.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp of when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id)
sym	nvarchar(30)		Identifies the name of the Session type.

Severity

CA

List of severity definitions used by CA SDM applications.

- **SQL Name** -- sevrty
- **Object** -- sev

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_ Table::enum	Identifies the Deleted flag as follows:**0** -- Active **1** -- Inactive/marked as deleted
enum	INTEGER		Primary key of this table.
id	INTEGER		Unique (to the table) numeric ID.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
nx_desc	nvarchar(40)		Describes the severity.
sym	nvarchar(12)		Identifies the symbolic name for this severity.

Service Level Agreement

This topic contains the following information:

SLA_Contract_Map Table

Maps a service type to a reference object; used by Service_Contract objects.

- **SQL Name** -- sdsc_map
- **Object** -- sdsc_map

Field	Data Type	Reference	Remarks
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
map_contract	INTEGER NOT_NULL	Service_Contract::id	
map_persid	STRING 60		
map_sdsc	STRING 30	Service_Desc::code	
persid	STRING 30		Persistent ID (SystemObjectName:id)

SLA_Template Table

Service Level Agreement Templates for Call and Change. Not related to service level agreements in problem manager. Links Service Descriptions with Events.

- **SQL Name** -- slatpl
- **Object** -- slatpl

Field	Data Type	Reference	Remarks
del	INTEGER NOT_NULL	Active_Boolean_Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
elapsed	DURATION NOT_NULL		Specifies the elapsed time to delay event.
event	STRING 30	Events::persid	Specifies the event.
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified
object_type	STRING 30		Specifies the object type for this SLA (cr, chg).
persid	STRING 30		Persistent ID (SystemObjectName:id)
service_type	STRING 30 NOT_NULL	Service_Desc::code	
sym	STRING 30 NOT_NULL S_KEY		Specifies the name of the SLA.

Spel Macro

This topic contains the following information:

Spell_Macro Table

This table stores proprietary Spel macros and macro fragments.

- **SQL Name** -- splmac
- **Object** -- macro

Field	Data Type	Reference	Remarks
description	STRING 80		Identifies the text description.
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Specifies the Deleted flag status, such as: **0** -- Active **1** -- Inactive/marked as deleted
fragment	STRING 4000		
id	INTEGER UNIQUE NOT_NULL KEY		Identifies the Numeric ID that is unique to the table.

last_mod_dt	LOCAL_TIME		Indicates the timestamp for when this record was last modified.
lock_object	INTEGER NOT_NULL		Specifies the boolean of the object.
msg_html	STRING 32768		Identifies the message used to keep the html template.
ob_type	STRING 30 NOT_NULL		
persid	STRING 30		Identifies the Persistent ID: (SystemObjectName:id)
sym	STRING 30 NOT_NULL UNIQUE NOT_NULL		
type	STRING 30 NOT_NULL	Spell_Macro_Type ::persid	
usr_integer1	INTEGER		
usr_integer2	INTEGER		
usr_integer3	INTEGER		
usr_string2	STRING 250		
usr_string3	STRING 125		
usr_string4	STRING 25		

Spell_Macro_Type Table

Stores type information for proprietary Spel macros and macro fragments .

- **SQL Name** -- splmactp
- **Object** -- macro_type

Field	Data Type	Reference	Remarks
description	STRING 200		Textual description
arg_list	STRING 80		
code	STRING 30 UNIQUE NOT_NULL NOT_NULL		
del	INTEGER NOT_NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
display_name	STRING 30		form name to display
execute_script	STRING 800		
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
lock_object_ flag	INTEGER		
persid	STRING 30		Persistent ID (SystemObjectName:id)

sym	STRING 30 NOT_NULL		
tech_desc	STRING 300		
validate_script	STRING 400		

Show Object

Show_Obj Table

Program control table used by Knowledge Management.

- **SQL Name** -- SHOW_OBJ
- **Object** -- SHOW_OBJ

Field	Data Type	Reference	Remarks
EXPIRE_DATE	LOCAL_TIME		
ID	INTEGER KEY		Unique (to the table) Numeric ID
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
OBJ_PERSID	STRING 255		Persistent ID (SystemObjectName:id)
PWD	STRING 255		

SQL Script

This article contains the following topics:

SQL_Script Table

Stores SQL scripts used by the CA SDM applications.

- **SQL Name** -- sql_tab

Field	Data Type	Reference	Remarks
del	INTEGER NOT_NULL		Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
sql_desc	STRING 240		user title for script
sql_name	STRING 30 S_KEY		user name for script
sql_script	STRING 500		actual script

Pcat_Loc Table

The following tables are added to support many-to-many relationships between Categories, Groups that can service the Categories, and Locations that groups are able to service. This is all used for the Auto-Assignment functionality. Used to build a list of Service Locations valid for the Category.

- **SQL Name** -- pcat_loc
- **Object** -- pcat_loc

Field	Data Type	Reference	Remarks
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
l_attr	STRING 30		left hand attribute name
l_persid	STRING 60		left hand key
l_sql	INTEGER		left hand sort order
r_attr	STRING 30		right hand attribute name
r_persid	STRING 60		right hand key
r_sql	INTEGER		right hand sort order

Person_Contacting Table

Reference table to denote the type of customer that made the contact. For example, the consumer, lawyer, media, and so on.

- **SQL Name** -- perscon
- **Object** -- perscnt

Field	Data Type	Reference	Remarks
del	INTEGER NOT_NULL	Active_Boolean_Table::enum	Deleted flag that indicates the following:**0** -- Active **1** -- Inactive/marked as deleted
id	INTEGER		Primary key of this table.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id)
sym	nvarchar(60)		The symbolic value for this Person_Contacting.

pr_trans Table

A transition object controls the current and next ticket status. The pr_trans table lists the status, new status, and actions that need to occur for a default transition.

- **SQL Name** -- pr_trans
- **Object** -- pr_trans

Label	Field	Description
id	INTEGER	Unique key.
status	SYMBOL	Specifies the current ticket status.
new_status	SYMBOL	Specifies the new ticket status
must_comment	INTEGER	Comment required when using a transition. On new default: 0
is_default	INTEGER	Default transition that appears when the Status field is empty. On new default: 0
condition	BOP_REF_STR_REF Macro	Site condition macro to approve transition.
condition error	STRING 255	Error message for site condition.
description	STRING 255	Description of this transition.
last_mod_dt	LOCAL_TIME	Timestamp of last update to this record.
last_mod_by	UUID REF ca_contact	User who last updated this.
del	INTEGER nn	Logical database delete status.
tenant	UUID REF ca_tenant	Reference to Tenant information.

Impact

Impact is a measure of the significance of an event by the user. It is used on Incident Reports.

- **SQL Name** -- impact
- **Object** -- imp

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
enum	INTEGER		Primary key for this table.
id	INTEGER		Specifies the unique (to the table) numeric ID.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
nx_desc	nvarchar(40)		Specifies the textual description of this impact.
sym	nvarchar(12)		Identifies the symbolic value for this impact.
value	INTEGER		Shows the numeric representation of this impact.

Web Screen Painter

This article contains the following topics:

wspcol Table

There are one to three rows in this table for every column created or updated by WSP. The columns table_name +column_name+status form a unique key. See also comments in wsp.maj.

- **SQL Name** -- wspcol
- **Object** -- wspcol

Field	Data Type	Reference	Remarks
description	STRING 300		Description
addl_info	STRING 500		Triggers, QREL stuff, and so on.
column_name	STRING 40 NOT_NULL		Column Majic name
dbms_name	STRING 40		DBMS schema name
display_name	STRING 80		Human-readable name
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
is_cluster	INTEGER		1 = DBMS cluster index
is_descending	INTEGER		1 = DBMS descending index
is_indexed	INTEGER		1 = DBMS-indexed column
is_local	INTEGER		1 = local column
is_not_null	INTEGER		1 = column cannot be null
is_order_by	INTEGER		1 = DBMS order by index
is_required	INTEGER		1 = column required
is_skey	INTEGER		1 = column is secondary key
is_unique	INTEGER		1 = DBMS unique index
is_write_new	INTEGER		1 = read-only after creation
is_wsp	INTEGER		1 = table created by WSP
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
on_ci_set	STRING 80		Value to set on update
on_new_default	STRING 80		Default value of attribute
persid	STRING 30		Persistent ID (SystemObjectName:id)
schema_name	STRING 40		CA SDM schema name
status	INTEGER NOT_NULL		Modification status
string_len	INTEGER		String len
table_name	STRING 40 NOT_NULL		Table Majic name

type	INTEGER		Column type
xrel_table	STRING 80		Operand of SRELBRELQREL

wsptbl Table

There are one to three rows in this table for every table created or updated by WSP. The columns table+name+status form a unique key. See also comments in wsp.maj for interpretation of integer codes.

- **SQL Name** -- wsptbl
- **Object** -- wsptbl

Field	Data Type	Reference	Remarks
description	STRING 300		Identifies the description of the table.
common_name	STRING 40		Specifies the name for the identification column.
dbms_name	STRING 40		Specifies the DBMS schema name.
display_group	STRING 40		Identifies the Grouping to display.
display_name	STRING 80		Specifies the user-readable name.
function_group	STRING 40		Identifies the Security function group.
id	INTEGER UNIQUE NOT_NULL KEY		Specifies the numeric ID, which is also unique to the table.
is_local	INTEGER		Identifies whether the table is local (1 = local table).
is_wsp	INTEGER		Identifies whether the table is created by WSP (1 = table created by WSP).
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Identifies the timestamp for when this record was last modified.
methods	STRING 500		Identifies Spell methods.
persid	STRING 30		Identifies the Persistent ID: (SystemObjectName:id)
rel_attr	STRING 40		Specifies the Foreign key column.
schema_name	STRING 40		Specifies the CA SDM schema name.
sort_by	STRING 150		Specifies the Default sort sequence.

status	INTEGER NOT_NULL		Identifies the Modification status.
table_name	STRING 40 NOT_NULL		Specifies the Table Majic name.
triggers	STRING 500		Identifies any Triggers for the table.

USP Window

This article contains the following topics:

usp_window Table

Contains information on the blackout and maintenance window.

- **SQL Name** -- usp_window
- **Object** -- window

Column	Type	Remarks
id	INTEGER	UNIQUE NOT_NULL KEY
sym	INTEGER	UNIQUE Name of window
window_type	INTEGER	REQUIRED References usp_window_type and specifies blackout or maintenance.
start_date	LOCAL_TIME	REQUIRED Start date of first recurrence
end_date	LOCAL_TIME	REQUIRED End date of first recurrence
final_end_date	LOCAL_TIME	REQUIRED End date of last recurrence
timezone	STRING 30	Timezone
icon	STRING 100	URL to icon
recurs	INTEGER	Specifies the following: **0** -- None **1** -- Daily **2** -- Weekly **3** -- Monthly **4** -- Annually
recurrence_interval	INTEGER	Days, weeks, months, or years
sunday	INTEGER	Restricts daily or weekly recurrence
monday	INTEGER	Restricts daily or weekly recurrence
tuesday	INTEGER	Restricts daily or weekly recurrence
wednesday	INTEGER	Restricts daily or weekly recurrence

thursday	INTEGER	Restricts daily or weekly recurrence
friday	INTEGER	Restricts daily or weekly recurrence
saturday	INTEGER	Restricts daily or weekly recurrence
occurrence	INTEGER	Restricts monthly or yearly recurrence **1** -- First **2** -- Second **3** -- Third **4** -- Fourth **5** -- Last
description	STRING 400	Text of window
legend	STRING 100	Legend on the Change Calendar
color	STRING 100	Web color of text
bgcolor	STRING 100	Web background color
style	STRING 100	Text style (italic, bold or normal)
last_mod_dt	LOCAL_TIME	Indicates the timestamp of when this record was last modified.
last_mod_by	UUID	Specifies the UUID of the contact who last modified this record.
del	INTEGER	
tenant	UUID	References the ca_tenant table.

The significance of recurrence_interval, occurrence, and the weekday attributes (sunday, monday, and so on) depend on the value of recurs:

- **1 (daily)**
  Specifies the days on which the event recurs.
  If the recurrence_interval is greater than one, the weekday attributes are ignored, and the event recurs at the interval you specified. Occurrence is ignored.
- **2 (weekly)**
  Specifies the number of weeks between recurrences, and the weekday attributes specify the days within the week the window recurs. Occurrence is ignored.
- **3 (monthly)**
  Specifies the number of months between occurrences.
  If the occurrence is zero, the recurrence is always on the day of the month of the start_date. If the occurrence is non-zero, it specifies that recurrence occurs in the first, second, third, fourth, or last week of the month on the weekday of the start_date (which will be the only non-zero weekday attribute).
- **4 (yearly)**
  Specifies the number of years between occurrences.
  If the occurrence is zero, the recurrence is always on the date of the start_date. If the occurrence is non-zero, it specifies that recurrence occurs in the first, second, third, fourth, or last week of the month on the weekday of the start_date (which will be the only non-zero weekday attribute).

usp_window_type Table

Contains information on blackout and maintenance windows.

- **SQL Name** -- usp_window_type
- **Object** -- window_type

Column	Type	Remarks
id	INTEGER	UNIQUE NOT_NULL KEY
del	INTEGER	NOT_NULL
sym	STRING 60	NOT_NULL S_KEY
description	STRING 100	
last_mod_by	UUID	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME	Indicates the timestamp of when this record was last modified.

Role Table

This article contains the following topics:

usp_role Table

Contains the information on the Roles.

- **SQL Name** -- usp_role
- **Object** -- role

Column	Type	Remarks
id	INTEGER	
admin	INTEGER	Admin Function Access
call_mgr	INTEGER	Requests Function Access
change_mgr	INTEGER	Change Orders Function Access
code	STRING 30	Code
data_partition	SREL dmn	
default_flag	INTEGER	Default
description	STRING (1024)	Description
form_group	SREL fmgrp	
grant_level	SREL acc_lvls	
help_view	SREL help_set	
initial_form	STRING (256)	Initial Form
interface_type	INTEGER	Interface Type
inventory	INTEGER	Inventory Function Access
issue_mgr	INTEGER	Issues Function Access
kcat	INTEGER	
kd	INTEGER	

kd_query_description	STRING (255)	
kd_query_id	SREL crsq	
kt_admin	INTEGER	Knowledge System Administrator
kt_analyst	INTEGER	Knowledge Analyst
kt_customer	INTEGER	Knowledge Customer
kt_engineer	INTEGER	Knowledge Engineer
kt_manager	INTEGER	Knowledge Manager
kt_type	INTEGER	kt type
name	STRING (80)	Role Name
notify	INTEGER	Notify Function Access
override_cnt_datapart	INTEGER	Override Contact Data Partition
pref_doc	INTEGER	Preferred Document
reference	INTEGER	Reference Function Access
sd_admin	INTEGER	CA SDM System Administrator
sd_analyst	INTEGER	CA SDM Analyst
sd_customer	INTEGER	CA SDM Customer
sd_employee	INTEGER	CA SDM Employee
security	INTEGER	Security Function Access
single_tenant	SREL tenant	
tenant	SREL tenant	
tenant_access	INTEGER	Tenant Access
tenant_group	SREL tenant_group	
tn_admin	INTEGER	
update_global	INTEGER	Update Global
view_internal	INTEGER	View Internal

usp_role_go_form Table

Links between Roles and Go Form type Web Forms.

- **SQL Name** -- usp_role_go_form
- **Object** -- role_go_form

Field	Data Type	Reference	Remarks
id	INTEGER		Specifies the unique (to the table) numeric ID.
is_default	INTEGER		Indicates if this is the default Go Form for the role.
role_obj	INTEGER	usp_role::id	Foreign key to the role_id field of the usp_role table.

web_form_obj	INTEGER	usp_web_form::id	Foreign key to the web form id field of the usp_web_form table.
menu_bar_obj	INTEGER	usp_menu_bar:: id	Foreign key to the menu bar id field of the usp_menu_bar table.

usp_role_tab Table

Links between Roles and their Tabs.

- **SQL Name** -- usp_role_tab
- **Object** -- role_tab

Field	Data Type	Reference	Remarks
id	INTEGER		Specifies the unique (to the table) numeric ID.
sequence	INTEGER		Used to order the role's tabs in the web interface.
role_obj	INTEGER	usp_role::id	Foreign key to the role_id field of the usp_role table.
tab_obj	INTEGER	usp_tab::id	Foreign key to the tab id field of the usp_tab table.

usp_role_web_form Table

Links between Roles and Report type Web Forms.

- **SQL Name** -- usp_role_web_form
- **Object** -- role_web_form

Field	Data Type	Reference	Remarks
id	INTEGER		Specifies the unique (to the table) numeric ID.
role_obj	INTEGER	usp_role::id	Foreign key to the role_id field of the usp_role table.
web_form_obj	INTEGER	usp_web_form::id	Foreign key to the web form id field of the usp_web_form table.

usp_acctyp_role Table

Links between Activity Types and Roles.

- **SQL Name** -- usp_acctyp_role
- **Object** -- acctyp_role

Field	Data Type	Reference	Remarks
access_type	INTEGER	Access_Type_v2::id	Foreign key to access type id in Access_Type_v2.
id	INTEGER		Specifies the unique (to the table) numeric ID.
is_default	INTEGER		Specifies if this is the default role for this access type. 1 = default
role_obj	INTEGER	usp_role::id	Foreign key to the role_id field of the usp_role table.

Contact

This article contains the following topics:

usp_contact Table

This table provides extensions to the ca_contact table that are used only by CA SDM products.

- **SQL Name** -- usp_contact
- **Object** -- cnt

Field	Data Type	Reference	Remarks
c_acctyp_id	INTEGER	Access_Type::id	Identifies the unique (to the table) numeric ID.
c_available	INTEGER		Displays as a check box to indicate that the analyst is available.
c_cm_id1	INTEGER	Contact_Method::id	Foreign key to the id field of the ct_mth table, this identifies the low priority of the contact method.
c_cm_id2	INTEGER	Contact_Method::id	Foreign key to the id field of the ct_mth table, this identifies the next level of low priority for the contact method.
c_cm_id3	INTEGER	Contact_Method::id	Foreign key to the id field of the ct_mth table, this identifies the standard level of priority for the contact method.
c_cm_id4	INTEGER	Contact_Method::id	Foreign key to the id field of the ct_mth table, this identifies the high level of priority for the contact method.

c_domain	INTEGER	Domain:id	Foreign key to the id field of the dmn table, this is the Data Partition.
c_email_service	nvarchar 30		(Not used by CA SDM) Identifies the pointer to the access type email service (such as, PROFS, and so on).
c_nx_ref_1	byte(16)		Foreign key to the contact_uuid field of the ca_contact table, this is a user-defined field.
c_nx_ref_2	byte(16)		Foreign key to the contact_uuid field of the ca_contact table, this is a user-defined field.
c_nx_ref_3	byte(16)		Foreign key to the contact_uuid field of the ca_contact table, this is a user-defined field.
c_nx_string1	nvarchar(40)		Identifies the emergency Workshift 4 Smag fields.
c_nx_string2	nvarchar(40)		Allows for a user-defined string field.
c_nx_string3	nvarchar(40)		Allows for a user-defined string field.
c_nx_string4	nvarchar(40)		Allows for a user-defined string field.
c_nx_string5	nvarchar(40)		Allows for a user-defined string field.
c_nx_string6	nvarchar(40)		Allows for a user-defined string field.
c_parent	byte(16)		(Not used in CA SDM) Foreign key to the contact_uuid field of the ca_contact table.
c_schedule	nvarchar 30	Bop_Workshift::persid	Foreign key to the persistent_id field of the bpwshft table, this is the Analyst's workshift for Auto Assignment.
c_service_type	nvarchar 30	Service_Desc::code	Foreign key to the code field of the srv_desc table, this identifies the Classic Service Type.
c_timezone	nvarchar(12)	Timezone::code	Foreign key to the code field of the tz table, this defines the Timezone.
c_val_req	INTEGER		Specifies a Force validation of the userid.
c_vendor	byte(16)	ca_company::company_uuid	Foreign key to the id filed of the ca_company table, this represents the Vendor.

c_ws_id1	nvarchar 30	Bop_Workshift::persid	Foreign key to the persistent_id field of the bpwshft table, this represents a workshift ID.
c_ws_id2	nvarchar 30	Bop_Workshift::persid	Foreign key to the persistent_id field of the bpwshft table, this represents a workshift ID.
c_ws_id3	nvarchar 30	Bop_Workshift::persid	Foreign key to the persistent_id field of the bpwshft table, this represents a workshift ID.
c_ws_id4	nvarchar 30	Bop_Workshift::persid	Foreign key to the persistent_id field of the bpwshft table, this represents a workshift ID.
contact_uuid	byte(16)		Primary key of this table.
global_queue_id	INTEGER	Global_Queue_Names::id	Specifies the pointer to the global queue.
ldap_dn	nvarchar(512)		Identifies the ldap dn value for this usp_contact.

usp_contact_handling Table

The usp_contact_handling table associates contacts with a special handling classification.

Attribute	Data Type	SREL References	Flags
contact	UUID	ca_contact	NOT_NULL
id	INTEGER		UNIQUE NOT_NULL KEY
special_handling	INTEGER	usp_special_handling	NOT NULL
tenant	UUID	ca_tenant	

usp_credentials_Table

The usp_credentials table contains all the UNC credential details to access UNC shared drive in the Advance availability configuration.

Attribute	Data Type	Reference	Description
id	INTEGER		Specifies the primary key of this table.
persid	STRING		Persistent ID.
del	INTEGER		Deleted flag 0 = Active 1 = Inactive or marked as deleted.
sym	INTEGER		60 NOT_NULL UNIQUE NOT_NULL S_KEY

description	STRING 500		Textual description of the UNC Credentials details.
unc userid	STRING		User name used for authentication to access the UNC shared drive.
unc password	STRING		Password in encrypted format to access UNC shared drive.
last_mod_dt	LOCAL_TIME		Timestamp of when the record was last modified.
last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
tenant	UUID	ca_tenant	

usp_rest_access Table

The following table lists the usp_rest_access attributes:

- **SQL Name** -- usp_rest_access
- **Object** -- rest_access

Field	Data Type	Reference	Remarks
id	INTEGER		Primary key to this table, it is a unique numeric ID.
access_key	INTEGER		UNIQUE
contact	UUID	ca_contact	Specifies the UUID of the contact.
secret_key	STRING 64		Encrypted shared secret key value.
expiration_date	LOCAL_TIME		Specifies the expiration date of the REST Access Key.

Resolution

This article contains the following topics:

usp_resolution_code Table

The usp_resolution_code table details the categorization of the incident/request resolution.

- **SQL Name** -- usp_resolution_code
- **Object** -- resocode

Field	Data Type	Reference	Remarks
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
persid	STRING 30		Persistent ID

del	INTEGER NOT NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- active **1** -- inactive/marked as deleted
sym	STRING 128 UNIQUE NOT_NULL S_KEY		
last_mod_dt	LOCAL_TIME		Timestamp of when this record was last modified.
last_mod_by	UUID REF	ca_contact	Specifies the UUID of the contact who last modified this record.
description	STRING 240		Textual description of the root cause.
tenant	UUID REF	ca_tenant	

usp_resolution_method Table

The usp_resolution_method table details how the incident was resolved or how the service was restored.

- **SQL Name** -- usp_resolution_method
- **Object** -- resomethod

Field	Data Type	Reference	Remarks
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
persid	STRING 30		Persistent ID
del	INTEGER NOT NULL	Active_Boolean_ Table::enum	Deleted flag **0** -- active **1** -- inactive/marked as deleted
sym	STRING 128 UNIQUE NOT_NULL S_KEY		
last_mod_dt	LOCAL_TIME		Timestamp of when this record was last modified.
last_mod_by	UUID REF	ca_contact	Specifies the UUID of the contact who last modified this record.
description	STRING 240		Textual description of the root cause.
tenant	UUID REF	ca_tenant	

USP Menu

This article contains the following topics:

usp_menu_bar Table

Contains the information on the menu bars used in the Role Based UI.

- **SQL Name** -- usp_menu_bar
- **Object** -- menu_bar

Column	Type	Remarks
id	INTEGER	
name	STRING (80)	Name of the role.
code	STRING 30	Code
del	INTEGER	Active_Boolean_Table::enum
description	STRING (255)	Description
htmpl_name	STRING (40)	Name of the HTMPL file containing the menu bar

usp_menu_tree Table

Each record is a node in a menu tree used in the Role Based UI.

- **SQL Name** -- usp_menu_tree
- **Object** -- menu_tree

Field	Data Type	Reference	Remarks
id	INTEGER		
caption	STRING (50)		Caption displayed for the menu tree node
description	STRING (255)		Description
has_children	INTEGER		0 indicates that this is a leaf node on the menu tree. 1 indicates that this node has child nodes.
parent_id	INTEGER		The id of the parent menu_tree node if this is a child node.
mt_resource	INTEGER	usp_menu_tree_res::id	The id of the usp_menu_tree_res that contains the resource for this node. Empty if this node is a label node only.
tree_name	INTEGER	usp_menu_tree_name::id	The id of the usp_menu_tree_name record for this menu_tree.

internal	INTEGER	Boolean_ Table::enum	Indicates if this menu tree is delivered out-of-box. If so, the menu tree is not customizable. User created menu trees are customizable.

usp_menu_tree_name Table

Contains the information on the named menu trees used in the Role Based UI.

- **SQL Name** -- usp_menu_tree_name
- **Object** -- menu_tree_name

Field	Data Type	Reference	Remarks
id	INTEGER		
name	STRING (80)		Name of the menu tree
code	STRING 30		Code
del	INTEGER	Active_ Boolean_ Table::enum	
description	STRING (255)		Description
internal	INTEGER	Boolean_ Table::enum	Indicates if this menu tree is delivered out-of-box. If so, the menu tree is not customizable. User created menu trees are customizable.

usp_menu_tree_res Table

Contains the information on the menu tree resources used by menu trees in the Role Based UI.

- **SQL Name** -- usp_menu_tree_res
- **Object** -- menu_tree_res

Field	Data Type	Reference	Remarks
id	INTEGER		
name	STRING (50)		Name of the menu tree resource
del	INTEGER	Active_ Boolean_ Table::enum	
description	STRING (255)		Description
mtr_resource	STRING (255)		The URL that displayed the web page

mtr_type	INTEGER		Internal flag used by CA Service Desk development. User created usp_menu_tree_res records will have mtr_type = 0.

USP Mailbox

This article contains the following topics:

usp_mb_rule_action_type Table

The usp_mb_rule_action_type table lists the types of filters in the usp_mailbox_rule table.

Field	Data Type	Description
del	INTEGER	
id	INTEGER	Specifies the REL_ATTR value of the table.
sym	STRING 60	Specifies the symbolic value for the action type.

The following values are the defaults for the usp_mb_rule_action_type table:

id	sym
1	Ignore Email
2	Ignore Email and Reply
3	Update Object
4	Create/Update Object

usp_mb_rule_filter_type Table

The usp_mb_rule_filter_type table lists the types of filters in the usp_mailbox_rule table.

Field	Data Type	Description
del	INTEGER	
id	INTEGER UNIQUE KEY	Specifies the REL_ATTR value of the table.
sym	STRING 60	Specifies the symbolic value for the filter type.

The following values are the defaults for the usp_mb_rule_filter_type table:

id	sym
1	Subject contains
2	Body contains
3	(Reserved for future use)
4	From Address contains

usp_mb_rule_subject_handling Table

The usp_mb_rule_subject_handling table lists how subjects are handled (prepend or append subjects) in the usp_mailbox_rule table.

Field	Data Type	Description
del	INTEGER	
id	INTEGER UNIQUE KEY	Specifies the REL_ATTR value of the table.
sym	STRING 60	Specifies the symbolic value for the subject handling type.

The following values are the defaults for the usp_mb_rule_subject_handling table:

id	sym
1	Prepend
2	Append

usp_mailbox Table

A mailbox object represents a connection to a mail server for a single inbox. The usp_mailbox table lists the filter, policy, and actions that must occur for an inbox.

Label	Field	Data Type	Reference	Description
id		INTEGER UNIQUE KEY		Specifies the primary key of this table.
	last_mod_dt	LOCAL_TIME		Specifies the time stamp of when this record was last modified.
	last_mod_by	UUID REF	ca_contact	Specifies the UUID of the contact who last modified this record.
Active	del	INTEGER NOT_NULL		Specifies if the mailbox is active. Inactive mailboxes are not polled.
Check Interval	check_interval	INTEGER		Specifies the number of seconds to wait between checks of the mailbox. 30 seconds is the default.
Name	name	STRING 60 UNIQUE S_KEY		Names the mailbox connection
Allow Anonymous	allow_anonymous	INTEGER		Allows messages to create or update tickets with sender addresses not attached to a known Contact.

Email Type	email_type	STRING 10		Specifies the email type: NONE, IMAP, or POP3. If NONE is selected, the mailbox is not polled.
Hostname	host_name	STRING 128		Specifies the host name or IP address of the mail server.
Port Override	host_port	INTEGER		Specifies a port number to override the default for Email Type. The default is 110 for POP3 and 143 for IMAP.
Userid	userid	STRING 64		Specifies the userid to log in to the mail server.
Password	password	STRING 64		Specifies the password for userid.
Security Level	security_lvl	INTEGER		Specifies a security level for encoding of login information: **0** -- Clear Text **1** - APOP (POP3 only) **2** - NTLM **3** - MD5 4 - OAUTH2.0 5 - LOGIN
Attachment Repository	attmnt_repository	STRING 60	Document_Repository	Specifies the repository for attachments. The repository must be local to pdm_maileater_nxd. If not specified, the mailbox is not polled.
Attach Entire Email	attach_email	INTEGER		This option will attach the entire email to a ticket. It overrides the default value of splitting out attachments.
Force Attachment Splitout	split_out_ attachment	INTEGER		Forces attachment split out if attach Entire Email is set. The entire message and individual attachments are all attached to the ticket.
Save Unknown Emails	save_unknown_ emails	INTEGER		Saves emails that are not able to be processed in NX_ROOT/site/unknown_mails.
Description	description	STRING 1000		Describes the mailbox.

Email Address/Hour	email_address_ per_hour	INTEGER		Specifies the maximum number of emails per email address per hour. You can specify the following values: **-1** -- No limit (default) **0** -- No emails allowed. **1** or more -- Maximum number of emails allowed.
Log Violation	log_policy_violation	INTEGER	usp_mailbox_violation_ log_type	Controls logging of policy violations to the standard log.You can specify the following values: Do not log First violation only (default) All violations
Inclusion List	inclusion_list	STRING 32768		Specifies email addresses or domains that are allowed to process emails -- only emails matching the list are allowed. You can specify multiple addresses or domains by delimiting them with a space character or new line.
Exclusion Lists	exclusion_list	STRING 32768		Specifies email addresses or domains that are not allowed to process emails. You can specify multiple addresses or domains by delimiting them with a space character or new line.
	tenant	UUID	ca_tenant	Specifies the tenant.

usp_mailbox_artifact_type Table

The usp_mailbox_artifact_type table lists the artifacts types referenced by the filter_min_artifact_type field in the usp_mailbox_rule table.

Field	Data Type	Description
id	INTEGER UNIQUE KEY	Specifies the REL_ATTR value of the table.

sym	STRING 60	Specifies the symbolic value for the artifact type.

The following values are the defaults for the usp_mailbox_artifact_type table:

id	sym	prefix
1	Protected	A
2	Secure	B

usp_mailbox_rule Table

The usp_mailbox_rule table lists the filter, the rule, and any action, any replies, or both that must occur when the filter matches.

Label	Field	Data Type	Reference	Description
id		INTEGER UNIQUE KEY		Specifies the primary key of this table.
last_mod_by		UUID REF	ca_contact	Specifies the UUID of the contact who last modified this record.
last_mod_dt		LOCAL_TIME		Specifies the time stamp of when this record was last modified.
Sequence	sequence	INTEGER		Specifies the sequence number of the rule. Rules are processed in sequence number order.
Mailbox	mailbox	INTEGER	usp_mailbox	Specifies the mailbox that this rule belongs to.
Active	del	INTEGER NOT_NULL		Specifies that the rule is active. A rules is not processed if it is inactive. Rules can be deleted, the active flags let you disable rules temporarily.
Description	description	STRING 1000		Describes the rule.
Filter	filter_type	INTEGER	usp_mb_rule_filter_type	Specifies the field to check against the Filter String.

Filter String	filter_string	STRING 255		Specifies a regular expression string to match. You can use the placeholder “{{object_id}}” to identify the object ID that the Text API can use with the Action Object field to identify the ticket.
Ignore Case	filter_ignore_ case	INTEGER		Specifies case-insensitive pattern matching for Filter String.
Action	action_ operation	INTEGER	usp_mb_rule_ action_type	Specifies the action to perform if the filter matches.
Action Object	action_object	STRING 30	usp_ticket_type	Specifies the type of ticket object to use for the action.
Write to Stdlog	action_write_ to_log	INTEGER		Writes the entire email text to the standard log (STDLOG) if the filter matches.
Log Entry Prefix	action_log_ prefix	STRING 30		Specifies a prefix for log entries. Lets you match rules to logs.
Add Subject Line	action_subject_handling	INTEGER	usp_mb_rule_ subject_handling	Adds a subject line to the message body before processing as follows: append, prepend, or null (do not add subject).
TextAPI Defaults	text_api_ defaults	STRING 1000		Specifies additional default commands for the Text API when the filter matches. Combines with the contents of the [EMAIL_DEFAULTS] section of the text_api.cfg file.
TextAPI Ignore Incoming	text_api_ignore_incoming	STRING 1000		Specifies additional Ignore Details for the Text API when the filter matches. Combines with the contents of the [EMAIL_IGNORE_INCOMING] section of the text_api.cfg file.

Reply	reply_method	INTEGER	Contact_Method	Specifies the notification method to use for automatic replies. If not set, no reply is created.
Reply Subject	reply_subject	STRING 200		Specifies a subject line to use for automatic replies.
Reply Success Text	reply_success_text	STRING 10000		Specifies the message body in plain text for the automatic response that is sent when the message is processed successfully.
Reply Success HTML	reply_success_html	STRING 10000		Specifies the message body in HTML for the automatic response that is sent when the message is processed successfully.
Reply Failure Text	reply_failure_ text	STRING 10000		Specifies the message body in plain text for the automatic response that is sent when a failure occurs processing the message.
Reply Failure HTML	reply_failure_ html	STRING 10000		Specifies the message body in HTML for the automatic response that is sent when a failure occurs processing the message.

usp_mailbox_violation_log_type Table

The usp_mailbox_violation_log_type table lists the violation log types referenced by the Log Violation drop-down list (log_policy_violation) in the usp_mailbox table.

Field	Description
id	Specifies the REL_ATTR value of the table.
sym	Specifies the symbolic value for the violation log type.

The following values are the defaults for the usp_mailbox_violation_log_type table:

id	sym
1	First violation only
2	All violations
3	Do not log

USP Relational Table - Macros

This article contains the following topics:

usp_lrel_false_action_act_f Table

Relates macros to events. Identifies the macros to execute upon a false evaluation of the event condition.

- **SQL Name** -- usp_lrel_false_action_act_f
- **Object** -- lrel_false_action_act_f

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
evt	STRING 30	Events	
macro	STRING 30	Spell_Macro	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	

usp_lrel_false_bhv_false Table

Relates macros to workflow task behaviors. Identifies the macros to execute upon a false evaluation of a behavior condition.

- **SQL Name** -- usp_lrel_false_bhv_false
- **Object** -- lrel_false_bhv_false

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
bhvtpl	INTEGER	Behavior_Template	
macro	STRING 30	Spell_Macro	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_true_action_act_t Table

Relates macros to events. Identifies the macros to execute upon a true evaluation of the event condition.

- **SQL Name** -- usp_lrel_true_action_act_t
- **Object** -- lrel_true_action_act_t

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
evt	STRING 30	Events	
macro	STRING 30	Spell_Macro	
last_mod_dt	LOCAL_TIME		

last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_true_bhv_true Table

Relates macros to workflow task behaviors. Identifies the macros to execute upon a true evaluation of a behavior condition.

- **SQL Name** -- usp_lrel_true_bhv_true
- **Object** -- lrel_true_bhv_true

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
bhvtpl	INTEGER	Behavior_Template	
macro	STRING 30	Spell_Macro	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

USP Relational Table - Stakeholders Notify List

This article contains the following topics:

usp_lrel_notify_list_cntchgntf Table

Relates contacts to change orders and supports the Stakeholders Notify List.

- **SQL Name** -- usp_lrel_notify_list_cntchgntf
- **Object** -- lrel_notify_list_cntchgntf

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
chg	INTEGER	Change_Request	
cnt	UUID	ca_contact	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_notify_list_cntissntf Table

Relates contacts to issues and supports the Stakeholders Notify List.

- **SQL Name** -- usp_lrel_notify_list_cntissntf
- **Object** -- lrel_notify_list_cntissntf

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
iss	STRING 30	Issue	
cnt	UUID	ca_contact	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_notify_list_cntntf Table

Relates contacts to requests and supports the Stakeholders Notify List.

- **SQL Name** -- usp_lrel_notify_list_cntntf
- **Object** -- lrel_notify_list_cntntf

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
cr	STRING 30	Call_Req	
cnt	UUID	ca_contact	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

USP Relational Table - Notification Rule

This article contains the following topics:

usp_lrel_ntfr_cntlist_att_ntfrlist Table

Relates contacts to a Notification Rule. For example, the Contacts tab on the Notification Rule Detail page uses this relational data.

- **SQL Name** -- usp_lrel_ntfr_cntl_att_ntfrl
- **Object** -- lrel_ntfr_cntlist_att_ntfrlist

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
ntfr	INTEGER	Notify_Rule	
cnt	UUID	ca_contact	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	

tenant	UUID	ca_ tenant	

usp_lrel_ntfr_ctplist_att_ntfrlist Table

Relates contact types to a notification rule. For example, the Contact Types tab on the Notification Rule Detail page uses this relational data.

- **SQL Name** -- lrel_ntfr_ctplist_att_ntfrl
- **Object** -- lrel_ntfr_ctplist_att_ntfrlist

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
ntfr	INTEGER	Notify_Rule	
ctp	INTEGER	ca_contact_type	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_ntfr_macrolist_att_ntfrlist Table

Relates a macro to a notification rule. Because this table contains internal data, do not change this data.

- **SQL Name** -- usp_lrel_ntfr_macrolist_att_ntfrllist
- **Object** -- lrel_ntfr_macrolist_att_ntfrllist

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
ntfr	INTEGER	Notify_Rule	
macro	STRING 30	Spell_Macro	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_ntfr_ntflist_att_ntfrlist Table

Relates object contacts to notification rules. For example, the Object Contacts tab of a Notification Rule Detail page uses this relational data.

- **SQL Name** -- usp_lrel_ntfr_ntflist_att_ntfrl
- **Object** -- lrel_ntfr_ntflist_att_ntfrlist

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY

ntfr	INTEGER	Notify_Rule	
ntfl	INTEGER	Object_Attr	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

USP Relational Table - Service Groups

This article contains the following topics:

usp_lrel_svc_grps_svc_chgcat Table

Relates service groups to a change category for auto-assignment.

- **SQL Name** -- usp_lrel_svc_grps_svc_chgcat
- **Object** -- lrel_svc_grps_svc_chgcat

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
chgcat	STRING 12	Change_Category	
cnt	UUID	ca_contact	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_svc_grps_svc_isscat Table

Relates service groups to an issue category for auto-assignment.

- **SQL Name** -- usp_lrel_svc_grps_svc_isscat
- **Object** -- lrel_svc_grps_svc_isscat

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
isscat	STRING 12	Issue_Category	
cnt	UUID	ca_contact	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_svc_grps_svc_pcat Table

Relates service groups to a request, problem, or incident area for auto-assignment.

- **SQL Name** -- usp_lrel_svc_grps_svc_pcat
- **Object** -- lrel_svc_grps_svc_pcat

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
pcat	STRING 30	Prob_Category	
cnt	UUID	ca_contact	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_svc_grps_svc_wftpl Table

Relates service groups to a classic workflow task template for auto-assignment.

- **SQL Name** -- usp_lrel_svc_grps_svc_wftpl
- **Object** -- lrel_svc_grps_svc_wftpl

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
wftpl	INTEGER	Workflow_Task_ Template	
cnt	UUID	ca_contact	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_svc_locs_svc_chgcat Table

Relates service locations to a change category for auto-assignment.

- **SQL Name** -- usp_lrel_svc_locs_svc_chgcat
- **Object** -- lrel_svc_locs_svc_chgcat

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
chgcat	STRING 12	Change_Category	
loc	UUID	ca_location	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_svc_locs_svc_groups Table

Relates service groups and locations for auto-assignment.

- **SQL Name** -- usp_lrel_svc_locs_svc_groups
- **Object** -- lrel_svc_locs_svc_groups

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
cnt	UUID	ca_contact	
loc	UUID	ca_location	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_svc_locs_svc_isscat Table

Relates service locations to an issue category for auto-assignment.

- **SQL Name** -- usp_lrel_svc_locs_svc_isscat
- **Object** -- lrel_svc_locs_svc_isscat

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
isscat	STRING 12	Issue_Category	
loc	UUID	ca_location	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_svc_locs_svc_pcat Table

Relates service locations to a request, problem, or incident area for auto-assignment.

- **SQL Name** -- usp_lrel_svc_locs_svc_pcat
- **Object** -- lrel_svc_locs_svc_pcat

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
pcat	STRING 30	Prob_Category	
loc	UUID	ca_location	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	

tenant	UUID	ca_tenant	

usp_lrel_svc_schedules_chgcat_svc Table

Relates workshifts to change categories and supports service schedules for auto-assignment.

- **SQL Name** -- usp_lrel_svc_sch_chgcat_svc
- **Object** -- lrel_svc_schedules_chgcat_svc

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
chgcat	STRING 12	Change_Category	
wrkshft	STRING 30	Bop_Workshift	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_svc_schedules_isscat_svc Table

Relates workshifts to issue categories and supports service schedules for auto-assignment.

- **SQL Name** -- usp_lrel_svc_sch_isscat_svc
- **Object** -- lrel_svc_schedules_isscat_svc

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
isscat	STRING 12	Issue_Category	
wrkshft	STRING 30	Bop_Workshift	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_svc_schedules_pcat_svc Table

Relates workshifts to request, problem, or incident areas and supports service schedules for auto-assignment.

- **SQL Name** -- usp_lrel_svc_sch_pcat_svc
- **Object** -- lrel_svc_schedules_pcat_svc

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
pcat	STRING 30	Prob_Category	

wrkshft	STRING 30	Bop_Workshift	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

USP Relational Table

This article contains the following topics:

usp_lrel_asset_chgnr Table

Relates CIs to Change Orders. For example, the Configuration Items tab on a Change Order Detail page uses this relational data.

- **SQL Name** -- usp_lrel_asset_chgnr
- **Object** -- lrel_asset_chgnr

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
chg	INTEGER	Change_Request	
nr	UUID		ca_owned_resource
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_asset_issnr Table

Relates CIs to Issues. For example, the Configuration Items tab on an Issue Detail page uses this relational data.

- **SQL Name --** usp_lrel_asset_issnr
- **Object** -- lrel_asset_issnr

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
iss	STRING 30	issue	
nr	UUID	ca_owned_ resource	
last_mod_ dt	LOCAL_TIME		
last_mod_ by	UUID	ca_contact contact_uuid	
tenant	UUID		

usp_lrel_att_cntlist_macro_ntf Table

Relates contacts to notification-type macros.

- **SQL Name** -- usp_lrel_att_cntlist_macro_ntf
- **Object** -- usp_lrel_att_cntlist_macro_ntf

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
macro	STRING 30	Spell_Macro	
cnt	UUID	ca_contact	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_att_ctplist_macro_ntf Table

Relates contact types to notification-type macros.

- **SQL Name** -- usp_lrel_att_ctplist_macro_ntf
- **Object** -- lrel_att_ctplist_macro_ntf

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
macro	STRING 30	Spell_Macro	
ctp	INTEGER	ca_contact_type	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_att_ntflist_macro_ntf Table

Relates object contacts to notification-type macros.

- **SQL Name** -- usp_lrel_att_ntflist_macro_ntf
- **Object** -- usp_lrel_att_ntflist_macro_ntf

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
macro	STRING 30	Spell_Macro	
ntfl	INTEGER	Notify_Object_Attr	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_aty_events Table

Relates events to activity notifications. For example, the Events tab of the Activity Notification Detail page uses this relational data.

- **SQL Name** -- usp_lrel_aty_events
- **Object** -- lrel_aty_events

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
aty	STRING 12	Act_Type	
evt	STRING 30	Events	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_cenv_cntref Table

Relates CIs to contacts. For example, the Environment tab on a Contact page uses this relational data.

- **SQL Name** -- usp_lrel_cenv_cntref
- **Object** -- lrel_cenv_cntref

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
cnt	UUID	ca_contact	
nr	UUID	ca_owned_resource	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_kwrds_crsolref Table

Relates key words to request solutions.

- **SQL Name** -- usp_lrel_kwrds_crsolref
- **Object** -- lrel_kwrds_crsolref

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
crsol	STRING 30	Call_Solution	
kwrd	INTEGER	Knowledge_Keywords	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_oenv_orgref Table

Relates CIs to an organization. For example, the Environment tab of the Organization Detail page uses this relational data.

SQL Name -- usp_lrel_oenv_orgref

Object -- lrel_oenv_orgref

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
org	UUID	ca_organization	
nr	UUID	ca_owned_resource	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_status_codes_tsktypes Table

Relates change order status to workflow task types. For example, the Status Codes tab on a Workflow Task Type Detail page uses this relational data.

- **SQL Name** -- usp_lrel_status_codes_tsktypes
- **Object** -- lrel_status_codes_tsktypes

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
tskty	STRING 12	Task_Type	
tskstat	STRING 12	Task_Status	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_bm_reps_assets Table

Relates CIs to a NSM repository.

- **SQL Name** -- usp_lrel_bm_reps_assets
- **Object** -- lrel_bm_reps_assets

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
nr	UUID	ca_owned_resource	
bmrep	INTEGER	Business_Management_ Repository	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

usp_lrel_bm_reps_bmhiers Table

Relates CI relationships to a NSM repository.

- **SQL Name** -- usp_lrel_bm_reps_bmhiers
- **Object** -- lrel_bm_reps_bmhiers

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
bmhier	INTEGER	Business_Management	
bmrep	INTEGER	Business_Management_Repository	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_ tenant	

USP Relational Table - Managed Surveys

This article contains the following topics:

usp_lrel_dist_cntlist_mgs_ntf Table

Relates contact lists to managed surveys.

- **SQL Name** -- usp_lrel_dist_cntlist_mgs_ntf
- **Object** -- lrel_dist_cntlist_mgs_ntf

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
cnt	UUID	ca_contact	
mgs	INTEGER	Managed_Survey	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_tenant	

usp_lrel_dist_ctplist_mgs_ntf Table

Relates contact types to managed surveys.

- **SQL Name** -- usp_lrel_dist_ctplist_mgs_ntf
- **Object** -- lrel_dist_ctplist_mgs_ntf

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
ctp	INTEGER	ca_contact_type	
mgs	INTEGER	Managed_Survey	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	

tenant	UUID	ca_tenant	

usp_lrel_dist_ntflist_mgs_ntf Table

Relates notifications to managed surveys.

- **SQL Name** -- usp_lrel_dist_ntflist_mgs_ntf
- **Object** -- lrel_dist_ntflist_mgs_ntfTable

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE KEY
ntfl	INTEGER	Notify_Object_Attr	
mgs	INTEGER	Managed_Survey	
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_tenant	

USP Tables - KPI

This article contains the following topics:

Kpi Table

Contains KPIs of Stored Query type, SQL type, and System type.

- **SQL Name** -- usp_kpi
- **Object** -- kc

Field	Data Type	Reference	Remarks
name	string		Defines the name of the KPI.
type	integer		Indicates where the KPI data is retrieved from, such as the crsq database table, BPVirtdb, or daemons in the BOP system.
status	integer		Indicates the on/off state of the KPI.
process_type	integer		Indicates the daemon process where the KPI data resides.
metric_type	integer		Indicates the type of metric the KPI produces (for example, count or sum).

stored_query_id	string		Provides a REF to the id attribute in the Crsq table. It is a SREL to REL_ATTR of object crsq, and contains the value of the code of a row in the crsq object.
user_context	string		Provides a UUID REF to the ca_contact table.
sql_query	string		Defines a SQL query.
description	string		Describes the measurement goal for the KPI.
refresh_time	integer		The time interval at which the KPI data is updated.
sys_name	string		The internal KPI name for sys type KPIs.
curr_kpi_time_stamp	integer		The last collection time of this KPI.
version_number	integer		Indicate the version of this KPI.
tenant	string		Provides a UUID REF to the ca_tenant table.
last_mod_by	string		Specifies the foreign key of the ca_contact table (UUID).
last_mod_dt	integer		Specifies the date and time of last update (integer value in UNIX ticks).

Kpi_Data Table

Contains the KPI data periodically retrieved from system daemons by the kpi_daemon. Provides raw data for web reporting.

- **SQL Name** -- usp_kpi_data
- **Object** -- kcd

Field	Data Type	Reference	Remarks
id	integer		Specifies the id of the record.
kpi_id	integer		Provides a REF to the id in the KPI.
kpi_time_stamp	integer		Indicates the time when a new KPI record is inserted into this table.
metric_type	integer		Indicates the type of metric the KPI produces (count, sum, max, or duration).

kpi_value	integer		Indicates the value of the metric produced by the KPI. The value matches the metric_type (count, sum, max, or duration).
duration_max	integer		Specifies the maximum value of duration for KPIs of duration metric_type. The time unit is millisecond.
duration_sum	integer		Specifies the total duration for KPIs of duration metric_type. The time unit is seconds.
duration_count	integer		Indicates the amount of duration data that has been collected in a period of time.
duration_average	integer		Indicates the average duration in a period of time for KPIs with duration metric_type. The time unit is milliseconds.
execute_time	integer		Indicates the total time for retrieving the KPI data, from the time the request is sent to time the result is received. The time unit is milliseconds.
version_number	integer		Indicate the version of this KPI.

Kpi_Ticket_Data Table

This table contains data retrieved from CA SDM tickets. The data can be used for reporting on ticket performance, such as how long a ticket was in each state.

- **SQL Name** -- usp_kpi_ticket_data
- **Object** -- ktd

> **NOTE**
> This table can collect data for custom fields when you manually add the UI_INFO flag to the custom fields in a majic file. For example, you can use the following attributes:

```
priority SREL pri REQUIRED { ON_NEW SET 0 ;
UI_INFO "KPI" ;} ;
urgency SREL urg { UI_INFO "KPI" ; };
```

Field	Data Type	Remarks
end_time	integer	Indicates the time when a ticket attribute is changed. It can also be used as a unique id for identifying a ticket object in which one or more attributes are changed.
prev_time	integer	Indicates the time stamp of the last change to this ticket object.

obj_name	string	Indicates ticket objects in majic files (cr, chg, iss,in and pr). Identifies a ticket in which an attribute is changed.
obj_id	integer	Indicates the id of a ticket object in which an attribute is changed.
obj_type	string	Contains type names for call request, such as Problem, Incident, or Request.
field_name	string	Indicates an attribute name in the ticket object in majic files of which the value is changed. They are assignee, priority, customer, etc.
field_value	string	Indicates the attribute value before current value. If it is a SREL, use a common name.
next_value	string	Indicates the attribute value after current value. If it is a SREL, use a common name.
operation	string	There are three kinds of operations: insert, delete and update. Insert or delete creates a usp_kpi_ticket_data record without setting the attribute fields.
attr_obj	string	The name of the object table the SREL links to.
attr_from_id	integer	The previous attr_obj id.
attr_from_uuid	uuid	The previous attr_obj uuid.
attr_to_id	integer	The next attr_obj id.
attr_to_uuid	uuid	The next attr_obj uuid.
user_context	uuid	Whenever the attribute of a ticket object has been changed, the user_context field needs to be filled. It is an SREL to cnt.
ktd_duration	integer	Indicates the calculated time duration between two changes. **Note:** The calculated duration is based on a change to values in a ticket in real time, not in business hours.
ktd_duration_wshft	integer	Indicates the time duration difference between two changes in workshift hours. This is a new column. **Note:** • If the workshift for the assignee is null, **ktd_duration_wshft** will be same as the **ktd_duration**. • if there is no assignee but the ticket has a group, the group workshift is considered. • If there is no group and no assignee for a ticket, **ktd_duration_wfsht** will be same as the **ktd_duration**

workshift	string	Indicates the workshift **persistent_id** for a record and currently active workshift on a ticket. The current active workshift of a ticket is derived from the current assignee's workshift (optionally, we can consider the ticket group if the assignee is empty).

Transition

This topic contains the following information:

Transition_Points Table

Lists the transitions of interest to the Notification System. E.g. Incident_report creation, or reassignment, TT creation TT closure and so on.

- **SQL Name** -- nottrn

Field	Data Type	Reference	Remarks
del	INTEGER NOT_NULL		Deleted flag **0** -- Active **1** -- Inactive/marked as deleted
enum	INTEGER NOT_NULL		Enumerated value for this entry - specifies ordering in lists and relative values
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
nx_desc	STRING 40		longer description of event
sym	STRING 30 UNIQUE NOT_NULL S_KEY		transition event symbol
tp_use_pri	INTEGER		flag indicating whether priority is meaningfull

Transition_Types Table

Transition types and their corresponding status transitions control when employees using self-service can close or reopen incidents or requests.

- **SQL Name** -- transition_type
- **Object** -- transition_type

Label	Field	Description
id	INTEGER	Unique key
sym	STRING 80 nn	
ss_flag	INTEGER nn	Specifies whether the status transition appears in the Employee Self-Service interface.
ss_button_text	STRING 80	Displays text on the button that performs the status transition.

ss_header_text	STRING 128	Used as the form header when the employee is prompted for comments after selecting a status transition button.
description	STRING 1000	Provides a description of the record.
del	INTEGER nn	Deleted flag: **0** -- Active **1** -- Inactive/marked as deleted
last_mod_dt	LOCAL_TIME	Indicates the timestamp of when this record was last modified.
last_mod_by	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.

Timespan

List of defined time-span elements used by applications to calculate hours-of-operation time calculations.

- **SQL Name** -- tspan
- **Object** -- tspan

Field	Data Type	Reference	Remarks
code	STRING 10 UNIQUE NOT_NULL		
end_day	STRING 5		
end_hour	STRING 5		
end_minute	STRING 5		
end_month	STRING 5		
end_year	STRING 5		
id	INTEGER UNIQUE KEY		Unique (to the table) Numeric ID
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
nx_desc	STRING 240		
start_day	STRING 5		
start_hour	STRING 5		
start_minute	STRING 5		
start_month	STRING 5		
start_year	STRING 5		
sym	STRING 30 UNIQUE NOT_NULL		
trigger_day	STRING 5		
trigger_hour	STRING 5		
trigger_minute	STRING 5		
trigger_month	STRING 5		

trigger_year	STRING 5		

Tasks

This topic contains the following information:

Task_Status Table

Workflow task states. Possible states include: Wait, pending, approve, reject, and so on.

- **SQL Name** -- tskstat
- **Object** -- tskstat

Field	Data Type	Reference	Remarks
allow_accumulate	INTEGER		Identifies the Allow Accumulate flag, as follows:**0 --** Do not accumulate**1 --** Accumulate
allow_task_update	INTEGER		Specifies the Allow Task Update flag, as follows:**0 --** Cannot update**1** -- Can update
code	nvarchar(12)		Primary key of this table.
del	INTEGER	Active_Boolean_ Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
description	nvarchar(500)		Provides a text description of the status.
do_next_task	INTEGER		Sets the Do Next Task flag, as follows:**0** -- No**1** -- Yes
hold	INTEGER		Sets the Hold flag, as follows:**0** -- Start events**1** -- Stop events
id	INTEGER		Specifies the unique (to the table) numeric ID.
is_internal	INTEGER		Specifies the Internal flag, as follows:**0** -- No**1** -- Yes (do not display in most status selections).
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp of when this record was last modified.
no_update_msg	nvarchar(500)		Sets the No Update Message flag to No.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
sym	nvarchar(30)		Identifies the Task Status name.

task_complete	INTEGER		Sets the Task Complete, as follows:**0** -- No**1** -- Yes

Task_Type Table

This table contains the list of task types used in the workflow used by CA SDM.

- **SQL Name** -- tskty
- **Object** -- tskty

Field	Data Type	Reference	Remarks
code	nvarchar(12)		Primary key of this table.
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
description	nvarchar(500)		Identifies the text description of the task.
id	INTEGER		Identifies the numeric ID, which is unique to the table.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp for when this record was last modified.
persid	nvarchar(30)		Identifies the Persistent ID: (SystemObjectName:id).
sym	nvarchar(30)		Identifies the name of task.

Service Type

This article contains the following topics:

target_tgttpls_srvtypes Table

Links the Target Template to Service Types.

- **SQL Name** -- target_tgttpls_srvtypes
- **Object** -- tgt_tgttpls_srvtypes

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE NOT_NULL KEY Unique (to the table) Numeric ID.
del	INTEGER		NOT_NULL Deleted flag: **0** -- Active **1** -- Inactive/marked as deleted

last_mod_dt	LOCAL_TIME		Indicates the timestamp when this record was last modified.
last_mod_by	UUID	ca_contact	UUID of the contact who last modified this record.
tenant	UUID	ca_tenant	UUID REF to the ca_tenant table.
tgt_time_tpl	INTEGER	target_time_tpl	NOT_NULL. The Service Target Template that is linked to a Service Type.
sdsc	STRING 30	Service_Desc	NOT_NULL.The Service Type to which the template is linked.
target_ duration	DURATION		NOT_NULL. The amount of time in which this target is reached.
set_actual	INTEGER	Boolean_Table	A flag that determines whether the Set Actual Service Target Action is available on the ticket.
reset_actual	INTEGER	Boolean_Table	A flag that determines whether the Reset Actual Service Target Action is available on the ticket.
cost	STRING 255		Text information to appear on the ticket.
work_shift	STRING 30	Bop_Workshift	The workshift used in time calculations.

target_time Table

Represents a Service Target that is attached to a ticket.

- **SQL Name** -- target_time
- **Object** -- tgt_time

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE NOT_NULL KEY Unique (to the table) Numeric ID.
del	INTEGER		NOT_NULL Deleted flag: **0** -- Active **1** -- Inactive/marked as deleted
sym	STRING 60		NOT_NULL Identifies the symbolic value for this target.
last_mod_dt	LOCAL_TIME		Indicates the timestamp when this record was last modified.
last_mod_by	UUID	ca_contact	UUID of the contact who last modified this record.
tenant	UUID	ca_tenant	UUID REF to the ca_tenant table.

target_duration	DURATION		NOT_NULL. The amount of time in which this target is reached.
condition	STRING 30	Macro	Identifies the condition to be detected in determining that this target has been reached.
condition_outcome	INTEGER	Boolean_Table	Specifies the required outcome of the evaluation of the condition.
service_type	STRING 30	Service_Desc	NOT_NULL. The Service Type to which this target belongs.
object_type	STRING 30		NOT_NULL Ticket type that this target is attached.
object_id	INTEGER	S_KEY	NOT_NULL id of the ticket that this target is attached.
set_actual	INTEGER	Boolean_Table	A flag that determines whether the Set Actual Service Target Action is available on the ticket.
reset_actual	INTEGER	Boolean_Table	A flag that determines whether the Reset Actual Service Target Action is available on the ticket.
lock_target	INTEGER	Boolean_Table	Prevents Service Target recalculations for tickets that transition to and from the Hold status.
cost	STRING 255		Text information to appear on the ticket.
target_time	LOCAL_TIME		Deadline for the Service Target.
actual_time	LOCAL_TIME		The time that the target completed.
time_left	DURATION		Number of remaining minutes.
_mapped_cr	STRING 30	Call_Req	Pointer to the Request, Incident, or Problem that uses this Service Target.
_mapped_chg	INTEGER	Change_ Request	Pointer to the Change Order that uses this Service Target.
_mapped_iss	STRING 30	Issue	Pointer to the Incident that uses this Service Target.
target_tpl	INTEGER	target_tpl	Identifies the target template for this target.
work_shift	STRING 30	Bop_Workshift	Identifies the workshift to use for calculating the target.

target_time_tpl Table

Contains the Service Target template defaults to link to a Service Type.

- **SQL Name** -- target_time_tpl
- **Object** -- tgt_time_tpl

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE NOT_NULL KEY Unique to the table Numeric ID.
del	INTEGER		NOT_NULL Deleted flag: **0** -- Active **1** -- Inactive/marked as deleted
sym	STRING 60		NOT_NULL Identifies the symbolic value for this target.
last_mod_dt	LOCAL_TIME		Indicates the timestamp when this record was last modified.
last_mod_by	UUID	ca_contact	UUID of the contact who last modified this record.
tenant	UUID	ca_tenant	UUID REF to the ca_tenant table.
target_duration	DURATION		NOT_NULL. The amount of time in which this target is reached.
condition	STRING 30	Macro	Identifies the condition to use to determine whether the target was met.
condition_outcome	INTEGER	Boolean_Table	Specifies the required outcome of the evaluation of the condition.
object_type	STRING 30		NOT_NULL. Identifies the valid ticket type for this template.
set_actual	INTEGER	Boolean_Table	A flag that determines whether the Set Actual Service Target Action is available on the ticket.
reset_actual	INTEGER	Boolean_Table	A flag that determines whether the Reset Actual Service Target Action is available on the ticket.
cost	STRING 255		Indicates the text information to appear on the ticket.
work_shift	STRING 30	Bop_Workshift	Indicates the workshift used for time calculations.

True and False Strings

True_False_Table Table

Contains localized True or False strings that display on the UI.

- **SQL Name** -- True_False_Table
- **Object** -- true_false

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE NOT_NULL KEY Unique to the table Numeric ID.
del	INTEGER		NOT_NULL Deleted flag: **0** -- Active **1** -- Inactive/marked as deleted
enum	INTEGER		NOT_NULL Enumerated value for this entry **0** -- False **1** -- True
sym	STRING 60		UNIQUE NOT_NULL S_KEY Identifies the symbolic value for this target
desc	STRING 40		Describes the enum

Type of Contact

Type_Of_Contact Table

Reference table to denote the type of issue. examples: complaint, complement and so on.

- **SQL Name** -- toc
- **Object** -- typecnt

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/marked as deleted
id	INTEGER		Primary key of this table, this is the unique numeric ID.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Species the timestamp of when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
sym	nvarchar(60)		The symbolic value for this Type_Of_Contact.

User Query

User_Query Table

User scoreboard queries.

- **SQL Name** -- usq
- **Object** -- usq

Field	Data Type	Reference	Remarks
expanded	INTEGER		
factory	STRING 30		
id	INTEGER UNIQUE NOT_NULL KEY		Unique (to the table) Numeric ID
label	STRING 80 NOT_NULL		
last_mod_by	UUID	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
obj_persid	STRING 60		Persistent ID (SystemObjectName:id)
parent	INTEGER	usq::id	
persid	STRING 30		Persistent ID (SystemObjectName:id)
query	STRING 30	Cr_Stored_Queries::code	
query_set	INTEGER		
query_type	INTEGER		
sequence	INTEGER NOT_NULL		

Urgency

List of urgency codes/descriptions used in CA SDM applications.

- **SQL Name** -- urgency
- **Object** -- urg

Field	Data Type	Reference	Remarks
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
enum	INTEGER		Primary key of this table.
id	INTEGER		Identifies the unique (to the table) numeric ID.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.

last_mod_dt	INTEGER		Indicates the timestamp of when this record was last modified.
nx_desc	nvarchar(40)		Provides the description of the urgency level.
sym	nvarchar(12)		Identifies the symbolic name of this urgency.
value	INTEGER		Indicates the numeric representation of this urgency.

Timezone

This table contains the Timezones used in CA SDM applications.

- **SQL Name** -- tz
- **Object** -- tz

Field	Data Type	Reference	Remarks
code	nvarchar(12)		Primary key of this table.
del	INTEGER	Active_Boolean_Table::enum	Specifies the Deleted flag, as follows:**0** -- Active**1** -- Inactive/ marked as deleted
description	nvarchar(500)		Identifies the text description of the timezone.
dst_delta	INTEGER		Specifies the delta seconds for daylight saving time.
end_abs_date	LOCAL_TIME		Identifies the *absolute* start date.
end_day	INTEGER		Represents the end day of the week, such as 0-6.
end_mon	INTEGER		Represents the ending month, such as 0-11 for the month of the year.
end_pos	INTEGER		Represents the ending position, such as 0 for "First" or 1 for "Last".
gmt_delta	INTEGER		Represents the delta seconds from GMT.
id	INTEGER		Identifies the numeric ID, which is unique to the table.
last_mod_by	byte(16)	ca_contact::uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	INTEGER		Identifies the timestamp for when this record was last modified.
persid	nvarchar(30)		Persistent ID (SystemObjectName:id).
start_abs_date	INTEGER		Identifies the *absolute* start date.
start_day	INTEGER		Represents the Start day used to calculate DST, for example, 0-6 day of the week.

start_mon	INTEGER		Represents the starting month for the timezone, such as 0-11 for the month of the year.
start_pos	INTEGER		Represents the starting position, such as 0 for "First" or 1 for "Last".
sym	nvarchar(30)		Specifies the name of the service type.

CA Process Automation

This article contains the following topics:

usp_caextwf_instances Table

Associates CA Process Automation process instances that launched from the CA SDM interface. Includes the CA SDM entity to which the instance belongs.

- **SQL Name** -- usp_caextwf_instances
- **Object** -- caextwf_inst

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE NOT_NULL KEY
inactive	INTEGER		NOT_NULL
instance_id	STRING 255		CA Process Automation instance ID
object_persid	STRING 60		The persistent id of the CA SDM entity from which a CA Process Automation process instance was launched.
procname	STRING 32768		The CA Process Automation process definition name and reference path.
starttime	LOCAL_TIME		Indicates CA Process Automation workflow start time.
endtime	LOCAL_TIME		Indicates CA Process Automation workflow end time.

usp_caextwf_start_forms Table

Stores CA SDM objects with the launchable CA Process Automation process definitions. The Start Request Form indicates the process that runs on the CA Process Automation server, but CA SDM stores the process information.

- **SQL Name** -- usp_caextwf_start_forms
- **Object** -- caextwf_sfrm

Field	Data Type	Reference	Remarks
id	INTEGER		UNIQUE NOT_NULL KEY

object_persid	STRING	60	UNIQUE NOT_NULL KEY The CA SDM object that provides the CA Process Automation process definition information for launching a new process instance.
caextwf_form	STRING	255	CA Process Automation Process Definition name.
caextwf_path	STRING	32768	The CA Process Automation reference path where this Process Definition is stored in the CA Process Automation library.

Outage Table

This article contains the following topics:

usp_outage_reason Table

The usp_outage_reason table associates an outage reason with a special handling classification.

- **SQL Name** -- usp_outage_reason
- **Object** -- None

Attribute	Data Type	SREL References	Flags
del	INTEGER		NOT_NULL
description	STRING 4000		
id	INTEGER		UNIQUE NOT_NULL KEY
last_mod_by	UUID	ca_contact	
last_mod_dt	LOCAL_TIME		
sym	STRING 60		NOT NULL
tenant	UUID	ca_tenant	

usp_outage_type Table

The usp_outage_type table associates outage types with a special handling classification.

- **SQL Name** -- usp_outage_type
- **Object** -- outage_type

Attribute	Data Type	SREL References	Flags
del	INTEGER		NOT_NULL
description	STRING 4000		
id	INTEGER		UNIQUE NOT_NULL KEY
last_mod_by	UUID	ca_contact	

last_mod_dt	LOCAL_TIME		
sym	STRING 60		NOT NULL
tenant	UUID	ca_tenant	

Comments

This table contains the following topics:

O_COMMENTS Table

Program control table used by CA SDM Knowledge Management.

- **SQL Name** -- O_Comments
- **Object** -- O_Comments

Field	Data Type	Reference	Remarks
ASSIGNEE	UUID		SREL to the Contacts table.
CLOSE_DATE	DATE		Represents the timestamp of when this flag was closed.
CLOSE_DESC	nvarchar(50)		Displays the description when the flag is closed.
COMMENT_TEXT	nvarchar(255)		Displays comment text.
COMMENT_ TIMESTAMP	DATE		Indicates the timestamp of when this record was last created.
DEADLINE_DATE	DATE		Displays the date when the flag should be fixed by the user.
DOC_ID	INTEGER	SKELETONS:: id	Identifies the unique doc ID.
EMAIL_ADDRESS	nvarchar(75)		Identifies the email address.
FLG_CODE	nvarchar(50)		Prevents duplicate flags from being created for the same broken link.
FLG_STATUS	INTEGER	KT_FLG_ STATUS	Identifies the flag status as active or inactive.
FLG_TYPE	INTEGER		SREL to the KT_Flg_Type Table.
ID	INTEGER KEY		Unique (to the table) Numeric ID.
last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
last_mod_dt	DATE		Indicates the timestamp of when this record was last modified.
tenant	UUID	ca_tenant	Specifies the UUID of the tenant.
USER_ID	UUID	ca_contact:: uuid	Identifies the unique user ID

USER_NAME	STRING 50		Identifies the user name.
VER_COUNT	INTEGER		Identifies the unique version ID and displays the number of versions created (1, 2, 3, 4...)

NR_Comment Table

Standard comments table.

- **SQL Name** -- nr_com
- **Object** -- nr_com

Field	Data Type	Reference	Remarks
attr_name	nvarchar(60)		For ITIL, this contains the Asset attribute that has changed.
com_comment	nvarchar(1000)		Identifies the comment text.
com_dt	INTEGER		Acts as the pointer to the parent row (nr_did), for the dt_ comment written.
com_par_id	byte(16)	ca_owned_ resource::uuid	Foreign key to the id field of the ca_owned_resource table, this is the Asset.
com_userid	nvarchar(40)		Identifies the userid of the commenting author.
id	INTEGER		Primary key of this table, this is a unique numeric ID.
new_value	nvarchar(1000)		For ITIL, this contains the new value of the Asset's attribute that has changed.
old_value	nvarchar(1000)		For ITIL, this contains the old value of the Asset's attribute that has changed.
writer_id	byte(16)		Primary key to the table, this is a unique identifier.

Technical Reference

This section lists all the CA Service Desk Manager commands:

- CA SDM Text API Interface
- The Configuration File
- View Field Descriptions
- RFC 2251 LDAP Result Codes
- pdm_configure--Open the Configuration Window
- pdm_key_refresh--Refresh Cached Key Information
- pdm_lexutil--Modify SDM Lexicons
- pdm_listconn--List Active Connections
- pdm_logfile--Change stdlog Cutover Size
- pdm_task--Set Environment Variables
- pdm_uconv--Convert Local Charset to UTF-8
- pdm_webstat--Return Web Usage Statistics
- pdm_mail Utility--Send Email Information
- CA SDM PDM Database Commands
- CA SDM Report Command
- CA SDM Form Groups
- Contents of the Samples Directory
- Schema Files Syntax
- Object Definition Syntax
- STANDARD_LISTS Optional Statement
- FACTORY Optional Statement
- Where Clauses
- Attribute Data Types
- Web Services Methods
- REST HTTP Methods
- Web Services Attachment-Related Methods
- Web Services Knowledge Attachment Methods
- Web Services Miscellaneous Methods
- Web Services Knowledge Management
- getCategory Method
- LREL Methods
- dbmonitor_nxd--Database Monitoring Daemon
- List/Query Methods
- Asset Management Methods
- Web Services Business Methods
- notifyContacts Method
- attachChangeToRequest Method
- createTicket Method
- Group Management Methods
- Contact Management Methods
- getPolicyInfo
- loginServiceManaged Method
- Using the Automated Tasks Editor
- How an Automated Task Runs
- Automated Task Elements
- Script Library Management
- Functions COM Object Methods
- WScript COM Object Methods
- EBR_DICTIONARY Table

- EBR_FULLTEXT Table
- EBR_INDEX Table
- EBR_SYNONYMS Table
- bop_sinfo--Display System Information
- FS_CONSTANTS Object

CA SDM Text API Interface

This article contains the following topics:

Overview of Text API

The *Text API* is an interface that lets you use text-based input to create and update objects in the CA SDM database, such as issues, requests, contacts, and assets. Using the Text API, you can assign values to most fields that are accessible to users.

> **WARNING**
> CA SDM requires that all input be in UTF-8 format, or data can be corrupted. The pdm_unconv utility lets you convert data from a local charset to UTF-8 and from UTF-8 to a local charset.

You can access the Text API by using the following interfaces:

- Command line
- Email

> **NOTE**
> You can use web services as the alternative to the Text API for cross-application integration.

Command Line Interface

Use the *pdm_text_cmd* command to activate the Text API command line interface. You can then specify information such as the table to process and the operation to perform by using parameters to the pdm_text_cmd command.

The input to the Text API is passed to the pdm_text_cmd command in the form of an input file, or directly from STDIN.

> **NOTE**
>
> When passing the parameters from command prompt, use Ctrl+Z in Windows and Ctrl+D in POSIX.
>
> **WARNING**
> You cannot use single or double quotation marks as parameters for the bop_cmd and pdm_text_nxd commands.

Input Format

Input to the Text API is specified in the following ways:

- In the command-line interface, input is typically specified in a text file passed to the *pdm_text_cmd* command.
- In the email interface, input is specified in the text of the email. You specify a regular expression to find the target object identifiers.

You format the Text API input in the same way no matter which interface you use.

The basic format for input is as follows:

```
%keyword=value
```

or

```
%PROPERTY={{property_label}}value
```

The normal behavior of Text API commands has the following exceptions, where the last-appearing of two or more conflicting commands takes precedence:

- When a message contains multiple valid ticket ID artifacts matching the mailbox rule filter string, or multiple Text API ticket ID commands, the first one encountered is used. Also, a ticket ID artifact, which is identified using the mailbox rule filter string, overrides any Text API ticket ID command, regardless of which appears first.
- When a message contains multiple log comment Text API commands, all comments are posted, although the order in which they appear in the ticket activity log can vary.

All ticket ID artifacts that match the filter, valid or otherwise, and Text API ticket ID commands within the message, applicable or otherwise, applied or otherwise, are commented out before the message is posted. The ticket ID artifacts identified through the mailbox rule filters appear as -((...))-. Leading percentage signs (%) in Text API ticket ID commands are converted to two opening parentheses ((, and two closing parentheses)) follow the command. If a Text API ticket ID command appears after another Text API command with a log comment (%LOG=…), the commented-out Text API ticket ID command is made into a separate log comment.

NOTE
The log comment is the only Text API command that can appear multiple times in one message and still have each occurrence applied. For any other commands, the Text API uses the last occurrence only, because multiple occurrences of other commands conflict with each other. Multiple log comment commands post separate log comment messages to the ticket, and not necessarily in any particular order.

In addition, if a Text API ticket ID command appears in the message either at the beginning of the message or in between two other Text API commands, it is converted into a log comment. If the previous command is a log comment (%LOG=…) or update description (%DESCRIPTION=…), it is appended to that command, rather than becoming a separate log comment.

How the Text API Uses Keywords

You can use two types of keywords as input to the Text API.

- Definitions in the [KEYWORDS] section of the *text_api.cfg* file -- This type is a group of keywords that are related directly to the fields for the various tables that you can update. For example, most of the fields on the Issue Detail form are defined the [KEYWORDS] section. Using these keywords, you can set values for fields in the record that you are updating or creating. For example, the following line sets the priority of the issue to 5:

  ```
  %PRIORITY=5
  ```

  The [KEYWORDS] section of the *text_api.cfg* file lists all keywords. You can define additional keywords (for example, to allow Text API access to fields that you have added when customizing your database schema).
- The following special keywords are always defined as follows, regardless of the contents of the *text_api.cfg* file:

Keyword	Description
ASSET	Used to attaches an item to a ticket (valid for requests, issues, and change orders). The value specified is the item name, which must already exist. You can specify this keyword multiple times, because a ticket can have multiple items attached to it.
ATTACHMENT	Used internally by the email interface to add email attachments to a ticket.
DESCRIPTION	Specifies the value to use for the ticket's description field. This keyword is assumed if input is sent to the Text API without an explicit keyword. This keyword is applied automatically by the Mail Eater when the message does not begin with a keyword but does contain a ticket ID artifact or keyword. You can change how the DESCRIPTION keyword is handled for updates using the following entry in the [OPTIONS] section of text_api.cfg: UPDATE_DESC_IS_LOG If this option is set to YES, the value is used to create a log comment. If the value is set to NO, the value overwrites the existing description field.

FROM_EMAILFROM_EMAIL_OVERRIDE	Used by the email interface to match against the Email Address field in the ca_contact record. It is also used as the log_agent for the ticket. If both are supplied, FROM_EMAIL is ignored. **Note:** FROM_EMAIL is set automatically by the Mail Eater with the sender address of the message.
FROM_PERSID	Used by the command line interface to define the log_agent for an operation (for example, when a ca_contact record does not have a User ID). This keyword is passed automatically by pdm_text_cmd if the -p parameter is specified. The value is matched to a ca_contact record persistent_id.
FROM_USERID	Used only in the command line interface to define the log_agent for an operation. This keyword is passed automatically by *pdm_text_cmd* if the -u parameter is specified. The value is matched to a contact's User ID.
LOG	Used to create a log entry (valid for requests, change orders, issues, and contacts). This keyword is applied automatically by the Mail Eater when the message does not begin with a keyword but does contain either a ticket ID artifact or keyword, or a DESCRIPTION keyword.
PROPERTY	Used to set the value of a property (valid only for requests, change orders, and issues). Unlike other keywords, which are followed by an equal sign and a value, the PROPERTY keyword syntax must include the property label, as follows: PROPERTY={{property_label}}value You must specify the *property_label* exactly as it appears in the database.
SEARCH	Used only in the command-line interface to supply a list of keywords for use in a query to update multiple tickets for an asset. The value is a list of keywords used in the search.

Keyword Input Conventions

The following conventions apply to keyword input formatting:

- Prefix every keyword (including PROPERTY) with a percent (%) sign. The percent sign must be in column position one. If the first nonempty line of the input does not have a percent sign at the start of the line, either *%DESCRIPTION= or %LOG=* is used as the prefix for the incoming data, depending on whether a ticket ID artifact or keyword was found. If *%DESCRIPTION* is set, the contents of the message up to the first Keyword is posted as a ticket description. If *%LOG= is* set, the contents of the message up to the first Keyword is posted as a log comment.
- Do not use any intervening spaces within the keyword between the percent sign and the keyword, or between the keyword and the equal (=) sign.
- Do not quote values; all data after the equal sign is assumed to be the value.
- Keywords are not case sensitive.
- If the input includes duplicate keywords, the last keyword is used; otherwise, the order in which you specify the keyword/value pairs is unimportant.
- Specify keyword values as you would for the corresponding field in the web interface. For example, to specify an Analyst contact type, you use *%CONTACT_TYPE=Analyst*, even though in the database this value is stored as an integer. The *CONTACT_TYPE* keyword is defined in *text_api.cfg* so that it converts the specified value to match the stored value.

NOTE
Whether the value is case sensitive depends on your underlying DBMS.

- You can extend string data across multiple lines.

Format an Email Message To Update a Ticket

A user can format an email message to create or update a ticket.

To format an email message to create or update a ticket, use the following fields:

- **To**
  Specifies the mailbox name assigned to the CA SDM contact set up for the privileged user.
- **From**
  Specifies the person sending the email. The person must be defined in the *ca_contact* table unless the **Allow Anonymous** option is specified in the applicable mailbox rule.

  NOTE
  The From address is typically part of your email program configuration, and it is not typically set on a per-message basis.
- **Attachments**
  Attaches documents and other files to the email to send attachments to the Text API.
- **Subject**
  Matches keywords in a mailbox rule filter string, particularly when creating a ticket.
- **Body**
  Specifies the message body of the email using the Text API. You can specify the keyword *ISSUE_ID*, *REQUEST_ID*, or *CHANGE_ID*, depending on the type of ticket to create or update a ticket.

Start and End Email Message Delimiters

Some email interfaces add information to the beginning or end of mail messages (for example, MIME encoding) that can cause the email interface to malfunction. If your email interface adds information, you can use the following delimiters: start-request and end-request. The email interface ignores information that is specified prior to start-request and subsequent to end-request.

NOTE

CA SDM Maileater now supports HTML and Rich-Text Format (RTF) emails for incoming mails.

Example: Use start-request and end-request Delimiters

```
"start-request"
message_body
"end-request"
```

How the Text API Uses Artifacts

The Text API processes the subject or body of email notifications. Mailbox rules let you identify artifacts and values that the Text API uses. For example, you can define the rule for incidents as Incident:{{object_id}}, so finding Incident:1234 translates to %INCIDENT_ID=1234 for the Text API. 1234 is the ref_num for the Incident. Because the artifact must be unique in the email and easy to find, you can make the artifact more distinctive such as %Incident:{{object_id}}%.

Follow the {{object_id}} keyword with a character which is not a letter, number, comma, forward-slash (/), plus sign (+), or equals (=) sign, because these characters can appear within an artifact. Otherwise, it is possible that characters which follow the artifact can be misinterpreted as part of the value of the artifact, or that a character within the value of the artifact can be misinterpreted as the character which follows the value.

Mail Eater does the following:

1. Finds the artifact within an email (such as Incident:1234) that maps to the appropriate ticket or other object supported by the Text API.
2. Translates the artifact to a Text API token (such as %INCIDENT_ID=1234).
3. Mail Eater submits the tagged message to the Text API. The Text API processes the email, applies the text, commands, or both which it contains to the appropriate ticket, and generates an automatic response email indicating whether the email message it received was successfully applied. Depending on the actions performed, a notification email message is also sent separately to indicate certain specific events, such as the creation of a ticket.

How to Set Up Notification Replies to Update Tickets

The Text API daemon (pdm_text_nxd) creates and updates tickets with information from external interfaces, such as the command line and email. You can set up mail to use the Text API so that users (contacts) can update tickets by replying to email notifications. The text of the reply is added as a log comment activity to the ticket.

To set up notification replies to update tickets, do the following:

1. Set the notification method that the contact uses to pdm_mail - T *reply_email_address* or pdm_mail - F *reply_email_address.* The *reply_email_address* specifies the incoming address for the mailbox. When the contact clicks reply on an email that address is filled in from the From or Reply-To address of the message to which they are replying.
   -T sets the Reply-To address. -F sets the From address, which is used as the reply address if a separate one is not specified.

   NOTE
   Some mail programs do not or cannot honor a Reply-To address.
2. Create or update a mailbox rule using a Text API keyword.
   The user-defined artifacts in the mailbox rule filters replace the following Text API keywords:

Object	Text API Keyword	Identifier
Incident	%INCIDENT_ID	Ref_num
Problem	%PROBLEM_ID	Ref_num
Request	%REQUEST_ID	Ref_num
Chg_ref_num	%CHANGE_ID	Chg_ref_num
Issue	%ISSUE_ID	Ref_num

1. Create or update a notification phrase that matches the rule.
2. Create or update a message template that uses the notification phrase.
3. Update the mailbox rule that you created in Step 2 to specify the message template that you created or updated in Step 4.

After the user receives the notification and replies to it, the following actions occur:

1. When the filter string is found, the relevant ticket ID keyword and value denoted by the placeholder, if any, are appended to the message.
2. If a matching ticket ID artifact is found, the corresponding ticket is updated, with either a log comment, a new description, or other values in accordance with the text, keywords, and commands in the message.
3. If a matching ticket ID artifact is not found, a ticket is created with a description and other parameters in accordance with the text, keywords, and commands in the message.

How to Set Up a Reply to an Incident Notification Example

This example shows how to set up a reply to an incident notification.

To set up a reply to an incident notification, do the following:

1. Create a mailbox rule using the following fields and values:
   - Filter -- Body contains
   - Filter String -- %Incident:{{object_id}}%
   - Ignore Case -- YES
   - Action -- Update Object
   - Action Object -- Incident
2. Create a notification phrase that includes the rule as follows:
   - Symbol -- Incident Reply
   - Code -- IncidentReply
   - Active -- Active
   - Description -- Comment that embeds the reply for an Incident/Problem/Request.
   - Phrase -- In order to add a comment to your @{call_req_id.type.sym}, just reply to this email or include the line below (on a line by itself):

     ```
     %Incident:{call_req_id.ref_num}%
     ```

     NOTE
     In auto-reply text of the mailbox rule, omit the call_req_id. prefix. This prefix applies a context which the mailbox rule text is already in, and such a context change is not valid when already acting within that context.
3. Create or update a message template that uses the notification phrase as follows:
   - Notification Message Body

     ```
     This is a simple notification.
     @{notification_phrase[IncidentURL1].phrase}
     ```
4. Update the mailbox rule that you created in Step 1 to specify the message template that you created in Step 3, as follows:
   Message Template -- *mailbox rule name*

How an End User Updates a Ticket Example

The following example demonstrates how an end user (John Smith) replies to an email notification to update an incident ticket.

The Body or Subject of the email includes the object identifier. The {{object_id}} placeholder within the filter string denotes the object identifier.

1. A notification is sent to John Smith and includes the following instructions:

   ```
   In order to add a comment to your incident, just reply to this email or include the line below (on a line
    by itself).
   %Incident:1234%
   ```
2. John Smith replies to the notification as follows:

   ```
   This is my response...
   ```
3. The Mail Eater receives the following text version of the John Smith's email:

   ```
   This is my response...
   From: Service Desk
   Sent: Wednesday, September 18, 2009 10:22 AM
   To: Smith, John
   Subject: Simple Notification
   This is a simple notification.
   In order to add a comment to your incident, just reply to this email or include the line below (on a line
    by itself).
   %Incident:1234%
   ```

4. The Mail Eater processes rules in order and finds the %Incident:1234% artifact:

```
This is my response...
From: Service Desk
Sent: Wednesday, September 18, 2009 10:22 AM
To: Smith, John
Subject: Simple Notification
This is a simple notification.
In order to add a comment to your incident, just reply to this email or include the line below (on a line
 by itself).
%INCIDENT_ID=1234
```

5. The Mail Eater adds the Text API keywords and the {{object_id}} value to an %INCIDENT_ID= statement and leaves a marker where the {{object_id}} value was found. The following text shows the data that is sent to the Text API. The bold text shows values added by the Mail Eater.

```
%LOG=This is my response...
From: Service Desk
Sent: Wednesday, September 18, 2009 10:22 AM
To: Smith, John
Subject: Simple Notification
This is a simple notification.
In order to add a comment to your incident, just reply to this email or include the line below (on a line
by itself).
 %Incident:-((...))-%
%FROM_EMAIL=john.smith@company.com
 %INCIDENT_ID=1234
```

6. The Text API add a log comment for Incident 1234.

Keyword Conversion Methods

Many of the keywords defined in text_api.cfg have an associated method to convert the value specified to a value that is appropriate for storage in the database. This feature lets users specify values just as they would in the web interface without having any knowledge of the underlying implementation.

The configuration file has several examples of this type of keyword definition, including ISSUE.PRIORITY and CONTACT.CONTACT_TYPE. If you need to define additional keywords (for example, to allow Text API access to fields that you have added when customizing your database schema), you can use one of the following predefined methods:

Method	Output Type
lookup_actbool	INTEGER
lookup_asset_by_name	UUID
lookup_asset_by_persid	UUID
lookup_chg_category	STRING
lookup_chg_status	STRING
lookup_cnt_by_email	UUID
lookup_cnt_by_last_first_middle	UUID
lookup_cnt_by_logonid	UUID
lookup_cnt_by_persid	UUID
lookup_cnt_meth	INTEGER
lookup_cnt_type	INTEGER

lookup_company	UUID
lookup_cr_status	STRING
lookup_cr_template	STRING
lookup_domain	INTEGER
lookup_grc	INTEGER
lookup_group	UUID
lookup_impact	INTEGER
lookup_iss_category	STRING
lookup_iss_status	STRING
lookup_loc	UUID
lookup_mfr_model	UUID
lookup_nr_family	INTEGER
lookup_org	UUID
lookup_person_contacting	INTEGER
lookup_position	INTEGER
lookup_priority	INTEGER
lookup_prob_category	STRING
lookup_product	INTEGER
lookup_resource_status	INTEGER
lookup_service_lvl	STRING
lookup_severity	INTEGER
lookup_state	INTEGER
lookup_timezone	STRING
lookup_type_of_contact	INTEGER
lookup_urgency	INTEGER
lookup_workshift	STRING

If the value you need to convert is not addressed by any of these predefined methods, you need to write a customized method. The method should take a STRING value as its input and return a value (either INTEGER, STRING or UUID) as its output. Return a value of -1 (or “-1”) to denote that the value cannot be determined and is therefore, not set. For UUID, return a “(uuid) NULL”.

For example, you might develop a method to convert a user ID to a ca_contact table reference. The incoming value, such as Administrator, would be passed to the method, and the method would return the ca_contact table id for the user ID of Administrator.

The manner in which you define keywords in the configuration file offers you the advantage of defining multiple keyword mappings to the same field, including different conversion methods, depending on the value being specified. For example, assignee can have several different keyword mappings to define how to set its value based on different input values. One input might be user ID, another might be last name, first name, middle name, and still another might be the actual ca_contact id (for example, 793ED69B4E87A545BD8E911834D829FC). Each keyword maps to a different conversion method, except the last one, which does not need to be converted.

The Configuration File

This article contains the following topics:

The text_api.cfg file defines the keywords that are related directly to the fields of the various tables that you can update. You use this file both as a reference to find certain predefined values, such as keywords, and as a mechanism for configuring the Text API, although the default configuration file works for most installations without modification.

The text_api.cfg file is located in the following directory:

- UNIX -- $NX_ROOT/site
- Windows -- *installation-directory*\site. For example: C:\Program Files\CA\Service Desk\site

The configuration file is divided into sections, with particular attributes defined within each section. Attribute definitions are of the following form:

```
keyword=value
```

None of the keywords are case-sensitive, whereas all values (except in the [OPTIONS] section) are case-sensitive.

> **NOTE**
> You can view and modify the text_api.cfg file using any text editor.

Options

The [OPTIONS] section of the text_api.cfg file defines processing options that may differ from one site to another. For example, there are options to determine the incoming date format, which fields allow linefeeds to be retained, and whether to allow issues, requests, or change orders to be updated using the email interface. All options in this section are configurable. Be aware that although you can remove table names from the VALID_TABLE_LIST, if you do not want to support Text API access to those tables, you cannot add table names to this list.

Defaults

Use the [XX_DEFAULTS] section provided in the text_api.cfg file for each interface using the Text API (for example, [EMAIL_DEFAULTS] for the email interface and [CMD_DEFAULTS] for the command line interface). The [XX_DEFAULTS] section defines the default values for fields and properties that are required in case the user does not supply them directly. XX refers to the interface type, such as CMD or EMAIL.

To set default values, use one of the following formats:

- table_name**.**keyword**=**value
  The *keyword* must be defined either in the [KEYWORDS] section or as properties in your database. Any method associated with the keyword is automatically applied to the *value*. For example:
  ```
  ISSUE.PRIORITY=1
  ```
  The PRIORITY keyword is defined in text_api.cfg so that it performs a lookup to convert the value you specify to match the corresponding value that is stored in the database. Here, value 1 is converted to 5, which is the underlying database value for the priority symbol 1. This feature lets users specify the value just as they would in the web interface.
- table_name**.PROPERTY={{**property_label**}}**value
  The *property_label* must be defined as a property in your database.

In both formats, the *table_name* must be one of the values defined by VALID_TABLE_LIST in the [OPTIONS] section, such as Issue, Request, or Contact.

Ignore Incoming

There are several [..._IGNORE_INCOMING] sections in the text_api.cfg file, one for each interface that uses the Text API (for example [EXT_IGNORE_INCOMING] for the external interface used by other CA products). These sections define

fields and properties that are ignored in the input (the format is the same as described in Defaults, except no "**=** *value*" is specified). This feature lets you prevent users from setting certain values, which in turn, provides you with more security for such times as letting customers use the email interface.

The IGNORE sections work well when used in conjunction with the corresponding [..._DEFAULTS] sections because you can prevent the user from setting a particular value and supply a default value at the same time. For example, if you want to prevent email interface users from setting the priority of an issue, you could set the following values:

```
[EMAIL_DEFAULTS]
ISSUE.PRIORITY=2
[EMAIL_IGNORE_INCOMING]
ISSUE.PRIORITY
```

In this case, any priority that the user specifies in the email message body is ignored, and all issues created by the email interface are automatically assigned a priority of 2.

Example Input

The following examples show input that you can use in the body of an email message or in a file serving as input to the command-line interface. **Example: First Line Does Not Include a Keyword** In this example, because the first line is missing a %keyword in the first column, the literal %DESCRIPTION= is added to the beginning of the message. This addition sets the description field to "This entire text goes to the description field" (with the line break intact, because ISSUE.DESCRIPTION is included in the list of fields for the LINEFEEDS_ALLOWED entry in the [OPTIONS] section of text_api.cfg).

```
This entire text goes
into the description field
%PRIORITY=None
```

Example: First Line Includes a Keyword In this example, the PRIORITY keyword is defined in text_api.cfg so that it performs a lookup to convert the value you specify to match the corresponding value that is stored in the database. Here, the value None is converted to 0, which is the underlying database value for the priority symbol 1. This feature lets users specify the value as they would in the web interface.

```
%description=This is my description
%priority=None
%CATEGORY=Upgrade.PC
%PROPERTY={{Current CPU}}266 mhz
%PROPERTY={{Current Harddrive}}1 gig
%PROPERTY={{Requested Upgrade}}4 gig harddrive
```

The specified values are used to set the description and priority fields for the ticket, similar to the previous example (notice that keyword case is unimportant). The value of Upgrade.PC is searched, and the category field for the ticket is set appropriately. Matching the following labels sets the three property values:

- Current CPU
- Current Hard drive
- Requested Upgrade

View Field Descriptions

This article contains the following topics:

You can use the field description information in the basic and advanced views that are supplied with CA SDM.

The following points apply to many of the tables:

- You must turn on audit logging, found in Administration, Options Manager, Audit Log, to see data in the advanced views.
- pdmtime refers to date/time fields that are in GMT format (the number of elapsed seconds since 1/1/1970).
- The terms change request and change order are used interchangeably.

View_Act_Log

The following is a basic view of the request activity log table. Activity type and the analyst's full name are also listed in the view. The activity log table (act_log) was joined with the activity type table (act_type) and the contact table (ca_contact) to give the actual activity type of each activity log entry, and the analyst who performed the activity. Extracted fields from the joins that might be useful are located at the end of this list.

Field	Remarks
id	act_log.id: The unique identifier for this record in the act_log table.
persid	act_log.persid: The unique identifier for this record in the act_log table, preceeded by the object identifier (alg for act_log) and a colon.
call_req_id	act_log.call_req_id: Pointer to call request persid to which this activity belongs. act_log.call_req_id = call_req.persid.
last_mod_dt	act_log.last_mod_dt: The last modify date/time (pdmtime).
time_spent	act_log.time_spent: The duration of time spent on this activity, stored as the total number of seconds. For example, 80 = 1 minute, 20 seconds.
time_stamp	act_log.time_stamp: User modifiable date/time of activity (pdmtime).
system_time	act_log.system_time: The date/time of record creation (pdmtime).
analyst	act_log.analyst: The uuid pointer to the contact uuid to get the analyst who performed the activity. act_log.analyst = ca_contact.contact_uuid.
description	act_log.description: The text description of this activity, which can be modified by the user.
action_desc	act_log.action_desc: The text description of automated action, which cannot be modified by the user.
type	act_log.type: The text pointer to a record in the activity type table. For example, act_log.type = act_type.code.
knowledge_session	act_log.knowledge_session: An identifier for a particular session of a particular user.
knowledge_tool	act_log.knowledge_tool: An indicator of the knowledge management tool used for the search, such as NLS_FAQ or EXPERT, etc.
internal	act_log.internal: An integer flag (1 or 0), which indicates if this log entry is intended for all to see or just for internal use.
activity_type	act_type.symActivity: The type derived from act_log.type = act_type.code.
analyst_lastname	View_Contact_Full.last_name: The analyst's last name, derived from act_log.analyst = ca_contact.contact_uuid.

analyst_firstname	View_Contact_Full.first_name: The analyst's first name.
analyst_middlename	View_Contact_Full.middle_name: The analyst's middle name.

View_Audit_Assignee

The following is an advanced view of the audit log where assignee is tracked. This view shows the duration of time between assignee changes for every request and change order. Requests or change orders that are changed from a particular assignee to a null assignee, and then from null assignee back to a particular assignee, do not have the duration of the null assignee listed in this view. This view lists the following fields for both requests and change orders. There may be more than one entry for each audobj_uniqueid (request or change order).

Field	Remarks
audobj_uniqueid	audit_log.audobj_uniqueid: The audit log object unique id representing the chg.id or the call_req.id.
from_val	audit_log.attr_after_val: The ‘changed from’ assignee value.
to_val	audit_log.attr_after_val: The ‘changed to’ assignee value.
from_time	audit_log.attr_from_time: The beginning time an assignee was assigned (pdmtime).
to_time	audit_log.attr_from_time: The ending time the same assignee was assigned (pdmtime).

View_Audit_Group

The following is an advanced view of the audit log where group is tracked. This view shows the duration of time between group changes for every request and change order. Requests or change orders that are changed from a particular group to null group, and then from null group back to a particular group, do not have the duration of the null group listed in this view. This view lists the following fields for both requests and change orders. There may be more than one entry for each audobj_uniqueid (request or change order).

Field	Remarks
audobj_uniqueid	audit_log.audobj_uniqueid: The audit log object unique id which represents chg.id or call_req.id.
from_val	audit_log.attr_after_val: The ‘changed from’ group value.
to_val	audit_log.attr_after_val: The ‘changed to’ group value.
from_time	audit_log.attr_from_time: The beginning time group was assigned (pdmtime).
to_time	audit_log.attr_from_time: The ending time same group was assigned (pdmtime).

View_Audit_Priority

The following is an advanced view of the audit log where priority is tracked. This view shows the duration of time between priority changes for every request and change order. This view lists the following fields for both requests and change orders. There may be more than one entry for each audobj_uniqueid (request or change order).

Field	Remarks
audobj_uniqueid	audit_log.audobj_uniqueid: The audit object unique id, represents a request call_req.id or change order chg.id.
from_val	audit_log.attr_after_val: The ‘changed from’ value of priority.
to_val	audit_log.attr_after_val: The ‘changed to’ value of priority.
from_time	audit_log.attr_from_time: The beginning time priority was in a particular state (pdmtime).
to_time	audit_log.attr_from_time: The ending time priority was in the same state (pdmtime).

View_Audit_Status

The following is an advanced view of the audit log where status is tracked. This view shows the duration of time between status changes for every request and change order. This view lists the following fields for both requests and change orders. There may be more than one entry for each audobj_uniqueid (request or change order).

Field	Remarks
audobj_uniqueid	audit_log.audobj_uniqueid: The audit object unique id, which represents a request call_req.id or change order chg.id.
from_val	audit_log.attr_after_val: The ‘changed from’ value of priority.
to_val	audit_log.attr_after_val: The ‘changed to’ value of priority.
from_time	audit_log.attr_after_time: The beginning time status was in a particular state (pdmtime).
to_time	audit_log.attr_after_time: The ending time status was in the same state (pdmtime).

View_Change_Act_Log

The following is a basic view of all change order activity logs. This is a view of the change request activity log table (chgalg) joined with the activity type table (act_type) and the contact table (ca_contact) to give more meaningful data, such as the actual activity type description and full name of the analyst who performed the activity.

Field	Remarks
id	chgalg.id: The unique identifier for this record in the chgalg table.
persid	chgalg.persid: The unique identifier for this record in the chgalg table, preceded by the object identifier (chgalg for chgalg) and a colon.
change_id	chgalg.change_id: The pointer to the change order id to which this activity belongs.chgalg.change_id = chgalg.id
last_mod_dt	chgalg.last_mod_dt: The last modify date/time (pdmtime).

time_spent	chgalg.time_spent: The duration of time spent on this activity, stored as the total number of seconds. For example, 80 = 1 minute, 20 seconds.
time_stamp	chgalg.time_stamp: The user modifiable date/time of activity (pdmtime).
system_time	chgalg.system_time: The date/time of record creation (pdmtime).
analyst	chgalg.analyst: The uuid pointer to the contact uuid to get the analyst who performed the activity.chgalg.analyst = ca_contact.contact_uuid
description	chgalg.description: The text description of this activity, which can be modified by the user.
action_desc	chgalg.action_desc: The text description of the automated action, which cannot be modified by the user.
type	chgalg.type: The text pointer to a record in the activity type table.chgalg.type = act_type.code
internal	chgalg.internal: The integer flag (1 or 0), which indicates if this log entry is intended for all to see or just for internal use.
knowledge_session	chgalg.knowledge_session: An identifier for a particular session of a particular user.
knowledge_tool	chgalg.knowledge_tool: An indicator of the knowledge management tool used for the search, such as NLS_FAQ or EXPERT, and so on.
analyst_lastname	View_Contact_Full.last_name: The analyst's last name, derived from chgalg.analyst = ca_contact.contact_uuid.
analyst_firstname	View_Contact_Full.first_name: The first name of the analyst.
analyst_middlename	View_Contact_Full.middle_name: The middle name of the analyst.
activity_type	act_type.sym: The activity type referenced by chgalg.type = act_type.code.

View_Change

The following is a basic view of all change orders, listing the status, priority, category, organizations, the affected end user's full name, the requester's full name, the assignee's full name, the group name and ID, and so on. Here, the change request table (chg) was joined with many other tables to give some more meaningful data about the change order.

Field	**Remarks**
id	chg.id: The unique identifier for this record in the chg table.
persid	chg.persid: The unique identifier for this record in the chg table, preceded by the object identifier (chg for table chg) and a colon.
chg_ref_num	chg.chg_ref_num: The change order reference number, which is used by analysts and customers to refer to a particular change order.
description	chg.description: The long description of a change order, as dictated by an analyst or customer.

status	chg.status: The unique identifier of a change order status, which is a pointer to the chgstat table.chg.status = chgstat.code.
active_flag	chg.active_flag: The integer flag to determine whether this change record is active or not (1 or 0).
start_date	chg.start_date: The date the first task goes to a pending status (pdmtime).
open_date	chg.open_date: The change order creation date (pdmtime).
last_mod_dt	chg.last_mod_dt: The last modified date (pdmtime).
last_mod_by	chg.last_mod_by: The pointer to the contact uuid who was the last contact to modify this change order.chg.last_mod_by = ca_contact.contact_uuid.
close_date	chg.close_date: The date the change order was set to inactive (pdmtime).
resolve_date	chg.resolve_date: The date the change order was set to a status configured to indicate the change was resolved (pdmtime).
rootcause	chg.rootcause: A pointer to a record in the rootcause table, which represents the original situation which required this change order to be executed.chg.rootcause = rootcause.id.
est_total_time	chg.est_total_time: The estimated total time (pdmtime) it will take to complete this change.
actual_total_time	chg.actual_total_time: The actual total time (pdmtime) it took to complete this change.
log_agent	chg.log_agent: A binary unique identifier referencing the ca_contact table, referencing the person who was the change's original creator.chg.log_agent = ca_contact.contact_uuid.
assignee	chg.assignee: The pointer to the contact uuid who is currently assigned to the change order.chg.assignee = ca_contact.contact_uuid.
organization	chg.organization: The pointer to internal organization uuid, which represents the organization to whom this change order belongs.chg.organization = ca_organization.organization_uuid.
group_id	chg.group_id: The pointer to contact uuid, which represents the group currently assigned to the change order.chg.group_id = ca_contact.contact_uuid
affected_contact	chg.affected_contact: The pointer to the contact uuid, which represents the affected contact for this change order.chg.affected_contact = ca_contact.contact_uuid
requestor	chg.requrestor: The pointer to the contact uuid, which represents the person who ordered the change.chg.requestor = ca_contact.contact_uuid
category	chg.category: The pointer to the change category code to get the category into which this change falls.chg.category = chgcat.code
priority	chg.priority: The pointer to priority enum, which represents the priority into which this change falls.chg.priority = pri.enum

need_by	chg.need_by: The date which indicates when the affected_end_user needs to have the change completed (pdmtime).
est_comp_date	chg.est_comp_date: The estimated completion date (pdmtime) for this Change Order.
actual_comp_date	chg.actual_comp_date:The actual completion date (pdmtime) of this change order.
est_cost	chg.est_cost: The estimated cost of this change order.
actual_cost	chg.actual_cost: The actual cost to implement this change order.
justification	chg.justification: A text field which allows a requestor to document the reason(s) this change is required.
backout_plan	chg.backout_plan: A text field that allows an analyst to document a backout plan for this change.
impact	chg.impact: A pointer to an impact table record, which indicates the scope of resources that this change affects.chg.impact = impact.enum
parent	chg.parent: A pointer to another change request id, which allows creation of a hierarchy of change orders.chg.parent = chg.id
effort	chg.effort: A text field which explains the plan for implementing this change order.
support_lev	chg.support_lev: A pointer to a service desc record, which automates some constraints for which this change must be completed.chg.support_lev = srv_desc.code
template_name	chg.template_name: The name of and pointer to a change order template.chg.template_name = chg_template.template_name
sla_violation	chg.sla_violation: The integer to count the number of times slas attached to this "change has been violated".
predicted_sla_viol	chg.predicted_sla_viol: (r5.5) Neugent related technology field.
macro_predict_viol	chg.macro_predict_viol: (r5.5) Neugent related technology field.
created_via	chg.created_via: A pointer to a record in the interface table, which indicates from which interface the change order originated. chg.created_via = interface.id
call_back_date	chg.call_back_date:A date/time field (pdmtime), which indicates a future date/time the requestor is to be contacted.
call_back_flag	chg.call_back_flag: A boolean indicator displayed as a checkbox to the user, indicating whether or not to notify the analyst at the chg.call_back_date.
string1	This is a user-definable text field.
string2	This is a user-definable text field.
string3	This is a user-definable text field.
string4	This is a user-definable text field.
string5	This is a user-definable text field.
string6	This is a user-definable text field.

service_date	chg.service_date: The Date/ Time (pdmtime) that an outside vendor is expected to spend to service this change order.
service_num	chg.service_num: The text field to document an outside vendor's service or purchase order number.
product	chg.product: A pointer to a record in the product table, which indicates the product that is affected by this change.chg.product = product.id
actions	chg.actions: A big text field for documenting actions.
type_of_contact	chg.type_of_contact: A pointer to a record in the toc table, which indicates a general categorization of the affected_end_user's perspective of the change order.chg.type_of_contact = toc.id
reporting_method	chg.reporting_method: A pointer to a record in the repmeth table, which classifies the origination of the change order, and is selected by the person creating the change order.chg.reporting_method = repmeth.id
person_contacting	chg.person_contacting: A pointer to a record in the perscon table, which indicates the role of the affected_end_user or requestor.chg.person_contacting = perscon.id
status_name	chgstat.sym: The description of the status as seen by a user.chg.status = chgstat.code
priority_num	pri.sym: The description of the priority as seen by a user.chg.priority = pri.enum
category_name	chgcat.sym: The name of the Change Category as viewed by a user.chg.category = chgcat.code
organization_name	ca_organization.org_name: The name of an organization as viewed by a user.chg.organization = ca_organization.organization_uuid
affected_end_user_lastname	ca_contact.last_name: The affected end user's last name.chg.affected_end_user = ca_contact.contact_uuid
affected_end_user_firstname	ca_contact.first_name: The affected end user's first name.chg.affected_end_user = ca_contact.contact_uuid
affected_end_user_middlename	ca_contact.middle_name: The affected end user's middle name.chg.affected_end_user = ca_contact.contact_uuid
requester_lastname	ca_contact.last_name: The requestor's last name.chg.requestor = ca_contact.contact_uuid
requester_firstname	ca_contact.first_name: The requestor's first name.chg.requestor = ca_contact.contact_uuid
requester_middlename	ca_contact.middle_name: The requestor's middle name.chg.requestor = ca_contact.contact_uuid
business	ca_organization.org_name: The name of the Requester's organization as seen by users.chg.requestor = ca_organization.organization_uuid
assignee_lastname	ca_contact.last_name: The assignee's last name.chg.assignee = ca_contact.contact_uuid

assignee_firstname	ca_contact.first_name: The assignee's first name.chg.assignee = ca_contact.contact_uuid
assignee_middlename	ca_contact.middle_name: The assignee's middle name.chg.assignee = ca_contact.contact_uuid
groupID	ca_contact.contact_uuid: A binary representation of the internal id used for the group assigned to this change order.chg.group_id = ca_contact.contact_uuid
group_name	ca_contact.last_name: The name of the group assigned to this change order.chg.group = ca_contact.contact_uuid
service_type	srv_desc.sym: The name of the service type applied to this change order.chg.support_lev = srv_desc.code
impact_num	impact.sym: The description of the impact as seen by users.chg.impact = impact.enum
product_sym	product.sym: The product description as seen by users.chg.product = product.id
type_of_contact_sym	toc.sym: The Type Of Contact description as seen by users.chg.type_of_contact = toc.id
rpting_method_sym	repmeth.sym: The Reporting method description as seen by users.chg.reporting_method = repmeth.id
person_contacting_sym	perscon.sym: The Person Contacting description as seen by users.chg.person_contacting = perscon.id

View_Change_to_Assets

The following list of fields is a basic view of change orders and their assets. The change request table (chg) is indirectly joined with the network resource table (ca_owned_resource) to get a list of each change order’s assets. This view may not list all change orders, particularly those that have no assets.

Field	**Remarks**
View_Change.*	All fields listed in the View_Change view defined earlier in this document.
assetID	ca_owned_resource.own_resource_uuid: The binary field which serves as the internal, unchanging unique identifier for an asset record.
asset_serial_num	ca_owned_resource.serial_number: The serial number for an asset record.
asset_class	ca_resource_class.name: A short description of the class to which an asset belongs.ca_owned_resource.resource_class = ca_resource_class.id
asset_family	ca_resource_family.name: The family of assets to which this asset belongs.ca_owned_resource.resource_class = ca_resource_class.id AND ca_resource_class.family_id = ca_resource_family.id
asset_name	ca_owned_resource.resource_name: The network name by which this asset is known.

View_Change_to_Change_Act_Log

The following is a basic view of all change orders and the activity logs that go with them. This view joins the View_Change view with the Change Order Activity Log (chgalg) to give detailed information about change orders and their activity logs.

Field	Remarks
View_Change.*	This shows all fields listed in the View_Change view defined earlier in this document.
chgalg_id	chgalg.id: The unique identifier for this record in the chgalg table.
chgalg_persid	chgalg.persid: The unique identifier for this record in the chgalg table, preceded by the object identifier (chgalg for chgalg) and a colon.
change_id	chgalg.change_id: This is a pointer to change the order id to which this activity belongs.chgalg.change_id = chgalg.id
chgalg_last_mod_dt	chgalg.last_mod_dt: The last modify date/time (pdmtime).
time_spent	chgalg.time_spent: The duration of time spent on this activity, stored as the total number of seconds. For example, 80 = 1 minute, 20 seconds.
time_stamp	chgalg.time_stamp: The user modifiable date/time of activity (pdmtime).
system_time	chgalg.system_time: The date/time of record creation (pdmtime).
analyst	chgalg.analyst: The uuid pointer to contact uuid to get the analyst who performed the activity.chgalg.analyst = ca_contact.contact_uuid
chgalg_description	chgalg.description: The text description of this activity, which can be modified by the user.
action_desc	chgalg.action_desc: The text description of automated action, which cannot be modified by the user.
type	chgalg.type: The text pointer to a record in the activity type table.chgalg.type = act_type.code
internal	chgalg.internal: The integer flag (1 or 0), which indicates if this log entry is intended for all to see or just for internal use.
knowledge_session	chgalg.knowledge_session: This is an identifier for a particular session of a particular user.
knowledge_tool	chgalg.knowledge_tool: This is an indicator of the knowledge management tool used for the search, such as NLS_FAQ or EXPERT, and so on.
chgalg_analyst_id	chgalg.analyst: This is the uuid pointer to contact uuid to get the analyst who performed the activity.chgalg.analyst = ca_contact.contact_uuid

View_Change_to_Change_WF

This view is a result of the View_Change view joined with the Workflow task table (wf) to give a basic view of change orders and their workflow tasks. This may not list all change orders, particularly when there are no workflow tasks assigned.

Field	Remarks
View_Change.*	This shows all fields listed in the View_Change view defined earlier in this document.
wf_id	wf.id: The unique identifier for a record in the wf table.
wf_persid	wf.persid: This is a unique identifier for this record in the wf table, preceded by the object identifier (wf for wf) and a colon.
del	wf.del: This is a boolean indicator. It specifies whether this record is to be displayed to the user.
object_type	wf.object_type: This is the factory name, which is used to identify the type of record (for example, chg) to which this workflow task is attached.
object_id	wf.object_id: This is the unique identifier used to identify the specific record to which this workflow task is attached.wf.object_id = chg.id
task	wf.task: This is an identifier, which references the type of task this record represents.wf.task = tskty.code
wf_template	wf.wf_template: This is an identifier, which references from which template this workflow task record was created.wf.wf_template = wftpl.id
sequence	wf.sequence: This is an integer, which indicates the order this particular workflow task record should be displayed and executed by CA SDM (for example, Ascending).
wf_status	wf.status: This is an identifier, which references a tskstat record that indicates the current status of this workflow task.wf.status = tskstat.code
group_task	wf.group_task: This is a Boolean, which indicates whether this task belongs to a group.
asset	wf.asset: This is a UUID (binary) identifier, which references a record in the ca_owned_resource table.wf.asset = ca_owned_resource.own_resource_uuid
creator	wf.creator: This is a UUID (binary) identifier, which references a record in the ca_contact table. It indicates the person who created this workflow task.wf.creator = ca_contact.contact_uuid
date_created	wf.date_created: This is the Date/timestampe this workflow task was created (pdmtime).
wf_assignee	wf.assignee: This is the UUID (binary) identifier, which references a record in the ca_contact table. It indicates the person who is currently assigned to this workflow task.wf.assignee = ca_contact.contact_uuid

done_by	wf.done_by: This is the UUID (binary) identifier, which references a record in the ca_contact table. It indicates the person who completed or approved this workflow task.wf.done_by = ca_contact.contact_uuid
wf_start_date	wf_start_date: This is the timestamp when the workflow task moved into an active status (pdmtime).
wf_est_comp_date	wf.est_comp_date: This is the timestamp (pdmtime) when users believe this task will be completed.
est_duration	wf.est_duration: This is the estimated duration for this workflow task.
completion_date	wf.completion_dat: This is the timestamp (pdmtime) when this workflow task was completed.
actual_duration	wf.actual_duration: This is the actual amount of time it took to complete this workflow task.
wf_est_cost	wf.est_cost: This is the estimated cost of this workflow task.
cost	wf.cost: This is the actual cost required to complete this workflow task.
wf_description	wf.description: This is the description of the workflow task.
wf_last_mod_dt	wf.last_mod_dt: This is the timestamp (pdmtime) when this workflow task was last changed.
wf_last_mod_by	wf.last_mod_by: This is the UUID (binary) unique identifier referencing a record in the contact table, which indicates the last person to make changes to this workflow task.wf.last_mod_by = ca_contact.contact_uuid

View_Change_to_Properties

This view is a result of the View_Change view joined with the Properties table (prp) to give a basic view of change orders and their assigned properties. This may not list all change orders, particularly when there are no properties assigned.

Field	**Remarks**
View_Change.*	This shows all fields listed in the View_Change view defined earlier in this document.
prp_id	prp.id: This is an integer unique identifier for the property record.
prp_persid	prp.persid: This is a unique identifier for this record in the wf table, preceded by the object identifier (prp for prp) and a colon.
object_type	prp.object_type: This is the factory name, which is used to identify the type of record (for example, chg) to which this property record is attached.
object_id	prp.object_id: This is the unique identifier used to identify the specific record to which this property is attached.prp.object_id = chg.id
sequence	prp.sequence: This is an integer that indicates the order for which this particular property record should be displayed by CA SDM (for example, Ascending).

property	prp.property: This is an identifier, which references a record in the prptpl table. It represents the template from which this property was created.prp.property=prptpl.code
value	prp.value: This is the value entered by the user in response to the prp_description and prp.label fields.
prp_last_mod_dt	prp.last_mod_dt: This is the timestamp (pdmtime) when this property was last modified.
prp_last_mod_by	prp.last_mod_by: This is a binary identifier, which references a record in the ca_contact table. It represents the person who last modified this record.prp.last_mod_by = ca_contact.contact_uuid
required	prp.required: This is a Boolean indicating whether this property must have a prp.value before the record is saved.
sample	prp.sample: This is a text field, which displays example values to guide the user in typing the most useful value in prp.value.
prp_description	prp.description: This is a text field, which explains what kind of value should be entered in prp.value.
label	prp.label: This is a short description of what should be placed in the prp.value field.

View_Change_to_Request

The following is a basic view of change orders that have assigned requests only. This view is a result of the View_Change view joined with the request table (call_req) to give details about the change order and its associated request.

Field	**Remarks**
View_Change.*	This shows all fields listed in the View_Change view defined earlier in this document.
cr_id	call_req.id: This is the unique identifier for this record in the call_req table.
ref_num	call_req.ref_num: This is the Request reference number, which is used by analysts and customers to refer to a particular Request.
cr_summary	call_req.summary: This is a brief description of the request for quick reference.
cr_persid	call_req.persid: This is a unique identifier for this record in the call_req table, preceded by the object identifier (cr for table call_req) and a colon.
cr_description	call_req.description: This is the long description of a request, as dictated by an analyst or customer.
cr_status	call_req.status: This is a unique identifier referencing a record in the cr_stat table. It indicates the status of this request:call_req.status = cr_stat.code
cr_active_flag	call_req.active_flag: This is the Integer flag used to determine whether this request record is active (1 or 0).
time_spent_sum	call_req.time_spent_sum: This is the derived total of all act_log records' time_spent fields, stored in seconds (i.e. 80 = 1 minute 20 seconds).

cr_open_date	call_req.open_date: This is the Request creation timestamp (pdmtime).
cr_last_mod_dt	call_req.last_mod_dt: This is the last modified timestamp (pdmtime).
cr_close_date	call_req.close_date: This is the Timstamp for when the request was set to inactive (pdmtime).
cr_log_agent	call_req.log_agent: This is a binary unique identifier referencing the ca_contact table. It references the person who was the request's original creator.call_req.log_agent = ca_contact.contact_uuid
cr_group_id	call_req.group_id: This is a binary unique identifier referencing a record in the ca_contact table. It represents the group currently assigned to the request.call_req.group_id = ca_contact.contact_uuid
cr_assignee	call_req.assignee: This is a binary unique identifier referencing a record in the ca_contact table. It represents the person currently assigned to the request.call_req.assignee = ca_contact.contact_uuid
customer	call_req.customer: This is a binary unique identifier referencing a record in the ca_contact table. It represents the affected end user for this request.call_req.customer = ca_contact.contact_uuid
charge_back_id	charge_back_id: This is a text field available for use as an indicator of accounting jargon for expensing this request to the appropriate cost center.
affected_rc	call_req.affected_rc: This is a binary unique identifier referencing a row in the ca_owned_resource table. It represents the asset to which this request applies.call_req.affected_rc = ca_owned_resource.own_resource_uuid.
cr_support_lev	call_req.support_lev: This is a pointer to a service desc record, which automates some constraints under which this request must be completed.call_req.support_lev = srv_desc.code
cr_category	call_req.category: This is a unique identifier referencing a record in the prob_ctg table,. It represents the category to which this request belongs.call_req.category = prob_ctg.persid
solution	call_req.solution: This is a pointer to call solution to get solution.call_req.solution = crsol.persid
cr_impact	call_req.impact: This is an integer unique identifier referencing a row in the impact table. It indicates the scope this request is affecting.call_req.impact = impact.enum
cr_priority	call_req.priority: This is an integer unique identifier referencing a record in the pri table. It indicates how analysts will prioritize the work associated with this request.call_req.priority = pri.enum
urgency	call_req.urgency: This is an integer unique identifier referencing a row in the urgncy table. It documents the user's feeling of urgency for having this request resolved.call_req.urgency = urgncy.enum

severity	call_req.severity: This is an integer unique identifier referencing a row in the severity table. It indicates the severity of the consequences of this unresolved request.call_req.severity = sevrty.enum
extern_ref	This specifies an associated ticket.
last_act_id	This is the id of the last activity.
cr_tticket	This is a pointer to a trouble ticket to get the associated ticket.
cr_parent	call_req.parent: This is a persid pointer to another request persid, which facilitates creation of a hierarchy of change orders.call_req.parent = call_req.persid
cr_template_name	call_req.template_name: This is a text value, which indicates this request is designated for and can be chosen from a list as a template for other similar requests.cr_template.template = call_req.persid
cr_sla_violation	call_req.sla_violation: This is an integer, which counts number of times slas attached to this request have been violated.
cr_predicted_sla_viol	call_req.predicted_sla_viol: (r5.5) Neugent related technology field.
cr_created_via	call_req.created_via: This is an integer pointer to a record in the interface table. It indicates from which interface the change order originated.call_req.created_via = interface.id
cr_call_back_date	call_req.call_back_date: This is a timestamp field (pdmtime), which indicates a future date/time the affected_end_user is to be contacted.
cr_call_back_flag	call_req.call_back_flag: This is a Boolean indicator displayed as a checkbox to the user, indicating whether to notify the analyst at the call_req.call_back_date.
event_token	call_req.event_token: This is used by CA NSM for message matching.
type	call_req.type: This is a text field referencing a record in the crt table. It indicates the ITIL type of this request.call_req.type = crt.code
cr_string1	This is a user-definable string.
cr_string2	This is a user-definable string.
cr_string3	This is a user-definable string.
cr_string4	This is a user-definable string.
cr_string5	This is a user-definable string.
cr_string6	This is a user-definable string.
change	call_req.change: This is an integer unique identifier, referencing a row in the chg table. It indicates the change order that was created as a result of this request.call_req.change = chg.id.

View_Contact_Full

The following of fields is a basic view of all contacts. This view lists all fields in the ca_contact table, plus referenced fields such as short descriptions for contact type, location name, organization names, and service type for each contact. This view has already been joined with the ca_location, ca_organization, srv_desc, and ca_contact_type tables to get the actual names and symbols for some of the fields in the ca_contact table. The actual names and symbols are located at the end of this view's field list.

Field	Remark
contact_uuid	ca_contact.contact_uuid: A binary-unique identifier for each ca_contact record.
middle_name	ca_contact. middle_name: The middle name of this contact.
alias	ca_contact.alias: The alternate, often informal name for this contact.
last_name	ca_contact.last_name: This contact's surname.
first_name	ca_contact.first_name: This contact's formal first name.
pri_phone_number	ca_contact.pri_phone_number: The contact's primary phone number.
alt_phone_number	ca_contact.alt_phone_number: The contact's alternate phone number.
fax_number	ca_contact.fax_number: The contact's facsimile number.
mobile_phone	ca_contact.mobile_phone: The contact's mobile phone number.
pager_number	ca_contact.pager_number: The number for issuing a page to the contact's page.
email_address	ca_contact.email_address: The contact's email address.
location_uuid	ca_contact.location_uuid: A binary-unique identifier referencing a record in the ca_location table, which indicates the contact's static location.ca_contact.location_uuid = ca_location.location_uuid
floor_location	ca_contact.floor_location: A contact's floor number.
pager_email_address	ca_contact.pager_email_address: A contact's email address for their pager.
room_location	ca_contact.room_location: The contact's specific room on the floor at the static location.
contact_type	ca_contact.contact_type: An unique integer-identifier, referring to a row in the ca_contact_type table, which indicates the general function of this contact within the service desk application.ca_contact.contact_type = ca_contact_type.id
inactive	ca_contact.inactive: A boolean indicator of the state of this record, determining its inclusion or exclusion from standard searches within service desk.
creation_user	ca_contact.creation_user: The userid of the contact who created this record.ca_contact.creation_user = ca_contact.userid
creation_date	ca_contact.creation_date: A timestamp (pdmtime), indicating the date and time this contact was created.

last_update_user	ca_contact.last_update_user: The userid of the contact who last updated this contact record.ca_contact.last_update_user = ca_contact.userid
last_update_date	ca_contact.last_update_date: A timestamp (pdmtime), indicating the date and time this record was last modified.
version_number	ca_contact.version number: The internal version indicator.
department	ca_contact.department: An integer-unique identifier, referencing a row in the ca_resource_department table, that indicates the contact's department.ca_contact.department = ca_resource_department.id
comment	ca_contact.comment: A freeform text comment field for analysts to document important facts that influence the handling of this particular contact.
company_uuid	ca_contact.company_uuid: A binary-unique identifier, which references a row in the ca_company table. It indicates this contact's affiliation with a company.ca_contact.company_uuid = ca_company.company_uuid
organization_uuid	ca_contact.organizaiton_uuid: A binary-unique identifier which references a row in the ca_organization table. It indicates this contact's working organization.ca_contact.organizaiton_uuid = ca_organization.organization_uuid
admin_organization_uuid	ca_contact.admin_organization_uuid: A binary-unique identifier, which references a row in the ca_organization table. It indicates this contact's administrative organization.ca_contact.admin_organization_uuid = ca_organization.organization_uuid
alternate_identifier	ca_contact.alternate_identifier: A user-defined identifier, usually an entity used by human resources to uniquely identify this contact.
job_title	ca_contact.job_title: An integer-unique identifier, which references the ca_job_title table. It indicates the standardized job title for this contact.ca_contact.job_title = ca_job_title.id
job_function	ca_contact.job_function: An integer-unique identifier, which references the ca_job_function table. It indicates a standardized general description of the contact's job function.ca_contact.job_function = ca_job_function.id
mail_stop	ca_contact.mail_stop: Mail Stop.
cost_center	ca_contact.cost_center: An integer-unique identifier, which references the ca_resource_cost_center table. It indicates this contact's primary cost center.ca_contact.cost_center = ca_cost_center.id
userid	ca_contact.userid: The user identifier this contact will use for logging into service desk.
supervisor_contact_uuid	ca_contact.supervisor_contact_uuid: A binary-unique identifier, referencing a row in the ca_contact table, that creates a hierarchy of contacts to indicate the reporting structure of each contact.ca_contact.supervisor_contact_uuid = ca_contact.contact_uuid

exclude_registration	ca_contact.exclude registration: An internal flag.
delete_time	ca_contact.delete_time: A timestamp indicating when the inactive flag was set to 1.
contact_type_name	ca_contact_type.name: A short description of this contact's ca_contact.contact_type.
location_name	ca_location.name: A short description of this contact's static location.
organization	ca_organization.org_name: A short description of this contact's ca_contact.organizaiton_uuid.
admin_organization	ca_organization.org_name: A short description of this contact's organization name.ca_contact.admin_organization_uuid
service_type	srv_desc.sym: A short description of this contact's usp_contact.c_service_type.ca_contact.contact_uuid = usp_contact.contact_uuid AND usp_contact.c_service_type = srv_desc.code
state_sym	ca_location.state: An integer-unique identifier, referencing a row in the ca_state_provice table, that indicates the State, Province, or other artificially defined geographic region.ca_contact.location_uuid = ca_location.location_uuid AND ca_location.state = ca_state_province.id

View_Contact_to_Environment

The following list of fields is a basic view of contacts and their environment (assets). This view is a view of the contact table (ca_contact), but is also joined with the owned resource table (ca_owned_resource) to get a list of all assets associated with a contact. This view may be joined with the View_Contact_Full view to get the contact type, service type, organizations and location of each contact. Or, this view may be joined with the individual tables to get that same information.

Field	Remarks
ca_contact.*	The ca_contact fields that are defined in the View_Contact_Full view definition.
asset_uuid	ca_owned_resource.own_resource_uuid: A binary-unique identifier for an asset in the ca_owned_resource table.
asset_name	ca_owned_resource.resource_name: The designated network name of this asset.

View_Group

The following list of fields is a basic view of the contact table, but lists only group contacts. Contact type id = 2308, which is for group types. You may want to join this view with other CA SDM tables to get more meaningful data for reporting. For example, you can join it with the Location (ca_location) table to find the name and address for the group’s location. You

can also join this view with the Organization (ca_organization) table to get the functional and administrative organization names for the group.

Field	Remarks
ca_contact.*	The ca_contact fields that are defined in the View_Contact_Full view definition.

The following example shows how joining tables works, and how reporting fields are extracted. The field from one table (on the right) is joined (->) with a field from another table (on the left). To join properly between tables and views, you need to understand the differences in joins for the database. The field defined in the parentheses, as follows, is what you may want to use in your reports if the previous tables are joined with the View_Group view:

- View_Group.contact_type -> ca_contact_type.id (ca_contact_type.sym)
- View_Group.location_uuid -> ca_location.location_uuid (ca_location.location_name)
- View_Group.organization_uuid -> ca_organization.organization_uuid (ca_organization.org_name)
- View_Group.admin_organization_uuid -> ca_organization(2).organization_uuid (ca_organization(2). org_name)

View_Group_to_Contact

The following list of fields is a basic view of all group contacts (members). It also includes managers. Here, View_Group is joined with the Group_Member table, and then joined with the ca_contact table. All fields in View_Group are listed, as well as the first, middle and last name from the ca_contact table. The Group_Member manager flag is also listed. The group member manager flag is 1 or 0, which means that the member is either (yes - 1) a manager, or (no - 0) not a manager. Most of the information in this view pertains to the group itself, not the actual members. This view is used to find information on a particular group, including the names of its members.

Field	Remarks
View_Group.*	The View_Group fields that are defined in the View_Group definition.
member_lastname	ca_contact.last_name: The surname of the group member.
member_firstname	ca_contact.first_name: The formal first name of the group member.
member_middlename	ca_contact.middle_name: The middle name of the group member.
grpmem_manager_flag	ca_contact.manager_flag: The Group Member Manager (1 or 0) indicator.

View_Issue

The following is a basic view of all issues, listing the status, priority, category, organizations, the requester's full name, the assignee's full name, the group name and ID, and so on. Here, the issue table is joined with many other tables to give some more meaningful data about the issue.

Field	Remarks
id	issue.id: The unique identifier for this record in the issue table.
persid	issue.persid: The unique identifier for this record in the issue table, preceded by the object identifier (iss for table issue) and a colon.
issue_ref_num	issue.iss_ref_num: The Issue reference number, which is used by analysts and customers to refer to a particular issue.

description	issue.description: The long description of an issue as dictated by an analyst or customer.
status	issue.status: The unique identifier of an issue status, which is a pointer to the issstat table.issue.status = issstat.code
active_flag	issue.active_flag: The integer flag used to determine whether this issue is active (1 or 0).
start_date	issue.start_date: The date the first task goes to a pending status (pdmtime).
open_date	issue.open_date: The issue creation date (pdmtime).
last_mod_dt	issue.last_mod_dt: The last modified date (pdmtime).
last_mod_by	issue.last_mod_by: The pointer to the contact uuid, which indicates the last contact to modify this issue.issue.last_mod_by = ca_contact.contact_uuid
close_date	issue.close_date: The date the issue was set to inactive (pdmtime).
resolve_date	issue.resolve_date: The date the issue was set to a status configured to indicate the issue was resolved (pdmtime).
rootcause	issue.rootcause: A pointer to a record in the rootcause table, which represents the original situation that required this issue to be logged.issue.rootcause = rootcause.id
est_total_time	issue.est_total_time: The estimated total time (pdmtime) it will take to complete this issue.
actual_total_time	issue.actual_total_time actual: The total time (pdmtime) it took to complete this issue.
log_agent	issue.log_agent: A binary-unique identifier, this references the ca_contact table, which in turn references the person who was the issue's original creator.issue.log_agent = ca_contact.contact_uuid
assignee	issue.assignee: A pointer to the contact uuid who is currently assigned to the change order.issue.assignee = ca_contact.contact_uuid
organization	issue.organization: A pointer to the internal organization uuid, which represents the organization to whom this issue belongs.issue.organization = ca_organization.organization_uuid
group_id	issue.group_id: A pointer to the contact uuid, which represents the group currently assigned to the issue.issue.group_id = ca_contact.contact_uuid
affected_contact	issue.affected_contact: A pointer to the contact uuid, which represents the affected contact for this issue.issue.affected_contact = ca_contact.contact_uuid
requestor	issue.requrestor: A pointer to the contact uuid, which represents the person who asked this issue to be logged.issue.requestor = ca_contact.contact_uuid
category	issue.category: A pointer to the issue category code to reference the category into which this issue falls.issue.category = isscat.code

priority	issue.priority: A pointer to priority enum, which represents the priority into which this issue falls.issue.priority = pri.enum
need_by	issue.need_by: The date, which indicates when the affected_end_user needs to have the issue completed (pdmtime).
est_comp_date	issue.est_comp_date: The estimated completion date (pdmtime) for this Issue.
actual_comp_date	issue.actual_comp_date: The actual completion date (pdmtime) of this issue.
est_cost	issue.est_cost: The estimated cost of this issue.
actual_cost	issue.actual_cost: The actual cost to implement this issue.
justification	issue.justification: A text field, which allows a requester to document the reason(s) this issue is required.
backout_plan	issue.backout_plan: A text field that allows an analyst to document a backout plan for this issue.
impact	issue.impact: A pointer to an impact table record, which indicates the scope of resources that this issue affects.issue.impact = impact.enum
parent	issue.parent: A pointer to another issue id, which allows creation of a hierarchy of issues.issue.parent = issue.id
effort	issue.effort: A text field which explains the plan for implementing this issue.
support_lev	issue.support_lev: A pointer to a service desc record, which automates some constraints under which this issue must be completed.issue.support_lev = srv_desc.code
template_name	issue.template_name: The name of and pointer to an issue template.issue.template_name = iss_template.template_name
sla_violation	issue.sla_violation: The integer to count the number of times slas attached to this issue have been violated.
predicted_sla_viol	issue.predicted_sla_viol: (r5.5) Neugent related technology field.
macro_predict_viol	issue.macro_predict_viol: (r5.5) Neugent related technology field.
created_via	issue.created_via: A pointer to a record in the interface table. This indicates from which interface the issue originated.issue.created_via = interface.id
call_back_date	issue.call_back_date: A date/time field (pdmtime), which indicates a future date/time the requester is to be contacted.
call_back_flag	issue.call_back_flag: A boolean indicator displayed as a checkbox to the user, to indicate whether to notify the analyst at the issue.call_back_date.
string1	This is a user-definable text field.
string2	This is a user-definable text field.
string3	This is a user-definable text field.
string4	This is a user-definable text field.
string5	This is a user-definable text field.

string6	This is a user-definable text field.
service_date	issue.service_date: The Date/ Time (pdmtime) that an outside vendor is expected to service this issue.
service_num	issue.service_num: The text field to document an outside vendor's service or purchase order number.
product	issue.product: A pointer to a record in the product table, which indicates the product that is affected by this issue.issue.product = product.id
actions	issue.actions: A big text field for documenting actions.
type_of_contact	issue.type_of_contact: A pointer to a record in the toc table, which indicates a general categorization of the affected_end_user's perspective of the issue.issue.type_of_contact = toc.id
reporting_method	issue.reporting_method: A pointer to a record in the repmeth table, which classifies the origination of the issue, and is selected by the person creating the issue.issue.reporting_method = repmeth.id
person_contacting	issue.person_contacting: A pointer to a record in the perscon table, which indicates the role of the affected_end_user or requester.issue.person_contacting = perscon.id
status_name	issstat.sym: The description of the status as seen by a user.issue.status = issstat.code
priority_num	pri.sym: The description of the priority as seen by a user.issue.priority = pri.enum
category_name	isscat.sym: The name of the issue category as viewed by a user.issue.category = isscat.code
organization_name	ca_organization.org_name: The name of an organization as viewed by a user.issue.organization = ca_organization.organization_uuid
affected_end_user_lastname	ca_contact.last_name: The affected End User's last name.issue.affected_end_user = ca_contact.contact_uuid
affected_end_user_firstname	ca_contact.first_name: The affected End User's first name.issue.affected_end_user = ca_contact.contact_uuid
affected_end_user_middlename	ca_contact.middle_name: The affected End User's middle name.issue.affected_end_user = ca_contact.contact_uuid
assignee_lastname	ca_contact.last_name: The assignee's last name.issue.assignee = ca_contact.contact_uuid
assignee_firstname	ca_contact.first_name: The assignee's first name.issue.assignee = ca_contact.contact_uuid
assignee_middlename	ca_contact.middle_name: The assignee's middle name.issue.assignee = ca_contact.contact_uuid
groupID	View_Group.contact_uuid: A binary representation of the internal id used for the group assigned to this issue.issue.group_id = ca_contact.contact_uuid
group_name	View_Group.last_name: The name of the group assigned to this issue.issue.group = ca_contact.contact_uuid

service_type	srv_desc.sym: The name of the service type applied to this issue.issue.support_lev = srv_desc.code
impact_num	impact.sym: The description of the impact as seen by users.issue.impact = impact.enum
product_sym	product.sym: The product description as seen by users.issue.product = product.id
type_of_contact_sym	toc.sym: The Type Of Contact description as seen by users.issue.type_of_contact = toc.id
rpting_method_sym	repmeth.sym: The Reporting method description as seen by users.issue.reporting_method = repmeth.id
person_contacting_sym	perscon.sym: The Person Contacting description as seen by users.issue.person_contacting = perscon.id
created_via_sym	interface.sym: issue.created_via = interface.id.
rootcause_sym	rootcause.sym: issue.rootcause = rootcause.id.

View_Issue_Act_Log

The following is a basic view of all issue activity logs. This is a view of the issue activity log table (issalg) joined with the activity type table (act_type) and the contact table (ca_contact) to give more meaningful data, such as the actual activity type and full name of the analyst who performed the activity.

Field	**Remarks**
id	issalg.id: The unique identifier for this record in the issalg table.
persid	issalg.persid: The unique identifier for this record in the issalg table, preceded by the object identifier (issalg for issalg) and a colon.
issue_id	issalg.issue_id: The pointer to issue id to which this activity belongs.issalg.issue_id = issalg.id
last_mod_dt	issalg.last_mod_dt: The last modify date/time (pdmtime).
time_spent	issalg.time_spent: The duration of time spent on this activity, stored as the total number of seconds. For example, 80 = 1 minute, 20 seconds.
time_stamp	issalg.time_stamp: The user modifiable date/time of activity (pdmtime).
system_time	issalg.system_time: The date/time of record creation (pdmtime).
analyst	issalg.analyst: A binary-unique identifier referring to the contact uuid to get the analyst who performed the activity.issalg.analyst = ca_contact.contact_uuid
description	issalg.description: The text description of this activity, which can be modified by the user.
action_desc	issalg.action_desc: The text description of the automated action, which cannot be modified by the user.
type	issalg.type: The text pointer to a record in the activity type table.issalg.type = act_type.code

internal	issalg.internal: The integer flag (1 or 0), which indicates if this log entry is intended for all to see or just for internal use.
knowledge_session	issalg.knowledge_session: An identifier for a particular session of a particular user.
knowledge_tool	issalg.knowledge_tool: An indicator of the knowledge management tool used for the search, such as NLS_FAQ or EXPERT, and so on.
analyst_lastname	View_Contact_Full.last_name: The Analyst's last name, derived from issalg.analyst = ca_contact.contact_uuid.
analyst_firstname	View_Contact_Full.first_name: The analyst's first name.
analyst_middlename	View_Contact_Full.middle_name: The middle name of the analyst.
activity_type	act_type.sym: The activity type referenced by issalg.type = act_type.code.

View_Issue_to_Assets

The following list of fields is a basic view of issues and their assets. The issue table (issue) is indirectly joined with the owned resource table (ca_owned_resource), and other asset-related tables, to get a list of each issue's assets. This may not list all issues, particularly those that have no assets.

Field	Remarks
View_Issue.*	The View_Issue view which defines all fields listed in the View_Issue view.
assetID	ca_owned_resource.own_resource_uuid: The binary field which serves as the internal, unchanging unique identifier for an asset record.
asset_serial_num	ca_owned_resource.serial_number: The serial number for an asset record.
asset_class	ca_resource_class.name: A short description of the class to which an asset belongs. ca_owned_resource.resource_class = ca_resource_class.id
asset_family	ca_resource_family.name: The family of assets to which this asset belongs. ca_owned_resource.resource_class = ca_resource_class.id AND ca_resource_class.family_id = ca_resource_family.id
asset_name	ca_owned_resource.resource_name: The network name by which this asset is known.

View_Issue_to_Issue_Act_Log

The following is a basic view of all issues and the activity logs that go with them. This view joins the View_Issue view with the View_Issue_Act_Log view to give detailed information about issues and their activity logs. Actual data is at the end of the fields list.

Field	Remarks
View_Issue.*	Please refer to the View_Issue view defined earlier in this document.

issalg_id	issalg.id: The unique identifier for this record in the issalg table.
issalg_persid	issalg.persid: The unique identifier for this record in the issalg table, preceded by the object identifier (issalg for issalg) and a colon.
issue_id	issalg.issue_id: The pointer to issue id to which this activity belongs.issalg.issue_id = issalg.id
issalg_last_mod_dt	issalg.last_mod_dt: The last modify date/time (pdmtime).
time_spent	issalg.time_spent: The duration of time spent on this activity, stored as the total number of seconds. For example, 80 = 1 minute, 20 seconds.
time_stamp	issalg.time_stamp: The user modifiable date/time of activity (pdmtime).
system_time	issalg.system_time: The date/time of record creation (pdmtime).
analyst	issalg.analyst: The unique binary pointer to the contact uuid to get the analyst who performed the activity.issalg.analyst = ca_contact.contact_uuid
issalg_description	issalg.description: The text description of this activity, which can be modified by the user.
action_desc	issalg.action_desc: The text description of the automated action, which cannot be modified by the user.
type	issalg.type: The text pointer to a record in the activity type table.issalg.type = act_type.code
internal	issalg.internal: The integer flag (1 or 0), which indicates if this log entry is intended for all to see or just for internal use.
knowledge_session	issalg.knowledge_session: An identifier for a particular session of a particular user.
knowledge_tool	issalg.knowledge_tool: An indicator of the knowledge management tool used for the search, such as NLS_FAQ or EXPERT, and so on.
issalg_analyst_id	issalg.analyst: The unique binary pointer to the contact uuid to get the analyst who performed the activity.issalg.analyst = ca_contact.contact_uuid

View_Issue_to_Issue_WF

This view is a result of the View_Issue view joined with the workflow task table (isswf) to give a basic view of issues and their workflow tasks. This may not list all issues, particularly if there are no workflow tasks assigned to them.

Field	**Remarks**
View_Issue.*	Please refer to the View_Issue definition earlier in this document for a description of each field.
wf_id	isswf.id: This is a unique identifier for a record in the isswf table.
wf_persid	isswf.persid: This is a unique identifier for this record in the isswf table, preceded by the object identifier (isswf) and a colon.

del	isswf.del: This is a boolean indicator of whether this record is to be displayed to the user.
object_type	isswf.object_type: This is the factory name, which is used to identify the type of record (for example, iss) to which this workflow task is attached.
object_id	isswf.object_id: This is a unique identifier used to identify the specific record to which this workflow task is attached.isswf.object_id = issue.id
task	isswf.task: This is an identifier, which references the type of task this record represents.isswf.task = tskty.code
wf_template	isswf.wf_template: This is an identifier, which references from which template this workflow task record was created.isswf.wf_template = wftpl.id
sequence	isswf.sequence: This is an integer that indicates the order for which this particular workflow task record should be displayed and executed by CA SDM (for example, Ascending).
wf_status	isswf.status: This is an identifier, which references a tskstat record. It indicates the current status of this workflow task.isswf.status = tskstat.code
group_task	isswf.group_task: This is a boolean, which indicates whether this task belongs to a group.
asset	isswf.asset: This is a unique binary identifier, which references a record in the ca_owned_resource table.isswf.asset = ca_owned_resource.own_resource_uuid
creator	isswf.creator: This is a unique binary identifier, which references a record in the ca_contact table. It indicates the person who created this workflow task.isswf.creator = ca_contact.contact_uuid
date_created	isswf.date_created: This is the date/timestamp that this workflow task was created (pdmtime).
wf_assignee	isswf.assignee: This is a unique binary identifier, which references a record in the ca_contact table. It indicates the person who is currently assigned to this workflow task.isswf.assignee = ca_contact.contact_uuid
done_by	isswf.done_by: This is a unique binary identifier, which references a record in the ca_contact table. It indicates the person who completed or approved this workflow task.isswf.done_by = ca_contact.contact_uuid
wf_start_date	wf_start_date: This is the timestamp when the workflow task moved into an active status (pdmtime).
wf_est_comp_date	isswf.est_comp_date: This is a timestamp (pdmtime) indicating when users believe this task will be completed.
est_duration	isswf.est_duration: This is the estimated duration for this workflow task.
completion_date	isswf.completion_date: This is a timestamp (pdmtime) indicating when this workflow task was completed.

actual_duration	isswf.actual_duration: This is the actual amount of time it took to complete this workflow task.
wf_est_cost	isswf.est_cost: This is the estimated cost of this workflow task
cost	isswf.cost: This is the actual cost required to complete this workflow task.
wf_description	isswf.description: This is a description of the workflow task.
wf_last_mod_dt	isswf.last_mod_dt: This is the timestamp (pdmtime) indicating when this workflow task was last changed.
wf_last_mod_by	isswf.last_mod_by: This is the unique binary identifier, which references a record in the contact table. It indicates the last person to make changes to this workflow task.isswf.last_mod_by = ca_contact.contact_uuid

View_Issue_to_Properties

This view is a result of the View_Issue view joined with the issue properties table (issprp) to give a basic view of issues and their assigned properties. This may not list all issues, particularly if there are no properties assigned to them.

Field	**Remarks**
View_Issue.*	Please refer to the View_Issue definition earlier in this document, for a description of each field.
prp_id	issprp.id: This is an integer-unique identifier for the property record.
prp_persid	issprp.persid: This is a unique identifier for this record in the prp table, preceded by the object identifier (prp) and a colon.
sequence	issprp.sequence: This is an integer that indicates the order for which this particular property record should be displayed by CA SDM (for example, Ascending).
label	issprp.label: This is a short description of what should be placed in the issprp.value field.
value	issprp.value: This is a value entered by the user in response to the prp_description and issprp.label fields.
prp_last_mod_dt	issprp.last_mod_dt: This is the timestamp (pdmtime) indicating when this property was last modified.
prp_last_mod_by	issprp.last_mod_by: This is a binary identifier, which references a record in the ca_contact table. It represents the person who last modified this record.issprp.last_mod_by = ca_contact.contact_uuid
required	issprp.required: This is a boolean, indicating whether this property must have a issprp.value before the record is saved.
sample	issprp.sample: This is a text field, which displays example values to guide the user in entering the most useful value in issprp.value.
owning_iss	issprp.owning_iss: This is a unique identifier used to identify the specific record to which this property is attached.issprp.object_id = issue.persid

prp_description	issprp.description: This is a text field, which explains the type of value that should be entered in issprp.value.

View_Request

The following is a basic view of all requests. Here, the Request table has been joined with other CA SDM tables to give you more specific information, such as the request service type, severity, urgency, category, and priority. There is some additional information about the request listed, as well. All fields from the Request (call_req) table are selected. The extracted fields that are a result of the joined tables are listed at the end of this fields list.

Field	Remarks
id	call_req.id: This is the unique identifier for this record in the call_req table.
persid	call_req.persid: This is a unique identifier for this record in the call_req table, preceded by the object identifier (cr for table call_req) and a colon.
ref_num	call_req.ref_num: This is a Request reference number, which is used by analysts and customers to refer to a particular Request.
summary	call_req.summary: This is a brief description of the request for quick reference.
description	call_req.description: This is the long description of a request, as dictated by an analyst or customer.
status	call_req.status: This is a unique identifier referencing a record in the cr_stat table. It indicates the status of this request. call_req.status = cr_stat.code
active_flag	call_req.active_flag: This is an integer flag to determine whether this request record is active (1 or 0).
open_date	call_req.open_date: This is the Request creation timestamp (pdmtime).
time_spent_sum	call_req.time_spent_sum: This is the derived total of all of the act_log records' time_spent fields, stored in seconds (i.e. 80 = 1 minute 20 seconds).
last_mod_dt	call_req.last_mod_dt: This is the last modified timestamp (pdmtime).
close_date	call_req.close_date: This is the timstamp when the request was set to inactive (pdmtime).
resolve_date	This is the date for when the request was resolved (pdmtime).
rootcause	This is a pointer to the rootcause.id.
log_agent	call_req.log_agent: This is a binary-unique identifier, referencing the ca_contact table. It references the person who was the request's original creator.call_req.log_agent = ca_contact.contact_uuid
assignee	call_req.assignee: This is a binary-unique identifier, referencing a record in the ca_contact table. It represents the person currently assigned to the request.call_req.assignee = ca_contact.contact_uuid

group_id	call_req.group_id: This is a binary-unique identifier, referencing a record in the ca_contact table. It represents the group currently assigned to the request.call_req.group_id = ca_contact.contact_uuid
customer	call_req.customer: This is a binary-unique identifier, referencing a record in the ca_contact table. It represents the affected end user for this request.call_req.customer = ca_contact.contact_uuid
charge_back_id	charge_back_id: This is a text field available for use as an indicator of accounting jargon for expensing this request to the appropriate cost center.
affected_rc	call_req.affected_rc: This is a binary-unique identifier, referencing a row in the ca_owned_resource table. It represents the asset to which this request applies.call_req.affected_rc = ca_owned_resource.own_resource_uuid.
support_lev	call_req.support_lev: This is a pointer to a service desc record, which automates some constraints under which this request must be completed.call_req.support_lev = srv_desc.code.
category	call_req.category: This is a unique identifier, referencing a record in the prob_ctg table. It represents the category to which this request belongs.call_req.category = prob_ctg.persid
solution	call_req.solution: This is a pointer to call solution to get solution.call_req.solution = crsol.persid
impact	call_req.impact: This is an integer-unique identifier, referencing a row in the impact table. It indicates the impact scope of the request.call_req.impact = impact.enum
priority	call_req.priority: This is an integer-unique identifier, referencing a record in the pri table. It indicates how analysts will prioritize the work associated with this request.call_req.priority = pri.enum
urgency	call_req.urgency: This is an integer-unique identifier, referencing a row in the urgency table. It indicates the user's feeling of urgency for having this request resolved.call_req.urgency = urgncy.enum
severity	call_req.severity: This is an integer-unique identifier, referencing a row in the severity table. It indicates the severity of the consequences of this unresolved request.call_req.severity = sevrty.enum
extern_ref	This is an external reference to an associated ticket.
last_act_id	This identifies the id of the last activity.
cr_tticket	This is a pointer to the trouble ticket to get the associated ticket.
parent	call_req.parent: This is a ersid pointer to another request persid, which facilitates creation of a hierarchy of change orders.call_req.parent = call_req.persid
template_name	call_req.template_name: This is a text value, which indicates this request is designated for and can be chosen from a list as a template for other similar requests.cr_template.template = call_req.persid

sla_violation	call_req.sla_violation: This is an integer, which counts the number of times that slas attached to this request have been violated.
predicted_sla_viol	This specifies that a request has been predicted by neugents to likely violate SLA.
macro_predicted_violation	This indicates that the request has been predicted by neugents to likely violate SLA.
created_via	call_req.created_via: This is an integer pointer to a record in the interface table that indicates from which interface the change order originated.call_req.created_via = interface.id
call_back_date	call_req.call_back_date: This is a timestamp field (pdmtime), which indicates a future date/time that the affected_end_user is to be contacted.
call_back_flag	call_req.call_back_flag: This is a boolean indicator, displayed as a checkbox to the user, indicating whether to notify the analyst at the call_req.call_back_date.
event_token	call_req.event_token: This is used by CA NSM for message matching.
sched_token	call_req.sched_token: This is used by CA NSM for message matching.
type	call_req.type: This is a text field referencing a record in the crt table. It indicates the ITIL type for this request.call_req.type = crt.code
string1	This is a user-definable string.
string2	This is a user-definable string.
string3	This is a user-definable string.
string4	This is a user-definable string.
string5	This is a user-definable string.
string6	This is a user-definable string.
problem	This is an ITIL problem.
incident_priority	This is an ITIL incident priority.
change	call_req.change: This is an integer-unique identifier, referencing a row in the chg table. It indicates the change order that was created as a result of this request.call_req.change = chg.id.
service_type	srv_desc.sym: This indicates the actual Service Type.call_req.support_lev = srv_desc.code
severity_num	sevrty.sym: This is the actual Severity number.call_req.severity = sevrty.enum
urgency_num	urgncy.sym: This indicates the actual Urgency number.call_req.urgency = urgncy.enum
category_name	prob_ctg.sym: This is the actual Request Area (problem category).call_req.category = prob_ctg.id
asset	ca_owned_resource.resource_name: This is the actual Asset name.call_req.affected_rc = ca_owned_resource.own_resource_uuid

impact_num	impact.sym: This is the actual Impact number.call_req.impact = impact.enum
assignee_lastname	ca_contact.last_name: This is the actual Assignee last name.call_req.assignee = ca_contact.contact_uuid
assignee_firstname	ca_contact.first_name: This is the actual Assignee first name.call_req.assignee = ca_contact.contact_uuid
assignee_middlename	ca_contact.middle_name: This is the actual Assignee middle name.call_req.assignee = ca_contact.contact_uuid
customer_lastname	ca_contact.last_name: This is the actual last name of the Affected End User.call_req.customer = ca_contact.contact_uuid
customer_firstname	ca_contact.first_name: This is the actual first name of the Affected End User.call_req.customer.ca_contact.contact_uuid
customer_middlename	ca_contact.middle_name: This is the actual middle name of the Affected End User.call_req.customer = ca_contact.contact_uuid
group_name	View_Group.last_name: This is the actual Group name.
GroupID	View_Group.contact_uuid: This is the actual Group key ID.
status_name	cr_stat.sym: This is the actual status.
priority_num	pri.sym: This is the actual Priority number.

View_Request_to_Act_Log

The following is a basic view of all requests with their activity logs. The View_Request view is joined with the View_Act_Log view to give more detailed information about each activity per request.

Field	Remarks
View_Request.*	Please refer to the field defined in the View_Request section earlier in this document.
View_Act_Log.*	Please refer to the fields defined in the View_Act_Log section earlier in this document.

View_Request_to_Properties

The following is a basic view of call requests and their properties. This view lists everything from the call request (call_req) table and the request property (cr_prp) table.

Field	Remarks
View_Request.*	Please refer to the fields defined in the View_Request section of this document.
crprp_id	cr_prp.id: This is an integer-unique identifier for the property record.
crprp_persid	cr_prp.persid: This is a unique identifier for this record in the cr_prp table, preceded by the object identifier (cr_prp) and a colon.
sequence	cr_prp.sequence: This is an integer that indicates the order for which this particular property record should be displayed by CA SDM (for example, Ascending).

label	cr_prp.label: This is a short description of what should be placed in the cr_prp.value field.
value	cr_prp.value: This is a value entered by the user in response to the prp_description and cr_prp.label fields.
crprp_last_mod_dt	cr_prp.last_mod_dt: This is the timestamp (pdmtime) identifying when this property was last modified.
crprp_last_mod_by	cr_prp.last_mod_by: This is a binary identifier, which references a record in the ca_contact table. It represents the person who last modified this record.cr_prp.last_mod_by = ca_contact.contact_uuid
required	cr_prp.required: This is the boolean, indicating whether this property must have a cr_prp.value before the record is saved.
sample	cr_prp.sample: This is a text field, which displays example values to guide the user in entering the most useful value in cr_prp.value.
owning_cr	cr_prp.owning_cr: This is the unique identifier used to identify the specific record to which this property is attached.cr_prp.object_id = call_req.persid
crprp_description	cr_prp.description: This is a text field, which explains the type of value that should be entered in cr_prp.value.

View_Request_to_Request_WF

NOTE

This feature will only be available if you apply a patch for CA Service Management Release 14.1.01. Find the patch and the download details from CA Support Online.

This view is a result of the View_Request view joined with the workflow task table (crwf) to give a basic view of requests and their workflow tasks. This may not list all requests, particularly if there are no workflow tasks assigned to them.

Field	Remarks
View_Request.*	Please refer to the View_Request definition earlier in this document for a description of each field.
wf_id	crwf.id: This is a unique identifier for a record in the crwf table.
wf_persid	crwf.persid: This is a unique identifier for this record in the crwf table, preceded by the object identifier (crwf) and a colon.
del	crwf.del: This is a boolean indicator of whether this record is to be displayed to the user.
object_type	crwf.object_type: This is the factory name, which is used to identify the type of record (for example, cr) to which this workflow task is attached.
object_id	crwf.object_id: This is a unique identifier used to identify the specific record to which this workflow task is attached.cr.object_id = issue.id
task	crwf.task: This is an identifier, which references the type of task this record represents.crwf.task = tskty.code

wf_template	crwf.wf_template: This is an identifier, which references from which template this workflow task record was created.crwf.wf_template = wftpl.id
sequence	crwf.sequence: This is an integer that indicates the order for which this particular workflow task record should be displayed and executed by CA SDM (for example, Ascending).
wf_status	crwf.status: This is an identifier, which references a tskstat record. It indicates the current status of this workflow task.crwf.status = tskstat.code
group_task	crwf.group_task: This is a boolean, which indicates whether this task belongs to a group.
creator	crwf.creator: This is a unique binary identifier, which references a record in the ca_contact table. It indicates the person who created this workflow task.crwf.creator = ca_contact.contact_uuid
date_created	crwf.date_created: This is the date/timestamp that this workflow task was created (pdmtime).
wf_assignee	crwf.assignee: This is a unique binary identifier, which references a record in the ca_contact table. It indicates the person who is currently assigned to this workflow task.crwf.assignee = ca_contact.contact_uuid
done_by	isswf.done_by: This is a unique binary identifier, which references a record in the ca_contact table. It indicates the person who completed or approved this workflow task.isswf.done_by = ca_contact.contact_uuid
wf_start_date	wf_start_date: This is the timestamp when the workflow task moved into an active status (pdmtime).
wf_est_comp_date	crwf.est_comp_date: This is a timestamp (pdmtime) indicating when users believe this task will be completed.
est_duration	crwf.est_duration: This is the estimated duration for this workflow task.
completion_date	crwf.completion_date: This is a timestamp (pdmtime) indicating when this workflow task was completed.
actual_duration	crwf.actual_duration: This is the actual amount of time it took to complete this workflow task.
wf_est_cost	crwf.est_cost: This is the estimated cost of this workflow task
cost	crwf.cost: This is the actual cost required to complete this workflow task.
wf_description	crwf.description: This is a description of the workflow task.
wf_last_mod_dt	crwf.last_mod_dt: This is the timestamp (pdmtime) indicating when this workflow task was last changed.
wf_last_mod_by	crwf.last_mod_by: This is the unique binary identifier, which references a record in the contact table. It indicates the last person to make changes to this workflow task.crwf.last_mod_by = ca_contact.contact_uuid

View_Request_to_Request_WF

RFC 2251 LDAP Result Codes

This article contains the following topics:

LDAP Return Codes

LDAP has a set of operation result codes that may be generated by the LDAP server in response to various LDAP requests. These codes indicate the status of the protocol operation and are categorized by server or client return code categories.

LDAP Server Return Codes

The following table lists the server return codes:

Hex	Decimal	Description
0x00	0	LDAP_SUCCESS Indicates the requested client operation completed successfully.
0x01	1	LDAP_OPERATIONS_ERROR Indicates an internal error occurred. The server is unable to respond with a more specific error and is also unable to properly respond to a request. It does not indicate that the client has sent an erroneous message.
0x02	2	LDAP_PROTOCOL_ERROR Indicates that the server has received an invalid or malformed request from the client.
0x03	3	LDAP_TIMELIMIT_EXCEEDED Indicates that the operation's time limit specified by either the client or the server has been exceeded. On search operations, incomplete results are returned.
0x04	4	LDAP_SIZELIMIT_EXCEEDED Indicates that in a search operation, the size limit specified by the client or the server has been exceeded. Incomplete results are returned.
0x05	5	LDAP_COMPARE_FALSE Does not indicate an error condition. Indicates that the results of a compare operation are false.
0x06	6	LDAP_COMPARE_TRUE Does not indicate an error condition. Indicates that the results of a compare operation are true.

0x07	7	LDAP_AUTH_METHOD_NOT_SUPPORTED Indicates that during a bind operation the client requested an authentication method not supported by the LDAP server.
0x08	8	LDAP_STRONG_AUTH_REQUIRED Indicates one of the following: In bind requests, the LDAP server accepts only strong authentication. In a client request, the client requested an operation, such as delete, that requires strong authentication. In an unsolicited notice of disconnection, the LDAP server discovers the security protecting the communication between the client and server has unexpectedly failed or been compromised.
0x09	9	Reserved.
0x0A	10	LDAP_REFERRAL Does not indicate an error condition. In LDAPv3, indicates that the server does not hold the target entry of the request, but that the servers in the referral field may.
0x0B	11	LDAP_ADMINLIMIT_EXCEEDED Indicates that an LDAP server limit set by an administrative authority has been exceeded.
0x0C	12	LDAP_UNAVAILABLE_CRITICAL_EXTENSION Indicates that the LDAP server was unable to satisfy a request because one or more critical extensions were not available. Either the server does not support the control or the control is not appropriate for the operation type.
0x0D	13	LDAP_CONFIDENTIALITY_REQUIRED Indicates that the session is not protected by a protocol, such as Transport Layer Security (TLS), which provides session confidentiality.
0x0E	14	LDAP_SASL_BIND_IN_PROGRESS Does not indicate an error condition, but indicates that the server is ready for the next step in the process. The client must send the server the same SASL mechanism to continue the process.
0x0F	15	Not used.

0x10	16	LDAP_NO_SUCH_ATTRIBUTE Indicates that the attribute specified in the modify or compare operation does not exist in the entry.
0x11	17	LDAP_UNDEFINED_TYPE Indicates that the attribute specified in the modify or add operation does not exist in the LDAP server's schema.
0x12	18	LDAP_INAPPROPRIATE_MATCHING Indicates that the matching rule specified in the search filter does not match a rule defined for the attribute's syntax.
0x13	19	LDAP_CONSTRAINT_VIOLATION Indicates that the attribute value specified in a modify, add, or modify DN operation violates constraints placed on the attribute. The constraint can be one of size or content (string only, no binary).
0x14	20	LDAP_TYPE_OR_VALUE_EXISTS Indicates that the attribute value specified in a modify or add operation already exists as a value for that attribute.
0x15	21	LDAP_INVALID_SYNTAX Indicates that the attribute value specified in an add, compare, or modify operation is an unrecognized or invalid syntax for the attribute.
	22-31	Not used.
0x20	32	LDAP_NO_SUCH_OBJECT Indicates that the target object cannot be found. This code is not returned on the following operations: Search operations that find the search base but cannot find any entries that match the search filter. Bind operations
0x21	33	LDAP_ALIAS_PROBLEM Indicates that an error occurred when an alias was dereferenced.
0x22	34	LDAP_INVALID_DN_SYNTAX Indicates that the syntax of the DN is incorrect. However, if the DN syntax is correct, but the LDAP server's structure rules do not permit the operation, the server returns the following: LDAP_UNWILLING_TO_PERFORM

0x23	35	LDAP_IS_LEAF Indicates that the specified operation cannot be performed on a leaf entry. (This code is not currently in the LDAP specifications, but is reserved for this constant.)
0x24	36	LDAP_ALIAS_DEREF_PROBLEM Indicates that during a search operation, either the client does not have access rights to read the aliased object's name or dereferencing is not allowed.
	37-47	Not used.
0x30	48	LDAP_INAPPROPRIATE_AUTH Indicates that during a bind operation, the client is attempting to use an authentication method that the client cannot use correctly. For example, either of the following causes this error: The client returns simple credentials when strong credentials are required. The client returns a DN and a password for a simple bind when the entry does not have a password defined.
0x31	49	LDAP_INVALID_CREDENTIALS Indicates that during a bind operation, one of the following occurred: The client passed either an incorrect DN or password. The password is incorrect because it has expired; intruder detection has locked the account, or some other similar reason.
0x32	50	LDAP_INSUFFICIENT_ACCESS Indicates that the caller does not have sufficient rights to perform the requested operation.
0x33	51	LDAP_BUSY Indicates that the LDAP server is too busy to process the client request at this time, but if the client waits and resubmits the request, the server may be able to process it then.
0x34	52	LDAP_UNAVAILABLE Indicates that the LDAP server cannot process the client's bind request, usually because it is shutting down.

0x35	53	LDAP_UNWILLING_TO_PERFORM Indicates that the LDAP server cannot process the request because of server-defined restrictions. This error is returned for the following reasons: The add entry request violates the server's structure rules. The modify attribute request specifies attributes that users cannot modify. Password restrictions prevent the action. Connection restrictions prevent the action.
0x36	54	LDAP_LOOP_DETECT Indicates that the client discovered an alias or referral loop, and is thus unable to complete this request.
	55-63	Not used.
0x40	64	LDAP_NAMING_VIOLATION Indicates that the add or modify DN operation violates the schema's structure rules. For example: The request places the entry subordinate to an alias. The request places the entry subordinate to a container that is forbidden by the containment rules. The RDN for the entry uses a forbidden attribute type.
0x41	65	LDAP_OBJECT_CLASS_VIOLATION Indicates that the add, modify, or modify DN operation violates the object class rules for the entry. For example, the following types of request return this error: The add or modify operation tries to add an entry without a value for a required attribute. The add or modify operation tries to add an entry with a value for an attribute which the class definition does not contain. The modify operation tries to remove a required attribute without removing the auxiliary class that defines the attribute, as required.

0x42	66	LDAP_NOT_ALLOWED_ON_NONLEAF Indicates that the requested operation is permitted only on leaf entries. For example, the following types of requests return this error: The client requests a delete operation on a parent entry. The client request a modify DN operation on a parent entry.
0x43	67	LDAP_NOT_ALLOWED_ON_RDN Indicates that the modify operation attempted to remove an attribute value that forms the entry's relative distinguished name.
0x44	68	LDAP_ALREADY_EXISTS Indicates that the add operation attempted to add an entry that already exists, or that the modify operation attempted to rename an entry with the name of an entry that already exists.
0x45	69	LDAP_NO_OBJECT_CLASS_MODS Indicates that the modify operation attempted to modify the structure rules of an object class.
0x46	70	LDAP_RESULTS_TOO_LARGE Reserved for CLDAP.
0x47	71	LDAP_AFFECTS_MULTIPLE_DSAS Indicates that the modify DN operation moves the entry from one LDAP server to another and thus requires more than one LDAP server.
	72-79	Not used.
0x50	80	LDAP_OTHER Indicates an unknown error condition. This is the default value for NDS error codes which do not map to other LDAP error codes.

LDAP Client Return Codes

The following table lists the client return codes:

Hex	Decimal	Description
0x51	81	LDAP_SERVER_DOWN Indicates that the LDAP libraries cannot establish an initial connection with the LDAP server. Either the LDAP server is down, or the specified host name or port number is incorrect.
0x52	82	LDAP_LOCAL_ERROR Indicates that the LDAP client has an error. This is usually a failed dynamic memory allocation error.
0x53	83	LDAP_ENCODING_ERROR Indicates that the LDAP client encountered errors when encoding an LDAP request intended for the LDAP server.
0x54	84	LDAP_DECODING_ERROR Indicates that the LDAP client encountered errors when decoding an LDAP response from the LDAP server.
0x55	85	LDAP_TIMEOUT Indicates that the time limit of the LDAP client was exceeded while waiting for a result.
0x56	86	LDAP_AUTH_UNKNOWN Indicates that the ldap_bind or ldap_bind_s function was called with an unknown authentication method.
0x57	87	LDAP_FILTER_ERROR Indicates that the ldap_search function was called with an invalid search filter.
0x58	88	LDAP_USER_CANCELLED Indicates that the user cancelled the LDAP operation.
0x59	89	LDAP_PARAM_ERROR Indicates that an LDAP function was called with an invalid parameter value (for example, the ID parameter is NULL).
0x5A	90	LDAP_NO_MEMORY: Indicates that a dynamic memory allocation function failed when calling an LDAP function.

0B	91	LDAP_CONNECT_ERROR Indicates that the LDAP client has lost either its connection or cannot establish a connection to the LDAP server.
0x5C	92	LDAP_NOT_SUPPORTED Indicates that the client does not support the requested functionality. For example, if the LDAP client is established as an LDAPv2 client, the libraries set this error code when the client requests LDAPv3 functionality.
0x5D	93	LDAP_CONTROL_NOT_FOUND Indicates that the client requested a control that the libraries cannot find in the list of supported controls sent by the LDAP server.
0x5E	94	LDAP_NO_RESULTS_RETURNED Indicates that the LDAP server sent no results. When the ldap_parse_result function is called, no result code is included in the server's response.
0x5F	95	LDAP_MORE_RESULTS_TO_RETURN Indicates that more results are chained in the result message. The libraries set this code when the call to the ldap_parse_result function reveals that additional result codes are available.
0x60	96	LDAP_CLIENT_LOOP Indicates the LDAP libraries detected a loop. Usually, this happens when following referrals.
0x61	97	LDAP_REFERRAL_LIMIT_EXCEEDED Indicates that the referral exceeds the hop limit. The hop limit determines how many servers the client can hop through to retrieve data. For example, suppose the following conditions: The hop limit is two. The referral is to server D which can be contacted only through server B (1 hop) which contacts server C (2 hops) which contacts server D (3 hops) With these conditions, the hop limit is exceeded and the LDAP libraries set this code.

LDAP-Associated RFC Standards

The following table describes the LDAP-associated RFC standards available for your use:

RFC	Description
1274	The COSINE and Internet X.500 Schema
1275	Replication Requirements to provide an Internet Directory using X.500
1276	Replication and Distributed Operations extensions to provide an Internet Directory using X.500
1308	Executive Introduction to Directory Services Using the X.500 Protocol
1309	Technical Overview of Directory Services Using the X.500 Protocol
1430	A Strategic Plan for Deploying an Internet X.500 Directory Service
1488	The X.500 String Representation of Standard Attribute Syntaxes
1558	A String Representation of LDAP Search Filters
1617	Naming and Structuring Guidelines for X.500 Directory Pilots
1777	Lightweight Directory Access Protocol v2
1778	The String Representation of Standard Attribute Syntaxes
1779	A String Representation of Distinguished Names
1804	Schema Publishing in X.500 Directory
1823	The LDAP Application Program Interface
1959	An LDAP URL Format
1960	A String Representation of LDAP Search Filters
2044	UTF -8, a transformation format of Unicode and ISO 10646
2164	Use of an X.500/LDAP Directory to support MIXER address mapping
2218	A Common Schema for the Internet White Pages Service
2247	Using Domains in LDAP/X.500 Distinguished Names
2251	Lightweight Directory Access Protocol (v3)
2252	Lightweight Directory Access Protocol (v3): Attribute Syntax Definitions
2253	Lightweight Directory Access Protocol (v3): UTF-8 String Representation of Distinguished Names
2254	The String Representation of LDAP Search Filters
2255	The LDAP URL Format
2256	A Summary of the X.500(96) User Schema for use with LDAPv3
2279	UTF-8, a transformation format of ISO 10646
2293	Representing Tables and Subtrees in the X.500 Directory
2294	Representing the O/R Address hierarchy in the X.500 Directory Information Tree
2307	An Approach for Using LDAP as a Network Information Service

2377	Naming Plan for Internet Directory-Enabled Applications
2531	Content Feature Schema for Internet Fax
2559	Internet X.509 Public Key Infrastructure Operational Protocols - LDAPv2
2587	Internet X.509 Public Key Infrastructure LDAPv2 Schema
2589	Lightweight Directory Access Protocol (v3): Extensions for Dynamic Directory Services
2596	Use of Language Codes in LDAP
2649	An LDAP Control and Schema for Holding Operation Signatures
2657	RFC 2657 - LDAPv2 Client vs. the Index Mesh
2696	LDAP Control Extension for Simple Paged Results Manipulation
2713	Schema for Representing Java(tm) Objects in an LDAP Directory
2714	Schema for Representing CORBA Object References in an LDAP Directory
2739	Calendar Attributes for vCard and LDAP
2798	Definition of the inetOrgPerson LDAP Object Class
2820	Access Control Requirements for LDAP
2829	Authentication Methods for LDAP
2830	Lightweight Directory Access Protocol (v3): Extension for Transport Layer Security
2849	The LDAP Data Interchange Format (LDIF) - Technical Specification
2879	Content Feature Schema for Internet Fax (V2)
2891	LDAP Control Extension for Server Side Sorting of Search Results
3045	Storing Vendor Information in the LDAP root DSE
3062	LDAP Password Modify Extended Operation
3112	LDAP Authentication Password Schema
3296	Named Subordinate References in Lightweight Directory Access Protocol Directories
3377	Lightweight Directory Access Protocol (v3): Technical Specification
3384	Lightweight Directory Access Protocol (version 3) Replication Requirements

pdm_configure--Open the Configuration Window

pdm_configure opens a window containing the CA SDM configuration window. Use this window to set the CA SDM configuration after installing CA SDM, or to change the CA SDM configuration after CA SDM is running. Following are the possible reasons to change the CA SDM configuration:

- Changing the database type or server
- Reload default data
- Change passwords for the CA SDM system accounts
- Rebuild schema files to incorporate changes
- Reconfigure the following server, depending on your CA SDM configuration:
  - Conventional: Primary or secondary server
  - Advanced availability: Application or background server

Syntax

This command has the following format:

```
pdm_configure
```

Restrictions

pdm_configure can be run on any CA SDM server or a CA SDM Linux client. You must be the privileged user or root to run pdm_configure.

pdm_key_refresh--Refresh Cached Key Information

The pdm_key_refresh utility refreshes cached key information from the key control table in the database. If key control information is updated, this utility can be executed instead of stopping and restarting the system to force CA SDM to use the new keyid base value.

> **WARNING**
> Changing the key control table may cause data corruption. You should not attempt to change the key control without specific instructions from CA Technical Support.

Syntax

This command has the following format:

```
pdm_key_refresh
```

> **WARNING**
>
> On Non-Windows, the LIBPATH must be set before running several CA SDM utilities. Use *pdm_task* to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

pdm_lexutil--Modify SDM Lexicons

The pdm_lexutil utility allows you to modify Service Desk lexicons, to add or delete words for the spell check dictionary.

Syntax

This command has the following format:

```
pdm_lexutil -a | -d [-f] [-l] wordlist
```

- **-a**
  Add words.
- **-d**
  Delete words.
- **-f**
  File or lexicon containing list of words to be added or deleted.
- **-l**

Lexicon name.
Default: userdict.tlx

- **wordlist**
  Words to be added or deleted.

  WARNING

  On Non-Windows, the LIBPATH must be set before running several CA SDM utilities. Use *pdm_task* to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

pdm_listconn--List Active Connections

The pdm_listconn utility can be used to list active connections for both clients and servers.

Syntax

This command has the following format:

```
pdm_listconn [-c] [-s] [ - s -c] [ - t nn] [proc1 [proc2...]]
```

The default command has the following format if no parameters are specified:

```
pdm_listconn -s -t 2
```

-c

Lists connections by client. The utility displays two lines for each client:

```
(n secs) client_type node|cproc
connected to alias (sproc) at time
```

n is the number of seconds it took for the client to respond.

client_type is "vbop," "animator daemon," or "web engine."

node is the IP address of the client's node.

cproc is the client's slump procname.

alias is the alias of the server the client is connected with.

sproc is the server's slump procname.

time is the connection time.

For example:

```
(0 secs) vbop client 141.202.211.34|vbop-0x40120000:anthill:0
         connected to CMD40120 on anthill (domsrvr) at 02/19/1999 10:44:16
```

-s

Lists connections by server (default). The utility displays two lines for each server:

```
(n secs) server  alias (node|sproc) willingness willingness
count connected clients (use pdm_listconn -c -s to list clients by server)
```

n is the number of seconds it took for the server to respond.

node is the IP address of the server's node.

alias is the alias of the server the client is connected with.

sproc is the server's slump procname.

willingness is the server's willingness to accept new clients (0 - 100).

count is the number of connected clients.

For example:

```
(0 secs) server CMD40120 on anthill (domsrvr) willingness 98
2 connected clients (use pdm_listconn -c -s to list clients by server)
   vbop client 141.202.211.34|vbop2 connected 02/19/1999 10:53:16
   vbop client 141.202.211.34|vbop-0x40120000:anthill:0 connected 02/19/1999 10:44:17
```

-s -c

Lists connections by server, including client detail for each server. The utility displays several lines for each server:

```
(n secs) server  alias (node|sproc) willingness willingness
count connected clients:
client_type node|cproc connected time
```

where:

n is the number of seconds it took for the server to respond.

node is the IP address of the server's node.

alias is the alias of the server the client is connected with.

sproc is the server's slump procname.

willingness is the server's willingness to accept new clients (0 - 100).

count is the number of connected clients.

client_type is "vbop," "animator daemon," or "web engine."

node is the IP address of the client's node.

cproc is the client's slump procname.

time is the connection time.

For example:

```
(0 secs) server CMD40120 on anthill (domsrvr) willingness 98
2 connected clients:
    vbop client 141.202.211.34|vbop2 connected 02/19/1999 10:53:16
    vbop client 141.202.211.34|vbop-0x40120000:anthill:0 connected 02/19/1999 10:44:17
```

-t nn

Specifies a timeout interval in seconds. Because pdm_listconn receives information from an unknown number of clients and servers, it terminates when the specified timeout interval elapses after the last message received.

Default: 2

proc1

Specifies one or more slump procnames, separated by spaces. The utility displays client or server information from them, as appropriate.

> **WARNING**
>
> On Non-Windows, the LIBPATH must be set before running several CA SDM utilities. Use *pdm_task* to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

pdm_logfile--Change stdlog Cutover Size

pdm_logfile lets you change your stdlog.*x* cutover size. The cutover can occur after a specified number of bytes are written. On UNIX, this value is reset with each pdm_init. On Windows, the settings are retained with each pdm_halt and pdm_init.

Syntax

This command has the following format:

```
pdm_logfile [-L|-h]
```

or

```
pdm_logfile [-g -h] [-b bytes]
```

Example

To change your stdlog.*x* files to cutover at 500,000 bytes, issue the following command:

```
pdm_logfile -f STD -b 500000
```

-L

Creates a listing of current cutovers.

-q

Runs pdm_logfile in quiet mode.

-b bytes

Specifies the number of bytes written before cutover occurs.

Restrictions

You can run pdm_load while CA SDM is active, but performance can become very slow. It is best to run pdm_load when no one is using CA SDM.

> **WARNING**
>
> On UNIX, the LIBPATH must be set before running several CA SDM utilities. Use *pdm_task* to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

pdm_task--Set Environment Variables

Applies to UNIX only

The pdm_task utility sets environment variables for commands that do not have wrappers. For example, the pdm_task command must precede the report command on the same command line **only** when the command is invoked through a script or the command line. If you issue the report command from a menu, you do not need to include pdm_task, because all environment variables are set by the application.

> **NOTE**
> Reports cannot be generated from the command line on a client.

Syntax

This command has the following format:

```
pdm_task command
```

command

Specifies a command that does not have a wrapper and, therefore, does not automatically set environment variables. See report--Generate Reports for more information about issuing pdm_task with a command.

pdm_uconv--Convert Local Charset to UTF-8

The pdm_uconv utility assists you in converting data from previous releases of CA SDM or integrations with other CA Technologies products. The most common usage of this utility is to convert from a local charset to UTF-8 and from UTF-8 to a local charset.

Syntax

This command has the following format:

```
pdm_uconv -h [-V] [-s] [-v] [-l | --list-code
| --default-code | -L] [--cannon] [-x] [--to-callback | -c] [--from-callback | -i] [--fallback | --no-
fallback] [-b] [-f] [--t] [--add-signature] [--remove-signature] [-o] [file ...]
```

- **-h**
  Opens the help menu.
- **-V**
  Prints the program version.
- **-s**
  Uses silent operation and suppresses messages.
- **-v**
  Displays the progress information of the utility.
- **-l**
  Lists all available encoding. The following are valid:
  - **--list-code**
    Lists only the given encoding.
  - **--default-code**
    Lists only the default encoding.
  - **-L**
    Lists all available transliterators.
- **--cannon**
  Prints the list in cnvrtrs.txt(5) format.
- **-x**
  Runs the progress through transliteration.
- **--to-callback *callback***
  Uses callback on destination encoding.
- **-c**
  Omits invalid characters from the output.
- **--from-callback *callback***
  Uses callback on original encoding.
- **-i**
  Ignores invalid sequences in the input.
- **--callback *callback***
  Uses callback on both encoding.s.
- **-b**
  Specifies the block size.
  Default: 4096
- **--fallback**

Uses fallback mapping.

- **--no-fallback**
  Does not use fallback mapping.
- **-f**
  Sets theoriginal encoding.
- **-t**
  Sets the destination encoding.
- **--add-signature**
  Adds U+FEFF Unicode signature character (BOM).
- **--remove-signature**
  Removes U+FEFF Unicode signature character (BOM)
- **-o**
  Writes output to file.

Examples:

- **From local charset to UTF-8**
  pdm_uconv -t utf-8 inputfile.txt > outputfile.txt
- **From specific charset (iso-2022-jp) to UTF-8**
  pdm_uconv -f iso-2022-jp -t utf-8 inputfile.txt > outputfile.txt
- **From UTF-8 to local charset**
  pdm_uconv -f utf-8 inputfile.txt > outputfile.txt
- **From UTF-8 to specific charset**
  pdm_uconv -f utf-8 -t iso-2022-jp inputfile.txt > outputfile.txt

The pdm_uconv utility has the following are valid callbacks:

- substitute
- skip
- stop
- escape
- escape-icu
- escape-java
- escape-c
- escape-xml
- escape-xml-hex
- escape-xml-dec
- escape-unicode

pdm_webstat--Return Web Usage Statistics

Use pdm_webstat to return CA SDM session and user statistics for one or more web engine processes. The pdm_webstat command shows cumulative sessions, most sessions at a time, and current active sessions. It can also provide information about the individual users.

Syntax

This command has the following format:

```
pdm_webstat [-r] [-d | -D] [-i] [-t timeout] [-p webengine process] [-n] [-h]
```

-r

Specifies raw text mode, without titles and other formatting. Within the output for a single web engine process there are no line breaks; however, the output for each web engine process always starts on a new line.

Raw text mode displays data in exactly the same order as when you use pdm_webstat without the -r option. Use raw text mode if you want to use the resulting data in a spreadsheet or other type of report. For example, the following syntax:

```
pdm_webstat -r
```

shows the following output:

```
10/11/2005 10:31:49 web:local:0 12 4 2
10/11/2005 10:31:49 web:local:1 9 2 2
```

-d

Specifies detailed output that displays user sessions. The -d option lists all current sessions in the form of *userid@IPaddress*. If a session is displayed without a user ID, it usually means that the session has not logged on yet. For example, the following syntax:

```
pdm_webstat -d
```

shows the following output:

```
PDM_Webstat: Invoked at 10/11/2005 10:27:31
========================================
Report from Webengine: web:local:0
========================================
Cumulative sessions so far = 12
Most sessions at a time    = 4
Currently active sessions  = 2
  @192.168.1.16
  usery@192.168.1.20
========================================
Report from Webengine: web:local:1
========================================
Cumulative sessions so far = 7
Most sessions at a time    = 2
Currently active sessions  = 2
  SrvcPlus@192.168.1.14
  userx@192.168.1.8
```

-D

Provides more detailed output for debugging purposes. This option should be specified only when specifically requested by CA support. It adds internal information about each session to the detailed output. For example, the following syntax:

```
pdm_webstat -D
```

shows the following output:

```
PDM_Webstat: Invoked at 10/11/2007 10:28:10
========================================
Report from Webengine: web:local:0
========================================
Cumulative sessions so far = 12
Most sessions at a time    = 4
Currently active sessions  = 2
  @192.168.1.16            SessionStat   1
  userx@192.168.1.20       SessionStat   5
```

```
========================================
Report from Webengine: web:local:1
========================================
Cumulative sessions so far = 7
Most sessions at a time    = 2
Currently active sessions  = 2
  SrvcPlus@192.168.1.14    SessionStat   7
  userx@192.168.1.8        SessionStat  13
```

- **-i**
  Specifies an interval in seconds between successive reports. When the -i argument is specified, pdm_webstat runs continuously. The pdm_webstat command outputs its report in the format requested by other arguments, waits for the interval specified, and outputs the report again. With - i, pdm_webstat terminates only when explicitly cancelled, normally with Ctrl+C. For example, the following syntax:

  ```
  pdm_webstat  - i 5  - p web:local  - r
  ```

  shows the following output:

  ```
  09/21/2007 16:27:25 web:local 14 10 6
  09/21/2007 16:27:30 web:local 17 10 9
  09/21/2007 16:27:35 web:local 18 10 10
  09/21/2007 16:27:41 web:local 21 13 13
  ```

-t timeout

Specifies a time-out value in seconds. This parameter causes pdm_webstat to wait for a number of seconds for a response before terminating. The default is 30 seconds.

-p webengine_process

Specifies the process name of the web engine for which you want to report. By default, all web engine processes are reported. The process name (also called slump-name) is the same name that would be viewed in an slstat output and always starts with "web:".

-n

Suppresses the normal log entry for each reported web engine. By default, a log entry summarizing each web engine is created.

> **NOTE**
> By default, if you do not specify any parameters, pdm_webstat displays summary data for all running processes.

pdm_webstat returns zero if the command completes successfully, and a nonzero value if errors occur (for example, a time-out or no active web engine processes).

Example: pdm_webstat Output When No Parameters Are Specified

The following example shows the output of pdm_webstat when it is executed with no parameters:

```
PDM_Webstat: Invoked at 10/11/2007 10:26:42
========================================
Report from Webengine: web:local:0
========================================
Cumulative sessions so far = 12
Most sessions at a time    = 4
Currently active sessions  = 2
========================================
Report from Webengine: web:local:1
========================================
Cumulative sessions so far = 7
```

```
Most sessions at a time    = 2
Currently active sessions  = 2
```

WARNING

On UNIX, the LIBPATH must be set before running several CA SDM utilities. Use *pdm_task* to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

pdm_mail Utility--Send Email Information

The pdm_mail utility is used in notifications to send emails by sending email information to the pdm_mail_nxd process. The pdm_mail utility can also be used for commands, but not to both. If no parameters are used, then the default behavior of using the NX_NTF_xxxx variable to pass parameters is in effect.

For email, the utility is invoked as follows:

```
pdm_mail [-i [-s subject] [-e email_address] [-q]]  [-p] [-M] [-F] [-T] [-B] [-H] [-N] [-R] [-h]
```

- **-i**
  Uses STDIN instead of NTF variables. The following parameters are used for STDIN email behavior only:
  - **-e**
    Specifies the email address (for recipient).
  - **-s**
    Specifies the subject for the email.
  - **-q**
    Disables the display prompt for STDIN.
- **-p**
  Utilizes pager logic. This option includes using the pager email address instead of the regular email address. Only the plain text version of the notification is used (no HTML).
- **-M**
  Uses plain text only (no MIME) in body.
- **-F**
  Specifies the From address of the email.
- **-T**
  Specifies the Reply-To address of the email.
- **-B**
  Specifies the body charset. This might be useful for pagers that do not support the UTF-8.
- **-H**
  Specifies the header charset. This might be useful for pagers that do not support the UTF-8.
- **-N**
  Specifies the Delivery Status Notification (DSN) Notify option.
- **-R**
  Specifies the Delivery Status Notification (DSN) Return option.
- **-h**
  Displays help on the utility.

NOTE
For commands, the utility is invoked as follows:

NOTE
Command to mail server

```
pdm_mail -c option [parameter]
```

NOTE
Command to mail eater

```
pdm_mail -x option [parameter]
```

- **check_interval**
  (-x only) Changes the check mail interval to a specified value (in seconds).
- **report_interval**
  Changes the report interval to a specified value (in seconds).
- **report_now**
  Forces a report to the logs. The counter is not reset.
- **send_q**
  (-c only) Sends local mail queue to remote mail server.
- **trace**
  Turns tracing on or off.

WARNING

On Non-Windows, the LIBPATH must be set before running several CA SDM utilities. Use *pdm_task* to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

4CA SDM PDM Database Commands

This article describes the following topics:

pdm_replace--Replace a Database Table

pdm_replace deletes a table in a CA SDM database and replaces it with a table from a temporary file you specify with the -f option; the data from the input file is the only data that is in that table after running pdm_replace. Back up your table before running pdm_replace.

NOTE
As part of its processing, pdm_replace first shuts down the daemons (Non-Windows) or services (Windows).

pdm_replace accepts a text file as input, which is the same file format used by pdm_userload. You can create an input file for pdm_replace using pdm_extract; however, you cannot use the output of pdm_backup as input to pdm_replace.

WARNING
Be sure to name your input file with a name different from the table name you are attempting to replace. For example, if you are replacing a table named ca_contacts and you name the input file ca_contacts.dat, after you execute the pdm_replace command to point to the input file (ca_contacts.dat), it deletes the file after execution because it has the same name as the table.

Restrictions

- pdm_replace can be run only on the following servers, depending on your CA SDM configuration:
  - Conventional: Primary server
  - Advanced availability: Background server

    WARNING
    Ensure that you have stopped all application and standby servers before running this command on the background server.
- Only the privileged user or root can run pdm_replace.
- Do not run pdm_replace when users are logged in to CA SDM.

Syntax

This command has the following format:

```
pdm_replace [-v] -f filename
```

-v

Specifies verbose mode.

-f *filename*

Specifies an ASCII file with the following format:

```
TABLE table_name
fieldname1 fieldname2 . . . . fieldnameN
{ "value11", "value12", . . . "value1N" }
{ "value21", "value22", . . . "value2N" }
.
.
.
{ "valueN1", "valueN2", . . . "valueNN" }
```

This format is the same file format used by pdm_userload. You can create an input file for pdm_replace using pdm_extract; however, you cannot use the output of pdm_backup as input to pdm_replace.

pdm_restore--Restore a Database

pdm_restore stops CA SDM and then deletes all records from a CA SDM database and replaces them with records from a file you specify with the -f option. The data from the input file is the only data that will be in the CA SDM database after running pdm_restore.

The input file must be created using pdm_extract or pdm_backup, or otherwise formatted for pdm_restore. pdm_backup can back up non-database data, and pdm_restore can restore this data also. pdm_backup and pdm_restore are not recommended when other backup and restoration tools are available.

> **NOTE**
> As part of its processing, pdm_restore first shuts down the daemons (Non-Windows) or services (Windows).

Syntax

This command has the following format:

```
pdm_restore [-d] [-g] [-n] [-w] [-v]  - f filename
```

Restrictions

pdm_restore can be run only on a CA SDM server. Only the privileged user or root can run pdm_restore. The following restrictions are applicable if you are using the advanced availability configuration:

If you are restoring the database, run the pdm_restore command only on the background server.

- Ensure that you have stopped all servers (application, background, and standby) before you run the pdm_restore command.

  > **WARNING**
  > Use pdm_restore only with a full database backup created by pdm_backup, because your current database is deleted and replaced by the backup file. Do not run pdm_restore when users are logged in to CA SDM.

- **-d**
  Specifies that only database data is restored.
- **-g**
  Specifies that only non-database data be restored. Only windows (forms) and other non-database data are restored.
- **-n**

Specifies that NX.env is restored if restoring non-database data. By default, NX.env is not restored. We recommend that the NX.env file not be restored unless the restore is to the same server the backup came from. Restoring an incorrect NX.env can affect unintended databases.

- **-w**
  Specifies that web.cfg is restored if restoring non-database data. By default, web.cfg is not restored.
- **-v**
  Specifies verbose mode.

-f *filename*

Specifies an input file that contains a full backup created by pdm_backup.

pdm_load--Add, Update, and Delete Database Records

> **WARNING**
> Using pdm_load can be destructive so always back up your database before you perform a pdm_load, and use pdm_userload unless instructed to use pdm_load.

pdm_load updates a CA SDM database using an input file you specify, up to a maximum of 112 attributes.

Whenever you upload tickets (such as requests or issues), your ticket number should include a unique prefix or suffix in its string. CA SDM views this number as a string of characters not as a sequential number, and thus cannot guarantee that it will assign a unique number to the uploaded tickets. As long as you assign a unique prefix or suffix using awk or another text processor, you can upload tickets without CA SDM writing over previously assigned numbers.

Syntax

This command has the following format:

```
pdm_load [-c] [-h] [-m] [-r] [-u] [-v]  - f filename
```

The input file entries follow this format:

```
TABLE table_name
fieldname1 fieldname2 . . . . fieldnameN
{ "value11", "value12", . . . "value1N" }
{ "value21", "value22", . . . "value2N" }
.
.
.
{ "valueN1", "valueN2", . . . "valueNN" }
```

table_name is the name of the table to be loaded, as listed in the CA SDM database schema file, which is located in $NX_ROOT/site/sch/schema.sch (Non-Windows) or *installation-directory*\site\sch\schema.sch (Windows). $NX_ROOT or *installation-directory* is the directory where you installed CA SDM.

-f *filename*

Specifies an input ASCII file.

-c

Checks the input file against the database and reports on the updates that would be made, but does not make the updates.

-m

Specifies mass update. Specify when you are using pdm_load to add or delete a large number of records. This option suppresses all client notifications of updates and sends a cache refresh message for a table when pdm_load finishes processing the table.

-r

Removes database records that match input records.

> **WARNING**
> Make a backup copy of the database before running pdm_load with this option. After old database records are removed, you must restore the CA SDM database with this backup copy if you want to recover any deleted records.

-u

Updates existing records, but does not insert new records into the database.

CA SDM PDM Daemon Commands

pdm_halt--Terminate Daemons or Stop Services

pdm_halt cleanly terminates all CA SDM daemons (Non-Windows) or the System Server Service (Windows) on the system from which pdm_halt is executed. The pdm_halt utility usually takes about 30 seconds to complete. If it hangs for more than two minutes, press Ctrl+C to stop pdm_halt, and try again.

Syntax

This command has the following format:

```
pdm_halt [-w] [-a] [time]
```

- **-w**
  Waits for the daemons to stop.
- **-a**
  Stops all proctors defined in the pdm_startup file.
- **[time]**
  Specifies the number of seconds to wait until the command executes.

Restrictions

pdm_halt can be run on a CA SDM server or a CA SDM Non-Windows client. You must be the privileged user to run pdm_halt.

pdm_init--Start Daemons

Applies to Non-Windows only

pdm_init starts all CA SDM automatic processes on the system from which pdm_init is executed. These automatic processes are called *daemons* and run continually in the background while you work. Not all of the daemons are started; some are applicable only to certain operating systems.

> **NOTE**
> Use pdm_d_refresh to start daemons that failed to start the first time after remedying the problem that caused them not to start initially. In most cases you do not need to terminate the daemons running to start ones that initially failed.

Syntax

```
pdm_init
```

Restrictions

pdm_init can be run on a CA SDM server or a CA SDM Non-Windows client. You must be the privileged user to run pdm_init.

pdm_d_refresh--Start Failed Daemons

The pdm_d_refresh command line utility is used primarily for remote daemon configurations. It tells the daemon manager to try to start daemons that have failed to start ten times and that the daemon manager has flagged as "not runnable.". Running this utility flags all daemons as able to run and resets the restart counter. Then, the daemon manager tries to start all stopped daemons.

Syntax

This command has the following format:

```
pdm_d_refresh
```

> **WARNING**
> On Non-Windows, the LIBPATH must be set before running several CA SDM utilities. Use *pdm_task* to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

pdm_status--Show Status of Daemons or Processes

pdm_status shows the status of all CA SDM daemons (Non-Windows) or processes (Windows) on the system from which the command is executed.

Output is displayed in this format:

```
DAEMON                          STATUS      HOST      PID   SLUMP CONNECT TIME
-------------------------------------------------------------------------------
Agent anthill                   Running     anthill   455   Tue Feb 17 17:55:12
Ddict_rd             (ddictrd)  Completed   anthill
Data Dictionary   (ddictbuild)  Completed   anthill
...
User Validation      (boplgin)  Running     anthill   456   Tue Feb 17 17:55:21
```

Syntax

This command has the following format:

```
pdm_status
```

CA SDM PDM Server Commands

pdm_proctor_init--Start Proctor on Secondary Servers

Applies to Non-Windows only

Use pdm_proctor_init to start the proctor on secondary servers. All secondary servers should be started prior to starting the daemons from the primary server. After all daemons have been stopped from the primary server, use pdm_halt on the secondary server to stop this proctor.

> **NOTE**
> Do not use pdm_proctor_init on the primary server.

Syntax

This command has the following format:

```
pdm_proctor_init
```

pdm_server_control Utility--Identify Servers

The pdm_server_utility commands identify the server as background or standby in an advance availability configuration.

Execute the following command from the command prompt:

```
pdm_server_control -h | -b | -q interval [-s server_name] | -t | -c [-server_name]
```

- **-h**
  Displays the help page.
- **-b**
  Notifies a local standby server to become the background server. The standby server must be running. If the server is not running, it is started but no failover is performed. To start a failover, run the command again.
- **-q interval [-s server_name]**
  Notifies a local or remote application server to quiesce in a specified time interval. This interval is the number of seconds before the server shutdowns. When using this option without a server_name, the local server is notified to quiesce. This option cannot be used for a background or a standby server.
- **-t**
  Displays the type of this CA SDM advance availability Server.
- **-c [-s server_name]**
  Notifies a local or remote application server to cancel the previous quiesce the request.

pdm_rest_util – Manage the CA SDM RESTful Web Services Application

CA SDM uses this utility automatically. You can run it manually if you require the utility, such as after an unexpected error occurs. This REST web services utility deploys the REST services web application. This utility deploys the REST application to the dedicated REST Tomcat instance. A batch file in the NX_ROOT\bin directory (pdm_rest_util.bat for Windows or ./pdm_rest_util.sh for Non-Windows) lets you invoke the utility.

This command has the following formats and options:

```
pdm_rest_util  - h | [-deploy] | [-undeploy]
```

- **-h**
  Prints command-line help.
- **-deploy**
  Generates, compiles, and deploys all Majic factories.
- **-undeploy**
  Undeploys REST Web Services on the local server.

Undeploy the REST Web Services Application

You can undeploy the REST Web Services application with the pdm_rest_util utility. For example, you want to perform CA SDM maintenance and prefer to undeploy REST during this operation.

Follow these steps:

1. Open a command prompt.
2. Execute the following command:
   ```
   pdm_rest_util -undeploy
   ```
   The REST Web Services application is undeployed.

   WARNING
   If you undeploy the REST application with this utility, REST can redeploy automatically when CA SDM restarts. We recommend that you use the CA SDM configuration to disable REST indefinitely.

pdm_k_reindex -- Knowledge Re-Index Utility

The Knowledge reindex utility, pdm_k_reindex.exe, is located under the Knowledge Management installation directory.

NOTE
Reindexing the documents in the knowledge base can be a time-consuming operation, depending upon the size of your database. We recommend that you run the Knowledge Reindex utility after all the changes have been added. For advanced availability configuration, you cannot execute the knowledge re-index utility during failover.

WARNING
On Non-Windows, the LIBPATH must be set before running several CA SDM utilities. Use pdm_task to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

Follow these steps:

1. Open the command prompt.
2. Enter the following command at the command prompt to run the knowledge reindex:

For example:

```
pdm_k_reindex
```

The following options are available with this command.

- **-D**
  Defines the debug mode, such as printing to the command window.
- **-v**
  Defines the verbose mode, such as printing to the stdlog file.
- **-i**
  Does not create table indexes in the reindex table after reindexing.

  NOTE
  Parameters with a dash as a prefix, such as " - D", must precede other parameters that do not have this prefix.

The other option is as follow:

- ***File*:reindex.txt**
  Documents are reindexed to the specified file.
- **+i**
  Creates the indexes of the reindexed table only, which is the search table after reindexing. The old indexes are dropped before reindexing.
- **+t**
  Switches the names of search and reindex tables only.

  NOTE
  A “+” prefix denotes only this parameter applies.

- **sdtout**
  Defines the frequency of statistic appearing in the command window. By default the knowledge reindex utility provides statistics into the command window for every 1000 documents processed. However, sometimes statistics are required to be provided more often. Use the following parameter:

  ```
  pdm_k_reindex -i sdtout:10
  ```

  In this case, statistics display in the command window for every ten documents.

The documents are reindexed in the knowledge base.

When to Use pdm_k_reindex

Run the pdm_k_reindex utility when one or more of the following search settings were changed:

- Noise words
- Synonyms
- Special terms
- Language
- Remove Similar Words
- Remove Noise Words
- Valid Character Range
- Recognize Special Terms

An appropriate message appears on Knowledge node of Administration tab when a change occurs.

Re-Index Tracking

While the re-index is running, you can view the status of the process in the Re-Index Tracking section in the lower half of the page. Each field is described as follows:

- **Document #**
  Specifies the number of documents already processed.
- **Average Size (Words)**
  Specifies the size of the current documents, calculated by the number of words minus the number of noise words.
- **Rate (Docs/Sec)**
  Specifies the number of documents processed, per second.
- **Time from Start**
  Indicates the duration of the re-index process since start.
- **Time Remaining**
  Specifies the estimated amount of time remaining for the process, based on the current rate and the remaining number of documents.
- **Failures #**
  Represents the number of failed documents (maximum=100). When the maximum number of failures is reached, the administrator is prompted to either continue or cancel the process.

  NOTE
  If changes have been made to Noise Words, Special Terms, Synonyms or Parse Settings and you do not re-index, you are prompted the next time you access the Knowledge node of the Administration tab. Changes will take effect only after the Knowledge Re-index utility is run.

Import and Re-Indexing

When pdm_kit.exe is invoked from the command line, the pdm_kit utility imports the documents into the database. After pdm_kit is completed, assuming document indexing or re-indexing has not been disabled using the command-line options, the re-index utility (pdm_k_reindex.exe) is automatically invoked. The status and output of the re-indexing operation is automatically written to the "EBR_REINDEX.LOG" in the $NX_ROOT\log directory.

Index and De-Index Queue Settings for Batch and Instant Processing

Indexing and De-Indexing both run a batch process to include a predefined number of documents at one run. These batch processes are used for performance optimization. If more documents are included in the batch, system performance increases.

The number of documents you can process is limited; the limit depends on the size of the documents and the linked attachments. The document size is calculated based on the pure text and its attachments. Imagine and format elements are not calculated.

NOTE
You can limit the size of attachments by navigating to Attachments Library, Repositories on the Administration tab and editing the repository to set the File Limit Size (KB).

The recommended batch maximum size is between 2-12 MB (per the EBR_MAX_INDEX_BATCH_SIZE parameter of the NX.env file and the average document size).

- If the average size of your document (including attachments) is approximately 0.1 MB, keep the default setting in NX.env:

  ```
  @EBR_MAX_INDEX_BATCH_SIZE=128
  @NX_EBR_INDEX_QUEUE_TIMEOUT=10
  @NX_EBR_REINDEX_QUEUE_TIMEOUT=1
  @NX_EBR_INDEX_QUEUE_ONLINE=Yes
  @NX_EBR_NON_KD_INDEX_QUEUE_ONLINE=Yes
  ```

  This setting means that one batch processes 128 documents, that batch executions have ten second intervals, and when in reindex, the wait interval between two batches is one second.
- If the average size of your document (including attachments) is approximately 0.5 MB, keep the default setting in NX.env

  ```
  @EBR_MAX_INDEX_BATCH_SIZE=25
  @NX_EBR_INDEX_QUEUE_TIMEOUT=10
  @NX_EBR_REINDEX_QUEUE_TIMEOUT=10
  @NX_EBR_INDEX_QUEUE_ONLINE=No
  @NX_EBR_NON_KD_INDEX_QUEUE_ONLINE=No
  ```

  This setting means that one batch processes 25 documents, that batch executions have ten second intervals, and when in reindex, the wait interval between two batches is ten seconds.

PDM-Discovered Asset Commands

pdm_discupd -- Discovered Asset Update

Batch update of non-CA SDM Discovered Assets. Use this utility to update assets that were imported by the pdm_discimp command.

Syntax

```
pdm_discupd [-t] [-v] [-d domsrvr]
```

Where

- **t**
  Test
- **v**
  Verbose/diagnostic mode.
- **d**
  Object manager (domsrvr) to use for processing.

  WARNING
  On Non-Windows, the LIBPATH must be set before running several CA SDM utilities. Use *pdm_task* to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

pdm_discimp -- Discovered Asset Import

Batch registration of non-CA SDM Discovered Assets. Use this to search the MDB for assets that were registered by other software products and register them as CA SDM assets, so they can be used in CA SDM.

Logic is similar to Discovered Assets dialog that can be launched from Asset Search/List web form. That is an interactive and batch process.

This program will query the ca_logical_asset, ca_asset, and ca_logical_asset_property tables, using various parameters, and attempt to register new CA SDM Assets from the discovered values.

Syntax

```
pdm_discimp [-l label] [-s serial number] [-t asset tag] [-n hostname] [-d dns name] [-m mac address] [-c
 asset class] [-v] [-r] [-o object manager]
```

Asset selection criteria (use % for wild card):

- **l**
  Match this asset label.
- **s**
  Match this serial number.
- **t**
  Match this asset tag.
- **n**
  Match this hostname.

Asset property selection criteria (use % for wild card):

- **d**
  Match this dns name.
- **m**
  Match this mac address.

Other options:

- **c**
  Asset class to assign when registering new owned assets defaults to Discovered Hardware.
- **v**
  Verbose/diagnostic mode.
- **r**
  Register assets, otherwise runs in simulate mode.
- **h**
  Displays this information.
- **o**
  Object manager (domsrvr) to use for processing.

NOTE
If processing results in a blank Asset Label, the value found for the host name or DNS Name will be used as the Asset Label. Assets must have at least a Label and Asset Class to be registered for use in CA SDM.

NOTE
Because of the structure of the MDB and CA SDM architecture limitations, two queries are performed to select the appropriate records to process. It is important to understand this as it could affect performance. The first query retrieves the rows from a join between the ca_logical_asset and ca_asset tables that match label, serial number, tag and hostname. Then for each resulting row, a query is performed against ca_logical_asset_property to match dns_name and mac_address. The asset from the first query is chosen for registration if the second query results in rows being returned.

WARNING
On Non-Windows, the LIBPATH must be set before running several CA SDM utilities. Use *pdm_task* to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

pdm_text_cmd--Text API Command Line Interface

Use the pdm_text_cmd command for the Text API, which you can use to create and update various objects such as requests, change orders, issues, assets, and contacts.

> **WARNING**
> You cannot use a single or double quote as the parameter of the pdm_text_nxd or bop_cmd commands.

Syntax

This command has the following format:

```
pdm_text_cmd  - t table {-u from_userid  - p from_persid} [-o operation] [-f input file] [-T timeout] [-h]
```

-t *table*

(Required) Specifies the table to process. The *table* name can be one of the following values (not case sensitive):

- Asset
- Contact
- Change
- Issue
- Request

> **NOTE**
> See the [OPTIONS] section of the text_api.cfg file for a complete list of valid table names.

-u *from_userid* **| -p** *from_persid*

(One option required) Identifies the contact for this operation:

- **-u *from_userid***
  Identifies the contact using the User ID value.
- **-p *from_persid***
  Identifies the contact using the unique object identifier for the contact record. *from_persid* must be of the form **cnt:** *xxxx. xxxx* is the persistent ID of the object.

  > **NOTE**
  > The value that you specify with this option is appended to the end of the input for the pdm_text_cmd command using the appropriate keyword, %FROM_USERID or %FROM_PERSID.

-o *operation*

Specifies the operation to perform. The *operation* must be one of the following values (not case sensitive):

- NEW -- Creates an object. This value is the default if no operation is specified.
- UPDATE | UPD -- Creates an object if not found or updates an existing object if found.
- UPDATE_ONLY | UPDO -- Updates object if found; otherwise, does nothing.

Both UPDATE and UPDATE_ONLY require the %SEARCH keyword in the command input. You can perform only one operation transaction with each invocation of pdm_text_cmd.

-f *input_file*

Specifies the full path of the file to process, which is a text file containing valid Text API commands. If you omit this parameter, commands are used from STDIN. The Text API uses the following basic format for input:

```
%keyword=value
```

You can issue multiple commands within the input by separating the command request by at least five percent signs (%%%%%).

NOTE
For more information about valid keywords and about formatting input to the Text API, see the file text_api.cfg.

-T *timeout*

Specifies the number of seconds to wait for a response from the server before timing out. The default is 30 seconds.

NOTE
pdm_text_cmd shows the text-based replies received back from the Text API, which include success or error messages, and the original text sent using the API for processing. pdm_text_cmd returns zero if the command completes successfully without warnings or errors or one if the command completes successfully, but with warnings. Any other return value indicates that an error occurred.

WARNING
On Non-Windows, the LIBPATH must be set before running several CA SDM utilities. Use *pdm_task* to set the LIBPATH before running a utility. For example, input "pdm_task pdm_clean_attachments ...".

WARNING
When passing the parameters from command prompt, use Ctrl+Z in Windows and Ctrl+D in POSIX.

Input Examples

pdm_text_cmd is the command line interface for the Text API that you can use to create and update various objects such as requests, change orders, issues, assets, and contacts.

Example: Use an Input File to Create an Issue

The following example demonstrates how to use a pdm_text_cmd input file to create an issue:

```
%DESCRIPTION=This is my Test.
%PRIORITY=3
```

To process this file, assuming its full path is c:\input.txt, issue the following command:

```
pdm_text_cmd -t Issue  - u user01 -f c:\input.txt
```

Example: Use an Input File to Update an Issue

The following example demonstrates an input file to update issue 123 to a priority of 2:

```
%SEARCH=ISSUE_ID
%ISSUE_ID=123
%PRIORITY=2
```

To process this file, assuming its full path is c:\update.txt, issue the following command:

```
pdm_text_cmd -t Issue -o UPDATE_ONLY  - u user01 -f c:\update.txt
```

Example: Use an Input File to Create Multiple Requests

The following example demonstrates creating multiple requests with one input file. This command can be helpful creating test data on a test system.

```
%DESCRIPTION=This is Test 1.
%PRIORITY=3
%%%%%
%DESCRIPTION=This is Test 2.
%PRIORITY=2
%%%%%
%DESCRIPTION=This is Test 3.
%PRIORITY=None
```

To process this file, assuming its full path is c:\testdata.txt, issue the following command:

```
pdm_text_cmd -t Request  - u user01-f c:\testdata.txt
```

pdm_log4j_config Utility--Modify the log4j properties File

The pdm_log4j_config.pl utility lets you configure the log4j2 properties file of CA SDM, web components, PDM_RPC,Rest, and CMDB Visualizer. Execute the utility batch script that is based on your environment. For Windows, execute pdm_log4j_config from the command line. For Non-Windows, execute the pdm_log4j_config.sh file.

This command has the following format:

```
pdm_log4j_config  - f <component> -d
pdm_log4j_config -h
pdm_log4j_config  - f <component>  [-a | -n <name>] [-l <log level>] [I <max # of log files>] [-s <max size of
 log files>] [-t <log level threshold>]
```

- **-f**
  Specifies the log4j and log4j2 configuration of CA SDM or the component of CA SDM that you want to change. Enter one of the following values:
  SDM_WEB, SDM_RPC, REST, RPT, or Viz.
  Note: Use the mandatory option along with the other options.
- **-d**
  Displays the current log4j.properties and log4j2 properties configuration.
- **-h**
  Displays help for the utility.
- **-a**
  Completes all changes to log4j.properties and log4j2 properties globally.
- **-n**
  Specifies that you only want to modify a specific class or package name.
  Specify a specific class name, such as bop_logging, or a complete package name, such as com.ca.ServicePlus.
- **-l**
  Specifies the log level that you want to set.

  NOTE

  For the -l, -l, -s, and -t parameters, specify the -a or -n option.
- **-i**
  Specifies the max file number index that you want to set.
  -s
  Specifies the max file size that you want to set.

  WARNING
  Change the appender in the log4j2 properties file of Visualizer to Rolling File Appender before you execute the command with this parameter. If you do not change the appender, MaxFileSize generates logs in the same file.
- **-t**
  Specifies the log level threshold.

Utility Usage Examples

The following list provides examples of using the pdm_log4j_config utility:

- Modify
  Modify the log verbosity level for all loggers configured in the properties file by using the -l and -a variables.
  For example, set all of the loggers configured in REST to a level of DEBUG:

```
pdm_log4j_config  - f REST -a -l DEBUG
```

- Modify the log verbosity level for a specific logger class or package name in the CA SDM log4j2 properties file by using the -l and -n variables.
  For example, set the logger for the pdm_rpc package to DEBUG using one of the following code samples:
  ```
  pdm_log4j_config - f SDM_RPC -n pdm_rpc -l DEBG
            pdm_log4j_config - f SDM_RPC -n com.ca.ServicePlus.pdm_rpc -l DEBUG
  ```
- Modify the maximum number of log files to create for all the appenders (MaxBackupIndex property) by using the -i and -a variables in the log4j2 properties file of REST.
  For example, set the maximum number of files for all appenders to 9.
  ```
  pdm_log4j_config  - f REST -a -i 9
  ```
- Modify the maximum number of log files configured in the CA SDM log4j2 properties file to be created for an appender for a specific class or set of classes (MaxBackupIndex property) by using the -i and -n variables.
  For example, set the maximum number of files for bop_logging to 7.
  ```
  pdm_log4j_config  - f SDM_WEB -n bop_logging -i 7
  ```
- Modify the maximum size of each log file configured in the REST log4j2 properties file to create for any appender (MaxFileSize property) by using the -s and -a variables.
  For example, set the maximum size of files for all appenders to 9 MB.
  ```
  pdm_log4j_config  - f REST -a -s 9MB
  ```
- Modify the maximum size of each log file configured in the CA SDM log4j2 properties file to create for an appender for a specific class or set of classes (MaxFileSize property) by using the -s and -n variables.
  For example, set the maximum size of files for bop_logging to 7 MB.
  ```
  pdm_log4j_config  - f SDM_WEB -n bop_logging -s 7MB
  ```
- Modify the log level threshold for all the appenders configured in the CA Reporting log4j2 properties file (Threshold property) by using the -t and -a variables.
  For example, set the log level threshold to DEBUG.
  ```
  pdm_log4j_config  - f RPT -a -t DEBUG
  ```
- Modify the log level threshold for all the appenders configured in the CA Visualizer log4j2 properties file (Threshold property) by using the -t and -a variables.
  For example, set the log level threshold to DEBUG.
  ```
  pdm_log4j_config  - f VIZ -a -t DEBUG
  ```
- Modify the log level threshold for an appender for a specific class or set of classes (Threshold property) configured in the CA SDM log4j2 properties file by using the -t and -n variables.
  For example, set the log level threshold to WARN.
  ```
  ppdm_log4j_config  - f SDM_WEB -n bop_logging -t WARN
  ```

  NOTE

  The -t parameter log level threshold overrides the -l parameter log level. If you modify the log level and the threshold level, the DEBUG logs from the servlet do not appear in the file.
- Execute the following command to view the current logger and appender configuration of the REST log4j2 properties file.
  ```
  pdm_log4j_config  - f REST -d
  ```

  WARNING
  Use -l, -i, -s, and -t variables together with one of the -a or -n options, Do not use both the options. The -f option is mandatory. The -h and -d options are mutually exclusive to any other option.

Modify the Log File Refresh Interval Manually

Administrators can modify how often CA SDM monitors the log42j.properties file for changes. By default, the monitor interval is set to 60 seconds(present as monitorInterval attribute in log4j2.properties file). Change this value if desired.

Modify the jsrvr.log Appender

By default, servlets such as pdmweb, UploadServlet, and PDMReport log INFO level messages to the jsrvr.log file. You change the threshold level of the jsrvr.log appender to log any messages under the INFO level.

Follow these steps:

1. Modify the log level of UploadServlet:
   ```
   logger.uploadservlet.level=debug
   ```
2. Open the jsrvr.log file.
   ```
   logger.uploadservlet.level=debug
   ```
3. Confirm that the DEBUG log messages of UploadServlet appear.

   NOTE

   Not all servlets have explicit loggers attached. For example, log4j2.properties file does include pdmweb, PDMExport, PDMReport, and pdm_cache, which are part of the pdmweb servlet. To see DEBUG logs from these servlets, modify the pdmweb log level.

Modify the jstd.log Appender

All logs from nonwebapp applications dump into the jstd.log file separately. You can display logs for any of these applications, such as pdm_rpc by changing the log level of that specific application.

Follow these steps:

1. Modify the following log level:
   ```
   logger.rpc.level = DEBUG
   ```
2. Open the jstd.log file to confirm that the log entries appear.

CA SDM Report Command

This article contains the following topics:

rpt_srv--Generate Reports

Valid for Windows only

The report program lets you generate a report from the command line on the server. To issue the report command at the command line or in a script, you must include pdm_task. The pdm_task command sets up environment variables for commands that do not have a wrapper. Enter pdm_task with the report command on the same command line only when the report is invoked through a script or the command line. If you issue the report command from a menu, you do not need to include pdm_task, because all environment variables are set by the application.

Syntax

This command has the following format:

```
rpt_srv  - m [-h] [-e] [-f] [-F ffstring] [-p pagelength] [-C] [-B] filename [ command line arguments]
```

- **-m**
  Signifies that the report is manually being run from the command line.
- **-e**
  Echoes compiled script (for debugging purposes).
- **-f**
  Uses form feeds between pages.
- **-F *ffstring***
  Sets the optional form-feed string.
- **-p *pagelength***

Sets the page length. The default page length is 66.

- **-C**
  Changes encoding from UTF-8 to another charset. The default output is UTF-8.
  Example: To convert the output to JIS, you would run "-C iso-2022-jp"
  Example: To encode to the operating system's native charset, use "DEFAULT" or "NATIVE".
- **-B**
  Suppresses the Byte Order Mark if the variable NX_ADD_UTF8_BYTE_ORDER_MARK is set.
  The NX_ADD_UTF8_BYTE_ORDER_MARK option is a signature into a file. It allows editors that support UTF-8 to maintain the UTF-8 integrity of the file.

  NOTE
  This is only needed for non-ASCII data. If this is not installed, the default behavior omits the Byte Order Mark (BOM). If installed, set it to "1" or "Yes".
- ***filename***
  Specifies the report template. If you are not running the report command from the directory in which the template file is located, include the complete file path name. The command sends the output as standard output (stdout).
- **command line arguments**
  Specifies parameters received by the report template. If the report is designed to accept command line arguments, you must enter a command line argument for each parameter in the report template. If the argument is empty, enter the null string.
  For example, the following command supplies the command line arguments Smith, Jane, and L. The report template requires these three parameters to generate the Affected Contacts Report.
  For example, enter the following command:

  ```
  rpt_srv  - m c:\reports\affected.rpt Smith Jane L
  ```

  In the next example, Jane Smith does not have a middle initial:

  ```
  rpt_srv  - m c:\reports\affected.rpt Smith Jane "
  ```

report--Generate Reports

Applies to UNIX only

The report program lets you generate a report from the command line on the server. To issue the report command at the command line or in a script, you must include pdm_task. The pdm_task command sets up environment variables for commands that do not have a wrapper. Enter pdm_task with the report command on the same command line *only* when the report is invoked through a script or the command line. If you issue the report command from a menu, you do not need to include pdm_task, because all environment variables are set by the application.

Syntax

This command has the following format:

```
pdm_task report [-h] [-e] [-f] [-F ffstring] [-p pagelength] filename [ command line arguments]
```

-e

Echoes compiled script (for debugging purposes).

-f

Uses form feeds between pages.

-F ffstring

Sets the optional form-feed string.

-p pagelength

Sets the page length. The default page length is 66.

filename

The report template. If you are not running the report command from the directory in which the template file is located, include the complete file path name. The command sends the output as standard output (stdout).

command line arguments

Specifies parameters received by the report template. If the report is designed to accept command line arguments, you must enter a command line argument for each parameter in the report template. If the argument is empty, enter the null string.

For example, the following command supplies the command line arguments Smith, Jane, and L. The report template requires these three parameters to generate the Affected Contacts Report.

For example, enter:

```
pdm_task report /opt/CAisd/samples/sdk/reports/affected.rpt
```

In the next example, Jane Smith does not have a middle initial:

```
pdm_task report /opt/CAisd/samples/sdk/reports/affected.rpt Smith Jane "
```

CA SDM Form Groups

This article contains the following topics:

Customer Forms Group

The following web forms are included in the Customer forms groups:

-
- about.htmpl
-
- bin_form_np.htmpl
-
- chg_lr.htmpl
-
- cr_lr.htmpl
-
- detail_iss.htmpl
-
- detail_issalg.htmpl
-
- detail_KD.htmpl
-
- generic.htmpl
-
- home.htmpl
-
- iss_lr.htmpl
-
- issue_status_change.htmpl
-
- list_iss.htmpl
-
- list_isscat.htmpl
-
- list_KD.htmpl
-
- menu_frames.htmpl
-
- std_body.htmpl
-
- std_body_site.htmpl
-
- std_footer.htmpl
-
- std_footer_site.htmpl
-
- std_head.htmpl
-
- std_header.htmpl
-
- std_head_site.htmpl

Employee Forms Group

The following web forms are included in the Employee forms groups:

-
- about.htmpl
-
- bin_form_np.htmpl
-
- buttons.htmpl
-
- change_status_change.htmpl
-
- chg_lr.htmpl
-
- cr_lr.htmpl
-
- detail_alg.htmpl
-
- detail_chg.htmpl
-
- detail_chgalg.htmpl
-
- detail_cr.htmpl
-
- detail_in.htmpl
-
- detail_KD.htmpl
-
- generic.htmpl
-
- home.htmpl
-
- iss_lr.htmpl
-
- list_chg.htmpl
-
- list_chgcat.htmpl
-
- list_cr.htmpl
-
- list_in.htmpl
-
- list_KD.htmpl
-
- list_pcat.htmpl
-
- list_pcat_cr.htmpl
-
- list_pcat_in.htmpl
-
- menu_frames.htmpl
-
- request_status_change.htmpl
-
- show_error.htmpl
-

Analyst Forms Group

The following web forms are included in the Analyst forms groups:

A

-
- about.htmpl
-
- acctyp_role_tab.htmpl
-
- acctyp_web_auth_tab.htmpl
-
- acctyp_wsp_tab.htmpl
-
- admin_empty.htmpl
-
- admin_main_role.htmpl
-
- admin_tab_dflt.htmpl
-
- admin_tree.htmpl
-
- attmnt_content_tab.htmpl
-
- attmnt_fields.htmpl
-
- attmnt_permissions_tab.htmpl
-
- attmnt_upload_popup.htmpl
-
- att_mgs_event.htmpl
-
- att_stype_event.htmpl
-
- aty_chg_ntfr_tab.htmpl
-
- aty_chg_svy_tab.htmpl
-
- aty_cr_ntfr_tab.htmpl
-
- aty_cr_svy_tab.htmpl
-
- aty_iss_ntfr_tab.htmpl
-
- aty_iss_svy_tab.htmpl
-
- aty_kdComment_ntfr_tab.htmpl
-
- aty_kd_ntfr_tab.htmpl
-
- aty_mgs_ntfr_tab.htmpl

B

-
- bhvtpl_todo_tab.htmpl
-
- bhvtpl_trans_info_tab.htmpl
-
- bin_form_np.htmpl

C

-
- cancel.htmpl
-
- cancel_empty.htmpl
-
- category_content_tab.htmpl
-
- category_permissions_tab.htmpl
-
- chgcat_auto_assignment_tab.htmpl
-
- chgcat_prptpl_tab.htmpl
-
- chgcat_wftpl_tab.htmpl
-
- chg_accumulate.htmpl
-
- chg_causedreq_tab.htmpl
-
- chg_close_all_child.htmpl
-
- chg_expedite.htmpl
-
- chg_lr.htmpl
-
- chg_relchg_tab.htmpl
-
- chg_relreq_tab.htmpl
-
- cia_bmhier_tab.htmpl
-
- cia_export_bmhier.htmpl
-
- cia_export_nr.htmpl
-
- cia_nr_tab.htmpl
-
- cia_pwd_tab.htmpl
-
- cia_sync_stat.htmpl
-
- cnote_tracker.htmpl
-
- cnt_addr_tab.htmpl
-
- cnt_auto_assignment_tab.htmpl
-
- cnt_env_tab.htmpl
-

- cnt_grp_tab.htmpl
-
- cnt_mem_tab.htmpl
-

D

-
- dcon_constraint_tab.htmpl
-
- dcon_sql_tab.htmpl
-
- detail.template
-
- detail_acctyp.htmpl
-
- detail_act_type_assoc.htmpl
-
- detail_ADMIN_TREE.htmpl
-
- detail_alg.htmpl
-
- detail_arcpur_rule.htmpl
-
- detail_asset.htmpl
-
- detail_atev.htmpl
-
- detail_atomic_cond.htmpl
-
- detail_attmnt_edit.htmpl
-
- detail_attmnt_error.htmpl
-
- detail_attmnt_folder.htmpl
-
- detail_attmnt_ro.htmpl
-
- detail_attr_alias.htmpl
-
- detail_aty.htmpl
-
- detail_audlog.htmpl
-
- detail_bhvtpl.htmpl
-
- detail_bmcls.htmpl
-
- detail_bmhier.htmpl
-
- detail_bmrep.htmpl
-
- detail_BU_TRANS.htmpl
-
- detail_ca_cmpny.htmpl
-

- detail_chg.htmpl
-
- detail_chgalg.htmpl
-

G

-
- generic.htmpl
-
- get_comment.htmpl
-
- g_profile_browser.htmpl
-
- g_profile_browser2.htmpl
-
- g_profile_browser3.htmpl
-
- g_profile_browser_frameset.htmpl
-
- g_profile_jump.htmpl
-
- g_profile_scratchpad.htmpl

H

-
- hierload_admin_tree.htmpl
-
- hierload_KCAT.htmpl
-
- hiersel_admin_tree.htmpl
-
- hiersel_KCAT.htmpl
-
- hourglass.htmpl
-
- html_editor_create_change_order.htmpl
-
- html_editor_create_ticket.htmpl
-
- html_editor_frames.htmpl
-
- html_editor_insert_image.htmpl
-
- html_editor_insert_link.htmpl
-
- html_editor_insert_table.htmpl
-
- html_editor_tabs.htmpl
-
- html_editor_toolbar.htmpl

I

-
- insert_iss_wf.htmpl
-
- insert_wf.htmpl
-
- in_relreq_tab.htmpl
-
- isscat_auto_assignment_tab.htmpl
-
- isscat_prptpl_tab.htmpl
-
- isscat_wftpl_tab.htmpl
-
- issue_status_change.htmpl
-
- iss_accumulate.htmpl
-
- iss_close_all_child.htmpl
-
- iss_custfld_tab.htmpl
-
- iss_expedite.htmpl
-
- iss_lr.htmpl
-
- iss_reliss_tab.htmpl
-
- iss_resol_tab.htmpl

K

- kd_action_forward.htmpl
- kd_action_publish.htmpl
- kd_action_reject.htmpl
- kd_action_unpublish.htmpl
- kd_action_unretire.htmpl
- kd_attachments_tab.htmpl
- kd_attributes_tab.htmpl
- kd_categories_tab.htmpl
- kd_content_tab.htmpl
- kd_file_prop_tab.htmpl
- kd_permissions_tab.htmpl
- kd_qa_attributes_tab.htmpl
- kd_qa_content_tab.htmpl
- keit_tmpl_export_fields_tab.htmpl
- keit_tmpl_export_filter_tab.htmpl
- keit_tmpl_import_settings_tab.htmpl
- keit_tmpl_name_tab.htmpl
- kt_admin_attachments.htmpl
- kt_admin_automated_policies.htmpl
- kt_admin_document_settings.htmpl
- kt_admin_faq_options.htmpl
- kt_admin_general_settings.htmpl
- kt_admin_integration.htmpl
- kt_admin_knowledge.htmpl
- kt_admin_parse_settings.htmpl
- kt_admin_report_card.htmpl
- kt_admin_search_config_ar.htmpl

L

-
- list.template
-
- list_acctyp.htmpl
-
- list_act_type_assoc.htmpl
-
- list_alg.htmpl
-
- list_all_fmgrp.htmpl
-
- list_all_lr.htmpl
-
- list_architect_KDs.htmpl
-
- list_architect_KDs_Pref.htmpl
-
- list_arcpur_hist.htmpl
-
- list_arcpur_rule.htmpl
-
- list_atev.htmpl
-
- list_atomic_cond.htmpl
-
- list_attmnt.htmpl
-
- list_attr_alias.htmpl
-
- list_aty.htmpl
-
- list_audlog.htmpl
-
- list_bmcls.htmpl
-
- list_bmhier.htmpl
-
- list_bmrep.htmpl
-
- list_bm_task.htmpl
-
- list_bool.htmpl
-
- list_ca_cmpny.htmpl
-
- list_ca_logical_asset.htmpl
-
- list_chg.htmpl
-

- list_chgalg.htmpl
-
- list_chgcat.htmpl
-

M

-
- macro_atomic_cond_tab.htmpl
-
- macro_cnt_tab.htmpl
-
- macro_ctp_tab.htmpl
-
- macro_ntfl_tab.htmpl
-
- macro_rrf_tab.htmpl
-
- mactyp_exescript_tab.htmpl
-
- mactyp_valscript_tab.htmpl
-
- mapped_contracts_tab.htmpl
-
- menubar.template
-
- menubar_admin.htmpl
-
- menubar_architect.htmpl
-
- menubar_chg_sched.htmpl
-
- menubar_dtbuilder.htmpl
-
- menubar_html_editor.htmpl
-
- menubar_kt.htmpl
-
- menubar_no.htmpl
-
- menubar_sd.htmpl
-
- menubar_sd_chg_manager.htmpl
-
- menubar_sd_cust_mgr.htmpl
-
- menubar_sd_cust_rep.htmpl
-
- menubar_sd_hd_manager.htmpl
-
- menubar_sd_inc_manager.htmpl
-
- menubar_sd_know_analyst.htmpl
-
- menubar_sd_know_manager.htmpl
-

- menubar_sd_l1_analyst.htmpl
-
- menubar_sd_l2_analyst.htmpl
-

N

-
- new_lr.htmpl
-
- nf.htmpl
-
- nosession.htmpl
-
- nr_bm_tab.htmpl
-
- nr_chg_tab.htmpl
-
- nr_contact_tab.htmpl
-
- nr_inc_tab.htmpl
-
- nr_inv_tab.htmpl
-
- nr_iss_tab.htmpl
-
- nr_loc_tab.htmpl
-
- nr_log_tab.htmpl
-
- nr_org_tab.htmpl
-
- nr_prb_tab.htmpl
-
- nr_projex_tab.htmpl
-
- nr_rel_tab.htmpl
-
- nr_reqitil_tab.htmpl
-
- nr_req_tab.htmpl
-
- nr_serv_tab.htmpl
-
- ntfl_ntfr_tab.htmpl
-
- ntfr_aty_tab.htmpl
-
- ntfr_cnt_tab.htmpl
-
- ntfr_ctp_tab.htmpl
-
- ntfr_ntfl_tab.htmpl

O

-
- order_status_change.htmpl
-
- org_address_tab.htmpl
-
- org_env_tab.htmpl

P

-
- pcat_auto_assignment_tab.htmpl
-
- pcat_prptpl_tab.htmpl
-
- pcat_wftpl_tab.htmpl
-
- power_user_tips.htmpl
-
- profile_browser.htmpl
-
- profile_browser2.htmpl
-
- profile_browser3.htmpl
-
- profile_browser_frameset.htmpl
-
- profile_envcnt.htmpl
-
- profile_envorg.htmpl
-
- profile_histcnt_chg.htmpl
-
- profile_histcnt_cr.htmpl
-
- profile_histcnt_in.htmpl
-
- profile_histcnt_iss.htmpl
-
- profile_histcnt_pr.htmpl
-
- profile_historg_chg.htmpl
-
- profile_historg_cr.htmpl
-
- profile_historg_in.htmpl
-
- profile_historg_iss.htmpl
-
- profile_historg_pr.htmpl
-
- profile_infocnt.htmpl
-
- profile_infoorg.htmpl
-
- profile_menu.htmpl
-
- profile_qtemplate.htmpl
-

- pr_attinc_tab.htmpl
-
- pr_relreq_tab.htmpl

R

-
- reports.htmpl
-
- reports.htmpl.tpl
-
- request_status_change.htmpl
-
- role_auth_tab.htmpl
-
- role_fnacc_tab.htmpl
-
- role_goform_tab.htmpl
-
- role_kt_ct_tab.htmpl
-
- role_kt_docs_tab.htmpl
-
- role_webform_tab.htmpl
-
- role_web_interface_tab.htmpl

S

-
- sapolicy_ac_tab.htmpl
-
- sapolicy_pt_tab.htmpl
-
- saprobtyp_dh_tab.htmpl
-
- saprobtyp_rd_tab.htmpl
-
- scoreboard.htmpl
-
- scratchpad.htmpl
-
- screen_reader_usage.htmpl
-
- sdsc_chg_slatpl_tab.htmpl
-
- sdsc_chg_wf_slatpl_tab.htmpl
-
- sdsc_cr_slatpl_tab.htmpl
-
- sdsc_iss_slatpl_tab.htmpl
-
- sdsc_iss_wf_slatpl_tab.htmpl
-
- sdsc_map_cnt_tab.htmpl
-
- sdsc_map_grp_tab.htmpl
-
- sdsc_map_nr_tab.htmpl
-
- sdsc_map_pri_tab.htmpl
-
- sdsc_map_urg_tab.htmpl
-
- sd_kt_admin.htmpl
-
- sd_main.htmpl
-
- sd_main_role.htmpl
-
- search_child_KCATs_filter.htmpl
-
- show_error.htmpl
-
- show_main_detail.htmpl
-
- std_body.htmpl
-

- std_body_site.htmpl
-
- std_footer.htmpl
-
- std_footer_site.htmpl

T

-
- tab_detail.template
-
- tenant_address_tab.htmpl
-
- tenant_groups_tab.htmpl
-
- tenant_group_members_tab.htmpl
-
- tskty_tskstat_tab.htmpl

U

-
- update_lrel_bmrep.htmpl
-
- update_lrel_chg.htmpl
-
- update_lrel_chgcat.htmpl
-
- update_lrel_cnt.htmpl
-
- update_lrel_cr.htmpl
-
- update_lrel_ctp.htmpl
-
- update_lrel_goform.htmpl
-
- update_lrel_help_content.htmpl
-
- update_lrel_in.htmpl
-
- update_lrel_iss.htmpl
-
- update_lrel_isscat.htmpl
-
- update_lrel_loc.htmpl
-
- update_lrel_macro.htmpl
-
- update_lrel_nr.htmpl
-
- update_lrel_ntfl.htmpl
-
- update_lrel_ntfr.htmpl
-
- update_lrel_org.htmpl
-
- update_lrel_pcat.htmpl
-
- update_lrel_pr.htmpl
-
- update_lrel_role.htmpl
-
- update_lrel_tab.htmpl
-
- update_lrel_tenant.htmpl
-
- update_lrel_tenant_group.htmpl
-
- update_lrel_tskstat.htmpl
-

- update_lrel_webform.htmpl
-
- update_lrel_wrkshft.htmpl
-
- upd_chg_sched.htmpl

V

-
- v30_date_helper.htmpl

W

-
- wfdef.htmpl
-
- wftpl_auto_assignment_tab.htmpl
-
- wftpl_bhvtpl_tab.htmpl
-
- working.htmpl
-
- workitems.htmpl
-
- wrkshft_auto_assignment_tab.htmpl
-
- wrkshft_schedule_tab.htmpl
-
- wspmain.htmpl

X

-
- xfer_esc_chg.htmpl
-
- xfer_esc_cr.htmpl
-
- xfer_esc_iss.htmpl
-
- xx_attmnt_tab.htmpl
-
- xx_candp_tab.htmpl
-
- xx_nr_tab.htmpl
-
- xx_prop_tab.htmpl
-
- xx_solnalg_tab.htmpl
-
- xx_stype_tab.htmpl
-
- xx_template_tab.htmpl
-
- xx_wf_tab.htmpl

Contents of the Samples Directory

This article contains the following topics:

You can modify several files in the $NX_ROOT/samples directory for use with various external interfaces. These files are grouped into various subdirectories. None of the files in the samples directory are executable as originally shipped.

How to Modify the Message Catalog

To modify the message catalog, complete the following steps:

1. Refer to the format of pdm.xml that is located in $nx_root\bin
2. Create a customized copy of pdm.xml and place it in the $nx_root\msg_catalog directory
3. Add, modify, or add and modify messages in the XML message files from the previous step.

call_mgt

Contains samples for customization in request management:

- **gencr.frg**
  This file can be used in conjunction with bop_cmd to create requests from a command line. All notifications and activity log entries will occur, however no Request Form will display on the server when created. You must use the -u parameter to execute gencr.frg with the bob_cmd utility. Be sure to read the gencr_readme.txt file to learn the syntax, and how to modify it if necessary. The file should be placed in $NX_ROOT/site/mods/interp, if the directory does not exist, you should create it. Example: bop_cmd -d domsrvr -u nsm -f gencr.frg "gencr('My Description')"
- **iss_site.mod**
  This file can be used to enable activity logging of site-adapted fields in issues. This file should be placed in $NX_ROOT/site/mods/majic after it has been changed for the site-adapted fields.
- **cr_site.mod**
  This file can be used to enable activity logging of site-adapted fields in requests. This file should be placed in $NX_ROOT/site/mods/majic after it has been changed for the site-adapted fields.
- **chg_site.mod**
  This file can be used to enable activity logging of site-adapted fields in change orders. This file should be placed in $NX_ROOT/site/mods/majic after it has been changed for the site-adapted fields.
- **gencr_readme.txt**
  This file contains instructions on how to use the gencr.frg file.
- **chg_site.spl**
  This file may be modified to change the mapping of attributes when creating a change order from a request. This file should be placed in $NX_ROOT/site/mods/majic after the appropriate changes have been made.
- **audlog_site.mod**
  This file can be used to enable audit logging of site-adapted fields. This file should be placed in $NX_ROOT/site/mods/majic after it has been changed for the site-adapted fields
- **Notify_add.spl (UNIX only)**
  This file can be used to add the request's log agent, assignee and group to the request notification list. This file should be placed in $NX_ROOT/site/mods/majic
- **Notify_replace.spl (UNIX only)**
  This file can be used to add the request's log agent, assignee and group to the request notification list when they are changed. This file should be placed in $NX_ROOT/site/mods/majic.

data

This directory contains files depicting the Logical Data Model for most database tables in CA SDM. This data directory also includes sample data for Knowledge Management.

macro_lock

This file contains a spel fragment that can be run using a bop_cmd to turn off locks that are being held by macros.

multi-tenancy

This directory contains sample files for administering multi-tenancy.

pdmconf

- **web.xml.tpl**
- **pdm_startup.tpl**
- **pdm_edit_usage_notes.htm**
- **alias_install.bat**
- **web.cfg.tpl**
- **pdm_startup.i.tpl**
- **pdm_edit.pl**
- **README_files** All of these files are used by pdm_edit.pl to create startup files for a primary server and secondary servers that are configured to run a variety of daemons.

reporting

This directory contains documentation and samples for configuring offline reporting.

sdk for Axis

This directory contains a sample file for making CA SDM SOAP web service calls.

TableOfContents.doc further explains what is available, found in the following directory:

$NX_ROOT\samples\sdk\websvc

PKI_loginServiceManaged_JAVA_steps.doc and PKI_loginServiceManaged_PERL_steps.doc explain how to configure ServiceDesk for digital certificate logins in the following directory:

$NX_ROOT\samples\sdk\websvc\java\test1_pki

The following lists PERL and JAVA samples, examples, and locations:

- PERL samples: $NX_ROOT\samples\sdk\websvc\perl\test1_pki
  Example: loginServiceManaged() web service call
- JAVA samples $NX_ROOT\samples\sdk\websvc\java\test1_pki
  Example: loginServiceManaged() and getBopsid() web service call
  $NX_ROOT\samples\sdk\websvc\java\test2_basic
  Example: Combined CreateRequest() and CreateChangeOrder() web service call

sdk for CXF

This directory contains a sample file for making CA SDM SOAP web service calls.

> **NOTE**
> Before executing the scripts present in the samples folder, follow these steps:
>
> 1. Download the apache-cxf-3.4.5.zip file from https://archive.apache.org/dist/cxf/3.4.5 and extract the contents of the zip file.
> 2. Navigate to $NX_ROOT\samples\sdk\cxfwebsvc\java.

3. Edit the run_java_test.bat file and search for CXFLIBPATH. The default folder is C:\apache-cxf-3.4.5\lib or you can place the file in a directory of your choice.

TableOfContents.doc further explains what is available, found in the following directory:

$NX_ROOT\samples\sdk\cxfwebsvc

PKI_loginServiceManaged_JAVA_steps.doc explains how to configure ServiceDesk for digital certificate logins in the following directory:

$NX_ROOT\samples\sdk\cxfwebsvc\java\test1_pki

The following are the JAVA samples, examples, and locations:

- JAVA samples $NX_ROOT\samples\sdk\cxfwebsvc\java\test1_pki
  Example: loginServiceManaged() and getBopsid() web service call
  $NX_ROOT\samples\sdk\cxfwebsvc\java\test2_basic
  Example: Combined CreateRequest() and CreateChangeOrder() web service call

views

This directory contains database scripts uses for migration, multi-tenancy administration, and other processes.

Load Supplemental Content - sd_content.dat

Supplemental content for CA SDM is available in sd_content.dat. This data file contains Change Category and Root Cause records. To load the data from a command window, go to $NX_ROOT/data and run the following command:

```
pdm_load  - f sd_content.dat
```

Schema Files Syntax

This article contains the following topics:

The CA SDM database schema is defined in multiple .sch files in the $NX_ROOT/site (UNIX) or *installation-directory*/site (Windows) directory. During configuration, these .sch files along with any customized .sch files you may create are merged together into a single file called $NX_ROOT/site /ddict.sch (UNIX) or *installation-directory*/site/ddict.sch (Windows).

Review Using the Web Screen Painter prior to making any changes to these files.

> **NOTE**
> Do not modify any .sch files in the $NX_ROOT/site (UNIX) or *installation-directory*/site (Windows) directory. Any changes you make to these files will be lost when upgrading to a new release or when certain patches are applied. If you want to make schema changes, you must create a file in the $NX_ROOT/site/mods (UNIX) or *installation-directory*/site/mods (Windows) directory with the file suffix .sch. Then add your changes to your .sch file, and your changes will then be merged with delivered schema during configuration. This is the only way to preserve your modifications when you upgrade to a new release.

TABLE Statement

Defines the logical tables in the CA SDM database schema and the logical columns (fields) in those tables. These logical tables and columns are then mapped to the physical tables and columns used by your database management system in a mapping statement that follows the TABLE statement.

> **NOTE**
> If you define a new table, you must define a mapping statement for that table. The Mapping Statement is illustrated at the end of this chapter, followed by an example that combines the TABLE, TABLE_INFO, and mapping statements.

Syntax

TABLE *table_name* {*field value_type field_attributes*; [...]}

Arguments

- **TABLE**
  Introduces the TABLE statement. Must be uppercase. You must have one TABLE statement for each logical table in the schema.
- ***table_name***
  The name of the database table, for example, Call_Req. If adding a database table, you can specify any name beginning with a lowercase letter z. (This avoids possible conflict with existing and future CA SDM table names.) If changing an existing table, find the table in one of the .sch files and use the same name.
- ***field***
  The name of a logical column in the table, for example, id or desc. You must identify each column by name. If adding a table or adding a column to an existing table, you can specify any name beginning with a lowercase letter z; however, field names must not end with the characters "_f." (This avoids possible conflict with existing and future CA SDM column names.) If changing an existing column, find the column in one of the .sch files and use the same name.
- ***value_type***
  The field's data type. Valid values are:

Value	Description
STRING *nn*	A string that is *nn* characters long.
INTEGER	A 32-bit number.
LOCAL_TIME	The number of seconds since January 1, 1970. CA SDM automatically reformats this data type to the designated date format, for example: *mm*/*dd*/*yy hh*:*mm*:*ss*.
DURATION	A period of time, measured in seconds.
REAL	A floating point number
UUID	A 16 byte binary value.

field_attributes

A description of the field. Valid values are:

Value	Description
KEY	Identifies this field as the primary key to be used for identifying records to be updated with pdm_load. This is used if the default primary key, id, is not specified. Must be specified if the field is the primary key in the table.
NOT_NULL	Indicates that the field must contain a value. Must be specified if the field is the primary key in the table. Optional if the field is not the primary key.
REF other_table_name	Indicates that the field references another table. Optional whether the field is the primary key or not.
S_KEY	Optionally identifies this field as the secondary key to be used for identifying records to be updated with pdm_load.

UNIQUE	Indicates that the values in the field must be unique. Must be specified if the field is the primary key in the table. Optional if the field is not the primary key.

Macros are synonyms that will be converted during configuration to the value the macro represents. You can use macros for either data types or attributes. If you wish to use macros, you must add in #include statement to include the file that defines the macro including the path name (usually relative to your schema file). The include statement must be defined prior to using the macro. Example of an include statement:

```
#include "../schema.mac"
```

The following are some of the macros defined in .mac files located in the $ NX_ROOT/site (Non-Windows) or *installation-directory*/site (Windows) directory.

Data Type	Equivalent
nn	NOT_NULL
uniq	UNIQUE NOT_NULL
ADDR_LINE	STRING 30
EMAILADDR	STRING 120
ENT_DESC	STRING 40
ENT_NAME	STRING 30
OSI_NAME	STRING 80
OSI_TYPE_STRING	STRING 60
USERID	STRING 85
PHONENUM	STRING 32
SYMBOL	STRING 12
HIER_SYM	STRING 60
LONG_SYM	STRING 30
COMMENT	STRING 1000
LONG_STR	STRING 500
LONG_DESC	STRING 240
BOOL	INTEGER

Examples

This TABLE statement in the database schema defines severities. The macro nn indicates that a value is required in the del field. The macro uniq indicates that values are required and must be unique:

```
#include :../schema.mac"
TABLE Severity {
   id        INTEGER uniq KEY;    // key id
   del       INTEGER nn;          // 0=present,1=gone
   sym       SYMBOL uniq S_KEY; // type symbol
   desc      ENT_DESC;            // non-OSI specified column
}
```

This modified TABLE statement makes the Priority field on the Request Detail window required:

```
TABLE Call_Req {priority INTEGER NOT_NULL;}
```

This TABLE statement adds a resolution code field to the Call_Req table. The content of the field is numeric and references the Resolution_Code table. This reference allows users to double-click the Resolution Code field on the Request Detail window to display the values in the Resolution_Code table:

```
TABLE Call_Req {zres_code INTEGER REF Resolution_Code;}
```

TABLE_INFO Statement

This instructs your database management system how to store and index data in the logical tables. The extent to which these instructions are followed depends on the database management system. If no instructions are provided, the database management system follows its own storage and indexing instructions.

Syntax

```
TABLE_INFO table_name {
```

[STORAGE *storage_mtd Field* ;] [INDEX *ndx_props field1* [*field2* ...];] ...}

Arguments

- **TABLE_INFO**
  Introduces the TABLE_INFO statement. Must be uppercase. The TABLE_INFO statement is optional, but if specified, you can have only one TABLE_INFO for each TABLE statement, and it must follow the TABLE statement.
- ***table_name***
  The name of the database table in the TABLE statement.
- **STORAGE *storage_mtd***
  Identifies the storage method. Valid values are listed as follows, but note that some database management systems ignore these values:

Value	Description
BTREE	Indicates to use the balanced tree storage method.
HASH	Indicates to use the hash table storage method. This is valid only if the field is the primary key.
HEAP	Indicates to use the heap storage method.

- ***field***
  Identifies the column that is to be stored according to the specified storage method (STORAGE *storage_mtd*). Must be specified the same way as the name of the column in the TABLE statement.
- **INDEX *ndx_props***
  Identifies one or more properties for an index that consists of the fields specified. Valid values are:

Value	Description
SORT ASCENDING | DESCENDING	Indicates whether to sort the data in the fields in ascending or descending order. Data is sorted in ascending order by default; therefore, only SORT DESCENDING need be specified.
PRIMARY	Indicates to use this index as the default sort order for the table.
CLUSTER	Identifies this as a clustering index.
UNIQUE	Indicates that values in the index must be unique.

- ***field1* [*field2 . . .*]**

Identifies the column or columns that are to be indexed according to the specified index properties (INDEX *ndx_props*). Must be specified the same way as the name of the columns in the TABLE statement.

Examples

This TABLE_INFO statement instructs the database management system to use a hash table to store values in the id field in the Contact_Type table, and to sort the table in descending order according to the values in the sym field. It also indicates that values must be unique:

```
TABLE_INFO Contact_Type {
```

STORAGE HASH id; INDEX SORT DESCENDING PRIMARY UNIQUE sym;}

Mapping Statement

Defines the correspondence between the logical tables and columns in the CA SDM database schema and the physical tables and columns used by your database management system. This statement follows each TABLE statement in a.sch file. You must define it when you define a new table.

Syntax

```
p1 logical_table_name -> CURR_PROV physical_table_name
   [{logical_field -> physical_field ...] ;
```

Arguments

- **p1**
  Introduces the mapping statement. Must be specified as p1.
- ***logical_table_name***
  The name of the database table in the TABLE statement, for example, zManufacturer.
- **CURR_PROV**
  A required keyword.
- ***physical_table_name***
  The name of the table used by your database management system, for example, man. Short names improve performance and are required by some database management systems.
- ***logical_field***
  The name of the column in the CA SDM database schema, for example, desc. Must be the same as *field* in the TABLE statement. Omit this when the logical columns and physical columns have identical names. When omitted, the semicolon follows *physical_table_name*.
- ***physical_field***

The name of the column used by your database management system, for example, nx_desc. Omit this when the logical columns and physical columns have identical names. When omitted, the semicolon follows *physical_table_name*.

Examples

This example illustrates how TABLE, mapping (p1), and TABLE_INFO statements define a zManufacturer table:

```
TABLE zManufacturer {
        id       INTEGER  uniq KEY;       // key id
        del      INTEGER  nn;             // 0=present,1=gone
        sym      HIER_SYM uniq S_KEY;     // manufacturer name
        desc     ENT_DESC;                // manufacturer description
}

p1 zManufacturer -> CURR_PROV man   // maps logical table "zManufacturer"
{                                  // to physical table "man"
        desc -> nx_desc;           // maps logical column "desc"
```

```
}                                   // to physical column "nx_desc"

TABLE_INFO zManufacturer {
```

STORAGE HASH id; INDEX SORT ASCENDING PRIMARY UNIQUE sym;}

Object Definition Syntax

This article contains the following topics:

Many of the components of CA SDM consist of business objects. These objects are defined in a metalanguage named Majic. You can use Majic statements to create new objects and modify existing objects, thus customizing these objects to meet your needs.

Directories

The Majic files are organized in two directories:

Directory	Description
bopcfg/majic (UNIX) or bopcfg\majic (Windows)	Contains the .maj files that have been used to create windows defined in the database. These files should not be changed because changes will be overwritten by new releases of CA SDM.
site/mods/majic (UNIX) or site\mods\majic (Windows)	Contains the .mod files you use to customize windows.

Types of Statements

The following Majic statements are used in screen painter and database customization procedures.

Statement	Description
OBJECT	Defines a business object
MODIFY	Changes existing object attributes
MODIFY FACTORY	Changes existing factories

MODIFY Statement

Changes the way attributes are defined on OBJECT statements. MODIFY statements are read after OBJECT statements.

Syntax

```
MODIFY obj_name att_name [status_type;]
   [ON_NEW DEFAULT|SET value|NOW ;]|
   [ON_CI DEFAULT|SET value|NOW ;]|
   [ON_DB_INIT DEFAULT|SET value|NOW ;]
```

Arguments

-
- ***obj_name*** Identifies the object whose attribute is being modified.
-
- ***att_name***
  Identifies the attribute being modified.
-
- ***status_type***

Modifies the properties of the attribute to allow or prohibit null values. There are two valid options for this keyword:

- –
- – **REQUIRED**
  Indicates that the attribute is required.
- –
- – **NOT_REQUIRED**
  Indicates that the attribute is not required.

-
- **ON Statements**
  See ON Statements for a description of these statements.

Example

The following example changes the salary attribute in the emp object so that it is now a required attribute:

```
MODIFY emp salary REQUIRED;
```

Example

The following example changes the address2 attribute in the emp object so that is now not required.

```
MODIFY emp address2 NOT_REQUIRED;
```

MODIFY FACTORY Statement

Changes the way factories are defined on OBJECT statements. MODIFY statements are read after OBJECT statements.

Syntax

```
MODIFY FACTORY fac_name {
   [FUNCTION_GROUP name ;]
   [DISPLAY_NAME name ;]
   [STANDARD_LISTS {
       [SORT_BY index_att ;]
       [FETCH fetch_att ;]
       [WHERE string ;]
       [MLIST ON|OFF;]
       [RLIST ON|OFF;] } ;] };
```

Arguments

-
- ***fac_name***
  Identifies the factory, if included on the original OBJECT statement.

Optional Statements

At least one of these optional statements must be specified.

-
- **FUNCTION_GROUP *name***
  Indicates which security access groups are permitted to access the object. For example:
  ```
  FUNCTION_GROUP "admin" ;
  ```
-
- **DISPLAY_NAME name**
  Defines an external name for the table.

DISPLAY_NAME "Call Request" ;

-
- **STANDARD_LISTS**
  Creates lists of objects that are kept in a cache. The parameters determine whether the lists are master lists or restricted lists, whether the objects included in the list must meet specified conditions, and how the lists can be sorted. Refer STANDARD_LISTS Optional Statements for a description of the syntax.

OBJECT Statement

Defines a business object.

Syntax

```
OBJECT obj_name {
    [ATTRIBUTES [table_name]{
        att_name [field_name] value_type [access_type[status_type]] [DISPLAY_NAME string][{
           [ON_NEW DEFAULT|SET value|NOW ;]
           [ON_CI DEFAULT|SET value|NOW ;]
           [ON_DB_INIT DEFAULT|SET value|NOW ;]} ;]};]


    [FACTORY [fac_name]{
        [REL_ATTR name ;]
        [COMMON_NAME name ;]
     [DISPLAY_NAME name ;]
        [FUNCTION_GROUP name ;]
        [STANDARD_LISTS {
           [SORT_BY index_att ;]
           [FETCH fetch_att ;]
           [WHERE string ;]
           [MLIST ON|OFF;]
           [RLIST ON|OFF;] } ;]};]
};
```

Arguments

-
- ***obj_name***
  The object's name (for example, cnt for contact or cr for request).

Optional Statements

Either ATTRIBUTES or FACTORY must be specified. Both can be specified.

-
- **ATTRIBUTES [*table_name*] { }**
  Defines the properties of the object. Most attributes map to a field (column) in a database table. The ATTRIBUTES Optional Statement describes its syntax.
-
- **FACTORY [*fac_name*] { }**
  Defines access to the object, like its relation attribute, a common name, the security group that can access it, the type of lists produced, and how those lists can be sorted. The FACTORY Optional Statement describes its syntax.

Example

This example defines an object named ctp. The ATTRIBUTES statement defines attributes named sym, delete_flag, and description whose values are stored in the Contact_Type table in the database. The FACTORY statement creates a master list of objects, sorted by values in the field that corresponds to the sym attribute, and specifies that the id attribute will represent ctp when it is referenced by an SREL:

```
OBJECT ctp {
   ATTRIBUTES Contact_Type {
       sym                 STRING REQUIRED ;
       delete_flag del     INTEGER {
          ON_NEW DEFAULT 0 ;
       } ;
       description desc    STRING ;
   } ;
   FACTORY {
       STANDARD_LISTS {SORT_BY "sym"} ;
       REL_ATTR  id ;
   };
};
```

STANDARD_LISTS Optional Statement

This article contains the following topics:

The optional statement on the FACTORY statement that defines the object's standard lists.

Syntax (STANDARD_LISTS Optional Statement)

```
STANDARD_LISTS {
    [SORT_BY index_att ;]
    [FETCH fetch_att ;]
    [WHERE string ;]
    [MLIST ON|OFF;]
    [RLIST ON|OFF;] } ;
```

Optional Statements (STANDARD_LISTS Optional Statement)

At least one of these optional statements must be specified:

-
- **SORT_BY *index_att***
  Defines the attributes that can be used to sort the standard lists. If specified, a master list is produced. Attributes must be enclosed in quotes and separated by commas. When displayed in a list or select window, the list is sorted by the first attribute, by default. For example:
  ```
  SORT_BY "sym, code" ;
  ```
-
- **FETCH *fetch_att***
  Specifies additional attributes to keep in the cache, besides those used to sort the list. They must be enclosed in quotes and separated by commas. For example:
  ```
  FETCH "description" ;
  ```
-
- **WHERE *string***
  Specifies a condition, in SQL format and surrounded by quotes, that must be met for an object to be included in a restricted list. If specified, a restricted list is produced. This example specifies that the restricted list contain only records that were not deleted:

```
    WHERE "delete_flag = 0" ;
```

-
- **MLIST ON|OFF**
  Indicates whether to produce a master list, which includes all objects, using one of the following values:

Value	Description
ON	Produces a master list (default if SORT_BY is specified)
OFF	Does not produce a master list (default if SORT_BY is not specified or has no value defined)

NOTE
CA SDM web engine only uses the MLIST to populate the data in the web forms. MLIST is created on the domsrvr cache.

-
- **RLIST ON|OFF**
  Indicates whether to also produce a restricted list, which includes only the objects that meet the criteria in the WHERE clause, using one of the following values:

Value	Description
ON	Produces a restricted list (default if WHERE is specified)
OFF	Does not produce a restricted list (default if WHERE is not specified or has no value defined.)

NOTE
RLISTs can speed up access and display but they use memory. They are usually used in select windows.

WARNING
MLIST OFF must be specified if you specify RLIST OFF.

Example (STANDARD_LISTS Optional Statement)

This example provides both a master list and a restricted list. Both lists contain the values defined for the sym, code, and description attributes. The records in the list can be sorted according to the values in the sym and code attributes. The restricted list contains only records that were not deleted:

```
STANDARD_LISTS {
    SORT_BY "sym,code" ;
    FETCH "description" ;
    WHERE "delete_flag = 0" ;
};
```

The STANDARD_LISTS statement alone does not necessarily create entries that are listed in a dropdown field. To create the dropdown list in a web interface form, you need to customize the statement.

FACTORY Optional Statement

This article contains the following topics:

Defines access to the object, like its relation attribute, a common name, the security group that can access it, the type of lists produced, and how those lists can be sorted. If omitted, the object is treated according to default specifications.

Syntax (FACTORY Optional Statement)

```
FACTORY [fac_name]{
```

```
    [REL_ATTR name ;]
    [COMMON_NAME name ;]
    [FUNCTION_GROUP name ;]
    [DISPLAY_NAME name ;]
    [STANDARD_LISTS {
        [SORT_BY index_att ;]
        [FETCH fetch_att ;]
        [WHERE string ;]
        [MLIST ON|OFF;]
        [RLIST ON|OFF;] } ;]
};
```

Arguments (FACTORY Optional Statement

- ***fac_name***
  The name of the factory that initiates the object. Specify this only if it is different from the name of the object. For example, the cnt object has four factories: cnt, cst, agt, grp.

Optional Statements (FACTORY Optional Statement)

At least one of these optional statements must be specified

- **REL_ATTR *name***
  Identifies the attribute that will represent this object when it is referenced (used as an SREL) by another object. Here is an example:
  ```
  REL_ATTR id ;
  ```
- **REL_ATTR name, srel_name (attr1,attr2,…)**
  Identifies the attributes that will represent this object when it is referenced (used as an “named” SREL) by another object. whe
  - **srel_name**
    matches the "named" SREL name
  - **attr1**
    is mapped to by the first attribute in the "named" SREL attribute list.
  - **attr2**
    is mapped to by the second attribute in the "named" SREL attribute list.
- DISPLAY_NAME name
  Defines an external name for the table.
  DISPLAY_NAME "Call Request" ;
- **COMMON_NAME *name***
  Defines the attribute to be displayed in drop-down lists or when the user double-clicks a field, as well as when the tag does not specify a complete attribute. In the first example, the value for sym appears on the window instead of the value for the REL_ATTR. The second example allows you to specify a tag as cr.customer instead of cr.customer.combo_name.
  ```
  COMMON_NAME sym ;
  COMMON_NAME combo_name ;
  ```
- **FUNCTION_GROUP *name***
  Indicates which security access group is permitted to access the object. For example:
  ```
  FUNCTION_GROUP "admin" ;
  ```
- **STANDARD_LISTS { }**
  Creates lists of objects that are kept in a cache and can be displayed on list or select windows. The parameters determine whether the lists are master lists or restricted lists, whether the objects included in the list must meet

specified conditions, how the lists can be sorted, and what additional attributes are stored. Refer STANDARD_LISTS Optional Statement for the description of the syntax.

Example (FACTORY Optional Statement)

This example defines access to the object. A master list is produced. It can be sorted according to the values in the object's sym and code attributes. When first displayed, it is sorted according to the values in the sym attribute by default. When referenced by another object, this object is represented by the code attribute. When displayed in a window, the value for sym appears instead of the value for code. Only users in the admin security group can accessed it:

```
FACTORY {
    STANDARD_LISTS {SORT_BY "sym,code"} ;
    REL_ATTR code ;
    COMMON_NAME sym ;
    FUNCTION_GROUP "admin" ;
};
```

ATTRIBUTES Optional Statement

The optional statement on the OBJECT statement that defines the properties of the object.

Syntax

```
ATTRIBUTES [table_name]{
   att_name [field_name] value_type [access_type[status_type][DISPLAY_NAME string]] {
       [ON_NEW DEFAULT|SET value|NOW ;]
       [ON_CI DEFAULT|SET value|NOW ;]
    [ON_DB_INIT DEFAULT|SET value|NOW ;];]};]
```

Arguments

- ***table_name***
  The name of the table in the database that stores the values associated with the attributes in the object. If the table name is not specified, the *obj_name* in the OBJECT statement is used.
- ***att_name***
  The name of the attribute. Each attribute usually maps to a field (column) in the database table.
- ***field_name***
  The name of the field in the database table or LOCAL if the attribute does not map to a field or DERIVED (derived-expr) if the attribute is derived from other attributes. If neither LOCAL, DERIVED nor a field name is specified, the name of the field is assumed to be the same as the name of the attribute.
  A variable declared as DERIVED is constructed only when its value is retrieved. The operand of DERIVED contains a list of attribute names and string constants separated by spaces. All attributes in a derived value must be simple values (that is, they cannot be xRELs), and should be declared prior to the derived variable. The derived attribute's value is the concatenation of the values of its constituent values.
  String constants within a derived expression may contain references to environment variables in the one of the forms:
  ${var}
  ${var#pattern}
  ${var#pattern#replacement}
  Such specifications are replaced with the value of the environment variable at domsrvr startup time. The #pattern operand is optional. If provided, it is treated as a regular expression, and replaced wherever it appears in the environment variable's value. The #replacement operand defaults to null if not specified. Because # is a fixed delimeter, the pattern cannot contain a # symbol. There are no restrictions on the use of derived attributes in other messages. They behave in same way as standard attributes. A hotlink for a derived attribute fires whenever any of the attributes from which it is built changes.
- ***value_type***

Identifies the data type of the attribute's value as:
- INTEGER
- DOUBLE
- STRING [*length*]
- DURATION
- UUID
- DATE
- SREL *obj2_name*
- SREL { ob2_name_name srel_name (name, name2, ...) }

If STRING is specified, the size can be specified in an integer following STRING. If no size is specified, the value in the database is used.
UUID is 16 bytes of binary data that is used as a unique identifier for certain database records.
SREL refers the attribute to another object. If SREL is specified, *obj2_name* must be specified to identify the object that the attribute refers to.
srel_name specifies a "named" SREL. Like a "simple" SREL, a "named" is a type of MAJIC OBJECT attribute that represents a single relation, which uniquely identifies a row in another table (ob2_name). A "simple" SREL attribute normally maps to the "id" field in another table, however a "named" SREL maps two or more attributes (name, name2, …) to two or more attributes in the referenced table that uniquely identify a row in the referenced table.

- ***access_type***
  Defines access to the attribute. Valid values are:

Value	Description
CONST	Cannot be changed
PRIVATE	Read-only
PUBLIC	Read/write access (the default)
WRITE_NEW	Can be written only when the object is created, before the object is saved

- ***status_type***
  Indicates the status of the attribute as:
  - REQUIRED
  - NOT_REQUIRED (the default)
- **DISPLAY_NAME string**
  Specifies a string to be used in place of the attribute name in messages concerning this attribute, such as"required attribute missing"

ON Statements

Use one of these only when *value_type* is INTEGER, STRING, DATE, or SREL.

- **ON_NEW DEFAULT|SET *value*|NOW**
  Indicates to set the value of an attribute when the object is being created for the first time:

Value	Description
DEFAULT	Changes a null current value to *value* or NOW.
SET	Changes any current value to *value* or NOW.
value	Specifies a numeric value or a string value, depending on the data type of the attribute.

NOW	Specify this if the attribute is of type DATE; it sets the attribute to the current date and time.

In the following example, 90 is the value set as a default when the object is created:

```
ON_NEW DEFAULT 90 ;
```

- **ON_CI DEFAULT|SET *value*|NOW**
  Indicates to set the value of an attribute when the attribute is being checked into the database. See the description of each parameter for ON_NEW.
- **ON_DB_INIT DEFAULT|SET *value*|NOW**
  Indicates to set the value of an attribute when the attribute is being instantiated from the database. See the description of each parameter for ON_NEW.

Example

This example defines attributes with names like start_date whose values are stored in fields like nlh_start in the Notify_Log_Header table in the database. The field names are followed by each attribute's data type. Optional parameters define access to some of the attributes, indicate that the attribute is required, and tell when to set the value of some of the attributes to the current date and time.

For example, an attribute named last_mod is defined; its value is set to the current date and time when the attribute is checked into the database. An attribute named contact is also defined; its value is a single relation stored in database field nlh_c_addressee. The object referred to is cnt:

```
ATTRIBUTES Notify_Log_Header {
   start_date         nlh_start          DATE WRITE_NEW {ON_NEW DEFAULT NOW;} ;
   last_mod                              DATE {ON_CI SET NOW ;} ;
   msg_hdr            nlh_hdr            STRING 20 WRITE_NEW ;
   msg_text           nlh_msg            STRING WRITE_NEW ;
   msg_ack            nlh_user_ack       STRING ;
   contact            nlh_c_addressee    SREL cnt WRITE_NEW ;
   notify_method      nlh_cm_method      INTEGER WRITE_NEW ;
   activity_notify    nlh_transition     INTEGER WRITE_NEW ;
   pri_event          nlh_pri            INTEGER WRITE_NEW ;
   notify_type        nlh_type           INTEGER WRITE_NEW ;
   ack_time           nlh_ack_time       DURATION ;
   status             nlh_status         INTEGER REQUIRED ;
   end_date           nlh_end            DATE {ON_NEW DEFAULT NOW ;} ;
```

};

Where Clauses

This article contains the following topics:

Several Web Services methods, such as doSelect() and doQuery(), require *the Where clauses* for searching by CA SDM and Knowledge Management. A Where clause is the string appearing after the 'WHERE' keyword in an SQL statement. For example, a where clause to find contacts (the 'cnt' object) by last name:

```
last_name = 'Jones'
```

or

```
last_name LIKE 'Jone%'
```

The second example finds all contacts with names beginning with 'Jone', while the first just finds the Jones'.

CA SDM supports only a subset of the standard SQL parameters for where clauses, and are listed as follows:

- Logical operators AND, OR, and NOT
- LIKE and IS
- NULL
- IN
- Wildcard characters '%' and '_' for string matches
- All comparison operators: <, >, <=, >=, !=, <>

NOTE

Parenthesis is used for grouping. Explicit joins, EXISTS, and GROUP BY elements are not supported by CA SDM. String value must be enclosed in quotes, for example, 'Jones'.

The column names denote the object attribute names. CA SDM data types,data and duration, are treated as integers. For example:

```
creation_date > 38473489389
```

NOTE

You must use the attribute names at the object Level. Do not use the actual DBMS column names.

Dot-notation is allowed in the Where clause to search through SREL (foreign key) types. For example, a query against the Request ('cr') object, returns all Requests assigned to contacts with a specific last name, as illustrated by the following example:

```
assignee.last_name like 'Martin%'
```

Dot-notation is very helpful in forming the where clauses, but you must ensure that the query is an efficient one. The query in the example *assignee.last_name like 'Martin%*' can be inefficient if the contact's last_name attribute is not indexed in the DBMS. To ensure indexes are used to their best advantage when searching through SRELs, make use of the ID attributes of the CA SDM objects. All tables in CA SDM have an index on the ID attribute.

The ID attribute of an object can be easily obtained from the object's handle. An object's handle is a string of the form "<*objectName*>:<*id*>", where <*id*> is the value of the ID attribute found in every CA SDM object. Extract the ID portion and use "<attributeName>.id" in the Where clause.

An object's ID is either an integer or a UUID. If it is an integer, simply use it as such. For example, to search for Requests with the *rootcause* pointing to a Root Cause object with handle, "*rc:1234*", the Where clause is:

```
rootcause.id = 1234
```

If the ID attribute of an object is a UUID type, you must format it as:

```
U'<uuid>'
```

The string representation of a UUID is enclosed in single quotes and prefixed with capital 'U'. This string is the <id> part of an object handle. For example, if you know that the handle for a particular contact is cnt:913B485771E1B347968E530276916387, you can form the query as:

```
assignee.id = U'913B485771E1B347968E530276916387'
```

Do not form the Where clauses by querying the 'persistent_id' attribute, as in the following example:

```
rootcause.persistent_id = 'rc:1234'
```

For more information about handles, see Default Handles.

IN Clause

The IN clause requires some special explanation. The two syntactic forms are:

```
SREL_attr_name.subq_WHERE_attr[.attr] IN ( value1 [, value2 [,...]]  )
SREL_attr_name.[subq_SELECT_attr]LIST_name.subq_WHERE_attr IN (value1, [,value2 [,...]]  )
```

The left side of the clause must begin with an SREL-type attribute of the table being queried, which is represented by *SREL_attr_name*. *subq_WHERE_attr* is an attribute of the foreign object, which itself may be another SREL pointer.

For example, a query against the request (‘cr’) object may be coded as follows:

```
category.sym IN ('Soft%', 'Email')
```

This translates to the following pseudo-SQL:

```
SELECT ... FROM cr WHERE cr.category IN (SELECT persistent_id FROM pcat WHERE sym LIKE 'Soft%' OR sym =
 'Email')
```

In the previous sub query , ‘pcat’ is the object name pointed to by cr.category.

The second form of the IN clause can search through BREL lists. For example, to find all requests assigned to an analyst in a specific group, the clause is as follows:

```
assignee.[member]group_list.group IN (U'913B485771E1B347968E530276916387')
```

The first part of the clause, assignee, is an SREL (foreign key) of the cr object, pointing to the cnt object. Next, group_list, which is an attribute of the cnt object, is a list of cnt objects that represent groups to which a contact belongs. The last part, group, forms the first part of the where clause for the IN sub query. ‘U’913B485771E1B347968E530276916387is the foreign key value to match on group. The sub query return is specified by [member]. This translates to the following pseudo-SQL statement:

```
SELECT ... FROM cr WHERE cr.assignee IN (SELECT member from grpmem WHERE group =
 U'913B485771E1B347968E530276916387')
```

You can specify multiple foreign keys for matching multiple objects by providing a comma-separated list:

```
assignee.[member]group_list.group IN (U'913B485771E1B347968E530276916387',
 U'913B485771E1B347968E530276916300')
```

You cannot extend the dot notation for this use of the IN clause, for example, the following is not valid:

```
assignee.[member]group_list.group.last_name IN ('Account Center')
```

One use of IN is to avoid Cartesian products. For example, the following query results in a Cartesian product and is very inefficient:

```
assignee.last_name LIKE 'MIS%' OR group.last_name LIKE 'MIS%'
```

Using IN, the query can be coded as follows:

```
assignee.last_name IN 'MIS%' OR group.last_name IN 'MIS%'
```

This query does not create a Cartesian product; in fact, it creates no joins at all.

> **NOTE**
> The parentheses that normally enclose the list of values on the right side of IN can be omitted if there is only one value in the list. Similarly, you should avoid joins by converting queries.

```
assignee.last_name LIKE 'Smith'
```

to:

```
assignee = U'913B485771E1B347968E530276916387'
```

This avoids the join with some loss in CA. Using IN, the same partition can be written as follows, with the CA of the first version and almost the same efficiency as the second version:

```
assignee.last_name IN 'Smith'
```

The ‘NOT’ keyword cannot be in conjunction with IN, for example, “NOT IN”.

Lists

Some Web Services methods return *lists*, represented by a unique integer handle. A list is simply a collection of same-type objects. Lists are especially useful when dealing with a large collection of objects (for example, all the contacts in the system) because you can retrieve information about items in a range of the list. The disadvantage is that you must make more method calls to obtain a list handle, retrieve information, and finally, free the list handle. If the expected number of list rows is small, use methods that do not involve list handles, such as doSelect().

The following describes more details about lists:

- **Lists are homogenous**
  List may only contain objects of a single type, for example, lists of contacts, list of organizations, and so on.
- **Lists are Static**
  For example, if a list object is obtained for all contacts and another contact is added to the system, the update is not reflected in the list. Another list handle must be obtained to get the most current data.
- **List Handles**
  A request for a list returns an integer handle representing the list of same-type objects. No other information is sent to the client. The client may query the list for specific information about its rows. When a client is finished with a list, the handle must be released with freeListHandles(). The CA SDM server maintains the list, consuming system resources. Therefore, it is important to free lists. Unlike object handles, list handles are not persistent across sessions.
- **Integer Index**
  Several methods require an integer index into a list. Lists are zero-based so the first element is at index = 0.

As previously mentioned, using list handles is most useful for larger sets of data that may be queried multiple times. For some operations, however, lists are excessive. Several methods are provided, but the most notable is doSelect(), as it returns requested information about a set of data without the overhead of list handles.

The decision to use list handles versus methods, such as doSelect(), is one of performance and convenience. For example, suppose your application does processing on all 15,000 Contacts in your system. The doSelect() method can retrieve all the contact data in one call, but the reply will be delayed and will negatively impact overall system performance while it assembles and returns a very large data set. The doQuery() method, in this case, will return a list reference very quickly. Ranges of data can be queried from the list to improve response times from the server. A good practice to follow is to use list references if the data set exceeds 250 items.

Sometimes it does not make sense to use list handles. For example, an issue has a list of Activity Logs. Depending on the installation, the number of logs can range from a few to several dozen. It is probably faster to request the data all at once instead of requesting a list reference, querying it for data, and then releasing the list.

Examples of methods that return data sets instead of list references include the following:

- doSelect()
- getRelatedListValues()
- getLrelValues()
- getTaskListValues()
- getValidTaskTransitions()

As previously stated, queries that return a large number of rows can severely impact the performance of the server. To protect against this, CA SDM limits the number of rows returned to 250. This affects all CA SDM Web Services methods that return lists of objects, including the following:

- doSelect()
- doSelectKD()
- getGroupMemberListValues()
- getListValues()
- getPropertyInfoForCategory()
- getRelatedListValues()
- getTaskListValues()
- getValidTaskTransitions()

This limit applies even if you request one of these methods to retrieve more than 250 rows.

To retrieve large numbers of rows, you should obtain a handle to the list of results and use getListValues() to retrieve chunks of 250 or fewer rows each. This strategy helps keep the server from becoming slow while serving huge amounts of data.

Attribute Data Types

This article contains the following topics:

Each attribute of an object is of a specific type that has meaning to the CA SDM application, such as string, date, or integer. Knowing the attribute type is essential to correctly retrieving and updating attribute values.

The CA SDM uses an enumeration to identify each data type. These enumeration values are returned in various web methods, as illustrated by the following table:

Data Type	Value
Integer	2001
String	2002
Duration	2003
Date	2004
SREL	2005
UNKNOWN	2006
List (QREL/BREL)	2007
Lrel (many-to-many)	2008
UUID	2009

Integer

The Integer data type represents a 32-bit signed integer. Null an integer attribute by passing the empty string.

String

The String data type represents a character string, where the maximum length is defined by the database storage allocated for a particular string attribute.

If you attempt to set a string attribute to a value that exceeds its length, the value is truncated and an error message is written to the CA SDM log.

Duration

The Duration data type is an integer representing time duration in seconds. For example, “90” represents one minute and 30 seconds. To set a duration type, use an integer representing the number of seconds for the duration. Negative values are not permitted. To make a duration attribute null, pass the empty string.

Date

The Date data type represents a date value. This is stored as a UNIX-like UTC value in the database (the number of seconds since 1-1-1970). When retrieving date values, the integer UTC is returned. Similarly, use a UTC value to set a date. Negative values are not permitted. Null a date attribute by passing the empty string.

SREL

The SREL data type represents an SREL (Single RELation), which is a pointer to another object. It is a foreign key to another table in a database. For example, an Issue object has a pointer attribute to a Contact representing the Assignee.

Most CA SDM Web Services methods permit dot-notation to retrieve information about objects to which an SREL points. For example, to specify the name of a Contact’s organization from the context of the Contact, use the following:

```
organization.name
```

You may expand to an arbitrary number of levels as shown in the following example:

```
organization.contact.first_name
```

Dot-notation can only be used to retrieve attribute values, such as using getObjectValues(), or in a Where clause. You cannot use dot-notation to set values.

To set an SREL attribute, such as with updateObject(), you can pass the persistent id of the object to which you want to point. In order to simplify this action, this release of CA SDM has been enhanced so that the REL_ATTR (foreign key) value may be used to set an SREL.

For example, as the REL_ATTR of the crt object (Request Type) is its code attribute, the values “R”, “I” and “P” can be used to set the type attribute of a cr object to specify that the ticket is a Request, Incident or Problem. The cr’s type attribute can be set to “R” instead of “crt:180”, “I” instead of “crt:182” and “P” instead of “crt:181”.

To set an SREL attribute to null, pass the empty string (“”).

List (QREL/BREL)

An object can have a list attribute that represents a one-to-many relationship. These are defined in majic files with the QREL or BREL keywords. A list exists at the object level -- it does not take any additional storage in the DBMS.

The CA SDM system handles list collections as abstract data types. The Web Services provides several methods to interact with lists -- for references and queries to lists defined in an object, use getRelatedList() and getRelatedListValues(). For more information about lists, see CA SDM Lists in this chapter.

LREL

The LREL data type represents a many-to-many relationship between two object types. An LREL has two names, one for each side of the relationship. The CA SDM Web Services provides special functions for interacting with LRELs.

UNKNOWN

The Unknown data type represents an unknown data type.

UUID

A UUID is a 128-bit integer (16 bytes) or a 32-byte character that can be used across all computers and networks wherever a unique identifier is required. Such an identifier has a very low probability of being duplicated (for example, a contact ID). UUIDs are used mostly with primary keys.

Web Services Methods

This section provides the details for using the Web Services methods. Each method explains the parameters, description, and returns.

addAssetLog

- Input Parameters: (int sid, String assetHandle, String contactHandle, String logText)
- Return Type [XML Root Element]: void
- Description: Adds a new log entry to an asset.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:addAssetLog>
<sid>7424993</sid>
<assetHandle>nr:1C1B4D8C331CCE4E9AF08AF2A07AC66F</assetHandle>
<contactHandle>DD392B8AC9B2994893F7886017ACC60B</contactHandle>
<logText>Adding entry in configuration Item</logText>
</ser:addAssetLog>
</soapenv:Body>
</soapenv:Envelope>
```

addBookmark

- Input Parameters: (int sid, String contactId, int docId)
- Return Type [XML Root Element]: String [UDSObject]
- Description: Adds a bookmark to a knowledge document.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:addBookmark>
<sid>486898605</sid>
<contactId>DD392B8AC9B2994893F7886017ACC60B</contactId>
<docId>400005</docId>
</ser:addBookmark>
</soapenv:Body>
</soapenv:Envelope>
```

addComment

- Input Parameters: (int sid, String comment, int docId, String email, String username, String contactId)
- Return Type [XML Root Element]: String [UDSObject]
- Description: Adds a comment to a knowledge document.
- Input parameter for SOAP Requests:

```
<soapenv:Envelopexmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:addComment>
<sid>1250747028</sid>
<comment>Addedcommentfordocumentid</comment>
<docId>400008</docId>
<email></email>
<username>ServiceDesk</username>
<contactId>DD392B8AC9B2994893F7886017ACC60B</contactId>
</ser:addComment>
</soapenv:Body>
</soapenv:Envelope>
```

addMemberToGroup

- Input Parameters: (int sid, String contactHandle, String groupHandle)
- Return Type [XML Root Element]: void
- Description: Adds a contact to a group.
- Input parameter for SOAP Requests:

```
<soapenv:Envelopexmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:addMemberToGroup>
<sid>7424993</sid>
<contactHandle>cnt:DD392B8AC9B2994893F7886017ACC60B</contactHandle>
<groupHandle>cnt:012E6144A6768941BA42BBA063E1CFE9</groupHandle>
</ser:addMemberToGroup>
</soapenv:Body>
</soapenv:Envelope>
```

attachChangeToRequest

- Input Parameters: (int sid, String creator, String requestHandle, String changeHandle, ArrayOfString changeAttrVals, String description)
- Return Type [XML Root Element]: String
- Description: Attaches a new or existing change order to a request.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
```

```
<ser:attachChangeToRequest>
<sid>354388306</sid>
<creator>ServiceDesk</creator>
<requestHandle>cr:400107</requestHandle>
<changeHandle>chg:400001</changeHandle>
<changeAttrVals>
<!--1ormorerepetitions:-->
<string>?</string>
</changeAttrVals>
<description>test</description>
</ser:attachChangeToRequest>
</soapenv:Body>
</soapenv:Envelope>
```

attachURLLink

- Input Parameters: (int sid, int docId, String url, String attmntName, String description)
- Return Type [XML Root Element]: int
- Description: Attaches a URL link to a knowledge document.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:attachURLLink>
<sid>410981252</sid>
<docId>400008</docId>
<url>http://google.com</url>
<attmntName>Google</attmntName>
<description>UpdatedGooglelink</description>
</ser:attachURLLink>
</soapenv:Body>
</soapenv:Envelope>
```

attmntFolderLinkCount

- Input Parameters: (int sid, int folderId)
- Return Type [XML Root Element]: int
- Description: Returns the number of attachment links for a folder.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:attmntFolderLinkCount>
<sid>936858701</sid>
<folderId>1001</folderId>
</ser:attmntFolderLinkCount>
</soapenv:Body>
</soapenv:Envelope>
```

callServerMethod

- Input Parameters: (int sid, String methodName, String factoryName, String formatList, ArrayOfString parameters)
- Return Type [XML Root Element]:String [ServerReturn]
- Description: Invokes any arbitrary server-side method.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:callServerMethod>
<sid>56997651</sid>
<methodName>get_env_var</methodName>
<factoryName>cr</factoryName>
<formatList>?</formatList>
<parameters>
<string>0</string>
</parameters>
</ser:callServerMethod>
</soapenv:Body>
</soapenv:Envelope>
```

changeStatus

- Input Parameters: (int sid, String creator, String objectHandle, String description, String newStatusHandle)
- Return Type [XML Root Element]: String [UDSObject (Handle only)]
- Description: Performs an update status activity on a ticket.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:changeStatus>
<sid>1100057841</sid>
<creator>ServiceDesk</creator>
<objectHandle>cr:400098</objectHandle>
<description>Updatingstatus</description>
<newStatusHandle></newStatusHandle>
</ser:changeStatus>
</soapenv:Body>
</soapenv:Envelope>
```

clearNotification

- Input Parameters: (int sid, String lrObject, String clearBy)
- Return Type [XML Root Element]: int
- Description: Clears a notification message.

closeTicket

- Input Parameters: (int sid, String description, String ticketHandle)
- Return Type [XML Root Element]: String [UDSObject (Handle only)]
- Description: Sets the status of the ticket to Closed.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:closeTicket>
<sid>331513514</sid>
<description></description>
<ticketHandle>cr:400054</ticketHandle>
</ser:closeTicket>
</soapenv:Body>
</soapenv:Envelope>
```

createActivityLog

- Input Parameters: (int sid, String creator, String objectHandle, String description, String logType, int timeSpent, boolean internal)
- Return Type [XML Root Element]: String [UDSObject (Handle only)]
- Description: Creates an activity log entry for a ticket.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:createActivityLog>
<sid>1284058108</sid>
<creator>ServiceDesk</creator>
<objectHandle>cr:400151</objectHandle>
<description>CXF</description>
<logType>LOG</logType>
<timeSpent>00</timeSpent>
<internal>1</internal>
</ser:createActivityLog>
</soapenv:Body>
</soapenv:Envelope>
```

createAsset

- Input Parameters: ((int sid, ArrayOfString attrVals, ArrayOfString attributes, StringHolder createAssetResult, StringHolder newAssetHandle, StringHolder newExtensionHandle, StringHolder newExtensionName)
- Return Type [XML Root Element]: void
- Output Parameters (Type: StringHolder) [XML Root Element]: createAssetResult [UDSObject] newAssetHandle newExtensionHandle newExtensionName
- Description: Creates a configuration item (asset).
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:createAsset>
<sid>591379952</sid>
<attrVals>
<string>class</string>
<string>grc:300110</string>
<string>is_asset</string>
<string>0</string>
<string>is_ci</string>
<string>1</string>
<string>system_name</string>
<string>AxistoCXF</string>
<string>name</string>
<string>AxistoCXF</string>
<string>description</string>
<string>AxistoCXF</string>
</attrVals>
<attributes>
<!--1ormorerepetitions:-->
<string>name</string>
</attributes>
<createAssetResult>?</createAssetResult>
<newAssetHandle>?</newAssetHandle>
<newExtensionHandle>?</newExtensionHandle>
<newExtensionName></newExtensionName>
</ser:createAsset>
</soapenv:Body>
</soapenv:Envelope>
```

createAssetParentChildRelationship

- Input Parameters: (int sid, String parentHandle, String childHandle)
- Return Type [XML Root Element]: String
- Description: Creates an asset parent-child relationship.
- Input parameter for SOAP Requests:

```
<soapenv:Envelopexmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:createAssetParentChildRelationship>
<sid>2043865190</sid>
<parentHandle>nr:1A8F01150F178E47B9846F91BA6C9561</parentHandle>
<childHandle>nr:94C7E556C8D6C647A04793BD3B8234CF</childHandle>
</ser:createAssetParentChildRelationship>
</soapenv:Body>
</soapenv:Envelope>
```

createAttachment

- Input Parameters: (int sid, String repositoryHandle, String objectHandle, String description, String fileName)
- Return Type [XML Root Element]: String
- Description: Uploads a file to the back-end server and attaches it to a ticket.

createAttmnt

- Input Parameters: (int sid, String repositoryHandle, int folderId, int objectHandle, String description, String fileName)
- Return Type [XML Root Element]: String
- Description: Uploads a file to the back-end server and attaches it to a knowledge document.
- Input parameter for SOAP Requests:

```
<soapenv:Envelopexmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:createAttmnt>
<sid>1451738759</sid>
<repositoryHandle>doc_rep:1001</repositoryHandle>
<folderId>1001</folderId>
<objectHandle>400001</objectHandle>
<description>test</description>
<fileName>c:\testing.txt</fileName>
</ser:createAttmnt>
</soapenv:Body>
</soapenv:Envelope>
```

createChangeOrder

- Input Parameters: (int sid, String creatorHandle, ArrayOfString attrVals, ArrayOfString propertyValues, String template, ArrayOfString attributes, StringHolder newChangeHandle, StringHolder newChangeNumber)
- Return Type [XML Root Element]: String [UDSObject]
- Output Parameters (Type: StringHolder) [XML Root Element]: newChangeHandle newChangeNumber
- Description: Creates a Change Order ticket.
- Input parameter for SOAP Requests:

```
<soapenv:Envelopexmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:createChangeOrder>
<sid>834772785</sid>
<creatorHandle>cnt:98E76E8F273CD3478FB4AEB981290869</creatorHandle>
<attrVals>
<string>requestor</string>
<string>cnt:98E76E8F273CD3478FB4AEB981290869</string>
<string>category</string>
<string>chgcat:6102</string>
<string>priority</string>
string>
<string>pri:504</string>
```

```
<string>assignee</string>
<string>cnt:98E76E8F273CD3478FB4AEB981290869</string>
</attrVals>
<propertyValues>
</propertyValues>
<template/>
<attributes>
</attributes>
</ser:createChangeOrder>
</soapenv:Body>
</soapenv:Envelope>
```

createDocument

- Input Parameters: (int sid, ArrayOfString kdAttributes)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Creates a knowledge document.
- Input parameter for SOAP Requests:

```
<soapenv:Envelopexmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:createDocument>
<sid>1451738759</sid>
<kdAttributes>
<!--1ormorerepetitions:-->
<string>TITLE</string>
<string>NewDocument</string>
<string>STATUS_ID</string>
<Integer>70</Integer>
</kdAttributes>
</ser:createDocument>
</soapenv:Body>
</soapenv:Envelope>
```

createFolder

- Input Parameters: (int sid, int parentFolderId, int repId, int folderType, String description, String folderName)
- Return Type [XML Root Element]: String [UDSObject]
- Description: Creates a folder in an attachment repository.
- Input parameter for SOAP Requests:

```
<soapenv:Envelopexmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:createFolder>
<sid>584240889</sid>
<parentFolderId>0</parentFolderId>
<repId>1001</repId>
<folderType>0</folderType>
```

```
<description>NewReporistory</description>
<folderName>NewReporistory</folderName>
</ser:createFolder>
</soapenv:Body>
</soapenv:Envelope>
```

createIssue

- Input Parameters: (int sid, String creatorHandle, ArrayOfString attrVals, ArrayOfString propertyValues, String template, ArrayOfString attributes, StringHolder newIssueHandle, StringHolder newIssueNumber)
- Return Type [XML Root Element]: String [UDSObject]
- Output Parameters (Type: StringHolder) [XML Root Element]: newIssueHandle newIssueNumber
- Description: Creates an Issue ticket.

createLrelRelationships

- Input Parameters: (int sid, String contextObject, String lrelName, ArrayOfString addObjectHandles)
- Return Type [XML Root Element]: void
- Description: Adds one or more many-to-many relationships.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
  <soapenv:Header/>
<soapenv:Body>
<ser:createLrelRelationships>
<sid>584240889</sid>
<contextObject>cnt:DD392B8AC9B2994893F7886017ACC60B</contextObject>
<lrelName>cenv</lrelName>
<addObjectHandles>
<string>nr:5B796D610A00BB4BA65C25440C46A56D</string>
<string>nr:7A51DF12B1EA8F4ABF77E2EA025AE9A1</string>
</addObjectHandles>
</ser:createLrelRelationships>
</soapenv:Body>
</soapenv:Envelope>
```

createObject

- Input Parameters: (int sid, String objectType, ArrayOfString attrVals, ArrayOfString attributes, StringHolder createObjectResult, StringHolder newHandle)
- Return Type [XML Root Element]: void
- Output Parameters (Type: StringHolder) [XML Root Element]: createObjectResult [UDSObject] newHandle
- Description: Creates any CA SDM object.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:createObject>
```

```
<sid>584240889</sid>
<objectType>loc</objectType>
<attrVals>
<!--1ormorerepetitions:-->
<string>name</string>
<string>INDIA</string>
</attrVals>
<attributes>
<!--1ormorerepetitions:-->
<string>persistent_id</string>
<string>name</string>
</attributes>
<createObjectResult>?</createObjectResult>
<newHandle>?</newHandle>
</ser:createObject>
</soapenv:Body>
</soapenv:Envelope>
```

createQuickTicket

- Input Parameters: (int sid, String customerHandle, String description, StringHolder newTicketHandle, StringHolder newTicketNumber)
- Return Type [XML Root Element]: String [UDSObject]
- Output Parameters (Type: StringHolder) [XML Root Element]: newTicketHandle newTicketNumber
- Description: Creates a ticket based on the preferred document type of the given end user.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk"
<soapenv:Header/>
<soapenv:Body>
<ser:createQuickTicket>
<sid>1322287124</sid>
<customerHandle>cnt:DD392B8AC9B2994893F7886017ACC60B</customerHandle>
<description>TestingSDMCXF</description>
<newTicketHandle>?</newTicketHandle>
<newTicketNumber>?</newTicketNumber>
</ser:createQuickTicket>
</soapenv:Body>
</soapenv:Envelope>
```

createRequest

- Input Parameters: (int sid, String creatorHandle, ArrayOfString attrVals, ArrayOfString propertyValues, String template, ArrayOfString attributes, StringHolder newRequestHandle, StringHolder newRequestNumber)
- Return Type [XML Root Element]: String [UDSObject]
- Output Parameters (Type: StringHolder) [XML Root Element]: newRequestHandle newRequestNumber
- Description: Creates a Request ticket.

createTicket

- Input Parameters: (int sid, String description, String problem_type, String userid, String asset, String duplication_id, StringHolder newTicketHandle, StringHolder newTicketNumber, StringHolder returnUserData, StringHolder returnApplicationData)
- Return Type [XML Root Element]: String [UDSObject (empty)]
- Output Parameters (Type: StringHolder) [XML Root Element]: newTicketHandle newTicketNumber returnUserData returnApplicationData
- Description: Creates a ticket based on the rules defined in the Service Aware Policy and the given Problem Type.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:createTicket>
<sid>139569419</sid>
<description>CXF</description>
<problem_type>I</problem_type>
<userid>cnt:DD392B8AC9B2994893F7886017ACC60B</userid>
<asset>nr:DFE72B341C93EC4DBA8F706535CBE8B3</asset>
<duplication_id>?</duplication_id>
<newTicketHandle>?</newTicketHandle>
<newTicketNumber>?</newTicketNumber>
<returnUserData>?</returnUserData>
<returnApplicationData>?</returnApplicationData>
</ser:createTicket>
</soapenv:Body>
</soapenv:Envelope>
```

createWorkFlowTask

- Input Parameters: (int sid, ArrayOfString attrVals, String objectHandle, String creatorHandle, String selectedWorkFlow, String taskType, ArrayOfString attributes, StringHolder createWorkFlowTaskResult, StringHolder newHandle)
- Return Type [XML Root Element]: void
- Output Parameters (Type: StringHolder) [XML Root Element]: createWorkFlowTaskResult [UDSObject] newHandle
- Description: Creates Workflow Task.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:createWorkFlowTask>
<sid>834772785</sid>
<attrVals>
<string>assignee</string>
<string>cnt:DD6AE042C307E147A320856C3B1716A0</string>
</attrVals>
<objectHandle>iss:400001</objectHandle>
<creatorHandle>cnt:DD6AE042C307E147A320856C3B1716A0</creatorHandle>
<selectedWorkFlow>iss_wf:400001</selectedWorkFlow>
<taskType>GRPSTART</taskType>
```

```
<attributes>
</attributes>
</ser:createWorkFlowTask>
</soapenv:Body>
</soapenv:Envelope>
```

deleteBookmark

- Input Parameters: (int sid, String contactId, int docId)
- Return Type [XML Root Element]: int
- Description: Deletes a bookmark from a knowledge document.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:deleteBookmark>
<sid>139569419</sid>
<contactId>DD392B8AC9B2994893F7886017ACC60B</contactId>
<docId>400005</docId>
</ser:deleteBookmark>
</soapenv:Body>
</soapenv:Envelope>
```

deleteComment

- Input Parameters: (int sid, int commentId)
- Return Type [XML Root Element]: int
- Description: Deletes a comment for a knowledge document.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:deleteComment>
<sid>331513514</sid>
<commentId>?</commentId>
</ser:deleteComment>
</soapenv:Body>
</soapenv:Envelope>
```

deleteDocument

- Input Parameters: (int sid, int docId)
- Return Type [XML Root Element]: int
- Description: Flags a knowledge document for deletion.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
```

```
<soapenv:Header/>
<soapenv:Body>
<ser:deleteDocument>
<sid>331513514</sid>
<docId>400008</docId>
</ser:deleteDocument>
</soapenv:Body>
</soapenv:Envelope>
```

deleteWorkFlowTask

- Input Parameters: (int sid, String workFlowHandle, String objectHandle)
- Return Type [XML Root Element]: void
- Description: Removes a workflow task from its associated ticket.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:deleteWorkFlowTask>
<sid>139569419</sid>
<workFlowHandle>cr_wf:400014</workFlowHandle>
<objectHandle>cr:400203</objectHandle>
</ser:deleteWorkFlowTask>
</soapenv:Body>
</soapenv:Envelope>
```

detachChangeFromRequest

- Input Parameters: (int sid, String creator, String requestHandle, String description)
- Return Type [XML Root Element]: String
- Description: Detaches a change order from a request.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:detachChangeFromRequest>
<sid>139569419</sid>
<creator>DD392B8AC9B2994893F7886017ACC60B</creator>
<requestHandle>cr:400106</requestHandle>
<description>test</description>
</ser:detachChangeFromRequest>
</soapenv:Body>
</soapenv:Envelope>
```

doQuery

- Input Parameters: (int sid, String objectType, String whereClause)
- Return Type [XML Root Element]: ListResult [listHandle, listLength]
- Description: Performs a SQL-like select on the specified object type.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:doQuery>
<sid>139569419</sid>
<objectType>chg</objectType>
<whereClause>priority=2</whereClause>
</ser:doQuery>
</soapenv:Body>
</soapenv:Envelope>
```

doSelect

- Input Parameters: (int sid, String objectType, String whereClause, int maxRows, ArrayOfString attributes)
- Return Type [XML Root Element]: String [UDSList]
- Description: Performs a SQL-like select on the specified object type.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:doSelect>
<sid>139569419</sid>
<objectType>cnt</objectType>
<whereClause>last_nameLIKE'Service%'</whereClause>
<maxRows>2</maxRows>
<attributes>
<!--1ormorerepetitions:-->
<string>last_name</string>
<string>first_name</string>
</attributes>
</ser:doSelect>
</soapenv:Body>
</soapenv:Envelope>
```

doSelectKD

- Input Parameters: (int sid, String whereClause, String sortBy, boolean desc, int maxRows, ArrayOfString attributes, int skip)
- Return Type [XML Root Element]: String[UDSObjectList]
- Description: Performs a SQL-like select on the knowledge document object.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:doSelectKD>
<sid>912138386</sid>
<whereClause>id=400001</whereClause>
<sortBy>id</sortBy>
<desc>TRUE</desc>
<maxRows>-1</maxRows>
<attributes>
<!--1ormorerepetitions:-->
</attributes>
<skip>0</skip>
</ser:doSelectKD>
</soapenv:Body>
</soapenv:Envelope>
```

escalate

- Input Parameters: (int sid, String creator, String objectHandle, String description, boolean setAssignee, String newAssigneeHandle, boolean setGroup, String newGroupHandle, boolean setOrganization, String newOrganizationHandle, boolean setPriority, String newPriorityHandle)
- Return Type [XML Root Element]: String [UDSObject (Handle only)]
- Description: Performs an escalate activity on a ticket.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:escalate>
<sid>834772785</sid>
<creator>cnt:98E76E8F273CD3478FB4AEB981290869</creator>
<objectHandle>cr:400354</objectHandle>
<description>escalating</description>
<setAssignee>false</setAssignee>
<newAssigneeHandle/>
<setGroup>false</setGroup>
<newGroupHandle/>
<setOrganization/>
<newOrganizationHandle/>
<setPriority>true</setPriority>
<newPriorityHandle>pri:504</newPriorityHandle>
</ser:escalate>
</soapenv:Body>
</soapenv:Envelope>
```

faq

- Input Parameters: (int sid, String categoryIds, int resultSize, String propertyList, String sortBy, boolean descending, String whereClause, int maxDocIDs)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Performs a faq search on knowledge documents.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:faq>
<sid>673016018</sid>
<categoryIds>1</categoryIds>
<resultSize>10</resultSize>
<propertyList>TITLE,PROBLEM,SUMMARY,RESOLUTION</propertyList>
<sortBy></sortBy>
<descending>1</descending>
<whereClause></whereClause>
<maxDocIDs>50</maxDocIDs>
</ser:faq>
</soapenv:Body>
</soapenv:Envelope>
```

findContacts

- Input Parameters: (int sid, String userName, String lastName, String firstName, String email, String accessType, int inactiveFlag)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Retrieves a list of contacts.

freeListHandles

- Input Parameters: (int sid, ArrayOfInt handles)
- Return Type [XML Root Element]: void
- Description: Frees the server-side resources for lists and invalidates the list handles.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:freeListHandles>
<sid>834772785</sid>
<handles>
<integer>-607747422</integer>
<integer>1995893622</integer>
</handles>
</ser:freeListHandles>
</soapenv:Body>
</soapenv:Envelope>
```

getAccessTypeForContact

- Input Parameters: (int sid, String contactHandle)
- Return Type [XML Root Element]: String
- Description: Returns a handle for the Access Type of a contact.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getAccessTypeForContact>
<sid>1871998466</sid>
<contactHandle>cnt:DD392B8AC9B2994893F7886017ACC60B</contactHandle>
</ser:getAccessTypeForContact>
</soapenv:Body>
</soapenv:Envelope>
```

getArtifact

- Input Parameters: (int sid, String contact, String password)
- Return Type [XML Root Element]: String
- Description: Returns an artifact for appending to URLs.

getAssetExtensionInformation

- Input Parameters: (int sid, String assetHandle, ArrayOfString attributes, StringHolder getAssetExtInfoResult, StringHolder extensionHandle, StringHolder extensionName)
- Return Type [XML Root Element]: void
- Output Parameters (Type: StringHolder) [XML Root Element]: getAssetExtInfoResult [UDSObject] extensionHandle extensionName
- Description: Returns extension information for an asset.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getAssetExtensionInformation>
<sid>308769167</sid>
<assetHandle>nr:665DE0917C402B41940904946E9A48EB</assetHandle>
<attributes>
<string>audit_userid</string>
<string>creation_user</string>
<string>version_number</string>
<string>creation_date</string>
<string>total_capacity</string>
<string>persistent_id</string>
</attributes>
</ser:getAssetExtensionInformation>
</soapenv:Body>
</soapenv:Envelope>
```

getAttmntInfo

- Input Parameters: (int sid, int attmntId)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Returns the attributes for an attachment.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getAttmntInfo>
<sid>1333090960</sid>
<attmntId>400019</attmntId>
</ser:getAttmntInfo>
</soapenv:Body>
</soapenv:Envelope>
```

getAttmntList

- Input Parameters: (int sid, int folderId, int repId)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Returns a list of attachments under a given attachment folder.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getAttmntList>
<sid>1333090960</sid>
<folderId>1001</folderId>
<repId>1001</repId>
</ser:getAttmntList>
</soapenv:Body>
</soapenv:Envelope>
```

getAttmntListPerKD

- Input Parameters: (int sid, int docId)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Returns a list of attachments for a given knowledge document.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getAttmntListPerKD>
<sid>331513514</sid>
<docId>400008</docId>
</ser:getAttmntListPerKD>
</soapenv:Body>
```

```
</soapenv:Envelope>
```

getBookmarks

- Input Parameters: (int sid, String contactId)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Retrieves bookmarks for a given contact.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getBookmarks>
<sid>520739344</sid>
<contactId>cnt:DD6AE042C307E147A320856C3B1716A0</contactId>
</ser:getBookmarks>
</soapenv:Body>
</soapenv:Envelope>
```

getBopsid

- Input Parameters: (int sid, String contact)
- Return Type [XML Root Element]: String
- Description: Returns a token for appending to URLs.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getBopsid>
<sid>139569419</sid>
<contact>ServiceDesk</contact>
</ser:getBopsid>
</soapenv:Body>
</soapenv:Envelope>
```

getCategory

- Input Parameters: (int sid, int catId, boolean getCategoryPaths)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Retrieves information for a knowledge category.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getCategory>
<sid>1333090960</sid>
<catId>400001</catId>
```

```
<getCategoryPaths>1</getCategoryPaths>
</ser:getCategory>
</soapenv:Body>
</soapenv:Envelope>
```

getComments

- Input Parameters: (int sid, String docIds)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Retrieves all comments from the list of knowledge documents.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getComments>
<sid>331513514</sid>
<docIds>400008</docIds>
</ser:getComments>
</soapenv:Body>
</soapenv:Envelope>
```

getConfigurationMode

- Input Parameters: (int sid)
- Return Type [XML Root Element]: String
- Description: Returns confirmation if CA SDM is running in ITIL mode.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getConfigurationMode>
<sid>929546997</sid>
</ser:getConfigurationMode>
</soapenv:Body>
</soapenv:Envelope>
```

getContact

- Input Parameters: (int sid, String contactId)
- Return Type [XML Root Element]: String [UDSObject]
- Description: Retrieves information for a given contact.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
```

```
<ser:getContact>
<sid>1871998466</sid>
<contactId>DD392B8AC9B2994893F7886017ACC60B</contactId>
</ser:getContact>
</soapenv:Body>
</soapenv:Envelope>
```

getDecisionTrees

- Input Parameters: (int sid, String propertyList, String sortBy, boolean descending)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Retrieves all Decision Tree knowledge documents.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getDecisionTrees>
<sid>929546997</sid>
<propertyList>id,DOC_TYPE_ID,BU_RESULT</propertyList>
<sortBy>id</sortBy>
<descending>0</descending>
</ser:getDecisionTrees>
</soapenv:Body>
</soapenv:Envelope>
```

getDependentAttrControls

- Input Parameters: (int sid, String handle, ArrayOfString attrVals)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Returns a list of locked and required attributes for the Status object.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getDependentAttrControls>
<sid>929546997</sid>
<handle>cr:400251</handle>
<attrVals>
<!--1ormorerepetitions:-->
</attrVals>
</ser:getDependentAttrControls>
</soapenv:Body>
</soapenv:Envelope>
```

getDocument

- Input Parameters: (int sid, int docId, String propertyList, boolean relatedDoc, boolean getAttmnt, boolean getHistory, boolean getComment, boolean getNotiList)
- Return Type [XML Root Element]: String [UDSObject]
- Description: Retrieves information for a knowledge document.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getDocument>
<sid>929546997</sid>
<docId>400001</docId>
<propertyList>id</propertyList>
<relatedDoc>0</relatedDoc>
<getAttmnt>0</getAttmnt>
<getHistory>0</getHistory>
<getComment>0</getComment>
<getNotiList>0</getNotiList>
</ser:getDocument>
</soapenv:Body>
</soapenv:Envelope>
```

getDocumentsByIDs

- Input Parameters: (int sid, String docIds, String propertyList, String sortBy, boolean descending)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Retrieves information for one or more knowledge documents.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getDocumentsByIDs>
<sid>929546997</sid>
<docIds>400001,400002,400003,400004</docIds>
<propertyList>id,BU_RESULT,DOC_TYPE_ID</propertyList>
<sortBy>id</sortBy>
<descending>0</descending>
</ser:getDocumentsByIDs>
</soapenv:Body>
</soapenv:Envelope>
```

getDocumentTypes

- Input Parameters: (int sid)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Returns a list of all knowledge document types.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getDocumentTypes>
<sid>627668377</sid>
</ser:getDocumentTypes>
</soapenv:Body>
</soapenv:Envelope>
```

getFolderInfo

- Input Parameters: (int sid, int folderId)
- Return Type [XML Root Element]: String [UDSObject]
- Description: Retrieves information for a given attachment folder.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getFolderInfo>
<sid>640797658</sid>
<folderId>1001</folderId>
</ser:getFolderInfo>
</soapenv:Body>
</soapenv:Envelope>
```

getFolderList

- Input Parameters: (int sid, int parentFolderId, int repId)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Returns a list of folders under a given parent folder.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getFolderList>
<sid>640797658</sid>
<parentFolderId>0</parentFolderId>
<repId>1001</repId>
</ser:getFolderList>
</soapenv:Body>
```

```
</soapenv:Envelope>
```

getGroupMemberListValues

- Input Parameters: (int sid, String whereClause, int numToFetch, ArrayOfString attributes)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Queries the group of the system and member relationship.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getGroupMemberListValues>
<sid>640797658</sid>
<whereClause>member.last_nameLIKE'Service%'</whereClause>
<numToFetch>1</numToFetch>
<attributes>
<!--1ormorerepetitions:-->
<string>persistent_id</string>
<string>manager_flag</string>
<string>notify_flag</string>
<string>group</string>
<string>member</string>
<string>id</string>
</attributes>
</ser:getGroupMemberListValues>
</soapenv:Body>
</soapenv:Envelope>
```

getHandleForUserid

- Input Parameters: (int sid, String userID)
- Return Type [XML Root Element]: String
- Description: Returns the persistent handle for a contact.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getHandleForUserid>
<sid>1882907549</sid>
<userID>ServiceDesk</userID>
</ser:getHandleForUserid>
</soapenv:Body>
</soapenv:Envelope>
```

getKDListPerAttmnt

- Input Parameters: (int sid, int attmntId)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Returns a list of knowledge documents with reference to a given attachment.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getKDListPerAttmnt>
<sid>1882907549</sid>
<attmntId>400001</attmntId>
</ser:getKDListPerAttmnt>
</soapenv:Body>
</soapenv:Envelope>
```

getListValues

- Input Parameters: (int sid, int attmntId)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Returns a list of knowledge documents with reference to a given attachment.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getListValues>
<sid>1242674495</sid>
<listHandle>-1724830848</listHandle>
<startIndex>0</startIndex>
<endIndex>-1</endIndex>
<attributeNames>
<string>priority</string>
</attributeNames>
</ser:getListValues>
</soapenv:Body>
</soapenv:Envelope>
```

getLrelLength

- Input Parameters: (int sid, String contextObject, String lrelName)
- Return Type [XML Root Element]: int
- Description: Returns the number of objects on one side of a many-to-many relationship.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
```

```
<ser:getLrelLength>
<sid>1333090960</sid>
<contextObject>iss:400001</contextObject>
<lrelName>asset</lrelName>
</ser:getLrelLength>
</soapenv:Body>
</soapenv:Envelope>
```

getLrelValues

- Input Parameters: (int sid, String contextObject, String lrelName, int startIndex, int endIndex, ArrayOfString attributes)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Returns attribute values for a range of objects in a many-to-many relationship.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getLrelValues>
<sid>1242674495</sid>
<contextObject>chg:3230</contextObject>
<lrelName>asset</lrelName>
<startIndex>0</startIndex>
<endIndex>-1</endIndex>
<attributes>
<string>priority</string>
</attributes>
</ser:getLrelValues>
</soapenv:Body>
</soapenv:Envelope>
```

getNotificationsForContact

- Input Parameters: (int sid, String contactHandle, int queryStatus)
- Return Type [XML Root Element]: ListResult [listHandle, listLength]
- Description: Returns a list handle of notifications (lr objects) for a given contact.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getNotificationsForContact>
<sid>1242674495</sid>
<contactHandle>cnt:98E76E8F273CD3478FB4AEB981290869</contactHandle>
<queryStatus></queryStatus>
</ser:getNotificationsForContact>
</soapenv:Body>
</soapenv:Envelope>
```

getObjectTypeInformation

- Input Parameters: (int sid, String factory)
- Return Type [XML Root Element]: String [UDSObject (special)]
- Description: Returns a list of all attribute names for a given object.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getObjectTypeInformation>
<sid>1197146496</sid>
<factory>cnt</factory>
</ser:getObjectTypeInformation>
</soapenv:Body>
</soapenv:Envelope>
```

getObjectValues

- Input Parameters: (int sid, String objectHandle, ArrayOfString attributes)
- Return Type [XML Root Element]: String [UDSObject]
- Description: Returns the attribute values for a given object handle.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getObjectValues>
<sid>1242674495</sid>
<objectHandle>cnt:98E76E8F273CD3478FB4AEB981290869</objectHandle>
<attributes>
<string>first_name</string>
<string>last_name</string>
</attributes>
</ser:getObjectValues>
</soapenv:Body>
</soapenv:Envelope>
```

getPendingChangeTaskListForContact

- Input Parameters: (int sid, String contactHandle)
- Return Type [XML Root Element]: ListResult [listHandle, listLength]
- Description: Returns all the pending change order workflow tasks assigned to a given contact.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getPendingChangeTaskListForContact>
```

```
<sid>982096980</sid>
<contactHandle>cnt:DD392B8AC9B2994893F7886017ACC60B</contactHandle>
</ser:getPendingChangeTaskListForContact>
</soapenv:Body>
</soapenv:Envelope>
```

getPendingIssueTaskListForContact

- Input Parameters: (int sid, String contactHandle)
- Return Type [XML Root Element]: ListResult [listHandle, listLength]
- Description: Returns all the pending issue workflow tasks assigned to a contact.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getPendingIssueTaskListForContact>
<sid>982096980</sid>
<contactHandle>cnt:DD392B8AC9B2994893F7886017ACC60B</contactHandle>
</ser:getPendingIssueTaskListForContact>
</soapenv:Body>
</soapenv:Envelope>
```

getPermissionGroups

- Input Parameters: (int sid, int groupId)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Retrieves info for a permission group.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getPermissionGroups>
<sid>982096980</sid>
<groupId>0</groupId>
</ser:getPermissionGroups>
</soapenv:Body>
</soapenv:Envelope>
```

getPolicyInfo

- Input Parameters: (int sid)
- Return Type [XML Root Element]: String [SAPolicy]
- Description: Returns information about the access policy for the current session.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getPolicyInfo>
<sid>935378745</sid>
</ser:getPolicyInfo>
</soapenv:Body>
</soapenv:Envelope>
```

getPriorities

- Input Parameters: (int sid)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Retrieves all the knowledge priorities.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getPriorities>
<sid>935378745</sid>
</ser:getPriorities>
</soapenv:Body>
</soapenv:Envelope>
```

getPropertyInfoForCategory

- Input Parameters: (int sid, String categoryHandle, ArrayOfString attributes)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Retrieves property information for a given category.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getPropertyInfoForCategory>
<sid>834772785</sid>
<categoryHandle>pcat:5103</categoryHandle>
<attributes>
<string>id</string>
<string>code</string>
<string>persistent_id</string>
<string>label</string>
<string>description</string>
</attributes>
</ser:getPropertyInfoForCategory>
</soapenv:Body>
</soapenv:Envelope>
```

getQuestionsAsked

- Input Parameters: (int sid, int resultSize, boolean descending)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Retrieves historical knowledge document search text.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getQuestionsAsked>
<sid>834772785</sid>
<resultSize>10</resultSize>
<descending>true</descending>
</ser:getQuestionsAsked>
</soapenv:Body>
</soapenv:Envelope>
```

getRelatedList

- Input Parameters: (int sid, String objectHandle, String listName)
- Return Type [XML Root Element]: ListResult [listHandle, listLength]
- Description: Returns a list handle for a list (QREL or BREL) of an object.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getRelatedList>
<sid>1242674495</sid>
<objectHandle>cr:400354</objectHandle>
<listName>act_log</listName>
</ser:getRelatedList>
</soapenv:Body>
</soapenv:Envelope>
```

getRelatedListValues

- Input Parameters: (int sid, String objectHandle, String listName, int numToFetch, ArrayOfString attributes, StringHolder getRelatedListValuesResult, IntHolder numRowsFound)
- Return Type [XML Root Element]: void
- Output Parameters (Type: StringHolder) [XML Root Element]: getRelatedListValuesResult [UDSObjectList] numRowsFound (IntHolder)
- Description: Returns values for lists of an object.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
```

```
<soapenv:Body>
<ser:getRelatedListValues>
<sid>1242674495</sid>
<objectHandle>cr:400354</objectHandle>
<listName>act_log</listName>
<numToFetch>5</numToFetch>
<attributes>
<string>action_desc</string>
<string>analyst</string>
<string>call_req_id</string>
<string>description</string>
</attributes>
</ser:getRelatedListValues>
</soapenv:Body>
</soapenv:Envelope>
```

getRepositoryInfo

- Input Parameters: (int sid, int repositoryId)
- Return Type [XML Root Element]: String [UDSObject]
- Description: Returns information of a repository.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getRepositoryInfo>
<sid>982096980</sid>
<repositoryId>1001</repositoryId>
</ser:getRepositoryInfo>
</soapenv:Body>
</soapenv:Envelope>
```

getStatuses

- Input Parameters: (int sid)
- Return Type [XML Root Element]: String [UDSObject]
- Description: Retrieves all the knowledge statuses.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getStatuses>
<sid>935378745</sid>
</ser:getStatuses>
</soapenv:Body>
</soapenv:Envelope>
```

getTaskListValues

- Input Parameters: (int sid, String objectHandle, ArrayOfString attributes)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Returns values for tasks associated with a given issue or change order.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getTaskListValues>
<sid>1242674495</sid>
<objectHandle>iss:400001</objectHandle>
<attributes>
<string>assignee</string>
<string>description</string>
</attributes>
</ser:getTaskListValues>
</soapenv:Body>
</soapenv:Envelope>
```

getTemplateList

- Input Parameters: (int sid)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Retrieves all the document templates.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getTemplateList>
<sid>935378745</sid>
</ser:getTemplateList>
</soapenv:Body>
</soapenv:Envelope>
```

getValidTaskTransitions

- Input Parameters: (int sid, String taskHandle, ArrayOfString attributes)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Returns all possible status transitions for a particular task.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getValidTaskTransitions>
<sid>1242674495</sid>
```

```
<taskHandle>iss_wf:400001</taskHandle>
<attributes>
</attributes>
</ser:getValidTaskTransitions>
</soapenv:Body>
</soapenv:Envelope>
```

getWorkflowTemplateList

- Input Parameters: (int sid)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Retrieves all the knowledge workflow templates.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getWorkflowTemplateList>
<sid>982096980</sid>
</ser:getWorkflowTemplateList>
</soapenv:Body>
</soapenv:Envelope>
```

getWorkFlowTemplates

- Input Parameters: (int sid, String objectHandle, ArrayOfString attributes)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Returns all the workflow templates associated with the category of a change order or issue.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getWorkFlowTemplates>
<sid>609535729</sid>
<objectHandle>iss:400001</objectHandle>
<attributes>
<!--1ormorerepetitions:-->
<string>object_attrname</string>
<string>est_cost</string>
<string>object_attrval</string>
<string>auto_assign</string>
</attributes>
</ser:getWorkFlowTemplates>
</soapenv:Body>
</soapenv:Envelope>
```

impersonate

- Input Parameters: (int sid, String userid)
- Return Type [XML Root Element]: int
- Description: Invalidates the old session and returns a new session ID for the new user.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:impersonate>
<sid>520739344</sid>
<userid>cnt:DD392B8AC9B2994893F7886017ACC60B</userid>
</ser:impersonate>
</soapenv:Body>
</soapenv:Envelope>
```

isAttmntLinkedKD

- Input Parameters: (int sid, int attmntId)
- Return Type [XML Root Element]: int
- Description: Returns the number of attachment links to all knowledge documents.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:isAttmntLinkedKD>
<sid>331513514</sid>
<attmntId>attmnt:400106</attmntId>
</ser:isAttmntLinkedKD>
</soapenv:Body>
</soapenv:Envelope>
```

logComment

- Input Parameters: (int sid, String ticketHandle, String comment, int internalFlag)
- Return Type [XML Root Element]: void
- Description: Performs a log comment activity on a ticket.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:logComment>
<sid>331513514</sid>
<ticketHandle>cr:400101</ticketHandle>
<comment>axis</comment>
<internalFlag>withoutinternalflag</internalFlag>
```

```
</ser:logComment>
</soapenv:Body>
</soapenv:Envelope>
```

login

- Input Parameters: ((String username, String password)
- Return Type [XML Root Element]: int
- Description: Authenticates a user and returns a unique session ID.

loginService

- Input Parameters: (String username, String password, String policy)
- Return Type [XML Root Element]: int
- Description: Authenticates a user and returns a unique session ID.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:loginService>
<username>ServiceDesk</username>
<password>N0tallowed</password>
<policy>DEFAULT</policy>
</ser:loginService>
</soapenv:Body>
</soapenv:Envelope>
```

loginServiceManaged

- Input Parameters: (String policy, String encrypted_policy)
- Return Type [XML Root Element]: String
- Description: Performs user authentication for PKI configurations and returns a session ID.

loginWithArtifact

- Input Parameters: (String userid, String artifact)
- Return Type [XML Root Element]: int
- Description: Authenticates a user with artifact and returns a unique session ID.
- Input parameter for SOAP Requests:

logout

- Input Parameters: (int sid)
- Return Type [XML Root Element]: void
- Description: Invalidates and frees a session ID.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:logout>
<sid>982096980</sid>
</ser:logout>
</soapenv:Body>
</soapenv:Envelope>
```

modifyDocument

- Input Parameters: (int sid, int docId, ArrayOfString kdAttributes)
- Return Type [XML Root Element]: String [UDSObject]
- Description: Updates a knowledge document.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:modifyDocument>
<sid>802643880</sid>
<docId>400102</docId>
<kdAttributes>
<!--1ormorerepetitions:-->
<string>TITLE</string>
<string>hello</string>
</kdAttributes>
</ser:modifyDocument>
</soapenv:Body>
</soapenv:Envelope>
```

notifyContacts

- Input Parameters: (int sid, String creator, String contextObject, String messageTitle, String messageBody, int notifyLevel, ArrayOfString notifyees, boolean internal)
- Return Type [XML Root Element]: String [UDSObject (Handle only)]
- Description: Sends a notification to one or more contacts.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:notifyContacts>
<sid>431549590</sid>
<creator>DD392B8AC9B2994893F7886017ACC60B</creator>
<contextObject>cr:400253</contextObject>
<messageTitle>UpdateCXF</messageTitle>
<messageBody>HelloCXFData</messageBody>
```

```
<notifyLevel>1</notifyLevel>
<notifyees>
<!--1ormorerepetitions:-->
<string>>22484CFF0C8F6A4E9CDA3B0E439249E9</string>
</notifyees>
<internal>1</internal>
```

rateDocument

- Input Parameters: (int sid, int docId, int rating, int multiplier, String ticketPerId, boolean onTicketAccept, boolean solveUserProblem, boolean isDefault)
- Return Type [XML Root Element]: String [UDSObjectList]
- Description: Rates a particular knowledge document.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:rateDocument>
<sid>1901545587</sid>
<docId>400103</docId>
<rating>0</rating>
<multiplier>2</multiplier>
<ticketPerId></ticketPerId>
<onTicketAccept>false</onTicketAccept>
<solveUserProblem>false</solveUserProblem>
<isDefault>false</isDefault>
</ser:rateDocument>
</soapenv:Body>
</soapenv:Envelope>
```

removeAttachment

- Input Parameters: (int sid, String attHandle)
- Return Type [XML Root Element]: int
- Description: Removes an attachment from a ticket.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:removeAttachment>
<sid>841253387</sid>
<attHandle>attmnt:400106</attHandle>
</ser:removeAttachment>
</soapenv:Body>
</soapenv:Envelope>
```

removeLrelRelationships

- Input Parameters: (int sid, String contextObject, String lrelName, ArrayOfString removeObjectHandles)
- Return Type [XML Root Element]: void
- Description: Removes one or more many-to-many relationships.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:removeLrelRelationships>
<sid>308769167</sid>
<contextObject>cnt:8E3616E86592644DBE43A87C6D3817D4</contextObject>
<lrelName>cenv</lrelName>
<removeObjectHandles>
<string>nr:A0FC4439B41FC64499990EED44A40A15</string>
<string>nr:C48848CF35918C4C8D58BA7C119074C2</string>
</removeObjectHandles>
</ser:removeLrelRelationships>
</soapenv:Body>
</soapenv:Envelope>
```

removeMemberFromGroup

- Input Parameters: (int sid, String contactHandle, String groupHandle)
- Return Type [XML Root Element]: void
- Description: Removes a contact from a group.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:removeMemberFromGroup>
<sid>1505146322</sid>
<contactHandle>cnt:AE7753EBA85EB84385F20BC497577475</contactHandle>
<groupHandle>cnt:CC02CB2CE1D21648A2B8895E689CD144</groupHandle>
</ser:removeMemberFromGroup>
</soapenv:Body>
</soapenv:Envelope>
```

search

- Input Parameters: (int sid, String problem, int resultSize, String properties, String sortBy, boolean descending, boolean relatedCategories, int searchType, int matchType, int searchField, String categoryPath, String whereClause, int maxDocIDs)
- Return Type [XML Root Element]: String [UDSObjectList (nested)]
- Description: Searches for knowledge document solutions.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:search>
<sid>174545534</sid>
<problem>mail</problem>
<resultSize></resultSize>
<properties>TITLE,PROBLEM,SUMMARY,RESOLUTION</properties>
<sortBy></sortBy>
<descending>true</descending>
<relatedCategories>false</relatedCategories>
<searchType></searchType>
<matchType></matchType>
<searchField></searchField>
<categoryPath></categoryPath>
<whereClause></whereClause>
<maxDocIDs>50</maxDocIDs>
</ser:search>
</soapenv:Body>
</soapenv:Envelope>
```

serverStatus

- Input Parameters: (int sid)
- Return Type [XML Root Element]: int
- Description: Returns the status (up or down) of the CA SDM server.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:serverStatus>
<sid>331513514</sid>
</ser:serverStatus>
</soapenv:Body>
</soapenv:Envelope>
```

transfer

- Input Parameters: (int sid, String creator, String objectHandle, String description, boolean setAssignee, String newAssigneeHandle, boolean setGroup, String newGroupHandle, boolean setOrganization, String newOrganizationHandle)
- Return Type [XML Root Element]: String [UDSObject (Handle only)]
- Description: Performs a transfer activity on a ticket.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
```

```
<soapenv:Body>
<ser:transfer>
<sid>834772785</sid>
<creator>cnt:98E76E8F273CD3478FB4AEB981290869</creator>
<objectHandle>cr:400354</objectHandle>
<description>transferringticket</description>
<setAssignee>true</setAssignee>
<newAssigneeHandle>cnt:F4C2A60100EA154695D233F90ABFFD21</newAssigneeHandle>
<setGroup>false</setGroup>
<newGroupHandle></newGroupHandle>
<setOrganization>false</setOrganization>
<newOrganizationHandle></newOrganizationHandle>
</ser:transfer>
</soapenv:Body>
</soapenv:Envelope>
```

updateObject

- Input Parameters: (int sid, String objectHandle, ArrayOfString attrVals, ArrayOfString attributes)
- Return Type [XML Root Element]: String [UDSObject]
- Description: Updates one or more attributes for a given object.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:updateObject>
<sid>431549590</sid>
<objectHandle>cnt:22484CFF0C8F6A4E9CDA3B0E439249E9</objectHandle>
<attrVals>
<!--1ormorerepetitions:-->
<string>userid</string>
<string>testuser1</string>
<string>last_name</string>
<string>testuserLastNameupdated</string>
</attrVals>
<attributes>
<!--1ormorerepetitions:-->
<string>userid</string>
</attributes>
</ser:updateObject>
</soapenv:Body>
</soapenv:Envelope>
```

updateRating

- Input Parameters: (int sid, int buId, int rate)
- Return Type [XML Root Element]: String [UDSObject]
- Description: Updates an existing rating of a knowledge document.
- Input parameter for SOAP Requests:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://www.ca.com/
UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:updateRating>
<sid>1901545587</sid>
<buId>400139</buId>
<rate>0</rate>
</ser:updateRating>
</soapenv:Body>
</soapenv:Envelope>
```

loginWithArtifact

- Input Parameters: (String userid, String artifact)
- Return Type [XML Root Element]: int
- Description: Authenticates a user with artifact and returns a unique session ID.

getArtifact

- Input Parameters: (int sid, String contact, String password)
- Return Type [XML Root Element]: String
- Description: Returns an artifact for appending to URLs.

XML Object Returns

Many of the Web Services methods return an XML representation of CA SDM objects. The Web Services uses a standard XML structure beginning with the following root element:

```
<UDSObject>
```

The format of the XML representation is described in the following table:

XML Element	Type	Description
<UDSObject>	N/A	Identifies the root node.
<Handle>	String	Identifies the object's handle.
<Attributes>	Sequence	Identifies the attribute values. This holds zero or more elements for the object's attribute values.

<*attrName0* DataType = *“typeEnum"*>	String	Identifies the *AttrName0,* which is an object attribute name as defined in the CA SDM majic (.maj) or mod (.mod) file. This name may use dot-notation depending on the web method used. The element’s value is the attribute’s value. An empty element indicates a null/empty value for this object’s attribute. The DataType attribute is an integer indicating the attribute’s data type in the CA SDM environment.

For example, a call to getObjectValues() can return information illustrated by the following:

```
<UDSObject>
    <Handle>cnt:555A043EDDB36D4F97524F2496B35E75</Handle>
    <Attributes>
        <Attribute DataType="2003">
            <AttrName>first_name</AttrName>
            <AttrValue>first name</AttrValue>
            <DisplayValue>Yaakov</DisplayValue>
        </Attribute>
        <Attribute DataType="2005">
            <AttrName>organization</AttrName>
                <AttrValue>342</AttrValue>
        <DisplayValue>Accounting Crew</DisplayValue>
    </Attribute>
</Attributes>
    <Lists>        <List name="mylist1">
            <UDSObject>...</UDSObject>
            <UDSObject>...</UDSObject>
        </List>
    </Lists>
</UDSObject>
```

Some methods, such as doSelect(), return a sequence of <UDSObject> elements contained inside a <UDSObjectList> element.

The <Lists> section holds zero or more <List> nodes. A <List> node holds zero or more <UDSObject> nodes. <List> elements are generally returned only when a specific request for list values is made.

When you want to return a list of values related to a specific object, you should use the *getRelatedListValues* method.

If a request is made just for a list with no attribute name, such as actlog, then the entire <UDSObject> is returned in the <List> section.

Specialized methods, like getDocument(), can of course be different. When a request is made for an attribute, the database value is returned. For SREL attributes, this may not be so useful. Requesting the assignee attribute of a Request returns an integer because the Contact REL_ATTR (foreign key) is its ID. For CA Service Desk Manager r11.0, the return data for attributes includes elements for the DBMS and common name value of SREL references.

REST HTTP Methods

This article contains the following topics:

The REST API supports the following HTTP methods to manipulate resources:

- POST (CREATE) creates a resource.
- GET (READ) returns a representation of a resource.
- PUT (UPDATE) updates an existing resource.
- DELETE deletes the resource.

Refer to this basic set of methods as CRUD. Each method works in the same manner on all CA SDM resources. You require an HTTP client library, available with most programming languages. Use the HTML client library to complete the following tasks:

- Access and modify associated (or related) resources using a multilevel URI path.
- Send an HTTP request to the server for the resource that you want to manipulate.
- Control the object attributes to be returned by using HTTP headers.

After you update Majic object definitions and recycle CA SDM, the product automatically regenerates and redeploys the corresponding Plain Old Java Objects (POJOs) into the REST Tomcat *webapps* directory.

NOTE
The pdm_rest_util command line utility lets you *manually* generate, compile, and deploy Java code that REST web services require.

WARNING
Requests for attributes that do *not* respond indicate a null attribute value. Modify your client code accordingly because REST does *not* display null attribute values in responses.

REST Web Services provide a scalable configuration and better flexibility to our users by letting you connect to a dedicated domsrvr on the local server. By default, CA SDM Release 12.9 provides the NX_REST_WEBSERVICE_DOMSRVR variable to domsrvr. You can edit NX.env to change this setting.

WARNING
The REST API does not support Majic attributes of type DOUBLE.

The following table shows how the REST API uses HTTP methods on resources.

Resource	CREATE	READ	UPDATE	DELETE
Collection URL For example: http://mywebsite.com/resources/	Creates an entry in the collection. Assigns a new entry URL automatically and returns it by the operation.	Lists the URLs and other details of the collection members.	N/A	N/A
Element URL For example: http://mywebsite.com/resources/item1	N/A	Retrieves a representation of the addressed member of the collection.	Updates the addressed member of the collection, and when the member does not exist, creates it.	Deletes the addressed member of the collection.

Sample URI Paths for CRUD Operations

The following table shows sample URI paths for CRUD operations:

CRUD Operation	HTTP Method	Noun	Sample URI Path
CREATE	Post	URI	/caisd-rest/chg
READ (Multiple)	GET	Collection URI	/caisd-rest/chg
Read (Multiple with Filter)	GET	Collection URI	/caisd-rest/chg?WC=status%3D6001

Read (Single ID)	GET	Entry URI	/caisd-rest/chg/400001
Read (Single COMMON_NAME)	GET	Entry URI	/caisd-rest/chg/COMMON_NAME-21
Read (Single REL_ATTR)	GET	Entry URI	/caisd-rest/chgstat/REL_ATTR-OP
Update	PUT	Entry URI	/caisd-rest/chg/40001
Delete	DELETE	Entry URI	/caisd-rest/grpmem/400001

WARNING
COMMON_NAME and REL_ATTR are case-sensitive. If you do not use upper case, REST returns the HTTP-404 error code.

NOTE
When specifying a contact (last name, first name, middle name), you can delimit the contact name with commas (,) or blank spaces, but not both. Commas are preferable because names often have embedded spaces, which can cause issues.

REST Considerations

Consider the following information about REST methods:

- POST operations create a resource from the provided representation, but CA SDM can add or modify attribute values internally. These values are based on business object logic that attribute triggers, SPEL code, methods, or both set. The result causes the POST operation to return the actual representation that it created to the client. This behavior lets clients reconcile the representation that they sent to the server with the actual representation that they created.
- For GET operations, enter **%20** instead of a space in the WHERE clause.

REST Limitations

The REST API does *not* support dotted attribute name references in its resource queries or attributes. REST does *not* expose any BREL, QREL or LREL attribute that contains a dotted attribute in its relationship query. For example, the chg object in base.maj contains the following QREL attributes:

```
workload_chg QREL chg DYNAMIC {
WHERE "assignee = ? and active = 1" ;
PARAM_NAMES { id } ;
DOMSET chg_list;
} ;
change_tasks QREL wf DYNAMIC {
WHERE "assignee = ? AND status.allow_task_update = 1 " ;
PARAM_NAMES { id } ;
DOMSET wf_list ;
} ;
```

In this example, workload_chg QREL attributes are available. The change_tasks QREL is unavailable because it contains a dotted attribute (status.allow_task_update) in its WHERE clause.

REST and Object Access

Many CA SDM components consist of objects. A metalanguage named Majic defines these objects. Majic statements let you create objects and modify existing objects, so you can customize these objects to meet your business requirements. You can customize the exposure of the CA SDM objects through REST to determine the following:

- Objects accessible through REST
- Permitted operations for each object

After you install CA SDM, all objects are accessible through REST. The REST_OPERATIONS keyword specifies the operations that you can perform on an object. By default, all objects allow the CREATE, READ, and UPDATE operations. Some default objects also permit the DELETE operation, as indicated in the REST_OPERATIONS list in their object definition. You do *not* need to use the REST_OPERATIONS keyword for the default setup. You can refine object access to perform the following tasks:

- Override the REST_OPERATIONS list by using a Majic MODIFY statement to remove or add operations, except for the DELETE operation.
  Note: You cannot add the DELETE operation to any default object.
- Remove an object from REST entirely by specifying NONE in the REST_OPERATIONS list.

rest_access resource

The rest_access resource contains REST API access information for the authenticated users. It is an administrative table and contains the list of users allowed through the REST API.

The following list shows the deviations from the default behavior:

- **POST**
  Creates a REST access object and returns access key, secret key and expiration date as part of default values.
- **GET**
  Retrieves REST access information (except for secret key).
- **DELETE**
  Deletes the REST access object.
- **PUT**
  Does not allow updates to secret_key, access_key, and contact due to the Majic WRITE_NEW property.

REST_OPERATIONS Keyword

The Majic keyword, REST_OPERATIONS, uses a parenthesized list of tokens. These tokens let you specify the REST operations that are permitted for an object. You use an object definition and a MODIFY clause to customize the keyword.

The REST_OPERATIONS keyword has the following syntax:

```
REST_OPERATIONS "[<OP>],[<OP>] | NONE";
```

- **OP**
  Specifies the CREATE, READ, UPDATE, and DELETE operations.

This keyword produces the following statements:

```
REST_OPERATIONS "CREATE, READ, UPDATE DELETE";
REST_OPERATIONS "NONE";
```

At run time, CA SDM allows overrides by using the following Majic MODIFY statement:

```
MODIFY FACTORY cr {
             REST_OPERATIONS "READ, UPDATE";
       };
```

REST_OPERATIONS Syntax Examples

The following example modifies the cr object to limit the operations to READ and UPDATE. Any attempt to create or delete the object returns an error of *HTTP error code 405 Method Not Allowed*.

WARNING
CA SDM sets the default value for a REST_OPERATIONS property as “CREATE READ UPDATE” for most Objects. For example, cr, iss, and chg. For some object, CA SDM sets the default REST_OPERATIONS value to “CREATE READ UPDATE DELETE.” For example, rest_access and KCAT.

```
MODIFY FACTORY cr {
REST_OPERATIONS "READ UPDATE";
};
```

The following example removes announcements from REST completely:

```
MODIFY FACTORY cnote {
REST_OPERATIONS "NONE";
};
```

Note: You cannot use DELETE in a MODIFY statement. The NONE operation is standalone and not allowed with the CREATE READ UPDATE combination in the MODIFY statement.

REST API Servlet Settings

You can configure additional REST API settings in the web deployment descriptor *web.xml* file. The file *$NX_ROOT/site/rest/web.xml.tpl* is provided as a template and should not be modified. When you first start CA Service Desk Manager, it creates a *web.xml* file from the template into the same directory. You can modify this file, which is included in the WAR file to be deployed.

Available Parameters

ExternalBaseURI

This optional setting allows you to provide a custom REST API address for hyperlinks included in the response body. This is useful for sites that have load balancers, reverse proxy servers, external hostnames, https addresses, etc. Make sure the REST API web app name such as “caisd-rest” is included in this value. The sample default value is: http://host:8050/caisd-rest.

Perform the following steps:

1. Stop all CA Service Desk Manager services.
2. Navigate to the $NX_ROOT/site/rest/ folder and edit the *web.xml* file.
3. Locate REST API external address section in the file and uncomment the following elements:
   <context-param>
   <param-name>ExternalBaseURI</param-name>
   <param-value>http://host:port_number/caisd-rest</param-value >
   </context-param>

   NOTE

   : Replace *host:port_number* with an appropriate external address and port number.
4. Save and close the file.
5. Restart the CA Service Desk Manager services.
   You can recreate the file if Tomcat fails to start or report errors.

NULL Value in Update Operation

You can set the HTTP header *X-AttrsToNull* and assign a NULL value to an attribute for the update operations. This header takes a list of comma-separated attribute names. The header value takes precedence over the attributes in the request body in case of duplicates. BREL and QREL type attributes do not apply, but SREL attributes do.

Example 1: Update the *call_back_date* attribute of a Request ticket to NULL. In this example, setting *call_back_date* to NULL requires that the *call_back_flag* is not equal to 1.

HTTP URL: PUT /caisd-rest/cr/400055HTTP Header X-AttrsToNull: call_back_datePayload (XML): <cr><call_back_flag>0</call_back_flag></cr>

Example 2: Update the email address to NULL for a contact.

HTTP URL: PUT /caisd-rest/cnt/U'ADE6396E2A3D9E4A958AC522521C5A35'HTTP Header X-AttrsToNull: email_addressPayload (XML): <cnt></cnt>

> **NOTE**
>
> The examples above do not show all HTTP headers.

Working with BLRELs

The BLREL keyword refers to Majic attributes that point to an LREL table. These attributes are part of a three-factory relationship. BLREL attributes were known as LREL attributes in previous releases of CA SDM, but they have changed to BREL attributes. This name helps distinguish BLRELs from the classic BREL attributes which are only a two-factory relationship.

An example of this three-factory relationship is the group and members relationship. This relationship consists of a grp object, a cnt object, and the LREL table object that the grpmem object represents. In this three-factory relationship, two BLREL attributes always point to a common LREL table. In the group/member relationship, both BLREL attributes are frequently part of the same object, but not always.

The following example shows how CA SDM defines BLREL attributes in Majic files.

```
OBJECT cnt {
    ...
    member_list         BREL  grpmem group DYNAMIC { LREL member; } ;
    group_list          BREL  grpmem member DYNAMIC { LREL group; } ;
}
```

You can always view the attributes of an object by executing the bop_sinfo command on an active system:

```
bop_sinfo -d grpmem
```

The LREL table record creates an association between records of two other tables. For the group/member relationship, the grpmem record is the LREL table that creates this association. It is a many-to-many relationship, where groups can have many members, and members can be part of many groups.

WHERE Clause Resource Search

REST Web Services let you use a *WC* (WHERE clause) query for the GET request. REST Web Services let you use a WC (WHERE clause) query for the GET request. REST supports the SQL query notation, and offers limited support to the Majic query notation. For example, the dotted attribute notation will pass if they have passed in other areas of the product (SOAP). If you do not add a WC parameter to the URL, all records return up to the page size and maximum size limits.

Consider the following information:

- Use percent-encoding for all requests before you send them to the server. For example, enter **%3D** instead of =, the equals sign.
- The WC parameter is part of the URI instead of a message header to allow bookmarking.
- The WC query parameter only applies to Collection GET requests.

Example: Search for Contact Records with a Last Name that Equals ServiceDesk

```
http://<host>:<REST port>/caisd-rest/cnt?WC=last_name%3D'ServiceDesk'
```

Example: Search for Priority 2 Incidents

```
/in?WC=priority%3D4
```

The 4 value in the example specifies the REL_ATTR value for the Priority 2 record.

> **WARNING**
> Use REL_ATTR values in the WC parameter when you reference SREL attributes such as priority.

Example: Search for Incidents that Contain test% in the Summary Field

```
/in?WC=summary%20LIKE%20'test%25'
```

Valid URI Path Patterns

Not all resource paths and subresource paths expose all four HTTP methods, depending on the URI path level and the resource. The following list shows the exposed methods:

- **/caisd-rest/<factory>**

Example: /caisd-rest/cr

GET searches for objects and returns a collection of objects.

POST creates an object and returns the object that you created.

- **/caisd-rest/<factory>/<object id>**

Example: /caisd-rest/cr/400001

GET returns details for one object.

PUT updates the object.

DELETE removes the object. If allowed.

- **/caisd-rest/<factory>/<object id>/<attribute name (SREL)>**

Example: /caisd-rest/chg/400001/status

GET returns Details for the SREL attribute object, same as going directly to the SREL object if you know the ID. For example, /caisd-rest/crs/5200

- **/caisd-rest/<object>/<object id>/<attribute name (BREL)>**

Example: /caisd-rest/chg/400001/act_log

GET returns a collection of objects of the type the BREL points to.

- **/caisd-rest/<factory>/<object id>/<attribute name (BLREL)>**

Example: /caisd-rest/chg/400001/attachments

GET returns a collection of objects of the type the BLREL points to (the LREL record).

- **/caisd-rest/<factory>/<object id>/<attribute name (QREL)>**

Example: /caisd-rest/chg/400001/workflow

GET returns a collection of objects of the type the QREL points to.

Search Result Sorting

CA SDM Web Services let you sort search results by using the SORT query parameter, included as part of the URI. You can specify a list of attributes as the sort order. You can also specify ASC for ascending order, or specify DESC for descending order.

For default behavior, the SORT parameter provides that same support that the DBMS provides for the ORDER BY clause. For example, if you omit ASC or DESC, the DBMS may use ASC automatically. If you do not provide a SORT parameter in the URI, the DBMS uses *id ASC* by default.

This behavior only applies to collection GET requests, such as the example:

```
GET http://<host>:<REST-port>/caisd-rest/<fac name>
GET http://myserver:8080/caisd-rest/cr?SORT=priority DESC, severity ASC
```

Consider the following information:

- You *must* encode all requests before sending them to the server. For example, enter **%20** instead of spaces.
- All object attributes specified in the SORT parameter add to the X-Obj-Attrs list of object attributes automatically. This behavior helps ensure that the attributes selected for sorting are part of the response.
- The SORT parameter is part of the URI instead of a message header to allow for bookmarking.

Note: If you do not provide the X-Obj-Attrs header, id, REL_ATTR, and COMMON_Name list by default, as they always appear as part of the response.

Search Result Navigation

Provide ATOM links to improve navigating search and list results. These links take the user to the previous or next step when more records become available. By default, the Collection GET result contains up to 25 records. Use the rest_webservice_list_page_length and rest_webservice_list_max_length options to configure the page length and maximum number of records to display.

The ATOM links appear only for available records for that purpose. REST does not provide the *previous* link if the current page contains the first record in the list. Similarly, REST only provides the *next* and *all* links when you have more available records.

Users can also provide their own list return size with the *size* parameter. If the user does not provide *size*, REST defaults to the value of rest_webservice_list_page_length. If you do not provide start, REST defaults to 1.

Example Request Returns a Specific Number of Records

In this example, you request a return of 10 records:

```
http://myserver:8050/caisd-rest/cnt?start=11&size=10
```

If this result set contained 50 records, the following links appear:

```
<link href="http://myserver:8050/caisd-rest/cnt?start=1&size=10" rel="previous"/>
<link href="http://myserver:8050/caisd-rest/cnt?start=21&size=10" rel="next"/>
<link href="http://myserver:8050/caisd-rest/chg?start=1&size=50" rel="all"/>
```

Note: When the result set contains more than the maximum number allowed, the all link does not appear.

HTTP Status and Error Codes

Every HTTP response contains an HTTP status code. Successful HTTP response code numbers range from 200 to 399. Standard HTTP error response code numbers range from 400 to 599.

The following list shows the successful and error code numbers that the REST API returns for each of the HTTP methods.

- **GET (single)**
  200, 400, 401, 404, 409
- **GET (collection)**
  200, 400, 401
- **PUT**

200, 400, 401, 404, 409

- **DELETE**
  204, 400, 401, 409
- **POST**
  201, 400, 401, 409

Note: For the PUT and DELETE operations, the API does *not* try to determine the validity of the ID initially. Instead, the API tries to run the update and delete queries directly. If an error occurs, the API returns the *409 Conflict* code. If the API returns a *404 Not Found* code in this situation, performance degrades.

In addition, the following error messages return under the following cases:

- If the HTTP request contains an invalid or inaccessible URI address, the server responds with a *404 Not Found* response code.
- If the HTTP request contains an unsupported HTTP method for a valid URI, the server responds with a *405 Method Not Allowed* response code.
- If the HTTP request requests an unsupported media type (Accept header), the server responds with a *406 Not Acceptable* response code.
- If the HTTP request sends an unsupported media type (Content-Type header), the server responds with a *415 Unsupported Media Type* response code.
- Various syntax or internal Web Server errors can return a 500 internal error.

Code Matching Limitations

Because of a limited number of available HTTP codes, not all CA SDM errors match to an HTTP error code. Matching errors return the corresponding HTTP error codes whenever possible, but most backend process errors return a generic *400 Bad Request* code. A message summarizes the error.

The following error messages match the corresponding HTTP error code. All others use error code 400.

- A request for a resource that does not exist in CA SDM returns a *404 Not Found* error code.
- A request for a resource that returns multiple matches, such as when using a nonunique COMMON_NAME value returns a *409 Conflict* error code.
- A request for an unaccessible resource due to Authentication Failure or Function Access security return a *401 Unauthorized* error code.

Known Status Codes

The following list describes the known status codes that the API returns. Other codes may exist from the web server or the CXF framework, but it depends on the type of error.

- **200**
  OK
  Indicates a successful return.
- **201**
  Created
  Indicates a new record.
- **204**
  No Content
  Indicates an empty response body.
- **304**
  Not Modified
  Indicates that the record did not update.
- **400**
  Bad Request

Indicates that an error occurred due to a user or backend server issue.

- **401**
  Unauthorized
  Indicates a function Access error or any authentication failure.
- **404**
  Not Found
  Indicates that a record is not found.
- **405**
  Method Not Allowed
  Indicates an unsupported HTTP method.
- **406**
  Not Acceptable
  Indicates an unsupported requested format.
- **409**
  Conflict
  Indicates that multiple records were found for the given identifier.
- **415**
  Unsupported Media Type
  Indicates that the provided format is not supported.
- **500**
  Internal Server Error
  Indicates an error on the server or CXF framework.

Atom Feeds

CA SDM supports Atom feeds through the collection GET URI path because Atom feeds represent a list of records. This implementation supports all query parameters that REST Web Services support, such as the WC, start, and size query parameters. This implementation supports two additional query parameters, EntryTitle and EntrySummary, which let you specify a mapping between the Atom entry Title and Summary elements to Majic attributes.

```
EntryTitle=<Majic attribute name>
EntrySummary=<Majic attribute name>
```

Note: If you do *not* specify these parameters, REST uses the COMMON_NAME as the default title and summary values.

The following sample displays these query parameters:

```
GET /caisd-rest/cnt?size=2&EntryTitle=userid&EntrySummary=notes HTTP/1.1
Accept: application/atom+xml
```

The following sample displays the Atom response:

```
<?xml version="1.0" encoding="UTF-8"?>
<feed xmlns="http://www.w3.org/2005/Atom">
  <author>
    <name>CA Service Desk Manager</name>
  </author>
  <title type="text">REST API Atom feed</title>
  <id>http://myserver:8050/caisd-rest/cnt</id>
  <updated>2012-01-17T17:56:04.301Z</updated>
  <link href="http://myserver:8050/caisd-rest/cnt?start=3&#38;size=2"
  rel="next"/>
  <link href="http://myserver:8050/caisd-rest/cnt?start=1&#38;size=12"
  rel="all"/>
  <link href="http://myserver:8050/caisd-rest/cnt"
```

```
  rel="self"/>
  <entry>
    <author>
      <name>CA Service Desk Manager</name>
    </author>
    <title type="text">System_AM_User</title>
    <id>http://myserver:8050/caisd-rest/cnt/U'16226C765005B94E957E0F477DEF1B1C'</id>
    <updated>1970-01-01T00:00:00.000Z</updated>
    <summary type="text"> User for Asset Management Integration</summary>
    <content type="application/xml">
      <cnt id="U'16226C765005B94E957E0F477DEF1B1C'" REL_ATTR="U'16226C765005B94E957E0F477DEF1B1C'"
      COMMON_NAME="System_AM_User" xmlns="">
        <link href="http://myserver:8050/caisd-rest/cnt/U'16226C765005B94E957E0F477DEF1B1C'"
        rel="self"/>
      </cnt>
    </content>
  </entry>
  <entry>
    <author>
      <name>CA Service Desk Manager</name>
    </author>
    <title type="text">cavizuser</title>
    <id>http://myserver:8050/caisd-rest/cnt/U'17DEA1027C7C3746B6F25DB6604EEE23'</id>
    <updated>1970-01-01T00:00:00.000Z</updated>
    <summary type="text">Username used by Visualizer for
            accessing CA Service Desk Manager using Web Services</summary>
    <content type="application/xml">
      <cnt id="U'17DEA1027C7C3746B6F25DB6604EEE23'" REL_ATTR="U'17DEA1027C7C3746B6F25DB6604EEE23'"
      COMMON_NAME="System_CMDB_Visualizer_User" xmlns="">
        <link href="http://myserver:8050/caisd-rest/cnt/U'17DEA1027C7C3746B6F25DB6604EEE23'"
        rel="self"/>
      </cnt>
    </content>
  </entry>
</feed>
```

Additional REST Support when Requesting Data Formats

REST Web Services also supports other ways of specifying the representation format using the URI. The Web Services work similarly specifying the representation format in the Accept HTTP header.

Example 1:

GET /caisd-rest/chg/400001.xml HTTP/1.1

GET /caisd-rest/chg/400001.json HTTP/1.1

GET /caisd-rest/chg.feed HTTP/1.1

Example 2:

GET /caisd-rest/chg/400001?_type=xml HTTP/1.1

GET /caisd-rest/chg/400001?_type=json HTTP/1.1

GET /caisd-rest/chg?_type=atom HTTP/1.1

Consider the following information:

- If the representation format is specified on the URI and through the “Accept” header, the one in the URI takes precedence over the Accept header.
- If a representation format is not specified in any way, the representation returns in the XML format by default.
- The *_type=* query parameter supersedes all parameters.

CA SDM Role Authorization

As with SOAP Web Services, part of the REST Web Services Access Key creation (login operation) includes verifying the user has authorization to access REST Web Services. In SOAP, the Web Service and API Role lookup field controls this verification in the Access Type detail form. In REST, a new lookup field named REST Web Service API Role controls the verification for REST. You can only associate one role to this field, and this field is the default role for the user. If this lookup field is empty, the users belonging to this Access Type do not have access to CA SDM through the REST Web Services interface.

In addition, REST Web Services supports the same list of Attached Roles that are part of the Web Client interface. A REST user can select a different role from the list of Attached Roles (including the roles in its Contact record) by passing in an additional message header as part of the request.

Example: Use the Administrator role for the request

```
POST /caisd-rest/cnt HTTP/1.1
Host: hostname
Date: Mon, 21 Apr 2011 19:37:58 +0000
X-Role: 10002
```

REST Java Sample Code

CA SDM provides REST Java sample code in the NX_ROOT\samples\sdk\rest\java directory. These sample files let you build custom Java client code in your environment, similar to SOAP web service samples. This directory contains instructions for compiling and executing the sample programs in the README.txt file. Each Java file documents additional instructions and sample input parameters. The following directories in \rest\java provide the following samples:

- **test1_basic**
- SampleBasicAuth.java
  Demonstrates how to get an Access Key using a username/password through the Basic Authentication scheme.
- SampleCRUDOperations.java
  Demonstrates how to perform simple CRUD operations (Create, Read, Update and Delete) on Incident tickets. The same can be used for any other object in CA SDM.
- **test2_auths**
- SampleBOPSIDAuth.java
  Demonstrates how to get an Access Key using a BOPSID token. This CA SDM token can be obtained from other CA SDM Interfaces such as SOAP Web Services.
- SampleEEMAuth.java
  Demonstrates how to get an Access Key using an CA EEM artifact/token. This token can be obtained from other applications that use CA EEM as their authentication server.
- SampleSDMAuth.java
  Demonstrates how to get an Access Key and a Secret Key using a username/password through the CA SDM custom authentication scheme.
- SampleUsingSecretKey.java

Demonstrates how to make a REST API request (get a list of Contacts) using the Secret Key. It uses the Secret Key obtained from the CA SDM custom authentication operation to encrypt a configurable amount of data. All requests using this scheme are verified against the Secret Key.

- **test3_attachments**
- SampleNewResourceWithAttachment.java
  Demonstrates how to create a new Change Order ticket with an attachment document all in one step.
- SampleAttachFileToResource.java
  Demonstrates how to attach a document to an existing Change Order ticket.
- **test4_xrels**
- SampleGetQRELDetails.java
  Demonstrates how to retrieve details on a QREL attribute. In this sample, details for chg.children are retrieved. The same approach works for BREL and BLREL attributes.
- SampleCreateBRELResource.java
  Demonstrates how to create a new record for a BREL attribute. In this sample, a new Log Comment activity is added to an existing Change Order. The same approach works for creating a new record for BLREL attributes.

Build and Execute the Sample Programs

You can build and execute the sample Java programs to test them in your environment.

Follow these steps:

1. Copy rest_java_test.bat.txt (Windows) or rest_java_test.sh.txt (UNIX) to the subdirectory where the sample program exists.
2. Rename the copied file by removing the .txt extension.
3. Edit the copied file and verify that the first three SET variables are correct for your CA SDM installation.
4. In addition, edit the Java file that you want to run and verify that the configurable variables are set for your CA SDM installation. Also, read the comments at the top of the page.

CA SDM Authentication Scheme

CA SDM provides sample code in the NX_ROOT/samples/sdk/rest/java directory. This directory also contains instructions about how to compile and execute the sample programs. REST Web Services support the following security authentication schemes:

- REST Secret Key Authentication that uses SSL and HMAC for login
- REST BOPSID Authentication that validates CA SDM BOPSIDs
- REST Basic Authentication that uses clear text encoded username and password
- External (CA EEM) Artifact Authentication that uses a CA EEM artifact token

REST Secret Key Authentication

REST Secret Key authentication uses a custom, secure HMAC mechanism. This authentication lets CA SDM verify the identity of a user and also verifies that the request came from a registered, verified user. To complete this authentication successfully, each request must provide information about the identity of the request sender.

HMAC_ALGORITHM supports the HmacSHA1, HmacSHA256, HmacSHA384, HmacSHA512, and HmacMD5 valid HMAC algorithms. By default, REQUEST_METHOD, REQUEST_URI, and QUERY_STRING (if present) are used to compute the signature. You can also use other header fields to compute the signature, such as date, x-obj-attrs, and content-type. To add these fields set the following NX variable in NX.env file:

```
@NX_STRING_TO_SIGN_FIELDS=date,x-obj-attrs,content-type
```

Note: When you compute the signature on the Client side, use the same fields exactly in the same order as specified the fields for the STRING_TO_SIGN_FIELDS option.

- **Access Key**

The Access Key is an assigned value to CA SDM clients after a successful login authentication. Requests use the Access Key to identify the client responsible for the request. However, because an Access Key is sent as a request parameter, the Access Key is not secret. A possibility exists that anyone could use the Access Key by sending a request to CA SDM. As a result, this authentication requires a Secret Key. REST uses the CA SDM session ID as the Access Key.

- **Secret Key**
  After you log in to CA SDM successfully, the product assigns a Secret Key and Access Key to the client. To protect users from impersonation, the client must provide additional information that CA SDM can use to verify the identity. CA SDM generates the 40-character Secret Key during the REST Access Key creation automatically.

The following steps describe the authentication process:

1. The client obtains the Access Key and Secret Key through the REST URI (POST /caisd-rest/rest_access) by providing user credentials using the basic authentication style over SSL.
2. For every subsequent HTTP request, the client uses the Secret Key, NX_STRING_TO_SIGN_FIELDS provides the header fields, and the NX_HMAC_ALGORITHM variable provides the hash function. This function calculates the request signature, a Keyed-Hash based Message Authentication Code (HMAC).
3. The client sends the request data, the signature, and the Access Key to CA SDM.
4. CA SDM uses the Access Key to look up the Secret Key from the persistence store.
5. CA SDM uses the request data and the Secret Key to generate the signature using the same hash algorithm that the client used.
6. If the signature that CA SDM generates matches the signature that the Client sent, CA SDM considers the request as authentic. Otherwise, CA SDM discards the request and returns an error response.

REST BOPSID Authentication

After a user logs in to CA SDM through one interface, they can access CA SDM from a different interface by using a BOPSID token. The BOPSID token provides a way for the invoked application to authenticate the user without requiring a login, such as Single Sign-On. After CA SDM authenticates a user, the application can request a BOPSID from boplgin. This single-use token identifies the user and session. Boplgin returns the userid and session when it receives the BOPSID. Boplgin also cancels the BOPSID if it does not receive a verification request within 5 minutes of token creation.

The following example shows the HTTP message header for passing the BOPSID token:

```
POST /caisd-rest/rest_access HTTP/1.1
Host: hostname
Date: Mon, 21 Apr 2012 19:37:58 +0000

X-BOPSID: <BOPSID token>
```

Note: To obtain a BOPSID token through REST Web Services, you send a POST request to /caisd-rest/bopsid.

REST Basic Authentication

The basic authentication scheme assumes that the credentials of a client contain a username and password, where the password is a secret known only to the client and server.

If the incoming request does not contain the client credentials, the server sends back a 401 response that contains an authentication challenge. This challenge consists of the "Basic" token and a name-value pair that specifies the name of the protected realm, such as the following example:

```
WWW-Authenticate: Basic realm="<USDK> 12.9"
```

After receiving the 401 response from the server, the client (such as a browser) prompts for the username and password associated with that realm. The Authentication header of the client follow-up request should contain the "Basic" token and the base64-encoded group of the username, password, and a colon.

```
POST /caisd-rest/rest_access HTTP/1.1
```

```
Host: hostname
Date: Mon, 21 Apr 2012 19:37:58 +0000

Authorization: Basic QWRtaW46Zm9vYmFy
```

REST decodes credentials using base64 and compares them against the username/password and validates the credentials through boplgin. If this validation succeeds, the server provides access to the requested resource.

If the user sends the BOPSID instead of the username and password, CA SDM uses the the boplgin method validate_bopsid(). If you are concerned about using basic authentication scheme, you can disable it by setting the Options Manager option NX_REST_WEBSERVICE_DISABLE_BASIC_AUTH to Yes.

Important! Basic authentication is not as secure as the Secret Key method. However, you can use Basic authentication over an SSL connection for increased security.

External CA EEM Artifact Authentication

You can use CA EEM Artifact Authentication for REST requests. Clients send the CA EEM Artifact with the username using predefined customer headers (X-ExtAuthArtifact, X-UserName). When this header entry appears in an incoming request, the security interceptor performs the login validation by sending a VALIDATE_ARTIFACT message to boplgin.

The following example shows how to use the CA EEM Artifact:

```
POST /caisd-rest/rest_access HTTP/1.1
Host: hostname
Date: Mon, 21 Apr 2012 19:37:58 +0000

X-ExtAuthArtifact: <EEM Artifact token>
X-UserName: <username>
```

CRUD Operations on Tickets

The REST API provides sample Java code for the user to work with tickets through the REST API. These files contain the following operations:

- Create a Change Order with hard-coded sample data.
- Update a recent Change Order with a status update.
- Create an Incident and display the Incident number on the console.
- Get a list of Incident numbers using a where clause and display on the console.

BREL, QREL, and BLREL Processing

The REST API provides the sample Java code for the user to work with tickets through the REST API with BREL and QREL attributes. The BREL, QREL, and BLREL attributes only support the Get operation. The files contain the following operations:

- Create BREL using the documented, multi-step process. For example, add activity logs to a Change Order. For QREL, create multiple Change Orders as children to a Change Order. For BLREL, add multiple CIs to a Change Order.
- Get a list of attributes that returns the URI for a BREL, QREL, or BLREL attribute, and display them on the console.
- Get the collection of the BREL, QREL, or BLREL attribute with the URI and display it on the console. For example, /caisd-rest/chg/40001/act_log.

Managing Attachments for Tickets

The REST API provides the sample Java code for the user to work with tickets through the REST API. The files contain the following operations:

- Create an Incident.
- Create an attachment.
- Add the attachment to the Incident (2 step).
- Create an Incident with the attachment (1 step).
- Delete the attachment from the Incident.
- Download an Attachment
  Perform the following to download an attachment file:

1. a. To download a file, issue the following HTTP request:

```
GET /caisd-rest/attmnt/<attmnt id>/file-resource
```

   Where 'file resource' is the plain text
   For example, to download a file associated with the attachment record 400015, issue the following request:

```
GET /caisd-rest/attmnt/400015/file-resource
```

   The response content type for file downloads is *application/octet-stream.* A client such as Apache Commons HTTP Client can easily handle this content type. For a complete working sample using Apache Commons HTTP Client, check *SampleDownloadAttachment.java* file located in the *NX_ROOT\samples\sdk\rest\java\test3_attachments* directory.

CA SDM Resource Examples

Examples demonstrate how REST API uses basic Create, Read, Update, and Delete (CRUD) HTTP operations on CA SDM objects. Use the examples to understand how each REST operation works in the same manner on all CA SDM objects.

> **NOTE**
> For readability, the examples do not show all HTTP message headers -- only the relevant information.

Example Create a Change Order With an Attachment

This REST API example provides a complex use case: create a change order ticket and an attachment.

> **NOTE**
> MIME types application/xml and text/xml are often used interchangeably. We recommend that you use application/xml. All text/*-MIME types have an us-ascii character set unless otherwise explicitly specified in the HTTP headers. In effect, any encoding defined in the XML prolog (for example, <?xml version=”1.0” encoding=”UTF-8”?>) is ignored.

The following example shows the request:

```
POST /caisd-rest/chg/?repositoryId=1002&AttachmentId=att1&serverName=HOSTNAME&mimeType=doc&description=Desc
 HTTP/1.1
Content-Type: multipart/form-data
X-AccessKey: 51461077
User-Agent: Jakarta Commons-HttpClient/3.0.1
Host: hostname:8050
Content-Length: 1045

--Vschb1Sy2JD93ODUnWVAkRxp3IoXIMgXd
Content-Disposition: form-data; name="chg"
Content-Type: application/xml; charset=US-ASCII
Content-Transfer-Encoding: 8bit
```

```
<chg><description>Attachments Testing</description><status COMMON_NAME="RFC" REL_ATTR="RFC" id="40020"><link
 rel="self" href="http://hostname:8050/caisd-rest/chgstat/40020"/></status><summary>Attachment
 test</summary><requestor COMMON_NAME="ServiceDesk" REL_ATTR="U'279B25DD051D0A47B54880D86700397F'"
 id="U'279B25DD051D0A47B54880D86700397F'"><link rel="self" href="http://hostname:8050/caisd-rest/cnt/
U'279B25DD051D0A47B54880D86700397F'"/></requestor></chg>
--Vschb1Sy2JD93ODUnWVAkRxp3IoXIMgXd
Content-Disposition: form-data; name="Test.txt"; filename="Test.txt"
Content-Type: application/octet-stream; charset=ISO-8859-1
Content-Transfer-Encoding: binary
```

The following example shows the response:

```
HTTP/1.1 201 Created
Content-Type: application/xml;charset=UTF-8
Location: http://hostname:8050/caisd-rest/chg/400202

<chg id="400202" REL_ATTR="400202" COMMON_NAME="1047">
    <link href="http://hostname:8050/caisd-rest/chg/400202" rel="self"/>
</chg>
```

Example Create a Resource

This REST API example demonstrates how to create a resource. In this example, the REST API creates a change order ticket.

The following example shows the request:

```
POST /caisd-rest/chg HTTP/1.1
Host: hostname
Accept: application/xml
Content-Type: application/xml;charset=UTF-8
X-Obj-Attrs: chg_ref_num
<chg>
  <summary>Created via REST API</summary>
  <requestor REL_ATTR="U'793ED69B4E87A545BD8E911834D829FC'"/>
</chg>
```

The following example shows the response:

```
HTTP/1.1 201 Created
Content-Type: application/xml;charset=UTF-8
Location: http://hostname:8050/caisd-rest/chg/400001

<?xml version="1.0" encoding="UTF-8"?>
<chg id="400003" REL_ATTR="400003" COMMON_NAME="23">
  <link href="http://hostname:8050/caisd-rest/chg/400003"
        rel="self"/>
  <chg_ref_num>23</chg_ref_num>
</chg>
```

Example Delete a Resource

This REST API example demonstrates how to delete a resource.

NOTE
Not all objects are available for deletion. Most objects only support updating the delete_flag to true or false, or status equal to active or inactive. You perform these actions using an UPDATE request rather than a DELETE request.

The following example shows the request:

```
DELETE /caisd-rest/grpmem/400001 HTTP/1.1
Host: hostname
```

The following example shows the response:

```
HTTP/1.1 204 No Content
Content-Type: application/xml;charset=UTF-8
Content-Length: 0
```

Example Delete an Access Key

This REST API example demonstrates how to delete a CA SDM access key.

The following example shows the request:

```
DELETE /caisd-rest/rest_access/1201703106 HTTP/1.1
Host: hostname
```

The following example shows the response:

```
HTTP/1.1 204 No Content
Content-Type: application/xml;charset=UTF-8
Content-Length: 0
```

Example Mark a Resource Inactive

This REST API example demonstrates how to mark a resource as inactive.

NOTE
Most CA SDM objects only support marking them inactive and not actually deleting them.

The following example shows the request:

```
PUT /caisd-rest/loc/U'0502D608F9122B48B7C9DAB9E0457F94' HTTP/1.1
Host: hostname
Content-Type: application/xml;charset=UTF-8
X-Obj-Attrs: delete_flag

<loc id="U'0502D608F9122B48B7C9DAB9E0457F94'">
  <delete_flag COMMON_NAME="Inactive"/>
</loc>
```

The following example shows the response:

```
HTTP/1.1 200 OK
Content-Type: application/xml;charset=UTF-8

<?xml version="1.0" encoding="UTF-8"?>
<loc id="U'0502D608F9122B48B7C9DAB9E0457F94'"
     REL_ATTR="U'0502D608F9122B48B7C9DAB9E0457F94'"
     COMMON_NAME="Stadium 1">
  <link href="http://hostname:8050/caisd-rest/loc/U'0502D608F9122B48B7C9DAB9E0457F94'"
```

```
        rel="self"/>
  <delete_flag id="4552" REL_ATTR="1"
               COMMON_NAME="Inactive">
    <link href="http://hostname:8050/caisd-rest/actbool/4552"
          rel="self"/>
  </delete_flag>
</loc>
```

Example Obtain a BOPSID Token

A BOPSID token is single use authentication token that provides a single sign-on capability. This REST API example demonstrates how to obtain a BOPSID token to use for logging in to the web client.

The following example shows the request:

```
POST /caisd-rest/bopsid HTTP/1.1
Host: hostname.ca.com
Accept: application/xml
<bopsid/>
```

The following example shows the response:

```
HTTP/1.1 201 OK
Content-Type: application/xml;charset=UTF-8

<bopsid>
  <bopsid_val>987982618</bopsid_val>
</bopsid>
```

Example Obtain an Access Key

This REST API example demonstrates how to obtain an access key (login) for userid ServiceDesk. Perform this operation using SSL to avoid risking an unauthorized user stealing your secret key. Keep the force_unique_userid Options Manager option enabled at all times. When you disable this option, and multiple contact records with the same login ID exist, problems with data partitions, multi-tenancy, security, and other functions can occur.

The following example shows the request:

```
POST /caisd-rest/rest_access HTTP/1.1
Host: hostname
Content-Type: application/xml;charset=UTF-8
Authorization: Basic QWxhZGRpbjpvcGVuIHNlc2FtZQ==
<rest_access/>
```

The following example shows the response:

```
HTTP/1.1 201 Created
Content-Type: application/xml

<?xml version="1.0" encoding="UTF-8"?>
<rest_access id="400001" REL_ATTR="400001" COMMON_NAME="770921656">
    <link href="http://hostname:8050/caisd-rest/rest_access/400001"
    rel="self"/>
    <access_key>770921656</access_key>
    <expiration_date>1335276895</expiration_date>
</rest_access>
```

Example Retrieve a Collection of Resources

This REST API example demonstrates how to retrieve a collection of resources. In this example, the REST API retrieves a collection of all change order ticket objects.

> **NOTE**
> The root node (for example, collection_chg) also has other link elements such as previous, next, and all for navigating lists.

The following example shows the request:

```
GET /caisd-rest/chg HTTP/1.1
Host: hostname
Accept: application/xml
```

The following example shows the response:

```
HTTP/1.1 200 OK
Content-Type: application/xml;charset=UTF-8

<?xml version="1.0" encoding="UTF-8"?>
<collection_chg>
  <chg id="400001" REL_ATTR="400001" COMMON_NAME="21">
    <link href="http://hostname:8050/caisd-rest/chg/400001"
          rel="self"/>
  </chg>
  <chg id="400002" REL_ATTR="400002" COMMON_NAME="22">
    <link href="http://hostname:8050/caisd-rest/chg/400002"
          rel="self"/>
  </chg>
</collection_chg>
```

Example Retrieve a Collection of Resources Using a Where Clause

This REST API example demonstrates how to retrieve a collection of resources using a where clause. In this example, the REST API retrieves a collection of all request (cr) ticket objects with the Closed status.

> **NOTE**
> BREL queries are supported.

The following examples show the request:

```
GET /caisd-rest/cr?WC=status%3D'cl'
GET /caisd-rest/cr?WC=status%3D'op'%20and%20active%3D0
Host: hostname
Accept: application/xml
X-Obj-Attrs: ref_num, priority
```

The following example shows the response:

```
HTTP/1.1 200 OK
Content-Type: application/xml;charset=UTF-8

<?xml version="1.0" encoding="UTF-8"?>
<collection_cr COUNT="3" TOTAL_COUNT="3">
  <cr id="2903" REL_ATTR="cr:2903" COMMON_NAME="AM:12">
    <link href="http://hostname:8050/caisd-rest/cr/2903"
          rel="self"/>
    <priority id="502" REL_ATTR="3" COMMON_NAME="3">
      <link href="http://hostname:8050/caisd-rest/pri/502"
```

```
            rel="self"/>
    </priority>
    <ref_num>AM:12</ref_num>
  </cr>
  <cr id="2907" REL_ATTR="cr:2907" COMMON_NAME="AM:14">
    <link href="http://hostname:8050/caisd-rest/cr/2907"
          rel="self"/>
    <priority id="502" REL_ATTR="3" COMMON_NAME="3">
      <link href="http://hostname:8050/caisd-rest/pri/502"
            rel="self"/>
    </priority>
    <ref_num>AM:14</ref_num>
  </cr>
  <cr id="3105" REL_ATTR="cr:3105" COMMON_NAME="UAPM:13">
    <link href="http://hostname:8050/caisd-rest/cr/3105"
          rel="self"/>
    <priority id="502" REL_ATTR="3" COMMON_NAME="3">
      <link href="http://hostname:8050/caisd-rest/pri/502"
            rel="self"/>
    </priority>
    <ref_num>UAPM:13</ref_num>
  </cr>
</collection_cr>
```

Example Retrieve a Specific Resource

This REST API example demonstrates how to retrieve a specific resource using ID, COMMON_NAME, or REL_ATTR. In this example, the REST API retrieves details about the unique change order ticket object ID 400001.

The following example shows the request when you use ID:

```
GET /caisd-rest/chg/400001 HTTP/1.1
Host: hostname.ca.com
Accept: application/xml
X-Obj-Attrs: chg_ref_num, summary, requestor, status
```

The following example shows the request when you use COMMON_NAME:

```
GET /caisd-rest/chg/COMMON_NAME-32 HTTP/1.1
```

The following example shows the request when you use REL_ATTR:

```
GET /caisd-rest/chg/REL_ATTR-chg:400001 HTTP/1.1
```

The following example shows the response:

```
HTTP/1.1 200 OK
Content-Type: application/xml;charset=UTF-8

<?xml version="1.0" encoding="UTF-8"?>
<chg id="400001" REL_ATTR="400001" COMMON_NAME="21">
  <link href="http://hostname:8050/caisd-rest/chg/400001"
        rel="self"/>
  <chg_ref_num>21</chg_ref_num>
  <requestor id="U'793ED69B4E87A545BD8E911834D829FC'"
             REL_ATTR="U'793ED69B4E87A545BD8E911834D829FC'"
             COMMON_NAME="System_AHD_generated">
```

```
    <link href="http://hostname:8050/caisd-rest/cnt/U'793ED69B4E87A545BD8E911834D829FC'"
          rel="self"/>
  </requestor>
  <status id="40020" REL_ATTR="RFC" COMMON_NAME="RFC">
    <link href="http://hostname:8050/caisd-rest/chgstat/40020"
          rel="self"/>
  </status>
  <summary>Testing</summary>
```

Example Retrieve a Subresource

This REST API example demonstrates how to retrieve a subresource.

The following example shows the request:

```
GET /caisd-rest/chg/400001/status HTTP/1.1
Host: hostname
Accept: application/xml
X-Obj-Attrs: code, sym, description
HTTP/1.1 200 OK
Content-Type: application/xml;charset=UTF-8

<?xml version="1.0" encoding="UTF-8"?>
<chgstat id="40020" REL_ATTR="RFC" COMMON_NAME="RFC">
  <link href="http://hostname:8050/caisd-rest/chgstat/40020"
        rel="self"/>
  <code>RFC</code>
  <description>Request for Change is in draft form</description>
  <sym>RFC</sym>
</chgstat>
```

Example Update a Resource

This REST API example demonstrates how to update a resource. In this example, the REST API updates the summary field for the change order ticket ID 400003.

The following example shows the request:

```
PUT /caisd-rest/chg/400003 HTTP/1.1
Host: hostname
Content-Type: application/xml;charset=UTF-8
X-Obj-Attrs: chg_ref_num, summary
<chg id="400003">
  <summary>Summary updated via REST API</summary>
</chg>
```

The following example shows the response:

```
HTTP/1.1 200 OK
Content-Type: application/xml

<?xml version="1.0" encoding="UTF-8"?>
<chg id="400003" REL_ATTR="400003" COMMON_NAME="23">
  <link href="http://hostname:8050/caisd-rest/chg/400003"
        rel="self"/>
  <chg_ref_num>23</chg_ref_num>
  <summary>Summary updated via REST API</summary>
```

```
</chg>
```

Example Get a BLREL Record

This example shows how to retrieve a BLREL record:

The following example shows the request:

```
GET /caisd-rest/grpmem HTTP/1.1
Host: localhost
Accept: application/xml
X-Obj-Attrs: *
```

The following example shows the response:

```
HTTP/1.1 200 OK
Content-Type: application/xml;charset=UTF-8
<?xml version="1.0" encoding="UTF-8"?>
<collection_grpmem COUNT="3" START="1" TOTAL_COUNT="3">
    <grpmem id="400001" REL_ATTR="grpmem:400001" COMMON_NAME="400001">
        <link href="http://localhost:8050/caisd-rest/grpmem/400001"
        rel="self"/>
        <group id="U'CFCC2DC94B3A66448C085B07E7286CAA'" REL_ATTR="U'CFCC2DC94B3A66448C085B07E7286CAA'"
        COMMON_NAME="Unicef">
            <link href="http://localhost:8050/caisd-rest/grp/U'CFCC2DC94B3A66448C085B07E7286CAA'"
            rel="self"/>
        </group>
        <manager_flag>0</manager_flag>
        <member id="U'3F05B7450203AD449BFB8088D991A03E'" REL_ATTR="U'3F05B7450203AD449BFB8088D991A03E'"
        COMMON_NAME="System_SD_User">
            <link href="http://localhost:8050/caisd-rest/cnt/U'3F05B7450203AD449BFB8088D991A03E'"
            rel="self"/>
        </member>
        <notify_flag>1</notify_flag>
        <persistent_id>grpmem:400001</persistent_id>
        <producer_id>grpmem</producer_id>
    </grpmem>
    <grpmem id="400002" REL_ATTR="grpmem:400002" COMMON_NAME="400002">
        <link href="http://localhost:8050/caisd-rest/grpmem/400002"
        rel="self"/>
        <group id="U'55E3CCE805756B4F8084D63E05E6C216'" REL_ATTR="U'55E3CCE805756B4F8084D63E05E6C216'"
        COMMON_NAME="Apache">
            <link href="http://localhost:8050/caisd-rest/grp/U'55E3CCE805756B4F8084D63E05E6C216'"
            rel="self"/>
        </group>
        <manager_flag>0</manager_flag>
        <member id="U'FCF9A8AC6381AA4386C9B10EE382E10B'" REL_ATTR="U'FCF9A8AC6381AA4386C9B10EE382E10B'"
        COMMON_NAME="System_MA_User">
            <link href="http://localhost:8050/caisd-rest/cnt/U'FCF9A8AC6381AA4386C9B10EE382E10B'"
            rel="self"/>
        </member>
        <notify_flag>1</notify_flag>
        <persistent_id>grpmem:400002</persistent_id>
        <producer_id>grpmem</producer_id>
    </grpmem>
```

```
    <grpmem id="400003" REL_ATTR="grpmem:400003" COMMON_NAME="400003">
        <link href="http://localhost:8050/caisd-rest/grpmem/400003"
        rel="self"/>
        <group id="U'55E3CCE805756B4F8084D63E05E6C216'" REL_ATTR="U'55E3CCE805756B4F8084D63E05E6C216'"
        COMMON_NAME="Apache">
            <link href="http://localhost:8050/caisd-rest/grp/U'55E3CCE805756B4F8084D63E05E6C216'"
            rel="self"/>
        </group>
        <manager_flag>0</manager_flag>
        <member id="U'16226C765005B94E957E0F477DEF1B1C'" REL_ATTR="U'16226C765005B94E957E0F477DEF1B1C'"
        COMMON_NAME="System_AM_User">
            <link href="http://localhost:8050/caisd-rest/cnt/U'16226C765005B94E957E0F477DEF1B1C'"
            rel="self"/>
        </member>
        <notify_flag>1</notify_flag>
        <persistent_id>grpmem:400003</persistent_id>
        <producer_id>grpmem</producer_id>
    </grpmem>
</collection_grpmem>
```

This sample returns three grpmem records that represent the following associations:

- Group Unicef <-> Member System_SD_User
- Group Apache <-> Member System_MA_User
- Group Apache <-> Member System_AM_User

Example Retrieve a List of LREL Records Associated with a Group

The following example returns two grpmem records associated with the Group Apache whose REL_ATTR value is U'55E3CCE805756B4F8084D63E05E6C216'. Note that

- Group Apache <-> Member System_MA_User
- Group Apache <-> Member System_AM_User

The following example shows the request:

```
GET /caisd-rest/grpmem?WC=group%3DU%2755E3CCE805756B4F8084D63E05E6C216%27 HTTP/1.1
Host: hostname
Accept: application/xml
X-Obj-Attrs: *
```

The following example shows the response:

```
HTTP/1.1 200 OK
Content-Type: application/xml;charset=UTF-8
<?xml version="1.0" encoding="UTF-8"?>
<collection_grpmem COUNT="2" START="1" TOTAL_COUNT="2">
    <grpmem id="400002" REL_ATTR="grpmem:400002" COMMON_NAME="400002">
        <link href="http://hostname:8050/caisd-rest/grpmem/400002"
        rel="self"/>
        <group id="U'55E3CCE805756B4F8084D63E05E6C216'" REL_ATTR="U'55E3CCE805756B4F8084D63E05E6C216'"
        COMMON_NAME="Apache">
            <link href="http://hostname:8050/caisd-rest/grp/U'55E3CCE805756B4F8084D63E05E6C216'"
            rel="self"/>
        </group>
        <manager_flag>0</manager_flag>
        <member id="U'FCF9A8AC6381AA4386C9B10EE382E10B'" REL_ATTR="U'FCF9A8AC6381AA4386C9B10EE382E10B'"
```

```
        COMMON_NAME="System_MA_User">
            <link href="http://hostname:8050/caisd-rest/cnt/U'FCF9A8AC6381AA4386C9B10EE382E10B'"
            rel="self"/>
        </member>
        <notify_flag>1</notify_flag>
        <persistent_id>grpmem:400002</persistent_id>
        <producer_id>grpmem</producer_id>
    </grpmem>
    <grpmem id="400003" REL_ATTR="grpmem:400003" COMMON_NAME="400003">
        <link href="http://hostname:8050/caisd-rest/grpmem/400003"
        rel="self"/>
        <group id="U'55E3CCE805756B4F8084D63E05E6C216'" REL_ATTR="U'55E3CCE805756B4F8084D63E05E6C216'"
        COMMON_NAME="Apache">
            <link href="http://hostname:8050/caisd-rest/grp/U'55E3CCE805756B4F8084D63E05E6C216'"
            rel="self"/>
        </group>
        <manager_flag>0</manager_flag>
        <member id="U'16226C765005B94E957E0F477DEF1B1C'" REL_ATTR="U'16226C765005B94E957E0F477DEF1B1C'"
        COMMON_NAME="System_AM_User">
            <link href="http://hostname:8050/caisd-rest/cnt/U'16226C765005B94E957E0F477DEF1B1C'"
            rel="self"/>
        </member>
        <notify_flag>1</notify_flag>
        <persistent_id>grpmem:400003</persistent_id>
        <producer_id>grpmem</producer_id>
    </grpmem>
</collection_grpmem>
```

Example Create a BLREL Record

The following example adds an existing contact System_SA_User as a member of group Apache. It uses the COMMON_NAME value of each of those records.

Group Apache <-> Member System_SA_User

The following example shows the request:

```
POST /caisd-rest/grpmem HTTP/1.1
Host: hostname
Accept: application/xml
Content-Type: application/xml;charset=UTF-8
X-Obj-Attrs: *
<grpmem>
    <group id="U'55E3CCE805756B4F8084D63E05E6C216'"/>
    <manager_flag>0</manager_flag>
    <member id="U'E70DFE4817614C06BE9E5991A96A6015'"/>
    <notify_flag>1</notify_flag>
</grpmem>
```

The following example shows the response:

```
HTTP/1.1 201 Created
Content-Type: application/xml;charset=UTF-8
<?xml version="1.0" encoding="UTF-8"?>
<grpmem id="400005" REL_ATTR="grpmem:400005" COMMON_NAME="400005">
    <link href="http://hostname:8050/caisd-rest3/grpmem/400005"
```

```
    rel="self"/>
    <group id="U'55E3CCE805756B4F8084D63E05E6C216'" REL_ATTR="U'55E3CCE805756B4F8084D63E05E6C216'"
    COMMON_NAME="Apache">
        <link href="http://hostname:8050/caisd-rest3/grp/U'55E3CCE805756B4F8084D63E05E6C216'"
        rel="self"/>
    </group>
    <manager_flag>0</manager_flag>
    <member id="U'E70DFE4817614C06BE9E5991A96A6015'" REL_ATTR="U'E70DFE4817614C06BE9E5991A96A6015'"
    COMMON_NAME="System_SA_User">
        <link href="http://hostname:8050/caisd-rest3/cnt/U'E70DFE4817614C06BE9E5991A96A6015'"
        rel="self"/>
    </member>
    <notify_flag>1</notify_flag>
    <persistent_id>grpmem:400005</persistent_id>
    <producer_id>grpmem</producer_id>
</grpmem>
```

Example Update a BLREL Record

The following example updates the relationship of the added record and sets member System_SA_User as a Manager.

The following example shows the request:

```
PUT /caisd-rest/grpmem/400005 HTTP/1.1
Host: hostname
Accept: application/xml
Content-Type: application/xml;charset=UTF-8
X-Obj-Attrs: *
<grpmem>
    <manager_flag>1</manager_flag>
</grpmem>
```

The following example shows the response:

```
HTTP/1.1 200 OK
Content-Type: application/xml;charset=UTF-8
<?xml version="1.0" encoding="UTF-8"?>
<grpmem id="400005" REL_ATTR="grpmem:400005" COMMON_NAME="400005">
    <link href="http://hostname:8050/caisd-rest3/grpmem/400005"
    rel="self"/>
    <group id="U'55E3CCE805756B4F8084D63E05E6C216'" REL_ATTR="U'55E3CCE805756B4F8084D63E05E6C216'"
    COMMON_NAME="Apache">
        <link href="http://hostname:8050/caisd-rest3/grp/U'55E3CCE805756B4F8084D63E05E6C216'"
        rel="self"/>
    </group>
    <manager_flag>1</manager_flag>
    <member id="U'E70DFE4817614C06BE9E5991A96A6015'" REL_ATTR="U'E70DFE4817614C06BE9E5991A96A6015'"
    COMMON_NAME="System_SA_User">
        <link href="http://hostname:8050/caisd-rest3/cnt/U'E70DFE4817614C06BE9E5991A96A6015'"
        rel="self"/>
    </member>
    <notify_flag>1</notify_flag>
    <persistent_id>grpmem:400005</persistent_id>
    <producer_id>grpmem</producer_id>
</grpmem>
```

Example Delete a BLREL Record

The following example removes member System_MA_User from group Apache. This example does not delete the member record or the group record. It only removes the association between them. It does physically remove the grpmem record.

The following example shows the request:

```
DELETE /caisd-rest/grpmem/400002 HTTP/1.1
Host: hostname
```

The following example shows the response:

```
HTTP/1.1 204 No Content
Content-Type: application/xml;charset=UTF-8
Content-Length: 0

Additional REST API settings can be configured via web deployment descriptor web.xml file.
File $NX_ROOT\site\rest\web.xml.tpl is provided as a template and should not be modified.
On first run, SDM will create a new file web.xml from template into the same directory.  Users can
modify this file, which will be included in the WAR file to be deployed.

Updating Servlet Configuration Parameters

To make updates, perform the following steps:
*    Shutdown SDM
*    Open file $NX_ROOT/site/rest/web.xml for editing
*    Find the parameter to be updated inside <param-name> tag
*    Carefully update the value inside corresponding <param-value> tag
*    Save file
*    Startup SDM
If for any reason, REST Tomcat fails to start or reports errors due to the above changes, you can start
fresh by creating a new web.xml file from template $NX_ROOT/site/rest/web.xml.tpl.

Available Parameters

ExternalBaseURI

This optional setting allows you to provide a custom REST API address for hyperlinks included in the
response body.  This is useful for sites that have load balancers, reverse proxy servers, external
hostnames, https addresses, etc.  Make sure the REST API web app name such as “caisd-rest” is included
in this value.
The sample default value is http://host:8050/caisd-rest
```

Web Services Attachment-Related Methods

This article contains the following topics:

This section describes the Web Services attachment-related methods. Only file attachments are handled by these methods; link type attachments are handled by generic methods, such as CreateObject(). In addition, file uploads through Web Services employ the Direct Internet Message Encapsulation (DIME) protocol. Your SOAP implementation must support DIME in order to use these methods.

NOTE

- This is valid only for Axis 1.4 based SOAP interface, and not recommended to be used in the latest implementations.
- The latest version of Axis2 and CXF frameworks dropped support for DIME encapsulation in favor of SOAP Message Transmission Optimization Mechanism(MTOM), and DIME was dropped.

createAttachment

The following parameters apply to the createAttachment method:

Parameter	Data Type	Description
SID	INTEGER	Identifies the session retrieved from logging in.
repositoryHandle	STRING	Identifies the object handle of a document repository.
objectHandle	STRING	Identifies the object handle of a call request, change order, or issue, to which this attachment is attached. This parameter can be NULL, however, you must manage the attachment ID that is returned because the attachment is not associated to a ticket when NULL is passed in.
description	STRING	Identifies the description for the attachment object.
fileName	STRING	Identifies the full path of the file to be uploaded.

Description

Uploads a file to the back-end server. An uploaded file is stored in a document repository specified by the repositoryHandle. An attachment object is then created and attached to a ticket object specified by the objectHandle. The attachment object has all the information for accessing the newly uploaded file in the repository.

The maximum file size for upload using Web Services has increased from 100MB to 500MB.

Perform the following steps:

1. Navigate to `$NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\cxf\WEB-INF` folder and edit **cxf-beans.xml**.
2. Search for the entry `<entry key="attachment-memory-threshold" value="102400000"/>`
3. Replace the above entry with `<entry key="attachment-memory-threshold" value="512000000"/>`
4. Restart the CA SDM Service.

Returns

createAttachment has the following returns:

Parameter	Type	Description
<Handle>	STRING	Identifies the object handle of the newly created attachment object.

Could not perform the operation, policy limit exceeded

NOTE
This is only valid for Axis 1.4 based SOAP web services.

- **Symptom:**

When using the createAttachment() web service method to attach a document to an existing request or incident, you receive this error, even when the web services policy setting for attachments is set to -1(unlimited), which is the out-of-the-box default setting. You may receive this error when calling the web service method when there are no attached files in the received SOAP message. You must attach at least one file to the SOAP message before calling the createAttachment() web service method.

- **Solution:**

Set your SOAP implementation to support Direct Internet Message Encapsulation (DIME), and attach the file to be uploaded manually using DIME support before calling the createAttachment() method.

You can refer to the examples on how to support DIME from the $NX_ROOT\samples\sdk\websvc\java\test3_attachments directory.

Example: Visual Basic .NET

This code example illustrates sample Visual Basic .NET code to attach a file using DIME support before calling the createAttachment() web service method.

```
Dim reqContext As SoapContext = objCA SDM_WS.RequestSoapContext
Dim dimeAttach As New DimeAttachment("image/gif",
TypeFormat.MediaType, "c:\test.txt")
reqContext.Attachments.Add(dimeAttach)

strResult = objCA SDM_WS.createAttachment(sid, "doc_rep:1002",
"cr:400001", "my desc", "c:\test.txt")
```

NOTE
For information about the DIME attachment methods used in the previous sample code, see the Microsoft website. For programs written in other languages, see your documentation for the SOAP implementation that supports DIME.

Example: Java

This code example illustrates sample Java code to attach a file using DIME support before calling the createAttachment() web service method.

```
FileDataSource fds = new FileDataSource(filename);
DataHandler dhandler = new DataHandler(fds);
CA SDM._setProperty(Call.ATTACHMENT_ENCAPSULATION_FORMAT,
Call.ATTACHMENT_ENCAPSULATION_FORMAT_DIME);
CA SDM.addAttachment(dhandler);

String handle = CA SDM.createAttachment(sid, repHandle, objHandle,
description, filename);
```

removeAttachment

The following parameters apply to the removeAttachment method:

Parameter	Type	Description
SID	Integer	Identifies the SID of the current login session.

attHandle	String	Identifies the object handle of an attachment to be removed.

Description

Removes an attachment from a ticket object. The attached file is then removed from the repository.

Returns

Nothing.

attachURLLinkToTicket

The following parameters apply to the attachURLLinkToTicket method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
docId	Integer	Identifies the unique ID of the ticket.
URL	String	Indicates the URL to attach to the ticket.
attmntName	String	Identifies the name of the attachment.
Description	String	Indicates the description of the attachment.

Description

Attaches a URL link to a ticket.

Example:

```
CA SDM.attachURLLinkToTicket(sid, "cr:400001", "http://<URL NAME>", "<URL NAME>", "Broadcom Technologies
 Website");
```

Web Services Knowledge Attachment Methods

This article contains the following topics:

This section describes the Web Services Knowledge attachment methods.

createAttmnt

The following parameters apply to the createAttmnt method:

Parameter	Data Type	Description
SID	Integer	Identifies the session retrieved from logging in.
repositoryHandle	String	Identifies the object handle of a document repository.
folderId	Integer	Identifies the folder handle ID.
objectHandle	Int	Identifies the object handle of a Knowledge document to which this attachment is attached.

description	String	Identifies the description for the attachment object.
fileName	String	Identifies the name of the full path of the file to be uploaded.

Description

Uploads a file to the back-end server. An uploaded file is stored in a document repository specified by the repositoryHandle. An attachment object is then created and attached to a document object specified by the objectHandle. The attachment object has all the information for accessing the newly uploaded file in the repository.

Returns

The following:

Parameter	Type	Description
<Handle>	String	Identifies the object handle of the newly created attachment object.

attmntFolderLinkCount

The following parameters apply to the attmntFolderLinkCount method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
folderId	Integer	Identifies the unique ID of the folder

Description

Describes the number of attachment links under a specific folder to be attached.

Returns

Returns the number of attachments found under the specific folder.

attachURLLink

The following parameters apply to the attachURLLink method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
docId	Integer	Identifies the unique ID of the Knowledge document.
URL	String	Indicates the URL to attach to the Knowledge document.
attmntName	String	Identifies the name of the attachment.
Description	String	Indicates the description of the attachment.

Description

Attaches a URL link to a Knowledge document.

Returns

Returns error codes only for *individual* errors. For additional information, see Error Codes.

getKDListPerAttmnt

The following parameters apply to the getKDListPerAttmnt method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
attmntId	Integer	Identifies the unique ID of the attachment.

Description

Returns a list of Knowledge documents with reference to a given attachment.

Returns

A <UDSObject> node with zero or more <UDSObject> nodes describing the Knowledge document with the following <Attributes> child nodes:

XML Element Value	Type	Description
id	Integer	Identifies the unique ID of the Knowledge document.

getAttmntListPerKD

The following parameters apply to the getAttmntListPerKD method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
docId	Integer	Identifies the unique ID of the Knowledge document.

Description

A list of attachments with reference to a given Knowledge document.

Returns

A <UDSObject> node with zero or more <UDSObject> nodes describing the attachment with the following <Attributes> child nodes:

XML Element Value	Type	Description
id	Integer	Identifies the unique ID of the attachment

isAttmntLinkedKD

The following parameters apply to the AttmntLinkedKD method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
attmntId	Integer	Identifies the unique ID of the attachment.

Description

Checks if an attachment is linked to any Knowledge document, and returns the number of links found.

Returns

Any number of links found.

createFolder

The following parameters apply to createFolder method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
parentFolderId	Integer	Identifies the unique ID of the parent folder (zero if no parent).
repId	Integer	Identifies the unique ID of the repository.
folderType	Integer	Identifies the type of folder to be created.
description	String	Identifies the description of the folder.
folderName	String	Identifies the name of the folder.

Description

Creates a new folder in the Service Desk repository that is accessed through the Attachments Library.

Note: Folder types other then 0 are hidden folders and can be viewed only by navigating to Attachments Library on the Administration tab. These folders are automatically created by the system during upload and should not be created by the user or Web Services, because their files are private and cannot be shared with other Knowledge Documents, QA or Knowledge Files.

Returns

A <UDSObject> describing the folder created, with some of the following child <Attributes> nodes:

XML Element Value	Type	Description
repository	SREL	Identifies the repository name.
parent_id	SREL	Identifies the unique ID of the parent folder.
folder_type	Integer	Identifies the type of folder created.
folder_name	String	Identifies the name of the folder.
description	String	Identifies the description of the folder.

getFolderList

The following parameters apply to getFolderList method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
parentFolderId	Integer	Identifies the unique ID of the parent folder (0 if no parent).
repId	Integer	Identifies the unique ID of the repository.

Description

Returns a list of folders under a given parent folder

Returns

A <UDSObjectList> with zero or more <UDSObject> describing the attachment folder, with some of the following child <Attributes> nodes:

XML Element Value	Type	Description
repository	SREL	Identifies the repository name.
parent_id	SREL	Identifies the unique ID of the parent folder.
folder_type	Integer	Identifies the type of folder created.
folder_name	String	Identifies the name of the folder.
description	String	Identifies the description of the folder.

getFolderInfo

The following parameters apply to the getFolderInfo method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
folderId	Integer	Identifies the unique ID of the attachment folder.

Description

Returns the attributes of a folder.

Returns

A <UDSObject> describing the attachment folder, with some of the following child <Attributes> nodes:

XML Element Value	Type	Description
repository	SREL	Identifies the repository name.
parent_id	SREL	Identifies the unique ID of the parent attachment folder.
folder_type	Integer	Identifies the type of attachment folder.

folder_name	String	Identifies the name of the attachment folder.
description	String	Identifies the description of the attachment folder.

getAttmntList

The following parameters apply to the getAttmntList method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
folderId	Integer	Identifies the unique ID of the attachment folder.
repId	Integer	Identifies the unique ID of the repository. This parameter is required only when the folder ID is zero, which indicates the root folder.

Description

Returns a list of attachments under a given attachment folder.

Returns

A <UDSObjectList> with zero or more <UDSObject> describing the attachment, with some of the following child <Attributes> nodes:

XML Element Value	Type	Description
repository	SREL	Identifies the repository name.
parent_id	SREL	Identifies the unique ID of the parent folder.
folder_type	Integer	Identifies the type of folder created.
folder_name	String	Identifies the name of the folder.
description	String	Identifies the description of the folder.

getAttmntInfo

The following parameters apply to the getAttmntInfo method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
attmntId	Integer	Identifies the unique ID of the attachment folder.

Description

Returns the attributes of an attachment.

Returns

A <UDSObject> describing the attachment, with some of the following child <Attributes> nodes:

XML Element Value	Type	Description
id	Integer	Identifies the unique ID of the attachment.
description	String	Identifies the description of the attachment.
attmnt_name	String	Identifies the name of the attachment.
file_name	String	Identifies the name of the file.
folder_id	Integer	Identifies the unique ID of the attachment folder.
repository	SREL	Identifies the unique ID of the repository.

getRepositoryInfo

The following parameters apply to the getRepositoryInfo method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
repositoryId	Integer	Identifies the unique ID of the repository.

Description

Returns the attributes of a repository.

Returns

A <UDSObject> describing the repository, with some of the following child <Attributes> nodes:

XML Element Value	Type	Description
repository	SREL	Identifies the repository name.
parent_id	SREL	Identifies the unique ID of the parent attachment folder.
folder_type	Integer	Identifies the type of attachment folder.
folder_name	String	Identifies the name of the attachment folder.
description	String	Identifies the description of the attachment folder.

Web Services Miscellaneous Methods

This article contains the following topics:

This section describes the Web Services Miscellaneous methods.

callServerMethod

The following parameters apply to the callServerMethod method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
methodName	String	Identifies the name of the method to call.
factoryName	String	Identifies the factory name of the object type containing the method.
formatList	String	Identifies the format list, a series of characters describing the intended data types for the incoming parameters. It is related to the Parameter() description in this table).
parameters	String[]	Indicates zero or more parameter values for the method.

Description

Use this method to invoke an arbitrary server-side method. These are methods defined in the proprietary “spel” scripting language.

Only factory methods can be called and the caller must be logged in with full administrative rights.

The format list is a series of zero or more characters that indicate (in order) the data types of the parameters to follow. The character codes are as follows:

- 0 S -- string
- 0 I -- integer (covers dates and duration)
- 0 N -- null

For example, suppose a spell method is defined as follows:

```
cr::DoStuff(int in_one, string in_two, string in_three);
```

Invoke it with the following:

```
callServerMethod("DoStuff", "cr", "ISS", [3, "a string", "another one"]);
```

This method is intended for CA Development and services for customizations only; it is not recommended for most sites.

Returns

Each return message component in its own XML element. The elements are all string representations of the value. The elements are ordered in the return order from the server using the following format:

```
<ServerReturn>
   <Paramx>
```

This call does not support object reference returns. If an object reference is returned by the spell method, the return data is the string, “OBJECT”. This is not an error and any other parameters are also returned.

XML Element	Type	Description
<ServerReturn>	N/A	Indicates the outer element, which contains zero or more <ParamX> elements for return values.
<Paramx>	String	Indicates zero or more for the return values, where *x* is an integer starting at zero and increments for each return element.

You can validate BOPSIDs using check_bopsid. This is invoked with callServerMethod as follows:

```
String bopsid; // somehow gets populated with the BOPSID value
String [] stuff  = new String [] { bopsid };

String ret = usd.callServerMethod
(sid, "check_bopsid", "api", "I", stuff);
```

If the BOPSID validation fails, a SOAP Fault is returned. If validation succeeds, the return value is a small XML structure of the following form:

```
<ServerReturn>
<Param0>CONTACT_PERSID</Param0>
<Param1>SESSION_TYPE</Param1>
<Param2>SESSION_ID</Param2>
</ServerReturn>
```

- **CONTACT_PERSID**
  Defines the unique persistent id of the validated/trusted contact. It is of the form, "cnt:<uuid>".
- **SESSION_TYPE**
  Defines a small integer id indicating the type of session that generated the BOPSID. This is typically not used by integrators
- **SESSION_ID**
  Defines an optional session id. This is the id of the session that generated the BOPSID. It may or may not be set. This is useful to maintain a user's logical session within CA SDM, especially if the user is "passed" back to CA SDM by another BOPSID.

  NOTE
  If the BOPSID validation returns a success, log into CA Support Automation bypassing the login screen, else the CA Support Automation Login screen will be displayed.

createObject

The following parameters apply to the createObject method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
ObjectType	String	Identifies the type of object to create (the majic factory name).

attrVals	String[]	Identifies a sequence of name-value pairs used to set the initial attribute values for the new issue. **Note:** Dotted names are not permitted.
attributes	String[]	Identifies a sequence of attribute names from the new object for which to return values. Dot-notation is permitted. If this field is empty, all attribute values are returned.
createObjectResult	String Holder	Identifies the String Holder object for capturing returned data. See the XML Element Return list below for details.
newHandle	String Holder	Identifies the String Holder object for capturing returned data. See the XML Element Return list below for details.

Description

Creates a CA SDM object. The caller is responsible for setting any required fields in the *attrVals* parameter. Dotted-names are not permitted.

- **ObjectType**
  Identifies the name for an object type (factory).
- **attrVals**
  Describes the array of name-value pairs used to initialize the new object. For example, the following pseudo-code shows how to create a contact and return a <UDSObject> element with values for all its attributes:

```
String [4] attrVals;
attrVals[0] = "first_name";    // attribute name
attrVals[1] = "Edgar";
attrVals[2] = "last_name";
attrVals[3] = "Martin";
string [0] emptyArray;
CreateObject(sid, "cnt", attrVals, emptyArray, createObjectReturn, newHandle);
```

> **NOTE**
> Do not use this method for new assets, issues, requests, or change orders. Use the specialized createXXX() methods for those object types. This comment also applies if you are using the ITIL methodology -- use the appropriate methods to create Configuration Items, Incidents, and Problems.

Returns

A <UDSObject> element containing the new objects handle, along with attribute values specified in the *attributes* parameter. If the *attributes* parameter is empty, *all* attribute values are returned. List and LREL types are also returned, but as empty elements.

XML Element	Type	Description
<createObjectResult>	N/A	Identifies the standard UDSObject element containing the handle and requested attribute values.
<newHandle>	String	Identifies the new objects handle

serverStatus

The following parameter applies to the serverStatus method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.

Description

Returns the status of the CA SDM server, that is, whether it is up and ready or shut down.

> **NOTE**
> This method executes rapidly on the server. Calling this method periodically is a good way to keep a SID active.

Returns

The following values apply:

- **1** -- Indicates the Service Desk server is not available
  Note: Any non-zero return code indicates that the server is unavailable, such as 1010.
- **0** -- Indicates the Service Desk server it is running

XML Element	Type	Description
<ServerStatus>	Integer	Identifies the value associated with the server status, zero or 1.

updateObject

The following parameters apply to the updateObject method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
objectHandle	String	Identifies the handle of a CA SDM object to update.
attrVals	String[]	Identifies the name-value pairs for the update.
attributes	String[]	Identifies the sequence of attribute names from the object for which to return values. Dot-notation is permitted. If this field is empty, all attribute values are returned.

Description

Updates one or more attributes for the specified object.

To set values for the object, the caller passes a single-dimensional array of attribute name-value pairs. The first half of the pair is an attribute name; the second is the actual value. Dotted-names are not permitted.

To update an attribute that is a Pointer type (for example, the customer field on a request) a handle must be used for the value. For Integer, Date, and Duration types, pass the string representation of an integer.

For example, to update a request with a new assignee, description and priority, the array would appear as follows:

```
[0]  -  "assignee"
[1]  -  "cnt:555A043EDDB36D4F97524F2496B35E75"  (a contact Handle)
[2]  -  "description"
[3]  -  "My new description"
[4]  -  "priority"
[5]  -  "pri:38903"  (a priority Handle)
```

If the update fails for any reason, the entire operation aborts and no changes occur.

NOTE
When updating a task, set the status value last in the attribute array.

Returns

A <UDSObject> element containing the updated object's handle, along with attribute values specified in the *attributes* parameter. If the *attributes* parameter is empty, *all* of the attribute values are returned. List and LREL types are also returned, but as empty elements.

Web Services Knowledge Management

This article contains the following topics:

To use the Web Services Knowledge Management, it is helpful if you are familiar with the database structure.

Table Types

Some of the more important tables are described as follows:

Table Type	Description
skeleton	Stores all information pertaining to documents with each row representing one document. Field names from this table can be used when passing the PropertyList and SortBy parameters to methods such as FAQ() and Search(). The field names are case-sensitive so make sure you pass them just as they are in the database.
o_indexes	Stores all information pertaining to categories with each row representing one catego

Knowledge Management General Methods

This section describes Knowledge Management general methods. Valid Knowledge document sorting properties (when available) are as follows:

- RELEVANCE
- AUTHOR_ID
- BU_RESULT
- CREATION_DATE
- DOC_TYPE_ID
- EXPIRATION_DATE
- HITS
- id
- MODIFY_DATE
- OWNER_ID
- PRIORITY_ID
- ACCEPTED_HITS
- ASSET_ID
- SD_ASSET_ID
- ASSIGNEE_ID
- PRODUCT_ID
- START_DATE
- STATUS_ID
- SUBJECT_EXPERT_ID

faq

The following parameters apply to the faq method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
categoryIds	String	Identifies the category ID used to perform the faq. Use 1 for the ‘Root’ category. **Note:** Multiple ids are supported, for example, "1, 2, 3”.
resultSize	Integer	Identifies the number of documents for which you want to retrieve detailed information. For the rest of the documents, only their IDs return. Detailed information for these documents can be accessed later using the getDocumentsByIDs() method. The default is 10.
propertyList	String	Identifies the comma-separated list of database fields from which you want to retrieve information. The following fields are always returned, regardless of the propertyList parameter: id DOC_TYPE_ID BU_RESULT

sortBy	String	Identifies the database field that you want to use for sorting the results. Multiple sort fields are not supported. The default is BU_RESULT, meaning that the faq rating sorts it. When ID is used as a secondary sort, it always sorts the results. A value of **HITS** in "sortBy" is also supported. **HITS** Identifies the number of times that the document has been viewed.
descending	Boolean	Identifies an indicator available for sorting the results in descending order.
whereClause	String	Use this to add your own 'SQL where clause' for filtering the results of the search.
maxDocIds	Integer	Identifies the maximum amount of document IDs to be returned (the default is 100). For example, if you specify a resultSize of 10 and a maxDocIds of 50, if there are 100 matching documents in the database, then 10 have their detailed information retrieved and 40 have just their IDs returned. The remaining 50 are not returned at all.

Description

Use to perform a faq search. Documents are retrieved based on the category ID that is passed. Any documents residing in that category or in any sub-category are returned. To improve performance, these methods only retrieve detailed information on a user-defined set of documents, which is controlled through the resultSize parameter. The rest of the documents return their IDs only. Using this method, you can for example, set up a paging mechanism where the user can click on 'Top', 'Previous', 'Next', and 'Bottom' links. When you need to retrieve the next set of information, you can use the getDocumentsByIDs() method. The maximum number of 100 IDs is returned.

Returns

A <UDSObjectList> node with the following sections:

<UDSObject> node from the <UDSObject> Node Description section of this chapter. There is a <UDSObject> node with all the given properties for the first *n* documents that the method finds where *n* equals the resultSize parameter.

For example, if the resultSize parameter is 10, the maxDocIDs parameter is 50, and the method finds 100 documents, then there are 10 <UDSObject> with <Attributes> nodes in the first<UDSObject> section with detail attribute information from propertyList parameter and 40 <UDSObject> nodes with only ID <Attributes> in the following section. If you want to retrieve detailed ID <AttrName> information for documents numbering 11-20, you have to make a call to the getDocumentsByIDs() method and pass it those IDs from <AttrValue>.

search

The following parameters apply to the search method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
problem	String	Identifies the problem description to which you want to find solutions.

resultSize	Integer	Identifies the number of documents for which you want to retrieve detailed information. The remaining documents have their IDs returned only. Detailed information for these documents can be accessed later using the getDocumentsByIDs() method. The default is 10.
propertyList	String	Identifies the comma-separated list of database fields from which you want to retrieve information. The following fields are always returned, regardless of the propertyList parameter: id DOC_TYPE_ID
sortBy	String	Identifies the database field that you want to use for sorting the results. Multiple sort fields are not supported. The default is RELEVANCE. When id is a secondary sort, it always sorts the results. For a valid sort property, see the faq method.
descending	Boolean	Identifies an indicator you can use for sorting the results in descending order.
relatedCategories	Boolean	Returns a list of all related categories for the documents found.
searchType	Integer	Type of search to perform: 1 = Natural Language Search (NLS) 2 = Knowledge Management search
matchType	Integer	Represents the type of match: 0 = OR type match 1 = AND type match 2 = Exact match **Note:** If NLS is selected for the searchType parameter, then only the OR and AND matchTypes are valid.

searchField	Integer	Represents the binary combination of fields in which to search: Title = 1 Summary = 2 Problem = 4 Resolution = 8 For example, to search all fields, specify 15 (1+2+4+8). To search in Summary and Problem only, specify 6 (2+4). **Note:** The default is to search Problem. If you set the searchType parameter to NLS, the searchFields parameter is ignored because NLS searches can only search the Problem field.
categoryPath	String	Limits the results of the search to a specific category or categories. You need to specify the full ID path to the category and separate multiple categories with commas. For example, 1-3-5, 1-4-8 to limit the search to categories 5 and 8 (and their sub-categories).
whereClause	String	Use this to add your own 'SQL where clause' for filtering the results of the search.
maxDocIds	Integer	Represents the maximum amount of document IDs allowed to be returned. For example, if you specify a resultSize of 10 and a maxDocIds of 50, if there are 100 matching documents in the database, then 10 have their detailed information retrieved, and 40 have just their IDs returned. The remaining 50 are not returned at all. The default is 100.

Description

Searches for solutions to a problem. Documents are retrieved based on the problem that is passed. Any documents matching the description of the problem or a similar description, are returned. To improve performance, these methods only retrieve detailed information on a user-defined set of documents, which is controlled through the resultSize parameter. The rest of the documents return their ids only. Using this method, you can for example, set up a paging mechanism, where the user can click on 'Top', 'Previous', 'Next', and 'Bottom' links. When you need to retrieve the next set of information, you can use the getDocumentsByIDs() method.

Returns

A <UDSObjectList> node with the following sections:

<UDSObject> node from the <UDSObject> Node Description section of this chapter. There will be a <UDSObject> node with all the given properties for the first *n* documents that the method finds, where *n* equals the resultSize parameter. For example, if the resultSize parameter is 10, the maxDocIds parameter is 50, and the method finds 100 documents, then there are 10 <UDSObject> nodes with all the properties requested in the <Attributes> section and 40 <UDSObject> with only the ID property in the <Attributes> section. If you want to retrieve detailed <UDSObject> information for documents 11-20, you need to make a call to the getDocumentsByIDs() method and pass it those IDs.

If the getRelatedCategories parameter is set to True, the <UDSObjectList> node is included in the <Attributes> section for related categories. Each <INDEX_DOC_LINKS> node contains the relational ID of the category, as shown by the following example:

```
1-70
```

doSelectKD

The following parameters apply to the doSelectKD method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
whereClause	String	(Optional) Identifies the where clause for the query.
sortBy	String	Identifies the database field that you want to use for sorting the results. Multiple sort fields are not supported. The default is BU_RESULT, meaning that the faq rating sorts it. When id is used as the secondary sort, it always sorts the results.
desc	Boolean	Identifies the indicator available for sorting the results in descending order. Use True for descending and False for ascending the document order.
maxRows	Integer	Indicates the maximum number of rows to return. Specify -1 to return all rows. **Note:** Regardless of the integer specified, CA SDM will return a maximum of 250 rows per call.
attributes	String[]	Identifies the attribute list from which to fetch values. Dotted-attributes are permitted. If this field is blank, all value-based attributes are returned. These attributes cannot be defined as LOCAL in the majic definition file. LOCAL attributes are temporal; they have no database storage.
skip	Integer	Identifies the number of knowledge documents to skip from the beginning. Enter zero (0) to return all documents.

Description

Performs an SQL-like select on a Knowledge Document table. Supply one or more attributes you want fetched from the objects that match the supplied where clause.

Returns

A sequence of <UDSObject> elements. The following format applies:

```
<UDSObjectList>
<UDSObject>
```

```
 <Handle>
 <Attributes>
<AttributeName0>
<AttributeName1>
```

XML Element	Type	Description
<UDSObject>	Sequence	Contains a <Handle> element and an <Attributes> sequence.
<UDSObjectList>		Signifies the outer element, which contains a sequence of <UDSObject> elements.

createDocument

The following parameters apply to the createDocument method:

Parameter	Type	Description
SID	String	Identifies the session retrieved from logging in.
kdAttributes	String[]	Identifies an array of name-value pairs used to set the initial attribute values for the new Knowledge document, as illustrated by the following: `"SUMMARY","Summary text",` "TITLE","Title text"

As part of the createDocument method, the following table reflects examples of valid, commonly used attribute values for a document. Data in the Type column reflect the actual type, which must be parsed to the method in string format in the attrVals string array.

Attribute Value	Type	Description
PRIMARY_INDEX	Integer	Identifies the category ID in which to create the document. Use 1 for the *Root* category.
USER_DEF_ID	String	Identifies any ID that you would like to use to represent the document.
TITLE	String	Identifies the title of the document.
SUMMARY	String	Identifies the summary of the document.
PROBLEM	String	Identifies the problem of the document.
RESOLUTION	String	Identifies the resolution of the document. This can contain html.
STATUS_ID	Integer	Identifies the status ID for the document. The default is 10 (Draft).
PRIORITY_ID	Integer	Identifies the priority ID for the document. The default is 20 (Normal).
CREATION_DATE	Date (String)	Identifies the date and time the document was created. Leave blank to assign current date.

MODIFY_DATE	Date (String)	Identifies the date and time the document was last modified. Leave blank to assign the current date.
START_DATE	Date (String)	Identifies the date the document becomes active and is used in conjunction with Expiration_Date. Leave blank to specify no start date and the document will be active as long as the expiration date has not been reached.
EXPIRATION_DATE	Date (String)	Identifies the date the document expires, and it is used in conjunction with Start_Date. Leave blank to specify no expiration date.
PUBLISHED_DATE	Date (String)	Identifies the date and time the document was published. Leave blank to assign current date if the status is Published. If the status is not Published, this parameter is ignored.
SD_PRODUCT_ID	Integer	Identifies the product ID from CA SDM with which to associate this document.
ASSIGNEE_ID	UUID	Identifies the unique assignee ID from CA SDM to which this document is assigned.
SD_ASSET_ID	UUID	Identifies the asset ID from CA SDM with which to associate this document.
SD_ROOTCAUSE_ID	Integer	Identifies the root cause ID from CA SDM with which to associate this document.
SD_PRIORITY_ID	Integer	Identifies the priority ID from CA SDM with which to associate this document.
SD_SEVERITY_ID	Integer	Identifies the severity ID from CA SDM with which to associate this document.
SD_IMPACT_ID	Integer	Identifies the impact ID from CA SDM with which to associate this document.
SD_URGENCY_ID	Integer	Identifies the urgency ID from CA SDM with which to associate this document.
AUTHOR_ID	UUID	Identifies the unique ID of the contact who authored this document. If the author is not an internal contact, you can set this field to zero and use the Author parameter instead.
OWNER_ID	UUID	Identifies the unique ID of the contact who owns this document.
SUBJECT_EXPERT_ID	UUID	Identifies the unique ID of the contact who is the subject expert for this document.
NOTES	String	Identifies the notes for the document.

READ_GROUP_LIST	String	Identifies the dash-separated list of group IDs that have read permission for this document (for example: 1-3-4). Use *A* to assign permission to everyone.
WRITE_GROUP_LIST	String	Identifies the dash-separated list of group IDs that have write permission for this document (for example: 1-3-4). Use *A* to assign permission to everyone.
INHERITPERMISSIONS	Boolean	Indicates the status of the flag to inherit permissions from the category in which the document is being created. Set to True if you want to inherit permissions, and then ReadPermissions and the WritePermissions parameters will be ignored.
DOC_TYPE_ID	Integer	Identifies the ID for the type of document that this document will be; a regular document or a tree document. The default is a regular document.
HITS	Integer	Identifies the number of times that the document has been viewed.
DOC_TEMPLATE_ID	Integer	Identifies the ID for the template you want to assign to this document.
WF_TEMPLATE	Integer	Identifies the ID for the workflow template you want to assign to this document.
CUSTOM1	String	Specifies a custom field.
CUSTOM2	String	Specifies a custom field.
CUSTOM3	String	Specifies a custom field.
CUSTOM4	String	Specifies a custom field.
CUSTOM5	String	Specifies a custom field.
CUSTOM_NUM1	Double	Specifies a numeric custom field.
CUSTOM_NUM2	Double	Specifies a numeric custom field.

Description

Creates a new document.

Returns

A <UDSObject> node describing the Knowledge Document created.

modifyDocument

The following parameters apply to the modifyDocument method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.

docId	Integer	Identifies the unique ID of the document you want to modify.
kdAttributes	String[]	Specifies the name-value pairs for the update, for example, “SUMMARY”, ”Summary text”, “TITLE”, and “Title text”.

As part of the modifyDocument method, the following table reflects examples of valid, commonly used attribute values for a document. Data in the Type column reflect the actual type, which must be parsed to the method in string format in the attrVals string array.

Parameter	Type	Description
MODIFY_DATE	(String)	Indicates a special field used for "record locking" purposes to make sure that someone else is not updating the document at the same time that you are. You must pass in the existing MODIFY_DATE of the document. If you leave the modify date blank, you receive an error that another user has updated the document.
USER_DEF_ID	String	Specifies any ID that you want to use to represent the document.
TITLE	String	Indicates the title of the document.
SUMMARY	String	Indicates the summary of the document.
PROBLEM	String	Indicates the problem of the document.
RESOLUTION	String	Indicates the resolution of the document. This can contain html.
STATUS_ID	Integer	Indicates the status ID for the document. The default is 10 (Draft).
PRIORTY_ID	Integer	Indicates the priority ID for the document. The default is 20 (Normal).
START_DATE	Date (String)	Indicates the date that the document becomes active, which is also used in conjunction with ExpirationDate. Leave blank to specify no start date and the document becomes active as long as the expiration date is not exceeded.
EXPIRATION_DATE	Date (String)	Indicates the date that the document expires, which is used in conjunction with StartDate. Leave blank to specify no expiration date.
SD_PRODUCT_ID	Integer	Indicates the product ID from CA SDM with which to associate this document.
ASSIGNEE_ID	UUID	Indicates the unique ID from CA SDM to which this document is assigned.
SD_ASSET_ID	UUID	Indicates the asset ID from CA SDM with which to associate this document.

SD_ROOTCAUSE_ID	Integer	Indicates the root cause ID from CA SDM with which to associate this document.
SD_PRIORITY_ID	Integer	Indicates the priority ID from CA SDM with which to associate this document.
SD_SEVERITY_ID	Integer	Indicates the severity ID from CA SDM with which to associate this document.
SD_IMPACT_ID	Integer	Indicates the impact ID from CA SDM with which to associate this document.
SD_URGENCY_ID	Integer	Specifies the urgency ID from CA SDM with which to associate this document.
AUTHOR_ID	UUID	Identifies the unique ID of the contact who authored this document. If the author is not an internal contact, you can set this filed to zero (0) and use the Author parameter instead.
OWNER_ID	UUID	Identifies the unique ID of the contact who owns this document.
SUBJECT_EXPERT_ID	UUID	Indicates the unique ID of the contact who is the subject expert for this document.
NOTES	String	Indicates notes for the document.
READ_GROUP_LIST	String	Indicates the dash-separated list of group ids that have read permission for this document (for example: 1-3-4). Use *A* to assign permission to everyone.
WRITE_GROUP_LIST	String	Indicates the dash-separated list of group IDs that have write permission for this document (for example: 1-3-4). Use *A* to assign permission to everyone.
INHERIT_PERMISSION	Boolean	Indicates the status of the inherit permissions flag. Set to True if you want to inherit permissions from the category in which the document is being created. If set to True, the ReadPermissions and WritePermissions parameters are ignored.
DOC_TYPE_ID	Integer	Identifies the ID for the type of document that this document is to become; a regular document or a tree document. The default is a regular document.
HITS	Integer	Indicates the number of times that the document has been viewed.
DOC_TEMPLATE_ID	Integer	Identifies the ID for the template you want to assign to this document.
WF_TEMPLATE	Integer	Identifies the ID for the workflow template you want to assign to this document.

WF_ACTION	String	Identifies the action for the workflow you want to assign to this document. For example, wf_unpublish lets the document become unpublished.
WF_COMMENT	String	Identifies the comment for the workflow you want to assign to this document. For example, unpublish.
WF_CONTACT_ID	UUID	Identifies the user id for the workflow you want to assign to this document.
WF_REJECT_TO_ID	Integer	Identifies the task id for the workflow you want to assign to this document, matching its approval process. Task ids are stored in the CI_ACTIONS table.
CUSTOM1	String	Indicates a custom field.
CUSTOM2	String	Indicates a custom field.
CUSTOM3	String	Indicates a custom field.
CUSTOM4	String	Indicates a custom field.
CUSTOM5	String	Indicates a custom field.
CUSTOMNUM1	Double	Indicates a numeric custom field.
CUSTOMNUM2	Double	Indicates a numeric custom field.

Description

Modifies a document.

Returns

A <UDSObject> node describing the Knowledge Document modified.

deleteDocument

The following parameters apply to the deleteDocument method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
docId	Integer	Identifies the unique ID of the document you want to delete.

Description

Flags a document for deletion. The Knowledge Management Windows Service permanently deletes the document.

Returns

Returns error codes only when there are *individual* errors. For additional information, see Error Codes.

Use the Knowledge Management Web Services

The login process and any error codes that may display for the Knowledge Management Web Services are the same as those found for the CA SDM Web Services. For additional information, see Login and Error Codes.

Access the Knowledge Management Web Services

The Knowledge Management Web Services uses Apache Axis and Apache CXF implementation of standards set forth by the W3C. Ideally, a client on any type of platform should be able to access the services, but vendor implementations vary. For example, Java and .NET both provide tools for generating proxy classes from a WSDL service description. If you experience any issues using the Web Services with a different technology, consult your platform vendor or Microsoft's knowledge base.

addComment

The following parameters apply to the addComment method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
comment	String	Identifies the comment to add.
docId	Integer	Identifies the document ID for the comment you want to add.
email	String	Indicates the email address of the person who submitted the comment. Leave blank if you want the email address retrieved from the database based on the user ID parameter.
username	String	Indicates the user name of the person who submitted the comment. Leave blank if you want the user name retrieved from the database based on the user id parameter.
contactId	String	Indicates the ID of the person submitting the comment. If this contact ID exists in the database, the associated email and user name are retrieved and placed in the email and user name fields. If this ID does not exist, the email and user name parameters are used instead, if supplied. Use zero (0) if you are not using this parameter. Contact ID is UUID in string format.

Description

Adds a comment to a particular document.

Returns

A <UDSObject> node with the following<Attributes> child nodes describing the comment most recently added:

XML Element Value	Data Type	Description
id	Integer	Identifies the unique identifier for the comment most recently added.
DOC_ID	Integer	Identifies the document ID for the comment most recently added.
USER_ID	Integer	Identifies the ID of the person who submitted the comment.

USER_NAME	String	Identifies the user name of the person who submitted the comment.
EMAIL_ADDRESS	String	Identifies the email address of the person who submitted the comment.
COMMENT_TEXT	String	Identifies the text for the comment recently added.
COMMENT_TIMESTAMP	Date	Identifies the date and time the comment was added.

deleteComment

The following parameters apply to the deleteComment method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
commentId	Integer	Identifies the unique ID for the comment you want to delete.

Description

Deletes a comment.

Returns

Returns error codes only for *individual* errors. For additional information, see Error Codes.

rateDocument

The following parameters apply to the rateDocument method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
docId	Integer	Identifies the document ID to rate.
rating	Integer	Identifies the rating to give the document (a scale of 0-4, where 0 is the worst and 4 is the best).
multiplier	Integer	Identifies the multiplier parameter. This parameter can be used to simulate many ratings at once. Use the default of 1 for a single rating and any other number for multiples. For example, if you submit 3, it acts as if you called the method 3 times. Three ratings with the value you supplied in the rating parameter is added to the database.
ticketPerId	String	Identifies the persistent ID of a ticket related to this Knowledge document.

onTicketAccept	Boolean	Identifies whether the document was accepted as a solution for the ticket.
solveUserProblem	Boolean	Identifies whether this document solved the user's problem. It signifies how the user responded to the question "Did this document solve your problem?" on the Solution Survey.
isDefault	Boolean	Indicates a default rating status. If you are setting the rating just because the user viewed the document and not because he actually rated it, set this to True. This is used for reporting reasons.

Description

Rates a particular document.

Returns

A <UDSObject> node with the following <Attributes> children nodes describing the rating BU_TRANS :

XML Element Value	Data Type	Description
id	Integer	Identifies the unique identifier for the rating most recently added. Use this with the updateRating() method if you want to modify the rating at a later time.
DOC_ID	Integer	Identifies the document ID.
INDEX_ID	Integer	Identifies the category ID.
BU_RATING	String	Identifies the rating given to the document.
HIT_NO_VOTE	Integer	Identifies the rating set because a user viewed the document and not actually rated it, or vice versa.

updateRating

The following parameters apply to the updateRating method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
buID	Integer	Identifies the unique ID of the rating you want to modify. This ID is returned by the rateDocument() method.
rate	Integer	Identifies the new rating to apply to the document (a scale of 0 -- 4, where 0 is the worst and 4 is the best).

Description

Updates one of the ratings of a particular document.

Returns

A <UDSObject> node describing BU_TRANS with the updated rating attribute.

getQuestionsAsked

The following parameters apply to the getQuestionsAsked method:

Parameter	Type	Description
SID	String	Identifies the session retrieved from logging in.
resultSize	Integer	Identifies the number of searched text for which you want to retrieve detailed information.
Descending	Boolean	Indicates an option available for sorting the results in descending order of the ASKED_DATE.

Description

Retrieves historical Knowledge document search text.

Returns

A <UDSObjectList> node with zero or more <UDSObject> nodes describing EBR_LOG with the following <Attributes> child nodes:

XML Element Value	Data Type	Description
id	Integer	Indicates the unique identifier of the question asked.
SEARCH_TEXT	Integer	Indicates the search text of the question asked.

getBookmarks

The following parameters apply to the getBookmarks method:

Parameter	Type	Description
SID	String	Identifies the session retrieved from logging in.
contactId	String	Identifies the unique ID of the contact for which you want to retrieve bookmarks. Contact ID is UUID in string format.

Description

Retrieves bookmarks for a particular contact.

Returns

A <UDSObjectList> node with zero or more <UDSObject> nodes describing CI_BOOKMARKS with the following <Attributes> child nodes:

XML Element Name	Type	Description
DOCUMENT_ID	Integer	Identifies the unique ID of the document.
id	Integer	Identifies the bookmark ID
USER_ID	String	Identifies the user ID for the owner of this bookmark.
BOOKMARK_TITLE	String	Identifies the bookmark title from the document.

addBookmark

The following parameters apply to the addBookmark method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
contactId	String	Identifies the unique ID of the contact for which you want to retrieve bookmarks. Contact ID is UUID in string format.
docId	Integer	Identifies the document ID you want to bookmark.

Description

Adds a bookmark for a particular contact.

Returns

A <UDSObject> node describing the newly created bookmark.

deleteBookmark

The following parameters apply to the deleteBookmark method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
contactId	String	Identifies the unique ID of the contact for which you want to delete a bookmark. Contact ID is UUID in string format
docId	Integer	Identifies the document ID of the bookmark you want to remove.

Description

Deletes a bookmark for a particular contact.

Returns

Returns error codes only for *individual* errors. For additional information, see Error Codes.

getStatuses

The following parameter applies to the getStatuses method:

Parameter	Type	Description
SID	String	Identifies the session retrieved from logging in.

Description

Retrieves the list of statuses.

Returns

A <UDSObjectList> node with zero or more <UDSObject> nodes describing CI_STATUSES with the following <Attributes> child nodes:

XML Element Values	Data Type	Description
id	Integer	Identifies the unique ID of the status.
STATUS	String	Identifies the name for the status.
STATUS_DESCRIPTION	String	Identifies the description for the status.
PREDEFINED	Integer	Indicates whether the status is predefined by the Knowledge Management system, meaning that it cannot be deleted.
STATUS_ORDER	Integer	Describes the order by which the status should appear in the Workflow task list. Workflows can only be created when they follow this order.

getPriorities

The following parameter applies to the getPriorities method:

Parameter	Type	Description
SID	String	Identifies the session retrieved from logging in.

Description

Retrieves the list of priorities.

Returns

A <UDSObjectList> node with zero or more <UDSObject> nodes describing CI_PRIORITIES with the following <Attributes> child nodes:

XML Element Values	Data Type	Description
id	Integer	Identifies the unique ID of the priority.
PRIORITY	String	Identifies the name for the priority.

getDocumentTypes

The following parameter applies to the getDocumentTypes method:

Parameter	Type	Description
SID	String	Identifies the session retrieved from logging in.

Description

Retrieves the list of document types.

Returns

A <UDSObjectList> node with zero or more <UDSObject> nodes describing CI_DOC_TYPES with the following <Attributes> child nodes:

XML Element Values	Data Type	Description
id	Integer	Identifies the unique ID of the document type.
DOC_TYPE_TXT	String	Identifies the name for the document type.

getTemplateList

The following parameter applies to the getTemplateList method:

Parameter	Type	Description
SID	String	Identifies the session retrieved from logging in.

Description

Retrieves the list of document templates

Returns

A <UDSObjectList> node with zero or more <UDSObject> nodes describing CI_DOC_TEMPLATES with the following <Attributes> child nodes:

XML Element Value	Type	Description
id	Integer	Identifies the unique ID of the document type.
TEMPLATE_NAME	String	Identifies the name for the document template.
IS_PREDEFINED	Integer	Indicates whether the template is predefined by the Knowledge Management system and cannot be deleted.
IS_DEFAULT	Integer	Indicates whether the template is the default that will be assigned to new documents.

getWorkflowTemplateList

The following parameter applies to the getWorkflowTemplateList method:

Parameter	Type	Description
SID	String	Identifies the session retrieved from logging in.

Description

Retrieves the list of workflow templates.

Returns

A <UDSObjectList> node with zero or more <UDSObject> nodes describing WF_TEMPLATE with the following <Attributes> child nodes:

XML Element Value	Type	Description
id	Integer	Identifies the unique ID of the workflow template.
WF_NAME	String	Identifies the name for the workflow template.
WF_DESCRIPTION	String	Identifies the description for the workflow template.
IS_DEFAULT	Integer	Indicates that the default template is to be assigned to new documents.

getCategory Method

This article contains the following topics:

The following parameters apply to the getCategory method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
categoryId	Integer	Identifies the category ID in which to create the document.
getCategoryPaths	Boolean	Indicates the path for which to get category information. It returns category information and the full text category path for each category.

Description

Retrieves information for a category, including a listing of all of its child categories.

Returns

A <UDSObjectList> node with an <UDSObject> node describing the category requested.

getPermissionGroups

The following parameters apply to the getPermissionGroups method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
groupID	Integer	Returns only the group with this ID. Pass zero (0) when you do not want to use this parameter. **Note:** This groupID relates to a knowledge category, and differs from the groupID for contacts.

Description

Retrieves the list of Permission Groups.

Returns

A <UDSObject> node with zero or more <UDSObject> nodes describing Permission Group with the following <Attributes> child nodes:

XML Element Value	Type	Description
id	Integer	Identifies the unique ID of the group.
GRP_LIST_KEY	String	Shows a list of the IDs of CA SDM groups, separated by commas.
GRP_LIST	String	Displays a field containing the entire group list.

getComments

The following parameters apply to the getComments method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
docIds	String	Identifies the document IDs for all the comments you want to retrieve. **Note:** Use commas to separate, for example, "400001,400002".

Description

Gets all the comments from documents.

Returns

A <UDSObjectList> node with zero or more <UDSObject> nodes describing O_COMMENTS with the following <Attributes> child nodes:

XML Element Value	Data Type	Description
id	Integer	Identifies the unique identifier for the comment most recently added.
DOC_ID	Integer	Identifies the document IDs for the comment recently added.
USER_ID	Integer	Identifies the ID of the person who submitted the comment.
USER_NAME	String	Identifies the user name of the person who submitted the comment.
EMAIL_ADDRESS	String	Identifies the email address of the person who submitted the comment.
COMMENT_TEXT	String	Identifies the text for the comment recently added.
COMMENT_TIMESTAMP	Date	Identifies the date and time the comment was added.

getDecisionTrees

The following parameters apply to the getDecisionTrees method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
propertyList	String	Identifies the comma-separated list of database fields for which you want to retrieve information. The following fields are always returned, regardless of the propertyList parameter: id DOC_TYPE_ID BU_RESULT
sortBy	String	Identifies the database field that you want to use for sorting the results. The default is id. Multiple sort fields are not supported. If you specify another field, id as a secondary sort always sorts the results.
descending	Boolean	Identifies an indicator available for sorting the results in descending order.

Description

Retrieves all Decision Trees. Decision trees are Knowledge Documents that provide users with resolutions after answering specific questions on the document.

Returns

A <UDSObjectList> node with the following sections:

```
<UDSObject> nodes with requested <Attributes> nodes
```

getDocument

The following parameters apply to the getDocument method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
docId	Integer	Identifies the document ID to retrieve.
propertyList	String	Identifies the comma-separated list of database fields from which you want to retrieve information. Leave blank to retrieve all fields.
relatedDoc	Boolean	Indicates whether to retrieve a list of documents that are related to this document.
getAttmnt	Boolean	Indicates whether to retrieve the list of attachments and URL links for the document.
getHistory	Boolean	Indicates whether you want to retrieve the complete history for the document.
getComment	Boolean	Indicates whether you want to retrieve all comments for the document.
getNotiList	Boolean	Indicates whether you want to retrieve the email notification list for the document.

Description

Retrieves information for a document.

Returns

A <UDSObject> node, as described in the <UDSObject> Node Description, with requested <Attributes> provided by the propertList parameter.

getDocumentsByIDs

The following parameters apply to the getDocumentsByIDs method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
docIds	String	Identifies the comma-separated list of document IDs to retrieve.

propertyList	String	Identifies the comma-separated list of database fields for which you want to retrieve information. The following fields are always returned, regardless of the propertyList parameter: id DOC_TYPE_ID BU_RESULT
sort By	String	Identifies the database field that you want to use for sorting the results. The default is id, but multiple sort fields are not supported. If you specify another field, id as a secondary sort always sorts the results.
descending	Boolean	Identifies an indicator available for sorting the results in descending order.

Description

Retrieves information on one or more documents by passing the document IDs for which you want to retrieve information. This is usually used after calling the faq() or search() methods. In order to improve performance, these methods only retrieve detailed information on a user-defined set of documents. The rest of the documents return their IDs only. For example, you can set up a paging mechanism, where the user can click on Top, Previous, Next, and Bottom links. When you need to retrieve the next set of information, you can use the getDocumentsByIDs() method.

Returns

A <UDSObjectList> node with the following section:

```
<UDSObject> nodes with requested <Attributes> nodes describing Knowledge Document
```

You should pass the IDs into the getDocumentsByIDs() docIds parameter in this same format.

getBopsid

The following parameters apply to the getBopsid method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
contact	String	Identifies the name of the user associated with the returned BOPSID. **Note:** This is the system login name, not the CA SDM contact name.

Description

Facilitate the building of Web Interface URLs, which may be used to launch the Web Interface in the context of a given user without a login challenge. The URL may look similar to the following:

```
http://host/CAisd/pdmweb.exe?BOPSID=nnnnn+OP=xxxx...
```

To launch the Web Interface in the context of a given user (for example, an analyst), a calling application must first construct a Web Interface URL, which includes a BOPSID token (a web-interface security token). Failure to provide a BOPSID token may result in an interactive login challenge when attempting to launch the Web Interface in the chosen

context (such as, a detail view of a given ticket). The getBopsid method allows the BOPSID to be generated in the context of the user provided by the Name parameter. If the Name parameter is not provided, it uses the user associated with the current Web Interface BOPSID.

> **NOTE**
> To prevent unauthorized elevation of privileges, the BOPSID of the current login must have equal or greater access rights than the name of the user entered.

Returns

A BOPSID based on the name of the user entered. You must use the BOPSID to launch the Web Interface within 30 seconds of it being generated.

> **NOTE**
> The BOPSID of the current login must have equal or greater access rights than the name of the user entered.

getConfigurationMode

The following parameters apply to the getConfigurationMode method:

Parameter	Type	Description
SID	Integer	Identifies the SID of the current login session.

Description

This method returns a string indicating if the CA SDM installation is in the ITIL mode.

Returns

A string "itil" if the installation is in ITIL mode. Otherwise, an empty string is returned.

getObjectValues

The following parameters apply to the getObjectValues method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
objectHandle	String	Identifies the handle of a CA SDM object to query.
attributes	String[]	Identifies the names of attributes to fetch. If this field is empty, all the attributes for the object are returned. **Note:** Dot-names for attributes are only supported for SREL type attributes, and not for LREL/BREL/QREL type attributes.

Description

This method returns the attribute values of an object. The caller passes one or more attribute names to fetch the object and dotted-names are permitted.

All values are returned as a string. Empty/null attributes are returned as empty strings.

Returns

A <UDSObject> element.

XML Element	Type	Description
<UDSObject>	N/A	Contains a <Handle> element and zero or more <AttributeNameX> elements.

getObjectTypeInformation

The following parameters apply to the getObjectTypeInformation method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
factory	String	Identifies the object type (known as 'factory') to query. This is the majic name of the object, for example: "cr" = Request

Description

A list of all attribute names for a given object type, along with type information for each attribute. Information returned for the attribute's type includes the Integer, String, Date, Pointer, List, and so on, if the attribute is required for back-storing its storage space requirements (if appropriate).

Callers should cache the type information requested per object type to avoid multiple, redundant (and expensive) calls. The attribute information can change only after modifications are performed on the CA SDM server and the service is recycled.

Returns

The following:

XML Element	Type	Description
<UDSObject>	N/A	Indicates the root node.
<Attributes>	Sequence	Indicates zero or more elements for each attribute.
<attrName DataType="dataType" Size="storageSize" Required="Boolean" Factory="factoryName">	EmptyElement	Indicates an element with a name matching an object attribute name. The element has several attributes: **DataType**Signifies the integer representation of the data type. **Size**Represents the maximum size needed to store this attribute in a string. **Required**Represents the flag status of True if this attribute must be set for the object to back store. **Factory**Represents the Type name of the object if the attribute is a List, Lrel or Pointer type. It is not written unless it is a Llist, Lrel, or Pointer type data type.

getArtifact

The following parameters apply to the getArtifact method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
contact	String	Identifies the name of the user associated with the returned ARTIFACT. **Note:** This is the system login name, not the contact name for CA Service Desk Manager.
password	String	Identifies the password.

Description

Facilitates the building of Web Interface URLs, which may be used to launch the Web Interface in the context of a given user without a login challenge. The URL may look similar to the following:

```
http://host/CAisd/pdmweb.exe?USERNAME=xxxxx+ARTIFACT=nnnnn+OP=xxxx....
```

To launch the Web Interface in the context of a given user (for example, an analyst), a calling application must first construct a Web Interface URL, which includes an CA EEM ARTIFACT token (a web-interface security token). Failure to provide a CA EEM ARTIFACT token may result in an interactive login challenge when attempting to launch the Web Interface in the chosen context (such as, a detail view of a given ticket). The getArtifact method allows the CA EEM ARTIFACT to be generated in the context of the user provided by the contact parameter.

> **NOTE**
> Service Desk needs to be integrated with CA EEM to perform this operation.

Returns

An CA EEM ARTIFACT based on the name of the user entered. You can use the ARTIFACT to launch the Web Interface only once.

LREL Methods

This article contains the following topics:

LREL methods supply information about object relationships. Objects with relationships have a left-hand side (lhs) and right-hand side (rhs) definition to describe many-to-many relationships. Some examples of many-to-many relationships include the following:

-
- Issues and configuration items
-
- Contacts and configuration items
-
- Task types and status codes

When working with LREL methods, the BREL or LREL attribute describe many-to-many relationships. The BREL attribute replaces the LREL attribute to define many-to-many relationships. However, the LREL attribute remains backward compatible with previous releases. You declare each relationship in a .maj file and then your code uses the LREL methods, such as *CreateLrelRelationship()*, with the existing web service client code.

NOTE
When working with group object management, you can use the special web methods to define a member that belongs to a group.

You declare the relationship using the BREL attribute to define relationships in majic files. For example, the following statement declares a many-to-many relationship for the Change Order (chg) object in change.maj:

```
asset BREL lrel_asset_chgnr chg {LREL nr;};
```

The *BREL* attribute for the change order named, "asset" is a list of all associated configuration items. The optional, LREL flag is an attribute that describes a set of owned resources. The corresponding majic definition of the Configuration Item (nr) includes the following:

```
chgnr BREL lrel_asset_chgnr nr {LREL chg;};
```

The *chgnr* attribute is a list of all change orders for a configuration item.

In your code, you can discover how many configuration items are associated with a change order, call *getLrelLength()* with the following parameters:

```
getLrelLength(sid, ChangeHandle, "asset")
```

You use the *sid* parameter for the Service ID from a login method. The *ChangeHandle* parameter is a string handle to a particular change order. Similarly, the following statement describes how to get the names of all configuration items that are related to a change order:

```
String attrs[] = {"name"};
getLrelValues(sid, ChangeHandle, "asset", 0, -1, attrs);
```

The *getLrelValues()* method provides the relationships in the *attrs* array.

NOTE
For more information about LREL tables and objects, see the Data Element Dictionary and Objects and Attributes.

getLrelLength

The following parameters apply to the getLrelLength method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
contextObject	String	Identifies the object on one side of the LREL.
LrelName	String	Identifies the LrelName. Use the name.

Description

Returns the number of objects on one side of a many-to-many relationship:

-
- **contextObject**
  Specifies it as a handle to an object on one side of the LREL relationship.
-
- **LrelName**
  Specifies it as the name of the side of the relationship identified by *ObjHandle*.

Returns

The following:

XML Element	Type	Description
<Length>	Integer	Specifies the number of objects.

getLrelValues

The following parameters apply to the getLrelValues method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
contextObject	String	Identifies the object on one side of the LREL.
LrelName	String	Identifies the Lrel Name. Use the name.
startIndex	Integer	Identifies the position in the “list” from which to begin fetching.
endIndex	Integer	Identifies the Last “list” position from which to fetch. Specify -1 to fetch all rows from startIndex.
attributes	String[]	Identifies an array of one or more attribute names for which to fetch values.

Description

Returns attribute values for a range of objects in an LREL relationship. Remember that items involved in an LREL relationship have no specific ordering. In fact, it is not really a “list”, as defined in this document.

The start and end index parameters are there to help throttle a large number of items. The format is as follows:

```
< UDSObjectList >
   <UDSObject>
   <Handle>
   <AttributeName0>
   <AttributeName1>
```

Returns

This method has the following returns:

XML Element	Type	Description
<UDSObjectList>	Array	Specifies the outer element, which contains a sequence of <UDSObject> elements.
<UDSObject>	Sequence	Contains a <Handle> element and zero or more <*AttributeNameX*> elements.

createLrelRelationships

The following parameters apply to the createLrelRelationships method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
contextObject	String	Identifies the object on one side of the LREL.
LrelName	String	Identifies the Lrel Name as seen by *contextObject*.
addObectHandles	String[]	Identifies the handles of objects for the other side of the LREL relationships.

Description

Adds one or more many-to-many relationships. *contextObject* is one side of the LREL relation. The caller passes one or more object handles for the other side.

If a relationship already exists between the two objects, no change is made and the system continues to process the *addObjectHandles* array. If an invalid object handle is passed, the entire operation is canceled.

The following example shows how to add several assets to a contact's environment:

```
createLrelRelationships(sid, ContactHandle, "cenv",
 ["nr:655A043EDDB36D4F97524F2496B35E75", "nr:755A043EDDB36D4F97524F2496B35E75"])

ContactHandle is a string handle to a contact, and
"nr:655A043EDDB36D4F97524F2496B35E75" and
 "nr:755A043EDDB36D4F97524F2496B35E75" are Asset handles.
```

Returns

Nothing.

removeLrelRelationships

The following parameters apply to the removeLrelRelationships method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
contextObject	String	Identifies the object on one side of the LREL.
LrelName	String	Identifies the Lrel Name (as seen by *contextObject).*
removeObjectHandles	String[]	Identifies the handles of objects to remove from the other side of the LREL relationships.

Description

Removes one or more many-to-many relationships. *contextObject* is one side of the LREL relationship. The caller passes one or more object handles for the other side.

It is not an error if no relationship existed between the two objects. If an invalid object handle is passed, the entire operation is canceled.

For a usage example, see createLrelRelationships() in this chapter.

Returns

Nothing.

dbmonitor_nxd--Database Monitoring Daemon

The Database Monitoring daemon (dbmonitor_nxd) provides a mechanism to allow CA SDM cache of specific database tables to be refreshed when changes are made externally from CA SDM.

The main function of dbmonitor_nxd is to generate CHANGE notifications for changes in specified tables that did not occur through CA SDM. In order to perform this function, the monitor periodically queries the database, determines what was changed externally and then sends CHANGE notifications to the bpvirtdb_nxd server. The bpvirtdb_nxd server notifies all domsrvr servers of the change, which causes each domsrvr to update its cache of specific database objects and then notify all other processes that subscribe for changes in the specified tables.

This mechanism works well for the occasional external change in tables that are monitored. However, in cases where mass updates are made externally a storm of CHANGE notifications are broadcast which leads to many database queries from various CA SDM processes which significantly impacts CA SDM's performance.

In order to eliminate this impact on CA SDM performance, dbmonitor_nxd has been updated for this release of the product. The Monitor supports a command line interface that allows the user to start and stop the monitoring of specified tables.

Syntax

This command has the following format:

```
dbmonitor_nxd  -c <command> -t <tables>
```

- **<command>**
  Enter start or stop.
- **<tables>**
  Specifies a table name or a comma delimited list of table names that must match one or more of the tables specified in the NX_DBMONITOR_TABLES environment variable.

Each request is sent to the dbmonitor_nxd daemon. The daemon takes the appropriate action and returns a message to the user indicating the action taken.

- If a start request is invoked for a table that is already started, no action is taken.
- If a stop request is made for a table that is already stopped, no action is taken.
- If monitoring is successfully stopped or started for a table, a log message is also written to stdlog.

  NOTE
  When the Monitor is paused for a table, all CA SDM processes that cache data from these tables may become out of date and no provision is made to update this cache.

For example, BOPLGIN caches Contact records (from the ca_contact and usp_contact tables) and this cache would not be updated if the Monitor was paused for the ca_contact table during the time external updates were loaded into the database. In the BOPLGIN case this will have little consequence because the essential Contact attributes cached in BOPLGIN are taken from the usp_contact table and not the ca_contact table.

NOTE
When the Monitor is paused for a table, Web Users will not be able to see changes in the table while viewing a detail form that were made externally while the Monitor was paused.

List/Query Methods

This article contains the following topics:

Two paradigms are available for working with lists. One paradigm uses a list handle for referring to and making queries on a server-side list and the other simply performs a SQL-like select.

If you need to maintain reference to a static list, use the methods that return list handles. These methods are especially useful when working with very large lists. For example, your application may need to perform operations using the entire table of 10,000 Contacts. Downloading values for all 10,000 at once could result in an unacceptable performance lag (this condition is actually prevented by the system -- see Where Clauses). With a list handle, however, you can select a range of rows upon which to query.

The primary drawback to using a list handle is the extra method calls it requires. At least two or three calls are necessary, as indicated by the following:

- One to get the handle
- A second (or third) to retrieve values
- A final call to free the list

You need to balance the amount of remote method calls versus the expected amount of data returned.

NOTE
CA SDM restricts the amount of data that can be returned from any one list. For more information, see Where Clauses.

doSelect

The following parameters apply to the doSelect method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
objectType	String	Identifies the object type (factory) to query.
whereClause	String	(Optional) Identifies the where clause for the query.
maxRows	Integer	Indicates the maximum number of rows to return. Specify -1 to return all rows. **Note:** Regardless of the integer specified, CA SDM will return a maximum of 250 rows per call.
attributes	String[]	Identifies the attribute list for which to fetch values. Dotted-attributes are permitted. If this field is blank, all value-based attributes are returned. These attributes cannot be defined as LOCAL in the majic definition file. LOCAL attributes are temporal; they have no database storage.

Description

Performs an SQL-like select on a specified object table. Supply one or more attributes you want fetched from the objects that match the supplied where clause.

Returns

A sequence of <UDSObject> elements. The following format applies:

```
<UDSObjectList>
   <UDSObject>
    <Handle>
    <Attributes>
   <AttributeNameA>
   <AttributeValueA0>
   <AttributeValueA1>
   <AttributeNameB>
   <AttributeValueB0>
   ...
```

XML Element	Type	Description
<UDSObject>	N/A	Specifies the standard UDSObject element containing the handle and requested attribute values.
<UDSObjectList>	Sequence	Contains a <Handle> element and an <Attributes> sequence.

For example, if the method used is the following:

```
String[] myArray = ["last_name", "first_name"]
doSelect(mySID, "cnt", "last_name LIKE 'J%'", 2, myArray)
```

The return could be the following:

```
<UDSObjectList>
   <UDSObject>
   <Handle>cnt:555A043EDDB36D4F97524F2496B35E75</Handle>
   <Attributes>
   <AttributeName>last_name</AttributeName>
   <AttributeValue>Johnson</AttributeValue>
   <AttributeName>first_name</AttributeName>
   <AttributeValue>Carol</AttributeValue>
   </Attributes>
</UDSObject>
<UDSObject>
   <Handle>cnt:555A043EDDB36D4F97524F2496B35E76</Handle>
   <Attributes>
   <AttributeName>last_name</AttributeName>
   <AttributeValue>Jones</AttributeValue>
   <AttributeName>first_name</AttributeName>
   <AttributeValue>Ron</AttributeValue>
   </Attributes>
</UDSObject>
</UDSObjectList>
```

doQuery

The following parameters apply to the doQuery method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
ObjectType	String	Identifies the object type (factory) to query.
WhereClause	String	(Optional) Identifies the where clause for the query.

Description

Performs an SQL-like select for the specified object type. It also returns a *list handle* that points to a list of the rows returned from the query, where each row represents a CA SDM object that matched the supplied where clause. The caller can fetch values for the list rows using getListValues ().

> **NOTE**
> For more information about where clauses, see Where Clauses.

> **WARNING**
> The object list is stored on the CA SDM server and consumes system resources. The caller is responsible for freeing the list with freeListHandles(). Leaving a list in memory may increase memory for the process beyond the 2GB limit, resulting in memory leaks and can cause system failure.

Lists created with this function are homogenous, meaning the objects are all the same type, and they are static, meaning the list never changes even if a data change to an object excludes it from the initial where clause.

Returns

A list handle that must be freed with freeListHandle().

XML Element	Type	Description
<listHandle>	Integer	Identifies the list handle.
<listLength>	Integer	Identifies the length of the list generated.

getListValues

The following parameters apply to the getListValues method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
ListHandle	Integer	Identifies the list handle obtained with a previous call.
StartIndex	Integer	Identifies the position in the list from which to begin fetching.

EndIndex	Integer	Identifies the last list position from which to fetch. Specify -1 to fetch all rows from StartIndex. **Note:** Regardless of the integer specified, Web Services will return a maximum of 250 rows per call.
AttributeNames	String[]	Identifies an array of one or more attribute names for which you want to fetch values.

Description

Returns attribute values for a range of objects in a list. For example:

```
< UDSObjectList >
   <UDSObject>
   <Handle>
   <Attributes>
       <AttributeName0>
       <AttributeName1>
```

Returns

This method has the following returns:

XML Element	Type	Description
<UDSObjectList>	Sequence	Identifies the outer Element, which contains a sequence of <UDSObject> elements.
<UDSObject>	N/A	Contains a <Handle> element and <Attributes> sequence.

freeListHandles

The following parameters apply to the freeListHandles method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
Handles	Integer[]	Identifies an array of list handles to free.

Description

Frees the server-side resources for a list and invalidates the list handles. This method should be called whenever a list reference is no longer needed.

Returns

Nothing.

getRelatedList

The following parameters apply to the getRelatedList method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
ObjectHandle	String	Identifies the object handle.
ListName	String	Identifies a list-type attribute name of the object.

Description

Returns a list handle for list (QREL or BREL) attribute of an object. For example, the request object has a related list named “children”, which is a list of its child requests. The Request’s Activity Log (“act_log” or “act_log_all”) is another example.

To retrieve information about an object’s list attributes, refer to the object schema (majic) documentation or use getObjectTypeInformation().

Returns

The following:

XML Element	Type	Description
<listHandle>	Integer	Identifies the list handle.
<listLength>	Integer	Identifies the length of the list generated.

getRelatedListValues

The following parameters apply to the getRelatedListValues method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
objectHandle	String	Identifies the object handle.
listName	String	Identifies a list-type attribute name for the object.
numToFetch	Integer	Signifies the maximum number of rows to return. Cannot be zero Specify -1 to return all rows **Note:** Regardless of the integer specified, Web Services can return a maximum of 250 rows per call.
attributes	String[]	Identifies an array of one or more attribute names for which to fetch values. Dotted names are permitted.

getRelatedListValuesResult	String Holder	Identifies the String Holder object for capturing returned data.
numRowsFound	Integer Holder	Identifies the Integer Holder object for capturing returned data.

Description

Returns values for lists related to a specific object. The lists must be defined as a QREL or BREL. Use the LREL methods to query LREL types.

For example, the request object has a related list named “children”, that is a list of its child requests. This method is a list handle-free alternative to getRelatedList(). The return format is similar to getListValues(), as indicated by the following:

```
<numRowsFound>
< UDSObjectList >
   <UDSObject>
   <Handle>
   <AttributeName0>
   <AttributeName1>
```

You can retrieve information for object list attributes using object schema (majic). An alternative method is to use getObjectTypeInformation().

Returns

This method has the following returns:

XML Element	Type	Description
<getRelatedListValuesResult>	N/A	Identifies the outer element, <UDSObjectList>, that contains a sequence of <UDSObject> elements. Each <UDSObject> element contains a <Handle> element and zero or more <AttributeNameX> elements.
<numRowsFound>	Integer	Indicates the total number of rows in the queried list. **Note:** The total number of rows is not necessarily the number of rows returned.

getPendingChangeTaskListForContact

The following parameters apply to the getPendingChangeTaskListForContact method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
contactHandle	String	Identifies the contact handle.

Description

Returns a list handle representing all the “pending” change order workflow tasks assigned to a contact. A “pending” task is an active workflow task with a status that permits task updates.

Returns

The following:

XML Element	Type	Description
<listHandle>	Integer	Identifies the list handle.
<listLength>	Integer	Identifies the length of the list generated.

getPendingIssueTaskListForContact

The following parameters apply to the getPendingIssueTaskListForContact method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
contactHandle	String	Identifies the contact handle.

Description

Returns a list handle representing all the “pending” Issue tasks assigned to a contact. A “pending” task is an active task with a status that permits task updates.

Returns

The following:

XML Element	Type	Description
<listHandle>	Integer	Identifies the list handle.
<listLength>	Integer	Identifies the length of list generated.

getNotificationsForContact

The following parameters apply to the getNotificationsForContact method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
contactHandle	String	Identifies the contact handle.
queryStatus	Integer	(Optional) Identifies the target of the notifications.

Description

Returns a list of notifications (‘lr’ objects) for a contact.

You can query on a specific status for the notifications with the *queryStatus* field, which is useful for returning, for example, only non-cleared messages. The possible *queryStatus* values are as follows:

- **-1** -- Fetch all notifications
- **0** -- Fetch non-cleared notifications (those with a status value of less than 7)
- **1** -- Fetch cleared notifications (those with a status value of 7, 8 or 9)

Returns

The following:

XML Element	Type	Description
<listHandle>	Integer	Identifies the list handle.
<listLength>	Integer	Identifies the length of the list generated.

Asset Management Methods

This article contains the following topics:

It is possible for a client site to enhance the asset object using *extensions*. Asset extensions are separate tables that hold extra attribute information for an asset. The extension table is linked to a particular asset by using the asset ID as a foreign key. CA SDM Web Services ships two predefined extensions, Computer and Software. For more information, read the text executed by the following command:

```
/bopcfg/majic/assetx.maj
```

The asset's family attribute determines if the asset has an extension. Setting the Class attribute generally sets the family at asset creation time. To determine if an asset has an extension, query the 'extension_name' attribute of the asset's family (for example, "family.extension_name").

To retrieve values from an extension object, query it like any other object by using the following method:

```
getObjectValues()
```

To get the handle for a particular extension object, use the following method:

```
getAssetExtensionInformation()
```

Update an extension object like you would any other object by using the following method:

```
updateObject()
```

We do not recommend that you create your own extension objects. One is created for you, if needed, when createAsset() is called. Because of this automatic creation, we recommend that you only use createAsset() to create asset objects.

> **NOTE**
> If you are using the ITIL methodology, remember that Asset and Configuration Item are interchangeable in this context.

createAsset

The following parameters apply to the createAsset method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
attrVals	String	Identifies the array of name-value pairs used to set the initial attribute values for the new asset.

attributes	String Holder	Identifies the String Holder object for caputring returned data.
createAssetResult	String	Identifies the name for the new asset's extension. If no extension was created, this field is empty.
newAssetHandle	String Holder	Identifies the String Holder object for capturing returned data.
newExtensionHandle	String Holder	Identifies the String Holder object for capturing returned data.
newExtensionName	String Holder	Identifies the String Holder object for capturing returned data.

Description

Describes the recommend method for creating an asset. If you intend to create an asset with an extension, be sure to set the Asset Class attribute in the *attrVals* section.

NOTE
If you are using the ITIL methodology, use this method to create a Configuration Item.

Returns

A <UDSObject> element containing the handle for the new object, with attribute values specified in the *attributes* parameter. If the *attributes* parameter is empty, *all* attribute values are returned. List and LREL types are also returned, but as empty elements.

XML Element	Type	Description
<createAssetResult>	N/A	Identifies the standard <UDSObject> element containing the handle and requested attribute values.
<newAssetHandle>	String	Identifies the new request's handle.
<newExtensionHandle>	String	Identifies the handle for the new Asset's extension. If no extension was created, this field is empty.
<newExtensionName>	String	Identifies the name for the new asset's extension. If no extension was created, this field is empty.

getAssetExtensionInformation

The following parameters apply to the getAssetExtensionInformation method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
assetHandle	String	Identifies the asset to query.

attributes	String[]	Identifies the standard array of attributes from the asset extension object from which to request values. If this value is empty, all attributes are returned.
getAssetExtInfoResult	String Holder	Identifies the String Holder object for capturing returned data.
extensionHandle	String Holder	Identifies the String Holder object for capturing returned data.
extensionName	String Holder	Identifies the String Holder object for capturing returned data.

Description

Returns extension information for an asset. If the asset does not have an extension, nothing is returned.

An asset has an extension if a value exists for its “family.extension_name” property. This property is empty if the asset does not have an extension.

Returns

The following elements return with empty values when the asset does not have an extension:

XML Element	Type	Description
<getAssetExtInfoResult>	String	Identifies all the attribute values for the extension.
<extensionHandle>	String	Identifies the extensions handle.
<extensionName>	String	Identifies the name for the assets extension.

addAssetLog

The following parameters apply to the addAssetLog method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
assetHandle	String	Identifies the asset handle.
contactHandle	String	(Required) Identifies the handle of the contact used for the log’s author.
logText	String	Identifies the text for the new asset log.

Description

Adds a new log entry for an asset. The log’s author is the user associated with the SID.

Returns

Nothing.

createAssetParentChildRelationship

The following parameters apply to the createAssetParentChildRelationship method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
parentHandle	String	Identifies the asset handle for the parent.
childHandle	String	Identifies the asset handle for the child.

Description

Makes *assetParent* a parent of *assetHandle*. Web Services creates a separate object (the *hier* object, which is the *Assignment* table) for parent-child relationships between assets. These are stored in related lists, *child_hier* and *parent_hier,* in the Asset (nr) object.

Returns

Handle of the new hier (Assignment) object.

Web Services Business Methods

This article contains the following topics:

This section describes the Web Services Business methods.

createIssue

The following parameters apply to the createIssue method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
creatorHandle	String	Identifies the handle of the contact responsible for the creation of the issue (the log agent). Pass an empty string to specify the default CA SDM user.
attrVals	String[]	Identifies the array of name-value pairs that is used to set the initial attribute values for the new issue. **Note:** Dotted names are not permitted.
propertyValues	String[]	(Optional) Identifies the array of values for any properties that are attached to the new issue.
template	String	(Optional) Identifies the handle of an issue template (iss_tpl object) from which to create the issue.
attributes	String[]	Identifies the sequence of attribute names from the new object for which to return values. Dot-notation is permitted. If this field is empty, all attribute values are returned.

newIssueHandle	String Holder	Identifies the String Holder object for capturing returned data. See the XML Element Return list below for details.
newIssueNumber	String Holder	Identifies the String Holder object for capturing returned data. See the XML Element Return list below for details.

Description

Creates a CA SDM Issue (iss) object. For more information about creating an Issue object with properties, see createRequest().

NOTE
You *must* use this function to create a Issue; do not use createObject().

Returns

Returns the new object handle, along with *all* of its attribute values. List and LREL types are also returned, but as empty elements.

XML Element	Type	Description
<UDSObject>	N/A	Identifies the standard UDSObject element containing the handle and requested attribute values.
<newIssueHandle>	String	Identifies the new issue's handle.
<newIssueNumber>	String	Identifies the new issue's number (its "ref_num" attribute).

createRequest

The following parameters apply to the createRequest method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
creatorHandle	String	Identifies the handle of the contact responsible for the creation of the request (the log agent). Pass an empty string to specify the default CA SDM user.
attrVals	String[]	Identifies an array of name-value pairs that is used to set the initial attribute values for the new request. **Note:** Dotted names are not permitted.
propertyValues	String[]	Identifies the array of values for any properties that are attached to the new request.
template	String	(Optional) Identifies the handle of the request template (cr_tpl) from which to create the request.

attributes	String[]	Specifies the sequence of attribute names from the new object for which to return values. Dot-notation is permitted. If this field is empty, all value-based attribute values are returned.
newRequestHandle	String Holder	Specifies the String Holder object for capturing returned data. See the XML Element Return list below for details.
newRequestNumber	String Holder	Specifies the String Holder object for capturing returned data. See the XML Element Return list below for details.

Description

Creates a CA SDM Request (cr) object. You *must* use this function to create a Request; do not use createObject().

propertyValues is a list of values for each Property object that will be attached to the new Request. The Properties that are attached are determined by the new Request's 'category' attribute value. All properties created from the CA SDM Web Services interface will have a default value (for more information, see Categories and Properties), which is important because a Request will not save until all of its Properties marked "required" have a value.

You may override the default by supplying values for any properties that will be attached when the Request is created. You must supply this information before the Request is created, since createRequest() attempts to back-store the object you most recently create. Use getPropertyInfoForCategory() to get a list of properties for a specific Category. This function returns the properties in order of their 'sequence' attribute, which is the expected order of the *propertyValues* array. For example, if the sequences and symbols of the properties are as follows:

```
100  -  Hard Drive Size
200  -  CPU
300  -  Memory
```

The *propertyValues* array, depending on the programming language, may appear as follows:

```
["40 GB", "Pentium 4 1.7 Ghz", "256"]
```

getPropertyInfoForCategory() indicates which Properties are marked *required*.

If you do not set the Request category or do not want to set any Property values, pass an empty string for *propertyValues.*

If you do not want to rely on the default property values, the following is the suggested order for creating a new Request (or Issue or Change Order):

1.
2. Retrieve a list of Categories/Areas. The object name for Request Area is 'pcat'.
3.
4. Call getPropertyInfoForCategory() and examine the list of properties for the category of the new Request/Issue/Change.
5.
6. Create a value array for each of the properties returned. This is identified by the *propertyValues* parameter for the create operation.
7.
8. Assemble the *attrVals* array and call the create method.

As an alternative to the previous procedure, you can retrieve the list of properties using getRelatedListValues() after createRequest() returns. Properties are stored in the 'properties' list for a Request.

Depending upon the application, it may be faster to at least cache the list of Categories, since this data does not change often at many client sites.

NOTE
By default, this method creates a Request. If you are using the ITIL methodology, you need to set the 'type' attribute in the attrVals array to define whether you are creating an Incident or a Problem ticket. For more information about ITIL procedures, see ITIL Methodology.

Returns

Returns the new objects handle with *all* of its attribute values. List and LREL types are also returned, but as empty elements.

XML Element	Type	Description
<UDSObject>	N/A	Identifies the standard UDSObject element containing the handle and requested attribute values.
<newRequestHandle>	String	Identifies the new request handle.
<newRequestNumber>	String	Identifies the new request's number (its "ref_num" attribute).

createChangeOrder

The following parameters apply to the createChangeOrder method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
creatorHandle	String	Identifies the handle of the contact responsible for the creation of the change order (the log agent). Pass an empty string to specify the default CA SDM user.
attrVals	String[]	Identifies an array of name-value pairs that is used to set the initial attribute values for the change order. **Note:** Dotted names are not permitted.
propertyValues	String[]	(Optional) Identifies the array of values for any properties that are attached to the new change order.
template	String	(Optional) Identifies the handle of the change template (chg_tpl object) from which to create the change order.
attributes	String[]	Specifies the sequence of attribute names from the new object for which to return values. Dot-notation is permitted. If this field is empty, all value-based attribute values are returned.

newChangeHandle	String Holder	Specifies the String Holder object for capturing returned data.
newChangeNumber	String Holder	Specifies the String Holder object for capturing returned data.

Description

Creates a CA SDM Change Order (chg) object. You *must* use this function to create a change order; do not use createObject().

Returns

The new object handle with *all* of its attribute values. List and LREL types are also returned, but as empty elements.

XML Element	Type	Description
<UDSObject>	N/A	Identifies the standard UDSObject element containing the handle and request attribute values.
<newChangeHandle>	String	Identifies the new change order handle.
<newChangeNumber>	String	Identifies the new change order number (its 'chg_ref_num' attribute).

createActivityLog

The following parameters apply to the createActivityLog method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
creator	String	Identifies the handle of the contact responsible for the activity.
objectHandle	String	Identifies the handle for a request, issue, or change order. Any other object type is rejected.
description	String	Identifies the description for the activity, which will appear in the activity log.
LogType	String	Identifies the type of log to create -- see the following Description.
TimeSpent	Integer	Sets the Time Spent field for the activity log, which is the duration of the activity. Pass zero for the default.
Internal	Boolean	Identifies the values that apply: True = Internal-only activity False = Non-internal activity that can be viewed by everyone.

Description

Creates an activity log for a specified request, issue or change order. This method corresponds to, “Activities - Log Comment/Research/Callback” on a Change/Request/Issue detail in the CA SDM interface. *LogType* is the code attribute for the activity type of the new log. The most common codes are as follows:

-
- “CB” (Callback)
-
- “RS” (Research)
-
- “LOG” (Log Comment)

The CA SDM Administrative Client also shows the code values. To access the code values, select from the Main Menu Administration, then select Notification, Activity Notifications.

Returns

The handle to the activity log object created.

XML Element	Type	Description
<LogHandle>	String	Identifies the handle for the new activity log.

transfer

The following parameters apply to the transfer method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
creator	String	Identifies the handle of the contact responsible for the activity.
objectHandle	String	Identifies the handle for a request, issue, or change order. Any other object type is rejected.
description	String	Identifies the description for the activity, which will appear in the activity log.
setAssignee	Boolean	Used to update the assignee field with the value in newAssignee, if the value is true.
newAssigneeHandle	String	Identifies the new assignee for the object.
setGroup	Boolean	Updates the group field, if true.
newGroupHandle	String	Identifies the new group for the object.
setOrganization	Boolean	Update the organization, if the value is true.
newOrgnizationHandle	String	(Issues and Change Orders only) Identifies the new organization for the object.

Description

Performs a transfer activity on an Issue, Request or Change Order. This method corresponds to the “Activities -- Transfer” command in the CA SDM interface. This method generates an activity log and optionally sets a new assignee, group, or organization.

The assignee, group or organization is not updated unless one or more of the corresponding *setAssignee/setGroup/ setOrganization* parameters is set to true.

If the companion parameter is false, then transfer will not attempt to update the field, even if a value is passed for that field. For example, if *setAssignee* is passed as false, transfer will not update the assignee even if *newAssignee* specifies a value. If the *setXXXX* parameter is true, then the field is updated. Pass the empty string to set a field to empty/null.

Returns

One or more handles to the activity log objects created. The returns are under a parent element named <Logs>.

XML Element	Type	Description
<LogHandle>	String	Identifies the handle for the new activity log (zero or more of these can be returned).

escalate

The following parameters apply to the escalate method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
creator	String	Identifies the handle of the contact responsible for the activity.
objectHandle	String	Identifies the handle for a request, issue, or change order. Any other object type is rejected.
description	String	Identifies the description for the activity, which will appear in the activity log.
setAssignee	Boolean	Updates the assignee field, if true.
newAssigneeHandle	String	Identifies the handle of the new assignee for the object.
setGroup	Boolean	Updates the group field with the value in newGroupHandle, if true.
newGroupHandle	String	Identifies the handle of the new group for the object.
setOrganization	Boolean	Sets the organization field with the value specified in the newOrganizationHandle, if true.
newOrgnizationHandle	String	(Issues and Change Orders only) Identifies the handle of the new organization for the object.
setPriorityHandle	Boolean	Updates the priority field with the value specified in newPriority, if true.
newPriority	String	Identifies the handle of the new priority for the object.

Description

Performs an escalate activity on an Issue, Request or Change Order. This method generates an activity log and optionally sets a new assignee, group, priority and/or organization.

It corresponds to the “Activities -- Escalate” command in the CA SDM interface.

The assignee, group, or organization is not updated unless one or more of the corresponding *setAssignee/setGroup/setOrganization* parameters is set to true. If the companion parameter is false, then escalate will not attempt to update the field, even if a value is passed for that field. For example, if *setAssignee* is passed as false, escalate will not update the assignee even if *newAssignee* specifies a value. If the *setXXXX* parameter is true, then the field is updated. Pass the empty string to set a field to empty/null.

NOTE
Organization is not used for Requests.

Returns

One or more handles to the activity log objects created.

XML Element	Type	Description
<LogHandle>	String	Identifies the handle for the new activity log (zero or more of these can be returned).

changeStatus

The following parameters apply to the changeStatus method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
creator	String	Identifies the handle of the contact responsible for the activity.
objectHandle	String	Identifies the handle for a request, issue, or change order. Any other object type is rejected.
description	String	Identifies the description for the activity, which will appear in the activity log.
newStatusHandle	String	Identifies the handle of the status for the object.

Description

Performs a status change activity on an issue, request, or change order. This method generates an activity log and optionally sets the status value. It corresponds to the “Activites -- Update Status” command in the CA SDM interface.

Returns

The handle to the activity log object created.

XML Element	Type	Description
<LogHandle>	String	Identifies the handle for the new activity log (zero or more of these can be returned).

getPropertyInfoForCategory

The following parameters apply to the getPropertyInfoForCategory method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
categoryHandle	String	Identifies a category handle.
attributes	String[]	Identifies the names of one or more attributes from the property template object for which to fetch values. If this is empty, all value-based attributes are fetched.

Description

Information about Properties for the specified category. This method is used to help pre-populate Request/Issue/Change Order properties on insert operations with user-defined data.

Depending on the category, this method queries either the ‘prptpl’ or the ‘cr_prptpl’ object types. Both types are nearly identical. The suggested attributes to fetch are ‘sequence’, ‘label’, ‘description’ and ‘required’.

Returns

A <UDSObject> element containing a sequence of <UDSObject> elements, in the order of the ‘sequence’ attribute.

logComment

The following parameters apply to the logComment method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
ticketHandle	String	Identifies the handle of the ticket where the activity log should be added.
Comment	String	Identifies the comment text.
internal_flag	String	Identifies the internal flag. Set to True if the new activity log should be marked as internal.

Description

Attaches a ‘Log Comment’ activity log to a ticket. It is a simplified version of createActivityLog().

Returns

Nothing.

notifyContacts Method

This article contains the following topics:

The following parameters apply to the notifyContacts method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
creator	String	Identifies the handle of the contact responsible for the activity.
contextObject	String	Identifies the handle for a request, issue, or change order. The following applies: It is the context for the notification. Any other object type is rejected. An activity log is added to the object to record the notification.
messageTitle	String	Identifies the title of the notification message.
messageBody	String	Identifies the body of the notification message.
notifyLevel	Integer	Indicates the notification level. Specify an integer from 1 (Low) to 4 (Emergency).
notifyees	String[]	Identifies the array of contact handles to notify.
Internal	Boolean	Indicates internal-only notification. Set to True to flag for an internal-only notification, which guarantees that the message is sent only to those who can view internal logs, and the resulting activity log is flagged as internal.

Description

Sends a notification to one or more contacts. This is equivalent to the Manual Notify activity on requests, issues, and change orders.

Returns

The following is returned:

XML Element	Type	Description
<LogHandle>	String	Identifies the handle for new activity log.

clearNotification

The following parameters apply to the clearNotification method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
lrObject	String	Identifies the handle to a Notify Log Header (lr) object.
clearBy	String	Identifies the name of the contact responsible for the clear operation.

Description

Clears a notification message.

Returns

The new status of the notification message.

attachChangeToRequest Method

This article contains the following topics:

The following parameters apply to the attachChangeToRequest method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
creator	String	Identifies the handle of the contact responsible for the activity.
requestHandle	String	Identifies the request to which you attach the change.
changeHandle	String	Identifies the handle of the change to add. If this is blank, a new change order is created.
changeAttrVals	String[]	Identifies the attribute-value pairs used to initialize a new change order if the changeHandle is blank.
description	String	(Optional) Identifies the description of the activity.

Description

Attaches a new or existing change order to a request. It corresponds with “Activities -- New Change Request” or “Activities -- Attach to Existing Request” on a Request detail in the CA SDM interface.

To create a new change order, pass the empty string in *changeHandle*. The system will create a new change order with values initialized from the request, including the change’s Requestor, Affected End User, Description, and Priority (you can see this effect in the CA SDM interface). You can override these or set additional values with *changeAttrVals,* which is a name-value array similar to what is passed for *createObject().*

To attach an existing change order, specify a handle in the *changeHandle* parameter. In this case, *changeAttrVals* is ignored.

If a new change order is created, *description* is used on the new change order’s activity log. If an existing change is attached, *description* is used on the request’s activity log.

If the request already has an attached change order, the following error is returned:

```
UDS_CREATION_ERROR
```

> **NOTE**
> This method works exactly the same for the ITIL methodology -- simply verify that you are passing the appropriate handle of an ITIL Incident or Problem to the method.

Returns

The following is returned:

XML Element	Type	Description
<changeHandle>	String	Identifies the handle for the change order, created or attached.

detachChangeFromRequest

The following parameters apply to the detachChangeFromRequest method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
creator	String	Identifies the handle of the contact responsible for the activity.
requestHandle	String	Identifies the handle of a request.
Description	String	(Optional) Identifies the description of the activity.

Description

Detaches a change order from a request. This method corresponds with “Activities -- Detach Change Order” on the CA SDM client. The change order is not deleted from the system.

There is no effect if the request did not have an attached change.

Returns

The handle of the request’s activity log marking the event.

createTicket Method

This article contains the following topics:

The following parameters apply to the createTicket method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
Description	String	(Optional) Identifies the description of the ticket.
problem_type	String	Specifies the code (not the persistent ID) for an existing problem type for the policy under which a Web Services application is running. If this is blank or contains a bad value, the default problem type is used.
Userid	String	(Optional) Specifies the user ID of the end user for the new ticket. If this is blank or the supplied user ID is not found, the proxy contact defined in the access policy is used and the ticket is still created.

Asset	String	(Optional) Identifies the handle of an asset to be attached to the ticket.
DuplicationID	String	(Optional) Allows callers to assist the duplication handling routines used for classifying tickets as being unique or different. If duplicate handling is on, this string is inspected after other duplicate handling criteria match to determine whether this is a unique or duplicate call to this method.
newTicketHandle	String Holder	Identifies the String Holder object for capturing returned data. See the XML Element Return list on this page for details.
newTicketNumber	String Holder	Identifies the String Holder object for capturing returned data. See the XML Element Return list on this page for details.
returnUserData	String Holder	Identifies the String Holder object for capturing returned data. See the XML Element Return list on this page for details.
returnApplicationData	String Holder	Identifies the String Holder object for capturing returned data. See the XML Element Return list on this page for details.

Description

Creates a ticket based on the rules defined in both the Service Aware Policy and the specified Problem Type. The Problem Type specifies the type of ticket created by a Request/Change/Issue template. The supplied description is copied in and the user is set from the access policy proxy contact.

If input problem_type does not exist, the policy default problem type is used. The problem type also defines how to handle duplicate ticket creation, and additional returned data.

Returns

Returns the new ticket handle, ticket number, and the user-specified return data defined in the problem type that is used to create the ticket.

XML Element	Type	Description
<UDSObject>	N/A	Returns the XML element so that it is consistent with the returns of these other methods: createRequest() createChange() createIssue() The body of this tag is empty.
<newTicketHandle>	String	Identifies the new ticket handle.
<newTicketNumber>	String	Identifies the new ticket number (its “ref_num” or “chg_ref_num” attribute).
<returnUserData>	String	Identifies the user-specified data for the problem type intended for display to the end user, or for log entries. You can set this value in the User Data Return field on the Returned Data tab of the Problem Type Detail window.

<returnApplicationData>	String	Identifies the user-specified data for the problem type intended for use within the application code, especially for actions. You can set this value in the Application Data Return field on the Returned Data tab of the Problem Type Detail window.

createQuickTicket

The following parameters apply to the createQuickTicket method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
customerHandle	String	Identifies the customer handle used to create the ticket.
description	String	(Optional) Identifies the description of the ticket.
newTicketHandle	String Holder	String Holder object for capturing returned data. See the XML Element Return list on this page for details.
newTicketNumber	String Holder	String Holder object for capturing returned data. See the XML Element Return list on this page for details.

Description

Creates a ticket based on the preferred document type of the user represented by customerHandle. Contact access rights are determined by an Access Type record, which also sets the contact preferred document type (Request, Incident, Problem, Issue or Change Order). If a contact document type is Issue, this method will create an Issue; if the document type is Request, a Request is created, and so on. The contact represented by customerHandle is used to set the end user/customer field of the new ticket. The ticket description is set to the input *description* value.

Returns

Returns the new ticket handle, ticket number, and a brief representation of the new ticket in <UDSObject> format.

XML Element	Type	Description
<UDSObject>	N/A	Returns a partial set of attributes because it is a high-level method designed to simplify the process. Their values come from the following methods, and the XML element is returned so it is consistent with the returns of these methods: createRequest() createChange() createIssue()
<newTicketHandle>	String	Identifies the new ticket handle.

<newTicketNumber>	String	Identifies the new ticket number (its “ref_num” or “chg_ref_num” attribute).

closeTicket

The following parameters apply to the closeTicket method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
description	String	(Optional) Identifies the description of the ticket, which can be used in the Close activity log.
ticketHandle	String	Identifies the ticket to close.

Description

Sets the status of the ticket to “Closed” and adds an activity log with the input description.

Returns

The handle to the activity log object created. It provides the same return as the changeStatus() method.

XML Element	Type	Description
<LogHandle>	String	Specifies the handle for new activity log (0 or more of these can be returned).

Group Management Methods

This article contains the following topics:

This section explains the Web Services Group Management Methods.

addMemberToGroup

The following parameters apply to the addMemberToGroup method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
ContactHandle	String	Identifies the handle of the contact to add as a member.
GroupHandle	String	Identifies the group to which you will add the contact.

Description

Adds a contact to a group. Nothing happens if the contact is already a member.

Returns

Nothing.

removeMemberFromGroup

The following parameters apply to the remove MemberFromGroup method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
ContactHandle	String	Identifies the handle of the contact to remove.
GroupHandle	String	Identifies the group from which you will remove the contact.

Description

Removes a contact from a group. Nothing happens if the contact is not a member.

Returns

Nothing.

getGroupMemberListValues

The following parameters apply to the getGroupMemberListValues method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
whereClause	String	Identifies the SQL where clause.
numToFetch	Integer	Determines the maximum number of records to return. This cannot be zero (0). Use ‘1’ to return all.
attributes	String[]	Identifies the array of attribute names for which to retrieve values.

Description

Functions similarly to getListValues(), except that it queries the system’s group and member relationships. The system uses a special ‘Group Member’ (grpmem) object for each group/member relationship. The CA SDM system administers grpmem objects behind the scenes (you do not manipulate them directly), and they are essential for certain queries.

The grpmem object simply contains two pointers, one to a Contact and the other to a Group, which in itself is a Contact object. The attribute names are ‘member’ and ‘group’, respectively. Because these are pointers, you must use dot-notation to form a query as normal. For example, to find all Contacts with a last name beginning with ‘B’ and in a group with the name, “Seattle”, you would use the following:

```
member.last_name LIKE 'B%' AND group.last_name = 'Seattle'
```

You may also use handles as normal, illustrated by the following:

```
member.last_name LIKE 'A%' AND group.id = U'555A043EDDB36D4F97524F2496B35E75'
```

It is important to note that this method can retrieve values from all members and groups, not just a single group. To simply get information about all the members of a specific group or member, just specify a handle in the where clause. For example, the following would retrieve values from a specific group:

```
group.id = U'555A043EDDB36D4F97524F2496B35E75'
```

The following concepts are important to remember:

-
- grpmem object contains two pointer attributes, one is to member and the other is to group
-
- grpmem object exists for each group/member relationship

The grpmem method actually queries the grpmem object table, thereby returning an object representing a relationship between two contacts. Therefore, the attribute values you want to fetch in attributes must use dot-notation from the grpmem object. To fetch values from the member, all your attribute names should be of the form, ‘member.*ATTRNAME*’, as illustrated by the following example:

```
'member.last_name'
```

To fetch values from the group, use ‘group.*ATTRNAME*’.

> **NOTE**
> For an example of efficient querying of groups and members, see Where Clauses.

Returns

Automatically returns no handles. The <Handle> element in the return is always empty. To request the member or group handle for each row, use one of the following in the attributes parameter described in the table

- “member.persistent_id”
- “group.persistent_id”

XML Element	Type	Description
<UDSObjectList>	N/A	Identifies the outer element, which contains an array of <UDSObject> elements. Each object is really a grpmem object.

Contact Management Methods

This article contains the following topics:

This section explains the Web Services Contact Management methods.

login

The following parameters apply to the login method:

Parameter	Type	Description
username	String	Identifies the user ID.
password	String	Identifies the password.

Description

Login validates a user with CA SDM login validation and returns a unique session ID that is required for most other web method calls. This key should be freed with logout(). A SID may expire if it is not used before a time out elapses.

The username/password is validated exactly the same as the CA SDM Web client; the contact's Access Type specifies the validation method. The default access policy will be applied to control and manage all subsequent accesses after a successful call of this function. The login user (not proxy contact specified in the default policy) is then responsible for subsequent web service activities. All function group security and data partition are enforced for the login user.

Returns

The following is returned:

Parameter	Type	Description
<SID>	Integer	Identifies the unique SID to use for all other Web Services calls.

loginService

The following parameters apply to the loginService method:

Parameter	Type	Description
username	String	Identifies the user ID.
password	String	Identifies the password.
policy	String	(Required) Identifies the policy code, which must be in plain text. Although it is required, it may be empty.

Description

Lets users log in with a conventional username/password authentication scheme where if valid, the system returns a unique session ID. This key should be freed with logout(). A SID may expire if it is not used before a timeout elapses.

User authentication is performed on the username and password while access control is applied based on the policy specified. The authentication is performed as described in login(). Empty policy will allow default policy to be applied automatically. The login user (not the proxy contact specified in the policy) is responsible for subsequent web service activities. All function group security and data partitions are enforced for the login user.

Returns

The following is returned:

Parameter	Type	Description
SID	Integer	Specifies the unique SID to use for all other Web Services calls.

impersonate

The following parameters apply to the impersonate method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
username	String	(Required) Identifies the user name of the user being impersonated.

Description

Lets an administrator switch the user responsible for all web services activities in a current web services session without additional user authentication. Invoking this method is allowed only if the current web services session is started by using the PKI access authentication scheme and the access policy is defined to allow impersonation.

The Access Type of the user to be impersonated is checked against the Access Type of the proxy user of the policy used in the current web services session. If the access_level of the new user's access type is less than or equal to the grant_level of the proxy user's access type, this method will replace the current user with the new user. A new web services session starts while the old session ends. A new SID is then returned. In addition, the new user is given the responsibility for all subsequent activities initiated in this new session. The function group security and data partition are enforced for the new user.

Returns

The following is returned:

Parameter	Type	Description
SID	Integer	Identifies the unique SID to use for all other Web Services calls.

logout

The following parameter applies to the logout method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.

Description

Invalidates a SID obtained from login(), loginService(), and loginServiceManaged().

Returns

Nothing.

loginWithArtifact

The following parameters apply to the loginWithArtifact method:

Parameter	Type	Description
userid	String	Identifies the user ID.
artifact	String	Identifies the Artifact obtained from CA EEM.

Description

Performs the user authentication by the provided CA EEM Artifact and opens a session with the back-end server. The Artifact can be used only once for authentication. The returned session ID (SID) can be used for subsequent web services method invocations. The Artifact can be acquired directly from CA EEM or can be obtained by the getArtifact method.

> **NOTE**
> Service Desk needs to be integrated with CA EEM to perform this operation.

The SID should be freed with logout(). A SID may expire if it is not used before a timeout elapses.

Returns

The following is returned:

Parameter	Type	Description
SID	String	Identifies the unique session ID (SID) to use for all other Web Services calls. It is in plain text format.

getPolicyInfo

This article contains the following topics:

The following parameters apply to the getPolicyInfo method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.

Description

Returns information about the access policy that is controlling and managing the current CA SDM Web Services session.

Returns

The following XML string:

XML Element	Type	Description
<SAPolicy>	N/A	Identifies the detailed information of this access policy and its related problem types.

The content of <SAPolicy> is shown as follows:

```
<SAPolicy>
<Name> name of policy </Name>
<Code> policy code </Code>
<ContactName> policy proxy contact's combo name </ContactName>
<ContactHandle> handle of policy's contact </ContactHandle>
<Access>
       <TicketCreation> limitation </TicketCreation>
       <ObjectCreation> limitation </ObjectCreation>
       <ObjectUpdate> limitation </ObjectUpdate>
       <Attachments> limitation </Attachments>
       <Queries> limitation </Queries>
       <Knowledge> limitation </Knowledge>
</Access>
<ProblemTypes> (zero or more <ProblemType> elements)
    <ProblemType>
        <Code>code of a problem type </Code>
        <Status>active or inactive </Status>
    </ProblemType>
</ProblemTypes>
</SAPolicy>
```

getTaskListValues

The following parameters apply to the getTaskListValues method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
objectHandle	String	Identifies the object handle for an issue or change order.
attributes	String	Identifies the sequence of attribute names for which to fetch values. Dotted-attributes are permitted. If this is blank, all attributes are fetched.

Description

Returns values for all the tasks associated with the specified issue or change order.

NOTE
This is a convenience method. The same list could be obtained using doSelect().

Returns

This method has the following returns:

XML Element	Type	Description
<UDSObjectList>	N/A	Outer element -- contains zero or more <UDSObject> elements. Each <UDSObject> node represents a Task. The nodes are ordered by the Task's 'sequence' attribute.

getValidTaskTransitions

The following parameters apply to the getValidTaskTransitions method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
taskHandle	String	Identifies the handle to a Workflow task (for an issue or change order).
attributes	String[]	Identifies the names of attributes to fetch from the 'tskstat' object. If this field is empty, all value-based attributes are returned.

Description

Returns all of the possible values for the 'status' attribute of a particular task. The Status codes to which a task may be set depend upon several factors, such as the current status of the task, and restrictions set by the administrator.

NOTE
taskHandle can be a task owned by either a change order or an issue. The objects returned are Task Status ('tskstat') objects used for both types of tasks.

Returns zero or more Status objects to which a task can be set.

Returns

This method has the following returns:

XML Element	Type	Description
<UDSObjectList>	N/A	Identifies the outer element, which contains zero or more <UDSObject> elements with the requested attribute values.

getValidTransitions

The getValidTransitions method lets you list the transitions for a ticket.

The following parameters apply:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
handle	String	Identifies the handle for a ticket or status. If a ticket handle is provided, the method retrieves the valid transitions for the current status of the ticket.
ticketFactory	String	Identifies the factory of the ticket. Valid values are only "cr" for Request, "in" for Incident, "pr" for Problem, "chg" for Change Order and "iss" for Issue.

Description

Returns all of the possible transition values for the status of a ticket. Administrators can configure valid status transitions for all ticket types.

Returns zero or more request/incident/problem/change/issue transition objects, depending on the values passed for handle and ticketFactory. If zero objects are returned, there are no transitions on the status of the ticket.

Returns

This method has the following return:

XML Element	Type	Description
<UDSObjectList>	N/A	Identifies the outer element, which contains zero or more <UDSObject> elements.

getDependentAttrControls

The getDependentAttrControls method lets you list the locked and required attributes for the Status record (either the ticket's current status or any status record). At this time, the only attribute supported is the Status attribute. If the attrVals parameter is passed as empty, the method returns the dependent attributes for the current status of the ticket (assuming

that a valid handle is used). Otherwise, you can request the dependent attributes for any status record by using the attrVals parameter.

Enter the attrVals parameter syntax as follows:

-
- Specify the word "status" for the first item.
-
- For the second item, specify the code of the status record, for example, WIP.

The following parameters apply:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
handle	String	Identifies the handle for a ticket.
attrVals	String	Identifies the name-value pairs of a ticket attribute for this method to get its dependent attribute controls.

Description

Returns all of the possible dependent attribute controls for the Status field of a ticket. Administrators can configure attribute restrictions.

Returns zero or more attribute control objects depending on the values specified. If zero objects are returned, there are no attribute restrictions on the object.

Returns

The following:

XML Element	Type	Description
<UDSObjectList>	N/A	Identifies the outer element, which contains zero or more <UDSObject> elements.

getHandleForUserid

The following parameters apply to the getHandleForUserid method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
userID	String	Identifies the user ID upon which to query.

Description

Returns the persistent handle for a Contact represented by userID.

Returns

The following is returned:

Parameter	Type	Description
<Handle>	String	Identifies the contact's handle.

getAccessTypeForContact

The following parameters apply to the getAccessTypeForContact method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
ContactHandle	String	Identifies the contact handle upon which to query.

Description

Returns a handle for the Access Type for a Contact.

Every Contact is assigned an Access Type object, which defines a Contact's permissions and security. Note the access_type field on a Contact is not required. To accommodate this, a single Access Type is marked as the default for Contacts who do not have an object specifically assigned.

This method returns the Access Type directly assigned to a Contact (that is, the information to which the access_type field points in the Contact record) or it returns the default. Typical value methods, such as getObjectValues() or getListValues(), may not return the correct Access type; therefore, these are not the accurate methods for retrieving the Contact's Access.

Returns

A string handle for an Access Type object.

getContact

The following parameter applies to the getContact method:

Parameter	Type	Description
SID	String	Identifies the session retrieved from logging in.
contactId	String	Identifies the unique ID of the contact to retrieve. contactId is UUID in string format.

Description

Retrieves information on all contacts.

Returns

A <UDSObject> node with a <UDSObject> node describing a contact with some of the following child <Attributes> nodes:

XML Element Value	Type	Description
contactid	String	Specifies a unique ID of the contact. contactId is UUID in string format.
userid	String	Indicates the user name of the contact.
last_name	String	Indicates the last name of the contact.

first_name	String	Specifies the first name of the contact.
middle_name	String	Specifies the middle name of the contact.
location	String	Indicates the location of the contact.
dept	String	Identifies the department of the contact.
organization	String	Identifies the organization of the contact.
email_address	String	Identifies the email address of the contact.
pemail_address	String	Signifies the alternate email address of the contact.
phone_number	String	Indicates the phone number of the contact.
alt_phone	String	Indicates the alternate phone number of the contact.
address1	String	Specifies the address of the contact.
address2	String	Specifies the alternate address of the contact.
city	String	Identifies the city of the contact.
state	String	Identifies the state of the contact.
zip	String	Indicates the ZIP code of the contact.
country	String	Indicates the country of the contact.
delete_flag	Integer	Indicates whether the contact is active: **0** -- Active **1** -- Inactive

findContacts

The following parameters apply to the findContacts method:

Parameter	Type	Description
SID	Integer	Identifies the session retrieved from logging in.
userName	String	Returns only the contacts with this user name. You can do wildcard searches by using the percent (%) sign. For example, to search for all contacts where the User Name begins with Smi, specify Smi%.
lastName	String	Returns only the contacts with this last name. You can do wildcard searches by using the percent (%) sign.
firstName	String	Returns only the contacts with this first name. You can do wildcard searches by using the percent (%) sign.
email	String	Returns only the contacts with this email address. You can do wildcard searches by using the percent (%) sign.

accessType	String	Returns only the contacts with this access type. You can specify multiple roles by separating them with commas. Specify the following: 10002 - Administration 10004 - Customer 10005 - Employee 10009 - IT Staff 10013 - Knowledge Management 10014 - Process Management 10010 - Service Desk Management 10008 - Service Desk Staff 10020 - Vendor Staff
inactiveFlag	Integer	Returns only the contacts that are inactive or active. Specify the following: 0 for active - 999 for all other for inactive

Description

Retrieves the list of contacts.

Returns

A <UDSObjectList> node with zero or more <UDSObject> nodes describing contacts with the following <Attributes> child nodes:

XML Element Value	Type	Description
id	UUID	Specifies the unique ID of the contact.
userid	String	Specifies the user name of the contact.
last_name	String	Identifies the last name of the contact.
first_name	String	Identifies the first name of the contact.
access_type	Integer	Specifies the Role ID of the contact.
delete_flag	Integer	Indicates whether the contact is active or inactive: 0 - Active 1 - Inactive

loginServiceManaged Method

This article contains the following topics:

The following parameters apply to the loginServiceManaged method:

Parameter	Type	Description
policy	String	(Required) Identifies the policy, which must be must be in plain text.

encrypted_policy	String	(Required) Identifies the digital signature of the policy code, encrypted with the policyholder's private key. It is in BASE64 text format.

Description

Performs the user authentication by locating the policy through the plain text policy code, finding the policyholder's public key associated with the policy, decrypting the encrypted policy code, matching decrypted content with the policy code, and finally, opening a session with a back-end server. The returned session ID (SID) can be used for subsequent web services method invocations. Proxy contact specified in the policy is responsible for all subsequent web service activities initiated. All function group security and data partition are enforced for the proxy contact defined in the policy.

It is also important to note that the encrypted_policy parameter is in the BASE64 text format, and it is necessary to perform proper conversion from the binary format. The SID should be freed with logout(). A SID may expire if it is not used before a timeout elapses.

Returns

The following is returned:

Parameter	Type	Description
SID	String	Identifies the unique session ID (SID) to use for all other Web Services calls. It is in plain text format.

Implement loginServiceManaged in Java

The following shows how to generate Certificates and then use these generated Certificates to access the CA SDM web services.

In the following example, the login process completes using the CA SDM Certificate and then performs two common web services calls. The getBopsid() web services method call allows you to obtain a token that is linked to a specific user. This token can be used to login to the CA SDM web interface as the linked user without being prompted for a password. This allows seamless integration to be enabled between different applications.

> **WARNING**
> The generated BOPSID token expires after 30 seconds, so it must be used promptly.

Important! There is a known issue when using the 1.4 version of the AXIS tool. For more information, see the *Release Notes*.

Follow these steps:

1. Generate the stub classes with AIXS Tool WSDL2Java. For more information, see the Generating Stub Classes with AXIS Tool WSDL2Java section from the PKI_loginServiceManaged_JAVA_steps file. Find the file in the following location:
   ```
   $NX_ROOT/samples/sdk/websvc/java/test1_pki
   ```
2. Start the CA SDM service.
3. Run pdm_pki -p DEFAULT.
   DEFAULT.p12 is created in the current directory. This policy will have the password equal to the policy name (in this case DEFAULT).

   > **NOTE**
   > This command will also add the Certificate's public key to the field pub_key field (public_key attribute) in the sapolicy table/object.

4. Log into CA SDM
5. Select **SOAP Web Services Policy**, **Policies** on the **Administration** tab.
   The **SOAP Web Services Access Policy List** page opens.
6. Click **DEFAULT**.
   The **SOAP Web Services Access Policy Detail** page opens.
7. Complete the **Proxy Contact** field (in this example, ServiceDesk) and confirm that the DEFAULT policy record Has Key field displays "Yes."
8. Copy DEFAULT.p12 (from the directory where command pdm_pki is executed), the JSP file called *pkilogin.jsp* and the HTML file called *pkilogin.htm* (from the $NX_ROOT/samples/sdk/websvc/java/test1_pki directory) to the following directory:
   ```
   $NX_ROOT/bopcfg/www/CATALINA_BASE/webapps/axis
   ```
9. Open the HTML form (from the axis directory). For example, http://localhost:8080/axis/pkilogin.htm
   Complete the appropriate fields.

   NOTE
   The Directory field identifies the location of the Certificate file. Modify the path to the correct location.
10. Click Log me in!
    The results page opens.
11. Click the BOPSID URL.

    WARNING
    Click this immediately! The BOPSID has a limited life token of about 30 seconds.

    The format of a URL using a BOPSID is as follows:
    http://<server name>:CA Portal/CAisd/pdmweb.exe?BOPSID=<BOPSID value>

    NOTE
    In order to use the loginServiceManaged method for a Java client program running on AIX, you may need to replace a pair of security policy files within your JAVA_HOME. Go to http://www.ibm.com and search for "developerworks java technology security information AIX". In the "developerWorks : Java technology : Security" document, follow the link to "IBM SDK Policy files". Download the unrestricted policy files, local_policy.jar and US_export_policy.jar. Use these files to replace the original files in your JAVA_HOME/lib/security directory."

Generate Stub Classes with WSDL2Java

You can generate the stub classes for the CA SDM web services.

To generate stub classes with WSDL2Java

1. Open a command prompt and navigate to the "<drive>:\program files\CA" directory.
   The directory appears.
2. Run the dir /x command.
   The short form of the CA SDM directory appears. For example, the short name is "SERVIC~1."
3. Search for javac.exe on all of the server's local drives. If you locate the file, make note of its location because you will need to reference it in a batch file.

   NOTE
   If you do not locate javac.exe, go to Http://java.sun.com and search for Java J2SE SDK to download. This may require a reboot.
4. Build a batch file called build_wsdl.bat and place it in the following directory:
   ```
   $NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\axis
   ```
5. Use the following code for the batch file, updating the bold items as appropriate:

```
@echo off
::#################################################################
```

```
::# Simple bat file to Build Unicenter Service Desk Version 11.0 USD Stub classes
::# Use it to create the required USD Unicenter Service Desk Version 11.0 Java Web Services classes
::#
::#  Usage: build_wsdl
::#################################################################
@REM Update this with the PATH to USD NX_ROOT location
@SET USD_SHORT_PATH=C:\Progra~1/CA/Servic~1/

@REM Update this with the PATH to the JDK javac.exe compiler
@REM (this is used in the 2nd part of this file)
@SET JAVAC_EXE="C:\j2sdk1.4.2_13\bin\javac.exe"
@REM Update this to the path to the USD <USDK> <rellevel> NX_ROOT/java/lib location
@SET USD_TOMCAT=%USD_SHORT_PATH%java/lib
@SET CP=%USD_TOMCAT%/axis.jar;%USD_TOMCAT%/commons-discovery.jar;%USD_TOMCAT%/commons-logging.jar;%USD_TOMCAT
%/jaxrpc.jar;%USD_TOMCAT%/saaj.jar;%USD_TOMCAT%/log4j-1.2.8.jar;%USD_TOMCAT%/xml-apis.jar;%USD_TOMCAT%/
xercesImpl.jar;%USD_TOMCAT%/wsdl4j.jar;%USD_TOMCAT%/axis-ant.jar
@REM Please specify the path to java.exe file below
@REM You can obtain this by reviewing the NX.env file use the @NX_JRE_INSTALL_DUR
@REM variable to derive this info
@SET JAVA_PATH=C:\Program Files(x86)\CA\SC\JRE\1.6.0_30
@SET JAVA_EXE="%JAVA_PATH%\bin\java.exe"
@cd WEB-INF\classes
%JAVA_EXE% -cp %CP% org.apache.axis.wsdl.WSDL2Java http://localhost:8080/axis/services/USD_Unicenter Service
 Desk Version 11.0_WebService?wsdl
@cd ..\..
::#################################################################
::# This next section compiles the Service Desk stub code
::# Once complete, you should recycle tomcat with the following
::# commands or by recycling Service Desk:
::#       pdm_tomcat_nxd -c STOP
::#       pdm_tomcat_nxd -c START
::#################################################################
@SET CP=".\classes;%CP%"
@SET STUBS_DIR=classes\com\ca\www\UnicenterServicePlus\ServiceDesk
@cd WEB-INF
%JAVAC_EXE% -classpath %CP% -deprecation -d classes %STUBS_DIR%\ArrayOfInt.java
%JAVAC_EXE% -classpath %CP% -deprecation -d classes %STUBS_DIR%\ArrayOfString.java
%JAVAC_EXE% -classpath %CP% -deprecation -d classes %STUBS_DIR%\ListResult.java
%JAVAC_EXE% -classpath %CP% -deprecation -d classes %STUBS_DIR%\USD_WebService.java
%JAVAC_EXE% -classpath %CP% -deprecation -d classes %STUBS_DIR%\USD_WebServiceLocator.java
%JAVAC_EXE% -classpath %CP% -deprecation -d classes %STUBS_DIR%\USD_WebServiceSoap.java
%JAVAC_EXE% -classpath %CP% -deprecation -d classes %STUBS_DIR%\USD_WebServiceSoapSoapBindingStub.java
@cd ..
```

After running the batch file from the command prompt, the stub classes are in place and compiled.

- Recycle Tomcat as follows:
  - pdm_tomcat_nxd -c STOP
  - pdm_tomcat_nxd -c START

loginServiceManaged Method for CXF

This article contains the following topics:

The following parameters apply to the loginServiceManaged method:

Parameter	Type	Description
policy	String	(Required) Identifies the policy, which must be must be in plain text.
encrypted_policy	String	(Required) Identifies the digital signature of the policy code, encrypted with the policyholder's private key. It is in BASE64 text format.

Description

Performs the user authentication by locating the policy through the plain text policy code, finding the policyholder's public key associated with the policy, decrypting the encrypted policy code, matching decrypted content with the policy code, and finally, opening a session with a back-end server. The returned session ID (SID) can be used for subsequent web services method invocations. Proxy contact specified in the policy is responsible for all subsequent web service activities initiated. All function group security and data partition are enforced for the proxy contact defined in the policy.

It is also important to note that the encrypted_policy parameter is in the BASE64 text format, and it is necessary to perform proper conversion from the binary format. The SID should be freed with logout(). A SID may expire if it is not used before a timeout elapses.

Returns

The following is returned:

Parameter	Type	Description
SID	String	Identifies the unique session ID (SID) to use for all other Web Services calls. It is in plain text format.

Implement loginServiceManaged in Java

The following shows how to generate Certificates and then use these generated Certificates to access the CA SDM web services.

In the following example, the login process completes using the CA SDM Certificate and then performs two common web services calls. The getBopsid() web services method call allows you to obtain a token that is linked to a specific user. This token can be used to login to the CA SDM web interface as the linked user without being prompted for a password. This allows seamless integration to be enabled between different applications.

> **WARNING**
> The generated BOPSID token expires after 30 seconds, so it must be used promptly.

Follow these steps:

1. Generate the stub classes with CXF Tool WSDL2Java. For more information, see the Generating Stub Classes with CXF Tool WSDL2Java section from the PKI_loginServiceManaged_JAVA_steps file. Find the file in the following location:
   ```
   $NX_ROOT/samples/sdk/cxfwebsvc/java/test1_pki
   ```
2. Start the CA SDM service.
3. Run pdm_pki -p DEFAULT.
   DEFAULT.p12 is created in the current directory. This policy will have the password equal to the policy name (in this case DEFAULT).

NOTE
This command will also add the Certificate's public key to the field pub_key field (public_key attribute) in the sapolicy table/object.

4. Log into CA SDM.
5. Select **SOAP Web Services Policy**, **Policies** on the **Administration** tab.
   The **SOAP Web Services Access Policy List** page opens.
6. Click **DEFAULT**.
   The **SOAP Web Services Access Policy Detail** page opens.
7. Complete the **Proxy Contact** field (in this example, ServiceDesk) and confirm that the DEFAULT policy record Has Key field displays "Yes."
8. Copy DEFAULT.p12 (from the directory where command pdm_pki is executed), the JSP file called *pkilogin.jsp* and the HTML file called *pkilogin.htm* (from the $NX_ROOT/samples/sdk/cxfwebsvc/java/test1_pki directory) to the following directory:
   ```
   $NX_ROOT/bopcfg/www/CATALINA_BASE/webapps/cxf
   ```
9. Open the HTML form (from the cxf directory). For example, http://localhost:8080/cxf/pkilogin.htm
   Complete the appropriate fields.

   NOTE
   The Directory field identifies the location of the Certificate file. Modify the path to the correct location.
10. Click Log me in!
    The results page opens.
11. Click the BOPSID URL.

    WARNING
    Click this immediately! The BOPSID has a limited life token of about 30 seconds.

    The format of a URL using a BOPSID is as follows:
    http://<server name>:CA Portal/CAisd/pdmweb.exe?BOPSID=<BOPSID value>

    NOTE
    In order to use the loginServiceManaged method for a Java client program running on AIX, you may need to replace a pair of security policy files within your JAVA_HOME. Go to http://www.ibm.com and search for "developerworks java technology security information AIX". In the "developerWorks : Java technology : Security" document, follow the link to "IBM SDK Policy files". Download the unrestricted policy files, local_policy.jar and US_export_policy.jar. Use these files to replace the original files in your JAVA_HOME/lib/security directory."

Using the Automated Tasks Editor

This article contains the following topics:

You can author, debug, test, and upload automated tasks in your support environment using the Automated Tasks Editor. You can create automated tasks by customizing default templates and components. Modifying these defaults requires minimal or no coding, but if any errors occur, the application displays the line of code reporting the error.

The editor uses web services to access automated tasks depending on provided credentials during CA SDM authentication. An Administrator can permit analysts to access the server with the Automated Task Editor, and allow analysts to modify and upload public or tenant-specific scripts. The editor provides dialogs to edit the basic automated task settings, security settings, and dependencies on libraries and static content items. The automated task framework implements three objects used by the automated task author: Task, Step, and Logger.

You can do the following with automated tasks

- Create and customize steps based on automated task step templates provided by CA Technologies.
- Customize an automated task by editing the step template properties in a property sheet, or by editing the code directly. The code editor supports syntax highlighting and automatic code completion functionality while typing.
- An Administrator can run reports that show whether automated tasks were successful or not. The definition of success varies considerably depending on what the automated task is trying to do. CA SDM provides the *Functions.SetExecutionState()* function call that enables the task to declare its execution state as being success, fail, or not setting it at all. The automated task execution state field filters the report.
- Send emails to end users or to an external system using the CA SDM mail daemon.

Localize automated tasks by declaring all strings to localize as automated task step properties. You can edit these step properties in the Automated Task Editor for each supported locale separately. The values of these properties are accessed from automated task step script code using Step.GetProperty() which returns the correct localized version of the specified property based on the localization ID of the customer running the automated task.

Edit an Automated Task

You can download automated tasks from the server and edit them in the Automated Task Editor. Any content that is newer in the version than the existing content on the server is imported into the database and made available to administrators of that tenant.

To edit an automated task

1.
2. Open the Automated Task Editor.
   The Support Automation Task Editor appears.
3.
4. On the toolbar, select the Upload Automated Task icon.
   The Open Automated Task from Server pane appears.
5.
6. Select the automated task that you want to download. **Note:** If you are a privileged user in a multi-tenancy environment, you can edit public or tenant-specific automated tasks.
7.
8. Click Open Task.
   The Support Automation Task Editor downloads the task to the client and opens it in the application.
   The download creates a text file on the computer of the task author that contains the dependent content.

Upload an Automated Task

You can upload automated tasks that you created in the application. When you select a task, all dependent content automatically uploads, such as libraries and static content.

To upload an automated task

1.
2. Open the Automated Task Editor.
   The Support Automation Task Editor appears.
3.
4. Select the automated task you want to upload.
5.
6. Select the classification where you want to upload the task.
   Note: If you are a privileged user in a multi-tenancy environment, select the appropriate tenant when uploading the automated task, or make the task public.
7.
8. On the toolbar, select the Upload Automated Task icon.

The Support Automation Task Editor uploads the task to the server.

How an Automated Task Runs

This article contains the following topics:

CA SDM provides a framework which is a block of script code responsible for the code that deals with presenting user interfaces and retrieving user input from user interface steps. The framework exists so that you can do complex things with automated tasks, interface-related, using simple code, rather than complex code that would otherwise be necessary. The framework includes two objects (Task and Step). You can use these objects to interact with the framework and control how automated tasks execute, including setting up and accessing the user interface.

Note: When running an automated task in the Automated Task Editor, the editor acts as both server and client.

The following process describes what happens when you execute an automated task:

1.
2. Use the framework to assemble the script code to execute together with the step code in the automated task definition, and any dependent library code.
3.
4. The server assembles and distributes the code to the end-user environments.
5.
6. Both the end user and the analyst receive the entire assembled automated task.
7.
8. The server initiates each step in turn at the relevant client.

How Analysts Receive Data

You can have an automated task gather data and store it in the Support Automation database associated to the specific task execution. The data can be picked up by a post session integration that can use the data or pass it to an external system. In such a case, typically you construct this data as an XML fragment.

Use Functions.SetAcquiredData() to store data (text), associated with a task execution. Support Automation allows one such text field of any length to be stored for each execution of an automated task. You can store data in one step and access it in a subsequent step as follows:

1.
2. The server sends the initial state of this data storage to the client with the instruction to execute a step.
3.
4. At the end of the step, the client sends the persisted data state back to the server so the server can send this data to whichever client is executing the next step.
5.
6. An analyst user interface step can gather input from the analyst and store it.
7.
8. The end-user user interface step can access the data that was supplied by the analyst.

Automated Task Elements

This article contains the following topics:

You create automated tasks by defining steps in an automated process. Automated tasks predefine routines with specific actions on the end-user computer without the need for the analyst or end user to do the process. Common routines include gathering telemetry information, diagnosing problems, and implementing resolutions.

An automated task consists of the following elements:

-
- **Task Settings**
  Specifies the title, description, dimensions of the step windows, HTML headers and footers, and a CSS style sheet applied to the end-user interface.
-
- **Dependencies**
  Specifies the libraries and static content items that the task depends on, such as images or CSS files.
-
- **Task Steps**
  Specifies the executed code for each step. You can author steps by selecting automated task step templates and modify them to fit your needs.
-
- **Security Settings**
  Specifies the appropriate security settings to run the automated task on an end-user computer, such as read/write permissions, registry privileges, and so on.
-
- **XSDF Files**
  Specifies automated task definitions saved to an XML schema in XSDF files (XML Script Definition Format). The XSDF file contains all automated task settings, dependencies, security settings, and step code information.
  If the automated task is dependent on a particular library, the full script code of that library is written to the XSDF file, not only the dependency information. Dependencies are also imported when XSDF files are loaded into the Automated Task Editor or imported into CA SDM, unless a later version of the same dependent library exists in the target environment. The same is true for dependent static content. Script libraries can also be saved on their own to XSDF files for distribution between systems.

The editor lets you define CSS styles for all automated task elements. You can control the appearance by using CSS Styles.

Example Default CSS Styles

The following CSS styles are defined by default.

NOTE
If you use a custom CSS file, you can override the default values.

```
#header {
                padding: 2px 8px;
                font-size: 16pt;
                font-weight: bold;
                border-bottom: 2px solid green;
}
#title {
                background-color: #eee;
                padding: 2px 8px;
                font-size: 12pt;
                font-weight: bold;
                border-bottom: 1px solid green;
}
#content {
                padding: 8px;
}
#buttons {
                background-color: #eee;
                border-top: 1px solid green;
```

```
                text-align: right;
                padding: 2px 8px;
}
#buttons input {
                font-family: Verdana;
                font-size: 10pt;
}
#footer {
                border-top: 2px solid green;
                padding: 2px 8px;
                font-size: 9pt;
}
```

The supplied HTML components declare a CSS Style for an HTML DIV that encloses the component. This declaration allows all elements of the components to be styled through CSS. The name of this style can usually be set through automated task step template properties for steps that use these HTML components.

Task Object Reference

The task object is directly referenced as Task [e.g., Task.End();]

Methods

Task	Description
End()	Ends the Task immediately.
GetNamedDataItem(key)	Returns the value previously stored with the specified key.
SetNamedDataItem(key, value)	Stores the value against the specified key. These values can be stored in one step and accessed in another using GetNamedDataItem(). The specified value can be JavaScript primitive type (number, boolean, string) or an object or an array of objects. In VBScript, only primitive types can be stored.
GetNextElementId()	Returns a unique identifier that can be used for HTML DOM objects.
GetPageBodyElementById(id)	Returns the HTML DOM object corresponding to the specified ID.
ReplaceDataItems(string)	Returns a copy of the specified string with all instances of stored data items in the format ${key} replaced by their stored values.
CloseIEWindow()	Closes the UI window if it is open.

Properties

Property	Description
Title	The title of the automated task.
HeaderHtml	The HTML placed in the header section of the UI window.
FooterHtml	The HTML placed in the footer section of the UI window.
Height	The height, in pixels, of the UI window.
Width	The width, in pixels, of the UI window.
NextStepIndex	The index (0-based) of the next step to be executed.

Step Object Reference

The step object is directly referenced as Step [e.g., Step.End();]

Methods

Step Object	Description
End()	Ends the step immediately. The next step will be invoked.
Controls.Add()	Adds a string or HTML component to the UI content. Strings are automatically converted to Label HTML components.
GetProperty(propertyKey)	Returns the value of the specified property key for the current localization. If no value exists for the current localization, the value of the property in the default localization will be returned.
SetProperty(localizationID, propertyKey, value)	Sets the value of the specified property to the specified value for the specified localization ID.
RegisterEventHandler(controlId, eventHandler, tag)	Registers the specified event handler as the function to be invoked when the IE window's raiseUIEvent(controlId) function is called. Any data passed as tag will be made available when the event handler is dispatched as Step.EventData.
IsUIStep()	Returns true if the current step is a UI step; false otherwise.

Properties

Property	Description
Title	The title of the step - displayed in the UI header section.
NextButtonLabel	The label for the "Next" button. Defaults to "Next".
NextButtonVisible	True to show the "Next" button. False to hide it.
PreviousButtonLabel	The label for the "Previous" button. Defaults to "Previous."
PreviousButtonVisible	True to show the "Previous" button. False to hide it.
FinishButtonLabel	The label for the "Finish" button. Defaults to "Finish."
FinishButtonVisible	True to show the "Finish" button. False to hide it.
EventData	When a UI event is raised (by calling raiseUIEvent() from HTML within the IE window) the event handler function previously registered with RegisterEvenHandler() is invoked. The data stored as the "tag" when the event handler was registered will be available as Step.EventData when the handler is dispatched.

Logger Object Reference

The Logger object is accessed as Trace.Framework.

Example:

Trace. Framework.Level = TraceLevels.Info

Trace.Framework.Error("An error has occurred")

Methods

Logger Object	Description
Fatal(msg)	Log the specified message if the current trace level is set at Fatal, Error, Warning, Info, Debug, or Verbose.
Error(msg)	Log the specified message if the current trace level is set at Error, Warning, Info, Debug, or Verbose.
Warning(msg)	Log the specified message if the current trace level is set at Warning, Info, Debug, or Verbose.
Info(msg)	Log the specified message if the current trace level is set at Info, Debug, or Verbose.
Debug(msg)	Log the specified message if the current trace level is set at Debug or Verbose.
Verbose(msg)	Log the specified message if the current trace level is set at Verbose.

Properties

Property	Description
Level	The current trace level. Set to one of the values in the TraceLevels enumeration (None, Fatal, Error, Warning, Info, Debug, or Verbose).

Global Functions

The following global functions are used in Support Automation procedures.

Function	Description
AnyArrayToVBArray(array)	Converts any array (VBScript or JavaScript) to a VBArray. Arrays in JavaScript are different objects to VBScript arrays so it is necessary to convert between them.
AnyArrayToJSArray(array)	Converts any array to JavaScript array.
CreateHashMap()	Returns an object that can be used as a hashmap for storing arbitrary data (key/value pairs). The returned object implements the following methods:
	Get(key) - Returns the value stored against the specified key.
	Set(key,value) - Stores the value for the specified key.
	GetAllKeys() - Returns an array of the keys.

Automated Task Step Templates

Automated task step templates are the script code for a given automated task step that can also define automated task step properties. You can customize the behavior of the automated task step template by changing the values of automated task step properties in the property sheet. You can accomplish more extensive customization by editing the code directly, but the intent is to minimize the amount of coding you do.

Steps are end-user action steps or user interface steps that show either to the end user or analyst. The automated task step templates are either end-user action, analyst interface, or end-user interface templates. CA Technologies provides automated task step templates, both action and user interface, to form the building blocks of tasks and enable you to create functional automated tasks with minimal coding.

For steps that define properties, you are prompted to provide values for these properties when the step is first created from the template. You can modify these values using the property sheet.

User Interface Steps

All user interface steps must implement an OnLoad function called by the framework before the step is displayed. The OnLoad function sets up the user interface. The step object exposes several properties and methods that you can use to set up the user interface.

You can control the visibility of the buttons on the step footer using Step.NextButtonVisible, Step.FinishButtonVisible, and Step.PreviousButtonVisible. When you set these properties to True, it makes the button appear. If a button is shown, a handler must be implemented for it. You can add the actual user interface controls to the page body using HTML components.

Example Text Input Box as an HTML Component

The step object has a controls array that is populated with a series of HTML components. These script objects render in HTML, and when the framework renders a user interface page, each HTML component in its controls array renders itself and adds the returned HTML fragment to the HTML body of the page.

HTML components let you write complex user interfaces without extensive HTML knowledge. Using an HTML component, you can create and configure commonly used controls with minimal coding.

Example: Text Input box as an HTML Component

An example of an HTML component is a text input box. The following example combines a label called PromptText and an HTML input box. If you want to prompt the customer or technician to supply a user name for example, the user interface page is constructed in an OnLoad function as follows:

```
Function OnLoad
 Dim myInputBox
 Set myInputBox = UI.MakeInputBox()
 myInputBox.PromptText = "Please enter your name:"
 Step.Controls.Add(myInputBox)
End Function
```

CA Technologies supplies samples, but you can also create custom HTML components. An HTML component is a script object that implements a method called GetHtml. Here, for example, is the implementation of that *MakeInputBox()* method call used in the previous example:

```
function InputBox() {
 // public properties
 this.PromptText = "";
 this.CssClass = "UIInputBox";
 this.InputMaxLength =50;
 this.InputSize = 50;

 // private member props
```

```
 var textboxId = Task.GetNextElementId();

 // privileged methods
 // (For example, publicly accessible with access to private props)
 this.GetHtml = function () {
  var inputHtml =
   "<DIV class = '" + this.CssClass + "'>\n";
  if(this.PromptText.length>0) {
   inputHtml +=
"<SPAN class ='PromptText'>" + this.PromptText + "</SPAN>\n" +
    "<BR>\n";
  }
  inputHtml +=
   "<INPUT type = text class = 'InputText' id = '"+textboxId+
   "' size = '"+this.InputSize+
"' maxlength='"+this.InputMaxLength+"'></INPUT>" +
   "<BR><BR>\n" +
   "</DIV>\n";
  return inputHtml;
 }

 this.GetUserInput = function() {
  var textBox = Task.GetPageBodyElementById(textboxId);
  if (textBox) {
   return textBox.value;
  }
  else {
   throw new Exception (Severity.Error,
"Could not read textbox value",
"Could not read textbox value");
  }
 }
}
return new InputBox();
```

This automated task library function defines a constructor for an object of type InputBox. The last line of the function constructs an instance of the InputBox class and returns it.

Object definitions differ in JavaScript and VBScript. In VBScript, the class keyword is used to declare a class. In JavaScript the function keyword declares the class, because in JavaScript functions are objects. Both languages let the class have public and private member variables and methods. A full description of how object syntax is handled in the two languages is beyond the scope of this chapter, but the internet can provide examples and tutorials on this subject.

In the example above, the code declares the InputBox class with some public and private properties and two public methods, GetHtml and GetUserInput.

The Framework calls GetHtml when rendering user interface steps. The InputBox control also exposes GetUserInput which you can use to access the value that the end user enters into the text box. Each HTML component can expose different methods for configuring it and accessing any user input values.

The HTML coding is abstracted from you by these HTML components so that creating user interface pages is simplified.

You can write your own HTML entirely for a particular automated task. In such a case, you have two choices: Write a new HTML component for the case you require, or construct the HTML code in a string variable and wrap it in a special HTML component called RawHTMLContainer as follows:

```
Function OnLoad
 Dim myRawHtmlContainer, myHtml
 myHtml = "<H1>Hello <B>World</B></H1>"
 Set myRawHtmlContainer = UI.MakeRawHtmlContainer(myHtml)
 Step.Controls.Add(myRawHtmlContainer)
End Function
```

Example End-User Input Handling

After the user interface page displays, in many cases it is necessary to handle end-user input when the user clicks one of the footer buttons, such as Next, Previous, or Finish. You implement handler functions to handle end-user input. You implement a function named *NextButtonHandler* to handle the clicking of the Next button. You can use the *PreviousButtonHandler* and *FinishButtonHandler* functions to handle the Previous and Finish buttons.

The handler performs the following actions:

-
- Retrieves input data from HTML components on the page
-
- Validates the end-user input
-
- If end-user input is invalid, display an error message and refresh the page
-
- If end-user input is valid, logs stores it for use in a later step
-
- Controls the sequence of step execution

Example: End-User Input Handling

HTML components that wrap input controls, such as the InputBox component, expose methods to access the values supplied by the end user. In this case, the *GeUserInput()* function is exposed.

You can control the next step that the framework presents as follows:

-
- Return True from the handler to present the default next step (currentStep+1 for NextButtonHandler, currentStep-1 for PreviousButtonHandler, end the automated task for FinishButtonHandler).
-
- Return False from the handler to refresh the current step.
-
- Set Step.NextStepIndex to the step index (1 based) to skip or repeat steps.

You can handle the Next button being clicked by supplying a NextButtonHandler. Because the NextButtonHandler has to access the HTML component that was created in the OnLoad, the declaration (Dim statement) of this variable is placed outside these functions. Thus, the entire step code would look as follows:

```
Dim myInputBox

Function OnLoad
 Set myInputBox = UI.MakeInputBox()
 myInputBox.PromptText = "Please enter your name:"
 Step.Controls.Add(myInputBox)
End Function
Function NextButtonHandler
 Dim userInputValue
```

```
 ' Retrieve the data supplied by the user
userInputValue = myInputBox.GetUserInput()

' Validate the data
If Len(userInputValue) = 0 Then
 ' Show error message
 Functions.ShowMessage "You must supply a name"

 ' Re-present current step
 NextButtonHandler=False
Else
 ' Log the data
 Functions.LogMessage "User name set to: " & userInputValue

 ' Store the data
 Task.SetNamedDataItem "UserName", userInputValue

' Proceed to next step
NextButtonHandler=True
End If
End Function
```

Script Library Management

This article contains the following topics:

You can use the same script functionality that gathers specific data or impacts certain changes on a remote machine in many various automated tasks. Support Automation provides the concept of script library to package such reusable code. You use script libraries to help you reuse and maintain code. These libraries contain functions written in JavaScript or VBScript that automated tasks can call. The automatic code completion feature of the Automated Task Editor provides a function description text and argument descriptions when you edit automated task step script code.

You can create your own libraries. The Automated Task Editor displays defined JavaScript and VBScript libraries on the left pane. Script libraries let you do the following:

-
- Create libraries and functions
-
- Provide descriptions of the Function arguments
-
- Export and Import from XSDF files or directly from the Support Automation server
-
- Define the function name, description, arguments, and script text

Return Objects from Library Functions

You can return an instance of an object from a library function, such as for library functions that manufacture HTML Components. Returning an object requires you to declare a class and instantiate an instance of that class. Class declarations are done differently in JavaScript and VBScript.

In JavaScript, you define classes by declaring a function that is actually the constructor for the class. Public properties and methods are added to the class using the *this* keyword. You declare private properties by using the *var* keyword. In JavaScript, you can declare a class inside a function definition.

In VBScript, class definitions are not permitted inside functions, however, you can provide the code that forms the implementation of a given function. You create class definitions using one of the following techniques:

-
- Constructing the class definition code in a string and executing it with *ExecuteGlobal*.
-
- Calling a second function and returning whatever it returns. End the function. Declare the class, then the second function.

Local File System/Registry Access

Many automated tasks require reading or modifying the file system or registry. The Windows Script Host (WSH) provides some scriptable COM components (such as FileSystemObject) to enable scripts to read or modify the file system or registry. Some antivirus programs implement script blocker utilities that interfere with those COM components, often flagging any script that tries to instantiate the relevant component as being malicious. Clearly, it is not desirable to have script blockers interfere with the activities of analysts who are trusted and trying to repair a problem. The CA SDM functions library contains functions for interacting with the file system and registry that enable much the same functionality that the Windows components expose, without being subject to the action of script blocker programs.

The standard WSH components are used when you write automated tasks for an internal corporate environment where the end-user computer state is known, and assumed that script blockers are not present. You can use the functionality in the CA SDM Functions COM component to implement any file system or registry activity for external environments (customers at home or on the public Internet) when the organization that is providing support does not control the end-user computer configuration.

Windows Management Instrumentation (WMI) Support

Automated tasks support Windows Management Instrumentation (WMI), a component of the Windows operating system that provides management information and control in an enterprise environment. By using industry standards, you can use WMI to query and set information about desktop systems, applications, networks, and other enterprise components.

WMI lets you access and modify the operating system. WMI is based upon SQL-style queries that allow an automated task to query information from the operating system and then, using objects, to manipulate that data. WMI is a powerful system and is often the preferred method for automated tasks to carry out their purposes.

Example: Script fragment that enumerates the currently configured Operating System Services.

```
strComputer = "."
Set objWMIService = GetObject("winmgmts:" _
    & "{impersonationLevel=impersonate}!\\" & strComputer & "\root\cimv2")

Set colRunningServices = objWMIService.ExecQuery("Select * from Win32_Service")

For Each objService in colRunningServices
    Wscript.Echo objService.DisplayName  & VbTab & objService.State
Next
```

Static Content Management

Automated tasks can present a user interface that requires you to use images or CSS files to appear correctly. This static content (images and CSS files) is associated with the automated task definition as the task is dependent on them. You can do the following:

-
- Import static content and associate it to automated tasks.
-
- Add, edit, delete, and upload to the Support Automation server.
-
- Select an image or text file to import and assign a globally unique identifier (GUID).
-
- Use the Automated Task Editor to assign static content items to automated tasks.

When you assign the static content item, you associate the automated task and it is saved in the XSDF with the task. The static content automatically deploys to the server when the dependent automated task deploys.

Static content is stored in the Support Automation database on the server. A servlet provides access to these images based on their ID. The Functions library call *Functions.GetStaticContentItemRef()* returns the URL to that servlet, which takes the ID as a parameter. The Automated Task Editor provides a toolbar button that brings up the Static Content Items dialog, which allows you to select the desired item. When you click OK, the *Functions.GetStaticContentItemRef()* call is automatically inserted into the code with the relevant ID.

NOTE
Status content has a limit of 50 KB per item.

Functions and WScript Usage

The framework provides two scriptable COM components that are accessible to all automated tasks running in CA SDM. These COM components are Functions and WScript.

Functions

The functions component is a collection of procedure calls that are used for the following:

-
- Accessing the local file system and or registry without triggering antivirus script blocker programs
-
- Accessing the Support Automation server functionality such as:
  -
  - Writing session log entries
  -
  - Storing automated task acquired data
  -
  - Setting automated task execution state
  -
  - Sending email
  -
  - Escalating between products
  -
  - Executing custom server routines

Functions COM Object Methods

This article contains the following topics:

The following Functions COM Object Methods are available for use in Support Automation automated tasks.

Function	Description
B64Encode (BSTR data, VARIANT *varResult)	Base64 encodes the specified data.
B64Decode (BSTR data, VARIANT *varResult)	Decodes the specified data.
CheckIfServerExists (BSTR URL, USHORT port, VARIANT *varResult)	Makes a request on the specified server to determine if it is accepting connections.
Eof (LONG fileno, VARIANT *varResult)	Determines if there is unread data in the specified text file
EscalateToLive (VARIANT *varResult)	Escalates the customers session from Self-Serve to Live Session
ExecuteRC (BSTR MethodName, BSTR ParamStr, int ParamCount, VARIANT *varResult)	Executes a RuleConduit call on the server allowing arbitrary server methods to be executed.
ExecuteWaitAndCaptureStream (BSTR strProg, int nStream, VARIANT *varResult)	Executes the specified command and waits for the process to exit, capturing the output from the process execution.
FileAddAttributes (BSTR szFilePath, LONG dwAttr, VARIANT *bvarResult)	Call to add attributes (Read-Only, Compressed, Archived, etc…) to a file without changing attributes already there.
FileCloseTextFile (LONG fileno)	Closes a text file previously opened with FileCreateTextFile or FileOpenTextFile.
FileCopyTo (BSTR szSourceFilePath, BSTR szDestFilePath, VARIANT_BOOL bcanOverwrite, VARIANT *bvarResult)	Call FileCopyTo to copy a file from one place to another.
FileCreateTextFile (BSTR szFilePath, BOOL overwrite, VARIANT *varResult)	Creates a new text file in the specified location returning a handle that can be used to interact with the file.
FileDelete (BSTR szFilePath, VARIANT *bvarResult)	Call to permanently delete a file.
FileExecute (BSTR szCommand, VARIANT *bvarResult)	Call to execute a file.
FileExecuteAndWait (BSTR szCommand, LONG dwMilliseconds, VARIANT *bvarResult)	Call to execute a file and wait for it to exit.
FileExists (BSTR szFilePath, VARIANT *bvarResult)	Checks whether a file exists or not.
FileGetAttributes (BSTR szFilePath, VARIANT *dwvarResult)	Call to retrieve the attributes (Read-only, archive, compressed, etc…) of a file.
FileGetDirectoryContents (BSTR pathName, BOOL showFiles, BOOL showDirs, VARIANT *varResult)	Returns the selected contents of the specified directory.
FileGetDrives (VARIANT *varResult)	Call to get a list of drives mounted.
FileGetDriveType (BSTR driveName, VARIANT *varResult)	Returns the type of the specified drive.
FileGetInfo (BSTR szFilePath, VARIANT *varResult)	Call to get the size and date information for a file.
FileGetINISectionContents (BSTR filename, BSTR sectionName, VARIANT *varResult)	Gets a pipe delimited list of the contents of the specified INI section.
FileGetINIValue (BSTR fileName, BSTR sectionName, BSTR keyName, BSTR defaultValue, VARIANT *varResult)	Gets the value of the specified item in the INI file.
FileGetInternetFile (BSTR szURL, BSTR szFilePath, VARIANT *bvarResult)	Call to download a file from the internet.
FileGetSize (BSTR szPath, VARIANT *varResult)	Gets the size of a file in bytes.
FileGetVersion (BSTR szFilePath, VARIANT *szvarResult)	Retrieves the version number of a file.
FileGetVolumeInfo (BSTR szVolumeRootPath, VARIANT *varResult)	Gets the information for a drive.
FileMakeDir (BSTR szDirPath, VARIANT *bvarResult)	Creates a directory.
FileMoveTo (BSTR szSourceFilePath, BSTR szDestFilePath, VARIANT_BOOL bcanOverwrite, VARIANT *bvarResult)	Moves a file from one place to another.

FileOpenTextFile (BSTR szFilePath, BSTR szMode, VARIANT *varResult)	Opens the specified text file returning a handle that can be used to interact with the file.
FileReadLine (LONG fileno, VARIANT *varResult)	Reads a line of text from the text file.
FileRecycle (BSTR szFilePath, VARIANT *bvarResult)	Recycles a file. Recycling places the file in the recycle bin and removes it from its original folder.
FileRenameTo (BSTR szSourceFilePath, BSTR szDestFilePath, VARIANT_BOOL bcanOverwrite, VARIANT *bvarResult)	Call FileRenameTo to change the name of a file.
FileSetAttributes (BSTR szFilePath, LONG dwAttr, VARIANT *bvarResult)	Sets the attributes (Read-only, archived, compressed, etc…) of a file.
FileSetINIData (BSTR filename, BSTR sectionName, BSTR keyName, BSTR data, VARIANT *varResult)	Sets the data for the specified key in the INI file.
FileWriteLine (LONG fileno, BSTR value, VARIANT *varResult)	Writes a line of text into the text file.
GetChildWindowCaption (int index, VARIANT *varResult)	Gets the caption for the specified child window.
GetChildWindowHandle (int index, VARIANT *varResult)	Gets the handle for the specified child window.
GetChildWindowProcessName (int index, VARIANT *varResult)	Gets the name of the process that owns the specified child window.
GetChildWindows (int hwnd, VARIANT *varResult)	Enumerates the windows that are children of the specified window storing the results for future querying, returning the count.
GetExecutionEnvironment (VARIANT *varResult)	Gets a string indicating where the script is being executed.
GetLocalization (VARIANT *varResult)	Gets the localization of the current user.
GetRootUrl (VARIANT *result)	Return the Server Root URL.
GetStaticContentUrl (VARIANT *result)	Return the Server Static Content URL.
GetStaticItemURL (BSTR itemGuid, VARIANT *varResult)	Retrieves the root URL of the given static content item.
GetSupportDeskOpen (VARIANT *varResult)	Gets the hours of operation status of the support desk.
GetWindowCaption (int index, VARIANT *varResult)	Gets the caption for the specified window.
GetWindowHandle (int index, VARIANT *varResult)	Gets the handle for the specified window.
GetWindowProcessName (int index, VARIANT *varResult)	Gets the name of the process that owns the specified window.
GetWindows (VARIANT *varResult)	Enumerates the system top level window handles, stores the results for future calls and returns the count.
HttpGet (BSTR szUrl, VARIANT *varResult)	Makes a request on the specified URL and returns the results.
HttpPost (BSTR szUrl, BSTR szBody)	Posts the specified request body to the specified URL.
InternetCheckConnection (VARIANT *varResult)	Determines if the customer machine appears to have a working internet connection.
LogMessage (BSTR szMessage)	Adds a message to the session log for the script execution.
OutlookGetPSTContents (BSTR szPSTPath, VARIANT *varResult)	Gets the contents of the specified Outlook PST file.
PromptUser (LONG Timeout, BOOL DefaultResult, BSTR Message, VARIANT *varResult)	Prompts the user with given message.
ReadLine (LONG fileno, VARIANT *varResult)	Reads a line of text from the text file. (deprecated, use FileReadLine instead).
RegCopyKey (BSTR szSourceRoot, BSTR szSourceKey, BSTR szDestRoot, BSTR szDestKey, VARIANT_BOOL bCanOverwrite, VARIANT *varResult)	Copies the entire contents of one key to another.
RegCopyValue (BSTR szSourceRoot, BSTR szSourceKey, BSTR szSourceValue, BSTR szDestRoot, BSTR szDestKey, BSTR szDestValue, VARIANT *bvarResult)	Copy the contents of one key/value to another. The user must have the ability to query keys for the root specified.

RegCreateKey (BSTR szRootPath, BSTR szKeyPath, VARIANT *bvarResult)	Creates a key entry in the user's registry.
RegDeleteKey (BSTR szRootPath, BSTR szKeyPath, VARIANT *bvarResult)	Removes a key from the registry. The user must have the ability to remove keys from the root specified.
RegDeleteValue (BSTR szRootPath, BSTR szKeyPath, BSTR szValueName, VARIANT *bvarResult)	Removes a key value from the registry. The user must have the ability to query keys for the root specified.
RegGetDWORD (BSTR szRootPath, BSTR szKeyPath, BSTR szValueName, VARIANT *dwvarResult)	Retrieves the a DWORD value from a specified key in the registry. The user must have the ability to query keys for the root specified.
RegGetString (BSTR szRootPath, BSTR szKeyPath, BSTR szValueName, VARIANT *szvarResult)	Retrieves a string value from a registry key. The user must have the ability to query keys for the root specified.
RegGetSubkeyByIndex (BSTR szRoot, BSTR szKey, LONG idxSubkey, VARIANT *varResult)	Retrieves the name of a subkey based on its index. The user must have the ability to query keys for the root specified.
RegGetSubkeysCount (BSTR szRoot, BSTR szKey, VARIANT *varResult)	Gets the number of keys underneath any given key. The user must have the ability to query keys for the root specified.
RegGetValueByIndex (BSTR szRoot, BSTR szKey, LONG idxValue, VARIANT *varResult)	Retrieves a key value given its index. The user must have the ability to query keys for the root specified.
RegGetValuesCount (BSTR szRoot, BSTR szKey, VARIANT *varResult)	Gets the number of values for a given key. The user must have the ability to query keys for the root specified.
RegGetValueType (BSTR szRootPath, BSTR szKeyPath, BSTR szValueName, VARIANT *dwvarResult)	Finds out the type of value for a given Registry key. The user must have the ability to query keys for the root specified.
RegKeyExists (BSTR szRootPath, BSTR szKeyPath, VARIANT *bvarResult)	Queries a key from the registry. The user must have the ability to query keys from the root specified.
RegMoveKey (BSTR szSourceRoot, BSTR szSourceKey, BSTR szDestRoot, BSTR szDestKey, VARIANT_BOOL bCanOverwrite, VARIANT *varResult)	Moves the entire contents of one key to another.
RegMoveValue (BSTR szSourceRoot, BSTR szSourceKey, BSTR szSourceValue, BSTR szDestRoot, BSTR szDestKey, BSTR szDestValue, VARIANT *bvarResult)	Moves the contents of one key value to another. The user must have the ability to query keys for the root specified and write to the destination.
RegRenameValue (BSTR szRootPath, BSTR szKeyPath, BSTR szOldValueName, BSTR szNewValueName, VARIANT *bvarResult)	Changes the name of a registry value. Note: The user must have the ability to query keys for the root specified.
RegRestoreKeyFromFile (BSTR szRootPath, BSTR szKeyPath, BSTR szFilePath, VARIANT *bvarResult)	Restores the entire contents of a key from a file into the registry. The user must have the ability to query and write keys for the root specified.
RegSaveKeyToFile (BSTR szRootPath, BSTR szKeyPath, BSTR szFileName, VARIANT *bvarResult)	Saves the entire contents of a key from the registry into a file. This file may then be restored using RegRestoreKeyFromFile. The user must have the ability to query keys from the root specified.
RegSetDWORD (BSTR szRootPath, BSTR szKeyPath, BSTR szValueName, LONG dwValue, VARIANT *bvarResult)	Sets a DWORD value in the registry. The user must have the ability to query keys for the root specified.
RegSetString (BSTR szRootPath, BSTR szKeyPath, BSTR szValueName, BSTR szValue, VARIANT *bvarResult)	Sets a string value for a key in the registry. The user must have the ability to query keys for the root specified.
RegValueExists (BSTR szRootPath, BSTR szKeyPath, BSTR szValueName, VARIANT *bvarResult)	Checks for the existence of a key/value pair in the registry. The user must have the ability to query keys for the root specified.
RunCommandAsImpersonatedUser (BSTR bapp, BSTR bargs, VARIANT_BOOL block, LONG timeout, VARIANT *varResult)	Runs an application with the same credentials the script has (deprecated). Returns 0 on success.
RunCommandAsUser (BSTR domain, BSTR user, BSTR password, BSTR app, BSTR args, VARIANT_BOOL block, LONG timout, VARIANT *varResult)	Runs an application as another user. Returns 0 on success.
SaveLog (BSTR fileName)	Saves the session log.

SendEmail (BSTR toAddress, BSTR emailSubject, BSTR emailBody)	Sends an email via CA Service Desk Manager mail daemon.
SetAcquiredData (BSTR data)	Save a chunk of script specific character data on the server for later use.
SetScriptSuccess (BOOL bSuccess, VARIANT *varResult)	Indicates whether the script executed successfully from the script's point of view.
ShowFileBrowseDlg (BSTR browseType, BSTR fileExt, VARIANT *varResult)	Shows the standard windows file browse dialog and returns the selected file.
ShowMessage (BSTR szMessage)	Displays the message to the user in a message box.
Sleep (LONG time)	Causes the execution thread to sleep.
TransferToQueue (BSTR queueID, BSTR queueName)	Sets the customer's current queue, this is the queue that the customer will enter if escalated to live.
UtilAttachDll (BSTR szDllPath, BSTR szHTTPLocation, VARIANT *varResult)	Loads a dll into the process space so that functions may be called from it.
UtilCloseTask (BSTR szProcName, VARIANT *bvarResult)	Closes a running task.
UtilDetachDll (LONG idDll, VARIANT *varResult)	Unloads a DLL from memory.
UtilEndProcess (BSTR szProcName, VARIANT *bvarResult)	Terminates the process with the specified name.
UtilExitWindows (LONG lAction, VARIANT_BOOL bForce, VARIANT_BOOL bRelogin)	Exits windows either by log off, restart, or shutdown.
UtilExternalMethodCall (LONG idDll, BSTR szFuncName, LONG lFuncType, BSTR szParams, VARIANT *varResult)	Executes a function from a loaded DLL.
UtilGetComputerName (VARIANT *varResult)	Gets the current computer name.
UtilGetEnvironmentVariable (BSTR szVarName, VARIANT *varResult)	Gets the value of a defined environment variable.
UtilGetOSVersion (VARIANT *varResult)	Gets the OS version.
UtilGetProcessByIndex (LONG idxProcess, VARIANT *varResult)	Retrieves process information for the specified process.
UtilGetProcesses (VARIANT *varResult)	Gets a list of all processes running on the customer machine.
UtilGetProcessesCount (VARIANT *varResult)	Gets the number of processes currently running on the customer machine.
UtilGetProcessMemory (LONG processID, VARIANT *varResult)	Gets the amount of memory used by the specified process.
UtilGetRunningTask (LONG idxWnd, VARIANT *varResult)	Gets information related to a task running on the customer machine. UtilGetRunningTasksCount should be called first to determine the number of tasks available.
UtilGetRunningTasksCount (VARIANT *varResult)	Gets a count of the tasks running on the customer machine.
UtilGetSystemDir (VARIANT *varResult)	Gets the windows system directory.
UtilGetSystemMetrics (LONG nIndex, VARIANT *varResult)	Gets the value of the specified system metric.
UtilGetUserName (VARIANT *varResult)	Gets the current user's logon name.
UtilGetWindowsDir (VARIANT *varResult)	Gets the directory for windows.
UtilGlobalMemoryStatus (VARIANT *varResult)	Gets information on global memory availability.

WScript

Many public domain script examples are available from sources such as Microsoft TechNet ScriptCenter, which are typically aimed at a system administrator audience, but are often useful for use in automated tasks. They often assume

that the existence of an object called WScript, which Microsoft provides as part of their command-line script execution engine, but not part of the Windows Script Host in itself.

CA SDM implements a simple version of the WScript object providing the Echo, Sleep, GetObject, and CreateObject methods, providing compatibility for the most commonly used WScript methods. WScript.Echo(), which usually writes to the command line, is implemented in CA SDM to write to the automated task log, and thus WScript.Echo() is synonymous with Functions.LogMessage().

WScript COM Object Methods

The following WScript COM Object Methods are available for use in Support Automation automated tasks.

Object Method	Description
CreateObject(objDescription)	Creates an instance of the specified COM object.
Echo(msg)	Logs the specified message. Equivalent to Functions.LogMessage(msg).
GetObject(objDescription)	Used to get an instance of an object from a string reference to it.
Quit()	Ends the executing task immediately.
Sleep(time)	Pauses execution for the specified number of milliseconds. Equivalent to Functions.Sleep(time).

EBR_DICTIONARY Table

This article contains the following topic:

-
- **SQL Name** -- EBR_DICTIONARY
-
- **Object** -- EBR_DICTIONARY

Field	Data Type	Reference	Remarks
id	INTEGER UNIQUE KEY		
WORD_ID	INTEGER		
WORD	STRING 50 NOT_NULL S_KEY		
WORD_TYPE	INTEGER		
WORD_TOTAL_COUNT	INTEGER		
DF	INTEGER		
WORD_IDF	INTEGER		
last_mod_dt	LOCAL_TIME		

EBR_DICTIONARY_ADM Table

-
- **SQL Name** -- EBR_DICTIONARY_ADM
-
- **Object** -- EBR_DICTIONARY_ADM

Field	Data Type	Reference	Remarks
id	INTEGER UNIQUE KEY		
WORD_ID	INTEGER		
WORD	STRING 50 NOT_NULL S_KEY		
WORD_TYPE	INTEGER		
WORD_TOTAL_COUNT	INTEGER		
DF	INTEGER		
WORD_IDF	INTEGER		
last_mod_dt	LOCAL_TIME		

EBR_FULLTEXT Table

This topic contains the following topics:

Program control table used by Knowledge Management.

-
- **SQL Name** -- EBR_FULLTEXT
-
- **Object** -- EBR_FULLTEXT

Field	Data Type	Reference	Remarks
DOC_TYPE	INTEGER		
ENTITY_ID	INTEGER		
FULL_WORD	STRING 50		
FULL_WORD_ REVERSE	STRING 50		
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
PERMISSION_INDEX_ID	INTEGER		
PRODUCT	STRING 50		
SHORT_WORD	STRING 50		
TABLE_ID	INTEGER		
WORD_COUNT	INTEGER		
WORD_COUNT_ PROBLEM	INTEGER		
WORD_COUNT_ RESOLUTION	INTEGER		
WORD_COUNT_ SUMMARY	INTEGER		
WORD_COUNT_ TITLE	INTEGER		
WORD_IDF	INTEGER		
WORD_ORDER	INTEGER		

WORD_TYPE	INTEGER		

EBR_FULLTEXT_ADM Table

Program control table used by Knowledge Management.

-
- **SQL Name** -- EBR_FULLTEXT_ADM
-
- **Object** -- EBR_FULLTEXT_ADM

Field	Data Type	Reference	Remarks
DOC_TYPE	INTEGER		
ENTITY_ID	INTEGER		
FULL_WORD	STRING 50		
FULL_WORD_ REVERSE	STRING 50		
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
PERMISSION_ INDEX_ID	INTEGER		
PRODUCT	STRING 50		
SHORT_WORD	STRING 50		
TABLE_ID	INTEGER		
WORD_COUNT	INTEGER		
WORD_COUNT_ PROBLEM	INTEGER		
WORD_COUNT_ RESOLUTION	INTEGER		
WORD_COUNT_ SUMMARY	INTEGER		
WORD_COUNT_ TITLE	INTEGER		
WORD_IDF	INTEGER		
WORD_ORDER	INTEGER		
WORD_TYPE	INTEGER		

EBR_FULLTEXT_SD Table

Program control table used by Knowledge Management.

-
- **SQL Name** -- EBR_FULLTEXT_SD
-
- **Object** -- EBR_FULLTEXT_SD

Field	Data Type	Reference	Remarks
ENTITY_ID	INTEGER		

FULL_WORD	INTEGER		
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
SHORT_WORD	STRING 50		
TABLE_ID	INTEGER		
WORD_COUNT	INTEGER		
WORD_COUNT_ PROBLEM	INTEGER		
WORD_COUNT_ RESOLUTION	INTEGER		
WORD_COUNT_ SUMMARY	INTEGER		
WORD_COUNT_ TITLE	INTEGER		
WORD_IDF	INTEGER		
WORD_ORDER	INTEGER		
WORD_TYPE	INTEGER		

EBR_FULLTEXT_SD_ADM Table

Program control table used by Knowledge Management.

-
- **SQL Name** -- EBR_FULLTEXT_SD_ADM
-
- **Object** -- EBR_FULLTEXT_SD_ADM

Field	Data Type	Reference	Remarks
ENTITY_ID	INTEGER		
FULL_WORD	INTEGER		
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
SHORT_WORD	STRING 50		
TABLE_ID	INTEGER		
WORD_COUNT	INTEGER		
WORD_COUNT_ PROBLEM	INTEGER		
WORD_COUNT_ RESOLUTION	INTEGER		
WORD_COUNT_ SUMMARY	INTEGER		
WORD_COUNT_TITLE	INTEGER		
WORD_IDF	INTEGER		
WORD_ORDER	INTEGER		
WORD_TYPE	INTEGER		

EBR_INDEX Table

This article contains the following topics:

Program control table used by Knowledge Management.

-
- **SQL Name** -- EBR_INDEX
-
- **Object** -- EBR_INDEX

Field	Data Type	Reference	Remarks
id	INTEGER UNIQUE KEY		
ENTITY_ID	INTEGER NOT_NULL S_KEY		
WORD_ID	INTEGER NOT_NULL S_KEY		
WORD_TYPE	INTEGER NOT_NULL S_KEY		
WORD_ORDER	INTEGER		
WORD_COUNT	INTEGER		
WORD_COUNT_ TITLE	INTEGER		
WORD_COUNT_ SUMMARY	INTEGER		
WORD_COUNT_ PROBLEM	INTEGER		
WORD_COUNT_ RESOLUTION	INTEGER		

EBR_INDEX_ADM Table

Program control table used by Knowledge Management.

-
- **SQL Name** -- EBR_INDEX_ADM
-
- **Object** -- EBR_INDEX_ADM

Field	Data Type	Reference	Remarks
id	INTEGER UNIQUE KEY		
ENTITY_ID	INTEGER NOT_NULL S_KEY		
WORD_ID	INTEGER NOT_NULL S_KEY		
WORD_TYPE	INTEGER NOT_NULL S_KEY		
WORD_ORDER	INTEGER		

WORD_COUNT	INTEGER		
WORD_COUNT_ TITLE	INTEGER		
WORD_COUNT_ SUMMARY	INTEGER		
WORD_COUNT_ PROBLEM	INTEGER		
WORD_COUNT_ RESOLUTION	INTEGER		

EBR_INDEXING_QUEUE Table

Program control table used by Knowledge Management.

-
- **SQL Name** -- EBR_INDEXING_QUEUE
-
- **Object** -- EBR_INDEXING_QUEUE

Field	Data Type	Reference	Remarks
ACTION	INTEGER		
ACTION_DATE	DATE		
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
INDEXED	INTEGER		
OBJ_PERSID	STRING 30		Persistent ID (SystemObjectName:id)
PRIORITY	INTEGER		
TEXT	STRING 32768		
TEXT	STRING 32768		

EBR_SYNONYMS Table

This article contains the following topics:

Program control table used by Knowledge Management.

-
- **SQL Name** -- EBR_SYNONYMS
-
- **Object** -- EBR_SYNONYMS

Field	Data Type	Reference	Remarks
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
KEYWORD1	STRING 50		
KEYWORD2	STRING 50		
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.

EBR_SYNONYMS_ADM Table

Program control table used by Knowledge Management.

-
- **SQL Name** -- EBR_SYNONYMS_ADM
-
- **Object** -- EBR_SYNONYMS_ADM

Field	Data Type	Reference	Remarks
ID	INTEGER NOT_NULL KEY		Unique (to the table) Numeric ID
KEYWORD1	STRING 50		
KEYWORD2	STRING 50		
LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.

bop_sinfo--Display System Information

The bop_sinfo utility displays information about a single Majic-defined object. You can run this utility on the objects listed under Technical Reference.

Syntax

This command has the following format:

```
bop_sinfo [-s server] [-p] [-l] [-d] [f] [-q] [-t] [-m] [-a] object [-h]
```

- **-s server**
  Specifies the server to query.
- **-p**
  Displays the producer information.
- **-l**
  Displays the domset list.
- **-d**
  Displays database object information, including the name and type of all attributes.
- **-f**
  Displays factory information, including the rel_attr and common_name.
- **-q**
  Displays the schema name of the associated table.
- **-t**
  Displays triggers on the object.
- **-m**
  Displays methods used by the object.
- **-a**
  Displays attribute details.
- **object**
  The name of the object to query
- **-h**
  Displays help on the utility.

Example: Display System Information for the dmn object

```
bop_sinfo -d dmn
Factory dmn
 Attributes
   id            INTEGER
   producer_id    LOCAL STRING(20)
   persistent_id    STRING(30)
   sym            STRING(60) REQUIRED
   delete_flag    SREL -> actbool.enum REQUIRED
   desc        STRING(40)
   tables        BREL <- dcon.dom_id {dom_id = ?}
   last_mod        DATE
   last_mod_by    SREL -> cnt.id
   audit_userid    LOCAL SREL -> cnt.id
```

ES_CONSTANTS Object

This article contains the following topics:

The object details are as follows:

1. Associated Table: ES_CONSTANTS
2. Factories: default
3. REL_ATTR: id
4. Common Name: NAME
5. Function Group:
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
Comments	COMMENTS	STRING		
id	ID	INTEGER		REQUIRED KEY
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
Name	NAME	STRING		
Property ID	PROPERTYID	INTEGER		
Property Value	PROPVALUE	INTEGER		

ES_NODES Object

The object details are as follows:

1. Associated Table: ES_NODES
2. Factories: default
3. REL_ATTR: id
4. Common Name: NODE_SHORT_DESC
5. Function Group:
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
DISPLAYED_TEXT	DISPLAYED_TEXT	STRING		

id	ID	INTEGER		REQUIRED KEY
last_mod_dt	LAST_MOD_DT	LOCAL_TIME		
LINK_ID	LINK_ID	INTEGER	ES_NODES id	
NODE_ID	NODE_ID	INTEGER		
NODE_SHORT_ DESC	NODE_SHORT_ DESC	STRING		
NODE_TYPE	NODE_TYPE	INTEGER		
PARENT_NODE_ID	PARENT_NODE_ID	INTEGER		
QUERY_RESP_ NUMBER	QUERY_RESP_ NUMBER	INTEGER		
QUERY_RESP_ TYPE	QUERY_RESP_ TYPE	STRING		
RESPLINKID1	RESPLINKID1	INTEGER		
RESPLINKID2	RESPLINKID2	INTEGER		
RESPLINKID3	RESPLINKID3	INTEGER		
RESPLINKID4	RESPLINKID4	INTEGER		
RESPLINKID5	RESPLINKID5	INTEGER		
RESPLINKID6	RESPLINKID6	INTEGER		
RESPLINKID7	RESPLINKID7	INTEGER		
RESPONSE1	RESPONSE1	STRING		
RESPONSE2	RESPONSE2	STRING		
RESPONSE3	RESPONSE3	STRING		
RESPONSE4	RESPONSE4	STRING		
RESPONSE5	RESPONSE5	STRING		
RESPONSE6	RESPONSE6	STRING		
RESPONSE7	RESPONSE7	STRING		
ROOT_ID	ROOT_ID	INTEGER	ES_NODES id	
TREE_ID	TREE_ID	INTEGER	SKELETONS id	

ES_RESPONSES Object

The object details are as follows:

1. Associated Table: ES_RESPONSES
2. Factories: default
3. REL_ATTR: id
4. Common Name: RESPONSE_LINK_TEXT
5. Function Group:
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	ID	INTEGER		REQUIRED KEY
last_mod_dt	LAST_MOD_ DT	LOCAL_TIME		

PARENT_ NODE_ID	PARENT_ NODE_ID	INTEGER	ES_NODES id	
RESPONSE_LINK_ID	RESPONSE_ LINK_ID	INTEGER	ES_NODES id	
RESPONSE_ LINK_ORDER	RESPONSE_ LINK_ORDER	INTEGER		
RESPONSE_ LINK_TEXT	RESPONSE_ LINK_TEXT	STRING		

ES_SESSIONS Object

The object details are as follows:

1. Associated Table: ES_SESSIONS
2. Factories: default
3. REL_ATTR: id
4. Common Name: EXTERNAL_ID
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
COMMENT_TEXT	COMMENT_TEXT	STRING		
EVALUATION	EVALUATION	INTEGER		
EXTERNAL_ID	EXTERNAL_ID	STRING		
id	ID	INTEGER		REQUIRED KEY
last_mod_dt	LAST_MOD_DT	LOCAL_TIME		
PATH_IDS	PATH_IDS	STRING		
PATH_QAS	PATH_QAS	STRING		
SESSION_ID	SESSION_ID	INTEGER		
TREE_ID	TREE_ID	INTEGER	ES_NODES id	

BSVC--func_access Object

This article contains the following topics:

The object details are as follows:

1. Associated Table: usp_functional_access
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: security
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
code	code	STRING 30		REQUIRED UNIQUE
type	type	SREL	func_access_type.id	REQUIRED

sym	sym	STRING 60		REQUIRED
description	description	STRING 1000		
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	UUID	cnt.id	

BSVC--func_access_level Object

The object details are as follows:

1. Associated Table: usp_functional_access_level
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: security
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
access_level	access_level	INTEGER		REQUIRED
type	type	SREL	func_access_type.id	REQUIRED
sym	sym	STRING 60		REQUIRED
description	description	STRING 1000		
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	UUID	cnt.id	

BSVC--func_access_role Object

The object details are as follows:

1. Associated Table: usp_functional_access_role
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: security
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
access_level	access_level	SREL	func_access_level.id	
func_access	func_access	SREL	func_access.code	
role	role	SREL	role.id	
description	description	STRING 1000		
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	UUID	cnt.id	

func_access_type Object

The object details are as follows:

1. Associated Table: usp_functional_access_type
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: security
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
sym	sym	STRING 60		
default_access	default_access	SREL	func_access_level.id	
description	description	STRING 1000		
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	UUID	cnt.id	

Table and Object Cross-References

This article contains the following topics:

This chapter provides several tables that allow you to easily cross-reference table names, SQL names, and object names. The chapter “Data Element Dictionary” lists a complete definition of the tables in the schema. The chapter “Objects and Attributes” lists the object definitions.

Cross-References between Table Name, SQL Name, and Object Name

This table provides a cross-reference of each table in the database schema to its corresponding SQL and object names:

Table	SQL Name (AKA)	Object
Access_Levels	acc_lvls	acc_lvls
Access_Type_v2	acctyp_v2	acctyp
Act_Log	act_log	alg
Act_Type	act_type	aty
Act_Type_Assoc	atyp_asc	act_type_assoc
Active_Boolean_Table	actbool	actbool
Active_Reverse_Boolean_Table	actrbool	actrbool
admin_tree	admin_tree	ADMIN_TREE
Am_Asset_Map	am_map	am_asset_map
Animator	anima	ANI
Archive_Purge_History	arcpur_hist	arcpur_hist
Archive_Purge_Rule	arcpur_rule	arcpur_rule
Asset_Assignment	hier	hier
Atomic_Condition	atomic_cond	atomic_cond

Attached_Events	att_evt	atev
Attached_SLA	attached_sla	attached_sla
Attachment	attmnt	attmnt
attmnt_folder	attmnt_folder	attmnt_folder
Attribute_Name	atn	
Audit_Log	audit_log	audlog
Behavior_Template	bhvtpl	bhvtpl
Boolean_Table	bool_tab	bool
Bop_Workshift	bpwshft	wrkshft
BU_TRANS	BU_TRANS	BU_TRANS
Business_Management	busmgt	bmhier
Business_Management_Class	buscls	bmcls
Business_Management_Repository	busrep	bmrep
Business_Management_Status	busstat	bms
ca_asset_type	ca_asset_type	
ca_company	ca_company	ca_cmpny
ca_company_type	ca_company_type	vpt
ca_contact	ca_contact	cnt
ca_contact_type	ca_contact_type	ctp
ca_country	ca_country	country
ca_job_function	ca_job_function	job_func
ca_job_title	ca_job_title	position
ca_location	ca_location	loc
ca_model_def	ca_model_def	mfrmod
ca_organization	ca_organization	org
ca_owned_resource	ca_owned_resource	nr
ca_resource_class	ca_resource_class	grc
ca_resource_cost_center	ca_resource_cost_center	cost_cntr
ca_resource_department	ca_resource_department	dept
ca_resource_family	ca_resource_family	nrf
ca_resource_gl_code	ca_resource_gl_code	gl_code
ca_resource_operating_system	ca_resource_operating_system	opsys
ca_resource_status	ca_resource_status	rss
ca_schema_info	ca_schema_info	
ca_site	ca_site	site
ca_state_province	ca_state_province	state
ca_tenant	ca_tenant	tenant
ca_tenant_group	ca_tenant_group	tenant_group

ca_tenant_group_member	ca_tenant_group_member	tenant_group_member
Call_Req	call_req	cr
Call_Req_Type	crt	crt
Call_Solution	crsol	crsol
Change_Act_Log	chgalg	chgalg
Change_Category	chgcat	chgcat
Change_Request	chg	chg
Change_Status	chgstat	chgstat
Chg_Template	chg_template	chg_tpl
CI_ACTIONS	CI_ACTIONS	CI_ACTIONS
CI_ACTIONS_ALTERNATE	CI_ACTIONS_ALTERNATE	CI_ACTIONS_ALTERNATE
CI_BOOKMARKS	CI_BOOKMARKS	CI_BOOKMARKS
CI_DOC_LINKS	CI_DOC_LINKS	CI_DOC_LINKS
CI_DOC_TEMPLATES	CI_DOC_TEMPLATES	CI_DOC_TEMPLATES
CI_DOC_TYPES	CI_DOC_TYPES	CI_DOC_TYPES
CI_PRIORITIES	CI_PRIORITIES	CI_PRIORITIES
CI_STATUSES	CI_STATUSES	CI_STATUSES
CI_WF_TEMPLATES	CI_WF_TEMPLATES	CI_WF_TEMPLATES
Column_Name	cn	
Contact_Method	ct_mth	cmth
Controlled_Table	ctab	ctab
Cr_Call_Timers	crctmr	ctimer
cr_heat_mapping	cr_heat_mapping	cr_heat_mapping
Cr_Status	cr_stat	crs
Cr_Stored_Queries	crsq	crsq
Cr_Template	cr_template	cr_tpl
D_PAINTER	D_PAINTER	
Delegation_Server	dlgtsrv	dlgsrvr
Document_Repository	doc_rep	doc_rep
Domain	dmn	dmn
Domain_Constraint	dcon	dcon
Domain_Constraint_Type	dcon_typ	dcon_typ
EBR_ACRONYMS	EBR_ACRONYMS	EBR_ACRONYMS
EBR_FULLTEXT	EBR_FULLTEXT	EBR_FULLTEXT
EBR_FULLTEXT_ADM	EBR_FULLTEXT_ADM	EBR_FULLTEXT_ADM
EBR_FULLTEXT_SD	EBR_FULLTEXT_SD	EBR_FULLTEXT_SD
EBR_FULLTEXT_SD_ADM	EBR_FULLTEXT_SD_ADM	EBR_FULLTEXT_SD_ADM
EBR_INDEXING_QUEUE	EBR_INDEXING_QUEUE	EBR_INDEXING_QUEUE

EBR_KEYWORDS	EBR_KEYWORDS	EBR_KEYWORDS
EBR_LOG	EBR_LOG	EBR_LOG
EBR_METRICS	EBR_METRICS	EBR_METRICS
EBR_NOISE_WORDS	EBR_NOISE_WORDS	EBR_NOISE_WORDS
EBR_PATTERNS	EBR_PATTERNS	EBR_PATTERNS
EBR_PREFIXES	EBR_PREFIXES	EBR_PREFIXES
EBR_PROPERTIES	EBR_PROPERTIES	EBR_PROPERTIES
EBR_SUBSTITS	EBR_SUBSTITS	EBR_SUBSTITS
EBR_SUFFIXES	EBR_SUFFIXES	EBR_SUFFIXES
EBR_SYNONYMS	EBR_SYNONYMS	EBR_SYNONYMS
EBR_SYNONYMS_ADM	EBR_SYNONYMS_ADM	EBR_SYNONYMS_ADM
ES_CONSTANTS	ES_CONSTANTS	ES_CONSTANTS
ES_NODES	ES_NODES	ES_NODES
ES_RESPONSES	ES_RESPONSES	ES_RESPONSES
ES_SESSIONS	ES_SESSIONS	ES_SESSIONS
Event_Delay	evt_dly	evtdly
Event_Delay_Type	evtdlytp	evtdlytp
event_log	event_log	event_log
event_type	event_type	event_type
Events	evt	evt
ext_appl	ext_appl	
External_Entity_Map	xent_map	ext_entity_map
Form_Group	frmgrp	fmgrp
Global_Change_Extension	g_chg_ext	g_chg_ext
Global_Change_Queue	g_chg_queue	g_chg_queue
Global_Contact	g_contact	g_cnt
Global_Issue_Extension	g_iss_ext	g_iss_ext
Global_Issue_Queue	g_iss_queue	g_iss_queue
Global_Location	g_loc	g_loc
Global_Organization	g_org	g_org
Global_Product	g_product	g_prod
Global_Queue_Names	g_queue_names	g_qname
Global_Request_Extension	g_req_ext	g_cr_ext
Global_Request_Queue	g_req_queue	g_cr_queue
Global_Servers	g_srvr	g_srvrs
Global_Table_Map	g_tbl_map	g_tblmap
Global_Table_Rule	g_tbl_rule	g_tblrule
Group_Member	grpmem	grpmem

Impact	impact	imp
INDEX_DOC_LINKS	INDEX_DOC_LINKS	INDEX_DOC_LINKS
Interface	interface	intfc
Iss_Template	iss_template	iss_tpl
Issue	issue	iss
Issue_Act_Log	issalg	issalg
Issue_Category	isscat	isscat
Issue_Property	issprp	iss_prp
Issue_Status	issstat	issstat
Issue_Workflow_Task	isswf	iss_wf
KD_ATTMNT	KD_ATTMNT	KD_ATTMNT
kdlinks	kdlinks	kdlinks
Key_Control	kc	
Knowledge_Keywords	km_kword	kwrd
KT_REPORT_CARD	KT_REPORT_CARD	KT_REPORT_CARD
LONG_TEXTS	LONG_TEXTS	LONG_TEXTS
Managed_Survey	managed_survey	mgs
Mgs_Act_Log	mgsalg	mgsalg
Mgs_Status	mgsstat	mgsstat
Note_Board	cnote	cnote
NOTIFICATION	NOTIFICATION	NOTIFICATION
Notification_Urgency	noturg	noturg
Notify_Log_Header	not_log	lr
Notify_Object_Attr	ntfl	ntfl
NR_Comment	nr_com	nr_com
Object_Promotion	object_promotion	object_promotion
O_COMMENTS	O_COMMENTS	O_COMMENTS
O_EVENTS	O_EVENTS	O_EVENTS
O_INDEXES	O_INDEXES	KCAT
Options	options	options
Promo_Hist	promo_hist	promo_hist
P_GROUPS	P_GROUPS	P_GROUPS
Pcat_Loc	pcat_loc	pcat_loc
Person_Contacting	perscon	perscnt
Priority	pri	pri
Prob_Category	prob_ctg	pcat
Product	product	prod
Property	prp	prp

Property_Template	prptpl	prptpl
Queued_Notify	not_que	notque
Quick_Template_Types	quick_tpl_types	quick_tpl_types
Remote_Ref	rem_ref	rrf
Reporting_Method	repmeth	rptmeth
Req_Property	cr_prp	cr_prp
Req_Property_Template	cr_prptpl	cr_prptpl
Response	response	response
Reverse_Boolean_Table	rbooltab	rev_bool
Rootcause	rootcause	rc
Rpt_Meth	rptmth	rptm
SA_Policy	sapolicy	sapolicy
SA_Prob_Type	saprobtyp	saprobtyp
Sequence_Control	seqctl	seq
Server_Aliases	srvr_aliases	srvr_aliases
Server_Zones	srvr_zones	srvr_zones
Service_Contract	svc_contract	svc_contract
Service_Desc	srv_desc	sdsc
session_log	session_log	session_log
session_type	session_type	session_type
Severity	sevrty	sev
SHOW_OBJ	SHOW_OBJ	SHOW_OBJ
SKELETONS	SKELETONS	KD
SLA_Contract_Map	sdsc_map	sdsc_map
SLA_Template	slatpl	slatpl
Spell_Macro	splmac	macro
Spell_Macro_Type	splmactp	macro_type
SQL_Script	sql_tab	
Survey	survey	survey
Survey_Answer	survey_answer	svy_ans
Survey_Answer_Template	survey_atpl	svy_atpl
Survey_Question	survey_question	svy_ques
Survey_Question_Template	survey_qtpl	svy_qtpl
Survey_Stats	survey_statistics	svystat
Survey_Template	survey_tpl	svy_tpl
Survey_Tracking	survey_tracking	svytrk
Table_Name	tn	
target_tgttpls_srvtypes	target_tgttpls_srvtypes	tgt_tgttpls_srvtypes

target_time	target_time	tgt_time
target_time_tpl	target_time_tpl	tgt_time_tpl
Task_Status	tskstat	tskstat
Task_Type	tskty	tskty
Timespan	tspan	tspan
Timezone	tz	tz
Transition_Points	nottrn	
True_False_Table	True_False_Table	true_false
Type_Of_Contact	toc	typecnt
Urgency	urgncy	urg
User_Query	usq	usq
usp_caextwf_instances	usp_caextwf_instances	caextwf_inst
usp_caextwf_start_forms	usp_caextwf_start_forms	caextwf_sfrm
usp_ci_window	usp_ci_window	ci_window
usp_contact	usp_contact	cnt
usp_functional_access	usp_functional_access	func_access
usp_functional_access_level	usp_functional_access_level	func_access_level
usp_functional_access_role	usp_functional_access_role	func_access_role
usp_functional_access_type	usp_functional_access_type	func_access_type
usp_kpi	usp_kpi	kc
usp_kpi_data	usp_kpi_data	kcd
usp_lrel_asset_chgnr	usp_lrel_asset_chgnr	lrel_asset_chgnr
usp_lrel_asset_issnr	usp_lrel_asset_issnr	lrel_asset_issnr
usp_lrel_att_cntlist_macro_ntf	usp_lrel_att_cntlist_macro_ntf	lrel_att_cntlist_macro_ntf
usp_lrel_att_ctplist_macro_ntf	usp_lrel_att_ctplist_macro_ntf	lrel_att_ctplist_macro_ntf
usp_lrel_att_ntflist_macro_ntf	usp_lrel_att_ntflist_macro_ntf	lrel_att_ntflist_macro_ntf
usp_lrel_attachments_changes	usp_lrel_attachments_changes	lrel_attachments_changes
usp_lrel_attachments_issues	usp_lrel_attachments_issues	lrel_attachments_issues
usp_lrel_attachments_requests	usp_lrel_attachments_requests	lrel_attachments_requests
usp_lrel_aty_events	usp_lrel_aty_events	lrel_aty_events
usp_lrel_bm_reps_assets	usp_lrel_bm_reps_assets	lrel_bm_reps_assets
usp_lrel_bm_reps_bmhiers	usp_lrel_bm_reps_bmhiers	lrel_bm_reps_bmhiers
usp_lrel_cenv_cntref	usp_lrel_cenv_cntref	lrel_cenv_cntref
usp_lrel_false_action_act_f	usp_lrel_false_action_act_f	lrel_false_action_act_f
usp_lrel_dist_cntlist_mgs_ntf	usp_lrel_dist_cntlist_mgs_ntf	lrel_dist_cntlist_mgs_ntf
usp_lrel_dist_ctplist_mgs_ntf	usp_lrel_dist_ctplist_mgs_ntf	lrel_dist_ctplist_mgs_ntf
usp_lrel_dist_ntflist_mgs_ntf	usp_lrel_dist_ntflist_mgs_ntf	lrel_dist_ntflist_mgs_ntf
usp_lrel_false_bhv_false	usp_lrel_false_bhv_false	lrel_false_bhv_false

usp_lrel_kwrds_crsolref	usp_lrel_kwrds_crsolref	lrel_kwrds_crsolref
usp_lrel_notify_list_cntchgntf	usp_lrel_notify_list_cntchgntf	lrel_notify_list_cntchgntf
usp_lrel_notify_list_cntissntf	usp_lrel_notify_list_cntissntf	lrel_notify_list_cntissntf
usp_lrel_notify_list_cntntf	usp_lrel_notify_list_cntntf	lrel_notify_list_cntntf
usp_lrel_ntfr_cntlist_att_ntfrlist	usp_lrel_ntfr_cntl_att_ntfrl	lrel_ntfr_cntlist_att_ntfrlist
usp_lrel_ntfr_ctplist_att_ntfrlist	usp_lrel_ntfr_ctplist_att_ntfrl	lrel_ntfr_ctplist_att_ntfrlist
usp_lrel_ntfr_macrolist_att_ntfrl	usp_lrel_ntfr_macrolist_att_ntfrl	lrel_ntfr_macrolist_att_ntfrl
usp_lrel_ntfr_ntflist_att_ntfrlist	usp_lrel_ntfr_ntflist_att_ntfrl	lrel_ntfr_ntflist_att_ntfrlist
usp_lrel_oenv_orgref	usp_lrel_oenv_orgref	lrel_oenv_orgref
usp_lrel_status_codes_tsktypes	usp_lrel_status_codes_tsktypes	lrel_status_codes_tsktypes
usp_lrel_svc_grps_svc_chgcat	usp_lrel_svc_grps_svc_chgcat	lrel_svc_grps_svc_chgcat
usp_lrel_svc_grps_svc_isscat	usp_lrel_svc_grps_svc_isscat	lrel_svc_grps_svc_isscat
usp_lrel_svc_grps_svc_pcat	usp_lrel_svc_grps_svc_pcat	lrel_svc_grps_svc_pcat
usp_lrel_svc_grps_svc_wftpl	usp_lrel_svc_grps_svc_wftpl	lrel_svc_grps_svc_wftpl
usp_lrel_svc_locs_svc_chgcat	usp_lrel_svc_locs_svc_chgcat	lrel_svc_locs_svc_chgcat
usp_lrel_svc_locs_svc_groups	usp_lrel_svc_locs_svc_groups	lrel_svc_locs_svc_groups
usp_lrel_svc_locs_svc_isscat	usp_lrel_svc_locs_svc_isscat	lrel_svc_locs_svc_isscat
usp_lrel_svc_locs_svc_pcat	usp_lrel_svc_locs_svc_pcat	lrel_svc_locs_svc_pcat
usp_lrel_svc_schedules_chgcat_svc	usp_lrel_svc_sch_chgcat_svc	lrel_svc_schedules_chgcat_svc
usp_lrel_svc_schedules_isscat_svc	usp_lrel_svc_sch_isscat_svc	lrel_svc_schedules_isscat_svc
usp_lrel_svc_schedules_pcat_svc	usp_lrel_svc_sch_pcat_svc	lrel_svc_schedules_pcat_svc
usp_lrel_true_action_act_t	usp_lrel_true_action_act_t	lrel_true_action_act_t
usp_lrel_true_bhv_true	usp_lrel_true_bhv_true	lrel_true_bhv_true
usp_kpi_ticket_data	usp_kpi_ticket_data	ktd
usp_organization	usp_organization	org
usp_owned_resource	usp_owned_resource	nr
usp_pri_cal	usp_pri_cal	
USP_PREFERENCES	USP_PREFERENCES	USP_PREFERENCES
USP_PROPERTIES	USP_PROPERTIES	USP_PROPERTIES
Workflow_Task	wf	wf
Workflow_Task_Template	wftpl	wftpl
wspcol	wspcol	wspcol
wsptbl	wsptbl	wsptbl
Request_Workflow_Task	crwf	cr_wf

CA SDM Object List Table for Multi-Tenancy

The following table lists all CA SDM objects and indicates whether each object has a tenant attribute, and if so, if it is optional or required.

NOTE
The Rel_attr must be unique, regardless of tenant.

Object	Rel_Attr	Description	Tenancy
acc_lvls	enum	Access Level	No
acctyp	id	Access Type	No
acctyp_role	id	Access Type Role	No
act_type_assoc	code	Activity Type Association	Optional
actbool	enum	Active Boolean	No
actlog_prod_list	sym	actlog_prod_list	No
actrbool	enum	Active Reverse Boolean	No
ADMIN_TREE	id	Administration Tree	No
agt	id	Analyst contacts	Required
alg	persistent_id	Request Activity Log	Required
all_fmgrp	id	all_fmgrp	No
all_lr	persistent_id	all_lr	Required
am_asset_map	persistent_id	AM Asset Map	No
ANI	persistent_id	Animator	Optional
api	persistent_id	api	No
app_extx	id	app_extx	Optional
app_inhx	id	app_inhx	Optional
arcpur_hist	persistent_id	Archive Purge History	No
arcpur_rule	persistent_id	Archive Purge Rule	No
arg_history	persistent_id	arg_history	No
asset	id	asset	No
assetx_prod_list	sym	assetx_prod_list	No
atev	persistent_id	Attached Event	Required
atomic_cond	id	Atomic Condition	Optional
attached_sla	id	Attached Service Type	Required
attmnt	id	Attachment	Optional
attmnt_folder	id	Attachments Folder	Optional
attmnt_lrel	persistent_id	Attachment LREL	No
attr_alias	id	Attribute Alias	No
aty	code	Activity Type	Optional
audlog	persistent_id	Audit Log	Required
bhvtpl	id	Behavior Template	Optional
bhvtpl_wftpl	id	bhvtpl_wftpl	Optional
bm_task	id	bm_task	Optional
bmcls	id	Business Management Class	No

bmhier	id	Business Management Hierarchy	Optional
bmlrel	persistent_id	Business Management LREL	No
bmrep	id	Business Management Repository	No
bms	status_no	Business Management Status	No
bool	enum	Boolean	No
BU_TRANS	id	Bubble Up Transaction	Optional
ca_application_registration	id	ca_application_registration	No
ca_asset	id	Asset	No
ca_asset_source	id	Asset Source	No
ca_asset_source_unrestricted	id	ca_asset_source_ unrestricted	No
ca_asset_subschema	id	Asset Subschema	No
ca_asset_type	id	Asset Type	No
ca_cmpny	id	Company	Optional
ca_logical_assst	id	Logical Asset	No
ca_logical_asset_property	id	Logical Asset Property	No
chg	id	chg	Required
chg_tpl	template_name	Change Template	Required
chgalg	id	Change Order Activity Log	Required
chgaty	code	chgaty	Optional
chgcat	code	Change Category	Optional
chgcat_grp	persistent_id	Change Category Group LREL	No
chgcat_loc	persistent_id	Change Category Location LREL	No
chgcat_workshift	persistent_id	Change Category Workshift LREL	No
chgstat	code	Change Status	Optional
chgtype	id	Change Type	No
CI_ACTIONS	id	Knowledge Workflow Task	Optional
CI_ACTIONS_ALTERNATE	id	Alternate Knowledge Workflow Task	Optional
CI_BOOKMARKS	id	Knowledge Document Bookmark	Optional
CI_DOC_LINKS	id	Knowledge Document Link	Optional
CI_DOC_TEMPLATES	id	Knowledge Document Template	Optional
CI_DOC_TYPES	id	Knowledge Document Type	No
ci_mdr_idmap	id	ci_mdr_idmap	Optional
ci_mdr_provider	id	ci_mdr_provider	Optional
CI_PRIORITIES	id	Approval Process Template	Optional

ci_rel_type	id	ci_rel_type	No
CI_STATUSES	id	Knowledge Status	No
CI_WF_TEMPLATES	id	Knowledge Workflow Template	No
cmth	id	Contact Method	No
cnote	persistent_id	Announcement	Optional
cnt	id	Contact	Required
cnt_role	id	Contact Role	Required
cntx	id	cntx	Optional
conx	id	conx	Optional
cost_cntr	id	Cost Center	Optional
country	id	Country	No
cr	persistent_id	Request	Required
cr_prp	id	Request Property	Required
cr_prptpl	id	Request Property Template	Optional
cr_tpl	template_name	Request Template	Required
craty	code	craty	Optional
crs	code	Request Status	Optional
crs_cr	code	crs_cr	Optional
crs_in	code	crs_in	Optional
crs_pr	code	crs_pr	Optional
crsol	persistent_id	Request Solution	No
crsq	code	Stored Query	Optional
crt	code	Request Type	No
cst	id	Customer contacts	Required
ctab	id	Controlled Table	No
ctimer	persistent_id	Request Timer	No
ctp	id	Contact Type	No
dat_basx	id	dat_basx	Optional
dcon	id	Data Partition Constraint	No
dcon_typ	enum	Data Partition Constraint Type	No
dept	id	Department	Optional
dlgsrvr	id	Delegation Server	No
dmn	id	Data Partition	No
doc_rep	persistent_id	Attachments Repository	Optional
DOC_VERSIONS	id	Document Version	Optional
docx	id	docx	Optional
EBR_ACRONYMS	id	EBR Acronym	No
EBR_DICTIONARY	id	EBR Dictionary	No

EBR_DICTIONARY_ADM	id	EBR Dictionary Alternate	No
EBR_FULLTEXT	id	EBR Knowledge Document Index	No
EBR_FULLTEXT_ADM	id	EBR Knowledge Document Index Alternate	No
EBR_KS	id	Knowledge Source	No
EBR_KS_ACCESS	id	Knowledge Source Access	No
EBR_KS_INDEXING_QUEUE	id	Knowledge Source Indexing Queue	No
EBR_LOG	id	EBR Log	No
EBR_METRICS	id	EBR Metric	No
EBR_NOISE_WORDS	id	EBR Noise Word	No
EBR_PATTERNS	id	EBR Pattern	No
EBR_PREFIXES	id	EBR Prefix	No
EBR_PROPERTIES	id	EBR Property	No
EBR_SUBSTITS	id	EBR Substitution	No
EBR_SUFFIXES	id	EBR Suffix	No
EBR_SYNONYMS	id	EBR Synonym	No
EBR_SYNONYMS_ADM	id	EBR Synonyms Alternate	No
edit_macros	persistent_id	edit_macros	No
entservx	id	entservx	Optional
enttx	id	enttx	Optional
ES_CONSTANTS	id	ES_CONSTANTS	No
ES_NODES	id	Decision Tree Node	Optional
ES_RESPONSES	id	ES_RESPONSES	Optional
ES_SESSIONS	id	ES_SESSIONS	Optional
event_log	id	Event Log	No
event_prod_list	sym	event_prod_list	No
event_type	id	Event Type	No
evt	persistent_id	Event	Optional
evtdly	persistent_id	Event Delay	Optional
evtdlytp	enum	Event Delay Type	No
ext_entity_map	persistent_id	External Entity Map	No
fac_acx	id	fac_acx	Optional
fac_firex	id	fac_firex	Optional
fac_furnx	id	fac_furnx	Optional
fac_othx	id	fac_othx	Optional
fac_upsx	id	fac_upsx	Optional
fmgrp	id	Form Group	No

g_chg_ext	id	Global Change Extension	Required
g_chg_queue	id	Global Change Queue	No
g_cnt	id	Global Contact	No
g_cr_ext	id	Global Request Extension	Required
g_cr_queue	id	Global Request Queue	No
g_iss_ext	id	Global Issue Extension	Required
g_iss_queue	id	Global Issue Queue	No
g_loc	id	g_loc	No
g_org	id	Global Organization	No
g_prod	id	Global Product	No
g_qname	id	Global Queue	No
g_srvrs	remote_sys_id	Global Server	No
g_tblmap	id	Global Table Map	No
g_tblrule	id	Global Table Rule	No
g_tenant	id	Global Tenant	No
gl_code	id	Resource Code	No
grc	id	Resource Class	No
grp	id	Group Contacts	Required
grpmem	persistent_id	Group Member	Required
har_lparx	id	har_lparx	Optional
har_maix	id	har_maix	Optional
har_monx	id	har_monx	Optional
har_othx	id	har_othx	Optional
har_prix	id	har_prix	Optional
har_serx	id	har_serx	Optional
har_stox	id	har_stox	Optional
har_virx	id	har_virx	Optional
har_worx	id	har_worx	Optional
help_content	name	help_content	No
help_item	id	help_item	No
help_lookup	id	help_lookup	No
help_set	id	help_set	No
hier	id	Asset Relation	Optional
ical_alarm	id	iCalendar Alarm	No
ical_event_prod_list	sym	ical_event_prod_list	No
ical_event_template	id	iCalendar Event	No
imp	enum	Impact	No
in	persistent_id	Incident	Required

INDEX_DOC_LINKS	id	Knowledge Category-Document Link	Optional
interface_type	id	interface_type	No
intfc	id	interface	No
invidex	id	invidex	Optional
invothx	id	invothx	Optional
invprjx	id	invprjx	Optional
iss	persistent_id	iss	Required
iss_prp	id	Issue Property	Required
iss_tpl	template_name	Issue Template	Required
iss_wf	id	Issue Workflow Task	Required
issalg	id	Issue Activity Log	Required
issaty	code	issaty	Optional
isscat	code	Issue Category	Optional
isscat_grp	persistent_id	Issue Category Group LREL	No
isscat_loc	persistent_id	Issue Category Location LREL	No
isscat_workshift	persistent_id	Issue Category Workshift LREL	No
issstat	code	Issue Status	Optional
job_func	id	Job Function	Optional
kc	id	KPI	Optional
KCAT	id	Knowledge Category	Optional
kcd	id	KPI Data	Optional
KD	id	Knowledge Document	Optional
KD_ALL	id	KD_ALL	Optional
KD_ATTMNT	id	Document-Attachment Link	Optional
KD_FILE	id	KD_FILE	Optional
KD_QA	id	KD_QA	Optional
KD_SAVE_AS	persistent_id	KD_SAVE_AS	No
KD_TASK	persistent_id	KD_TASK	Optional
kdaty	code	kdaty	Optional
kdlinks	persistent_id	Document-Ticket Link	Optional
KEIT_IMPORT_PACKAGES	id	Knowledge Import Packages	No
KEIT_TEMPLATES	id	Knowledge Export/Import Template	Optional
KEIT_TRANSACTION_ STATUSES	id	Knowledge Transaction Statuses	No
KEIT_TRANSACTIONS	id	Knowledge Import Transactions	Optional
kmlrel	persistent_id	Knowledge LREL	No
KT_ACT_CONTENT	id	Action Content	Optional

KT_BLC	id	Recommended Document	Optional
KT_BLC_TYPE	id	KT_BLC_TYPE	No
KT_FILE_TYPE	id	File Type	No
KT_FLG_STATUS	id	Flag Status	No
KT_FLG_TYPE	id	Comment Type	Optional
KT_FREE_TEXT	persistent_id	Free Text	No
KT_KCAT_NTF	id	Category Notification	Optional
KT_LIFE_CYCLE_REP	id	KT Life Cycle Report	Optional
KT_QA_RESP_TYPE	id	Forum Response Type	No
KT_QA_STATUS	id	Forum Status	No
KT_REPORT_CARD	id	Knowledge Report Card	No
KT_STATUS_ROLE	id	Status Role	No
ktd	id	KPI Ticket Data	Required
kwrd	id	Keyword	No
ldap	id	LDAP	No
ldap_group	id	LDAP Group	No
loc	id	Location	Optional
locx	id	locx	Optional
LONG_TEXTS	id	Forum Reply	Optional
lr	persistent_id	Notification Log	Required
macro	persistent_id	Macro	Optional
macro_prod_list	sym	macro_prod_list	No
macro_type	code	Macro Type	No
menu_bar	id	Menu Bar	No
menu_tree	id	Menu Tree Node	No
menu_tree_name	id	Menu Tree	No
menu_tree_res	id	Menu Tree Resource	No
mfrmod	id	Model Definition	Optional
mgs	id	Managed Survey	Optional
mgsalg	id	Managed Survey Activity Log	Optional
mgsaty	code	mgsaty	Optional
mgsstat	code	Managed Survey Status	No
MSysConf	Config	MSysConf	No
net_brix	id	net_brix	Optional
net_clux	id	net_clux	Optional
net_conx	id	net_clux	Optional
net_frox	id	net_frox	Optional
net_gatx	id	net_gatx	Optional

net_hubx	id	net_hubx	Optional
net_nicx	id	net_nicx	Optional
net_othx	id	net_othx	Optional
net_perx	id	net_perx	Optional
net_porx	id	net_porx	Optional
net_rgrpx	id	net_rgrpx	Optional
net_roux	id	net_roux	Optional
net_rsrcx	id	net_rsrcx	Optional
no_contract_sdsc	code	no_contract_sdsc	Optional
node_prod_list	sym	node_prod_list	No
NOTIFICATION	id	Notification	Optional
notque	persistent_id	Queued Notification	No
noturg	enum	Notification Urgency	No
nr	id	Configuration Item	Optional
nr_com	id	Asset comment	Optional
nrf	id	Resource Family	No
ntfl	id	Notification Log	No
ntfm	persistent_id	Notification Message Template	Optional
ntfm_prod_list	sym	ntfm_prod_list	No
ntfr	id	Notification Rule	Optional
ntfr_prod_list	sym	ntfr_prod_list	No
O_COMMENTS	id	Knowledge Document Comment	Optional
O_EVENTS	id	Document History	Optional
OA_COLUMNS	COLUMN_NAME	OA_COLUMNS	No
OA_FKEYS	PKCOLUMN_NAME	OA_FKEYS	No
OA_INFO	INFO_NAME	OA_INFO	No
OA_STATISTICS	COLUMN_NAME	OA_STATISTICS	No
OA_TABLES	TABLE_NAME	OA_TABLES	No
OA_TYPES	TYPE_NAME	OA_TYPES	No
object_notify_prod_list	sym	object_notify_prod_list	No
opsys	id	Operating System	Optional
opsysx	id	opsysx	Optional
options	persistent_id	Option	No
org	id	Organization	Required
orgx	id	orgx	Optional
P_GROUPS	id	KT Permission Group	Optional
pcat	persistent_id	Request Area	Optional
pcat_cr	persistent_id	pcat_cr	Optional

pcat_in	persistent_id	pcat_in	Optional
pcat_loc	persistent_id	Request Area Location LREL	No
pcat_pr	persistent_id	pcat_pr	Optional
pcat_workshift	persistent_id	Request Area Workshift LREL	No
perscnt	id	Person Contacting	Optional
position	id	Position	Optional
pr	persistent_id	Problem	Required
pri	enum	Priority	No
prio_service_type	id	Priority Service Type	Required
prod	id	Product	Optional
prod_list	sym	Object Name	No
projex	id	projex_detail	Optional
prp	id	Property	Required
prptpl	id	Property Template	Optional
prptpl_chgcat	id	prptpl_chgcat	Optional
prptpl_isscat	id	prptpl_isscat	Optional
prpval	id	Property Value	No
prpval_rule	id	Property Validation Rule	No
prpval_type	id	Property Validation Type	No
QUERY_POLICY	id	Query Policy	Optional
QUERY_POLICY_ACTIONS	id	Query Policy Actions	Optional
quick_tpl_types	enum	Quick Template Type	No
rc	id	Root Cause	Optional
response	id	Personalized Response	Optional
rev_bool	enum	Reverse Boolean	No
role	id	Role	No
role_go_form	id	Role Go Form	No
role_tab	id	Role Tab	No
role_web_form	id	Role Web Form	No
rptm	persistent_id	Report Method	No
rptmeth	id	Reporting Method	No
rrf	code	Remote Reference	No
rss	id	Resource Status	No
sapolicy	id	Web Services Access Policy	Optional
saprobtyp	id	Web Services Error Type	Optional
sd_chg_map	persistent_id	sd_chg_map	No
sd_cr_map	persistent_id	sd_cr_map	No
sdsc	code	Service Type	Optional

sdsc_map	id	Service Type Map	Required
secx	id	secx	Optional
seq	id	Sequence	Optional
serx	id	serx	Optional
session_log	id	Session Log	No
session_type	id	Session Type	No
sev	enum	Severity	No
SHOW_OBJ	id	Show Object	No
site	id	Site	Optional
slatpl	id	SLA Template	Required
slax	id	slax	Optional
sqchg	code	sqchg	Optional
sqcr	code	sqcr	Optional
sqiss	code	sqiss	Optional
srvr_aliases	id	Server Alias	No
srvr_zones	id	Server Zone	No
state	id	State	No
stored_query_prod_list	sym	stored_query_prod_list	No
survey	id	Survey	Optional
svc_contract	id	Service Contract	Required
svy_ans	id	Survey Answer	Optional
svy_atpl	id	Survey Answer Template	Optional
svy_qtpl	id	Survey Question Template	Optional
svy_ques	id	Survey Question	Optional
svy_tpl	id	Survey Template	Optional
svystat	id	Survey Statistic	Optional
svytrk	id	Survey Tracking	Required
tab	id	Tab	No
tel_cirx	id	tel_cirx	Optional
tel_othx	id	tel_othx	Optional
tel_radx	id	tel_radx	Optional
tel_voix	id	tel_voix	Optional
tel_wirx	id	tel_wirx	Optional
tenant	id	Tenant	Required
tenant_group	id	Tenant Group	No
tenant_group_member	persistent_id	Tenant Group Member	Required
text_api	persistent_id	text_api	No
tgm_groups	persistent_id	Tenant Groups	Required

tgm_members	persistent_id	Tenant Group Members	Required
tkt	persistent_id	tkt	Required
tskstat	code	Task Status	Optional
tskty	code	Task Type	Optional
tspan	sym	Time Span	No
ttv_slas	id	ttv_slas	Required
typecnt	id	Reason	No
tz	code	Time Zone	No
urg	enum	Urgency	No
url	persistent_id	URL	No
USP_PREFERENCES	id	Preference	No
USP_PROPERTIES	id	Property	Optional
usp_servers	id	usp_servers	No
usq	id	User Stored Query	Optional
vis_configuration	persistent_id	vis_configuration	Optional
vis_graph_metadata	persistent_id	vis_graph_metadata	Optional
vis_object_store_criteria	persistent_id	vis_object_store_criteria	Optional
vis_object_store_master	persistent_id	vis_object_store_master	Optional
vpt	id	Company Type	No
web_form	id	Web Form	No
web_form_pref	id	Web Form Preference	Required
wf	id	Workflow Task	Required
wftpl	id	Workflow Task Template	Optional
wftpl_chgcat	id	wftpl_chgcat	Optional
wftpl_isscat	id	wftpl_isscat	Optional
workflow_prod_list	sym	workflow_prod_list	No
wrkshft	persistent_id	Workflow Template Workshift LREL	No
wspcol	id	Web Screen Painter Column	No
wspdomset	id	wspdomset	No
wsptbl	id	Web Screen Painter Table	No
cr_wf	id	Request Workflow Task	Required

CMDB Technical Reference

This article contains the following topics:

Introduction

This section is intended for implementers who perform the following configuration management database (CMDB) tasks:

- Map your data into the CMDB.
- Manage CMDB configuration items.
- Use the Advantage Data Transformer (ADT) to write a Federation Adapter.
- Use the CMDBf Web Services to interact with CMDB.

The information in this page can assist you as you plan your CMDB implementation. You can print and have the printouts handy while performing these tasks.

CI Families and Classes

Configuration item (CI) *families* categorize your business assets by type and assign meaningful attributes for each CI in the family. Families are general categories of CIs, such as hardware, software, and services CIs.

CI *classes* are specific categories within the family categories. For example, the Hardware family contains CI classes such as modem, router, repeater, and bridge.

You can organize your CIs into families and classes to make them easier to manage. For example, you can generate a list of CIs that belong to a particular family or class.

You use the following sequence to categorize your business assets:

1. Define CI families.
2. Define CI classes.
3. Define CIs.

List Configuration Item Families

You can list CMDB CI families and view their descriptions.

To list configuration item families

1. Log into CA SDM as an administrator.
   The web interface appears.
2. Click Administration.
   The Administration tree appears.
3. Navigate the folder structure by clicking CMDB, CI Families.
   The CI families and their descriptions are listed.
4. (Optional) Click a CI family name.
   CI family details appear.

Generate a Configuration Item Families Summary

You can list CMDB CI families and view their descriptions in a report format.

To list configuration item families

1. Log into CA SDM as an administrator.
   The web interface appears.
2. Click Administration.
   The Administration tree appears.
3. Navigate the folder structure by clicking CA SDM, CI Families.
   The CI families and their descriptions are listed.
4. Click Reports, Summary.
   A summary report appears in a separate window.
5. (Optional) Click Print to select a printer and print the report.
   The report prints.

MDB Extension Tables

Each CI family has a set of family-specific attributes that reside in an *extension table* in the MDB. The family-specific attributes describe the unique characteristics of each type of CI. For example, a CI in the Hardware.Server family has attributes that represent the following:

- swap_size -- The size of the disk space allocated on a hardware or network device to store the state of a process that has been swapped out.
- mem_capacity -- The total amount of memory that can be installed and made available.
- slot_total_mem -- The total amount of memory available on memory cards in a hardware or network device.

When you implement CA SDM, you can determine the types of CIs that you want to manage and the attributes that you can track for them.

Common Attributes

The following attributes are common to various families.

Object Name	Description
acquire_date	Date the resource was acquired.
alarm_id	IP address. (hardware only)
asset_count	Resource quantity.
asset_num	Alternate resource identifier, for example, an alternate ID located on sticker placed on a computer.
class	In the object, this is the name of the class. In the table, this is a foreign key to a record in the ca_resource_class table (SREL integer to grc).
company_bought_for_uuid	In the object, this is the name of the company for which the CI was bought. In the table, this is a foreign key to the ca_company table (SREL uuid to ca_cmpny).
contact_1	In the object, this is a user-defined contact field. In the table, this is a foreign key to the ca_contact table (SREL uuid to cnt).
contact_2	In the object, this is a user-defined contact field. In the table, this is a foreign key to the ca_contact table (SREL uuid to cnt).
contact_3	In the object, this is a user-defined contact field. In the table, this is a foreign key to the ca_contact table (SREL uuid to cnt).
creation_date	Timestamp (pdmtime) indicating the date and time that the CI was created.
creation_user	User ID of the contact who created the CI
delete_flag	Active|FALSE|0 (zero)|No: CI is active and displays in display lists (the default). Inactive|TRUE|1 (one)|Yes: CI is not active and does not appear in display lists.

department	In the object, this is the name of the department. In the table, this is a foreign key to the ca_resource_department table (SREL integer to dept).
description	Longer name or description of the resource.
dns_name	The name by which this device is know in the domain name server.
exclude_registration	Exclude Registration.
expense_code	In the object, this is the CI cost center. In the table, this is a foreign key to the ca_resource_cost_center table (SREL integer to cost_cntr).
expiration_date	Date the license, lease, and so on, expires.
family	In the object, this is the name of the family. In the table, this is a foreign key to a record in the ca_resource_family table (SREL integer to nrf). Used to extend at a high level, for example, hardware.server, network.router, software.database.
financial_num	Financial number.
install_date	Date resource was installed in organization or network.
is_asset	Boolean flag that can be set to categorize an Asset for filtering purposes and to control display in CA CMDB or other products such as CA Asset Portfolio Management. CA CMDB does not allow the Asset flag to be changed to NO when an asset is managed by CA Asset Portfolio Management.
is_ci	Boolean flag that can be set to categorize a CI for filtering purposes and to control display in CA CMDB or other products such as CA Asset Portfolio Management. By default, a CI created by CA CMDB is flagged as a CI but not as an Asset.
last_mod_by	User ID of the contact who last modified the CI
license_number	License Information.
loc_cabinet	Cabinet location.
loc_floor	Floor location.
loc_room	Room location.
loc_shelf	Shelf location.
loc_slot	Slot location.
location	In the object, this is the name of the location. In the table, this is a foreign key to a record in the location table (SREL uuid to loc).
mac_address	MAC address. (hardware only)
manufacturer	In the object, this is the name of the company who manufactured the CI. In the table, this is a foreign key to a record in the ca_company table (SREL uuid to ca_cmpny).

model	In the object, this is the model name for the CI. In the table, this is a foreign key to the ca_model_def table (SREL uuid to mfrmod).
name	The name of the resource.
name_type	Foreign key to the ca_asset_type table to represent Hardware, Software, and so on.
org_bought_for_uuid	In the object, this is the name of the organization for which the CI was bought. In the table, this is a foreign key to the ca_organization table (SREL uuid to org).
priority	Enumerated value for this entry, it specifies ordering in lists and relative values (SREL integer to pri).
product_version	Product release.
repair_org	In the object, this is the name of the organization responsible for maintenance of the CI. In the table, this is a foreign key to the ca_organization table (SREL uuid to org).
resource_alias	Resource alias.
resource_contact	In the object, this is the name of the contact responsible for the CI. In the table, this is a foreign key to the CA contact table (SREL uuid to cnt).
resource_owner_uuid	In the object, this is the name of the owner for the CI. In the table, this is a foreign key to the ca_contact table (SREL uuid to cnt).
serial_number	Serial number.
service_org	In the object, this is the name of the organization ultimately responsible for the resource. In the table, this is a foreign key to the ca_organization table (SREL uuid to org).
service_type	Noneditable enum (SREL string to no_contract_sdsc).
sla	The SLA value for this usp_owned_resource.
smag_1	User-defined string field.
smag_2	User-defined string field.
smag_3	User-defined string field.
smag_4	User-defined string field.
smag_5	User-defined string field.
smag_6	User-defined string field.
standard_ci	Standard configuration for comparison.
status	In the object, this is the status indicator for the CI. In the table, this is a foreign key to the ca_resource_status table (SREL integer to rss).

supplier	In the object, this is the name of the vendor responsible for supplying the CI. In the table, this is a foreign key to the ca_company table (SREL uuid to ca_cmpny).
system_name	Computer name. (hardware only)
tenant	Tenant assignment for the CI
vendor_repair	In the object, this is the name of the vendor providing maintenance for the CI. In the table, this is a foreign key to the ca_company table (SREL uuid to ca_cmpny).
vendor_restore	In the object, this is the name of the company ultimately responsible for the resource. In the table, this is a foreign key to the ca_company table (SREL uuid to ca_cmpny).
warranty_end	Warranty end date.
warranty_start	Warranty start date.

Relationship Types

Relationships are *directional* connections between CIs.

Provider/Dependent	Dependent/Provider	Relationship Description
administers	is administered by	A responsible entity, usually a person, performs day-to-day administration of other entities.
approves	is approved by	A responsible entity grants approval for another entity to proceed with a planned or desired activity.
authorizes	is authorized by	A responsible entity ratifies activities of other entities.
authors	is authored by	A responsible person writes/creates document CIs.
backs up	is backed up by	For data recovery and preservation, one entity's critical information is stored upon another entity.
communicates with	communicates with	A peer-to-peer relationship where two entities which have a logical or physical connection convey data or information back and forth.
complies to	is complied to by	One entity abides by regulations (COBIT, SOX, and so on) set forth by another entity.
connects to	connects to	A peer-to-peer relationship where two entities have a logical or physical connection.

contains	is contained by	If one entity physically or logically houses another entity, then it contains that entity. The contained entity provides a service to the container.
controls	is controlled by	One entity, typically an SLA, specifies the levels of service that another entity is expected to provide.
defines	is defined by	If one entity describes another's actual or desired state, then it defines the other.
deploys	is deployed by	A responsible entity assembles and distributes other entities.
documents	is documented by	One entity, usually a document, describes the operation or other aspects of another entity. The 'documents' relationship is primarily descriptive instead of normative.
fails over	fails over	A peer-to-peer relationship between two entities where one entity can replace the other, usually in response to a disastrous interruption in service.
fronts	is fronted by	An entity is responsible for accepting and responding to requests for another physical entity. For example, a web server fronts an application.
governs	is governed by	A governing body (NIST, SOX PCAOB, SEC) typically issues regulations and rulings to which a governed entity, usually a service, must comply.
has an assignee	is assigned to	An entity, usually a person, has been designated responsible for another entity.
hosts	is hosted by	One entity hosts another entity which is continuous. The hosted entity uses services provided by the host entity.
is business owner of	is owned by	An entity, usually a person, has been designated as the responsible business contact for another entity.
is gateway for	has for gateway	An entity, a hardware (computer) or network component, allows or controls access to another management device.
is high availability server for	has for high availability server	Uses clustering and database mirroring to provide very rapid recovery from system failures.
is location for	located at	An entity, in this case a physical location, has been designated as the place where another entity resides.
is primary contact for	has primary contact of	One entity is the primary contact for another entity.

is proxy for	is proxied by	An entity serves as a substitute pathway for connection to a network or remote storage device. For example, this gateway is a proxy for the clients on this LAN.
is recovery server of	has for recovery server	A service or application and a server that is configured to restore the specific service or application. Generally, recovery servers are an alternative to a cluster and are used when slower recovery is acceptable.
is required by	requires	An entity that cannot function properly without another entity.
is server of	is client of	A server-client relationship where the server responds to requests from the client. Alternative for "serves - is served by" relationship.
is source code for	source code is from	An entity, application code or an application library, provides the instructions that are executable in another entity.
is subscribed to by	subscribes to	An entity, either a group of users or a single user, "signs up" to have access to or use of another entity.
is the parent of	is the child of	One entity is the parent of another entity if the other entity cannot exist without the parent entity.
manages	is managed by	One entity manages another entity.
monitors	is monitored by	One entity monitors another entity if it tracks aspects of the other entity.
notifies	is notified by	An entity advises another entity that pertinent information of specific interest is now available.
provides to	is provided by	An entity is responsible for making another entity, usually a service, available to customers. For example, user, organization, or other entity provides a service.
regulates	is regulated by	One entity periodically adjusts some parameter of another entity. A time server which periodically regulates the time on other devices is an example.
runs	runs on	One entity runs another transient entity.
secures	is secured by	An entity guards another entity against risks.
serves	is served by	Alternative for "is server of - is client of" relationship.
services	is serviced by	An entity, typically a maintenance organization or vendor, is responsible for responding to service calls for a physical entity.

supports	is supported by	An entity, usually an organization, is responsible for responding to incidents that emanate from another entity, usually a service.
updates	is updated by	An entity brings another entity's data up-to-date.
is used by	uses	An entity consumes data or services from another entity.

List Relationship Types

You can list CMDB relationship types to see the *directional* connections between CIs.

To list relationship types

1. Log into CA SDM as an administrator.
   The web interface appears.
2. Click Administration.
   The Administration tree appears.
3. Navigate the folder structure by clicking CMDB, CI Relationship Types.
   The relationship types are listed in columns: Provider To Dependent, Dependent to Provider, and Peer-to-Peer.
4. (Optional) Click a relationship type.
   Relationship type details appear in a separate window, and you can edit the relationship type.

Families and Classes

This section contains the following articles:

- Base Families
- Cluster Families
- Contact Family
- Contract Family
- Document Family
- Enterprise Families
- Facilities Family
- Hardware Families
- Investment Families
- Location Family
- Network Families
- Organization Family
- Security Family
- Service Family
- Service Level Agreement (SLA) Family
- Software Families
- Storage Area Network (SAN) Families
- Telecom Families

Base Families

NOTE
The following CA SDM and CA APM base families do not have their own CMDB extension tables:

- Computer
- Hardware
- Other
- Projects (includes a CA Service Desk extension table)
- Software

In CMDB, CIs in these base families receive CMDB CI Detail pages with some extraneous fields and lacking an Attributes tab. You can use the Change Family and Class capability to convert these CIs to CMDB families to take advantage of CMDB advanced features such as the ability to track family-specific attributes, versioning, snapshots, and baselines.

Cluster Families

The Cluster families include the following:

- **Cluster**
  Identifies multiple servers linked together to handle variable workloads or if one or more devices fail, to provide continued operation.
- **Cluster.Resource**
  Identifies a member of a cluster resource group.
- **Cluster.Resource Group**
  Identifies a group of devices in a cluster.

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name	Description
Cluster	Cluster	net_clux	ci_network_cluster	Failover Cluster
Cluster.Resource	Resource	net_rsrcx	ci_network_resource	Resource Cluster
Cluster.Resource Group	Resource Group	net_rgrpx	ci_network_resource_group	Resource Group Cluster

Cluster Attributes

The Cluster family includes the following attributes that correspond to the net_clux extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
channel_address	Channel Address	The tag used to identify a channel on a port.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
gateway_id	Gateway ID	The unique identifier for a network point that acts as an entrance (gateway) to another network.

last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_effective_date	Lease Effective Date	The date on which a lease begins.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI (for example, vendor or in-house).
network_address	Network Address	The IP address at which this CI resides (for example, 192.168.0.4)
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
os_version	OS Version	The version number of a CIs operating system.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
quorum	Quorum	The name of the definitive repository for all configuration information relating to a cluster.
retire_date	Retire Date	The date on which a CI is no longer active.

SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
virtual_ip	Virtual IP Address	The designation of the IP address that is shared among multiple domain names or multiple servers.

Cluster.Resource Attributes

The Cluster.Resource family includes the following attributes that correspond to the net_rsrcx extension table:

Object Name	Label	Description
resource_disk	Resource Disk	The identifier for a shared disk to which access can be requested by a server or cluster node.
resource_file	Resource File	The identifier for a file folder whose subfolders can be shared among cluster resources.
resource_group_type	Resource Group Type	The type of recovery domain for a cluster (for example, data resiliency, application resiliency, or device resiliency).
resource_mount_point	Resource Mount Point	The name of the directory where the device must be mounted.
resource_type	Resource Type	The categorization of a cluster resource (for example, physical disk, print spooler, file share, network name, local quorum, and so on).

Cluster.Resource Group Attributes

The Cluster.Resource Group family include the following attribute that correspond to the net_rgrpx extension table:

Object Name	Label	Description
resource_group_type	Resource Group Type	The type of recovery domain for a cluster (for example, data resiliency, application resiliency, or device resiliency).

Contact Family

The Contact family identifies a person or role that is active in the IT infrastructure.

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name	Description
Contact	Executive	cntx	ci_contact	A company executive

Contact	External Contact	cntx	ci_contact	A person or role from outside
Contact	Managerial	cntx	ci_contact	A manager
Contact	Other Contact	cntx	ci_contact	Miscellaneous person or role
Contact	Technical	cntx	ci_contact	A technician

Contact Attributes

The Contact family includes the following that correspond to the cntx extension table:

- **base_contact**
  Specifies the person or group that the CI represents (SREL uuid to cnt). Represents an exclusive relationship where only one CI represents a contact in the Contact family.

Object Name	Label
access_type	Access Type
available	Available
bm_status	Operational Status
contact_num	Contact ID
domain	Data Partition
first_name	First Name
global_queue_id	Global Queue
last_name	Group Name
last_name	Last Name
middle_name	Middle Name
position	Job Title
schedule	Work Schedule
service_type	Service Type
status	Configured Status
timezone	Time Zone
type	Contact Type
userid	User ID

Contract Family

The Contract family identifies a legally binding business document signed between two parties.

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name	Description
Contract	License Agreement	conx	ci_contract	License Agreement Contract
Contract	Other Contract	conx	ci_contract	Miscellaneous Contract

Contract	Warranty/Maintenance Contract	conx	ci_contract	Warranty/Maintenance Contract

Contract Attributes

The Contract family includes the following attributes that correspond to the conx extension table:

Object Name	Label	Description
con_comments	Comments	Free-form text to more fully describe the particular CI.
con_num	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
con_end_date	End Date	The date on which a contract, warranty, or other legal agreement expires.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
con_ref	Contract Reference	The name or number of another document that is related to a specified contract.
con_renewal_date	Renewal Date	The date on which an existing contract, warranty, or other legal agreement is put into effect for an additional period of time.
con_start_date	Start Date	The date on which a contract, document, service, or SLA becomes active.
con_status	Status	An indication of the status of an Application, Contract, Document, Service, or SLA CI (development, review, active, retired, and so on).
con_type	Type	A description of the kind of Application, Contract, Document, Service, or SLA CI.

Document Family

The Document family identifies printed or electronically stored text which is human-readable.

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name	Description
Document	Admin Guide	docx	ci_document	Administration Guide
Document	Application Test Plan	docx	ci_document	Application Test Plan Document
Document	Business Continuity Plan	docx	ci_document	Business Continuity Plan Document
Document	Other Document	docx	ci_document	Miscellaneous Document
Document	Policies and Standards	docx	ci_document	Policies and Standards Document

Document	Training Class Collateral	docx	ci_document	Training Class Collateral Document
Document	User Guide	docx	ci_document	User Guide Document

Document Attributes

The Document family includes the following attributes that correspond to the docx extension table:

Object Name	Label	Description
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
doc_category	Category	The high-level type designation for an application, service, SLA, or document.
doc_end_date	End Date	The date on which a document expires or is no longer valid.
doc_id	Document ID	The name or number that identifies a particular document.
doc_start_date	Start Date	The date on which a contract, document, service, or SLA becomes active.
doc_status	Status	An indication of the status of an Application, Contract, Document, Service, or SLA CI (development, review, active, retired, and so on).
doc_type	Type	A description of the kind of Application, Contract, Document, Service, or SLA CI.
doc_version	Version	A number or other identifier that indicates the current level (version) of an Application, Document, Service, or SLA CI.
priority	Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.

Enterprise Families

The Enterprise families include the following:

- **Enterprise.Service**
  Identifies a combination of people, processes, and information technology that directly or indirectly supports enterprise business processes.
- **Enterprise.Transaction**

Identifies a single transaction in a transactional application.

- **Enterprise Transaction Context**
  Identifies an entity describing the flow of information for a specific application or the details of a complex transaction.

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name
Enterprise.Service	Business Service	entsrvx	ci_enterprise_service
Enterprise.Service	Infrastructure Service	entsrvx	ci_enterprise_service
Enterprise.Service	Other Service	entsrvx	ci_enterprise_service
Enterprise.Transaction	Business Transactions	enttx	ci_enterprise_transaction
Enterprise Transaction Context	TransactionContext	trn_ctx	ci_transaction_ctx

Enterprise.Service Attributes

The Enterprise.Service family includes the following classes:

- Business Service
- Infrastructure Service
- Other Service

The Enterprise.Service family includes the following attributes that correspond to the entsrvx extension table:

NOTE
(R) indicates that the attribute can be stored as a relationship to other CIs.

Object Name	Label	Description
availability_end	Availability End	End of next anticipated service availability period for an intermittent service
availability_start	Availability Start	Start of next anticipated service availability period for an intermittent service
business_contacts (R)	Business Contacts	Business persons to contact with questions about service
business_impact	Business Impact	Magnitude of the effect on business if service is stopped or impaired
business_owner (R)	Business Owner	Person or persons who own the service
business_priority	Business Priority	Importance of the service to business
business_risk	Business Risk	Risk to business implied by the service
business_unit (R)	Business Unit	Business unit or units that receive the service
cancel_date	Cancel Date	Date service cancelled or terminated
category	Service Category	Service category
charge_code	Charge Code	Accounting code used to track service expenses
cobit_objective	Cobit Objective	Applicable COBIT control objective
description	Service Description	Service description
design_end_date	Design End Date	End date of design life cycle phase

design_start_date	Design Start Date	Start date of design life cycle phase
escalation_contacts (R)	Escalation Contacts	Persons to contact when escalating service issues
lifecycle_state	Service Lifecycle State	Conforms to ITIL v3. For example: design, transition, production, terminated.
lifecycle_status	Service Lifecycle Status	Status within lifecycle_state: Pending approval, Pending funding
operation_end_date	Operation End Date	End date of operations life cycle phase
operation_start_date	Operation Start Date	Start date of operations life cycle phase
portfolio (R)	Portfolio	Service portfolio holding service
service_alignment	Service Alignment	How well is service aligned to corporate goal? HIGH-MEDIUM-LOW
service_goal	Service Goal	Describe corporate strategy supported by the service
service_hours	Service Hours	Hours when service is normally available
service_manager (R)	Service Manager	Person or persons who manage the service
site (R)	Site	Primary location where service is maintained
SLA (R)	SLA	Brief description of applicable SLAs
transition_end_date	Transition End Date	End date of transition life cycle phase
transition_start_date	Transition Start Date	Start date of transition life cycle phase
unavailability_end	Unavailability End	End of next anticipated service blackout for an intermittent service
unavailability_start	Unavailability Start	Start of next anticipated service blackout for an intermittent service
version	Service Version	Current release of the service

Enterprise.Transaction Attributes

The Enterprise.Transaction family includes the following attributes that correspond to the enttx extension table:

NOTE
(R) indicates that the attribute can be stored as a relationship to other CIs.

Object Name	Label	Description
availability_end	Availability End	End of next anticipated service availability period for an intermittent service
availability_start	Availability Start	Start of next anticipated service availability period for an intermittent service
business_contacts (R)	Business Contacts	Business persons to contact with questions about service
business_impact	Business Impact	Magnitude of the effect on business if service is stopped or impaired
business_owner (R)	Business Owner	Person or persons who own the service

business_priority	Business Priority	Importance of the service to business
business_unit (R)	Business Unit	Business unit or units that receive the service
cancel_date	Cancel Date	Date service cancelled or terminated
category	Transaction Category	Service category
description	Transaction Description	Service description
design_end_date	Design End Date	End date of design life cycle phase
design_start_date	Design Start Date	Start date of design life cycle phase
escalation_contacts (R)	Escalation Contacts	Persons to contact when escalating service issues
lifecycle_state	Transaction Lifecycle State	DESIGN-TRANSITION-PRODUCTION-TERMINATED
lifecycle_status	Transaction Lifecycle Status	Status within lifecycle_state: Pending approval, Pending funding
operation_end_date	Operation End Date	End date of operations life cycle phase
operation_start_date	Operation Start Date	Start date of operations life cycle phase
site (R)	Site	Primary location where service is maintained
transaction_alignment	Transaction Alignment	Transaction Alignment
transaction_goal	Transaction Goal	Transaction Goal
transaction_manager	Transaction Manager	Transaction Manager
transition_end_date	Transition End Date	End date of transition life cycle phase
transition_start_date	Transition Start Date	Start date of transition life cycle phase
unavailability_end	Unavailability End	End of next anticipated service blackout for an intermittent service
unavailability_start	Unavailability Start	Start of next anticipated service blackout for an intermittent service
version	Transaction Version	Current release of the service

Enterprise TransactionContext Attributes

The Enterprise.Transaction Context family includes the following attribute that corresponds to trn_ctx extension table:

Object Name	Label	Description
ContextType	Context Type	Type of context, whether a kind of application or a business context.

NOTE

Facilities Family

The Facilities families include the following:

- **Facilities.Air Conditioning**

Identifies air conditioning, heating, ventilation, humidity control, or general environment management systems.

- **Facilities.Fire Control**
  Identifies equipment for fire suppression.
- **Facilities.Furnishings**
  Identifies furnishings used to store important IT items.
- **Facilities.Other**
  Identifies miscellaneous facilities equipment or supplies.
- **Facilities.Uninterruptible Power Supply**
  Identifies uninterruptible power supplies, and other power conditioning and regulation systems.

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name	Description
Facilities.Air Conditioning	Air Conditioning	fac_acx	ci_fac_ac	Air Conditioning Facilities
Facilities.Fire Control	Fire Control	fac_firex	ci_fac_fire_control	Fire Control
Facilities.Furnishings	Equipment Rack	fac_furx	ci_fac_furnishings	Equipment Rack
Facilities.Furnishings	File Cabinet	fac_furx	ci_fac_furnishings	File Cabinet
Facilities.Other	Other Facilities	fac_othx	ci_fac_other	Miscellaneous Facilities
Facilities.Uninterruptible Power Supply	Uninterruptible Power Supply	fac_upsx	fac_upsx	Uninterruptible Power Supply

Facilities.Air Conditioning Attributes

The Facilities.Air Conditioning family includes the following attributes that correspond to the fac_acx extension table:

Object Name	Label	Description
active_date	Activation Date	The date on which the CI was put into active status.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_end_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_start_date	Lease Effective Date	The date on which a lease begins.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.

maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.

Facilities.Fire Control Attributes

The Facilities.Fire Control family includes the following attributes that correspond to the fac_firex extension table:

Object Name	Label	Description
active_date	Activation Date	The date on which the CI was put into active status.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_end_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.

lease_start_date	Lease Effective Date	The date on which a lease begins.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.

Facilities.Furnishings Attributes

The Facilities.Furnishings family includes the following attributes that correspond to the fac_furx extension table:

Object Name	Label	Description
active_date	Activation Date	The date on which the CI was put into active status.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.

lease_end_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_start_date	Lease Effective Date	The date on which a lease begins.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
warehouse_loc	Warehouse Location	The physical location of a warehouse or other storage facility where a CI resides after it has been received and is in "in stock" status.

Facilities.Other Attributes

The Facilities.Other family includes the following attributes that correspond to the fac_othx extension table:

Object Name	Label	Description
active_date	Activation Date	The date on which the CI was put into active status.

contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_end_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_start_date	Lease Effective Date	The date on which a lease begins.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.

SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
warehouse_loc	Warehouse Location	The physical location of a warehouse or other storage facility where a CI resides after it has been received and is in "in stock" status.

Facilities.Uninterruptible Power Supply Attributes

The Facilities.Uninterruptible Power Supply family includes the following attributes that correspond to the fac_upsx extension table:

Object Name	Label	Description
active_date	Activation Date	The date on which the CI was put into active status.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_end_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_start_date	Lease Effective Date	The date on which a lease begins.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.

mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
warehouse_loc	Warehouse Location	The physical location of a warehouse or other storage facility where a CI resides after it has been received and is in "in stock" status.

Hardware Families

The Hardware families include the following:

- **Hardware.Logical Partition**
  Identifies Logical Partitions (LPAR) that are a mainframe architecture that segments a single system into several independent logical systems.
- **Hardware.Mainframe**
  Identifies large central computing devices, traditionally manufactured by IBM and running z/OS, OS/390, and so on.
- **Hardware.Monitor**
  Identifies computer, video, and surveillance displays. Includes CRT's, LCD's, and plasma monitors.
- **Hardware.Other**
  Identifies miscellaneous IT hardware.
- **Hardware.Printer**
  Identifies a device typically connected to a computing system which converts electronic documents to visual physical media, usually paper.
- **Hardware.Server**
  Identifies computers on a network whose main function is to respond to requests from other computers rather than provide a display and keyboard to an individual user.
- **Hardware.Storage**
  Identifies units designed to store electronic data. Tape drives, optical disks, and SANs are all included.
- **Hardware.Virtual Machine**

Identifies servers running on a system simulated in software, for example, VMWare, MSVM.

- **Hardware.Workstation**
  Identifies computers primarily used by end-users rather than serving other computers.
- **Hardware.EnvironmentalSensor**
  Identifies environmental sensor, measures physical quantity and converts it to a numeric value.
- **Hardware.File**
  Identified a file, directory or volume/filesystem that is being watched, on an executing hardware device, or that holds relevant management data.
- **Hardware.DiskPartition**
  Identifies the logical division of a physical hard drive to treat the drive as though it were multiple, independent disks.
- **Hardware.Memory**
  Identifies the physical or paging memory in an executing hardware device.
- **Hardware.PowerSupply**
  Identifies the hardware providing the current/voltages needed to operate a device.
- **Hardware.Processor**
  Identifies a processor (such as a CPU or math processor) in an executing hardware device.
- **Hardware.StoragePool**
  Identifies the grouping of storage capacity based on various criteria, such as location, cost or hardware ownership. Pools may consist of other pools or be assembled across MediaDrives.
- **Hardware.StorageVolume**
  Identifies the storage on an array backed by a StoragePool, MediaDrive or built on other lower-level volumes. A StorageVolume is published for use outside of its hosting system/array.
- **Hardware.VMDataStore**
  Identifies storage location for virtual machine files.

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name	Description
Hardware.Logical Partition	Logical Partition	har_lparx	ci_hardware_lpar	Mainframe Logical Partition
Hardware.Mainframe	Cray	har_maix	ci_hardware_mainframe	Cray Mainframe
Hardware.Mainframe	Group 80	har_maix	ci_hardware_mainframe	Group 80 Mainframe
Hardware.Mainframe	MVS	har_maix	ci_hardware_mainframe	MVS Mainframe
Hardware.Mainframe	OS/390	har_maix	ci_hardware_mainframe	OS/390 Mainframe
Hardware.Mainframe	Other Hardware Mainframe	har_maix	ci_hardware_mainframe	Miscellaneous Mainframe Hardware
Hardware.Mainframe	System 390	har_maix	ci_hardware_mainframe	System 390 Hardware
Hardware.Mainframe	System Z	har_maix	ci_hardware_mainframe	System Z Hardware
Hardware.Mainframe	Tandem - Mainframe	har_maix	ci_hardware_mainframe	Tandem Hardware
Hardware.Mainframe	Unisys.Mainframe	har_maix	ci_hardware_mainframe	Unisys Mainframe Hardware
Hardware.Mainframe	VAX - Mainframe	har_maix	ci_hardware_mainframe	VAX Hardware
Hardware.Mainframe	Virtual Storage Array	har_maix	ci_hardware_mainframe	Virtual Storage Array Hardware
Hardware.Mainframe	z/OS	har_maix	ci_hardware_mainframe	z/OS Hardware
Hardware.Monitor	CRT	har_monx	ci_hardware_monitor	Cathode Ray Tube Monitor
Hardware.Monitor	Flat Screen	har_monx	ci_hardware_monitor	Flat Screen Monitor

Hardware.Monitor	Other Monitor	har_monx	ci_hardware_monitor	Miscellaneous Display Hardware
Hardware.Monitor	Terminal	har_monx	ci_hardware_monitor	Terminal Hardware
Hardware.Other	Barcode Reader	har_othx	ci_hardware_other	Barcode Reader Hardware
Hardware.Other	Copier	har_othx	ci_hardware_other	Copier Hardware
Hardware.Other	Digital Camera	har_othx	ci_hardware_other	Digital Camera
Hardware.Other	Electronic Whiteboard	har_othx	ci_hardware_other	Electronic Whiteboard
Hardware.Other	Other Hardware	har_othx	ci_hardware_other	Miscellaneous Hardware
Hardware.Other	Projector	har_othx	ci_hardware_other	Projector Hardware
Hardware.Other	Shredder	har_othx	ci_hardware_other	Shredder Hardware
Hardware.Other	Television	har_othx	ci_hardware_other	Television Hardware
Hardware.Other	VCR/DVD	har_othx	ci_hardware_other	VCR/DVD Hardware
Hardware.Other	Video Camera	har_othx	ci_hardware_other	Video Camera Hardware
Hardware.Printer	Bubble Jet	har_prix	ci_hardware_printer	Bubble Jet
Hardware.Printer	Ink Jet	har_prix	ci_hardware_printer	Ink Jet
Hardware.Printer	Laser	har_prix	ci_hardware_printer	Laser Printer
Hardware.Printer	Microfiche	har_prix	ci_hardware_printer	Microfiche Printer
Hardware.Printer	Other Printer	har_prix	ci_hardware_printer	Miscellaneous Printer Hardware
Hardware.Printer	Plotter	har_prix	ci_hardware_printer	Plotter Printer
Hardware.Server	AIX	har_serx	ci_hardware_server	Server using AIX
Hardware.Server	HP UX	har_serx	ci_hardware_server	Server using HP-UX
Hardware.Server	Linux	har_serx	ci_hardware_server	Server using Linux
Hardware.Server	Other Operating System	har_serx	ci_hardware_server	Server using miscellaneous OS
Hardware.Server	Server	har_serx	ci_hardware_server	Server Hardware
Hardware.Server	Sun	har_serx	ci_hardware_server	Server using Sun
Hardware.Server	Tandem	har_serx	ci_hardware_server	Server using Tandem
Hardware.Server	Unisys	har_serx	ci_hardware_server	Server using Unisys
Hardware.Server	UNIX	har_serx	ci_hardware_server	Server using UNIX
Hardware.Server	VAX	har_serx	ci_hardware_server	Server using VAX
Hardware.Server	VM	har_serx	ci_hardware_server	Server using VM
Hardware.Server	Windows	har_serx	ci_hardware_server	Server using Windows
Hardware.Storage	CD-Rom Drive	har_stox	ci_hardware_storage	CD-Rom Drive
Hardware.Storage	Disk Array	har_stox	ci_hardware_storage	Disk Array
Hardware.Storage	DVD	har_stox	ci_hardware_storage	DVD Storage
Hardware.Storage	File System	har_stox	ci_hardware_storage	File System Storage
Hardware.Storage	Hard Drive	har_stox	ci_hardware_storage	Hard Drive
Hardware.Storage	Network Attached Storage	har_stox	ci_hardware_storage	Network Attached Storage
Hardware.Storage	Optical	har_stox	ci_hardware_storage	Optical Hardware

Hardware.Storage	Other Hardware Storage	har_stox	ci_hardware_storage	Miscellaneous Storage Hardware
Hardware.Storage	Silo	har_stox	ci_hardware_storage	Storage Silo Hardware
Hardware.Storage	Storage Area Network	har_stox	ci_hardware_storage	Storage Area Network (SAN) Hardware
Hardware.Storage	Tape Array	har_stox	ci_hardware_storage	Tape Storage Array
Hardware.Storage	Tape Library	har_stox	ci_hardware_storage	Tape Storage Library
Hardware.Storage	Virtual Tape System	har_stox	ci_hardware_storage	Virtual Tape System
Hardware.Storage	Zip Drive	har_stox	ci_hardware_storage	Zip Drive Hardware
Hardware.Virtual Machine	ESX Server	har_virx	ci_hardware_virtual	ESX Server
Hardware.Virtual Machine	GSX Server	har_virx	ci_hardware_virtual	GSX Server
Hardware.Virtual Machine	Microsoft Virtual Server	har_virx	ci_hardware_virtual	Microsoft Virtual Server
Hardware.Virtual Machine	Other Hardware Virtual Machine	har_virx	ci_hardware_virtual	Miscellaneous Virtual Machines
Hardware.Workstation	Workstation	har_worx	ci_hardware_workstation	Workstation Hardware
Hardware.Environmental Sensor	Temperature	har_comp	ci_hardware_component	Environmental Sensor
Hardware.File	File	har_file	ci_hardware_file	File, Directory or Volume/ Filesystem
Hardware.DiskPartition	BFS	har_dpar	ci_hardware_dpar	Logical division of a physical hard drive
Hardware.Memory	Physical	har_mem	ci_hardware_memory	Memory (physical or paging)
Hardware.PowerSupply	PowerSupply	har_comp	ci_hardware_component	
Hardware.Processor	x86	har_prcr	ci_hardware_processor	Processor
Hardware.StoragePool	StoragePool	har_stgpl	ci_hardware_storagepool	Storage Capacity
Hardware.StorageVolume	StorageVolume	har_stgvol	ci_hardware_storagevolume	Storage on an Array
Hardware.VMDataStore	NetworkFileSystem	har_vmds	ci_hardware_vmdatastore	Storage location for virtual machine files

Hardware.Logical Partition Attributes

The Hardware.Logical Partition family includes the following attributes that correspond to the har_lparx extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.

contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
current_memory	Current Memory Used	An indication of how much memory is used, as opposed to the total amount available.
current_processors	Current Processors Used	An indication of how many of the processors are in use compared to the number available.
desired_memory	Desired Amount of Memory	The amount of memory to be allocated to a logical partition as long as the memory on the managed resource is not overcommitted.
desired_processors	Desired Number of Processors	The number of processors to be allocated to a logical partition as long as the processors on the managed resource are not overcommitted.
disk_type	Disk Type	The type of disk drive that resides on a workstation or server.
hard_drive_capacity	Disk Capacity	The amount of hard drive capacity that is available for use on a Hardware CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_start_date	Lease Effective Date	The date on which a lease becomes effective (also known as the lease start date).
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_end_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
max_memory	Maximum Memory Size	The maximum amount of memory available in an LPAR.
max_processors	Maximum Number of Processors	The maximum number of processors available in an LPAR.

mem_capacity	Memory Capacity	The total amount of memory that can be installed and made available.
min_memory	Minimum Amount of Memory	The minimum amount of memory required for an LPAR.
min_processors	Minimum Number of Processors	The minimum number of processors required for an LPAR.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
number_mips	MIPS	An indication of the processing speed and capacity of a hardware or network device.
panel_display	Panel Display	The operator console used to manage logical partition configurations and booting, starting, and stopping of system or individual partitions.
phys_mem	Memory Installed	The physical amount of memory installed on a hardware device.
priority	Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proc_speed	Processor Speed	A measurement of the rate at which a computer performs its operations.
proc_type	Processor Type	The kind of CPU in a hardware device.
profile	Profile	The configuration name for a logical partition which indicates the desired system resource allocations.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.

Hardware.Mainframe Attributes

The Hardware.Mainframe family includes the following attributes that correspond to the har_maix extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.

contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
disk_type	Disk Type	The type of disk drive that resides on a workstation or server.
hard_drive_capacity	Disk Capacity	The amount of hard drive capacity that is available for use on a Hardware CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease becomes effective (also known as the lease start date).
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_end_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mem_capacity	Memory Capacity	The total amount of memory that can be installed and made available.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
number_mips	MIPS	An indication of the processing speed and capacity of a hardware or network device.
phys_mem	Memory Installed	The physical amount of memory installed on a hardware device.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proc_speed	Processor Speed	A measurement of the rate at which a computer performs its operations.
proc_type	Processor Type	The kind of CPU in a hardware device.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.

purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.

Hardware.Monitor Attributes

The Hardware.Monitor family includes the following attributes that correspond to the har_monx extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease becomes effective (also known as the lease start date).
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.

proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.

Hardware.Other Attributes

The Hardware.Other family includes the following attributes that correspond to the har_othx extension table:

Object Name	Label	Description
active_date	Activation Date	The date on which the CI was put into active status.
array_name	Storage Array Name	The identifier for an enterprise storage system that contains multiple disk drives and performs functions like RAID and virtualization.
array_serial_num	Storage Array Serial Number	The manufacturer's serial number for an enterprise storage system that contains multiple disk drives and performs functions like RAID and virtualization.
bios_ver	BIOS Version	The version number of the BIOS - the code that's run when a personal computer starts up.
cd_rom_type	CD ROM Type	The type of CD ROM drive that resides on a workstation or server.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
disk_type	Disk Type	The type of disk drive that resides on a workstation or server.
graphics_card	Graphics Card Model	The model designation for an expansion card that is installed in an available slot in a device for enhanced graphics capabilities.
hard_drive_capacity	Disk Capacity	The amount of hard drive capacity that is available for use on a Hardware CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease becomes effective (also known as the lease start date).

lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_end_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
media_drive_num	Media Drive Capacity	The capacity of the hardware device that consolidates multiple memory cards into one unit.
media_type	Media Type	The kind of storage media on a hardware device, for example, disk, CD ROM.
mem_cache_proc	Processor Cache	The identifier of the hardware device that processes the high-speed memory storage between memory and the CPU.
mem_capacity	Memory Capacity	The total amount of memory that can be installed and made available.
memory_shares	Number of Memory Shares	The specified memory share granted to this virtual machine.
modem_card	Modem Card	The identifier of a card in a workstation or network device that enables a faster connection to a network or the Internet.
modem_type	Modem Type	The classification/speed of a modem used by a workstation for a faster connection to a network or the Internet.
monitor	Monitor Model	The type of display unit connected to a hardware, network, or telecom device.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
net_card	Network Card	The designation for an expansion card that is installed in an available slot in a computer or network device so that it can connect and communicate to another networked component.
number_mips	MIPS	An indication of the processing speed and capacity of a hardware or network device.

number_net_card	Number of Network Cards	The number of expansion cards that have been installed in the available slots in a computer.
number_net_port	Number of Network Ports	The total number of ports in use on a device.
number_net_port_conn	Number of Network Port Connections	The total number of ports on a server.
number_proc_inst	Number of Processors Installed	The total number of processors installed on a hardware or network device.
number_slot_proc	Processor Capacity	The total number of processor slots on a hardware device.
phys_mem	Memory Installed	The physical amount of memory installed on a hardware device.
printer	Printer	The type or model of printer attached to a hardware or network device.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proc_speed	Processor Speed	A measurement of the rate at which a computer performs its operations.
proc_type	Processor Type	The kind of CPU in a hardware device.
processor_count	Processor Capacity	The number of CPU's or microprocessors available on a Hardware CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
role	Role	The business function supported by a hardware or network device, for example, production, test.
security_patch_level	Security Patch Level	An indication of the current security patch version for this CI.
server_type	Server Type	The kind of server, for example, application, mail, web, proxy, FTP.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
slot_mem_used	Number of Memory Slots Used	The amount of memory in use from the available memory cards in a hardware or network device.
slot_total_mem	Number of Memory Slots	The total amount of memory available on memory cards in a hardware or network device.

swap_size	Swap Size	The size of the disk space allocated on a hardware or network device to store the state of a process that has been swapped out.
technology	Technology	The technology, TCP/IP, Ethernet, FDDI, and so on, employed by a hardware or network device.
total_capacity	Total Disk Capacity	The total amount of storage available on a hardware device.
type_net_conn	Type of Network Connection	An indication of the kind of network connection used by a hardware or network device.
used_space	Total Disk Used	The amount of available disk storage space that is in use by a CI.

Hardware.Printer Attributes

The Hardware.Printer family includes the following attributes that correspond to the har_prix extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease becomes effective (also known as the lease start date).
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_end_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.

mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.

Hardware.Server Attributes

The Hardware.Server family includes the following attributes that correspond to the har_serx extension table:

Object Name	Label	Description
active_date	Activation Date	The date on which the CI was put into active status.
bios_ver	BIOS Version	The version number of the BIOS - the code that's run when a personal computer starts up.
cd_rom_type	CD ROM Type	The type of CD ROM drive that resides on a workstation or server.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
hard_drive_capacity	Disk Capacity	The amount of hard drive capacity that is available for use on a Hardware CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease becomes effective (also known as the lease start date).
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.

lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mem_capacity	Memory Capacity	The total amount of memory that can be installed and made available.
monitor	Monitor Model	The type of display unit connected to a hardware, network, or telecom device.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
net_card	Network Card	The designation for an expansion card that is installed in an available slot in a computer or network device so that it may connect and communicate to another networked component.
number_mips	MIPS	An indication of the processing speed and capacity of a hardware or network device.
number_net_card	Number of Network Cards	The number of expansion cards that have been installed in the available slots in a computer.
number_net_port	Number of Network Ports	The total number of ports in use on a device.
number_net_port_conn	Number of Network Port Connections	The total number of ports on a server.
number_proc_inst	Number of Processors Installed	The total number of processors installed on a hardware or network device.
number_slot_proc	Processor Capacity	The total number of processor slots on a hardware device.
panel_display	Panel Display	The operator console used to manage logical partition configurations and booting, starting, and stopping of system or individual partitions.
phys_mem	Memory Installed	The physical amount of memory installed on a hardware device.
printer	Printer	The type or model of printer attached to a hardware or network device.

ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proc_speed	Processor Speed	A measurement of the rate at which a computer performs its operations.
proc_type	Processor Type	The kind of CPU in a hardware device.
profile	Profile	The configuration name for a logical partition which indicates the desired system resource allocations.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
role	Role	The business function supported by a hardware or network device, for example, production, test.
security_patch_level	Security Patch Level	An indication of the current security patch version for this CI.
server_type	Server Type	The kind of server, for example, application, mail, web, proxy, FTP.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
slot_mem_used	Number of Memory Slots Used	The amount of memory in use from the available memory cards in a hardware or network device.
slot_total_mem	Number of Memory Slots	The total amount of memory available on memory cards in a hardware or network device.
swap_size	Swap Size	The size of the disk space allocated on a hardware or network device to store the state of a process that has been swapped out.
technology	Technology	The technology, TCP/IP, Ethernet, FDDI, and so on, employed by a hardware or network device.
type_net_conn	Type of Network Connection	An indication of the kind of network connection used by a hardware or network device.

Hardware.Storage Attributes

The Hardware.Storage family includes the following attributes that correspond to the har_stox extension table:

Object Name	Label	Description
active_date	Activation Date	The date on which the CI was put into active status.
array_name	Storage Array Name	The identifier for an enterprise storage system that contains multiple disk drives and performs functions like RAID and virtualization.
array_serial_num	Storage Array Serial Number	The manufacturer's serial number for an enterprise storage system that contains multiple disk drives and performs functions like RAID and virtualization.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
disk_type	Disk Type	The type of disk drive that resides on a workstation or server.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease becomes effective (also known as the lease start date).
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
media_drive_num	Media Drive Capacity	The capacity of the hardware device that consolidates multiple memory cards into one unit.
media_type	Media Type	The kind of storage media on a hardware device, for example, disk, CD ROM.

mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
total_capacity	Total Disk Capacity	The total amount of storage available on a hardware device.
used_space	Total Disk Used	The amount of available disk storage space that is in use by a CI.

Hardware.Virtual Machine Attributes

The Hardware.Virtual Machine family includes the following attributes that correspond to the har_virx extension table:

Object Name	Label	Description
active_date	Activation Date	The date on which the CI was put into active status.
bios_ver	BIOS Version	The version number of the BIOS - the code that's run when a personal computer starts up.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
cpu_shares	Number of CPU Shares	The specified CPU shares granted to this virtual machine.
disk_type	Disk Type	The type of disk drive that resides on a workstation or server.
hard_drive_capacity	Disk Capacity	The amount of hard drive capacity that is available for use on a Hardware CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.

lease_effective_date	Lease Effective Date	The date on which a lease becomes effective (also known as the lease start date).
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
media_type	Media Type	The kind of storage media on a hardware device, for example, disk, CD ROM.
mem_capacity	Memory Capacity	The total amount of memory that can be installed and made available.
memory_shares	Number of Memory Shares	The specified memory share granted to this virtual machine.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
number_mips	MIPS	An indication of the processing speed and capacity of a hardware or network device.
phys_mem	Memory Installed	The physical amount of memory installed on a hardware device.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proc_speed	Processor Speed	A measurement of the rate at which a computer performs its operations.
proc_type	Processor Type	The kind of CPU in a hardware device.
processor_affinity	Processor Affinity	An indicator of the preferred processor on which a task should be scheduled to run.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.

security_patch_level	Security Patch Level	An indication of the current security patch version for this CI.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
virtual_processors	Number of Virtual Processors	The number of virtual processors, the representations of physical processors to the operating system of a logical partition that uses the shared processor pool.

Hardware.Workstation Attributes

The Hardware.Workstation family includes the following attributes that correspond to the har_worx extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
bios_ver	BIOS Version	The version number of the BIOS - the code that's run when a personal computer starts up.
cd_rom_type	CD ROM Type	The type of CD ROM drive that resides on a workstation or server.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
disk_type	Disk Type	The type of disk drive that resides on a workstation or server.
graphics_card	Graphics Card Model	The model designation for an expansion card that is installed in an available slot in a device for enhanced graphics capabilities.
hard_drive_capacity	Disk Capacity	The amount of hard drive capacity that is available for use on a Hardware CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease becomes effective (also known as the lease start date).
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.

leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
max_memory	Maximum Memory Size	The maximum amount of memory available in an LPAR.
max_processors	Maximum Number of Processors	The maximum number of processors available in an LPAR.
media_drive_num	Media Drive Capacity	The capacity of the hardware device that consolidates multiple memory cards into one unit.
media_type	Media Type	The kind of storage media on a hardware device, for example, disk, CD ROM.
mem_cache_proc	Processor Cache	The identifier of the hardware device that processes the high-speed memory storage between memory and the CPU.
mem_capacity	Memory Capacity	The total amount of memory that can be installed and made available.
modem_card	Modem Card	The identifier of a card in a workstation or network device that enables a faster connection to a network or the Internet.
modem_type	Modem Type	The classification/speed of a modem used by a workstation for a faster connection to a network or the Internet.
monitor	Monitor Model	The type of display unit connected to a hardware, network, or telecom device.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
net_card	Network Card	The designation for an expansion card that is installed in an available slot in a computer or network device so that it may connect and communicate to another networked component.
number_proc_inst	Number of Processors Installed	The total number of processors installed on a hardware or network device.
number_slot_proc	Processor Capacity	The total number of processor slots on a hardware device.
phys_mem	Memory Installed	The physical amount of memory installed on a hardware device.

printer	Printer	The type or model of printer attached to a hardware or network device.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proc_speed	Processor Speed	A measurement of the rate at which a computer performs its operations.
proc_type	Processor Type	The kind of CPU in a hardware device.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retired Date	The date on which a CI is no longer active.
scsi_card	SCSI Card Model	The model identifier for a card that provides a standard interface and command set for transferring data between internal and external peripheral devices.
security_patch_level	Security Patch Level	An indication of the current security patch version for this CI.
svclvl	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for the IT component.
slot_mem_used	Number of Memory Slots Used	The amount of memory in use from the available memory cards in a hardware or network device.
slot_total_mem	Number of Memory Slots	The total amount of memory available on memory cards in a hardware or network device.

Hardware.EnvironmentalSensor Attributes

The Hardware.EnvironmentalSensor family includes the following attributes that corresponds to the har_comp extension table:

Object Name	Label	Description
DeviceAssetNumber	Device Asset Number	Number or other designator assigned to hardware by asset management, often the finance department, that is used for tracking ownership of the resource.
DeviceBiosSystemID	Device BIOS System ID	The identifier reported from the BIOS for the device.
DeviceDnsName	Device DNS Name	The fully qualified DNS name of the device.

DeviceSysName	Device System Name	The system name of the device, as defined in the system block of SNMP's MIB-II. Other information (than the MIB-II sysName) should NOT be used for this element.
DevicePhysSerialNumber	Device Physical Serial Number	An identifying string assigned by the hardware manufacturer and printed on a tag attached to the component.
DeviceMacAddress	Device MAC Address	A MAC address for the entity. The address is expressed as 6 groups of 2 hex digits (using only uppercase), separated by dashes.
DeviceIPV4Address	Device IPV4 Address	An IPv4 address for the device. The address is expressed using typical dotted decimal notation (4 groups of up to 3 decimal digits, separated by periods).
DeviceIPV4AddressWithDomain	Device IPV4 Address With Domain	An IPv4 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
DeviceIPV6Address	Device IPV6 Address	An IPv6 address for the device. The address is expressed using the formal, complete IPv6 notation (8 groups of up to 4 hex digits, using only uppercase where applicable, separated by colons).
DeviceIPV6AddressWithDomain	Device IPV6 Address With Domain	An IPv6 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
OSNumeric	OS Numeric	A numeric value either assigned by or calculated from the operating system.
ContainingIndex	Containing Index	A numeric value defining the index of the containing entity, as specified by the value of entPhysicalContainedIn in the Entity MIB.
IsPhysical	Is Physical	A boolean indicating whether the hardware is physical (if true) or logical/simulated (if false).
ContextID	Context ID	The ContextID element is used for identification.

Hardware.File Attributes

The Hardware.File family includes the following attributes that corresponds to the har_file extension table:

Object Name	Label	Description
DeviceAssetNumber	Device Asset Number	Number or other designator assigned to hardware by asset management, often the finance department, that is used for tracking ownership of the resource.
DeviceBiosSystemID	Device BIOS System ID	The identifier reported from the BIOS for the device.
DeviceDnsName	Device DNS Name	The fully qualified DNS name of the device.
DeviceSysName	Device System Name	The system name of the device, as defined in the system block of SNMP's MIB-II. Other information (than the MIB-II sysName) should NOT be used for this element.
DevicePhysSerialNumber	Device Physical Serial Number	An identifying string assigned by the hardware manufacturer and printed on a tag attached to the component.
DeviceMacAddress	Device MAC Address	A MAC address for the entity. The address is expressed as 6 groups of 2 hex digits (using only uppercase), separated by dashes.
DeviceIPV4Address	Device IPV4 Address	An IPv4 address for the device. The address is expressed using typical dotted decimal notation (4 groups of up to 3 decimal digits, separated by periods).
DeviceIPV4AddressWithDomain	Device IPV4 Address With Domain	An IPv4 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
DeviceIPV6Address	Device IPV6 Address	An IPv6 address for the device. The address is expressed using the formal, complete IPv6 notation (8 groups of up to 4 hex digits, using only uppercase where applicable, separated by colons).
DeviceIPV6AddressWithDomain	Device IPV6 Address With Domain	An IPv6 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
FilePathUrl	File Path URL	A fully qualified path and file name (if applicable). Wild card characters are not allowed.

Hardware.DiskPartition Attributes

The Hardware.DiskPartition family includes the following attributes that corresponds to the har_dpar extension table:

Object Name	Label	Description
DeviceAssetNumber	Device Asset Number	Number or other designator assigned to hardware by asset management, often the finance department, that is used for tracking ownership of the resource.
DeviceBiosSystemID	Device BIOS System ID	The identifier reported from the BIOS for the device.
DeviceDnsName	Device DNS Name	The fully qualified DNS name of the device.
DeviceSysName	Device System Name	The system name of the device, as defined in the system block of SNMP's MIB-II. Other information (than the MIB-II sysName) should NOT be used for this element.
DevicePhysSerialNumber	Device Physical Serial Number	An identifying string assigned by the hardware manufacturer and printed on a tag attached to the component.
DeviceMacAddress	Device MAC Address	A MAC address for the entity. The address is expressed as 6 groups of 2 hex digits (using only uppercase), separated by dashes.
DeviceIPV4Address	Device IPV4 Address	An IPv4 address for the device. The address is expressed using typical dotted decimal notation (4 groups of up to 3 decimal digits, separated by periods).
DeviceIPV4AddressWithDomain	Device IPV4 Address With Domain	An IPv4 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
DeviceIPV6Address	Device IPV6 Address	An IPv6 address for the device. The address is expressed using the formal, complete IPv6 notation (8 groups of up to 4 hex digits, using only uppercase where applicable, separated by colons).
DeviceIPV6AddressWithDomain	Device IPV6 Address With Domain	An IPv6 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
OSNumeric	OS Numeric	A numeric value either assigned by or calculated from the operating system.
ContainingIndex	Containing Index	A numeric value defining the index of the containing entity, as specified by the value of entPhysicalContainedIn in the Entity MIB.
IsPhysical	Is Physical	A boolean indicating whether the hardware is physical (if true) or logical/simulated (if false).

CapacityInMB	Capacity In MB	Maximum capacity of this Partition (in binary mebi-bytes).
ContextID	Context ID	The ContextID element is used for identification.

Hardware.Memory Attributes

The Hardware.Memory family includes the following attributes that corresponds to the har_mem extension table:

Object Name	Label	Description
DeviceAssetNumber	Device Asset Number	Number or other designator assigned to hardware by asset management, often the finance department, that is used for tracking ownership of the resource.
DeviceBiosSystemID	Device BIOS System ID	The identifier reported from the BIOS for the device.
DeviceDnsName	Device DNS Name	The fully qualified DNS name of the device.
DeviceSysName	Device System Name	The system name of the device, as defined in the system block of SNMP's MIB-II. Other information (than the MIB-II sysName) should NOT be used for this element.
DevicePhysSerialNumber	Device Physical Serial Number	An identifying string assigned by the hardware manufacturer and printed on a tag attached to the component.
DeviceMacAddress	Device MAC Address	A MAC address for the entity. The address is expressed as 6 groups of 2 hex digits (using only uppercase), separated by dashes.
DeviceIPV4Address	Device IPV4 Address	An IPv4 address for the device. The address is expressed using typical dotted decimal notation (4 groups of up to 3 decimal digits, separated by periods).
DeviceIPV4AddressWithDomain	Device IPV4 Address With Domain	An IPv4 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
DeviceIPV6Address	Device IPV6 Address	An IPv6 address for the device. The address is expressed using the formal, complete IPv6 notation (8 groups of up to 4 hex digits, using only uppercase where applicable, separated by colons).
DeviceIPV6AddressWithDomain	Device IPV6 Address With Domain	An IPv6 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").

OSNumeric	OS Numeric	A numeric value either assigned by or calculated from the operating system.
ContainingIndex	Containing Index	A numeric value defining the index of the containing entity, as specified by the value of entPhysicalContainedIn in the Entity MIB.
IsPhysical	Is Physical	A boolean indicating whether the hardware is physical (if true) or logical/simulated (if false).
ContextID	Context ID	The ContextID element is used for identification.
SizeInMB	Size In MB	Maximum size of this File (file system, directory or individual file).

Hardware.Processor Attributes

The Hardware.Processor family includes the following attributes that corresponds to the har_prcr extension table:

Object Name	Label	Description
DeviceAssetNumber	Device Asset Number	Number or other designator assigned to hardware by asset management, often the finance department, that is used for tracking ownership of the resource.
DeviceBiosSystemID	Device BIOS System ID	The identifier reported from the BIOS for the device.
DeviceDnsName	Device DNS Name	The fully qualified DNS name of the device.
DeviceSysName	Device System Name	The system name of the device, as defined in the system block of SNMP's MIB-II. Other information (than the MIB-II sysName) should NOT be used for this element.
DevicePhysSerialNumber	Device Physical Serial Number	An identifying string assigned by the hardware manufacturer and printed on a tag attached to the component.
DeviceMacAddress	Device MAC Address	A MAC address for the entity. The address is expressed as 6 groups of 2 hex digits (using only uppercase), separated by dashes.
DeviceIPV4Address	Device IPV4 Address	An IPv4 address for the device. The address is expressed using typical dotted decimal notation (4 groups of up to 3 decimal digits, separated by periods).
DeviceIPV4AddressWithDomain	Device IPV4 Address With Domain	An IPv4 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").

DeviceIPV6Address	Device IPV6 Address	An IPv6 address for the device. The address is expressed using the formal, complete IPv6 notation (8 groups of up to 4 hex digits, using only uppercase where applicable, separated by colons).
DeviceIPV6AddressWithDomain	Device IPV6 Address With Domain	An IPv6 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
OSNumeric	OS Numeric	A numeric value either assigned by or calculated from the operating system.
ContainingIndex	Containing Index	A numeric value defining the index of the containing entity, as specified by the value of entPhysicalContainedIn in the Entity MIB.
IsPhysical	Is Physical	A boolean indicating whether the hardware is physical (if true) or logical/simulated (if false).
ContextID	Context ID	The ContextID element is used for identification.
SpeedInGHz	Speed In GHz	The speed (in giga-hertz) of the Processor.

Hardware.StoragePool Attributes

The Hardware.StoragePool family includes the following attributes that corresponds to the har_stgpl extension table:

Object Name	Label	Description
DeviceAssetNumber	Device Asset Number	Number or other designator assigned to hardware by asset management, often the finance department, that is used for tracking ownership of the resource.
DeviceBiosSystemID	Device BIOS System ID	The identifier reported from the BIOS for the device.
DeviceDnsName	Device DNS Name	The fully qualified DNS name of the device.
DeviceSysName	Device System Name	The system name of the device, as defined in the system block of SNMP's MIB-II. Other information (than the MIB-II sysName) should NOT be used for this element.
DevicePhysSerialNumber	Device Physical Serial Number	An identifying string assigned by the hardware manufacturer and printed on a tag attached to the component.
DeviceMacAddress	Device MAC Address	A MAC address for the entity. The address is expressed as 6 groups of 2 hex digits (using only uppercase), separated by dashes.

DeviceIPV4Address	Device IPV4 Address	An IPv4 address for the device. The address is expressed using typical dotted decimal notation (4 groups of up to 3 decimal digits, separated by periods).
DeviceIPV4AddressWithDomain	Device IPV4 Address With Domain	An IPv4 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
DeviceIPV6Address	Device IPV6 Address	An IPv6 address for the device. The address is expressed using the formal, complete IPv6 notation (8 groups of up to 4 hex digits, using only uppercase where applicable, separated by colons).
DeviceIPV6AddressWithDomain	Device IPV6 Address With Domain	An IPv6 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
IsHAEnabled	Is HA Enabled	A numeric value either assigned by or calculated from the operating system. Boolean indicating whether the group (typically a Cluster instance or a GroupType="ResourceCluster") supports high availability/failover (if true) or not (if false).
IsMonitoringMembers	Is Monitoring Members	Boolean indicating whether the group (typically a Cluster instance or a GroupType="ResourceCluster") currently is using a heartbeat or other individual monitoring technique to determine if availability is lost (if true) or whether this capability does not exist or is not currently enabled (if false).
MaxFailures	Max Failures	Number of failures that can be supported by the Group, by sparing or other load balancing means.
MemberCriteria	Member Criteria	Defines a comma-separated list of QNames, representing the constraints on the constituency of the Group. Instances referenced as the Target element in the HasMember relationship semantic should have one of the types specified.
GroupType	Group Type	An enumerated value describing the primary type or category of the Group - such as an "authorization group", a "virtual resource pool" or a "data center".
HomePage	Home Page	The URL of the website home page.
BusinessRelevance	Business Relevance	Description of the relevance of the Entity, to the business.

CapacityInGB	Capacity In GB	The total capacity (in gibi-bytes) available for allocation to StorageVolumes or child StoragePools.
RaidLevel	Raid Level	The RAID level used for the Pool.

Hardware.StorageVolume Attributes

The Hardware.StorageVolume family includes the following attributes that corresponds to the har_stgvol extension table:

Object Name	Label	Description
DeviceAssetNumber	Device Asset Number	Number or other designator assigned to hardware by asset management, often the finance department, that is used for tracking ownership of the resource.
DeviceBiosSystemID	Device BIOS System ID	The identifier reported from the BIOS for the device.
DeviceDnsName	Device DNS Name	The fully qualified DNS name of the device.
DeviceSysName	Device System Name	The system name of the device, as defined in the system block of SNMP's MIB-II. Other information (than the MIB-II sysName) should NOT be used for this element.
DevicePhysSerialNumber	Device Physical Serial Number	An identifying string assigned by the hardware manufacturer and printed on a tag attached to the component.
DeviceMacAddress	Device MAC Address	A MAC address for the entity. The address is expressed as 6 groups of 2 hex digits (using only uppercase), separated by dashes.
DeviceIPV4Address	Device IPV4 Address	An IPv4 address for the device. The address is expressed using typical dotted decimal notation (4 groups of up to 3 decimal digits, separated by periods).
DeviceIPV4AddressWithDomain	Device IPV4 Address With Domain	An IPv4 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
DeviceIPV6Address	Device IPV6 Address	An IPv6 address for the device. The address is expressed using the formal, complete IPv6 notation (8 groups of up to 4 hex digits, using only uppercase where applicable, separated by colons).
DeviceIPV6AddressWithDomain	Device IPV6 Address With Domain	An IPv6 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").

OSNumeric	OS Numeric	A numeric value either assigned by or calculated from the operating system.
ContainingIndex	Containing Index	A numeric value defining the index of the containing entity, as specified by the value of entPhysicalContainedIn in the Entity MIB.
IsPhysical	Is Physical	A boolean indicating whether the hardware is physical (if true) or logical/simulated (if false).
LogicalUnitNumber	Logical Unit Number	The LUN (logical unit number) used to access the Volume.
PortID	Port ID	The system/array port through which the StorageVolume is accessed.
PortWWName	Port WW Name	The World-Wide Name of the system/array port through which the StorageVolume is accessed.
CapacityInMB	Capacity In MB	Maximum capacity of this Volume (in binary mebi-bytes).
IsThinlyProvisioned	Is Thinly Provisioned	Boolean indicating if the Volume is allocated on-demand ("thinly provisioned"), if true, or strictly allocated, if false.
IsDeDupeEnabled	Is DeDuplication Enabled	Boolean indicating if the Volume is deduplication enabled, if true, or if redundant data is persisted, if false.
IsMasked	Is Masked	When this boolean is true, the system/array validates the World-Wide Names of ports accessing the Volume, to ensure they are masked for that volume (present in the list, MaskedWWNames).
MaskedWWNames	Masked WW Names	A comma-separated list of WWNames of the allowed ports, permitted access by the StorageVolume.
RaidLevel	Raid Level	The RAID level used for the Volume. Values (both standard and non-standard) are defined by the open enumeration, RaidLevelEnum.

Hardware.VMDataStore Attributes

The Hardware.VMDataStore family includes the following attributes that corresponds to the har_vmds extension table:

Object Name	Label	Description
DeviceAssetNumber	Device Asset Number	Number or other designator assigned to hardware by asset management, often the finance department, that is used for tracking ownership of the resource.

DeviceBiosSystemID	Device BIOS System ID	The identifier reported from the BIOS for the device.
DeviceDnsName	Device DNS Name	The fully qualified DNS name of the device.
DeviceSysName	Device System Name	The system name of the device, as defined in the system block of SNMP's MIB-II. Other information (than the MIB-II sysName) should NOT be used for this element.
DevicePhysSerialNumber	Device Physical Serial Number	An identifying string assigned by the hardware manufacturer and printed on a tag attached to the component.
DeviceMacAddress	Device MAC Address	A MAC address for the entity. The address is expressed as 6 groups of 2 hex digits (using only uppercase), separated by dashes.
DeviceIPV4Address	Device IPV4 Address	An IPv4 address for the device. The address is expressed using typical dotted decimal notation (4 groups of up to 3 decimal digits, separated by periods).
DeviceIPV4AddressWithDomain	Device IPV4 Address With Domain	An IPv4 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
DeviceIPV6Address	Device IPV6 Address	An IPv6 address for the device. The address is expressed using the formal, complete IPv6 notation (8 groups of up to 4 hex digits, using only uppercase where applicable, separated by colons).
DeviceIPV6AddressWithDomain	Device IPV6 Address With Domain	An IPv6 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
FilePathUrl	File Path URL	A fully qualified path and file name (if applicable). Wild card characters are not allowed.
CapacityInMB	Capacity In MB	Maximum capacity of this datastore (in binary mebi-bytes).
IsMultiHost	Is Multi Host	Indicates whether (or not) more than one hosting system has been configured with access to the data store.

Investment Families

The Investment families include the following:

- **Investment.Idea**

Identifies the initial stage of creating new opportunities for investment such as projects, assets, applications, products, services, and other work. Ideas are containers for pertinent information that become the foundation for specific investments.

- **Investment.Other**
  Identifies a broad category to include investments in Application, Asset, Product, Service, and Other Work.
- **Investment.Project**
  Identifies a set of activities designed to achieve a specific objective. Labor, time, and budget constraints guide the projects.

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name	Description
Investment.Idea	Portfolio Idea	invidex	ci_investment_idea	Ideas are the initial stage of creating new opportunities for investment such as projects, assets, applications, products, services, and other work. Ideas lay the foundation for a specific type of investment by serving as a container for pertinent information.
Investment.Other	Portfolio Application	invothx	ci_investment_other	Captures data specific to applications running or being implemented in an organization.
Investment.Other	Portfolio Asset	invothx	ci_investment_other	Captures data specific to assets which incur costs and benefits for an organization.
Investment.Other	Portfolio Product	invothx	ci_investment_other	Captures data specific to products produced or owned by an organization.
Investment.Other	Portfolio Service	invothx	ci_investment_other	Captures data specific to services provided by an organization.
Investment.Other	Portfolio Work	invothx	ci_investment_other	Captures data specific to steady-state work performed. Other work can represent overhead tasks such as management and maintenance.
Investment.Project	Portfolio Program	invprjx	ci_investment_project	A Program is a top-level project that serves as the parent or "umbrella" project to one or more child projects.

Investment.Project	Portfolio Project	invprjx	ci_investment_project	A Project is a set of activities designed to achieve a specific objective. Projects are guided by labor, time and budget constraints.

Investment.Idea Attributes

The Investment.Idea family includes the following class:

- Portfolio Idea

The Investment.Idea family includes the following attributes that correspond to the invidex extension table:

Object Name	Label
business_unit	Business Unit
dependencies	Dependencies
est_finish_date	Estimated Finish Date
est_start_date	Estimated Start Date
existing_initiative_impact	Existing Initiative Impact
general_notes	General Notes
idea_priority	Idea Priority
owner	Owner
risks	Risks
subject	Subject
target_manager	Target Manager

Investment.Other Attributes

The Investment.Other family includes the following classes:

- Portfolio Application
- Portfolio Asset
- Portfolio Product
- Portfolio Service
- Portfolio Work

The Investment.Other family includes the following attributes that correspond to the invothx extension table:

Object Name	Label
active	Investment Active?
alignment	Alignment
charge_code	Charge Code
currency	Currency
finish_date	Finish Date

goal	Goal
investment_priority	Investment Priority
investment_status	Investment Status
manager	Manager
progress	Progress
risk	Risk
stage	Stage
start_date	Start Date
status_comment	Investment Status Comment
status_indicator	Investment Status Indicator
total_cost	Total Cost
total_effort	Total Effort
type	Investment Type

Investment.Project Attributes

The Investment.Project family includes the following classes:

- Portfolio Program
- Portfolio Project

The Investment.Project family includes the following attributes that correspond to the invprjx extension table:

Object Name	Label
active	Project Active?
alignment	Alignment
charge_code	Charge Code
currency	Currency
finish_date	Finish Date
goal	Goal
manager	Manager
progress	Progress
project_priority	Project Priority
project_status	Project Status
risk	Risk
stage	Stage
start_date	Start Date
status_comment	Project Status Comment
status_indicator	Project Status Indicator
total_cost	Total Cost
total_effort	Total Effort

Location Family

The Location family includes the following:

- **Location**
  Identifies a physical position or site.

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name	Description
Location	Building	locx	ci_location	A site contained within a single physical structure
Location	Campus	locx	ci_location	A group of buildings
Location	City	locx	ci_location	A governmental designation of a relatively large and populous area
Location	Country	locx	ci_location	An area comprised of multiple cities, regions, or states
Location	Datacenter	locx	ci_location	A site dedicated to IT operations
Location	Floor	locx	ci_location	A segment of a site on a single floor

Location Attributes

The Location family includes the following class:

- base_location
  Specifies the location that the CI represents (SREL uuid to loc). Represents an exclusive relationship where only one CI represents a location in the Location family.

The Location family includes the following attributes that correspond to the locx extension table:

Object Name	Label
address1	Address
city	City
country	Country
description	Description
site	Site
state	State/Province
zip	ZIP/Postal Code

Network Families

The Network families include the following:

- **Network.Bridge**
  Identifies an abstract device that connects multiple network segments along the data link layer.
- **Network.Controller**

Identifies miscellaneous devices that throttle or manage bandwidth use.

- **Network.Frontend**
  Identifies a network front-end device that handles communication with host computers such as mainframes.
- **Network.Hub**
  Identifies a network device that connects together network devices by repeating the signal received at one port to others.
- **Network.Network Interface Card**
  Identifies a Network Interface Card (NIC) using any network communications protocol. Ethernet LAN and FDDI Ring network cards are NICs.
- **Network.Other**
  Identifies unclassified network components.
- **Network.Peripheral**
  Identifies network peripherals that are appliances such as printers and FAX machines that contain their own NIC cards.
- **Network.Port**
  Identifies a port on a network hub or switch that is used to connect to other hubs and switches instead of an end station.
- **Network.Router**
  Identifies any device in a network that routes messages between computers.
- **Network.Switch**
  Identifies a network device that intelligently forwards packets received at one port to other ports.

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name	Description
Network.Bridge	Bridge	net_brix	ci_network_bridge	Network Bridge
Network.Controller	Controller	net_conx	ci_network_controller	Controller
Network.Frontend	3270 Terminal	net_frox	ci_network_frontend	3270 Terminal
Network.Frontend	Network Terminal	net_frox	ci_network_frontend	Network Terminal
Network.Frontend	X Terminal	net_frox	ci_network_frontend	X Terminal
Network.Hub	Network Hub	net_hubx	ci_network_hub	Hub on a network
Network.Network	Interface Card	net_nicx	ci_network_nic	Network Interface Card (NIC)
Network.Other	Other Network Device	net_othx	ci_network_other	Other Network Device
Network.Peripheral	Fax Machine	net_perx	ci_network_peripheral	Fax Machine
Network.Port	Port	net_porx	ci_network_port	Network Port
Network.Router	Router	net_roux	ci_network_router	Ethernet Router
Network.Switch	Network Switch	net_gatx	ci_network_gateway	Network Switch

Network.Bridge Attributes

The Network.Bridge family includes the following attributes that correspond to the net_brix extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.

addr_class	Address Class	IP address values are arranged in Address Classes (A, B, and C). The Address Classes determine how many workstations can possibly exist on the network.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
gateway_id	Gateway ID	The unique identifier for a network point that acts as an entrance (gateway) to another network
ip_mgmt_addr	Management IP Address	The IP address assigned to a station (PC or workstation) that is authorized for either manager- or operator-level access to a switch.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease begins
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.

number_net_card	Number of Network Cards	The number of expansion cards that have been installed in the available slots in a computer.
number_net_port	Number of Network Ports	The total number of ports in use on a server.
number__net_port_conn	Number of Network Port Connections	The total number of network port connections.
number_ports	Number of Ports	The total number of ports on a network device.
number_ports_used	Number of Ports Used	The total number of ports in use on a network device.
os_version	OS Version	The version number of a CI's operating system.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
protocol	Protocol	The communication method employed by a network device.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
role	Role	The business function supported by a hardware or network device, for example, production, test.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
subnet_mask	Subnet Mask	The identifier of the subnet into which a CI falls. Expressed in the same format as an IP address, for example, 255.128.0.0.
type_net_conn	Type of Network Connection	An indication of the kind of network connection used by a hardware or network device.

Network.Controller Attributes

The Network.Controller family includes the following attributes that correspond to the net_conx extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.

addr_class	Address Class	IP address values are arranged in Address Classes (A, B, and C). The Address Classes determine how many workstations can possibly exist on the network.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
ip_mgmt_addr	Management IP Address	The IP address assigned to a station (PC or workstation) that is authorized for either manager- or operator-level access to a switch.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease begins
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
number_mips	MIPS	An indication of the processing speed and capacity of a hardware or network device.

number_net_card	Number of Network Cards	The number of expansion cards that have been installed in the available slots in a computer.
number_net_port	Number of Network Ports	The total number of ports in use on a server.
number_net_port_conn	Number of Network Port Connections	The total number of network port connections.
number_ports	Number of Ports	The total number of ports on a network device.
number_ports_used	Number of Ports Used	The total number of ports in use on a network device.
number_proc_inst	Number of Processors Installed	The total number of processors installed on a hardware or network device.
number_smips	Number of SMIPS	The toal number of SMIPS.
os_version	OS Version	The version number of a CI's operating system.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
role	Role	The business function supported by a hardware or network device, for example, production, test.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
subnet_mask	Subnet Mask	The identifier of the subnet into which a CI falls. Expressed in the same format as an IP address, for example, 255.128.0.0.
technology	Technology	The technology (TCP/IP, Ethernet, FDDI, and so on) employed by a hardware or network device.
type_net_conn	Type of Network Connection	An indication of the kind of network connection used by a hardware or network device.

Network.Frontend Attributes

The Network.Frontend family includes the following attributes that correspond to the net_frox extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
addr_class	Address Class	IP address values are arranged in Address Classes (A, B, and C). The Address Classes determine how many workstations can possibly exist on the network.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
ip_mgmt_addr	Management IP Address	The IP address assigned to a station (PC or workstation) that is authorized for either manager- or operator-level access to a switch.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease begins
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.

network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
number_mips	MIPS	An indication of the processing speed and capacity of a hardware or network device.
number_net_card	Number of Network Cards	The number of expansion cards that have been installed in the available slots in a computer.
number_net_port	Number of Network Ports	The total number of ports in use on a server.
number_net_port_conn	Number of Network Port Connections	The total number of network port connections.
number_ports	Number of Ports	The total number of ports on a network device.
number_ports_used	Number of Ports Used	The total number of ports in use on a network device.
number_proc_inst	Number of Processors Installed	The total number of processors installed on a hardware or network device.
number_smips	Number of SMIPS	The total number of SMIPS.
os_version	OS Version	The version number of a CI's operating system.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
role	Role	The business function supported by a hardware or network device, for example, production, test.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
subnet_mask	Subnet Mask	The identifier of the subnet into which a CI falls. Expressed in the same format as an IP address, for example, 255.128.0.0.
technology	Technology	The technology (TCP/IP, Ethernet, FDDI, and so on) employed by a hardware or network device.
type_net_conn	Type of Network Connection	An indication of the kind of network connection used by a hardware or network device.

Network.Hub Attributes

The Network.Hub family includes the following attributes that correspond to the net_hubx extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
addr_class	Address Class	IP address values are arranged in Address Classes (A, B, and C). The Address Classes determine how many workstations can possibly exist on the network.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
gateway_id	Gateway ID	The unique identifier for a network point that acts as an entrance (gateway) to another network
ip_mgmt_addr	Management IP Address	The IP address assigned to a station (PC or workstation) that is authorized for either manager- or operator-level access to a switch.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease begins
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.

mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
net_card	Network Card	The designation for an expansion card that is installed in an available slot in a computer or network device so that it may connect and communicate to another networked component.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
number_net_card	Number of Network Cards	The number of expansion cards that have been installed in the available slots in a computer.
number_net_port	Number of Network Ports	The total number of ports in use on a server.
number_net_port_conn	Number of Network Port Connections	The total number of network port connections.
number_ports	Number of Ports	The total number of ports on a network device.
number_ports_used	Number of Ports Used	The total number of ports in use on a network device.
os_version	OS Version	The version number of a CI's operating system.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
role	Role	The business function supported by a hardware or network device, for example, production, test.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
subnet_mask	Subnet Mask	The identifier of the subnet into which a CI falls. Expressed in the same format as an IP address, for example, 255.128.0.0.

technology	Technology	The technology (TCP/IP, Ethernet, FDDI, and so on) employed by a hardware or network device.
type_net_conn	Type of Network Connection	An indication of the kind of network connection used by a hardware or network device.

Network.Network Interface Card Attributes

The Network.Network Interface Card family includes the following attributes that correspond to the net_nicx extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
addr_class	Address Class	IP address values are arranged in Address Classes (A, B, and C). The Address Classes determine how many workstations can possibly exist on the network.
gateway_id	Gateway ID	The unique identifier for a network point that acts as an entrance (gateway) to another network
ip_mgmt_addr	Management IP Address	The IP address assigned to a station (PC or workstation) that is authorized for either manager- or operator-level access to a switch.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
line_speed	Line Speed	The rate at which information is transmitted on a network connection.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
number_net_card	Number of Network Cards	The number of expansion cards that have been installed in the available slots in a computer.
number_net_port	Number of Network Ports	The total number of ports in use on a server.
number_net_port_conn	Number of Network Port Connections	The total number of network port connections.

number_ports	Number of Ports	The total number of ports on a network device.
number_ports_used	Number of Ports Used	The total number of ports in use on a network device.
os_version	OS Version	The version number of a CI's operating system.
protocol	Protocol	The communication method employed by a network device.
retire_date	Retire Date	The date on which a CI is no longer active.
role	Role	The business function supported by a hardware or network device, for example, production, test.
subnet_mask	Subnet Mask	The identifier of the subnet into which a CI falls. Expressed in the same format as an IP address, for example, 255.128.0.0.
technology	Technology	The technology (TCP/IP, Ethernet, FDDI, and so on) employed by a hardware or network device.
type_net_conn	Type of Network Connection	An indication of the kind of network connection used by a hardware or network device.

Network.Other Attributes

The Network.Other family includes the following attributes that correspond to the net_othx extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
bios_ver	BIOS Version	The version number of the BIOS - the code that is run when a personal computer starts up.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
gateway_id	Gateway ID	The unique identifier for a network point that acts as an entrance (gateway) to another network
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease begins

lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
os_version	OS Version	The version number of a CI's operating system.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
subnet_mask	Subnet Mask	The identifier of the subnet into which a CI falls. Expressed in the same format as an IP address, for example, 255.128.0.0.

Network.Peripheral Attributes

The Network.Peripheral family includes the following attributes that correspond to the net_perx extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
addr_class	Address Class	IP address values are arranged in Address Classes (A, B, and C). The Address Classes determine how many workstations can possibly exist on the network.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
gateway_id	Gateway ID	The unique identifier for a network point that acts as an entrance (gateway) to another network.
ip_mgmt_addr	Management IP Address	The IP address assigned to a station (PC or workstation) that is authorized for either manager- or operator-level access to a switch.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease begins
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.

mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
number_net_card	Number of Network Cards	The number of expansion cards that have been installed in the available slots in a computer.
number_net_port	Number of Network Ports	The total number of ports in use on a server.
number_net_port_conn	Number of Network Port Connections	The total number of network port connections.
number_ports	Number of Ports	The total number of ports on a network device.
number_ports_used	Number of Ports Used	The total number of ports in use on a network device.
os_version	OS Version	The version number of a CI's operating system.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
role	Role	The business function supported by a hardware or network device, for example, production, test.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
subnet_mask	Subnet Mask	The identifier of the subnet into which a CI falls. Expressed in the same format as an IP address, for example, 255.128.0.0.
technology	Technology	The technology (TCP/IP, Ethernet, FDDI, and so on) employed by a hardware or network device.
type_net_conn	Type of Network Connection	An indication of the kind of network connection used by a hardware or network device.

Network.Port Attributes

The Network.Port family includes the following attributes that correspond to the net_porx extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
addr_class	Address Class	IP address values are arranged in Address Classes (A, B, and C). The Address Classes determine how many workstations can possibly exist on the network.
channel_address	Channel Address	The tag used to identify a channel on a port.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
domain	Domain	The identifier of the logical grouping (domain) to which a network or telecom device is assigned.
gateway_id	Gateway ID	The unique identifier for a network point that acts as an entrance (gateway) to another network.
ip_mgmt_addr	Management IP Address	The IP address assigned to a station (PC or workstation) that is authorized for either manager- or operator-level access to a switch.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease begins
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
line_speed	Line Speed	The rate at which information is transmitted on a network connection.
line_type	Line Type	The categorization of a network communication line, for example, ISDN.

maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
number_net_card	Number of Network Cards	The number of expansion cards that have been installed in the available slots in a computer.
number_net_port	Number of Network Ports	The total number of ports in use on a server.
number_net_port_conn	Number of Network Port Connections	The total number of network port connections.
number_ports	Number of Ports	The total number of ports on a network device.
number_ports_used	Number of Ports Used	The total number of ports in use on a network device.
os_version	OS Version	The version number of a CI's operating system.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
protocol	Protocol	The communication method employed by a network device.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
role	Role	The business function supported by a hardware or network device, for example, production, test.

SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
subnet_mask	Subnet Mask	The identifier of the subnet into which a CI falls. Expressed in the same format as an IP address, for example, 255.128.0.0.
technology	Technology	The technology (TCP/IP, Ethernet, FDDI, and so on) employed by a hardware or network device.
type_net_conn	Type of Network Connection	An indication of the kind of network connection used by a hardware or network device.

Network.Router Attributes

The Network.Router family includes the following attributes that correspond to the net_roux extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
addr_class	Address Class	IP address values are arranged in Address Classes (A, B, and C). The Address Classes determine how many workstations can possibly exist on the network.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
flow	Flow	The amount of network traffic that can be handled by a router.
ip_mgmt_addr	Management IP Address	The IP address assigned to a station (PC or workstation) that is authorized for either manager- or operator-level access to a switch.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease begins
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.

lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mem_cache_proc	Processor Memory Cache	The identifier of the hardware device that processes the high-speed memory storage between memory and the CPU.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
number_net_card	Number of Network Cards	The number of expansion cards that have been installed in the available slots in a computer.
number_net_port	Number of Network Ports	The total number of ports in use on a server.
number_net_port_conn	Number of Network Port Connections	The total number of network port connections.
number_ports	Number of Ports	The total number of ports on a network device.
number_ports_used	Number of Ports Used	The total number of ports in use on a network device.
number_proc_inst	Number of Ports Installed	The total number of processors installed on a hardware or network device.
os_version	OS Version	The version number of a CI's operating system.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.

protocol	Protocol	The communication method employed by a network device.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
role	Role	The business function supported by a hardware or network device, for example, production, test.
rout_prot	Router Protocol	The communication method employed by a network router.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
slot_mem_used	Number of Memory Slots Used	The amount of memory in use from the available memory cards in a hardware or network device.
slot_total_mem	Number of Memory Slots	The total amount of memory available on memory cards in a hardware or network device.
subnet_mask	Subnet Mask	The identifier of the subnet into which a CI falls. Expressed in the same format as an IP address, for example, 255.128.0.0.
technology	Technology	The technology (TCP/IP, Ethernet, FDDI, and so on) employed by a hardware or network device.
type_net_conn	Type of Network Connection	An indication of the kind of network connection used by a hardware or network device.

Network.Switch Attributes

The Network.Switch family includes the following attributes that correspond to the net_gatx extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
addr_class	Address Class	IP address values are arranged in Address Classes (A, B, and C). The Address Classes determine how many workstations can possibly exist on the network.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.

gateway_id	Gateway ID	The unique identifier for a network point that acts as an entrance (gateway) to another network
graphics_card	Graphics Card Model	The model designation for an expansion card that is installed in an available slot in a device for enhanced graphics capabilities.
ip_mgmt_addr	Management IP Address	The IP address assigned to a station (PC or workstation) that is authorized for either manager- or operator-level access to a switch.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease begins
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
number_net_card	Number of Network Cards	The number of expansion cards that have been installed in the available slots in a computer.

number_net_port	Number of Network Ports	The total number of ports in use on a server.
number_net_port_conn	Number of Network Port Connections	The total number of network port connections.
number_ports	Number of Ports	The total number of ports on a network device.
number_ports_used	Number of Ports Used	The total number of ports in use on a network device.
os_version	OS Version	The version number of a CI's operating system.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
role	Role	The business function supported by a hardware or network device, for example, production, test.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
subnet_mask	Subnet Mask	The identifier of the subnet into which a CI falls. Expressed in the same format as an IP address, for example, 255.128.0.0.
technology	Technology	The technology (TCP/IP, Ethernet, FDDI, and so on) employed by a hardware or network device.
type_net_conn	Type of Network Connection	An indication of the kind of network connection used by a hardware or network device.

Organization Family

The Organization family includes the following:

- **Organization**
  Identifies an entity representing a structured group of persons.

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name	Description
Organization	External	orgx	ci_organization	An organization not part of the company

Organization	Internal	orgx	ci_organization	An organizational segment of the company

Organization Attributes

The Organization family contains the following class:

- **base_organization**
  Specifies the organization that the CI represents (SREL uuid to org). Represents an exclusive relationship where only one CI represents an organization in the Organization family.

The Organization family includes the following attributes that correspond to the net_orgx extension table:

Object Name	Label
alt_phone	Alternate Phone Number
billing_code	Billing Code
contact	Organization Contact
description	Description
email_addr	Email Address
fax_phone	Fax Number
location	Location
org_num	Organization Code
owning_contract	Service Contract
pemail_addr	Pager Email Address
phone_number	Primary Phone Number
service_type	Service Type
status	Configured Status

Security Family

Contents

The Security family includes the following:

- **Security**
  Identifies security systems that protect data, software, and hardware against unauthorized access or manipulation. These systems include digital certificates, directory services, and biomechanical or key-based systems.

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name	Description
Security	Application Security	secx	ci_security	Application Security
Security	Building Security	secx	ci_security	Building Security
Security	Data Security	secx	ci_security	Data Security
Security	Other Security	secx	ci_security	Miscellaneous Security

Security Attributes

The Security family includes the following attributes that correspond to the secx extension table:

Object Name	Label	Description
appl	Applies To	The designation of the domain of this security CI.
avail	Availability	An indication of when access to a security-related CI is offered.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
confidentiality_level	Confidentiality Level	The level of confidentiality (for example, view-only, high, medium, low) for a security-related CI.
integrity_level	Integrity Level	The level of integrity (for example, high, medium, low) for a particular security-related CI.
priority	Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
security_id	SecurityID	Number or other identifier for a security-related CI.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.

Service Family

WARNING
The Service family has been deprecated in CA CMDB Release 12.9. Use the Enterprise Service family instead.

The Service family comprises:

- **Service**
  Identifies an entity that delivers or performs a consistent set of tasks to a consumer. Services can be high-level business services or lower level IT technical services. Support, email, accounting are often delivered as services.

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name	Description
Service	Component	serx	ci_service	CA NSM Business Process View (BPV) Component
Service	Document	serx	ci_service	CA NSM BPV Document
Service	Person	serx	ci_service	CA NSM BPV Person
Service	Practice	serx	ci_service	CA NSM BPV Practice

Service	Process	serx	ci_service	CA NSM BPV Process
Service	Role	serx	ci_service	CA NSM BPV Role
Service	Service	serx	ci_service	CA NSM BPV Service

Service Attributes

The Service family includes the following attributes that correspond to the serx extension table:

Object Name	Label	Description
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
end_date	End Date	The date on which a service expires or is no longer valid.
portfolio	Service Portfolio	The name or identifier for a grouping of related services.
priority	Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
service_id	Service ID	The name or other unique identifier for a Service CI.
site	Site	A designation to describe the location of a CI.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
start_date	Start Date	The date on which a contract, document, service, or SLA becomes active.
type	Type	A description of the kind of Application, Contract, Document, Service, or SLA CI.
version	Version	A number or other identifier that indicates the current level (version) of an Application, Document, Service, or SLA CI.

Service Level Agreement (SLA) Family

The Service Level Agreement family includes the following classes that identify agreements between a service provider and consumer:

- Operational Level Agreement
- Other Service Level Agreement
- Service Level Agreement
- Underpinning Contract

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name
Service Level Agreement	Operational Level Agreement	slax	ci_sla
Service Level Agreement	Other Service Level Agreement	slax	ci_sla
Service Level Agreement	Service Level Agreement	slax	ci_sla
Service Level Agreement	Underpinning Contract	slax	ci_sla

Service Level Agreement Attributes

The Service Level Agreement family includes the following attributes that correspond to the slax extension table:

Object Name	Label	Description
sla_category	Service Level Agreement Category	The high-level type designation for an application, service, SLA, or document.
sla_date_active	Service Level Agreement Activation Date	The date on which the Configuration Item was made available to users.
sla_end_date	Service Level Agreement End Date	The date on which an SLA expires or is no longer valid.
sla_id	Service Level Agreement ID	The unique name or other identifier for a Service Level Agreement CI.
sla_start_date	Service Level Agreement Start Date	The date on which a contract, document, service, or SLA becomes active.
sla_status	Service Level Agreement Status	An indication of the status of an Application, Contract, Document, Service, or SLA CI (development, review, active, retired, and so on).
sla_type	Service Level Agreement Type	A description of the kind of Application, Contract, Document, Service, or SLA CI.
sla_version	Service Level Agreement Version	A number or other identifier that indicates the current level (version) of an Application, Document, Service, or SLA CI.

Software Families

The Software families include the following:

- **Software.Application System**
  Identifies a group of related applications that perform a high-level business function, such as an SAP Financials system.
- **Software.Application Component**

Identifies an aspect of a software program, application or application system, that is managed as part of a business transaction, and is responsible for a specific operation. Examples include a database connection or a web service.

- **Software.Application**
  Identifies programmatic components of the IT infrastructure.
- **Software.Application Server**
  Identifies a software engine that delivers client applications to client computers, typically through the Internet and using HTTP (Hypertext Transfer Protocol).
- **Software.Bespoke**
  Identifies software customized or constructed to order.
- **Software.COTS**
  Identifies software that was purchased, or leased and is manufactured outside of the owning company.
- **Software.Database**
  Identifies database management systems (DBMS) such as Oracle, DB2, and MS SQL.
- **Software.In-House**
  Identifies software applications developed by the company using the application.
- **Software.Operating System**
  Identifies system Software installed on a computer or similar device that provides basic services and enables other software to run.
- **Software.MessageServer**
  Identifies the software that processes incoming and outgoing mail message sent to/from authorized users.
- **Software.ESXHypervisor**
  Identifies the VMware software running on the OS of a VM hosting machine.
- **Software.HyperVHypervisor**
  Identifies the hyper-V software running on the OS of a VM hosting machine.
- Software.NetworkServer
  Identifies a server that is solely focused on providing protocol-specific functionality.
- Software.ResourceServer
  Identifies a system/software providing storage and query of resources - such as media - accessed via HTTP requests/responses.
- **Software.Schema**
  Identifies a named collection of data tables, stored procedures, and so on, that are managed and manipulated in a database management system (server).
- **Software.Tablespace**
  Identifies the description of the logical storage of a Database. A Tablespace bridges between the structure of the database (tables, indices, etc.) and a system's file system.
- **Software.VirtualManager**
  Identifies an application that manages a virtualization environment, the VM definitions, host machines and VMs running on the hosts.
- **Software.Website**
  Identifies a specialization of Group, representing a collection of files, designed for access using the World Wide Web, with a starting location/URI (also known as a "home page").

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name
Software.Application System	Application System	app_extx	ci_app_ext
Software.Application Component	SoftwareComponent	app_extx	ci_app_ext
Software.Application	Application	app_extx	ci_app_ext
Software.Application	Application Instance	app_extx	ci_app_ext

Software.Application Server	Application Server	app_extx	ci_app_ext
Software.Application Server	Application Server Instance	app_extx	ci_app_ext
Software.Bespoke	Bespoke	app_extx	ci_app_ext
Software.COTS	Batch	app_extx	ci_app_ext
Software.COTS	CICS	app_extx	ci_app_ext
Software.COTS	COTS	app_extx	ci_app_ext
Software.COTS	Network Software	app_extx	ci_app_ext
Software.COTS	Security	app_extx	ci_app_ext
Software.COTS	STC	app_extx	ci_app_ext
Software.COTS	TSO	app_extx	ci_app_ext
Software.COTS	WebSphere MQ	app_extx	ci_app_ext
Software.COTS	BackgroundProcess	app_extx	ci_app_ext
Software.COTS	BusinessProcessServer	app_extx	ci_app_ext
Software.COTS	Bootsoftware	app_extx	ci_app_ext
Software.COTS	ManagementAgent	app_extx	ci_app_ext
Software.Database	CA-Datacom	dat_basx	ci_database
Software.Database	CA-IDMS	dat_basx	ci_database
Software.Database	DB2	dat_basx	ci_database
Software.Database	IMS	dat_basx	ci_database
Software.Database	Ingres	dat_basx	ci_database
Software.Database	Oracle	dat_basx	ci_database
Software.Database	Other Software Database	dat_basx	ci_database
Software.Database	SQL	dat_basx	ci_database
Software.Database	Sybase	dat_basx	ci_database
Software.In-House	In-House	app_inhx	ci_app_inhouse
Software.Operating System	AIX OS	opsysx	ci_operating_system
Software.Operating System	HP UX OS	opsysx	ci_operating_system
Software.Operating System	Linux OS	opsysx	ci_operating_system
Software.Operating System	MVS OS	opsysx	ci_operating_system
Software.Operating System	OS/390 OS	opsysx	ci_operating_system
Software.Operating System	Other Software	opsysx	ci_operating_system
Software.Operating System	Sun OS	opsysx	ci_operating_system
Software.Operating System	Tandem OS	opsysx	ci_operating_system
Software.Operating System	Unisys OS	opsysx	ci_operating_system
Software.Operating System	UNIX OS	opsysx	ci_operating_system
Software.Operating System	Vax OS	opsysx	ci_operating_system
Software.Operating System	VM OS	opsysx	ci_operating_system
Software.Operating System	Windows OS	opsysx	ci_operating_system
Software.Operating System	z/OS OS	opsysx	ci_operating_system
Software.Schema	DatabaseSchema	app_extx	ci_app_ext
Software.DirectoryServer	LDAP	app_extx	ci_app_ext

Software.MessageServer	CommunicationServer	app_extx	ci_app_ext
Software.MessageServer	MailServer	app_extx	ci_app_ext
Software.ESXHypervisor	ESXHypervisor	app_esx	ci_app_esxhypervisor
Software.HyperVHypervisor	HyperVHypervisor	app_hyp	ci_app_hypervhypervisor
Software.NetworkServer	NetworkServer	app_netsvr	ci_app_netsvr
Software.ResourceServer	ResourceServer	app_ressvr	ci_app_ressvr
Software.ResourceServer	SecurityServer	app_ressvr	ci_app_ressvr
Software.ResourceServer	TransactionServer	app_ressvr	ci_app_ressvr
Software.ResourceServer	MessageServer	app_ressvr	ci_app_ressvr
Software.Tablespace	Tablespace	app_extx	ci_app_ext
Software.VirtualManager	VirtualizationManager	app_virmgr	ci_app_virtualmgr
Software.Website	Website	app_website	ci_app_website

Software Attributes

The following attributes correspond to the app_extx extension table and apply to the following families:

- Software.Application
- Software.Application Server
- Software.Bespoke
- Software.COTS

Object Name	Label	Description
app_id	Application ID	Application name or other unique identifier.
category	Category	The high-level type designation for an application.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on of a CI or group of CIs.
date_installed	Date Installed	The date on which the physical installation of a Configuration Item was completed.
environment	Environment	The application environment (for example, development, test, production) or project environment (for example, mainframe, distributed).
highavail_appl_resources	High Availability Resource	The name of the resource that provides high availability capability for an Application CI.
highly_avail	Under High Availability?	An indication (Yes/No) that an Application CI operates in a high availability production scenario.

inhouse_or_vendor	External Vendor	The internal department responsible for development/maintenance of this software.
install_dir	Installation Directory	The directory where an application stores its program files.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_end_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_start_date	Lease Effective Date	The date on which a lease begins.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific timeframe or was purchased.
main_process	Main Process	The designation of the main thread of an application process.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The timeframe for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
portfolio	Portfolio	A grouping of projects into a unit for management and tracking purposes.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
response_time	Response Time	The desired time measurement between the time a transaction is entered and the application returns a response.
server	Server	The name of the server on which an application runs.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
storage_used	Storage Used	The amount of available storage that is in use.

support_end_date	Support End Date	The date on which support for an application is no longer provided.
support_start_date	Support Start Date	The beginning date on which support for an application is provided.
support_type	Support Type	The kind of support that is provided for this CI, for example, gold/silver/bronze.
type	Type	A description of the kind of Application, Contract, Document, Service, or SLA CI.
uptime	Uptime	The desired "availability" that indicates the proportion of time a component is in a fully functioning condition.
version	Version	A number or other identifier that indicates the current level (version) of an Application, Document, Service, or SLA CI.

Software.Database Attributes

The Software.Database family includes the following attributes that correspond to the dat_basx extension table:

Object Name	Label	Description
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on of a CI or group of CIs.
db_id	Database ID	A name that uniquely identifies a database.
environment	Environment	The application environment (for example, development, test, production) or project environment (for example, mainframe, distributed).
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific timeframe or was purchased.
portfolio	Portfolio	A grouping of projects into a unit for management and tracking purposes.
priority	Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
server	Server	The name of the server on which an application runs.

SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
support_end_date	Support End Date	The date on which support for an application is no longer provided.
support_start_date	Support Start Date	The beginning date on which support for an application is provided.
support_type	Support Type	The kind of support that is provided for this CI, for example, gold/silver/bronze.
type	Type	A description of the kind of Application, Contract, Document, Service, or SLA CI.
version	Version	A number or other identifier that indicates the current level (version) of an Application, Document, Service, or SLA CI.

Software.In-House Attributes

The Software.In-House family includes the following attributes that correspond to the app_inhx extension table:

Object Name	Label	Description
app_id	Application ID	Application name or other unique identifier.
category	Category	The high-level type designation for an application, service, SLA, or document.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
date_installed	Date Installed	The date on which the physical installation of a Configuration Item was completed.
environment	Environment	The application environment (for example, development, test, production) or project environment (for example, mainframe, distributed).
highavail_appl_resources	High Availability Resource	The name of the resource that provides high availability capability for an Application CI.
highly_avail	Under High Availability?	An indication (Yes/No) that an Application CI operates in a high availability production scenario.
inhouse_or_vendor	Inhouse Department	The internal department responsible for development/maintenance of this software.
install_dir	Installation Directory	The directory where an application stores its program files.

leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific timeframe or was purchased.
main_process	Main Process	The designation of the main thread of an application process.
portfolio	Portfolio	A grouping of projects into a unit for management and tracking purposes.
priority	Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost of a CI.
response_time	Response Time	The desired time measurement between the time a transaction is entered and the application returns a response.
server	Server	The name of the server on which an application runs.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
storage_used	Storage Used	The amount of available storage that is in use.
support_end_date	Support End Date	The date on which support for an application is no longer provided.
support_start_date	Support Start Date	The beginning date on which support for an application is provided.
support_type	Support Type	The kind of support that is provided for this CI, for example, gold/silver/bronze.
type	Type	A description of the kind of Application, Contract, Document, Service, or SLA CI.
uptime	Uptime	The desired "availability", indicating the proportion of time a component is in a fully functioning condition.
version	Version	A number or other identifier that indicates the current level (version) of an Application, Document, Service, or SLA CI.

Software.Operating System Attributes

The Software.Operating System family includes the following attributes that correspond to the opsysx extension table:

Object Name	Label	Description
os_id	Operating System ID	Operating system name or other unique identifier.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on of a CI or group of CIs.
date_installed	Date Installed	The date on which the physical installation of a Configuration Item was completed.
environment	Environment	The application environment (for example, development, test, production) or project environment (for example, mainframe, distributed).
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_end_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_start_date	Lease Effective Date	The date on which a lease begins.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific timeframe or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The timeframe for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
server	Server	The name of the server on which an application runs.

SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
support_end_date	Support End Date	The date on which support for an application is no longer provided.
support_start_date	Support Start Date	The beginning date on which support for an application is provided.
support_type	Support Type	The kind of support that is provided for this CI, for example, gold/silver/bronze.
type	OS Type	A description of the kind of Application, Contract, Document, Service, or SLA CI.
version	Version	A number or other identifier that indicates the current level (version) of an Application, Document, Service, or SLA CI.

Software.ESXHypervisor Attributes

The Software.ESXHypervisor family includes the following attributes that corresponds to the app_esx extension table:

Object Name	Label	Description
DeviceAssetNumber	Device Asset Number	Number or other designator assigned to hardware by asset management, often the finance department, that is used for tracking ownership of the resource.
DeviceBiosSystemID	Device BIOS System ID	The identifier reported from the BIOS for the device.
DeviceDnsName	Device DNS Name	The fully qualified DNS name of the device.
DeviceSysName	Device System Name	The system name of the device, as defined in the system block of SNMP's MIB-II. Other information (than the MIB-II sysName) should NOT be used for this element.
DevicePhysSerialNumber	Device Physical Serial Number	An identifying string assigned by the hardware manufacturer and printed on a tag attached to the component.
DeviceMacAddress	Device MAC Address	A MAC address for the entity. The address is expressed as 6 groups of 2 hex digits (using only uppercase), separated by dashes.
DeviceIPV4Address	Device IPV4 Address	An IPv4 address for the device. The address is expressed using typical dotted decimal notation (4 groups of up to 3 decimal digits, separated by periods).

DeviceIPV4AddressWithDomain	Device IPV4 Address With Domain	An IPv4 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
DeviceIPV6Address	Device IPV6 Address	An IPv6 address for the device. The address is expressed using the formal, complete IPv6 notation (8 groups of up to 4 hex digits, using only uppercase where applicable, separated by colons).
DeviceIPV6AddressWithDomain	Device IPV6 Address With Domain	An IPv6 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
IsMigrationEnabled	Is Migration Enabled	Boolean indicating whether (or not) migration of VM's between hosts is enabled.
ComputeResourceIndex	Compute Resource Index	Identifier for a single host, acting as a "compute resource". Either a single system or a cluster can be a "compute resource".
HostIndex	Host Index	An index generated by VMware for the host, for example, host-6746.
DatacenterPath	Datacenter Path	Although hosts are managed by vCenter and its data centers, this element is provided for query purposes. It indicates the data center to which the host belongs.
FTVersion	FT Version	The version of Fault Tolerance running on the host. Only hosts with the same version of Fault Tolerance are compatible.
NumberOfPrimaryVMs	Number Of Primary VMs	The total number of primary VMs configured to this host, supported by fault tolerance.
NumberOfSecondaryVMs	Number Of Secondary VMs	The total number of secondary VMs configured to this host, to support fault tolerance.

Software.HyperVHypervisor Attributes

The Software.HyperVHypervisor family includes the following attributes that corresponds to the app_hyp extension table:

Object Name	Label	Description
DeviceAssetNumber	Device Asset Number	Number or other designator assigned to hardware by asset management, often the finance department, that is used for tracking ownership of the resource.
DeviceBiosSystemID	Device BIOS System ID	The identifier reported from the BIOS for the device.

DeviceDnsName	Device DNS Name	The fully qualified DNS name of the device.
DeviceSysName	Device System Name	The system name of the device, as defined in the system block of SNMP's MIB-II. Other information (than the MIB-II sysName) should NOT be used for this element.
DevicePhysSerialNumber	Device Physical Serial Number	An identifying string assigned by the hardware manufacturer and printed on a tag attached to the component.
DeviceMacAddress	Device MAC Address	A MAC address for the entity. The address is expressed as 6 groups of 2 hex digits (using only uppercase), separated by dashes.
DeviceIPV4Address	Device IPV4 Address	An IPv4 address for the device. The address is expressed using typical dotted decimal notation (4 groups of up to 3 decimal digits, separated by periods).
DeviceIPV4AddressWithDomain	Device IPV4 Address With Domain	An IPv4 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
DeviceIPV6Address	Device IPV6 Address	An IPv6 address for the device. The address is expressed using the formal, complete IPv6 notation (8 groups of up to 4 hex digits, using only uppercase where applicable, separated by colons).
DeviceIPV6AddressWithDomain	Device IPV6 Address With Domain	An IPv6 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
IsMigrationEnabled	Is Migration Enabled	Boolean indicating whether (or not) migration of VM's between hosts is enabled.
DefaultExternalDataRoot	Default External Data Root	The fully-qualified path to the default location for the Hyper-V files.
DefaultVhdPath	Default VHD Path	The fully-qualified path to the default location for the Hyper-V Virtual Hard Disk files.
MinimumMacAddress	Minimum MAC Address	The minimum MAC address for dynamically generated MAC addresses. The address is expressed as 6 groups of 2 hex digits (using only uppercase), separated by dashes.
MaximumMacAddress	Maximum MAC Address	The maximum MAC address for dynamically generated MAC addresses. The address is expressed as 6 groups of 2 hex digits (using only uppercase), separated by dashes.

Software.NetworkServer Attributes

The Software.NetworkServer family includes the following attributes that corresponds to the app_netsvr extension table:

Object Name	Label	Description
DeviceAssetNumber	Device Asset Number	Number or other designator assigned to hardware by asset management, often the finance department, that is used for tracking ownership of the resource.
DeviceBiosSystemID	Device BIOS System ID	The identifier reported from the BIOS for the device.
DeviceDnsName	Device DNS Name	The fully qualified DNS name of the device.
DeviceSysName	Device System Name	The system name of the device, as defined in the system block of SNMP's MIB-II. Other information (than the MIB-II sysName) should NOT be used for this element.
DevicePhysSerialNumber	Device Physical Serial Number	An identifying string assigned by the hardware manufacturer and printed on a tag attached to the component.
DeviceMacAddress	Device MAC Address	A MAC address for the entity. The address is expressed as 6 groups of 2 hex digits (using only uppercase), separated by dashes.
DeviceIPV4Address	Device IPV4 Address	An IPv4 address for the device. The address is expressed using typical dotted decimal notation (4 groups of up to 3 decimal digits, separated by periods).
DeviceIPV4AddressWithDomain	Device IPV4 Address With Domain	An IPv4 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
DeviceIPV6Address	Device IPV6 Address	An IPv6 address for the device. The address is expressed using the formal, complete IPv6 notation (8 groups of up to 4 hex digits, using only uppercase where applicable, separated by colons).
DeviceIPV6AddressWithDomain	Device IPV6 Address With Domain	An IPv6 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
ProcessID	Process ID	The PID of the running software, as defined by the operating system. This value is important to distinguish between multiple running instances of the same ProvisionedSoftware.

AccessedViaTcpPort	Accessed Via TCP Port	The TCP port number to use when communicating with the software.
ProcessDistinguishingID	Process Distinguishing ID	A string providing an additional identifier/ distinguisher for RunningSoftware when the ProcessID and TCP port data are not available or are not sufficient to distinguish the instances (for example, for an Application where little information can be discovered via the current access mechanisms).
Protocol	Protocol	Defines the protocol supported by the NetworkServer - such as DHCP and DNS.
ContextID	Context ID	The ContextID element is used for identification.

Software.ResourceServer Attributes

The Software.ResourceServer family includes the following attributes that corresponds to the app_ressvr extension table:

Object Name	Label	Description
DeviceAssetNumber	Device Asset Number	Number or other designator assigned to hardware by asset management, often the finance department, that is used for tracking ownership of the resource.
DeviceBiosSystemID	Device BIOS System ID	The identifier reported from the BIOS for the device.
DeviceDnsName	Device DNS Name	The fully qualified DNS name of the device.
DeviceSysName	Device System Name	The system name of the device, as defined in the system block of SNMP's MIB-II. Other information (than the MIB-II sysName) should NOT be used for this element.
DevicePhysSerialNumber	Device Physical Serial Number	An identifying string assigned by the hardware manufacturer and printed on a tag attached to the component.
DeviceMacAddress	Device MAC Address	A MAC address for the entity. The address is expressed as 6 groups of 2 hex digits (using only uppercase), separated by dashes.
DeviceIPV4Address	Device IPV4 Address	An IPv4 address for the device. The address is expressed using typical dotted decimal notation (4 groups of up to 3 decimal digits, separated by periods).
DeviceIPV4AddressWithDomain	Device IPV4 Address With Domain	An IPv4 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").

DeviceIPV6Address	Device IPV6 Address	An IPv6 address for the device. The address is expressed using the formal, complete IPv6 notation (8 groups of up to 4 hex digits, using only uppercase where applicable, separated by colons).
DeviceIPV6AddressWithDomain	Device IPV6 Address With Domain	An IPv6 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
ProcessID	Process ID	The PID of the running software, as defined by the operating system. This value is important to distinguish between multiple running instances of the same ProvisionedSoftware.
AccessedViaTcpPort	Accessed Via TCP Port	The TCP port number to use when communicating with the software.
ProcessDistinguishingID	Process Distinguishing ID	A string providing an additional identifier/ distinguisher for RunningSoftware when the ProcessID and TCP port data are not available or are not sufficient to distinguish the instances (for example, for an Application where little information can be discovered via the current access mechanisms).
capabilities	capabilities	A comma-separated list of the capabilities of the Server.
ContextID	Context ID	The ContextID element is used for identification.

Software.VirtualManager Attributes

The Software.VirtualManager family includes the following attributes that corresponds to the app_virmgr extension table:

Object Name	Label	Description
DeviceAssetNumber	Device Asset Number	Number or other designator assigned to hardware by asset management, often the finance department, that is used for tracking ownership of the resource.
DeviceBiosSystemID	Device BIOS System ID	The identifier reported from the BIOS for the device.
DeviceDnsName	Device DNS Name	The fully qualified DNS name of the device.
DeviceSysName	Device System Name	The system name of the device, as defined in the system block of SNMP's MIB-II. Other information (than the MIB-II sysName) should NOT be used for this element.

DevicePhysSerialNumber	Device Physical Serial Number	An identifying string assigned by the hardware manufacturer and printed on a tag attached to the component.
DeviceMacAddress	Device MAC Address	A MAC address for the entity. The address is expressed as 6 groups of 2 hex digits (using only uppercase), separated by dashes.
DeviceIPV4Address	Device IPV4 Address	An IPv4 address for the device. The address is expressed using typical dotted decimal notation (4 groups of up to 3 decimal digits, separated by periods).
DeviceIPV4AddressWithDomain	Device IPV4 Address With Domain	An IPv4 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
DeviceIPV6Address	Device IPV6 Address	An IPv6 address for the device. The address is expressed using the formal, complete IPv6 notation (8 groups of up to 4 hex digits, using only uppercase where applicable, separated by colons).
DeviceIPV6AddressWithDomain	Device IPV6 Address With Domain	An IPv6 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
ProcessID	Process ID	The PID of the running software, as defined by the operating system. This value is important to distinguish between multiple running instances of the same ProvisionedSoftware.
AccessedViaTcpPort	Accessed Via TCP Port	The TCP port number to use when communicating with the software.
ProcessDistinguishingID	Process Distinguishing ID	A string providing an additional identifier/ distinguisher for RunningSoftware when the ProcessID and TCP port data are not available or are not sufficient to distinguish the instances (for example, for an Application where little information can be discovered via the current access mechanisms).
ApiVersion	API Version	The version identifier for the APIs applicable to the Manager.
ContextID	Context ID	The ContextID element is used for identification.

Software.Website Attributes

The Software.Website family includes the following attributes that corresponds to the app_website extension table:

Object Name	Label	Description
DeviceAssetNumber	Device Asset Number	Number or other designator assigned to hardware by asset management, often the finance department, that is used for tracking ownership of the resource.
DeviceBiosSystemID	Device BIOS System ID	The identifier reported from the BIOS for the device.
DeviceDnsName	Device DNS Name	The fully qualified DNS name of the device.
DeviceSysName	Device System Name	The system name of the device, as defined in the system block of SNMP's MIB-II. Other information (than the MIB-II sysName) should NOT be used for this element.
DevicePhysSerialNumber	Device Physical Serial Number	An identifying string assigned by the hardware manufacturer and printed on a tag attached to the component.
DeviceMacAddress	Device MAC Address	A MAC address for the entity. The address is expressed as 6 groups of 2 hex digits (using only uppercase), separated by dashes.
DeviceIPV4Address	Device IPV4 Address	An IPv4 address for the device. The address is expressed using typical dotted decimal notation (4 groups of up to 3 decimal digits, separated by periods).
DeviceIPV4AddressWithDomain	Device IPV4 Address With Domain	An IPv4 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
DeviceIPV6Address	Device IPV6 Address	An IPv6 address for the device. The address is expressed using the formal, complete IPv6 notation (8 groups of up to 4 hex digits, using only uppercase where applicable, separated by colons).
DeviceIPV6AddressWithDomain	Device IPV6 Address With Domain	An IPv6 address for the device, prefixed by a contextual domain name, where the domain name is separated from the address by a dash ("-").
IsHAEnabled	Is HA Enabled	Boolean indicating whether the group (typically a Cluster instance or a GroupType="ResourceCluster") supports high availability/failover (if true) or not (if false).

IsMonitoringMembers	Is Monitoring Members	Boolean indicating whether the group (typically a Cluster instance or a GroupType="ResourceCluster") currently is using a heartbeat or other individual monitoring technique to determine if availability is lost (if true) or whether this capability does not exist or is not currently enabled (if false).
MaxFailures	Max Failures	Number of failures that can be supported by the Group, by sparing or other load balancing means.
MemberCriteria	Member Criteria	Defines a comma-separated list of QNames, representing the constraints on the constituency of the Group. Instances referenced as the Target element in the HasMember relationship semantic should have one of the types specified.
GroupType	Group Type	An enumerated value describing the primary type or category of the Group, such as an "authorization group", a "virtual resource pool" or a "data center".
HomePage	Home Page	The URL of the web site's home page.
BusinessRelevance	Business Relevance	Description of the relevance of the Entity, to the business.

J2EE Conventions

CA CMDB includes the following families of Software CIs:

- Software.COTS extension: ci_app_ext
- Software.Application extension: ci_app_ext
- Software.Application Server extension: ci_app_ext
- Software.Bespoke extension: ci_app_ext
- Software.In-House extension: ci_app_inhouse
- Software.Database extension: ci_database
- Software.Operating System extension: ci_operating_system

The existing CA CMDB/CA Cohesion ACM integration uses the family Software.COTS for all software CIs including both J2EE applications and J2EE application servers. Software CIs are reconciled by their system_name attribute which is composed in the following format:

HostName|AppName|Version|Qualifier

In addition, these CIs are named using a similar format:

AppName|Version|Qualifier

Families, Classes, and Reconciliation for J2EE

The following reconciliation considerations apply if you use J2EE Application or J2EE Application Server CIs:

- If you have J2EE Application and Application Server CIs, and you do *not* have CA Wily products or other MDR sources for J2EE CIs, you can use your existing reconciliation strategy. CA Cohesion ACM provides an export capability that you can customize to behave in its original mode.
- If you intend to use CA Wily products, and you already have CIs discovered by CA Cohesion ACM, you can write a script to mark your existing J2EE CIs as Inactive. Then the CIs can be rediscovered using the new CA CMDB integrations.

For future integrations with CA CMDB, use the following Families, Classes, and reconciliation key for J2EE Application CIs:

Objects	Values
Family	Software.Application
Extension	ci_app_ext
Class	Application
Attributes	Category and Type **Note:** These attributes distinguish J2EE applications from other kinds of applications.
Reconciliation attributes/key	Name: AppName|Port system_name: HostName|AppName|Port

For future integrations with CA CMDB, use the following Families, Classes, and reconciliation key for J2EE Application Server CIs:

Objects	Values
Family	Software.Application Server
Extension	ci_app_ext
Class	Application Server
Attributes	Category and Type **Note:** These attributes distinguish J2EE applications from other kinds of applications.
Reconciliation attributes/key	Name: *Host*Name|Port system_name: HostName|Port

Storage Area Network (SAN) Families

The Storage Area Network (SAN) families include the following:

- **SAN.Interface**
  Identifies a fiber channel interface, similar to a network interface card, used in a SAN fabric.
- **SAN.Switch**
  Identifies a fiber channel switch, similar to a network switch, used in a SAN fabric.

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name	Description
SAN.Interface	Interface	net_nicx	ci_network_nic	Interface to a SAN
SAN.Switch	Hub	net_hubx	ci_network_hub	Hub on a SAN
SAN.Switch	Switch	net_hubx	ci_network_hub	Switch on a SAN

SAN.Interface Attributes

The SAN.Interface family includes the following attributes that correspond to the net_nicx extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
addr_class	Address Class	IP address values are arranged in Address Classes (A, B, and C). The Address Classes determine how many workstations can exist on the network.
gateway_id	Gateway ID	The unique identifier for a network point that acts as an entrance (gateway) to another network
ip_mgmt_addr	Management IP Address	The IP address assigned to a station (PC or workstation) that is authorized for either manager- or operator-level access to a switch.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
line_speed	Line Speed	The rate at which information is transmitted on a network connection.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
number_net_card	Number of Network Cards	The number of expansion cards that have been installed in the available slots in a computer.
number_net_port	Number of Network Ports	The total number of ports in use on a server.
number_net_port_conn	Number of Network Port Connections	The total number of ports on a server.
number_ports	Number of Ports	The total number of ports on a network device.
number_ports_used	Number of Ports Used	The total number of ports in use on a network device.
os_version	OS Version	The version number of a CI's operating system.

protocol	Protocol	The communication method employed by a network device.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
role	Role	The business function supported by a hardware or network device, for example, production, test.
subnet_mask	Subnet Mask	The identifier of the subnet into which a CI falls. Expressed in the same format as an IP address, for example, 255.128.0.0.
technology	Technology	The technology (TCP/IP, Ethernet, FDDI, and so on) employed by a hardware or network device.
type_net_conn	Type of Network Connection	An indication of the kind of network connection used by a hardware or network device.

SAN.Switch Attributes

The SAN.Switch family includes the following attributes that correspond to the net_hubx extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
addr_class	Address Class	IP address values are arranged in Address Classes (A, B, and C). The Address Classes determine how many workstations can exist on the network.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and son on, of a CI or group of CIs.
gateway_id	Gateway ID	The unique identifier for a network point that acts as an entrance (gateway) to another network
ip_mgmt_addr	Management IP Address	The IP address assigned to a station (PC or workstation) that is authorized for either manager- or operator-level access to a switch.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.

lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_effective_date	Lease Effective Date	The date on which a lease begins.
leased_or_owned_status	Leased or Owned	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
number_net_card	Number of Network Cards	The number of expansion cards that have been installed in the available slots in a computer.
number_net_port	Number of Network Ports	The total number of ports in use on a server.
number_net_port_conn	Number of Network Port Connections	The total number of ports on a server.
number_ports	Number of Ports	The total number of ports on a network device.
number_ports_used	Number of Ports Used	The total number of ports in use on a network device.
os_version	OS Version	The version number of a CI's operating system.
priority	Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
project code	Project Code	
protocol	Protocol	The communication method employed by a network device.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.

role	Role	The business function supported by a hardware or network device, for example, production, test.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
subnet_mask	Subnet Mask	The identifier of the subnet into which a CI falls. Expressed in the same format as an IP address, for example, 255.128.0.0.
technology	Technology	The technology (TCP/IP, Ethernet, FDDI, and so on) employed by a hardware or network device.
type_net_conn	Type of Network Connection	An indication of the kind of network connection used by a hardware or network device.

Telecom Families

The Telecom families include the following:

- **Telecom.Circuit**
  Identifies a dedicated connection between two nodes of a telecommunications network.
- **Telecom.Other**
  Identifies miscellaneous telecom components.
- **Telecom.Radio**
  Identifies an RF receiver or transmitter.
- **Telecom.Voice**
  Identifies a multiplexed connection supporting multiple voice lines on the same circuit.
- **Telecom.Wireless**
  Identifies telecom devices that do not rely on land lines, such as mobile or cellular phones, or wireless handsets or headsets.

Family	Class	Extension Table / Logical Name	Extension Table / Physical Name
Telecom.Circuit	Communication Circuit	tel_cirx	ci_telcom_circuit
Telecom.Circuit	Other Telecom Circuit	tel_cirx	ci_telcom_circuit
Telecom.Circuit	Satellite Link	tel_cirx	ci_telcom_circuit
Telecom.Other	ACD	tel_othx	ci_telcom_other
Telecom.Other	IVR	tel_othx	ci_telcom_other
Telecom.Other	Other Telecom	tel_othx	ci_telcom_other
Telecom.Other	PDA	tel_othx	ci_telcom_other
Telecom.Other	Video Conferencing Unit	tel_othx	ci_telcom_other
Telecom.Radio	Other Telecom Radio	tel_radx	ci_telcom_radio
Telecom.Radio	Radio Data Modem	tel_radx	ci_telcom_radio
Telecom.Radio	Radio Handsets	tel_radx	ci_telcom_radio

Telecom.Voice	Centrex	tel_voix	ci_telcom_voice
Telecom.Voice	Conference Bridge Line	tel_voix	ci_telcom_voice
Telecom.Voice	Desk Phone	tel_voix	ci_telcom_voice
Telecom.Voice	Other Telecom Voice	tel_voix	ci_telcom_voice
Telecom.Voice	PBX	tel_voix	ci_telcom_voice
Telecom.Voice	Phone Card	tel_voix	ci_telcom_voice
Telecom.Wireless	Mobile Phone	tel_wirx	ci_telcom_wireless
Telecom.Wireless	Other Telecom Wireless	tel_wirx	ci_telcom_wireless
Telecom.Wireless	Pager	tel_wirx	ci_telcom_wireless

Telecom.Circuit Attributes

The Telecom.Circuit family includes the following attributes that correspond to the tel_cirx extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
bandwidth	Bandwidth	The amount of data that can be carried in a given time period over a wired or wireless communications link. Usually specified as bits per second, KB per second, MB per second, and so on.
carrier	Carrier	A company that provides telecommunication services, such as AT&T, Cingular, Sprint, Verizon, and so on.
circuit_number	Circuit Number	The number issued by the phone company that uniquely identifies a circuit.
circuit_type	Circuit Type	The high-level type designation for a telecommunication circuit.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease begins.
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.

leased_or_owned_status	Leased or Ownedv	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
server_id	Server ID	The name or other unique identifier for a server.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.

Telecom.Other Attributes

The Telecom.Other family includes the following attributes that correspond to the tel_othx extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.

bandwidth	Bandwidth	The amount of data that can be carried in a given time period over a wired or wireless communications link. Usually specified as bits per second, KB per second, MB per second, and so on.
bios_ver	BIOS Version	The version number of the BIOS - the code that's run when a personal computer starts up.
carrier	Carrier	A company that provides telecommunication services, such as AT&T, Cingular, Sprint, Verizon, and so on.
circuit_number	Circuit Number	The number issued by the phone company that uniquely identifies a circuit.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
cpu_type	CPU Type	The type (and speed) of the central processor in a telecom device.
domain	Domain	The identifier of the logical grouping (domain) to which a network or telecom device is assigned.
frequency	Frequency	The wavelength at which a telecom signal is transmitted to a wireless or radio device.
gateway_id	Gateway ID	The unique identifier for a network point that acts as an entrance (gateway) to another network
harddrive_capacity	Hard Drive Capacity	The amount of hard drive capacity that is available for use on a Telecom CI.
harddrive_used	Hard Drive Space Used	The amount of hard drive capacity that is being used.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease begins.
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Ownedv	An indication of whether a particular CI has been leased for a specific time frame or was purchased.

license_expiration_date	License Expiration Date	The date on which a hardware or software license expires.
ci_license_number	CI License Number	The valid license number for a hardware or software CI.
line_id	Line ID	The designation that uniquely identifies a telecommunication line.
main_extension	Main Extension	The primary telephone number for a business.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
memory_available	Memory Available	The amount of memory that is still available for use.
memory_used	Memory Used	The amount of the available memory that is in use.
monitor	Monitor Model	The type of display unit connected to a hardware, network, or telecom device.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
nic_card	NIC Card	Each device (Node) on a network has a Network Interface Card (NIC). The NIC can be Ethernet, Token Ring, RF, or other. The NIC is installed inside the device and provides a real-time dedicated connection to the network.
phone_number	Phone Number	The number issued by the phone company that uniquely identifies a land line or cellular telephone connection.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.

retire_date	Retire Date	The date on which a CI is no longer active.
server_id	Server ID	The name or other unique identifier for a server.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
subnet_mask	Subnet Mask	The identifier of the subnet into which a CI falls. Expressed in the same format as an IP address for example, 255.128.0.0.

Telecom.Wireless Attributes

The Telecom.Wireless family includes the following attributes that correspond to the tel_wirx extension table:

Object Name	Label	Description
active_date	Activation Date	The date on which the CI was put into active status.
bandwidth	Bandwidth	The amount of data that can be carried in a given time period over a wired or wireless communications link. Usually specified as bits per second, KB per second, MB per second, and so on.
bios_ver	BIOS Version	The version number of the BIOS - the code that's run when a personal computer starts up.
carrier	Carrier	A company that provides telecommunication services, such as AT&T, Cingular, Sprint, Verizon, and so on.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
cpu_type	CPU Type	The type (and speed) of the central processor in a telecom device.
domain	Domain	The identifier of the logical grouping (domain) to which a network or telecom device is assigned.
frequency	Frequency	The wavelength at which a telecom signal is transmitted to a wireless or radio device.
gateway_id	Gateway ID	The unique identifier for a network point that acts as an entrance (gateway) to another network
harddrive_capacity	Hard Drive Capacity	The amount of hard drive capacity that is available for use on a Telecom CI.

harddrive_used	Hard Drive Space Used	The amount of hard drive capacity that is being used.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease begins.
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Ownedv	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
license_expiration_date	License Expiration Date	The date on which a hardware or software license expires.
ci_license_number	CI License Number	The valid license number for a hardware or software CI.
line_id	Line ID	The designation that uniquely identifies a telecommunication line.
main_extension	Main Extension	The primary telephone number for a business.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
memory_available	Memory Available	The amount of memory that is still available for use.
memory_used	Memory Used	The amount of the available memory that is in use.
monitor	Monitor Model	The type of display unit connected to a hardware, network, or telecom device.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.

network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
nic_card	NIC Card	Each device (Node) on a network has a Network Interface Card (NIC). The NIC can be Ethernet, Token Ring, RF, or other. The NIC is installed inside the device and provides a real-time dedicated connection to the network.
phone_number	Phone Number	The number issued by the phone company that uniquely identifies a land line or cellular telephone connection.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
subnet_mask	Subnet Mask	The identifier of the subnet into which a CI falls. Expressed in the same format as an IP address for example, 255.128.0.0.

Telecom.Radio Attributes

The Telecom.Radio family includes the following attributes that correspond to the tel_radx extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
bandwidth	Bandwidth	The amount of data that can be carried in a given time period over a wired or wireless communications link. Usually specified as bits per second, KB per second, MB per second, and so on.
bios_ver	BIOS Version	The version number of the BIOS - the code that's run when a personal computer starts up.

contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
cpu_type	CPU Type	The type (and speed) of the central processor in a telecom device.
domain	Domain	The identifier of the logical grouping (domain) to which a network or telecom device is assigned.
frequency	Frequency	The wavelength at which a telecom signal is transmitted to a wireless or radio device.
gateway_id	Gateway ID	The unique identifier for a network point that acts as an entrance (gateway) to another network
harddrive_capacity	Hard Drive Capacity	The amount of hard drive capacity that is available for use on a Telecom CI.
harddrive_used	Hard Drive Space Used	The amount of hard drive capacity that is being used.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease begins.
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Ownedv	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
license_expiration_date	License Expiration Date	The date on which a hardware or software license expires.
ci_license_number	CI License Number	The valid license number for a hardware or software CI.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
memory_available	Memory Available	The amount of memory that is still available for use.
memory_used	Memory Used	The amount of the available memory that is in use.

monitor	Monitor Model	The type of display unit connected to a hardware, network, or telecom device.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.
mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
nic_card	NIC Card	Each device (Node) on a network has a Network Interface Card (NIC). The NIC can be Ethernet, Token Ring, RF, or other. The NIC is installed inside the device and provides a real-time dedicated connection to the network.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.
subnet_mask	Subnet Mask	The identifier of the subnet into which a CI falls. Expressed in the same format as an IP address for example, 255.128.0.0.

Telecom.Voice Attributes

The Telecom.Voice family includes the following attributes that correspond to the tel_voix extension table:

Object Name	Label	Description
active_date	Active Date	The date on which the CI was put into active status.
carrier	Carrier	A company that provides telecommunication services, such as AT&T, Cingular, Sprint, Verizon, and so on.

circuit_number	Circuit Number	The number issued by the phone company that uniquely identifies a circuit.
contract_number	Contract Number	The unique identifier for a legal contract covering the purchase, lease, warranty, maintenance, and so on, of a CI or group of CIs.
cpu_type	CPU Type	The type (and speed) of the central processor in a telecom device.
harddrive_capacity	Hard Drive Capacity	The amount of hard drive capacity that is available for use on a Telecom CI.
harddrive_used	Hard Drive Space Used	The amount of hard drive capacity that is being used.
last_mtce_date	Last Maintenance Date	The latest date on which maintenance was performed on a CI.
lease_cost_per_month	Monthly Lease Cost	The dollar amount owed to the vendor each month for a lease.
lease_effective_date	Lease Effective Date	The date on which a lease begins.
lease_renewal_date	Lease Renewal Date	The date on which a lease must be renewed for the next time period, or the affected CIs must be returned to the vendor.
lease_termination_date	Lease Termination Date	The date on which a lease ends and the affected CIs must be returned to the vendor.
leased_or_owned_status	Leased or Ownedv	An indication of whether a particular CI has been leased for a specific time frame or was purchased.
main_extension	Main Extension	The primary telephone number for a business.
maintenance_fee	Maintenance Fee	The amount of money paid to cover the cost of maintenance services over a specified time period.
maintenance_period	Maintenance Period	The time frame for which a maintenance contract is active.
memory_available	Memory Available	The amount of memory that is still available for use.
memory_used	Memory Used	The amount of the available memory that is in use.
monitor	Monitor Model	The type of display unit connected to a hardware, network, or telecom device.
mtce_contract_number	Maintenance Contract Number	The number that uniquely identifies a maintenance contract.
mtce_level	Maintenance Level	An indication of the current patch version for this CI.

mtce_type	Maintenance Type	The kind of maintenance that is provided for this CI, for example, vendor or in-house.
network_address	Network Address	The IP address at which this CI resides, for example, 192.168.0.4.
network_name	Network Name	The unique name or identifier for a communications system that connects two or more computers and their peripheral devices.
nic_card	NIC Card	Each device (Node) on a network has a Network Interface Card (NIC). The NIC can be Ethernet, Token Ring, RF, or other. The NIC is installed inside the device and provides a real-time dedicated connection to the network.
phone_number	Phone Number	The number issued by the phone company that uniquely identifies a land line or cellular telephone connection.
ci_priority	CI Priority	The service level designation that is assigned to indicate the priority for restoration of this CI.
proj_code	Project Code	The ID or other unique identifier for the project to which a CI is assigned.
purchase_amountc	Purchase Amount	The cost incurred to buy a CI.
retire_date	Retire Date	The date on which a CI is no longer active.
SLA	Service Level Agreement	The name or identifier of the contract between IT and the customer that governs the level of service and support options that are expected and acceptable for this IT component.

General Resource Loader - GRLoader

GRLoader Considerations

Review the following considerations before you use GRLoader:

- GRLoader Release 12.9 is compatible with earlier releases of CA CMDB, but early releases of GRLoader are incompatible with CA CMDB Release 12.9. For example, a site has an existing CA CMDB r11.2 installation and later installs CA CMDB Release 12.9, so the site has two installations. GRLoader Release 12.9 works with both installed systems, but GRLoader r11.2 only works with the r11.2 installation. You can use the - s parameter to specify which one of the multiple installations for GRLoader to use.
- We recommend that you migrate all your MDR scripts to use the latest version of GRLoader. If you want to insert new CIs, migration requires altering existing scripts to specify -n.

NOTE

If you do not specify -n or -a, GRLoader does not insert or update CIs and relationships.

- You can specify GRLoader options in a configuration file.
- Whenever you update a CI with the -a option, Last Change Date and user displayed in the Configuration Item List are updated even if no attributes were changed. This update occurs whether a CI was edited in the user interface (and saved without changes) or updated by using GRLoader.
- If GRLoader generates a warning, the error log records the CI information, an error, or a skip. For example, when GRLoader loads CIs, but -a is not specified to allow updates, the information appears in the error log with an appropriate message.
- When you run GRLoader from a batch file, specify the *-ad attr=value* as *-ad attr{value}* to get past the Windows command parser which may remove the equal "=" symbols.
- Do *not* use curly brackets "{ }" as work area delimiter characters when you want GRLoader to insert CIs into the CMDB. TWA delimiters work properly when inserting CIs from general notes, CSV and Excel files, and JDBC/ODBC databases.
- GRLoader does not support smart quotes. Use double quote characters, or any other single character delimiter instead.
- GRLoader does not let you create NULL or empty relationships between CIs. The CA SDM web interface lets you create these relationships, but GRLoader displays the *ERROR:Relationship type is required* message in the generated error XML file.
- GRLoader does not let you create duplicate relationships between the same two CIs. Instead, GRLoader tries to update the existing relationship between the same two CIs that have the same relationship type.
  For example, the CMDB contains a relationship such as *ci_1 manages ci_2*. When you try to insert the same relationship through GRLoader, the utility tries to update the existing relationship, instead of creating a new relationship. The CA SDM web interface lets you create a duplicate relationship between the same two CIs, or update the existing relationship.
- You cannot use the *SELECT ** syntax on UNIX systems.

Use Database Queries to Verify Correct Data

We recommend that you use database queries before executing GRLoader. These queries help you debug imports to SQL Server. For example, you want to specify Server as the class in your SQL statement.

Follow these steps:

1. Create a configuration file named sqlServer.cfg with the following code:
   ```
   grloader.jdbc.driver=com.microsoft.sqlserver.jdbc.SQLServerDriver
   grloader.jdbc.url=jdbc:sqlserver://sqlserverhostname:1433;databaseName=mdb;
   grloader.jdbc.user=userid
   grloader.jdbc.password=password
   ```
2. Save the file.
3. Execute the following query in SQL Server:
   ```
   select ca_owned_resource.resource_name as name,
          ca_resource_class.name as class
          from ca_owned_resource, ca_resource_class
          where ca_owned_resource.resource_class = ca_resource_class.id
   ```
   The query executes successfully because you specified the correct class name.
4. Open GRLoader.
5. Execute the following command:

```
GRLoader -cfg sqlserverdb.cfg -u userid -p password -E -s http://sdmhostname:8080 -E -dbstmt "select
 ca_owned_resource.resource_name as name, ca_resource_class.name as class from ca_owned_resource,
 ca_resource_class where ca_owned_resource.resource_class = ca_resource_class.id" -a -e "c:
\errorjdbc1.xml"
```

The data is imported.

The GRLoader Command

The General Resource Loader (GRLoader) imports CI information into CA SDM. GRLoader uses XML documents as input, which lets you import data that originates in different data sources. Run GRLoader from a command prompt or by using a .bat or .cmd file. The CA SDM installation adds the GRLoader to the path during installation, so it runs from any directory.

Results from an import show counts for all processed CIs and Relationships, including the amount of Read, Skipped, Inserts, Updates, Errors, and Warnings. GRLoader logs all processing details and errors in the *nx_root*/log/grloader.log file, where *nx_root* specifies the CA SDM installation directory.

Syntax

```
C:\WINDOWS>GRLoader -?
```

The GRLoader command uses the following parameters:

- **-u *userid***
  (Required) Specifies the user ID that runs the GRLoader process.
- **-p *password***
  (Required) Specifies the password for the user ID. If you run GRLoader without the -p parameter, the utility prompts the console for the password.
- **-s http[s]://*cmdb_servername*:*port***
  (Required) Specifies the server URL including the port number that runs the web services. For running GRLoader on the primary server or application server in a default installation, you can use the following command:
  ```
  -s http://localhost:8080
  ```
  NOTE
  If you specify the optional -C parameter, GRLoader ignores the -s parameter.
- **-i *input_file***
  (Required) Specifies a full path name or a relative path name. If the filename contains a .xls or .xlsx suffix, GRLoader considers the file as a spreadsheet, otherwise it considers it as an XML file.
- **-n**
  (Optional) Allows new CI insertion into the CMDB. Without -n, CIs write to the XML error file (see the -e parameter). Relationships are only added if either -n or -a is specified. If neither is specified, no updates are performed. Updating CIs also require the -a parameter.
- **-a**
  (Optional) Allows updates to configuration items (by default, updates are not allowed if the CI exists in the CMDB). The -n flag also is required to add new CIs.
- **-D**
  (Optional) Specifies a name prefix for relations (defaults to "GRLoader"). Use the prefix for the sym field in new relationships. The sym file must be unique, so a datetime field and a number is appended to this prefix to make it unique.
  Default Prefix: GRLoader.
- **-e *XML_err_file***

(Optional) Produces an XML error file when GRLoader detects errors or warnings. By default, the error file name uses the name of the input file, appended with _err.xml. For example, using the input file as abc.xml creates the error file as abc_err.xml. Use the -e parameter to override this default name.

- **-E**
  (Optional) Lets you overwrite the XML error file. By default, the error file is not overwritten.
- **-I**
  (Optional) Ignores case. When you use this parameter, GRLoader is not case-sensitive when comparing the input value of a lookup field with the actual value stored in the database. By default, lookups are case-sensitive.
- **-ua**
  Always updates the CMDB.
- **-lftwa [-chg *nnnn*]**
  (Optional) Loads TWA transactions into the CMDB. If used with -chg, the load selects only those transactions associated with change order *nnnn*.

  NOTE
  The Change Order string is not validated when loaded into the CMDB.
- **-lftwai [-chg *nnnn*]**
  (Optional) Runs TWA transactions to update the CMDB. Transactions that run successfully are set to Inactive so that they do not appear in lists. If you use -chg, the load selects only those transactions associated with change order *nnnn*.
- **-lttwa**
  (Optional) Loads XML into the transaction work area (TWA) instead of directly into the CMDB. After data has been loaded into the TWA, it can be edited, changed and verified. After the data modification process completes, individual transactions can load into the CMDB (see - lftwa).
- **-lttwar**
  (Optional) Loads XML into the initial state in the transaction work area (TWA) instead of directly into the CMDB. Transaction data in the TWA can be edited, changed, and verified (see -simci and -simrel). After the data modification process completes, individual transactions can load into the CMDB (see - lftwai).
- **-nospinner (-spinner)**
  (Optional) Turns off the spinner that displays CI and relationship progress. Use -spinner to enable the progress display.
- **-P**
  (Optional) Specifies preload data to improve performance for large processing. For large input files, supplying the - P parameter preloads a few tables into memory so that they can be processed more quickly. For smaller inputs (< 50 entries), preload is not necessary.
- **-rs**
  (Optional) Replaces symbolic values that CA SDM includes in the XML input file. If you enable this parameter, corresponding values replace the following symbolic values:
  - *now* -- Replaced by a unique date/time string appended with a sequence number to help ensure uniqueness.
  - *userid* -- Indicates the userid specified in the -u parameter.
  - *inputfile* -- Indicates the filename specified in the -I parameter.
  - *relationcount* -- Specifies the number of relationships processed so far in this GRLoader run.
  - *lastciuuid* -- Specifies the UUID of the most recently processed CI.
  - *cicount* -- Specifies the number of CIs processed so far in this GRLoader run.

Examples: Use the -rs Parameter

With -rs enabled, the following example creates 100 CIs named ci1, ci2, …, ci100

```
<GRLoader>
<ci><name>ci*cicount*</name><class>Server</class></ci>
[...repeated 100 times...]
</GRLoader>
```

With -rs enabled, the following example updates the CI description with information about the most recent update.

```
<GRLoader>
<ci>
<name>server1</name>
<description>updated by *userid* on *now* using input file *inputfile*</description>
</ci>
</GRLoader>
```

- **-simci**
  (Optional) Simulates CI operations to predetermine whether a set of transactions creates CIs, and therefore possible ambiguities for other CIs.
- **-simrel**
  (Optional) Simulates relationship operations to predetermine whether a relationship transaction creates a relationship or updates a relationship.
- **-T *trace_level***
  (Optional) Specifies the tracing level. Known tracing levels are 0 (off, the default), 1 (low), 5 (medium) and 10 (verbose). We recommend only using this setting when necessary because much output can result.
- **-tf *filename***
  (Optional) Runs GRLoader using translation rules. *filename* specifies the name of the file that contains the translation rule set.
- **-slump**
  (Optional) Specifies the slump.jar file. This parameter can provide better performance than web services. **Important:** -slump only can be used with the -s parameter to target the following server:
  – Conventional: Primary server
  – Advanced Availability: Application server.

  NOTE
  If another CA product is installed, for example, CA Coheshion ACM or CMDB service pack is installed, verify that the slump.jar file is identical to the one that is installed on the target CA SDM system.
- **-C**
  (Optional) Validates the XML input file without any additional processing. This argument only validates the XML tags, not field values.
- **-h (or -?)**
  (Optional) Displays online help.
- **-v**
  (Optional) Displays the GRLoader product version and build date.
- **-maxerror *number***
  (Optional) Specifies the maximum number of errors that can occur before remaining CIs or relationships are skipped.
- **-maxwarn *number***
  (Optional) Specifies the maximum number of warnings that can occur before remaining CIs or relationships are skipped.
- **-chg *nnnn***
  Used with - lftwa and -lftwar. Loads only those transactions associated with change order *nnnn*.

  NOTE
  The Change Order string is not validated when loaded into the CMDB.
- **-cfg *myconfigfile*.cfg**
  (Optional) Specifies the name of the input configuration file.
- **-dt t*enant***
  (Optional) Specifies the tenant assignment for the CI/relationship. You must enable multi-tenancy to use this parameter. You can use PUBLIC to indicate that the object is public. If the tenant access of the user does not authorize creating public objects, the object is created using the default tenant.
- **-sc classname**

NOTE
Lists the attributes of CIs in the class you specify.

- **-scx classname**

  NOTE
  Lists the attributes of CIs in the class you specify in XML format. The XML is stored in a file named classname.xml. Any special characters are removed from the class name.

Example: Load CI and Relationship Data

The following example loads the CI and relationship data contained in the filehardware_servers.xml (in the current directory) into the CMDB that resides on the server located on the local computer on port 8080.

```
grloader -u CMDBAdmin -p password -s http://localhost:8080 -i hardware_servers.xml -n
```

Input Options

JDBC Database Input Options

The following table lists the options that you use in the configuration file and the corresponding command line options when you input JDBC databases.

NOTE

If you use SQL Server or Oracle databases, CA SDM includes the required JAR files. If you use other database types, you must use the -addjar option to add support for those databases to GRLoader dynamically. Consult your database vendor documentation for the name and location of the necessary JAR files to use JDBC. You can also consult your database vendor or database administrator for specific values and credentials.

GRLoader Option	Command Line Option	Description
grloader.jdbc.driver=*name*	-dbdriver *name*	Specifies the JDBC driver name. **Note:** This driver must be available on the classpath, similar to -addjar. Consult the database vendor for specific values.
grloader.jdbc.url=*URL*	-dburl *URL*	Specifies the JDBC database URL that describes the location of the database which contains the table you want to load.
grloader.jdbc.user=*name*	-dbuser *name*	Specifies the user ID for the JDBC database.
grloader.jdbc.password=*password*	-dbpswd *password*	Specifies the password for the user ID for the JDBC database.
grloader.jdbc.statement=*statement*	-dbstmt *statement*	Specifies the JDBC statement that describes the columns and selection criteria for the data you want to import. **Note:** The column names used in the query statement must match CMDB attribute names. If these names differ, use the SQL *AS* keyword to map database column names to CMDB attributes.

Spreadsheet Input Options

The following table lists the options that you use in the configuration file and the corresponding command line options when you input spreadsheets. You can also usethe {} symbols as delimiters surrounding the lookup field as well as using the keyword EMPTY.

- **boolean**
  Specifies a value from one the following pairs: 1/0, YES/NO, or TRUE/FALSE.

  NOTE

  If the input file name ends in .xls or .xlsx when you use the *grloader.inputfile=name* GRLoader option or the *-i* command line argument, GRLoader assumes that it is a spreadsheet.

GRLoader Option	Command Line Option	Description
grloader.spreadsheet.filename=*name*	-ssf	Specifies the Excel spreadsheet file name when it does not contain the .XLS or .XLSX file extension.
grloader.spreadsheet.sheetname=*name*	-sss	Specifies the sheet name. **Default:** The first sheet in the spreadsheet.
grloader.spreadsheet.firstrow=*n*	-ssfr	Set this value to skip over the first *n-1* rows in the spreadsheet.
grloader.spreadsheet.lastrow=*n*	-sslr	Ignores *rows > n (greater than)* in the spreadsheet.
grloader.spreadsheet.firstcol=*x*	-ssfc	Specifies to start processing on this column. You can express this column as a letter or number, depending on your spreadsheet options.
grloader.spreadsheet.lastcol=*x*	-sslc	Ignores *columns > x (greater than)* in the spreadsheet. You can express this column as a letter or number, depending on your spreadsheet options.
grloader.spreadsheet.embeddedseparator	-sses	Specifies a character to separate multiple values contained in a single cell. This option only applies to the relationship type column in a row with multiple embedded relationships. **Default:** semi-colon ":"
grloader.attributedefault.atrname=*value*	-ad attrname=*value*	Provides a default value if you did not specify one in the input source. **Note:** These values do not undergo attribute name or data value translation. **Note:** When you run GRLoader from a batch file, specify the *-ad attr=value* as *-ad attr{value}* to get past the Windows command parser which may remove the equal "=" symbols.

	-dt	Deprecated. Instead, use the -ad tenant=*name* option.
grloader.maxci=*n*	-maxci	Specifies the maximum number of CIs to import before skipping further CI imports.
grloader.maxrel=*n*	-maxrel	Specifies the maximum number of relationships to import before skipping further relationship imports.
N/A	-sc *classname*	Lists the attirbutes of CIs in the class you specify.
N/A	-scx *classname*	Lists the attributes of CIs in the class you specify in XML format.
grloader.workarea.changeorderrequired =*boolean*	-cor	If set to yes, TWA transactions that do not contain a non-blank change order number are ignored.
grloader.ignoreinvalidattributes=*boolean*	-iia	Specifies if you want to ignore invalid attributes. This command suppresses all warning messages about invalid attribute names.

CSV Input Options

The following table lists the options that you use in the configuration file and the corresponding command line options when you input CSV files.

NOTE

If required, consult your database vendor or database administrator for specific values and credentials.

GRLoader Option	Command Line Option	Description
-i		GRLoader assumes that a file ending in .csv is a CSV file.
grloader.csv.filename=*name*	-csvf *name*	Specifies the filename when it does not end in .csv.
grloader.csv.separator=*x*	-csvsep *x*	Specifies when the CSV file uses other than comma delimiters. You can specify a single character, such as a Tab (\t) or a semicolon.
grloader.csv.escape=*x*	-csvesc *x*	Specifies when the CSV file uses an escape character (\).
grloader.csv.comment=*x*	-csvcom *x*	Specifies when the CSV file uses the comment character (#).
grloader.csv.quote=*x*	-csvquote *x*	Specifies when the CSV file uses the quote character (").

TWA Input Options

The following table lists the options that you use in the configuration file and the corresponding command line options when you input from the TWA.

GRLoader Option	Command Line Option	Description
grloader.workarea.changeorderrequired=yes/no	-cor	GRLoader ignores TWA transactions that do not contain a non-blank Change Order number.

General Options

The following table lists the options that you use in the configuration file and the corresponding command line options when you input from all inputs.

NOTE

If necessary, consult your database vendor or database administrator for specific values and credentials.

GRLoader Option	Command Line Option	Description
grloader.system.addjar=*xx;yy;zz*	-addjar *xx*	Adds JAR files to the GRLoader classpath. You can use this option with - jdbcdriver. **Note:** The grloader.system.addjar option can only appear once in the configuration file. You can add multiple jars through grloader.system.addjar by separating the filenames with a semi-colon. This option only specifies a single jar file, and you can specify it as many times as required.
grloader.attributedefault.attrname=*value*	-ad attrname=*value*	Provides a default value if you did not specify a value in the input source. **Note:** These values do not undergo attribute name or data value translation. When you run GRLoaderfrom from a batch file, specify -ad attr=value as *-ad attr{value}.* The Windows command parser can delete the equals symbol.
N/A	-ad tenant=*name*	Specifies the tenant name.
grloader.maxci=*n*	-maxci *n*	Specifies the maximum number of CIs to import before skipping further CI imports.
grloader.maxrel=*n*	-maxrel *n*	Specifies the maximum number of relationships to import before skipping further relationship imports.
N/A	-sc *xx*	Lists attributes of CIs in the class *xx*.
N/A	-scx *xx*	Lists attributes of CIs in class *xx* in XML format.

grloader.reader.allowembeddedrelationships=yes/no	-aer	Allows embedded relationships. **Default:** Yes For backward compatibility only
grloader.ignoreinvalidattributes=yes/no	-iia	Ignores invalid attributes by suppressing all warning messages about invalid attribute names.
grloader.updatealways=yes/no		

Example Display CI Class Attributes

When you create input for GRLoader, list the attributes associated with a specific class. To list the attributes, execute the following command:

```
grloader  - u username  - p password  - s http://sdm-host:8080 -sc [class name]
```

- **sc**
  Lists attributes of CIs in a class that you specify.
- **class name**
  Specifies any valid CA CMDB class name.

Example: List Attributes for Class Server

```
grloader  - u username  - p password  - s http://sdm-host:8080 -sc Server

10:33:01.997 CI and Relationship Loader for CA Service Desk Manager

List of attributes in class(Server) extension(har_serx)

ATTRIBUTE NAME                    DATA TYPE
acquire_date                      Date
active_date                       Date
alarm_id                          STRING(64)
ambiguity                         Integer
asset_count                       Integer
asset_num                         STRING(64)
audit_userid                      SREL(cnt.combo_name)
```

Example Display CI Class Attributes in XML Format

When you create input for GRLoader, list the attributes associated with a specific class in XML format. The command creates the file [class name].xml with the result. To list the attributes in XML format, execute the following command:

```
grloader  - u username  - p password  - s http://sdm-host:8080 -scx [class name]
```

- **scx**
  Lists attributes of CIs in XML format in a class that you specify.
- **class name**
  Specifies any valid CA CMDB class name.

Example: List Attributes for Class Server

```
grloader  - u username  - p password  - s http://sdm-host:8080 -scx Server

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<GRLoader>
 <ci>
    <acquire_date></acquire_date>                      <!-- Date -->
```

```
<active_date></active_date>                          <!-- Date -->
<alarm_id></alarm_id>                                <!-- String(64) -->
<ambiguity></ambiguity>                              <!-- Integer -->
<asset_count></asset_count>                          <!-- Integer -->
<asset_num></asset_num>                              <!-- String(64) -->
<audit_userid lookup="combo_name"></audit_userid>    <!-- SREL cnt -->
```

NOTE

If the class name contains special characters, they are removed.

Data Error Handling

If GRLoader finds errors, the failing CI or Relation node writes to an error file. We recommend that you edit the XML file to correct the problem, and then run GRLoader against the edited file. The -e parameter lets you name the error file. The -maxerror and -maxwarn parameters let you specify the maximum number of errors or warnings that can occur until GRLoader skips the remaining CIs or relationships.

Consider the following information about error destinations:

- If you use the TWA as the input source, the TWA also becomes the error destination.
- If you use the JDBC database as the input source, $NX_ROOT\log\grloader_err.xml becomes the error destination.
- If you use filename.xls, filename.xlsx, filename.xml, or filename.csv as the input source, filename_err.xml becomes the error destination.

Note: When you troubleshoot errors or you perform simulations, review these files and the standard CA SDM logs.

GRLoader Configuration File

You can specify GRLoader options in a configuration file. This approach provides the following advantages:

- You no longer enter passwords in the command line.
- The command prompt window (Windows) does not display the password, or in the results of the -ps command (UNIX).
- Standardized commands reduce errors.

  WARNING
  We recommend that you keep configuration files in a secure location. Avoid specifying passwords in command lines and use configuration files, especially in Linux environments.

The following input specifies the format of the configuration file parameter:

- **-cfg *myconfigfile*.cfg**
  Specifies the name of the input configuration file. You can specify the -cfg parameter at any location in the GRLoader parameter string.

  NOTE
  If the command line and the configuration file conflict, GRLoader uses the last value you entered.

Example: Specify a Configuration File

A configuration file command uses the following syntax:

```
GRLoader -cfg myconfigfile.cfg -i myinputfile.xml
```

Instead of specifying the more complex command:

```
GRLoader -u userid -p password -i myinputfile.xml -a -n -E -maxerror 10 -maxwarn 10 -dt IBM -nomn
```

Example: Last Password Value is Used

The configuration file GRLoader.cfg specifies the following passwords:

```
GRLoader.password=password1
GRLoader.password=password2
```

The command line specifies the following password:

```
GRLoader -p password3 -cfg GRLoader.cfg
```

The password that is used is password2.

If the command line was changed to specify:

```
GRLoader -cfg GRLoader.cfg -p password3
```

The password that is used is password3 because it was the last one specified.

Configuration File Options

The following table lists the GRLoader options that you can use in the configuration file and the corresponding command line options.

- **boolean**
  Specifies a value from one the following pairs: 1/0, YES/NO, or TRUE/FALSE.

GRLoader Option	Command Line Option	Description
grloader.userid=*userid*	-u	Specifies the user name of the administrator.
grloader.password=*password*	-p	Specifies the administrator password.
grloader.server=*server*	-s	Specifies the URL of the CA SDM server.
grloader.inputfile=*name*	-i **Note:** If the input file name ends in .XLS or .XLSX, GRLoader assumes it is a spreadsheet.	Specifies the file you want to import with GRLoader.
grloader.errorfile=*name*	-e	
grloader.nxroot=*name*	-N	
grloader.casesensitive=boolean	-I	
grloader.loadfromtwa=yes	-lftwa [-chg *nnnn*]	
grloader.loadfromtwa.inactivatesuccessful=yes	-lftwai [-chg *nnnn*]	
grloader.loadtotwa=yes	-lttwa	
grloader.loadtotwa.ready=yes	-lttwar	
grloader.simulateloadci=boolean	-simci	
grloader.simulateloadrelation=boolean	-simrel	
grloader.emptyvalue=EMPTY		
grloader.workarea.delimiters={ }		Provides alternate lookups for the dependent CI.
grloader.workarea.ignore_transaction_dates=yes		
grloader.normalizemac=boolean	-nm/nonm	

grloader.maxerror=*number*	-maxerror	
grloader.maxwarn=*number*	-maxwarn	
grloader.defaulttenant= *tenant* | PUBLIC	-dt *tenant* | PUBLIC **Note:** The multi-tenancy option must be *setup* or *on* to use these options.	
grloader.allowupdate=boolean	-a	
grloader.allowinsert=boolean	-n	
grloader.overwriteerrorxml=boolean	-E	
grloader.slump=boolean	-slump (primary or application server only)	
grloader.preload=boolean	-P	
grloader.replacesymbols=boolean	-rs	
grloader.translationfile=*filename*	-tf	
grloader.tracelevel=number	-T	
grloader.spinner=boolean	-spinner/-no (equivalent to -nospinner)	
grloader.cmdbversion=11.0	(no equivalent)* *Required for CA CMDB r11.0 only. GRLoader is compatible with all later releases.	
grloader.spreadsheet.filename=*name*	-ssf	Specifies the Excel spreadsheet file name when it does not contain the .XLS or .XLSX file extension.
grloader.spreadsheet.sheetname=*name*	-sss	Specifies the sheet name. **Default:** The first sheet in the spreadsheet.
grloader.spreadsheet.firstrow=*n*	-ssfr	Set this value to skip over the first *n-1* rows in the spreadsheet.
grloader.spreadsheet.lastrow=*n*	-sslr	Ignores *rows > n (greater than)* in the spreadsheet.
grloader.spreadsheet.firstcol=*x*	-ssfc	Specifies to start processing on this column. You can express this column as a letter or number, depending on your spreadsheet options.
grloader.spreadsheet.lastcol=*x*	-sslc	Ignores *columns > x (greater than)* in the spreadsheet. You can express this column as a letter or number, depending on your spreadsheet options.
grloader.spreadsheet.embeddedseparator	-sses	Separates multiple values contained in a single cell. This option only applies to the relationship type column in a row with multiple embedded relationships. **Default:** semi-colon ":"

grloader.attributedefault.atrname=*value*	-ad attrname=*value*	Provides a default value if you did not specify one in the input source. **Note:** These values do not undergo attribute name or data value translation. **Note:** When you run GRLoader from a batch file, specify the *-ad attr=value* as *-ad attr{value}* to get past the Windows command parser which may remove the equal "=" symbols.
	-dt	Deprecated. Instead, use the -ad tenant=*name* option.
grloader.maxci=*n*	-maxci	Specifies the maximum number of CIs to import before skipping further CI imports.
grloader.maxrel=*n*	-maxrel	Specifies the maximum number of relationships to import before skipping further relationship imports.
N/A	-sc *classname*	Lists the attributes of CIs in the class you specify.
N/A	-scx *classname*	Lists the attributes of CIs in the class you specify in XML format.
grloader.workarea.changeorderrequired =*boolean*	-cor	If set to yes, TWA transactions that do not contain a non-blank change order number are ignored.
grloader.ignoreinvalidattributes=*boolean*	-iia	Specifies if you want to ignore invalid attributes. This command suppresses all warning messages about invalid attribute names.

NOTE

If the input file name ends in .xls or .xlsx when you use the grloader.inputfile=name GRLoader option or the -i command line argument, GRLoader assumes that it is a spreadsheet.

GRLoader XML

This article contains the following topics:

GRLoader requires XML document input that consists of a document header followed by enclosing <GRLoader> XML elements tags with one or more <ci> tags (for CI definitions) or <relation> tags (for relationships).

Specify the XML document header as follows:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes" ?>
```

Update the encoding attribute as needed to handle the appropriate character encoding requirements. For example, specify “ISO-8859-1” to handle special Norwegian characters.

Example: Format a GRLoader XML File

The following template presents the format for a GRLoader XML file:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes" ?>
```

```
<GRLoader>

<ci>
[define a CI: common and family-specific attributes, versioning, reconciliation, MDR]
</ci>
[repeat as necessary for each CI]

<relation>
        <type>relationship_type</type>
        <delete_flag>active_state</delete_flag>
        <provider>
                <name>resource name</name>
                <serial_number>serial number</serial_number>
                <system_name>host name</system_name>
                <asset_num>resource tag</asset_num>
                <mac_address>mac address</mac_address>
                <dns_name>dns name</dns_name>
                <id>ci_uuid</id>
        </provider>
        <dependent>
                <name>resource name</name>
                <serial_number>serial number</serial_number>
                <system_name>host name</system_name>
                <asset_num>resource tag</asset_num>
                <mac_address>mac address</mac_address>
                <dns_name>dns name</dns_name>
                <id>ci_uuid</id>
        </dependent>
</relation>
[repeat as necessary for each relationship]
</GRLoader>
```

XML Content The CI Tag

GRLoader uses the CI XML definition to load a CIs attribute values and relationships. The CI definition must include a minimum set of required attributes to be created or updated by using <*ci*> XML element tags.

You define the XML for a CI by specifying values for the following attributes:

- Class identification (required)
- Reconciliation attributes (required)
- Common attributes
- Family-specific attributes
- MDR identification attributes
- Versioning attributes

The CI Tag Family and Class Identification

Class identification must be specified for each CI to associate the proper family and class with the CI.

Specify the family and class attributes using the following XML tags:

- **<family>**
  (Optional) Specifies a collection of CIs that have similar attributes.
- **<class>**

(Required) Specifies a subset of CIs within a family.

NOTE
If GRLoader cannot find family or class, the CI is not created or updated.

Example: Identify a CI by Family and Class

The following example shows a CI named ServerCI that is identified by the family Hardware.Server and class Windows.

```
<ci>
        <name>ServerCI</name>
        <family>Hardware.Server</family>
        <class>Windows</class>
...
</ci>
```

The CI Tag Reconciliation Attributes (Required)

One or more reconciliation attributes are required when creating, updating, or referencing a CI. GRLoader uses these attributes to uniquely identify the CI to be created or updated. Reconciliation attributes are also used to identify a provider/dependent relationship between two CIs.

Specify the reconciliation attributes using the following XML element tags:

- <name> -- The name of the CI or resource (required when creating the CI for first time)
- <serial_number> -- The manufacture unique identifier
- <asset_num> -- Alternate resource identifier, for example, an alternate ID located on sticker placed on computer
- <system_name> -- Computer name (hardware only)
- <dns_name> -- The name by which this device is known in the domain name server
- <mac_address> -- MAC address. (hardware only)
- <id> -- UUID of the CI, used for direct updates when ID is known

The name attribute is required when creating a CI for the first time. If GRLoader cannot resolve the specified reconciliation attributes, an existing CI is not updated. Reconciliation attributes are special purpose Common Attributes that are used for identification purposes.

Example: Identify a CI When Creating or Updating It

In the following example, the CI definition uses name, serial_number, dns_name, mac_address and system_name to uniquely identify the CI when creating or updating it.

```
<ci>
    <name>ServerCI</name>
    <serial_number>HMVV081</serial_number>
    <dns_name>serverci.myco.com</dns_name>
    <mac_address>00:12:3F:48:F0:95</mac_address>
    <system_name>ServerCI</system_name>
...
</ci>
```

The CI Tag Common Attributes

In general, common attributes are attributes that can be used in any CMDB family or class. The XML element tag used for the attribute is the same as the attribute object name. The attribute value depends on its type, which can be a constant or an SREL value that indicates a foreign key reference to another table.

Example: Specify Common Attributes

In the following example, the CI definition named ServerCI specifies the following common attributes: manufacturer, model, and alarm_id (IP Address). The ServerCI name is also a common attribute.

```
<ci>
    <name>ServerCI</name>
...
    <manufacturer>Dell Inc.</manufacturer>
    <model>OptiPlex GX280</model>
    <alarm_id>130.200.19.220</alarm_id>
...
</ci>
```

The CI Tag Family-Specific Attributes

Class attributes are unique to a specific CI family or class. The XML element tag used for the class attribute is the same as the attribute object name found in the family/class specific tables.

Example: Specify Family-Specific Attributes

In the following example, the CI definition named ServerCI specifies the attributes specific to the Hardware.Server family that include bios_ver, cd_rom_type, hard_drive_capacity, and so on.

```
<ci>
    <name>ServerCI</name>
...
    <bios_ver>A04</bios_ver>
    <cd_rom_type>DVD+-RW DVD8701</cd_rom_type>
    <hard_drive_capacity>90 MB</hard_drive_capacity>
    <number_net_card>3</number_net_card>
    <number_proc_inst>1</number_proc_inst>
    <phys_mem>2048 MB</phys_mem>
    <proc_speed>2793 MHz</proc_speed>
   <swap_size>4959 MB</swap_size>
...
</ci>
```

The CI Tag MDR Identification

A management data repository (MDR) identifies the data provider for a CI and how the CIs maps back to the corresponding MDR.

CA SDM uses MDR information to perform the following tasks:

- Launch in context from the CI log directly to the MDR data provider.
- Tracks CI attribute changes back to the source MDR.
- Detects when more than one MDR updates a CI attribute. This situation occurs when multiple MDRs contribute data independently to a CI definition.
- Identifies which MDR acts as the authoritative source.

  NOTE
  For more information about MDRs, see this topic.

Use the following XML element tags to specify MDR attributes:

- **<mdr_class>**
  Specifies the MDR class to group MDRs that CA SDM processed similarly.
- **<mdr_name>**

Specifies the MDR name that an MDR uses to reference itself. Verify that the mdr_name and mdr_class value combination is unique within your enterprise.

- **<federated_asset_id>**
  Specifies the Federated asset ID that indicates the unique identifier of an MDR for a CI.

If GRLoader cannot resolve the specified mdr_class and mdr_name to an existing MDR, GRLoader does not import the CI. A CI with no associated federated_asset_id mapping is not federated.

Example: Identify a CI in the MDR

In the following example, the CI definition named ServerCI specifies mdr_class and mdr_name to uniquely identify the MDR and federated asset id, and thus identify the CI in the MDR.

> **NOTE**
> CA SDM uses the mdr_class string value *Cohesion* when federating data from the [assign the value for acm in your book] product.

```
<ci>
    <name>ServerCI</name>
...
    <federated_asset_id>1001118</federated_asset_id>
    <mdr_class>Cohesion</mdr_class>
    <mdr_name>CohesionServer</mdr_name>
...
</ci>
```

The CI Tag Versioning Attributes

You can use GRLoader to set versioning attributes for a CI.

> **NOTE**
> For more information about versioning, see the Versioning section.

Specify the Versioning attributes using the following XML element tags:

- **<milestone>**
  Specifies the label associated with that milestone that displays in the Versioning tab.
- **<standard_ci>**
  Specifies the name of the standard CI to use for baseline comparisons in the Versioning tab.

The CI that you specified for the standard_ci attribute must already exist in the CMDB or be specified before you specify the CI definition in the XML file. The milestone generated records the state of the CI at the time that GRLoader executes.

Example: Specify Baseline Comparisons

In the following example, the CI definition named ServerCI specifies the standard CI named standard server config for baseline comparisons with ServerCI (the focal CI). This example assumes that the standard CI already exists in CA SDM. In addition, a milestone named Fiscal year end 2008 is also created to preserve the state of the CI at the time that GRLoader imports the XML.

```
<ci>
     <name>ServerCI</name>
     <class>Server</class>
     <standard_ci>standard server config</standard_ci>
     <milestone>Fiscal year end 2008</milestone>
...
</ci>
```

XML Content The Relation Tag

GRLoader can create or update relationships between configuration items by using the <relation> XML element tag. Relationships are many-to-many, and the relationship type specifies how two provider/dependent configuration items relate to one another in CMDB.

Specify the relation attributes using the following XML element tags:

- **<type>**
  (Optional) Specifies the name of the relationship type.
- **<delete_flag>**
  Designates a relationship as inactive or active. Specify 1 (one), yes, or true to make the relationship inactive. Specify 0 (zero), no or false to make the relationship active again. Setting the delete_flag to true leaves the existing relationship intact but marks it as inactive.
- **<provider>**
  (Required) Identifies the provider CI for the relationship, which contains one or more of the CI reconciliation attributes.
- **<dependent>**
  (Required) Identifies the dependent CI for the relationship, which contains one or more of the CI reconciliation attributes.

  NOTE
  If GRLoader cannot find a specified type, provider CI, or dependent CI, the relationship is created or updated.

Example: Define a Relationship Between CIs

The following example defines a relationship between the CIs named ServerCI (provider) and ServerCI|NetworkAdaptor-0 (dependent). The relationship type is contains. The example assumes that both CIs have already been defined in the CMDB or are specified preceding the relationship definition in an XML file. In addition, both the provider and dependent CIs must match all reconciliation attributes for the relationship to be created.

```
<relation>
    <type>contains</type>

    <provider>
        <name>ServerCI</name>
        <serial_number>HMVV081</serial_number>
        <dns_name>serverci.myco.com</dns_name>
        <mac_address>00:12:3F:48:F0:95</mac_address>
        <system_name>ServerCi</system_name>
    </provider>
    <dependent>
        <name>ServerCI|NetworkAdaptor-0</name>
    </dependent>
</relation>
```

XML Content Special Values

Special-purpose XML attributes can modify how a CI value is set or updated when imported by GRLoader. You can use these attributes to perform special processing or formatting when setting the value; for example, to format a date value or use the result of a lookup.

Examples of special XML values include the following ones:

- **lookup**
  Specifies a CI by an attribute other than combo_name (lastname, firstname, middle). Examples include: userid,
- **update_if_null**

Specifies the update_if_null option for GRLoader to use to distinguish between values that are blank and those which are not supplied in the XML. By default, update_if_null is set to "", which means that blank or missing values are ignored by GRLoader.
The following attribute descriptions for serial number are equivalent:

```
<serial_number></serial_number>
<serial_number/>
<serial_number update_if_null="">
```

If you want to remove the serial number from a CI that has one, the previous XML does *not* work, because GRLoader ignores blank or missing values. Instead, code xml for the serial number as follows:

```
<serial_number update_if_null="true"></serial_number>
```

This syntax always updates the attribute, even if the value is blank or missing.

- **dateformat=[utc | localtime]**
  Sets the *dateformat* attribute for the date field to be either “utc” or “localtime”. Required when the format of the date is in UNIX Time Code (UTC) format. If dateformat is not set, the default is “localtime".

Date Formats

CMDB supports the following localtime date formats:

- yyyy.mm.dd
- yyyy.mm.dd hh:mm:ss

If the value does not match either of these formats, the parser tries to resolve the date as a UTC time. If the date format is not UTC, CMDB uses the system locale setting: for US English, the 12-hour format of “mm/dd/yyyy” or “mm/dd/yyyy hh:mm:ss *a*” where *a* specifies either AM or PM).

Contact and Other Lookup Fields

The Contact object combines first name, middle initial, and last name. The object has the following format:

```
<resource_contact>Lastname, Firstname MiddleInitial</resource_contact>
```

If you want to use a different field for a lookup field, you can supply a lookup attribute. For example, if you wanted to look up John Q. Doe by userid, use the following entry:

```
<resource_contact lookup="userid">doejo04</resource_contact>
```

Fields Validated Against Data in Existing Tables (SREL)

Common attributes accept only a specific set of values that must be defined in related tables in CMDB. These attributes can also have additional restrictions and exceptions that must be met for the assignment to occur. For example, a class attribute specified in XML must match one of the existing class names (CMDB default or user-defined). Otherwise, the CI is not created or updated. In addition, the value cannot be set to null, and the class must be Active for the assignment to occur.

The following fields validate data against data in existing tables:

- audit_userid
- bm_rep
- bm_status
- class
- company_bought_for_uuid
- contact_1
- contact_2
- contact_3
- delete_flag
- department
- expense_code
- family
- location
- manufacturer
- model
- operating_system
- org_bought_for_uuid
- priority
- repair_org
- resource_contact
- resource_owner_uuid
- service_org
- service_type
- status
- supplier
- vendor_repair
- vendor_restore

XML Input

When importing CI data, format the data in a supported format, such as XML or a spreadsheet (XLS or XLSX).

Consider the following XML format example:

XML Document	Notes

`<?xml version="1.0"`
`encoding="UTF-8"`
`standalone="yes" ?>`
`<GRLoader>`
	These headers are required.

`<ci>`	Include zero or more <ci> nodes to define the CIs.

`<name>value</name>`
`<mac_address>value</mac_address>`
`<dns_name>value</dns_name>`
`<asset_num>value</asset_num>`
`<serial_number>value</serial_number>`
`<system_name>value</system_name>`
	These six characteristics uniquely identify a CI in a CI or Relations definition. At least one must be specified.

<class>value</class> <family>value</family> <manufacturer>value</manufacturer> <model>value</model>	These four values determine the class and family of a CI. You specify either (class) or (manufacturer/model).
<mem_capacity>value</mem_capacity> <number_net_card>value </number_net_card> <phys_mem>value</phys_mem_update> <proc_speed>value</proc_speed> <proc_type>value</proc_type> <server_type>value</server_type> </ci>	Family-specific values. Zero or more family-specific values can be provided when defining a CI.
<relation> <type>relation_type</type>	Include zero or more <relation> nodes to define relationships. Specify the relationship type.
<provider> <name>value</name> <mac_address>value</mac_address> <dns_name>value</dns_name> <asset_num>value</asset_num> <serial_number>value</serial_number> </provider>	Identify the provider CI with at least one attribute.
<dependent> <name>value</name> <mac_address>value</mac_address> <dns_name>value</dns_name> <asset_num>value</asset_num> <serial_number>value</serial_number> </dependent> </relation>	Identify the dependent CI with at least one attribute.
</GRLoader>	

Example: XML Input

```
<?xml version="1.0" encoding="UTF-8" standalone="yes" ?>
<GRLoader>
    <ci>
        <name>Host1</name>
        <class>Server</class>
    </ci>
    <ci>
        <name>Host2</name>
        <class>Server</class>
    </ci>
    <relation>
        <type>connects to</type>
        <provider>
            <name>host1</name>
        </provider>
        <dependent>
            <name>host2</name>
```

```
        </dependent>
    </relation>
</GRLoader>
```

MAC Address Normalization

Previous releases of GRLoader normalized the MAC address of CIs by removing the ":" and "-" delimiters from the MAC address. This normalization resulted in a MAC address of: aa:bb:cc:dd:ee storing as aabbccddee.

Consider the following MAC address behavior:

- The default is no MAC address normalization.
- CIs created with no normalization in CMDB reconcile with CIs that were created without normalization in CMDB r11.x.
- Invalid MAC addresses are treated as simple strings and are stored unmodified.

The following GRLoader parameters let you enable or disable MAC normalization:

- **-mn**
  Removes the ":" and "-" delimiters from MAC addresses (MAC normalization).
- **-nomn**
  Does not remove the ":" and "-" delimiters from MAC addresses.

  WARNING
  Installing an earlier version if CMDB enables MAC address normalization automatically. You can override normalization by using the *-nomn* parameter

Because options are processed sequentially on the command line, the order of the options is important in the syntax.

Contents

Complete the following steps when preparing to load data from a JDBC database table into CA SDM:

1. Identify the database, database type, userid, password and the JDBC drivers used to access the database.
2. Identify the table and columns that you want to load.
3. If necessary, map column names using the SQL *AS* keyword to CMDB attribute names.
4. If necessary, map data values using a translation file or a SQL join.
5. Identify portions of the table that you want GRLoader to ignore or skip.

   Example JDBC Attribute Mapping

   The following example shows how to use the SQL AS clause for JDBC attribute mapping:

```
GRLoader  - dbdriver sun.jdbc.odbc.JdbcOdbcDriver
-dburl "jdbc:odbc:Driver={Microsoft Access Driver (*.mdb, *.accdb)};DBQ=filename"
-dbuser administrator
-dbpswd adminpassword
-s http://hostname:8080
-n  - a -E
 - dbstmt "SELECT 'ci'    AS objecttype, resource_name AS name , ip_address AS alarm_id     FROM
 my_table"
```

   Consider the following information about the previous example:

   - The objecttype column is required when you import CIs without specifying a class column in the SELECT statement.
     Note: Use the SQL “AS” keyword when you use input from database tables to create views.
   - You use the *ci* value in the obecttype column as a constant value that does not require a physical presence in the database.

Example Load Data from a Microsoft Access Database Using ODBC CA SDM includes the generic JDBC-ODBC driver named sun.jdbc.odbc.JdbcOdbcDriver in the rt.jar file. If you want to use a database-specific driver, specify additional JDBC JAR files, as required by your database vendor. Use the -addjar facility to add any necessary JARs to the classpath dynamically. In this example, you load data from a Microsoft Access database. **Follow these steps:**

1. Open GRLoader.
2. Execute the following command:

```
GRLoader
 - dbdriver sun.jdbc.odbc.JdbcOdbcDriver
-dburl "jdbc:odbc:Driver={Microsoft Access Driver (*.mdb, *.accdb)};DBQ=filename"
-dbuser administrator
-dbpswd adminpassword
-s http://hostname:8080
-n  - a -E
-u userid -p password
-dbstmt "SELECT ciname AS name, ciclass AS class,
 FROM table1"
-e jdbc_err.xml
```

> **WARNING**
> When you use values that contain spaces, enclose the values in double quotations, such as with dbdriver and dbstmt.

Example Load Data from a Microsoft Access Database Using ODBC and a Configuration File In this example, you include the database connection information in a separate configuration file. Using this file reduces the command line length in GRLoader. **Note:** You do not have to include quotes in the configuration file. Input statements on single lines and do not continue the input across multiple lines. **Follow these steps:**

1. Create a file named table1.cfg and add the following code:

```
grloader.jdbc.driver=sun.jdbc.odbc.JdbcOdbcDriver
grloader.jdbc.url=jdbc:odbc:Driver={Microsoft Access Driver (*.mdb, *.accdb)};DBQ=filename
grloader.jdbc.user=administrator
grloader.jdbc.password=adminpassword
```

   Note: In this example file, the grloader.jdbc.url appears on one line, as each option in the configuration file also appear.
2. Save the file.
3. Open GRLoader.
4. Execute the following command:

```
GRLoader
 - cfg table1.cfg
-s http://hostname:8080
-n  - a -E
-u userid -p password
-dbstmt "SELECT ciname AS name, ciclass AS class FROM table1"
```

Example Load Data from a SQL Server Database Table In this example, you load data from a SQL Server database using drivers supplied by the database vendor. **Follow these steps:**

1. Open GRLoader.
2. Execute the following command:

```
GRLoader
-dbdriver com.microsoft.sqlserver.jdbc.SQLServerDriver
-dburl jdbc:sqlserver://localhost:1433;databaseName=mdb;
-dbuser servicedesk -dbpswd password
```

```
-s http://hostname:8080  -a -n -E
-u userid -p password
-dbstmt "SELECT name, 'Server' AS class
         FROM ca_owned_resource"
-e jdbc_err.xml
```

Example Load Data from an Unsupported MySQL Database In this example, you load data from an unsupported MySQL Server database to the TWA using drivers supplied by the database vendor. You specify the database user and password in the dburl option. **Follow these steps:**

1. Open GRLoader.
2. Execute the following command:

```
GRLoader
-dbdriver com,mysql.jdbc.Driver
-dburl jdbc:mysql://hostname/test?user=abed&password=pwd
-s http://hostname:8080  -a -n -E
-lttwa
-u userid -p password
-dbstmt "SELECT name, class from mytable3"
-e jdbc_err.xml
```

Example Import Relationships from a Database Table In this example, you load relationships from a database table. The provider and dependent attribute names are prefixed with *provider_* and *dependent_*, as in the TWA. **Follow these steps:**

1. Open GRLoader.
2. Execute the following command:

```
GRLoader
-dbdriver com.microsoft.sqlserver.jdbc.SQLServerDriver
-dburl jdbc:sqlserver://localhost:1433;databaseName=mdb;
-dbuser servicedesk -dbpswd password
-s http://hostname:8080  -a -n -E
-u userid -p password
-dbstmt "SELECT provider_name, type, dependent_name,
 FROM mytable4"
```

Example Display the WHERE Clause in the SELECT Statement In this example, you display the WHERE clause in the SELECT statement to filter input data by location. **Follow these steps:**

1. Open GRLoader.
2. Execute the following command:

```
GRLoader
-dbdriver com.microsoft.sqlserver.jdbc.SQLServerDriver
-dburl jdbc:sqlserver://localhost:1433;databaseName=mdb;
-dbuser servicedesk -dbpswd password
-s http://hostname:8080  -a -n -E
-u userid -p password
-dbstmt "SELECT name, class,
 FROM mytable5 WHERE location='Brooklyn' "
```

Example Set Default Values for Attributes that Do Not Appear in the Input Table In this example, you can specify any attribute with the -ad option. You can apply this example to spreadsheets and XML files.

> **WARNING**
> We recommend that you do not specify the tenant with the -ad option. Instead, GRLoader should inherit the tenant from the user ID that runs GRLoader. The Service Provider assigns

each tenant a tenant-specific user ID, and then each tenant uses that user ID when creating objects that belong to that tenant.

Follow these steps:

1. Open GRLoader.
2. Execute the following command to set a default IP address:
   ```
   grloader  - i myspreadsheet.xlsx  - ad alarm_id="Unknown"
   ```
   Note: If a row in the spreadsheet contains a value for alarm_id, then GRLoader does not use the default value.
3. Use the -iia option to ignore an invalid attribute name to hide the warning message.
   For example, the spreadsheet contains both CIs and relationships, and the relationships would receive a warming message because alarm_id is not a valid relationship attribute.

How to Prepare for Loading CSV File Data

Contents

GRLoader can read CSV files to load CIs and relationships, similar to how it processes JDBC database tables and spreadsheets. GRLoader treats any rows that begin with # as comments.

Note: The first row in the CSV file must contain column headings so that GRLoader translates the CI and relationship attribute names correctly.

Complete the following steps when preparing to load data from a CSV file:

1. Identify the CSV file that you want to load.
2. If necessary, add a header row.
3. If necessary, map attribute names and data values using a translation file.
4. Identify parts of the CSV file that GRLoader *must* ignore or skip. If necessary, write corresponding translation rules.

Example Load Data from a CSV File

In this example, you load data from a file named sample.csv and the example.rul translation file. **Follow these steps:**

1. Open the CSV file that contains the following information:
   ```
   "name","class","alarm_id","disk space","carrier","phys_mem"
   "server 1","Server","1","5","","7"
   "server2","Server","2","6","",""
   "acd1","ACD","3","","Vendor1",""
   "server3","Discovered Hardware","4","","","",""
   ```
2. Open the translation file that contains the following information:
   ```
   <ruleset>
   <rule><attribute>attributename</attribute>
         <from>disk space</from><to>hard_drive_capacity</to>
         <rulename>rule12</rulename></rule>
   </ruleset>
   ```
3. Execute the following GRLoader command:
   ```
   grloader  - u username -p password  - s  http://sdmhost:8080  - i example13.csv  - tf
    example13.rul -lttwa.
   ```
   GRLoader loads 4 CIs to the TWA.

How to Prepare for Loading Spreadsheet Data

This article contains the following topics:

If the spreadsheet includes instructions, comments, and data that you do *not* want load into CA SDM, specify the exact area you want to import into GRLoader. To specify the area of the spreadsheet to use as input, use GRLoader to provide the spreadsheet file name, the spreadsheet sheet name, the first and last rows, and the first and last columns, as set by the - i (or -ssf), -sss, -ssfr, -sslr, -ssfc, and - sslc options.

Perform the following tasks before loading CI data from a spreadsheet with GRLoader.

1. Identify the general area of the spreadsheet to use as an input to GRLoader.
2. If necessary, map attribute names and data values using a translation file.
3. Identify parts of the spreadsheet that GRLoader *must* ignore or skip.

The input examples in the following sections use a fictitious spreadsheet named *grloader_spreadsheet_example.xls*. This spreadsheet contains nine subsheets, named Sheet1, Sheet2, Sheet3, and so on.

Spreadsheet Data Loading Support

GRLoader supports loading CI and CI relationship data from spreadsheets in Microsoft® Excel XLS and XLSX format, but it does *not* support XLSB files. By default, GRLoader attempts to process the first sheet of the spreadsheet file. In general, GRLoader attempts to match the column names in the spreadsheet to attribute names in the CI or relationship. GRLoader uses rules similar to when importing data from the TWA.

For example, a spreadsheet contains the following cells:

name	class	phys_mem
server1	Server	1gb
server2	Server	2gb

The following XML matches the spreadsheet cells exactly:

```
<GRLoader>
<ci><name>server1</server><class>Server</class>
    <phys_mem>1gb</phys_mem></ci>
<ci><name>server2</server><class>Server</class>
    <phys_mem>2gb</phys_mem></ci>
</GRLoader>
```

Note: Like when loading data to the TWA, if the GRLoader import results in an error, a file ending in **_err.XML* appears in the input file directory with information about the error. For example, if your spreadsheet file name is cidata.xlsx, the error file is named cidata_err.xml. As usual, logging information is also directed to the GRLoader.log file in the nxroot\log directory.

Spreadsheet Column Rules

A single spreadsheet can contain both CIs and relationships. When you load spreadsheet data, the presence of specific columns implies the target object type, as described by the following rules:

- If a column *class* is present with a nonblank value, the object type is assumed as *CI*.
- If any column prefixed with *provider_* is present with a nonblank value, the object type is assumed as *relation*.
- If data exists in both the *class* and *provider_* columns in the spreadsheet, you can specify *objecttype*. Use *ci* or *relation* as the value of the objecttype.
- A default class specified on the command line with *-ad class=class name*.
- A default provider attribute is specified on the command line with *-ad provider_xxxx*=nonblank.

By applying these rules, you cannot determine if the following example input data represents a CI or a relationship, which can represent an update to a CI:

name	hard_drive_capacity
Server1	10 GB

If you did not specify class, set the *objecttype* on the command line, or set it explicitly in the spreadsheet table as shown in the following example:

objecttype	name	hard_drive_capacity
ci	Server1	10 GB

The objecttype attribute takes precedence over the existence of class or provider_ attribute names when GRLoader determines if the row represents a CI or a relationship.

When you load relationship data from spreadsheets or databases, prefix the CI-identifying attribute with either *provider_* or *dependent_*, to match how they were specified in the TWA. For example, specify the name of a relationship provider by naming a column as *provider_name* in the database table.

Spreadsheet Rows with Embedded Relationships

For convenience of entering data, a single row in a spreadsheet can contain embedded relationships. A row with an embedded relationship contains both a CI and its associated relationships. The following example shows a CI with an embedded relationship.

objecttype	name	hard_drive_capacity	is used by
ci	Server1	10 GB	Server2; Server3

How GRLoader Converts Spreadsheet Data Types

When you use spreadsheets to load data, GRLoader performs data type conversions automatically:

1. Converts *DATES* to standard CA SDM dates in the *yyyy.mm.dd hh:mm:ss* format.
   Note: Excel stores these dates as milliseconds past the epoch.
2. Converts *NUMERIC* values to strings.
3. Converts *BOOLEAN* values to 0 or 1.
4. *STRINGS* are not altered.
5. GRLoader ignores *BLANK* values.
6. Evaluates *FORMULAS* first and then applies the previously specified rules to the cell.

Translation Rule Spreadsheet Attribute Mapping

Column names typically define the same CI or CI relationship attribute names. If the names differ, use a translation file to convert a column heading to an attribute name.

> **WARNING**
> The first row of the selected input area is significant because it contains column headings.

You map attribute names by including a rule which specifies a rule for the new keyword attribute name named *attributename*. GRLoader processes rules for *attributename* before applying value maps. A rule which specifies *<attribute>attributename</attribute>* is an attribute map, not a value map, so it changes the attribute name of an attribute, not the value of the attribute data.

For example, CA SDM requires that you load the IP address of a CI into a field named *alarm_id*. If the spreadsheet column name is *ip address*, use a translation rule to map *ip address* to *alarm_id* by using the following translation rule:

```
<rule>
   <attribute>attributename</attribute>
   <from>ip address</from>
   <to>alarm_id</to>
</rule>
```

Mapping an attribute name to “” (blank) is equivalent to ignoring that attribute. In the following example, GRLoader ignores all *owner* data:

```
<rule>
   <attribute>attributename</attribute>
   <from>owner</from>
   <to></to>
</rule>
```

Setting Default Attribute Values

In a typical business environment, you use an existing spreadsheet as an input to GRLoader. We recommend that you do not change the content in the spreadsheet. If that spreadsheet is incomplete and does *not* contain information necessary to import the data, you *must* specify the - ad (attribute default) option on the command line, as shown in the following example:

```
grloader  - u username  - p password  - s http://sdm-host:8080  - i myspreadsheet.xlsx  - ad tenant=tenanta -
ad mdr_class=spreadsheet -ad mdr_name=mdr1 -n
```

You can specify any attribute with the *- ad* option. For example, set a default IP address with the following command:

```
grloader  - u username  - p password  - s http://sdm-host:8080  - i myspreadsheet.xlsx  - ad
 alarm_id="Unknown" -n
```

If a row in the spreadsheet contains a value for alarm_id, then the default value is not used.

You can also specify attribute defaults for relationships. If myspreadsheet.xlsx contains both CIs and relationships, the relationships receive a warning message because alarm_id is not a valid relationship attribute. In this case, use the *-iia* parameter in GRLoader to ignore the invalid attribute name because this option suppresses the warning message.

Spreadsheet Considerations

The spreadsheet input area contains two areas, column headings and data. The GRLoader options bind the square input area for the start row, end row, start column, and end column. If you do not supply these options, then the bounds define the natural limits of the spreadsheet.

If you want to import CI data and CI relationships from spreadsheets, consider the following information:

- GRLoader considers the first populated row in the input area as column headings, and *must* contain CI attribute names or relationship attribute names, as XML attribute names appear for XML-based input.
- GRLoader does *not* process comments from Excel spreadsheets.
  Note: A red triangle in the upper right corner of the cell indicates a comment in Excel.
- GRLoader ignores the entire column when the heading row value is prefixed with the comment character (#).
- If a heading line consists entirely of comment, GRLoader considers it as a blank line and skips it.
- GRLoader can view hidden rows and columns in spreadsheets.
- GRLoader ignores embedded objects within spreadsheets.
- GRLoader identifies embedded relationships by an attribute name when it is an actual relationship type.
- GRLoader supports embedded relationships with the following limitations:

  - The CI identified by name, serial number, mac address, and so on, represents the provider CI.
  - In the relationship column, specify the CI as the dependent, when you use a provider/dependent relationship type.
  - Using a dependent/provider relationship type reverses the roles of the two CIs.
  - Multiple embedded relationships can occur in a single relationship cell. You separate each partner name from other partners by the *grloader.spreadsheet.embeddedseparator* character specified in the config file.
  - If you *must* specify more than one identifying attributes for a dependent or provider in a relationship, then you *must* fall back to the fully qualified style of creating relationships. In this case, you specify provider_name, provider_mac_address, type, dependent_name, dependent_mac_address, and so on. If you want to specify more than one identifying attribute, you cannot use embedded relationships.
- GRLoader can evaluate most functions in a spreadsheet, but not all functions. The following functions are known to be incompatible with GRLoader:
  - edate()
  - text()

    WARNING
    Any cell that directly or indirectly references an incompatible function generates an error message and GRLoader ignores the cell.

Error Handling

When loading data into integer fields (such as hardware_server.purchase_amount) previous versions of GRLoader would attempt to load the data resulting in partial and incorrect data. The current GRLoader generates a warning message indicating the data is invalid. The only valid characters in an integer field are [0-9].

Because using the *- ad* option often results in an attribute warning message when both CI and relationships are imported in the same run, referring to an invalid attribute in a CI or Relationship definition results in a warning instead of an error. The Ignore Invalid Attribute *- iia* option can be used to suppress these warnings.

Example Load CI Data from a Simple Spreadsheet

Use GRLoader to load a simple spreadsheet with CI data. The column headings match CI attribute names in the following example in *Sheet1* of *grloader_sample_spreadsheet.xls*:

name	**class**
server1	Server
server2	Server
acd1	ACD

To load the data with GRLoader, execute the following command:

```
grloader  - u username  - p password  - s http://sdm-host:8080 -i grloader_sample_spreadsheet. -n
```

GRLoader loads three CIs with no errors.

Example Load CI Data from the Extension Table

Use GRLoader to load a spreadsheet with CI data, including data from the extension table. The column headings match CI attribute names in the following example in *Sheet2* of *grloader_sample_spreadsheet*.xls:

name	**class**	**alarm_id**	**hard_drive_capacity**	**carrier**	**phys_mem**
server1	Server	1	5		7
server2	Server	2	6		
acd1	ACD	3		carrier1	
server3	Discovered Hardware	4			

To load the data with GRLoader, execute the following command:

```
grloader  - u username  - p password  - s http://sdm-host:8080 -i grloader_sample_spreadsheet.xls -
sss Sheet2 -n
```

GRLoader loads four CIs with no errors, including family-specific attributes. If the data is offset from the A1 cell in *Sheet3* in the spreadsheet, and the data has blank rows and columns, load the data by executing the following command:

```
grloader  - u username  - p password  - s http://sdm-host:8080 -i grloader_sample_spreadsheet.xls -
sss Sheet3 -n
```

GRLoader ignores blank rows, columns, and cells, and loads four CIs with no errors. If the data in *Sheet4* contains missing (or blank) column headings, such as E5, execute the following command:

```
grloader  - u username  - p password  - s http://sdm-host:8080 -i grloader_sample_spreadsheet.xls -
sss Sheet4 -n
```

GRLoader ignores the empty Column E (E5) and loads four CIs with no errors. If the data in *Sheet5* contains four rows that you want to ignore, and the CI data begins on the C5, execute the following command:

```
grloader  - u username  - p password  - s http://sdm-host:8080 -i grloader_sample_spreadsheet.xls -
sss Sheet5 -ssfr 5 -n
```

GRLoader ignores the first four rows and loads four CIs with no errors.

Example Load a Spreadsheet with Column Names that Contain Invalid CI Attribute Names

Use GRLoader and translation file to load a spreadsheet with column names that contain invalid CI attribute names. The name column begins on row C5. The column headings match CI attribute names in the following example in *Sheet7* of *grloader_sample_spreadsheet.xls*:

name	class		hard_drive_capacity	carrier	phys_mem
server1	Server	ignore	ok		ok
server2	Server	ignore	ok		ok
acd1	ACD	ignore		ok	ok
server3	Discovered Hardware	ignore			ok

Before you run GRLoader, create a translation file named *Sheet7.rul* that contains the following XML:

```
<ruleset>
<rule>
   <attribute>attributename</attribute>
   <from>ip address</from>
   <to>alarm_id</to>
</rule>
</ruleset>
```

After you create this XML file, load the data with GRLoader by executing the following command:

```
grloader  - u username  - p password  - s http://sdm-host:8080  - i grloader_sample_spreadsheet.xls
  - sss Sheet7  - ssfr 5  - tf Sheet7.rul -n
```

GRLoader ignores the first four rows and column E because E5 is empty, and loads four CIs with no errors.

Load a Spreadsheet that Does Not Contain mdr_name or tenant

Use GRLoader to load a spreadsheet that does not contain mdr_name or tenant. You can set attributes from the command line. The name column begins on C5. The column headings match CI attribute names in the following example in *Sheet8* of *grloader_sample_spreadsheet.xls*:

name	class	tenant	federated_asset_id
server1	Server		f1
server2	Server		f2
acd1	ACD	tenantb	f3
server3	Discovered Hardware	tenantb	f4

To load the data with GRLoader, execute the following command:

```
grloader  - u username  - p password  - s http://sdm-host:8080  - i grloader_sample_spreadsheet.xls  - sss
 Sheet8  - ssfr 5  - ad tenant=tenanta  - ad mdr_name=ACMserver1  - ad mdr_class=ACM12 -n
```

GRLoader assigns server1 and server2 to tenanta, assigns all CIs with the mdr_name as ACMserver1, assigns all CIs with the mdr_class as ACM12, and loads the four CIs with no errors.

Example Load a Spreadsheet that Rejects Bad Data

Use GRLoader to load a spreadsheet that rejects bad data by using a translation rule. In the following example, the name column begins on row C5 in *Sheet9* of *grloader_sample_spreadsheet.xls*:

name	class	skip	ip address
server1	Server		
server2	Server	yes	
server3	ACD	1	
server4	ACD		bad
server5	ACD		bad
server6	ACD		

Before you run GRLoader, create a translation file named *Sheet9.rul* that contains the following XML:

```
<ruleset>
<rule><attribute>skip</attribute><from>1</from>
      <reject>yes</reject><rulename>rule1</rulename></rule>
<rule><attribute>skip</attribute><from>yes</from>
      <reject>yes</reject><rulename>rule2</rulename></rule>
<rule><attribute>alarm_id</attribute><from>bad</from>
      <reject>yes</reject><rulename>rule3</rulename></rule>
<rule><attribute>attributename</attribute>
      <from>ip address</from><to>alarm_id</to>
      <rulename>rule4</rulename></rule>
</ruleset>
```

After you create this XML file, load the data with GRLoader by executing the following command:

```
grloader  - u username  - p password  - s http://sdm-host:8080  - i grloader_sample_spreadsheet.xls
  - sss Sheet9  - ssfr 6  - tf Sheet9.rul -n
```

Based on the translation rule, GRLoader completes the following actions:

1. GRLoader loads two CIs: server1 and server6.

2. GRLoader rejects server2 because the skip column is set to yes (rulename rule2).
3. GRLoader rejects server3 because the skip column is set to 1 (rulename rule1).
4. GRLoader rejects server4 and server5 because of bad IP addresses for these CIs (rulename rule3).

Note: The attribute name translation (rule4) occurs before value translation, so the value translation rules *must* specify the translated attribute name. Even though the spreadsheet contained a column heading of *ip address*, the rule to reject the *bad* ip addresses (rule 3) *must* specify alarm_id.

Example Load a Spreadsheet Using Embedded Relationships

Use GRLoader to load a spreadsheet that contains CIs and relationships using embedded relationships. If you want to specify more than one identifying attribute, you cannot use embedded relationships. In the following example, the name column begins on row A1 in *Sheet10* of *grloader_sample_spreadsheet.xls*:

name	class	connects to	hosts
accounting	Service		
server1	Server		accounting
server2	AIX	server1	accounting
server3	UNIX		GIS

Load the data with GRLoader by executing the following command:

```
grloader  - u username  - p password  - s http://sdm-host:8080  - i grloader_sample_spreadsheet.xls
 -sss Sheet10 -aer yes -n
```

GRLoader creates the following CIs:

- accounting
- server1
- server2
- server3

GRLoader creates the following relationships:

- server1 hosts accounting
- server2 connects to server1
- server2 hosts accounting
- server3 hosts GIS

Loading Relationship Data from Spreadsheets

When you load relationship data from spreadsheets, specify the attributes for the provider and dependent CIs in the same way as they were in the TWA. As a result, you prefix the CI-identifying attribute with either *provider_* or *dependent_*.

For example, to specify the name of a relationship provider, specify *provider_name* in the spreadsheet column heading.

Example Load CI Relationships

Use GRLoader to load a spreadsheet that contains CI relationships. In the following example, the dependent_name column begins on row A5 in *Sheet6* of *grloader_sample_spreadsheet.xls*:

dependent_name	type	provider_name
service1	uses	server2
service1	uses	server3

To load the spreadsheet with CI relationships, execute the following command:

```
grloader  - u username  - p password  - s http://sdm-host:8080  - i grloader_sample_spreadsheet.xls  - sss
 Sheet6  - ssfr 5 -n
```

After you execute the command, GRLoader ignores the first four rows and loads two relationships with no errors.

Example Load a CI with Multiple Embedded Relationships

Use GRLoader to load a spreadsheet that contains CIs and relationships using embedded relationships. If you want to specify more than one identifying attribute, you cannot use embedded relationships. In the following example, the name column begins on row A1 in *Sheet11* of *grloader_sample_spreadsheet.xls*:

name	class	manages	is managed by
chief	manager		
agent99	agent		
agent86	agent		
chief	manager	agent86;agent99	CEO

Load the data with GRLoader by executing the following command:

```
grloader  - u username  - p password  - s http://sdm-host:8080  - i grloader_sample_spreadsheet.xls
 -sss Sheet11 -aer yes -n
```

GRLoader creates the following CIs and relationships:

- The first three rows create three CIs.
- Row 4, processed after rows 1-3, creates the following CI relationships:
  - chief *manages* agent86
  - chief *manages* agent99
  - chief *is managed by* CEO
- As with XML input, the direction of the relationship is significant. In the *manages* column labeled, GRLoader creates two relationships, because chief is the *provider* for both dependents, agent86 and agent99.
- Because the *is managed by* attribute is a dependent/provider relationship type, the provider and dependents are switched in the *is managed by* column, because chief is a dependent and CEO is the provider.

Example Create a CI and a Relationship by Specifying the UUID

Use GRLoader to load a spreadsheet that contains CIs and relationships by specifying the UUID. In the following example, the *# transaction* column begins on row A1 in *Sheet12* of *grloader_sample_spreadsheet.xls*:

# transaction	objecttype	name	class	provider_id	type	dependent_name
1	ci	server1	Server			
2	relation			12345678901 23456789012 3456789012	uses	server2

Load the data with GRLoader by executing the following command:

```
grloader  - u username  - p password  - s http://sdm-host:8080  - i grloader_sample_spreadsheet.xls
 -sss Sheet12 -n
```

GRLoader creates the following CIs and relationships:

- The transaction creates the server1 CI.
- The transaction creates a relationship between server1 and server2.

Example Load a Spreadsheet that Contains Change Specifications

Use GRLoader to load a Change Order with multiple change specifications. In the following example, the # *transaction* column begins on row A1 in *Sheet13* of *grloader_sample_spreadsheet.xls*:

objecttype	chg	ci	attribute_name	attribute_value_planned
change specification	12345	server1	ip address	1.2.3.4
change specification	12345	server2	ip address	1.2.3.5
change specification	12345	server3	ip address	1.2.3.6
change specification	12345	server4	ip address	1.2.3.7

Load the data with GRLoader by executing the following command:

```
grloader  - u username  - p password  - s http://sdm-host:8080  - i grloader_sample_spreadsheet.xls
  - sss Sheet13
```

GRLoader creates four CIs and inserts the CIs into Change Order 12345.

Data Translation

This article contains the following topics:

Data values provided by an MDR may not meet the requirements of CMDB because of the following reasons:

- The country or language for an MDR can differ from the country or language selected for CMDB server installation. For example: A CA Cohesion MDR that uses American English transfers data to a CMDB in France. When CA Cohesion creates server CIs, it specifies the CI family as "Server". However, in France the CI family must be specified as "Serveur". An inbound "Server" value must be translated to the required "Serveur" value whenever the American-based MDR communicates with the French CMDB installation.
- Data inconsistencies can occur in lookup (SREL) fields. For example: An MDR contains CIs with a manufacturer of "Dell Inc", "Dell Corporation", or simply "Dell". If the CMDB manufacturer table requires "Dell Inc", other values are rejected with warning messages. The invalid inbound "Dell Corporation" and "Dell" values must be translated to the standard "Dell Inc" value for the manufacturer attribute.
- Data inconsistencies in non-SREL fields. For example, some MDRs report data in units, while others report data in bytes or gigabytes. You can standardize the format of the data stored in the CMDB.

To satisfy these requirements, GRLoader can translate any incoming value to another value, by using an XML-based lookup file when GRLoader is run.

> **WARNING**
> The pre-edit translation and validation step occurs when CI and relationship XML is read, before normal GRLoader processing occurs (for example, update_if_null, lookup, dateformat) and before data is transmitted to the CMDB server.

Because each MDR can have specific translation requirements, the data translation file is specified for each GRLoader invocation. For standardization purposes, we recommend that this file is located on a common file system and shared among the CMDB data providers.

Create Translation Rules

To use the GRLoader data translation and validation feature, create a set of rules to specify the data to be translated. Rules are required for each attribute and value being translated. Data translation rules are applied to the GRLoader input

XML using the - tf *filename* parameter. The rules in *filename* are applied to all input submitted to GRLoader using the -i parameter. You can also use translation rules when importing CI data from spreadsheets.

NOTE

For information about creating translation rules for importing spreadsheet data, see this topic.

To create translation rules, use a text editor to create and save rules in the GRLoader input XML like the following:

```
1. <ruleset>
2. <rule>
3. <attribute>class</attribute>
4. <from>Server</from>
5. <to>Serveur</to>
6. </rule>
7. <rule>
8. <attribute>manufacturer</attribute>
9. <from>Dell Corporation</from>
10. <to>Dell Inc</to>
11. </rule>
12. <rule>
13. <attribute>manufacturer</attribute>
14. <from>Dell</from>
15. <to>Dell Inc</to>
16. </rule>
17. </ruleset>
```

The translation rules are created.

Notes:

Lines 2-6 specify that whenever GRLoader encounters a line specifying <class>Server</class>: replace Server with Serveur (French) before sending the data to CMDB.

Lines 7-11 specify that a manufacturer of Dell Corporation should be replaced by Dell Inc. A single set of XML rules can be used to redefine several different attributes.

Lines 12-16 translate any input specifying Dell to the standard Dell Inc. The single set of XML contains multiple rules. When taken together, the rules specify multiple from/to values.

Data Transformation Example

This example shows a sample subset of the required rules for sharing data between MDRs using different languages. The GRLoader input XML example translates three classes from English to their French equivalents.

```
1. <?XML version="1.0" encoding="UTF-8"?>
2. <ruleset>
3. <rule>
4. <attribute>class</attribute>
5. <from>Server</from>
6. <to>Serveur</to>
7. </rule>
8. <rule>
9. <attribute>class</attribute>
10. <from>Printer</from>
11. <to>Imprimante</to>
```

```
12. </rule>
13. <rule>
14. <attribute>class</attribute>
15. <from>Contract</from>
16. <to>Contrat</to>
17. </rule>
18. </ruleset>
```

Data Validation

Often the values that are accepted into an attribute must be validated against a list of acceptable values before the CI is stored. The relationship between an attribute and its set of acceptable values (stored in a separate table) is named a *single relationship* (SREL).

When you want to validate data even when an SREL is not created for it, data translation rules can enforce standardization of data values.

Example: Convert Units of Data Storage

In the following GRLoader input XML, the MDR provides data in gigabytes (GB), but we want to store the total number of bytes in the CMDB.

```
1. <ruleset>
2. <rule>
3. <attribute>phys_mem</attribute>
4. <from>1 GB</from>
5. <to>1,073,741,824</to>
6. </rule>
7. <rule>
8. <attribute>phys_mem</attribute>
9. <from>2 GB</from>
10. <to>2,147,483,648</to>
11. </rule>
12. ...
13. </ruleset>
```

Unmatched or Non-Standard Input Values

When validating data, you can reject unacceptable values and replace them with new values. When input data does not match a rule, it proceeds unchanged to the next GRLoader phase.

Example: Validate Primary Colors

In the following example, if the GRLoader input specifies <color>hot pink</color>, the color data is unaffected by any translation.

```
1. <ruleset>
2. <rule>
3. <attribute>color</attribute>
4. <from>red</from>
5. <to>red</to>
6. </rule>
7. <rule>
8. <attribute>color</attribute>
9. <from>blue</from>
```

```
10. <to>blue</to>
11. </rule>
12. <rule>
13. <attribute>color</attribute>
14. <from>yellow</from>
15. <to>yellow</to>
16. </rule>
17. </ruleset>
```

In the previous example, the "from" and "to" values are the same. The following example shows a shortened form of the rule definition that does not include the "to" value:

```
1.  <ruleset>
2.  <rule>
3.  <attribute>color</attribute>
4.  <from>red</from>
5.  </rule>
6.  <rule>
7.  <attribute>color</attribute>
8.  <from>blue</from>
9.  </rule>
10. <rule>
11. <attribute>color</attribute>
12. <from>yellow</from>
13. </rule>
14. </ruleset>
```

Using the shortened form of the rule definition, line 16 is more apparent. Line 16 specifies that if there is no matching "from" value for an attribute, that whatever value is specified, it is replaced by the "to" value.

```
1.  <ruleset>
2.  <rule>
3.  <attribute>color</attribute>
4.  <from>red</from>
5.  </rule>
6.  <rule>
7.  <attribute>color</attribute>
8.  <from>blue</from>
9.  </rule>
10. <rule>
11. <attribute>color</attribute>
12. <from>yellow</from>
13. </rule>
14. <rule>
15. <attribute>color</attribute>
16. <to>unknown color</to>
17. <unmatched>yes</unmatched>
18. </rule>
19. </ruleset>
```

The GRLoader input includes a rule that matches "hot pink" (the "unmatched" rule on line 16). If color specifies an attribute other than red, blue or yellow (as indicated on lines 4, 8 and 12 respectively), that color is changed to the "to" value. For example, <color>hot pink</color> is recoded to <color>unknown color</color>.

If only lines 14-18 appear in the rule set (that is, no matches are possible), all colors in the GRLoader input XML file are set to "unknown color". This technique forces all values of a specific attribute to a single value.

> **WARNING**
> The editing process cannot create new XML when none exists. If the input XML does not include information about <widgets>, all rules about <widgets> are ignored.

Example: Change All Unmatched "owner" Attributes to "Pete"

The following GRLoader input XML sets the value Pete for unmatched owners.

1. <ruleset>
2. <rule>
3. <attribute>owner</attribute>
4. <unmatched>yes</unmatched>
5. <to>Pete</to>
6. </rule>
7. </ruleset>

Consider how the following GRLoader input XML uses the previous ruleset:

1. <GRLoader>
2. <ci>
3. <name>server1</name>
4. <owner>John</owner>
5. </ci>
6. </GRLoader>

If the attribute "owner" has a rule, the rule attempts to match the value "John". Because no rule for the value "John" exists, GRLoader looks for an unmatched rule for the attribute "owner". If one exists, the translated input results in the following:

1. <GRLoader>
2. <ci>
3. <name>server1</name>
4. <owner>Pete</owner>
5. </ci>
6. </GRLoader>

Now consider the following GRLoader input XML file:

1. <GRLoader>
2. <ci>
3. <name>server2</name>
4. </ci>
5. </GRLoader>

The ruleset results in the following:

1. <GRLoader>
2. <ci>
3. <name>server2</name>
4. </ci>

5. </GRLoader>

The CI "server2" does not set the owner to Pete because no owner tag exists in the original XML.

Specify an Empty String

When an empty string must be specified as either the "from" value or the "to" value, always include the <from> or <to> value in the rule set.

> **WARNING**
> Specifying <to></to> or not specifying <to> in the XML have very different XML meanings!

1. <ruleset>
2. <rule>
3. <attribute>size</attribute>
4. <from>XXL</from>
5. </rule>
6. <rule>
7. <attribute>manufacturer</attribute>
8. <from>General Motors</from>
9. <to></to>
10. </rule>
11. </ruleset>

Lines 2-5 specify that a size of XXL is possible. Because <to> is not specified, no recoding is performed on a size=XXL. This kind of rule is only useful when an unmatched rule appears later in the rule set for the same attribute.

Lines 6-10 examine all input data for a manufacturer="General Motors". Whenever this rule is found, because "<to></to>" is specified in the rule on line 9, the value of "General Motors" is replaced by "".

If you want to blank out the manufacturer, specify the update_if_null="YES" keyword in the GRLoader input XML.

> **NOTE**
> For more information about the use of the "update_if_null" option to blank out values in the database, see this topic.

Alternative Comparison Methods

The default method uses "equals" for comparison. That is, when the <from> value is compared with the value in the GRLoader input, the two are considered to match when they are equal. The <comparetype> tag specifies alternative forms of comparison.

The comparetype tag accepts one of the following values:

- startswith
- endswith
- contains
- equals
- equalsignorecase

Example: Standardize a Company Name

In the following example, all manufacturer names beginning with "Dell" (such as "Dell Corp", "Dell Inc", "Dell Corporation") are reset to "Dell".

1. <ruleset>
2. <rule>
3. <attribute>manufacturer</attribute>
4. <from>Dell Corp</from>

5. <to>Dell</to>
6. </rule>
7. <rule>
8. <attribute>manufacturer</attribute>
9. <from>Dell Inc</from>
10. <to>Dell</to>
11. </rule>
12. <rule>
13. <attribute>manufacturer</attribute>
14. <from>Dell Corporation</from>
15. <to>Dell</to>
16. </rule>
17. </ruleset>

Alternatively, the following rule produces the same result:

1. <ruleset>
2. <rule>
3. <attribute>manufacturer</attribute>
4. <from>Dell</from>
5. <comparetype>startswith</comparetype>
6. <to>Dell</to>
7. </rule>
8. </ruleset>

Input Rejection

To reject input from an MDR before loading data into the CMDB, use the <reject> tag.

Example: Rejecting Input Data

The<reject tag> can be used with the <comparetype> tag, as shown in following example.

1. <ruleset>
2. <rule>
3. <attribute>name</attribute>
4. <from>test</from>
5. <comparetype>startswith</comparetype>
6. <reject>yes</reject>
7. </rule>
8. </ruleset>

When a reject rule is matched, the corresponding CI or relationship is rejected and the CMDB is not updated or created for that entire object. The transaction is skipped, and the XML is written to the _err file with an error message indicating that it was rejected.

Rule Syntax

The following table describes the XML tags that are used in a data translation rule set.

Tag	Description
<?XML version="1.0" encoding="codepage"?>	Enables different code pages for GRLoader.
<ruleset>	Begins a rule set. A ruleset can contain many rules.

<rule>	Begins a rule
<attribute>*attr_name*</attribute>	Specifies an attribute that the rest of the rule applies to. *attribute* must be a valid CMDB attribute name.
<from>*value*</from>	Specifies a value to be changed. The <from> tag is modified by the <comparetype> tag.
<to>*value*</to>	Specifies the replacement value
<comparetype>*value*</comparetype>	(Optional) Specifies one of the following values: equals startswith endswith contains equalsignorecase If not specified, "equals" is the default.
<reject>yes</reject>	Specifies that GRLoader reject the CI or relationship. Yes can be specified as "yes" or one (1). No can be specified as "no" or zero (0). If not specified in a rule, the default is "no" (reject).
<rulename>*rule_name*</rulename>	(Optional) *name* assigned to identify this rule. This name appears in debugging messages.
</rule>	Ends a rule
</ruleset>	Ends a rule set

Running GRLoader with Translation Enabled

To run GRLoader using translation/transformation, run GRLoader with the **- tf** *filename* option. *filename* specifies the file which that contains the translation rule set.

NOTE
Alternatively, you can specify grloader.translationfile=*filename* in the configuration file.

Logging

Input modifications are logged in the stdlog.n and the GRLoader log messages, which reflect the data values after translation rules have run.

You can run GRLoader with the - T option set to five (5) or greater to display additional debugging information.

Localized CMDB Considerations

When implementing a localized CMDB, you can translate the class and family names from one language to another. Translation rules are provided in the $nxroot/java/lib/GRLoader directory. These rules are named xlate_*xx*_to_*yy*.RUL. *xx* and *yy* represent the language codes (en, fr, es, dm, and so on).

You can expand these rules to accommodate any additional SREL fields.

The XML Header

Following XML coding standards, if the XML content in the rule set contains non-UTF-8 characters, you may require a line at the beginning of the XML translation file that is similar to the following:

<?XML version="1.0" encoding="*codepage*"?>

codepage defines the code page.

NOTE
For more information about GRLoader XML, see the CMDB Technical Reference .

Test the Rules

Before running the XML input file through GRLoader, test the rules and view the translation results.

To test the translation rules, run GRLoader without the " - a" or " - n" options.

Running without inserts and updates effectively writes the translated and validated XML to the _err.XML file, where the results of the rule translation can be reviewed.

Run GRLoader from a Remote MDR

You can use GRLoader to copy data from a remote MDR to the CMDB in either of two ways:

- Copy the XML data from the remote system that runs the MDR to the system running CMDB, and then execute GRLoader on the CMDB system.
- Execute GRLoader on the remote MDR system itself.

To prepare to execute GRLoader from a remote system that does not have CMDB installed

1. Verify that the Java Runtime Environment (JRE) version 11.0.18 or higher is installed and available.
2. Copy the contents of the %NX_ROOT%\java\lib directory from the CMDB system to a directory on the remote system where you want to run it. This remote directory is called %ROOT%.
3. Create a file called NX.ENV in the %ROOT% directory:
   @NX_LOG=path_which_will_contain log files
4. Create directory %ROOT%\site\cfg
5. Create directory %ROOT%\log

To run GRLoader from the remote system, execute the following command:

```
java -Xmx512M -cp %ROOT% -jar %ROOT%/GRLoader.jar -N %ROOT% -u [userid]  - s [server] -i [other GRLoader
 options]
```

where %ROOT% is the fully qualified path containing the files that were copied in Step 2.

GRLoader and Multi-Tenancy

Multi-tenancy allows multiple independent tenants to share hardware and application support resources in a single implementation of CMDB. You can use the tenant attribute (<tenant>) in XML so that GRLoader assigns tenants for multi-tenancy use in CMDB. All changes that you make to the tenant attribute are reflected in the CMDB Versioning tab.

The tenant attribute is as follows:

- **<tenant>**
  Specifies the tenant assignment for the CI/Relationship. You can use PUBLIC to specify that the object is public. The Tenant may or may not be set in the object, depending on your default roles tenant access.

Consider the following tenant assignment behavior *before* you implement multi-tenancy using GRLoader:

- Tenants can only be assigned during the creation of a CI or relationship.
- All CIs that GRLoader loads are assigned either a default tenant or a specific one from the XML file.
- GRLoader XML lets you specify the <tenant> attribute or a default tenant for a CI or a relationship.
- If you do not specify <tenant> or a default tenant, the tenant is assumed to be blank and the tenant assignment is based on the logged on users default role. This default role assignment is used primarily for Cohesion and other MDRs that do not specify a tenant when creating CIs.
- GRLoader sets the tenant of a CI or relationship based on input from the following sources. When the default role lets you select the choice of tenant in the objects created, you can specifically set the tenant for an object. The multi-tenancy option must be set to setup or on to use <tenant>.
  - Including <tenant> in the xml.
  - Use of the - dt command line option when invoking GRLoader.
  - Use of the grloader.defaulttenant option in the configuration file.
  - The default tenant associated with the contact.

Example: Set the Tenant for an Object

Your default access allows you to create CIs for a specific tenant and for public use. You want to create several public CIs.

Run grloader with a default tenant of PUBLIC to indicate the tenant of the new objects specifically.

Bulk Loading Change Specifications with GRLoader

Contents

After a user creates a Change Order, you can use GRLoader to bulk load change specifications, instead of using the web interface. Use the *object* node with the *objecttype* attribute.

You can use the following attributes with GRLoader:

- **objecttype**
  (Required) Identifies the Change Specification.
- **chg**
  (Required) Specifies the Change Order ticket number.
- **ci**
  Specifies the CI name.
- **attribute_name**
  (Required) Specifies the CI attribute that you want to change.
- **attribute_value_planned**
  Specifies the new value of the attribute_name you entered.
- **status**
  Specifies the initial verify status. By default, GRLoader uses the managed attribute initial verify status.
- **description**
  Specifies descriptive text of the change.

Note: Change specifications are not reconciled. If you run GRLoader multiple times with identical input, CACF creates duplicate change specifications. Use the web interface to edit and remove duplicates that CACF creates accidentally. Duplicates are independent of each other.

Change Specification Example XML

The following example XML adds change specifications for a Change Order numbered 24. The changeorder24.xml file specifies that alarm_id becomes server1, and location becomes 2.4.6.8.

```
<GRLoader>
     <object>
```

```
        <objecttype>ci_planned_change</objecttype>
        <description>created by grloader *now* </description>
        <attribute_name>alarm_id</attribute_name>
        <chg>24</chg>
        <ci>server1</ci>
        <attribute_value_planned>2.4.6.8</attribute_value_planned>
    </object>
    <object>
        <objecttype>ci_planned_change</objecttype>
  <description>created by grloader *now* </description>
        <attribute_name>location</attribute_name>
        <chg>24</chg>
        <ci>server1</ci>
        <attribute_value_planned>NY</attribute_value_planned>
    </object>
</GRLoader>
```

Execute the following command:

```
grloader -u ServiceDesk -p password -s http://hostname:8080 -i changeorder24.xml
```

Change Specification Example Spreadsheet

The following example spreadsheet adds Change Specification for a Change Order numbered 24, which assigns new IP addresses and locations to three CIs:

objecttype	chg	CI	attribute_name	attribute_value_planned	description
ci_planned_change	24	server1	alarm_id	1.1.1.1	loaded by grloader at *now*
ci_planned_change	24	server2	alarm_id	1.1.1.2	loaded by grloader at *now*
ci_planned_change	24	server3	alarm_id	1.1.1.3	loaded by grloader at *now*
ci_planned_change	24	server1	location	NY	loaded by grloader at *now*
ci_planned_change	24	server2	location	NY	loaded by grloader at *now*
ci_planned_change	24	server3	location	NY	loaded by grloader at *now*

CI Reconciliation

This article describes the following:

- CI Reconciliation Attributes
- Transaction Work Area Attributes

CI Reconciliation Attributes

Reconciliation associates imported CI data with CIs in the CMDB.

Reconciliation uses the following CI identifying attributes:

- Name
- Serial Number
- Asset Number
- System Name
- DNS Name
- MAC Address

You must specify at least one of these values when you create or reference an existing CI.

The following table shows the results of the reconciliation process:

Name	Serial Number	Asset Number	System Name	DNS Name	MAC Address	Result
Unique	Null	Null	Null	Null	Null	CI Created
Null	Unique	Null	Null	Null	Null	CI Created
Null	Null	Unique	Null	Null	Null	CI Created
Null	Null	Null	Unique	Null	Null	CI Created
Null	Null	Null	Null	Unique	Null	CI Created
Null	Null	Null	Null	Null	Unique	CI Created
Duplicate	Duplicate	Duplicate	Unique	Duplicate	Duplicate	CI Created
Unique	Duplicate	Duplicate	Duplicate	Duplicate	Duplicate	Recognized as Duplicate CI
Null	Null	Null	Null	Unique	Unique	CI Created
Null	Null	Null	Null	Duplicate	Unique	Recognized as Duplicate CI
Null	Null	Null	Null	Unique	Duplicate	Recognized as Duplicate CI
Duplicate	Duplicate	Unique	Duplicate	Duplicate	Duplicate	CI Created
Duplicate	Unique	Duplicate	Duplicate	Duplicate	Duplicate	CI Created
Duplicate	Duplicate	Duplicate	Duplicate	Duplicate	Unique	Recognized as Duplicate CI
Duplicate	Duplicate	Duplicate	Duplicate	Unique	Duplicate	Recognized as Duplicate CI
Duplicate	Duplicate	Duplicate	Duplicate	Unique	Unique	Recognized as Duplicate CI

Transaction Work Area Attributes

Contents

The Transaction Work Area (twa) tables are:

- **ci_twa_ci**
  A single table that includes all attributes across all CA CMDB families. Table data is stored in denormalized form to enable customers and services to understand and manipulate the content more easily.
- **ci_twa_relation**
  Complements the ci_twa_ci table. Contains relationship information.
- **ci_twa_statusnames**

Descriptive labels for row status.

External processes update these tables and GRLoader reads them during transaction processing. When processing is complete, GRLoader updates the row_status and tran_message columns to indicate whether the transaction has completed successfully.

NOTE
If multiple errors or warnings occur, the messages are concatenated.

ci_twa_ci Attributes

The ci_twa_ci table contains attributes for CI transactions.

Column Name	Notes
id	Transaction identifier
last_mod_dt	Sets the current date every time the row is added or updated.
tran_dt	Sets the current date and time if no value is supplied when row is added.
creation_date	Sets the current date and time when row is added.
delete_flag	Sets to zero (0) if no other value is supplied when row is added.
tran_status	Sets to zero (0) if no other value is supplied when row is added.

ci_twa_relation Attributes

The ci_twa_relation table contains attributes for relationship transactions.

Column Name	Notes
id	Transaction identifier
last_mod_dt	Sets the current date every time the row is added or updated.
tran_dt	Sets the current date and time if no value is supplied when row is added.
creation_date	Sets the current date and time when row is added.
delete_flag	Sets to zero (0) if no other value is supplied when row is added.
tran_status	Set to zero (0) if no other value is supplied when row is added.

CMDB Web Services

CA CMDB provides a set of high-level web services that supports CMDBf Web Services Standard version 1.0. These services allow external CMDBf-aligned applications and also registered Management Data Repositories (MDRs) to interact with CA CMDB, including federated MDRs in accordance with the CMDBf/DMTF standard.

For more information, see the CMDBf specification standards.

Web Services Deployment

CMDB installation automatically deploys CMDB web services. If you want, you can redeploy the web services.

To redeploy CMDB Web Services

1. Execute the following command:
   : install-dir \sdk\websvc\cmdbf
2. Deploy the following files:
   – deploy.wsdd
   – cmdbf.jar
3. Execute the following file:
   deploy_cmdbws.bat
   The CMDB web services are deployed.

Web Service Components

CA CMDB Web Services consists of two defined services:

- **Registration Service**
  Allows clients to create\update CIs and Relationships. The endpoint can be located at the following address:
  `http://<servername>:< port >/axis/services/RegistrationPort`
  `For CA Service Management 17.3.0.18 and above versions, http://<servername>:< port >/cxf/services/RegistrationPort`
- **Query Service**
  Allows clients to query for CIs and Relationships. The endpoint can be located at the following address:
  `http://<servername>:< port >/axis/services/QueryPort`
  `For CA Service Management 17.3.0.18 and above versions, http://<servername>:< port >/cxf/services/QueryPort`

Registration Service

The Registration service uses push mode federation. The fundamentals of push mode federation are as follows:

- The client invokes the Register operation for configuration items or relationships that it wants to register. Each item or relationship must be associated with at least one record type supported by the Registration service.

  NOTE
  The CMDBf Register web service either creates a new CI or, if that CI already exists, updates it. If the CI already exists and is Inactive, the CI is set to Active and all attributes passed to it are updated. To prevent updates to Inactive CIs, send Inactive as one of the attributes.
- The Registration service responds with the registration status for each item or relationship named in the Register operation. The status is either accepted or declined.

The management data repository (MDR) also uses the Register operation to update registered data. An update can consist of any combination of the following:

- Changes to existing data, such as a property value change
- Deregistering a previously registered record type for this configuration item or relationship

MDR Registration

You must do the following to register an MDR:

1. Create a valid MDR manually before using CMDBf web services to register a CI.
2. Set the MdrName to your MDR Name.
3. Set the MdrClass to “cmdbf” (a static value).

Query Service

The Query service contains a GraphQuery operation that can be used for anything from a simple instance query to a much more complex topological query. A GraphQuery request describes items and relationships of interest in the form of a graph. Constraints can be applied to the nodes (items) and edges (relationships) in that graph to further refine them.

The GraphQuery response contains the items and relationships that, in combination, compose a graph that satisfies the constraints of the graph in the query. A graph query is only one level deep.

Login

You log in by passing credential information either through Java or a SOAP message. The following examples show you how you can log in.

Example: Java

```
QueryBindingStub binding;

binding = (QueryBindingStub) new QueryServiceLocator().getQueryPort(new URL(Endpoint));

SOAPHeaderElement Header = new SOAPHeaderElement("http://schemas.xmlsoap.org/soap/envelope/",
 "securityHeader");

Header.setPrefix("sec");

javax.xml.soap.SOAPElement Element = null;

Element = Header.addChildElement("username");
Element.addTextNode("CMDBAdmin");
Element = Header.addChildElement("password");
Element.addTextNode("password");
binding.setHeader(Header);
```

Example: SOAP Message

```
<soapenv:Header>
        <sec:securityHeader  xmlns:sec="http://schemas.xmlsoap.org/soap/enveloe/">
        <sec:username>CMDBAdmin</sec:username>
        <sec:password>password</sec:password>
        </sec:securityHeader>
        </soapenv:Header>
```

CMDB Web Services Access

You can access CMDB web services by using one of the following methods:

- Create your own message for the SOAP interface to call your endpoint.
- Write your own Java program to access the CMDB endpoint.

Code to Return All CIs From Every Family in CMDB

You can use the following code to return all CIs from every Family in CMDB:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
   <soapenv:Header>
```

```
        <sec:securityHeader xmlns:sec="http://schemas.xmlsoap.org/soap/envelope/">
                <sec:username>cmdbadmin</sec:username>
                <sec:password>miramar</sec:password>
        </sec:securityHeader>
   </soapenv:Header>
   <soapenv:Body>
      <dat:query>
        <itemTemplate suppressFromResult="false" id="All">
[constraints go here]
         </itemTemplate>
      </dat:query>
   </soapenv:Body>
</soapenv:Envelope>
//*********************************************//
```

Sample Java Program

To assist you with web services client application development, CMDB provides the following sample Java programs:

- RegistrationServiceTestCase.java shows you how to create two CIs and a Relationship.
- QueryServiceTestCase.java shows you how to query your CMDB for all CIs.

You can find sample programs at the following location:

```
%NX_ROOT%\sdk\websvc\cmdbf\
```

WSDL Document

The location of the Web Services Description Language (WSDL) document that you need depends on the function of CMDB you intend to use. The following locations are available:

- The default location of the WSDL for the CMDBf Web Services for Registration is the following URL:
  `http://<servername>:<port>/axis /services/RegistrationPort?wsdl`
  `CA Service Management 17.3.0.18 and above versions, http://<servername>:<port>/cxf/services/RegistrationPort?wsdl`
- The default location of the WSDL for the CMDBf Web Services for graphQuery is the following URL:
  `http://<servername>:<port>/axis/services/QueryPort?wsdl`
  `CA Service Management 17.3.0.18 and above versions, http://<servername>:<port>/cxf/services/QueryPort?wsdl`

> **NOTE**
> Many servlet containers use a port number different from 80. For example, Tomcat defaults to port 8080, which is established during installation.

Security Considerations

There are important security considerations when deploying web services. The default configuration when using HTTP is insecure, as it is for all information in web service calls sent between the client and the server in plain text over the network using the HTTP protocol. This configuration includes application data and login methods, and it can include passwords. Administrators who deploy web services are encouraged to consider security carefully and to take additional configuration steps at the application and network levels to secure your web service environment.

> **WARNING**
> The default web service configuration used with HTTP is insecure and vulnerable to security threats that can include password discovery, session fixation, and data spying, and so on.

CMDBf Implementation CMDB Limitations

CMDB has limitations in its CMDBf implementation. The following CMDBf pseudo-schema highlights some limitation areas:

```
<query>
        <itemTemplate id="xs:ID" suppressFromResult="xs:boolean">
                (<contentSelector ...>...</contentSelector> ?
                <instanceIdConstraint>...</instanceIdConstraint> ?
                <recordConstraint>
                        <recordType ... /> *
                        <propertyValue ...>...</propertyValue> *
                </recordConstraint> *)
                |
                (<xpathExpression...>...</xpathExpression> *)
                xs:any
        </itemTemplate> *
        <relationshipTemplate id="xs:ID" suppressFromResult="xs:boolean">
                (<contentSelector ...>...</contentSelector> ?
                <instanceIdConstraint>...</instanceIdConstraint> ?
                <recordConstraint>
                        <recordType>...</recordType> *
                        <propertyValue>...</propertyValue> *
                </recordConstraint> *)
                |
                (<xpathExpression ...>...</xpathExpression> *)
                <sourceTemplate ref="xs:IDREF" minimum="xs:int"?
                 maximum="xs:int"?/>
                <targetTemplate ref="xs:IDREF" minimum="xs:int"?
                 maximum="xs:int"?/>
                <depthLimit ... /> ?
                xs:any
        </relationshipTemplate> *
</query>
```

Item Template Limitations

The CMDB implementation uses the CMDBf specification with the following item template limitations:

- Multiple RecordConstraints under one ItemTemplate are handled as Logical OR, not Logical AND.
- suppressFromResult="xs:boolean" is not supported.
  CMDB always displays the results.
- <contentSelector matchedRecords="xs:*boolean*">:matchedRecords="true" is supported, but matchedRecords="false" is not.
  CMDB only supports one selectedRecordType per contentSelector.
- <recordConstraint> only supports one <recordType ... /> expression; for example <recordType namespace="http://cmdb.ca.com/Hardware" localName="Hardware.Server"/>
  The localName expression must identify a valid CMDB family.
  Replace each space in a family name that includes spaces with a dash (-). For example, replace Software.Application Server with Software.Application-Server.
- <propertyValue namespace="xs:anyURI" localName="xs:NCName" recordMetadata="xs:boolean" matchAny="xs:boolean">
  recordMetadata is not supported.

matchAny is set to the default value of false; this value allows "Logical AND" and "Logical OR" CMDBf queries on all the property values. "Logical AND" queries are not supported.
“like” operators are not supported.
“equals” does not support caseSensitive, or negate.

- <xpathExpression...>...</xpathExpression>: is not implemented.

Registration Limitations

Registration has the following limitations:

- Multiple record elements under one Item element or Relationship element
- Additional RecordType

Relationship Template Support and Limitations

CMDB supports the following relationshipTemplate features:

- contentSelector
- relationshipTemplate ID
- recordConstraint -- can use propertySelectors
- sourceTemplate
- targetTemplate

CMDB *does not* support the following relationshipTemplate features:

- Source/Target Template @minimum
- Source/Target Template @maximum
- Depth Limit with @MaxIntermediateItems
- Depth Limit with @intermediateItem Template
- instanceIdConstraint
- xpathExpression
- suppressFromResult in relationship Template

Example: Register Request relationshipTemplate Using a contentSelector and propertySelectors Under the recordConstraint

```
<relationshipTemplate  id="rels">
            <contentSelector>
                    <selectedRecordType namespace=" http://cmdb.ca.com/r1" localName="is-deployed-by">
                            <selectedProperty namespace=" http://cmdb.ca.com/r1" localName="last_mod_by"/>
                            <selectedProperty namespace=" http://cmdb.ca.com/r1" localName="last_mod_dt"/>
                            <selectedProperty namespace=" http://cmdb.ca.com/r1" localName="child"/>
                            <selectedProperty namespace=" http://cmdb.ca.com/r1" localName="parent"/>
                    </selectedRecordType>
              </contentSelector>
                             <sourceTemplate ref="Linux1" />
                             <targetTemplate ref="Linux2" />
                             <recordConstraint>
                                  <recordType namespace=" http://cmdb.ca.com/r1" localName="is-deployed-by"/
>
                                      <propertyValue namespace=" http://cmdb.ca.com/r1" localName="parent"
 matchAny="true">
                                      <equal>test</equal>
                                      </propertyValue>
                              </recordConstraint>
```

```
</relationshipTemplate>
```

Example: Response (edges portion)

```
<edges templateId="rels">
         <relationship xsi:type="ns3:RelationshipType" xmlns:ns3="http://cmdbf.org/schema/1-0-0/datamodel">
               <source>
                    <mdrId xsi:type="xsd:string">http://cmdb.ca.com/r1</mdrId>
                    <localId xsi:type="xsd:string">nr:C2B975A96C03934BA61080C0F79C8BD2</localId>
               </source>
               <target>
                    <mdrId xsi:type="xsd:string">http://cmdb.ca.com/r1</mdrId>
                    <localId xsi:type="xsd:string">nr:B985B5297C46224283D0E5F2632A2A44</localId>
               </target>
               <record xsi:type="ns3:RecordType">
                    <recordMetadata>
                        <recordId xsi:type="xsd:string">bmhier:400004</recordId>
                    </recordMetadata>
                  <is-deployed-by xmlns="http://cmdb.ca.com/r1/is-deployed-by">
                     <child>ali5</child>
                     <last_mod_dt>6 Oct 2008 16:34:48 GMT</last_mod_dt>
                     <parent>ali</parent>
                     <last_mod_by>ServiceDesk</last_mod_by>
                  </is-deployed-by>
               </record>
               <instanceId xsi:type="ns3:MdrScopedIdType">
                  <mdrId xsi:type="xsd:string">http://cmdb.ca.com/r1</mdrId>
                  <localId xsi:type="xsd:string">bmhier:400004</localId>
               </instanceId>
        </relationship>
</edges>
```

Generic Limitations

The following generic limitations apply:

- <recordMetadata>
  <recordId>...</recordId>
  <lastModified>...</lastModified> ?
  <baselineId>...</baselineId> ?
  <snapshotId>...</snapshotId> ?
  xs:any </recordMetadata>:

  NOTE
  recordMetadata only returns recordId, and the xs:any.

  Other values have no meaning to CMDB.
- CMDB does not support case-sensitivity for the equal, contains, and like operators.
- CMDB does not support escape sequences as unique wild card characters.
- CMDBf supports XSD date and XSD dateTime formats:
  "YYYY-MM-DD" -XSD date
  "YYYY-MM-DDThh:mm:ss" -XSD dateTime

Date Data Type

The Date Data type is specified in the following form:

YYYY-MM-DD

where:

- **YYYY**
  Specifies the year.
- **MM**
  Specifies the month.
- **DD**
  Specifies the day of the month.

  NOTE
  All components are required.

DateTime Data Type

The DateTime data type is used to specify both a date and a time on that date.

dateTime is specified in the following form:

YYYY-MM-DDThh:mm:ss

where:

- **YYYY**
  Specifies the year.
- **MM**
  Specifies the month.
- **DD**
  Specifies the day.
- **T**
  Specifies the start of the required time section.
- **hh**
  Specifies the hour.
- **mm**
  Specifies the minute.
- **ss**
  Specifies the second.

  NOTE
  All components are required.

Multi-Tenancy and CIs

This article describes the following:

- How Multi-Tenancy Affects CIs
- CI Lists and Multi-Tenancy Relationships
- CI Creation and Multi-Tenancy Relationships
- CI Update and Multi-Tenancy Relationships

How Multi-Tenancy Affects CIs

The following CA CMDB objects are *tenanted*:

- CIs and their associated extension tables
- CI relationships
- Management Data Repository (MDR) providers
- MDR mappings

To create, edit, and list CIs effectively, you must understand how multi-tenancy affects CIs. When you create, list, or update CIs, consider the following:

- When multi-tenancy is installed, the Role Detail form includes Tenant Access and Tenant Write Access drop-down lists on its Authorization tab that contains the following options:
  - Contact's Tenant
  - Single Tenant
  - Tenant Group
  - All Tenants

  The specified Tenant Access affects how you can work with CIs in the CA CMDB-related applications.
- If the Tenant Access or Tenant Write Access type is not specified for the contact, the default role is used.
- On the Role Detail form, the Update Public check box controls whether a user in the role is authorized to create or update public data. This check box is effective only for users associated with the service provider, as tenant users are restricted to read-only access to data not belonging to their tenant.

  WARNING
  Users associated with a tenant other than the service provider can only create or update objects associated with their own tenant unless authorized by their role. Users associated with the service provider are permitted to create or update objects belonging to tenants other than their own.

CI Lists and Multi-Tenancy Relationships

The following table shows the results of listing CIs in CA CMDB-related applications with multi-tenancy enabled.

This table is a partial listing of the many possible combinations of role access options, and how they affect the various applications.

Role Tenant Access Option	Web UI	CA APM	Visualizer	GRLoader	CA Cohesion ACM
Contact's Tenant	Lists CIs in the same tenant and public	Lists all CIs	Lists CIs in the same tenant and public	Lists CIs in the same tenant and public	Lists CIs in the same tenant and public
Single Tenant	Lists CIs in the same tenant and public	Lists all CIs	Lists CIs in the same tenant and public	Lists CIs in the same tenant and public	Lists CIs in the same tenant and public
Tenant Group	Lists CIs in the all tenants in a tenant group and public	Lists all CIs	Lists CIs in the all tenants in a tenant group and public	Lists CIs in the all tenants in a tenant group and public	Lists CIs in the all tenants in a tenant group and public
All Tenants	Lists all CIs	Lists all CIs	Lists all CIs	Lists all CIs	Lists all CIs

CI Creation and Multi-Tenancy Relationships

The following table shows the results of creating CIs in CA CMDB-related applications with multi-tenancy enabled.

This table is a partial listing of the many possible combinations of role access options, and how they affect the various applications.

Role Tenant Access Option	Web UI	CA APM	Visualizer	GRLoader	CA Cohesion ACM
Contact's Tenant	CI tenant is the one assigned to the signed-on user	CI tenant is created as public	Uses the Web UI to create CIs	CI tenant is the default tenant associated with the signed-on user **Note:** We recommend that you set up a distinct contact for every data source. The contact definition should specify a role that is Contact's Tenant.	CI tenant is the one assigned to the signed-on user
Single Tenant	CI tenant is the one assigned to the signed-on user's proxy	CI tenant is created as public	Uses the Web UI to create CIs	CI tenant is the default tenant associated with the signed-on user **Note:** We recommend that you set up a distinct contact for every data source. The contact definition should specify a role which is contact-tenant.	CI tenant is the one assigned to the signed-on user's proxy
Tenant Group	CI tenant can be selected from list in the UI that includes only those tenants in the group **Note:** The default tenant can be overridden by using the GRLoader -dt option.	CI tenant is created as public	Uses the Web UI to create CIs	Can use the <tenant> option to assign a tenant. **Note:** We recommend that you set up a distinct contact for every data source. The contact definition should specify a role which is contact-tenant.	CI is created as public unless -dt is specified

All Tenants	CI tenant can be selected from a list in the UI	CI tenant is created as public	Uses the Web UI to create CIs	Can use the <tenant> option to assign a tenant. If a tenant is not specified, the default is public. **Note:** We recommend that you set up a distinct contact for every data source. The contact definition should specify a role that is Contact's Tenant.	CI is created as public unless -dt is specified

CI Update and Multi-Tenancy Relationships

The following table shows the results of updating CIs in CA CMDB-related applications with multi-tenancy enabled. Consider the following when you update CIs:

- Only CIs that can be listed can be updated.
- The tenant attribute can only be changed by using the command line.
- CA Cohesion ACM does not populate the <tenant> attribute.

This table is a partial listing of the many possible combinations of role access options, and how they affect the various applications.

Role Tenant Access Option	Web UI	CA APM	Visualizer	GRLoader	CA Cohesion ACM
Contact's Tenant	Can update a CI in same tenant Role and service provider determine the public read/write access	Can update any CI	Uses the Web UI to update CIs	Can update a CI in same tenant Role and service provider determine the public read/write access	Can update a CI in same tenant Role and service provider determine the public read/write access
Single Tenant	Can update a CI in same tenant Role and service provider determine the public read/write access	Can update any CI	Uses the Web UI to update CIs	Can update a CI in same tenant Role and service provider determine the public read/write access	Can update a CI in same tenant Role and service provider determine the public read/write access

Tenant Group	Can update a CI in same tenant Cannot update CIs in the tenant group Role and service provider determine the public read/write access CI relationship updates across tenants requires that the contact must be a service provider tenant	Can update any CI	Uses the Web UI to update CIs	Can update a CI in same tenant Cannot update CIs in the tenant group Role and service provider determine the public read/write access CI relationship updates across tenants requires that the contact must be a service provider tenant	Can update a CI in same tenant Cannot update CIs in the tenant group Role and service provider determine the public read/write access CI relationship updates across tenants requires that the contact must be a service provider tenant
All Tenants	Can update any CI	Can update any CI	Uses the Web UI to update CIs	Can update any CI	Can update any CI

API Documentation for RESTful Services

The API documentation for the out-of-the-box CA SDM RESTful services are now available and no longer required to be generated.

Complete the following step to view the API documentation for RESTful services:

1. Back up your custom API documentation for RESTful services.
   The backup is mandatory and ensures documentation recovery in case of any problem.
2. Navigate to `$NX_ROOT\doc` folder.
3. Extract the
   `rest-doc.zip`
   file content to
   `$NX_ROOT\doc\rest-doc`
   folder.
4. Navigate to `$NX_ROOT\doc\rest-doc\pojo` folder.
5. Open the `overview-summary.html` file in a browser to view the API documentation for RESTful services.

Objects and Attributes

This page lists the objects and attributes that define CA SDM. The system uses objects and attributes to build notification text, scoreboard queries, and data partition constraints. The *FACTORY Optional* statement that accompanies each object, defines access to the object, including its relation attribute, a common name, the security group that can access it, the type of lists produced, and how those lists can be sorted. If omitted, the object is treated according to default specifications.

> **NOTE**
> When an object is documented as a containing a LOCAL *DB Field*, the attribute maps to a local variable instead of a database column. Objects that are documented as a BREL *Data Type*, refer to a backward single relationship (SREL). For more information about default specifications and detailed lists of the attributes of each object, see the chapter "Object Definition Syntax."

This article contains the following topics:

attached_sla Object

The object details are as follows:

1. Associated Table: Attached_SLA
2. Factories: default, ttv_slas
3. REL_ATTR: id
4. Common Name: ticket_type
5. Function Group:
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
_mapped_chg	_mapped_chg	INTEGER	chg id	
_mapped_cr	_mapped_cr	STRING	call_req persid	
_mapped_iss	_mapped_iss	STRING	issue persistent_id	
_mapped_iss_wf	_mapped_iss_wf	INTEGER	isswf id	
_mapped_wf	_mapped_wf	INTEGER	wf id	
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
last_ttv_upd	last_ttv_upd	LOCAL_TIME		
map_sdsc	map_sdsc	STRING	srv_desc code	REQUIRED
persistent_id	persid	STRING		
sla_viol_status	sla_viol_status	INTEGER		
ticket_id	ticket_id	INTEGER		REQUIRED S_KEY
ticket_type	ticket_type	STRING		REQUIRED
time_to_violation	time_to_violation	LOCAL_TIME		
ttv_event	ttv_event	STRING	att_evt persid	

attr_control Object

The object details are as follows:

1. Associated Tables: dependent_control and usp_attr_control
2. Factories: default
3. REL_ATTR: id
4. Common Name: attrname

Function Group: admin

REST Operations: CREATE READ UPDATE

Attribute	Data Type	SREL References	Flags
control		SREL dependent_control	Dependent control
attrname	STRING		Name of controlled attribute

locked	INTEGER		1 = attribute is read-only
required	INTEGER		1= attribute is required
delete_flag	INTEGER	SREL actbool	Required; on new default: 0
last_mod_by		SREL cnt	On new default user; on CI set user
last_mod_dt	DATE		On CI set now

auto_close Object

The object details are as follows:

1. Associated Table: usp_auto_close
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: admin
6. Tenant: optional
7. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
sys	symbol	STRING		Required
cr_ach	request	INTEGER		Required On new default: 0
in_ach	incident	INTEGER		Required On new default: 0
pr_ach	problem	INTEGER		Required On new default: 0
chg_ach	change order	INTEGER		Required On new default: 0
iss_ach	issue	INTEGER		Required On new default: 0
delete_flag		INTEGER	actbool	On CI set now
description	description	STRING		
last_mod_dt		DATE		On CI set now
last_mod_by		DATE	cnt	On CI set user On new default user

aty Object

The object details are as follows:

1. Associated Table: Act_Type
2. Factories: default, chgaty, craty, issaty, mgsaty
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: admin

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
chg_default_ survey	chg_default_survey	INTEGER	survey_tpl id	
chg_notify_info	chg_notify_info	STRING	splmac persid	
chg_send_ survey	chg_send_survey	INTEGER		
chg_survey_ method	chg_survey_ method	INTEGER	ct_mth id	
chg_survey_ msgbody	chg_survey_ msgbody	STRING		
chg_survey_ msgtitle	chg_survey_msgtitl	STRING		
code	code	STRING		UNIQUE REQUIRED S_KEY
cr_default_ survey	cr_default_survey	INTEGER	survey_tpl id	
cr_notify_info	cr_notify_info	STRING	splmac persid	
cr_send_survey	cr_send_survey	INTEGER		
cr_survey_method	cr_survey_method	INTEGER	ct_mth id	
cr_survey_msgbody	cr_survey_msgbody	STRING		
cr_survey_msgtitle	cr_survey_msgtitle	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
description	description	STRING		
cr_flag	flag1	INTEGER	bool_tab enum	
chg_flag	flag2	INTEGER	bool_tab enum	
iss_flag	flag3	INTEGER	bool_tab enum	
mgs_flag	flag4	INTEGER	bool_tab enum	
kd_flag	flag5	INTEGER	bool_tab enum	
id	id	INTEGER		UNIQUE REQUIRED KEY
internal	internal	INTEGER		
iss_default_survey	iss_default_survey	INTEGER	survey_tpl id	
iss_notify_info	iss_notify_info	STRING	splmac persid	
iss_send_ survey	iss_send_survey	INTEGER		
iss_survey_ method	iss_survey_method	INTEGER	ct_mth id	
iss_survey_ msgbody	iss_survey_ msgbody	STRING		
iss_survey_ msgtitle	iss_survey_msgtitl	STRING		
kd_notify_info	kd_notify_info	STRING	splmac persid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
mgs_notify_info	mgs_notify_info	STRING	splmac persid	
notify_msg_ack	notify_ack	STRING		
persistent_id	persid	STRING		
sym	sym	STRING		REQUIRED S_KEY

audlog Object

The object details are as follows:

1. Associated Table: Audit_Log
2. Factories: default
3. REL_ATTR: code
4. Common Name: audobj_persid
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
analyst	analyst	UUID	ca_contact uuid	
attr_after_val	attr_after_val	STRING		
attr_before_val	attr_before_val	STRING		
attr_name	attr_name	STRING		
aud_opr	aud_opr	INTEGER		
audobj_name	audobj_name	STRING		
audobj_persid	audobj_persid	STRING		
audobj_trkid	audobj_trkid	STRING		
audobj_uniqueid	audobj_uniqueid	STRING		
change_date	change_date	LOCAL_TIME		
id	id	INTEGER		UNIQUE REQUIRED KEY
int1_rsrved	int1_rsrved	INTEGER		
int2_rsrved	int2_rsrved	INTEGER		
persistent_id	persid	STRING		
str1_rsrved	str1_rsrved	STRING		

bhvtpl Object

The object details are as follows:

1. Associated Table: Behavior_Template
2. Factories: default
3. REL_ATTR: id
4. Common Name: description
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
context_attrname	context_attrname	STRING		
context_attrval	context_attrval	INTEGER		
context_type	context_type	STRING		REQUIRED
delete_flag	del	INTEGER	actbool enum	REQUIRED

description	description	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
modified_date	last_mod_dt	LOCAL_TIME		
macro_condition	macro_condition	STRING	splmac persid	
object_id	object_id	INTEGER		REQUIRED
object_type	object_type	STRING		REQUIRED
persistent_id	persid	STRING		
transition_errmsg	transition_errmsg	STRING		
transition_test	transition_test	STRING	splmac persid	

BU_TRANS Object

The object details are as follows:

1. Associated Table: BU_TRANS
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group:
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
Document Hit Date	BU_DATE	LOCAL_TIME		
Processed by Bubble Up Flag	BU_PROCESSED	INTEGER		
Document Rating	BU_RATING	REAL		
Document ID	DOC_ID	INTEGER	SKELETONS id	
No Rating Flag	HIT_NO_VOTE	INTEGER		
Hit Origin	HIT_ORIGIN	INTEGER		
id	ID	INTEGER		REQUIRED KEY
Category Id	INDEX_ID	INTEGER	O_INDEXES id	
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
Contact ID	USER_ID	UUID	ca_contact uuid	

ADMIN_TREE Object

The object details are as follows:

1. Associated Table: admin_tree
2. Factories: default
3. REL_ATTR: id
4. Common Name: CAPTION
5. Function Group: reference

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
CAPTION	CAPTION	STRING		
DESCRIPTION	DESCRIPTION	STRING		
HAS_CHILDREN	HAS_CHILDREN	INTEGER		
id	ID	INTEGER		KEY
KT_ADMIN	KT_ADMIN	INTEGER		
KT_KS_CAPTION	KT_KS_CAPTION	STRING		
KT_KS_FLAG	KT_KS_FLAG	INTEGER		
KT_MANAGER	KT_MANAGER	INTEGER		
last_mod_dt	last_mod_dt	LOCAL_TIME		
PARENT_ID	PARENT_ID	SREL		
RESOURCE	RESOURCE	STRING		
SD_ADMIN	SD_ADMIN	INTEGER		

alg Object

The object details are as follows:

1. Associated Table: Act_Log
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: description
5. Function Group: call_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
action_desc	action_desc	STRING		
analyst	analyst	UUID	ca_contact uuid	
call_req_id	call_req_id	STRING	call_req persid	
description	description	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
internal	internal	INTEGER		
session	knowledge_ session	STRING		
k_tool	knowledge_ tool	STRING		
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
system_time	system_time	LOCAL_TIME		
time_spent	time_spent	DURATION		
time_stamp	time_stamp	LOCAL_TIME		

type	type	STRING	act_type code	

am_asset_map Object

The object details are as follows:

1. Associated Table: Am_Asset_Map
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: dmuuid
5. Function Group:
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
dmuuid	am_dmuuid	STRING		
domain_id	am_domain_id	INTEGER		
server	am_server	STRING		
type	am_type	INTEGER		
unit_domain_id	am_unit_domain_id	INTEGER		
unit_id	am_unit_id	INTEGER		
id	id	INTEGER		UNIQUE REQUIRED KEY
int1_rsrved	int1_rsrved	INTEGER		
int2_rsrved	int2_rsrved	INTEGER		
ob_persid	ob_persid	STRING		
ob_type	ob_type	STRING		
persistent_id	persid	STRING		
str1_rsrved	str1_rsrved	STRING		
str2_rsrved	str2_rsrved	STRING		
schema_ver	version	INTEGER		REQUIRED

arcpur_rule Object

The object details are as follows:

1. Associated Table: Archive_Purge_Rule
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: name
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
add_query	add_query	STRING		
arc_file_name	arc_file_name	STRING		

conf_obj_name	conf_obj_name	STRING		
days_inactive	days_inactive	INTEGER		REQUIRED
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
name	name	STRING		REQUIRED UNIQUE REQUIRED S_KEY
oper_type	oper_type	INTEGER		REQUIRED
persistent_id	persid	STRING		
reoccur_interv	reoccur_interv	DURATION		
sched	sched	STRING	bpwshft persid	

atev Object

The object details are as follows:

1. Associated Table: Attached_Events
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: user_smag
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
cancel_time	cancel_time	LOCAL_TIME		
event_tmpl	event_tmpl	STRING	evt persid	REQUIRED S_KEY
fire_time	fire_time	LOCAL_TIME		
first_fire_time	first_fire_time	LOCAL_TIME		
group_name	group_name	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_dt	last_mod_dt	LOCAL_TIME		
num_fire	num_fire	INTEGER		
obj_id	obj_id	STRING		REQUIRED
owning_ast	owning_ast	INTEGER	attached_sla id	
persistent_id	persid	STRING		
start_time	start_time	LOCAL_TIME		
status_flag	status_flag	INTEGER		
user_smag	user_smag	STRING		
wait_time	wait_time	DURATION		

atomic_cond Object

The object details are as follows:

1. Associated Table: Atomic_Condition
2. Factories: default
3. REL_ATTR: id
4. Common Name: description
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
cond_code	cond_code	STRING		
connecter	connecter	INTEGER		REQUIRED
delete_flag	del	INTEGER	actbool enum	REQUIRED
description	description	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
l_paran	l_paran	INTEGER		REQUIRED
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
lval	lval	STRING	atyp_asc code	REQUIRED
operator	operator	INTEGER		REQUIRED
owning_macro	owning_macro	STRING	splmac persid	
persistent_id	persid	STRING		
r_paran	r_paran	INTEGER		REQUIRED
rval	rval	STRING		
rval_assoc	rval_assoc	STRING	atyp_asc code	
sequence	sequence	INTEGER		REQUIRED

act_type_assoc Object

The object details are as follows:

1. Associated Table: Act_Type_Assoc
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
act_type	act_type	STRING	act_type code	
code	code	STRING		UNIQUE REQUIRED S_KEY

description	description	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
internal	int1_rsrved	INTEGER		
int2_rsrved	int2_rsrved	INTEGER		
int3_rsrved	int3_rsrved	INTEGER		
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
log_me_f	log_me_f	INTEGER		
ob_type	ob_type	STRING		
ob_type_attr	ob_type_attr	STRING		
persistent_id	persid	STRING		
str1_rsrved	str1_rsrved	STRING		
str2_rsrved	str2_rsrved	STRING		
str3_rsrved	str3_rsrved	STRING		
sym	sym	STRING		REQUIRED

ca_cmpny Object

The object details are as follows:

1. Associated Table: ca_company
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
alias	alias	STRING		
authentication_password	authentication_password	STRING		
authentication_user_name	authentication_user_name	STRING		
bbs	bbs	STRING		
Company Name	company_name	STRING		
company_type	company_type	INTEGER	ca_company_type id	
id	company_uuid	UUID		UNIQUE REQUIRED KEY
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		
description	description	STRING		

exclude_registration	exclude_registration	integer		
delete_flag	inactive	integer	actbool enum	
last_update_date	last_update_date	LOCAL_TIME		
last_update_user	last_update_user	STRING		
location_uuid	location_uuid	UUID	ca_location location_uuid	
month_fiscal_year_ends	month_fiscal_year_ends	integer		
parent_company_uuid	parent_company_uuid	UUID	ca_company company_uuid	
primary_contact_uuid	primary_contact_uuid	UUID	ca_contact uuid	
version_number	version_number	integer		
web_address	web_address	STRING		

ca_tou Object

The object details are as follows:

1. Associated Table: ca_tou
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: security
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
name	name	STRING		REQUIRED
description	description	STRING		
terms_of_usage_text		STRING		
version_number	version_number	INTEGER		DEFAULT 0
creation_user	creation_user		cnt	DEFAULT USER
creation_date	creation_date	DATE		
last_mod_dt	last_mod_dt	DATE		
last_update_date	last_update_date	DATE	cnt	
delete_flag	del	INTEGER	actbool enum	REQUIRED

caextwf_inst Object

The object details are as follows:

1. Associated Table: usp_caextwf_instances
2. Factories: default
3. REL_ATTR: id
4. Common Name: procname
5. Function Group: reference

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag		actbool.enum		REQUIRED
instance_id	instance_id	STRING(255)		
object_persid	object_persid	STRING(60)		
procname	procname	STRING(32768)		
starttime	starttime	DATE		
endtime	endtime	DATE		

caextwf_sfrm Object

The object details are as follows:

1. Associated Table: usp_caextwf_start_forms
2. Factories: default
3. REL_ATTR: id
4. Common Name: procname
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
object_persid	object_persid	STRING(60)		
caextwf_form	caextwf_form	STRING(255)		
caextwf_path	caextwf_path	STRING(32768)		
procname		STRING(0)		

closure_code Object

The object contains the following:

1. Associated Table: usp_closure_code
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	Data Type	SREL References	Flags
id	Integer		Unique
producer_id	String 20		
persistent_id	String 60		
delete_flag	SREL	actbool.enum	Standard active indicator Required
sym	String 40		Required
code	String 40		Required

last_mod_by	SREL	cnt.id	
last_mod_date	DATE		
description	String 40		
tenant	SREL		

cmth Object

The object details are as follows:

1. Associated Table: Contact_Method
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: notification_reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
cm_template	cm_template	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		DATE ON_CI {NOW}
description	nx_desc	STRING (40)		
producer_id		LOCAL STRING (20)		
persistent_id	persid	STRING (30)		
supports_smtp	supports_smtp	SREL	bool.enum	
sym	sym	STRING		UNIQUE REQUIRED S_KEY
write_file	write_file	INTEGER		

cnote Object

The object details are as follows:

1. Associated Table: Note_Board
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: text
5. Function Group: announcement
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
active	active_flag	INTEGER	bool_tab enum	REQUIRED

close_date	close_date	LOCAL_TIME		
cnote_type	cnote_type	INTEGER		
control_group	control_group	UUID		
del	del	INTEGER		
id	id	INTEGER		UNIQUE REQUIRED KEY
internal	internal	INTEGER		
location	loc_id	UUID	ca_location location_uuid	
organization	organization	UUID	ca_organization uuid	
persistent_id	persid	STRING		
posted_by	posted_by	UUID	ca_contact uuid	
posted_date	posted_date	LOCAL_TIME		
text	text	STRING		

cnt Object

The object details are as follows:

1. Associated Table: ca_contact, usp_contact
2. Factories: default, agt, cst, grp
3. REL_ATTR: id
4. Common Name: combo_name
5. Function Group: contact
6. ca_contact Table
7. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
access_type	access_type	SREL	acctyp.id	
admin_org	admin_organization_uuid	UUID	ca_organization uuid	
alias	alias	STRING		
alt_phone	alt_phone_ number	STRING		
cnthandling_list		BREL	contact_handling	
contact_num	alternate_ identifier	STRING		
notes	comment	STRING		
company	company_uuid	UUID	ca_company company_uuid	
type	contact_type	integer	ca_contact_type id	
id	contact_uuid	UUID		UNIQUE REQUIRED KEY
billing_code	cost_center	INTEGER	ca_resource_cost_center id	
creation_date	creation_date	LOCAL_TIME		

creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		
dept	department	INTEGER	ca_resource_department id	
email_address	email_address	STRING		
exclude_ registration	exclude_ registration	integer		
fax_phone	fax_number	STRING		
first_name	first_name	STRING		
floor_location	floor_location	STRING		
delete_flag	inactive	integer	actbool enum	
job_function	job_function	integer		
position	job_title	integer	ca_job_title id	
last_name	last_name	STRING		
last_mod	last_update_date	LOCAL_TIME		
last_mod_by	last_update_user	STRING		
location	location_uuid	UUID	ca_location location_uuid	
mail_stop	mail_stop	STRING		
middle_name	middle_name	STRING		
mobile_phone	mobile_phone_ number	STRING		
organization	organization_ uuid	UUID	ca_organization uuid	
pemail_address	pager_email_ address	STRING		
beeper_phone	pager_number	STRING		
phone_number	pri_phone_ number	STRING		
roles	roles	LIST		
room_location	room_location	STRING		
supervisor_ contact_uuid	supervisor_ contact_uuid	UUID	ca_contact uuid	
userid	userid	STRING		
version_number	version_number	integer		

cnt_role Object

The object details are as follows:

1. Associated Table: usp_cnt_role
2. Factories: default
3. REL_ATTR: id
4. Common Name: id
5. Function Group: security
6. REST Operations: CREATE READ UPDATE

Attributes	Data Type	Related Object	Flags
id	INTEGER		UNIQUE

contact	SREL	cnt	
role_obj	SREL	role	
is_default	INTEGER		
last_mod_dt	DATE		

cost_cntr Object

The object details are as follows:

1. Associated Table: ca_resource_cost_center
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: inventory
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		
exclude_registration	exclude_registration	integer		
id	id	INTEGER		UNIQUE REQUIRED KEY
delete_flag	inactive	integer	actbool enum	
last_mod	last_update_date	LOCAL_TIME		
last_update_user	last_update_user	STRING		
name	name	STRING		
version_number	version_number	integer		

country Object

The object details are as follows:

1. Associated Table: ca_country
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: inventory
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		

description	description	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
delete_flag	inactive	integer	actbool enum	
last_mod	last_update_date	LOCAL_TIME		
last_update_user	last_update_user	STRING		
name	name	STRING		
version_number	version_number	integer		

cr Object

The object details are as follows:

Associated Table: Call_Req

Factories: default

REL_ATTR: persistent_id

Common Name: ref_num

Function Group: call_mgr

REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
active	active_flag	INTEGER	bool_tab enum	REQUIRED
active_prev	LOCAL	SREL	bool.enum	
affected_ resource	affected_rc	UUID	ca_owned_ resource uuid	
assignee	assignee	UUID		
assignee_prev	LOCAL	SREL	agt.id	
call_back_date	call_back_date	LOCAL_ TIME		
call_back_flag	call_back_flag	INTEGER		
category	category	STRING	prob_ctg persid	
category_prev	LOCAL	SREL	pcat.persistent_id	
caused_by_chg	caused_by_chg	SREL	chg.id	
change	change	INTEGER	chg id	
charge_back_id	charge_back_ id	STRING		
close_date	close_date	LOCAL_ TIME		
cr_tticket	cr_tticket	INTEGER		
created_via	created_via	INTEGER	interface id	
customer	customer	UUID	ca_contact uuid	REQUIRED
event_token	event_token	STRING		
external_system_ticket		STRING (4000)		

extern_ref	extern_ref	STRING		
group	group_id	UUID		
group_prev	LOCAL	SREL	grp.id	
id	id	INTEGER		UNIQUE REQUIRED KEY
impact	impact	INTEGER	impact enum	
impact_prev	LOCAL	SREL	imp.enum	
incident_priority	incident_priority	INTEGER		
incorrectly_assigned		SREL	actbool	
last_act_id	last_act_id	STRING		
last_mod_dt	last_mod_dt	LOCAL_TIME		
log_agent	log_agent	UUID	ca_contact uuid	REQUIRED
macro_ predicted_ violation	macro_predict_ viol	INTEGER		
major_incident		SREL	actbool	
open_date	open_date	LOCAL_ TIME	org	
orig_user_admin_org		SREL	org	
orig_user_cost_center		SREL	cost_cntr	
orig_user_dept		SREL	dept	
orig_user_organization		SREL	org	
outage_detail_what		STRING (4000)		
outage_detail_who		STRING (4000)		
outage_detail_why		STRING (4000)		
outage_end_time	outage_end_time	LOCAL_TIME		
outage_reason		SREL	outage_reason	
outage_start _time	outage_start _time	LOCAL_TIME		
outage_type		SREL	outage_type	
parent	parent	STRING	call_req persid	
pct_service_restored		INTEGER		
persistent_id	persid	STRING		
predicted_sla_violation	predicted_sla_viol	INTEGER		
priority	priority	INTEGER	pri enum	REQUIRED
priority_prev	LOCAL	SREL	pri.enum	
problem	problem	STRING		
ref_num	ref_num	STRING		UNIQUE REQUIRED S_KEY
remote_control_used		SREL	actbool	
requested_by		SREL	cnt	
resolvable_at_lower		SREL	actbool	

resolve_date	resolve_date	LOCAL_ TIME		
return_to_service		SREL	actbool	
rootcause	rootcause	INTEGER	rootcause id	
extern_token	sched_token	STRING		
severity	severity	INTEGER	sevrty enum	
severity_prev	LOCAL	SREL	sev.enum	
sla_violation	sla_violation	INTEGER		
base_template	solution	STRING	call_req persid	
status	status	STRING	cr_stat code	
status_prev	LOCAL	SREL	crs.code	
string1	string1	STRING		
string2	string2	STRING		
string3	string3	STRING		
string4	string4	STRING		
string5	string5	STRING		
string6	string6	STRING		
children		QREL	cr	
sla_events		QREL	tev	
attached_slas		BREL	attached_sla._mapped_ cr	
add_sla_persids	add_sla_persids	STRING		
site_sla_attrs	site_sla_attrs	STRING		
audit_userid	audit_userid	SREL	cnt.id	
fldchange_log	fldchange_log	STRING		
affected_service	affected_service	SREL	nr	
init_urg		LOCAL INTEGER		
init_imp		LOCAL INTEGER		
cap_reason		LOCAL STRING		
auto_urg		LOCAL INTEGER		
auto_imp		LOCAL INTEGER		
man_urg		LOCAL INTEGER		
man_imp		LOCAL INTEGER		
summary	summary	STRING		
support_lev	support_lev	STRING	srv_desc code	
symptom_code		SREL	symptom_code	
template_name	template_name	STRING	cr_template template_name	
time_spent_ sum	time_spent_ sum	DURATION		
type	type	STRING	crt code	

urgency	urgency	INTEGER	urgncy enum	
urgency_prev	LOCAL	SREL	urg.enum	
target_times	target_times	BREL	tgt_time_mapped_cr	
target_start_last	target_start_last	DATE		
target_hold_last	target_hold_last	DATE		
target_hold_count	target_hold_count	INTEGER		
target_resolved_last	target_resolved_last	DATE		
target_resolved_count	target_resolved_count	INTEGER		
target_closed_last	target_closed_last	DATE		
target_closed_count	target_closed_count	INTEGER		
close_date_prev	close_date_prev	LOCAL DATE		
resolve_date_prev	resolve_date_prev	LOCAL DATE		
target_hold_count_prev	target_hold_count_prev	INTEGER		
target_resolved_count_prev	target_resolved_count_prev	LOCAL INTEGER		
target_closed_count_prev	target_closed_count_prev	LOCAL INTEGER	tgt_time_mapped_iss	

Ir Object

The object details are as follows:

1. Associated Table: Notify_Log_Header
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: msg_hdr
5. Function Group: notify
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
cmth_used	cmth_used	INTEGER	ct_mth id	
cntxt_obj	cntxt_obj	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod	last_mod	LOCAL_TIME		
nlh_ack_by	nlh_ack_by	LOCAL_TIME		
ack_time	nlh_ack_time	DURATION		
contact	nlh_c_addressee	UUID	ca_contact uuid	
nlh_c_alias	nlh_c_alias	UUID		
notify_method	nlh_cm_method	INTEGER	noturg enum	
email_address	nlh_email	STRING		

end_date	nlh_end	LOCAL_TIME		
msg_hdr	nlh_hdr	STRING		
msg_text	nlh_msg	STRING		
msg_html	nlh_msg_html	STRING		
pri_event	nlh_pri	INTEGER		
start_date	nlh_start	LOCAL_TIME		REQUIRED
status	nlh_status	INTEGER		
activity_notify	nlh_transition	INTEGER		
notify_type	nlh_type	INTEGER		
msg_ack	nlh_user_ack	STRING		

symptom_code Object

The object details are as follows:

1. Associated Table: usp_symptom_code
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Field	Data type	Reference	Flags
delete_flag	SREL	actbool.enum	ON_NEW
description	STRING (4000)		
id	INTEGER UNIQUE		
producer_id	LOCAL STRING (20)		
persistent_id	LOCAL STRING (60)		
last_mod_by	SREL	cnt	ON_NEW {USER} ON_CI {USER}
last_mod_dt	DATE		ON_CI {NOW}
sym	STRING 60		

state Object

The object details are as follows:

1. Associated Table: ca_state_province
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		
exclude_registration	exclude_registration	integer		
id	id	INTEGER		UNIQUE REQUIRED KEY
delete_flag	inactive	integer	actbool enum	
last_mod_dt	last_update_date	LOCAL_TIME		
last_mod_by	last_update_user	STRING		
sym	symbol	STRING		
version_number	version_number	integer		

crt Object

The object details are as follows:

1. Associated Table: Call_Req_Type
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: call_mgr_reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
code	code	nvarchar(10)		UNIQUE REQUIRED S_KEY
delete_flag	del	INTEGER	actbool enum	REQUIRED
display_name	display_name	nvarchar 30		
id	id	INTEGER		UNIQUE REQUIRED KEY
description	nx_desc	nvarchar 30		
persistent_id	persid	nvarchar 30		
sym	sym	nvarchar 30		REQUIRED

ctab Object

The object details are as follows:

1. Associated Table: Controlled_Table
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym

5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
desc	nx_desc	nvarchar(40)		
obj_name	obj_name	nvarchar 30		
persistent_id	persid	nvarchar 30		
sym	sym	nvarchar 30		UNIQUE REQUIRED S_KEY

ctimer Object

The object details are as follows:

1. Associated Table: Cr_Call_Timers
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: color
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
beep	beep	INTEGER		
color	color	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
text	text	STRING		
threshold	threshold	DURATION		REQUIRED

ctp Object

The object details are as follows:

1. Associated Table: ca_contact_type
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		
description	description	STRING		
exclude_registration	exclude_registration	integer		
id	id	INTEGER		UNIQUE REQUIRED KEY
delete_flag	inactive	integer	actbool enum	
last_mod_dt	last_update_date	LOCAL_TIME		
last_mod_by	last_update_user	STRING		
sym	name	STRING		
user_uuid	user_uuid	UUID	ca_contact uuid	
version_number	version_number	integer		
view_internal	view_internal	integer		

dblocks Object

The object details are as follows:

1. Associated Table: Virtual internal table
2. Factories: default
3. REL_ATTR: id
4. Common Name: Id
5. Function Group: dblock
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
name		STRING 0		
table_name		STRING 0		
factory_name		SREL	prod_list.sym	
lock_hash		INTEGER		
lock_uuid		UUID		
lock_string		STRING 0		
lock_owner		STRING 0		
when_taken		DATE		
process		STRING 0		
request		INTEGER		
delete_flag		SREL	actbool.enum	

dcon Object

The object details are as follows:

1. Associated Table: Domain_Constraint
2. Factories: default
3. REL_ATTR: id
4. Common Name: constraint_majic
5. Function Group: security
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE REQUIRED KEY
persistent_id	persid	STRING		REQUIRED
delete_flag	del	INTEGER	actbool enum	REQUIRED
alias	alias	INTEGER		
last_mod_dt	last_mod_dt	LOCAL_TIME		
last_mod_by	last_mod_by	STRING	ca_contact uuid	REQUIRED
dom_id	dom_id	INTEGER	Domain id	REQUIRED
tbl_id	tbl_id	INTEGER	Controlled_Table id	REQUIRED
type	type	INTEGER	Domain_Constraint_Type	REQUIRED
error_msg	error_msg	STRING	enum	
constraint_majic	constraint_majic	STRING		
constraint_SQL	constraint_sql	STRING		

dcon_typ Object

The object details are as follows:

1. Associated Table: Domain_Constraint_Type
2. Factories: default
3. REL_ATTR: enum
4. Common Name: sym
5. Function Group: security
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
enum	enum	INTEGER		REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
description	nx_desc	nvarchar(40)		
persistent_id	persid	nvarchar 30		
sym	sym	nvarchar(12)		UNIQUE REQUIRED S_KEY

dept Object

The object details are as follows:

1. Associated Table: ca_resource_department
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: inventory
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		
exclude_registration	exclude_registration	integer		
id	id	INTEGER		UNIQUE REQUIRED KEY
delete_flag	inactive	integer	actbool enum	
last_mod	last_update_date	LOCAL_TIME		
last_update_user	last_update_user	STRING		
name	name	STRING		
version_number	version_number	integer		

dlgsrvr Object

The object details are as follows:

1. Associated Table: Delegation_Server
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
anon_userid	anon_userid	STRING		
appl_addr	appl_addr	STRING		
default_assignee	default_assignee	UUID	ca_contact uuid	
default_userid	default_userid	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
metafile	metafile	STRING		

description	nx_desc	STRING		
password	password	STRING		
server	server	STRING		
sym	sym	STRING		UNIQUE REQUIRED S_KEY
transport	transport	INTEGER		

dmn Object

The object details are as follows:

1. Associated Table: Domain
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: security
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod	last_mod_dt	LOCAL_TIME		
desc	nx_desc	STRING		
persistent_id	persid	STRING		
sym	sym	STRING		UNIQUE REQUIRED S_KEY

doc_rep Object

The object details are as follows:

1. Associated Table: Document_Repository
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
archive_path	archive_path	STRING		
archive_type	archive_type	INTEGER		
cgi_path	cgi_path	STRING		
default_rep	default_rep	INTEGER		

delete_flag	del	INTEGER	actbool enum	REQUIRED
file_limit_size	file_limit_size	INTEGER		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
prohibited_file_ext	prohibited_ext	STRING		
protocol	protocol	STRING		
repository_type	repository_type	INTEGER		
retrieve_path	retrieve_path	STRING		
server	server	STRING		
servlet_path	servlet_path	STRING		
sym	sym	STRING		REQUIRED S_KEY
upload_path	upload_path	STRING		

ext_entity_map Object

The object details are as follows:

1. Associated Table: External_Entity_Map
2. Factories: default, sd_chg_map
3. REL_ATTR: persistent_id
4. Common Name: xschema_code
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE REQUIRED KEY
int1_rsrved	int1_rsrved	INTEGER		
int2_rsrved	int2_rsrved	INTEGER		
int3_rsrved	int3_rsrved	INTEGER		
int4_rsrved	int4_rsrved	INTEGER		
int5_rsrved	int5_rsrved	INTEGER		
int6_rsrved	int6_rsrved	INTEGER		
lstr1_rsrved	lstr1_rsrved	STRING		
lstr2_rsrved	lstr2_rsrved	STRING		
ob_persid	ob_persid	STRING		
ob_type	ob_type	STRING		
persistent_id	persid	STRING		
str1_rsrved	str1_rsrved	STRING		

str2_rsrved	str2_rsrved	STRING		
xentity_id	xentity_id	STRING		REQUIRED
xschema_code	xschema_code	STRING		REQUIRED
xschema_ver	xschema_ver	INTEGER		REQUIRED

fmgrp Object

The object details are as follows:

1. Associated Table: Form_Group
2. Factories: default, all_fmgrp
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
sym	sym	STRING		REQUIRED S_KEY

gl_code Object

The object details are as follows:

1. Associated Table: ca_resource_gl_code
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: inventory
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		
exclude_ registration	exclude_ registration	integer		
id	id	INTEGER		UNIQUE REQUIRED KEY
delete_flag	inactive	integer	actbool enum	
last_mod	last_update_date	LOCAL_TIME		

last_update_user	last_update_user	STRING		
name	name	STRING		
version_number	version_number	integer		

grc Object

The object details are as follows:

1. Associated Table: ca_resource_class
2. Factories: default
3. REL_ATTR: id
4. Common Name: type
5. Function Group: inventory
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		
exclude_ registration	exclude_ registration	integer		
family	family_id	INTEGER	ca_resource_ family id	
id	id	INTEGER		UNIQUE REQUIRED KEY
delete_flag	inactive	integer	actbool enum	
last_mod	last_update_date	LOCAL_TIME		
last_update_user	last_update_user	STRING		
type	name	STRING		
parent_id	parent_id	SREL		
nsm_class	usp_nsm_class	INTEGER	buscls id	
version_number	version_number	integer		

grpmem Object

The object details are as follows:

1. Associated Table: Group_Member
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name:
5. Function Group:
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
group	group_id	UUID		REQUIRED

id	id	INTEGER		UNIQUE REQUIRED KEY
manager_flag	manager_flag	INTEGER		
member	member	UUID	ca_contact uuid	REQUIRED
notify_flag	notify_flag	INTEGER		

hier Object

The object details are as follows:

1. Associated Table: Asset_Assignment
2. Factories: default
3. REL_ATTR: id
4. Common Name: license_num
5. Function Group: inventory
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
child	hier_child	byte(16)	ca_owned_ resource uuid	REQUIRED S_KEY
license_num	hier_license_num	nvarchar(40)		
log_date	hier_log_date	INTEGER		REQUIRED
parent	hier_parent	byte(16)	ca_owned_ resource uuid	REQUIRED S_KEY
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	byte(16)	ca_contact uuid	
last_mod_dt	last_mod_dt	INTEGER		
persistent_id	persid	nvarchar(30)		

ical_alarm Object

The object details are as follows:

1. Associated Table: usp_ical_alarm
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	Data Type	Remarks
sym	STRING	Required
alarm_value	STRING	

ical_event_template Object

The object details are as follows:

1. Associated Table: usp_ical_event_template
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	Data Type	Remarks
code	STRING	Required
sym	INTEGER	Required
alarm	SREL ical_alarm	
start_date	STRING	
end_date	STRING	
obj_type	SREL actlog_prod_list	
categories	STRING	
summary	STRING	
description	STRING	
url	STRING	
extra_entries	STRING	

imp Object

The object details are as follows:

1. Associated Table: Impact
2. Factories: default
3. REL_ATTR: enum
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
enum	enum	INTEGER		REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
description	nx_desc	STRING		
sym	sym	STRING		UNIQUE REQUIRED S_KEY

value	value	INTEGER		

in Object

The object details are as follows:

1. Associated Table: Call_Req
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: ref_num

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
active	active_flag	INTEGER	bool_tab enum	REQUIRED
active_prev	LOCAL	SREL	bool.enum	
affected_ resource	affected_rc	UUID	ca_owned_ resource uuid	
assignee	assignee	UUID		
assignee_prev	LOCAL	SREL	agt.id	
call_back_date	call_back_date	LOCAL_TIME		
call_back_flag	call_back_flag	INTEGER		
category	category	STRING	prob_ctg persid	
category_prev	LOCAL	SREL	pcat.persistent_id	
caused_by_chg	caused_by_chg	SREL	chg.id	
change	change	INTEGER	chg id	
charge_back_id	charge_back_id	STRING		
close_date	close_date	LOCAL_TIME		
cr_tticket	cr_tticket	INTEGER		
created_via	created_via	INTEGER	interface id	
customer	customer	UUID	ca_contact uuid	REQUIRED
event_token	event_token	STRING		
extern_ref	extern_ref	STRING		
group	group_id	UUID		
group_prev	LOCAL	SREL	grp.id	
id	id	INTEGER		UNIQUE REQUIRED KEY
impact	impact	INTEGER	impact enum	
impact_prev	LOCAL	SREL	imp.enum	
incident_priority	incident_priority	INTEGER		
last_act_id	last_act_id	STRING		
last_mod_dt	last_mod_dt	LOCAL_TIME		
log_agent	log_agent	UUID	ca_contact uuid	REQUIRED

macro_ predicted_ violation	macro_predict_ viol	INTEGER		
open_date	open_date	LOCAL_TIME		
outage_end_ time	outage_end_ time	LOCAL_TIME		
outage_start_ time	outage_start_ time	LOCAL_TIME		
parent	parent	STRING	call_req persid	
persistent_id	persid	STRING		
predicted_sla_ violation	predicted_sla_ viol	INTEGER		
priority	priority	INTEGER	pri enum	REQUIRED
priority_prev	LOCAL	SREL	pri.enum	
problem	problem	STRING		
ref_num	ref_num	STRING		UNIQUE REQUIRED S_KEY
resolve_date	resolve_date	LOCAL_ TIME		
rootcause	rootcause	INTEGER	rootcause id	
extern_token	sched_token	STRING		
severity	severity	INTEGER	sevrty enum	
severity_prev	LOCAL	SREL	sev.enum	
sla_violation	sla_violation	INTEGER		
base_template	solution	STRING	call_req persid	
status	status	STRING	cr_stat code	
status_prev	LOCAL	SREL	crs.code	
string1	string1	STRING		
string2	string2	STRING		
string3	string3	STRING		
string4	string4	STRING		
string5	string5	STRING		
string6	string6	STRING		
summary	summary	STRING		
support_lev	support_lev	STRING	srv_desc code	
template_name	template_name	STRING	cr_template template_name	
time_spent_ sum	time_spent_ sum	DURATION		
type	type	STRING	crt code	
urgency	urgency	INTEGER	urgncy enum	
urgency_prev	LOCAL	SREL	urg.enum	

in_trans Object

The object details are as follows:

1. Associated Tables: in_trans
2. Factories: default
3. REL_ATTR: id
4. Common Name: condition_error
5. Function Group: call_mgr_reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
status	status	STRING	crs	Specifies the current ticket status.
new_status	new status	STRING	crs	Specifies the new ticket status.
is default		INTEGER		Default transition that appears when the Status field is empty. On new default: 0
must_comment		INTEGER		Comment required when using a transition. On new default: 0
delete_flag	del		actbool	Required. On new default: 0
condition			macro	Site condition macro to approve transition.
condition_error		STRING		Error message for site condition.
description		STRING		Description of this transition.
last_mod_by			cnt	On new default user; on CI set user
last_mod_dt		DATE		On CI set user now

INDEX_DOC_LINKS Object

The object details are as follows:

1. Associated Table: INDEX_DOC_LINKS
2. Factories: default
3. REL_ATTR: id
4. Common Name: RELATIONAL_ID
5. Function Group:
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SRE References	Flags
Document ID	DOC_ID	INTEGER	SKELETONS id	

id	ID	INTEGER		REQUIRED KEY
Category ID	INDEX_ID	INTEGER	O_INDEXES id	
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
Relational ID	RELATIONAL_ID	STRING		

intfc Object

The object details are as follows:

1. Associated Table: Interface
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
code	code	STRING		UNIQUE REQUIRED
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
desc	nx_desc	STRING		
persistent_id	persid	STRING		
sym	sym	STRING		UNIQUE REQUIRED

job_func Object

The object details are as follows:

1. Associated Table: ca_job_function
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: inventory
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		
description	description	STRING		
exclude_ registration	exclude_ registration	integer		
id	id	INTEGER		UNIQUE REQUIRED KEY

delete_flag	inactive	integer	actbool enum	
last_mod	last_update_date	LOCAL_TIME		
last_update_user	last_update_user	STRING		
name	name	STRING		
version_number	version_number	integer		

kc Object

The object details are as follows:

1. Associated Table: usp_kpi
2. Factories: default
3. REL_ATTR: id
4. Common Name: kpi_name
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
kpi_name	name	STRING		REQUIRED
sys_name	sys_name	STRING		
kpi_type	type	INTEGER		REQUIRED
delete_flag	status		actbool	REQUIRED
process_type	process_type	INTEGER		
stored_query_id	stored_query_id	STRING	crsq	
user_context	user_context	UUID	cnt	
sql_query	sql_query	STRING		
description	description	STRING		
refresh_time	refresh_time	INTEGER		REQUIRED
metric_type	metric_type	INTEGER		REQUIRED
curr_kpi_time_ stamp	curr_kpi_time_ stamp	INTEGER		
version_number	version_number	INTEGER		
tenant	tenant	STRING		
last_mod_by	last_mod_by	UUID	cnt	
last_mod_dt	last_mod_dt	DATE		

KCAT Object

The object details are as follows:

1. Associated Table: O_INDEXES
2. Factories: default
3. REL_ATTR: id
4. Common Name: CAPTION
5. Function Group:

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
DESCRIPTION	DESCRIPTION	STRING		
AUTHOR_ID	AUTHOR_ID	UUID	ca_contact uuid	
CAPTION	CAPTION	STRING		REQUIRED
DOC_TEMPLATE	DOC_TEMPLATE	INTEGER	CI_DOC_ TEMPLATES id	
HAS_CHILDREN	HAS_CHILDREN	INTEGER		
HAS_DOCS	HAS_DOCS	INTEGER		
id	ID	INTEGER		KEY
KEYWORDS	KEYWORDS	STRING		
ON_COPY_PASTE	ON_COPY_PASTE	INTEGER		
ON_CUT_PASTE	ON_CUT_PASTE	INTEGER		
last_mod_dt	LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
last_mod_by	LAST_MOD_BY	SREL	cnt	Specifies the UUID of the contact who modified this record.
OWNER_ID	OWNER_ID	UUID	ca_contact uuid	
PARENT_ID	PARENT_ID	SREL	O_INDEXES id	
ALLOW_QA	ALLOW_QA	INTEGER		
PERMISSION_ INDEX_ID	PERMISSION_ INDEX_ID	INTEGER	O_INDEXES id	
READ_PGROUP	READ_PGROUP	INTEGER	P_GROUPS id	
persistent_id	persid	STRING		
PGROUP TYPE	PGROUP_TYPE	INTEGER		Indicates if the P Groups is based on Roles or Groups: **1** -- Groups (Default) **2** -- Roles
producer_id	producer_id	STRING		
RELATIONAL_ID	RELATIONAL_ID	STRING		
SUBJECT_EXPERT_ID	SUBJECT_EXPERT_ID	UUID	ca_contact uuid	
WF_TEMPLATE	WF_TEMPLATE	SREL	CI_WF_ TEMPLATES id	
WRITE_PGROUP	WRITE_PGROUP	INTEGER	P_GROUPS id	

kcd Object

The object details are as follows:

1. Associated Table: usp_kpi_data
2. Factories: default

3. REL_ATTR: id
4. Function Group: admin
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
kpi_id	kpi_id	INTEGER	kc	
kpi_time_ stamp	kpi_time_ stamp	DATE		
metric_type	type	INTEGER		REQUIRED
kpi_value	kpi_value	INTEGER		REQUIRED
duration_count	duration_count	INTEGER		
duration_sum	duration_sum	INTEGER		
duration_max	duration_max	INTEGER		
duration_ average	duration_ average	INTEGER		
execute_time	execute_time	INTEGER		
version_number	version_number	INTEGER		

kdlinks Object

The object details are as follows:

1. Associated Table: kdlinks
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: sd_obj_type
5. Function Group:
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
Request Linked	cr	STRING	call_req persid	
id	ID	INTEGER		KEY
Issue Linked	iss	STRING	issue persistent_id	
Document	kd	INTEGER	SKELETONS id	
Analyst	last_mod_by	UUID	ca_contact uuid	
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
Link Type	link_type	INTEGER		
Ticket ID	sd_obj_id	INTEGER		
Ticket Type	sd_obj_type	STRING		

KT_REPORT_CARD Object

The object details are as follows:

1. Associated Table: KT_REPORT_CARD
2. Factories: default

3. REL_ATTR: id
4. Common Name: SUBJECT_ID
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	ID	INTEGER		KEY
SUBJECT_ID	SUBJECT_ID	nvarchar(40)		
PAST_DAYS	PAST_DAYS	INTEGER		
ORG_STATISTICS	ORG_STATISTICS	INTEGER		
DOCUMENTS_ SUBMITTED	DOCUMENTS_ SUBMITTED	INTEGER		
DOCUMENTS_ PUBLISHED	DOCUMENTS_ PUBLISHED	INTEGER		
TOTAL_HITS	TOTAL_HITS	INTEGER		
AVERAGE_ EFFECTIVENESS_ RATING	AVERAGE_ EFFECTIVENESS_ RATING	INTEGER		
TOTAL_ SOLUTION_ COUNT	TOTAL_ SOLUTION_ COUNT	INTEGER		
creation_user	creation_user	nvarchar(64)		
creation_date	creation_date	INTEGER		
last_update_user	last_update_user	nvarchar(64)		
last_update_date	last_update_date	INTEGER		
TOTAL_VOTES	TOTAL_VOTES	INTEGER		
AVG_RATING	AVG_RATING	DOUBLE		
USER_SLV_CNT	USER_SLV_CNT	INTEGER		
LINKED_ KNOWLEDGE_BY_OTHERS	LINKED_ KNOWLEDGE_BY_OTHERS	INTEGER		
LINKED_ KNOWLEGE_BY_ ME	LINKED_ KNOWLEGE_BY_ ME	INTEGER		
CLOSED_TICKETS	CLOSED_TICKETS	INTEGER		
TICKETS_WITH_ KNOWLEDGE	TICKETS_WITH_ KNOWLEDGE	INTEGER		
TICKETS_HAD_ SEARCH_ ACTIVITIES	TICKETS_HAD_ SEARCH_ ACTIVITIES	INTEGER		
KNOWLEDGE_ SUBMIT_FROM_ TICKET	KNOWLEDGE_ SUBMIT_FROM_ TICKET	INTEGER		
TIME_TO_PUBLISH	TIME_TO_PUBLISH	INTEGER		
MY_COMMENTS	MY_COMMENTS	INTEGER		
DOCUMENTS_ RETIRED	DOCUMENTS_ RETIRED	INTEGER		

TIME_TO_FIX	TIME_TO_FIX	INTEGER		
FLAGS_FIXED	FLAGS_FIXED	INTEGER		
COMMENTS_ON_MY_DOCUMENTS	COMMENTS_ON_MY_DOCUMENTS	INTEGER		
FIRST_CALL_RES_WITH_KNOW	FIRST_CALL_RES_WITH_KNOW	INTEGER		
FIRST_CALL_RES_WITHOUT_KNOW	FIRST_CALL_RES_WITHOUT_KNOW	INTEGER		
CONTRIBUTOR_ UUID	CONTRIBUTOR_ UUID	SREL	cnt	TENANCY_UNRESTRICTED
IS_SUPERVISOR	IS_SUPERVISOR	INTEGER		
SUPERVISOR_ID	SUPERVISOR_ID	SREL	cnt	TENANCY_UNRESTRICTED
MY_ORG_REF_ID	MY_ORG_REF_ID	SREL	KT_REPORT_CARD	
PAST_DAYS_TEXT	PAST_DAYS_TEXT	nvarchar(20)		Populates from the resource file.
LAST_MOD_DT_ TEXT	LAST_MOD_DT_ TEXT	nvarchar(20)		Specifies the KRC calculation date.
DOCUMENTS_PUBLISHED_ PERCENT	DOCUMENTS_PUBLISHED_ PERCENT	nvarchar(4)		
AVG_HITS	AVG_HITS	nvarchar(20)		
AVG_RATING_ TEXT	AVG_RATING_ TEXT	nvarchar(100)		
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt	

ktd Object

The object details are as follows:

1. Associated Table: usp_kpi_ticket_data
2. Factories: default
3. REL_ATTR: id
4. Function Group: admin
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
end_time	end_time	INTEGER		REQUIRED
prev_time	prev_time	INTEGER		
obj_name	obj_name	STRING		REQUIRED
obj_id	obj_id	INTEGER		REQUIRED
obj_type	obj_type	STRING		
field_name	field_name	STRING		
field_value	field_value	STRING		
next_value	next_value	STRING		

operation	operation	STRING		
attr_obj	attr_obj	STRING		
attr_from_id	attr_from_id	INTEGER		
attr_to_id	attr_to_id	INTEGER		
attr_from_uuid	attr_from_uuid	UUID		
attr_to_uuid	attr_to_uuid	UUID		
ktd_duration	duration	INTEGER		REQUIRED
user_context	user_context	UUID	cnt	

kwrd Object

The object details are as follows:

1. Associated Table: Knowledge_Keywords
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
sym	sym	STRING		REQUIRED S_KEY

loc Object

The object details are as follows:

1. Associated Table: ca_location
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: inventory
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
address1	address_1	STRING		
address2	address_2	STRING		
address3	address_3	STRING		
address4	address_4	STRING		
address5	address_5	STRING		
address6	address_6	STRING		

city	city	STRING		
description	comment	STRING		
contact_address_ flag	contact_address_ flag	integer		
country	country	integer	ca_country id	
county	county	STRING		
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		
exclude_ registration	exclude_ registration	integer		
fax_number	fax_number	STRING		
geo_coord_type	geo_coord_type	integer		
geo_coords	geo_coords	STRING		
delete_flag	inactive	integer	actbool enum	
last_mod	last_update_date	LOCAL_TIME		
last_update_user	last_update_user	STRING		
name	location_name	STRING		
id	location_uuid	UUID		UNIQUE REQUIRED KEY
mail_address_1	mail_address_1	STRING		
mail_address_2	mail_address_2	STRING		
mail_address_3	mail_address_3	STRING		
mail_address_4	mail_address_4	STRING		
mail_address_5	mail_address_5	STRING		
mail_address_6	mail_address_6	STRING		
pri_phone_number	pri_phone_number	STRING		
primary_contact_uuid	primary_contact_uuid	UUID		
site	site_id	integer	ca_site id	
state	state	integer	ca_state_ province id	
version_number	version_number	integer		
zip	zip	STRING		

LONG_TEXTS Object

The object details are as follows:

1. Associated Table: LONG_TEXTS
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: REF_PERSID
5. Function Group:

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
Chunk	ACTUAL_TEXT	STRING		
Cunk Order	CNT_ORDER	INTEGER		
id	ID	INTEGER		KEY
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
Document ID	REF_PERSID	STRING		

mfrmod Object

The object details are as follows:

1. Associated Table: ca_model_def
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: ci_reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
abbreviation	abbreviation	STRING		
resource_class	class_id	INTEGER	ca_resource_ class id	
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
current_as_of_ date	current_as_of_ date	integer		
delete_time	delete_time	LOCAL_TIME		
exclude_ registration	exclude_ registration	integer		
family_id	family_id	INTEGER		
delete_flag	inactive	integer	actbool enum	
last_mod	last_update_ date	LOCAL_TIME		
last_update_ user	last_update_ user	STRING		
manufacturer	manufacturer_ uuid	UUID	ca_company company_uuid	
id	model_uuid	UUID		UNIQUE REQUIRED KEY
sym	name	STRING		
operating_ system	operating_ system	integer		
preferred_seller_uuid	preferred_seller_uuid	UUID		
version_number	version_number	integer		
model_name	description	STRING		

mgsalg Object

The object details are as follows:

1. Associated Table: Mgs_Act_Log
2. Factories: default
3. REL_ATTR: id
4. Common Name: description
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
action_desc	action_desc	STRING		
analyst	analyst	UUID	ca_contact uuid	
id	id	INTEGER		UNIQUE REQUIRED KEY
internal	internal	INTEGER		
last_mod_dt	last_mod_dt	LOCAL_TIME		
mgs_id	mgs_id	INTEGER	managed_survey id	
persistent_id	persid	STRING		
system_time	system_time	LOCAL_TIME		
time_spent	time_spent	DURATION		
time_stamp	time_stamp	LOCAL_TIME		
type	type	STRING	act_type code	

mgsstat Object

The object details are as follows:

1. Associated Table: Mgs_Status
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
active	active	INTEGER		
code	code	STRING		UNIQUE REQUIRED S_KEY
delete_flag	del	INTEGER	actbool enum	REQUIRED
hold	hold	INTEGER		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	

modified_date	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
sym	sym	STRING		REQUIRED

nr Object

The object details are as follows:

1. Associated Table: ca_owned_resource, usp_owned_resource
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: inventory
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
backup_ services_ contact_uuid		SREL	cnt	Specifies the name of the contact responsible for backup services.
billing_contact_ uuid		SREL	cnt	Specifies the name of the contact responsible for billing.
disaster_ recovery_ contact_uuid		SREL	cnt	Specifies the name of the contact responsible for disaster recovery services.
id	id	UUID		
network_contact_uuid		SREL	cnt	Specifies the name of the contact responsible for network operations.
producer_id	producer_id	LOCAL STRING 30		
persistent_id	persid	LOCAL STRING 60		
chgnr	chgnr	LREL	chg asset	
issnr	issnr	LREL	iss asset	
cntref	cntref	LREL	cnt:PDM cenv	
orgref	orgref	LREL	org:PDM oenv	
acquire_date	acquire_date	LOCAL_TIME		
asset_source_ uuid	asset_source_ uuid	UUID		
loc_cabinet	cabinet_location	STRING		
company_bought_ for_uuid	company_bought_for_uu id	UUID	ca_company company_uuid	
expense_code	cost_center	INTEGER	ca_resource_ cost_center id	
creation_date	creation_date	LOCAL_TIME		
creation_system	creation_system	STRING		

creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		
department	department	INTEGER	ca_resource_ department id	
exclude_ registration	exclude_ registration	INTEGER		
expense_code	expense_code	SREL	cost_cntr	
loc_floor	floor_location	STRING		
gl_code	gl_code	integer		
support_contact1_uuid		SREL	cnt	Specifies the name of the first contact responsible for support services.
support_contact2_uuid		SREL	cnt	Specifies the name of the second contact responsible for support services.
support_contact3_uuid		SREL	cnt	Specifies the name of the third contact responsible for support services.
system_name	host_name	STRING		
delete_flag	inactive	integer	actbool enum	
install_date	installation_date	LOCAL_TIME		
alarm_id	ip_address	STRING		
last_mod	last_update_date	LOCAL_TIME		
last_mod_by	last_update_user	STRING		
license_number	license_ information	STRING		
license_uuid	license_uuid	UUID		
location	location_uuid	UUID	ca_location location_uuid	
mac_address	mac_address	STRING 64		
dns_name	dns_name	STRING 100		
repair_org	maintenance_ org_uuid	UUID	ca_organization uuid	
vendor_repair	maintenance_ vendor_uuid	UUID	ca_company company_uuid	
manufacturer	manufacturer_ uuid	UUID	ca_company company_uuid	
model	model_uuid	UUID	ca_model_def model_uuid	
product_version	product_version	STRING 16		
operating_system	operating_system	INTEGER		
org_bought_for_ uuid	org_bought_for_ uuid	UUID	ca_organization uuid	
id	own_resource_ uuid	UUID		UNIQUE REQUIRED KEY
product_version	product_version	STRING		

purchase_order_id	purchase_order_id	STRING		
requisition_id	requisition_id	STRING		
resource_alias	resource_alias	STRING		
class	resource_class	INTEGER	ca_resource_ class id	
resource_contact	resource_contact_uuid	UUID	ca_contact uuid	
description	resource_ description	STRING		
ufam	ufam	INTEGER		
is_selected	is_selected	LOCAL INTEGER		
family	resource_family	INTEGER	ca_resource_ family id	
name	resource_name	STRING		
resource_owner_ uuid	resource_owner_ uuid	UUID	ca_contact uuid	
asset_count	resource_quantity	INTEGER		
status	resource_status	INTEGER	ca_resource_ status id	
asset_num	resource_tag	STRING		
child_hier	child_hier	BREL	hier.parent	
parent_hier	parent_hier	BREL	hier.child	
asset_log	asset_log	BREL	nr_com.asset_ id	
assoc_cr	assoc_cr	QREL	cr	
all_creq	all_creq	BREL	cr.affected_ resource	
all_open_creq	all_open_creq	QREL	cr	
bm_child_hier	bm_child_hier	QREL	bmhier	
bm_parent_hier	bm_parent_hier	QREL	bmhier	
assoc_reqs	assoc_reqs	QREL	cr	
service_org	responsible_org_ uuid	UUID	ca_organization uuid	
vendor_restore	responsible_ vendor_uuid	UUID	ca_company company_uuid	
loc_room	room_location	STRING		
serial_number	serial_number	STRING		
loc_shelf	shelf_location	STRING		
loc_slot	slot_location	STRING		
status_date	status_date	LOCAL_TIME		
supplier	supply_vendor_ uuid	UUID	ca_company company_uuid	
version_number	version_number	INTEGER		

nr_com Object

The object details are as follows:

1. Associated Table: NR_Comment
2. Factories: default

3. REL_ATTR: id
4. Common Name: writer_name
5. Function Group: inventory
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
attr_name	attr_name	STRING		
log	com_comment	STRING		
log_date	com_dt	LOCAL_TIME		REQUIRED
asset_id	com_par_id	UUID	ca_owned_resource uuid	REQUIRED S_KEY
writer_name	com_userid	STRING		REQUIRED S_KEY
id	id	INTEGER		UNIQUE REQUIRED KEY
new_value	new_value	STRING		
old_value	old_value	STRING		
writer_id	writer_id	UUID		

nrf Object

The object details are as follows:

1. Associated Table: ca_resource_family
2. Factories: default
3. REL_ATTR: enum
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE REQUIRED KEY
producer_id	producer_id	LOCAL STRING 20		
persistent_id	persid	LOCAL STRING 60		
description	description	STRING		
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		
exclude_registration	exclude_registration	INTEGER		
delete_flag	inactive	SREL	actbool enum	REQUIRED
include_reconciliation	include_reconciliation	INTEGER		
last_update_date	last_update_date	LOCAL_TIME		
last_update_user	last_update_user	STRING		
sym	name	STRING 255		

extension_name	table_extension_name	STRING		
version_number	version_number	INTEGER		
physical_table_name	physical_table_name	STRING 30		

O_COMMENTS Object

The object details are as follows:

1. Associated Table: O_COMMENTS
2. Factories: default
3. REL_ATTR: id
4. Common Name: COMMENT_TEXT
5. Function Group: kd
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		KEY
producer_id	producer_id	LOCAL STRING(20)		
persistent_id	persistent_id	LOCAL STRING(60)		
DOC_ID	DOC_ID	SREL	KD	TENANCY_ UNRESTRICTED
SUPPRESS_ OEVENTS	SUPPRESS_ OEVENTS	INTEGER		
VER_COUNT	VER_COUNT	INTEGER		
USER_NAME	USER_NAME	STRING 50		
USER_ID	USER_ID	SREL	cnt	
COMMENT_ TEXT	COMMENT_ TEXT	STRING 255		REQUIRED
COMMENT_ TIMESTAMP	COMMENT_ TIMESTAMP	DATE		
EMAIL_ ADDRESS	EMAIL_ ADDRESS	STRING 75		
TICKET		LOCAL STRING		
FLG_TYPE	FLG_TYPE	SREL	KT_FLG_ TYPE	
ASSIGNEE	ASSIGNEE	SREL	cnt	
DEADLINE_ DATE	DEADLINE_ DATE	DATE		
CLOSE_DATE	CLOSE_DATE	DATE		
FLG_STATUS	FLAG_STATUS	SREL	KT_FLG_ STATUS	
CLOSE_DESC	CLOSE_DESC	STRING 2000		
FLG_CODE	FLG_CODE	STRING 50		
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt	
tenant	tenant	UUID	ca_tenant	

O_EVENTS Object

The object details are as follows:

1. Associated Table: O_EVENTS
2. Factories: default
3. REL_ATTR: id
4. Common Name: EVENT_NAME
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
Change details	ACTION	STRING		
Document ID	ENTITY_ID	INTEGER	SKELETONS id	
Change Type	EVENT_NAME	STRING		
Date and Time	EVENT_TIMESTAMP	LOCAL_TIME		
id	ID	INTEGER		KEY
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
For future use	VER_COUNT	INTEGER		
Change Type Enum	WF_ACTION_ID	INTEGER		
User ID	WF_USER_ID	UUID	ca_contact uuid	

opsys Object

The object details are as follows:

1. Associated Table: ca_resource_operating_system
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: inventory
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		
exclude_registration	exclude_registration	integer		
id	id	INTEGER		UNIQUE REQUIRED KEY
delete_flag	inactive	integer	actbool enum	
last_mod	last_update_date	LOCAL_TIME		
last_update_user	last_update_user	STRING		
name	name	STRING		

version_number	version_number	integer		

options Object

The object details are as follows:

1. Associated Table: Options
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
action	action	INTEGER		
action_status	action_status	STRING		
app_name	app_name	STRING		
default_value	default_value	STRING		
deinstall_script	deinstall_script	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
error_msg	error_msg	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
install_script	install_script	STRING		
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
option_name	option_name	STRING		
persistent_id	persid	STRING		
readme	readme	STRING		
sequence	sequence	INTEGER		
sym	sym	STRING		REQUIRED
validation	validation	STRING		
value	value	STRING		
value_active	value_active	INTEGER		

org Object

The object details are as follows:

1. Associated Table: Internal_Organization
2. Factories: default
3. REL_ATTR: id
4. Common Name: name

5. Function Group: organization

ca_organization Table

REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
org_num	abbreviation	STRING		
alt_phone_cc	alt_phone_cc	integer		
alt_phone	alt_phone_number	STRING		
comment	comment	STRING		
company	company_uuid	UUID	ca_company company_uuid	
contact	contact_uuid	UUID	ca_contact uuid	
billing_code	cost_center	integer	ca_resource_cost_center id	
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		
email_addr	email_address	STRING		
exclude_registration	exclude_registration	integer		
fax_cc	fax_cc	integer		
fax_phone	fax_number	STRING		
delete_flag	inactive	integer	actbool enum	
last_mod	last_update_date	LOCAL_TIME		
last_update_user	last_update_user	STRING		
location	location_uuid	UUID	ca_location location_uuid	
name	org_name	STRING		
id	organization_uuid	UUID		UNIQUE REQUIRED KEY
pemail_addr	pager_email_address	STRING		
parent_org_uuid	parent_org_uuid	UUID		
pri_phone_cc	pri_phone_cc	integer		
phone_number	pri_phone_number	STRING		
version_number	version_number	integer		

usp_organization Table

Attribute	DB Field	Data Type	SREL References	Flags
iorg_assigned_svr	iorg_assigned_svr	INTEGER		
service_type	iorg_service_type	STRING	srv_desc code	

last_mod	last_mod	LOCAL_TIME		
id	organization_uuid	UUID		
owning_contract	owning_contract	INTEGER	svc_contract id	

outage_type Object

The object details are as follows:

1. Associated Table: usp_outage_type
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	Data Type	Reference	Flags
delete_flag	SREL actbool.enum	del	ON_NEW
description	STRING (4000)		
id	INTEGER		UNIQUE
producer_id	LOCAL STRING (20)		
persistent_id	LOCAL STRING (60)		
last_mod_by	SREL	cnt.id	ON_NEW {USER} ON_CI {USER}
last_mod_dt	DATE		ON_CI {NOW}
sym	STRING (60)		

P_GROUPS Object

The object details are as follows:

1. Associated Table: P_GROUPS
2. Factories: default
3. REL_ATTR: id
4. Function Group:
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
GRP_LIST	GRP_LIST	STRING		
GRP_LIST_KEY	GRP_LIST_KEY	STRING		
GROUP_LIST_N	(local attr)	STRING		
TYPE	TYPE	INTEGER		Indicates if the P Groups is based on Roles or Groups: **1** -- Groups (Default) **2** -- Roles
id	ID	INTEGER		KEY

contained_ groups	contained_ groups	BREL	usp_group_ pgroup	
contained_roles	contained_roles	BREL	usp_role_ pgroup	
last_mod_dt	last_mod_dt	DATE		Indicates the timestamp of when this record was last modified.
last_mod_by	last_mod_by	UUID	cnt	Specifies the UUID of the contact who last modified this record
producer_id	producer_id	STRING		
persistent_id	persid	STRING		

perscnt Object

The object details are as follows:

1. Associated Table: Person_Contacting
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
sym	sym	STRING		UNIQUE REQUIRED S_KEY

position Object

The object details are as follows:

1. Associated Table: ca_job_title
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		

exclude_registration	exclude_registration	integer		
id	id	INTEGER		UNIQUE REQUIRED KEY
delete_flag	inactive	integer	actbool enum	
last_mod_dt	last_update_date	LOCAL_TIME		
last_mod_by	last_update_user	STRING		
sym	name	STRING		
version_number	version_number	integer		

pr Object

The object details are as follows:

1. Associated Table: Call_Req
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: ref_num
5. Function Group: call_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
active	active_flag	INTEGER	bool_tab enum	REQUIRED
active_prev	LOCAL	SREL	bool.enum	
affected_ resource	affected_rc	UUID	ca_owned_ resource uuid	
assignee	assignee	UUID	agt.id	
assignee_prev	LOCAL	SREL	pcat.persistent_id	
call_back_date	call_back_date	LOCAL_ TIME		
call_back_flag	call_back_flag	INTEGER		
category	category	STRING	prob_ctg persid	
category_prev	LOCAL	SREL		
caused_by_chg	caused_by_chg	SREL	chg.id	
change	change	INTEGER	chg id	
charge_back_id	charge_back_ id	STRING		
close_date	close_date	LOCAL_ TIME		
cr_tticket	cr_tticket	INTEGER		
created_via	created_via	INTEGER	interface id	
customer	customer	UUID	ca_contact uuid	REQUIRED
event_token	event_token	STRING		
extern_ref	extern_ref	STRING		
group	group_id	UUID		

group_prev	LOCAL	SREL	grp.id	
id	id	INTEGER		UNIQUE REQUIRED KEY
impact	impact	INTEGER	impact enum	
impact_prev	LOCAL	SREL	imp.enum	
incident_priority	incident_priority	INTEGER		
last_act_id	last_act_id	STRING		
last_mod_dt	last_mod_dt	LOCAL_TIME		
log_agent	log_agent	UUID	ca_contact uuid	REQUIRED
macro_ predicted_ violation	macro_predict_ viol	INTEGER		
open_date	open_date	LOCAL_ TIME		
outage_end_time	outage_end_ time	LOCAL_TIME		
outage_start_ time	outage_start_ time	LOCAL_TIME		
parent	parent	STRING	call_req persid	
persistent_id	persid	STRING		
predicted_sla_ violation	predicted_sla_viol	INTEGER		
priority	priority	INTEGER	pri enum	REQUIRED
priority_prev	LOCAL	SREL	pri.enum	
problem	problem	STRING		
ref_num	ref_num	STRING		UNIQUE REQUIRED S_KEY
resolve_date	resolve_date	LOCAL_ TIME		
rootcause	rootcause	INTEGER	rootcause id	
extern_token	sched_token	STRING		
severity	severity	INTEGER	sevrty enum	
severity_prev	LOCAL	SREL	sev.enum	
sla_violation	sla_violation	INTEGER		
base_template	solution	STRING	call_req persid	
status	status	STRING	cr_stat code	
status_prev	LOCAL	SREL	crs.code	
string1	string1	STRING		
string2	string2	STRING		
string3	string3	STRING		
string4	string4	STRING		
string5	string5	STRING		
string6	string6	STRING		
summary	summary	STRING		
support_lev	support_lev	STRING	srv_desc code	

template_name	template_name	STRING	cr_template template_name	
time_spent_ sum	time_spent_ sum	DURATION		
type	type	STRING	crt code	
urgency	urgency	INTEGER	urgncy enum	
urgency_prev	LOCAL	SREL	urg.enum	

pr_trans Object

The object details are as follows:

1. Associated Tables: pr_trans
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
status	status	STRING	crs	Specifies the current ticket status.
new_status	new status	STRING	crs	Specifies the new ticket status.
is default		INTEGER		Default transition that appears when the Status field is empty. On new default: 0
must_comment		INTEGER		Comment required when using a transition. On new default: 0
delete_flag	del		actbool	Required. On new default: 0
condition			macro	Site condition macro to approve transition.
condition_error		STRING		Error message for site condition.
description		STRING		Description of this transition.
last_mod_by			cnt	On new default user; on CI set user
last_mod_dt		DATE		On CI set user now

prod Object

The object details are as follows:

1. Associated Table: Product

2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
sym	sym	STRING		UNIQUE REQUIRED S_KEY

quick_tpl_types Object

The object details are as follows:

1. Associated Table: Quick_Template_Types
2. Factories: default
3. REL_ATTR: enum
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
enum	enum	INTEGER		UNIQUE REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
description	nx_desc	STRING		
persistent_id	persid	STRING		
sym	sym	STRING		

rc Object

The object details are as follows:

1. Associated Table: Rootcause
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
sym	sym	STRING		UNIQUE REQUIRED S_KEY

resocode Object

This object is a reference object to denote the resolution code of the request or incident.

The object details are as follows:

1. Associated Table: usp_resolution_code
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	Data Type	Reference	Flags
delete_flag	SREL	actbool.enum	ON_NEW
description	STRING		
id	INTEGER		UNIQUE
persistent_id	STRING		
last_mod_by	SREL	cnt	ON_NEW {USER} ON_CI {USER}
last_mod_dt	DATE		ON_CI {NOW}
sym	STRING		REQUIRED

resomethod Object

This object is a reference object to denote the resolution method of the request or incident.

The object details are as follows:

1. Associated Table: usp_resolution_method
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym

5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	Data Type	Reference	Flags
delete_flag	SREL	actbool.enum	ON_NEW
description	STRING		
id	INTEGER		UNIQUE
persistent_id	STRING		
last_mod_by	SREL	cnt	ON_NEW {USER} ON_CI {USER}
last_mod_dt	DATE		ON_CI {NOW}
sym	STRING		REQUIRED

response Object

The object details are as follows:

1. Associated Table: Response
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
chg_flag	chg_flag	INTEGER		S_KEY
cr_flag	cr_flag	INTEGER		S_KEY
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
in_flag	in_flag	INTEGER		S_KEY
iss_flag	iss_flag	INTEGER		S_KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
pr_flag	pr_flag	INTEGER		S_KEY
response	response	STRING		
response_owner	response_owner	UUID	ca_contact uuid	S_KEY
sym	sym	STRING		REQUIRED S_KEY

rest_access Object

The object details are as follows:

1. Associated Table: usp_rest_access
2. Factories: rest_access
3. REL_ATTR: id
4. Common Name: access_key
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE DELETE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		KEY
producer_id		LOCAL STRING 20		
persistent_id		LOCAL STRING 60		
access_key	access_key	INTEGER		REQUIRED
contact	contact	SREL	cnt.id	REQUIRED
secret_key	secret_key	STRING 64		
expiration_date	expiration_date	DATE		
suppress_trigger		LOCAL INTEGER		

rrf Object

The object details are as follows:

1. Associated Table: Remote_Ref
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
arch_type	arch_type	STRING		
code	code	STRING		UNIQUE REQUIRED S_KEY
delete_flag	del	INTEGER	actbool enum	REQUIRED
exec_str	exec_str	STRING		
function_group	function_group	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
pcexec_str	pcexec_str	STRING		

sym	sym	STRING		REQUIRED

rss Object

The object details are as follows:

1. Associated Table: ca_resource_status
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: ci_reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		
exclude_registration	exclude_registration	integer		
id	id	INTEGER		UNIQUE REQUIRED KEY
delete_flag	inactive	integer	actbool enum	
last_update_date	last_update_date	LOCAL_TIME		
last_update_user	last_update_user	STRING		
sym	name	STRING		
version_number	version_number	integer		

seq Object

The object details are as follows:

1. Associated Table: Sequence_Control
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
code	code	STRING		UNIQUE REQUIRED S_KEY
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY

last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
prefix	prefix	STRING		
suffix	suffix	STRING		
sym	sym	STRING		REQUIRED

sev Object

The object details are as follows:

1. Associated Table: Severity
2. Factories: default
3. REL_ATTR: enum
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
enum	enum	INTEGER		REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
description	nx_desc	STRING		
sym	sym	STRING		UNIQUE REQUIRED S_KEY

SHOW_OBJ Object

The object details are as follows:

1. Associated Table: SHOW_OBJ
2. Factories: default
3. REL_ATTR: id
4. Common Name: OBJ_PERSID
5. Function Group:
6. REST Operations: CREATE READ UPDATE

Attributes	DB Field	Data Type	SREL References	Flags
Expiration Date	EXPIRE_DATE	LOCAL_TIME		
id	ID	INTEGER		KEY
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
Object ID	OBJ_PERSID	STRING		
Password	PWD	STRING		

site Object

The object details are as follows:

1. Associated Table: ca_site
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: inventory
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
alias	alias	STRING		
contact	contact_uuid	UUID	ca_contact uuid	
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		
exclude_registration	exclude_registration	integer		
id	id	INTEGER		UNIQUE REQUIRED KEY
delete_flag	inactive	integer	actbool enum	
last_mod	last_update_date	LOCAL_TIME		
last_update_user	last_update_user	STRING		
name	name	STRING		
version_number	version_number	integer		

slatpl Object

The object details are as follows:

1. Associated Table: SLA_Template
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
elapsed	elapsed	DURATION		REQUIRED
event	event	STRING	evt persid	
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	

last_mod_dt	last_mod_dt	LOCAL_TIME		
object_type	object_type	STRING		
persistent_id	persid	STRING		
service_type	service_type	STRING	srv_desc code	REQUIRED
sym	sym	STRING		REQUIRED S_KEY

special_handling Object

The object details are as follows:

1. Associated Table: usp_special_handling
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: inventory
6. REST Operations: CREATE READ UPDATE

Field	Data Type	Reference	Flags
id	INTEGER UNIQUE		
producer_id	LOCAL STRING (20)		
persistent_id	LOCAL STRING (60)		
delete_flag	SREL	actbool	REQUIRED. ON_NEW
sym	STRING (60)		REQUIRED
description	STRING (4000)		
alert_icon_url	STRING (1000)		
alert_text	STRING (60)		
autodisplay_notes	SREL	bool.enum	ON_NEW
escalate_urgency	SREL	bool.enum	ON_NEW
cnthandling_list	BREL	contact_handling.special_handling	DYNAMIC
last_mod_by	SREL	cnt	ON_NEW {USER} ON_CI SET {USER}
last_mod_dt	DATE		ON_CI_SET {NOW}

svc_contract Object

The object details are as follows:

1. Associated Table: Service_Contract
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: admin

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
active	active	INTEGER	actbool enum	
contract_num	contract_num	STRING		
delete_flag	del	INTEGER	actbool enum	
dflt_chgcat_st	dflt_chgcat_st	STRING	srv_desc code	
dflt_cnt_st	dflt_cnt_st	STRING	srv_desc code	
dflt_isscat_st	dflt_isscat_st	STRING	srv_desc code	
dflt_nr_st	dflt_nr_st	STRING	srv_desc code	
dflt_pcat_st	dflt_pcat_st	STRING	srv_desc code	
dflt_pri_st	dflt_pri_st	STRING	srv_desc code	
expiration	expiration	LOCAL_TIME		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
description	nx_desc	STRING		
service_type	org_svc_type	STRING	srv_desc code	
persistent_id	persid	STRING		
svc_advocate	svc_advocate	UUID	ca_contact uuid	
svc_owner	svc_owner	UUID	ca_contact uuid	
sym	sym	STRING		

typecnt Object

The object details are as follows:

1. Associated Table: Type_Of_Contact
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		

sym	sym	STRING		UNIQUE REQUIRED S_KEY

tz Object

The object details are as follows:

1. Associated Table: Timezone
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: timezone
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
code	code	STRING		UNIQUE REQUIRED S_KEY
delete_flag	del	INTEGER	actbool enum	REQUIRED
dst_delta	dst_delta	INTEGER		
end_abs_date	end_abs_date	LOCAL_TIME		
end_day	end_day	INTEGER		
end_mon	end_mon	INTEGER		
end_pos	end_pos	INTEGER		
gmt_delta	gmt_delta	INTEGER		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
start_abs_date	start_abs_date	LOCAL_TIME		
start_day	start_day	INTEGER		
start_mon	start_mon	INTEGER		
start_pos	start_pos	INTEGER		
sym	sym	STRING		REQUIRED S_KEY

tspan Object

The object details are as follows:

1. Associated Table: Timespan
2. Factories: default
3. REL_ATTR: sym
4. Common Name: sym
5. Function Group: admin

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
code	code	STRING		UNIQUE REQUIRED
end_day	end_day	STRING		
end_hour	end_hour	STRING		
end_minute	end_minute	STRING		
end_month	end_month	STRING		
end_year	end_year	STRING		
id	id	INTEGER		UNIQUE KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
desc	nx_desc	STRING		
start_day	start_day	STRING		
start_hour	start_hour	STRING		
start_minute	start_minute	STRING		
start_month	start_month	STRING		
start_year	start_year	STRING		
sym	sym	STRING		UNIQUE REQUIRED
trigger_day	trigger_day	STRING		
trigger_hour	trigger_hour	STRING		
trigger_minute	trigger_minute	STRING		
trigger_month	trigger_month	STRING		
trigger_year	trigger_year	STRING		

tab Object

The object details are as follows:

1. Associated Table: usp_tab
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: Security
6. REST Operations: CREATE READ UPDATE

Attribute Name	Data Type	Relationship Object	Flags
id	INTEGER		UNIQUE
name	STRING		REQUIRED
code	STRING		UNIQUE; REQUIRED
display_name	STRING		REQUIRED
delete_flag	SREL	actbool	REQUIRED

description	STRING		
menu_bar_obj	SREL	menu_bar	
web_form_obj	SREL	web_form	
last_mod_dt	DATE		
last_mod_by	SREL	cnt	

transition_type Object

The object details are as follows:

1. Associated Table: transition_type
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: admin
6. Tenant: optional
7. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
sym		STRING		Required
ss_flag		INTEGER		Required On new default: 0
ss_button_text		STRING		Required
ss_header_text		STRING		Required
delete_flag	del		actbool	Required On new default: 0
description		STRING		
last_mod_dt		DATE		On CI set now
last_mod_by			cnt	On CI set user On new default user

urg Object

The object details are as follows:

1. Associated Table: Urgency
2. Factories: default
3. REL_ATTR: enum
4. Common Name: sym
5. Function Group: prioritization
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
enum	enum	INTEGER		REQUIRED

id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
description	nx_desc	STRING		
sym	sym	STRING		UNIQUE REQUIRED S_KEY
value	value	INTEGER		

USP_PREFERENCES Object

The object details are as follows:

1. Associated Table: USP_PREFERENCES
2. Factories: default
3. REL_ATTR: id
4. Common Name: id
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
Contact ID	ANALYST_ID	UUID	ca_contact uuid	
Knowledge Categories - Documents Per Page	ARC_DOCS_TO_DISPLAY	INTEGER		
Assignee	ASSIGNEE	INTEGER		
Author	AUTHOR	INTEGER		
FAQ Rating	BU_RESULT	INTEGER		
Mouseover Menus	CLASSIC_RESULTSET_CONTEXT	INTEGER		
Created Via	CREATED_VIA	INTEGER		
Creation Date	CREATION_DATE	INTEGER		
Current Task	CURRENT_ACTION	INTEGER		
Custom 1	CUSTOM1	INTEGER		
Custom 2	CUSTOM2	INTEGER		
Custom 3	CUSTOM3	INTEGER		
Custom 4	CUSTOM4	INTEGER		
Custom 5	CUSTOM5	INTEGER		
Custom Num 1	CUSTOM_NUM1	INTEGER		
Custom Num 2	CUSTOM_NUM2	INTEGER		
Document ID	DOC_ID	INTEGER		
Document Template	DOC_TEMPLATE	INTEGER		
Document Type	DOC_TYPE	INTEGER		
Document Version	DOC_VERSION	INTEGER		

Expiration Date	EXPIRATION_DATE	INTEGER		
GLOBALSD_ACTIVE_ZONE	GLOBALSD_ACTIVE_ZONE	INTEGER		
Hits	HITS	INTEGER		
id	ID	INTEGER		REQUIRED KEY
INBOX_COUNTER	INBOX_COUNTER	INTEGER		
Initiator	INITIATOR	INTEGER		
Item	ITEM	INTEGER		
KT_REPORT_CARD_APAST_DAYS	KT_REPORT_CARD_PAST_DAYS	INTEGER		
KT_REPORT_CARD_SCREEN_DEFAULT	KT_REPORT_CARD_SCREEN_DEFAULT	INTEGER		
Last Accepted Date	LAST_ACCEPTED_DATE	INTEGER		
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
Modify Date	MODIFY_DATE	INTEGER		
Knowledge Documents View Mode	ONE_B_DOC_VIEW_MODE	INTEGER		
Knowledge Documents Per Page	ONE_B_DOCS_TO_DISPLAY	INTEGER		
Knowledge Documents Show Details Flag	ONE_B_HIDE_DETAILS	INTEGER		
EBR Match Type	ONE_B_MATCH_TYPE	INTEGER		
EBR Search Fields	ONE_B_SEARCH_FIELDS	INTEGER		
EBR Search Order	ONE_B_SEARCH_ORDER	STRING		
EBR Search Type	ONE_B_SEARCH_TYPE	INTEGER		
EBR Word Parts	ONE_B_WORD_PARTS	INTEGER		
Owner	OWNER	INTEGER		
Primary Category	PRIMARY_INDEX	INTEGER		
Workflow Priority	PRIORITY	INTEGER		
Product	PRODUCT	INTEGER		
Published Date	PUBLISHED_DATE	INTEGER		
Review Date	REVIEW_DATE	INTEGER		
Solution Count	SD_ACCEPTED_HITS	INTEGER		
Impact	SD_IMPACT	INTEGER		
Priority	SD_PRIORITY	INTEGER		
Root Cause	SD_ROOTCAUSE	INTEGER		
SD_SEARCH_FIELDS_CR	SD_SEARCH_FIELDS_CR	INTEGER		

SD_SEARCH_FIELDS_ISS	SD_SEARCH_FIELDS_ISS	INTEGER		
Severity	SD_SEVERITY	INTEGER		
Urgency	SD_URGENCY	INTEGER		
Start Date	START_DATE	INTEGER		
Status	STATUS	INTEGER		
Subject Expert	SUBJECT_EXPERT	INTEGER		
User Defined ID	USER_DEF_ID	INTEGER		
WEB_LAST_LOGIN	WEB_LAST_LOGIN	LOCAL_TIME		
WEB_POPUP1_HEIGHT	WEB_POPUP1_HEIGHT	INTEGER		
WEB_POPUP1_WIDTH	WEB_POPUP1_WIDTH	INTEGER		
WEB_POPUP2_HEIGHT	WEB_POPUP2_HEIGHT	INTEGER		
WEB_POPUP2_WIDTH	WEB_POPUP2_WIDTH	INTEGER		
WEB_POPUP3_HEIGHT	WEB_POPUP3_HEIGHT	INTEGER		
WEB_POPUP3_WIDTH	WEB_POPUP3_WIDTH	INTEGER		
WEB_POPUP4_HEIGHT	WEB_POPUP4_HEIGHT	INTEGER		
WEB_POPUP4_WIDTH	WEB_POPUP4_WIDTH	INTEGER		
WEB_ROLE_ID	WEB_ROLE_ID	STRING		
WEB_PREFERENCES	WEB_PREFERENCES	INTEGER		
WEB_SUPPRESS_TOUR	WEB_SUPPRESS_TOUR	INTEGER		
WEB_TOOLBAR_TAB	WEB_TOOLBAR_TAB	INTEGER		
WF_TEMPLATE	WF_TEMPLATE	INTEGER		

usp_exlist_format Object

The object details are as follows:

1. Associated Table: usp_exlist_format
2. Factories: default
3. REL_ATTR: id
4. Common Name: file_extension
5. Function Group: admin
6. Tenant: optional

Attribute	DB Field	Data Type	SREL References	Flags
file_extension		STRING		REQUIRED
mime_type		STRING		REQUIRED
xslt_name		STRING		
delete_flag		del	actbool enum	REQUIRED. On new default: 0

USP_PROPERTIES Object

The object details are as follows:

1. Associated Table: USP_PROPERTIES
2. Factories: default
3. REL_ATTR: id
4. Common Name: PROPERTY_NAME
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	ID	INTEGER		REQUIRED KEY
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
Property Default	PROPERTY_DEFAULT	STRING		
Property Description	PROPERTY_DESCRIPTION	STRING		
Property Name	PROPERTY_NAME	STRING		S_KEY
Property Type	PROPERTY_TYPE	STRING		
Property Value	PROPERTY_VALUE	STRING		

usp_session_ticket Object

The object details are as follows:

1. Associated Table: usp_session_ticket
2. REL_ATTR: id
3. Common Name: session_persid
4. Function Group: change_reference
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE REQUIRED KEY
producer_id		LOCAL_STRING 20		
persistent_id	persid	STRING 30		
delete_flag	del	SREL	actbool.enum	REQUIRED
sym	sym	STRING 60		REQUIRED
description	description	STRING 240		
modified_date	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt.id	

usq Object

The object details are as follows:

1. Associated Table: User_Query
2. Factories: default
3. REL_ATTR: id
4. Common Name: label
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
expanded	expanded	INTEGER		
factory	factory	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
label	label	STRING		REQUIRED
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
obj_persid	obj_persid	STRING		
parent	parent	INTEGER	usq id	
persistent_id	persid	STRING		
role_persid	role_persid	STRING		
query	query	STRING	crsq code	
query_set	query_set	INTEGER		
query_type	query_type	INTEGER		
sequence	sequence	INTEGER		REQUIRED

vpt Object

The object details are as follows:

1. Associated Table: ca_company_type
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: ci_reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
description	description	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY

delete_flag	inactive	integer	actbool enum	
last_mod	last_update_date	LOCAL_TIME		
last_update_user	last_update_user	STRING		
sym	name	STRING		
version_number	version_number	integer		

wrkshft Object

The object details are as follows:

1. Associated Table: Bop_Workshift
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: sym
5. Function Group: workshifts
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
description	description	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
sched	sched	STRING		
sym	sym	STRING		REQUIRED

usq Object

The object details are as follows:

1. Associated Table: User_Query
2. Factories: default
3. REL_ATTR: id
4. Common Name: label
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
expanded	expanded	INTEGER		
factory	factory	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
label	label	STRING		REQUIRED
last_mod_by	last_mod_by	UUID	ca_contact uuid	

last_mod_dt	last_mod_dt	LOCAL_TIME		
obj_persid	obj_persid	STRING		
parent	parent	INTEGER	usq id	
persistent_id	persid	STRING		
role_persid	role_persid	STRING		
query	query	STRING	crsq code	
query_set	query_set	INTEGER		
query_type	query_type	INTEGER		
sequence	sequence	INTEGER		REQUIRED

Access

This article contains the following topics:

acctyp_role Object

The object details are as follows:

1. Associated Table: usp_acctyp_role
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: Security
6. REST Operations: CREATE READ UPDATE

Attributes	Data Type	Related Object	Flags
id	INTEGER		
access_type	SREL	acctyp	
role_obj	SREL	role	
is_default	INTEGER		
last_mod_dt	DATE		

acctyp Object

The object details are as follows:

1. Associated Table: Access_Type_v2
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: security
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
access_level	access_level	INTEGER	acc_lvls enum	REQUIRED
default_flag	default_flag	INTEGER		

delete_flag	del	INTEGER	actbool enum	REQUIRED
description	description	STRING		
external_auth	external_auth	INTEGER		REQUIRED
grant_level	grant_level	INTEGER	acc_lvls enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
ldap_access_grou	ldap_access_group	STRING	ldap_group_id	
pin_field	pin_field	STRING		
persistent_id	persid	LOCAL STRING 60		
roles	roles	LIST		
sym	sym	STRING		REQUIRED S_KEY
user_auth	User_auth	INTEGER		REQUIRED
view_internal	view_internal	INTEGER		REQUIRED
wsp	wsp	INTEGER		
cmdlind_role	cmdline_role	INTEGER	role id	
reporting_role	reporting_role	INTEGER	role id	
web_service_role	web_service_role	INTEGER	role id	
rest_web_service_role	rest_web_service_role	SREL	role.id	

acc_lvls Object

The object details are as follows:

1. Associated Table: Access_Levels
2. Factories: default
3. REL_ATTR: enum
4. Common Name: sym
5. Function Group: security
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
enum	enum	INTEGER		UNIQUE REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
description	nx_desc	STRING		
persistent_id	persid	STRING		
sym	sym	STRING		UNIQUE REQUIRED S_KEY

Attachment Objects

This topic contains the following information:

attmnt Object

The object details are as follows:

1. Associated Table: Attachment
2. Factories: default
3. REL_ATTR: id
4. Common Name: created_dt
5. Function Group: call_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
attmnt_name	attmnt_name	STRING		
attmnt_uuid	attmnt_uuid	UUID		
created_by	created_by	UUID	ca_contact uuid	
created_dt	created_dt	LOCAL_TIME		
delete_flag	del	INTEGER	actbool enum	REQUIRED
description	description	STRING		
exec_cmd	exec_cmd	STRING	rem_ref code	
file_date	file_date	LOCAL_TIME		
file_name	file_name	STRING		
file_size	file_size	INTEGER		
file_type	file_type	STRING		
folder_id	folder_id	INTEGER	attmnt_folder id	
folder_path_ ids	folder_path_ ids	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
inherit_ permission_id	inherit_ permission_id	INTEGER		
KDS_ ATTACHED	KDS_ ATTACHED	INTEGER		
last_mod_by	last_mod_by	UUID	ca_contact uuid	Specifies the UUID of the contact who last modified this record.
last_mod_dt	last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
link_only	link_only	INTEGER	bool_tab enum	
link_type	link_type	STRING 255		
orig_file_name	orig_file_name	STRING		
persistent_id	persid	STRING		

pgroup_type	pgroup_type	INTEGER		Indicates if the P Groups is based on Roles or Groups: **1** -- Groups (Default) **2** -- Roles
read_pgroup	read_pgroup	INTEGER	P_GROUPS id	
rel_file_path	rel_file_path	STRING		
repository	repository	SREL	doc_rep persid	
sec_uuid	sec_uuid	UUID		
status	status	STRING		
tenant	tenant	UUID	ca_tenant	Specifies the UUID of the tenant.
write_pgroup	write_pgroup	INTEGER	P_GROUPS id	
zip_flag	zip_flag	INTEGER		

attmnt_folder Object

The object details are as follows:

1. Associated Table: attmnt_folder
2. Factories: default
3. REL_ATTR: id
4. Common Name: folder_name
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
folder_name	folder_name	STRING		
folder_path_ids	folder_path_ids	STRING		
folder_type	folder_type	INTEGER		
has_children	has_children	INTEGER		
id	id	INTEGER		REQUIRED KEY
inherit_ permission_id	inherit_ permission_id	INTEGER	attmnt_folder id	
last_mod_date	last_mod_date	LOCAL_TIME		
parent_id	parent_id	SREL	attmnt_folder id	
read_pgroup	read_pgroup	INTEGER	P_GROUPS id	
repository	repository	SREL	doc_rep persid	
write_pgroup	write_pgroup	INTEGER	P_GROUPS id	

Boolean Objects

This topic contains the following information:

rev_bool Object

The object details are as follows:

1. Associated Table: Reverse_Boolean_Table
2. Factories: default
3. REL_ATTR: enum
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER		REQUIRED
enum	enum	INTEGER		REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
desc	nx_desc	STRING		
sym	sym	STRING		UNIQUE REQUIRED S_KEY

actbool Object

The object details are as follows:

1. Associated Table: Active_Boolean_Table
2. Factories: default
3. REL_ATTR: enum
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER		REQUIRED
description	description	STRING		
enum	enum	INTEGER		REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_dt	last_mod_dt	LOCAL_TIME		
sym	sym	STRING		S_KEY

actrbool Object

The object details are as follows:

1. Associated Table: Active_Reverse_Boolean_Table
2. Factories: default
3. REL_ATTR: enum
4. Common Name: sym
5. Function Group: admin

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER		REQUIRED
description	description	STRING		
enum	enum	INTEGER		REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_dt	last_mod_dt	LOCAL_TIME		
sym	sym	STRING		S_KEY

Business Management

This topic contains the following information:

bmcls Object

The object details are as follows:

1. Associated Table: Business_Management_Class
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
description	nx_desc	STRING		
persistent_id	persid	STRING		
sym	sym	STRING		UNIQUE REQUIRED S_KEY

bmhier Object

The object details are as follows:

1. Associated Table: Business_Management
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: inventory

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
bm_rep	bm_rep	INTEGER	busrep id	
cost	cost	INTEGER		
child	hier_child	UUID	ca_owned_resource uuid	REQUIRED
parent	hier_parent	UUID	ca_owned_resource uuid	
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
description	nx_desc	STRING		
persistent_id	persid	STRING		
sym	sym	STRING		UNIQUE REQUIRED S_KEY

bmrep Object

The object details are as follows:

1. Associated Table: Business_Management_Repository
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
hostname	hostname	STRING		UNIQUE REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
description	nx_desc	STRING		
password	password	STRING		
persistent_id	persid	STRING		
sym	sym	STRING		UNIQUE REQUIRED S_KEY
userid	userid	STRING		

bms Object

The object details are as follows:

1. Associated Table: Business_Management_Status

2. Factories: default
3. REL_ATTR: status_no
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
description	nx_desc	STRING		
persistent_id	persid	STRING		
status_no	status_no	INTEGER		UNIQUE REQUIRED S_KEY
sym	sym	STRING		UNIQUE REQUIRED S_KEY

Change Request Objects

This article contains the following topics:

chg Object

The object details are as follows:

1. Associated Table: Change_Request
2. Factories: default
3. REL_ATTR: id
4. Common Name: chg_ref_num
5. Function Group: change_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
actions	actions	STRING		
active	active_flag	INTEGER	bool_tab enum	REQUIRED
active_prev	LOCAL	SREL	bool.enum	
actual_comp_ date	actual_comp_ date	LOCAL_ TIME		
cost	actual_cost	INTEGER		
actual_total_ time	actual_total_ time	DURATION		
affected_contact	affected_contact	UUID	ca_contact uuid	REQUIRED
assignee	assignee	UUID		
assignee_prev	LOCAL	SREL	agt.id	
backout_plan	backout_plan	STRING		

business_case	business_case	STRING 4000		
call_back_date	call_back_date	LOCAL_ TIME		
call_back_flag	call_back_flag	INTEGER		
category	category	STRING	chgcat code	
category_prev	LOCAL	SREL	chgcat.code	
cawf_procid	cawf_procid	STRING		
chg_ref_num	chg_ref_num	STRING		UNIQUE REQUIRED S_KEY
chgtype	chgtype	SREL	chgtype.id	
close_date	close_date	LOCAL_TIME		
closure_code	closure_code	SREL		
created_via	created_via	INTEGER	interface id	
effort	effort	STRING 4000		
est_comp_date	est_comp_date	LOCAL_TIME		
est_cost	est_cost	INTEGER		
est_total_time	est_total_time	DURATION		
requested_by		SREL	cnt.id	
external_system_ticket		STRING (4000)		
orig_user_dept		SREL	dept.id	
orig_user_organization		SREL	org.id	
orig_user_admin_org		SREL	org.id	
orig_user_cost_center		SREL	cost_cntr.id	
flag1	flag1	INTEGER		
flag2	flag2	INTEGER		
flag3	flag3	INTEGER		
flag4	flag4	INTEGER		
flag5	flag5	INTEGER		
flag6	flag6	INTEGER		
group	group_id	UUID		
group_prev	LOCAL	SREL	grp.id	
id	id	INTEGER		UNIQUE REQUIRED KEY
impact	impact	INTEGER	impact enum	
impact_prev	LOCAL	SREL	imp.enum	
justification	justification	STRING 4000		Business Justification
last_mod_by	last_mod_by	UUID	ca_contact uuid	
modified_date	last_mod_dt	LOCAL_TIME		
log_agent	log_agent	UUID	ca_contact uuid	REQUIRED

macro_predicted_violation	macro_predict_ viol	INTEGER		
need_by	need_by	LOCAL_TIME		
open_date	open_date	LOCAL_TIME		
organization	organization	UUID	ca_organization uuid	
orig_user_admin_org		SREL	org	
orig_user_cost_center		SREL	cost_cntr	
orig_user_dept		SREL	dept	
orig_user_organization		SREL	org	
parent	parent	INTEGER	chg id	
persistent_id	persid	STRING		
person_ contacting	person_ contacting	INTEGER	perscon id	
predicted_sla_ violation	predicted_sla_ viol	INTEGER		
priority	priority	INTEGER	pri enum	REQUIRED
priority_prev	LOCAL	SREL	pri.enum	
product	product	INTEGER	product id	
reporting_ method	reporting_ method	INTEGER	repmeth id	
requestor	requestor	UUID	ca_contact uuid	REQUIRED
resolve_date	resolve_date	LOCAL_TIME		
risk	risk	SREL	risk_level	
rootcause	rootcause	INTEGER	rootcause id	
sched_start_date	sched_start_date	LOCAL_TIME		
sched_end_date	sched_end_date	LOCAL_TIME		
sched_duration	sched_duration	LOCAL_TIME		
service_date	service_date	LOCAL_TIME		
service_num	service_num	STRING		
sla_violation	sla_violation	INTEGER		
start_date	start_date	LOCAL_TIME		
status	status	STRING	chgstat code	
status_prev	LOCAL	SREL	chgstat.code	
string1	string1	STRING		
string2	string2	STRING		
string3	string3	STRING		
string4	string4	STRING		
string5	string5	STRING		
string6	string6	STRING		
submittedSurvey		BREL	risk_svy	
summary	summary	STRING		
support_lev	support_lev	STRING	srv_desc code	

template_name	template_name	STRING	chg_template template_name	
type_of_contact	type_of_contact	INTEGER	toc id	
user1	user1	STRING		
user2	user2	STRING		
user3	user3	STRING		
cab	Cab	UUID		
closure_code	Closure_code	INTEGER	Closure_code.id	
cab_approval	Cab_approval	INTEGER	Boolean.id	
target_times	target_times	BREL	tgt_time_mapped_chg	
target_start_last	target_start_last	DATE		
target_hold_last	target_hold_last	DATE		
target_hold_count	target_hold_count	INTEGER		
target_resolved_last	target_resolved_last	DATE		
target_resolved_count	target_resolved_count	INTEGER		
target_closed_last	target_closed_last	DATE		
target_closed_count	target_closed_count	INTEGER		
close_date_prev	close_date_prev	LOCAL DATE		
resolve_date_prev	resolve_date_prev	LOCAL DATE		
target_hold_count_prev	target_hold_count_prev	LOCAL INTEGER		
target_resolved_count_prev	target_resolved_count_prev	LOCAL INTEGER		
target_closed_count_prev	target_closed_count_prev	LOCAL INTEGER		

chg_tpl Object

The object details are as follows:

1. Associated Table: Change_Template
2. Factories: default
3. REL_ATTR: template_name
4. Common Name: template_name
5. Function Group: change_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		

persistent_id	persid	STRING		
quick_tmpl_type	quick_tmpl_type	INTEGER	quick_tpl_types enum	REQUIRED
template	template	INTEGER	chg id	
template_class	template_class	STRING		
template_name	template_name	STRING		UNIQUE REQUIRED S_KEY

chg_trans Object

The object details are as follows:

1. Associated Tables: chg_trans
2. Factories: default
3. REL_ATTR: id
4. Common Name: condition_error
5. Function Group: change_reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
status	status	STRING	crs	Specifies the current ticket status.
new_status	new status	STRING	crs	Specifies the new ticket status.
is default		INTEGER		Default transition that appears when the Status field is empty. On new default: 0
must_comment		INTEGER		Comment required when using a transition. On new default: 0
delete_flag	del		actbool	Required. On new default: 0
condition			macro	Site condition macro to approve transition.
condition_error		STRING		Error message for site condition.
description		STRING		Description of this transition.
last_mod_by			cnt	On new default user; on CI set user
last_mod_dt		DATE		On CI set user now

chgalg Object

The object details are as follows:

1. Associated Table: Change_Act_Log

2. Factories: default
3. REL_ATTR: id
4. Common Name: description
5. Function Group: change_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
Description	description	STRING		
action_desc	action_desc	STRING		
analyst	analyst	UUID	ca_contact uuid	
change_id	change_id	INTEGER	chg id	
id	id	INTEGER		UNIQUE REQUIRED KEY
internal	internal	INTEGER		
knowledge_session	knowledge_session	STRING		
knowledge_tool	knowledge_tool	STRING		
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
system_time	system_time	LOCAL_TIME		
time_spent	time_spent	DURATION		
time_stamp	time_stamp	LOCAL_TIME		
type	type	STRING	act_type code	

chgcat Object

The object details are as follows:

1. Associated Table: Change_Category
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
assignee	assignee	UUID		
auto_assign	auto_assign	INTEGER		
cab	cab	UUID		
cawf_defid	cawf_defid	STRING		
chgtype	chgtype	SREL	chgtype.id	
children_ok	children_ok	INTEGER		REQUIRED

code	code	STRING		UNIQUE REQUIRED S_KEY
delete_flag	del	INTEGER	actbool enum	REQUIRED
group	group_id	UUID		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
modified_date	last_mod_dt	LOCAL_ TIME		
organization	organization	UUID	ca_organization uuid	
owning_ contract	owning_ contract	INTEGER	svc_contract id	
persistent_id	persid	STRING		
risk_survey	risk_survey	SREL	risk_svy_tpl	
schedule	schedule	INTEGER		
ss_sym		STRING		
ss_include		INTEGER	bool	REQUIRED On new default: 0
service_type	service_type	STRING	srv_desc code	
survey	survey	INTEGER	survey_tpl id	
sym	sym	STRING 1000		REQUIRED S_KEY

chgcnf chg Object

The object details are as follows:

1. Associated Table: usp_conflict_chg
2. Factories: default
3. REL_ATTR: code
4. Common Name: last_mod_dt
5. Function Group: chg_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE KEY
last_mod_by	last_mod_by	UUID	ca_contact	User who last updated this.
last_mod_dt	last_mod_dt	LOCAL_TIME		timestamp of last update to this record.
change	change	INTEGER	chg	Pointer to change order involved in conflict.
conflict_change	conflict_change	INTEGER	chg	Pointer to change order involved in conflict.
is_cause	is_cause	INTEGER		Change in change column caused conflict.

chgcnf Object

The object details are as follows:

1. Associated Table: usp_conflict, usp_conflict_status, usp_conflict_type, usp_conflict_chg
2. Factories: default
3. REL_ATTR: id
4. Common Name: id
5. Function Group: change_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	delete_flag	INTEGER	actbool enum	**0** -- Active **1** -- Inactive
comments	comments	STRING		Reference to comment information.
conflict_begin	conflict_begin	LOCAL_TIME		Start of time period of the conflict.
source_change	source_change	INTEGER	chg	Pointer to change order involved in conflict.
conflict_change	conflict_change	INTEGER	chg	Pointer to change order involved in conflict.
conflict_ci	conflict_ci	UUID	ca_owned_ resource	Pointer to Configuration Item involved in conflict.
conflict_end	conflict_end	LOCAL_TIME		End of time period of the conflict.
conflict_status	conflict_status	STRING	chgcnf_ status	Pointer to usp_conflict_status that indicates the status of this conflict. For example: Resolved, Unresolved, Researching.
conflict_type	conflict_type	STRING	chgcnf_type	Pointer to usp_conflict_type that classifies the type of conflict represented by this record. For example: Schedule Collision.
creation_dt	creation_dt	LOCAL_TIME		Date conflict was last detected.
id	id	INTEGER		UNIQUE KEY

is_resolved	is_resolved	INTEGER		Flag that indicates whether this conflict should be considered resolved. The entry in usp_conflict_status pointed to by conflict_status has an is_resolved flag that is reflected in this column.
last_mod_dt	last_mod_dt	LOCAL_TIME		Last modified timestamp
last_mod_by	last_mod_by	UUID	ca_contact	Last modified by
persistent_id	persid	STRING		Persistent ID.
suppress_log		INTEGER		LOCAL
tenant	tenant	UUID	ca_contact	Reference to Tenant information.

chgcnf status Object

The object details are as follows:

1. Associated Table: usp_conflict_status
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
code	code	STRING		
delete_flag	delete_flag	INTEGER		Logical database delete status.
description	description	STRING		Textual description of the meaning of the status.
id	id	INTEGER		UNIQUE KEY
is_resolved	is_resolved	INTEGER		Used in stored queries to distinguish Resolved and Unresolved conflicts.
last_mod_by	last_mod_by	UUID	ca_contact	User who last updated this
last_mod_dt	last_mod_dt	LOCAL_TIME		Indicates the timestamp of last update to this record.
sym	sym	STRING		Text value displayed to the user.

chgcnf type Object

The object details are as follows:

1. Associated Table: usp_conflict_type
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: change_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
code	code	STRING	actbool	
delete_flag	delete_flag	INTEGER		Logical database delete status.
description	description	STRING		Textual description of the meaning of the status.
icon	icon	STRING		
id	id	INTEGER		UNIQUE KEY
last_mod_by	last_mod_by	UUID	ca_contact	User who last updated this
last_mod_dt	last_mod_dt	LOCAL_TIME		Indicates the timestamp of last update to this record.
sym	sym	STRING		Text value displayed to the user.

chgstat Object

The object details are as follows:

1. Associated Table: Change_Status
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: change_reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
active	active	INTEGER		
code	code	STRING		UNIQUE REQUIRED S_KEY
delete_flag	del	INTEGER	actbool enum	REQUIRED
hold	hold	INTEGER		
id	id	INTEGER		UNIQUE REQUIRED KEY

last_mod_by	last_mod_by	UUID	ca_contact uuid	
modified_date	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
resolved	resolved	INTEGER		
sym	sym	STRING		REQUIRED

BSVC--chgtype Object

The object details are as follows:

1. Associated Table: usp_change_type
2. REL_ATTR: id
3. Common Name: sym
4. Function Group: change_reference
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE REQUIRED KEY
producer_id		LOCAL_STRING 20		
persistent_id	persid	STRING 30		
delete_flag	del	SREL	actbool.enum	REQUIRED
sym	sym	STRING 60		REQUIRED
description	description	STRING 240		
modified_date	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt.id	

Configuration Item

This article contains the following topics:

CI_ACTIONS Object

The object details are as follows:

1. Associated Table: CI_ACTIONS
2. Factories: default
3. REL_ATTR: id
4. Common Name: ACTION_TITLE
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
Task Order	ACTION_ORDER	INTEGER		
Task Title	ACTION_TITLE	STRING		
Contact ID	ANALYST_ID	UUID	ca_contact uuid	

Group ID	GROUP_ID	UUID	ca_contact uuid	
id	ID	INTEGER		REQUIRED KEY
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
Predefined	PREDEFINED	INTEGER		
Current Status ID	STATUS_CURRENT_ID	INTEGER	CI_STATUSES id	
Unpublish Task	UNPUBLISH	INTEGER		
Unretire Task	UNRETIRE	INTEGER		
Workflow Template ID	WF_TEMPLATE_ID	INTEGER	CI_WF_TEMPLATES id	

CI_ACTIONS_ALTERNATE Object

The object details are as follows:

1. Associated Table: CI_ACTIONS_ALTERNATE
2. Factories: default
3. REL_ATTR: id
4. Common Name: id
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
Task ID	ACTION_ID	INTEGER	CI_ACTIONS id	
Contact ID	CONTACT_ID	UUID	ca_contact uuid	
Contact Type	CONTACT_TYPE	INTEGER		
id	ID	INTEGER		REQUIRED KEY
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		

CI_BOOKMARKS Object

The object details are as follows:

1. Associated Table: CI_BOOKMARKS
2. Factories: default
3. REL_ATTR: id
4. Common Name: BOOKMARK_TITLE
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
Bookmark Title	BOOKMARK_TITLE	STRING		
Document ID	DOCUMENT_ID	INTEGER	SKELETONS id	
id	ID	INTEGER		REQUIRED KEY
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
Contact ID	USER_ID	UUID	ca_contact uuid	

CI_DOC_LINKS Object

The object details are as follows:

1. Associated Table: CI_DOC_LINKS
2. Factories: default
3. REL_ATTR: id
4. Common Name: id
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
DOC_ID1	DOC_ID1	INTEGER	SKELETONS id	
DOC_ID2	DOC_ID2	INTEGER	SKELETONS id	
SUPPRESS_OEVENTS	SUPPRESS_OEVENTS	INTEGER		
parent_child	parent_child	INTEGER		
id	ID	INTEGER		REQUIRED KEY
last_mod_dt	last_mod_dt	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
last_mod_by	last_mod_by	UUID	ca_contact	Specifies the UUID of the contact who last modified this record.
tenant	tenant	UUID	ca_tenant	Specifies the UUID of the tenant.

CI_DOC_TEMPLATES Object

The object details are as follows:

1. Associated Table: CI_DOC_TEMPLATES
2. Factories: default
3. REL_ATTR: id
4. Common Name: TEMPLATE_NAME
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	ID	INTEGER		REQUIRED KEY
Default Template Flag	IS_DEFAULT	INTEGER		
Predefined Flag	IS_PREDEFINED	INTEGER		
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
Template HTML	PAGE_HTML	STRING		
Template Name	TEMPLATE_NAME	STRING		

CI_DOC_TYPES Object

The object details are as follows:

1. Associated Table: CI_DOC_TYPES
2. Factories: default
3. REL_ATTR: id
4. Common Name: DOC_TYPE_TXT
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
Document Type	DOC_TYPE_TXT	STRING		
id	ID	INTEGER		KEY
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		

ci_managed_attribute Object

The object details are as follows:

1. Associated Table: ci_managed_attribute
2. Factories: default
3. REL_ATTR: id
4. Common Name: attribute_label
5. Function Group: ci
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		KEY
producer_id		LOCAL STRING 20		
persistent_id		LOCAL STRING 60		
attribute_name	attribute_name	STRING 128		Required
attribute_label	attribute_label	STRING 50		Required
description	description	STRING 255		
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt.id	
delete_flag	del	SREL	actbool.enum	
initial_status	initial_status	SREL	ci_planned_change_status.sym	
case_sensitive	case_sensitive	SREL	bool.enum	
attribute_type	attribute_type	INTEGER		
attribute_length	attribute_length	INTEGER		
srel_factory	srel_factory	STRING 26		
srel_rel_attr	srel_rel_attr	STRING 26		
srel_common_name_attr	srel_common_name_attr	STRING 26		

srel_show_dropdown	srel_show_dropdown	SREL	bool.enum	
mdr_name	N/A	LOCAL STRING		
mdr_class	N/A	LOCAL STRING		

ci_managed_chgstat Object

The object details are as follows:

1. Associated Table: ci_verification_policy_act
2. Factories: default
3. REL_ATTR: code
4. Common Name: persistent_id
5. Function Group: ci
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		KEY
producer_id		LOCAL STRING 20		
persistent_id		LOCAL STRING 60		
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt.id	
delete_flag	del	SREL	actbool.enum	Required
code				Required
can_edit_criteria	can_edit_criteria	SREL	bool.enum	Required
verification_active	verification_active	SREL	bool.enum	Required
is_implementation	is_implementation	SREL	bool.enum	Required
autopromote_chg	autopromote_chg	SREL	bool.enum	Required
show_override_buttons	show_override_buttons	SREL	bool.enum	Required
mdr_name	N/A	LOCAL STRING		
mdr_class	N/A	LOCAL STRING		

ci_planned_change Object

The object details are as follows:

1. Associated Table: ci_planned_change
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: persistent_id
5. Function Group: ci
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		KEY
producer_id		LOCAL STRING 20		

persistent_id		LOCAL STRING 60		
attribute_name	attribute_name	SREL	ci_managed_attribute.attribute_name	Required
attribute_value_original	attribute_value_original	STRING 255		
attribute_value_planned	attribute_value_planned	STRING 255		
attribute_value_discovered	attribute_value_discovered	STRING 255		
chg	chg	SREL	chg.id	Required
ci	ci	SREL	nr.id	
delete_flag	del	SREL	actbool.enum	
description	description	STRING 255		
status	status	SREL	ci_planned_change_status.sym	
incident	incident	in.persistent_id		
last_verification_policy	last_verification_policy	SREL	ci_verification_policy_id	
ci_twa_ci	ci_twa_ci	SREL	ci_twa_ci.id	
ci_twa_relation	ci_twa_relation	SREL		
verification_msg	verification_msg	STRING 255		
attribute_value_internal	attribute_value_internal	STRING 255		
attribute_value_disc_internal	attribute_value_disc_internal	STRING 255		
attribute_value_orig_internal	attribute_value_orig_internal	STRING 255		
mdr_name	N/A	LOCAL STRING		
mdr_class	N/A	LOCAL STRING		
attribute_value_planned_sp	N/A	LOCAL SREL	cnt.id	
attribute_value_planned_nonsp	N/A	LOCAL SREL	org.id	
allow_update	N/A	LOCAL INTEGER		
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt.id	

ci_planned_change_status Object

The object details are as follows:

1. Associated Table: ci_planned_change_status
2. Factories: default
3. REL_ATTR: sym
4. Common Name: name
5. Function Group: admin

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		KEY
producer_id		LOCAL STRING 20		
persistent_id		LOCAL STRING 60		
name	name	STRING 128		
sym	sym	STRING 10		Required
description	description	STRING 255		Required
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt.id	
delete_flag	del	SREL	actbool.enum	
isinitial	isinitial	INTEGER	bool.enum	
isfinal	isfinal	INTEGER	bool.enum	
isselectable	isselectable	INTEGER	bool.enum	

CI_PRIORITIES Object

The object details are as follows:

1. Associated Table: CI_PRIORITIES
2. Factories: default
3. REL_ATTR: id
4. Common Name: PRIORITY
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	ID	INTEGER		KEY
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
Workflow Priority	PRIORITY	STRING		

CI_STATUSES Object

The object details are as follows:

1. Associated Table: CI_STATUSES
2. Factories: default
3. REL_ATTR: id
4. Common Name: STATUS
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	ID	INTEGER		KEY
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		

Predefined Flag	PREDEFINED	INTEGER		
Status Name	STATUS	STRING		
Status Description	STATUS_DESCRIPTION	STRING		
Status Order	STATUS_ORDER	INTEGER		

ci_verification_policy Object

The object details are as follows:

1. Associated Table: ci_verification_policy
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: ci
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		KEY
producer_id		LOCAL STRING 20		
persistent_id		LOCAL STRING 60		
attribute_name	attribute_name	STRING 128		Required
ci_name_pattern	ci_name_pattern	STRING 255		
class_pattern	class_pattern	STRING 255		
delete_flag	del	SREL	actbool.enum	
description	description	STRING 255		
mdr_name_pattern	mdr_name_pattern	STRING 255		
mdr_class_pattern	mdr_class_pattern	STRING 255		
rolename_pattern	rolename_pattern	STRING 255		Required
priority	priority	SREL	pri.enum	
sequence	sequence	INTEGER		Required
service_type	service_type	SREL	no_contract_sdsc.code	
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt.id	
location_pattern	location_pattern	STRING 255		
isrogue_insert	isrogue_insert	SREL	bool.enum	Required
isvariance	isvariance	SREL	bool.enum	Required
isrogue_update	isrogue_update	SREL	bool.enum	Required
isnotverifiable	isnotverifiable	SREL	bool.enum	Required
action	action	SREL	ci_verification_policy_act.sym	Required
write_twa	write_twa	SREL	ci_verification_twa_act.id	Required
write_incident	write_incident	SREL	bool.enum	Required

incident_template	incident_template	SREL	cr_tpl.template_name	
autoclose_incident	autoclose_incident	SREL	bool.enum	Required
log_only_mode	log_only_mode	SREL	bool.enum	Required
start_date	start_date	DATE		
end_date	end_date	DATE		
current_date	N/A	LOCAL DATE		
mdr_name	N/A	LOCAL STRING		
mdr_class	N/A	LOCAL STRING		

ci_verification_policy_act Object

The object details are as follows:

1. Associated Table: ci_verification_policy_act
2. Factories: default
3. REL_ATTR: sym
4. Common Name: name
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		KEY
producer_id		LOCAL STRING 20		
persistent_id		LOCAL STRING 60		
description	description	STRING 255		
name	name	STRING 50		Required
sym	sym	STRING 30		Required
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt.id	

ci_verification_twa_act Object

The object details are as follows:

1. Associated Table: ci_verification_twa_act
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: ci
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		KEY
producer_id		LOCAL STRING 20		
persistent_id		LOCAL STRING 60		

description	description	STRING 255		
name	name	STRING 50		Required
sym	sym	STRING 32		Required
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt.id	

CI_WF_TEMPLATES Object

The object details are as follows:

1. Associated Table: CI_WF_TEMPLATES
2. Factories: default
3. REL_ATTR: id
4. Common Name: WF_NAME
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	ID	INTEGER		KEY
Default Flag	IS_DEFAULT	INTEGER		
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
Workflow Template Description	WF_DESCRIPTION	STRING		
Workflow Template Name	WF_NAME	STRING		

Call Request (cr) Object

The object details for cr Object are as follows:

- **Associated Table**: Call_Req
- **Factories**: default
- **REL_ATTR**: persistent_id
- **Common Name**: ref_num
- **Function Group**: call_mgr
- **REST Operations**: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
active	active_flag	INTEGER	bool_tab enum	REQUIRED
active_prev	LOCAL	SREL	bool.enum	
affected_resource	affected_rc	UUID	ca_owned_resource uuid	
assignee	assignee	UUID		
assignee_prev	LOCAL	SREL	agt.id	
call_back_date	call_back_date	LOCAL_TIME		
call_back_flag	call_back_flag	INTEGER		

category	category	STRING	prob_ctg persid	
category_prev	LOCAL	SREL	pcat.persistent_id	
caused_by_chg	caused_by_chg	SREL	chg.id	
change	change	INTEGER	chg id	
charge_back_id	charge_back_id	STRING		
close_date	close_date	LOCAL_TIME		
cr_tticket	cr_tticket	INTEGER		
created_via	created_via	INTEGER	interface id	
customer	customer	UUID	ca_contact UUID	REQUIRED
event_token	event_token	STRING		
external_system_ticket		STRING (4000)		
extern_ref	extern_ref	STRING		
group	group_id	UUID		
group_prev	LOCAL	SREL	grp.id	
id	id	INTEGER		UNIQUE REQUIRED KEY
impact	impact	INTEGER	impact enum	
impact_prev	LOCAL	SREL	imp.enum	
incident_priority	incident_priority	INTEGER		
incorrectly_assigned		SREL	actbool	
last_act_id	last_act_id	STRING		
last_mod_dt	last_mod_dt	LOCAL_TIME		
log_agent	log_agent	UUID	ca_contact_UUID	REQUIRED
macro_predicted violation	macro_predicted_viol	INTEGER		
major_incident		SREL	actbool	
open_date	open_date	LOCAL_TIME	org	
orig_user_admin_org		SREL	org	
orig_user_cost_center		SREL	cost_cntr	
orig_user_dept		SREL	dept	
orig_user_organization		SREL	org	
outage_detail_what		STRING(4000)		
outage_detail_who		STRING(4000)		
outage_detail_why		STRING(4000)		
outage_end_time	outage_end_time	LOCAL_TIME		
outage_type		SREL	outage_type	
parent	parent	STRING	call_req persid	
pct_service_restored		INTEGER		
persistent_id	persid	STRING		
predicted_sla_violation	predicted_sla_viol	INTEGER		
priority	priority	INTEGER	pri enum	REQUIRED
priority_prev	LOCAL	SREL	pri.enum	
Problem	problem	STRING		

ref_num	ref_num	STRING		UNIQUE REQUIRED KEY
remote_control_used		SREL	actbool	
requested_by		SREL	cnt	
resolvable_at_lower		SREL	actbool	
resolve_date	resolve_date	LOCAL TIME		
return_to_service		SREL	actbool	
rootcause	rootcause	INTEGER	rootcause id	
extern_token	sched_token	STRING		
severity	severity	INTEGER	severity enum	
severity_prev	LOCAL	SREL	sev.enum	
sla_violation	sla_violation	INTEGER		
base_template	solution	STRING	call_req persid	
status	status	STRING	cr_stat code	
status_prev	LOCAL	SREL	crs.code	
string1	string1	STRING		
string2	string2	STRING		
string3	string3	STRING		
string4	string4	STRING		
string5	string5	STRING		
string6	string6	STRING		
children		QREL	cr	
sla_events		QREL	tev	
attached_slas		BREL	attached_sla_mapped_cr	
add_sla_persids	add_sla_persids	STRING		
site_sla_attrs	site_ sla_attrs	STRING		
audit_userid	audit_userid	SREL	cnt.id	
fldchange_log	fldchange_log	STRING		
affected_service	affected_service	SREL	nr	
init_urg		LOCAL INTEGER		
init_imp		LOCAL INTEGER		
cap_reason		LOCAL STRING		
auto_EGERurg		LOCAL INTEGER		
auto_imp		LOCAL INTEGER		
man_urg		LOCAL INTEGER		
man_imp		LOCAL INTEGER		
summary	summary	STRING		
support_lev	support_lev	STRING	srv_desc_code	
symptom_code		SREL	symptom_code	
template_name	template_name	STRING	cr_template template_name	
time_spent_sum	time_spent_sum	DURATION		

type	type	STRING	crt code	
urgency	urgency	INTEGER	urgency enum	
urgency_prev	LOCAL	SREL	urg.enum	
target_times	target_times	BREL	tgt_time_mapped_cr	
target_start_last	target_start_last	DATE		
target_hold_last	target_hold_last	DATE		
target_resolved_last	target_resolved_last	DATE		
target_resolved_count	target_resolved_count	INTEGER		
target_closed_last	target_closed_last	DATE		
target_closed_count	target_closed_count	INTEGER		
close_date_prev	close_date_prev	LOCAL DATE		
resolve_date_prev	resolve_date_prev	LOCAL DATE		
target_hold_count_prev	resolve_date_prev	INTEGER		
target_resolved_count_prev	target_hold_count_prev	LOCAL INTEGER		
target_closed_count_prev	target_closed_count_prev	LOCAL INTEGER	tgt_time_mapped_iss	

cr_prp Object

cr_prp Object

The object details are as follows:

- Associated Table: Req_Property
- Factories: default
- REL_ATTR: id
- Common Name: label
- Function Group: call_mgr
- REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	nvarchar (240)		
error_msg	error_msg	STRING		
id	id	INTEGER		primary key of this table
label	label	nvarchar (80)		REQUIRED
last_mod_by	last_mod_by	byte(16)	ca_contact uuid	
last_mod_dt	last_mod_dt	INTEGER		
owning_cr	owning_cr	nvarchar (30)	call_req persid	REQUIRED
owning_macro	owning_macro	BOP_REF_SR	macro	REQUIRED
persistent_id	persid	STRING		
required	required	INTEGER		REQUIRED
sample	sample	nvarchar (240)		

sequence	sequence	INTEGER		REQUIRED
value_description	value_description	STRING		
validation_rule	validation_rule	BOP_REF_STR	prpval_rule	REQUIRED
validation_type	validation_type	BOP_REF_STR	prpval_type	REQUIRED
value	value	nvarchar (240)		

cr_prptpl Object

The object details are as follows:

- **Associated Table**: Req_Property_Template
- **Factories**: default
- **REL_ATTR**: id
- **Common Name**: label
- **Function Group**: call_mgr_reference
- **REST Operations**: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
code	code	STRING		UNIQUE REQUIRED S_KEY
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
label	label	STRING		REQUIRED
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
owning_area	owning_area	STRING	prob_ctg_persid	REQUIRED
persistent_id	persid	STRING		
producer_id		LOCAL STRING 20		
required	required	INTEGER		REQUIRED
sample	sample	STRING		
sequence	sequence	INTEGER		REQUIRED
validation_rule	validation_rule	BOP_REF_STR	prpval_rule	REQUIRED

cr_tpl Object

The object details are as follows:

- **Associated Table**: Cr_Template
- **Factories**: default
- **REL_ATTR**: template_name
- **Common Name**: template_name
- **Function Group**: call_mgr_template
- **REST Operations**: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
delete_flags	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact_uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
quick_tmpl_type	quick_tmpl_type	INTEGER	quick_tpl_types enum	REQUIRED
template	template	STRING		call_req persid
template_class	template_class	STRING		
template_name	template_name	STRING		UNIQUE REQUIRED S_KEY

cr_trans Object

The object details are as follows:

- **Associated Tables:** cr_trans
- **Factories**: default
- **REL_ATTR**: id
- **Common Name**: condition_error
- **Function Group**: call_mgr_reference
- **REST Operations**: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
status	status	STRING	crs	Specifies the current ticket status.
new_status	new status	STRING	crs	Specifies the new ticket status.
t_type		INTEGER	transition type	Specifies the transition type.
is default		INTEGER		Default transition that appears when the Status field is empty. On new default: 0
must_comment		INTEGER		Comments required when using a transition. On new default: 0

delete_flag	del		actbool	Required. On new default: 0
condition			macro	Site condition macro to approve transition.
condition_error		STRING		Error message for site condition.
description		STRING		Description of this transition.
last_mod_by			cnt	On new default user; on CI set user
last_mod_dt		DATE		On CI set user now

crs Object

The object details are as follows:

- Associated Table: Cr_Status
- Factories: default
- REL_ATTR: code
- Common Name: sym
- Function Group: call_mgr_reference
- REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
active	active	INTEGER		
code	code	STRING		UNIQUE REQUIRED S_KEY
cr_flag	cr_flag	INTEGER		
delete flag	del	INTEGER		
hold	hold	INTEGER		
id	id	INTEGER		UNIQUE REQUIRED KEY
in_flag	in_flag	INTEGER		
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
pr_flag	pr_flag	INTEGER		
resolved	resolved	INTEGER		
sym	sym	STRING		REQUIRED

crsol Object

The object details are as follows:

- **Associated Table**: Call_Solution
- **Factories**: default
- **REL_ATTR**: persistent_id
- **Common Name**: sym
- **Function Group**: call_mgr_reference
- **REST Operations**: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flag
description	description	STRING		
cr_count	cr_count	INTEGER		
cr_flag	cr_flag	INTEGER		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
in_flag	in_flag	INTEGER		
last_mod_dt	last_mod_dt	LOCAL_TIME		
desc	nx_desc	STRING		
persistent_id	persid	STRING		
pr_flag	pr_flag	INTEGER		
solution_approved	sapproved	INTEGER	bool_tab enum	
solution_name	sname	STRING		
solution	solution	STRING		
sym	sym	STRING		REQUIRED S_KEY
tcode	tcode	INTEGER		

crsq Object

The object details are as follows:

- **Associated Table**: Cr_Stored_Queries
- **Factories**: default, sqchg, sqcr, sqiss
- **REL_ATTR**: code
- **Common Name**: label
- **Function Group**: stored_queries
- **REST Operations**: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
code	code	STRING		UNIQUE REQUIRED S_KEY
count_url	count_url	STRING		
where_clause	criteria	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY

label	label	STRING		
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
obj_type	obj_type	STRING		
persistent_id	persid	STRING		

crt Object

The object details are as follows:

- **Associated Table**: Call_Req_Type
- **Factories**: default
- **REL_ATTR**: code
- **Common Name**: sym
- **Function Group**: call_mgr_reference
- **REST Operations**: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
code	code	nvarchar (10)		UNIQUE REQUIRED S_KEY
delete_flag	del	INTEGER	actbool enum	REQUIRED
display_name	display_name	nvarchar 30		
id	id	INTEGER		UNIQUE REQUIRED KEY
description	nx_desc	nvarchar 30		
persistent_id	persid	nvarchar 30		
sym	sym	nvarchar 30		REQUIRED

ctab Object

The object details are as follows:

- **Associated Table**: Controlled_Table
- **Factories**: default
- **REL_ATTR**: id
- **Common Name**: sym
- **Function Group**: admin
- **REST Operations**: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool_enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY

desc	nx_desc	nvarchar (40)		
obj_name	obj_name	nvarchar 30		
persistent_id	persid	nvarchar 30		
sym	sym	nvarchar 30		UNIQUE REQUIRED S_KEY

ctimer Object

The object details are as follows:

- **Associated Table**: Cr_Call_Timers
- **Factories**: default
- **REL_ATTR**: persistent_id
- **Common Name:** color
- **Function Group**: admin
- **REST Operations**: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
beep	beep	INTEGER		
color	color	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
text	text	STRING		
threshold	threshold	DURATION		REQUIRED

ctp Object

The object details are as follows:

- Associated Table: ca_contact_type
- Factories: default
- REL_ATTR: id
- Common Name: sym
- Function Group: reference
- REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
creation_date	creation_date	LOCAL_TIME		
creation_user	creation_user	STRING		
delete_time	delete_time	LOCAL_TIME		

description	description	STRING		
exclude_registration	exclude_registration	integer		
id	id	INTEGER		UNIQUE REQUIRED KEY
delete_flag	inactive	integer	actbool enum	
last_mod_dt	last_update_date	LOCAL TIME		
last_mod_by	last_update_user	STRING		
sym	name	STRING		
user_uuid	user_uuid	UUID	ca_contact uuid	
version_number	version_number	integer		
view_internal	view_inernal	integer		

EBR

This article contains the following topics:

EBR_ACRONYMS Object

The object details are as follows:

1. Associated Table: EBR_ACRONYMS
2. Factories: default
3. REL_ATTR: id
4. Common Name: ACRONYM
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
Acronym	ACRONYM	STRING		
id	ID	INTEGER		REQUIRED KEY
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		

EBR_FULLTEXT Object

The object details are as follows:

1. Associated Table: EBR_FULLTEXT
2. Factories: default
3. REL_ATTR: id
4. Common Name: FULL_WORD
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
DOC_TYPE	DOC_TYPE	INTEGER		

ENTITY_ID	ENTITY_ID	INTEGER		
FULL_WORD	FULL_WORD	STRING		
FULL_WORD_REVERSE	FULL_WORD_ REVERSE	STRING		
id	ID	INTEGER		REQUIRED KEY
PERMISSION_ INDEX_ID	PERMISSION_ INDEX_ID	INTEGER		
PRODUCT	PRODUCT	STRING		
SHORT_WORD	SHORT_WORD	STRING		
TABLE_ID	TABLE_ID	INTEGER		
WORD_COUNT	WORD_COUNT	INTEGER		
WORD_COUNT_PROBLEM	WORD_COUNT_ PROBLEM	INTEGER		
WORD_COUNT_RESOLUTION	WORD_COUNT_ RESOLUTION	INTEGER		
WORD_COUNT_SUMMARY	WORD_COUNT_ SUMMARY	INTEGER		
WORD_COUNT_TITLE	WORD_COUNT_ TITLE	INTEGER		
WORD_IDF	WORD_IDF	INTEGER		
WORD_ORDER	WORD_ORDER	INTEGER		
WORD_TYPE	WORD_TYPE	INTEGER		

EBR_FULLTEXT_ADM Object

The object details are as follows:

1. Associated Table: EBR_FULLTEXT_ADM
2. Factories: default
3. REL_ATTR: id
4. Common Name: FULL_WORD
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
DOC_TYPE	DOC_TYPE	INTEGER		
ENTITY_ID	ENTITY_ID	INTEGER		
FULL_WORD	FULL_WORD	STRING		
FULL_WORD_REVERSE	FULL_WORD_ REVERSE	STRING		
id	ID	INTEGER		REQUIRED KEY
PERMISSION_ INDEX_ID	PERMISSION_ INDEX_ID	INTEGER		
PRODUCT	PRODUCT	STRING		
SHORT_WORD	SHORT_WORD	STRING		

TABLE_ID	TABLE_ID	INTEGER		
WORD_COUNT	WORD_COUNT	INTEGER		
WORD_COUNT_PROBLEM	WORD_COUNT_ PROBLEM	INTEGER		
WORD_COUNT_RESOLUTION	WORD_COUNT_ RESOLUTION	INTEGER		
WORD_COUNT_SUMMARY	WORD_COUNT_ SUMMARY	INTEGER		
WORD_COUNT_TITLE	WORD_COUNT_ TITLE	INTEGER		
WORD_IDF	WORD_IDF	INTEGER		
WORD_ORDER	WORD_ORDER	INTEGER		
WORD_TYPE	WORD_TYPE	INTEGER		

EBR_FULLTEXT_SD Object

The object details are as follows:

1. Associated Table: EBR_FULLTEXT_SD
2. Factories: default
3. REL_ATTR: id
4. Common Name: FULL_WORD
5. Function Group:

Attribute	DB Field	Data Type	SREL References	Flags
DOC_TYPE	DOC_TYPE	INTEGER		
ENTITY_ID	ENTITY_ID	INTEGER		
FULL_WORD	FULL_WORD	STRING		
FULL_WORD_ REVERSE	FULL_WORD_ REVERSE	STRING		
id	ID	INTEGER		REQUIRED KEY
PERMISSION_ INDEX_ID	PERMISSION_INDEX_ID	INTEGER		
PRODUCT	PRODUCT	STRING		
SHORT_WORD	SHORT_ WORD	STRING		
TABLE_ID	TABLE_ID	INTEGER		
WORD_COUNT	WORD_ COUNT	INTEGER		
WORD_COUNT_PROBLEM	WORD_ COUNT_ PROBLEM	INTEGER		
WORD_COUNT_RESOLUTION	WORD_COUNT_ RESOLUTION	INTEGER		
WORD_COUNT_SUMMARY	WORD_COUNT_ SUMMARY	INTEGER		
WORD_COUNT_TITLE	WORD_COUNT_ TITLE	INTEGER		
WORD_IDF	WORD_IDF	INTEGER		

WORD_ORDER	WORD_ORDER	INTEGER		
WORD_TYPE	WORD_TYPE	INTEGER		

EBR_FULLTEXT_SD_ADM Object

The object details are as follows:

1. Associated Table: EBR_FULLTEXT_SD_ADM
2. Factories: default
3. REL_ATTR: id
4. Common Name: FULL_WORD
5. Function Group:

Attribute	DB Field	Data Type	SREL References	Flags
DOC_TYPE	DOC_TYPE	INTEGER		
ENTITY_ID	ENTITY_ID	INTEGER		
FULL_WORD	FULL_WORD	STRING		
FULL_WORD_ REVERSE	FULL_WORD_ REVERSE	STRING		
id	ID	INTEGER		REQUIRED KEY
PERMISSION_ INDEX_ID	PERMISSION_ INDEX_ID	INTEGER		
PRODUCT	PRODUCT	STRING		
SHORT_WORD	SHORT_WORD	STRING		
TABLE_ID	TABLE_ID	INTEGER		
WORD_COUNT	WORD_COUNT	INTEGER		
WORD_COUNT_PROBL EM	WORD_COUNT_PROBL EM	INTEGER		
WORD_COUNT_RESOL UTION	WORD_COUNT_RESOL UTION	INTEGER		
WORD_COUNT_SUMM ARY	WORD_COUNT_SUMM ARY	INTEGER		
WORD_COUNT_TITLE	WORD_COUNT_ TITLE	INTEGER		
WORD_IDF	WORD_IDF	INTEGER		
WORD_ORDER	WORD_ORDER	INTEGER		
WORD_TYPE	WORD_TYPE	INTEGER		

EBR_INDEXING_QUEUE Object

The object details are as follows:

1. Associated Table: EBR_INDEXING_QUEUE

2. Factories: default
3. REL_ATTR: id
4. Common Name: OBJ_PERSID
5. Function Group:

Attribute	DB Field	Data Type	SREL References	Flags
ACTION	ACTION	INTEGER		
ACTION_DATE	ACTION_DATE	DATE		
id	ID	INTEGER		REQUIRED KEY
INDEXED	INDEXED	INTEGER		
OBJ_PERSID	OBJ_PERSID	STRING		
PRIORITY	PRIORITY	INTEGER		
TEXT	TEXT	STRING		

EBR_KEYWORDS Object

The object details are as follows:

1. Associated Table: EBR_KEYWORDS
2. Factories: default
3. REL_ATTR: id
4. Common Name: FULL_WORD
5. Function Group:

Attribute	DB Field	Data Type	SREL References	Flags
ENTITY_ID	ENTITY_ID	INTEGER		
EXT_TABLE_ID	EXT_TABLE_ID	INTEGER		
FULL_WORD	FULL_WORD	STRING		
id	ID	INTEGER		REQUIRED KEY

EBR_LOG Object

The object details are as follows:

1. Associated Table: EBR_LOG
2. Factories: default
3. REL_ATTR: id
4. Common Name: SEARCH_TEXT
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
ASKED_DATE	ASKED_DATE	LOCAL_TIME		

BEST_IDS	BEST_IDS	STRING		
EXTERNAL_ID	EXTERNAL_ID	STRING		
FILTER_DATA	FILTER_DATA	STRING		
FUZZINESS	FUZZINESS	INTEGER		
id	ID	INTEGER		KEY
KEYWORDS	KEYWORDS	STRING		
MATCH_TYPE	MATCH_TYPE	INTEGER		
METHOD_ PERFORMANCE	METHOD_ PERFORMANCE	INTEGER		
METHOD_TYPE	METHOD_TYPE	INTEGER		
NUM_MATCHES	NUM_MATCHES	INTEGER		
ORDER_DIRECTION	ORDER_DIRECTION	INTEGER		
PRIMARY_ORDER	PRIMARY_ORDER	STRING		
ROWS_FOUND	ROWS_FOUND	INTEGER		
ROWS_TO_FETCH	ROWS_TO_FETCH	INTEGER		
SEARCH_IN	SEARCH_IN	INTEGER		
SEARCH_QUALITY	SEARCH_QUALITY	INTEGER		
SEARCH_TEXT	SEARCH_TEXT	STRING		
SEARCH_TYPE	SEARCH_TYPE	INTEGER		
SECONDARY_ORDER	SECONDARY_ORDER	INTEGER		
SESSION_ID	SESSION_ID	INTEGER		
SQL_TEXT	SQL_TEXT	STRING		
TOP_MATCH_ID	TOP_MATCH_ID	INTEGER		
UNIQUE_WORD_ COUNT	UNIQUE_WORD_ COUNT	INTEGER		
USER_ID	USER_ID	STRING		
WORD_COUNT	WORD_COUNT	INTEGER		

EBR_METRICS Object

The object details are as follows:

1. Associated Table: EBR_METRICS
2. Factories: default
3. REL_ATTR: id
4. Common Name: METRIC
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
Comments	COMMENTS	STRING		

id	ID	INTEGER		REQUIRED KEY
Metric	METRIC	STRING		
Weight	WEIGHT	REAL		

EBR_NOISE_WORDS Object

The object details are as follows:

1. Associated Table: EBR_NOISE_WORDS
2. Factories: default
3. REL_ATTR: id
4. Common Name: NOISE_WORD
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	ID	INTEGER		REQUIRED KEY
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
Noise Word	NOISE_WORD	STRING		

EBR_PATTERNS Object

The object details are as follows:

1. Associated Table: EBR_PATTERNS
2. Factories: default
3. REL_ATTR: id
4. Common Name: PATTERN_NAME
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	ID	INTEGER		REQUIRED KEY
PATTERN_DEFAULT	PATTERN_DEFAULT	STRING		
PATTERN_NAME	PATTERN_NAME	STRING		
PATTERN_VALUE	PATTERN_VALUE	STRING		
PATTERN_VALUE_ADM	PATTERN_VALUE_ADM	STRING		

EBR_PREFIXES Object

The object details are as follows:

1. Associated Table: EBR_PREFIXES

2. Factories: default
3. REL_ATTR: id
4. Common Name: PREFIX
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	ID	INTEGER		REQUIRED KEY
Prefix	PREFIX	STRING		

EBR_PROPERTIES Object

The object details are as follows:

1. Associated Table: EBR_PROPERTIES
2. Factories: default
3. REL_ATTR: id
4. Common Name: PROPERTY_NAME
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	ID	INTEGER		REQUIRED KEY
PROPERTY_ADMIN	PROPERTY_ADMIN	INTEGER		
PROPERTY_DEFAULT	PROPERTY_DEFAULT	STRING		
PROPERTY_NAME	PROPERTY_NAME	STRING		S_KEY
PROPERTY_TYPE	PROPERTY_TYPE	STRING		
PROPERTY_VALUE	PROPERTY_VALUE	STRING		
PROPERTY_VALUE_ADM	PROPERTY_VALUE_ADM	STRING		

EBR_SUBSTITS Object

The object details are as follows:

1. Associated Table: EBR_SUBSTITS
2. Factories: default
3. REL_ATTR: id
4. Common Name: SYMBOL1
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	ID	INTEGER		REQUIRED KEY

SYMBOL1	SYMBOL1	STRING		
SYMBOL2	SYMBOL2	STRING		

EBR_SUFFIXES Object

The object details are as follows:

1. Associated Table: EBR_SUFFIXES
2. Factories: default
3. REL_ATTR: id
4. Common Name: SUFFIX
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	ID	INTEGER		REQUIRED KEY
Suffix	SUFFIX	STRING		

EBR_SYNONYMS Object

The object details are as follows:

1. Associated Table: EBR_SYNONYMS
2. Factories: default
3. REL_ATTR: id
4. Common Name: KEYWORD1
5. Function Group:
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	ID	INTEGER		REQUIRED KEY
Keyword 1	KEYWORD1	STRING		
Keyword 2	KEYWORD2	STRING		
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		

EBR_SYNONYMS_ADM Object

The object details are as follows:

1. Associated Table: EBR_SYNONYMS_ADM
2. Factories: default
3. REL_ATTR: id
4. Common Name: KEYWORD1
5. Function Group:

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	ID	INTEGER		REQUIRED KEY
Keyword 1	KEYWORD1	STRING		
Keyword 2	KEYWORD2	STRING		
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		

Event Objects

This topic contains the following information:

event_log Object

The object details are as follows:

1. Associated Table: event_log
2. Factories: default
3. REL_ATTR: id
4. Common Name: sd_obj_type
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
event	event	INTEGER	event_type id	
id	id	INTEGER		UNIQUE REQUIRED KEY
kd	kd_id	INTEGER	SKELETONS id	
log_time	log_time	LOCAL_TIME		
millitime	millitime	INTEGER		
numdata1	numdata1	INTEGER		
numdata2	numdata2	INTEGER		
sd_obj_id	sd_obj_id	INTEGER		
sd_obj_type	sd_obj_type	STRING		
session	session	INTEGER	session_log id	
textdata1	textdata1	STRING		
textdata2	textdata2	STRING		

event_type Object

The object details are as follows:

1. Associated Table: event_type
2. Factories: default
3. REL_ATTR: id
4. Common Name: description

5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
code	code	STRING		UNIQUE REQUIRED S_KEY
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		

evt Object

The object details are as follows:

1. Associated Table: Events
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
condition	condition	STRING	splmac persid	
delete_flag	del	INTEGER	actbool enum	REQUIRED
delay_time	delay_time	DURATION		REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_dt	last_mod_dt	LOCAL_TIME		
modulo_time	modulo_time	DURATION		REQUIRED
obj_type	obj_type	STRING		
on_done_flag	on_done_flag	INTEGER		REQUIRED
persistent_id	persid	STRING		
sym	sym	STRING		REQUIRED unique S_KEY
urgency	urgency	INTEGER		
user_settime	user_settime	INTEGER		REQUIRED
user_smag	user_smag	STRING		
violate_on_ false	violate_on_ false	INTEGER		
violate_on_ true	violate_on_ true	INTEGER		
work_shift	work_shift	STRING	bpwshft persid	

evtdly Object

The object details are as follows:

1. Associated Table: Event_Delay
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: group_name
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
act_delay	act_delay	DURATION		
cancel_time	cancel_time	LOCAL_TIME		
create_time	create_time	LOCAL_TIME		
delay_type	delay_type	INTEGER	evtdlytp enum	
eff_delay	eff_delay	DURATION		
group_name	group_name	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_dt	last_mod_dt	LOCAL_TIME		
obj_id	obj_id	STRING		REQUIRED S_KEY
persistent_id	persid	STRING		
start_time	start_time	LOCAL_TIME		
start_userid	start_userid	UUID	ca_contact uuid	
status_flag	status_flag	INTEGER		
stop_time	stop_time	LOCAL_TIME		
stop_userid	stop_userid	UUID	ca_contact uuid	
support_lev	support_lev	STRING	srv_desc code	

evtdlytp Object

The object details are as follows:

1. Associated Table: Event_Delay_Type
2. Factories: default
3. REL_ATTR: enum
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
enum	enum	INTEGER		REQUIRED

id	id	INTEGER		UNIQUE REQUIRED KEY
description	nx_desc	STRING		
sym	sym	STRING		UNIQUE REQUIRED S_KEY

Global Objects

This article contains the following topics:

g_chg_ext Object

The object details are as follows:

1. Associated Table: Global_Change_Extension
2. Factories: default
3. REL_ATTR: id
4. Common Name: chg_ref_num
5. Function Group: change_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
active	active_flag	INTEGER	actrbool enum	REQUIRED
affected_contact	affected_contact	UUID	ca_contact uuid	REQUIRED
assignee	assignee	UUID	ca_contact uuid	
category	category	STRING		
chg_ref_num	chg_ref_num	STRING		REQUIRED
close_date	close_date	LOCAL_TIME		
global_queue_id	global_queue_id	INTEGER	g_queue_names id	
group	group_id	UUID		
id	id	INTEGER		UNIQUE REQUIRED KEY
impact	impact	INTEGER	impact enum	
last_mod_dt	last_mod_dt	LOCAL_TIME		REQUIRED
open_date	open_date	LOCAL_TIME		REQUIRED
priority	priority	INTEGER	pri enum	REQUIRED
remote_id	remote_id	INTEGER		REQUIRED S_KEY
requestor	requestor	UUID	ca_contact uuid	REQUIRED
status	status	STRING	chgstat code	REQUIRED
summary	summary	STRING		

g_chg_queue Object

The object details are as follows:

1. Associated Table: Global_Change_Queue
2. Factories: default
3. REL_ATTR: id
4. Common Name: chg_ref_num
5. Function Group: change_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
active	active_flag	INTEGER	actrbool enum	REQUIRED
affected_contact	affected_contact	UUID		REQUIRED
assignee	assignee	UUID		
category	category	STRING		
chg_ref_num	chg_ref_num	STRING		REQUIRED
close_date	close_date	LOCAL_TIME		
global_queue_id	global_queue_id	INTEGER	g_queue_names id	
group	group_id	UUID		
id	id	INTEGER		UNIQUE REQUIRED KEY
impact	impact	INTEGER	impact enum	
last_mod_dt	last_mod_dt	LOCAL_TIME		REQUIRED
open_date	open_date	LOCAL_TIME		REQUIRED
priority	priority	INTEGER	pri enum	REQUIRED
remote_id	remote_id	INTEGER		REQUIRED S_KEY
remote_sys_id	remote_sys_id	INTEGER	g_srvr remote_sys_id	REQUIRED S_KEY
requestor	requestor	UUID		REQUIRED
status	status	STRING	chgstat code	REQUIRED
summary	summary	STRING		

g_cnt Object

The object details are as follows:

1. Associated Table: Global_Contact
2. Factories: default
3. REL_ATTR: id
4. Common Name: combo_name
5. Function Group: multisite_reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
contact_num	contact_num	STRING		

delete_flag	del	INTEGER	actbool enum	REQUIRED
email_address	email_address	STRING		
first_name	first_name	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_dt	last_mod_dt	LOCAL_TIME		
last_name	last_name	STRING		
location	loc_id	UUID		
middle_name	middle_name	STRING		
organization	org_id	UUID		
pri_phone_number	pri_phone_number	STRING		
remote_id	remote_id	UUID		REQUIRED S_KEY
remote_sys_id	remote_sys_id	INTEGER	g_srvr remote_sys_id	REQUIRED S_KEY
userid	userid	STRING		

g_cr_ext Object

The object details are as follows:

1. Associated Table: Global_Request_Extension
2. Factories: default
3. REL_ATTR: id
4. Common Name: ref_num
5. Function Group: call_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
active	active_flag	INTEGER	actrbool enum	REQUIRED
assignee	assignee	UUID	ca_contact uuid	
category	category	STRING		
close_date	close_date	LOCAL_TIME		
customer	customer	UUID	ca_contact uuid	REQUIRED
global_queue_ id	global_queue_ id	INTEGER	g_queue_names id	
group	group_id	UUID		
id	id	INTEGER		UNIQUE REQUIRED KEY
impact	impact	INTEGER	impact enum	
last_mod_dt	last_mod_dt	LOCAL_TIME		REQUIRED
open_date	open_date	LOCAL_TIME		REQUIRED
priority	priority	INTEGER	pri enum	REQUIRED
ref_num	ref_num	STRING		REQUIRED
remote_id	remote_id	INTEGER		REQUIRED S_KEY

status	status	STRING	cr_stat code	REQUIRED
summary	summary	STRING		
type	type	STRING	crt code	

g_cr_queue Object

The object details are as follows:

1. Associated Table: Global_Request_Queue
2. Factories: default
3. REL_ATTR: id
4. Common Name: ref_num
5. Function Group: call_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
active	active_flag	INTEGER	actrbool enum	REQUIRED
assignee	assignee	UUID		
category	category	STRING		
close_date	close_date	LOCAL_TIME		
customer	customer	UUID		REQUIRED
global_queue_id	global_queue_id	INTEGER	g_queue_names id	
group	group_id	UUID		
id	id	INTEGER		UNIQUE REQUIRED KEY
impact	impact	INTEGER	impact enum	
last_mod_dt	last_mod_dt	LOCAL_TIME		REQUIRED
open_date	open_date	LOCAL_TIME		REQUIRED
priority	priority	INTEGER	pri enum	REQUIRED
ref_num	ref_num	STRING		REQUIRED
remote_id	remote_id	INTEGER		REQUIRED S_KEY
remote_sys_id	remote_sys_id	INTEGER	g_srvr remote_sys_id	REQUIRED S_KEY
status	status	STRING	cr_stat code	REQUIRED
summary	summary	STRING		
type	type	STRING	crt code	

g_iss_ext Object

The object details are as follows:

1. Associated Table: Global_Issue_Extension
2. Factories: default
3. REL_ATTR: id
4. Common Name: ref_num
5. Function Group: issue_mgr

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
active	active_flag	INTEGER	actrbool enum	REQUIRED
assignee	assignee	UUID	ca_contact uuid	
category	category	STRING		
close_date	close_date	LOCAL_TIME		
global_queue_ id	global_queue_ id	INTEGER	g_queue_names id	
group	group_id	UUID		
id	id	INTEGER		UNIQUE REQUIRED KEY
impact	impact	INTEGER	impact enum	
last_mod_dt	last_mod_dt	LOCAL_TIME		REQUIRED
open_date	open_date	LOCAL_TIME		REQUIRED
priority	priority	INTEGER	pri enum	REQUIRED
product	product	INTEGER	product id	
ref_num	ref_num	STRING		REQUIRED
remote_id	remote_id	INTEGER		REQUIRED S_KEY
requestor	requestor	UUID	ca_contact uuid	REQUIRED
status	status	STRING	issstat code	REQUIRED
summary	summary	STRING		

g_iss_queue Object

The object details are as follows:

1. Associated Table: Global_Issue_Queue
2. Factories: default
3. REL_ATTR: id
4. Common Name: ref_num
5. Function Group: issue_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
active	active_flag	INTEGER	actrbool enum	REQUIRED
assignee	assignee	UUID		
category	category	STRING		
close_date	close_date	LOCAL_TIME		
global_queue_id	global_queue_id	INTEGER	g_queue_names id	
group	group_id	UUID		
id	id	INTEGER		UNIQUE REQUIRED KEY
impact	impact	INTEGER	impact enum	

last_mod_dt	last_mod_dt	LOCAL_TIME		REQUIRED
open_date	open_date	LOCAL_TIME		REQUIRED
priority	priority	INTEGER	pri enum	REQUIRED
product	product	INTEGER		
ref_num	ref_num	STRING		REQUIRED
remote_id	remote_id	INTEGER		REQUIRED S_KEY
remote_sys_id	remote_sys_id	INTEGER	g_srvr remote_sys_id	REQUIRED S_KEY
requestor	requestor	UUID		REQUIRED
status	status	STRING	issstat code	REQUIRED
summary	summary	STRING		

g_loc Object

The object details are as follows:

1. Associated Table: Global_Location
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: inventory
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
del	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_dt	last_mod_dt	LOCAL_TIME		
loc_name	loc_name	STRING		
remote_id	remote_id	UUID		REQUIRED S_KEY
remote_sys_id	remote_sys_id	INTEGER	g_srvr remote_sys_id	REQUIRED S_KEY

g_org Object

The object details are as follows:

1. Associated Table: Global_Organization
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: inventory
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY

last_mod_dt	last_mod_dt	LOCAL_TIME		
name	org_name	STRING		
remote_id	remote_id	UUID		REQUIRED S_KEY
remote_sys_id	remote_sys_id	INTEGER	g_srvr remote_sys_id	REQUIRED S_KEY

g_prod Object

The object details are as follows:

1. Associated Table: Global_Product
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: multisite_reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_dt	last_mod_dt	LOCAL_TIME		
remote_id	remote_id	INTEGER		REQUIRED S_KEY
remote_sys_id	remote_sys_id	INTEGER	g_srvr remote_sys_id	REQUIRED S_KEY
sym	sym	STRING		

g_qname Object

The object details are as follows:

1. Associated Table: Global_Queue_Names
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: multisite_reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
sym	sym	STRING		UNIQUE REQUIRED

g_tblmap Object

The object details are as follows:

1. Associated Table: Global_Table_Map
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
map_definition	map_definition	STRING		REQUIRED
sym	sym	STRING		UNIQUE REQUIRED

g_srvrs Object

The object details are as follows:

1. Associated Table: Global_Servers
2. Factories: default
3. REL_ATTR: remote_sys_id
4. Common Name: sym
5. Function Group: multisite_admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
chg_prefix	chg_prefix	STRING		
cr_prefix	cr_prefix	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
global_name	global_name	STRING		UNIQUE REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
is_master	is_master	INTEGER	bool_tab enum	
iss_prefix	iss_prefix	STRING		
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
remote_sys_id	remote_sys_id	INTEGER		UNIQUE REQUIRED

slump_addr	slump_addr	STRING		
sym	sym	STRING		UNIQUE REQUIRED
web_protocol	web_protocol	STRING		
web_server	web_server	STRING		
web_server_port	web_server_port	STRING		
web_url	web_url	STRING		

g_tblrule Object

The object details are as follows:

1. Associated Table: Global_Table_Rule
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
addl_query	addl_query	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
last_sync_dt	last_sync_dt	LOCAL_TIME		
reoccur_interv	reoccur_interv	DURATION		
sched	sched	STRING	bpwshft persid	
sym	sym	STRING		UNIQUE REQUIRED S_KEY
table_map	table_map	INTEGER	g_tbl_map id	

Issue Objects

This article contains the following topics:

iss Object

The object details are as follows:

1. Associated Table: Issue
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: ref_num
5. Function Group: issue_mgr

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
actions	actions	STRING		
active_flag	active_flag	INTEGER		REQUIRED
active_prev	LOCAL	SREL	bool.enum	
actual_comp_date	actual_comp_date	LOCAL_TIME		
actual_cost	actual_cost	INTEGER		
actual_total_time	actual_total_time	DURATION		
affected_contact	affected_contact	UUID	ca_contact uuid	
assignee	assignee	UUID		
assignee_prev	LOCAL	SREL	agt.id	
backout_plan	backout_plan	STRING		
call_back_date	call_back_date	LOCAL_TIME		
call_back_flag	call_back_flag	INTEGER		
category	category	STRING	isscat code	
category_prev	LOCAL	SREL	pcat.persistent_id	
cawf_procid	cawf_procid	STRING		
close_date	close_date	LOCAL_TIME		
created_via	created_via	INTEGER	interface id	
effort	effort	STRING		
est_comp_date	est_comp_date	LOCAL_TIME		
est_cost	est_cost	INTEGER		
est_total_time	est_total_time	DURATION		
external_system_ticket		STRING		
flag1	flag1	INTEGER		
flag2	flag2	INTEGER		
flag3	flag3	INTEGER		
flag4	flag4	INTEGER		
flag5	flag5	INTEGER		
flag6	flag6	INTEGER		
group	group_id	UUID		
group_prev	LOCAL	SREL	grp.id	
id	id	INTEGER		UNIQUE REQUIRED KEY
impact	impact	INTEGER	impact enum	
impact_prev	LOCAL	SREL	imp.enum	
justification	justification	STRING		

last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
log_agent	log_agent	UUID	ca_contact uuid	REQUIRED
macro_predicted_violation	macro_predict_ viol	INTEGER		
need_by	need_by	LOCAL_TIME		
open_date	open_date	LOCAL_TIME		
organization	organization	UUID	ca_organization uuid	
orig_user_admin_org		SREL	org	
orig_user_cost_ center		SREL	cost_cntr	
orig_user_dept		SREL	dept	
orig_user_ organization		SREL	org	
parent	parent	STRING	issue persistent_id	
persistent_id	persid	STRING		
person_contacting	person_contacting	INTEGER	perscon id	
predicted_sla_ violation	predicted_sla_viol	INTEGER		
priority	priority	INTEGER	pri enum	REQUIRED
priority_prev	LOCAL	SREL	pri.enum	
product	product	INTEGER	product id	
ref_num	ref_num	STRING		UNIQUE REQUIRED S_KEY
reporting_method	reporting_method	INTEGER	repmeth id	
requestor	requestor	UUID	ca_contact uuid	REQUIRED
requested_by		UUID	ca_contact	
resolve_date	resolve_date	LOCAL_TIME		
rootcause	rootcause	INTEGER	rootcause id	
service_date	service_date	LOCAL_TIME		
service_num	service_num	STRING		
sla_violation	sla_violation	INTEGER		
start_date	start_date	LOCAL_TIME		
status	status	STRING	issstat code	
status_prev	LOCAL	SREL	crs.code	
string1	string1	STRING		
string2	string2	STRING		
string3	string3	STRING		
string4	string4	STRING		
string5	string5	STRING		
string6	string6	STRING		
summary	summary	STRING		

support_lev	support_lev	STRING	srv_desc code	
template_name	template_name	STRING	iss_template template_name	
type_of_contact	type_of_contact	INTEGER	toc id	
user1	user1	STRING		
user2	user2	STRING		
user3	user3	STRING		
target_times	target_times			
target_start_last	target_start_last	DATE		
target_hold_last	target_hold_last	DATE		
target_hold_count	target_hold_count	INTEGER		
target_resolved_ last	target_resolved_ last	DATE		
target_resolved_ count	target_resolved_ count	INTEGER		
target_closed_last	target_closed_last	DATE		
target_closed_ count	target_closed_ count	INTEGER		
close_date_prev	close_date_prev	LOCAL DATE		
resolve_date_prev	resolve_date_prev	LOCAL DATE		
target_hold_ count_prev	target_hold_ count_prev	LOCAL INTEGER		
target_resolved_ count_prev	target_resolved_ count_prev	LOCAL INTEGER		
target_closed_ count_prev	target_closed_ count_prev	LOCAL INTEGER		

iss_prp Object

The object details are as follows:

1. Associated Table: Issue_Property
2. Factories: default
3. REL_ATTR: id
4. Common Name: label
5. Function Group: issue_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
error_msg	error_msg	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
label	label	STRING		
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
owning_iss	owning_iss	STRING	issue persistent_id	REQUIRED

owning_macro	owning_macro	BOP_REF_STR	macro	REQUIRED
persistent_id	persid	STRING		
required	required	INTEGER	bool_tab enum	
sample	sample	STRING		
sequence	sequence	INTEGER		REQUIRED
value_description	value_description	STRING		
validation_rule	validation_rule	BOP_REF_STR	prpval_rule	REQUIRED
validation_type	validation_type	BOP_REF_STR	prpval_type	REQUIRED
value	value	STRING		

iss_tpl Object

The object details are as follows:

1. Associated Table: Iss_Template
2. Factories: default
3. REL_ATTR: template_name
4. Common Name: template_name
5. Function Group: issue_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
quick_tmpl_ type	quick_tmpl_ type	INTEGER	quick_tpl_ types enum	REQUIRED
template	template	STRING	issue persistent_id	
template_class	template_class	STRING		
template_name	template_name	STRING		UNIQUE REQUIRED S_KEY

iss_trans Object

The object details are as follows:

1. Associated Tables: iss_trans
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: issue_reference

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
status	status	STRING	crs	Specifies the current ticket status.
new_status	new status	STRING	crs	Specifies the new ticket status.
is default		INTEGER		Default transition that appears when the Status field is empty. On new default: 0
must_comment		INTEGER		Comment required when using a transition. On new default: 0
delete_flag	del		actbool	Required. On new default: 0
condition			macro	Site condition macro to approve transition.
condition_error		STRING		Error message for site condition.
description		STRING		Description of this transition.
last_mod_by			cnt	On new default user; on CI set user
last_mod_dt		DATE		On CI set user now

iss_wf Object

The object details are as follows:

1. Associated Table: Issue_Workflow_Task
2. Factories: default
3. REL_ATTR: id
4. Common Name: description
5. Function Group: issue_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
actual_duration	actual_duration	DURATION		
asset	asset	UUID	ca_owned_resource uuid	
assignee	assignee	UUID	ca_contact uuid	
completion_date	completion_date	LOCAL_TIME		
cost	cost	INTEGER		
creator	creator	UUID	ca_contact uuid	

date_created	date_created	LOCAL_TIME		
del	del	INTEGER		REQUIRED
done_by	done_by	UUID	ca_contact uuid	
est_completion_date	est_comp_date	LOCAL_TIME		
est_cost	est_cost	INTEGER		
est_duration	est_duration	DURATION		
group	group_id	UUID		
group_task	group_task	INTEGER		REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
object_id	object_id	STRING		REQUIRED
object_type	object_type	STRING		REQUIRED
persistent_id	persid	STRING		
sequence	sequence	INTEGER		REQUIRED
start_date	start_date	LOCAL_TIME		
status	status	STRING	tskstat code	
support_lev	support_lev	STRING	srv_desc code	
task	task	STRING	tskty code	REQUIRED
wf_template	wf_template	INTEGER	wftpl id	

issalg Object

The object details are as follows:

1. Associated Table: Issue_Act_Log
2. Factories: default
3. REL_ATTR: id
4. Common Name: description
5. Function Group: issue_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
action_desc	action_desc	STRING		
analyst	analyst	UUID	ca_contact uuid	
id	id	INTEGER		UNIQUE REQUIRED KEY
internal	internal	INTEGER		
issue_id	issue_id	STRING	issue persistent_id	
knowledge_ session	knowledge_ session	STRING		

knowledge_tool	knowledge_tool	STRING		
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
system_time	system_time	LOCAL_TIME		
time_spent	time_spent	DURATION		
time_stamp	time_stamp	LOCAL_TIME		
type	type	STRING	act_type code	

isscat Object

The object details are as follows:

1. Associated Table: Issue_Category
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
assignee	assignee	UUID		
auto_assign	auto_assign	INTEGER		
cawf_defid	cawf_defid	STRING		
children_ok	children_ok	INTEGER		REQUIRED
code	code	STRING		UNIQUE REQUIRED S_KEY
del	del	INTEGER		REQUIRED
group_id	group_id	UUID		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
organization	organization	UUID	ca_organization uuid	
owning_contract	owning_contract	INTEGER	svc_contract id	
persistent_id	persid	STRING		
ss_sym		STRING		
ss_include			bool	REQUIRED On new default: 0
schedule	schedule	INTEGER		
service_type	service_type	STRING	srv_desc code	
survey	survey	INTEGER	survey_tpl id	

sym	sym	STRING 1000		REQUIRED S_KEY

issstat Object

The object details are as follows:

1. Associated Table: Issue_Status
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
active	active	INTEGER		
code	code	STRING		UNIQUE REQUIRED S_KEY
del	del	INTEGER		REQUIRED
hold	hold	INTEGER		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
resolved	resolved	INTEGER		
sym	sym	STRING		REQUIRED

Knowledge Documents Object

This topic contains the following information:

KD Object

The object details are as follows:

1. Associated Table: SKELETONS
2. Factories: default
3. REL_ATTR: id
4. Common Name: TITLE
5. Function Group: kd
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
AVG_RATING	AVG_RATING	DOUBLE		
Document State	ACTIVE_STATE	INTEGER		

Document State Date	ACTIVE_STATE_ DATE	LOCAL_TIME		
Assignee ID	ASSIGNEE_ID	UUID	ca_contact uuid	
Author ID	AUTHOR_ID	UUID	ca_contact uuid	
FAQ Rating	BU_RESULT	REAL		
Created Via	CREATED_VIA	INTEGER		
Creation Date	CREATION_DATE	LOCAL_TIME		
Task ID	CURRENT_ACTION_ID	INTEGER	CI_ACTIONS id	
Custom 1	CUSTOM1	STRING		
Custom 2	CUSTOM2	STRING		
Custom 3	CUSTOM3	STRING		
Custom 4	CUSTOM4	STRING		
Custom 5	CUSTOM5	STRING		
Custom Num 1	CUSTOM_NUM1	REAL		
Custom Num 2	CUSTOM_NUM2	REAL		
Document Template ID	DOC_TEMPLATE_ ID	INTEGER	CI_DOC_ TEMPLATES id	
Document Type ID	DOC_TYPE_ID	INTEGER	CI_DOC_ TYPES id	
Document Version	DOC_VERSION	STRING		
Expiration Date	EXPIRATION_DATE	LOCAL_TIME		
EXPIRE_ NOTIFICATION_ SENT	EXPIRE_ NOTIFICATION_ SENT	INTEGER		
Decision Tree Root ID	EXT_DOC_ID	INTEGER		
ebr_search_text	ebr_search_text	STRING		
ebr_min_relevance	ebr_min_relevance	STRING		
ebr_fuzziness	ebr_fuzziness	STRING		
ebr_search_type	ebr_search_type	STRING		
ebr_match_type	ebr_match_type	STRING		
ebr_search_in	ebr_search_in	STRING		
ebr_primary_order	ebr_primary_order	STRING		
ebr_secondary_ order	ebr_secondary_ order	STRING		
ebr_order_direction	ebr_order_direction	STRING		
ebr_custom_filter	ebr_custom_filter	STRING		
ebr_sd_persid	ebr_sd_persid	STRING		
ebr_filter_data	ebr_flter_data	STRING		
ebr_url_text	ebr_url_text	STRING		
ebr_ks_source	ebr_ks_source	STRING		
ebr_kcat_id	ebr_kcat_id	STRING		
ebr_ad_blc	ebr_ad_blc	STRING		
ebr_serial_num	ebr_serial_num	STRING		

ebr_relevance	ebr_relevance	STRING		
ebr_ks_teaser	ebr_ks_teaser	STRING		
ebr_ks_concepts	ebr_ks_concepts	STRING		
FLG_COUNT	FLG_COUNT	QREL		
FULLWORDS	FULLWORDS	STRING		
Hits	HITS	INTEGER		
id	ID	INTEGER		KEY
INDEXED	INDEXED	INTEGER		
Inherit Permissions Flag	INHERIT_ PERMISSION	INTEGER		
Initiator	INITIATOR	STRING		
Initiator ID	INITIATOR_ID	UUID	ca_contact uuid	
Inherit Permissions from Category ID	KD_PERMISSION_ INDEX_ID	INTEGER	O_INDEXES id	
KS_TYPE	KS_TYPE	INTEGER		
Last Accepted Date	LAST_ACCEPTED_DAT E	LOCAL_TIME		
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		Indicates the timestamp of when this record was last modified.
Locked By ID	LOCKED_BY_ID	UUID	ca_contact uuid	
Modify Date	MODIFY_DATE	LOCAL_TIME		
Notes	NOTES	STRING		
Owner ID	OWNER_ID	UUID	ca_contact uuid	
Parent Request	PARENT_CR	STRING	call_req persid	
Parent Issue	PARENT_ISS	STRING	issue persistent_id	
persistent_id	persid	STRING		
Primary Category	PRIMARY_INDEX	INTEGER	O_INDEXES id	
Workflow Priority ID	PRIORITY_ID	INTEGER	CI_PRIORITIES id	
Problem	PROBLEM	STRING		
producer_id	producer_id	STRING		
Published Date	PUBLISHED_DATE	LOCAL_TIME		
READ_PGROUP	READ_PGROUP	INTEGER	P_GROUPS id	
PGROUP TYPE	PGROUP_TYPE	INTEGER		Indicates if the P Groups is based on Roles or Groups: 1 -- Groups (Default) 2 -- Roles
QA_STATUS	QA_STATUS	SREL	kt_qa_status.id	
Resolution Text	RESOLUTION	STRING		
Resolution Length	RESOLUTION_ LENGTH	INTEGER		

Short Resolution	RESOLUTION_ SHORT	STRING		
Review Date	REVIEW_DATE	LOCAL_TIME		
Solution Count	SD_ACCEPTED_ HITS	INTEGER		
Asset ID	SD_ASSET_ID	UUID	ca_owned_ resource uuid	
Impact ID	SD_IMPACT_ID	INTEGER	impact enum	
PRIMARY_KCAT_ CHANGE_ONKCAT_ DELETE	PRIMARY_KCAT_ CHANGE_ONKCAT_ DELETE	INTEGER		
Priority ID	SD_PRIORITY_ID	INTEGER	pri enum	
PROBLEM_SHORT	PROBLEM_SHORT	STRING 255		
Product ID	SD_PRODUCT_ID	INTEGER	product id	
RESOLUTION_ TEXT	RESOLUTION_ TEXT	STRING		
Root Cause ID	SD_ROOTCAUSE_ ID	INTEGER	rootcause id	
Severity ID	SD_SEVERITY_ID	INTEGER	sevrty enum	
Urgency ID	SD_URGENCY_ID	INTEGER	urgncy enum	
USER_SLV_CNT	USER_SLV_CNT	INTEGER		
SHORTWORDS	SHORTWORDS	STRING		
Start Date	START_DATE	LOCAL_TIME		
Status ID	STATUS_ID	INTEGER	CI_STATUSES id	
Subject Expert ID	SUBJECT_EXPERT_ID	UUID	ca_contact uuid	
SUBMIT_ KNOWLEDGE_ GUEST_USER_ NAME	SUBMIT_ KNOWLEDGE_ GUEST_USER_ NAME	STRING		
SUBMIT_ KNOWLEDGE_ GUEST_MAIL	SUBMIT_ KNOWLEDGE_ GUEST_MAIL	STRING		
Summary	SUMMARY	STRING		
ticket_avoided_cnt	ticket_avoided_cnt	INTEGER		
Title	TITLE	STRING		
User Defined ID	USER_DEF_ID	STRING		
Version Comment	VER_COMMENT	STRING		
Version Count	VER_COUNT	INTEGER		
Version Cross Reference ID	VER_CROSS_REF_ ID	INTEGER		
Workflow Template	WF_TEMPLATE	INTEGER	CI_WF_ TEMPLATES id	
WORD_COUNT_ TOTAL	WORD_COUNT_ TOTAL	INTEGER		
WORDCOUNT	WORDCOUNT	INTEGER		
WORDCOUNTS	WORDCOUNTS	STRING		
WORDORDERS	WORDORDERS	STRING		

WORDPLACES	WORDPLACES	STRING		
WORDSPANS	WORDSPANS	STRING		
WRITE_PGROUP	WRITE_PGROUP	INTEGER	P_GROUPS id	

KD_APPROVAL_METHODS Object

The object details are as follows:

REST Operations: CREATE READ UPDATE

Name	Type	Description	
id	Long		
KD	Long	Knowledge Document's SREL	Tenant implying attribute
CI_WF_TEMPLATES	Long	SREL to CI_WF_TEMPLATES	Tenant implying attribute
KD_ACTION	String	On of: Forward, reject, publish, retire and unretire	
ACTION_ID	Long	SREL to ci_actions	
ASSIGNEE	UUID	User assigned the Knowledge Document	

KD_ATTMNT Object

The object details are as follows:

1. Associated Table: KD_ATTMNT
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group:
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
ATTMNT_ID	ATTMNT_ID	INTEGER	attmnt id	
DOC_ID	DOC_ID	INTEGER	SKELETONS id	
id	ID	INTEGER		KEY
last_mod_dt	LAST_MOD_DT	LOCAL_TIME		

Layout

This article contains the following topics:

menu_bar Object

The object details are as follows:

1. Associated Table: usp_menu_bar
2. Factories: default
3. REL_ATTR: id

4. Common Name: name
5. Function Group: Security
6. REST Operations: CREATE READ UPDATE

Attribute Name	Data Type	Relationship Object	Flags
id	INTEGER		UNIQUE
name	STRING		REQUIRED
code	STRING		UNIQUE; REQUIRED
delete_flag	SREL	actbool	REQUIRED
description	STRING		
htmpl_name	STRING		
last_mod_by	SREL	cnt	
last_mod_dt	DATE		

menu_tree Object

The object details are as follows:

1. Associated Table: usp_menu_tree
2. Factories: default
3. REL_ATTR: id
4. Common Name: caption
5. Function Group: security
6. REST Operations: CREATE READ UPDATE

Attributes	Data Type	Related Object	Flags
id	INTEGER		UNIQUE
caption	STRING		REQUIRED
description	STRING		
has_children	INTEGER		
parent_id	INTEGER		
resource	SREL	menu_tree_res	
tree_name	SREL	menu_tree_name	
last_mod_dt	DATE		
last_mod_by	SREL	cnt	

menu_tree_name Object

The object details are as follows:

1. Associated Table: usp_menu_tree_name
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: security

6. REST Operations: CREATE READ UPDATE

Attributes	Data Type	Related Object	Flags
id	INTEGER		UNIQUE
name	STRING		REQUIRED
code	STRING		UNIQUE; REQUIRED
delete_flag	SREL	actbool	REQUIRED
description	STRING		
internal	SREL	bool	REQUIRED
last_mod_dt	DATE		
last_mod_by	SREL	cnt	

menu_tree_res Object

The object details are as follows:

1. Associated Table: usp_menu_tree_res
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: Security
6. REST Operations: CREATE READ UPDATE

Attributes	Data Type	Related Object	Flags
id	INTEGER		UNIQUE
name	STRING		REQUIRED
delete_flag	SREL	actbool	REQUIRED
description	STRING		
resource	STRING		
type	INTEGER		
last_mod_dt	DATE		
last_mod_by	SREL	cnt	

Macro

This topic contains the following information:

macro Object

The object details are as follows:

1. Associated Table: Spell_Macro
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: sym
5. Function Group: admin

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
usr_string1	fragment	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_dt	last_mod_dt	LOCAL_TIME		
lock_object	lock_object	INTEGER		REQUIRED
msg_html	msg_html	STRING		
ob_type	ob_type	STRING		REQUIRED
persistent_id	persid	STRING		
sym	sym	STRING		REQUIRED UNIQUE
type	type	STRING	splmactp persid	REQUIRED
usr_integer1	usr_integer1	INTEGER		
usr_integer2	usr_integer2	INTEGER		
usr_integer3	usr_integer3	INTEGER		
usr_string2	usr_string2	STRING		
usr_string3	usr_string3	STRING		
usr_string4	usr_string4	STRING		

macro_type Object

The object details are as follows:

1. Associated Table: Spell_Macro_Type
2. Factories: default, edit_macros
3. REL_ATTR: code, persistent_id
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
arg_list	arg_list	STRING		
code	code	STRING		UNIQUE REQUIRED
delete_flag	del	INTEGER	actbool enum	REQUIRED
display_name	display_name	STRING		
execute_script	execute_script	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_dt	last_mod_dt	LOCAL_TIME		

lock_object_flag	lock_object_flag	INTEGER		
persistent_id	persid	STRING		
sym	sym	STRING		REQUIRED
tech_desc	tech_desc	STRING		
validate_script	validate_script	STRING		

pdmMacroControlType Object

This object contains design view control types. The object details are as follows:

1. Associated Table: usp_pdmMacroControlType
2. Factories: default
3. REL_ATTR: enum
4. Common Name: name
5. UI_INFO: NOLOOKUP
6. Function Group: admin

Attribute	DB Field	Data Type	SREL References	Flags
enum	enum	INTEGER		
sym	sym	STRING		
description	description	STRING		
delete_flag	del	INTEGER		
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt	

pdmMacroParam Object

This object describes an argument of a web macro. The object details are as follows:

1. Associated Table: usp_pdmMacroParam
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. UI_INFO: NOLOOKUP
6. Function Group: admin

Attribute	DB Field	Data Type	SREL References	Flags
macro	macro	SREL	pdmMacro	
name	name	STRING		REQUIRED
caption	caption	STRING		REQUIRED
description	description	STRING		
tooltip	tooltip	STRING		
help_form	help_form	STRING		
is_required	is_required	INTEGER		
is_advanced	is_advanced	INTEGER		

default_value	default_value	STRING		
type	type	SREL	pdmMacroParamType	REQUIRED
size_textbox	size_textbox	INTEGER		
value_list	value_list	STRING		
delete_flag	del	SREL	actbool	REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt	

pdmMacroParamType Object

This object contains design view control types. The object details are as follows:

1. Associated Table: usp_pdmMacroParamType
2. Factories: default
3. REL_ATTR: enum
4. Common Name: name
5. UI_INFO: NOLOOKUP
6. Function Group: admin

Attribute	DB Field	Data Type	SREL References	Flags
enum	enum	INTEGER		
sym	sym	STRING		
description	description	STRING		
delete_flag	del	INTEGER		
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt	

pdmMacroType Object

This object contains macro form types (detail or list). The object details are as follows:

1. Associated Table: usp_pdmMacroType
2. Factories: default
3. REL_ATTR: enum
4. Common Name: name
5. UI_INFO: NOLOOKUP
6. Function Group: admin

Attribute	DB Field	Data Type	SREL References	Flags
enum	enum	INTEGER		
sym	sym	STRING		
description	description	STRING		
delete_flag	del	INTEGER		
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt	

Notification Objects

This article contains the following topics:

Notification

The object details are as follows:

1. Associated Table: Notification
2. Factories: default
3. REL_ATTR: id
4. Common Name: ALT_EMAIL
5. Function Group:
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
Email Address	ALT_EMAIL	STRING		
Contact ID	ANALYST_ID	UUID	ca_contact uuid	
Document ID	DOC_ID	INTEGER		
id	ID	INTEGER		REQUIRED KEY
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
KT Notification Level	NTF_LEVEL	INTEGER		

notque Object

The object details are as follows:

1. Associated Table: Queued_Notify
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: msg_title
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
cmth_override	cmth_override	INTEGER		
context_persid	context_persid	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
internal	internal	INTEGER		
msg_ack	msg_ack	STRING		
msg_body	msg_body	STRING		
msg_body_html	msg_body_html	STRING		
msg_title	msg_title	STRING		
notify_level	notify_level	INTEGER		

persistent_id	persid	STRING		
transition_pt	transition_pt	INTEGER		

noturg Object

The object details are as follows:

1. Associated Table: Notification_Urgency
2. Factories: default
3. REL_ATTR: enum
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
enum	enum	INTEGER		REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
description	nx_desc	STRING		
sym	sym	STRING		UNIQUE REQUIRED S_KEY

ntfl Object

The object details are as follows:

1. Associated Table: Notify_Object_Attr
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE REQUIRED KEY
producer_id	producer_id	LOCAL STRING 20		
persistent_id	persid	STRING 30	persid	
mgs_ntf	mgs_ntf	LREL	mgs dist_ntflist	
delete_flag	del	INTEGER	actbool enum	REQUIRED
sym	sym	STRING 60		REQUIRED
last_mod_by	last_mod_by	UUID	ca_contact uuid	
modified_date	last_mod_dt	LOCAL_TIME		
description	description	STRING 240		
object_attr	object_attr	STRING		

object_type	object_type	STRING		
macro_ntf	macro_ntf	LREL	att_ntflist	
att_ntfrlist	att_ntfrlist	LREL	ntfr_ntflist	

ntfm Object

The object details are as follows:

1. Associated Table: Notify_Msg_Tpl
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: sym
5. Function Group: notification_reference
6. REST Operations: CREATE READ UPDATE

Attribute	Data Type	SREL References	Flags
id	INTEGER		UNIQUE REQUIRED KEY
producer_id	LOCAL STRING 20		
persistent_id	STRING 30		
delete_flag	SREL	actbool	REQUIRED
sym	STRING 128		REQUIRED
notify_flag	INTEGER		
notify_level	SREL	noturg	
notify_msg_title	STRING 80		REQUIRED
notify_msg_body	STRING 4000		
notify_msg_body_html	STRING 32768		
notify_msg_body_html_real	LOCAL STRING 0		
obj_type	SREL	ntfm_prod_list.sym	REQUIRED
last_mod_dt			
last_mod_by	SREL	cnt	
tenant	SREL	tenant.id	

ntfr Object

The object details are as follows:

1. Associated Table: Notify_Rule
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: notification _reference

6. REST Operations: CREATE READ UPDATE

Attribute	Data Type	SREL References	Flags
id	INTEGER		UNIQUE REQUIRED KEY
producer_id	LOCAL STRING 20		
persistent_id	STRING 30		
delete_flag	SREL	actbool	REQUIRED
sym	STRING 128		REQUIRED
description	STRING 500		
condition	SREL	macro.persistent_id	
obj_type	SREL	ntfr_prod_list	REQUIRED
last_mod_dt	DATE		
last_mod_by	SREL	cnt	
default_rule	INTEGER		
cr_notify_info	SREL	ntfm.persistent_id	
chg_notify_info	SREL	ntfm.persistent_id	
iss_notify_info	SREL	ntfm.persistent_id	
mgs_notify_info	SREL	ntfm.persistent_id	
kd_notify_info	SREL	ntfm.persistent_id	
kd_comment_notify_info	SREL	ntfm.persistent_id	
krc_notify_info	SREL	ntfm.persistent_id	
sa_notify_info	SREL	ntfm.persistent_id	
cnt_notify_info	SREL	ntfm.persistent_id	
ci_notify_info	SREL	ntfm.persistent_id	
ntfr_ntflist	LREL	ntfl att_ntfrlist	
ntfr_cntlist	LREL	cnt:PDM att_ntfrlist	
ntfr_ctplist	LREL	ctp:PDM att_ntfrlist	
ntfr_macrolist	LREL	marco att_ntfrlist	
tenant	SREL	tenant.id	

Priority

This article contains the following topics:

pri Object

The object details are as follows:

1. Associated Table: Priority
2. Factories: default
3. REL_ATTR: enum
4. Common Name: sym
5. Function Group: reference

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
enum	enum	INTEGER		REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
description	nx_desc	STRING		
service_type	service_type	STRING	srv_desc code	
sym	sym	STRING		UNIQUE REQUIRED S_KEY

pri_cal Object

The object details are as follows:

1. Associated Table: usp_pri_cal
2. REL_ATTR: id
3. Common Name: name
4. Function Group: admin
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE REQUIRED KEY
producer_id		LOCAL_STRING 20		
persistent_id	persid	STRING 60		
delete_flag	del	SREL	actbool.enum	REQUIRED
name	name	STRING 80		REQUIRED
description	description	STRING 240		
in_flag	in_flag	SREL	bool.enum	
pr_flag	pr_flag	SREL	bool.enum	
imp_def	imp_flag	SREL	imp.enum	
urg_def	urg_def	SREL	urg.enum	
ci_imp	ci_imp	SREL	bool.enum	
cat_urg	bk_window	SREL	bool.enum	
bk_window	cnt_vip	SREL		
cnt_vip	cnt_vip	SREL		
pri_5_4	pri_5_4	SREL	pri_enum	
pri_5_3	pri_5_3	SREL	pri_enum	

pri_5_2	pri_5_2	SREL	pri_enum	
pri_5_1	pri_5_1	SREL	pri_enum	
pri_5_0	pri_5_0	SREL	pri_enum	
pri_5_x	pri_5_x	SREL	pri_enum	
pri_4_4	pri_4_4	SREL	pri_enum	
pri_4_3	pri_4_3	SREL	pri_enum	
pri_4_2	pri_4_2	SREL	pri_enum	
pri_4_1	pri_4_1	SREL	pri_enum	
pri_4_0	pri_4_0	SREL	pri_enum	
pri_4_x	pri_4_x	SREL	pri_enum	
pri_3_4	pri_3_4	SREL	pri_enum	
pri_3_3	pri_3_3	SREL	pri_enum	
pri_3_2	pri_3_2	SREL	pri_enum	
pri_3_1	pri_3_1	SREL	pri_enum	
pri_3_0	pri_3_0	SREL	pri_enum	
pri_3_x	pri_3_x	SREL	pri_enum	
pri_2_4	pri_2_4	SREL	pri_enum	
pri_2_3	pri_2_3	SREL	pri_enum	
pri_2_2	pri_2_2	SREL	pri_enum	
pri_2_1	pri_2_1	SREL	pri_enum	
pri_2_0	pri_2_0	SREL	pri_enum	
pri_2_x	pri_2_x	SREL	pri_enum	
pri_1_4	pri_1_4	SREL	pri_enum	
pri_1_3	pri_1_3	SREL	pri_enum	
pri_1_2	pri_1_2	SREL	pri_enum	
pri_1_1	pri_1_1	SREL	pri_enum	
pri_1_0	pri_1_0	SREL	pri_enum	
pri_1_x	pri_1_x	SREL	pri_enum	
pri_0_4	pri_0_4	SREL	pri_enum	
pri_0_3	pri_0_3	SREL	pri_enum	
pri_0_2	pri_0_2	SREL	pri_enum	
pri_0_1	pri_0_1	SREL	pri_enum	
pri_0_0	pri_0_0			
pri_0_x	pri_0_x	SREL	pri_enum	
pri_x_4	pri_x_4	SREL	pri_enum	
pri_x_3	pri_x_3	SREL	pri_enum	
pri_x_2	pri_x_2	SREL	pri_enum	
pri_x_1	pri_x_1	SREL	pri_enum	

pri_x_0	pri_x_0	SREL	pri_enum	
pri_x_x	pri_x_x	SREL	pri_enum	
cap_reason	cap_reason	SREL	bool.enum	
template_flag	template_flag	SREL	bool.enum	
last_mod_dt	last_mod_dt	LOCAL_TIME		
last_mod_by	last_mod_by	SREL	cnt.id	
tenant	tenant	UUID	ca_tenant	

Problem Category

This article contains the following topics:

pcat Object

The object details are as follows:

1. Associated Table: Prob_Category
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE REQUIRED KEY
producer_id		STRING 20		
persistent_id	persid	STRING 30		
delete_flag		SREL	actbool.enum	REQUIRED
sym	sym	STRING 1000		REQUIRED S_KEY
last_mod_dt	last_mod_dt	LOCAL_TIME		
last_mod_by		SREL	cnt.id	
description	description	STRING 500		
organization		SREL	org.id	
assignee	assignee	SREL	agt.id	
group	group_id	UUID	group.id	
properties		BREL	cr_prptpl.owning_are	
survey		SREL	svy_tpl.id	
auto_assign	auto_assign	INTEGER		
service_type		SREL	sdsc.code	
category_urgency		SREL	urg.enum	
owning_contract		SREL	svc_contract.id	
cr_flag	cr_flag	INTEGER		

in_flag	in_flag	INTEGER		
pr_flag	pr_flag	INTEGER		
suggest_knowledge	suggest_knowledge	INTEGER		
assignable_ci_attr	assignable_ci_attr	STRING 60		
service_grps		BREL	lrel_svc_grps_svc_pcat.pcat	
service_locs		BREL	lrel_svc_locs_svc_pcat.pcat	
service_schedules		BREL	lrel_svc_schedules_pcat_svc.pcat	
ss_include		SREL	bool.enum	REQUIRED On new default: 0
ss_sym	ss_sym	STRING 128		
flow_flag	flow_flag	INTEGER		

pcat_loc Object

The object details are as follows:

1. Associated Table: Pcat_Loc
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: lpid
5. Function Group:

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE REQUIRED KEY
lattr	l_attr	STRING		
lpid	l_persid	STRING		
lseq	l_sql	INTEGER		
rattr	r_attr	STRING		
rpid	r_persid	STRING		
rseq	r_sql	INTEGER		

Property Objects

This article contains the following topics:

prp Object

The object details are as follows:

1. Associated Table: Property
2. Factories: default

3. REL_ATTR: id
4. Common Name: value
5. Function Group: change_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
error_msg	error_msg	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
label	label	STRING		
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
object_id	object_id	INTEGER	chg id	REQUIRED
object_type	object_type	STRING		REQUIRED
owning_macro	owning_macro	BOP_REF_STR	macro	REQUIRED
persistent_id	persid	STRING		
property	property	INTEGER	prptpl id	
required	required	INTEGER	bool_tab enum	
sample	sample	STRING		
sequence	sequence	INTEGER		REQUIRED
value_description	value_description	STRING		
validation_rule	validation_rule	BOP_REF_STR	prpval_rule	REQUIRED
validation_type	validation_type	BOP_REF_STR	prpval_type	REQUIRED
value	value	STRING		

prptpl Object

The object details are as follows:

1. Associated Table: Property_Template
2. Factories: default
3. REL_ATTR: id
4. Common Name: label
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
code	code	STRING		UNIQUE REQUIRED S_KEY
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY

label	label	STRING		REQUIRED
last_mod_by	last_mod_by	UUID	ca_contact uuid	
modified_date	last_mod_dt	LOCAL_TIME		
object_attrname	object_attrname	STRING		REQUIRED
object_attrval	object_attrval	INTEGER		
object_type	object_type	STRING		REQUIRED
persistent_id	persid	STRING		
producer_id		LOCAL STRING 20		
required	required	INTEGER		REQUIRED
required_sym	required_sym	SREL	bool.enum	
sample	sample	STRING		
sequence	sequence	INTEGER		REQUIRED
validation_rule	validation_rule	BOP_REF_STR	prpval_rule	REQUIRED

prpval Object

The object details are as follows:

1. Associated Table: Property_Value
2. REL_ATTR: id
3. Common Name: value
4. Function Group: admin
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE REQUIRED KEY
producer_id		LOCAL_STRING 20		
persistent_id	persid			
delete_flag	del	SREL	actbool.enum	REQUIRED
description	description	STRING 240		
owning_rule	owning_rule	SREL	prpval_rule.id	
value	value	STRING 240		REQUIRED
is_default	is_default	INTEGER		
modified_date	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt.id	

prpval_rule Object

The object details are as follows:

1. Associated Table: Property_Validation_Rule
2. REL_ATTR: id
3. Common Name: sym

4. Function Group: admin
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE REQUIRED KEY
producer_id		LOCAL_STRING 20		
persistent_id	persid	STRING 30		
delete_flag	del	SREL	actbool.enum	REQUIRED
sym	sym	STRING 60		REQUIRED
description	description	STRING 240		
validation_type	validation_type	SREL	prpval_type.id	
values		BREL	prpval.owning_rule	
modified_date	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt.id	

prpval_type Object

The object details are as follows:

1. Associated Table: Property_Validation_Type
2. REL_ATTR: id
3. Common Name: sym
4. Function Group: admin
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE REQUIRED KEY
producer_id		LOCAL_STRING 20		
persistent_id	persid			
delete_flag	del	SREL	actbool.enum	REQUIRED
sym	sym	STRING 60		REQUIRED
description	description	STRING 240		
modified_date	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt.id	

Relational Information

This topic contains the following information:

lrel_asset_chgnr Object

The object details are as follows:

1. Associated Table: usp_lrel_asset_chgnr
2. Factories: default
3. REL_ATTR: id
4. Common Name:id
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE DELETE

Attribute	DB Field	Data Type	SREL References	Flags
nr	nr	BREL	nr.id	LREL REQUIRED
chg	chg	BREL	chg.id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_asset_issnr Object

The object details are as follows:

1. Associated Table: usp_lrel_asset_issnr
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE DELETE

Attribute	DB Field	Data Type	SREL References	Flags
nr	nr	BREL	nr.id	LREL REQUIRED
iss	iss	BREL	iss.persistent_id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_att_cntlist_macro_ntf Object

The object details are as follows:

1. Associated Table: usp_lrel_att_cntlist_macro_ntf
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
cnt	cnt	BREL	cnt.id	LREL REQUIRED
macro	macro	BREL	macro.persistent_ id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_att_ctplist_macro_ntf Object

The object details are as follows:

1. Associated Table: usp_lrel_att_ctplist_macro_ntf
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
ctp	ctp	BREL	ctp.id	LREL REQUIRED
macro	macro	BREL	macro.persistent_id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_att_ntflist_macro_ntf Object

The object details are as follows:

1. Associated Table: usp_lrel_att_ntflist_macro_ntf
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
ntfl	ntfl	BREL	ntfl.id	LREL REQUIRED
macro	macro	BREL	macro.persistent_id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		

last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_attachments_changes Object

The object details are as follows:

1. Associated Table: usp_lrel_attachments_changes
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE DELETE

Attribute	DB Field	Data Type	SREL References	Flags
attmnt	attmnt	BREL	attmnt id	LREL
chg	chg	BREL	chg id	LREL
last_mod_by	last_mod_by	BREL	ca_contact contact_uuid	LREL
last_mod_dt	last_mod_dt	BREL	cnt.id	LREL

lrel_attachments_issues Object

The object details are as follows:

1. Associated Table: usp_lrel_attachments_issues
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE DELETE

Attribute	DB Field	Data Type	SREL References	Flags
attmnt	attmnt	BREL	attmnt.id	LREL REQUIRED
iss	iss	BREL	iss.persistent_id	LREL REQUIRED
login_id		LOCAL STRING(0)		
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_attachments_requests Object

The object details are as follows:

1. Associated Table: usp_lrel_attachments_requests
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference

6. REST Operations: CREATE READ UPDATE DELETE

Attribute	DB Field	Data Type	SREL References	Flags
attmnt	attmnt	BREL	attmnt.id	LREL REQUIRED
cr	cr	BREL	cr.persistent_id	LREL REQUIRED
login_id		LOCAL STRING(0)		
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_aty_events Object

The object details are as follows:

1. Associated Table: usp_lrel_aty_events
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Remarks
aty	aty	BREL	aty.code	LREL REQUIRED
evt	evt	BREL	evt.persistent_id	LREL REQUIRED
last_mod_dt	last_mod_dt	LOCAL_TIME		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_bm_reps_assets Object

The object details are as follows:

1. Associated Table: usp_lrel_bm_reps_assets
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference

Attribute	DB Field	Data Type	SREL References	Flags
bmrep	bmrep	BREL	bmrep.id	LREL REQUIRED
nr	nr	BREL	nr.id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		

last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_bm_reps_bmhiers Object

The object details are as follows:

1. Associated Table: usp_lrel_bm_reps_bmhiers
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference

Attribute	DB Field	Data Type	SREL References	Flags
bmrep	bmrep	BREL	bmrep.id	LREL REQUIRED
bmhier	bmhier	BREL	bmhier.id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_cenv_cntref Object

The object details are as follows:

1. Associated Table: usp_lrel_cenv_cntref
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE DELETE

Attribute	DB Field	Data Type	SREL References	Flags
nr	nr	BREL	nr.id	LREL REQUIRED
cnt	cnt	BREL	cnt.id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_dist_cntlist_mgs_ntf Object

The object details are as follows:

1. Associated Table: usp_lrel_dist_cntlist_mgs_ntf
2. Factories: default

3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
cnt	cnt	BREL	cnt.id	LREL REQUIRED
mgs	mgs	BREL	mgs.id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_dist_ctplist_mgs_ntf Object

The object details are as follows:

1. Associated Table: usp_lrel_dist_cntlist_mgs_ntf
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
ctp	ctp	BREL	ctp.id	LREL REQUIRED
mgs	mgs	BREL	mgs.id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	

lrel_dist_ntflist_mgs_ntf Object

The object details are as follows:

1. Associated Table: usp_lrel_dist_ntflist_mgs_ntf
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
ntfl	ntfl	BREL	ntfl.id	LREL REQUIRED
mgs	mgs	BREL	mgs.id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		

last_mod_by	last_mod_by	BREL	cnt.id	

lrel_false_action_act_f Object

The object details are as follows:

1. Associated Table: usp_lrel_false_action_act_f
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
macro	macro	BREL	macro.persistent_id	LREL REQUIRED
evt	evt	BREL	evt.persistent_id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_false_bhv_false Object

The object details are as follows:

1. Associated Table: lrel_false_bhv_false
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
macro	macro	BREL	macro.persistent_id	LREL REQUIRED
bhvtpl	bhvtpl	BREL	bhvtpl.id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_kwrds_crsolref Object

The object details are as follows:

1. Associated Table: usp_lrel_kwrds_crsolref
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
kwrd	kwrd	BREL	kwrd.id	LREL REQUIRED
crsol	crsol	BREL	crsol.persistent_id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_notify_list_cntchgntf Object

The object details are as follows:

1. Associated Table: usp_lrel_notify_list_cntchgntf
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE DELETE

Attribute	DB Field	Data Type	SREL References	Flags
cnt	cnt	BREL	cnt.id	LREL REQUIRED
chg	chg	BREL	chg.id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_notify_list_cntissntf Object

The object details are as follows:

1. Associated Table: usp_lrel_notify_list_cntissntf
2. Factories: default
3. REL_ATTR: id
4. Common Name:

5. Function Group: reference
6. REST Operations: CREATE READ UPDATE DELETE

Attribute	DB Field	Data Type	SREL References	Flags
cnt	cnt	BREL	cnt.id	LREL REQUIRED
iss	iss	BREL	iss.persistent_id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_notify_list_cntntf Object

The object details are as follows:

1. Associated Table: usp_lrel_notify_list_cntntf
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE DELETE

Attribute	DB Field	Data Type	SREL References	Flags
cnt	cnt	BREL	cnt.id	LREL REQUIRED
cr	cr	BREL	cr.persistent_id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_ntfr_cntlist_att_ntfrlist Object

The object details are as follows:

1. Associated Table: usp_lrel_ntfr_cntlist_att_ntfrlist
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
cnt	cnt	BREL	cnt.id	LREL REQUIRED
ntfr	ntfr	BREL	ntfr.id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		

last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_ntfr_ctplist_att_ntfrlist Object

The object details are as follows:

1. Associated Table: usp_lrel_ntfr_ctplist_att_ntfrlist
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
ctp	ctp	BREL	ctp.id	LREL REQUIRED
ntfr	ntfr	BREL	ntfr.id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_ntfr_macrolist_att_ntfrlist Object

The object details are as follows:

1. Associated Table: usp_lrel_ntfr_macrolist_att_ntfrlist
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
macro	macro	BREL	macro.persistent_id	LREL REQUIRED
ntfr	ntfr	BREL	ntfr.id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_ntfr_ntflist_att_ntfrlist Object

The object details are as follows:

1. Associated Table: usp_lrel_ntfr_ntflist_att_ntfrlist
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
macro	macro	BREL	macro.persistent_id	LREL REQUIRED
ntfl	ntfl	BREL	ntfl.id	LREL REQUIRED
ntfr	ntfr	BREL	ntfr.id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_oenv_orgref Object

The object details are as follows:

1. Associated Table: usp_lrel_oenv_orgref
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE DELETE

Attribute	DB Field	Data Type	SREL References	Flags
nr	nr	BREL	nr.id	LREL REQUIRED
org	org	BREL	org.id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_status_codes_tsktypes Object

The object details are as follows:

1. Associated Table: usp_lrel_status_codes_tsktypes

2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
tskstat	tskstat	BREL	tskstat.code	LREL REQUIRED
tskty	tskty	BREL	tskty.code	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_svc_grps_svc_chgcat Object

The object details are as follows:

1. Associated Table: usp_lrel_svc_grps_svc_chgcat
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
cnt	cnt	BREL	cnt.id	LREL REQUIRED
chgcat	chgcat	BREL	chgcat.code	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_svc_grps_svc_isscat Object

The object details are as follows:

1. Associated Table: usp_lrel_svc_grps_svc_isscat
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
cnt	cnt	BREL	cnt.id	LREL REQUIRED
isscat	isscat	BREL	isscat.code	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_svc_grps_svc_pcat Object

The object details are as follows:

1. Associated Table: usp_lrel_svc_grps_svc_pcat
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
cnt	cnt	BREL	cnt.id	LREL REQUIRED
pcat	pcat	BREL	pcat.persistent_id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_svc_grps_svc_wftpl Object

The object details are as follows:

1. Associated Table: usp_lrel_svc_grps_svc_wftpl
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
cnt	cnt	BREL	cnt.id	LREL REQUIRED

wftpl	wftpl	BREL	wftpl.id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_svc_locs_svc_chgcat Object

The object details are as follows:

1. Associated Table: usp_lrel_svc_locs_svc_chgcat
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
loc	loc	BREL	loc.id	LREL REQUIRED
chgcat	chgcat	BREL	chgcat.code	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_svc_locs_svc_groups Object

The object details are as follows:

1. Associated Table: usp_lrel_svc_locs_svc_groups
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
loc	loc	BREL	loc.id	LREL REQUIRED
cnt	cnt	BREL	cnt.id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_svc_locs_svc_isscat Object

The object details are as follows:

1. Associated Table: usp_lrel_svc_locs_svc_isscat
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
loc	loc	BREL	loc.id	LREL REQUIRED
isscat	isscat	BREL	isscat.code	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_svc_locs_svc_pcat Object

The object details are as follows:

1. Associated Table: usp_lrel_svc_locs_svc_pcat
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
loc	loc	BREL	loc.id	LREL REQUIRED
pcat	pcat	BREL	pcat.persistent_id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_svc_schedules_chgcat_svc Object

The object details are as follows:

1. Associated Table: usp_lrel_svc_schedules_chgcat_svc
2. Factories: default
3. REL_ATTR: id
4. Common Name:

5. Function Group: security
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
wrkshft	wrkshft	BREL	wrkshft.persistent_id	LREL REQUIRED
chgcat	chgcat	BREL	chgcat.code	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_svc_schedules_isscat_svc Object

The object details are as follows:

1. Associated Table: usp_lrel_svc_schedules_isscat_svc
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: security
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
isscat	isscat	BREL	isscat code	LREL
wrkshft	wrkshft	BREL	wrkshft.persistent_id	LREL REQUIRED
isscat	isscat	BREL	isscat.code	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_svc_schedules_pcat_svc Object

The object details are as follows:

1. Associated Table: usp_lrel_svc_schedules_pcat_svc
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: security

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
wrkshft	wrkshft	BREL	wrkshft.persistent_id	LREL REQUIRED
pcat	pcat	BREL	pcat.persistent_id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_true_action_act_t Object

The object details are as follows:

1. Associated Table: usp_lrel_true_action_act_t
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
macro	macro	BREL	macro.persistent_id	LREL REQUIRED
evt	evt	BREL	evt.persistent_id	LREL REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

lrel_true_bhv_true Object

The object details are as follows:

1. Associated Table: usp_lrel_true_bhv_true
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
macro	macro	BREL	macro.persistent_id	LREL REQUIRED
bhvtpl	bhvtpl	BREL	bhvtpl.id	LREL REQUIRED

last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	BREL	cnt.id	LREL

Reporting Method

This topic contains the following information:

rptm Object

The object details are as follows:

1. Associated Table: Rpt_Meth
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
default_page_length	def_pg_len	STRING		
default_out	default_out	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
is_default	is_default	INTEGER		
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
script	script	STRING		
sym	sym	STRING		REQUIRED

rptmeth Object

The object details are as follows:

1. Associated Table: Reporting_Method
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY

last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
sym	sym	STRING		UNIQUE REQUIRED S_KEY

Request

This topic contains the following information:

cr Object

The object details are as follows:

1. Associated Table: Call_Req
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: ref_num
5. Function Group: call_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
active	active_flag	INTEGER	bool_tab enum	REQUIRED
active_prev	LOCAL	SREL	bool.enum	
affected_ resource	affected_rc	UUID	ca_owned_ resource uuid	
assignee	assignee	UUID		
assignee_prev	LOCAL	SREL	agt.id	
call_back_date	call_back_date	LOCAL_ TIME		
call_back_flag	call_back_flag	INTEGER		
category	category	STRING	prob_ctg persid	
category_prev	LOCAL	SREL	pcat.persistent_id	
caused_by_chg	caused_by_chg	SREL	chg.id	
change	change	INTEGER	chg id	
charge_back_id	charge_back_ id	STRING		
close_date	close_date	LOCAL_ TIME		
cr_tticket	cr_tticket	INTEGER		
created_via	created_via	INTEGER	interface id	
customer	customer	UUID	ca_contact uuid	REQUIRED
event_token	event_token	STRING		
external_system_ticket		STRING (4000)		
extern_ref	extern_ref	STRING		

group	group_id	UUID		
group_prev	LOCAL	SREL	grp.id	
id	id	INTEGER		UNIQUE REQUIRED KEY
impact	impact	INTEGER	impact enum	
impact_prev	LOCAL	SREL	imp.enum	
incident_priority	incident_priority	INTEGER		
incorrectly_assigned		SREL	actbool	
last_act_id	last_act_id	STRING		
last_mod_dt	last_mod_dt	LOCAL_TIME		
log_agent	log_agent	UUID	ca_contact uuid	REQUIRED
macro_ predicted_ violation	macro_predict_ viol	INTEGER		
major_incident		SREL	actbool	
open_date	open_date	LOCAL_ TIME	org	
orig_user_admin_org		SREL	org	
orig_user_cost_center		SREL	cost_cntr	
orig_user_dept		SREL	dept	
orig_user_organization		SREL	org	
outage_detail_what		STRING (4000)		
outage_detail_who		STRING (4000)		
outage_detail_why		STRING (4000)		
outage_end_time	outage_end_time	LOCAL_TIME		
outage_reason		SREL	outage_reason	
outage_start _time	outage_start _time	LOCAL_TIME		
outage_type		SREL	outage_type	
parent	parent	STRING	call_req persid	
pct_service_restored		INTEGER		
persistent_id	persid	STRING		
predicted_sla_violation	predicted_sla_viol	INTEGER		
priority	priority	INTEGER	pri enum	REQUIRED
priority_prev	LOCAL	SREL	pri.enum	
problem	problem	STRING		
ref_num	ref_num	STRING		UNIQUE REQUIRED S_KEY
remote_control_used		SREL	actbool	
requested_by		SREL	cnt	
resolvable_at_lower		SREL	actbool	
resolve_date	resolve_date	LOCAL_ TIME		

return_to_service		SREL	actbool	
rootcause	rootcause	INTEGER	rootcause id	
extern_token	sched_token	STRING		
severity	severity	INTEGER	sevrty enum	
severity_prev	LOCAL	SREL	sev.enum	
sla_violation	sla_violation	INTEGER		
base_template	solution	STRING	call_req persid	
status	status	STRING	cr_stat code	
status_prev	LOCAL	SREL	crs.code	
string1	string1	STRING		
string2	string2	STRING		
string3	string3	STRING		
string4	string4	STRING		
string5	string5	STRING		
string6	string6	STRING		
children		QREL	cr	
sla_events		QREL	tev	
attached_slas		BREL	attached_sla._mapped_cr	
add_sla_persids	add_sla_persids	STRING		
site_sla_attrs	site_sla_attrs	STRING		
audit_userid	audit_userid	SREL	cnt.id	
fldchange_log	fldchange_log	STRING		
affected_service	affected_service	SREL	nr	
init_urg		LOCAL INTEGER		
init_imp		LOCAL INTEGER		
cap_reason		LOCAL STRING		
auto_urg		LOCAL INTEGER		
auto_imp		LOCAL INTEGER		
man_urg		LOCAL INTEGER		
man_imp		LOCAL INTEGER		
summary	summary	STRING		
support_lev	support_lev	STRING	srv_desc code	
symptom_code		SREL	symptom_code	
template_name	template_name	STRING	cr_template template_name	
time_spent_ sum	time_spent_ sum	DURATION		
type	type	STRING	crt code	
urgency	urgency	INTEGER	urgncy enum	

urgency_prev	LOCAL	SREL	urg.enum	
target_times	target_times	BREL	tgt_time_mapped_cr	
target_start_last	target_start_last	DATE		
target_hold_last	target_hold_last	DATE		
target_hold_count	target_hold_count	INTEGER		
target_resolved_last	target_resolved_last	DATE		
target_resolved_count	target_resolved_count	INTEGER		
target_closed_last	target_closed_last	DATE		
target_closed_count	target_closed_count	INTEGER		
close_date_prev	close_date_prev	LOCAL DATE		
resolve_date_prev	resolve_date_prev	LOCAL DATE		
target_hold_count_prev	target_hold_count_prev	INTEGER		
target_resolved_count_prev	target_resolved_count_prev	LOCAL INTEGER		
target_closed_count_prev	target_closed_count_prev	LOCAL INTEGER	tgt_time_mapped_iss	

cr_prp Object

The object details are as follows:

1. Associated Table: Req_Property
2. Factories: default
3. REL_ATTR: id
4. Common Name: label
5. Function Group: call_mgr
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	nvarchar(240)		
error_msg	error_msg	STRING		
id	id	INTEGER		Primary key of this table.
label	label	nvarchar(80)		REQUIRED
last_mod_by	last_mod_by	byte(16)	ca_contact uuid	
last_mod_dt	last_mod_dt	INTEGER		
owning_cr	owning_cr	nvarchar 30	call_req persid	REQUIRED
owning_macro	owning_macro	BOP_REF_STR	macro	REQUIRED
persistent_id	persid	STRING		
required	required	INTEGER		REQUIRED
sample	sample	nvarchar(240)		
sequence	sequence	INTEGER		REQUIRED

value_description	value_description	STRING		
validation_rule	validation_rule	BOP_REF_STR	prpval_rule	REQUIRED
validation_type	validation_type	BOP_REF_STR	prpval_type	REQUIRED
value	value	nvarchar(240)		

cr_prptpl Object

The object details are as follows:

1. Associated Table: Req_Property_Template
2. Factories: default
3. REL_ATTR: id
4. Common Name: label
5. Function Group: call_mgr_reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
code	code	STRING		UNIQUE REQUIRED S_KEY
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
label	label	STRING		REQUIRED
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
owning_area	owning_area	STRING	prob_ctg persid	REQUIRED
persistent_id	persid	STRING		
producer_id		LOCAL STRING 20		
required	required	INTEGER		REQUIRED
sample	sample	STRING		
sequence	sequence	INTEGER		REQUIRED
validation_rule	validation_rule	BOP_REF_STR	prpval_rule	REQUIRED

cr_trans Object

The object details are as follows:

1. Associated Tables: cr_trans
2. Factories: default
3. REL_ATTR: id
4. Common Name: condition_error

Function Group: call_mgr_reference

REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
status	status	STRING	crs	Specifies the current ticket status.
new_status	new status	STRING	crs	Specifies the new ticket status.
t_type		INTEGER	transition type	Specifies the transition type.
is default		INTEGER		Default transition that appears when the Status field is empty. On new default: 0
must_comment		INTEGER		Comment required when using a transition. On new default: 0
delete_flag	del		actbool	Required. On new default: 0
condition			macro	Site condition macro to approve transition.
condition_error		STRING		Error message for site condition.
description		STRING		Description of this transition.
last_mod_by			cnt	On new default user; on CI set user
last_mod_dt		DATE		On CI set user now

cr_tpl Object

The object details are as follows:

1. Associated Table: Cr_Template
2. Factories: default
3. REL_ATTR: template_name
4. Common Name: template_name
5. Function Group: call_mgr_template
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED

id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
quick_tmpl_type	quick_tmpl_type	INTEGER	quick_tpl_types enum	REQUIRED
template	template	STRING	call_req persid	
template_class	template_class	STRING		
template_name	template_name	STRING		UNIQUE REQUIRED S_KEY

crs Object

The object details are as follows:

1. Associated Table: Cr_Status
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: call_mgr_reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
active	active	INTEGER		
code	code	STRING		UNIQUE REQUIRED S_KEY
cr_flag	cr_flag	INTEGER		
delete_flag	del	INTEGER	actbool enum	REQUIRED
hold	hold	INTEGER		
id	id	INTEGER		UNIQUE REQUIRED KEY
in_flag	in_flag	INTEGER		
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
pr_flag	pr_flag	INTEGER		
resolved	resolved	INTEGER		
sym	sym	STRING		REQUIRED

crsol Object

The object details are as follows:

1. Associated Table: Call_Solution
2. Factories: default
3. REL_ATTR: persistent_id
4. Common Name: sym
5. Function Group: call_mgr_reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
cr_count	cr_count	INTEGER		
cr_flag	cr_flag	INTEGER		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
in_flag	in_flag	INTEGER		
last_mod_dt	last_mod_dt	LOCAL_TIME		
desc	nx_desc	STRING		
persistent_id	persid	STRING		
pr_flag	pr_flag	INTEGER		
solution_approved	sapproved	INTEGER	bool_tab enum	
solution_name	sname	STRING		
solution	solution	STRING		
sym	sym	STRING		REQUIRED S_KEY
tcode	tcode	INTEGER		

crsq Object

The object details are as follows:

1. Associated Table: Cr_Stored_Queries
2. Factories: default, sqchg, sqcr, sqiss
3. REL_ATTR: code
4. Common Name: label
5. Function Group: stored_queries
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
code	code	STRING		UNIQUE REQUIRED S_KEY

count_url	count_url	STRING		
where_clause	criteria	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
label	label	STRING		
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
obj_type	obj_type	STRING		
persistent_id	persid	STRING		

Risk

This article contains the following topics:

risk_level Object

The object details are as follows:

1. Rel Attr: value
2. Function Group: change_reference
3. Common Name: sym
4. Associated Table: usp_risk_level
5. Index Key: enum, sym
6. Factories: risk_level
7. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE NOT_NULL KEY
delete_flag	del	INTEGER		
sym	sym	STRING 60		UNIQUE NOT_NULL S_KEY
value	enum	INTEGER		
description	desc	STRING 40		
last_mod_dt	last_mod_dt	LOCAL_TIME		
last_mod_by	last_mod_by	UUID	ca_contact	

risk_range Object

The object details are as follows:

1. Rel Attr: id
2. Function Group: change_reference
3. Associated Table: usp_risk_range
4. Factories: risk_range

5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE KEY
persistent_id	persid	BOP_REF_STR		Persistent id
delete_flag	del	INTEGER		Active: 0 Inactive: 1
risk_lvl	risk_lvl	INTEGER	risk_level	Reference to Risk Level
max_val	max_val	INTEGER		Maximum value of Risk Level
min_val	min_val	INTEGER		Minimum value of Risk Level
owning_survey	owning_survey	INTEGER	risk_svy_ tpl	Reference to owning Risk Survey Template
last_mod_dt		LOCAL_TIME		Last modified by
last_mod_by		UUID	ca_contact	Reference to Contact information
tenant		UUID	ca_tenant	Reference to Tenant information

risk_svy Object

The object details are as follows:

1. Rel Attr: id
2. Common Name: description
3. Function Group: change_reference
4. Associated Table: usp_risk_svy
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE KEY
persistent_id	persid	STRING 30		
delete_flag	del	SREL		REQUIRED
description	description	STRING 400		
sym	sym	STRING 60		REQUIRED
include_comment	include_comment	INTEGER		REQUIRED
comment_label	comment_label	STRING 80		
comment	comment	STRING 200		
chg_id	chg_id	SREL	chg	
survey_template	survey_template	SREL		
total_weightage	total_weightage	INTEGER		

last_mod_dt	last_mod_dt	DATE		Last modified by
last_mod_by	last_mod_by	SREL	cnt.id	
questions		QREL	risk_svy_ question	

risk_svy_answer Object

The object details are as follows:

1. Rel Attr: id
2. Common Name: text
3. Function Group: change_reference
4. Reference Table: usp_risk_svy_answer
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		
persistent_id	persid	STRING 30		
delete_flag	del	SREL	actbool.enum	
text	txt	STRING 400		
sequence	sequence	INTEGER		REQUIRED
weightage	weightage	INTEGER		
selected	selected	INTEGER		
owning_survey_ question	owning_survey_ question	SREL	risk_svy_ question.id	REQUIRED
last_mod_dt	last_mod_dt	LOCAL_TIME		
last_mod_by	last_mod_by	UUID	ca_contact	
tenant		UUID	ca_tenant	

risk_svy_atpl Object

The object details are as follows:

1. Rel Attr: id
2. Common Name: text
3. Function Group: change_reference
4. Reference Table: usp_risk_svy_atpl
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE KEY
persistent_id	persid	BOP_REF_STR		Persistent id
delete_flag	del	SREL	act_bool.enum	
txt	txt	STRING 400		Answer Text

sequence	sequence	INTEGER		Display order
weightage	weightage	INTEGER		Weightage of answer
owning_survey_ question	owning_survey_question	INTEGER	risk_svy_qtpl	Reference to owning Risk Survey question template
last_mod_dt	last_mod_dt	LOCAL_TIME		Last modified by
last_mod_by	last_mod_by	UUID	ca_contact	
tenant		UUID	ca_tenant	Reference to Tenant information

risk_svy_question Object

The object details are as follows:

1. Rel Attr: id
2. Common Name: text
3. Function Group: change_reference
4. Reference Table: usp_risk_svy_question
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		
persistent_id	persid			
delete_flag	del	SREL	actbool.enum	
text	txt	STRING 400		
sequence	sequence	INTEGER		REQUIRED
include_ qcomment	include_ qcomment	INTEGER		
qcomment_ label	qcomment_ label	STRING 80		
qcomment	qcomment	STRING 2000		
mult_resp_flag	mult_resp_flag	INTEGER		
owning_ survey	owning_survey	SREL	risk_svy.id	REQUIRED
last_mod_dt	last_mod_dt	LOCAL_TIME		
last_mod_by	last_mod_by	UUID	ca_contact	
answers		QREL	risk_svy_answer	
tenant		UUID	ca_tenant	

risk_svy_qtpl Object

The object details are as follows:

1. Rel Attr: id
2. Common Name: text
3. Function Group: change_reference

4. Reference Table: usp_risk_svy_qtpl
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE KEY
persistent_id	persid	BOP_REF_STR		Persistent id
delete_flag	del	SREL	actbool.enum	
text	txt	STRING 400		Question Text
sequence	sequence	INTEGER		Display order
include_qcomment	include_qcomment	INTEGER		Set to 1 to include comment box for this question.
qcomment_label	qcomment_label	STRING 80		Label for Comment field
mult_resp_flag	mult_resp_flag	INTEGER		0: Choose 1 response (radio) 1: Choose "n" (checkboxes)
owning_survey	owning_survey	INTEGER	risk_svy_tpl	Reference to owning Risk Survey template
last_mod_dt	last_mod_dt	LOCAL_TIME		Last modified by
last_mod_by	last_mod_by	UUID	ca_contact	Reference to Contact information
tenant		UUID	ca_tenant	Reference to Tenant information

risk_svy_tpl Object

The object details are as follows:

1. Rel Attr: id
2. Common Name: sym
3. Function Group: change_reference
4. Reference Table: usp_risk_svy_tpl
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE NOT_NULL KEY
persistent_id	persid	BOP_REF_STR		
delete_flag	del	SREL	actbool.enum	
description	description	STRING 400		
sym	sym	STRING 60		UNIQUE S_KEY
include_comment	include_comment	INTEGER		

comment_label	comment_label	STRING 80		
last_mod_dt	last_mod_dt	LOCAL_TIME		
last_mod_by	last_mod_by	UUID	ca_contact	
question_ templates		QREL	risk_svy_qtpl	
assocCategories		BREL	risk_survey	
assocRiskRanges		QREL	risk_range	
tenant		UUID	ca_tenant	

Role

This article contains the following topics:

role Object

The object details are as follows:

1. Associated Table: usp_role
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: security
6. REST Operations: CREATE READ UPDATE

Attributes	Data Type	Related Object	Flags
id	INTEGER		UNIQUE
producer_id	LOCAL STRING 20		
persistent_id	LOCAL STRING 60		
name	STRING		REQUIRED
code	STRING		REQUIRED; UNIQUE
delete_flag	SREL	actbool	REQUIRED
default_flag	INTEGER		
description	STRING		
interface_type	INTEGER		
pref_doc	INTEGER		REQUIRED
form_group	SREL	fmgrp	
help_view	SREL	help_set	
initial_form	STRING		REQUIRED
grant_level	SREL	acc_lvls	REQUIRED
view_internal	INTEGER		
data_partition	SREL	dmn	
override_cnt_datapart	INTEGER		REQUIRED

update_global	INTEGER		
tenant_access	INTEGER		
single_tenant	SREL	tenant	
tenant_group	SREL	tenant_group	
call_mgr	INTEGER		
change_mgr	INTEGER		
issue_mgr	INTEGER		
inventory	INTEGER		
references	INTEGER		
notify	INTEGER		
admin	INTEGER		
security	INTEGER		
sd_admin	INTEGER		
sd_analyst	INTEGER		
sd_emloyee	INTEGER		
sd_customer	INTEGER		
kt_admin	INTEGER		
kt_manager	INTEGER		
kt_engineer	INTEGER		
kt_analyst	INTEGER		
kt_customer	INTEGER		
tn_admin	INTEGER		
kt_type	INTEGER		
kd	INTEGER		
kcat	INTEGER		
kd_query_id	SREL	crsq	
kd_query_description	STRING 255		
wsp	INTEGER		
wsp_editForm	INTEGER		
wsp_publishForm	INTEGER		
wsp_updPreview	INTEGER		
wsp_modifySchema	INTEGER		
wsp_publishSchema	INTEGER		
Tabs	BREL	role_tab	
web_forms	BREL	role_web_form	
go_forms	BREL	role_go_form	
last_mod_dt	DATE		
last_mod_by	SREL	cnt	

role_web_form Object

The object details are as follows:

1. Associated Table: usp_role_web_form
2. Factories: default
3. REL_ATTR: id
4. Common Name: id
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attributes	Data Type	Related Object	Flags
id	INTEGER		UNIQUE
role_obj	SREL	role	
web_form_obj	SREL	web_form	
last_mod_dt	DATE		

role_go_form Object

The object details are as follows:

1. Associated Table: usp_role_go_form
2. Factories: default
3. REL_ATTR: id
4. Common Name: id
5. Function Group: security
6. REST Operations: CREATE READ UPDATE

Attributes	Data Type	Related Object	Flags
id	INTEGER		UNIQUE
role_obj	SREL	role	
web_form_obj	SREL	web_form	
menu_bar_obj	SREL	menu_bar	
is_default	INTEGER		
last_mod_dt	DATE		

role_tab Object

The object details are as follows:

1. Associated Table: usp_role_tab
2. Factories: default
3. REL_ATTR: id
4. Common Name: id
5. Function Group: Security

6. REST Operations: CREATE READ UPDATE

Attributes	Data Type	Related Object	Flags
id	INTEGER		UNIQUE
sequence	INTEGER		
role_obj	SREL	role	
tab_obj	SREL	tab	
last_mod_dt	DATE		

Server Objects

This article contains the following topics:

srvr_aliases Object

The object details are as follows:

1. Associated Table: Server_Aliases
2. Factories: default
3. REL_ATTR: id
4. Common Name: alias_name
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
alias_name	alias_name	STRING		REQUIRED
delete_flag	del	INTEGER	actbool enum	REQUIRED
host_addr	host_addr	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
zone_id	zone_id	INTEGER	srvr_zones id	REQUIRED

srvr_zones Object

The object details are as follows:

1. Associated Table: Server_Zones
2. Factories: default
3. REL_ATTR: id
4. Common Name: zone_name

5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
is_default	is_default	INTEGER	bool_tab enum	
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
zone_name	zone_name	STRING		REQUIRED

Service Level and Service Contract Mapping Objects

This article contains the following topics:

sdsc Object

The object details are as follows:

1. Associated Table: Service_Desc
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: service_level
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
code	code	STRING		UNIQUE REQUIRED S_KEY
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
owning_contract	owning_contract	INTEGER	svc_contract id	
persistent_id	persid	STRING		
rank	rank	INTEGER		
schedule	schedule	STRING	bpwshft persid	
sym	sym	STRING		REQUIRED S_KEY
violation_cost	violation_cost	INTEGER		
tgttpls	tgttpls	BREL	tgt_tgttpls_srvtypes sdsc	

sdsc_map Object

The object details are as follows:

1. Associated Table: SLA_Contract_Map
2. Factories: default
3. REL_ATTR: id
4. Common Name: map_persid
5. Function Group: service_level
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
map_contract	map_contract	INTEGER	svc_contract id	REQUIRED
map_persid	map_persid	STRING		
map_sdsc	map_sdsc	STRING	srv_desc code	
persistent_id	persid	STRING		

Session Objects

This topic contains the following information:

session_log Object

The object details are as follows:

1. Associated Table: session_log
2. Factories: default
3. REL_ATTR: id
4. Common Name: id
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
contact	contact	UUID	ca_contact uuid	
id	id	INTEGER		UNIQUE REQUIRED KEY
login_time	login_time	LOCAL_TIME		
logout_time	logout_time	LOCAL_TIME		
policy	policy	INTEGER	sapolicy id	
session_id	session_id	INTEGER		
session_type	session_type	INTEGER	session_type id	
status	status	INTEGER		

session_type Object

The object details are as follows:

1. Associated Table: session_type
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
sym	sym	STRING		REQUIRED S_KEY

Survey Objects

This article contains the following topics:

svy_qtpl Object

The object details are as follows:

1. Associated Table: Survey_Question_Template
2. Factories: default
3. REL_ATTR: id
4. Common Name: text
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
include_qcomment	include_qcomment	INTEGER		
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
mult_resp_flag	mult_resp_flag	INTEGER		
owning_survey	owning_survey	INTEGER	survey_tpl id	
persistent_id	persid	STRING		
qcomment_label	qcomment_label	STRING		

resp_required	resp_required	INTEGER		
sequence	sequence	INTEGER		REQUIRED
text	txt	STRING		

svy_ques Object

The object details are as follows:

1. Associated Table: Survey_Question
2. Factories: default
3. REL_ATTR: id
4. Common Name: text
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
include_qcomment	include_qcomment	INTEGER		
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
mult_resp_flag	mult_resp_flag	INTEGER		
owning_survey	owning_survey	INTEGER	survey id	
persistent_id	persid	STRING		
qcomment	qcomment	STRING		
qcomment_label	qcomment_label	STRING		
resp_required	resp_required	INTEGER		
response	response	INTEGER		REQUIRED
sequence	sequence	INTEGER		REQUIRED
text	txt	STRING		

svy_tpl Object

The object details are as follows:

1. Associated Table: Survey_Template
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: admin

6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
comment_label	comment_label	STRING		
Survey Completion Message	conclude_text	STRING		
cycle_counter	cycle_counter	INTEGER		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
include_comment	include_comment	INTEGER		REQUIRED
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
Submit Cycle	submit_cycle	INTEGER		
Survey Name	sym	STRING		UNIQUE REQUIRED S_KEY
tracking_flag	tracking_flag	INTEGER		
Survey Introduction	description	STRING		

svystat Object

The object details are as follows:

1. Associated Table: Survey_Stats
2. Factories: default
3. REL_ATTR: id
4. Common Name:
5. Function Group: admin

Attribute	DB Field	Data Type	SREL References	Flags
cyc_counter	cyc_counter	INTEGER		
cycle	cycle	INTEGER		
delete_flag	del	INTEGER	actbool enum	REQUIRED
eval_counter	eval_counter	INTEGER		
id	id	INTEGER		UNIQUE REQUIRED KEY
persistent_id	persid	STRING		
sub_counter	sub_counter	INTEGER		
tplid	tplid	INTEGER	survey_tpl id	

svy_atpl Object

The object details are as follows:

1. Associated Table: Survey_Answer_Template
2. Factories: default
3. REL_ATTR: id
4. Common Name: text
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
owning_ survey_ question	own_srvy_question	INTEGER	survey_qtpl id	
persistent_id	persid	STRING		
sequence	sequence	INTEGER		REQUIRED
text	txt	STRING		

svy_ans Object

The object details are as follows:

1. Associated Table: Survey_Answer
2. Factories: default
3. REL_ATTR: id
4. Common Name: text
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
owning_survey_question	own_srvy_question	INTEGER	survey_question id	
persistent_id	persid	STRING		
selected	selected	INTEGER		
sequence	sequence	INTEGER		REQUIRED
text	txt	STRING		

svytrk Object

The object details are as follows:

1. Associated Table: Survey_Tracking
2. Factories: default
3. REL_ATTR: id
4. Common Name: object_type
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
cntid	cntid	UUID	ca_contact uuid	
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
notif_dt	notif_dt	LOCAL_TIME		
object_id	object_id	INTEGER		
object_type	object_type	STRING		
persistent_id	persid	STRING		
recv_dt	recv_dt	LOCAL_TIME		
status	status	INTEGER		
tplid	tplid	INTEGER	survey_tpl id	

tgt_tgttpls_srvtypes Object

The object details are as follows:

1. Associated Table: target_tgttpls_srvtypes
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: service_level
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	SREL	actbool.enum	REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt.id	
tgt_time_tpl	tgt_time_tpl	SREL	tgt_time_tpl	REQUIRED
sdsc	sdsc	SREL	sdsc.code	REQUIRED
target_duration	target_duration	DURATION		REQUIRED

sa_survey Object

The object details are as follows:

1. Associated Table: sa_survey
2. Factories: default
3. REL_ATTR: id
4. Common Name: surveyName
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE NOT_NULL KEY
surveyName	surveyName	STRING 32		REQUIRED
question	question	STRING 512		REQUIRED
responseType	responseType	INTEGER		REQUIRED
isDeleted	isDeleted	INTEGER		
last_mod_by	last_mod_by	UUID	ca_contact	
last_mod_dt	last_mod_dt	LOCAL_TIME		

sa_survey_localized Object

The object details are as follows:

1. Associated Table: sa_survey_localized
2. Factories: default
3. REL_ATTR: id
4. Common Name: question
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE NOT_NULL KEY
surveyID	surveyID	SREL	sa_survey	REQUIRED
localizationID	localizationID	SREL	sa_localization	REQUIRED
question	question	STRING 512		REQUIRED
last_mod_by	last_mod_by	UUID	ca_contact	
last_mod_dt	last_mod_dt	LOCAL_TIME		
tenant	tenant	UUID	ca_tenant	

sa_survey_result Object

The object details are as follows:

1. Associated Table: sa_survey_result
2. Factories: default
3. REL_ATTR: id
4. Common Name: SurveyComment

5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE NOT_NULL KEY
surveyID	surveyID	SREL	sa_survey	REQUIRED
selfServe SessionID	selfServe SessionID	SREL	sa_self_serve_ session	REQUIRED
scriptID	scriptID	SREL	sa_script	
response	response	INTEGER		
completion	completion	INTEGER		REQUIRED
SurveyComment	SurveyComment	STRING 512		
surveyEpoch	surveyEpoch	LOCAL_TIME		
last_mod_by	last_mod_by	UUID	ca_contact	
last_mod_dt	last_mod_dt	LOCAL_TIME		
tenant	tenant	UUID	ca_tenant	

mgs Object

The object details are as follows:

1. Associated Table: Managed_Survey
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
open_date	create_date	LOCAL_TIME		
active	del	INTEGER	bool_tab enum	REQUIRED
close_date	end_date	LOCAL_TIME		
group	group_id	UUID		
id	id	INTEGER		UNIQUE REQUIRED KEY
initial_method	initial_method	INTEGER	ct_mth id	
initial_msgbody	initial_msgbody	STRING		
initial_msgtitle	initial_msgtitle	STRING		
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
assignee	owner	UUID	ca_contact uuid	
persistent_id	persid	STRING		
reminder_method	reminder_method	INTEGER	ct_mth id	

reminder_ msgbody	reminder_ msgbody	STRING		
reminder_msgtitle	reminder_msgtitle	STRING		
start_dt	start_date	LOCAL_TIME		
status	status	STRING	mgsstat code	
sym	sym	STRING		UNIQUE REQUIRED S_KEY
tplid	tplid	INTEGER	survey_tpl id	

survey Object

The object details are as follows:

1. Associated Table: Survey
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
comment_label	comment_label	STRING		
conclude_text	conclude_text	STRING		
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
include_comment	include_comment	INTEGER		REQUIRED
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
comment	nx_comment	STRING		
object_id	object_id	INTEGER		
object_type	object_type	STRING		
persistent_id	persid	STRING		
sym	sym	STRING		S_KEY

Target Time

This topic contains the following information:

tgt_time Object

The object details are as follows:

1. Associated Table: target_time
2. Factories: default
3. REL_ATTR: id

4. Common Name: sym
5. Function Group: service_level
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	INTEGER	actbool	REQUIRED
sym	sym	STRING 60		REQUIRED
last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt.id	
target_duration	target_duration	DURATION		REQUIRED
condition	condition	SREL	macro.persistent	
condition_ outcome	condition_ outcome	SREL	true_false.enum	
service_type	service_type	STRING	sdsc.code	REQUIRED
object_type	object_type	STRING	event_prod_list. sym	REQUIRED
object_id	object_id	INTEGER		REQUIRED
set_actual	set_actual	SREL	bool.enum	
reset_actual	reset_actual	SREL	bool.enum	
lock_target	lock_target	SREL	bool.enum	
cost	cost	STRING 255		
target_time	target_time	DATE		
actual_time	actual_time	DATE		
time_left	time_left	DURATION		
work_shift	work_shift	SREL	wrkshft.persistent_id	
_mapped_cr	_mapped_cr	SREL	cr.persistent_id	
_mapped_chg	_mapped_chg	SREL	chg.persistent_id	
_mapped_iss	_mapped_iss	SREL	iss.persistent_id	
target_tpl	target_tpl	SREL	tgt_time_tpl.id	
suppress_log	INTEGER			

tgt_time_tpl Object

The object details are as follows:

1. Associated Table: target_time_tpl
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: service_level
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
delete_flag	del	SREL	actbool.enum	REQUIRED
sym	sym	STRING 60		REQUIRED

last_mod_dt	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt.id	
target_duration	target_duration	DURATION		REQUIRED
condition	condition	SREL	macro.persistent_id	
condition_outcome	condition_outcome	SREL	true_false.enum	
object_type	object_type	SREL	event_prod_list.sym	REQUIRED
set_actual	set_actual	SREL	bool.enum	
reset_actual	reset_actual	SREL	bool.enum	
cost	cost	STRING 255		
work_shift	work_shift	SREL	wrkshft.persistent_id	
srvtypes	srvtypes	BREL	tgt_tgttpls_srvtypes.tgt_time_tpl	

Task Objects

This article contains the following topics:

tskstat Object

The object details are as follows:

1. Associated Table: Task_Status
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
allow_accumulate	allow_accumulate	INTEGER		REQUIRED
allow_task_update	allow_task_update	INTEGER		REQUIRED
code	code	STRING		UNIQUE REQUIRED S_KEY
delete_flag	del	INTEGER	actbool enum	REQUIRED
do_next_task	do_next_task	INTEGER		REQUIRED
hold	hold	INTEGER		REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
is_internal	is_internal	INTEGER		REQUIRED
last_mod_by	last_mod_by	UUID	ca_contact uuid	
modified_date	last_mod_dt	LOCAL_TIME		
no_update_msg	no_update_msg	STRING		

persistent_id	persid	STRING		
sym	sym	STRING		REQUIRED
task_complete	task_complete	INTEGER		REQUIRED

tskty Object

The object details are as follows:

1. Associated Table: Task_Type
2. Factories: default
3. REL_ATTR: code
4. Common Name: sym
5. Function Group: workflow_reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
code	code	STRING		UNIQUE REQUIRED S_KEY
delete_flag	del	INTEGER	actbool enum	REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
modified_date	last_mod_dt	LOCAL_TIME		
persistent_id	persid	STRING		
sym	sym	STRING		REQUIRED S_KEY

Tenant Objects

This topic contains the following information:

tenant Object

The object details are as follows:

Associated Table: ca_tenant

Factories: default

REL_ATTR: id

Common Name: name

Function Group: tenant_admin

REST Operations: CREATE READ UPDATE

Attributes	Data Type	SREL References	Flags
id	UUID		

producer_id	STRING 20		
persistent_id	STRING 60		
name	STRING 255		
tenant_ number	STRING 30		
service_provider	INTEGER		
contact	SREL	cnt.id	
logo	STRING 255		
description	STRING 1024		
phone_number	STRING 255		
fax_number	STRING 255		
alt_phone	STRING 255		
location	SREL	loc.id	
delete_flag	SREL	actbool.enum	
version_number	INTEGER		
creation_user	STRING 64		
creation_date	LOCAL_ TIME		
last_update_user	LOCAL_ TIME		
last_update_date	LOCAL_ TIME		
ldap_tenant_group	SREL	ldap_group.id	
groups	BREL	tgm_groups. tenant_id	
tenant	SREL	tenant.id	REQUIRED
terms_of_usage	SREL	ca_tou	

tenant_group Object

The object details are as follows:

Associated Table: ca_tenant_group

Factories: default

REL_ATTR: id

Common Name: name

Function Group: admin

REST Operations: CREATE READ UPDATE

Attributes	Data Type	SREL References	Flags
id	UUID		
producer_id	STRING 20		
persistent_id	STRING 60		
name	STRING 255		
description	STRING 1024		
delete_flag	SREL	actbool.enum	
version_number	INTEGER		
creation_user	STRING 64		
creation_date	LOCAL_ TIME		
last_update_user	STRING 64		
last_update_date	LOCAL_TIME		
members	BREL	tgm_members. tenant_group	

tenant_group_member Object

The object details are as follows:

Associated Table: ca_tenant_group_member

Factories: default

REL_ATTR: persistent_id

Common Name: creation_user

Function Group: admin

REST Operations: CREATE READ UPDATE

Attributes	Data Type	SREL References	Flags
id	UUID		
producer_id	STRING 20		
persistent_id	STRING 60		
tenant_id	SREL	tenant.id	
tenant_group	SREL	tenant_group. id	
creation_date	LOCAL_ TIME		
creation_user	STRING 64		
tenant	SREL	tenant.id	REQUIRED

USP

This article contains the following topics:

USP_PREFERENCES Object

The object details are as follows:

1. Associated Table: USP_PREFERENCES
2. Factories: default
3. REL_ATTR: id
4. Common Name: id
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
Contact ID	ANALYST_ID	UUID	ca_contact uuid	
Knowledge Categories - Documents Per Page	ARC_DOCS_TO_DISPLAY	INTEGER		
Assignee	ASSIGNEE	INTEGER		
Author	AUTHOR	INTEGER		
FAQ Rating	BU_RESULT	INTEGER		
Mouseover Menus	CLASSIC_RESULTSET_CONTEXT	INTEGER		
Created Via	CREATED_VIA	INTEGER		
Creation Date	CREATION_DATE	INTEGER		
Current Task	CURRENT_ACTION	INTEGER		
Custom 1	CUSTOM1	INTEGER		
Custom 2	CUSTOM2	INTEGER		
Custom 3	CUSTOM3	INTEGER		
Custom 4	CUSTOM4	INTEGER		
Custom 5	CUSTOM5	INTEGER		
Custom Num 1	CUSTOM_NUM1	INTEGER		
Custom Num 2	CUSTOM_NUM2	INTEGER		
Document ID	DOC_ID	INTEGER		
Document Template	DOC_TEMPLATE	INTEGER		
Document Type	DOC_TYPE	INTEGER		
Document Version	DOC_VERSION	INTEGER		
Expiration Date	EXPIRATION_DATE	INTEGER		
GLOBALSD_ACTIVE_ZONE	GLOBALSD_ACTIVE_ZONE	INTEGER		
Hits	HITS	INTEGER		
id	ID	INTEGER		REQUIRED KEY

INBOX_COUNTER	INBOX_COUNTER	INTEGER		
Initiator	INITIATOR	INTEGER		
Item	ITEM	INTEGER		
KT_REPORT_CARD_APAST_DAYS	KT_REPORT_CARD_PAST_DAYS	INTEGER		
KT_REPORT_CARD_SCREEN_DEFAULT	KT_REPORT_CARD_SCREEN_DEFAULT	INTEGER		
Last Accepted Date	LAST_ACCEPTED_DATE	INTEGER		
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
Modify Date	MODIFY_DATE	INTEGER		
Knowledge Documents View Mode	ONE_B_DOC_VIEW_MODE	INTEGER		
Knowledge Documents Per Page	ONE_B_DOCS_TO_DISPLAY	INTEGER		
Knowledge Documents Show Details Flag	ONE_B_HIDE_DETAILS	INTEGER		
EBR Match Type	ONE_B_MATCH_TYPE	INTEGER		
EBR Search Fields	ONE_B_SEARCH_FIELDS	INTEGER		
EBR Search Order	ONE_B_SEARCH_ORDER	STRING		
EBR Search Type	ONE_B_SEARCH_TYPE	INTEGER		
EBR Word Parts	ONE_B_WORD_PARTS	INTEGER		
Owner	OWNER	INTEGER		
Primary Category	PRIMARY_INDEX	INTEGER		
Workflow Priority	PRIORITY	INTEGER		
Product	PRODUCT	INTEGER		
Published Date	PUBLISHED_DATE	INTEGER		
Review Date	REVIEW_DATE	INTEGER		
Solution Count	SD_ACCEPTED_HITS	INTEGER		
Impact	SD_IMPACT	INTEGER		
Priority	SD_PRIORITY	INTEGER		
Root Cause	SD_ROOTCAUSE	INTEGER		
SD_SEARCH_FIELDS_CR	SD_SEARCH_FIELDS_CR	INTEGER		
SD_SEARCH_FIELDS_ISS	SD_SEARCH_FIELDS_ISS	INTEGER		
Severity	SD_SEVERITY	INTEGER		
Urgency	SD_URGENCY	INTEGER		
Start Date	START_DATE	INTEGER		

Status	STATUS	INTEGER		
Subject Expert	SUBJECT_EXPERT	INTEGER		
User Defined ID	USER_DEF_ID	INTEGER		
WEB_LAST_LOGIN	WEB_LAST_LOGIN	LOCAL_TIME		
WEB_POPUP1_HEIGHT	WEB_POPUP1_HEIGHT	INTEGER		
WEB_POPUP1_WIDTH	WEB_POPUP1_WIDTH	INTEGER		
WEB_POPUP2_HEIGHT	WEB_POPUP2_HEIGHT	INTEGER		
WEB_POPUP2_WIDTH	WEB_POPUP2_WIDTH	INTEGER		
WEB_POPUP3_HEIGHT	WEB_POPUP3_HEIGHT	INTEGER		
WEB_POPUP3_WIDTH	WEB_POPUP3_WIDTH	INTEGER		
WEB_POPUP4_HEIGHT	WEB_POPUP4_HEIGHT	INTEGER		
WEB_POPUP4_WIDTH	WEB_POPUP4_WIDTH	INTEGER		
WEB_ROLE_ID	WEB_ROLE_ID	STRING		
WEB_PREFERENCES	WEB_PREFERENCES	INTEGER		
WEB_SUPPRESS_TOUR	WEB_SUPPRESS_TOUR	INTEGER		
WEB_TOOLBAR_TAB	WEB_TOOLBAR_TAB	INTEGER		
WF_TEMPLATE	WF_TEMPLATE	INTEGER		

usp_conflict

Change order conflicts.

Attribute	Data Type	SREL References	Flags
id	INTEGER		UNIQUE NOT_NULL KEY
persid	STRING 30		
del	INTEGER		NOT_NULL
comments	STRING 1000		
conflict_begin	LOCAL_TIME		
conflict_change	INTEGER	Change_Request	NOT_NULL
conflict_ci	UUID	ca_owned_resource	
conflict_end	LOCAL_TIME		
conflict_status	STRING 30	usp_conflict_status	NOT_NULL
conflict_type	STRING 30	usp_conflict_type	NOT_NULL
creation_dt	LOCAL_TIME		
is_resolved	INTEGER		NOT_NULL
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
source_change	INTEGER	Change_Request	NOT_NULL

tenant	UUID	ca_tenant	

usp_conflict_chg

Change order conflict change.

Attribute	Data Type	SREL References	Flags
id	INTEGER		UNIQUE NOT_NULL KEY
change	INTEGER	Change_Request	NOT_NULL
conflict	INTEGER	usp_conflict	NOT_NULL
conflict_change	INTEGER	Change_Request	
is_cause	INTEGER		NOT_NULL
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
tenant	UUID	ca_tenant	

usp_conflict_status

Change order conflict status.

Attribute	Data Type	SREL References	Flags
id	INTEGER		UNIQUE NOT_NULL KEY
persid	STRING 30		
del	INTEGER		NOT_NULL
code	STRING 30		UNIQUE NOT_NULL
description	STRING 1000		
is_resolved	INTEGER		NOT_NULL
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
sym	STRING 60		UNIQUE NOT_NULL S_KEY
tenant	UUID	ca_tenant	

usp_conflict_type

Change order conflict type.

Attribute	Data Type	SREL References	Flags
id	INTEGER		UNIQUE NOT_NULL KEY
persid	STRING 30		

del	INTEGER		NOT_NULL
code	STRING 30		UNIQUE NOT_NULL
description	STRING 1000		
icon	STRING 1000		
last_mod_dt	LOCAL_TIME		
last_mod_by	UUID	ca_contact	
sym	STRING 60		UNIQUE NOT_NULL S_KEY
tenant	UUID	ca_tenant	

usp_exlist_format Object

The object details are as follows:

1. Associated Table: usp_exlist_format
2. Factories: default
3. REL_ATTR: id
4. Common Name: file_extension
5. Function Group: admin
6. Tenant: optional

Attribute	DB Field	Data Type	SREL References	Flags
file_extension		STRING		REQUIRED
mime_type		STRING		REQUIRED
xslt_name		STRING		
delete_flag		del	actbool enum	REQUIRED. On new default: 0

USP_PROPERTIES Object

The object details are as follows:

1. Associated Table: USP_PROPERTIES
2. Factories: default
3. REL_ATTR: id
4. Common Name: PROPERTY_NAME
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	ID	INTEGER		REQUIRED KEY
Last Modified Date	LAST_MOD_DT	LOCAL_TIME		
Property Default	PROPERTY_DEFAULT	STRING		
Property Description	PROPERTY_DESCRIPTION	STRING		

Property Name	PROPERTY_NAME	STRING		S_KEY
Property Type	PROPERTY_TYPE	STRING		
Property Value	PROPERTY_VALUE	STRING		

usp_session_ticket Object

The object details are as follows:

1. Associated Table: usp_session_ticket
2. REL_ATTR: id
3. Common Name: session_persid
4. Function Group: change_reference
5. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
id	id	INTEGER		UNIQUE REQUIRED KEY
producer_id		LOCAL_STRING 20		
persistent_id	persid	STRING 30		
delete_flag	del	SREL	actbool.enum	REQUIRED
sym	sym	STRING 60		REQUIRED
description	description	STRING 240		
modified_date	last_mod_dt	DATE		
last_mod_by	last_mod_by	SREL	cnt.id	

window Object

The object details are as follows:

1. Associated Table: usp_window
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References
sym	sym	STRING 60	
window_type	window_type	INTEGER	usp_window_type
start_date	start_date	LOCAL_TIME	
end_date	end_date	LOCAL_TIME	
final_end_date	final_end_date	LOCAL_TIME	
timezone	timezone	STRING 30	
non_global	is_non_global	INTEGER	
icon	icon	INTEGER	

recurs	recurs	INTEGER	
recurrence_ interval	recurrence_interval	INTEGER	
sunday	sunday	INTEGER	
monday	monday	INTEGER	
tuesday	tuesday	INTEGER	
wednesday	wednesday	INTEGER	
thursday	thursday	INTEGER	
friday	friday	INTEGER	
saturday	saturday	INTEGER	
occurrence	occurrence	INTEGER	
description	description	STRING 400	
legend	legend	STRING 100	
color	color	STRING 100	
bgcolor	bgcolor	STRING 100	
style	style	STRING 100	
last_mod_dt	last_mod_dt	LOCAL_TIME	
last_mod_by	last_mod_by	UUID	ca_contact
delete_flag	del	INTEGER	
tenant	tenant	UUID	ca_tenant

window_type Object

The object details are as follows:

1. Associated Table: usp_window_type
2. Factories: default
3. REL_ATTR: id
4. Common Name: sym
5. Function Group: reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
sym	sym	STRING 60		
description	description	STRING 100		
last_mod_dt	last_mod_dt	LOCAL_TIME		
last_mod_by	last_mod_by	UUID	ca_contact	
delete_flag	del	INTEGER	actbool.enum	REQUIRED

Web Form

This article contains the following topics:

web_form Object

The object details are as follows:

1. Associated Table: usp_web_form
2. Factories: default
3. REL_ATTR: id
4. Common Name: name
5. Function Group: Security
6. REST Operations: CREATE READ UPDATE

Attribute Name	Data Type	Relationship Object	Flags
id	INTEGER		UNIQUE
name	STRING		REQUIRED
code	STRING		UNIQUE; REQUIRED
delete_flag	SREL	actbool	REQUIRED
description	STRING		
resource	STRING		
wf_type	INTEGER		
dflt_for_obj	STRING		
last_mod_by	SREL	cnt	
last_mod_dt	DATE		

web_form_pref Object

The object details are as follows:

1. Associated Table: usp_web_form_pref
2. Factories: default
3. REL_ATTR: id
4. Common Name: frame_name
5. Function Group: Reference
6. REST Operations: CREATE READ UPDATE

Attribute Name	Data Type	Relationship Object	Flags
id	INTEGER		UNIQUE
contact	SREL	cnt	
web_form_obj	SREL	web_form	
role_tab_obj	SREL	role_tab	
frame_name	STRING		
last_mod_dt	DATE		

Web Screen Painter Objects

This topic contains the following information:

wspcol Object

The object details are as follows:

1. Associated Table: wspcol
2. Factories: default
3. REL_ATTR: id
4. Common Name: column_name
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
addl_info	addl_info	STRING		
column_name	column_name	STRING		REQUIRED
dbms_name	dbms_name	STRING		
display_name	display_name	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
is_cluster	is_cluster	INTEGER		
is_descending	is_descending	INTEGER		
is_indexed	is_indexed	INTEGER		
is_local	is_local	INTEGER		
is_REQUIRED	is_REQUIRED	INTEGER		
is_order_by	is_order_by	INTEGER		
is_required	is_required	INTEGER		
is_skey	is_skey	INTEGER		
is_unique	is_unique	INTEGER		
is_write_new	is_write_new	INTEGER		
is_wsp	is_wsp	INTEGER		
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
on_ci_set	on_ci_set	STRING		
on_new_default	on_new_default	STRING		
persistent_id	persid	STRING		
schema_name	schema_name	STRING		
status	status	INTEGER		REQUIRED
string_len	string_len	INTEGER		
table_name	table_name	STRING		REQUIRED
type	type	INTEGER		
xrel_table	xrel_table	STRING		

wsptbl Object

The object details are as follows:

1. Associated Table: wsptbl
2. Factories: default
3. REL_ATTR: id
4. Common Name: table_name
5. Function Group: admin
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
common_name	common_name	STRING		
dbms_name	dbms_name	STRING		
display_group	display_group	STRING		
display_name	display_name	STRING		
function_group	function_group	STRING		
id	id	INTEGER		UNIQUE REQUIRED KEY
is_local	is_local	INTEGER		
is_wsp	is_wsp	INTEGER		
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_TIME		
methods	methods	STRING		
persistent_id	persid	STRING		
rel_attr	rel_attr	STRING		
schema_name	schema_name	STRING		
sort_by	sort_by	STRING		
status	status	INTEGER		REQUIRED
table_name	table_name	STRING		REQUIRED
triggers	triggers	STRING		

Workflow Objects

This topic contains the following information:

wf Object

The object details are as follows:

1. Associated Table: Workflow_Task
2. Factories: default
3. REL_ATTR: id

4. Common Name: description
5. Function Group: change_mgr
6. REST Operations: CREATE READ UPDATE DELETE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
actual_duration	actual_duration	DURATION		
asset	asset	UUID	ca_owned_ resource uuid	
assignee	assignee	UUID	ca_contact uuid	
completion_ date	completion_ date	LOCAL_ TIME		
comments	comments	STRING		
cost	cost	INTEGER		
creator	creator	UUID	ca_contact uuid	
date_created	date_created	LOCAL_ TIME		
delete_flag	del	INTEGER	actbool enum	REQUIRED
done_by	done_by	UUID	ca_contact uuid	
est_completion_date	est_comp_date	LOCAL_ TIME		
est_cost	est_cost	INTEGER		
est_duration	est_duration	DURATION		
group	group_id	UUID		
group_task	group_task	INTEGER		REQUIRED
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
last_mod_dt	last_mod_dt	LOCAL_ TIME		
chg	object_id	INTEGER	chg id	REQUIRED
object_type	object_type	STRING		REQUIRED
persistent_id	persid	STRING		
sequence	sequence	INTEGER		REQUIRED
start_date	start_date	LOCAL_ TIME		
status	status	STRING	tskstat code	
support_lev	support_lev	STRING	srv_desc code	
task	task	STRING	tskty code	REQUIRED
wf_template	wf_template	INTEGER	wftpl id	

wftpl Object

The object details are as follows:

1. Associated Table: Workflow_Task_Template
2. Factories: default

3. REL_ATTR: id
4. Common Name: id
5. Function Group: workflow_reference
6. REST Operations: CREATE READ UPDATE

Attribute	DB Field	Data Type	SREL References	Flags
description	description	STRING		
assignee	assignee	UUID	ca_contact uuid	
auto_assign	auto_assign	INTEGER		
delete_flag	del	INTEGER	actbool enum	REQUIRED
deleteable	deleteable	INTEGER		REQUIRED
est_cost	est_cost	INTEGER		
est_duration	est_duration	DURATION		
group	group_id	UUID		
id	id	INTEGER		UNIQUE REQUIRED KEY
last_mod_by	last_mod_by	UUID	ca_contact uuid	
modified_date	last_mod_dt	LOCAL_TIME		
object_attrname	object_attrname	STRING		REQUIRED
object_attrval	object_attrval	INTEGER		REQUIRED
object_type	object_type	STRING		REQUIRED
persistent_id	persid	STRING		
sequence	sequence	INTEGER		REQUIRED
service_type	service_type	STRING	srv_desc code	
task	task	STRING	tskty code	REQUIRED

CA Service Catalog Glossary

accounts

Accounts are part of a business unit. Accounts are used to subscribe and request services. Charges are applied at the account level in Service Accounting Component.

action

An *action* is the smallest unit of work in the Rule Management Engine. This task is executed when a rule gets notified. Examples include sending an email, running a script, and running Java code.

Activity Based Costing (ABC)

Activity Based Costing is a costing methodology that is used to trace overhead costs directly to cost objects. The cost objects can be products, processes, services, or customers. This costing methodology helps managers to make the right decisions regarding product mix and competitive strategies.

Activity Based Management (ABM)

Activity Based Management uses Activity Based Cost information to allocate resources more effectively during budgeting and planning processes.

adjustments

Adjustments are credits or debits applied to services, individual charges of a Service Option Group, and SLA violations. These adjustments are either a fixed dollar amount or a percentage. Several types of general and SLA violation adjustments are available.

batch print job

A *batch print job* is a collection of invoices that have been grouped according to the specified credentials. After the collection has been created, it remains unchanged. The invoices can then be printed as a whole or individually at any time.

business unit

The *business unit* is a branch in the organizational structure. This organizational unit has some characteristics of the service provider and all characteristics of a sub-department.

CA Embedded Entitlements Manager (CA EEM)

CA Embedded Entitlements Manager provides enterprises with the ability to manage the identities of their employees, partners, and customers. CA EEM can also provision web services, and applications to the users, and define authentication and authorization policies.

cost allocation

Cost allocation is a method to determine the cost of services that are provided to users of that service. This method does not determine the price of the service, but rather it determines the cost of providing the service.

cost element

A *cost element* is a resource that is used to subdivide costs corresponding to the consumption of a particular service. Cost Element is the amount that is paid for a resource that an activity consumes and included in the activity cost pool. Such information helps validating that the overhead calculated at the beginning of the process is the same as the overhead that is assigned to each individual product using ABC.

cost pool

A *cost pool* is a grouping of all cost elements that are associated with an activity.

Dashboard Library

A *Dashboard Library* is a hierarchical tree structure of name spaces facilitating the sharing of information. The Dashboard Library does not hold actual data. Instead, it stores information about how the data can be accessed.

data collector (DC)

A *data collector* collects data for particular metrics.

data mediation profile

The *data mediation profile* is a definition of the data feed and structure of the external data. Data Mediation Profile also provides features to manipulate and normalize usage event data.

data object

A *data object* defines the data for a chart or table. The data object source can be a SQL database, comma-separated file, or any data source that a Java plug-in can access. Data objects are managed in folders to help categorize them by purpose.

data view

A *data view* formats the data that a data object produces. Data can be presented in tabular or chart format or both. Data views are managed in folders to help categorize them by purpose.

direct costs

Direct costs are those costs that can be easily traced and assigned to activities.

dynamic invoice group

Dynamic invoice groups are invoice groups that contain account lists that are generated dynamically. The invoice groups are generated based on the specified criteria that are stored as a data object.

event filter

An *event filter* can be specified as part of a rule and is used to refine the event. The event filter allows actions that are associated with a rule to be invoked only if certain event conditions are met.

failover

Failover is the process of maintaining an up-to-date copy of a database on an alternate system for backup. The traditional failover architecture is comprised of one system running the application and the other is idle in standby mode. The standby system is ready to take over if the primary system fails. An alternative architecture includes clustering.

invoice group

An *invoice group* is comprised of a set of accounts that can be run together in one instance.

IT Infrastructure Library (ITIL)

ITIL is a set of practices that is used to aid the implementation of a framework for IT Service Management. This customizable framework defines how service management is applied within an organization. ITIL is used across the world as the de-facto standard for best practice in the provision of IT Service.

Although ITIL covers a number of areas, its main focus is on IT Service Management.

Java Runtime Environment (JRE)

Java Runtime Environment or Java Runtime provides the minimum requirements for executing a Java application. JRE consists of the Java Virtual Machine (JVM), core classes, and supporting files. JRE is part of the Java Development Kit (JDK), a set of programming tools for developing Java applications.

Management Database (MDB)

The *Management Database (MDB)* is the CA Management Database schema. The MDB is a common enterprise data repository that integrates CA product suites. The MDB provides a unified database schema for the management data that all CA products store. The CA products can be distributed or mainframe. Use of the MDB with CA products enables full integration for managing your IT infrastructure. The MDB integrates management data from all IT disciplines and CA products. Customers can extend the MDB Schema to include other IT management data from non-CA software products and tools.

pool worksheets

Pool worksheets are used to trace cost elements to activities to obtain dollar values for each related activity cost pool. Formulas can be used to adjust costs spanning across fiscal periods and sets.

process definition

A *process definition* is a representation of one of your business processes. A process definition is comprised of nodes, events, roles, actors, work, and the criteria for process logic. Running a process definition creates a process instance. You can create multiple process instances of the same process definition.

process instance

A *process instance* represents what is actually happening. By running a process definition, you create a process instance. You can create multiple process instances of the same process definition. Process instances are sometimes called a process definition instance.

proration

Proration is the process of dividing the cost of a subscription over the billing cycle period of an account.

report layout

A *report layout* is used to present multiple report elements as one report. With report layouts, you can design customized reports by using objects such as text, URLs, and multiple data view objects. You can choose sizes, colors, borders/styles, set up snap to grid, and other tools to obtain the desired result. Layouts are managed in folders to help categorize them by purpose.

routing nodes

Routing nodes define the branching of control flow in a Workflow process definition. Choice routing nodes allow one route for the next activity. Parallel routing nodes allow more than one route to be taken.

rule

A *rule* is associated with an event. A rule can have a set of filter conditions that define when the rule is applied. When the filter conditions are satisfied and the rule is enabled, the rule actions are launched.

service

A *service* is a product or an application that your clients can access through a request or a subscription. A service is defined by associating service option groups to the service before it can be subscribed to or requested.

service catalog

A *service catalog* consists of services that a business unit or an entire enterprise publishes. Services are built of one or more service option groups that describe IT services and how to charge for them. The service catalog enables an organization to model its business units and manage the users accounts contained within those units. The service catalog also defines the inclusion, inheritance, dependencies, and associative relationships between each of the published items in the catalog. The published items can be services, service option groups, and service option group definitions. Accounts and users subscribe to or request catalog services. Services in the service catalog can be organized into folders. Each service contains detailed information about the price of the service. Services can represent one or more metrics and include service level agreements.

A *service catalog* is also known as a *catalog*.

Service Level Agreement (SLA)

A *Service Level Agreement* is a contract. This contract specifies the level of service to be provided while the agreement is in effect.

Service Level Objective (SLO)

A *Service Level Objective* is the building block for an SLA. The SLO provides the business objectives or business rules that set the warning and violation threshold levels. The SLO thus determines the success or failure of the overall SLA.

service option element

A *service option element* is the smallest unit of the catalog. This unit defines a piece of text, fee, or an application in a service. Service option elements are grouped or classified in service option groups.

service option group

A *service option group* contains the cost that is associated with subscribing to a service. The cost can be based on transaction, usage-based billing, or rate. Service Option Group can be composed of many types of chargeable and non-chargeable items. For example, rates, applications and agreements.

service worksheets

Service worksheets contain services, service option groups, and service option elements. Service worksheets are used to apply applicable costs to associated services. These costs can be direct or derived from associated activity cost pools. Formulas can be used to adjust costs spanning across fiscal periods and sets.

Simple Object Access Protocol (SOAP)

SOAP is a lightweight, XML-based protocol for exchanging information in a decentralized, distributed environment. SOAP uses HTTP as the transport protocol and XML as the payload encoding scheme. Developers can use any programming language to call the exposed methods of CA Service Catalog, using standard SOAP call syntax.

worksheets

Worksheets enable you to define costs that are related to business activities. Service Worksheet and Pool Worksheet are the two types of worksheets.

WSDL

WSDL (Web Services Description Language) is an XML format that is published for describing web services.

CA Service Management Common Data Object Field Level Mapping Details

This section describes the Common Data Object reference field level mapping details for implementers who are using Web Services with CA ITAM, CA SDM, and/or CA Service Catalog. As a CA Service Management solution implementer, you might need field-level mapping details between point products like CA SDM, CA ITAM, CA Service Catalog in order to leverage the mapping information.

This section contains the following topics:

- Field Level Reference Mapping for ca_company
- Field Level Reference Mapping for ca_site
- Field Level Reference Mapping for ca_organization
- Field Level Reference Mapping for ca_owned_resource
- Field Level Reference Mapping for ca_contact
- Field Level Reference Mapping for ca_resource_cost_center
- Field Level Reference Mapping for ca_resource_family
- Field Level Reference Mapping for ca_resource_class
- Field Level Reference Mapping for ca_resource_department

Field Level Reference Mapping for ca_company

Contents

The ca_company table describes the CA ITAM and CA Service Desk Manager reference mapping details.

CA ITAM Department	CA SDM Department	Database Column Name	Data Type
companyid	id	company_uuid	UUID
value	Company Name	company_name	string
companytypekey	company_type	company_type	Number
alias	alias	alias	String
bbs	bbs	bbs	String
inactiveflag	delete_flag	inactive	boolean
description	description	description	string
creationdate	creatioin_date	creation_date	date
creationuser	creation_user	creation_user	string
lastupdatedate	last_mod	last_update_date	date
lastupdateuser	last_update_user	last_update_user	String
description	description	description	String
defaultlocationid	location_uuid	location_uuid	UUID
fiscalmonthkey	month_fiscal_year_ends	month_fiscal_year_ends	Number
parentcompanyid	parent_company_uuid	parent_company_uuid	UUID
primarycontactid	primary_contact_uuid	primary_contact_uuid	UUID
website	web_address	web_address	String
Assetcount	(Not Applicable)	Asset_count	Number

Divisionid	(Not Applicable)	Division_id	Number
Domainuuid	(Not Applicable)	Domain_uuid	UUID
Duplicateid	(Not Applicable)	Duplicate_with_uuid	UUID
Employeecount	(Not Applicable)	Employee_count	Number
Fastflag	(Not Applicable)	Fast_flag	Number
siiaflag	(Not Applicable)	siia_flag	Number
Sourcetypeid	(Not Applicable)	Source_type_id	Number
Userpriorityflag	(Not Applicable)	User_priority_flag	Number
(Not Applicable)	authentication_password	authentication_password	String
(Not Applicable)	authentication_user_name	authentication_user_name	number
(Not Applicable)	delete_time	delete_time	number
(Not Applicable)	exclude_registration	exclude_registration	Integer

Field Level Reference Mapping for ca_location

The ca_location table describes the field level reference mapping details for CA ITAM, CA Service Desk Manager, and CA Service Catalog.

CA ITAM Location	CA SDM Location	CA Service Catalog Location	DB Column Name	Data Type
Locationid	Id	(Not Applicable)	Location_uuid	UUID
Location Name	name	Name	Location_name	String
address1	address1	Address	address_1	String
address2	address2	Address	address_2	String
address3	address3	Address	address_3	String
address4	address4	Address	address_4	String
address5	address5	Address	address_5	String
address6	address6	Address	address_6	String
city	city	City	city	String
stateprovincekey	state	State/Province	state	Number
zip	zip	ZIP/Postal Code	zip	String
telephone	pri_phone_number	Primary Phone	pri_phone_number	String
fax	fax_number	Fax Number	fax_number	String
inactiveflag	delete_flag	(Not Applicable)	inactive	Boolean (1 or 0)
creationuser	(Not Applicable)	(Not Applicable)	Creation_user	String
creationdate	(Not Applicable)	(Not Applicable)	Creation_date	Long Number
lastupdateuser	last_update_user	(Not Applicable)	Last_update_user	String
lastupdatedate	last_mod	(Not Applicable)	Last_update_date	Long Number
locationtypekey	(Not Applicable)	(Not Applicable)	location_type_id	Number
countrykey	Country	Country	country	Number
regionid	(Not Applicable)	(Not Applicable)	region_id	Number
mailaddress1	mail_address1	(Not Applicable)	mail_address_1	String
mailaddress2	mail_address2	(Not Applicable)	mail_address_2	String

mailaddress3	mail_address3	(Not Applicable)	mail_address_3	String
mailaddress4	mail_address4	(Not Applicable)	mail_address_4	String
mailaddress5	mail_address5	(Not Applicable)	mail_address_5	String
mailaddress6	mail_address6	(Not Applicable)	mail_address_6	String
county	county	(Not Applicable)	county	String
siteid	Site	(Not Applicable)	site_id	Number
comments	description	Description	comments	String
organizationid	(Not Applicable)	(Not Applicable)	organization_uuid	UUID
individualid	primary_contact_uuid	(Not Applicable)	primary_contact_uuid	UUID
parentlocationid	(Not Applicable)	(Not Applicable)	parent_location_uuid	UUID

Field Level Reference Mapping for ca_site

The ca_site table describes the CA ITAM and CA SDM field level reference mapping details.

CA ITAM Site	CA SDM Site	Database Column Name	Data Type
siteid	id	Site_id	Number
sitename	name	name	String
inactiveflag	delete_flag	inactive	Boolean (1 or 0)
contactid	contact	contact_uuid	UUID
alias	alias	alias	String
description	description	description	String
lastupdatedate	last_mod	last_update_date	Number
lastupdateuser	last_update_user	last_update_user	String
creationdate	creation_date	creation_date	Number
creationuser	creation_user	creation_user	String

Field Level Reference Mapping for ca_organization

The ca_organization table describes the field reference mapping details CA ITAM and CA Service Desk Manager.

CA ITAM Organization	CA SDM Organization	Column Name	DataType
Organizationid	id	Organization_uuid	UUID
Name	name	name	String
abbreviation	org_num	abbreviation	String
inactiveflag	delete_flag	inactive	Boolean
primaryphonenumber	phone_number	pri_phone_number	String
altphonenumber	alt_phone	alt_phone_number	String
faxnumber	fax_phone	fax_number	String
contactid	contact	contact_uuid	UUID
emailaddress	email_addr	email_address	String
pageremailaddress	pemail_addr	pager_email_address	String

costcenterkey	billing_code	cost_center	Number
locationid	location	location_uuid	UUID
description	Description	description	String
lastupdatedate	last_mod	last_update_date	Number
lastupdateuser	Last Modified By	last_update_user	String
parentorganizationid	(Not Applicable)	parent_org_uuid	UUID
creationdate	(Not Applicable)	creation_date	Number
creationuser	(Not Applicable)	creation_user	String
(Not Applicable)	iorg_service_type	Table: usp_organization column: iorg_service_type	String
(Not Applicable)	owning_contract	Table: usp_organization column: owning_contract	Number

Field Level Reference Mapping for ca_owned_resource

The ca_owned_resource table describes the CA ITAM and CA Service Desk Manager reference mapping details.

CA ITAM Asset	CA SDM Configuration Item	Data Type	Ca_owned_resource Column	Reference Database Table
assetid	id	UUID	Own_resource_uuid	(Not Applicable)
assetname	name	String	Resource_name	(Not Applicable)
serialnumber	serial_number	String	Serial_number	(Not Applicable)
altassetid	alt ci id	String	resource_tag	(Not Applicable)
hostname	system_name	String	host_name	(Not Applicable)
dnsname	dns_name	String	dns_name	(Not Applicable)
macaddress	Mac_address	String	mac_address	(Not Applicable)
costcenterkey	cost center	Number	cost_center	Ca_resource_cost_center
ci	ci	Boolean	is_ci	(Not Applicable)
asset	issnr	Boolean	is_asset	(Not Applicable)
inactive	delete_flag	Boolean	inactive	(Not Applicable)
floorlocation	loc_floor	UUID	floor_location	Ca_location
roomlocation	loc_room	UUID	room_location	Ca_location
cabinetlocation	loc_cabinet	UUID	cabinet_location	Ca_location
shelflocation	loc_shelf	UUID	shelf_location	Ca_location
slotlocation	loc_slot	UUID	slot_location	Ca_location
assettypekey	family	Number	resource_family	Ca_resource_family
class	class	Number	resource_class	Ca_resource_class
quantity	Asset_count	Number	resource_quantity	(Not Applicable)
itemid	model	UUID	model_uuid	Ca_model_def
contactid	resource_contact	UUID	resource_contact_uuid	ca_contact
locationid	location	UUID	Location_uuid	Ca_location
sellercompanyid	supplier	UUID	supply_vendor_uuid	Ca_company

statuskey	status	Number	resource_status	(Not Applicable)
description	description	String	Resource_description	(Not Applicable)
productversion	Product_version	String	product_version	(Not Applicable)
acquiredate	Acquire_date	Date	acquire_date	(Not Applicable)
installationdate	install_date	Date	installation_date	(Not Applicable)
billingcontactid	billing_contact_ uuid	UUID	billing_contact_uuid	ca_contact
supportcontact1id	support_contact1_uuid	UUID	support_contact1_uuid	ca_contact
Supportcontact2id	support_contact2_uuid	UUID	support_contact2_uuid	ca_contact
maintenancevendorid	vendor_repair	UUID	maintenance_vendor_uuid	Ca_company
manufacturerid	manufacturer	UUID	manufacturer_uuid	Ca_company
responsibleorganizationid	service_org	UUID	responsible_org_uuid	ca_organization
servicetypeid	(Not Applicable)	Number	Usp_owned_resource.service_type	(Not Applicable)
priorityid	(Not Applicable)	Number	usp_owned_resource.nr_pr_id	(Not Applicable)
ipaddress	alarm_id	String	ip_address	(Not Applicable)
licenseinformation	license_number	String	license_information	(Not Applicable)
supportcontact3id	support_contact3_uuid	UUID	support_contact3_uuid	ca_contact
disasterrecoverycontactid	disaster_ recovery_ contact_uuid	UUID	disaster_recovery_contact_uuid	ca_contact
backupservicescontactid	backup_ services_ contact_uuid	UUID	backup_services_contact_uuid	ca_contact
networkcontactid	network_contact_uuid	UUID	network_contact_uuid	ca_contact
responsiblevendorid	vendor_restore	UUID	responsible_vendor_uuid	(Not Applicable)
maintenancevendorid	vendor_repair	UUID	maintenance_org_uuid	(Not Applicable)
assetalias	resource_alias	String	resource_alias	(Not Applicable)
companyboughtforid	company_bought_ for_uuid	UUID	company_bought_for_uuid	(Not Applicable)
organizationboughtforid	org_bought_for_ uuid	UUID	org_bought_for_uuid	(Not Applicable)
processorcount	(Not Applicable)	Number	processor_count	(Not Applicable)
glcodekey	Gl_code	Number	gl_code	Not Applicable)
auditdate	(Not Applicable)	Date	audit_date	(Not Applicable)
operatingsystemkey	operating_system	Number	operating_system	(Not Applicable)
capacity	(Not Applicable)	Number	resource_capacity	(Not Applicable)
capacityunitskey	(Not Applicable)	Number	resource_capacity_unit	(Not Applicable)
departmentkey	department	Number	department	(Not Applicable)
requisitionid	requisition_id		requisition_id	(Not Applicable)
purchaseorderid	purchase_order_id	Number	purchase_order_id	(Not Applicable)
lastupdatedate	last_mod	Date	last_update_date	(Not Applicable)
lastupdateuser	Last_mod_by	String	last_update_user	(Not Applicable)
statusdate	Status_date	Date	status_date	(Not Applicable)
deploymentstatuskey	(Not Applicable)	Number	(Not Applicable)	(Not Applicable)

lifecyclestatuskey	(Not Applicable)	Number	lifecycle_status	(Not Applicable)
lifecyclestatusdate	(Not Applicable)	Date	lifecycle_status_date	(Not Applicable)
Subclass	(Not Applicable)	Number	resource_subclass	(Not Applicable)
contactid	resource_contact	UUID	resource_contact_uuid	(Not Applicable)
excludereconciliationkey	(Not Applicable)	Boolean	exclude_reconciliation	(Not Applicable)
managedkey	ufam	Number	ufam	(Not Applicable)
processortype	(Not Applicable)	Number	processor_type	(Not Applicable)
discoverylastrundate	(Not Applicable)	Date	discovery_last_run_date	(Not Applicable)
reconciliationdate	(Not Applicable)	Date	reconciliation_date	(Not Applicable)
creationdate	(Not Applicable)	Date	creation_date	(Not Applicable)
creationuser	(Not Applicable)	String	creation_user	(Not Applicable)

Field Level Reference Mapping for ca_contact

The ca_contact table describes the field level reference mapping details for CA ITAM, CA Service Desk Manager, and CA Service Catalog.

CA ITAM Contact	CA SDM Contact	CA Service Catalog User	Database Column Name	Data Type
lastname	last_name	last name	last_name	string
firstname	first_name	first name	first_name	string
middlename	middle_name	middle name	middle_name	string
companyid	company	(Not Applicable)	company_uuid	uuid
inactiveflag	delete_flag	(Not Applicable)	inactive	boolean (1 or 0)
altcontactid	contact_num	(Not Applicable)	alternate_identifier	string
userid	userid	user id	userid	string
contacttypekey	type	(Not Applicable)	contact_type	number
jobtitlekey	position	job title	job_title	number referring to id column of ca_job_title table
alias	alias	alias	alias	string
lastupdatedate	Last_mod	(Not Applicable)	last_update_date	long number
lastupdateuser	last_mod_by	(Not Applicable)	last_update_user	string
telephone	phone_number	primary phone	pri_phone_number	string
fax	fax_phone	fax number	fax_number	string
pagertelephone	beeper_phone	pager number	pager_number	string
alttelephone	alt_phone	Secondary Phone	alttelephone	String
emailid	email_address	Email	email_address	String
pageremailaddress	pemail_address	(Not Applicable)	pager_email_address	String
locationid	location	Location	location_uuid	UUID
organizationid	organization	(Not Applicable)	organization_uuid	UUID
adminorganizationid	admin_org	(Not Applicable)	adminorganizationid	UUID
supervisorid	supervisor_ contact_uuid	Manager	supervisor_contact_uuid	UUID

departmentkey	dept	(Not Applicable)	department	Number
costcenterkey	billing_code	(Not Applicable)	cost_center	Number
comments	notes	Description	comments	String
creationdate	(Not Applicable)	(Not Applicable)	Creation_date	Long number
creationuser	(Not Applicable)	(Not Applicable)	Creation_user	String

Field Level Reference Mapping for ca_resource_cost_center

The ca_resource_cost_center table describes the CA ITAM and CA SDM field level reference mapping details.

CA ITAM Cost Center	CA SDM Cost Center	Database Column Name	Data Type
id	id	id	number
value	name	name	string
inactiveflag	delete_flag	inactive	boolean
description	description	description	string
creationdate	creation_date	creation_date	date
creationuser	creation_user	creation_user	string
lastupdatedate	last_mod	last_update_date	date
lastupdateuser	last_update_user	last_update_user	string
(Not Applicable)	delete_time	delete_time	number
(Not Applicable)	exclude_registration	exclude_registration	integer

Field Level Reference Mapping for ca_resource_family

The ca_resource_family table describes the CA APM and CA SDM field level reference mapping details.

CA APM Asset Family	CA SDM Family	Database Column Name	Data Type
id	id	id	Number
Value	sym	name	String
inactiveflag	delete_flag	inactive	Boolean
description	description	description	String
lastupdatedate	last_update_date	last_update_date	Number
lastupdateuser	last_update_user	last_update_user	String
extensiontablename	extension_name	table_extension_name	String
physicaltablename	physical_table_name	physical_table_name	String
includereconciliationkey	include_reconciliation	include_reconciliation	Boolean
Is ITAM	(Not Applicable)	is_itam	Boolean
creationdate	(Not Applicable)	creation_date	Number
creationuser	(Not Applicable)	creation_user	String
(Not Applicable)	exclude_registration	exclude_registration	Number

Field Level Reference Mapping for ca_resource_class

The ca_resource_class table describes the CA ITAM and CA Service Desk Manager field level reference mapping details.

CA ITAM Asset Class	CA SDM Class	Database Column Name	Data Type
Value	type	name	String
inactiveflag	delete_flag	inactive	Boolean
description	description	description	String
lastupdatedate	last_mod	last_update_date	Number
lastupdateuser	last_update_user	last_update_user	String
creationuser	creation_user	creation_user	String
creationdate	creation_date	creation_date	Number
familyid	family	id	Number
(Not Applicable)	exclude_ registration	exclude_ registration	(Not Applicable)

Field Level Reference Mapping for ca_resource_department

The ca_resource_department table describes the CA ITAM and CA Service Desk Manager field level reference mapping details.

CA ITAM Department	CA SDM Department	Database Column Name	Data Type
name	id	name	string
value	name	name	string
inactiveflag	delete_flag	inactive	boolean
description	description	description	string
creationdate	creation_date	creation_date	date
creationuser	creation_user	creation_user	string
lastupdatedate	last_mod	last_update_date	date
lastupdateuser	last_update_user	last_update_user	string
(Not Applicable)	delete_time	delete_time	number
(Not Applicable)	exclude_registration	exclude_registration	integer

USM Data Mapping for CA Service Desk Manager Connector

When connectors import services and CIs from domain managers, they normalize the classes, properties, relationships, and severities in the domain manager to adhere to the USM schema. This section lists the CA SDM classes, severities, and relationships and their USM mapping after the import.

NOTE
For more information about CI property mapping, see the CA SDM Connector policy file located at the policy registry location.

More Information:

- Device Properties Calculation
- Relationship Mapping
- Severity Mapping
- Type Mapping

Device Properties Calculation

Certain CIs, such as ProvisionedSoftware and NIC Card, are mapped to USM CI types that require device properties defined by USM import standards in Catalyst. When device properties are not defined for the CI, the connector uses this property to obtain the information from the system on which the CI is hosted. One or more of the following CI device properties may be required:

- DeviceAssetNumber
- DeviceBiosSystemID
- DeviceDnsName
- DeviceSysName
- DevicePhysSerialNumber
- DeviceMacAddress
- DeviceIPV4Address
- DeviceIPV4AddressWithDomain
- DeviceIPV6Address
- DeviceIPV6AddressWithDomain

CA SDM Connector queries CMDB for the device properties of the CI as follows:

1. Based on the system name, the connector queries CMDB for the CI device properties, and the CI is populated when found. For application discovered CIs in CMDB, the System Name field appears in the following format:
   ```
   SNMP_CI_name| CI_name | file_path
   ```
2. If the SNMP_CI_name does not exist in CMDB and the query fails, the connector uses the DeviceSysName attribute to query CMDB for the device properties of the CI.
3. If the queries described in step 1 and 2 are unsuccessful, the connector uses the DeviceXXXFromRelationship attribute to query CMDB for the CI device properties based on the relationship type.

Device Properties

The CA SDM Connector maps to the USM types based on the values of the following CA SDM properties.

Device (USM)	CA SDM Attributes	User Interface
DeviceAssetNumber	Calculated by the connector.	N/A

DeviceBiosSystemID	Calculated by the connector.	N/A
DeviceDnsName	Calculated by the connector.	N/A
DeviceSysName	Calculated by the connector.	System Name
DevicePhysSerialNumber	Calculated by the connector.	N/A
DeviceMacAddress	Calculated by the connector.	N/A
DeviceIPV4Address	Calculated by the connector.	N/A
DeviceIPV4AddressWithDomain	N/A	N/A
DeviceIPV6Address	Calculated by the connector.	N/A
DeviceIPV6AddressWithDomain	N/A	N/A

Relationship Mapping

This topic contains the following information:

Relationship Mapping

The CA SDM classes are transformed into USM types as shown in the following table:

BinaryRelationship	CA SDM Attributes	User Interface /CMDB Relationship Details
SourceMdrProduct	CA:00020	n/a
SourceMdrProdInstance	<primary server host name>	n/a
SourceMdrElementID	nr.persid/chg.persid/cr.persid	<bop_dump>
SourceCorrelatedKey	n/a	n/a
TargetMdrProduct	CA:00020	n/a
TargetMdrProdInstance	<primary server host name>	n/a
TargetMdrElementID	nr.persid/chg.persid/cr.persid	<bop_dump>
ScopeMdrProduct	CA:00020	n/a
ScopeMdrProdInstance	<primary server host name>	n/a
ScopeMdrElementID	Configured by the connector	<bop_dump>
Significance	n/a	n/a
Semantic	Bmhier:ci_rel_type	CMDB Relationships tab; Relationship

Relationship Semantic Mapping

The relationship semantic mapping is shown in the following table:

CA SDM <Property>	USM Type
Change, Problem, Incident, Request	IsImpactedBy
IsRequiredBy	HasRequirementFor
Connects To	IsConnectedTo
Contains	IsPartOf

Is Source Code For	IsInstanceOf
Hosts	IsHostedBy
Manages	IsManagedBy
IsUsedBy	HasAccessTo
Monitors	IsDiscoveredBy
Supports	IsAffectedBy
Runs	HasMember
Is The Child Of	HasDetail
Is Location For	IsResidentOf
Administers	HasContact
Is Primary Contact For	HasContact
Is Business Owner Of	HasContact
Approves	HasContact
Has As Assignee	HasContact
Financially Belongs To	IsComposedOf
Is Bound To	IsBoundTo
Is Clone Of	IsCloneOf
Is Composed Of	IsComposedOf
Is Evolution Of	IsEvolutionOf
Is Result Of	IsResultOf
Is Successor Of	IsSuccessorOf
Is Triggered By	IsTriggeredBy
Is Affected By	IsAffectedBy
Has Member	HasMember
Is Discovered By	IsDiscoveredBy
Manages	IsManagedBy
Has Access To	HasAccessTo
Is Required By	HasRequirementFor
Is Triggered By	Is Triggered By
Has Detail	HasDetail
Is Successor Of	IsSuccessorOf
Is Source Code For	IsInstanceOf
Is Result Of	IsResultOf
Is Primary Contact For	HasContact
Is Location For	IsResidentOf
Is Evolution Of	IsEvolutionOf
Is Composed Of	IsComposedOf
Is Clone Of	IsCloneOf

Is Business Owner Of	HasContact
Is Bound To	IsBoundTo
Hosts	IsHostedBy
Has As Assignee	HasContact

Severity Mapping

The following table shows how the CA SDM Connector maps alert severities to USM severities:

CA SDM Connector Alerts	USM Severity
Too Many Open Incidents	Critical
SLA to Expire	Major
SLA Violated	Critical
Upcoming Maintenance Window	Critical
Upcoming Blackout Window	Critical

NOTE
In CA SDM, blackout windows are not configured with CIs, therefore, they do not appear in the Service Console. You can, however, view blackout windows for CIs in the REST interface by disabling the useAlertFilter through the Connector Controls panel.

Type Mapping

When connectors import services and CIs from domain managers, they normalize the classes in the domain manager to the standard USM types. This section lists the CA SDM classes and their mapping after the import.

The CMDB classes are transformed into USM types for CIs as shown in the following table:

USM Type(entitytype = CI)	User interfaceCI detail where CMDB family = ...
Application	Software.Application
ApplicationServer	Software.Application Server
Cluster	Cluster
ComputerSystem	Hardware.Server
DatabaseInstance	Software.Database
GenericIPDevice	Network.Hub
InterfaceCard	Network.Network Interface Card
MediaDrive	Hardware.Storage
OperatingSystem	Software.Operating System
Port	Network.Port
ProvisionedSoftware	Software.COTS
Printer (Network)	Hardware.Printer
Router	Network.Router
RunningHardware	Hardware
Service	Enterprise.Service/Service
Switch	Network.Switch

VirtualSystem	Hardware.VirtualMachine

The CA SDM classes are transformed into USM types for other elements as shown in the following table:

USM type(entitytype = other elements)	CA SDM CMDB Definition	User Interface
Change Order	SDM: chg	Change Order
Incident	SDM: cr WHERE cr.type='I'	Incident
Location	SDM:loc CMDB: ci_location	Location
OrganizationEntity	SDM: org CMDB: ci_organization	Organization
Person	SDM: cnt WHERE cnt.type!=2308 CMDB: ci_contact	Contact
Problem	SDM: cr WHERE cr.type='P'	Problem
Request	SDM: cr WHERE cr.type='R'	Request

Application

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

NOTE

You can define the Vendor and Model properties through the CA SDM connector policy to manage your configuration items. For more information about these properties, see Post-Installation: Manufacturers and Models.

Application (USM)	CA Service Management Attribute	User Interface (CI detail form)
Description	nr.description	
Vendor	nr.manufacturer.xym [ca_cmpny]	Inventory tab; Manufacturer (Configure this property in the connector policy to the requirement of your site)
ProductName	nr.name	Name
Version	app_extx.version	Attributes tab version
MajorVersion	n/a	n/a
BuildNumber	n/a	n/a
ServiceLevel	n/a	n/a
Device properties required	n/a	n/a
ProcessID	n/a	n/a
AccessedViaTcpPort	n/a	n/a
ProcessDistinguishingID	app_extx. app_id	Attributes tab; Application ID

ApplicationServer

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

NOTE

You can define the Vendor and Model properties through the CA SDM connector policy to manage your configuration items. For more information about these properties, see Post-Installation: Manufacturers and Models.

ApplicationServer (USM)	CA SDM Attribute	User Interface (CI detail form)
Vendor	nr.manufacturer.xym [ca_cmpny]	Inventory tab; Manufacturer (Configure this property in the connector policy to the requirement of your site)
ProductName	nr.name	Name
Version	app_extx.version	Attributes tab version
MajorVersion	n/a	n/a
BuildNumber	n/a	n/a
ServiceLevel	n/a	n/a
Device properties required	n/a	n/a
ProcessID	n/a	n/a
AccessedViaTcpPort	n/a	n/a
ProcessDistinguishingID	app_extx. app_id	Attributes tab; Application ID
AppServerType	app_extx.type	Attributes tab; Type

Change Order

The CA SDM Connector maps to the USM types based on the values of the following CA SDM properties.

ChangeOrder (USM Type)	CA SDM Attributes	User Interface
AssignedID	chg.chg_reg_num	Text on button bar (left side)
IsTemplate	n/a	n/a
TemplateUsed	n/a	n/a
Severity	n/a	n/a
IsActive	issue 157	n/a
Urgency	n/a	n/a
Impact	chg.mpact; imp.sym	Impact
Priority	chg.priority; pri.sym	Priority
Open Timestamp	chg.target_start_last	Open Date (above tabs)
CallbackTimestamp	chg.call_back_date	Call Back Date/Time
Resolution Timestamp	chg.resolve_date	Resolve Date (above tabs)
Closure Timestamp	chg.close_date	Close Date (above tabs)
ChargebackID	n/a	n/a
RootCause	chg.rc	Root Cause
SolutionUrls	n/a	n/a
NeedByDate	chg.need_by	Need By Date
COType [COTypeEnum]	chg.chgtype; chgtype.sym	Type
RequiresCabApproval	chg.cab_approval	CAB Approval

COStatus [COStatusEnum]	chg.chgtype; chgtype.sym	Status
CompletionCode	chg.closure_code; closure_code.sym	Closure Code
FulfillmentData	n/a	config.items
ChangeBackedOut [ChageReversonEnum]	n/a	n/a
COCategory [CoCategoryEnum]	chg.category; chgcat.code	Category
EstimatedCost	chg.est_cost	Costs/Plans tab; Est Cost
ActualCost	chg.cost	Costs/Plans tab; Actual Cost
Currency [CurrencyEnum]	n/a	n/a
EstimatedStartDate	chg.sched_start_date	Schedule Start Date
Estimated TotalTime	chg.sched_duration	Schedule Duration
ActualStartDate	chg.actual_start_date	Costs/Plans tab; Actual Implementation Start Date
ActualTotalTime	chg.actual_total_time	Costs/Plans tab; Actual Duration
ProjectID	chg.project; nr.family.extension_name; projex.CA_id (if null, then use the chg.project.name)	n/a
BusinessCase	chg.business_case	Costs/Plans tab; Business Case
Effort	chg.effort	Costs/Plans tab; Implementation Plan

Cluster

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Cluster (USM Type)	**CA SDM Attributes**	**User Interface / CI Detail Form**
PrimaryIPV4Address	nr.alarm_id if IPV4 is formatted	Inventory tab; IP Address (V4)
PrimaryIPV6AddressWithDomain	n/a	n/a
PrimaryIPV6Address	nr.alarm_id if IPV6 is formatted	Inventory tab; IP Address (V6)
PrimaryIPV4AddressWithDomain	n/a	n/a
GroupName	nr.name	Attributes Network Name
PrimaryIPV4AddressWithDomain	n/a	n/a
MemberCriteria	net_clux.quorum	Attributes; Quorum
PrimaryDnsName	nr.dns name	DNS Name
LoadBalancingType	n/a	n/a
usm-core2: MemberStatus	n/a	
InstanceName	<<AUTO-GENERATED BY GLOBAL Policy>>	
NamedAliases	<<AUTO-GENERATED BY GLOBAL Policy>>	
Label	<<AUTO-GENERATED BY GLOBAL Policy>>	

Tags	<<AUTO-GENERATED BY GLOBAL Policy>>	
Description	nr.description	
CreationTimestamp	nr.creation_user	
CreationUserName	nr.creation_date	
LastModTimestamp	nr.last_mod	
UrlParams	"nr"	

Common CI Properties

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Common CI Properties (USM)	CA SDM Attributes	User Interface / CI Detail Form
MdrProduct	ConnectorConfigMdrProduct	n/a
MdrProdInstance	ConnectorConfigMdrProdInstance	n/a
MdrElementID	nr.persid (Outbound)	<bop_dump>
UrlParams	nr	n/a
NamedAliases	nr.name, nr.system_name, nr.dns_name	Name, Host Name, Dns Name
Label	Auto generated by GLOBAL policy	n/a
Description	nr.description	Notes
TenantID	nr.tenant	Tenant ID
Tags	Auto generated by GLOBAL policy	n/a
CreationTimestamp	nr.creation_date	<bop_dump>
CreationUserName	nr.creation.user	<bop_dump>
LastModTimestamp	nr.last_mod	<bop_dump>
LastModActivity	Generated by connector.	n/a
LastModUserName	nr.last_mod_by	<bop_dump>
Deletion Timestamp	This is the timestamp when the CI became inactive in SDM.	<bop_dump>
DeletionUserName	nr.last_mod_by (When it is deleted)	n/a
InstanceName	Auto generated by GLOBAL policy	n/a
TypeName	n/a	n/a
AdministrativeStatus	nr.status.sym [rss]	Inventory tab; Service Status (Ensure that AdministrativeStatus corresponds with AdministrativeStatusEnum in the connector policy)
Retirement Timestamp	nr.retire_date	Attributes tab; Retire Date
IsInMaintenance	nr.status.sym [rss]	Inventory tab; Service Status
MaintStartTimestamp	n/a	n/a
MainExpectedDuration	n/a	n/a
Vendor	nr.manufacturer	
Model	nr.model	
GroupName	nr.name	Name

WebSite	nr.class	Class
Software.WebSite	nr.family	Family
usm-core2:HomePage|usm-core2:DeviceBiosSystemID	nr.system_name	Hostname
usm-core2:DeviceAssetNumber	app_website.DeviceAssetNumber	Device Asset Number
usm-core2:DeviceBiosSystemID	app_website.DeviceBiosSystemID	Device Bios System ID
usm-core2:DeviceDnsName	app_website.DeviceDnsName	Device DNS Name
usm-core2:DevicePhysSerialNumber	app_website.DevicePhysSerialNumber	Device Physical Serial Number
usm-core2:DeviceMacAddress	app_website.DeviceMacAddress	Device MAC Address
usm-core2:DeviceIPV4Address	app_website.DeviceIPV4Address	Device IPv4 Address
usm-core2:DeviceIPV4AddressWithDomain	app_website.DeviceIPV4AddressWithDomain	Device IPv4 Address With Domain
usm-core2:DeviceIPV6Address	app_website.DeviceIPV6Address	Device IPv6 Address
usm-core2:DeviceIPV6AddressWithDomain	app_website.DeviceIPV6AddressWithDomain	Device IPv6 Address With Domain
usm-core2:IsHAEnabled	app_website.IsHAEnabled	Is HA Enabled
usm-core2:IsMonitoringMembers	app_website.IsMonitoringMembers	Is Monitoring Members
usm-core2:MaxFailures	app_website.MaxFailures	Max Failures
usm-core2:MemberCriteria	app_website.MemberCriteria	Member Criteria
usm-core2:GroupType	app_website.GroupType	Group Type
usm-core2:HomePage	app_website.HomePage	Home Page
usm-core2:BusinessRelevance	app_website.BusinessRelevance	Business Relevance

ComputerSystem

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

NOTE

: You can define the Vendor and Model properties through the CA SDM connector policy to manage your configuration items. For more information about these properties, see Post-Installation: Manufacturers and Models.

ComputerSystem (USM Type)	CA SDM Attributes	User Interface / CI Detail Form
Vendor	nr.manufacturer	Inventory tab; Manufacturer (Configure this property in the connector policy to the requirement of your site)
Model	nr.model	Inventory tab; Model (Configure this property in the connector policy to the requirement of your site)
PhysSerialNumber	nr.serial_number	Serial Number
VendorSerialNumber	n/a	n/a
AssetNumber	nr.asset_num	Alternate CI ID
PrimaryDnsName	nr.dns_name	DNS Name
OtherDnsNames	n/a	n/a
SysName	n/a	n/a

PrimaryMacAddress	nr.mac_address	MAC Address
OtherMacAddresses	n/a	n/a
PrimaryIPV4Address	nr.alarm_id	Inventory tab; IP Address
PrimaryIPV6Address	nr.alarm_id (if IPV6 is formatted)	n/a
OtherIPAddresses	n/a	n/a
ComputerName	nr.system_name	Name
SystemType	nr.family	Family
MemoryInGB	har_serx.phys_mem	Attributes tab; Memory Installed
StorageInGB	har_serx.hard_drive_capacity	Attributes tab; Disk Capacity
ProcessorType	har_serx.proc_type	Attributes tab; Processor Type
NumberOfCores	har_serx.number_proc_inst	Attributes tab; Number of Processors Installed
ProcessorSpeedInGHz	har_serx.proc_speed	Attributes tab; Processor Speed
PrimaryOSType	nr.class	Class
PrimaryOSVersion	n/a	n/a
OtherOSDetails	n/a	n/a
Description	nr.description	
usm-core2:OSName	n/a	
PrimaryOSVersion	n/a	
usm-core2:BiosVersion	har_serx.bios_ver	
ProcessorType	har_serx.proc_type	
ProcessorSpeedInGHz	har_serx.proc_speed	
NumberOfCores	har_serx.number_proc_inst	
CreationUserName	nr.creation_user	
CreationTimestamp	nr.creation_date	
PrimaryOSType	nr.class	
LastModTimestamp	nr.last_mod	
MdrElementID	nr.persid (Outbound)	
usm-core2: BusinessRelevance	n/a	
LastModTimeStamp	nr.last_mod	
AdministrativeStatus(New, managed, unmanged)	nr.status.sym [rss]	
UrlParams	"nr"	
usm-core2:NumberOf InterfaceCards	har_serx.number_net_card	
usm-core2:BiosDate	n/a	
usm-core2:OSPatchLevel	har_serx.security_patch_level	
StorageInGB	har_serx.hard_drive_capacity	
usm-core2:NumberOfDiskPartitions	n/a	
usm-core2:NumberOfPhysicalDrives	n/a	
usm-core2:NumberOfMemorySlots	har_serx.slot_total_mem	

DatabaseInstance

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

DatabaseInstance (USM Type)	CA SDM Attributes	User Interface / CI Detail Form
DBInstanceName	dat_basx.db_id If db_id is found, then set to nr.system_name	Attributes tab; Database ID or System Name
ProductName	nr.name	Name
DBServerType	nr.class	Class
ProcessDistinguishingID	dat_basx.db_id	Attributes tab; Database ID
Device properties required	n/a	n/a

DiskPartition

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Common CI Properties (USM)	CA SDM Attributes	User Interface / CI Detail Form
PartitionType|usm-core2:OSDriveName	nr.name	Name
FAT32	nr.class	Class
Hardware.DiskPartition	nr.family	Family
PartitionType|OSNumeric|usm-core2:OSDriveName|DeviceDnsName	nr.system_name	Hostname
usm-core2:DeviceAssetNumber	har_dpar.DeviceAssetNumber	Device Asset Number
usm-core2:DeviceBiosSystemID	har_dpar.DeviceBiosSystemID	Device Bios System ID
usm-core2:DeviceDnsName	har_dpar.DeviceDnsName	Device DNS Name
usm-core2:DeviceSysName	har_dpar.DeviceSysName	Device System Name
usm-core2:DevicePhysSerialNumber	har_dpar.DevicePhysSerialNumber	Device Physical Serial Number
usm-core2:DeviceMacAddress	har_dpar.DeviceMacAddress	Device MAC Address
usm-core2:DeviceIPV4Address	har_dpar.DeviceIPV4Address	Device IPv4 Address
usm-core2:DeviceIPV4AddressWithDomain	har_dpar.DeviceIPV4AddressWithDomain	Device IPv4 Address With Domain
usm-core2:DeviceIPV6Address	har_dpar.DeviceIPV6Address	Device IPv6 Address
usm-core2:DeviceIPV6AddressWithDomain	har_dpar.DeviceIPV6AddressWithDomain	Device IPv6 Address With Domain
OSNumeric	har_dpar.OSNumeric	OS Numeric
ContainingIndex	har_dpar.ContainingIndex	Containing Index
IsPhysical	har_dpar.IsPhysical	Is Physical
usm-core2:OSDriveName	har_dpar.OSDriveName	OS Drive Name
IsBootable	har_dpar.IsBootable	Is Bootable
IsPrimary	har_dpar.IsPrimary	Is Primary
usm-core2:CapacityInMB	har_dpar.CapacityInMB	Capacity In MB
ContextID	har_dpar.ContextID	Context ID

EnvironmentalSensor

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Common CI Properties (USM)	CA SDM Attributes	User Interface / CI Detail Form
SensorType|OS Numeric |Containing Index| DeviceDnsName	nr.name	Name
Vibration	nr.class	Class
Hardware.EnvironmentalSensor	nr.family	Family
SensorType|OS Numeric |Containing Index| DeviceDnsName	nr.system_name	Hostname
usm-core2:DeviceAssetNumber	har_compr.DeviceAssetNumber	Device Asset Number
usm-core2:DeviceBiosSystemID	har_compr.DeviceBiosSystemID	Device Bios System ID
usm-core2:DeviceDnsName	har_compr.DeviceDnsName	Device DNS Name
usm-core2:DeviceSysName	har_compr.DeviceSysName	Device System Name
usm-core2:DevicePhysSerialNumber	har_compr.DevicePhysSerialNumber	Device Physical Serial Number
usm-core2:DeviceMacAddress	har_compr.DeviceMacAddress	Device MAC Address
usm-core2:DeviceIPV4Address	har_compr.DeviceIPV4Address	Device IPv4 Address
usm-core2:DeviceIPV4AddressWithDomain	har_compr.DeviceIPV4AddressWithDomain	Device IPv4 Address With Domain
usm-core2:DeviceIPV6Address	har_compr.DeviceIPV6Address	Device IPv6 Address
usm-core2:DeviceIPV6AddressWithDomain	har_compr.DeviceIPV6AddressWithDomain	Device IPv6 Address With Domain
OSNumeric	har_compr.OSNumeric	OS Numeric
ContainingIndex	har_compr.ContainingIndex	Containing Index
IsPhysical	har_compr.IsPhysical	Is Physical
ContextID	har_compr.ContextID	Context ID

ESXHypervisor

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Common CI Properties (USM)	**CA SDM Attributes**	**User Interface / CI Detail Form**
ProductName	nr.name	Name
ESXHypervisor	nr.class	Class
Software.ESXHypervisor	nr.family	Family
ProductName|ProcessID	nr.system_name	Hostname
usm-core2:DeviceAssetNumber	app_esx.DeviceAssetNumber	Device Asset Number
usm-core2:DeviceBiosSystemID	app_esx.DeviceBiosSystemID	Device Bios System ID
usm-core2:DeviceDnsName	app_esx.DeviceDnsName	Device DNS Name
usm-core2:DeviceSysName	app_esx.DeviceSysName	Device System Name
usm-core2:DevicePhysSerialNumber	app_esx.DevicePhysSerialNumber	Device Physical Serial Number
usm-core2:DeviceMacAddress	app_esx.DeviceMacAddress	Device MAC Address
usm-core2:DeviceIPV4Address	app_esx.DeviceIPV4Address	Device IPv4 Address
usm-core2:DeviceIPV4AddressWithDomain	app_esx.DeviceIPV4AddressWithDomain	Device IPv4 Address With Domain
usm-core2:DeviceIPV6Address	app_esx.DeviceIPV6Address	Device IPv6 Address

usm-core2:DeviceIPV6AddressWithDomain	app_esx.DeviceIPV6AddressWithDomain	Device IPv6 Address With Domain
IsMigrationEnabled	app_esx.IsMigrationEnabled	Migration Enabled
usm-core2:ComputeResourceIndex	app_esx.ComputeResourceIndex	Compute Resource Index
usm-core2:HostIndex	app_esx.HostIndex	Host Index
usm-core2:DatacenterPath	app_esx.DatacenterPath	Datacenter Path
usm-core2:FTVersion	app_esx.FTVersion	FT Version
usm-core2:NumberOfPrimaryVMs	app_esx.NumberOfPrimaryVMs	Number Of Primary VMs
usm-core2:NumberOfSecondaryVMs	app_esx.NumberOfSecondaryVMs	Number Of Secondary VMs
ProcessID	app_esx.ProcessID	Process ID
AccessedViaTcpPort	app_esx.AccessedViaTcpPort	Accessed Via Tcp Port
ProcessDistinguishingID	app_esx.ProcessDistinguishingID	Process Distinguishing ID

File

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Common CI Properties (USM)	CA SDM Attributes	User Interface / CI Detail Form
FilePathUrl|FileType |DeviceDnsName	nr.name	Name
File	nr.class	Class
Hardware.File	nr.family	Family
FilePathUrl|FileType |DeviceDnsName	nr.system_name	Hostname
usm-core2:DeviceAssetNumber	har_file.DeviceAssetNumber	Device Asset Number
usm-core2:DeviceBiosSystemID	har_file.DeviceBiosSystemID	Device Bios System ID
usm-core2:DeviceDnsName	har_file.DeviceDnsName	Device DNS Name
usm-core2:DeviceSysName	har_file.DeviceSysName	Device System Name
usm-core2:DevicePhysSerialNumber	har_file.DevicePhysSerialNumber	Device Physical Serial Number
usm-core2:DeviceMacAddress	har_file.DeviceMacAddress	Device MAC Address
usm-core2:DeviceIPV4Address	har_file.DeviceIPV4Address	Device IPv4 Address
usm-core2:DeviceIPV4AddressWithDomain	har_file.DeviceIPV4AddressWithDomain	Device IPv4 Address With Domain
usm-core2:DeviceIPV6Address	har_file.DeviceIPV6Address	Device IPv6 Address
usm-core2:DeviceIPV6AddressWithDomain	har_file.DeviceIPV6AddressWithDomain	Device IPv6 Address With Domain
FilePathUrl	har_file.FilePathUrl	File Path Url

GenericIPDevice

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

GenericIPDevice (USM)	**CA SDM Attribute**	**User Interface (CI detail form)**
PrimaryDnsName	nr.dns_name	DNS Name
OtherDnsNames	n/a	n/a
SysName	n/a	n/a
PrimaryMacAddress	nr.mac_address	MAC Address
OtherMacAddresses	n/a	n/a

PrimaryIPV4Address	nr.alarm_id	Inventory tab; IP Address
PrimaryIPV4AddressWithDomain	n/a	n/a
PrimaryIPV6Address	nr.alarm_id (if IPv6 is formatted)	Inventory tab; IP Address (if IPv6 is formatted)
PrimaryIPV6AddressWithDomain	n/a	n/a
OtherIPAddresses	n/a	n/a
OtherIPAddressesWithDomains	n/a	n/a

HyperVHypervisorManager

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Common CI Properties (USM)	CA SDM Attributes	User Interface / CI Detail Form
ProductName	nr.name	Name
HyperVHypervisor	nr.class	Class
Software.HyperVHypervisor	nr.family	Family
ProductName|ProcessID	nr.system_name	Hostname
usm-core2:DeviceAssetNumber	app_hyp.DeviceAssetNumber	Device Asset Number
usm-core2:DeviceBiosSystemID	app_hyp.DeviceBiosSystemID	Device Bios System ID
usm-core2:DeviceDnsName	app_hyp.DeviceDnsName	Device DNS Name
usm-core2:DeviceSysName	app_hyp.DeviceSysName	Device System Name
usm-core2:DevicePhysSerialNumber	app_hyp.DevicePhysSerialNumber	Device Physical Serial Number
usm-core2:DeviceMacAddress	app_hyp.DeviceMacAddress	Device MAC Address
usm-core2:DeviceIPV4Address	app_hyp.DeviceIPV4Address	Device IPv4 Address
usm-core2:DeviceIPV4AddressWithDomain	app_hyp.DeviceIPV4AddressWithDomain	Device IPv4 Address With Domain
usm-core2:DeviceIPV6Address	app_hyp.DeviceIPV6Address	Device IPv6 Address
usm-core2:DeviceIPV6AddressWithDomain	app_hyp.DeviceIPV6AddressWithDomain	Device IPv6 Address With Domain
IsMigrationEnabled	app_hyp.IsMigrationEnabled	Migration Enabled
AccessedViaTcpPort	app_hyp.DefaultExternalDataRoot	Accessed Via Tcp Port
usm-core2:DefaultExternalDataRoot	app_hyp.DefaultVhdPath	Default External Data Root
usm-core2:DefaultVhdPath	app_hyp.MinimumMacAddress	Default Vhd Path
usm-core2:MinimumMacAddress	app_hyp.MaximumMacAddress	Minimum Mac Address
usm-core2:MaximumMacAddress	app_hyp.ProcessID	Maximum Mac Address
ProcessDistinguishingID	app_hyp.AccessedViaTcpPort	Process Distinguishing ID
ProcessID	app_hyp.ProcessDistinguishingID	Process ID

Idea

The CA SDM Connector maps to the USM types based on the values of the following CA SDM properties.

Idea (USM)	CA SDM Attributes	User Interface
usm-udm:IdeaName	nr.name	Name
usm-udm:IsActive	nr.delete_flag	Active?

usm-udm:Manager	invidex.target_manager	Target Manager
usm-udm:GeneralNotes	invidex.general_notes	General Notes
usm-core2:URI	nr.zinvestment_uri	n/a
usm-udm:Priority	invidex.zudm_idea_priority	SPM Idea Priority
usm-udm:IdeaStatus	invidex.zudm_idea_lifecycle_status	SPM Idea Lifecycle Status
usm-udm:OriginatingRequestor	invidex.zoriginating_requestor	Originating Requestor
usm-udm:RequestID	invidex.zrequest_id	Request ID
usm-core2:CompleteUrl	invidex.zcomplete_url	n/a

Incident

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Incident (USM Type)	**CA SDM Attributes**	**User Interface**
AssignedID	cr.ref_num	Text on button bar (left side)
Urgency	n/a	Urgency
Impact	cr.impact; imp.sym	Impact
CalculatedPriority	cr.priority; pri.sym	Priority
Open Timestamp	cr.target_start	<bop_dump>
CallbackTimestamp	cr.call_back_date	Call Back Date/Time
Resolution Timestamp	cr.resolve_date	Resolve Date (above tabs)
Closure Timestamp	cr.close_date	Close Date (above tabs)
ChargebackID	n/a	n/a
RootCause	cr.rc	n/a
SolutionUrls	soln_log	n/a
IncidentCategory	cr.category	Incident Area
IsMajor	cr.major_incident	Major Incident
IncidentStatus	cr.status; cr.sym	Status
CreationDocUrl	n/a	n/a
OutageStartTimestamp	cr.outage_start_time	Outage tab; Start Time
OutageEndTimestamp	cr.outage_end_time	Outage tab; End Time
OutageType	cr.outage_type	Outage tab; Type
IsReturnedToService	cr.return_to_service	Outage tab; Return to Service
SymptomCodes	cr.symptom code	Symptom

InterfaceCard

The CA SDM Connector maps to the USM types based on the values of the following CA SDM properties.

NOTE

: You can define the Vendor and Model properties through the CA SDM connector policy to manage your configuration items. For more information about these properties, see Post-Installation: Manufacturers and Models.

InterfaceCard (USM)	CA SDM Attributes	User Interface
Vendor	nr.manufacturer.xym [ca_cmpny]	Inventory tab; Manufacturer (Configure this property in the connector policy to the requirement of your site)
Model	nr.model.sym [mfrmod]	Inventory tab; Model (Configure this property in the connector policy to the requirement of your site)
PhysSerialNumber	nr.serial_number	Serial Number
AssetNumber	nr.asset_num	Alternate CI ID
Device properties required	n/a	n/a
OSNumeric	n/a	n/a
ContainingIndex	n/a	n/a
IsPhysical	n/a	n/a

Location

The CA SDM Connector maps to the USM types based on the values of the following CA SDM properties.

Location (USM)	CA SDM Attributes	User Interface
LocationName	loc.name	Name
Uri	n/a	n/a
CountryCode	loc.country	Address tab; Country
CountryName	loc.country	Address tab; Country
PostalCode	loc.zip	Addresss tab; ZIP/Postal Code
StateOrProvince	loc.state	Address tab; State/Province
City	loc.city	Address tab; City
County	loc.county	n/a
AddressLine1	loc.address1	Address; Address (1)
AddressLine2	loc.address2	Address; Address (2)
AddressLine3	loc.address3	Address; Address (3)
Building	n/a	n/a
Floor	n/a	n/a
Room	n/a	n/a
Description	nr.description, loc.description	
PostalCode	loc.zip	
StateOrProvince	loc.state	
City	loc.city	
County	loc.county	
AddressLine1	loc.address1	

AddressLine2	loc.address2	
AddressLine3	loc.address3	
Floor	nr.loc_floor	
Room	nr.loc_room	

MediaDrive

The CA SDM Connector maps to the USM types based on the values of the following CA SDM properties.

NOTE

: You can define the Vendor and Model properties through the CA SDM connector policy to manage your configuration items. For more information about these properties, see Post-Installation: Manufacturers and Models.

MediaDrive (USM)	CA SDM Attributes	User Interface
Vendor	nr.manufacturer.xym [ca_cmpny]	Inventory tab; Manufacturer (Configure this property in the connector policy to the requirement of your site)
Model	nr.model.sym [mfrmod]	Inventory tab; Model (Configure this property in the connector policy to the requirement of your site)
PhysSerialNumber	nr.serial_number	Serial Number
AssetNumber	nr.asset_num	Alternate CI ID
Device properties required	n/a	n/a
OSNumeric	n/a	n/a
ContainingIndex	n/a	n/a
IsPhysical	n/a	n/a
DriveType	nr.class	Class
CapacityInMB	har_stox. media_drive_num	Attributes Tab memory capacity
SupportsRemovableMedia	n/a	n/a
SupportsWrite	n/a	n/a
Description	nr.description	
usm-core2: DriveInterfaceType	har_stox. Disk_type	

Memory

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Common CI Properties (USM)	CA SDM Attributes	User Interface / CI Detail Form
MemoryType|SizeInMB |ContainingIndex|DeviceDnsName	nr.name	Name
Physical	nr.class	Class
Hardware.Memory	nr.family	Family
ClassName|MemoryType|SizeInMB | ContainingIndex|DeviceDnsName	nr.system_name	Hostname
usm-core2:DeviceAssetNumber	har_mem.DeviceAssetNumber	Device Asset Number

usm-core2:DeviceBiosSystemID	har_mem.DeviceBiosSystemID	Device Bios System ID
usm-core2:DeviceDnsName	har_mem.DeviceDnsName	Device DNS Name
usm-core2:DeviceSysName	har_mem.DeviceSysName	Device System Name
usm-core2:DevicePhysSerialNumber	har_mem.DevicePhysSerialNumber	Device Physical Serial Number
usm-core2:DeviceMacAddress	har_mem.DeviceMacAddress	Device MAC Address
usm-core2:DeviceIPV4Address	har_mem.DeviceIPV4Address	Device IPv4 Address
usm-core2:DeviceIPV4AddressWithDomain	har_mem.DeviceIPV4AddressWithDomain	Device IPv4 Address With Domain
usm-core2:DeviceIPV6Address	har_mem.DeviceIPV6Address	Device IPv6 Address
usm-core2:DeviceIPV6AddressWithDomain	har_mem.DeviceIPV6AddressWithDomain	Device IPv6 Address With Domain
OSNumeric	har_mem.OSNumeric	OS Numeric
ContainingIndex	har_mem.ContainingIndex	Containing Index
IsPhysical	har_mem.IsPhysical	Is Physical
SizeInMB	har_mem.SizeInMB	Size In MB
ContextID	har_mem.ContextID	Context ID

NetworkServer

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Common CI Properties (USM)	CA SDM Attributes	User Interface / CI Detail Form
ProductName	nr.name	Name
NetworkServer	nr.class	Class
Software.NetworkServer	nr.family	Family
Protocol|ProcessDistinguishingID|DeviceDnsName	nr.system_name	Hostname
usm-core2:DeviceAssetNumber	app_netsvr.DeviceAssetNumber	Device Asset Number
usm-core2:DeviceBiosSystemID	app_netsvr.DeviceBiosSystemID	Device Bios System ID
usm-core2:DeviceDnsName	app_netsvr.DeviceDnsName	Device DNS Name
usm-core2:DeviceSysName	app_netsvr.DeviceSysName	Device System Name
usm-core2:DevicePhysSerialNumber	app_netsvr.DevicePhysSerialNumber	Device Physical Serial Number
usm-core2:DeviceMacAddress	app_netsvr.DeviceMacAddress	Device MAC Address
usm-core2:DeviceIPV4Address	app_netsvr.DeviceIPV4Address	Device IPv4 Address
usm-core2:DeviceIPV4AddressWithDomain	app_netsvr.DeviceIPV4AddressWithDomain	Device IPv4 Address With Domain
usm-core2:DeviceIPV6Address	app_netsvr.DeviceIPV6Address	Device IPv6 Address
usm-core2:DeviceIPV6AddressWithDomain	app_netsvr.DeviceIPV6AddressWithDomain	Device IPv6 Address With Domain
ProcessID	app_netsvr.ProcessID	Process ID
AccessedViaTcpPort	app_netsvr.AccessedViaTcpPort	Accessed Via Tcp Port
ProcessDistinguishingID	app_netsvr.ProcessDistinguishingID	Process Distinguishing ID
Protocol	app_netsvr.Protocol	Protocol
ContextID	app_netsvr.ContextID	Context ID

OperatingSystem

The CA SDM Connector maps to the USM types based on the values of the following CA SDM properties.

NOTE

You can define the Vendor and Model properties through the CA SDM connector policy to manage your configuration items. For more information about these properties, see Post-Installation: Manufacturers and Models.

OperatingSystem (USM Type)	CA SDM Attributes	User Interface / CI Detail Form
Vendor	nr.manufacturer	Inventory tab; Manufacturer (Configure this property within the connector policy to the requirement of your site)
ProductName	nr.name	Name
Version	nr.version	Attributes tab; Version
MajorVersion	nr.product_version	n/a
MinorVersion	nr.product_version	n/a
BuildNumber	n/a	n/a
Device properties required	n/a	n/a
NamedAliases	nr.name	
Description	nr.description	
OSType	nr.class	

OrganizationEntity

The CA SDM Connector maps to the USM types based on the values of the following SDM properties.

OrganizationEntity (USM Type)	SDM Attributes	User Interface
GroupName	n/a	Name
MemberCriteria	n/a	n/a
Category	n/a	n/a
OrgType	n/a	n/a
PrimaryPhoneNumber	org.phone_number	Primary Phone Number
OtherPhoneNumbers	org.phone	Alternate Phone Number
EmailAddresses	org.email_addr	Email Address
WebSites	n/a	n/a

Person

The CA SDM Connector maps to the USM types based on the values of the following CA SDM properties.

Person (USM)	CA SDM Attributes	User Interface
FirstName	cnt.first_name	First Name
MiddleNames	cnt.middle_name	Middle Name
FamilyName	cnt.last_name	Last Name

FullName	cnt.combo_name	n/a
EmployeeID	cnt.userid	Contact ID
UserName	cnt.userid	User ID
JobTitle	cnt.position	Job Title
CompanyName	n/a	n/a
IsActive	cnt.available	n/a
PrimaryPhoneNumber	cnt.phone_number	Notification tab; Telephone Number
OtherPhoneNumbers	n/a	n/a
EmailAddresses	cnt.email_address	Notification tab; Email Address

Port

The CA SDM Connector maps to the USM types based on the values of the following CA SDM properties.

NOTE

: You can define the Vendor and Model properties through the CA SDM connector policy to manage your configuration items. For more information about these properties, see Post-Installation: Manufacturers and Models.

Port (USM)	CA SDM Attributes	User Interface
Vendor	nr.manufacturer.xym [ca_cmpny]	Inventory tab; Manufacturer (Configure this property in the connector policy to the requirement of your site)
Model	nr.model.sym [mfrmod]	Inventory tab; Model (Configure this property in the connector policy to the requirement of your site)
PhysSerialNumber	nr.serial_number	Serial Number
AssetNumber	nr.asset_num	Alternate CI ID
Device properties required	n/a	n/a
OSNumeric	n/a	n/a
ContainingIndex	n/a	n/a
IsPhysical	n/a	n/a
PortID	n/a	n/a
CardOSNumeric	n/a	n/a
CardContainingIndex	n/a	n/a
ifindex	n/a	n/a
IfType	net_porx.technology	Attributes tab Technology
IfTypeExtension	n/a	n/a
NomSpeedInBitsPerSec	n/a	n/a
PrimaryMacAddress	nr.mac_address	MAC Address
OtherMacAddresses	n/a	n/a
PrimaryIPV4Address	nr.alarm_id	Inventory tab; IP Address
PrimaryIPV6Address	nr.alarm_id (if IPV6 is formatted)	Inventory tab; IP Address
OtherIPAddresses	n/a	n/a

OtherIPAddressesWithDomains	n/a	n/a
Description	nr.description	
PrimaryIPV4Address, PrimaryIPV6Address or OtherIPAddresses	nr.alarm_id	
PrimaryMacAddress or OtherMacAddresses	nr.mac_address	
usm-core2:IsFullDuplex	n/a	
usm-core2:DuplexIsNegotiated	n/a	

PortfolioApplication

The CA SDM Connector maps to the USM types based on the values of the following CA SDM properties.

PortfolioApplication (USM)	CA SDM Attributes	User Interface
usm-udm:PortfolioAppName	nr.name	Name
usm-udm:IsActive	nr.delete_flag	Active?
usm-udm:AvailabilityStart	invothx.start_date	Start Date
usm-udm:AvailabilityEnd	invothx.finish_date	Finish Date
usm-udm:Manager	invothx.manager	Manager
usm-core2:URI	nr.zinvestment_uri	n/a
usm-udm:OpenForTimeEntry	invothx.zopen_for_time_entry	Open for Time Entry?
usm-udm:Stage	invothx.zudm_stage	SPM Stage

Printer

The CA SDM Connector maps to the USM types based on the values of the following CA SDM properties.

NOTE

: You can define the Vendor and Model properties through the CA SDM connector policy to manage your configuration items. For more information about these properties, see Post-Installation: Manufacturers and Models.

Network Printer (USM Type)	CA SDM Attributes	User Interface / CI Detail Form
Vendor	nr.manuafacturer	Inventory tab; Manufacturer (Configure this property within the connector policy to the requirement of your site)
Model	nr.model.sym [mfrmod]	Inventory tab; Model (Configure this property within the IE Policy file to the requirement of your site)
RetirementTimestamp	nr.retire_date	Attributes tab; Retire Date
SerialNumber	nr.serial_number	Serial Number
VendorSerialNumber	n/a	n/a
AssetNumber	nr.asset_num	Alt CI ID
PrimaryDnsName	nr.dns_name	DNS Name
OtherDnsNames	n/a	n/a

LocalPrintername	nr.system_name	n/a
PrimaryMacAddress	nr.mac_address	MAC Address
OtherMacAddresses	n/a	n/a
PrimaryIPV4Address	nr.alarm_id (if IPV4 format)	Inventory tab; IP Address (V4)
PrimaryIPV6Address	nr.alarm_id (if IPV6 format)	Inventory tab; IP Address (V6)
OtherIPAddresses	n/a	n/a
SupportsColor	n/a	n/a
IsPhotoPrinter	n/a	n/a
SupportsDoubleSided	n/a	n/a
SupportsStapling	n/a	n/a
IsScanner	n/a	n/a
IsCopier	n/a	n/a
IsFax	n/a	n/a
FaxNumber	n/a	n/a

Problem

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Problem (USM Type)	**CA SDM Attributes**	**User Interface**
AssignedID	cr.ref_num	Text on button bar (left side)
IsActive [transform to "true" or "false"]	n/a	n/a
Urgency [IssueUrgencyEnum]	n/a	n/a
Impact	cr.impact; imp.sym	Impact
CalculatedPriority [IssuePriorityEnum]	cr.priority; pri.sym	Priority
Open Timestamp	cr.target_start_last	<bop_dump>
CallbackTimestamp	cr.call_back_date	Call Back Date/Time
Resolution Timestamp	cr.resolve_date	Resolve Date (above tabs)
Closure Timestamp	cr.close_date	Close Date (above tabs)
ChargebackID	n/a	n/a
RootCause [Issue Root Cause Enum]	cr.rc	Root Cause (Configure this property in the connector policy to the requirement of your site)
SolutionUrls	n/a	No Knowledge on Problem
ProblemCategory	cr.category	Problem Area
IsMajor	n/a	n/a
Status [Incident Status Enum]	cr.status; cr.sym	Status
CreationDocUrl	n/a	n/a

Processor

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Common CI Properties (USM)	CA SDM Attributes	User Interface / CI Detail Form
ProcessorType|SpeedInGHz|DeviceDnsName	nr.name	Name
x86	nr.class	Class
Hardware.Processor	nr.family	Family
ProcessorType|OSNumeric|ContainingIndex|DeviceDnsName	nr.system_name	Hostname
usm-core2:DeviceAssetNumber	har_prcr.DeviceAssetNumber	Device Asset Number
usm-core2:DeviceBiosSystemID	har_prcr.DeviceBiosSystemID	Device Bios System ID
usm-core2:DeviceDnsName	har_prcr.DeviceDnsName	Device DNS Name
usm-core2:DeviceSysName	har_prcr.DeviceSysName	Device System Name
usm-core2:DevicePhysSerialNumber	har_prcr.DevicePhysSerialNumber	Device Physical Serial Number
usm-core2:DeviceMacAddress	har_prcr.DeviceMacAddress	Device MAC Address
usm-core2:DeviceIPV4Address	har_prcr.DeviceIPV4Address	Device IPv4 Address
usm-core2:DeviceIPV4AddressWithDomain	har_prcr.DeviceIPV4AddressWithDomain	Device IPv4 Address With Domain
usm-core2:DeviceIPV6Address	har_prcr.DeviceIPV6Address	Device IPv6 Address
usm-core2:DeviceIPV6AddressWithDomain	har_prcr.DeviceIPV6AddressWithDomain	Device IPv6 Address With Domain
OSNumeric	har_prcr.OSNumeric	OS Numeric
ContainingIndex	har_prcr.ContainingIndex	Containing Index
IsPhysical	har_prcr.IsPhysical	Is Physical
Speed In GHz	har_prcr.SpeedInGHz	Speed In GHz
ContextID	har_prcr.ContextID	Context ID

ProvisionedSoftware

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

NOTE

: You can define the Vendor and Model properties through the CA SDM connector policy to manage your configuration items. For more information about these properties, see Post-Installation: Manufacturers and Models.

ProvisionedSoftware (USM)	**CA SDM Attribute**	**User Interface (CI detail form)**
Vendor	nr.manufacturer.xym [ca_cmpny]	Inventory tab; Manufacturer (Configure this property in the connector policy to the requirement of your site)
ProductName	nr.name	Name
Version	app_extx.version	Attributes tab version
MajorVersion	n/a	n/a
BuildNumber	n/a	n/a
ServiceLevel	n/a	n/a

Locales	n/a	n/a
ProcessorEnvironments	n/a	n/a
OSEnvironments	n/a	n/a
VirtualizationEnvironments	n/a	n/a
ReleaseType	n/a	n/a
SoftwareCategories	app_extx.category	Attributes tab category
IsLocal		
Device properties required	n/a	n/a
DeviceIPV6AddressWithDomain	n/a	n/a
ProvisionedForItem	n/a	n/a
SoftwarePathUrl	app_extx. install_dir	Attributes tab Install Directory
ProvisioningMethod	n/a	n/a
MdrElementID	nr.persid (Outbound)	
SoftwarePathUrl	app_extx. install_dir	
CreationTimestamp	nr.creation_date	
AdministrativeStatus = "Missing-InSubseqDiscover"	nr.status.sym [rss]	
usm-core2: BlueprintName	nr.family.sym + nr.class.sym	
SoftwareCategories	app_extx.category	

Request (USM Type)

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Request (USM Type)	CA SDM Attributes	User Interface
DeletionUserName	n/a	n/a
InstanceName	n/a	n/a
TypeName	n/a	n/a
AssignedID	cr.ref_num	Text on button bar (left side)
IsTemplate	n/a	n/a
TemplateUsed	n/a	n/a
Severity	n/a	Severity
Urgency [RequestUrgencyEnum]	n/a	Urgency
Priority	cr.priority; pri.sym	Priority
OpenTimestamp	cr.target_start	<bop_dump>
CallbackTimestamp	cr.call_back_date	Call Back Date/Time
ResolutionTimestamp	cr.resolve_date	Resolve Date (above tabs)
ClosureTimestamp	cr.close_date	Close Date (above tabs)
ChargebackID	n/a	n/a
RootCause	cr.rc	Root Cause
SolutionUrls	soln_log	n/a
RequestStatus	cr.status; cr.sym	Status

RequestCategory	cr.category	Request Area
UserAspect	n/a	n/a
CreationDocUrl	n/a	n/a
IsReturnedToService	cr.return_to_service	Outage tab; Return to Service

ResourceServer

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Common CI Properties (USM)	CA SDM Attributes	User Interface / CI Detail Form
ProductName	nr.name	Name
ResourceServer	nr.class	Class
Software.ResourceServer	nr.family	Family
ProductName|ProcessID	nr.system_name	Hostname
usm-core2:DeviceAssetNumber	app_ressvr.DeviceAssetNumber	Device Asset Number
usm-core2:DeviceBiosSystemID	app_ressvr.DeviceBiosSystemID	Device Bios System ID
usm-core2:DeviceDnsName	app_ressvr.DeviceDnsName	Device DNS Name
usm-core2:DeviceSysName	app_ressvr.DeviceSysName	Device System Name
usm-core2:DevicePhysSerialNumber	app_ressvr.DevicePhysSerialNumber	Device Physical Serial Number
usm-core2:DeviceMacAddress	app_ressvr.DeviceMacAddress	Device MAC Address
usm-core2:DeviceIPV4Address	app_ressvr.DeviceIPV4Address	Device IPv4 Address
usm-core2:DeviceIPV4AddressWithDomain	app_ressvr.DeviceIPV4AddressWithDomain	Device IPv4 Address With Domain
usm-core2:DeviceIPV6Address	app_ressvr.DeviceIPV6Address	Device IPv6 Address
usm-core2:DeviceIPV6AddressWithDomain	app_ressvr.DeviceIPV6AddressWithDomain	Device IPv6 Address With Domain
ProcessID	app_ressvr.ProcessID	Process ID
AccessedViaTcpPort	app_ressvr.AccessedViaTcpPort	Accessed Via Tcp Port
ProcessDistinguishingID	app_ressvr.ProcessDistinguishingID	Process Distinguishing ID
Resources	app_ressvr.capabilities	Capabilities
ContextID	app_ressvr.ContextID	Context ID

Router

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

NOTE

You can define the Vendor and Model properties through the CA SDM connector policy to manage your configuration items. For more information about these properties, see Post-Installation: Manufacturers and Models.

Router (USM Type)	**CA SDM Attributes**	**User Interface / CI Detail Form**
Vendor	nr.manuafacturer	Inventory tab; Manufacturer (Configure this property in the connector policy to the requirement of your site)

Model	nr.model.sym [mfrmod]	Inventory tab; Model (Configure this property in the connector policy to the requirement of your site)
SerialNumber	nr.serial_number	Serial Number
VendorSerialNumber	n/a	n/a
AssetNumber	nr.asset_num	Alternate CI ID
PrimaryDnsName	nr.dns_name	DNS name
OtherDnsNames	n/a	n/a
SysName	n/a	n/a
PrimaryMacAddress	nr.mac_address	MAC Address
OtherMacAddresses	n/a	n/a
PrimaryIPV4Address	nr.alarm_id	Inventory tab; IP Address (V4)
PrimaryIPV6Address	nr.alarm_id	IP Inventory tab; IP Address (V6)
OtherIPAddresses	n/a	n/a
FirmwareVersion	net_roux.rout_prot	Attributes tab; Router Protocol
Redundancy Type	n/a	n/a
FirmwareVersion	net_roux.os_version	
RoutingProtocolTypes	net_roux.rout_prot	
RoutingRedundancyType	n/a	

RunningHardware

The CA SDM Connector maps to the USM types based on the values of the following CA SDM properties.

NOTE

: You can define the Vendor and Model properties through the CA SDM connector policy to manage your configuration items. For more information about these properties, see Post-Installation: Manufacturers and Models.

RunningHardware (USM)	CA SDM Attributes	User Interface
Vendor	nr.manufacturer.xym [ca_cmpny]	Inventory tab; Manufacturer (Configure this property in the connector policy to the requirement of your site)
Model	nr.model.sym [mfrmod]	Inventory tab; Model (Configure this property in the connector policy to the requirement of your site)
PhysSerialNumber	nr.serial_number	Serial Number
AssetNumber	nr.asset_num	Alternate CI ID
Device properties required	n/a	n/a
OSNumeric	n/a	n/a
ContainingIndex	n/a	n/a
IsPhysical	n/a	n/a
Description	nr.description	

Service (USM Type)

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Service (USM Type)	CA SDM Attributes	User Interface / CI Detail Form
ServiceName	nr.name	Name
ServiceVersion	serx.version or entservx	Attributes tab; Version
AccessedViaTCPPort	n/a	n/a
BusinessRisk	entservx.business_risk or servx.business_risk	Attributes tab; Business Risk
BusinessImpact	n/a	Attributes tab; Business Impact
ImpactDescription	entservx.business_impact or serx.business_impact	n/a
AvailabilityStart	entservx.availability_start or serx.availability_start	Attributes tab; Availability Start
AvailabilityEnd	entservx.availability_end or serx.lifecycle_state	Attributes tab; Availability End
MdrElementID	nr.persid (OutBound)	
Description	nr.description	
CreationUserName	nr.creation_user (SREL --> cnt.user_name)	
CreationTimestamp	nr.creation_date	
LastModTimestamp	nr.last_mod	
UrlParams	"nr"	
ServiceCapabilities	nr.class	
usm-udm:IsActive	nr.delete_flag	Active?
usm-udm:ServiceManager	entservx.service_manager	Service Manager
usm-core2:URI	nr.zinvestment_uri	n/a
usm-udm:OpenForTimeEntry	entservx.zopen_for_time_entry	Open for Time Entry?
ServiceLifecycleState	entservx.zudm_lifecycle_state	SPM Service Lifecycle State

entservx.availability_start or serx.availability start

StoragePool

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Common CI Properties (USM)	CA SDM Attributes	User Interface / CI Detail Form
GroupName	nr.name	Name
StoragePool	nr.class	Class
Hardware.StoragePool	nr.family	Family
GroupName|GroupType|usm-core2:Device DnsName	nr.system_name	Hostname
usm-core2:DeviceAssetNumber	har_stgpl.DeviceAssetNumber	Device Asset Number
usm-core2:DeviceBiosSystemID	har_stgpl.DeviceBiosSystemID	Device Bios System ID
usm-core2:DeviceDnsName	har_stgpl.DeviceDnsName	Device Dns Name

usm-core2:DeviceSysName	har_stgpl.DeviceSysName	Device Sys Name
usm-core2:DevicePhysSerialNumber	har_stgpl.DevicePhysSerialNumber	Device Physical Serial Number
usm-core2:DeviceMacAddress	har_stgpl.DeviceMacAddress	Device Mac Address
usm-core2:DeviceIPV4Address	har_stgpl.DeviceIPV4Address	Device IPv4 Address
usm-core2:DeviceIPV4AddressWithDomain	har_stgpl.DeviceIPV4AddressWithDomain	Device IPv4 Address With Domain
usm-core2:DeviceIPV6Address	har_stgpl.DeviceIPV6Address	Device IPv6 Address
usm-core2:DeviceIPV6AddressWithDomain	har_stgpl.DeviceIPV6AddressWithDomain	Device IPv6 Address With Domain
usm-core2:IsHAEnabled	har_stgpl.IsHAEnabled	Is HA Enabled
usm-core2:IsMonitoringMembers	har_stgpl.IsMonitoringMembers	Is Monitoring Members
usm-core2:MaxFailures	har_stgpl.MaxFailures	Max Failures
usm-core2:MemberCriteria	har_stgpl.MemberCriteria	Member Criteria
usm-core2:GroupType	har_stgpl.GroupType	Group Type
usm-core2:BusinessRelevance	har_stgpl.BusinessRelevance	Business Relevance
usm-core2:CapacityInGB	har_stgpl.CapacityInGB	Capacity In GB
usm-core2:RaidLevel	har_stgpl.RaidLevel	Raid Level

StorageVolume

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Common CI Properties (USM)	CA SDM Attributes	User Interface / CI Detail Form
usm-core2:StorageID	nr.name	Name
StorageVolume	nr.class	Class
Hardware.StorageVolume	nr.family	Family
usm-core2:StorageID|usm-core2:PortWWName|usm-core2:PortID	nr.system_name	Hostname
usm-core2:DeviceAssetNumber	har_stgvol.DeviceAssetNumber	Device Asset Number
usm-core2:DeviceBiosSystemID	har_stgvol.DeviceBiosSystemID	Device Bios System ID
usm-core2:DeviceDnsName	har_stgvol.DeviceDnsName	Device DNS Name
usm-core2:DeviceSysName	har_stgvol.DeviceSysName	Device System Name
usm-core2:DevicePhysSerialNumber	har_stgvol.DevicePhysSerialNumber	Device Physical Serial Number
usm-core2:DeviceMacAddress	har_stgvol.DeviceMacAddress	Device MAC Address
usm-core2:DeviceIPV4Address	har_stgvol.DeviceIPV4Address	Device IPv4 Address
usm-core2:DeviceIPV4AddressWithDomain	har_stgvol.DeviceIPV4AddressWithDomain	Device IPv4 Address With Domain
usm-core2:DeviceIPV6Address	har_stgvol.DeviceIPV6Address	Device IPv6 Address
usm-core2:DeviceIPV6AddressWithDomain	har_stgvol.DeviceIPV6AddressWithDomain	Device IPv6 Address With Domain
usm-core2:OSNumeric	har_stgvol.OSNumeric	OS Numeric
usm-core2:ContainingIndex	har_stgvol.ContainingIndex	Containing Index
IsPhysical	har_stgvol.IsPhysical	Is Physical
usm-core2:LogicalUnitNumber	har_stgvol.LogicalUnitNumber	Logical Unit Number
usm-core2:PortID	har_stgvol.PortID	Port ID
usm-core2:PortWWName	har_stgvol.PortWWName	Port WW Name
usm-core2:CapacityInMB	har_stgvol.CapacityInMB	Capacity In MB

usm-core2:IsThinlyProvisioned	har_stgvol.IsThinlyProvisioned	Is Thinly Provisioned
usm-core2:IsDeDupeEnabled	har_stgvol.IsDeDupeEnabled	Is DeDupe Eanbled
usm-core2:IsMasked	har_stgvol.IsMasked	Is Masked
usm-core2:MaskedWWNames	har_stgvol.MaskedWWNames	Masked WW Names
usm-core2:RaidLevel	har_stgvol.RaidLevel	Raid Level

Switch

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

NOTE

You can define the Vendor and Model properties through the CA SDM connector policy to manage your configuration items. For more information about these properties, see Post-Installation: Manufacturers and Models.

Switch (USM Type)	CA SDM Attributes	User Interface / CI Detail Form
Vendor	nr.manuafacturer	Inventory tab; Manufacturer (Configure this property in the connector policy to the requirement of your site)
Model	nr.model.sym [mfrmod]	Inventory tab; Model (Configure this property in the connector policy to the requirement of your site)
SerialNumber	nr.serial_number	Serial Number
VendorSerialNumber	n/a	n/a
AssetNumber	nr.asset_num	Alt CI ID
PrimaryDnsName	nr.dns_name	DNS Name
OtherDnsNames	n/a	n/a
PrimaryMacAddress	n/a	n/a
PrimaryIPV4Address	nr.alarm_id	Inventory tab; IP Address
PrimaryIPV6Address	nr.alarm_id (if IPV6 format)	Inventory tab; IP Address
OtherIPAddresses	n/a	n/a
FirmwareVersion	net_nubx.os_version	Attributes tab; OS version

TimeReporting

The CA SDM Connector maps to the USM types based on the values of the following CA SDM properties.

usm-core2:TimeReporting (USM)	CA SDM Attributes	User Interface
usm-core2:ReportingID	alg.persistent_id or chgalg.persistent_id	n/a
usm-core2:IssueType	cr.type or chg.type	n/a
usm-core2:TimeSpentInHours	alg.time_spent or chgalg.time_spent	Time Spent
usm-core2:StartingDate	alg.time_stamp or chgalg.time_stamp	Date of Activity
Description	alg.description or chgalg.description	User Description
usm-core2:IssueID	alg.call_req_id or chg.persistent_id	n/a

usm-udm:InvestmentURI	nr.zinvesement_uri or chg.zproject_id	n/a
usm-udm:InvestmentType	based on the selection from nr.zinvestment_uri or chg.zproject_id	n/a
usm-core2:ProjectTaskID	chg.zproject_task_id	n/a
usm-udm:OriginatingURL	n/a	n/a
usm-core2:UserID	cnt.userid	Analyst

TransactionContext

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Common CI Properties (USM)	CA SDM Attributes	User Interface / CI Detail Form
ContextName	nr.name	Name
TransactionContext	nr.class	Class
Enterprise TransactionContext	nr.family	Family
ContextName|ContextType	nr.system_name	Hostname
ContextType	trn_ctx.ContextType	ContextType

VirtualManager

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Common CI Properties (USM)	CA SDM Attributes	User Interface / CI Detail Form
ProductName	nr.name	Name
VirtualManager	nr.class	Class
Software.VirtualManager	nr.family	Family
ProductName|ProcessID	nr.system_name	Hostname
usm-core2:DeviceAssetNumber	app_virmgr.DeviceAssetNumber	Device Asset Number
usm-core2:DeviceBiosSystemID	app_virmgr.DeviceBiosSystemID	Device Bios System ID
usm-core2:DeviceDnsName	app_virmgr.DeviceDnsName	Device DNS Name
usm-core2:DeviceSysName	app_virmgr.DeviceSysName	Device System Name
usm-core2:DevicePhysSerialNumber	app_virmgr.DevicePhysSerialNumber	Device Physical Serial Number
usm-core2:DeviceMacAddress	app_virmgr.DeviceMacAddress	Device MAC Address
usm-core2:DeviceIPV4Address	app_virmgr.DeviceIPV4Address	Device IPv4 Address
usm-core2:DeviceIPV4AddressWithDomain	app_virmgr.DeviceIPV4AddressWithDomain	Device IPv4 Address With Domain
usm-core2:DeviceIPV6Address	app_virmgr.DeviceIPV6Address	Device IPv6 Address
usm-core2:DeviceIPV6AddressWithDomain	app_virmgr.DeviceIPV6AddressWithDomain	Device IPv6 Address With Domain
ProcessID	app_virmgr.ProcessID	Process ID
AccessedViaTcpPort	app_virmgr.AccessedViaTcpPort	Accessed Via Tcp Port
ProcessDistinguishingID	app_virmgr.ProcessDistinguishingID	Process Distinguishing ID
ApiVersion	app_virmgr.ApiVersion	Api Version
ContextID	app_virmgr.ContextID	Context ID

VirtualSystem

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

NOTE

: You can define the Vendor and Model properties through the CA SDM connector policy to manage your configuration items. For more information about these properties, see Post-Installation: Manufacturers and Models.

VirtualSystem (USM Type)	CA SDM Attributes	User Interface / CI Detail Form
Vendor	nr.manufacturer.xym [ca_cmpny]	Inventory tab; Manufacturer (Configure this property in the connector policy to the requirement of your site)
Model	nr.model.sym [mfrmod]	Inventory tab; Model (Configure this property in the connector policy to the requirement of your site)
PhysSerialNumber	nr.serial_number	Serial Number
AssetNumber	nr.asset_num	Alternate CI ID
PrimaryDnsName	dns_name	Virtual Host Name
OtherDnsNames	n/a	n/a
SysName	n/a	n/a
PrimaryMacAddress	nr.mac_address	MAC Address
OtherMacAddresses	n/a	n/a
PrimaryIPV4Address	nr.alarm_id	Inventory tab; IP Address
PrimaryIPV6Address	nr.alarm_id (if IPV6 format)	Inventory tab; IP Address (if IPV6 format)
OtherIPAddresses	n/a	n/a
ComputerName	nr.name	Name
BiosSystemID	har_virx.bios_ver	Attributes tab; Bios Version
Type	n/a	n/a
MemoryInGB	har_virx.phys_mem (SDM no float type)	Attributes tab; Memory Installed
StorageInGB	har_virx.hard_drive_capacity (SDM no float type)	Attributes tab; Disk Capacity
ProcessorType	har_virx.proc_type	Attributes tab; Processor Type
NumberOfCores	har_virx.virtual_processors	Attributes tab; No of Process Installed
ProcessorSpeedInGHz	har_virx.proc_speed	Attributes tab; Processor Speed
PrimaryOSType	nr.class	Class
PrimaryOSVersion	n/a	n/a
OtherOSDetails	n/a	n/a
usm-core2:VirtualizationEnvironment	nr.class	
usm-core2:IsAutomaticallyStarted	n/a	

Attributes tab; No of Processs Installed

VMDataStore

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Common CI Properties (USM)	CA SDM Attributes	User Interface / CI Detail Form
StoreName	nr.name	Name
NetworkFileSystem	nr.class	Class
Hardware.VMDataStore	nr.family	Family
StoreName|FilePathUrl	nr.system_name	Hostname
usm-core2:DeviceAssetNumber	har_vmds.DeviceAssetNumber	Device Asset Number
usm-core2:DeviceBiosSystemID	har_vmds.DeviceBiosSystemID	Device Bios System ID
usm-core2:DeviceDnsName	har_vmds.DeviceDnsName	Device DNS Name
usm-core2:DeviceSysName	har_vmds.DeviceSysName	Device System Name
usm-core2:DevicePhysSerialNumber	har_vmds.DevicePhysSerialNumber	Device Physical Serial Number
usm-core2:DeviceMacAddress	har_vmds.DeviceMacAddress	Device MAC Address
usm-core2:DeviceIPV4Address	har_vmds.DeviceIPV4Address	Device IPv4 Address
usm-core2:DeviceIPV4AddressWithDomain	har_vmds.DeviceIPV4AddressWithDomain	Device IPv4 Address With Domain
usm-core2:DeviceIPV6Address	har_vmds.DeviceIPV6Address	Device IPv6 Address
usm-core2:DeviceIPV6AddressWithDomain	har_vmds.DeviceIPV6AddressWithDomain	Device IPv6 Address With Domain
FilePathUrl	har_vmds.FilePathUrl	File Path Url
CapacityInMB	har_vmds.CapacityInMB	Capacity In MB
IsMultiHost	har_vmds.IsMultiHost	Is Multi Host

Website

The CA SDM Connector maps to USM types based on the values of the following CA SDM properties.

Common CI Properties (USM)	CA SDM Attributes	User Interface / CI Detail Form
GroupName	nr.name	Name
WebSite	nr.class	Class
Software.WebSite	nr.family	Family
usm-core2:HomePage|usm-core2:DeviceBiosSystemID	nr.system_name	Hostname
usm-core2:DeviceAssetNumber	app_website.DeviceAssetNumber	Device Asset Number
usm-core2:DeviceBiosSystemID	app_website.DeviceBiosSystemID	Device Bios System ID
usm-core2:DeviceDnsName	app_website.DeviceDnsName	Device DNS Name
usm-core2:DevicePhysSerialNumber	app_website.DevicePhysSerialNumber	Device Physical Serial Number
usm-core2:DeviceMacAddress	app_website.DeviceMacAddress	Device MAC Address
usm-core2:DeviceIPV4Address	app_website.DeviceIPV4Address	Device IPv4 Address
usm-core2:DeviceIPV4AddressWithDomain	app_website.DeviceIPV4AddressWithDomain	Device IPv4 Address With Domain
usm-core2:DeviceIPV6Address	app_website.DeviceIPV6Address	Device IPv6 Address
usm-core2:DeviceIPV6AddressWithDomain	app_website.DeviceIPV6AddressWithDomain	Device IPv6 Address With Domain

usm-core2:IsHAEnabled	app_website.IsHAEnabled	Is HA Enabled
usm-core2:IsMonitoringMembers	app_website.IsMonitoringMembers	Is Monitoring Members
usm-core2:MaxFailures	app_website.MaxFailures	Max Failures
usm-core2:MemberCriteria	app_website.MemberCriteria	Member Criteria
usm-core2:GroupType	app_website.GroupType	Group Type
usm-core2:HomePage	app_website.HomePage	Home Page
usm-core2:BusinessRelevance	app_website.BusinessRelevance	Business Relevance
usm-core2:DeviceSysName	app_website.DeviceSysName	Device System Name

Default Port Numbers and Connectivity - SOI Integration with SDM

The following list shows all of the default port numbers available for each of the components in the CA SDM-CA SOI integration:

NOTE
The list describes only the default ports. Some of the ports may or may not be applicable for this integration.

CA Catalyst

- Server Container: 7000
- Connector Container: 8080
- Search Server: 7443 (HTTPS)
- Persistence Store (Microsoft SQL database): 1433
- ActiveMQ Server: 61616
- Registry shutdown: 8005
- Registry AJP: 8009
- Registry: 8081, 8443
- Persistence Store (Oracle Database Server): 1521
- Administration UI: 8082
- Administration UI AJP: 8010
- Administration UI shutdown: 8006

CA SDM Server

- Server Tomcat Port: 8080
- SA Tomcat Port: 8070
- SA Tomcat Shutdown Port: 8075
- Shutdown Port: 8085
- CA CMDB Visualizer: 9080
- CA CMDB Visualizer Shutdown Port: 9085

CA SOI

- Shutdown Port: 7095 (Manager)
- Shutdown Port: 7005 (UI Server)
- Tomcat Server Port: 7090 (Manager)
- Tomcat Server Port: 7070 (UI Server)
- SSL Port: 7493 (Manager)
- SSL Port: 7403 (UI Server)
- JDBC Port: 1433
- CMS Server Port: 6400

Relationship and Service Mapping

This topic includes the following information about Relationship and Service Mapping for CA CMDB and CA Configuration Automation integration:

Supported CIs and Relationships

The ServiceDeskManagerConnector_policy.xml and ServiceDeskManager_policySB.xml files contain complete mapping information of the CI and CI relationship data for the integration.

The following table lists the Relationship mapping for the integration:

Parent	Relationship Type	Child Type	CMDB Support?
Service			Yes
	IsHostedBy	ComputerSystem	Yes
	HasDetail	ComplianceStatus	No
	HasContact	Person	Yes
ComputerSystem			Yes
	IsHostFor	ProvisionedSoftware	Yes
	IsHostFor	ComputerSystem	Yes
	IsManagerFor	ComputerSystem	Yes
	HasDetail	ComplianceStatus	No
	HasContact	Person	Yes
	HasDetail	IPConfig	No
	IsComposedOf	Port	Yes
	IsComposedOf	OperatingSystem	Yes
	IsComposedOf	RunningHardware	Yes
	IsComposedOf	Memory	No
	IsComposedOf	MediaDrive	Yes
	IsComposedOf	Processor	No
ProvisionedSoftware			Yes
	HasAccessTo	ComputerSystem	Yes
	HasRequirementFor	ComputerSystem	Yes
Compliance Status			No
Location			Yes
	IsLocationFor	ComputerSystem	Yes
	IsLocationFor	Router	Yes
	IsLocationFor	Service	Yes
VirtualSystem			Yes
Router			Yes

Port			Yes
	HasDetail	IPConfig	No
Cluster			Yes
	HasMember	ComputerSystem	Yes
MediaDrive			Yes
	IsHostFor	File	Yes
Application			Yes
	HasRequirementFor	Port	Yes
	IsConnectedTo	Port	Yes
	IsConnectedTo	Application	Yes
BackgroundProcess			No
	HasRequirementFor	Port	No
	IsConnectedTo	BackgroundProcess	No
RunningHardware			Yes
Memory			No
OperatingSystem			Yes
Processor			No
DiskPartition			No
File			Yes
IPConfig			No
BinaryRelationship			Yes
ApplicationServer			Yes
DatabaseInstance			Yes
GenericIPDevice			Yes
InterfaceCard			Yes
Printer			Yes
Switch			Yes
Person			Yes
OrganizationEntity			Yes
Incident			Supported (Only Outbound)
	IsImpactedBy	(Any Supported CMDB CI)	Supported (Only Outbound)
Problem			Supported (Only Outbound)
	IsImpactedBy	(Any Supported CMDB CI)	Supported (Only Outbound)
Request			Supported (Only Outbound)
	IsImpactedBy	(Any Supported CMDB CI)	Supported (Only Outbound)
ChangeOrder			Supported (Only Outbound)
	IsImpactedBy	(Any Supported CMDB CI)	Supported (Only Outbound)

Service Mapping

The following table lists the mapping for the Services:

Services (CA Configuration Automation)	CA Configuration Automation Schema [acm_svc]	USM:Service	CA SDM Business Service
service uuid	svc_uuid	MdrElementID	nr.persid (Outbound)
service name	svc_name	ServiceName	nr.name
description	descr	Description	nr.description
created by	created_by	CreationUserName	nr.creation_user (SREL cnt.user_name)
creation time	creation_tm	CreationTimestamp	nr.creation_date
business process	business_process	usm-core2:BusinessRelevance **Note:** All CIs contain this field, not only for USM:Service	nr.service_goal
modification time	modification_tm	LastModTimestamp	nr.last_mod
Launch-in-context URL	[acm_svc].svc_name	UrlParams	nr **Note:** This integration only uses the CA SDM object factory for backward compatibility with SSA/Catalyst 2.5.
N/A	N/A	ServiceVersion	serx.version or entservx.version
N/A	N/A	AccessedViaTCPPort	N/A
N/A	N/A	BusinessRisk	entservx.business_risk or servx.business_risk
N/A	N/A	ImpactDescription	entservx.business_impact or serx.business_impact
N/A	N/A	AvailabilityStart	entservx.availability_start or serx.availability start
N/A	N/A	AvailabilityEnd	entservx.availability_start or serx.availability start
N/A	N/A	usm-core2: CommonlyKnownNames	nr.asset_num
N/A	N/A	ServiceCapabilities	nr.class Note: For Services, you can also use Alt CI Id
N/A	N/A	ServiceLifecycleState	entservx.lifecycle_state

Troubleshooting

The section contains the following articles:

- Troubleshooting CA Service Management
- Troubleshooting CA Service Desk Manager
- Troubleshooting xFlow Interface
- Troubleshooting Service Point
- Troubleshooting CA Service Management Administration
- Troubleshooting Insights (CABI JasperReports)
- Troubleshooting CA Service Catalog
- Troubleshooting CA Asset Portfolio Management
- Troubleshooting CA Service Desk Manager Connector
- Troubleshooting CA CMDB and CA Configuration Automation Integration
- Troubleshooting CA Virtual Analyst Integration with MS Teams

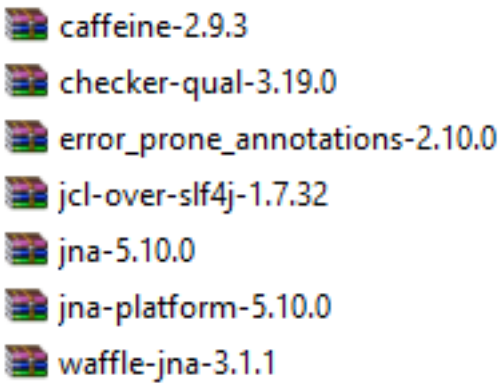
Troubleshooting CA Service Management

Common issues encountered while implementing CA Service Management:
3

SDM Login Page Not Accessible After Upgrading to 17.4 in NTLM Environment

If the CA Service Management login page is not accessible after upgrading to 17.4 in NTLM environment, follow these steps:

1. Download the latest waffle from https://jar-download.com/artifacts/com.github.waffle/waffle-jna/3.1.1.
2. Navigate to SDM Server and delete all existing waffle dependent jars from `$NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF\lib.`
3. Place the downloaded **waffle 3.1.1 jars** in the `$NX_ROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF\lib` location and delete **slf4j-api-1.7.32.jar** file.



4. Remove below line present in **$NX_ROOT\bopcfg\www\CATALINA_BASE\conf\context.xml** (if present).

```
<Context>
    <Valve className="waffle.apache.NegotiateAuthenticator" principalFormat="fqn" roleFormat="both" />
    <Realm className="waffle.apache.WindowsRealm" />
</Context>
```

5. Restart the SDM Tomcat Services.

After implementing , fuzzy text is displayed on the web interface

CA Sans is a new font used in CA Service Management 17.4 release. This font is difficult for few users to read and understand. As a workaround, users can override the default font for the web interface.

Firefox

1. Click the menu button and choose **Options**.
2. In the General panel, scroll down to **Language and Appearance**.
3. Choose a default font and size in the **Fonts & Colors** section.
4. Uncheck the **Allow pages to choose their own fonts, instead of my selections above** option.

Internet Explorer

1. Click on **Tools**, **Internet Options**, and **Accessibility**.
2. Check the **Ignore font styles specified on webpages** option.
3. Click **Ok**.
4. You can specify default fonts for different styles, including serif, sans-serif, and monospace by clicking the Fonts section.
5. Click **Apply**.

Google Chrome

Enable the Font Changer with Google Web Fonts extension . You can use this extension to pick and choose the web interface font.

> **TIP**
>
> CA makes no representations or warranties regarding these third party sites or any content that may be linked to through a third party hyperlink

Unable to Install

If the CA Service Management installation fails, check the log files, take corrective actions, and retry the installation. If the installation still fails, proceed to integrate the required products or common components manually.

Post Installation Steps for CA Service Desk Manager Fail

Complete the following steps when the Post Installation steps for CA Service Desk Manager Updates(s):17.4 task fails during the patch installation:

1. Do not close the common installer window when you get the above error.
2. Check the .HIS file; if language combo and language patch information are missing, run Step 3.
3. Manually install language combo and language patch using the applyPTF tool.
4. Close the error popup window in the common installer.
5. Click on the "Retry Install" button in the common installer.

The above workaround steps will complete the "Post Installation steps for CA Service Desk Manager Updates(s):17.4" task without any errors.

Error Messsage: Failed to grant service user 'NT AUTHORITY\LocalService' write permissions to log path 'C:\Windows\system32\LogFiles\Apache' due to error '19: The media is write protected.'

If you view this error message, do one of the following:

- Set the Apache Tomcat 9.0 jasperreportsTomcat service manually under the Log On tab as Local system account.
- Run the following command:

```
sc config jasperreportsTomcat obj=LocalSystem
```

Integrate Service Desk Manager with Asset Portfolio Management Manually

WARNING

Review the following section only if you want to manually integrate CA Service Desk Manager with CA Asset Portfolio Management when the automatic integration between the two has failed.

To manually integrate CA APM and CA SDM, follow these steps:

1. Enable Single Sign-On (SSO) from CA Asset Portfolio Management to CA Service Desk Manager
2. Verify Single Sign-On
3. Launch CA Asset Portfolio Management in Context From CA Service Desk Manager

Enable Single Sign-On (SSO) from CA Asset Portfolio Management to CA Service Desk Manager

CA Service Management provides the following ways to enable SSO from CA Asset Portfolio Management (CA APM) to CA Service Desk Manager.

- Use the Web Service Access Policy File
- Use the CA Service Desk Manager Administrator Credentials

Both the mechanisms internally use the CA Service Desk Manager SOAP Web Services to achieve the SSO from CA APM to CA SDM. You can configure either one or both the mechanisms to achieve SSO from CA APM to CA SDM.

WARNING
Ensure that CA APM and CA Service Desk Manager are installed on the same database.

NOTE
CA APM uses the fallback mechanism during the Single-Sign On process. If both the mechanisms are configured, the Policy File mechanism is used. If the Policy file does not work, then CA Service Desk Manager Administrator credentials are used. If neither of these are successful, then the CA Service Desk Manager login screen is displayed on clicking the respective hyperlink in CA APM. If the CA APM user does not have valid access rights (role/group) in CA Service Desk Manager, the Single Sign-On login screen is displayed on clicking the CA Service Desk Manager links on the CA APM page

Use the CA Service Desk Manager Administrator Credentials

The CA Service Desk Manager admin credentials are used from CA APM to login to CA Service Desk Manager through the CA Service Desk Manager Web Services.

Follow these steps:

1. Log in to CA APM as an Administrator. Default user name is *uapmadmin*.
2. Click Administration, System Configuration, Service Desk Manager.
3. Enter the complete URL for CA Service Desk Manager SOAP Web Services in the following format:
   `http://<ServiceDeskHostName>:<PortNum>/axis/services/USD_R11_WebService`
4. Enter the CA Service Desk Manager Administrator login user name and password.
5. Click Save.

NOTE
Do not perform IISReset on the CA APM Web Server or CA APM Application Server.

Use the Web Service Access Policy File

You create the Access Policy file in CA Service Desk Manager and then use this file in CA APM. The Access Policy file contains encrypted policy details that are assigned to a CA Service Desk Manager user who has access to various capabilities in the CA SDM application.

Follow these steps:

1. Log in to the CA Service Desk Manager as an Administrator. Use the Administration tab to perform the following tasks:
   a. Create the Web Services Access Policy using the SOAP Web Services Policy option. Complete the following:
      a. Assign a name. The default name and code are CASM_POLICY; however, you can use any name as per your requirements.
      b. Assign a proxy user to this policy.

         NOTE

         This proxy user must have administrative access to CA Service Desk Manager.
      c. Verify that the Allow Impersonate property of CA Service Desk Manager is set to Yes.
         CA Service Desk Manager checks the access type of the CA APM user to be impersonated against the access type of the proxy user of the policy. If the access level of the CA APM user access type is less than or equal to the grant level of the proxy user access type, CA SDM replaces the CA APM user with the proxy user.
   b. Generate the policy key file with the help of pdm_pki tool for the Web Services policy that you created.
      In the Service Desk environment, open the Command Prompt and enter:
      ```
      C:>pdm_pki -p CASM_POLICY -f
      ```
      The policy file CASM_POLICY.p12 is generated in the same folder.
      The default/recommended name is CASM_POLICY.p12, but you can use any name as per your requirement.
   c. Verify that the Has Key property is set to Yes in the newly created Web Services access policy. The system typically configures this setting automatically when you generate the policy key file.
   d. Copy and save the policy key file to the CA APM default installation directory of the CA APM Web Server.
      ```
      C:\Program Files (x86)\CA\ITAM
      ```

      NOTE
      If you are using application clustering, copy the policy key file to the CA APM installation directory of every CA APM Webserver Component cluster node. If you re-configure the Policy in CA SDM, then you must re-generate the policy file and copy it to CA APM Web Server installed directory.
2. Log in to CA APM as an Administrator.
   Default: *uapmadmin*.
3. Click Administration, System Configuration, Service Desk Manager.
4. Enter the complete URL for CA Service Desk Manager SOAP Web Services in the following format:
   ```
   http://<ServiceDeskHostName>:<PortNum>/axis/services/USD_R11_WebService.
   ```
5. Copy the CA Service Desk Manager SOAP Policy file from CA Service Desk Manager Server to the CA APM web server installation folder. Default installation directory :
   ```
   C:\Program Files (x86)\CA\ITAM
   ```
6. Enter the file name of CA Service Desk Manager policy file that is copied on the CA APM Web server component installation folder.
7. Enter the CA Service Desk Manager policy code that is the policy name as configured in the CA Service Desk Manager Administration SOAP Web Services policy configuration page.
8. Click Save.

   NOTE
   Do not perform IISReset on the CA APM Web Server or CA APM Application Server.

Verify Single Sign-On

To verify that Single Sign-On from CA APM to CA Service Desk Manager has been enabled, click on any *one* of the following CA Service Desk Manager hyperlinks in the CA APM page:

- Log in to CA APM, Asset Details page and click on the CA Service Desk Manager link on the top
- Click Asset Relationships (menu on the left), CA Service Desk Manager Tickets links. A Criteria and Results page with Service Desk Ticket ID Column values in hyperlink is displayed in the results section. The hyperlinks are the entry point to CA SDM and display ticket details after you have successfully logged into CA SDM.
- Click on the Launch CA Service Desk Manager link in the Asset Viewer page. This launches the CA Service Desk Manager application.

View the Contact Update Details

You can view the contact update details for existing or new assets (known as configuration items in CA Service Desk Manager) in the following ways:

- Contact Update Details for existing Contacts in CA APM
  Ensure that you have enabled Single Sign-On from CA APM to CA Service Desk Manager for existing contact details to be updated and reflected in CA Service Desk Manager Contact's Quick Profile.
- Contact Update Details for new Assets in CA APM
  In this release, if you have integrated CA APM and CA Service Desk Manager, new Assets that are created have the Contact details automatically saved and reflected in CA Service Desk Manager Contact's Quick Profile.

Launch CA Asset Portfolio Management in Context From CA Service Desk Manager

To launch CA APM in context from CA SDM, you need to add a Management Data Repository (MDR) in CA SDM to point to the proper CA APM URL.

Follow these steps:

1. Create an MDR in CA SDM with the following properties:
   a. Button Name: CA APM
   b. MDR Name: CA APM
   c. MDR Class: GLOBAL
   d. Hostname: CA APM Server Name
   e. Path: ITAM/Pages/Asset.aspx
   f. Parameters: assetid={federated_asset_id}
2. Click the CA APM button.
   The corresponding asset record opens in CA APM.

Integrate CA Service Catalog with CA Service Desk Manager Manually

WARNING

Review the following section only if you want to manually integrate CA Service Catalog with CA Service Desk Manager when the automatic integration between the two has failed.

CA Service Catalog integration with CA Service Desk Manager is achieved through catalog request fulfillment, which can be configured to automatically open CA Service Desk Manager requests or change orders.

If you are integrating CA Service Catalog and CA Asset Portfolio Management, the associated assets become configuration items on the related change order. The CA Service Desk Manager request and change orders can be viewed in the CA Service Catalog request's Related Tickets column.

Using the integration with CA Service Desk Manager, when a user requests a service from the catalog and it is approved, the fulfillment process could include identifying the correct existing asset and a CA Service Desk Manager change order could be opened assigning a configuration and delivery task to a technician for the asset.

Finally, you can also integrate CA Service Catalog and CA Service Desk Manager by configuring them to use the same common multi-tenant administration.

Set up the CA Service Catalog - CA Service Desk Manager Integration

The major tasks to integrate CA Service Catalog and CA Service Desk Manager follow. These tasks are explained in detail in the sections that follow.

1. Verify the Prerequisites for CA Service Catalog - CA Service Desk Manager Integration
2. Understand the Key Terms
3. Configure communication between CA Service Catalog and CA Service Desk Manager by ensuring that a CA Service Desk Manager primary server is properly defined to CA Service Catalog.
   A CA Service Desk Manager primary server is always required to view or open tickets or change orders. To configure CA Service Catalog so that a user can view a CA Service Desk Manager ticket (change order or request) while viewing a CA Service Catalog request, define the CA Service Desk Manager host specified when you select Administration, Configuration, Service Desk.
   In those configuration options, you must also update configuration details for the CA Service Desk Manager primary server.
4. You can optionally bypass user login by configuring CA Service Catalog and the CA products with which it integrates to use single sign-on.
   To configure CA Service Desk Manager to use single sign-on, configure it to use either CA SiteMinder or NTLM authentication on Windows.
   Similarly, to configure CA Service Catalog to use single sign-on, configure it to use either CA SiteMinder or NTLM authentication on Windows.
5. (Optional) Configure the integrating products to open CA Service Desk Manager change orders during request fulfillment.
6. (Optional) Configure the integrating products to open CA Service Desk Manager service requests during request fulfillment.
7. (Optional) Synchronize notes and attachments.

CA Service Catalog integration with CA Service Desk Manager is achieved through catalog request fulfillment, which can be configured to automatically open CA Service Desk Manager requests or change orders.

The CA Service Desk Manager request and change orders can be viewed in the CA Service Catalog request's Related Tickets column.

Using the integration with CA Service Desk Manager, when a user requests a service from the catalog and it is approved, the fulfillment process could include identifying the correct existing asset and a CA Service Desk Manager change order could be opened assigning a configuration and delivery task to a technician for the asset.

Finally, you can also integrate CA Service Catalog and CA Service Desk Manager by configuring them to use the same common multi-tenant administration.

{"URL":["/usm/wpf?Node=iclaunchpad.tools_configuration"],"customLabelGetStarted":"Service Desk","description":"IPCE_Service_Desk"}

Step 1 - Verify the Prerequisites for CA Service Catalog - CA Service Desk Manager Integration

To enable the integration between CA Service Desk Manager and CA Service Catalog, meet the following prerequisites:

- You must be familiar with the basic functions and administration of CA Service Desk Manager and CA Service Catalog.

WARNING
Ensure that CA Service Catalog and CA Service Desk Manager share the same installation of the MDB, CA EEM, and CA Process Automation (if you are using CA Process Automation).

- Both CA Service Catalog and CA Service Desk Manager must be installed, configured, and running.
  Verify that CA Service Desk Manager has the latest service packs and patches applied.
- If you are using CA Process Automation as the process automation tool for CA Service Catalog and CA Service Desk Manager, verify that you:
  - Install and configure CA Process Automation.
    Verify that CA Process Automation has the latest service packs and patches applied.
  - Install and configure a CA Process Automation-CA Service Desk Manager connector on the CA Process Automation domain orchestrator computer. For more information, see your CA Process Automation administrator.
  - Integrate CA Service Catalog with CA Process Automation Manually. If your implementation of CA Service Desk Manager uses CA Process Automation and the ITIL content pack, review the CA Process Automation Workflow documentation.
  - Integrate CA Service Desk Manager with CA Process Automation manually.

Step 2 - Understand the Key Terms

This section describes several key terms that you should know when implementing an integration between CA Service Desk Manager and CA Service Catalog. In CA Service Desk Manager, requests, change orders, and related items are the types of tickets that you can open. CA Service Desk Manager defines requests, change orders, and related terms from both an ITIL and non-ITIL perspective.

For a CA Service Desk Manager customer who is *not* running ITIL, the possible ticket types to open up are as follows:

- Requests are requests for service from internal customers, or in the non-ITIL case could also represent a failure.
- Issues are same as the above except they are meant for external customers.
- Change orders are a request for some type of change to the environment.

For a CA Service Desk Manager customer who *is* running ITIL, the possible ticket types to open up are as follows:

- Requests are requests for service for internal customers.
- Issues are requests for service from external customers.
- Change orders are a request for some type of change to the environment.
- Incident is a disruption in service, meant to be restored as quickly as possible.
- Problem is for getting to the root cause.

The following is a typical scenario to distinguish between these ticket types on an ITIL installation:

1. An end user opens up an incident with the helpdesk because of problems connecting to email.
2. At the same time, other end users also open up incidents complaining of the same.
3. To restore service as quickly as possible, the email server is rebooted.
   End users can now connect to email. The incident(s) is resolved. Although rebooting the email server got the end users up and running, we still need to determine and fix the root cause.
   From here a problem is opened and attached to the incident(s) for additional research.

After research, the root cause looks to be that there is not enough RAM on the email server, so a change order is opened for the proper approvals to go through to add the additional RAM, this change order is then linked to the problem.

Step 3 - Open Change Orders During Request Fulfillment

During the fulfillment phase of the request life cycle, CA Service Catalog can open a CA Service Desk Manager change order. Be careful when activating rules to understand the full implications of your changes. For more information about the CA Service Catalog request life cycle, see the section Request Management from an Administrator Perspective.

To configure CA Service Desk Manager so that CA Service Catalog request fulfillment can open CA Service Desk Manager change orders, complete the CAfollowing tasks:

1. If you are using CA Process Automation, update CA Process Automation-related settings.
2. Enable rule actions.

Update CA Process Automation Settings

Update these CA Process Automation-related settings to complete the process of configuring CA Process Automation to support the opening of change orders in CA Service Desk Manager during request fulfillment.

Follow these steps:

1. In CA Process Automation, perform these steps:
   a. Access the Service Desk SRF folder (CA SDM/SRF).
   b. Edit the HWSW_FilledFromInventory SRF.
   c. Add the <chgcat> tag to this SRF.
   This step updates the SRF to process change orders in CA Service Desk Manager.
2. In CA Process Automation, perform these steps:
   a. Click Configuration, Domain, Modules, and click Lock.
   b. Double click Edit CA ServiceDesk Module.
   c. Enter the ServiceDeskWebService URL, for example, http://hyderabad129:8080/axis/services/USD_R11_WebService.
   d. Enter the user name and password of the CA Service Desk Manager administrator.
   e. Save and click Unlock.
3. In CA Service Desk Manager, perform these steps:
   a. Select Options Manager, Change order Manager, Category_Defaults.
   b. Install the Category_Defaults option.
   c. Restart CA Service Desk Manager.
   This step updates CA Service Desk Manager to process change orders from CA Service Catalog according to your specifications.

You have updated the CA Process Automation-related settings.

Enable Rule Actions

To cause CA Service Catalog to open a CA Service Desk Manager change order during request fulfillment, you must enable several rule actions that are disabled by default.

Follow these steps:

1. Select **Administration**, **Events-Rules-Actions**.
2. Click the Event Type named **Request/Subscription Item Change**.
3. Click the rule named **When Category is Hardware and Status is Filled from Inventory**.
   The list shows the CA Process Automation actions that you can use with this rule. Use the remaining steps as a model to modify your start request forms (SRFs) and processes.
4. To enable the CA Service Desk Manager change orders, disable the **Launch FilledFromInventory SRF for Hardware** action and enable the **Launch HWSWFilledFromInv_SDM SRF** action.
   a. Click the Edit icon for the action named **Launch FilledFromInventory SRF for Hardware**.

b. Select **Disabled** from the **Status** drop-down list and click OK.
c. Click the Edit icon for the action named **Launch HWSWFilledFromInv_SDM SRF**.
d. Select **Enabled** from the **Status** drop-down list and click OK.

The Rule Action screen is displayed showing the **Launch FilledFromInventory SRF for Hardware** action disabled and the **Launch HWSWFilledFromInv_SDM SRF** action enabled.

5. Repeat the previous steps for the rule named **When Category is Software and Status is Filled from Inventory**.

Activate all except **When Status is Pending Fulfillment** for the entire complex request life cycle involving CA Service Desk Manager to work correctly.

Step 4 - Synchronize Notes and Attachments

If you have configured CA Service Catalog to create a ticket (change order or incident) in CA Service Desk Manager when a catalog user submits a request for certain services, each ticket is associated with a specific service option in the requested service. You can enhance the integration by synchronizing notes and attachments to be copied from a service option its related ticket. You can specify this synchronization to occur both before and after the ticket is created. This synchronization helps provide the latest information about the request to the CA Service Desk Manager staff (for example, the analysts) who process the related ticket.

Follow these steps:

1. Enable the CA SDM options, as follows:
   a. Log in to the CA SDM web UI from the following server, depending on your CA SDM configuration:
      Conventional: Primary server
      Advanced availability: Background server
   b. Select Options Manager, CA Service Catalog from the Administration tab.
   c. To enable the integration, install the following options:
      casc_aty_sync

   Specifies the type of activity logs to be synchronized from CA SDM to CA Service Catalog. If this field is left blank, Log Comment activity logs are synchronized. To synchronize enter the respective log type codes, separated by comma. Out of the box integration between CA SDM and CA Service Catalog syncs the CA SDM Escalate/Closed status.

   casc_endpoint

   Specifies the CA Service Catalog Web Services URL.

   Format: http://<CA_Service_Catalog_hostname>:<CA_Service_Catalog_portnum>/usm/services.

   (Optional) **casc_session_timeout**

   Specifies the time in minutes for which the CA Service Catalog web services session will be cached.

   Range: 20 to 60 mins.

   casc_user

   Specifies the CA Service Catalog administrator username that is responsible for making the web service calls to CA Service Catalog. This user must exist in EEM against which CA SDM is configured.

   casc_user_password

   Specifies the password of the CA Service Catalog user as entered in the casc_user option.

   (Optional) **casc_ws_retry**

   Specifies the number of retries to synchronize a CA SDM ticket update (activity logs/ attachments/status) with CA Service Catalog, if there is a failure. Enter -1 to try indefinitely, until the synchronization succeeds. As a system administrator you can define the time interval after which you want the failed ticket update to be retried again.

   Run the following command to define the time interval:

   ```
   pdm_options_mgr -c -a pdm_option.inst -a option.inst -s NX_CASC_RETRY_INTERVAL -v [time interval in minutes]
   ```

By default this time interval is set to 15 minutes. If you do not set any value or provide a non-numeric value to this variable, the failed update is retried according to the default time interval (15 mins.).

Range: -1 to 50

casc_integrated

Enables the self-service integration.

WARNING

Restart CA SDM services for the options to work.

The options are installed successfully.

2. From CA Service Catalog, click the Event Type named Request/Subscription Item Change, and proceed as follows:
   a. Click the rule named When Category is Hardware and Status is Filled from Inventory.
   b. Select the action named Launch HWSWFilledFromInv_SDM SRF and click the Disable button on the Actions bar.
      This step disables the default fulfillment action, which does *not* support the synchronization of notes and attachments.
   c. Select the action named Launch HWSWFilledFromInv_SDM_SYNC SRF and click the Enable button on the Actions bar.
      This action synchronizes both notes and attachments between service options and tickets *before* the ticket is created.
   d. Click Done.
      You return to the Event Details page.
3. Perform these actions, using the previous step as a model:
   a. Open the Document Create Event Type.
   b. Enable the rule named When attachment is added to Service Catalog request.
   c. Enable the action named Launch PAM SRF to sync attachments.

   This action synchronizes attachments between service options and tickets *after* the ticket is created. That is, if a catalog user adds an attachment to a service option after submitting the request, this action passes the attachment to the related ticket.
4. Perform these actions:
   a. Open the Notes Create Event Type.
   b. Enable the rule named When note is added to Service Catalog request.
   c. Enable the action named Launch PAM SRF to sync notes.

   This action synchronizes notes between service options and tickets *after* the ticket is created. That is, if a catalog user adds a note to a service option after submitting the request, this action passes the note to the related ticket.

Integrate CA Service Desk Manager with Common Components Manually

This article contains the following topics:

- Integrate CA Service Desk Manager with CA Embedded Entitlements Manager Manually
- Integrate CA Service Desk Manager with CA Process Automation Manually
- How to Troubleshoot CA Embedded Entitlements Manager Installation
- EEM Application is Frequently Down

Integrate CA Service Desk Manager with CA Embedded Entitlements Manager Manually

Complete the following steps if you are not able to integrate CA SDM and CA EEM through the CA Service Management installer:

1. Ensure that you have installed CA EEM.
2. Depending on the CA SDM configuration, log in to the following server:

- Conventional: Primary server
- Advanced availability: Background server

3. Install the following options:
   - **use_eiam_authentication**
     Allows use of EEM authentication when users log in to CA SDM / Knowledge Management and the access type authentication is set to OS.
   - **use_eiam_artifact**
     Allows EEM artifact to be used when users log in to CA SDM / Knowledge Management using URL.
   - **eiam_hostname**
     Specifies the hostname of the server where EEM is running. Required for EEM user name and password or artifact authentication.

     NOTE
     It's recommended that you provide a FQDN name as the host name. Multiple host names are not supported and only a single hostname must be provided.

At the end of this topic, you have successfully integrated CA EEM with CA SDM.

Integrate CA Service Desk Manager with CA Process Automation Manually

CA Process Automation is a stand-alone CA product with features for automating and tracking hardware and software administration tasks in enterprise IT environments. CA Process Automation automates tasks and manages user interactions, such as approvals and notifications for compliance and accuracy within production environments.

When you integrate CA SDM and CA Process Automation, you can leverage the benefits of CA Process Automation workflow capabilities from key points within CA SDM. An effective integration between CA SDM and CA Process Automation requires you to understand both products.

CA Process Automation Components

CA Process Automation offers multiple capabilities and structures that facilitate a wide range of activities as part of CA Process Automation process management. For the integration with CA SDM, however, only the following CA Process Automation components are critical for CA Process Automation integration:

- **Process Definition --** Identifies a collective series of tasks, steps, and conditions that are structured in a specific order to be initiated, and completed by various individuals or parties. This component is the central building block of all CA Process Automation content.
- **Start Request Form --** An object containing descriptive information for end users. The Start Request Form presents a process definition to users while hiding the technical details of the process definition.
- **Keywords --** A list of pre-defined words or phrases to attach to Start Request Forms.
- **Automation Library --** An area within CA Process Automation that stores and shows Process Definitions and Start Request Forms.
- **Library Path or Reference Path --** A folder structure that organizes and describes Process Definitions and Start Request Forms within the automation library.
- **Process Instance --** An active entity that executes the rules that are defined in a process definition. The process instance progresses until the process definition state is complete.
- **Process Instance Log Messages --** A configurable, running record that details the progression of activities of the process instance. Log message categories are useful to the CA SDM integration with CA Process Automation.

Note: The scope of the definitions of CA Process Automation components is limited to the usage within CA SDM. For information about CA Process Automation components and capabilities, see the CA Process Automation documentation.

Complete the following steps only if you are not able to integrate CA Process Automation and CA SDM through the CA Service Management installer:

Verify the Integration Requirements

You can integrate CA Process Automation and CA SDM to coexist on a single server when the server architecture supports both products. When CA Process Automation or CA SDM components cannot integrate on the same server, consider installing each product on separate servers.

> **WARNING**
> If CA Process Automation is configured in FIPS mode, you must also configure EEM server and EEM SDK in FIPS mode. For more information about configuring the EEM SDK in FIPS mode, see the CA EEM documentation.

Before you configure CA Process Automation and CA SDM, confirm that both products are installed and are working independently.

Follow these steps:

1. Open a browser on the server that hosts CA SDM and verify that a CA Process Automation user can log in to CA Process Automation. Change the place holders to match the target CA Process Automation installation.
   ```
   http(s)://<server>:CA Portal/itpam
   ```
2. Enter the following URL. Change the place holders to match the target CA Process Automation installation.
   ```
   http(s)://<server>:CA Portal/itpam/JNLPRequestProcessor?processType=startUI
   ```
   The CA Process Automation product is accessible from the CA SDM host.

Install the CA Process Automation Workflow Options

When you install the CA Process Automation Workflow options, you specify connectivity between CA SDM and CA Process Automation. If you are using CA EEM for authentication, you also specify the CA EEM host name.

Follow these steps:

1. On the Administration Tab, select Options Manager, CA IT PAM Workflow.
   The Option List appears.
2. Right-click the name of each option and select Edit from the context menu. Install the following options:
   - **caextwf_eem_hostname**
     Specifies the name of the CA EEM server. For example, *<wf_hostname>* identifies the authentication host. You install caextwf_eem_hostname only if you configured CA Process Automation to use CA EEM as an authentication server. CA SDM uses this value to transform a user name and password into a CA EEM token. Then, the user name and password do not pass in plain text over HTTP.

     > **NOTE**
     > If the CA Process Automation installation is not using CA EEM, do not place a value in the caextwf_eem_hostname option, and do not install the caextwf_eem_hostname. Placing a false value or installing caextwf_eem_hostname when it is not necessary causes the integration to fail.
   - **caextwf_endpoint**
     Specifies the URL that points to the CA Process Automation web services by including the CA Process Automation host name, port, and the mandatory /itpam/soap path. For example, http://<wf_hostname>:<wf_tomcat_port>/itpam/soap identifies the endpoint. If your implementation uses CA EEM, installing the caextwf_eem_hostname option is required for the integration between CA Process Automation and CA SDM to operate properly.
   - **caextwf_log_categories**
     Specifies a comma-separated list of CA Process Automation process instance log category names to appear on the CA SDM Request, Change Order, and Issue Workflow Tasks tab. For example, *Operator,Response,MyOwnCategory* supplies three log categories.
     You install caextwf_log_categories based on business decisions from the CA SDM and CA Process Automation process design personnel. This option adjusts the default data that appears on the Workflow Tasks tab for requests, change orders, and issues.

When you install the caextwf_log_categories option, all CA Process Automation process instance log messages from the Process category and the categories that you specify appear on the Workflow Tasks tab. When you do not install caextwf_log_categories, only the CA Process Automation process instance log messages from the Process category appear on the Workflow Tasks tab.

- **caextwf_processdisplay_url**
  Specifies how to launch a graphical snapshot of a CA Process Automation process instance by supplying the host name and the mandatory /itpam/Web.jsp?page=runtimeeditor&ROID path. For example, http://<wf_hostname>:<wf_tomcat_port>/itpam/Web.jsp?page=runtimeeditor&ROID= launches a snapshot of a process instance. On the Workflow Tasks tab of a request, change order or issue, the user selects View Process to see the snapshot.
  Installing the caextwf_processdisplay_url option is required for the integration between CA Process Automation and CA SDM to operate appropriately.
- **caextwf_worklist_url**
  Specifies the process instance path by supplying the host name and the mandatory /itpam?page=tasklist path. For example, http://<wf_hostname>:<wf_tomcat_port>/itpam?page=tasklist enables CA SDM users to see a list of CA Process Automation process instances that require attention. The list appears in CA Process Automation when the CA SDM user selects a link associated with any listed task in the request, change order, or issue Workflow Tasks tab.
  Installing the caextwf_worklist_url option is required for the integration between CA Process Automation and CA SDM to operate properly.
- **caextwf_ws_password**
  Specifies the administrative password associated with the CA Process Automation user name from the caextwf_ws_user option. CA SDM uses the user name and password to access the CA Process Automation web service functions to perform integration activities such as selecting start request forms, process definition information, and process instance information.
  Installing the caextwf_ws_password option is required for the integration between CA Process Automation and CA SDM. The password and user name that you specify requires the appropriate access to CA Process Automation. However, it is not necessary the CA Process Automation user name and password to exist within the CA SDM contact records.
- **caextwf_ws_user**
  Specifies the CA Process Automation administrative user name associated with the CA Process Automation user name from the caextwf_ws_password option. CA SDM uses the user name and password to access the CA Process Automation web service functions. These services perform integration activities such as selecting start request forms, selecting process definition information, selecting process instance information, or launching process instances.
  Installing the caextwf_ws_user option is required for the integration between CA Process Automation and CA SDM to operate. The user name and password that you specify requires the appropriate access to CA Process Automation. However, it is not necessary the CA Process Automation user name and password to exist within the CA SDM contact records.
- **caextwf_retry_count**
  Specifies the number of times an event is triggered if CA Process Automation is unavailable. CA Process Automation Workflows can be attached to CA SDM tickets (for example,Change Orders) through CA SDM Events and Macros. The attached events are triggered by CA SDM at the specified retry interval duration. During retry, if the CA Process Automation is unavailable, the attached event is marked as Unknown and the process is not executed. CA SDM retry mechanism automatically re-triggers the attached Unknown events when CA Process Automation is unavailable. Default value is 3 and can be set in the range [1 – 20].
- **caextwf_retry_interval**
  Specifies time interval after which the event is again triggered in case CA Process Automation is unavailable.

3. Click Install.
4. Restart the CA SDM service.

The CA SDM and CA Process Automation can communicate even though there is no process instance data. CA SDM and CA Process Automation are ready for you to create CA Process Automation process definitions and CA Process Automation start request forms.

Set Up SSL Communications with CA Process Automation

For security reasons, CA Process Automation implementers may have chosen to install or reconfigure CA Process Automation to require SSL communications. If CA Process Automation is configured to require SSL communications, integrated applications such as CA SDM require a certificate from the CA Process Automation keystore for communication.

Follow these steps:

1. Configure CA SDM options to use the CA Process Automation HTTPS address.
2. Export the CA Process Automation keystore certificate to a file and copy the file to CA SDM.
3. Load the certificate file into CA SDM using the CA SDM pdm_keystore_mgr utility.
4. If applicable to your CA SDM architecture, update the version control files to deliver the CA SDM keystore to all secondary servers.
5. Restart CA SDM.

Enable Communications When CA Process Automation is SSL Enabled

When CA Process Automation communicates with SSL, configure the CA SDM servers to communicate with CA Process Automation.

To enable communications when CA Process Automation is SSL enabled, do the following tasks:

1. Verify that you can use CA Process Automation in a browser, without launching CA SDM. Record the CA Process Automation URL and use it for reference when you configure the CA Process Automation Workflow options in Options Manager.
2. Log in to CA SDM and install or modify the CA Process Automation Workflow options in Options Manager. For each of the following options, use the syntax *https://server:8443* instead of http://server:8080 for reaching the SSL enabled CA Process Automation application. If the CA Process Automation installation uses another port instead of the 8443 SSL port, specify the appropriate port number.
   - caextwf_endpoint
   - caextwf_processdisplay_url
   - caextwf_worklist_url

   NOTE
   If the values do not match the actual CA Process Automation installation values, CA SDM cannot communicate with CA Process Automation. A runtime error occurs. Verify that the values match the actual CA Process Automation installation values, because the CA Process Automation installer might have selected a different port instead of port 8443.
3. On the CA Process Automation server, locate the KEYSTOREID and itpam.web.keystorealias entries in the following file:

   ```
   C:\Progra~1\ITPAM\server\c2o\.config\OasisConfig.properties
   ```

4. Copy the KEYSTOREID. Be prepared to paste the KEYSTOREID value as the password after you issue the keytool command.
5. On the CA Process Automation server, issue the following keytool command as one line on the command line:

   ```
   C:\Progra~1\ca\sc\jre\1.6.0_00\bin\keytool.exe -keystore C:\Progra~1\ITPAM\server\c2o\.config\c2okeystore
    -export -alias <keystorealias> -file itpam.cer
   ```

   The keytool utility prompts you for a password.
6. Paste or type the KEYSTOREID value as the password.

The keytool utility uses the final parameter (-file itpam.cer) to create a file that is named *itpam.cer*. The *itapm.cer* file contains the necessary certificate information for communications with CA SDM.

7. Move the *itpam.cer* file to one of the following locations on the CA SDM server:
   - (Windows) %NX_ROOT%\bin
   - (UNIX) $NX_ROOT/bin
8. Import the CA Process Automation certificate information into CA SDM by entering the following command:

   ```
   (Windows) pdm_perl %NX_ROOT%\bin\pdm_keystore_mgr.pl -import %NX_ROOT%\bin\itpam.cer
   (UNIX) pdm_perl $NX_ROOT/bin/pdm_keystore_mgr.pl -import $NX_ROOT/bin itpam.cer
   ```

   The *pdm_keystore_mgr.pl* script generates the keystore file in the following locations:
   - (Windows) %NX_ROOT%\pdmconf\nx.keystore
   - (UNIX) $NX_ROOT/pdmconf/nx.keystore
9. The *nx.keystore* must be delivered to the following CA SDM servers, depending on the CA SDM configuration:
   - Conventional:Secondary server.
   - Advanced Availability: Application and standby server.

   Open the *server_secondary.ver* file from one of the following locations:
   - (Windows) %NX_ROOT%\site\server_secondary.ver
   - (UNIX) $NX_ROOT/site/server_secondary.ver
10. Modify the *server_secondary.ver* for version control by adding the following information:

    ```
    [SSL_Keystore]
    filename = "nx.keystore"
    directory = "$NX_ROOT/pdmconf"
    component_type = "file"
    O_mode = "RW"
    g_mode = "RW"
    w_mode = "RW"
    file_ctl
    ```

11. Restart CA SDM.
    The CA SDM server can communicate with the SSL enabled CA Process Automation application.

Configure the Integration

Both CA Process Automation and CA SDM, as stand-alone products, have individual requirements for authentication and authorization. To support a unified Service Oriented Architecture (SOA) strategy, you can configure both products to use CA EEM for authentication.

When you install CA Process Automation with CA EEM as the authentication server, the installer creates several policies and eight essential entities by default:

- Four application users: pamadmin, pamuser, pamproduser, and pamdesigner.
- Four application groups: PAMAdmins, PAMUsers, Designers, and Production Users.

CA SDM users who also use CA Process Automation can be divided between PAMAdmins and PAMUsers as follows:

- CA SDM analysts must be members of PAMUsers when their duties entail:
  - Approving, rejecting, or otherwise responding to CA Process Automation Interaction Request Forms.
  - Listing CA Process Automation process instances assigned to the user.
  - Viewing the graphical display by clicking the View Process button of CA Process Automation's process status screen. The CA Process Automation PAMUsers group requires an additional CA Process Automation policy to grant access the graphic.
- CA SDM analysts are members of PAMAdmins when their duties entail:

  - Creating and checking in CA Process Automation process definitions and/or start request forms.
  - Terminating process instances directly within CA Process Automation. Terminating process instances are an administrative exception to expected integration procedures.
  - Delegating CA Process Automation process instance tasks.
  - If the user is the user name defined in CA SDM Options Manager.
- CA SDM users require no access to CA Process Automation when their duties entail:
  - Creating requests, change orders, and issues that launch CA Process Automation instances.
  - Reviewing the Workflow tab which shows CA Process Automation process instance status and task information.
  - Changing the status of a request, change order, or issue which causes the termination of a CA Process Automation process (such as canceling a change order).
  - Selecting a CA Process Automation process definition on a CA SDM request area, change category, issue category.

Finalize the Integration

If you have CA SDM and CA Process Automation integration, set up the Single Sign-On (SSO).

Follow these steps:

1. Verify that the following requirements have been met:
   - CA SDM and CA Process Automation are configured to use the same CA EEM installation.
   - The user that logs in to CA SDM is also a user in CA Process Automation.
   - When CA EEM uses the internal database as a user store, the users must have either global permissions or belong to the same folder. Otherwise, if CA EEM references an external user store like an external directory or CA Siteminder, the users must be of the same store to access single sign-on.
2. Install CA EEM from CA Service Management installation or use any existing CA EEM install (for example, CA EEM for CA Process Automation).
3. On the CA SDM Administration tab, install the following options from under the Options Manager, Security folder:
   - eiam_hostname
   - use_eiam_authentication

     NOTE
     You do not need to install the option caextwf_eem_hostname under Options Manager, CA Process Automation folder. But if you do install it, the value must be the same as eiam_hostname option.
4. Restart CA SDM.
5. To create a user in CA Process Automation, perform the following:
   a. Log in to CA EEM using the CA Process Automation application context using the EiamAdmin userid or any other administration user.
   b. Select the Manage Identities tab and click the icon next to the Users folder.
   c. On the New User page, the Name field at the top is the userid that must match the userid in the CA SDM contact table.
   d. Click the Add Application User Details button and complete the following:
      - Add any of the groups that are listed. Add at least one of these groups in order to be able to log in to CA Process Automation.
      - Complete the New User fields, such as First Name, Last Name, Display, and Password.
6. Create a user in the CA SDM contact table with the same userid. Verify that the Access Type Validation Type field for the user is set to CA EEM.
   You can log in to CA EEM and CA SDM with this user and the password specified in CA EEM.

CA Process Automation Integration with CA SDM at Run Time

When you enable the integration, CA SDM users experience the following:

- On a new Request, Change Order, or Issue, a CA Process Automation process instance initiates based on the ticket category or area. Summary information immediately appears on the Workflow Tasks tab.
- When a Request Area, Change Category, or Issue Category changes, an attached CA Process Automation process instance terminates and a new process instance initiates.
- When a CA SDM user attempts to close a Request, Change Order, or Issue where the CA Process Automation process instance is not yet complete, the user cannot close the ticket. Instead, the user must first cancel the ticket. The Cancel status terminates the CA Process Automation process instance before the ticket closes.
- When a user wants to understand the state of the process instance without navigating away from the ticket, the user can click the ticket Workflow Tasks tab. The Workflow Tasks tab shows the process instance start date, end date, current state, and a current audit trail of messages indicating the path of the process instance.
- When a user wants to see the current path of the process instance relative to the entire process, the user selects the View Process button on the Workflow Tasks tab. The View Process button launches a graphical snapshot of the entire process instance, and shows the current path.
- When a user wants to see CA Process Automation interaction request forms that are waiting for user action, the user can select any entry in the Workflow Tasks tab. The Workflow Tasks tab contains an audit trail of process instance messages that appear on the CA Process Automation task list.

  NOTE
  When a user selects the CA SDM View Process button or CA Process Automation process instance messages, the system prompts for a CA Process Automation user name and password for a single browser session. After the initial prompt, the system does not prompt the user again until the CA SDM browser closes.

How to Create a Process Definition

When you create the process definition, you populate CA Process Automation with content to appear in CA SDM. You use the CA Process Automation graphical process designer to create, test, and check in a process definition. From CA SDM, you can also create macros to initiate CA Process Automation processes. To create a process definition in CA Process Automation, perform the following:

Follow these steps:

1. Log in to the CA Process Automation client as an administrative user.
2. Use the CA Process Automation graphical process designer to create, test, and check in a process definition. When you work with the process definition, use the instructions in the CA Process Automation user documentation.

   NOTE
   If you fail to check in the process definition before attempting to use it, the workflow cannot operate properly in CA SDM.

   The following process definition items are available in CA SDM:
   - **Process Name**
     Appears on the Request Area, Change Category, and Issue Detail page. The process name also appears on the Workflow Tasks tab of a ticket. The process name describes the process definition to CA SDM to Analysts and other users who manage tickets.
   - **Process Reference Path**
     Appears on the Request Area, Change Category, and Issue Detail pages. The Process Reference Path also appears on the ticket Workflow Tasks tab. The Process Reference Path can be useful to describe the purpose of the process to end users. For example, "/Processes/Approval" is not helpful. Instead, a reference path like "/Office Supplies/Approvals/Over-200-CA SDM" describes the workflow to manage orders that exceed $200 US dollars.
   - **Process Log Messages**
     Appears on the CA SDM ticket Workflow Tasks tab. By default, a record of activities stores with the process instance. Process log messages have the Process category. A process designer can create custom messages to appear on the Workflow Tasks tab of a CA SDM ticket.

Create a Start Request Form

When you create a Start Request Form, you associate it with the process definition and check it in. You include the appropriate keywords in the Start Request Form properties. If the appropriate keyword is missing from the Start Request Form properties, the Start Request Form and its associated process definition fail to appear in CA SDM.

Follow these steps:

1. Log in to the CA Process Automation client as an administrative user.
2. Open the CA Process Automation Library and navigate to the path Start Request Form.
   The Start Request Form appears in the right pane of CA Process Automation library.
3. Select the Start Request Form from the list.
   A shortcut menu appears.
4. Select Properties.
   The Library Object Properties page appears.
5. (Optional) Click the General tab and modify the description of the Start Request Form. Add a description that identifies the proper usage of the Start Request Form and the associated Process Definition to the CA SDM Administrator.
6. Click the Keywords tab.
   The Keywords tab is active.
7. Click the ab+ icon.
   A row adds to the empty list.
8. Click the row.
   A blinking cursor highlights the row and indicates the row is ready for typing.
   a. Enter one of the following values to associate a keyword to the appropriate ticket area or category.
      For example, to make a Start Request Form available for a CA SDM request area, enter the pcat keyword.
   b. Add a row to the list for each applicable keyword.
      For example, to make the Start Request Form appear on both request areas and change categories, add one row for the chgcat keyword and another row for the pcat keyword.
   c. Click OK.
      CA Process Automation saves the keywords and description and closes the Library Object Properties dialog.
   d. Check in the Start Request Form.

Ticket	Use Keyword
request area	pcat
change category	chgcat
issue category	isscat

NOTE

If you fail to check in the Start Request Form, the form fails to appear in CA SDM.

The CA Process Automation Start Request Form information appears on the CA SDM Start Request Form List. The CA SDM administrator can associate the CA SDM Start Request Form with the Process Definition, on a Request Area, Change Category, or Issue Category Detail page.
The following Start Request Form items appear in CA SDM:

1. – **Start Request Form Name**
   Appears on the Request Areas, Change Categories, and Issue Categories list pages.
   – **Start Request Form Reference Path**
   Appears on the Request Areas, Change Categories, and Issue Categories list pages.
   – **Start Request Form Description**

Appears on the Request Areas, Change Categories, and Issue Categories list pages. The text in this field describes how the Start Request Form is appropriate for selection on a particular Request Area, Change Category, or Issue Category.

How to Troubleshoot CA Embedded Entitlements Manager Installation

When installing CA Embedded Entitlements Manager, you may encounter the following error - *Failed to install Microsoft run time libraries*

To resolve the error, relaunch the installation wizard.

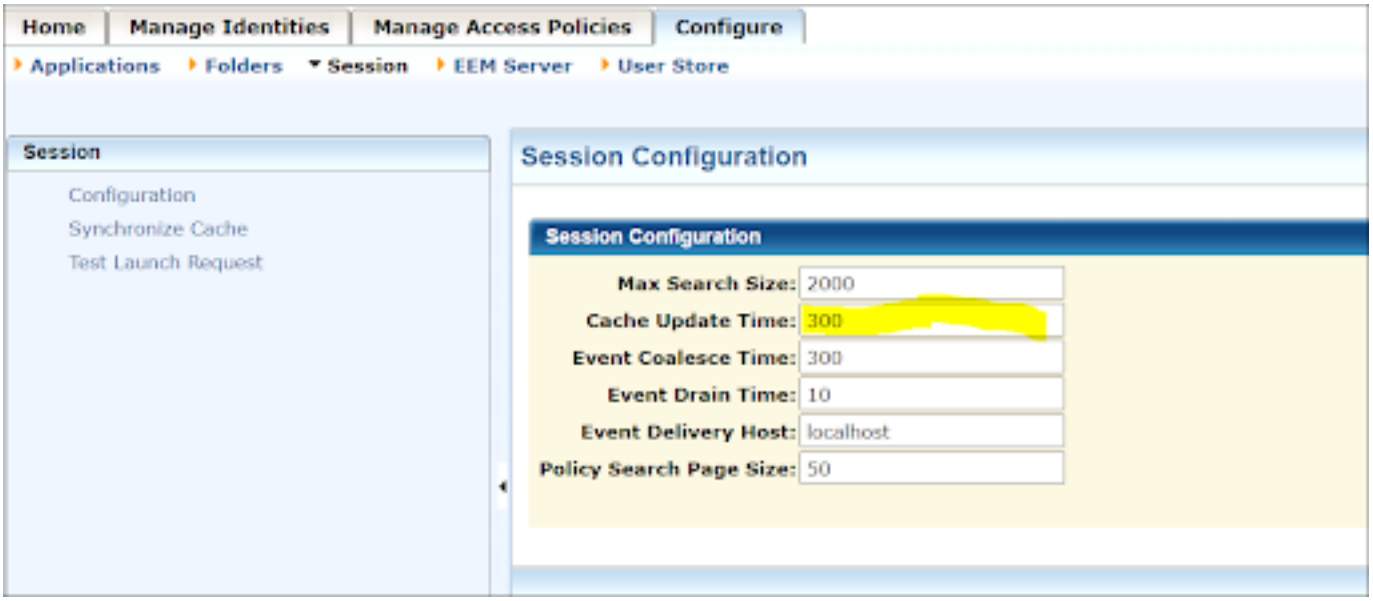
EEM Application is Frequently Down

Users are unable to log in to EEM or SDM using the EEM user accounts intermittently. This might be due to session cache timeout mismatch between EEM and SDM..

Starting from 17.3 RU13, this issue is fixed by introducing a new NX variable "NX_EEM_CACHE_UPDATE_TIME" in SDM to address EEM hang issue due to a huge number of Service Desk Manager sessions being handled by EEM.

To resolve the error, follow these steps:

Keep the value the same as the EEM cache timeout configuration value mentioned in the following screenshot.



To set the EEM cache update time to 300 seconds, perform the following steps:

To add the variable in the **nx.env** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst
```

To add the variable in the **nxenv_nt.tpl** file, run the following command:

```
pdm_options_mgr -c -s  EEM_CACHE_UPDATE_TIME  -v 300 -a pdm_option.inst -t
```

Integrate CA Service Catalog with Common Components Manually

This section contains the following articles:

- Integrate CA Service Catalog with CA EEM Manually
- Integrate CA Service Catalog with CA Business Intelligence Manually
- Integrate CA Service Catalog with CA Process Automation Manually

Integrate CA Service Catalog with CA EEM Manually

WARNING

Review the following section only if you want to manually integrate CA Service Catalog with CA EEM when the automatic integration between the two has failed.

Ensure that both the CA EEM server and CA Service Catalog server are up and running.

Follow these steps:

1. Log in to the setup utility and click Security on the left menu.
2. Perform *one* of the following actions, whichever applies:
   - If CA EEM is installed on one computer only, enter its host name.
   - If CA EEM is installed on multiple computers as a cluster *without* a Proxy, enter the host names of the CA EEM computers. Separate the names with commas.
     (*Proxy* here means load balancer.)
   - If CA EEM is installed on multiple computers as a cluster *with* a Proxy, enter the host name of the Proxy computer.
3. For the CA EEM application instance name, perform *one* of the following actions, whichever applies:
   - If you upgraded CA EEM during this CA Service Catalog upgrade or if you are using an existing version of CA EEM, specify the existing application instance name.
   - If you installed CA EEM for the first time during this CA Service Catalog upgrade, specify the application instance name of your choice.
4. Enter the CA EEM administrator (EiamAdmin) password.
5. Click Save.
   The utility creates the required CA Service Catalog objects in CA EEM. Examples include the *Service Catalog* application, policies, and users (including spadmin).

Integrate CA Service Catalog with CA Business Intelligence Manually

WARNING

Review the following section only if you want to manually integrate CA Service Catalog with CA Business Intelligence when the automatic integration between the two has failed. If you have any other earlier version of CA Business Intelligence, install CA Business Intelligence and migrate the data from your earlier version for the reports to work.

CA Business Intelligence is a set of reporting and analytic software. You can use CA Business Intelligence to integrate, analyze, and then present through various reporting options, vital information that is required for effective enterprise IT management. CA Business Intelligence is a flexible, scalable, and reliable business intelligence reporting system that can be integrated into your information technology infrastructure.

CA Service Catalog supplies the data that you require to get started with CA Business Intelligence reports. After you have installed CA Service Catalog and CA Business Intelligence, you perform required setup tasks before running reports.

Follow these best practices when maintaining and using CA Business Intelligence:

- Install and maintain one universe per CA product.
- Do *not* modify the default universe. Instead, copy it and modify the copy. Otherwise, your changes can be erased when you apply service packs, patches, and other updates. Back up all your changes and then apply the patches to your customized universe.
- Reports:
  - Verify that the services in Central Configuration Manager (CCM) is running, when the reports stop running.
  - Do not overwrite predefined reports.
  - Always use a predefined report as a base to build a custom report and maintain consistent formatting in all reports.
  - Administrators *can modify all the reports and can create* new reports that are based on the existing universe. However, administrators must not add any reports to the existing folders.
  - Both administrators and end users *must not* change pre-defined reports. Any changes to those reports are applied to all other users using the same CA Business Intelligence instance. Instead, both administrators and end users must create their own custom folders, copy the reports there, rename them, and customize them.
  - Both administrators and end users must add new reports that they create to their custom folders.

Follow these steps:

NOTE

If you plan to use Secure Socket Layer between CA Service Catalog and CA Business Intelligence, ensure that you:

- Configure CA Business Intelligence for Secure Socket Layer (SSL)
- Configure CA Business Intelligence to communicate with CA Service Catalog using SSL

Step 1 - Verify that CA Business Intelligence Is Installed

You obtain best system performance by installing CA Business Intelligence and CA Service Catalog components on separate computers.

For more information about installing CA Business Intelligence, see CA Business Intelligence 4.1 SP3 documentation. To verify that CA Business Intelligence is installed and running, access CA Business Intelligence and start it.

Step 2 - Import the BIAR File

Importing the Business Intelligence Archive Resource (BIAR) file is a required task to set up CA Business Intelligence reports for CA Service Catalog.

For more information, see how to import the BIAR file.

Step 3 - Set Up the User Database in CA Business Intelligence

CA Service Catalog users can access *only* the CA Business Intelligence reports that their role authorizes them to.

Follow these steps:

1. Verify that a matching user name exists in CA Business Intelligence for each CA Service Catalog user.

   NOTE

   The BIAR file automatically provides the CA Service Catalog default administrative user named spadmin and CASMAdmin.

   Optionally, import the users from your CA Service Catalog user database into your CA Business Intelligence user database. For example, in LDAP or Active Directory. Alternatively, create the users with matching user names directly in your CA Business Intelligence user database.
2. Assign those users to the CA Business Intelligence user groups that provide the access rights you want.

Typically, you assign the users to groups that match their roles in CA Service Catalog. For example, you typically assign CA Service Catalog administrators to administrator groups in CA Business Intelligence.

WARNING
All CA Service Catalog users who need access to CA Business Intelligence reports must exist in the CA Business Intelligence user database. Also, these users must belong to the CA Business Intelligence user group that provides the appropriate level of access rights.

The CA Service Catalog Administrator groups are as follows:
- SLCM Catalog Administrators
- SLCM Administrators
- SLCM Service Managers
- SLCM Super Business Unit Administrators
- SLCM Service Delivery Administrators

The CA Service Catalog End User groups are as follows:
- SLCM End Users
- SLCM Catalog End Users
- SLCM Request Managers

Step 4 - Set the Administration Configuration Parameters

If you are running multiple instances of CA Business Intelligence, the configuration settings apply to all instances.

Follow these steps:

1. Click **Administration**, **Configuration** in the CA Service Catalog section.
2. Click the Modify icon to next to each property that you want to update:
   - **CMS Host Name**
     Specifies the computer name on which Central Management Server is installed.
   - **CMS Port Number (1 - 65535)**
     Specifies the port number on which the Central Management Server is running.
   - **Enable HTTPS**
     Specifies a web protocol, as follows:
     Select No (the default) to use HTTP to communicate with CA Business Intelligence.
     Select Yes to use HTTPS to communicate with the CA Business Intelligence.

     WARNING
     If you select Yes, verify that CA Business Intelligence is also using HTTPS. For more information about how to configure CA Business Intelligence to use HTTPS, see the CA Business Intelligence documentation.
   - **Host Name**
     Specifies the computer name on which CA Business Intelligence web application server is hosted.
   - **Port Number** (1 - 65535)
     Specifies the port number on which CA Business Intelligence web application server is running.
3. Click the **Launch** button to test the connection between CA Service Catalog and Launch Pad.
4. Recycle Catalog Component.

The CA Business Intelligence configuration information is updated with the values that you specified.

Step 5 - Configure Trusted Authentication

Configure CA Service Catalog and CA Business Intelligence to use trusted authentication for the integration between the two products. Trusted authentication provides Single Sign-On. Single Sign-On enables CA Service Catalog users to access the Launch Pad application of CA Business Intelligence directly from the CA Service Catalog GUI.

CA Service Catalog uses standard CA Business Intelligence login if trusted authentication is not set up or if it is not working properly.

NOTE

The following login methods do *not* apply to CA Business Intelligence: CA EEM token, CA SiteMinder Integration, and NTLM authentication on Windows.

Follow these steps:

1. Log on to the CA Business Intelligence Central Management Console as a user with administrative rights.
2. Go to **Manage**, **Authentication** area of the Central Management Console.
3. Click the **Enterprise** tab.
4. Check **Trusted Authentication is Enabled** to enable trusted authentication.
5. Click **New shared secret** and you see that **Download Shared Secret** is enabled.

   NOTE
   The CA Business Intelligence client and the Central Management Console use the shared secret password to create a trusted authentication password. The value of this password and the frequency with which you update it meet your password security standards.
6. Click **Download Shared Secret** to generate TrustedPrincipal.conf file.
7. Perform the remaining steps on *every* Catalog Component computer.
8. Open the file that is named USM_HOME\reporting\CABI\TrustedPrincipal.conf using a text editor.
9. Scroll to the following line and specify the same shared secret password that you downloaded on the CA Business Intelligence Central Management Console.

   ```
   SharedSecret=password
   ```
10. Save the TrustedPrincipal.conf file.
11. Restart the Catalog Component computer.

WARNING

Whenever you update the password on the CA Business Intelligence Central Management Console, make the same change to the password in the TrustedPrincipal.conf file on every Catalog Component computer.

Step 6 - Run Pre-Defined Reports

The **BI Launch Pad** button enables you to run the pre-defined reports quickly and easily from within CA Service Catalog.

Optionally view the pre-defined reports in localized format. To do so, set the language configuration in the **Preferences**, **Locales and Time Zone**, **Preferred Viewing Locale** section of CA Business Intelligence.

NOTE

Some fields remain in English even when you view reports in localized format. These fields include Request Status, Billing Status, Account Status, Account Type, and fields related primarily to payment and adjustment. In addition, in both English and localized reports, custom status values do appear in the reports; however, their descriptions do not.

Follow these steps:

1. Select **Home**, **DashBoards**, **Administration Quick Start**, **BI Launch Pad** in CA Service Catalog.
2. Click **Documents**, **Folders**, **Public Folders**, **CA Reports**, **CA Service Management, CA Service Catalog**.
3. Click **Admin Reports** or **User Reports**, as required.
   The Admin Reports folder and its reports are visible only to users who are members of the Administrator groups.
   The User Reports folder and its reports are visible only to users who are members of either the Administrator groups or the End User groups.
4. Double-click the report which you want to run.

5. Specify the parameters for your report.
6. Click Run Query.
7. View the report.

NOTE

In a multi-tenant (business unit) implementation, your reports display only the data of tenants to which you have access.

If you run a report for an integrated product (such as CA Service Desk Manager or CA Asset Portfolio Management), but that product has not been integrated with CA Service Catalog, then the report typically fails.

Update CA Business Intelligence in CA Service Catalog Configuration.

If you are running multiple instances of CA Business Intelligence, the configuration settings apply to all instances.

Follow these steps:

1. Click **Administration**, **Configuration** in the CA Service Catalog section.
2. Click the Modify icon to next to each property that you want to update:
   - **CMS Host Name**
     Specifies the computer name on which Central Management Server is installed.
   - **CMS Port Number (1 - 65535)**
     Specifies the port number on which the Central Management Server is running.
   - **Enable HTTPS**
     Specifies a web protocol, as follows:
     Select No (the default) to use HTTP to communicate with CA Business Intelligence.
     Select Yes to use HTTPS to communicate with the CA Business Intelligence.

     WARNING
     If you select Yes, verify that CA Business Intelligence is also using HTTPS. For more information about how to configure CA Business Intelligence to use HTTPS, see the CA Business Intelligence documentation.
   - **Host Name**
     Specifies the computer name on which CA Business Intelligence web application server is hosted.
   - **Port Number** (1 - 65535)
     Specifies the port number on which CA Business Intelligence web application server is running.
3. Click the **Launch** button to test the connection between CA Service Catalog and Launch Pad.
4. Recycle Catalog Component.

The CA Business Intelligence configuration information is updated with the values that you specified.

{"URL":["/usm/wpf?Node=iclaunchpad.tools_configuration"],"customLabelGetStarted":"CA Business Intelligence","description":"IPCE_CA_Business_Intelligence"}

Import the BIAR File

This article contains the following topics:

The Business Intelligence Archive Resource (BIAR) file includes several required components and CA Service Catalog-specific content for use with CA Business Intelligence. The file includes CA Service Catalog universe, default objects, pre-defined reports, user groups, database schemas, locale strings, and data that helps you customize your CA Business Intelligence implementation, especially your reports.

To import the BIAR file, verify that a data source name (DSN) and ODBC connection exists for the DBMS used by the MDB. If necessary, create a Data Source Name and ODBC Connection for the MDB on SQL Server or Oracle. Then you use the biconfig utility to Import the BIAR file

Create a Data Source Name and ODBC Connection for the MDB on SQL Server

If you are using Microsoft SQL Server as your DBMS for the MDB, a data source name (DSN) and ODBC connection for the MDB on SQL Server are prerequisites for setting up your CA Business Intelligence implementation. If necessary, create them now.

Follow these steps:

1. From the Windows Control Panel, open ODBC.
2. Click Add on the System DSN tab.
3. Select SQL Server as the driver and click Finish.
4. Enter the following information and click Next:
   - Data source name (DSN)

     NOTE

     Ensure that you create the DSN with the name "caslcm_cabi_dsn".
   - Description
   - Name of the SQL Server server
5. Perform the following actions and click Next:
   a. Select SQL Server Authentication
   b. Select the option to Connect to SQL Server to obtain default settings for configuration options. Enter the user ID and password to connect to the database.
6. Perform the following actions and click Next:
   a. Change the default database to the SQL Server database.
   b. Select the option to use ANSI quoted identifiers.
   c. Select the option to use ANSI nulls, paddings, and warning.
7. (For localized environments *only*) Check the option to use regional settings for currency, numbers, dates, and times.
8. Click Finish.

You have created the DSN and ODBC connection for SQL Server.

NOTE

For more information about creating DSNs and ODBC connections, see your Windows or SQL Server documentation.

Create a Data Source Name and ODBC Connection for the MDB on Oracle

If you are using Oracle as your DBMS for the MDB, a data source name (DSN) and ODBC connection for the MDB on Oracle are prerequisites for setting up your CA Business Intelligence implementation. If necessary, create them now.

Follow these steps:

1. From the Windows Control Panel, open ODBC.
2. Click Add on the System DSN tab.
3. Select Oracle client /server for Oracle as the driver and click Finish.
4. Enter the following information and click OK:
   - Data source name (DSN)

NOTE

Ensure that you create the DSN with the name "caslcm_cabi_dsn".

- Description
- DBMS user name
- One of the following values:
  - SID of the Oracle database, if you are using a standalone Oracle setup.
  - Local Net Service Name, if you are using a distributed Oracle setup.

You have created the (DSN) and ODBC connection for Oracle.

Use biconfig to Import the BIAR File

Importing the BIAR file is a required task to set up your CA Business Intelligence implementation. We recommend that you use the CA Business Intelligence biconfig batch utility to import the BIAR file.

NOTE

Delete the CA SLCM Connection from Central Management Console before you import the BIAR file.

Follow these steps:

1. Go to the CA Service Management Install path and locate the \filestore\BOXI\biconfig folder.
2. Copy the contents of the biconfig folder on to the CA Business Intelligence computer.
3. Create an XML file (with the contents of Example file for SQL server or Oracle) and update the appropriate file, according to your requirements.
   Set the following parameters in your BIAR file:
   - Networklayer: ODBC
   - RDBMS:
     - For Microsoft SQL Server (SQL Server): MS SQL Server 2008 or MS SQL Server 2012
     - For Oracle: Generic ODBC datasource
   - User name and password parameters for your database:
     - For SQL Server: Use *sa* for the user name.
     - For Oracle: Use *mdbadmin* for the user name.

       NOTE
       These users have access to all required tables in the MDB.

     - Path name of the SLCM_universe.biar file on the CA Business Intelligence computer
     - Name of the DSN created previously in the datasource parameters. **Value:** caslcm_cabi_dsn
     - Server parameter: Computer name of your Oracle or SQL Server server
4. Open a Windows command prompt and navigate to the biconfig folder on the CA Business Intelligence computer and enter the following command:

   ```
   biconfig -h "host" [-n "port"] -u "user" [-p "password"] -f "XML-config-file-name" [--help]
   ```

   The quotation marks (" ") *are* required as shown in the command. However, the brackets [ ] are *not* required; the brackets signify optional parameters.
   The following parameters are required, unless noted otherwise.
   - **host**
     Specifies the CA Business Intelligence Central Management Server (CMS) host.
   - **port**
     (Optional) Specifies the CA Business Intelligence CMS port. The default is 6400.
   - **user**

Specifies the CA Business Intelligence CMS user.

– **password**
Optional. Specifies the CA Business Intelligence CMS password.

– **XML-config-file-name**
Specifies the path name of the XML file you modified.
If the file resides in the biconfig folder, you can specify the file name only. Otherwise, specify the complete path name.

– **help**
Optional. Displays the help for the biconfig utility.
Enter *two* dashes to display the help.

NOTE
For more information about using biconfig, see the biconfig-readme.txt file in the biconfig folder.

5. (Only for Oracle and if both your CA Service Catalog and CA Service Desk Manager are configured to use the same instance of CA Business Intelligence server, and the date format in odbc.prm file is changed to the CA SDM date format) Update the oracle.prm file at <CABI_Home>\SAP BusinessObjects Enterprise XI 4.0\dataAccess \connectionServer\odbc\extensions\qt as follows:
6. a. Replace <Parameter Name="USER_INPUT_DATE_FORMAT">'dd-Mmm-yyyy'</Parameter> with existing USER_INPUT_DATE_FORMAT tag.

NOTE

The CA SDM date format is **mm/dd/yyyy hh:mm:ss AM/PM**

b. Restart the Apache Tomcat for BI 4 and Server Intelligence Agent services in Central Configuration Manager.
7. Verify the import, as follows:
   a. Review the biconfig.log file in the biconfig folder.
   This file lists the status of the import. This file also includes error messages if the BIAR file is not imported successfully. A return code of zero (0) indicates a successful import.
   b. Log in to BI launch pad as a CA Business Intelligence administrator.
   c. Verify that you can view the CA Service Catalog Reports under Public Folders/CA Reports/CA Service Management/CA Service Catalog.
8. Specify a password for the spadmin user in CA Business Intelligence. This password is empty by default.

NOTE

If you must import the BIAR file again, delete the following elements from the CA Business Intelligence Central Management Console *before* you import the BIAR file again:

- CA SLCM folder
- CA SLCM Universe
- CA SLCM Groups
- SLCM Connection

Example XML File (SQL Server) to Import BIAR

```
<?xml version="1.0"?>
<biconfig version="1.0">
<!-- Import BIAR file -->
<step priority="1">
<add>
<biar-file name="Specify path of the SLCM_universe.biar file on the CA Business Intelligence computer">
<networklayer>ODBC</networklayer>
<rdbms><Specify the MS SQL Server version></rdbms>
<username>sa</username>
<password><Specify the sa password></password>
```

```
<datasource>caslcm_cabi_dsn</datasource>
<server><Specify the Database Server name></server>
</biar-file>
</add>
</step>
<step priority="2">
<add-if-missing>
<user name="spadmin">
<password mode="normal"><Specify the password for the spadmin user in CA Business Intelligence></password>
<description>SLCM Admin User</description>
<password-expiry>false</password-expiry>
<can-change-password>true</can-change-password>
<change-password-on-next-logon>false</change-password-on-next-logon>
</user>
</add-if-missing>
</step>
<step priority="3">
<add-if-missing><membership>
<group>SLCM Service Delivery Administrators</group>
<user>spadmin</user>
</membership>
</add-if-missing>
</step>
</biconfig>
```

Example XML File (Oracle) to Import BIAR

WARNING

If both your CA Service Catalog and CA Service Desk Manager are configured to use the same instance of CA Business Intelligence server, and the date format in odbc.prm file is changed to the CA SDM date format, specify your version of Oracle within the <rdbms> and </rdbms> tags.

```
<?xml version="1.0"?>
<biconfig version="1.0">
<!-- Import BIAR file -->
<step priority="1">
<add>
<biar-file name="Specify path of the SLCM_universe.biar file on the CA Business Intelligence computer">
<networklayer>ODBC</networklayer>
<rdbms>Generic ODBC datasource</rdbms>
<username>mdbadmin</username>
<password><Specify the mdbadmin password></password>
<datasource><caslcm_cabi_dsn></datasource>
<server>abcdef-xp</server>
</biar-file>
</add>
</step>
<step priority="2">
<add-if-missing>
<user name="spadmin">
<password mode="normal"><Specify the password for the spadmin user in CA Business Intelligence></password>
<description>SLCM Admin User</description>
<password-expiry>false</password-expiry>
```

```
<can-change-password>true</can-change-password>
<change-password-on-next-logon>false</change-password-on-next-logon>
</user>
</add-if-missing>
</step>
<step priority="3">
<add-if-missing>
<membership>
<group>SLCM Service Delivery Administrators</group>
<user>spadmin</user>
</membership>
</add-if-missing>
</step>
</biconfig>
```

Integrate CA Service Catalog with CA Process Automation Manually

WARNING

Review the following section only if you want to manually integrate CA Service Catalog with CA Process Automation when the automatic integration between the two has failed.

CA Process Automation uses graphical processes that system administrators create to execute operational processes automatically. CA Process Automation supports an integrated development and administrative environment to manage, create, and configure CA Process Automation components on your system. CA Process Automation also supports client applications that allow operators and other personnel to schedule and monitor automated processes. CA Process Automation includes efficient graphical tools for viewing, monitoring, and debugging process at run-time.

In CA Process Automation, operators define policies and rules for self-managed processes. These processes include configuration, monitoring, optimization, repair, and protection. By integrating the people, processes, and technology of an organization in an end-to-end, automated fashion, organizations can reduce operational expenses, increase staff productivity, and can deliver higher service level metrics. Examples of processes that CA Process Automation automates and manages include:

- Applications Monitoring and Restart
- Disaster Recovery
- Virtual Infrastructure Management
- ITIL compliance
- Security

For ITIL-based implementations, see CA Process Automation Workflow. Such implementations use *both* the basic predefined CA Process Automation that is supplied with CA Service Catalog and other ITIL-based CA Process Automation content.

To configure your implementation of CA Service Catalog, CA Process Automation, and any other integrated CA Technologies products primarily for an ITIL-based environment, review and follow all applicable instructions in the CA Process Automation Workflow CA Process Automation Workflow. Use this section as a secondary reference tool for any information.

To configure your implementation of the CA Service Catalog, CA Process Automation, and any other integrated CA Technologies products primarily for an environment that is not ITIL-based, review all applicable instructions in this section. Use CA Process Automation Workflow section as a secondary reference tool if any ITIL-related questions arise.

Do *not* combine, substitute, or interchange procedures from this section and the CA Process Automation Workflow for performing a specific task. Doing so can produce unpredictable or inconsistent results.

Set Up the CA Process Automation - CA Service Catalog Integration

WARNING

Review the following section only if you want to manually integrate CA Process Automation with CA Service Catalog.

To Integrate CA Service Catalog with CA Process Automation, follow this process:

- Step 1 - Perform Setup Tasks
- Step 2 - Set Up Rules, Actions, and Processes for Approval and Fulfillment
- Step 3 - Assign Users to Required CA Process Automation User Groups

Step 1 - Perform Setup Tasks

Perform these tasks after installing CA Service Catalog and CA Process Automation:

1. Become familiar with CA Process Automation so that you become a knowledgeable administrator and user of it. You must also be able to write basic CA Process Automation processes that include reusable operators. These operators enable communication between CA Process Automation and the products with which it integrates, including CA Service Catalog.
2. Verify that CA Process Automation is running. Install CA Process Automation either before or after you install CA Service Catalog.

   WARNING
   Do *not* install the CA Process Automation domain orchestrator and CA Service Catalog components on the same computer.
3. Verify that you have:
   - Installed the DBMS client (Oracle client or SQL Server client) locally, unless the local computer is the DBMS server. For more information about how to install the DBMS client, see your DBMS documentation.
   - Implemented CA Service Catalog clustering, if you have installed two or more instances of CA Service Catalog. For more information about how to implement CA Service Catalog clustering, see the Implement Clustering section.
   - Met the following naming requirements:
     - The CA Service Catalog computer name must *not* begin with a number.
     - If you are using a load balancer for CA Service Catalog or CA Process Automation, the computer name of the load balancer must *not* begin with a number.

     If you do not meet these requirements, web service calls for the integration can result in errors, for example:

     ```
     Caused by: com.sun.xml.messaging.saaj.util.JaxmURI$MalformedURIException: Host is not a well formed
      address!
     ```
4. If you plan to use Secure Socket Layer, perform the following tasks:
   - Configure CA Process Automation for Secure Socket Layer (SSL).
   - Configure CA Process Automation to communicate with CA Service Catalog using SSL.
5. Perform one of the following actions to specify your login method:
   - If you plan to log in using standard login credentials, skip the next step.
   - If you plan to log in using a CA EEM token instead of standard login credentials, set up certificate-based login.
6. Set and test the administration configuration parameters for CA Process Automation.
7. Load and configure content in CA Service Catalog.
8. Configure content in CA Process Automation.

Set Up Certificate-Based Login in CA Service Catalog

To enhance security, you can set up CA Service Catalog to use certificate-based login to communicate with CA Process Automation. Otherwise, CA Service Catalog uses user name and password login to communicate with CA Process Automation.

The CA Process Automation integration with CA Service Catalog requires CA EEM certificates for certificate-based login. If you are upgrading CA Service Catalog and you already have a .p12 certificate for connecting to CA Process Automation, convert it to .pem format. Similarly, if you are installing CA Service Catalog for the first time, generate a new .pem file.

Follow these steps:

1. Configure CA Service Catalog and CA Process Automation to use SSL, if other CA products integrated with CA Service Catalog are configured with SSL. These tasks include creating and merging keystore files and adding self-signed certificates to the keystore.
2. Edit the USM_HOME\build.xml file, as follows:
   ```
   <sysproperty key="javax.net.ssl.trustStore" value="${env.USM_HOME}/keystore/.keystore" />
   ```
3. Verify that the file name and path name of the keystore file are correct. If necessary, update them.
   For example, if the keystore is located in USM_HOME and the keystore file name is .mykeystore, update this line as follows:
   ```
   <sysproperty key="javax.net.ssl.trustStore" value="${env.USM_HOME}/.mykeystore" />
   ```
4. Locate the following line:
   ```
   <sysproperty key="javax.net.ssl.trustPass" value="changeit" />
   ```
5. Verify that the keystore password is correct. If necessary, update it.
   For example, if the keystore password is mykeystorepw, update this line as follows:
   ```
   <sysproperty key="javax.net.ssl.trustPass" value="mykeystorepw" />
   ```
6. Verify that both CA Service Catalog and CA Process Automation are using the same CA EEM server. Typically, you completed this action when you set up the integration. However, see the eiam.backend property in USM_HOME \config.properties file in CA Service Catalog to verify the name of the CA EEM server.
7. Access the USM_HOME\scripts\eiam folder on any Catalog Component computer and edit the file that is named issue ITPAMCertificatePEM.xml, as follows:
   a. Replace __USMHOME_ (a placeholder value) with the actual value of USM_HOME, for example, C:\Program_files \CA\CA Service Catalog.
   b. Replace ITPAMContext (a placeholder value) with the actual value of the application context name that CA Process Automation uses to connect to CA EEM. For example, ITPAM.
   c. Comment or Uncomment the GroupMembership command line depending on the CA Process Automation version.
8. Locate the following Safex section and the group membership lines.
   For example:
   ```
   <Safex>
   <Attach label="ITPAM" />
   <IssueCertificate certtype="pem" certfile="C:/Progra~1/ca/servic~1/ITPAMCertfile.pem" keyfile="C:/
   Progra~1/ca/servic~1/ITPAMCertfile.key"/>
     <AddOrModify>
       <!--Add user and assign them to appropriate groups-->
       <User folder="/Users" name="CERT-ITPAM">
         <GroupMembership>ITPAMAdmins</GroupMembership>
         <GroupMembership>ITPAMUsers</GroupMembership>
     </User>
     </AddorModify>
   <Detach/>
   </Safex>
   ```
9. Update the values in the group membership lines, as follows:

```
<GroupMembership>PAMAdmins</GroupMembership>
<GroupMembership>PAMUsers</GroupMembership>
```

10. Save and close the file.
11. Verify that *filename* is the same in *filename*.key and *filename*.pem, for example, ITPAMCertfile.key and ITPAMCertfile.pem.
12. (Upgrades) Perform the following actions to convert your existing .p12 file (if you have one) to a .pem file:
    a. Open the Windows command prompt on the computer where CA EEM is installed. Change to the folder named *drive*:\Program Files\CA\SharedComponents\iTechnology.
    b. Run the following command:

```
igwCertUtil -version 4.6.0.0 -conv  -cert "<Certificate><certType>p12</
certType><certURI>ITPAMCertfile.p12</certURI><certPW>ca</certPW></Certificate>" -target
 "<Certificate><certType>pem</certType><certURI>ITPAMCertfile.pem</certURI><keyURI>ITPAMCertfile.key</
keyURI></Certificate>"
```

    This action converts the existing .p12 file to a .pem file.
13. (New Installations) Perform the following actions to create a .pem file:
    a. Open the CA Service Catalog command prompt by entering USM_HOME\usm.cmd at the Windows command prompt.
    b. Change to the USM_HOME\bin\safex folder, and run the following command at the CA Service Catalog command prompt:

```
safex.exe -u EiamAdmin -p <password> -h <eemserver> -f USM_HOME\scripts\eiam
\issueitpamcertificatepem.xml -sdkconfig  USM_HOME\eiam.config
```

    This action creates the ITPAMCertfile.pem and ITPAMCertfile.key files. Both files are necessary for the .pem file authentication to work.
14. Log in to CA Service Catalog. Click Administration, Configuration, CA Process Automation and do the following actions:
    a. Enter the name of the .pem file in the PEM Certificate File Name column. The .pem file is stored in USM_HOME.
    b. Select Yes for the option that is named *Use Certificate for Authentication*.
    c. Set the other configuration parameters for the properties.
15. Click Test to test the two-way web service connection between CA Service Catalog and CA Process Automation.
16. Load and Configure Content in CA Service Catalog.
17. Click the Launch button to start the CA Process Automation Server web URL with the updated properties.
18. Update the user password for CA Service Catalog in the CA Process Automation SLCM_GlobalDataset.

You have set up certificate-based login between CA Service Catalog and CA Process Automation.

Set Up Certificate-Based Login in CA Process Automation

To enhance security, you can set up CA Process Automation to use certificate-based login to communicate with CA Service Catalog. If you do not use this feature, CA Process Automation uses user name and password login to communicate with CA Service Catalog.

> **WARNING**
> This procedure applies *only* if you are using a CA Process Automation version that also supports certificate-based login for CA Service Catalog.

Follow these steps:

1. Access the USM_HOME folder of the CA Service Catalog computer and verify that the following certificate files exist:
   – USMcertfile.key
   – USMcertfile.pem
   – USMcertfile.p12

   The CA Service Catalog installation program creates these files automatically.

2. Decide the files to use with CA Process Automation:
   – If CA EEM is using FIPS mode, select USMcertfile.pem.
   – If CA EEM is using non-FIPS mode, select USMcertfile.p12.

   To verify whether CA EEM is using FIPS mode, log in to CA EEM. Click the About link on the top right corner of the page. An information dialog appears. Review the FIPS-related details on this dialog.
3. Log in as an administrator to CA Process Automation from the CA Service Catalog computer.
4. Perform the following actions:
   a. Click Configuration.
   b. Click Manage User Resources at the left bottom portion of the page.
      The User Resource folder is selected in the Repository tree, and the User Resource page appears on the right pane.
5. Perform the following actions for the file or files that you selected in Step 2. If you selected USMcertfile.key and USMcertfile.pem, perform these actions individually, once for each file.
   a. Click New.
   b. Click Browse in the Resource File field and navigate to the USM_HOME folder on the CA Service Catalog computer.
   c. Select the file and specify the resource name. You can specify an arbitrary name of your choice.
   d. Record the path names of the file or files. You require this information when you set the related administration configuration parameters.
6. Save your changes.
7. If you are using multiple CA Process Automation domain servers with CA Service Catalog, repeat these steps on each one.

You have set up CA Process Automation to use certificate-based login to communicate with CA Service Catalog.

Set Administration Configuration Parameters

If you are running multiple instances of CA Process Automation, the following settings apply to all instances.

Follow these steps:

1. Log in to CA Service Catalog.
2. Click **Administration**, **Configuration**.
3. Click **CA Process Automation** and complete the configuration options, as follows:
   – **Certificate File Path on CA Process Automation**
     Applies only when you are using certificate-based login in CA Process Automation with USMcertfile.pem (for FIPS mode) or USMcertfile.p12 (for non-FIPS mode).
     If this condition exists, specify the path name of the appropriate file on the CA Process Automation server.
   – **Enable Automatic Retry**
     Applies in cases when a request is created in CA Service Catalog that is not processed in CA Process Automation due to connectivity issues between CA Service Catalog and CA Process Automation. The request is placed in queued state. Retry manually to process the request to CA Process Automation. Specify Yes for automatic retry or No for manual retry.
     When the Automatic Retry is selected as Yes, the CA Service Catalog connection is established. The ITPAM_QUEUE_FAILURE automatically gets deleted from the Home, Messages, Alerts, List All Failed Actions list. When the message disappears from the List All Failed Actions, verify that the request has changed to the appropriate status. For example, if the original status was Submitted (Queued) then the updated status must be Pending Approval.
   – **Key File Path on CA Process Automation**
     Applies only when you are using certificate-based login in CA Process Automation with the USMcertfile.key file.
     If this condition exists, specify the path name of the USMcertfile.key file on the CA Process Automation server.
   – **Password for Certificate on CA Process Automation**

Applies only when you are using certificate-based login in CA Process Automation with the USMcertfile.p12 file. If you are using CA Process Automation 4.0 SP1 CP2, obtain the password from the USM_HOME\scripts\EIAM \issueCertificateP12.xml file. The password is a random-generated number that is specified in the IssueCertificate tag of the password attribute, as follows:

```
<IssueCertificate certfile="C:/Program Files/CA/Service Catalog/USMcertfile.p12" password="password"/>
```

- **Enable HTTPS**
  Indicates whether CA Service Catalog uses HTTPS when communicating with CA Process Automation. Specify Yes or No.
- **Host Name**
  Specifies the host name of CA Process Automation computer (the Domain Orchestrator).

  NOTE
  After the integration has been configured and in use, do *not* change the host name of the CA Process Automation computer or the CA Service Catalog computer. The only exception occurs when the two host computers use the same MDB. For information about the MDB, see the MDB documentation included on the CA Service Catalog installation media.
- **PEM Certificate File Name**
  Applies only when you are using a CA EEM certificate for authentication. That is, when the setting named *Use Certificate for Authentication* is Yes. If that setting is Yes, then PEM Certificate File Name specifies the name that you specified earlier when set up certificate-based login for CA Process Automation.
- **Port Number**
  Specifies the port number on the CA Process Automation computer that listens for incoming calls from CA Service Catalog.
- **Retry Count**
  Specifies the number of retry attempts. This value indicates the number of times for CA Service Catalog to try again to perform a failed CA Process Automation action. The default retry count value is 3.
  If all retry attempts fail, CA Service Catalog logs a queue alert as ITPAM_FAILED, and ITPAM_QUEUE_FAILURE in Home, Messages, Alerts. You can manually retry the failed action by viewing the queue alerts in List All Failed Actions. Click the Retry button.
- **Retry Interval**
  Specifies the number of seconds between the retry attempts described for the previous item, Retry Count. The default retry interval is 15 seconds.
- **Use Certificate for Authentication**
  Indicates whether to use a CA EEM certificate for authentication. This certificate is used for communication from CA Service Catalog to CA Process Automation. Specify Yes or No.
- **User ID and User Password**
  Specifies the user ID and password for accessing the CA Process Automation computer. This user ID and password apply *only* when you are using standard login credentials. It does *not* apply when you are using a CA EEM certificate for authentication.

  WARNING
  Verify that the user ID you specify has administrative access to *all* CA Process Automation processes. The user ID requires the right to execute a CA Process Automation process.

The new parameter settings are in effect. Verify that the request flow for a service that uses a CA Process Automation process works properly from approval to fulfillment.

Load and Configure Content in CA Service Catalog

The CA Process Automation *content* refers to the processes, SRFs, rules, actions, and related items that are supplied with CA Service Catalog. You load and configure this content in CA Service Catalog to activate it. Afterwards, you configure this content in CA Process Automation.

Follow these steps:

1. Log in to CA Service Catalog.
2. Click **Administration**, **Configuration**.
3. Click **CA Process Automation** in the Options section.
4. Click **Load**.
   The Load Configuration Dialog appears with two options.
5. Select one or both options (as applicable) and click **Load** in the Load Configuration Dialog.
   The loading process typically requires a few minutes. This process loads the CA Process Automation content for use with CA Service Catalog.
   If the loading process completes successfully, a confirmation message appears. A successful load imports the CA Process Automation content, including processes, and related CA Service Catalog rules, actions, and services.
   If an error message appears, verify that the CA Process Automation server is available and that the configuration parameters are valid.
6. Click **Configure**.
   The CA Service Catalog configuration properties SLCM_GlobalDataset and SDM_GlobalDataset are configured.

You have loaded and configured the content in CA Service Catalog.

Configure Content in CA Process Automation

The CA Process Automation content refers to the processes, SRFs, rules, actions, and so forth, supplied with CA Service Catalog.

Follow these steps:

1. Log in to your CA Process Automation client.
2. Navigate to the CA SLCM folder in the Library Browser, and open the data set named SLCM_GlobalDataset.
3. Open the Login Parameters tab In the Value Definition palette of the data set and perform the following actions:
   a. Verify that the values of the following options are correct for your implementation. Update these values as needed.
      - userID__ - Specifies an administrative user ID with the Service Delivery administrator role.
        As a best practice, create an administrative user in CA Service Catalog exclusively for this integration.
      - Password__ - Specifies the password for this administrative user.

        WARNING
        This field is empty by default; therefore, enter a valid value.
   b. Record these values for use in a later step.
   c. Open the APP URLs tab in the global data set. Verify that the values of the following options are correct for your implementation. Update these values as needed.
      - SLCM_URL - Specifies the URL to launch CA Service Catalog, in the format http:*//hostname:portnumber*.
        If CA Service Catalog runs in a clustered environment, enter the host name and port number of the load balancer.
      - SLCM_CONTEXT - Specifies the URL of the CA Service Catalog context, in the format */usm*.
        For example, http:*//m:port/usm*

        NOTE
        You can deploy CA Service Catalog in Tomcat web server, where the context name would be usm. If you deploy CA Service Catalog on other supported web servers, you can rename the context.
   d. Record these values for use in a later step.
4. To save your changes to SLCM_GlobalDataset, perform the following actions:
5. Click Save.
6. Click the checkin icon to update and save your changes to SDM_GlobalDataset.
7. Perform the following steps, If, you have integrated CA Service Catalog with service desk:
   a. Select the CA SDM folder in the Library Browser, and open the data set named SDM_GlobalDataset.

b. Open the Login Parameters tab In the Value Definition palette of the data set. Verify that the values for its parameters match the values you specified earlier for the Login Parameters tab of the SLCM_GlobalDataset. Update these values as needed.
c. Open the SoapCallPaths tab In the Value Definition palette of the data set. Verify that the values for its parameters match the values you specified earlier for the APP URLs tab of the SLCM_GlobalDataset. Update these values as needed.

8. Save your changes to SDM_GlobalDataset.

You are now ready to set up rules, actions, and processes for approval and fulfillment.

Test the Web Service Connection to CA Process Automation

You can test the connection at any time when an issue is detected with processing CA Service Catalog requests. If the test connection fails, verify the view.log or error.log information for the possible reasons for failure.

Follow these steps:

1. Log in to CA Service Catalog.
2. Click **Administration**, **Configuration**.
3. Click **CA Process Automation** in the Options section.
4. Click **Test**.
   The two-way connection is tested between CA Service Catalog and CA Process Automation, using the CheckSLCMConnectivity SRF and the new values specified. If the connection fails, verify the configuration values provided.

Add Multiple CA Process Automation Instances in CA Service Catalog

There can be many rules that are written between departments in a business unit depending on the various services offered. When a rule is created, an action is allocated to the rule for the CA Process Automation instances to process. Multiple CA Process Automation instances can be added in CA Service Catalog for different services and different users. Add multiple CA Process Automation instances in CA Service Catalog to each domain for approval and fulfillment phases based on the category id and domain id. Adding multiples CA Process Automation instances support you to share work between multiple departments.

Follow these steps:

1. Log in to CA Service Catalog.
2. Click **Administration**, **Configuration**.
3. Click **CA Process Automation** in the Options list.
4. Click **Add**.
   The Explorer User Prompt appears:
5. Enter a configuration name for the new CA Process Automation instance.

   NOTE
   You can only enter alphanumeric characters to add a CA Process Automation instance. All the properties that are associated with the default CA Process Automation instance are imported into the new CA Process Automation instance. Configure the property values.

6. Click OK.
7. (Optional) Click the Modify icon and update the values of each property.

   NOTE
   Each instance that you add accepts the same configuration parameters as the original instance. Typically, for best performance, configure a maximum of ten CA Process Automation instances.

You have added a CA Process Automation instance to CA Service Catalog.

You can also delete a CA Process Automation instance. For example, if the instance it is corrupted, or if you want to replace the existing instance with a different instance. All requests must be in a canceled or completed state before deleting a CA Process Automation instance.

> **WARNING**
> The default CA Process Automation instance cannot be deleted.

Update CA Process Automation in CA Service Catalog Configuration.

If you are running multiple instances of CA Process Automation, the following settings apply to all instances.

Follow these steps:

1. Log in to CA Service Catalog.
2. Click **Administration**, **Configuration**.
3. Click **CA Process Automation** and complete the configuration options, as follows:
   - **Certificate File Path on CA Process Automation**
     Applies only when you are using certificate-based login in CA Process Automation with USMcertfile.pem (for FIPS mode) or USMcertfile.p12 (for non-FIPS mode).
     If this condition exists, specify the path name of the appropriate file on the CA Process Automation server.
   - **Enable Automatic Retry**
     Applies in cases when a request is created in CA Service Catalog that is not processed in CA Process Automation due to connectivity issues between CA Service Catalog and CA Process Automation. The request is placed in queued state. Retry manually to process the request to CA Process Automation. Specify Yes for automatic retry or No for manual retry.
     When the Automatic Retry is selected as Yes, the CA Service Catalog connection is established. The ITPAM_QUEUE_FAILURE automatically gets deleted from the Home, Messages, Alerts, List All Failed Actions list. When the message disappears from the List All Failed Actions, verify that the request has changed to the appropriate status. For example, if the original status was Submitted (Queued) then the updated status must be Pending Approval.
   - **Enable HTTPS**
     Indicates whether CA Service Catalog uses HTTPS when communicating with CA Process Automation. Specify Yes or No.
   - **Host Name**
     Specifies the host name of CA Process Automation computer (the Domain Orchestrator).

     > **NOTE**
     > After the integration has been configured and in use, do *not* change the host name of the CA Process Automation computer or the CA Service Catalog computer. The only exception occurs when the two host computers use the same MDB. For information about the MDB, see the MDB documentation included on the CA Service Catalog installation media.
   - **Key File Path on CA Process Automation**
     Applies only when you are using certificate-based login in CA Process Automation with the USMcertfile.key file.
     If this condition exists, specify the path name of the USMcertfile.key file on the CA Process Automation server.
   - **Password for Certificate on CA Process Automation**
     Applies only when you are using certificate-based login in CA Process Automation with the USMcertfile.p12 file.
     If you are using CA Process Automation 4.0 SP1 CP2, obtain the password from the USM_HOME\scripts\EIAM \issueCertificateP12.xml file. The password is a random-generated number that is specified in the IssueCertificate tag of the password attribute, as follows:

     ```
     <IssueCertificate certfile="C:/Program Files/CA/Service Catalog/USMcertfile.p12" password="password"/>
     ```
   - **PEM Certificate File Name**

Applies only when you are using a CA EEM certificate for authentication. That is, when the setting named *Use Certificate for Authentication* is Yes. If that setting is Yes, then PEM Certificate File Name specifies the name that you specified earlier when set up certificate-based login for CA Process Automation.

- **Port Number**
  Specifies the port number on the CA Process Automation computer that listens for incoming calls from CA Service Catalog.
- **Retry Count**
  Specifies the number of retry attempts. This value indicates the number of times for CA Service Catalog to try again to perform a failed CA Process Automation action. The default retry count value is 3.
  If all retry attempts fail, CA Service Catalog logs a queue alert as ITPAM_FAILED, and ITPAM_QUEUE_FAILURE in Home, Messages, Alerts. You can manually retry the failed action by viewing the queue alerts in List All Failed Actions. Click the Retry button.
- **Retry Interval**
  Specifies the number of seconds between the retry attempts described for the previous item, Retry Count. The default retry interval is 15 seconds.
- **Use Certificate for Authentication**
  Indicates whether to use a CA EEM certificate for authentication. This certificate is used for communication from CA Service Catalog to CA Process Automation. Specify Yes or No.
- **User ID and User Password**
  Specifies the user ID and password for accessing the CA Process Automation computer. This user ID and password apply *only* when you are using standard login credentials. It does *not* apply when you are using a CA EEM certificate for authentication.

  WARNING
  Verify that the user ID you specify has administrative access to *all* CA Process Automation processes. The user ID requires the right to execute a CA Process Automation process.

The new parameter settings are in effect. Verify that the request flow for a service that uses a CA Process Automation process works properly from approval to fulfillment.

{"URL":["/usm/wpf?Node=iclaunchpad.tools_configuration"],"customLabelGetStarted":"CA Process Automation","description":"IPCE_CA_Process_Automation"}

Step 2 - Set Up Rules, Actions, and Processes for Approval and Fulfillment

This article contains the following topics:

To use CA Process Automation for approval and fulfillment of CA Service Catalog requests, you set up rules and actions in CA Service Catalog. You also set up CA Process Automation processes for the actions that you select.

1. Verify that your service is set for workflow-driven approval.
2. Review the important information about CA Service Catalog events, rules, and actions in the Manage Events-Rules-Actions section. Events are always enabled. For events, certain rules are enabled by default, and other rules are disabled by default. If enabled, rules are triggered and executed when the event occurs. You specify whether to keep or change the default settings for which rules are enabled or disabled. For rules, certain actions are enabled by default, and other actions are disabled by default. If enabled, actions are triggered and executed when the conditions specified by the rule occur.
3. The rules and actions that you use depend on the CA Process Automation content that you load for use with CA Service Catalog. CA Process Automation actions apply only to the rules for the event named Request/Subscription Item Change. By default, for the event *Request/Subscription Item Change*, for the rule *When Status is Canceled*, an action is enabled that terminates CA Process Automation instances when a request is canceled. For best product performance, keep this default setting.
4. Review and decide the rules, actions, and CA Process Automation process that you want to use for *approval*. For more information, see CA Process Automation-Driven Approval.

5. Review and decide the rules, actions, and CA Process Automation processes that you want to use for *fulfillment*. For more information, see CA Process Automation-Driven Fulfillment.
6. In addition, you can create or customize rules or processes that are used in approval and fulfillment. Some customization is permitted for built-in rules, but you can optionally create and customize new rules.
   Similarly, you can optionally create or customize CA Process Automation processes. For more information, see your CA Process Automation documentation.

Processes and Start Request Forms

Each CA Process Automation action that you use with a CA Service Catalog rule invokes a start request form (SRF). The SRF starts a CA Process Automation process that aligns to one or more business process or processes. For more information about processes and SRFs, see your CA Process Automation documentation.

CA Process Automation-Driven Approval

CA Service Catalog supplies a pre-defined rule that fires once for each service when a request is submitted for approval.

In each rule, you specify a CA Process Automation action to initiate an approval CA Process Automation start request form . Each start request form or an SRF initiates a CA Process Automation process that manages approval of requests.

Approval Rule, Condition, and CA Process Automation Action

The approval-related rule is associated with the Request/Subscription Item Change event. This event is launched when any service option element in any service in a request changes.

Because approvals are done at the service level, assign a rule to "act on" the status changes for every service in a request. You specify the conditions in the rule to help ensure that the rule actions are performed only for the first service option element in the first row of the first service option group in each service in the request. Thus, the rule launches only once for each service, regardless of the number of service options and service option elements in the service.

The rule that is associated with the approval process is as follows:

When Status is Submitted and Approval Process is driven by Workflow is launched when these conditions are satisfied:

- 100 <= old status < 200 - The old status is in the not submitted range.
- 200 <= new status < 400 - The new status is in the submitted range.
- approval_process = 2 - The approval process for the service is workflow-driven approval. Workflow drives a service approval.
- sd_row = 1 - This service option element is the first service option element in the service.
- rate_item_col = 0 - This service option element is the first service option element in its row of service option elements.

  NOTE
  This rule is launched regardless of the category of any of the service option elements in a service.

Define a CA Process Automation action for approval and associate it with this rule. This action uses the following values:

- Type - CA Process Automation
- Start Request Form (SRF) - click the Search icon and select your approval SRF from the list.
- Configuration Name - All the CA Process Automation multiple instances are listed. Select the CA Process Automation and create the process instance.
- Parameters - When you select an SRF, the required input parameters are automatically listed. Either enter one or more custom values or select one or more values from the list.

The action launches the SRF, passing the information necessary to assign an approval task to the approvers defined in your CA Process Automation approval process. Typically, you use the CA Process Automation process that the SRF calls to set the status of the requested item to Pending Approval (400).

CA Process Automation-Driven Fulfillment

CA Service Catalog supplies rules, conditions, and actions that you can associate with CA Process Automation start request forms (SRFs). These SRFs initiate CA Process Automation processes that handle various fulfillment processes. While approval is handled at the service level, fulfillment is handled at the service option level. The characteristics of the first service option element in a service option determine the fulfillment process to use. The distributed rule conditions are based on the service option element Category field with separate rules for Hardware, Software, None, and all other Category values.

A set of rules is associated with a complex fulfillment process and a simple fulfillment process. Another set of rules opens a CA Service Desk Manager request. The fulfillment process is then left to the CA Service Desk Manager workflow processes.

You can alter the distributed components to fit your business processes more accurately.

> **NOTE**
> Enabling all the rules that are related to all three options is not recommended, because the three fulfillment options overlap.

- **Complex Fulfillment Overview**
  The complex fulfillment process takes into account the current service option element status value to determine the next step in the fulfillment process. The major fulfillment phases related to:
  - Checking availability of the requested item when initially requested or when the item status is Received.
  - Optionally opening a CA Service Desk Manager Change Order (for Hardware or Software). Or notifying appropriate fulfillment personnel when the item is found in available inventory.
  - Notifying the Procurement personnel of the requested item when it is not found in available inventory. The Procurement personnel marks the item as Received when it arrives.
- **Simple Fulfillment Overview**
  The simple fulfillment process notifies the appropriate fulfillment personnel that the item must be fulfilled with no special regard for fulfillment phases.

CA Process Automation Fulfillment Rules, Conditions, and Actions

The fulfillment-related rules are associated with the Request/Subscription Item Change event. This event is launched when any service option element in any service in a request changes.

Use the rule condition to help ensure that the rule actions are performed only at the appropriate time.

The fulfillment process takes different paths that are based on the category and status of the first service option element in the service option. The categories are Software (0), Hardware (1), None (-1), and any custom categories that you define.

The list of categories and their associated numeric values are defined in the following line:

```
USM_HOME\view\webapps\usm\locale\icusen\billing\category.xml
```

The list of statuses and their associated numeric values are defined in the following line:

```
USM_HOME\view\webapps\usm\locale\icusen\request\requestshared.xml
```

CA Process Automation Simple Fulfillment

Simple fulfillment consists of notifying the appropriate department that the requested item must be fulfilled. The assigned fulfillment users change the status of the requested item as appropriate until the status is changed to Fulfilled or Fulfillment Cancelled.

CA Process Automation Simple Fulfillment Rule and Actions

The rule that is associated with the simple fulfillment process used for CA Process Automation is *When Status is Pending Fulfillment*.

WARNING
If you enable this rule, verify that *all other* fulfillment rules are disabled. Otherwise requests pending action can be duplicated.

- **When Status is Pending Fulfillment Rule**
  This rule is launched when a request item status change and these conditions are satisfied:
  - 800 <= old status <= 999 - The old status is in the approved range.
  - new status = 1000 - The new status is Pending Fulfillment (1000)
  - rate_item_col = 0 - This service option element is the first service option element in its row of service option elements.

  This rule condition ensures that the rule is used only when a request first enters the fulfillment phase.
  Define CA Process Automation Action
  Define a CA Process Automation action for simple fulfillment and associate it with this rule.
  This action uses the following values:
  - Type - CA Process Automation
  - Start Request Form (SRF) - click the Search icon and select your simple fulfillment SRF from the list.
  - Configuration Name - All the CA Process Automation multiple instances are listed. Select the CA Process Automation to create the process instance.
  - Parameters - When you select an SRF, the required input parameters are automatically listed. Either enter one or more custom values or select one or more values from the list.

  When this action starts the Fulfillment, it passes the information necessary to assign a fulfillment needed task to the appropriate fulfillment user or group. Typically, you also use the CA Process Automation process that the SRF calls to set the status of the requested item to Pending Fulfillment (1000).

CA Process Automation Complex Fulfillment

The first step of complex fulfillment is for the appropriate department to check the availability of the requested item. For example, for hardware and software this department can be the IT Services department.

For hardware and software items, if the requested item is found in available inventory, then you can optionally perform one of the following actions:

- Notify the IT Services department to fulfill the request.
- Integrate with another product, such as CA Service Desk Manager. You can now open CA Service Desk Manager change order.

For items that are neither hardware nor software, if the requested item is found in available inventory, then the appropriate department is notified to fulfill the request.

For all items, if the requested item is not found in available inventory, then the Procurement department is notified that the requested item must be purchased. After the status of the requested item is set to Received, it is assumed that the item is then available in the inventory. The appropriate department must check the inventory.

If the appropriate department is notified that fulfillment is needed, the user assigned the fulfillment task sets the request item status to Fulfilled. When all request item statuses are set to Fulfilled, the system changes their statuses to Completed, as appropriate. This action allows billing from Service Accounting Component.

If you have integrated with another product, then you can configure an associated CA Process Automation process in that product to manage the rest of the fulfillment process. For example, if you configure the CA Process Automation process to create a CA Service Desk Manager change order, then, after the change order is closed, the associated CA Process Automation process in CA Service Desk Manager sets the request item status to Fulfilled.

If for some reason, the requested item cannot be fulfilled, then the status is set to Fulfillment Cancelled. The system changes the status to Cancelled when all of the request item statuses are set to Fulfilled or Fulfillment Cancelled. This status change marks the end of the fulfillment phase.

CA Process Automation Complex Fulfillment Rules and Actions

The following rules are associated with the complex fulfillment process:

Check Availability Rules for CA Process Automation

The Check Availability rules for use with CA Process Automation are as follows:

- **When Category is Hardware and Status is Pending Fulfillment** is launched when these conditions are satisfied:
  - 800 <= old status < = 999 - The old status is in the approved range.
  - new status = 1000 - The new status is pending fulfillment (1000)
  - category = 1 - The service option element category is Hardware (1)
  - rate_item_col = 0 - This service option element is the first service option element in its row of service option elements.
- **When Category is Software and Status is Pending Fulfillment** is launched when these conditions are satisfied:
  - 800 <= old status < = 999 - The old status is in the approved range.
  - new status = 1000 - The new status is pending fulfillment (1000)
  - category = 0 - The service option element category is Software (0)
  - rate_item_col = 0 - This service option element is the first service option element in its row of service option elements.
- **When Category is neither Hardware nor Software and Status is Pending Fulfillment** is launched when these conditions are satisfied:
  - 800 <= old status < = 999 - The old status is in the approved range.
  - new status = 1000 - The new status is pending fulfillment (1000)
  - category > 1 - The service option element category is not Hardware (1) or Software (0) or None (-1)
  - rate_item_col = 0 - This service option element is the first service option element in its row of service option elements.

Define CA Process Automation Action

For each Check Availability rule, define a CA Process Automation action and associate it with the rule.

Each action uses the following values:

- Type - CA Process Automation
- Start Request Form (SRF) - click the Search icon and select your Check Availability SRF from the list
- Configuration Name - All the CA Process Automation multiple instances are listed. Select the CA Process Automation to create the process instance
- Parameters - When you select an SRF, the required input parameters are automatically listed. Either enter one or more custom values or select one or more values from the list.

When this action starts the Check Availability SRF, it passes the information necessary to assign a check availability task to the appropriate fulfillment user or group. Typically, you use the CA Process Automation process that the SRF calls to set the status of the requested item to Check Availability (1001).

Fulfillment Rules and Actions for CA Process Automation

The fulfillment rules are as follows:

- **When Category is Hardware and Status is Filled From Inventory** is launched when these conditions are satisfied:
  - old status <> 1002 - The old status is not Filled From Inventory (1002)
  - new status = 1002 - The new status is Filled From Inventory (1002)
  - category = 1 - The service option element category is Hardware (1)
  - rate_item_col = 0 - This service option element is the first service option element in its row of service option elements

Define CA Process Automation Action

If you use the When Category is Hardware and Status is Filled From Inventory rule, define a CA Process Automation action and associate it with the rule. Define this action to pass the information to doone or more tasks, for example:

- Assign a fulfillment needed task to the appropriate fulfillment user.
- Integrate with another product; for example, open a CA Service Desk Manager change order.

Each action uses the following values:

- Type - CA Process Automation
- Start Request Form (SRF) - click the Search icon and select your Fulfillment SRF from the list
- Configuration Name - All the CA Process Automation multiple instances are listed. Select the CA Process Automation to create the process instance.
- Parameters - When you select an SRF, the required input parameters are automatically listed. Either enter one or more custom values or select one or more values from the list.

Typically, you use the CA Process Automation process the SRF calls to set the status of the requested item to Notified IT Services (1015).

- **When Category is Software and Status is Filled From Inventory** is launched when these conditions are satisfied
  - old status <> 1002 - The old status is not Filled From Inventory (1002)
  - new status = 1002 - The new status is Filled From Inventory (1002)
  - category = 0 - The service option element category is Software (0)
  - rate_item_col = 0 - This service option element is the first service option element in its row of service option elements

Define CA Process Automation Action

If you use the When Category is Software and Status is Filled From Inventory rule, define a CA Process Automation action and associate it with the rule. Define this action to pass the information to doone or more tasks, for example:

- Assign a fulfillment needed task to the appropriate fulfillment user.
- Integrate with another product; for example, open a CA Service Desk Manager change order, associating any assigned assets.

Each action uses the following values:

- Type - CA Process Automation
- Start Request Form (SRF) - click the Search icon and select your Fulfillment SRF from the list
- Configuration Name - All the CA Process Automation multiple instances are listed. Select the CA Process Automation to create the process instance.
- Parameters - When you select an SRF, the required input parameters are automatically listed. Either enter one or more custom values or select one or more values from the list.

Typically, you use the CA Process Automation process called by the SRF to set the status of the requested item to Notified IT Services (1015).

- **When Category is neither Hardware nor Software and Status is Filled From Inventory** is launched when these conditions are satisfied:
  - old status <> 1002 - The old status is not filled from inventory (1002)
  - new status = 1002 - The new status is filled from inventory (1002)
  - category > 1 - The service option element category is not None (-1), Software (0) or Hardware (1)
  - rate_item_col = 0 - This service option element is the first service option element in its row of service option elements

Define CA Process Automation Action

If you use the When Category is neither Hardware or Software and Status is Filled From Inventory rule, If you use the When Category is Software and Status is Filled From Inventory rule, define a CA Process Automation action and associate it with the rule. Define this action to pass the information to doone or more tasks, for example:

- Assign a fulfillment needed task to the appropriate fulfillment user.
- Integrate with another product; for example, open a CA Service Desk Manager change order, associating any assigned assets.

Each action uses the following values:

- Type - CA Process Automation
- Start Request Form (SRF) - click the Search icon and select your Fulfillment SRF from the list
- Configuration Name - All the CA Process Automation multiple instances are listed. Select the CA Process Automation to create the process instance.
- Parameters - When you select an SRF, the required input parameters are automatically listed. Either enter one or more custom values or select one or more values from the list.

Typically, you use the CA Process Automation process called by the SRF to set the status of the requested item to Notified IT Services (1015).

- **When Category is None and Status is Pending Fulfillment** is launched when these conditions are satisfied:
  - 800 <= old status <=999 - The old status is in the approved range
  - new status = 1000 - The new status is Pending Fulfillment (1000)
  - category = -1 - The service option element category is None (-1)
  - rate_item_col = 0 - This service option element is the first service option element in its row of service option elements

Define CA Process Automation Action

If you use the When Category is None and Status is Pending Fulfillment rule, Define a CA Process Automation action for simple fulfillment and associate it with this rule.

This action uses the following values:

- Type - CA Process Automation
- Start Request Form (SRF) - click the Search icon and select your simple fulfillment SRF from the list
- Configuration Name - All the CA Process Automation multiple instances are listed. Select the CA Process Automation to create the process instance.
- Parameters - When you select an SRF, the required input parameters are automatically listed. Either enter one or more custom values or select one or more values from the list.

When this action starts the Fulfillment (SRF), it passes the information necessary to assign a fulfillment needed task to the appropriate fulfillment user or group. Typically, you also use the CA Process Automation process called by the SRF to set the status of the requested item to Pending Fulfillment (1000).

Procurement Rules for CA Process Automation

The procurement rules for use with CA Process Automation are as follows:

- **When Category is Hardware and Status is Not Filled From Inventory** rule is launched when these conditions are satisfied:

  - old status <> 1003 - The old status is Not Filled From Inventory (1003)
  - new status = 1003 - The new status is Not Filled From Inventory (1003)
  - category = 1 - The service option element category is Hardware (1)
  - rate_item_col = 0 - This service option element is the first service option element in its row of service option elements
- **When Category is Software and Status is Not Filled From Inventory** rule is launched when these conditions are satisfied:
  - old status <> 1003 - The old status is not Not Filled From Inventory (1003)
  - new status = 1003 - The new status is Not Filled From Inventory (1003)
  - category = 0 - The service option element category is Software (0)
  - rate_item_col = 0 - This service option element is the first service option element in its row of service option elements
- **When Category is neither Hardware nor Software and Status is Not Filled From Inventory** rule is launched when these conditions are satisfied
  - old status <> 1003 - The old status is Not Filled From Inventory (1003)
  - new status = 1003 - The new status is Not Filled From Inventory (1003)
  - category > 1 - The service option element category is not None (-1), Software (0) or Hardware (1)
  - rate_item_col = 0 - This service option element is the first service option element in its row of service option elements

Define CA Process Automation Action

For each procurement rule, define a CA Process Automation action and associate it with the rule.

Each action uses the following values:

- Type - CA Process Automation
- Start Request Form (SRF) - click the Search icon and select your procurement SRF from the list
- Configuration Name - All the CA Process Automation multiple instances are listed. Select the CA Process Automation to create the process instance
- Parameters - When you select an SRF, the required input parameters are automatically listed. Either enter one or more custom values or select one or more values from the list.

When this action starts the Check Availability SRF, it passes the information necessary to assign a procurement task to the appropriate procurement user or group. Typically, you use the CA Process Automation process called by the SRF to set the status of the requested item to Pending Procurement (1012).

Received or Order Cancelled Rules

The Received or Order Cancelled rules for use with CA Process Automation are as follows:

- **When Category is Hardware and Status is Received or Order Canceled** rule is launched when these conditions are satisfied:
-
  - old status <> 1007 - The old status is not Received (1007)
  - old status <> 1008 - The old status is not Order Cancelled (1008)
  - new status = 1007 OR new status = 1008 - The new status is Received (1007) or Order Cancelled (1008)
  - category = 0 - The service option element category is Software (0)
  - rate_item_col = 0 - This service option element is the first service option element in its row of service option elements.
- **When Category is Software and Status is Received or Order Canceled** rule is launched when these conditions are satisfied

  - old status <> 1007 - The old status is not Received (1007)
  - old status <> 1008 - The old status is not Order Cancelled (1008)
  - new status = 1007 OR new status = 1008 - The new status is Received (1007) or Order Cancelled (1008)
  - category = 0 - The service option element category is Hardware (1)
  - rate_item_col = 0 - This service option element is the first service option element in its row of service option elements
- **When Category is neither Hardware nor Software and Status is Received or Order Canceled** rule is launched when these conditions are satisfied
  - old status <> 1007 - The old status is not Received (1007)
  - old status <> 1008 - The old status is not Order Cancelled (1008)
  - new status = 1007 OR new status = 1008 - The new status is Received (1007) or Order Cancelled (1008)
  - category <> 0 and category <> 1 - The service option element category is neither Hardware (1) nor Software (0)
  - rate_item_col = 0 - This service option element is the first service option element in its row of service option elements

Define CA Process Automation Action

For each Received or Order Cancelled rule, define a CA Process Automation action and associate it with the rule.

Each action uses the following values:

- Type - CA Process Automation
- Start Request Form (SRF) - Click the Search icon and select your Check Availability SRF from the list
- Configuration Name - All the CA Process Automation multiple instances are listed. Select the CA Process Automation to create the process instance
- Parameters - When you select an SRF, the required input parameters are automatically listed. Either enter one or more custom values or select one or more values from the list.

When this action starts the Check Availability SRF, it passes the information necessary to assign a check availability task to the appropriate fulfillment user or group. Typically, you use the CA Process Automation process the SRF calls to set the status of the requested item to Check Availability (1001).

Step 3 - Assign Users to Required CA Process Automation User Groups

You can launch the CA Process Automation client from the CA Service Catalog request tracking page. To do so, assign users to the appropriate CA Process Automation groups in CA EEM. Launch the client in context to an active CA Process Automation process.

When you install CA Process Automation, it creates the following user groups in CA EEM:

- PAMADMIN - the administrators user group
  By default, this group contains one member, named pamadmin, but you can optionally add others.
  Users in this group have complete administrative access to CA Process Automation.
- PAMUser - the user group for non-administrative users
  By default, this group contains one member, named pamuser, but you can optionally add others.

Assign the users to one of these groups. Verify that the users can launch the CA Process Automation client from the CA Service Catalog request tracking page.

Troubleshooting CA Service Desk Manager

This section contains the following articles:

- Why are undelivered e-mail notifications from CA Service Desk Manager (SDM) saved in the NX_ROOT\\site\ \mail_undeliverable directory?
- Unable to Synchronize Notes and Attachments from CA SDM to CA Service Catalog
- CA Service Desk Manager Advanced Availability Shows Errors
- Troubleshoot Search Server Issues
- How to Identify Performance Problems in CA SDM
- How to Monitor LDAP Using Trace Logging
- Tomcat Logging
- How to Hide Apache Tomcat Version Number from Error Pages in CA Service Desk Manager?
- REST Logging
- Troubleshooting LDAP Configuration with CA SDM
- Troubleshooting for Email-Related Issues
- Migration and Upgrade Failing on Standby Server
- CA SDM Performance Problems - Quick Checklist
- CA SDM Migration Fails with exitCode:231 Error
- Unable to Install Any Options in CA SDM with Status as 'In Progress'
- How to Increase Tomcat Max Threads on CA Service Desk Manager
- How to extend the SOAP Web Services method 'doQuery' limitation of 250 rows?
- How to modify the CA SDM Web Session timeout?
- CA Service Desk Manager Installation Fails
- Delayed CA SDM Server Response After Upgrading to Oracle 12c
- CA SDM Conventional Configuration - The Secondary Server Tomcat Engine is Not Responding
- CA SDM Authentication Issues
- How to extend or disable the \"Delayed Server Response\" timeout in CA SDM?
- How to mask out the \"None\" Impact value?
- Unable to Set a Default Value for a Field in Web Screen Painter
- \"i18nMsg.c Can't Construct Translator\" -stdlog Error Message
- How to Log a Comment on a Closed CA SDM Ticket?
- How to set the Initial status of a Copied Incident / Requests to Open?
- How can we import a vendor supplied certificate into CA Service Desk Manager's Tomcat?
- Secondary Tomcat may not start
- Invalid character is found in the request target

Why are undelivered e-mail notifications from CA Service Desk Manager (SDM) saved in the NX_ROOT\site\mail_undeliverable directory?

Problem: It is noticed that when an e-mail notification is sent from CA Service Desk Manager (CA SDM), it is saved in the "NX_ROOT\site\mail_undeliverable" folder. If an e-mail notification is sent from CA SDM, it will try to relay the email notification to the email server which is configured in the Options Manager. There are two different scenarios which can occur where the email notification is not delivered, but only one will result in a saved version of the email notification being saved in the NX_ROOT\site\mail_undeliverable folder. Consider the following two possible causes, if you are facing this issue in your environment :

1. The e-mail is successfully relayed to the email server, but the email server is unsuccessful in routing the email notification to the specific mailbox.
   In this case, since the email is relayed to the email server and from a CA SDM perspective, the email notification is considered to have been delivered, no error messages is logged in the the STDLOG. No saved version of the email message is created in the "**mail_undeliverable**" directory.

2. The email is not successfully relayed to the email server. This may be due to incorrect email settings within the Options Manager, incorrect email address, communication problems, and so on.

   NOTE
   In such cases, since CA SDM is unable to relay the email notification to the email server, this will be considered an undelivered email. A message is logged in the STDLOG, and the email is saved the "**mail_undeliverable**" directory.

Unable to Synchronize Notes and Attachments from CA SDM to CA Service Catalog

This article describes how to troubleshoot the SSL Handshake issues when synchronizing notes and attachments from CA Service Desk Manager (SDM) to CA Service Catalog:

Ensure that CA Service Desk Manager and CA Service Catalog are SSL Enabled. For more information, see How to Configure SSL Authentication.

Problem: Synchronizing notes and attachments from CA SDM to CA Service Catalog is failing. The below error message is shown:

```
ERROR CatalogServiceProcessor 119 Error synching Log comment to Catalog. ; nested exception is:
        javax.net.ssl.SSLHandshakeException: PKIX path building failed:
 sun.security.provider.certpath.SunCertPathBuilderException: unable to find valid certification path to
 requested target
```

Solution: To resolve the SSL Handshake exception, navigate to *\Program Files\CA\SC\JRE\11.0.3\bin*, and perform the following steps:

Export CA Certificate to the Keystore

Open the command prompt, and run the following command

```
keytool -export -alias <alias_name> -storepass <password> -file <filename.cer> -keystore
 <path_and_file_specification_for_keystore>
```

CA certificate is exported to keystore. Self-signed certificates work successfully now.

Add CA Certificate to the Keystore

After creating the keystore file, you must add a CA certificate to the keystore. Without a CA certificate present in keystore, the self-signed certificates do not work.

```
keytool - import - alias <alias_name> - file <certfile> - keystore %USM_HOME%\embedded\jdk\lib\security
\cacerts
```

After completing the above steps, restart the CA SDM Services.

CA Service Desk Manager Advanced Availability Shows Errors

While Installing CA Service Desk Manager Advanced Availability via the CA Service Management Installer 17.4, error messages are seen as shown below:

```
web:wsp 14198 SEVERE_ERROR value.c 424 Bad value type67108864
web:wsp 14198 SEVERE_ERROR value.c 424 Bad value type-2113929216
slump_nxd 13864 SIGNIFICANT list.c 559 Node(10.131.121.220) Slump ID(12263) has disconnected
slump_nxd 13864 SIGNIFICANT list.c 512 Node(10.131.121.220) Slump ID(12263) has connected
pdm_ver_nxd 14215 ERROR pdm_version.c 1049 Message received with invalid checksum: 4539741/65536
```

Ignore the error message and continue with the installation. The installation completes successfully.

Troubleshoot Search Server Issues

Troubleshoot the following search server issues:

Search Server service is missing after rolling back to 17.3 GA (by uninstallation of the latest patch)

While unistalling the latest patches to 17.3 GA, Search server service is missing.

Perform the following steps to generate the Search Server service "CA Service Management Search Server":

1. Set the **JAVA_HOME** using command prompt.
   Example:
   ```
   SET JAVA_HOME=C:/PROGRA~1/CA/SC/JRE/1.8.0_74
   ```
2. Run the following commands from command prompt:
   ```
   CMD /C CALL "C:\Program Files\CA\SC\CASearchServer\service.bat" "C:\Program Files\CA\SC\CASearchServer
   \elasticsearch-2.1.1\bin\service.bat" install 1.8.0_74
   CMD /C CALL "C:\Program Files\CA\SC\CASearchServer\service.bat" "C:\Program Files\CA\SC\CASearchServer
   \elasticsearch-2.1.1\bin\service.bat" start 1.8.0_74
   ```

Getting Logged out while entering search key words in the Global Search Bar

While entering search key words in the **xFlow Interface** Global Search bar, you may sometimes get logged out. Increasing the Timeout value to the desired value as per environment requirements helps to resolve this issue:

Perform the following steps to increase the Timeout value:

1. Log into a system where you have installed **xFlow Interface** .
2. For **Windows** Environment, perform the following:
   a. Navigate to the install directory (*<C:\Program Files\CA\xFlow\Apps>*) and edit the *ServiceConfig.xml* file to add "*MsTimeout=120000*" to the Root install directory:
   <C:\Program Files\CA\xFlow\APPS" JreVersion="1.8.0_74>
   For example:
   ```
   <Root InstallDir="C:\Program Files\CA\xFlow\APPS" JreVersion="1.8.0_74" MsTimeout="120000"/>
   ```
   Note that the *MsTImeout* Value is in Milliseconds (Windows Environment)
3. For **Non-Windows** Environment, navigate to:
   ```
   /opt/CA/xFlow/APPS
   ```
   a. Edit the xFlow Server file and search for the parameter "*Sleep*", increase it to a value of your choice.
   For example: *120 Seconds.*
4. Restart the CA **xFlow Interface** Service.

   NOTE

   Ensure to wait for some time before performing any activities as the Timeout value is now increased and it may take some time for the xFlow microservices to start successfully.

Unable to start the elastic search server service

Unable to start the elastic search server service when Elastic Search Server is installed in a non-default location. For example: \test\CA\SC\JRE\11.0.3

Workaround: To resolve this issue, perform the following steps:

1. Move the SC\JRE\11.0.3 folder to the default Elastic Search Server Install location. For example:
   C:\Program Files\CA\ folder
   Linux: Defualt location of JRE: /opt/CA/SC/JRE/11.0.3

Windows: Defualt location of JRE: C:\Program Files\CA\SC\JRE\11.0.3

2. Try to start the Elastic Search Server service.
   The service starts without any errors.

Unable to Start the Search Server Service With the Process Terminated Error Message

Unable to start the Search Server Service with the process terminated error message "Windows could not start the CA Service Management Search Server service on Local Computer.Error 1067: The process terminated unexpectedly."

This issue occurs when the **elasticsearch** folder does not exist or not accessible. You can see this message in the Search Server log file for the startup attempt.

Workaround: Creating a directory named **elasticsearch** in the **temp** directory resolves the issue.

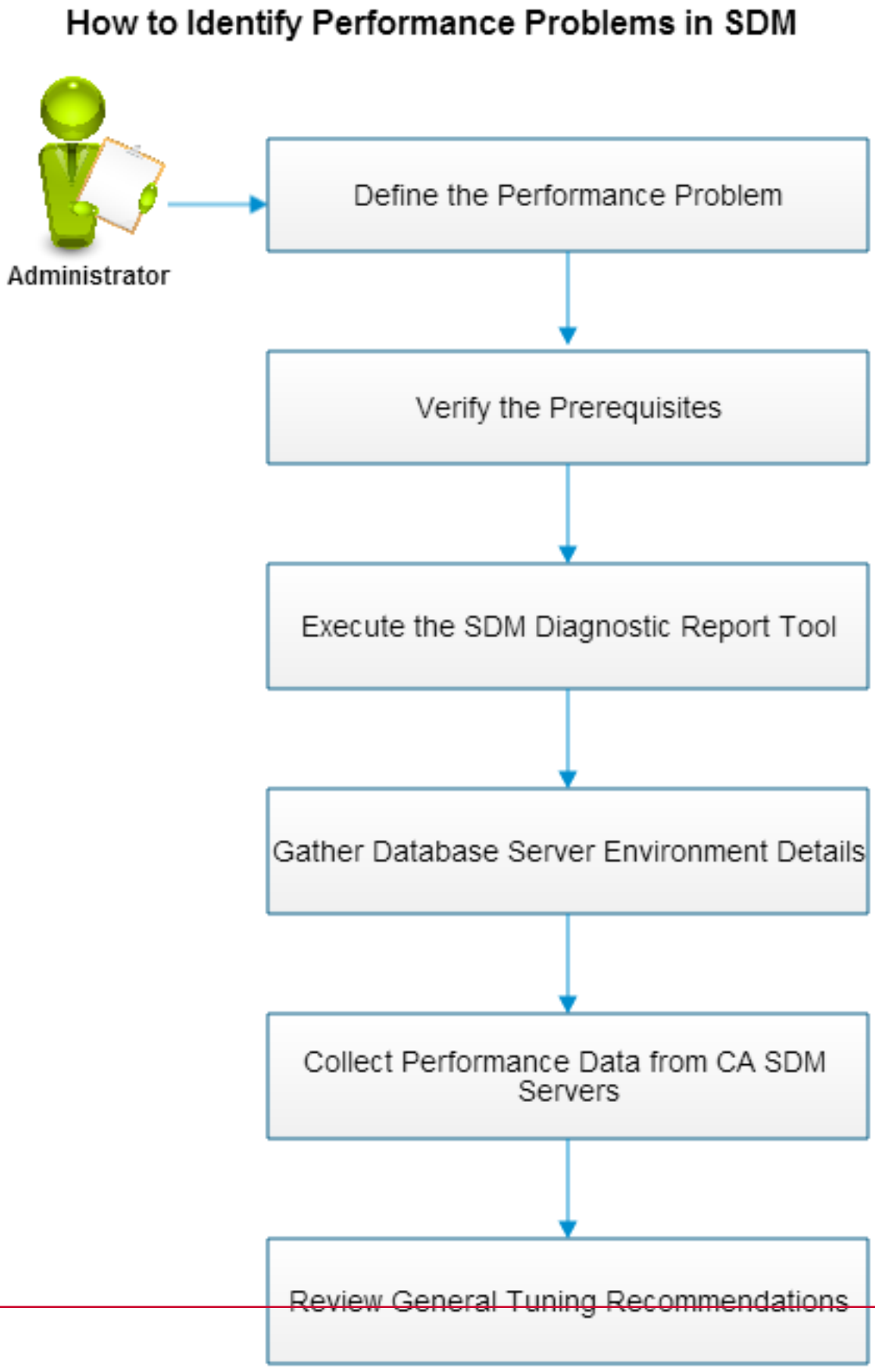
For more information about this issue, refer to the CA Search Server Failed To Start KB article.

How to Identify Performance Problems in CA SDM

As a system administrator, you gather information about your installation environment, resource usage, configuration details, and other available system resources. For example, understand the computer configuration to determine the Operating System type, version, and available system resources. This data helps you to identify and diagnose a performance or memory-related issue in CA SDM. The CA Diagnostic Report tool and the Interval Logging utility help you collect diagnostic information about your CA SDM environment.

The following diagram describes how to gather diagnostic information and identify performance problems in CA SDM:

Figure 42: Identifying Performance Problems in CA SDM Servers



Follow these steps:

1. Define the Performance Problem.
2. Verify the Prerequisites
   - Install PS Tools
3. Execute the SDM Diagnostic Report Tool.
   - Collect Information from the CA Diagnostic Report Tool.
   - Verify Windows Report
   - Verify UNIX Report
   - Verify Collected Diagnostic Report
4. Gather database server environment details.
5. Collect Performance Data from CA SDM Servers.
   - Collect Usage Data using Interval Logging.
6. Review general tuning recommendations.

Define the Performance Problem

To define performance problems, begin by collecting the system information. Then consider the following example questions for relevance:

- What did users experience to disrupt their tasks?
- When did users first discover the problem?
- Has your environment changed recently, such as hardware upgrades?
- What functionality of the product experiences issues?
- How many users does the issue impact, and what type of users?
- What types of users are not impacted?
- What is the geographic location of the users that see the problem?
- What is the Access Level of the impacted users?
- Do you host CA SDM on a VMware ESX server or other virtualized environment?
- What other software are you running on the ESX server or host machine?
- What are the specifications of this environment?
- How many CPUs do you have on each computer?
- How much memory did you configure for each computer?

Verify the Prerequisites

1. For Windows operating system, install pslist.exe tools and add its directory path variable on each CA SDM server.
2. Install and configure the CA SDM servers before running the diagnostic tool.

> **NOTE**
> We recommend that you contact CA Support Online before you use the diagnostic tool.

Install PS Tools for Windows

> **NOTE**
> For a Windows installation, install the pslist.exe tool and add its directory path variable to the system path variable.

Follow these steps:

1. Download PS Tools Suite from Microsoft.
2. Extract pstools.zip to any directory of your choice and add the directory to the Windows %PATH% environment variable.

3. Execute pslist.exe once manually from the command prompt as the Local System user and accept the license agreement. To execute pslist.exe as the Local System user, run the following command:

```
psexec.exe -s -i pslist.exe
```

NOTE
You can start the CA SDM Windows service under a different user account other than the Local System account. For that service, execute pslist.exe under that account and accept the license agreement. To add pslist.exe after installing and configuring CA SDM, accept the PsList license agreement and restart the CA SDM services.

Execute the SDM Diagnostic Report Tool

The CA Diagnostic tool creates a .CAZ file on the Windows OS and .tar.gz on UNIX in the $NX_ROOT\diag\rpt directory. You upload the file that the CA Diagnostic tool creates with your issue at Broadcom Support.

Follow these steps:

1. Execute supp_diag.cmd on Windows, or execute supp_diag.sh on UNIX.
   The diagnostic tool can take five to 10 minutes to complete.
2. If the data collection process does not complete, navigate to the $NX_ROOT\diag\<host_name>_supp_diag.log log file. To determine the errors that occurred, examine the log file.

   NOTE
   If you want to cancel the background batch job, use CTRL-C to cancel the batch file. Some processes still run in the background, such as MSINFO32.exe. If you have any questions about using this diagnostic tool, contact the CA Support Online.
3. The script directory structure displays the location of the script files, diagnostic zip files, and log files:
   - The $NX_ROOT\diag\bin directory contains script files.
   - The $NX_ROOT\diag\rpt directory contains the diagnostics zip file (in .caz format on Windows systems and in .tar.gz format on UNIX systems).
   - The $NX_ROOT\diag\misc_logs directory contains the log files that can be automatically included in the zip file.
4. To unzip the gathered files, complete the steps that are appropriate for the operating system:
   - **Windows**
   - Open a command prompt.
   - Cd to $NX_ROOT\diag\rpt or any directory where the .CAZ file is located.
   - Execute the following command:

     ```
     $NX_ROOT\diag\bin\cazipxp -u <package_name>.CAZ
     ```
   - **UNIX**
   - Open a command prompt.
   - Cd to $NX_ROOT/diag/rpt or any directory where the .tar or .tar.gz file is located.
   - Uncompress and untar the file:

     ```
     gunzip -d <package_name>.tar.gz
     tar -xvf <package_name>.tar
     ```

Collect Information from CA Diagnostic Report Tool

The CA SDM installation media includes the Diagnostic Report Tool to help collecting information for diagnosing performance problems. You can use the diagnostic tool to collect information relevant to the operating system. You can use the diagnostic tool to determine the commands for collecting data from CA SDM.

Configure the CA SDM server before running the diagnostic tool.

Follow these steps:

- **Windows**
- Set $NX_ROOT as your CA SDM installation root directory.
  The default for $NX_ROOT is the C:\Program Files\CA\Service Desk\ on Windows. You can change the default when you want to change the default directory during the installation process.
- Verify that the system path variable includes $NX_ROOT\bin.
- **UNIX/Linux**
- Set $NX_ROOT as your CA SDM installation root directory.
  The default for $NX_ROOT is -- /opt/CAisd/ on a Unix or Linux operating system. You can change the default when you want to change the default directory during the installation.
- Verify that your $PATH includes $NX_ROOT/bin.

Verify Windows Report

The following list describes the Windows report files that are created and included in the CAZ\tar file package.

- **Ca.olf**
  The Ca.olf report file specifies the CA licensing information from ca_lic directory.
- **Lic98.log**
  The Lic98.log report file specifies the log file that is related to CA licensing from ca_lic directory.
- **Lic98version.log**
  The Lic98version.log report file specifies the log file that is related to CA licensing from ca_lic directory.
- **Licdebug.log**
  The Licdebug.log report file specifies the file that is related to CA licensing from ca_lic directory.
- **Drwatsoninfo.txt**
  The Drwatsoninfo.text report file specifies the Dr. Watson configuration of the computer.
- **<host name>_env.txt**
  The <host name>_env.txt report file specifies the environment variables that are set on the computer.
- **<host name>_slstat.txt**
  The <host name>_sistat.txt report file specifies the output of slstat command.
- **<host name>_pdm_status.txt**
  The <host name>_pdm_status.txt file specifies the output of slstat command.
- **<host name>_dir_listing.txt**
  The <host name>_dir_listing.txt report file specifies the Service Desk install directory listing.
- **<host name>_pslist.txt**
  The <host name>_splist.txt report file specifies the process listing when the PsList Microsoft tool is installed.
- **<host name>_MSINFO32.NFO**
  The <host name>_MSINFO32.NFO report file specifies the MSINFO output gathering system information.
- **<host name>_SYSTEMINFO.TXT**
  The <host name>_SYSTEMINFO.TXT report file specifies the system information.
- **<host name>_appevents.csv**
  The <host name>_appevents.csv report file specifies the application event logs created in the past seven days.
- **<host name>_sysevents.csv**
  The <host name>_sysevents.csv report file specifies the application event logs created in the past seven days.
- **<host name>_hostinfo.txt**
  The <host name>_hostinfo.txt report file specifies the computer information.
- **<host name>_prodinstallinfo.txt**
  The <host name>_prodinstallinfo.txt report file specifies installation information for CA products.
- **<host name>_caprod_registry.txt**
  The <host name>_caprod_registry.txt report file specifies the Registry information of installed CA products.
- **<host name>_softfeatures.txt**

The <host name>_softfeatures.txt report file specifies the list of software features that are installed for Service Desk.

- **<host name>_ipconfig.txt**
  The <host name>_ipconfig.txt report file specifies the IP configuration information.
- **<host name>_supp_diag.log**
  The <host name>_supp_diag.log report file specifies the log created for running the supp_diag tool.

Verify UNIX Report

The following list describes the UNIX report files that are created and included in the CAZ\tar file.

- **Ca.olf**
  The Ca.olf report file specifies the CA licensing information from ca_lic directory.
- **Lic98.log**
  The Lic98.log report file specifies the log file that is related to the CA licensing from ca_lic directory.
- **<host name>_env.txt**
  The <host name>_env.txt report file specifies the environment variables that are set on the computer.
- **<host name>_slstat.txt**
  The <host name>_slstat.txt report file specifies the output of slstat command.
- **<host name>_pdm_status.txt**
  The <host name>_pdm_status.txt report file specifies the output of the pdm status command.
- **<host name>_dir_listing.txt**
  The <host name>_dir_listing.txt report file specifies the Service Desk install directory listing.
- **<host name>_pslist.txt**
  The <host name>_pslist.txt report file specifies the process listing when the pslist Microsoft tool is installed.
- **<host name>_uname.txt**
  The <host name>_uname.txt report file specifies the output of the uname operating system command.
- **<host name>_diskinfo.txt**
  The <host name>_diskinfo.txt report file specifies the output of df operating system command.
- **<host name>_freemem.txt**
  The <host name>_freemem.txt report file specifies the output of memory information.
- **<host name>_supp_diag.log**
  The <host name>_supp_diag.log report file specifies the log created for running the supp_diag tool.
- **<host name>_prtconf.txt**
  The <host name>_prtconf.txt report file specifies the output of the prtconf operating system command on Solaris and the AIX computers.
- **<host name>_solrev.txt**
  The <host name>_solrev.txt report file specifies the OS version and patches Information on Solaris computers.
- **<host name>_netconf.txt**
  The <host name>_netconf.txt report file specifies the IP configuration Information on AIX computers.

Verify Collected Diagnostic Report

The default CA SDM documentation file\directory list includes the following reports in the CAZ\tar file. Add files or directories that you include in the CAZ\tar file by placing it in the $NX_ROOT\diag\misc_logs directory.

- $NX_ROOT/GENLEVEL or $NX_ROOT/.GENLEVEL
- $NX_ROOT/<COMPUTERNAME>.his
- $NX_ROOT/NX.env
- $NX_ROOT/NX.env.last
- $NX_ROOT/log\
- $NX_ROOT/pdmconf\
- $NX_ROOT/pdmconf\version
- $NX_ROOT/bopcfg\www*.cfg
- $NX_ROOT/site\mods\
- $NX_ROOT/site\ddict.sch
- $NX_ROOT/site\eh\
- $NX_ROOT/bopcfg\www\CATALINA_BASE\logs\
- $NX_ROOT/bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF\web.xml
- $NX_ROOT/bopcfg\www\CATALINA_BASE\webapps*.xml

Gather Database Server Environment Details

You can gather details about your database server to help identify performance problems in CA SDM.

Follow these steps:

1. Determine the location of your database server, such as local or remote.
2. Determine the DBMS version, Operating System version, and patch level.
3. Complete the appropriate steps for your database type:
   - Execute the following queries for SQL Server and note the results:
     ```
     select @@version
     SELECT SERVERPROPERTY('productversion'), SERVERPROPERTY ('productlevel')
     ```
   - Execute the following query for Oracle:
     ```
     select * from v$version where banner like 'Oracle%';
     ```
4. Confirm the version of the database client that you installed on the application server.
5. If available, gather information about Environmental Data, such as the Operating System and other databases.
6. Gather network topology and topology information or other products that you integrate with CA SDM.
   For example, locate information about the products from available PDF files or diagrams.

Collect Performance Data from the CA SDM Servers

You can collect resource usage data on each SDM server in a multiple server configuration to determine the problems. The collected data sets are used to analyze and troubleshoot any performance or memory-related issues in the CA SDM servers. The interval logging utility allows you to start or stop the diagnostic data collection using the CA SDM web interface. Share the collected data set with CA Support Online. CA Support can help identify and resolve the CA SDM server problems.

Interval logging utility runs on pdm_intrvlog_nxd daemon that collects log data from a server. The daemon automatically starts when a Service Desk server start.

- To stop the interval logging daemon manually, remove the entry of interval logging utility from pdm_startup.
- To start the interval logging daemon manually, double-click the pdm_intrvlog_nxd.bat file in the $NX_ROOT/bin folder.

Collect Usage Data using Interval Logging

To collect the resource usage data on the CA SDM servers, configure the server for interval logging. You can select the type of data you want to collect from each server. You can change the logging options for the servers at any time.

For example, if you select only the CPU usage option, the utility collects only CPU usage data on the server.

Follow these steps:

1. Log in to CA SDM.
2. Select Systems, Interval Logging under the Administration tab.
   The Interval Logging Configurations List opens.
3. Click Create New to add an interval log configuration.
4. Complete the following fields:
   - **Server Name**
     The Server Name field specifies the server for which you want to collect the log data. Clicking Search displays the list of servers you can configure for the interval logging.
   - **Recurrence Interval**
     The Recurrence Interval field specifies the time interval during which the log data is collected for the server. For example, if the recurrence interval is set to 3 minutes, the log is collected for every 3 minutes.
     - **Default: 3 minutes**
     - **Minimum: 2 minutes**
   - **Enabled**
     The Enabled field indicates whether the interval logging is enabled or disabled for the server. If enabled, interval log is collected for the server.

     NOTE
     To stop the interval logging before the scheduled end date, change the Enabled status to NO.
   - **Record Status**
     The Record Status field indicates whether the interval log configuration is active or inactive.
   - **Scheduled Start Date**
     The Scheduled Start Date field specifies the start date and time for collecting the log data. If a start date is not provided, interval logging starts immediately or whenever you activate and enable the configuration.
   - **Scheduled End Date**
     The Scheduled End Data field specifies the end date and time for collecting the log data. If you do not enter an end date, interval logging runs until you inactivate or disable the configuration.
5. Select the appropriate log option that you want to collect from the server.
   For example, select CPU Usage if you want to capture the log for CPU usage data.
   - **CPU Usage**
     The CPU Usage log option collects CPU usage statistics for the server. This log option executes "pslist - x" on Windows and “ps” on UNIX. Depending on your configuration type, you can collect the diagnostic data on the following servers:
     - Conventional: Primary server.
     - Advanced Availability: All servers.
   - **Memory Usage**
     The Memory Usage log option collects memory usage data for the server. This log option executes the "pslist - m" command on Windows and the “ps” command on UNIX. Depending on your configuration type, you can collect the diagnostic data on the following servers:
     - Conventional: Primary server.
     - Advanced Availability: All servers.
   - **Network Status**
     The Network Status log option collects information on all active connections. This log option collects network statistics by executing the "netstat /b" command or the "netstat /a" command. Depending on your configuration type, you can collect the diagnostic data on the following servers:
     - Conventional: Primary server.
     - Advanced Availability: All servers.
   - **Task List**

The Task List log option collects application and services information for all tasks running on the server. Depending on your configuration type, you can collect the diagnostic data on the following servers:
  - Conventional: Primary server.
  - Advanced Availability: All servers.
- **Web Session Counts**
  The Web Session Counts log option collects CA SDM sessions and user statistics for the web engine processes. This log option executes the "pdm_webstat" command. You can collect the data for any CA SDM servers.
- **Server Status**
  The Server Status log option collects information about all the CA SDM daemons or processes on the server. This log option executes the "pdm_status" command. You can collect the data for any CA SDM servers.
- **DB Report**
  The DB Report log option collects information of database record types by executing the db_report command. You can collect the data for any CA SDM servers.
- **Virtual DB Info**
  The Virtual DB Info log option collects information that is related to the queued database requests. This log option executes the "pdm_vdbinfo" command. You can collect the data for any CA SDM servers.
- **List Server Connections**
  The List Server Connections log option collects information on active connections for the server by executing the pdm_listconn command. You can collect the data for any CA SDM servers.
- **List Slump Processes**
  The List Slump Processes log option collects information about slump connections and processes. This log option executes the "slstat" command. You can collect the data for any CA SDM servers.

6. Click Save.
   The server is configured for the interval logging.
7. (Optional) Repeat Steps 1-5 to create more interval log configurations.
8. Go to the $NX_ROOT\log directory and view the generated log files. The output files that are generated depend on the Interval Logging options you select, where files include:
   - cpu_usage_hostname
   - db_report_hostname
   - memory_usage_hostname
   - netstat_hostname
   - pdm_listconn_hostname
   - pdm_vdbinfo_hostname
   - server_status_hostname
   - session_counts_hostname
   - tasklist_hostname

     WARNING
     The maximum limit of the output file size defaults to 30 KB. You can modify the file size by changing the @NX_LOGFILE_LIMIT value in NX_env file. If the generated output file exceeds the maximum file size limit, a new file is created. The new file name is appended with the suffix of 1. Files that are generated later have the suffix of the number incrementally.

     WARNING
     You can share the collected log data with CA Support to help identify performance problems in your SDM installation.

Review General Tuning Recommendations

As a best practice, we recommend you to monitor the key performance indicators and the resource consumption regularly. Also, ensure that routine maintenance is applied to identify small problems before they become serious.

If you suspect an issue or users complain about slow performance, open a CA Support Online issue. Include information about the problem. For example, send the Interval Logging data, CA SDM Diagnostic Tool Report, and CA SDM standard logs. Send the information to CA Support from each CA SDM server.

The following list describes common signs of performance problems:

- The long messages in the CA SDM standard logs (stdlog.x files)
- Search for the messages in the stdlogs that state "The following query took #### milliseconds...".
- Queuing on database agents in pdm_vdbinfo output
- Search for *Queued Requests(#)* in the pdm_vdbinfo output. You can execute this command at the OS prompt manually. The Interval Logging script also executes this command.
- A large number of connected users to each web engine. For example, more than 200 users.

  NOTE

  To display the number of concurrent users per web engine, execute pdm_webstat. You can execute this command manually from the OS command prompt. The Interval Logging script also executes this command.
- User complaints
- When CA SDM is up and running, it is recommended to check the process memory. For optimal performance, we recommend the following method:
  a. Set a notification when the process memory exceeds 1.25 GB and begin a check on the processes that are running.
  b. Set a warning notification when the process memory exceeds 1.5 GB and take corrective actions to check the memory usage.

How to Monitor LDAP Using Trace Logging

This article contains the following topics:

To turn on trace logging to monitor LDAP use within CA SDM, use the pdm_logstat utility.

The pdm_logstat command has the following syntax:

```
pdm_logstat -f ldap_virtdb.c 1000
```

The following stdlog messages help you understand the status of the connection process.

Determine if ldap_virtdb Process Has Started

The first line to look for when analyzing stdlogs for LDAP messages is the startup of the ldap_virtdb process. The LDAP awareness for CA SDM begins only when this process starts.

NOTE
Even if LDAP integration options are not installed or set up, this process still runs.

```
06/03 17:00:18.27 cpasd1   bopLDAP       1964 SIGNIFICANT  ldap_virtdb.c
680 STARTUP of LDAP_virtdb
```

Determine if All Required Options are Installed

If any of the required LDAP options have not been defined, the stdlog shows the missing options. The following example shows the output:

```
06/03 17:00:18.72 cpasd1   bopLDAP  1964 SEVERE_ERROR ldap_virtdb.c  1023 LDAP Server port id missing
06/03 17:00:18.78 cpasd1   bopLDAP  1964 SEVERE_ERROR ldap_virtdb.c  1023 LDAP Server distinguished name
 missing
```

```
06/03 17:00:18.78 cpasd1   bopLDAP  1964 SEVERE_ERROR ldap_virtdb.c  1023 LDAP Server distinguished name
 password missing
```

Determine if the LDAP Connection is Successful

You can identify whether the LDAP connection is successful by looking at the entries in the stdlog. The entries indicate that a connection is successfully established with the LDAP server. The following example shows the output:

```
06/05 12:35:10.41 cpasd1   bopLDAP  1912 SIGNIFICANT  ldap_virtdb.c  958 LDAP_SRVR connecting to
 host(Francisco.us.danconia.net) port(389)
06/05 12:35:11.01 frisco   bopLDAP  1912 SIGNIFICANT  ldap_virtdb.c 1002 LDAP_SRVR binding with
 username(simon)
```

Determine if the LDAP Connection is not Available

If you lose the LDAP server connection, the *LDAP Entries*, *Merge LDAP*, or any other LDAP functionality is disconnected and returns no results. In such circumstances, the stdlog displays messages as the following:

```
06/03 17:00:32.25 cpasd1   bopLDAP  1964 SIGNIFICANT  ldap_virtdb.c    219 LDAP server not available;
 'register_producer' not processed
06/05 10:52:57.63 cpasd1   bopLDAP  1896 SIGNIFICANT  ldap_virtdb.c    219 LDAP server not available;
 'select_full' not processed
06/05 10:52:57.66 cpasd1   web:local 1868 ERROR   sel_data_cache.  611 Error in ldap Select_Cache method
 got_initial_count: LDAP server not available; 'select_full' not processed
06/05 10:52:57.66 cpasd1   bopLDAP  1896 SIGNIFICANT  ldap_virtdb.c    219 LDAP server not available;
 'select_cancel' not processed
```

Determine Actual Filter Used

CA SDM fetches records from the LDAP Directory as per the search base, user search criteria, and the Options Manager filter definition. To determine the actual filter that the search request generates, look for the following message:

```
06/24 14:18:28.32 mcxxx04- bopLDAP    3844 TRACE   ldap_virtdb.c    853 Starting select full:
 base=DC=kirklandsd,DC=ca,DC=com; filter=(&(objectCategory=person)(|(sn=Jones)(pzLastName=Jones)));
 attributes=(uid,sAMAccountName,pzUserName,distinguishedName)
```

Determine Attributes Fetched

CA SDM fetches records from the LDAP Directory according to the search base and the Options Manager filter definition. The SEARCH_BASE and attribute mapping as defined in ldap.maj and ldap_group.maj is checked for accuracy by the following message:

```
06/24 14:18:28.39 mcxxx04- bopLDAP       3844 TRACE     ldap_virtdb.c    766 Starting select short:
 base=CN=John D. Jones,CN=Users,DC=kirklandsd,DC=ca,DC=com; filter=(&(objectCategory=person));
 attributes=(modifyTimestamp,sn,pzLastName,givenName,pzFirstName,initials,pzMiddleName,uid,sAMAccountName,pzUserName,te
```

Determine Which LDAP Data is Available and Not Available

```
06/24 14:18:28.41 mclda04- bopLDAP       3844 TRACE   ldap_virtdb.c   1396 Value not available for
 'modifyTimestamp'
06/24 14:18:28.41 mclda04- bopLDAP       3844 TRACE   ldap_virtdb.c   1396 Value not available for
 'telephoneNumber,pzWorkPhoneNumber'
```

When CA SDM maps successfully to the LDAP Object ID, the attributes are retrieved for each entry. These attributes are defined in $NX_ROOT/bopcfg/majic/ldap.maj. If the mapping is unsuccessful, a message is logged to indicate that an attribute is not defined.

Tomcat Logging

Tomcat Logging for CA Service Desk Manager.

CA SDM uses a separate **log4j.properties** file for its web components such as Servlets and PDM_RPC. Support Automation, CMDB Visualizer, and REST also have a separate l**og4j.properties** file. Starting or stopping the Tomcat logging does not require you to recycle the Tomcat daemons.

The following list describes how the **log4j.properties** and **log4j2.properties** file for CA SDM components differ:

- CA SDM monitors the **log4j2.properties** in the NX_ROOT\site\cfg and in the NX_ROOT\bopcfg\www \CATALINA_BASE\webapps\CAisd\WEB-INF for changes.
- REST monitors the rest.log4j2.properties in the NX_ROOT\site\cfg.

Use the pdm_log4j_config utility for changing variables in the log4j2.properties, logback.properties, and rest.log42j.properties files. You cannot use the pdm_log4j_config utility to modify the polling interval. CA SDM checks the log4j2.properties file for any changes periodically using the monitorInterval attribute.

Servlet Defaults

The following servlets log Info level messages to the **jsrvr.log** file by default:

- **PDMweb**: Log entry from operations on the user interface.
- **UploadServlet**: Log entry generated when you attach files to a ticket.
- **PDMExport**: Log entry generated when you click Export on list forms.
- **PDMReport**: Log entry generated when you click the Report menu on list forms.
- **Page_cache**: Log entry that is generated from operations on the user interface.

Enable Additional Logging for Visualizer

To enable additional logging for Visualizer, follow these steps:

1. Increase the log level for visualizer_slump.log and visualizer_bop.log under the NX_ROOT\bopcfg\www \CATALINA_BASE_VIZ\logs directory. directory.
   a. Edit the NX_ROOT\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\WEB-INF\classes \log4j2.properties file.
   b. For rootLogger.level, change from INFO to DEBUG and for logger.bop_logging.level, change from INFO to DEBUG
      NOTE
      Example: rootLogger.level=DEBUG logger.bop_logging.level=DEBUG
   c. You can also increase the log level for the slump package and bop package by changing the level to debug for logger.slump.name and logger.bop.name packages.
      NOTE
      Example: logger.slump.level=DEBUG logger.bop.level = INFO
2. Increase the log level for VisualizerMS_debug.log under NX_ROOT\log directory
   a. Edit the `NX_ROOT\bopcfg\www\CATALINA_BASE_VIZ\webapps\CMDBVisualizer\WEB-INF\classes \logback.properties` file.
   b. For **rootLogger.level**, change from INFO to DEBUG. For example:

      ```
      rootLogger.level=DEBUG
      ```

   c. Save the changes and restart the Visualizer tomcat using following commands:

      ```
      pdm_tomcat_nxd -c stop -t VIZ
      pdm_tomcat_nxd -c start -t V
      ```

How to Hide Apache Tomcat Version Number from Error Pages in CA Service Desk Manager?

Problem: Due to security issues, there is a need to hide the Apache Tomcat version when the HTTP 404 error page appears.
Solution: To hide Apache Tomcat Version Number from Error Pages in CA Service Desk Manager, perform the following steps:

1. Navigate to where you have installed the CA SDM Server and locate the Program Files (x86)\CA\SC\tomcat\8.5.6\lib directory.
2. Create the following path under the lib subdirectory - org\apache\catalina\util (Example: C:\Program Files (x86)\CA\SC\tomcat\8.5.6\lib\org\apache\catalina\util).
3. Create a file with the name '**ServerInfo.properties**' in the UTIL folder.
4. Edit the file and type the message you want to appear. For example: *server.info=Apache Tomcat*.
5. Save the changes.
6. Restart the CA Service Desk Manager service.
7. Delete the web browser cache.

Now when an Apache Tomcat page is viewed, the version information is replaced with the text entered in the **ServerInfo.properties** file in Step 3.

REST Logging

REST uses the pdm_rest_util.jar and rest-core.jar packages, which contain log4j logging support. These components do not write any messages to the standard logs (stdlog.*), but through the log4j component. By default, these packages log INFO, ERROR, and WARN messages. Because the Java packages use the same log4j configuration properties file, each logs messages to the same output file. You view the rest.log4j.properties file in the $NX_ROOT\site\cfg\ directory.

To increase the log level to trace or debug, update the following section in the *rest.log4j.properties* file:

```
log4j.rootCategory=debug, jrestlog
```

For CA Service Management 17.3.0.16 and above versions, increase the log level to trace or debug, update the following section in the *rest.log4j2.properties* file:

```
rootLogger.level=DEBUG
```

Locate the output file, as defined in the configuration properties file in the following directory:

```
$NX_ROOT\log\jrest.log
```

Enable CXF Logging

CA SDM disables CXF logging by default because it can affect performance on a production environment. If your environment requires logging for debugging purposes, the administrator can modify the beans.xml file to enable CXF logging.

Follow these steps:

1. Locate the beans.xml file in the following directory:
   ```
   NX_ROOT\bopcfg\www\CATALINA_BASE_REST\webapps\caisd-rest\WEB-INF
   ```
2. Locate the following section of the XML file:
   ```
   <cxf:bus>
       <cxf:features>
       </cxf:features>
   ```

```
</cxf:bus>
```

3. Add <cxf:logging/> to the section, as shown in the following example:

```
<cxf:bus>
    <cxf:features>
       <cxf:logging/>
    </cxf:features>
</cxf:bus>
```

4. Save the XML file.

Troubleshooting LDAP Configuration with CA SDM

This article contains the following topics:

The primary consideration when troubleshooting communications with an LDAP server is that seldom are any two LDAP implementations identical. The CA SDM utilities can verify that LDAP integration is working correctly.

> **NOTE**
>
> CA SDM is preconfigured for integration with Microsoft Active Directory, eTrust, and iPlanet only. Integrating with other LDAP servers often requires changes and accommodations on both sides.
>
> To monitor LDAP using the pdm_logstat utility, see How to Monitor LDAP Using Trace Logging.

Show Status of Daemons or Processes

The ldap_virtdb process manages interactions between CA SDM and the LDAP virtual database.

To show the status of all CA SDM daemons (UNIX) or processes

1. Execute pdm_status at the command line with no parameters:

   ```
   pdm_status
   ```

   The pdm_status command shows the status of all CA SDM daemons (UNIX) or processes (Windows). The following output is an example:

   ```
   DAEMON                          STATUS      HOST      PID   SLUMP CONNECT TIME
   ------------------------------------------------------------------------------
   Agent antfarm                   Running     antfarm   455   Tue Feb 17 17:55:12
   Ddict_rd             (ddictrd)  Completed   antfarm
   Data Dictionary   (ddictbuild)  Completed   antfarm
   ...
   User Validation      (boplgin)  Running     antfarm   456   Tue Feb 17 17:55:21
   ```

2. Examine the command output for the status of the ldap_virtdb process.

slstat Command

Run the following command without parameters for verifying whether the bopLDAP is connected:

```
slstat
```

To see the status of bopLDAP, examine the command output.

NX.env File

To verify that the basic LDAP options are correctly installed, review the $NX_ROOT/NX.env file.

For example, based on the LDAP options that are installed, the NX.env file may contain the following parameters:

```
@NX_LDAP_DN=<distinguished_name>
```

```
@NX_LDAP_ENABLE=Yes
@NX_LDAP_ENABLE_AUTO=Yes
@NX_LDAP_HOST=myserver
@NX_LDAP_PORT=389
@NX_LDAP_PWD=OBUNQXo7CmgbThZlCiMKIwJlA3UXdVNAOjUpHjstfDt2LBIDPgwtWA==
@NX_LDAP_SEARCH_BASE=dc=mycontroller, dc=xyz, dc=com
@NX_LDAP_SERVICE_TYPE=Active Directory
@NX_LDAP_SYNC_ON_NULL=Yes
@NX_LDAP_USER_OBJECT_CLASS=person
```

WARNING
The Sun Java System Directory Server and Novell directory servers does not support paged searching. Hence, LDAP search, import, and sync are limited to the value of NX_LDAP_MAX_FETCH records per invocation. The default value is 100. To specify the maximum number of LDAP records, add the NX_LDAP_MAX_FETCH to your NX.env file. You can set NX_LDAP_MAX_FETCH to any value less than the value of LDAP_SIZELIMIT_EXCEEDED or LDAP_ADMINLIMIT_EXCEEDED on your LDAP server.

Troubleshooting for Email-Related Issues

Frequently, end users tend to think of mail functionality within Service Desk as one significant operation. Mail functionality in Service Desk comprises of two distinct and separate entities, Maileater and outgoing mail notification functionalities. The email-related issues, can be either with the incoming Maileater functionality or the outgoing mail notification functionality.

Troubleshooting Email Related Issues

The following scenarios are specific to concerns with the email notifications.

- Are your end-users not receiving specific notifications?

Find out if anyone is receiving notifications at all, even test with a simple manual notification to yourself on a support ticket. Investigate further if some notifications are getting through and under which circumstances the notifications are affected.

- Is the incoming mail being processed, but the response sent out is incorrect?

Suppose that the incoming message is processed, but the end users are not receiving the resultant message. In that case, the problem is with mail notifications and not Maileater.

- Is the received mail formatted incorrectly?

This issue requires some finessing of the message templates to see what content appears to end users. Also, find out if other external components, such as a mail filter, may influence the message by disabling links or image content.

The following scenarios are specific to concerns with the Maileater.

- Are incoming mails not being processed correctly?

Find out how the mail is being processed, if any are processing correctly. If some are processed correctly in the same mailbox, look for patterns. There may be an invalid rule that the message is triggering.

- Is the assigned mailbox piling up in its Inbox with no processing taking place?

There can be errors connecting to the mailbox in the maileater_nxd.log in the log directory of SDM Server.

- Do you need the mail to be processed in a specific manner?

Review your Maileater rules to see what rules should be present and which rules should take the precedence. Remember that Maileater always operates on the first rule that matches the mail message. You cannot apply multiple rules across the given mail message during processing.

Troubleshooting Maileater Connectivity with Office 365 Server

If the Maileater process is experiencing issues with connecting with Office 365 server, replace the certificates with the latest in CA Service Desk Manager.

Follow these steps:

1. Download and save the certificate in the *\certificates* folder.
   For example, save the certificate in *C:\certificates\O365Cert.cer*

   NOTE

   DigiCert Global Root CA has signed the Office 365 certificate, as of December 2016. For more information about this certificate, see DigiCert Global Root CA. If outlook.com gets signed by a different root certificate in the future, obtain the latest root certificate from Microsoft or the vendor providing the certificate.
2. Log in to CA Service Desk Manager and navigate to **Administration**, **Email**, **Mailboxes.** Revert to the settings before the issue occurred.
3. Update the latest certificate details:
   - Navigate to **Administration**, **Email**, **Mailboxes, Selected Mailbox** and define the path to the latest certificate.
     For example, define the path as: *C:\certificates\O365Cert.cer*
   - Save the configuration.
4. In the CA Service Desk Manager server, kill the *pdm_maileater_nxd* process, if necessary.

Outlook 365 Configuration Mail Fails SendAsDenied

When binding Service Desk Manager to an Outlook 365 mail server, the following error may be seen:

```
DEBUG [mailNxdActorSystem-akka.actor.default-dispatcher-7]
c.c.S.m.c.JavaMailSMTPClient - com.sun.mail.smtp.SMTPSendFailedException: 554 5.2.0
STOREDRV.Submission.Exception:SendAsDeniedException.MapiExceptionSendAsDenied; Failed to
process message due to a permanent exception with message Cannot submit message.
```

SendAsDeniedException occurs in Office365 since July of 2019 when your From Address is different to your User ID, unless the administrator manually adds that address to your aliases. To resolve the issue, set the From Address and User ID to the same values.

Maileater Certificate Errors With Office 365 Appearing Without Warning

Maileater has been connecting to Office 365/Outlook.Com for some time now, but as of Jul or Aug 2020, experiencing errors in the maileater_nxd.log.

```
2020-09-01 13:01:01:883 ERROR  [ForkJoinPool-1-
worker-1] c.c.S.m.c.JavaMailIMAPClient - Failed to connect to the Store.
javax.mail.MessagingException: sun.security.validator.ValidatorException: PKIX path building fai
```

Also seeing:

```
2020-09-03 09:20:07:638 ERROR  [ForkJoinPool-1-
worker-3] c.c.S.m.c.JavaMailIMAPClient - Failed to make connection with STARTTLS to server outlo
2020-09-03 09:20:08:393 ERROR  [ForkJoinPool-1-
worker-3] c.c.S.m.c.JavaMailIMAPClient - Failed to connect to the Store.
javax.mail.MessagingException: PKIX path building failed: sun.security.provider.certpath.SunCert
```

On Jul or Aug 2020, a new root certificate was issued for SSL connections to Microsoft using Office 365 or Outlook.Com. Any existing certificates are no longer valid and must be replaced.

On or around Jan 2021, Microsoft started using a new vendor for their Root CA needs. The existing Root CA must be replaced in SDM.

The recommended approach to obtain the root CA is to use OpenSSL. OpenSSL does not need to be installed on the same server as Service Desk. One can install OpenSSL on any computer with internet access to access the Outlook Mail server on the Internet using OpenSSL and acquire the Root CA.

With OpenSSL, one can run a specific command that accesses the given Office 365 server being used for IMAP connections. You can look up the necessary Root CA Certificate specification which can change per Microsoft specifications.

Download OpenSSL from the following location:

https://sourceforge.net/projects/openssl/

Extract the "openssl-1.0.2j-fips-x86_64.zip" file and then locate/run the following command in the command prompt:

```
openssl s_client -starttls imap -connect outlook.office365.com:143 -showcerts -debug
```

Another variation that you can try when you are using Direct SSL (port 993) or for POP (port 995):

```
openssl s_client -connect outlook.office365.com:993 -showcerts -debug
openssl s_client -connect outlook.office365.com:995 -showcerts -debug
```

> **NOTE**
> The same Root CA should be generated regardless of which of the above three OpenSSL commands are tried here. They are supplied here to use OpenSSL and obtain the Root CA block. That it is the same Root CA when connecting using POP (port 995), IMAP/StartTLS (port 143) or IMAP/Direct SSL (Port 993).

openssl.exe is located in the **openssl-1.0.2j-fips-x86_64\OpenSSL\bin** location (where you would have unzipped the **openssl-1.0.2j-fips-x86_64.zip** file).

Look for a text line that reads "**Root CA**". Most likely, you can see "DigiCert Global Root CA".

This line is where you can locate the root certificate which you can then copy into a cer file.

Copy the content between "-----BEGIN CERTIFICATE-----" and "-----END CERTIFICATE-----". Make sure to include the "-----BEGIN CERTIFICATE-----" and "-----END CERTIFICATE-----" header/footer and include a single carriage return at the end of the file. You are writing/saving the "cer" file in Notepad, that is "outlook_root.cer."

Sample certificate file can be found here.

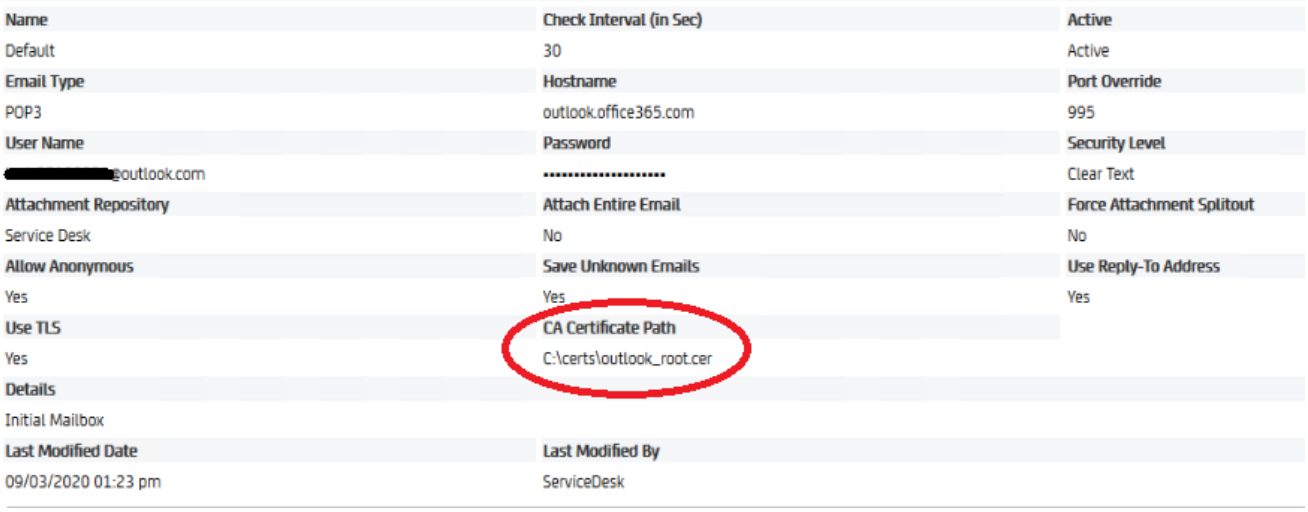
You must follow this approach as Microsoft may change their certificate requirements at will. You can see a snapshot into their latest certificate file usage.

NOTE
A previous method to download the Root Certificate directly from GlobalSign is no longer valid as Microsoft has moved to using DigiCert for their Root CA solution.

To install the new certificate:

1. Create a "outlook_root.cer" file in Notepad and paste the contents of the given certificate into the file, then save the file to a given location on your SDM Server. In this case, the folder is "C:\certs."

   NOTE
   Ensure that sure you are using a new certificate file name. Do not use an existing certificate file as the keystore rejects the updated certificate due to a filename conflict.
2. Go to Maileater settings. Log in to SDM as an Administrative user. Navigate to the Administration tab, then select Email -> Mailboxes, and select the specific Mailbox that is connecting to Outlook/Office 365.
3. Edit the Mailbox settings. Under "CA Certificate Path" change the entry to this certificate file and path:



4. Save this Mailbox setting. You must also update the Mail Notifications in Options Manager with the location of the new certificate file when Office 365 is also used here - **option: mail_ca_cert_path.**
5. Recycle SDM Services to have the changes take effect.
   Alternatively, instead of recycling SDM, you may also try bouncing Maileater and mail processes by running:

```
pdm_bounce maileater_nxd
pdm_bounce mail_nxd
```

You may also copy this certificate file to any other locations within your SDM environment, such as a Standby or App Server for Advance Availability.

In addition, **make sure your Outlook/Office 365 hostname and port configurations are correct.** There have been some reports of accesses to "outlook.com" or imap-mail.outlook.com" being affected. The specific instructions to verify that your IMAP/POP/SMTP Server configurations are available here:

Connecting Maileater to Office 365 Mail

Additional Information:

The rootCA certificate file attached to this tech doc previously worked for POP and IMAP protocols and the outgoing mail notifications with Office 365. However, the certificate was issued by GlobalSign while Microsoft had moved to using DigiCert since at least Jan 2021.

Make sure that you are using a Root CA file. A way to confirm that you are importing a root certificate file is to use the keytool.exe command (supplied with all JRE implementations):

An example command that you can run:

```
"C:\Program Files (x86)\Java\jre1.8.0_261\bin\keytool.exe" -printcert -v -file test-new.cer
```

The "test-new.cer" file is a text-based file that contains a certificate block. Running keytool on the root certificate results in text similar to this output:

```
Owner: CN=DigiCert Cloud Services CA-1, O=DigiCert Inc, C=US
Issuer: CN=DigiCert Global Root CA, OU=www.digicert.com, O=DigiCert Inc, C=US
```

Do not place multiple certificate entries in the same certificate file, including appending the certificate content into an existing certificate file. To verify that the certificate is present:

- Go to a command prompt on a running SDM Server where you have copied the certificate file.
- Run the `nxcd bin` command to access the bin directory of SDM install directory.
- Run the `pdm_perl pdm_keystore_mgr.pl -list -v` command.

You should see a result such as:

```
Alias name: outlook_root.cer
Creation date: Sep 3, 2020
Entry type: trustedCertEntry

Owner: CN=GlobalSign Root CA, OU=Root CA, O=GlobalSign nv-sa, C=BE
Issuer: CN=GlobalSign Root CA, OU=Root CA, O=GlobalSign nv-sa, C=BE
Serial number: 40000000001154b5ac394
Valid from: Tue Sep 01 08:00:00 EDT 1998 until: Fri Jan 28 07:00:00 EST 2028
Certificate fingerprints:
         SHA1: B1:BC:96:8B:D4:F4:9D:62:2A:A8:9A:81:F2:15:01:52:A4:1D:82:9C
         SHA256:
 EB:D4:10:40:E4:BB:3E:C7:42:C9:E3:81:D3:1E:F2:A4:1A:48:B6:68:5C:96:E7:CE:F3:C1:DF:6C:D4:33:1C:99
Signature algorithm name: SHA1withRSA
Subject Public Key Algorithm: 2048-bit RSA key
Version: 3
```

If this certificate is not present in the nx.keystore, or any problem with the certificate import attempt, you must import the certificate file manually into the nx.keystore. You can import the certificate file manually in the nx.keystore by following these steps.

Perform these steps in an Admin Command Prompt on the SDM Server.

1. Run the `nxcd bin` command to access the bin directory of SDM install path.
2. Run the `pdm_perl pdm_keystore_mgr.pl -list -v` command.
   This command lists all certificates in the keystore; write down the existing email certificate ALIAS NAME exactly as displayed
3. Run the `pdm_perl pdm_keystore_mgr.pl -delete [enter alias name here no brackets]` command.
4. Run the `pdm_perl pdm_keystore_mgr.pl -import [path and location of newly updated .cer file]`
   Example: pdm_perl pdm_keystore_mgr.pl -import C:\certs\outlook_root.cer
5. Restart **pdm_maileater_nxd** and **pdm_mail_nxd.**

If manually adding the root CA certificate to this nx.keystore is unsuccessful, refer to the following KB article to rebuild the nx.keystore:

https://knowledge.broadcom.com/external/article?articleId=103456

Migration and Upgrade Failing on Standby Server

The pdm_server_control, confirm_db, sql_check_db utilities fail to detect the standby server resulting in the migration and upgrade failure on CA Service Desk Manager.

To ensure successful migration on the standby server, perform the following steps:

1. Log in to the background server URL.
2. Navigate to **Administration**, **System**, **Servers** and, identify and edit the standby server on which the migration failed.
3. Ensure that the standby server status is active.
4. Retry the migration.

CA SDM Performance Problems - Quick Checklist

This article provides a following checklist for quick verification in situations where CA Service Desk Manager is having performance issues.

This article provides the following checklist for quick verification in situations where CA Service Desk Manager is having performance issues. Sometimes there may be a delay in getting response from CA SDM by editing, visualizing, saving, creating information or navigating the tool. This behavior can produce an hourglass running response or a message (such as "Delayed Response from the Server") when performing the operation.

Run Periodic Archive and Purge for the Session Log and Activity Notification Tables

Session log and activity Notification tables can grow very quickly. If they are highly populate, you should not perform the archive and purge in a single operation. It is recommended to schedule short archive periods, say, once in a month. You can set a permanent archive job in Archive and Purge for Auditing, Event_Log, Knowledge Search Log, Notify and Session Log tables. The Attachments table can be put in this maintenance as well. The frequency depends on how fast these tables grow or on the availability to run the archive purge operation (must not be performed during peak and working hours).

Period Maintenance of MDB Tables and Index

If running Microsoft SQL Server, ensure to shrink the transaction log files. For For other types of verification and maintenance to be done on the MDB, you should contact your database administrator.

Avoid Contacts with Duplicate User IDs

The following query can be used to find duplicate User IDs in the ca_contact table:\

```
SELECT userid,
COUNT(userid) AS NumOccurrences
FROM ca_contact
GROUP BY userid
HAVING ( COUNT(userid) > 1 )
```

After you identify the duplicate user IDs, ensure that you do not delete these duplicate IDs as this may cause inconsistencies in your environment. You can deactivate one of the duplicate User IDs. Recommended is that you must change one of the duplicate User IDs to as deactivated User IDs can be found through search.

Avoid Incorrect Email Addresses Set for Users

This results in an overheard for CA SDM processors, if there are many contacts hitting this condition.

Avoid a Notification Method Set without an Email

This results in an overheard for CA SDM processors, if there are many contacts hitting this condition.

CA SDM Monitor Joins

In case, if you recently upgraded your CA SDM environment, ensure to check the Monitor Joins variable. This variable can adversely impact the performance in your environment. Uninstall this option for optimal performance.

The **monitor_joins** variable specifies whether updates to dynamic lists for joined tables are monitored. This option is delivered out of the box as installed. For better performance, dsiable this option through Options Manager (**Administration** tab, **Options Manager**,**Request Manager**) Right-click and select **Edit** or click the option and click the **Edit** button. Click **Deinstall**. Restart the CA SDM services to make the changes effective.

WARNING

- Do not update the **Nx.env** file manually.
- Do not update the **Nx.env_nt.TPL** manually.
- Do not set the value of the option to **NO**.

SREL_BLOCKS_TIMEOUT

Verify if the SREL blocks timeout table is installed (**Administration** tab, **System**, **Current Locks**). If this option is installed, users can experience delay when running basic activities. To resolve this situation, the **SREL_BLOCKS_TIMEOUT** option needs to be installed. This option can be installed by running the following command:

```
pdm_options_mgr -c -s SREL_BLOCKS_TIMEOUT -v 30 -a pdm_option.inst
```

To avoid losing the change, when you run the *pdm_configure* command, run the above command with the '*-t*' flag as follows:

```
pdm_options_mgr -c -s SREL_BLOCKS_TIMEOUT -v 30 -a pdm_option.inst -t
```

NOTE

You must run both **pdm_options_mgr** commands described above for a successful installation of the option. The option is set to "*30*" by default.

NOTE

- For CA SDM Conventional Configuration, the variable must be installed on the primary server machine.
- For CA SDM Advanced Availability (AA), the variable must be installed on the background server.

Maintaining the CA SDM Knowledge Base

Run the *pdm_k_reindex* command periodically to keep the CA SDM knowledge base in good shape.

CA SDM Migration Fails with exitCode:231 Error

CA SDM Migration fails with exitCode:231 error if the registry file does not have an appropriate value. The *install.log* file is updated with the following:

```
 <Date> <Time> DEBUG [stderr( C:\PROGRA~2\CA\SERVIC~1\bin\pdm_perl.exe):Thread-125] [MigrateSDM] Cannot
 Continue Migration Process.
<Date> <Time> INFO [DeployThread: Migrating CA Service Desk Manager] [ActiveProcessHolder] Exit Monitor.
 Process ID (PID): 4776
<Date> <Time> INFO [DeployThread: Migrating CA Service Desk Manager] [ProcessUtilities] Process C:\PROGRA~2\CA
\SERVIC~1\bin\pdm_perl.exe completed with exit code 231
2<Date> <Time> DEBUG [DeployThread: Migrating CA Service Desk Manager] [ProgressPage] ProgressPage - Task
 Changed Migrating CA Service Desk Manager
2<Date> <Time> DEBUG [DeployThread: Migrating CA Service Desk Manager] [ProgressPage] Task State Changed.
 TaskID: task.migrate_sdm,Migrating CA Service Desk Manager, Old State: running, New State:failed
```

Perform the following steps to resolve this error:

1. Open the command prompt and run the following command:
   ```
   regedit
   ```
2. Set the value 0 (Zero) to the **NtfsDisable8dot3NameCreation** registry in *HKLM\SYSTEM\CurrentControlSet\Control \FileSystem*
3. Save it.
4. Retry migrating SDM.

Unable to Install Any Options in CA SDM with Status as 'In Progress'

CA SDM installation shows error while trying to install any options in CA SDM. The status of these options remains in the "In Progress" state. If CA SDM is installed on a Windows environment with Windows 8.3 file notation disabled, various components of CA SDM may fail. CA Service Desk Manager command line utilities such as pdm_load or pdm_options_mgr may also fail to run. This happens when Windows 8.3 file notation is disabled. The Windows 8.3 file notation is a requirement for CA SDM and its components.

Perform the following steps to enable the Windows 8.3 file notation for WIndows 2008 and Windows 2012 Servers respectively:

To enable 8.3 File notation on a Windows 2008 server:

1. Click **Start, Run.**
2. Run the following command:
   ```
   Regedit
   ```
3. Navigate to *HKLM\SYSTEM\CurrentControlSet\Control\FileSystem.*
4. Change the value for **NtfsDisable8dot3NameCreation** registry to "**0**"
5. Restart the server.
6. Verify that the Windows 8.3 notation is available.
   For example: C:\Program Files (x86)\CA>dir /X
   Volume in drive C has no label.
   Volume Serial Number is 24BF-3A18
   Directory of *C:\Program Files (x86)\CA*
   ```
   02/26/2013 04:50 PM <DIR> .
   02/26/2013 04:50 PM <DIR> ..
   11/20/2010 12:01 AM <DIR> DCS
   06/18/2012 03:30 PM <DIR> DIRECT~1 Directory
   08/28/2007 07:11 PM <DIR> ETRUST~1 eTrust Antivirus
   02/26/2013 04:11 PM <DIR> SC
   03/14/2013 01:16 PM <DIR> SERVIC~1 Service Desk Manager
   03/02/2011 11:22 AM <DIR> SHARED~1 SharedComponents
   0 File(s) 0 bytes
   8 Dir(s) 2,206,724,096 bytes free
   ```
7. If you do not see the Windows 8.3 notation available for a directory, you must reinstall CA SDM if this is a new or fresh installation.
   Verify that the *%WINDIR%\paradigm.ini* file includes the Windows 8.3 notation.
   For example:
   NX_ROOT=C:/PROGRA~1/CA/SERVIC~1

NX_LOCAL=C:/PROGRA~1/CA/SERVIC~1

8. Start the CA Service Desk Manager Server service.
9. Verify you can install the options in Options Manager.

To enable 8.3 File notation on a Windows 2012 server

Perform the following steps:

1. Use the *FSUTIL.EXE* tool to query the state of a volume. Enable the Windows 8.3 name creation (default for Windows Server 2012 is set to **Off**).

```
C:\Users\Admin>fsutil 8dot3name set C: 0
Successfully set 8dot3name behavior
```

2. Verify that the setting is now enabled as the following:

```
C:\Users\Admin>fsutil 8dot3name query C:
```

3. The volume state for *Disable8dot3* is 0 (8dot3 name creation is enabled).
4. The registry state of *NtfsDisable8dot3NameCreation* is 2, the default (Volume level setting).
   Based on the above two settings, 8dot3 name creation is enabled.
   If you do not see the Windows 8.3 notation available for a directory, you may need to reinstall CA SDM if this is a new or fresh installation.
5. Restart the server.
6. Verify that the "%WINDIR%\paradigm.ini" file has a 8.3 notation available.
   For example:
   C:\Program Files\CA>dir /X
   Volume in drive C has no label.
   Volume Serial Number is 24BF-3A18
   Directory of C:\Program Files\CA

```
  02/26/2013 04:50 PM <DIR> .
02/26/2013 04:50 PM <DIR> ..
11/20/2010 12:01 AM <DIR> DCS
06/18/2012 03:30 PM <DIR> DIRECT~1 Directory
08/28/2007 07:11 PM <DIR> ETRUST~1 eTrust Antivirus
02/26/2013 04:11 PM <DIR> SC
03/14/2013 01:16 PM <DIR> SERVIC~1 Service Desk Manager
03/02/2011 11:22 AM <DIR> SHARED~1 SharedComponents
0 File(s) 0 bytes
8 Dir(s) 2,206,724,096 bytes free
```

7. Verify the "%WINDIR%\paradigm.ini" file includes the 8.3 notation.
   For example:
   NX_ROOT=C:/PROGRA~1/CA/SERVIC~1
   NX_LOCAL=C:/PROGRA~1/CA/SERVIC~1
8. Start the CA Service Desk Manager Server service.
9. Verify you can install the options in Options Manager

How to Increase Tomcat Max Threads on CA Service Desk Manager

The Max Threads for Tomcat must be increased from inception as it is originally set to a low value (75) by default. The recommended size is 150 to 300 and it is changed in the server.xml file. The instructions in this article applies to both CA SDM 12.9 and CA SDM 14.1 Conventional and Advanced Availability modes.

Perform the following steps to increase the number of Tomcat threads:

1. Take a backup copy of the server.xml file from *NX_ROOT/bopcfg/www/CATALINA_BASE/conf*.
2. Open **server.xml** file from *NX_ROOT/bopcfg/www/CATALINA_BASE/conf* with a text editor.
3. Find the appropriate **connector** section and set maxThreads to 300:

```
<Connector connectionTimeout="20000" port="8080" protocol="HTTP/1.1"
redirectPort="8443"/>
```

4. After increasing the thread count to 300, the changes will be as following:

```
<Connector connectionTimeout="20000" port="8080"
protocol="HTTP/1.1"maxThreads="300" redirectPort="8443"/>
```

5. Restart the Tomcat Server.
6. To stop the Tomcat server, run the following command:

```
pdm_tomcat_nxd -c stop
```

7. To start the Tomcat Server, run the following command:

```
pdm_tomcat_nxd -c start
```

8. To check the status of the Tomcat server, run the following command:

```
pdm_tomcat_nxd -c status
```

How to extend the SOAP Web Services method 'doQuery' limitation of 250 rows?

CA SDM Web Services method *doQuery* only returns 250 rows. This article describes how to extend the 250 row limitation.

To extend you can use a different method using the CA SDM Web Services *doSelect(). The method "doSelect()" will return a "list handle".* The "listHandle" can be passed to another call - "getListValues()". This method takes a start index and end index argument. This retrieves 250 rows at a time. However, it is possible to retrieve the next 250 rows and so on, until the entire list is retrieved.

The examples here shows the XML representation of the SOAP Web Services calls doSelect():

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" x
mlns:ser="http://www.ca.com/UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:doQuery>
<sid>163624873</sid>
<objectType>cnt</objectType>
<whereClause>last_name IS NOT NULL</whereClause>
</ser:doQuery>
</soapenv:Body>
</soapenv:Envelope>
```

It produces a very large list of 13798 rows:

```
<soapenv:Envelope xmlns:soapenv=http://schemas.xmlsoap.org/soap/envelope/
xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance">
<soapenv:Body>
```

```
<doQueryResponse xmlns="http://www.ca.com/UnicenterServicePlus/ServiceDesk">
<doQueryReturn xmlns="">
<listHandle>-1610501534</listHandle>
<listLength>13798</listLength>
</doQueryReturn>
</doQueryResponse>
</soapenv:Body>
</soapenv:Envelope>
```

You can then call *getListValues* and pass a *startIndex* and *endIndex* to retrieve multiple or group of results at a time. For example:

```
 <soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" x
mlns:ser="http://www.ca.com/UnicenterServicePlus/ServiceDesk">
<soapenv:Header/>
<soapenv:Body>
<ser:getListValues>
<sid>163624873</sid>
<listHandle>-1610501534</listHandle>
<startIndex>251</startIndex>
<endIndex>500</endIndex>
<attributeNames>
<!--1 or more repetitions:-->
<string>last_name</string>
</attributeNames>
</ser:getListValues>
</soapenv:Body>
</soapenv:Envelope>
```

You can continue to retrieve the rest of the list in groups or chunks.

NOTE

It is important to also call the *freeListHandles* when after you have completed extending the rows in order to flush out or remove the list from the memory on the server to avoid memory growth.

How to modify the CA SDM Web Session timeout?

By default, CA SDM has a one (1) hour session timeout value. The session timeout for CA SDM is defined in the *WEB.CFG* file located in *NX_ROOT\BOPCFG\WWW.*

- For **Conventional Configuration**, if you have secondary servers in the CA SDM environment, you must change the values in the *NX_ROOT\BOPCFG\<hostname>WEB1.cfg* file on the CA SDM secondary server(s).
- For **Advanced Availability**, you must apply the changes in *WEB.cfg* on all the server where this file exists.
- To avoid losing changes during pdm_configure, ensure that the same changes are made to the *WEB.CFG.TPL* file located in the *NX_ROOT\BOPCFG\WWW* directory on the CA SDM server(s).

The web session timeout value has the following parameters:

WARNING
We strongly advise to consider the implications of setting the Timeout value, particularly when a longer value is considered. Record locks and performance degradation may occur if a given session is allowed to remain active for an extended period

NOTE
CA SDM Web session timeout notifications are displayed under Windows Action Center also. For more information about setting up notification in a system (pop-up notifications or handled via Windows Action Center), see Windows Notifications.

```
!Timeout:60
```

Timeout: specifies the maximum number of minutes a user's session can be inactive before it is cancelled and all resources are released. After this period has elapsed, the user must start a new session by logging in again.

To modify the Timeout value, perform the following steps:

1. First remove the **'!'** (exclamation mark) as follows:
   ```
   Timeout 60
   ```
2. Modify the value (in minutes) as required.
   For example, to reduce the timeout by half an hour, the value must be:
   ```
   Timeout 30
   ```
   After modifying the timeout value, restart CA SDM services for the changes to take effect.
3. **TimeoutWarning**: 5
   TimeoutWarning specifies the number of minutes prior to session timeout that the web client should pop up a warning alert box. The default value is five minutes; specify **TimeoutWarning=0** to suppress this feature
4. **BrowserNotifications**: true
   Notifications are enabled for all the browsers when this value is set to true. Notifications are enabled for chromium-based browsers and disabled for rest of the browsers when this value is set to false.

   NOTE
   Notifications APIs will work only with HTTPS (SSL enabled) connections.

   Web Notification for session time out is enabled only after allowing browser notification. While logging into to CA Service Desk Manager, a popup is displayed to Allow/Block the notifications on the browser. Click 'Allow' and the notifications are triggered for the Session timeout value configured in the web.cfg file.

   WARNING
   Web Notification are not displayed if a user blocks the browser notification. If a user clicks 'Block' notification when asked for 'Allow/Block' notifications, user is logged out when the timeout value ends or is over.

   NOTE
   Recent versions of Chrome and Edge browsers do not support the Javascript timeout popups. These browsers only provide support for the new Browser Notifications method with only HTTPS. Other upported browsers (other than recent versions of Chrome and Edge) continue to support Javascript timeout popups on HTTP and HTTPS.

   See the table below for the current list of browsers supporting these Browser Notifications:

Web Browser	BrowserNotifications (true)	BrowserNotifications (false)
Microsoft Internet Explorer 11.0.9600 and above	Timeout Popups	Timeout Popups
Microsoft Edge 41.16299 and above on Windows 10	Timeout Notifications if HTTPS is enabled	Timeout Notifications if HTTPS is enabled
Mozilla Firefox ESR 60.3.0 and above	Timeout Notifications if HTTPS is enabled	Timeout Popups
Google Chrome 68.0 and above	Timeout Notifications if HTTPS is enabled	Timeout Notifications if HTTPS is enabled

Apple Safari 12.0.3 and above	Timeout Notifications if HTTPS is enabled	Timeout Popups

CA Service Desk Manager Installation Fails

CA Service Desk Manager install fails at **Integrate the Products** step of the CA Service Management Installation Wizard. This issue is observed while connecting to the Microsoft SQL Server Database.

The installation fails as the Microsoft SQL Native Client layer cannot connect to Microsoft SQL Server after the TLS 1.2 protocol is enabled. The Microsoft SQL Native Client layer does not support the TLS 1.2 restrictions as compared to the Microsoft SQL Server layer. For more information, see Microsoft Documentation.

To resolve this issue, follow the steps in this Microsoft Article. Download and install the Microsoft Native SQL Client fix that is listed in the Clients Component section of this article.

What is the Correct Registry Setting to Enable TLS 1.2 for Microsoft SQL Server Communication?

The correct registry settings are as follows:

```
[HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.2]

[HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.2\Client]
"DisabledByDefault"=dword:00000000 "Enabled"=dword:00000001

[HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.2\Server]
"DisabledByDefault"=dword:00000000 "Enabled"=dword:00000001
```

Delayed CA SDM Server Response After Upgrading to Oracle 12c

Upgrading to Oracle to 12c results in a high CPU load. CA SDM shows delayed server responses due to slow Oracle DB. The new feature in Oracle 12c called **Adaptive Query Optimization** helps in better performance and is enabled by default. Due to the high number of small queries that is parsed by this new feature, it adds overhead with high CPU usage on the Oracle database server.

To disable the **Adaptive Query Optimization** feature, run the following commands:

```
alter system set optimizer_adaptive_features=FALSE scope = both;
alter system set optimizer_adaptive_reporting_only=TRUE scope = both;
```

CA SDM Conventional Configuration - The Secondary Server Tomcat Engine is Not Responding

After restarting the CA Service Desk Conventional Configuration, the Secondary server Tomcat engines stop responding and gives an error message: "*AHD4409 pdmweb is unable to connect with the web engine*". This may happen if the server (domsrvr) is not fully initialized before Tomcat starts.

To resolve this issue, perform the following:

1. Change NX.env and NX.env template file (NX_ROOT\pdmconf\nx.env_nt.tpl) of Primary and Secondary servers from:

```
! NX_SLUMP_FIXED_SOCKETS=1
! NX_SLUMP_SECONDARY_SOCKET=2100
```

To the following:

```
@NX_SLUMP_FIXED_SOCKETS=1
@NX_SLUMP_SECONDARY_SOCKET=2100
```

2. Set a time delay so that the tomcat server will try to connect to slump a little after the primary server settles down. You can achieve this in two ways:

   NOTE

   When a primary server is restarted, it becomes busy and starts a lot of components. So the machine (not slump, but Windows) may not be able to respond quickly to the incoming TCP requests. That is why the Secondary Tomcat server encounters issue where it cannot get a connection successfully established in time.

   a. First, you can perform the following on the secondary server:
      a. Run the following command:

         NOTE

         pdm_tomcat_nxd -c stop
         pdm_tomcat_nxd -c status

      Ensure the status is shown as **stopped**.
      Wait for 30-60 minutes before running the following command:

      ```
      pdm_tomcat_nxd -c start
      ```

   b. A long-term solution would be to update the following 2 files on the primary server:

      ```
      <Service Desk>\pdmconf\pdm_startup
      <Service Desk>\pdmconf\pdm_startup.tpl
      ```

      a. Change the section for TOMCAT in the secondary servers. In the below examples *SERVERSEC01* and *SERVERSEC02* are secondary servers.

         ```
         PDM_TOMCAT_CONTROLLER (_pdm_tomcat_SERVERSEC01,"SERVERSEC01", pdm_tomcat-SERVERSEC01)
         after domsrvr connects + 120
         after _ver_ctl_SERVERSEC01 finishes
         after _msgcat_build_SERVERSEC01 finishes

         PDM_TOMCAT_CONTROLLER (_pdm_tomcat_SERVERSEC02,"SERVERSEC02", pdm_tomcat-SERVERSEC02)
         after domsrvr connects + 120
         after _ver_ctl_SERVERSEC02 finishes
         after _msgcat_build_SERVERSEC02 finishes
         ```

      +120 in the above code implies that delay time is 120 seconds (2 minutes) to startup TOMCAT on secondary servers. You can adjust this value according as per your requirements.

3. Restart the Service Desk Services in the following order:
   a. Start on Primary
   b. Start on Secondary servers
   c. Stop on Secondary servers
   d. Stop on Primary

CA SDM Authentication Issues

The Data Protection Operation for CA SDM SAML Authentication on IIS was Unsuccessful

Symptom:

After enabling the SAML Authentication for CA SDM on IIS, I am unable to launch CA SDM web interface though the user credentials are provided correctly. The below error message is shown:

"The data protection operation was unsuccessful. This may have been caused by not having the user profile loaded for the current thread's user context, which may be the case when the thread is impersonating."

Solution:

This error occurs, if the value of Load User Profile parameter is set to False.

Follow these steps:

1. Open IIS Manager and do the following:
2. Expand the Server where CA Service Desk Manager is installed.
3. Go to the Sites node, right click on CAisd, select Manage Applicaton and click Advanced Settings. Note the Application Pool used by CAisd.
4. Go to the Application Pools node, right click on the CAisd used Application Pool and click Advanced Settings.
5. Go to Process Model settings.
6. Set the Load User Profile value to "True".
7. Click OK.
8. Restart the IIS Service.

SAML login for CA Service Desk Managerfails with 500 error

Symptom: SAML login for ServiceDesk fails with 500 error.

Solution: Perform the following steps to resolve this issue:

1. Login to CA SDM.
2. Stop the CA SDM Services.
3. Go to following location $NX_ROOT/bopcfg/www/CATALINA_BASE/shared/resources/abc.def. Open this file and edit the following line in

   ```
   abc.def=https://xflowpoc.YourCo.lab:8443/CAisd/pdmweb.exe
   ```

   to

   ```
   abc.def=https://xflowpoc.YourCo.lab:8443/CAisd/pdmweb.exe|https://xflowpoc.YourCo.lab:8443/CAisd/
   pdmweb.exe
   ```
4. After the above modification, restart the CA SDM Services.

How to extend or disable the "Delayed Server Response" timeout in CA SDM?

By default a "Delayed Server Response" message is displayed by CA SDM after 20 seconds to inform users about the delay. Recommendation is to remove all issues and roadblocks from the said environment to help reduce the number of times this message is displayed. This document provides a solution to extend the time frame for the popup to appear or to disable it completely so that users will see it less often in environments where delays are to be expected.

This message occurs when it takes longer than the expected 20 seconds for CA SDM to complete a task. An example of delay might be a database query taking longer than expected to return results.

Resolution

Add the value "NX_EXPECTED_WEB_RESPONSE" in the NX.ENV file to either increase the timeout or disable it completely.

NOTE

You must add the variable to the *nx.env* template file so that when pdm_configure is run, the value in the nx.env will not be reset or overwritten.

To increase the threshold to 1 minute perform the following steps:

1. To add the variable to the *nx.env* run the following command:
   ```
   pdm_options_mgr -c -s EXPECTED_WEB_RESPONSE -v 60 -a pdm_option.inst
   ```

2. To add the variable to the *nxenv_nt.tpl* file run the following command:
   ```
   pdm_options_mgr -c -s EXPECTED_WEB_RESPONSE -v 60 -a pdm_option.inst -t
   ```

To disable the threshold perform the following steps:

1. To add the variable to the nx.env run the following command:
   ```
   pdm_options_mgr -c -s EXPECTED_WEB_RESPONSE -v 0 -a pdm_option.inst
   ```

2. To add the variable to the nxenv_nt.tpl file run the following command:
   ```
   pdm_options_mgr -c -s EXPECTED_WEB_RESPONSE -v 0 -a pdm_option.inst -t
   ```

Considerations for CA SDM Conventional and Advanced Availability Architecture

If are using a conventional architecture, or you are using an advanced availability architecture, with a background, standby, and multiple application servers, you need to run both commands (to add the variable to the nx.env and nx.env template files) on all systems in a environment where CA SDM is up and running. After running the commands on all servers, you must recycle the CA Service Desk Manager services as a whole to invoke the changes:

Additional Information

The addition of "NX_EXPECTED_WEB_RESPONSE in NX.ENV variable and associated value must never be used to mask a real performance problem in a CA SDM environment. Reviewed your system environment and architecture properly to ensure that everything else is functioning as expected and that there is nothing else that can be done to improve the performance of the CA SDM environment (especially with regards to network connection speeds). This option should only be used if there is a true environmental limitation that cannot be overcome.

Things to consider and perform while reviewing the performance of an environment:

- Run regular archive/purge rules on items such as tickets, notification log, and so on.
- Avoid allowing users have the ability to search on fields such as description, activity logs, or any other field where there is a large amount of text to parse.
- Perform standard maintenance on all database servers at scheduled intervals.

How to mask out the "None" Impact value?

When trying to mask out the ***None*** Impact value for end users while creating an incident, the data partitioning on the Impact table becomes ineffective. Rendering the entry as ***Inactive*** is unsupported. This article describes how you can mask out the ***None*** impact value from the end-user display.

Resolution:

The solution is to create a data partition and a web page customization: Perform the following steps:

1. Create the following Data Partition Constraint (removing any existing constraints in place for ***View*** on a table, say, Impact:
   a. Table: Impact
   b. Type: View
   c. Code: enum in (1,2,3,4,5)
2. Locate the *detail_in.htmpl* file (make sure there is a copy in CA SDM directory site/mods) and find the following command:

```
<PDM_MACRO name=dtlDropdown hdr="Impact" attr=impact
 evt="onBlur='manual_modify();detailSyncEditForms(this)'" lookup=no>
```

3. Modify the command in Step 2 as:

```
<PDM_MACRO name=dtlDropdown hdr="Impact" attr=impact
evt="onBlur='manual_modify();detailSyncEditForms(this)'" lookup=no initial="">
```

   You have now masked out the None Impact Value.

Unable to Set a Default Value for a Field in Web Screen Painter

Setting the default value in Web Screen Painter (WSP) for a field on a form does not actually work. While creating a new object you notice that it is not defaulting to the values you specified in WSP. Two such fields where this behavior is seen are, for example, **Impact** and **Priority for Requests**. When the default value is set and a new request is created these values default to **None** instead of the value you defined in the WSP form.

This is expected as some attributes have a default value set on the majic layer. The default attribute defined on the Web Screen Painter macro is only used when the domsrvr/majic layer is defaulting to **NULL** as the domsrvr/majic layer takes precedence over the macro.

Resolution

The workaround is to default these attributes to NULL. Perform the following to achieve this:

1. Find the attribute you are trying to default in the Schema Designer.
   For example, a Request's Impact would be found under **cr** (Request), **impact** (Impact))
2. After selecting the attribute click the **Advanced** tab on the right pane.
3. Under Site-Defined Triggers put: "**ON_NEW DEFAULT "NULL";**" without the external quotes.
4. Repeat steps 1-3 for all attributes that you want to set your own default value for.
5. Click **File, Save,** and **Publish**
6. Stop CA Service Desk Manager, and run the following command:

```
pdm_publish
```

7. Start CA Service Desk Manager.
   The default values that you set are now visible.

"i18nMsg.c Can't Construct Translator" -stdlog Error Message

CA SDM Secondary Server stdlog file displays the following error messages:

```
10/25 17:38:16.16 <server-name> domsrvr:21 7196 ERROR i18nMsg.c 382 Can't Construct Translator
10/25 17:38:22.52 <server-name> domsrvr:21 7196 ERROR i18nMsg.c 490 Unable to open language library
```

The above error messages are displayed if the $NX_ROOT\bin\pdm.dll file is corrupted or missing on the CA SDM Secondary :

Resolution

Perform the following:

- Copy the *pdm.dll* file from a working/active (CA SDM Primary or Secondary) server to the non-working/inactive server. Restarting of CA Service Desk services is not required.

How to Log a Comment on a Closed CA SDM Ticket?

CA SDM does not allow you to log a comment on, or edit a ticket that is already closed. An option in Options Manager "**edit_inactive**" if set to **INSTALLED** prevents users from editing tickets that are in a closed state.

Resolution

To allow users to be able to log a comment or edit a closed ticket, you must ensure that the "**edit_inactive**" option is not installed in Options Manager. Perform the following steps:

1. Log into CA Service Desk Manager as an Administrator. Navigate to the **Administration**, **Options Manager, Request-Change.**
2. Click the "**edit_inactive**" option and select **Edit**. Click "**Deinstall**".
3. Restart CA Service Desk Manger services on all servers
   You are now able to edit inactive tickets in the system.

How to set the Initial status of a Copied Incident / Requests to Open?

When an Incident / Request is Copied, by design, the Initial status for the Copied ticket shows the following behavior:

- If the status is 'Closed', the Copied ticket status is set to 'Open'
- If the status is different from 'Closed', the Copied ticket status is the same as the source ticket.

Perform the following steps:

1. Add the following variable towards the end of the files: *$NX_ROOT\NX.env* and *$NX_ROOT\pdmconf\NX.env_nt.tpl*
   ```
   @NX_RIP_COPY_INITIAL_STATUS=OP
   ```
2. Restart SDM Services.

How can we import a vendor supplied certificate into CA Service Desk Manager's Tomcat?

This article provides a sample list of steps for importing VeriSign certificate into CA Service Desk Manager's tomcat server. The following is an example and may vary slightly depending on the product/certificate purchased from VeriSign.

Perform the following steps depending on the type of VeriSign purchase that you made ("Secure Site", "Secure Site Pro", and so on). When prompted for the certificate type, be sure to request the Microsoft PKCS7 format. The java "keytool" reads this without an issue.

1. Import the certificate received from VeriSign by running the following command:
   ```
   keytool -import -trustcacerts -file c:\newcert.crt -keystore "c:\.keystore" -alias tomcat
   ```

NOTE

the Microsoft PKCS 7 format should have the VeriSign intermediate certificates.

2. To edit the server.xml file in a text editor, navigate to *Service Desk\bopcfg\www\CATALINA_BASE\conf\server.xml*. Locate the SSL connector section and uncomment it as appropriate. Edit as shown in the example below where the location of the ".keystore" file is in "c:\cert\":

```
<!-- Define a SSL Coyote HTTP/1.1 Connector on port 8443 -->
<Connector className="org.apache.coyote.tomcat4.CoyoteConnector"
port="8443" minProcessors="5" maxProcessors="75"
enableLookups="true"
acceptCount="100" debug="0" scheme="https" secure="true"
useURIValidationHack="false" disableUploadTimeout="true">
<Factory className="org.apache.coyote.tomcat4.CoyoteServerSocketFactory"
clientAuth="false" protocol="TLS" keystoreFile="C:\cert\.keystore" keystorePass="changeit" />
</Connector>
```

3. Restart CA Service Desk Manager Tomcat by performing the following steps:

```
pdm_tomcat_nxd -c stop
pdm_tomcat_nxd -c start
```

NOTE

Information: For additional information, see the How to Configure SSL Authentication.

Secondary Tomcat server may not start

The secondary tomcat server URL may not start in certain conditions because Secondary server proctor services are running during the patch installation.

Solution:

1. Stop the Secondary Server proctor services.
2. Stop The Primary Server SDM Services.
3. Remove the older version of tomcat (9.0.71)from the Secondary server \CA\SC\tomcat location.
4. Start the Secondary server Services.
5. Start Primary Server services.

Invalid character is found in the request target

Problem:
The presence of backslash in the URL is resulting in a Bad Request and Tomcat is throwing Bad Request: "Invalid character found in the request target. The valid characters are defined in RFC 7230 and RFC 3986"

Solution:

1. Update the Tomcat Connector with *relaxedPathChars* and *relaxedQueryChars* in the ***$NXROOT\bopcfg\www\CATALINA_BASE\conf\server.xml*** file.
   For instance: Add the following to the HTTP and HTTPS connector tags:

```
relaxedPathChars="[]|{}^&#x5c;&#x60;&quot;&lt;&gt;" relaxedQueryChars="[]|{}^&#x5c;&#x60;&quot;&lt;&gt;"
```

   The BOServlet used for legacy CABI has been removed from the code but the web.xml entry still exists.
   An example of a URL is:

https://helpdeskxyz.abc.com:8443/CAisd/BOServlet?BOPSID=
To overcome this, navigate to *$NXROOT\bopcfg\www\CATALINA_BASE\webapps\CAisd\WEB-INF\web.xml* and remove the servlet and servlet-mapping related to the BOServlet.

2. Repeat Step 1 to update *web.xml.tpl* file in the *NX_ROOT\samples\pdmconf* folder to avoid losing these changes when pdm_configure is run at a later point of time.
3. Stop the Tomcat using pdm_tomcat_nxd -c STOP
4. Wait for two minutes.
5. Start the Tomcat using pdm_tomcat_nxd -c START

Troubleshooting xFlow Interface

Content

Unable to Log in to xFlow Interface

Perform the below steps to resolve this issue:

1. Stop the xFlow Interface services.
2. Stop the CA Service Desk Manager services.
3. Now clear the contents of CasmPlayAPi along with the **playtemp*** folders cache present *under/temp* folder -
   (Windows): Navigate to \Windows\Temp\CasmPlayApi and remove all **playtemp*** instances.
   (Linux): Navigate to /tmp/CasmPlayApi and remove all **playtemp*** instances.
   In CA Service Management 17.4 Windows OS, navigate to \Program Files\CA\xFlow\APPS\temp and remove all **playtemp*** instances.
   In CA Service Management 17.4 LinuxOS, navigate to /opt/CA/xFlow/APPS/temp and remove all **playtemp*** instances.
4. Start the CA SDM Services.
5. Start the xFlow Interface Services.

Unable to load the login screen from the xFlow Interface

1. Log in to CA SDM as an administrator.
2. Restart the CA xFlow Interface service.
3. Restart the *CA Service Management Search Server* service.

Unable to search Users in the xFlow Interface.

1. Ensure that the following services are running:
   a. CA Service Management Search Server
   b. CA Service Desk Manager service
2. Open command prompt on the server where the CA Service Management Search Server is installed.
3. Navigate to where you have installed the search server. For example: *C:\Program Files\CA\SC\CASearchServer \search\bin.*
4. Execute the following command:

```
pdm_es_rebuild_index
```

5. Restart the *CA Service Management Search Server* service.

Broken Images in xFlow Interface

Configure MIME Types to use the xFlow Interface:

1. Start the Internet Information Services (IIS) Manager.

2. From the left navigation pane, select **Sites** and then select xFlow Interface.
3. Double-click **MIME Types.**
4. In the Actions pane, click **Add**.
5. In the **Add MIME Type** dialog box, type ***.svg*** as the filename extension and ***image/svg+xml*** as the MIME type.
6. In the **Add MIME Type** dialog box, type ***.svgz*** as the filename extension and ***image/svg+xml*** as the MIME type.
7. Click **OK**.
8. Execute the following command to reset IIS:

```
iisreset
```

The browser is redirected to CA SDM CGI upon accessing the xFlow Interface.

Remove any HTTP redirect URLs that are configured on the IIS web server:

1. Start the Internet Information Services (IIS) Manager.
2. From the left navigation pane, click **Web Server.**
3. From the right navigation pane, double-click **HTTP Redirect**.
4. Clear the redirect URL.
5. Click **Apply**.

After upgrading to xFlow Interface 17.4, xFlow Interface Custom Changes are Lost

After installing or upgrading to CA Service Management 17.4 the casm.conf.js file overwrites the modifications you made in the previously installed CA Service Management version. So the custom changes you made to xFlow Interface in the previously used CA Service Management version are lost.

You can manually update the casm.conf.js file under the below CA Service Management 17.4 installation location:

<xFLOW_Installed Directory>\APPS\Services\incidentmicroservice-17.0.479\public\conf

For your reference, a back-up of the casm.conf.js file from the previously used CA Service Management solution version is saved under the path, <xFlow_Installed Directory>\REPLACED\xFlow-CUM4-17.4.0.0\APPS\Services \incidentmicroservice-0.1-SNAPSHOT\public\conf

xFlow - Activity Feed: system-description missing

From 17.4 RU5, a new configuration parameter, ***showSystemDescription***, has been introduced in the *casm.conf.do-not-change.js* file to control the display of system descriptions in the All Feed section.

- Default value: **false**

  NOTE
  This is applicable for the descriptions given during status change or transfer.
- When set to **true**, system descriptions are displayed in the feed based on the configurations shown in the table below.

This update improves how user and system descriptions appear in the activity feed.

Configuration value in xFlow :showSystemDescription	User Description	System Description	
false	(non- empty)	(any value)	Only **userDes**
false	(empty or not provided)	(any value)	**systemDescr** only one entry Only **systemD**
true	(non- empty)	(any value)	Both **userDes**

Configuration value in xFlow :showSystemDescription	User Description	System Description	
true	(empty or not provided)	(any value)	**systemDescr** only one entry Only **systemD**

Enable Secure Socket Layer for xFlow Interface

To configure and enable SSL for xFlow Interface and Search Server manually, perform the following steps:

> **NOTE**
> CA Service Management provides the SSL Configurator Utility to enable SSL for CA Service Desk Manager and its components, xFlow Interface/Service Point, and CA Service Catalog. For more information, see SSL Configurator Utility.

Create a Self-Signed Certificate

Perform the following steps:

1. Create a document with names that you intend to use for the java keystore - the **client certificate alias**, the **client certificate request file**, the **server certificate alias**, the **server certificate request file**, the **server certificate name**, the **OpenSSL private key name**, and the **OpenSSL certificate name**. Enter the following values in the document:
   - Java keystore = hostname_keystore.jks
   - Client certificate alias = hostname_clientcert
   - Client certificate request file = hostname_clientcert.crt
   - Server certificate alias = hostname_certname
   - Server certificate request file= hostname_certname.csr
   - Server certificate name = hostname_certname.crt
   - OpenSSL private key name = hostname_SSL.key
   - OpenSSL certificate = hostname_SSL.crt
2. Navigate to *<xFlowInstallation Home Dir>\jre\bin*, launch the command line interface and run the following commands to generate key and certificates:

```
keytool -genkeypair -v -alias hostname_clientcert -dname "CN=<common name>, OU=<Organization Unit>,
 O=<Organization>, L=<Location>, ST=<State>, C=<Country>" -keystore hostname_keystore.jks -keyalg RSA -
keysize 4096 -ext KeyUsage:critical="keyCertSign" -ext BasicConstraints:critical="ca:true" -validity 9999
keytool -exportcert -v -alias hostname_clientcert -file hostname_clientcert.crt -keystore
 hostname_keystore.jks -rfc
keytool -genkeypair -v -alias hostname_certname -dname "CN=<common name>, OU=<Organization Unit>,
 O=<Organization>, L=<Location>, ST=<State>, C=<Country>" -keystore hostname_keystore.jks -keyalg RSA -
keysize 2048 -validity 385
keytool -certreq -v -alias hostname_certname -keystore hostname_keystore.jks -file hostname_certname.csr
keytool -gencert -v -alias hostname_clientcert -keystore hostname_keystore.jks
 -infile hostname_certname.csr -outfile hostname_certname.crt -ext
 KeyUsage:critical="digitalSignature,keyEncipherment" -ext EKU="serverAuth" -ext SAN="DNS:hostname" -rfc
keytool -importcert -v -alias hostname_certnameimport -file hostname_certname.crt -keystore
 hostname_keystore.jks -storetype JKS
keytool -list -v -keystore hostname_keystore.jks
```

   For example:

```
keytool -genkeypair -v -alias <hostname>_clientcert -dname "<hostname>, OU=sdm, O=ca, L=hyd, ST=ts, C=in"
 -keystore <hostname>_keystore.jks -keyalg RSA -keysize 4096 -ext KeyUsage:critical="keyCertSign" -ext
 BasicConstraints:critical="ca:true" -validity 9999
keytool -exportcert -v -alias <hostname>_clientcert -file <hostname>_clientcert.crt -keystore
 <hostname>_keystore.jks -rfc
keytool -genkeypair -v -alias <hostname>_certname -dname "CN=<hostname>, OU=sdm, O=ca, L=hyd, ST=ts, C=in"
 -keystore <hostname>_keystore.jks -keyalg RSA -keysize 2048 -validity 385
keytool -certreq -v -alias <hostname>_certname -keystore <hostname>_keystore.jks -file
 <hostname>_certname.csr
keytool -gencert -v -alias <hostname>_clientcert -keystore <hostname>_keystore.jks
 -infile <hostname>_certname.csr -outfile <hostname>_certname.crt -ext
 KeyUsage:critical="digitalSignature,keyEncipherment" -ext EKU="serverAuth" -ext SAN="DNS:<hostname>" -rfc
keytool -importcert -v -alias <hostname>_certnameimport -file <hostname>_certname.crt -keystore
 <hostname>_keystore.jks -storetype JKS
keytool -list -v -keystore <hostname>_keystore.jks
```

3. Copy the **.crt** and **.jks** files in one location.

   NOTE

   We recommend that you create a folder called **certificates** on the root of your local drive, for example, ***C:\certificates*** on **Windows** *, and* ***/opt/certificates*** on **Linux**. You can now see the files under the following paths:

   Windows

   - *C:\certificates\hostname_keystore.jks*
   - *C:\certificates\hostname_certname.crt*

   Linux

   - /opt/certificates/*hostname_keystore.jks*
   - */opt/certificates/*hostname_certname.crt

   This article here will refer to these paths, and if you decide to use a different path, you need to change the paths as per your local drive.

4. Do the following:
   a. **Windows**
   Navigate to the Command Line Interface, change the directory <xFlowInstallation Home Dir>\APPS\Services\ and run the below commands:

```
echo -Dhttps.port=9444 -Dplay.server.https.keyStore.path=C:\certificates
\hostname_keystore.jks -Dplay.server.https.keyStore.password=(keystore password) >
 incidentmicroservice-17.0.479\INCIDENTMICROSERVICE_config.txt
echo -Dhttps.port=9446 -Dplay.server.https.keyStore.path=C:\certificates
\hostname_keystore.jks -Dplay.server.https.keyStore.password=(keystore password) >
 collabmicroservice-17.0.479\COLLABMICROSERVICE_config.txt


echo -Dhttps.port=9448 -Dplay.server.https.keyStore.path=C:\certificates
\hostname_keystore.jks -Dplay.server.https.keyStore.password=(keystore password) >
 insightmicroservice-17.1.706\INSIGHTMICROSERVICE_config.txt
echo -Dhttps.port=9450 -Dplay.server.https.keyStore.path=C:\certificates
\hostname_keystore.jks -Dplay.server.https.keyStore.password=(keystore password) >
 pushmicroservice-17.0.479\PUSHMICROSERVICE_config.txt
```

```
echo -Dhttps.port=9452 -Dplay.server.https.keyStore.path=C:\certificates
\hostname_keystore.jks -Dplay.server.https.keyStore.password=(keystore password) >
 searchmicroservice-17.0.479\SEARCHMICROSERVICE_config.txt
```

b. **Linux**
Go to Command Line Interface, change the directory *<xFlowInstallation Home Dir>\APPS\Services* and run the below commands:

```
echo -Dhttps.port=9444 -Dplay.server.https.keyStore.path=/opt/certificates/hostname_keystore.jks
 -Dplay.server.https.keyStore.password=(keystore password) > incidentmicroservice-17.0.479/
INCIDENTMICROSERVICE_config.txt
echo -Dhttps.port=9446 -Dplay.server.https.keyStore.path=/opt/certificates/hostname_keystore.jks
 -Dplay.server.https.keyStore.password=(keystore password) > collabmicroservice-17.0.479/
COLLABMICROSERVICE_config.txt
echo -Dhttps.port=9448 -Dplay.server.https.keyStore.path=/opt/certificates/hostname_keystore.jks
 -Dplay.server.https.keyStore.password=(keystore password) > insightmicroservice-17.1.706/
INSIGHTMICROSERVICE_config.txt
echo -Dhttps.port=9450 -Dplay.server.https.keyStore.path=/opt/certificates/hostname_keystore.jks
 -Dplay.server.https.keyStore.password=(keystore password) > pushmicroservice-17.0.479/
PUSHMICROSERVICE_config.txt
echo -Dhttps.port=9452 -Dplay.server.https.keyStore.path=/opt/certificates/hostname_keystore.jks
 -Dplay.server.https.keyStore.password=(keystore password) > searchmicroservice-17.0.479/
SEARCHMICROSERVICE_config.txt
```

(Optional) Configuring the xFlow Interface to Connect to SSL based Micro Services

NOTE

Optional steps provided here are required when you are using HTTPS.

Perform the following steps:

1. Backup the **casm.conf.js** and **casm.conf.do-not-change.js** files . The default location to access these files is, *C:\Program Files\CA\xFlow\APPS*\Services\incidentmicroservice-<xxxx>\public\conf
2. Verify that SSL is enabled for Tomcat in CA Service Desk Manager.
3. Open the **casm.conf.do-not-change.js** file and do the following:
   a. Find the text ***api : {server : 'https://localhost:8080/'},*** and provide the xFlow SSL port number in place of 8080.
   b. Save and close the file.
4. Open casm.conf.js file and configure the microservices that are required for your environment as below:
   a. Find the text ***api : {server : 'https://<hostname>:9004'},*** and modify the port number as ***9444*** .
   b. Find the text ***search : {server : 'https://<hostname>:9006'},*** and modify the port number as ***9452.***
   c. Find the text ***websocket : {server : 'wss://<hostname>:9008'},*** and modify the port number as ***9450.***
   d. Find the text ***sdm : {server : 'https://<hostname>/CAisd/pdmweb. exe'}*** and add the port number ***8443*** to host name. For example, http://sdmhostname:8443/CAisd/pdmweb.exe.
   e. Find the text ***insights: {server : 'https://<hostname>:port'}*** , and add the port number **9448** to the host name. For example, https://insights:9448.
5. Save and close the file.
6. Restart the xFlow Interface Service.
7. Navigate to `C:\Program Files\CA\xFlow\APPS\Services\insightmicroservice-17.1.706\conf`
8. Open the
   `application.conf`
   file in a text editor.
9. Find the parameter `Security headers.`

10. Search for `play.filters.headers.frameOptions = "ALLOW-FROM <http://hostname:9002>"`
11. Replace it with `play.filters.headers.frameOptions = "ALLOW-FROM <https://hostname:9444>>"`
12. Search for
    ```
    play.filters.headers.contentSecurityPolicy = "child-src 'self' 'unsafe-inline' 'unsafe-eval' <http://
    hostname:9002>"
    ```
13. Replace it with `play.filters.headers.contentSecurityPolicy = "child-src 'self' 'unsafe-inline' 'unsafe-eval' <https://hostname:9444>"`
14. Verify you are able to access the xFlow Interface <https://hostname:9444/>

Enable SSL on JasperSoft Tomcat Server:

To enable HTTPS on Jasper Tomcat server follow these steps:

1. Go to the Tomcat server where Jasper is installed.
2. Navigate to %JAVA_HOME%\bin and open command prompt. From this directory run the below command:

   ```
   keytool -genkey -alias alias -keyalg RSA -keysize 2048 -keystore <jasperkeystoreName>.keystore -sigalg
    "SHA1withRSA"
   ```
3. Enter and confirm keystore and certificate passwords. When it prompts for details, enter <jaster installed Tomcat server hostname> as Common Name (CN). Fill up the rest of the details such as Organization, etc.
4. Export the certificate from the keystore using the following command:

   ```
   keytool -export -storepass <keystorepassword> -alias alias -keystore <jasperkeystoreName>.keystore -file
    filename.crt
   ```
5. Once the certificate and key are saved in the Tomcat keystore, you need to configure your secure socket in the $CATALINA_BASE/conf/server.xml file, where $CATALINA_BASE represents the base directory for the Tomcat instance. Comment out the <Connector> element and add the following code below it. Replace <keystorepassword> with password chosen above.

   ```
   <Connector protocol="org.apache.coyote.http11.Http11Protocol" port="8443" SSLEnabled="true"
    maxThreads="150" scheme="https" secure="true" clientAuth="false" sslProtocol="TLS" keyAlias="alias"
    keystoreFile="%JAVA_HOME%\bin\<jasperkeystoreName>.keystore" keystorePass=<keystorepassword> />
   ```
6. You must configure the web application to enforce SSL as the only protocol allowed. Otherwise, requests coming through HTTP are still serviced.
   Edit the file **<js-webapp>/WEB-INF/web.xml**. At the end of the file, make the following changes inside the first **<security-constraint>** tag.
   a. Comment out the line **<transport-guarantee>NONE</transport-guarantee>**
   b. Uncomment the line **<transport-guarantee>CONFIDENTIAL</transport-guarantee>**
7. Login to Jasperserver through **https://hostname:8443/jasperserver-pro/** to verify changes.

For more information about the Jasper Soft application security, see https://community.jaspersoft.com/documentation/jasperreports-server-ultimate-guide/v561/using-ssl-web-server.

Enable SSL for Insights Microservices

After configuring JasperSoft Reports Server for SSL, perform the following steps to enable SSL for Insights Microservices:

1. Login to classic **CA SDM, Administration**, **xFlow Interface**, **General**. Select **insights.domain** from the General Configuration List.
2. Change **hostname:8080** (default) to **JasperHostname:8443** (for example, if 8443 is your JasperSoft server's SSL port).
3. Save the changes.

4. Now, change the insights protocol. From the xFlow General Configuration List (**Step 1**), select **insights.protocol** and edit to change HTTP to HTTPS.
5. Save the changes.
6. Restart the xFlow Analyst Interface Services.

Adding Jasper Certificate to Insights Trust Store

For insights to connect to HTTPS enabled jasper, the jasper certificate you exported above should be imported into the Insights trust store.

1. Navigate to <xFlowInstallation Home Dir>\jre\bin and run the following command:

```
keytool -importcert -alias alias -file filename.crt -keystore c:\\<Insightkeystorename>.jks -keypass
 <JasperCertificatePassword> -storepass <InsightsKeyStorePassword>
```

2. Navigate to *<xFlow_server>\APPS\Services\insightmicroservice-version\conf\application.conf* file and add the java keystore path (jks):

```
https configurations
play.ws.ssl { trustManager = { stores = [ { path = "C:\.jks", password: "<Password for
 InsightKeystoreName.jks" } ] } }
```

3. Restart the Insights server.

Enable SSL for EBL and Search Server

The search servers do not have SSL/TLS enabled, by default. For the search microservices and event-based load to interact with the search server securely, you can configure SSL. SSL encrypts any communication with the search server.

Step 1: Install and Configure Reverse-Proxy on the ngnix Server

This procedure explains how to configure the *nginx-1.10.0* server as a reverse-proxy on Windows. You can follow similar steps to configure reverse-proxy on Apache or other similar servers.

Prerequisite

- Download and install the *nginx-1.10.0* server on each search server.

  NOTE

  Configure nginx server as reverse-proxy on all the search servers in the cluster.

Follow these steps:

1. Navigate to the *<nginx_install>\conf* folder and define the search server details in the *nginx.conf* file.
   - Locate the **server** section in the file and perform the following actions:
     - (Optional) Edit the **listen** port number, if necessary.
       For example, if you have IIS running, which listens to port 80, then change to a different port number in the **listen** property.
     - Edit the **proxy_pass** value in the **location** section and define the search server port number.
       Default Port Number: 9012

```
server {



listen 80;
server_name localhost;
#charset koi8-r;
```

```
#access_log logs/host.access.log main;



location / {
    proxy_pass http://localhost:9012;
    proxy_read_timeout 90;
}
```

- Save the file.

2. Verify the configuration:
   - From the command prompt, navigate to the *nginx* folder and execute *nginx.exe* script to start the ngnix server.
   - Access the search server: *http://<hostname>:<listen-port>/*
     For example, access http://localhost:80/
     A message appears displaying the search server details, such as cluster name and version, in JSON format.

NOTE

To troubleshoot, view the *logs\error.log* file.

Step 1.1: Create an SSL Certificate

Prerequisites

- Ensure that you have installed OpenSSL 0.9.x in your environment.
  CA Service Management is certified with OpenSSL 0.9.8zh.

NOTE

You can create a certificate on any system that has OpenSSL installed. However, create the SSL certificate for each search server in the cluster.

Perform the following steps:

1. From the command prompt, execute the following command to generate the self-signed certificate:

```
openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout <hostname>.key -out <hostname>.crt
```

NOTE

: In this command, replace the <hostname> with the search server hostname.

2. Define the following details:
   Country; state; city; organization name; organization unit name; common name; email address

NOTE

: Define the common name as the hostname of the nginx server.

   The command generates these files: *<hostname>.crt* and *<hostname> .key*
3. Repeat step 1 and 2 to create SSL certificates for each search server.

Step 1.2: Configure the Certificates in the ngnix Server

NOTE

: Define the certificate details in the *ngnix.conf* file in each search server.

Follow these steps:

1. In the search server, navigate to the *<nginx_install>\conf* folder and edit the *ngnix.conf* file.
2. Disable the existing listen port.
   For example, *#listen 80*

```
server {
#listen 80;
server_name localhost;



#charset koi8-r;
#access_log logs/host.access.log main;


location / {
   proxy_pass http://localhost:9012/;
   proxy_read_timeout 90;
}
```

3. Define the new listen port.
   For example, define the new listen port as 443.

```
server {
#listen 80;
server_name localhost;



#charset koi8-r;
#access_log logs/host.access.log main;




listen 443 ssl;


location / {
   proxy_pass http://localhost:9012/;
   proxy_read_timeout 90;
}
```

4. Define the search server certificate details.

```
server {
#listen 80;
server_name localhost;



#charset koi8-r;
#access_log logs/host.access.log main;




listen 443 ssl;
```

```
ssl_certificate ./cert/<hostname>.crt;
ssl_certificate_key ./cert/<hostname>.key;

location / {
   proxy_pass http://localhost:9012/;
   proxy_read_timeout 90;
}
```

5. Navigate to the nginx folder and execute *nginx.exe* to restart the nginx server.
6. Verify if you can access the URL: https://
   For example, access https://<hostname>:443
7. Install the certificates when prompted.

Step 2: Configure Certificates on the xFlow Interface Server and Event Based Load

Step 2.1: Configure the Certificates on the xFlow Interface Server

Configuring the search server certificates on the xFlow Interface server enables secure communication between the search microservices and search servers.

> **NOTE**
>
> : Configure certificates of each search server on each xFlow Interface server. For example, if there are two search servers and two xFlow Interface servers, configure the certificates of both the search servers on each xFlow Interface server.

Follow these steps:

1. In the xFlow Interface server, copy the *<hostname>.crt* file.
2. Navigate to the path that stores keytool.
   For example, navigate to *C:\Program Files\CA\SC\JRE\1.8.0_74\bin.*
3. Execute the following command:

```
keytool -importcert -alias "<hostname>" -file <hostname>.crt -keystore c:\\es_new_keystore.jks -keypass
 <password> -storepass <password>
```

4. Repeat step 1 and 2 for each search server.

   > **NOTE**
   >
   > Replace the <hostname> values with the search server host names.

5. Navigate to the <xFlow_server>\APPS\Services*searchmicroservice- 0.1-SNAPSHOT*\conf\application.conf file and add the java keystore path (jks):

```
#https configurations



 play.ws.ssl {
    trustManager = {
        stores = [
            { path = "C:\\es_new_keystore.jks"}
        ]
    }
}
```

Step 2.2: Configure the HTTPS NX Variables for Event Based Load

1. Ensure that you have the path to the java keystore file created in the procedure *Step 2.1: Configure the Certificates on the xFlow Interface Server.*
2. Generate an encrypted password by executing the following command:

```
$<sdm_install>\bin\pdm_pen <password>
```

NOTE

: Password is the same as the one you used in *Step 2.1: Configure the Certificates on the xFlow Interface Server* procedure.

3. Execute the **pdm_options_mgr** command to configure the jks file and the encrypted password in the *https* NX variables.
   For more information about configuring the NX variables, see Configure Event Based Load.

Step 2.3: Update the Search server port number and protocol Details in CA SDM.

For example, in CA SDM update the port number as *443* and protocol as *https*. For more information about editing the server details, see Configure Search Servers in CA SDM.

Step 2.4: (Optional) If your search server is secure by a firewall, update the HTTP port from 9012 to 443.

Step 2.5: Restart the CA SDM services.

Step 2.6: Restart the CA Service Management xFlow Interface Server service.

Enable Rate Limiting Configuration

NOTE
This is applicable from CASM 17.4.5.0 and above.

Each incoming request to the backend passes through multiple filters before being fulfilled. To protect against potential misuse or malicious activity, a **Rate Limiter Filter** has been introduced.
The Rate Limiter Filter is a comprehensive API protection mechanism that safeguards against API abuse by enforcing configurable request limits. It supports both *token-based* and *IP-based* identification strategies to ensure secure and controlled access.

To enable or disable the rate- limiting functionality:

- Locate the file ***application.conf*** at the default location. *(Default location: **C:\Program Files\CA\xFlow\APPS\Services\incidentmicroservice-17.0.479\conf**).*
- After updating the below configurations, restart xFlow service.

Rate Limitation Configuration

Configuration	Description
ratelimit.enabled = false (OOTB)	Enables or disables rate limiting for API endpoints.
ratelimit.defaultPerUserLimit = 50	Number of requests allowed per API endpoint per X-Access Token within the defined time window. Applies individually to each API Example: 50 for /api/tickets, 50 for /api/emailtemplate.
ratelimit.timeWindowMinutes = 1	Duration of the rate-limiting time window (in minutes). Users can make the configured number of requests within this window. The cleanup interval is automatically calculated as max(2 * timeWindow, 5 minutes)

Configuration	Description
ratelimit.customIpHeader = ""	Specifies a custom header for extracting the client IP address for rate limiting. If configured, this header is checked before standard headers (X-Forwarded-For, X-Real-IP, Remote-Addr). Example: ratelimit.customIpHeader = "X-Custom-Client-IP"

Troubleshooting Service Point

If you experience issues, check these troubleshooting articles and try the recommended actions in the order in which they are described:

Unable to Search Tickets and View Service Offerings

The following error message is displayed in the `initial_load.log` file that is available in the `<Search Server Installation Drive>:\search\logs` folder:
`index_name has more than one indices associated with it index_name could be any index name like sdm_index_w`

Follow these steps:

1. Install any Elastic Search Client web browser plugin (for example: ElasticSearch Head for Chrome).
2. Connect to the Elastic Search server and delete all indices manually.
3. Rebuild index from where you have installed the Elastic Search server: `\search\bin \pdm_es_rebuild_index.bat` (.sh for UNIX)

Smart Search Results are Too Large

Symptom:

When the smart search results are too large, the following error message is displayed: `Result window is too large, from + size must be less than or equal to: [10000]`

Solution: The 10000 limit can be modified by changing the [index.max_result_window] index level parameter. To change the limit, follow these steps:

1. Download and install an Elasticsearch client.
2. Run the following command:

   ```
   PUT <<INDEX_NAME>>/_settings{   "max_result_window" : 500000 }
   ```

   For example:

   ```
   PUT sdm_index_cr/_settings{   "max_result_window" : 500000 }
   PUT catalog_index_service_offering/_settings{   "max_result_window" : 500000 }
   ```
3. You should see the following success response:

   ```
   {
    "acknowledged": true
   }
   ```
4. Perform the steps every time you refresh the index and data.

Issues Related to CA Service Catalog Featured Offerings

If you are unable to view the **More** button option in Service Point UI or CA Service Catalog Featured Offerings and Requests, perform the following:

Validate CA Service Catalog Integration and Configurations

Perform the following:

1. Ensure that CA Service Catalog is installed and running.
2. Navigate to **Administration**, **Options Manager**, **CA Service Catalog**.
3. Check that variables are installed and that **casc_endpoint** has a valid Catalog Web Interface URL and **casc_integrated** is set to **Yes**.
   Or
   Alternatively,you can also verify the Integration from the CA Service Catalog Interface. Login to CA Service Catalog. Navigate to **Administration**, **Configuration, CA Service Desk.** Verify the CA Service Desk integration properties and values. Click **Test** to test your integration with CA Service Catalog/CA SDM.
4. Verify the configuration for Service Point. Navigate to **Administration, xFlow Interface, ServicePoint, Configurations**.
   a. Open the Default configuration or the tenant-specific configuration.
   b. Verify **Service Catalog** is set to **Yes** to view CA Service Catalog Featured Offerings in Service Point UI.
5. If after performing the above steps, you are unable to view CA Service Catalog Featured Offerings, restart the xFlow Analyst Interface services.

Validate xFlow Interface and CA EEM Integration

Perform the following steps:

1. Run **Integrate pre-installed solution components** from CASM DVD on all servers where you have installed the xFlow Analyst Interface.
2. Verify the following:
   a. From *xFlow_Home\APPS\Services* location, ensure that **USMcertfile.key** and **USMcertfile.pem** files are present.
   b. Open **eiam.config** and **eiam.log4j.config** file from *xFlow_Home\APPS\Services\searchmicroservice-<release_version>.479\conf*
      It must show the complete file path. For example, the **LoggerConfiguration** file attribute in **eiam.config** must show complete file path.
3. If you are unable to view CA Service Catalog offerings and requests after performing the above steps, restart xFlow Analyst Interface services.

Validate Elastic Search Server Installation

Perform the following steps:

1. Verify that the Search server is installed and running.
2. Install **ElasticSearch Head** chrome plugin and provide search server "**hostname:9012**" to connect.
3. Verify if you can view data for CA Service Catalog Offerings and Requests on the browser tab.
4. If you are unable to view the data, do initial load or rebuild index from search server. Perform Initial Load and Refresh Index and Data on the Search Server.
5. Verify the search server logs.

Access Type Configuration for Employee User

Employee access type in CA Service Desk Manager must have the authentication type set to "**EEM**". If the authentication type is set to other types, EEM artifact cannot be fetched and you may see an I**NTERNAL_SERVER_ERROR** in the **getEEMAuth** API call.

Created tickets are not displayed in Service Point

Problem:

When a ticket is created in Service Point, it is displayed as expected. If a user logs out and logs back into it, no ticket is displayed in Service Point, but it is displayed in Service Desk Manager User interface and xFlow Analyst interface.

Solution:

You need to install the option "xflow_installed", restart Service Desk Manager and xFlow Analyst interface services, and run re-index command for the Search server as mentioned:

1. Log into Service Desk Manager (SDM) using the Service Desk administrator account and navigate to **Options Manager** > **General.**
2. Open **xflow_installed Update Options**, and set **Option Value** to "**Yes**". Click **Install** and save it.
3. Restart the CA Service Desk Manager server service.
4. Restart the xFlow Analyst Interface service.
5. Ensure that the above two services and Service Management Search Server service are up and running.
6. Run re-index, **pdm_es_rebuild_index** command located under (Install drive)\Program Files\CA\SC\CASearchServer \search\bin folder.
7. Restart the CA Service Management Search Server service.

Unable to locate Service catalog offering under Service Point.

Problem:

After Upgrade of CA Servicedesk Manager and Service Catalog, unable to locate Service Catalog Offerings under Service Point.

Solution:

Follow these steps:

1. Access Elasitc Search server through Chrome Addon and deleted all the indexes.
2. Stop Search Server.
3. Take a backup of al_cdb_configurationparameters
4. Add the missing table entry:

```
USE [mdb]
GO

INSERT INTO [dbo].[al_cdb_configurationparameters]
([componentkey]
,[machinename]
,[configkey]
,[configvalue]
,[productcode]
,[last_update_date])
VALUES
('ITSM','common','slcm.hostname','Catalog_Real_hostName',9,1552577972583)
GO
```

5. Build meta data:

```
pdm_es_build_index_metadata.bat C:\Program Files\CA\SC\CASearchServer\search\bin>
 pdm_es_build_index_metadata.bat
```

6. Build Initial Load: C:\Program Files\CA\SC\CASearchServer\search\bin>pdm_es_initial_load.bat

Service Point does not reflect the changes made in Service Desk

Problem:Changes made in Service Desk do not reflect in Service Point.

Solution:

The option **xflow_installed** is not installed in **NX.env** but shows as installed in CA SDM.

If xflow_installed option is not installed, then es_ebl Search Daemon process will not run. Follow these steps to install it:

- Log into **CA SDM** > **Administration** > **Options Manager** > **General**. Now:
  - If the xflow_installed option is set to **NO**, set to **YES** > **Save Changes** > **Restart SDM Services**.
  - If the xflow_installed option is set to **YES** but the NX.env does not show it: **Uninstall this option** > **Install It again** > **Restart SDM Services**.

Cannot find new tickets (request/incident) when searching in xFlow/ Service Point

Problem: When searching for request/incident in xFlow or Service Point, they do not appear in the ticket list.

Solution:

1. Backup es_events table by running the following commands:
   ```
   select *
   into es_events_bkp
   from es_events
   ```
2. Confirm that there is backup for the es_events table and then run:
   ```
   truncate table es_events
   ```
   This action deletes all data from es_events.
3. Restart Search Daemon by running the following in cmd:
   ```
   pdm_kill es_ebl
   ```
4. Ensure CA Service Management Search Server is up and running, so es_ebl can send the data to Search Server.
5. Create a ticket (or change status, etc.) in CA SDM so a row is added to es_events, processed by es_ebl and sent to CA Service Management Search Server.

Chrome Browser shows errors in Service Point

Chrome browser version v80 or later is used with Service Point, Service Point Home Page repeatedly throws login pop-ups. This is observed only when CA Service Catalog is integrated with NTLM. In non-NTLM environments, CA Service Catalog request submission failure is seen due to CORS error.

> **NOTE**
> When ServicePoint and Service Catalog are installed on separate systems, it has been observed that in ServicePoint, the Service Catalog service dropdown fields are not displaying, and request submissions are failing due to a CORS error. To resolve this issue, follow the steps outlined below.

There are two ways of resolving this:

- In SSL

- In non-SSL

In SSL

To resolve this issue, perform the following steps:

1. Enable SSL for CA Service Management solution products. For more information on how to configure SSL using the SSL Configurator Utility, see SSL Configurator Utility.
2. Ensure that you have installed CA Service Management 17.3.0.1 and above in your environment.
3. Add the following lines under the '**context**' element of **server.xml** file in USM_HOME/view/conf:

```
<CookieProcessor sameSiteCookies="none" />
```

For example:

```
<Context path="/usm" docBase="\Program Files\CA\Service Catalog\view\webapps\usm" debug="0"
reloadable="false" ><CookieProcessor sameSiteCookies="none" />
<Manager pathname="" />
<Loader delegate="true" />
</Context>
```

4. Restart CA Service Catalog services.

(Optional) Disable Browser Flags

> **NOTE**
> This workaround is applicable only for Chrome version 90 and earlier Chrome versions. Chrome version 91 and later versions do not support the below flags.

Try disabling your Browser flags to fix this issue.

> **NOTE**
> If you have performed the steps mentioned earlier in this article, the below steps are optional.

Perform the following steps:

1. In Microsoft Edge Browser, enter '**edge://flags**' in the browser. In Chrome Browser, enter '**chrome://flags**' in the browser.
2. Search for the below options:
   - **SameSite by default cookies**: Select **Disable** from the drop-down list.
   - **Cookies without sameSite must be secure**: Select **Disable** from the drop-down list.

In non-SSL

Follow these steps:

1. Configure F5 load balancer for both CA Service Catalog and CA Service Point with the same Hostname.
2. Navigate to Service Catalog> Administration >Configuration> Service Information >Update Hostname(F5 load balancer) & Port Number.
3. Navigate to Catalog Server and open **web.xml from C:\Program Files\CA\ServiceCatalog\view\webapps\usm \WEB-INF**, and update the load balancer URL of Service Point under "**param-value**" of "cors.allowed.origins"
4. Open **server.xml** under USM_HOME/view/conf and remove the entry(if you already added) "<CookieProcessor sameSiteCookies="none" />" (add this only if you are enabling SSL)
5. Restart the Service Catalog services.
6. Login to CA Service Desk Manager and navigate to Administration >Options Manager >CA Service Catalog >update the casc_endpoint with the load balancer URL.
7. In conventional CA SDM, recycle services on both the Primary and Secondary servers and, in AA setup Recycle services on BG and all App servers.
8. Restart the xFlow Service.
9. Access the Service Point URL using Chrome or any other latest browser and you can submit the request without an issue.

Troubleshooting CA Service Management Administration

CASM Administration troubleshooting steps.

If you experience issues, check these troubleshooting articles and try the recommended actions in the order in which they are described:

CA Service Management Admin (CASMAdmin) service is missing after uninstalling rollup patch.

Problem:

CA Service Management Admin (CASMAdmin) service is missing after uninstalling rollup patch.

Solution: Follow these steps:

1. Go to Service Management Administration installation directory.
2. Navigate to Scripts ->Utilities ->Service folder.
3. Run batch file "Tomcat_latest_service_creation.bat" to resolve the issue.

Troubleshooting Insights (CABI JasperReports)

This article includes information that can help you troubleshoot Insights issues.

Failed Test Connections for sdm_ds from Jasper Reports Server 8.1.1

Test connection fails for sdm_ds from Jasper Reports Server 8.1.1 but connection succeeds from Jasper Studio.

This issue happens when Jasper Server is installed on an IPV6 enabled environment.

To troubleshoot this issue, follow the instructions in the KB article- https://knowledge.broadcom.com/external/article/228560.

Cannot Use Fields with Local Attributes (CA SDM) as a Filter or in Cross-Tab

Any fields that contain local attributes (CA SDM) cannot be used as a filter or in a cross-tab.
For example, ‘a**ssignee_combo_name**’ is a local field in **"chg"** object. An Ad-Hoc view can be created using this field in Table format but, the field cannot be used in CrossTab format or as a filter to filter information.

To troubleshoot this issue, construct the local attribute using the direct fields in derived Tables.
For example, '**Assignee Combo Name**' was constructed using '**assignee_first_name**’, ‘a**ssignee_last_name**’ and '**assignee_middle_name**’ direct fields in derived tables.

JasperReports Server 8.1.1 Out-of-the Box Transactional Reports Error.

Problem: In case of SDM installed with Oracle database and integrated with Jasper 8.1.1, you may see an issue while running some of the reports which has **bigint** datatype.
Solution: Perform the following steps:

1. To resolve this issue, update and modify the **applicationcontext-semanticLayer.xml** file. Open this file in Jasper installed location: (for example: C:\Jaspersoft\jasperreports-server-7.9.0\apache-tomcat\webapps\jasperserver-pro \WEB-INF) and search for the following:

```
    switch (args[0]?.javaType)
{
  case 'java.lang.Integer':
  case 'java.lang.Short':
  return String.format("SUM(cast(%s as bigint))", sqlArgs[0])
```

```
default:
return String.format("SUM(%s)", sqlArgs[0])
                    }
```

2. Comment the following lines from the above step 1:

```
<!--case 'java.lang.Integer':
case 'java.lang.Short':
return String.format("SUM(cast(%s as bigint))", sqlArgs[0])-->
```

3. Save the file and restart JasperReports Tomcat services.

How can I troubleshoot issues with Datamart installation?

To troubleshoot issues with Datamart installation, review the `.log` file that is located in `<Datamart Installed Directory>\_CA_Datamart_Installation\Logs\CA_Datamart_Install_MM_DD_YYYY_Hours_Minutes_Seconds.log` folder.

How can I troubleshoot issues with data load failure during ETL?

Follow these steps:

1. Review the following files and take corrective measures:
   a. `config.properties` file located in `<Datamart Installed Directory>\config` folder.
   b. `datamart_std.log` file located in `<Datamart Installed Directory>\logs` folder.
   c. `Jobmetrics.log` file located in `<Datamart Installed Directory>\logs` folder
2. Re-initiate the ETL job.

The integration of xFlow Interface with CABI JasperReports Server fails even after following the integration steps. How can I troubleshoot the issue?

Follow these steps:

1. Review the `install log` file displayed on the Installation Summary screen of the CASM installer. Take corrective measures, and then select the **Retry** option to proceed with integration.
2. Verify that the following files are located in `<CA Business Intelligence Installation Directory>\apache-tomcat\webapps\jasperserver-pro\WEB-INF\config` folder:
   - `casm_insights.jks`
   - `casm_insights.properties`
3. Review the `jasperserver.log` file located in `<CA Business Intelligence Installation Directory>\apache-tomcat\webapps\jasperserver-pro\WEB-INF\logs` folder.
4. Review the `install log` file located in `%temp%\casm\install.log` folder.

Why do I encounter the following error message?

`'Could not retrieve the document due to internal server error. Please contact your administrator.'`

Follow these steps:

1. Review the following file and take corrective measures:
   a. `insightsMS.log` file located in `<xFlow Interface Installation Directory>\Apps` folder

Aggregate Reports

How do I verify the Status of the ETL Jobs Related to Aggregate (Datamart)?

If you want to verify the status of the jobs which perform the data extract, transform and load (ETL) from CMDB to Aggregate .CA Service Management maintains the log of the jobs which perform the data ETL from CMDB to Aggregate.

Navigate to the server where Aggregate is installed and navigate to <Datamart Installation Folder>\pdi-ce-7.1.0.0-12\data-integration\logs. The folder includes the following log files:

- **jobmetrics.log**
  Includes the log entries related to ETL jobs specifying whether each job is run successfully or failed.
- **spoon.log**
  Includes the detailed log entries related to the ETL jobs which run from Spoon GUI.
- **output.log**
  Includes the detailed log entries related to the ETL jobs which are run from STD-Load.bat using command line interface. The folder includes the latest file named as output.log along with 10 roll-up files. Each roll-up file is named with a incremented number used as suffix to the file name. For example, output.log.1. Each file can have the log data of a maximum 10MB in size.
  The STD-Load.bat file is typically available under the path <Datamart Installation Folder>\bin\ on the Datamart installed server.

Why is the Microsoft SQL Server Enterprise Edition the only supported SQL Server version for Insights Datamart?

CA Service Management 17.4 Supportability Matrix shows Microsoft SQL Server Enterprise Edition (2016, 2017, 2019 and 2022) as the only Microsoft SQL Server version supported/certified with Insights - Aggregate (Datamart). There are some online index operations that the Insight Aggregate jobs depend on (Pentaho API) which the Microsoft SQL Server Standard Edition does not support. Since the online index operations can only be successfully executed with Microsoft SQL Server Enterprise Edition, the SQL Server Enterprise Edition (2016, 2017, 2019 and 2022) is the only SQL version supported/certified with Insights Aggregate.

JasperSoft Studio shows a JAVA Heap Space Error

JasperSoft Studio 8.x shows a JAVA Heap Space error similar to the following:

```
java.lang.OutOfMemoryError: Java heap space
at java.util.Arrays.copyOf(Unknown Source)
at java.io.ByteArrayOutputStream.write(Unknown Source)
at sun.nio.cs.StreamEncoder.writeBytes(Unknown Source)
at sun.nio.cs.StreamEncoder.implWrite(Unknown Source)
at sun.nio.cs.StreamEncoder.write(Unknown Source)
at sun.nio.cs.StreamEncoder.write(Unknown Source)
at java.io.OutputStreamWriter.write(Unknown Source)
at java.io.Writer.write(Unknown Source)
```

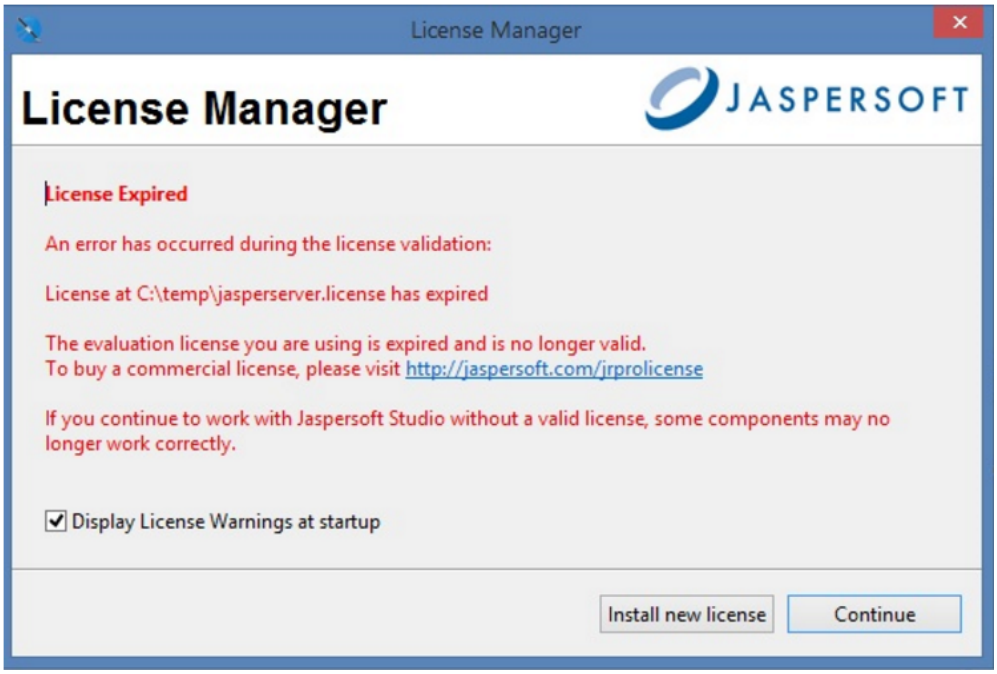
Perform the following to resolve this issue:

1. Increase the JVM Heap Size allocated to Jaspersoft Studio.
2. On the machine where Jaspersoft Studio is installed, modify the *Jaspersoft* Studio Professional.ini file.
   This file is located in the root of the JasperStudio install directory (example: C:\Program Files\TIBCO\Jaspersoft Studio Professional-8.0 final),
3. Modify the **-Xms** and **-Xmx** lines to the following (assuming you have enough RAM available):

   ```
   -Xms1024m
   -Xmx4096m
   ```
4. If JasperSoft Studio is open when the above changes are made, a restart of JasperSoft Studio is required.

JasperSoft Studio License Expired

After installing Jasper Studio Professional, if the license file is not properly applied, a "License Expired" message might be displayed by the "License Manager" similar to the following:



The out of the box reports provided in Jasper Reports Server for CA Service Management solution does not fulfill customer expectations. Users are required to develop custom reports in Jasper Studio Professional to import them into the Jasper Reports server. After installing the Jasper Studio, users cannot use it. A "License Expired" message is displayed.

The Jasper Studio license is included in the JasperSoft install media.

1. Launch Jaspersoft Studio Professional and navigate to Help/License Manager to install the license key.
2. Click Install new license.
3. Search and select the jasperserver.licence from the CASM DVD.
4. Click Open to install the license file.
5. Wait until the message “The new license for Jaspersoft Studio Professional was successfully installed.” displays. Click **OK**.

Datasource connectivity failure message

When editing the JRS 9.0.0 OOTB domains, we get the data source connectivity error alert. At this time, SDM and Jasper are up and running and integration works fine.

In such a case, restart the Jasper tomcat, and it works again.

An error with a message "The data source connection failed" message is displayed.

Error in jasperserver.log

If the jasperserver.log shows the error: "error com.ca.sdm.bop.BPMessagingException: 10 register_session_user failed; cannot find session with id", run the following steps:

1. Run the following commands in the CA SDM command prompt:

```
REPORTS_SESSION_OBJECT_DELETE_DELAY.

pdm_options_mgr -c -s REPORTS_SESSION_OBJECT_DELETE_DELAY -v 3600 -a pdm_option.inst
pdm_options_mgr -c -s REPORTS_SESSION_OBJECT_DELETE_DELAY -v 3600 -a pdm_option.inst -t
```

2. After executing the commands mentioned above, check if **NX_REPORTS_SESSION_OBJECT_DELETE_DELAY=3600** entry is added in the NX.env file. If not, add it manually in the NX.env file.
3. Restart SDM Services.
4. Restart the JasperReports Server services.

Jaspersoft Scheduled Reports are not Running

Jaspersoft Scheduled Reports Issue

Problem: Login to Jaspersoft server and create a report, schedule the report to run on a daily basis.
Jaspersoft Scheduled Reports are not running as expected.
Solution: Perform the following steps to resolve this issue:

1. Open View menu and click on **Schedules**.
2. Edit the schedule report which is failing.
3. Click the output options on the right side.
4. Select the check box beside **Sequential File Names by Timestamp**.
5. Click**Save**.
   Scheduled Reports works successfully.

Troubleshooting CA Service Catalog

Review the following to resolve the following issues:

- CA Service Catalog Performance Tuning
- Maintain Log Files
- Install or Upgrade Issues
- CA Service Catalog Request Management Issues
- Browser Issues
- Integration Issues
- Miscellaneous Issue

CA Service Catalog Performance Tuning

This article contains the following topics:

Review and verify the following configuration files to optimize CA Service Catalog performance requirements.

> **NOTE**
> USM_HOME refers to CA Service Catalog installation directory.

CONFIG.PROPERTIES

- **Location**: *USM_Home*
- **Purpose**: The maximum number of active connections that can be allocated from this pool at the same time is negative or no limit.
- **Parameters**:
  DBSource.maxtotal

DBSource.removeabandonedtimeout

- **Usage**:
  DBSource.maxtotal: Set the value to **2000.**
  DBSource.removeabandonedtimeout: Set the value to **900**.

SERVER.XML

- **Location**: USM_HOME/view/conf
- **Purpose**: The maximum number of request processing threads to be created by this Connector, which determines the maximum number of simultaneous requests that can be handled.
- **Parameter**: *maxThreads* attribute in HTTP/HTTPS/AJP Connector
- **Usage**:<Connector port="__USMPORT__" enableLookups="false" redirectPort="8443" maxThreads=" **2000** ">

VIEWSERVICE.CONF

- **Location**: USM_HOME/view/conf
- **Purpose**: Sets the initial memory in megabytes (allocated by JVM at startup).
- **Parameter**:
  wrapper.java.initmemory
  wrapper.java.maxmemory
- **Usage**:
  wrapper.java.initmemory : Set the value to **4096**.
  wrapper.java.maxmemory : Set the value to **6144**

APPLICATIONCONTEXT.XML

- **Location**: USM_HOME/view/webapps/usm/WEB-INF
- **Purpose**: Number of threads that will process requests present in the RequestQueue.
- **Parameter**: *requestListenerCount*
- **Usage**: <property name="requestListenerCount" value="**10**"/>

ehcache.xml

- **Location**: $USM_HOME/view/conf
- **Purpose**: Number of threads that will process requests present in the RequestQueue.
- **Parameter**:
  system.form.cache
  tenant.cache
- **Usage**:
  system.form.cache: Set the value of maxBytesLocalHeap to **50M**, timeToIdleSeconds to **28800**, and timeToLiveSeconds to **36000.**
  tenant.cache: Set the values of timeToIdleSeconds to **28800** and timeToLiveSeconds to **36000**.

Maintain Log Files

CA Service Catalog includes log files, each of which is helpful for researching a specific component or a function. To maintain log files, follow these steps:

1. Review the names and locations of all log files.
2. Review the descriptions of the most frequently used log files.
3. Become familiar with the process of setting log levels.

4. Review the process of configuring rollover settings for selected log files.
5. To keep older rolled over log files from consuming excessive disk space, delete them manually at regular intervals. Verify that you comply with the file retention policy of your organization. By default, certain log files are automatically rolled over and are automatically deleted periodically. Both processes occur according to your specifications when you configure rollover settings for selected log files. However, *no other* rolled over log files are automatically deleted; therefore, delete them manually.

Names and Locations of All Log Files

The following log files are located in the USM_HOME\logs folder:

- view.log
- view.log.1, view.log.2, view.log.3, and so forth
  You can configure rollover settings for these log files.

The following log files are located in the USM_HOME\logs\accounting folder:

- accounting.log
- AccountingService.log

The following log files are located in the USM_HOME\logs\install folder:

- createFiscalPeriod.log
- createRootOffering.log
- DeployServices.log
- importCommonReports.log
- importPlanningContent.log
- importPortalContent.log
- ixerr.log
- ixutil.log
- seeddata.log
- sqlUtil.log

All active MQ messages are logged in: *USM_HOME\jms-data*

WARNING

Warning! Do not delete the folder ending with *RequestQueue* as it contains submitted request data. For example: *queue#3a#2f#2fRequestQueue*

The following log file is located in the USM_HOME\logs\LDAPImporter folder:

- LDAPImporter.log

The following log file is located in the USM_HOME\logs\view folder:

- postEvent.log
- tomcat_view.log
- ViewService.log

The following log files are located in the USM_HOME\logs\repagent folder:

- repagent.log
- RepositoryAgentService.log

Most Frequently Used Log Files

If a file is marked with an asterisk (*), you can set the logging levels for the file.

- **accounting.log***
  Stores log messages that Service Accounting Component generates.
  This file is located in the USM_HOME\logs\accounting folder.
- **AccountingService.log**
  Stores log messages that CA Service Accounting Windows service generates.
  This file is located in the USM_HOME\logs\accounting folder.
- **ixutil.log***
  Stores log messages that the IXUtil import/export command-line utility generates.
  This file is located in the USM_HOME\logs folder.
- **repagent.log***
  Stores log messages that the CA repository agent generates.
  This file is located in the USM_HOME\logs\repagent folder.
- **RepositoryAgentService.log**
  Stores log messages that the Windows service of the repository agent generates.
  This file is located in the USM_HOME\logs\repagent folder.
- **seeddata.log***
  Stores log messages that the database containing the initial data that is supplied with CA Service Catalog generates.
  This initial data is required to get CA Service Catalog running immediately after installation.
  This file is located in the USM_HOME\logs folder.
- **view.log*, view.log.1*, view.log.2*, view.log.3*, and so forth**
  Store log messages that the Catalog Component component generates. The logs also contain rolled over log files from previous days.
  These files are located in the USM_HOME\logs folder.
- **ViewService.log**
  Stores log messages that the Windows service of Catalog Component generates.
  This file is located in the USM_HOME\logs\view folder.
- **usmInstall.log**
  Stores log messages that the CA Service Catalog installation program generates.
  This file is located in the USM_HOME\logs\install folder.
- **tomcat_view.log***
  Stores log messages that Apache Tomcat hosting Catalog Component and the Service Accounting Component services generates.
  This file is located in the USM_HOME\logs\view folder.

Set Log Levels

Configuring the log level helps you maintain the log files and troubleshoot any problems that occur in your system.

Follow these steps:

1. Determine the log file whose output you want to control.
2. Determine the log4j.xml file that controls the output of the log file whose log level you want to set.
3. Edit that log4j file and set the new logging level.
4. Review the contents of the affected log file after approximately 60 seconds. If necessary, reset the log level and review the contents again until you are satisfied with the level of detail.

   NOTE
   For more information about the logging features that CA Service Catalog uses, see the information about log4j2 at https://logging.apache.org/log4j/2.x/index.html”

Log Files Controlled by Each Log4j.xml File

CA Service Catalog includes several log4j.xml files that control the output of the most frequently used log files. Each log4j.xml file controls the log file of CA Service Catalog services, as summarized in the following table:

Service Logged	Location of Log4j File	Log Files
Catalog Component	USM_HOME\view\conf	seeddata.l
Service Accounting Component	USM_HOME\accounting\conf	accounting
Repository Agent	USM_HOME\repagent\config	repagent.lo
IXUtil import\export utility	USM_HOME\scripts	ixutil.log

Logging messages that are generated by a CA Service Catalog component, such as the Event Manager, are written to the log file of the calling service. Example: If the Catalog Component service calls the Event Manager, the log for the call is written in the view.log file. Calls from the Service Accounting Component service to the Event Manager are written to the accounting.log file.

Set the Log Level of a Service

You can set the log level to a higher level to investigate a problem. After you are finished investigating, you can reduce the log level for efficiency reasons.

Follow these steps:

1. Open the log4j.xml file of the CA Service Catalog service whose logging level you want to set.
2. Find the following section:

```
<Root level="log-level">
    <AppenderRef ref="accounting"/>
</Root>
```

3. Specify one of the following values for the *log-level*:
   - **Fatal**
     Logs only errors that shut down a CA Service Catalog component. This log level typically provides the least detail. This log level requires the least amount of disk space for rolled-over log files and hence the least maintenance.
   - **Error**
     Logs all of the messages from the previous level and failed actions. For example, a user submitted a request, but the system did not present the request for approval.
   - **Warn**
     Logs all of the messages from the previous levels and warning messages. For example, a user creates a service with a name that is not unique.
   - **Informational**
     Logs all of the messages from the previous levels and messages that are informational only. For example, the total number of open database connections.
   - **Debug**
     Logs all of the messages from the previous levels and more detailed messages to help troubleshoot a problem. For example, debug messages can include every step of a multiple-step process.
   - **Trace**
     Logs every action and displays the final XML. This log level typically provides the most detail. This log level requires the most amount of disk space for rolled-over log files and the hence most maintenance.
4. Save and close the log4j.xml file.
5. Recycle the CA Service Catalog service whose logging level you updated.

WARNING
To make any *other* changes to any log4j file, consult CA Technologies.

Configure Rollover Settings for Selected Log Files

To maintain maximum efficiency, CA Service Catalog automatically rolls over selected frequently used log files, according to default settings. You can configure both the rollover size and the number of rolled over log files retained. Doing so helps you customize the log files to match the needs of your organization more closely.

Follow these steps:

1. Open the log4j.xml file that controls the output for the log file that you want to update. Use an editor of your choice.
2. Find the following section:

```
<RollingFile
    name="view"
    fileName="${sys:usm.home}/logs/view.log"
    filePattern="${sys:usm.home}/logs/view.log.%i">
        <Policies>
        <SizeBasedTriggeringPolicy size="size" />
    </Policies>
    <DefaultRolloverStrategy max="backup-index" />
</RollingFile>
```

3. Specify the following settings:
   - ***size***
     Specifies the rollover size, in MB. For example, specify 5MB to make the log file roll over when its size reaches 5 MB. .
     Default: 20MB
   - ***backup-index***
     Specifies the number of recently rolled over log files that you want to keep on the disk. For example, 50.
     Default: 100
4. Save and close the log4j.xml file.
5. Verify that the log file output matches your needs. If necessary, reconfigure the log4j file or files to match your needs more closely.
   You have configured the rollover settings for selected log files.

Install or Upgrade Issues

This section contains the following issues:

- CA Service Catalog - Install/Upgrade Task Fails
- CA Service Catalog Fails while Importing Content Packs
- Troubleshoot Installation or Upgrade Issues

CA Service Catalog - Install/Upgrade Task Fails

When installing or upgrading CA Service Catalog, few tasks may fail based on environment and other considerations as shown below for the following scenarios:

Configuring CA Service Catalog - MDB Task Fails

Perform the following steps to resolve this error:

1. The Configuring CA Service Catalog - MDB Task fails due to mdbadmin schema missing from the database. The log file is available in *USM_HOME\logs\view.log.* For example, the following error is displayed:

```
ERROR [main] [ConfigureSQLServer] The specified schema name "mdbadmin" either does not exist or you do not
 have permission to use it.
ERROR [main] [CmdDeploymentManager] Deployment of the module 'mdb' failed
com.ca.catalog.deploy.exceptions.OperationFailedException
at
 com.ca.catalog.deploy.database.ConfigureSQLServer.createSynonymForUsmFnProductCode(ConfigureSQLServer.java:659)
```

2. To resolve this error, execute the SQL Query:

```
Create Schema 'mdbadmin'
```

3. Click Retry Install.

Configuring CA Service Catalog - Component Task Fails

Perform the following steps to resolve this error:

1. The Configuring CA Service Catalog - Component MDB task fails when the MDB deployment conflicts with DB constraint. The log file is available in *USM_HOME\logs\install\seeddata.log*
   An example of such an error is:

```
INFO Reading: C:\Program Files\CA\Service Catalog\view\scripts\MDB\SQL\usm_task.xml
ERROR Exception while communicating with the database:The INSERT statement conflicted with the FOREIGN KEY
 constraint "$usm_t_r00002c7f00000000".
The conflict occurred in database "mdb", table "dbo.usm_runtime_component", column 'runtime_component_id'.
com.microsoft.sqlserver.jdbc.SQLServerException: The INSERT statement conflicted with the FOREIGN
 KEY constraint "$usm_t_r00002c7f00000000". The conflict occurred in database "mdb", table
 "dbo.usm_runtime_component", column 'runtime_component_id'.
at com.microsoft.sqlserver.jdbc.SQLServerException.makeFromDatabaseError(SQLServerException.java:217)
```

2. To resolve this error, execute the SQL Query:

```
insert into usm_runtime_component VALUES ('Scheduler',NULL,NULL,NULL,NULL,NULL)
```

3. Click Retry Install.

CA Service Catalog Fails while Importing Content Packs

When CA Service Management installation has reached the "**importing Content Pack**" stage, installation fails and a popup message "error Importing Content Packs" is displayed. This issue is observed because of lack of required heap space while importing content packs.

To resolve this issue, perform the following steps:

1. Open the file ‘contentpack.bat’ file located under *USM_HOME\scripts.**
2. Now, locate the following lines:

```
SET JAVA_OPTS=%JAVA_OPTS% -DEiamTraceLevel=WARNING -Xms1024m -Xmx2048m -Dfile.encoding=utf-8 -noverify
And now, you may update it as follows:
SET JAVA_OPTS=%JAVA_OPTS% -DEiamTraceLevel=WARNING -Xms1024m -Xmx4096m -Dfile.encoding=utf-8 -noverify
```

3. Save the changes and resume installation.
4. If you still face the same problem, you may try increasing the *-Xmx* parameter in the above code appropriately, and resume the installation.

Troubleshoot Installation or Upgrade Issues

This article contains the following topics:

During the installation or upgrade of CA Service Catalog, CA MDB is automatically installed or upgraded

Example: CA MDB 1.0.4 is automatically upgraded to CA MDB 1.5 during the CA Service Catalog installation or upgrade. However, suppose CA MDB 1.0.4 is shared with other CA products. Then, the CA MDB upgrade can fail if duplicate records exist where the CA MDB tables require unique constraints. Examples include the following tables: ca_resource_class, ca_resource_family, ca_discovered_software, and cr_stat.

Solution: Back up these tables, evaluate them and clean up any duplicate records in them. Then try again to install or upgrade CA Service Catalog.

The BU Name is not auto-populated in the Installer Configuration Details Screen when CA Service Catalog is upgraded

Symptom: When CA Service Catalog is upgraded from a previous version to CA Service Catalog 17.3 using the traditional upgrade model, the BU Name is not auto-populated in the CA Service Catalog Installer Details screen.

Solution: Follow these steps:

1. Connect to the database server where the MDB is restored, and execute the following Microsoft SQL query to locate the BU Name:
   ```
   select tenant_id from usm_tenant where domain_type='SP'
   ```
2. Update the BU Name in the CA Service Catalog Installer Configuration Details screen.

CA Service Catalog service fails to start on port 7777 due to port conflict with JMS

Symptom: CA Service Catalog service fails to start on port 7777 due to port conflict with JMS

Solution: We do not recommend using port 7777 as the CA Service Catalog startup or shutdown port. Port 7777 is reserved for Java Messaging Service (JMS). If you must use port 7777 as the startup or shutdown port, then reset the JMS port number after you have finished running the setup utility. Otherwise, port conflicts occur, and the product does not function correctly.

Follow these steps:

1. Open the file that is named USM_HOME/config.properties in a text editor.
2. Update the value of the jms.port property to a new value.
3. Restart the Windows service named CA Service Catalog.

CA Service Catalog Request Management Issues

This section contains the following issues:

- Pending My Action page Is Not Refreshed Until User Logs Off
- Request Approval or Fulfillment Pending Action Is Not Assigned
- Requests Are Assigned to Multiple Users and Groups
- Requests Do Not Move to the Next Status

Pending My Action page Is Not Refreshed Until User Logs Off

Symptom:

After you add or remove a user from a group in EEM, the Pending My Action page is not refreshed.

Solution:

Log off and log in again to refresh the Pending My Action page.

Request Approval or Fulfillment Pending Action Is Not Assigned

Symptom:

An approval or fulfillment task for a request is assigned to a user or group. However, the request does not appear in the pending action list for the user.

Solution:

Use a user ID or group name not longer than 50 characters.

Requests Are Assigned to Multiple Users and Groups

Symptom:

Service Catalog assigns requests pending action to multiple users and groups.

Solution:

Service Catalog assigns requests pending action to *both* an Application group and a Global group that is defined in EEM when both the following conditions exist:

- Both groups have the same name in EEM.
- Both groups are configured to approve requests.

Consider renaming one of the groups or configuring one of them *not* to approve requests.

Requests Do Not Move to the Next Status

Symptom:

Requests do not move to the next status of the request life cycle as expected in CA Process Automation and/or Automic. Example: From **Submitted** to **Pending Approval**.

Solution:

To move requests to the next status of the request life cycle as expected, verify the related date settings.

> **NOTE**
> The below steps are applicable for both CA Process Automation and Automic Automation.

1. Identify the status change where requests are getting stuck.
2. Verify that the related processes are activated.
3. Verify the following:
   - Verify that you have enabled the rule actions for updating the request status of interest.
   - Perform the following steps if you are using CA Service Catalog to manage the request status change of interest:
     - Verify that the host name, port number, and other connection parameters are configured correctly.
     - (Applicable for CA Process Automation) Verify that the Global Data Set parameter points to the correct Catalog Component computer.
   - Verify whether the assignees for the request are members of CA EEM groups whose names include special characters.

If the assignees are members of such groups, the request statuses can fail to update. If necessary, rename the applicable groups in CA EEM so that their names do *not* include special characters.

4. Restart the CA Service Catalog Windows services (CA Service Catalog and CA Service Accounting).

Browser Issues

This section contains the following issues:

- TelemetryAdminRole users are unable to login into Service Catalog
- Pages Do Not Appear to Be Refreshing Properly

TelemetryAdminRole users are unable to login into Service Catalog

Symptom:

In Service Catalog, when user with **TelemetryAdminRole** permissions tries to log into the application gets the following message on browser

```
Error Occurred: You do not have enough privileges to access this feature, please contact your administrator
```

Solution:

User must close the browser and relaunch the browser to login with valid user.

> **NOTE**
> For more information, see the documentation for your web browser.

Pages Do Not Appear to Be Refreshing Properly

Symptom:

Service Catalog pages do not appear as expected in my web browser.

Solution:

Set your browser cache settings to retrieve an updated page from the server on every visit to the page.

> **NOTE**
> For more information, see the documentation for your web browser.

Integration Issues

This section contains the following issues:

- CA Service Catalog Integration Fails if PKI is Enabled
- Errors for Service Catalog integration with CA APM
- Integration Fails
- Integration of SDM and Service Catalog stops working with SSL Self-Signed certificate for 17.4RU2
- Issue with ITPAM Integration

CA Service Catalog Integration Fails if PKI is Enabled

Log in to CA Service Catalog as an Administrator, for example, **spadmin**, Navigate to the **Administration** tab and select the configuration link, under the **Options** list. Select CA Service Desk and on the right pane and select the **Test** button. Following message is displayed when the integration fails:

```
Error occurred while trying to connect. Please check your configuration and server availability
```

This issue can be identified in the error in the log file if the PKI certificate file is not found in the location *..\Program Files \CA\Service Catalog\XXX_Policy.p12:*

```
ERROR [WR_56513232] [http-bio-8080-exec-23] [Configuration] CCONF.TESTCONF.EXCPT
com.ca.usm.webservices.WebserviceException: ..\Program Files\CA\Service Catalog\XXX_Policy.p12 (The system
 cannot find the file specified)
```

NOTE

Enabling PKI is mandatory. If PKI is disabled, the integration with CA SDM in CA Service Catalog fails.

In order to properly integrate CA SDM and CA Service Catalog, the policy code name and keystore file must be placed in the correct location on all servers, If it is used as different file or if the name is incorrect, the integration fails.

Ensure to complete the following for the integration to work successfully:

1. Ensure that the same policy code name and keystore file is used for both CA SDM and CA Service Catalog. The default value for the keystore is: **CASM_POLICY.p12** and for the Policy Code: **CASM_POLICY.** The default location of the certificate files in CA Service Catalog is *..\CA\Service Catalog*\
2. The default location of the certificate files in CA SDM is *..\CA\Service Desk Manager.*

Errors for Service Catalog integration with CA APM

Symptom:

I am unable to integrate CA Service Catalog with CA Asset Portfolio Management.

Solution:

If CA Service Catalog is installed first and CA Asset Portfolio Management is installed later, you must recycle the CA Service Catalog Windows service and then proceed with the integration.

Symptom:

I get an error when trying to assign CA Asset Portfolio Management assets to a CA Service Catalog request item or to view associated assets. The error text is similar to the following: “Warning: A connection to the database could not be established.”

Solution:

The user does not have a role in CA Service Catalog. If the user is meant to be able to assign assets, then the user must use a role in CA Asset Portfolio Management that allows update access to the Asset Fulfillment security object. If the user is only meant to be able to view assigned assets, the user must use a role in CA Asset Portfolio Management which allows view access to the Asset Fulfillment security object, but not update access.

Symptom:

When trying to access CA Asset Portfolio Management, I receive a pop-up message saying "please wait" and CA Asset Portfolio Management opens very slowly or not at all. This error occurs when I use the Microsoft Internet Explorer browser. This error commonly occurs when I attempt to assign a CA Asset Portfolio Management model to a service option.

Solution:

Change your Internet Explorer settings to avoid the pop-up and the delay.

To change Internet Explorer settings:

1. Open the Internet Explorer browser.
2. In the browser bar, select Tools, Internet Options.
   The Internet Options dialog appears.
3. Click the Security tab.
   The security options appear.
4. Click the Local Intranet zone to select it, and click Custom Level.
   The security settings dialog for this zone appears.
5. Scroll to the Miscellaneous group and locate the option named Access data sources across domains.
6. Click Enable or click Prompt; do not click Disable.
7. Click OK to close the local intranet security settings dialog.
8. Click OK to close the Internet Options dialog.

Symptom:

I get an error when trying to assign CA Asset Portfolio Management models to a CA Service Catalog service option.

Solution:

One reason for this error is that the user does not have a role in CA Service Catalog. The user must use a role in CA Asset Portfolio Management that allows access to the Model security object.

Integration Fails

Symptom:

The integration fails between CA Service Catalog and other products. The other products can be CA Service Desk Manager, CA Business Intelligence, and CA Asset Portfolio Management.

Solution:

Verify that every computer with CA Service Catalog or an integrating product is set to the exact date and time. The date and time must also be adjusted for the time zone. This requirement applies to even the CA Service Catalog DBMS Server and integrating products DBMS Servers.

Example: Consider one of the computers is set to November 11, 6:27 p.m. Eastern USA. Now, all computers must be set to November 11, 6:27 p.m. Eastern USA or its equivalent, regardless of the computer's physical or geographic location. Some examples of equivalent dates and times for November 11, 6:27 p.m. Eastern USA are as follows:

- November 11, 3:27 p.m. Pacific USA
- November 11, 5:27 p.m. Central USA
- November 11, 9:27 p.m. Eastern Brazil
- November 11, 11:27 p.m. GMT
- November 12, 5:07 a.m. India Standard
- November 12, 7:27 a.m. China Standard

Methods for synchronizing the time and time zone on all computers include, but are not limited to, the following methods:

- Setting the time and time zone manually on each computer. For more information, see your operating system documentation.
- Configuring the Windows Time service to use an external time source or an authoritative time server. For more information, see the Microsoft web site.

Integration of SDM, Service Catalog, and ITPAM stops working with SSL Self-Signed certificate for 17.4RU2

Integration between CA SDM, Service Catalog and, ITPAM stops working after the upgrade to 17.4 RU2 when configured with SSL Self-Signed certificate.

Symptom:

view.log shows the following error:

```
com.ca.usm.webservices.WebserviceException: org.apache.cxf.service.factory.ServiceConstructionException:
 Failed to create service.
```

Solution:

In CA Service Catalog:

1. Regenerate a new custom key-store file for CA Service Catalog using the SAN details.
   ```
   keytool -genkey -alias <aliasname> -keyalg RSA -keystore "<keystore path>" -ext san=dns:<Service Catalog
    Server hostname>
   ```
   The default validity of the certificate is 90 days. You can extend the certificate duration by using the <-validity number of days> parameter while creating the certificate.
2. After generating the above certificate, run the SSL Configurator Utility by keeping the above generated keystore in the *Configurator Utility* location and perform the *Enable HTTPS for Service Management* (Service Desk Manager, xFlow Interface, and Service Catalog) section to enable Https for Service Catalog.
3. Restart CA Service Catalog.
4. Open Service Catalog Web Interface and ensure the newly created certificate is reflected in the browser.
5. Export the certificate as catalog.crt format and save it.
6. Import the sdm.crt (created in step 5 in CA Service Desk Manager into Service Catalog java keystore)
   ```
   keytool -importcert -file c:\sdm.crt -alias <aliasname> -keystore "C:\Program Files\CA\Service Catalog
   \embedded\jdk\lib\security\cacerts"
   ```
   The default password of keystore is *"changeit"*.
7. After successful import of certificate, restart Service Catalog.

In CA Service Desk Manager:

1. Regenerate a new custom key-store file for CA SDM using the SAN details.
   ```
   keytool -genkey -alias "itpam" -keyalg RSA -keystore "<keystore path>" -ext san=dns:<SDM Server hostname>
   ```
   The default validity of the certificate is 90 days. You can extend the certificate duration by using the <-validity number of days> parameter while creating the certificate.
2. After generating the above certificate, run the SSL Configurator Utility by keeping the above generated keystore in the *Configurator Utility* location and perform the *Enable HTTPS for Service Management* (Service Desk Manager, xFlow Interface, and Service Catalog) section to enable Https for CA SDM and XFlow.
3. Restart the CA SDM and xFlow services.
4. Open CA SDM Web Interface and ensure the newly created certificate is reflected in the browser.
5. Export the certificate as sdm.crt and save it.
6. Import the catalog.crt (created in step 5 in CA Service Catalog into SDM java keystore)
   ```
   keytool -importcert -file c:\catalog.crt -keystore "C:\Program Files (x86)\CA\SC\JRE\11.0.18\lib\security
   \cacerts"
   ```
   Default Password: changeit
   ```
   keytool -importcert -file c:\catalog.crt -keystore "C:\Program Files\CA\SC\JRE\11.0.18\lib\security
   \cacerts"
   ```
   Default Password: changeit
7. After successful import of certificate, restart CA SDM services.

For ITPAM

Problem:

When configured with SSL Self-Signed certificate, ITPAM Integration with SDM and Service Catalog stops working after the upgrade to 17.4 RU2.

Solution:

Steps to be performed for successful integration in the above scenario:

In ITPAM:

1. Regenerate a new custom key-store file for ITPAM using the SAN details.
   a. keytool -genkey -alias "itpam" -keyalg RSA -keystore "<keystore path>" -ext san=dns:<ITPAM Server hostname>
   The default validity of the certificate is 90 days. You can extend the certificate duration this using the -validity <number of days> parameter when creating the certificate.
2. Update the above generated certificate in ITPAM server.
   For 04.3.05 release:
   a. Navigate to <install_loc>\server\c2o\.config\OasisConfig.properties
      a. Add three entries to point to the custom certificate.
      b. itpam.custom.web.keystorepath, itpam.custom.web.password and itpam.custom.web.keystorealias
         NOTE
         itpam.custom.web.password value can be encrypted using PasswordEncryption.bat file available in the <install_loc>\server\c2o location.
   b. Navigate to <install_loc>\server\c2o\deploy\jbossweb.sar\server.xml
      a. Modify the file to use the custom entries mentioned in oasis config file.
      b. For e.g.:

```
<Connector protocol="org.apache.coyote.http11.Http11Protocol" SSLEnabled="true"
     port="${tomcat.secure.port}" address="${jboss.bind.address}"
     maxThreads="100" strategy="ms" maxHttpHeaderSize="8192"
     emptySessionPath="true"
     scheme="https" secure="true" clientAuth="false"
     keystoreFile="${itpam.custom.web.keystorepath}"
     keyAlias="${itpam.custom.web.keystorealias}"
     keystorePass="${itpam.custom.web.keystore.password}"
     sslProtocol = "${SSL_PROTOCOL}" algorithm = "${X509_ALGORITHM}" ciphers="${jboss.ssl.ciphers}"
     useBodyEncodingForURI="true" maxPostSize="12582912"/>
```

      c. Restart the ITPAM server.
      d. Validate the SAN details in the certificate using the browser. (Modify the SDM side only after validating the SAN details in the browser).
3. **For 04.4 release**
   a. Follow the techdocs for replacing above generated custom certificate in ITPAM 04.4 release mentioned in Configure Custom Certificates and Password Vault (Wildfly).
   b. Restart the ITPAM server.
   c. Validate the SAN details in the certificate using the browser. After the validation is successful, start modifying the SDM /SLCM side.

In CA SDM

1. ?wsdl should be appended to the PAM url for the integration (if SDM is on 17.4 GA)
2. Export the PAM customer certificate from browser and import it into SDM key-store file
   a. keytool -importcert -file c:\itpamcert.crt -keystore "C:\Program Files (x86)\CA\SC\JRE\11.0.18\lib\security\cacerts"
   Default password: ***changeit***

   b. keytool -importcert -file c:\itpamcert.crt -keystore "C:\Program Files\CA\SC\JRE\11.0.18\lib\security\cacerts" Default password: ***changeit***
3. Restart the SDM windows service (a complete service restart is needed to force to load the key-store file for SDM)
4. Validate ITPAM + SDM integration.

In SLCM

1. Export the PAM customer certificate from browser and import it into SLCM key-store file.
   a. keytool -importcert -file c:\itpamcert.crt -keystore "C:\Program Files\CA\SC\JRE\11.0.3\lib\security\cacerts" Default password: ***changeit***
2. Restart SLCM services.
3. Validate ITPAM + SLCM integration.

Issue with ITPAM Integration

ITPAM integration might fail with Service Catalog due to incorrect username, when EEM configure with multiple AD domains. This was caused by improper handling of backslashes (\) in domain-based user IDs. The PAM Global Dataset(/CA SLCM/SLCM_GlobalDataset) configuration now correctly interprets escaped backslashes.

Problem:

When EEM is configured with multi-AD, ITPAM integration with Catalog might fail due to incorrect user name.

Solution:

Ensure you update the username value with formate of "\\\\" in PAM global dataset (/CA SLCM/SLCM_GlobalDataset).

For exmaple:

```
<<DomainName>>\\\\<<username>>
domain1\\\\testuser1
```

Miscellaneous Issues

This section contains the following issues:

- Cannot Add or Update a User Because of Duplicate User ID
- Cannot Connect to a Trusted Computer
- Cannot Delete an Account
- Cannot Email a Request
- Compilation Errors After Customization
- How to Limit the Size of the Attachments in CA Service Catalog
- IXUTIL Out-of-Memory Error Occurs
- Service Catalog Single Sign-on Using NTLM Not Working For All Users
- Sorting of Services by Selection Type
- Windows Service Does Not Start
- Submenu is missing under Catalog Tab with Administrator Privileged Role
- CA Service Catalog Application Logs out Repeatedly

Cannot Add or Update a User Because of Duplicate User ID

Symptom:

When I try to add a new user or update an existing user ID, I receive the following error message:

```
Error - User with this user id already exists. Choose a different user id.
```

Solution:

A duplicate user profile exists. This problem can occur even if that user has been deleted. A deleted user is marked as inactive but the user is retained in the database.

Before assigning the user ID to a different user, remove the user ID from the inactive user.

Follow these steps:

1. Click Administration, Users.
2. Click the View Advanced icon option of the Search Users options.
3. Select User ID from the list of fields and select Contains from the list of operators.
4. Enter the user ID in the value field and click the Add link.
5. Select Inactive from the list of fields and select Equals from the list of operators.
6. Enter 1 for “true” in the value field and click the Add link.
7. Click Search.
   The list of users that match the selection criteria appears.
8. Edit the user: Change, or clear the User ID field, and click OK.

Cannot Connect to a Trusted Computer

Symptom:

I have integrated CA Service Catalog with CA Service Desk Manager. The two products use Secure Socket Layer (SSL) to connect to each other using a trusted relationship. However, when I test the connection, it fails. I also receive error messages that state there are no trusted relationship exists.

Solution:

- Verify that you have configured CA Service Catalog to use SSL.
- Verify especially that you have added self-signed certificates to the keystore, if applicable.
  When you use *self-signed* certificates for any computer that connects directly to CA Service Catalog, add these certificates to the keystore. **Example:** Suppose you want to use clustering with load balancing for CA Service Catalog. In that case, if you are using a self-signed certificate for the load balancing computer, add it to the keystore.
  Moreover, verify the computer to be trusted. Example: Suppose you want to integrate CA Service Catalog with CA Service Desk Manager through a load balancing computer. In that case, CA Service Catalog connects directly to the load balancer (not CA Service Desk Manager). Therefore, the computer to be trusted is the load balancer and not the CA Service Desk Manager computer.

Cannot Delete an Account

Symptom:

When I try to delete an account, a message appears indicating that I cannot delete this account.

Solution:

You cannot delete an account with an active subscription. Before you delete an account, prepare to delete it.

Follow these steps:

1. Cancel the subscriptions that are associated with the account.
2. Close the account by setting the status of the account to Closed.
3. Verify that all the billing runs affecting the account are completed.

Cannot Email a Request

Symptom:

I have CA EEM configured to use an external directory containing many groups. When emailing a request, I get the following error:

```
Cannot send email, please contact your administrator.
```

Solution:

To process the number of groups that are defined in your external directory, increase the value of CA EEM Max Search Size.

Follow these steps:

1. Click the embedded **CA EEM** link in the **Administration Quick Start** menu.
2. Log in using the EiamAdmin user name and password.
3. Click **Configure**, **Session**, **Configuration** menu option.
4. Change the **Max Search Size** to a value larger than the number of groups in your external directory.
5. Click Save.

Cannot Log In to CA Service Catalog

Symptom:

I cannot log in to a CA Service Catalog computer.

Solution:

- Verify that your user name and password are valid.
- Verify that the CA Service Catalog services are running on the computer that you are trying to log in to.
- Verify that the CA Service Catalog computer name does not contain an underscore, if you are using Internet Explorer to access CA Service Catalog. If an underscore appears in the computer name, perform one of the following steps:
  - Remove the underscore in the CA Service Catalog computer name. Update all references to the computer name throughout your environment. You can now use Internet Explorer to access CA Service Catalog.
    For more information about this Internet Explorer issue, see the related knowledge base article on the Microsoft web site.
  - Use Mozilla Firefox or Safari instead of Internet Explorer and then access CA Service Catalog.

Compilation Errors After Customization

Symptom:

After I customize files, multiple compilation-related error messages appear in the logs of Catalog Component computers.

```
...
yyyy/mm/dd computer-name-or-address ERROR [TP-Processor21] [DomProcessor] Error Generating Document
javax.xml.transform.TransformerConfigurationException: Could not compile stylesheet
...
```

Solution:

- Review the instructions for customizing XSL, XML, JavaScript, and image files. Verify that you have followed all instructions.
- Verify that the filestore is set up correctly. Especially verify that *all* Catalog Component computers share the filestore.

Data Not Uploaded If CA Repository Agent Service Is Configured with Active Directory

Symptom:

Data sets are not uploaded if CA Service Catalog is configured for use with Active Directory. Exceptions appear in the RepositoryAgentService.log file.

Solution:

Install a secondary view server and do *not* configure this view server for use with Active Directory. Configure the CA Repository Agent services and files on this secondary view server. Use this secondary view server to upload the data in CA Service Catalog.

Dual List Selection in IE Browser

Symptom:
In IE Browser, items in the dual list are not getting selected using the Up and Down arrow keys.

Solution:

Use the **Shift+F8** Windows Keyboard Shortcut to turn on the '*Selection Mode'*. Once the selection mode is turned on, you can use the Up and Down arrow keys to navigate between the list items and use the Space Bar to select an individual item or unselect an item.

How to Limit the Size of the Attachments in CA Service Catalog

You can limit the size of an attachment in CA Service Catalog by editing the *%USM_HOME%/view/webapps/usm/web-inf/web.xml* file:

```
  <init-param>
 <param-name>maxPostSize</param-name>
 <param-value>5242880</param-value>
 </init-param>
```

By default, the value is set to 5,242,880 bytes (5 MB). After modifying this value, you can limit the attachment file size as per your requirement. You must restart the CA Service Catalog service for changes to take effect.

IXUTIL Out-of-Memory Error Occurs

Symptom:

When I use the IXUTIL utility to import or export Service Catalog data, the attempt fails. I receive an error message, such as a "generic" SAXException error.

Solution:

Open the ixutil.bat file. Increase the size of the JVM parameter in the ixutil.bat file to 1024 or higher.

NOTE

The ixutil.bat file is located in the USM_HOME\scripts folder.

Service Catalog Single Sign-on Using NTLM Not Working For All Users

Symptom:
We have enabled single sign-on using NTLM for Service Catalog, but it doesn't work for all users.

Solution:
When configuring single sign-on using NTLM for Catalog users, EEM has to be configured to use a Windows domain. Some users log in to Windows using this domain while some log in to a sub domain.

All users must log in to Windows using the same domain as the one which is configured as the user store in EEM. If users log in to another domain, even a sub domain of the main domain, single sign-on will not work.

Sorting of Services by Selection Type

Symptom:

When I view services in the catalog, they are not sorted by Selection Type as I expect them to be. Even the Featured Services are not sorted by Selection Type.

Solution:

When you create a service, you specify a Selection Type. The sorting of services according to Selection Type is controlled internally. The sort order is *not* based on the name of the Selection Type. Instead, the sort order is based on the *numeric value* of the Selection Type, as follows:

Numeric Value	Selection Type
0	No Selection
1	Single Selection
3	Multiple Selection

Windows Service Does Not Start

Symptom:

Windows services fail to start for Service Catalog.

Solution:

The Windows services are set by default to start automatically after a reboot or shutdown. If the services fail to start automatically, start the services manually.

Submenu is missing under Catalog Tab with Administrator Privileged Role

Symptom

Symptom:

Submenu is missing under Catalog Tab with Administrator privileged Role

Solution:

Follow these steps to resolve the issue:

1. Login to EEM using Eiamadmin super user account and select **Service Catalog** as an **Application** from the dropdown.
2. Click Manage Access Policies Tab - under Access Policies.
3. Click on "Explicit Denies" Tab and update fro Links Dashboard, Launchpads, Guinode
4. Updated "ACL_restrict_r11_launchpads", with values 17.0,17.1,17.2,17.3,17.4
5. Updated "ACL_restrict_unwanted_Launchpads", with values 17.0,17.1,17.2,17.3,17.4

   NOTE
   Ensure there are no spaces between the values.

CA Service Catalog Application Logs out Repeatedly

Symptom

Symptom:

Application logs out after every navigation

Solution:

If CA Service Catalog application is configured to run on HTTP, “samesitecookies” entry must not be present in server.xml:

1. Find the following lines under the 'context' element in USM_HOME/view/conf/server.xml file.
   ```
   <CookieProcessor sameSiteCookies="none" />
   ```
2. Remove the entry and save the file.
3. Restart CA Service Catalog services.

Troubleshooting CA Asset Portfolio Management

This section contains the following articles:

- Unable to Find Multi-Tenancy Service in the Services Console
- Data Importer is Failing while Creating a New Asset
- Tenancy Management Does Not Appear
- Tenancy Management Cannot be Displayed
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- Log In Fails with a User Name Containing Extended Characters
- Missing Operating System Message Appears in Message Queue
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- CA Service Management Installer Does Not Recognize .NET Framework
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Unable to Find Multi-Tenancy Service in the Services Console

After successful installation of CA Asset Portfolio Management, CA CASM (Multi-Tenancy) service is missing from the services console.

Solution:

Perform the following steps to resolve this issue:

1. Open the **install.log** under *%temp%/casm* folder.
2. Lookup for the logging text starting with **CASMPostInstallAction**
3. If the service is not installed as per the log, copy the command text for CASM service installation found in the log. For example:
   “ %temp%/casm_itam/AppLauncher.exe” -AppName=CASMr125.exe -installOption=-f -paramFile=installer.properties -ProductName=CASM -InstallDir="<<ITAM_installDrive>>\ITAM" -SubstituteTextCASM='-DBTYPE=<<mssql/oracle>>,-DBSERVER=<<dbServer>>,-DBNAME=<<dbName>>,-ORACLESERVICENAME=<<oracleNetServiveName>>,-DBPORT=<<dbPort>>,-DBUSER=mdbadmin,-DBPWD=<<mdbadmin_pwd>
4. Open command prompt and run as an Administrator.
5. Paste the copied command from **Step 3** in the command window and press **Enter**.
6. Wait for the cursor to return.

CA CASM Service is now installed successfully.

NOTE
If the installation command is not found in the log, please raise a case with Broadcom Support.

Data Importer is Failing while Creating a New Asset

Data Importer is Failing while Creating a New Asset when subclass is not present in the mappings in Oracle database.This issue is observed when you create Asset, Model, Family, Class, and Subclass at the same time.

Resolution:

To resolve this issue, provide the following mapping information in Data Importer:

- Asset Name
- Asset family
- Asset Class
- Asset Subclass
- Model name
- Model family
- Model class
- Model Subclass

Tenancy Management Page Does Not Appear

Symptom:

When I click the Administration tab, I do not see a Tenancy Management option.

Solution:

The CA APM administrator has not assigned you to a role in which Tenancy Administration Access is enabled. If you need access to Tenancy Management, contact your CA APM administrator.

Tenancy Management Page Cannot Be Displayed

Symptom:

CA APM Multi-Tenancy page is not displayed. Browser Error Appears

The following browser error message appears on clicking Administration, Tenancy Management:

```
Page cannot be displayed.
```

Solution:

Verify that the CASM service is started.

Symptom:

CA APM Multi-Tenancy page is not displayed and shows the 'You do not have required privileges to perform the action'. This is observed in the following cases:

- CA Asset Portfolio Management - Registration Service is stopped on ITAM application server.
- CORA configuration setting is set to default.
- On CA APM Database server, database service is restarted.

Solution:

Restart the CA CASM Service on CA APM Application server.

Web Servers Named with Underscore Characters

Valid on all supported operating environments.

Symptom:

Using underscore characters in web server host names can cause problems. You see the problems when logging in to the product or when you use Embedded Entitlements Manager for user configuration.

Solution:

If you use a virtual or ghosted system, configure a new host name by creating another image without the underscore. For a production system, add a new host name to your internal Domain Name System (DNS). Adding a host name lets you access the product with a different URL.

Log In Fails with a User Name Containing Extended Characters

Symptom:

When using Embedded Entitlements Manager with Single DB Login authentication, I cannot log in to the Unicenter APM web interface.

Solution:

Select a User Name that does not contain extended characters (that is, Japanese or German characters).

Missing Operating System Message Appears in Message Queue

Symptom:

I receive one of the following error messages in the message queue during the Reconciliation Engine processing of normalization rules:

- The following discovered operating systems are missing from the Public Operating System list:
  Missing Operating System: *operating system name*
- The following discovered operating systems are missing from the Operating System list and must be added to the Public Operating System list or the list for tenant: *tenant name*
  Missing Operating System: *operating system name*

  NOTE
  The Reconciliation Engine writes messages to the message queue in the database. To see these error messages in the message queue, set the Engine Debug Level in the Hardware Reconciliation Engine Configuration Settings to Fatal (or a higher level of detail). For more information about the message queue and the configuration settings, see Hardware Reconciliation.

Solution:

The normalization rules apply to all tenants and public data and can be used across tenants. If an operating system value assigned through the normalization list does not exist for a tenant, the Reconciliation Engine produces an error message.

NOTE
For more information about normalization rules, see Data Normalization.

If one or two operating systems are missing, resolve the problem by adding the operating systems manually to normalization rules.

If numerous operating systems are missing, complete the following steps to resolve the problem:

1. Log in to Unicenter APM and click Administration, Reconciliation Management.
2. On the left, click Reconciliation Message Search.

The message queue displays reconciliation log messages in the Search Results.

3. Search to find the missing operating system normalization rule error messages.
   The message queue displays all missing operating system normalization rule error messages.
4. Verify that the system administrator email address in the product is correct and click Export to CSV.
   The operating systems that are missing are exported to a CSV file. The system administrator receives an email message with a link to the CSV file.
5. Edit the content of the CSV file to prepare the file for the ITAM Data Importer. For example, you can remove duplicate operating systems or extraneous words from the file.

   NOTE
   For more information about using the ITAM Data Importer, see Import Data section.
6. Log in to the CA APM.
7. Click Administration, ITAM Data Importer, and select the tenant or public data that is missing the operating systems.
8. Import the CSV file.
   The missing operating systems are imported into the Management Database and are available for use during Reconciliation Engine normalization processing.

Import Data to CA APM Installed on Oracle Database

You may encounter issues when you import data to CA APM that is installed on Oracle database. This article explains the various scenarios and the recommended tasks you must perform to import data successfully.

Terminologies you Must Know

Onboarding

Onboarding refers to the activity of importing data to the database for the first time. This is a one-time activity and does not repeat.

Ongoing

Ongoing refers to the activity of importing data to the database on a frequent basis. The frequency of data import may be daily, weekly, monthly, scheduled, or on-demand.

Prerequisites

- Upgrade to CA APM 17.4

Recommendations for Data Import

The recommendation steps for the various scenarios only apply to the following environments:

- Oracle Database 19c, 21c
- Standalone installation of CA APM or CA APM integrated with CA SDM, CA Service Catalog, or CA ITCM.

Scenario 1: Onboarding Data More than 100 Thousand Records

To onboard data of more than 100 thousand records on a pristine CA APM Oracle Database, complete the following steps:

1. Stop the following Windows services:
   - CA APM Event ServiceCA APM
   - HW Reconciliation Engine
   - CA APM Registration Service
2. Verify that the CA APM Data Importer Engine Service is running.

3. Start the data import process.
4. After the data import is complete, start CA APM Registration Service.
5. After the registration is complete, start the following services:
   – CA APM HW Reconciliation Engine
   – CA APM Event Service

Scenario 2: Ongoing Data More than 25 Thousand Records

- Existing data on the database - 100 thousand Assets and/or 150 thousand Service Desk Configuration Items.
- Ongoing data load - More than 25 thousand records.

Follow these steps:

1. Stop the following Windows services:
   – CA APM Event Service
   – CA APM HW Reconciliation Engine
   – CA APM Registration Service
2. Verify that the CA APM Data Importer Engine Service is running.
3. Start the data import process.
4. After the data import is complete, start the following Windows services:
   – CA APM Registration Service
   – CA APM HW Reconciliation Engine
   – CA APM Event Service

Scenario 3: Ongoing Data Less than 25 Thousand Records

- Existing data on the database – Less than 25 thousand Assets/Configuration Item data records (record count of the table ca_owned_resource).
- Ongoing data load - Less than 25 thousand records.

No special steps are recommended. All the Windows services may continue to run.

CA Service Management Installer Does Not Recognize .NET Framework

Symptom

I have installed .Net Framework 2.0 and .Net Framework 4.5.2 prior to installing CA Asset Portfolio Management. But the Installation Prerequisite screen of the installer shows as .Net Framework 2.0 and .Net Framework 4.5.2 are not available though I have the administrator privileges on the server where I am installing CA Asset Portfolio Management.

Reason

This issue occurs, if "Prevent access to registry editing tools" policy is enabled.

Solution

You must disable the "Prevent access to registry editing tools" policy.

Follow these steps:

1. Go to Start, enter **gpedit.msc** in the Search programs and files box.
   The Local Group Policy Editor window opens.
2. Expand the User Configuration, Administrative Templates, and select the System folder.
3. Double click the "Prevent access to registry editing tools" setting.
4. Select the Disabled radio-button and click OK.
5. Restart the server.

Event Notifications are Non-Functional for Reference Type Fields in the Oracle Database

Symptom:

Event notification fails for the Asset reference field and the Contact reference field created based on other objects.

Solution:

Follow these steps:

1. Stop the Event service.
2. Navigate to Application Server. For example: C:\Program Files (x86)\CA\ITAM\MDB\PostInstallSteps.
3. Open **CreateAuditTableAndTriggers** file in notepad and Launch the SQL developer. Open a new SQL Worksheet and execute the script.
4. Click **Commit** to save the transaction.
5. Reset IIS.
6. Start Event Services.

Incorrect Oracle Database Version is Displayed

Symptom:

Oracle server version is displayed as Oracle 11g instead of Oracle 19c and Oracle 21c.

Solution:

Follow these steps:

1. Launch SQL developer and connect to MDB.
2. Open a new SQL Worksheet and execute the following query.

```
select * from AL_CDB_CONFIGURATIONPARAMETERS  where CONFIGVALUE='Oracle 11g' and componentkey='Database'
 and configkey='OracleVersion';
```

3. Run the following query as per your database version.
   - If the search result has a record finding and using the Oracle 19c database, run the following query.

```
update AL_CDB_CONFIGURATIONPARAMETERS set  CONFIGVALUE='Oracle 19c' where CONFIGVALUE='Oracle 11g' and
 componentkey='Database' and configkey='OracleVersion';
```

   - If the search result has a record finding and using the Oracle 21c database, run the following query.

```
update AL_CDB_CONFIGURATIONPARAMETERS set  CONFIGVALUE='Oracle 21c' where CONFIGVALUE='Oracle 11g' and
 componentkey='Database' and configkey='OracleVersion';
```

4. Click **Commit** to save the transaction.
5. Reset IIS.

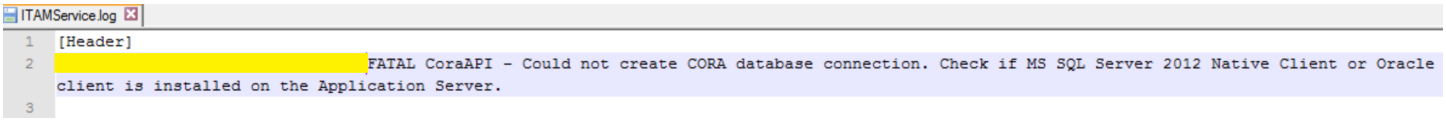
CORA Troubleshooting

CORA might run into certain issues, if the pre-requisite conditions are not met. Refer Installation of pre-requisites of CA APM for more information.

Symptom:

The following error message is shown in the CA APM while creating a new asset or a list under list management in CA APM:
The application encountered an error during asset registration in CORA.
CORA not initialized or prerequisites missing. See CORA troubleshooting in techdocs.

Figure 43: The following Error is seen in log file:



Solution:

If CA APM is installed on a SQL Server database, install the 64-bit SQL Native Client 11.0 on the APM Application server to ensure CORA functionality works, as CORA uses the SQLNCLI11 provider to connect to the database.
Download the 64-bit SQL Native Client 11", named as Microsoft® SQL Server® 2012 Native Client from: https://www.microsoft.com/en-us/download/confirmation.aspx?id=50402

Troubleshooting CA Service Desk Manager Connector

This topic contains the following information:

Verify that the CA SDM Connector Installed Successfully

Symptom:

I want to verify that the CA SDM Connector is installed successfully.

Solution:

Follow these steps:

1. Check if there are any errors in the Catalyst_SDMConnector_InstallDebug.log file or CatalystInstallDebug.log file.

   NOTE
   These files are located in the %TEMP% directory.
2. If either of the log files has errors, open a support issue.
3. If there are no errors in either of the log file, complete the following steps:
   a. Open the CA Catalyst Registry Server UI using the following URL:
      `https://<registryserver:port>/registry/carbon/admin/login.jsp`
   b. Log in to the CA Catalyst Registry UI and browse to the following directory:
      `\topology\physical\<SDM Connector NODE>`
   c. Check if connector-modules.xml and startup.properties files are present in the directory.
   d. If either connector-modules.xml or startup.properties file is missing from the directory, then the CA SDM Connector installation has failed.
      - Uninstall and reinstall the CA SDM Connector.
4. Verify that the CA SDM Connector installer has completed the steps to configure the CA Catalyst server. Complete the following steps to verify:
   a. Verify reconciledViews.xml content.
   b. Verify plannerpolicy.xml content.
   c. Verify plancontrol.xml content.
   d. Verify correlation.properties content.

Verify that the CA SDM Connector Started Properly

Symptom:

I want to verify that the CA SDM Connector started properly.

Solution:

Follow these steps:

1. Open the CA Catalyst Admin UI using the following URL:
   `http://<<rose>-Server:port>/adminui`
2. Log in to the CA Catalyst Admin UI.
3. Click CA SDM Container node from the left panel.
   The CA SDM Container node is expanded to display the CA SDM Connector and status of the CA SDM Connector.
   If the status of the CA SDM Connector is RUNNING, then the CA SDM Connector has started properly.

   NOTE
   If you have just started the CA SDM Connector, wait for the CA SDM Connector nodes to move to the Running state before you perform any operation.

CA SDM Connector Fails to Start

Symptom:

The CA SDM Connector fails to start.

Solution:

The CA SDM Connector failed to start because of one of the following reasons:

- CA SDM Server is Down
- Login Failure

Server is Down

Symptom:

If the CA SDM server is down and you try to start the CA SDM Connector, the CA SDM Connector attempts to connect to the server a few times and shuts down. The following line appears in the ServiceDeskManagerConnector.log file located in the CATALYST_HOME/container/data/logs/ directory:

```
Unable to connect to Service Desk Manager.
```

Solution:

Complete the following steps to start the CA SDM Connector:

1. Ensure that the CA SDM Server is up and running.
2. Restart CA SDM Connector Container.

Login Failure

Symptom:

If the password of the logged-in user has changed in CA SDM, the CA SDM Connector fails to log in to CA SDM. The following line appears in the ServiceDeskManagerConnector.log file located in the CATALYST_HOME/container/data/logs/ directory:

```
Login failure
```

Solution:

Complete the following steps to start the CA SDM Connector:

1. Open the following directory from the command prompt:
   `CATALYST_HOME\tools\encrypt`

   NOTE
   Ensure that java.exe is in the PATH.

2. Run encrypter.bat <New Password> command and copy the encrypted string output from the command.
3. Browse to the following directory from the https://registryserver:port/registry/carbon/admin/login.jsp URL:
   `\topology\physical\<SDM Connector Server>\modules\configuration\ServiceDeskManagerConnector.xml`
4. Click Edit as Text.
5. Replace the value of the password with the encrypted string output that is copied from encrypter utility.
6. Click Save Content.
7. Restart the CA SDM Connector Container.

CA SDM Connector Container Fails to Connect to CA Catalyst Registry or CA Catalyst Server Container

Symptom:

The CA SDM Connector Container failed to connect to the CA Catalyst Registry or the CA Catalyst Server Container.

Solution:

The connection has failed because of one of the following reasons:

- CA Catalyst Registry or CA Catalyst Server Container Service is Down
- CA Catalyst Registry Database Server or CA Catalyst Server Container Database Server is Down
- Firewall Restrictions

CA Catalyst Registry or CA Catalyst Server Container Service is Down

Symptom:

The CA Catalyst Registry or the CA Catalyst Server Container service is down.

Solution:

Complete the following steps to start the connection:

1. Start the CA Catalyst Registry or the CA Catalyst Server Container.
2. Restart the CA SDM Connector Container.

CA Catalyst Registry Database Server or CA Catalyst Server Container Database Server is Down

Symptom:

The database server used by the CA Catalyst Registry or the CA Catalyst Server Container is down.

Solution:

Complete the following steps to start the connection:

1. Start the database server.
2. Start the CA Catalyst Registry or the CA Catalyst Server Container.
3. Restart the CA SDM Connector Container.

Firewall Restrictions

Symptom:

If the CA SDM Connector and the CA Catalyst Registry are on different computers, then firewall restricts the connection.

Solution:

Ensure that the firewall is not blocking any port that is used for communication between the CA Catalyst Registry and the CA SDM Connector container.

Verify the Data Published By the CA SDM Connector

Symptom:

I want to verify the data published by the CA SDM Connector.

Solution:

Follow these steps:

1. Launch the following URL:
   `http://<<rose>-server:port>/ca-rest/home`
2. Log in with the credentials of the CA Catalyst server.
3. Click Browse by CI Type.
   A new page with the data source button is displayed.
4. Select ServiceDesk-CMDB and look for any CI or relationship information.

CIs are Not Displayed in CMDB

Symptom:

I do not see the CIs in the CMDB, even after running the Catalyst job or the Management profile with the enabled integration.

Solution:

Follow these steps:

- If the CIs appear in the Catalyst, then verify that you applied the correct registry configuration steps and recycled the CA Catalyst container service in the CA Catalyst server. When the CA Catalyst Server and the CA SDM Connector are online, perform the full synchronization using the Priming Utility. For more information, see the *CA Catalyst documentation*.
- If the CIs do not appear in the Catalyst, then ensure that the CA Catalyst server, the CA Configuration Automation Connector, and the CA SDM Connector are up and running. Then rerun the CA Configuration Automation Catalyst Job.

Catalyst UI Login Failure

Symptom:

I am unable to log in to the Catalyst UIs such as the Admin UI or USM Web View even though my credentials are correct. However, I am able to log in to the Registry UI with the same credentials.

Solution:

Follow these steps:

1. Ensure that CA Embedded Entitlements Manager is up and running.

   NOTE
   The Registry UI does not use CA Embedded Entitlements Manager for its user validation.
2. Restart the CA Catalyst container service in the CA Catalyst server.
3. Restart CA Catalyst Admin UI service in the CA Catalyst server.

Failed to Launch In-Context to a CI in CA Configuration Automation

Symptom:

I am unable to launch in-context to a CI in CA Configuration Automation while viewing the CI in CA SDM.

Solution:

If the in-context button (CCA$) does not appear for the CIs, ensure that the MDR for CA Configuration Automation was configured. For more information, see Create an MDR Definition in CA SDM.

Relationships are not Displayed in CMDB

Symptom:

I am unable to view the relationships in CMDB.

Solution:

Check the CA Catalyst Profile in CA Configuration Automation to find out the relationship types that were chosen for the export. If you enable TWA, then the relationships appear in TWA after the CI transactions are loaded to CMDB from TWA.

CIs are Not Displayed in the ServiceDesk-CMDB Data Repository

Symptom:

I am unable to view the CIs in the ServiceDesk-CMDB data repository. However, I can view the CIs listed for the default CA Catalyst data repository in the USM Web View.

Solution:

You are unable to view the CIs in the ServiceDesk-CMDB data repository because you have enabled TWA while installing the CA SDM Connector. Complete the following steps to disable the TWA:

1. Browse to the following directory from the https://<registryserver:port>/registry/carbon/admin/login.jsp URL:
   `\topology\physical\<SDM Connector Server>\modules\configuration\ServiceDeskManagerConnector.xml`
2. Click Edit as Text.
3. Replace the value for enable_twa property to **false**.
4. Click Save Content.
5. Restart the CA Catalyst Connector container service on the CA SDM server.

> **NOTE**
> If you do not want to disable TWA, follow the CMDB process of verifying and loading the CIs to CMDB using GRloader utility, provided by CMDB.

Limit the Data Exported to CMDB

Symptom:

I want to limit the data exported to CMDB.

Solution:

Follow these steps:

1. Use the CA Catalyst Profiles in CA Configuration Automation to determine which attributes are exported. The attributes include Network Discovery Gateway CIs, Hardware or Storage or Network Details components, and Relationship types. The default profile exports all the NDG CIs, each of the Hardware or Software or Network Details components, and all relationship types.
2. Use the Catalyst Jobs or Management Profiles to determine which servers or services, and blueprints are exported.

Troubleshooting CA CMDB and CA Configuration Automation Integration

This topic contains the following information:

Verify that the CA Configuration Automation Connector Installed Successfully

Symptom:

I want to verify that the CA Configuration Automation Connector started properly.

Solution:

Follow these steps:

1. Check if there are any errors in the Catalyst_CCAConnector_InstallDebug.log file or CatalystInstallDebug.log file.

   NOTE
   These files are located in the %TEMP% directory.
2. If either of the log files has errors, open a support issue.
3. If there are no errors in either of the log file, complete the following steps:
   a. Open the CA Catalyst Server Registry UI using the following URL:
      `https://<registryserver:port>/registry/carbon/admin/login.jsp`
   b. Log in to the CA Catalyst Registry UI and browse to the following directory:
      `\topology\physical\<<ACM>_Connector NODE>`
   c. Check if connector-modules.xml and startup.properties files are present in the directory.
   d. If either connector-modules.xml or startup.properties file is missing from the directory, then the CA Configuration Automation Connector installation has failed.
      - Uninstall and reinstall the CA Configuration Automation Connector.

Verify that the CA Configuration Automation Connector Started Properly

Symptom:

I want to verify that the CA Configuration Automation Connector started properly.

Solution:

Follow these steps:

1. Open the CA Catalyst Admin UI using the following URL:
   `http://<CA_Catalyst_Server>:<port>/adminui`
2. Log in to the CA Catalyst Admin UI.
3. Click CA Configuration Automation Container node from the left panel.
   The CA Configuration Automation Container node is expanded to display the CA Configuration Automation Connector and status of the CA Configuration Automation Connector.
   – If the status of the CA Configuration Automation Connector is RUNNING, then the CA Configuration Automation Connector has started properly.

   NOTE
   If you have just started the CA Configuration Automation Connector, wait for the CA Configuration Automation Connector nodes to move to the Running state before you perform any operation.

cca.log File Displays Exception Traces

Symptom:

The cca.log file shows errors with exception traces.

Solution:

Review and identify a resolution for the exceptions that appear in the cca.log file. The following possible exceptions may appear in the cca.log file:

- CA Configuration Automation Server Connection Exception
- Database Exception
- JVM Port Binding Exception

Server Connection Exception

Complete the following steps to remove the CA Configuration Automation Server Connection Exception found in the cca.log file:

1. Check if the CA Configuration Automation server is online and can be pinged from the CA Configuration Automation Connector Server.
2. Check if there is any firewall that is blocking access to the necessary CA Configuration Automation server port from the CA Configuration Automation Connector server.

   WARNING
   If you have restarted the CA Configuration Automation server while starting the CA Configuration Automation Connector, restart the CA Catalyst container service on the CA Configuration Automation Connector server.
3. If the password of the CA Configuration Automation user has changed since the installation of the CA Configuration Automation Connector, update the password parameter in the CCAConnector.xml file from the CA Catalyst server Registry UI. Complete the following steps:
   a. Go to the following directory:
      `<ROSE>_HOME\ tools\encrypt`

      NOTE
      Ensure that java.exe in present in the path.
   b. Run encrypter.bat <New Password> command and copy the encrypted string output from the command.
   c. Open the following URL:
      `https://<registryserver:port>/registry/carbon/admin/login.jsp`
   d. Browse to the following directory:
      `\topology\physical\<Configuration_Automation_Connector_Server>\modules\configuration\CCAConnector.xml`
   e. Click Edit as Text.
   f. Replace the value for db.password with the value copied from the encryptor utility.
   g. Click Save Content.
   h. Restart the CA Catalyst Container service and verify it again.

Database Exception

Complete the following steps to remove the Database Exception found in the cca.log file:

1. Check if the database server is online and can be pinged from the CA Configuration Automation Connector server.
2. Check if there is any firewall that is blocking access to the necessary database ports from the CA Configuration Automation Connector server.
3. If the DB password has changed since the installation of the CA Configuration Automation Connector, update the db.password parameter in the CCAConnector.xml file from the CA Catalyst server Registry UI. Complete the following steps:
   a. Go to the following directory:
      `CATALYST_HOME\ tools\encrypt`

      NOTE
      Ensure that java.exe in present in the path.
   b. Run encrypter.bat <New Password> command and copy the encrypted string output from the command.
   c. Open the following URL:
      `https://<registryserver:port>/registry/carbon/admin/login.jsp`
   d. Browse to the following directory:

```
\topology\physical\<CA_Configuration_Automation_Connector_Server>\modules\configuration
\CCAConnector.xml
```

e. Click Edit as Text.
f. Replace the value for db.password with the value copied from the encryptor utility.
g. Click Save Content.
h. Restart the CA Catalyst Container service and verify it again.

JVM Port Binding Exception

Complete the following steps to remove the JVM Port Binding Exception found in the cca.log file:

1. Check if any process is using the port that is used for the notification_listen_port parameter during the CA Configuration Automation Connector installation.
2. If a port conflict exists after the installation, update the parameter in the CCAConnector.xml file from the CA Catalyst server Registry UI. Complete the following steps:
   a. Open the following URL:
      ```
      https://<registryserver:port>/registry/carbon/admin/login.jsp
      ```
   b. Browse to the following directory:
      ```
      \topology\physical\<CA_Configuration_Automation_Connector_Server>\modules\configuration
      \CCAConnector.xml
      ```
   c. Click Edit as Text.
   d. Replace the value of the notification_listen_port with the value that does not conflict with other ports on the CA Configuration Automation server.
   e. Click Save Content.
   f. Restart the CA Catalyst Container service and verify it again.

Verify that the CA Configuration Automation Connector Sent Data to the CA Catalyst Server Successfully

Symptom:

I want to verify that the CA Configuration Automation Connector sent data to the CA Catalyst Server successfully.

Solution:

Follow these steps:

1. Open the Catalyst USM web view using the following URL:
   ```
   http://catalystserver:8080/ca-rest/browse/type?mdr=all
   ```
2. Select CMDB-View from Data Source and check if the CIs are listed on the page.
3. If the CIs are not listed in the CMDB-View, complete the following steps:
   a. Ensure that the Catalyst Container Service of the CA Configuration Automation Connector is restarted when you started the CA Configuration Automation server.
   b. Check if the time settings on the CA Catalyst server and the CA Configuration Automation Connector server nodes are same to avoid connectivity issues.
   c. If you used the Catalyst Job to push the data from CA Configuration Automation Connector, verify completion of the Catalyst Job. This verification is applicable for the Management Profile based integration. If the Catalyst Job is complete, the log view on the CA Configuration Automation server displays the Job started and Job finished messages.
   d. Match the Check Sum Table with the projections of the CA Configuration Automation Connector.

Match the Check Sum Table

Match the Check Sum table with the projections of the CA Configuration Automation Connector. The CA Catalyst Persistence Store maintains the projections of the CA Configuration AutomationConnector for every CI sent to the CA Catalyst server.

Follow these steps:

1. Run queries to find the failed CIs.

   NOTE
   You must adapt the queries with the corresponding database names, passwords, and so on. The Database may also need to configure cross computer DB access for CA Configuration Automation and CA Catalyst databases.

   – Use the following query in SQL to identify the failed CIs:

```
select ci_id = CASE CHARINDEX( '|', ci_typ)
WHEN '0' THEN ci_id
ELSE (ci_id + substring( ci_typ, CHARINDEX( '|', ci_typ) + 1, len(ci_typ)))
END collate sql_latin1_general_cp1_ci_as from acm_catlst_ci_cksum
EXCEPT
select c_mdrelementid from [catalystdb].[dbo].t_ci_detail where c_mdrproduct = 'CA:00033'
```

   – Use the following query in ORACLE to identify the failed CIs:

```
CREATE PUBLIC DATABASE LINK catalystdb CONNECT TO catalystadmin IDENTIFIED BY <Password> USING '//
<DBServer>/<DB SID>';
select CASE instr( ci_typ, '|')
HEN 0 THEN cast( ci_id as NVARCHAR2(36) )
ELSE concat( cast( ci_id as NVARCHAR2(36) ) , substr( ci_typ, instr( ci_typ, '|') + 1, length(ci_typ)))
END as c_mdrelementid
from acm_catlst_ci_cksum
minus
select c_mdrelementid from t_ci_detail@catalystdb where c_mdrproduct = 'CA:00033'
```

2. Check the invalidCIs.log file from the CATALYST_HOME\container\data\log directory.

   NOTE
   For each CI ID output that is received from the query, there exists a corresponding dump for the CI in the invalidCIs.log file.

   – If either the invalidCIs.log file is empty or the CI ID is not present in the file, then the CI synchronization from CA Configuration Automation Connector to the CA Catalyst server is still in progress.

   NOTE
   The synchronization and reconciliation activities on the CA Catalyst server are not instantaneous and results in the CIs to take time to get across to CMDB.

Failed to Select the Items to Synchronize Between CA Configuration Automation and CMDB

Symptom:
I am unable to choose the items to synchronize between CA Configuration Automation and CMDB. In the old CA Configuration Automation or CMDB CA Business Intelligence-boxed integration I could choose the CI types.
Solution:
A combination of the attribute profile and Catalyst Job or management profile helps you filter data. This filtering mechanism has no similarity with the previous CA Business Intelligence integration where the filter was using CI types.

Troubleshooting CA Virtual Analyst Integration with MS Teams

Troubleshooting CA Virtual Analyst Integration with MS Teams.

No chat responses in MS Teams chatbot

Problem: No chat responses in MS Teams chatbot.
Solution: Chrome provides the following flags to disable this constraint:

1. Ensure all passwords in the .env file are encoded.
2. If the Reverse proxy/Load balancer is configured for the integration, ensure it is receiving messages and redirecting them to the Bot server.
3. Check if the MongoDB server is running and configured in the .env file.
4. The allowApiImpersonation flag should be set in SDM under **Administration**> **Xflow Interface**> **General configuration.**

Users are unable to receive SDM notifications in MS Teams chatbot

Problem: Users are unable to receive SDM notifications in MS Teams chatbot.
Solution: Check availability of ActiveMQ server and Webhook servers.

Unable to receive SDM notifications in chatbot

Problem: Receiving chat responses like "Failed to initiate Chat Session" or "Please contact your Administrator".
Solution: Configure the VA_SERVER_TIMEOUT in the .env file to a greater value (like 15 seconds).

Additional Resources

This section contains additional resources that you help you with using CA Service Management.

- Education and Social Collaboration
- Content Pack for ITIL CA Service Desk Manager
- CA SDM Connector Glossary
- Pre-Built CA Process Automation Workflows

Education and Social Collaboration

Learn about education and available training materials, featured case studies, Video tutorials from CA Technologies and our partners. Take advantage of social media channels (Community, blogs, webcasts).

Learning Paths

The CA Service Management Learning Paths contains recommended courses, by role,. These roles include Implementer, Administrator, and the End User. Courses are updated based on impact from release and demand, which is why you may see courses from previous releases included in the current learning path.

For more information, see Learning Subscription and Contact Education.

Social Media Channels

These social media channels, including the Community, blogs, webcasts, BrightTalk, Twitter and YouTube videos are available to help you self-serve, collaborate, and connect:

Community

The CA Service Management Community is the place to share ideas, tips, information, insights, and more with your business peers and CA Technologies experts. The community provides a unique opportunity to network and help you maximize your software investment by tapping into a community of expertise, open 24/7.

Twitter (@Broadcom)

Follow Broadcom on Twitter for product updates and insights.

YouTube Videos

Subscribe to our YouTube channel to watch CA Service Management Videos.

Content Pack for ITIL CA Service Desk Manager

The Information Technology Infrastructure Library (ITIL®) is a set of books developed by the United Kingdom's Office of Government Commerce (OGC). The books describe an integrated, process based, best practice framework for managing IT services. The current version of ITIL is 2011.

The Content Pack for ITIL for the current release of CA Service Desk Manager provides additional content for CA Service Desk Manager (CA SDM) to fully support the following 15 IT Service Management ITIL 2011 edition processes:

- Availability Management
- Capacity Management
- Change Management
- Event Management
- Financial Management
- Incident Management
- IT Service Continuity Management
- Knowledge Management
- Problem Management
- Release & Deployment Management
- Request Fulfillment
- Service Asset & Configuration Management
- Service Catalog Management
- Service Level Management
- Service Portfolio Management

This document records the content and modifications that are included in the Content Pack. This package includes:

- Schema and object modifications
- New and updated analyst web forms (htmpl files)
- Updated help files (html)
- New CA Process Automation Manager workflows
- ITIL Reports
- New WebI and Crystal reports

> **WARNING**
>
> Review the list of schema and data updates included in this package to ensure that your CA SDM environment does not have any conflicts.

For example, if you have already created a “z” table or “z” field with the same name that the installer will configure, it is not recommended to run the installer. See the Schema Updates for information on the schema delivered with this package. Although CA SDM will ignore any schema conflicts deployed with the package, the forms delivered with the package will reference them. See New and Updated HTMPL Forms and Updated Help Files sections for a list of modified forms.
If you already have data in your system as specified in the ITIL Content Data section, the installer may create duplicates. See the ITIL Content Data section for information on what data will be loaded into the server to identify any possible duplication.

Schema Updates

The MDB will be modified to support additional data capturing for ITIL process support. Several fields were added to existing tables and new tables were added to the MDB. The installer will put the updated schema in place for you automatically. When the installer script is run, a schema_done file will be created. The schema_done file is used to identify whether the ITIL schema was added to the server. Flag files will be created under the $NX_ROOT\samples \ITIL_Content_17_4 folder.
A vanilla or out of the box installation or upgrade of CA SDM will not have any schema conflicts. If you have modified your environment, ensure that there are not conflicts with what you have created and what the content pack delivers.
The installer will automatically create schema modifications to the MDB. If matching schema is found in the form of a table or field name, the installer will ignore the attempted update by the content installer and will not modify the site defined field, table or schema definition.

The following updates to the MDB database schema were made. Note that the ‘TABLE’ name column represents the table name as defined in $NX_ROOT/site/ddict.sch. For more information on TABLE name, SQL Name, and Object Name, see the Data Element Dictionary.

TABLE	FIELD	TYPE	Summary of Modification
Change_Request	zcategory_init	STRING 12 REF Change_Category	Initial change category
	zchg_bus_cls	INTEGER REF zchg_bus_cls	Business Classification
	zisRelease	INTEGER REF Boolean_Table	RFC authorization for Release
	zowner	UUID REF ca_contact	Change Owner
	zscope	STRING 1024	Change Order Scope
	zurgency	INTEGER REF Urgency	Urgency of change
Call_Req	zcategory_init	STRING 30 REF Prob_Category	Initial category on the incident, problem or request.
usp_owned_resource	zDML_path	STRING 1024	DML Path
	zSPMLifecycleStage	INTEGER REF zSPMLifecycleStage	SPM Lifecycle Stage
	zactual_budget	INTEGER	Actual Budget
	zbiz_proc_supported	STRING 255	Business processes supported
	zbusiness_approver_group	UUID REF ca_contact	
	zbusiness_impact	INTEGER REF zbusiness_impact	Business Impact
	zbusiness_owners	UUID REF ca_contact	Business Owner
	zbusiness_users	UUID REF ca_contact	Business User
	zcharges_service	INTEGER	CI service charges
	zchg_bus_cls	INTEGER REF zchg_bus_cls	
	zci_cost_category	INTEGER REF zci_cost_category	CI cost category
	zcost_biz_units	STRING 255	Business units where costs should be allocated
	zcost_hw_maintenance	INTEGER	Hardware maintenance cost
	zcost_hw_purchases	INTEGER	Hardware purchases cost
	zcost_personnel	INTEGER	Personnel costs
	zcost_service	INTEGER	Service Cost
	zcost_sw_licenses	INTEGER	Software licenses cost
	zcost_sw_maintenance	INTEGER	Software maintenance cost
	zcost_utility	INTEGER	Utility costs
	zcurrent_monthly_service_cost	INTEGER	Current Monthly Service Cost
	zcurrentAvailabiity	INTEGER	Current availability
	zcurrentCapacity	INTEGER	Current capacity
	zcurrentPerformance	INTEGER	Current performance

	zcurrent_monthly_service_cost	INTEGER	Current monthly service cost
	zest_budget	INTEGER	Estimated budget
	zpost_program_ROI	INTEGER	Post-Program ROI
	zpre_program_ROI	INTEGER	Pre-Program ROI
	zprevAvailability	INTEGER	Previous availability
	zprevCapacity	INTEGER	Previous capacity
	zprevPerformance	INTEGER	Previous performance
	zprice_service	INTEGER	Service price
	zrevenue_service	INTEGER	Service revenue
	zrelease_package_num	INTEGER	Release package number
	zservice_description	STRING 1024	Service description
	zstakeholders_group	UUID REF ca_contact;	Stakeholders group
	ztechnical_approver_group	UUID REF ca_contact;	Technical approver group
	zvalue_prop	STRING 1024;	Value proposition
usp_organization	zcost_center	SREL cost_cntr	Cost Center
Service_Desc	zKPI1	STRING 255	KPI
	zKPI2	STRING 255	KPI
	zKPI3	STRING 255	KPI
	zagreement_date	LOCAL_TIME	Agreement Date
	zfield_date	LOCAL_TIME	Agreement Start
	zfield_date2	LOCAL_TIME	Agreement End
	zfield_int	INTEGER	Additional field
	zfield_long	STRING 1024	Scope
	zfield_small	STRING 255	Additional field
	zquestion1	STRING 255	Locations Included
	zquestion2	STRING 255	Definitions
	zquestion3	STRING 255	Responsibilities and Dependencies
	zquestion4_short	STRING 50	Additional field
	zquestion5_short	STRING 50	Additional field
	ztype	INTEGER REF zservice_type_class	Service Type Classification
zSPMLifecycleStage	description	STRING 255	Description
	sym	STRING 40	Symbol
zbusiness_impact	description	STRING 255	Description
	delete_flag	INTEGER REF Active_Boolean_Table	Active/Inactive
	sym	STRING 60	Symbol
zci_cost_category	description	STRING 255	Description

	sym	STRING 40	Symbol
zservice_type_class	description	STRING 255	Description
	Sym	STRING 60	Symbol
zchg_bus_cls	Description	STRING 255	Description
	Delete_flag	INTEGER REF Active_Boolean_Table	Active/Inactive
	Sym	STRING 60	Symbol

New and Updated HTMPL Forms

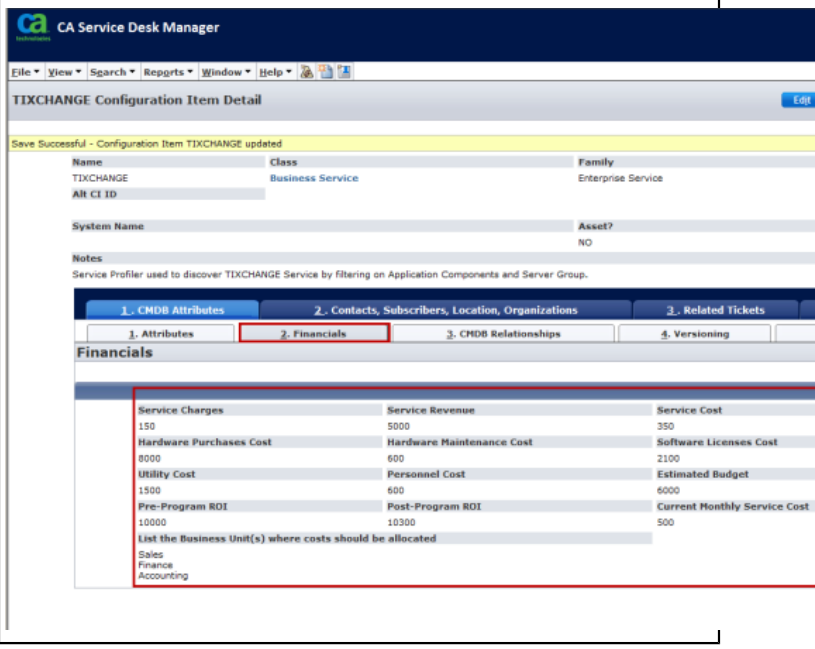
Several HTMPL forms were modified to support the ITIL Content. The installer will automatically take a backup of your existing site defined htmpl files before doing a copy and replace of the files noted in the table below. Any matching site defined htmpl files located in $NX_ROOT\site\mods\www\htmpl\web\analyst will be copied to $NX_ROOT\samples \ITIL_Content_17_4\<platform>\backups\<date-time>, where <platform> is either nt or unix. If you do not want to use the updated htmpl files that the installer will provide, you will need to delete the ITIL Content version from $NX_ROOT\site \mods\www\htmpl\web\analyst and replace the it with your original file located in the backups folder. Alternately, you can use the Web Screen Painter to update the ITIL Content version to include your previous updates. The original out of the box htmpl files will not be altered.
The following HTMPL forms were modified as part of the extended ITIL content support.

Form Name	Modification Made
cmdb_detail.htmpl	Field added: Business Criticality (Zbusiness_impact)
cmdbNotebook.htmpl	Contacts tab renamed to “Contacts, Subscribers”
detail_chg.htmpl	New function on form to copy category to zcategory_init Fields added: • Business Classification (zchg_bus_cls), • Urgency (zurgency) • Authorized for Release (zisRelease)
detail_cr.htmpl	New function on form to copy category to zcategory_init
detail_in.htmpl	Exposed Created Via (created_via.sym) ootb field New function on form to copy category to zcategory_init
detail_org.htmpl	Field added: Cost Center (zcost_center)
detail_pr.htmpl	Exposed Symptom (symptom_code), Major Problem (major_incident) and Created Via (created_via) ootb fields on form New function on form to copy category to zcategory_init
detail_sdsc.htmpl	Fields added: • Record Type (ztype) • Agreement Start Date (zfield_date) • Agreement End Date (zfield_date2) Tab added with three sub tabs: • Related Contacts (displays all contacts with this ST) • Related Configuration Items (displays all CIs with the ST) • Additional Content: link to zsdsc_tab.htmpl
detail_zbusiness_impact.htmpl	New form with fields: Symbol, Description fields
detail_zchg_bus_cls.htmpl	New form with fields: Symbol, Description fields, last modified by/ date

in_relreq_tab.htmpl	New button: Link Incidents, Updated list to show cr and not in
list_chg.htmpl	Search Fields added: • Business Classification (zchg_bus_cls), • Urgency (zurgency) • Authorized for Release (zisRelease) List updated to include Authorized for Release
list_nr.htmpl	Search Fields added: • Business Criticality (zbusiness_impact) • SPM Lifecycle Stage List updated to include Business Criticality
list_sdsc.htmpl	Search Field added: • Record Type (ztype) List updated to include Type
list_zbusiness_impact.htmpl	New form: Symbol, Description fields
list_zchg_bus_cls.htmpl	New form: Symbol, Description fields
menubar_sd.htmpl	Added to File menu item: New Knowledge Document
nr_cmdb_invothx_tab.htmpl	Fields added: • SPM Lifecycle Stage (zSPMLifecycleStage)
nr_contact_tab.htmpl	Fields added: • Business Owner Group (zbusiness_owners) • Business Users Group (zbusiness_users)
nr_financials_tab.htmpl	New form with fields: zcharges_service, zrevenue_service, zcost_service, zprice_service, zcost_hw_purchases, zcost_hw_maintenance, zcost_sw_licenses, zcost_sw_maintenance, zcost_utility, zcost_personnel, zest_budget, zactual_budget, zpre_program_ROI, zpost_program_ROI, zcurrent_monthly_service_cost, zci_cost_category, zcost_biz_units
nr_inv_tab.htmpl	Field added: • Release Package Number (zrelease_package_num)
nr_loc_tab.htmpl	Field added: • Definitive Media Library Path (zDML_path)
nr_serv_tab.htmpl	Fields added: • Previous Availability (zprevAvailability) • Previous Capacity (prevCapacity) • Previous Performance (prevPerformance) • Current Availability (zcurrentAvailability) • Current Capacity (zcurrentCapacity) • Current Performance (zcurrentPerformance) • Service Description (zservice_description) • Service Value Proposition (zvalue_prop) • Business Processes Supported (zbiz_proc_supported)
xx_prop_tab.htmpl	Field added: • zscope

zsdsc_tab.htmpl	Fields added: • Location included (zquestion1) • Definitions (zquestion2) • Responsibilities and Dependencies (zquestion3) • Scope (zfield_long) • KPI (zKPI1) • KPI (zKPI2) • KPI (zKPI3)

Screen Shots

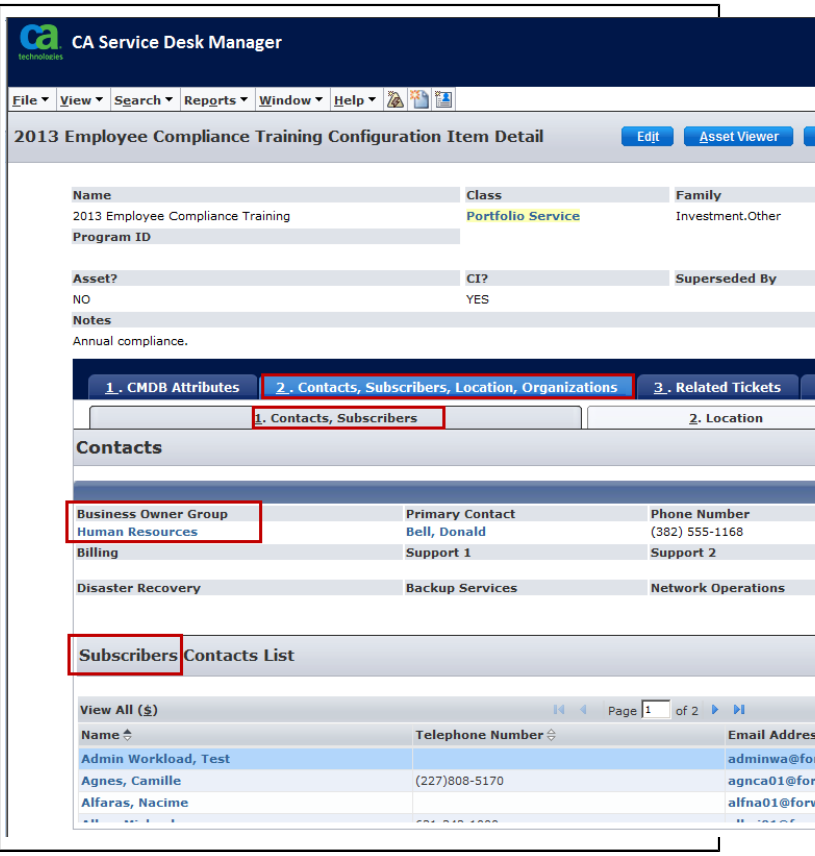
Name	Form View
Configuration Item (cmdb_detail.htmpl) New Field Added: • Business Criticality New Tab Added: • Financials (all fields on Financials tab are new)	

Configuration Item Notebook Tab 2 (cmdbNotebook.htmpl)
Rename Tab 2

- Renamed Contacts to Contacts, Subscribers

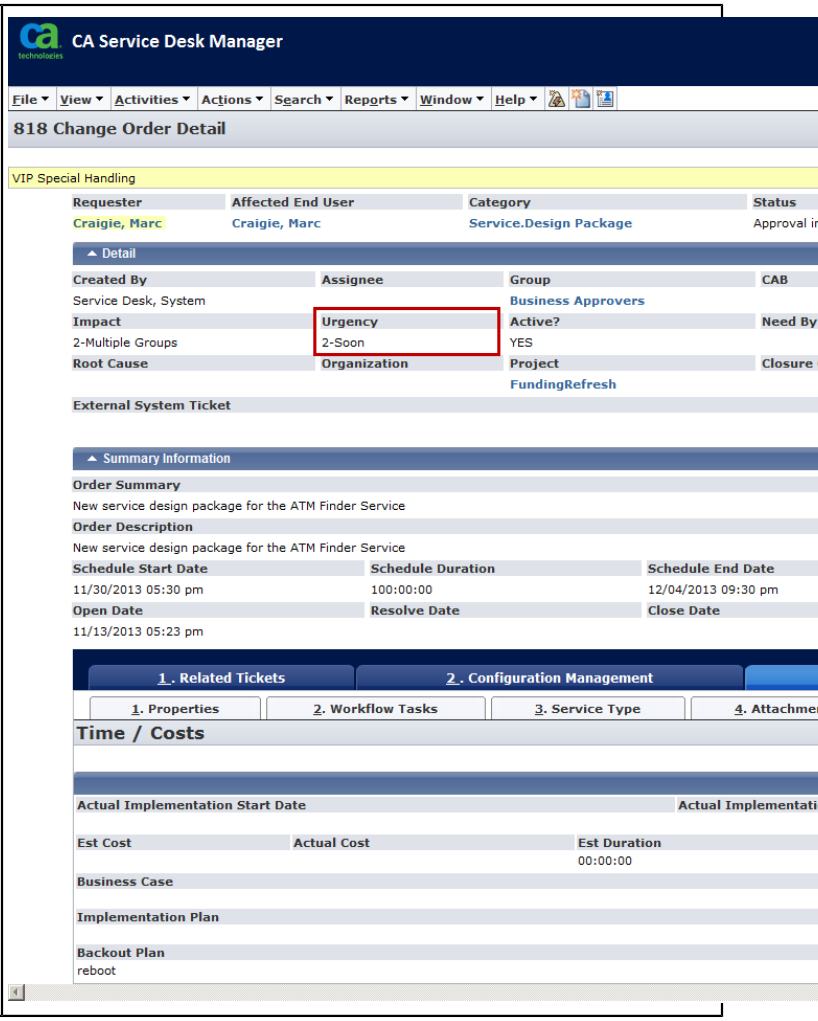
New Fields Added:

- Business Owner Group
- Business Users Group



Change Order Detail (detail_chg.htmpl)
New Fields Added:

- Business Classification
- Urgency
- Authorized for Release



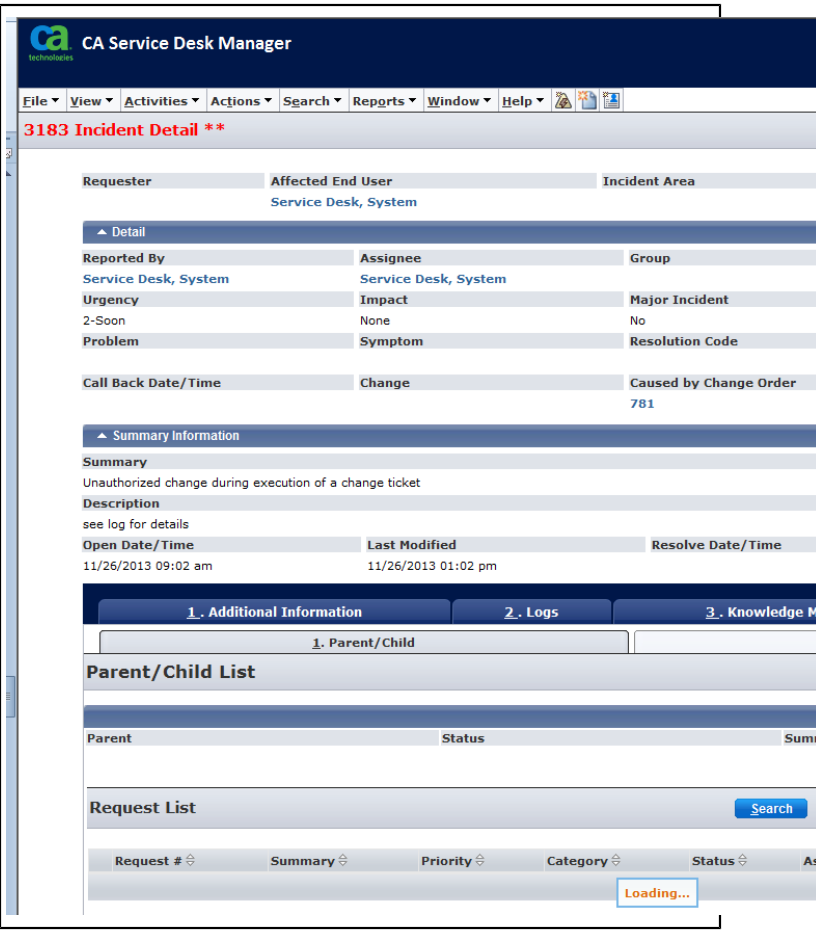
Incident Detail (detail_in.htmpl)
Existing Field Exposed:

- Created Via

New button:

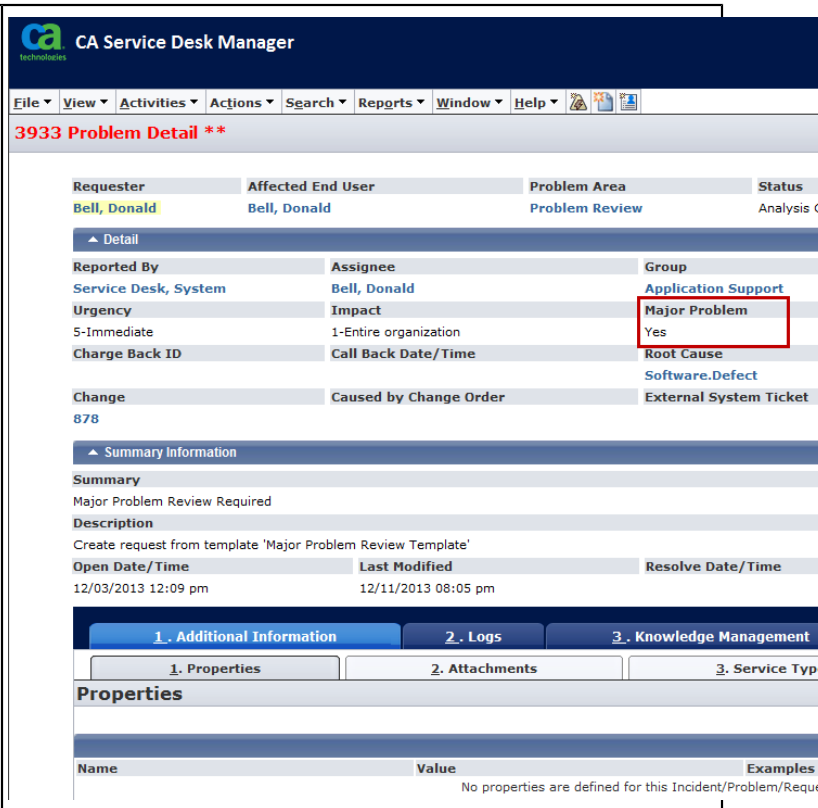
- Link Requests

Updated list query to pull the Request List to show both requests and incidents



Problem Detail (detail_pr.htmpl)
New Field added:

- Symptom
- Created Via
- Major Problem

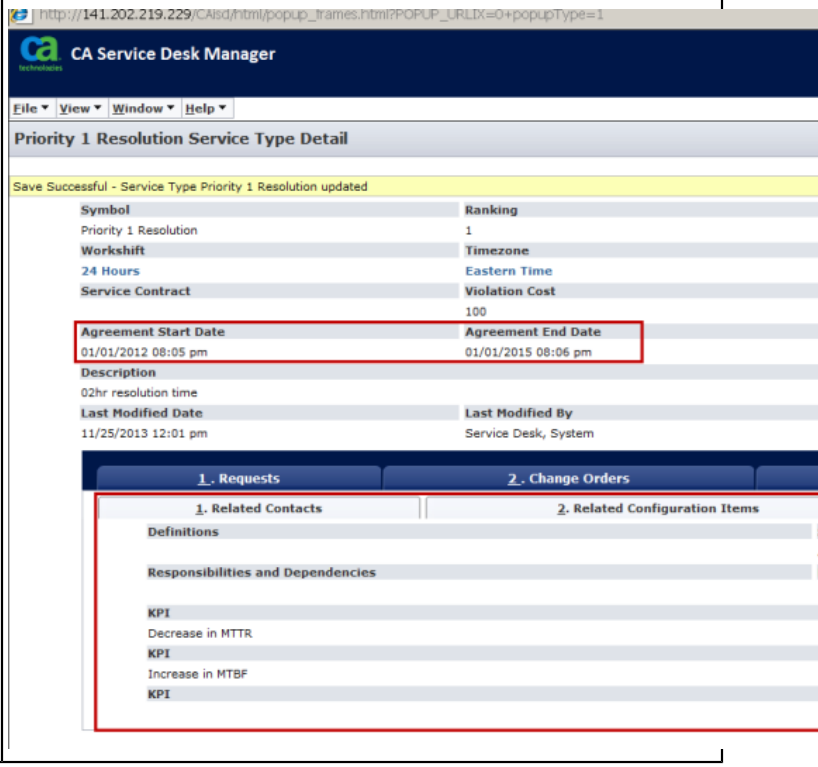


Service Type Detail (detail_sdsc.htmpl)
New Fields added:

- Record Type Agreement Start Date
- Agreement End Date

New Scope Tab Added:

- Related Contacts Tab - displays all contacts with this ST
- Related Configuration Items Tab - displays all CIs with the ST
- Additional Content Tab: link to zsdsc_tab.htmpl

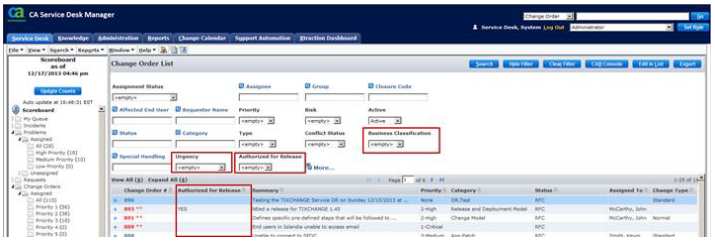


Change Order List (list_chg.htmpl) Search Fields added:

- Business Classification
- Urgency

Authorized for Release Search List updated:

- Authorized for Release

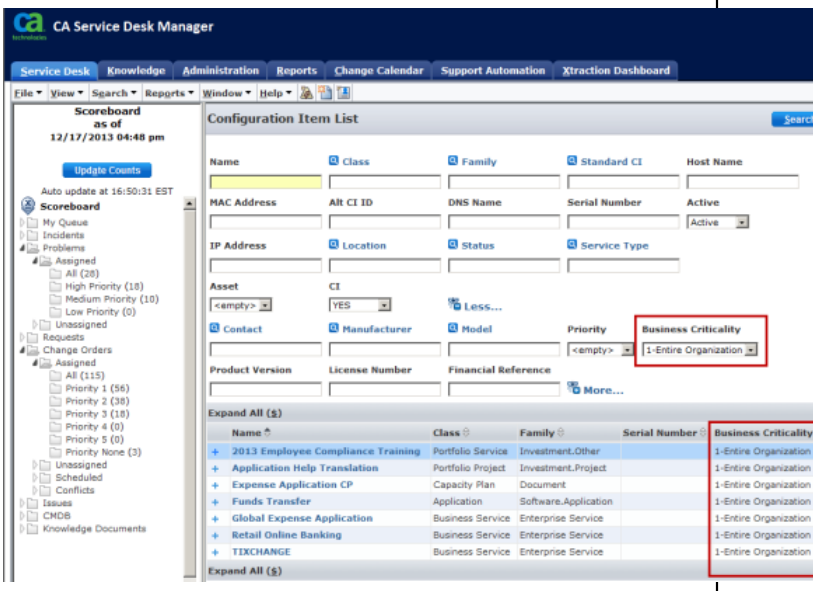


Configuration Item List:
Search Fields added:

- Business Criticality

Search List updated:

- Business Criticality
- Status

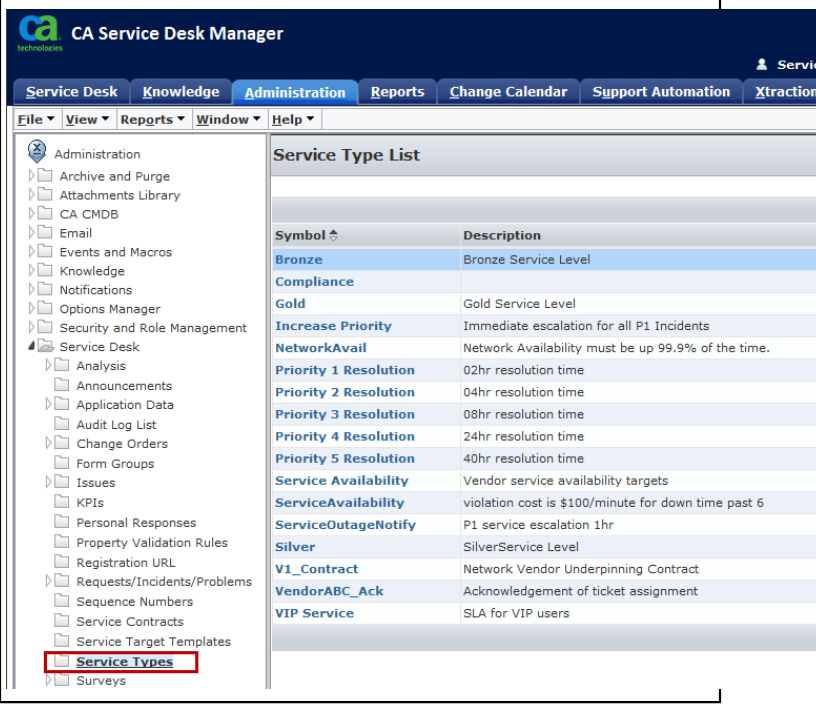


Service Type List (list_sdsc.htmpl) Search Field added:

- Record Type

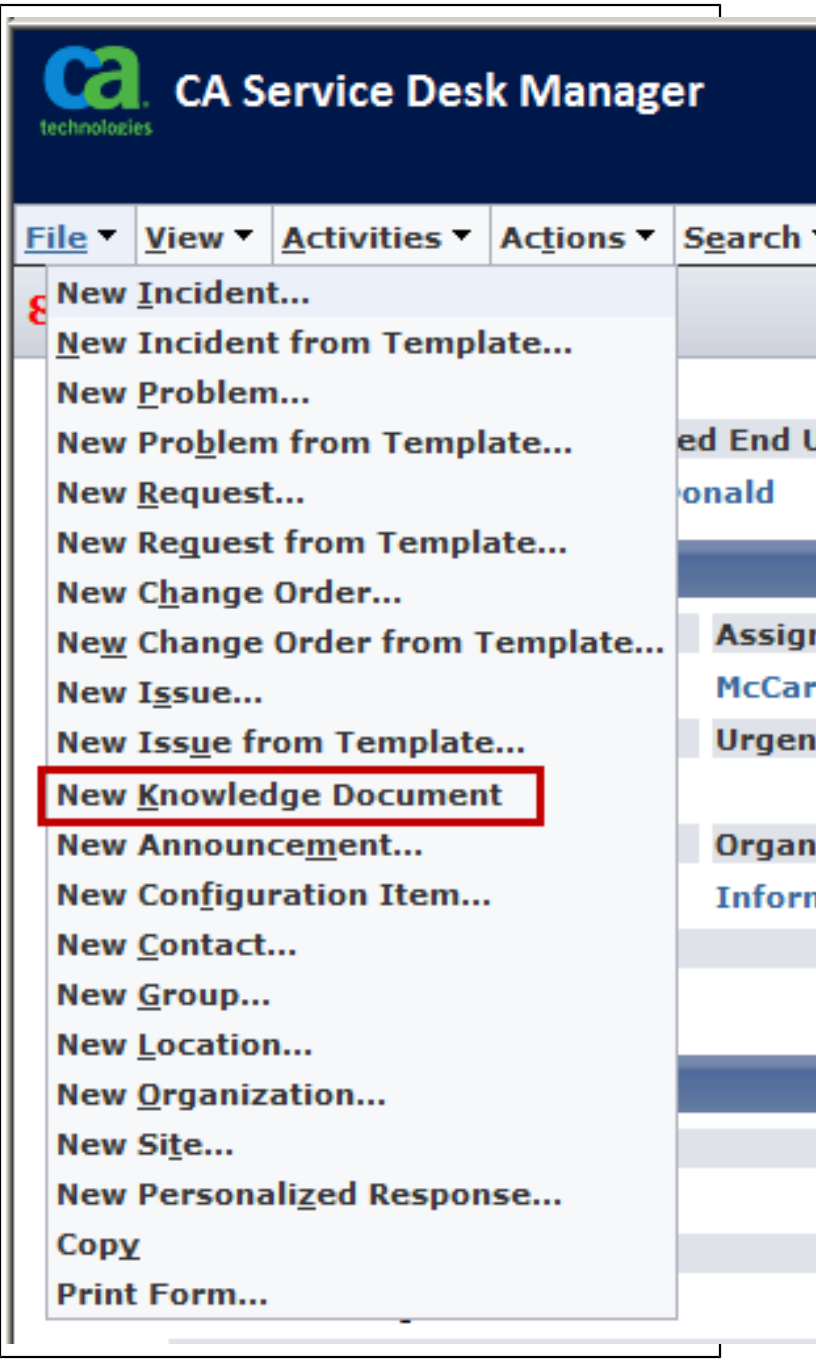
Search List updated:

- Type



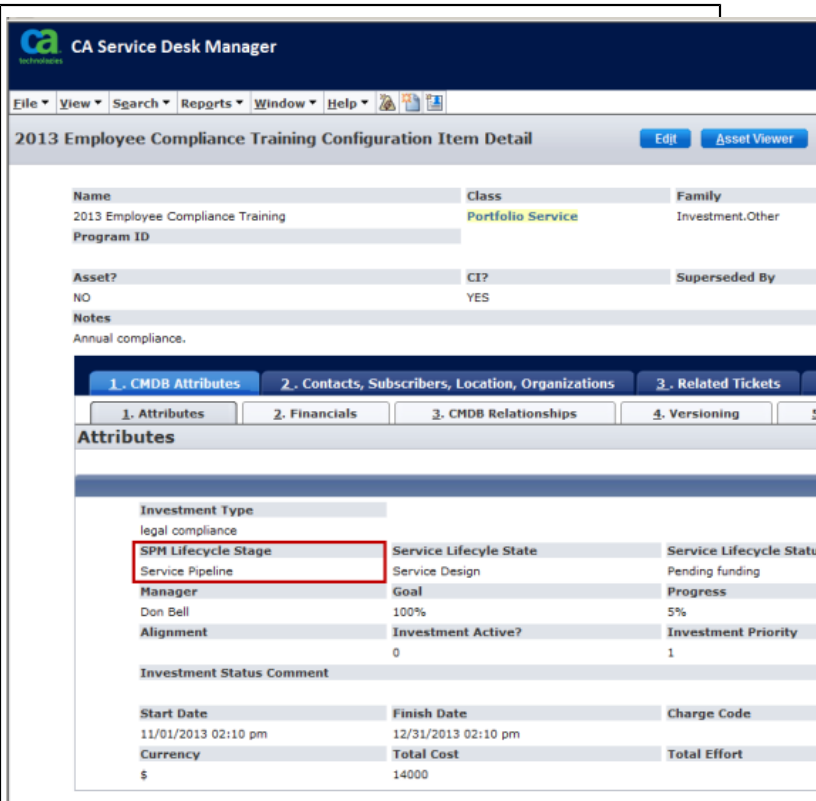
Menu Bar (menubar_sd.htmpl)
New File Menu:

- New Knowledge Document



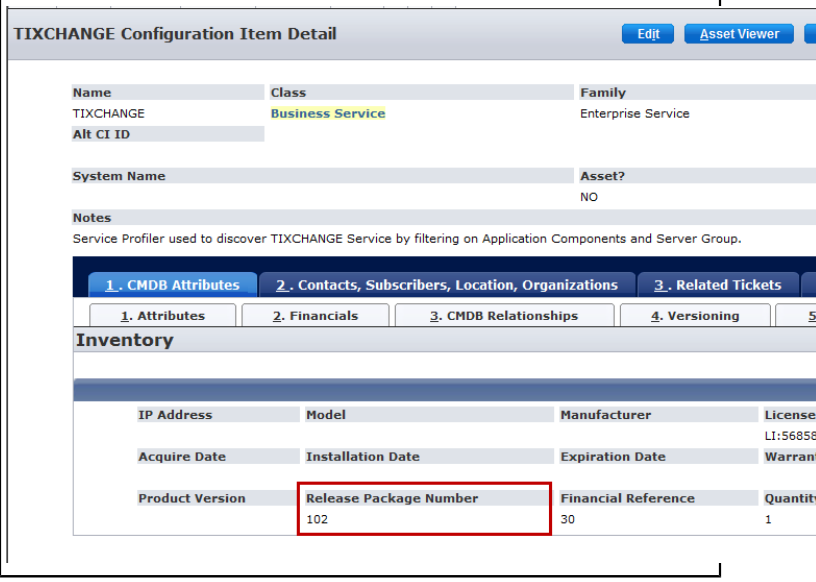
Investment.Other Attributes Tab (nr_cmdb_invothx_tab.htmpl)
Field Added:

- SMP Lifecycle Stage



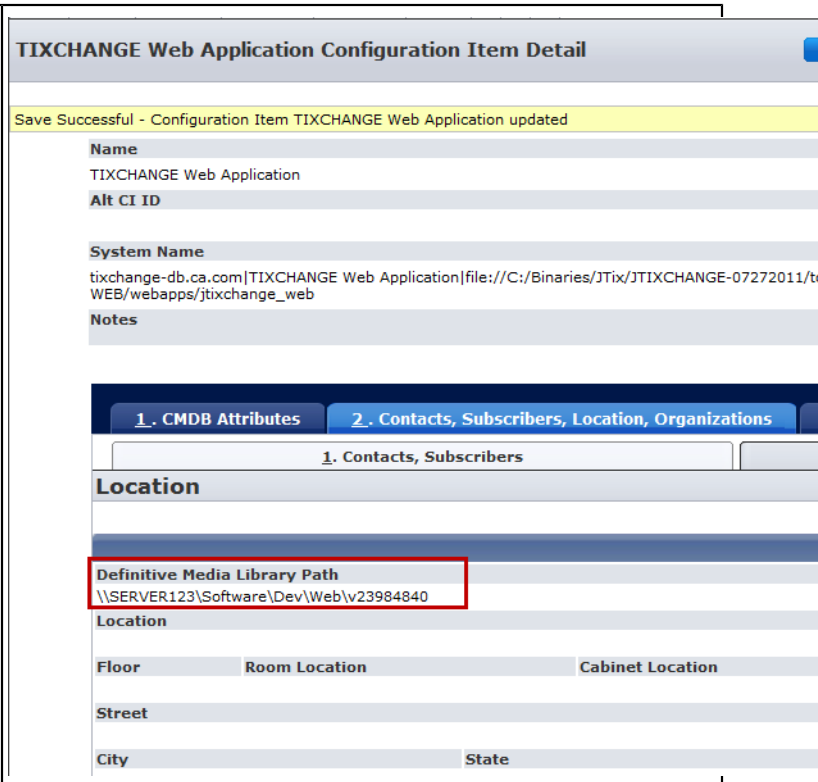
CI Inventory Tab (nr_inv_tab.htmpl)
Field added:

- Release Package Number



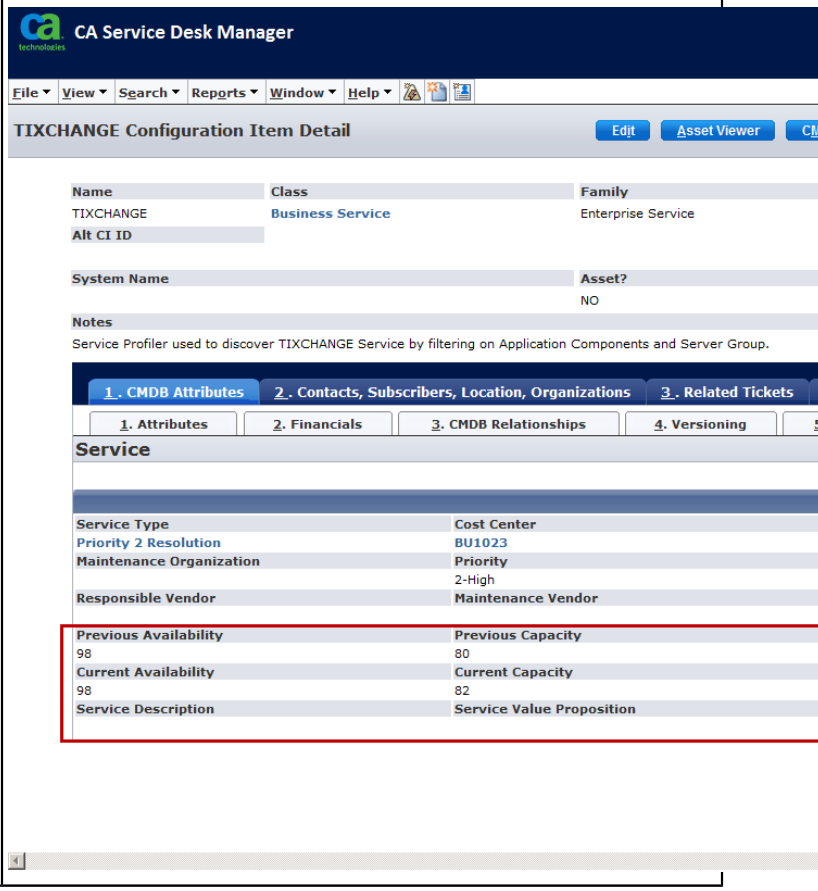
CI Location Tab (nr_loc_tab.htmpl) Field added:

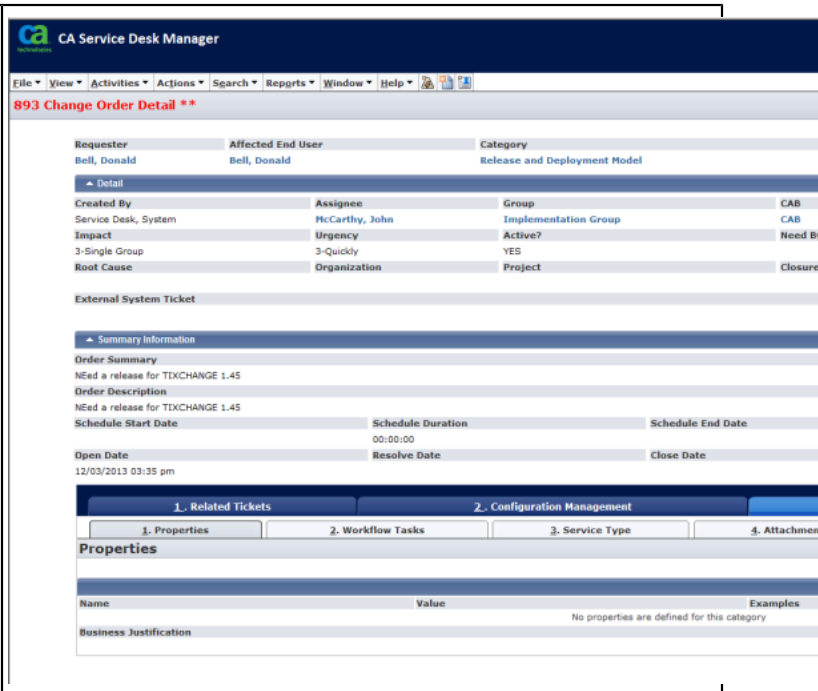
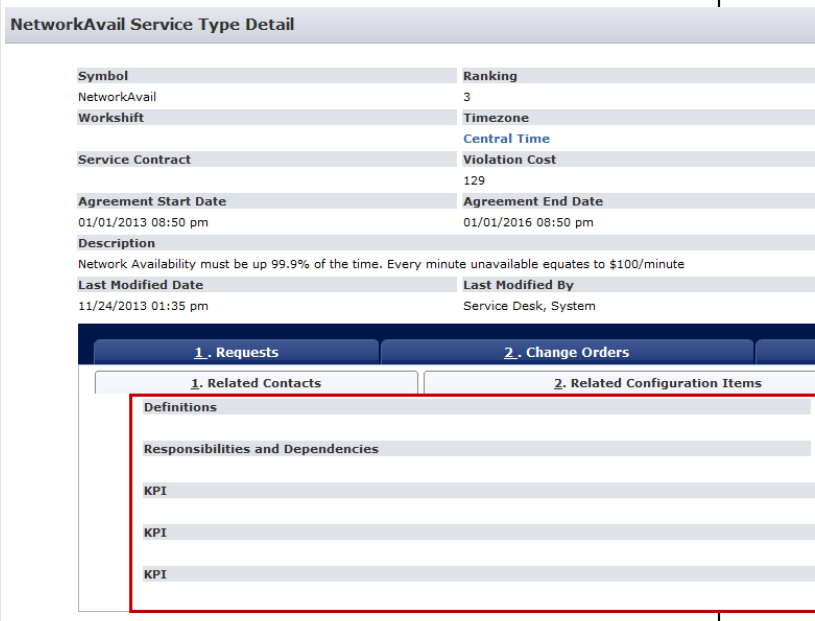
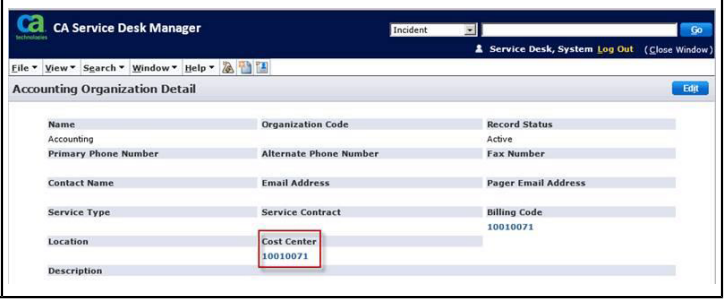
- Definitive Media Library Path



CI Service Tab (nr_serv_tab.htmpl)
Fields added:

- Previous Availability
- Current Availability
- Previous Capacity
- Current Capacity
- Previous Performance
- Current Performance
- Service Description
- Service Value Proposition
- Business Processes Supported



Properties Tab xx_prop_tab.htmpl Field added: zscope	
Service Type Additional Content Tab (zsdsc_tab.htmpl) Fields added: • Locations included • Definitions • Responsibilities and Dependencies • Scope • 3 KPI fields	
Organization Detail (detail_ort.htmpl) Field Added: Cost Center	

ITIL Content Data

Content Data

When the installer script is run, a dataLoad_done file will be created. The dataLoad_done file identifies if the ITIL content was loaded on the server. This will ensure that if the script is re-executed that duplicate data will not be loaded into the system. It also allows CA Support to track if the content exists or not. Flag files will be created under the $NX_ROOT \samples\ITIL_Content_17_4 folder.

The following table shows what tables will have data loaded into them and what the values of that data will be. Before running the install script, ensure that the values listed below do not already exist in your environment. For example, if you already have a Change Category titled “Service.Improvement Plan” you may not want to create a duplicate value in your database when the install script is run. You can manually modify the data that gets loaded into the application by editing the data files located in the ITIL_Content_17_4\files\data folder.

Description: Table	Data File(s)	Values
Activity Type Association : Act_Type_Assoc	Act_Type_Assoc.dat	RFC Authorized for Release
Aliases attr_alias	attribute_alias.dat	map_sdsc_servicetype_class_description map_sdsc_servicetype_class_symbol map_sdsc_violation_cost request_created_via request_sla_macro_predicted category_init (for chg, cr, in, pr) zServiceType configuration_item_financial_number
Macro Conditions: Atomic_Condition	Atomic_Condition.dat Atomic_Cond_Deref.dat	Must have a backout plan to go from Approval in Progress -> Approved status. RFC is Authorized for Release
Categories Change_Category	Change_Category.dat	Change Model DR.Test Release and Deployment Model Service.Design Package Service.Design Package.New Service.Improvement Plan Service.Quality Plan
Change Status Change_Status	Change_Status.dat	Build (zBLD) Test (zTST) Deployed (zDPLY) Incomplete (zINCPL)
Change Status Transitions chg_trans	chg_trans_input.dat chg_trans_upd.dat chg_trans_Deref.dat	Approval in progress -> Approved (updated) Build->Test Test->Deploy Deploy->Closed Implemented->Closed Implementation in progress->Implemented Implementation in progress->Deployed RFC->Build Build->Rejected Build->Incomplete Test->Approval in progress

CI Status Codes: ca_resource_status	ca_resource_status.dat	Analyzed Approved Chartered Defined In Development Live Pipeline Retired
Incident, Problem, and/or Request Status: Cr_Status	cr_status.dat	Pending Major Problem Review Pending Fulfillment
Request Status Transition cr_trans	cr_trans.dat	Approved (PRBAPP)-->Pending Fulfillment (zPendRul)
Knowledge Categories: O_INDEXES	O_INDEXES_deref.dat O_INDEXEX_input.dat O_INDEXES_load.dat	Availability Plans Capacity Plans Compliance CSI Register Disaster Recovery Plans DML Known Errors Release and Deployment Service Lifecycle Knowledge Articles Service Quality Plan
Problem Status Transition pr_trans	pr_trans_input.dat pr_trans_Deref.dat	Pending Major Problem Review to Closed Pending Major Problem Review to Cancelled Pending Major Problem Review to Fixed Pending Major Problem Review to Known Error Pending Major Problem Review to Problem Closed Fixed to Pending Major Problem Review Analysis Complete to Pending Major Problem Review Resolved to Pending Major Problem Review
Indicent, Problem, Request Categories: Problem_Category	Prob_Category.dat	Incident Model Knowledge.Broken Link Problem Model Problem Review User.Complaint User.Compliment Service.Improvement Plan Service.Quality Plan Service.Development Service.Outage Service.Portfolio
Macros: Spell_Macro	spell_macro.dat	'Backout Plan <> NULL' RFC Authorized for Release
Survey Answers : Survey_Answer_Template	ITIL_Survey_Answer.dat Survey_Answers_Deref.dat	view ITIL_Survey_Answer.dat file for details

Survey Questions : Survey_Question_Template	ITIL_Survey_Question.dat Survey_Questions_Deref.dat	view ITIL_Survey_Question.dat file for details
Survey: Survey_Template	Survey_Step1_pdm_load.dat	Request Fulfillment Survey Service Desk Quality Survey
Role: Usp_role	usp_role.dat	Release and Deployment Management Role
Link Roles and Reports: usp_role_web_form	usp_role_web_form_input.dat web_form_obj_Deref.dat	inc_priority_detail inc_priority
Web Reports: usp_web_form	usp_web_form.dat	Active Incident by Priority - Detail Active incident by priority
Business Impact: zbusiness_impact	zbusiness_impact.dat	5-One person 4-Small group 3-single group 2-multiple groups 1-entire organization
Business Classification: zchg_bus_cls	zchg_bus_cls.dat	Minor Significant Major Business Classification
CI Cost Category: zci_cost_category	zci_cost_category.dat	capital operational direct, indirect fixed, variable unit costs
Type of Service Agreement: zservice_type_class	zServiceTypeClassData.dat	SLA (Service Level Agreement) OLA (Operational Level Agreement) UC (Underpinning Contract)
Portfolio lifecycle stages: zSPMLifecycleStage	zSPMLifecycleStage.dat	Service Catalog Service Pipeline Retired Service
Menu Tree Resource: usp_menu_tree_res	usp_menu_tree_res.dat	Business Classification Business Impact

Additional Content Data

Templates

The following data is not supplied automatically by the ITIL Content installer, however it is recommended that you manually add the following templates to your CA SDM application to further support the incident, request, problem, change, and release and deployment ITIL processes

Object	Suggested Name/Title	Suggested Default Data	Description	How-to Documentation
Incident Template	Incident Model for <insert type of Incident> For example: Incident Model for Printer Repair	Incident Area = <any>	Contains the pre-defined steps that should be taken for dealing with a particular type of Incident. Further supported through incident status transitions.	CA SDM online help, Section on ‘Create an Incident Template’

Incident Template	User Compliment	Incident Area = User.Compliment	Used to record user compliments received from interaction with analyst.	CA SDM online help, Section on ‘Create an Incident Template’
Incident Template	User Complaint	Incident Area = User.Complaint	Used to record user complaints received from interaction with analyst.	CA SDM online help, Section on ‘Create an Incident Template’
Request Template	Knowledge Document Update	Request Area = Knowledge.BrokenLink	Request to update knowledge documents that have been flagged with a broken link.	CA SDM online help, Section on ‘Create a Request Template’
Request Template	New Service Request	Request Area = <any>	Contains the pre-defined steps that should be taken for dealing with a service request. Further supported through request status transitions.	CA SDM online help, Section on ‘Create a Request Template’
Problem Template	Problem Model for <insert type of Problem> For example: Problem Model for Known Error	Problem Area = <any>	Use for the resolution of dormant and underlying problems. Further supported through problem status transitions.	CA SDM online help, Section on ‘Create a Problem Template’
Problem Template	Major Problem Review	Problem Area = Problem Review	Major Problem Review Required	CA SDM online help, Section on ‘Create a Problem Template’
Change Order Template	Change Model for <insert type of change> For example: Change Model for New Hires	Category = <any> Change Type = Normal	Defines specific pre-defined steps that will be followed to manage a Change. Further supported through change status transitions.	CA SDM online help, Section on ‘Create a Change Order Template’
Change Order Template	Release and Deployment Model	Category = Release and Deployment Model	Defines specific pre-defined steps that will be followed to manage a Release. Further supported through the CA Process Automation Workflow, Change and Release Management, when that workflow is linked to the Release and Deployment Model Change Category.	CA SDM online help, Section on ‘Create a Change Order Template’

Change Category	Release and Deployment Model	Properties: • Is this a known or scheduled disaster recovery test? • Does the count towards the yearly failover test requirements? • Was the failover plan followed? • How long did the failover test occur for? • Was the failover successful? • Was failback implemented?	Properties added to support the gathering of data.	CA SDM online help, Section on 'Define Change Order Categories'

Administrative Data

The following data is not supplied automatically by the ITIL Content installer, however it is recommended that you manually add the following Administrative Data to your CA SDM application to further support the incident, request, problem, change, and release and deployment ITIL processes

Object	Suggested Name/Title	Suggested Default Data	Description	How-to Documentation
Change Category	Release and Deployment Model	Properties: • Is this a known or scheduled disaster recovery test? • Does the count towards the yearly failover test requirements? • Was the failover plan followed? • How long did the failover test occur for? • Was the failover successful? • Was failback implemented?	Properties added to support the gathering of data.	CA SDM online help, Section on 'Define Change Order Categories'
Knowledge Document	Test Checklist	Knowledge Category = Disaster Recovery Plans	Test checklists should be created and stored under relevant knowledge categories.	CA SDM online help, Section on 'Create Knowledge Documents'
Knowledge Document	Technical and Service Information	Knowledge Category = Service Quality Plan	Record technical and service related information under relevant knowledge categories.	CA SDM online help, Section on 'Create Knowledge Documents'

Knowledge Document	Plans	Knowledge Categories = Availability Plans, Capacity Plans, Disaster Recovery Plans	Record plans under relevant knowledge categories.	CA SDM online help, Section on ‘Create Knowledge Documents’
Knowledge Document	Procedures	Knowledge Categories = Compliance	Record procedures under relevant knowledge categories.	CA SDM online help, ’Section on ‘Create Knowledge Documents’
CI Class	Design Package	Family = Service	Service Design Package	CA SDM online help, Section on ‘Configuration Item Classes’
Menu Tree Resource	Business Impact	Sample Business Impacts are provided with the Content Pack for ITIL for CA SDM.	If you would like to add to the content we provided, we recommend doing so via a menu tree resource for ‘Business Impact’, under the Administration Tab->Service Desk->Application Data -> Codes	‘Menu Trees’, as well as the CA SDM online help, section on ‘Create a Menu Tree Resource’
Menu Tree Resource	Business Classification	Sample Business Classifications are provided with the Content Pack for ITIL for CA SDM.	If you would like to add to the content we provided, we recommend doing so via a menu tree resource for ‘Business Classification, under the Administration Tab->Service Desk->Application Data -> Codes	‘Menu Trees’, as well as the CA SDM online help, section on ‘Create a Menu Tree Resource’

Updated Help Files

Several help files were updated to support the extended ITIL content. Help data is in standard HTML files, not HTMPL templates, so the web engine cannot dynamically change file references. Since there is no /CAisd/site/mods/help subdirectory, all modified help files will be copied to /CAisd/help. For example: $NX_ROOT\bopcfg\www\wwwroot\help \web. The installer will back up the original html files to $NX_ROOT\samples\ ITIL_Content_17_4\files\help.
If you have modified any of the following help files you will want to review the Summary of Modifications column below to understand what was modified and decide whether to use the out of the box version or the supplied modified file. The original files will be backed up under the $NX_ROOT\samples\ITIL_Content_17_4\Backups folder and the modified version will be to /CAisd/help by the installer.
Due to the structure of help files, any patch or future version of CA SDM that delivers any of the same help files included in the content pack will be over ridden with the version included in the patch or new release.

HTML HELP FILE	Summary of Modifications
add_act_to_co.html	Updated summary text
add_act_to_inc.html	Updated summary text
add_act_to_prob.html	Updated summary text
add_act_to_req.html	Updated summary text
chg_order_fields.html	Added Authorized for Release

CI_inv_flds.html	Added Release Package Number
CI-svc_flds.html	Also added Service Description and Business Processes Supported
ci_tabs.html	Update tab name to Contacts, Subscribers Added Financials Tab
CI-flds.html	Field documented: Business Criticality
CI-loc_info.html	Field documented: DML Path
CI-srch_flds.html	Field documented: Business Criticality, SPM Lifecycle Stage
CI-svc_flds.html	Field documented: SPM Lifecycle Stage, Service Value Proposition field, Previous Availability, Previous Capacity, Previous Performance, Current Availability, Current Capacity, Current Performance
CO-use_temp.html	Description updated
CO-srch_flds.html	Added Authorized for Release
co_tabs.html	Properties updated, Cost/Plans updated
create_act_not.html	Added created_via
create_quick_close_ticket.html	Note sample quick close templates for complaint and compliment.
create_svc_type.html	Add instructions for how to Edit a Service Type
def_CO_cat.html	Document categories: Service.Improvement Plan, Service.Quality Plan
define_knowfiles_fields.html	Field documented: Doc ID
define_status_transitions.html	Added pending major problem review note
from_inc.html	Documented the Create Incident button
Inc_def_areas.html	Document categories: Service.Improvement Plan, Service.Quality Plan
INC-use-temp.html	Documented template: Create Incident Model Template Note that templates can be used to log events from monitoring tools.
incident_fields.html	Field documented: created_via, open date/time fields added
manage_CI_classes.html	Document new CI Classes: Design Package, Portfolio Service
manage_co_chg_win.html	
manage_know_cat.html	Data documented: knowledge categories of Availability Plans, Capacity Plans, Known Errors, business cases, Compliance, CSI Register, Service Improvement and Quality Plans, Release & Deployment, Workaround
organization_fields.html	Help page added for organization fields
prob_fields.html	Fields documented: symptom, created via, Major Problem
PROB-use_temp.html	Documented templates
REQ-flds.html	Field documented: RFC open date/time field, Service Development listed as a request area example.
req-inc-prob-area_flds.html	Field documented: symbol
role_list.html	Removed reference to “ITIL v3” and just used “ITIL”

search_for_documents.html	Added summary sentence
survey_templates.html	Note 2 OOTB surveys: Service Desk Quality Survey and Request Fulfillment Survey
svc_type_flds.html	Fields documented: Record Type, Agreement Start Date, Agreement End Date, and Scope tab
service_type_tabs.html	Document the new Scope tab
view_req.html	Button documented: ‘Create Incident’ button

New Reports

Find the following additional reports under Folder Name (Public Folders-> CA Reports -> CA Service Desk).

Report Name	Report Description	Report Type
Business Impact	Ticket list per service CI showing business impact.	Web Intelligence Report
Incident Closure Category Summary Report	Incident report showing the initial categorization and the closing categorization. This report shows the total count of Incidents with pie charts showing categories with the highest number of re-classifications.	Web Intelligence Report
Problem Closure Category Summary Report	Problem report showing the initial categorization and the closing categorization. This report shows the total count of problems with pie charts showing categories with the highest number of re-classifications.	Web Intelligence Report
Service Request Closing Category Summary Report	Service Request report showing the initial categorization and the closing categorization. This report shows the total count of problems with pie charts showing categories with the highest number of re-classifications.	Web Intelligence Report
RFC Closure Category Summary Report	Request for Change report showing the initial categorization and the closing categorization. This report shows the total count of RFCs with pie charts showing categories with the highest number of re-classifications.	Web Intelligence Report
SLA Management Report	Service level management report showing different trends based on ticket type and service.	Web Intelligence Report
SLA Management Report Previous 12 months	Service level management report showing different trends based on ticket type and service for the last 12 months.	Web Intelligence Report
SLA Management Report Previous 3 Months	Service level management report showing different trends based on ticket type and service for the last 3 months.	Web Intelligence Report

SLA Management Report Previous 7 Days	Service level management report showing different trends based on ticket type and service for the last 7 days.	Web Intelligence Report
SLAs To be validated Today	Service levels that will be violated today.	Web Intelligence Report
SLAs Violating Today Summary Report	Summary report showing SLAs with a validation date of today.	Web Intelligence Report
Violations Report	Service type violations report. Filter available for selecting OLA, SLA, or UC.	Crystal Report
SLA Monitoring Chart	Service Level Agreement Monitoring chart to show service level performance history.	Crystal Report
Change Order Closure Category	Detailed report to track the initial and closing categorization on change requests.	Crystal Report
Incident Closure Category Report	Detailed report to track the initial and closing categorization on incidents.	Crystal Report
Incidents Closure Category Count Report	High level summary to view the total count of categorization differences between initial and closing categorization on incidents.	Crystal Report
Problems Closure Category Report	Detailed report to track the initial and closing categorization on problems.	Crystal Report
Request Closure Category Report	Detailed report to track the initial and closing categorization on requests.	Crystal Report
Service Requests by Requests Area	Graph showing the number of requests over a period of time grouped by request area.	Crystal Report
Trend: RFCs and Affected CIs	Report to show Change Orders with associated Configuration Items. Sort filters are by earliest Open Date, Latest Open Date, CI, CI Class Type or CI Family.	Crystal Report
Service Availability and Performance Violation Report	Service type violations report for service availability and performance tickets. Filter available for created via, category, and affected end user.	Crystal Report

Options Manager

Options Manager settings can be found under the Administrative tab in the Options Manager folder. It is recommended to install the following Options Manager Options when using the ITIL Content Pack for CA SDM:

- employee_intf_incident_support
- catalog_server

CA Process Automation Workflows

When the installer script is run, a subdirectory is created called ‘workflows’. It contains two files.
1.) An XML file with all of the Processes, SRF’s, IRF’s, and Custom Operators required for implementation of the process.
2.) A PDF file with setup, configuration, integration, and Best Practices information. It also contains a screenshot demo of all Process Definitions ‘In Action’. Note that this content builds upon the existing ‘CA Process Automation Sample Process Definitions for CA SDM’. The chart below outlines those process definitions that are new with the Content Pack for ITIL

for CA SDM, the ones that are updated for the Content Pack for ITIL for CA SDM, and those that existed previously in an OOB install of CA SDM.

Process Name	New/Updated	What it does
Request Fulfillment	New	The goal of this process is to provide a foundation for Request Fulfillment which aligns to ITIL 2011.
Change and Release Management	Updated	This process definition adds Release and Deployment Management to the existing Change Management process included in the 'CA Process Automation Sample Process Definitions for CA SDM'. The goal of this process is to provide a foundation for Change and Release and Deployment Management which aligns to ITIL 2011. Provides the ability to manage a Change Management process for all Standard, Normal or Emergency Changes from the initial Request for Change through to its deployment and Post Implementation Review. It analyzes the 'Type' of Change Order and associated 'Risk' Level to determine what level of approvals are necessary for Change implementation. Should be called from a Change Category.
Change Management	Existing, no updates	Replaced with Change and Release Management Process Definition
Order PC	Existing, no updates	Sample building block which walks you through approval steps for and end user to raise the request and fulfill a PC Order.
Problem Analysis	Existing, no updates	The goal of this process definition is to make a decision regarding if an RFC/ Change should be created for a given problem. Once the problem assignee has finished researching the problem and has a suggested solution, they will fill out this problem survey and provide a recommendation. The survey should give the approver a good idea of the impact of the problem and if it is necessary to resolve it through Change Management. The idea of a Problem Pain/Value Analysis comes from ITIL v3 and should help CA better align to ITIL.

Install the ITIL Content for CA Service Desk Manager

Run the installer for either fresh installation of CA SDM ITIL content or when you upgrade to CA SDM 17.4, the ITIL content is migrated, but some schema files may be missing. To rectify this issue, you need to run the installer.

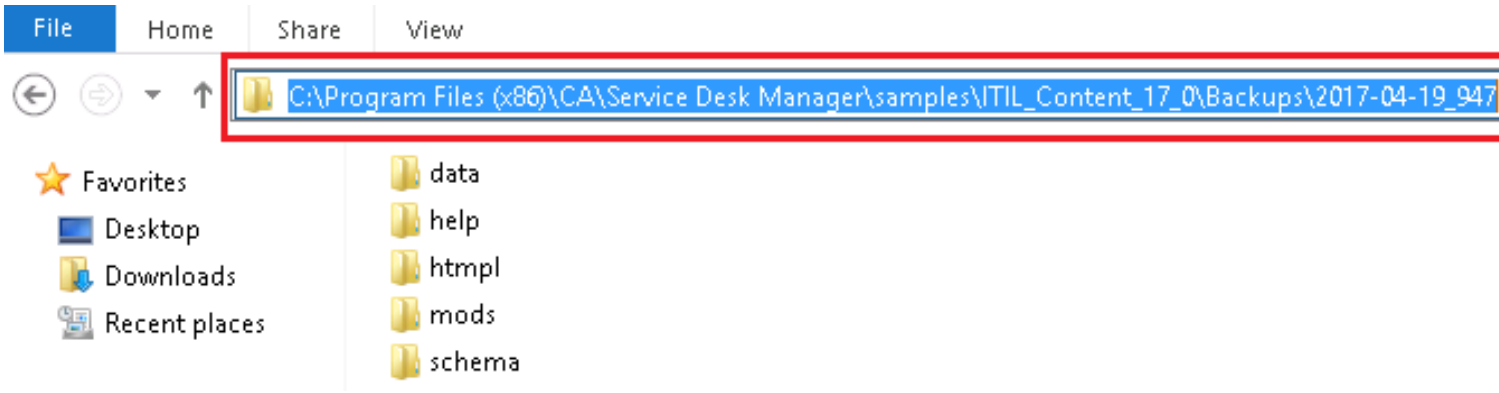
Run the Installer

NOTE
If you are updating from the 17.3 ITIL environment to 17.4, this is the only step you must follow:

- Replace "list_nr.htmpl" from <download_location>/ITIL_Content_17_4\files\htmpl from CONTENT PACK FOR ITIL FOR CA SERVICE DESK MANAGER 17.4.iso to <NX_ROOT>/site\mods\www\htmpl\web\analyst

Run these steps if you are integrating ITIL content for the first time:

1. From the Broadcom Support portal, download the CONTENT PACK FOR ITIL FOR CA SERVICE DESK MANAGER 17.4.iso
2. Extract the DVD content.
3. Unzip `ITIL_Content_17_4.zip to the $NX_ROOT\samples` folder on CA SDM primary server and all CA SDM standby servers if using Advanced Availability.
4. Backups will be made under $NX_ROOT \samples\ITIL_Content_17_4\<platform>\backups\<date-time> folder

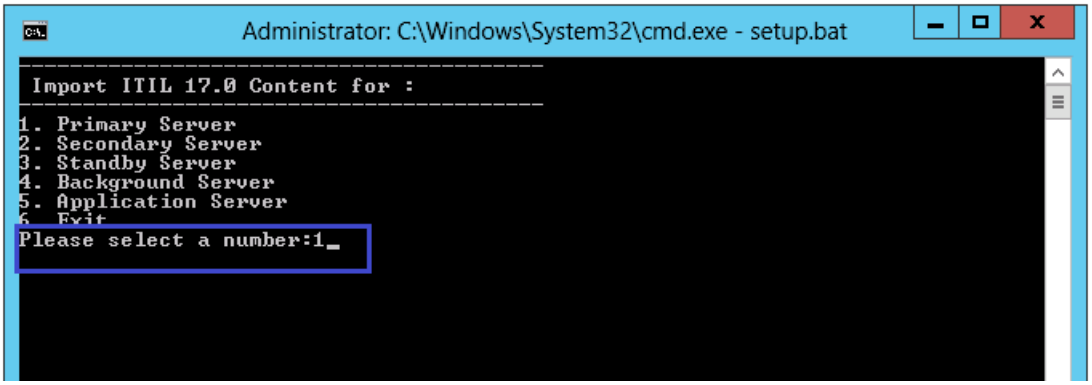


5. Review the content included with the installer as defined throughout this document.
6. Log files will be created under the $NX_ROOT\logs folder.

Install the Content on Windows

For Conventional Environment – Primary Server Install

1. Login to the Primary CA SDM Server.
2. Navigate to $NX_ROOT\Samples\ITIL_Content_17_4\nt and execute setup.bat
3. Select Run
4. Select the appropriate option based on your server environment. For example, select 1 if you are installing onto your Primary Server:

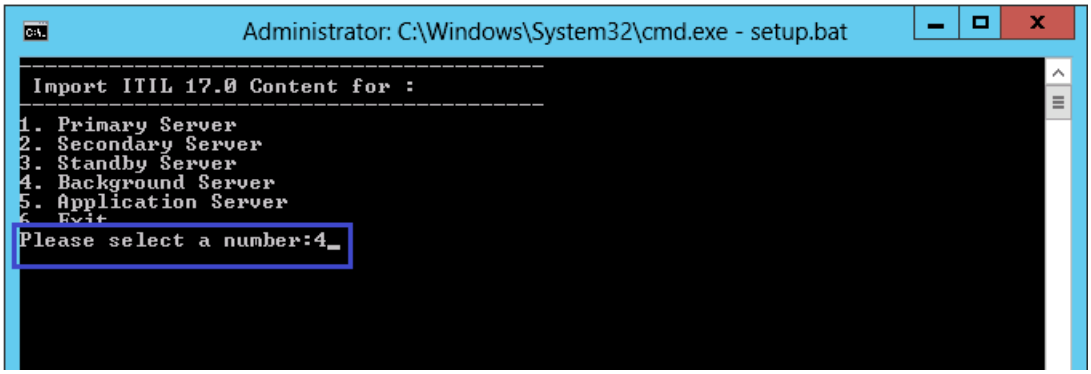


The installation will initiate and start CA SDM if necessary.

For Advanced Availability Environment

ITIL Data will be loaded only if the Primary or Background options are selected and executed only once.
Complete the following steps on the original Standby Server:

1. Login to the Standby Server. After deploying the ITIL Content pack on this machine, a failover will be performed, and this server will become the new Background server.
2. Navigate to $NX_ROOT/samples/ITIL_Content_17_4/nt and execute setup.bat
3. When prompted, select option 4 for Background Server



4. Click "Y" to do failover during pdm_publish.
5. On the original Background Server, now Standby Server.
   a. to the “new” Standby Server (Former Background)
   b. Navigate to $NX_ROOT/samples/ITIL_Content_17_4/nt and execute setup.bat
   c. When prompted, select option 3 for Standby Server.
6. On the Application Servers,
   a. Login to the Application Server
   b. Navigate to $NX_ROOT/samples/ITIL_Content_17_4/nt and execute setup.bat
   c. When prompted, select option 5 for Application Server.

Install the Content on LINUX

> **NOTE**
>
> CA SDM services will restart during the deployment process; ensure that the environment variables (PATH - Oracle executables, ORACLE variables) are set in the terminal/console before executing the setup.sh file

For Conventional Setup:

1. Login to the Primary Server
2. Unzip the `ITIL_Content_17_4.zip` to the `$NX_ROOT\samples` folder on your CA SDM primary server and all CA SDM standby servers if using Advanced Availability.
3. Navigate to /opt/CAisd/samples/ITIL_Content_17_4/unix and execute ./setup.sh
4. Select the appropriate option based on the server.

For Advanced Availability Setup:

1. Login to Standby Server and select Option as Background server, as during pdm_publish, we failover and make this server as Background.
2. Navigate to /opt/CAisd/samples/ITIL_Content_17_4/unix and execute ./setup.sh
3. Click "Y" to do failover during pdm_publish
4. Login to Standby Server (Former Background, Now standby after failover) or Application Server
5. Select the appropriate option based on the server.

NOTE

- Ensure that the ITIL Content pack is deployed on the new background server. If not, perform a manual failover using the command: pdm_server_control -b
- ITIL Data will be loaded only in the Primary and Background option. So it has to execute only once.
- Data Extract and Load will happen only on Background and Primary Server only. Choose the option carefully.

Install CA Process Automation Workflows

CA Process Automation workflows are located under $NX_ROOT\Samples\ITIL_Content_17_4\files\workflows\

Follow these steps:

1. Import CA SDM CA Process Automation Process Definitions
2. Configure SDM Dataset for Sample Process Definitions
3. Increase 'Maximum Number of Log Messages' in CA Process Automation
4. Verify CA SDM Options Manager Options

Install ITIL Reports

NOTE

- For new installations ensure to Integrate JasperReports Server 8.1.1 with CA Service Management.
  If you have already integrated ITIL Reports with JasperReports Server 7.9 and upgraded to JasperReports Server 8.1.1, you must again integrate using 17.4 and later. Perform steps 5 to 8, in this section.
- Access ITIL reports from JasperReports Server.

NOTE

- For new installations, ensure to Integrate JasperReports Server 9.0.0 with CA Service Management.
  If you have already integrated ITIL Reports with JasperReports Server 8.1.1 and upgraded to JasperReports Server 9.0.0, you must again integrate using 17.4 RU2 and later. Perform steps 5 to 8, in this section.
- Access ITIL reports from JasperReports Server.

Follow these steps:

1. From the Broadcom Support portal, download the CONTENT PACK FOR ITIL FOR CA SERVICE DESK MANAGER 17.4.iso
2. Extract the DVD content and verify the **ITIL_Content_17_4.zip** file is available.
3. Log in to the primary CA SDM server or one of the standby servers in the event of advanced availability.
4. Navigate to the `$NX_ROOT\samples` folder and unzip **ITIL_Content_17_4.zip**.
5. Copy the **ca-customCASMMethods.jar** file from the **files** folder in step 4.
6. From the location where you installed JasperReports Server, navigate to the `apache-tomcat\webapps\jasperserver-pro\WEB-INF\lib` folder and paste the copied jar file.
7. Restart the Tomcat service for JasperReports Server.
8. Log in to JasperReports Server as a superuser.
   `http://<machine_name>:< jaspereportsserver_port>/jasperserver-pro/login.html`
9. From the home page, click **Manage** and select **Organizations**.
10. Select **Add Organization** and provide the following details.
    a. Organization name: servicemanagement
    b. Organization ID: servicemanagement
    c. Organization Alias: servicemanagement
11. Click **Add Organization**.
12. From the home page, click **Manage** and select **Server Settings**.

13. Select **Import** and click the **Choose File** button.
14. Navigate to the `$NX_ROOT\samples\ITIL_Content_17_4\files\reports_CABI_8.x_JasperReports` folder and select the **ITIL_Reports.zip** file.
15. Click **Open** to import the file.
16. From the home page, click **View** and select **Repository**.
17. From the left navigation select **Organizations**, **servicemanagement**, **capability**, and **datasources**.
18. Right-click the file **sdm_ds** and select **Edit**.
19. In Edit Data Source Properties Screen, enter the properties required by the data adapter type. For example, select the JDBC driver for a database JDBC connection, set the URL, and provide the CA SDM database username and password.
    a. For JDBC driver drop-down:
        a. **For Jasper 9.0.0**:
        If **Other** option is not present, go to Manage > Server Settings> General > and enable the **Allow .jar** file updload button.
        b. **If APAR No. 99112250 for Jasper 9.0.0 is applied:**
        If **Other** option is not present then in *`apache-tomcat\webapps\jasperserver-pro\WEB-INF\js.config.properties`* path, set the *jdbc.driver.upload.enabled* property to ***true.***
        Restart the jasperserver service from services console.
    b. JDBC Driver: **com.ca.sdm.dal.sql.jdbc.driver.DalJdbcDriver**
    c. JDBC URL: **jdbc:sdm:bop:@<SDM Host Name>:2100;impersonateUserUuid={attribute('contactUUID')}**
    d. Username: **ServiceDesk**
    e. Password: Use the encrypted password. Perform the following steps to encrypt the password.
    - Navigate to `CASM DVD\filestore\Jasper\Utils` folder and execute the following command:

        IMPORTANT
        Ensure that java is added to PATH env. variable to run the below command.

        ```
        gen_passwd.bat <service desk username> <space> <service desk password>
        ```

        WARNING
        Do not test the connection at this stage.
20. Click **Save**.
21. Restart JasperReports Server services.
22. Test the connection.
    a. Log in to JasperReports Server.
    b. From the home page, click **View** and select **Repository**.
    c. From the left navigation select **Organizations**, **servicemanagement**, **capability**, and **datasources**.
    d. Right-click the file **sdm_ds** and select **Edit**.
    e. Click **Test connection**.
23. Verify the import.
    a. From the home page, click **View** and select **Repository**.
    b. From the left navigation select **Organizations**, **servicemanagement**, **capability**, **reports**, **CA Service Desk**, and **Additional ITIL Reports**.
24. Create a user and add the contactUUID attribute.
    a. From the home page, click **Manage** and select **Users**.
    b. From the left navigation select **servicemanagement.**
    c. Click **Add User**.
    d. On the **Add User** form, fill in the fields. Ensure to add servicedesk or srvcdesk as your username.
    e. Click **Add User to servicemanagement**.
    f. Select the newly created User.

g. From the **Properties** pane, click **Edit**.
h. From the **Attributes** tab, click **Add new attribute**.
i. In the Name field, enter contactUUID.
j. Click **OK** and **Save**.

25. Once the ITIL content is imported to Jasper follow these steps based on your requirement to disable the Allow.jar settings:
    a. For **Jasper 9.0.0**, **disable** the **Allow.jar** file.
    b. If hotfix **T5UG571.caz** is applied, navigate to *apache-tomcat\webapps\jasperserver-pro\WEB-INF\js.config.properties* path, and set the *jdbc.driver.upload.enabled* property to ***false.***
    Restart the jasperserver service from services console.

Reports

The following table lists the name, description, option name, input controls, and outcome of these reports.

S.No	Report Name	Report Description	Option Name	Input Control	Default Input Control	Outcome
1	Agreement/ Contract Renewal Review Report	This report presents a list of ticket having an SLA or Contract about to expire. This report works for all ticket types (Incidents/ Problems/ Requests/ Change Orders/ Releases).		• Number of Days before Contract Violation: • Ticket Type • Categories • SLM Type • Agreement/ Contract Name	• Number of Days before Contract Violation: 7 • Ticket Type: ALL • Categories: ALL • SLM Type: ALL • Agreement/ Contract Name: ALL	Detailed ticket list by ticket type, summary by ticket types, and contracts (column chart and crosstab)
1a	Agreement/ Contract Renewal Review Report	This report presents a list of ticket having an SLA or Contract about to expire. This report works for all ticket types (Incidents/ Problems/ Requests/ Change Orders/ Releases).	24hours || All Tickets	• Number of Days before Contract Violation: • Ticket Type • Categories • SLM Type • Agreement/ Contract Name	• Number of Days before Contract Violation: 1 • Ticket Type: ALL • Categories: ALL • SLM Type: ALL • Agreement/ Contract Name: ALL	Detailed ticket list by ticket type, summary by ticket types, and contracts (column chart and crosstab)

1b	Agreement/ Contract Renewal Review Report	This report presents a list of ticket having an SLA or Contract about to expire. This report works for all ticket types (Incidents/ Problems/ Requests/ Change Orders/ Releases).	7Days || SLAs || Change Orders	• Number of Days before Contract Violation: • Ticket Type • Categories • SLM Type • Agreement/ Contract Name	• Number of Days before Contract Violation: 7 • Ticket Type: Change Orders • Categories: ALL • SLM Type: SLA • Agreement/ Contract Name: ALL	Detailed ticket list by ticket type, summary by ticket types, and contracts (column chart and crosstab)
1c	Agreement/ Contract Renewal Review Report	This report presents a list of ticket having an SLA or Contract about to expire. This report works for all ticket types (Incidents/ Problems/ Requests/ Change Orders/ Releases).	7Days || SLAs || Incidents	• Number of Days before Contract Violation: • Ticket Type • Categories • SLM Type • Agreement/ Contract Name	• Number of Days before Contract Violation: 7 • Ticket Type: Incidents • Categories: ALL • SLM Type: SLA • Agreement/ Contract Name: ALL	Detailed ticket list by ticket type, summary by ticket types, and contracts (column chart and crosstab)
1d	Agreement/ Contract Renewal Review Report	This report presents a list of ticket having an SLA or Contract about to expire. This report works for all ticket types (Incidents/ Problems/ Requests/ Change Orders/ Releases).	7Days || SLAs || Problems	• Number of Days before Contract Violation: • Ticket Type • Categories • SLM Type • Agreement/ Contract Name	• Number of Days before Contract Violation: 7 Ticket Type: Problems • Categories: ALL • SLM Type: SLA • Agreement/ Contract Name: ALL	Detailed ticket list by ticket type, summary by ticket types, and contracts (column chart and crosstab)
1e	Agreement/ Contract Renewal Review Report	This report presents a list of ticket having an SLA or Contract about to expire. This report works for all ticket types (Incidents/ Problems/ Requests/ Change Orders/ Releases).	7Days || SLAs || Releases	• Number of Days before Contract Violation: • Ticket Type • Categories • SLM Type • Agreement/ Contract Name	• Number of Days before Contract Violation: 7 • Ticket Type: Releases • Categories: ALL • SLM Type: SLA • Agreement/ Contract Name: ALL	Detailed ticket list by ticket type, summary by ticket types, and contracts (column chart and crosstab)

1f	Agreement/ Contract Renewal Review Report	This report presents a list of ticket having an SLA or Contract about to expire. This report works for all ticket types (Incidents/ Problems/ Requests/ Change Orders/ Releases).	7Days || SLAs || Requests	• Number of Days before Contract Violation: • Ticket Type • Categories • SLM Type • Agreement/ Contract Name	• Number of Days before Contract Violation: 7 • Ticket Type: Requests • Categories: ALL • SLM Type: SLA • Agreement/ Contract Name: ALL	Detailed ticket list by ticket type, summary by ticket types, and contracts (column chart and crosstab)
2	Business Impact	This report illustrates Business Impact for Enterprise Services by ticket types and categories. This report is available for Incident, Request and Problem tickets.		• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) categories	• Time period: Current Year • Ticket Type: ALL • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status): ALL Categories: ALL	For each ticket type selected: • Volume of tickets by month for enterprise service (stacked column chart) • Volume of ticket by weekday for Enterprise Service (stacked column chart) • Volume of ticket for enterprise service by Category and by • Detailed ticket list by enterprise service

2a	Business Impact	This report illustrates Business Impact for Enterprise Services by ticket types and categories. This report is available for Incident, Request and Problem tickets.	Business Impact || Incidents || Current Year	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories	• Time period: Current Year • Ticket Type: Incident • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status): ALL • Categories: ALL	• Volume of tickets by month for enterprise service (stacked column chart) • Volume of ticket by weekday for Enterprise Service (stacked column chart) • Volume of ticket for enterprise service by Category and by • Detailed ticket list by enterprise service
2b	Business Impact	This report illustrates Business Impact for Enterprise Services by ticket types and categories. This report is available for Incident, Request and Problem tickets.	Business Impact || Problems || Current Year	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories	• Time period: Current Year • Ticket Type: Problems • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status): ALL • Categories: ALL	• Volume of tickets by month for enterprise service (stacked column chart) • Volume of ticket by weekday for Enterprise Service (stacked column chart) • Volume of ticket for enterprise service by Category and by • Detailed ticket list by enterprise service

2c	Business Impact	This report illustrates Business Impact for Enterprise Services by ticket types and categories. This report is available for Incident, Request and Problem tickets.	Business Impact || Requests || Current Year	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories	• Time period: Current Year • Ticket Type: Requests • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status): ALL • Categories: ALL	• Volume of tickets by month for enterprise service (stacked column chart) • Volume of ticket by weekday for Enterprise Service (stacked column chart) • Volume of ticket for enterprise service by Category and by • Detailed ticket list by enterprise service
3	Business Impact Change Orders and Releases	This report illustrates Business Impact for Enterprise Services by ticket types and categories. This report is available for Change Order and Release tickets.		• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Change/ Releases Categories	• Time period: Current Year • Ticket Type: ALL • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Change/ Releases Categories: ALL	• Volume of tickets by month for enterprise service (stacked column chart) • Volume of ticket by weekday for Enterprise Service (stacked column chart) • Volume of ticket for enterprise service by Category and by • Detailed ticket list by enterprise service

3a	Business Impact Change Orders and Releases	This report illustrates Business Impact for Enterprise Services by ticket types and categories. This report is available for Change Order and Release tickets.	Business Impact || Change Orders || Current Year	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Change/ Releases Categories	• Time period: Current Year • Ticket Type: Change Orders • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Change/ Releases Categories: ALL	• Volume of tickets by month for enterprise service (stacked column chart) • Volume of ticket by weekday for Enterprise Service (stacked column chart) • Volume of ticket for enterprise service by Category and by • Detailed ticket list by enterprise service
3b	Business Impact Change Orders and Releases	This report illustrates Business Impact for Enterprise Services by ticket types and categories. This report is available for Change Order and Release tickets.	Business Impact || Releases || Current Year	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Change/ Releases Categories	• Time period: Current Year • Ticket Type: Releases • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Change/ Releases Categories: ALL	• Volume of tickets by month for enterprise service (stacked column chart) • Volume of ticket by weekday for Enterprise Service (stacked column chart) • Volume of ticket for enterprise service by Category and by • Detailed ticket list by enterprise service

4	Closure Category Report for Changes and Releases	Summary and detail report for Change Orders and Releases showing a comparison of the initial categorization and the closing categorization.		• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Change/ Releases Categories	• Time period: Current Year • Ticket Type: ALL (Change Order and Release) • Change/ Release Categories: ALL	By ticket types: • Pie Chart on re-categorized ticket by closure category • Stacked column chart: re-categorized/ non recategorized tickets by closure categories. • Pie Chart on re-categorized ticket by initial category • Stacked column chart: re-categorized/ non recategorized tickets by initial categories. • Pie Chart on re-categorized ticket by customer organization • Stacked column chart: re-categorized/ non re-categorized tickets by customer organization • Stacked column chart: re-categorized/ non recategorized tickets by initial categories. • Pie Chart on re-categorized ticket by ticket template • Stacked column chart: re-categorized/ non recategorized tickets by ticket template • Crosstab

4a	Closure Category Report for Changes and Releases	Summary and detail report for Change Orders and Releases showing a comparison of the initial categorization and the closing categorization.	Closure Category || Change Orders || Current Year	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Change/ Releases Categories	• Time period: Current Year • Ticket Type: Change Orders • Change/ Release Categories: ALL	By ticket types: • Pie Chart on re-categorized ticket by closure category • Stacked column chart: re-categorized/ non recategorized tickets by closure categories. • Pie Chart on re-categorized ticket by initial category • Stacked column chart: re-categorized/ non recategorized tickets by initial categories. • Pie Chart on re-categorized ticket by customer organization • Stacked column chart: re-categorized/ non re-categorized tickets by customer organization • Stacked column chart: re-categorized/ non recategorized tickets by initial categories. • Pie Chart on re-categorized ticket by ticket template • Stacked column chart: re-categorized/ non recategorized tickets by ticket template • Crosstab

4b	Closure Category Report for Changes and Releases	Summary and detail report for Change Orders and Releases showing a comparison of the initial categorization and the closing categorization.	Closure Category || Releases || Current Year	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Change/ Releases Categories	• Time period: Current Year • Ticket Type: Releases • Change/ Release Categories: ALL	By ticket types: • Pie Chart on re-categorized ticket by closure category • Stacked column chart: re-categorized/ non recategorized tickets by closure categories. • Pie Chart on re-categorized ticket by initial category • Stacked column chart: re-categorized/ non recategorized tickets by initial categories. • Pie Chart on re-categorized ticket by customer organization • Stacked column chart: re-categorized/ non re-categorized tickets by customer organization • Stacked column chart: re-categorized/ non recategorized tickets by initial categories. • Pie Chart on re-categorized ticket by ticket template • Stacked column chart: re-categorized/ non recategorized tickets by ticket template • Crosstab

5	Closure Category Report for Incidents, Problems and Requests	Summary and detail report for Incidents, Problems and Requests showing a comparison of the initial categorization and the closing categorization.		• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories	• Time period: Current Year • Ticket Type: All (Incidents, Requests and Problems) • Categories: ALL	By ticket types: • Pie Chart on re-categorized ticket by closure category • Stacked column chart: re-categorized/ non recategorized tickets by closure categories. • Pie Chart on re-categorized ticket by initial category • Stacked column chart: re-categorized/ non recategorized tickets by initial categories. • Pie Chart on re-categorized ticket by customer organization • Stacked column chart: re-categorized/ non re-categorized tickets by customer organization • Stacked column chart: re-categorized/ non recategorized tickets by initial categories. • Pie Chart on re-categorized ticket by ticket template • Stacked column chart: re-categorized/ non recategorized tickets by ticket template • Crosstab

5a	Closure Category Report for Incidents, Problems and Requests	Summary and detail report for Incidents, Problems and Requests showing a comparison of the initial categorization and the closing categorization.	Closure Category || Incidents || Current Year	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories	• Time period: Current Year • Ticket Type: Incidents • Categories: ALL	• Pie Chart on re-categorized ticket by closure category • Stacked column chart: re-categorized/ non recategorized tickets by closure categories. • Pie Chart on re-categorized ticket by initial category • Stacked column chart: re-categorized/ non recategorized tickets by initial categories. • Pie Chart on re-categorized ticket by customer organization • Stacked column chart: re-categorized/ non re-categorized tickets by customer organization • Stacked column chart: re-categorized/ non recategorized tickets by initial categories. • Pie Chart on re-categorized ticket by ticket template • Stacked column chart: re-categorized/ non recategorized tickets by ticket template • Crosstab counting initial

5b	Closure Category Report for Incidents, Problems and Requests	Summary and detail report for Incidents, Problems and Requests showing a comparison of the initial categorization and the closing categorization.	Closure Category || Problems || Current Year	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories	• Time period: Current Year • Ticket Type: Incidents • Categories: ALL	• Pie Chart on re-categorized ticket by closure category • Stacked column chart: re-categorized/ non recategorized tickets by closure categories. • Pie Chart on re-categorized ticket by initial category • Stacked column chart: re-categorized/ non recategorized tickets by initial categories. • Pie Chart on re-categorized ticket by customer organization • Stacked column chart: re-categorized/ non re-categorized tickets by customer organization • Stacked column chart: re-categorized/ non recategorized tickets by initial categories. • Pie Chart on re-categorized ticket by ticket template • Stacked column chart: re-categorized/ non recategorized tickets by ticket template • Crosstab counting initial

5c	Closure Category Report for Incidents, Problems and Requests	Summary and detail report for Incidents, Problems and Requests showing a comparison of the initial categorization and the closing categorization.	Closure Category || Requests || Current Year	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories	• Time period: Current Year • Ticket Type: Requests • Categories: ALL	• Pie Chart on re-categorized ticket by closure category • Stacked column chart: re-categorized/ non recategorized tickets by closure categories. • Pie Chart on re-categorized ticket by initial category • Stacked column chart: re-categorized/ non recategorized tickets by initial categories. • Pie Chart on re-categorized ticket by customer organization • Stacked column chart: re-categorized/ non re-categorized tickets by customer organization • Stacked column chart: re-categorized/ non recategorized tickets by initial categories. • Pie Chart on re-categorized ticket by ticket template • Stacked column chart: re-categorized/ non recategorized tickets by ticket template • Crosstab counting initial

6	Configuration Item Card	This report displays Configuration Item main attributes, direct relationship CIs and tickets (Incidents, Requests, Problems, Change Orders and Releases) associated to the CI.		• Configuration Item Families • Audit History (Yes/No)	• Configuration Item Families: ALL • Audit History (Yes/No): Yes	• Main CI attributes • Direct relationship • Ticket list associated to the CI.
6a	Configuration Item Card	This report displays Configuration Item main attributes, direct relationship CIs and tickets (Incidents, Requests, Problems, Change Orders and Releases) associated to the CI.	Enterprise Services || Audit	• Configuration Item Families • Audit History (Yes/No)	• Configuration Item Families: Enterprise Services • Audit History (Yes/No): Yes	• Main CI attributes • Direct relationship • Ticket list associated to the CI.
6b	Configuration Item Card	This report displays Configuration Item main attributes, direct relationship CIs and tickets (Incidents, Requests, Problems, Change Orders and Releases) associated to the CI.	HardwareServer || Audit	• Configuration Item Families • Audit History (Yes/No)	• Configuration Item Families: Hardware Server • Enterprise Services • Audit History (Yes/No): Yes	• Main CI attributes • Direct relationship • Ticket list associated to the CI.

7	SLA Management By Enterprise Services	This report illustrates SLA management by Enterprise Service and ticket types (Incidents, Requests, Problems, Change Orders, Releases). A summary is provided for each ticket type and Enterprise Service.		• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • SLM Types (OLA, SLA, UC) • Agreements/ contract name • Enterprise Service	• Time period: Current Year • Ticket Type: ALL • Categories (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • SLM Types (OLA, SLA, UC): ALL • Agreements/ contract name: ALL • Enterprise Service: ALL	• Detailed ticket list by Enterprise Service and ticket type including SLM performance. • Summary by ticket types and contracts (column chart and crosstab) • Summary by enterprise service (violated agreements and number of tickets) • Count of violations per enterprise service and agreements (crosstab)
7a	SLA Management By Enterprise Services	This report illustrates SLA management by Enterprise Service and ticket types (Incidents, Requests, Problems, Change Orders, Releases). A summary is provided for each ticket type and Enterprise Service.	Enterprise Services || UCs || Current Year || Incidents/ Request/ Problems	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems): • Categories Changes and Release • SLM Types (OLA, SLA, UC) • Agreements/ contract name • Enterprise Service	• Time period: Current Year • Ticket Type: Incidents/ Request/ Problems • Categories (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • SLM Types (OLA, SLA, UC): UC • Agreements/ contract name: ALL • Enterprise Service: ALL	• Detailed ticket list by Enterprise Service and ticket type including SLM performance. • Summary by ticket types and contracts (column chart and crosstab) • Summary by enterprise service (violated agreements and number of tickets) • Count of violations per enterprise service and agreements (crosstab)

7b	SLA Management By Enterprise Services	This report illustrates SLA management by Enterprise Service and ticket types (Incidents, Requests, Problems, Change Orders, Releases). A summary is provided for each ticket type and Enterprise Service.	Enterprise Services || UCs || Current Year || Change Orders / Releases	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • SLM Types (OLA, SLA, UC): • Agreements/ contract name • Enterprise Service	• Time period: Current Year • Ticket Type: Changes Orders / Releases • Categories (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • SLM Types (OLA, SLA, UC): UC • Agreements/ contract name: ALL • Enterprise Service: ALL	• Detailed ticket list by Enterprise Service and ticket type including SLM performance. • Summary by ticket types and contracts (column chart and crosstab) • Summary by enterprise service (violated agreements and number of tickets) • Count of violations per enterprise service and agreements (crosstab)
7c	SLA Management By Enterprise Services	This report illustrates SLA management by Enterprise Service and ticket types (Incidents, Requests, Problems, Change Orders, Releases). A summary is provided for each ticket type and Enterprise Service.	Enterprise Services || OLAs || Current Year || Change Orders / Releases	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • SLM Types (OLA, SLA, UC) • Agreements/ contract name • Enterprise Service	• Time period: Current Year • Ticket Type: Change Orders/ Releases • Categories (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • SLM Types (OLA, SLA, UC): OLA • Agreements/ contract name: ALL • Enterprise Service: ALL	• Detailed ticket list by Enterprise Service and ticket type including SLM performance. • Summary by ticket types and contracts (column chart and crosstab) • Summary by enterprise service (violated agreements and number of tickets) • Count of violations per enterprise service and agreements (crosstab)

7d	SLA Management By Enterprise Services	This report illustrates SLA management by Enterprise Service and ticket types (Incidents, Requests, Problems, Change Orders, Releases). A summary is provided for each ticket type and Enterprise Service.	Enterprise Services || OLAs || Current Year || Incidents/ Request/ Problems	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • SLM Types (OLA, SLA, UC) • Agreements/ contract name • Enterprise Service	• Time period: Current Year • Ticket Type: Incidents/ Request/ Problems • Categories (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • SLM Types (OLA, SLA, UC): OLA • Agreements/ contract name: ALL • Enterprise Service: ALL	• Detailed ticket list by Enterprise Service and ticket type including SLM performance. • Summary by ticket types and contracts (column chart and crosstab) • Summary by enterprise service (violated agreements and number of tickets) • Count of violations per enterprise service and agreements(crosstab)
7e	SLA Management By Enterprise Services	This report illustrates SLA management by Enterprise Service and ticket types (Incidents, Requests, Problems, Change Orders, Releases). A summary is provided for each ticket type and Enterprise Service.	Enterprise Services || SLAs || Current Year || Change Orders/ Releases	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • SLM Types (OLA, SLA, UC) • Agreements/ contract name • Enterprise Service	• Time period: Current Year • Ticket Type: Change Orders/ Releases • Categories (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • SLM Types (OLA, SLA, UC): SLA • Agreements/ contract name: ALL • Enterprise Service: ALL	• Detailed ticket list by Enterprise Service and ticket type including SLM performance. • Summary by ticket types and contracts (column chart and crosstab) • Summary by enterprise service (violated agreements and number of tickets) • Count of violations per enterprise service and agreements(crosstab)

7f	SLA Management By Enterprise Services	This report illustrates SLA management by Enterprise Service and ticket types (Incidents, Requests, Problems, Change Orders, Releases). A summary is provided for each ticket type and Enterprise Service.	Enterprise Services || SLAs || Current Year || Incidents/ Requests/ Problems	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • SLM Types (OLA, SLA, UC) • Agreements/ contract name • Enterprise Service	• Time period: Current Year • Ticket Type: Incidents/ Requests/ Problems • Categories (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • SLM Types (OLA, SLA, UC): SLA • Agreements/ contract name: ALL • Enterprise Service: ALL	• Detailed ticket list by Enterprise Service and ticket type including SLM performance. • Summary by ticket types and contracts (column chart and crosstab) • Summary by enterprise service (violated agreements and number of tickets) • Count of violations per enterprise service and agreements(crosstab)
7g	SLA Management By Enterprise Services	This report illustrates SLA management by Enterprise Service and ticket types (Incidents, Requests, Problems, Change Orders, Releases). A summary is provided for each ticket type and Enterprise Service.	Enterprise Services || SLAs || Last 30 days || Incidents	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • SLM Types (OLA, SLA, UC) • Agreements/ contract name • Enterprise Service	• Time period: Last 30 days • Ticket Type: Incidents • Categories (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • SLM Types (OLA, SLA, UC): SLA • Agreements/ contract name: ALL • Enterprise Service: ALL	• Detailed ticket list by Enterprise Service and ticket type including SLM performance. • Summary by ticket types and contracts (column chart and crosstab) • Summary by enterprise service (violated agreements and number of tickets) • Count of violations per enterprise service and agreements(crosstab)

8	SLA Management By Enterprise Services Charts and Summary	This report illustrates SLA management by Enterprise Service and ticket type (Incidents, Requests, Problems, Change Orders, Releases). A summary is provided for each ticket type and Enterprise Service.		• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • SLM Types (OLA, SLA, UC) • Agreements/ contract name: • Enterprise Service	• Time period: Current Year • Ticket Type: ALL • Categories (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • SLM Types (OLA, SLA, UC): ALL • Agreements/ contract name: ALL • Enterprise Service: ALL	• Summary by ticket types and contracts (column chart and crosstab) • Summary by enterprise service (violated agreements and number of tickets) • Count of violations per enterprise service and agreements(crosstab)
9	SLA Management by Ticket Type and Categories	This report illustrates SLA management by ticket types and categories. Summary calculations are provided for each category. This report works for Incidents, Requests, Problems, Change Orders and Releases		• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • SLM Types (OLA, SLA, UC) • Agreements/ contract name: • Enterprise Service	• Time period: Current Year • Ticket Type: ALL • Categories (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • SLM Types (OLA, SLA, UC): ALL • Agreements/ contract name: ALL • Enterprise Service: ALL	• Detailed ticket list with violation cost and violation status for each agreement. • Summary for each category calculating violation cost, violated tickets, percentage of violations. • Crosstab at summary section with violated tickets per agreement and ticket types

9a	SLA Management by Ticket Type and Categories	This report illustrates SLA management by ticket types and categories. Summary calculations are provided for each category. This report works for Incidents, Requests, Problems, Change Orders and Releases	Categories || OLAs || Current Year || Change Orders/ Releases	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • SLM Types (OLA, SLA, UC) • Agreements/ contract name • Enterprise Service	• Time period: Current Year • Ticket Type: Change Orders/ Releases • Categories (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • SLM Types (OLA, SLA, UC): OLA • Agreements/ contract name: ALL • Enterprise Service: ALL	• Detailed ticket list with violation cost and violation status for each agreement. • Summary for each category calculating violation cost, violated tickets, percentage of violations. • Crosstab at summary section with violated tickets per agreement and ticket types
9b	SLA Management by Ticket Type and Categories	This report illustrates SLA management by ticket types and categories. Summary calculations are provided for each category. This report works for Incidents, Requests, Problems, Change Orders and Releases	Categories || OLAs || Current Year || Incidents/ Requests/ Problems	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • SLM Types (OLA, SLA, UC) • Agreements/ contract name • Enterprise Service	• Time period: Current Year • Ticket Type: Incidents/ Requests/ Problems • Categories (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • SLM Types (OLA, SLA, UC): OLA • Agreements/ contract name: ALL • Enterprise Service: ALL	• Detailed ticket list with violation cost and violation status for each agreement. • Summary for each category calculating violation cost, violated tickets, percentage of violations. • Crosstab at summary section with violated tickets per agreement and ticket types

9c	SLA Management by Ticket Type and Categories	This report illustrates SLA management by ticket types and categories. Summary calculations are provided for each category. This report works for Incidents, Requests, Problems, Change Orders and Releases	Categories || SLAs || Current Year || Change Orders/ Releases	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • SLM Types (OLA, SLA, UC) • Agreements/ contract name • Enterprise Service	• Time period: Current Year • Ticket Type: Change Orders/ Releases • Categories (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • SLM Types (OLA, SLA, UC): SLA • Agreements/ contract name: ALL • Enterprise Service: ALL	• Detailed ticket list with violation cost and violation status for each agreement. • Summary for each category calculating violation cost, violated tickets, percentage of violations. • Crosstab at summary section with violated tickets per agreement and ticket types
9d	SLA Management by Ticket Type and Categories	This report illustrates SLA management by ticket types and categories. Summary calculations are provided for each category. This report works for Incidents, Requests, Problems, Change Orders and Releases	Categories || SLAs || Current Year ||Incidents/ Requests/ Problems	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • SLM Types (OLA, SLA, UC) • Agreements/ contract name • Enterprise Service	• Time period: Current Year • Ticket Type: Incidents/ Requests/ Problems • Categories (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • SLM Types (OLA, SLA, UC): SLA • Agreements/ contract name: ALL • Enterprise Service: ALL	• Detailed ticket list with violation cost and violation status for each agreement. • Summary for each category calculating violation cost, violated tickets, percentage of violations. • Crosstab at summary section with violated tickets per agreement and ticket types

9e	SLA Management by Ticket Type and Categories	This report illustrates SLA management by ticket types and categories. Summary calculations are provided for each category. This report works for Incidents, Requests, Problems, Change Orders and Releases	Categories || UCs || Current Year || Change Orders/ Releases	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • SLM Types (OLA, SLA, UC) • Agreements/ contract name • Enterprise Service	• Time period: Current Year • Ticket Type: Change Orders/ Releases • Categories (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • SLM Types (OLA, SLA, UC):UC • Agreements/ contract name: ALL • Enterprise Service: ALL	• Detailed ticket list with violation cost and violation status for each agreement. • Summary for each category calculating violation cost, violated tickets, percentage of violations. • Crosstab at summary section with violated tickets per agreement and ticket types
9f	SLA Management by Ticket Type and Categories	This report illustrates SLA management by ticket types and categories. Summary calculations are provided for each category. This report works for Incidents, Requests, Problems, Change Orders and Releases	Categories || UCs || Current Year || Incidents/ Requests/ Problems	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • SLM Types (OLA, SLA, UC) • Agreements/ contract name • Enterprise Service	• Time period: Current Year • Ticket Type: Incidents/ Requests/ Problems • Categories (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • SLM Types (OLA, SLA, UC): UC • Agreements/ contract name: ALL • Enterprise Service: ALL	• Detailed ticket list with violation cost and violation status for each agreement. • Summary for each category calculating violation cost, violated tickets, percentage of violations. • Crosstab at summary section with violated tickets per agreement and ticket types

10	Surveys By Survey Template	This report illustrates Survey Results for all ticket types by Survey Template.		• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • Survey Template	• Time period: Current Year • Ticket Type: ALL • Categories: (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • Survey Template: ALL	By ticket type: • Column stacked graph by survey template • Crosstab with answer count
10a	Surveys By Survey Template	This report illustrates Survey Results for all ticket types by Survey Template.	Survey Templates || 30 DAYS || Change Orders	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • Survey Template	• Time period: Current Year • Ticket Type: Change Orders • Categories: (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • Survey Template: ALL	By ticket type: • Column stacked graph by survey template. • Crosstab with answer count.
10b	Surveys By Survey Template	This report illustrates Survey Results for all ticket types by Survey Template.	Survey Templates || 30 DAYS || Incidents	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • Survey Template	• Time period: Current Year • Ticket Type: Incidents • Categories: (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • Survey Template: ALL	By ticket type: • Column stacked graph by survey template. • Crosstab with answer count.

10c	Surveys By Survey Template	This report illustrates Survey Results for all ticket types by Survey Template.	Survey Templates || 30 DAYS || Problems	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type: • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • Survey Template	• Time period: Current Year • Ticket Type: Problems • Categories: (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • Survey Template: ALL	By ticket type: • Column stacked graph by survey template. • Crosstab with answer count.
10d	Surveys By Survey Template	This report illustrates Survey Results for all ticket types by Survey Template.	Survey Templates || 30 DAYS || Releases	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • Survey Template	• Time period: Current Year • Ticket Type: Releases • Categories: (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • Survey Template: ALL	By ticket type: • Column stacked graph by survey template. • Crosstab with answer count.
10e	Surveys By Survey Template	This report illustrates Survey Results for all ticket types by Survey Template.	Survey Templates || 30 DAYS || Requests	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • Survey Template:	• Time period: Current Year • Ticket Type: Requests • Categories: (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • Survey Template: ALL	By ticket type: • Column stacked graph by survey template. • Crosstab with answer count.

10f	Surveys By Survey Template	This report illustrates Survey Results for all ticket types by Survey Template.	Survey Templates || Current Year Survey	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • Survey Template	• Time period: Current Year • Ticket Type: Incidents/ Requests/ Problems • Categories: (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • Survey Template: ALL	By ticket type: • Column stacked graph by survey template. • Crosstab with answer count.
10g	Surveys By Survey Template	This report illustrates Survey Results for all ticket types by Survey Template.	Survey Templates || Last Year Survey	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release • Survey Template	• Time period: Last Year • Ticket Type: Incidents/ Requests/ Problems • Categories: (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL • Survey Template: ALL	By ticket type: • Column stacked graph by survey template. • Crosstab with answer count.
11	Surveys By Ticket Categories	This report illustrates Survey Results for all ticket types by ticket Categories		• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Categories (Incidents/ Requests/ Problems) • Categories Changes and Release	• Time period: Last Year • Ticket Type: Incidents/ Requests/ Problems • Categories: (Incidents/ Requests/ Problems): ALL • Categories Changes and Release: ALL	By ticket type: • Column stacked graph by survey template. • Crosstab with answer count.

12	Top Categories	This report lists the top categories by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.		• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Incidents/ Requests/ Problems • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* categories • Detailed ticket list by category ordered by priority
12a	Top Categories	This report lists the top categories by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.	Categories || Incidents || Current Year || 10	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Incidents • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* categories • Detailed ticket list by category ordered by priority

12b	Top Categories	This report lists the top categories by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.	Categories || Problems || Current Year || 10	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Problems • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* categories • Detailed ticket list by category ordered by priority
12c	Top Categories	This report lists the top categories by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.	Categories || Requests || Current Year || 10	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Problems • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* categories • Detailed ticket list by category ordered by priority

13	Top Configuration Items	This report lists the top Configuration Items by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.		• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Incidents/ Requests/ Problems • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* Configuration Items • Detailed ticket list by Configuration Item ordered by priority
13a	Top Configuration Items	This report lists the top Configuration Items by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.	CIs || Incidents || Current Year || 10	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	Time period: Current Year Ticket Type: Incidents Enterprise Service: ALL Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) Categories: ALL Top: 10	• Graph representing top *n* Configuration Items • Detailed ticket list by Configuration Item ordered by priority

13b	Top Configuration Items	This report lists the top Configuration Items by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.	CIs || Problems || Current Year || 10	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	Time period: Current Year Ticket Type: Problems Enterprise Service: ALL Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) Categories: ALL Top: 10	• Graph representing top *n* Configuration Items • Detailed ticket list by Configuration Item ordered by priority
13c	Top Configuration Items	This report lists the top Configuration Items by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.	CIs || Requests || Current Year || 10	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Problems • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* Configuration Items • Detailed ticket list by Configuration Item ordered by priority

14	Top Configuration Items Classes	This report lists the top Configuration Items by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.		• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Incidents/ Requests/ Problems • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* Configuration Item Classes • Detailed ticket list by Configuration Item Class ordered by priority
14a	Top Configuration Item Classes	This report lists the top Configuration Item Classes by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.	CI Classes || Incidents || Current Year || 10	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Incidents • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* Configuration Item Classes • Detailed ticket list by Configuration Item Class ordered by priority

14b	Top Configuration Item Classes	This report lists the top Configuration Item Classes by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.	CI Classes || Problems || Current Year || 10	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Problems • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* Configuration Item Classes • Detailed ticket list by Configuration Item Class ordered by priority
14c	Top Configuration Item Classes	This report lists the top Configuration Item Classes by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.	CI Classes || Requests || Current Year || 10	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Problems • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* Configuration Item Classes • Detailed ticket list by Configuration Item Class ordered by priority

15	Top Configuration Item Families	This report lists the top Configuration Item Families by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.		• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Incidents/ Requests/ Problems • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* Configuration Item Families • Detailed ticket list by Configuration Item Family ordered by priority
15a	Top Configuration Item Families	This report lists the top Configuration Item Families by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.	CI Families || Incidents || Current Year ||10	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Incidents • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* Configuration Item Families • Detailed ticket list by Configuration Item Family ordered by priority

15b	Top Configuration Item Families	This report lists the top Configuration Item Families by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.	CI Families || Problems || Current Year || 10	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Problems • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* Configuration Item Families • Detailed ticket list by Configuration Item Family ordered by priority
15c	Top Configuration Item Families	This report lists the top Configuration Item Families by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.	CI Families || Requests || Current Year || 10	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Problems • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* Configuration Item Families • Detailed ticket list by Configuration Item Family ordered by priority

16	Top Customer Organizations	This report lists the top Customer Organizations by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.		• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Incidents/ Requests/ Problems • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* Customer Organizations • Detailed ticket list by Customer Organization ordered by priority
16a	Top Customer Organizations	This report lists the top Customer Organizations by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.	Customer Org || Incidents || Current Year || 10	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Incidents • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* Customer Organizations • Detailed ticket list by Customer Organization ordered by priority

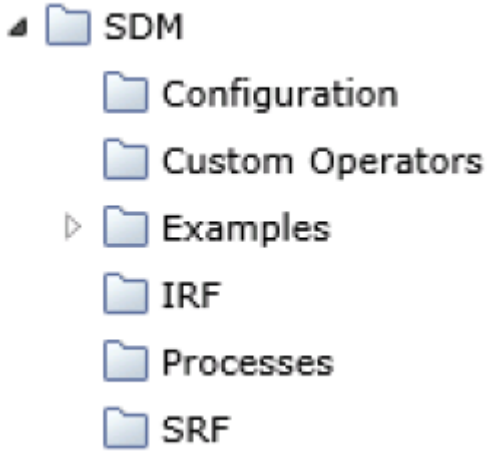
16b	Top Customer Organizations	This report lists the top Customer Organizations by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.	Customer Org || Problems || Current Year || 10	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Problems • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* Customer Organizations • Detailed ticket list by Customer Organization ordered by priority
16c	Top Customer Organizations	This report lists the top Customer Organizations by priority for Incident, Request and Problem tickets. Output is in the form of a chart and detailed ticket list.	Customer Org || Requests || Current Year || 10	• Time period: (Current Year, Last Year, Last 60 days, Last 30 days or specific date range) • Ticket Type • Enterprise Service • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories • Top	• Time period: Current Year • Ticket Type: Problems • Enterprise Service: ALL • Status Mode (All Statuses, All Statuses except Resolved and Closed, Specific Status) • Categories: ALL • Top: 10	• Graph representing top *n* Customer Organizations • Detailed ticket list by Customer Organization ordered by priority

Import CA SDM CA Process Automation Process Definitions

NOTE

These process definitions are imported to the CA SDM directory off of the root directory in PAM, as shown in the screenshot below. This is the same place that the CA SDM CA Process Automation Sample Process Definitions available in $NX_ROOT/samples/CA_PAM_Workflows, are imported to. The Process Definitions that are provided with the Content Pack for ITIL for CA SDM build upon and enhance those previously provided in the directory below, if you have already imported the CA SDM CA Process Automation Sample Process Definitions. Be sure to take a backup prior to importing the process definitions provided with the Content Pack for ITIL for CA SDM.

NOTE
After installing CASM 17.4.5.0, re-import the ITIL content pack XML file from the following location: ***<17.4 RU5 DVD>\filestore\ITILContent\ContentPackITILCASDM174PAMWF.xml***



Before we begin: The XML File we will be importing into PAM in the instructions to follow includes the following components to ensure that these flows will function properly. This XML file can be found at: $NX_ROOT/samples/ITIL_Content_17_4/files/workflows/ContentPackITILCASDM170PAMWF.XML

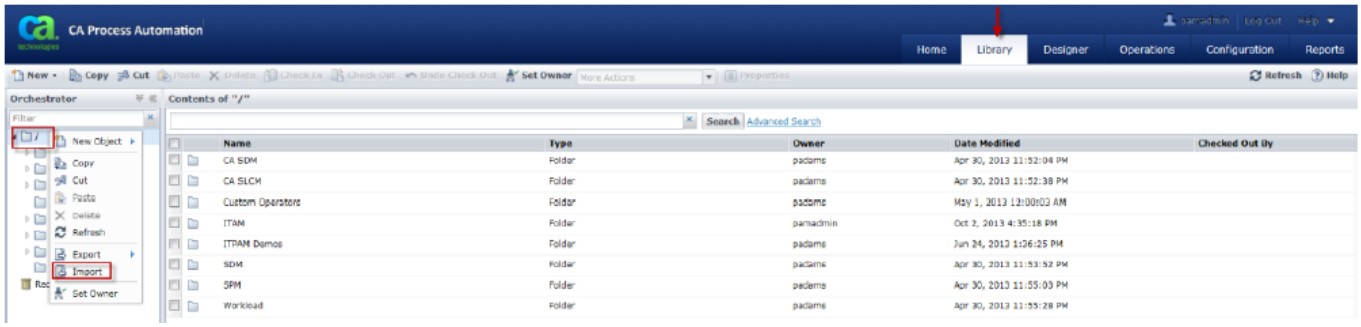
NOTE

From CASM 17.4.5.0, the ITIL content pack XML file must be imported from: ***<17.4 RU5 DVD>\filestore\ITILContent\ContentPackITILCASDM174PAMWF.xml***

CA Process Automation Object	What it Does
Custom Operators	Pre-configure parameters for repeated use in many different processes
Global Datasets	Define and group variables used as parameters for module and job invocations. Datasets can define common parameters-such as application locations, passwords, and profile names-required by modules and applications on a network. These variables can be easily configured so that Processes and scheduling can be efficiently updated to reflect changes in an application environment.
Interaction Request Forms (IRF's)	Provides field and other UI elements with which users can enter information required to execute an Operator or continue a process.
Processes	Graphically describe the ordering and dependencies for Operations Performed by modules. Operators are represented in a Process graphically and by Operators with links and dependencies that define the order and logic for performing tasks in a Process. These processes are samples with which you can use as-is, or modify to meet your business needs.
Start Request Forms (SRF's)	Define shortcuts to allow an operator to invoke processes manually using the Management Console. Start Request Form objects define custom dialogs that prompt operators for the values of parameters required by the associated project. These are also used by CA SDM to launch CA Process Automation Process definitions by attaching a definition to a Request/Incident/Problem Area, or Change Order/Issue Category.

Follow these steps:

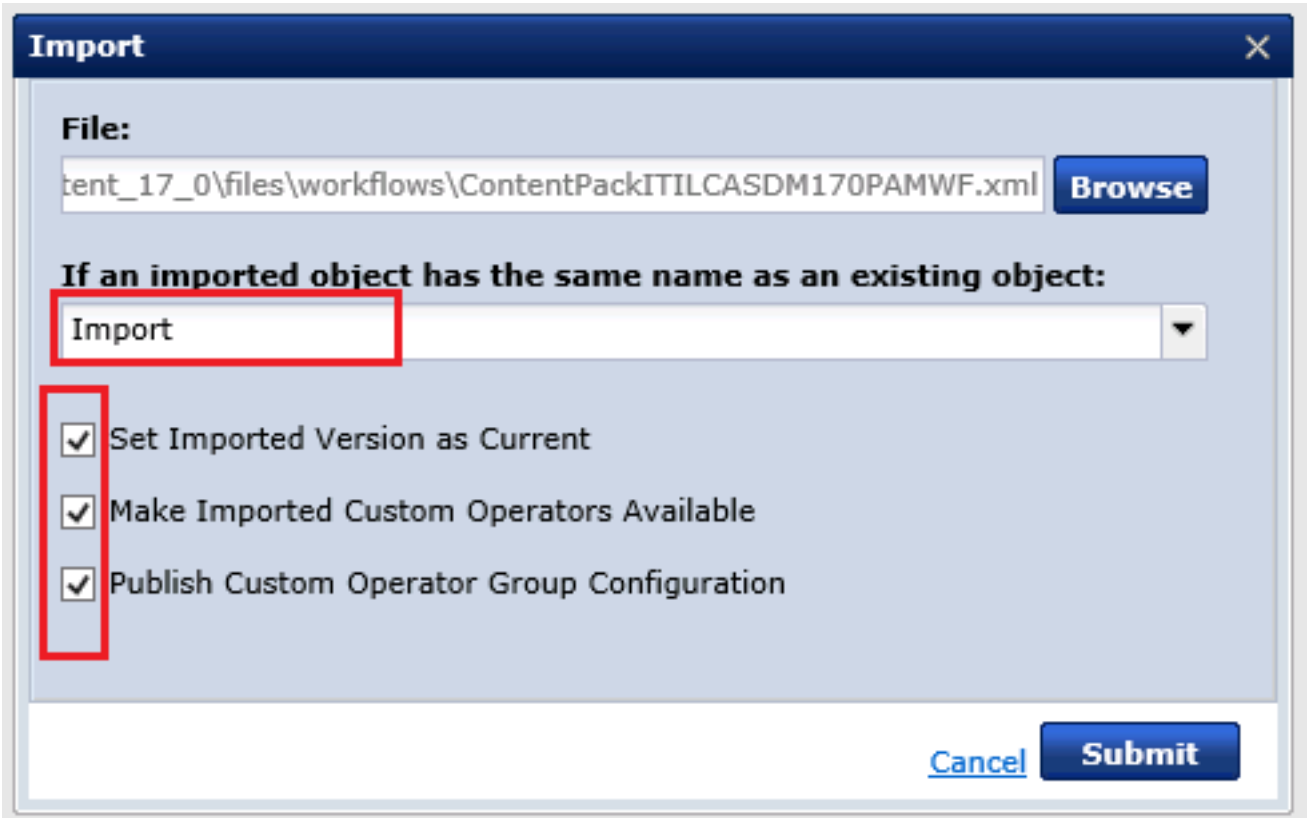
1. To manually import the sample process definitions do the following. Login to CA Process Automation by either navigating to Start, Programs, CA, <*CA Process Automation*>, Start CA Process Automation locally on the CA Process Automation Server, or via the direct URL if you are launching it from a server remote from CA SDM
   a. Login as ‘itpamadmin’ into CA Process Automation.
2. Import the Process Definitions into CA Process Automation:
   a. Click on the ‘Library’ Tab
   b. RMC on the root directory, and select ‘Import:



   c. Browse to the XML File which contains the CA Process Automation Process Definitions, select it, then select all remaining checkboxes, then hi ‘submit’. In my example I have logged into CA Process Automation via url from the CA SDM Server. I have browsed to the ContentPackITILCASDM170PAMWF.XML file under $NX_ROOT/samples/ITIL_Content_17_4/files/workflows/

NOTE

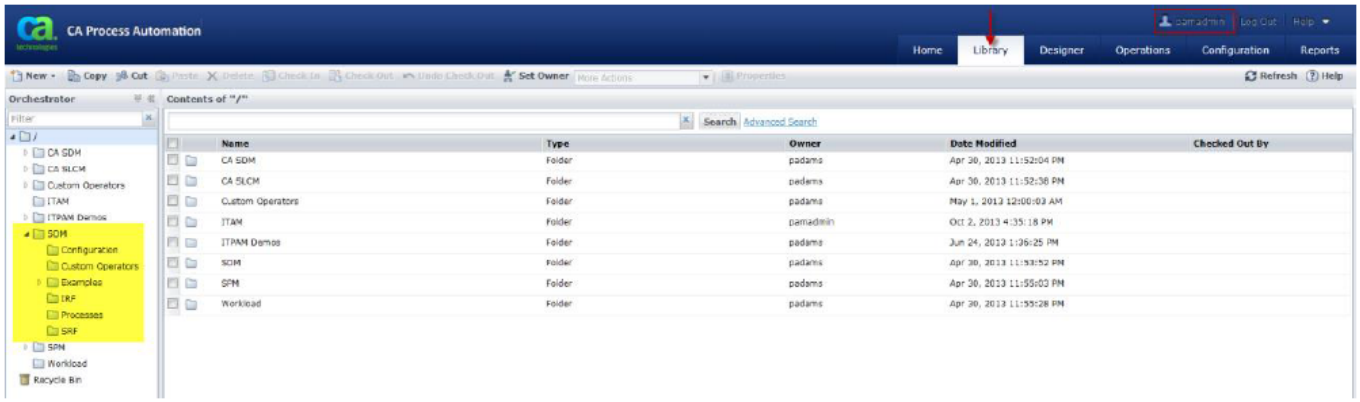
From CASM 17.4.5.0, the ITIL content pack XML file must be imported from: ***<17.4 RU5 DVD>\filestore\ITILContent\ContentPackITILCASDM174PAMWF.xml***



d. You will now see a new folder, CA SDM, as shown in the screenshot below.

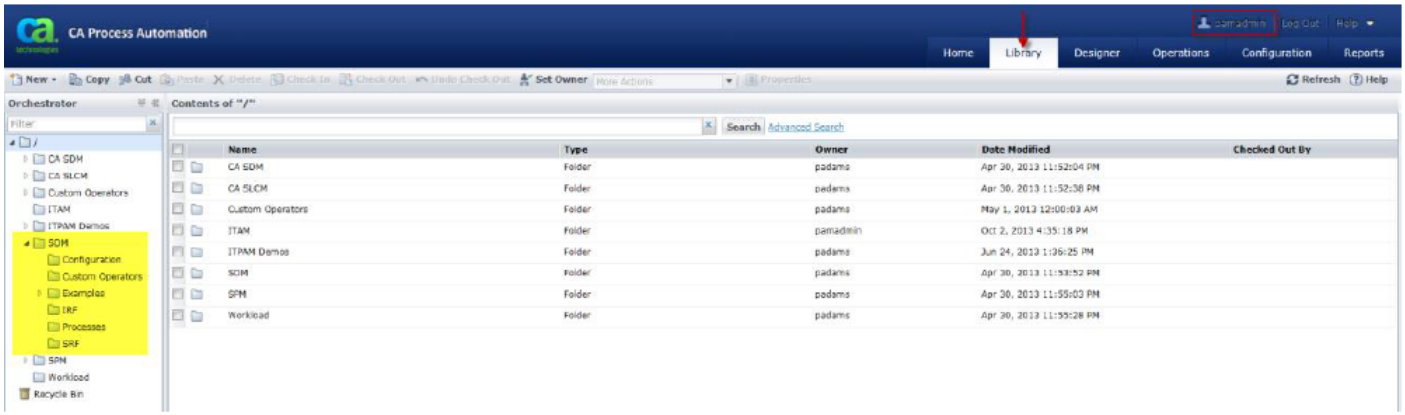
NOTE

If you have already imported the process definitions available in $NX_ROOT/samples/CA_PAM_Workflows then this will overwrite the content already in the 'SDM' directory, however, depending on the flag you specified above during the import, it will also create a new 'version'.

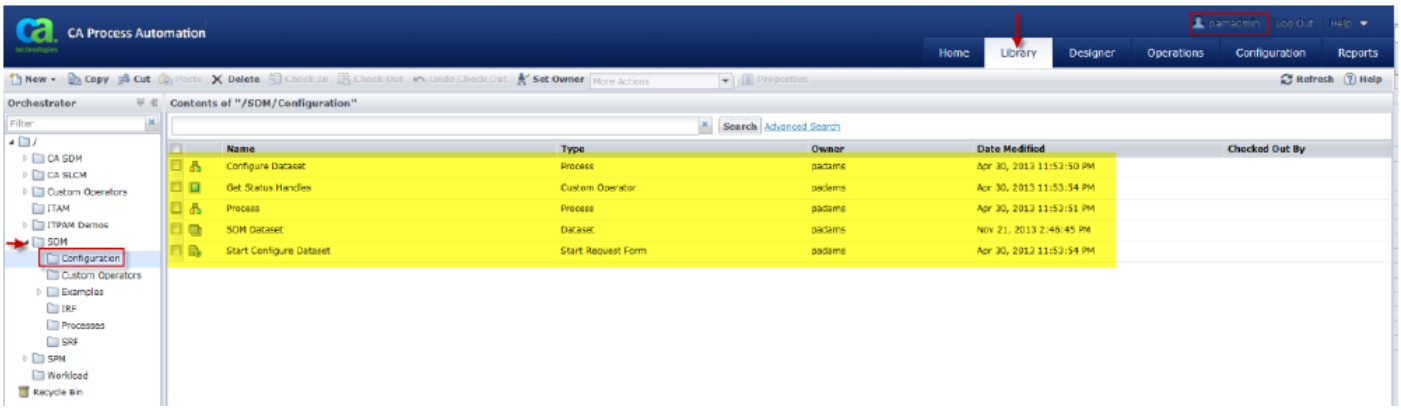


View Imported Process Definitions

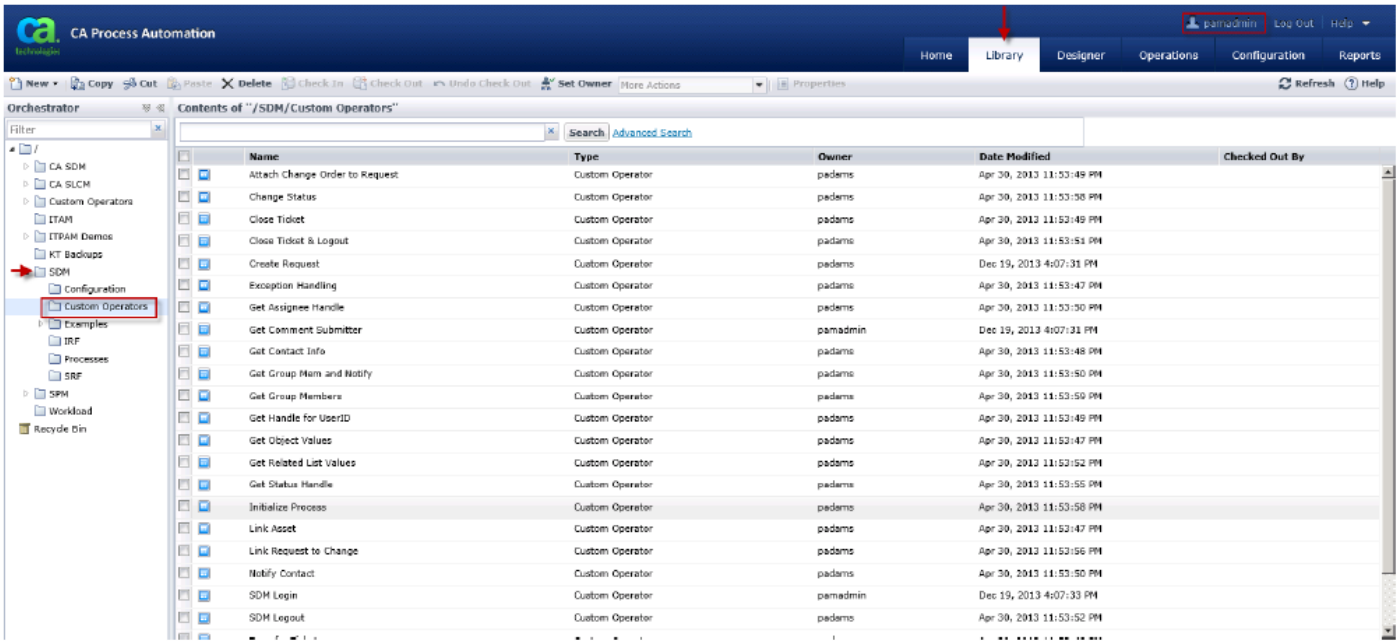
1. Once you have imported the Process Definitions, a new folder called 'SDM' appears, which includes Configuration, Custom Operators, Examples, IRF's, Processes, and SRF's:



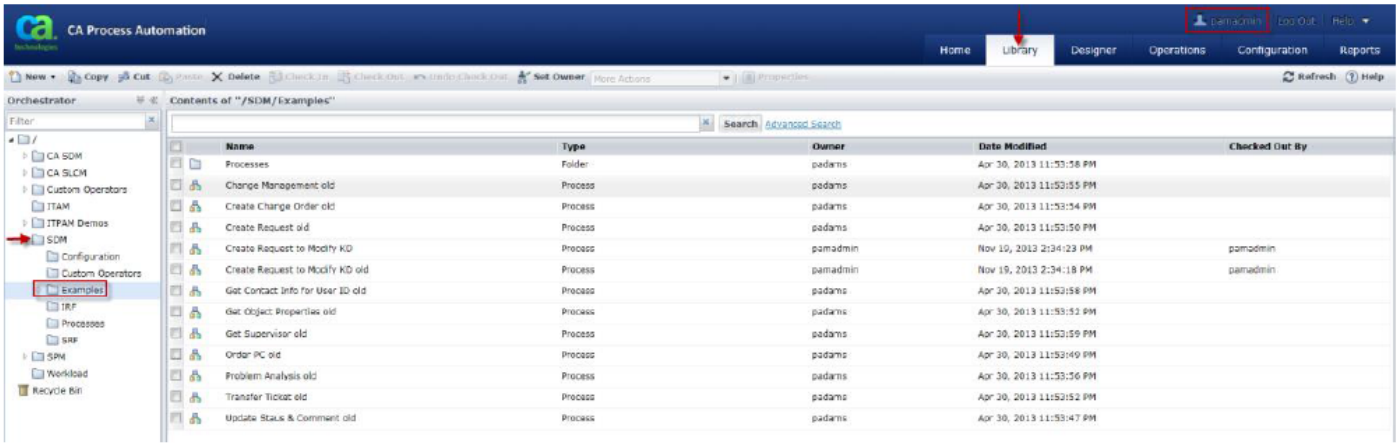
2. Configuration:



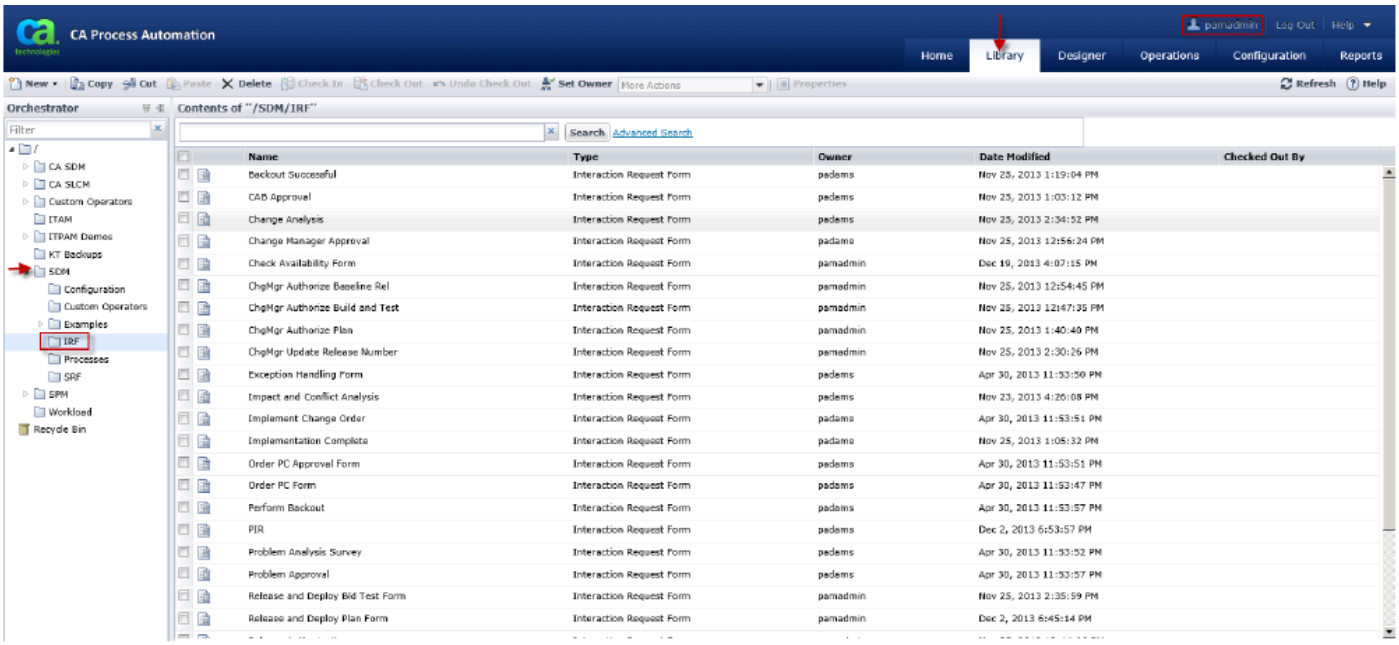
3. Custom Operators:



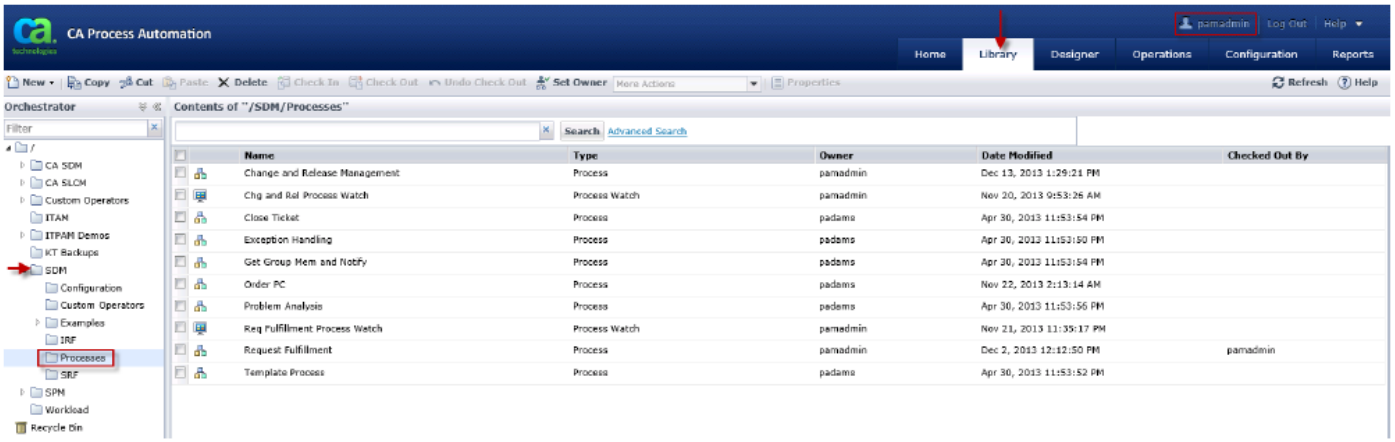
4. Examples:



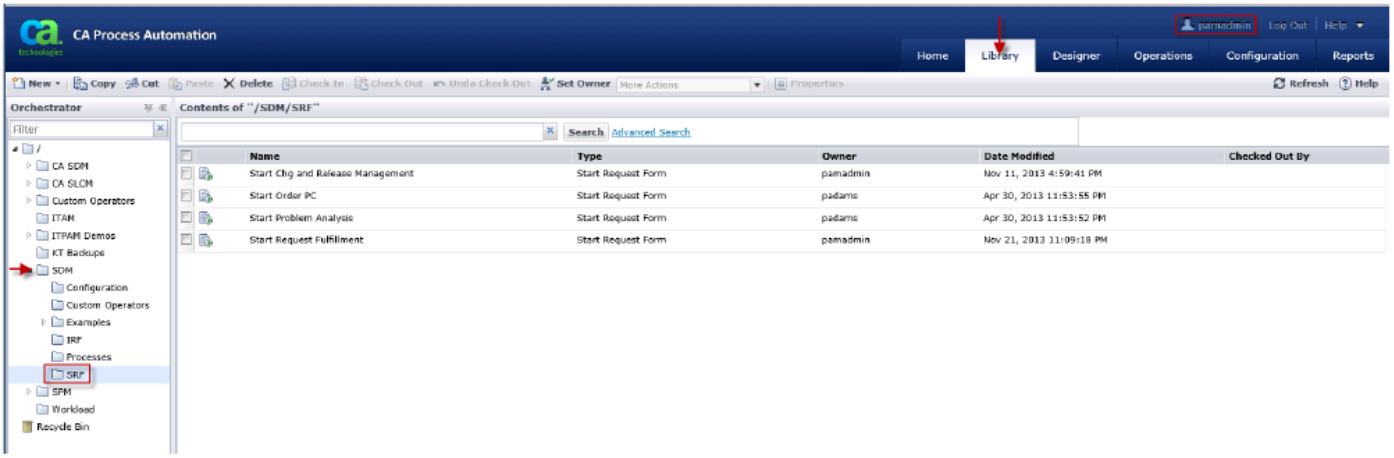
5. IRF's:



6. Processes:

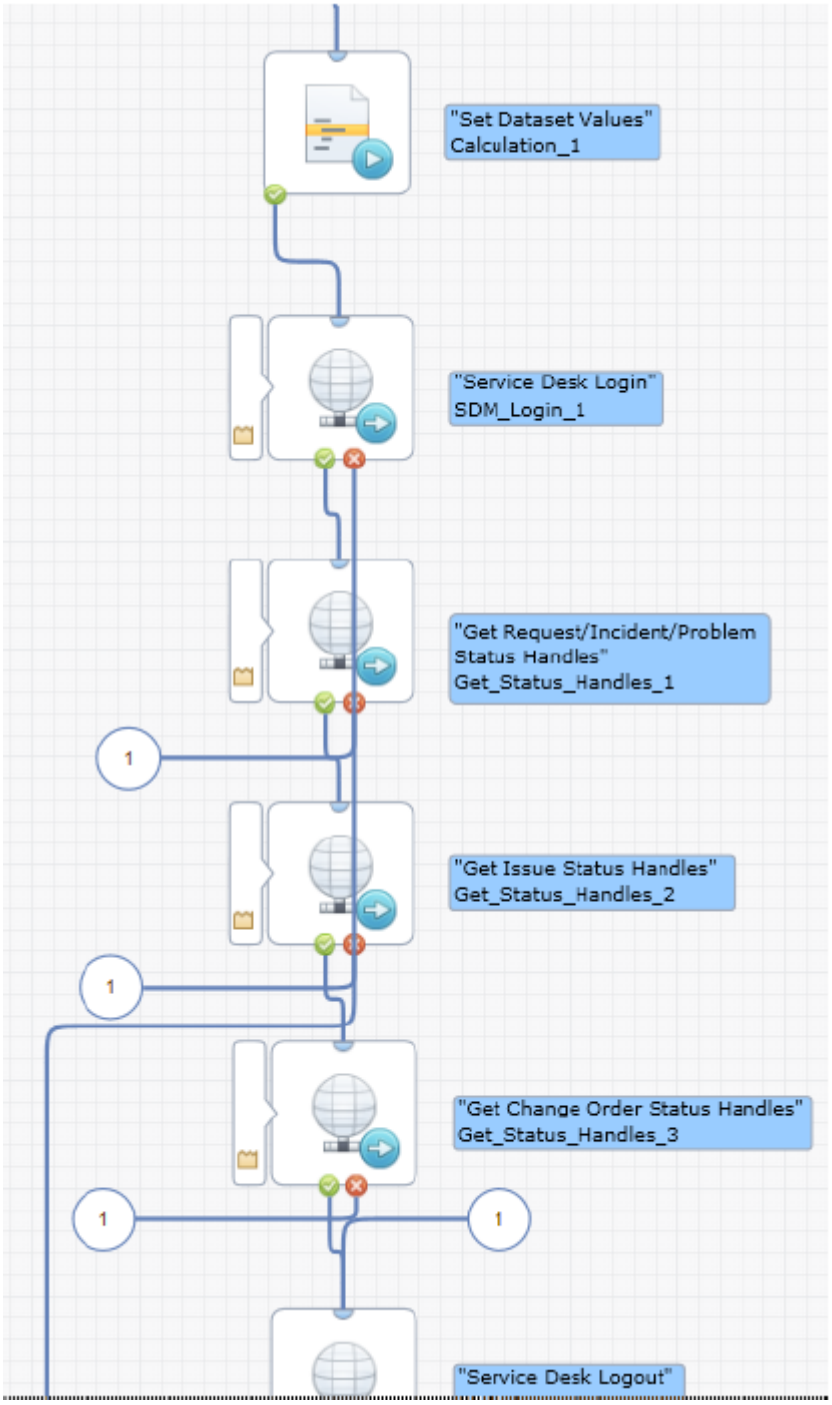


7. SRF's:

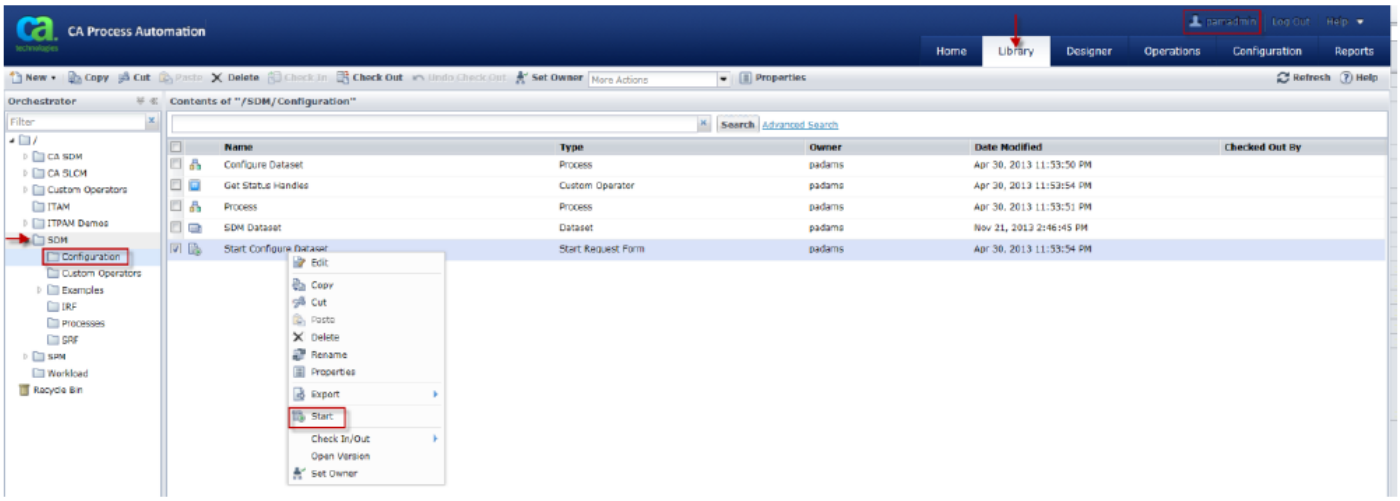


Configure SDM Dataset for Sample Process Definitions

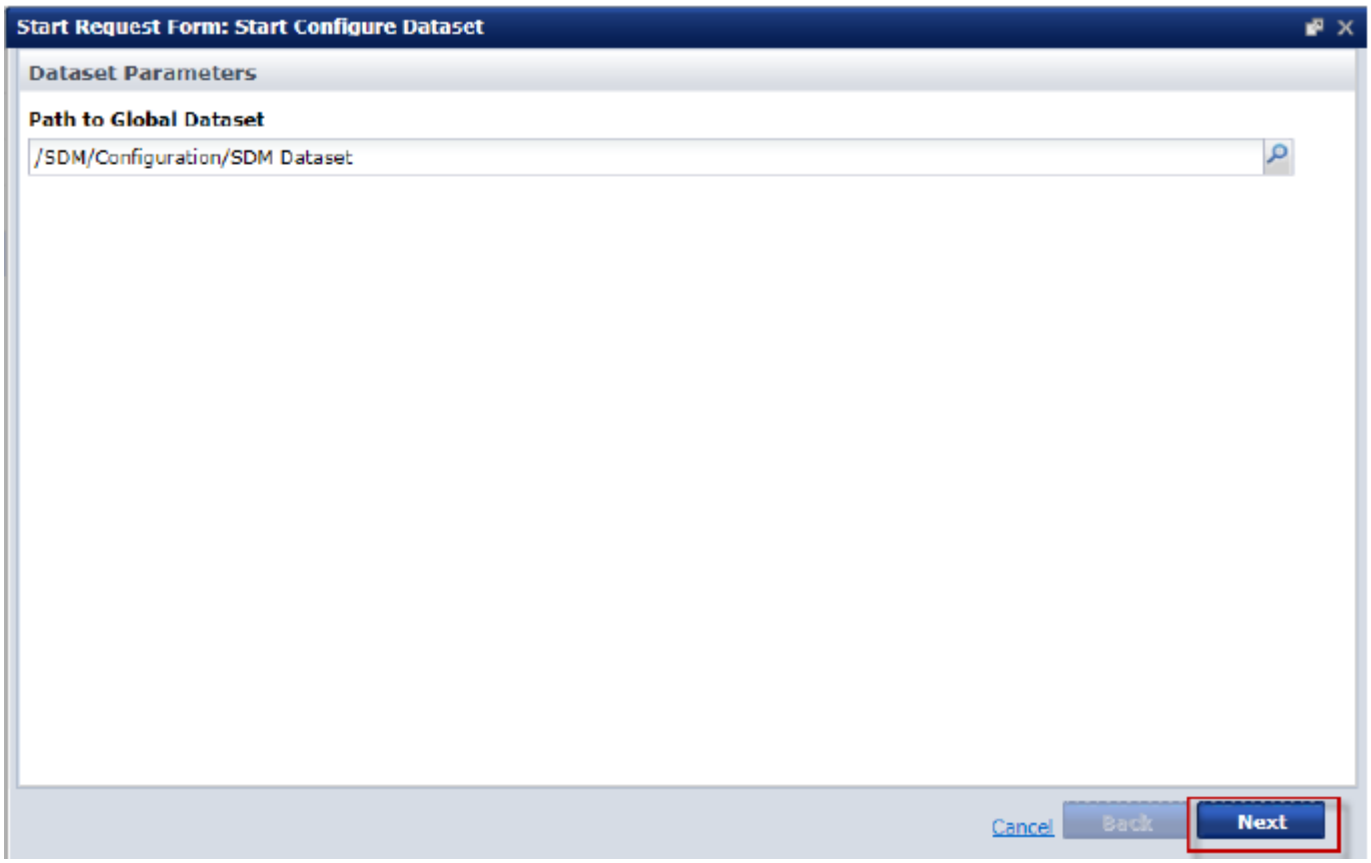
1. In the CA Process Automation Client, navigate back to SDM, Configuration. Here is where the dataset is configured for the sample process definitions. There is an SRF, ‘Start Configure Dataset’, which instantiates a ‘Configure Dataset’ Process which initializes the CA SDM Server, CA Process Automation Server, Exception Handling, and also obtains all handles (persid’s) for all ticket statuses:



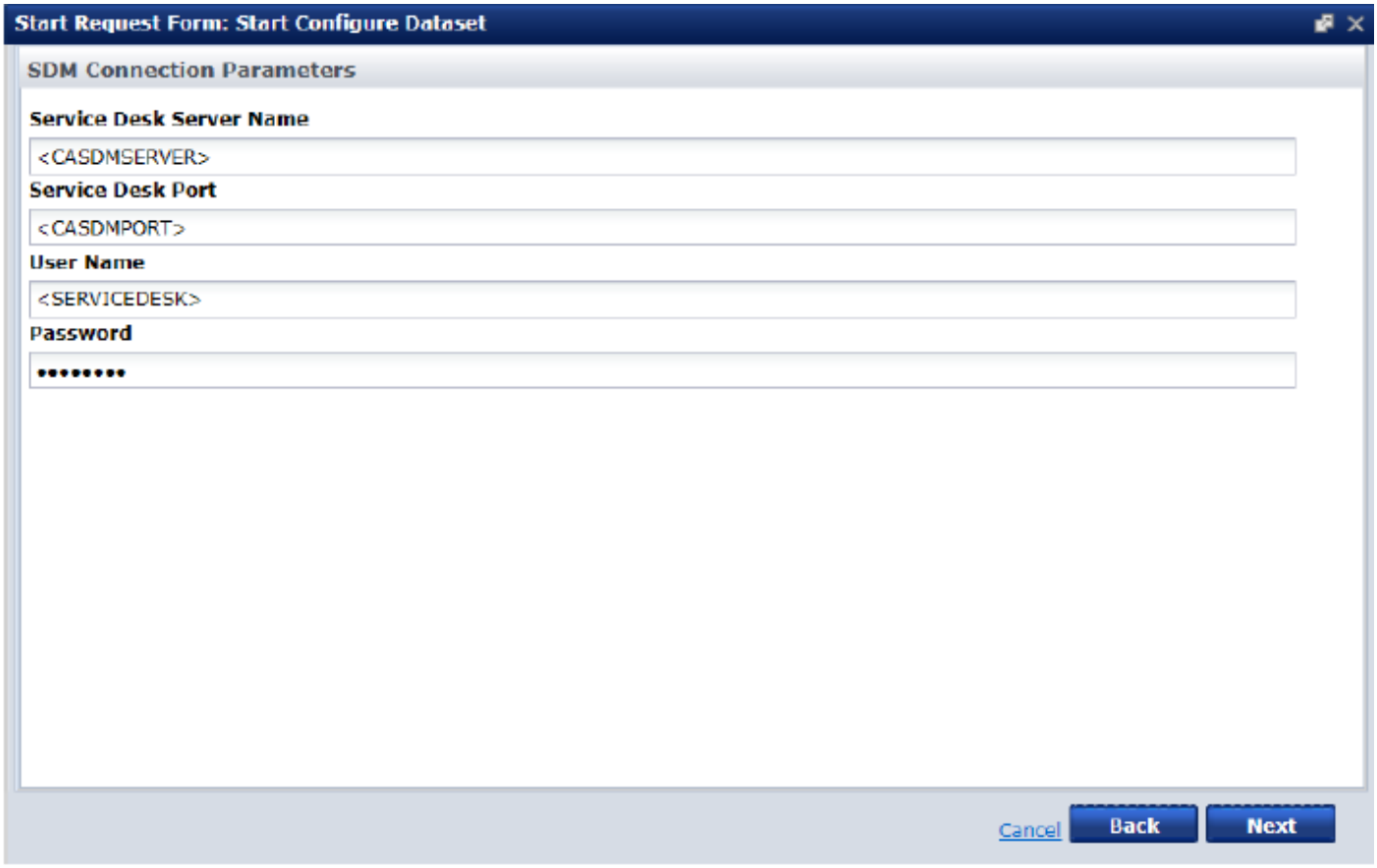
2. Launch the ‘Start Request Form’ for ‘Start Configure Dataset’:



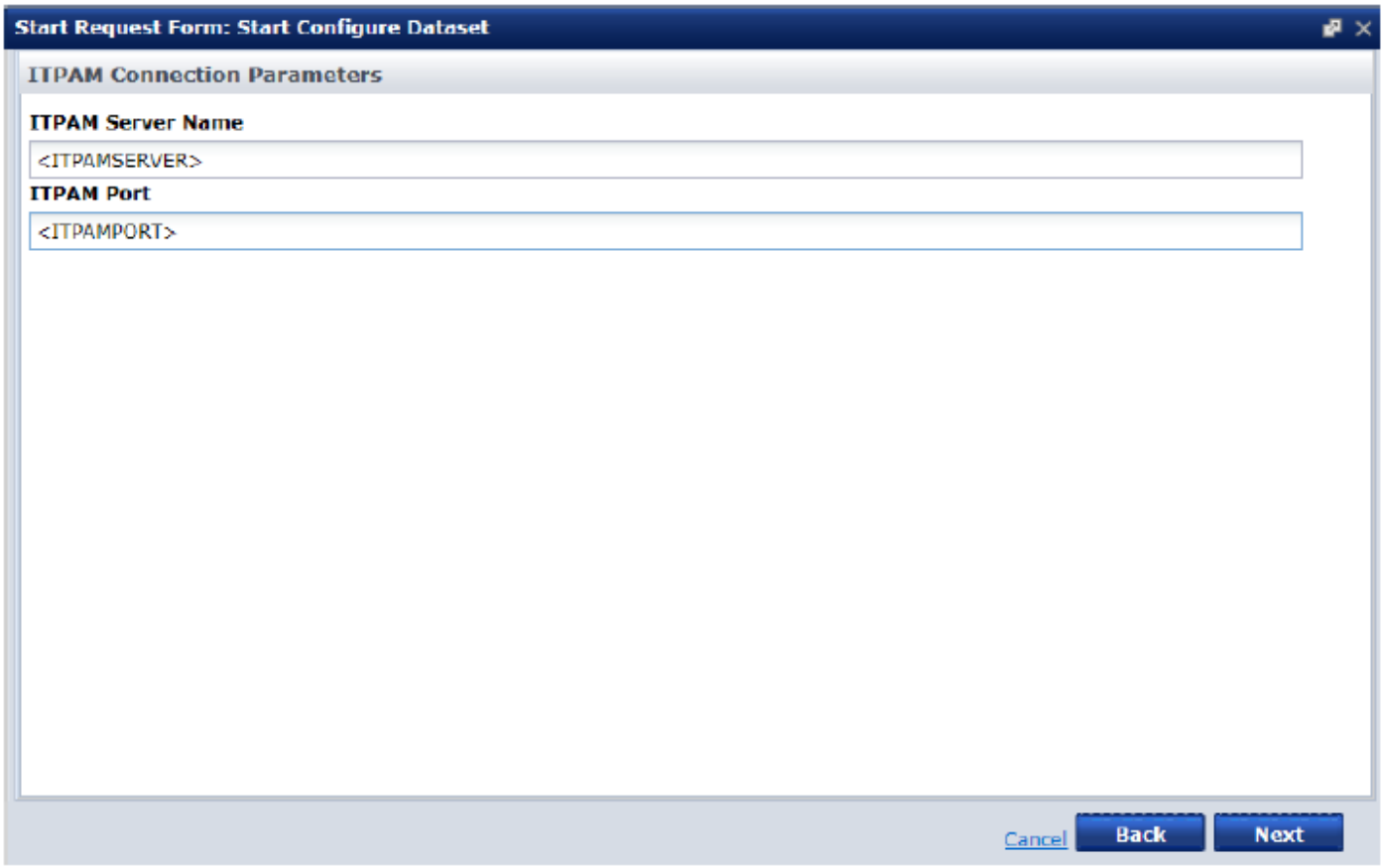
3. Verify the path to the Global Dataset (where all of the following data will be saved for other sample processes to leverage), and hit 'Next'.



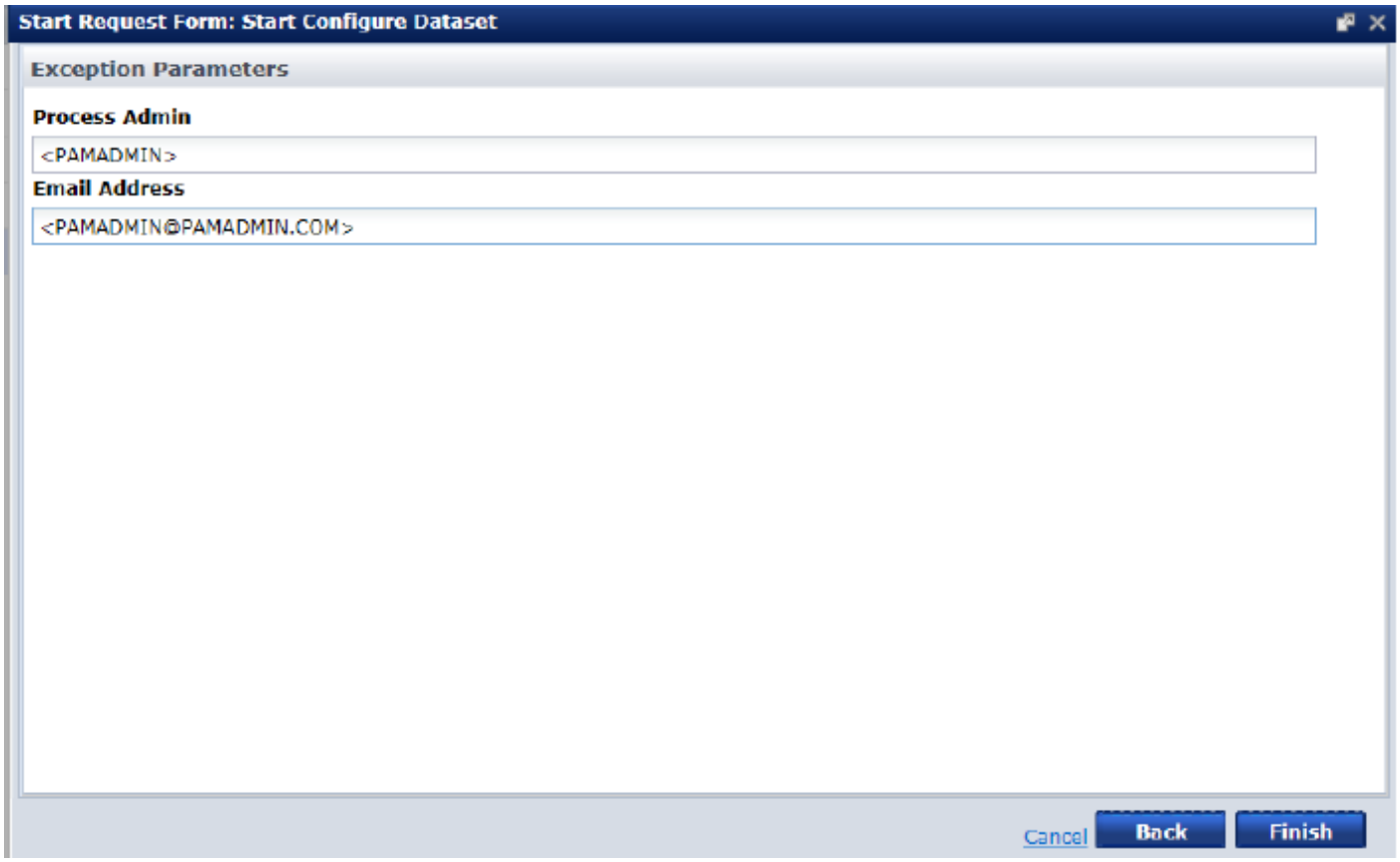
4. Enter in the values for the CA SDM server, port, Administrative Username and Password (replace what is shown here with your own data):



5. Enter in the values for the CA Process Automation server, and port (replace what is shown here with your own data):



6. Enter in the value for Process Admin and an email address (replace what is shown here with your own data):

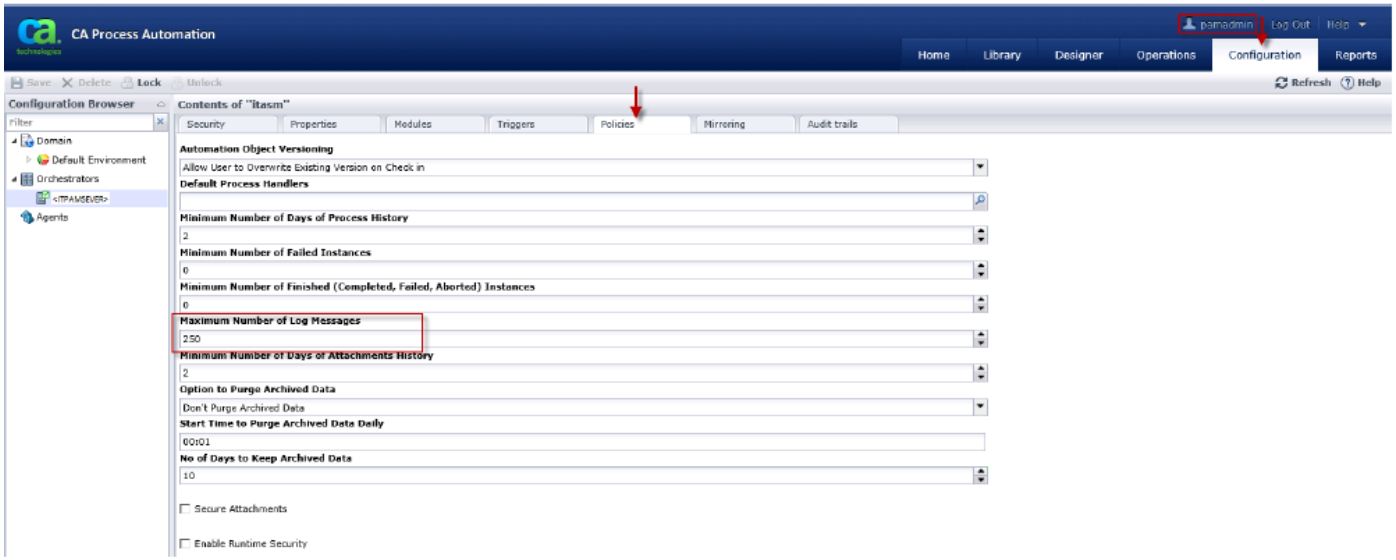


7. Hit Finish. Your CA Process Automation integration settings are now complete. All data resulting from this process definition are saved to the CA SDM Dataset.

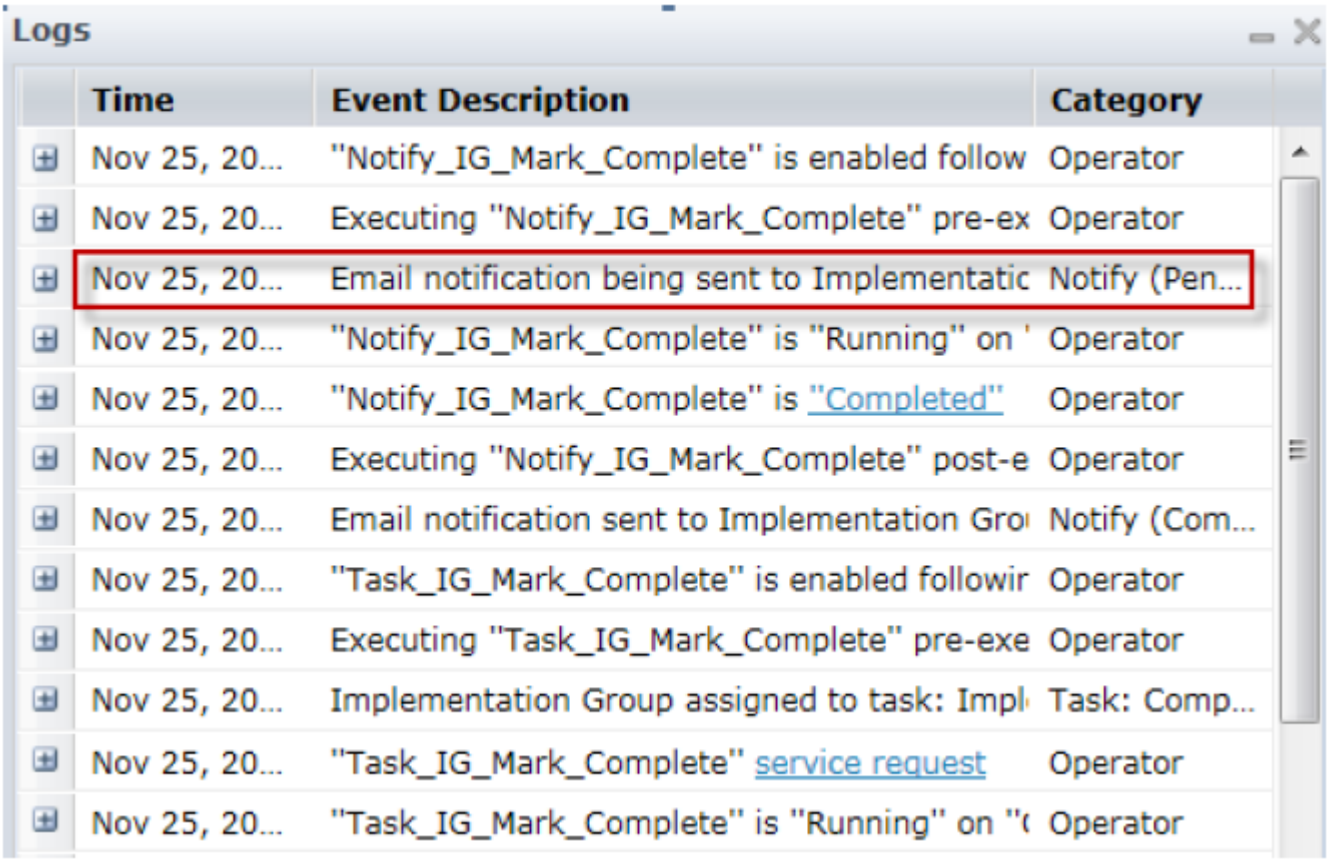
Increase 'Maximum Number of Log Messages' in CA Process Automation

Follow these steps:

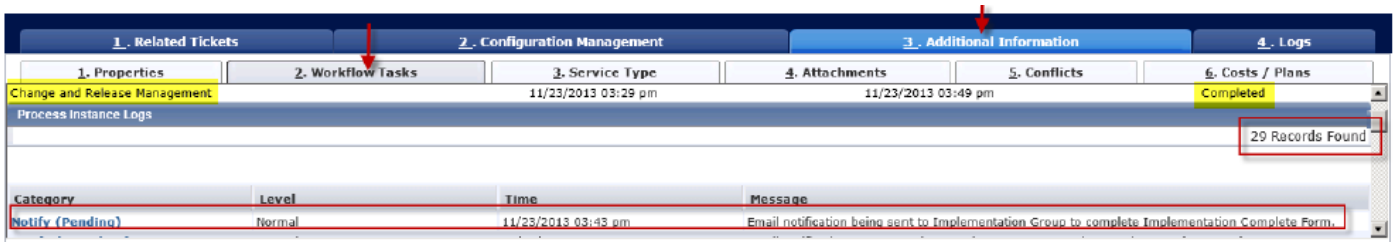
1. In the CA Process Automation Client, navigate to Configuration, select your Orchestrator and hit the ‘Policies’ Tab
2. Lock the Orchestrator for the change to be made and then unlock it, to save the changes.
3. Increase ‘Maximum number of Log Messages’ to 250.



4. This number will ensure that all Process Instance Logs in CA Process Automation...



5. are carried over to the Workflow Tasks Tab in a CA SDM Problem, Request, or Change Order. If you do not increase this number, you will find that the Workflow Tasks Tab will not show the full list of Process Instance Logs, only the last 29 will be saved in cache and displayed:



Verify CA SDM Options Manager Options

Configure the CA Process Automation Workflow options in Options Manager to enable connectivity between CA SDM and CA Process Automation.

Follow these steps:

1. Log in to CA SDM as an Administrator, using one of the following methods:
   a. Navigate to Start> Programs> CA> Service Desk Manager> Service Desk Web Client.
   b. Access the following URL:
   http://<casdmserver>:<casdmport>/CAisd/pdmweb.exe
2. Navigate to Administration, Options Manager, CA IT PAM Workflow.

You can view the list of options.

CASDM Options Manager Option	Description	Mandatory?	Value
Caextwf_eem_hostname	Specifies the name of the CA EEM server. The server name must match the host name that is returned by running the hostname command in the command prompt on your CA EEM Server. This option identifies the authentication host. You install caextwf_eem_hostname only if you configured CA Process Automation to use CA EEM as the authentication server. CA SDM uses this value to transform a user name and password into a CA EEM token, so that a username and password are not passed in plain text over HTTP. **Note:** If the CA Process Automation installation is using CA EEM for authentication, then place a value in the caextwf_eem_hostname option and install caextwf_eem_hostname. Entering a false value or installing caextwf_eem_hostname when it is not necessary causes the integration to fail.	No	<CAEEMSERVER>
Caextwf_endpoint	Specifies the URL that points to the CA Process Automation web services by including the CA Process Automation host name, port, and the mandatory /itpam/soap path. This option identifies the endpoint.	Yes	http://<CAITPAMSERVER>:<CAITPAMPORT>/itpam/soap

Caextwf_log_categories	When you install the caextwf_log_categories option, all CA Process Automation Process instance log messages from the Process and Operator categories appear on the Workflow Tasks tab within a CA SDM Ticket. You can also add in your own custom categories. Leveraging this capability, you can tailor the types of messages that are displayed and can make them more user-friendly and relative to your business. When you do not install caextwf_log_categories, only the CA Process Automation process instance log messages from the Process category appear on the Workflow Tasks tab. By default, the messages that are displayed on the Workflow Tasks tab within a CA SDM ticket have a 1:1 relationship with the messages that are displayed in the CA Process Automation Process Instance logs.	No	The default value is: Process, Operator Other values that you can define are: Notify (Pending), Notify (Complete), Task: Complete Form (Pending), Task: Complete Form (Complete), Task: Approve/Reject (Pending), Task: Approve/ Reject (Complete), Status Change (Pending), Status Change (Complete), Task: Other (Pending), Task: Other (Complete), Closure Code Change (Pending), Closure Code Change (Complete), Create Incident (Pending), Create Incident (Complete), Create Change Order (Pending), Create Change Order (Complete), Process Form Data
Caextwf_processdisplay_url	Specifies how to launch a graphical snapshot of a CA Process Automation Process instance from the Workflow Tasks tab in a ticket by supplying the host name and the mandatory /itpam/ JNLPRequestProcessor? processType=startUI&roid path. On the Workflow Tasks tab of a ticket, the user selects the View Process button to see the graphical snapshot.	Yes	http://< CAITPAMSERVER>:<ITPAMPORT>/ itpam/ JNLPRequestProcessor? processType=startUI&roid=

Caextwf_worklist_url	Specifies the process instance path by supplying the host name and the mandatory /itpam?webPage=mytaskfilter&view=tasklist path. This option enables CA SDM users to see a list of CA Process Automation Process instances that require attention, through the CA Process Automation Task list. The list appears in CA Process Automation when the CA SDM user selects a link that is associated with any pending task in the Workflow Tasks tab of a ticket.	Yes	http://< CAITPAMSERVER >:<ITPAMPORT>/itpam?webPage=mytaskfilter&view=tasklist "http://< CAITPAMSERVER >:/itpam?page=tasklist"
Caextwf_ws_password	Specifies the encrypted administrative password that is associated with the CA Process Automation user name from the caextwf_ws_user option. CA SDM uses the user name and password to access the CA Process Automation web service functions to perform integration activities. These activities are selecting start request forms, process definition information, process instance information, or launching process instances. The user name and password that you specify requires the appropriate access to CA Process Automation. However, it is not necessary that the CA Process Automation user name is defined as a CA SDM contact record.	Yes	<Encrypted password>

Caextwf_ws_use	Specifies the CA Process Automation administrative user name that is associated with the CA Process Automation user name. CA SDM uses the user name and password to access the CA Process Automation web service functions to perform integration activities. These activities are selecting Start Request Forms, process definition information, process instance information, or launching process instances. Typically this user would be pamadmin for install of CA Process Automation. The user must have appropriate access to CA Process Automation. However, it is not necessary that the CA Process Automation user name to exist as a CA SDM contact record.	Yes	Itpamadmin <for an upgraded instance of CA Process Automation, 'pamadmin' on a fresh install

3. Configure the required options, install and save.
4. Restart the CA SDM Services for changes to Options Manager Options to take effect.

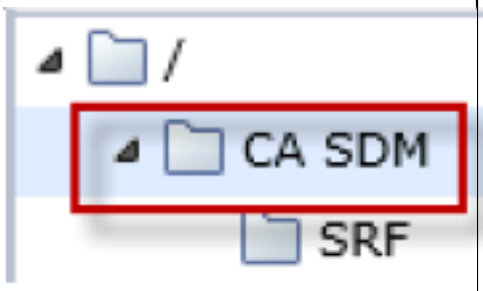
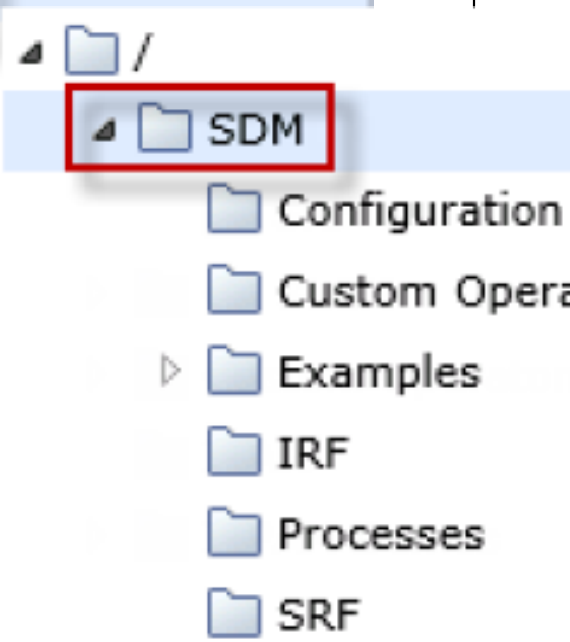
CA Process Automation Process Definitions for CA Service Desk Manager

This topic covers steps to import and configure the sample CA SDM CA Process Automation process definitions provided with the Content Pack for ITIL for the current release of CA SDM. Before we get started, let's discuss some of the CA Process Automation content that either is currently available, or has been available with previous versions of CA SDM, and how that content relates to the Content Pack for ITIL for the current release of CA SDM which we are focusing in on in this document.

Other CA Process Automation Content for CA SDM

This section describes additional CA Process Automation Content that either is currently available, or has been available with previous versions of CA SDM, and how that content relates to the Content Pack for ITIL for the CA SDM Release 17.4.

	ITIL Content Pack	CA Process Automation Sample Process Definitions for CA SDM
Available Out-of-box in CA SDM Release?	Yes, with a caveat. The original ITIL Content Pack was shipped with CA Process Automation Process Definitions and documentation, however what we provide out-of-box in the current release of CA SDM is the corresponding documentation only. The Process Definitions are no longer supported or available for download.	Yes. These have also been referred to as the 'CA SDM Out-of-box CA Workflow Samples Converted to CA Process Automation.' The Process Definitions are available out-of-box in the directory below, however they need to be manually imported into CA Process Automation.
Where to get it	$NX_ROOT/samples/Sample_ITIL_Content	$NX_ROOT/samples/CA_PAM_Workflows

What it includes	Documentation Only. The original ITIL Content Pack was shipped with Process Definitions and documentation, however what we provide out-of-box in CA SDM 14.1 is the corresponding documentation only. The Process Definitions are no longer supported or available for download.	The following CA Process Automation (Full) Process Definitions and corresponding documentation: - Order PC - Problem Analysis - Change Management
Where to check if you have imported this content into CA Process Automation?		Note that this is the same directory that the CA Process Automation Process Definitions in the Content Pack for ITIL for CA SDM are imported to, because it builds upon and enhanced this content. More information on this to follow later in this document: 
Supported?	No	Yes…but is replaced with the Content Pack for ITIL for CA SDM.
Replaced with Content Pack for ITIL for CA SDM CA Process Automation Process Definitions?	Yes, however note that the Content Pack for ITIL for CA SDM is not related to the ITIL Content Pack.	Yes, builds upon and enhances this content.

Prerequisites

This document is meant to be used in addition to the information provided in the Install CA Process Automation for CA Service Management.

This document assumes that the following applications have been installed and are operational:

- CA EEM 12.6.6 and above
- CA SDM 17.4

  - Email is configured and tested for CA SDM outbound email notifications
- CA Process Automation 4.4 CP02 and above
- This document also assumes that CA EEM, CA Service Desk Manager, and CA Process Automation have been integrated according to the instructions provided in the Install CA Process Automation for CA Service Management.

WARNING

This document will walk you through importing the CA Process Automation Process Definitions which are available with the 'Content Pack for ITIL for CA SDM 17.4'. In order for these process definitions to run successfully, you must have already installed the 'Content Pack for ITIL for CA SDM', as outlined in Install the ITIL Content for CA Service Desk Manager, which installs schema and HTMPL modifications, as well additional Administrative Data. The process definitions which we will walk though throughout this document rely heavily on this content.

Where to Find Future Updates to the CA SDM CA Process Automation Process Definitions

You can find any available updates on the CA SDM Homepage on support.broadcom.com, please be sure to check this location regularly.

CA SDM CA Process Automation Process Definitions Defined

Following are the process names:

Process Name	Building Block or Full Process Definition	New with Content Pack for ITIL for CA SDM, Updated, or Existing?	What it does
Close Ticket	Building Block	Existing	Closes a Request, Incident, Problem, Change Order or Issue.
Exception Handling	Building Block	Existing	Exception handling template which can be called from within any CA Process Automation process definition
Get Group Mem and Notify	Building Block	Existing	This building block is passed a group UUID and notify flags as input, and as a result returns a list of all group members, group manager, and notifications to these users dependent upon flags passed.
Template Process	Building Block	Existing	Template which can be used to create your own CA Process Automation Process definition.
Order PC	Full Process Definition	Existing	The goal of this process is to walk through an end user ordering a new PC and the subsequent approval steps in order to fulfill or reject this order. Should be called from a Change Category.

Problem Analysis	Full Process Definition	Existing	The goal of this process is to decide if an RFC/Change should be created for a given Problem. Once the problem assignee has finished researching the problem and has a suggested solution, they will fill out this problem survey and provide a recommendation. The survey should give the approver a good idea of the impact of the problem and if it is necessary to resolve it through Change Management. The idea of Problem Pain/Value Analysis comes from ITIL 2011. Should be called from within a Problem Area.
Request Fulfillment	Full Process Definition	New	The goal of this process is to provide a foundation for Request Fulfillment which aligns to ITIL 2011.
Change and Release Management	Full Process Definition	Updated	This process definition adds Release and Deployment Management to the existing Change Management process included in the 'CA Process Automation Sample Process Definitions for CA SDM'. The goal of this process is to provide a foundation for Change and Release and Deployment Management which aligns to ITIL 2011. Provides the ability to manage a Change Management process for all Standard, Normal or Emergency Changes from the initial Request for Change through to its deployment and Post Implementation Review. It analyzes the 'Type' of Change Order and associated 'Risk' Level to determine what level of approvals are necessary for Change implementation. Should be called from a Change Category.

NOTE

The difference between a 'full process definition' and a 'building block' is that full process definition can is designed to be attached to a change category or macro directly to be instantiated. A building block is designed to be invoked from within another process definition.

CA SDM CA Process Automation Custom Operators Defined

Following are the process names:

Process Name	New with Content Pack for ITIL for CA SDM, Updated, or Existing?	Process or Soap Operator?	CA SDM Web Service Invoked	Input Parameters	What it does
Attach Change Order to Request	Existing	Soap	attachChangeToRequest	Assignee Handle Request Handle Description	Attaches a Change Order to a Request
Change Status	Existing	Soap	changeStatus	Creator Handle Ticket Handle New Status Code Comments	Updates the status of a ticket
Close Ticket	Existing	Soap	closeTicket	Persid Description	Closes a ticket
Close Ticket & Logout	Existing	Process	Via Close Ticket Process	Ticket Handle Initial Delay Comments Ticket Number	Invokes Close Ticket Process which closes a ticket and logs out CA SDM Web Services
Create Request	Existing	Soap	createRequest	Creator Handle Parent Change Handle Request Type Description Customer Handle Parent Request Handle Template	Creates a new Request/Incident/ Problem, dependent upon the ‘type’ parameter passed.
Exception Handling	Existing	Process	Via Exception Handling Process	Ticket Handle Exception Type Process Name Process ID Node Name Touchpoint Name Agent Name Error Message Operation Name Fault Response	Dependent upon the error message received, the exception handler will determine if the system should skip the operation, try and obtain a new Session ID (SID), try the operation again, or notify the System Administrator.
Get Assignee Handle	Existing	Soap	doSelect	Ticket Type Ticket Handle	Gets handle of ticket assignee
Get Contact Info	Existing	Soap	getContact	Contact ID	Retrieves attributes of a contact

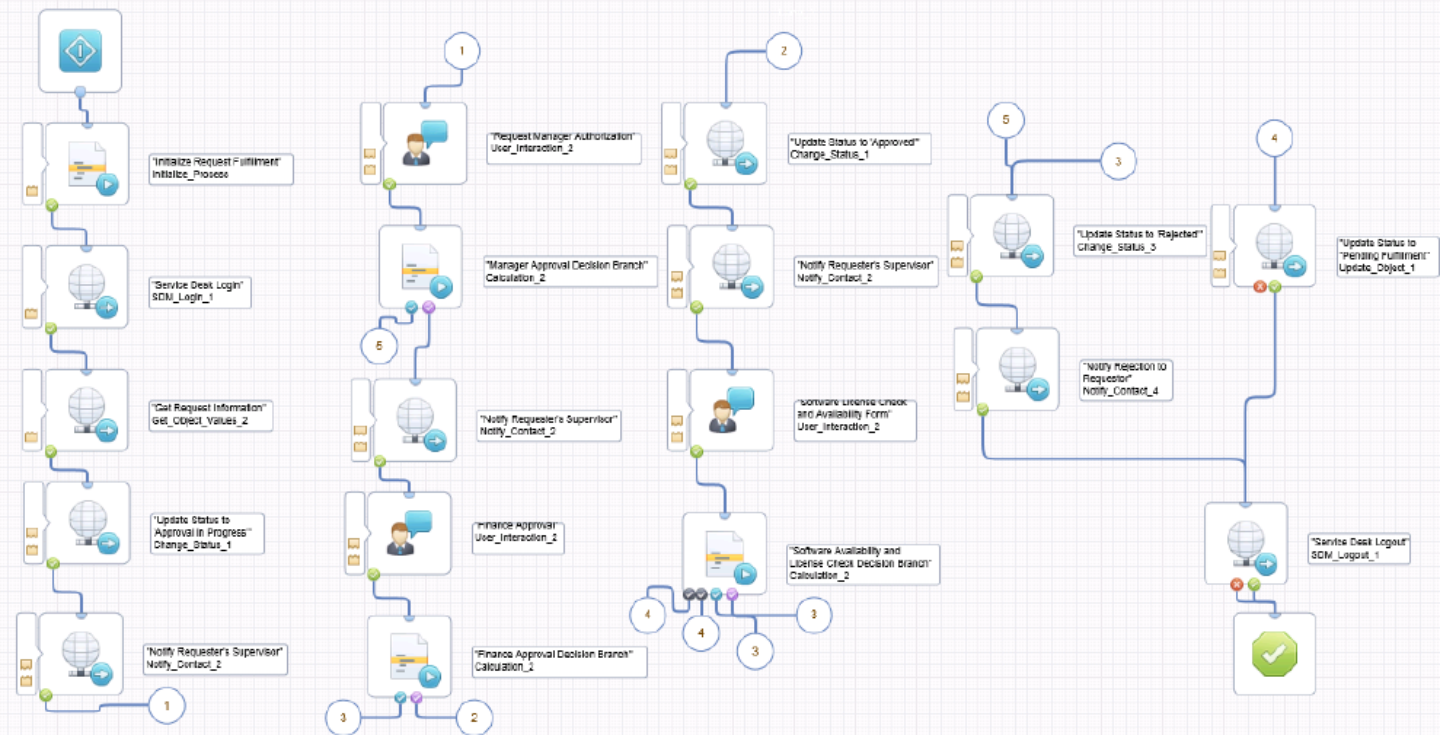
Get Group Mem and Notify	Existing	Process	Via Get Group Mem and Notify Process	Creator Handle Ticket Handle Group Handle Notify List Message Title Message Body Urgency Internal Only	Invokes the Get Group Mem and Notify Process, which is passed a group UUID and notify flags as input, and as a result returns a list of all group members, group manager, and notifications to these users dependent upon flags passed.
Get Group Members	Existing	Soap	getGroupMemberListValues	Group Handle	Retrieves members of a group.
Get Handle for UserID	Existing	Soap	getGroupMemberListValues	User ID	Retrieves handle for a given user ID.
Get Object Values	Existing	Soap	getObjectValues	Object Handle Attributes	Retrieves all attributes for a Request/Incident/ Problem/Change Order. The return is a UDSObject (XML); retrieve data with XPATH.
Get Related List Values	Existing	Soap	getRelatedListValues	Object Handle List Name	Returns values for lists related to a specific object. The lists must be defined as a QREL or BREL. Use the LREL methods to query LREL types.
Get Status Handle	Existing	Soap	doSelect	Ticket Type New Status Code	Retrieves the handle for a given status code, to be passed to the Update Status Method.
Initialize Process	Existing	Process	Via SDM Dataset		Initializes variables in the SDM Dataset to be used in the sample process definitions.
Link Asset	Existing	Soap	createLrelRelationships	Object Handle Asset Handle	Links an Asset to a Request, Incident, Problem, Change Order or Issue.
Link Request to Change	Existing	Soap	updateObject	Request Handle Change Handle	Links a Request/ Incident/Problem to a Change Order.

Notify Contact	Existing	Soap	notifyContact	Creator Handle Ticket Handle Contact Handle Message Title Message Body Urgency Internal Only	Sends a notification.
SDM Login	Existing	Soap	login	Soap Request File Soap Response File	Performs a Service Desk Web Service login and returns the SID
SDM Logout	Existing	Soap	Transfer	Creator Handle Ticket Handle Activity Log New Assignee Handle New Group Handle New Organization Handle	Transfers ownership of a Request, Incident, Problem, Change Order or Issue.
Transfer Ticket	Existing	Soap	Transfer	Creator Handle Ticket Handle Activity Log New Assignee Handle New Group Handle New Organization Handle	Transfers ownership of a Request, Incident, Problem, Change Order or Issue.
Update Object	Existing	Soap	updateObject	Ticket Handle Attribute New Value	Update any single attribute in an object. This supports fields that are NOT LREL's such as asset (CI). If you want to update an SREL field like priority ensure to pass the handle (persid).
Update Status & Comment	Existing	Process	Via Update Status	Creator Handle Ticket Handle New Status Code Comments	Updates ticket status and logs a comment in the ticket.

Request Fulfillment Workflow

This topic contains the following information:

Design View in CA Process Automation



How to configure the Request Fulfillment Workflow

There is data which is a pre-requisite for running this Process Definition, which you must manually configure. It is listed in the following table. Note that there is ‘Example Data’ specified in the middle column of the chart below which indicates the name of the Contacts/Request Areas, etc that we used as a part of our walk-through, these of course can be changed to Contacts/Request Areas, etc for your own environment.

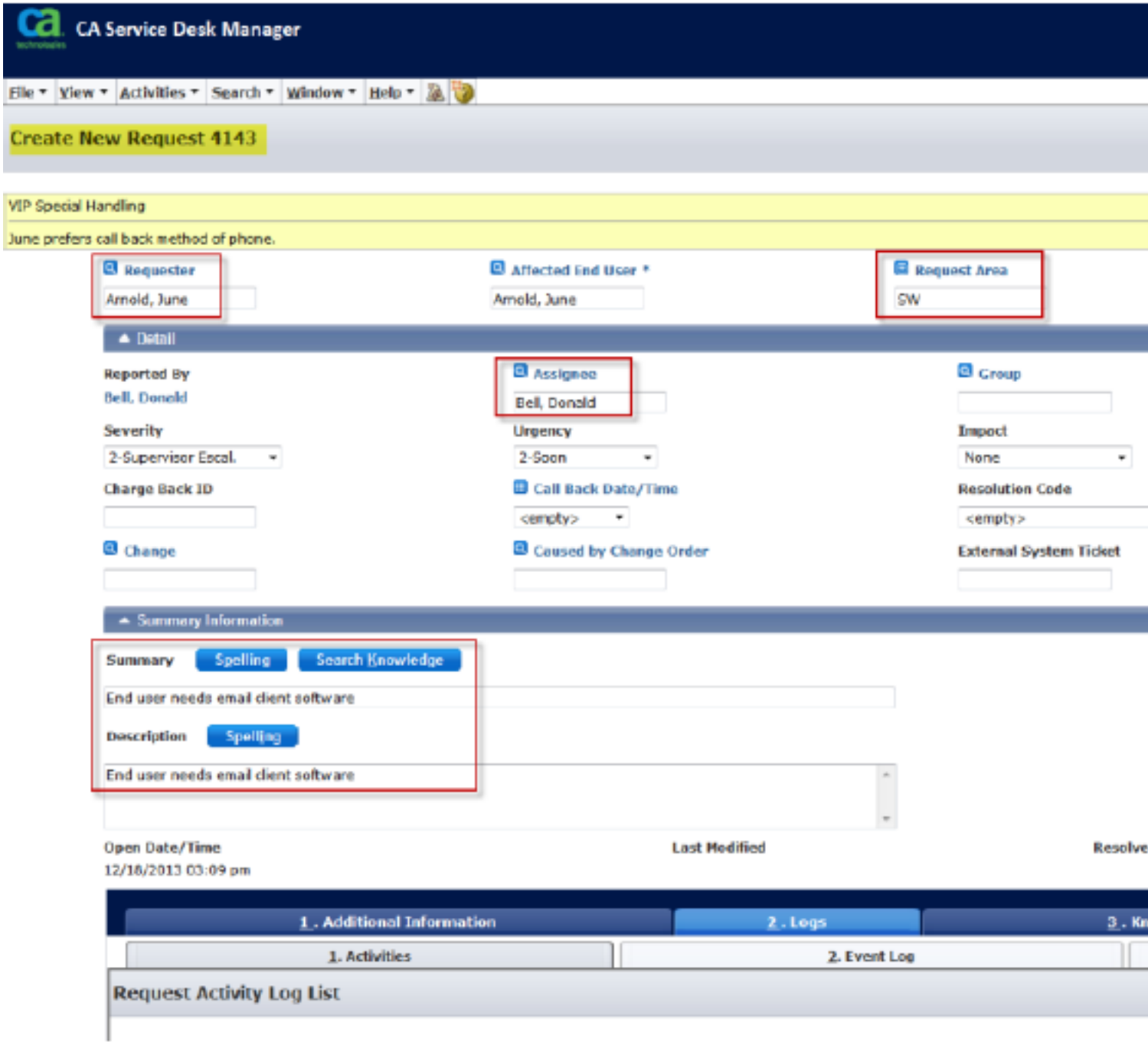
CA SDM Object/Option	Example Data	Notes
Add contact (Analyst)	June Arnold	June will be the Analyst creating/assigned to the Request. 1. Must be defined as a contact in CA SDM as well as CA EEM, if you are leveraging EEM for authentication. 2. Must have an email address defined in the CA SDM contact record. 3. Must have a notification method set to ‘Email’ in the CA SDM contact record.

Add contact (Analyst)	Donald Bell	Donald will be June's supervisor, and as such, will be the approver of the Request. 1. Must be defined as a contact in CA SDM as well as CA EEM, if you are leveraging EEM for authentication. 2. Must have an email address defined in the CA SDM contact record. 3. Must have a notification method set to 'Email' in the CA SDM contact record. 4. Assign Donald as June's Supervisor by going to June's' contact record, and navigating to the Organization Tab, and entering in Donald as June's supervisor.
Create Request Area	SW	This Request Area will have the following values set: 1. Assignee: June Arnold 2. <in the 'Workflow' Tab> CA IT PAM Process Name: 'Request Fulfillment'
Create CI	Email Service	Create a new configuration item, (Service Desk Tab->File->New Configuration Item), which we will assign to the Problem as we go through the demo. 1. Email Service
Install 'Category_Defaults' Option.	Yes	This option needs to be installed so that the information from the SW Request Area is pulled over to the Request once it is created. Recycle CA SDM services for changes to take effect

Request Fulfillment Workflow in Action

Follow these steps:

1. Login to CA SDM as Donald Bell, create a new Request.
2. Fill out all of the fields highlighted in red below, then hit save.



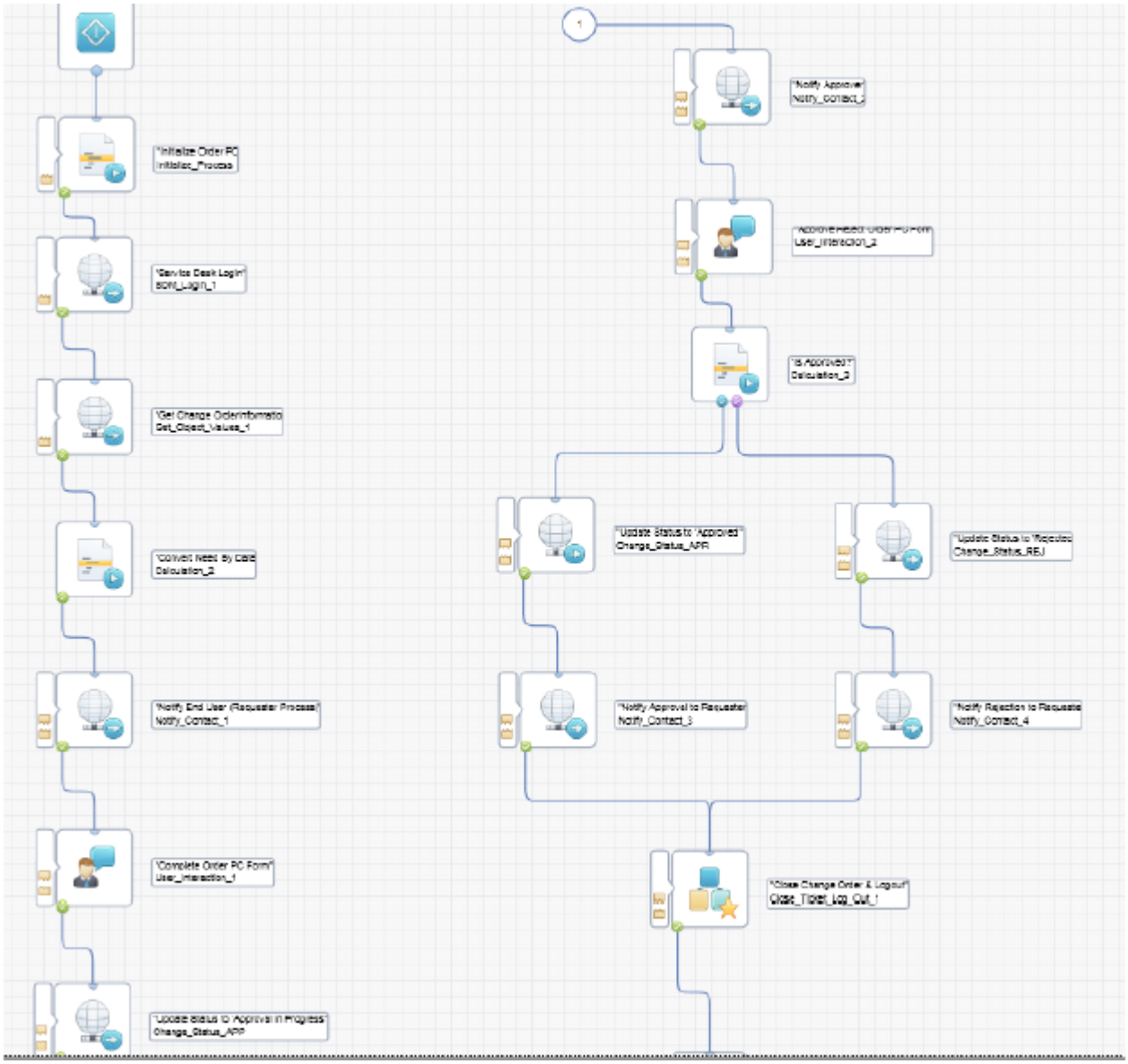
3. Click on the Relationships tab, then drill down to the Workflow Tasks Tab. Scroll down through the tasks. You will see the status has changed to ‘Approval in Progress’.
4. Continue to scroll through the tasks. You will see a notification has gone out to the approver (Donald Bell) to authorize end user access. To complete this task Donald can either click on the link provided via email, or alternatively, he can scroll to the next pending task in the Workflow and click on the hyperlink for ‘Task: Approve/Reject (Pending)’.
5. Once logged in to CA Process Automation as Donald, you should see one pending task to complete the ‘Request Manager Authorization Form’. RMC on the task, and select ‘Reply’.
6. Complete the information in the Request Manager Authorization Form. In this case, approve the request.

7. Navigate back to the Request. Click on the Relationships Tab, and then drill down to the Workflow Tasks Tab. Scroll down through the tasks. You will see a notification has gone out to the assignee (June Arnold's) supervisor for Financial Approval, which in our case is Donald Bell. To complete this task Donald can either click on the link provided via email, or alternatively, he can scroll to the next pending task in the Workflow and click on the hyperlink for 'Task: Approve/Reject (Pending)'.
8. Once logged in to CA Process Automation as Donald, you should see one pending task to complete the 'Finance Approval Form'. RMC on the task, and select 'Reply'.
9. Complete the information in the Financial Approval Form. In this case, approve the request.
10. Navigate back to the Request. The status of the Request is now 'Approved'. Click on the Relationships Tab, and then drill down to the Workflow Tasks Tab. Scroll down through the tasks. You will see a notification has gone out to the requestor (June Arnold's) supervisor (Donald Bell) to confirm license check and availability of software. To complete this task Donald can either click on the link provided via email, or alternatively, he can scroll to the next pending task in the Workflow and click on the hyperlink for 'Task: Approve/Reject (Pending)'.
11. Once logged in to CA Process Automation as Donald, you should see one pending task to 'Confirm software license and availability'. RMC on the task, and select 'Reply'.
12. Confirm software license and availability, and hit 'Finish':
13. Navigate back to the Request. The status of the Request is now 'Pending Fulfillment'.
14. The workflow is now complete.

Order PC Workflow

This topic contains the following information

Design View in Process Automation



How Configure the Order PC Workflow

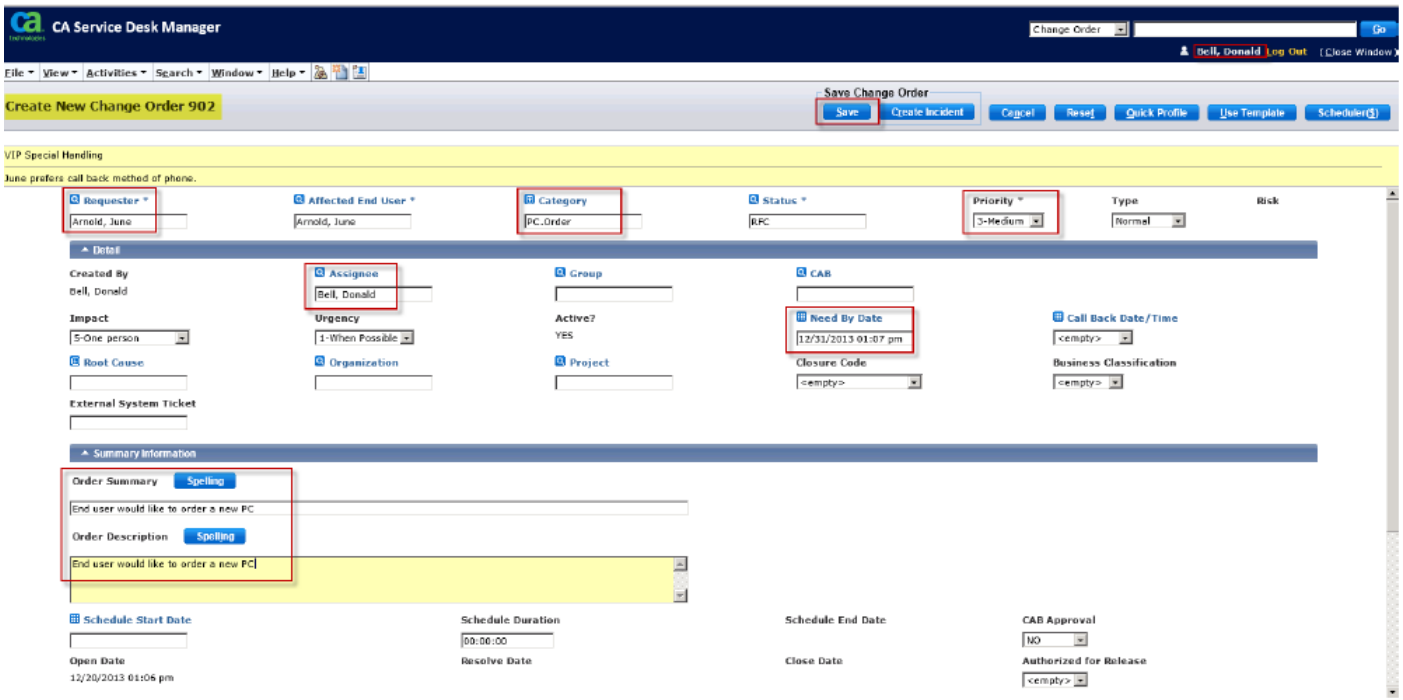
There is data which is a pre-requisite for running this Process Definition, which you must manually configure. It is listed in the following table. Note that there is ‘Example Data’ specified in the middle column of the chart below which indicates

the name of the Contacts/Categories, etc that we used as a part of our walk-through, these of course can be changed to Contacts/Categories, etc for your own environment.

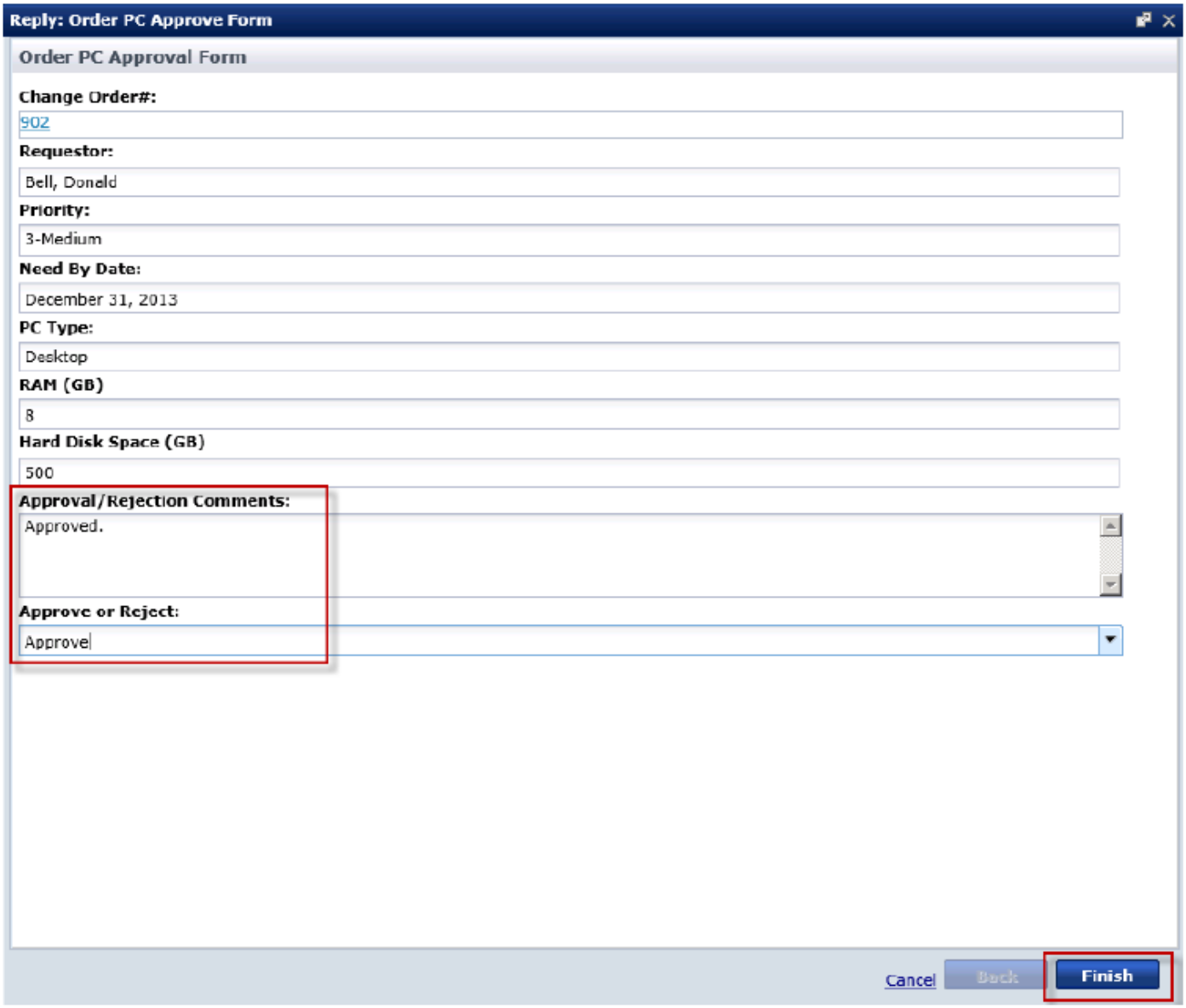
CA SDM Object/Option	Example Data	Notes
Add contact (Employee)	June Arnold	June will be the end user who requests the PC. 1. Must be defined as a contact in CA SDM as well as CA EEM, if you are leveraging EEM for authentication. 2. Must have an email address defined in the CA SDM contact record. 3. Must have a notification method set to 'Email' in the CA SDM contact record.
Add contact (Analyst)	Donald Bell	Donald will be the Analyst creating the Change Order. He will also be the Approver assigned to the Change Order Category, and thus assigned to the Change Order (through Category_Defaults Option, see below for details) 1. Must be defined as a contact in CA SDM as well as CA EEM, if you are leveraging EEM for authentication. 2. Must have an email address defined in the CA SDM contact record. 3. Must have a notification method set to 'Email' in the CA SDM contact record.
Add contact (Analyst)	John McCarthy	John will be the approver assigned to the Change Order Category, and thus assigned to the Change Order (through Category_Defaults Option, see below for details) 1. Must be defined as a contact in CA SDM as well as CA EEM, if you are leveraging EEM for authentication. 2. Must have an email address defined in the CA SDM contact record. 3. Must have a notification method set to 'Email' in the CA SDM contact record.
Create Change Order Category	PC.Order	This category will have the following values set: 1. Assignee: Donald Bell 2. <in the 'Workflow' Tab> CA IT PAM Process Name: 'Order PC'
Install 'Category_Defaults' Option	Yes	This option needs to be installed so that the information from the PC.Order Change Category is pulled over to the Change Order once it is created. Recycle CA SDM services for changes to take effect.

Order PC Workflow in Action

1. Login to CA SDM as Donald Bell, create a new Change Order, either from scratch or based on an existing Incident.
2. Fill out all of the fields highlighted in red below, also including ‘Backout Plan’ on the ‘Cost/Plans’ Tab. (There is a status transition in place such that this field must not be NULL in order for Change Order to move from status ‘Approval in Progress’ to ‘Approved’.) Then hit save.



3. Click on the Additional Information Tab, and then drill down to the Workflow Tasks Tab. Scroll down through the tasks. You will see a notification has gone out to the requester (June Arnold) to complete the ‘Order PC’ Form. To complete this task June can either click on the link provided via email, or alternatively, she can scroll to the next pending task in the Workflow and click on the hyperlink for ‘Task: Complete Form (Pending)’:
4. Once logged in to CA Process Automation as June, you should see one pending task to ‘Complete the ‘Order PC Form’. RMC on the task, and select ‘Reply’:
5. Complete the information in the Order PC Form. Hit ‘Finish’,
6. Navigate back to the Change Order. The status of the Change Order is now ‘Approval in Progress’. Click on the Additional Information Tab, and then drill down to the Workflow Tasks Tab. Scroll down through the tasks. You will see a notification has gone out to the approver (assignee of the Change Category) to Approve/Reject, which in our case is John McCarthy. To complete this task John can either click on the link provided via email, or alternatively, he can scroll to the next pending task in the Workflow and click on the hyperlink for ‘Task: Approve/Reject (Pending)’:
7. Once logged in as John, you should see one pending task to ‘Approve/reject the PC Order’. RMC on the task, and select ‘Reply’.
8. All of the information that had been initially filled out by Donald remains in this form for John to review. Either approve, reject, or mark as incomplete. In this example we will be approving this order. Hit finish.

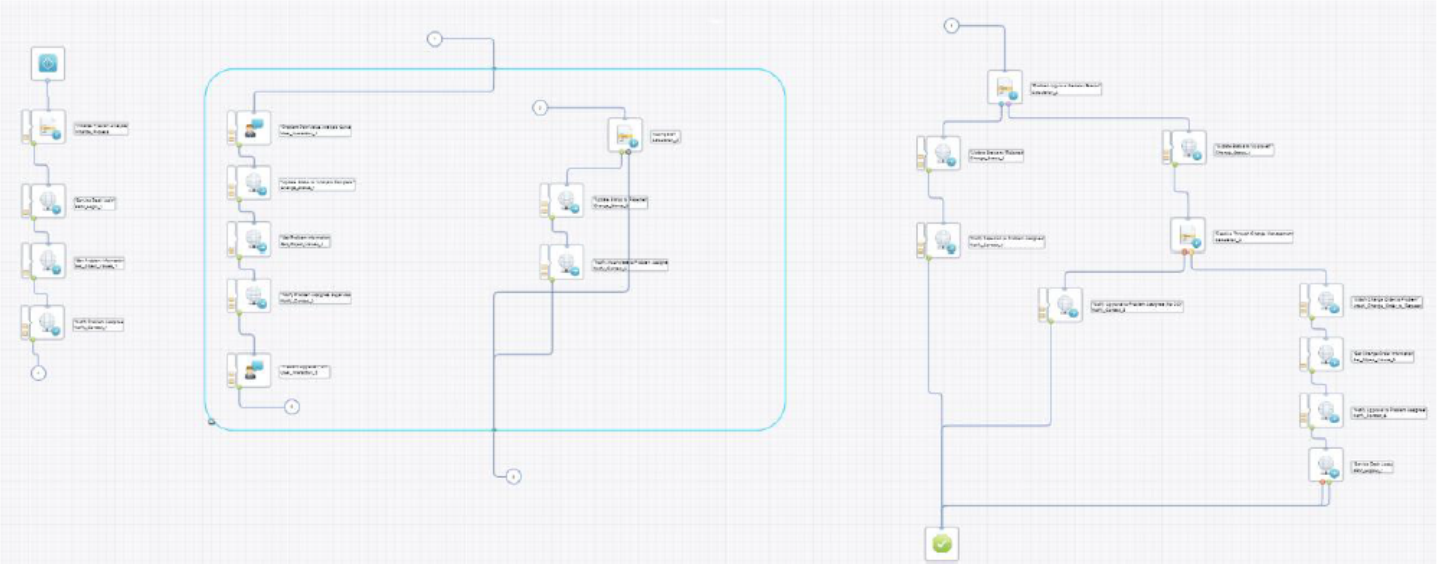


9. Navigate back to the Change Order. The status of the Change is now ‘Approved’. Click on the Additional Information Tab, and then drill down to the Workflow Tasks Tab. Scroll down through the tasks. You can see that an email notification has one out to the Requester saying it has been approved.
10. Refresh the Change Order. Scroll further down in the Workflow Tasks. You will see that the status has changed to ‘Closed’.

Problem Analysis Workflow

This topic contains the following information:

Design view in CA Process Automation



How to Configure the Problem Analysis Workflow

There is data which is a pre-requisite for running this Process Definition, which you must manually configure. It is listed in the following table. Note that there is ‘Example Data’ specified in the middle column of the chart below which indicates the name of the Contacts/Problem Areas, etc that we used as a part of our walk-through, these of course can be changed to Contacts/Problem Areas, etc for your own environment.

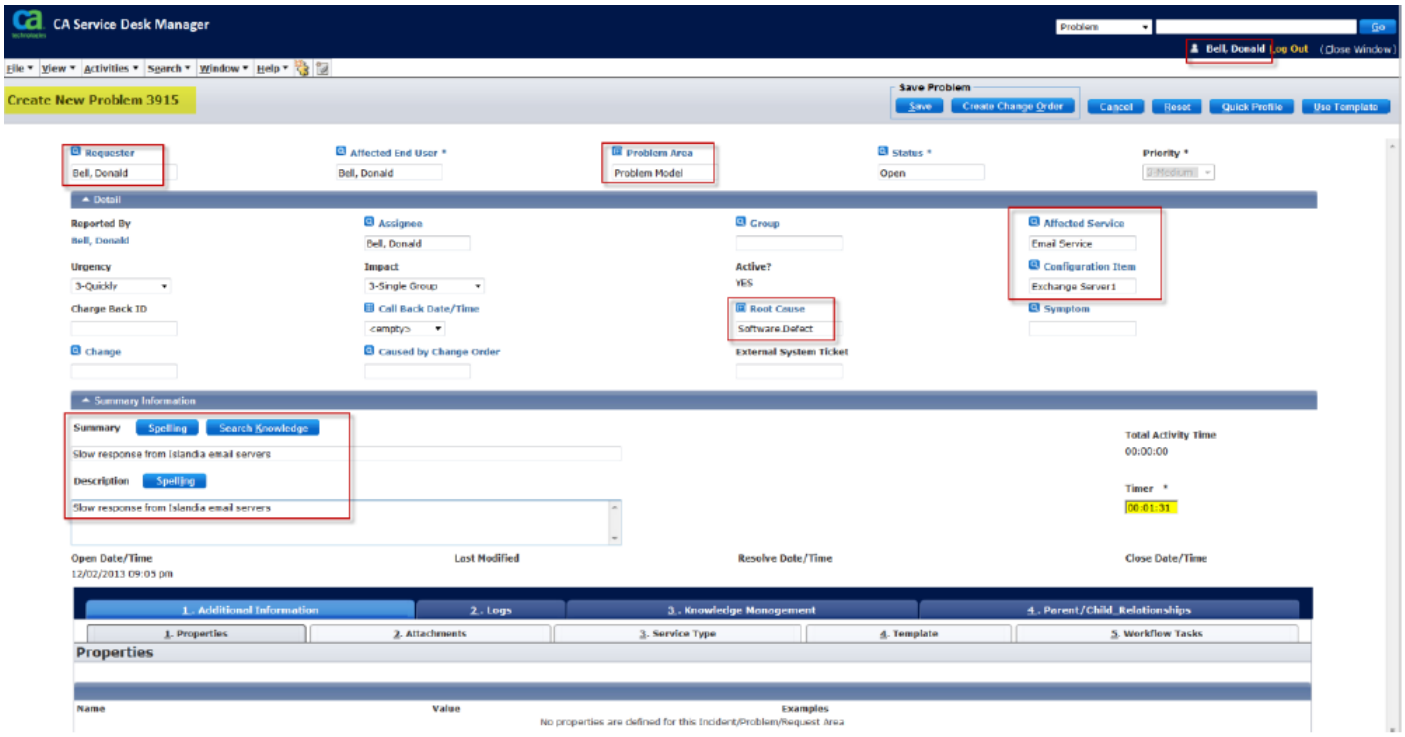
CA SDM Object/Option	Example Data	Notes
Add contact (Employee)	June Arnold	June will be the end user who creates the initial Incident, that leads to the creation of the Problem. If a Problem is created directly, she will be the end user requesting the problem. 1. Must be defined as a contact in CA SDM as well as CA EEM, if you are leveraging EEM for authentication. 2. Must have an email address defined in the CA SDM contact record. 3. Must have a notification method set to ‘Email’ in the CA SDM contact record.
Add contact (Analyst)	Donald Bell	Donald will be the Analyst creating/ assigned to the Change Order resulting from the Problem. 1. Must be defined as a contact in CA SDM as well as CA EEM, if you are leveraging EEM for authentication. 2. Must have an email address defined in the CA SDM contact record. 3. Must have a notification method set to ‘Email’ in the CA SDM contact record.

Add contact (Analyst)	Marc Craigie	Marc will be Donald's supervisor, and as such, will be the approver of the Problem. 1. Must be defined as a contact in CA SDM as well as CA EEM, if you are leveraging EEM for authentication. 2. Must have an email address defined in the CA SDM contact record. 3. Must have a notification method set to 'Email' in the CA SDM contact record. 4. Assign Marc as Donald's Supervisor by going to Donald's contact record, and navigating to the Organization Tab, and entering in Marc as Donald's supervisor.
Create Problem Area	Problem Model	This Problem Area will have the following values set: 1. Assignee: Donald Bell 2. <in the 'Workflow' Tab> CA IT PAM Process Name: 'Problem Analysis'
Create CI's	Email Service Exchange Server1	Create two new configuration items (Service Desk Tab->File->New Configuration Item), which we will assign to the Problem as we go through the demo: 1. Email Service 2. Exchange Server
Create root cause	Software Defect	Create a root cause (Administration Tab->Service Desk->Application Data->Codes->Root Cause), which we will assign to the Problem as we go through the demo: 1. Software Defect
Install 'Category_Defaults' Option	Yes	This option needs to be installed so that the information from the 'Problem Model' Problem Area is pulled over to the Problem once it is created. Recycle CA SDM services for changes to take effect.

Problem Analysis Workflow in Action

Follow these steps:

1. Login to CA SDM as Donald Bell, create a new Problem, either from scratch or based on an existing Incident.
2. Fill out all of the fields highlighted in red below, then hit save.



3. Click on the Additional Information Tab, and then drill down to the Workflow Tasks Tab. Scroll down through the tasks. You will see a notification has gone out to the assignee (Donald Bell) to complete the Problem Pain/Value Analysis Form. To complete this task Donald can either click on the link provided via email, or alternatively, he can scroll to the next pending task in the Workflow and click on the hyperlink for ‘Task: Complete Form (Pending)’:
4. Once logged in to CA Process Automation as Donald, you should see one pending task to complete the ‘Problem Pain/Value Analysis Survey’. You will also see an ‘I’ flashing in the lower right hand side of the screen, indicating user interaction is pending. RMC on the task, and select ‘Reply’:
5. Complete the information in the Problem Analysis Survey. Be sure to respond to the last question ‘Should this Problem be resolved through Change Management?’ as ‘Yes’. Hit ‘Finish’,
6. Navigate back to the Problem. The status of the Problem is now ‘Analysis Complete’. Click on the Additional Information Tab, and then drill down to the Workflow Tasks Tab. Scroll down through the tasks. You will see a notification has gone out to the assignee (Donald Bell’s) supervisor to Approve/Reject , which in our case is Mark Craigie. To complete this task Marc can either click on the link provided via email, or alternatively, he can scroll to the next pending task in the Workflow and click on the hyperlink for ‘Task: Approve/Reject (Pending)’:
7. Once logged in as Marc, you should see one pending task to ‘Approve/reject the Problem Pain/ Value Analysis Survey’. RMC on the task, and select ‘Reply’:
8. All of the information that had been initially filled out by Donald remains in this form for Marc to review. Scroll down to the bottom of the form. Either approve, reject, or mark as incomplete. In this example we will be approving this Problem to be resolved through Change Management. Hit finish, and then ‘sign out’ of CA Process Automation as Marc.
9. Navigate back to the Problem. The status of the Problem is now ‘Approved’. Click on the Additional Information Tab, and then drill down to the Workflow Tasks Tab. Scroll down through the tasks. You will see that a Change Order has been opened and attached to the Problem, an email notification has gone out to the assignee (Donald Bell) indicating the same.
10. Click on the link for the Change Order to view details.

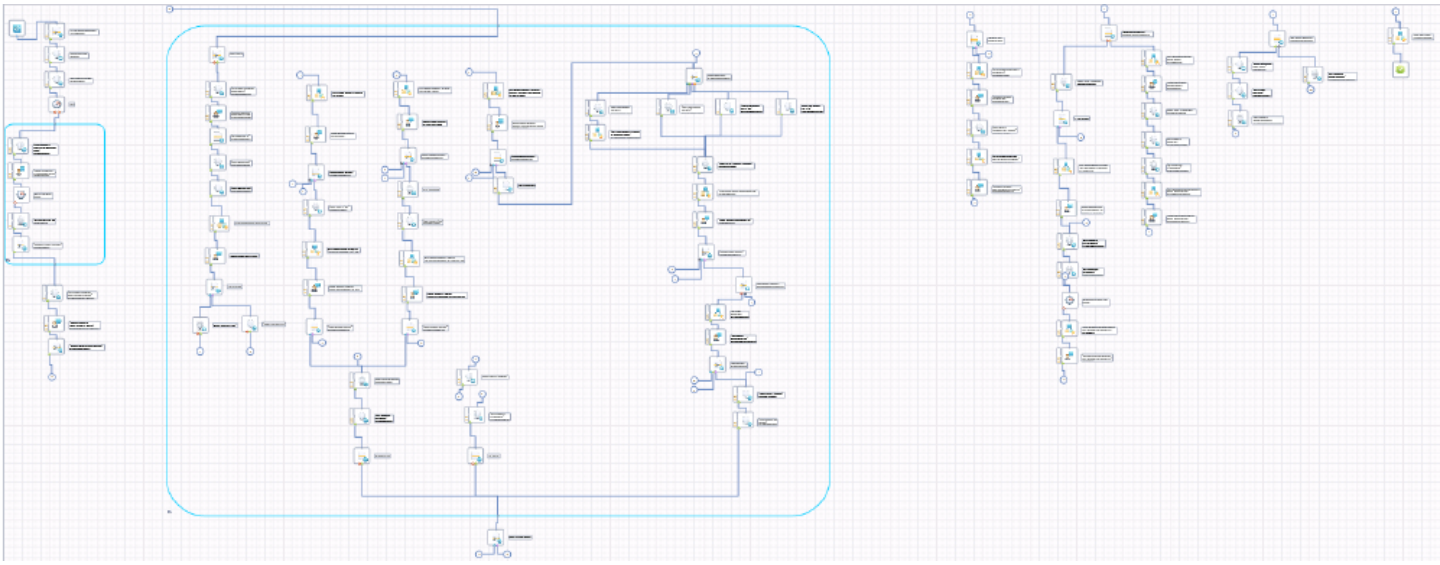
Change and Release Management Workflow

This topic contains the following information:

WARNING

If you are using only Change Management, and not Release Management, you can leverage this workflow process from a Change Management perspective. The process looks for a user-defined flag indicating whether the Change is a release. If the RFC is solely managing Change, then the process moves from Change Analysis directly to Change Manager and CAB Approvals. If it is a Release, then the process moves from Change Analysis through to Release approvals, then merges with Change at the Change Manager Approvals step. Any task that is specific to RDM only is delimited in the following steps with 'RDM'.

Design View in CA Process Automation



How to configure the Change and Release Management Workflow

The following table lists the prerequisites for running this Process Definition.

NOTE

Example Data indicates the examples of Contacts or Change Categories. You can change according to your organization.

Note that there is

CA SDM Object/Option	Example Data	Notes
Add contact (Analyst)	Donald Bell	Donald is the Requester of the Change Order and also a member of both the CAB and Implementation Groups. 1. Must be defined as a contact in CA SDM and also in CA EEM, if you are leveraging CA EEM for authentication. 2. Must have an email address defined in the CA SDM contact record. 3. Must have a notification method set to ‘Email’ in the CA SDM contact record.
Add contact (Analyst)	Kevin Smith	Kevin is a member of both the CAB and Implementation Groups. 1. Must be defined as a contact in CA SDM and also in CA EEM, if you are leveraging EEM for authentication. 2. Must have an email address defined in the CA SDM contact record. 3. Must have a notification method set to ‘Email’ in the CA SDM contact record.
Add contact (Analyst)	John McCarthy	John is a member and manager of both the CAB and Implementation Groups. 1. Must be defined as a contact in CA SDM and also in CA EEM, if you are leveraging EEM for authentication. 2. Must have an email address defined in the CA SDM contact record. 3. Must have a notification method set to ‘Email’ in the CA SDM contact record.
Add an Implementation Group	Implementation Group	Following is the group that implements the Change Order. 1. Group must be defined in CA SDM and also in CA EEM, if you are leveraging EEM for authentication. 2. When creating this group in EEM, the ‘Group Name’ in CA SDM must match the ‘Principal Name’ in EEM.

Add users to Implementation Group	John McCarthy Donald Bell Kevin Smith	1. This group must have more than one member 2. The members must be added to the group in CA SDM, and CA EEM if using EEM for authentication. 3. To add the user to the Group in CA EEM, go to Manage Identities->Users. Run a search. Highlight the user to view details. Navigate to the 'Global Group Membership' box. Under 'Available Global User Groups', search, and highlight the group that you would like to add the user to. Hit the right arrow, so that the group name is populated under 'Selected Global User Groups.'
Make one of the users previously added to the Implementation Group the Manager	John McCarthy	The group must have only one manager, this contact must be defined in CA SDM group record as the manager. This user is the Change Manager. 1. To make a contact a manager in the group record, navigate to the 'Members, Service Contracts, Auto Assignment' Tab, then drill down to 'Members' Tab
Add CAB Group	CAB	The CAB (Change Advisory Board) serves as approval group for the Change Order. This group implements the Change Order. 1. Group must be defined in CA SDM and CA EEM, if you are leveraging EEM for authentication. 2. When creating this group in EEM, the 'Group Name' in CA SDM must match the 'Principal Name' in EEM.
Add users to CAB Group	John McCarthy Donald Bell Kevin Smith	1. This group must have more than one member 2. The members must be added to the group in CA SDM and CA EEM, if using EEM for authentication. 3. To add the user to the Group in CA EEM, go to Manage Identities->Users. Run a search. Highlight the user to view details. Navigate to the 'Global Group Membership' box. Under 'Available Global User Groups', search, and highlight the group that you would like to add the user to. Hit the right arrow, so that the group name is populated under 'Selected Global User Groups.'

Make one of the users previously added to the CAB Group the Manager	John McCarthy	This user is the CAB Manager. 1. The group must have only one manager, this contact must be defined in CA SDM group record as the manager. This user is the Change Manager. 2. To make a contact a manager in the group record, navigate to the 'Members, Service Contracts, Auto Assignment' Tab, then drill down to 'Members' Tab
Create Change Order Category	Change and Release Mgmt	This Change Category has the following values set: 1. Assignee: John McCarthy 2. Risk Survey: General 3. CAB: CAB 4. Implementation Group: Implementation Group 5. <in the 'Workflow' Tab> CA IT PAM Process Name: 'Change and Release Management'
Create at least two new CI's, that have pre-existing relationships established and Change Orders already opened against them during the same scheduled timeframe.	TIXCHANGE TIXCHANGE Web Application	1. Create at least two new Configuration Items (**Service Desk, File, New Configuration Item**) which we associate to the Change Order. 2. These CIs must all have relationships established with other CIs, so that a good view of the CA CMDB Visualizer can be displayed. 3. These CIs must have pre-existing Change Orders opened against them which can be shown as a part of Conflict Analysis. In order for pre-existing Change Orders to appear during Conflict Analysis step in the Workflow, they must be opened against the same CI, during the same scheduled timeframe.
Install Category_Defaults Option	Yes	This option must be installed so that the information from the Change and Release Model Category is pulled over to the Change Order once it is created. Recycle CA SDM services for changes to take effect
Modify Change Order Status	Check 'Make Change Order Active'	Use Administration, ServiceDesk, Change Orders, Status to check 'Make Change Order Active', so that when a Change Order enters the Backed Out status as a part of this workflow, it remains active.

Change and Release Management Workflow in Action

WARNING

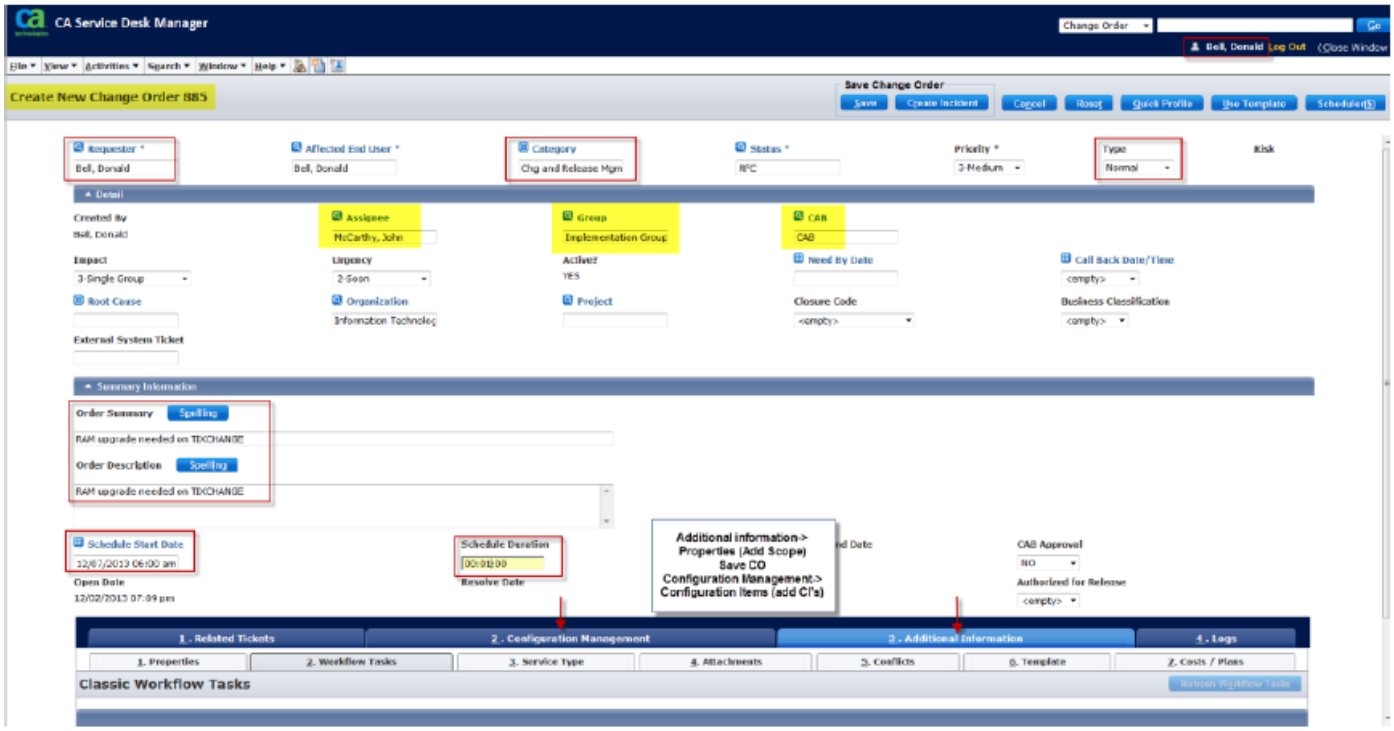
Logic built into many of the IRFs (Interaction Request Forms) such that a response of 'no' when a 'yes' is expected result in an automatic 'rejection' of the Change Order or Release.

Follow these steps:

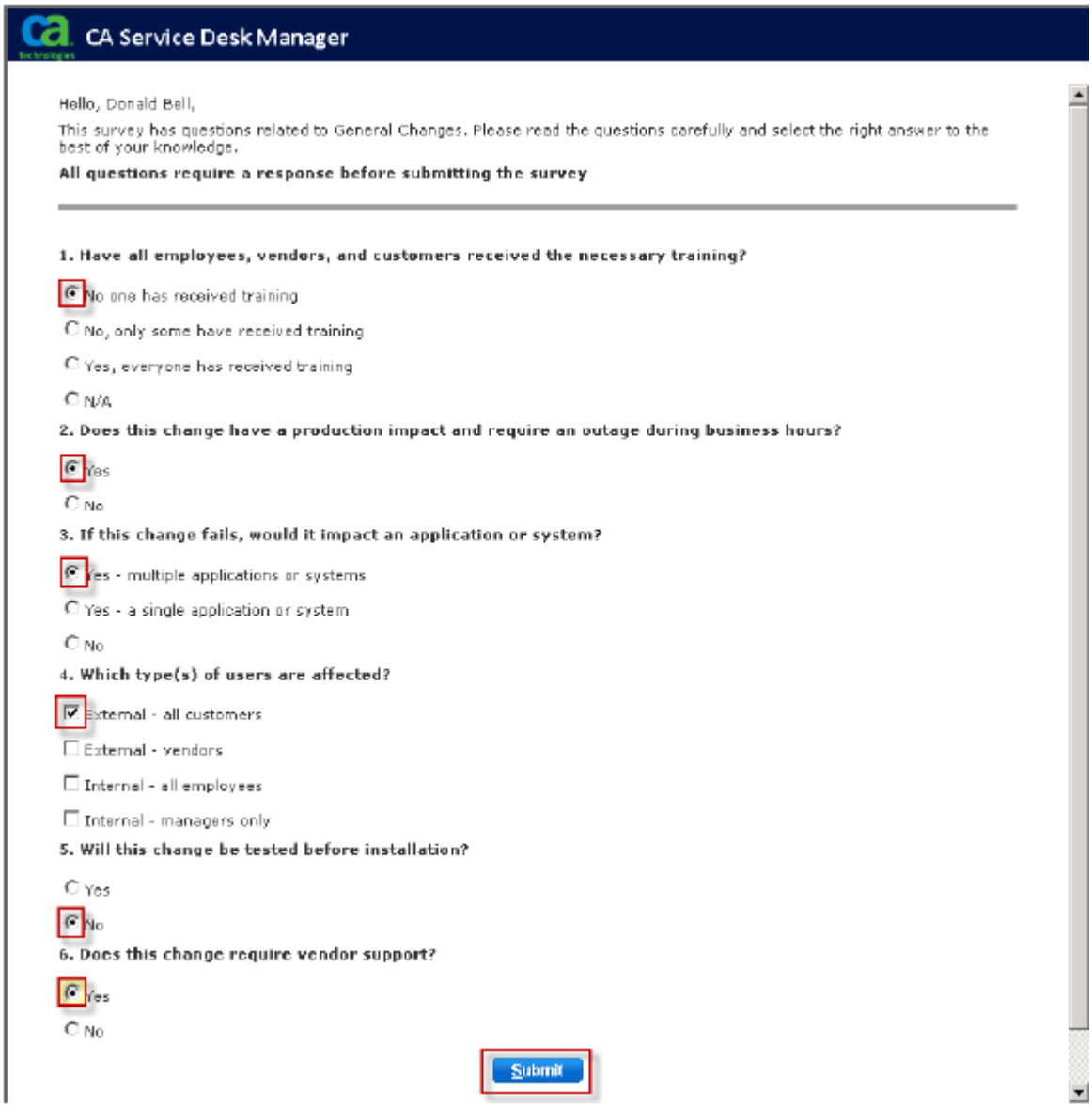
1. Log in to CA SDM as Donald Bell. Create a Change Order.
2. Fill out all of the highlighted fields including ‘Backout Plan’ on the ‘Cost/Plans’ Tab. For the Change Order to move from status ‘Approval in Progress’ to ‘Approved' status transition ensures that this field must not be NULL.

   NOTE

   When you select the Category, and tab out of the field, the ‘Assignee’, ‘Implementation Group’ and ‘CAB’ are filled in, because the category_defaults option is turned on. You will not be able to add CI until after you save the Change order.



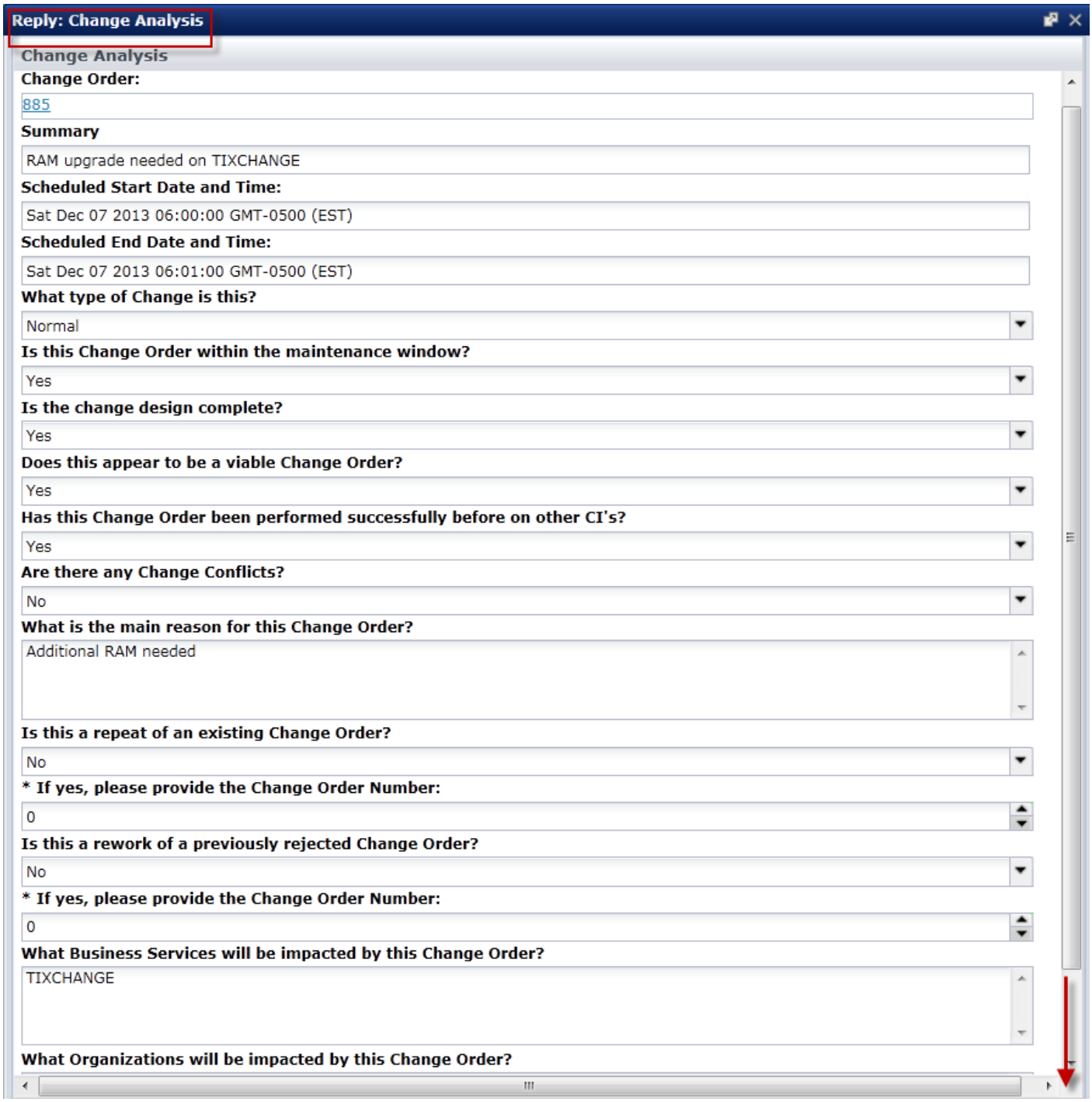
3. Select Additional Information, Workflow Tasks and scroll down through the tasks.
   You will see a notification has gone out to the requester (Donald Bell) to complete the Risk Assessment Survey. There is currently no Risk value assigned to the Change Order.
4. To complete this task, Donald can either click on the link provided in the email, or can scroll to the next pending task in the Workflow and click on the Task: Other (Pending) hyperlink.
   Clicking on the hyperlink will automatically log Donald into his tasklist.
5. RMC on the task and then click Reply.
   Donald is prompted to complete the Risk Assessment Survey. He clicks on the Risk Assessment Survey link. The link brings Donald in context to the Risk Assessment Survey. The survey which we have assigned to the Change Category is the default ‘General’ Risk Survey. Complete the Risk Survey, answering questions in such a way that it will generate high risk:



6. Once the Risk Survey is complete, navigate to CA Process Automation in context of the task and click Finish on the Risk Assessment task. You can remain logged into CA Process Automation as Donald.
7. Navigate back to the Change Order. Select Additional Information, Workflow Tasks and scroll down through the tasks. You will see a notification has gone out to the requester (Donald Bell) to complete Impact and Conflict Analysis. To complete this task Donald can either click on the link provided via email or scroll to the next pending task in the Workflow and click on the Task: Other (Pending) hyperlink.
8. Navigate back into CA Process Automation as Donald.
   You will see one pending task for Impact and Conflict Analysis.
   RMC on the task and click reply.
9. Clicking on the link to Impact and Conflict Analysis brings you in context to the Change Order where Impact and Conflict Analysis must be instantiated manually. The first step is to perform Impact Analysis. Impact Analysis will show which CI is impacted if the TIXCHANGE Servers are temporarily down for the RAM upgrade. There are two ways to run Impact Analysis. You can either run it through Impact Explorer which is accessible using the Additional Information, Configuration Items tab within a Change Order. Or you can use CA CMDB Visualizer. You can filter the Visualizer for Impact Analysis. Once Impact Analysis is complete, the next step is Conflict Analysis, which is also run within context

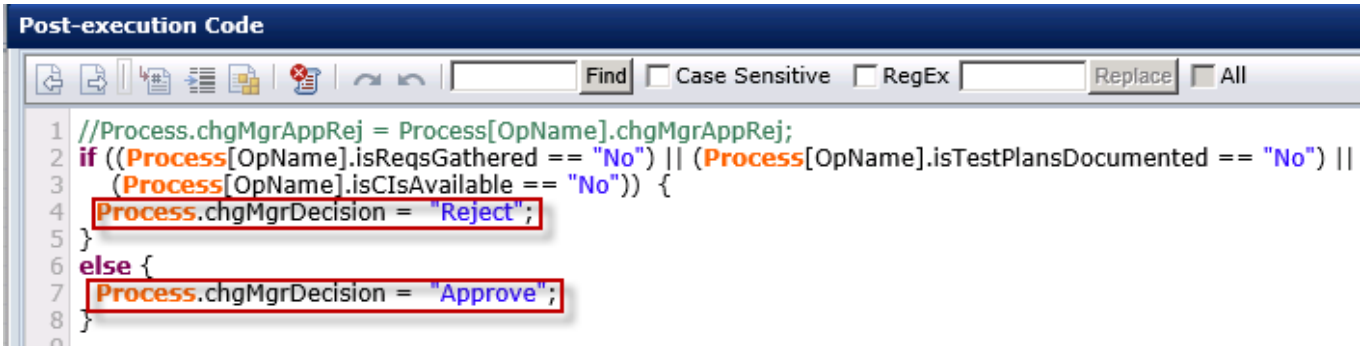
of a Change Order, from Additional Information, Conflicts Tab. Conflict Analysis shows if there are any Change Orders open for the same CI in the same timeframe, thus generating conflict. If there are any conflicts they must be resolved before proceeding further with the Change Order.

10. Finish the CA Process Automation task. You can remain logged in as Donald Bell.
11. Navigate to the Change Order. Click Additional Information, Workflow Tasks and scroll down through the tasks. You will see a notification has gone out to the requester (Donald Bell) to complete Change Analysis. To complete this task Donald can either click on the link provided in the email or scroll to the next pending task in the Workflow and click on the Task: Complete Form (Pending) hyperlink.
12. Navigate to CA Process Automation as Donald Bell. You will see one pending task for Change Analysis. If any of the form data indicates that any step is not complete or not done, the workflow takes different paths. We assume everything is complete.
13. RMC on the task and click Reply.
14. Complete the information in the Change Analysis form click finish. Following is an example of an IRF where an answer of ‘no’ to questions where a ‘yes’ is expected result in an automatic rejection of the Change Order:



15. Navigate to the Change Order. Select Additional Information and Workflow Tasks.
    You will see that the information from the Change Analysis form has been saved to the Change Order.
16. Scroll down through the tasks.
    You will see a notification has been sent to the Change Manager, Donald Bell. Donald must verify if this Change is going to be a Release. To complete this task Donald can either click on the link provided in the email or scroll to the next pending task in the Workflow and click on the Task: Approve/Reject (Pending) hyperlink.
17. Navigate to CA Process Automation as Donald Bell.
    You will see one pending task for Release Verification.

18. RMC on the task and click Reply.
19. Donald specifies that this is a Release, and then click Finish. Setting this to ‘Yes’ guides the user through the Release portion of this process.
    a. Navigate to the Change Order.
       The ‘Authorized for Release’ Flag is set to ‘Yes’.
    b. Scroll through the tasks.
       You will see a notification has been sent to the Change Manager, Donald Bell. Donald must authorize Plan and Build Release. To complete this task Donald can either click on the link provided in the email or scroll to the next pending task in the Workflow and click on the Task: Complete Form (Pending) hyperlink.
    c. Navigate to CA Process Automation as Donald Bell.
       You will see one pending task for Authorize Plan and Build Release.
    d. RMC on the task and click Reply. Complete the information in the Authorize Plan and Build Release form, scroll down and click finish. If any tasks are ‘Rejected’ or marked ‘Incomplete’, the workflow takes different paths. We will approve all tasks.
    e. Navigate to the Change Order.
       The status has changed to ‘Build’.
    f. Scroll through the tasks.
       You will see a notification has been sent to the Change Manager, Donald Bell. Donald must complete the Release and Deployment Plan and Build form. To complete this task Donald can either click on the link provided in the email or scroll to the next pending task in the Workflow and click on the Task: Complete Form (Pending) hyperlink.
    g. Navigate to CA Process Automation as Donald Bell. You will see one pending task for to complete the Release and Deployment Plan form. RMC on the task and click Reply. Complete the information in the Plan and Build Release form. responding ‘no’ to some of the questions automatically result in the Change Order being rejected. Following is an example of the Operator’s Post Execution Code.



    h. Complete the form, scroll down and click finish.
    i. Navigate to the Change Order and scroll through the tasks.
       You will see a notification has one out to the Change Manager, Donald Bell. Donald must Authorize Build and Test Release. To complete this task Donald can either click on the link provided in the email or can scroll to the next pending task in the Workflow and click on the Task: Complete Form (Pending) hyperlink.
    j. Navigate to CA Process Automation as Donald Bell. You will see one pending task to Authorize Build and Test Release. RMC on the task and click Reply. Complete the information in the Build and Test Release form and click finish.
    k. Navigate to the Change Order.
       The status has changed to ‘Test’.
    l. Scroll through the tasks.
       You will see a notification has one out to the Change Manager, Donald Bell. Donald must complete the Release and Deployment Test form. To complete this task Donald can either click on the link provided in the email or scroll to the next pending task in the Workflow and click on the Task: Complete Form (Pending) hyperlink.

m. Navigate to CA Process Automation as Donald Bell.
You will see one pending task to complete Release and Deployment Build and Test form.
n. RMC on the task and click Reply. Complete the information in the Release and Deployment Build and Test form and click finish.
o. Navigate to the Change Order and scroll through the tasks.
You will see a notification has one out to the Change Manager, Donald Bell. Donald must authorize checkin of baselined release package into the DML. To complete this task Donald can either click on the link provided in the email or scroll to the next pending task in the Workflow and click on the Task: Approve/Reject (Pending) hyperlink.
p. Navigate to CA Process Automation as Donald Bell.
You will see one pending task to authorize checkin of baselined release package into DML. RMC on the task and hit 'Reply'. Authorize checkin of baselined release package into DML by SACM, scroll down and hit finish.

20. Navigate to the Change Order. Change Manager and CAB Approval is required as the 'type' and level of 'risk' is associated with the Change. The status has changed to 'Approval in Progress' and the 'CAB Approval' flag is set to yes. Scroll through the tasks.
You will see a notification has one out to the Change Manager, who is the manager of the Implementation Group, Donald Bell. Donald must approve or reject the Change Order. To complete this task Donald can either click on the link provided in the email or scroll to the next pending task in the Workflow and click on the Task: Approve/Reject (Pending) hyperlink.
21. Navigate to CA Process Automation as Donald Bell.
You will see one pending task to approve/reject Change Order.
22. RMC on the task and click Reply.
23. Complete the Approval/Rejection comments. Dependent upon whether the Change Manager Approves, Rejects, or marks the Change Order Incomplete, the Workflow takes different paths. Mark it as 'Approved'.
24. Navigate to the Change Order. Select Additional Information, Workflow Tasks, and scroll down through the tasks. You will see a notification has gone out to the CAB, including the CAB Manager, who incidentally is also Donald Bell. The CAB Manager, Donald, needs to Approve or Reject the Change Order on behalf of the CAB. To complete this task John can either click on the link provided in the email or scroll to the next pending task in the Workflow and click on the Task: Approve/Reject (Pending) hyperlink.
25. As Donald, you see one pending task for CAB Approval. RMC on the task and hit 'Reply'.
26. Complete CAB Date/CAB Chair, and the CAB Approval/Rejection comments. Dependent upon whether the CAB Manager Approves, Rejects, or marks the Change Order Incomplete, the Workflow will take different paths. In this case mark it as 'Approved'.
27. Navigate back to the Change Order. The status of the Change Order is now 'Approved'. Click on the WF Tasks Tab. Scroll down through the tasks. You will see a notification has gone out to the Implementation Group, of which John McCarthy is a member. The Implementation Group is now responsible for implementing the RAM upgrade on TIXCHANGE.
28. To complete this task Donald can either click on the link provided via email, or alternatively, he can scroll to the next pending task in the Workflow and click on the hyperlink for 'Task: Other (Pending)'. As Donald, you will need to click on 'Group Tasks' on the left hand side of the pane since this task of Implementing the Change Order is assigned to the full Implementation Group. RMC on the Implementation task and hit 'Reply'.
29. The link in this task will bring you directly into CA SDM in context of the Change Order. Once the Implementation Group begins the RAM upgrade, Donald can hit Finish to complete the task.
30. Navigate back to the Change Order. The status has changed from 'Approved' to 'Implementation in Progress'. Click on the WF Tasks Tab. Scroll down through the tasks. You will see a notification has gone out to all of the members of the Implementation Group, of which Donald Bell is a member. The Implementation Group is now responsible for confirming the successful implementation of the patches. To complete this task Donald can either click on the link provided via email, or alternatively, he can scroll to the next pending task in the Workflow and click on the hyperlink for 'Task: Complete Form (Pending)'.
31. As Donald, you will again need to click on 'Group Tasks' on the left hand side of the pane since this task of confirming implementation of the Change Order is assigned to the full Implementation Group. RMC on the Implementation

Complete task and hit ‘Reply’. Implementation Status as ‘Complete’ or ‘Incomplete’ and as a result the Workflow will take different paths. In our case we will mark it as ‘Complete’.

32. Navigate back to the Change Order. Navigate back to the Change Order. The status has changed to Deployed.

    NOTE

    The workflow will proceed to this step if you answered ‘Yes’ to Release Verification. Otherwise skip to step 33.

    a. Click on the WF Tasks Tab. Scroll down through the tasks. You will see a notification has gone out to the Change Manager, Donald Bell, who needs to update the release number of all records of all Service CI’s associated with this Release. To complete this task Donald can either click on the link provided via email, or alternatively, he can scroll to the next pending task in the Workflow and click on the hyperlink for ‘Task: Approve/Reject (Pending)’.
    b. As Donald, you will again need to click on ‘Group Tasks’ on the left hand side of the pane since this task of confirming implementation of the Change Order is assigned to the full Implementation Group. RMC on the Implementation Complete task and hit ‘Reply’. Donald updates the release number of all Service CI’s associated with the Release.
    c. Once complete he can hit ‘Finish’ on the task.
33. Navigate back to the Change Order. The closure code has been set to ‘successful’. Click on the WF Tasks Tab. Scroll down through the tasks. You will see a notification has gone out to all of the members of the Implementation Group, of which Donald Bell is a member. The Implementation Group is now responsible for doing a Post Implementation Review (PIR). To complete this task John can either click on the link provided via email, or alternatively, he can scroll to the next pending task in the Workflow and click on the hyperlink for ‘Task: Complete Form (Pending)’.
34. As John, you will again need to click on ‘Group Tasks’ on the left hand side of the pane since this task of confirming implementation of the Change Order is assigned to the full Implementation Group. RMC on the Post Implementation Review task and hit ‘Reply’.
35. Complete the Post Implementation Review Form, and once complete hit ‘Finish’.
36. Navigate back to the Change Order. The status of the CO is set to ‘Closed’ as all steps in the Workflow are now complete and the RAM upgrade was successful.

CA SDM Connector Glossary

The following list contains concepts and terms that are useful if you are integrating a CA Catalyst Connector with CA Catalyst.

- **Connectors**
  Connectors are the links from products that consume data from external products, referred to in this document as domain managers. Each connector retrieves information from its domain manager and transmits the information through the connector framework to the consuming product for visualization and analysis. Connectors can also enact inbound operations on data in the source domain manager, such as object creation.
- **USM**
  The *Unified Service Model* (USM) is a schema of common object types and properties to which data from all connectors is converted. The USM schema enables data analysis from all domain managers on a common interface with identical formatting.
- **Configuration Items (CIs)**
  Configuration items (CIs) represent IT elements that the domain manager manages. Each CI belongs to a *type* (defined in the USM schema) such as ComputerSystem, Database, Process, BinaryRelationship, and so on. Connectors transform managed objects from domain managers to adhere to the USM schema and import the objects to the consuming products as CIs.
- **Services**
  Services represent discrete business functions that can contain configuration items that multiple domain managers manage. For example, a payroll service contains an Active Directory database managed by which the Microsoft SCOM manages, user store manages by security product, batch jobs managed by mainframe product, router managed by

network product, applications managed by application management product, and so on. You can do the following actions in consuming products, such as CA SOI:
  - Detect the root cause of service degradation quickly and navigate to the appropriate product to resolve problems
  - Model services based on imported CIs or import existing service models from integrated products to construct a comprehensive and service-centric model of your enterprise
- **Alerts**
  Alerts are the CA SOI mechanism for reporting fault conditions and service degradation. *Infrastructure alerts* are fault conditions originally reported by one of the domain managers (such as a CA Spectrum alarm). An alert is associated with a corresponding CI and associated alert severities determine CI condition, and ultimately, service impact. *Service alerts* are conditions generated by CA Spectrum SA based on analysis of a modeled service. Service alerts result when the condition of one or more CIs combines to impact the overall quality or risk level associated with the service.
- **Outbound from connector operations**
  Outbound from connector operations are operations that a connector invokes to import data from domain managers into consuming products such as CA Catalyst and CA Spectrum SA. All connectors support outbound operations.
- **Inbound to connector operations**
  Inbound to connector operations invoke changes to the domain manager data store as a result of changes to the imported data in the consuming product. For example, CI reconciliation in CA Catalyst can change the values of CI properties. Connectors that support inbound operations can then enact that change in the source domain manager so that its data matches the reconciled data. Or if a CI is deleted in a domain manager that CA Catalyst defines as a source of truth, connectors that support inbound operations can delete the CI in other domain managers with a record of that CI.
- **ServiceDeskManagerConnector.conf**
  ServiceDeskManagerConnector.conf file is the configuration file for CA Catalyst Container r3.1 and r3.2. This file stores the configuration information (node and module information) needed for the CA Catalyst server and other peers. CA SDM Connector r2.5 framework does not have the ServiceDeskManagerConnector.conf file.
- **ServiceDeskManagerConnector.xml**
  ServiceDeskManagerConnector.xml is the configuration file for CA SDM Connector r3.1 and r3.2. For CA SDM Connector r2.5, the equivalent configuration file is ServiceDeskManager_<ComputerName>.xml where ComputerName is the name of the computer server where the CA SDM Connector resides. These files contain the configuration properties to help the connector operations. For example, toggling the Use Transaction Work Area, changing the configured username and password, excluding CI or relationship types, and so on.
- **USM-CMDB.xml**
  USM-CMDB.xml file contains the mapping information of the USM entities and CMDB families. Based upon this mapping the CA SDM Connector performs necessary actions.
  Example: The ServiceDeskObject.xml object lists the attributes to query while querying the CIs from CMDB. This information helps the CA SDM Connector to query CMDB with only those attributes which are mapped to USM attributes rather than querying all the attributes.
- **ServiceDeskManager_policy.xml**
  ServiceDeskManager_policy.xml file contains the outbound USM transformation policy information which transforms the incoming CIs from CMDB to USM entity.
- **ServiceDeskManager_policy.en**
  ServiceDeskManager_policy.en is the localization file for outbound policy. Change this file information to map with new values of CMDB families or attributes if you want to support CA SDM in a language other than English.
- **ServiceDeskManager_policySB.xml**
  ServiceDeskManager_policySB.xml file contains the inbound transformation policy information which transforms the incoming USM entity to CMDB CI before providing it to the CA SDM Connector.
- **ServiceDeskManager_policySB.en**
  ServiceDeskManager_policySB.en is the localization file for inbound policy. Change this file information to map with new values of CMDB families or attributes if you want to support CA SDM in a language other than English.

Pre-Built CA Process Automation Workflows

This section provides the steps required to integrate the pre-built CA Process Automation workflows with CA Service Catalog. It is assumed that each CA Service Management product in the solution has been appropriately installed in a certified architecture.

Ensure that the following products are installed before you proceed with the integration:

- CA EEM
- CA Process Automation
- CA Service Catalog
- Java Runtime Environment

> **WARNING**
> Take a backup of the custom process files before you upgrade with the latest CA Process Automation Content files.

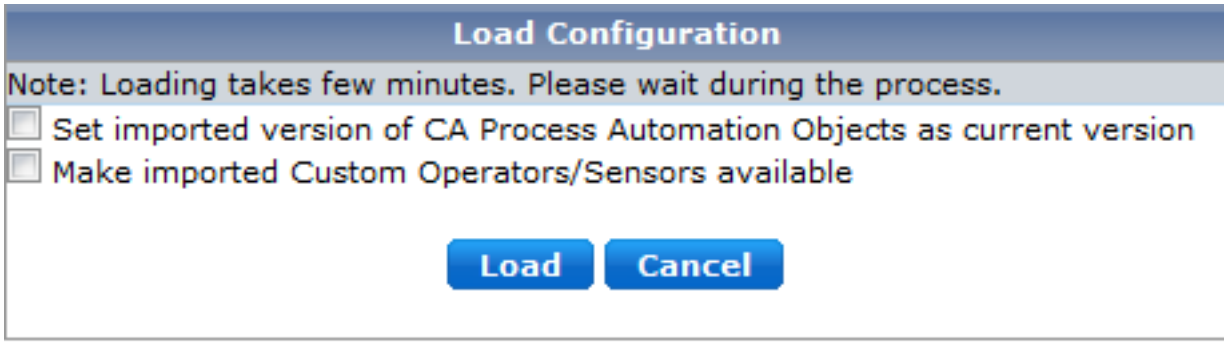
Follow these steps:

- Step 1 - Load Pre-Built CA Process Automation Workflows for CA Service Catalog
- Step 2 - Configure the pre-built CA Process Automation Workflows for CA Service Catalog
- Step 3 - Copy and Modify Actions
- Step 4 - Add Members to User Defined Group
- Step 5 - Configure CA Service Desk Manager

Step 1 - Load Pre-Built CA Process Automation Workflows for CA Service Catalog

Follow these steps:

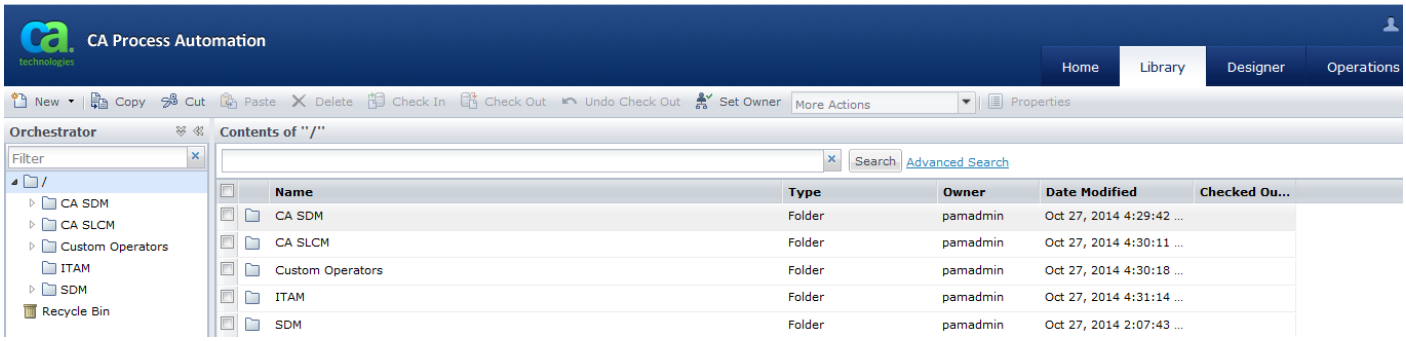
1. Log in to CA Service Catalog as the Service Delivery Administrator (spadmin)
2. Click **Administration**, **Configuration**.
3. Click the link for CA Service Desk and edit each property with the appropriate value.
4. Click **Test**. If you do not receive a “Connection Test is Successful” message, modify your settings again.
5. Click the link for the CA Process Automation option.
6. Edit each property with the appropriate value, and then click **Test**.
   If you do not receive a “Connection Test is Successful” message, modify your settings again.
7. Click **Load** and select both check boxes that appear.



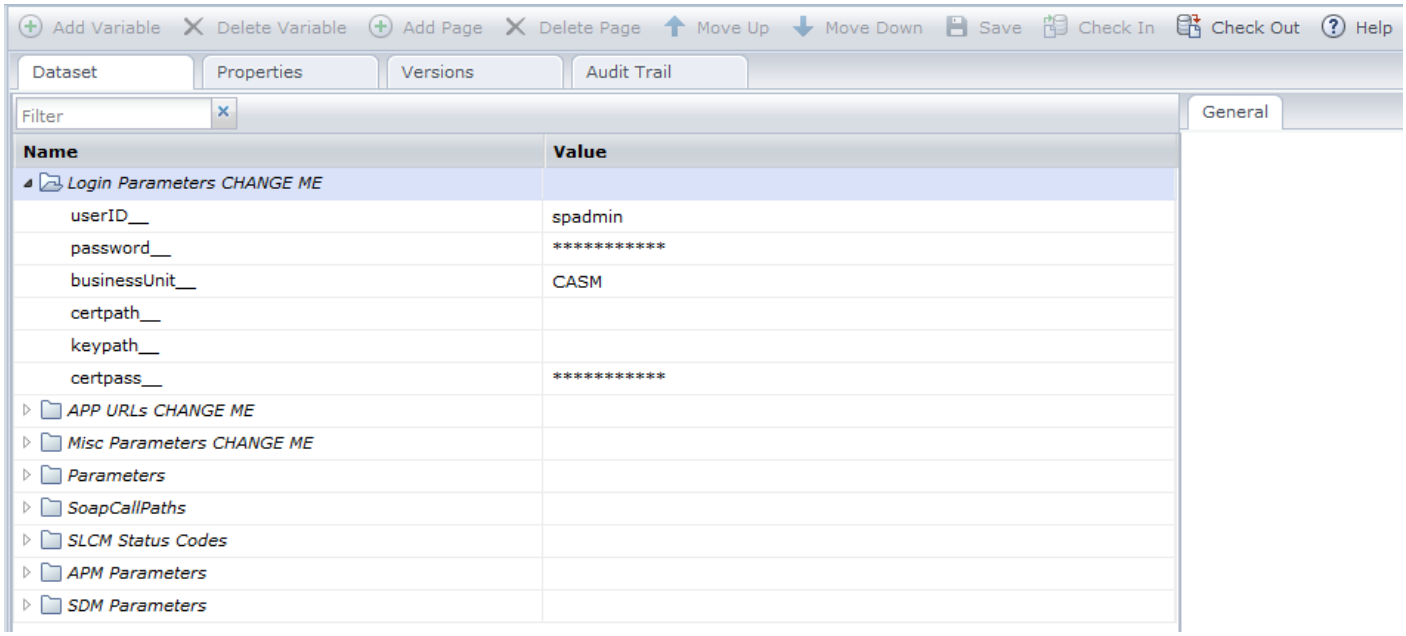
8. Click **Load** button.
9. Click **Configure**, once the load completes.

Step 2 - Configure the pre-built CA Process Automation Workflows for CA Service Catalog

1. Log in to CA Service Catalog as the Service Delivery Administrator (spadmin).
2. Click **Administration**, **Configuration**.
3. Go to **CA Process Automation** section, and then click the **Launch** button to initiate CA Process Automation.
4. Enter the CA Process Automation Administrator username and password, and click **Log in**.
5. Click the **Library** tab.
   The following screen appears:



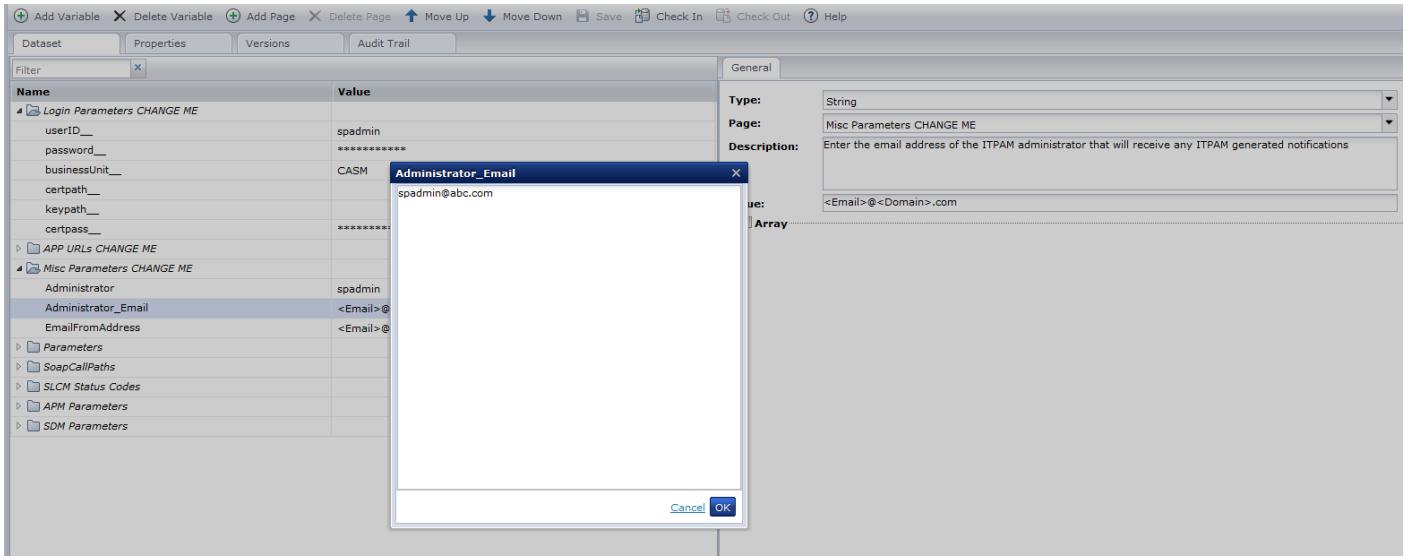
6. Click the CA SLCM folder to open it.
7. Double-click SLCM_GlobalDataset.



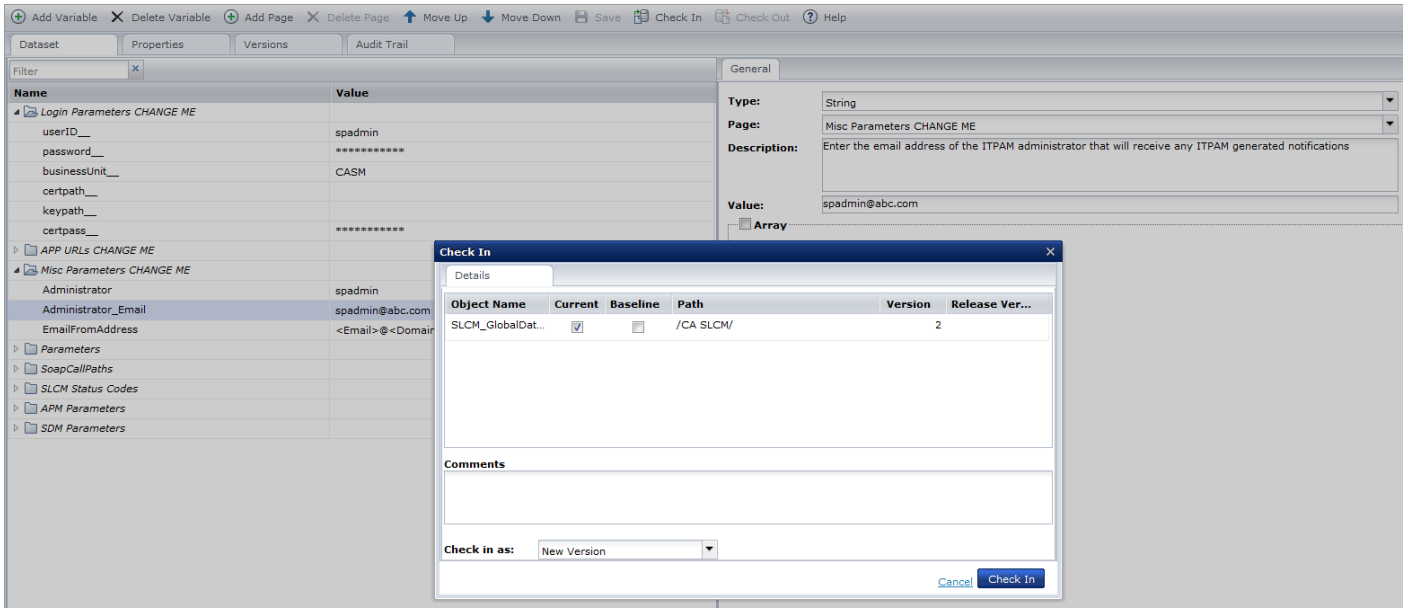
8. Click the Checkout icon

   in the SLCM_GlobalDataset screen.
   In the Login Parameters Value Definition screen, provide the Service Delivery Administrator user ID, password, and business Unit from the CA Service Catalog installation.
9. Click the CA SLCM folder to open it.
10. Double-click SLCM_GlobalDataset.

11. Click the Misc Parameters CHANGE ME section and modify the Administrator_Email and EmailFromAddress.



12. Click the **Save** icon, then click the **Check in** icon.
    The following screen appears:



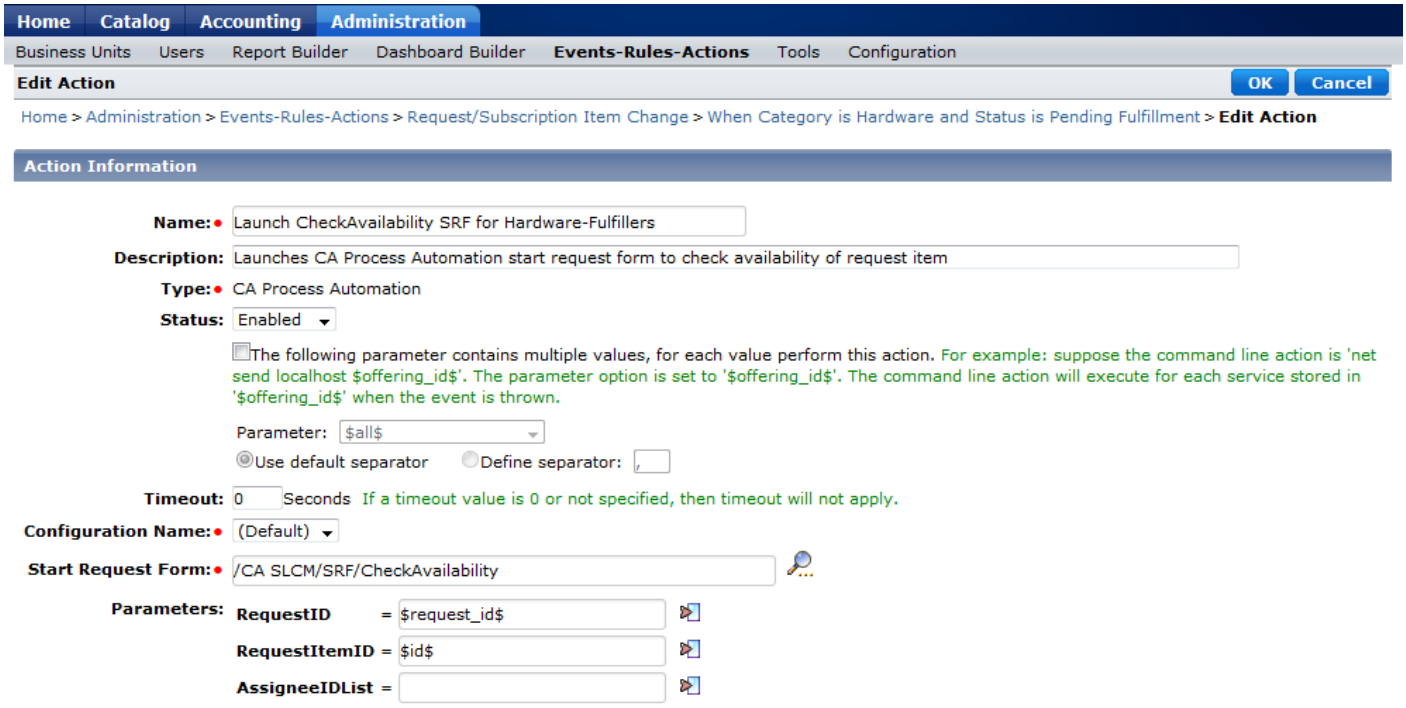
13. Enter any appropriate check-in comments and click OK.
14. Follow steps 4-8 for the SDM_GlobalDataset located in the CA SDM folder. Note that Administrator_Email and EmailFromAddress is located in the Parameters section.
15. Expand the CA SDM folder and click the SRF folder.
16. Right click the Service_Publishing Start Request Form and select Properties.
17. Click the **Tags** tab, then the **Edit** button.
18. Click in the text area and type chgcat then click **Save**.
19. Follow steps 12-14 for the HWSW_FilledFromInventory Start Request Form.
20. Click the Configuration Browser tab
21. Click the Modules tab and double click the CA Service Desk Module.
22. Right-click and select Lock.

23. Update the Service Desk WebService URL, Service Desk Administrator User ID (ServiceDesk), and the Service Desk Administrator password. Click Apply.
24. Click the Modules tab again and double click the Alert Module.
25. Update the SMTP Server for Outgoing e-mail and the From Address for Outgoing e-mail. Click Apply.
26. Right-click and select Unlock. Click Ok to the confirmation popup.
27. Close CA Process Automation.

Step 3 - Copy and Modify Actions

Follow these steps:

1. Log into CA Service Catalog as an administrator.
2. Click **Administration**, **Events-Rules-Actions**.
3. Click **Request/Subscription Item Change**.
4. Click on the **When Category is Hardware and Status is Pending Fulfillment** rule.
5. Click the Copy icon for the **Launch CheckAvailability SRF for Hardware** action.
6. Type in the name **Launch CheckAvailability SRF for Hardware–Fulfillers Group** and click **OK**.
7. Check the action **Launch CheckAvailability SRF for Hardware** and click the Disable button.
8. Click the Edit icon for the **Launch CheckAvailability SRF for Hardware–Fulfillers Group** action.
9. Select Enabled from the **Status** drop-down box in the Action Information section.



10. Type in Fulfillers in the text field for AssigneeIDList.
11. Click the **OK** button to save your changes.
12. Click the **Done** button.
13. Repeat Steps 1-12 for the all additional rules and actions.

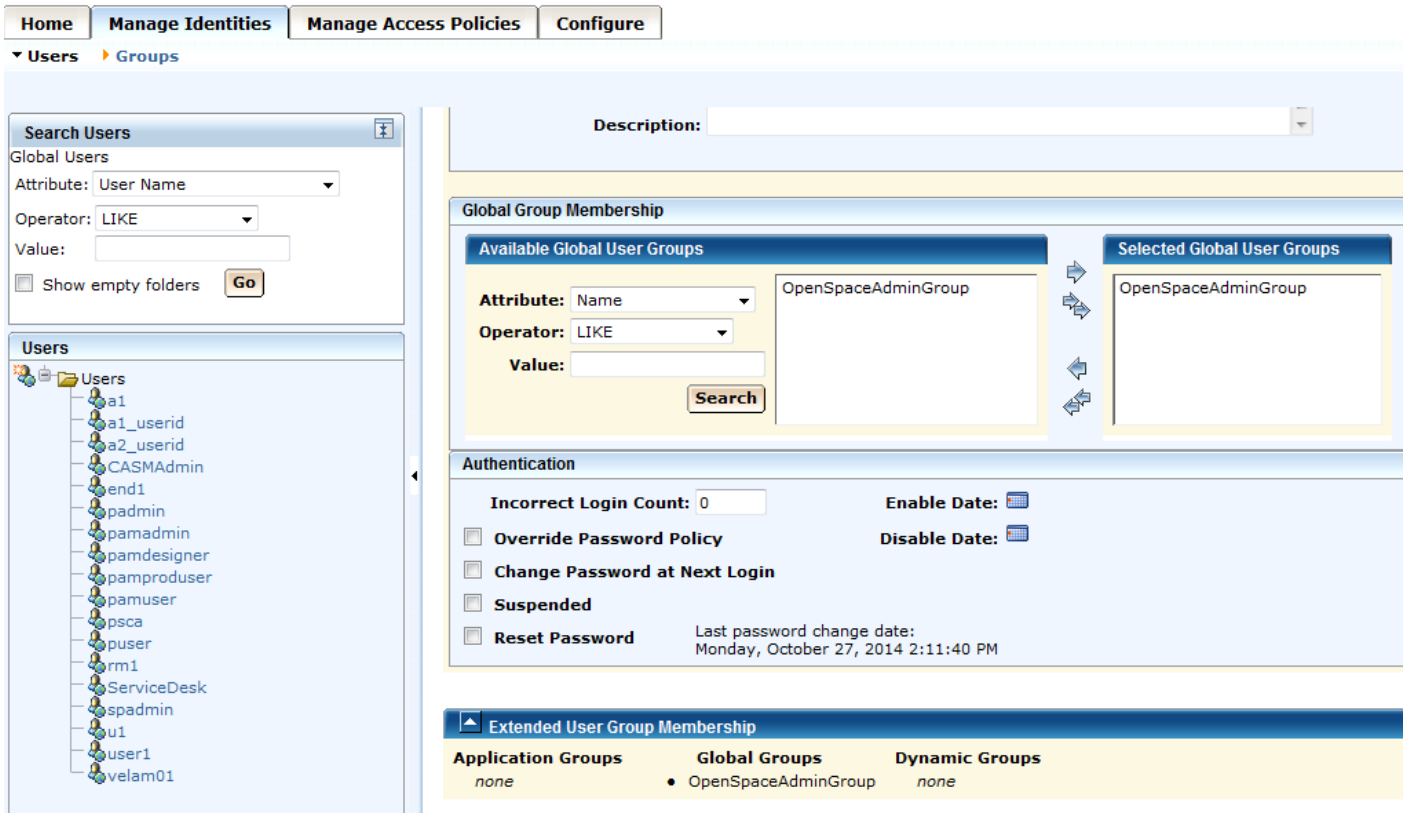
Use the following table as a reference to add additional rules and actions:

Rule Name	Action to Copy and Disable	New Enabled Action Name	How to Modify the Action
When Category is Software and Status is Pending Fulfillment	Launch CheckAvailability SRF for Software Availability	Launch CheckAvailability SRF for Software Availability–Fulfillers Group	AssigneeIDList = Fulfillers
When Category is Hardware and Status is Not Filled From Inventory	Launch SLCM_Fulfillment SRF	Launch SLCM_Fulfillment SRF–Procurement Group	AssigneeIDList = Procurement
When Category is Software and Status is Not Filled From Inventory	Launch SLCM_Fulfillment SRF	Launch SLCM_Fulfillment SRF–Procurement Group	AssigneeIDList = Procurement
When Category is Hardware and Status is Filled From Inventory	Launch HWSWFilledFromInv_SDM SRF	Launch HWSWFilledFromInv_SDM SRF–Change Analyst	AssigneeIDList = <SDM change analyst user id> SDM_Category = <tenanted category, ex. SLCM.HWFFI-A>
When Category is Software and Status is Filled From Inventory	Launch HWSWFilledFromInv_SDM SRF	Launch HWSWFilledFromInv_SDM SRF–Change Analyst	AssigneeIDList = <SDM change analyst user id> SDM_Category = <tenanted category, ex. SLCM.HWFFI-A>
When Category is Hardware and Status is Received or Order Cancelled	Launch Hardware CheckAvailability SRF	Launch Hardware CheckAvailability SRF–Fulfillers Group	AssigneeIDList = Fulfillers
When Category is Software and Status is Received or Order Cancelled	Launch CheckAvailability SRF for Software	Launch CheckAvailability SRF for Software–Fulfillers Group	AssigneeIDList = Fulfillers

Step 4 - Add Members to User Defined Group

Follow these steps:

1. Click **Manage Identities** tab, click the **Users** link in the CA EEM Interface.
2. Enter appropriate search criteria in the **Search Users** section, and click the Go button.
   Users appear in the Users section.
3. Select the user to add to the EEM group in the Users folder.
   The user details appear.
4. Search for the available global user groups in the Global Group Membership section.
5. Select the group(s) in the Global Group Membership section, to which the user must be added and move the selected groups into the **Selected Global User Groups** multi-select box.



6. Click **Save**.
7. Repeat steps 2 - 5 for each user that requires group membership.

Step 5 - Configure CA Service Desk Manager

Follow these steps to configure CA Service Desk Manager:

- Step 5a - Configure Options Manager
- Step 5b - Configure Web Screen Painter
- Step 5c - Add CA Service Desk Manager Groups
- Step 5d - Add Status
- Step 5e - Add Change Categories

Step 5a - Configure Options Manager

You first configure the Options Manager in CA Service Desk Manager.

Follow these steps:

1. Log into the CA Service Desk Manager as a privileged user. For example, ServiceDesk
2. Click the **Administration** tab and navigate to **Options Manager**. Edit the following:

Sub Node	Options to Configure	Option Value / Action
CA Process Automation Workflow	caextwf_eem_hostname	CA EEM Server host name. Click Install.
	caextwf_endpoint	URL of the load balancer for the CA Process Automation server. Click Install.

	caextwf_log_categories	Enter **Process**. Click Install.
	caextwf_processdisplay_url	Replace <default hostname>: <port number> portion of the URL in the Option Value (in front of the “/itpam/JNLP”) with the actual host name and port number of the load balancer for CA Process Automation server. Click Install.
	caextwf_worklist_url	Replace <default hostname>: <port number> portion of the URL in the Option Value (in front of the /itpam?webPage”) with the actual host name and port number of the load balancer for CA Process Automation server. Click Install
	caextwf_ws_password	itpamadmin user’s password in the Option Value field. Click Install.
	caextwf_ws_user	itpamadmin username in the Option Value field. Click Install.
Change Order Manager	Category Defaults	Click Install.
Security	eiam_hostname	CA EEM Server host name. Click Install.
	Use_eiam_authentication	Click Install.

3. Close the CA Service Desk Manager GUI.
4. Restart the CA Service Desk Manager service.

Step 5b - Configure Web Screen Painter

You configure the Web Screen Painter now.

Follow these steps:

1. Open the Web Screen Painter and log in as a privileged user. For example, ServiceDesk
2. Open the Schema Designer and navigate to chg (Change Order).
3. Click **Add Column**.
4. Type usmrequestid in the pop up.
5. Click OK and set the Field Type to INTEGER.
6. Repeat steps 1-5 for usmrequestitemid, and then click the **Save** button.
7. Navigate to chgcat (Change Category).
8. Click **Add Column** and type implementors in the popup.
9. Click **OK** and set the Field Type to SREL. Then set the SRel Table to grp (Group Contacts).
10. Repeat steps 8 and 9 for change_manager.
11. Click Save and Publish. Then select **Yes** to the next 2 popups.
12. Stop the Service Desk services.
13. Run pdm_publish on the Service Desk server.
14. Start the Service Desk Manager services.
15. Open the Web Screen Painter and log in as a privileged user (e.g. ServiceDesk)
16. Click the **Open** icon.
17. Navigate to the detail_chgcat.htmpl file.
18. Click **Open**.

19. Right click the second row that has the field “Self-Service Include” and select Insert Row.
20. Right click the added row and select Insert Lookup.
21. Double-click the newly created field, click the Attribute row and select zchange_manager.
22. Repeat steps 20 and 21 for the Implementers field.
23. Click **File** and select **Save**.
24. Click **File** and select **Publish** and accept all confirmation popups.
25. Type pdm_webcache from a command prompt on the Service Desk server.

Step 5c - Add CA Service Desk Manager Groups

Use the following table to configure CA Service Desk Manager Groups:

Group Name	Group Manager	Group Members
Change Manager Group		*
IT Asset Management		*
Catalog Administrators		*

*Each group must contain at least one contact. Configure the contact as follows:

- Access type must be Analyst.
- Valid email address.
- Must exist in CA EEM.
- Belong to ITPAMUsers group within CA EEM.
- Select Email for all the Notification Methods.

Follow these steps to configure the CA Service Desk Manager Groups:

1. Open the CA Service Desk Manager and log in as a privileged user. For example, ServiceDesk
2. Click the **Administration** tab and navigate to **Security and Role Management**, **Groups**.
3. Click the **Create New** button.
4. Type Change Manager Group in the Group Name field.
5. Click the **Members** tab and click **Update Members** to add users to this group.
6. Repeat steps 3-5 for each group in the table above.

Step 5d - Add Status

Follow these steps:

1. Open CA Service Desk Manager and log in as a privileged user. For example, ServiceDesk
2. Click the **Administration** tab and navigate to **Service Desk**, **Change Orders**, **Status**.
3. Click the **Create New** button.
4. Type Complete in the Symbol field.
5. Type COMP in the Code field.
6. Click the **Save** button.
7. Click the **Update Transitions** button and check the following Statuses:

- Cancelled
- Closed
- Customer Hold
- Implementation in progress
- Vendor Hold
- Verification in progress

8. Click **Save**.

Step 5e - Add Change Categories

Use the following table to configure CA Service Desk Manager Change Categories:

Symbol	Code	Groups	Workflow SRF
SLCM.Hardware Filled From Inventory	SLCM.HWFFI	**Group**: IT Asset Manager	HWSW_FilledFromInventory
SLCM.Software Filled From Inventory	SLCM.SWFFI	**Group**: IT Asset Manager	HWSW_FilledFromInventory
SLCM.Service Publishing	SLCM.CN0	**Group**: IT Asset Manager **Change Manager**: Change Manager Group **Implementers**: Catalog Administrators	Service_Publishing

Follow these steps:

1. Open the CA Service Desk Manager and log in as a privileged user. For example, ServiceDesk
2. Click the **Administration** tab and navigate to **Service Desk**, **Change Orders**, **Categories**.
3. Click the **Create New** button.
4. Type SLCM.Hardware Filled From Inventory in the Symbol field.
5. Type SLCM.HWFFI in the Code field.
6. Select IT Asset Manager in the Group field.
7. Click the 2. Workflow tab.
8. Click the **Use CA IT PAM** button.
9. Choose **HWSW_FilledFromInventory Start Request** Form.
10. Click the **Save** button.
11. Repeat steps 3-10 for each change category in the table above.

Usage Data (Telemetry)

You can configure your product to collect and send telemetry data — product usage and system configuration data to Broadcom. Use the information on this page to learn how to send usage data to Broadcom. If you are licensed to use this product under a Portfolio License Agreement (PLA) subscription, you must configure it to collect and send product-specific usage data. If you are licensed to use this product under a standard license, you can consent to having this product collect and send usage data. By default, this product does not collect and send usage data.

For your product, the reporting options are:

What License Usage Metric Data we Collect?

The product license consumption metrics captured for CA Service Management will vary depending on the number of installation/upgrade instances, number of users, number of authentications, and so on.

Customers are required to upgrade product versions within a time frame as specified in their PLA agreement. Telemetry services will collect license consumption metrics and report it on a quarterly basis to Broadcom via Cloud Services. If product license consumption metrics cannot be sent due to Firewall or other similar considerations, manual reports must be collected and submitted every quarter.

CA Service Management product instance (s) will report on a daily basis the set of Stock Keeping Units (SKUs) that are deployed and the quantities that is used (For example, Max entry of web concurrent license count for a day in CA SDM). If there are exceptions for products that does not have a component running daily, then, weekly license usage metrics is considered for reporting.

> **NOTE**
>
> Data Transmission is securely done via SSL (port 443) and is only one-way. No Personally Identifiable Information (PII) is transferred at any time other than the product license usage metric.

All license usage metrics related to Production/Non-Production environments is tracked and considered while calculating the license consumption metric. Data that has already been reported for consumption metrics will not be deleted even if you decide to opt out of the PLA program at a later stage.

How License Metrics are Calculated?

- Products calculate monthly usage as follows:
  a. **CA Service Desk**: "**Concurrent Analysts**" is the metric considered for usage calculation. Daily max usage, i.e., "**no of Concurrent Analysts**" is written to the **ca_telemetry** table and subsequently picked by telemetry service to calculate and report the monthly max "**Concurrent Analysts**".
  b. **CA Service Catalog**: Number of users" that are provisioned or registered with CA Service Catalog is the metric considered for usage calculation. Daily maximum count of users is written to the **ca_telemetry** table and subsequently picked by telemetry service to calculate and report the monthly maximum count of users.
  c. **CA IT Asset Manager (ITAM)**: For CA ITAM, Telemetry Services collects the number of managed assets. CA ITAM collects the number of managed assets, and provides the data to "**ca_telemetry**" table once a day, which is subsequently collected via cloud services. Telemetry services collects product consumption metrics for CA ITAM on a daily basis from the **ca_owned_resource** table and calculates the monthly max number of qualified assets.
- For Products that have multiple licensing metrics, different SKUs will exist for each component. These products must report each of the metric types for monthly usage while using the different SKUs in the service invocation.
- For customers with multiple deployments, each instance of the product will report the monthly usage (using a unique "**instance_id**" in the service invocation).
- The monthly usage metric reported does not include any concept of pricing.
- You can also manually collect and submit product usage metrics on a quarterly basis.

Usage data includes actual usage of the product based on how it was licensed. We value your privacy and we do not collect any personally identifiable information (PII) data. Our privacy policy is located here.

How To Report Data Usage Automatically

CA Service Management Administrators must enable and configure Telemetry Services using the CA Service Management Common Installer for collecting product-specific consumption metrics. For more information, see Enable and Configure Telemetry Services.

You also have the option of viewing product usage metrics via the User Interface. Navigate to the following product UI location to view the Telemetry data for CA Service Management:

- **CA Service Desk Manager (SDM)**: Login to CA SDM and navigate to **Administration**, **System**, **Telemetry Data** to view your monthly usage and the date when the telemetry Data was collected.
- **CA Service Catalog**: Login to CA Service Catalog and navigate to **Administration**, **Configuration**, **Telemetry Data** to view your monthly usage and the date when the telemetry Data was collected.
- **CA IT Asset Manager (ITAM)**: Login to CA ITAM and navigate to **Administration**, **Telemetry**, **Telemetry Details** to view your monthly usage and the date when the telemetry Data was collected.

How to Report Usage Data Manually?

Use the manual collection method to send usage data on a Quarterly basis, if your organization decides to not enable the automatic usage reporting. Following are the recommended ways to collect and submit product usage metrics manually to Broadcom:

1. Collect product usage data from the **ca_telemetry table** and **al_cdb_configurationparameters** table (filter with productcode=20) as follows:
   a. **al_cdb_configurationparameters**: The **al_cdb_configurationparameters** table contains few encrypted passwords and proxy host. You must filter the configkey while fetching data. Sample al_cdb_configurationparameters Example:

```
/**** MS SQL Query for al_cdb_configurationparameters******/SELECT
 componentkey,machinename,configkey,configvalue
 FROM al_cdb_configurationparameters  where productcode = 20 and configkey in
('telemetry.chargeback.id',
'telemetry.domain.name',
'telemetry.environment.id',
'telemetry.pla.agreement.flag',
'telemetry.segment.push.flag',
'telemetry.site.id',
'telemetry.sku.sdm.description',
'telemetry.sku.sdm.id',
'telemetry.sku.slcm.standalone.description',
'telemetry.sku.slcm.standalone.id',
'telemetry.sku.itam.standalone.description',
'telemetry.sku.itam.standalone.id',
telemetry.period.days,
telemetry.product.name.sdm,
telemetry.product.name.slcm,
telemetry.product.name.itam,
telemetry.payload.type
)
```

NOTE
If the *telemetry.segment.push.flag* is set to false, product usage metrics is not reported to Broadcom via cloud services. Usage data metrics is captured and stored in the ca_telemetry table in MDB. You can collect this and report the usage metrics to Broadcom manually.

In the above sample table, the following fields need explanation:

Fields	Field Description
chargeback.id	Specifies charge back ID.
domain.name	Specifies domain name of customer.
environment.id	Each combo value has comma-separated values **<key>:<environment id>**. Display value of Environment ID is stored insrc/main/java/com/ca/smsi/installui/resources/ui/configui.properties file with pattern as telemetry.env.type.<Environment ID>=<Value>.
pla.agreement.flag	Specifies whether license agreement is of PLA type. This flag stores the PLA agreement flag as True, if the corresponding checkbox is enabled.
segment.push.flag	Note that by default this flag is set to **true**. If you want to disable it, navigate to where you have downloaded the CA Service Management Installer and update the **config\common_config.properties:** *telemetry.segment.push.flag=false* If this flag is set to false, usage metrics is reported via cloud services. However usage data metrics is captured and stored in the ca_telemetry table in MDB.
site.id	Specifies customer site ID
SKU iD	Specifies SKU ID of CA SDM, CA Service Catalog, and CA Asset Portfolio Management (ITAM).
period_days	The approximate number of days the metrics in this ping represent,1 for daily, 7 for weekly, 30 for monthly, currently we are getting monthly_usage.
telemetry.product.name.sdm	Specifies the CA Service Desk Manager product name.
telemetry.product.name.slcm	Specifies the CA Service Catalog product name.
telemetry.product.name.itam	Specifies the CA Asset Portfolio Management (ITAM) product name.
telemetry.payload.type	Specifies the type of the payload provided by CA Service Management.

b. **ca_telemetry**: Manually collect the complete database details from this MDB table and submit it to Broadcom on a quarterly basis.

2. Collect product usage data from the **telemetrydata.log** file on a quarterly basis:
   a. **CA Service Desk**: Collect the product usage data from *NX_ROOT\site\telemetrydata.log*
   b. **CA Service Catalog**: Collect the product usage data from *USM_HOME\view\telemetry\telemetrydata.log*
   c. **CA Asset Portfolio Management (CA ITAM)**: Collect the product usage data from ITAM_HOME\Export Service \ITAMTelemetry.log

Additionally, you can also download and install the Product Usage Reporter to submit usage data manually.

How product usage metrics is collected from CA Service Management?

CA Service Desk Manager, CA Service Catalog, and CA IT Asset Manager (ITAM)) will use the following key metrics to collect product usage metrics:

CA Service Desk Manager

The **webConcurrentTotalLicenseCt** KPI for CA SDM gives the concurrent analyst count for CA SDM along with the xFlow Interface, REST, and SOAP Web Services. The data is pushed to the **usp_kpi_data** table every one hour by default on a daily basis. Ensure that the **webConcurrentTotalLicenseCt** is not disabled. By default, it is in an enabled state.Telemetry services picks product consumption metrics on a daily basis and calculates the monthly max value to perform the following:

- **Write to Database** – ca_telemetry table
- **Write to File** - *NX_ROOT\site\telemetrydata.log*
- **Upload to Telemetry Cloud Services**The above is performed by the **sdmtelemetry** process on the Primary Server (Conventional) and on the Background server (Advanced Availability).

CA Service Catalog

Telemetry Services collects the number of employees that are provisioned or registered with CA Service Catalog and provides the monthly maximum count of the users to CA MDB '**ca_telemetry**' table on a daily basis. Telemetry services picks product consumption metrics on a daily basis from the **ca_contact table** and calculates the monthly max value to perform the following:

- **Write to Database** – ca_telemetry table
- **Write to File** - *USM_HOME\view\telemetry\telemetrydata.log*
- **Upload to Telemetry Cloud Services**

CA IT Asset Manager (ITAM)

For CA ITAM, Telemetry Services collects the number of qualified assets. Criteria for qualified assets (asset families which can be considered as qualified asset) must be manually entered by Administrators. CA ITAM collects the number of assets, provides the data to "**ca_telemetry**" table once a day, which is subsequently collected via cloud services.Telemetry services collects product consumption metrics for CA ITAM on a daily basis from the **ca_owned_resource** table and calculates the monthly max number of qualified assets to perform the following:

- **Write to Database** – **ca_telemetry** table
- **Write to File** - *<install_dir>\ITAM\Export Service\ITAMTelemetry.log*
- **Upload to Telemetry Cloud Services**

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